Notes on Quantum Field Theory

GPT 5.5 under the supervision of Xi Yin

Draft monograph

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Volume Dependency Guide

This monograph is organized by logical dependency rather than by historical order or course sequence. A topic first appears where its primitive objects are needed, but full development is placed where the required mathematical machinery is available. The list below is a reader-facing map for such deferrals.

Scattering before perturbative amplitudes
: Particles, Haag–Ruelle wave operators, LSZ, and analytic scattering theory are developed in Volume II. Perturbative Feynman rules for S-matrix elements are to be read only after that nonperturbative construction.

Gauge theory and effective actions
: Gauge-fixed perturbation theory, BRST, BV, lattice gauge regulators, anomalies, QCD, jets, energy correlators, and the hybrid Standard Model framework are developed in Volume IV. Wilsonian or 1PI claims for gauge theories in Volume III are therefore conditional on either finite-cutoff BV data or a lattice gauge-theory construction.

Conformal fixed points
: The core CFT volume is Volume V. Volume III discusses RG fixed points and critical phenomena only to the extent needed for renormalization and scaling limits. Representation theory, radial quantization, OPE convergence, crossing, conformal blocks, and light-ray observables belong to Volume V.

Two-dimensional exact QFT
: Factorized scattering, Yang–Baxter consistency, form factors, thermodynamic Bethe ansatz, exact finite-volume data, sine–Gordon, Gross–Neveu, sigma-model families, and bridges to nonintegrable two-dimensional theories are developed in Volume VI.

Supersymmetry
: Supersymmetry algebras, particle versus field multiplets, superspace, supersymmetric gauge theories, holomorphy, Wilsonian scheme questions, Seiberg–Witten theory, localization caveats, and supersymmetric examples in various dimensions are developed in Volume VII.

Topological and cohomological theories
: Metric independence, bordism functoriality, extended TQFT, BF theory, Chern–Simons theory, cohomological field theory, topological sigma models, twists of supersymmetric theories, and Donaldson–Seiberg–Witten comparison problems are developed in Volume VIII.

Global structure and generalized symmetry
: Global forms of gauge groups, higher-form symmetries, line and surface operators, con-

finement diagnostics, discrete theta terms, anomaly inflow, defects, categorical symmetry, noninvertible defects, gauging, and SymTFT viewpoints are developed in Volume IX.

Thermal and hydrodynamic QFT
KMS states, finite-temperature path integrals, Schwinger–Keldysh theory, spectral functions, Kubo formulae, hydrodynamic constitutive relations, thermal gauge theory, kinetic theory, anomalous transport, nonequilibrium states, and long-time tails are developed in Volume X.

Constructive, lattice, and numerical routes
Constructive scalar models, lattice reflection positivity, continuum scaling windows, Wilson lattice gauge theory, Monte Carlo, sign problems, rigorous RG, lattice-to-continuum local QFT, stochastic quantization, Hamiltonian truncation, DLCQ, and lattice fermion/chiral-symmetry questions are developed in Volume XI.

Curved backgrounds
Locally covariant QFT, Hadamard states, point-splitting stress tensors, trace anomalies, Unruh and Hawking effects, index-theoretic anomalies, eta-invariants, cosmological spacetimes, microlocal spectrum conditions, perturbative algebraic QFT on curved backgrounds, and semiclassical backreaction are developed in Volume XII.

When a chapter uses a result from a later volume, the local statement should identify the object being imported and its theorem status. This guide only locates the development; it does not prove a claim by itself.

Source Assembly Map

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Part I

Foundations of Local Quantum Field Theory

Chapter 1

Starting Data for Local Quantum Field Theory

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

1.1 Spacetime And Operator Conventions

Throughout the opening volume, D denotes the spacetime dimension. The symbol $\mathbb{M}^D$ denotes D-dimensional Minkowski affine space. After a choice of inertial coordinates, $\mathbb{M}^D$ is identified with $\mathbb{R}^D$. The metric tensor has components

$$\eta_{\mu\nu} = \mathrm{diag}(-1, 1, \ldots, 1),$$

and units are chosen so that $\hbar = c = 1$. In the quantum-mechanical path-integral derivations $\hbar$ is displayed until the semiclassical weighting and Wick rotation have been derived; after that point $\hbar = 1$ is restored unless a dimensional check explicitly reintroduces it. The vacuum vector is denoted by Ω; the same printed letter may occur with subscripts for reconstructed cyclic vectors, but unsubscripted Ω always denotes the vacuum of the theory under discussion. A vector $x \in \mathbb{M}^D$ is spacelike when

$$x^2 := \eta_{\mu\nu} x^\mu x^\nu > 0.$$

A test function is a smooth compactly supported complex-valued function on $\mathbb{M}^D$, an element of $C_c^\infty(\mathbb{M}^D)$, unless another test-function space is explicitly declared. A pointlike quantum field

$\widehat{\Phi}_A(x)$ is an operator-valued distribution: its primary operator expression is the smeared field

$$\widehat{\Phi}_A(f) = \int_{\mathbb{M}^D} \mathrm{d}^D x\, f(x)\widehat{\Phi}_A(x), \qquad f \in C_c^\infty(\mathbb{M}^D).$$

The integral sign is distributional notation for evaluating the field on the test function f. When $\widehat{\Phi}_A(f)$ is unbounded, its common dense domain is part of the field framework being used. The symbol $\mathcal{B}(\mathcal{H})$ denotes the algebra of bounded linear operators on a Hilbert space $\mathcal{H}$, and $\mathcal{U}(\mathcal{H})$ denotes its unitary group. The label A denotes a component in a finite-dimensional representation space for Lorentz and internal indices, or a composite-operator label, as specified in the chapter where the field is introduced.

This notation distinguishes mathematical object classes. In Hilbert-space formulas a hat on a field symbol denotes an operator-valued distribution: $\widehat{\phi}$ for a scalar field and $\widehat{\Phi}_A$ for a multiplet. Unhatted fields such as ϕ, φ, or Φ_E denote classical variables, path-integral variables, or Euclidean Schwinger/random fields unless a chapter explicitly declares a different use. Fourier transforms of such variables are written with a tilde, for example $\widetilde{\phi}(k)$.

The test-function space is part of the framework data. Locality statements use C_c^∞ because support controls localization in bounded spacetime regions. Spectral, Wightman, and Euclidean reconstruction statements often use the Schwartz space $\mathcal{S}$, because temperedness and Fourier analysis are then built into the topology. A chapter may use either declared nuclear test-function space, provided all field and correlation distributions are continuous for that topology. A tempered field on $\mathcal{S}$ restricts to compactly supported test functions for local statements; a compactly supported-test distribution is treated as tempered only after a stated growth or continuity hypothesis supplies a continuous extension to $\mathcal{S}$.

Euclidean correlation distributions are called Schwinger functions when they are being used as reconstruction data and are then denoted S_n. In path-integral perturbation chapters the notation $G_{E,n}$ denotes the same kind of Euclidean n-point distribution before the OS reconstruction hypotheses have been imposed; G_E without an n-index is reserved for a two-point kernel or its Fourier transform.

The connected proper orthochronous Lorentz group is denoted $SO^+(1, D-1)$, and the connected Poincare group is

$$\mathcal{P}_+^\uparrow = \mathbb{R}^D \rtimes SO^+(1, D-1).$$

Two subsets $\mathcal{O}_1, \mathcal{O}_2 \subset \mathbb{M}^D$ are spacelike separated when $(x-y)^2 > 0$ for every $x \in \mathcal{O}_1$ and $y \in \mathcal{O}_2$.

1.2 Local Continuum Theories And Effective Presentations

The organizing structure of quantum field theory is quantum dynamics together with spacetime localization. The word "localization" has a precise content: observables, or field coordinates for observables, are attached to spacetime regions or test functions; spacetime symmetries move these localized objects covariantly; and observables localized in spacelike separated regions obey the appropriate commutativity law. The exact algebraic, field-theoretic, or Euclidean form of this structure must always be specified.

Two different mathematical statuses will appear throughout the monograph. They share locality, but they are not the same assertion.

Definition 1.1 (Continuum local quantum field theory)**.** A continuum local quantum field theory is a local quantum theory whose declared correlation functions, local algebras, or Hilbert-space operators exist after the ultraviolet regulator has been removed, in a stated topology and within a stated spacetime and state-sector framework. In a Minkowski vacuum presentation this includes Poincare symmetry, a spectrum condition, local observables or local field distributions, and microcausality. The phrase "infinitely many degrees of freedom" will be used in the operational sense that the local data are indexed by infinite-dimensional test-function or region data and are not specified by a finite list of canonical coordinates.

Definition 1.2 (Effective field theory presentation)**.** An effective field theory presentation is a regulated or renormalized local prescription for computing observables in a specified range of length, energy, and external-state scales. Its defining data include the fields or local variables, the symmetries, the regulator or renormalization prescription, the expansion parameters, and the class of observables for which the expansion is claimed. When it is perturbative, it defines an asymptotic expansion of Green functions, and of scattering quantities only after the nonperturbative scattering framework and LSZ interpretation have been supplied.

Power-counting renormalizability is a property of an effective presentation near a chosen fixed point or scaling regime: only finitely many local operators are needed as independent counterterms at a fixed order in the declared expansion. Existence of an ultraviolet continuum theory is a separate question, requiring constructive Wilsonian, lattice, or comparable nonperturbative input. Power-counting nonrenormalizable presentations are also local effective theories when the infinitely many allowed operators are organized by scale and symmetry.

Remark 1.3 (Conformal perturbation as a controlled construction method)*.* A deformation of a conformal field theory is a precise construction problem only after the conformal theory has already been supplied as local QFT data. The input consists of a stress tensor, a Hilbert space or Euclidean reconstruction datum, local operators O_A with specified normalization and contact-term convention, and a regulator or subtraction prescription for insertions of

$$\exp\left(-\sum_A g^A \int \mathrm{d}^D x\, O_A(x)\right).$$

When these data exist and the deformation parameters are controlled, conformal perturbation theory produces a local expansion of deformed correlation functions and beta functions. Its hypotheses are therefore:

1. an already constructed or otherwise supplied fixed point;
2. identified operator insertions, including their coincident-point renormalization;
3. a domain in coupling space where the expansion or nonperturbative deformation is defined;
4. a reconstruction or regulator argument connecting the deformed correlators to local QFT data.

Without these inputs, the phrase "perturb a UV CFT" is only a description of a possible route. Asymptotically free gauge theories, constructive scalar models, lattice scaling limits, stochastic-quantization constructions, and effective presentations of the Standard Model may each possess local QFT content in their own frameworks; none is thereby defined by conformal perturbation theory. This monograph uses CFT perturbations when their operator-algebra and regulator hypotheses are declared, for example in Wilson–Fisher and two-dimensional integrable-flow problems, and does not use them as the definition of quantum field theory.

The opening chapter will use two mathematical statuses: continuum local QFT data after regulator removal, and effective presentations valid in a declared scale window. Definitions 1.1 and 1.2 are the operative definitions; comparison maps among Wightman, OS, local-net, and regulated presentations are registered below only after their hypotheses have been stated.

1.3 Opening Framework

The first construction takes place in a flat-spacetime vacuum sector. The following definition gives the data used in the opening volume.

Definition 1.4 (Vacuum Minkowski local quantum theory)**.** In the opening framework, a vacuum Minkowski local quantum theory consists of:

1. a complex Hilbert space $\mathcal{H}$;
2. a spacetime symmetry convention. Let G_{Lor} be $SO^+(1, D-1)$, or the appropriate double cover when spinor fields are included, and let $\pi : G_{\mathrm{Lor}} \to SO^+(1, D-1)$ be the covering projection. The connected Poincare group used here is
$$\mathcal{P} = \mathbb{R}^D \rtimes G_{\mathrm{Lor}}, \qquad (a, \Lambda)(b, \Gamma) = (a + \pi(\Lambda)b, \Lambda\Gamma).$$
It acts on $\mathbb{M}^D$ from the left by
$$gx = \pi(\Lambda)x + a, \qquad g = (a, \Lambda) \in \mathcal{P},$$
and $g\mathcal{O} := \{gx : x \in \mathcal{O}\}$ for a region $\mathcal{O} \subset \mathbb{M}^D$;
3. a strongly continuous unitary representation
$$U : \mathcal{P} \to \mathcal{U}(\mathcal{H}), \qquad g \mapsto U(g),$$
as a left representation, $U(g_1 g_2) = U(g_1)U(g_2)$, where $\mathcal{U}(\mathcal{H})$ is the unitary group of $\mathcal{H}$;
4. self-adjoint translation generators P^μ, equivalently $P_\mu = \eta_{\mu\nu}P^\nu$, defined by
$$U(a, 1) = \exp(\mathrm{i}a_\mu P^\mu) = \exp(\mathrm{i}a^\mu P_\mu).$$
Thus a positive time translation gives $\exp(-\mathrm{i}a^0 P^0)$, with P^0 the Hamiltonian. Let E_P be the joint projection-valued spectral measure of the contravariant energy-momentum $P = (P^0, \ldots, P^{D-1})$. The spectrum condition is
$$\operatorname{supp} E_P \subset \overline{V}_+ = \{p \in \mathbb{R}^D : p^0 \geq 0,\ \eta_{\mu\nu}p^\mu p^\nu \leq 0\};$$
5. a Poincare-invariant unit vector $\Omega \in \mathcal{H}$,
$$U(g)\Omega = \Omega, \qquad g \in \mathcal{P}.$$
In this pure-vacuum-sector framework, vacuum uniqueness is part of the datum: the zero-momentum spectral projection is one-dimensional,
$$E_P(\{0\})\mathcal{H} = \mathbb{C}\Omega;$$

6. an assignment
$$\mathcal{O} \longmapsto \mathcal{A}(\mathcal{O}) \subset \mathcal{B}(\mathcal{H})$$
from bounded open regions $\mathcal{O} \subset \mathbb{M}^D$ to unital concrete $*$-subalgebras of the bounded operators on $\mathcal{H}$, satisfying covariance, isotony, and locality:
$$U(g)^{-1}\mathcal{A}(\mathcal{O})U(g) = \mathcal{A}(g\mathcal{O}), \qquad g \in \mathcal{P},$$
$$\mathcal{O}_1 \subset \mathcal{O}_2 \quad \Longrightarrow \quad \mathcal{A}(\mathcal{O}_1) \subset \mathcal{A}(\mathcal{O}_2),$$
and
$$[A, B] = 0 \quad \text{for } A \in \mathcal{A}(\mathcal{O}_1),\ B \in \mathcal{A}(\mathcal{O}_2)$$
whenever $\mathcal{O}_1$ and $\mathcal{O}_2$ are spacelike separated, with the graded commutator
$$[A, B]_{\mathrm{gr}} = AB - (-1)^{|A||B|}BA$$
used for homogeneous odd operators after a fermion parity grading and the parities $|A|, |B| \in \mathbb{Z}/2\mathbb{Z}$ have been specified;

7. vacuum cyclicity for the bounded local observable content. Let $\mathcal{A}_{\mathrm{loc}}^{\mathrm{alg}}$ denote the unital $*$-algebra generated in $\mathcal{B}(\mathcal{H})$ by $\bigcup_{\mathcal{O}} \mathcal{A}(\mathcal{O})$, where the union is over bounded open regions. Then
$$\overline{\mathcal{A}_{\mathrm{loc}}^{\mathrm{alg}}\Omega}^{\|\cdot\|_{\mathcal{H}}} = \mathcal{H};$$

8. when unbounded field coordinates are part of the presentation, a specified field-domain datum
$$(\mathcal{D}_\Phi, \{\widehat{\Phi}_A\}_{A \in I}).$$
Here I is the field-label set, $\mathcal{D}_\Phi \subset \mathcal{H}$ is a dense linear subspace with $\Omega \in \mathcal{D}_\Phi$, $U(g)\mathcal{D}_\Phi \subset \mathcal{D}_\Phi$ for all $g \in \mathcal{P}$, and each smeared field is a linear operator
$$\widehat{\Phi}_A(f) : \mathcal{D}_\Phi \to \mathcal{D}_\Phi, \qquad f \in \mathcal{T}_A,$$
where $\mathcal{T}_A$ is the declared test-function space for that field component. Finite products of such smeared fields are defined on $\mathcal{D}_\Phi$. If adjoints are used, then $\mathcal{D}_\Phi \subset \mathrm{Dom}(\widehat{\Phi}_A(f)^*)$ and the adjoint fields preserve $\mathcal{D}_\Phi$. For every $\psi, \chi \in \mathcal{D}_\Phi$, the matrix elements of finite smeared products are continuous multilinear functionals of their test functions. Field covariance, field locality, and any relation between $\widehat{\Phi}_A(f)$ with $\operatorname{supp} f \subset \mathcal{O}$ and $\mathcal{A}(\mathcal{O})$ are imposed on this same domain or by a specified affiliation to the chosen local observable algebra.

The uniqueness clause is a spectral assumption on the chosen representation. It identifies the projection onto translation-invariant vectors with the rank-one projection $|\Omega\rangle\langle\Omega|$. Consequently, after subtracting the vacuum expectation value of a scalar local field, the vector $\widehat{\phi}(f)\Omega$ has no residual $p = 0$ component. This is the assumption used in the Källén–Lehmann analysis when the $(\Omega, \cdot)$ projection removes the vacuum atom before the invariant-mass decomposition is formed. If a theory has several infinite-volume vacua or a mixed representation of phases, one must first choose a pure vacuum sector or else keep the additional $p = 0$ spectral summands explicitly; they sit at $\mu^2 = 0$ as zero-momentum atoms, distinct from massless one-particle states whose momenta lie on the nonzero light cone.

The covariance convention has the following infinitesimal form. With $U(a,1) = \exp(\mathrm{i}a^\mu P_\mu) = \exp(\mathrm{i}a_\mu P^\mu)$, a coordinate-labelled scalar field obeys

$$U(a,1)^{-1}\widehat{\phi}(x)U(a,1) = \widehat{\phi}(x+a), \qquad [P_\mu, \widehat{\phi}(x)] = \mathrm{i}\,\partial_\mu \widehat{\phi}(x)$$

as an identity of distributions on a common invariant domain. Equivalently, conjugation by $U(a,1)$ shifts the coordinate label by $-a$. The local-net covariance law above uses the same passive coordinate convention: conjugating an observable by $U(g)^{-1}$, with $g = (a, \Lambda)$, sends its localization region to $g\mathcal{O} = \pi(\Lambda)\mathcal{O} + a$.

Vacuum projection and translation convention. The vacuum-uniqueness clause in Definition 1.4 says $E_P(\{0\})\mathcal{H} = \mathbb{C}\Omega$, with $\|\Omega\| = 1$. Hence the projection onto translation-invariant vectors is $|\Omega\rangle\langle\Omega|$. If a scalar field satisfies

$$U(a,1)^{-1}\widehat{\phi}(x)U(a,1) = \widehat{\phi}(x+a),$$

then differentiating $e^{-\mathrm{i}a^\mu P_\mu}\widehat{\phi}(x)e^{\mathrm{i}a^\mu P_\mu} = \widehat{\phi}(x+a)$ at $a = 0$, as an identity of smeared distributions on the common invariant domain, gives

$$[P_\mu, \widehat{\phi}(x)] = \mathrm{i}\,\partial_\mu \widehat{\phi}(x).$$

In this opening convention $\mathcal{A}(\mathcal{O})$ is a concrete algebra of bounded observables on the chosen Hilbert space and is not assumed to be norm closed or weakly closed unless this is stated. If a C^*-algebra or von Neumann algebra is needed, the relevant closure or bicommutant is included in the datum or constructed explicitly. This convention is sufficient for the first dependency layer: localization, covariance, the spectrum condition, vacuum matrix elements, and comparison with field coordinates.

This framework describes a vacuum representation on flat spacetime. Thermal states, boundary conditions, curved spacetimes, and non-vacuum superselection sectors are introduced by changing the stated data: the state space, spacetime category, covariance group, region assignment, or representation sector. Existing Wightman, Euclidean, algebraic, constructive, perturbative, and factorization frameworks will be introduced as precise frameworks with their own domains, beginning with the Wightman and algebraic formulations.

Open Problem 1.5 (Four-dimensional constructive local QFT). A constructive solution of a four-dimensional local quantum field theory is a construction of the correlation functions, Hilbert space, local fields or observable net, symmetry representation, and positivity structure in a topology strong enough to verify the stated framework. In a Wightman presentation this means tempered vacuum distributions satisfying covariance, the spectrum condition, locality, positivity, and a reconstruction theorem. In an Osterwalder–Schrader presentation it means Euclidean Schwinger distributions with reflection positivity, Euclidean covariance, symmetry, clustering or the declared vacuum-sector condition, and the regularity hypotheses needed for OS reconstruction. In either presentation, an interacting model means that the truncated correlation functions and local operator content are not those of a Gaussian or generalized-free field alone, and that the local stress-tensor, charge, or gauge-invariant observables required in the later physical discussion are part of the constructed data.

Completed constructive examples occupy special regimes: $P(\phi)_2$ models, including ϕ^4_2, and massive ϕ^4_3 give non-Gaussian scalar models after their model-specific Wick ordering and local renormalizations; other superrenormalizable low-dimensional scalar–fermion models are constructed under similarly explicit hypotheses; and two-dimensional Yang–Mills has rigorous gauge-theory constructions through holonomy and heat-kernel measures. Four-dimensional physics presents a different constructive problem. Free and generalized-free Wightman fields are available, but the usual interacting scalar and gauge theories require separate continuum-limit theorems. Standard positive $\lambda\phi^4_D$ scaling-limit families in $D \geq 4$ are constrained by triviality theorems, with the four-dimensional marginal case governed by the Aizenman–Duminil-Copin triviality theorem for the nearest-neighbor Ising/ϕ^4_4 scaling problem. Four-dimensional pure Yang–Mills, QCD, and other asymptotically free gauge theories still require a construction of the continuum gauge-invariant Schwinger functions or Wightman theory with the desired spectral properties. Perturbative renormalizability and asymptotic freedom are mechanisms inside a specified expansion; they are not by themselves constructive existence theorems for the nonperturbative continuum theory.

1.4 Framework Presentations And Comparison Maps

The opening data admit several mathematically precise presentations. Each presentation emphasizes a different class of objects, and each comparison map carries hypotheses.

Definition 1.6 (Wightman field presentation)**.** A Wightman field presentation consists of Hilbert-space data

$$(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_A\}_{A\in I})$$

where I is an index set for fields, $\mathcal{D} \subset \mathcal{H}$ is a dense common invariant domain containing Ω, and each $\widehat{\Phi}_A$ is an operator-valued distribution on a declared invariant test-function space $\mathcal{T}$, with common domain $\mathcal{D}$. In the classical Wightman case $\mathcal{T} = \mathcal{S}(\mathbb{M}^D)$, so the vacuum distributions below are tempered. The domain assumptions are

$$\Omega \in \mathcal{D}, \qquad U(g)\mathcal{D} \subset \mathcal{D} \quad (g \in \mathcal{P}), \qquad \widehat{\Phi}_A(f)\mathcal{D} \subset \mathcal{D} \quad (A \in I,\ f \in \mathcal{T}),$$

and, for every $n \geq 1$,

$$\widehat{\Phi}_{A_1}(f_1) \cdots \widehat{\Phi}_{A_n}(f_n)\mathcal{D} \subset \mathcal{D}, \qquad A_i \in I, \quad f_i \in \mathcal{T}.$$

Matrix elements of finite products are required to be continuous multilinear functionals of the test functions:

$$(f_1, \dots, f_n) \longmapsto \left\langle \psi, \widehat{\Phi}_{A_1}(f_1) \cdots \widehat{\Phi}_{A_n}(f_n)\chi \right\rangle \in \mathbb{C}, \qquad \psi, \chi \in \mathcal{D}.$$

Let $g = (a, \Lambda) \in \mathcal{P}$, $gx = \pi(\Lambda)x + a$, and $f_g(y) = f(g^{-1}y)$. If the fields carry a finite-dimensional Lorentz representation R of G_{Lor}, covariance is the smeared identity

$$U(g)^{-1}\widehat{\Phi}_A(f)U(g) = R(\Lambda)_A{}^B\, \widehat{\Phi}_B(f_g),$$

with summation over repeated component labels. In point-field notation this is

$$U(g)^{-1}\widehat{\Phi}_A(x)U(g) = R(\Lambda)_A{}^B\widehat{\Phi}_B(gx).$$

The representation U satisfies the spectrum condition of Definition 1.4: if E_P is the joint spectral measure of the translation generators, then

$$\operatorname{supp} E_P \subset \overline{V}_+.$$

Locality is imposed after smearing:

$$[\widehat{\Phi}_A(f), \widehat{\Phi}_B(h)]_{\mathrm{gr}} = 0$$

whenever $\operatorname{supp} f$ and $\operatorname{supp} h$ are spacelike separated, with the ordinary commutator in the purely bosonic case.

Its primary correlation data are the vacuum distributions whose distribution kernels are written

$$W_{A_1\cdots A_n}(x_1,\ldots,x_n) = \langle\Omega|\widehat{\Phi}_{A_1}(x_1)\cdots\widehat{\Phi}_{A_n}(x_n)|\Omega\rangle.$$

Equivalently,

$$W_{A_1\cdots A_n}(f_1,\ldots,f_n) = \left\langle\Omega, \widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)\Omega\right\rangle.$$

Translation invariance gives a distribution in difference variables. With

$$\xi_j = x_{j+1} - x_j, \qquad \mathcal{W}_{A_1\cdots A_n}(\xi_1,\ldots,\xi_{n-1}) = W_{A_1\cdots A_n}(0,\xi_1,\xi_1+\xi_2,\ldots,\xi_1+\cdots+\xi_{n-1}),$$

the Wightman spectral condition is the support equation

$$\operatorname{supp}\widehat{\mathcal{W}}_{A_1\cdots A_n} \subset \overline{V}_+^{\,n-1}.$$

Positivity of the hierarchy means the following. For every finite sequence of test tensors in the declared test-function space

$$F = \{F^{(n)}_{A_1\cdots A_n}\}_{0\le n\le N}, \qquad F^{(0)} \in \mathbb{C},$$

define

$$\Psi_F = F^{(0)}\Omega + \sum_{n=1}^{N}\sum_{A_1,\ldots,A_n}\int F^{(n)}_{A_1\cdots A_n}(x_1,\ldots,x_n)\widehat{\Phi}_{A_1}(x_1)\cdots\widehat{\Phi}_{A_n}(x_n)\Omega\,\mathrm{d}^D x_1\cdots\mathrm{d}^D x_n.$$

The condition is $\langle\Psi_F,\Psi_F\rangle \ge 0$, with the inner product evaluated by the distributions W_n:

$$\langle\Psi_F,\Psi_F\rangle \ge 0.$$

Vacuum cyclicity is the density equation

$$\overline{\operatorname{span}\{\widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)\Omega : n\ge 0,\ f_i\in\mathcal{T},\ A_i\in I\}}^{\|\cdot\|_{\mathcal{H}}} = \mathcal{H}.$$

Definition 1.7 (Local-net presentation)**.** A local-net presentation consists of a specified class $\mathcal{K}(\mathbb{M}^D)$ of open spacetime regions, a quasilocal $*$-algebra or C^*-algebra $\mathcal{A}_{\mathrm{qloc}}$, and an assignment

$$\mathcal{O} \longmapsto \mathcal{A}(\mathcal{O}) \subset \mathcal{A}_{\mathrm{qloc}}$$

for $\mathcal{O}\in\mathcal{K}(\mathbb{M}^D)$. The assignment satisfies isotony, locality, covariance under the declared spacetime symmetry group, and any additional structural conditions imposed in the problem under study, such as additivity, Haag duality, or the time-slice property. A state is a normalized positive linear functional $\omega : \mathcal{A}_{\mathrm{qloc}} \to \mathbb{C}$; it gives a Hilbert-space representation by the GNS construction.

Definition 1.8 (Euclidean correlation presentation)**.** Let $E = \mathbb{R}^D$ be Euclidean space and let V be the finite-dimensional complex field-label space, with homogeneous basis labels a and parity $|a| \in \mathbb{Z}/2\mathbb{Z}$ when a grading is present. A Euclidean correlation presentation consists of a test-function space $\mathcal{T}_E$, usually $\mathcal{S}(E)$ in the tempered OS framework, an adjoint antilinear involution $a \mapsto a^\dagger$ on labels, a Euclidean rotation representation R_E on V, and a hierarchy

$$S_n \in \mathcal{T}'_{E,\circ}(E^n; (V^*)^{\otimes n}), \qquad n \geq 0, \qquad S_0 = 1.$$

Here $\mathcal{T}_{E,\circ}(E^n; V^{\otimes n})$ denotes the corresponding $V^{\otimes n}$-valued zero-diagonal test-function space on E^n: test functions vanish with all derivatives on every partial diagonal $x_i = x_j$. The prime denotes its continuous dual, and $S_n(F)$ denotes the distributional pairing. A contact-term extension of S_n from $\mathcal{T}_{E,\circ}$ to the full $\mathcal{T}_E$ is additional renormalization data and is not required for OS reconstruction. The component notation

$$S_{a_1 \cdots a_n}(x_1, \ldots, x_n)$$

is kernel notation for the distributional pairing of S_n with $V^{\otimes n}$-valued test functions. If $g = (b, r) \in \mathbb{R}^D \rtimes O(D)$, $gx = rx + b$, define the action on test tensors by

$$(gF)^{a_1 \cdots a_n}(x_1, \ldots, x_n) = R_E(r)^{a_1}{}_{b_1} \cdots R_E(r)^{a_n}{}_{b_n} \, F^{b_1 \cdots b_n}(g^{-1}x_1, \ldots, g^{-1}x_n).$$

Euclidean covariance is the distributional identity

$$S_n(gF) = S_n(F), \qquad g \in \mathbb{R}^D \rtimes O(D).$$

For a permutation $\sigma \in \mathfrak{S}_n$, graded permutation symmetry is

$$S_{a_{\sigma(1)} \cdots a_{\sigma(n)}}(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = \epsilon(\sigma; a_1, \ldots, a_n) \, S_{a_1 \cdots a_n}(x_1, \ldots, x_n),$$

as an equality of distributions, where

$$\epsilon(\sigma; a_1, \ldots, a_n) = \prod_{\substack{i<j \\ \sigma(i)>\sigma(j)}} (-1)^{|a_i||a_j|}.$$

In the ungraded case all parities are zero. Hermiticity is

$$\overline{S_{a_1 \cdots a_n}(x_1, \ldots, x_n)} = S_{a_n^\dagger \cdots a_1^\dagger}(x_n, \ldots, x_1),$$

again as a distributional statement.

Choose Euclidean coordinates $x = (\tau, \mathbf{x})$ and let

$$\theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x})$$

be time reflection. Set $E_+ = \{(\tau, \mathbf{x}) \in E : \tau > 0\}$. Reflection positivity is the requirement that for every finite sequence of test tensors

$$F = \{F^{(n)}_{a_1 \cdots a_n}\}_{0 \leq n \leq N}, \qquad F^{(n)} \in \mathcal{T}_E(E^n; V^{\otimes n}), \qquad \operatorname{supp} F^{(n)} \subset E^n_+,$$

the quadratic form

$$\langle F,F\rangle_{\mathrm{OS}} = \sum_{m,n=0}^{N} \sum_{\substack{a_1,\ldots,a_m\\ b_1,\ldots,b_n}} \int \overline{F^{(m)}_{a_1\cdots a_m}(x_1,\ldots,x_m)} F^{(n)}_{b_1\cdots b_n}(y_1,\ldots,y_n)$$
$$\times S_{a_m^\dagger\cdots a_1^\dagger b_1\cdots b_n}(\theta x_m,\ldots,\theta x_1,y_1,\ldots,y_n) \prod_{i=1}^{m} \mathrm{d}^D x_i \prod_{j=1}^{n} \mathrm{d}^D y_j$$

is nonnegative. The convention $S_0 = 1$ is used in the terms with $m = 0$ or $n = 0$.

The regularity and growth condition is also part of the data. The corrected OS-II reconstruction theorem requires more than ordinary temperedness at each fixed n: one needs uniform control on the growth of the distributional order with n. In a full-Schwartz presentation this includes constants $A, B, \gamma > 0$ and a seminorm order $N(n)$ growing at most linearly in n, such that every $F \in \mathcal{S}_\circ(E^n; V^{\otimes n})$ obeys a Schwartz-seminorm bound, where $\mathcal{S}_\circ$ denotes the Schwartz subspace of functions flat on the partial diagonals, using any fixed norm on the finite-dimensional space $V^{\otimes n}$:

$$|S_n(F)| \le A\,B^n (n!)^\gamma \sum_{|\alpha|,|\beta|\le N(n)} \sup_{X\in E^n} \left| X^\beta \partial^\alpha F(X) \right|.$$

If a different nuclear test-function space is chosen, the corresponding continuity seminorms and the growth assumptions needed for OS reconstruction must be stated with the presentation.

Under the full Osterwalder–Schrader hypotheses, quotienting by the null space of this form and completing reconstructs the Lorentzian Hilbert space; analytic continuation then gives Wightman boundary values.

The comparison from fields to a local net is obtained, when the needed operator-domain and affiliation statements hold, by assigning to a region $\mathcal{O}$ the von Neumann algebra generated by bounded functions, resolvents, or other bounded functional-calculus representatives of smeared fields $\widehat{\Phi}_A(f)$ with $\operatorname{supp} f \subset \mathcal{O}$. The comparison from a local net to a Hilbert-space theory is the GNS construction applied to a state, followed by the representation of the local algebras on the GNS Hilbert space. Recovering point-field coordinates from a represented net requires additional phase-space, regularity, or affiliated-field hypotheses; it is not a formal consequence of the net axioms alone. The comparison from Euclidean correlations to Lorentzian fields is the corrected OS-II reconstruction theorem followed by analytic continuation to Wightman boundary values. Conversely, Wightman fields give Euclidean Schwinger functions by tube analyticity and restriction to noncoincident imaginary-time configurations.

The theorem-level comparison register is the actual content of the comparison. Each entry names the source presentation, the target presentation, and the hypotheses that are not automatic from notation alone:

- **Wightman hierarchy to Wightman fields.** Theorem 2.4 reconstructs the cyclic field presentation from a positive Wightman hierarchy with covariance, spectrum, Hermiticity, locality, and tempered continuity.
- **Wightman fields to AQFT.** Theorem 3.10 constructs a represented local net once the relevant smeared Hermitian fields have essentially self-adjoint closures and strong locality of their spectral projections. The comparison is an observable-net identification only after the

additional datum of Definition 3.13 is supplied: the generated coordinate net must be additive in bounded functional calculus, and it must equal the represented observable net, either directly or through a specified fixed-point or gauge-invariant subnet operation.

- **AQFT to Wightman fields.** Remark 3.16 records the converse comparison: a represented positive-energy local net gives Wightman field coordinates only after affiliated operator-valued tempered distributions on a common invariant domain have been supplied.
- **OS data to Wightman fields.** Theorem 12.16 reconstructs a Lorentzian presentation from OS-admissible Euclidean data. Its analytic input is the OS-II positive-time semigroup and linear-growth package, not reflection positivity alone.
- **Path integral to Wightman fields.** The path-integral-to-Wightman paragraph after OS reconstruction records the composition: a regulated path integral determines a Wightman presentation only after its continuum Schwinger limits exist and satisfy the OS hypotheses.
- **Wightman fields to OS data.** Tube analyticity from the Wightman spectrum condition gives Euclidean Schwinger functions by restriction to noncoincident imaginary-time configurations; this is the reverse Wick-rotation direction discussed in Chapter 2.

These definitions make the foundational objects explicit. A later calculation may use fields, path integrals, local algebras, or Euclidean correlators; the framework in which the calculation lives and the comparison maps needed to interpret it must be stated.

1.5 Status of Axiom Systems and Constructive Examples

The presentations above have a precise logical role. They are theorem-bearing frameworks for local quantum field theory: each one specifies objects, topologies, positivity conditions, covariance conditions, and reconstruction or comparison theorems. A result proved in one framework has the hypotheses of that framework as part of its statement. Thus a theorem proved for Wightman fields is a theorem for theories admitting the declared Wightman presentation; a theorem proved for a represented local net is a theorem under the local-net hypotheses and any additional phase-space or modular hypotheses used in the proof; and a theorem proved from OS data is a theorem after the OS-II regularity and analytic-continuation input has been verified.

The physical expectation is broader and more constructive. A Poincare-invariant ultraviolet-complete local QFT in a vacuum sector is expected, when it has point-field coordinates of the appropriate regularity, to produce Wightman distributions; when it has Euclidean reflection-positive correlators with the required growth, to produce OS data; and when it is organized by bounded localized observables, to produce a Haag–Kastler type net. These are expected properties of a completed local quantum theory in the corresponding sector. Comparisons among the frameworks require explicit hypotheses: charged superselection sectors, gauge-invariant observable algebras, fermionic field variables in path integrals, curved-background formulations, and functorial or extended geometric structures may be primary in one presentation and derived or hidden in another.

The path-integral aspiration can be stated as a gluing problem. One seeks an assignment of amplitudes and correlation distributions to a declared category of backgrounds with sources,

$$(M, g, \text{bundles}, J, \dots) \longmapsto Z[M, g, \text{bundles}, J, \dots],$$

compatible with cutting and gluing, locality of source variation, reflection or unitarity, and covariance under background isomorphisms. The Kontsevich–Segal formulation makes this aspiration mathematically sharper: it asks for a functorial or multilinear assignment on admissible complex Riemannian bordisms, with Lorentzian metrics appearing as boundary values and reflection positivity/unitarity represented by structure on the target spaces. It is the most promising current framework for the path-integral and OPE aspects of the subject, precisely because sewing is a primitive operation. Chapter 13 develops the axioms, the finite-dimensional allowability condition, and the current construction status. In the opening construction we still use Wightman, OS, and local-net frameworks because they give the spectral theory, Källén–Lehmann representation, Haag–Ruelle scattering theory, LSZ comparison, and analytic continuation theorems in the form needed for the later chapters.

Constructive results calibrate these frameworks. Table 136.1 in Chapter 136 records representative examples and open problems. The completed non-Gaussian constructions are concentrated in special low-dimensional regimes, such as $P(\phi)_2$, massive ϕ^4_3, selected two-dimensional scalar–fermion models, massive Gross–Neveu type models, and two-dimensional Yang–Mills in holonomy or heat-kernel formulations. The reflection-positive Ising/$\lambda\phi^4$ route in $D \geq 4$ is constrained by triviality theorems, with the four-dimensional marginal case requiring the precise qualification stated in Table 136.1. A four-dimensional UV-complete interacting QFT agreeing with formal ϕ^4_4 perturbation theory to all orders is not ruled out by the positive-lattice triviality theorem as a purely logical matter; it is a distinct open construction problem. Four-dimensional Yang–Mills with mass gap remains the central constructive gauge-theory problem.

Open Problem 1.9 (Axiomatic comparison and physical examples). Give sharp hypotheses, and prove the resulting comparison theorems, relating the principal nonperturbative frameworks for local QFT:

$$\begin{array}{cc} \text{Wightman fields,} & \text{Osterwalder–Schrader data,} \\ \text{Haag–Kastler local nets,} & \text{functorial/path-integral data.} \end{array}$$

Some directions are established under substantial hypotheses: OS-II reconstruction gives Wightman boundary values from OS-admissible Schwinger data; Wightman fields give local nets after the relevant self-adjointness, bounded-functional-calculus, and strong-locality conditions are imposed; and positive states on local algebras give Hilbert-space representations by GNS. The converse directions, the precise functorial completion, and the verification of all required hypotheses in physically central interacting four-dimensional theories remain open. The problem has two parts: prove the comparison maps with minimal usable assumptions, and construct substantial examples for which the assumptions are actually verified.

1.6 Analytic Status of the Basic Objects

The opening framework deliberately separates several levels of mathematical status. This separation is part of the subject, because the same formula may denote different objects in different constructions.

Definition 1.10 (Regulated theory)**.** A regulated theory is a finite-dimensional or cutoff-defined construction with a specified state space or configuration space, algebra of variables, covariance or symmetry action, and expectation functional $\langle\cdot\rangle_\Lambda$, where Λ denotes the regulator data. Examples include a finite lattice, a finite spatial volume with momentum cutoff, or a finite-dimensional time-sliced integral. All integrals and operator products in a regulated theory are ordinary objects in the regulator-dependent category.

Definition 1.11 (Continuum correlation theory)**.** A continuum correlation theory is a hierarchy of distributions, Schwinger functions, Wightman functions, or time-ordered distributions obtained as a limit of regulated theories or specified directly by axioms. The topology of convergence is part of the datum: distributional convergence of correlators, convergence of moments, weak convergence of measures, or another declared topology.

Definition 1.12 (Contact terms and local source-coordinate systems)**.** Let M be the spacetime manifold and let

$$\Delta_n = \bigcup_{1\le i<j\le n} \{(x_1,\dots,x_n)\in M^n \mid x_i = x_j\}$$

be the collision locus. A separated-point n-point distribution is a distribution $T^\circ \in \mathcal{D}'(M^n\setminus\Delta_n)$, with values in the appropriate tensor product of operator-label bundles. A contact-term extension of T° is a distribution $T\in\mathcal{D}'(M^n)$ whose restriction to $M^n\setminus\Delta_n$ equals T°. A contact term is a distribution supported on Δ_n, or on one of its partial collision diagonals. Locally near a smooth partial diagonal, with coordinates z on the diagonal and normal coordinates y, such a distribution has finite normal order,

$$C = \sum_{|\alpha|\le N} \partial_y^\alpha \delta(y)\, c_\alpha(z), \qquad c_\alpha \in \mathcal{D}'(\Delta),$$

after choosing local coordinates and a density convention. Thus contact terms are ordinary distributional data on collision strata.

A local source-coordinate system consists of source fields $J^A \in C_c^\infty(M, E_A)$, a source functional $W[J]$ in the chosen category, and distributional source derivatives

$$G_{A_1\cdots A_n}(x_1,\dots,x_n) = \left.\frac{\delta^n W[J]}{\delta J^{A_1}(x_1)\cdots\delta J^{A_n}(x_n)}\right|_{J=0}.$$

Changing the chart by a local functional $\int_M \mathcal{L}(x, j_x^k J)\mathrm{d}^D x$, or by a local source-coordinate change whose nonlinear part is supported through jets at the same point, changes these derivatives by finite sums of distributions supported on collision diagonals. With the linear source normalization fixed, the restriction of $G_{A_1\cdots A_n}$ to $M^n\setminus\Delta_n$ is unchanged by such purely local finite choices, while the contact-term extension changes.

The source-coordinate system is bookkeeping for a concrete operation: differentiate a specified generating functional with respect to specified compactly supported backgrounds. It is not an additional invariant structure on the QFT. Its invariant content is the separated-point distribution together with the declared equivalence relation on local counterterms and local redefinitions of the sources. Whenever the monograph says that a contact term depends on this choice, the statement means exactly that two coordinate systems agree away from collision diagonals and differ on the diagonals by local distributions of the form displayed above.

Local source-coordinate changes. Fix the linear source normalization in Definition 1.12. If the source functional is changed by a local functional of a finite jet of the source,

$$W[J] \longmapsto W[J] + \int_M \mathcal{L}(x, j_x^k J)\mathrm{d}^D x,$$

then the induced change in each n-fold source derivative at $J = 0$ is a finite sum of distributions supported on collision diagonals in M^n. Consequently the separated-point distribution on $M^n \setminus \Delta_n$ is unchanged.

Every dependence of the added functional on the sources occurs through $\partial^\alpha J^A(x)$ at one integration point x, with $|\alpha| \leq k$. A functional derivative with respect to $J^A(y)$ therefore produces a finite linear combination of derivatives of $\delta(x - y)$. After n source derivatives, the x-integration leaves finite sums of products of delta functions and their derivatives imposing $y_i = y_j$ along partial or total diagonals. These distributions have support contained in Δ_n, hence vanish after restriction to $M^n \setminus \Delta_n$.

The Wightman and Osterwalder–Schrader frameworks assign distributional correlators to the selected basic or named fields. This assignment gives the distributions specified by those frameworks, including their declared domains and positivity properties. It does not canonically choose composite fields, time-ordered products, nonlinear source couplings, or the response of correlators to repeated local source differentiation on collision diagonals. Those choices are additional local source or operator-product data. Ward identities, current algebra, stress-tensor products, anomaly formulae, and deformations of a theory therefore carry contact-term conventions in addition to their separated-point correlation functions.

Renormalization is one construction in which these local extension choices are produced and compared. The concept is broader: a contact term is coincident-point/source-response data, while a counterterm is a local term in a chosen regulated or perturbative action functional. Perturbative Epstein–Glaser constructions, perturbative algebraic QFT, microlocal extension methods, Hollands–Wald local Wick-polynomial and time-ordered product constructions, and Costello–Gwilliam factorization algebras give precise comparison frameworks for such extensions under their stated hypotheses. They will be used as technical models where their hypotheses match the problem under discussion, without being treated as a complete nonperturbative axiom system for all QFTs.

Definition 1.13 (Hilbert-space or algebraic completion)**.** A Hilbert-space or algebraic completion is a reconstruction of operators, states, local algebras, and symmetries from correlation data or from an abstract local net. Positivity is the bridge: Hilbert spaces arise from positive inner products, while local observable algebras arise from positive states and their GNS representations.

Perturbation theory has its own status. A perturbative statement is an asymptotic expansion in a specified parameter, with coefficients computed from regulated distributions and then renormalized. The existence of the perturbative coefficients and the existence of a nonperturbative continuum object are distinct assertions. A perturbative assertion is specified when the observable being expanded, the expansion parameter, the renormalization prescription, and the error class or asymptotic meaning have been stated.

These levels of status will be tracked separately in later chapters. Spectral measures, scattering states, renormalization-group flows, anomalies, conformal data, and gauge-invariant observables are different constructions related by theorems, controlled approximations, or comparison problems.

1.7 Fields As Local Distributional Data

A field presentation assigns localized distributional operators to test functions. For a test function $f \in C_c^\infty(\mathbb{M}^D)$ with $\operatorname{supp} f \subset \mathcal{O}$,

$$\widehat{\Phi}_A(f) = \int_{\mathbb{M}^D} \mathrm{d}^D x \, f(x) \widehat{\Phi}_A(x)$$

is localized in $\mathcal{O}$, in the sense that the smeared operator is affiliated with the local algebra $\mathcal{A}(\mathcal{O})$ or belongs to a common dense invariant field domain associated with that algebra. The precise operator-domain statement will be made in the field framework of the next chapter.

The field presentation is used in two constructions. First, it gives concrete generators or affiliated operators for examples. Second, it gives correlation functions:

$$W_{A_1 \cdots A_n}(x_1, \ldots, x_n) = \langle \Omega | \widehat{\Phi}_{A_1}(x_1) \cdots \widehat{\Phi}_{A_n}(x_n) | \Omega \rangle,$$

again as distributions. Under suitable positivity, covariance, spectral, and domain assumptions, such correlation functions can reconstruct the Hilbert space and fields. Reconstruction theorems will be used later as comparison results with stated hypotheses.

1.8 Additional Constructions And Hypotheses

Several familiar objects are constructed from the framework after adding further hypotheses.

Definition 1.14 (First derived objects and their extra hypotheses)**.** A stable one-particle sector is a spectral subspace of the translation generators associated with an isolated positive-energy mass hyperboloid $p^2 = -m^2$, $p^0 > 0$, possibly with finite internal or spin multiplicity. An S-matrix is constructed from incoming and outgoing asymptotic multiparticle states and wave operators comparing asymptotic Hilbert spaces with the interacting Hilbert space. Surjectivity of those wave operators onto the physical sector is the stronger asymptotic-completeness property and must be supplied as a separate theorem or hypothesis. LSZ reduction is a theorem relating already-defined S-matrix elements to residues or pole parts of time-ordered correlation functions, after the relevant asymptotic states and field-particle overlaps have been identified.

A path-integral presentation is a construction or computational scheme for correlation functions after a regulator, contour, measure, or perturbative prescription has been specified. The symbol $\int D\phi \, \mathrm{e}^{\mathrm{i}S[\phi]}$ receives its meaning from that additional specification.

This dependency order is part of the construction. Perturbative Feynman graphs may be introduced as an expansion for Green functions before scattering is discussed. Diagrams for S-matrix elements acquire that interpretation after asymptotic states and LSZ reduction have been established.

1.9 The First Spectral Question

Translation invariance permits simultaneous spectral analysis of the energy-momentum generators P^μ. Let $E_P(\Delta)$ denote the joint spectral projection of $P = (P^0, \ldots, P^{D-1})$ associated with a Borel set $\Delta \subset \mathbb{R}^D$. The joint spectrum $\mathrm{Spec}(P)$ is the support of this projection-valued measure. The spectrum condition is

$$\mathrm{Spec}(P) \subset \overline{V}_+ = \{p \in \mathbb{R}^D : p^0 \geq 0,\ p^2 \leq 0\}.$$

For $\mu^2 \geq 0$, write $\Delta_F(x;\mu^2)$ for the free scalar Feynman distribution with mass parameter μ:

$$\Delta_F(x;\mu^2) = -\mathrm{i} \lim_{\epsilon \downarrow 0} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2 + \mu^2 - \mathrm{i}\epsilon},$$

where the limit is taken in the sense of tempered distributions. The prefactor $-\mathrm{i}$ and the denominator $k^2 + \mu^2 - \mathrm{i}\epsilon$ are part of the mostly-plus Lorentzian Feynman convention used throughout this opening volume, with Fourier factor $\mathrm{e}^{\mathrm{i}k\cdot x}$. With these conventions Δ_F is the vacuum time-ordered two-point distribution of a free scalar field. This is only a notation-setting definition at this point; its relation to spectral measures is derived in Chapter 14. For a scalar Wightman field in the vacuum Hilbert-space framework, positivity of the Hilbert-space inner product, translation invariance, and the spectrum condition imply that the two-point function admits a positive spectral measure on invariant mass. Let $\mathrm{d}\rho(\mu^2)$ denote this positive Borel measure on $[0,\infty)$. For two scalar fields at separated times, $T\{\widehat{\phi}(x)\widehat{\phi}(y)\}$ means

$$\theta(x^0 - y^0)\widehat{\phi}(x)\widehat{\phi}(y) + \theta(y^0 - x^0)\widehat{\phi}(y)\widehat{\phi}(x),$$

with the extension across equal-time coincidences fixed by the distributional Feynman prescription. The time-ordered two-point function has the form

$$\langle \Omega | T\{\widehat{\phi}(x)\widehat{\phi}(0)\} | \Omega \rangle = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2)\, \Delta_F(x;\mu^2).$$

When this measure is absolutely continuous with respect to Lebesgue measure, one may write

$$\mathrm{d}\rho(\mu^2) = \rho_{\mathrm{ac}}(\mu^2)\,\mathrm{d}\mu^2, \qquad \rho_{\mathrm{ac}}(\mu^2) \geq 0 \quad \text{for almost every } \mu^2.$$

This formula will be derived from the stated Hilbert-space assumptions in Chapter 14. It is the first result in the text that relates local two-point data to particle content: isolated atoms of the spectral measure correspond to stable one-particle states, while continuous spectral support corresponds to multi-particle and other continuum contributions.

1.10 Logical Order Of The Opening Construction

The opening construction will proceed as follows:

Hilbert space, symmetry, spectrum, vacuum, locality
$\longrightarrow$ local fields and Wightman functions $\longrightarrow$ path-integral constructions of Green functions
$\longrightarrow$ Källén–Lehmann spectral representation $\longrightarrow$ perturbative Green functions
$\longrightarrow$ Haag–Ruelle asymptotic states $\longrightarrow$ LSZ $\longrightarrow$ perturbative scattering amplitudes.

Each arrow indicates an added construction or theorem. The construction will use each arrow only after its assumptions have been stated.

The first detailed computations follow three dependency bands. First, finite and regulated Lagrangian quantum mechanics supplies path integrals, regularization, perturbation theory for correlation functions, diagrammatics, and local counterterms. Second, relativistic scalar fields supply the first local field examples, Green functions, spectral representations, asymptotic states, the nonperturbative scattering operator, and LSZ. Third, spin and gauge fields require representation theory, fermionic and gauge-field quantization, gauge fixing, ghosts, BRST cohomology, and applications such as QED precision observables. Perturbative Feynman rules for scattering amplitudes belong after the second band has supplied their interpretation.

Chapter 2

Wightman Fields and Reconstruction

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The first four chapters of this volume separate four ways of presenting the same physical theory, but they are not interchangeable levels of structure. This chapter starts from Lorentzian Wightman field data and reconstructs the cyclic field presentation carried by its vacuum n-point distributions. Chapter 12 starts instead from Euclidean Schwinger data and proves, under the corrected OS hypotheses, that such Wightman data are obtained as boundary values of analytic continuations. Chapter 3 then treats bounded local algebras as the invariant observable object: a Wightman field family may coordinatize a represented local net only after closure, affiliation, spectral-projection locality, additivity, and observable-subnet identification have been supplied. Finally Chapter 4 uses that bounded net as the input for DHR sectors, Doplicher–Roberts reconstruction, Tomita–Takesaki theory, split inclusions, and Bisognano–Wichmann modular structure.

Thus the foundational route is a chain of theorem boundaries rather than a single dictionary:

(a) Wightman distributions reconstruct a cyclic unbounded-field presentation;

(b) OS-positive Euclidean data reconstruct such Wightman distributions away from the contact-term choices on diagonals;

(c) passage from fields to an observable net requires additional bounded operator and localization input;

(d) sector, modular, split, and type-theoretic conclusions require their own AQFT hypotheses after the net has been constructed.

Statements later in the volume are meant to name which link is being used, not to slide silently from one presentation to another.

2.1 Field-Theoretic Data

Let $M = \mathbb{M}^D$ be D-dimensional Minkowski space with translation coordinates x^μ and metric $\eta = \mathrm{diag}(-1, 1, \ldots, 1)$. The proper orthochronous Poincare group is denoted by

$$\mathcal{P}_+^\uparrow = \mathbb{R}^D \rtimes SO^+(1, D-1).$$

With $p^2 = \eta_{\mu\nu} p^\mu p^\nu$, define the open and closed forward cones by

$$V_+ = \{p \in \mathbb{R}^D \mid p^0 > 0,\ p^2 < 0\}, \qquad \overline{V}_+ = \{p \in \mathbb{R}^D \mid p^0 \geq 0,\ p^2 \leq 0\}.$$

The framework-level definition in Definition 1.6 records the same object as one possible presentation of a QFT. The present chapter refines that entry by fixing the cyclic domain, field-label, adjunction, and reconstruction conventions needed for theorem-level work.

Definition 2.1 (cyclic Wightman field presentation)**.** A cyclic Wightman field presentation on M consists of

$$\mathfrak{W} = (\mathcal{H}, \Omega, U, \mathcal{D}, E_I, J, |\cdot|, \{\widehat{\Phi}_\alpha\}_{\alpha \in I})$$

with the following properties.

(i) $\mathcal{H}$ is a complex Hilbert space, $\Omega \in \mathcal{H}$ is a unit vector, and $U : \mathcal{P}_+^\uparrow \to \mathcal{U}(\mathcal{H})$ is a strongly continuous unitary representation with

$$U(g)\Omega = \Omega, \qquad g \in \mathcal{P}_+^\uparrow.$$

(ii) $\mathcal{D} \subset \mathcal{H}$ is a dense common invariant domain:

$$\Omega \in \mathcal{D}, \qquad U(g)\mathcal{D} = \mathcal{D}, \qquad \widehat{\Phi}_{\alpha_1}(f_1) \cdots \widehat{\Phi}_{\alpha_n}(f_n)\mathcal{D} \subset \mathcal{D}$$

for all $g \in \mathcal{P}_+^\uparrow$, all $n \geq 1$, all labels α_j, and all test functions f_j in the declared local test-function space. In this chapter the test-function space is the Schwartz space $\mathcal{S}(M)$; compactly supported local smearings are obtained by restriction.

(iii) For each finite multiplet under discussion,

$$E_I = \bigoplus_{\alpha \in I} \mathbb{C} e_\alpha$$

is the finite field-label space, $J : E_I \to E_I$ is an antilinear involution, and $|\cdot| : I \to \mathbb{Z}/2\mathbb{Z}$ is the parity assignment. We write

$$J e_\alpha = e_{\alpha^\dagger}, \qquad J^2 = 1,$$

allowing J to contain the finite-dimensional conjugation matrix when the chosen basis is not closed under componentwise conjugation.

(iv) Each $\widehat{\Phi}_\alpha$ is an operator-valued tempered distribution with common domain $\mathcal{D}$. Thus, for $f \in \mathcal{S}(M)$,

$$\widehat{\Phi}_\alpha(f) = \int_M \mathrm{d}^D x\, f(x)\widehat{\Phi}_\alpha(x)$$

means distributional pairing, and $\widehat{\Phi}_\alpha(f) : \mathcal{D} \to \mathcal{D}$. Matrix elements of finite products between vectors in $\mathcal{D}$ are continuous multilinear functionals of the test functions.

(v) For an E_I-valued test function $h = \sum_\alpha h^\alpha e_\alpha$, set

$$\widehat{\Phi}(h) = \sum_\alpha \widehat{\Phi}_\alpha(h^\alpha), \qquad h^\dagger(x) = Jh(x) = \sum_\alpha \overline{h^\alpha(x)}\, e_{\alpha^\dagger}.$$

The field adjunction is the domain inclusion

$$\widehat{\Phi}(h)^* \supset \widehat{\Phi}(h^\dagger). \tag{2.1}$$

Equivalently,

$$\widehat{\Phi}_\alpha(f)^* \supset \widehat{\Phi}_{\alpha^\dagger}(\overline{f}).$$

Here $\widehat{\Phi}(h)^*$ is the Hilbert-space adjoint of the densely defined operator $\widehat{\Phi}(h) : \mathcal{D} \to \mathcal{D}$; the inclusion means that $\mathcal{D} \subset \mathrm{Dom}(\widehat{\Phi}(h)^*)$ and that the two operators agree on $\mathcal{D}$.

(vi) If the field multiplet transforms in a finite-dimensional Lorentz representation R, covariance is the smeared identity

$$U(a,\Lambda)^{-1}\widehat{\Phi}_\alpha(f)U(a,\Lambda) = R(\Lambda)_\alpha{}^\beta\, \widehat{\Phi}_\beta(f_{a,\Lambda}), \qquad f_{a,\Lambda}(x) = f\big(\Lambda^{-1}(x-a)\big).$$

For pure translations we write $U(a) := U(a,1)$, and with the convention of Chapter 1,

$$U(a) = \exp(\mathrm{i}a_\mu P^\mu) = \exp(\mathrm{i}a^\mu P_\mu).$$

In particular $U(a^0, \vec{0}) = \exp(-\mathrm{i}a^0 P^0)$.

(vii) If E_P is the joint spectral measure of the translation generators $P = (P^0, \ldots, P^{D-1})$, then the spectrum condition is

$$\mathrm{Spec}(P) \subset \overline{V}_+.$$

(viii) If the supports of f and g are spacelike separated, then

$$[\widehat{\Phi}_\alpha(f), \widehat{\Phi}_\beta(g)]_{\mathrm{gr}} = 0$$

on $\mathcal{D}$, where for homogeneous fields

$$[A,B]_{\mathrm{gr}} = AB - (-1)^{|A||B|}BA.$$

For purely bosonic fields this is the ordinary commutator; when both fields are odd it is the anticommutator.

(ix) The vacuum is cyclic for the polynomial field algebra:

$$\mathrm{span}\{\widehat{\Phi}_{\alpha_1}(f_1)\cdots\widehat{\Phi}_{\alpha_n}(f_n)\Omega \mid n \geq 0\} \quad \text{is dense in } \mathcal{H}.$$

Definition 2.1 is the Wightman object used throughout this chapter. It is the cyclic, tempered, Minkowski-space specialization of the framework-level Wightman presentation introduced in Definition 1.6: the present chapter fixes the Schwartz test space, the mostly-plus Poincare convention, the field-label adjunction J, the parity grading, the spectrum cone, and the cyclicity condition needed for reconstruction. A Hermitian field smeared with a real test function is symmetric on $\mathcal{D}$, but essential self-adjointness of its closure is an additional analytic assertion. The spectral, covariance, locality, Hermiticity, and cyclicity requirements are independent conditions on the same field presentation. The reconstruction theorem below is the construction step that builds a cyclic field presentation from a Wightman distribution hierarchy.

A pure vacuum Wightman sector is a cyclic Wightman field presentation for which, writing E_P for the joint spectral measure of the translation generators, the zero-momentum spectral projection satisfies

$$E_P(\{0\})\mathcal{H} = \mathbb{C}\Omega.$$

This chapter does not build that purity condition into the base definition: Theorem 2.3 proves that it is equivalent, under the Wightman locality and spectrum assumptions, to the weak cluster property for polynomial local fields.

2.2 Fields and Local Observable Content

The objects $\widehat{\Phi}_\alpha(f)$ in Definition 2.1 are generally unbounded operators. Their algebraic products are defined first on the common invariant domain $\mathcal{D}$, and their locality statement is a statement about graded commutators on that domain. A local algebra of bounded observables is a further object constructed from, or compared with, these field coordinatizations.

For an open region $\mathcal{O} \subset M$, let

$$\mathfrak{P}_\Phi(\mathcal{O})$$

denote the unital $*$-algebra on $\mathcal{D}$ generated by smeared fields $\widehat{\Phi}_\alpha(f)$ with $\operatorname{supp} f \subset \mathcal{O}$, together with their adjoints on $\mathcal{D}$. The subscript Φ names the chosen field family; the individual Hilbert-space fields in the family remain the hatted operator-valued distributions $\widehat{\Phi}_\alpha$. This is an algebra of generally unbounded operators on a common domain, not a C^*-algebra. If the relevant smeared Hermitian fields are essentially self-adjoint and if the closures and bounded Borel functions have the required localization properties, one can form a represented local von Neumann algebra

$$\mathcal{R}_\Phi(\mathcal{O}) = \{\text{bounded functions of closures of fields localized in } \mathcal{O}\}'', \tag{2.2}$$

or, more invariantly, require the unbounded fields to be affiliated with a local algebra $\mathcal{R}(\mathcal{O})$. The double prime denotes weak operator closure. The affiliation formulation is often the correct one: spectral projections of a self-adjoint operator affiliated with $\mathcal{R}(\mathcal{O})$ belong to $\mathcal{R}(\mathcal{O})$.

Figure 2.1 compares two logically different routes to local observable algebras. The upper route starts from unbounded Wightman field coordinates and needs closure, functional-calculus, or affiliation hypotheses. The lower route starts from a local-net datum. The two right-hand algebras are identified only after a comparison theorem is proved; equality is not a notation convention.

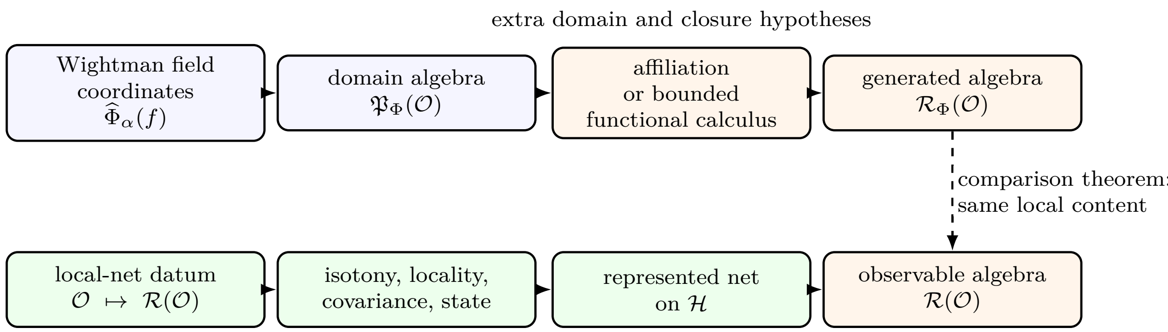


Figure 2.1: Two routes to local observable algebras. Field coordinates generate a local algebra only after closure and affiliation input; a local net may instead be supplied directly, and equivalence of the two descriptions is an additional theorem.

Several different field families may coordinatize the same local observable content. In this monograph a field coordinatization is treated as part of the presentation, while the local observable net, when it has been constructed, is the invariant local object. Thus the Wightman hierarchy for a chosen field family is complete for reconstructing that chosen field presentation, whereas comparisons between different field presentations require an additional equivalence statement: one must show that they generate the same represented local algebras, or a specified subtheory, in the relevant vacuum sector.

2.3 Wightman Functions

The vacuum n-point distributions are

$$W_{\alpha_1\cdots\alpha_n}(x_1,\ldots,x_n) = \langle\Omega|\widehat{\Phi}_{\alpha_1}(x_1)\cdots\widehat{\Phi}_{\alpha_n}(x_n)|\Omega\rangle. \tag{2.3}$$

They are tempered distributions on M^n. Translation invariance allows one to pass to difference variables

$$\xi_j = x_{j+1} - x_j, \qquad j = 1,\ldots,n-1.$$

The spectral condition implies that the Fourier transform in the difference variables has support in $\overline{V}_+^{\,n-1}$. This is the same positive spectral input that will lead in Chapter 14 to the Källén–Lehmann representation of two-point functions.

The Wightman hierarchy also records the adjunction operation on fields. On one-field test functions the adjunction is the map $h \mapsto h^\dagger$ fixed in (2.1). On test functions valued in field-label tensor powers, one must also reverse the order of insertions because the Hilbert-space adjoint reverses products. For $f_n \in \mathcal{S}(M^n, E_I^{\otimes n})$, define the antilinear tensor reversal $R_n : E_I^{\otimes n} \to E_I^{\otimes n}$ by

$$R_n(v_1 \otimes \cdots \otimes v_n) = Jv_n \otimes \cdots \otimes Jv_1$$

and set

$$f_n^\dagger(x_1,\ldots,x_n) = R_n\big(f_n(x_n,\ldots,x_1)\big).$$

Thus $f_n^\dagger$ simultaneously reverses the order of insertions, complex conjugates the coefficient functions, and applies the field-label adjunction to each tensor factor.

The Wightman hierarchy satisfies five structural conditions that are directly read from the Hilbert-space data:

Covariance. The distribution W_n transforms under $\mathcal{P}_+^\uparrow$ by the n-fold field representation on its labels and test functions.

Spectrum. In translation-difference variables,

$$\operatorname{supp} \widehat{W}_n \subset \overline{V}_+^{\,n-1}.$$

Hermiticity. For all labels and test functions,

$$\overline{W_{\alpha_1\cdots\alpha_n}(f_1 \otimes \cdots \otimes f_n)} = W_{\alpha_n^\dagger\cdots\alpha_1^\dagger}(\overline{f}_n \otimes \cdots \otimes \overline{f}_1).$$

This is the vacuum distribution form of the operator-domain adjoint condition $\widehat{\Phi}(h)^* \supset \widehat{\Phi}(h^\dagger)$, with product order reversed after taking adjoints.

Locality. W_n has the graded permutation property when adjacent field insertions are smeared with spacelike separated test functions.

Positivity. For every finite sequence $F = (f_0, f_1, f_2, \ldots)$ of test functions,

$$\sum_{m,n} W_{m+n}(f_m^\dagger \otimes f_n) \geq 0.$$

In the positivity condition, f_n denotes a compactly supported or Schwartz test function valued in the appropriate finite direct sum of field labels. Hermiticity is the compatibility condition that makes the reconstructed left-multiplication fields have the intended adjoints on the common domain.

2.4 Clustering and the Vacuum Sector

The vacuum vector should represent one pure infinite-volume phase. In the Wightman setting this is expressed by a cluster property. For two polynomial local operators A and B, formed from fields smeared in bounded regions, define

$$B(a) = U(a)BU(a)^{-1}.$$

The weak cluster property in the vacuum sector is

$$\lim_{\lambda\to+\infty} \langle \Omega, A\, B(\lambda a)\Omega\rangle = \langle \Omega, A\Omega\rangle\langle \Omega, B\Omega\rangle \tag{2.4}$$

for every spacelike vector a. In terms of Wightman functions, this says that widely spacelike-separated groups of insertions factorize.

The spectral argument is as follows. Insert the spectral resolution of translations between A and $B(a)$. The vacuum atom at $p = 0$ gives the product on the right side of (2.4). If the vacuum is the only translation-invariant vector, then no other atom at $p = 0$ contributes. The remaining continuous spectral part has oscillatory phase $\exp(\mathrm{i}\lambda a \cdot p)$; after smearing, its contribution vanishes in the large-spacelike-separation limit because locality and the spectrum condition make the relevant one-parameter spectral measure a Rajchman measure on every spacelike translation line. The analytic mechanism is the lemma below. With a mass gap and suitable regularity, the same argument gives stronger decay estimates. Selecting a pure vacuum sector is the condition that the zero-momentum projection is the one-dimensional projection onto $\mathbb{C}\Omega$.

Lemma 2.2 (Jost cluster step for local vectors)**.** *Let A and B be homogeneous polynomial field operators obtained by smearing fields in bounded regions $\mathcal{O}_A$ and $\mathcal{O}_B$, and let*

$$\psi = A^{\dagger}\Omega, \qquad \chi = (1 - P_0)B\Omega.$$

For every spacelike vector a,

$$\lim_{\lambda\to+\infty} \langle \psi, U(\lambda a)\chi \rangle = 0. \tag{2.5}$$

Proof. Set

$$F_+(x) = \langle \Omega, A\, B(x)\Omega \rangle - \langle \Omega, A\Omega \rangle\langle \Omega, B\Omega \rangle, \qquad B(x) = U(x)BU(x)^{-1}.$$

Equivalently,

$$F_+(x) = \langle A^{\dagger}\Omega, U(x)(1 - P_0)B\Omega \rangle. \tag{2.6}$$

The joint spectral theorem gives a finite complex measure

$$\mathrm{d}\mu_{\psi,\chi}(p) = \langle \psi, E_P(\mathrm{d}p)\chi \rangle, \qquad \operatorname{supp} \mu_{\psi,\chi} \subset \overline{V}_+,$$

and F_+ is its Fourier transform in the translation variable, with the Fourier sign convention fixed by $U(x) = \exp(\mathrm{i}x \cdot P)$. The subtraction by P_0 is exactly the assertion

$$\mu_{\psi,\chi}(\{0\}) = 0. \tag{2.7}$$

Let

$$\mathcal{C}_{A,B} = \{x \in M \mid \mathcal{O}_A \text{ is spacelike separated from } \mathcal{O}_B + x\}.$$

This is an open real set, and for every spacelike a there is λ_0 such that $\lambda a \in \mathcal{C}_{A,B}$ for $\lambda > \lambda_0$. On $\mathcal{C}_{A,B}$, locality gives

$$F_+(x) = \varepsilon_{AB}F_-(x), \qquad F_-(x) = \langle \Omega, B(x)A\Omega \rangle - \langle \Omega, B\Omega \rangle\langle \Omega, A\Omega \rangle, \tag{2.8}$$

where $\varepsilon_{AB} = (-1)^{|A||B|}$. The spectral condition gives a holomorphic function $\mathcal{F}_+$ in the forward tube whose boundary value is F_+. The same condition applied to the reversed matrix element gives a holomorphic function $\mathcal{F}_-$ in the backward tube whose boundary value is F_-. Equation (2.8) identifies the two boundary values on the real edge $\mathcal{C}_{A,B}$. The edge-of-the-wedge theorem therefore gives a single holomorphic continuation through a complex neighborhood of $\mathcal{C}_{A,B}$.

It remains to spell out the spectral consequence of this continuation along the line $\mathbb{R}a$. Let $H_a = a \cdot P$, and let E_a be its spectral measure. Suppose first that $E_a(\{q\})\chi$ contributes to the local matrix coefficient for some real q. Replacing χ by this spectral component in (2.6) gives a boundary value whose restriction to the a-line has the factor $\exp(\mathrm{i}\lambda q)$. The reversed boundary value has the opposite one-parameter spectral support. Since the two boundary values are restrictions of the same holomorphic function across the Jost edge, the Fourier support of the isolated component must lie simultaneously in the forward and backward translation spectra. Hence its full energy-momentum support is contained in

$$\overline{V}_+ \cap (-\overline{V}_+) = \{0\}.$$

For $q = 0$ this is precisely the removed projection $P_0\chi = 0$; for $q \neq 0$ it is impossible. Thus the one-parameter spectral measure of χ, tested against every local vector $\psi = A^\dagger\Omega$, has no point mass.

The continuous part is controlled by the same Jost-domain theorem in its one-parameter form. In the present notation, the theorem says that a tempered local matrix coefficient F_+ satisfying:

(a) forward-tube spectral support for F_+;

(b) backward-tube spectral support for the reversed ordering F_-;

(c) equality of the two boundary values on a nonempty Jost edge; and

(d) absence of the zero-momentum spectral atom after vacuum subtraction

has the following Rajchman property on every spacelike ray:

$$\lim_{\lambda \to +\infty} \int_{\mathbb{R}} e^{i\lambda q}\, d\nu_a(q) = 0, \qquad d\nu_a = (a \cdot p)_* d\mu_{\psi,\chi}. \tag{2.9}$$

The proof of this one-parameter theorem is a carrier-cone argument. Localize the edge to a compact set of spacelike points and apply the edge-of-the-wedge continuation to the forward and backward tubes. The Paley–Wiener–Schwartz carrier theorem then identifies the part of $\mu_{\psi,\chi}$ that could fail to have the decay (2.9) with a spectral component carried by a proper real hyperplane and invariant under the corresponding spacelike translations. Applying the preceding atom argument to that component puts its full energy-momentum carrier in $\overline{V}_+ \cap (-\overline{V}_+) = \{0\}$. Equation (2.7) removes this last possibility. Hence the push-forward measure is Rajchman, and (2.5) follows. □

Theorem 2.3 (Cluster decomposition and vacuum uniqueness)**.** *Let $(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_\alpha\})$ be the Hilbert-space and field part of a cyclic Wightman field presentation in the sense of Definition 2.1, with joint translation spectral measure E_P, and let*

$$P_0 = E_P(\{0\})$$

be the joint spectral projection of the translation generators onto zero momentum. The following conditions are equivalent.

(i) *The zero-momentum subspace is one-dimensional:*

$$P_0\mathcal{H} = \mathbb{C}\Omega.$$

(ii) *The vacuum is the unique translation-invariant vector up to scalar multiple.*

(iii) *The weak cluster property (2.4) holds for all polynomial field insertions A and B and for every spacelike translation direction a.*

Proof. The equivalence of (i) and (ii) is the joint spectral theorem for the translation representation: translation-invariant vectors are precisely the vectors in $\operatorname{Ran} P_0$. For (i) $\Rightarrow$ (iii), write

$$\langle \Omega, A\, B(\lambda a)\Omega\rangle = \langle A^\dagger\Omega, U(\lambda a)B\Omega\rangle,$$

first for polynomial fields whose domain adjoints are defined on $\mathcal{D}$, and then by linearity. The Wightman spectral condition and temperedness make this matrix element the Fourier transform, in the weak large-spacelike-separation sense, of the spectral distribution between $A^\dagger\Omega$ and $B\Omega$. Its zero-momentum atom contributes

$$\langle A^\dagger\Omega, P_0 B\Omega\rangle = \langle \Omega, A\Omega\rangle\,\langle \Omega, B\Omega\rangle,$$

because $P_0 = |\Omega\rangle\langle\Omega|$. The remaining spectral part has no zero-momentum atom. Lemma 2.2, applied to $(1 - P_0)B\Omega$, gives the required weak large-separation limit.

Conversely, assume the weak cluster property. The same spectral decomposition shows that the large-separation limit of the non-oscillatory part is $\langle A^\dagger\Omega, P_0 B\Omega\rangle$. Comparing with (2.4) gives

$$\langle A^\dagger\Omega, P_0 B\Omega\rangle = \langle A^\dagger\Omega, \Omega\rangle\,\langle \Omega, B\Omega\rangle$$

for all polynomial A, B. Since the polynomial field orbit of Ω is dense, this identity implies

$$P_0 B\Omega = \Omega\,\langle \Omega, B\Omega\rangle$$

on a dense set of vectors $B\Omega$. Hence $P_0 = |\Omega\rangle\langle\Omega|$, which is (i). ☐

The finite algebra in this proof is worth isolating. Let $\mathcal{E}$ be the linear span of polynomial vectors $B\Omega$. The weak cluster identity tests the sesquilinear form

$$(X, Y) \longmapsto \langle X, P_0 Y\rangle$$

on the dense subspace $\mathcal{E}$. If the identity

$$\langle X, P_0 Y\rangle = \langle X, \Omega\rangle\langle \Omega, Y\rangle, \qquad X, Y \in \mathcal{E},$$

holds, continuity of P_0 extends it to $\overline{\mathcal{E}} = \mathcal{H}$, so P_0 is the rank-one projection onto $\mathbb{C}\Omega$. Conversely, when $P_0 = |\Omega\rangle\langle\Omega|$, the zero-momentum atom in the spectral decomposition is exactly the product term. Thus the analytic content of the cluster theorem is not this projection algebra; it is Lemma 2.2, which proves that no other spectral piece survives along a spacelike ray after the vacuum atom is removed.

2.5 Reconstruction From Correlation Functions

The Wightman reconstruction theorem says that the correlation hierarchy is itself a complete encoding of a chosen Wightman field presentation, provided the hierarchy satisfies the Wightman conditions with the required regularity. The source trace for the reconstruction theorem in this form is Wightman's vacuum-expectation-value formulation and the Streater–Wightman presentation [1, 2]; the proof below is included because the null-space quotient, common domain, and uniqueness statements are load-bearing for the later comparison with local nets.

Theorem 2.4 (Wightman reconstruction). *Let $\{W_n\}_{n\geq 0}$ be a hierarchy with $W_0 = 1$, where W_n is a continuous linear functional on $\mathcal{S}(M^n, E_I^{\otimes n})$, and where E_I carries the label adjunction and parity data used in Hermiticity and locality. Equivalently, on every finite field-label subspace, each component distribution is bounded by finitely many Schwartz seminorms: after choosing scalar components F_j of a test function F, there are $C, N < \infty$ such that*

$$|W_n(F)| \leq C \sum_j \sum_{|a|,|b|\leq N} \sup_{x\in M^n} |x^a \partial^b F_j(x)|.$$

Suppose the hierarchy satisfies Poincare covariance, the spectral support condition, Hermiticity, locality, and positivity. Then there exists a cyclic Wightman field presentation in the sense of Definition 2.1,

$$(\mathcal{H}, \Omega, U, \mathcal{D}, E_I, J, |\cdot|, \{\widehat{\Phi}_\alpha\}_{\alpha\in I}),$$

with the following properties:

(i) *Its vacuum correlation functions are exactly $\{W_n\}$.*

(ii) *The dense domain is the polynomial field orbit of the vacuum,*

$$\mathcal{D} = \operatorname{span}\{\widehat{\Phi}_{\alpha_1}(f_1)\cdots\widehat{\Phi}_{\alpha_n}(f_n)\Omega \mid n \geq 0,\ \alpha_j \in I,\ f_j \in \mathcal{S}(M)\}.$$

In particular, Ω is cyclic for the reconstructed polynomial field algebra.

(iii) *U is a strongly continuous unitary representation of $\mathcal{P}_+^\uparrow$, the fields are covariant operator-valued tempered distributions on the common domain $\mathcal{D}$, and the spectrum, adjoint, and locality properties specified above hold on $\mathcal{D}$.*

The cyclic reconstruction satisfying these properties is unique up to the canonical unitary induced by identifying finite test-function sequences with the same Wightman inner products.

Proof. The construction begins with formal vectors

$$[\alpha_1, f_1; \ldots; \alpha_n, f_n]$$

and defines an inner product by the Wightman distributions. Positivity makes this a positive semidefinite form. Dividing by zero-norm vectors and completing gives $\mathcal{H}$. Fields act by left multiplication on the dense subspace of finite sequences, and the Poincare action is induced by covariance of the hierarchy.

More explicitly, let $\mathcal{V}$ be the vector space of finite sequences

$$F = (f_0, f_1, f_2, \ldots),$$

where f_n is a Schwartz test function on M^n valued in the n-fold field-label space and only finitely many components are nonzero. Define

$$\langle F, G\rangle_W = \sum_{m,n} W_{m+n}(f_m^\dagger \otimes g_n). \tag{2.10}$$

Let

$$\mathcal{N}_W = \{F \in \mathcal{V} \mid \langle F, F\rangle_W = 0\}.$$

The dense domain $\mathcal{D}$ is the image of $\mathcal{V}/\mathcal{N}_W$ inside its Hilbert-space completion. For an E_I-valued one-field test function h, the reconstructed smeared field is

$$\widehat{\Phi}(h)[F] = [h \otimes F],$$

where $h \otimes F$ means left insertion of the one-field test function before the sequence F. In components, $\widehat{\Phi}_\alpha(k)[F] = [(ke_\alpha) \otimes F]$. Positivity and the Wightman identities ensure that this action is well defined on the null quotient and that the vacuum vector is the class of $(1, 0, 0, \ldots)$. The Poincare representation acts by transforming all test functions; covariance of $\{W_n\}$ makes this action unitary.

The same finite-sequence construction gives cyclicity. Starting with the vacuum class $\Omega = [(1, 0, 0, \ldots)]$, repeated left insertion gives

$$\widehat{\Phi}(h_1) \cdots \widehat{\Phi}(h_n)\Omega = [h_1 \otimes \cdots \otimes h_n],$$

and finite sums of such vectors are precisely the image of $\mathcal{V}$ in the quotient. Since $\mathcal{D}$ is defined as this image, $\mathcal{D}$ is the polynomial field orbit of Ω, and its Hilbert-space closure is all of $\mathcal{H}$.

The same formula verifies the adjoint inclusion on the reconstructed common domain. For an E_I-valued one-field test function h and finite sequences F, G,

$$\langle \widehat{\Phi}(h)F, G\rangle_W = \langle F, \widehat{\Phi}(h^\dagger)G\rangle_W.$$

Indeed, inserting h on the left of F and then applying the dagger in (2.10) moves h to the right place with the label adjunction and complex conjugation fixed above. After quotienting by $\mathcal{N}_W$, this identity says precisely that

$$\widehat{\Phi}(h)^* \supset \widehat{\Phi}(h^\dagger)$$

on the dense finite-sequence domain.

Covariance of $\{W_n\}$ makes the induced Poincare action on $\mathcal{V}/\mathcal{N}_W$ isometric. Its completion gives the strongly continuous unitary representation U. Strong continuity follows because the Poincare action is strongly continuous on Schwartz test functions and the finite-sequence vectors are dense. The spectral condition for U follows from the stated support condition on the Fourier transforms of the Wightman distributions: if F, G are finite sequences, the distribution $a \mapsto \langle F, U(a)G\rangle_W$ has Fourier support in the closed forward cone after passing to translation-difference variables, hence the joint spectral measure of U is supported in $\overline{V}_+$.

Locality is also inherited at the finite-sequence level. If the supports of two one-field test functions are spacelike separated, the locality axiom for the hierarchy equates the corresponding Wightman functionals after graded interchange. Therefore the difference between the two graded-ordered field products has zero matrix element against every finite sequence. Positivity then puts that difference in the null subspace, so the graded commutator vanishes as an operator on $\mathcal{D}$. The same argument applies to polynomial products by iterating the adjacent interchange.

Finally, if two cyclic reconstructions are built from the same hierarchy, the map sending a finite test-function sequence in the first reconstruction to the class of the same sequence in the second preserves all inner products by (2.10). It descends through the same null quotient, has dense range, intertwines the fields and the Poincare action on finite sequences, and therefore extends uniquely to the canonical unitary equivalence. □

For a hierarchy satisfying the reconstruction hypotheses, the reconstructed translation representation has a zero-momentum projection P_0. Adding the weak cluster condition (2.4) selects exactly the case $P_0\mathcal{H} = \mathbb{C}\Omega$ by Theorem 2.3. Thus clustering is the correlation-function condition for reconstructing a pure vacuum sector.

The uniqueness statement is internal to the specified hierarchy and its field labels. It says that two cyclic reconstructions of the same $\{W_n\}$ have the same pre-Hilbert space of finite test-function sequences modulo null vectors, hence the same vacuum matrix elements for every finite product of field insertions. This is a uniqueness statement about the Wightman hierarchy and the cyclic field presentation it generates. It does not identify different choices of field coordinatization whose local observable algebras may agree only after additional closure, affiliation, or quotient constructions. Those comparisons belong to the relation between Wightman fields and local nets described above and in Chapter 3.

Figure 2.2 organizes the reconstruction proof: finite test-function sequences are quotiented by null vectors, completed to $\mathcal{H}_W$, and the fields act by left insertion on the common dense domain.

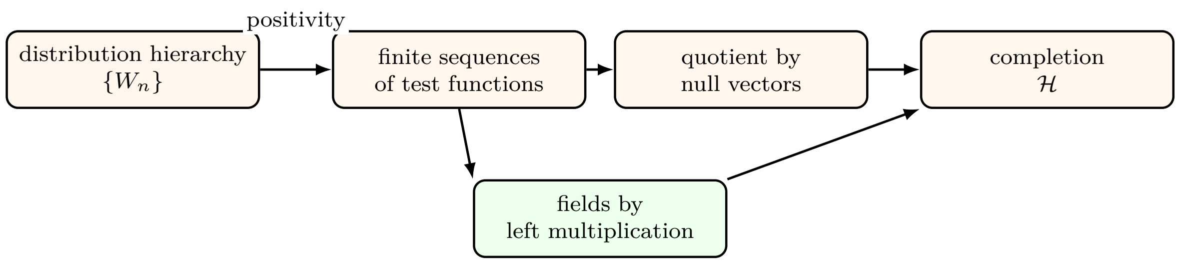


Figure 2.2: Wightman reconstruction builds the Hilbert space and fields from the positive correlation hierarchy.

2.6 Analytic Continuation From Spectral Positivity

For scalar fields, write the Wightman function in difference variables as

$$W_n(\xi_1, \ldots, \xi_{n-1}).$$

The support condition for the Fourier transform implies that W_n is the boundary value of a holomorphic function on the forward tube

$$T_{n-1} = \{z_j = \xi_j - \mathrm{i}\eta_j \in \mathbb{C}^D \mid \eta_j \in V_+,\ j = 1, \ldots, n-1\}.$$

Lorentz covariance enlarges the domain to the extended tube obtained by the complex Lorentz group. Locality then identifies the analytic continuations associated with permutations at Jost configurations, where all relevant real linear combinations are spacelike.

Proposition 2.5 (Wightman tube boundary-value package)**.** *Fix a component of an n-point Wightman distribution in a cyclic Wightman field presentation, and write it in difference variables $\xi = (\xi_1, \ldots, \xi_{n-1})$ as*

$$w_n(\xi) \in \mathcal{S}'(M^{n-1}).$$

Use the Fourier convention

$$w_n(\xi) = \int \frac{\mathrm{d}^{D(n-1)}p}{(2\pi)^{D(n-1)}} \exp\left(\mathrm{i}\sum_{j=1}^{n-1} p_j \cdot \xi_j\right) \widetilde{w}_n(p)$$

in distribution notation. The spectrum condition gives

$$\operatorname{supp} \widetilde{w}_n \subset \overline{V}_+^{\,n-1}.$$

For $z_j = \xi_j - \mathrm{i}\eta_j$, $\eta_j \in V_+$, the Fourier–Laplace transform

$$\mathcal{W}_n(z) = \frac{1}{(2\pi)^{D(n-1)}} \left\langle \widetilde{w}_n(p), \exp\left(\mathrm{i}\sum_{j=1}^{n-1} p_j \cdot z_j\right)\right\rangle$$

is holomorphic in T_{n-1} and has w_n as its distributional boundary value. If the fields transform in finite-dimensional Lorentz representations that extend holomorphically to the complexified spin group, Lorentz covariance continues $\mathcal{W}_n$ to the extended tube. Finally, if two adjacent orderings differ only by exchanging homogeneous fields whose coordinates are spacelike separated on a Jost real edge, their tube boundary values agree there up to the Koszul sign, and the distributional edge-of-the-wedge theorem gives a common holomorphic continuation across that edge.

Proof. Insert the joint spectral resolution of translations between successive fields. In the relative-coordinate matrix element, the Fourier variables are the momenta transported by the intermediate states between adjacent field insertions. The Wightman spectrum condition places every such momentum in $\overline{V}_+$, giving the displayed support inclusion for $\widetilde{w}_n$.

For $p \in \overline{V}_+$ and $\eta \in V_+$, the mostly-plus metric gives $p \cdot \eta \leq 0$. Hence

$$\exp\bigl(\mathrm{i}p \cdot (\xi - \mathrm{i}\eta)\bigr) = \exp(\mathrm{i}p \cdot \xi + p \cdot \eta)$$

is exponentially damped on closed subcones inside the tube direction. For a tempered distribution carried by $\overline{V}_+^{\,n-1}$, multiplication by the damping factor is defined by inserting a smooth conic cutoff equal to one near the support; independence of the cutoff follows because two such cutoffs differ away from the support. Differentiating with respect to any z_j only inserts polynomial factors in p_j, still controlled by the same damping on proper subtubes. This proves holomorphy of $\mathcal{W}_n$. The same finite-order Schwartz seminorm bound for $\widetilde{w}_n$, applied after the conic cutoff and differentiated finitely many times, gives the standard polynomial normal-growth estimate on each proper subtube; this is the growth hypothesis needed for distributional boundary values and for the edge-of-the-wedge step below.

To identify the boundary value, pair $\mathcal{W}_n(\xi - \mathrm{i}\eta)$ with a Schwartz test function $f(\xi)$. In Fourier variables this is the pairing of $\widetilde{w}_n$ with the Schwartz function $\check{f}(p)$ multiplied by $\exp(\sum_j p_j \cdot \eta_j)$ and by the same harmless conic cutoff. As $\eta \to 0$ within V_+^{n-1}, these multipliers converge to one in the C^∞ multiplier topology on every conic neighborhood of the support after the cutoff. Continuity of the tempered distribution therefore gives the limit $\langle w_n, f\rangle$.

For the covariance enlargement, first take real proper orthochronous Lorentz transformations. Covariance of the fields identifies the transformed boundary distributions with the original distribution acted on by the finite field-label matrices. Both sides have holomorphic tube continuations.

Since the finite-dimensional spin/tensor representation matrices are polynomial or holomorphic functions of the complexified Lorentz parameters, the identity extends by the identity theorem along the connected complexified Lorentz orbit as long as the transformed point remains in the holomorphy envelope. This is the extended-tube construction.

It remains to identify where locality enters. Let O be a real open set of noncoincident configurations on which the exchanged adjacent coordinates are spacelike separated. A test function supported in O can be written, by a partition of unity, as a finite sum of product-smearing pieces whose two exchanged one-field supports are spacelike separated. Microcausality on the common Wightman domain gives equality of the two smeared matrix elements, with sign $(-1)^{|a||b|}$ for homogeneous odd/even labels. Thus the two permuted tube functions have equal distributional boundary values on the Jost edge. The analytic theorem used at this last step is the distributional edge-of-the-wedge theorem stated and proved in Theorem 16.6: polynomial tube bounds come from the preceding Fourier–Laplace construction, and the boundary equality is the microcausality statement just verified. Therefore the permuted holomorphic functions glue across the edge. □

Figure 2.3 sketches a two-dimensional slice of the complexified difference variable. The real edge carries the boundary values, the forward tube is obtained by writing the difference variables as $z = \xi - \mathrm{i}\eta$ with $\eta \in V_+$, and the Jost configurations are real spacelike boundary points at which locality identifies different permutation-ordered boundary values.

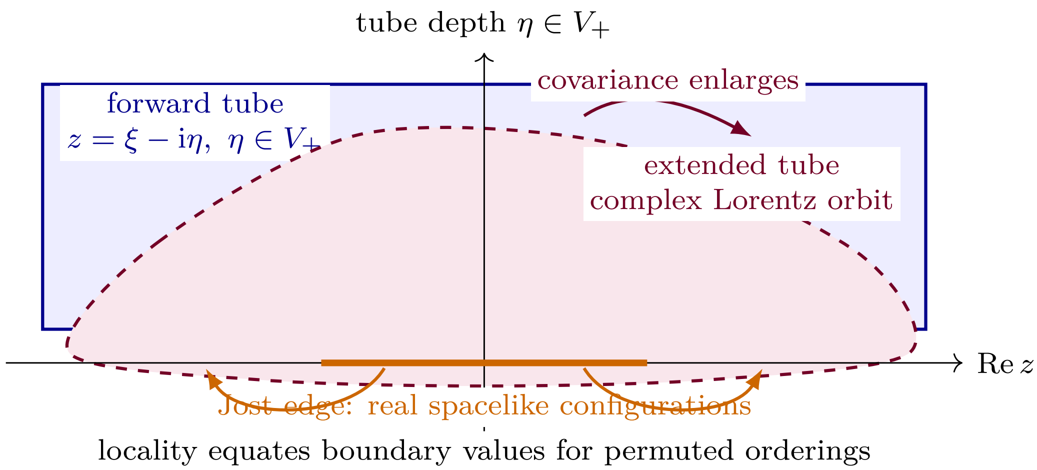


Figure 2.3: A slice of Wightman tube analyticity. Spectral support gives the mostly-plus forward tube $z = \xi - \mathrm{i}\eta$, $\eta \in V_+$, covariance enlarges it to the extended tube, and locality identifies the boundary values reached at real Jost configurations.

The Euclidean Schwinger functions are restrictions of these analytic functions to noncoincident imaginary-time configurations. The converse passage from Euclidean data to Lorentzian Wightman data requires reflection positivity and growth hypotheses. That converse is the subject of the next chapter.

2.7 The PCT Theorem

The theorem usually called CPT in particle language is most naturally a PCT theorem in the Wightman setting: it is an antiunitary implementation of the total spacetime inversion $x \mapsto -x$, together with the corresponding conjugation of field labels. Charge conjugation is not assumed as a separate unitary symmetry. It appears because the field adjunction sends a charged field to the field carrying the conjugate charge.

Let $D = 4$ in this section, and let the finite field-label space be closed under the adjoint operation fixed in (2.1). For each covariant field multiplet, the finite-dimensional representation of $\mathrm{Spin}^+(1,3)$ and the adjoint label map determine, after a choice of basis, an antilinear test-function operation

$$h \longmapsto h^\Theta, \qquad h^\Theta(x) = C_{\mathrm{PCT}}\,\overline{h(-x)}.$$

Here C_{PCT} is a finite matrix on field labels. It contains the Lorentz spinor/tensor matrix required by the representation and the internal conjugation sending a field to its adjoint charged multiplet. The displayed formula is a definition of notation: if the original field list is not closed under this operation, one first enlarges it by adjoining the conjugate fields.

Theorem 2.6 (Wightman PCT theorem)**.** Proof references.[1] *Let*

$$(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_\alpha\})$$

be the Hilbert-space and field part of a four-dimensional cyclic Wightman field presentation satisfying Definition 2.1. Assume also that the field-label family is stable under the PCT test-function operation $h \mapsto h^\Theta$ just described. Then there exists an antiunitary operator

$$\Theta_{\mathrm{PCT}} : \mathcal{H} \to \mathcal{H}$$

with the following properties.

(i) *Vacuum invariance:*

$$\Theta_{\mathrm{PCT}}\Omega = \Omega.$$

(ii) *Poincare covariance extended by spacetime inversion:*

$$\Theta_{\mathrm{PCT}}\, U(a,\Lambda)\, \Theta_{\mathrm{PCT}}^{-1} = U(-a,\Lambda), \qquad (a,\Lambda) \in \mathbb{R}^4 \rtimes SO^+(1,3). \tag{2.11}$$

Equivalently, $\Theta_{\mathrm{PCT}} P_\mu \Theta_{\mathrm{PCT}}^{-1} = P_\mu$, while the antiunitarity of Θ_{PCT} sends $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ to $U(-a)$.

(iii) *Distributional field covariance:*

$$\Theta_{\mathrm{PCT}}\, \widehat{\Phi}(h)\, \Theta_{\mathrm{PCT}}^{-1} = \widehat{\Phi}(h^\Theta) \tag{2.12}$$

as an identity of operators on the transformed common domain. In point-field notation this says, distributionally,

$$\Theta_{\mathrm{PCT}}\, \widehat{\Phi}_\alpha(x)\, \Theta_{\mathrm{PCT}}^{-1} = C_{\mathrm{PCT},\alpha}{}^{\beta}\widehat{\Phi}^\dagger_\beta(-x),$$

with the finite spin and charge-conjugation matrix appropriate to the field multiplet.

[1]The theorem is due to Jost and to Pauli–Luders in the Wightman framework. Standard mathematical proofs are given in R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That*, Benjamin, 1964, and in R. Jost, *The General Theory of Quantized Fields*, American Mathematical Society, 1965 [2, 3].

(iv) *Vacuum correlation PCT identity. For*

$$W_n(h_1,\dots,h_n)=\langle\Omega,\widehat{\Phi}(h_1)\cdots\widehat{\Phi}(h_n)\Omega\rangle,$$

one has

$$W_n(h_1,\dots,h_n)=\overline{W_n(h_1^\Theta,\dots,h_n^\Theta)}=W_n((h_n^\Theta)^\dagger,\dots,(h_1^\Theta)^\dagger). \tag{2.13}$$

The second equality is the adjoint reversal already present in the Wightman Hermiticity condition.

No mass gap is required for this theorem. A mass gap is an additional hypothesis for particle sectors, Haag–Ruelle scattering, and many DHR superselection statements, but not for the PCT antiunitary itself.

Proof. The proof has two logically distinct parts: first a distributional identity for the Wightman functions, and then the construction of the antiunitary operator from that identity.

Write the n-point functions in translation-difference variables. The spectrum condition makes them boundary values of holomorphic functions in the forward tube. Proper Lorentz covariance extends these functions to the extended tube, obtained by acting with the complex Lorentz group. In four dimensions the complex Lorentz group contains a path which implements the total spacetime inversion on Jost configurations, with the corresponding finite spin matrix acting on the field labels. A Jost configuration is a real configuration for which all relevant convex linear combinations of differences are spacelike; in such a configuration the fields may be reordered by graded locality. Applying locality to reverse the order at the Jost edge, and then using the edge-of-the-wedge theorem, gives the analytic identity whose boundary value is precisely (2.13). This is the Jost form of the PCT theorem: weak local commutativity at Jost points and the PCT identity are equivalent under the remaining Wightman analyticity hypotheses.

Now pass from the identity of distributions to Hilbert space. On the dense finite-sequence domain used in the reconstruction proof, define an antilinear map by

$$\Theta_0\,[h_1;\dots;h_n]=[h_1^\Theta;\dots;h_n^\Theta],\qquad \Theta_0[(1,0,0,\dots)]=(1,0,0,\dots),$$

and extend by antilinearity to finite sums. Equation (2.13), together with the Wightman inner product (2.10), implies

$$\langle\Theta_0F,\Theta_0G\rangle_W=\overline{\langle F,G\rangle_W}.$$

Thus Θ_0 maps null vectors to null vectors and descends to an anti-isometry of the quotient pre-Hilbert space. Its range contains the same finite polynomial domain because $h\mapsto h^\Theta$ is an involutive operation on the enlarged field-label space. Hence it extends uniquely to an antiunitary Θ_{PCT} on $\mathcal{H}$.

The vacuum vector is the class of the empty sequence, so it is fixed. The field identity (2.12) follows from left insertion:

$$\Theta_0\,\widehat{\Phi}(h)[F]=\Theta_0[h;F]=[h^\Theta;\Theta_0F]=\widehat{\Phi}(h^\Theta)\Theta_0[F].$$

The translation identity follows from $(U(a)h)^\Theta=U(-a)h^\Theta$ on test functions, with antiunitarity accounting for complex conjugation of the phase. The proper Lorentz identity follows in the same way from the covariance of C_{PCT} with the finite-dimensional spin representation. These identities extend from the dense domain to the Hilbert-space operators wherever the latter are bounded, and as domain identities for the unbounded field operators. □

2.8 Observable Subpresentations in Gauge Theories

A gauge construction enters the positive Hilbert-space Wightman framework through a specified observable sector. The object to verify is the following positive local field presentation.

Definition 2.7 (observable Wightman subpresentation)**.** Suppose a gauge construction supplies a positive physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$, a vacuum vector Ω_{phys}, a unitary Poincare representation U_{phys}, a common dense domain $\mathcal{D}_{\mathrm{phys}}$, and a local net of physical von Neumann algebras $\mathcal{R}_{\mathrm{phys}}(\mathcal{O})$. A family of gauge-invariant local distributions $\{O_\gamma\}_{\gamma\in J}$ is an observable Wightman subpresentation if:

(i) each $O_\gamma(f)$, $f \in \mathcal{S}(M)$, is an operator on $\mathcal{D}_{\mathrm{phys}}$ and the map $f \mapsto O_\gamma(f)$ is a tempered operator-valued distribution;

(ii) the data $(\mathcal{H}_{\mathrm{phys}}, \Omega_{\mathrm{phys}}, U_{\mathrm{phys}}, \mathcal{D}_{\mathrm{phys}}, \{O_\gamma\})$, with its observable label adjunction and parities, satisfy Definition 2.1 on $\mathcal{H}_{\mathrm{phys}}$;

(iii) for every open region $\mathcal{O}$, all $O_\gamma(f)$ with $\operatorname{supp} f \subset \mathcal{O}$ are affiliated with $\mathcal{R}_{\mathrm{phys}}(\mathcal{O})$, and the local von Neumann algebra generated by their bounded functional calculi is the specified observable subnet under discussion.

In a BRST realization, the auxiliary space $\mathcal{K}$ carries an odd nilpotent operator Q and the physical Hilbert space is obtained, after the required completion and quotient by null vectors, from

$$\mathcal{H}_{\mathrm{phys}} = \ker Q \,/\, \overline{\operatorname{im} Q}.$$

A local auxiliary operator B descends to a physical local observable when it preserves $\ker Q$ and $\overline{\operatorname{im} Q}$; a sufficient algebraic condition on a common invariant domain is

$$[Q, B]_{\mathrm{gr}} = 0.$$

Operators differing by a graded commutator $[Q, C]_{\mathrm{gr}}$ induce the same operator on BRST cohomology, provided the same domain and closure hypotheses hold. The Wightman functions to which the reconstruction theorem applies are therefore the vacuum matrix elements of the descended observable fields $\{O_\gamma\}$ on $\mathcal{H}_{\mathrm{phys}}$; auxiliary representatives enter through their cohomology classes.

Thus the Wightman framework supplies precise theorems for local observable fields, spectral positivity, analytic continuation, and reconstruction. Its output is Hilbert-space QFT data with operator-valued distributions and a distinguished vacuum representation; comparison between gauges is a comparison of the resulting observable nets or observable Wightman subpresentations.

Chapter 3

Algebraic QFT: Local Nets and States

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter uses local nets as the invariant operator-algebraic object and treats fields, Euclidean measures, perturbative functionals, and gauge variables as different ways of presenting or constructing that object. The logical order matters. First one specifies an abstract C^*-net, before a state has selected a Hilbert-space representation and before weak operator closure has any meaning. A state then gives a represented von Neumann net by the GNS construction. Only after that represented bounded net is in place can one compare it with unbounded Wightman coordinates, field nets, gauge fixed-points, or constructive model estimates. Thus the chapter does not use "field" and "local algebra" as interchangeable words: the former is a coordinate presentation, while the latter is the bounded localization datum whose equality, duality, split, and sector properties have to be proved.

The proof obligations are staged accordingly. The free Weyl net is a direct bounded-algebra benchmark: locality, additivity, and the time-slice property are verified inside the C^*-net itself. The Wightman comparison is conditional: self-adjoint closures, spectral-projection locality, and operator-theoretic additivity are additional inputs needed to promote domain-level field relations into a represented local net. Gauge or charged coordinates introduce a second comparison, from a field net to the observable fixed-point or gauge-invariant subnet, where fixed points and additivity must be checked rather than inferred from the presence of neutral expressions. Constructive examples such as $P(\phi)_2$ and the Schwinger model are used as estimate-to-property checkpoints: they show which analytic estimates or exact bosonizations actually supply the bounded local net, and which stronger AQFT properties, such as nuclearity, split inclusions, Haag duality, and DHR reconstruction, remain separate theorem boundaries.

3.1 Local Observable Nets

Let $M = \mathbb{M}^D$ be Minkowski spacetime, and let $\mathcal{K}(M)$ be a chosen class of open spacetime regions, for example double cones or bounded causally complete regions. The connected Poincare group is denoted

$$\mathcal{P}_+^\uparrow = \mathbb{R}^D \rtimes SO^+(1, D-1).$$

If $\mathcal{O}_1, \mathcal{O}_2 \in \mathcal{K}(M)$, the notation $\mathcal{O}_1 \perp \mathcal{O}_2$ means that every point of $\mathcal{O}_1$ is spacelike separated from every point of $\mathcal{O}_2$. If $N \subset M$, $D(N)$ denotes its domain of dependence.

Definition 3.1 (Haag–Kastler C^*-net)**.** A Haag–Kastler C^*-net on M, indexed by $\mathcal{K}(M)$, consists of the following data.

(i) For each $\mathcal{O} \in \mathcal{K}(M)$, a unital C^*-algebra $\mathcal{A}(\mathcal{O})$.

(ii) For every inclusion $\mathcal{O}_1 \subset \mathcal{O}_2$, an injective unital $*$-homomorphism $\iota_{\mathcal{O}_2\mathcal{O}_1} : \mathcal{A}(\mathcal{O}_1) \to \mathcal{A}(\mathcal{O}_2)$ compatible with composition. After using these embeddings, we write $\mathcal{A}(\mathcal{O}_1) \subset \mathcal{A}(\mathcal{O}_2)$.

(iii) A strongly continuous action $g \mapsto \alpha_g$ of $\mathcal{P}_+^\uparrow$ by $*$-automorphisms of the quasilocal algebra

$$\mathcal{A}_{\mathrm{qloc}} = \overline{\bigcup_{\mathcal{O} \in \mathcal{K}(M)} \mathcal{A}(\mathcal{O})}^{\|\cdot\|}.$$

The data obey:

(HK1) *Isotony:* the inclusions are the embeddings in (ii).

(HK2) *Einstein causality:* if $\mathcal{O}_1 \perp \mathcal{O}_2$, then

$$[A, B] = 0, \qquad A \in \mathcal{A}(\mathcal{O}_1), \quad B \in \mathcal{A}(\mathcal{O}_2).$$

(HK3) *Poincare covariance:*

$$\alpha_g(\mathcal{A}(\mathcal{O})) = \mathcal{A}(g\mathcal{O}), \qquad g \in \mathcal{P}_+^\uparrow.$$

(HK4) *Time-slice property:* whenever $N \in \mathcal{K}(M)$ contains a Cauchy surface for $D(N)$ and $D(N) \in \mathcal{K}(M)$, the inclusion

$$\mathcal{A}(N) \longrightarrow \mathcal{A}(D(N))$$

is an isomorphism.

(HK5) *Additivity:* if $\mathcal{O} = \bigcup_i \mathcal{O}_i$ with $\mathcal{O}_i, \mathcal{O} \in \mathcal{K}(M)$, then $\mathcal{A}(\mathcal{O})$ is the C^*-subalgebra generated by the images of the $\mathcal{A}(\mathcal{O}_i)$.

The definition is the C^*-algebraic form of the Haag–Kastler locality framework [18]. Its axioms are stated here as data, not as a consequence of a field presentation; the relation to Wightman fields is a separate conditional construction below.

Throughout this chapter $\mathcal{A}(\mathcal{O})$ denotes the abstract C^*-local algebra generated by observables localized inside $\mathcal{O}$, before a Hilbert-space representation has been chosen. The norm closure in $\mathcal{A}_{\mathrm{qloc}}$ is part of the abstract C^*-algebraic presentation. Weak operator closures belong to represented von Neumann algebras and are introduced only after a state has selected a representation.

Figure 3.1 records the distinction made in this definition: the abstract local net assigns C^*-algebras to regions before any Hilbert-space state or weak operator closure has been chosen.

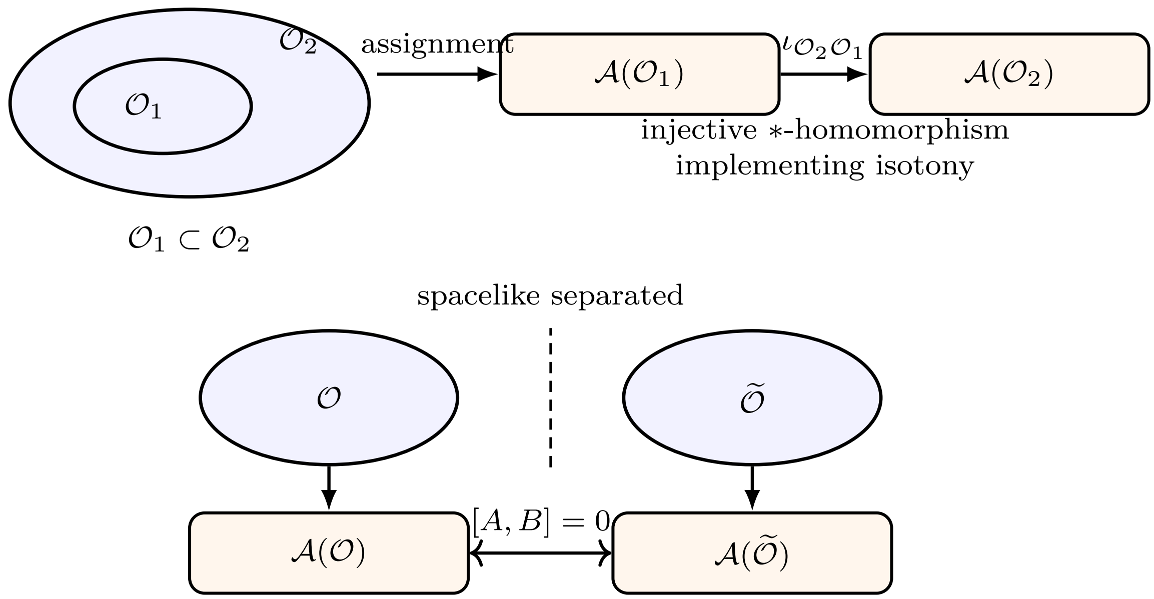


Figure 3.1: The local net is an isotone assignment from suitable spacetime regions to abstract C^*-algebras. Inclusion of regions is represented by the isotony morphism $\iota_{\mathcal{O}_2\mathcal{O}_1}$, while spacelike separation is represented algebraically by commutation of the corresponding observable algebras.

Benchmark model: the massive scalar Weyl net. Before entering perturbative or constructive interacting examples, it is useful to have one completely explicit local net in view. Let

$$P_m = \partial^\mu \partial_\mu - m^2$$

be the mostly-plus Klein–Gordon operator on $M = \mathbb{M}^D$, with $m > 0$, and let

$$\Delta_m^{\mathrm{ret}}, \qquad \Delta_m^{\mathrm{adv}}$$

be the retarded and advanced fundamental solutions:

$$P_m \Delta_m^{\mathrm{ret/adv}} f = f, \qquad \Delta_m^{\mathrm{ret/adv}} P_m f = f, \qquad \operatorname{supp} \Delta_m^{\mathrm{ret/adv}} f \subset J^\pm(\operatorname{supp} f)$$

for every $f \in C_c^\infty(M;\mathbb{R})$. The causal propagator is

$$\Delta_m = \Delta_m^{\mathrm{ret}} - \Delta_m^{\mathrm{adv}}.$$

Green hyperbolicity gives the exact sequence

$$C_c^\infty(M;\mathbb{R}) \xrightarrow{P_m} C_c^\infty(M;\mathbb{R}) \xrightarrow{\Delta_m} C_{\mathrm{sc}}^\infty(M;\mathbb{R}),$$

where C^∞_{sc} denotes smooth solutions with spatially compact support. Exactness at the middle term says

$$\ker \Delta_m = P_m C^\infty_c(M;\mathbb{R}).$$

Thus the real vector space

$$\mathsf{V}_m := C^\infty_c(M;\mathbb{R})\big/P_m C^\infty_c(M;\mathbb{R}) \tag{3.1}$$

is the test-function coordinatization of the classical solution space. For $[f],[g] \in \mathsf{V}_m$ define

$$\sigma_m([f],[g]) = \int_M f(x)\,(\Delta_m g)(x)\,\mathrm{d}^D x. \tag{3.2}$$

This is well defined because P_m is formally self-adjoint and $P_m\Delta_m = \Delta_m P_m = 0$ on test functions. It is antisymmetric since $\Delta_m(x,y) = -\Delta_m(y,x)$. It is nondegenerate on the quotient: if $\sigma_m([f],[g]) = 0$ for all g, then the distribution $\Delta_m f$ pairs trivially with all compactly supported test functions, so $\Delta_m f = 0$, hence $f \in P_m C^\infty_c(M;\mathbb{R})$ by exactness.

The Weyl C^*-algebra

$$\mathfrak{W}(\mathsf{V}_m, \sigma_m)$$

is the universal C^*-algebra generated by symbols $W(u)$, $u \in \mathsf{V}_m$, subject to

$$W(0) = 1, \qquad W(u)^* = W(-u), \qquad W(u)W(v) = \mathrm{e}^{-\frac{\mathrm{i}}{2}\sigma_m(u,v)}W(u+v). \tag{3.3}$$

The sign in (3.3) matches the field normalization of Volume I: if $W([f]) = \exp(\mathrm{i}\widehat{\phi}(f))$ in the Fock representation, then

$$[\widehat{\phi}(f), \widehat{\phi}(g)] = \mathrm{i}\sigma_m([f],[g])\,1,$$

and the Baker–Campbell–Hausdorff formula gives the displayed Weyl relation. For an open region $\mathcal{O}$, let

$$\mathsf{V}_m(\mathcal{O}) = \mathrm{image}\big(C^\infty_c(\mathcal{O};\mathbb{R}) \longrightarrow \mathsf{V}_m\big),$$

and define

$$\mathcal{A}_m(\mathcal{O}) = C^*\big(W(u) \mid u \in \mathsf{V}_m(\mathcal{O})\big) \subset \mathfrak{W}(\mathsf{V}_m, \sigma_m). \tag{3.4}$$

The Haag–Kastler properties can now be checked without appealing to unbounded fields. Isotony follows immediately from $C^\infty_c(\mathcal{O}_1) \subset C^\infty_c(\mathcal{O}_2)$. If $\mathcal{O}_1 \perp \mathcal{O}_2$, then $\operatorname{supp}\Delta_m g \subset J(\operatorname{supp} g)$ is disjoint from $\operatorname{supp} f$ for $f \in C^\infty_c(\mathcal{O}_1)$, $g \in C^\infty_c(\mathcal{O}_2)$. Hence $\sigma_m([f],[g]) = 0$, and the Weyl relations give

$$W([f])W([g]) = W([g])W([f]).$$

Since the local algebras are C^*-closures of the algebras generated by such Weyl elements, spacelike separated local algebras commute. Poincare covariance is induced by

$$\alpha_g\big(W([f])\big) = W([f \circ g^{-1}]), \qquad g \in \mathcal{P}^\uparrow_+,$$

where g acts on test functions by pullback. The covariance of P_m, $\Delta_m^{\mathrm{ret/adv}}$, and the volume form preserves the quotient and the symplectic form.

Additivity follows from compact support. Suppose $\mathcal{O} = \bigcup_i \mathcal{O}_i$, and let $f \in C^\infty_c(\mathcal{O};\mathbb{R})$. The compact set $\operatorname{supp} f$ is covered by finitely many $\mathcal{O}_i$'s. Choose a smooth partition of unity $\{\rho_j\}_{j=1}^r$ subordinate to this finite cover on a neighborhood of $\operatorname{supp} f$. Then

$$f = \sum_{j=1}^r f_j, \qquad f_j = \rho_j f, \qquad \operatorname{supp} f_j \subset \mathcal{O}_{i_j}.$$

In $\mathsf{V}_m(\mathcal{O})$ this gives $[f] = \sum_j [f_j]$. Iterating the Weyl relation (3.3) expresses $W([f])$, up to an explicit phase determined by the pairings $\sigma_m([f_j],[f_k])$, as a product of the $W([f_j])$'s. Hence every generator of $\mathcal{A}_m(\mathcal{O})$ belongs to the C^*-algebra generated by the $\mathcal{A}_m(\mathcal{O}_i)$'s, and the opposite inclusion is isotony. The phase just mentioned is not an additional analytic hypothesis. In any finite-dimensional symplectic vector space $(V_{\mathrm{fin}}, \sigma)$ with Weyl relations

$$W(u)W(v) = \exp\left(-\frac{\mathrm{i}}{2}\sigma(u,v)\right) W(u+v),$$

one has

$$W(u)W(v) = \exp[-\mathrm{i}\sigma(u,v)]\, W(v)W(u), \qquad W(u)W(v)W(u)^*W(v)^* = \exp[-\mathrm{i}\sigma(u,v)]\, 1.$$

Thus vanishing of the symplectic pairing is exactly the finite algebraic mechanism behind local commutation. Likewise, for a finite decomposition $u = \sum_{j=1}^r u_j$,

$$W(u) = \exp\left(\frac{\mathrm{i}}{2} \sum_{1\le j<k\le r} \sigma(u_j, u_k)\right) W(u_1)\cdots W(u_r).$$

The continuum additivity proof above supplies the support-theoretic decomposition $[f] = \sum_j [f_j]$ in the Green-hyperbolic quotient; the displayed finite cocycle identity is the bounded-algebra step that converts that decomposition into containment of local C^*-algebras.

The time-slice property is the only axiom in this model that uses a real hyperbolic argument. Let $U = D(N)$, and assume the chosen region class is regular enough that $N \subset U$ contains a Cauchy-surface collar: there is a smooth Cauchy time function τ on U and numbers $\tau_- < \tau_+$ for which $\tau^{-1}([\tau_-,\tau_+]) \subset N$. Choose $\chi \in C^\infty(U)$ with $\chi = 0$ for $\tau < \tau_-$ and $\chi = 1$ for $\tau > \tau_+$. For $f \in C_c^\infty(U;\mathbb{R})$, set

$$\varphi = \Delta_m f, \qquad f_N = P_m(\chi\varphi).$$

Because $P_m\varphi = 0$, the support of f_N lies where derivatives of χ are nonzero, hence in N. To compare the quotient classes, note that $\chi\varphi$ vanishes in the far past and solves

$$P_m(\chi\varphi) = f_N;$$

by uniqueness of the retarded solution it is $\Delta_m^{\mathrm{ret}} f_N$. Likewise $(\chi - 1)\varphi$ vanishes in the far future and solves

$$P_m\big((\chi-1)\varphi\big) = f_N;$$

by uniqueness of the advanced solution it is $\Delta_m^{\mathrm{adv}} f_N$. Therefore

$$\Delta_m f_N = \Delta_m^{\mathrm{ret}} f_N - \Delta_m^{\mathrm{adv}} f_N = \chi\varphi - (\chi-1)\varphi = \varphi = \Delta_m f.$$

Exactness of the Green-hyperbolic sequence then gives $[f] = [f_N]$ in the quotient associated with U. Since $\mathsf{V}_m(N)$ and $\mathsf{V}_m(U)$ were defined as image subspaces of the global quotient, this proves equality of image subspaces,

$$\mathsf{V}_m(N) = \mathsf{V}_m(U).$$

The generated Weyl algebras are therefore identical:

$$\mathcal{A}_m(N) = \mathcal{A}_m(D(N)).$$

The usual vacuum representation is a state on this abstract net, not part of the net definition. The positive-frequency two-point distribution Δ_+ defines a real covariance

$$\mu_m([f],[g]) = \mathrm{Re}\int_{M\times M} f(x)\Delta_+(x-y)g(y)\,\mathrm{d}^D x\,\mathrm{d}^D y$$

satisfying

$$|\sigma_m(u,v)|^2 \le 4\,\mu_m(u,u)\mu_m(v,v).$$

Consequently

$$\omega_0(W(u)) = \exp\left[-\frac{1}{2}\mu_m(u,u)\right]$$

is a regular quasi-free state, and its GNS representation is the scalar Fock representation. This final step illustrates the structural separation: $\mathcal{O} \mapsto \mathcal{A}_m(\mathcal{O})$ is the abstract observable net, while $(\mathcal{H}_{\omega_0}, \pi_{\omega_0}, \Omega_{\omega_0})$ is selected only after a state is chosen.

3.2 Perturbative Algebraic QFT and Interacting Examples

Perturbative algebraic QFT is the framework in which Epstein–Glaser causal renormalization, local time-ordered products, and algebraic locality are put in the same language. Its algebras are formal power series in the interaction and in $\hbar$; at every fixed formal order the locality and renormalization statements are statements about distributions and their extensions.

Definition 3.2 (Microcausal functionals and the BDF star product)**.** For a scalar field on a globally hyperbolic spacetime M, let $\mathcal{E} = C^\infty(M)$ be the classical configuration space. A smooth functional $F : \mathcal{E} \to \mathbb{C}$ belongs to $\mathcal{F}_{\mu c}$, the space of microcausal functionals, if every functional derivative $F^{(n)}(\varphi)$ is a compactly supported distribution on M^n and

$$\mathrm{WF}\big(F^{(n)}(\varphi)\big) \cap \big(\overline{V}^n_+ \cup \overline{V}^n_-\big) = \emptyset. \tag{3.5}$$

Here $\overline{V}^n_\pm$ is the set of n-tuples of nonzero covectors all lying in the closed future or all lying in the closed past light cone. If $H(x,y)$ is a Hadamard bidistribution for the free theory, the Brunetti–Duetsch–Fredenhagen product is

$$F \star_H G = m \circ \exp\left[\hbar \int_{M^2} H(x,y)\frac{\delta}{\delta\varphi(x)} \otimes \frac{\delta}{\delta\varphi(y)}\,\mathrm{d}x\,\mathrm{d}y\right](F\otimes G), \tag{3.6}$$

where $m(F\otimes G)(\varphi) = F(\varphi)G(\varphi)$. The exponential is expanded as a formal power series and every term is defined by the Hörmander wavefront-set product criterion applied to (3.5) and the Hadamard wavefront set of H.

Peierls bracket from the pAQFT star commutator. Assume that the antisymmetric part of H is $(\mathrm{i}/2)\Delta$, where Δ is the causal propagator of the free equation of motion. For regular functionals, and then by the microcausal extension wherever the products are defined,

$$[F,G]_{\star_H} = \mathrm{i}\hbar\,\{F,G\}_{\text{Peierls}} + O(\hbar^2). \tag{3.7}$$

The order-$\hbar$ part of (3.6) is

$$\hbar \int_{M^2} H(x,y)\frac{\delta F}{\delta\varphi(x)}\frac{\delta G}{\delta\varphi(y)}\,\mathrm{d}x\,\mathrm{d}y.$$

Subtracting the same expression with F and G interchanged replaces $H(x,y)$ by $H(x,y) - H(y,x) = \mathrm{i}\Delta(x,y)$. The resulting bilinear form is exactly $\mathrm{i}\hbar\{F,G\}_{\mathrm{Peierls}}$ in the free theory. Higher terms in the exponential contribute $O(\hbar^2)$.

Proposition 3.3 (Perturbative interacting local net)**.** *Let $T_n : \mathcal{F}_{\mathrm{loc}}^{\otimes n} \to \mathcal{F}_{\mu c}$ be Epstein–Glaser time-ordered products satisfying symmetry, covariance, microlocal spectrum bounds, unitarity, field independence, and causal factorization. Define*

$$\mathcal{S}(F) = 1 + \sum_{n\geq 1} \frac{\mathrm{i}^n}{\hbar^n n!} T_n(F^{\otimes n}).$$

Causal factorization is the identity

$$\mathcal{S}(F+G) = \mathcal{S}(F) \star_H \mathcal{S}(G) \quad \textit{if} \quad \operatorname{supp} F \cap J^-(\operatorname{supp} G) = \emptyset. \tag{3.8}$$

We shall also use its three-functional form

$$\mathcal{S}(F+G+H) = \mathcal{S}(F+G) \star_H \mathcal{S}(G)^{-1}_{\star_H} \star_H \mathcal{S}(G+H), \qquad \operatorname{supp} F \cap J^-(\operatorname{supp} H) = \emptyset, \tag{3.9}$$

which is part of the causal-factorization axiom for the formal S-matrix. For an interaction functional V, set

$$\mathcal{S}_V(F) := \mathcal{S}(V)^{-1}_{\star_H} \star_H \mathcal{S}(V+F), \tag{3.10}$$

where the inverse is taken in the completed formal $\star_H$-algebra. Let $\mathfrak{A}_V^{\mathrm{pert}}(\mathcal{O})$ be the formal $*$-algebra generated by $\mathcal{S}_V(F)$ with $\operatorname{supp} F \subset \mathcal{O}$, modulo the causal-factorization relations and the time-slice ideal. Then

$$\mathcal{O} \longmapsto \mathfrak{A}_V^{\mathrm{pert}}(\mathcal{O})$$

is an isotonic, time-slice local net of formal algebras. Covariance has the following precise form: a spacetime symmetry g sends $\mathfrak{A}_V^{\mathrm{pert}}(\mathcal{O})$ to $\mathfrak{A}_{gV}^{\mathrm{pert}}(g\mathcal{O})$; if V is invariant under the symmetry, this is covariance of a single net. Locality means that generators with spacelike separated supports commute.

Proof. The formal inverse in (3.10) exists because $\mathcal{S}(V) = 1 + O(V)$ in the completed formal algebra. The net maps for inclusions are the homomorphisms induced by the identity on generators: if $\mathcal{O}_1 \subset \mathcal{O}_2$, every generator with support in $\mathcal{O}_1$ is also one with support in $\mathcal{O}_2$, and the relations imposed in the smaller algebra are restrictions of the relations imposed in the larger algebra. This proves isotony. The time-slice property is exactly the quotient by the time-slice ideal: if $\mathcal{N} \subset \mathcal{O}$ contains a Cauchy surface for $\mathcal{O}$, the ideal identifies the generators in $\mathcal{O}$ with the algebra generated by representatives supported in $\mathcal{N}$, using the free equation of motion and causal propagation in the Epstein–Glaser construction.

For covariance, let g be a spacetime symmetry preserving the background data used in H. Covariance of the time-ordered products gives

$$\alpha_g(\mathcal{S}(F)) = \mathcal{S}(gF), \qquad \alpha_g(A \star_H B) = \alpha_g(A) \star_H \alpha_g(B),$$

where gF is the pushforward of the local functional. Applying this to (3.10) gives

$$\alpha_g\big(\mathcal{S}_V(F)\big) = \mathcal{S}_{gV}(gF), \qquad \operatorname{supp}(gF) = g(\operatorname{supp} F).$$

Thus g sends the algebra of $\mathcal{O}$ for V to the algebra of $g\mathcal{O}$ for the transformed interaction gV. If $gV = V$, this is a covariance automorphism of the same net.

It remains to prove locality. We first record the relative causal factorization identity. If

$$\operatorname{supp} F \cap J^-(\operatorname{supp} G) = \emptyset,$$

then (3.9), applied with the interaction V as the middle functional, gives

$$\mathcal{S}(V + F + G) = \mathcal{S}(V + F) \star_H \mathcal{S}(V)^{-1}_{\star_H} \star_H \mathcal{S}(V + G).$$

Multiplying on the left by $\mathcal{S}(V)^{-1}_{\star_H}$ yields

$$\mathcal{S}_V(F + G) = \mathcal{S}_V(F) \star_H \mathcal{S}_V(G). \tag{3.11}$$

For spacelike separated supports both causal orderings are allowed:

$$\operatorname{supp} F \cap J^-(\operatorname{supp} G) = \emptyset, \qquad \operatorname{supp} G \cap J^-(\operatorname{supp} F) = \emptyset.$$

Using (3.11) in the two orders and using symmetry of the time-ordered products, $\mathcal{S}_V(F + G) = \mathcal{S}_V(G + F)$, gives

$$\mathcal{S}_V(F) \star_H \mathcal{S}_V(G) = \mathcal{S}_V(G) \star_H \mathcal{S}_V(F).$$

Since the generators of $\mathfrak{A}^{\mathrm{pert}}_V(\mathcal{O}_1)$ and $\mathfrak{A}^{\mathrm{pert}}_V(\mathcal{O}_2)$ are of this form, and the quotient relations are compatible with $\star_H$, this proves the locality relation for spacelike separated regions. $\square$

Renormalization in this framework is the freedom in extending T_n to the diagonals. The ambiguity is local: two choices are related by a Stueckelberg–Petermann redefinition of the interaction by local functionals and their derivatives. For gauge theory the same construction must be combined with the BV complex and the quantum master equation; the physical perturbative algebra is then the BRST/BV cohomology of the interacting algebra. These facts make perturbative AQFT a natural mathematical home for local counterterms, contact terms, and gauge-theoretic Ward identities. It will not be treated only as a dictionary between path integrals and local nets. Later chapters must develop the perturbative algebra itself: the classification of time-ordered-product extensions, the renormalization group as the Stueckelberg–Petermann group, interacting fields by Bogoliubov's formula, the time-slice axiom in perturbation theory, BV cohomology for gauge theories, and the relation between these constructions and the 1PI and Wilsonian effective actions.

The algebraic formulation must also be developed through substantial interacting examples. A persistent weakness of much abstract AQFT literature is that the axioms, sector theory, and local-algebra technology are developed with too few nontrivial models in view. In this monograph, whenever an interacting construction is available or partially available, the associated algebraic questions should be asked explicitly: what are the local algebras or formal local algebras, what is the state space, which sectors are present, how does scattering or infraparticle structure appear, how are scaling limits and operator product structures represented, and which steps are theorem-level rather than expected structure? Constructive scalar models, perturbatively constructed Epstein–Glaser/pAQFT models, factorizing two-dimensional models, lattice-continuum scaling limits, conformal nets, and gauge theories with BRST/BV control are therefore not examples after the axioms; they are part of the theory development and tests of whether the algebraic formulation captures the physics it is meant to organize.

The purpose of including this material is not only to collect known constructions. The comparison among Wightman, Euclidean, Wilsonian, perturbative algebraic, BV, and operator-algebraic formulations exposes genuine gaps: gauge-theory local nets beyond perturbation theory, charged sectors requiring nonlocal dressing, infraparticle scattering, contact-term control in deformations, and the passage from constructive or lattice scaling limits to detailed algebraic data. These gaps should be formulated as precise mathematical and physical problems and, where feasible, developed within the monograph itself. Existing results supply boundary conditions for this development; they are not the endpoint of the project. No axiomatic framework is taken here as sacred or final. A framework earns its place by clarifying nontrivial physics and the mathematics behind it: which observables are defined, which limits exist, which locality and positivity statements are actually proven, how sectors and scattering are encoded, and where the formulation fails to see structures that are physically present. The comparison of frameworks is therefore part of the subject matter, not a preliminary choice of doctrine.

3.3 States and GNS Representations

Definition 3.4 (State on a C^*-algebra)**.** A state on a unital C^*-algebra $\mathcal{A}$ is a linear functional

$$\omega : \mathcal{A} \to \mathbb{C}$$

such that

$$\omega(A^*A) \geq 0, \qquad \omega(1) = 1.$$

The source trace for the operator-algebra form of the construction used here is the Gelfand–Naimark representation theorem together with Segal's state-to-representation construction [20, 21]; Takesaki's account is the modular-theory reference used later in this volume [26].

Theorem 3.5 (Gelfand–Naimark–Segal construction)**.** *Let ω be a state on a unital C^*-algebra $\mathcal{A}$. There are a Hilbert space $\mathcal{H}_\omega$, a unital $*$-representation $\pi_\omega : \mathcal{A} \to B(\mathcal{H}_\omega)$, and a cyclic unit vector $\Omega_\omega \in \mathcal{H}_\omega$ such that*

$$\omega(A) = \langle \Omega_\omega, \pi_\omega(A)\Omega_\omega \rangle_\omega, \qquad A \in \mathcal{A}. \tag{3.12}$$

If $(\mathcal{H}', \pi', \Omega')$ is another cyclic representation satisfying (3.12), there is a unique unitary $W : \mathcal{H}_\omega \to \mathcal{H}'$ with

$$W\Omega_\omega = \Omega', \qquad W\pi_\omega(A) = \pi'(A)W.$$

Proof. Let

$$\mathcal{N}_\omega = \{A \in \mathcal{A} \mid \omega(A^*A) = 0\}.$$

The positivity of ω gives the Cauchy–Schwarz inequality

$$|\omega(C^*D)|^2 \leq \omega(C^*C)\,\omega(D^*D), \qquad C, D \in \mathcal{A}.$$

Hence $C \in \mathcal{N}_\omega$ implies $\omega(C^*D) = \omega(D^*C) = 0$ for all $D \in \mathcal{A}$, so $\mathcal{N}_\omega$ is a linear subspace. It is also a left ideal. If $C \in \mathcal{N}_\omega$ and $A \in \mathcal{A}$, then the C^*-algebra order inequality $0 \leq A^*A \leq \|A\|^2 1$ gives

$$0 \leq \omega((AC)^*(AC)) = \omega(C^*A^*AC) \leq \|A\|^2 \omega(C^*C) = 0.$$

Thus $AC \in \mathcal{N}_\omega$. The quotient $\mathcal{A}/\mathcal{N}_\omega$ therefore has the well-defined inner product

$$\langle [A], [B] \rangle_\omega = \omega(A^* B).$$

Let $\mathcal{H}_\omega$ be its completion. Define

$$\pi_\omega(A)[B] = [AB], \qquad \Omega_\omega = [1].$$

The representation is well-defined because $\mathcal{N}_\omega$ is a left ideal, and

$$\|\pi_\omega(A)[B]\|_\omega^2 = \omega(B^* A^* AB) \le \|A\|^2 \omega(B^* B),$$

so $\pi_\omega(A)$ extends to a bounded operator on $\mathcal{H}_\omega$. Equation (3.12) follows from the definition of the inner product, and Ω_ω is cyclic because $\pi_\omega(\mathcal{A})\Omega_\omega$ contains all quotient classes $[A]$.

For uniqueness, define W on the dense subspace $\pi_\omega(\mathcal{A})\Omega_\omega$ by

$$W \pi_\omega(A)\Omega_\omega = \pi'(A)\Omega'.$$

This is well-defined and isometric because both cyclic vector states equal ω:

$$\langle \pi_\omega(A)\Omega_\omega, \pi_\omega(B)\Omega_\omega \rangle = \omega(A^* B) = \langle \pi'(A)\Omega', \pi'(B)\Omega' \rangle.$$

Cyclicity of Ω' makes the range dense, hence W extends to a unitary. The intertwining identity and $W\Omega_\omega = \Omega'$ follow on the dense cyclic subspace. $\square$

Figure 3.2 summarizes the GNS construction: a state defines the null ideal, the quotient pre-Hilbert space, the cyclic vector, and the represented algebra, uniquely up to unitary equivalence.

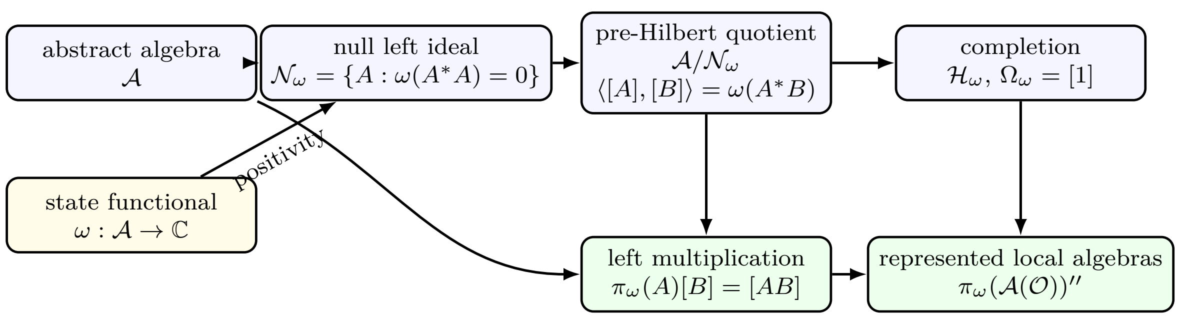


Figure 3.2: The GNS representation is obtained by quotienting the abstract algebra by the null ideal determined by the state, completing the resulting inner-product space, and representing the algebra by left multiplication. Local C^*-algebras become represented von Neumann algebras only after this state-dependent Hilbert space has been constructed.

3.4 Vacuum Representations

Definition 3.6 (Vacuum state and represented vacuum net)**.** Let $\mathcal{O} \mapsto \mathcal{A}(\mathcal{O})$ be a Poincare-covariant Haag–Kastler C^*-net. A state ω_0 on $\mathcal{A}_{\text{qloc}}$ is a vacuum state when:

(i) ω_0 is Poincare invariant:

$$\omega_0(\alpha_g(A)) = \omega_0(A), \qquad A \in \mathcal{A}_{\mathrm{qloc}}, \quad g \in \mathcal{P}_+^{\uparrow};$$

(ii) in the GNS representation $(\mathcal{H}_{\omega_0}, \pi_{\omega_0}, \Omega_{\omega_0})$, translations are implemented by a strongly continuous unitary representation whose joint spectrum lies in $\overline{V}_+$;

(iii) Ω_{ω_0} is the unique invariant vector up to phase.

The represented von Neumann local algebra of a region is

$$\mathcal{R}_{\omega_0}(\mathcal{O}) = \pi_{\omega_0}(\mathcal{A}(\mathcal{O}))''.$$

The double commutant indicates weak operator closure.

Proposition 3.7 (Unitary implementation of an invariant state)**.** *Let ω_0 be invariant under $\mathcal{P}_+^{\uparrow}$. In its GNS representation there is a unique strongly continuous unitary representation U_{ω_0} of $\mathcal{P}_+^{\uparrow}$ satisfying*

$$U_{\omega_0}(g)\pi_{\omega_0}(A)U_{\omega_0}(g)^{-1} = \pi_{\omega_0}(\alpha_g(A)), \qquad U_{\omega_0}(g)\Omega_{\omega_0} = \Omega_{\omega_0}. \tag{3.13}$$

Proof. On the algebraic cyclic subspace $\mathcal{D}_0 = \pi_{\omega_0}(\mathcal{A}_{\mathrm{qloc}})\Omega_{\omega_0}$, define

$$U_{\omega_0}(g)\pi_{\omega_0}(A)\Omega_{\omega_0} = \pi_{\omega_0}(\alpha_g(A))\Omega_{\omega_0}.$$

This is well defined on the GNS quotient. If $\pi_{\omega_0}(A)\Omega_{\omega_0} = 0$, then $\omega_0(A^*A) = 0$; invariance gives

$$\omega_0(\alpha_g(A)^*\alpha_g(A)) = \omega_0(\alpha_g(A^*A)) = \omega_0(A^*A) = 0,$$

so $\pi_{\omega_0}(\alpha_g(A))\Omega_{\omega_0} = 0$. The same calculation shows preservation of inner products:

$$\langle \pi_{\omega_0}(\alpha_g(A))\Omega_{\omega_0}, \pi_{\omega_0}(\alpha_g(B))\Omega_{\omega_0}\rangle = \omega_0(\alpha_g(A^*B)) = \omega_0(A^*B).$$

Thus $U_{\omega_0}(g)$ extends from $\mathcal{D}_0$ to an isometry on $\mathcal{H}_{\omega_0}$. Applying the same construction to g^{-1} gives the inverse isometry, so $U_{\omega_0}(g)$ is unitary. The group law holds on $\mathcal{D}_0$:

$$U_{\omega_0}(g)U_{\omega_0}(h)\pi_{\omega_0}(A)\Omega_{\omega_0} = \pi_{\omega_0}(\alpha_{gh}(A))\Omega_{\omega_0},$$

and hence holds on the Hilbert-space completion. The covariance relation in (3.13) is checked on $\mathcal{D}_0$ and extended by boundedness.

For strong continuity, it is enough to test vectors in the dense $\mathcal{D}_0$, uniformly boundedness then extends the result to all vectors. For $A \in \mathcal{A}_{\mathrm{qloc}}$,

$$\|U_{\omega_0}(g)\pi_{\omega_0}(A)\Omega_{\omega_0} - \pi_{\omega_0}(A)\Omega_{\omega_0}\|^2 = \omega_0\left((\alpha_g(A) - A)^*(\alpha_g(A) - A)\right),$$

which tends to zero by the assumed strong continuity of the automorphic action in the GNS norm. Finally, if a unitary representation satisfying (3.13) and fixing Ω_{ω_0} exists, its value on every vector $\pi_{\omega_0}(A)\Omega_{\omega_0}$ is forced by the covariance relation; density gives uniqueness. $\square$

When the vacuum representation is fixed, we write $\mathcal{R}(\mathcal{O})$ for $\mathcal{R}_{\omega_0}(\mathcal{O})$. Thus, in the remainder of this chapter, $\mathcal{A}(\mathcal{O})$ denotes the abstract local C^*-algebra, whereas $\mathcal{R}(\mathcal{O})$ denotes its vacuum-representation von Neumann completion $\pi_{\omega_0}(\mathcal{A}(\mathcal{O}))''$. These are related by the chosen representation and are distinct mathematical objects.

3.5 Correlation Functions in a Representation

The algebraic presentation produces correlation functions once a state and a representation have been chosen. If

$$A_j \in \mathcal{A}(\mathcal{O}_j), \qquad j = 1, \ldots, n,$$

are bounded local observables, their vacuum correlation function is

$$\omega_0(A_1 \cdots A_n) = \langle \Omega_{\omega_0}, \pi_{\omega_0}(A_1) \cdots \pi_{\omega_0}(A_n)\Omega_{\omega_0} \rangle.$$

Covariance gives spacetime dependence by translating the local observables:

$$A_j(x_j) = \alpha_{x_j}(A_j).$$

Thus bounded local observables give well-defined functions or distributions after smearing in the spacetime parameters x_j, depending on the continuity and phase-space properties imposed on the net.

Point fields require additional structure. A field $\Phi(f)$ localized in $\mathcal{O}$ should be affiliated with the represented algebra $\mathcal{R}(\mathcal{O})$, or should arise as a limit of bounded local observables in a topology specified by the construction. When this is available, the Wightman distributions are recovered as vacuum matrix elements of these affiliated unbounded operators. The algebraic net therefore supplies the local observable content, while field coordinatizations specify distributional generators for that content.

3.6 Haag Duality and the Dual Net

Definition 3.8 (Causal complement and Haag duality)**.** For a represented net, the causal complement of a region $\mathcal{O}$ is

$$\mathcal{O}^\perp = \{x \in M : \ (x-y)^2 > 0 \text{ for all } y \in \mathcal{O}\}.$$

The net satisfies Haag duality for $\mathcal{O}$ if

$$\mathcal{R}(\mathcal{O}) = \mathcal{R}(\mathcal{O}^\perp)'. \tag{3.14}$$

It says that every bounded operator commuting with all observables localized outside $\mathcal{O}$ is already generated by observables localized in $\mathcal{O}$.

Let $\mathcal{R}$ be a local represented net indexed by causally complete regions, so $\mathcal{O} = \mathcal{O}^{\perp\perp}$. Define

$$\mathcal{R}^d(\mathcal{O}) := \mathcal{R}(\mathcal{O}^\perp)'.$$

Then $\mathcal{R}^d$ is isotonic and

$$\mathcal{R}(\mathcal{O}) \subset \mathcal{R}^d(\mathcal{O}).$$

The equality $\mathcal{R} = \mathcal{R}^d$ is precisely Haag duality. The locality of $\mathcal{R}^d$ is an additional property, often called essential duality. Einstein causality gives

$$\mathcal{R}(\mathcal{O}) \subset \mathcal{R}(\mathcal{O}^\perp)' = \mathcal{R}^d(\mathcal{O}).$$

If $\mathcal{O}_1 \subset \mathcal{O}_2$, then $\mathcal{O}_2^\perp \subset \mathcal{O}_1^\perp$, hence $\mathcal{R}(\mathcal{O}_2^\perp) \subset \mathcal{R}(\mathcal{O}_1^\perp)$. Taking commutants reverses the inclusion and gives $\mathcal{R}^d(\mathcal{O}_1) \subset \mathcal{R}^d(\mathcal{O}_2)$. This is isotony.

The inclusion $\mathcal{R} \subset \mathcal{R}^d$ measures the possible gap between the chosen local observable generators and the maximal algebra determined by causal commutation. In superselection theory, Haag duality is the condition that permits localized representations to be represented cleanly by localized endomorphisms of the observable algebra.

Dual-net algebra. The algebra $\mathcal{R}^d(\mathcal{O})$ is the algebra detected by all causal-commutation tests outside $\mathcal{O}$. Thus an operator $T \in B(\mathcal{H})$ belongs to $\mathcal{R}^d(\mathcal{O})$ precisely when

$$[T, B] = 0, \qquad B \in \mathcal{R}(\mathcal{O}^\perp).$$

This criterion is intrinsic to the represented net, but it is weaker than membership in the originally assigned algebra $\mathcal{R}(\mathcal{O})$ unless Haag duality has been established. The dual net is therefore not a decorative reformulation of locality. It records the maximal bounded operator algebra which the exterior observables cannot distinguish from a local operator in $\mathcal{O}$.

Two elementary consequences are used repeatedly later. First, if $\mathcal{O}_1 \subset \mathcal{O}_2$, then

$$\mathcal{R}^d(\mathcal{O}_1) = \mathcal{R}(\mathcal{O}_1^\perp)' \subset \mathcal{R}(\mathcal{O}_2^\perp)' = \mathcal{R}^d(\mathcal{O}_2),$$

so the dual assignment is isotonic. Second, if $A \in \mathcal{R}(\mathcal{O})$ then locality gives

$$[A, B] = 0, \qquad B \in \mathcal{R}(\mathcal{O}^\perp),$$

hence $A \in \mathcal{R}^d(\mathcal{O})$. Haag duality is exactly the assertion that these two tests are equivalent for the chosen represented net.

Load-bearing use in localized sectors. Let ρ be a normal endomorphism of the represented quasilocal von Neumann algebra which is localized in a causally complete region $\mathcal{O}_0$, in the sense that

$$\rho(B) = B, \qquad B \in \mathcal{R}(\mathcal{O}_0^\perp).$$

Let $\mathcal{O}_1$ be causally complete with $\mathcal{O}_0 \subset \mathcal{O}_1$. For $A \in \mathcal{R}(\mathcal{O}_1)$ and $B \in \mathcal{R}(\mathcal{O}_1^\perp)$, one has $\mathcal{O}_1^\perp \subset \mathcal{O}_0^\perp$, so $\rho(B) = B$. Since A and B commute,

$$\rho(A)B = \rho(A)\rho(B) = \rho(AB) = \rho(BA) = \rho(B)\rho(A) = B\rho(A).$$

Therefore

$$\rho(A) \in \mathcal{R}(\mathcal{O}_1^\perp)' = \mathcal{R}^d(\mathcal{O}_1).$$

If Haag duality holds for $\mathcal{O}_1$, this improves to

$$\rho(\mathcal{R}(\mathcal{O}_1)) \subset \mathcal{R}(\mathcal{O}_1).$$

This is the concrete algebraic step behind the DHR passage from a sector that is indistinguishable from the vacuum outside $\mathcal{O}_0$ to an endomorphism of the local observable net. Without the equality $\mathcal{R} = \mathcal{R}^d$, the same calculation gives an endomorphism into the dual net; additional input is then required to identify that image with the originally specified observable algebra.

The same point controls intertwiners. If ρ and σ are localized in $\mathcal{O}$ and T satisfies

$$T\rho(A) = \sigma(A)T$$

for all quasilocal A, then for every $B \in \mathcal{R}(\mathcal{O}^\perp)$,

$$TB = T\rho(B) = \sigma(B)T = BT.$$

Thus $T \in \mathcal{R}^d(\mathcal{O})$, and Haag duality is precisely what places the intertwiner in the local algebra $\mathcal{R}(\mathcal{O})$. The tensor-category construction of bounded localized charges therefore uses Haag duality as a structural hypothesis, not as a terminology choice.

3.7 Point Fields as Coordinatizations

If a Wightman field $\Phi(f)$ is available, bounded functions of smeared fields can generate local algebras. Conversely, in many algebraic settings point fields arise as operator-valued distributions affiliated with the local algebras. The invariant object is the local observable content:

$$\mathcal{O} \mapsto \mathcal{A}(\mathcal{O}) \quad \text{or} \quad \mathcal{O} \mapsto \mathcal{R}(\mathcal{O}).$$

Fields are then coordinatizations of local observables, organized by symmetry, scaling, and the operator product expansion when those structures are available.

The comparison in this section has four distinct layers. The Wightman layer contains vacuum matrix elements and unbounded smeared fields on a common domain. The closure layer asks whether the Hermitian smearings have self-adjoint closures whose spectral projections satisfy spacelike strong commutation. The bounded-net layer asks whether those spectral projections generate an additive represented von Neumann net. The observable layer asks whether that coordinate-generated net is the actual observable net, or instead only a field net whose fixed points, gauge-invariant subalgebra, or BRST/BV observable sector must still be identified. Confusing any two adjacent layers is the usual source of false shortcuts: cyclic Wightman data do not identify von Neumann algebras, affiliation does not imply generation, and a fixed-point formula does not by itself prove Haag–Kastler additivity.

The passage from unbounded Wightman fields to a Haag–Kastler net is not a formal consequence of the Wightman axioms in their bare domain formulation. For real test functions the smeared fields are symmetric operators on a common invariant domain, but this alone neither supplies self-adjoint closures nor implies strong commutativity of the spectral measures of spacelike separated closures. The construction below records the standard route under the extra closure and strong-locality hypotheses needed to make it a theorem. The source traces for the two sides of this comparison are the Wightman and Haag–Kastler frameworks [1, 2, 18]; the theorem below isolates the extra functional-analytic hypotheses rather than folding them into either named framework.

Definition 3.9 (Affiliation of an unbounded operator)**.** For a self-adjoint operator T on $\mathcal{H}$, affiliation with a von Neumann algebra $\mathcal{M} \subset B(\mathcal{H})$ means that the spectral projections $E_T(\Delta)$ of T belong to $\mathcal{M}$ for every Borel set $\Delta \subset \mathbb{R}$. Equivalently, every unitary in the commutant $\mathcal{M}'$ preserves the graph of T and commutes with T on its domain.

Affiliation is the precise way in which an unbounded field can be localized in a bounded local algebra.

The equivalence in Definition 3.9 is often the practical test for locality of unbounded field coordinates, so we spell it out. Let E_T be the spectral measure of T. If $E_T(\Delta) \in \mathcal{M}$ for every Borel Δ, then every $B \in \mathcal{M}'$ commutes with every bounded Borel function of T. In particular,

$$B(1+T^2)^{-1/2} = (1+T^2)^{-1/2}B, \qquad BT(1+T^2)^{-1/2} = T(1+T^2)^{-1/2}B.$$

The first identity implies that $B \operatorname{Dom} T \subset \operatorname{Dom} T$: indeed

$$\operatorname{Dom} T = (1+T^2)^{-1/2}\mathcal{H}.$$

The second identity then gives $BT\xi = TB\xi$ for $\xi \in \operatorname{Dom} T$, by applying it to $\xi = (1+T^2)^{-1/2}\eta$. Thus commutant operators preserve the graph and commute with T on the graph domain.

Conversely, suppose every unitary $V \in \mathcal{M}'$ preserves $\operatorname{Dom} T$ and satisfies $VT\xi = TV\xi$ for $\xi \in \operatorname{Dom} T$. Then $VTV^{-1} = T$ as self-adjoint operators, so uniqueness in the spectral theorem gives

$$VE_T(\Delta)V^{-1} = E_T(\Delta)$$

for every Borel Δ. Hence $E_T(\Delta)$ commutes with all unitaries of $\mathcal{M}'$. The unitary group of a von Neumann algebra generates it ultraweakly, and the bicommutant theorem gives

$$(\mathcal{M}')' = \mathcal{M}.$$

Therefore $E_T(\Delta) \in \mathcal{M}$. This argument is the bounded operator-algebra content behind the phrase "the unbounded field is localized in $\mathcal{M}$"; the difficult QFT input is proving that the closed field operators in question have such spectral projections.

Theorem 3.10 (Wightman fields generate a represented local net under strong closure hypotheses)**.** *Let*

$$(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_\alpha\}_{\alpha \in I})$$

be a Wightman field presentation on M. Enlarge the field-label space, if needed, so that the smeared generators used below are Hermitian on $\mathcal{D}$. Suppose that for every real Hermitian labelled test function h the symmetric operator $\widehat{\Phi}(h)$ is essentially self-adjoint on $\mathcal{D}$; write $X(h)$ for its self-adjoint closure and

$$E_h(\Delta)$$

for the spectral projection of $X(h)$ associated with the Borel set $\Delta \subset \mathbb{R}$. Define, for every region $\mathcal{O} \in \mathcal{K}(M)$,

$$\mathcal{R}_\Phi(\mathcal{O}) := \{E_h(\Delta) \mid \operatorname{supp} h \subset \mathcal{O}, \quad h = h^\dagger, \quad \Delta \subset \mathbb{R} \text{ Borel}\}''. \tag{3.15}$$

Equivalently, by the spectral theorem,

$$\mathcal{R}_\Phi(\mathcal{O}) = \{\exp(\mathrm{i}tX(h)) \mid \operatorname{supp} h \subset \mathcal{O}, \quad h = h^\dagger, \quad t \in \mathbb{R}\}''. \tag{3.16}$$

Since t may be absorbed into h, this is the standard displayed construction

$$\mathcal{R}_\Phi(\mathcal{O}) = \{\exp(\mathrm{i}X(h)) \mid \operatorname{supp} h \subset \mathcal{O}, \quad h = h^\dagger\}''.$$

For every such h, the closed operator $X(h)$ is affiliated with $\mathcal{R}_\Phi(\mathcal{O})$. If $\mathcal{O}_1 \subset \mathcal{O}_2$, the same $X(h)$ is also affiliated with $\mathcal{R}_\Phi(\mathcal{O}_2)$. Assume strong locality of these closures: if $\mathcal{O}_1 \perp \mathcal{O}_2$, then every spectral projection entering $\mathcal{R}_\Phi(\mathcal{O}_1)$ commutes with every spectral projection entering $\mathcal{R}_\Phi(\mathcal{O}_2)$. Then

$$\mathcal{O} \longmapsto \mathcal{R}_\Phi(\mathcal{O})$$

is an isotonic, Poincare-covariant represented local net of von Neumann algebras on $\mathcal{H}$. The vector state

$$A \longmapsto \langle \Omega, A\Omega \rangle$$

is invariant under the represented Poincare action.

Proof. For a fixed h, the spectral theorem identifies the von Neumann algebra generated by the projection-valued measure E_h with the von Neumann algebra generated by the one-parameter unitary group $\{\exp(\mathrm{i}tX(h))\}_{t\in\mathbb{R}}$. Indeed $\exp(\mathrm{i}tX(h)) = \int \exp(\mathrm{i}t\lambda)\,\mathrm{d}E_h(\lambda)$, so the unitaries lie in $\{E_h(\Delta) \mid \Delta \subset \mathbb{R} \text{ Borel}\}''$; conversely, the spectral projections are bounded Borel functions of $X(h)$, hence lie in the von Neumann algebra generated by the unitary group by the Borel functional calculus for a self-adjoint operator. This proves the equivalence of (3.15) and (3.16), and also proves the affiliation statement because all spectral projections of $X(h)$ are generators of $\mathcal{R}_\Phi(\mathcal{O})$.

If $\mathcal{O}_1 \subset \mathcal{O}_2$, then the test functions with support in $\mathcal{O}_1$ are among those with support in $\mathcal{O}_2$, hence the generating spectral projections for $\mathcal{R}_\Phi(\mathcal{O}_1)$ are contained in $\mathcal{R}_\Phi(\mathcal{O}_2)$. This proves isotony. Wightman covariance gives

$$U(g)X(h)U(g)^{-1} = X(gh),$$

where gh is the transformed labelled test function. The spectral theorem then gives

$$U(g)E_h(\Delta)U(g)^{-1} = E_{gh}(\Delta),$$

so

$$U(g)\mathcal{R}_\Phi(\mathcal{O})U(g)^{-1} = \mathcal{R}_\Phi(g\mathcal{O}).$$

Strong locality is exactly the commutation of the spectral-projection generators for spacelike separated regions; taking double commutants preserves commutation of von Neumann algebras. The vacuum invariance follows from $U(g)\Omega = \Omega$. □

The constructive checkpoint below records a useful sufficient mechanism for the strong-locality hypothesis: Lemma 3.15 shows how joint analytic-vector estimates upgrade domain-level Wightman commutators to commutation of spectral projections. This is an additional functional-analytic input beyond the abstract Wightman distributional axioms.

Domain commutation and spectral locality. The word "strong" in the preceding hypothesis records a real operator-theoretic passage. Wightman locality gives an identity $[X,Y]\xi = 0$ on a common invariant field domain. The represented local algebra, however, is generated by the bounded Borel functional calculi of the self-adjoint closures. The needed implication is therefore

$$[X,Y]\mathcal{D} = 0 \quad \Longrightarrow \quad \exp(\mathrm{i}t\overline{X})\exp(\mathrm{i}s\overline{Y}) = \exp(\mathrm{i}s\overline{Y})\exp(\mathrm{i}t\overline{X}) \quad \text{for all } s,t \in \mathbb{R}, \tag{3.17}$$

or, equivalently, commutation of the two projection-valued spectral measures. Equation (3.17) is supplied by extra hypotheses such as the joint analytic-vector estimate in Lemma 3.15, by a direct resolvent commutation theorem, or by a constructive finite-regulator limit proving strong convergence of the relevant spectral windows. The finite companion check `calculation-checks/wightman_net_bridge_checks.py` includes a rational three-dimensional witness for the algebraic failure mode: a commutator can vanish on a stable test sector while the spectral projection of one self-adjoint generator fails to commute with a spectral projection of the other. The continuum theorem has a denser domain and hence requires stronger analysis, but the bounded-algebra target is the same spectral projection statement.

There is an analogous point for additivity. The formula (3.15) is isotonic by inclusion of its generating spectral projections, but Haag–Kastler additivity for a cover is not a formal consequence of writing a test function as a sum of smaller test functions. Suppose $\mathcal{O} \subset \bigcup_{j=1}^{r} \mathcal{O}_j$ and h is Hermitian with $\operatorname{supp} h \subset \mathcal{O}$. A partition of unity gives

$$h = h_1 + \cdots + h_r, \qquad \operatorname{supp} h_j \subset \mathcal{O}_j.$$

On the Wightman domain one has $\widehat{\Phi}(h) = \sum_j \widehat{\Phi}(h_j)$, but the bounded local algebra requires the spectral projections of the self-adjoint closure $X(h)$. A sufficient additivity condition is therefore the closure/Trotter statement

$$X(h) = \overline{X(h_1) + \cdots + X(h_r)}$$

on a common core, together with the strong Trotter product formula

$$\exp(\mathrm{i}tX(h)) = \operatorname*{s-lim}_{n\to\infty} \left(\exp\left(\frac{\mathrm{i}t}{n} X(h_1)\right) \cdots \exp\left(\frac{\mathrm{i}t}{n} X(h_r)\right)\right)^{n}. \tag{3.18}$$

The right side belongs to $\bigvee_j \mathcal{R}_\Phi(\mathcal{O}_j)$. The spectral projections of $X(h)$, being bounded Borel functions of the unitary group $\{\exp(\mathrm{i}tX(h))\}_{t\in\mathbb{R}}$, then belong to the same generated von Neumann algebra. Thus additivity of a Wightman-generated bounded net is a closure and functional-calculus statement about the self-adjoint field coordinates; it should not be imported from the distributional linearity of the fields alone. The finite check `calculation-checks/wightman_net_bridge_checks.py` contains the corresponding matrix-algebra shadow: the spectral projection of a sum of two noncommuting local self-adjoint generators lies in the von Neumann algebra generated by the two smaller algebras only after the sum has been specified as an operator in that generated algebra.

Remark 3.11 (Status of the Wightman-to-net direction). The theorem is conditional. Wightman commutativity is initially a statement about commutators of unbounded smeared fields on a common dense domain; it does not automatically imply commutativity of the spectral projections of self-adjoint closures. Thus the displayed Weyl-unitary formula should not be read as saying that arbitrary Wightman fields automatically span a local Haag–Kastler algebra. In constructive scalar models obtained from Euclidean measures, the route is model-specific: one proves the Schwinger functions, OS reconstruction, Wightman properties, and the required local-net properties with estimates particular to the construction. That is a proof of the Haag–Kastler axioms for those models, not a general implication from the abstract Wightman axioms alone.

Bounded orbit criterion for equality of local algebras. The equality condition in a complete Wightman coordinatization is an equality of represented von Neumann algebras. Cyclic correlator formulae belong to the earlier field-reconstruction layer; the following bounded criterion is often the operator-algebraic step that one needs after the local fields, closures, and functional calculi have already been constructed. Let $\mathcal{M}, \mathcal{N} \subset B(\mathcal{H})$ be von Neumann algebras and let $\Omega \in \mathcal{H}$ be separating for the generated von Neumann algebra $\mathcal{M} \vee \mathcal{N}$. Suppose that for every $A \in \mathcal{M}$ there is a uniformly bounded net $\{B_i\} \subset \mathcal{N}$ such that

$$B_i\Omega \longrightarrow A\Omega \qquad \text{in } \mathcal{H}.$$

Then $A \in \mathcal{N}$, and hence $\mathcal{M} \subset \mathcal{N}$. Indeed, after replacing B_i by a subnet, weak operator compactness of bounded balls in a von Neumann algebra gives a weak-operator limit $B \in \mathcal{N}$. The vector convergence above implies $B\Omega = A\Omega$. Since $A - B \in \mathcal{M} \vee \mathcal{N}$ and Ω is separating for $\mathcal{M} \vee \mathcal{N}$, one obtains $A = B$. Applying the same criterion in both directions gives $\mathcal{M} = \mathcal{N}$.

The hypotheses in this criterion are not decorative. Without separatingness, orbit agreement at a vector can be blind to operators that vanish on that vector. Without uniformly bounded approximants, equality of Hilbert-space orbit closures does not specify a bounded element of the candidate von Neumann algebra. Thus equality of Wightman polynomial matrix elements, or even equality of a dense set of cyclic vectors, does not by itself identify local algebras. One must first construct the bounded algebras and then prove a bounded orbit or strong-operator density statement of the type above. In QFT, the separating input is usually obtained from a Reeh–Schlieder theorem for an appropriate complementary region together with locality, as in Corollary 3.20; it is not a formal consequence of polynomial field cyclicity alone.

Definition 3.12 (Wightman-to-net comparison structure)**.** Let $(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_\alpha\}_{\alpha\in I})$ be a Wightman field presentation. A Wightman-to-net comparison structure for a chosen region class $\mathcal{K}(M)$ consists of the following additional objects and hypotheses.

(i) For every $\mathcal{O} \in \mathcal{K}(M)$, a real vector space

$$\mathcal{H}_\Phi(\mathcal{O}) \subset \mathrm{span}_{\mathbb{R}}\{\text{Hermitian labelled test functions supported in } \mathcal{O}\}$$

such that $\mathcal{O}_1 \subset \mathcal{O}_2$ implies $\mathcal{H}_\Phi(\mathcal{O}_1) \subset \mathcal{H}_\Phi(\mathcal{O}_2)$, and

$$U(g)\widehat{\Phi}(h)U(g)^{-1} = \widehat{\Phi}(gh), \qquad gh \in \mathcal{H}_\Phi(g\mathcal{O})$$

whenever $h \in \mathcal{H}_\Phi(\mathcal{O})$.

(ii) A self-adjoint closure $X(h) = \overline{\widehat{\Phi}(h)}$ for every $h \in \mathcal{H}_\Phi(\mathcal{O})$, with the covariance identity

$$U(g)X(h)U(g)^{-1} = X(gh)$$

understood as equality of self-adjoint operators on their closed graph domains.

(iii) Strong spacelike locality: if $\mathcal{O}_1 \perp \mathcal{O}_2$, $h_i \in \mathcal{H}_\Phi(\mathcal{O}_i)$, and E_{h_i} is the spectral measure of $X(h_i)$, then

$$E_{h_1}(\Delta_1)E_{h_2}(\Delta_2) = E_{h_2}(\Delta_2)E_{h_1}(\Delta_1) \tag{3.19}$$

for all Borel sets $\Delta_1, \Delta_2 \subset \mathbb{R}$.

(iv) An observable-subnet prescription. If the fields include charged, gauge-variant, or otherwise non-observable coordinates, the field-generated net

$$\mathcal{F}_\Phi(\mathcal{O}) = \{E_h(\Delta) \mid h \in \mathcal{H}_\Phi(\mathcal{O})\}''$$

is distinct from the neutral observable net. One must specify an observable subnet, for example a fixed-point net

$$\mathcal{R}_\Phi(\mathcal{O}) = \mathcal{F}_\Phi(\mathcal{O})^G$$

for an implemented compact internal group G, or a separately constructed gauge-invariant subnet.

This comparison structure is deliberately stronger than the Wightman distributional axioms. The Wightman hierarchy determines the cyclic field representation and its vacuum matrix elements. The comparison structure adds the bounded functional-calculus and observable-identification data needed to speak about a Haag–Kastler net. The resulting field-generated net is the represented von Neumann net

$$\mathcal{F}_\Phi(\mathcal{O}) = \{E_h(\Delta) \mid h \in \mathcal{H}_\Phi(\mathcal{O}), \quad \Delta \subset \mathbb{R} \text{ Borel}\}'',$$

or, when only observable coordinates are present, the same formula with $\mathcal{R}_\Phi$ in place of $\mathcal{F}_\Phi$. The remaining datum is whether this coordinate net exhausts the local observable content.

Definition 3.13 (Complete observable Wightman coordinatization)**.** Let

$$\mathcal{O} \longmapsto \mathcal{R}(\mathcal{O}) \subset B(\mathcal{H})$$

be a represented observable net on the same Hilbert space and with the same vacuum vector and spacetime symmetry representation as the Wightman field presentation. A Wightman-to-net comparison structure is a complete observable coordinatization of $\mathcal{R}$ on the region class $\mathcal{K}(M)$ if the following additional conditions hold.

(i) The covariance and vacuum actions agree:

$$U(g)\mathcal{R}(\mathcal{O})U(g)^{-1} = \mathcal{R}(g\mathcal{O}), \qquad U(g)\Omega = \Omega,$$

and the Wightman covariance in Definition 3.12 uses this same unitary representation U.

(ii) The bounded coordinate net is additive in the operator-theoretic sense appropriate to the chosen field coordinates. For every finite cover $\mathcal{O} \subset \bigcup_{j=1}^{r} \mathcal{O}_j$, every $h \in \mathcal{H}_\Phi(\mathcal{O})$, and every decomposition

$$h = h_1 + \cdots + h_r, \qquad h_j \in \mathcal{H}_\Phi(\mathcal{O}_j),$$

used to prove additivity, the closure and functional-calculus data satisfy

$$X(h) = \overline{X(h_1) + \cdots + X(h_r)}$$

on a common core and the unitary group of $X(h)$ is obtained from the unitary groups of the $X(h_j)$ by the strong Trotter limit (3.18). Equivalently, the spectral projections of $X(h)$ lie in $\bigvee_j \mathcal{F}_\Phi(\mathcal{O}_j)$, with $\mathcal{F}_\Phi$ replaced by the observable coordinate net when no charged field net is used.

(iii) The observable prescription is part of the comparison. If all chosen coordinates are observable, then the generated net itself exhausts the represented observable net:

$$\mathcal{R}_\Phi(\mathcal{O}) = \mathcal{R}(\mathcal{O}) \qquad \text{for all } \mathcal{O} \in \mathcal{K}(M).$$

If the chosen coordinates generate a charged or field net $\mathcal{F}_\Phi$, then an observable operation must be specified and must recover the represented observable net. For an implemented compact internal group G, this condition is

$$\mathcal{F}_\Phi(\mathcal{O})^G = \mathcal{R}(\mathcal{O}) \qquad \text{for all } \mathcal{O} \in \mathcal{K}(M),$$

with the fixed-point additivity and charged-intertwiner generation issues verified in the model rather than inferred from the equality of vector-space test functions.

(iv) The equality in the preceding item is equality of von Neumann algebras in the given representation. It requires equality of bounded operator closures, with the strong or weak operator topology built into the represented local algebra. Thus every bounded local observable in $\mathcal{R}(\mathcal{O})$ is obtained as a strong-operator limit of bounded functions of the specified local field coordinates, after applying the stated observable-subnet operation.

With this definition the statement that exponentials of smeared Wightman fields generate the local algebra has a precise meaning: it asserts Definition 3.13 for a specified region class, field-coordinate space, closure choice, strong locality statement, additivity mechanism, and observable-subnet prescription. In scalar theories with a complete neutral field system this may reduce to $\mathcal{R}_\Phi(\mathcal{O}) = \mathcal{R}(\mathcal{O})$. In a theory with charged fields, gauge variables, or a global internal symmetry, the same bounded functions first generate a field or coordinate net, and the observable net is obtained only after the fixed-point, gauge-invariant, BRST/BV, lattice physical-Hilbert-space, or other explicitly specified observable operation has been applied. Conversely, affiliated fields inside a given represented net provide local coordinates for a subnet unless the generation equality in Definition 3.13 is separately proved.

The finite algebra behind the last point is simple but important. Let $\Gamma = \mathbb{Z}/2\mathbb{Z}$ act on a two-dimensional Hilbert space by the parity unitary $P = \mathrm{diag}(1, -1)$. The self-adjoint operator $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is odd: $PXP^{-1} = -X$. Its spectral projections generate bounded field coordinates, but they are exchanged by Γ and therefore are not observables of the fixed-point algebra. The invariant subalgebra is the diagonal algebra generated by 1 and P, while adding the spectral projections of X generates the full matrix algebra. This toy model is a bounded-algebra diagnostic; it supplies no spacetime localization, domain, or spectrum theorem. Its role is to check the field-net versus observable-net distinction that the Wightman-to-net comparison structure must retain in every genuine model.

Affiliation is not generation. The converse comparison is equally important. Suppose a represented observable net $\mathcal{R}(\mathcal{O})$ is already given, and suppose one has a family of self-adjoint field coordinates $T_i(f)$ affiliated with $\mathcal{R}(\mathcal{O})$ whenever $\operatorname{supp} f \subset \mathcal{O}$. Affiliation means that all spectral projections of these coordinates are bounded local observables. It gives a canonical inclusion

$$\mathcal{R}_{\mathrm{coord}}(\mathcal{O}) := \{E_{T_i(f)}(\Delta) \mid \operatorname{supp} f \subset \mathcal{O}, \Delta \subset \mathbb{R} \text{ Borel}\}'' \subset \mathcal{R}(\mathcal{O}), \tag{3.20}$$

not an equality. Equality is a separate generation, or coordinatization completeness, hypothesis:

$$\mathcal{R}_{\mathrm{coord}}(\mathcal{O}) = \mathcal{R}(\mathcal{O}) \qquad \text{for every region } \mathcal{O}$$

in the region class under consideration. This hypothesis may hold for a chosen generating field system, may fail for a distinguished but incomplete field such as a current or stress tensor, and may be inappropriate when the physical observable net is a fixed-point or gauge-invariant subnet of a larger field net. Thus the statement that local fields are affiliated with local algebras is weaker than the statement that their bounded functional calculus generates all local observables.

The finite diagonal model makes the distinction visible without domains. In the algebra $D_3 \subset M_3(\mathbb{C})$ of diagonal matrices, the self-adjoint coordinate

$$T = \mathrm{diag}(0, 0, 1)$$

is affiliated with D_3, because its spectral projections $\mathrm{diag}(1,1,0)$ and $\mathrm{diag}(0,0,1)$ lie in D_3. They generate only the two-dimensional subalgebra $\mathbb{C}\,\mathrm{diag}(1,1,0)\oplus\mathbb{C}\,\mathrm{diag}(0,0,1)$, not the full diagonal algebra. A second coordinate $S=\mathrm{diag}(0,1,0)$ refines the spectral partition and, together with T, generates all three minimal diagonal projections. The finite check `calculation-checks/wightman_net_bridge_checks.py` verifies exactly this coarse-spectral-generation obstruction. In QFT the analogous issue is not finite-dimensional degeneracy but whether the chosen local fields separate the local bounded observable content with the topology relevant to the represented net.

Fixed points and additivity. Suppose the comparison structure has produced a represented field net $\mathcal{F}(\mathcal{O})$, and suppose a compact internal group G is implemented by strongly continuous unitaries $V(g)$ such that

$$V(g)\Omega=\Omega,\qquad V(g)U(a,\Lambda)=U(a,\Lambda)V(g),\qquad V(g)\mathcal{F}(\mathcal{O})V(g)^{-1}=\mathcal{F}(\mathcal{O}).$$

The fixed-point candidate for the observable net is

$$\mathcal{R}(\mathcal{O})=\mathcal{F}(\mathcal{O})^G=\{A\in\mathcal{F}(\mathcal{O})\mid V(g)AV(g)^{-1}=A\text{ for all }g\in G\}.\tag{3.21}$$

The Haar integral gives a normal conditional expectation

$$E_{\mathcal{O}}(A)=\int_G V(g)AV(g)^{-1}\,\mathrm{d}g,\qquad A\in\mathcal{F}(\mathcal{O}),\tag{3.22}$$

where the integral is taken in the ultraweak topology. It satisfies

$$E_{\mathcal{O}}^2=E_{\mathcal{O}},\qquad E_{\mathcal{O}}(B_1AB_2)=B_1E_{\mathcal{O}}(A)B_2,\quad B_1,B_2\in\mathcal{R}(\mathcal{O}),$$

and its range is $\mathcal{R}(\mathcal{O})$. Hence each $\mathcal{R}(\mathcal{O})$ is a von Neumann algebra, isotony and Poincare covariance are inherited from $\mathcal{F}$, and locality follows because $\mathcal{R}(\mathcal{O})\subset\mathcal{F}(\mathcal{O})$.

The previous paragraph is only the bounded fixed-point mechanism. It does not by itself prove all Haag–Kastler properties for the observable net. Additivity is the point where the distinction is visible. For two regions with field algebras carrying charged coordinates X_1 and X_2, a neutral composite such as X_1X_2 may be invariant under the diagonal action of G on the algebra generated by the two field algebras, while neither X_1 nor X_2 belongs to the separate fixed-point algebras. Thus

$$\bigvee_i\mathcal{F}(\mathcal{O}_i)^G\subset\left(\bigvee_i\mathcal{F}(\mathcal{O}_i)\right)^G$$

can be a proper inclusion. An observable-net additivity theorem must explain which neutral composites are generated locally, often using field-algebra reconstruction, charged intertwiners, split inclusions, or a model-specific constructive argument. The formula (3.21) supplies the bounded fixed-point algebras; DHR/DR reconstruction or a constructive model proof supplies the additional charged-sector and generation information needed for the full observable Haag–Kastler axioms.

Constructive interacting checkpoint: massive $P(\phi)_2$. The massive $P(\phi)_2$ models developed constructively in Chapter 137 give a useful test of the previous theorem because every word in the local-net construction has to be earned from estimates. Start with the infinite-volume

Schwinger hierarchy of Quoted Theorem 137.8, in a clustering single-vacuum phase. OS reconstruction gives a Hilbert space $\mathcal{H}_P$, vacuum Ω_P, translation representation U_P, and a scalar Wightman field $\Phi_P(f)$ on the reconstructed polynomial domain $\mathcal{D}_P$. The polynomial domain is generated by positive-time Euclidean fields modulo the OS null space; after analytic continuation it is the common domain for finite products of smeared Lorentzian fields.

The associated bounded local field net is obtained from the spectral projections of real smeared fields,

$$\mathcal{F}_P(\mathcal{O}) = \{E_{\overline{\Phi_P(f)}}(\Delta) \mid f \in C_c^\infty(\mathcal{O}, \mathbb{R}),\ \Delta \subset \mathbb{R} \text{ Borel}\}'', \tag{3.23}$$

where $\overline{\Phi_P(f)}$ denotes the self-adjoint closure obtained in the constructive Hilbert-space realization. This displayed formula is a definition of the net only after three analytic inputs have been supplied. First, the Hamiltonian or Euclidean construction must prove essential self-adjointness of the real smeared fields on a common invariant core. Second, spacelike separated closures must commute strongly, meaning that their spectral projections commute; the finite-cutoff equal-time tensor-factor locality and the thermodynamic/ultraviolet limiting estimates are the model mechanism for this statement. Third, the limiting bounded functions must be compatible with Poincare covariance and the time-slice dynamics. With these inputs, isotony follows from support inclusion, covariance from the reconstructed U_P, and locality from strong commutation of the spectral projections.

Lemma 3.14 (Strong-limit route from regulated locality to bounded locality)**.** *Let $\{A_\lambda\}_{\lambda\in\Lambda}$ and $\{B_\lambda\}_{\lambda\in\Lambda}$ be two nets of bounded operators on a Hilbert space, indexed by a directed set. Assume that*

$$\sup_\lambda \|A_\lambda\| < \infty, \qquad \sup_\lambda \|B_\lambda\| < \infty, \qquad A_\lambda B_\lambda = B_\lambda A_\lambda,$$

and that $A_\lambda \to A$, $B_\lambda \to B$ strongly. Then

$$AB = BA.$$

Consequently, suppose a constructive regulator supplies self-adjoint regulated local fields $\Phi_\lambda(f)$ whose spectral projections $E_{\Phi_\lambda(f)}(\Delta)$ converge strongly to $E_{\overline{\Phi(f)}}(\Delta)$ for every interval Δ in the chosen spectral-window class. If the finite-regulator projections commute whenever $\operatorname{supp} f$ *and* $\operatorname{supp} g$ *are spacelike separated, then the limiting spectral projections commute for the same windows.*

Proof. For $\xi \in \mathcal{H}$,

$$\|(A_\lambda B_\lambda - AB)\xi\| \le \|A_\lambda\|\,\|(B_\lambda - B)\xi\| + \|(A_\lambda - A)B\xi\|.$$

The first term tends to zero by strong convergence of B_λ and the uniform bound on A_λ. The second tends to zero by strong convergence of A_λ, applied to the fixed vector $B\xi$. Hence $A_\lambda B_\lambda \to AB$ strongly. The same argument gives $B_\lambda A_\lambda \to BA$ strongly. Since $A_\lambda B_\lambda = B_\lambda A_\lambda$ for every λ, their strong limits are equal.

Apply this bounded statement to

$$A_\lambda = E_{\Phi_\lambda(f)}(\Delta), \qquad B_\lambda = E_{\Phi_\lambda(g)}(\Delta')$$

for spacelike separated supports. Spectral projections have norm at most one, so the uniform boundedness hypothesis is automatic once the strong spectral-window convergence is supplied. The conclusion is the commutation of the limiting spectral projections. This proves bounded locality of the local von Neumann algebras generated by those projections on the declared spectral windows. ∎

This lemma is not a replacement for constructive estimates. It identifies the bounded convergence statement those estimates must produce. A convergence result only for vacuum expectation values of unbounded smeared fields, or only for formal commutators on a polynomial domain, does not by itself imply commutation of the spectral projections that define $\mathcal{F}_P(\mathcal{O})$. A model proof may use the strong-limit route above, the analytic-vector route of Lemma 3.15, or another functional calculus argument; it must supply one of them before the Wightman polynomial field presentation has become a bounded Haag–Kastler net.

If the polynomial P has a finite internal symmetry group Γ, for example $\Gamma = \mathbb{Z}_2$ when P is even and the chosen phase is $\mathbb{Z}_2$-invariant, the observable net is the fixed-point net

$$\mathcal{R}_P(\mathcal{O}) = \mathcal{F}_P(\mathcal{O})^{\Gamma}. \tag{3.24}$$

Thus the same constructive model distinguishes the field net, the observable net, and possible sector questions. In broken phases, kink sectors are localized by asymptotic boundary conditions and therefore sit at the boundary between ordinary local endomorphism sectors and more global charge data. Massive clustering gives a spectral gap through the argument in Chapter 137; nuclearity, split inclusions, Haag duality, and DHR sector reconstruction require separate model estimates. The example is therefore an interacting AQFT object: it supplies concrete local von Neumann algebras and also identifies which algebraic properties require additional theorems beyond the OS axioms.

Gauge-theory checkpoint: the local Schwinger-model observable net. The massless Schwinger model of Chapter 52 supplies a different test: the starting variables are a gauge potential and a charged Dirac field, but the physical local algebra is generated by gauge-invariant fields. Let $e > 0$ be the dynamical charge, let $m_{\mathrm{Sch}}^2 = e^2/\pi$, and fix a local electric-flux branch and a value of the theta parameter as in Lemma 52.5. The exact local bosonization result gives the operator identity

$$E = -\frac{e}{\sqrt{\pi}}\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right), \qquad (\partial^\mu\partial_\mu - m_{\mathrm{Sch}}^2)E = 0, \tag{3.25}$$

on the gauge-invariant local sector. The additive constant in (3.25) shifts smeared fields by a scalar multiple of the identity and therefore does not change the von Neumann algebras generated by their spectral projections. Consequently the local electric-field net is the massive scalar Weyl net of the benchmark model above at mass m_{Sch}:

$$\mathcal{R}_{\mathrm{Sch}}(\mathcal{O}) = \left\{E_{\overline{E(f)}}(\Delta) \mid f \in C_c^\infty(\mathcal{O},\mathbb{R}),\ \Delta \subset \mathbb{R} \text{ Borel}\right\}'' \cong \mathcal{A}_{m_{\mathrm{Sch}}}(\mathcal{O}). \tag{3.26}$$

This is an equality of represented local nets after the local bosonization map has been fixed; it is not a statement that the gauge representative a_μ or the charged fermion ψ is an observable field. The mechanism behind (3.26) is concrete. For $f \in C_c^\infty(\mathcal{O};\mathbb{R})$,

$$E(f) = -\frac{e}{\sqrt{\pi}}\,\varphi(f) - \frac{e\theta}{2\pi}\int f\,\mathrm{d}^2x. \tag{3.27}$$

Thus, for every Borel set $\Delta \subset \mathbb{R}$,

$$E_{\overline{E(f)}}(\Delta) = E_{\overline{\varphi(f)}}\left(-\frac{\sqrt{\pi}}{e}\left(\Delta + \frac{e\theta}{2\pi}\int f\,\mathrm{d}^2x\right)\right),$$

where the right side means the spectral projection of $\overline{\varphi(f)}$ for the indicated affine image of Δ. Since $-e/\sqrt{\pi} \neq 0$, the spectral projections generated by $E(f)$ and by $\varphi(f)$ are the same. Gauge-invariant currents do not add an independent local coordinate. If $h = h_\mu \mathrm{d}x^\mu$ is compactly supported in $\mathcal{O}$, then the bosonization formula (52.8) and integration by parts give

$$J(h) := \int h_\mu J^\mu \, \mathrm{d}^2 x = -\frac{1}{\sqrt{\pi}} \, \varphi(\partial_\nu(\epsilon^{\mu\nu} h_\mu)) \,, \tag{3.28}$$

with the same support control. The Gauss-law relation $\partial_x E + eJ^0 = 0$ is then an identity among these scalar coordinates, not a second local generator. Closed Wilson loops whose support bounds a compact region reduce, by Stokes' theorem and the chosen local flux branch, to bounded functions of smeared E. Open Wilson lines, line-dressed charged fields, external fractional probes, and flux-changing operators are different data: they carry endpoint, path, or asymptotic flux information and are therefore not elements of $\mathcal{R}_{\mathrm{Sch}}(\mathcal{O})$ for a bounded region unless an additional construction supplies a gauge-invariant local composite.

There is also a global boundary. On a spatial circle, or on a line after imposing boundary conditions at infinity, the electric field may have a harmonic or asymptotic flux coordinate. That coordinate is not produced by compactly supported smearings inside a bounded region. In a fixed flux branch it is a number in the representation, and (3.26) describes the local observable net in that branch. If several flux branches are represented at once, the spectral projections of the global flux are central operators for the represented global algebra; they commute with every bounded local algebra but are not elements of any proper bounded-region algebra. Thus flux-sector projections belong to the superselection or boundary datum of the representation, not to the local Schwinger-model net generated by $E(f)$ with $\operatorname{supp} f \Subset \mathcal{O}$. The finite check `calculation-checks/wightman_net_bridge_checks.py` includes the bounded shadow of this distinction: a flux-sector projection in an auxiliary tensor factor commutes with all local electric coordinates while remaining outside the local electric-coordinate algebra.

This example isolates the AQFT bridge in a solvable gauge theory. The Haag–Kastler properties of $\mathcal{R}_{\mathrm{Sch}}$ follow from the massive scalar Weyl net once the Schwinger-model derivation has proved the local bosonization identity and the Gauss-law constraint on physical states. External fractional probes, compact zero modes on a circle, and flux sectors are not defects of the local net; they are additional sector data attached to the same observable algebra. Thus the example separates three notions that are often conflated: the gauge representative used in a Lagrangian calculation, the gauge-invariant local observable net, and the charged or line-dressed sectors on which nonlocal probes act.

Lemma 3.15 (Analytic-vector route to strong locality)**.** *Let X and Y be symmetric operators on a Hilbert space $\mathcal{H}$, with a common dense invariant domain $\mathcal{D}$. Assume:*

(i) *X and Y are essentially self-adjoint on $\mathcal{D}$;*

(ii) *every vector in $\mathcal{D}$ is jointly entire analytic for the ordered pair (X, Y), meaning that, for every $\xi \in \mathcal{D}$ and every $r, s \geq 0$,*

$$\sum_{m,n \geq 0} \frac{r^m s^n}{m! \, n!} \, \|X^m Y^n \xi\| < \infty;$$

(iii) *$[X, Y]\xi = 0$ for every $\xi \in \mathcal{D}$.*

Then the self-adjoint closures $\overline{X}$ and $\overline{Y}$ strongly commute, equivalently all spectral projections of $\overline{X}$ commute with all spectral projections of $\overline{Y}$.

Proof. Because $\mathcal{D}$ is invariant under X and Y, the commutator assumption implies $X^mY^n\xi = Y^nX^m\xi$ for all $m, n \geq 0$ and $\xi \in \mathcal{D}$, by induction. The joint analyticity assumption implies separate analyticity for X and Y, and also gives absolute convergence of the double series below. Let $U(t) = \exp(\mathrm{i}t\overline{X})$ and $V(s) = \exp(\mathrm{i}s\overline{Y})$. For $\xi \in \mathcal{D}$, the joint analytic-vector hypothesis gives norm-convergent power series

$$U(t)\xi = \sum_{m\geq 0} \frac{(\mathrm{i}t)^m}{m!} X^m\xi, \qquad V(s)\xi = \sum_{n\geq 0} \frac{(\mathrm{i}s)^n}{n!} Y^n\xi.$$

The Cauchy product is legitimate because

$$\sum_{m,n\geq 0} \frac{|t|^m|s|^n}{m!\,n!} \|X^mY^n\xi\| < \infty,$$

and gives

$$U(t)V(s)\xi = \sum_{m,n\geq 0} \frac{(\mathrm{i}t)^m(\mathrm{i}s)^n}{m!\,n!} X^mY^n\xi = \sum_{m,n\geq 0} \frac{(\mathrm{i}t)^m(\mathrm{i}s)^n}{m!\,n!} Y^nX^m\xi = V(s)U(t)\xi.$$

Since $\mathcal{D}$ is dense and $U(t), V(s)$ are bounded unitaries, the equality holds on all of $\mathcal{H}$. The spectral theorem then implies commutation of the projection-valued measures of $\overline{X}$ and $\overline{Y}$. □

For the constructive net $\mathcal{O} \mapsto \mathcal{F}_P(\mathcal{O})$, this lemma states exactly which analytic estimate must accompany Wightman locality. If $f, g \in C_c^\infty(M, \mathbb{R})$ have spacelike separated supports, Wightman locality gives

$$[\Phi_P(f), \Phi_P(g)]\xi = 0, \qquad \xi \in \mathcal{D}_P.$$

The bounded local algebras in (3.23) are local only after the specific construction supplies a common core on which the smeared fields are essentially self-adjoint and analytic-vector estimates hold. In Hamiltonian constructions these estimates usually come from comparison of smeared fields with powers of the finite-volume Hamiltonian, followed by uniform bounds in the infinite-volume and ultraviolet limits. In Euclidean constructions the same information is encoded in hypercontractive or Nelson-type bounds for polynomial fields on the OS Hilbert space. The resulting strong commutation is a bounded-algebra statement; it is stronger than the formal vanishing of a commutator on the polynomial domain.

The structural information obtained from the model should therefore be read in the following order. The OS hierarchy first gives the Hilbert space, vacuum, translation representation, Wightman distributions, and polynomial field domain. Essential self-adjointness and analytic-vector estimates then turn real smeared fields into spectral projection generators. Strong commutation of these generators gives locality of $\mathcal{F}_P$. Internal fixed points give the observable net $\mathcal{R}_P = \mathcal{F}_P^\Gamma$. Further AQFT properties are separate: nuclearity is an estimate on

$$\Theta_{\beta,\mathcal{O}} \colon \mathcal{F}_P(\mathcal{O}) \to \mathcal{H}_P, \qquad A \longmapsto \exp(-\beta H_P)A\Omega_P,$$

split inclusions follow only after such maps are trace-class or nuclear with the appropriate local bounds, Haag duality is a statement about equality of a local algebra with a commutant of an exterior algebra, and DHR reconstruction requires a tensor category of transportable localized endomorphisms. None of these conclusions is contained in the symbol $P(\phi)_2$ itself. The constructive example is valuable precisely because each assertion can be attached to a named estimate or representation-theoretic construction.

For later use it is helpful to make this estimate-to-property map explicit. Let $\mathcal{O}$ be a double cone and let

$$\Theta_{\beta,\mathcal{O}} \colon \mathcal{F}_P(\mathcal{O})_1 \longrightarrow \mathcal{H}_P, \qquad A \longmapsto \mathrm{e}^{-\beta H_P} A \Omega_P$$

denote the energy-damped image of the unit ball of the local field algebra. Buchholz–Wichmann nuclearity is a uniform approximation statement for this entire image. For each $\beta > 0$ and each fixed local region one must produce vectors $\xi_j \in \mathcal{H}_P$ and bounded linear functionals $\ell_j \in \mathcal{F}_P(\mathcal{O})^*$ such that

$$\mathrm{e}^{-\beta H_P} A \Omega_P = \sum_{j \geq 1} \ell_j(A) \xi_j, \qquad \sum_{j \geq 1} \|\ell_j\| \, \|\xi_j\| < \infty, \qquad A \in \mathcal{F}_P(\mathcal{O})_1,$$

with a bound on the infimum of the last sum that is stable under the thermodynamic and ultraviolet limits. A typical useful form is

$$\|\Theta_{\beta,\mathcal{O}}\|_1 \leq \exp\big(C_{\mathcal{O}} \beta^{-s}\big)$$

for small β, with $C_{\mathcal{O}}$ controlled by the size of the region and with s below the phase-space exponent allowed by the split criterion used in Chapter 4. Only after such a nuclearity estimate has been proved may one invoke the split theorem for a positive collar $\mathcal{O}_1 \Subset \mathcal{O}_2$. The output is then a type-I factor $\mathcal{N}$ satisfying

$$\mathcal{F}_P(\mathcal{O}_1) \subset \mathcal{N} \subset \mathcal{F}_P(\mathcal{O}_2).$$

Mass-gap information and finite low-energy counting may help organize the estimate, but the nuclearity statement itself concerns the full local unit ball after energy damping. The type-I factor is a positive-collar interpolant; sharp-algebra type classification requires the separate modular-spectrum and factoriality analysis discussed in Chapter 4.

Haag duality and sector reconstruction require different information. Haag duality for a double cone is the equality

$$\mathcal{R}_P(\mathcal{O}) = \bigcap_{\widetilde{\mathcal{O}} \perp \mathcal{O}} \mathcal{R}_P(\widetilde{\mathcal{O}})',$$

so a constructive proof must control operators commuting with every exterior algebra and show that no boundary-condition or disorder operator has been left in the dual algebra. This is a duality statement about the continuum net, independent of the nuclear decomposition of $\Theta_{\beta,\mathcal{O}}$. DHR reconstruction is still another layer: one must construct transportable endomorphisms localized in bounded regions, prove conjugates and finite statistics, and check the symmetric exchange structure before the Doplicher–Roberts compact group theorem can be applied. In massive $P(\phi)_2$ these are concrete model questions. In a symmetric phase they ask whether charged fields in the field net generate all finite observable sectors; in a broken phase they ask whether kink Hilbert spaces with different spatial boundary conditions can be represented by bounded local endomorphisms of the vacuum observable net. The model name specifies the arena; the estimates and category constructions above supply the content.

This separation is also the right language for phase structure. In an unbroken finite-symmetry phase the fixed-point observable net has sectors created by localized charged fields in the larger field net, when such fields are available. In a broken phase the natural kink Hilbert spaces are built with different asymptotic boundary conditions at spatial infinity. A kink cannot be treated as an ordinary DHR charge in the vacuum representation unless one proves an additional localization/transportability statement that turns the boundary-condition sector into an endomorphism localized in bounded regions. Thus the example exhibits both the power and the limitations of the standard local-endomorphism formulation of superselection theory.

Remark 3.16 (Field coordinates as a Wightman presentation). Let

$$\mathcal{O} \longmapsto \mathcal{R}(\mathcal{O})$$

be a Poincare-covariant represented local net on a Hilbert space $\mathcal{H}$, with unitary implementation U and invariant unit vector Ω. Assume the translation spectrum of U is contained in $\overline{V}_+$. Suppose there are a dense subspace $\mathcal{D} \subset \mathcal{H}$, a label set I, and operators $\widehat{\Phi}_\alpha(f)$ such that:

(i) $\Omega \in \mathcal{D}$, $U(g)\mathcal{D} \subset \mathcal{D}$, and $\widehat{\Phi}_\alpha(f)\mathcal{D} \subset \mathcal{D}$;

(ii) for each α, the map $f \mapsto \widehat{\Phi}_\alpha(f)$ is an operator-valued tempered distribution on the common domain $\mathcal{D}$;

(iii) if $\operatorname{supp} f \subset \mathcal{O}$, then $\widehat{\Phi}_\alpha(f)$ is affiliated with $\mathcal{R}(\mathcal{O})$;

(iv) the fields satisfy the Wightman covariance and adjunction identities on $\mathcal{D}$;

(v) fields affiliated with spacelike separated regions have vanishing graded commutator on $\mathcal{D}$;

(vi) the polynomial field orbit of Ω is dense in $\mathcal{H}$.

Then

$$(\mathcal{H}, \Omega, U, \mathcal{D}, \{\widehat{\Phi}_\alpha\}_{\alpha \in I})$$

is a Wightman field presentation, and its vacuum distributions are the matrix elements

$$W_{\alpha_1 \cdots \alpha_n}(f_1, \ldots, f_n) = \langle \Omega, \widehat{\Phi}_{\alpha_1}(f_1) \cdots \widehat{\Phi}_{\alpha_n}(f_n)\Omega \rangle.$$

Indeed, the Hilbert space, invariant vacuum vector, unitary Poincare representation, dense common domain, and operator-valued tempered distributions are the data just listed. The spectral condition is the assumed joint spectrum condition for translations; covariance, adjunction, locality, and cyclicity are items (iv), (v), and (vi). Continuity of the vacuum matrix elements is part of the operator-valued tempered-distribution hypothesis on the common invariant domain. Thus the listed AQFT-plus-field-coordinate data are exactly a Wightman field presentation in the sense used in Chapter 2.

This relation is important for gauge theories. The algebraic observable net can be defined from gauge-invariant observables without choosing gauge potentials as physical operators. If charged fields are introduced, they belong to additional data: an enlarged field algebra, charged representations, or sector endomorphisms. They are not elements of the neutral observable algebra unless the construction explicitly identifies them as observables.

3.8 Reeh–Schlieder and Local Cyclicity

Definition 3.17 (Weak additivity in a vacuum representation)**.** Let Ω be the vacuum vector in a Poincare-covariant representation, and let

$$\mathcal{R}_{\mathrm{glob}} := \bigvee_{\mathcal{O} \in \mathcal{K}(M)} \mathcal{R}(\mathcal{O}).$$

Here $\bigvee$ denotes the von Neumann algebra generated by the displayed family. The represented net is weakly additive when, for every nonempty open region $\mathcal{U}$ with nontrivial causal completion,

$$\bigvee_{a \in \mathbb{R}^D} \mathcal{R}(\mathcal{U} + a) = \mathcal{R}_{\mathrm{glob}}.$$

Weak additivity is the translated-generation condition used in the proof below. Finite-cover additivity identifies the algebra of one region with the algebra generated by a cover of that same region; weak additivity is a global condition on translates of one fixed local algebra.

Lemma 3.18 (Positive-energy tube uniqueness for translation matrix coefficients)**.** *Let $M = \mathbb{R}^{1,D-1}$ carry the mostly-plus metric, and let μ be a finite complex Borel measure on momentum space with*

$$\operatorname{supp}\mu \subset \overline{V}_+.$$

Define, for $a \in M$,

$$F_0(a) = \int_{\overline{V}_+} \mathrm{e}^{\mathrm{i}a\cdot p}\,\mathrm{d}\mu(p).$$

For $b \in V_+$, set

$$F(a - \mathrm{i}b) = \int_{\overline{V}_+} \mathrm{e}^{\mathrm{i}(a-\mathrm{i}b)\cdot p}\,\mathrm{d}\mu(p).$$

Then F is holomorphic in the tube $M - \mathrm{i}V_+$ and has continuous boundary value F_0 as $b \downarrow 0$. If F_0 vanishes on a nonempty open subset of the real edge M, then F_0 vanishes on all of M.

Proof. For each compact subcone $K \Subset V_+$ there is a constant $c_K > 0$ such that

$$b \cdot p \le -c_K|p|, \qquad b \in K, \quad p \in \overline{V}_+.$$

Here $|p|$ is any fixed Euclidean norm after choosing inertial coordinates; changing this norm only changes c_K. Consequently

$$\left|\mathrm{e}^{\mathrm{i}(a-\mathrm{i}b)\cdot p}\right| = \mathrm{e}^{b\cdot p} \le \mathrm{e}^{-c_K|p|},$$

and the same exponential bound, multiplied by any fixed polynomial in p, dominates the derivatives obtained by differentiating under the integral. This proves holomorphy in $M - \mathrm{i}V_+$. Since μ is finite and $\mathrm{e}^{b\cdot p} \le 1$, dominated convergence gives the continuous boundary value F_0 as $b \downarrow 0$.

Let $O \subset M$ be a nonempty open set on which $F_0 = 0$. On the opposite tube $M + \mathrm{i}V_+$ take the holomorphic function 0. The boundary values of F and 0 agree on O. Applying the distributional edge-of-the-wedge theorem, Theorem 16.6, with cones $-V_+$ and V_+, gives a connected complex neighborhood $\mathcal{N}$ of a smaller open ball $B \Subset O$ and a holomorphic function on $\mathcal{N}$ agreeing with F on $\mathcal{N} \cap (M - \mathrm{i}V_+)$ and with 0 on $\mathcal{N} \cap (M + \mathrm{i}V_+)$. The latter set is open in $\mathcal{N}$, so the identity theorem makes the glued holomorphic function identically zero on $\mathcal{N}$. Hence F vanishes on the nonempty open set $\mathcal{N} \cap (M - \mathrm{i}V_+)$. Since the tube $M - \mathrm{i}V_+$ is connected, the identity theorem again gives $F = 0$ on the whole tube. Taking the continuous boundary value proves $F_0(a) = 0$ for every real translation a. □

The source trace for Reeh–Schlieder local cyclicity is [19, 2]. The proof is included in the represented-net form because the spectral condition and weak additivity are the inputs needed later in modular and scattering arguments.

Theorem 3.19 (Reeh–Schlieder theorem for a weakly additive vacuum net)**.** *Let $\mathcal{R}$ be a Poincare-covariant represented local net on $\mathcal{H}$. Assume:*

(i) *the vacuum vector Ω is translation invariant;*

(ii) *the joint translation spectrum lies in $\overline{V}_+$;*

(iii) Ω *is cyclic for* $\mathcal{R}_{\mathrm{glob}}$*;*

(iv) *the represented net is weakly additive in the sense of Definition 3.17.*

Then for every nonempty open region $\mathcal{O}$ *containing the closure of some nonempty open* $\mathcal{U} \in \mathcal{K}(M)$*,*

$$\overline{\mathcal{R}(\mathcal{O})\Omega} = \mathcal{H}.$$

Proof. Let ψ be orthogonal to $\mathcal{R}(\mathcal{O})\Omega$. Choose a nonempty open region $\mathcal{U}$ whose closure is contained in $\mathcal{O}$, and choose $A \in \mathcal{R}(\mathcal{U})$. For translations a in a neighborhood of 0, the inclusion $\mathcal{U} + a \subset \mathcal{O}$ and covariance give $U(a)AU(a)^{-1} \in \mathcal{R}(\mathcal{O})$, hence

$$\langle \psi,\, U(a)A\Omega \rangle = 0$$

there. Let E be the joint spectral measure of the translation generators, so that $U(a) = \int e^{ia\cdot p}\, dE(p)$ and $\operatorname{supp} E \subset \overline{V}_+$. The finite complex measure

$$\mu_{\psi,A}(\Delta) := \langle \psi, E(\Delta)A\Omega \rangle$$

gives, for real a,

$$F_A(a) := \langle \psi, U(a)A\Omega \rangle = \int_{\overline{V}_+} e^{ia\cdot p}\, d\mu_{\psi,A}(p).$$

For $z = a - ib$ with $b \in V_+$, the same spectral integral

$$F_A(z) = \int_{\overline{V}_+} e^{iz\cdot p}\, d\mu_{\psi,A}(p)$$

is holomorphic in z by Lemma 3.18. The same lemma applies because $F_A(a)$ vanishes for a in a neighborhood of 0. Hence

$$\langle \psi, U(a)A\Omega \rangle = 0$$

for every real translation a. Since A was arbitrary in $\mathcal{R}(\mathcal{U})$, weak additivity says that ψ is orthogonal to $\mathcal{R}_{\mathrm{glob}}\Omega$. Global cyclicity then gives $\psi = 0$. The orthogonal complement of $\mathcal{R}(\mathcal{O})\Omega$ is zero, which proves the density. □

Corollary 3.20 (Separatingness from Reeh–Schlieder and locality)**.** *Under the hypotheses of Theorem 3.19, let* $\mathcal{O}$ *be a nonempty open region to which the theorem applies. If* $B \in \mathcal{B}(\mathcal{H})$ *commutes with* $\mathcal{R}(\mathcal{O})$ *and* $B\Omega = 0$*, then* $B = 0$*. Consequently* Ω *is separating for every von Neumann algebra* $\mathcal{M} \subset \mathcal{R}(\mathcal{O})'$*.*

In particular, let W *be a wedge and define* $\mathcal{R}(W)$ *as the von Neumann algebra generated by double-cone algebras contained in* W*. If the causal complement* W' *contains a nonempty open region* $\mathcal{O}$ *to which Theorem 3.19 applies, then* Ω *is separating for* $\mathcal{R}(W)$*.*

Proof. Let B commute with $\mathcal{R}(\mathcal{O})$ and satisfy $B\Omega = 0$. For every $A \in \mathcal{R}(\mathcal{O})$,

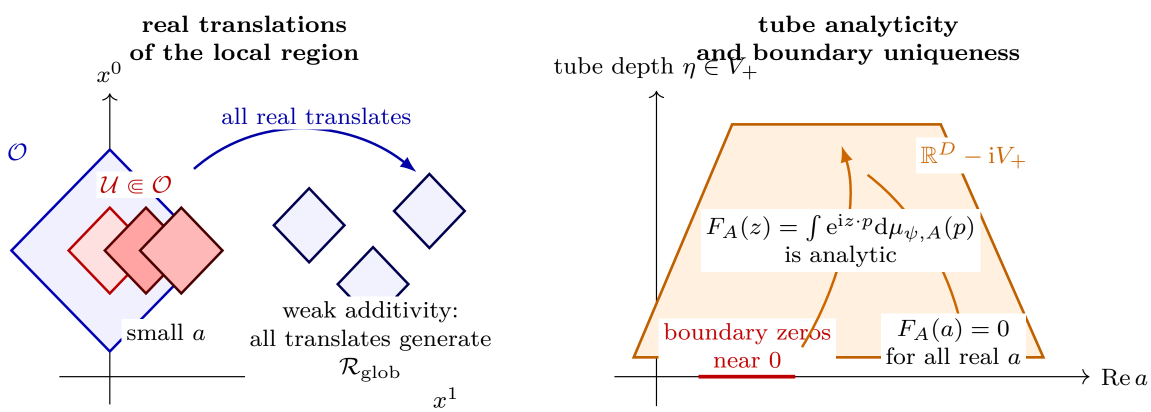
$$BA\Omega = AB\Omega = 0.$$

Theorem 3.19 says that $\mathcal{R}(\mathcal{O})\Omega$ is dense in $\mathcal{H}$. Since B is bounded, B vanishes on all of $\mathcal{H}$. If $\mathcal{M} \subset \mathcal{R}(\mathcal{O})'$, the same argument applies to each $B \in \mathcal{M}$ with $B\Omega = 0$, which is precisely separatingness of Ω for $\mathcal{M}$.

For the wedge statement, locality gives $\mathcal{R}(W) \subset \mathcal{R}(\mathcal{O})'$ whenever $\mathcal{O} \subset W'$. The preceding paragraph then applies with $\mathcal{M} = \mathcal{R}(W)$. No Haag duality is used here; Haag duality would identify a larger commutant with an algebra assigned to the causal complement, whereas the separatingness conclusion only needs the inclusion forced by locality. $\square$

The theorem and its separatingness corollary are used as inputs for modular theory, superselection analysis, and scattering theory in algebraic form.

Figure 3.3 emphasizes the strong content of the theorem. Operators localized in a fixed small region $\mathcal{O}$ generate, from the vacuum alone, a set of vectors dense in the whole vacuum Hilbert space. The conclusion is a Hilbert-space density statement controlled by positive-energy analytic continuation: weak additivity and analyticity leave no nonzero vector orthogonal to all vectors $A\Omega$ with $A \in \mathcal{R}(\mathcal{O})$.



$$\boxed{\psi \perp \mathcal{R}(\mathcal{O})\Omega \Rightarrow F_A(a) = 0 \text{ near } 0 \Rightarrow F_A(a) = 0 \text{ for all } a \Rightarrow \psi \perp \mathcal{R}_{\mathrm{glob}}\Omega \Rightarrow \psi = 0}$$

Figure 3.3: Reeh–Schlieder local cyclicity. The geometric input is the existence of $\mathcal{U} \Subset \mathcal{O}$, so that small real translations of $\mathcal{U}$ remain inside $\mathcal{O}$. If $\psi \perp \mathcal{R}(\mathcal{O})\Omega$, then $F_A(a) = \langle \psi, U(a)A\Omega\rangle$ vanishes on a real neighborhood of 0 for every $A \in \mathcal{R}(\mathcal{U})$. The spectral condition analytically continues F_A to the forward spectral tube $M - \mathrm{i}V_+$; the boundary-value uniqueness lemma for such tube-holomorphic functions propagates this vanishing to all real translations. Weak additivity then turns the translated local algebras into $\mathcal{R}_{\mathrm{glob}}$, and global cyclicity forces $\psi = 0$.

3.9 Locally Covariant Form

Definition 3.21 (Locally covariant C^*-algebraic QFT functor)**.** Let **Loc** be the category whose objects are oriented, time-oriented globally hyperbolic Lorentzian spacetimes X, and whose morphisms $\chi : X \to Y$ are isometric embeddings preserving orientation and time orientation, with causally convex image. Let $\mathbf{C^*Alg}_{\mathrm{inj}}$ be the category of unital C^*-algebras and injective unital $*$-homomorphisms. A locally covariant C^*-algebraic QFT is a functor

$$\mathfrak{A} : \mathbf{Loc} \longrightarrow \mathbf{C^*Alg}_{\mathrm{inj}}.$$

For a morphism $\chi : X \to Y$, write

$$\alpha_\chi := \mathfrak{A}(\chi) : \mathfrak{A}(X) \to \mathfrak{A}(Y).$$

The functor satisfies the time-slice property when α_χ is an isomorphism for every Cauchy morphism, meaning every χ whose image contains a Cauchy surface of Y. It satisfies Einstein causality when, for two morphisms $\chi_i : X_i \to Y$ with causally disjoint images,

$$[\alpha_{\chi_1}(A_1), \alpha_{\chi_2}(A_2)] = 0, \qquad A_i \in \mathfrak{A}(X_i).$$

This formulation preserves the local content of QFT in spacetimes where symmetry selects no distinguished global vacuum. On Minkowski spacetime, restricting $\mathfrak{A}$ to embeddings of suitable regions into M recovers a Haag–Kastler-type net.

For the present monograph, the algebraic framework supplies a precise language for local observables, states, sectors, and structural locality theorems. Its connection with path integrals, perturbative expansions, and point-field descriptions is a comparison problem to be tracked explicitly case by case.

Chapter 4

Superselection Sectors and Locality Properties

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

4.1 Representations as Sectors

Let $M = \mathbb{M}^D$ be Minkowski spacetime, let $\mathcal{O} \mapsto \mathcal{A}(\mathcal{O})$ be a local observable net, and let $\mathcal{A}_{\mathrm{qloc}}$ be its quasilocal algebra. Let

$$\pi_0 : \mathcal{A}_{\mathrm{qloc}} \to \mathcal{B}(\mathcal{H}_0)$$

be a vacuum representation. A representation

$$\pi : \mathcal{A}_{\mathrm{qloc}} \to \mathcal{B}(\mathcal{H}_\pi)$$

defines a representation sector. Two representations belong to the same sector when they are unitarily equivalent:

$$\pi_1 \simeq \pi_2 \quad \iff \quad \exists\, V : \mathcal{H}_{\pi_1} \to \mathcal{H}_{\pi_2} \text{ unitary with } V\pi_1(A) = \pi_2(A)V.$$

For irreducible representations, inequivalent sectors have no nonzero observable intertwiner between them. More generally, disjoint representations have no unitarily equivalent nonzero subrepresentations; this is the representation-theoretic content of superselection by the observable algebra.

In massive relativistic theories, the sectors relevant for ordinary charges are often captured by the Doplicher–Haag–Roberts localization criterion. A representation π is localized in a bounded region $\mathcal{O}$ relative to the vacuum if

$$\pi|_{\mathcal{A}(\mathcal{O}^\perp)} \simeq \pi_0|_{\mathcal{A}(\mathcal{O}^\perp)}, \tag{4.1}$$

where $\mathcal{O}^\perp$ denotes the spacelike complement and $\mathcal{A}(\mathcal{O}^\perp)$ is the algebra generated by local algebras $\mathcal{A}(\mathcal{U})$ with $\mathcal{U} \subset \mathcal{O}^\perp$. It is transportable if an equivalent representation can be localized in any other suitable bounded region. A transportable localized sector is a charge class whose localization region may be changed without changing its sector.

Figure 4.1 records the geometric content of the DHR localization condition in a $1+1$-dimensional slice. The relevant exterior region is the spacelike complement $\mathcal{O}^\perp$, not the set-theoretic complement of $\mathcal{O}$: observables supported there have no causal contact with the bounded localization region, and the sector representation restricts to the vacuum representation on the algebra generated by those observables.

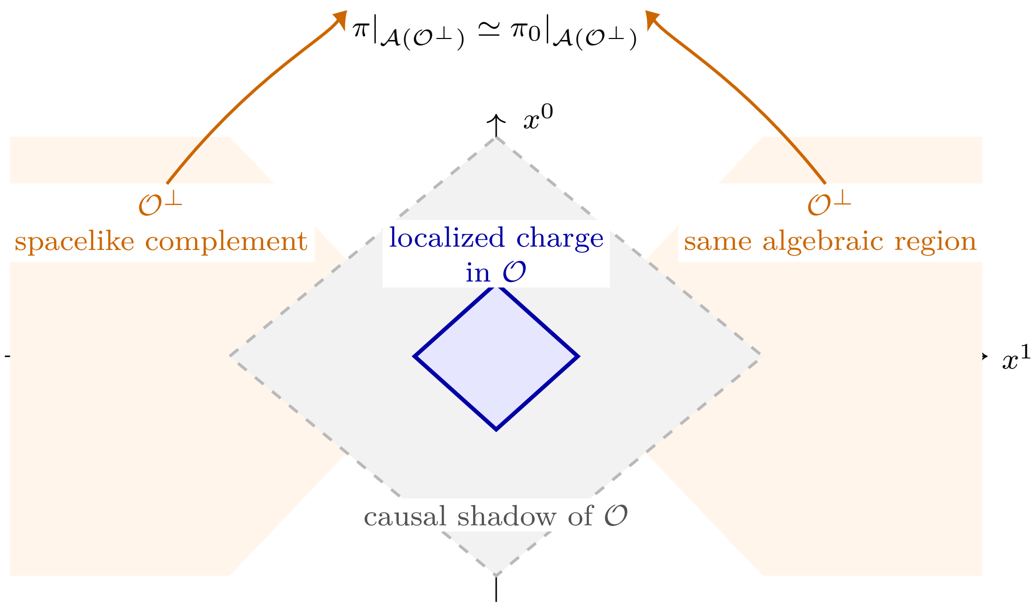


Figure 4.1: DHR localization in a spacetime slice. The sector may differ from the vacuum inside a bounded localization region $\mathcal{O}$, while on the algebra generated by observables in the spacelike complement $\mathcal{O}^\perp$ its representation is unitarily equivalent to the vacuum representation.

4.2 Endomorphisms, Fusion, and Intertwiners

Assume Haag duality in the vacuum representation,

$$\mathcal{R}(\mathcal{O}) = \mathcal{R}(\mathcal{O}^\perp)',$$

and assume the representation π satisfies the localization and transportability condition above. After identifying $\pi|_{\mathcal{A}(\mathcal{O}^\perp)}$ with $\pi_0|_{\mathcal{A}(\mathcal{O}^\perp)}$, and after replacing $\mathcal{A}_{\mathrm{qloc}}$ by its image in the vacuum representation, the action of π may be represented by an endomorphism

$$\rho : \mathcal{A}_{\mathrm{qloc}} \to \mathcal{A}_{\mathrm{qloc}}$$

localized in $\mathcal{O}$, meaning that

$$\rho(A) = A, \qquad A \in \mathcal{A}(\mathcal{O}^\perp).$$

The reason Haag duality appears in this sentence is the containment argument of Chapter 3. If $\mathcal{O}_0$ localizes ρ and $\mathcal{O}_0 \subset \mathcal{O}_1$, then for $A \in \mathcal{R}(\mathcal{O}_1)$ and $B \in \mathcal{R}(\mathcal{O}_1^\perp)$,

$$\rho(A)B = \rho(A)\rho(B) = \rho(AB) = \rho(BA) = B\rho(A).$$

Hence $\rho(A) \in \mathcal{R}(\mathcal{O}_1^\perp)'$. Haag duality turns this exterior-commutation conclusion into $\rho(A) \in \mathcal{R}(\mathcal{O}_1)$. Similarly, an intertwiner $T\rho(\,\cdot\,) = \sigma(\,\cdot\,)T$ between endomorphisms localized in $\mathcal{O}$ commutes with $\mathcal{R}(\mathcal{O}^\perp)$, and Haag duality places T in $\mathcal{R}(\mathcal{O})$. Without this step the same construction naturally lands in the dual net $\mathcal{R}^d(\mathcal{O}) = \mathcal{R}(\mathcal{O}^\perp)'$. Composition of localized endomorphisms defines fusion:

$$\rho_1 \otimes \rho_2 = \rho_1 \circ \rho_2.$$

We use ordinary function-composition order:

$$(\rho_1 \circ \rho_2)(A) = \rho_1(\rho_2(A)), \qquad A \in \mathcal{A}_{\mathrm{qloc}}.$$

Thus ρ_2 acts first and ρ_1 acts second. An intertwiner from ρ to σ is an observable $T \in \mathcal{A}_{\mathrm{qloc}}$ such that

$$T\rho(A) = \sigma(A)T, \qquad A \in \mathcal{A}_{\mathrm{qloc}}.$$

The intertwiners are morphisms between charge sectors. With direct sums, subobjects, conjugates, and a tensor product, transportable localized endomorphisms form a tensor category. Locality gives the exchange isomorphism

$$\varepsilon_{\rho,\sigma} : \rho \circ \sigma \to \sigma \circ \rho$$

when the localization regions are spacelike separated and then transported back to common reference regions. In spacetime dimension at least three, this exchange is symmetric when the DHR hypotheses used above are supplemented by transportability, Haag duality for the vacuum net, finite statistics, and the path-connectedness of the relevant spacelike localization configurations. The last condition is the topological input that makes the two exchanges ρ around σ homotopic. In two dimensions and for lower-codimension extended sectors, the corresponding configuration spaces need not have this symmetry; the exchange maps are then represented by braid group actions rather than by the symmetric group.

4.3 Statistics Operators and Conjugates

The exchange isomorphism contains the statistics of the sector. For a localized endomorphism ρ, choose a transported copy ρ' localized spacelike to ρ. Locality gives an intertwiner

$$\varepsilon_{\rho,\rho} : \rho \circ \rho \to \rho \circ \rho$$

after transporting ρ' back to the reference region. In spacetime dimension at least three, the DHR exchange operators satisfy symmetric tensor category identities, so irreducible sectors have Bose, Fermi, or more general parastatistics encoded by finite-dimensional permutation-group representations.

For objects η, θ, the morphism space $\mathrm{Hom}(\eta, \theta)$ is the space of intertwiners from η to θ. The symbol 1_η denotes the identity morphism in $\mathrm{Hom}(\eta, \eta)$; in the endomorphism realization it is the unit observable viewed as an intertwiner $\eta \to \eta$. A sector ρ has a conjugate $\bar{\rho}$ when there are intertwiners

$$R : \mathrm{id} \to \bar{\rho} \circ \rho, \qquad \bar{R} : \mathrm{id} \to \rho \circ \bar{\rho}$$

satisfying the conjugate equations

$$\bar{R}^* \rho(R) = 1_\rho, \qquad R^* \bar{\rho}(\bar{R}) = 1_{\bar{\rho}}.$$

The minimal value of $\|R\| \, \|\bar{R}\|$ defines the statistical dimension

$$d(\rho).$$

In the Doplicher–Haag–Roberts setting, saying that a sector has finite statistics is shorthand for a categorical finiteness package, not only for a choice of exchange sign. In the form used below it means that the sector lies in a C^*-tensor category with finite-dimensional intertwiner spaces, admits a conjugate satisfying the conjugate equations, and has finite statistical dimension $d(\rho) < \infty$. For reducible sectors, the statement is applied after decomposing into irreducible subobjects and requiring only finitely many irreducible summands in each object under consideration. For sectors reconstructed from a compact internal group, $d(\rho)$ is the ordinary dimension of the corresponding irreducible group representation. Thus charge conjugation, fusion multiplicities, and statistics are encoded in the tensor category of localized endomorphisms before any charged field coordinates are chosen.

4.4 Observable Algebra, Field Algebra, and Compact Internal Group

The observable algebra $\mathcal{A}_{\mathrm{qloc}}$ contains neutral operators. Charged fields, when they are introduced as operators, can be organized in an enlarged algebra

$$\mathcal{F}$$

with a compact internal group G acting by automorphisms. The observable algebra is the fixed-point algebra

$$\mathcal{A}_{\mathrm{qloc}} = \mathcal{F}^G.$$

Quoted Theorem 4.1 (Doplicher–Roberts reconstruction)**.** Proof references.[1] *Let $D \geq 3$. Let the vacuum representation of the observable net be irreducible and satisfy Haag duality for double cones. Let $\mathcal{C}$ be the category of DHR sectors represented by transportable localized endomorphisms of $\mathcal{A}_{\mathrm{qloc}}$. Equivalently, their associated representations satisfy the localization criterion* (4.1) *in some bounded double cone and are transportable to every other such double cone; in the endomorphism realization they act trivially on the algebra of the spacelike complement. Assume that $\mathcal{C}$ has the following structure:*

[1] The DHR sector analysis is developed in S. Doplicher, R. Haag, and J. E. Roberts, "Local observables and particle statistics I," *Communications in Mathematical Physics* **23** (1971), 199–230, and "Local observables and particle statistics II," *Communications in Mathematical Physics* **35** (1974), 49–85. The categorical compact-group duality is proved in S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Inventiones Mathematicae* **98** (1989), 157–218. The field-algebra reconstruction in the AQFT setting is proved in S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Communications in Mathematical Physics* **131** (1990), 51–107 [22, 23, 24, 25].

(i) *it is a symmetric C^*-tensor category with simple tensor unit;*

(ii) *each intertwiner space* $\mathrm{Hom}(\rho, \sigma)$ *is finite-dimensional;*

(iii) *the category is closed under finite direct sums and subobjects;*

(iv) *every irreducible object has a conjugate object satisfying the conjugate equations and has finite statistical dimension $d(\rho) < \infty$;*

(v) *every object of $\mathcal{C}$ is a finite direct sum of irreducible subobjects of the preceding kind.*

Then there is a compact group

$$G = \mathrm{Aut}^{\otimes,*}(\omega_{\mathrm{fib}})$$

for a faithful symmetric tensor $*$-functor $\omega_{\mathrm{fib}} : \mathcal{C} \to \mathrm{Hilb}_{\mathrm{fd}}$, where $\mathrm{Hilb}_{\mathrm{fd}}$ *denotes finite-dimensional Hilbert spaces, and*

$$\mathcal{C} \simeq \mathrm{Rep}_{\mathrm{fd}}(G)$$

as symmetric C^*-tensor categories. Moreover, after the standard Doplicher–Roberts field-algebra reconstruction, there is a C^*-algebra $\mathcal{F}$ carrying a continuous action of G by automorphisms such that

$$\mathcal{A}_{\mathrm{qloc}} \simeq \mathcal{F}^G.$$

The irreducible DHR sectors correspond to irreducible finite-dimensional unitary representations of G, and charged field multiplets in $\mathcal{F}$ transform according to these representations.

In this quoted theorem, the word gauge means a reconstruction of charged field coordinates from the neutral observable category; physical observables remain the G-invariant elements. The compact group description is therefore not an input at this stage. It is a conclusion from locality, transportability, Haag duality, symmetric exchange, and finite statistics.

Reconstruction logic and theorem boundary. The reconstruction has two logically separate components. The first is the QFT-specific passage from localized representations to a rigid symmetric C^*-tensor category. Haag duality is the step that permits a DHR representation localized in $\mathcal{O}$ to be represented by an endomorphism ρ of the vacuum quasilocal algebra acting trivially on $\mathcal{A}(\mathcal{O}^\perp)$. Transportability makes the localization region a choice of representative rather than part of the sector. Locality and the path-connectedness of spacelike configurations in $D \geq 3$ give a symmetric exchange isomorphism, while finite statistics supplies conjugates and finite statistical dimension. These are the hypotheses that must be verified in a model before the theorem applies.

The second component is categorical duality. Once the category $\mathcal{C}$ has the properties listed in Theorem 4.1, the compact group

$$G = \mathrm{Aut}^{\otimes,*}(\omega_{\mathrm{fib}})$$

is obtained as the group of unitary tensor automorphisms of a faithful fiber functor ω_{fib}. Its topology is the compact topology inherited from the product of the unitary groups $\prod_{\rho \in \mathrm{Obj}(\mathcal{C})} U(\omega_{\mathrm{fib}}(\rho))$, with the closed conditions expressing naturality and tensor compatibility. The equivalence $\mathcal{C} \simeq \mathrm{Rep}_{\mathrm{fd}}(G)$ is therefore a Tannakian output of the observable sector category.

The field algebra reconstructed from $\mathcal{C}$ can be recognized locally by its charged multiplets. For an irreducible sector ρ with statistical dimension $d(\rho)$, which is the dimension of the corresponding irreducible representation after the Doplicher–Roberts equivalence, the reconstructed algebra contains a finite-dimensional Hilbert space of charged operators

$$\mathcal{H}_\rho^{\mathrm{ch}} = \mathrm{span}\{\psi_{\rho,1}, \ldots, \psi_{\rho,d(\rho)}\} \subset \mathcal{F}$$

whose elements satisfy the intertwining relation

$$\rho(A)\psi_{\rho,i} = \psi_{\rho,i}A, \qquad A \in \mathcal{A}_{\mathrm{qloc}}. \tag{4.2}$$

After choosing an orthonormal multiplet, one has the Cuntz-type normalization

$$\psi_{\rho,i}^*\psi_{\rho,j} = \delta_{ij}\mathbf{1}, \qquad \sum_{i=1}^{d(\rho)} \psi_{\rho,i}\psi_{\rho,i}^* = \mathbf{1}, \tag{4.3}$$

in the finite-statistics case, and the observable endomorphism is recovered from the charged fields by

$$\rho(A) = \sum_{i=1}^{d(\rho)} \psi_{\rho,i}A\psi_{\rho,i}^*. \tag{4.4}$$

The G-action rotates the multiplet by the representation corresponding to ρ:

$$\alpha_g(\psi_{\rho,i}) = \sum_{j=1}^{d(\rho)} \psi_{\rho,j}\, D_\rho(g)_{ji}, \qquad g \in G. \tag{4.5}$$

Equations (4.2)–(4.5) are not additional assumptions on observables; they are the operator form of the reconstructed charged coordinates. The observable algebra remains the fixed algebra $\mathcal{F}^G$, and the physical sector information is already present in $\mathcal{C}$.

Local charged-field core. The preceding formulae become a local field net only after the localization regions of the DHR endomorphisms are used. Fix a double cone $\mathcal{O}$ and choose, for each irreducible DHR sector ρ localizable in $\mathcal{O}$, a charged multiplet

$$\mathcal{H}_\rho^{\mathrm{ch}}(\mathcal{O}) = \mathrm{span}\{\psi_{\rho,1}^{\mathcal{O}}, \ldots, \psi_{\rho,d(\rho)}^{\mathcal{O}}\}$$

satisfying (4.2) with ρ localized in $\mathcal{O}$. The algebraic local core is the finite linear span

$$\mathcal{F}_{\mathrm{alg}}(\mathcal{O}) = \mathrm{span}\{A\psi_{\rho,i}^{\mathcal{O}} : A \in \mathcal{R}(\mathcal{O}),\ \rho \in \mathrm{Irr}(\mathcal{C}),\ 1 \le i \le d(\rho)\}. \tag{4.6}$$

This is a core, not yet a completed local C^*-algebra. Its multiplication is forced by the intertwining relation. If $A, B \in \mathcal{R}(\mathcal{O})$, then

$$(A\psi_{\rho,i}^{\mathcal{O}})(B\psi_{\sigma,j}^{\mathcal{O}}) = A\,\rho(B)\,\psi_{\rho,i}^{\mathcal{O}}\psi_{\sigma,j}^{\mathcal{O}}. \tag{4.7}$$

Haag duality and the containment calculation above are used here in a load-bearing way: since ρ is localized in $\mathcal{O}$, $\rho(B) \in \mathcal{R}(\mathcal{O})$. The product $\psi_{\rho,i}^{\mathcal{O}}\psi_{\sigma,j}^{\mathcal{O}}$ transforms in the sector $\rho \circ \sigma$, because

$$(\rho \circ \sigma)(A)\psi_{\rho,i}^{\mathcal{O}}\psi_{\sigma,j}^{\mathcal{O}} = \psi_{\rho,i}^{\mathcal{O}}\sigma(A)\psi_{\sigma,j}^{\mathcal{O}} = \psi_{\rho,i}^{\mathcal{O}}\psi_{\sigma,j}^{\mathcal{O}}A.$$

After decomposing $\rho\circ\sigma$ into irreducible sectors, the Clebsch–Gordan intertwiners in $\mathrm{Hom}(\tau, \rho\circ\sigma)$ rewrite the right side of (4.7) as a finite sum of generators of the form (4.6). Thus fusion in the DHR category is the multiplication law of local charged coordinates.

The involution uses the conjugate sector. Finite statistics supplies standard conjugate data

$$R : \mathrm{id} \to \bar{\rho} \circ \rho, \qquad \bar{R} : \mathrm{id} \to \rho \circ \bar{\rho},$$

and the adjoint of a ρ-charged generator is expanded, through this conjugate data and local intertwiners, as a finite linear combination of $\bar{\rho}$-charged generators with coefficients in $\mathcal{R}(\mathcal{O})$. Finite statistics therefore supplies the adjoint-closing structure required by the field-algebra theorem, in addition to the exchange data.

The reconstructed compact group acts on the local core by

$$\alpha_g(A) = A, \qquad \alpha_g(\psi^{\mathcal{O}}_{\rho,i}) = \sum_j \psi^{\mathcal{O}}_{\rho,j} D_\rho(g)_{ji}.$$

The Haar conditional expectation on the local core is

$$E_{\mathcal{O}}(F) = \int_G \alpha_g(F)\,\mathrm{d}g. \tag{4.8}$$

By Schur orthogonality it kills every nontrivial irreducible charged block and keeps the trivial block. Consequently

$$E_{\mathcal{O}}(\mathcal{F}_{\mathrm{alg}}(\mathcal{O})) = \mathcal{R}(\mathcal{O}), \qquad \mathcal{F}_{\mathrm{alg}}(\mathcal{O})^G = \mathcal{R}(\mathcal{O})$$

at the algebraic core level. The Doplicher–Roberts field-algebra theorem is the statement that this core can be completed, with compatible choices as $\mathcal{O}$ varies, to a local charged-field net whose G-fixed subnet is the original observable net. The nontrivial analytic work is not the formal multiplication above; it is the proof that the transported charged multiplets, exchange operators, adjoints, C^*-norm, and locality relations can be chosen coherently across all regions.

The word conditional in $E_{\mathcal{O}}$ is also load-bearing. On the algebraic core, Haar invariance gives

$$E_{\mathcal{O}}^2 = E_{\mathcal{O}}, \qquad E_{\mathcal{O}}(\alpha_g(F)) = E_{\mathcal{O}}(F), \qquad E_{\mathcal{O}}(A_1 F A_2) = A_1 E_{\mathcal{O}}(F) A_2$$

for $A_1, A_2 \in \mathcal{R}(\mathcal{O})$. Positivity is inherited from averaging $*$-automorphisms. Thus the fixed-point statement has the algebraic form of a faithful expectation from charged coordinates back to neutral observables, with charge-zero projection as its representation-ring shadow. The theorem-level completion must preserve this expectation in the local C^*-norms.

Charged locality and observable locality. For spacelike separated double cones, the charged field generators need not commute as ordinary neutral observables. Their exchange is controlled by the DHR symmetry

$$\varepsilon_{\rho,\sigma} : \rho \circ \sigma \to \sigma \circ \rho.$$

In dimension $D \geq 3$, under the symmetric finite-statistics hypotheses of Theorem 4.1, this exchange reduces to the Bose/Fermi/parastatistical symmetry encoded by the reconstructed compact group and its central grading. The invariant subnet is local because the exchange factors cancel in the trivial representation channel. Thus the locality statement for observables is recovered as the invariant part of the charged-field exchange mechanism.

Neutral-channel locality in the reconstructed core. The preceding cancellation is a categorical statement before it is an operator-closure theorem. Let V_ρ and V_σ be the finite charged multiplet spaces carrying the compact-group representations D_ρ and D_σ. The projection onto the neutral component of the product multiplet is

$$P_{\mathbf{1}}^{\rho,\sigma} = \int_G D_\rho(g) \otimes D_\sigma(g)\,\mathrm{d}g \in \mathrm{End}(V_\rho \otimes V_\sigma). \tag{4.9}$$

For irreducible ρ, σ, Schur orthogonality says that $P_{\mathbf{1}}^{\rho,\sigma} = 0$ unless $\sigma \simeq \bar{\rho}$; in the conjugate case its image is the one-dimensional tensor-unit summand

$$\mathrm{Hom}_G(\mathbb{C}, V_\rho \otimes V_{\bar{\rho}}).$$

Applying $E_{\mathcal{O}}$ in (4.8) to a product of charged local generators is exactly the insertion of these projectors on the charged indices after the fusion decomposition. Thus an observable local element is not an arbitrary product of charged fields; it is the tensor-unit component of such a product, with all nontrivial charged blocks removed by Haar projection.

Now take $F_1 \in \mathcal{F}_{\mathrm{alg}}(\mathcal{O}_1)^G$ and $F_2 \in \mathcal{F}_{\mathrm{alg}}(\mathcal{O}_2)^G$ with $\mathcal{O}_1 \perp \mathcal{O}_2$. The charged-field exchange operation first transports the charged tensor factors by $\varepsilon_{\rho,\sigma}$. But F_1 and F_2 lie in tensor-unit subobjects after the projections (4.9). Naturality of the symmetric braiding and simplicity of the tensor unit give

$$\varepsilon_{\mathbf{1},\mathbf{1}} = \mathrm{id}_{\mathbf{1}}\,.$$

Hence no charge matrix remains on the invariant channel: the categorical exchange reduces to the ordinary equality required for observable locality. This proves only the algebraic neutral-channel mechanism in the DR core. The hard local-QFT theorem remains the coherent C^*-completion and regionwise choice of charged multiplets for which the charged exchange relations and the invariant local observable net survive completion.

Finite pointed check of the compact group output. A useful local model of the categorical part is the following pointed symmetric C^*-tensor category. Fix an integer $N \geq 2$. Let the simple objects be X_q, $q \in \mathbb{Z}/N\mathbb{Z}$, with

$$X_q \otimes X_r \simeq X_{q+r}, \qquad X_0 = \mathbf{1}, \qquad \mathrm{Hom}(X_q, X_r) = \begin{cases} \mathbb{C}, & q = r, \\ 0, & q \neq r. \end{cases}$$

Take the fiber functor $\omega_{\mathrm{fib}}(X_q) = \mathbb{C}$. A unitary tensor automorphism $u \in \mathrm{Aut}^{\otimes,*}(\omega_{\mathrm{fib}})$ is then a choice of phases $u_q \in U(1)$ satisfying

$$u_0 = 1, \qquad u_{q+r} = u_q u_r.$$

Thus $u_q = u_1^q$ and $u_1^N = 1$. The reconstructed compact group is

$$G \simeq \mu_N = \{\zeta \in U(1) : \zeta^N = 1\}, \tag{4.10}$$

and X_q is the one-dimensional representation

$$\zeta \longmapsto \zeta^q.$$

If the corresponding algebraic charged-field core is written as a finite grading

$$\mathcal{F}_{\mathrm{alg}} = \bigoplus_{q \in \mathbb{Z}/N\mathbb{Z}} \mathcal{F}_q, \qquad \mathcal{F}_q \mathcal{F}_r \subset \mathcal{F}_{q+r}, \qquad \alpha_\zeta(F_q) = \zeta^q F_q, \quad F_q \in \mathcal{F}_q, \tag{4.11}$$

then the fixed algebra is exactly the degree-zero summand:

$$\mathcal{F}_{\mathrm{alg}}^{\mu_N} = \mathcal{F}_0. \tag{4.12}$$

Indeed, averaging F_q over G multiplies it by

$$\frac{1}{N} \sum_{\zeta^N = 1} \zeta^q = \delta_{q,0}.$$

This computation functions as a finite-dimensional diagnostic for the theorem's output: the compact group is recovered from the tensor compatibility of the sector fiber functor, and invariants project precisely onto neutral degree.

Finite nonabelian check of the fiber functor. The compact-group output also has genuinely nonabelian content. Take the finite nonabelian group S_3. Its irreducible representations are the trivial representation $\mathbf{1}$, the sign representation ϵ, and the two-dimensional standard representation V. With conjugacy classes ordered as

$$(e), \qquad (12), \qquad (123),$$

their characters are

	e	(12)	(123)
$\mathbf{1}$	1	1	1
ϵ	1	-1	1
V	2	0	-1.

The inner product

$$\langle \chi, \psi \rangle_{S_3} = \frac{1}{6} \sum_{g \in S_3} \chi(g) \overline{\psi(g)}$$

gives the tensor-product rules

$$\epsilon \otimes \epsilon \simeq \mathbf{1}, \qquad \epsilon \otimes V \simeq V, \qquad V \otimes V \simeq \mathbf{1} \oplus \epsilon \oplus V. \tag{4.13}$$

For each group element g, evaluation gives a unitary tensor automorphism of the forgetful fiber functor

$$\omega_{\mathrm{fib}} : \mathrm{Rep}_{\mathrm{fd}}(S_3) \to \mathrm{Hilb}_{\mathrm{fd}}, \qquad \eta_g(W) = D_W(g).$$

Tensor compatibility is the identity

$$D_{W_1 \otimes W_2}(g) = D_{W_1}(g) \otimes D_{W_2}(g),$$

and naturality says that every intertwiner, in particular the projection maps realizing (4.13), commutes with η_g. The standard representation is faithful, so the six automorphisms η_g are distinct. The converse statement, that every unitary tensor natural automorphism of the forgetful functor is evaluation at a unique $g \in S_3$, is the finite Tannakian part of the Doplicher–Roberts theorem in this example. The calculation above makes visible what the theorem reconstructs: a nonabelian group is encoded by the compatibility of the multiplet transformations with all fusion and intertwiner maps, not by an externally assigned charge label.

The fixed-point projection is likewise representation-theoretic rather than degree-theoretic. If a finite charged-field core decomposes into matrix coefficient spaces for irreducible S_3-representations, Haar averaging is

$$E_{S_3}(F) = \frac{1}{6}\sum_{g\in S_3} \alpha_g(F).$$

For a nontrivial irreducible representation W, Schur orthogonality gives

$$\frac{1}{6}\sum_{g\in S_3} D_W(g)_{ab} = 0,$$

whereas the trivial representation is fixed. Thus averaging projects onto the $\mathbf{1}$-isotypic, observable part. In the pointed $\mathbb{Z}/N\mathbb{Z}$ model this projection is the elementary root-of-unity sum; in the nonabelian S_3 model it is the vanishing of nontrivial matrix coefficients.

Finite nonabelian charged-coordinate core. The preceding S_3 calculation can be made into a finite field-core model, with the observable algebra suppressed to $\mathbb{C}$, by taking

$$\mathcal{F}_{\mathrm{ch}} = \mathrm{Fun}(S_3)$$

with pointwise multiplication. The reconstructed compact group acts on the right:

$$(\alpha_h f)(g) = f(gh), \qquad f \in \mathrm{Fun}(S_3), \quad g, h \in S_3. \tag{4.14}$$

The fixed algebra is the constants, since right translations act transitively on S_3. For an irreducible finite-dimensional representation W, its matrix coefficients

$$f^W_{ab}(g) = D_W(g)_{ab}$$

span a right S_3-multiplet:

$$\alpha_h(f^W_{ab}) = \sum_c f^W_{ac}\, D_W(h)_{cb}. \tag{4.15}$$

Thus the column index is the charged multiplet index, while the row index records the multiplicity with which W appears in the finite regular field-core model. Haar projection is the finite conditional expectation

$$E_{S_3}(f) = \frac{1}{6}\sum_{h\in S_3} \alpha_h(f), \qquad E_{S_3}(f^W_{ab}) = \begin{cases} 1, & W = \mathbf{1}, \\ 0, & W \neq \mathbf{1}, \end{cases} \tag{4.16}$$

with the second line understood after choosing the constant matrix coefficient for $W = \mathbf{1}$. Multiplication of charged coordinates is the matrix-coefficient form of tensor product:

$$f^U_{ab}(g) f^V_{cd}(g) = D_{U\otimes V}(g)_{(a,c),(b,d)}. \tag{4.17}$$

Decomposing $U \otimes V$ by intertwiners gives the fusion rules such as (4.13). This finite model is not a local QFT field algebra; its observable part is only $\mathbb{C}$. Its purpose is to show, without the analytic burden of DHR localization, how nonabelian charged coordinates, Haar expectation, and fusion multiplication are the same data that the Doplicher–Roberts theorem reconstructs from a symmetric finite-statistics sector category.

Finite Tannakian converse through the regular algebra. The finite regular algebra also makes the converse direction of the Tannakian reconstruction visible. Let G be a finite group and let

$$\mathcal{R} = \mathrm{Fun}(G)$$

with point idempotents δ_x, $x \in G$. The left regular action is

$$(L_a f)(g) = f(a^{-1}g), \qquad L_a\delta_x = \delta_{ax}. \tag{4.18}$$

Every unital $*$-automorphism T of the commutative finite-dimensional algebra $\mathcal{R}$ permutes the minimal projections:

$$T\delta_x = \delta_{\phi(x)}$$

for a unique permutation ϕ of G. If T commutes with every left translation, then

$$\delta_{\phi(ax)} = TL_a\delta_x = L_aT\delta_x = \delta_{a\phi(x)}.$$

Hence $\phi(ax) = a\phi(x)$. Setting $x = e$ gives

$$\phi(a) = ah, \qquad h = \phi(e),$$

and therefore $\phi(x) = xh$ for all $x \in G$. Thus every left-equivariant algebra automorphism of $\mathrm{Fun}(G)$ is a right translation:

$$(T_h f)(g) = f(gh^{-1}). \tag{4.19}$$

For a matrix coefficient $f^W_{\alpha i}(g) = D_W(g)_{\alpha i}$, (4.19) gives

$$T_h f^W_{\alpha i}(g) = D_W(gh^{-1})_{\alpha i} = \sum_j f^W_{\alpha j}(g)\, D_W(h^{-1})_{ji}. \tag{4.20}$$

Thus the corresponding tensor natural automorphism of the forgetful fiber functor is

$$u_W = D_W(h^{-1})$$

on every finite-dimensional representation W. Conversely, a unitary tensor natural automorphism u_W acts on matrix coefficients by the right side of (4.20); tensor compatibility makes this action multiplicative under (4.17), naturality makes it commute with the intertwiners that decompose tensor products, and the Peter–Weyl decomposition of $\mathrm{Fun}(G)$ gives a left-equivariant $*$-automorphism of $\mathcal{R}$. The preceding paragraph then supplies a unique $h \in G$. For finite groups this proves

$$\mathrm{Aut}^{\otimes,*}(\omega_{\mathrm{fib}}) \simeq G$$

in the convention $u_W = D_W(h^{-1})$. The compact-group Doplicher–Roberts theorem uses the same Peter–Weyl mechanism with the dense Hopf $*$-algebra of representative functions and the C^*-norm closure; the local-QFT burden remains the DHR construction of the symmetric finite-statistics sector category and the field-algebra realization of this categorical duality.

Compact Peter–Weyl coordinate algebra. The compact version of the preceding finite calculation is the representative-function Hopf $*$-algebra. Let G be a compact group and let W range over finite-dimensional continuous unitary representations. For $v \in W$ and $\ell \in W^*$, write

$$f^W_{\ell,v}(g) = \ell(D_W(g)v).$$

The algebra

$$\mathrm{Pol}(G) = \mathrm{span}\{f^W_{\ell,v}\} \subset C(G)$$

is closed under multiplication because

$$f^{W_1}_{\ell_1,v_1}(g)\, f^{W_2}_{\ell_2,v_2}(g) = f^{W_1\otimes W_2}_{\ell_1\otimes\ell_2,\, v_1\otimes v_2}(g), \tag{4.21}$$

and it is closed under $*$ because complex conjugation is the matrix coefficient of the conjugate representation. The Hopf operations are the group-law operations written on functions; in this paragraph write Δ_{H} for the coproduct, to avoid confusing it with the modular operator Δ used elsewhere in the chapter:

$$(\Delta_{\mathrm{H}} f)(g,h) = f(gh), \quad \epsilon(f) = f(e), \quad (Sf)(g) = f(g^{-1}). \tag{4.22}$$

Associativity of group multiplication gives coassociativity of Δ_{H}; the identity element gives the counit identities; inversion gives the antipode identities

$$m(S\otimes \mathrm{id})\Delta_{\mathrm{H}} f = m(\mathrm{id}\otimes S)\Delta_{\mathrm{H}} f = \epsilon(f)\mathbf{1}.$$

Peter–Weyl says that $\mathrm{Pol}(G)$ is uniformly dense in $C(G)$, and Schur orthogonality decomposes it as the algebraic direct sum of matrix coefficient spaces

$$\mathrm{Pol}(G) \simeq \bigoplus_{\widehat{G}} W^* \otimes W.$$

Haar integration is the unique bi-invariant positive normalized functional

$$h_G(f) = \int_G f(g)\,\mathrm{d}g,$$

and it kills every nontrivial irreducible matrix-coefficient block while keeping the trivial block.

This compact Peter–Weyl algebra is the coordinate side of the compact group appearing in Doplicher–Roberts reconstruction. The tensor category supplies the representations W, tensor products supply multiplication in (4.21), and tensor natural automorphisms of the fiber functor are evaluated by the same matrices $D_W(g)$. The representative functions separate points of G, so the compact group is recoverable from how all charged multiplets transform; equivalently, after the C^*-completion $C(G)$ is supplied, the character space recovers G with its compact topology. None of this proves that a given observable net has the required DHR category; it identifies the compact-group algebra that appears after those local-QFT hypotheses have already been established.

Compact abelian completion: the $U(1)$ charge lattice. The finite pointed example also has a compact limit that contains no finite-group artifact. Let $\mathrm{Rep}_{\mathrm{fd}}(U(1))$ denote the category of finite-dimensional continuous unitary representations of $U(1)$. Its simple objects are the one-dimensional characters L_n, $n \in \mathbb{Z}$, with

$$L_m \otimes L_n \simeq L_{m+n}, \qquad L_0 = \mathbf{1}, \qquad \mathrm{Hom}(L_m, L_n) = \begin{cases} \mathbb{C}, & m = n, \\ 0, & m \neq n. \end{cases}$$

For the forgetful fiber functor $\omega_{\mathrm{fib}} : \mathrm{Rep}_{\mathrm{fd}}(U(1)) \to \mathrm{Hilb}_{\mathrm{fd}}$, a unitary tensor natural automorphism is a family of phases

$$u_n \in U(1), \qquad n \in \mathbb{Z},$$

satisfying

$$u_0 = 1, \qquad u_{m+n} = u_m u_n, \qquad u_{-n} = \overline{u_n}.$$

The first two equations give $u_n = u_1^n$ for $n > 0$ and then, by unitarity, for all $n \in \mathbb{Z}$. Conversely every $\lambda \in U(1)$ gives such a family by $u_n = \lambda^n$. With the topology of pointwise convergence on objects, the evaluation map

$$U(1) \longrightarrow \mathrm{Aut}^{\otimes,*}(\omega_{\mathrm{fib}}), \qquad \lambda \longmapsto (u_n = \lambda^n)_{n \in \mathbb{Z}},$$

is therefore a homeomorphism: continuity is pointwise continuity of λ^n, and the inverse reads off u_1.

The corresponding representative-function algebra is the Laurent polynomial $*$-algebra

$$\mathbb{C}[z, z^{-1}], \qquad z^m z^n = z^{m+n}, \qquad (z^n)^* = z^{-n}.$$

Its C^*-completion in the supremum norm on the unit circle is

$$C(U(1)),$$

because Laurent polynomials are a unital self-adjoint subalgebra separating points of $U(1)$. Haar expectation is the projection onto the trivial isotypic component:

$$E_{U(1)}(z^n) = \int_{U(1)} z^n \, \mathrm{d}z = \delta_{n0}. \tag{4.23}$$

Thus a $U(1)$-graded charged algebraic core

$$\mathcal{F}_{\mathrm{alg}} = \bigoplus_{n \in \mathbb{Z}} \mathcal{F}_n, \qquad \mathcal{F}_m \mathcal{F}_n \subset \mathcal{F}_{m+n}, \qquad \alpha_\lambda(F_n) = \lambda^n F_n$$

has invariant part

$$\mathcal{F}_{\mathrm{alg}}^{U(1)} = \mathcal{F}_0$$

on every finite Laurent-polynomial subspace. This compact abelian calculation is the charge-lattice version of the finite $\mathbb{Z}/N\mathbb{Z}$ diagnostic above. It supplies the categorical and representative-function completion, not the local-QFT theorem: a concrete QFT still has to produce the localized symmetric finite-statistics category, the field algebra, and the action whose fixed points are the observable net.

Example 4.2 (Pointed field algebra as a crossed product)**.** The same pointed category has an explicit charged-field algebra when the sector X_1 is represented by an order-N localized automorphism

$$\rho \in \mathrm{Aut}(\mathcal{A}), \qquad \rho^N = \mathrm{id}_{\mathcal{A}},$$

of a neutral algebra $\mathcal{A}$. Introduce a unitary charged generator u with

$$u^N = 1, \qquad uAu^* = \rho(A), \quad A \in \mathcal{A},$$

and define the algebraic field core

$$\mathcal{F}_{\mathrm{alg}} = \bigoplus_{q \in \mathbb{Z}/N\mathbb{Z}} \mathcal{A}u^q. \tag{4.24}$$

The multiplication and involution are

$$(Au^q)(Bu^r) = A\,\rho^q(B)\,u^{q+r}, \qquad (Au^q)^* = \rho^{-q}(A^*)u^{-q}. \tag{4.25}$$

Associativity follows from the group law $\rho^{q+r} = \rho^q\rho^r$:

$$\begin{aligned}((Au^q)(Bu^r))(Cu^s) &= A\rho^q(B)\rho^{q+r}(C)u^{q+r+s},\\ (Au^q)((Bu^r)(Cu^s)) &= A\rho^q\big(B\rho^r(C)\big)u^{q+r+s}.\end{aligned}$$

The two expressions agree because ρ^q is an algebra automorphism. The involution is anti-multiplicative because

$$\begin{aligned}\big((Au^q)(Bu^r)\big)^* &= \rho^{-q-r}\big(\rho^q(B^*)A^*\big)u^{-q-r}\\ &= \rho^{-r}(B^*)\,\rho^{-q-r}(A^*)u^{-q-r} = (Bu^r)^*(Au^q)^*.\end{aligned}$$

The reconstructed compact group $\mu_N = \{\zeta \in U(1) : \zeta^N = 1\}$ acts by

$$\alpha_\zeta(Au^q) = \zeta^q Au^q. \tag{4.26}$$

Its fixed algebra is exactly the neutral algebra:

$$\mathcal{F}_{\mathrm{alg}}^{\mu_N} = \mathcal{A}.$$

Indeed, the Haar average over the finite group is

$$E_{\mu_N}\left(\sum_q A_q u^q\right) = \frac{1}{N}\sum_{\zeta^N=1}\sum_q \zeta^q A_q u^q = A_0.$$

Thus the charge creator u implements the localized sector by

$$\rho(A) = uAu^*,$$

while the observable algebra is recovered as the invariant subalgebra. This example is the DR field-algebra output in the invertible pointed case. It is an algebraic output construction. To interpret it as a local charged-field net, the QFT must first supply the DHR input package: locality, transportability, Haag duality, and finite statistics for the sector endomorphisms.

This theorem therefore applies to charges whose sector representatives are localized in bounded regions and satisfy the finite-statistics DHR hypotheses. Charged sectors controlled by long-range Gauss laws, by Wilson-line or spacelike-cone localization, by infraparticle clouds, or by confining dynamics require different localization categories and additional analytic input. Such sectors are not failures of the reconstruction theorem; they lie outside the DHR category to which the theorem is attached. For them one must first define the relevant charged operators and localization structure before asking for an analogue of field-algebra reconstruction or of Haag–Ruelle scattering.

Constructive kink-sector diagnostic. The boundary of the DHR category is visible in broken two-dimensional constructive scalar models. Suppose a massive model has pure phases indexed by σ, vacuum representations π_σ, and a bounded local order-parameter observable B with distinct phase values m_σ. More precisely, choose spatial averaging functions supported in the far left and far right tails and let $B_L(R)$, $B_R(R)$ denote the corresponding bounded observables, with supports moving to $x^1 < -R$ and $x^1 > R$. A solitonic representation

$$\pi_{\sigma_-,\sigma_+}$$

interpolating from phase σ_- at $x^1 = -\infty$ to phase σ_+ at $x^1 = +\infty$ is characterized, when the constructive limits have been proved, by weak-operator tail limits

$$\operatorname*{w-lim}_{R\to\infty} \pi_{\sigma_-,\sigma_+}(B_L(R)) = m_{\sigma_-}\mathbf{1}, \qquad \operatorname*{w-lim}_{R\to\infty} \pi_{\sigma_-,\sigma_+}(B_R(R)) = m_{\sigma_+}\mathbf{1}. \tag{4.27}$$

These limits are macroscopic phase data, not local charged-field data. If π_{σ_-,σ_+} were DHR-localized in a bounded double cone $\mathcal{O}$ relative to a vacuum phase π_{σ_0}, then for sufficiently large R both $B_L(R)$ and $B_R(R)$ would belong to $\mathcal{A}(\mathcal{O}^\perp)$. The DHR equivalence

$$\pi_{\sigma_-,\sigma_+}|_{\mathcal{A}(\mathcal{O}^\perp)} \simeq \pi_{\sigma_0}|_{\mathcal{A}(\mathcal{O}^\perp)}$$

would therefore carry the weak limits of the two tails to the corresponding tail limits in the phase σ_0, namely $m_{\sigma_0}\mathbf{1}$ on both sides. Hence bounded-region DHR localization forces

$$m_{\sigma_-} = m_{\sigma_0} = m_{\sigma_+}.$$

Whenever the left and right asymptotic phases differ, the soliton sector is therefore not a DHR sector relative to a single vacuum representation. It is a sector controlled by asymptotic boundary conditions, equivalently by morphisms between phase representations or by half-line/spacelike-cone localization data. The diagnostic does not assert the existence of such soliton Hilbert spaces; that is a model theorem. It shows which representation-theoretic statement must be proved before a constructive kink can be brought into the DHR local-endomorphism framework, and why the Doplicher–Roberts theorem above is a bounded-charge theorem rather than a classification of all superselection sectors in low-dimensional models.

4.5 Split Property and Local Tensor Factorization

Let $\mathcal{O}_1 \Subset \mathcal{O}_2$ mean that the closure of $\mathcal{O}_1$ is contained in $\mathcal{O}_2$ with a positive spacetime collar. The split property is the existence of a type I factor $\mathcal{N}$ satisfying

$$\mathcal{R}(\mathcal{O}_1) \subset \mathcal{N} \subset \mathcal{R}(\mathcal{O}_2). \tag{4.28}$$

A type I factor is isomorphic to $\mathcal{B}(\mathcal{K})$ for some Hilbert space $\mathcal{K}$. The split inclusion gives a controlled tensor-product separation between observables well inside $\mathcal{O}_1$ and observables outside $\mathcal{O}_2$. It is a precise algebraic substitute for a finite-dimensional subsystem factorization at nonzero separation.

Figure 4.2 records the split inclusion $\mathcal{R}(\mathcal{O}_1) \subset \mathcal{N} \subset \mathcal{R}(\mathcal{O}_2)$ and its role as a type-I tensor-product separation at positive distance.

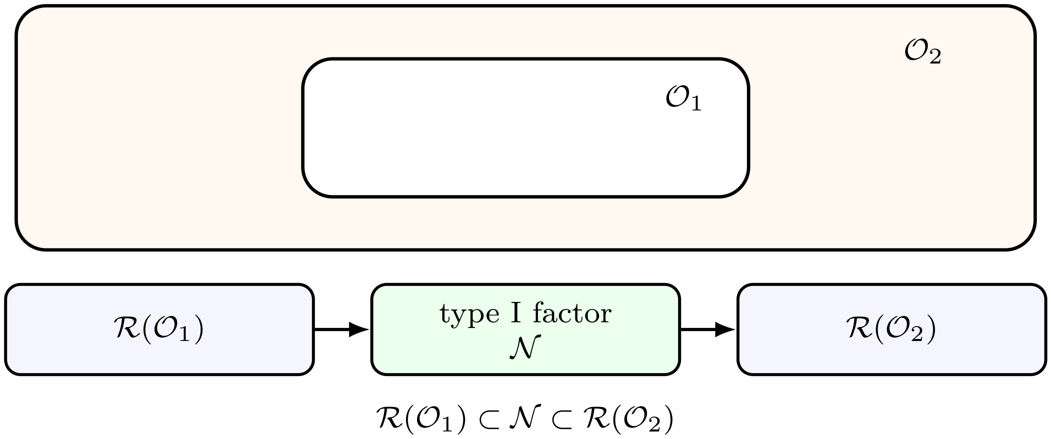


Figure 4.2: The split property inserts a type I factor between nested local algebras separated by a collar.

The split property is tied to phase-space and nuclearity conditions. Such conditions control the number of local degrees of freedom below a given energy scale and make local thermodynamic behavior mathematically tractable.

The operational meaning of the type-I interpolant is normal statistical independence at positive separation. If $\mathcal{M}_1 = \mathcal{R}(\mathcal{O}_1)$, $\mathcal{M}_2 = \mathcal{R}(\mathcal{O}_2)$, and $\mathcal{M}_1 \subset \mathcal{N} \subset \mathcal{M}_2$ is split, choose a spatial representation

$$U\mathcal{N}U^* = \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}, \qquad U\mathcal{N}'U^* = \mathbf{1} \otimes \mathcal{B}(\mathcal{L}).$$

Then $U\mathcal{M}_1U^* \subset \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}$ and $U\mathcal{M}_2'U^* \subset \mathbf{1} \otimes \mathcal{B}(\mathcal{L})$. Hence normal states φ_1 on $\mathcal{M}_1$ and φ_2 on $\mathcal{M}_2'$ extend to normal states $\widetilde{\varphi}_1$ on $\mathcal{B}(\mathcal{K})$ and $\widetilde{\varphi}_2$ on $\mathcal{B}(\mathcal{L})$. The normal product state $\widetilde{\varphi}_1 \bar{\otimes} \widetilde{\varphi}_2$ restricts to a normal state on the von Neumann algebra generated by the two commuting algebras, and on products it satisfies

$$\Phi(AB') = \varphi_1(A)\varphi_2(B'), \qquad A \in \mathcal{M}_1, \quad B' \in \mathcal{M}_2'.$$

The construction uses the ordinary tensor product after conjugating by U; the positive collar is what permits this tensor product to be represented normally inside the vacuum Hilbert space. As the collar is removed, the interpolating type-I factor is no longer part of the sharp local-algebra data, which is why split limits and relative entropy are the intrinsic continuum objects used later.

4.6 Modular Data and Locality

Let $\mathcal{R}$ be a von Neumann algebra on a Hilbert space with a vector Ω that is cyclic and separating for $\mathcal{R}$. Tomita–Takesaki theory defines an antilinear operator S by

$$SA\Omega = A^*\Omega, \qquad A \in \mathcal{R},$$

and its polar decomposition

$$S = J\Delta^{1/2}.$$

Here J is antiunitary and Δ is positive self-adjoint. The operator Δ and its square root may be unbounded, but the Borel functional calculus for positive self-adjoint operators gives Δ^{it} as a bounded unitary operator for every $t \in \mathbb{R}$, forming a strongly continuous one-parameter unitary group. The modular automorphism group is

$$\sigma_t(A) = \Delta^{\mathrm{i}t} A \Delta^{-\mathrm{i}t}.$$

Quoted Theorem 4.3 (Tomita–Takesaki modular automorphisms)**.** Proof references.[2] *For a cyclic and separating vector Ω for $\mathcal{R}$, the modular operators constructed above satisfy, for every $t \in \mathbb{R}$,*

$$\Delta^{\mathrm{i}t} \mathcal{R} \Delta^{-\mathrm{i}t} = \mathcal{R}.$$

Consequently

$$\sigma_t = \mathrm{Ad}_{\Delta^{\mathrm{i}t}}\big|_{\mathcal{R}}$$

is a one-parameter group of $*$-automorphisms of $\mathcal{R}$. Moreover, the modular conjugation identifies the algebra with its commutant:

$$J\mathcal{R}J = \mathcal{R}',$$

where $\mathcal{R}'$ denotes the commutant of $\mathcal{R}$ in $\mathcal{B}(\mathcal{H})$.

Example 4.4 (Finite standard form of a faithful matrix state)**.** Let $K \simeq \mathbb{C}^n$, let $\mathcal{R} = \mathcal{B}(K)$, and let $\rho \in \mathcal{B}(K)$ be a positive definite matrix with $\mathrm{Tr}_K \rho = 1$. Put

$$\mathcal{H}_{\mathrm{HS}} = \mathcal{B}(K), \qquad \langle X, Y \rangle_{\mathrm{HS}} = \mathrm{Tr}_K(X^* Y), \qquad \pi(A)X = AX, \qquad \Omega = \rho^{1/2}.$$

Then Ω is cyclic and separating for $\pi(\mathcal{R})$. The Tomita operator, modular conjugation, and modular operator are

$$S(X) = \rho^{-1/2} X^* \rho^{1/2}, \qquad J(X) = X^*, \qquad \Delta(X) = \rho X \rho^{-1}. \tag{4.29}$$

Consequently

$$J\pi(\mathcal{R})J = \pi(\mathcal{R})', \qquad \sigma_t(A) = \rho^{\mathrm{i}t} A \rho^{-\mathrm{i}t}.$$

With the boundary-value convention used below for KMS states, $F(t) = \omega(A\alpha_t(B))$ and $F(t+\mathrm{i}) = \omega(\alpha_t(B)A)$, the vector state

$$\omega(A) = \langle \Omega, \pi(A)\Omega \rangle_{\mathrm{HS}} = \mathrm{Tr}_K(\rho A)$$

is a 1-KMS state for $\alpha_t = \sigma_{-t}$.

Verification of the finite model. Every $X \in \mathcal{H}_{\mathrm{HS}}$ has the form

$$X = (X\rho^{-1/2})\Omega,$$

so Ω is cyclic. If $A\Omega = 0$, then $A\rho^{1/2} = 0$, and invertibility of $\rho^{1/2}$ gives $A = 0$; hence Ω is separating. For $X = A\Omega$,

$$S(X) = S(A\rho^{1/2}) = A^* \rho^{1/2} = \rho^{-1/2} X^* \rho^{1/2}.$$

[2] A complete proof of Tomita's theorem and the modular automorphism theorem is given in M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and Its Applications*, Lecture Notes in Mathematics **128**, Springer, 1970. See also M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979, Chapter VI, for the von Neumann algebra formulation used here [26].

The Hilbert-Schmidt basis of matrix units diagonalizing ρ shows that

$$(J\Delta^{1/2})(X) = J(\rho^{1/2}X\rho^{-1/2}) = \rho^{-1/2}X^*\rho^{1/2} = S(X),$$

with $J(X) = X^*$ and $\Delta = L_\rho R_{\rho^{-1}}$, where L and R denote left and right multiplication. The commutant of all left multiplications is the algebra of right multiplications, and

$$J\pi(A)J(X) = J(AX^*) = XA^*,$$

so $J\pi(\mathcal{R})J = \pi(\mathcal{R})'$. Conjugating a left multiplication by $\Delta^{\mathrm{i}t}$ gives

$$\Delta^{\mathrm{i}t}\pi(A)\Delta^{-\mathrm{i}t}X = \rho^{\mathrm{i}t}A\rho^{-\mathrm{i}t}X,$$

which is the displayed modular automorphism.

For this boundary-value convention, define

$$F_{A,B}(z) = \mathrm{Tr}_K\left(\rho A\rho^{-\mathrm{i}z}B\rho^{\mathrm{i}z}\right), \qquad 0 \le \operatorname{Im} z \le 1.$$

This is an entire finite sum after diagonalizing ρ. The lower boundary is $F_{A,B}(t) = \omega(A\alpha_t(B))$. At the upper boundary, using $\alpha_t(B) = \rho^{-\mathrm{i}t}B\rho^{\mathrm{i}t}$ and cyclicity of the finite trace,

$$\begin{aligned} F_{A,B}(t+\mathrm{i}) &= \mathrm{Tr}_K\left(\rho A\rho^{-\mathrm{i}t+1}B\rho^{\mathrm{i}t-1}\right) \\ &= \mathrm{Tr}_K\left(A\rho\rho^{-\mathrm{i}t}B\rho^{\mathrm{i}t}\right) = \mathrm{Tr}_K(\rho\alpha_t(B)A) = \omega(\alpha_t(B)A). \end{aligned}$$

Thus ω satisfies the 1-KMS boundary condition for $\alpha_t = \sigma_{-t}$.

Proof mechanism and local role. The theorem is a statement about a standard von Neumann algebra $(\mathcal{R}, \mathcal{H}, \Omega)$, where standard means that Ω is cyclic and separating for $\mathcal{R}$. The proof begins with the second antilinear operator

$$F_0(B\Omega) = B^*\Omega, \qquad B \in \mathcal{R}'.$$

The commutation of $\mathcal{R}$ and $\mathcal{R}'$ gives the adjoint identity for antilinear operators on the two algebraic cores,

$$\langle S_0(A\Omega), B\Omega\rangle = \langle A^*\Omega, B\Omega\rangle = \langle B^*\Omega, A\Omega\rangle = \langle F_0(B\Omega), A\Omega\rangle, \qquad A \in \mathcal{R},\ B \in \mathcal{R}',$$

using $AB = BA$. Hence $S_0 \subset F_0^*$ and $F_0 \subset S_0^*$. Cyclicity of Ω for $\mathcal{R}$ and $\mathcal{R}'$ supplies dense domains, and the two inclusions imply closability. Denote the closures by S and F. The same adjoint identity upgrades to

$$S^* = F, \qquad F^* = S.$$

The polar decomposition of the closed antilinear operator S gives

$$S = J\Delta^{1/2}, \qquad F = J\Delta^{-1/2}, \qquad J\Delta J = \Delta^{-1}.$$

The substantive step is the Tomita invariance theorem for the left Hilbert algebra $\mathfrak{A} = \mathcal{R}\Omega$. The product and involution on the core are

$$(A\Omega)(B\Omega) = AB\Omega, \qquad (A\Omega)^\# = A^*\Omega.$$

Left multiplication by $A \in \mathcal{R}$ is bounded on the Hilbert-space completion. The closure of the involution is precisely S. Tomita's analytic argument shows that the closed left Hilbert algebra is stable under the modular unitaries $\Delta^{\mathrm{i}t}$, and therefore $\Delta^{\mathrm{i}t} A \Delta^{-\mathrm{i}t}$ is again a bounded left multiplier of $\mathfrak{A}$. Bounded left multipliers of the closed algebra are exactly the elements of $\mathcal{R}$. This proves

$$\Delta^{\mathrm{i}t} \mathcal{R} \Delta^{-\mathrm{i}t} \subset \mathcal{R}.$$

Applying the same argument at time $-t$ gives equality. The commutant identity follows from the polar part of the same closed involution: JAJ acts as right multiplication by A^* on the core and hence commutes with all left multipliers, while the symmetric argument starting from $\mathcal{R}'$ gives the reverse inclusion.

For the QFT applications in this chapter the theorem supplies an abstract modular flow for any local algebra whose vacuum vector is standard. The QFT work begins after this theorem: one must prove Reeh–Schlieder or another standardness criterion for the local algebra, and one must separately identify the abstract modular flow with a geometric flow when such an identification is claimed. The Bisognano–Wichmann theorem boundary below is precisely that additional QFT input for wedges.

4.7 KMS States and Modular Equilibrium

The modular group is not only an algebraic automorphism group. It is the canonical source of the operator-algebraic equilibrium condition. We first state the condition for an arbitrary time evolution, because this is the form used in finite-temperature QFT and in infinite-volume statistical mechanics.

Definition 4.5 (KMS state)**.** Let $\mathfrak{A}$ be a C^*-algebra or a von Neumann algebra, let $\alpha : \mathbb{R} \to \mathrm{Aut}(\mathfrak{A})$ be a strongly continuous one-parameter automorphism group, and let ω be a state on $\mathfrak{A}$. For $\beta > 0$, ω is a β-KMS state for α when, for every pair $A, B \in \mathfrak{A}$, there is a function

$$F_{A,B} : \{z \in \mathbb{C} : 0 \leq \operatorname{Im} z \leq \beta\} \to \mathbb{C}$$

which is bounded and continuous on the closed strip, holomorphic in its interior, and satisfies the boundary conditions

$$F_{A,B}(t) = \omega(A\, \alpha_t(B))\,, \qquad F_{A,B}(t + \mathrm{i}\beta) = \omega(\alpha_t(B)\, A)\,, \qquad t \in \mathbb{R}. \tag{4.30}$$

The order of the two boundary values is the content of the condition. It is not periodicity of an ordinary function of real time; it is an analytic continuation statement whose upper boundary has the operators cyclically permuted. This distinction is essential in infinite volume, where the trace of $\mathrm{e}^{-\beta H}$ usually does not exist but the local state may still satisfy (4.30).

Example 4.6 (Gibbs traces satisfy KMS)**.** Let H be self-adjoint and suppose that $\mathrm{e}^{-\beta H}$ is trace class. On the algebra of bounded operators for which the following expressions are defined by trace-class products, set

$$\omega_\beta(A) = Z_\beta^{-1} \operatorname{Tr}\left(\mathrm{e}^{-\beta H} A\right), \qquad Z_\beta = \operatorname{Tr} \mathrm{e}^{-\beta H}, \qquad \alpha_t(A) = \mathrm{e}^{\mathrm{i}tH} A \mathrm{e}^{-\mathrm{i}tH}.$$

Then ω_β is a β-KMS state for α.

Verification of the trace-class KMS strip. For bounded A, B, define

$$F_{A,B}(z) = Z_\beta^{-1} \operatorname{Tr}\left(\mathrm{e}^{-\beta H} A \mathrm{e}^{\mathrm{i}zH} B \mathrm{e}^{-\mathrm{i}zH}\right).$$

In finite dimension this is entire. Under the stated trace-class hypothesis the same formula is holomorphic in the open strip and bounded on the closed strip by the standard trace-norm estimates for products with e^{-sH}, $0 \le s \le \beta$; finite-dimensional spectral truncations give the same identity and converge in trace norm. At the lower boundary,

$$F_{A,B}(t) = \omega_\beta(A\alpha_t(B)).$$

At the upper boundary,

$$\begin{aligned} F_{A,B}(t+\mathrm{i}\beta) &= Z_\beta^{-1} \operatorname{Tr}\left(\mathrm{e}^{-\beta H} A \mathrm{e}^{\mathrm{i}tH} \mathrm{e}^{-\beta H} B \mathrm{e}^{-\mathrm{i}tH} \mathrm{e}^{\beta H}\right) \\ &= Z_\beta^{-1} \operatorname{Tr}\left(A \mathrm{e}^{\mathrm{i}tH} \mathrm{e}^{-\beta H} B \mathrm{e}^{-\mathrm{i}tH}\right) \\ &= Z_\beta^{-1} \operatorname{Tr}\left(\mathrm{e}^{-\beta H} \mathrm{e}^{\mathrm{i}tH} B \mathrm{e}^{-\mathrm{i}tH} A\right) = \omega_\beta(\alpha_t(B)A). \end{aligned}$$

The first equality is the analytic continuation. The second uses cyclicity of the trace to move the final $\mathrm{e}^{\beta H}$ to the front, where it cancels one $\mathrm{e}^{-\beta H}$. The third uses cyclicity again and the commutation of $\mathrm{e}^{-\beta H}$ with $\mathrm{e}^{\mathrm{i}tH}$. These two trace-cyclicity moves are precisely what the KMS boundary condition retains after the literal Gibbs trace ceases to exist.

Theorem 4.7 (Modular KMS property)**.** *Let $\mathcal{R}$ be a von Neumann algebra and let Ω be cyclic and separating for $\mathcal{R}$. Let*

$$\omega(A) = \langle \Omega, A\Omega \rangle, \qquad \sigma_t(A) = \Delta^{\mathrm{i}t} A \Delta^{-\mathrm{i}t}$$

be the vector state and the Tomita–Takesaki modular automorphism group. Then ω satisfies the unit-width strip KMS condition for the modular flow. Equivalently, for every $A, B \in \mathcal{R}$ there is a bounded strip function $G_{A,B}$, analytic for $0 < \operatorname{Im} z < 1$, such that

$$G_{A,B}(t) = \omega(\sigma_t(A)B), \qquad G_{A,B}(t+\mathrm{i}) = \omega(B\sigma_t(A)). \tag{4.31}$$

In the convention of Definition 4.5, this says that ω is a 1-KMS state for the reversed modular dynamics $\alpha_t = \sigma_{-t}$.

Proof. The nontrivial input is Tomita–Takesaki theory: the closable antilinear operator $S_0 A\Omega = A^*\Omega$ has polar decomposition $S = J\Delta^{1/2}$, the unitary group $\Delta^{\mathrm{i}t}$ implements automorphisms of $\mathcal{R}$, and the analytic vectors for this unitary group form a strong-operator dense $*$-subalgebra of $\mathcal{R}$. For analytic A, the vector-valued function $z \mapsto \sigma_z(A)\Omega$ is analytic in the strip. Pairing it with $B^*\Omega$ gives the candidate $G_{A,B}(z)$. The lower boundary is the definition of σ_t. The upper boundary follows from

$$S\,\sigma_t(A)\Omega = \sigma_t(A)^*\Omega, \qquad S = J\Delta^{1/2},$$

or, equivalently, from moving a vector through the closed graph of S: the operation that is trace cyclicity in Example 4.6 is replaced by the Tomita relation between $\mathcal{R}$ and $\mathcal{R}'$. Thus

$$\langle \Omega, \sigma_{t+\mathrm{i}}(A)B\Omega \rangle = \langle \Omega, B\sigma_t(A)\Omega \rangle$$

for analytic A. The general case follows by Gaussian smearing in modular time,

$$A_\epsilon = \frac{1}{\sqrt{\pi\epsilon}} \int_{\mathbb{R}} \mathrm{e}^{-s^2/\epsilon} \sigma_s(A)\, \mathrm{d}s,$$

which produces entire analytic elements converging strongly to A and uniformly controls the strip functions by $\|A\|\,\|B\|$. Taking the limit gives (4.31). $\square$

The theorem is the precise sense in which a cyclic separating vector is an equilibrium state for its own modular time. The modular Hamiltonian

$$K_{\mathrm{mod}} := -\log \Delta$$

is generally not the physical Hamiltonian and need not be an integral of a local density. When a geometric theorem identifies modular flow with a spacetime symmetry, the abstract KMS condition becomes a physical thermal statement for the corresponding observer or subregion.

4.8 Relative Modular Operators and Connes Cocycles

The modular data of a single pair $(\mathcal{R}, \Omega)$ compare an algebra with its commutant. The modular data of two states compare two normal statistical preparations on the same algebra. This relative form is the operator-algebraic replacement for density-matrix quotients in local QFT, where local algebras are typically type III and no local density matrix exists.

Definition 4.8 (Relative Tomita operator)**.** Let $\mathcal{R} \subset \mathcal{B}(\mathcal{H})$ be a von Neumann algebra, and let $\Psi, \Omega \in \mathcal{H}$ be cyclic and separating for $\mathcal{R}$. Define the antilinear operator $S^{(0)}_{\Psi|\Omega}$ on the dense domain $\mathcal{R}\Omega$ by

$$S^{(0)}_{\Psi|\Omega} A\Omega = A^*\Psi, \qquad A \in \mathcal{R}.$$

It is closable. Its closure is denoted $S_{\Psi|\Omega}$, and

$$\Delta_{\Psi|\Omega} := S^*_{\Psi|\Omega} S_{\Psi|\Omega}$$

is the relative modular operator. The polar decomposition is

$$S_{\Psi|\Omega} = J_{\Psi|\Omega} \Delta^{1/2}_{\Psi|\Omega},$$

where $J_{\Psi|\Omega}$ is a partial antiunitary with initial and final closures determined by the supports of the two normal vector states. Under the cyclic-and-separating hypothesis the supports are 1, so $J_{\Psi|\Omega}$ is antiunitary.

The order in the subscript is part of the convention. The operator $\Delta_{\Psi|\Omega}$ has the state represented by Ψ on the left and the state represented by Ω on the right. In a finite type-I algebra this statement becomes literal. Let $\mathcal{R} = \mathcal{B}(\mathcal{K})$ act by left multiplication on the Hilbert–Schmidt space

$$\mathcal{H}_{\mathrm{st}} = \{X : \mathcal{K} \to \mathcal{K} : \mathrm{Tr}(X^*X) < \infty\}, \qquad \langle X, Y\rangle_{\mathrm{st}} = \mathrm{Tr}(X^*Y).$$

A faithful normal state with density matrix $\rho > 0$ is represented by $\Omega_\rho = \rho^{1/2}$. For faithful density matrices ρ_ψ, ρ_ω,

$$\Delta_{\psi|\omega}(X) = \rho_\psi X \rho_\omega^{-1}. \tag{4.32}$$

Indeed, for $X = A\rho_\omega^{1/2}$,

$$S_{\psi|\omega}X = A^*\rho_\psi^{1/2}.$$

The adjoint relation in the Hilbert–Schmidt inner product gives

$$S^*_{\psi|\omega}S_{\psi|\omega}(A\rho_\omega^{1/2}) = \rho_\psi A\rho_\omega^{-1/2} = \rho_\psi(A\rho_\omega^{1/2})\rho_\omega^{-1},$$

which is (4.32) on a dense set and hence by closure. This calculation fixes the convention used below.

Definition 4.9 (Araki relative entropy)**.** Let ψ, ω be faithful normal states represented by cyclic separating vectors Ψ, Ω. The Araki relative entropy of ψ with respect to ω is

$$S_{\mathcal{R}}(\psi\|\omega) = -\langle\Psi, \log\Delta_{\Omega|\Psi}\Psi\rangle, \tag{4.33}$$

with the value $+\infty$ if the logarithmic quadratic form is not finite or if the support of ψ is not dominated by the support of ω.

In the finite type-I model, (4.32) turns (4.33) into the ordinary Umegaki formula:

$$\begin{aligned} S_{\mathcal{B}(\mathcal{K})}(\psi\|\omega) &= -\operatorname{Tr}\Big[\rho_\psi^{1/2}\big(\log\rho_\omega - \log\rho_\psi\big)\rho_\psi^{1/2}\Big] \\ &= \operatorname{Tr}\rho_\psi(\log\rho_\psi - \log\rho_\omega). \end{aligned}$$

The algebraic definition is therefore not a new thermodynamic convention; it is the continuation of the density-matrix expression to algebras for which no density matrix exists.

Quoted Theorem 4.10 (Connes cocycle derivative)**.** Proof references.[3] *Let ω and ψ be faithful normal states on a von Neumann algebra $\mathcal{R}$. In a standard representation, define*

$$u_t = [D\psi : D\omega]_t = \Delta^{\mathrm{i}t}_{\psi|\omega}\Delta^{-\mathrm{i}t}_\omega, \qquad t \in \mathbb{R},$$

where $\Delta_\omega = \Delta_{\Omega|\Omega}$. Then $u_t \in \mathcal{R}$, u_t is unitary, and

$$u_{t+s} = u_t\,\sigma_t^\omega(u_s), \qquad t, s \in \mathbb{R}. \tag{4.34}$$

Moreover the modular flows of the two states are related by

$$\sigma_t^\psi(A) = u_t\,\sigma_t^\omega(A)\,u_t^*, \qquad A \in \mathcal{R}. \tag{4.35}$$

Proof mechanism and local role. The cocycle theorem is the Radon–Nikodym theorem for faithful normal states in standard form. The standard-form data are

$$(\mathcal{R}, \mathcal{H}, J, \mathcal{P}),$$

where J is the modular conjugation and $\mathcal{P}$ is the natural positive cone. Each faithful normal state has a unique vector representative in $\mathcal{P}$; write these representatives as Ω for ω and Ψ for ψ. The relative Tomita operator is initially defined on the dense core $\mathcal{R}\Omega$ by

$$S_{\Psi|\Omega}(A\Omega) = A^*\Psi, \qquad A \in \mathcal{R}.$$

[3] The cocycle derivative and Radon–Nikodym theorem for faithful normal weights are proved in A. Connes, "Une classification des facteurs de type III," *Annales scientifiques de l'Ecole Normale Superieure* **6** (1973), 133–252, and in Takesaki, *Theory of Operator Algebras II*, Chapter VIII [27].

Its closure has polar data

$$S_{\Psi|\Omega} = J_{\Psi|\Omega}\Delta^{1/2}_{\psi|\omega}.$$

The positive operator $\Delta_{\psi|\omega}$ is the noncommutative Radon–Nikodym derivative at the modular level. The crucial assertion is that the relative modular unitaries differ from the reference modular unitaries by a unitary belonging to the algebra:

$$\Delta^{\mathrm{i}t}_{\psi|\omega}\Delta^{-\mathrm{i}t}_{\omega} \in \mathcal{R}.$$

One proves this by testing the product against $\mathcal{R}'\Omega$, using the relative Tomita adjoint relation to show that it commutes with the commutant, and then applying $(\mathcal{R}')' = \mathcal{R}$. The same relative-modular calculation gives

$$\Delta^{\mathrm{i}(t+s)}_{\psi|\omega}\Delta^{-\mathrm{i}(t+s)}_{\omega} = \big(\Delta^{\mathrm{i}t}_{\psi|\omega}\Delta^{-\mathrm{i}t}_{\omega}\big)\Delta^{\mathrm{i}t}_{\omega}\big(\Delta^{\mathrm{i}s}_{\psi|\omega}\Delta^{-\mathrm{i}s}_{\omega}\big)\Delta^{-\mathrm{i}t}_{\omega},$$

which is exactly the cocycle relation (4.34). Conjugating $A \in \mathcal{R}$ by the relative modular unitary and inserting the preceding factorization gives (4.35).

The theorem's role in QFT is to compare modular Hamiltonians associated with different states while staying inside the same local algebra. Relative entropy, modular energy, and the algebraic proof of monotonicity use this cocycle as the replacement for a density-matrix quotient in type-III local algebras. The finite type-I calculation below is a normalization check; the operator-algebraic theorem supplies the type-III object used by local QFT.

Example 4.11 (Finite standard-form Connes cocycle)**.** Let ρ_ω, ρ_ψ be positive definite density matrices on $\mathcal{K}$, and represent $\mathcal{R} = \mathcal{B}(\mathcal{K})$ on the Hilbert–Schmidt standard space by left multiplication. With the convention (4.32), the operator

$$\Delta^{\mathrm{i}t}_{\psi|\omega}\Delta^{-\mathrm{i}t}_{\omega}$$

is left multiplication by

$$u_t = \rho^{\mathrm{i}t}_{\psi}\rho^{-\mathrm{i}t}_{\omega} \in \mathcal{B}(\mathcal{K}). \tag{4.36}$$

The family u_t is unitary in $\mathcal{R}$, satisfies the Connes cocycle identity

$$u_{t+s} = u_t\,\sigma^{\omega}_t(u_s), \qquad \sigma^{\omega}_t(A) = \rho^{\mathrm{i}t}_{\omega}A\rho^{-\mathrm{i}t}_{\omega},$$

and implements the change of modular flow,

$$\sigma^{\psi}_t(A) = u_t\sigma^{\omega}_t(A)u^*_t, \qquad \sigma^{\psi}_t(A) = \rho^{\mathrm{i}t}_{\psi}A\rho^{-\mathrm{i}t}_{\psi}.$$

Verification of the finite cocycle. On the Hilbert–Schmidt space, the relative and reference modular unitaries act by

$$\Delta^{\mathrm{i}t}_{\psi|\omega}X = \rho^{\mathrm{i}t}_{\psi}X\rho^{-\mathrm{i}t}_{\omega}, \qquad \Delta^{-\mathrm{i}t}_{\omega}X = \rho^{-\mathrm{i}t}_{\omega}X\rho^{\mathrm{i}t}_{\omega}.$$

Their product is therefore

$$\Delta^{\mathrm{i}t}_{\psi|\omega}\Delta^{-\mathrm{i}t}_{\omega}X = \rho^{\mathrm{i}t}_{\psi}\rho^{-\mathrm{i}t}_{\omega}X = u_tX,$$

which is left multiplication by the matrix in (4.36). Since the factors $\rho^{\mathrm{i}t}_{\psi}$ and $\rho^{-\mathrm{i}t}_{\omega}$ are unitary, u_t is a unitary element of $\mathcal{R}$.

The cocycle law keeps the density matrices in their noncommuting order:

$$\begin{aligned} u_t\sigma_t^\omega(u_s) &= \rho_\psi^{\mathrm{i}t}\rho_\omega^{-\mathrm{i}t}\rho_\omega^{\mathrm{i}t}(\rho_\psi^{\mathrm{i}s}\rho_\omega^{-\mathrm{i}s})\rho_\omega^{-\mathrm{i}t} \\ &= \rho_\psi^{\mathrm{i}(t+s)}\rho_\omega^{-\mathrm{i}(t+s)} = u_{t+s}. \end{aligned}$$

Likewise,

$$u_t\sigma_t^\omega(A)u_t^* = \rho_\psi^{\mathrm{i}t}\rho_\omega^{-\mathrm{i}t}\rho_\omega^{\mathrm{i}t}A\rho_\omega^{-\mathrm{i}t}\rho_\omega^{\mathrm{i}t}\rho_\psi^{-\mathrm{i}t} = \rho_\psi^{\mathrm{i}t}A\rho_\psi^{-\mathrm{i}t}.$$

This is exactly $\sigma_t^\psi(A)$. The finite calculation shows the algebraic object that the general theorem constructs: an algebra unitary whose cocycle law records the change of state and keeps the implementer inside $\mathcal{R}$.

For local QFT this quoted theorem is a calculational device. Changing the state in a bounded region is encoded not by dividing density matrices, but by a unitary cocycle inside the local algebra. In perturbative algebraic QFT this is the operator-algebraic structure behind relative Cauchy evolution and local source deformations; in constructive models it is the object on which monotonicity, convexity, and energy inequalities are formulated.

4.9 Type III Local Algebras and Phase-Space Bounds

The split property introduced above is an inclusion $\mathcal{R}(\mathcal{O}_1) \subset N \subset \mathcal{R}(\mathcal{O}_2)$ through a type-I factor for separated regions, while sharply localized tensor factorization would be a stronger and generally unavailable structure. For a proper bounded region $\mathcal{O}$, the physically expected local algebra $\mathcal{R}(\mathcal{O})$ is not a type-I factor. Under the standard structural hypotheses satisfied by known relativistic models, it is a type III_1 factor: it has no nonzero finite projections and no faithful normal trace. The absence of a trace is why local thermal and entanglement quantities must be formulated with modular theory rather than with local density matrices.

Definition 4.12 (Buchholz–Wichmann nuclearity map)**.** Let $U(t) = \mathrm{e}^{\mathrm{i}tH}$ be the vacuum time-translation group with $H \geq 0$, and let $\mathcal{R}(\mathcal{O})$ be a bounded-region algebra. For $\beta > 0$, define

$$\Theta_{\beta,\mathcal{O}} : \mathcal{R}(\mathcal{O}) \to \mathcal{H}, \qquad \Theta_{\beta,\mathcal{O}}(A) = \mathrm{e}^{-\beta H}A\Omega.$$

The net satisfies the Buchholz–Wichmann nuclearity condition on $\mathcal{O}$ if $\Theta_{\beta,\mathcal{O}}$ is a nuclear map for every $\beta > 0$ and its nuclear norm obeys a local phase-space growth estimate, for example

$$\|\Theta_{\beta,\mathcal{O}}\|_1 \leq \exp\bigl(C_{\mathcal{O}}\beta^{-s_{\mathcal{O}}}\bigr) \quad \text{as } \beta \downarrow 0$$

for some finite constants $C_{\mathcal{O}}, s_{\mathcal{O}}$.

Here nuclear means that

$$\Theta_{\beta,\mathcal{O}}(A) = \sum_{n=1}^{\infty} \ell_n(A)\xi_n, \qquad \sum_n \|\ell_n\|\,\|\xi_n\| < \infty,$$

for bounded linear functionals ℓ_n on $\mathcal{R}(\mathcal{O})$ and vectors $\xi_n \in \mathcal{H}$. The condition is an infinite-dimensional replacement for saying that only finitely many independent localized states are visible below energy scale $1/\beta$. It is not an ultraviolet cutoff: it is a phase-space compactness estimate in the continuum theory.

Free Fock phase-space benchmark. The growth form in Definition 4.12 is visible already in the free bosonic Fock counting problem. Let the spatial dimension be $s = D-1 \geq 1$, put a scalar field of mass $m > 0$ in a spatial torus of side length L, and write the one-particle energies as

$$\epsilon_n = \sqrt{\left(\frac{2\pi}{L}\right)^2 |n|_2^2 + m^2}, \qquad n \in \mathbb{Z}^s.$$

On the bosonic Fock space the finite-volume Hamiltonian is $H_L = d\Gamma(h_L)$. For finitely many one-particle modes with energies $\epsilon_1, \ldots, \epsilon_M$, the occupation-number basis gives

$$\begin{aligned}\operatorname{Tr}_{\mathcal{F}_B} \exp[-\beta\, d\Gamma(h_M)] &= \sum_{k_1,\ldots,k_M \geq 0} \exp\left[-\beta \sum_{j=1}^{M} k_j \epsilon_j\right] \\ &= \prod_{j=1}^{M} \frac{1}{1 - \mathrm{e}^{-\beta\epsilon_j}}.\end{aligned}$$

Passing to all torus modes is justified for $\beta > 0$ by convergence of

$$\log Z_B(\beta, L) = \sum_{n \in \mathbb{Z}^s} f(\beta\epsilon_n), \qquad f(x) := -\log(1 - \mathrm{e}^{-x}).$$

Indeed $f(x) \leq C\mathrm{e}^{-x/2}$ for large x, while the number of lattice points with $|n|_\infty = k$ is

$$N_s(k) = (2k+1)^s - (2k-1)^s \leq 2s(2k+1)^{s-1}.$$

For $n \neq 0$,

$$\epsilon_n \geq \frac{2\pi}{L}|n|_\infty.$$

Set $\tau = 2\pi\beta/L$. The remaining estimate is elementary but it is important to separate the logarithmic singularity of f near the origin from its exponential decay at infinity. For $0 < x \leq 1$,

$$f(x) \leq C(1 + |\log x|),$$

while for $x \geq 1$,

$$f(x) \leq C\mathrm{e}^{-x/2}.$$

Therefore, for $0 < \tau \leq 1$,

$$\sum_{1 \leq k \leq \tau^{-1}} (1+k)^{s-1} f(\tau k) \leq C_s \tau^{-s} \int_0^1 u^{s-1}(1 + |\log u|)\, \mathrm{d}u,$$

and

$$\sum_{k > \tau^{-1}} (1+k)^{s-1} f(\tau k) \leq C_s \sum_{k > \tau^{-1}} k^{s-1} \mathrm{e}^{-\tau k/2} \leq C_s' \tau^{-s} \int_1^\infty u^{s-1} \mathrm{e}^{-u/2}\, \mathrm{d}u.$$

The two integrals are finite, so the nonzero-mode sum is bounded by $C_s\tau^{-s}$. Hence

$$\log Z_B(\beta, L) \leq f(\beta m) + C_s \left(\frac{L}{2\pi\beta}\right)^s.$$

Since $s \geq 1$, the zero-mode term $f(\beta m) \sim \log(\beta^{-1})$ is absorbed into $C_{s,m,L}\beta^{-s}$ on $0 < \beta \leq 1$. Thus

$$\log Z_B(\beta, L) \leq C_{s,m,L}\,\beta^{-s}, \qquad 0 < \beta \leq 1. \tag{4.37}$$

Equivalently,

$$Z_B(\beta, L) \leq \exp(C_{s,m,L}\beta^{-s}).$$

The local nuclearity map replaces the global box trace by the localized map $A \mapsto \mathrm{e}^{-\beta H}A\Omega$. The theorem below is precisely the operator-algebraic statement that a comparable phase-space estimate for localized vectors is strong enough to create a split inclusion between separated local algebras. The free Fock calculation supplies the basic mode-counting scale; the local theorem requires the sharper localization and normality arguments described after the quoted theorem.

Quoted Theorem 4.13 (Buchholz–Wichmann nuclearity implies split inclusions)**.** Proof references.[4] *Let a vacuum local net satisfy isotony, locality, covariance, positivity of energy, uniqueness of the vacuum, Reeh–Schlieder for double cones, and the Buchholz–Wichmann nuclearity condition. Then every pair of double cones $\mathcal{O}_1 \Subset \mathcal{O}_2$ is split:*

$$\mathcal{R}(\mathcal{O}_1) \subset \mathcal{N} \subset \mathcal{R}(\mathcal{O}_2)$$

for some type-I factor $\mathcal{N}$. In the Doplicher–Longo split criterion, this conclusion is equivalent to normal extendability of the multiplication representation

$$\mathcal{R}(\mathcal{O}_1) \odot \mathcal{R}(\mathcal{O}_2)' \longrightarrow \mathcal{B}(\mathcal{H}), \qquad A \otimes B' \longmapsto AB',$$

to the spatial von Neumann tensor product.

Functional-analytic mechanism of the split conclusion. The split part of the theorem converts an energy-damped compactness estimate into a spatial tensor-product representation for separated algebras. Let

$$\mathcal{M}_1 = \mathcal{R}(\mathcal{O}_1), \qquad \mathcal{M}_2 = \mathcal{R}(\mathcal{O}_2), \qquad \mathcal{O}_1 \Subset \mathcal{O}_2.$$

Because $\mathcal{M}_1$ and $\mathcal{M}_2'$ commute, there is an algebraic multiplication representation

$$m : \mathcal{M}_1 \odot \mathcal{M}_2' \longrightarrow \mathcal{B}(\mathcal{H}), \qquad m(A \otimes B') = AB'.$$

If the inclusion is split, the normal tensor-product representation can be seen directly. A type-I factor $\mathcal{N}$ with $\mathcal{M}_1 \subset \mathcal{N} \subset \mathcal{M}_2$ is spatial: there are Hilbert spaces $\mathcal{K}, \mathcal{L}$ and a unitary $U : \mathcal{H} \to \mathcal{K} \otimes \mathcal{L}$ such that

$$U\mathcal{N}U^* = \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}.$$

Then

$$U\mathcal{M}_1U^* \subset \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}, \qquad U\mathcal{M}_2'U^* \subset U\mathcal{N}'U^* = \mathbf{1} \otimes \mathcal{B}(\mathcal{L}).$$

Therefore the product map is the restriction of the spatial normal representation

$$\mathcal{M}_1\bar{\otimes}\mathcal{M}_2' \longrightarrow \mathcal{B}(\mathcal{K} \otimes \mathcal{L}), \qquad A \otimes B' \longmapsto UAU^*\,UB'U^*. \tag{4.38}$$

[4]The phase-space condition and split property are developed in D. Buchholz and E. H. Wichmann, "Causal independence and the energy-level density of states in local quantum field theory," *Communications in Mathematical Physics* **106** (1986), 321–344, and in C. D'Antoni and R. Longo, "Interpolation by type I factors and the flip automorphism," *Journal of Functional Analysis* **51** (1983), 361–371 [29, 28].

The converse implication, from normal extendability of the multiplication representation to the existence of an interpolating type-I factor, is the Doplicher–Longo split criterion. It is part of the operator-algebraic input in Quoted Theorem 4.13, not a consequence of algebraic commutativity alone. When the criterion applies, the unitary identification can be chosen so that the two commuting algebras act as

$$\mathcal{M}_1 \subset \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}, \qquad \mathcal{M}_2' \subset \mathbf{1} \otimes \mathcal{B}(\mathcal{L}),$$

and the interpolating type-I factor is

$$\mathcal{N} = U^{-1}(\mathcal{B}(\mathcal{K}) \otimes \mathbf{1})\, U.$$

This criterion is the functional-analytic content behind the phrase "approximate tensor factorization": it is a normal tensor-product representation for algebras separated by a positive collar.

Split states are positive-collar states. The split inclusion also produces a precise replacement for the informal operation of assigning independent states to a region and its exterior. Fix a spatial implementation

$$U : \mathcal{H} \to \mathcal{K} \otimes \mathcal{L}, \qquad U\mathcal{N}U^* = \mathcal{B}(\mathcal{K}) \otimes \mathbf{1},$$

so that $U\mathcal{M}_1U^* \subset \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}$ and $U\mathcal{M}_2'U^* \subset \mathbf{1} \otimes \mathcal{B}(\mathcal{L})$. Let $\varphi_1 \in (\mathcal{M}_1)_*^+$ and $\varphi_2 \in (\mathcal{M}_2')_*^+$ be normal positive functionals. Choose normal positive extensions $\widetilde{\varphi}_1$ and $\widetilde{\varphi}_2$ to $\mathcal{B}(\mathcal{K})$ and $\mathcal{B}(\mathcal{L})$. This is a pure von Neumann algebra extension step: a normal positive functional on a von Neumann subalgebra of $\mathcal{B}(\mathcal{K})$ has a normal positive extension to $\mathcal{B}(\mathcal{K})$, with the same norm. Their product normal functional on $\mathcal{B}(\mathcal{K})\bar{\otimes}\mathcal{B}(\mathcal{L})$ restricts to a normal positive functional on the von Neumann algebra generated by $\mathcal{M}_1$ and $\mathcal{M}_2'$:

$$\omega^{\mathcal{O}_1 \Subset \mathcal{O}_2}_{\varphi_1,\varphi_2}(AB') = \varphi_1(A)\varphi_2(B'), \qquad A \in \mathcal{M}_1, \quad B' \in \mathcal{M}_2'. \tag{4.39}$$

The right side shows independence of the chosen normal extensions on the algebraic products AB', and normality gives the extension to the generated von Neumann algebra. Equation (4.39) is the exact mathematical meaning of statistical independence at a positive separation. The superscript $\mathcal{O}_1 \Subset \mathcal{O}_2$ is part of the datum: the collar and the interpolating type-I factor are not discarded. If the collar is shrunk, the theorem supplies no canonical limiting product state on $\mathcal{R}(\mathcal{O}) \vee \mathcal{R}(\mathcal{O}^\perp)$. In a type-III sharp local algebra there is no density matrix for $\mathcal{R}(\mathcal{O})$ and no literal Hilbert-space tensor factor $\mathcal{H}_\mathcal{O} \otimes \mathcal{H}_{\mathcal{O}^\perp}$ from which such a state could be obtained. Split states are therefore regulator-independent local QFT objects only after the positive collar has been specified.

The nuclearity hypothesis supplies the normality in (4.38). A nuclear decomposition of $\Theta_{\beta,\mathcal{O}}$ has the form

$$\mathrm{e}^{-\beta H} A\Omega = \sum_{n \geq 1} \ell_n(A)\xi_n, \qquad \sum_n \|\ell_n\|\,\|\xi_n\| < \infty.$$

Thus the set of vectors obtained by acting with the local unit ball and then damping energy by $\mathrm{e}^{-\beta H}$ is contained in the absolutely convex hull of a summable family. This Banach-space statement is only the first ingredient. Since $\mathcal{M}_1$ and $\mathcal{M}_2'$ are von Neumann algebras, the split criterion asks for normality, not only norm convergence. Equivalently, for every vector pair $\eta, \zeta \in \mathcal{H}$, the functional

$$A \otimes B' \longmapsto \langle \eta, AB'\zeta \rangle \qquad (A \in \mathcal{M}_1,\ B' \in \mathcal{M}_2')$$

must be represented, after the required limiting argument, by a normal functional on $\mathcal{M}_1 \bar{\otimes} \mathcal{M}_2'$. Finite sums of separated normal functionals,

$$\sum_{j=1}^{N} \varphi_j(A)\psi_j(B'), \qquad \varphi_j \in (\mathcal{M}_1)_*, \quad \psi_j \in (\mathcal{M}_2')_*,$$

are the elementary normal functionals on the spatial tensor product; a summable infinite version is the analytic form in which the split proof detects normal extendability of the multiplication representation.

The nuclear decomposition is used to produce such summable separated forms. For a fixed B' and a fixed vector η, the energy-damped matrix coefficient

$$A \longmapsto \langle \eta, B' \mathrm{e}^{-\beta H} A\Omega \rangle$$

has the absolutely convergent expansion

$$\sum_{n \geq 1} \ell_n(A) \, \langle B'^* \eta, \xi_n \rangle.$$

The displayed formula by itself does not prove splitness: the coefficient functionals ℓ_n in an arbitrary Banach-space nuclear decomposition need not be normal, and the undamped vector functional $(A, B') \mapsto \langle \eta, AB'\zeta \rangle$ is not obtained by algebraically removing $\mathrm{e}^{-\beta H}$. These two points are precisely where the local QFT hypotheses enter. The collar $\mathcal{O}_1 \Subset \mathcal{O}_2$ permits small imaginary time translations whose tube domains do not cross the spacelike complement of $\mathcal{O}_2$. Locality then moves operators in $\mathcal{M}_2'$ past the translated $\mathcal{M}_1$-operators, and the spectral condition supplies the energy damping that makes the nuclear expansion summable. The remaining analytic estimates identify the resulting bilinear vector functionals with limits, in the predual $(\mathcal{M}_1 \bar{\otimes} \mathcal{M}_2')_*$, of separated normal sums. This is the functional-analytic content hidden behind the short phrase "nuclearity implies the split property."

The finite-dimensional shadow is simple but useful for keeping the objects straight. If $\mathcal{M}_1 \subset \mathcal{B}(\mathcal{K}) \otimes \mathbf{1}$ and $\mathcal{M}_2' \subset \mathbf{1} \otimes \mathcal{B}(\mathcal{L})$, then any density matrices r_1 on $\mathcal{K}$ and r_2 on $\mathcal{L}$ give a normal product functional

$$A \otimes B' \longmapsto \mathrm{Tr}_{\mathcal{K}}(r_1 A) \ \mathrm{Tr}_{\mathcal{L}}(r_2 B') = \mathrm{Tr}_{\mathcal{K}\otimes\mathcal{L}}\big[(r_1 \otimes r_2)(A \otimes B')\big] \, .$$

For a finite-rank map $\Theta(A) = \sum_j \ell_j(A)\xi_j$, every coefficient $\langle \eta, B'\Theta(A) \rangle$ is correspondingly a finite separated sum. The infinite theorem is the statement that, under relativistic phase-space and collar hypotheses, this finite separated normality survives the continuum limit for the actual local multiplication representation.

This proof mechanism also explains the status of the constants in the phase-space bound. The exponential estimate

$$\|\Theta_{\beta,\mathcal{O}}\|_1 \leq \exp(C_{\mathcal{O}} \beta^{-s_{\mathcal{O}}})$$

is stronger than mere nuclearity and is used to make the construction stable as the collar, region, or thermodynamic comparison is varied. The existence of a single nuclear map for a fixed $(\beta, \mathcal{O})$ gives compactness at one scale; the uniform small-β bound is the continuum phase-space statement needed in applications.

Type classification is a separate sharp-region theorem. The split theorem and the type-classification theorem concern different objects. Let $\mathcal{M}$ be a von Neumann algebra. Its center is

$$Z(\mathcal{M}) = \mathcal{M} \cap \mathcal{M}',$$

and $\mathcal{M}$ is a factor when $Z(\mathcal{M}) = \mathbb{C}\mathbf{1}$. A projection $p \in \mathcal{M}$ is minimal when

$$p\mathcal{M}p = \mathbb{C}p.$$

A type-I factor has minimal projections and is isomorphic to $\mathcal{B}(\mathcal{K})$ for a Hilbert space $\mathcal{K}$. A factor is type III when it has no nonzero finite projections, equivalently when it has no nonzero normal semifinite trace. For a type-III factor the finer Connes invariant may be described by

$$S(\mathcal{M}) = \bigcap_{\varphi} \mathrm{Spec}(\Delta_\varphi),$$

where the intersection is over faithful normal semifinite weights φ and Δ_φ is the corresponding modular operator in the standard form. The type III_1 case is characterized by

$$S(\mathcal{M}) = \mathbb{R}_{\geq 0}.$$

These definitions show exactly why the type statement cannot be read off from the split inclusion alone. The split inclusion supplies, for $\mathcal{O}_1 \Subset \mathcal{O}_2$, an auxiliary type-I factor $\mathcal{N}$ between $\mathcal{R}(\mathcal{O}_1)$ and $\mathcal{R}(\mathcal{O}_2)$. It does not say that either endpoint algebra is type I. Conversely, a finite-regulator lattice system has type-I local algebras and may satisfy an exact tensor factorization at positive lattice separation; that finite statement is not the continuum sharp-region conclusion. The theorem-boundary problem for local QFT is to prove, in the continuum representation, the factor property, absence of finite local projections, and the modular spectrum needed for type-III_1. Those inputs use locality, covariance, the spectrum condition, Reeh–Schlieder standardness, additivity or scaling regularity, and phase-space control in a way that is not captured by the existence of a positive-collar type-I interpolant.

The quoted theorem just stated has one local-QFT message: a positive collar between regions, together with a nuclear phase-space estimate strong enough to make the multiplication representation normal, allows a type-I interpolant and hence a collar-dependent tensor-product representation. It does not classify the sharp endpoint algebras. The type-III statement is a different sharp-region theorem boundary. It says, under additional factoriality, additivity or scaling-regularity, and modular-spectrum hypotheses, that the collar cannot be shrunk to zero while retaining a type-I description. In particular, a formal notation such as

$$\mathcal{H} \overset{\text{formal}}{\neq} \mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_{\mathcal{O}^\perp}$$

is not a technical inconvenience; it is the local algebraic structure of a continuum relativistic QFT. Local entropy therefore has to be understood through split limits, relative entropy, or other regulator-dependent procedures whose limiting content is stated explicitly.

The type-III_1 identification is a further sharp-local-algebra statement. Its input is not the split property alone but the combination of relativistic locality, vacuum cyclicity and separatingness, additivity or scaling regularity, and the absence of finite-dimensional local summands. Under those hypotheses the local algebra has no minimal projections, no faithful normal trace, and no type-I density-matrix description. The split property supplies type-I factors only at positive separation; the sharp region algebra belongs to the type-III class.

4.10 Modular Hamiltonians and Energy Bounds

Let ω be a faithful normal state on $\mathcal{R}$, represented by a cyclic separating vector Ω. The modular Hamiltonian of the pair $(\mathcal{R}, \omega)$ is

$$K_\omega = -\log \Delta_\omega.$$

It is an operator on the GNS Hilbert space of the algebra-state pair. When $\mathcal{R} = \mathcal{B}(\mathcal{K}_A) \otimes 1$ in a finite bipartite system and ω has reduced density matrix ρ_A^ω, the restriction of K_ω to the left standard representation is $-\log \rho_A^\omega$ up to the right-action term that cancels in left expectation differences. Thus the finite-dimensional identity

$$S(\rho_A^\psi \| \rho_A^\omega) = \operatorname{Tr} \rho_A^\psi (\log \rho_A^\psi - \log \rho_A^\omega)$$

can be rewritten as

$$S(\rho_A^\psi \| \rho_A^\omega) = \Delta_\psi \langle K_A^\omega \rangle - \Delta_\psi S_A, \qquad K_A^\omega = -\log \rho_A^\omega. \tag{4.40}$$

Here

$$\Delta_\psi \langle K_A^\omega \rangle = \operatorname{Tr}(\rho_A^\psi - \rho_A^\omega) K_A^\omega, \qquad \Delta_\psi S_A = -\operatorname{Tr} \rho_A^\psi \log \rho_A^\psi + \operatorname{Tr} \rho_A^\omega \log \rho_A^\omega.$$

Equation (4.40) is often called the entropy–modular-energy identity. In a type-III local algebra the two terms on the right are not separately defined without a chosen split inclusion or regulator, but the left side is the Araki relative entropy (4.33). The mathematically primary object is therefore

$$S_{\mathcal{R}}(\psi \| \omega),$$

not an unqualified local von Neumann entropy.

Finite-dimensional first variation of relative entropy. Let $\rho_\lambda = \rho_0 + \lambda \dot\rho + O(\lambda^2)$ be a differentiable family of faithful density matrices with $\operatorname{Tr} \dot\rho = 0$. Let $K_0 = -\log \rho_0$. Then

$$\left.\frac{\mathrm{d}}{\mathrm{d}\lambda}\right|_{\lambda=0} S(\rho_\lambda \| \rho_0) = 0$$

is equivalent to

$$\left.\frac{\mathrm{d}}{\mathrm{d}\lambda}\right|_{\lambda=0} S_{\mathrm{vN}}(\rho_\lambda) = \operatorname{Tr}(\dot\rho K_0). \tag{4.41}$$

The calculation uses

$$S(\rho_\lambda \| \rho_0) = \operatorname{Tr} \rho_\lambda \log \rho_\lambda - \operatorname{Tr} \rho_\lambda \log \rho_0.$$

The Fréchet derivative of $X \mapsto \operatorname{Tr} X \log X$ at a faithful matrix ρ_0 is

$$\operatorname{Tr} \dot\rho (\log \rho_0 + 1).$$

The 1-term vanishes because $\operatorname{Tr} \dot\rho = 0$. Therefore

$$\left.\frac{\mathrm{d}}{\mathrm{d}\lambda}\right|_0 S(\rho_\lambda \| \rho_0) = \operatorname{Tr} \dot\rho \log \rho_0 - \operatorname{Tr} \dot\rho \log \rho_0 = 0.$$

On the other hand,

$$\left.\frac{\mathrm{d}}{\mathrm{d}\lambda}\right|_0 S_{\mathrm{vN}}(\rho_\lambda) = -\operatorname{Tr} \dot\rho \log \rho_0 = \operatorname{Tr} \dot\rho K_0,$$

which proves (4.41). The same identity in local QFT is obtained by applying this finite calculation to split inclusions and then passing only those quantities with regulator-independent limits to the sharp algebra.

The inequality

$$S_{\mathcal{R}}(\psi\|\omega) \geq 0$$

is the modular-theoretic source of many energy bounds. In a regulated type-I approximation it gives

$$\Delta_\psi S_A \leq \Delta_\psi \langle K_A^\omega \rangle.$$

When $\mathcal{R}$ is a wedge algebra and ω is the vacuum, Bisognano–Wichmann identifies

$$K_{W_R,\omega} = 2\pi K_R, \qquad U(\Lambda_R(s)) = \mathrm{e}^{\mathrm{i}sK_R},$$

because $\Delta_{W_R}^{\mathrm{i}t} = U(\Lambda_R(-2\pi t))$. If the theory has a stress tensor normalized so that K_R is the boost charge, then formally on the $x^0 = 0$ Cauchy surface

$$K_{W_R,\omega} = 2\pi \int_{x^1>0} x^1\, T_{00}(0, \mathbf{x})\, \mathrm{d}^{D-1}\mathbf{x}. \tag{4.42}$$

Equation (4.42) is not a definition of the local algebra; it is the consequence of the geometric modular-flow theorem plus the independently normalized stress tensor. The standard derivations of the Bekenstein-type bound for half-spaces and of the averaged null energy condition use three inputs: positivity of Araki relative entropy, monotonicity of relative entropy under inclusion of von Neumann algebras, and the geometric form of the vacuum modular Hamiltonian for wedges or balls. Whenever one of these inputs is absent, the claimed bound must be restated as a hypothesis or as a regulated theorem.

4.11 Half-Sided Modular Inclusions

Modular theory can also generate spacetime symmetry from inclusions of algebras. This is the mechanism behind the use of modular inclusions in chiral CFT, wedge-local constructions, and several reconstruction theorems.

Definition 4.14 (Half-sided modular inclusion)**.** Let $\mathcal{N} \subset \mathcal{M}$ be von Neumann algebras on $\mathcal{H}$, and let Ω be cyclic and separating for both. Let $\sigma_t^{\mathcal{M}}$ be the modular automorphism group of $(\mathcal{M}, \Omega)$. The inclusion is right half-sided modular if

$$\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}, \qquad t \geq 0,$$

and left half-sided modular if the same inclusion holds for $t \leq 0$.

Quoted Theorem 4.15 (Borchers–Wiesbrock structure)**.** Proof references.[5] *Let $\mathcal{N} \subset \mathcal{M}$ be a standard right half-sided modular inclusion, meaning here that Ω is cyclic and separating for $\mathcal{M}$, for $\mathcal{N}$, and for the relative commutant $\mathcal{N}' \cap \mathcal{M}$, and that*

$$\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}, \qquad t \geq 0.$$

[5]Borchers' commutation relations and Wiesbrock's reconstruction theorem are proved in H.-J. Borchers, "The CPT theorem in two-dimensional theories of local observables," *Communications in Mathematical Physics* **143** (1992), 315–332, and H.-W. Wiesbrock, "Half-sided modular inclusions of von Neumann algebras," *Communications in Mathematical Physics* **157** (1993), 83–92. See also Guido–Longo–Wiesbrock for the connection with conformal nets [30, 32, 31].

Then there is a strongly continuous one-parameter unitary group. With the Stone sign convention used below for physical inward light-ray translations, write it as

$$T(a) = \mathrm{e}^{-\mathrm{i}aP}, \qquad P \geq 0,$$

such that $T(a)\Omega = \Omega$, $T(a)\mathcal{M}T(a)^* \subset \mathcal{M}$ *for* $a \geq 0$, *and the modular group of* $\mathcal{M}$ *acts by dilations:*

$$\Delta_{\mathcal{M}}^{\mathrm{i}t} T(a) \Delta_{\mathcal{M}}^{-\mathrm{i}t} = T(\mathrm{e}^{+2\pi t} a). \tag{4.43}$$

Equivalently, with $K_{\mathcal{M}} = -\log \Delta_{\mathcal{M}}$,

$$\Delta_{\mathcal{M}}^{\mathrm{i}t} = \mathrm{e}^{-\mathrm{i}tK_{\mathcal{M}}}, \qquad [K_{\mathcal{M}}, P] = +2\pi\mathrm{i}P$$

on the common invariant core of analytic vectors. For a left half-sided modular inclusion the same statement is obtained after reversing the modular parameter; equivalently the dilation factor in (4.43) *is replaced by* $\mathrm{e}^{-2\pi t}a$, *and the infinitesimal commutator has the opposite sign.*

The modular-inclusion mechanism. The theorem is not the differentiated commutator below. Its content is the construction of the translation unitary $T(a)$ and the positivity of its generator from inclusion data alone. The standardness assumption provides three standard pairs:

$$(\mathcal{M}, \Omega), \qquad (\mathcal{N}, \Omega), \qquad (\mathcal{N}' \cap \mathcal{M}, \Omega).$$

Equivalently, in standard-subspace language, the closed real subspaces

$$H_{\mathcal{M}} = \overline{\{A\Omega : \ A = A^* \in \mathcal{M}\}}^{\mathcal{H}}, \qquad H_{\mathcal{N}} = \overline{\{A\Omega : \ A = A^* \in \mathcal{N}\}}^{\mathcal{H}}$$

are standard real subspaces: $H \cap \mathrm{i}H = \{0\}$ and $H + \mathrm{i}H$ is dense. Their Tomita operators

$$s_{\mathcal{M}}(h + \mathrm{i}k) = h - \mathrm{i}k, \qquad h, k \in H_{\mathcal{M}},$$

and similarly for $H_{\mathcal{N}}$, have polar decompositions with the same modular objects as the von Neumann algebras. The half-sided condition $\Delta_{\mathcal{M}}^{\mathrm{i}t} H_{\mathcal{N}} \subset H_{\mathcal{N}}$ for $t \geq 0$ is therefore a geometric statement about the position of two standard real subspaces under the modular dilations of the larger one.

The finite-dimensional comparison is rigid. Suppose, for comparison, that $\mathcal{M}$ is finite dimensional and that $\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}$ for all $t \geq 0$. The automorphism $\sigma_t^{\mathcal{M}}$ is a linear isomorphism of the finite-dimensional vector space $\mathcal{M}$. Its restriction maps the finite-dimensional subspace $\mathcal{N}$ injectively into itself, and hence onto itself:

$$\sigma_t^{\mathcal{M}}(\mathcal{N}) = \mathcal{N}, \qquad t \geq 0.$$

Applying the same equality to $-t$ by the group law gives invariance for all $t \in \mathbb{R}$. Thus the half-sided condition becomes ordinary modular invariance in finite dimension. The strict inclusion phenomenon in Quoted Theorem 4.15 depends on the infinite-dimensional standard-subspace geometry of local algebras: an injective continuous motion can move a closed real subspace properly inside itself without contradicting finite-dimensional rank.

Borchers–Wiesbrock theory proves that such a pair of standard real subspaces carries a positive-energy representation of the affine group of the line. In the Stone sign convention of the theorem statement, this representation is generated by the modular dilation group of $\mathcal{M}$ and a translation group $T(a) = \mathrm{e}^{-\mathrm{i}aP}$, $P \geq 0$, satisfying

$$\Delta_{\mathcal{M}}^{\mathrm{i}t} T(a) \Delta_{\mathcal{M}}^{-\mathrm{i}t} = T(\mathrm{e}^{+2\pi t} a).$$

The positivity $P \geq 0$ is the analytic half-sidedness input: it is the operator-algebraic analogue of the spectrum condition for the translation reconstructed from the inclusion. Once the unitary group has been obtained on the standard subspaces, standardness of the relative commutant lifts the standard-subspace inclusion back to the von Neumann algebra inclusion and gives

$$T(a)\mathcal{M}T(a)^* \subset \mathcal{M}, \qquad a \geq 0.$$

Thus the theorem reconstructs a translation-dilation semigroup from modular data; the Lie-algebra commutator is only the infinitesimal shadow of this stronger statement.

Derivation of the displayed commutator. The existence of $T(a)$ and the positivity $P \geq 0$ require the full half-sided modular inclusion result, Quoted Theorem 4.15. Once (4.43) is known, the commutator follows by differentiating twice. For the right half-sided convention displayed in (4.43), first differentiate in a at $a = 0$:

$$\Delta_{\mathcal{M}}^{\mathrm{i}t} P \Delta_{\mathcal{M}}^{-\mathrm{i}t} = \mathrm{e}^{+2\pi t} P.$$

Since $\Delta_{\mathcal{M}}^{\mathrm{i}t} = \mathrm{e}^{-\mathrm{i}tK_{\mathcal{M}}}$, the derivative at $t = 0$ gives

$$-\mathrm{i}[K_{\mathcal{M}}, P] = +2\pi P,$$

and hence $[K_{\mathcal{M}}, P] = +2\pi \mathrm{i} P$. In the opposite half-sided convention the Borchers relation has $\mathrm{e}^{-2\pi t} a$ in place of $\mathrm{e}^{+2\pi t} a$, so the differentiated commutator carries the opposite sign.

For the standard right wedge, take $\mathcal{M} = \mathcal{R}(W_R)$ and let $\mathcal{N}$ be the algebra of the wedge translated inward along a lightlike direction. The inclusion is half-sided because the Lorentz boost rescales that lightlike translation without leaving the wedge family. The sign is convention-sensitive, and the convention used in this chapter is the following one.

Example 4.16 (Right-wedge light-ray inclusion and the half-sided sign)**.** Let

$$e_+ = (1, 1, 0, \dots, 0), \qquad W_R(a) = W_R + ae_+, \qquad a > 0.$$

Then $W_R(a) \subset W_R$. Indeed, if $(x^0, x^1) \in W_R$ and $y = x + ae_+$, then

$$y^1 - y^0 = x^1 - x^0 > 0, \qquad y^1 + y^0 = x^1 + x^0 + 2a > 0,$$

so $y^1 > |y^0|$. Moreover

$$b \geq a \quad \Longrightarrow \quad W_R(b) \subset W_R(a), \tag{4.44}$$

because translating farther along e_+ moves the wedge deeper inside itself. Assume the Bisognano–Wichmann identification in the sign convention used below,

$$\sigma_t^{W_R} = \operatorname{Ad} \Delta_{W_R}^{\mathrm{i}t} = \operatorname{Ad} U(\Lambda_R(-2\pi t)).$$

Since $\Lambda_R(s)e_+ = \mathrm{e}^s e_+$, isotony and covariance give

$$\sigma_t^{W_R}\big(\mathcal{R}(W_R(a))\big) = \mathcal{R}\big(W_R(\mathrm{e}^{-2\pi t} a)\big). \tag{4.45}$$

Combining (4.44) and (4.45), this translated-wedge inclusion is left half-sided in Definition 4.14:

$$\sigma_t^{W_R}\big(\mathcal{R}(W_R(a))\big) \subset \mathcal{R}(W_R(a)) \quad \Longleftrightarrow \quad t \leq 0.$$

Changing the modular-flow sign convention, or using the corresponding oppositely directed light-ray inclusion, changes the word "left" to "right". The calculation verifies only the geometric hypothesis in this example. The Borchers–Wiesbrock theorem supplies the converse structural content: from the half-sided modular inclusion it reconstructs the translation unitary and its positive generator.

Physical light-ray translations in a covariant net. If the net already carries a Poincare representation, the abstract affine translation in the preceding paragraph has a visible candidate. With the translation convention of Definition 5.7,

$$U(x, \mathbf{1}) = \int_{\overline{V}_+} \mathrm{e}^{\mathrm{i}x^\mu p_\mu}\, \mathrm{d}E(p), \qquad p_\mu = \eta_{\mu\nu} p^\nu,$$

define the positive light-ray momentum

$$P_+^{\mathrm{phys}} := P^0 - P^1 = \int_{\overline{V}_+} (p^0 - p^1)\, \mathrm{d}E(p).$$

The spectrum condition gives $P_+^{\mathrm{phys}} \geq 0$, since $p^0 \geq |p^1|$ on $\overline{V}_+$. For $e_+ = (1, 1, 0, \ldots, 0)$, however,

$$e_+^\mu p_\mu = -p^0 + p^1 = -(p^0 - p^1),$$

so the physical inward light-ray translation is

$$U(ae_+, \mathbf{1}) = \mathrm{e}^{-\mathrm{i}aP_+^{\mathrm{phys}}}, \qquad a \geq 0.$$

This is the reason for the Stone sign convention in Quoted Theorem 4.15. Isotony and covariance give

$$U(ae_+, \mathbf{1})\mathcal{R}(W_R)U(ae_+, \mathbf{1})^* = \mathcal{R}(W_R(a)) \subset \mathcal{R}(W_R), \qquad a \geq 0,$$

and the Bisognano–Wichmann modular action gives the affine covariance

$$\Delta_{W_R}^{\mathrm{i}t} U(ae_+, \mathbf{1}) \Delta_{W_R}^{-\mathrm{i}t} = U(\Lambda_R(-2\pi t)ae_+, \mathbf{1}) = U(\mathrm{e}^{-2\pi t} ae_+, \mathbf{1}).$$

Thus, in a Poincare-covariant net satisfying Bisognano–Wichmann, the physical light-ray translations realize the left half-sided version of the positive-energy affine representation reconstructed by the Borchers–Wiesbrock theorem. The theorem's converse content is not this sign calculation: it is that a standard half-sided modular inclusion by itself already forces such a positive translation-dilation representation.

Thus the algebraic inclusion knows the translation-dilation subgroup before one has written down a point field. This is one reason modular theory is not only a structural theorem: it reconstructs geometric motion from the order structure of local algebras.

Let

$$W_R = \{x \in \mathbb{M}^D : \; x^1 > |x^0|\}$$

be the standard right wedge, and let $\Lambda_R(s)$ be the boost in the (x^0, x^1)-plane,

$$\Lambda_R(s) : \begin{pmatrix} x^0 \\ x^1 \end{pmatrix} \longmapsto \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix}.$$

Remark 4.17 (Bisognano–Wichmann theorem boundary for wedges). Let a Poincare-covariant quantum field theory on Minkowski space be generated by Wightman fields satisfying the Wightman domain, covariance, locality, vacuum, and spectrum hypotheses. Let

$$\mathcal{R}(W)$$

be the von Neumann algebra generated by fields smeared in a wedge W, and let Ω be the unique invariant vacuum. Then Ω is cyclic and separating for every wedge algebra. For the standard right wedge W_R, let (J_{W_R}, Δ_{W_R}) be the Tomita–Takesaki modular data of $(\mathcal{R}(W_R), \Omega)$. Then

$$\Delta_{W_R}^{\mathrm{i}t} = U(\Lambda_R(-2\pi t)), \qquad t \in \mathbb{R}.$$

If $W = gW_R$, then the modular group of $(\mathcal{R}(W), \Omega)$ is

$$\Delta_W^{\mathrm{i}t} = U\big(g\Lambda_R(-2\pi t)g^{-1}\big), \qquad t \in \mathbb{R}.$$

Moreover the modular conjugation implements the corresponding wedge reflection j_W on the local net:

$$J_W \mathcal{R}(\mathcal{O}) J_W = \mathcal{R}(j_W \mathcal{O})$$

for every local region $\mathcal{O}$ in the indexing class. On a charged field algebra reconstructed from the observable net, the same antiunitary operation includes the charge-conjugation action on charged field coordinates. This is a theorem boundary in the present monograph, with its scope recorded explicitly. The proof in the Wightman setting has two logically separate components. The first component is the polynomial-core KMS strip proved in Theorem 4.18 below from the spectrum condition, complex boosts, and wedge locality. The second component is the operator-domain closure theorem that identifies the closed Tomita operator of $(\mathcal{R}(W_R), \Omega)$, including the modular conjugation and the passage from field-polynomial boundary values to the full wedge von Neumann algebra. That closure theorem is the part supplied by the original Bisognano–Wichmann work and by the AQFT treatments of modular covariance.

Theorem 4.18 (Wightman polynomial-core wedge KMS mechanism)**.** *Let Φ_a be Wightman fields on the common invariant domain $\mathcal{D}$, transforming covariantly under the proper orthochronous Poincare group and satisfying the spectrum condition and graded locality. Let $\mathfrak{P}(W_R)$ be the unital $*$-algebra generated on $\mathcal{D}$ by smeared fields $\Phi_a(f)$ with $\operatorname{supp} f \subset W_R$, and let*

$$\alpha_s^R(A) = U(\Lambda_R(s)) A U(\Lambda_R(s))^{-1}.$$

For

$$A = \Phi_{a_1}(f_1) \cdots \Phi_{a_m}(f_m), \qquad B = \Phi_{b_1}(g_1) \cdots \Phi_{b_n}(g_n), \qquad \operatorname{supp} f_i, \operatorname{supp} g_j \subset W_R.$$

there is a function $F_{A,B}$, holomorphic for $0 < \operatorname{Im} z < 2\pi$ and with tempered boundary values on the two horizontal edges, such that

$$F_{A,B}(s) = \langle \Omega, A\alpha_s^R(B)\Omega \rangle, \qquad F_{A,B}(s + 2\pi\mathrm{i}) = \langle \Omega, \alpha_s^R(B) A\Omega \rangle. \tag{4.46}$$

Thus the vacuum functional satisfies the boost 2π-KMS boundary relation on the Wightman polynomial core of the wedge. For a fermionic field algebra the statement is the graded KMS relation on field coordinates; after restriction to the observable algebra the grading signs cancel.

Proof. The theorem is an analytic-domain statement: the boost generator supplies the real flow, while the KMS boundary relation requires holomorphic continuation and locality. The initial Tomita operator on polynomial wedge vectors is

$$S_{W_R,0}A\Omega = A^*\Omega, \qquad A \in \mathfrak{P}(W_R).$$

Reeh–Schlieder gives density of $\mathfrak{P}(W_R)\Omega$, and Corollary 3.20 supplies separatingness of Ω for $\mathcal{R}(W_R)$ from locality and cyclicity in the opposite wedge. These are the standardness inputs that make the initial Tomita operator closable in the full Bisognano–Wichmann theorem. The polynomial KMS strip itself is obtained directly from Wightman analyticity. Suppressing finite spin matrices in the covariance law, define the smeared boundary value

$$\begin{aligned} F_{A,B}(z) = \int \prod_i \mathrm{d}x_i\, f_i(x_i) \prod_j \mathrm{d}y_j\, g_j(y_j) \\ \times W_{a_1\ldots a_m b_1\ldots b_n}\big(x_1,\ldots,x_m,\Lambda_R(z)y_1,\ldots,\Lambda_R(z)y_n\big), \end{aligned} \tag{4.47}$$

with the usual component matrices inserted for non-scalar fields. The spectrum condition supplies the tube holomorphy as follows. After rewriting an ordered Wightman distribution in relative variables, its Fourier transform is supported where every partial momentum sum lies in the closed future cone. If a relative coordinate is shifted as $\xi \mapsto \xi - \mathrm{i}\eta$ with $\eta \in V_+$, the Fourier factor $\exp(-\mathrm{i}p\cdot(\xi - \mathrm{i}\eta))$ contains $\exp(p\cdot\eta)$; in the mostly-plus convention $p\cdot\eta \le 0$ for $p,\eta \in \overline{V}_+$. Thus the spectral support gives exponential damping in the tube selected by the ordering, and the Wightman distribution is recovered as a boundary value. For $z = s + \mathrm{i}u$ and $0 < u < \pi$, the imaginary part of $\Lambda_R(z)y$, $y \in W_R$, lies in the future timelike cone in the mostly-plus convention. Therefore the differences between each earlier real argument x_i and each complex-boosted later argument have imaginary part in the past tube, so the ordered distribution in (4.47) is holomorphic in the first half-strip and has tempered boundary values.

At $u = \pi$, $\Lambda_R(s + \mathrm{i}\pi)W_R = W_L$, the left wedge. Points of W_L are spacelike to points of W_R. Graded locality therefore permits the boosted fields in B to pass through the fields in A, with the sign determined by the field grading. On the observable subalgebra this sign is one. After this reordering, the same tube analyticity gives the continuation through the second half-strip $\pi < u < 2\pi$. The lower and upper boundary values are consequently

$$F_{A,B}(s + 2\pi\mathrm{i}) = \omega\big(\alpha_s^R(B)A\big), \qquad F_{A,B}(s) = \omega\big(A\alpha_s^R(B)\big), \tag{4.48}$$

where $\omega(C) = \langle \Omega, C\Omega\rangle$. This is (4.46). The full modular theorem adds the closure statement

$$\overline{S_{W_R,0}} = J_{W_R}\Delta_{W_R}^{1/2}.$$

The operator $\Delta_{W_R}^{1/2}$ is then obtained by analytic continuation of boosts to imaginary value $\mathrm{i}\pi$, and the remaining antiunitary maps the wedge to its geometric reflection. That closed-operator identification is the additional domain theorem named in Remark 4.17. □

Corollary 4.19 (PCT from wedge modular covariance)**.** *Let $D = 4$. Let a vacuum representation of a local observable net satisfy isotony, locality, Poincare covariance, the spectrum condition, uniqueness of the invariant vacuum, wedge cyclicity and separatingness, wedge duality, and the Bisognano–Wichmann modular-covariance conclusion*

$$\Delta_{W_R}^{\mathrm{i}t} = U(\Lambda_R(-2\pi t)), \qquad J_{W_R}\mathcal{R}(\mathcal{O})J_{W_R} = \mathcal{R}(j_R\mathcal{O})$$

for the standard right wedge. Let $R_\perp(\pi)$ be the spatial rotation by π in the (x^2, x^3)-plane, so that

$$R_\perp(\pi) j_R x = -x.$$

Then

$$\Theta_{\mathrm{PCT}} := U(R_\perp(\pi)) J_{W_R} \tag{4.49}$$

is an antiunitary operator satisfying

$$\Theta_{\mathrm{PCT}}\Omega = \Omega, \qquad \Theta_{\mathrm{PCT}} U(a,\Lambda)\Theta_{\mathrm{PCT}}^{-1} = U(-a,\Lambda),$$

and

$$\Theta_{\mathrm{PCT}}\mathcal{R}(\mathcal{O})\Theta_{\mathrm{PCT}}^{-1} = \mathcal{R}(-\mathcal{O}). \tag{4.50}$$

For a Doplicher–Roberts charged field algebra reconstructed from the observable category, the induced antiunitary sends each charged field multiplet to the conjugate charged multiplet, in agreement with the Wightman PCT theorem of Theorem 2.6.

Proof. The antiunitarity of Θ_{PCT} follows from antiunitarity of the modular conjugation J_{W_R} and unitarity of $U(R_\perp(\pi))$. The displayed hypotheses identify J_{W_R} with the geometric wedge reflection j_R on the net. Conjugating afterward by the proper spatial rotation $R_\perp(\pi)$ changes the geometric action from j_R to the total inversion $x \mapsto -x$, proving (4.50). Both J_{W_R} and the rotation preserve the vacuum, so $\Theta_{\mathrm{PCT}}\Omega = \Omega$.

The covariance identity follows by applying the same geometric conjugation to translations and proper Lorentz transformations. Since the total inversion commutes with every proper Lorentz transformation and sends a to $-a$, the automorphism of the Poincare group is

$$(a,\Lambda) \longmapsto (-a,\Lambda).$$

Antiunitarity then gives the same translation-generator convention as in (2.11). Finally, in the Doplicher–Roberts reconstruction the modular conjugation maps a localized endomorphism to its conjugate localized endomorphism; on the reconstructed field algebra this is exactly the passage from a charged multiplet to the conjugate charged multiplet. □

The cyclic and separating hypotheses needed for the modular data of wedge algebras are supplied in the Wightman setting by Theorem 3.19 and Corollary 3.20: cyclicity is the positive-energy local-density theorem, while separatingness for a wedge uses locality with the opposite wedge. In a purely algebraic formulation not assumed to arise from Wightman fields, the conclusion of Remark 4.17 is an explicit modular-covariance hypothesis on the vacuum net unless it has been proved from the chosen AQFT assumptions. Under that theorem or hypothesis, modular flow is geometric Lorentz boost flow. The result connects locality, positivity of energy, the KMS structure of Definition 4.5 for accelerated observers, and the PCT antiunitary stated in Corollary 4.19.

Indeed, define the boost automorphism of the standard wedge algebra by

$$\alpha_s^R(A) = U(\Lambda_R(s)) A U(\Lambda_R(s))^{-1}, \qquad A \in \mathcal{R}(W_R).$$

Bisognano–Wichmann says $\sigma_t = \alpha^R_{-2\pi t}$, so the modular KMS property implies that $\omega(A) = \langle \Omega, A\Omega\rangle$, restricted to $\mathcal{R}(W_R)$, is a 2π-KMS state for the boost dynamics α_s^R:

$$F_{A,B}(s) = \omega(A\alpha_s^R(B)), \qquad F_{A,B}(s+2\pi\mathrm{i}) = \omega(\alpha_s^R(B)A).$$

Thus the vacuum is thermal with respect to the boost generator, even though it is a pure state on the global algebra. Along a uniformly accelerated orbit whose proper time is related to boost parameter by $s = a\tau$, this becomes the Unruh inverse temperature $2\pi/a$ in units $\hbar = c = k_B = 1$.

Figure 4.3 shows the Bisognano–Wichmann wedge geometry: modular flow is Lorentz boost flow, and its KMS period becomes the Unruh temperature along an accelerated orbit.

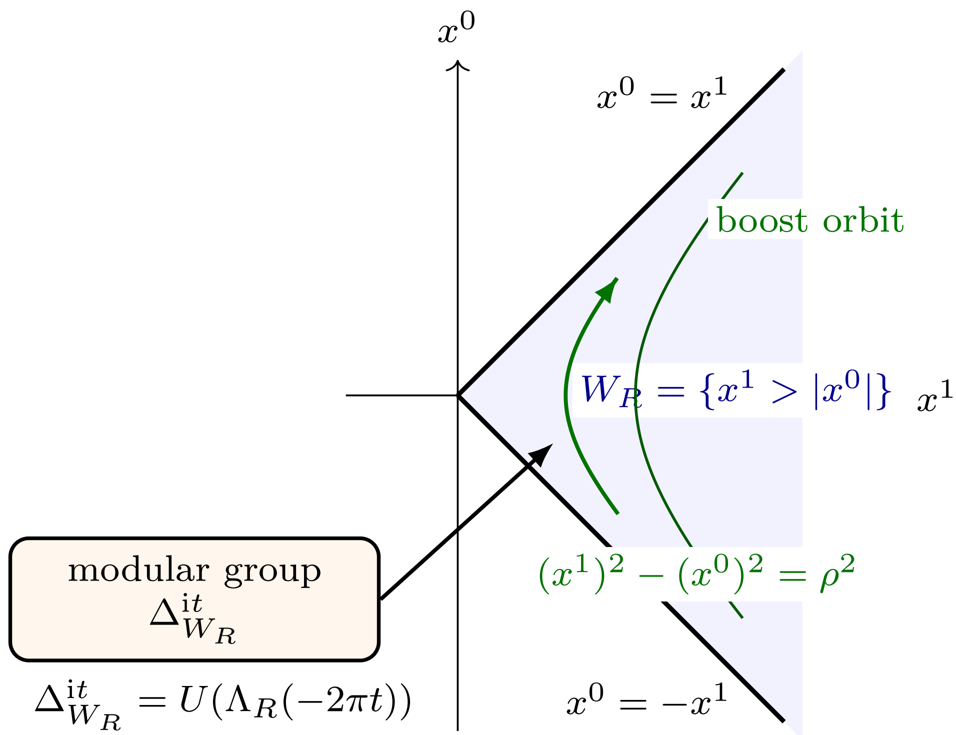


Figure 4.3: By the Bisognano–Wichmann theorem, or by the corresponding explicit modular-covariance hypothesis in an AQFT net, wedge modular flow is geometric Lorentz boost flow; its flow lines inside the right wedge are hyperbolae $(x^1)^2 - (x^0)^2 = \rho^2$.

4.12 Status of the Structural Inputs Used Above

The preceding sections use several deep operator-algebraic and local-QFT results. Their logical statuses differ. The distinction matters because an abstract von Neumann algebra theorem may be used after standardness and normality have been verified, whereas a local-QFT theorem usually requires covariance, spectrum, locality, additivity, phase-space control, or wedge-domain estimates in the concrete representation.

Doplicher–Roberts reconstruction. The purely categorical part says that a rigid symmetric C^*-tensor category with simple unit, finite-dimensional morphism spaces, conjugates, and finite statistical dimensions is equivalent to $\mathrm{Rep}_{\mathrm{fd}}(G)$ for a compact group G, after choosing a faithful symmetric tensor fiber functor. The local-QFT part is earlier: one must first prove that the sectors are transportable bounded-region DHR endomorphisms, that Haag duality places localized images and intertwiners in the assigned local observable algebra rather than only in the dual net, and that locality gives a symmetric exchange structure. The local charged-field core (4.6)–(4.8) proves the algebraic form of multiplication, fusion, conjugation, and fixed-point projection once those hypotheses are present. Charged fields for Gauss-law sectors, spacelike-cone sectors, infraparticles, or confining theories require different localization categories before a field-algebra theorem can even be formulated.

Tomita–Takesaki theory. This is pure operator-algebra infrastructure. Its input is a standard pair $(\mathcal{R}, \Omega)$, meaning Ω is cyclic and separating for $\mathcal{R}$. The chapter displays the S_0/F_0 adjoint-core argument, the closed Tomita operator, and the left-Hilbert-algebra mechanism leading to $\Delta^{it}\mathcal{R}\Delta^{-it} = \mathcal{R}$ and $J\mathcal{R}J = \mathcal{R}'$. In QFT applications, the theorem may be used only after standardness of the relevant local or wedge algebra has been proved, typically through Reeh–Schlieder and locality. It supplies an abstract modular flow. Identification of that flow with a spacetime symmetry is an additional local-QFT theorem.

Connes cocycle and Araki relative entropy. This is also operator-algebraic infrastructure, formulated for faithful normal states or weights in a common standard representation. The relative Tomita operator and the product $[D\psi : D\omega]_t = \Delta^{it}_{\psi|\omega}\Delta^{-it}_{\omega}$ give an algebra unitary implementing the change of modular flow. The finite standard-form calculation fixes the density-matrix normalization. In local QFT the required work is to specify the von Neumann algebra, the normal states on it, and the faithfulness/support condition. Those data provide the intrinsic type-III replacement for a formal density-matrix quotient.

Nuclearity, split inclusions, and type classification. Buchholz–Wichmann nuclearity is a local phase-space hypothesis on $\Theta_{\beta,\mathcal{O}} : A \mapsto e^{-\beta H}A\Omega$. The operator-algebraic split criterion asks for normal extendability of the multiplication representation $\mathcal{M}_1 \odot \mathcal{M}_2' \to \mathcal{B}(\mathcal{H})$ to $\mathcal{M}_1 \bar{\otimes} \mathcal{M}_2'$. The text above explains how summable separated normal functionals arise from energy damping, locality, and the positive collar $\mathcal{O}_1 \Subset \mathcal{O}_2$. The type III or type III_1 classification of sharp local algebras is a separate theorem boundary: a type-I interpolant at positive separation describes collar-separated independence, while type-III behavior concerns the sharp zero-collar algebra. Concrete models require their own phase-space estimates, factoriality input, and modular-spectrum control.

Borchers–Wiesbrock half-sided inclusions. The abstract theorem starts from a standard inclusion $\mathcal{N} \subset \mathcal{M}$ and a one-sided modular invariance condition. Its content is the construction of a positive-generator translation group with affine modular covariance. The finite-dimensional rank argument above isolates the content as infinite-dimensional standard-subspace geometry. In a Poincare-covariant QFT net, an inward translated wedge gives the geometric half-sided inclusion only after wedge standardness, isotony, covariance, and the relevant Bisognano–Wichmann modular identification are available. The light-ray calculation then matches the abstract generator to the positive physical momentum $P^0 - P^1$, while the converse reconstruction remains the Borchers–Wiesbrock theorem itself.

Bisognano–Wichmann and modular PCT. The Wightman polynomial-core theorem proved above establishes the boost KMS strip from tube analyticity and locality. The full wedge-modular theorem additionally identifies the closure of the wedge Tomita operator with geometric Lorentz boosts and the wedge reflection. That closed-operator/domain statement is the Bisognano–Wichmann QFT theorem boundary unless the net is assumed to satisfy modular covariance. Corollary 4.19 is therefore conditional on the explicit modular-covariance hypotheses stated there.

Thus each structural conclusion in this chapter is tied to a visible input: DHR localization and duality for charge reconstruction; standardness for modular theory; normal state data for relative entropy; phase-space bounds and collars for split inclusions; half-sided standardness for Borchers–Wiesbrock; and wedge-domain control for Bisognano–Wichmann.

4.13 Algebraic Scattering Interface

The Haag–Ruelle construction can be formulated in the algebraic setting using almost-local operators whose energy-momentum transfer is concentrated near an isolated mass shell. Acting on the vacuum, such operators create single-particle states. Time-dependent translated operators with separated velocity supports have limits that define incoming and outgoing scattering states:

$$|\psi_1, \ldots, \psi_n\rangle_{\mathrm{in/out}} = \lim_{t \to \mp\infty} A_1(t) \cdots A_n(t)\Omega.$$

This gives the same conceptual order established earlier: first the nonperturbative asymptotic states, then the scattering operator, and only afterward any perturbative representation of matrix elements.

The algebraic language records the dependence of scattering on local observables, spectral isolation, and phase-space properties. Infraparticle theories, gauge theories with long-range fields, and theories without a mass gap require modified asymptotic constructions whose hypotheses must be stated separately.

Chapter 5

Relativistic Quantum Mechanics and Local Operator Structure

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

5.1 Hilbert-Space Quantum Data

Definition 5.1 (Hilbert-space quantum datum)**.** Let $\mathcal{H}$ be a complex Hilbert space. A pure state is a ray in $\mathcal{H}$. A mixed state is a positive trace-class operator ρ on $\mathcal{H}$ satisfying

$$\rho \geq 0, \qquad \mathrm{Tr}_{\mathcal{H}}\, \rho = 1.$$

For a bounded operator $A \in \mathcal{B}(\mathcal{H})$, the expectation value in the state ρ is

$$\langle A \rangle_\rho = \mathrm{Tr}_{\mathcal{H}}(\rho A).$$

For an unbounded operator, the operator domain and the domain of every product appearing in a matrix element are part of the data.

Definition 5.2 (Hamiltonian time evolution)**.** A Hamiltonian time evolution is specified by a self-adjoint operator H and the unitary one-parameter group

$$U(t) = \mathrm{e}^{-\mathrm{i}tH}, \qquad A(t) = U(t)^{-1} A U(t).$$

Stone's theorem gives the converse: every strongly continuous unitary one-parameter group has a unique self-adjoint generator. When H is first given as a formal differential expression, choosing its self-adjoint realization is part of the definition of the quantum system. In a stable sector H is bounded below. Spacetime localization is additional structure: a local relativistic theory assigns operators or algebras to spacetime regions.

5.2 Rigged Hilbert Spaces and Continuous Spectra

The Hilbert space contains the normalizable states. The momentum kets, energy kets, position kets, and plane-wave scattering states used throughout the monograph are generalized vectors. Their precise home is a rigged Hilbert space, also called a Gelfand triple.

Definition 5.3 (Rigged Hilbert space)**.** A rigged Hilbert space consists of a dense subspace $\Phi \subset \mathcal{H}$, equipped with a locally convex topology finer than the Hilbert norm topology, together with the continuous inclusions

$$\Phi \subset \mathcal{H} \subset \Phi^{\times}.$$

Here $\Phi^{\times}$ is the continuous antilinear dual of Φ. The second inclusion sends $h \in \mathcal{H}$ to the functional

$$\phi \longmapsto \langle \phi, h\rangle_{\mathcal{H}}, \qquad \phi \in \Phi.$$

Thus every Hilbert vector defines a generalized vector, but $\Phi^{\times}$ is larger than $\mathcal{H}$. Distributional eigenvectors live in $\Phi^{\times}$.

The basic model is

$$\mathcal{S}(\mathbb{R}^d) \subset L^2(\mathbb{R}^d, \mathrm{d}^d x) \subset \mathcal{S}'(\mathbb{R}^d),$$

where $\mathcal{S}$ is the Schwartz space and $\mathcal{S}'$ is the space of tempered distributions. The position and momentum kets are the tempered distributions defined by

$$\langle x|\psi\rangle = \psi(x), \qquad \langle p|\psi\rangle = \widetilde{\psi}(p), \qquad \psi \in \mathcal{S}(\mathbb{R}^d),$$

with the Fourier convention chosen in the relevant chapter. The identities

$$\langle x|x'\rangle = \delta^{(d)}(x-x'), \qquad \langle p|p'\rangle = (2\pi)^d \delta^{(d)}(p-p'),$$

and

$$\int \mathrm{d}^d x\, |x\rangle\langle x| = \mathbf{1}, \qquad \int \frac{\mathrm{d}^d p}{(2\pi)^d}\, |p\rangle\langle p| = \mathbf{1}$$

are weak identities: after pairing with test vectors in Φ on the left and right, they become ordinary identities for functions or distributions. Point evaluation and Fourier evaluation are continuous linear functionals on $\mathcal{S}(\mathbb{R}^d)$, hence define elements of $\mathcal{S}'(\mathbb{R}^d)$. Pairing the displayed resolution of the identity with $\varphi, \psi \in \mathcal{S}(\mathbb{R}^d)$ gives respectively $\int \overline{\varphi(x)}\psi(x)\mathrm{d}^d x$ and the same inner product written after Fourier inversion. The delta distributions are precisely the kernels which reproduce these pairings, so the displayed bra-ket identities are kernel statements, not equalities between Hilbert vectors.

Definition 5.4 (Generalized eigenvector in a rigged Hilbert space)**.** For a self-adjoint operator A with spectral measure E_A, a number λ is an eigenvalue in the Hilbert-space sense precisely when

$$E_A(\{\lambda\})\mathcal{H} \neq 0.$$

If $E_A(\{\lambda\}) = 0$ but λ lies in the continuous spectrum, there is no normalizable eigenvector with that eigenvalue. A generalized eigenvector at λ is instead a functional

$$F_\lambda \in \Phi^\times$$

such that, on a common invariant test domain,

$$F_\lambda(A\phi) = \lambda F_\lambda(\phi), \qquad \phi \in \Phi.$$

This is the mathematical meaning of a plane wave of sharp momentum or an energy eigenstate in a continuous spectrum. It is a distributional object used to write kernels. Physical Hilbert vectors are obtained by smearing such kernels with square-integrable wave packets.

Definition 5.5 (Direct-integral coordinate ket)**.** Equivalently, the spectral theorem may be written as a direct integral

$$\mathcal{H} \simeq \int_X^\oplus \mathcal{H}_\lambda \, \mathrm{d}\nu(\lambda), \qquad A = \int_X^\oplus \lambda \, \mathbf{1}_{\mathcal{H}_\lambda} \, \mathrm{d}\nu(\lambda).$$

The label λ indexes fibers over the spectrum. A generalized ket $|\lambda, a\rangle$ is not an element of $\mathcal{H}$. It denotes the distributional coordinate functional that extracts the a-component in the fiber $\mathcal{H}_\lambda$ of the direct integral. Delta-normalization records the kernel of the identity relative to the measure $\mathrm{d}\nu$. Changing $\mathrm{d}\nu$ rescales these generalized kets and therefore changes the displayed delta factors; the Hilbert-space vector obtained after smearing is unchanged.

Definition 5.6 (Relativistic one-particle rigging)**.** For a relativistic one-particle mass shell this gives, for example,

$$\Phi_1 = C_c^\infty(\Sigma_m^+; V) \subset \mathcal{H}_1 = L^2(\Sigma_m^+, \mathrm{d}\mu_m; V) \subset \Phi_1^\times,$$

where V is the finite-dimensional little-group representation space. The little-group label convention is fixed by choosing an orthonormal basis $\{e_\sigma\}$ of V and writing

$$\langle e_\tau, e_\sigma \rangle_V = \delta_{\tau\sigma}.$$

Thus, in a bra-ket expression, the first Kronecker index is the label carried by the bra and the second is the label carried by the ket; since the Kronecker delta is symmetric, $\delta_{\tau\sigma} = \delta_{\sigma\tau}$. The symbol $|p, \sigma\rangle$ is an element of $\Phi_1^\times$, and

$$\langle p, \tau | p', \sigma \rangle = (2\pi)^{D-1} 2p^0 \, \delta^{(D-1)}(\vec{p} - \vec{p}')\delta_{\tau\sigma}$$

is the distribution kernel for the identity in the covariant mass-shell measure. All scattering kernels written later with sharp external momenta are to be read in this sense. The wave-packet formulation is the primary Hilbert-space statement; the plane-wave formula is the corresponding distributional kernel.

5.3 Poincare Symmetry

Definition 5.7 (Poincare-covariant vacuum Hilbert datum)**.** Let

$$\mathcal{P}_+^\uparrow = \mathbb{R}^D \rtimes SO^+(1, D-1)$$

be the connected Poincare group of $\mathbb{M}^D$. A relativistic quantum theory in the opening framework carries a strongly continuous unitary representation

$$U : \mathcal{P}_+^\uparrow \to \mathcal{U}(\mathcal{H}), \qquad (a, \Lambda) \mapsto U(a, \Lambda),$$

or the corresponding representation of the double cover when spinorial representations are present. The group law is

$$(a', \Lambda')(a, \Lambda) = (a' + \Lambda' a, \Lambda'\Lambda),$$

so a genuine unitary representation satisfies

$$U(a', \Lambda')U(a, \Lambda) = U(a' + \Lambda' a, \Lambda'\Lambda).$$

Infinitesimally,

$$U(a, \Lambda) = 1 + \mathrm{i} a_\mu P^\mu + \frac{\mathrm{i}}{2}\omega_{\mu\nu} J^{\mu\nu} + O(a^2, \omega^2, a\omega),$$

where P^μ are the self-adjoint energy-momentum generators, $P_\mu = \eta_{\mu\nu} P^\nu$, and $J^{\mu\nu} = -J^{\nu\mu}$ are the Lorentz generators on a common invariant domain. The commutation relations are

$$[P^\mu, P^\nu] = 0,$$

$$[P^\mu, J^{\rho\sigma}] = -\mathrm{i}(\eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho),$$

and

$$[J^{\mu\nu}, J^{\rho\sigma}] = -\mathrm{i}\left(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}\right).$$

Equivalently, $[J^{\rho\sigma}, P^\mu] = \mathrm{i}(\eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho)$. This is the Lorentzian Poincare subalgebra convention used again in the conformal chapters. There the Hermitian charge Q_X associated to a conformal Killing field X obeys $[Q_X, Q_Y] = -\mathrm{i} Q_{[X,Y]}$; see Chapter 57, (57.10). Taking $X = M^{\rho\sigma}$ and $Y = P^\mu$, then raising indices with η, gives the displayed $[J^{\rho\sigma}, P^\mu]$ bracket. The radial-quantization real form is obtained later by the explicit generator map (59.27). The spectrum condition is

$$\mathrm{Spec}(P) \subset \overline{V}_+.$$

In a vacuum sector there is a unit vector $\Omega \in \mathcal{H}$ satisfying

$$U(a, \Lambda)\Omega = \Omega.$$

5.4 Translation Spectrum and Mass Sectors

Proposition 5.8 (Joint translation spectrum)**.** *The restriction of the Poincare representation to translations has a unique joint spectral measure*

$$E : \{\textit{Borel subsets of } \mathbb{R}^D\} \longrightarrow \{\textit{orthogonal projections on } \mathcal{H}\}$$

such that

$$P^\mu = \int_{\mathbb{R}^D} p^\mu \, \mathrm{d}E(p), \qquad U(a, \mathbf{1}) = \int_{\mathbb{R}^D} \mathrm{e}^{\mathrm{i} a^\mu p_\mu} \, \mathrm{d}E(p).$$

The one-parameter Stone generators P^μ are therefore the coordinate operators of this single joint spectral measure. In particular the translation generators are strongly commuting self-adjoint *operators: their spectral projections $E(\Delta)$ commute. For a vector $\psi \in \mathcal{H}$, its energy-momentum spectral measure is*

$$\mu_\psi(\Delta) = \|E(\Delta)\psi\|^2, \qquad \Delta \subset \mathbb{R}^D \text{ Borel.}$$

The spectral support of ψ is the closed support of this measure. The spectrum condition in the preceding section is the statement that

$$E(\mathbb{R}^D \setminus \overline{V}_+) = 0.$$

Proof. Restrict U in Definition 5.7 to the translation subgroup. The subgroup is abelian and locally compact, and the restriction is strongly continuous and unitary. The Stone–Naimark spectral theorem therefore gives the single projection-valued measure E displayed above. Explicitly, the equality

$$U(a, \mathbf{1}) = \int_{\mathbb{R}^D} \mathrm{e}^{\mathrm{i}a^\mu p_\mu} \, \mathrm{d}E(p)$$

is an equality of bounded operators for every $a \in \mathbb{R}^D$, and uniqueness follows because Fourier transforms of finite complex Borel measures on $\mathbb{R}^D$ determine those measures. Differentiating the one-parameter subgroup $s \mapsto U(se_\mu, \mathbf{1})$ in Stone's sense gives the coordinate multiplication operator p^μ in this joint spectral representation, namely

$$P^\mu \psi = \int_{\mathbb{R}^D} p^\mu \, \mathrm{d}E(p) \, \psi$$

on its natural domain

$$\{\psi \in \mathcal{H} : \int_{\mathbb{R}^D} |p^\mu|^2 \, \mathrm{d}\|E(p)\psi\|^2 < \infty\}.$$

All spectral projections obtained from E commute because they are projections in the same countably additive projection-valued measure. This strong commutativity is stronger than the formal infinitesimal relation $[P^\mu, P^\nu] = 0$ on a common invariant domain: for unbounded operators, commutation on a core does not by itself imply that the spectral projections commute. The displayed Lie-algebra commutator is therefore the differential shadow of the joint functional calculus. Finally, the operator-valued support statement $E(\mathbb{R}^D \setminus \overline{V}_+) = 0$ is equivalent to saying that every scalar spectral measure $\mu_\psi(\Delta) = \|E(\Delta)\psi\|^2$ is supported in $\overline{V}_+$; this is the precise spectral-measure form of the spectrum condition. □

Definition 5.9 (Invariant mass spectral measure)**.** The invariant mass operator is defined by functional calculus:

$$M^2 = -P_\mu P^\mu = \int_{\mathbb{R}^D} (-p_\mu p^\mu) \, \mathrm{d}E(p).$$

For a Borel set $I \subset [0, \infty)$, the projection onto the corresponding mass range is

$$E_M(I) = E\big(\{p \in \overline{V}_+ : -p_\mu p^\mu \in I\}\big).$$

The vacuum subspace is the zero-momentum subspace

$$E(\{0\})\mathcal{H}.$$

In a pure vacuum sector this subspace is one-dimensional and is spanned by Ω.

Definition 5.10 (Mass sector and particle shell)**.** A mass sector of mass $m \geq 0$ is the spectral subspace of M^2 at the value m^2, namely $E_M(\{m^2\})\mathcal{H}$. For $m > 0$, the corresponding positive-energy mass shell is the smooth hyperboloid

$$\Sigma_m^+ = \{p \in \mathbb{R}^D : p^0 = \sqrt{\vec{p}^2 + m^2},\ p_\mu p^\mu = -m^2\}.$$

For $m = 0$, the nonzero massless particle shell is the punctured future light cone

$$\Sigma_0^+ = \{p \in \mathbb{R}^D : p^0 = |\vec{p}|,\ p_\mu p^\mu = 0,\ p \neq 0\}.$$

It is isolated when there is an open neighborhood $N \subset \mathbb{R}^D$ of the shell under discussion such that

$$\operatorname{Spec}(P) \cap N = \Sigma_m^+$$

in the massive case, and analogously with Σ_0^+ in a massless sector when such a global isolation statement is true. In a vacuum representation the massless cone usually accumulates at the vacuum point $p = 0$, so the global isolation condition is a genuine additional spectral hypothesis beyond the invariant-mass equation $M^2 = 0$.

Isolated shell as a one-particle subrepresentation. An isolated massive particle in a local quantum theory is the closed spectral subspace

$$\mathcal{H}_1 = E(\Sigma_m^+)\mathcal{H}$$

inside the physical Hilbert space. Its Wigner representation is the restriction of the Poincare representation to this subspace. Later, the Källén–Lehmann representation detects the same object as an atom in a two-point spectral measure, and Haag–Ruelle theory uses the isolation hypothesis to construct asymptotic multi-particle states. The set Σ_m^+ is Borel, so $E(\Sigma_m^+)$ is an orthogonal projection. The spectral covariance relation $U(a,\Lambda)E(\Delta)U(a,\Lambda)^{-1} = E(\Lambda\Delta)$ follows from conjugating the translation representation by $U(a,\Lambda)$. Since Σ_m^+ is a positive-energy Lorentz orbit, the range of $E(\Sigma_m^+)$ is invariant under the Poincare representation. Restricting the representation to this closed range gives the one-particle Wigner subrepresentation.

Remark 5.11 (Infraparticle sectors)*.* The word "isolated" is a spectral hypothesis in addition to the mass equation $M^2 = m^2$. A theory with a massless long-range gauge field can have charged sectors whose charged threshold is the lower edge of continuous energy-momentum support. In four-dimensional QED, Gauss's law ties nonzero electric charge to an electromagnetic field at spatial infinity; charged sector states are therefore represented by operators carrying noncompact dressing data. A charged dressing carries a Coulomb tail and admits arbitrarily soft photon excitations. The resulting charged particle is an infraparticle: its spectral support accumulates at the charged threshold in place of an isolated shell with a nonzero projection $E(\Sigma_m^+)$. The Haag–Ruelle and LSZ constructions below apply to sectors satisfying the isolated-shell hypothesis; the infrared replacement in QED is treated in Chapters 42 and 44.

Figure 5.1 displays the spectral separation assumed in the one-particle subrepresentation construction: an isolated positive mass shell inside the future cone, separated from the multiparticle continuum.

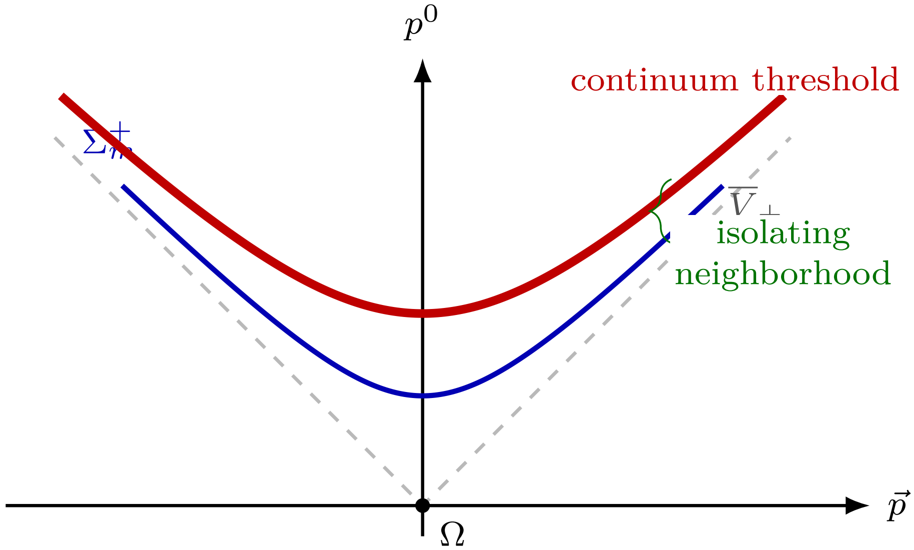


Figure 5.1: The translation spectrum lies in the closed future cone. A stable massive particle corresponds to an isolated positive-energy mass shell, separated from the continuum in a neighborhood of the shell.

5.5 One-Particle Representations

Definition 5.12 (Massive scalar Wigner one-particle space)**.** The spectral subspace $E(\Sigma_m^+)\mathcal{H}$ carries a unitary representation of the Poincare group. The corresponding abstract one-particle representation is described by Wigner's little-group construction. For a massive spinless particle, the one-particle Hilbert space is realized as

$$\mathcal{H}_1 = L^2\left(\Sigma_m^+, \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}2E_{\vec{p}}}\right), \qquad E_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}.$$

The measure displayed here is the Lorentz-invariant measure on Σ_m^+ in the chosen normalization. General massive particles carry a finite-dimensional unitary representation of the double cover of the massive little group $SO(D-1)$. Massless finite-helicity particles in four spacetime dimensions are classified by helicity representations of the massless little group.

Definition 5.13 (Four-dimensional Wigner orbit datum)**.** For later reference we record the corresponding four-dimensional Wigner classification in one place. Let

$$\widetilde{\mathcal{P}_+^\uparrow} = \mathbb{R}^{1,3} \rtimes SL(2,\mathbb{C})$$

be the universal-cover version of the connected Poincare group. An irreducible unitary representation whose translation spectrum is supported on a single Lorentz orbit is obtained by choosing a representative momentum $\bar{p}$, an irreducible unitary representation of its stabilizer

$$G_{\bar{p}} = \{\Lambda \in SL(2,\mathbb{C}) : \Lambda\bar{p} = \bar{p}\},$$

and inducing to $\widetilde{\mathcal{P}_+^\uparrow}$. The spectrum condition keeps only orbits contained in the closed future cone, but the representation-theoretic list is useful because it makes explicit which cases are being assumed away.

Orbit type	Representative and invariant	Little group	Irreducible little-group data	Role in this monograph
Vacuum orbit	$\bar{p} = 0$	$SL(2, \mathbb{C})$	The pure vacuum sector uses the trivial representation.	This is the zero-momentum state, not a particle shell.
Massive positive-energy orbit	$\bar{p} = (m, 0, 0, 0)$, $m > 0$, $\bar{p}^2 = -m^2$	$SU(2)$	Spin $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$, with $\dim V_j = 2j + 1$.	Isolated massive particles and their higher-spin tower; developed in Chapters 20 and 40.
Massless finite-helicity orbit	$\bar{p} = (\kappa, 0, 0, \kappa)$, $\kappa > 0$, $\bar{p}^2 = 0$	double cover of $ISO(2)$	The two null-translation generators act trivially; the rotation generator has helicity $h \in \frac{1}{2}\mathbb{Z}$.	Photons, gravitons, and finite-helicity external states; developed in Chapters 21 and 41.
Massless continuous-spin orbit	Same lightlike orbit, $\bar{p}^2 = 0$, with nontrivial little-group translations	double cover of $ISO(2)$	Nonzero transverse-translation orbit $N_1^2 + N_2^2 = \rho^2 > 0$, giving an infinite helicity tower mixed by boosts.	A legitimate positive-energy Wigner class, but not generated by point-local covariant fields under the usual locality assumptions; see Remark 21.3.
Spacelike, or tachyonic, orbit	$\bar{p} = (0, \mu, 0, 0)$, $\mu > 0$, $\bar{p}^2 = \mu^2$	double cover of $SO^+(1, 2)$	Unitary little-group representations are representations of a noncompact group, typically infinite-dimensional.	Excluded here by the spectrum condition: the full Lorentz orbit is not contained in $\overline{V}_+$ and has no invariant lower energy bound.

Table 5.1: Four-dimensional Wigner orbit classes for irreducible one-particle representations of the universal cover of the connected Poincare group. A general Hilbert space representation may contain direct sums or direct integrals of such irreducibles; a stable particle species is an isolated copy of one of the particle orbits satisfying the spectral and localization hypotheses used later.

This construction supplies one-particle state spaces and their Poincare transformation law. A local quantum theory also supplies a localization assignment for observables.

5.6 Free Fock Space From One-Particle Data

Definition 5.14 (Free bosonic and fermionic Fock spaces)**.** From a one-particle Hilbert space $\mathcal{H}_1$, define the bosonic Fock space

$$\mathcal{F}_s(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} \operatorname{Sym}^n \mathcal{H}_1$$

and the fermionic Fock space

$$\mathcal{F}_a(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} \wedge^n \mathcal{H}_1.$$

Definition 5.15 (Bosonic creation-operator tensor normalization)**.** The symmetric tensor powers use the creation-operator normalization. Let

$$\Pi_{s,n} = \frac{1}{n!} \sum_{\sigma \in S_n} U_\sigma$$

be the orthogonal projection from $\mathcal{H}_1^{\otimes n}$ to its symmetric subspace, where

$$U_\sigma(\psi_1 \otimes \cdots \otimes \psi_n) = \psi_{\sigma^{-1}(1)} \otimes \cdots \otimes \psi_{\sigma^{-1}(n)}.$$

For $\psi_1, \dots, \psi_n \in \mathcal{H}_1$ define

$$\psi_1 \odot \cdots \odot \psi_n := \sqrt{n!}\, \Pi_{s,n}(\psi_1 \otimes \cdots \otimes \psi_n) \in \mathrm{Sym}^n \mathcal{H}_1. \tag{5.1}$$

Bosonic Fock inner product in creation-operator normalization. With this convention

$$\big\langle \psi_1 \odot \cdots \odot \psi_n,\, \chi_1 \odot \cdots \odot \chi_m \big\rangle = \delta_{mn} \sum_{\sigma \in S_n} \prod_{j=1}^{n} \langle \psi_j, \chi_{\sigma(j)} \rangle_{\mathcal{H}_1}. \tag{5.2}$$

Different tensor powers in the direct sum defining $\mathcal{F}_s(\mathcal{H}_1)$ are orthogonal, giving the factor δ_{mn}. For $m = n$, Definition 5.15 gives

$$\psi_1 \odot \cdots \odot \psi_n = \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \psi_{\sigma^{-1}(1)} \otimes \cdots \otimes \psi_{\sigma^{-1}(n)}$$

and the analogous formula for the χ's. Hence

$$\big\langle \psi_1 \odot \cdots \odot \psi_n,\, \chi_1 \odot \cdots \odot \chi_n \big\rangle = \frac{1}{n!} \sum_{\sigma, \tau \in S_n} \prod_{i=1}^{n} \langle \psi_{\sigma^{-1}(i)}, \chi_{\tau^{-1}(i)} \rangle_{\mathcal{H}_1}.$$

For each fixed σ, set $j = \sigma^{-1}(i)$ and $\rho = \tau^{-1}\sigma$. The product becomes $\prod_{j=1}^{n} \langle \psi_j, \chi_{\rho(j)} \rangle_{\mathcal{H}_1}$, and as τ runs over S_n, so does ρ. The sum over σ cancels the prefactor $1/n!$, leaving the displayed permutation sum.

Definition 5.16 (Mass-shell Fock test domain and kernels)**.** For the massive scalar one-particle space $\mathcal{H}_1 = L^2(\Sigma_m^+, \mathrm{d}\mu_m)$, identify the mass shell with $\mathbb{R}^{D-1}$ by

$$\vec{p} \longmapsto (E_{\vec{p}}, \vec{p}), \qquad E_{\vec{p}} = \sqrt{\vec{p}^{\,2} + m^2},$$

and define

$$\mathcal{S}_m = \mathcal{S}(\Sigma_m^+)$$

to be the wave packets whose pullback to the $\vec{p}$-chart is a Schwartz function. The space $\mathcal{S}_m$ is dense in $\mathcal{H}_1$, is contained in the domain of every polynomial in the one-particle energy-momentum operators, and is invariant under the one-particle Poincare representation. The common algebraic Fock domain used below is

$$\mathcal{D}_0 = \mathrm{span}\{a^\dagger(\psi_1) \cdots a^\dagger(\psi_n)\Omega \mid n \geq 0,\ \psi_1, \dots, \psi_n \in \mathcal{S}_m\}.$$

It is dense in $\mathcal{F}_s(\mathcal{H}_1)$ and invariant under creation and annihilation operators smeared by elements of $\mathcal{S}_m$. For a free scalar species, the momentum-space annihilation and creation operators $a(\vec{p})$ and $a^\dagger(\vec{p})$ are operator-valued distributions characterized by

$$[a(\vec{p}), a^\dagger(\vec{q})] = (2\pi)^{D-1} 2E_{\vec{p}}\, \delta^{(D-1)}(\vec{p} - \vec{q}),$$

with all other commutators vanishing. The free Hamiltonian is

$$H_0 = \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}2E_{\vec{p}}}\, E_{\vec{p}}\, a^\dagger(\vec{p})a(\vec{p}),$$

after the vacuum energy has been normalized. The spatial momentum is

$$\vec{P}_0 = \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}2E_{\vec{p}}}\, \vec{p}\, a^\dagger(\vec{p})a(\vec{p}).$$

Write

$$\mathrm{d}\mu_m(\vec{p}) = \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}2E_{\vec{p}}}.$$

For a one-particle test wave packet $\psi \in \mathcal{S}_m$, the smeared creation and annihilation operators on $\mathcal{D}_0$ are

$$a^\dagger(\psi) = \int \mathrm{d}\mu_m(\vec{p})\, \psi(\vec{p})a^\dagger(\vec{p}), \qquad a(\psi) = \int \mathrm{d}\mu_m(\vec{p})\, \overline{\psi(\vec{p})}a(\vec{p}).$$

The closed Fock operators can be defined for all $\psi \in \mathcal{H}_1$. The operator-valued distribution notation and the common-domain commutator calculation use the dense test class $\mathcal{S}_m$. For $\psi, \chi \in \mathcal{S}_m$, the operators on $\mathcal{D}_0$ satisfy

$$[a(\psi), a^\dagger(\chi)] = \int \mathrm{d}\mu_m(\vec{p})\, \overline{\psi(\vec{p})}\chi(\vec{p}) = \langle \psi, \chi\rangle_{\mathcal{H}_1}.$$

Definition 5.17 (Noncovariant creation and momentum-ket normalization)**.** There is a second common normalization, useful for displaying the elementary Fock construction. Define operator-valued distributions $\mathfrak{a}_{\vec{p}}$ by

$$a(\vec{p}) = \sqrt{(2\pi)^{D-1}2E_{\vec{p}}}\, \mathfrak{a}_{\vec{p}}.$$

Then

$$[\mathfrak{a}_{\vec{p}}, \mathfrak{a}^\dagger_{\vec{q}}] = \delta^{(D-1)}(\vec{p}-\vec{q}), \qquad [\mathfrak{a}_{\vec{p}}, \mathfrak{a}_{\vec{q}}] = [\mathfrak{a}^\dagger_{\vec{p}}, \mathfrak{a}^\dagger_{\vec{q}}] = 0.$$

The corresponding sharp-momentum kets are distributional vectors,

$$|\vec{p}\rangle = \mathfrak{a}^\dagger_{\vec{p}}\Omega, \qquad |\vec{p}_1, \ldots, \vec{p}_n\rangle = \mathfrak{a}^\dagger_{\vec{p}_1} \cdots \mathfrak{a}^\dagger_{\vec{p}_n}\Omega,$$

and

$$\langle \vec{p}|\vec{q}\rangle = \delta^{(D-1)}(\vec{p}-\vec{q}).$$

In this normalization

$$H_0 = \int \mathrm{d}^{D-1}\vec{p}\, E_{\vec{p}}\, \mathfrak{a}^\dagger_{\vec{p}}\mathfrak{a}_{\vec{p}}, \qquad \vec{P}_0 = \int \mathrm{d}^{D-1}\vec{p}\, \vec{p}\, \mathfrak{a}^\dagger_{\vec{p}}\mathfrak{a}_{\vec{p}}.$$

Definition 5.18 (Free multi-species Fock representation)**.** For a finite collection of free stable species, let B be the set of bosonic species and F the set of fermionic species, with one-particle spaces $\mathcal{H}_{1,b}$ and $\mathcal{H}_{1,f}$. The free multi-species Hilbert space is the completed tensor product

$$\mathcal{H}_{\mathrm{Fock}} = \bigotimes_{b\in B} \mathcal{F}_s(\mathcal{H}_{1,b}) \otimes \bigotimes_{f\in F} \mathcal{F}_a(\mathcal{H}_{1,f}).$$

The vacuum is the tensor product of the factor vacua, and the free Poincare representation is the tensor product of the corresponding second-quantized one-particle representations. Equivalently, $\bigotimes_{b\in B} \mathcal{F}_s(\mathcal{H}_{1,b}) \simeq \mathcal{F}_s(\bigoplus_{b\in B} \mathcal{H}_{1,b})$, and similarly for fermions with $\mathcal{F}_a$. The operator algebra on the fermionic factors uses the graded tensor-product convention, so homogeneous odd creation or annihilation operators associated with distinct fermion species anticommute.

The Fock construction gives a free multi-particle representation. A local QFT also specifies which operators are localized in each bounded spacetime region.

5.7 Locality As A Constraint On Interacting Particle Dynamics

At a cutoff or algebraic bookkeeping level, an interaction written in free Fock variables has the form

$$H = H_0 + H_{\text{int}},$$

where H_{int} is a sum of normally ordered monomials such as

$$\int K_{2,1}\, a^\dagger a^\dagger a + \int K_{1,2}\, a^\dagger a a + \cdots ,$$

with kernels $K_{r,s}$ and the mass-shell measures suppressed in the displayed shorthand. This expression is not yet a definition of a relativistic local theory. A local-theory datum also includes translation generators P_μ, Lorentz generators $J^{\mu\nu}$, a Poincare-invariant vacuum, and local observables or fields satisfying covariance and microcausality. In an interacting theory the boost generators, the Hamiltonian, and the local field operators are tied together by the Poincare algebra and locality.

The causal input is spacetime-local operation. A local operation near a point or region can influence only its causal future. The algebraic expression of this statement is locality for spacelike separated regions.

Figure 5.2 fixes the light-cone convention used here: the local operation sits at the vertex, the allowed influence lies in the causal future, and spacelike separated regions remain outside this causal domain.

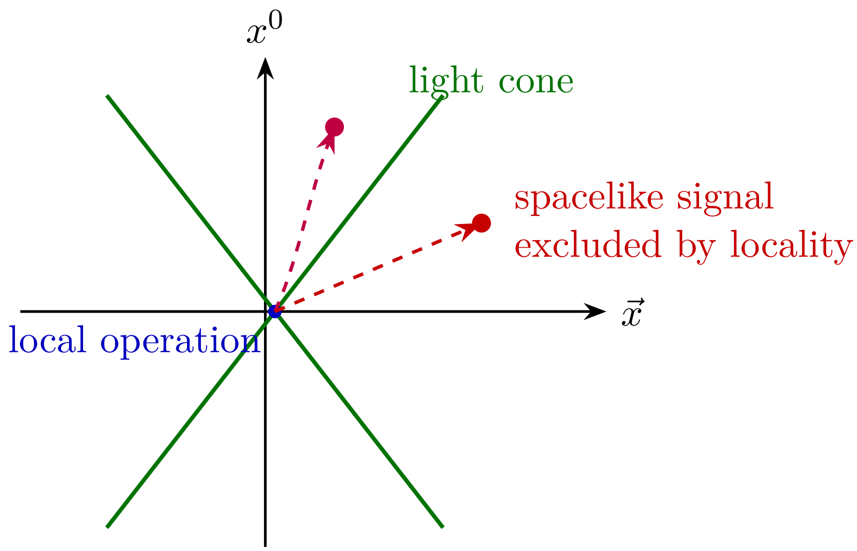


Figure 5.2: Causal localization in Minkowski space. The field-theoretic condition corresponding to the exclusion of spacelike signaling is commutativity, or graded commutativity, of observables localized in spacelike separated regions.

5.8 Locality Data

Definition 5.19 (Local observable assignment)**.** For bounded open regions $\mathcal{O} \subset \mathbb{M}^D$, the opening framework assigns algebras $\mathcal{A}(\mathcal{O}) \subset \mathcal{B}(\mathcal{H})$. If $\mathcal{O}_1$ and $\mathcal{O}_2$ are spacelike separated, locality is

$$[A, B] = 0, \qquad A \in \mathcal{A}(\mathcal{O}_1), \quad B \in \mathcal{A}(\mathcal{O}_2),$$

with the graded commutator after a fermion parity grading has been specified.

This condition states compatibility of spacelike separated local observables: their order gives the same matrix elements in every vector for which the matrix elements are defined.

5.9 Free Scalar Field As A Derived Example

Definition 5.20 (Free scalar field from mass-shell creation kernels)**.** Use the dense domain $\mathcal{D}_0 \subset \mathcal{F}_s(\mathcal{H}_1)$ generated from the mass-shell Schwartz test space $\mathcal{S}_m$. On $\mathcal{D}_0$, define the free scalar field as the operator-valued distribution

$$\widehat{\phi}(x) = \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2E_{\vec{p}}} \left(a(\vec{p}) \mathrm{e}^{\mathrm{i}p\cdot x} + a^\dagger(\vec{p}) \mathrm{e}^{-\mathrm{i}p\cdot x} \right),$$

where $p^0 = E_{\vec{p}}$ and

$$p \cdot x = \eta_{\mu\nu} p^\mu x^\nu.$$

For $f \in C_c^\infty(\mathbb{M}^D)$, the smeared field is

$$\widehat{\phi}(f) = \int \mathrm{d}^D x\, f(x) \widehat{\phi}(x).$$

In the noncovariant normalization this is equivalently

$$\widehat{\phi}(x) = \int \frac{\mathrm{d}^{D-1}\vec{p}}{\sqrt{(2\pi)^{D-1} 2E_{\vec{p}}}} \left(\mathfrak{a}_{\vec{p}} \mathrm{e}^{\mathrm{i}p\cdot x} + \mathfrak{a}^\dagger_{\vec{p}} \mathrm{e}^{-\mathrm{i}p\cdot x} \right).$$

Covariance of the free scalar field. The free scalar field is covariant. Let $g = (b, \Lambda) \in \mathcal{P}_+^\uparrow$, and let $U_1(g)$ be the one-particle scalar representation on $\mathcal{H}_1$. The free Fock representation is its second quantization $U_0(g) = \Gamma(U_1(g))$, so

$$U_0(g) a^\dagger(\psi) U_0(g)^{-1} = a^\dagger(U_1(g)\psi)$$

for $\psi \in \mathcal{S}_m$ on $\mathcal{D}_0$. Since $U_1(g)\mathcal{S}_m \subset \mathcal{S}_m$, this identity preserves the common test domain; the closed operators give the corresponding Hilbert-space identity for arbitrary $\psi \in \mathcal{H}_1$. The adjoint identity gives the annihilation part. Applying these identities to the distributional creation and annihilation kernels in Definition 5.20, followed by the invariant change of variables on Σ_m^+, gives

$$U_0(b, \Lambda) \widehat{\phi}(x) U_0(b, \Lambda)^{-1} = \widehat{\phi}(\Lambda x + b).$$

Thus the calculation checks compatibility of the free kernel definition with the chosen one-particle representation.

Proposition 5.21 (Free scalar microcausality)**.** *The commutator is*

$$[\widehat{\phi}(x), \widehat{\phi}(y)] = \mathrm{i}\Delta(x-y),$$

where the Pauli–Jordan distribution is

$$\Delta(x) = \frac{1}{\mathrm{i}} \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2E_{\vec{p}}} \left(\mathrm{e}^{\mathrm{i}p\cdot x} - \mathrm{e}^{-\mathrm{i}p\cdot x}\right).$$

The distribution Δ is Lorentz invariant and its restriction to the spacelike region vanishes. Consequently

$$[\widehat{\phi}(f), \widehat{\phi}(g)] = 0$$

when the supports of f and g are spacelike separated.

Proof. The canonical commutation relation in Definition 5.16 gives the displayed Pauli–Jordan commutator on the common domain $\mathcal{D}_0$. In smeared form, for $f, g \in C_c^\infty(\mathbb{M}^D)$,

$$[\widehat{\phi}(f), \widehat{\phi}(g)] = \mathrm{i} \int \mathrm{d}^D x\, \mathrm{d}^D y\, f(x) g(y) \Delta(x-y)$$

as a scalar multiple of the identity on $\mathcal{D}_0$. The mass-shell measure and the two exponentials are invariant under the connected Lorentz group, so $\Delta(\Lambda x) = \Delta(x)$ as a distribution. On the open spacelike region $x^2 > 0$, the mass-shell Fourier integral defines a smooth solution of the Klein–Gordon equation. For each spacelike x choose a proper orthochronous Lorentz transformation sending it to $(0, \vec{r})$. At equal time the two exponentials in the defining integral are exchanged by the measure-preserving map $\vec{p} \mapsto -\vec{p}$, hence the smooth restriction of Δ to that point is zero. Since the spacelike region is open and every point in it is covered by this argument, Δ vanishes there as a distribution. If the supports of f and g are spacelike separated, then $x - y$ remains in this open spacelike region on $\operatorname{supp} f \times \operatorname{supp} g$; the displayed smeared commutator therefore vanishes. □

Remark 5.22. The cancellation above uses the antisymmetric combination $\Delta_+(x-y) - \Delta_+(y-x)$, equivalently the ordinary commutator for an even scalar field. If the same scalar two-point function were assigned odd locality, the vacuum matrix element of the required anticommutator would contain

$$\Delta_+(x-y) + \Delta_+(y-x).$$

At equal time and spacelike separation this becomes

$$2 \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2E_{\vec{p}}} \cos(\vec{p}\cdot\vec{r}),$$

which has spacelike support. Thus the sign in the Pauli–Jordan distribution is already the local free-field manifestation of the spin-statistics connection. The theorem-level statement, with its positivity, covariance, and domain hypotheses, is discussed in Chapter 40.

This calculation establishes locality for the free scalar field from the stated data: the free scalar Fock representation, the Lorentz-invariant mass-shell measure, the distributional field definition, and smearing by test functions.

5.10 Constructed Structures

The chapter has constructed five distinct structures. Hilbert-space quantum mechanics supplies states and operators. Translation spectral theory supplies energy-momentum support, invariant mass, and particle subspaces inside the physical Hilbert space. Poincare representation theory supplies the transformation law of one-particle state spaces. Fock space supplies free multi-particle state spaces. A local quantum theory supplies local observable assignments satisfying covariance and microcausality. The free scalar field realizes these structures explicitly and will serve as the first model for the general field framework.

Chapter 6

Local Fields, Covariance, and Microcausality

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter works in the Wightman field presentation introduced in Chapter 2: local fields are operator-valued distributions on a common invariant dense domain, and every point-field formula is shorthand for a continuous smeared matrix element.

6.1 Field Multiplets As Operator-Valued Distributions

Definition 6.1 (Field multiplet as an operator-valued distribution)**.** Let $\mathcal{D} \subset \mathcal{H}$ be a dense linear subspace. In this chapter $\mathrm{End}(\mathcal{D})$ denotes the algebra of linear maps $\mathcal{D} \to \mathcal{D}$. A field multiplet is a collection $\widehat{\Phi}_A$, with A in a finite index set, such that each component defines a continuous linear map

$$\widehat{\Phi}_A : C_c^\infty(\mathbb{M}^D) \to \mathrm{End}(\mathcal{D}), \qquad f \mapsto \widehat{\Phi}_A(f).$$

Continuity is meant in the usual locally convex LF topology on $C_c^\infty(\mathbb{M}^D)$. This is the localization version of the test-function convention in Chapter 1; when a later spectral statement uses $\mathcal{S}(\mathbb{M}^D)$, the additional tempered-continuity hypothesis is part of that statement. The notation

$$\widehat{\Phi}_A(f) = \int_{\mathbb{M}^D} \mathrm{d}^D x\, f(x) \widehat{\Phi}_A(x)$$

is the distributional notation for this linear map. The symbol x in $\widehat{\Phi}_A(x)$ is a spacetime label, not an operator on $\mathcal{H}$. The operator is obtained only after smearing the distributional field against a test function. This convention separates spacetime localization data from the Hilbert-space operators that act on states.

Definition 6.2 (Localization of a smeared field)**.** The support of the smeared field $\widehat{\Phi}_A(f)$ is controlled by $\operatorname{supp} f$. If $\operatorname{supp} f \subset \mathcal{O}$, where $\mathcal{O} \subset \mathbb{M}^D$ is a bounded open region, the field is localized in $\mathcal{O}$. In the local-algebra framework of Chapter 1, this means that $\widehat{\Phi}_A(f)$ is affiliated with $\mathcal{A}(\mathcal{O})$ or belongs to the unbounded field algebra whose bounded functions generate observables localized in $\mathcal{O}$.

Definition 6.3 (Common invariant field domain)**.** A common invariant field domain for a family of fields is a dense subspace $\mathcal{D} \subset \mathcal{H}$ such that every smeared field $\widehat{\Phi}_A(f)$ maps $\mathcal{D}$ into itself, every finite product of smeared fields is defined on $\mathcal{D}$, the vacuum vector belongs to $\mathcal{D}$, and the Poincare representation maps $\mathcal{D}$ into itself. If adjoints are used, the adjoint field is required to preserve the same domain.

The domain condition is what makes a product such as $\widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)\Omega$ an actual vector before any distributional limit is taken. Matrix elements

$$(f_1,\dots,f_n) \longmapsto \left\langle \psi, \widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)\chi \right\rangle$$

are required to be continuous multilinear functionals of the test functions for ψ, χ in a stated dense domain. This continuity is the analytic content behind point-field notation.

Remark 6.4 (Distributional matrix elements). Suppose the smeared products above are defined on a common invariant field domain and the displayed matrix elements are continuous multilinear functionals on $C_c^\infty(\mathbb{M}^D)^n$. The Schwartz kernel theorem identifies the completed projective tensor product $C_c^\infty(\mathbb{M}^D)^{\widehat{\otimes} n}$ with $C_c^\infty((\mathbb{M}^D)^n)$. Thus the displayed matrix element is equivalently a distribution $W_{\psi,\chi}^{A_1\cdots A_n} \in \mathcal{D}'((\mathbb{M}^D)^n)$ satisfying

$$W_{\psi,\chi}^{A_1\cdots A_n}(f_1\otimes\cdots\otimes f_n) = \left\langle \psi, \widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)\chi \right\rangle.$$

This is the functional-analytic meaning of the stated continuity hypothesis.

Remark 6.5 (Direction of point-field notation). The preceding remark fixes the direction of logic: point-dependent expressions are compact notation for continuous smeared matrix elements. Later analytic continuation, spectral representation, and LSZ extraction all act on these distributions or on their Fourier transforms.

6.2 Poincare Covariance Of Fields

Let

$$g = (a, \Lambda) \in \mathcal{P}_+^\uparrow$$

act on spacetime by $gx = \Lambda x + a$. Let S be a finite-dimensional representation of $SO^+(1, D-1)$, or of its double cover for spinor fields, acting on the field components:

$$S(\Lambda)_A{}^B.$$

Definition 6.6 (Poincare pullback on test functions)**.** For $f \in C_c^\infty(\mathbb{M}^D)$, define the transformed test function

$$f_g(y) = f(g^{-1}y) = f(\Lambda^{-1}(y-a)).$$

The assignment $f \mapsto f_g$ is the pullback by g^{-1}. With the group law $hg : x \mapsto h(gx)$, it satisfies

$$(f_g)_h = f_{hg}.$$

Definition 6.7 (Covariant field multiplet)**.** A field multiplet $\widehat{\Phi}_A$ is Poincare covariant with component representation S when

$$U(g)^{-1}\widehat{\Phi}_A(f)U(g) = S(\Lambda)_A{}^B\,\widehat{\Phi}_B(f_g)$$

for every $g = (a, \Lambda)$, every test function f, and every component index A, with summation over repeated component indices.

In point-field notation this covariance law is written

$$U(a,\Lambda)^{-1}\widehat{\Phi}_A(x)U(a,\Lambda) = S(\Lambda)_A{}^B\,\widehat{\Phi}_B(\Lambda x + a).$$

The smeared equation is the definition; the point-field equation is its compact distributional form. With $U(a,\mathbf{1}) = \exp(\mathrm{i}a_\mu P^\mu) = \exp(\mathrm{i}a^\mu P_\mu)$, conjugation with U^{-1} on the left is the convention in which the coordinate label is translated by $+a$.

For a scalar field $S(\Lambda) = 1$. The infinitesimal translation law is

$$[P_\mu, \widehat{\Phi}(f)] = -\mathrm{i}\,\widehat{\Phi}(\partial_\mu f),$$

equivalently

$$[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\,\partial_\mu\widehat{\Phi}(x)$$

in distributional notation. The two signs are the same distributional statement: for compactly supported f,

$$\int \mathrm{d}^D x\, f(x)\,\partial_\mu\widehat{\Phi}(x) = -\int \mathrm{d}^D x\,(\partial_\mu f)(x)\widehat{\Phi}(x),$$

so moving the derivative from the field distribution to the test function produces the minus sign in the smeared commutator. Lorentz generators act by an orbital term on the spacetime argument and a spin term on the component index.

Infinitesimal translation covariance. Let $\widehat{\Phi}$ be a scalar covariant field and assume that $a \mapsto U(a,\mathbf{1})$ is generated on the common domain by $U(a,\mathbf{1}) = \exp(\mathrm{i}a^\mu P_\mu)$. Then, as an identity of smeared operators on $\mathcal{D}$,

$$[P_\mu, \widehat{\Phi}(f)] = -\mathrm{i}\,\widehat{\Phi}(\partial_\mu f).$$

Equivalently, in distributional point notation,

$$[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\,\partial_\mu\widehat{\Phi}(x).$$

For a pure translation $g = (a,\mathbf{1})$, Definition 6.7 gives

$$U(a)^{-1}\widehat{\Phi}(f)U(a) = \widehat{\Phi}(f_a), \qquad f_a(y) = f(y-a).$$

Differentiate at $a = 0$ in the μ-direction. The left side gives

$$\frac{d}{dt}\Big|_{t=0} e^{-\mathrm{i}tP_\mu}\widehat{\Phi}(f)e^{\mathrm{i}tP_\mu} = -\mathrm{i}[P_\mu, \widehat{\Phi}(f)],$$

while the right side gives

$$\frac{d}{dt}\Big|_{t=0} \widehat{\Phi}(f(\,\cdot - te_\mu)) = -\widehat{\Phi}(\partial_\mu f).$$

Equating the two derivatives gives the smeared formula. Pairing the point-field identity with f and integrating by parts gives the distributional version.

This is the infinitesimal form of the covariance definition; the displayed signs are part of the convention fixed by putting $U(a)^{-1}$ on the left in the finite transformation law.

Covariance of localization regions. If $\operatorname{supp} f \subset \mathcal{O}$, then $\operatorname{supp} f_g \subset g\mathcal{O}$. Therefore the covariant transform $U(g)^{-1}\widehat{\Phi}_A(f)U(g)$ is localized in $g\mathcal{O}$ whenever $\widehat{\Phi}_A(f)$ is localized in $\mathcal{O}$. The support identity $\operatorname{supp} f_g = g \operatorname{supp} f$ follows from $f_g(y) = f(g^{-1}y)$ and continuity of the homeomorphism g. The covariance law then rewrites the transformed operator as a linear combination of fields smeared with test functions supported in $g\mathcal{O}$.

Covariance of Wightman distributions. Assume that $\widehat{\Phi}_A$ is a covariant field multiplet on the common invariant domain of Definition 6.3, and that the vacuum is Poincare invariant:

$$U(g)\Omega = \Omega, \qquad g \in \mathcal{P}_+^\uparrow.$$

Then the n-point Wightman distributions satisfy

$$\begin{aligned} W_{A_1\cdots A_n}(f_1, \dots, f_n) = {} & S(\Lambda)_{A_1}{}^{B_1} \cdots S(\Lambda)_{A_n}{}^{B_n} \\ & \times W_{B_1\cdots B_n}\big((f_1)_g, \dots, (f_n)_g\big), \qquad g = (a, \Lambda), \end{aligned}$$

where $(f_i)_g(y) = f_i(g^{-1}y)$. In point-field notation this is

$$W_{A_1\cdots A_n}(x_1, \dots, x_n) = S(\Lambda)_{A_1}{}^{B_1} \cdots S(\Lambda)_{A_n}{}^{B_n} W_{B_1\cdots B_n}(gx_1, \dots, gx_n),$$

with the same distributional meaning as the smeared formula. The derivation uses only the stated field covariance and vacuum invariance. By vacuum invariance and unitarity,

$$\langle\Omega|\widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)|\Omega\rangle = \langle\Omega|U(g)^{-1}\widehat{\Phi}_{A_1}(f_1)\cdots\widehat{\Phi}_{A_n}(f_n)U(g)|\Omega\rangle.$$

Since $U(g)\mathcal{D} \subset \mathcal{D}$, conjugation by $U(g)$ may be applied to each factor on the common domain. Using Definition 6.7 for every field gives

$$U(g)^{-1}\widehat{\Phi}_{A_i}(f_i)U(g) = S(\Lambda)_{A_i}{}^{B_i}\widehat{\Phi}_{B_i}((f_i)_g).$$

Multiplying these identities in the given operator order yields the smeared covariance formula. The point-field formula is exactly the same identity paired with tensor-product test functions.

6.3 Spacelike Separation Of Supports

Definition 6.8 (Spacelike separated supports)**.** Two subsets $K_1, K_2 \subset \mathbb{M}^D$ are spacelike separated when

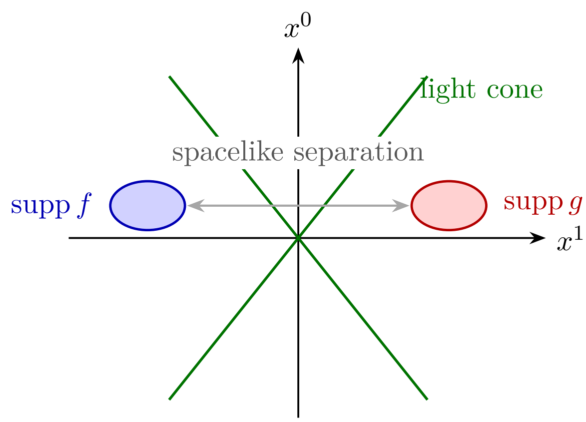
$$(x-y)^2 > 0 \qquad \text{for every } x \in K_1,\ y \in K_2.$$

Thus two test functions $f, g \in C_c^\infty(\mathbb{M}^D)$ have spacelike separated supports when $K_1 = \operatorname{supp} f$ and $K_2 = \operatorname{supp} g$ satisfy this condition.

Poincare invariance of spacelike support separation. Let $g = (a, \Lambda) \in \mathcal{P}_+^\uparrow$. If $K_1, K_2 \subset \mathbb{M}^D$ are spacelike separated, then gK_1 and gK_2 are spacelike separated: for $x \in K_1$ and $y \in K_2$,

$$(gx - gy)^2 = (\Lambda(x-y))^2 = (x-y)^2 > 0,$$

because translations cancel in differences and Λ preserves the Minkowski quadratic form. For test functions, $\operatorname{supp}(f_g) = g \operatorname{supp} f$: indeed $f_g(y) = f(g^{-1}y)$ can be nonzero only when $g^{-1}y \in \operatorname{supp} f$, equivalently $y \in g \operatorname{supp} f$, and the reverse inclusion follows after taking closures. Therefore Poincare transformation preserves spacelike separation of supports.



The figure displays one spatial dimension. The condition is the set-theoretic one written above: every pair of points, one from each support, has spacelike separation.

6.4 Microcausality

Definition 6.9 (Homogeneous parity and graded commutator)**.** Suppose the field multiplets carry a fermion parity $|\widehat{\Phi}_A| \in \{0, 1\}$. For homogeneous operators X, Y, define the graded commutator by

$$[X, Y]_{\text{gr}} = XY - (-1)^{|X||Y|} YX.$$

For bosonic fields both parities are zero, so this is the ordinary commutator.

Definition 6.10 (Microcausality for smeared fields)**.** The field multiplets satisfy microcausality on $\mathcal{D}$ when

$$[\widehat{\Phi}_A(f), \widehat{\Psi}_B(g)]_{\text{gr}} \psi = 0$$

for every $\psi \in \mathcal{D}$, whenever $\operatorname{supp} f$ and $\operatorname{supp} g$ are spacelike separated.

In point-field notation the bosonic case is written

$$[\widehat{\Phi}_A(x), \widehat{\Psi}_B(y)] = 0 \qquad (x-y)^2 > 0.$$

The smeared statement is the definition. It is the local compatibility condition for spacelike separated field insertions.

Adjacent spacelike exchange in Wightman functions. Assume microcausality in the sense of Definition 6.10. Let $\widehat{\Phi}_A$ and $\widehat{\Psi}_B$ be homogeneous fields of parities $|A|$ and $|B|$. If f and g have spacelike separated supports, then for any fields placed to the left and right whose products are defined on the common domain,

$$\begin{aligned}&\langle\Omega|\widehat{X}\,\widehat{\Phi}_A(f)\widehat{\Psi}_B(g)\,\widehat{Y}|\Omega\rangle\\ &\qquad = (-1)^{|A||B|}\langle\Omega|\widehat{X}\,\widehat{\Psi}_B(g)\widehat{\Phi}_A(f)\,\widehat{Y}|\Omega\rangle.\end{aligned}$$

Equivalently, adjacent entries of a Wightman distribution may be exchanged across a spacelike support separation with the Koszul sign $(-1)^{|A||B|}$.

Microcausality gives, on $\mathcal{D}$,

$$\widehat{\Phi}_A(f)\widehat{\Psi}_B(g) - (-1)^{|A||B|}\widehat{\Psi}_B(g)\widehat{\Phi}_A(f) = 0.$$

Let $\widehat{Y}|\Omega\rangle \in \mathcal{D}$; this is part of the common-domain assumption for finite products of smeared fields. Applying the displayed operator identity to this vector and then pairing with $\langle\Omega|\widehat{X}$ gives the stated matrix-element identity. Since Wightman distributions are the continuous multilinear extensions of these smeared matrix elements, the same exchange relation is their distributional support-separated boundary value.

6.5 Field Coordinates And Local Algebras

Definition 6.11 (Local polynomial field algebra on a common domain)**.** For a bounded open region $\mathcal{O}$, let

$$\mathcal{P}(\mathcal{O})$$

denote the polynomial $*$-algebra on $\mathcal{D}$ generated by finite sums and products of smeared fields $\widehat{\Phi}_A(f)$ with $\operatorname{supp} f \subset \mathcal{O}$, together with their adjoints when the adjoints preserve the chosen domain. This algebra records local unbounded field coordinates.

Isotony and graded locality for polynomial field algebras. If $\mathcal{O}_1 \subset \mathcal{O}_2$, then $\mathcal{P}(\mathcal{O}_1) \subset \mathcal{P}(\mathcal{O}_2)$. If $\mathcal{O}_1$ and $\mathcal{O}_2$ are spacelike separated and the fields satisfy Definition 6.10, then homogeneous monomials $X \in \mathcal{P}(\mathcal{O}_1)$ and $Y \in \mathcal{P}(\mathcal{O}_2)$ satisfy

$$XY = (-1)^{|X||Y|}YX$$

on the common invariant domain, after expanding X and Y into ordered products of homogeneous smeared fields. The isotony statement follows directly from support inclusion: every test function supported in $\mathcal{O}_1$ is also supported in $\mathcal{O}_2$. For locality, every generating smeared field from $\mathcal{P}(\mathcal{O}_1)$ has spacelike separated support from every generating smeared field from $\mathcal{P}(\mathcal{O}_2)$. Microcausality therefore permits each generator in X to be moved past each generator in Y. Each exchange of homogeneous factors contributes the sign $(-1)^{|A||B|}$. Multiplying these signs over all crossings gives the total Koszul sign $(-1)^{|X||Y|}$, where $|X|$ and $|Y|$ are the sums of the parities of their constituent fields modulo 2.

The observable algebra $\mathcal{A}(\mathcal{O}) \subset \mathcal{B}(\mathcal{H})$ is the bounded local algebra used in the opening framework. The relation between $\mathcal{P}(\mathcal{O})$ and $\mathcal{A}(\mathcal{O})$ depends on the analytic framework. For free scalar fields, Weyl operators are bounded functions of smeared fields and generate a local algebra. In Wightman language one works directly with operator-valued distributions on a common domain. In algebraic language one may take the bounded local net as primary and use fields as coordinates, charged fields, or affiliated unbounded operators when available. The precise comparison is the Wightman-to-net datum of Definition 3.12, strengthened to a complete observable coordinatization by Definition 3.13 when the generated bounded coordinate net is identified with the represented observable net. Thus a statement about polynomial fields, a statement about bounded functional calculus, and a statement about the neutral observable net are separate pieces of structure unless a construction proves that they coincide.

The monograph will use whichever of these frameworks is required by the construction at hand, with the framework declared before theorem-level claims are made.

6.6 Wightman Functions

Definition 6.12 (Wightman distribution of a field multiplet)**.** Let $\widehat{\Phi}_{A_1}, \ldots, \widehat{\Phi}_{A_n}$ be fields whose products on the vacuum vector are defined on the common domain. The n-point Wightman distribution is the multilinear functional

$$W_{A_1 \cdots A_n}(f_1, \ldots, f_n) = \langle \Omega | \widehat{\Phi}_{A_1}(f_1) \cdots \widehat{\Phi}_{A_n}(f_n) | \Omega \rangle.$$

Point notation writes this distribution as

$$W_{A_1 \cdots A_n}(x_1, \ldots, x_n) = \langle \Omega | \widehat{\Phi}_{A_1}(x_1) \cdots \widehat{\Phi}_{A_n}(x_n) | \Omega \rangle.$$

Field axioms inherited by Wightman distributions. Assume the common-domain and continuity hypotheses of Remark 6.4. If the vacuum is Poincare invariant and the field multiplet is covariant, inserting $U(g)U(g)^{-1}$ between neighboring smeared factors and using vacuum invariance gives the covariance law displayed in the covariance paragraph above. If the field multiplet satisfies microcausality, applying the vanishing graded commutator to adjacent factors with spacelike separated supports gives the graded exchange relation displayed above. The continuity of the smeared matrix elements then extends these identities from tensor products of test functions to the corresponding Wightman distributions.

Poincare covariance of the fields gives covariance of the Wightman distributions. Microcausality gives graded exchange relations when adjacent insertions have spacelike separated supports. The spectrum condition will give the analytic and spectral properties used in the Källén–Lehmann representation.

6.7 Position In The Construction

The data introduced in this chapter are local field data: test functions, smeared operators, covariance, and microcausality. They prepare the next stage, where Lagrangian and path-integral methods are introduced first as constructions of Green functions. Spectral representation will then connect two-point functions to particle content before scattering theory is defined.

Chapter 7

Hamiltonian Evolution and Time-Sliced Path Integrals

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

7.1 Classical Lagrangian and Hamiltonian Data

Definition 7.1 (Regular classical Lagrangian datum)**.** Let Q be a smooth d-dimensional configuration manifold. Local coordinates on Q are denoted by q^a, $a = 1, \ldots, d$. A tangent vector is written $(q, \dot{q}) \in TQ$, and a cotangent vector is written $(q, p) \in T^*Q$, with $p = p_a \, \mathrm{d}q^a$. A classical Lagrangian is a function

$$L : TQ \to \mathbb{R}.$$

For a path $q : [t_i, t_f] \to Q$, the action is

$$S[q] = \int_{t_i}^{t_f} L(q(t), \dot{q}(t)) \, \mathrm{d}t.$$

The variational principle with fixed endpoints gives the Euler–Lagrange equations

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}^a} = \frac{\partial L}{\partial q^a}.$$

Thus a Lagrangian description uses the pair $(q, \dot{q})$ as local Cauchy data for the classical second-order equation.

Assume now that the fiber derivative of L,

$$(q, \dot{q}) \longmapsto \left(q, p_a = \frac{\partial L}{\partial \dot{q}^a} \right),$$

is invertible on the region under consideration. The Hamiltonian associated to this regular Lagrangian is the Legendre transform

$$H(q, p) = p_a \dot{q}^a - L(q, \dot{q}) \quad \text{with } \dot{q} = \dot{q}(q, p).$$

The canonical Poisson bracket on T^*Q is

$$\{f, g\}_{\mathrm{P}} = \frac{\partial f}{\partial q^a} \frac{\partial g}{\partial p_a} - \frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q^a},$$

where repeated indices are summed. Hamiltonian time evolution is

$$\dot{f} = \{f, H\}_{\mathrm{P}},$$

and, in particular,

$$\dot{q}^a = \frac{\partial H}{\partial p_a}, \qquad \dot{p}_a = -\frac{\partial H}{\partial q^a}.$$

The Lagrangian and Hamiltonian descriptions are equivalent only after the regularity hypothesis above has been imposed; without it, constraints must be included as part of the classical data.

7.2 Canonical Quantum-Mechanical Data

Definition 7.2 (Schrodinger representation datum)**.** Let d be the number of degrees of freedom and let $\mathcal{Q} = \mathbb{R}^d$ be the configuration space. The Hilbert space is

$$\mathcal{H}_{\mathcal{Q}} = L^2(\mathcal{Q}, \mathrm{d}^d q).$$

The coordinate operators $\widehat{q}^a$, $a = 1, \ldots, d$, act by multiplication. The momentum operators are

$$\widehat{p}_a = -\mathrm{i}\hbar \frac{\partial}{\partial q^a}$$

first on the common invariant test domain $\mathcal{S}(\mathbb{R}^d) \subset \mathcal{H}_{\mathcal{Q}}$, and satisfy

$$[\widehat{q}^a, \widehat{p}_b] = \mathrm{i}\hbar\, \delta^a_b.$$

Closures or self-adjoint extensions are specified when a Hamiltonian is chosen. The Schwartz-domain convention is enough for the distributional kernel manipulations below; the operator limit is then stated at the Hilbert-space level.

Let $\widehat{H}$ be a self-adjoint Hamiltonian on $\mathcal{H}_{\mathcal{Q}}$. This is an analytic hypothesis on the chosen Hilbert-space realization of the formal classical Hamiltonian. By Stone's theorem, self-adjointness of $\widehat{H}$ is exactly the condition that the following formula defines a strongly continuous unitary one-parameter group with generator $-\mathrm{i}\widehat{H}/\hbar$. The time-evolution operator over time $T \in \mathbb{R}$ is

$$U(T) = \exp\left(-\frac{\mathrm{i}}{\hbar} T \widehat{H} \right).$$

In the Schrödinger picture, a vector evolves as

$$|\psi\rangle \longmapsto U(t)|\psi\rangle.$$

In the Heisenberg picture, an operator $\widehat{O}$ evolves as

$$\widehat{O}(t) = U(t)^{-1}\widehat{O}\,U(t), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\widehat{O}(t) = \frac{\mathrm{i}}{\hbar}[\widehat{H}, \widehat{O}(t)]$$

on vectors for which the commutator expression is defined. The path integral constructed below gives a regulated representation of matrix elements of this Hamiltonian time evolution.

Definition 7.3 (Position-momentum rigging and kernel notation)**.** The position and momentum generalized eigenstates are denoted by $|q\rangle$ and $|p\rangle$. They are elements of $\mathcal{S}'(\mathbb{R}^d)$ in the rigged Hilbert space

$$\mathcal{S}(\mathbb{R}^d) \subset L^2(\mathbb{R}^d, \mathrm{d}^d q) \subset \mathcal{S}'(\mathbb{R}^d),$$

with the $\hbar$-dependent Fourier convention

$$\widetilde{\psi}(p) = \int_{\mathbb{R}^d} \mathrm{d}^d q\, \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}p_a q^a}\psi(q).$$

They are normalized by

$$\langle q \mid p\rangle = \mathrm{e}^{\frac{\mathrm{i}}{\hbar}p_a q^a},$$

and used through the resolutions

$$\int_{\mathbb{R}^d} \mathrm{d}^d q\, |q\rangle\langle q| = 1, \qquad \int_{\mathbb{R}^d} \frac{\mathrm{d}^d p}{(2\pi\hbar)^d}\, |p\rangle\langle p| = 1.$$

These are weak identities: after inserting test vectors on the left and right they become the ordinary Fourier inversion formula. The transition amplitude is the distributional kernel

$$K(q_f, T; q_i, 0) = \langle q_f|U(T)|q_i\rangle.$$

The continuum Lagrangian path-integral notation for this same kernel is

$$\langle q_f|U(T)|q_i\rangle = \int_{q(0)=q_i}^{q(T)=q_f} [Dq]\, \exp\left(\frac{\mathrm{i}}{\hbar}S[q]\right).$$

This displayed equation is a compact name for the regulated construction below. The bracketed symbols $[Dq]$ and, in phase-space formulas, $[Dp]$ are determined by the regulator, limiting procedure, operator symbol, and boundary conditions; at fixed regulator the object is a finite-dimensional oscillatory integral. Unbracketed products such as $\mathrm{d}^d q_n$ and $\mathrm{d}^d p_n$ are reserved for the finite-dimensional measures that define the regulated expression.

7.3 Finite Time Slicing

Finite phase-space time-sliced construction. Fix the data of Definition 7.3 and an operator symbol $h(q,p)$ for $\widehat{H}$ in a specified ordering convention. The finite-N phase-space expression K_N is the regulated oscillatory integral obtained by inserting position and momentum resolutions in the product formula for $U(T)$. The theorem-level analytic input enters later, through the operator convergence theorem for the limiting procedure. We record the construction explicitly. Choose an integer $N \geq 1$, set $\epsilon = T/N$, and define

$$t_n = n\epsilon, \qquad n = 0, \ldots, N,$$

with $q_0 = q_i$ and $q_N = q_f$. Inserting $N-1$ position resolutions and

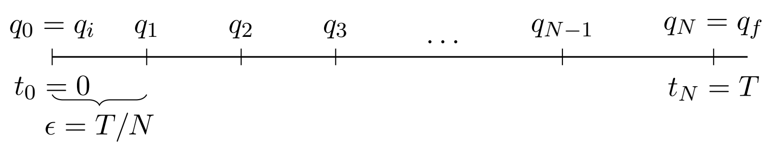
$$U(T) = \left[\exp\left(-\frac{\mathrm{i}}{\hbar}\epsilon\widehat{H}\right)\right]^N$$

gives first the position-space finite product

$$K(q_f, T; q_i, 0) = \int \prod_{n=1}^{N-1} \mathrm{d}^d q_n \prod_{n=0}^{N-1} \langle q_{n+1}| \exp\left(-\frac{\mathrm{i}}{\hbar}\epsilon\widehat{H}\right) |q_n\rangle.$$

Inserting N momentum resolutions gives

$$\begin{aligned} K(q_f, T; q_i, 0) = \int \prod_{n=1}^{N-1} \mathrm{d}^d q_n \prod_{n=0}^{N-1} \frac{\mathrm{d}^d p_n}{(2\pi\hbar)^d} \\ \times \prod_{n=0}^{N-1} \langle q_{n+1}|p_n\rangle\langle p_n| \exp\left(-\frac{\mathrm{i}}{\hbar}\epsilon\widehat{H}\right) |q_n\rangle. \end{aligned}$$



Suppose a symbol $h(q,p)$ for $\widehat{H}$ has been fixed by an operator ordering convention, for example by

$$\langle p|\widehat{H}|q\rangle = h(q,p)\langle p \mid q\rangle$$

on the chosen domain. Replacing each short-time matrix element by the first terms of its symbol expansion gives

$$\langle q_{n+1}|p_n\rangle\langle p_n|\mathrm{e}^{-\mathrm{i}\epsilon\widehat{H}/\hbar}|q_n\rangle = \exp\left[\frac{\mathrm{i}}{\hbar}\left(p_{n,a}(q^a_{n+1} - q^a_n) - \epsilon h(q_n, p_n) + O(\epsilon^2) + O(\hbar\epsilon)\right)\right]$$

for this convention, with the error understood in the regulated operator-symbol expansion rather than as a pointwise statement about an undefined continuum integral. The associated finite-N phase-space approximant is

$$K_N(q_f, T; q_i, 0) = \int \prod_{n=1}^{N-1} \mathrm{d}^d q_n \prod_{n=0}^{N-1} \frac{\mathrm{d}^d p_n}{(2\pi\hbar)^d} \exp\left(\frac{\mathrm{i}}{\hbar} S_N[p,q]\right),$$

where the discrete phase-space action is

$$S_N[p,q] = \sum_{n=0}^{N-1} \left[p_{n,a}(q^a_{n+1} - q^a_n) - \epsilon\, h(q_n, p_n)\right].$$

The continuum notation

$$\int [Dq]\,[Dp]\,\exp\left[\frac{\mathrm{i}}{\hbar}\int_0^T \big(p_a\dot q^a - h(q,p)\big)\mathrm{d}t\right]$$

denotes this regulated construction together with the specified limiting procedure. In the present chapter, the finite-N expression is the defined object.

7.4 Trotter Limits, Wiener Measure, and Kernel Convergence

Theorem 7.4 (Trotter product formula in the form used here)**.** *The passage from the finite product to a continuum path integral is a convergence statement about operators before it is a statement about kernels. Let $\widehat{A}$ and $\widehat{B}$ be self-adjoint operators on $\mathcal{H}_Q$. For the real-time statement, assume that* $\operatorname{Dom}\widehat{A}\cap\operatorname{Dom}\widehat{B}$ *is dense and that the algebraic sum*

$$\widehat{A}+\widehat{B}$$

on this domain is essentially self-adjoint, with self-adjoint closure $\widehat{H}$. Then

$$\exp\left(-\frac{\mathrm{i}T}{\hbar}\widehat{H}\right) = \operatorname*{s-lim}_{N\to\infty}\left[\exp\left(-\frac{\mathrm{i}\epsilon}{\hbar}\widehat{A}\right)\exp\left(-\frac{\mathrm{i}\epsilon}{\hbar}\widehat{B}\right)\right]^N, \qquad \epsilon=\frac{T}{N},$$

where the limit is in the strong operator topology on $\mathcal{H}_Q$. For the Euclidean statement, assume instead that $\widehat{A}$ and $\widehat{B}$ are lower-semibounded and that their closed quadratic forms $\mathfrak{a}$ and $\mathfrak{b}$ have dense common form domain

$$\mathcal{Q}_{\mathfrak{h}} = \operatorname{Dom}\mathfrak{a}\cap\operatorname{Dom}\mathfrak{b},$$

with $\mathfrak{h}=\mathfrak{a}+\mathfrak{b}$ closed and lower-semibounded on $\mathcal{Q}_{\mathfrak{h}}$. If $\widehat{H}=\widehat{A}\dotplus\widehat{B}$ is the self-adjoint form sum associated to $\mathfrak{h}$, then

$$\exp\left(-\frac{\tau}{\hbar}\widehat{H}\right) = \operatorname*{s-lim}_{N\to\infty}\left[\exp\left(-\frac{\delta}{\hbar}\widehat{A}\right)\exp\left(-\frac{\delta}{\hbar}\widehat{B}\right)\right]^N, \qquad \delta=\frac{\tau}{N},$$

again in the strong operator topology.

Proof. The product formula is a statement in $\mathcal{H}_Q$, not a pointwise identity between formal path-integral densities. In the bounded case the mechanism can be seen without domain issues. If A and B are bounded self-adjoint operators and t is fixed, Taylor expansion in operator norm gives, uniformly for $0\le\delta\le t$,

$$\mathrm{e}^{-\delta A}\mathrm{e}^{-\delta B} = 1-\delta(A+B)+O(\delta^2), \qquad \mathrm{e}^{-\delta(A+B)} = 1-\delta(A+B)+O(\delta^2).$$

Hence

$$\big\|\mathrm{e}^{-\delta A}\mathrm{e}^{-\delta B}-\mathrm{e}^{-\delta(A+B)}\big\| \le C_t\delta^2.$$

The telescoping identity

$$X^N - Y^N = \sum_{j=0}^{N-1} X^{N-1-j}(X-Y)Y^j$$

with $X = \mathrm{e}^{-tA/N}\mathrm{e}^{-tB/N}$ and $Y = \mathrm{e}^{-t(A+B)/N}$ then gives

$$\|X^N - Y^N\| \le N\,\|X - Y\| \max_{j,k\le N} \|X^j\|\,\|Y^k\| = O(N^{-1}).$$

Thus the noncommutativity of A and B first appears in terms of order δ^2, and the accumulated error over $N = t/\delta$ time slices vanishes.

For Schrödinger Hamiltonians $T + V$ the same calculation cannot be read as a proof, because T and V are unbounded and may not have a useful operator-domain intersection. The semibounded Euclidean statement is instead form-theoretic. After adding constants to make A and B nonnegative if necessary–which only multiplies the product formula by the corresponding scalar exponential factors–one starts from closed semibounded quadratic forms

$$\mathfrak{a}[\psi] = \langle A^{1/2}\psi, A^{1/2}\psi\rangle, \qquad \mathfrak{b}[\psi] = \langle B^{1/2}\psi, B^{1/2}\psi\rangle,$$

with common form domain $\operatorname{Dom}\mathfrak{a} \cap \operatorname{Dom}\mathfrak{b}$. If $\mathfrak{h} = \mathfrak{a} + \mathfrak{b}$ is closed and semibounded, the representation theorem for closed forms defines the self-adjoint form sum $H = A \dotplus B$ by

$$\mathfrak{h}(\varphi, \psi) = \langle \varphi, H\psi\rangle, \qquad \psi \in \operatorname{Dom} H, \quad \varphi \in \operatorname{Dom}\mathfrak{h}.$$

The Trotter–Kato form theorem proves that the alternating bounded contractions

$$\left(\mathrm{e}^{-tA/N}\mathrm{e}^{-tB/N}\right)^N$$

converge strongly to e^{-tH}. Its proof replaces the norm Taylor argument by resolvent and form convergence: the resolvent of H is characterized by the variational equation

$$\mathfrak{h}(\varphi, u) + \lambda\langle\varphi, u\rangle = \langle\varphi, f\rangle, \qquad \varphi \in \operatorname{Dom}\mathfrak{h},$$

and the product approximants have resolvents whose variational problems converge to this one on the common form domain. Strong resolvent convergence then gives strong semigroup convergence by the spectral theorem.

For the real-time statement, define

$$F(s) = \exp\left(-\frac{\mathrm{i}s}{\hbar}A\right)\exp\left(-\frac{\mathrm{i}s}{\hbar}B\right).$$

On the dense algebraic core $\operatorname{Dom} A \cap \operatorname{Dom} B$,

$$\lim_{s\to 0}\frac{F(s)\psi - \psi}{s} = -\frac{\mathrm{i}}{\hbar}(A + B)\psi.$$

The factors are unitary, hence $\|F(s)\| \le 1$. Chernoff's product theorem then identifies the strong limit of $F(T/N)^N$ with the unitary group whose generator is the closure of $-\mathrm{i}(A+B)/\hbar$, namely $\exp(-\mathrm{i}TH/\hbar)$. This real-time conclusion is therefore a theorem about self-adjoint generators; it should not be inferred by simply substituting $t \mapsto it$ into an uncontrolled kernel expression. $\square$

Definition 7.5 (Kato Schrodinger Euclidean datum)**.** For Schrödinger operators the words "suppose the hypotheses hold" can be replaced by concrete analytic criteria. Work on $L^2(\mathbb{R}^d, \mathrm{d}^d q)$, let

$$\widehat{p}_a = -\mathrm{i}\hbar\partial_a, \qquad T = \frac{1}{2m}\widehat{p}^2 = -\frac{\hbar^2}{2m}\Delta,$$

and let $V(\widehat{q})$ be multiplication by a real measurable function V. For a measurable function W, define the Kato class $\mathcal{K}_d$ by

$$\lim_{t\downarrow 0}\sup_{x\in\mathbb{R}^d}\int_0^t \big(\mathrm{e}^{s\Delta}|W|\big)(x)\,\mathrm{d}s = 0,$$

where $\mathrm{e}^{s\Delta}$ is the heat semigroup on $\mathbb{R}^d$. Define the local Kato class by requiring $\chi_K W \in \mathcal{K}_d$ for every compact $K \subset \mathbb{R}^d$. A standard sufficient Euclidean hypothesis is the Kato decomposition

$$V = V_+ - V_-, \qquad V_\pm \geq 0, \qquad V_+ \in \mathcal{K}_{d,\mathrm{loc}}, \qquad V_- \in \mathcal{K}_d,$$

together with a lower bound when the literal multiplication factors $\mathrm{e}^{-\delta V/\hbar}$ are used as bounded operators. Under these hypotheses the negative part is infinitesimally form-bounded relative to the free kinetic form: for every $a > 0$ there is $b_a < \infty$ such that

$$\int_{\mathbb{R}^d} V_-(q)|\psi(q)|^2\,\mathrm{d}^d q \leq a\int_{\mathbb{R}^d}|\nabla\psi|^2\,\mathrm{d}^d q + b_a\int_{\mathbb{R}^d}|\psi(q)|^2\,\mathrm{d}^d q.$$

Therefore the KLMN theorem applies. The quadratic form

$$Q[\psi] = \frac{\hbar^2}{2m}\int_{\mathbb{R}^d}|\nabla\psi|^2\,\mathrm{d}^d q + \int_{\mathbb{R}^d} V(q)|\psi(q)|^2\,\mathrm{d}^d q$$

on $H^1(\mathbb{R}^d)\cap L^2(V_+(q)\mathrm{d}^d q)$ is closed and semibounded, and it defines the Friedrichs Hamiltonian $H = T + V$. The Euclidean semigroup has the Trotter–Kato product formula, and its kernel is represented by the Feynman–Kac construction whenever the stated Kato hypotheses hold.

Theorem 7.6 (Wiener measure and the Feynman–Kac formula)**.** *Let $V : \mathbb{R}^d \to \mathbb{R}$ be real, locally Kato, and bounded from below. Let H be the Friedrichs Hamiltonian associated to*

$$Q[\psi] = \frac{\hbar^2}{2m}\int_{\mathbb{R}^d}|\nabla\psi|^2\,\mathrm{d}^d q + \int_{\mathbb{R}^d} V(q)|\psi(q)|^2\,\mathrm{d}^d q.$$

For each $x \in \mathbb{R}^d$ and $\tau > 0$, there is a countably additive probability measure $\mathbb{W}_x^\tau$ on

$$C([0,\tau],\mathbb{R}^d)$$

whose finite-dimensional distributions are

$$\begin{aligned}&\int F(\omega(t_1),\dots,\omega(t_n))\,\mathrm{d}\mathbb{W}_x^\tau(\omega)\\ &\quad= \int_{\mathbb{R}^{dn}} F(x_1,\dots,x_n)\prod_{j=1}^n k^0_{t_j - t_{j-1}}(x_{j-1},x_j)\,\mathrm{d}^d x_1\cdots\mathrm{d}^d x_n,\end{aligned}$$

for $0 < t_1 < \cdots < t_n \leq \tau$, with $t_0 = 0$, $x_0 = x$, and

$$k_s^0(x,y) = \Big(\frac{m}{2\pi\hbar s}\Big)^{d/2}\exp\left[-\frac{m|x-y|^2}{2\hbar s}\right].$$

This is Wiener measure with diffusion constant $\hbar/m$. For bounded Borel f, and then by extension for $f \in L^2(\mathbb{R}^d)$,

$$\big(\mathrm{e}^{-\tau H/\hbar}f\big)(x) = \int_{C([0,\tau],\mathbb{R}^d)}\exp\left[-\frac{1}{\hbar}\int_0^\tau V(\omega(s))\,\mathrm{d}s\right] f(\omega(\tau))\,\mathrm{d}\mathbb{W}_x^\tau(\omega)$$

for almost every x. Disintegrating $\mathbb{W}_x^\tau$ with respect to the endpoint $\omega(\tau) = y$ gives the Brownian-bridge probability measure $\mathbb{W}_{x,y}^\tau$, and the Euclidean transition kernel is

$$K_E(y, \tau; x, 0) = k_\tau^0(x, y) \int \exp\left[-\frac{1}{\hbar}\int_0^\tau V(\omega(s))\,\mathrm{d}s\right] \mathrm{d}\mathbb{W}_{x,y}^\tau(\omega).$$

Proof. The functions $k_s^0(x, y)$ are nonnegative, integrate to one in y, and satisfy the convolution identity

$$\int_{\mathbb{R}^d} k_s^0(x, z) k_t^0(z, y)\,\mathrm{d}^d z = k_{s+t}^0(x, y).$$

The displayed finite-dimensional distributions are therefore consistent. Kolmogorov extension gives a probability measure on paths with values in $(\mathbb{R}^d)^{[0,\tau]}$. The Gaussian increment moments obey

$$\mathbb{E}_{\mathbb{W}_x^\tau}|\omega(t) - \omega(s)|^{2r} = C_{d,r}\left(\frac{\hbar}{m}|t - s|\right)^r,$$

and Kolmogorov's continuity criterion gives a version supported on continuous paths. This proves the existence of the Wiener measure on $C([0, \tau], \mathbb{R}^d)$.

First assume that V is bounded and continuous. Let M_g denote multiplication by g, and let

$$L = \frac{\hbar}{2m}\Delta.$$

The heat semigroup e^{sL} has kernel k_s^0. Iterating the Markov property gives

$$\begin{aligned}&\left(\mathrm{e}^{\delta L} M_{\exp(-\delta V/\hbar)}\right)^N f(x)\\ &\quad = \mathbb{E}_x\left[\exp\left(-\frac{\delta}{\hbar}\sum_{j=1}^N V(B_{j\delta})\right) f(B_{N\delta})\right],\end{aligned}$$

where B is the coordinate process under $\mathbb{W}_x^\tau$ and $\delta = \tau/N$. Since Brownian paths are continuous and V is bounded continuous, the Riemann sums converge almost surely to $\int_0^\tau V(B_s)\,\mathrm{d}s$, and dominated convergence gives the Wiener expectation. The Trotter product formula identifies the same operator limit with $\mathrm{e}^{-\tau H/\hbar}$, because

$$-\frac{H}{\hbar} = L - \frac{1}{\hbar}V$$

as a form sum.

If V is continuous and bounded below, add a constant so that $V \geq 0$, prove the formula for $V_M = \min(V, M)$, and then let $M \to \infty$. The path weights decrease pointwise to the displayed weight, and the quadratic forms for $T + V_M$ increase to the quadratic form for $T + V$; monotone convergence of closed forms gives strong convergence of the semigroups. The locally Kato bounded-below case is obtained by approximating in the local Kato topology and using the Kato small-time estimate for the negative part. Finally, the bridge formula is the conditional-distribution form of the preceding identity with respect to the endpoint density $k_\tau^0(x, y)$. □

Thus in ordinary Euclidean quantum mechanics the continuum path integral is a genuine Wiener expectation on continuous paths, multiplied by the potential weight, under the hypotheses of Theorem 7.6. The time-sliced Lebesgue products are the approximating construction that converges to this measure-theoretic object. The Trotter product formula is still important, but its role is to identify the operator semigroup represented by the Wiener integral; it is weaker than, and should not be substituted for, the measure-level Feynman–Kac statement. This theorem governs bosonic Schrödinger quantum mechanics, and by extension supplies an important model for constructive scalar field theories when a positive Euclidean measure exists. The general path-integral framework is a typed assignment of mathematical objects by a regulator and limiting prescription. Fermionic fields use Berezin integration over Grassmann variables rather than countably additive Borel measures; gauge theories require quotient, gauge-fixing, BRST/BV, or lattice data in addition to any measure on fields; and topological terms such as a four-dimensional Yang–Mills θ-angle generally make the Euclidean weight complex. Every use of path-integral notation must therefore state the mathematical object being used: a measure, an oscillatory integral, a Berezin functional, a gauge-fixed/BV construction, a lattice regularization, a holonomy construction, or a formal perturbative expansion.

Theorem 7.7 (Faris–Lavine criterion)**.** *Essential self-adjointness on the algebraic core $C_c^\infty(\mathbb{R}^d)$ is a separate issue. One convenient sufficient theorem is the Faris–Lavine criterion, often cited as Reed–Simon Theorem X.41: if V is real, C^∞, and satisfies*

$$V(q) \geq -C(1+|q|^2)$$

for some $C < \infty$, then $-\Delta + V$ is essentially self-adjoint on $C_c^\infty(\mathbb{R}^d)$. Every smooth potential bounded below satisfies this growth-from-below hypothesis, and polynomial oscillator potentials with positive leading part do as well. These cases have a unique self-adjoint closure from the natural differential operator core, and the Trotter formula applies to that closure.

Proof. The criterion is a statement about the closure of a symmetric differential operator determined on the algebraic core $C_c^\infty(\mathbb{R}^d)$. Write $p_j = -\mathrm{i}\partial_j$, $p^2 = -\Delta$, and first reduce by rescaling and adding a constant to the model lower bound

$$V(x) \geq -|x|^2.$$

Let

$$A_0 = p^2 + V(x), \qquad N_0 = p^2 + V(x) + 2|x|^2$$

on $C_c^\infty(\mathbb{R}^d)$. Since $V+2|x|^2 \geq |x|^2 \geq 0$, the Kato theorem for nonnegative locally square-integrable potentials gives a positive self-adjoint closure N of N_0. The comparison with N is the point of the proof. On $C_c^\infty(\mathbb{R}^d)$,

$$N_0 - A_0 = 2|x|^2 \geq 0, \qquad N_0 + A_0 = 2p^2 + 2(V+|x|^2) \geq 0$$

as quadratic forms, hence

$$-N_0 \leq A_0 \leq N_0$$

in form order. The oscillator estimate, obtained by integration by parts from $p^2 + |x|^2$,

$$\||x|^2\psi\| \leq C\big(\|N_0\psi\| + \|\psi\|\big), \qquad \psi \in C_c^\infty(\mathbb{R}^d),$$

then implies that the N-graph norm controls A_0. Thus A_0 extends continuously from the N-graph domain, and one may apply Nelson's commutator theorem with comparison operator N.

It remains to check the commutator bound required by that theorem. The only noncommuting part is the added oscillator term:

$$\mathrm{i}[N_0, A_0] = \mathrm{i}[2|x|^2, p^2].$$

Using $[x_j, p_k] = \mathrm{i}\delta_{jk}$, one obtains, on the common algebraic core,

$$\mathrm{i}[N_0, A_0] = -4\sum_{j=1}^{d}(x_j p_j + p_j x_j).$$

The form bound follows from completing the square:

$$(p_j \pm x_j)^*(p_j \pm x_j) = p_j^2 + x_j^2 \pm (p_j x_j + x_j p_j) \geq 0,$$

and therefore

$$\pm\sum_{j=1}^{d}(x_j p_j + p_j x_j) \leq p^2 + |x|^2 \leq N_0$$

as quadratic forms. Hence

$$\big|\langle\psi, \mathrm{i}[N_0, A_0]\psi\rangle\big| \leq 4\,\langle\psi, N_0\psi\rangle.$$

Nelson's theorem says that a symmetric operator A_0 controlled in graph norm by a positive self-adjoint N, with the commutator bounded by $\langle\psi, N\psi\rangle$, is essentially self-adjoint on every N-core contained in its domain. Its proof uses the bounded regularized operators

$$A_\epsilon = (1 + \epsilon N)^{-1} A_0 (1 + \epsilon N)^{-1}$$

and the commutator inequality to obtain a uniform Gronwall bound on the $N^{1/2}$-energy of $\mathrm{e}^{\mathrm{i}tA_\epsilon}\psi$; after passing $\epsilon \downarrow 0$, the deficiency equations $(A_0^* \mp \mathrm{i})\eta = 0$ admit no nonzero L^2 solutions. This is the operator-theoretic reason that the quadratic lower bound prevents a boundary condition at infinity from being part of the Hamiltonian data. □

Example 7.8 (A limit-circle failure of formal Hamiltonian data)**.** The lower-growth hypothesis in the Faris–Lavine criterion is substantive. In one dimension, the symmetric differential operator

$$-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^3$$

on $\mathcal{S}(\mathbb{R})$ is a concrete failure mode. At $q \to -\infty$, the deficiency equations

$$\left(-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^3\right)\psi_\pm = \pm\mathrm{i}\,\psi_\pm$$

have two local solutions whose standard Liouville–Green asymptotics have leading behavior

$$\psi_\pm(q) \sim |q|^{-3/4}\exp\left(\pm\mathrm{i}\,\frac{2}{5}|q|^{5/2}\right).$$

Both branches are square-integrable at that end because the squared amplitude is asymptotic to $|q|^{-3/2}$ and $\int^\infty r^{-3/2}\mathrm{d}r < \infty$. By the Weyl limit-point/limit-circle classification theorem for one-dimensional Sturm–Liouville operators, this places the endpoint $q = -\infty$ in the limit-circle case. The symmetric operator is therefore not essentially self-adjoint on the Schwartz core: a self-adjoint Hamiltonian requires an additional boundary condition at this singular endpoint at infinity, and different self-adjoint extensions give different unitary groups. Thus the formal expression $\widehat{p}^2/2m + V(\widehat{q})$ does not by itself define the operator appearing in $\exp(-\mathrm{i}T\widehat{H}/\hbar)$.

For the standard Hamiltonian

$$\widehat{H} = \frac{1}{2m}\widehat{p}^2 + V(\widehat{q}),$$

the product formula says that the operator obtained by alternating free propagation and multiplication by the potential converges to the exact evolution operator. Inserting position resolutions between the factors gives a finite-dimensional integral at each N. Thus the time-sliced path integral is a representation of a product formula.

The distributional kernel

$$\langle q_f | \exp\left(-\frac{\mathrm{i}T}{\hbar}\widehat{H}\right) |q_i\rangle$$

requires an additional topology. The useful topologies include convergence after smearing q_i and q_f with test functions, convergence after analytic continuation to Euclidean time, and pointwise kernel convergence under stronger regularity assumptions on V. Therefore the continuum path-integral notation in this monograph always carries one of the following meanings: a finite regulator, a strong operator limit, a smeared distributional limit, a Euclidean measure or semigroup construction, or a perturbative asymptotic expansion.

7.5 Operator Ordering and Local Counterterms

Definition 7.9 (Regulated phase-space path-integral datum)**.** The function $h(q,p)$ depends on the ordering convention used to represent $\widehat{H}$. For example, using left endpoints q_n in the short-time matrix element and using right endpoints q_{n+1} define different symbols for the same operator. Their difference is of order $\hbar$ in the continuum action and appears as a local counterterm in the time-sliced description.

At fixed N, two representative choices are

$$\cdots\langle q_{n+1}|p_n\rangle\langle p_n|\mathrm{e}^{-\mathrm{i}\epsilon\widehat{H}/\hbar}|q_n\rangle\cdots, \qquad \langle p_n|\widehat{H}|q_n\rangle = h(q_n,p_n)\langle p_n \mid q_n\rangle,$$

and

$$\cdots\langle p_{n+1}|q_{n+1}\rangle\langle q_{n+1}|\mathrm{e}^{-\mathrm{i}\epsilon\widehat{H}/\hbar}|p_n\rangle\cdots, \qquad \langle q_n|\widehat{H}|p_n\rangle = h'(q_n,p_n)\langle q_n \mid p_n\rangle.$$

They discretize the kinetic term respectively as

$$\sum_{n=0}^{N-1} p_{n,a}(q^a_{n+1} - q^a_n), \qquad \sum_{n=0}^{N-1} p_{n+1,a}(q^a_{n+1} - q^a_n),$$

after relabelling endpoints where necessary. The two symbols h and h' represent the same operator only with their specified ordering conventions, and they can differ by terms of order $\hbar$. Such terms survive in the exponent as local regulator-dependent additions to the effective Lagrangian.

Thus the data of the regulated path integral are

$$(\text{partition},\ \text{measure},\ \text{operator symbol},\ \text{boundary conditions}).$$

A formula involving $[Dq]\,[Dp]$ is meaningful as notation for a regulated construction once these data have been declared.

7.6 Lagrangian Form for Quadratic Momentum Dependence

Gaussian elimination of quadratic momenta. Assume the Hamiltonian symbol has the form

$$h(q,p) = \frac{1}{2}\,G^{ab}(q)p_a p_b + V(q),$$

where $G^{ab}(q)$ is a positive-definite symmetric matrix and

$$G^{ab}(q)G_{bc}(q) = \delta^a_c.$$

At fixed time slice, the momentum integral is Gaussian:

$$\int \frac{\mathrm{d}^d p}{(2\pi\hbar)^d} \exp\left[\frac{\mathrm{i}}{\hbar}\left(p_a\Delta q^a - \frac{\epsilon}{2}G^{ab}(q)p_a p_b\right)\right] = \frac{\sqrt{\det G_{ab}(q)}}{(2\pi\mathrm{i}\hbar\epsilon)^{d/2}} \exp\left[\frac{\mathrm{i}\epsilon}{\hbar}\frac{1}{2}G_{ab}(q)v^a v^b\right],$$

where

$$\Delta q^a = q^a_{n+1} - q^a_n, \qquad v^a = \frac{\Delta q^a}{\epsilon}.$$

The Lagrangian function obtained by Legendre transform is

$$L(q,\dot{q}) = \frac{1}{2}G_{ab}(q)\dot{q}^a\dot{q}^b - V(q).$$

Conversely, if a classical Lagrangian in generalized coordinates has this metric form, as in the rigid-body configuration space example, then the corresponding quantum Hamiltonian symbol is

$$h(q,p) = \frac{1}{2}G^{ab}(q)p_a p_b + V(q) + O(\hbar)$$

until an ordering convention and measure convention have been fixed. The $O(\hbar)$ term records the same local regulator dependence described above. The finite-N configuration-space expression is therefore

$$\begin{aligned} K_N(q_f,T;q_i,0) = \int \prod_{n=1}^{N-1} \mathrm{d}^d q_n \prod_{n=0}^{N-1} \frac{\sqrt{\det G_{ab}(q_n)}}{(2\pi\mathrm{i}\hbar\epsilon)^{d/2}} \\ \times \exp\left[\frac{\mathrm{i}}{\hbar}\sum_{n=0}^{N-1} \epsilon\, L\left(q_n, \frac{q_{n+1}-q_n}{\epsilon}\right)\right]. \end{aligned}$$

For $G_{ab} = m\delta_{ab}$, the Lagrangian in this finite-dimensional time-sliced kernel is

$$L(q,\dot{q}) = \frac{m}{2}\delta_{ab}\dot{q}^a\dot{q}^b - V(q).$$

The calculation is the following finite-dimensional Gaussian completion. At fixed q_n, one rewrites the exponent as

$$p_a\Delta q^a - \frac{\epsilon}{2}G^{ab}p_a p_b = -\frac{\epsilon}{2}G^{ab}\left(p_a - \frac{1}{\epsilon}G_{ac}\Delta q^c\right)\left(p_b - \frac{1}{\epsilon}G_{bd}\Delta q^d\right) + \frac{1}{2\epsilon}G_{ab}\Delta q^a\Delta q^b.$$

The oscillatory Gaussian determinant gives $\sqrt{\det G_{ab}}(2\pi\mathrm{i}\hbar\epsilon)^{-d/2}$, and the remaining exponent is $\epsilon G_{ab}v^a v^b/2$. Multiplying over slices and including $-\epsilon V(q_n)$ gives the displayed configuration-space kernel.

7.7 Time-Ordered Insertions and Sources

Definition 7.10 (Finite-slice time-ordered insertions and source kernel)**.** Let $F(q(t_1), \ldots, q(t_r))$ be a product of coordinate insertions with

$$0 < t_1 < \cdots < t_r < T.$$

At finite N, choose the time slices nearest to the insertion times and insert the corresponding coordinate functions into the finite-dimensional integral. The resulting regulated expression represents

$$\langle q_f | T\{\widehat{q}^{a_1}(t_1) \cdots \widehat{q}^{a_r}(t_r)\} U(T) | q_i \rangle,$$

where

$$\widehat{q}^a(t) = U(t)^{-1} \widehat{q}^a U(t)$$

is the Heisenberg operator. The time ordering is inherited from the ordered product of short-time evolution operators.

For a smooth compactly supported source $J_a(t)$, define the finite-N source factor by

$$\exp\left[\frac{\mathrm{i}}{\hbar} \sum_{n=0}^{N-1} \epsilon\, J_a(t_n) q_n^a\right].$$

The source-dependent kernel $K_N[J]$ generates regulated time-ordered coordinate insertions by differentiation with respect to J. This is the finite-dimensional ancestor of QFT generating functionals for Green functions.

7.8 Euclidean Traces and Periodic Boundary Conditions

Trace and periodic Euclidean boundary conditions. Assume that $\widehat{H}$ is semibounded and that $\mathrm{e}^{-\beta_{\mathrm{T}} \widehat{H}/\hbar}$ is trace class for $\beta_{\mathrm{T}} > 0$. The Euclidean thermal trace is

$$Z(\beta_{\mathrm{T}}) = \mathrm{Tr}_{\mathcal{H}_Q}\, \mathrm{e}^{-\beta_{\mathrm{T}} \widehat{H}/\hbar}.$$

This trace-class hypothesis is a finite-volume or confining quantum-mechanical hypothesis. It is the correct assumption for a harmonic oscillator, for stable anharmonic oscillators with discrete spectrum growing fast enough, and for field theories after both an ultraviolet regulator and a finite spatial-volume regulator have been imposed. It is not the Hilbert-space structure of an infinite-volume translation-invariant QFT. In infinite spatial volume, translating a finite-energy local excitation through disjoint spatial regions produces the usual volume divergence; correspondingly $\mathrm{Tr}\, \mathrm{e}^{-\beta H}$ is infinite even when local thermal correlators exist.

Thermal QFT is therefore formulated by one of two equivalent local procedures. One may put the theory in a finite spatial box Λ, form

$$\rho_{\beta,\Lambda} = Z_{\beta,\Lambda}^{-1} \mathrm{e}^{-\beta H_\Lambda}, \qquad Z_{\beta,\Lambda} = \mathrm{Tr}_{\mathcal{H}_\Lambda}\, \mathrm{e}^{-\beta H_\Lambda},$$

and then take thermodynamic limits of local observables. Or one may define the infinite-volume thermal state directly as a KMS state on the quasilocal observable algebra, in the sense of Definition 4.5. Formal traces per unit volume, or traces with the translation volume divided out, compute densities rather than the literal trace over the full infinite-volume Hilbert space. For a trace-class quantum-mechanical Hamiltonian, the trace of the heat operator is obtained by integrating its diagonal position kernel:

$$Z(\beta_{\mathrm{T}}) = \int_{\mathbb{R}^d} \mathrm{d}^d q_0 \, \langle q_0 | \mathrm{e}^{-\beta_{\mathrm{T}} \widehat{H}/\hbar} | q_0 \rangle.$$

The same time-sliced construction as above, now with Euclidean time $\tau \in [0, \beta_{\mathrm{T}}]$, therefore has

$$q_N = q_0.$$

Thus the trace of a bosonic quantum-mechanical system is represented by the Euclidean path integral over periodic paths

$$q(\tau + \beta_{\mathrm{T}}) = q(\tau).$$

This conclusion is a statement about the boundary condition produced by the trace. When Theorem 7.6 applies, the corresponding continuum object is the Feynman–Kac weight integrated over Brownian loops, obtained by disintegrating Wiener measure with the endpoint constraint

$$q(\beta_{\mathrm{T}}) = q(0).$$

Outside such hypotheses, the trace path integral remains a regulated or formal expression whose convergence must be specified separately.

7.9 Vacuum Projection

Euclidean long-time projection to a gapped ground state. Assume $\widehat{H}$ has a normalized ground state $|0\rangle$, with energy E_0, and that a vector $|\psi\rangle \in \mathcal{H}_Q$ satisfies $\langle 0 \mid \psi \rangle \neq 0$. If the spectrum above E_0 is separated by a gap, then

$$\lim_{\tau \to \infty} \frac{\mathrm{e}^{-\tau(\widehat{H}-E_0)/\hbar} |\psi\rangle}{\left\| \mathrm{e}^{-\tau(\widehat{H}-E_0)/\hbar} |\psi\rangle \right\|} = |0\rangle$$

up to the phase of $\langle 0 \mid \psi \rangle$. Vacuum correlation functions can therefore be obtained from long Euclidean-time evolution:

$$\langle 0 | T\{\widehat{q}^{a_1}(t_1) \cdots \widehat{q}^{a_r}(t_r)\} | 0 \rangle$$

is the normalized long-time limit of the corresponding transition amplitude with insertions, after the Euclidean projection has been applied. This is the elementary spectral projection mechanism behind Euclidean vacuum boundary conditions. By the spectral theorem for $\widehat{H}$, write $|\psi\rangle = c_0|0\rangle + \eta$ with $c_0 = \langle 0 \mid \psi \rangle \neq 0$ and η orthogonal to the ground-state subspace. The gap assumption gives

$$\|\mathrm{e}^{-\tau(\widehat{H}-E_0)/\hbar} \eta\| \leq \mathrm{e}^{-\tau \Delta/\hbar} \|\eta\|$$

for some $\Delta > 0$. Therefore the normalized Euclidean evolution tends to $\mathrm{e}^{\mathrm{i} \arg c_0}|0\rangle$. Applying the same spectral projection to initial and final wave packets in a transition amplitude with time-ordered insertions selects the vacuum matrix element after the common exponential normalization is removed. The substantive hypotheses are the existence of the normalized ground state, the nonzero overlap of the boundary vector with it, and the gap. Without a gap the same formula can fail in norm, and one must specify the weaker spectral or distributional limit being used.

The next chapter will analyze the analytic continuation and Gaussian integrals that make this construction computationally useful. The present chapter has introduced the regulated path-integral data and its relation to Hamiltonian time evolution.

Chapter 8

Euclidean Correlation Functions and Gaussian Perturbation Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

8.1 Spectral Correlators and Euclidean Time

Let $\widehat{H}$ be a self-adjoint Hamiltonian on a Hilbert space $\mathcal{H}$. Assume in this section that $\widehat{H}$ has a normalized ground state $|0\rangle$ with energy E_0, and that the example under discussion admits a complete orthonormal basis $\{|n\rangle\}$ of energy eigenvectors,

$$\widehat{H}|n\rangle = E_n|n\rangle, \qquad E_n \geq E_0.$$

For a self-adjoint coordinate operator $\widehat{q}$, define the real-time Heisenberg operator

$$\widehat{q}(t) = \exp\left(\frac{\mathrm{i}}{\hbar}t\widehat{H}\right)\widehat{q}\exp\left(-\frac{\mathrm{i}}{\hbar}t\widehat{H}\right), \qquad t \in \mathbb{R}.$$

The ordered two-point function with the later insertion on the left is

$$W(t) = \langle 0|\widehat{q}(t)\widehat{q}(0)|0\rangle.$$

Inserting the energy basis gives

$$\begin{aligned} W(t) &= \sum_n \langle 0|\widehat{q}(0)|n\rangle\langle n|\widehat{q}(0)|0\rangle \, \exp\left[-\frac{\mathrm{i}}{\hbar}(E_n - E_0)t\right] \\ &= \sum_n |\langle n|\widehat{q}(0)|0\rangle|^2 \exp\left[-\frac{\mathrm{i}}{\hbar}(E_n - E_0)t\right]. \end{aligned}$$

Definition 8.1 (Spectral measure of a vacuum coordinate)**.** Let $P_{\widehat{H}}$ be the projection-valued spectral measure of $\widehat{H}$. For a vector

$$\eta_q := \widehat{q}|0\rangle,$$

assumed to lie in $\mathcal{H}$, define the positive finite Borel measure

$$\mu_q(B) = \langle \eta_q, P_{\widehat{H}}(B)\eta_q\rangle$$

for Borel sets $B \subset \mathbb{R}$. Its shifted version is the pushforward of μ_q under $E \mapsto E - E_0$. This is the one-dimensional prototype of the spectral measures that appear later in the Källén–Lehmann representation.

Lemma 8.2 (Lower-half-plane continuation from the spectral theorem)**.** *With the hypotheses above, the ordered two-point function admits the spectral representation*

$$W(t) = \int_{[E_0,\infty)} \exp\left[-\frac{\mathrm{i}}{\hbar}(E - E_0)t\right] \, \mathrm{d}\mu_q(E).$$

The same formula defines a holomorphic function for $\operatorname{Im} t < 0$*. For* $\tau > 0$*, its Euclidean boundary value at* $t = -\mathrm{i}\tau$ *is the Laplace transform*

$$G_E(\tau) = \int_{[E_0,\infty)} \exp\left[-\frac{1}{\hbar}(E - E_0)\tau\right] \, \mathrm{d}\mu_q(E).$$

If the spectral measure is pure point, the integral reduces to the displayed sum over energy eigenvectors.

Proof. The spectral theorem gives

$$\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}t(\widehat{H}-E_0)} = \int_{[E_0,\infty)} \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}t(E-E_0)} \, \mathrm{d}P_{\widehat{H}}(E)$$

in the functional calculus. Sandwiching this identity between $\eta_q = \widehat{q}|0\rangle$ and itself gives the stated formula for real t. The support is contained in $[E_0, \infty)$ because E_0 is the bottom of the spectrum, and the total mass is $\|\widehat{q}|0\rangle\|^2$.

If $t = x + \mathrm{i}y$ with $y < 0$, then

$$\left|\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}t(E-E_0)}\right| = \mathrm{e}^{\frac{1}{\hbar}y(E-E_0)} \leq 1.$$

The finite measure μ_q therefore gives a locally dominated integral. The integrand is entire as a function of t, so differentiation under the integral is justified on compact subsets of the lower half-plane by the same estimate with one additional power of $E - E_0$ after inserting a compact vertical margin $y \leq -\epsilon < 0$. Hence the integral is holomorphic there. Setting $t = -\mathrm{i}\tau$ gives the Laplace transform. In the pure-point case $\mu_q = \sum_n |\langle n|\widehat{q}|0\rangle|^2 \delta_{E_n}$, so the integral is the sum already obtained by inserting the eigenbasis. □

For $\tau > 0$, define

$$G_E(\tau) = W(-\mathrm{i}\tau) = \int_{[E_0,\infty)} \exp\left[-\frac{1}{\hbar}(E - E_0)\tau\right] \mathrm{d}\mu_q(E),$$

which in the pure-point case is

$$G_E(\tau) = \sum_n |\langle n|\widehat{q}(0)|0\rangle|^2 \exp\left[-\frac{1}{\hbar}(E_n - E_0)\tau\right].$$

Thus Euclidean time converts the spectral decomposition into a sum of decaying exponentials. The decay rates are

$$\omega_n = \frac{E_n - E_0}{\hbar}.$$

For $\tau_1, \ldots, \tau_r \in \mathbb{R}$, let T_E denote Euclidean ordering: operators with larger τ are placed to the left. The Euclidean r-point function of a single coordinate operator is

$$G_E^{(r)}(\tau_1, \ldots, \tau_r) = \langle 0|T_E\{\widehat{q}_E(\tau_1) \cdots \widehat{q}_E(\tau_r)\}|0\rangle,$$

where

$$\widehat{q}_E(\tau) = \exp\left(\frac{1}{\hbar}\tau\widehat{H}\right)\widehat{q}\exp\left(-\frac{1}{\hbar}\tau\widehat{H}\right)$$

is the analytic continuation of $\widehat{q}(t)$ to $t = -\mathrm{i}\tau$.

Definition 8.3 (Euclidean ordered source functional in quantum mechanics)**.** For a smooth compactly supported function $J(\tau)$, define the Euclidean ordered source functional

$$Z_E[J] = \langle 0|T_E \exp\left[\frac{1}{\hbar}\int_{\mathbb{R}} J(\tau)\widehat{q}_E(\tau)\,\mathrm{d}\tau\right]|0\rangle.$$

The exponential is understood by its power series in the source, with Euclidean ordering applied term by term. Its functional derivatives recover the Euclidean ordered correlation functions:

$$G_E^{(r)}(\tau_1, \ldots, \tau_r) = \hbar^r \frac{\delta}{\delta J(\tau_1)} \cdots \frac{\delta}{\delta J(\tau_r)} Z_E[J]\bigg|_{J=0}.$$

8.2 Wick Rotation of the Path Integral

Consider a Lagrangian of the form

$$L(q, \dot{q}) = \frac{1}{2}G_{ab}(q)\dot{q}^a\dot{q}^b - V(q),$$

where $G_{ab}(q)$ is positive definite. Under the analytic continuation

$$t = -\mathrm{i}\tau, \qquad \dot{q}^a = \frac{\mathrm{d}q^a}{\mathrm{d}t} = \mathrm{i}\,\frac{\mathrm{d}q^a}{\mathrm{d}\tau},$$

Definition 8.4 (Euclidean Lagrangian associated with a mechanical action)**.** For a Lorentzian mechanical Lagrangian $L(q, \dot{q})$ whose expression is analytic in $\dot{q}$ near the real axis, the Euclidean Lagrangian in the coordinate chart q^a is

$$L_E(q, q') = -L(q, \mathrm{i}q'), \qquad q'^a = \frac{\mathrm{d}q^a}{\mathrm{d}\tau}.$$

For the displayed metric kinetic energy, this gives

$$L_E(q, q') = \frac{1}{2}G_{ab}(q)q'^a q'^b + V(q).$$

Sign of the Wick-rotated weight. Under $t = -\mathrm{i}\tau$, with the orientation of the integration contour rotated accordingly, the Lorentzian phase transforms formally as

$$\exp\left[\frac{\mathrm{i}}{\hbar}\int L(q,\dot q)\,\mathrm{d}t\right] \longmapsto \exp\left[-\frac{1}{\hbar}\int L_E(q,q')\,\mathrm{d}\tau\right].$$

Indeed, along the rotated contour $\mathrm{d}t = -\mathrm{i}\,\mathrm{d}\tau$ and $\dot q = \mathrm{d}q/\mathrm{d}t = \mathrm{i}\,\mathrm{d}q/\mathrm{d}\tau = \mathrm{i}q'$. Therefore

$$\frac{\mathrm{i}}{\hbar}L(q,\dot q)\,\mathrm{d}t = \frac{\mathrm{i}}{\hbar}L(q,\mathrm{i}q')(-\mathrm{i}\,\mathrm{d}\tau) = \frac{1}{\hbar}L(q,\mathrm{i}q')\,\mathrm{d}\tau = -\frac{1}{\hbar}L_E(q,q')\,\mathrm{d}\tau.$$

Exponentiating the integrated identity gives the stated weight. This is an identity of the formal action density after choosing the contour; analytic existence of the path-integral limit is a separate assertion supplied only in the regulated cases below.

Lemma 8.5 (Euclidean long-time projection onto the vacuum)**.** *Let $|\psi\rangle \in \mathcal{H}$ satisfy $\langle 0 \mid \psi\rangle \neq 0$, and assume that the ground state is separated from the rest of the spectrum by a gap. More precisely, suppose that the spectrum of $\widehat H - E_0$ on $\{|0\rangle\}^\perp$ lies in $[\Delta,\infty)$ with $\Delta > 0$. Then*

$$\frac{\exp[-T(\widehat H - E_0)/\hbar]|\psi\rangle}{\left\|\exp[-T(\widehat H - E_0)/\hbar]|\psi\rangle\right\|} \longrightarrow \frac{\langle 0 \mid \psi\rangle}{|\langle 0 \mid \psi\rangle|}|0\rangle$$

in Hilbert norm as $T \to \infty$. Consequently Euclidean matrix elements with endpoints $|\psi\rangle$ and fixed insertions separated from the endpoints converge to vacuum Euclidean ordered correlation functions, after division by the corresponding endpoint partition function.

Proof. Write

$$|\psi\rangle = c_0|0\rangle + |\psi_\perp\rangle, \qquad c_0 = \langle 0 \mid \psi\rangle, \qquad \langle 0 \mid \psi_\perp\rangle = 0,$$

and let the spectrum of $\widehat H - E_0$ on the closed subspace

$$\{|0\rangle\}^\perp = \{\eta \in \mathcal{H} : \langle 0 \mid \eta\rangle = 0\}$$

lie in $[\Delta,\infty)$ with $\Delta > 0$. Spectral calculus gives

$$\mathrm{e}^{-T(\widehat H - E_0)/\hbar}|\psi\rangle = c_0|0\rangle + O(\mathrm{e}^{-T\Delta/\hbar})$$

in Hilbert norm. Therefore

$$\frac{\exp[-T(\widehat H - E_0)/\hbar]|\psi\rangle}{\left\|\exp[-T(\widehat H - E_0)/\hbar]|\psi\rangle\right\|} \longrightarrow \frac{c_0}{|c_0|}|0\rangle$$

as $T \to \infty$. The same estimate applies independently to the bra endpoint. The phase $c_0/|c_0|$ cancels between bra and ket in normalized matrix elements. For $0 < \tau_1 < \cdots < \tau_r < T$, inserting the semigroup factors between consecutive operators writes the endpoint matrix element as a bounded product of fixed-time operators and exterior factors $\exp[-T(\widehat H - E_0)/\hbar]|\psi\rangle$ and its adjoint. The norm convergence of the exterior factors gives the vacuum Euclidean ordered correlator. If the inserted operators are unbounded, this last sentence is read on a common invariant domain or after smearing/truncating the insertions; the regulated path-integral constructions below implement precisely that domain control. □

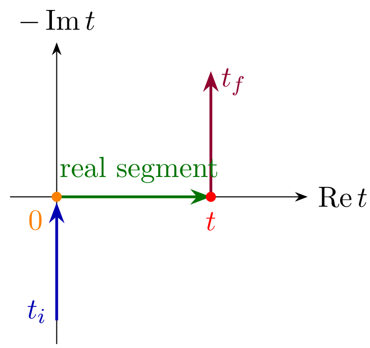
The same limit may be represented before Wick rotation by a regulated path integral on a complex-time contour. Let $\Gamma_{T,t}$ run from $t_i = \mathrm{i}T$ to 0, then along the real segment from 0 to t, and then to $t_f = t - \mathrm{i}T$, so that

$$\operatorname{Im} t_i \to +\infty, \qquad \operatorname{Im} t_f \to -\infty$$

as $T \to \infty$. If $[Dq]_{\Gamma,\psi}$ denotes the time-sliced measure on this contour with endpoint wavefunctions of a vector $|\psi\rangle$ included, then

$$\langle 0|\widehat{q}(t)\widehat{q}(0)|0\rangle = \lim_{T\to\infty} \frac{\int [Dq]_{\Gamma_{T,t},\psi} \exp\left[\frac{\mathrm{i}}{\hbar}\int_{\Gamma_{T,t}} L(q,\dot{q})\,\mathrm{d}t\right] q(t)q(0)}{\int [Dq]_{\Gamma_{T,t},\psi} \exp\left[\frac{\mathrm{i}}{\hbar}\int_{\Gamma_{T,t}} L(q,\dot{q})\,\mathrm{d}t\right]},$$

under the same ground-state overlap and gap hypotheses. This formula is a contour version of the operator projection above; the regulator and endpoint data are part of $[Dq]_{\Gamma,\psi}$.



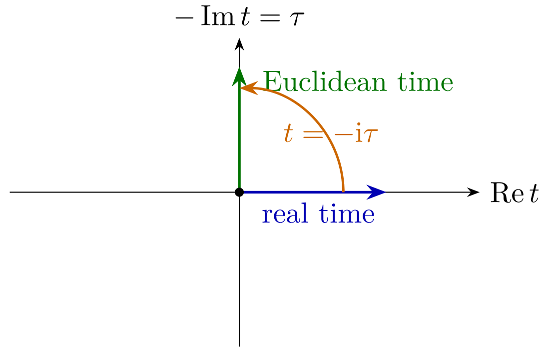
Convention 8.6 (Euclidean path-integral brackets in this chapter)**.** After the Wick rotation of the contour to Euclidean time, the Lagrangian path-integral notation

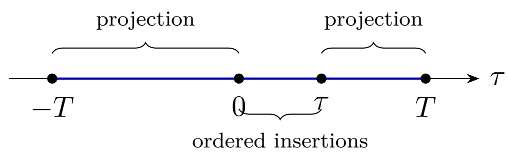
$$\langle 0|T_E\{\widehat{q}_E(\tau_1)\cdots\widehat{q}_E(\tau_r)\}|0\rangle = \lim_{T\to\infty} \frac{\int [Dq]\, \exp[-S_E[q]/\hbar]\, q(\tau_1)\cdots q(\tau_r)}{\int [Dq]\, \exp[-S_E[q]/\hbar]},$$

means the value produced by a specified regulator and limiting procedure, not an independently given translation-invariant measure on an infinite-dimensional space. The Euclidean action is

$$S_E[q] = \int_{-T}^{T} L_E(q,q')\,\mathrm{d}\tau.$$

For a bosonic Schrödinger operator satisfying the hypotheses of Theorem 7.6, this sentence can be read as a Wiener-measure statement: finite-interval transition amplitudes are Brownian bridge expectations with the potential weight, and vacuum correlators are obtained by the spectral projection limit described above. In the rest of this chapter the same bracket notation is also used for finite-dimensional Gaussian regulators and perturbative asymptotic expansions. These are different mathematical objects, and the text will specify which one is meant.





Definition 8.7 (Mode cutoff regulator)**.** A mode cutoff regulator consists of a finite set of functions $f_1, \ldots, f_{N_{\max}}$, a finite-dimensional coordinate map

$$q(\tau) = \sum_{n=1}^{N_{\max}} q_n f_n(\tau),$$

and a regulated measure defined by a finite product of Lebesgue measures in the coefficients q_n, together with the normalization and boundary conditions dictated by the chosen time-sliced construction. The limit $N_{\max} \to \infty$ is a further analytic operation and is not part of the symbol $[Dq]$ until its existence has been established for the observables under discussion.

The harmonic oscillator provides the basic Gaussian example.

Controlled Approximation 8.8 (Removing a mode cutoff)**.** In this chapter the phrase "remove the mode cutoff" has two distinct meanings. In the free Gaussian oscillator it denotes an actual limit of finite-dimensional Gaussian moments; the covariance kernels below converge after the finite mode projection is removed. In the interacting examples it denotes a coefficientwise operation on a perturbative expansion first defined at finite cutoff. No nonperturbative continuum measure, and no convergence of the formal power series in the coupling, is inferred from the notation.

8.3 The Harmonic Oscillator

Let

$$\widehat{H} = \frac{1}{2}(\widehat{p}^2 + \omega^2 \widehat{q}^2), \qquad L(q, \dot{q}) = \frac{1}{2}\dot{q}^2 - \frac{1}{2}\omega^2 q^2,$$

with $\omega > 0$. The Euclidean action on the interval $[-T, T]$ is

$$S_E[q] = \frac{1}{2} \int_{-T}^{T} \left[(q')^2 + \omega^2 q^2\right] \mathrm{d}\tau.$$

Impose Dirichlet boundary conditions $q(-T) = q(T) = 0$ and expand

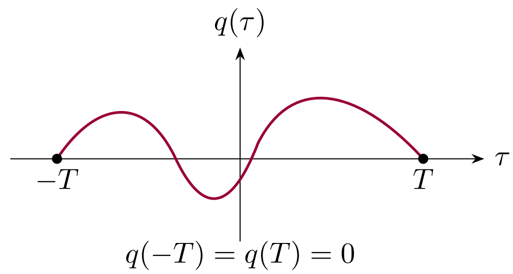
$$q(\tau) = \sum_{n=1}^{\infty} q_n f_n(\tau), \qquad f_n(\tau) = \frac{1}{\sqrt{T}} \sin \frac{n\pi(\tau + T)}{2T}.$$

The functions f_n obey

$$\int_{-T}^{T} f_n(\tau) f_m(\tau)\, \mathrm{d}\tau = \delta_{nm}, \qquad -\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} f_n = \left(\frac{n\pi}{2T}\right)^2 f_n.$$

Consequently

$$S_E[q] = \frac{1}{2} \sum_{n=1}^{\infty} \left[\left(\frac{n\pi}{2T}\right)^2 + \omega^2 \right] q_n^2.$$



Definition 8.9 (Dirichlet harmonic-oscillator Gaussian cutoff)**.** For $N \in \mathbb{N}$, let

$$q^{(N)}(\tau) = \sum_{n=1}^{N} q_n f_n(\tau), \qquad S_{E,T,N}(q_1, \ldots, q_N) = \frac{1}{2} \sum_{n=1}^{N} \left[\left(\frac{n\pi}{2T}\right)^2 + \omega^2 \right] q_n^2.$$

The normalized finite-T, finite-mode expectation value is

$$\langle F \rangle_{T,N} = \frac{\int_{\mathbb{R}^N} \prod_{n=1}^{N} \mathrm{d}q_n \, \exp[-S_{E,T,N}/\hbar]\, F(q^{(N)})}{\int_{\mathbb{R}^N} \prod_{n=1}^{N} \mathrm{d}q_n \, \exp[-S_{E,T,N}/\hbar]}.$$

Proposition 8.10 (Harmonic-oscillator Euclidean covariance from the cutoff)**.** *For the cutoff of Definition 8.9, the finite-mode covariance converges at fixed T. For fixed insertion times τ_1, τ_2 and then $T \to \infty$, the limiting Euclidean two-point function on the line is*

$$\langle q(\tau_1) q(\tau_2) \rangle = \frac{\hbar}{2\omega} \mathrm{e}^{-\omega|\tau_1 - \tau_2|}.$$

Proof. This is an ordinary finite-dimensional Gaussian integral. In this free subsection, $\langle F \rangle_T$ denotes the limit $\lim_{N\to\infty} \langle F \rangle_{T,N}$ only for those observables for which that limit is being proved or has already been proved. The independent Gaussian integrals give, for $1 \le n, m \le N$,

$$\langle q_n q_m \rangle_{T,N} = \delta_{nm} \frac{\hbar}{\left(\frac{n\pi}{2T}\right)^2 + \omega^2}.$$

Therefore the finite-mode covariance kernel is

$$\langle q^{(N)}(\tau_1) q^{(N)}(\tau_2) \rangle_{T,N} = \sum_{n=1}^{N} \frac{\hbar\, f_n(\tau_1) f_n(\tau_2)}{\left(\frac{n\pi}{2T}\right)^2 + \omega^2}.$$

Since $\sum_{n\geq 1}((n\pi/2T)^2+\omega^2)^{-1}<\infty$, this sequence has a pointwise limit for every fixed $\tau_1,\tau_2\in[-T,T]$, and the finite-T Gaussian covariance is

$$\langle q(\tau_1)q(\tau_2)\rangle_T=\sum_{n=1}^{\infty}\frac{\hbar\, f_n(\tau_1)f_n(\tau_2)}{\left(\frac{n\pi}{2T}\right)^2+\omega^2}.$$

The passage to the infinite Euclidean line is not a formal replacement of a sum by an integral; it is a limit at fixed insertion points. Set

$$k_n=\frac{n\pi}{2T},\qquad \Delta k=\frac{\pi}{2T}.$$

Using the explicit sine basis,

$$\begin{aligned}\langle q(\tau_1)q(\tau_2)\rangle_T&=\sum_{n=1}^{\infty}\frac{\hbar}{T}\frac{\sin k_n(\tau_1+T)\sin k_n(\tau_2+T)}{k_n^2+\omega^2}\\&=\frac{2\hbar}{\pi}\sum_{n=1}^{\infty}\Delta k\,\frac{\sin k_n(\tau_1+T)\sin k_n(\tau_2+T)}{k_n^2+\omega^2}.\end{aligned}$$

To state the limit as an operation on the regulated path integral, insert temporarily an Abel factor $\mathrm{e}^{-\varepsilon k_n}$, $\varepsilon>0$, in the mode sum and the corresponding factor $\mathrm{e}^{-\varepsilon k}$ in the continuum integral. The Riemann limit is then taken at fixed ε, and the limit $\varepsilon\downarrow 0$ is taken after the Fourier integral has been evaluated. Suppressing this auxiliary factor from the notation, the continuum expression is

$$\frac{2\hbar}{\pi}\int_0^{\infty}\mathrm{d}k\,\frac{\sin k(\tau_1+T)\sin k(\tau_2+T)}{k^2+\omega^2}.$$

Writing $2\sin a\sin b=\cos(a-b)-\cos(a+b)$, equivalently

$$\begin{aligned}\langle q(\tau_1)q(\tau_2)\rangle_T=\frac{\hbar}{\pi}\int_{-\infty}^{\infty}\frac{\mathrm{d}k}{k^2+\omega^2}\Bigg[&\frac{1}{4}\mathrm{e}^{\mathrm{i}k(\tau_1-\tau_2)}+\frac{1}{4}\mathrm{e}^{-\mathrm{i}k(\tau_1-\tau_2)}\\&-\frac{1}{4}\mathrm{e}^{\mathrm{i}k(\tau_1+\tau_2+2T)}-\frac{1}{4}\mathrm{e}^{-\mathrm{i}k(\tau_1+\tau_2+2T)}\Bigg]+o(1),\end{aligned}$$

where $o(1)$ denotes the regulator error in the infinite-volume limit. Thus the finite interval produces, besides the translation-invariant term, an image term tied to the Dirichlet endpoints.

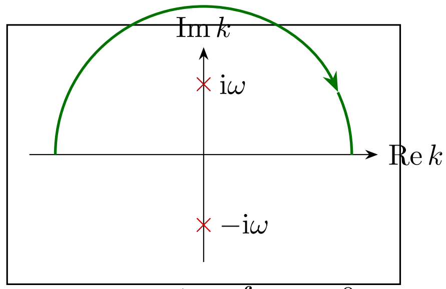
The elementary Fourier integral needed here is

$$I(\tau)=\int_{-\infty}^{\infty}\frac{\mathrm{e}^{\mathrm{i}k\tau}}{k^2+\omega^2}\,\mathrm{d}k.$$

For $\tau>0$, $|\mathrm{e}^{\mathrm{i}k\tau}|=\mathrm{e}^{-(\operatorname{Im}k)\tau}$, so the large semicircle is suppressed in the upper half-plane and the pole enclosed is $k=\mathrm{i}\omega$. Therefore

$$I(\tau)=\frac{\pi}{\omega}\mathrm{e}^{-\omega\tau}.$$

For $\tau<0$, close the contour in the lower half-plane and obtain $\pi\omega^{-1}\mathrm{e}^{\omega\tau}$.



upper contour for $\tau > 0$

For fixed τ_1, τ_2, the quantity $\tau_1 + \tau_2 + 2T$ is positive for large T, and the preceding integral gives

$$\langle q(\tau_1)q(\tau_2)\rangle_T = \frac{\hbar}{2\omega}\mathrm{e}^{-\omega|\tau_1-\tau_2|} - \frac{\hbar}{2\omega}\mathrm{e}^{-\omega(\tau_1+\tau_2+2T)} + o(1).$$

The second term is the endpoint image contribution and vanishes as $T \to \infty$. Hence the Euclidean two-point function of the harmonic oscillator on the line is

$$\boxed{\langle q(\tau_1)q(\tau_2)\rangle = \frac{\hbar}{2\omega}\mathrm{e}^{-\omega|\tau_1-\tau_2|}}.$$

This agrees with the spectral expression, since the first operator $\widehat{q}$ connects the ground state of the harmonic oscillator to the first excited state. □

8.4 Gaussian Integrals and Wick Pairings

For the remainder of the chapter set $\hbar = 1$, unless $\hbar$ is displayed explicitly.

Definition 8.11 (Finite-dimensional real Gaussian datum)**.** Let A be a real symmetric positive-definite $N \times N$ matrix, let $J \in \mathbb{R}^N$, and write repeated finite indices as summed. Define

$$Z_A[J] = \int_{\mathbb{R}^N} \mathrm{d}^N q\ \exp\left(-\frac{1}{2}q_i A_{ij} q_j + J_i q_i\right),$$

and the normalized Gaussian expectation

$$\langle F(q)\rangle_A = \frac{1}{Z_A[0]}\int_{\mathbb{R}^N} \mathrm{d}^N q\ \exp\left(-\frac{1}{2}q_i A_{ij} q_j\right) F(q).$$

The covariance matrix is $C = A^{-1}$.

Proposition 8.12 (Gaussian generating functional and Wick theorem)**.** *For the datum of Definition 8.11,*

$$Z_A[J] = (2\pi)^{N/2}(\det A)^{-1/2}\exp\left(\frac{1}{2}J_i(A^{-1})_{ij}J_j\right).$$

Consequently all odd moments vanish, and

$$\langle q_{i_1}\cdots q_{i_{2r}}\rangle_A = \sum_{\textit{complete pairings } P}\ \prod_{(a,b)\in P}(A^{-1})_{i_a i_b}.$$

Proof. Completing the square gives the displayed formula for $Z_A[J]$. Indeed positive-definiteness of A permits the shift $q \mapsto q - A^{-1}J$, and the remaining Gaussian integral is $(2\pi)^{N/2}(\det A)^{-1/2}$. Differentiating with respect to J gives

$$\langle F(q)\rangle_A = F\left(\frac{\partial}{\partial J}\right) \frac{Z_A[J]}{Z_A[0]}\bigg|_{J=0}.$$

Let $C = A^{-1}$. Then

$$\langle q_i\rangle_A = \frac{\partial}{\partial J_i} \exp\left(\frac{1}{2} J_a C_{ab} J_b\right)\bigg|_{J=0} = 0$$

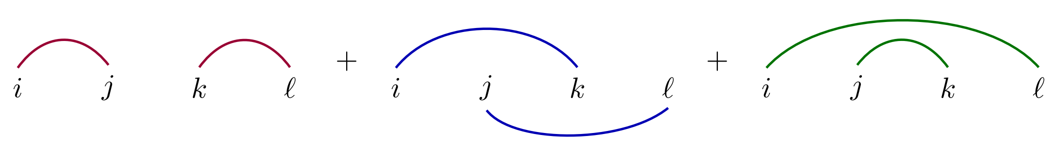
and

$$\langle q_i q_j\rangle_A = \frac{\partial}{\partial J_i}\frac{\partial}{\partial J_j} \exp\left(\frac{1}{2} J_a C_{ab} J_b\right)\bigg|_{J=0} = C_{ij}.$$

Write $\partial_i = \partial/\partial J_i$. For four variables,

$$\begin{aligned}
\langle q_i q_j q_k q_\ell\rangle_A &= \partial_i\partial_j\partial_k\partial_\ell \exp\left(\frac{1}{2} J_a C_{ab} J_b\right)\bigg|_{J=0} \\
&= \partial_i\partial_j\partial_k\partial_\ell \frac{1}{2}\left(\frac{1}{2} J_a C_{ab} J_b\right)^2\bigg|_{J=0} \\
&= C_{ij}C_{k\ell} + C_{ik}C_{j\ell} + C_{i\ell}C_{jk}.
\end{aligned}$$

Only the quadratic term squared in the exponential can survive after four source derivatives and then setting $J = 0$. The four derivatives can be assigned to the four source factors in $4!$ ways; these assignments group into the three displayed pairings, and the numerical factor $\frac{1}{2}(\frac{1}{2})^2 = \frac{1}{8}$ cancels the repeated orientations and the exchange of the two covariance factors inside each pairing.



A Wick contraction is an assigned pairing of two labels with value

$$(q_i q_j)_{\text{contr}} = C_{ij} = (A^{-1})_{ij}.$$

The same differentiation argument proves Wick's theorem for $2r$ variables: only the degree r term in $\exp(\frac{1}{2}J_a C_{ab} J_b)$ contributes after setting $J = 0$, and each surviving derivative assignment partitions the $2r$ labels into r unordered pairs. Thus the Gaussian expectation of $2r$ variables is the sum over all complete pairings, with one covariance factor for each pair:

$$\langle q_{i_1} \cdots q_{i_{2r}}\rangle_A = \sum_{\text{complete pairings } P} \prod_{(a,b)\in P} (A^{-1})_{i_a i_b}.$$

For an odd number of derivatives no monomial of odd total degree appears in the source exponential, so the moment vanishes. $\square$

8.5 Gaussian Functional Integrals

Definition 8.13 (Gaussian functional integral as a regulated limit)**.** Let A be the positive operator

$$A = -\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + \omega^2$$

on a function space with boundary conditions for which A is invertible. The Green function $G(\tau, \tau')$ is the integral kernel of A^{-1}:

$$(A^{-1}f)(\tau) = \int G(\tau, \tau') f(\tau') \, \mathrm{d}\tau',$$

and therefore satisfies

$$\left(-\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + \omega^2\right) G(\tau, \tau') = \delta(\tau - \tau').$$

With the inner product

$$(f, g) = \int f(\tau) g(\tau) \, \mathrm{d}\tau,$$

the Gaussian Euclidean action is

$$S_0[q] = \frac{1}{2}(q, Aq).$$

The source-dependent Gaussian functional integral is defined by a regulator, for instance a finite-mode cutoff, and has the finite-dimensional form in the cutoff variables. Its continuum notation is

$$Z_0[J] = \int [Dq] \, \exp\left[-\frac{1}{2}(q, Aq) + (J, q)\right].$$

This notation denotes the finite-dimensional Gaussian construction followed by a limit of covariance kernels in the specified topology; it does not assert the existence of a translation-invariant Lebesgue measure on the space of paths.

Gaussian functional source identity. For the regulated Gaussian construction of Definition 8.13, the normalization-independent source ratio is

$$\frac{Z_0[J]}{Z_0[0]} = \exp\left[\frac{1}{2} \int J(\tau) G(\tau, \tau') J(\tau') \, \mathrm{d}\tau \, \mathrm{d}\tau'\right],$$

and the two-point function is $G(\tau_1, \tau_2)$.

At finite cutoff this is exactly Proposition 8.12 with the matrix A_{ij} equal to the quadratic form of the cutoff action and with the source vector $J_i = (J, f_i)$. The inverse matrix has kernel equal to the cutoff Green function. Passing to the limit gives the displayed kernel identity whenever the cutoff Green functions converge against the chosen test sources.

The ratio independent of normalization is

$$\frac{Z_0[J]}{Z_0[0]} = \exp\left[\frac{1}{2} \int J(\tau) G(\tau, \tau') J(\tau') \, \mathrm{d}\tau \, \mathrm{d}\tau'\right].$$

Consequently

$$\langle q(\tau_1) q(\tau_2) \rangle_0 = G(\tau_1, \tau_2).$$

Gaussian integration by parts. In the same regulated Gaussian construction,

$$\langle q(\tau_1)q(\tau_2)\rangle_0 = G(\tau_1,\tau_2)$$

also follows from finite-dimensional integration by parts. Equivalently, the formal identity

$$q(\tau_1)\mathrm{e}^{-\frac{1}{2}(q,Aq)} = -\left(A^{-1}\frac{\delta}{\delta q}\right)(\tau_1)\mathrm{e}^{-\frac{1}{2}(q,Aq)}$$

is the continuum notation for that cutoff identity.

The same result can be obtained without tracking the normalization of the functional measure under discretization. At finite cutoff this is ordinary integration by parts in finitely many variables; the continuum notation is the limit of that identity. The functional derivative is normalized by

$$\frac{\delta q(\tau_2)}{\delta q(\tau')} = \delta(\tau'-\tau_2).$$

Since

$$\frac{\delta}{\delta q(\tau')}\exp\left[-\frac{1}{2}(q,Aq)\right] = -(Aq)(\tau')\exp\left[-\frac{1}{2}(q,Aq)\right],$$

one has

$$q(\tau_1)\exp\left[-\frac{1}{2}(q,Aq)\right] = -\left(A^{-1}\frac{\delta}{\delta q}\right)(\tau_1)\exp\left[-\frac{1}{2}(q,Aq)\right],$$

where

$$\left(A^{-1}\frac{\delta}{\delta q}\right)(\tau_1) = \int G(\tau_1,\tau')\frac{\delta}{\delta q(\tau')}\,\mathrm{d}\tau'.$$

The regulated Gaussian has no boundary term in this finite-dimensional integration by parts. Therefore

$$\begin{aligned}
&\int [Dq]\,\exp\left[-\frac{1}{2}(q,Aq)\right] q(\tau_1)q(\tau_2)\\
&= \int [Dq]\,\exp\left[-\frac{1}{2}(q,Aq)\right]\left(A^{-1}\frac{\delta}{\delta q}\right)(\tau_1)q(\tau_2)\\
&= \int [Dq]\,\exp\left[-\frac{1}{2}(q,Aq)\right]\int G(\tau_1,\tau')\delta(\tau'-\tau_2)\,\mathrm{d}\tau'.
\end{aligned}$$

After division by $Z_0[0]$, this gives again $\langle q(\tau_1)q(\tau_2)\rangle_0 = G(\tau_1,\tau_2)$.

On the infinite Euclidean line, write

$$q(\tau) = \int_{-\infty}^{\infty}\frac{\mathrm{d}k}{2\pi}\,\widetilde{q}(k)\mathrm{e}^{\mathrm{i}k\tau}, \qquad \widetilde{q}(k)^* = \widetilde{q}(-k).$$

Then

$$S_0[q] = \frac{1}{2}\int_{-\infty}^{\infty}\frac{\mathrm{d}k}{2\pi}\,\widetilde{q}(-k)(k^2+\omega^2)\widetilde{q}(k),$$

and

$$\langle\widetilde{q}(k_1)\widetilde{q}(k_2)\rangle_0 = \frac{2\pi\delta(k_1+k_2)}{k_1^2+\omega^2}.$$

Thus

$$G(\tau,\tau') = \int_{-\infty}^{\infty}\frac{\mathrm{d}k}{2\pi}\,\frac{\mathrm{e}^{\mathrm{i}k(\tau-\tau')}}{k^2+\omega^2} = \frac{1}{2\omega}\mathrm{e}^{-\omega|\tau-\tau'|}.$$

8.6 The Anharmonic Oscillator

Set $\omega = 1$ in this section. The anharmonic oscillator is defined by

$$\widehat{H} = \frac{1}{2}\widehat{p}^2 + \frac{1}{2}\widehat{q}^2 + \frac{g}{4!}\widehat{q}^4.$$

For real $g \geq 0$, this differential operator on $C_c^\infty(\mathbb{R})$ has a unique self-adjoint closure by the Faris–Lavine criterion stated in Chapter 7: the potential

$$V(q) = \frac{1}{2}q^2 + \frac{g}{4!}q^4$$

is smooth, bounded below, locally Kato, and has zero negative part. Hence the Euclidean semigroup and its time-sliced approximants are defined before the perturbative expansion in g is formed. The Euclidean Lagrangian is

$$L_E(q, q') = \frac{1}{2}(q')^2 + \frac{1}{2}q^2 + \frac{g}{4!}q^4.$$

Let

$$S_0[q] = \int \left[\frac{1}{2}(q')^2 + \frac{1}{2}q^2\right] \mathrm{d}\tau, \qquad S_{\mathrm{int}}[q] = \frac{g}{4!}\int q(\tau)^4 \,\mathrm{d}\tau.$$

Definition 8.14 (Cutoff perturbative expansion of the anharmonic oscillator)**.** At finite cutoff, perturbation theory is the coefficientwise expansion of $\exp[-S_{\mathrm{int}}]$ inside a finite-dimensional Gaussian integral:

$$\mathrm{e}^{-S_{\mathrm{int}}[q]} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{g}{4!}\int q(\tau)^4 \,\mathrm{d}\tau\right)^n.$$

No convergence of the infinite-dimensional integral or of the resulting power series in g is assumed at this point; the coefficients are first defined in the regulated Gaussian integral. Removing the mode cutoff in this interacting section means taking $N_{\max} \to \infty$, or equivalently $\Lambda \to \infty$, separately for each coefficient after its regulator dependence has been displayed. If the coefficient has a finite limit, that limit is part of the perturbative definition used here; if it does not, one must specify the local counterterm or subtraction coordinate before a finite coefficient is defined.

For the partition function,

$$Z_g = \int [Dq] \,\exp[-S_0[q] - S_{\mathrm{int}}[q]],$$

one obtains

$$\frac{Z_g}{Z_0} = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{g}{4!}\right)^n \int \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_n \,\left\langle q(\tau_1)^4 \cdots q(\tau_n)^4\right\rangle_0.$$

Here Z_0 is the Gaussian partition function and $\langle \cdots \rangle_0$ is evaluated with

$$G_0(\tau, \tau') = \frac{1}{2}\mathrm{e}^{-|\tau-\tau'|}.$$

At first order in g,

$$\begin{aligned}\left(\frac{Z_g}{Z_0} - 1\right)_{O(g)} &= -\frac{g}{4!}\int \mathrm{d}\tau \,\langle q(\tau)^4\rangle_0 \\ &= -\frac{g}{8}\int \mathrm{d}\tau \,G_0(\tau,\tau)^2.\end{aligned} \tag{8.1}$$

The factor 3 in $\langle q^4\rangle_0 = 3G_0(\tau,\tau)^2$ is the Wick count $(4-1)!! = 3$ for pairing four identical fields at the same Euclidean time among themselves. The 4! assignment count appears only when four labeled vertex half-edges are attached to four labeled insertions; that is a different combinatorial problem from the coincident self-contraction here. Since $G_0(\tau,\tau) = G_0(0)$ is independent of τ, this term is proportional to the Euclidean time interval. Its interpretation as a correction to the ground-state energy is derived below from the spectral decomposition of the Euclidean transition kernel.

At second order,

$$\frac{1}{2}\left(-\frac{g}{4!}\right)^2 \int \mathrm{d}\tau_1\, \mathrm{d}\tau_2\, \left\langle q(\tau_1)^4 q(\tau_2)^4\right\rangle_0$$

contains disconnected contractions and connected contractions. The connected sector already has distinct topologies: the two vertices may be joined by two propagators with one tadpole at each vertex, or they may be joined by four propagators. Figure 8.1 records this separation between disconnected vacuum products and the connected two-vertex topologies that enter the linked-cluster proof.

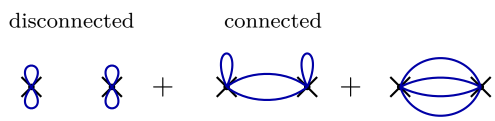


Figure 8.1: Second-order Wick topologies in the anharmonic oscillator expansion. The left pair is a product of disconnected one-vertex vacuum bubbles. The two right diagrams are connected two-vertex vacuum contractions: two inter-vertex lines with one tadpole at each vertex, and four inter-vertex lines.

Proposition 8.15 (Linked-cluster formula for the cutoff expansion)**.** *At every fixed cutoff and at the level of formal power series in g, $\log(Z_g/Z_0)$ is the sum of connected vacuum diagrams.*

Proof. Let $G^{(\ell)}$ be the sum of connected vacuum Wick contractions involving ℓ interaction vertices, including the coupling factors and integration over the ℓ vertex positions. If m_ℓ is the number of connected components of size ℓ, and if K is the total number of connected components, then

$$\sum_{\ell\geq 1} m_\ell = K, \qquad \sum_{\ell\geq 1} \ell\, m_\ell = n.$$

The number of ways to partition n labelled vertices into this pattern is

$$\frac{n!}{\prod_{\ell\geq 1}(\ell!)^{m_\ell} m_\ell!}.$$

For this fixed component pattern, the corresponding contribution to Z_g/Z_0 is therefore

$$\frac{1}{n!}\,\frac{n!}{\prod_{\ell\geq 1}(\ell!)^{m_\ell} m_\ell!}\prod_{\ell\geq 1}(G^{(\ell)})^{m_\ell} = \prod_{\ell\geq 1}\frac{1}{m_\ell!}\left(\frac{1}{\ell!}G^{(\ell)}\right)^{m_\ell}.$$

Summing over all nonnegative integers m_ℓ gives the linked-cluster formula

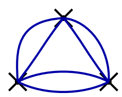
$$\frac{Z_g}{Z_0} = \sum_{\{m_\ell\}}\prod_{\ell\geq 1}\frac{1}{m_\ell!}\left(\frac{1}{\ell!}G^{(\ell)}\right)^{m_\ell} = \exp\left[\sum_{\ell\geq 1}\frac{1}{\ell!}G^{(\ell)}\right].$$

Equivalently, $\log Z_g$ is the sum of connected vacuum diagrams. The factor $1/\ell!$ is often absorbed into the symmetry factor assigned to the connected graph. □

For example, the connected three-vertex topology with two propagators on each edge has

$$N_{\triangle} = \binom{4}{2}^3 (2!)^3 = 1728$$

Wick-contraction assignments: at each vertex one chooses which two of the four half-edges go to one neighboring vertex, and along each of the three double edges one pairs the two half-edges. Its contribution to $\frac{1}{3!}G^{(3)}$ is therefore



$$\begin{aligned}\frac{1}{3!}\left(-\frac{g}{4!}\right)^3 N_{\triangle} \int \mathrm{d}\tau_1\,\mathrm{d}\tau_2\,\mathrm{d}\tau_3\, G_0(\tau_1,\tau_2)^2 G_0(\tau_2,\tau_3)^2 G_0(\tau_3,\tau_1)^2 \\ = -\frac{g^3}{48}\int \mathrm{d}\tau_1\,\mathrm{d}\tau_2\,\mathrm{d}\tau_3\, G_0(\tau_1,\tau_2)^2 G_0(\tau_2,\tau_3)^2 G_0(\tau_3,\tau_1)^2,\end{aligned}$$

where the last equality uses the normalization 4! in the interaction vertex. Translation invariance leaves one unrestricted center-of-time integral, so every connected vacuum graph is proportional to the total Euclidean time interval.

On a finite Euclidean interval of length T_{tot}, with endpoint values q_i and q_f, the path integral is the transition kernel

$$Z_g(T_{\mathrm{tot}}; q_f, q_i) = \langle q_f | \mathrm{e}^{-T_{\mathrm{tot}}\widehat{H}} | q_i \rangle.$$

Proposition 8.16 (Ground-state energy from the Euclidean kernel)**.** *Assume that the anharmonic oscillator Hamiltonian has discrete spectrum $E_0(g) < E_1(g) \le \cdots$, with nondegenerate ground state ψ_0. If the endpoint data used in the transition kernel have nonzero overlap with ψ_0, then*

$$E_0(g) = -\lim_{T_{\mathrm{tot}}\to\infty} \frac{1}{T_{\mathrm{tot}}} \log Z_g, \qquad E_0(g) - E_0(0) = -\lim_{T_{\mathrm{tot}}\to\infty} \frac{1}{T_{\mathrm{tot}}} \log \frac{Z_g}{Z_0}.$$

The first perturbative correction is

$$\Delta E_0 = \frac{g}{32} + O(g^2)$$

for $\omega = 1$.

Proof. For the anharmonic oscillator the spectrum is discrete and the ground state is nondegenerate. If the endpoint wavefunctions have nonzero overlap with the ground state, the spectral theorem gives

$$Z_g(T_{\mathrm{tot}}; q_f, q_i) = \psi_0(q_f)\overline{\psi_0(q_i)}\mathrm{e}^{-E_0(g)T_{\mathrm{tot}}} + O(\mathrm{e}^{-E_1(g)T_{\mathrm{tot}}}),$$

and hence

$$E_0(g) = -\lim_{T_{\mathrm{tot}}\to\infty} \frac{1}{T_{\mathrm{tot}}} \log Z_g.$$

The endpoint wavefunction factors contribute only $O(1)$ to $\log Z_g$, so they disappear after multiplication by T_{tot}^{-1}. Taking the difference between the interacting and free kernels gives

$$E_0(g) - E_0(0) = -\lim_{T_{\text{tot}}\to\infty} \frac{1}{T_{\text{tot}}} \log \frac{Z_g}{Z_0}.$$

The first-order term in (8.1) is connected, hence it is also the first-order term in $\log(Z_g/Z_0)$. On an interval of length T_{tot},

$$\frac{1}{T_{\text{tot}}} \log \frac{Z_g}{Z_0} = -\frac{g}{8} G_0(0)^2 + O(g^2).$$

The sign of the energy shift therefore comes from the minus sign in $E_0 = -T_{\text{tot}}^{-1} \log Z$ together with the per-vertex factor $-g/4!$ in (8.1). Thus

$$\Delta E_0 = \frac{g}{8} G_0(0)^2 + O(g^2) = \frac{g}{32} + O(g^2),$$

where the second equality uses $\omega = 1$; with general oscillator frequency the same term is $g(2\omega)^{-2}/8$. This agrees with first-order Hamiltonian perturbation theory:

$$\langle \psi_0 | \frac{g}{4!} \widehat{q}^4 | \psi_0 \rangle = \frac{g}{4!}\, 3 \left(\frac{1}{2}\right)^2 = \frac{g}{32}.$$

$\square$

8.7 Connected Two-Point Functions and Self-Energy

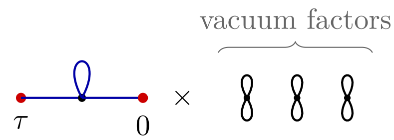
The normalized Euclidean two-point function is

$$G(\tau) = \langle q(\tau) q(0) \rangle_g = \frac{1}{Z_g} \int [Dq]\, \mathrm{e}^{-S_0[q] - S_{\text{int}}[q]}\, q(\tau) q(0).$$

Equivalently, at the regulated level,

$$G(\tau) = \frac{1}{Z_g} \left\langle q(\tau) q(0) \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{g}{4!} \int q(\tau')^4 \, \mathrm{d}\tau' \right)^n \right\rangle_0 .$$

Vacuum components in normalized two-point functions. At fixed cutoff and as a formal power series in g, the normalized two-point function is the sum of Wick diagrams whose connected component contains both external insertions. At fixed cutoff every coefficient of the formal series is a finite Gaussian moment, hence Wick's theorem expresses it as a finite sum over pairings. View a pairing as a graph whose vertices are the two external insertions and the interaction insertions, with an edge for each Wick contraction. Every such graph decomposes uniquely into connected components. The component containing the two external vertices carries all dependence on the external points; the remaining components contain interaction vertices only and are vacuum graphs. Conversely, a connected two-point graph together with an unordered finite collection of vacuum components determines a unique disconnected graph of the normalized numerator, with the exponential generating-function factors. Summing over all vacuum components therefore produces the same formal power series Z_g that appears in the denominator. Division by Z_g cancels this vacuum factor coefficientwise, leaving exactly the Wick graphs whose connected component contains both external insertions.



Consequently the normalized correlator is represented by connected diagrams with the two external insertions fixed at τ and 0. Figure 8.2 displays precisely the connected topologies that survive the division by $Z[0]$, so the vacuum factors drawn in the preceding bookkeeping figure are no longer present: The same path integral, with the same Euclidean ordering convention, computes

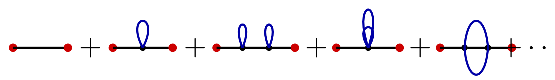


Figure 8.2: First connected topologies contributing to the normalized two-point function. Red endpoints mark the fixed insertions $q(\tau)$ and $q(0)$; black vertices are interaction insertions; blue arcs are contracted oscillator fields. Vacuum components are absent because they cancel against Z_g.

$\langle 0|\widehat{q}_E(0)\widehat{q}_E(\tau)|0\rangle$ when $\tau < 0$. Thus

$$G(\tau) = \begin{cases} \langle 0|\widehat{q}_E(\tau)\widehat{q}_E(0)|0\rangle, & \tau > 0, \\ \langle 0|\widehat{q}_E(0)\widehat{q}_E(\tau)|0\rangle, & \tau < 0. \end{cases}$$

At zeroth order,

$$G_0(\tau) = \frac{1}{2}\mathrm{e}^{-|\tau|} = \int_{-\infty}^{\infty} \frac{\mathrm{d}k}{2\pi}\,\frac{\mathrm{e}^{\mathrm{i}k\tau}}{k^2+1}.$$

At order g, the connected tadpole correction is

$$-\frac{g}{2}\int \mathrm{d}\tau'\, G_0(\tau-\tau')G_0(\tau')G_0(0).$$

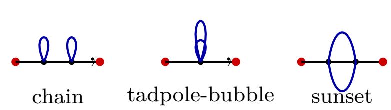
In momentum space this is

$$\begin{aligned} -\frac{g}{2}\int \mathrm{d}\tau' \int \frac{\mathrm{d}k}{2\pi}\frac{\mathrm{d}k'}{2\pi}\frac{\mathrm{d}k''}{2\pi}\,\widetilde{G}_0(k)\widetilde{G}_0(k')\widetilde{G}_0(k'')\,\mathrm{e}^{\mathrm{i}k(\tau-\tau')+\mathrm{i}k'\tau'} \\ = -\frac{g}{2}\left(\int \frac{\mathrm{d}k''}{2\pi}\,\widetilde{G}_0(k'')\right)\int \frac{\mathrm{d}k}{2\pi}\,\frac{\mathrm{e}^{\mathrm{i}k\tau}}{(k^2+1)^2}. \end{aligned}$$

Since $\int \mathrm{d}k''\,\widetilde{G}_0(k'')/(2\pi) = G_0(0) = 1/2$, this becomes

$$-\frac{g}{4}\int_{-\infty}^{\infty} \frac{\mathrm{d}k}{2\pi}\,\frac{\mathrm{e}^{\mathrm{i}k\tau}}{(k^2+1)^2}.$$

At second order the connected two-point graphs include



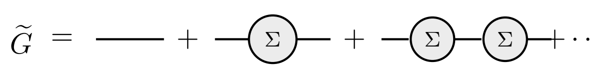
The first diagram is a chain of two one-loop insertions; the other two are the second-order amputated insertions that enter the self-energy below.

Definition 8.17 (Euclidean 1PI two-point insertion)**.** A connected two-point diagram is one-particle-irreducible, abbreviated 1PI, when it remains connected after cutting any one internal propagator. In this chapter the term is graph-theoretic and refers to Euclidean Green-function diagrams. Let $\Sigma(k)$ be the sum of amputated 1PI two-point insertions in momentum space, with the external free propagators removed.

With the sign convention of Definition 8.17, the full two-point function is the geometric series

$$\widetilde{G}(k) = \widetilde{G}_0(k)\sum_{r=0}^{\infty}\big(\widetilde{G}_0(k)\Sigma(k)\big)^r = \frac{1}{k^2+1-\Sigma(k)}, \qquad \widetilde{G}_0(k) = \frac{1}{k^2+1}.$$

Every connected two-point graph is obtained uniquely by arranging a finite chain of amputated 1PI two-point insertions along the external line and inserting one free propagator between consecutive amputated insertions. Cutting the propagators between adjacent insertions gives the unique chain decomposition; conversely gluing such a chain gives a connected two-point graph. Translation invariance diagonalizes the convolution in momentum space, so the chain with r insertions contributes $\widetilde{G}_0(k)(\widetilde{G}_0(k)\Sigma(k))^r$. Summing the formal geometric series gives the displayed denominator.



For the anharmonic oscillator, the two non-chain order-g^2 contributions are as follows. The perturbative coefficient before Wick contraction is $g^2/(2(4!)^2)$. For the tadpole-bubble topology, both external fields are contracted to one interaction vertex, two lines connect the two vertices, and the remaining two fields at the second vertex contract with each other. The number of such contractions is

$$2\cdot(4\cdot 3)\cdot\binom{4}{2}\,2! = 288,$$

so the net symmetry factor is $288/[2(4!)^2] = 1/4$. With

$$G_0(0) = \int_{-\infty}^{\infty}\frac{\mathrm{d}p}{2\pi}\,\widetilde{G}_0(p) = \frac{1}{2}, \qquad \int_{-\infty}^{\infty}\frac{\mathrm{d}p}{2\pi}\,\widetilde{G}_0(p)^2 = \frac{1}{4},$$

this topology contributes

$$\Sigma^{(2)}_{\mathrm{tb}}(k) = \frac{g^2}{4}\,G_0(0)\int_{-\infty}^{\infty}\frac{\mathrm{d}p}{2\pi}\,\widetilde{G}_0(p)^2 = \frac{g^2}{32} = \frac{g^2}{8}\,\frac{1}{4}.$$

For the sunset topology, the two external fields attach to different vertices and the remaining three legs at one vertex are paired with the remaining three legs at the other. The contraction count is

$$2\cdot 4\cdot 4\cdot 3! = 192,$$

so the symmetry factor is $192/[2(4!)^2] = 1/6$. Its momentum integral is the three-fold convolution

$$\begin{aligned}I_{\mathrm{sun}}(k) &= \int\frac{\mathrm{d}p}{2\pi}\frac{\mathrm{d}q}{2\pi}\,\widetilde{G}_0(p)\widetilde{G}_0(q)\widetilde{G}_0(k-p-q)\\ &= \int_{-\infty}^{\infty}\mathrm{d}\tau\,\mathrm{e}^{-\mathrm{i}k\tau}G_0(\tau)^3 = \frac{1}{8}\int_{-\infty}^{\infty}\mathrm{d}\tau\,\mathrm{e}^{-\mathrm{i}k\tau}\mathrm{e}^{-3|\tau|} = \frac{3}{4}\,\frac{1}{k^2+9}.\end{aligned}$$

Thus

$$\Sigma^{(2)}_{\text{sun}}(k)=\frac{g^2}{6}I_{\text{sun}}(k)=\frac{g^2}{8}\,\frac{1}{k^2+9}.$$

The momentum-independent term $\frac{g^2}{32}$ is the tadpole-bubble, equivalently the double-bubble insertion after external amputation. The term proportional to $(k^2+9)^{-1}$ is the sunset insertion. Its denominator is not an arbitrary algebraic remnant: the three-fold convolution is the Fourier transform of $\frac{1}{8}\mathrm{e}^{-3|\tau|}$, so its nearest singularities are $k=\pm 3\mathrm{i}$, or $k^2=-9=-(3\omega)^2$ with $\omega=1$. In the spectral language of this oscillator computation, this is the intermediate three-quantum energy carried by the sunset topology. Combining the one-loop term with these two order-g^2 contributions gives

$$\Sigma(k)=-\frac{g}{4}+\frac{g^2}{8}\left(\frac{1}{4}+\frac{1}{k^2+9}\right)+O(g^3).$$

The external free propagators are not part of Σ; in this convention the order-g term is the amputated tadpole, while the two displayed order-g^2 terms come from the two non-chain second-order topologies shown above after their external legs are amputated. The chain of two one-loop insertions is instead produced by the geometric series in $\widetilde{G}_0(k)\Sigma(k)$. Consequently

$$\widetilde{G}(k)=\frac{1}{k^2+1+\frac{g}{4}-\frac{g^2}{8}\left(\frac{1}{4}+\frac{1}{k^2+9}\right)+O(g^3)}.$$

Proposition 8.18 (Pole extraction of the first energy gap)**.** *For the anharmonic oscillator in the convention above, the pole of $\widetilde{G}(k)$ near $k=\mathrm{i}$ gives*

$$E_1-E_0=1+\frac{g}{8}-\frac{g^2}{32}+O(g^3).$$

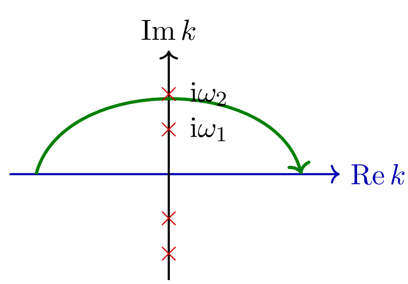
Proof. The spectral expansion says that for $\tau>0$,

$$G(\tau)=\sum_n |\langle n|\widehat{q}|0\rangle|^2\,\mathrm{e}^{-(E_n-E_0)\tau}.$$

The Fourier representation

$$G(\tau)=\int_{-\infty}^{\infty}\frac{\mathrm{d}k}{2\pi}\,\widetilde{G}(k)\mathrm{e}^{\mathrm{i}k\tau}$$

is evaluated for $\tau>0$ by closing the contour in the upper half-plane, where $\mathrm{e}^{\mathrm{i}k\tau}$ decays on the large semicircle.



If $\widetilde{G}(k)$ has simple poles at $k=\mathrm{i}\omega_n$, then

$$G(\tau)=\mathrm{i}\sum_n \operatorname*{Res}_{k=\mathrm{i}\omega_n}\widetilde{G}(k)\,\mathrm{e}^{-\omega_n\tau}.$$

Thus the pole positions encode the energy gaps $\omega_n = E_n - E_0$ in the $\hbar = 1$ convention.

It remains to solve the pole equation near $k = \mathrm{i}$. Write $\omega = 1 + ag + bg^2 + O(g^3)$ and set $k = \mathrm{i}\omega$. The denominator of $\widetilde{G}(k)$ is

$$-\omega^2 + 1 + \frac{g}{4} - \frac{g^2}{8}\left(\frac{1}{4} + \frac{1}{9 - \omega^2}\right) + O(g^3).$$

Since $9 - \omega^2 = 8 + O(g)$, the $O(g^2)$ part of the last factor is $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$. Also $\omega^2 = 1 + 2ag + (a^2 + 2b)g^2 + O(g^3)$. Vanishing of the denominator gives

$$-2a + \frac{1}{4} = 0, \qquad -(a^2 + 2b) - \frac{3}{64} = 0.$$

Thus $a = \frac{1}{8}$ and $b = -\frac{1}{32}$, proving the formula. □

8.8 Derivative Interactions, Regulators, and Counterterms

Consider the Lagrangian

$$L(q, \dot{q}) = \frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2 - \frac{g}{8}q^2\dot{q}^2.$$

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{q}} = \dot{q}\left(1 - \frac{g}{4}q^2\right),$$

and the corresponding Hamiltonian function is

$$H(q, p) = \frac{1}{2}\left(1 - \frac{g}{4}q^2\right)^{-1} p^2 + \frac{1}{2}q^2.$$

Already at the classical level this Hamiltonian shows the point that will matter quantum mechanically: the product of the q-dependent coefficient and p^2 has no unique operator meaning until an ordering prescription, or equivalently a regulated path-integral prescription, has been specified.

Remark 8.19 (Classical field redefinition check). Define a new coordinate y, to first order in g, by

$$\frac{\mathrm{d}y}{\mathrm{d}q} = \left(1 - \frac{g}{4}q^2\right)^{1/2}, \qquad y = q - \frac{g}{24}q^3 + O(g^2).$$

Then $q = y + \frac{g}{24}y^3 + O(g^2)$, and the Lorentzian Lagrangian becomes

$$L = \frac{1}{2}\dot{y}^2 - \left(\frac{1}{2}y^2 + \frac{g}{24}y^4\right) + O(g^2).$$

Thus the derivative-coupled oscillator is classically equivalent, through order g, to an ordinary anharmonic oscillator. This is a classical coordinate calculation: The defining equation gives $\dot{y} = (1 - \frac{g}{4}q^2)^{1/2}\dot{q}$, hence

$$\frac{1}{2}\dot{q}^2 - \frac{g}{8}q^2\dot{q}^2 = \frac{1}{2}\left(1 - \frac{g}{4}q^2\right)\dot{q}^2 = \frac{1}{2}\dot{y}^2 + O(g^2).$$

Inverting $y = q - \frac{g}{24}q^3 + O(g^2)$ gives $q = y + \frac{g}{24}y^3 + O(g^2)$. Therefore

$$-\frac{1}{2}q^2 = -\frac{1}{2}y^2 - \frac{g}{24}y^4 + O(g^2),$$

which proves the displayed Lagrangian. The quantum computation below keeps the q-coordinate and shows where the ordering or path-integral prescription enters.

The Euclidean Lagrangian is

$$L_E(q,q') = \frac{1}{2}(q')^2 + \frac{1}{2}q^2 - \frac{g}{8}q^2(q')^2.$$

Using

$$q^2(q')^2 = \frac{1}{3}\frac{\mathrm{d}}{\mathrm{d}\tau}(q^3q') - \frac{1}{3}q^3q'',$$

the interaction may also be represented, up to endpoint terms, by

$$\frac{g}{24}q^3q''.$$

At first order in g, the two-point function contains contractions in which derivatives act on propagators at coincident points. A representative term is

$$\frac{g}{4}\int \mathrm{d}\tau'\, G_0(\tau-\tau')G_0(\tau')\ \left[-\partial_{\tau'}^2 G_0(\tau'-\tau'')\right]\big|_{\tau''=\tau'}.$$

The derivative is taken with respect to one endpoint of the propagator before the coincident-point limit $\tau'' \to \tau'$ is imposed. In momentum space this contains

$$\int \frac{\mathrm{d}k'}{2\pi}\frac{k'^2}{k'^2+1},$$

which depends on the high-frequency definition of the path integral. Other contractions in the same Wick expansion are finite or vanish by parity. For example, with $G_0(k) = 1/(k^2+1)$, one obtains terms of the form

$$\frac{g}{4}\int \frac{\mathrm{d}k}{2\pi}\frac{\mathrm{e}^{\mathrm{i}k\tau}k^2}{(k^2+1)^2}\int\frac{\mathrm{d}k'}{2\pi}\frac{1}{k'^2+1},$$

and

$$-\frac{g}{2}\int \frac{\mathrm{d}k}{2\pi}\frac{\mathrm{e}^{\mathrm{i}k\tau}\mathrm{i}k}{(k^2+1)^2}\int\frac{\mathrm{d}k'}{2\pi}\frac{\mathrm{i}k'}{k'^2+1} = 0.$$

The divergent term is local in the external momentum expansion; this is why a local counterterm can absorb it.

Figure 8.3 separates the derivative contractions according to their ultraviolet role: loop-derivative contractions produce the local divergent term, and the parity-odd loop contraction vanishes.

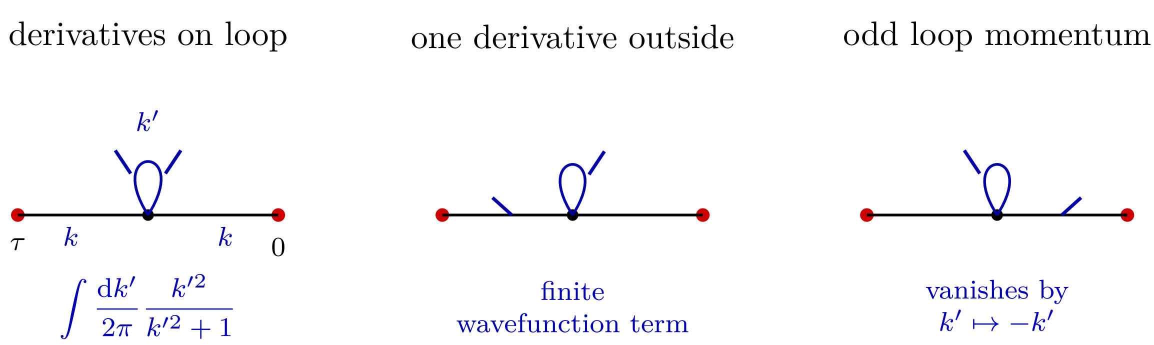


Figure 8.3: Derivative interactions produce derivatives acting on Wick contractions. When both derivatives act on the internal loop, the coincident-point contraction is sensitive to the high-frequency regulator. After cutoff regularization, the loop momentum k' in the first panel is restricted by $|k'| < \Lambda$. Other first-order contractions are finite or vanish by parity.

The divergent expression is therefore not a value assigned by an already defined continuum measure. It is a signal that the formal product $[Dq] = \prod_k \mathrm{d}\widetilde{q}(k)$ must first be replaced by a regulated finite-dimensional measure, and that the limiting prescription must be specified together with local counterterms.

Definition 8.20 (Frequency cutoff for the derivative interaction)**.** Define a frequency cutoff by

$$q(\tau) = \int_{|k|<\Lambda} \frac{\mathrm{d}k}{2\pi}\, \widetilde{q}(k) \mathrm{e}^{\mathrm{i}k\tau}, \qquad [Dq]_\Lambda = \prod_{|k|<\Lambda} \mathrm{d}\widetilde{q}(k),$$

with the real-field condition $\widetilde{q}(k)^* = \widetilde{q}(-k)$. After the finite-dimensional real slice has been chosen, $[Dq]_\Lambda$ denotes the corresponding finite product Lebesgue measure. The interaction is expanded before $\Lambda \to \infty$.

Local counterterm for the derivative interaction. After restoring $\hbar$, the first-order self-energy has the form

$$\Sigma_\Lambda(k) = \frac{\hbar g}{4}\left(\frac{\Lambda}{\pi} + \frac{k^2}{2} - \frac{1}{2} + O(\Lambda^{-1})\right) + O(\hbar^2 g^2).$$

A local Euclidean counterterm

$$\Delta L_E = C\, g\, \hbar\, \frac{q^2}{2}$$

adds $-Cg\hbar$ to $\Sigma_\Lambda(k)$, so that

$$\Sigma_\Lambda(k) = \hbar g\left(\frac{k^2}{8} + \frac{\Lambda}{4\pi} - \frac{1}{8} - C + O(\Lambda^{-1})\right) + O(g^2).$$

Choosing

$$C = \frac{\Lambda}{4\pi} + C_{\mathrm{fin}}$$

defines a finite limiting two-point function as $\Lambda \to \infty$. The finite constant C_{fin} is part of the quantum definition of the Hamiltonian represented by this regulated Lagrangian path integral. The contraction with two derivatives acting on the loop gives

$$\frac{g}{4}\int \frac{\mathrm{d}k}{2\pi}\frac{\mathrm{e}^{\mathrm{i}k\tau}}{(k^2+1)^2}\int_{|k'|<\Lambda}\frac{\mathrm{d}k'}{2\pi}\frac{k'^2}{k'^2+1}.$$

The cutoff integral is

$$\int_{-\Lambda}^{\Lambda}\frac{\mathrm{d}k'}{2\pi}\frac{k'^2}{k'^2+1} = \frac{1}{\pi}\int_0^\Lambda\left(1 - \frac{1}{k'^2+1}\right)\mathrm{d}k' = \frac{\Lambda - \arctan\Lambda}{\pi}.$$

The remaining first-order contractions are the finite even momentum term and the odd momentum term displayed above. Combining them and expanding $\arctan\Lambda = \pi/2 + O(\Lambda^{-1})$ gives the stated self-energy. The counterterm contributes a local two-point insertion $-Cg\hbar$ in the denominator convention used for Σ. Hence it shifts $\Sigma_\Lambda(k)$ by $-Cg\hbar$. Choosing $C = \Lambda/(4\pi) + C_{\mathrm{fin}}$ removes the linearly divergent local part. The remaining finite family is the one displayed above.

The pole of the regulated two-point function is read from the denominator of

$$\widetilde{G}_\Lambda(k) = \frac{\hbar}{k^2 + 1 - \Sigma_\Lambda(k)}.$$

Finite gap after choosing the finite local term. After the counterterm choice $C = \Lambda/(4\pi) + C_{\text{fin}}$, the first energy gap is

$$\omega_1 = 1 + \hbar g \left(\frac{C_{\text{fin}}}{2} + \frac{1}{8} \right) + O(g^2).$$

Setting $k = \mathrm{i}\omega_1$ in the denominator of $\widetilde{G}_\Lambda(k)$ and expanding to first order in g gives

$$\omega_1 = 1 + \hbar g \left(-\frac{\Lambda}{8\pi} + \frac{C}{2} + \frac{1}{8} \right) + O(g^2) = 1 + \hbar g \left(\frac{C_{\text{fin}}}{2} + \frac{1}{8} \right) + O(g^2).$$

Thus the finite part of the local counterterm is not a removable notation choice; it fixes which quantum Hamiltonian, within the ordering ambiguity of the derivative-coupled classical expression, the path integral represents. Once that quantum definition is fixed, the pole position is an ordinary observable.

Fixing C_{fin} requires physical input, for instance a renormalization condition on the energy gap or an independently specified operator ordering of the Hamiltonian. In a relativistic field theory the analogous finite local terms are constrained by Poincare covariance, locality/microcausality, internal symmetries, and the chosen normalization conditions on observables.

The structural statement carried forward to field theory is this: a path integral with interactions is a regulated construction of Green functions, and local counterterms are part of the data specifying the regulated construction and its limiting prescription. The next chapter replaces the single coordinate $q(\tau)$ by fields $\phi(x)$ and imposes relativistic covariance and locality on the resulting Green functions.

Chapter 9

Relativistic Scalar Fields and Canonical Quantization

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

9.1 Fields and Variational Equations

Definition 9.1 (Classical scalar Lagrangian field datum)**.** Let $\mathbb{M}^D$ be D-dimensional Minkowski spacetime with metric

$$\eta_{\mu\nu} = \mathrm{diag}(-1,+1,\dots,+1).$$

Write

$$x^\mu = (x^0, \vec{x}), \qquad x^0 = t, \qquad \vec{x} \in \mathbb{R}^{D-1}.$$

A real classical scalar field is a real-valued function

$$\phi : \mathbb{M}^D \to \mathbb{R}.$$

The field-theoretic analogue of finitely many coordinates $q^i(t)$ is the family of coordinates $\phi(t, \vec{x})$, labelled continuously by $\vec{x}$. A Lagrangian field theory is specified by an action functional

$$S[\phi] = \int_{\mathbb{M}^D} \mathrm{d}^D x\, \mathcal{L}\big(\phi(x), \partial_\mu \phi(x)\big),$$

where $\mathcal{L}$ is a Lorentz scalar Lagrangian density in the examples below.

For a compactly supported variation $\delta\phi$,

$$\begin{aligned}\delta S &= \int \mathrm{d}^D x \left[\frac{\partial \mathcal{L}}{\partial \phi}\,\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\,\partial_\mu \delta\phi\right] \\ &= \int \mathrm{d}^D x\,\delta\phi \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right].\end{aligned}$$

The Euler–Lagrange equation is therefore

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0.$$

The first equality is the chain rule applied pointwise to $\mathcal{L}(\phi, \partial\phi)$. The second equality follows by integrating $\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\partial_\mu \delta\phi$ by parts. Compact support of $\delta\phi$, or the boundary condition class chosen for the variational problem, removes the boundary term. Since $\delta\phi$ is otherwise arbitrary, the fundamental lemma of the calculus of variations gives the displayed Euler–Lagrange equation as a distributional identity.

9.2 The Free Massive Scalar

Definition 9.2 (Free real Klein–Gordon field)**.** The free massive real scalar field of mass $m \geq 0$ has Lagrangian density

$$\mathcal{L} = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i \phi)^2 - \frac{1}{2}m^2\phi^2.$$

Equivalently,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \phi\,\partial^\mu \phi - \frac{1}{2}m^2\phi^2, \qquad \partial^\mu = \eta^{\mu\nu}\partial_\nu.$$

The Euler–Lagrange equation is the Klein–Gordon equation

$$(\partial^\mu \partial_\mu - m^2)\phi = 0.$$

Use the Fourier convention

$$\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\,\widetilde{\phi}(k)\,\mathrm{e}^{\mathrm{i}k\cdot x}, \qquad k \cdot x = \eta_{\mu\nu}k^\mu x^\nu = -k^0 x^0 + \vec{k}\cdot\vec{x}.$$

The equation of motion implies

$$(k^2 + m^2)\widetilde{\phi}(k) = 0, \qquad k^2 = \eta_{\mu\nu}k^\mu k^\nu.$$

Thus $\widetilde{\phi}(k)$ is supported on the mass shell

$$k^2 + m^2 = 0, \qquad k^0 = \pm\omega_{\vec{k}}, \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^{\,2} + m^2}.$$

With $k \cdot x = -k^0 x^0 + \vec{k}\cdot\vec{x}$, one has

$$\partial^\mu \partial_\mu \mathrm{e}^{\mathrm{i}k\cdot x} = -\big(k^2\big)\mathrm{e}^{\mathrm{i}k\cdot x}.$$

The Klein–Gordon equation therefore becomes $-(k^2+m^2)\widetilde{\phi}(k)=0$ in Fourier space. In distribution language this says that $\widetilde{\phi}$ is supported where $k^2+m^2=0$. Since $k^2=-(k^0)^2+\vec{k}^{\,2}$, this hypersurface is $(k^0)^2=\vec{k}^{\,2}+m^2$, hence the two branches $k^0=\pm\omega_{\vec{k}}$.

Figure 9.1 displays the two branches of this support condition and fixes the positive-frequency branch used in the canonical expansion below.

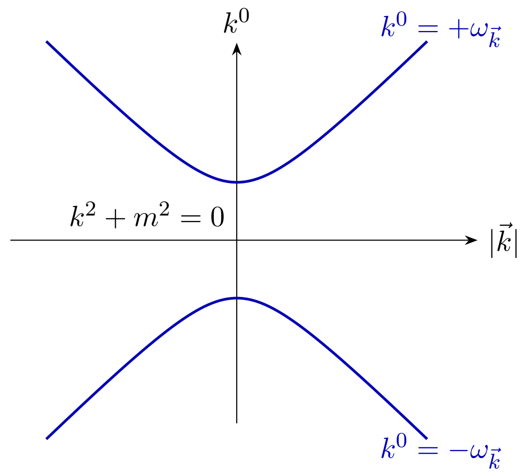


Figure 9.1: Mass shell for a real scalar solution. The Fourier transform $\widetilde{\phi}(k)$ is supported on the two branches $k^0=\pm\omega_{\vec{k}}$ of $k^2+m^2=0$.

Cauchy data and positive-frequency coefficients. A real solution can be written as

$$\phi(t,\vec{x})=\int\frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}}\left[A(\vec{k})\,\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}-\mathrm{i}\omega_{\vec{k}}t}+A(\vec{k})^*\,\mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x}+\mathrm{i}\omega_{\vec{k}}t}\right],$$

with a complex coefficient function $A(\vec{k})$ subject to whatever regularity conditions are imposed on the classical solution. If $m=0$, the spatial zero mode $\vec{k}=0$ is an infrared mode and is treated separately by the chosen boundary condition, volume regulator, or test-function class; the displayed division by $\omega_{\vec{k}}$ applies on the modes with $\omega_{\vec{k}}>0$. The Cauchy data

$$\phi(0,\vec{x}),\qquad \partial_t\phi(0,\vec{x})$$

determine $A(\vec{k})$ and $A(\vec{k})^*$, hence determine the free solution for all t. Explicitly, write

$$\phi(0,\vec{x})=\int\frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}}\,\widetilde{\phi}_0(\vec{k})\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}},\qquad \partial_t\phi(0,\vec{x})=\int\frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}}\,\widetilde{\pi}_0(\vec{k})\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}}.$$

Then

$$\widetilde{\phi}_0(\vec{k})=A(\vec{k})+A(-\vec{k})^*,\qquad \widetilde{\pi}_0(\vec{k})=-\mathrm{i}\omega_{\vec{k}}\big(A(\vec{k})-A(-\vec{k})^*\big),$$

so that

$$A(\vec{k})=\frac{1}{2}\left(\widetilde{\phi}_0(\vec{k})+\frac{\mathrm{i}}{\omega_{\vec{k}}}\widetilde{\pi}_0(\vec{k})\right).$$

Evaluate the positive- and negative-frequency expansion at $t=0$ and also differentiate it once in time before setting $t=0$. Reality of ϕ identifies the coefficient of $\mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x}}$ with the complex conjugate coefficient at momentum $\vec{k}$, equivalently $A(-\vec{k})^*$ in the $e^{i\vec{k}\cdot\vec{x}}$ coefficient. This gives the two displayed linear equations for $A(\vec{k})$ and $A(-\vec{k})^*$. Adding $\widetilde{\phi}_0(\vec{k})$ to $i\widetilde{\pi}_0(\vec{k})/\omega_{\vec{k}}$ solves the system and gives the stated formula. The Klein–Gordon equation is second order in time, so these two Cauchy data determine the solution on every time slice.

The displayed coefficient formula is a Fourier-coordinate expression for the free Cauchy problem. Questions about interacting fields and the domain of the quantum Hamiltonian belong to later constructions.

There are two constructions to keep distinct. Canonical quantization starts from the equal-time phase space (ϕ, Π) and a Hamiltonian. The Lagrangian path-integral construction is defined by first choosing a regulator that replaces the field by finitely many coordinates or modes:

$$\int_{\mathcal{C}_R} d\mu_R(\phi_R)\, \exp\left(\frac{i}{\hbar} S_R[\phi_R]\right).$$

Here R labels the chosen cutoff, $\mathcal{C}_R$ is the regulated configuration space, $d\mu_R$ is the corresponding finite-dimensional or lattice measure, and S_R is the regulated action. With the Lorentzian phase $\exp(iS_R/\hbar)$, this regulated object is an oscillatory integral or oscillatory distribution. For quadratic actions it is a Fresnel integral with a determinant magnitude and a phase fixed by the signature of the quadratic form; for general actions, stationary phase and distributional regularization are part of the definition. The continuum symbol $\int [D\phi] \exp(iS[\phi]/\hbar)$ denotes a specified limit or asymptotic expansion of these regulated oscillatory objects. This chapter develops the canonical construction; the scalar field path integral is introduced later.

9.3 Canonical Hamiltonian Formalism

Construction 9.3 (Canonical phase-space variables and Hamiltonian density)**.** Let

$$L(t) = \int d^{D-1}\vec{x}\, \mathcal{L}.$$

The canonical momentum density conjugate to $\phi(t, \vec{x})$ is

$$\Pi(t, \vec{x}) = \frac{\partial \mathcal{L}}{\partial(\partial_t \phi(t, \vec{x}))}.$$

For the free scalar,

$$\Pi(t, \vec{x}) = \partial_t \phi(t, \vec{x}).$$

This agrees with the covariant form of the Lagrangian. Treating the components $\partial_\mu \phi$ as independent variables in $-\frac{1}{2}\eta^{\alpha\beta}\partial_\alpha \phi\, \partial_\beta \phi$, one finds

$$\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = -\eta^{0\nu}\partial_\nu \phi = -\partial^0 \phi = \partial_0 \phi = \partial_t \phi,$$

because the mostly-plus metric gives $\partial^0 = \eta^{00}\partial_0 = -\partial_0$. The two minus signs, one from the covariant kinetic term and one from raising the time index, therefore cancel. The Hamiltonian density is

$$\mathcal{H} = \Pi\, \partial_t \phi - \mathcal{L},$$

and therefore

$$\mathcal{H} = \frac{1}{2}\Pi^2 + \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2 + \frac{1}{2}m^2\phi^2$$

for the free scalar field. The Hamiltonian is

$$H = \int \mathrm{d}^{D-1}\vec{x}\,\mathcal{H}.$$

This is a classical Hamiltonian density. After quantization $\widehat{\phi}(t,\vec{x})$ and $\widehat{\Pi}(t,\vec{x})$ are operator-valued distributions, so the pointwise products in $\frac{1}{2}(\widehat{\Pi}^2+(\nabla\widehat{\phi})^2+m^2\widehat{\phi}^2)$ are first understood with a spatial regulator or after smearing. The free Hamiltonian used below is the regulator limit of the normal-ordered quadratic Hamiltonian, equivalently the operator obtained by subtracting the vacuum scalar term from the regulated quadratic expression. We write this operator as $\widehat{H}_0^{\mathrm{ren}} =: \widehat{H}_0 :$ in the flat-space free theory.

Definition 9.4 (Equal-time scalar Poisson bracket)**.** For functionals $F[\phi,\Pi]$ and $G[\phi,\Pi]$, use the equal-time Poisson bracket

$$\{F,G\} = \int \mathrm{d}^{D-1}\vec{x}\left[\frac{\delta F}{\delta\phi(\vec{x})}\frac{\delta G}{\delta\Pi(\vec{x})} - \frac{\delta F}{\delta\Pi(\vec{x})}\frac{\delta G}{\delta\phi(\vec{x})}\right].$$

Then

$$\{\phi(\vec{x}),\Pi(\vec{y})\} = \delta^{(D-1)}(\vec{x}-\vec{y}),$$

with the ϕ-ϕ and Π-Π brackets equal to zero. Hamilton's equation is

$$\partial_t F = \{F,H\}.$$

Hamilton equations for the scalar field. Using the functional derivatives of H,

$$\frac{\delta H}{\delta\Pi(\vec{x})} = \Pi(\vec{x}), \qquad \frac{\delta H}{\delta\phi(\vec{x})} = -\sum_i \partial_i^2\phi(\vec{x}) + m^2\phi(\vec{x}),$$

where the spatial derivative term has been integrated by parts. Hence

$$\partial_t\phi = \{\phi,H\} = \Pi, \qquad \partial_t\Pi = \{\Pi,H\} = \sum_i \partial_i^2\phi - m^2\phi.$$

Eliminating Π gives $\partial_t^2\phi - \nabla^2\phi + m^2\phi = 0$, equivalently $(\partial^\mu\partial_\mu - m^2)\phi = 0$ with the mostly-plus convention.

9.4 Equal-Time Quantization

Definition 9.5 (Equal-time canonical commutation relations)**.** Canonical quantization promotes the classical fields to operator-valued distributions

$$\widehat{\phi}(t,\vec{x}), \qquad \widehat{\Pi}(t,\vec{x})$$

on a common dense domain. Their equal-time commutation relations are

$$[\widehat{\phi}(t,\vec{x}),\widehat{\Pi}(t,\vec{y})] = \mathrm{i}\hbar\,\delta^{(D-1)}(\vec{x}-\vec{y}),$$

and

$$[\widehat{\phi}(t,\vec{x}),\widehat{\phi}(t,\vec{y})] = 0 = [\widehat{\Pi}(t,\vec{x}),\widehat{\Pi}(t,\vec{y})].$$

All pointwise products are distributional notation. Smeared fields, for test functions $f, g \in C_c^\infty(\mathbb{R}^{D-1})$, are

$$\widehat{\phi}_t(f) = \int \mathrm{d}^{D-1}\vec{x}\, f(\vec{x})\widehat{\phi}(t,\vec{x}), \qquad \widehat{\Pi}_t(g) = \int \mathrm{d}^{D-1}\vec{x}\, g(\vec{x})\widehat{\Pi}(t,\vec{x}).$$

The smeared commutator is

$$[\widehat{\phi}_t(f),\widehat{\Pi}_t(g)] = \mathrm{i}\hbar \int \mathrm{d}^{D-1}\vec{x}\, f(\vec{x})g(\vec{x}).$$

For the rest of this chapter set $\hbar = 1$. With $\hbar$ retained, the oscillator algebra below reads

$$[b_{\vec{k}}, b^\dagger_{\vec{q}}] = \hbar(2\pi)^{D-1}\delta^{(D-1)}(\vec{k}-\vec{q}).$$

Equivalently, the $\hbar = 1$ variables are obtained from the $\hbar$-full variables by $b^{(\hbar=1)}_{\vec{k}} = \hbar^{-1/2} b^{(\hbar)}_{\vec{k}}$.

9.5 Oscillator Variables and the Hamiltonian

Equal-time oscillator variables and the CCR. Let $\nu_{\vec{k}} > 0$ be a positive even function of $\vec{k}$. Define operator-valued distributions $b_{\vec{k}}$ and $b^\dagger_{\vec{k}}$ by the equal-time expansion

$$\widehat{\phi}(0,\vec{x}) = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}} \frac{1}{\sqrt{2\nu_{\vec{k}}}} \left(b_{\vec{k}} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}} + b^\dagger_{\vec{k}} \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x}} \right),$$

$$\widehat{\Pi}(0,\vec{x}) = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}} (-\mathrm{i}) \sqrt{\frac{\nu_{\vec{k}}}{2}} \left(b_{\vec{k}} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}} - b^\dagger_{\vec{k}} \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x}} \right).$$

The equal-time commutation relations are equivalent to

$$[b_{\vec{k}}, b_{\vec{q}}] = 0 = [b^\dagger_{\vec{k}}, b^\dagger_{\vec{q}}], \qquad [b_{\vec{k}}, b^\dagger_{\vec{q}}] = (2\pi)^{D-1}\delta^{(D-1)}(\vec{k}-\vec{q}).$$

Substitute the mode expansions into the smeared equal-time commutator and use

$$\int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}} \mathrm{e}^{\mathrm{i}\vec{k}\cdot(\vec{x}-\vec{y})} = \delta^{(D-1)}(\vec{x}-\vec{y}).$$

The coefficient $(2\nu_{\vec{k}})^{-1/2}(-\mathrm{i})(\nu_{\vec{k}}/2)^{1/2}$, together with the term obtained by commuting $b^\dagger$ past b, gives exactly $\mathrm{i}\delta^{(D-1)}(\vec{x}-\vec{y})$ when the oscillator commutator has the displayed normalization. Conversely, Fourier transforming the canonical commutator against plane waves recovers the oscillator commutator. The ϕ-ϕ and Π-Π commutators vanish because the b-b and $b^\dagger$-$b^\dagger$ commutators vanish and the remaining terms cancel by $\vec{k} \mapsto -\vec{k}$.

Bogoliubov reparametrizations of oscillator variables. This algebra holds for every positive choice of $\nu_{\vec k}$. The notation $b_{\vec k}$, $b^\dagger_{\vec k}$ should therefore not yet be read as a particle interpretation. If $\lambda_{\vec k} > 0$ is another even function, define $c_{\vec k}$, $c^\dagger_{\vec k}$ by

$$c_{\vec k} + c^\dagger_{-\vec k} = \lambda_{\vec k}\big(b_{\vec k} + b^\dagger_{-\vec k}\big), \qquad c_{\vec k} - c^\dagger_{-\vec k} = \lambda^{-1}_{\vec k}\big(b_{\vec k} - b^\dagger_{-\vec k}\big).$$

Equivalently,

$$c_{\vec k} = \frac{\lambda_{\vec k} + \lambda^{-1}_{\vec k}}{2}\, b_{\vec k} + \frac{\lambda_{\vec k} - \lambda^{-1}_{\vec k}}{2}\, b^\dagger_{-\vec k}.$$

The c's obey the same canonical oscillator algebra. If a vector is annihilated by every $b_{\vec k}$, it is not annihilated by every $c_{\vec k}$ unless $\lambda_{\vec k} = 1$. The Hamiltonian is the datum that selects the oscillator normalization. Write $u_{\vec k} = (\lambda_{\vec k} + \lambda^{-1}_{\vec k})/2$ and $v_{\vec k} = (\lambda_{\vec k} - \lambda^{-1}_{\vec k})/2$. Evenness of λ gives $u_{-\vec k} = u_{\vec k}$ and $v_{-\vec k} = v_{\vec k}$. The only nonzero commutators in $[c_{\vec k}, c^\dagger_{\vec q}]$ are

$$u_{\vec k} u_{\vec q} [b_{\vec k}, b^\dagger_{\vec q}] - v_{\vec k} v_{\vec q} [b_{-\vec q}, b^\dagger_{-\vec k}],$$

and the delta functions set $\vec k = \vec q$. Since $u^2_{\vec k} - v^2_{\vec k} = 1$, the original oscillator normalization is preserved. The c-c and $c^\dagger$-$c^\dagger$ commutators vanish by the same cancellation. Finally, if $b_{\vec k}\Omega_b = 0$, then

$$c_{\vec k}\Omega_b = v_{\vec k} b^\dagger_{-\vec k}\Omega_b,$$

which vanishes for all $\vec k$ only when $v_{\vec k} = 0$, equivalently $\lambda_{\vec k} = 1$.

Free Hamiltonian diagonalization. Let

$$\omega_{\vec k} = \sqrt{\vec k^{\,2} + m^2}.$$

Substituting the mode expansion into the free Hamiltonian gives

$$\widehat H_0 = \int \frac{\mathrm{d}^{D-1}\vec k}{(2\pi)^{D-1}} \left[\frac{\nu^2_{\vec k} + \omega^2_{\vec k}}{2\nu_{\vec k}} \left(b^\dagger_{\vec k} b_{\vec k} + \frac{1}{2}(2\pi)^{D-1}\delta^{(D-1)}(0) \right) \right.$$
$$\left. + \frac{\omega^2_{\vec k} - \nu^2_{\vec k}}{4\nu_{\vec k}} \left(b_{\vec k} b_{-\vec k} + b^\dagger_{\vec k} b^\dagger_{-\vec k} \right) \right].$$

The off-diagonal terms vanish precisely for

$$\nu_{\vec k} = \omega_{\vec k}.$$

Insert the equal-time oscillator expansion into

$$H_0 = \frac{1}{2}\int \mathrm{d}^{D-1}\vec x\, \big(\Pi^2 + (\nabla\phi)^2 + m^2\phi^2\big).$$

The spatial integral pairs momenta $\vec k$ and $-\vec k$. The Π^2 term contributes $\nu_{\vec k}/2$ to $b^\dagger_{\vec k} b_{\vec k}$ and $-\nu_{\vec k}/4$ to the $b_{\vec k} b_{-\vec k} + b^\dagger_{\vec k} b^\dagger_{-\vec k}$ coefficient, while the gradient and mass terms together contribute $\omega^2_{\vec k}/(2\nu_{\vec k})$ to $b^\dagger_{\vec k} b_{\vec k}$ and $\omega^2_{\vec k}/(4\nu_{\vec k})$ to the off-diagonal coefficient. Adding the terms gives the displayed Hamiltonian. Since $\nu_{\vec k} > 0$, the off-diagonal coefficient vanishes if and only if $\nu^2_{\vec k} = \omega^2_{\vec k}$, hence $\nu_{\vec k} = \omega_{\vec k}$.

Definition 9.6 (Free scalar Fock representation)**.** With this choice,

$$\widehat{H}_0 = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}}\, \omega_{\vec{k}} \left(b^\dagger_{\vec{k}} b_{\vec{k}} + \frac{1}{2}(2\pi)^{D-1}\delta^{(D-1)}(0) \right).$$

At finite spatial volume and finite momentum cutoff, the second term is a finite scalar multiple of the identity. A constant multiple of the identity does not change Heisenberg evolution, but the quantum Hamiltonian must be a well-defined self-adjoint operator after the regulator is removed. We therefore choose the additive energy normalization so that the invariant vacuum has zero energy. This is a convention of the flat-spacetime framework with gravity decoupled. If the metric is dynamical, the vacuum expectation of the stress tensor contributes to the cosmological constant, and its finite renormalized value is additional physical data rather than a consequence of normal ordering. With this normalization the free Hamiltonian is

$$\widehat{H}_0^{\mathrm{ren}} = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}}\, \omega_{\vec{k}} b^\dagger_{\vec{k}} b_{\vec{k}}.$$

The free vacuum Ω is defined by

$$b_{\vec{k}}\Omega = 0 \qquad \text{for all } \vec{k},$$

and satisfies $\widehat{H}_0^{\mathrm{ren}}\Omega = 0$.

Heisenberg field and invariant mass-shell normalization. Time evolution generated by $\widehat{H}_0^{\mathrm{ren}}$ gives

$$\widehat{\phi}(x) = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}} \frac{1}{\sqrt{2\omega_{\vec{k}}}} \left(b_{\vec{k}} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x} - \mathrm{i}\omega_{\vec{k}} t} + b^\dagger_{\vec{k}} \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{x} + \mathrm{i}\omega_{\vec{k}} t} \right).$$

The invariant mass-shell normalization used earlier is obtained by defining

$$a(\vec{k}) = \sqrt{2\omega_{\vec{k}}}\, b_{\vec{k}}, \qquad a^\dagger(\vec{k}) = \sqrt{2\omega_{\vec{k}}}\, b^\dagger_{\vec{k}}.$$

Then

$$[a(\vec{k}), a^\dagger(\vec{q})] = (2\pi)^{D-1} 2\omega_{\vec{k}}\, \delta^{(D-1)}(\vec{k} - \vec{q}),$$

and

$$\widehat{\phi}(x) = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1} 2\omega_{\vec{k}}} \left(a(\vec{k}) \mathrm{e}^{\mathrm{i}k\cdot x} + a^\dagger(\vec{k}) \mathrm{e}^{-\mathrm{i}k\cdot x} \right), \qquad k^0 = \omega_{\vec{k}}.$$

This is the convention used for the free scalar field in the earlier Fock-space construction.

The commutator with the normal-ordered Hamiltonian gives

$$[\widehat{H}_0^{\mathrm{ren}}, b_{\vec{k}}] = -\omega_{\vec{k}} b_{\vec{k}}, \qquad [\widehat{H}_0^{\mathrm{ren}}, b^\dagger_{\vec{k}}] = \omega_{\vec{k}} b^\dagger_{\vec{k}},$$

so Heisenberg evolution multiplies $b_{\vec{k}}$ by $\mathrm{e}^{-\mathrm{i}\omega_{\vec{k}} t}$ and $b^\dagger_{\vec{k}}$ by $\mathrm{e}^{+\mathrm{i}\omega_{\vec{k}} t}$. This gives the displayed field. The rescaling $a(\vec{k}) = \sqrt{2\omega_{\vec{k}}}\, b_{\vec{k}}$ transforms the Lebesgue momentum measure into the invariant one-particle mass-shell measure $\mathrm{d}^{D-1}\vec{k}/((2\pi)^{D-1} 2\omega_{\vec{k}})$ and gives the stated commutator for $a, a^\dagger$.

9.6 Covariance and Microcausality

Definition 9.7 (Free scalar field on the finite-particle domain)**.** Let $\mathcal{D}_0$ be the finite-particle domain generated by applying finitely many smeared creation operators to Ω. On $\mathcal{D}_0$, $\widehat{\phi}(x)$ is an operator-valued distribution. For $f \in C_c^\infty(\mathbb{M}^D)$,

$$\widehat{\phi}(f) = \int \mathrm{d}^D x\, f(x)\widehat{\phi}(x)$$

is the corresponding smeared operator on $\mathcal{D}_0$.

Proposition 9.8 (Covariance and microcausality of the free scalar field)**.** *The free Fock representation carries the unitary Poincare representation induced from the one-particle mass shell. With the above normalization, the field transforms as a scalar:*

$$U(a,\Lambda)^{-1}\widehat{\phi}(x)U(a,\Lambda) = \widehat{\phi}(\Lambda x + a).$$

The commutator is

$$[\widehat{\phi}(x), \widehat{\phi}(y)] = \mathrm{i}\,\Delta(x-y),$$

where the Pauli–Jordan distribution is

$$\Delta(x) = \frac{1}{\mathrm{i}}\int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}2\omega_{\vec{k}}}\left(\mathrm{e}^{\mathrm{i}k\cdot x} - \mathrm{e}^{-\mathrm{i}k\cdot x}\right), \qquad k^0 = \omega_{\vec{k}}.$$

This distribution is Lorentz invariant. If x is spacelike, a Lorentz transformation sends it to $(0,\vec{r})$. At equal time,

$$\Delta(0,\vec{r}) = \frac{1}{\mathrm{i}}\int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}2\omega_{\vec{k}}}\left(\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{r}} - \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{r}}\right) = 0,$$

because the two terms are exchanged by $\vec{k} \mapsto -\vec{k}$. Therefore

$$[\widehat{\phi}(x), \widehat{\phi}(y)] = 0 \qquad \text{when } (x-y)^2 > 0.$$

Equivalently,

$$[\widehat{\phi}(f), \widehat{\phi}(g)] = 0$$

when $\operatorname{supp} f$ *and* $\operatorname{supp} g$ *are spacelike separated. This verifies microcausality for the canonically quantized free scalar field.*

Proof. The one-particle Hilbert space is

$$L^2\left(\{k^2 + m^2 = 0,\ k^0 > 0\}, \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1}2\omega_{\vec{k}}}\right),$$

and the positive-energy mass-shell measure is Lorentz invariant. The induced unitary representation acts by translating phases and by pulling back the mass-shell momentum under Lorentz transformations; second quantization gives the Fock-space representation. Substituting this action into the mass-shell expansion of $\widehat{\phi}(x)$ gives the displayed scalar covariance law.

The commutator computation uses only $[a(\vec{k}), a^\dagger(\vec{q})] = (2\pi)^{D-1}2\omega_{\vec{k}}\delta^{(D-1)}(\vec{k}-\vec{q})$, and yields $[\widehat{\phi}(x), \widehat{\phi}(y)] = \mathrm{i}\Delta(x-y)$. The distribution Δ is Lorentz invariant because the mass-shell measure and the condition $k^0 > 0$ are invariant under proper orthochronous Lorentz transformations, and the difference between the two exponentials gives the antisymmetric solution of the Klein–Gordon equation. If $x - y$ is spacelike, a Lorentz transformation sends it to an equal-time vector $(0, \vec{r})$. The equal-time integral vanishes by the change of variables $\vec{k} \mapsto -\vec{k}$. Thus $\Delta(x - y) = 0$ for spacelike separation. After smearing, the commutator is the pairing of $\Delta(x - y)$ with $f(x)g(y)$, so it vanishes whenever the two supports are spacelike separated. $\square$

Remark 9.9. For the scalar field, locality is carried by the commutator. The Pauli–Jordan distribution is the difference of the positive-frequency two-point distribution and its reversed argument, and this difference is supported on the causal cone. The anticommutator has the symmetric vacuum distribution

$$\Delta_+(x - y) + \Delta_+(y - x),$$

whose equal-time spacelike kernel is proportional to

$$\int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1} 2\omega_{\vec{k}}} \cos(\vec{k} \cdot \vec{r}).$$

This distribution is not supported in the causal cone and generally has nonzero spacelike matrix elements. The graded sign in the locality bracket is therefore part of the structure fixed by spin and positivity, rather than an optional convention. The corresponding spin-statistics theorem-level hypotheses are stated in Chapter 40.

The next step in the construction is to identify the local currents and charges attached to spacetime symmetries. That is the role of Noether's theorem in the following chapter.

Chapter 10

Symmetries, Noether Currents, and Stress Tensors

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

10.1 Variational Framework

Definition 10.1 (First-derivative classical field datum)**.** Work on D-dimensional Minkowski spacetime $\mathbb{M}^D$, with mostly-plus metric $\eta_{\mu\nu}$. Let A range over a finite set of bosonic field components. A classical field configuration is a smooth map

$$\phi : M \longrightarrow \mathbb{R}^N, \qquad x \longmapsto \{\phi^A(x)\},$$

on a spacetime region $M \subset \mathbb{M}^D$. In this chapter a Lagrangian density means a first-derivative local expression

$$\mathcal{L} = \mathcal{L}(\phi^A, \partial_\mu \phi^A)$$

with no dependence on derivatives of order two or higher. The corresponding action over M is

$$S_M[\phi] = \int_M \mathrm{d}^D x\, \mathcal{L}(\phi, \partial\phi).$$

A variation is a smooth one-parameter family ϕ_s with $\phi_0 = \phi$; its tangent is

$$\delta\phi^A = \frac{\mathrm{d}}{\mathrm{d}s}\phi^A_s\bigg|_{s=0}.$$

When the boundary term is not part of the question, $\delta\phi^A$ is assumed to have compact support in M°, the interior of M.

First variational identity. For the datum of Definition 10.1, define the Euler–Lagrange expression

$$E_A(\mathcal{L}) = \frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}.$$

Then every smooth variation satisfies

$$\delta\mathcal{L} = E_A(\mathcal{L})\,\delta\phi^A + \partial_\mu \theta^\mu(\delta\phi), \qquad \theta^\mu(\delta\phi) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\delta\phi^A,$$

where repeated component and spacetime indices are summed. Consequently

$$\delta S_M = \int_M \mathrm{d}^D x\, E_A(\mathcal{L})\,\delta\phi^A + \int_{\partial M} \mathrm{d}\Sigma_\mu\, \theta^\mu(\delta\phi),$$

where $\mathrm{d}\Sigma_\mu$ is the oriented hypersurface measure on ∂M.

The chain rule gives

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi^A}\delta\phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\partial_\mu \delta\phi^A.$$

Apply the product rule to the second term:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\partial_\mu \delta\phi^A = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\delta\phi^A \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\delta\phi^A.$$

Combining the two non-divergence terms gives $E_A(\mathcal{L})\delta\phi^A$. Integrating the pointwise identity over M and applying Stokes' theorem gives the displayed formula for δS_M.

Definition 10.2 (Classical solution)**.** A classical solution of the first-derivative Lagrangian problem is a field configuration obeying

$$E_A(\mathcal{L}) = 0$$

for every component A, together with the boundary or asymptotic conditions specified by the variational problem. The Euler–Lagrange equation alone is not a complete boundary-value or Cauchy problem; the boundary data are part of the definition of the classical problem being solved.

10.2 Infinitesimal Symmetries

Definition 10.3 (Infinitesimal variational symmetry)**.** An infinitesimal field-space transformation is a rule

$$\delta_R \phi^A = R^A[\phi],$$

where $R^A[\phi]$ is a local expression in x, the fields, and finitely many derivatives of the fields. The transformation is a variational symmetry of the action if there exists a local vector K_R^μ such that

$$\delta_R \mathcal{L} = \partial_\mu K_R^\mu.$$

This equation is the definition of the symmetry in the present classical Lagrangian framework. It is an off-shell identity of local functions of the fields; no Euler–Lagrange equation has yet been imposed.

Noether identity for a variational symmetry. Let R be a variational symmetry in the sense of Definition 10.3. The local current

$$j^\mu_R = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} R^A - K^\mu_R$$

obeys the off-shell identity

$$E_A(\mathcal{L})R^A + \partial_\mu j^\mu_R = 0. \tag{10.1}$$

This identity is an equality in the algebra of local functions on the finite jet bundle used by the Lagrangian; it is not an equality only after integrating over spacetime. The current is defined modulo an identically conserved improvement $\partial_\nu B^{[\nu\mu]}$, and the identity is unchanged by such a replacement. In particular, on every classical solution,

$$\partial_\mu j^\mu_R = 0.$$

The first variational identity from the preceding section is the load-bearing input. It separates the variation of the Lagrangian density into the Euler–Lagrange part and a total derivative before imposing any equation of motion. For the variation $\delta_R \phi^A = R^A$, it gives the pointwise jet identity

$$\delta_R \mathcal{L} = E_A(\mathcal{L})R^A + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} R^A \right).$$

The variational-symmetry condition is the second independent input: there exists a local K^μ_R such that $\delta_R \mathcal{L} = \partial_\mu K^\mu_R$ as an off-shell local identity. Subtracting this total derivative from the preceding display gives

$$0 = E_A(\mathcal{L})R^A + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} R^A - K^\mu_R \right),$$

which is the stated Noether identity. On a classical solution the Euler–Lagrange term vanishes componentwise, and the current is conserved. The improvement statement follows because $\partial_\mu \partial_\nu B^{[\nu\mu]} = 0$ by antisymmetry and commutativity of ordinary partial derivatives.

Localized-parameter form of Noether's theorem. Let $\epsilon \in C^\infty_c(M)$, and localize the transformation by

$$\delta_{\epsilon R} \phi^A = \epsilon R^A.$$

For a first-derivative Lagrangian,

$$\delta_{\epsilon R} \mathcal{L} = \partial_\mu(\epsilon K^\mu_R) + (\partial_\mu \epsilon) j^\mu_R.$$

Thus, for compactly supported ϵ,

$$\delta_{\epsilon R} S_M = \int_M \mathrm{d}^D x \, (\partial_\mu \epsilon) \, j^\mu_R.$$

If the field is a classical solution, this identity is equivalent, as a distributional statement, to $\partial_\mu j^\mu_R = 0$. The verification is the local variation formula with the parameter left unintegrated. Using the chain rule,

$$\delta_{\epsilon R} \mathcal{L} = \epsilon \, \delta_R \mathcal{L} + (\partial_\mu \epsilon) \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} R^A.$$

Since $\delta_R\mathcal{L} = \partial_\mu K_R^\mu$,

$$\epsilon\,\partial_\mu K_R^\mu = \partial_\mu(\epsilon K_R^\mu) - (\partial_\mu\epsilon)K_R^\mu.$$

Combining the two $\partial_\mu\epsilon$ terms gives $(\partial_\mu\epsilon)j_R^\mu$. The integral of $\partial_\mu(\epsilon K_R^\mu)$ vanishes because ϵ has compact support in M. On a solution the first variation of the action by a compactly supported variation vanishes, hence

$$0 = \int_M \mathrm{d}^D x\,(\partial_\mu\epsilon)j_R^\mu = -\int_M \mathrm{d}^D x\,\epsilon\,\partial_\mu j_R^\mu.$$

The fundamental lemma of distributions then gives $\partial_\mu j_R^\mu = 0$. The converse follows by integrating by parts.

Definition 10.4 (Current improvement)**.** The Noether current is a representative of a local equivalence class. If

$$B_R^{\nu\mu} = -B_R^{\mu\nu},$$

then

$$j_R^\mu \longmapsto j_R^\mu + \partial_\nu B_R^{\nu\mu}$$

is called a current improvement. It has the same local divergence because $\partial_\mu\partial_\nu B_R^{\nu\mu} = 0$. The corresponding integrated charge is unchanged when the surface integral of B_R at spatial boundary vanishes in the stated boundary condition or asymptotic domain.

10.3 Charges and Generators

Definition 10.5 (Classical charge of a conserved current)**.** Let $\Sigma_t = \{x^0 = t\}$ be a constant-time slice. If j_R^μ is conserved and the spatial integral exists, the charge on Σ_t is

$$Q_R(t) = \int_{\Sigma_t} \mathrm{d}^{D-1}\vec{x}\,j_R^0(t,\vec{x}).$$

For a spatial region $V \subset \mathbb{R}^{D-1}$, the finite-region charge is

$$Q_R^V(t) = \int_V \mathrm{d}^{D-1}\vec{x}\,j_R^0(t,\vec{x}).$$

The existence of $Q_R(t)$ is an analytic condition on the solution and on the asymptotic domain; it is not implied by the local equation $\partial_\mu j_R^\mu = 0$ alone.

Remark 10.6 (Charge balance in a spacetime slab). Let $V \subset \mathbb{R}^{D-1}$ be a spatial region with piecewise smooth boundary, and assume j_R^μ is smooth on $[t_1, t_2] \times V$ with $\partial_\mu j_R^\mu = 0$. Then

$$Q_R^V(t_2) - Q_R^V(t_1) = -\int_{t_1}^{t_2} \mathrm{d}x^0 \int_{\partial V} \mathrm{d}S_i\,j_R^i(x^0,\vec{x}),$$

where $\mathrm{d}S_i = n_i\,\mathrm{d}S$, n_i is the outward unit normal to ∂V, and $\mathrm{d}S$ is the Euclidean surface measure on ∂V. Hence $Q_R(t)$ is time independent whenever the boundary flux tends to zero in the limit $V \nearrow \mathbb{R}^{D-1}$. It follows by integrating $\partial_0 j_R^0 + \partial_i j_R^i = 0$ over the slab and applying the ordinary divergence theorem. The sign follows from using the outward spatial normal: positive outward flux reduces the charge remaining inside V. For an exhaustion $V_R \nearrow \mathbb{R}^{D-1}$, the infinite-volume charge is conserved provided $Q_R^{V_R}(t)$ has a limit and the corresponding boundary-flux integral tends to zero.

Figure 10.1 records the slab geometry behind the charge-conservation identity: time-slice charges differ exactly by flux through the spatial boundary.

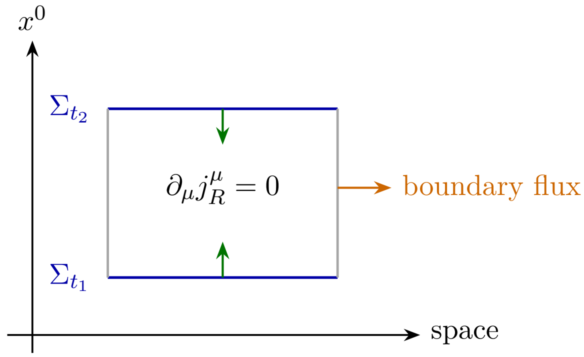


Figure 10.1: Current conservation in a spacetime slab. The difference between charges on Σ_{t_1} and Σ_{t_2} is the flux through the spatial boundary.

Definition 10.7 (Equal-time Poisson bracket and Hamiltonian generator)**.** Let Π_A denote the canonical momentum conjugate to ϕ^A. On the phase-space functionals whose variational derivatives and boundary terms are defined, the equal-time Poisson bracket is

$$\{F, G\} = \int \mathrm{d}^{D-1}\vec{x} \left[\frac{\delta F}{\delta \phi^A(\vec{x})} \frac{\delta G}{\delta \Pi_A(\vec{x})} - \frac{\delta F}{\delta \Pi_A(\vec{x})} \frac{\delta G}{\delta \phi^A(\vec{x})} \right].$$

In particular,

$$\{\phi^A(\vec{x}), \Pi_B(\vec{y})\} = \delta^A{}_B \, \delta^{(D-1)}(\vec{x} - \vec{y}).$$

A charge Q_R generates the infinitesimal transformation R on canonical observables if

$$\delta_R F = \{F, Q_R\}$$

for every functional F in the stated domain. Applied to the coordinate field this condition reads

$$R^A(t, \vec{x}) = \{\phi^A(t, \vec{x}), Q_R\}.$$

First-order Noether charges as phase-space generators. Assume $K^0_R = 0$, and assume R^A depends on the canonical coordinates and their spatial derivatives but not on the velocities or momenta. If the Noether charge is represented on phase space by

$$Q_R = \int_{\Sigma_t} \mathrm{d}^{D-1}\vec{x} \, \Pi_A R^A$$

up to spatial-divergence improvements whose boundary integrals vanish, then

$$\{\phi^A(t, \vec{x}), Q_R\} = R^A(t, \vec{x}).$$

More generally, if R^A or K^0_R contains velocities, the current obtained from the Lagrangian calculation must first be expressed as a functional on phase space before the generator statement is tested.

Under the stated hypotheses the functional derivative of Q_R with respect to $\Pi_A(\vec{x})$ is $R^A(\vec{x})$. The equal-time bracket in Definition 10.7 therefore gives

$$\{\phi^A(\vec{x}), Q_R\} = \frac{\delta Q_R}{\delta \Pi_A(\vec{x})} = R^A(\vec{x}).$$

Spatial-divergence improvements do not change the result under the assumed boundary condition. If velocities occur, the expression $\int j_R^0$ is not yet a function on canonical phase space; the Legendre map must be used first, and the displayed derivative calculation is then performed on the resulting phase-space functional.

Example 10.8 (Time translation)**.** For a time translation by τ,

$$R^A = -\tau\dot{\phi}^A, \qquad Q_R = -\tau H.$$

Hamilton's equation yields

$$\{\phi^A, -\tau H\} = -\tau\dot{\phi}^A.$$

Thus the generator statement is a phase-space statement with the Noether charge expressed in canonical variables.

10.4 Poincare Transformations of Scalar Fields

Definition 10.9 (Scalar representation of the Poincare group)**.** For a scalar field, a finite proper orthochronous Poincare transformation is described by

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu, \qquad \phi'(x') = \phi(x),$$

where $\Lambda^T\eta\Lambda = \eta$. Equivalently,

$$\phi'(x) = \phi\big(\Lambda^{-1}(x-a)\big).$$

Write $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu + O(\omega^2)$, with $\omega_{\mu\nu} = -\omega_{\nu\mu}$. The fixed-coordinate infinitesimal variation is

$$\delta\phi(x) = -\left(a^\mu + \omega^\mu{}_\nu x^\nu\right)\partial_\mu\phi(x).$$

The translation and Lorentz currents below are the Noether currents for the two independent parameters a^μ and $\omega_{\mu\nu}$.

10.5 Translation Symmetry and the Stress Tensor

Canonical stress tensor from translations. Assume that $\mathcal{L}$ has no explicit dependence on x. For a constant vector ξ^ν, use the fixed-coordinate field variation

$$\delta_\xi\phi^A(x) = -\xi^\nu\partial_\nu\phi^A(x).$$

Then the translation Noether current can be written

$$j_\xi^\mu = \xi^\nu T^\mu{}_\nu,$$

where

$$T^{\mu}{}_{\nu} = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\nu \phi^A + \delta^{\mu}{}_{\nu}\, \mathcal{L}.$$

On solutions,

$$\partial_\mu T^{\mu}{}_{\nu} = 0.$$

Equivalently, for compactly supported $\xi^\nu(x)$,

$$\delta_\xi S_M = \int_M \mathrm{d}^D x\, (\partial_\mu \xi^\nu)\, T^{\mu}{}_{\nu},$$

after the compactly supported total derivative has been separated. Thus $T^{\mu}{}_{\nu}$ is the response coefficient for a localized translation.

For constant ξ^ν, the density has no explicit coordinate dependence. Translation invariance of the scalar local density gives

$$\delta_\xi \mathcal{L} = -\xi^\nu \partial_\nu \mathcal{L} = \partial_\mu K^\mu_\xi, \qquad K^\mu_\xi = -\xi^\mu \mathcal{L}.$$

Substituting $R^A = -\xi^\nu \partial_\nu \phi^A$ and this K^μ_ξ into the identity (10.1) gives

$$j^\mu_\xi = -\xi^\nu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\nu \phi^A + \xi^\mu \mathcal{L} = \xi^\nu T^{\mu}{}_{\nu}.$$

Since ξ^ν is arbitrary, conservation of j^μ_ξ for every constant ξ^ν is equivalent to $\partial_\mu T^{\mu}{}_{\nu} = 0$.

For local $\xi^\nu(x)$, compute directly:

$$\delta_\xi \mathcal{L} = -\xi^\nu \partial_\nu \mathcal{L} - (\partial_\mu \xi^\nu) \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\nu \phi^A.$$

The first term is $-\partial_\mu(\xi^\mu \mathcal{L}) + (\partial_\mu \xi^\mu)\mathcal{L}$. The coefficient of $\partial_\mu \xi^\nu$ is therefore precisely $T^{\mu}{}_{\nu}$. Integration over M gives the localized-translation formula because the total derivative has compact support.

Definition 10.10 (Translation charges)**.** When the spatial integrals exist, the momentum covector and vector are

$$P_\nu = \int \mathrm{d}^{D-1}\vec{x}\, T^{0}{}_{\nu}, \qquad P^\nu = \eta^{\nu\rho} P_\rho = \int \mathrm{d}^{D-1}\vec{x}\, T^{0\nu}.$$

With the mostly-plus metric, P^0 is the energy. Equivalently, $P_0 = -P^0$.

Stress tensor of a real scalar field. For

$$\mathcal{L} = -\frac{1}{2} \partial_\rho \phi\, \partial^\rho \phi - V(\phi),$$

the canonical stress tensor is

$$T^{\mu}{}_{\nu} = \partial^\mu \phi\, \partial_\nu \phi - \delta^{\mu}{}_{\nu} \left(\frac{1}{2} \partial_\rho \phi\, \partial^\rho \phi + V(\phi) \right).$$

Raising the second index,

$$T^{\mu\nu} = \eta^{\nu\rho} T^{\mu}{}_{\rho},$$

gives a symmetric tensor in this example. Its energy density is

$$T^{00} = \frac{1}{2}(\partial_0\phi)^2 + \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2 + V(\phi).$$

For the free scalar field, $V(\phi) = \frac{1}{2}m^2\phi^2$, this is the Hamiltonian density obtained by canonical quantization.

The derivative of the Lagrangian with respect to $\partial_\mu\phi$ is

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = -\partial^\mu\phi.$$

Substitution into the translation-current formula above gives the displayed $T^{\mu}{}_{\nu}$. Since

$$T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\left(\frac{1}{2}\partial_\rho\phi\,\partial^\rho\phi + V(\phi)\right),$$

the tensor is symmetric. Finally, with $\eta_{00} = -1$,

$$\partial_\rho\phi\,\partial^\rho\phi = -(\partial_0\phi)^2 + \sum_{i=1}^{D-1}(\partial_i\phi)^2,$$

and the displayed expression for T^{00} follows.

10.6 Metric Variation and Improvement

Definition 10.11 (Lorentzian Hilbert stress tensor)**.** Suppose the flat-space action is obtained by restricting a generally covariant background-metric action $S_g[\phi]$ to $g_{\mu\nu} = \eta_{\mu\nu}$. With the Lorentzian mostly-plus action convention used in this volume, the Hilbert stress tensor is defined by

$$T_{\mathrm{H}}^{\mu\nu}(x) = \frac{2}{\sqrt{|g|}}\frac{\delta S_g}{\delta g_{\mu\nu}(x)}\bigg|_{g=\eta}.$$

Equivalently,

$$\delta_g S_g = \frac{1}{2}\int \mathrm{d}^D x\,\sqrt{|g|}\,T_{\mathrm{H}}^{\mu\nu}\,\delta g_{\mu\nu}.$$

Using $\delta g^{\mu\nu} = -g^{\mu\rho}g^{\nu\sigma}\delta g_{\rho\sigma}$, this is the same as

$$T_{\mathrm{H}}^{\mu\nu}(x) = -\frac{2}{\sqrt{|g|}}g^{\mu\rho}g^{\nu\sigma}\frac{\delta S_g}{\delta g^{\rho\sigma}(x)}\bigg|_{g=\eta}.$$

The Euclidean source-functional chapters state their Euclidean sign convention separately. The chain-rule conversion between covariant and contravariant metric variations is the same; the sign attached to the stress tensor depends on whether one is varying the Lorentzian action or the Euclidean $\exp(-S_E)$ source functional.

Minimal scalar Hilbert tensor. For the minimally coupled scalar action

$$S_g[\phi] = \int \mathrm{d}^D x \, \sqrt{|g|} \left(-\frac{1}{2} g^{\rho\sigma} \partial_\rho \phi \, \partial_\sigma \phi - V(\phi) \right),$$

Definition 10.11 gives

$$T_{\mathrm{H}}^{\mu\nu} = \partial^\mu \phi \, \partial^\nu \phi - g^{\mu\nu} \left(\frac{1}{2} \partial_\rho \phi \, \partial^\rho \phi + V(\phi) \right),$$

which agrees in flat space with the canonical tensor computed above.

Vary with respect to the contravariant metric. The determinant identity is

$$\delta \sqrt{|g|} = -\frac{1}{2} \sqrt{|g|} \, g_{\mu\nu} \delta g^{\mu\nu}.$$

Therefore

$$\delta_g S_g = -\frac{1}{2} \int \mathrm{d}^D x \, \sqrt{|g|} \left[\partial_\mu \phi \, \partial_\nu \phi + g_{\mu\nu} \left(-\frac{1}{2} g^{\rho\sigma} \partial_\rho \phi \, \partial_\sigma \phi - V \right) \right] \delta g^{\mu\nu}.$$

Comparing with the contravariant form of Definition 10.11 gives

$$T_{\mathrm{H}\,\mu\nu} = \partial_\mu \phi \, \partial_\nu \phi + g_{\mu\nu} \left(-\frac{1}{2} \partial_\rho \phi \, \partial^\rho \phi - V \right).$$

Raising both indices gives the displayed expression.

Definition 10.12 (Stress-tensor improvement)**.** Stress tensors that differ by

$$T^{\mu\nu} \longmapsto T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}, \qquad B^{\rho\mu\nu} = -B^{\mu\rho\nu},$$

have the same local conservation law. They define the same translation charges when the resulting surface integrals vanish. The choice of representative matters for local operator identities and for couplings to background geometry, so the representative is part of the framework.

10.7 Lorentz Currents

Lorentz current from a symmetric stress tensor. For a scalar field, an infinitesimal Lorentz transformation is specified by $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and

$$\delta_\omega \phi(x) = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi(x).$$

If $T^{\mu\nu} = T^{\nu\mu}$ and $\partial_\mu T^{\mu\nu} = 0$, then

$$M^{\mu,\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}$$

is conserved:

$$\partial_\mu M^{\mu,\rho\sigma} = 0.$$

The Lorentz current for parameter $\omega_{\rho\sigma}$ is

$$j^\mu_\omega = \frac{1}{2} \omega_{\rho\sigma} M^{\mu,\rho\sigma},$$

and the corresponding charges, when the spatial integrals exist, are

$$J^{\rho\sigma} = \int \mathrm{d}^{D-1}\vec{x}\, \left(x^\rho T^{0\sigma} - x^\sigma T^{0\rho}\right).$$

The current is obtained by contracting the antisymmetric Lorentz parameter with the antisymmetric tensor $M^{\mu,\rho\sigma}$. The factor 1/2 compensates for the double counting of the ordered pairs (ρ, σ) and (σ, ρ). To verify conservation, differentiate $M^{\mu,\rho\sigma}$ as an ordinary distribution:

$$\partial_\mu M^{\mu,\rho\sigma} = T^{\rho\sigma} + x^\rho \partial_\mu T^{\mu\sigma} - T^{\sigma\rho} - x^\sigma \partial_\mu T^{\mu\rho}.$$

Stress-tensor conservation removes the terms proportional to x, and symmetry of $T^{\mu\nu}$ removes the remaining difference. Therefore $\partial_\mu M^{\mu,\rho\sigma} = 0$, and contraction with the constant $\omega_{\rho\sigma}$ gives $\partial_\mu j^\mu_\omega = 0$. If the spatial integral of $M^{0,\rho\sigma}$ exists after the boundary conditions or smearing prescription have been specified, this conserved current defines the charge $J^{\rho\sigma}$.

Remark 10.13 (Spin terms and Belinfante improvement). For fields carrying Lorentz or spinor indices, the Lorentz current includes an additional spin current. In such cases one either keeps the canonical spin current explicitly or passes to a Belinfante-improved symmetric stress tensor, provided the improvement is defined and the boundary conditions make the charges equivalent. The spinor chapter fixes the corresponding representation matrices before using these currents.

10.8 Quantum Currents and Charges

Definition 10.14 (Quantum current)**.** In a quantum field theory, a current is a vector of operator-valued distributions j^μ_R on a common dense domain $\mathcal{D} \subset \mathcal{H}$. For $f \in C_c^\infty(\mathbb{M}^D)$, the smeared current is denoted

$$j^\mu_R(f) = \int \mathrm{d}^D x\, f(x) j^\mu_R(x).$$

Current conservation is a distributional statement:

$$\partial_\mu j^\mu_R = 0.$$

Equivalently, for all $\psi, \chi \in \mathcal{D}$ and all $f \in C_c^\infty(\mathbb{M}^D)$,

$$\sum_{\mu=0}^{D-1} \langle \psi, j^\mu_R(\partial_\mu f)\chi \rangle = 0,$$

with the derivative acting on the test function. This is the operator-valued distributional version of integration by parts.

Definition 10.15 (Regulated Ward identity and contact terms)**.** In correlation functions with local insertions, the quantum statement is a Ward identity of distributions and includes contact terms on collision diagonals. Its precise form is part of the chosen quantum construction. In a regulated source-functional presentation, let

$$X = \prod_{i=1}^{n} \mathcal{O}_i(x_i)$$

be a product of renormalized local insertions and let the local symmetry parameter be $\epsilon(x)$. The finite-regulator change of variables gives

$$\int \mathrm{d}^D x\,(\partial_\mu\epsilon)(x)\,\left\langle j^\mu_R(x)X\right\rangle_{\mathrm{reg}} = \langle\delta_{\epsilon R}X\rangle_{\mathrm{reg}} + \langle\mathcal{A}_{\epsilon,R}X\rangle_{\mathrm{reg}}\,. \tag{10.2}$$

Here $\mathcal{A}_{\epsilon,R}$ is the local term produced by the variation of the regulated integration density and by the local counterterms in the source-coordinate system; in a fermionic path integral this density is the finite Berezinian density described in Chapter 40. If the symmetry is implemented without anomaly in the chosen regularization and source-coordinate system, $\mathcal{A}_{\epsilon,R}=0$. The first term on the right of (10.2) gives contact distributions supported at $x=x_i$:

$$\delta_{\epsilon R}X = \sum_i \epsilon(x_i)\,\mathcal{O}_1(x_1)\cdots(\delta_R\mathcal{O}_i)(x_i)\cdots\mathcal{O}_n(x_n),$$

with the corresponding spin or derivative-of-delta terms when the inserted operators carry spacetime indices or derivatives. After the regulator is removed, (10.2) becomes an identity in the renormalized source-functional contact-term chart. The fixed-point version used later is formulated in Section 56.6, with the status of the source brackets specified in Definition 56.5.

Definition 10.16 (Quantum charge as a smeared limit)**.** A quantum charge is defined as a limit of smeared local charges when that limit exists. Choose a compactly supported time smearing $f_t(x^0)$ concentrated near t, normalized by $\int \mathrm{d}x^0\, f_t(x^0)=1$, and spatial cutoff functions $\chi_R(\vec{x})$ with $\chi_R\to 1$ as $R\to\infty$. The regulated charge is

$$Q_R[f_t,\chi_R] = \int \mathrm{d}^D x\, f_t(x^0)\chi_R(\vec{x})j^0_R(x).$$

If these operators converge on a specified domain, the limit is the charge operator Q_R. If Q_R is essentially self-adjoint or has a chosen self-adjoint extension, it exponentiates to a one-parameter unitary group by Stone's theorem.

Definition 10.17 (Quantum generator convention)**.** With the fixed-coordinate Noether convention, the infinitesimal quantum action on an operator $\widehat{\mathcal{O}}$ is

$$\delta_R\widehat{\mathcal{O}} = \frac{\mathrm{i}}{\hbar}[Q_R,\widehat{\mathcal{O}}]$$

on the common domain where the commutator is defined. For spacetime translations the covariance convention of the earlier chapters is

$$U(a)^{-1}\widehat{\Phi}_A(x)U(a) = \widehat{\Phi}_A(x+a), \qquad U(a) = \exp\left(\frac{\mathrm{i}}{\hbar}a^\nu P_\nu\right).$$

Translation commutator convention. The convention in Definition 10.17 is equivalent to

$$[P_\nu,\widehat{\Phi}_A(x)] = \mathrm{i}\hbar\,\partial_\nu\widehat{\Phi}_A(x)$$

in distributional notation. Differentiate

$$U(a)^{-1}\widehat{\Phi}_A(x)U(a) = \widehat{\Phi}_A(x+a)$$

with respect to a^ν at $a=0$, using $U(a)=\exp(\mathrm{i}a^\rho P_\rho/\hbar)$. The derivative of the left hand side is

$$-\frac{\mathrm{i}}{\hbar}P_\nu\widehat{\Phi}_A(x) + \frac{\mathrm{i}}{\hbar}\widehat{\Phi}_A(x)P_\nu = -\frac{\mathrm{i}}{\hbar}[P_\nu,\widehat{\Phi}_A(x)].$$

The derivative of the right hand side is $\partial_\nu\widehat{\Phi}_A(x)$. Multiplying by $\mathrm{i}\hbar$ gives the claimed commutator.

Definition 10.18 (Stress-tensor charges in the quantum theory)**.** When a renormalized stress tensor exists as a local operator-valued distribution, the Poincare generators are represented by

$$P^\nu = \int \mathrm{d}^{D-1}\vec{x}\, T^{0\nu}(x),$$

and, for a symmetric stress tensor in the scalar case,

$$J^{\rho\sigma} = \int \mathrm{d}^{D-1}\vec{x}\, \left(x^\rho T^{0\sigma}(x) - x^\sigma T^{0\rho}(x)\right),$$

with the same regulator-and-limit interpretation as for general charges. The quantum theory must specify the operator domains, renormalization prescription, conservation law, and boundary behavior under which the classical Noether densities define these charges.

Once those data are fixed, the stress tensor is the local bridge between spacetime symmetry and the generators that act on states and fields.

Chapter 11

Scalar Path Integrals and Green Functions

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

11.1 Field-Configuration Kernels

Convention 11.1 (Units)**.** Set $\hbar = 1$ throughout this chapter.

Definition 11.2 (Regulated equal-time configuration representation)**.** Let $\widehat{\phi}(t, \vec{x})$ be the canonical real scalar field at time $t = x^0$. The notation involving a spatial field configuration is defined through finite regulators. Let Λ be a spatial regulator and let E_Λ be the resulting finite-dimensional real vector space of spatial field variables. Its elements are denoted by φ_Λ, and a linear functional $\ell \in E_\Lambda^\vee$ evaluates the regulated field by $\ell(\varphi_\Lambda)$. The regulated configuration Hilbert space is

$$\mathcal{H}_\Lambda = L^2(E_\Lambda, \mathrm{d}_\Lambda\varphi),$$

where $\mathrm{d}_\Lambda\varphi$ is the Lebesgue measure determined by the chosen coordinates, lattice cell factors, or Fourier-mode normalization. The regulated equal-time field operator associated with ℓ acts by multiplication:

$$(\widehat{\phi}_\Lambda(\ell)\Psi)(\varphi_\Lambda) = \ell(\varphi_\Lambda)\Psi(\varphi_\Lambda).$$

A field-configuration generalized vector at time t and regulator Λ is the distributional vector

$|\varphi_\Lambda; t\rangle_\Lambda$ defined by evaluation:

$$_\Lambda\langle\varphi_\Lambda; t|\Psi\rangle = \Psi_t(\varphi_\Lambda), \qquad \Psi \in \mathcal{S}(E_\Lambda) \subset \mathcal{H}_\Lambda.$$

Here $\mathcal{S}(E_\Lambda)$ is the Schwartz test space after choosing a linear coordinate system on E_Λ. The resolution of identity is the weak identity

$$\int_{E_\Lambda} \mathrm{d}_\Lambda\varphi \, |\varphi; t\rangle_{\Lambda\Lambda}\langle\varphi; t| = 1$$

on $\mathcal{S}(E_\Lambda)$. Accordingly,

$$_\Lambda\langle\varphi; t_i|\varphi_i; t_i\rangle_\Lambda = \delta_\Lambda(\varphi - \varphi_i),$$

where δ_Λ is the Dirac delta distribution relative to $\mathrm{d}_\Lambda\varphi$. The continuum symbols $|\varphi; t\rangle$, $\Psi_t[\varphi]$, and $\delta[\varphi - \varphi_i]$ mean a compatible family of these regulated objects together with a stated limiting topology.

Definition 11.3 (Regulated scalar integration conventions)**.** For a finite-dimensional bosonic scalar regulator, the canonical reference density is the coordinate density

$$D_\Lambda^{\mathrm{ref}}\phi = \mathrm{d}_\Lambda\phi,$$

including the lattice cell factors or Fourier-mode normalization chosen with the regulator. If the Euclidean quadratic part is

$$S_{\mathrm{kin},\Lambda}[\phi] = \frac{1}{2}\langle\phi, C_\Lambda^{-1}\phi\rangle_\Lambda$$

with positive covariance C_Λ, the normalized Gaussian reference measure is

$$\mathrm{d}\mu_{C_\Lambda}(\phi) = Z_{0,\Lambda}^{-1} \exp\{-S_{\mathrm{kin},\Lambda}[\phi]\} \, D_\Lambda^{\mathrm{ref}}\phi. \tag{11.1}$$

If $L_\Lambda[\phi]$ denotes the remaining interaction, the unnormalized full Euclidean density is

$$\mathrm{d}\rho_{\Lambda,S}(\phi) = \exp\{-L_\Lambda[\phi]\} \, \mathrm{d}\mu_{C_\Lambda}(\phi) = Z_{0,\Lambda}^{-1} \exp\{-S_{\mathrm{kin},\Lambda}[\phi] - L_\Lambda[\phi]\} \, D_\Lambda^{\mathrm{ref}}\phi. \tag{11.2}$$

Thus $D_\Lambda^{\mathrm{ref}}\phi$, $\mathrm{d}\mu_{C_\Lambda}$, and $\mathrm{d}\rho_{\Lambda,S}$ are distinct named objects related by explicit Radon–Nikodym factors. In formulas where the older compact notation $[D\phi]_\Lambda$ appears, it denotes the reference density $D_\Lambda^{\mathrm{ref}}\phi$ in the finite-dimensional bosonic scalar setting. Gaussian-measure formulas must use $\mathrm{d}\mu_{C_\Lambda}$, and full source-functional formulas must display the action weight or the density $\mathrm{d}\rho_{\Lambda,S}$.

For fermionic, gauge, BV, Lorentzian oscillatory, dimensional regularization, and other schemes beyond finite Lebesgue densities, a chapter must specify the corresponding regulated integration object and its algebraic or analytic status separately.

Finite lattice or finite spin system. The object is a finite product of site variables, link variables, or spins. The measure is an ordinary finite product measure, counting measure, compact Haar measure, or finite-dimensional density. Thermodynamic and continuum limits are separate questions.

Finite Fourier-mode cutoff. At finite volume and finite mode number, this is a finite-dimensional field space with a reference density and, after the quadratic part is chosen, a Gaussian measure. Sharp momentum cutoffs are useful perturbatively but need not preserve locality or reflection positivity.

Smooth covariance or Wilsonian cutoff. A profile such as $C_\Lambda(k) = \chi(k^2/\Lambda^2)/(k^2 + m^2)$ defines a regulated covariance. With a genuine finite regulator it gives a Gaussian measure or density; in continuum perturbation theory it is a covariance assignment for graphs and Wilsonian pushforwards.

Heat-kernel, spectral, or point-splitting cutoff. These are smoothing kernels, finite-rank projectors, or separated insertion prescriptions. They regulate kernels, traces, determinants, or composite operators; they are not automatically measures on field configurations.

Pauli–Villars auxiliary-field or determinant regulator. This is a massive auxiliary-field or determinant-ratio prescription with specified statistics, coefficients, masses, and symmetry action. It is not a measure on the original field space. If written as a path integral, the enlarged field space, its density or integration cycle, and the sign or indefinite-metric structure of the auxiliary sector must be part of the definition. The monograph does not use Pauli–Villars as a default path-integral construction.

Dimensional regularization. This is a meromorphic assignment to perturbative graph distributions, with specified tensor, spinor, polarization, and external-state algebra in $D = d - \varepsilon$. It is not a Borel measure, not a Berezin measure, and not a measure on a noninteger-dimensional field-configuration space.

BPHZ, momentum subtraction, MS. These are subtraction and coordinate schemes imposed on already specified amplitudes or graph distributions. They are not by themselves regulators of a field measure.

Table 11.1: Regulator and subtraction classes used in the monograph. The catalog names the object that exists before a limit is taken or before a formal expansion is interpreted, and separates cutoff measures from regulated covariances, operator-insertion regulators, Pauli–Villars auxiliary determinants, dimensional regularization, and subtraction prescriptions.

Table 11.1 controls the shorthand used later. A formula written with a lattice spacing, a Fourier cutoff, or a smooth covariance cutoff may denote an actual finite-regulator integral only after the finite field space and density have been specified. A formula written in dimensional regularization denotes a perturbative meromorphic graph assignment; it is not an integral over a noninteger-dimensional space of fields. A formula described as Pauli–Villars regulated must display the auxiliary field or determinant data; without that data it is only the name of a possible subtraction mechanism, not a path-integral measure. A subtraction prescription is a further coordinate choice on regulated or meromorphic data, not a replacement for stating what was regulated.

Definition 11.4 (Finite-regulator time-sliced scalar kernel)**.** At fixed finite-dimensional Λ, the field theory is represented by a quantum-mechanical system with finitely many coordinates. Choose a time partition

$$t_i = t_0 < t_1 < \cdots < t_N = t_f.$$

With boundary fields $\varphi_0 = \varphi_i$ and $\varphi_N = \varphi_f$, the finite time-sliced kernel has the form

$$K_{\Lambda,N}(\varphi_f, t_f; \varphi_i, t_i) = \int_{E_\Lambda^{N-1}} \prod_{n=1}^{N-1} \mathrm{d}_\Lambda\varphi_n \, \mathcal{N}_{\Lambda,N} \, \exp\bigl[\mathrm{i}S_{\Lambda,N}(\varphi_0, \ldots, \varphi_N)\bigr].$$

The normalization factor $\mathcal{N}_{\Lambda,N}$, the discrete action $S_{\Lambda,N}$, and possible local measure factors are part of the chosen operator-symbol and time-slicing convention, exactly as in the finite-dimensional construction of Chapter 7.

Finite-dimensional Schrödinger regulators and Trotter–Kato. The finite-dimensional Trotter–Kato and Feynman–Kac theorems recalled in Chapter 7 apply to a spatial regulator only after the regulator has actually reduced the theory to a finite-dimensional Schrödinger problem. The regulator must have the following properties. The configuration space is a finite-dimensional real vector space $E_\Lambda \simeq \mathbb{R}^{M_\Lambda}$, the kinetic energy is a positive constant-coefficient quadratic form in the conjugate momenta, and the Hamiltonian on $L^2(E_\Lambda, \mathrm{d}_\Lambda\varphi)$ is the Friedrichs Hamiltonian associated to

$$Q_\Lambda[\Psi] = \frac{1}{2}\int_{E_\Lambda} G_\Lambda^{AB}\partial_A\overline{\Psi}\,\partial_B\Psi\,\mathrm{d}_\Lambda\varphi + \int_{E_\Lambda} U_\Lambda(\varphi)|\Psi(\varphi)|^2\,\mathrm{d}_\Lambda\varphi,$$

where G_Λ^{AB} is positive definite and $U_\Lambda : E_\Lambda \to \mathbb{R}$ is locally Kato and bounded from below. Under these hypotheses the fixed-Λ transition kernels are exactly the objects governed by the finite-dimensional theorem: the Euclidean transition kernel is a Wiener or Brownian-bridge expectation on $C([0,\tau], E_\Lambda)$ with potential weight $\exp[-\int_0^\tau U_\Lambda(\Phi(s))\mathrm{d}s]$, after the linear change of variables that puts G_Λ^{AB} in Euclidean form.

This hypothesis is satisfied, for example, by a finite spatial lattice in a finite box with stable real polynomial potential energy, and by a genuine finite-mode truncation with a stable real polynomial potential. It is not satisfied by a continuum smooth cutoff that leaves infinitely many spatial degrees of freedom, nor by a formal rule that only modifies the propagator in a functional integral. In those cases the phrase "the Trotter limit for H_Λ" has no content until the Hilbert space, domain, closed quadratic form, and limiting topology have been specified.

Under the finite-dimensional Schrödinger-regulator hypotheses just stated, the Lorentzian time-sliced kernels of Definition 11.4 have the corresponding strong operator or distributional limits after smearing boundary configurations. In that case

$$\lim_{N\to\infty} K_{\Lambda,N}(\varphi_f, t_f; \varphi_i, t_i) = {}_\Lambda\langle\varphi_f; t_i|\exp[-\mathrm{i}H_\Lambda(t_f - t_i)]|\varphi_i; t_i\rangle_\Lambda$$

as a distributional kernel after smearing the boundary configurations. The continuum expression

$$\langle\varphi_f; t_f|\varphi_i; t_i\rangle = \int_{\phi(t_i)=\varphi_i}^{\phi(t_f)=\varphi_f} [D\phi]\,\exp\left(\mathrm{i}S[\phi]\right)$$

is a compact notation for this two-stage construction: first a finite regulator and time slicing, then a stated continuum or perturbative limiting procedure. If the regulator is instead a Euclidean covariance cutoff, a Euclidean spacetime lattice action introduced directly, or a purely perturbative cutoff, the construction requires its own definition: a Gaussian measure with an interaction, a finite lattice integral, a transfer-matrix or Hamiltonian construction, a constructive limit, or a formal asymptotic expansion. At fixed finite-dimensional Λ, the same hypotheses identify the time-sliced kernels with the Trotter products for the self-adjoint Hamiltonian H_Λ. The strong operator convergence acts on square-integrable wavefunctions. Pairing the resulting operator kernel with Schwartz test functions in the boundary variables gives the distributional kernel displayed above. Passing from the regulated family to a continuum symbol is therefore notation for the specified continuum or perturbative limiting prescription.

Definition 11.5 (Compact scalar path-integral notation)**.** The compact path-integral notation

$$Z = \int [D\phi]\,\mathrm{e}^{\mathrm{i}S[\phi]}$$

is used only as an abbreviation for such regulated limits or for asymptotic expansions derived from them. Its meaning is part of the chosen regularization and limiting prescription. A positive Borel measure is one possible mathematical object for some bosonic Euclidean scalar theories; it is not the definition of a general QFT path integral and is not implied by the symbol $[D\phi]$.

Definition 11.6 (Lorentzian oscillatory path-integral notation)**.** Fix a finite spatial regulator Λ, a finite time partition, and coordinates $u = (u^1, \ldots, u^M)$ on all intermediate field variables. With an insertion F, the regulated Lorentzian expression has the form

$$I_{\Lambda,N}(F) = \int_{\mathbb{R}^M} \exp(\mathrm{i}S_{\Lambda,N}(u))\, a_{\Lambda,N}(u; F)\, \mathrm{d}^M u.$$

This formula is classified as an oscillatory integral or oscillatory distribution. One concrete definition is the Abel boundary value

$$I_{\Lambda,N}(F) = \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^M} \exp(\mathrm{i}S_{\Lambda,N}(u) - \epsilon Q(u))\, a_{\Lambda,N}(u; F)\, \mathrm{d}^M u,$$

whenever the limit exists in the stated distribution topology; here Q is a positive definite quadratic form used only to approach the boundary value. Equivalent finite-dimensional formulations use the Fourier-transform definition of oscillatory integrals with amplitudes in a specified symbol class.

For a nondegenerate quadratic phase

$$S(u) = \frac{1}{2} u^T A u + J^T u, \qquad A = A^T, \quad \det A \neq 0,$$

the normalized Fresnel value is

$$\lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^M} \exp\left(\frac{\mathrm{i}}{2} u^T A u + \mathrm{i} J^T u - \frac{\epsilon}{2} u^T u\right) \mathrm{d}^M u = (2\pi)^{M/2} \mathrm{e}^{\mathrm{i}\pi \,\mathrm{sgn}(A)/4} |\det A|^{-1/2} \exp\left(-\frac{\mathrm{i}}{2} J^T A^{-1} J\right),$$

where $\mathrm{sgn}(A) = n_+(A) - n_-(A)$ is the signature. The phase $\mathrm{e}^{\mathrm{i}\pi\,\mathrm{sgn}(A)/4}$ is part of the finite-regulator definition; in families of quadratic phases its changes are recorded by the Maslov index. For a nonlinear phase, perturbation theory is the stationary phase expansion of the finite-regulator oscillatory distribution around the chosen critical configurations, with Hessian signatures and contour or boundary-value data included in the definition.

The continuum Lorentzian symbol

$$\int [D\phi]\, \mathrm{e}^{\mathrm{i}S[\phi]}$$

is therefore an oscillatory pseudo-integral: a shorthand for compatible finite-regulator oscillatory distributions, their Abel or Fourier boundary values, or their stationary-phase asymptotic expansions. This classification is separate from the Euclidean positive-measure construction used in constructive scalar examples, from Berezin integration for fermionic variables, and from gauge-fixed, BRST/BV, lattice, or holonomy constructions for gauge theories. The relevant analytic frameworks are finite-dimensional oscillatory-integral theory and stationary phase in the sense of Hörmander, the Maslov-index refinement of the Fresnel phase, and the Albeverio–Høegh-Krohn theory of infinite-dimensional Fresnel integrals when such infinite-dimensional oscillatory objects are constructed directly.

Regime	Rigorous status	Scope of the statement
$P(\phi)_2$, including ϕ_2^4	Constructed non-Gaussian Euclidean and Wightman models for bounded-below polynomial interactions after Wick ordering and the required local renormalizations.	This is the basic scalar positive-measure constructive class in two spacetime dimensions. Reflection positivity, locality, and the Wightman/OS structures can be verified in the constructed models.
ϕ_3^4	Constructed non-Gaussian massive scalar models with stable quartic interaction after mass, coupling, and vacuum-energy counterterms.	The construction is substantially harder than $P(\phi)_2$ and is tied to superrenormalizability in three spacetime dimensions.
Selected low-dimensional superrenormalizable scalar–fermion models	Constructive results exist for specific Yukawa-type and related models after the ultraviolet cutoff, Wick ordering, fermion determinant/Pfaffian conventions, and local renormalizations are fixed.	Each theorem is model-specific. The label of a Lagrangian family does not define a uniform constructive theorem; the required hypotheses must be cited when such a model is used.
Two-dimensional Yang–Mills	Constructed continuum gauge theory for compact gauge group through heat-kernel/holonomy measures on generalized connections or lattice projective limits.	This is a gauge-theory construction, not a scalar Borel measure on ordinary fields. The theory has no local propagating gluon degrees of freedom, and its status should not be extrapolated to $D = 4$.
ϕ_D^4, $D \geq 5$	For the standard reflection-positive lattice/Ising/$\lambda\phi^4$ critical scaling problem with positive quartic coupling, the scaling limits covered by the triviality theorems are Gaussian.	The theorem concerns that constructive scaling-limit family. It is a nonperturbative obstruction to the standard scalar route to an interacting continuum limit above four dimensions.
ϕ_4^4	For the standard reflection-positive lattice/Ising/$\lambda\phi^4$ critical scaling problem, marginal triviality theorems show Gaussian scaling limits.	This is sharper than the perturbative Landau-pole argument. It still leaves open the logically different question whether another UV-complete four-dimensional QFT could agree with formal ϕ_4^4 perturbation theory to all orders while differing nonperturbatively.
Four-dimensional pure Yang–Mills	Finite Wilson lattices are rigorous compact-group integrals, and asymptotic freedom gives the perturbative ultraviolet mechanism. Continuum existence with mass gap remains a separate constructive problem.	The finite-regulator theory is not the desired continuum QFT. A continuum limit must produce gauge-invariant Schwinger functions satisfying positivity, locality, covariance, and the spectral requirements used for reconstruction.
Four-dimensional interacting local QFT of scalar/gauge type	Open constructive problem once one asks for non-Gaussian local fields or gauge-invariant observables with Wightman/OS positivity, covariance, locality, and the spectral properties used later in the monograph.	Perturbative formal power series, finite regulators, and generalized-free fields do not supply this construction. The status of each four-dimensional model must be stated separately: free, perturbative, finite-regulator, conditional continuum limit, trivial scaling limit, or open problem.

Table 11.2: Selected rigorous existence and triviality regimes relevant to the symbol $[D\phi]$ and to continuum-limit claims. The table records theorem classes and open construction problems; it is not a general definition of path integrals.

References for Table 11.2. Standard entry points are E. Nelson's construction of the two-dimensional quartic interaction, B. Simon's *The $P(\phi)_2$ Euclidean (Quantum) Field Theory*, and J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, for $P(\phi)_2$ and ϕ_3^4 constructive scalar models; B. Driver, A. Sengupta, and T. Lévy for rigorous two-dimensional Yang–Mills constructions; M. Aizenman and J. Fröhlich for scalar triviality above four dimensions; and M. Aizenman and H. Duminil-Copin, "Marginal triviality of the scaling limits of critical 4D Ising and ϕ_4^4 models," *Ann. of Math.* **194** (2021) 163–235, for the four-dimensional marginal triviality theorem. The Glimm–Jaffe functional-integral construction is used here as a reference for the positive-measure scalar sector. The organizing framework of this monograph is broader: a path integral is specified by the mathematical object assigned by the chosen regulator and limiting prescription, which may be a Borel measure, an oscillatory distribution, a Berezin functional, a gauge-fixed or BV integral, a lattice or holonomy construction, or a formal asymptotic expansion. A broader catalog of constructive routes and theorem status appears in Table 136.1 of the constructive volume; the present table records only the examples needed for the first path-integral distinctions.

Vacuum correlation functions avoid direct use of field-configuration eigenvectors. For points $x_a = (x_a^0, \vec{x}_a)$, the basic Lorentzian operator distributions are

$$\langle\Omega|\widehat{\phi}(x_1)\cdots\widehat{\phi}(x_n)|\Omega\rangle.$$

These are distributions in the spacetime variables, and their analytic continuations require spectral and growth assumptions.

11.2 Complex Time Contours

Definition 11.7 (Complex-time contour with ordered insertions)**.** Let z^0 be a complex time coordinate. If a contour C in the z^0-plane is used for the time variable, the corresponding action is

$$S_C[\phi] = \int_C \mathrm{d}z^0 \int \mathrm{d}^{D-1}\vec{x}\,\mathcal{L}(\phi, \partial_\mu\phi).$$

Insertions are ordered along the contour. For vacuum correlation functions the useful contour has insertion times whose imaginary parts are ordered:

$$\operatorname{Im} z_1^0 < \operatorname{Im} z_2^0 < \cdots < \operatorname{Im} z_n^0.$$

The real parts are free variables. The ordering in imaginary time provides the analytic separation that will become time ordering after a uniform rotation to real time.

Figure 11.1 records the contour-ordering convention: the contour carries the action, and the imaginary parts of insertion times impose the analytic ordering used for later time ordering.

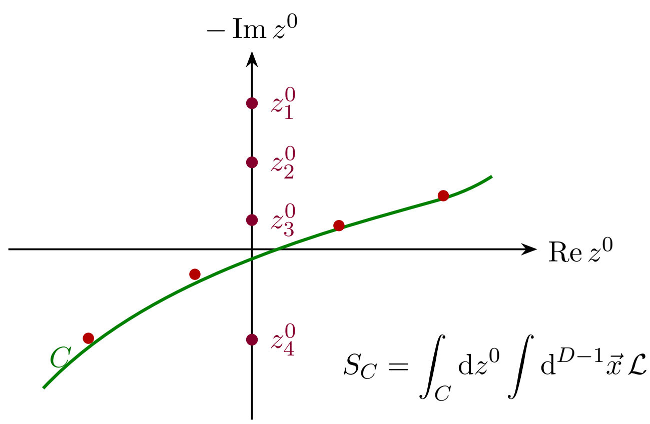


Figure 11.1: Complex-time contour data for ordered vacuum insertions. The ordering of imaginary parts determines the analytic domain that rotates to Lorentzian time ordering.

The figure records only the ordering data. The analytic statement is the existence of a holomorphic continuation of the relevant correlation function in the domain where the imaginary parts are ordered.

Hypothesis 11.8 (Ordered tube analyticity for scalar vacuum correlators)**.** For the scalar vacuum n-point distribution under consideration, assume there exists a holomorphic function

$$\mathcal{W}_n(z_1, \ldots, z_n), \qquad z_a = (z_a^0, \vec{x}_a),$$

on the ordered domain

$$\operatorname{Im} z_1^0 < \operatorname{Im} z_2^0 < \cdots < \operatorname{Im} z_n^0$$

for fixed real spatial points, with distributional boundary value equal to the Lorentzian Wightman distribution in the corresponding operator order. The hypothesis includes the growth condition needed for the boundary value to exist as a tempered or otherwise specified distribution. In a Wightman theory this is a consequence of the spectral condition and locality in the usual tube-domain construction; in a path-integral construction it is an assumption or theorem of the chosen reconstruction framework.

11.3 Euclidean Functional Expression

Definition 11.9 (Euclidean continuation of a scalar Lagrangian)**.** Put the complex time on the Euclidean axis by writing

$$z^0 = -\mathrm{i}\tau, \qquad x_E = (\tau, \vec{x}) \in \mathbb{R}^D.$$

When the analytic continuation of vacuum correlators exists, Euclidean field-insertion notation is defined inside ordered correlation functions by

$$\widehat{\phi}(z^0, \vec{x})\big|_{z^0=-\mathrm{i}\tau}.$$

This is a shorthand for the analytic continuation of Lorentzian Hilbert-space matrix elements; the Euclidean functional variable in $S_E[\phi]$ remains the unhatted field ϕ. For Euclidean times ordered as

$$\tau_{i_1} > \tau_{i_2} > \cdots > \tau_{i_n},$$

this notation records the same analytic boundary value that rotates to the Lorentzian time-ordered product with $x_{i_1}^0 > x_{i_2}^0 > \cdots > x_{i_n}^0$. For a Lorentzian Lagrangian density

$$\mathcal{L}(\phi, \partial_0\phi, \partial_i\phi),$$

define the Euclidean Lagrangian density by

$$\mathcal{L}_E(\phi, \partial_\tau\phi, \partial_i\phi) = -\mathcal{L}(\phi, \mathrm{i}\partial_\tau\phi, \partial_i\phi),$$

when this analytic substitution is meaningful for the regulated theory. The Euclidean action is

$$S_E[\phi] = \int \mathrm{d}^D x_E\, \mathcal{L}_E(\phi, \partial_E\phi).$$

Euclidean scalar action. For the real scalar field with

$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi),$$

this gives

$$\mathcal{L}_E = \frac{1}{2}(\partial_\tau\phi)^2 + \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2 + V(\phi).$$

With mostly-plus Lorentzian metric,

$$-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi = \frac{1}{2}(\partial_0\phi)^2 - \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2.$$

Substitution of $\partial_0 = \mathrm{i}\partial_\tau$ gives

$$\mathcal{L}(\phi, \mathrm{i}\partial_\tau\phi, \partial_i\phi) = -\frac{1}{2}(\partial_\tau\phi)^2 - \frac{1}{2}\sum_{i=1}^{D-1}(\partial_i\phi)^2 - V(\phi).$$

The definition $\mathcal{L}_E = -\mathcal{L}|_{\partial_0=\mathrm{i}\partial_\tau}$ then gives the displayed positive kinetic terms.

Definition 11.10 (Regulated Euclidean scalar Green function)**.** With a regulator in place, the normalized Euclidean n-point function is

$$\langle\phi(x_{E,1})\cdots\phi(x_{E,n})\rangle_{E,\Lambda} = \frac{1}{Z_{E,\Lambda}}\int D_\Lambda^{\mathrm{ref}}\phi\,\mathrm{e}^{-S_{E,\Lambda}[\phi]}\,\phi(x_{E,1})\cdots\phi(x_{E,n}),$$

where

$$Z_{E,\Lambda} = \int D_\Lambda^{\mathrm{ref}}\phi\,\mathrm{e}^{-S_{E,\Lambda}[\phi]}.$$

Here the Euclidean action $S_{E,\Lambda}$ includes its quadratic part and $D_\Lambda^{\mathrm{ref}}\phi$ is the reference density of Definition 11.3. If

$$S_{E,\Lambda} = S_{\mathrm{kin},\Lambda} + L_\Lambda,$$

the same normalized correlation function may equivalently be written with the Gaussian measure $\mathrm{d}\mu_{C_\Lambda}$ and interaction weight $\mathrm{e}^{-L_\Lambda[\phi]}$. The choice of notation therefore records which factor carries the quadratic kinetic form. The continuum Euclidean Green function is the limit of these distributions when the limit exists in the chosen topology and with the chosen renormalization conditions.

11.4 Uniform Wick Rotation

Definition 11.11 (Uniform Wick-rotation path for ordered times)**.** Fix real numbers

$$t_1 > t_2 > \cdots > t_n.$$

For $0 < \alpha \leq \pi/2$, set

$$z_a^0(\alpha) = \mathrm{e}^{-\mathrm{i}\alpha}t_a.$$

At $\alpha = \pi/2$, the insertion lies on the Euclidean axis:

$$z_a^0 = -\mathrm{i}t_a.$$

As $\alpha \to 0^+$, the insertion approaches the Lorentzian time t_a. Since

$$\operatorname{Im} z_a^0(\alpha) = -t_a \sin\alpha,$$

the ordering

$$\operatorname{Im} z_1^0(\alpha) < \cdots < \operatorname{Im} z_n^0(\alpha)$$

is maintained throughout the rotation.

Figure 11.2 displays the uniform rotation of all insertion times and the preserved imaginary-time ordering along the rotation path.

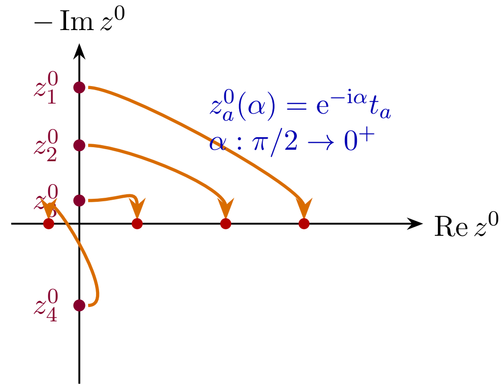


Figure 11.2: Uniform Wick rotation of ordered insertion times. The deformation $z_a^0(\alpha) = \mathrm{e}^{-\mathrm{i}\alpha} t_a$ preserves the imaginary-time ordering until the Lorentzian boundary is reached.

Euclidean boundary value and the time-ordered function. Under Hypothesis 11.8, the Euclidean correlation function continued along this common rotation becomes the Lorentzian time-ordered correlator with $t_1 > \cdots > t_n$:

$$\langle \phi(x_{E,1}) \cdots \phi(x_{E,n}) \rangle_E \longrightarrow \langle \Omega | \, \mathrm{T}\{\widehat{\phi}(x_1) \cdots \widehat{\phi}(x_n)\} | \Omega \rangle.$$

Here T denotes time ordering of the operator-valued distributions.

By Definition 11.11, the points $z_a^0(\alpha) = \mathrm{e}^{-\mathrm{i}\alpha} t_a$ stay inside the ordered domain of Hypothesis 11.8 for every $0 < \alpha \leq \pi/2$. At $\alpha = \pi/2$, the points lie on the Euclidean axis and give the Euclidean boundary value. As $\alpha \to 0^+$, the ordered tube boundary value is the Lorentzian Wightman distribution in the operator order $1, \ldots, n$. Since the real times already obey $t_1 > \cdots > t_n$, that Wightman order is exactly the time-ordered product.

11.5 The Free Euclidean Gaussian Measure

Definition 11.12 (Finite-regulator Euclidean Gaussian datum)**.** At finite Euclidean regulator, the free scalar path integral is an ordinary Gaussian integral. Let F_Λ be the finite-dimensional real vector space of regulated Euclidean fields, let $\mathrm{d}_\Lambda\phi$ be the regulator Lebesgue measure, and let

$$A_\Lambda : F_\Lambda \longrightarrow F_\Lambda^\vee$$

be a symmetric positive-definite operator. The pairing between $F_\Lambda^\vee$ and F_Λ is denoted by $\langle J, \phi\rangle_\Lambda$. The regulated free action is

$$S_{E,0,\Lambda}[\phi] = \frac{1}{2}\langle A_\Lambda\phi, \phi\rangle_\Lambda.$$

The corresponding normalized Gaussian expectation is

$$\langle F(\phi)\rangle_{0,\Lambda} = \frac{1}{Z_{0,\Lambda}}\int_{F_\Lambda} \mathrm{d}_\Lambda\phi\, \mathrm{e}^{-\frac{1}{2}\langle A_\Lambda\phi,\phi\rangle_\Lambda} F(\phi),$$

with

$$Z_{0,\Lambda} = \int_{F_\Lambda} \mathrm{d}_\Lambda\phi\, \mathrm{e}^{-\frac{1}{2}\langle A_\Lambda\phi,\phi\rangle_\Lambda}.$$

The covariance is the inverse map

$$C_\Lambda = A_\Lambda^{-1} : F_\Lambda^\vee \to F_\Lambda.$$

Finite-dimensional Gaussian source functional. For sources $J \in F_\Lambda^\vee$, completing the square gives

$$-\frac{1}{2}\langle A_\Lambda\phi, \phi\rangle_\Lambda + \langle J, \phi\rangle_\Lambda = -\frac{1}{2}\langle A_\Lambda(\phi - C_\Lambda J), \phi - C_\Lambda J\rangle_\Lambda + \frac{1}{2}\langle J, C_\Lambda J\rangle_\Lambda.$$

Therefore

$$\frac{Z_{0,\Lambda}[J]}{Z_{0,\Lambda}[0]} = \exp\left(\frac{1}{2}\langle J, C_\Lambda J\rangle_\Lambda\right), \qquad Z_{0,\Lambda}[J] = \int_{F_\Lambda} \mathrm{d}_\Lambda\phi\, \mathrm{e}^{-S_{E,0,\Lambda}[\phi]+\langle J,\phi\rangle_\Lambda}.$$

Differentiating at $J = 0$ gives, for $f, g \in F_\Lambda^\vee$,

$$\big\langle \langle f, \phi\rangle_\Lambda \langle g, \phi\rangle_\Lambda \big\rangle_{0,\Lambda} = \langle f, C_\Lambda g\rangle_\Lambda,$$

and all higher moments are sums over pairings with this covariance. This is Wick's theorem as a finite-dimensional Gaussian identity.

The completion-of-the-square identity follows from $A_\Lambda C_\Lambda = \mathrm{id}_{F_\Lambda^\vee}$. Translation invariance of the Lebesgue density gives

$$\int_{F_\Lambda} \mathrm{d}_\Lambda\phi\, \mathrm{e}^{-\frac{1}{2}\langle A_\Lambda\phi,\phi\rangle_\Lambda + \langle J,\phi\rangle_\Lambda} = \mathrm{e}^{\frac{1}{2}\langle J, C_\Lambda J\rangle_\Lambda} \int_{F_\Lambda} \mathrm{d}_\Lambda\phi\, \mathrm{e}^{-\frac{1}{2}\langle A_\Lambda\phi,\phi\rangle_\Lambda},$$

after the shift $\phi \mapsto \phi + C_\Lambda J$. Dividing by $Z_{0,\Lambda}[0]$ gives the source functional. Applying $\partial_s\partial_t$ to the identity with $J = sf + tg$ and setting $s = t = 0$ gives the two-point formula. Repeated differentiation gives the sum over pairings because the exponent is quadratic in J; odd derivatives vanish at $J = 0$, and even derivatives contract derivatives in pairs. This finite-dimensional calculation is the regulator-level meaning of the free Euclidean Gaussian path integral. The nontrivial continuum questions are the choice of regulator, the topology in which C_Λ converges, and the existence of limiting distributions after the test functions have been fixed.

Definition 11.13 (Continuum free scalar Euclidean covariance)**.** For the continuum free scalar field of mass $m \geq 0$,

$$V(\phi) = \frac{1}{2}m^2\phi^2,$$

and the Euclidean action is

$$S_{E,0}[\phi] = \frac{1}{2}\int \mathrm{d}^D x_E\, \phi(x_E)(-\partial_E^2 + m^2)\phi(x_E),$$

after integration by parts with the chosen boundary conditions. The operator

$$A_E = -\partial_E^2 + m^2$$

is the continuum counterpart of A_Λ. If $m > 0$, or if boundary conditions or an infrared regulator remove the constant zero mode when $m = 0$, the inverse covariance is represented by the Green function Δ_E defined by

$$A_E \Delta_E(x_E - y_E) = \delta^{(D)}(x_E - y_E).$$

For test functions f, g, the continuum covariance is the distributional limit

$$\lim_\Lambda \langle f_\Lambda, C_\Lambda g_\Lambda\rangle_\Lambda = \int \mathrm{d}^D x_E\, \mathrm{d}^D y_E\, f(x_E)\Delta_E(x_E - y_E)g(y_E),$$

where f_Λ, g_Λ are the regulated images of f, g. On $\mathbb{R}^D$, translation invariance gives the momentum-space kernel

$$\Delta_E(x_E - y_E) = \int \frac{\mathrm{d}^D k_E}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k_E\cdot(x_E - y_E)}}{k_E^2 + m^2},$$

where

$$k_E^2 = (k_\tau)^2 + \vec{k}^2, \qquad k_E \cdot x_E = k_\tau \tau + \vec{k}\cdot\vec{x}.$$

In point notation, after the distribution has been paired with test functions, the free two-point function is written

$$\langle \phi(x_E)\phi(y_E)\rangle_E = \Delta_E(x_E - y_E)$$

for the free theory. This equation is precisely the continuum form of the finite-dimensional covariance identity above.

11.6 Spectral Zeta Determinants

The finite-dimensional Gaussian normalization contains the determinant of the quadratic operator:

$$\int_{\mathbb{R}^N} \mathrm{d}^N\phi\, \exp\left[-\frac{1}{2}\phi^i A_{ij}\phi^j\right] = (2\pi)^{N/2}(\det A)^{-1/2}$$

for a positive-definite symmetric matrix A. In field theory the analogous quantity is a determinant of a differential operator and requires its own regularized definition. A spectral zeta determinant is one such definition for operators with discrete positive spectrum. It regulates the determinant; it does not by itself choose the continuum theory or the subtraction coordinates of an interacting path integral.

Definition 11.14 (Spectral zeta determinant)**.** Let A be a positive self-adjoint elliptic operator of order $r > 0$ on a compact d-dimensional Euclidean manifold, with boundary condition chosen so that A has compact resolvent. Let

$$0 < \lambda_1 \leq \lambda_2 \leq \cdots$$

be the positive eigenvalues of A, repeated with multiplicity. Zero modes, if present, are omitted from this list and must be integrated or treated as collective coordinates separately. For $\operatorname{Re} s > d/r$, set

$$\zeta_A(s) = \sum_j \lambda_j^{-s}. \tag{11.3}$$

Equivalently, in the same half-plane,

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty \mathrm{d}t\, t^{s-1} \operatorname{Tr} \mathrm{e}^{-tA}. \tag{11.4}$$

Assume that the heat trace has the usual small-t asymptotic expansion and that the meromorphic continuation of $\zeta_A(s)$ is regular at $s = 0$. For a reference scale μ of mass dimension 1, define

$$\log \det_\zeta \left(\frac{A}{\mu^r} \right) = - \left. \frac{\mathrm{d}}{\mathrm{d}s} \right|_{s=0} \sum_j \left(\frac{\lambda_j}{\mu^r} \right)^{-s} = -\zeta_A'(0) - \zeta_A(0) \log \mu^r. \tag{11.5}$$

Scale dependence of a zeta determinant. In the setting of Definition 11.14, suppose

$$\operatorname{Tr} \mathrm{e}^{-tA} \sim \sum_{j \geq 0} a_j(A)\, t^{(j-d)/r} \qquad (t \downarrow 0)$$

with no logarithmic term at order t^0, and suppose zero modes have been removed. Then

$$\zeta_A(0) = a_d(A).$$

Consequently a change $\mu \mapsto \mathrm{e}^\sigma \mu$ changes the determinant by

$$\log \det_\zeta \left(\frac{A}{(\mathrm{e}^\sigma \mu)^r} \right) - \log \det_\zeta \left(\frac{A}{\mu^r} \right) = -r\sigma\, a_d(A).$$

Thus the μ-dependence of the one-loop determinant is controlled by the local heat-kernel coefficient $a_d(A)$. The Mellin representation

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty \mathrm{d}t\, t^{s-1} \operatorname{Tr} \mathrm{e}^{-tA}$$

continues meromorphically by subtracting the small-t heat-kernel asymptotic series. The coefficient $a_d(A)$ multiplies t^0, hence contributes $a_d(A)/(s\Gamma(s))$ to $\zeta_A(s)$ near $s = 0$. Since $\Gamma(s)^{-1} = s + O(s^2)$, this contribution has value $a_d(A)$ at $s = 0$. The remaining subtracted terms are either regular and vanish at $s = 0$ after multiplication by $\Gamma(s)^{-1}$, or have poles away from $s = 0$ under the stated no-log hypothesis. This gives $\zeta_A(0) = a_d(A)$. Substituting $(\mathrm{e}^\sigma \mu)^r = \mathrm{e}^{r\sigma} \mu^r$ into the defining formula gives

$$\log \det_\zeta \left(\frac{A}{(\mathrm{e}^\sigma \mu)^r} \right) = -\zeta_A'(0) - \zeta_A(0)(\log \mu^r + r\sigma),$$

and subtracting

$$\log \det_\zeta \left(\frac{A}{\mu^r}\right) = -\zeta_A'(0) - \zeta_A(0)\log \mu^r$$

leaves $-r\sigma\zeta_A(0) = -r\sigma a_d(A)$.

This is the determinant counterpart of the regulator/subtraction distinction recorded in Table 11.1: changing the determinant scale changes a one-loop effective action by a local heat-kernel coefficient, not by a new nonlocal observable.

One-loop determinant from a quadratic fluctuation. Let $\phi = \phi_{\mathrm{cl}} + \eta$ be a background split for a bosonic Euclidean path integral, and suppose that the quadratic fluctuation operator

$$A_{\phi_{\mathrm{cl}}} = S_E''[\phi_{\mathrm{cl}}]$$

is positive and satisfies the hypotheses of Definition 11.14. With the Gaussian normalization chosen by spectral zeta regularization, the one-loop contribution of the nonzero modes to the effective action is

$$\Gamma_1[\phi_{\mathrm{cl}}] = \frac{1}{2}\log \det_\zeta \left(\frac{A_{\phi_{\mathrm{cl}}}}{\mu^r}\right). \tag{11.6}$$

For fermionic Gaussian variables the corresponding Pfaffian or determinant appears with the opposite statistics sign and with the spinor and reality structure stated as part of the regulator. At finite regulator,

$$S_E[\phi_{\mathrm{cl}} + \eta] = S_E[\phi_{\mathrm{cl}}] + \frac{1}{2}\langle \eta, A_{\phi_{\mathrm{cl}},\Lambda}\eta\rangle_\Lambda + O(\eta^3)$$

when ϕ_{cl} obeys the regulated equation of motion. The Gaussian integral over the nonzero fluctuation modes gives

$$Z_\Lambda^{1-\mathrm{loop}}[\phi_{\mathrm{cl}}] = \exp[-S_E[\phi_{\mathrm{cl}}]](2\pi)^{N_\Lambda/2}\big(\det A_{\phi_{\mathrm{cl}},\Lambda}\big)^{-1/2}$$

up to the finite-dimensional zero-mode and collective-coordinate factors that were excluded. The factor $(2\pi)^{N_\Lambda/2}$ is independent of the background in a fixed bosonic coordinate density and is part of the vacuum normalization. Replacing the divergent determinant by $\det_\zeta(A_{\phi_{\mathrm{cl}}}/\mu^r)$ gives the logarithm (11.6). The argument uses only the quadratic Gaussian identity; interactions beyond quadratic order are additional vertices in perturbation theory.

Example 11.15 (Thermal harmonic oscillator determinant)**.** Let $S^1_\beta = \mathbb{R}/\beta\mathbb{Z}$ and

$$A_\omega = -\frac{\mathrm{d}^2}{\mathrm{d}\tau^2} + \omega^2, \qquad \omega > 0,$$

acting on periodic functions. The eigenvalues are

$$\lambda_n = \left(\frac{2\pi n}{\beta}\right)^2 + \omega^2, \qquad n \in \mathbb{Z}.$$

For this one-dimensional second-order operator $\zeta_{A_\omega}(0) = 0$, so the determinant is independent of μ. Differentiate the logarithm before integrating it:

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\log \det_\zeta A_\omega = 2\omega \sum_{n\in\mathbb{Z}} \frac{1}{(2\pi n/\beta)^2 + \omega^2}.$$

The partial-fraction identity

$$\sum_{n\in\mathbb{Z}} \frac{1}{n^2+a^2} = \frac{\pi}{a}\coth(\pi a), \qquad a>0,$$

with $a=\beta\omega/(2\pi)$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\log\det\nolimits_\zeta A_\omega = \beta\coth\frac{\beta\omega}{2}.$$

Therefore

$$\log\det\nolimits_\zeta A_\omega = 2\log\left(2\sinh\frac{\beta\omega}{2}\right)+C.$$

The constant is fixed by the small-ω behavior: $\det_\zeta A_\omega \sim \omega^2 \det'_\zeta(-\mathrm{d}^2/\mathrm{d}\tau^2)$ and $\det'_\zeta(-\mathrm{d}^2/\mathrm{d}\tau^2) = \beta^2$. Hence

$$\det\nolimits_\zeta A_\omega = 4\sinh^2\frac{\beta\omega}{2}, \qquad Z_{\mathrm{osc}}(\beta,\omega) = (\det\nolimits_\zeta A_\omega)^{-1/2} = \frac{1}{2\sinh(\beta\omega/2)}. \tag{11.7}$$

The last expression is the canonical thermal partition function of the oscillator,

$$\operatorname{Tr}\mathrm{e}^{-\beta\omega(a^\dagger a+1/2)} = \frac{\mathrm{e}^{-\beta\omega/2}}{1-\mathrm{e}^{-\beta\omega}}.$$

Example 11.16 (Circle Casimir finite part)**.** For a massless real scalar on a spatial circle of circumference L, remove the constant zero mode and write the oscillator frequencies as

$$\omega_n = \frac{2\pi|n|}{L}, \qquad n\in\mathbb{Z}\setminus\{0\}.$$

The zeta-regularized finite part of the vacuum energy is

$$E_0^\zeta(L) = \frac{1}{2}\sum_{n\neq 0}\omega_n = \frac{2\pi}{L}\zeta_{\mathrm{R}}(-1) = -\frac{\pi}{6L}, \tag{11.8}$$

where $\zeta_{\mathrm{R}}(-1)=-1/12$ is the analytic continuation of the Riemann zeta function. The zero mode and any additive local vacuum-energy normalization are separate data. In two-dimensional CFT language the result is the $c=1$ case of $E_0=-\pi c/(6L)$ on the cylinder.

11.7 The Feynman Prescription

Definition 11.17 (Wick-rotated free propagator poles)**.** Let

$$\omega_{\vec{k}} = \sqrt{\vec{k}^2+m^2}.$$

For a Euclidean separation $(\tau,\vec{r})$, with $\tau>0$, the free propagator is

$$\Delta_E(\tau,\vec{r}) = \int \frac{\mathrm{d}k_\tau\,\mathrm{d}^{D-1}\vec{k}}{(2\pi)^D}\frac{\mathrm{e}^{\mathrm{i}k_\tau\tau+\mathrm{i}\vec{k}\cdot\vec{r}}}{k_\tau^2+\omega_{\vec{k}}^2}.$$

Continue to complex time by setting $z^0 = \mathrm{e}^{-\mathrm{i}\alpha} t$, with $0 < \alpha \le \pi/2$, and rotate the momentum variable by

$$k_\tau = \mathrm{i}\mathrm{e}^{\mathrm{i}\alpha} k^0.$$

Then

$$\Delta_E = -\mathrm{i}\mathrm{e}^{\mathrm{i}\alpha} \int \frac{\mathrm{d}k^0\, \mathrm{d}^{D-1}\vec{k}}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{r} - \mathrm{i}k^0 t}}{-\mathrm{e}^{2\mathrm{i}\alpha}(k^0)^2 + \vec{k}^2 + m^2},$$

with the contour orientation chosen to match the original Euclidean k_τ-contour. The denominator vanishes at

$$k^0 = \pm \mathrm{e}^{-\mathrm{i}\alpha} \omega_{\vec{k}}.$$

As $\alpha \to 0^+$, these poles approach the real axis as

$$-\omega_{\vec{k}} + \mathrm{i}0, \qquad +\omega_{\vec{k}} - \mathrm{i}0.$$

Figure 11.3 shows how the Euclidean poles rotate toward the Lorentzian contour and become the Feynman $\mathrm{i}\epsilon$ placement.

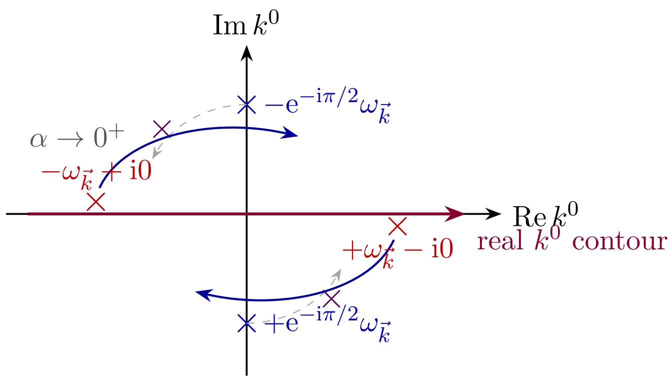


Figure 11.3: Pole motion under Wick rotation. The Euclidean poles rotate to the Lorentzian Feynman positions $-\omega_{\vec{k}} + \mathrm{i}0$ and $+\omega_{\vec{k}} - \mathrm{i}0$.

Feynman boundary value from pole rotation. The limiting Lorentzian distribution is therefore

$$\Delta_F(x; m^2) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2 + m^2 - \mathrm{i}\epsilon}, \qquad \epsilon > 0,$$

where

$$k \cdot x = -k^0 x^0 + \vec{k}\cdot\vec{x}, \qquad k^2 = -(k^0)^2 + \vec{k}^2.$$

The limit $\epsilon \to 0^+$ is taken after the distribution has been paired with test functions or after the contour integral has been evaluated. Definition 11.17 shows that the positive-frequency pole approaches the real k^0-axis from below and the negative-frequency pole approaches it from above. The denominator $k^2 + m^2 - \mathrm{i}\epsilon$, with $k^2 = -(k^0)^2 + \vec{k}^2$, has precisely these pole placements:

$$k^2 + m^2 - \mathrm{i}\epsilon = -(k^0)^2 + \omega_{\vec{k}}^2 - \mathrm{i}\epsilon.$$

Solving to first order in ϵ gives

$$k^0 = +\omega_{\vec{k}} - \mathrm{i}0, \qquad k^0 = -\omega_{\vec{k}} + \mathrm{i}0.$$

The prefactor $-\mathrm{i}$ is the Jacobian and orientation factor obtained from the rotation $k_\tau = \mathrm{i}e^{\mathrm{i}\alpha}k^0$. The limit is a boundary value of meromorphic functions in k^0, hence is interpreted only after pairing with test functions or after an equivalent contour evaluation.

11.8 Three Green Functions

Definition 11.18 (Wightman, time-ordered, and Euclidean Green functions)**.** Let $\widehat{\phi}$ be a scalar operator-valued distribution with vacuum Ω. The Wightman n-point distribution is

$$W_n(x_1, \ldots, x_n) = \langle \Omega | \widehat{\phi}(x_1) \cdots \widehat{\phi}(x_n) | \Omega \rangle.$$

The order displayed in the formula is part of the definition.

For distinct time coordinates, the time-ordered Green function is

$$G_{T,n}(x_1, \ldots, x_n) = \langle \Omega | \, \mathrm{T}\{\widehat{\phi}(x_1) \cdots \widehat{\phi}(x_n)\} | \Omega \rangle.$$

If

$$x^0_{i_1} > x^0_{i_2} > \cdots > x^0_{i_n},$$

then

$$G_{T,n}(x_1, \ldots, x_n) = W_n(x_{i_1}, \ldots, x_{i_n}).$$

Coincident-time prescriptions are contact-term data and must be fixed in the chosen operator or path-integral framework.

The Euclidean Green function is a distribution in Euclidean points $x_{E,a} = (\tau_a, \vec{x}_a)$. It is denoted $G_{E,n}$ in this path-integral construction. When the same distributions are used as OS reconstruction data, they are denoted S_n and called Schwinger functions. When the analytic continuation from Wightman functions exists and

$$\tau_{i_1} > \tau_{i_2} > \cdots > \tau_{i_n},$$

it is given by

$$G_{E,n}(x_{E,1}, \ldots, x_{E,n}) = W_n(x_{i_1}, \ldots, x_{i_n})\big|_{x^0_a = -\mathrm{i}\tau_a}.$$

These three distributions encode different boundary values of the same analytic structure when the required analyticity and spectral assumptions are available. The next chapter uses this structure in the two-point function to derive the Källén–Lehmann spectral representation.

Chapter 12

Osterwalder–Schrader Reconstruction

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

12.1 Euclidean Correlation Data

Let $E = \mathbb{R}^D$ be Euclidean space with coordinates

$$x = (\tau, \mathbf{x}), \qquad \mathbf{x} \in \mathbb{R}^{D-1}.$$

Let V be the complex vector space of field labels. For scalar fields $V = \mathbb{C}$. For tensor, spinor, or internal multiplets, V carries the corresponding Euclidean rotation and internal representations. An n-point Euclidean test insertion is an element of

$$\mathcal{S}(E^n; V^{\otimes n}),$$

or of $C_c^\infty(E^n; V^{\otimes n})$ if compactly supported test functions are the chosen test class. The test class is part of the data because it determines the topology in which the Schwinger functions are distributions. Let

$$\Delta_n = \bigcup_{1 \le i < j \le n} \{(x_1, \ldots, x_n) \in E^n \mid x_i = x_j\}$$

be the union of partial diagonals. The corrected OS reconstruction theorem is most honestly stated on the zero-diagonal Schwartz space

$$\mathcal{S}_\circ(E^n; V^{\otimes n}) = \{f \in \mathcal{S}(E^n; V^{\otimes n}) \mid \partial^\alpha f|_{\Delta_n} = 0 \text{ for every multi-index } \alpha\}.$$

Pairing with $\mathcal{S}_\circ$ treats the Schwinger functions as distributions away from coincident points. A choice of contact-term extension to the full Schwartz space is extra structure and is not part of the OS reconstruction input used here.

Definition 12.1 (Schwinger hierarchy)**.** A Schwinger hierarchy with field-label space V is a family

$$S_n \in \mathcal{S}'_\circ(E^n; (V^*)^{\otimes n}), \qquad n \geq 0,$$

with $S_0 = 1$. The value $S_n(f_n)$ denotes the distributional pairing of S_n with the $V^{\otimes n}$-valued test function f_n. In component notation this is written as

$$S_{\alpha_1 \cdots \alpha_n}(x_1, \ldots, x_n), \qquad \alpha_j \in I,$$

where I is a basis of labels for V. The component notation is a kernel notation for the distributional pairing; the pairing with test functions is the primary object.

In a construction by a Euclidean measure, these distributions are moments of random distributions. In an axiomatic Euclidean formulation, the hierarchy $\{S_n\}_{n\geq 0}$ is primary data.

Let θ be time reflection:

$$\theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x}).$$

The Euclidean group is

$$\mathrm{Euc}(D) = \mathbb{R}^D \rtimes O(D).$$

It acts on test insertions by pullback on E^n and by the declared label representation on V. The reflection also acts on labels. We denote by $J : V \to V$ the antilinear label map induced by field adjunction and by the Euclidean reflection of tensor or spinor indices. For a scalar Hermitian field, J is ordinary complex conjugation. For a test function f_n supported in the ordered positive-time region

$$E^n_> = \{(x_1, \ldots, x_n) : 0 < \tau_1 < \cdots < \tau_n\},$$

define the reflected adjoint test function Θf_n by reversing the order, reflecting all arguments, and applying J to the labels. For a scalar test function this reads

$$(\Theta f_n)(x_1, \ldots, x_n) = \overline{f_n(\theta x_n, \ldots, \theta x_1)}.$$

With labels, the same formula is composed with $J^{\otimes n}$. Let Θ denote the same antilinear operation on arbitrary n-point test functions; positive-time support is needed only when we form the OS pre-Hilbert space below. Let $\mathcal{E}_+$ be the vector space of finite sequences

$$F = (f_0, f_1, f_2, \ldots),$$

where only finitely many f_n are nonzero and every f_n is a labelled zero-diagonal test function supported in positive Euclidean time with a chosen ordering of its time variables. Reflection positivity is the requirement that the sesquilinear form

$$\langle F, G \rangle_{\mathrm{OS}} = \sum_{m,n} S_{m+n}(\Theta f_m, g_n) \tag{12.1}$$

satisfy

$$\langle F,F\rangle_{\mathrm{OS}}\geq 0 \tag{12.2}$$

for every $F\in\mathcal{E}_+$.

The notation $S_{m+n}(\Theta f_m,g_n)$ means that the reflected m negative-time insertions are followed by the n positive-time insertions in the argument of the distribution. Thus the object required to be positive is the reflected positive-time sesquilinear form on $\mathcal{E}_+$.

Definition 12.2 (OS-admissible Euclidean data)**.** An OS-admissible Euclidean field presentation is a Schwinger hierarchy $\{S_n\}_{n\geq 0}$, together with the label reflection J, satisfying:

(i) normalization: $S_0=1$;

(ii) Euclidean covariance: $S_n(gf_n)=S_n(f_n)$ for $g\in\mathrm{Euc}(D)$;

(iii) permutation symmetry: S_n is invariant under simultaneous permutation of arguments and labels;

(iv) Hermiticity: for every $n\geq 0$ and every labelled n-point test function f_n,

$$S_n(\Theta f_n)=\overline{S_n(f_n)}. \tag{12.3}$$

(v) reflection positivity: $\langle F,F\rangle_{\mathrm{OS}}\geq 0$ for all $F\in\mathcal{E}_+$;

(vi) positive-time semigroup regularity: the positive-time translation operators defined on the OS quotient form a strongly continuous symmetric contraction semigroup, as stated in (12.5)–(12.8);

(vii) corrected OS-II analytic regularity: the difference-variable hierarchy satisfies a many-variable continuation and growth theorem strong enough to give the Lorentzian boundary-value package of Definition 12.8. A standard sufficient hypothesis is the OS-II linear-growth condition (12.4), together with the OS-II several-complex-variable construction described below.

For fermionic or graded fields, the permutation line is replaced by graded permutation symmetry with the Koszul sign determined by the declared parity of each label. Clustering is added when the target Lorentzian theory is required to have a unique vacuum vector.

The last line is a load-bearing analytic hypothesis, not a cosmetic regularity clause. Let S_n^{red} denote the translation-reduced Schwinger distribution in $n-1$ difference variables, acting on the corresponding zero-diagonal difference-test space. With a fixed family of Schwartz seminorms

$$q_{n,N}(F)=\sum_{|\alpha|,|\beta|\leq N}\sup_{\Xi\in E^{n-1}}\left|\Xi^\beta\partial^\alpha F(\Xi)\right|,\qquad F\in\mathcal{S}_\circ(E^{n-1};V^{\otimes n}),$$

one convenient OS-II linear-growth condition is the existence of constants

$$C_{\mathrm{OS}},N_0\in\mathbb{N},\qquad A,B,\gamma>0,$$

independent of n, such that

$$|S_n^{\mathrm{red}}(F)|\leq A\,B^n\,(n!)^\gamma\,q_{n,N_0+C_{\mathrm{OS}}n}(F) \tag{12.4}$$

for all F in that test space. Equivalent OS-II formulations package the same content by using a fixed seminorm family $|\cdot|_{s,n}$ whose definition already contains the n-dependence, together with constants $\sigma_n \leq AB^n(n!)^\gamma$. The word "linear" refers to the allowed growth of the distributional seminorm order with the number of insertions. It is not a statement that a fixed n-point function grows linearly in spacetime.

Together with the positive-time semigroup hypotheses (12.5)–(12.8), this condition has a specific OS-II role. The positive-time Hilbert space and the one-variable semigroup continuation are constructed before the linear-growth estimate is used. The linear-growth estimate controls the many-variable analytic continuation and the polynomial boundary estimates needed to obtain tempered Lorentzian Wightman distributions. The correction from the first Osterwalder–Schrader reconstruction paper to the second is precisely that reflection positivity, Euclidean covariance, symmetry, clustering, and ordinary temperedness do not by themselves prove that the separately obtained one-variable Laplace continuations assemble into a joint many-variable Fourier–Laplace transform with tempered boundary values. Thus an application of OS reconstruction must either prove an OS-II growth condition such as (12.4), or replace it by a different theorem that supplies the same Lorentzian boundary-value estimates. [1]

Remark 12.3 (CFT reconstruction is a different route). For a Euclidean CFT, one should not silently replace the OS-II growth package by conformal terminology. Conformal kinematics gives additional structure: radial ordering, the inversion pairing, positivity of the state-operator inner product, and conformal covariance of operator insertions. In that setting a Lorentzian theory can be reconstructed by proving the convergence and distributional boundary-value properties of conformal correlation functions directly from the conformal Hilbert-space and OPE data. This is a different theorem from OS reconstruction. Kravchuk, Qiao, and Rychkov prove the relevant Lorentzian distribution and OPE-convergence estimates for CFT correlators using conformal kinematics and radial variables;[2] the CFT volume uses that route only after the state-operator Hilbert space and conformal representation theory have been stated as hypotheses.

Figure 12.1 fixes the geometry used in the OS reflection form. In this Euclidean reconstruction figure the reflection time τ is drawn horizontally, so that the fixed plane $\tau = 0$ is the vertical dashed line.

[1]See K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's Functions," Commun. Math. Phys. **31** (1973) 83–112, and "Axioms for Euclidean Green's Functions II," Commun. Math. Phys. **42** (1975) 281–305. The second paper states the corrected reconstruction theorem with the linear-growth input. Modern checks of this analytic input are model dependent. For example, Gubinelli and Hofmanová, "A PDE Construction of the Euclidean Φ^4_3 Quantum Field Theory," Commun. Math. Phys. **384** (2021) 1–75, construct limit measures with reflection positivity, translation invariance, non-Gaussianity, and stretched exponential integrability estimates strong enough for the OS distributional regularity axiom in their formulation; rotation invariance and clustering remain separate inputs in that construction [7, 8, 9].

[2]See P. Kravchuk, J. Qiao, and S. Rychkov, "Distributions in CFT II. Minkowski Space," JHEP **08** (2021) 094, arXiv:2104.02090 [13].

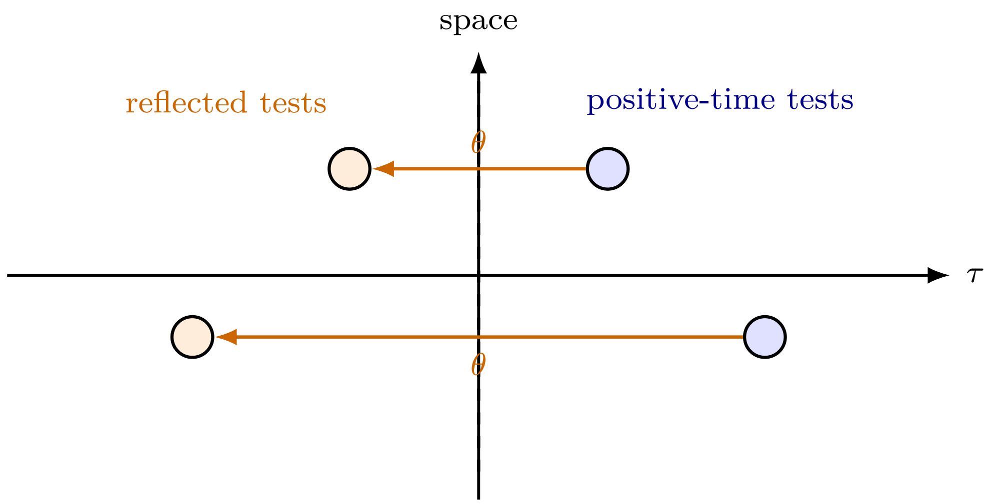


Figure 12.1: Reflection positivity turns Euclidean reflection through $\tau = 0$ into Hilbert-space positivity after reconstruction. The positive quadratic form $\sum S(\Theta f, f) \geq 0$ descends, after quotient by null vectors, to the reconstructed Lorentzian inner product.

12.2 Hilbert Space From Positive Time

Use the positive-time vector space $\mathcal{E}_+$ and the form (12.1). Reflection positivity says that this form is nonnegative. Let

$$\mathcal{N} = \{F \in \mathcal{E}_+ \mid \langle F, F\rangle_{\mathrm{OS}} = 0\}.$$

The reconstructed Hilbert space is

$$\mathcal{H}_{\mathrm{OS}} = \overline{\mathcal{E}_+/\mathcal{N}}.$$

The vector represented by $F = (1, 0, 0, \dots)$ is the vacuum. The quotient is taken before completion. Since (12.1) is positive semidefinite, the Cauchy–Schwarz inequality holds on $\mathcal{E}_+$, and $\mathcal{N}$ is exactly the subspace of zero-norm vectors. Therefore $\mathcal{E}_+/\mathcal{N}$ is a pre-Hilbert space with inner product induced by $\langle \cdot, \cdot\rangle_{\mathrm{OS}}$.

Euclidean time translations by $s \geq 0$ preserve positive-time support and therefore act on $\mathcal{E}_+$. Write $a_s(\tau, \mathbf{x}) = (\tau + s, \mathbf{x})$, and let R_s be the induced action on test sequences, so that each component is translated by a_s. The OS regularity hypotheses needed at this point are the following seminorm statements on $\mathcal{E}_+$:

$$R_0 = \mathbf{1}, \qquad R_{s+t} = R_s R_t, \tag{12.5}$$

$$\|R_s F\|_{\mathrm{OS}} \leq \|F\|_{\mathrm{OS}}, \tag{12.6}$$

$$\langle R_s F, G\rangle_{\mathrm{OS}} = \langle F, R_s G\rangle_{\mathrm{OS}}, \tag{12.7}$$

$$\lim_{s\downarrow 0} \|R_s F - F\|_{\mathrm{OS}} = 0, \tag{12.8}$$

for all $F, G \in \mathcal{E}_+$, where $\|F\|^2_{\mathrm{OS}} = \langle F, F\rangle_{\mathrm{OS}}$. The inequality (12.6) is the contraction statement needed to obtain a nonnegative Hamiltonian. In the classical scalar OS proof it is derived from reflection positivity, Euclidean covariance, the semigroup law, and a polynomial bound on translated matrix elements by the iterative Cauchy–Schwarz argument written below. In the abstract formulation of OS-admissible data we record the contraction and strong continuity explicitly, so that every later use of the Hamiltonian has a named hypothesis.

Remark 12.4 (Failure mode of the first OS reconstruction theorem). The famous error in the first Osterwalder–Schrader reconstruction paper is not the positive-time Hilbert-space quotient and not the one-variable semigroup continuation. Those parts are the reliable core of the construction. The faulty step was the passage from separately controlled one-variable Laplace continuations to a joint many-variable Fourier–Laplace transform. A function on $(0,\infty)^2$ may be a Laplace transform of a tempered distribution in each variable separately without being a joint Laplace transform; the model obstruction is $F(x,y) = \exp(-xy)$. The corrected OS-II proof replaces that step by a several-complex-variable continuation ladder and then uses a linear-growth estimate to prove the polynomial bounds that make the boundary values tempered distributions.[3]

Proposition 12.5 (Contraction from the OS semigroup estimate)**.** *Let $R_t : \mathcal{E}_+ \to \mathcal{E}_+$, $t \geq 0$, satisfy the semigroup law $R_{t+s} = R_t R_s$, the OS symmetry relation*

$$\langle R_t F, G\rangle_{\mathrm{OS}} = \langle F, R_t G\rangle_{\mathrm{OS}}, \qquad F, G \in \mathcal{E}_+,$$

and, for each $F \in \mathcal{E}_+$, a polynomial growth bound

$$|\langle F, R_u F\rangle_{\mathrm{OS}}| \leq C_F (1+u)^{N_F}, \qquad u \geq 0.$$

Then

$$\|R_t F\|_{\mathrm{OS}} \leq \|F\|_{\mathrm{OS}}, \qquad t \geq 0.$$

In particular $R_t \mathcal{N} \subset \mathcal{N}$, so R_t descends to a contraction on the OS quotient.

Proof. Assume first that $\|F\|_{\mathrm{OS}} > 0$. Put

$$a_j = \langle F, R_{2^j t} F\rangle_{\mathrm{OS}}, \qquad j = 0, 1, 2, \ldots.$$

For $j \geq 1$, the semigroup law and OS symmetry give

$$a_j = \langle R_{2^{j-1}t} F, R_{2^{j-1}t} F\rangle_{\mathrm{OS}} \geq 0.$$

The Cauchy–Schwarz inequality in the positive semidefinite OS form gives, for $j \geq 1$,

$$a_j^2 = |\langle F, R_{2^j t} F\rangle_{\mathrm{OS}}|^2 \leq \|F\|_{\mathrm{OS}}^2 \|R_{2^j t} F\|_{\mathrm{OS}}^2 = a_0\, a_{j+1}.$$

Induction therefore yields

$$a_1 \leq a_0^{1-2^{-N}} a_N^{2^{-N}}, \qquad N \geq 1.$$

The polynomial bound gives

$$a_N^{2^{-N}} \leq C_F^{2^{-N}} (1 + 2^N t)^{N_F 2^{-N}} \longrightarrow 1.$$

Letting $N \to \infty$ gives $a_1 \leq a_0$. Since

$$a_1 = \langle F, R_{2t} F\rangle_{\mathrm{OS}} = \langle R_t F, R_t F\rangle_{\mathrm{OS}} = \|R_t F\|_{\mathrm{OS}}^2,$$

the contraction estimate follows. If $\|F\|_{\mathrm{OS}} = 0$, Cauchy–Schwarz already gives $\langle F, R_u F\rangle_{\mathrm{OS}} = 0$ for all $u \geq 0$, and the same identity gives $\|R_t F\|_{\mathrm{OS}} = 0$. Thus null vectors are preserved, and the quotient operator is a contraction. □

[3]OS II states explicitly that Lemma 8.8 of OS I is wrong and gives the $F(x,y) = \exp(-xy)$ obstruction. See K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's Functions II," Commun. Math. Phys. **42** (1975) 281–305, Introduction and Chapters IV–VI [8].

Contractivity implies $R_s\mathcal{N} \subset \mathcal{N}$, so R_s descends to a bounded operator $T_E(s)$ on the quotient and then to $\mathcal{H}_{\mathrm{OS}}$. Equations (12.5)–(12.8) say precisely that $\{T_E(s)\}_{s\geq 0}$ is a strongly continuous semigroup of self-adjoint contractions. Indeed, (12.7) gives, on the dense quotient,

$$\langle [F], T_E(s)[G]\rangle_{\mathrm{OS}} = \langle [F], [R_sG]\rangle_{\mathrm{OS}} = \langle [R_sF], [G]\rangle_{\mathrm{OS}} = \langle T_E(s)[F], [G]\rangle_{\mathrm{OS}}.$$

The equality descends to the quotient because $R_s\mathcal{N} \subset \mathcal{N}$, and it extends to $\mathcal{H}_{\mathrm{OS}}$ because $T_E(s)$ is bounded by (12.6). Thus the self-adjointness of the Euclidean time semigroup is not inferred from a formal reflection manipulation outside positive time; it is precisely the OS-II symmetry condition on the reflection-positive quotient.

Let A be its infinitesimal generator,

$$A\psi = \lim_{s\downarrow 0} \frac{T_E(s)\psi - \psi}{s},$$

on the vectors for which the limit exists in $\mathcal{H}_{\mathrm{OS}}$. The self-adjoint contraction-semigroup theorem gives $A = A^*$ and $\sigma(A) \subset (-\infty, 0]$. Setting $H = -A$, the spectral theorem gives

$$T_E(s) = \mathrm{e}^{-sH}, \qquad H = H^*, \qquad H \geq 0.$$

Stone's theorem then applies to the strongly continuous unitary group $U_{\mathrm{time}}(t) = \mathrm{e}^{-\mathrm{i}tH}$: its named hypotheses are the unitary group law, strong continuity in t, and self-adjointness of the generator. This H is the Lorentzian Hamiltonian in the reconstructed vacuum sector.

Proposition 12.6 (One-gap continuation from the OS semigroup)**.** *Let $F, G \in \mathcal{E}_+$ be two positive-time test sequences whose supports are arranged so that, for $s > 0$, R_sG remains later than the insertions in F. Let*

$$\psi_F = [F], \qquad \psi_G = [G] \in \mathcal{H}_{\mathrm{OS}}.$$

Then the Euclidean matrix element with distinguished time gap s is

$$\langle \psi_F, T_E(s)\psi_G\rangle_{\mathrm{OS}} = \langle \psi_F, \mathrm{e}^{-sH}\psi_G\rangle_{\mathcal{H}_{\mathrm{OS}}}.$$

For $z \in \mathbb{C}$ with $\Re z > 0$, define

$$M_{F,G}(z) = \langle \psi_F, \mathrm{e}^{-zH}\psi_G\rangle_{\mathcal{H}_{\mathrm{OS}}}.$$

Then $M_{F,G}$ is holomorphic on the right half-plane and satisfies

$$|M_{F,G}(z)| \leq \|\psi_F\|\,\|\psi_G\|, \qquad \Re z > 0.$$

Moreover, for every $t \in \mathbb{R}$,

$$\lim_{\epsilon\downarrow 0} M_{F,G}(\epsilon + \mathrm{i}t) = \langle \psi_F, \mathrm{e}^{-\mathrm{i}tH}\psi_G\rangle_{\mathcal{H}_{\mathrm{OS}}}.$$

Proof. The first identity is the definition of the descended positive-time translation semigroup:

$$T_E(s)[G] = [R_sG], \qquad T_E(s) = \mathrm{e}^{-sH}.$$

For the holomorphic extension, let E_H be the spectral resolution of the nonnegative self-adjoint operator H, and let

$$\mu_{F,G}(\Delta) = \langle \psi_F, E_H(\Delta)\psi_G\rangle$$

be the finite complex spectral measure associated to ψ_F, ψ_G. Its total variation is bounded by $|\mu_{F,G}|([0,\infty)) \le \|\psi_F\| \, \|\psi_G\|$. The spectral theorem gives

$$M_{F,G}(z) = \int_{[0,\infty)} \mathrm{e}^{-z\lambda} \, \mathrm{d}\mu_{F,G}(\lambda), \qquad \Re z > 0.$$

On compact subsets of the right half-plane, derivatives with respect to z are obtained by differentiating under the integral:

$$\partial_z^k M_{F,G}(z) = \int_{[0,\infty)} (-\lambda)^k \mathrm{e}^{-z\lambda} \, \mathrm{d}\mu_{F,G}(\lambda),$$

because $\lambda^k \mathrm{e}^{-(\Re z)\lambda}$ is bounded on $[0,\infty)$ uniformly for $\Re z$ bounded below by a positive constant. This proves holomorphy. The norm bound follows from $|\mathrm{e}^{-z\lambda}| \le 1$. Finally, $\mathrm{e}^{-(\epsilon+\mathrm{i}t)\lambda} \to \mathrm{e}^{-\mathrm{i}t\lambda}$ pointwise and is bounded by 1; dominated convergence for the finite measure $|\mu_{F,G}|$ gives the stated boundary value. $\square$

This proposition is the sound one-variable part of the OS construction. It does not by itself produce an n-variable Wightman distribution. The next QFT-specific task is to show that all one-gap continuations coming from different cuts of the same Schwinger function assemble into a joint many-variable analytic function with the spectral support and boundary-value estimates required below. That assembly is the content of the corrected OS-II analytic theorem.

Spatial translations and rotations act unitarily on the quotient. Together with the analytic continuation of Euclidean rotations, the full OS theorem gives a strongly continuous unitary representation of the connected Poincare group with joint translation spectrum contained in the closed forward cone.

The passage from Euclidean covariance to Lorentzian covariance occurs here. Reflection positivity constructs the positive-time quotient and Euclidean time semigroup. The corrected OS-II analytic theorem, with its growth hypothesis, then supplies the joint Lorentzian boundary values and the Poincare representation. These are distinct steps: the one-variable semigroup argument above is necessary, but the many-variable analytic continuation is the additional QFT-specific content.

12.3 Fields on the Reconstructed Domain

Let $\mathcal{D}_{\mathrm{OS}} \subset \mathcal{H}_{\mathrm{OS}}$ be the image of $\mathcal{E}_+$ in the quotient. This dense subspace is the natural common domain for the reconstructed fields before closure. If h is a Euclidean test function valued in V, with support at positive time and ordered before the supports appearing in $F \in \mathcal{E}_+$, define

$$\Phi_E(h)[F] = [h \otimes F],$$

where $h \otimes F$ denotes the sequence obtained by inserting one additional Euclidean field smeared with h. This formula is only meaningful after one checks that it descends through the null quotient. The check is a reflection-positive adjointness argument, not a formal manipulation with point fields.

Proposition 12.7 (Well-definedness and closability of ordered insertions)**.** *Assume the operator-domain part of the OS data supplies the following adjoint-insertion identity. For every ordered* *positive-time test insertion h there is an ordered test insertion $h^\sharp$, with adjoint label and with supports arranged by a common positive time translation, such that for all $F, G \in \mathcal{E}_+$*

$$\langle [G], [h \otimes F] \rangle_{\mathrm{OS}} = \langle [h^\sharp \otimes G], [F] \rangle_{\mathrm{OS}}. \tag{12.9}$$

Then $F \in \mathcal{N}_{\mathrm{OS}}$ implies $h \otimes F \in \mathcal{N}_{\mathrm{OS}}$. Hence $\Phi_E(h)[F] = [h \otimes F]$ is a well-defined operator on the quotient domain $\mathcal{D}_{\mathrm{OS}}$. This operator is closable.

Proof. Let $F \in \mathcal{N}_{\mathrm{OS}}$. In a positive semidefinite Hermitian form, Cauchy–Schwarz gives, for every $K \in \mathcal{E}_+$,

$$\langle K, F \rangle_{\mathrm{OS}} = 0.$$

Therefore, for every $G \in \mathcal{E}_+$,

$$\langle [G], [h \otimes F] \rangle_{\mathrm{OS}} = \langle [h^\sharp \otimes G], [F] \rangle_{\mathrm{OS}} = 0.$$

Taking $G = h \otimes F$, with the common positive time translation already included in the ordered-support hypothesis, gives $\|h \otimes F\|_{\mathrm{OS}}^2 = 0$. Thus $h \otimes F \in \mathcal{N}_{\mathrm{OS}}$, and the quotient operator is well defined.

Let $\psi_k \in \mathcal{D}_{\mathrm{OS}}$ satisfy $\psi_k \to 0$ and $\Phi_E(h)\psi_k \to \eta$ in $\mathcal{H}_{\mathrm{OS}}$. To prove closability it suffices to show $\eta = 0$. For a dense test vector $[G] \in \mathcal{D}_{\mathrm{OS}}$, the adjoint-insertion identity gives

$$\langle [G], \eta \rangle_{\mathrm{OS}} = \lim_k \langle [G], \Phi_E(h)\psi_k \rangle_{\mathrm{OS}} = \lim_k \langle [h^\sharp \otimes G], \psi_k \rangle_{\mathrm{OS}} = 0,$$

because $[h^\sharp \otimes G]$ is a fixed vector in $\mathcal{D}_{\mathrm{OS}}$ and $\psi_k \to 0$. Since $\mathcal{D}_{\mathrm{OS}}$ is dense, η is orthogonal to all of $\mathcal{H}_{\mathrm{OS}}$, hence $\eta = 0$. This is the standard closability criterion, with the QFT-specific content isolated in the matrix-element identity (12.9). □

Lorentzian fields are obtained by continuing these matrix elements on the reconstructed Hilbert space. For ordered imaginary parts

$$\epsilon_1 > \epsilon_2 > \cdots > \epsilon_n > 0,$$

the Wightman distributions are boundary values of the Schwinger functions continued to

$$\tau_j = \epsilon_j + \mathrm{i}t_j, \qquad j = 1, \ldots, n. \tag{12.10}$$

Equivalently, the complex Lorentzian time coordinate is

$$z_j^0 = -\mathrm{i}\tau_j = t_j - \mathrm{i}\epsilon_j.$$

Thus for an ordered adjacent difference,

$$z_j - z_{j+1} = (x_j - x_{j+1}) - \mathrm{i}(\epsilon_j - \epsilon_{j+1})e_0, \qquad e_0 = (1, \mathbf{0}).$$

The tube depth is the future vector $(\epsilon_j - \epsilon_{j+1})e_0 \in V_+^\circ$, while the literal imaginary part of $z_j - z_{j+1}$ is its negative. Throughout this chapter the "forward tube" in mostly-plus signature means

$$z = \xi - \mathrm{i}\eta, \qquad \eta \in V_+^\circ,$$

matching the Wightman convention of Proposition 2.5. This sign is not notation: for $p, \eta \in \overline{V}_+$, the mostly-plus pairing obeys $p \cdot \eta \leq 0$, so the factor $\exp(\mathrm{i}p \cdot (\xi - \mathrm{i}\eta)) = \exp(\mathrm{i}p \cdot \xi + p \cdot \eta)$ is damped. The reconstructed operator $\widehat{\Phi}_\alpha(t, \mathbf{f})$ is the operator-valued distribution whose vacuum matrix elements are these boundary values and whose time dependence is implemented by the unitary group

$$\mathrm{e}^{-\mathrm{i}tH}.$$

Thus the OS construction produces the vacuum vector, the Hilbert space, the positive Hamiltonian, and the field domains from the same Euclidean correlation hierarchy.

As in the Wightman presentation, these reconstructed fields are generally unbounded distributional coordinates. A represented local observable algebra $\mathcal{R}(\mathcal{O})$ is obtained by specifying a closure, affiliation, or bounded functional-calculus construction, or by building a local net directly and proving that the reconstructed fields are affiliated with it. The OS theorem reconstructs a Lorentzian Wightman field presentation; comparison with AQFT local observable content is a further construction with its own hypotheses.

12.4 Analytic Continuation Input

The OS-II regularity assumptions are used through the following boundary-value package. It is stated separately because this is the part of the reconstruction theorem where the first OS paper needed correction.

Definition 12.8 (OS Lorentzian boundary-value package)**.** For each n, the Schwinger distribution S_n is said to satisfy the OS Lorentzian boundary-value package if the following data and estimates are available.

(i) For every Euclidean time ordering there is a holomorphic function

$$\mathcal{W}_n(z_1, \ldots, z_n)$$

on the corresponding mostly-plus forward tube in the difference variables

$$z_j - z_{j+1} = \xi_j - \mathrm{i}\eta_j, \qquad \eta_j \in V_+^\circ,$$

equivalently $-\operatorname{Im}(z_j - z_{j+1}) \in V_+^\circ$. Here V_+° is the open Lorentzian forward cone. The restriction of $\mathcal{W}_n$ to ordered imaginary times agrees with S_n away from coincident points.

(ii) The holomorphic functions obey tempered tube bounds: for every compact subcone $K \subset (V_+^\circ)^{n-1}$ and every positive lower bound on the distance of the tube-depth variables η_j from the boundary of the cone, the boundary pairings with Schwartz tests are bounded by finitely many Schwartz seminorms with polynomial dependence on the reciprocal distance to the cone boundary.

(iii) The permuted Euclidean orderings give holomorphic functions on the corresponding permuted tubes that agree on their common Euclidean real submanifolds. For spacelike separated adjacent insertions, the two permuted tubes have a connected continuation through the corresponding Jost domain by the distributional edge-of-the-wedge theorem 16.6, with the Koszul sign included for graded fields.

Theorem 12.9 (Tempered boundary values from polynomial tube bounds). *Let $C \subset \mathbb{R}^d$ be an open convex cone and let*

$$T_C = \mathbb{R}^d + \mathrm{i}C$$

be the corresponding tube. Let F be holomorphic on T_C. Suppose that for every closed subcone $K \Subset C$ there are integers M, N and a constant A_K such that

$$|\langle F(\cdot + \mathrm{i}y), f\rangle| \leq A_K \ \mathrm{dist}(y, \partial C)^{-M} \sum_{|\alpha|,|\beta| \leq N} \sup_{x \in \mathbb{R}^d} |x^\beta \partial^\alpha f(x)| \tag{12.11}$$

for all $y \in K$ and all $f \in \mathcal{S}(\mathbb{R}^d)$, whenever the left hand side is defined by integration over x. Then F has a unique tempered boundary value

$$\mathrm{bv}\, F \in \mathcal{S}'(\mathbb{R}^d)$$

in the sense that $F(\cdot + \mathrm{i}y) \to \mathrm{bv}\, F$ in $\mathcal{S}'(\mathbb{R}^d)$ as $y \to 0$ inside any proper subcone of C. If, in addition, F is the Fourier–Laplace transform of a distribution carried by the dual closed cone

$$C^* = \{p \in \mathbb{R}^d \mid p \cdot y \geq 0 \textit{ for all } y \in C\},$$

then the Fourier transform of $\mathrm{bv}\, F$ *is carried by C^*.*

The cone-support statement used for the spectral condition has a direct Fourier–Laplace proof, recorded first because it is the part of the theorem that enters the QFT interpretation most explicitly.

Lemma 12.10 (Fourier–Laplace boundary and cone support). *Let $C \subset \mathbb{R}^d$ be an open convex cone whose dual closed cone C^* has the following proper-separation property: for every $K \Subset C$ there is $c_K > 0$ such that*

$$p \cdot y \geq c_K |p|\,|y|, \qquad p \in C^*, \ y \in K. \tag{12.12}$$

Let $u \in \mathcal{S}'(\mathbb{R}^d)$ have support contained in C^*. For $y \in C$, put

$$u_y = \mathrm{e}^{-p \cdot y} u$$

as a product of a distribution with a smooth multiplier, defined in a conic neighborhood of $\operatorname{supp} u$, *and let*

$$F(\cdot + \mathrm{i}y) = \mathcal{F}^{-1} u_y.$$

Then F is holomorphic in the tube $T_C = \mathbb{R}^d + \mathrm{i}C$, and

$$F(\cdot + \mathrm{i}y) \longrightarrow \mathcal{F}^{-1} u \quad \textit{in } \mathcal{S}'(\mathbb{R}^d)$$

as $y \to 0$ inside every proper subcone of C. Consequently the Fourier transform of the boundary value is u, hence is carried by C^*.

Proof. The only point needing care is the multiplication by $\mathrm{e}^{-p \cdot y}$, because this function may grow outside the dual cone. Fix $K \Subset C$. Choose a closed conic neighborhood U of $C^* \setminus \{0\}$ so small that

$$p \cdot y \geq \frac{c_K}{2} |p|\,|y|, \qquad p \in U, \ y \in K, \tag{12.13}$$

and choose a smooth conic cutoff χ which is equal to 1 on a neighborhood of $C^* \setminus \{0\}$ and whose support is contained in U outside a compact ball. Near the origin modify χ smoothly so that it is identically 1 on a neighborhood of $\operatorname{supp} u \cap B_R$ for a fixed large R. Since distributions are tested only in a neighborhood of their support, the product $\mathrm{e}^{-p\cdot y}u$ is the same as $(\chi(p)\mathrm{e}^{-p\cdot y})u$; a different cutoff with the same property gives the same distribution because the two multipliers agree near $\operatorname{supp} u$.

For $y \in K$, (12.13) implies that every derivative of $\chi(p)\mathrm{e}^{-p\cdot y}$ is bounded by a polynomial in $|y|$ times an exponentially decaying factor on the support of χ, apart from a compact set. Hence multiplication by $\chi\mathrm{e}^{-p\cdot y}$ is a continuous operation on $\mathcal{S}'(\mathbb{R}^d)$, uniformly for y in compact subsets of K. If $f \in \mathcal{S}(\mathbb{R}^d)$, then

$$\langle u_y, \widehat{f}\rangle = \langle u, \chi(p)\mathrm{e}^{-p\cdot y}\widehat{f}(p)\rangle.$$

As $y \to 0$ in K, the Schwartz functions

$$\chi(p)\big(\mathrm{e}^{-p\cdot y} - 1\big)\widehat{f}(p)$$

converge to zero in $\mathcal{S}(\mathbb{R}^d)$. Indeed, after applying any derivative, each term contains either a factor tending uniformly to zero on compact sets or a positive power of y; on the conic part the rapid decay of $\widehat{f}$ and (12.13) dominate all polynomial weights. Therefore $u_y \to u$ in $\mathcal{S}'(\mathbb{R}^d)$, and continuity of the inverse Fourier transform on tempered distributions gives

$$\mathcal{F}^{-1}u_y \to \mathcal{F}^{-1}u.$$

Holomorphy follows from the same estimates. Differentiating the pairing $\langle u, \chi\mathrm{e}^{-p\cdot y}\mathrm{e}^{-\mathrm{i}p\cdot x}\rangle$ with respect to the complex variable $z = x + \mathrm{i}y$ brings down polynomial factors in p, which preserve the preceding tempered estimates. The Cauchy–Riemann relations hold at the level of pairings with Schwartz tests, hence F is a holomorphic $\mathcal{S}'$-valued function on the tube. The boundary value has Fourier transform u by construction, so its Fourier support is contained in C^*. □

Proof of Theorem 12.9. The theorem is an analytic statement about holomorphic functions with values in a finite-order dual of the Schwartz space. We spell out the mechanism because OS reconstruction uses exactly the distinction between polynomial tube growth and spectral cone support.

Let

$$p_N(f) = \sum_{|\alpha|,|\beta|\le N} \sup_{x\in\mathbb{R}^d} |x^\beta \partial^\alpha f(x)|$$

be a Schwartz seminorm. The hypothesis says that, for y in every proper subcone $K \Subset C$, the distribution

$$F_y : f \longmapsto \langle F(\cdot + \mathrm{i}y), f\rangle$$

belongs to the Banach dual of the completion of $\mathcal{S}(\mathbb{R}^d)$ in the norm p_N, with norm bounded by a fixed power of $\operatorname{dist}(y, \partial C)^{-1}$. Cauchy's integral formula in small polydiscs contained in the tube gives the same type of bound for all complex derivatives of F, with the exponent increased by the derivative order. Thus $y \mapsto F_y$ is a weakly holomorphic, locally bounded finite-order distribution-valued function on every proper subtube.

Fix a vector $\eta \in C$ and choose $K \Subset C$ containing the ray $\mathbb{R}_{>0}\eta$ near the origin. For $0 < s < t$ and $f \in \mathcal{S}(\mathbb{R}^d)$, apply Cauchy's formula to the scalar holomorphic function obtained by translating the test function:

$$z \longmapsto \big\langle F(\cdot + \mathrm{i}z\eta), f(\cdot - a)\big\rangle, \qquad a \in \mathbb{R}^d.$$

After integrating the resulting identity against a compactly supported approximate identity in a, and then moving derivatives from the Cauchy kernel onto the test function, one obtains an estimate of the form

$$|\langle F_{t\eta} - F_{s\eta}, f\rangle| \le C_{K,m}\, \omega_{f,m}(s,t), \tag{12.14}$$

where m is chosen larger than the polynomial-growth exponent and $\omega_{f,m}(s,t) \to 0$ as $s,t \downarrow 0$. The point of choosing m larger than the growth order is that the normal antiderivative supplied by the Cauchy kernel is locally integrable at the edge; all singular powers are absorbed by finitely many derivatives of the test function. Hence $\{F_{t\eta}\}_{t>0}$ is Cauchy in $\mathcal{S}'(\mathbb{R}^d)$ and defines a tempered distribution b_η.

The limit is independent of the proper approach cone. If $\eta_0, \eta_1 \in C$, the convexity of C puts the segment $(1-r)\eta_0 + r\eta_1$ in a proper subcone after shrinking to small t. Applying the preceding Cauchy estimate in the two real variables (t,r) to the holomorphic function in the corresponding two-dimensional complex slice shows that

$$\lim_{t\downarrow 0} F_{t\eta_0} = \lim_{t\downarrow 0} F_{t\eta_1}$$

in $\mathcal{S}'(\mathbb{R}^d)$. We denote the common limit by $\operatorname{bv} F$. The same argument with $y \to 0$ inside an arbitrary proper subcone, using a polygonal chain of rays inside that subcone, gives the full boundary-value convergence stated in the theorem. Uniqueness is immediate: a tempered distribution is determined by its pairings with Schwartz tests, and any two candidate limits agree with the radial limit just constructed.

If F is given as a Fourier–Laplace transform of a distribution carried by C^*, Lemma 12.10 identifies the boundary value with the inverse Fourier transform of that distribution. Therefore the Fourier transform of the boundary value is carried by C^*. □

Thus the QFT-specific work in OS reconstruction is not the analytic boundary-value theorem or the Fourier–Laplace cone-support argument above, but the proof that the holomorphic functions constructed from the Schwinger hierarchy obey the polynomial tube estimates to which the boundary-value theorem applies.

Proposition 12.11 (OS-II analytic route to the boundary-value package)**.** *Fix $k = n-1$ and write the ordered difference variables as*

$$\xi_j = x_{j+1} - x_j, \qquad j = 1, \dots, k.$$

Assume the following OS-II analytic data have been constructed from the Schwinger hierarchy.

(i) There are scalar analyticity domains $C_k^{(N)}$ in the complexified time-difference variables and mixed Hilbert-vector domains $D_m^{(N)}$ for the OS vectors Ψ_m, with

$$\bigcup_{N\ge 0} C_k^{(N)} = \{(\zeta_1, \dots, \zeta_k) : \Re\zeta_j > 0 \text{ for all } j\}.$$

(ii) On each $C_k^{(N)}$ there is a holomorphic representative $S_k^{(N)}$ of the difference-variable Schwinger function, and the representatives agree on overlaps.

(iii) The Hilbert-vector fields $\Psi_m(x,\zeta)$ are holomorphic on $D_m^{(N)}$ and satisfy the OS scalar-product formula

$$\langle \Psi_r(x,\zeta), \Psi_s(y,\eta)\rangle_{\mathcal{H}_{\mathrm{OS}}} = S^{(N)}_{r+s-1}\big(-\overline{\zeta},\ x+y,\ \eta\big) \tag{12.15}$$

whenever both sides are defined. The block $-\overline{\zeta}$ is the reflected left block, $x+y$ is the real bridge time, and η is the right block.

(iv) The local step $A_N \Rightarrow P_N$ is obtained by Hilbert-space Taylor reconstruction: the squared norm of the vector Taylor remainder is the scalar Taylor remainder of the function on the right hand side of (12.15). The step $P_N \Rightarrow A_{N+1}$ is obtained from the scalar products of these vector fields and extension to the envelope of holomorphy defining $C_k^{(N+1)}$.

(v) The continued scalar functions obey a uniform polynomial tube estimate: for every closed subcone $K \Subset (V_+^\circ)^k$ and every $\delta > 0$, if the Lorentzian difference variables are written as $\xi_j - \mathrm{i}\eta_j$ with tube-depth variables $\eta = (\eta_1, \ldots, \eta_k)$ lying in K at distance at least δ from its boundary, then

$$|\langle S_k(\cdot - \mathrm{i}\eta), f\rangle| \le C_{k,K}\, \delta^{-N_k} \sum_{|\alpha|,|\beta|\le N_k} \sup_\xi |\xi^\beta \partial^\alpha f(\xi)| \tag{12.16}$$

for all Schwartz tests f in the reduced difference variables.

(vi) Euclidean covariance identifies the product right-half-plane continuation with the standard forward-tube continuation in Lorentzian difference variables, and the corresponding statements hold for every permutation of the field insertions, with the Koszul sign for graded labels.

Then the Schwinger hierarchy satisfies the OS Lorentzian boundary-value package of Definition 12.8.

Proof. Items (i)–(iii) provide a single holomorphic function in the ordered difference variables: the functions $S_k^{(N)}$ agree on overlaps, and the union of their domains is the full product right half-plane in the ordered time gaps. Item (vi) transports this function by Euclidean covariance to the mostly-plus forward tube $z = \xi - \mathrm{i}\eta$, $\eta \in V_+^\circ$, in Lorentzian difference variables. This gives part (i) of Definition 12.8.

The estimate (12.16) is exactly the tempered tube-bound input in part (ii) of the boundary-value package. The right hand side is a finite Schwartz seminorm, and the dependence on the distance to the cone boundary is polynomial. Therefore the family of functionals $f \mapsto \langle S_k(\cdot - \mathrm{i}\eta), f\rangle$ is bounded in a fixed finite-order tempered-distribution topology on each closed subtube. Theorem 12.9 then gives tempered boundary values as $\eta \to 0$ inside the cone, after the sign conversion $y = -\eta$ explained below.

For a permutation of insertions, item (vi) gives the corresponding permuted tube. Euclidean permutation symmetry says that the permuted holomorphic functions have the same values on their common Euclidean ordered real submanifolds, after inserting the Koszul sign for odd labels. The identity theorem and the distributional edge-of-the-wedge theorem 16.6 applied at Jost configurations therefore give the connected continuation through the common real edge. This is part (iii) of Definition 12.8. □

Theorem 12.12 (Corrected OS-II analytic theorem)**.** *Let the Schwinger hierarchy satisfy normalization, Euclidean covariance, graded permutation symmetry, Hermiticity, reflection positivity, the positive-time semigroup regularity stated above, and the linear-growth condition (12.4). Work* *componentwise in a fixed finite field-label basis; graded signs enter only when two labelled insertions are permuted. Then the OS-II analytic data listed in Proposition 12.11 are obtained from the Schwinger hierarchy. Hence the hierarchy satisfies the OS Lorentzian boundary-value package.*

Proof. We write the proof for scalar components. Since V is finite dimensional in the present field presentation, labelled components are handled by the same argument applied to matrix-valued distributions; the label reflection J and the Koszul signs only change the component bookkeeping.

Let $k = n - 1$. Translation invariance lets us write the ordered Euclidean n-point function in reduced variables

$$\xi = (\xi_1, \ldots, \xi_k), \qquad \xi_j = x_{j+1} - x_j \in E.$$

On a fixed ordered component of the complement of the partial diagonals, write

$$\xi_j = (t_j, \mathbf{x}_j), \qquad t_j > 0,$$

after a Euclidean transformation if necessary. The complex variables used below are the time gaps

$$\zeta = (\zeta_1, \ldots, \zeta_k), \qquad C_+^k = \{\zeta \in \mathbb{C}^k : \Re\zeta_j > 0\}.$$

Spatial difference variables are kept as real parameters until the final Euclidean covariance step; this is only a notational separation, not a restriction on covariance.

Step 1: local one-direction continuation. Choose a split of the insertions into a reflected left block and a positive right block, and separate the two blocks by one positive Euclidean time gap $s > 0$. Reflection positivity gives OS vectors ψ_F, ψ_G and Proposition 12.6 gives

$$s \longmapsto S_{r+s-1}(-\xi_L, s, \xi_R) = \langle \psi_F, \mathrm{e}^{-sH}\psi_G\rangle_{\mathcal{H}_{\mathrm{OS}}}$$

as the boundary value on $s > 0$ of a function holomorphic for $\Re z > 0$. Euclidean covariance lets us rotate the distinguished positive direction. Thus every noncoincident real configuration has finitely many one-complex-variable continuations in directions spanning the complexified difference-variable space.

The passage from these one-variable continuations to a local many-variable continuation is an application of the Malgrange–Zerner extension theorem for compatible holomorphic continuations from flat wedges. Explicitly, choose D real directions e_μ which form a Euclidean basis and a radius $\rho(\xi) > 0$ small enough that every displacement

$$\xi_j + \sum_{\mu=1}^{D} z_{j\mu} e_\mu, \qquad |z_{j\mu}| < \rho(\xi),$$

stays in the same noncoincident ordered component after the relevant rotations. The one-variable continuations agree on their common real slices because they all restrict to the same Schwinger distribution. The Malgrange–Zerner theorem therefore gives a holomorphic function in the polydisc $|z_{j\mu}| < c\rho(\xi)$, whose real restriction is the original Schwinger function. This proves real analyticity away from the partial diagonals and gives the initial assertion A_0.

Step 2: the A_N/P_N induction. We now define the induction precisely. $A_N(k)$ means that the reduced k-gap Schwinger function has a holomorphic representative

$$S_k^{(N)}(\zeta \mid \mathbf{x})$$

for $\zeta \in C_k^{(N)}$, continuous in the real spatial differences $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_k)$, and agreeing with the previous representatives on overlaps. The initial domain $C_k^{(0)}$ is the union of the local polydiscs supplied by Step 1 around positive real time gaps.

Given $C_\ell^{(N)}$, define the vector domain $D_m^{(N)}$ as the set of pairs (x, ζ), with $x > 0$ a real bridge time and $\zeta = (\zeta_1, \dots, \zeta_{m-1})$, for which the reflected norm argument

$$(-\overline{\zeta}_{m-1}, \dots, -\overline{\zeta}_1,\ 2x,\ \zeta_1, \dots, \zeta_{m-1})$$

lies in $C_{2m-1}^{(N)}$. $P_N(m)$ means that there is a holomorphic $\mathcal{H}_{\mathrm{OS}}$-valued function $\Psi_m(x, \zeta)$ on this domain such that, whenever the displayed scalar argument lies in $C_{r+s-1}^{(N)}$,

$$\big\langle \Psi_r(x, \zeta), \Psi_s(y, \eta) \big\rangle_{\mathcal{H}_{\mathrm{OS}}} = S_{r+s-1}^{(N)}(-\overline{\zeta},\ x + y,\ \eta). \tag{12.17}$$

This is the scalar-product formula (12.15).

The implication $A_N \Rightarrow P_N$ is the Hilbert Taylor argument. Fix a real point $(x, \xi) \in D_m^{(N)}$ and a polydisc $P \Subset D_m^{(N)}$. At the real point the vector $\Psi_m(x, \xi)$ is the OS class of the ordered test insertion. Its strong derivatives are defined by differentiating the test insertion before passing to the OS quotient. For $\zeta \in P$ set the L-th vector Taylor remainder

$$R_L(\zeta) = \Psi_m(x, \zeta) - \sum_{|\alpha| \le L} \frac{(\zeta - \xi)^\alpha}{\alpha!}\, \partial_\xi^\alpha \Psi_m(x, \xi).$$

Using (12.17) for real ξ and then $A_N(2m-1)$, the squared norm of this remainder is the scalar Taylor remainder of the holomorphic function

$$S_{2m-1}^{(N)}(-\overline{\zeta}, 2x, \zeta)$$

about $(-\xi, 2x, \xi)$. The scalar Taylor remainder tends to zero uniformly on smaller polydiscs. Hence $R_L(\zeta) \to 0$ in $\mathcal{H}_{\mathrm{OS}}$, and the Taylor series defines a holomorphic vector $\Psi_m(x, \zeta)$. Polarizing the norm identity gives (12.17), so P_N follows.

Conversely, assume $P_N(r)$ and $P_N(s)$ for all lower vector lengths. For $k = r + s - 1$ the scalar product

$$(x, \zeta; y, \eta) \longmapsto \langle \Psi_r(x, \zeta), \Psi_s(y, \eta) \rangle_{\mathcal{H}_{\mathrm{OS}}}$$

is holomorphic on the mixed vector domain and agrees with S_k on the real ordered slice. Let $C_k^{(N+1)}$ be the envelope of holomorphy of the union of all scalar domains generated in this way, together with $C_k^{(N)}$. The identity theorem gives a single extension $S_k^{(N+1)}$. Thus $P_N \Rightarrow A_{N+1}$. Starting from A_0, induction gives

$$A_0 \Rightarrow P_0 \Rightarrow A_1 \Rightarrow P_1 \Rightarrow \cdots .$$

Step 3: exhaustion of the product half-plane. The induction just defined enlarges only the time-gap domains. To see that it reaches all of C_+^k, use logarithmic coordinates

$$\zeta_j = \mathrm{e}^{u_j + \mathrm{i} v_j}, \qquad -\frac{\pi}{2} < v_j < \frac{\pi}{2}.$$

Let $c_k^{(N)}$ be the closed set of allowed argument vectors $(v_1, \dots, v_k)$ for $C_k^{(N)}$, and let $d_m^{(N)}$ be the corresponding set for $D_m^{(N)}$. The envelope-of-holomorphy theorem for tubes says that the passage to $C_k^{(N+1)}$ replaces the union of the argument bases by its convex hull. In these variables the recurrence is

$$c_k^{(N+1)} = \operatorname{conv}\{(-v, \vartheta, w) : (0, v) \in d_r^{(N)},\ (0, w) \in d_s^{(N)},\ r + s - 1 = k,\ |\vartheta| \le \frac{\pi}{2}\}, \tag{12.18}$$

with the evident deletion of empty blocks. The vector domains satisfy the same recurrence with one distinguished real bridge argument fixed to 0.

The exhaustion needed here follows from an explicit insertion induction, not from a qualitative appeal to convexity. For $0 < a < \pi/2$ put

$$B_k(a) = [-a, a]^k.$$

We claim that for every k there is an integer $N_k(a)$ such that

$$B_k(a) \subset c_k^{(N_k(a))}.$$

For $k = 1$, $c_1^{(1)}$ contains the whole interval $[-\pi/2, \pi/2]$, because the single time gap may be chosen as the bridge variable in the one-gap OS semigroup continuation; hence $N_1(a) = 1$. Now assume $B_{k-1}(a) \subset c_{k-1}^{(N)}$. In the recurrence (12.18), choose the split with empty left block and a right block carrying the remaining $k - 1$ gap arguments. In the notation of the recurrence this is the term obtained after deleting the empty v-block:

$$(\vartheta, w), \qquad |\vartheta| \le \frac{\pi}{2}, \qquad w \in B_{k-1}(a).$$

The vector-domain recurrence with the distinguished real bridge argument fixed to 0 identifies this right block with an element of the corresponding d-domain. Therefore every point of $B_k(a) = [-a, a] \times B_{k-1}(a)$ lies in the generating set whose convex hull is $c_k^{(N+1)}$. Thus

$$B_k(a) \subset c_k^{(N+1)}.$$

Induction gives $N_k(a) \le k$. Since every point of C_+^k has arguments strictly between $-\pi/2$ and $\pi/2$, choose $a < \pi/2$ larger than the maximum absolute argument of the point. This proves

$$[-\frac{\pi}{2} + \epsilon, \frac{\pi}{2} - \epsilon]^k \subset c_k^{(k)} \tag{12.19}$$

for every $\epsilon > 0$, and hence

$$\bigcup_{N \ge 0} C_k^{(N)} = C_+^k.$$

Step 4: real-domain polynomial bounds from linear growth. The preceding steps prove holomorphic continuation but not tempered boundary values. The linear-growth condition enters here. Fix a real noncoincident configuration ξ and let $\rho(\xi)$ be the local analyticity radius from Step 1, chosen so that

$$\rho(\xi) \ge c_k (1 + \|\xi\|)^{-a_k} \operatorname{dist}(\xi, \Delta)^{b_k} \tag{12.20}$$

on each ordered component; the exponents depend only on k and the dimension. Choose a nonnegative C^∞ radial kernel g_ρ on $\mathbb{C}^D$ with support in $|z| < \rho/8$, total integral 1, and derivative bounds

$$\|\partial^\alpha g_\rho\|_\infty \le C_\alpha \rho^{-2D-|\alpha|}.$$

Let $k_\rho = g_\rho * g_\rho$, so k_ρ is supported in $|z| < \rho/4$ and has total integral 1. Applying the mean-value theorem for holomorphic functions blockwise and inserting k_ρ, one obtains a regularized function $T_{k,\rho}$ such that for $|\eta_j| < \rho/2$

$$S_k(\xi+\eta) = \int T_{k,\rho}(\xi+\eta+\mathrm{i}y)\,\mathrm{d}\nu_\rho(y), \tag{12.21}$$

where ν_ρ is a positive probability measure supported in $|y_j| < \rho/4$. Consequently

$$|S_k(\xi+\eta)| \le \sup_{|y_j|<\rho/4} |T_{k,\rho}(\xi+\eta+\mathrm{i}y)|. \tag{12.22}$$

The regularized function is designed so that OS positivity still applies. For each split $k = r+s-1$ and each of the finitely many Euclidean basis directions, the value of $T_{k,\rho}$ after a real displacement in that direction can be written as

$$T_{k,\rho}(\xi+ue_\mu) = \langle \Phi_{L,\rho}(\xi,u), \Phi_{R,\rho}(\xi,u)\rangle_{\mathcal{H}_{\mathrm{OS}}}, \tag{12.23}$$

where the two OS vectors are obtained by smearing the left and right ordered blocks with the kernels g_ρ and k_ρ. This is an identity of test functions before quotienting: the reflected left test and the right test have exactly the convolution kernel appearing in $T_{k,\rho}$. The semigroup continuation in the distinguished direction gives

$$|T_{k,\rho}(\xi+we_\mu)| \le \|\Phi_{L,\rho}(\xi,u)\|_{\mathcal{H}_{\mathrm{OS}}}\,\|\Phi_{R,\rho}(\xi,u)\|_{\mathcal{H}_{\mathrm{OS}}}, \qquad \Re w > 0. \tag{12.24}$$

The norm of, say, $\Phi_{L,\rho}$ is an OS quadratic form $S(\Theta F, F)$ with an explicitly known test function F. If the left vector has r ordered insertions, this quadratic form is a $(2r-1)$-point reduced Schwinger distribution, in the notation of (12.17). The linear-growth hypothesis gives a seminorm of order

$$N_{\mathrm{OS}}(2r-1) = N_0 + C_{\mathrm{OS}}(2r-1). \tag{12.25}$$

Each derivative landing on a smearing kernel g_ρ or k_ρ costs a power of ρ^{-1}; each polynomial weight in the seminorm is evaluated on a point whose coordinates are bounded by a constant times $1+\|\xi\|+\|u\|+\rho$. Since $\rho \le 1$ after shrinking the local polydisc, (12.4) gives

$$\|\Phi_{L,\rho}(\xi,u)\|^2_{\mathcal{H}_{\mathrm{OS}}} \le A_r\,B_r\,\rho^{-M_r}(1+\|\xi\|+\|u\|)^{M_r}, \qquad M_r = M_0 + M_1\,N_{\mathrm{OS}}(2r-1), \tag{12.26}$$

with constants depending on the fixed kernel family, the dimension, and the field labels in the finite component under consideration. The right vector obeys the same estimate with r replaced by s. Because the scalar block has $k = r+s-1$ gaps, both $2r-1$ and $2s-1$ are bounded by $2k+1$. Thus the product of the square-root estimates is controlled by a single exponent affine in the number of gaps. In particular there are constants A_k, B_k, M_k, with at most the factorial growth inherited from (12.4), such that

$$\|\Phi_{L,\rho}(\xi,u)\|_{\mathcal{H}_{\mathrm{OS}}}\,\|\Phi_{R,\rho}(\xi,u)\|_{\mathcal{H}_{\mathrm{OS}}} \le A_kB_k\,\rho^{-M_k}(1+\|\xi\|+\|u\|)^{M_k}. \tag{12.27}$$

Combining (12.24) and (12.27) gives polynomial bounds in each direction. Passing to logarithmic variables turns the directional wedge domains into flat tubes. The Malgrange–Zerner theorem glues the directional continuations, and the maximum principle on the envelope of holomorphy propagates the same bound to the whole local polydisc. Returning through (12.22) gives a real-domain estimate of the form

$$|S_k(\xi)| \le A'_k\, \rho(\xi)^{-M'_k}(1+\|\xi\|)^{M'_k}. \tag{12.28}$$

Together with (12.20), this is a polynomial growth estimate on every ordered real component, with finite distributional order controlled linearly in k by (12.4).

Step 5: continuation of the estimates to C_+^k. It remains to carry the real-domain estimate through the A_N/P_N ladder. Let $\mathcal{B}_{k,t}$ be the Banach space of continuous functions of the k spatial difference variables with norm

$$\|f\|_{k,t} = \sup_{\mathbf{x}}(1+\max_j |\mathbf{x}_j|)^{-t}|f(\mathbf{x})|.$$

For t larger than the exponent in (12.28), the functions $\zeta \mapsto S_k^{(N)}(\zeta \mid \cdot)$ are $\mathcal{B}_{k,t}$-valued holomorphic functions on $C_k^{(N)}$. Introduce a regulator $\epsilon > 0$ and divide by a positive scalar majorant $Q_{k,\epsilon}(\zeta)$ whose logarithm is convex in the variables $\Re\zeta_j$:

$$S_{k,\epsilon}^{(N)}(\zeta) = Q_{k,\epsilon}(\zeta)^{-1} S_k^{(N)}(\zeta + \epsilon\mathbf{1} \mid \cdot).$$

The majorant is chosen so that the real-domain estimate gives

$$\|S_{k,\epsilon}^{(0)}(\zeta)\|_{k,t} \le A_k \qquad (\zeta \in C_k^{(0)}), \tag{12.29}$$

uniformly in ϵ.

Assume the bound is known on $C_\ell^{(N)}$ for all relevant lower scalar lengths. On a mixed domain generated by P_N, the scalar-product formula and Cauchy–Schwarz give

$$\|S_{r+s-1}^{(N)}(-\overline{\zeta}, x+y, \eta)\|_{r+s-1,t} \le \|S_{2r-1}^{(N)}(-\overline{\zeta}, 2x, \zeta)\|_{2r-1,t}^{1/2} \|S_{2s-1}^{(N)}(-\overline{\eta}, 2y, \eta)\|_{2s-1,t}^{1/2}. \tag{12.30}$$

The convexity of $\log Q_{k,\epsilon}$ is precisely the algebraic statement that the denominator for the left side is dominated by the product of the two denominators on the right. Therefore

$$\|S_{k,\epsilon}^{(N)}\|_{k,t} \le A_k$$

on the generating mixed region. Since $C_k^{(N+1)}$ is the envelope of holomorphy of that region and the functions take values in the Banach space $\mathcal{B}_{k,t}$, the Banach-valued maximum principle extends the same bound to all of $C_k^{(N+1)}$. Induction gives the estimate on every $C_k^{(N)}$.

Finally choose $N = N(\zeta)$ using the exhaustion (12.19); the required N is bounded by a constant times a logarithm of the inverse angular distance of ζ to ∂C_+^k. Undoing the harmless regulator ϵ and the majorant $Q_{k,\epsilon}$ yields the tube estimate

$$|\langle S_k(\cdot - \mathrm{i}\eta), f\rangle| \le C_{k,K}\, \delta^{-N_k} \sum_{|\alpha|,|\beta| \le N_k} \sup_\xi |\xi^\beta \partial^\alpha f(\xi)|$$

for η in any closed subcone $K \Subset (V_+^\circ)^k$ at distance δ from the cone boundary. This is (12.16). Euclidean covariance transports the product right-half-plane continuation to the mostly-plus Lorentzian forward tubes, and permutation symmetry supplies the permuted tubes with the graded signs. Thus all hypotheses of Proposition 12.11 have been constructed from the stated Euclidean data and the linear-growth condition. $\square$

Proposition 12.13 (Fourier–Laplace boundary values from spectral support)**.** *Let $d = D(n-1)$, let $\Gamma \subset \mathbb{R}^d$ be a closed convex cone, and let $\Gamma^{\circledast}$ be its open dual cone:*

$$\Gamma^{\circledast} = \{y \in \mathbb{R}^d \mid p \cdot y > 0 \textit{ for every } 0 \neq p \in \Gamma\}.$$

Let $W \in \mathcal{S}'(\mathbb{R}^d)$ have Fourier transform $\widehat{W}$ supported in Γ. For $y \in \Gamma^{\circledast}$ define, by using any smooth cutoff equal to 1 on a conic neighbourhood of Γ on which $p \cdot y$ is nonnegative,

$$\mathcal{W}(x + \mathrm{i}y) = (2\pi)^{-d} \big\langle \widehat{W}(p), \exp(-\mathrm{i}p \cdot x) \exp(-p \cdot y) \big\rangle. \tag{12.31}$$

Then $\mathcal{W}$ is holomorphic on the tube $\mathbb{R}^d + \mathrm{i}\Gamma^{\circledast}$. If $K \Subset \Gamma^{\circledast}$ is a closed subcone and $\operatorname{dist}(y, \partial\Gamma^{\circledast}) \geq \delta > 0$*, then for every Schwartz test function f*

$$\left| \int_{\mathbb{R}^d} \mathcal{W}(x + \mathrm{i}y) f(x) \, \mathrm{d}^d x \right| \leq C_{K,N} \, \delta^{-N} \sum_{|\alpha|, |\beta| \leq N} \sup_p |p^\beta \partial_p^\alpha \widehat{f}(p)|$$

for some N determined by the distributional order of $\widehat{W}$. Moreover,

$$\mathcal{W}(\cdot + \mathrm{i}y) \longrightarrow W \qquad \textit{in } \mathcal{S}'(\mathbb{R}^d)$$

as $y \to 0$ inside K.

Proof. Because $\widehat{W}$ is a tempered distribution, there are C, N such that

$$|\langle \widehat{W}, \psi \rangle| \leq C \sum_{|\alpha|, |\beta| \leq N} \sup_p |p^\beta \partial_p^\alpha \psi(p)| \qquad \psi \in \mathcal{S}(\mathbb{R}^d).$$

For $y \in \Gamma^{\circledast}$, the multiplier $\exp(-p \cdot y)$ is exponentially decreasing on the support cone Γ. To make the pairing literally a Schwartz pairing, choose a smooth conic cutoff χ_y equal to 1 near Γ and supported where $p \cdot y \geq c_y |p| - C_y$; then $\chi_y \exp(-p \cdot y) \exp(-\mathrm{i}p \cdot x)$ is a Schwartz multiplier after testing in x. Different choices of χ_y give the same value, because their difference vanishes in a neighbourhood of $\operatorname{supp} \widehat{W}$. Derivatives in $x + \mathrm{i}y$ only multiply the cutoff multiplier by polynomials in p, so the same estimate proves holomorphy by differentiation under the distributional pairing.

If y stays in a closed subcone at distance δ from the boundary, one cutoff may be chosen uniformly on that subcone, and cone duality gives

$$p \cdot y \geq c_K \delta |p| \qquad (p \in \Gamma)$$

with $c_K > 0$. Hence every derivative of $\exp(-p \cdot y) \widehat{f}(p)$ is bounded by a finite linear combination of Schwartz seminorms of $\widehat{f}$ times a power of δ^{-1}. This gives the displayed tube bound.

For the boundary value, test against f. The pairing equals

$$(2\pi)^{-d} \langle \widehat{W}(p), \mathrm{e}^{-p \cdot y} \widehat{f}(p) \rangle.$$

More explicitly, the left side is computed with the same cutoff convention, and this convention is invisible to $\widehat{W}$. The cutoff functions $\chi \, \mathrm{e}^{-p \cdot y} \widehat{f}(p)$ converge to $\chi \, \widehat{f}(p)$ in the Schwartz topology on every conic neighbourhood of Γ on which $\chi = 1$ near Γ: the exponential is bounded by 1 on Γ, the derivative terms are controlled by the rapid decrease of $\widehat{f}$, and the factors $\mathrm{e}^{-p \cdot y} - 1$ converge uniformly on compact p-sets while the tails are uniformly small. Continuity of $\widehat{W}$ gives convergence to $(2\pi)^{-d} \langle \widehat{W}, \widehat{f} \rangle = W(f)$. ☐

The linear-growth estimate (12.4) is one standard route to this package in the OS-II theorem: the controlled growth of distributional seminorm order is what keeps the Fourier–Laplace continuation inside the tempered distribution class as the tube-depth variables approach the Lorentzian boundary. In a concrete constructive model, this route is a separate analytic estimate to be proved, not a formal consequence of reflection positivity alone.

Applied to reconstruction, Proposition 12.13 is used with $d = D(n-1)$, spectral cone $\Gamma = (\overline{V}_+)^{n-1}$, literal imaginary variable $y \in (-V_+)^{n-1}$, and tube depth $\eta = -y \in (V_+)^{n-1}$. The proposition is written for an abstract tube $x + \mathrm{i}y$ with $p \cdot y > 0$ on the spectral cone. In the mostly-plus Wightman convention the same point is written as $\xi - \mathrm{i}\eta$, and the damping factor is $\exp(p \cdot \eta)$ because $p \cdot \eta \leq 0$. The distribution W is the translation-reduced Wightman distribution. The OS-II linear-growth condition is the Euclidean-side estimate that makes the reconstructed matrix elements have the required finite distributional order uniformly in the number of insertions. The spectral support itself comes from the nonnegative Hamiltonian, spatial translation covariance, and the OS-continued Lorentz transformations. Thus the corrected OS theorem has two distinct analytic duties: it must construct the positive-energy translation representation, and it must prove the polynomial tube bounds needed for the boundary-value theorem above.

Warning 12.14 (Tube-sign conversion). The abstract Fourier–Laplace theorem above uses the variable $x + \mathrm{i}y$ with $p \cdot y > 0$ on the spectral cone. The physical Wightman tube in the mostly-plus convention is instead written $\xi - \mathrm{i}\eta$, with $\eta \in V_+$ and $p \cdot \eta \leq 0$ for $p \in \overline{V}_+$. Therefore the substitution in OS reconstruction is $y = -\eta$. Replacing y by η flips the damping factor and would reverse the spectral-support statement. This is a convention-sensitive point of the corrected OS-II boundary-value construction, not a typographical preference.

Proposition 12.15 (Consequences of the OS boundary-value package)**.** *Assume Definition 12.8. Then:*

(a) The limits of $\mathcal{W}_n(\xi - \mathrm{i}\eta)$ as $\eta \to 0$ within the ordered tube exist in $\mathcal{S}'((\mathbb{M}^D)^n)$ and define tempered Wightman distributions.

(b) Permutation symmetry of the Schwinger hierarchy gives graded commutativity of the boundary-value fields at spacelike separation.

(c) Continuing the Wightman distributions back to ordered imaginary time recovers the original S_n away from the partial diagonals.

Proof. For (a), fix one ordering and write the reduced difference variables as $\xi = (\xi_1, \ldots, \xi_{n-1})$. The tube bound in Definition 12.8 says that, for each smaller cone of tube depths, the function $\xi \mapsto \mathcal{W}_n(\xi - \mathrm{i}\eta)$ defines a continuous functional on $\mathcal{S}(\mathbb{R}^{D(n-1)})$ with seminorm constants polynomial in the distance of η to the cone boundary. Proposition 12.13 displays the model estimate after the sign conversion $y = -\eta$: after Fourier transformation, the tube depth inserts exponential damping against a distribution whose spectral support lies in the forward cone. Hence, for every Schwartz test f, the pairings $\langle \mathcal{W}_n(\cdot - \mathrm{i}\eta), f \rangle$ have a limit as $\eta \to 0$ within the ordered cone. The limiting functional is continuous in the same finite collection of Schwartz seminorms and is therefore a tempered distribution. This distribution is the ordered Wightman distribution.

For (b), first treat unsmeared configurations away from diagonals. In reduced variables, a real point ξ is a Jost point if every nonzero positive linear combination of the successive differences $\xi_j = x_j - x_{j+1}$ is spacelike. Such points are precisely the real edges at which the forward tube for one ordering and the forward tube for a permuted ordering meet after complex Lorentz transformations. Euclidean permutation symmetry gives equality, with the Koszul sign, of the two holomorphic functions on the Euclidean real slice where both orderings are defined. By the identity theorem this equality holds in the common analytic continuation region. The edge-of-the-wedge theorem applies because the two tempered boundary values have a common real edge at the Jost set and agree there as distributions. Therefore their analytic continuations coincide in the envelope of holomorphy of the union of the two tubes.

Now smear the adjacent fields with Lorentzian test functions whose supports are spacelike separated. A partition of unity reduces the support of the smearing to coordinate neighbourhoods whose relative configurations lie in Jost neighbourhoods. The boundary-value equality obtained above can then be paired with these test functions. The result is

$$\langle \Omega, \cdots \widehat{\Phi}_\alpha(f)\widehat{\Phi}_\beta(g) \cdots \Omega\rangle = (-1)^{|\alpha||\beta|}\langle \Omega, \cdots \widehat{\Phi}_\beta(g)\widehat{\Phi}_\alpha(f) \cdots \Omega\rangle$$

for every surrounding finite string of test insertions. Since the finite-insertion vectors are dense in $\mathcal{D}_{\mathrm{OS}}$, this is exactly local graded commutativity on the reconstructed common domain.

For (c), let $0 < \tau_1 < \cdots < \tau_n$ be an ordered Euclidean configuration away from Δ_n. The original Schwinger function and the function obtained by first taking the Lorentzian boundary value and then substituting $z_j^0 = -\mathrm{i}\tau_j$ are restrictions of the same holomorphic function $\mathcal{W}_n$ on the connected ordered tube. If two such holomorphic functions had the same Lorentzian boundary value but differed in the ordered Euclidean interior, their difference would have zero distributional boundary value on a nonempty real edge and hence, by the same edge-of-the-wedge uniqueness theorem, would vanish in the connected tube. Thus the continuation back to ordered imaginary time recovers the original S_n away from the partial diagonals. Contact-term extensions on the diagonals are additional Euclidean data and are not recovered from off-diagonal analytic continuation alone. □

12.5 Reconstruction Theorem

The source trace for the corrected reconstruction theorem is the Osterwalder–Schrader pair of papers, with the linear-growth correction in the second paper [7, 8].

Theorem 12.16 (Osterwalder–Schrader reconstruction)**.** *Let $\{S_n\}_{n\geq 0}$ be Euclidean distributions satisfying Euclidean covariance, permutation symmetry, reflection positivity, Hermiticity, normalization, the OS-II positive-time semigroup conditions* (12.5)–(12.8)*, and the linear-growth condition* (12.4)*. Assume also the ordered-insertion adjoint identity* (12.9) *for the field labels to be reconstructed. By the corrected OS-II analytic theorem, Theorem 12.12, these hypotheses supply the analytic route of Proposition 12.11. Then there exists Wightman field data*

$$(\mathcal{H}_{\mathrm{OS}}, \Omega_{\mathrm{OS}}, U, \mathcal{D}_{\mathrm{OS}}, \{\widehat{\Phi}_\alpha\}_{\alpha\in I})$$

whose Schwinger functions are $\{S_n\}$ at noncoincident Euclidean points. The reconstructed Wightman functions are obtained as boundary values of analytic continuations of the S_n. The symbol Ω_{OS} denotes the vacuum vector produced by this reconstruction; it is a subscripted instance of the global vacuum notation Ω.

Proof. Step 1 constructs the Hilbert space. Let $\mathcal{E}_+$ be the vector space of finite labelled Euclidean insertions whose time coordinates are strictly positive. Reflection positivity makes

$$\langle F, G\rangle_{\mathrm{OS}} := S_{m+n}(\Theta F, G)$$

a positive semidefinite Hermitian form on $\mathcal{E}_+$. Its null space is

$$\mathcal{N}_{\mathrm{OS}} = \{F \in \mathcal{E}_+ : \langle F, F\rangle_{\mathrm{OS}} = 0\}.$$

The Cauchy–Schwarz inequality for positive semidefinite forms implies that $\langle F, G\rangle_{\mathrm{OS}} = 0$ whenever $F \in \mathcal{N}_{\mathrm{OS}}$; therefore the form descends to $\mathcal{E}_+/\mathcal{N}_{\mathrm{OS}}$. Completing gives $\mathcal{H}_{\mathrm{OS}} = \overline{\mathcal{E}_+/\mathcal{N}_{\mathrm{OS}}}$. The class of the empty insertion is Ω_{OS}, and finite positive-time insertions are dense by construction.

Step 2 constructs the positive-energy time evolution. For $s \geq 0$, Euclidean time translation by s maps $\mathcal{E}_+$ to itself. The semigroup law, OS symmetry, and polynomial matrix-element growth imply contractivity by Proposition 12.5; strong continuity is the hypothesis (12.8). Hence the descended operators $T_E(s)$ form a strongly continuous semigroup of self-adjoint contractions on $\mathcal{H}_{\mathrm{OS}}$. The self-adjoint contraction-semigroup theorem gives a nonnegative self-adjoint generator

$$H, \qquad T_E(s) = e^{-sH}, \qquad \sigma(H) \subset [0, \infty).$$

Stone's theorem gives Lorentzian time translations

$$U_{\mathrm{time}}(t) = \mathrm{e}^{-\mathrm{i}tH}.$$

This is the part of the reconstruction theorem that uses reflection positivity directly; it does not use the flawed separate-Laplace step of the first OS paper.

Step 3 constructs the spacetime symmetry representation. Spatial translations and rotations preserve the positive-time half-space after a common positive time shift. Euclidean covariance and the same null-space argument used in Step 1 therefore give unitary operators on $\mathcal{H}_{\mathrm{OS}}$. Proposition 12.11 and Theorem 12.12 supply the missing many-variable analytic continuation: Euclidean rotations mixing time and space continue to real Lorentzian boosts, and the continued matrix elements obey the polynomial tube estimates required for tempered boundary values. The resulting unitary representation U of the connected Poincare group has joint translation spectrum in the closed forward light cone. The forward spectral support follows from $H \geq 0$, spatial covariance, and the Lorentzian boost covariance just obtained.

Step 4 constructs the field-valued distributions on the common dense domain. For a labelled test function f supported at positive Euclidean time, define, before taking Lorentzian boundary values,

$$\Phi_E(\alpha, f)[F] = [(\alpha, f) \otimes F],$$

after translating all supports by a common positive Euclidean time if necessary so that the concatenated configuration is strictly ordered. Proposition 12.7 proves that this operation descends through the OS null quotient and gives closable operators on the dense domain $\mathcal{D}_{\mathrm{OS}}$. The continuity estimates in the OS data make all vacuum and finite-domain matrix elements distributions in the test function. Their Lorentzian boundary values define the Wightman distributions and the corresponding operator-valued distributions $\widehat{\Phi}_\alpha$ on $\mathcal{D}_{\mathrm{OS}}$.

Step 5 verifies the Wightman properties. Hermiticity of the Schwinger hierarchy gives the adjoint relation on $\mathcal{D}_{\mathrm{OS}}$, and Euclidean covariance gives the Poincare covariance of the reconstructed fields. Proposition 12.15 gives temperedness, the edge-of-the-wedge/Jost-domain proof of local graded commutativity at spacelike separation, and the reverse continuation back to the original ordered Euclidean Schwinger functions away from the partial diagonals. The restriction to noncoincident Euclidean points is essential: contact extensions on the diagonals are part of the declared Schwinger hierarchy and are not recovered from off-diagonal analytic continuation alone. □

Output ledger for AQFT use. Theorem 12.16 is the Euclidean-to-Wightman bridge used later in the local-net chapters. Its output has a precise boundary. It constructs

$$\mathcal{H}_{\mathrm{OS}}, \qquad \Omega_{\mathrm{OS}}, \qquad U, \qquad \mathcal{D}_{\mathrm{OS}}, \qquad \widehat{\Phi}_\alpha(f),$$

with positive-energy Poincare covariance, tempered Wightman distributions, domain-level adjoints, and graded locality on the common finite-insertion domain. If a cluster condition is added, it also identifies the vacuum sector as in Corollary 12.19. These are field-presentation outputs, not yet a Haag–Kastler net in the sense of Definition 3.1.

To use an OS-constructed model as an AQFT model one must supply the bounded local-algebra layer separately. A typical route is the comparison structure of Definition 3.12: prove self-adjoint closures for the relevant Hermitian smeared fields, prove strong commutation of their spectral measures for spacelike separated supports, form the represented von Neumann algebras generated by the spectral projections, and prove isotony, covariance, additivity, and time-slice or causal propagation properties for those bounded algebras. If the OS fields are charged, gauge-fixed, or otherwise larger than the observable content, one must then pass to the observable subnet by fixed points, BRST/BV physical cohomology, Gauss-law constraints, or another declared operation and verify that operation preserves the local-net properties. Equality with a proposed observable net is the completeness condition of Definition 3.13, not a formal consequence of equality of Schwinger functions.

The same boundary applies to stronger structural properties. OS reconstruction by itself does not prove Reeh–Schlieder for the bounded net, Haag duality, nuclearity, split inclusions, type-III local factor classification, DHR transportability, Doplicher–Roberts reconstruction, or Bisognano–Wichmann modular covariance. Some of these properties can be proved later from the Wightman spectral condition plus weak additivity, as in the Reeh–Schlieder theorem of Chapter 3; others require model-specific phase-space estimates, duality arguments, sector constructions, or wedge-domain theorems. Thus the correct dependency chain is

$$\begin{aligned} \text{OS Schwinger data} &\Longrightarrow \text{Wightman field presentation} \\ &\Longrightarrow \text{bounded local-net comparison} \\ &\Longrightarrow \text{additional AQFT structure.} \end{aligned}$$

Skipping the middle two arrows is precisely the hidden proof debt that the AQFT chapters keep separate.

Proposition 12.17 (Uniqueness of OS reconstruction)**.** *Two OS reconstructions from the same Schwinger hierarchy are unitarily equivalent on the reconstructed Wightman field presentation.*

Proof. Let

$$(\mathcal{H}_1, \Omega_1, U_1, \mathcal{D}_1, \widehat{\Phi}^{(1)}_\alpha), \qquad (\mathcal{H}_2, \Omega_2, U_2, \mathcal{D}_2, \widehat{\Phi}^{(2)}_\alpha)$$

be two reconstructions from the same hierarchy. Both start from the same positive-time vector space $\mathcal{E}_+$ and the same OS sesquilinear form, hence from the same null space $\mathcal{N}$. On the dense quotient define

$$V_0[F]_1 = [F]_2.$$

This is well defined because $[F]_1 = 0$ iff $\langle F, F\rangle_{\mathrm{OS}} = 0$ iff $[F]_2 = 0$. It is isometric because both inner products are the same OS form:

$$\langle V_0[F]_1, V_0[G]_1\rangle_{\mathcal{H}_2} = \langle F, G\rangle_{\mathrm{OS}} = \langle [F]_1, [G]_1\rangle_{\mathcal{H}_1}.$$

Since the quotient is dense in both completions, V_0 extends uniquely to a unitary $V : \mathcal{H}_1 \to \mathcal{H}_2$. The empty insertion is mapped to the empty insertion, so $V\Omega_1 = \Omega_2$. Euclidean translations, rotations, and the continued Poincare transformations are all defined on the dense quotient by the same operations on test sequences, hence

$$VU_1(g) = U_2(g)V$$

on a dense domain and therefore on the Hilbert spaces. Field insertions are also defined by the same operation on the common quotient:

$$V\widehat{\Phi}^{(1)}_\alpha(f)[F]_1 = V[(\alpha, f), F]_1 = [(\alpha, f), F]_2 = \widehat{\Phi}^{(2)}_\alpha(f)V[F]_1,$$

with equality on the common invariant domain. Thus the two reconstructions are unitarily equivalent as Wightman field presentations. □

Path-integral Schwinger limits as Wightman data. Let $\{\mathcal{E}_\Lambda, \langle\cdot\rangle_\Lambda\}$ be a regulated Euclidean path-integral presentation, where Λ denotes the regulator and $\langle\cdot\rangle_\Lambda$ is the regulated expectation functional on a declared algebra of Euclidean field insertions. Assume that for every n and every labelled test insertion f_n in the chosen test class the regulated moments

$$S_{n,\Lambda}(f_n) := \langle f_n(\Phi_E, \ldots, \Phi_E)\rangle_\Lambda$$

converge to a distributional limit $S_n(f_n)$, and that the limiting hierarchy $\{S_n\}_{n\geq 0}$ satisfies the OS-admissibility hypotheses of Definition 12.2. Then the continuum path-integral presentation determines a Wightman field presentation by applying Theorem 12.16 to the limiting Schwinger hierarchy. If the limiting hierarchy also satisfies the Euclidean cluster condition in Corollary 12.19, the reconstructed zero-momentum subspace is $\mathbb{C}\Omega_{\mathrm{OS}}$.

The convergence assumption defines a family $\{S_n\}_{n\geq 0}$ of continuous linear functionals on the declared test-function spaces; $S_0 = 1$ is the normalization of the regulated expectation functional after taking the limit. The OS-admissibility assumption supplies Euclidean covariance, permutation symmetry, Hermiticity, reflection positivity, and the OS-II semigroup and linear-growth estimates used in Theorem 12.16. Applying that theorem constructs

$$(\mathcal{H}_{\mathrm{OS}}, \Omega_{\mathrm{OS}}, U, \mathcal{D}_{\mathrm{OS}}, \{\widehat{\Phi}_\alpha\})$$

whose Wightman boundary values have the limiting Schwinger hierarchy as their Euclidean continuation at noncoincident points. The last sentence is exactly Corollary 12.19.

Remark 12.18 (Constructive status of OS inputs). OS reconstruction is a bridge theorem. It turns a Schwinger hierarchy with reflection positivity, Euclidean covariance, permutation symmetry, Hermiticity, clustering when desired, and the OS-II analytic bounds into a Lorentzian Wightman presentation. It does not by itself construct the Schwinger hierarchy. Completed constructions verifying such inputs are model-dependent and concentrated in low-dimensional regimes, including $P(\phi)_2$, massive ϕ^4_3, selected two-dimensional scalar–fermion models, and two-dimensional Yang–Mills in holonomy or heat-kernel formulations. Standard reflection-positive Ising/$\lambda\phi^4$ routes in $D \geq 4$ are constrained by triviality theorems, while four-dimensional Yang–Mills, QCD-like theories, and the full Standard Model remain open as complete constructive continuum theorems. Chapter 136 gives the status table and route-by-route discussion.

Corollary 12.19 (OS reconstruction with and without clustering)**.** *Let P_0^{OS} be the joint spectral projection of the reconstructed Lorentzian translation representation onto zero momentum. The reconstruction theorem by itself produces an invariant cyclic vector Ω_{OS} and a positive-energy representation, but it leaves the dimension of $P_0^{\mathrm{OS}}\mathcal{H}_{\mathrm{OS}}$ to the Schwinger hierarchy. Thus, in the absence of a cluster condition, the reconstructed representation may contain several translation-invariant vacuum vectors.*

If the Euclidean hierarchy clusters, then for finite polynomial Euclidean field insertions A_E and B_E separated by a large spatial Euclidean translation,

$$\lim_{\lambda\to+\infty} \langle A_E\, B_E(\lambda a)\rangle_E = \langle A_E\rangle_E\, \langle B_E\rangle_E, \qquad a \in \mathbb{R}^{D-1},\ a \neq 0,$$

then its Lorentzian reconstruction satisfies the Wightman weak cluster property. Consequently

$$P_0^{\mathrm{OS}}\mathcal{H}_{\mathrm{OS}} = \mathbb{C}\Omega_{\mathrm{OS}}$$

by Theorem 2.3. Conversely, a one-dimensional reconstructed zero-momentum subspace gives Wightman clustering, and its restriction back to noncoincident Euclidean points gives the corresponding Euclidean cluster property.

When $P_0^{\mathrm{OS}}\mathcal{H}_{\mathrm{OS}}$ has dimension larger than one, the Schwinger functional is a translation-invariant state that is not extremal. Its central decomposition is a direct integral, or in finite mixtures a convex sum, of pure vacuum sectors. OS positivity and covariance still reconstruct a valid Wightman presentation, while clustering is the additional condition that selects a single pure infinite-volume phase.

The theorem has three distinct pieces. First, reflection positivity constructs the Hilbert space, vacuum vector, and positive Hamiltonian by the positive-time quotient. Second, Euclidean covariance and the OS regularity assumptions construct the Lorentzian Poincare representation and its spectrum condition. Third, growth and analyticity assumptions control the distributional boundary values and produce Wightman functions in the intended test-function class. In the tempered case, this target class is $\mathcal{S}'(M^n)$. In larger analytic frameworks, such as hyperfunction versions of reconstruction, the target distribution class is correspondingly larger and the growth hypotheses are changed.

The uniqueness statement is internal to the specified Euclidean hierarchy, label space, reflection map, and test-function class. Two reconstructions of the same OS-admissible data are related by a unitary equivalence preserving the vacuum, the reconstructed Poincare representation, and the smeared fields. Equivalence of two different Euclidean presentations, or of the local nets generated after taking closures and affiliations, is an additional comparison problem.

12.6 Two-Point Spectral Positivity

For a scalar field, OS positivity gives a direct Euclidean form of spectral positivity. Let $f(\mathbf{x})$ be a spatial test function and let $\Phi_E(\tau, f)$ denote the reconstructed Euclidean field smeared at positive time τ. The notation $\Phi_E(0, f)$ below denotes the boundary value from positive Euclidean time when this domain limit exists. For $\tau > 0$, time translation in the reconstructed Hilbert space gives

$$\langle \Omega_{\mathrm{OS}}, \Phi_E(\tau, f)\Phi_E(0, g)\Omega_{\mathrm{OS}}\rangle_{\mathrm{OS}} = \langle \Phi_E(0, f)^\dagger \Omega_{\mathrm{OS}}, \mathrm{e}^{-\tau H}\Phi_E(0, g)\Omega_{\mathrm{OS}}\rangle_{\mathrm{OS}}.$$

Applying the spectral theorem to the nonnegative operator H, this is

$$\int_{[0,\infty)} \mathrm{e}^{-\tau E}\, \mathrm{d}\nu_{f,g}(E),$$

where

$$\nu_{f,g}(\Delta) = \langle \Phi_E(0, f)^\dagger \Omega_{\mathrm{OS}}, E_H(\Delta)\Phi_E(0, g)\Omega_{\mathrm{OS}}\rangle_{\mathrm{OS}}$$

and E_H is the spectral measure of H. For $f = g$, the measure is positive. Including spatial translations refines this to the joint spectral measure of $(H, \mathbf{P})$. Lorentz covariance and the spectrum condition then repackage the scalar two-point measure by invariant mass, giving the Källén–Lehmann representation used in Chapter 14.

This calculation identifies the precise bridge: reflection positivity builds the Hilbert-space inner product, the Euclidean time semigroup gives the nonnegative Hamiltonian, and the spectral theorem supplies the positive energy-momentum measure.

12.7 Euclidean Measures and Generating Functionals

Suppose a Euclidean theory is given by a probability measure μ on a specified measurable space of scalar distributions φ, with generating functional

$$Z_E[J] = \int \exp\left(\int_E J(x)\varphi(x)\,\mathrm{d}^D x\right) \mathrm{d}\mu(\varphi)$$

for real test functions J in a neighborhood of the origin. The Schwinger functions are moments:

$$S_n(x_1, \ldots, x_n) = \int \varphi(x_1)\cdots\varphi(x_n)\,\mathrm{d}\mu(\varphi).$$

Reflection positivity is then a property of the measure together with the chosen time reflection. It is stable under lattice approximations only when the regulator action, observable insertions, and continuum limit preserve the positive-time quadratic form.

Remark 12.20. Ordinary positivity of the probability measure μ means $\int |F(\varphi)|^2\,\mathrm{d}\mu(\varphi) \geq 0$ for measurable functions F. OS positivity is the reflected statement $\int \overline{F(\theta\varphi)}F(\varphi)\,\mathrm{d}\mu(\varphi) \geq 0$ for positive-time cylinder functions F, with the label adjunction included. It is this reflected positivity that becomes the Lorentzian Hilbert-space inner product.

12.8 Reflection Positivity for Generating Functionals

When the Euclidean moments are encoded by $Z_E[J]$, reflection positivity has an equivalent source form. Let J_r be finitely many sources supported in positive Euclidean time and let $c_r \in \mathbb{C}$. Define the reflected source by

$$(\Theta J_r)(x) = \overline{J_r(\theta x)}$$

with the field adjunction included when the source carries labels. The generating functional is reflection positive when

$$\sum_{r,s} \bar{c}_r c_s \, Z_E[\Theta J_r + J_s] \geq 0. \tag{12.32}$$

Expanding Z_E in powers of the sources recovers (12.2) for all Schwinger functions. Conversely, if the moment expansion is valid in a neighborhood of the origin, the Schwinger-function inequalities imply (12.32).

For a Gaussian scalar measure with covariance $C(x-y)$, the criterion reduces to positivity of the kernel reflected across $\tau = 0$:

$$\int_{\tau_x>0} \mathrm{d}^D x \int_{\tau_y>0} \mathrm{d}^D y \, \overline{f(x)} \, C(\theta x - y) \, f(y) \geq 0$$

for all positive-time test functions f. The free massive covariance

$$C(x-y) = \int \frac{\mathrm{d}^D k_E}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k_E\cdot(x-y)}}{k_E^2 + m^2}$$

satisfies this condition because its Fourier representation in spatial momentum is

$$C(\tau, \mathbf{x}) = \int \frac{\mathrm{d}^{D-1}\mathbf{p}}{(2\pi)^{D-1}} \frac{\mathrm{e}^{\mathrm{i}\mathbf{p}\cdot\mathbf{x}} \mathrm{e}^{-\omega_{\mathbf{p}}|\tau|}}{2\omega_{\mathbf{p}}}, \qquad \omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2},$$

so the reflected quadratic form is an integral of absolute squares with positive weight. This computation is the Euclidean shadow of the positive one-particle Hilbert space of the free scalar field.

In this free example, covariance gives the Schwinger functions, reflection positivity gives the inner product, and the positive-frequency kernel gives the one-particle space. For an interacting Euclidean measure the same sentences become obligations: one must verify the full OS positivity, covariance, regularity, and growth requirements before invoking reconstruction.

12.9 Relation To Wick Rotation

For time-ordered Green functions in perturbation theory, Wick rotation was used earlier as a contour and boundary-value operation. OS reconstruction is the nonperturbative counterpart at the level of complete Euclidean correlation data. It supplies the Hilbert space, Hamiltonian, vacuum, and Lorentzian Wightman functions from Euclidean data satisfying precise positivity and growth conditions.

The framework identifies the Euclidean information that reconstructs a Lorentzian theory: a full compatible hierarchy of Schwinger distributions together with covariance, reflection positivity, growth, regularity, and normalization conditions.

Chapter 13

Kontsevich–Segal Functorial Quantum Field Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

13.1 Functorial Data and the Path-Integral Problem

The Wightman, Osterwalder–Schrader, and Haag–Kastler frameworks encode different shadows of locality. Wightman fields encode distributional Lorentzian local operators on a Hilbert space; OS data encode Euclidean reflection-positive correlation distributions; Haag–Kastler nets encode bounded observable algebras localized in spacetime regions. The Kontsevich–Segal viewpoint begins from the geometric operation that a path integral is expected to satisfy before any choice of coordinates on fields is made: cut a background along hypersurfaces, assign state spaces to the cuts, and recover the amplitude of the glued background by composing the amplitudes of the pieces. The source trace for this chapter's functorial and allowability language is Segal's conformal-field-theory sewing framework and the Kontsevich–Segal Wick-rotation criterion [35, 36].

Let D be the spacetime dimension. The intended datum is a symmetric monoidal assignment

$$Z : \mathrm{Bord}_D^{\mathrm{adm}} \longrightarrow \mathrm{TVect}_{\mathbb{C}},$$

where $\mathrm{Bord}_D^{\mathrm{adm}}$ is a category of $(D-1)$-manifolds and D-dimensional bordisms with admissible complex metrics and auxiliary background fields, and $\mathrm{TVect}_{\mathbb{C}}$ is a category of complex topological vector spaces and continuous linear maps. The target cannot be restricted in advance to Hilbert spaces. Hilbert spaces arise from a reflection-positive pairing on suitable real slices; before that pairing is imposed and completed, the natural state spaces are locally convex spaces or distribution spaces.

The word "admissible" carries the physical content of positive energy. The complex metric is allowed to move inside a domain containing Riemannian metrics and having Lorentzian metrics on its boundary. Analytic dependence inside this domain replaces the upper-half-plane analyticity of $\exp(\mathrm{i}tH)$ in quantum mechanics. The objective is therefore sharper than a generic functorial field theory: the functor must be holomorphic in admissible complex metrics, compatible with sewing, and equipped with a reflection structure whose Lorentzian boundary values produce unitary evolution and microcausal fields when the required analytic hypotheses hold.

13.2 The One-Dimensional Semigroup Model

The smallest model of the Kontsevich–Segal idea is ordinary positive-energy quantum mechanics written without choosing a preferred real time coordinate. Let $\mathcal{H}$ be a Hilbert space and let H be a self-adjoint operator bounded below. After subtracting the infimum of the spectrum, assume $H \geq 0$. The Euclidean time evolution is

$$T(\tau) = \exp(-\tau H), \qquad \tau > 0.$$

It is a strongly continuous semigroup of self-adjoint contractions. The Lorentzian evolution is the boundary value

$$U(t) = \lim_{\epsilon \downarrow 0} T(\epsilon + \mathrm{i}t) = \exp(-\mathrm{i}tH), \qquad t \in \mathbb{R},$$

in the strong operator topology. Thus positive energy is equivalently the existence of a bounded holomorphic semigroup

$$z \longmapsto T(z) = \exp(-zH), \qquad \operatorname{Re} z > 0,$$

whose boundary values on the imaginary axis are unitary.

Proposition 13.1 (Holomorphic semigroup and positive generator)**.** *Let $T(z)$, $\operatorname{Re} z > 0$, be a holomorphic family of bounded operators on a Hilbert space $\mathcal{H}$. Assume:*

1. *$T(z + w) = T(z)T(w)$ for $\operatorname{Re} z, \operatorname{Re} w > 0$;*
2. *$T(\tau)$ is self-adjoint and positive for every real $\tau > 0$;*
3. *$T(\tau)$ is strongly continuous in τ and $\|T(\tau)\| \leq 1$;*
4. *$T(\tau) \to \mathbf{1}$ strongly as $\tau \downarrow 0$.*

Then there is a unique self-adjoint operator $H \geq 0$ such that

$$T(\tau) = \exp(-\tau H), \qquad \tau > 0.$$

If the non-tangential strong boundary values $\lim_{\epsilon \downarrow 0} T(\epsilon + \mathrm{i}t)$ exist for all t, they are the unitary group $\exp(-\mathrm{i}tH)$.

Proof. The real-time family $\tau \mapsto T(\tau)$ is a strongly continuous contraction semigroup, so the Hille–Yosida theorem gives a closed densely defined generator $-H$ with $T(\tau) = \exp(-\tau H)$. Because each $T(\tau)$ is self-adjoint, the generator is self-adjoint. Positivity of $T(\tau)$ and the spectral mapping theorem give $\operatorname{spec}(T(\tau)) \subset [0,1]$, hence $\operatorname{spec}(H) \subset [0,\infty)$. Equivalently, the spectral calculus gives $T(\tau) = \int e^{-\tau\lambda}\, dE_H(\lambda)$ with $\lambda \geq 0$. The holomorphic semigroup is then uniquely determined by this spectral formula:

$$T(z) = \int e^{-z\lambda}\, dE_H(\lambda), \qquad \operatorname{Re} z > 0.$$

Letting $z = \epsilon + \mathrm{i}t$ and using dominated convergence in the spectral integral gives the strong boundary value

$$\lim_{\epsilon \downarrow 0} T(\epsilon + \mathrm{i}t) = \int e^{-\mathrm{i}t\lambda}\, dE_H(\lambda) = e^{-\mathrm{i}tH},$$

which is unitary by the spectral theorem. □

In the functorial language, an interval of complex length z with $\operatorname{Re} z > 0$ is a one-dimensional admissible bordism. Sewing intervals gives the semigroup law. Reflection positivity gives positivity of the real-length operators. The Lorentzian line is not an interior part of the domain; it is a boundary where the holomorphic contraction semigroup has unitary boundary values. The higher-dimensional Kontsevich–Segal condition is designed so that every time-thickness and every local quadratic field mode has this same analytic structure.

13.3 Segal Sewing in Two-Dimensional CFT

The two-dimensional conformal case is the prototype. Let $\mathrm{Bord}_2^{\mathrm{conf}}$ denote the category whose objects are finite disjoint unions of analytically parametrized oriented circles and whose morphisms are compact Riemann surfaces with incoming and outgoing parametrized boundary components. Composition is sewing along boundary circles with inverse parametrizations. A Segal CFT assigns a topological vector space $\mathcal{H}_S$ to each object S and a continuous linear map

$$Z(\Sigma) : \mathcal{H}_{S_{\mathrm{in}}} \longrightarrow \mathcal{H}_{S_{\mathrm{out}}}$$

to a Riemann surface $\Sigma : S_{\mathrm{in}} \to S_{\mathrm{out}}$, with

$$Z(\Sigma_2 \circ \Sigma_1) = Z(\Sigma_2) Z(\Sigma_1), \qquad Z(\Sigma \sqcup \Sigma') = Z(\Sigma) \otimes Z(\Sigma').$$

The sewing map is required to be continuous or holomorphic with respect to the relevant moduli of surfaces and boundary parametrizations. Reflection of a Riemann surface across a parametrized boundary circle gives an antilinear involution on the state space, and reflection positivity requires the doubled surface to define a positive semidefinite pairing.

The operator product in a Segal CFT is a sewing statement. A pair of punctures contained in a disk determines a state on the boundary circle of that disk. Expanding this state in the spectral decomposition of the annulus semigroup gives the local fields that may replace the pair of insertions when the disk is sewn into a larger surface. Thus convergence of the OPE is not an extra mnemonic; it is the convergence of a state-vector expansion in the boundary state space chosen by the sewing theory.

13.4 Allowable Complex Metrics

Let V be a real vector space of dimension D. A complex metric on V is a nondegenerate complex-valued symmetric bilinear form

$$g \in \mathrm{Sym}^2(V^*) \otimes_{\mathbb{R}} \mathbb{C}.$$

Let $|\wedge^D V^*|$ be the real density line. A choice of square root of $\det g$ defines a complex density

$$\mathrm{vol}_g = (\det g)^{1/2} |dy^1 \cdots dy^D| \in |\wedge^D V^*| \otimes_{\mathbb{R}} \mathbb{C}$$

in a basis $y^1, \ldots, y^D$. The sign of the square root is fixed by requiring the 0-form positivity condition below.

Definition 13.2 (Kontsevich–Segal allowability)**.** A complex metric g on V is allowable when, for every $q = 0, \ldots, D$, the quadratic map

$$Q_{q,g} : \wedge^q V^* \longrightarrow |\wedge^D V^*| \otimes_{\mathbb{R}} \mathbb{C}, \qquad \alpha \longmapsto \alpha \wedge *_g \alpha$$

has positive real part on every nonzero real q-form α. For $q = 0$, this says that $\mathrm{Re}\,\mathrm{vol}_g$ is a positive real density.

For a q-form gauge field strength F, the Euclidean quadratic action is

$$I_q(F; g) = \frac{1}{2q!} \int_M \sqrt{\det g}\, g^{i_1 j_1} \cdots g^{i_q j_q} F_{i_1 \cdots i_q} F_{j_1 \cdots j_q}\, \mathrm{d}^D x.$$

Definition 13.2 is precisely the pointwise condition that the real part of this quadratic form be positive for every real q-form field strength, simultaneously for all q. Imposing all degrees is essential because a field theory may contain scalars, gauge fields, higher-form fields, dual fields, and quantized flux sectors whose integration cycles cannot all be repaired by a degree-dependent phase rotation.

Theorem 13.3 (Argument criterion for allowability)**.** *Let g be a complex metric on V. It is allowable if and only if there is a real basis in which*

$$g = \sum_{a=1}^{D} \lambda_a (dy^a)^2, \qquad \lambda_a \in \mathbb{C} \setminus (-\infty, 0],$$

and, with the principal argument $\theta_a = \mathrm{Arg}\,\lambda_a \in (-\pi, \pi)$,

$$\sum_{a=1}^{D} |\theta_a| < \pi. \tag{13.1}$$

Proof. Assume first that g is allowable. The $q = 0$ condition chooses the branch of $(\det g)^{1/2}$ with positive real part. The $q = 1$ condition states that the complex symmetric matrix

$$W^{ij} = (\det g)^{1/2} g^{ij}$$

has positive-definite real part. Write $W = A + \mathrm{i}B$ with A, B real symmetric and $A > 0$. A real change of basis puts A in the identity form. A further orthogonal change of basis diagonalizes B.

Hence W is diagonal in a real basis, so W^{-1} and then g are diagonal in the same basis. Write the diagonal entries as λ_a.

For a subset $S \subset \{1, \dots, D\}$, the real basis q-form $\mathrm{d}y^S = \wedge_{a\in S}\mathrm{d}y^a$ has

$$\mathrm{d}y^S \wedge *_g \mathrm{d}y^S = A_S\, |dy^1 \cdots dy^D|, \qquad A_S = (\det g)^{1/2} \prod_{a\in S} \lambda_a^{-1}. \tag{13.2}$$

The q-form positivity conditions are therefore equivalent to $\operatorname{Re} A_S > 0$ for all subsets S. If $\theta_a = \operatorname{Arg}\lambda_a$, then

$$\operatorname{Arg} A_S = \frac{1}{2}\sum_{a=1}^{D}\theta_a - \sum_{a\in S}\theta_a = \frac{1}{2}\left(\sum_{a\notin S}\theta_a - \sum_{a\in S}\theta_a\right).$$

Let $P = \{a : \theta_a \geq 0\}$. Positivity for $S = P$ gives

$$-\frac{\pi}{2} < -\frac{1}{2}\sum_{a=1}^{D}|\theta_a| < \frac{\pi}{2},$$

and hence (13.1). Conversely, if (13.1) holds, then for every subset S,

$$|\operatorname{Arg} A_S| \leq \frac{1}{2}\sum_{a=1}^{D}|\theta_a| < \frac{\pi}{2}.$$

Thus every A_S has positive real part. Since the form coefficients are diagonal in the real basis, every nonzero real q-form has $\operatorname{Re} Q_{q,g} > 0$. This is Definition 13.2. □

Example 13.4 (Euclidean, Lorentzian, and multi-time signatures)**.** A positive Riemannian metric has all $\theta_a = 0$, hence is allowable. A Lorentzian metric with diagonal entries $(-1, 1, \dots, 1)$ has one argument equal to π, so it lies on the boundary of the allowable domain. The regularized metrics

$$-(1 - \mathrm{i}\epsilon)(dy^1)^2 + \sum_{a=2}^{D}(dy^a)^2, \qquad -(1 + \mathrm{i}\epsilon)(dy^1)^2 + \sum_{a=2}^{D}(dy^a)^2, \qquad \epsilon > 0,$$

lie in the two adjacent allowable components approaching that Lorentzian boundary point. A real metric with two negative directions has two arguments equal to π and does not lie on this boundary. The criterion therefore distinguishes Lorentzian signature from signatures with more than one time direction.

13.5 The Admissible Bordism Category

Definition 13.5 (Admissible complex-metric bordism)**.** An admissible D-bordism is a smooth compact D-manifold M with boundary decomposed into incoming and outgoing collars, together with:

$$g \in \Gamma(\mathrm{Sym}^2\, T^*M \otimes \mathbb{C}), \qquad \text{background bundles and sources,}$$

where g_x is allowable for every $x \in M$. Boundary collars are part of the datum so that gluing is defined by identifying collars with matching induced complex metrics and background fields. The category is symmetric monoidal under disjoint union.

The collar is not decorative. Without a collar, equality of boundary metrics does not determine a smooth glued metric; with a collar, composition is ordinary gluing of germs. A morphism is therefore an equivalence class of collared backgrounds, where two representatives are identified only when there is a collar-preserving diffeomorphism respecting the complex metric and all background fields. This convention keeps sewing as an exact categorical composition, not as a limiting operation.

The target category must remember topology. A useful minimal target is the symmetric monoidal category of complete locally convex complex vector spaces with continuous linear maps and completed projective tensor product $\widehat{\otimes}_\pi$. In examples with trace or nuclear sewing estimates, one may restrict to nuclear spaces; in Hilbert-space boundary sectors one passes only after reflection positivity and null-state quotient. Thus the symbols $E(Y_1)\widehat{\otimes}_\pi E(Y_2)$ and $E(Y_1 \sqcup Y_2)$ are identified as part of the monoidal datum, not by an algebraic tensor product completion chosen after the fact.

The state space assigned to a boundary germ Y will be denoted

$$E(Y).$$

The notation emphasizes that Y is a germ of a hypersurface together with its collar metric and background fields, rather than only the underlying $(D-1)$-manifold. For a bordism $M : Y_0 \to Y_1$, the amplitude is a continuous linear map

$$Z(M) : E(Y_0) \longrightarrow E(Y_1).$$

For a closed M, $Z(M) \in \mathbb{C}$ is the partition function. For marked points or extended insertions, $E(Y)$ is replaced by a multilinear observable bundle over the configuration space of insertions.

Definition 13.6 (K-S functorial QFT datum)**.** A Kontsevich–Segal functorial QFT on a specified class of backgrounds consists of:

1. a symmetric monoidal functor Z from admissible complex bordism categories to complex topological vector spaces;
2. holomorphic dependence of $Z(M, g)$ on the admissible complex metric and on complexified sources in local coordinate charts on the space of background data;
3. collar locality: if $M = M_2 \circ_Y M_1$, then
$$Z(M) = Z(M_2)Z(M_1)$$
as a continuous map after the matching state-space identifications along Y;
4. a reflection operation $Y \mapsto \bar{Y}$ and $M \mapsto \bar{M}$ compatible with complex conjugation of metric and sources;
5. reflection positivity on real Riemannian reflection doubles, as stated below.

13.6 Reflection Positivity and Hilbert Spaces

Let Y be a real Riemannian boundary germ fixed by an orientation-reversing reflection ρ. Suppose M_+ is a bordism from the empty object to Y, and let $\bar{M}_+$ be its reflected conjugate from Y to the empty object. The double

$$\bar{M}_+ \circ_Y M_+$$

is a closed Riemannian background. If $u, v \in E(Y)$ arise from states created by such half-bordisms with insertions, define

$$\langle u, v\rangle_Y = Z(\bar{M}_u \circ_Y M_v).$$

Reflection positivity requires this sesquilinear form to be positive semidefinite on the subspace generated by positive-side states. The physical Hilbert space associated to Y is then

$$\mathcal{H}_Y = \overline{E_+(Y)/\{u : \langle u, u\rangle_Y = 0\}}, \tag{13.3}$$

where the bar denotes Hilbert completion.

Theorem 13.7 (Positive-energy cylinder from K-S reflection positivity)**.** *Let Y be a real Riemannian boundary germ for which the reflection-positive construction above gives a Hilbert space $\mathcal{H}_Y$. Suppose the K-S functor assigns to each real positive cylinder*

$$C_\tau = [0, \tau] \times Y, \qquad \tau > 0,$$

a continuous operator $T_Y(\tau)$ on the positive-side state space and that:

1. *sewing gives $T_Y(\tau + \sigma) = T_Y(\tau)T_Y(\sigma)$;*
2. *reflection of the cylinder gives $\langle u, T_Y(\tau)v\rangle_Y = \langle T_Y(\tau)u, v\rangle_Y$;*
3. *the doubled cylinder form gives $\langle u, T_Y(\tau)u\rangle_Y \geq 0$;*
4. *$T_Y(\tau)$ descends to a strongly continuous contraction semigroup on $\mathcal{H}_Y$, after the vacuum-energy normalization of the cylinder.*

Then there is a unique self-adjoint $H_Y \geq 0$ on $\mathcal{H}_Y$ such that

$$T_Y(\tau) = e^{-\tau H_Y}.$$

If the functor is holomorphic for the corresponding admissible complex cylinders and has strong boundary values at imaginary cylinder length, those boundary values are the unitary Lorentzian propagators

$$U_Y(t) = e^{-\mathrm{i}tH_Y}.$$

Proof. Let $N_Y = \{u : \langle u, u\rangle_Y = 0\}$. Since the doubled form is positive semidefinite, Cauchy–Schwarz gives $u \in N_Y$ precisely when $\langle u, w\rangle_Y = 0$ for every positive-side state w. We first check that the cylinder operator is compatible with quotienting by this null space. If $u \in N_Y$, then reflection and sewing give

$$\|T_Y(\tau)u\|_Y^2 = \langle T_Y(\tau)u, T_Y(\tau)u\rangle_Y = \langle u, T_Y(2\tau)u\rangle_Y = 0.$$

Thus $T_Y(\tau)N_Y \subset N_Y$, and $T_Y(\tau)$ descends to $E_+(Y)/N_Y$. The contraction hypothesis is part of the analytic input: after the vacuum-energy normalization,

$$\|T_Y(\tau)[u]\|_{\mathcal{H}_Y} \leq \|[u]\|_{\mathcal{H}_Y},$$

so the descended operator extends uniquely to a bounded contraction on the Hilbert completion. Sewing descends to the semigroup law $T_Y(\tau + \sigma) = T_Y(\tau)T_Y(\sigma)$. Reflection gives symmetry on the dense quotient:

$$\langle [u], T_Y(\tau)[v]\rangle_Y = \langle T_Y(\tau)[u], [v]\rangle_Y.$$

Boundedness then extends this symmetry to all of $\mathcal{H}_Y$, so $T_Y(\tau)$ is self-adjoint. Doubled-cylinder positivity gives $\langle\psi, T_Y(\tau)\psi\rangle \geq 0$ first on the dense quotient and then by continuity on the Hilbert space.

We now apply the Hilbert-space semigroup statement isolated in Proposition 13.1. A strongly continuous self-adjoint contraction semigroup with positive operators has a unique self-adjoint generator H_Y satisfying

$$T_Y(\tau) = e^{-\tau H_Y}.$$

The spectrum of H_Y lies in $[0, \infty)$, because otherwise $\|e^{-\tau H_Y}\| \leq 1$ would fail for $\tau > 0$. If the K-S functor is holomorphic on the corresponding complex cylinder semigroup, spectral calculus gives the strong boundary values

$$\lim_{\epsilon \downarrow 0} e^{-(\epsilon+\mathrm{i}t)H_Y} = e^{-\mathrm{i}tH_Y}.$$

These boundary values are unitary since $|e^{-\mathrm{i}t\lambda}| = 1$ for $\lambda \in \mathrm{spec}(H_Y)$. □

This theorem is the precise point at which functorial gluing becomes positive energy. It also explains why reflection positivity alone is insufficient: one still needs domain control for the quotient, strong continuity of the cylinder semigroup, and the boundedness or vacuum-energy normalization needed to apply the semigroup theorem. These analytic requirements are part of the K-S construction problem in every infinite-dimensional QFT example.

In a real-analytic globally hyperbolic Lorentzian spacetime, a Lorentzian time-slab appears as a boundary limit of admissible complex slabs. Under the holomorphic and boundedness hypotheses needed to take this limit, the K-S construction gives unitary propagation between the Hilbert spaces of time-symmetric hypersurfaces. Local field operators on a time slice are obtained as limits of insertion bordisms pinched to the slice; operators at spacelike separated points commute because both insertions can be represented on one common slice and the Euclidean sewing map is symmetric under exchanging their order.

13.7 Comparison with OS, Wightman, and Local Nets

Flat-space comparison with OS Schwinger functions. Assume a K-S functorial QFT is defined on Euclidean $\mathbb{R}^D$ with translation-invariant vacuum state, local observable bundles $\mathcal{O}_x$, and tempered boundary values for point insertions. For local observables $A_1, \ldots, A_n$, define

$$S_n(A_1, x_1; \ldots; A_n, x_n) = Z(\mathbb{R}^D; A_1(x_1), \ldots, A_n(x_n))$$

for distinct Euclidean points, extended as a distribution by the assumed tempered boundary value. Then Euclidean covariance, symmetry at separated points, reflection positivity, and clustering of the Schwinger functions follow from the functorial axioms plus the corresponding covariance and vacuum uniqueness assumptions on Z.

Euclidean covariance is functoriality under background isomorphisms. Symmetry follows by moving separated point insertions inside a Riemannian background without crossing. The configuration space with diagonals removed records the possible permutation monodromy; ordinary bosonic local observables have trivial monodromy. Reflection positivity is the positivity of the doubled half-space bordism. Clustering is not a pure sewing identity: it is the assertion that the translation semigroup on a long cylinder has a unique zero-energy vector and that the remaining spectrum is separated or decays in the topology used for the state spaces. Under that spectral hypothesis, inserting a long cylinder between two groups of observables projects to the vacuum state, giving the OS cluster property.

Construction 13.8 (K-S Lorentzian boundary problem)**.** Let $M_L : \Sigma_0 \to \Sigma_1$ be a real-analytic globally hyperbolic Lorentzian bordism with real-analytic Cauchy collars. A K-S Lorentzian boundary construction for M_L consists of the following data and estimates.

(i) An admissible family of complex metrics g_ϵ, $\epsilon > 0$, on the same collared bordism, with g_ϵ in the K-S allowable domain pointwise and $g_\epsilon \to g_L$ as a boundary value of complex metrics on compact subsets of the collars and interior.

(ii) Hilbert spaces $\mathcal{H}_{\Sigma_i}$ obtained from the reflection positive pairing on the time-symmetric boundary germs Σ_i, including the null-space quotient and completion.

(iii) Continuous amplitude maps
$$Z(M, g_\epsilon) : \mathcal{H}^{\mathrm{core}}_{\Sigma_0} \longrightarrow \mathcal{H}_{\Sigma_1}$$
on a dense reflection-positive core, together with a topology in which the limit
$$U(M_L) = \lim_{\epsilon \downarrow 0} Z(M, g_\epsilon)$$
exists on that core and extends to a bounded operator $\mathcal{H}_{\Sigma_0} \to \mathcal{H}_{\Sigma_1}$.

(iv) Independence of $U(M_L)$ from the admissible approach g_ϵ to g_L, within a specified approach class.

(v) Compatibility with sewing: if $M_L = M_{2,L} \circ_\Sigma M_{1,L}$, then the boundary limits obey
$$U(M_L) = U(M_{2,L})U(M_{1,L})$$
on the common dense domain and hence on Hilbert completions.

(vi) Reflection compatibility:
$$U(\bar{M}_L) = U(M_L)^*.$$
For a slab whose reflected reverse is the inverse Lorentzian evolution, this identity and sewing imply that $U(M_L)$ is unitary.

(vii) Local insertion limits. If two insertion germs have spacelike separated Lorentzian supports and can be represented on a common admissible complex slice without crossing singular diagonals, then the two boundary operators commute, with the Koszul sign for graded fields.

Construction 13.8 is deliberately not stated as a theorem about all K-S functors. The finite-dimensional metric allowability criterion is Theorem 13.3; the product-cylinder positive-energy statement is Theorem 13.7. The passage from an arbitrary admissible complex slab to a Lorentzian globally hyperbolic slab requires additional analysis: uniform bounds for the amplitudes, existence and approach-independence of boundary values, stability of the null-space quotient under the limiting maps, and sewing estimates strong enough to pass through Hilbert completion.

The mechanism is nevertheless precise. Reflection positivity supplies the Hilbert spaces on time-symmetric cuts. Holomorphic dependence on admissible complex metrics gives a candidate family $Z(M, g_\epsilon)$. If the limit exists and is independent of the approach, reflection gives
$$U(M_L)^* = U(\bar{M}_L).$$

When $\bar{M}_L \circ M_L$ and $M_L \circ \bar{M}_L$ are the identity cylinders after Lorentzian boundary limiting, sewing gives

$$U(M_L)^* U(M_L) = \mathbf{1}_{\mathcal{H}_{\Sigma_0}}, \qquad U(M_L) U(M_L)^* = \mathbf{1}_{\mathcal{H}_{\Sigma_1}},$$

so the boundary evolution is unitary. For insertions, spacelike commutativity is the same sewing statement applied to two punctures: if the punctures can be exchanged inside the admissible complex configuration space without crossing the diagonal, functoriality identifies the two sewn amplitudes. Thus the open analytic work is not the algebraic conclusion once the limits exist; it is proving the boundary limits and their stability for a given infinite-dimensional QFT.

Combining this flat-space compatibility statement with OS reconstruction gives the route

$$\text{K-S functorial data} \longrightarrow \text{OS Schwinger functions} \longrightarrow \text{Wightman fields.}$$

After Wightman fields have been reconstructed, local nets are obtained by taking von Neumann algebras generated by bounded functions of smeared fields, provided the self-adjointness and strong-locality hypotheses needed for those bounded functions are verified. Thus the arrow from K-S to Haag–Kastler is a composite arrow with analytic and operator-domain assumptions at each stage.

The reverse direction requires additional structure. OS Schwinger functions on flat space do not by themselves assign state spaces to arbitrary hypersurface germs, nor do they define amplitudes for all admissible complex bordisms. A reconstruction of a K-S functor from OS data therefore requires sewing estimates, collar factorization, and control of metric dependence. Those hypotheses are available in selected two-dimensional constructions and topological theories; they are open in physically central higher-dimensional interacting theories.

13.8 OPE from Sewing

Let $B_r(x) \subset M$ be a small ball whose boundary sphere is $S_r(x)$. A collection of local insertions inside $B_r(x)$ defines, by the K-S amplitude of the punctured ball, a vector

$$\Psi_{A_1,\ldots,A_k}(r; \xi_1, \ldots, \xi_k) \in E(S_r(x)),$$

where ξ_i are coordinates of the insertion points relative to x. Suppose the annular evolution identifying $E(S_r(x))$ with $E(S_1(x))$ has a discrete generalized eigenspace decomposition

$$E(S_1(x)) = \widehat{\bigoplus_{\Delta,s}} E_{\Delta,s}$$

with nuclear convergence for the expansion of punctured-ball states. Choose a basis of local operators $\mathcal{O}_{\Delta,s,a}$ corresponding to the dual basis of these eigenspaces. Then sewing the punctured ball into the complement $M \setminus B_r(x)$ gives

$$A_1(x+\xi_1) \cdots A_k(x+\xi_k) \sim \sum_{\Delta,s,a} C^{\Delta,s,a}_{A_1 \ldots A_k}(\xi_1, \ldots, \xi_k)\, \mathcal{O}_{\Delta,s,a}(x), \tag{13.4}$$

with convergence in the topology in which the boundary-state expansion converges. In a CFT, annular evolution is generated by dilations, so the coefficients have homogeneous leading behavior determined by scaling dimensions and spin. In a massive theory the same construction gives a local short-distance expansion when the small-ball state admits the corresponding asymptotic spectral expansion.

This derivation identifies the analytic hypotheses behind OPE convergence: state spaces on spheres, trace or nuclear bounds for annuli, spectral control of the annular semigroup, and continuity of sewing. Without these hypotheses, (13.4) is a formal expansion rather than a theorem.

13.9 Stoyanovsky-Style Wick Rotation

Stoyanovsky's axiomatic Wick-rotation program may be viewed as a hypersurface-evolution formulation of the same problem. The basic object is a bundle of state spaces over a space of embedded hypersurfaces in a complexified spacetime, equipped with flat evolution maps for deformations of the hypersurface. Local fields act by insertion when the hypersurface passes the insertion point. Flatness expresses independence of the chosen interpolating family of hypersurfaces; positivity and boundary-value conditions select the Lorentzian real slice.

The K-S metric formulation supplies a concrete admissible domain for those hypersurface deformations. Rather than complexifying the time coordinate alone, it complexifies the metric and imposes (13.1) pointwise. This gives a finite-dimensional linear-algebra condition at each tangent space and a gluing category of admissible metric bordisms. The two viewpoints should agree when both are defined: hypersurface evolution is the local expression of the functor on thin bordisms.

13.10 Examples and Construction Status

Topological theories are the cleanest functorial examples. If Z is independent of the metric and satisfies Atiyah–Segal sewing, then it is a K-S-type theory restricted to the topological subcategory: the admissible metric dependence is constant. Finite gauge theory, Dijkgraaf–Witten theory, and Reshetikhin–Turaev theories provide theorem-level examples in this topological sector, with their anomaly and extended-boundary data recorded in Volume VIII.

Two-dimensional CFT supplies the deepest non-topological testing ground. Segal's sewing axioms have been verified in important representation-theoretic and probabilistic settings, including free-fermionic and loop-group examples in the operator-algebraic literature and Liouville-theory sewing results in the probabilistic conformal-bootstrap program. Each such result is a two-dimensional Segal-CFT theorem with its own topology on state spaces and moduli dependence. Passing from these theorems to a general-dimensional K-S construction is a separate problem.

Two-dimensional Yang–Mills has exact heat-kernel formulas on surfaces and a rigorous measure-theoretic construction in several formulations. These formulas obey cutting identities at the level of class functions and holonomy kernels. A full continuum Segal functor with all state spaces, collar topologies, and boundary field insertions specified is a further structure that must be supplied before calling it a completed K-S example.

Free Gaussian theory illustrates both what is clear and what remains unfinished. Let A be a positive self-adjoint operator on a Hilbert space of spatial modes, and first replace it by a finite-dimensional positive matrix. The Euclidean cylinder kernel is the harmonic-oscillator Mehler kernel

$$K_\tau(\varphi_0,\varphi_1) = \det\left(\frac{A^{1/2}}{2\pi\sinh(\tau A^{1/2})}\right)^{1/2} \exp\left[-\frac{1}{2}\begin{pmatrix}\varphi_0\\ \varphi_1\end{pmatrix}^T B_\tau \begin{pmatrix}\varphi_0\\ \varphi_1\end{pmatrix}\right],$$

where

$$B_\tau = \begin{pmatrix} A^{1/2}\coth(\tau A^{1/2}) & -A^{1/2}\operatorname{csch}(\tau A^{1/2}) \\ -A^{1/2}\operatorname{csch}(\tau A^{1/2}) & A^{1/2}\coth(\tau A^{1/2}) \end{pmatrix}.$$

For $\operatorname{Re}\tau > 0$ this kernel is holomorphic and composes by Gaussian integration. Reflection positivity follows because the same kernel is the integral kernel of $e^{-\tau H_A}$ for a positive harmonic-oscillator Hamiltonian H_A. Thus every finite collection of modes is a genuine one-dimensional K-S semigroup.

The continuum free scalar in $D \geq 3$ is obtained formally by replacing the finite determinant and finite Gaussian integral by zeta, heat-kernel, or Fock-space constructions. A complete K-S construction still has to specify state spaces for arbitrary hypersurface germs, prove trace or nuclear sewing for admissible collars, control point-field insertions as distributions or hyperfunctions, prove reflection positivity for the chosen completions, and show holomorphic dependence on the admissible metric. The monograph therefore treats the free scalar in $D \geq 3$ as a principal construction problem rather than as an already completed K-S theorem.

13.11 Open Problems and Construction Targets

Open Problem 13.9 (Free fields as K-S functors). Construct the full K-S functor for the free massive scalar field in $D \geq 3$. The construction must specify hypersurface state spaces, Gaussian amplitudes for admissible collars, holomorphic metric dependence, reflection positivity, distributional field insertions, and sewing estimates.

Open Problem 13.10 ($P(\phi)_2$ and constructive scalar models). Starting from constructive OS data for $P(\phi)_2$, build state spaces on curves, prove collar factorization, and extend the Schwinger functions to a Segal/K-S functor on the relevant admissible two-dimensional bordism category.

Open Problem 13.11 (Liouville and nonrational CFT). Determine the exact relation between probabilistic Liouville correlators, the DOZZ/bootstrap construction, Segal sewing, and a complete functorial state-space assignment. The required output is a functor with specified boundary state spaces and sewing maps, not only closed-surface correlators.

Open Problem 13.12 (Two-dimensional Yang–Mills). Upgrade the heat-kernel and holonomy-cutting formulas of two-dimensional Yang–Mills to a continuum Segal functor with boundary Hilbert spaces, defect insertions, and compatibility with the admissible complex metric domain.

Open Problem 13.13 (Three-dimensional constructive models). For ϕ^4_3, Gross–Neveu type models, and related constructive theories, identify the additional estimates needed to pass from OS correlators to state spaces on arbitrary surfaces and to admissible complex-metric sewing.

Open Problem 13.14 (Four-dimensional gauge theory). Construct four-dimensional Yang–Mills with mass gap as a K-S functor. This is at least as hard as the corresponding OS, Wightman, and Haag–Kastler existence problems and additionally requires functorial gluing and holomorphic admissible-metric dependence.

Open Problem 13.15 (Fermions and gauge fields in K-S form). Develop K-S functorial constructions for fermionic and gauge theories in a regularization that preserves the relevant reflection positivity, anomaly bookkeeping, and sewing identities. For gauge theory the BV formulation of regularized path integrals and the lattice formulation are the currently available frameworks with a precise gauge-consistency condition.

Open Problem 13.16 (Extended K-S structures). Formulate and construct extended K-S theories assigning data to corners and higher-codimension strata, analogous to extended TQFT but with non-topological metric dependence and analytic sewing. The target should record defects, boundary conditions, and local operator categories.

Chapter 14

Källén–Lehmann Spectral Representation

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

14.1 Spectral Data of a Scalar Field

Definition 14.1 (Centered scalar two-point spectral datum)**.** Let $\mathcal{H}$ carry a strongly continuous unitary representation of spacetime translations with commuting self-adjoint generators P^μ. Let

$$E(\Delta)$$

be the joint projection-valued spectral measure of P^μ, where $\Delta \subset \mathbb{R}^D$ is Borel. Assume the spectrum condition

$$\mathrm{Spec}(P) \subset \overline{V}_+ = \{p \in \mathbb{R}^D : p^0 \geq 0,\ p^2 \leq 0\},$$

with $p^2 = \eta_{\mu\nu} p^\mu p^\nu$. Let Ω be a translation-invariant vacuum vector in the pure vacuum sector,

$$E(\{0\})\mathcal{H} = \mathbb{C}\Omega.$$

Let $\widehat{\phi}$ be a scalar operator-valued distribution. If

$$\langle \Omega | \widehat{\phi}(x) | \Omega \rangle$$

is nonzero, replace $\widehat{\phi}$ by $\widehat{\phi} - \langle\Omega|\widehat{\phi}|\Omega\rangle$. This removes the vacuum atom from the two-point function and keeps the connected spectral data: because the zero-momentum spectral projection is $|\Omega\rangle\langle\Omega|$, there is no remaining translation-invariant component after this subtraction. Point-field notation below is distributional notation; after smearing, the matrix elements are ordinary Hilbert-space pairings.

Proposition 14.2 (Two-point spectral measure of a scalar field)**.** *In the datum of Definition 14.1, the centered two-point distribution determines a unique positive tempered Borel measure ν_ϕ on $\mathbb{R}^D$ by*

$$W_2(f) = \int_{\mathbb{R}^D} \widetilde{f}(-p)\,\mathrm{d}\nu_\phi(p), \qquad f \in \mathcal{S}(\mathbb{M}^D), \tag{14.1}$$

where

$$\widetilde{f}(p) = \int \mathrm{d}^D x\, f(x)\mathrm{e}^{-\mathrm{i}p\cdot x}.$$

Equivalently, for smeared fields,

$$\langle\widehat{\phi}(f)\Omega, \widehat{\phi}(g)\Omega\rangle = \int_{\mathbb{R}^D} \overline{\widetilde{f}(p)}\,\widetilde{g}(p)\,\mathrm{d}\nu_\phi(p), \qquad f, g \in \mathcal{S}(\mathbb{M}^D). \tag{14.2}$$

The measure is supported in $\overline{V}_+$. If $\widehat{\phi}$ is Lorentz scalar and the vacuum is Lorentz invariant, ν_ϕ is invariant under $SO^+(1, D-1)$. In formal point-field notation one writes

$$\nu_\phi(\Delta) = \langle\Omega|\widehat{\phi}(0)E(\Delta)\widehat{\phi}(0)|\Omega\rangle,$$

but this line is shorthand for the smeared characterization above, not a literal assertion that $\widehat{\phi}(0)\Omega$ is a Hilbert-space vector. The Wightman two-point distribution is the Fourier–Stieltjes transform

$$W_2(x-y) = \int_{\overline{V}_+} \mathrm{e}^{\mathrm{i}p\cdot(x-y)}\,\mathrm{d}\nu_\phi(p).$$

Proof. The centered Wightman distribution

$$W_2(x-y) = \langle\Omega|\widehat{\phi}(x)\widehat{\phi}(y)|\Omega\rangle$$

is a tempered distribution of positive type. Indeed, for every $f \in \mathcal{S}(\mathbb{M}^D)$,

$$\int \mathrm{d}^D x\,\mathrm{d}^D y\,\overline{f(x)}\,f(y)\,W_2(x-y) = \|\widehat{\phi}(f)\Omega\|^2 \geq 0.$$

The Bochner–Schwartz theorem for positive-type tempered distributions therefore gives a unique positive tempered Borel measure ν_ϕ whose Fourier–Stieltjes transform is W_2, namely (14.1). Positivity is thus a Hilbert-space positivity statement before it is a statement about a measure.

The support is fixed by the spectrum condition. In the present translation convention,

$$U(a)^{-1}\widehat{\phi}(0)U(a) = \widehat{\phi}(a), \qquad U(a) = \mathrm{e}^{\mathrm{i}a^\mu P_\mu},$$

so $W_2(x)$ has spectral Fourier transform supported in $\mathrm{Spec}(P)$. Since $E(\mathbb{R}^D \setminus \overline{V}_+) = 0$, this support lies in $\overline{V}_+$. Equivalently, after smearing, the spectral theorem gives the same measure by the identity (14.2); the unsmeared expression $\langle\Omega|\widehat{\phi}(0)E(\Delta)\widehat{\phi}(0)|\Omega\rangle$ is the customary notation for the corresponding spectral-measure matrix element.

Finally suppose the field is a scalar and the vacuum is Lorentz invariant. Then $W_2(\Lambda x) = W_2(x)$ for every $\Lambda \in SO^+(1, D-1)$. Applying this equality to test functions and using uniqueness in the Bochner–Schwartz representation gives $\nu_\phi(\Lambda\Delta) = \nu_\phi(\Delta)$ for every Borel set Δ. $\square$

Equivalently, if one writes the same statement in generalized energy-momentum eigenvector notation, with

$$P^\mu|\alpha\rangle = p_\alpha^\mu|\alpha\rangle,$$

then

$$W_2(x-y) = \int \mathrm{d}\alpha\, \mathrm{e}^{\mathrm{i}p_\alpha\cdot(x-y)} \left|\langle\alpha|\widehat{\phi}(0)|\Omega\rangle\right|^2.$$

This formula is only notation for the spectral measure above; the measure $d\nu_\phi$ is the invariant object.

14.2 Invariant-Mass Decomposition

For a scalar field and invariant vacuum, ν_ϕ is invariant under the proper orthochronous Lorentz group. The Lorentz-invariant parameter on the forward cone is the invariant mass squared

$$\mu^2 = -p^2 \geq 0.$$

Let

$$\pi : \overline{V}_+ \to [0,\infty), \qquad \pi(p) = -p^2,$$

which is a Borel map between standard Borel spaces. The spectral density measure below is a quotient measure for the Lorentz orbits of this map, not the literal pushforward $\pi_*\nu_\phi$: a single massive shell has infinite Lorentz-invariant volume, so the raw pushforward would assign infinite mass to an isolated one-particle shell. The normalization of the invariant measure on each shell must be fixed first.

Proposition 14.3 (Invariant mass-shell disintegration)**.** *Let ν be a positive locally finite Borel measure on $\overline{V}_+$ that is invariant under $SO^+(1,D-1)$. Assume that ν has no atom at $p=0$. Then there is a positive Borel measure ρ on $[0,\infty)$ such that for every $F \in C_c(\overline{V}_+)$,*

$$\int_{\overline{V}_+} F(p)\,\mathrm{d}\nu(p) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2) \int \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\,\theta(p^0)\delta(p^2+\mu^2)F(p).$$

The measure ρ is the spectral density measure after the normalization of the invariant measure on each shell has been fixed.

Proof. Let

$$G = SO^+(1,D-1), \qquad X = \overline{V}_+ \setminus \{0\}.$$

The excluded point has zero ν-measure by hypothesis, so it may be removed throughout the proof. The space X is locally compact, second countable, σ-compact, and standard Borel. The group G is a locally compact second countable Lie group, and its action on X is continuous.

We first verify the regular-orbit hypotheses in the Mackey–Effros decomposition theorem. The G-orbits in X are

$$\Sigma_s^+ = \{p \in \overline{V}_+ : -p^2 = s\}, \qquad s > 0,$$

together with the punctured future light cone

$$\Sigma_0^{+\times} = \{p \in \overline{V}_+ : p^2 = 0,\ p \neq 0\}.$$

For $s > 0$, a representative is $k_s = (\sqrt{s}, 0, \ldots, 0)$, and the stabilizer is the compact group $SO(D-1)$. For the null orbit, a representative is $\ell = (1, 1, 0, \ldots, 0)$, and the stabilizer is the Euclidean group

$$ISO(D-2) = \mathbb{R}^{D-2} \rtimes SO(D-2)$$

embedded as the usual massless little group. These stabilizers are closed subgroups of G, and the orbit maps identify the orbits with the homogeneous spaces

$$\Sigma_s^+ \simeq G/SO(D-1), \qquad \Sigma_0^{+\times} \simeq G/ISO(D-2).$$

The orbit space is the standard Borel space $[0, \infty)$, with orbit map $\pi(p) = -p^2$, after the origin has been removed from the fiber over 0. A Borel cross-section is

$$c(0) = \ell, \qquad c(s) = k_s \quad (s > 0).$$

Thus the action is smooth in Mackey's sense: the orbit equivalence relation is regularly embedded, the quotient is standard Borel, and a Borel section exists. Since G, $SO(D-1)$, and $ISO(D-2)$ are unimodular, each homogeneous orbit carries a nonzero G-invariant Radon measure, unique up to a positive scalar. The Mackey–Effros disintegration theorem for smooth Borel actions of locally compact second countable groups therefore applies to the σ-finite invariant Radon measure $\nu|_X$.

Consequently there are conditional invariant Radon measures ν_s, supported on the orbit $\pi^{-1}(s) \cap X$, and a positive Borel measure λ on $[0, \infty)$ such that

$$\int F(p)\, \mathrm{d}\nu(p) = \int_{[0,\infty)} \left(\int_{\pi^{-1}(s)} F(p)\, \mathrm{d}\nu_s(p) \right) \mathrm{d}\lambda(s)$$

for all $F \in C_c(X)$. This λ is an orbit-space measure; it is determined by the chosen normalization of conditional orbit measures. A single massive orbit has infinite invariant volume, so this normalization is part of the data entering the quotient measure.

It remains to identify the orbit measures and fix their normalization. For $s > 0$, the standard invariant measure on the positive mass shell is

$$\mathrm{d}\sigma_s(p) = \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\, \theta(p^0) \delta(p^2 + s).$$

Indeed, the distribution $\theta(p^0)\delta(p^2 + s)\mathrm{d}^D p$ is Lorentz invariant, and in coordinates $p = (\omega_{\vec{p}}, \vec{p})$, $\omega_{\vec{p}} = (\vec{p}^2 + s)^{1/2}$, it is

$$\mathrm{d}\sigma_s(p) = \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2\omega_{\vec{p}}}.$$

For $s = 0$ the same formula gives the invariant measure on the punctured future light cone,

$$\mathrm{d}\sigma_0(p) = \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2|\vec{p}|}, \qquad \vec{p} \neq 0,$$

which is locally finite on X. The origin would give a separate orbit and a possible atom; it was removed by hypothesis.

The uniqueness of invariant Radon measure on each homogeneous orbit gives

$$\nu_s = c(s)\sigma_s$$

for λ-almost every s, with $c(s) \geq 0$. The family $s \mapsto \sigma_s$ is measurable: for $F \in C_c(X)$,

$$s \longmapsto \int_{\pi^{-1}(s)} F(p)\, \mathrm{d}\sigma_s(p)$$

is Borel, and it is continuous on $(0, \infty)$ by the displayed $\vec{p}$-coordinate formula and dominated convergence on compact supports; the $s = 0$ value is the corresponding null-shell limit whenever the support stays away from the origin. The measurable scalar $c(s)$ is then fixed λ-almost everywhere by testing ν_s against any countable measure-determining family of compactly supported continuous functions on X. Absorbing $c(s)$ into $\mathrm{d}\lambda(s)$ defines the positive Borel measure $\mathrm{d}\rho(s) := c(s)\mathrm{d}\lambda(s)$. Substituting $\nu_s = c(s)\sigma_s$ gives the stated formula for every $F \in C_c(X)$, and hence for every $F \in C_c(\overline{V}_+)$ because the removed origin has zero ν-measure. □

Definition 14.4 (Källén–Lehmann spectral measure)**.** Applying Proposition 14.3 to ν_ϕ, with the vacuum atom removed as above, gives

$$\mathrm{d}\nu_\phi(p) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2)\, \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\, \theta(p^0)\delta(p^2 + \mu^2).$$

Here $d\rho$ is a positive Borel measure on $[0, \infty)$. When it is absolutely continuous one writes $\mathrm{d}\rho(\mu^2) = \rho(\mu^2)\mathrm{d}\mu^2$; when it has atoms, the atomic weights are part of the same measure. In the special case in which $d\nu_\phi$ has a density with respect to $\mathrm{d}^D p$, scalar covariance and the spectrum condition say equivalently that the density is of the form

$$\frac{\theta(p^0)}{(2\pi)^{D-1}}\rho(-p^2),$$

with support on $-p^2 \geq 0$. The measure notation is retained because the one-particle contribution is an atom, not an ordinary function.

Definition 14.5 (Positive-frequency massive scalar distribution)**.** Define the positive-frequency scalar two-point function of mass μ by

$$\Delta_+(x; \mu^2) = \int \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\, \theta(p^0)\delta(p^2 + \mu^2)\, \mathrm{e}^{\mathrm{i}p\cdot x}.$$

Equivalently,

$$\Delta_+(x; \mu^2) = \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}}\, \frac{\mathrm{e}^{\mathrm{i}\vec{p}\cdot\vec{x} - \mathrm{i}\sqrt{\vec{p}^2+\mu^2}\, x^0}}{2\sqrt{\vec{p}^2 + \mu^2}}.$$

Proposition 14.6 (Källén–Lehmann representation for W_2)**.** *The Wightman two-point function is then*

$$W_2(x) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2)\, \Delta_+(x; \mu^2).$$

This is the Källén–Lehmann representation for the Wightman two-point function.

Proof. Let $f \in \mathcal{S}(\mathbb{M}^D)$. The spectral Fourier representation of Proposition 14.2 gives

$$W_2(f) = \int_{\overline{V}_+} \mathrm{d}\nu(p)\, \widetilde{f}(-p),$$

with ν the positive translation-spectral measure carried by the vacuum two-point vector. By Definition 14.4, ν disintegrates over the invariant mass coordinate $s = -p^2 \geq 0$ as

$$\mathrm{d}\nu(p) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2)\, \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\, \theta(p^0)\delta(p^2 + \mu^2).$$

Pairing with $\widetilde{f}(-p)$ is legitimate because f is Schwartz and the spectral measure is tempered by the Wightman distribution hypothesis. The positive measure permits Tonelli for nonnegative test majorants, and the Schwartz decay supplies the finite signed/complex integrals by decomposing $\widetilde{f}$ into real and imaginary parts. Hence

$$W_2(f) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2) \int \frac{\mathrm{d}^D p}{(2\pi)^{D-1}}\, \theta(p^0)\delta(p^2 + \mu^2)\, \widetilde{f}(-p).$$

The inner integral is precisely the distributional pairing $\Delta_+(\,\cdot\,; \mu^2)(f)$ in Definition 14.5. Therefore the displayed formula holds as an identity on every Schwartz test function, hence as an identity of tempered distributions. □

Spacelike symmetry of the scalar two-point function. For each μ^2, the distribution $\Delta_+(x; \mu^2)$ is even at spacelike separation:

$$\Delta_+(x; \mu^2) = \Delta_+(-x; \mu^2), \qquad x^2 > 0,$$

and hence $W_2(x) = W_2(-x)$ for spacelike x. No such equality follows from the spectral representation for timelike x. For $x^2 > 0$, choose a proper orthochronous Lorentz transformation carrying x to $(0, \vec{r})$. The mass-shell measure in Definition 14.5 is Lorentz invariant, so $\Delta_+(x; \mu^2) = \Delta_+(0, \vec{r}; \mu^2)$. In that frame the integral is

$$\int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}} \frac{\mathrm{e}^{\mathrm{i}\vec{p}\cdot\vec{r}}}{2\sqrt{\vec{p}^2 + \mu^2}},$$

which has an even denominator. Changing variables $\vec{p} \mapsto -\vec{p}$ gives $\Delta_+(0, \vec{r}; \mu^2) = \Delta_+(0, -\vec{r}; \mu^2)$, which is $\Delta_+(-x; \mu^2)$. Integrating this equality against the positive measure $d\rho(\mu^2)$ gives the statement for W_2.

14.3 Time-Ordered and Euclidean Forms

Definition 14.7 (Time-ordered scalar two-point distribution)**.** The time-ordered two-point function is

$$G_T(x) = \langle \Omega |\, \mathrm{T}\{\widehat{\phi}(x)\widehat{\phi}(0)\} | \Omega \rangle.$$

Using the definition of time ordering,

$$G_T(x) = \theta(x^0) W_2(x) + \theta(-x^0) W_2(-x).$$

Definition 14.8 (Massive Feynman two-point kernel)**.** For $\mu^2 \geq 0$, define

$$\Delta_F(x; \mu^2) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2 + \mu^2 - \mathrm{i}\epsilon}.$$

Proposition 14.9 (Time-ordered Källén–Lehmann representation)**.** *The time-ordered two-point function satisfies*

$$G_T(x) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2)\, \Delta_F(x;\mu^2)$$

as a distribution. Equivalently,

$$\Delta_F(x;\mu^2) = \theta(x^0)\Delta_+(x;\mu^2) + \theta(-x^0)\Delta_+(-x;\mu^2).$$

Proof. The first equality follows from Definition 14.7, Proposition 14.6, and the second displayed identity for Δ_F. It remains to prove that the iϵ kernel in Definition 14.8 has this time-ordered boundary value.

Set

$$\omega_{\vec{k},\mu} = \sqrt{\vec{k}^{\,2} + \mu^2}.$$

Since $k^2 + \mu^2 = -(k^0)^2 + \omega^2_{\vec{k},\mu}$, the k^0-integral in Δ_F is

$$I(x^0,\vec{k};\mu^2) = \int_{-\infty}^{\infty} \frac{\mathrm{d}k^0}{2\pi\mathrm{i}} \frac{\mathrm{e}^{-\mathrm{i}k^0x^0}}{-(k^0)^2 + \omega^2_{\vec{k},\mu} - \mathrm{i}\epsilon}.$$

The two poles are displaced to

$$k^0 = +\omega_{\vec{k},\mu} - \mathrm{i}0, \qquad k^0 = -\omega_{\vec{k},\mu} + \mathrm{i}0.$$

For $x^0 > 0$, the exponential damps in the lower half-plane and the closed contour has clockwise orientation. For $x^0 < 0$, the exponential damps in the upper half-plane. The residue theorem gives

$$I(x^0,\vec{k};\mu^2) = \theta(x^0)\frac{\mathrm{e}^{-\mathrm{i}\omega_{\vec{k},\mu}x^0}}{2\omega_{\vec{k},\mu}} + \theta(-x^0)\frac{\mathrm{e}^{+\mathrm{i}\omega_{\vec{k},\mu}x^0}}{2\omega_{\vec{k},\mu}}.$$

After the remaining spatial Fourier transform this is exactly

$$\Delta_F(x;\mu^2) = \theta(x^0)\Delta_+(x;\mu^2) + \theta(-x^0)\Delta_+(-x;\mu^2).$$

□

Figure 14.1 displays the two contour closures used in the residue calculation, with the pole placement matching the time-ordering terms in Δ_F.

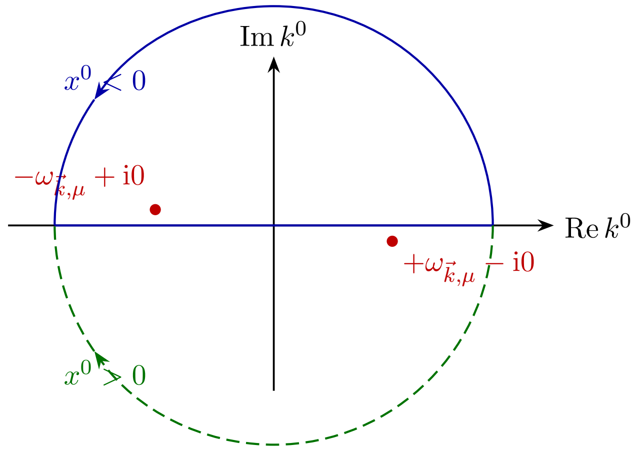


Figure 14.1: Contour closures for the Feynman two-point kernel. Closing below ($x^0 > 0$) is drawn dashed, while closing above ($x^0 < 0$) is drawn solid. Together the residues produce the time-ordered combination of positive-frequency two-point functions.

Equivalently,

$$G_T(x) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \mathrm{e}^{\mathrm{i}k\cdot x} \int_{[0,\infty)} \frac{\mathrm{d}\rho(\mu^2)}{k^2 + \mu^2 - \mathrm{i}\epsilon}.$$

Proposition 14.10 (Euclidean Källén–Lehmann representation)**.** *Let G_E be the Euclidean two-point distribution obtained from the ordered Wightman boundary values by $x^0 = -\mathrm{i}\tau_x$, $y^0 = -\mathrm{i}\tau_y$. Then*

$$G_E(x_E) = \int_{[0,\infty)} \mathrm{d}\rho(\mu^2) \, \Delta_E(x_E; \mu^2),$$

where

$$\Delta_E(x_E; \mu^2) = \int \frac{\mathrm{d}^D k_E}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k_E\cdot x_E}}{k_E^2 + \mu^2}. \tag{14.3}$$

Thus the same positive measure $d\rho$ controls the Wightman, time-ordered, and Euclidean scalar two-point functions.

Proof. The Euclidean function is obtained by analytic continuation of the Wightman functions in the time variable. For

$$z = x - y, \qquad z^0 \in \mathbb{C}, \qquad \mathrm{Im}\, z^0 < 0,$$

the factor $\exp(-\mathrm{i}p^0 z^0)$ is damped by the spectrum condition $p^0 \geq 0$, so $W_2(z)$ is analytic in this tube after smearing in the spatial variables. Set $x^0 = -\mathrm{i}\tau_x$, $y^0 = -\mathrm{i}\tau_y$. If $\tau_x > \tau_y$, then $z^0 = -\mathrm{i}(\tau_x - \tau_y)$ lies in the lower half plane and the Euclidean two-point function is the continuation of $W_2(x-y)$. If $\tau_x < \tau_y$, the time-ordered function instead continues $W_2(y-x)$. This gives a single symmetric Euclidean distribution. For each fixed μ^2, the same continuation of $\Delta_+(x;\mu^2)$ gives the free Euclidean kernel (14.3); this is the one-particle Wick rotation already derived in Chapter 11. Substituting the spectral integral of Proposition 14.6 and testing against compactly supported Euclidean test functions permits the continuation under the positive spectral integral. The result is the displayed formula for G_E. The three Green functions are therefore different boundary values or analytic continuations of the same positive spectral data. $\square$

14.4 Stieltjes Positivity of the Euclidean Propagator

Definition 14.11 (Stieltjes datum of a scalar Euclidean two-point function)**.** The Euclidean momentum-space two-point function contains the same information in a form that is often the most robust for constructive and lattice arguments. In this section z denotes the complex continuation of the Euclidean invariant Q^2. On the physical Euclidean axis set

$$Q^2 = k_E^2 \geq 0.$$

If the spectral measure obeys the Stieltjes integrability condition

$$\int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{1+s} < \infty, \tag{14.4}$$

then, after Fourier transform,

$$\widetilde{G}_E(Q^2) = \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{Q^2+s}. \tag{14.5}$$

Equation (14.4) is automatic if $\rho([0,\infty)) < \infty$, as in the canonically normalized elementary scalar case discussed below. It is not automatic for arbitrary local composite operators; for them the two-point function may require subtractions, and polynomial ambiguities in Q^2 are precisely the momentum-space form of contact terms at coincident Euclidean points.

Proposition 14.12 (Stieltjes positivity and spectral recovery)**.** *Under the integrability condition* (14.4)*, the function*

$$F(z) = \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{z+s}$$

is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, restricts to $\widetilde{G}_E(Q^2)$ on $Q^2 > 0$, obeys

$$(-1)^r \frac{\mathrm{d}^r}{\mathrm{d}(Q^2)^r} \widetilde{G}_E(Q^2) \geq 0, \qquad Q^2 > 0, \quad r = 0, 1, 2, \ldots,$$

and recovers the absolutely continuous spectral density by the discontinuity formula

$$\rho_{\mathrm{ac}}(s) = -\frac{1}{2\pi\mathrm{i}} \operatorname{Disc}_{z=-s} \widetilde{G}_E(z).$$

Atoms of $d\rho$ are simple poles on the cut with positive residues.

Proof. Under (14.4), $\widetilde{G}_E$ is a Stieltjes transform of a positive measure. More precisely,

$$F(z) = \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{z+s}$$

converges absolutely and uniformly on compact subsets of $\mathbb{C} \setminus (-\infty, 0]$: for every compact $K \subset \mathbb{C} \setminus (-\infty, 0]$ there is $c_K > 0$ such that $|z + s| \geq c_K(1 + s)$ for $z \in K$ and $s \geq 0$. Dominated convergence then permits differentiation under the integral sign, so F is holomorphic on this cut plane and its restriction to $z = Q^2 > 0$ is the Euclidean function in (14.5). The endpoint $z = 0$ is part of the cut if the spectrum has support arbitrarily near $s = 0$; no continuation through that endpoint is being assumed. If the spectral support begins at a positive mass gap $s = m_0^2 > 0$, the cut instead begins at $z = -m_0^2$, and the function is analytic in a neighborhood of $z = 0$. Consequently, for every $Q^2 > 0$,

$$(-1)^r \frac{\mathrm{d}^r}{\mathrm{d}(Q^2)^r} \widetilde{G}_E(Q^2) = r! \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{(Q^2+s)^{r+1}} \geq 0, \qquad r = 0, 1, 2, \ldots.$$

These inequalities are not perturbative statements. They are Hilbert-space positivity written in Euclidean momentum variables.

The measure can be recovered from the discontinuity across the negative real z-axis. The continuation is performed by treating z as a complex variable on the cut plane and approaching the point $z = -s$ from the two sides of the cut. If the absolutely continuous part has density $\rho_{\mathrm{ac}}(s)$, then

$$\rho_{\mathrm{ac}}(s) = -\frac{1}{2\pi \mathrm{i}} \operatorname{Disc}_{z=-s} \widetilde{G}_E(z), \qquad s > 0,$$

with atoms appearing as simple poles at $z = -s_0$ of positive residue. This is the Euclidean form of the statement that stable particles are isolated positive atoms of the spectral measure and multiparticle states are continuous spectral support. □

Proposition 14.13 (Subtracted Stieltjes form and contact terms)**.** *If the spectral measure has tempered growth in the sense that there is $N \geq 0$ with*

$$\int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{(1+s)^{N+1}} < \infty,$$

then the $(N+1)$-st derivative of the nonlocal Euclidean momentum-space two-point function is the convergent Stieltjes transform

$$F^{(N+1)}(z) = (-1)^{N+1}(N+1)! \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{(z+s)^{N+2}}.$$

Integrating back determines F only up to a polynomial of degree N, which is the momentum-space ambiguity of contact terms at coincident points. Under inverse Fourier transform this polynomial is precisely a finite linear combination of derivatives of the delta distribution supported at the coincident diagonal.

Proof. For a more singular local field, the spectral measure may have only tempered growth, by which we mean here that there is an integer $N \geq 0$ such that

$$\int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{(1+s)^{N+1}} < \infty.$$

The unsubtracted Stieltjes integral need not define an ordinary function. The nonlocal part of the Euclidean momentum-space distribution is nevertheless characterized by its $(N+1)$-st derivative,

$$F^{(N+1)}(z) = (-1)^{N+1}(N+1)! \int_{[0,\infty)} \frac{\mathrm{d}\rho(s)}{(z+s)^{N+2}}.$$

The right-hand side is holomorphic on the same cut plane. If F_1 and F_2 are two distributional momentum-space representatives with this same $(N+1)$-st derivative, their difference $P = F_1 - F_2$ satisfies $\partial_z^{N+1} P = 0$ on the cut plane. On every simply connected subdomain this implies that

P is a polynomial of degree at most N; the polynomial pieces agree on overlaps, so the same polynomial ambiguity is global. In momentum space a polynomial in p_E^2 and, in the nonscalar case, polynomial tensor structures, Fourier-transform to finite linear combinations of derivatives of $\delta(x)$. These are exactly coincident-point contact terms. They have no discontinuity across the cut, and therefore do not change the spectral measure away from coincident-point ambiguities. $\square$

14.5 Canonical Normalization and Spectral Weight

Definition 14.14 (Canonical equal-time normalization of a scalar field)**.** The measure $d\rho$ depends on the normalization of the local field. A canonical equal-time normalization consists of a regularity datum on the operator-valued distribution together with a smeared commutator identity. Let $d = D - 1$. Assume there is a dense common invariant domain $\mathcal{D} \subset \mathcal{H}$, with $\Omega \in \mathcal{D}$, such that for every $f \in \mathcal{S}(\mathbb{R}^d)$ the smeared equal-time fields

$$\widehat{\phi}_t(f) = \int \mathrm{d}^d\vec{x}\, f(\vec{x})\widehat{\phi}(t,\vec{x}), \qquad \widehat{\pi}_t(f) = \int \mathrm{d}^d\vec{x}\, f(\vec{x})\widehat{\pi}(t,\vec{x})$$

preserve $\mathcal{D}$, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\widehat{\phi}_t(f)\psi = \widehat{\pi}_t(f)\psi, \qquad \psi \in \mathcal{D},$$

where the derivative is taken in the Hilbert norm. This is C^1 distributional time-regularity of $\widehat{\phi}$. The canonical normalization is the domain identity

$$[\widehat{\phi}_t(f), \widehat{\pi}_t(g)]\psi = \mathrm{i}\int_{\mathbb{R}^d} f(\vec{x})g(\vec{x})\,\mathrm{d}^d\vec{x}\,\psi, \qquad f, g \in \mathcal{S}(\mathbb{R}^d), \quad \psi \in \mathcal{D}.$$

The formal notation

$$[\widehat{\phi}(t,\vec{x}), \partial_t\widehat{\phi}(t,\vec{y})] = \mathrm{i}\,\delta^{(d)}(\vec{x}-\vec{y})\mathbf{1}$$

means precisely this smeared statement with $\widehat{\pi}_t = \partial_t\widehat{\phi}_t$ as an operator-valued distribution. An interacting local field used as an interpolating operator need not satisfy this canonical identity. If an analogous equal-time commutator exists with a central normalization $C_\phi \geq 0$,

$$[\widehat{\phi}_t(f), \widehat{\pi}_t(g)]\psi = \mathrm{i}C_\phi \int_{\mathbb{R}^d} f(\vec{x})g(\vec{x})\,\mathrm{d}^d\vec{x}\,\psi,$$

then the same argument below gives $\int \mathrm{d}\rho = C_\phi$. If no such equal-time normalization is part of the field data, the Källén–Lehmann representation by itself gives positivity and support, but not a total-weight sum rule.

Proposition 14.15 (Canonical spectral-weight sum rule)**.** *Assume the canonical equal-time normalization of Definition 14.14, and assume that the spectral truncations converge after applying $\partial_t|_{t=0}$ in $\mathcal{S}'(\mathbb{R}^{D-1})$. Then*

$$\int_{[0,\infty)} \mathrm{d}\rho(s) = 1.$$

If the equal-time commutator has central normalization C_ϕ, then

$$\int_{[0,\infty)} \mathrm{d}\rho(s) = C_\phi.$$

Proof. The commutator two-point distribution is

$$C(x) := \langle\Omega|[\widehat{\phi}(x), \widehat{\phi}(0)]|\Omega\rangle = \int_{[0,\infty)} \mathrm{d}\rho(s)\,\big(\Delta_+(x;s) - \Delta_+(-x;s)\big).$$

The sign of the equal-time derivative is fixed by commutator antisymmetry. The canonical identity gives, in smeared form, $[\widehat{\phi}_0(\vec{r}), \widehat{\pi}_0(\vec{0})] = \mathrm{i}\delta^{(d)}(\vec{r})\mathbf{1}$. Since the derivative $\partial_t C(t,\vec{r})|_{t=0}$ differentiates the first field in $[\widehat{\phi}(t,\vec{r}), \widehat{\phi}(0,\vec{0})]$, it produces

$$[\widehat{\pi}_0(\vec{r}), \widehat{\phi}_0(\vec{0})] = -[\widehat{\phi}_0(\vec{0}), \widehat{\pi}_0(\vec{r})] = -\mathrm{i}\delta^{(d)}(\vec{r})\mathbf{1},$$

where the last equality uses the symmetry of the spatial delta distribution. The C^1 hypothesis gives the equal-time derivative as a spatial tempered distribution:

$$\partial_t C(t,\vec{r})|_{t=0} = \langle\Omega|[\widehat{\pi}_0(\vec{r}), \widehat{\phi}_0(\vec{0})]|\Omega\rangle = -\mathrm{i}\,\delta^{(d)}(\vec{r}). \tag{14.6}$$

It remains to justify the spectral calculation of the same derivative. Let ρ_R be the restriction of ρ to $[0,R]$, and let

$$C_R(t,\vec{r}) = \int_{[0,R]} \mathrm{d}\rho(s)\,\big(\Delta_+(t,\vec{r};s) - \Delta_+(-t,-\vec{r};s)\big).$$

The measure ρ_R is finite by local finiteness. For a spatial test function $h \in \mathcal{S}(\mathbb{R}^d)$, set

$$F_s(t;h) = \int_{\mathbb{R}^d} \mathrm{d}^d\vec{r}\, h(\vec{r})\big(\Delta_+(t,\vec{r};s) - \Delta_+(-t,-\vec{r};s)\big).$$

Writing $\omega_{\vec{p},s} = \sqrt{\vec{p}^2 + s}$, the spatial Fourier representation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}F_s(t;h) = -\frac{\mathrm{i}}{2}\int \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d}\left(\widetilde{h}(\vec{p})\mathrm{e}^{-\mathrm{i}\omega_{\vec{p},s}t} + \widetilde{h}(-\vec{p})\mathrm{e}^{+\mathrm{i}\omega_{\vec{p},s}t}\right),$$

where

$$\widetilde{h}(\vec{p}) = \int_{\mathbb{R}^d} \mathrm{d}^d\vec{r}\, h(\vec{r})\mathrm{e}^{\mathrm{i}\vec{p}\cdot\vec{r}}.$$

The bound

$$\left|\frac{\mathrm{d}}{\mathrm{d}t}F_s(t;h)\right| \le \int \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d}\,|\widetilde{h}(\vec{p})|$$

is uniform in $s \ge 0$ and t. Hence, for every finite R, dominated convergence permits the derivative to pass through the s-integral. At $t = 0$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}F_s(t;h)\right|_{t=0} = -\mathrm{i}h(\vec{0}).$$

Therefore

$$\langle\, \partial_t C_R(t,\cdot)|_{t=0}\,, h\rangle = -\mathrm{i}\,\rho([0,R])\,h(\vec{0}).$$

The final limiting step is part of the equal-time regularity assumption: the spectral truncations C_R converge to C after applying the operation $\partial_t|_{t=0}$ in $\mathcal{S}'(\mathbb{R}^d)$. Comparing with (14.6) gives

$$\rho([0,R]) \longrightarrow 1 \qquad (R \to \infty),$$

and monotone convergence yields the canonical sum rule

$$\int_{[0,\infty)} \mathrm{d}\rho(s) = 1. \tag{14.7}$$

It fixes the chosen field normalization; changing the field normalization rescales the spectral measure. If the equal-time commutator has central normalization C_ϕ, the right hand side of (14.6) is multiplied by C_ϕ, and the same comparison gives $\int_{[0,\infty)} \mathrm{d}\rho(s) = C_\phi$. □

One-particle residue bound for a canonically normalized field. If a canonically normalized scalar field has a one-particle atomic contribution $Z\,\delta_{m^2}(\mathrm{d}s)$, then

$$0 \le Z \le 1.$$

More generally, if the equal-time commutator has central normalization C_ϕ, then $0 \le Z \le C_\phi$. Without such a finite equal-time normalization, the atom weight Z is positive for the chosen field but has no universal upper bound.

If the same canonically normalized field has a one-particle atomic contribution $Z\,\delta_{m^2}(\mathrm{d}s)$, where δ_{m^2} is the unit point measure at $s = m^2$, positivity of $d\rho$ and (14.7) imply

$$0 \le Z \le 1.$$

The remainder $1 - Z$ is the total spectral weight carried by all continuum states and by any other atoms in this normalization. For a noncanonical interpolating field, a composite operator, or a renormalized field in a different convention, the same Källén–Lehmann representation holds, but (14.7) is replaced, when an equal-time commutator normalization exists, by

$$\int_{[0,\infty)} \mathrm{d}\rho(s) = C_\phi.$$

Then positivity gives $0 \le Z \le C_\phi$. Without such a finite normalization, the coefficient Z is still a positive spectral weight of the chosen interpolating field, but it has no universal upper bound; rescaling the field rescales Z. The canonical free field used later in LSZ is the asymptotic field on the one-particle Fock space, normalized separately from the interpolating local field whose correlators contain the pole.

Remark 14.16 (Support hypothesis for a single massive scalar channel). The lower endpoint 0 in the Källén–Lehmann integral is conventional: it is the possible lower endpoint of invariant mass squared under the spectrum condition, not a claim that the chosen field has spectral weight at zero. In the special situation used by the elementary scalar LSZ discussion, assume:

(i) the theory has no massless particle seen by $\widehat{\phi}$, so $d\rho_\phi(\{0\}) = 0$;

(ii) the field has nonzero overlap with a stable scalar one-particle mass $m > 0$, giving the atom $Z_\phi \delta_{m^2}(\mathrm{d}s)$;

(iii) there are no lighter massive particles or bound-state atoms in this channel; and

(iv) the non-one-particle part created by $\widehat{\phi}$ begins with states containing at least two particles of mass m.

Then the non-vacuum spectral measure has the support decomposition

$$\operatorname{supp} d\rho_\phi \subset \{m^2\} \cup [4m^2, \infty), \qquad \mathrm{d}\rho_\phi(s) = Z_\phi \delta_{m^2}(\mathrm{d}s) + \mathrm{d}\rho_{\phi,\mathrm{c}}(s),$$

with $\operatorname{supp} d\rho_{\phi,\mathrm{c}} \subset [4m^2, \infty)$. This gap between the atom and the rest of the support is the spectral input used later by Haag–Ruelle and LSZ. If massless particles, infraparticle thresholds, or additional bound-state atoms are present, the displayed support decomposition must be replaced by the corresponding spectral statement.

Remark 14.17 (Existence of an isolated mass shell is a spectral statement). The Källén–Lehmann representation does not imply the existence of a one-particle state. For a fixed scalar local field $\widehat{\phi}$, a term

$$\mathrm{d}\rho_\phi(s) = Z_\phi \, \delta_{m^2}(\mathrm{d}s) + \mathrm{d}\rho_{\phi,\mathrm{c}}(s), \qquad Z_\phi > 0,$$

is exactly the statement that $\widehat{\phi}\Omega$ has nonzero spectral projection onto the positive mass shell of invariant mass m. A positive measure whose support begins at m^2 but has no atom at m^2 gives a threshold, not a normalizable Wigner one-particle vector created by that field.

The field-independent Hilbert-space assertion needed by Haag–Ruelle theory is stronger and has two parts. With E the joint spectral measure of P^μ and

$$\Sigma_m^+ = \{p : p^0 > 0,\ p^2 = -m^2\},$$

one must have

$$E(\Sigma_m^+)\mathcal{H} \neq \{0\}, \qquad \operatorname{Spec}(P) \cap N = \Sigma_m^+$$

for some open neighborhood N of Σ_m^+. The first condition says that the one-particle subspace is nonzero; the second says that the shell is isolated from the rest of the energy-momentum spectrum. If the first condition fails, the notation $E(\Sigma_m^+)\mathcal{H}$ denotes the zero subspace and the corresponding scattering construction is vacuous. If the second condition fails, the sharp Wigner particle hypothesis required by ordinary Haag–Ruelle estimates is absent even when spectral weight accumulates at the same invariant mass.

An atom may be invisible to a particular interpolating field if that field has zero matrix element to the corresponding superselection sector or internal channel. Conversely, absence of an atom in every physical local interpolating field is evidence for absence of that stable particle sector, but the invariant statement is the vanishing of the spectral projection on the physical Hilbert space. The nonzero isolated-shell property is a theorem in free massive fields and in those constructive massive models for which the Wightman theory, the mass gap, and a separated one-particle spectral component have actually been proved. Formal perturbation theory alone supplies candidate pole data rather than this Hilbert-space spectral theorem.

14.6 Scalar One-Particle States and Normalization

Definition 14.18 (Scalar massive one-particle normalization datum)**.** Assume now that the Hilbert space contains an isolated positive-mass one-particle subspace with mass $m > 0$. In this section "scalar particle" means that the corresponding massive Wigner representation has trivial little-group representation and no internal degeneracy. This is a property of the one-particle representation; it is not the same statement as saying that a particular local field is a Lorentz scalar.

Let

$$k_R^\mu = (m, \vec{0}), \qquad k^\mu = (\omega_{\vec{k}}, \vec{k}), \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^{\,2} + m^2}.$$

Choose, for every $\vec{k}$, a standard boost $L(\vec{k}) \in SO^+(1, D-1)$ such that

$$L(\vec{k})k_R = k.$$

With generalized momentum eigenvectors normalized by

$${}_\delta\langle \vec{k}\,' | \vec{k}\rangle_\delta = \delta^{(D-1)}(\vec{k}\,' - \vec{k}),$$

write

$$|\vec{k}\rangle_\delta = N(\vec{k})\, U(L(\vec{k}))|\vec{0}\rangle_\delta, \qquad N(\vec{0}) = 1.$$

Figure 14.2 fixes the geometric meaning of the chosen standard boost $L(\vec{k})$: it carries the rest momentum k_R to the point k on the positive mass shell.

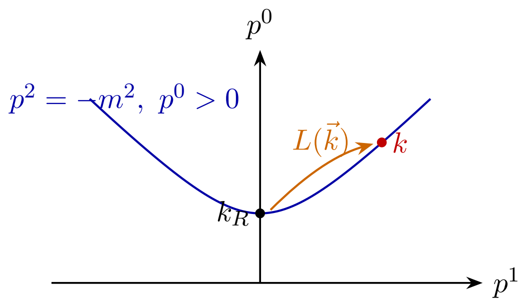


Figure 14.2: Standard boost from the rest momentum to a point on the positive mass shell. The choice of $L(\vec{k})$ fixes the Wigner-rotation convention used for one-particle states.

Lemma 14.19 (Delta-normalized and invariant scalar momentum states)**.** *For the scalar massive one-particle representation,*

$$N(\vec{k}) = \sqrt{\frac{m}{\omega_{\vec{k}}}},$$

up to a phase convention. The invariantly normalized states

$$|\vec{k}\rangle_{\text{inv}} = \sqrt{(2\pi)^{D-1} 2\omega_{\vec{k}}}\, |\vec{k}\rangle_\delta$$

obey

$${}_{\text{inv}}\langle \vec{k}\,' | \vec{k}\rangle_{\text{inv}} = (2\pi)^{D-1} 2\omega_{\vec{k}}\, \delta^{(D-1)}(\vec{k}\,' - \vec{k}), \qquad U(\Lambda)|\vec{k}\rangle_{\text{inv}} = |\Lambda\vec{k}\rangle_{\text{inv}}.$$

Proof. For a Lorentz transformation Λ, define the Wigner rotation

$$W(\Lambda, k) = L(\Lambda k)^{-1} \Lambda L(k).$$

It fixes k_R, hence lies in the little group $SO(D-1)$. For the scalar one-particle representation, $U(W(\Lambda, k))$ acts trivially on the rest state. Therefore

$$U(\Lambda)|\vec{k}\rangle_\delta = \frac{N(\vec{k})}{N(\Lambda\vec{k})} |\Lambda\vec{k}\rangle_\delta,$$

where $\Lambda\vec{k}$ denotes the spatial part of Λk. Unitarity fixes N. Since the on-shell measure

$$\frac{\mathrm{d}^{D-1}\vec{k}}{2\omega_{\vec{k}}}$$

is Lorentz invariant,

$$\delta^{(D-1)}(\Lambda\vec{k}\,' - \Lambda\vec{k}) = \frac{k^0}{(\Lambda k)^0}\delta^{(D-1)}(\vec{k}\,' - \vec{k}).$$

Comparing

$${}_\delta\langle\vec{k}\,'|\vec{k}\rangle_\delta = {}_\delta\langle\vec{k}\,'|U(\Lambda)^\dagger U(\Lambda)|\vec{k}\rangle_\delta$$

gives

$$|N(\Lambda\vec{k})|^2 = |N(\vec{k})|^2\,\frac{k^0}{(\Lambda k)^0}.$$

Taking $k = k_R$ and $\Lambda = L(\vec{k})$, together with $N(\vec{0}) = 1$, yields

$$N(\vec{k}) = \sqrt{\frac{m}{\omega_{\vec{k}}}},$$

after fixing the harmless phase convention.

It is often better to use invariantly normalized states

$$|\vec{k}\rangle_{\mathrm{inv}} = \sqrt{(2\pi)^{D-1}2\omega_{\vec{k}}}\,|\vec{k}\rangle_\delta,$$

for which

$${}_{\mathrm{inv}}\langle\vec{k}\,'|\vec{k}\rangle_{\mathrm{inv}} = (2\pi)^{D-1}2\omega_{\vec{k}}\,\delta^{(D-1)}(\vec{k}\,' - \vec{k}), \qquad U(\Lambda)|\vec{k}\rangle_{\mathrm{inv}} = |\Lambda\vec{k}\rangle_{\mathrm{inv}}.$$

□

Proposition 14.20 (Scalar field matrix element on an isolated scalar shell)**.** *Work in a selected spin-zero isolated one-particle channel with the delta-normalized and invariantly normalized states of Lemma 14.19. If $\widehat{\phi}$ is a scalar local field and the vacuum is invariant, then the matrix element*

$$A(\vec{k}) = {}_\delta\langle\vec{k}|\widehat{\phi}(0)|\Omega\rangle$$

has the same $\vec{k}$-dependence as $N(\vec{k})$. Equivalently, there is a nonnegative number Z, depending on the chosen field, such that

$${}_\delta\langle\vec{k}|\widehat{\phi}(0)|\Omega\rangle = \left[\frac{Z}{(2\pi)^{D-1}2\omega_{\vec{k}}}\right]^{1/2},$$

or equivalently

$${}_{\mathrm{inv}}\langle\vec{k}|\widehat{\phi}(0)|\Omega\rangle = Z^{1/2}.$$

Proof. The point notation is distributional: equivalently one may smear $\widehat{\phi}$ in a small neighborhood of the origin and then read the coefficient of the isolated shell. The isolated-shell projection removes continuum contributions, and the assumption of a selected spin-zero channel means that the little group acts trivially on the rest vector up to the phase already fixed in Lemma 14.19.

The scalar covariance equation $U(\Lambda)\widehat{\phi}(0)U(\Lambda)^{-1} = \widehat{\phi}(0)$, vacuum invariance, and Lemma 14.19 imply the transformation law

$$A(\vec{k}) = \frac{N(\vec{k})}{N(\Lambda\vec{k})} A(\Lambda\vec{k}).$$

Taking $\Lambda = L(\vec{k})^{-1}$ reduces the matrix element to its rest-frame value and gives $A(\vec{k}) = N(\vec{k})A(\vec{0})$. If $A(\vec{0}) = 0$, then the field has zero overlap with this channel and $Z = 0$. Otherwise the remaining phase freedom of the one-particle states fixes $A(\vec{0}) \geq 0$. Writing $Z = (2\pi)^{D-1}2m\,|A(\vec{0})|^2$ gives the delta-normalized formula, and the invariant normalization removes the kinematic factor. If several spin-zero channels of the same mass are present, the same argument applies to the vector of rest-frame overlaps in the finite or direct-integral multiplicity space; choosing a normalized component gives the scalar formula above, while Z for the full field is the squared norm of that overlap vector. □

14.7 One-Particle Atoms

Proposition 14.21 (One-particle atom in the scalar spectral measure)**.** *Assume the spectrum contains an isolated scalar one-particle mass $m > 0$, and assume the local field $\widehat{\phi}$ has nonzero projection onto that one-particle subspace. With the invariant normalization from Lemma 14.19, define $Z \geq 0$ by*

$${}_{\text{inv}}\langle\vec{p}|\widehat{\phi}(0)|\Omega\rangle = Z^{1/2}.$$

Then the one-particle contribution to the spectral measure is

$$\mathrm{d}\rho_1(\mu^2) = Z\,\delta_{m^2}(\mathrm{d}\mu^2),$$

so

$$\mathrm{d}\rho(\mu^2) = Z\,\delta_{m^2}(\mathrm{d}\mu^2) + \mathrm{d}\rho_{\mathrm{c}}(\mu^2)$$

with $\mathrm{d}\rho_{\mathrm{c}}$ positive.

Proof. With the invariant normalization from the one-particle construction,

$$\langle\vec{p}\,'|\vec{p}\rangle = (2\pi)^{D-1}2\omega_{\vec{p}}\,\delta^{(D-1)}(\vec{p}\,' - \vec{p}), \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^{\,2} + m^2},$$

define $Z \geq 0$ by

$${}_{\text{inv}}\langle\vec{p}|\widehat{\phi}(0)|\Omega\rangle = Z^{1/2}$$

for scalar one-particle states. Lorentz covariance makes the right-hand side independent of $\vec{p}$ in this normalization. Let δ_{m^2} denote the unit point measure at $\mu^2 = m^2$. The one-particle contribution to the spectral measure is

$$\mathrm{d}\rho_1(\mu^2) = Z\,\delta_{m^2}(\mathrm{d}\mu^2).$$

Thus

$$\mathrm{d}\rho(\mu^2) = Z\,\delta_{m^2}(\mathrm{d}\mu^2) + \mathrm{d}\rho_{\mathrm{c}}(\mu^2),$$

where $d\rho_{\mathrm{c}}$ is the remaining positive measure. At the level of the momentum-space measure, using the delta-normalized states in the previous section, the same statement is the calculation

$$\begin{aligned}\mathrm{d}\nu_{\phi,1}(p) &= \int \mathrm{d}^{D-1}\vec{k}\, \left|{}_{\delta}\langle \vec{k}|\widehat{\phi}(0)|\Omega\rangle\right|^2 \delta^{(D)}(p-k)\,\mathrm{d}^D p \\ &= \int \mathrm{d}^{D-1}\vec{k}\, \frac{Z}{(2\pi)^{D-1}2\omega_{\vec{k}}}\,\delta^{(D)}(p-k)\,\mathrm{d}^D p = \frac{Z\,\mathrm{d}^D p}{(2\pi)^{D-1}}\,\theta(p^0)\delta(p^2+m^2).\end{aligned}$$

Comparison with the invariant-mass decomposition then gives $\mathrm{d}\rho_1(\mu^2) = Z\,\delta_{m^2}(\mathrm{d}\mu^2)$. □

The spectral decomposition of $\widehat{\phi}(f)\Omega$ used in the preceding measure calculation is the orthogonal sum of the vacuum atom $E(\{0\})\widehat{\phi}(f)\Omega$, the isolated one-particle atom $E(\mu^2 = m^2)\widehat{\phi}(f)\Omega$, and the continuum projection $E_{\mathrm{c}}\widehat{\phi}(f)\Omega$. After subtracting the vacuum expectation value of $\widehat{\phi}$, the vacuum projection is absent. The one-particle atom remains precisely when the field has nonzero overlap with the isolated mass shell.

Under the same assumptions,

$$G_T(x) = Z\,\Delta_F(x;m^2) + \int_{[0,\infty)} \mathrm{d}\rho_{\mathrm{c}}(\mu^2)\,\Delta_F(x;\mu^2).$$

If the theory has a single stable particle of mass m, no lighter particles, and the continuum part is generated by states containing at least two such particles, then

$$\operatorname{supp} d\rho_{\mathrm{c}} \subset [4m^2,\infty).$$

When $d\rho_{\mathrm{c}}$ has a density $\sigma(\mu^2)$ in that region,

$$G_T(x) = Z\,\Delta_F(x;m^2) + \int_{4m^2}^{\infty} \mathrm{d}\mu^2\,\sigma(\mu^2)\Delta_F(x;\mu^2).$$

Figure 14.3 displays this threshold structure in the spectral measure: the isolated atom at m^2 and the continuum support beginning at $4m^2$ under the stated assumption.

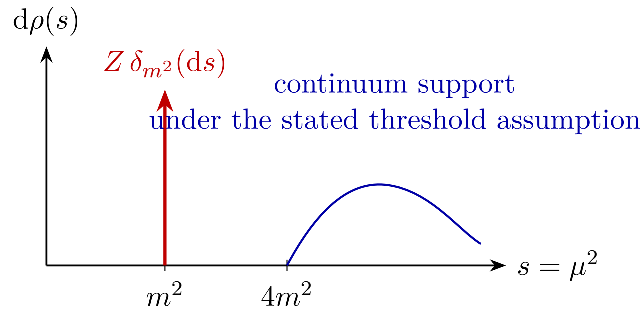


Figure 14.3: One-particle pole and continuum part of the Källén–Lehmann measure. Under the stated threshold assumption the atom at m^2 is separated from continuum support beginning at $4m^2$.

The number Z is attached to the chosen local field. Rescaling the field rescales Z. Once the field normalization is fixed, Z is a definite matrix element and the one-particle pole residue of its two-point function. It is therefore not an unphysical datum in the following precise sense: it is not invariant under changing the operator whose two-point function is being studied, but for a fixed operator it is an unambiguous spectral weight.

14.8 Role in the Construction

The Källén–Lehmann representation is nonperturbative. It states that Hilbert-space positivity and the spectrum condition force the scalar two-point function to be a positive superposition of free massive two-point functions. Isolated atoms of the measure identify stable one-particle mass shells seen by the local field. Continuous spectral support records the remaining invariant-mass spectrum.

The local field $\widehat{\phi}$ in this chapter is an interpolating local field. Its time-ordered two-point function contains the full spectral measure

$$G_T^{\phi}(x) = Z\,\Delta_F(x; m^2) + \int_{[0,\infty)} \mathrm{d}\rho_{\mathrm{c}}(s)\,\Delta_F(x; s)$$

whenever an isolated one-particle atom of mass m is present. The asymptotic scalar field associated with that atom is a different object: it is the canonical free scalar field on the asymptotic Fock space generated by the one-particle Hilbert space $\mathcal{H}_1$, and its two-point function is

$$G_T^{\mathrm{as}}(x) = \Delta_F(x; m^2)$$

with unit one-particle residue and no continuum part. The relation between the two fields is the one-particle projection and normalization

$$P_1\widehat{\phi}(f)\Omega = Z^{1/2}\psi_f,$$

where, in the Fourier convention

$$f_-(p) = \int \mathrm{d}^D x\, f(x)\mathrm{e}^{\mathrm{i}p\cdot x},$$

$$\psi_f = \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, f_-(p)|p\rangle \in \mathcal{H}_1.$$

Haag–Ruelle constructs the asymptotic Fock space from such vectors, and LSZ extracts the corresponding pole coefficient from correlation functions of the interpolating local field.

Remark 14.22 (Infraparticle spectrum in charged QED). The word "atom" here means a literal point mass in the positive spectral measure, for example $Z\,\delta_{m^2}(\mathrm{d}s)$ with $Z > 0$. This is exactly the input needed later by Haag–Ruelle theory and by LSZ pole-coefficient extraction. The charged sector of QED with a massless photon realizes a different spectral structure. It contains the charged threshold together with states differing by photons of arbitrarily small energy; the energy-momentum spectral support therefore accumulates at the threshold instead of leaving an open neighborhood containing only Σ_m^+. Equivalently, the charged particle is an infraparticle: threshold spectral support replaces the simple one-particle pole, and the charged sector has no normalizable one-particle Wigner subspace obtained from $P_1 = E(\Sigma_m^+)$.

Gauge-fixed perturbative notation must be interpreted through this physical spectral statement. A gauge-dependent electron propagator is not, by itself, the positive Källén–Lehmann measure of a compactly localized physical observable in the Hilbert space. Physical charged asymptotic data require either inclusive probabilities or dressed charged states. The later infrared chapters develop the corresponding Bloch–Nordsieck cancellation and Faddeev–Kulish-type dressing constructions in Chapter 42 and Chapter 44.

Remark 14.23 (Confinement and physical spectral atoms). In a confining nonabelian gauge theory the positive Hilbert-space spectral measure is attached to fields or observables that act on the physical Hilbert space. Gauge-invariant color-singlet interpolating fields have Källén–Lehmann measures supported on hadronic states; when such a measure has an isolated atom, the atom corresponds to a stable hadron or glueball seen by that interpolating field. Gauge-dependent quark and gluon fields are not local physical observables with positive spectral measures on the physical Hilbert space, and confinement excludes isolated quark or gluon mass-shell atoms from the physical asymptotic spectrum. Consequently the Haag–Ruelle and LSZ constructions later in this volume apply in a confining theory only to stable color-singlet particles, after the corresponding physical interpolating operators and isolated hadronic atoms have been identified.

The next step in the construction is the perturbative computation of Green functions. The construction of scattering states and the S-matrix comes later, after the spectral one-particle data have been separated from the local two-point functions.

Chapter 15

Perturbative Green Functions and Feynman Graphs

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

15.1 Regulated Euclidean Scalar Theory

Definition 15.1 (Regulated Euclidean ϕ^4 datum)**.** Let $D \geq 1$. In this chapter $x \in \mathbb{R}^D$ denotes a Euclidean spacetime point and $k \in \mathbb{R}^D$ denotes the corresponding Euclidean momentum. The field variable is a real scalar function ϕ. For the perturbative Euclidean construction in this chapter, henceforth assume $m > 0$; this is a stricter infrared convention than the nonnegative invariant mass parameter allowed in the spectral discussion. The Lorentzian model whose Green functions are studied perturbatively has classical density, in the mostly-plus convention,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{4!}\phi^4.$$

The perturbative construction in this chapter is made first for Euclidean Green functions. Its Euclidean density is

$$\mathcal{L}_E = \frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{g}{4!}\phi^4,$$

so that the interaction part of the Euclidean action is

$$S_{\mathrm{int},E}[\phi] = \frac{g}{4!}\int \mathrm{d}^D x\,\phi(x)^4,$$

where $g \in \mathbb{R}$ is the quartic coupling. This expression is a regulated interaction coordinate. With the finite-dimensional cutoff introduced below, $\phi_\Lambda(x)^4$ is an ordinary function on the regulated configuration space and the integral is a well-defined polynomial. The continuum symbol $\phi(x)^4$ is not, by itself, a defined local operator: products of continuum fields at coincident points require a renormalization prescription. In this chapter the symbol denotes the regulated monomial together with the counterterm and renormalization conditions used to define Green functions order by order. Renormalized composite insertions, obtained by coupling local sources and differentiating with respect to renormalized source coordinates, are constructed later in Chapter 34.

The free covariance is the Euclidean propagator

$$\Delta_E(x-y;m^2) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k\cdot(x-y)}}{k^2+m^2},$$

with $m > 0$.

The status of the regulator-removal problem depends sharply on D. For $D = 2$ and $D = 3$, the renormalized ϕ^4 models have constructive non-Gaussian continuum realizations: in $D = 2$ this belongs to the $P(\phi)_2$ construction, and in $D = 3$ the ϕ^4_3 construction is obtained after the required local counterterms. For the standard reflection-positive lattice $\lambda\phi^4$ model with $\lambda \geq 0$ in $D = 4$, the continuum scaling limits selected by the usual critical or near-critical tuning are Gaussian. This is the four-dimensional marginal triviality theorem completed by the Aizenman–Duminil-Copin analysis, together with the earlier triviality results of Aizenman and Fröhlich. The theorem is a statement about that constructive scaling-limit problem. The broader question whether some ultraviolet-complete QFT can have a small-coupling expansion agreeing with formal ϕ^4_4 perturbation theory to all orders is a separate nonperturbative existence problem. Thus the four-dimensional perturbative expansion below should be read as a finite-cutoff and renormalized asymptotic expansion, and as a valid effective-field-theory expansion below a cutoff, unless an additional nonperturbative ultraviolet construction is supplied.

The corresponding constructive question for pure Yang–Mills theory in $D = 4$ has a different status. The Clay Millennium problem called "Yang–Mills existence and mass gap" asks, for every compact simple gauge group G, for a continuum quantum Yang–Mills theory on $\mathbb{R}^4$ whose gauge-invariant Schwinger functions satisfy the structural requirements needed for Euclidean reconstruction and whose reconstructed physical Hamiltonian has a unique vacuum at energy 0 and spectrum contained in $\{0\} \cup [\Delta, \infty)$ for some $\Delta > 0$. Perturbative asymptotic freedom, lattice regularizations, strong-coupling expansions in finite cutoff systems, and numerical evidence all belong to the surrounding mathematical and physical picture; the continuum existence theorem together with the positive physical mass gap is the open problem. This is logically distinct from the ϕ^4_4 triviality statement above: the latter is a proved obstruction for the standard reflection-positive scalar scaling-limit family, while the former is an open constructive existence and spectral problem for a gauge theory.

The functional notation is used with a regulator. One concrete choice is a finite Euclidean box with periodic boundary conditions and only finitely many Fourier modes retained. Equivalently, in infinite-volume notation, choose a smooth cutoff function $\chi_\Lambda(k)$ which is 1 for $|k| \ll \Lambda$ and rapidly decays for $|k| \gg \Lambda$, and replace the free covariance by

$$C_\Lambda(x-y) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\chi_\Lambda(k)\,\mathrm{e}^{\mathrm{i}k\cdot(x-y)}}{k^2+m^2}.$$

At finite regulator the Gaussian expectation $\langle\cdots\rangle_{0,\Lambda}$ is an ordinary finite-dimensional Gaussian expectation, or the limit of such expectations. Formulas written without Λ are coefficientwise perturbative formulas or shorthand for regulator-dependent formulas before the limit is studied.

Remark 15.2 (Constructive status of the scalar example). The constructive comments above are not used as hidden axioms in the perturbative derivations below. They delimit the status of the continuum limit: every formula in this chapter is first an identity at finite regulator or a coefficient of a regulated formal power series; the additional question of removing Λ is a separate renormalization or constructive existence problem.

Definition 15.3 (Finite Fourier cutoff Gaussian scalar datum)**.** With a sharp Fourier cutoff this finite-dimensional statement is represented by

$$\int [\mathrm{d}\phi]_\Lambda \quad \rightsquigarrow \quad \int \prod_{|k|\le\Lambda} \mathrm{d}\widetilde{\phi}(k), \qquad \widetilde{\phi}(-k) = \overline{\widetilde{\phi}(k)}$$

after choosing independent real coordinates for one representative of each pair $k, -k$ and fixing the resulting finite-dimensional Gaussian normalization. More explicitly, put the theory first on a Euclidean torus of side L, let $\mathcal{K}_{\Lambda,L}$ be a finite symmetric set of allowed momenta, and set

$$E_{\Lambda,L} = \left\{\widetilde{\phi} : \mathcal{K}_{\Lambda,L} \to \mathbb{C} : \widetilde{\phi}(-k) = \overline{\widetilde{\phi}(k)}\right\}.$$

After choosing real coordinates on $E_{\Lambda,L}$, the free expectation is the centered Gaussian probability measure with covariance

$$\langle\widetilde{\phi}(k)\widetilde{\phi}(k')\rangle_{0,\Lambda,L} = L^D \,\frac{\delta_{k+k',0}}{k^2+m^2}.$$

Equivalently, for a real source J whose Fourier transform is supported in $\mathcal{K}_{\Lambda,L}$,

$$Z_{0,\Lambda,L}[J] = \exp\left[\frac{1}{2}\int_{T_L^D} \mathrm{d}^D x\, \mathrm{d}^D y\, J(x) C_{\Lambda,L}(x-y) J(y)\right].$$

The infinite-volume smooth-cutoff formulas in this chapter are shorthand for the corresponding limit or for the directly regulated covariance C_Λ. The symbol $[\mathrm{d}\phi]$ is not used here as an unregulated Lebesgue measure on an infinite-dimensional vector space.

Definition 15.4 (Normalized regulated Euclidean Green functions)**.** The normalized Euclidean n-point function is

$$G_{E,n,\Lambda,L}(x_1,\ldots,x_n) = \frac{\langle\phi(x_1)\cdots\phi(x_n)\exp[-S_{\mathrm{int},E}[\phi]]\rangle_{0,\Lambda,L}}{\langle\exp[-S_{\mathrm{int},E}[\phi]]\rangle_{0,\Lambda,L}}.$$

In infinite-volume shorthand the subscript L is suppressed. It is a distribution in the insertion points after the regulator is removed, when that limit exists or has been renormalized. Perturbation theory is the coefficientwise expansion of this expression in powers of the coupling g, with the regulator kept fixed before renormalized limits are considered.

Coefficientwise perturbation at fixed regulator. At any finite regulator for which $S_{\mathrm{int},E}$ is a polynomial on a finite-dimensional Gaussian space, the coefficient of g^V in $G_{E,n,\Lambda}$ is obtained by expanding numerator and denominator in powers of $S_{\mathrm{int},E}$, taking ordinary Gaussian moments, and then dividing the resulting formal power series by the denominator series. This is finite-dimensional formal power-series algebra. At finite regulator the Gaussian expectation is an ordinary normalized finite-dimensional integral with positive covariance. The exponential $\exp[-S_{\mathrm{int},E}]$ is treated either as an honest integrable function for values of g where the integral converges, or as the formal series

$$\sum_{V\geq 0}\frac{(-1)^V}{V!}S_{\mathrm{int},E}[\phi]^V.$$

Since $S_{\mathrm{int},E}$ is polynomial, each coefficient is a finite sum of Gaussian moments. The denominator has constant term 1, so it has a unique inverse as a formal power series in g. Multiplication by that inverse gives the displayed normalized Green function coefficient by coefficient. This algebraic step uses only finite-dimensional Gaussian moments and the formal inverse of a series with constant term 1; regulator removal and renormalization enter later.

15.2 Wick Expansion and Connected Functions

Wick graph expansion and connected cumulants. For the centered Gaussian expectation,

$$\langle\phi(x_1)\cdots\phi(x_{2r})\rangle_{0,\Lambda}=\sum_{\text{pairings }P}\prod_{\{a,b\}\in P}C_\Lambda(x_a-x_b),$$

and odd moments vanish. Expanding the interaction gives

$$\begin{aligned}&\langle\phi(x_1)\cdots\phi(x_n)\exp[-S_{\mathrm{int},E}[\phi]]\rangle_{0,\Lambda}\\&=\sum_{V=0}^{\infty}\frac{(-g)^V}{V!(4!)^V}\int\prod_{a=1}^{V}\mathrm{d}^Dz_a\,\langle\phi(x_1)\cdots\phi(x_n)\prod_{a=1}^{V}\phi(z_a)^4\rangle_{0,\Lambda}.\end{aligned}$$

Each Gaussian moment on the right is a sum over pairings. A pairing is drawn as an edge; an integration point z_a is drawn as a four-valent vertex; an insertion x_i is an external labelled endpoint. This is the origin of the Feynman graphs in this chapter.

The denominator in $G_{E,n,\Lambda}$ cancels components that have no external endpoint. Therefore a normalized n-point function is a sum over graphs whose every connected component contains at least one external label. The connected n-point function is the part in which all external labels belong to one connected component. The first displayed formula is Wick's theorem for the finite-dimensional Gaussian vector obtained by evaluating the regulated field at the indicated test functions or points. Substituting the fixed-regulator formal exponential expansion expresses each coefficient as a finite sum over pairings of the $n+4V$ field factors. Every pair in a Wick pairing supplies one covariance C_Λ; grouping the paired half-edges incident to the same interaction point gives precisely the graph description.

The cancellation of vacuum components is the exponential formula for labelled structures applied to Wick pairings. Equivalently, write every pairing graph as the disjoint union of connected components. Components with no external label are generated by the same series that appears in $Z_\Lambda[0]$, so division by $Z_\Lambda[0]$ removes them. The connected part is then selected by requiring that the external labels lie in one component.

Figure 15.1 records this component selection: vacuum components cancel against $Z_\Lambda[0]$, and surviving components contain external insertions.

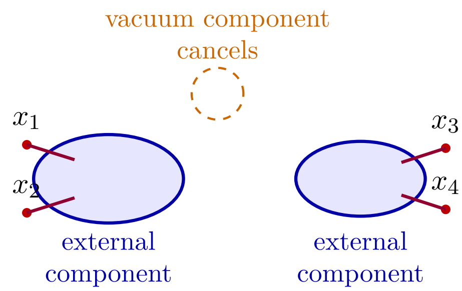


Figure 15.1: Component selection in normalized Euclidean correlators. Vacuum components cancel against the denominator, while every surviving component contains at least one external insertion.

Introduce a source J and define

$$Z_\Lambda[J] = \langle \exp\left[-S_{\mathrm{int},E}[\phi] + \int \mathrm{d}^D x\, J(x)\phi(x)\right]\rangle_{0,\Lambda}.$$

The connected generating functional is

$$W_\Lambda[J] = \log Z_\Lambda[J].$$

For $n \geq 1$,

$$G^{\mathrm{conn}}_{E,n,\Lambda}(x_1, \ldots, x_n) = \left.\frac{\delta^n W_\Lambda[J]}{\delta J(x_1)\cdots\delta J(x_n)}\right|_{J=0}.$$

This identity is the cumulant identity for the regulated probability measure. At finite regulator $Z_\Lambda[J]$ is the moment generating functional of a finite-dimensional random vector after adjoining the polynomial weight $\exp[-S_{\mathrm{int},E}]$. The logarithm of a moment generating function is the cumulant generating function. Differentiating $W_\Lambda[J]$ at $J = 0$ therefore gives the joint cumulant of the n inserted fields. The moment-cumulant relation is equivalent to the connected-component expansion of the preceding proposition, so the functional derivatives of W_Λ are exactly the connected regulated Green functions.

15.3 Momentum-Space Graph Weights

Definition 15.5 (Euclidean graph weight)**.** In translation-invariant infinite-volume notation, write

$$G^{\mathrm{conn}}_{E,n}(x_1, \ldots, x_n) = \int \prod_{i=1}^{n} \frac{\mathrm{d}^D k_i}{(2\pi)^D}\, \mathrm{e}^{\mathrm{i}\sum_i k_i\cdot x_i}\, \widetilde{G}^{\mathrm{conn}}_{E,n}(k_1, \ldots, k_n).$$

A connected graph Γ with labelled external legs contributes a distribution of the form

$$\mathcal{A}_E(\Gamma; k_1, \ldots, k_n) = (2\pi)^D \delta^{(D)}\left(\sum_{i=1}^{n} k_i\right) I_E(\Gamma; k_1, \ldots, k_n).$$

The reduced integrand I_E is computed as follows. Assign momentum to every edge, impose momentum conservation at every vertex, integrate over one independent momentum for each loop, and multiply by

$$\frac{(-g)^{V(\Gamma)}}{|\operatorname{Aut}\Gamma|}\prod_{e\in E(\Gamma)}\frac{1}{p_e^2+m^2}.$$

Here $V(\Gamma)$ is the number of quartic vertices, $E(\Gamma)$ is the set of internal and external propagator edges, p_e is the momentum carried by edge e, and $\operatorname{Aut}\Gamma$ is the group of graph automorphisms preserving all external labels. This convention includes the factor 1/4! in the action and the Wick-contraction multiplicities in the single symmetry factor.

For a connected graph with $I(\Gamma)$ internal edges and $V(\Gamma)$ vertices, the number of independent loop momenta is

$$L(\Gamma) = I(\Gamma) - V(\Gamma) + 1.$$

This is the first Betti number of the graph. Momentum conservation at all vertices gives $V(\Gamma)$ linear constraints, but the sum of those constraints is the external momentum-conservation delta function already displayed, so only $V(\Gamma) - 1$ vertex constraints eliminate independent internal edge momenta.

Figure 15.2 fixes the two momentum-space rules used below: the regulated scalar propagator on each edge and the quartic vertex with momentum conservation.

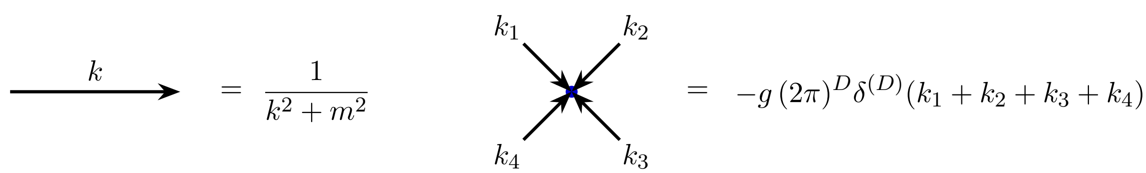


Figure 15.2: Momentum-space Feynman rules for Euclidean ϕ^4 theory with labelled external legs. The propagator is $(k^2+m^2)^{-1}$, and the quartic vertex includes momentum conservation.

The displayed vertex rule is the labelled-leg rule. If instead one expands directly from the action, each vertex gives $-g/4!$ and the 4! is recovered by counting Wick contractions.

15.4 Symmetry Factors from Wick Pairings

Automorphism denominator from Wick pairings. The coefficient $|\operatorname{Aut}\Gamma|^{-1}$ in the graph rule is the residual count left by the regulated Wick expansion. Consider a connected graph Γ with V quartic vertices and labelled external endpoints. Let $\operatorname{Aut}\Gamma$ denote the automorphism group preserving the external labels and the incidence relation between vertices and half-edges. The V-th term of the interaction expansion carries the prefactor

$$\frac{(-g)^V}{V!(4!)^V}.$$

If the abstract graph Γ is fixed, the number of choices of ordered interaction vertices, ordered half-edges at each vertex, and Wick pairings which realize this same labelled graph is

$$\frac{V!(4!)^V}{|\operatorname{Aut}\Gamma|}.$$

Multiplying the two factors gives precisely

$$\frac{(-g)^V}{|\operatorname{Aut}\Gamma|}.$$

Thus the graph automorphism factor is the quotient between the arbitrary labellings introduced by the expansion and the relabellings leaving the graph unchanged.

Start with the V-th term of the regulated expansion, where the integration variables $z_1, \ldots, z_V$ are ordered and the four field factors at each vertex are ordered by their position in $\phi(z_a)^4$. These choices give $V!(4!)^V$ auxiliary labels for the same abstract incidence graph. The Wick pairing then identifies half-edges in pairs to form internal edges or connects half-edges to the fixed external labels. For a fixed labelled external graph Γ, the group $\operatorname{Aut}\Gamma$ is exactly the stabilizer of this labelling data: it consists of the vertex and half-edge relabellings that leave the incidence relations and all external labels unchanged. By the orbit-stabilizer identity, the number of labelled Wick contractions giving the same abstract Γ is $V!(4!)^V/|\operatorname{Aut}\Gamma|$. Multiplying by the expansion prefactor gives $(-g)^V/|\operatorname{Aut}\Gamma|$.

For example, the order-g tadpole correction to the two-point function has one quartic vertex. The two external labels are fixed, while the two remaining half-edges form the loop and may be interchanged. Hence $|\operatorname{Aut}\Gamma| = 2$, and the coefficient is $-g/2$. In the unlabelled-vertex convention this same count is

$$\left(-\frac{g}{4!}\right)\left(4 \cdot 3 \cdot \frac{2!}{2}\right) = -\frac{g}{2}.$$

Here the 4 chooses the vertex half-edge contracted with the first external endpoint, and the 3 chooses the half-edge contracted with the second external endpoint. The remaining two half-edges form the loop. If they are temporarily ordered this gives a factor 2!, and the quotient by 2 identifies the interchange of the two ends of the same loop, leaving one loop pairing.

For the two-vertex sunset insertion in the two-point function, the expansion prefactor and the Wick-pairing count give

$$\frac{1}{2!}\left(-\frac{g}{4!}\right)^2 \left(2 \cdot 4^2 \cdot 3!\right) = 4 \cdot 4! \left(-\frac{g}{4!}\right)^2 = \frac{g^2}{6}.$$

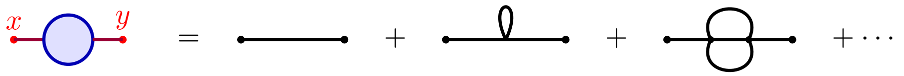
The first factor 2 assigns the two external endpoints to the two interaction vertices, each 4 chooses the external half-edge at one vertex, and 3! pairs the remaining half-edges between the vertices. The single quartic contribution to the connected four-point function with four labelled external insertions has trivial automorphism group, and its coefficient is $-g$. Vacuum components have already been removed by normalization; the logarithm of the generating functional then selects the component in which all external labels are connected.

15.5 Two-Point Function and Self-Energy

The connected Euclidean two-point function has Fourier transform

$$G_E(x, y) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \mathrm{e}^{\mathrm{i}k\cdot(x-y)}\, \widetilde{G}_E(k).$$

Its perturbative expansion begins as



Definition 15.6 (Two-point one-particle irreducibility)**.** A connected two-point graph is one-particle reducible if it contains an internal edge whose removal disconnects the two external endpoints. It is one-particle irreducible, abbreviated 1PI, if no such edge exists. The amputated reduced value of a 1PI two-point graph is obtained by omitting the two external propagator factors attached to the external endpoints while keeping the loop integrals, vertex factors, and graph automorphism factor.

This is a statement about connected Green-function graphs. Its role here is only to organize the two-point function. The systematic functional definition of the one-particle-irreducible effective action is developed later in Chapter 30.

Definition 15.7 (Euclidean self-energy)**.** Define the Euclidean self-energy $\Sigma_E(k)$ by the Dyson form

$$\widetilde{G}_E(k) = \frac{1}{k^2 + m^2 - \Sigma_E(k)}. \tag{15.1}$$

Equivalently,

$$\widetilde{G}_E = G_{0,E} + G_{0,E}\Sigma_E G_{0,E} + G_{0,E}\Sigma_E G_{0,E}\Sigma_E G_{0,E} + \cdots , \qquad G_{0,E}(k) = \frac{1}{k^2 + m^2}.$$

With this sign convention Σ_E is the sum of amputated connected one-particle-irreducible two-point insertions.

Proposition 15.8 (Dyson decomposition of the connected two-point function)**.** *As a formal power series at fixed regulator, every connected two-point graph decomposes uniquely into a chain of* 1PI *two-point blocks joined by free propagator edges. If* $\Sigma_E(k)$ *denotes the sum of amputated* 1PI *two-point insertions with the sign convention of Definition 15.7, then*

$$\widetilde{G}_E(k) = G_{0,E}(k) + G_{0,E}(k)\Sigma_E(k)G_{0,E}(k) + G_{0,E}(k)\Sigma_E(k)G_{0,E}(k)\Sigma_E(k)G_{0,E}(k) + \cdots .$$

Equivalently,

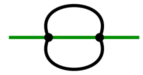
$$\widetilde{G}_E(k) = \frac{1}{G_{0,E}(k)^{-1} - \Sigma_E(k)} = \frac{1}{k^2 + m^2 - \Sigma_E(k)}.$$

Proof. Work at finite regulator and at a fixed order in g, so only finitely many Wick graphs occur. In a connected two-point graph, call an internal edge a separating edge if deleting it disconnects the two external endpoints. The set of separating edges on any path from one external endpoint to the other is linearly ordered by separation: an edge e_1 precedes e_2 if the component containing the left external endpoint after deleting e_1 is contained in the corresponding component after deleting e_2. Cutting all separating edges decomposes the graph into maximal subgraphs with two distinguished attachment half-edges and no further separating internal edge between them. These maximal subgraphs are exactly the 1PI two-point blocks; the cut separating edges are free propagator factors.

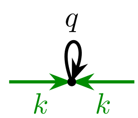
The decomposition is unique because the separating edges are intrinsic to the incidence graph. Conversely, any ordered chain of 1PI two-point blocks joined by free propagators gives a connected two-point graph, and the automorphism factors multiply correctly once the two attachment half-edges of each block are distinguished by their position in the chain. Summing over chains therefore gives the displayed geometric series. Since $G_{0,E}(k) = (k^2 + m^2)^{-1}$, the formal geometric series is

$$G_{0,E}(k) \sum_{r \geq 0} \big(\Sigma_E(k) G_{0,E}(k)\big)^r = \frac{1}{G_{0,E}(k)^{-1} - \Sigma_E(k)},$$

as an identity of formal power series. □

$\Sigma_E(k) =$  $+$  $+$  $+ \cdots .$

One-loop tadpole self-energy and cutoff growth. The order-g tadpole is



$$= \quad -\frac{g}{2} \int \frac{\mathrm{d}^D q}{(2\pi)^D} \frac{1}{q^2 + m^2}.$$

The factor $1/2$ is the symmetry factor of the tadpole graph. This insertion is independent of the external momentum k, because no external momentum flows through the closed loop. In the continuum notation the loop integral is ultraviolet divergent for $D \geq 2$. With a sharp momentum cutoff used only to display the degree of divergence,

$$C_1(\Lambda, m) = \frac{1}{2} \int_{|q| \leq \Lambda} \frac{\mathrm{d}^D q}{(2\pi)^D} \frac{1}{q^2 + m^2}$$

equals

$$\frac{1}{2} \frac{\mathrm{vol}(S^{D-1})}{(2\pi)^D} \int_0^\Lambda \mathrm{d}r \, \frac{r^{D-1}}{r^2 + m^2}, \qquad \mathrm{vol}(S^{D-1}) = \frac{2\pi^{D/2}}{\Gamma(D/2)}.$$

Thus

$$C_1(\Lambda, m) \sim \begin{cases} \dfrac{1}{4\pi} \log \dfrac{\Lambda}{m}, & D = 2, \\ \dfrac{1}{2} \dfrac{\mathrm{vol}(S^{D-1})}{(2\pi)^D} \dfrac{\Lambda^{D-2}}{D-2}, & D \geq 3, \end{cases}$$

up to finite terms and convention-dependent cutoff terms. The first line uses

$$\int_0^\Lambda \frac{r \, \mathrm{d}r}{r^2 + m^2} = \frac{1}{2} \log \frac{\Lambda^2 + m^2}{m^2} = \log \frac{\Lambda}{m} + O(\Lambda^{-2})$$

inside the radial expression. The second line uses $r^{D-1}(r^2 + m^2)^{-1} = r^{D-3} + O(r^{D-5})$ at large r. Hence

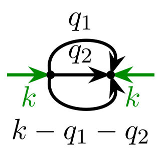
$$\Sigma_E^{(1)}(k) = -g \, C_1(\Lambda, m).$$

The coefficient $-g/2$ is the symmetry factor from the Wick-pairing automorphism count above: the two unattached half-edges at the quartic vertex form one loop and may be interchanged. No external momentum enters the loop propagator, so the amputated insertion is momentum independent and equals the displayed regulated integral. In spherical coordinates the cutoff integral is

$$\frac{1}{2}\frac{\mathrm{vol}(S^{D-1})}{(2\pi)^D}\int_0^{\Lambda}\frac{r^{D-1}\,\mathrm{d}r}{r^2+m^2}.$$

For $D=2$ the integral is $\frac{1}{2}\log((\Lambda^2+m^2)/m^2)=\log(\Lambda/m)+O(\Lambda^{-2})$. For $D\geq 3$, subtract the leading asymptotic term r^{D-3}; the remainder is lower degree at infinity, so the leading growth is $\Lambda^{D-2}/(D-2)$ with the displayed prefactor. The Dyson sign convention of Definition 15.7 then gives $\Sigma_E^{(1)}=-gC_1$.

Two-loop sunset insertion. The order-g^2 sunset insertion is



$$= \frac{g^2}{6}\int\frac{\mathrm{d}^Dq_1\,\mathrm{d}^Dq_2}{(2\pi)^{2D}}$$

$$\times\,\frac{1}{(q_1^2+m^2)(q_2^2+m^2)\big((k-q_1-q_2)^2+m^2\big)}.$$

This formula is a Euclidean integral. Its Lorentzian boundary values are obtained by analytic continuation with contour-deformation control; for timelike external momentum the three-particle Landau threshold of this graph prevents a naive independent Wick rotation of the loop energies.

The graph has two quartic vertices and three internal lines joining them. After the two external half-edges are fixed, the remaining three internal edges may be permuted, giving the automorphism factor 3!. Thus the coefficient is $g^2/6$, in agreement with the Wick-pairing count in the preceding section. Momentum conservation at the two vertices leaves two independent loop momenta, chosen as q_1, q_2; the third internal momentum is then $k-q_1-q_2$. Multiplication of the three Euclidean propagators gives the displayed integral. The final statement follows because the Euclidean integral is defined on real D-momenta, while Lorentzian boundary values are obtained by continuing external energy and deforming loop-energy contours without crossing singularities. At the three-particle threshold the Landau equations allow simultaneous on-shell internal lines, so this deformation cannot be represented as an independent rotation of each loop energy.

15.6 Pole Mass and Counterterm Bookkeeping

Definition 15.9 (Pole mass of an interpolating scalar field)**.** As derived in Chapter 14, the spectral representation of the Euclidean two-point function has the form

$$\widetilde{G}_E(k)=\frac{Z}{k^2+m_*^2}+\int_{[0,\infty)}\frac{\mathrm{d}\rho_{\mathrm{c}}(s)}{k^2+s},$$

with an isolated atom at $s=m_*^2$ when the chosen field has nonzero overlap with a stable one-particle state. The parameter m_* is the pole mass read from the two-point function. Equivalently, singularities of the Euclidean function in the complex k^2-plane lie at $k^2=-s$, and an isolated one-particle state gives the nearest such pole when no lighter state couples to the field. At fixed spatial momentum, the same pole appears in the complex Euclidean energy plane at

$$k_D=\pm\mathrm{i}\sqrt{\vec{k}^{\,2}+m_*^2}.$$

The field variable ϕ appearing in the regulated Lagrangian and in the source J is the perturbative representative of an interpolating local field. Its renormalized two-point function is the full function $\widetilde{G}_E(k)$, with pole residue Z and continuum spectral contributions. The asymptotic scalar field used in scattering theory is the free field on the isolated one-particle asymptotic Fock space with pole mass m_* and unit residue. Perturbation theory in this chapter locates the pole and residue of the interpolating-field Green function; the LSZ theorem later uses those data to project external legs onto the asymptotic one-particle states.

One-loop pole shift and mass-coordinate prescription. Using

$$\widetilde{G}_E(k) = \frac{1}{k^2 + m^2 - \Sigma_E(k)}$$

and the tadpole value above, the pole position satisfies, to first order,

$$m_*^2 = m^2 + gC_1(\Lambda, m) + O(g^2)$$

in this cutoff scheme. The parameter m is therefore a parameter of the regulated quadratic action. Its relation to the spectral pole mass depends on the regulator and on the prescription used to define the finite theory. At fixed regulator, g is a bookkeeping parameter for the coefficientwise expansion. The coefficient $C_1(\Lambda, m)$ grows with the cutoff in $D \geq 2$, so the product $gC_1(\Lambda, m)$ is not uniformly small as $\Lambda \to \infty$. Thus the distinction between a cutoff-dependent quadratic action coordinate and a mass defined by spectral data is operational in the cutoff-dependent calculation.

A perturbative mass prescription introduces a renormalized parameter m_R and a counterterm δm^2:

$$m^2 = m_R^2 + \delta m^2, \qquad \delta m^2 = \sum_{r \geq 1} g^r (\delta m^2)^{(r)}.$$

The Euclidean Lagrangian density is written as

$$\mathcal{L}_E = \frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m_R^2\phi^2 + \frac{g}{4!}\phi^4 + \frac{1}{2}\delta m^2\phi^2.$$

The counterterm coefficient is part of this same regulated Lagrangian, written in the renormalized coordinate m_R. In perturbative bookkeeping it is treated as part of the interaction. At order g, the two-valent counterterm vertex contributes

 $\qquad\longleftrightarrow\qquad -(\delta m^2)^{(1)} g.$

The condition $m_* = m_R + O(g^2)$, written at the level of squared masses, requires

$$(\delta m^2)^{(1)} = -C_1(\Lambda, m_R) + \text{finite prescription term}.$$

Different finite terms correspond to different renormalization prescriptions. The split into m_R^2 and δm^2 is a prescription-dependent reorganization of the regulated perturbative calculation. Its role is to make the cancellation of cutoff-dependent local terms occur coefficient by coefficient in the expansion in g. The physical statement is made after Green functions are expressed in terms of the chosen finite parameters, such as the pole mass m_*. In the on-shell mass prescription the finite terms are chosen so that $m_R = m_*$ order by order.

To first order, insert $\Sigma_E^{(1)}(k) = -gC_1(\Lambda, m)$ from the tadpole calculation above into the Dyson denominator:

$$k^2 + m^2 - \Sigma_E(k) = k^2 + m^2 + gC_1(\Lambda, m) + O(g^2).$$

The pole in the variable k^2 is therefore at $k^2 = -m^2 - gC_1(\Lambda, m) + O(g^2)$, equivalently $m_*^2 = m^2 + gC_1(\Lambda, m) + O(g^2)$. If the quadratic coordinate of the regulated action is rewritten as $m_R^2 + \delta m^2$, the counterterm is a two-valent interaction vertex with the sign displayed above. Requiring the pole to sit at m_R^2 through order g forces the order-g part of δm^2 to cancel $C_1(\Lambda, m_R)$, up to the chosen finite renormalization convention.

15.7 Connected Four-Point Function

Remark 15.10 (Four-point moment-cumulant decomposition). For four insertions, the normalized Green function decomposes into pairwise products of two-point functions and a connected part:

$$\begin{aligned} &G_{E,4}(x_1, x_2, x_3, x_4) \\ &\quad = G_E(x_1, x_2)G_E(x_3, x_4) + G_E(x_1, x_3)G_E(x_2, x_4) + G_E(x_1, x_4)G_E(x_2, x_3) \\ &\qquad + G_{E,4}^{\text{conn}}(x_1, x_2, x_3, x_4). \end{aligned}$$

The identity is not a separate theorem: applying the moment-cumulant identity to four fields gives it directly. If the theory is $\mathbb{Z}_2$-invariant, as the regulated ϕ^4 theory is, all odd cumulants vanish. The partitions of four labels therefore contribute either one four-label cumulant or one of the three partitions into two pairs. The pair cumulants are the full two-point functions G_E, and the four-label cumulant is, by definition, $G_{E,4}^{\text{conn}}$.

Figure 15.3 displays the three pair partitions and the connected four-point cumulant that appear in the moment-cumulant identity.

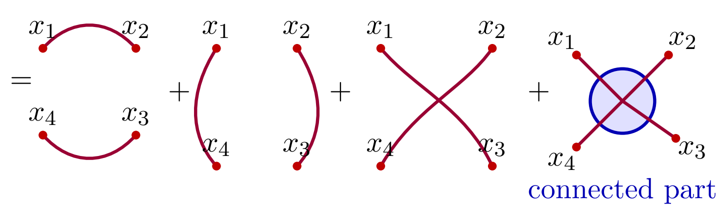


Figure 15.3: Disconnected and connected parts of the Euclidean four-point function. The three Wick pairings are separated from $G_{E,4}^{\text{conn}}$.

The Fourier transform of the connected part is defined by

$$G_{E,4}^{\text{conn}}(x_1, \ldots, x_4) = \int \prod_{i=1}^{4} \frac{\mathrm{d}^D k_i}{(2\pi)^D} \, \mathrm{e}^{\mathrm{i} \sum_i k_i \cdot x_i} \, \widetilde{G}_{E,4}^{\text{conn}}(k_1, \ldots, k_4).$$

To leading order,

$$\begin{aligned} \widetilde{G}_{E,4}^{\text{conn}}(k_1, \ldots, k_4) &= (2\pi)^D \delta^{(D)}(k_1 + k_2 + k_3 + k_4) \\ &\qquad \times \left[-g \prod_{i=1}^{4} \frac{1}{k_i^2 + m^2} + O(g^2) \right]. \end{aligned}$$

This is a connected Green function with external propagators included. Later, after scattering states and LSZ have been constructed, the operation of removing the external propagators will acquire its scattering interpretation. The order-g connected four-point graph has one quartic vertex and four labelled external endpoints. The $1/4!$ in the interaction is cancelled by the $4!$ assignments of the four labelled external fields to the four half-edges of the vertex, so the vertex coefficient is $-g$. Translation invariance supplies the momentum-conserving delta function. Each external field remains connected to the vertex by a Euclidean free propagator, giving $\prod_i(k_i^2 + m^2)^{-1}$. Since this is a Green function rather than an amputated scattering amplitude, those external propagators are part of the coefficient.

Definition 15.11 (Euclidean amputation of connected four-point Green functions)**.** Away from the free Euclidean pole hypersurfaces $k_i^2 + m^2 = 0$, define the amputated connected Euclidean four-point kernel by the algebraic operation

$$\widetilde{\Gamma}_{E,4}(k_1, k_2, k_3, k_4) := \left(\prod_{i=1}^{4}(k_i^2 + m^2)\right) \widetilde{G}^{\text{conn}}_{E,4}(k_1, k_2, k_3, k_4).$$

This definition removes the external free propagator factors from the connected Green function. It is not a definition of a scattering amplitude: scattering requires the nonperturbative one-particle subspace, asymptotic states, and the LSZ limiting operation developed later.

Tree-level four-point kernel. With the convention of Definition 15.11, the leading amputated connected four-point kernel is

$$\widetilde{\Gamma}_{E,4}(k_1, k_2, k_3, k_4) = -(2\pi)^D g\, \delta^{(D)}(k_1 + k_2 + k_3 + k_4) + O(g^2).$$

This is obtained by multiplying the leading connected Green function in the formula above by $\prod_i(k_i^2 + m^2)$. The four free external propagators cancel coefficientwise, leaving the momentum-conserving delta distribution and the quartic vertex coefficient $-g$.

Chapter 16

Lorentzian Green Functions and Analytic Continuation

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

16.1 Time-Ordered Boundary Values

Definition 16.1 (Lorentzian scalar vacuum-sector datum)**.** Let $x^\mu = (x^0, \vec{x}) \in \mathbb{R}^{1,D-1}$ and use the mostly-plus metric

$$\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+), \qquad k^2 = \eta_{\mu\nu}k^\mu k^\nu.$$

Write $M = \mathbb{R}^{1,D-1}$. The framework in this chapter is a scalar Wightman field presentation in a vacuum sector: a Hilbert space $\mathcal{H}$, a vacuum vector Ω, a strongly continuous unitary representation of translations with self-adjoint generators P^μ whose joint spectrum lies in the closed forward cone

$$\overline{V}_+ = \{p \in M : p^0 \geq 0,\ p^2 \leq 0\},$$

and a scalar operator-valued distribution $\widehat{\phi}$ defined on a common invariant dense domain. Point-field notation denotes the kernel notation for smeared operator-valued distributions.

For $r \geq 1$, $\mathcal{S}(M^r)$ denotes the Schwartz space on M^r, and $\mathcal{S}'(M^r)$ its continuous dual. This chapter works in the tempered distribution category:

$$W_r,\ G_r \in \mathcal{S}'(M^r).$$

Every compactly supported smearing used to define a local field matrix element is also a Schwartz test function, so this convention is compatible with the local operator-valued-distribution notation. After translation invariance is used, analytic boundary values are paired with test functions in the relative variables. Thus a holomorphic function F on a tube over M^{r-1} has boundary value $T \in \mathcal{S}'(M^{r-1})$ when, for every $f \in \mathcal{S}(M^{r-1})$,

$$\lim_{y \to 0} \int_{M^{r-1}} \mathrm{d}^{D(r-1)}\xi \, F(\xi - \mathrm{i}y) f(\xi) = \langle T, f \rangle,$$

with y approaching the real edge inside the specified tube cone.

Definition 16.2 (Wightman and time-ordered distributions)**.** The Wightman n-point distribution is

$$W_n(x_1, \ldots, x_n) = \langle \Omega | \widehat{\phi}(x_1) \cdots \widehat{\phi}(x_n) | \Omega \rangle \in \mathcal{S}'(M^n).$$

For a permutation $\pi \in S_n$, set

$$M^n_\pi = \{(x_1, \ldots, x_n) : x^0_{\pi(1)} > \cdots > x^0_{\pi(n)}\}.$$

A vacuum time-ordered Green function for this field is a distribution

$$G_n \in \mathcal{S}'(M^n)$$

whose restriction to M^n_π is

$$G_n(x_1, \ldots, x_n)\big|_{M^n_\pi} = \langle \Omega | \widehat{\phi}(x_{\pi(1)}) \cdots \widehat{\phi}(x_{\pi(n)}) | \Omega \rangle.$$

Equivalently, on M^n_π away from coincident insertion points,

$$G_n(x_1, \ldots, x_n) = \langle \Omega | \widehat{\phi}(x_{\pi(1)}) \cdots \widehat{\phi}(x_{\pi(n)}) | \Omega \rangle.$$

Local commutativity makes the restrictions compatible across spacelike equal-time boundaries. A choice of G_n is therefore an extension across partial diagonals $x_{a_1} = \cdots = x_{a_r}$. The possible differences between two such extensions are local distributions supported on these diagonals. In perturbation theory this extension datum is fixed together with the renormalization prescription for local products. The compact operator notation

$$G_n(x_1, \ldots, x_n) = \langle \Omega | \, \mathrm{T}\{\widehat{\phi}(x_1) \cdots \widehat{\phi}(x_n)\} | \Omega \rangle$$

refers to this specified distribution.

Proposition 16.3 (Compatibility of time-ordered Wightman pieces)**.** *Let G°_n denote the piecewise distribution on the complement of the partial diagonals $x_a = x_b$, obtained by assigning the Wightman ordering corresponding to each strict time-ordering chamber M^n_π. If $\widehat{\phi}$ obeys scalar microcausality,*

$$[\widehat{\phi}(f), \widehat{\phi}(g)] = 0$$

whenever $\operatorname{supp} f$ *and* $\operatorname{supp} g$ *are spacelike separated, then the definitions on adjacent chambers agree across a common spacelike boundary. Thus G°_n is a well-defined distribution on*

$$M^n \setminus \bigcup_{a<b} \{x_a = x_b\}.$$

Proof. Let $U \subset M^n \setminus \bigcup_{a<b}\{x_a = x_b\}$ be a sufficiently small product coordinate neighborhood. The equal-time hyperplanes cut U into finitely many strict time-ordering chambers. If two such chambers share a codimension-one wall in U, then only two labels, say a, b, have been exchanged; on the wall $x_a^0 = x_b^0$, while $x_a \neq x_b$ because partial diagonals have been removed. After shrinking U, the projections of U to the two M-factors keep the supports of the corresponding localized test functions spacelike separated. For such test functions, microcausality gives

$$\widehat{\phi}(f_a)\widehat{\phi}(f_b) = \widehat{\phi}(f_b)\widehat{\phi}(f_a)$$

on the common invariant domain of local-field products. All remaining field operators in the smeared product are unchanged, so inserting this equality at the adjacent positions in the Wightman matrix element shows that the two chamber formulae have the same restriction to U.

Higher-codimension walls, where several time coordinates coincide but no two points coincide in spacetime, are handled by the same local argument. The permutations labeling the chambers meeting such a wall form a parabolic subgroup of the symmetric group, generated by adjacent transpositions of labels whose equal-time points remain spacelike separated in U. Since each generator leaves the smeared vacuum matrix element unchanged, any two chamber prescriptions meeting U agree on their common restriction. Thus the Wightman pieces form a compatible family on the chamber cover of the off-diagonal open set. Distributions form a sheaf: a compatible family of local distributions has a unique glued distribution. The glued object is the claimed G_n°. □

Proposition 16.4 (Contact-term freedom of the time-ordered extension)**.** *Let G_n and G_n' be two tempered extensions of the same off-diagonal time-ordered distribution G_n° to M^n. Then*

$$G_n - G_n'$$

is supported on the union of partial diagonals

$$\bigcup_{\mathcal{P}} \{x_a = x_b \text{ whenever } a, b \text{ lie in the same block of } \mathcal{P}\},$$

where $\mathcal{P}$ ranges over partitions of $\{1, \dots, n\}$ with at least one block of size ≥ 2. Locally near a fixed diagonal, this difference is a finite sum of derivatives of delta distributions in the normal coordinates, with distributional coefficients along the diagonal. In a translation-invariant local renormalization prescription these coefficients are local polynomials in derivatives acting on delta functions, with finite coefficients fixed by the chosen extension convention.

Proof. The two extensions agree by hypothesis on the open set

$$M^n \setminus \bigcup_{a<b}\{x_a = x_b\}.$$

Let $T = G_n - G_n'$. If a test function is supported in this open set, both extensions evaluate it by G_n°, hence T vanishes on such test functions. Therefore

$$\operatorname{supp} T \subset \bigcup_{a<b}\{x_a = x_b\}.$$

Refining this closed cover by the coincidence pattern of the labels gives the displayed union over partitions $\mathcal{P}$: the block structure records exactly which variables are forced to coincide.

For the local normal form, fix one such stratum and choose coordinates (u, v), where u are coordinates along the diagonal and v are normal relative coordinates. A distribution supported on $v = 0$ has finite transverse order on every compact coordinate chart. Equivalently, there are an integer N and distributions T_α along the diagonal such that locally

$$\sum_{|\alpha|\leq N} T_\alpha(u)\, \partial_v^\alpha \delta(v).$$

The finite order follows from the seminorm estimate defining continuity of a distribution on compactly supported tests; Taylor expansion in the normal variables reduces the action of T to finitely many normal derivatives at $v = 0$. Translation invariance removes dependence on the common center-of-mass coordinate in flat space. Locality of the renormalization prescription further restricts the coefficient distributions to derivatives of delta functions in relative diagonal variables, with numerical coupling-dependent coefficients. These supported delta-derivative contributions are precisely the contact terms. □

Free scalar Feynman boundary value. For the free scalar field of mass $m > 0$, the two-point time-ordered Green function has Fourier representation

$$\Delta_F(x; m^2) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2 + m^2 - \mathrm{i}\epsilon}, \qquad \epsilon > 0.$$

At fixed $\vec{k}$, with

$$\omega_{\vec{k}} = \sqrt{\vec{k}^{\,2} + m^2},$$

the poles in the complex k^0-plane are

$$k^0 = -\omega_{\vec{k}} + \mathrm{i}0, \qquad k^0 = +\omega_{\vec{k}} - \mathrm{i}0.$$

This is the Feynman boundary value of the meromorphic function of k^0. The positive-frequency Wightman two-point function is

$$\Delta_+(x; m^2) = \int \frac{\mathrm{d}^{D-1}\vec{k}}{(2\pi)^{D-1} 2\omega_{\vec{k}}}\, \mathrm{e}^{-\mathrm{i}\omega_{\vec{k}} x^0 + \mathrm{i}\vec{k}\cdot\vec{x}}.$$

The time-ordered distribution is $\theta(x^0)\Delta_+(x) + \theta(-x^0)\Delta_+(-x)$, with the value on $x^0 = 0$ fixed as a distributional extension. For $x^0 > 0$, close the k^0-contour of

$$-\mathrm{i} \int \frac{\mathrm{d}k^0}{2\pi} \frac{\mathrm{e}^{-\mathrm{i}k^0 x^0}}{-(k^0)^2 + \omega_{\vec{k}}^2 - \mathrm{i}0}$$

in the lower half-plane; the pole at $+\omega_{\vec{k}} - \mathrm{i}0$ gives $\mathrm{e}^{-\mathrm{i}\omega_{\vec{k}} x^0}/(2\omega_{\vec{k}})$. For $x^0 < 0$, close the contour in the upper half-plane; the pole at $-\omega_{\vec{k}} + \mathrm{i}0$ gives $\mathrm{e}^{+\mathrm{i}\omega_{\vec{k}} x^0}/(2\omega_{\vec{k}})$. Combining the two time regions yields the displayed Fourier representation. The pole placement is therefore not an independent rule: it is the boundary-value encoding of time ordering and positive energy.

16.2 Tube Analyticity From the Spectrum Condition

Proposition 16.5 (Forward-tube analyticity from the spectrum condition)**.** *The* $i\epsilon$ *prescription is the two-point perturbative expression of a more general analytic fact. Let*

$$W_n(x_1,\ldots,x_n) = \langle\Omega|\widehat{\phi}(x_1)\cdots\widehat{\phi}(x_n)|\Omega\rangle$$

be a Wightman distribution. By translation invariance write it in the relative variables

$$\xi_j = x_j - x_{j+1}, \qquad j = 1,\ldots,n-1.$$

Let

$$w_n(\xi_1,\ldots,\xi_{n-1}) \in \mathcal{S}'(M^{n-1})$$

be the corresponding relative-coordinate distribution. Its Fourier transform $\widetilde{w}_n$ *is defined with the convention*

$$w_n(\xi) = \int \frac{\mathrm{d}^{D(n-1)}p}{(2\pi)^{D(n-1)}}\,\mathrm{e}^{\mathrm{i}\sum_j p_j\cdot\xi_j}\,\widetilde{w}_n(p)$$

in distribution notation. Equivalently, for $f \in \mathcal{S}(M^{n-1})$*,*

$$\langle w_n, f\rangle = \left\langle \widetilde{w}_n, \check{f}\right\rangle, \qquad \check{f}(p) = \int \frac{\mathrm{d}^{D(n-1)}\xi}{(2\pi)^{D(n-1)}}\,\mathrm{e}^{\mathrm{i}\sum_j p_j\cdot\xi_j}\,f(\xi).$$

The spectrum condition gives the support inclusion

$$\operatorname{supp}\widetilde{w}_n \subset \overline{V}_+^{\,n-1}.$$

For $\eta = (\eta_1,\ldots,\eta_{n-1}) \in V_+^{n-1}$*, the function* $\exp(\sum_j p_j\cdot\eta_j)$ *decreases exponentially on every closed subcone of* $\overline{V}_+^{\,n-1}$ *which contains the support of* $\widetilde{w}_n$*. Multiplication of* $\widetilde{w}_n$ *by this function is therefore a well-defined tempered distribution after inserting any smooth cutoff equal to one on a conic neighborhood of the support; the result is independent of the cutoff. The inverse Fourier transform is a holomorphic function of*

$$z_j = \xi_j - \mathrm{i}\eta_j, \qquad \eta_j \in V_+, \qquad j = 1,\ldots,n-1, \tag{16.1}$$

where V_+ *is the open future light cone. In distribution notation this Fourier-Laplace transform is written as*

$$\mathcal{W}_n(z) = \frac{1}{(2\pi)^{D(n-1)}}\left\langle \widetilde{w}_n(p), \exp\left\{\mathrm{i}\sum_{j=1}^{n-1} p_j\cdot z_j\right\}\right\rangle.$$

Indeed, for $p \in \overline{V}_+$ *and* $\eta \in V_+$*, the mostly-plus inner product satisfies*

$$p\cdot\eta \le 0,$$

so the factor

$$\exp\{\mathrm{i}p\cdot(\xi - \mathrm{i}\eta)\} = \exp\{\mathrm{i}p\cdot\xi + p\cdot\eta\}$$

is damped rather than enlarged. The boundary value is w_n *in* $\mathcal{S}'(M^{n-1})$*:*

$$\lim_{\eta\to 0,\ \eta_j\in V_+} \int_{M^{n-1}} \mathrm{d}^{D(n-1)}\xi\,\mathcal{W}_n(\xi - \mathrm{i}\eta)f(\xi) = \langle w_n, f\rangle, \qquad f \in \mathcal{S}(M^{n-1}).$$

Proof. Insert a complete spectral resolution of the translation generators between successive fields. In relative coordinates the Fourier variables are the successive momentum transfers carried by these intermediate states. The joint spectral condition therefore implies $\operatorname{supp}\widetilde{w}_n \subset \overline{V}_+^{n-1}$. If $\eta_j \in V_+$ and $p_j \in \overline{V}_+$, the mostly-plus metric gives $p_j \cdot \eta_j \le 0$, with strict negativity away from the boundary unless one vector is null and parallel to the other. Hence the factor $\exp(\sum_j p_j \cdot \eta_j)$ supplies exponential damping on every closed subcone inside the tube direction.

For fixed $\eta \in V_+^{n-1}$, multiplication of the tempered distribution $\widetilde{w}_n$ by the smooth function $\exp(\sum_j p_j \cdot \eta_j)$ is defined by first inserting a conic cutoff equal to one near the support; two such cutoffs differ away from the support and therefore give the same product. Differentiation with respect to z_j brings down polynomial factors in p_j, which remain valid multipliers of tempered distributions after the same exponential damping. Thus the Fourier-Laplace transform is holomorphic in the forward tube. Letting $\eta \downarrow 0$ inside the tube removes the damping multiplier and recovers the original Fourier transform in the topology of tempered distributions, which is exactly the stated boundary value. $\square$

Different operator orderings give holomorphic functions in corresponding permuted tubes. Local commutativity identifies their boundary values on real regions where the exchanged neighboring points are spacelike separated.

Theorem 16.6 (Distributional edge-of-the-wedge statement used here)**.** *Let $U \subset \mathbb{R}^N$ be open, and let C_+ and C_- be open convex cones in $\mathbb{R}^N$ whose convex hull is all of $\mathbb{R}^N$. Suppose $F_\pm$ are holomorphic on the tubes $U + \mathrm{i}C_\pm$. Assume the following polynomial boundary growth: for every compact $K \subset U$ and every closed cone $C'_\pm$ whose nonzero vectors lie in $C_\pm$, there are constants $A, N < \infty$ such that*

$$|F_\pm(x + \mathrm{i}y)| \le A(1 + |x|)^N |y|^{-N}, \qquad x \in K, \quad y \in C'_\pm, \quad 0 < |y| < 1.$$

These estimates define distributional boundary values

$$b_\pm \in \mathcal{D}'(U).$$

If $b_+ = b_-$ on a nonempty open real subset $O \subset U$, then there is a connected complex neighborhood $\mathcal{N}$ of O and a holomorphic function F on $\mathcal{N}$ whose restrictions agree with F_+ and F_- wherever the original tubes meet $\mathcal{N}$.

Proof. This is analytic infrastructure from several complex variables and distribution theory; the proof is included here to make clear exactly which part is analytic and which part is supplied by Wightman locality. The convex-hull hypothesis is part of the full-neighborhood form of the theorem. In the more general wedge theorem the continuation lies in the wedge with imaginary cone $\operatorname{conv}(C_+ \cup C_-)$; a genuine complex neighborhood of the real edge is obtained when this cone contains all imaginary directions. The permuted Wightman tubes used at a Jost configuration supply precisely this two-sided normal data after restricting to the relevant spacelike edge.

First, the polynomial tube bounds put each normal translate

$$F_{\pm,\epsilon v} : x \longmapsto F_\pm(x + \mathrm{i}\epsilon v), \qquad v \in C_\pm,$$

in a fixed finite-order distribution class on every $K \Subset U$. The point is not the pointwise bound itself, which may diverge as $\epsilon \downarrow 0$, but the combination of that bound with Cauchy's formula.

If m is larger than the normal growth order, Cauchy's integral formula in a polydisc of radius comparable to ϵ expresses $F_{\pm,\epsilon v}$ paired with $\varphi \in C_c^\infty(K)$ as a finite sum of pairings of $F_\pm(\cdot + \mathrm{i}\epsilon' v')$ with derivatives of φ, where ϵ' remains comparable to ϵ and v' remains in a proper subcone. Integrating by parts transfers the singular normal powers to finitely many derivatives of φ. Thus, for some integer M_K,

$$\left|\langle F_{\pm,\epsilon v}, \varphi\rangle\right| \le C_{K,v} \sum_{|\alpha|\le M_K} \sup_{x\in K} |\partial^\alpha \varphi(x)| \qquad (0 < \epsilon < \epsilon_0).$$

Weak compactness in this finite-order dual gives distributional accumulation points. Holomorphy makes the accumulation point unique: if $v_0, v_1 \in C_\pm$, integrate $\bar\partial F_\pm = 0$ over the truncated two-real-dimensional surface $\{x + \mathrm{i}\epsilon((1-r)v_0 + rv_1) : 0 < \epsilon < \epsilon_0,\ 0 < r < 1\}$, paired with a test density in x. The side contributions vanish as the truncation shrinks, and the two normal boundary terms are the two candidate limits. Hence the limit is independent of the approach direction and defines the boundary distribution $b_\pm \in \mathcal{D}'(U)$.

The gluing step is local. Choose a ball $B \Subset O$. In one complex variable, take a test $(1,1)$-form supported in a small disc crossing the real edge and define a distribution by using F_+ in the upper wedge and F_- in the lower wedge. Pairing its $\bar\partial$ with the test form is the limit of the contour integral over a rectangle whose horizontal sides lie at heights $\pm\epsilon$. The holomorphic interiors contribute zero. The two horizontal boundary terms converge to $\langle b_+, \varphi\rangle - \langle b_-, \varphi\rangle$, with the orientation of the lower side supplying the minus sign. Since $b_+ = b_-$ on B, the glued distribution has $\bar\partial = 0$. The Weyl lemma for the $\bar\partial$-operator, equivalently Morera's theorem in distribution form, therefore represents it by a holomorphic function across the edge.

For $N > 1$, apply this one-variable argument to complex lines whose real directions are tangent to B and whose imaginary directions lie in $\mathrm{conv}(C_+ \cup C_-)$. The same boundary-jump cancellation gives holomorphic continuation on each line. Separate holomorphy on the resulting family of line slices and Hartogs' theorem give joint holomorphy on the local wedge generated by the convex hull. Bochner's tube theorem then identifies the envelope of holomorphy of this local tube with the tube over that convex hull. Under the hypothesis $\mathrm{conv}(C_+ \cup C_-) = \mathbb{R}^N$, the envelope contains a complex neighborhood of B. Gluing the local holomorphic extensions over overlapping balls in O gives the connected neighborhood $\mathcal{N}$.

The pure analytic inputs in this last paragraph are Hartogs' theorem, the $\bar\partial$ Weyl lemma, and Bochner's tube theorem. The QFT-specific burden in this chapter is the verification of the hypotheses that enter the analytic theorem: tube growth from the spectrum condition and equality of boundary distributions from local commutativity. □

Verification of the real-edge equality in the Wightman case. Fix two adjacent field labels a, b and let $O_{ab} \subset M^n$ be an open real set on which $x_a - x_b$ is spacelike and no partial diagonal $x_i = x_j$ is met. Let $W_{\dots ab \dots}$ and $W_{\dots ba \dots}$ denote the two Wightman distributions that differ only by interchanging the adjacent fields $\widehat{\phi}(x_a)$ and $\widehat{\phi}(x_b)$. If $f \in C_c^\infty(O_{ab})$, a partition of unity subordinate to sufficiently small product neighborhoods writes the smeared matrix element as a finite sum in which the a- and b-smearings have spacelike separated supports. Scalar microcausality gives

$$\widehat{\phi}(f_a)\widehat{\phi}(f_b) = \widehat{\phi}(f_b)\widehat{\phi}(f_a)$$

on the common invariant domain of the field products appearing in the vacuum matrix element. All other smeared fields are unchanged, hence

$$\langle W_{\cdots ab\cdots} - W_{\cdots ba\cdots}, f\rangle = 0, \qquad f \in C_c^\infty(O_{ab}).$$

The boundary values of the two corresponding tube functions therefore agree as distributions on O_{ab}. For fields with a $\mathbb{Z}_2$-grading, the same argument gives the Koszul sign determined by moving the two homogeneous fields past one another. This is the real-edge input used by the distributional edge-of-the-wedge theorem at Jost configurations.

The theorem therefore supplies the common holomorphic continuation of the permuted tube functions across those spacelike real regions. Time-ordered Green functions are boundary values obtained by approaching real Lorentzian points from imaginary time orderings compatible with the chosen operator order. For two points, this general construction is exactly the pole displacement

$$\frac{1}{k^2 + m^2 - \mathrm{i}\epsilon}$$

appearing in the Feynman propagator.

16.3 Boundary-Value Dictionary

Definition 16.7 (Ordered Lorentzian and Euclidean boundary values)**.** The preceding tube statement gives a precise dictionary among Wightman, time-ordered, and Euclidean functions. Fix an ordering π and real Lorentzian points with

$$x^0_{\pi(1)} > x^0_{\pi(2)} > \cdots > x^0_{\pi(n)}.$$

Choose positive numbers

$$\epsilon_1 > \epsilon_2 > \cdots > \epsilon_n > 0$$

and set

$$z_{\pi(j)} = x_{\pi(j)} - \mathrm{i}\epsilon_j e_0, \qquad e_0 = (1, \vec{0}).$$

Then

$$z_{\pi(j)} - z_{\pi(j+1)} = x_{\pi(j)} - x_{\pi(j+1)} - \mathrm{i}(\epsilon_j - \epsilon_{j+1})e_0$$

lies in the forward tube for the j-th relative variable. The time-ordered boundary value on the region M^n_π is

$$G_n(x_1, \ldots, x_n)\big|_{M^n_\pi} = \lim_{\epsilon_1 > \cdots > \epsilon_n \downarrow 0} \mathcal{W}_{n,\pi}(z_{\pi(1)}, \ldots, z_{\pi(n)}),$$

where $\mathcal{W}_{n,\pi}$ denotes the holomorphic continuation of the Wightman ordering displayed by π. The limit is a distributional boundary value: it is taken after pairing with a Schwartz test function in the relative variables.

Euclidean Schwinger functions arise from the same holomorphic functions by restriction to imaginary time. For Euclidean points

$$x_{E,a} = (\tau_a, \vec{x}_a),$$

with

$$\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(n)},$$

define

$$z^0_{\pi(j)} = -\mathrm{i}\tau_{\pi(j)}, \qquad \vec{z}_{\pi(j)} = \vec{x}_{\pi(j)}.$$

Then

$$\begin{aligned} &S_n(x_{E,1}, \ldots, x_{E,n})\big|_{\tau_{\pi(1)} > \cdots > \tau_{\pi(n)}} \\ &\quad = \mathcal{W}_{n,\pi}(z_{\pi(1)}, \ldots, z_{\pi(n)}). \end{aligned}$$

Source sign under Wick rotation. The same dictionary fixes the sign in source functionals. Let

$$Z_L^{(+)}[J] = \langle\Omega|\, \mathrm{T}\exp\left(\mathrm{i}\int_{\mathbb{R}^{1,D-1}} J(x)\widehat{\phi}(x)\,\mathrm{d}^D x\right)|\Omega\rangle$$

denote the Lorentzian source convention in which functional derivatives produce factors of i. Its opposite-source convention is

$$Z_L^{(-)}[J] = Z_L^{(+)}[-J].$$

To compare with a Euclidean source one must specify the contour, because a real Euclidean test function and a real Lorentzian test function live on different real slices. Let C_- be the imaginary-time contour

$$z^0 = -\mathrm{i}\tau, \qquad \mathrm{d}z^0 = -\mathrm{i}\,\mathrm{d}\tau,$$

with the orientation inherited from increasing τ, and let J_C be a source density defined on this contour. Term by term in the source expansion, or in a regulated theory where the contour-ordered source exponential is an operator-valued analytic function of the source, the source part of the exponent is

$$\mathrm{i}\int_{C_-} \mathrm{d}z^0\,\mathrm{d}^d\vec{x}\,J_C(z^0,\vec{x})\widehat{\phi}(z^0,\vec{x}) = \int_{\mathbb{R}} \mathrm{d}\tau\,\mathrm{d}^d\vec{x}\,J_C(-\mathrm{i}\tau,\vec{x})\widehat{\phi}_E(\tau,\vec{x}).$$

Thus the Euclidean convention

$$Z_E[J_E] = \langle\Omega|T_E\exp\left(\int J_E(\tau,\vec{x})\widehat{\phi}_E(\tau,\vec{x})\,\mathrm{d}\tau\,\mathrm{d}^d\vec{x}\right)|\Omega\rangle$$

is the C_- boundary value of $Z_L^{(+)}$ with

$$J_E(\tau,\vec{x}) = J_C(-\mathrm{i}\tau,\vec{x}).$$

No additional minus sign is produced in this convention; the factor i in the Lorentzian source and the Jacobian $-\mathrm{i}$ of the lower-half-plane Wick contour multiply to $+1$. If one instead rotates through the opposite contour C_+, $z^0 = +\mathrm{i}\tau$, then $\mathrm{d}z^0 = \mathrm{i}\,\mathrm{d}\tau$ and the same Lorentzian convention gives

$$\mathrm{i}\int_{C_+} \mathrm{d}z^0\,J_C\widehat{\phi} = -\int \mathrm{d}\tau\,J_C(\mathrm{i}\tau)\widehat{\phi}_E.$$

In that contour convention the Euclidean functional with $+\int J_E\phi_E$ is represented by $Z_L^{(+)}[-J_C]$, equivalently by $Z_L^{(-)}[J_C]$, after setting $J_E(\tau,\vec{x}) = -J_C(\mathrm{i}\tau,\vec{x})$. The two formulas describe different contour/source-sign conventions, not two different Green functions. Expand the contour-ordered exponential in powers of the source. Each Lorentzian source insertion carries the factor $\mathrm{i}\,\mathrm{d}z^0$. On the lower imaginary-time contour $z^0 = -\mathrm{i}\tau$, the one-dimensional volume form is $\mathrm{d}z^0 = -\mathrm{i}\,\mathrm{d}\tau$, and therefore $\mathrm{i}\,\mathrm{d}z^0 = +\mathrm{d}\tau$. The source term becomes $+\int J_E\phi_E$ after the identification

$J_E(\tau, \vec{x}) = J_C(-\mathrm{i}\tau, \vec{x})$. On the upper contour $z^0 = +\mathrm{i}\tau$, one instead has $\mathrm{i}\,\mathrm{d}z^0 = -\mathrm{d}\tau$, so the same Lorentzian source convention produces the opposite Euclidean sign unless the source is replaced by its negative. The calculation is termwise in the regulated source expansion, hence fixes all Euclidean n-point signs.

Conversely, an OS-admissible Schwinger hierarchy reconstructs Wightman boundary values by the positive-time quotient and analytic-continuation theorem of Chapter 12. Thus the boundary-value dictionary has two controlled entrances: spectral Wightman data with tube analyticity, and Euclidean data satisfying the OS positivity, covariance, regularity, and growth hypotheses.

Two-point Feynman prescription from ordered tubes. For $n = 2$, the dictionary gives the Feynman prescription. If $x^0 > 0$, the ordered boundary value is obtained with the positive-energy pole below the real k^0-axis and the negative-energy pole above it:

$$k^0 = +\omega_{\vec{k}} - \mathrm{i}0, \qquad k^0 = -\omega_{\vec{k}} + \mathrm{i}0.$$

If $x^0 < 0$, the contour closure selects the opposite residue in the same single distribution Δ_F. This two-point pole rule is the momentum space encoding of the ordered tube boundary value above. The ordered tube prescription for $x^0 > 0$ approaches the first field from a larger negative imaginary time than the second. In Fourier variables this is the same damping choice as in the free two-point boundary-value calculation above, so the positive-energy pole is passed below the real axis. If $x^0 < 0$, the operator order is reversed and the contour closure selects the negative-energy pole above the axis. The two cases are precisely the two boundary values of the single meromorphic denominator $k^2 + m^2$, hence the denominator is $k^2 + m^2 - \mathrm{i}0$.

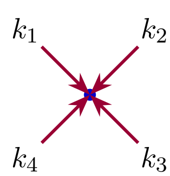
16.4 Lorentzian Feynman Rules for Green Functions

Definition 16.8 (Lorentzian labelled graph rules for Green functions)**.** Consider the same scalar ϕ^4 interaction as in the Euclidean chapter, with Lorentzian interaction density

$$\mathcal{L}_{\text{int}} = -\frac{g}{4!}\phi^4.$$

The perturbative rules below compute time-ordered Green functions. They use Lorentzian momenta on every line.

$$= \frac{-\mathrm{i}}{k^2 + m^2 - \mathrm{i}\epsilon}$$



$$= \quad -\mathrm{i}g\,(2\pi)^D \delta^{(D)}(k_1 + k_2 + k_3 + k_4)$$

This is the labelled-leg convention matching the Euclidean convention of the previous chapter. Direct expansion of the interaction density instead assigns

$$-\frac{\mathrm{i}g}{4!}\,(2\pi)^D \delta^{(D)}(k_1 + k_2 + k_3 + k_4)$$

to an unlabelled four-field insertion and recovers the labelled rule by counting contractions. In the labelled-leg convention used below, the factor 4! associated with assigning the four incident half-edges to a quartic vertex has already been absorbed into the vertex rule $-\mathrm{i}g$. Therefore a connected graph Γ with labelled external legs carries only the residual factor

$$\frac{1}{|\operatorname{Aut}_{\mathrm{ext}}\Gamma|},$$

where $\operatorname{Aut}_{\mathrm{ext}}\Gamma$ is the automorphism group of the graph that fixes every external label and preserves the incidence relations. For the tadpole graph below this residual group has order 2, coming from the interchange of the two internal half-edges forming the loop; this is why the tadpole coefficient is $-\mathrm{i}g/2$, not $-\mathrm{i}g$.

Lorentzian–Euclidean graph factor under simultaneous rotation. Consider a connected ϕ^4 graph contributing to a time-ordered n-point Green function with external propagators included. Suppose the external momenta lie in the first-sheet Euclidean domain and the internal loop energy contours can be simultaneously deformed to imaginary axes without crossing or pinching singularities. Let I be the number of internal propagators, V the number of quartic vertices, and

$$L = I - V + 1$$

the number of loop momenta. With the labelled conventions of this chapter and Chapter 15, the Lorentzian graph boundary value at $k_a^0 = \mathrm{i}k_{E,a}^D$ is

$$(-1)^n \mathrm{i}^{\,n+1}$$

times the corresponding Euclidean connected Green-function graph. In particular, a connected two-point graph gives the factor $-\mathrm{i}$, while a connected four-point graph gives the factor $+\mathrm{i}$. The Lorentzian labelled graph contains one factor $-\mathrm{i}g$ for each quartic vertex, one factor

$$\frac{-\mathrm{i}}{p_e^2 + m^2 - \mathrm{i}0}$$

for each propagator edge, and one real loop measure for each independent loop momentum. In a connected n-point Green function with external propagators included, the total number of propagator edges is $I + n$, because each external insertion is attached to the interaction graph by one external propagator. Rotating each independent loop energy by $\ell^0 = \mathrm{i}\ell_E^D$ gives one Jacobian factor i for each loop. The denominator of every propagator becomes the corresponding Euclidean denominator by the assumed contour deformation.

Thus the total phase multiplying the Euclidean denominators and the factor g^V is

$$(-\mathrm{i})^V (-\mathrm{i})^{I+n} \mathrm{i}^L.$$

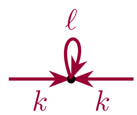
The Euclidean graph with the same labelled convention has vertex factor $(-g)^V$, the same symmetry factor, and no propagator numerator phase. Dividing by the Euclidean vertex phase gives the ratio

$$\frac{(-\mathrm{i})^{V+I+n}\mathrm{i}^{I-V+1}}{(-1)^V} = (-1)^n \mathrm{i}^{\,n+1},$$

where $L = I - V + 1$ has been used. The result is independent of the number of vertices and internal lines. The assumption on contour deformation is what excludes threshold pinches; without it the graph is defined by boundary value on the cut rather than by this direct simultaneous rotation.

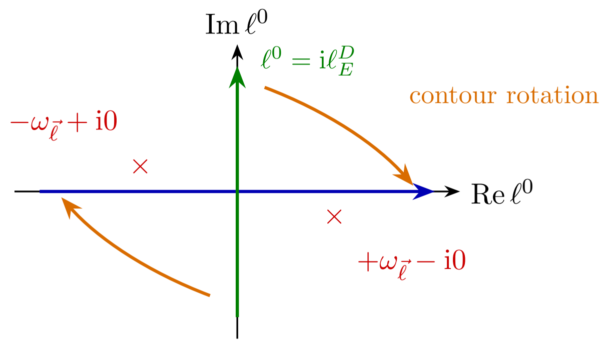
16.5 Loop Rotation and the Tadpole

Tadpole contour rotation and self-energy matching. The one-loop tadpole contribution to the Lorentzian two-point self-energy insertion is



$$= \quad -\frac{\mathrm{i}g}{2}\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\,\frac{-\mathrm{i}}{\ell^2+m^2-\mathrm{i}\epsilon}.$$

The factor $1/2$ is the same tadpole symmetry factor as in Euclidean signature. This is a combinatorial statement: the Wick contraction count depends only on the polynomial ϕ^4 and on the residual graph automorphism, while the signature-dependent factors enter separately through the labelled Lorentzian vertex $-\mathrm{i}g$ and propagator numerator $-\mathrm{i}$. The loop energy ℓ^0 is integrated along the real axis with the Feynman pole displacement. The Euclidean integral is obtained by rotating the contour to $\ell^0 = \mathrm{i}\ell_E^D$, with the contour passing between the two displaced poles.



With this orientation,

$$\ell^0 = \mathrm{i}\ell_E^D, \qquad \mathrm{d}\ell^0 = \mathrm{i}\,\mathrm{d}\ell_E^D, \qquad \mathrm{d}^D\ell = \mathrm{i}\,\mathrm{d}^D\ell_E,$$
$$\ell^2+m^2-\mathrm{i}\epsilon = -(\ell^0)^2+\vec{\ell}^{\,2}+m^2-\mathrm{i}\epsilon \longrightarrow \ell_E^2+m^2.$$

Therefore the measure and propagator numerator together give

$$\mathrm{d}^D\ell\,\frac{-\mathrm{i}}{\ell^2+m^2-\mathrm{i}\epsilon} \longrightarrow \mathrm{i}\,\mathrm{d}^D\ell_E\,\frac{-\mathrm{i}}{\ell_E^2+m^2} = \mathrm{d}^D\ell_E\,\frac{1}{\ell_E^2+m^2}.$$

Including the vertex factor, this gives

$$-\frac{\mathrm{i}g}{2}\int\frac{\mathrm{d}^D\ell}{(2\pi)^D}\frac{-\mathrm{i}}{\ell^2+m^2-\mathrm{i}\epsilon} = \mathrm{i}\left[-\frac{g}{2}\int\frac{\mathrm{d}^D\ell_E}{(2\pi)^D}\frac{1}{\ell_E^2+m^2}\right].$$

Thus the Lorentzian two-point insertion is $\mathrm{i}\Sigma(k)$, where $\Sigma(k)$ is the first-sheet analytic continuation of the Euclidean self-energy $\Sigma_E(k_E)$ with the sign convention of Chapter 15. More explicitly, in the same regulator and the same local two-point subtraction convention, the denominator self-energy obeys

$$\Sigma\big(k^0 = \mathrm{i}k_E^D, \vec{k} = \vec{k}_E\big) = \Sigma_E(k_E), \tag{16.2}$$

where Σ_E is defined by Eq. (15.1). The factor i displayed above belongs to the Lorentzian two-point insertion in the Dyson series; it is not an extra sign in the scalar function Σ appearing in the denominator of $\widetilde{G}(k)$. The coefficient $-\mathrm{i}g/2$ follows from Definition 16.8: the labelled quartic vertex supplies $-\mathrm{i}g$, and the loop made from the two remaining half-edges has residual automorphism group of order 2. The Feynman pole displacement places the positive-energy pole below and the negative-energy pole above the real loop-energy axis. Rotating the real ℓ^0-cycle to $\ell^0 = \mathrm{i}\ell_E^D$ crosses no pole in this tadpole because no external energy flows through the loop. The Jacobian gives $\mathrm{d}^D\ell = \mathrm{i}\mathrm{d}^D\ell_E$, while the propagator numerator gives $-\mathrm{i}$; their product is $+1$. The remaining vertex factor is exactly i times the Euclidean tadpole insertion. Since the Lorentzian two-point Green function is written with a global propagator numerator $-\mathrm{i}$, this extra i belongs to the insertion in the Dyson series and not to the denominator self-energy scalar itself. This gives (16.2).

Proposition 16.9 (Loop rotation and pinch obstruction)**.** *The contour rotation just used is a special case with no external energy flowing through the loop. The same direct rotation is also legitimate for loop integrals with Euclidean external momenta, where the external energies lie on the imaginary axis and the Feynman poles can be separated continuously from the rotating contours. For general Lorentzian timelike external momenta, one must instead analyze the full contour deformation in the complex loop energy variables. The obstruction is a pinch singularity: poles from different propagators can trap the integration cycle so that it cannot be deformed to the Euclidean cycle without crossing singularities.*

For example, the Lorentzian continuation of the order-g^2 sunset self-energy contains the denominator product

$$\frac{1}{(q_1^2 + m^2 - \mathrm{i}0)(q_2^2 + m^2 - \mathrm{i}0)\big((k - q_1 - q_2)^2 + m^2 - \mathrm{i}0\big)}.$$

At $\vec{k} = 0$, the Landau equations have a solution with all three internal lines on shell and carrying positive energy when

$$(k^0)^2 \geq (3m)^2.$$

At and above this three-particle threshold the q_1^0, q_2^0 contours are pinched. Thus the Euclidean sunset integral defines a first-sheet analytic function away from its Landau singularities, and the physical Lorentzian value above threshold is the specified boundary value on the cut, not the result of a naive independent rotation of each loop energy. The discontinuity across such cuts is computed by the Cutkosky rules, which are the perturbative expression of the same spectral and unitarity structure developed in Chapter 28.

Proof. For Euclidean external energies, every external energy entering a propagator has imaginary value. For fixed spatial loop momenta, the pole locations in each loop-energy variable remain a positive distance away from the rotating contour on compact subsets of the Euclidean external domain. The Feynman $i0$ prescription places the positive- and negative-energy poles on opposite sides of the original real contour, and the imaginary external energies prevent a pole from crossing the homotopy from the real contour to the imaginary contour. Cauchy's theorem for the regularized meromorphic integrand identifies the Lorentzian integral with the Euclidean integral on this first-sheet domain. Equivalently, insert a rapidly decreasing regulator, perform the contour deformation with vanishing large arcs, and remove the regulator inside the common analytic domain.

For real timelike k, the sunset denominators admit the simultaneous on-shell solution

$$q_1^2 = q_2^2 = (k - q_1 - q_2)^2 = -m^2, \qquad q_1^0, q_2^0, k^0 - q_1^0 - q_2^0 > 0$$

when $k^0 \geq 3m$ in the rest frame. At threshold the three internal spatial momenta may be zero; above threshold they may share the total energy k^0. The Landau stationarity condition says that the gradients of the three on-shell denominators are linearly dependent with nonnegative coefficients, so the singular hypersurfaces meet the integration cycle in a trapping configuration. In the q_1^0, q_2^0 variables, poles approach from opposite sides as the external energy is continued to the physical value. No deformation to the imaginary cycle exists without crossing a pole. The Lorentzian value above threshold is therefore a specified boundary value on the cut of the first-sheet analytic function, not the result of an independent Euclidean rotation of each loop energy. □

16.6 Full Two-Point Function and First Sheet

Definition 16.10 (First-sheet two-point continuation)**.** The perturbative time-ordered two-point function is written as

$$G(x,y) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \mathrm{e}^{\mathrm{i}k\cdot(x-y)} \widetilde{G}(k),$$

where

$$\widetilde{G}(k) = \frac{-\mathrm{i}}{k^2 + m^2 - \mathrm{i}\epsilon - \Sigma(k)}.$$

The function $\Sigma(k)$ is initially computed for Euclidean external momenta, where loop integrals have Euclidean denominators. Using

$$k^0 = \mathrm{i}k_E^D,$$

the Euclidean external energy lies on the imaginary k^0-axis. The matching identity (16.2) gives

$$\widetilde{G}(\mathrm{i}k_E^D, \vec{k}_E) = -\mathrm{i}\, \widetilde{G}_E(k_E),$$

because $k^2 = k_E^2$ at $k^0 = \mathrm{i}k_E^D$ with the mostly-plus convention. The Lorentzian value is the boundary value obtained by analytic continuation of k^0 from this axis to the prescribed side of the real axis without crossing the first-sheet cuts.

Spectral thresholds and first-sheet cuts. The spectral representation fixes the location of the first singular thresholds. Suppose the continuum part $d\rho_{\mathrm{c}}$ in the two-point spectral measure has lowest invariant mass M, meaning

$$M^2 = \inf \operatorname{supp} d\rho_{\mathrm{c}}$$

in the Källén–Lehmann variable $s = \mu^2$. Under the two-particle threshold assumption stated in Chapter 14, namely $\operatorname{supp} d\rho_{\mathrm{c}} \subset [4m^2, \infty)$ with the $\phi\phi$ continuum saturating the lower endpoint, this notation gives

$$M^2 = 4m^2, \qquad M = 2m.$$

If the first continuum channel has a different threshold, M denotes that actual square-root threshold. At fixed spatial momentum $\vec{k}$, define

$$E_M(\vec{k}) = \sqrt{\vec{k}^{\,2} + M^2}.$$

The continuum denominators have singular support when

$$(k^0)^2 = \vec{k}^{\,2} + s, \qquad s \geq M^2.$$

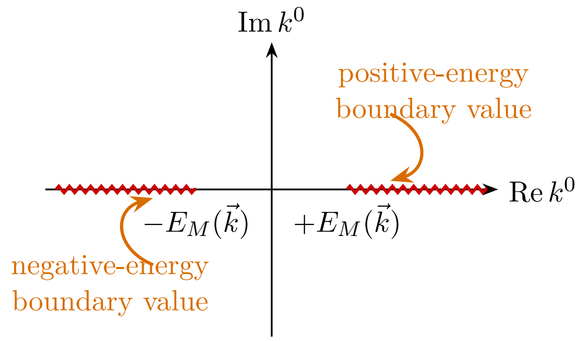
Consequently the first sheet has branch cuts beginning at

$$k^0 = +E_M(\vec{k}), \qquad k^0 = -E_M(\vec{k}).$$

The Feynman boundary value on the positive-energy cut is reached from the upper half-plane. Thus the value of $\Sigma(k)$ at real positive k^0 is the boundary value approached from $\operatorname{Im} k^0 > 0$ on the first sheet. The negative-energy cut is reached with the conjugate placement. The reason is direct but important. The Källén–Lehmann representation writes the connected two-point function as an integral of free denominators over the positive spectral measure:

$$\int_{[M^2,\infty)} \frac{\mathrm{d}\rho_{\mathrm{c}}(s)}{k^2 + s - \mathrm{i}0}.$$

At fixed $\vec{k}$, each value of s has branch points or poles at $k^0 = \pm\sqrt{\vec{k}^{\,2} + s}$. The union of these singular loci over $s \geq M^2$ is exactly the pair of cuts beginning at $\pm E_M(\vec{k})$. The Feynman prescription puts the positive-energy denominator below the real axis, so the physical positive-energy boundary value is obtained by approaching the cut from $\operatorname{Im} k^0 > 0$. The negative-energy side is the conjugate boundary value imposed by the same real spectral measure.



The output of the continuation is a family of Lorentzian time-ordered Green functions with specified singularity prescriptions. The spectral poles and cuts identify one-particle and continuum invariant masses seen by the field. The next construction concerns asymptotic states: it asks for Hilbert-space limits in which separated wave packets behave as freely propagating stable particles in the far past or far future. That construction is logically additional to the analytic continuation of Green functions.

Part II

Particles, Scattering, and Analyticity

Chapter 17

Haag–Ruelle Scattering Theory

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

17.1 Scattering States as Hilbert-Space Limits

The previous chapters constructed local fields, their spectral content, and Lorentzian time-ordered Green functions. We now add a new kind of object: asymptotic particle states. These are vectors in the physical Hilbert space obtained as large-time limits of localized wave packets. Once these limits are constructed, the scattering operator is a comparison between two Hilbert-space identifications: one made in the far past and one made in the far future.

The construction in this chapter is for a massive stable bosonic particle. It uses a Hilbert space $\mathcal{H}$, a vacuum vector Ω, local operators, the Poincare representation, the spectrum condition, and an isolated one-particle mass shell. These hypotheses are stated explicitly because the large-time limit is a theorem under hypotheses, not an algebraic manipulation of Green functions.

This chapter is the introductory construction layer. It works with scalar local fields and displays the velocity-support geometry, spectral filtering, and Cook estimates that make the large-time limits plausible and usable in physics applications. The theorem-level algebraic statement, formulated for vacuum local nets and almost-local operators, is Theorem 22.1 in Chapter 22. The present chapter should therefore be read as the point-field model and estimate layer feeding the later theorem, not as a second independent foundation of scattering theory.

Let spacetime have dimension D, and write $d = D - 1$. The mostly-plus metric is

$$\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +), \qquad p^2 = \eta_{\mu\nu} p^\mu p^\nu.$$

Let $U(a, \Lambda)$ be a strongly continuous unitary representation of the proper orthochronous Poincare group on $\mathcal{H}$. Its translation subgroup is written

$$U(a, \mathbf{1}) = \mathrm{e}^{\mathrm{i} a_\mu P^\mu},$$

where the commuting self-adjoint generators P^μ are the energy-momentum operators. The spectrum condition is

$$\mathrm{Spec}(P) \subset \overline{V}_+ = \{p \in \mathbb{R}^D : p^0 \geq 0,\ p^2 \leq 0\}.$$

The invariant mass operator is

$$M^2 = -P_\mu P^\mu.$$

Definition 17.1 (Massive one-particle sector)**.** Fix $m > 0$. A massive one-particle sector of mass m is a closed subspace $\mathcal{H}_1 \subset \mathcal{H}$ such that:

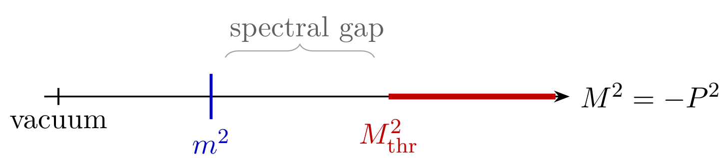
1. $\mathcal{H}_1$ reduces the Poincare representation;
2. $M^2 = m^2$ on $\mathcal{H}_1$;
3. the spectrum of P^μ on $\mathcal{H}_1$ lies on

$$\Sigma_m^+ = \{(\omega_{\vec{p}}, \vec{p}) : \vec{p} \in \mathbb{R}^d\}, \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^{\,2} + m^2};$$

4. there is an open interval around m^2 in the spectrum of M^2 whose only spectral point is m^2.

Let P_1 denote the orthogonal projection from $\mathcal{H}$ onto $\mathcal{H}_1$.

The last condition is the mass-gap form of stability used in this chapter. It says that the one-particle shell is separated from the continuum. In a scalar theory with a lightest stable particle, the picture is:



Here M_{thr} is the lowest invariant mass in the continuum channel under discussion.

For the scalar multiplicity-one sector used below, the abstract Hilbert space $\mathcal{H}_1$ is represented concretely as wave functions on the positive mass shell. Let

$$\mathrm{d}\mu_m(p) = \frac{\mathrm{d}^d \vec{p}}{(2\pi)^d 2\omega_{\vec{p}}}, \qquad p = (\omega_{\vec{p}}, \vec{p}).$$

After fixing the relativistic normalization of one-particle states, choose a unitary map

$$W_1 : L^2(\Sigma_m^+, \mathrm{d}\mu_m) \longrightarrow \mathcal{H}_1$$

which intertwines translations by

$$U(a, \mathbf{1}) W_1 f = W_1\big(p \mapsto \mathrm{e}^{\mathrm{i} a_\mu p^\mu} f(p)\big).$$

In generalized momentum notation this is

$$W_1 f = \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, f(p)|p\rangle, \qquad \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, |p\rangle\langle p| = \mathbf{1}_{\mathcal{H}_1}.$$

The kets in this display are distributional symbols for the spectral representation; Hilbert vectors are obtained after L^2-smearing. With an internal multiplicity space or nontrivial spin, the scalar model is replaced by

$$\mathcal{H}_1 \simeq L^2(\Sigma_m^+, \mathrm{d}\mu_m; \mathfrak{k}),$$

where $\mathfrak{k}$ carries the little-group/internal representation and the Poincare action contains the corresponding Wigner rotations. The present chapter takes $\mathfrak{k} = \mathbb{C}$.

Remark 17.2 (When the isolated shell is a theorem). The definition above is an assumption on the Hilbert-space spectrum. It is automatic for a free massive scalar field, and it is a proved property in constructive massive scalar models only when the construction supplies both the Wightman theory and the relevant mass-gap/one-particle spectral theorem. In particular, low-dimensional $P(\phi)_2$ and massive $\lambda\phi_3^4$ models provide the standard constructive setting for such statements in stable massive sectors. By contrast, a four-dimensional perturbative $\lambda\phi^4$ calculation by itself does not prove the existence of a UV-complete Hilbert-space theory with an isolated mass shell. In QED, long-range soft photons destroy the sharp charged-particle mass hyperboloid expected by this definition; in QCD, colored quark and gluon shells are absent from the gauge-invariant observable spectrum. Stable hadrons, if isolated below decay thresholds, may satisfy the hypothesis, while resonances and unstable particles do not.

17.2 Local Operators and One-Particle Wave Packets

To state the large-time construction without domain distractions, use bounded local operators. A bounded operator A is localized in a bounded spacetime region O when it belongs to the local algebra $\mathcal{A}(O)$. Locality means that if O_1 and O_2 are spacelike separated, then

$$[A_1, A_2] = 0, \qquad A_i \in \mathcal{A}(O_i).$$

For unbounded point fields, this is the corresponding statement after smearing and restricting to the common invariant domain on which the products are defined.

Definition 17.3 (Almost-local bounded operator)**.** Let $\mathcal{O}_R$ be a cofinal family of double cones whose radii tend to infinity; for example, one may take

$$\mathcal{O}_R = \{(x^0, \vec{x}) : |x^0| + |\vec{x}| < R\}, \qquad R \geq 1.$$

A bounded operator $B \in \mathcal{B}(\mathcal{H})$ is *almost local* if there are bounded operators $B_R \in \mathcal{A}(\mathcal{O}_R)$ such that, for every integer $N \geq 1$, there is a constant $C_N < \infty$ with

$$\|B - B_R\| \leq C_N R^{-N} \qquad (R \geq 1).$$

The particle must be creatable from the vacuum by local operations. In the present scalar setting this means that there is a local operator A such that

$$P_1 A\Omega \neq 0.$$

Definition 17.4 (Spectrally filtered interpolating creator)**.** Choose a Schwartz test function $\chi \in \mathcal{S}(\mathbb{R}^D)$ whose Fourier transform is supported in a small neighborhood of the positive mass shell Σ_m^+ and avoids the rest of the energy-momentum spectrum. Define

$$B = A(\chi) = \int_{\mathbb{R}^D} \mathrm{d}^D x\, \chi(x)\, A(x), \qquad A(x) = U(x)AU(x)^{-1}.$$

Proposition 17.5 (Filtered local creators are almost local)**.** *If $A \in \mathcal{A}(O)$ is bounded and $\chi \in \mathcal{S}(\mathbb{R}^D)$, then the filtered operator $B = A(\chi)$ of Definition 17.4 is almost local. If the support of $\widehat{\chi}$ meets the joint spectrum of P^μ only on the isolated one-particle shell in Definition 17.1, then $B\Omega$ has no spectral component outside $\mathcal{H}_1$ in that support window.*

Proof. Let O be contained in a double cone of radius R_0. Choose smooth cutoff functions θ_R on spacetime, equal to 1 on $\{x : |x^0| + |\vec{x}| \le R\}$, supported in $\{x : |x^0| + |\vec{x}| \le 2R\}$, and with derivatives bounded uniformly after rescaling. Define

$$B_R = \int \mathrm{d}^D x\, \theta_R(x)\chi(x)A(x).$$

The integral is a norm-convergent Bochner integral because $\chi \in \mathcal{S}(\mathbb{R}^D)$ and $\|A(x)\| = \|A\|$. Since $\operatorname{supp}\theta_R$ is bounded and $A(x)$ is localized in $O + x$, the operator B_R belongs to the local algebra of a double cone whose radius is $2R + R_0$. Reindexing the cofinal family of double cones is irrelevant in the definition of almost locality.

The tail is bounded in operator norm by

$$\|B - B_R\| \le \|A\| \int_{|x^0|+|\vec{x}|\ge R} \mathrm{d}^D x\, |\chi(x)|.$$

For every N, the Schwartz seminorm $\sup_x (1+|x^0|+|\vec{x}|)^{N+D+1}|\chi(x)|$ is finite, and the last integral is bounded by $C_N R^{-N}$. Hence B is almost local.

For the spectral statement, write the joint spectral measure of P^μ as $E(\cdot)$ and use the Fourier convention

$$\widehat{\chi}(p) = \int \mathrm{d}^D x\, \chi(x)\mathrm{e}^{\mathrm{i}x_\mu p^\mu}.$$

Because $U(x)\Omega = \Omega$,

$$B\Omega = \int \mathrm{d}^D x\, \chi(x)U(x)A\Omega = \int_{\mathrm{Spec}(P)} \widehat{\chi}(p)\, \mathrm{d}E(p)A\Omega,$$

where the last equality is the joint spectral theorem applied to the vector $A\Omega$. Thus $B\Omega$ has spectral support contained in $\operatorname{supp}\widehat{\chi} \cap \mathrm{Spec}(P)$. If this set is contained in the isolated mass shell Σ_m^+, the corresponding vector lies in $E(\Sigma_m^+)\mathcal{H} = \mathcal{H}_1$, and all components outside $\mathcal{H}_1$ in the chosen support window vanish. $\square$

Let $h \in C_c^\infty(\mathbb{R}^d)$ be a smooth compactly supported momentum wave packet. Define the positive-energy Klein–Gordon solution

$$h_t(\vec{x}) = \int \frac{\mathrm{d}^d \vec{p}}{(2\pi)^d}\, h(\vec{p})\, \mathrm{e}^{\mathrm{i}\omega_{\vec{p}}t - \mathrm{i}\vec{p}\cdot\vec{x}}.$$

The associated Haag–Ruelle operator is

$$B_t(h) = \int_{\mathbb{R}^d} \mathrm{d}^d\vec{x}\, h_t(\vec{x})\, B(t,\vec{x}), \qquad B(t,\vec{x}) = U(t,\vec{x}) B U(t,\vec{x})^{-1}.$$

The one-particle vector created by this family is

$$\psi_h := P_1 B_t(h)\Omega.$$

It is independent of t. The role of the phase in h_t is precisely to cancel the free time evolution on the mass shell.

Spectral filter and one-particle time independence. Let B be an almost-local operator whose energy-momentum transfer is contained in a neighborhood of Σ_m^+ disjoint from the rest of $\mathrm{Spec}(P)$. Write the one-particle spectral wavefunction of $B\Omega$ as

$$P_1 B\Omega = W_1 b, \qquad b \in L^2(\Sigma_m^+, \mathrm{d}\mu_m).$$

For $h \in C_c^\infty(\mathbb{R}^d)$, the Haag–Ruelle approximant satisfies

$$P_1 B_t(h)\Omega = W_1(bh), \qquad P_1 \frac{\mathrm{d}}{\mathrm{d}t} B_t(h)\Omega = 0.$$

In particular, the one-particle vector represented by $B_t(h)\Omega$ is independent of t. The verification is the mass-shell phase cancellation: because $U(t,\vec{x})\Omega = \Omega$ and P_1 commutes with translations, the mass-shell component of $B(t,\vec{x})\Omega$ is

$$P_1 B(t,\vec{x})\Omega = W_1\left(p \mapsto \mathrm{e}^{-\mathrm{i}\omega_{\vec{p}} t + \mathrm{i}\vec{p}\cdot\vec{x}} b(p)\right),$$

using $a_\mu p^\mu = -t\omega_{\vec{p}} + \vec{x}\cdot\vec{p}$. Substitution in the definition of $B_t(h)$ gives

$$\begin{aligned}
P_1 B_t(h)\Omega &= W_1\left(p \mapsto b(p) \int_{\mathbb{R}^d} \mathrm{d}^d\vec{x}\, h_t(\vec{x}) \mathrm{e}^{-\mathrm{i}\omega_{\vec{p}} t + \mathrm{i}\vec{p}\cdot\vec{x}}\right) \\
&= W_1\left(p \mapsto b(p) \int \frac{\mathrm{d}^d\vec{q}}{(2\pi)^d}\, h(\vec{q}) \mathrm{e}^{\mathrm{i}(\omega_{\vec{q}} - \omega_{\vec{p}})t} \int_{\mathbb{R}^d} \mathrm{d}^d\vec{x}\, \mathrm{e}^{\mathrm{i}(\vec{p}-\vec{q})\cdot\vec{x}}\right) \\
&= W_1(bh).
\end{aligned}$$

The last expression is independent of t. Differentiating the first line under the smooth compactly supported momentum integral gives $P_1 \frac{\mathrm{d}}{\mathrm{d}t} B_t(h)\Omega = 0$.

17.3 Point-Field Smearing and Propagation

The bounded-operator formulation is the cleanest theorem statement. In applications one often starts from a scalar Wightman field $\hat{\phi}(x)$. The same construction is then expressed by smearing $\hat{\phi}$ against spacetime test functions and by choosing the Fourier support so that only the isolated one-particle shell is selected.

Proposition 17.6 (Point-field projection to the isolated shell). *Assume that $\hat{\phi}$ is a scalar local field whose matrix element between the vacuum and the one-particle sector is nonzero. The kets below are generalized vectors in the rigged-Hilbert-space sense of Section 5.2. With relativistically normalized generalized one-particle kets,*

$$\langle q|p\rangle = (2\pi)^d 2\omega_{\vec{p}}\,\delta^{(d)}(\vec{q}-\vec{p}), \qquad \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\,|p\rangle\langle p| = \mathbf{1}_{\mathcal{H}_1},$$

where

$$\mathrm{d}\mu_m(p) = \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d 2\omega_{\vec{p}}}, \qquad p = (\omega_{\vec{p}}, \vec{p}),$$

the one-particle part of $\hat{\phi}(x)\Omega$ has the form

$$P_1\hat{\phi}(x)\Omega = Z_\phi^{1/2}\int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\,\mathrm{e}^{-\mathrm{i}p\cdot x}|p\rangle.$$

Here $Z_\phi > 0$ is the residue of the two-point function at this particular field's one-particle pole. All remaining spectral components of $\hat{\phi}(x)\Omega$ have invariant mass separated from m by the assumed gap.

Let

$$f(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\,\tilde{f}(k)\,\mathrm{e}^{\mathrm{i}k\cdot x}, \qquad \hat{\phi}_f = \int \mathrm{d}^D x\, f(x)\hat{\phi}(x).$$

If $\tilde{f}$ is supported in a small neighborhood of Σ_m^+ and away from the rest of the energy-momentum spectrum, then

$$\hat{\phi}_f\Omega = Z_\phi^{1/2}\int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\,\tilde{f}(p)|p\rangle,$$

with no continuum contribution.

Proof. The point-field expression is a distributional statement. The actual Hilbert vectors are obtained after smearing in x. In the scalar multiplicity-one sector, define the one-particle form-factor distribution by

$$F_x(p) = \langle p|\hat{\phi}(x)\Omega\rangle.$$

Translation covariance and $U(x)\Omega = \Omega$ give

$$F_x(p) = \langle p|U(x)\hat{\phi}(0)\Omega\rangle = \mathrm{e}^{-\mathrm{i}p\cdot x}F_0(p),$$

because $U(x)|p\rangle = \mathrm{e}^{\mathrm{i}p\cdot x}|p\rangle$. Scalar Lorentz covariance gives $F_0(\Lambda p) = F_0(p)$ on the positive mass shell. Since Σ_m^+ is a single $SO^+(1,d)$-orbit and the multiplicity space is $\mathbb{C}$, F_0 is a constant distribution on the shell. After fixing the phase of the interpolating field, write this constant as $Z_\phi^{1/2} > 0$; equivalently $Z_\phi = |F_0|^2$ is the coefficient of the one-particle delta mass in the Källén–Lehmann measure of this field.

Therefore, in the rigged-Hilbert notation,

$$P_1\hat{\phi}(x)\Omega = Z_\phi^{1/2}\int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\,\mathrm{e}^{-\mathrm{i}p\cdot x}|p\rangle.$$

For a Schwartz smearing function with

$$f(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \tilde{f}(k) \mathrm{e}^{\mathrm{i}k\cdot x},$$

pairing this distribution with f gives

$$P_1 \hat{\phi}_f \Omega = Z_\phi^{1/2} \int_{\Sigma_m^+} \mathrm{d}\mu_m(p) \left(\int \mathrm{d}^D x\, f(x) \mathrm{e}^{-\mathrm{i}p\cdot x} \right) |p\rangle = Z_\phi^{1/2} \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, \tilde{f}(p) |p\rangle.$$

The remaining spectral part of the smeared vector $\hat{\phi}_f \Omega$ is obtained by applying the joint spectral measure of P^μ to the part of the field-created distribution supported away from the isolated shell. If $\operatorname{supp} \tilde{f}$ is contained in the chosen isolated-shell neighborhood and avoids the rest of the spectrum, that spectral projection is zero. The displayed mass-shell integral is then the full vector $\hat{\phi}_f \Omega$. □

A Gaussian packet centered near a spacetime point $(t_*, \vec{x}_*)$ and a momentum $(E_*, \vec{P}_*)$ may be used as an approximation,

$$f(x) = \exp\left[-\frac{(x^0 - t_*)^2 + |\vec{x} - \vec{x}_*|^2}{L^2} + \mathrm{i}\vec{P}_* \cdot (\vec{x} - \vec{x}_*) - \mathrm{i}E_*(x^0 - t_*) \right], \qquad E_*^2 = \vec{P}_*^{\,2} + m^2.$$

For exact spectral separation it is better to choose $\tilde{f}$ smooth and compactly supported in momentum space; the spacetime function is then rapidly decreasing rather than compactly supported. This is the analytic setting used by Haag–Ruelle theory.

Define the translated family $f^{(T)}$ by

$$\tilde{f}^{(T)}(k) = \mathrm{e}^{\mathrm{i}(k^0 - \omega_{\vec{k}})T} \tilde{f}(k), \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^{\,2} + m^2}.$$

On the positive mass shell the extra phase is equal to 1. Consequently

$$P_1 \hat{\phi}_{f^{(T)}} \Omega = P_1 \hat{\phi}_f \Omega$$

for all T, while the spacetime profile of the smearing function moves. In position space,

$$\begin{aligned} f^{(T)}(x^0, \vec{x}) &= \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \mathrm{e}^{\mathrm{i}k\cdot x + \mathrm{i}(k^0 - \omega_{\vec{k}})T} \tilde{f}(k) \\ &= \int \mathrm{d}^d \vec{y}\, K(\vec{x} - \vec{y}; T)\, f(x^0 - T, \vec{y}), \end{aligned}$$

where

$$K(\vec{x}; T) = \int \frac{\mathrm{d}^d \vec{k}}{(2\pi)^d}\, \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x} - \mathrm{i}\omega_{\vec{k}} T}.$$

The kernel satisfies the Klein–Gordon equation in the variables $(T, \vec{x})$,

$$\left(\partial_T^2 - \Delta_{\vec{x}} + m^2 \right) K(\vec{x}; T) = 0, \qquad K(\vec{x}; 0) = \delta^{(d)}(\vec{x})$$

in the distributional sense. If $\tilde{f}$ is concentrated near $\vec{P}_*$, stationary phase shows that the packet center moves as

$$(t_*, \vec{x}_*) \longmapsto (t_* + T, \vec{x}_* + \vec{v}_* T), \qquad \vec{v}_* = \nabla_{\vec{p}} \omega_{\vec{p}} \big|_{\vec{p} = \vec{P}_*} = \frac{\vec{P}_*}{E_*}.$$

Thus the same one-particle vector may be represented by smearing a local field over different large spacetime regions. The equality is only after projection to $\mathcal{H}_1$; the parts outside the isolated shell are removed by the spectral support condition.

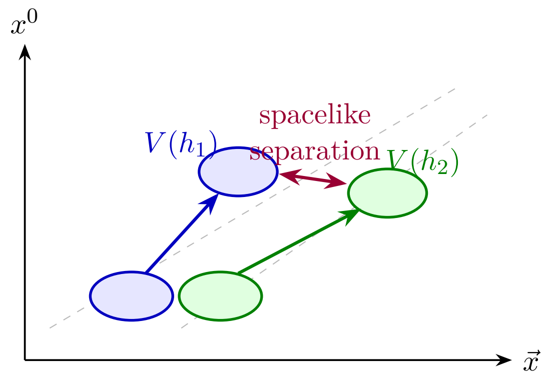
The velocity support of h is

$$V(h) = \left\{ \frac{\vec{p}}{\omega_{\vec{p}}} : \vec{p} \in \operatorname{supp} h \right\} \subset \mathbb{R}^d.$$

Stationary phase gives the spacetime localization of h_t: for large $|t|$, the wave packet is concentrated near rays

$$\vec{x} = t\, \frac{\vec{p}}{\omega_{\vec{p}}}, \qquad \vec{p} \in \operatorname{supp} h.$$

Thus two wave packets with disjoint velocity supports become spacelike separated as $|t| \to \infty$.



17.4 Velocity Support Separation

The phrase "disjoint velocity supports" has a precise geometric content. Let $K_i = \operatorname{supp} h_i$ and set

$$V_i = \left\{ \vec{p}/\omega_{\vec{p}} : \vec{p} \in K_i \right\}.$$

If $V_i \cap V_j = \varnothing$, compactness gives a number

$$\delta_{ij} := \operatorname{dist}(V_i, V_j) > 0.$$

Choose $0 < \epsilon < \delta_{ij}/4$ and define the equal-time tubes

$$\Gamma_i(t, \epsilon) = \{\vec{x} \in \mathbb{R}^d : \operatorname{dist}(\vec{x}/t, V_i) < \epsilon\} \qquad (t \neq 0).$$

For $\vec{x} \in \Gamma_i(t, \epsilon)$ and $\vec{y} \in \Gamma_j(t, \epsilon)$,

$$|\vec{x} - \vec{y}| \geq |t|(\delta_{ij} - 2\epsilon) \geq \frac{1}{2}|t|\delta_{ij}.$$

The spacetime points $(t, \vec{x})$ and $(t, \vec{y})$ are therefore spacelike separated with a separation growing linearly in $|t|$.

The part of $h_{i,t}$ outside $\Gamma_i(t,\epsilon)$ is rapidly decreasing. Indeed, away from the stationary set $\vec{x}/t = \vec{p}/\omega_{\vec{p}}$, the phase

$$\omega_{\vec{p}}t - \vec{p}\cdot\vec{x}$$

has no stationary point on K_i, and repeated integration by parts in $\vec{p}$ gives a bound $O(|t|^{-N})$ for every N. Thus the Haag–Ruelle operators can be replaced, up to rapidly decreasing errors, by almost-local operators supported in mutually spacelike tubes. Locality then turns geometric separation into the commutator estimate used in the existence proof.

Proposition 17.7 (Velocity-tube separation and off-tube decay)**.** *Let $h \in C_c^\infty(\mathbb{R}^d)$, and let*

$$V(h) = \{\nabla_{\vec{p}}\omega_{\vec{p}} : \vec{p} \in \operatorname{supp} h\}.$$

For every $\epsilon > 0$ and every $N \geq 1$, there is $C_{N,\epsilon}$ such that

$$\sup_{\operatorname{dist}(\vec{x}/t, V(h)) \geq \epsilon} |h_t(\vec{x})| \leq C_{N,\epsilon}(1+|t|)^{-N}, \qquad |t| \geq 1.$$

If h_i, h_j have compact velocity supports with $\delta_{ij} = \operatorname{dist}(V(h_i), V(h_j)) > 0$, then for $0 < \epsilon < \delta_{ij}/4$ and $\vec{x} \in \Gamma_i(t,\epsilon)$, $\vec{y} \in \Gamma_j(t,\epsilon)$,

$$|\vec{x} - \vec{y}| \geq \frac{1}{2}\delta_{ij}|t|.$$

Thus the equal-time spacetime points $(t,\vec{x})$ and $(t,\vec{y})$ are spacelike separated by a distance growing linearly in $|t|$.

Proof. Write

$$h_t(\vec{x}) = \int \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d} h(\vec{p})\, \mathrm{e}^{\mathrm{i}t\Phi_{\vec{x}/t}(\vec{p})}, \qquad \Phi_{\vec{u}}(\vec{p}) = \omega_{\vec{p}} - \vec{p}\cdot\vec{u}.$$

On the compact set $\operatorname{supp} h$, the gradient is

$$\nabla_{\vec{p}}\Phi_{\vec{u}}(\vec{p}) = \frac{\vec{p}}{\omega_{\vec{p}}} - \vec{u}.$$

If $\operatorname{dist}(\vec{u}, V(h)) \geq \epsilon$, then $|\nabla_{\vec{p}}\Phi_{\vec{u}}| \geq \epsilon$ on $\operatorname{supp} h$. Apply repeatedly the adjoint of

$$L_{\vec{u}} = \frac{1}{\mathrm{i}t}\frac{\nabla_{\vec{p}}\Phi_{\vec{u}}\cdot\nabla_{\vec{p}}}{|\nabla_{\vec{p}}\Phi_{\vec{u}}|^2}, \qquad L_{\vec{u}}\mathrm{e}^{\mathrm{i}t\Phi_{\vec{u}}} = \mathrm{e}^{\mathrm{i}t\Phi_{\vec{u}}}.$$

Derivatives of $\Phi_{\vec{u}}$ are uniformly bounded for $\vec{p}$ in the fixed compact support and for the relevant $\vec{u}$. Each integration by parts gives a factor $|t|^{-1}$, proving the displayed off-tube estimate.

For the geometric separation, choose $\vec{v}_i \in V(h_i)$ and $\vec{v}_j \in V(h_j)$ with

$$|\vec{x}/t - \vec{v}_i| < \epsilon, \qquad |\vec{y}/t - \vec{v}_j| < \epsilon.$$

Then

$$|\vec{x} - \vec{y}| = |t|\,|\vec{x}/t - \vec{y}/t| \geq |t|(\delta_{ij} - 2\epsilon) \geq \frac{1}{2}\delta_{ij}|t|.$$

Equal-time separation is spacelike when $\vec{x} \neq \vec{y}$, since the Minkowski squared separation is $|\vec{x} - \vec{y}|^2$. ∎

17.5 Multi-Particle Limits

Let $B_1, \dots, B_n$ be almost-local operators obtained as above, and let $h_1, \dots, h_n \in C_c^\infty(\mathbb{R}^d)$ have pairwise disjoint velocity supports. Define the time-dependent vector

$$\Psi_t(h_1, \dots, h_n) = B_{1,t}(h_1) \cdots B_{n,t}(h_n)\Omega.$$

Proposition 17.8 (Cook estimate form of the Haag–Ruelle limits)**.** *Assume the locality, spectrum, isolated-shell, and regularity hypotheses stated above. Suppose, in addition, that the Haag–Ruelle product $\Psi_t(h_1, \dots, h_n)$ is strongly differentiable on the large-time tails and that there are integrable functions $R_\pm$ such that*

$$\left\| \frac{\mathrm{d}}{\mathrm{d}t} \Psi_t(h_1, \dots, h_n) \right\| \le R_+(t) \quad (t \ge T_0), \qquad \left\| \frac{\mathrm{d}}{\mathrm{d}t} \Psi_t(h_1, \dots, h_n) \right\| \le R_-(t) \quad (t \le -T_0).$$

Suppose also that the recursive scalar-product contraction estimates hold: large-time scalar products of velocity-separated products are obtained by pairing one-particle vectors, up to errors that vanish on the large-time tails. Finally, suppose the same Cook and contraction estimates hold after replacing one creator by the difference of two creators with the same one-particle vector. Then the limits

$$\Psi^{\mathrm{out}}(h_1, \dots, h_n) = \lim_{t \to +\infty} \Psi_t(h_1, \dots, h_n),$$

$$\Psi^{\mathrm{in}}(h_1, \dots, h_n) = \lim_{t \to -\infty} \Psi_t(h_1, \dots, h_n)$$

exist in the norm topology of $\mathcal{H}$. The limits depend only on the one-particle vectors

$$\psi_j = P_1 B_{j,t}(h_j)\Omega \in \mathcal{H}_1, \qquad j = 1, \dots, n,$$

and not on the auxiliary operators B_j used to interpolate them.

Proof. For $t' > t \ge T_0$,

$$\|\Psi_{t'} - \Psi_t\| \le \int_t^{t'} \left\| \frac{\mathrm{d}}{\mathrm{d}s} \Psi_s \right\| \mathrm{d}s \le \int_t^{t'} R_+(s)\, \mathrm{d}s.$$

Since $R_+ \in L^1([T_0, \infty))$, the right side tends to zero when $t, t' \to +\infty$. Hence Ψ_t is Cauchy in the Hilbert norm and has an outgoing limit. The incoming proof is identical with R_- on the negative tail.

If B_j and B_j' have the same one-particle vector, multilinearity writes the difference of the two n-particle approximants as a finite sum in which one factor is $D_j = B_j - B_j'$. For that factor

$$P_1 D_{j,t}(h_j)\Omega = 0.$$

The assumed Cook estimate for such zero-one-particle replacements implies that each difference term has a norm limit. Its limiting Fock inner product with all velocity-separated Haag–Ruelle vectors is zero, because the recursive contraction of asymptotic scalar products contains no one-particle contraction for the D_j entry. Density of the velocity-separated algebraic Fock domain then gives that this limit vector is zero. Thus the large-time vector depends only on $\psi_j = P_1 B_{j,t}(h_j)\Omega$. □

The proposition has two analytic inputs. First, disjoint velocity supports imply that the effective localization regions of $B_{j,t}(h_j)$ separate spacelike at large $|t|$. Locality then gives rapid decay of commutators

$$[B_{i,t}(h_i), B_{j,t}(h_j)]$$

for $i \neq j$. Second, spectral isolation of the mass shell controls the part of $B_{j,t}(h_j)\Omega$ orthogonal to $\mathcal{H}_1$. In Cook's method one estimates $\frac{\mathrm{d}}{\mathrm{d}t}\Psi_t$; the commutator terms and the orthogonal spectral-error terms are integrable in t, hence Ψ_t is a Cauchy family at $+\infty$ and at $-\infty$.

For $n = 2$, the essential estimate is visible from the norm difference

$$\|\Psi_{t'} - \Psi_t\|^2 = \langle \Psi_{t'} - \Psi_t, \Psi_{t'} - \Psi_t \rangle.$$

The scalar products entering this expression are vacuum correlation functions of four almost-local operators. At large positive or negative times, the two velocity clusters are separated by spacelike distances growing linearly in $|t|$. The cluster property and locality reduce the large-time scalar product to the product of the corresponding one-particle scalar products.

In the point-field notation above, for two packets with disjoint velocity supports one may write

$$\Psi_T = \hat{\phi}_{f_1^{(T)}} \hat{\phi}_{f_2^{(T)}} \Omega.$$

For fixed ΔT,

$$\langle \Psi_T, \Psi_{T+\Delta T} \rangle = \langle \Omega | \hat{\phi}^*_{f_2^{(T)}} \hat{\phi}^*_{f_1^{(T)}} \hat{\phi}_{f_1^{(T+\Delta T)}} \hat{\phi}_{f_2^{(T+\Delta T)}} | \Omega \rangle.$$

As $T \to +\infty$ or $T \to -\infty$, the two spacetime clusters separate spacelike. The vacuum cluster property gives

$$\langle \Psi_T, \Psi_{T+\Delta T} \rangle - \langle \hat{\phi}_{f_1}\Omega, \hat{\phi}_{f_1}\Omega \rangle \langle \hat{\phi}_{f_2}\Omega, \hat{\phi}_{f_2}\Omega \rangle \longrightarrow 0,$$

and the same limit is obtained for $\Delta T = 0$. Therefore

$$\|\Psi_{T+\Delta T} - \Psi_T\| \longrightarrow 0.$$

Under the smooth spectral separation hypotheses of Haag–Ruelle theory, the error decreases faster than any inverse power of $|T|$. This explicit two-packet calculation is the concrete model for the general Cook estimate.

17.6 The Estimate Behind the Limit

The preceding proposition is a norm-convergence statement. Its proof is governed by an estimate of the following form. Let $B_{i,t}(h_i)$ and $B_{j,t}(h_j)$ be Haag–Ruelle operators with disjoint velocity supports. For every $N \geq 1$, almost locality and stationary phase give constants C_N such that

$$\|[B_{i,t}(h_i), B_{j,t}(h_j)]\| \leq C_N (1 + |t|)^{-N} \tag{17.1}$$

for sufficiently large $|t|$. The estimate says more than that the operators eventually commute: it gives an integrable error.

The time derivative of a single approximant has no one-particle component:

$$P_1 \frac{\mathrm{d}}{\mathrm{d}t} B_{j,t}(h_j)\Omega = 0.$$

The remaining spectral support lies away from the isolated mass shell. By choosing the energy-momentum support of the smearing function χ inside the mass-gap neighborhood, that remainder is controlled by integration by parts in the energy variable. In the multi-particle product, $\frac{\mathrm{d}}{\mathrm{d}t}\Psi_t$ is a sum of terms in which either a one-particle remainder appears or two factors must be commuted past each other. The one-particle remainders are integrable by the spectral gap, and the commutator terms are integrable by (17.1). Therefore

$$\int_T^\infty \left\| \frac{\mathrm{d}}{\mathrm{d}t}\Psi_t \right\| \mathrm{d}t \longrightarrow 0 \qquad (T \to \infty),$$

which is Cook's Cauchy criterion for the outgoing limit. The incoming limit is the same argument with $t \to -\infty$.

This estimate is the point at which locality, spectrum, and particle stability meet. Without locality the commutator decay is unavailable; without the spectral gap the one-particle remainder need not be integrable; without positive energy the large-time stationary-phase localization has no stable direction of propagation.

17.7 Fock Structure and Wave Operators

Definition 17.9 (Scalar asymptotic Fock space)**.** Let

$$\mathcal{F}_s(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} \mathcal{H}_1^{\odot n}$$

be the bosonic Fock space over $\mathcal{H}_1$, where $\mathcal{H}_1^{\odot n}$ denotes the symmetric n-fold tensor product and $\mathcal{H}_1^{\odot 0} = \mathbb{C}\Omega_F$. The vector Ω_F is the Fock vacuum.

The construction is intrinsic: it is the Hilbert completion of the algebraic finite symmetric tensor algebra over $\mathcal{H}_1$. The mass-shell model above turns this intrinsic object into the familiar wave-function representation

$$\mathcal{F}_s(\mathcal{H}_1) \simeq \mathbb{C}\Omega_F \oplus \bigoplus_{n\geq 1} L^2_{\mathrm{sym}}\big((\Sigma_m^+)^n, \mathrm{d}\mu_m(p_1)\cdots \mathrm{d}\mu_m(p_n)\big).$$

Under this identification a decomposable n-particle vector is represented by

$$(f_1 \odot \cdots \odot f_n)(p_1, \dots, p_n) = \frac{1}{\sqrt{n!}} \sum_{\sigma\in S_n} \prod_{j=1}^{n} f_j(p_{\sigma(j)}),$$

which gives

$$\langle f_1 \odot \cdots \odot f_n, g_1 \odot \cdots \odot g_n\rangle = \sum_{\sigma\in S_n} \prod_{j=1}^{n} \langle f_j, g_{\sigma(j)}\rangle_{L^2(\Sigma_m^+, \mathrm{d}\mu_m)}.$$

This is the Fock-space inner product that the Haag–Ruelle limits reproduce inside the physical Hilbert space.

Remark 17.10 (Statistics of the asymptotic Fock space). The notation $\mathcal{F}_s(\mathcal{H}_1)$ is appropriate here because the one-particle sector under discussion is the scalar bosonic sector constructed earlier. For several stable species the asymptotic Fock space is the graded tensor product determined by their statistics: integer-spin bosonic species contribute symmetric factors, while half-integer-spin fermionic species contribute antisymmetric factors. The spin-statistics theorem gives this assignment under the Wightman hypotheses stated in Chapter 40; the present Haag–Ruelle construction uses the scalar instance.

Definition 17.11 (Haag–Ruelle wave operators on the separated algebraic domain)**.** The asymptotic identifications from the free Fock space $\mathcal{F}_s(\mathcal{H}_1)$ to the physical Hilbert space are called the incoming and outgoing Haag–Ruelle wave operators. On the dense algebraic domain generated by velocity-separated decomposable vectors, each wave operator assigns to an abstract collection of asymptotic one-particle wave packets the physical Hilbert-space vector obtained as a large-time Haag–Ruelle limit.

The topology in this statement is pointwise Hilbert-norm convergence on the asymptotic Fock domain. Choose, for a dense algebraic subspace $\mathcal{F}^{\mathrm{alg}}_{s,\mathrm{sep}}(\mathcal{H}_1) \subset \mathcal{F}_s(\mathcal{H}_1)$, Haag–Ruelle creators and velocity-separated wave packets representing each decomposable vector F. The time-dependent identification map $\Omega^{\mathrm{HR}}_{\mathrm{out},t}$ is defined on that vector by the corresponding product of Haag–Ruelle operators applied to the vacuum. The Cook-limit statement says

$$\lim_{t\to+\infty} \left\|\Omega^{\mathrm{HR}}_{\mathrm{out},t}F - \Omega_{\mathrm{out}}F\right\|_{\mathcal{H}} = 0, \qquad F \in \mathcal{F}^{\mathrm{alg}}_{s,\mathrm{sep}}(\mathcal{H}_1), \tag{17.2}$$

and analogously for $t \to -\infty$ and Ω_{in}. Packaged as maps from the asymptotic Fock domain to $\mathcal{H}$, this is strong operator convergence on that domain: convergence holds after applying the approximating map to each fixed vector F. The weak convergence of all matrix elements $\langle \eta, \Omega^{\mathrm{HR}}_{\mathrm{out},t}F\rangle$ follows immediately from (17.2), while the converse weak statement does not by itself give the isometric wave operator. Operator-norm convergence,

$$\sup_{\|F\|_{\mathcal{F}_s}=1} \left\|\Omega^{\mathrm{HR}}_{\mathrm{out},t}F - \Omega_{\mathrm{out}}F\right\|_{\mathcal{H}} \longrightarrow 0,$$

is not asserted in Haag–Ruelle theory. The isometric extension of $\Omega_{\mathrm{out/in}}$ to all of $\mathcal{F}_s(\mathcal{H}_1)$ is obtained after the Fock inner product has been reproduced on the dense algebraic domain.

Proposition 17.12 (Definition and isometry of the scalar Haag–Ruelle maps)**.** *Under the Cook and scalar-product contraction hypotheses of Proposition 17.8, the following assignments on velocity-separated decomposable vectors are well-defined:*

$$\Omega_{\mathrm{out}}(\psi_1 \odot \cdots \odot \psi_n) = \Psi^{\mathrm{out}}(h_1, \dots, h_n),$$
$$\Omega_{\mathrm{in}}(\psi_1 \odot \cdots \odot \psi_n) = \Psi^{\mathrm{in}}(h_1, \dots, h_n),$$

where $\psi_j = P_1 B_{j,t}(h_j)\Omega$. The Cook-limit statement above makes these definitions independent of the choice of interpolating operators. By linearity and continuity they extend from such vectors to all of $\mathcal{F}_s(\mathcal{H}_1)$.

The Fock inner product is reproduced by the Hilbert-space inner product of asymptotic states:

$$\begin{aligned}&\left\langle \Psi^{\mathrm{out}}(\psi_1, \dots, \psi_n), \Psi^{\mathrm{out}}(\varphi_1, \dots, \varphi_m)\right\rangle_{\mathcal{H}} \\ &= \delta_{nm} \sum_{\sigma\in S_n} \prod_{j=1}^{n} \langle \psi_j, \varphi_{\sigma(j)}\rangle_{\mathcal{H}_1},\end{aligned}$$

and the same formula holds for incoming states. Therefore

$$\Omega_{\rm out}, \Omega_{\rm in} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{H}$$

are isometric wave operators. Their ranges are the outgoing and incoming scattering subspaces,

$$\mathcal{H}_{\rm out} = \operatorname{Ran}\Omega_{\rm out}, \qquad \mathcal{H}_{\rm in} = \operatorname{Ran}\Omega_{\rm in}.$$

Proof. Suppose two choices of almost-local creators and wave packets give the same one-particle vector ψ_j. The Cook-limit hypothesis gives large-time limits for both choices, and the independence part of Proposition 17.8 says that the difference of the two limiting vectors has Hilbert norm zero. Hence the assignment depends only on $\psi_1, \ldots, \psi_n$, not on the interpolating coordinates used to produce them.

On the velocity-separated algebraic domain, the scalar-product contraction estimate of Proposition 17.8 gives the permanent formula displayed above. That formula is precisely the inner product of the symmetric Fock tensor $\psi_1 \odot \cdots \odot \psi_n$ with $\varphi_1 \odot \cdots \odot \varphi_m$. Thus the map preserves inner products on a dense subspace of $\mathcal{F}_s(\mathcal{H}_1)$. A norm-preserving linear map on a dense subspace extends uniquely to an isometry on the Hilbert completion, and the range of an isometry is closed because a Cauchy sequence of image vectors has a Cauchy preimage sequence. $\square$

The Poincare group acts on $\mathcal{F}_s(\mathcal{H}_1)$ by the free Fock representation U_F induced from its action on $\mathcal{H}_1$. The Haag–Ruelle wave operators intertwine the physical and free actions:

$$U(a,\Lambda)\Omega_{\rm out/in} = \Omega_{\rm out/in} U_F(a,\Lambda).$$

This equation states that an asymptotic n-particle state transforms as the corresponding symmetrized tensor product of one-particle Wigner states.

Equivalently, the wave operators may be described by generalized momentum kernels. For bosons,

$$\begin{aligned} &{}^{\rm out/in}\langle q_1, \ldots, q_m | p_1, \ldots, p_n \rangle^{\rm out/in} \\ &\qquad = \delta_{mn} \sum_{\sigma \in S_n} \prod_{j=1}^{n} (2\pi)^d 2\omega_{\vec{p}_j}\, \delta^{(d)}(\vec{q}_j - \vec{p}_{\sigma(j)}). \end{aligned}$$

For smooth wave packets,

$$\Psi^{\rm out/in}(f_1, \ldots, f_n) = Z_\phi^{n/2} \int \prod_{j=1}^{n} \mathrm{d}\mu_m(p_j) \prod_{j=1}^{n} \tilde{f}_j(p_j)\, |p_1, \ldots, p_n\rangle^{\rm out/in},$$

whenever the f_j are chosen with separated velocity supports. This is the precise form of the statement that either the incoming or the outgoing asymptotic basis labels states by free particle momenta, while the comparison between the two bases is the scattering matrix.

$$\begin{tikzcd} \mathcal{F}_s(\mathcal{H}_1) \arrow[r, "\Omega_{\rm in}"] \arrow[d, "S"'] & \mathcal{H}_{\rm in} \subset \mathcal{H} \arrow[d, dashed, leftrightarrow] \\ \mathcal{F}_s(\mathcal{H}_1) \arrow[r, "\Omega_{\rm out}"'] & \mathcal{H}_{\rm out} \subset \mathcal{H} \end{tikzcd}$$

17.8 The Scattering Operator

Definition 17.13 (Scattering operator in a massive Haag–Ruelle sector)**.** If

$$\mathcal{H}_{\mathrm{in}} = \mathcal{H}_{\mathrm{out}},$$

then incoming scattering states evolve into outgoing scattering states in the sector under consideration. The scattering operator on the asymptotic Fock space is

$$S = \Omega_{\mathrm{out}}^* \Omega_{\mathrm{in}} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{F}_s(\mathcal{H}_1).$$

Unitarity and matrix elements of S. When the incoming and outgoing ranges coincide, the operator

$$S = \Omega_{\mathrm{out}}^* \Omega_{\mathrm{in}}$$

is unitary. The matrix element between an incoming n-particle wave packet $\psi_1 \odot \cdots \odot \psi_n$ and an outgoing m-particle wave packet $\varphi_1 \odot \cdots \odot \varphi_m$ is

$$\begin{aligned}
&\langle \varphi_1 \odot \cdots \odot \varphi_m,\, S(\psi_1 \odot \cdots \odot \psi_n)\rangle_{\mathcal{F}_s(\mathcal{H}_1)} \\
&\qquad = \langle \Omega_{\mathrm{out}}(\varphi_1 \odot \cdots \odot \varphi_m),\, \Omega_{\mathrm{in}}(\psi_1 \odot \cdots \odot \psi_n)\rangle_{\mathcal{H}}\,.
\end{aligned}$$

This equation is the nonperturbative definition of the scattering matrix elements in the massive sector. The right hand side is an inner product in the physical Hilbert space; the left hand side records the same number in the free asymptotic particle basis. Let $\mathcal{R} = \operatorname{Ran}\Omega_{\mathrm{in}} = \operatorname{Ran}\Omega_{\mathrm{out}}$. Because the wave operators are isometries from $\mathcal{F}_s(\mathcal{H}_1)$ onto $\mathcal{R}$,

$$\Omega_{\mathrm{in}}^*\Omega_{\mathrm{in}} = \Omega_{\mathrm{out}}^*\Omega_{\mathrm{out}} = I_{\mathcal{F}_s(\mathcal{H}_1)}, \qquad \Omega_{\mathrm{in}}\Omega_{\mathrm{in}}^* = \Omega_{\mathrm{out}}\Omega_{\mathrm{out}}^* = P_{\mathcal{R}}.$$

Consequently

$$S^*S = \Omega_{\mathrm{in}}^*\Omega_{\mathrm{out}}\Omega_{\mathrm{out}}^*\Omega_{\mathrm{in}} = \Omega_{\mathrm{in}}^* P_{\mathcal{R}} \Omega_{\mathrm{in}} = I_{\mathcal{F}_s(\mathcal{H}_1)}$$

using $S^* = \Omega_{\mathrm{in}}^*\Omega_{\mathrm{out}}$, and

$$SS^* = \Omega_{\mathrm{out}}^*\Omega_{\mathrm{in}}\Omega_{\mathrm{in}}^*\Omega_{\mathrm{out}} = \Omega_{\mathrm{out}}^* P_{\mathcal{R}} \Omega_{\mathrm{out}} = I_{\mathcal{F}_s(\mathcal{H}_1)}.$$

Thus S is unitary. For asymptotic vectors $\Phi, \Psi \in \mathcal{F}_s(\mathcal{H}_1)$, the defining property of the Hilbert adjoint gives

$$\langle \Phi, S\Psi\rangle = \langle \Phi, \Omega_{\mathrm{out}}^*\Omega_{\mathrm{in}}\Psi\rangle = \langle \Omega_{\mathrm{out}}\Phi, \Omega_{\mathrm{in}}\Psi\rangle_{\mathcal{H}},$$

which is the displayed matrix-element formula after substituting the symmetrized tensor wave packets.

Definition 17.14 (Range and completeness statements)**.** There are three logically distinct range statements.

(a) *Range coincidence for the chosen asymptotic species.* The equality

$$\mathcal{H}_{\mathrm{in}} = \mathcal{H}_{\mathrm{out}} =: \mathcal{H}_{\mathrm{sc}}$$

says that the incoming and outgoing Haag–Ruelle constructions based on the selected isolated one-particle subspace have the same physical range. This is exactly the hypothesis needed for S above to be a unitary operator on the corresponding asymptotic Fock space.

(b) *Completeness in a specified massive scattering sector.* Let $\mathcal{H}_{\text{sec}} \subset \mathcal{H}$ be a closed reducing subspace selected by superselection charge, mass gap, and a finite list of stable isolated particle species. Replacing $\mathcal{F}_s(\mathcal{H}_1)$ by the appropriate graded Fock space over the direct sum of those one-particle spaces, sector completeness is the statement

$$\mathcal{H}_{\text{in}} = \mathcal{H}_{\text{out}} = \mathcal{H}_{\text{sec}}.$$

This is the level used by later cluster and partial-wave arguments for a closed massive sector: all states in that sector are accounted for by the listed asymptotic particles, while other superselection sectors or other parts of the full theory may remain outside the discussion.

(c) *Full asymptotic completeness.* After including every stable particle species and superselection sector in the physical theory, full asymptotic completeness is the statement

$$\mathcal{H}_{\text{in}} = \mathcal{H}_{\text{out}} = \mathcal{H}$$

on the relevant finite-energy physical Hilbert space.

Thus ordinary asymptotic completeness is the last statement: it includes the range coincidence in (a) and asserts that the resulting scattering sector is all of $\mathcal{H}$. The definition of S on a chosen scattering sector requires (a). The later closed-channel unitarity formulae require the sector statement (b) for the sector under analysis. Full asymptotic completeness asserts

$$\mathcal{H} = \mathcal{H}_{\text{in}} = \mathcal{H}_{\text{out}}$$

for the whole chosen physical Hilbert space. This is an additional dynamical theorem or hypothesis. For nontrivial interacting massive local Wightman theories in four spacetime dimensions, full asymptotic completeness is presently an open problem rather than a known consequence of the axioms. Proofs exist only in restricted model classes, notably constructive or integrable models in lower spacetime dimensions and related weakened-locality settings.

17.9 Relation to Local Correlation Functions

The construction above produces the S-operator from Hilbert-space limits. It also identifies what a later reduction formula must compute. If $\psi_j = P_1 B_{j,t}(h_j)\Omega$ and $\varphi_i = P_1 C_{i,t}(g_i)\Omega$, then an m-out, n-in matrix element is

$$\begin{aligned}
&\langle \Omega_{\text{out}}(\varphi_1 \odot \cdots \odot \varphi_m),\, \Omega_{\text{in}}(\psi_1 \odot \cdots \odot \psi_n)\rangle_{\mathcal{H}} \\
&\quad = \lim_{T\to+\infty} \langle \Omega | C_{m,T}(g_m)^* \cdots C_{1,T}(g_1)^* B_{1,-T}(h_1) \cdots B_{n,-T}(h_n) | \Omega \rangle.
\end{aligned}$$

For large T, the outgoing packets are localized near future timelike directions and the incoming packets near past timelike directions. Locality permits these vacuum matrix elements to be reorganized into time-ordered correlation functions up to terms that vanish in the scattering limit. The next chapter turns this statement into LSZ reduction: one extracts the above matrix elements from the one-particle poles of time-ordered Green functions.

17.10 Scope of the Massive Construction

The massive Haag–Ruelle construction gives a precise scattering theory for stable particles whose one-particle mass shell is isolated and whose sector is generated by the local creators used in the theorem. Gauge theories require an additional localization check. A local gauge-invariant observable cannot change an unscreened Gauss-law surface charge, so charged particles require charged field algebras or noncompact gauge-invariant dressings rather than ordinary bounded-region observable creators; the algebraic formulation is developed in Chapter 22. Massless particles, long-range gauge forces, infraparticle sectors, and confinement then add further asymptotic questions with their own hypotheses and limiting procedures.

The logical output of this chapter is therefore fixed. A scattering operator has been defined before any LSZ formula and before any perturbative interpretation of Feynman graphs as scattering amplitudes. Perturbative scattering calculations, when available, compute matrix elements of this operator after the reduction theorem has connected the Hilbert-space limits to Green functions.

Chapter 18

LSZ Reduction

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

18.1 The Reduction Problem

Haag–Ruelle theory defines incoming and outgoing scattering states by large-time Hilbert-space limits. LSZ reduction is the theorem that computes matrix elements of that already-defined scattering operator from the singular part of time-ordered Green functions at the one-particle mass shell.

Definition 18.1 (Massive scalar LSZ datum)**.** This chapter keeps the same massive scalar particle as in the Haag–Ruelle chapter. The assumptions are:

1. the particle has mass $m > 0$ and an isolated one-particle subspace $\mathcal{H}_1$;
2. the Haag–Ruelle construction gives isometric wave operators

   $$\Omega_{\mathrm{in}}, \Omega_{\mathrm{out}} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{H}$$

   from the bosonic asymptotic Fock space over $\mathcal{H}_1$;
3. the incoming and outgoing scattering subspaces for this particle coincide, so the scattering operator

   $$S = \Omega_{\mathrm{out}}^* \Omega_{\mathrm{in}}$$

   is unitary on $\mathcal{F}_s(\mathcal{H}_1)$;

4. the scalar local field $\widehat{\phi}$ has nonzero one-particle matrix element;
5. the time-ordered products used below have been defined as distributions.

Definition 18.2 (Relativistic external-state normalization)**.** Let generalized one-particle momentum states, understood as distributional vectors as in Section 5.2, be normalized by

$$\langle \vec{q}|\vec{p}\rangle = (2\pi)^d \, 2\omega_{\vec{p}} \, \delta^{(d)}(\vec{q}-\vec{p}), \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^2+m^2},$$

where $d = D-1$. The Lorentz-invariant mass-shell measure is

$$\mathrm{d}\mu_m(\vec{p}) = \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d \, 2\omega_{\vec{p}}}.$$

The corresponding positive-energy mass shell is

$$\Sigma_m^+ = \{p = (\omega_{\vec{p}}, \vec{p}) : \vec{p} \in \mathbb{R}^d\}.$$

The field normalization on this shell is recorded by the positive number Z_ϕ defined by

$$\langle \Omega|\widehat{\phi}(0)|\vec{p}\rangle = Z_\phi^{1/2}.$$

Kernel notation such as

$${}_{\mathrm{out}}\langle \vec{q}_1, \ldots, \vec{q}_m|\vec{p}_1, \ldots, \vec{p}_n\rangle_{\mathrm{in}}$$

means the distributional kernel obtained after smearing all momenta with smooth compactly supported wave packets.

Definition 18.3 (Wave-packet scattering matrix element)**.** For wave packets $f_i, g_j \in C_c^\infty(\mathbb{R}^d)$, set

$$F_{\mathrm{out}} = f_1 \odot \cdots \odot f_m, \qquad F_{\mathrm{in}} = g_1 \odot \cdots \odot g_n$$

as vectors in $\mathcal{F}_s(\mathcal{H}_1)$, where a momentum-space packet f denotes the one-particle vector

$$\psi_f = \int \mathrm{d}\mu_m(\vec{p}) \, f(\vec{p})|\vec{p}\rangle.$$

The scattering matrix element is the Hilbert-space number

$$\langle F_{\mathrm{out}}, SF_{\mathrm{in}}\rangle_{\mathcal{F}_s(\mathcal{H}_1)} = \langle \Omega_{\mathrm{out}}F_{\mathrm{out}}, \Omega_{\mathrm{in}}F_{\mathrm{in}}\rangle_{\mathcal{H}}.$$

Its connected component is obtained by subtracting the contributions assigned to proper partitions of the incoming and outgoing labels. The partition formula is stated in Section 18.10; the theorem below identifies each connected component with an external one-particle residue of a time-ordered Green function.

18.2 The LSZ Theorem as a Wave-Packet Statement

Let

$$q_i = (q_i^0, \vec{q}_i), \qquad p_j = (p_j^0, \vec{p}_j)$$

be external momentum variables. Their physical boundary values lie on Σ_m^+:

$$q_i^0 = \omega_{\vec{q}_i}, \qquad p_j^0 = \omega_{\vec{p}_j}.$$

Introduce all-incoming Green-function momenta

$$k_i = q_i \quad (1 \le i \le m), \qquad k_{m+j} = -p_j \quad (1 \le j \le n).$$

Let G^{conn}_{m+n} be the connected time-ordered distribution and let $\widetilde{G}^{\mathrm{conn}}_{m+n}$ denote its Fourier transform in the convention of Section 18.8.

For off-shell external variables in a neighborhood of the physical mass shells, define the amputated external-residue distribution

$$\begin{aligned} &\mathcal{L}_{m,n}(q_1, \dots, q_m | p_1, \dots, p_n) \\ &\quad = \left[\prod_{a=1}^{m+n} Z_\phi^{-1/2}\, \mathrm{i}(k_a^2 + m^2) \right] \widetilde{G}^{\mathrm{conn}}_{m+n}(k_1, \dots, k_{m+n}). \end{aligned} \tag{18.1}$$

The notation $\mathcal{L}^{\mathrm{on}}_{m,n}$ denotes the distributional boundary value of (18.1) obtained after smearing the spatial momenta with wave packets, taking the simultaneous external one-particle residues, and then setting $q_i^0 = \omega_{\vec{q}_i}$, $p_j^0 = \omega_{\vec{p}_j}$. Here "residue" means the coefficient of the isolated Feynman denominator in each external variable after the boundary value has been taken. It is not a complex residue in a one-dimensional variable k_a^2; for fixed spatial momentum the meromorphic variable is k_a^0.

Definition 18.4 (LSZ external boundary-value extraction)**.** Let $H(k)$ be a distribution of one external momentum variable, with all other variables already smeared, and suppose that in a neighborhood of the positive or negative mass shell selected by the external wave packet it has the decomposition

$$H(k) = \frac{-\mathrm{i} Z_\phi\, H_{\mathrm{red}}(k)}{k^2 + m^2 - \mathrm{i}0} + H_{\mathrm{less}}(k),$$

where H_{red} has a boundary value on that shell and $(k^2 + m^2) H_{\mathrm{less}}(k)$ has zero boundary value there. The LSZ extraction of that external leg is

$$\mathrm{LSZ}_{\phi,k}\, H := \operatorname*{bv}_{\Sigma_m} \left[Z_\phi^{-1/2}\, \mathrm{i}(k^2 + m^2) H(k) \right].$$

Under the displayed decomposition,

$$\mathrm{LSZ}_{\phi,k}\, H = Z_\phi^{1/2} \operatorname*{bv}_{\Sigma_m} H_{\mathrm{red}}.$$

For several external legs the operation is the iterated distributional boundary value after all spatial momenta have been paired with wave packets.

Theorem 18.5 (LSZ reduction for a massive scalar)**.** *Under the assumptions above,*

$$\begin{aligned} &\big(\langle F_{\mathrm{out}}, S F_{\mathrm{in}} \rangle_{\mathcal{F}_s(\mathcal{H}_1)} \big)_{\mathrm{conn}} \\ &\quad = \int \prod_{i=1}^{m} \mathrm{d}\mu_m(\vec{q}_i)\, f_i(\vec{q}_i)^* \prod_{j=1}^{n} \mathrm{d}\mu_m(\vec{p}_j)\, g_j(\vec{p}_j) \\ &\qquad\quad \times \mathcal{L}^{\mathrm{on}}_{m,n}(q_1, \dots, q_m | p_1, \dots, p_n). \end{aligned} \tag{18.2}$$

The left side is the connected component in the partition expansion of the scattering kernel. The right side is the simultaneous external one-particle residue of the connected time-ordered distribution, paired with the same incoming and outgoing wave packets.

Proof. The theorem compares two already constructed objects. The left side is the Haag–Ruelle scattering matrix element. The right side is obtained from time-ordered boundary-value distributions by connected truncation and external pole extraction. The residue factor $Z_\phi^{-1/2}$ is attached to the chosen interpolating field $\widehat{\phi}$; the resulting scattering matrix element is a Hilbert-space matrix element of S on the asymptotic Fock space.

We give the proof at the level at which the theorem is stated: all external variables are paired with compactly supported wave packets in the isolated mass-shell neighborhood, and the time-ordered products are assumed to be defined distributions with the stated one-particle pole.

First replace the incoming and outgoing Fock wave packets by the Haag–Ruelle approximants from Chapter 17. For large $T > 0$, the connected part of

$$\langle\Omega|\hat{\phi}^*_{f_m^{(T)}}\cdots\hat{\phi}^*_{f_1^{(T)}}\hat{\phi}_{g_1^{(-T)}}\cdots\hat{\phi}_{g_n^{(-T)}}|\Omega\rangle$$

has the same limit as the connected scattering matrix element on the left of (18.2). This is the definition of the Haag–Ruelle in/out vectors together with the connected partition subtraction described in Section 18.10.

Second use locality and the velocity separation of the Haag–Ruelle packets. The outgoing packets lie in future velocity tubes and the incoming packets lie in past velocity tubes. Reordering the corresponding fields into the time-ordered product changes the large-T expression by a term whose norm goes to zero by the same commutator estimates used in the Haag–Ruelle construction. Thus the connected scattering matrix element is the large-time limit of the connected time-ordered distribution smeared with the translated external packets.

Third take the Fourier transform in the external variables. Each outgoing external packet contributes an oscillatory factor selecting the positive-energy pole $k^0 = \omega_{\vec{k}} - \mathrm{i}0$, and each incoming packet, written with all Green-function momenta incoming, contributes the oscillatory factor selecting the negative-energy pole $k^0 = -\omega_{\vec{k}} + \mathrm{i}0$. The Riemann–Lebesgue estimate, equivalently repeated integration by parts away from the selected pole, kills the part regular at the selected shell. The only surviving contribution is the coefficient of

$$\frac{-\mathrm{i}Z_\phi}{k^2 + m^2 - \mathrm{i}0}$$

in every external variable. Definition 18.4 is exactly this coefficient extraction, with one factor $Z_\phi^{-1/2}$ for each external interpolating field.

Finally, after all external residues are extracted, the remaining variables are the physical on-shell momenta $q_i \in \Sigma_m^+$ and $p_j \in \Sigma_m^+$, with all-incoming Green-function momenta $k_i = q_i$ and $k_{m+j} = -p_j$. Pairing this on-shell distribution with the same wave packets gives the right side of (18.2). The disconnected two-point pole factors have been removed by connected truncation on both sides; the full kernel is recovered from connected kernels by the partition formula in Section 18.10. □

18.3 The One-Particle Pole

Definition 18.6 (Interpolating-field pole datum)**.** The matrix element defining Z_ϕ says that $\widehat{\phi}$ is an interpolating scalar field for the particle species under consideration. With the normalization above, the Källén–Lehmann measure of $\widehat{\phi}$ contains the atom

$$\mathrm{d}\rho(s) = Z_\phi \, \delta(s - m^2) \, \mathrm{d}s + \mathrm{d}\rho_{\mathrm{cont}}(s).$$

Consequently the Fourier transform of the time-ordered two-point function has the pole

$$\widetilde{G}_2(k) = \frac{-\mathrm{i}Z_\phi}{k^2 + m^2 - \mathrm{i}0} + \widetilde{G}_{2,\mathrm{less}}(k),$$

where $\widetilde{G}_{2,\mathrm{less}}$ has no pole at the isolated one-particle mass shell. The number Z_ϕ depends on the chosen local field. The particle state and the S-operator are invariant data; changing the interpolating field changes the residue factor appearing in the reduction formula.

The preceding display identifies the coefficient of the invariant Feynman denominator. For fixed $\vec{k}$, the actual meromorphic variable is k^0, and

$$k^2 + m^2 - \mathrm{i}0 = -\big(k^0 - \omega_{\vec{k}} + \mathrm{i}0\big)\big(k^0 + \omega_{\vec{k}} - \mathrm{i}0\big).$$

Consequently

$$\frac{-\mathrm{i}Z_\phi}{k^2 + m^2 - \mathrm{i}0} = \frac{\mathrm{i}Z_\phi}{2\omega_{\vec{k}}} \left[\frac{1}{k^0 - \omega_{\vec{k}} + \mathrm{i}0} - \frac{1}{k^0 + \omega_{\vec{k}} - \mathrm{i}0} \right].$$

Thus the complex residues at the two linear k^0-poles are $\mathrm{i}Z_\phi/(2\omega_{\vec{k}})$ and $-\mathrm{i}Z_\phi/(2\omega_{\vec{k}})$, whereas $-\mathrm{i}Z_\phi$ is the coefficient of the invariant denominator $k^2 + m^2 - \mathrm{i}0$. LSZ uses the latter coefficient, together with the mass-shell measure, to match the Haag–Ruelle one-particle normalization.

The operation

$$Z_\phi^{-1/2} \, \mathrm{i}(k^2 + m^2)$$

removes an external one-particle propagator with the conventions of the Lorentzian Green-function chapter:

$$Z_\phi^{-1/2} \, \mathrm{i}(k^2 + m^2) \, \frac{-\mathrm{i}Z_\phi}{k^2 + m^2 - \mathrm{i}0} \longrightarrow Z_\phi^{1/2}$$

as a mass-shell boundary-value coefficient. Applying one such operation to every external field is the analytic core of LSZ reduction.

Invariant denominator coefficient and linear pole residues. With the mostly-plus convention $k^2 = -(k^0)^2 + \vec{k}^{\,2}$ and $\omega_{\vec{k}} = (\vec{k}^{\,2} + m^2)^{1/2}$,

$$k^2 + m^2 - \mathrm{i}0 = -\big(k^0 - \omega_{\vec{k}} + \mathrm{i}0\big)\big(k^0 + \omega_{\vec{k}} - \mathrm{i}0\big).$$

Consequently the one-particle part of the Feynman two-point function obeys

$$\frac{-\mathrm{i}Z_\phi}{k^2 + m^2 - \mathrm{i}0} = \frac{\mathrm{i}Z_\phi}{2\omega_{\vec{k}}} \left[\frac{1}{k^0 - \omega_{\vec{k}} + \mathrm{i}0} - \frac{1}{k^0 + \omega_{\vec{k}} - \mathrm{i}0} \right],$$

so the coefficient extracted by invariant-denominator LSZ is $-\mathrm{i}Z_\phi$, while the linear k^0-residues are $\mathrm{i}Z_\phi/(2\omega_{\vec{k}})$ and $-\mathrm{i}Z_\phi/(2\omega_{\vec{k}})$. Since

$$k^2 + m^2 = -(k^0)^2 + \omega_{\vec{k}}^2 = -(k^0 - \omega_{\vec{k}})(k^0 + \omega_{\vec{k}}),$$

the displayed Feynman prescription assigns the positive-energy pole below the real k^0-axis and the negative-energy pole above it. The partial-fraction identity

$$\frac{1}{(k^0-\omega_{\vec{k}}+\mathrm{i}0)(k^0+\omega_{\vec{k}}-\mathrm{i}0)}=\frac{1}{2\omega_{\vec{k}}}\left[\frac{1}{k^0-\omega_{\vec{k}}+\mathrm{i}0}-\frac{1}{k^0+\omega_{\vec{k}}-\mathrm{i}0}\right]$$

then gives the second display after multiplying by the overall minus sign in the invariant denominator and by $-\mathrm{i}Z_\phi$. The listed residues are the coefficients of the two linear factors.

18.4 The Pole as a Spectral Projection

The preceding discussion can be stated without treating the pole as a formal singularity. Let P_1 denote the spectral projection of the translation representation onto the isolated positive-energy mass shell

$$\Sigma_m^+=\{p\in\mathbb{R}^{1,d}:p^0>0,\ p^2=-m^2\}.$$

For a test function $f\in C_c^\infty(\mathbb{R}^{1,d})$, define its two mass-shell restrictions by

$$f_+(\vec{p})=\int \mathrm{d}^D x\, f(x)e^{-ip\cdot x},\qquad f_-(\vec{p})=\int \mathrm{d}^D x\, f(x)e^{ip\cdot x},\qquad p^0=\omega_{\vec{p}}.$$

With the translation convention $U(a)=\exp(\mathrm{i}a_\mu P^\mu)$, the matrix element above gives

$$P_1\widehat{\phi}(f)\Omega=Z_\phi^{1/2}\int \mathrm{d}\mu_m(\vec{p})\, f_-(\vec{p})|\vec{p}\rangle.$$

Equivalently, for another test function g,

$$\langle\Omega|\widehat{\phi}(f)P_1\widehat{\phi}(g)|\Omega\rangle=Z_\phi\int \mathrm{d}\mu_m(\vec{p})\, f_+(\vec{p})g_-(\vec{p}).$$

This identity is the Hilbert-space origin of the pole in the two-point function. The remaining spectral projection $1-P_1$ has spectral support away from Σ_m^+ in a neighborhood of the isolated shell, so its Fourier transform has no one-particle pole there. Thus the residue of the two-point function is the mass-shell kernel induced by P_1 and by the chosen field-state matrix element.

Lemma 18.7 (External pole as one-particle spectral projection)**.** *Let $f,g\in C_c^\infty(\mathbb{R}^{1,d})$, and let*

$$f_\pm(\vec{p})=\int \mathrm{d}^D x\, f(x)\mathrm{e}^{\mp\mathrm{i}p\cdot x},\qquad g_\pm(\vec{p})=\int \mathrm{d}^D x\, g(x)\mathrm{e}^{\mp\mathrm{i}p\cdot x},\qquad p^0=\omega_{\vec{p}}.$$

Then the isolated one-particle projection of the smeared field is

$$P_1\widehat{\phi}(f)\Omega=Z_\phi^{1/2}\int \mathrm{d}\mu_m(\vec{p})\, f_-(\vec{p})|\vec{p}\rangle,$$

and

$$\langle\Omega|\widehat{\phi}(f)P_1\widehat{\phi}(g)|\Omega\rangle=Z_\phi\int \mathrm{d}\mu_m(\vec{p})\, f_+(\vec{p})g_-(\vec{p}).$$

The coefficient of the external LSZ pole is therefore exactly the kernel of the projection P_1, multiplied by the field-state matrix-element factors.

Proof. Translation covariance gives

$$\widehat{\phi}(x) = U(x)\widehat{\phi}(0)U(x)^{-1}, \qquad U(x)\Omega = \Omega.$$

On the one-particle subspace, the generalized momentum ket satisfies

$$U(x)|\vec{p}\rangle = \mathrm{e}^{\mathrm{i}x_\mu p^\mu}|\vec{p}\rangle.$$

The normalization $\langle\Omega|\widehat{\phi}(0)|\vec{p}\rangle = Z_\phi^{1/2}$ and hermiticity of the scalar interpolating field give the one-particle component of $\widehat{\phi}(x)\Omega$ as

$$P_1\widehat{\phi}(x)\Omega = Z_\phi^{1/2}\int \mathrm{d}\mu_m(\vec{p})\,\mathrm{e}^{\mathrm{i}x_\mu p^\mu}|\vec{p}\rangle.$$

Smearing with $f(x)$ gives the first displayed identity, because the definition of f_- uses $\mathrm{e}^{+\mathrm{i}p\cdot x}$. Taking the inner product of the corresponding f- and g-smeared vectors, and using the relativistic delta-function normalization of $|\vec{p}\rangle$, gives the second display. Since the complement $1 - P_1$ has spectral support separated from the isolated shell in the chosen neighborhood, it contributes no one-particle pole there. □

For an N-point time-ordered function, LSZ extraction applies this one-particle projection in each external variable. After the external projections have been taken, connected truncation removes vacuum-disconnected factors, and the result is compared with the Haag–Ruelle wave operators. This is the precise sense in which the reduction formula converts time-ordered correlation functions into scattering matrix elements.

18.5 Connected Truncation and External Amputation

Definition 18.8 (Connected Lorentzian source convention)**.** The pole extracted by LSZ is an external one-particle pole. It is useful to separate this operation from two other operations that are often compressed in notation. First, the connected time-ordered functions are obtained from the generating functional by taking cumulants. If

$$Z[J] = \langle\Omega|\,\mathrm{T}\exp\left(\mathrm{i}\int J(x)\widehat{\phi}(x)\,\mathrm{d}^Dx\right)|\Omega\rangle,$$

then

$$G_N^{\mathrm{conn}}(x_1,\ldots,x_N) = \mathrm{i}^{-N}\frac{\delta^N \log Z[J]}{\delta J(x_1)\cdots\delta J(x_N)}\bigg|_{J=0}. \tag{18.3}$$

With the sign convention for J inherited from the exponential above, this is the cumulant part of the time-ordered distribution: it contains precisely the vacuum-connected components. External one-particle propagators remain until the residue operation is applied. The prefactor is part of the source convention. If instead one defines

$$Z_-[J] = \langle\Omega|\,\mathrm{T}\exp\left(-\mathrm{i}\int J(x)\widehat{\phi}(x)\,\mathrm{d}^Dx\right)|\Omega\rangle,$$

then the same connected distribution is recovered by

$$G_N^{\mathrm{conn}}(x_1,\ldots,x_N) = (-\mathrm{i})^{-N}\frac{\delta^N \log Z_-[J]}{\delta J(x_1)\cdots\delta J(x_N)}\bigg|_{J=0} = \mathrm{i}^N\frac{\delta^N \log Z_-[J]}{\delta J(x_1)\cdots\delta J(x_N)}\bigg|_{J=0}.$$

Thus the Jacobian between source derivatives and time-ordered fields changes with the sign chosen in the source coupling; the Green function itself does not.

The Euclidean source convention is compatible with this normalization. Let

$$Z_E[J_E] = \langle\Omega|T_E \exp\left(\int J_E(\tau,\vec{x})\widehat{\phi}_E(\tau,\vec{x})\,\mathrm{d}\tau\,\mathrm{d}^d\vec{x}\right)|\Omega\rangle$$

denote the corresponding Euclidean ordered source functional. Along the imaginary-time contour used in Chapter 16,

$$x^0 = -\mathrm{i}\tau, \qquad \mathrm{d}x^0 = -\mathrm{i}\,\mathrm{d}\tau,$$

the Lorentzian source term transforms as

$$\mathrm{i}\int \mathrm{d}x^0\,\mathrm{d}^d\vec{x}\,J_L(x^0,\vec{x})\widehat{\phi}(x^0,\vec{x}) \longmapsto \int \mathrm{d}\tau\,\mathrm{d}^d\vec{x}\,J_L(-\mathrm{i}\tau,\vec{x})\widehat{\phi}_E(\tau,\vec{x}).$$

Thus $Z_E[J_E]$ is the imaginary-time restriction of $Z[J_L]$ with $J_E(\tau,\vec{x}) = J_L(-\mathrm{i}\tau,\vec{x})$, in the convention where the Euclidean exponential contains $+\int J_E\phi_E$. Before this contour rotation is made, differentiating the Lorentzian exponential N times with respect to the real-time source produces a factor i^N; the prefactor i^{-N} in Eq. (18.3) is exactly the compensation that leaves the connected time-ordered distribution rather than the source-normalized derivative.

Definition 18.9 (LSZ external amputation map)**.** Second, amputation in LSZ is extraction of the coefficient of the isolated external denominator, not a residue in the complex variable k^2. More precisely,

$$\operatorname*{bv}_{\Sigma_m}\left[(k^2+m^2)\widetilde{G}_2(k)\right] = -\mathrm{i}Z_\phi.$$

The symbol bv_{Σ_m} denotes the Feynman mass-shell boundary value: one first regards the two-point function, for fixed $\vec{k}$, as a boundary value of a meromorphic function of k^0, then multiplies by the invariant denominator and restricts to the appropriate positive- or negative-energy shell selected by the external wave packet. Multiplication by $Z_\phi^{-1/2}\mathrm{i}(k^2+m^2)$ is meaningful after smearing in a neighborhood of the mass shell and taking the boundary value. It removes only the external pole associated with the chosen stable particle. Internal propagators, threshold singularities, bound-state poles in other channels, and contact terms remain part of the amputated connected kernel.

Thus the reduction map has three logically distinct inputs:

$$\text{time-ordered distributions} \longrightarrow \text{connected distributions}$$
$$\longrightarrow \text{external one-particle residues} \longrightarrow \text{scattering matrix elements.}$$

The Haag–Ruelle construction supplies the last target independently of perturbation theory; the Green functions supply a way to compute its matrix elements when the one-particle pole hypotheses hold.

18.6 Contact Terms and Interpolating Fields

The LSZ operation depends only on the simultaneous external one-particle pole. This fact gives two useful stability properties.

First, different definitions of time-ordered products may differ by contact terms supported on partial diagonals, for example on $x_i = x_j$. In momentum space such terms are polynomial, or more generally local, in the momenta associated with the coincident insertions. They have no one-particle pole in each external variable. After multiplication by

$$\prod_a (k_a^2 + m^2)$$

and taking the on-shell boundary value with separated wave packets, these local terms do not contribute to the connected scattering kernel. They still matter for Ward identities, composite operators, and renormalized local products; they are local insertions rather than external one-particle residues.

Second, the local field used in the reduction is an interpolating field. Let $\widehat{\phi}'$ be another scalar local field with

$$\langle \Omega | \widehat{\phi}'(0) | \vec{p} \rangle = a\, Z_\phi^{1/2}, \qquad a \neq 0,$$

on the same isolated one-particle subspace. Its two-point function has one-particle residue $Z_{\phi'} = |a|^2 Z_\phi$. The LSZ factors $Z_{\phi'}^{-1/2}$ remove the changed residue, and the resulting S-matrix element is the same. Terms in $\widehat{\phi}'$ whose vacuum-to-one-particle matrix elements vanish, such as equation-of-motion terms of the form $(\Box - m^2)\widehat{\chi}$ in these sign conventions, do not create the external particle pole and therefore drop out of the reduction formula. The scattering operator belongs to the asymptotic Hilbert space; the interpolating local fields are coordinates used to access its matrix elements.

External-pole stability. Let $C(k_1, \ldots, k_N)$ be a distribution that is polynomial in one external momentum k_a in a neighborhood of the isolated mass shell, after all other variables have been smeared. Then

$$\operatorname*{bv}_{\Sigma_m} \left[(k_a^2 + m^2) C(k_1, \ldots, k_N) \right] = 0.$$

Consequently local contact terms in time-ordered products do not contribute to the simultaneous LSZ external residue. If another interpolating field $\widehat{\phi}'$ has

$$\langle \Omega | \widehat{\phi}'(0) | \vec{p} \rangle = a\, \langle \Omega | \widehat{\phi}(0) | \vec{p} \rangle, \qquad a \neq 0,$$

then its pole residue is $Z_{\phi'} = |a|^2 Z_\phi$, and the LSZ-normalized scattering matrix element is unchanged after the corresponding external field-coordinate factors are included.

The first assertion follows directly from the definition of the boundary-value coefficient. A polynomial in k_a has no factor $(k_a^2 + m^2 - \mathrm{i}0)^{-1}$. Multiplication by $k_a^2 + m^2$ gives a distribution whose restriction to $k_a^2 = -m^2$ is zero in the external coefficient sense used by Definition 18.4. Such terms can affect local Ward identities or composite-operator contact relations, but they are not external one-particle poles.

For the field-coordinate statement, the two-point one-particle term for $\widehat{\phi}'$ is multiplied by $a^* a$, hence

$$Z_{\phi'} = |a|^2 Z_\phi.$$

An N-leg Green function with external fields ϕ'_a has its simultaneous one-particle pole coefficient multiplied by the product of the corresponding one-particle overlap factors. The LSZ normalization contributes the inverse square roots of the same positive residues. These factors cancel in the pairing with normalized Haag–Ruelle one-particle wave packets, leaving the same Hilbert-space matrix element of S. If a component of $\widehat{\phi}'$ has zero vacuum-to-one-particle matrix element, then Lemma 18.7 gives zero projection to $\mathcal{H}_1$, so that component contributes no external LSZ pole.

18.7 Large-Time Oscillations and External Residues

The wave-packet statement above can be made more explicit. Let $\hat{\phi}_{f_i^{(T)}}$ create the outgoing packets at large positive time and $\hat{\phi}_{g_j^{(-T)}}$ create the incoming packets at large negative time, as in the Haag–Ruelle construction. The scattering matrix element is obtained from

$$\lim_{T\to+\infty}\langle\Omega|\hat{\phi}^*_{f_m^{(T)}}\cdots\hat{\phi}^*_{f_1^{(T)}}\hat{\phi}_{g_1^{(-T)}}\cdots\hat{\phi}_{g_n^{(-T)}}|\Omega\rangle.$$

For large T, locality permits this expression to be written as the corresponding time-ordered vacuum distribution, up to errors that vanish in the Haag–Ruelle limit. After Fourier transform each outgoing external variable carries an oscillatory phase of the form

$$\mathrm{e}^{-\mathrm{i}(k^0-\omega_{\vec{k}})T},$$

and each incoming external variable carries a phase of the form

$$\mathrm{e}^{\mathrm{i}(-k^0-\omega_{\vec{k}})(-T)}=\mathrm{e}^{\mathrm{i}(k^0+\omega_{\vec{k}})T},$$

where the Green-function momentum k is taken as entering the time-ordered correlator. If the Fourier-transformed Green function is regular at $k^0=\omega_{\vec{k}}$ or $k^0=-\omega_{\vec{k}}$, the large-T integral vanishes by repeated integration by parts. A nonzero limit is produced by the one-particle pole.

The following contour integral displays the residue. With the denominator convention used in the Lorentzian Green functions,

$$\begin{aligned}\lim_{T\to+\infty}&\int\frac{\mathrm{d}^Dk}{(2\pi)^D}\,\frac{\widetilde{g}(-k)\,\mathrm{e}^{\mathrm{i}(k^0+\omega_{\vec{k}})T}}{k^2+m^2-\mathrm{i}0}\\&=\int\frac{\mathrm{d}^d\vec{k}}{(2\pi)^d}\,\widetilde{g}(\omega_{\vec{k}},-\vec{k})\,\frac{\mathrm{i}}{2\omega_{\vec{k}}}.\end{aligned}$$

The pole enclosed by the large-time prescription is

$$k^0=-\omega_{\vec{k}}+\mathrm{i}0.$$

For an outgoing external variable the conjugate calculation selects

$$k^0=\omega_{\vec{k}}-\mathrm{i}0.$$

Large-time pole selector. Let $a\in\mathcal{S}(\mathbb{R}^{1,d})$. With the Feynman prescription above,

$$\begin{aligned}\lim_{T\to+\infty}&\int\frac{\mathrm{d}^Dk}{(2\pi)^D}\,\frac{a(k)\,\mathrm{e}^{\mathrm{i}(k^0+\omega_{\vec{k}})T}}{k^2+m^2-\mathrm{i}0}\\&=\int\frac{\mathrm{d}^d\vec{k}}{(2\pi)^d}\,a(-\omega_{\vec{k}},\vec{k})\,\frac{\mathrm{i}}{2\omega_{\vec{k}}},\end{aligned}$$

provided the spatial support is kept in the isolated mass-shell neighborhood. The conjugate large-time factor selects the positive-energy pole with the opposite boundary prescription. For fixed $\vec{k}$, factor the denominator as in the linear-residue calculation above. The phase $\mathrm{e}^{\mathrm{i}(k^0+\omega_{\vec{k}})T}$ has a nonzero large-T limit only against the singular distribution at

$$k^0 = -\omega_{\vec{k}} + \mathrm{i}0.$$

Equivalently, close the k^0-contour in the upper half-plane for $T > 0$. The enclosed pole is the negative-energy Feynman pole, and the residue of

$$\frac{1}{k^2 + m^2 - \mathrm{i}0}$$

there is $\mathrm{i}/(2\omega_{\vec{k}})$ with the Fourier normalization used in the display. The part of $a(k)$ multiplied by a function regular at the selected shell gives zero by repeated integration by parts in k^0. The positive-energy statement is the complex-conjugate contour calculation.

If the full Feynman two-point singularity $-\mathrm{i}Z_\phi/(k^2+m^2-\mathrm{i}0)$ is used instead of the denominator in the displayed integral, the extra factor $-\mathrm{i}Z_\phi$ combines with the residue above to give the mass-shell measure and the external factor $Z_\phi^{1/2}$. The LSZ operator $Z_\phi^{-1/2}\mathrm{i}(k^2+m^2)$ is the distributional way of extracting this same residue from every external leg at once.

This also explains the identity term in the S-matrix. A disconnected two-point factor

$$(2\pi)^D \delta^{(D)}(k+k') \, \frac{-\mathrm{i}Z_\phi}{k^2+m^2-\mathrm{i}0}$$

has a simultaneous one-particle pole and gives, after the same limiting procedure,

$$Z_\phi \int \mathrm{d}\mu_m(\vec{p}) \, f(\vec{p})^* g(\vec{p}),$$

which is the one-particle inner product created by the chosen field. In a multi-particle matrix element these two-point factors reproduce the free permutation terms; the remaining part is the connected scattering kernel.

18.8 Momentum-Space Kernel

The time-ordered N-point function is

$$G_N(x_1,\ldots,x_N) = \langle\Omega|\,\mathrm{T}\{\widehat{\phi}(x_1)\cdots\widehat{\phi}(x_N)\}|\Omega\rangle.$$

Its Fourier transform is defined by

$$G_N(x_1,\ldots,x_N) = \int \prod_{a=1}^{N} \frac{\mathrm{d}^D k_a}{(2\pi)^D} \, \mathrm{e}^{\mathrm{i}\sum_a k_a\cdot x_a} \, \widetilde{G}_N(k_1,\ldots,k_N).$$

Translation invariance gives

$$\widetilde{G}_N(k_1,\ldots,k_N) = (2\pi)^D \delta^{(D)}(k_1+\cdots+k_N) \, \widetilde{\mathcal{G}}_N(k_1,\ldots,k_{N-1}),$$

where $\widetilde{\mathcal{G}}_N$ denotes the reduced distribution after removing the overall momentum-conservation delta function.

For m outgoing and n incoming scalar particles, set

$$k_i = q_i \quad (1 \le i \le m), \qquad k_{m+j} = -p_j \quad (1 \le j \le n).$$

The word scalar fixes the statistics used in this section: the external asymptotic particles lie in the symmetric Fock space constructed in Chapter 17. For fermionic external particles the same Hilbert-space and distributional logic is applied in the graded asymptotic Fock space, and permutations carry the Koszul signs fixed by the spin-statistics theorem; the free spinor verification and the theorem-level hypotheses are recorded in Chapter 40. The wave-packet theorem is abbreviated by the distributional kernel notation

$$\begin{aligned}
&{}_{\mathrm{out}}\langle \vec{q}_1, \dots, \vec{q}_m | \vec{p}_1, \dots, \vec{p}_n \rangle_{\mathrm{in,conn}} \\
&\quad = \mathcal{L}^{\mathrm{on}}_{m,n}(q_1, \dots, q_m | p_1, \dots, p_n).
\end{aligned}$$

This equality is not a pointwise evaluation of generalized momentum states. It means that both sides give the same result after smearing with test wave packets and taking the external boundary value specified in Theorem 18.5.

Disconnected terms are treated by the same rule, but one-particle two-point factors reproduce the identity contribution to the scattering operator. For two identical scalar particles,

$$\begin{aligned}
&{}_{\mathrm{out}}\langle \vec{q}_1, \vec{q}_2 | \vec{p}_1, \vec{p}_2 \rangle_{\mathrm{in}} \\
&\quad = (2\pi)^d 2\omega_{\vec{p}_1} \delta^{(d)}(\vec{q}_1 - \vec{p}_1)\, (2\pi)^d 2\omega_{\vec{p}_2} \delta^{(d)}(\vec{q}_2 - \vec{p}_2) \\
&\qquad + (2\pi)^d 2\omega_{\vec{p}_1} \delta^{(d)}(\vec{q}_2 - \vec{p}_1)\, (2\pi)^d 2\omega_{\vec{p}_2} \delta^{(d)}(\vec{q}_1 - \vec{p}_2) \\
&\qquad + {}_{\mathrm{out}}\langle \vec{q}_1, \vec{q}_2 | \vec{p}_1, \vec{p}_2 \rangle_{\mathrm{in,conn}}.
\end{aligned}$$

Remark 18.10 (Identity kernel and statistics). The plus sign between the two delta-function products is the identity kernel on the two-particle symmetric tensor product. For two identical fermions the corresponding identity kernel has the antisymmetric sign. Thus the permutation sums in LSZ are part of the asymptotic Fock-space structure, not an additional convention imposed after the reduction formula.

18.9 Invariant Amplitudes

Definition 18.11 (Invariant scalar amplitude convention)**.** The connected kernel contains the overall conservation law

$$q_1 + \cdots + q_m = p_1 + \cdots + p_n.$$

It is conventional to define the invariant amplitude $\mathcal{M}$ by

$$\begin{aligned}
&{}_{\mathrm{out}}\langle \vec{q}_1, \dots, \vec{q}_m | \vec{p}_1, \dots, \vec{p}_n \rangle_{\mathrm{in,conn}} \\
&\quad = \mathrm{i}(2\pi)^D \delta^{(D)}(q_1 + \cdots + q_m - p_1 - \cdots - p_n)\, \mathcal{M}(q_1, \dots, q_m | p_1, \dots, p_n).
\end{aligned}$$

With this convention the operator identity $S = \mathbf{1} + \mathrm{i}T$ gives

$${}_{\mathrm{out}}\langle q | T | p \rangle = (2\pi)^D \delta^{(D)}(P_{\mathrm{out}} - P_{\mathrm{in}})\, \mathcal{M}(q|p).$$

Here $P_{\text{out}} = \sum_i q_i$ and $P_{\text{in}} = \sum_j p_j$.

This convention uses relativistically normalized external momentum states. If one instead uses δ-normalized states

$${}_\delta\langle \vec{q} | \vec{p} \rangle_\delta = \delta^{(d)}(\vec{q} - \vec{p}),$$

related to the relativistic states by

$$|\vec{p}\rangle = \left[(2\pi)^d 2\omega_{\vec{p}}\right]^{1/2} |\vec{p}\rangle_\delta,$$

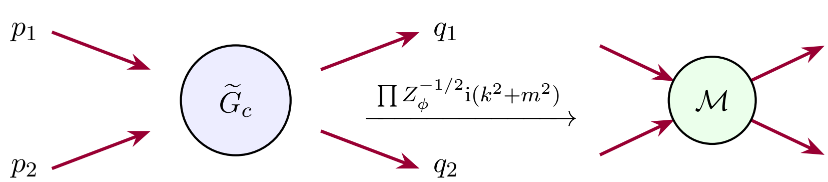
then the connected kernel is often written

$${}_{\delta,\text{out}}\langle q|p\rangle_{\delta,\text{in,conn}} = \mathrm{i}(2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) M_\delta(q|p),$$

where

$$M_\delta(q|p) = \left[\prod_{i=1}^{m} \left((2\pi)^d 2\omega_{\vec{q}_i}\right)^{-1/2} \prod_{j=1}^{n} \left((2\pi)^d 2\omega_{\vec{p}_j}\right)^{-1/2}\right] \mathcal{M}(q|p).$$

The invariant object is $\mathcal{M}$; M_δ carries the normalization factors attached to a nonrelativistic delta-function basis.



The diagram displays the operation. The left object is the connected time-ordered Green function with its external one-particle poles. The right object is the on-shell invariant amplitude obtained after the pole residues have been extracted.

18.10 Cluster Decomposition of the Scattering Kernel

Partition formula for connected scattering kernels. Let $S(\beta|\alpha)$ denote the distributional kernel of the scattering operator on the Haag–Ruelle asymptotic Fock space, where α is the ordered list of incoming external labels and β is the ordered list of outgoing external labels. Connected scattering kernels S^{conn} are defined by the following moment–cumulant relation:

$$S(\beta|\alpha) = \sum_{\mathcal{P}} \prod_{I \in \mathcal{P}} S^{\text{conn}}(\beta_I|\alpha_I).$$

Here $\mathcal{P}$ runs over simultaneous partitions

$$\alpha = \bigsqcup_{I \in \mathcal{P}} \alpha_I, \qquad \beta = \bigsqcup_{I \in \mathcal{P}} \beta_I,$$

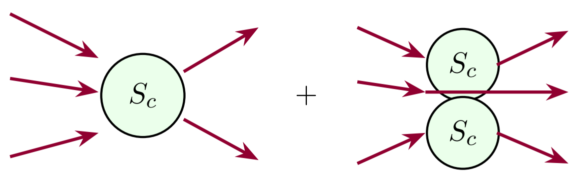
and the singleton identity kernels of the identity operator in $S = \mathbf{1} +$ interaction are included as allowed one-particle blocks. Equivalently, S^{conn} is obtained from S by Möbius inversion on this finite partition lattice. With this definition, Haag–Ruelle cluster decomposition says that S^{conn}

is the part of the distribution that cannot be written as a product of scattering kernels for mutually separated spacetime clusters. The statement is not perturbative: perturbation theory later represents the connected kernels by connected Feynman graphs, but the connected/disconnected separation is already defined on the asymptotic Hilbert space.

The scattering kernel is a distributional matrix coefficient of S on the asymptotic Fock space. For each finite set of external labels the set of simultaneous partitions is finite, so the displayed relation can be solved recursively for $S^{\rm conn}$, beginning with the one-block partition. Explicitly,

$$S^{\rm conn}(\beta|\alpha) = S(\beta|\alpha) - \sum_{\mathcal{P}\neq\{\text{all}\}} \prod_{I\in\mathcal{P}} S^{\rm conn}(\beta_I|\alpha_I),$$

which is the Möbius inversion formula written in recursive form. The identity operator contributes the one-particle pass-through kernel $(2\pi)^d 2\omega_{\vec{p}}\delta^{(d)}(\vec{q}-\vec{p})$ in the corresponding singleton block; if one instead works with $T = S - \mathbf{1}$, these identity blocks are removed before taking connected interaction kernels. Finally, the Haag–Ruelle cluster property states that translating one group of wave-packet creation operators far from another makes the Hilbert-space matrix coefficient factor into the product of the corresponding cluster matrix coefficients. The recursive inversion is precisely the subtraction of all such factorized contributions, hence it isolates the non-factorizing connected kernel.



18.11 Perturbative Evaluation After Reduction

Tree-level ϕ^4 amplitude after LSZ. Perturbation theory supplies an expansion of the time-ordered Green function. LSZ then turns that expansion into an expansion of matrix elements of the scattering operator. This is the point where Feynman graphs acquire their scattering-amplitude interpretation.

For the scalar interaction

$$\mathcal{L}_{\rm int} = -\frac{g}{4!}\phi^4,$$

the leading connected four-point Green function is

$$\begin{aligned}&\widetilde{G}_4^{\rm conn}(q_1,q_2,-p_1,-p_2)\\ &\quad = (2\pi)^D\delta^{(D)}(q_1+q_2-p_1-p_2)(-\mathrm{i}g)\prod_{a=1}^{4}\frac{-\mathrm{i}}{k_a^2+m^2-\mathrm{i}0} + O(g^2),\end{aligned}$$

with

$$k_1 = q_1,\quad k_2 = q_2,\quad k_3 = -p_1,\quad k_4 = -p_2,\qquad Z_\phi = 1 + O(g^2).$$

The four numerator factors in the external propagators are

$$\prod_{a=1}^{4}(-\mathrm{i}) = (-\mathrm{i})^4 = 1.$$

Equivalently, applying the four LSZ multipliers gives, for each external leg,

$$\mathrm{i}(k_a^2 + m^2)\,\frac{-\mathrm{i}}{k_a^2 + m^2 - \mathrm{i}0} \longrightarrow 1,$$

after the mass-shell boundary value. Thus the external-leg factors contribute

$$\prod_{a=1}^{4}\left[\mathrm{i}(k_a^2 + m^2)\frac{-\mathrm{i}}{k_a^2 + m^2 - \mathrm{i}0}\right] \longrightarrow \big(\mathrm{i}(-\mathrm{i})\big)^4 = 1.$$

For an N-point connected Green function with N external scalar propagators the same bookkeeping is $\mathrm{i}^N(-\mathrm{i})^N = 1$; the remaining nontrivial factor is the connected amputated kernel. Applying LSZ gives

$${}_{\mathrm{out}}\langle \vec{q}_1, \vec{q}_2 | \vec{p}_1, \vec{p}_2 \rangle_{\mathrm{in,conn}} = (2\pi)^D \delta^{(D)}(q_1 + q_2 - p_1 - p_2)\left[-\mathrm{i}g + O(g^2)\right].$$

Therefore

$$\mathcal{M}(q_1, q_2 | p_1, p_2) = -g + O(g^2)$$

with the amplitude convention above. Higher orders are obtained from the same Green-function expansion together with the external residue factors $Z_\phi^{-1/2}$.

The leading connected four-point Green function has one local quartic vertex and four external scalar Feynman denominators. The LSZ theorem has already identified the physical scattering matrix element with the simultaneous external pole coefficient of this connected Green function. Applying $\mathrm{i}(k_a^2 + m^2)$ to each external scalar denominator gives one for each external leg in the displayed convention, and $Z_\phi = 1 + O(g^2)$ at this order. After the four external poles are removed, the remaining connected kernel is the vertex factor $-\mathrm{i}g$ times the momentum-conservation delta function. Comparing with Definition 18.11, where the connected kernel equals $\mathrm{i}(2\pi)^D\delta^{(D)}(\sum q - \sum p)\mathcal{M}$, gives $\mathcal{M} = -g + O(g^2)$.

18.12 Scope of the Formula

Remark 18.12 (Scope of scalar LSZ reduction). The LSZ theorem in this chapter applies to stable massive particles represented by isolated one-particle poles in local-field Green functions. It is a statement about boundary values of distributions after wave-packet smearing. Its hypotheses include the existence of the asymptotic states constructed in Haag–Ruelle theory and the existence of the relevant time-ordered products.

Massless gauge theories, infraparticle sectors, confined degrees of freedom, unstable resonances, and conformal theories require different asymptotic objects or different observables. In particular, an unstable resonance is not an external LSZ particle: it is represented by pole data of analytically continued amplitudes between stable asymptotic states, as discussed in Chapter 26. The common principle remains the same: first construct the asymptotic Hilbert-space or detector data, then relate that data to correlation functions in the regime where a reduction theorem is available.

There is a separate gauge-theoretic localization issue before the infrared question is reached. In a theory with an unscreened Gauss-law charge, a bounded gauge-invariant observable cannot create a charged vector from the vacuum. A charged LSZ formula must therefore be formulated, when it exists, for charged field-algebra operators or for gauge-invariant noncompact dressed operators such as Wilson-line or Coulomb dressings. The external factor is then the residue or threshold coefficient of the dressed correlator, including the renormalization of the dressing itself. The scalar local-field formula above contains none of this dressing data.

In a confining gauge theory such as QCD, this means that LSZ is applied, when its hypotheses hold, to gauge-invariant color-singlet interpolating operators whose Källén–Lehmann measures contain isolated hadronic atoms. There are no physical quark or gluon one-particle atoms for Haag–Ruelle to generate and for LSZ to amputate; gauge-fixed quark and gluon propagators therefore cannot be read as physical scattering amplitudes by the scalar-particle formula above. High-energy QCD observables involving jets are therefore not colored-particle S-matrix elements; they are color-singlet, detector-defined observables, such as inclusive hadronic tensors and energy-flow correlators, whose nonperturbative definitions are given in terms of physical states and the stress tensor.

Chapter 19

Cross Sections, Phase Space, and Unitarity

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

19.1 Transition Probabilities

Definition 19.1 (Transition probability for an outgoing event class)**.** Let $\mathcal{H}_{\mathrm{as}}$ be the asymptotic Hilbert space in a sector where the incoming and outgoing Haag–Ruelle ranges coincide. The scattering operator

$$S : \mathcal{H}_{\mathrm{as}} \to \mathcal{H}_{\mathrm{as}}$$

is then unitary. Write

$$S = \mathbf{1} + \mathrm{i}T.$$

For a normalized incoming vector $\Psi_{\mathrm{in}} \in \mathcal{H}_{\mathrm{as}}$ and an outgoing class of states $\mathcal{D}$, let $\Pi^{\mathrm{out}}_{\mathcal{D}}$ be the orthogonal projection onto the closed subspace spanned by $\mathcal{D}$. The transition probability into $\mathcal{D}$ is

$$P_{\Psi\to\mathcal{D}} = \left\| \Pi^{\mathrm{out}}_{\mathcal{D}} S \Psi_{\mathrm{in}} \right\|^2 .$$

This is the primary definition. Sharp-momentum formulas are distributional kernels obtained from this expression after smearing with wave packets and then taking a narrow-packet limit.

Definition 19.2 (Invariant amplitude convention)**.** For positive-energy momenta in the mostly-plus metric,

$$p_i = (E_i, \vec{p}_i), \qquad q_a = (E_a, \vec{q}_a),$$

with

$$p_i^2 = -m_i^2, \qquad q_a^2 = -M_a^2,$$

set

$$P_{\text{in}} = \sum_i p_i, \qquad P_{\text{out}} = \sum_a q_a.$$

The invariant amplitude convention inherited from LSZ is

$$\langle q_1, \ldots, q_m | T | p_1, \ldots, p_n \rangle = (2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) \mathcal{M}(q_1, \ldots, q_m | p_1, \ldots, p_n).$$

The bra and ket are generalized basis vectors in $\mathcal{H}_{\text{as}}$. They are elements of the appropriate dual test-vector space, as in Section 5.2. Equivalently, the same distribution is obtained from the connected part of the Haag–Ruelle kernel

$$_{\text{out}}\langle q_1, \ldots, q_m | p_1, \ldots, p_n \rangle_{\text{in}}$$

after subtracting the identity contribution. The connected contribution to S is therefore

$$\mathrm{i}(2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) \mathcal{M}.$$

Throughout this chapter, $\mathcal{M}(f|i)$ means the T-matrix element with ket label i and bra label f, namely the transition $i \to f$. The notation is ordered; $\mathcal{M}(f|i)$ and $\mathcal{M}(i|f)$ are different matrix elements unless an additional reciprocity symmetry is assumed.

19.2 Sharp-Momentum Limit and Delta-Squares

Example 19.3 (Sharp-momentum density limit)**.** The wave-packet definition is primary, but the conventional plane-wave formula is useful as a check on all factors. Temporarily use δ-normalized one-particle states in $d = D - 1$ spatial dimensions,

$$_\delta\langle \vec{q} | \vec{p} \rangle_\delta = \delta^{(d)}(\vec{q} - \vec{p}),$$

and write the connected kernel as

$$S^{\text{conn}}_{\beta\alpha} = \mathrm{i}(2\pi)^D \delta^{(D)}(P_\beta - P_\alpha) M_\delta(\beta|\alpha).$$

The subscript in M_δ is part of the notation: it records the $\delta^{(d)}$-normalized sharp-momentum basis used in this temporary calculation. It is neither the invariant amplitude $\mathcal{M}$ nor a partial-wave coefficient. Later in this chapter the invariant amplitude is expanded in partial waves with coefficients $a_\ell(s)$, while $S_\ell(s)$ denotes an eigenvalue of the scattering operator on a partial-wave channel:

$M_\delta(\beta\|\alpha)$	$\delta^{(d)}$-normalized connected sharp-momentum kernel,
$\mathcal{M}(f\|i)$	invariant amplitude after external normalization factors,
$a_\ell(s)$	coefficient in the ordered 16π partial-wave expansion,
$S_\ell(s)$	elastic partial-wave S-matrix eigenvalue.

Let $\mathcal{D}$ be a final region that does not contain the identity no-scattering contribution. Squaring the distributional kernel gives

$$P_{\alpha\to\mathcal{D}} = (2\pi)^D\delta^{(D)}(0)\int_{\mathcal{D}} \mathrm{d}\beta_\delta\,(2\pi)^D\delta^{(D)}(P_\beta - P_\alpha)|M_\delta(\beta|\alpha)|^2,$$

where $\mathrm{d}\beta_\delta$ is the product of the final $\mathrm{d}^d\vec{q}$ measures, with the resolution-of-identity factors appropriate to the outgoing Fock sector. A finite box regularization of spatial volume V and observation time $\mathcal{T}$ gives

$$(2\pi)^D\delta^{(D)}(0) = V\mathcal{T}, \qquad \delta^{(d)}(0) = \frac{V}{(2\pi)^d}.$$

For two sharp incoming particles, the normalized box state carries the factor $\delta^{(d)}(0)^{-2}$. Hence

$$P_{\alpha\to\mathcal{D}} = \frac{(2\pi)^{2d}\mathcal{T}}{V}\int_{\mathcal{D}} \mathrm{d}\beta_\delta\,(2\pi)^D\delta^{(D)}(P_\beta - P_\alpha)|M_\delta(\beta|\alpha)|^2.$$

If the two beams have relative speed $v_{\mathrm{rel}} = |\vec{v}_1 - \vec{v}_2|$ in the frame used for the box, the geometric probability to scatter into $\mathcal{D}$ during time $\mathcal{T}$ is

$$P_{\alpha\to\mathcal{D}} = \frac{\sigma_{\mathcal{D}} v_{\mathrm{rel}}\mathcal{T}}{V}.$$

Therefore

$$\sigma_{\mathcal{D}} = \frac{(2\pi)^{2D-2}}{v_{\mathrm{rel}}}\int_{\mathcal{D}} \mathrm{d}\beta_\delta\,(2\pi)^D\delta^{(D)}(P_\beta - P_\alpha)|M_\delta(\beta|\alpha)|^2.$$

Converting M_δ to the invariant amplitude $\mathcal{M}$ by the external normalization factors stated in the LSZ chapter gives the covariant phase-space formula below.

The connected sharp kernel contains a single total momentum-conservation distribution. In the probability, the square of this distribution is defined by putting the theory in a spatial box of volume V and evolving for an observation time $\mathcal{T}$. In that regulator,

$$\left[(2\pi)^D\delta^{(D)}(P_\beta - P_\alpha)\right]^2 = (2\pi)^D\delta^{(D)}(0)\,(2\pi)^D\delta^{(D)}(P_\beta - P_\alpha)$$

with $(2\pi)^D\delta^{(D)}(0) = V\mathcal{T}$. A two-particle incoming δ-normalized box state contributes $\delta^{(d)}(0)^{-2} = [(2\pi)^d/V]^2$. Multiplying these factors gives the displayed probability density $(2\pi)^{2d}\mathcal{T}/V$ times the final-state phase-space integral. Dividing by the geometric probability $v_{\mathrm{rel}}\mathcal{T}/V$ for a unit transverse area gives the sharp cross-section density. The last conversion is exactly the external normalization conversion between M_δ and $\mathcal{M}$ fixed in Definition 19.2.

19.3 Wave-Packet Origin of Cross Sections

Cross sections are transition probabilities per incoming transverse flux. Let $\psi_1, \psi_2 \in \mathcal{H}_1$ be normalized incoming one-particle packets with narrow momentum supports and distinct group velocities. In the center-of-mass frame let $\vec{p}_*$ be the central momentum of the first packet, and let

$$\mathbb{R}^{D-2}_\perp = \{b \in \mathbb{R}^{D-1} : b\cdot\vec{p}_* = 0\}$$

be the transverse impact-parameter plane. If τ_b denotes the spatial translation acting on the one-particle Hilbert space, define

$$\Psi_b = \psi_1 \odot \tau_b\psi_2.$$

Here $\odot$ denotes the two-particle product in the asymptotic Fock space, with the species labels and Bose or Fermi statistics appropriate to the sector under consideration. For an outgoing channel $\mathcal{D}$ whose projection removes the identity no-scattering component, the wave-packet cross section is

$$\sigma_{\psi_1,\psi_2}(\mathcal{D}) = \int_{\mathbb{R}_\perp^{D-2}} \mathrm{d}^{D-2}b \, \left\| \Pi_{\mathcal{D}}^{\mathrm{out}} S\Psi_b \right\|^2 .$$

The dimension is length to the power $D-2$, the dimension of transverse area in D spacetime dimensions.

The plane-wave formula is the narrow-packet density of this Hilbert-space quantity. The Fourier transform in b enforces transverse momentum conservation. The long-time Haag–Ruelle localization enforces energy and longitudinal momentum conservation in the scattering limit. Together these produce the single overall distribution

$$(2\pi)^D \delta^{(D)}(P_{\mathrm{out}} - P_{\mathrm{in}})$$

appearing in the invariant amplitude convention. The incoming current through the transverse impact-parameter plane is the invariant relative speed multiplied by the one-particle normalizations. Thus the denominator in the sharp-momentum two-particle formula is the invariant flux

$$\mathcal{F}(p_1, p_2) = 4E_1 E_2 v_{\mathrm{rel}}.$$

The next section expresses the same factor in covariant form.

19.4 Invariant Phase Space and Flux

Definition 19.4 (Invariant phase space)**.** For one on-shell final particle of mass M, the Lorentz-invariant positive-energy measure is

$$\mathrm{d}\mu_M(q) = \frac{\mathrm{d}^D q}{(2\pi)^{D-1}} \, \theta(q^0)\delta(q^2 + M^2) = \frac{\mathrm{d}^{D-1}\vec{q}}{(2\pi)^{D-1} 2E_{\vec{q}}},$$

where

$$E_{\vec{q}} = \sqrt{\vec{q}^{\,2} + M^2}.$$

For m final particles with total incoming momentum P, define the m-body phase-space measure

$$\mathrm{d}\Phi_m(P; q_1, \ldots, q_m) = (2\pi)^D \delta^{(D)}\left(P - \sum_{a=1}^m q_a \right) \prod_{a=1}^m \mathrm{d}\mu_{M_a}(q_a).$$

This formula is written first for labeled outgoing particles. The identical particle factors come from the resolution of the identity on the asymptotic Fock space. For one bosonic species, the r-particle space is $\mathcal{H}_1^{\odot r}$. If

$$|\vec{q}_1, \ldots, \vec{q}_r\rangle_{\mathrm{sym}} = a^\dagger(\vec{q}_1) \cdots a^\dagger(\vec{q}_r)\Omega_F,$$

with the relativistic one-particle normalization used above, then the identity on this symmetric sector is

$$\mathbf{1}_{\mathcal{H}_1^{\odot r}} = \frac{1}{r!} \int \prod_{j=1}^r \mathrm{d}\mu_m(q_j) \, |\vec{q}_1, \ldots, \vec{q}_r\rangle_{\mathrm{sym}\,\mathrm{sym}}\langle \vec{q}_1, \ldots, \vec{q}_r|.$$

The factor $1/r!$ compensates for the $r!$ labeled tuples that represent the same unordered Bose state. For several species, the physical phase-space measure carries the product

$$\prod_{\lambda} \frac{1}{n_\lambda!},$$

where n_λ is the number of identical outgoing particles of species λ. Fermionic sectors have the corresponding antisymmetric identity resolution; the same factorial appears, with signs carried by the antisymmetric external wavefunctions and matrix elements.

Invariant two-particle flux. For a two-particle initial state, the invariant flux factor used in the cross-section normalization is

$$\mathcal{F}(p_1, p_2) = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}.$$

Equivalently,

$$\mathcal{F} = 4E_1 E_2 v_{\text{rel}}, \qquad v_{\text{rel}} = \frac{\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}}{E_1 E_2}.$$

In the center-of-mass frame,

$$p_1 = (E_1, \vec{p}_*), \qquad p_2 = (E_2, -\vec{p}_*),$$

so

$$\mathcal{F} = 4(E_1 + E_2)|\vec{p}_*|.$$

The verification is the center-of-mass evaluation of the invariant. The quantity $(p_1 \cdot p_2)^2 - m_1^2 m_2^2$ is Lorentz invariant, so it suffices to evaluate it in the center-of-mass frame. With mostly-plus metric,

$$p_1 \cdot p_2 = -E_1 E_2 - \vec{p}_*^{\,2}.$$

Using $E_i^2 = \vec{p}_*^{\,2} + m_i^2$,

$$\begin{aligned} (p_1 \cdot p_2)^2 - m_1^2 m_2^2 &= (E_1 E_2 + \vec{p}_*^{\,2})^2 - (E_1^2 - \vec{p}_*^{\,2})(E_2^2 - \vec{p}_*^{\,2}) \\ &= \vec{p}_*^{\,2}(E_1 + E_2)^2. \end{aligned}$$

Thus $\mathcal{F} = 4(E_1 + E_2)|\vec{p}_*|$. Since $\vec{v}_1 = \vec{p}_*/E_1$ and $\vec{v}_2 = -\vec{p}_*/E_2$, the relative speed is

$$v_{\text{rel}} = |\vec{v}_1 - \vec{v}_2| = \frac{(E_1 + E_2)|\vec{p}_*|}{E_1 E_2},$$

which proves the equivalent form $\mathcal{F} = 4E_1 E_2 v_{\text{rel}}$.

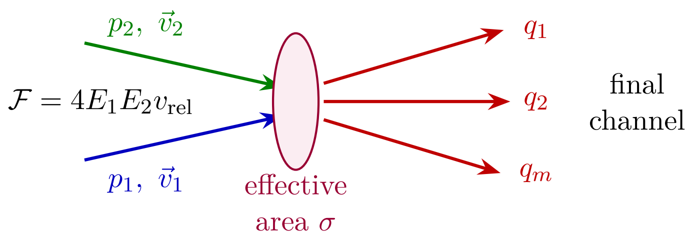
Definition 19.5 (Differential cross section from invariant data)**.** The differential cross section for

$$p_1 + p_2 \longrightarrow q_1 + \cdots + q_m$$

is

$$\mathrm{d}\sigma = \frac{1}{\mathcal{F}(p_1, p_2)} \left|\mathcal{M}(q_1, \ldots, q_m | p_1, p_2)\right|^2 \mathrm{d}\Phi_m(P_{\text{in}}; q_1, \ldots, q_m).$$

All factors in this expression have now been fixed: the asymptotic state normalization, the amplitude convention, the final-state measure, and the incoming flux.



19.5 Two-Body Final States in Four Dimensions

Set $D = 4$. For a two-body final state, define

$$s = -(p_1 + p_2)^2 = P_{\text{in}}^{0\,2} - \vec{P}_{\text{in}}^{\,2}.$$

In the center-of-mass frame,

$$P_{\text{in}} = (\sqrt{s}, \vec{0}).$$

The Källén polynomial is

$$\lambda_K(s, a, b) = s^2 + a^2 + b^2 - 2sa - 2sb - 2ab.$$

For incoming masses m_1, m_2 and final masses M_1, M_2,

$$|\vec{p}_*| = \frac{\sqrt{\lambda_K(s, m_1^2, m_2^2)}}{2\sqrt{s}}, \qquad |\vec{q}_*| = \frac{\sqrt{\lambda_K(s, M_1^2, M_2^2)}}{2\sqrt{s}}.$$

The next calculation performs the two-body phase-space integral over the energy-conserving delta function. Here $\mathrm{d}\Omega$ is the solid-angle measure of one final momentum in the center-of-mass frame.

Two-body phase-space reduction in four dimensions. With the normalization of Definition 19.4, the two-body final-state measure in the center-of-mass frame is

$$\mathrm{d}\Phi_2 = \frac{1}{16\pi^2} \frac{|\vec{q}_*|}{\sqrt{s}} \, \mathrm{d}\Omega.$$

Consequently

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 s} \frac{|\vec{q}_*|}{|\vec{p}_*|} |\mathcal{M}(s, \Omega)|^2.$$

In the center-of-mass frame the spatial delta function in $\mathrm{d}\Phi_2$ sets $\vec{q}_2 = -\vec{q}_1$. Writing $q = |\vec{q}_1|$, the remaining radial integral contains

$$\delta(\omega_1(q) + \omega_2(q) - \sqrt{s}), \qquad \omega_a(q) = \sqrt{q^2 + M_a^2}.$$

At the root $q = |\vec{q}_*|$,

$$\frac{\mathrm{d}}{\mathrm{d}q}(\omega_1 + \omega_2) = q \left(\frac{1}{\omega_1} + \frac{1}{\omega_2} \right) = \frac{q\sqrt{s}}{\omega_1 \omega_2}.$$

Therefore

$$\int \frac{q^2 \mathrm{d}q \, \mathrm{d}\Omega}{(2\pi)^2 \, 4\omega_1\omega_2} \, \delta(\omega_1 + \omega_2 - \sqrt{s}) = \frac{|\vec{q}_*|}{16\pi^2 \sqrt{s}} \, \mathrm{d}\Omega.$$

Dividing by the center-of-mass flux $\mathcal{F} = 4\sqrt{s}\,|\vec{p}_*|$ gives the displayed differential cross-section formula.

For two identical final particles, the symmetric two-particle identity resolution supplies the factor 1/2 when the integral is written over the full labeled solid angle.

For elastic scattering of identical scalar particles the preceding formula already gives the total cross section. Since $|\vec{q}_*| = |\vec{p}_*|$ and $s = E_{\rm cm}^2$,

$$\sigma^{\rm id}_{2\to2} = \frac{1}{2}\int \mathrm{d}\Omega\,\frac{|\mathcal{M}(E_{\rm cm},\Omega)|^2}{64\pi^2 s} = \frac{1}{2^7\pi^2 E_{\rm cm}^2}\int \mathrm{d}\Omega\,|\mathcal{M}(E_{\rm cm},\Omega)|^2.$$

The direct delta-function integration gives the same expression and is a useful normalization check. In the center-of-mass frame the spatial delta function sets $\vec{q}_2 = -\vec{q}_1$. Writing $q = |\vec{q}_1|$, the radial energy-conservation integral is

$$\int q^2 \mathrm{d}q\,\delta(\omega_1(q)+\omega_2(q)-E_{\rm cm}) = \frac{q\,\omega_1\omega_2}{E_{\rm cm}}.$$

Together with $E_1E_2v_{\rm rel} = |\vec{p}_*|E_{\rm cm}$, this reduces the labeled two-body phase-space integral to the same result above, with the factor 1/2 supplied by the symmetric two-particle resolution of the identity.

For the tree-level ϕ^4 example with a single scalar mass m, the elastic kinematics obey

$$|\vec{q}_*| = |\vec{p}_*|, \qquad \sqrt{s} = E_{\rm cm}.$$

If the tree-level ϕ^4 invariant amplitude is

$$\mathcal{M} = -g + O(g^2),$$

then

$$\sigma_{2\to2} = \frac{g^2}{32\pi s} + O(g^3)$$

for identical final particles.

19.6 Unitarity and the Optical Theorem

Theorem 19.6 (Sharp-kernel unitarity and the optical theorem)**.** *Let $S = \mathbf{1} + \mathrm{i}T$ be unitary on the asymptotic Hilbert space. With the amplitude convention of Definition 19.2, the connected sharp kernels obey*

$$-\mathrm{i}\,[\mathcal{M}(f|i) - \mathcal{M}(i|f)^*] = \sum_X \int \mathrm{d}\Phi_X\,\mathcal{M}(X|f)^*\mathcal{M}(X|i),$$

where X runs over a complete orthogonal resolution of outgoing asymptotic channels, including the corresponding phase-space measures and identical-particle symmetry factors. In the forward case $f = i$,

$$2\ \mathrm{Im}\,\mathcal{M}(i|i) = \sum_X \int \mathrm{d}\Phi_X\,|\mathcal{M}(X|i)|^2.$$

For two incoming particles,

$$\sigma_{\rm tot} = \frac{2\ \mathrm{Im}\,\mathcal{M}(i|i)}{\mathcal{F}(p_1,p_2)}.$$

Proof. The operator identity

$$S^\dagger S = \mathbf{1}, \qquad S = \mathbf{1} + \mathrm{i}T,$$

is equivalent to

$$-\mathrm{i}(T - T^\dagger) = T^\dagger T.$$

Let i and f denote incoming and outgoing asymptotic momentum labels, including all discrete quantum numbers. Insert a complete set of outgoing asymptotic states between $T^\dagger$ and T. Removing the common overall momentum-conservation delta function gives the first displayed sharp-kernel identity. The conjugated factor on the right is fixed by Hilbert-space adjunction:

$$\langle f|T^\dagger|X\rangle = \langle X|T|f\rangle^*.$$

Thus it is $\mathcal{M}(X|f)^*$, not in general $\mathcal{M}(f|X)^*$. Replacing one by the other would require an additional reciprocity statement, for example from a time-reversal or detailed-balance symmetry after its antiunitary phases have been fixed.

Setting $f = i$ gives the optical theorem. No time-reversal invariance is used in this forward specialization: $\langle i|T^\dagger|X\rangle = \langle X|T|i\rangle^*$ by the definition of the adjoint, so each term is the norm square of the outgoing transition amplitude from the fixed incoming state. Dividing the complete transition probability per unit time by the incoming flux gives the total cross section. The amplitude on the left is the forward amplitude: the outgoing labels are the same as the incoming labels. □

Remark 19.7 (Complete sums and detector resolution)*.* The symbol X in the optical theorem denotes a complete orthogonal resolution of the outgoing scattering Hilbert space in the ideal theory under discussion. A detector with finite resolution is represented more accurately by a positive outgoing effect $E_{\mathcal{C}}$, $0 \le E_{\mathcal{C}} \le \mathbf{1}$, associated with an observed event class $\mathcal{C}$. The corresponding measured transition probability is

$$\langle i|T^\dagger E_{\mathcal{C}} T|i\rangle,$$

or, in sharp-kernel notation, the same phase-space sum as above with the weights and integration region specified by $E_{\mathcal{C}}$. The forward imaginary part $2\,\mathrm{Im}\,\mathcal{M}(i|i)$ corresponds to the special choice $E_{\mathcal{C}} = \mathbf{1}$, namely the complete inclusive sum over all exact outgoing states of the theory. Angular cuts, energy cuts, jet definitions, and unresolved soft radiation define different effects $E_{\mathcal{C}}$ and hence different cross sections.

In a theory with massless long-range fields the complete sharp fixed-particle-number terms are infrared-regulator dependent. The operational quantity is obtained by first keeping an infrared regulator, summing over the states that the detector cannot distinguish, and only then removing the regulator. Chapter 44 carries this out for charged QED, where real unresolved photons and virtual soft photons combine into a finite resolution-inclusive probability.

19.7 Partial-Wave States in Four Dimensions

Partial waves are most transparently defined as generalized eigenstates of the free two-particle asymptotic representation. In the center-of-mass frame use the δ-normalized two-particle basis and write, for scalar particles,

$$_\delta\langle \vec{k}_1, \vec{k}_2|E, \vec{P} = \vec{0}, \ell, m\rangle = \delta^{(3)}(\vec{k}_1 + \vec{k}_2)\,\delta(\omega_1 + \omega_2 - E)\,\mathcal{N}(E) Y_{\ell m}(\widehat{k}_1).$$

The angular momentum labels satisfy

$$J^2 = \ell(\ell+1), \qquad J_3 = m.$$

For identical spinless bosons, only even ℓ occurs. The normalization is chosen so that

$$\langle E', \vec{P}', \ell', m' | E, \vec{P}, \ell, m\rangle = \delta(E'-E)\delta^{(3)}(\vec{P}'-\vec{P})\delta_{\ell'\ell}\delta_{m'm}.$$

Using

$$\int k^2 \mathrm{d}k\, \delta(\omega_1(k)+\omega_2(k)-E) = \frac{k\,\omega_1\omega_2}{E},$$

and using the 1/2 in the symmetric two-particle resolution of the identity, one finds

$$\mathcal{N}(E) = \left(\frac{2E}{k\,\omega_1\omega_2}\right)^{1/2} \qquad \text{for identical scalar bosons.}$$

For distinguishable particles the same formula lacks the factor of 2 under the square root.

Rotational invariance in the center-of-mass frame makes the elastic partial waves diagonal:

$$S(E', \vec{P}', \ell', m' | E, \vec{P}, \ell, m) = \delta(E'-E)\delta^{(3)}(\vec{P}'-\vec{P})\delta_{\ell'\ell}\delta_{m'm}\, S_\ell(E)$$

in an elastic single-channel region. If inelastic channels are present, this equation describes the elastic matrix element of a larger unitary matrix, so

$$|S_\ell(E)| \le 1,$$

with equality in a purely elastic partial wave.

In the same δ-normalized convention, the connected elastic sharp-momentum kernel is obtained by inserting the resolution of the identity in partial waves. For identical scalar bosons in the center-of-mass frame,

$$\mathrm{i}(2\pi)^4 M_\delta(\widehat{q}|\widehat{p}) = \frac{\mathcal{N}(E)^2}{4\pi} \sum_{\substack{\ell=0\\ \ell\ \text{even}}}^{\infty} (2\ell+1)P_\ell(\widehat{q}\cdot\widehat{p})\big(S_\ell(E)-1\big).$$

This formula is the direct bridge between the partial-wave diagonalization of the S-operator and the sharp-momentum amplitude. Multiplying by the external normalization factors converts M_δ into the invariant amplitude $\mathcal{M}$ used in the cross-section formula.

19.8 Partial-Wave Amplitudes in Four Dimensions

For scalar $2 \to 2$ scattering in the center-of-mass frame, rotational invariance makes the amplitude a function of s and

$$x = \cos\theta = \widehat{p}_* \cdot \widehat{q}_*.$$

For distinguishable particles, or for an ordered pair of external labels, use the partial-wave expansion

$$\mathcal{M}(s,x) = 16\pi \sum_{\ell=0}^{\infty} (2\ell+1) a_\ell(s) P_\ell(x),$$

where P_ℓ is the Legendre polynomial. The inverse formula is

$$a_\ell(s) = \frac{1}{32\pi}\int_{-1}^{1} \mathrm{d}x\, P_\ell(x)\mathcal{M}(s,x).$$

This section fixes the ordered 16π normalization. For an unordered identical-boson channel the physical amplitude is the Bose symmetrization of the ordered amplitude; the corresponding even-ℓ 32π convention is displayed below. For equal incoming masses in the elastic region, define

$$\beta(s) = \frac{2|\vec{p}_*|}{\sqrt{s}}.$$

This $\beta(s)$ is a scalar two-body kinematic factor. The letter ρ is reserved in this monograph for spectral measures and their densities, such as the Källén–Lehmann measure $d\rho$. The partial-wave S-matrix eigenvalue is

$$S_\ell(s) = 1 + 2\mathrm{i}\beta(s)a_\ell(s).$$

Elastic unitarity is

$$|S_\ell(s)| = 1.$$

Equivalently, there is a real phase shift $\delta_\ell(s)$ such that

$$S_\ell(s) = \mathrm{e}^{2\mathrm{i}\delta_\ell(s)}$$

and

$$a_\ell(s) = \frac{\mathrm{e}^{2\mathrm{i}\delta_\ell(s)} - 1}{2\mathrm{i}\beta(s)} = \frac{\mathrm{e}^{\mathrm{i}\delta_\ell(s)}\sin\delta_\ell(s)}{\beta(s)}.$$

The same condition can be written

$$\operatorname{Im} a_\ell(s) = \beta(s)|a_\ell(s)|^2.$$

With inelastic channels open,

$$|S_\ell(s)| \le 1,$$

with equality precisely in a purely elastic partial wave.

For distinguishable scalar particles, the elastic total cross section is

$$\sigma_{\mathrm{el}} = \frac{\pi}{|\vec{p}_*|^2}\sum_{\ell=0}^{\infty}(2\ell+1)|S_\ell(s)-1|^2 = \frac{4\pi}{|\vec{p}_*|^2}\sum_{\ell=0}^{\infty}(2\ell+1)\sin^2\delta_\ell(s).$$

For identical scalar bosons, the physical two-particle amplitude on unordered states is Bose symmetric. With the convention

$$\mathcal{M}_{\mathrm{id}}(s,x) = 32\pi \sum_{\substack{\ell=0\\ \ell\ \mathrm{even}}}^{\infty} (2\ell+1)a_\ell(s)P_\ell(x),$$

and the full-solid-angle final-state factor $1/2$, the elastic cross section is

$$\sigma_{\mathrm{el}}^{\mathrm{id}} = \frac{2\pi}{|\vec{p}_*|^2} \sum_{\substack{\ell=0\\ \ell\ \mathrm{even}}}^{\infty} (2\ell+1)|S_\ell(s)-1|^2.$$

This is the convention used later for identical-particle high-energy bounds.

19.9 Partial-Wave Unitarity Geometry

Partial-wave unitarity circle. Define

$$b_\ell(s) = \beta(s)a_\ell(s), \qquad S_\ell(s) = 1 + 2\mathrm{i}b_\ell(s).$$

In an elastic partial wave, $|S_\ell| = 1$. Writing

$$b_\ell = x_\ell + \mathrm{i}y_\ell,$$

this condition becomes

$$x_\ell^2 + \left(y_\ell - \frac{1}{2}\right)^2 = \frac{1}{4}.$$

Thus the elastic amplitude lies on the circle of radius $1/2$ centered at $\mathrm{i}/2$ in the complex b_ℓ-plane. Equivalently,

$$y_\ell = |b_\ell|^2.$$

The phase-shift form

$$b_\ell = \mathrm{e}^{\mathrm{i}\delta_\ell} \sin \delta_\ell$$

parametrizes this circle.

When inelastic channels are open, the elastic partial wave is a matrix element of a unitary matrix on a larger channel space. It is convenient to write

$$S_\ell = \eta_\ell \mathrm{e}^{2\mathrm{i}\delta_\ell}, \qquad 0 \le \eta_\ell \le 1.$$

Then $b_\ell = (S_\ell - 1)/(2\mathrm{i})$ lies inside the same circle:

$$x_\ell^2 + \left(y_\ell - \frac{1}{2}\right)^2 \le \frac{1}{4}, \qquad y_\ell \ge |b_\ell|^2.$$

The difference

$$y_\ell - |b_\ell|^2 = \frac{1 - \eta_\ell^2}{4}$$

is the contribution of inelastic final states in that angular-momentum channel. This is the partial-wave form of the optical theorem.

Substitute $S_\ell = 1 + 2\mathrm{i}b_\ell$ with $b_\ell = x_\ell + \mathrm{i}y_\ell$. Then

$$S_\ell = (1 - 2y_\ell) + 2\mathrm{i}x_\ell.$$

The elastic condition $|S_\ell| = 1$ is therefore

$$(1 - 2y_\ell)^2 + 4x_\ell^2 = 1,$$

which is equivalent to the displayed circle equation and to $y_\ell = |b_\ell|^2$. If $S_\ell = \eta_\ell \mathrm{e}^{2\mathrm{i}\delta_\ell}$ with $0 \le \eta_\ell \le 1$, the same substitution gives

$$|1 + 2\mathrm{i}b_\ell|^2 = \eta_\ell^2.$$

Rearranging yields

$$y_\ell - |b_\ell|^2 = \frac{1 - \eta_\ell^2}{4} \ge 0,$$

and the inequality form of the unitarity disk follows. The right side is the deficit $1 - \eta_\ell^2$ in the norm of the elastic component, expressed in the $b_\ell = (S_\ell - 1)/(2\mathrm{i})$ normalization. This is the partial-wave optical theorem.

19.10 Thresholds, Poles, and Resonances

The same variables organize the analytic structure of scattering. A threshold is the value of s at which a new on-shell intermediate channel becomes kinematically available. Across such a threshold, unitarity gives the discontinuity of amplitudes in terms of phase-space integrals over lower amplitudes.

A stable bound state in a two-particle channel appears as a pole of the partial-wave amplitude on the physical, or first, sheet below threshold. A resonance is represented by a pole reached after analytic continuation through a unitarity cut to an adjacent sheet. The sheet terminology and the distinction between a physical real region and the physical analytic sheet are fixed in Section 28.1; the organizing data are already present here: the S-operator, phase space, unitarity, and partial waves.

Chapter 20

Massive Particles and Spin

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

20.1 Massive One-Particle Representations

Definition 20.1 (Massive one-particle spin datum)**.** Work in $D = 4$. A massive one-particle datum consists of a Hilbert space $\mathcal{H}_1$ carrying an irreducible positive-energy unitary representation of the Poincare group whose translation generators P^μ have joint spectrum supported on a single positive-energy mass shell of mass $m > 0$. Equivalently, it is the massive row of Table 5.1. The mass shell is

$$\Sigma_m^+ = \{p = (p^0, \vec{p}) : p^0 = \omega_{\vec{p}},\ p^2 = -m^2\}, \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}.$$

Generalized momentum states, understood as distributional vectors in the rigged-Hilbert-space sense of Section 5.2, are normalized by

$$\langle \vec{q}, \tau | \vec{p}, \sigma \rangle = (2\pi)^3\, 2\omega_{\vec{p}}\, \delta^{(3)}(\vec{q} - \vec{p})\, \delta_{\tau\sigma}.$$

The labels σ, τ span a finite-dimensional internal vector space which is fixed by Lorentz symmetry.

Remark 20.2 (Spacelike Wigner orbits). The full representation theory of the Poincare group also contains spacelike orbits $p^2 = \mu^2 > 0$, sometimes called tachyonic orbits. They are absent from this chapter and the following massless chapter because the spectrum condition assumed in Chapter 5 requires $\mathrm{Spec}(P) \subset \overline{V}_+$, while a spacelike orbit lies outside the closed forward lightcone.

Definition 20.3 (Standard rest momentum and normalization frames)**.** Choose the reference momentum

$$p_R = (m, \vec{0}).$$

For every $p \in \Sigma_m^+$, choose a standard Lorentz transformation

$$L(p) \in SO^+(1,3), \qquad L(p)p_R = p, \qquad L(p_R) = 1.$$

It is sometimes useful to compare this covariant normalization with the delta-function normalization

$${}_\delta\langle \vec{q}, \tau | \vec{p}, \sigma \rangle_\delta = \delta^{(3)}(\vec{q} - \vec{p})\delta_{\tau\sigma}.$$

In that convention one may define

$$|\vec{p}, \sigma\rangle_\delta = N(\vec{p})\, U(L(p))|\vec{0}, \sigma\rangle_\delta, \qquad N(\vec{p}) = \left(\frac{m}{\omega_{\vec{p}}}\right)^{1/2}.$$

The factor $N(\vec{p})$ is the Jacobian needed to keep the measure $d^3\vec{p}$ compatible with Lorentz transformations in a delta-normalized basis. The covariantly normalized states used below absorb this factor into the invariant mass-shell measure.

Definition 20.4 (Massive Wigner rotation)**.** The stabilizer of p_R in $SO^+(1,3)$ is the rotation group $SO(3)$. Whenever a Wigner rotation is inserted into a spin representation below, it is understood through its lift to the universal cover of $SO(3)$; the reason for this lift is explained in Section 20.2. The same letter denotes the projected Lorentz transformation when it acts on four-vectors.

For a Lorentz transformation Λ, define the Wigner rotation

$$W(\Lambda, p) = L(\Lambda p)^{-1}\Lambda L(p).$$

It obeys

$$W(\Lambda, p)p_R = p_R,$$

and is therefore an element of the massive little group. If the particle has spin j, with

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots,$$

let $D^{(j)}$ be the corresponding irreducible unitary representation of $SU(2)$ on a vector space V_j of dimension $2j+1$. For a four-vector r, write $\mathrm{sp}(r)$ for its spatial part, so

$$p = (\omega_{\vec{p}}, \vec{p}), \qquad \mathrm{sp}(p) = \vec{p}.$$

Massive Wigner transformation law. The covariantly normalized one-particle states transform as

$$U(\Lambda)|\vec{p}, \sigma\rangle = \sum_{\tau=-j}^{j} D^{(j)}_{\tau\sigma}(W(\Lambda, p))\,|\,\mathrm{sp}(\Lambda p), \tau\rangle,$$

where $\sigma, \tau = -j, -j+1, \dots, j$. The covariant normalization above absorbs the Jacobian of the mass-shell measure, so no additional $\sqrt{(\Lambda p)^0/p^0}$ factor appears. In the delta-normalized convention the same computation gives

$$U(\Lambda)|\vec{p}, \sigma\rangle_\delta = \left(\frac{(\Lambda p)^0}{p^0}\right)^{1/2} \sum_{\tau=-j}^{j} D^{(j)}_{\tau\sigma}(W(\Lambda, p))\,|\,\mathrm{sp}(\Lambda p), \tau\rangle_\delta.$$

This is the same representation written in a basis adapted to $d^3\vec{p}$ rather than to $d^3\vec{p}/(2p^0)$. Keeping both formulas visible prevents the normalization factor from being mistaken for a property of spin. By definition $L(p)p_R = p$ and $L(\Lambda p)p_R = \Lambda p$. Therefore

$$W(\Lambda, p)p_R = L(\Lambda p)^{-1}\Lambda L(p)p_R = p_R,$$

so $W(\Lambda, p)$ belongs to the little group of the rest momentum. The restriction of the Poincare representation to the little group at p_R is an irreducible $SU(2)$ representation $D^{(j)}$. Acting first by $L(p)$, then by Λ, and finally comparing with the chosen standard transformation $L(\Lambda p)$ gives exactly the displayed spin matrix $D^{(j)}(W(\Lambda, p))$.

The absence or presence of the square-root factor is a statement about the basis measure. The invariant mass-shell measure satisfies

$$\frac{d^3\vec{p}}{2p^0} = \frac{d^3\,\mathrm{sp}(\Lambda p)}{2(\Lambda p)^0}.$$

Equivalently,

$$d^3\,\mathrm{sp}(\Lambda p) = \frac{(\Lambda p)^0}{p^0}\,d^3\vec{p}.$$

Thus the covariantly normalized basis, whose resolution of the identity uses $d^3\vec{p}/(2p^0)$, transforms without a scalar Jacobian. A basis normalized against $d^3\vec{p}$ must instead carry the unitary half-density factor $((\Lambda p)^0/p^0)^{1/2}$. This proves both formulas.

20.2 Projective Rotations and the Covering Group

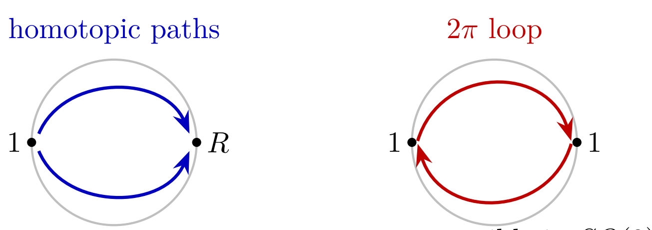
The appearance of $SU(2)$ rather than only $SO(3)$ is forced by the quantum implementation of rotations. A physical pure state is a ray, so the operator associated with a rotation $R \in SO(3)$ is defined on Hilbert space only up to a phase if no continuity convention has been chosen. Thus a projective representation may obey

$$U(R_2)U(R_1) = \mathrm{e}^{\mathrm{i}\alpha(R_2,R_1)}U(R_2R_1)$$

for a real phase function α. The phase ambiguity is removed by lifting rotations to paths. If $\gamma : [0, 1] \to SO(3)$ is a continuous path with $\gamma(0) = 1$ and $\gamma(1) = R$, continuity of the infinitesimal generators determines an operator U_γ starting from $U_{\gamma(0)} = 1$. If γ and γ' are homotopic with endpoints fixed, then

$$U_\gamma = U_{\gamma'}.$$

Consequently the correct group acting linearly is the universal cover of $SO(3)$, whose points are endpoint-fixed homotopy classes of such paths.



not contractible in $SO(3)$
but its square is

Massive spin labels from the covering group. The universal cover of the massive rotation little group is

$$\widetilde{SO(3)} \simeq SU(2).$$

Its finite-dimensional irreducible unitary representations are indexed by

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,$$

with representation space V_j of dimension $2j+1$. The nontrivial central element of $SU(2)$, namely the lift of a 2π rotation, acts by

$$D^{(j)}(-1) = (-1)^{2j}\mathbf{1}_{V_j}.$$

The lift of a 4π rotation is the identity. Hence an irreducible massive one-particle representation carries exactly one such spin label j. The group $SO(3)$ is $\mathbb{RP}^3$. Its fundamental group is therefore

$$\pi_1(SO(3)) \simeq \mathbb{Z}_2.$$

The nontrivial class is represented by a 2π rotation loop, while the double loop representing a 4π rotation is contractible. The simply connected compact Lie group with Lie algebra $\mathfrak{so}(3)$ is $SU(2)$, and the covering map $SU(2) \to SO(3)$ has kernel $\{\pm 1\}$. The highest-weight classification of irreducible unitary $SU(2)$ representations gives one representation $D^{(j)}$ for each nonnegative half-integer j, of dimension $2j+1$. In the spin-j representation the central element -1 is the exponential of a 2π rotation and acts by $(-1)^{2j}$. Irreducibility of the one-particle Poincare representation means that its little-group representation is one irreducible summand, not a direct sum. With column vectors the matrices obey $D(W_2W_1) = D(W_2)D(W_1)$; row-index conventions on bras or matrix elements transpose this order and do not alter the representation.

20.3 Spin Frames on the Mass Shell

Definition 20.5 (Massive spin frame)**.** The choice of standard transformations $L(p)$ is a choice of spin frame over the mass shell. To make this statement precise, let

$$d\mu_m(p) = \frac{d^3p}{(2\pi)^3\, 2\omega_{\vec{p}}}$$

be the invariant measure on Σ_m^+. After choosing $L(p)$, the one-particle Hilbert space is represented as

$$\mathcal{H}_1 \simeq L^2(\Sigma_m^+, d\mu_m; V_j).$$

The generalized basis associated to this frame is the distributional map

$$F_L(p) : V_j \longrightarrow \mathcal{H}_1, \qquad F_L(p)e_\sigma = |\vec{p}, \sigma\rangle,$$

where $\{e_\sigma\}_{\sigma=-j}^{j}$ is an orthonormal basis of V_j.

Lemma 20.6 (Wigner cocycle and spin-frame change)**.** *The Lorentz action is*

$$U(\Lambda)F_L(p) = F_L(\Lambda p)D^{(j)}(W(\Lambda, p)).$$

This formula is a representation of the Lorentz group because the Wigner rotation obeys the cocycle identity

$$W(\Lambda_2\Lambda_1, p) = W(\Lambda_2, \Lambda_1 p)W(\Lambda_1, p).$$

If $Q : \Sigma_m^+ \to SU(2)$ *is a smooth little-group-valued function, then*

$$L'(p) = L(p)Q(p)$$

is another choice of standard transformations. The corresponding Wigner rotation is

$$W'(\Lambda, p) = Q(\Lambda p)^{-1}W(\Lambda, p)Q(p).$$

The associated generalized basis is

$$F_{L'}(p) = F_L(p)D^{(j)}(Q(p)).$$

Thus a wavefunction written as a column $\psi_L(p) \in V_j$ *in the L frame is written in the* L' *frame as*

$$\psi_{L'}(p) = D^{(j)}(Q(p))^{-1}\psi_L(p).$$

The inner product

$$\langle \psi, \chi \rangle = \int_{\Sigma_m^+} d\mu_m(p)\, \psi_L(p)^\dagger \chi_L(p)$$

is independent of the spin frame because $D^{(j)}(Q(p))$ *is unitary. In this language, a scattering amplitude with external massive spin states is a multilinear pairing of spin-frame components and dual spin-frame components whose value is invariant under the simultaneous transformations above.*

Proof. The action formula is the massive Wigner transformation law above written as a map $V_j \to \mathcal{H}_1$. The cocycle identity follows from a direct factorization:

$$\begin{aligned}
W(\Lambda_2\Lambda_1, p) &= L(\Lambda_2\Lambda_1 p)^{-1}\Lambda_2\Lambda_1 L(p) \\
&= L(\Lambda_2\Lambda_1 p)^{-1}\Lambda_2 L(\Lambda_1 p)\, L(\Lambda_1 p)^{-1}\Lambda_1 L(p) \\
&= W(\Lambda_2, \Lambda_1 p)W(\Lambda_1, p).
\end{aligned}$$

If $L'(p) = L(p)Q(p)$, then

$$\begin{aligned}
W'(\Lambda, p) &= L'(\Lambda p)^{-1}\Lambda L'(p) \\
&= Q(\Lambda p)^{-1}L(\Lambda p)^{-1}\Lambda L(p)Q(p) \\
&= Q(\Lambda p)^{-1}W(\Lambda, p)Q(p).
\end{aligned}$$

The basis relation

$$F_{L'}(p) = F_L(p)D^{(j)}(Q(p))$$

therefore forces the component relation $\psi_{L'}(p) = D^{(j)}(Q(p))^{-1}\psi_L(p)$. Since $D^{(j)}$ is unitary,

$$\psi_{L'}(p)^\dagger \chi_{L'}(p) = \psi_L(p)^\dagger \chi_L(p)$$

pointwise on the mass shell, and the displayed inner product is spin-frame independent. Dual spin-frame components transform by the inverse dual matrices, so any contracted scattering amplitude is independent of the choice of $L(p)$. □

20.4 Covariant Fields as Intertwiners

Proposition 20.7 (Vacuum-to-particle intertwiners)**.** *Let $\widehat{\Phi}_A(x)$ be a local field whose index A transforms in a finite-dimensional representation R of the double cover* $\mathrm{Spin}^+(1,3) \simeq SL(2,\mathbb{C})$*:*

$$U(\Lambda)\widehat{\Phi}_A(x)U(\Lambda)^{-1} = R_A{}^B(\Lambda)\widehat{\Phi}_B(\Lambda x).$$

The vacuum-to-particle matrix element is

$$\langle\Omega|\widehat{\Phi}_A(0)|\vec{p},\sigma\rangle = u_A{}^\sigma(p).$$

Translation covariance gives the spacetime dependence

$$\langle\Omega|\widehat{\Phi}_A(x)|\vec{p},\sigma\rangle = \mathrm{e}^{\mathrm{i}p\cdot x}u_A{}^\sigma(p).$$

Lorentz covariance and the Wigner transformation law imply

$$\sum_{\tau=-j}^{j} u_A{}^\tau(\Lambda p)D^{(j)}_{\tau\sigma}(W(\Lambda,p)) = R_A{}^B(\Lambda^{-1})u_B{}^\sigma(p).$$

This is the equivariance condition that connects the covariant field index A to the spin-frame index σ.

At the reference momentum p_R, the matrix

$$u_A{}^\sigma(p_R) : V_j \longrightarrow R|_{SU(2)}$$

is an intertwiner of $SU(2)$ representations:

$$R(W)_A{}^B u_B{}^\sigma(p_R) = \sum_{\tau=-j}^{j} u_A{}^\tau(p_R)D^{(j)}_{\tau\sigma}(W), \qquad W \in SU(2).$$

Thus a covariant field can have nonzero overlap with a spin-j particle exactly through the spin-j part contained in the restriction of R to the massive little group. The boosted wavefunctions are

$$u_A{}^\sigma(p) = R_A{}^B(L(p)^{-1})\, u_B{}^\sigma(p_R),$$

with the convention for $L(p)$ fixed above. Different choices of standard boost are spin-frame changes of the type described in the previous section, and the field intertwiners are used with the corresponding transformed spin-frame components.

Proof. Translation covariance gives

$$\widehat{\Phi}_A(x) = \mathrm{e}^{-\mathrm{i}P\cdot x}\widehat{\Phi}_A(0)\mathrm{e}^{\mathrm{i}P\cdot x}$$

in the convention used throughout this volume. Since $P^\mu|\vec{p},\sigma\rangle = p^\mu|\vec{p},\sigma\rangle$ and $\langle\Omega|P^\mu = 0$, the spacetime dependence is $\mathrm{e}^{\mathrm{i}p\cdot x}u_A{}^\sigma(p)$.

For Lorentz covariance, compute the same matrix element in two ways. The Wigner transformation law gives

$$\langle\Omega|\widehat{\Phi}_A(0)U(\Lambda)|\vec{p},\sigma\rangle = \sum_{\tau=-j}^{j} u_A{}^\tau(\Lambda p)D^{(j)}_{\tau\sigma}(W(\Lambda,p)).$$

Vacuum invariance and field covariance give

$$\begin{aligned}\langle\Omega|\widehat{\Phi}_A(0)U(\Lambda)|\vec{p},\sigma\rangle &= \langle\Omega|U(\Lambda)U(\Lambda)^{-1}\widehat{\Phi}_A(0)U(\Lambda)|\vec{p},\sigma\rangle\\ &= R_A{}^B(\Lambda^{-1})u_B{}^\sigma(p).\end{aligned}$$

Equating these expressions proves the equivariance condition. If $\Lambda = W$ fixes p_R, and the standard section has $L(p_R) = 1$, the equivariance equation becomes precisely the displayed $SU(2)$-intertwiner condition at the rest momentum. Conversely, a rest intertwiner satisfying this condition determines the boosted functions by

$$u_A{}^\sigma(p) = R_A{}^B(L(p)^{-1})u_B{}^\sigma(p_R),$$

and substituting the cocycle identity of Lemma 20.6 verifies the equivariance equation for arbitrary Λ. The possible nonzero intertwiners are therefore exactly the copies of V_j appearing in the restriction of R to the massive little group. □

20.5 Spin One Half

For spin $j = \frac{1}{2}$, the smallest Lorentz-covariant field space is a Dirac spinor. Let the gamma matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-,+,+,+).$$

The global spinor and gamma-matrix conventions are collected in Section 40.16; the present section uses the same symbols and fixes the particle-intertwiner normalizations. One concrete realization, useful for checking signs, is

$$\gamma^0 = -\mathrm{i}\begin{pmatrix}0 & I_2\\ I_2 & 0\end{pmatrix}, \qquad \gamma^i = -\mathrm{i}\begin{pmatrix}0 & \sigma_i\\ -\sigma_i & 0\end{pmatrix}, \qquad i = 1,2,3,$$

where

$$\sigma_1 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_2 = \begin{pmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{pmatrix}, \quad \sigma_3 = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

The following formulas are basis-independent statements about the Clifford module. Define

$$S^{\mu\nu} = -\frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu], \qquad R(\Lambda) = \exp\left(-\frac{\mathrm{i}}{2}\omega_{\mu\nu}S^{\mu\nu}\right)$$

for

$$\Lambda = \exp(\omega), \qquad \omega_{\mu\nu} = -\omega_{\nu\mu}.$$

The generators satisfy

$$[S^{\mu\nu}, \gamma^\rho] = -\mathrm{i}\gamma^\mu\eta^{\nu\rho} + \mathrm{i}\gamma^\nu\eta^{\mu\rho},$$

and therefore

$$R(\Lambda)\gamma^\mu R(\Lambda)^{-1} = \Lambda^\mu{}_\nu\gamma^\nu.$$

On the rest subspace, a spatial rotation by angle θ about the 3-axis acts in the spin-$\frac{1}{2}$ representation as

$$D(R_{12}(\theta)) = \exp\left(-\frac{\mathrm{i}\theta}{2}\sigma_3\right) = \begin{pmatrix}\mathrm{e}^{-\mathrm{i}\theta/2} & 0\\ 0 & \mathrm{e}^{\mathrm{i}\theta/2}\end{pmatrix}.$$

Thus a 2π rotation acts by -1 on a spinor, while a 4π rotation acts by $+1$. This is the infinitesimal form of the covering-group discussion above.

Proposition 20.8 (Massive Dirac eigenspaces, projectors, and spin sums). *Write*

$$\not p = p_\mu \gamma^\mu.$$

For $p_R = (m, \vec{0})$,

$$\not p_R = -m\gamma^0, \qquad \not p_R^2 = -m^2.$$

Choose two rest spinors u_R^σ *and two rest spinors* v_R^σ, $\sigma = 1, 2$, *by*

$$\not p_R u_R^\sigma = \mathrm{i}m\, u_R^\sigma, \qquad \not p_R v_R^\sigma = -\mathrm{i}m\, v_R^\sigma.$$

Their boosts are

$$u^\sigma(p) = R(L(p)^{-1})u_R^\sigma, \qquad v^\sigma(p) = R(L(p)^{-1})v_R^\sigma.$$

Using covariance of γ^μ, *these spinors obey*

$$(\not p - \mathrm{i}m)u^\sigma(p) = 0, \qquad (\not p + \mathrm{i}m)v^\sigma(p) = 0.$$

These are the momentum-space Dirac equations associated with the two eigenspaces of $\not p$ *on the mass shell.*

Let

$$\beta = \mathrm{i}\gamma^0, \qquad \bar{u} = u^\dagger\beta, \qquad \bar{v} = v^\dagger\beta.$$

The hermiticity identities in the above signature are

$$(\gamma^0)^\dagger = -\gamma^0, \qquad (\gamma^i)^\dagger = \gamma^i, \qquad R(\Lambda)^\dagger = \beta R(\Lambda^{-1})\beta.$$

They imply that the β*-pairing is Lorentz invariant:*

$$(R(\Lambda)s)^\dagger\beta R(\Lambda)t = s^\dagger\beta t.$$

The projectors onto the two eigenspaces of $\not p$ *are*

$$\Pi_+(p) = \frac{m - \mathrm{i}\not p}{2m}, \qquad \Pi_-(p) = \frac{m + \mathrm{i}\not p}{2m}.$$

For the delta-normalized one-particle states it is useful to include the same Jacobian factor as in the state normalization and define

$$\mathcal{U}^\sigma(p) = \left(\frac{m}{p^0}\right)^{1/2} R(L(p)^{-1})\mathcal{U}_R^\sigma, \qquad \mathcal{V}^\sigma(p) = \left(\frac{m}{p^0}\right)^{1/2} R(L(p)^{-1})\mathcal{V}_R^\sigma.$$

Choose rest spinors so that

$$\mathcal{U}_R^{\sigma\dagger}\beta\mathcal{U}_R^\tau = \delta^{\sigma\tau}, \qquad \mathcal{V}_R^{\sigma\dagger}\beta\mathcal{V}_R^\tau = -\delta^{\sigma\tau}, \qquad \mathcal{U}_R^{\sigma\dagger}\beta\mathcal{V}_R^\tau = 0.$$

Then

$$\mathcal{U}^\sigma(p)^\dagger\beta\mathcal{U}^\tau(p) = \frac{m}{p^0}\delta^{\sigma\tau}, \qquad \mathcal{V}^\sigma(p)^\dagger\beta\mathcal{V}^\tau(p) = -\frac{m}{p^0}\delta^{\sigma\tau}.$$

The corresponding spin sums, with ordinary matrix multiplication in spinor indices, are

$$\sum_{\sigma=1}^{2}\mathcal{U}^\sigma(p)\mathcal{U}^\sigma(p)^\dagger = \frac{m - \mathrm{i}\not p}{2p^0}\,\beta, \qquad \sum_{\sigma=1}^{2}\mathcal{V}^\sigma(p)\mathcal{V}^\sigma(p)^\dagger = \frac{-m - \mathrm{i}\not p}{2p^0}\,\beta.$$

Equivalently,

$$\sum_{\sigma=1}^{2}\mathcal{U}^\sigma(p)\overline{\mathcal{U}}^\sigma(p)=\frac{m-\mathrm{i}\not{p}}{2p^0},\qquad \sum_{\sigma=1}^{2}\mathcal{V}^\sigma(p)\overline{\mathcal{V}}^\sigma(p)=-\frac{m+\mathrm{i}\not{p}}{2p^0}.$$

The minus sign in the second identity is fixed by the sign of the β-norm on the negative Dirac eigenspace; the particle Hilbert-space inner product remains positive.

Proof. The Clifford relation gives

$$\not{p}^2=p_\mu p_\nu\gamma^\mu\gamma^\nu=p^2\mathbf{1}=-m^2\mathbf{1}$$

on the mass shell, because the antisymmetric commutator part drops out after contraction with $p_\mu p_\nu$. Hence the minimal polynomial of $\not{p}$ divides λ^2+m^2, and the two spectral projectors are

$$\Pi_+(p)=\frac{m-\mathrm{i}\not{p}}{2m},\qquad \Pi_-(p)=\frac{m+\mathrm{i}\not{p}}{2m}.$$

They satisfy $\Pi_\pm^2=\Pi_\pm$, $\Pi_+\Pi_-=0$, and $\Pi_++\Pi_-=1$. The defining equations for u_R^σ and v_R^σ say that they span the $\Pi_+(p_R)$ and $\Pi_-(p_R)$ eigenspaces.

The covariance identity

$$R(\Lambda)\gamma^\mu R(\Lambda)^{-1}=\Lambda^\mu{}_\nu\gamma^\nu$$

implies

$$\not{p}\,R(L(p)^{-1})=R(L(p)^{-1})\not{p}_R,$$

with $p=L(p)p_R$. Applying this identity to u_R^σ and v_R^σ proves the two momentum-space Dirac equations.

The relation

$$R(\Lambda)^\dagger=\beta R(\Lambda^{-1})\beta$$

is equivalent to $R(\Lambda)^\dagger\beta R(\Lambda)=\beta$, so the β-pairing is Lorentz invariant. Multiplication by $(m/p^0)^{1/2}$ in the delta-normalized spinors therefore changes the rest norms only by the factor m/p^0, giving the displayed β-inner-products.

It remains to identify the matrix-valued completeness relations. At fixed p, the vectors $\mathcal{U}^\sigma(p)$ form a β-orthogonal basis of the positive eigenspace, while $\mathcal{V}^\sigma(p)$ form a β-orthogonal basis of the negative eigenspace with negative β-norm. Hence

$$\sum_{\sigma=1}^{2}\mathcal{U}^\sigma(p)\overline{\mathcal{U}}^\sigma(p)=\frac{m}{p^0}\Pi_+(p),\qquad \sum_{\sigma=1}^{2}\mathcal{V}^\sigma(p)\overline{\mathcal{V}}^\sigma(p)=-\frac{m}{p^0}\Pi_-(p).$$

Substituting the projectors gives the two spin sums with bars. Since $\beta^2=1$, the identities without bars follow by multiplying on the right by β. This proves all displayed projector and spin-sum formulas. $\square$

For the explicit gamma matrices displayed above, an adapted rest basis is

$$\mathcal{U}_R^1=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\\0\end{pmatrix},\qquad \mathcal{U}_R^2=\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\0\\1\end{pmatrix},$$

$$\mathcal{V}_R^1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \qquad \mathcal{V}_R^2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

If

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix},$$

then

$$\mathcal{V}_R^\sigma = \gamma_5 \mathcal{U}_R^\sigma.$$

These projector identities are the algebraic source of the external spinor factors in spinorial LSZ reduction.

20.6 Particle Spin and Field Indices

The spin j labels an irreducible unitary representation of the massive little group on the one-particle Hilbert space. A field index A labels a finite-dimensional Lorentz representation used by a local operator. The two are related by the intertwiner

$$u_A{}^\sigma(p) = \langle \Omega | \widehat{\Phi}_A(0) | \vec{p}, \sigma \rangle.$$

This distinction is part of the basic architecture of relativistic quantum theory. The same spin-j particle may be created by different local operators, and a single covariant field representation may contain several little-group spins when restricted to $SU(2)$. LSZ for spinning particles uses the pole of the relevant two-point function together with the intertwiner $u_A{}^\sigma(p)$ to convert field indices into external particle spin labels.

The next chapter constructs spinor field operators and their functional integral calculus. The representation-theoretic data established here will then become the spinor completeness relations, propagator numerators, and external-line factors used in perturbative calculations.

Chapter 21

Massless Particles, Helicity, and Gauge Redundancy

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

21.1 The Massless Orbit

Work in $D = 4$ with mostly-plus metric

$$\eta_{\mu\nu} = \mathrm{diag}(-, +, +, +).$$

The positive-energy massless orbit is

$$\Sigma_0^+ = \{k = (k^0, \vec{k}) : k^2 = 0,\ k^0 = |\vec{k}|,\ k^0 > 0\}.$$

Together with the massive positive-energy shells of Chapter 20, this is one of the orbit types compatible with the spectrum condition; the complete four-dimensional orbit classification is summarized in Table 5.1. Spacelike Wigner orbits $k^2 > 0$ are excluded by $\mathrm{Spec}(P) \subset \overline{V}_+$, as stated there. Choose a reference momentum

$$q = (\kappa, 0, 0, \kappa), \qquad \kappa > 0.$$

For each $k \in \Sigma_0^+$, choose a standard Lorentz transformation

$$L(k) \in SO^+(1, 3), \qquad L(k)q = k.$$

As in the massive case, Hilbert-space formulas use the universal cover of $SO^+(1,3)$ when spinorial phases are involved, while the same symbols denote the projected Lorentz transformations acting on four-vectors. The little group of q is

$$G_q = \{W \in SO^+(1,3) : Wq = q\}.$$

Definition 21.1 (Massless one-particle orbit and standard section)**.** The positive-energy massless orbit is the homogeneous space $\Sigma_0^+ \simeq SO^+(1,3)/G_q$, where G_q is the stabilizer of $q = (\kappa, 0, 0, \kappa)$. A standard section is a choice of Lorentz transformation $L(k)$ for each $k \in \Sigma_0^+$ such that $L(k)q = k$. It is not canonical: replacing $L(k)$ by $L(k)Q(k)$, with $Q(k) \in G_q$, changes the one-particle helicity frame.

The non-uniqueness of $L(k)$ is the origin of helicity-frame phases and of the representative freedom of vector polarizations. It is the choice of local trivialization of a line bundle over the null orbit.

For vector indices, write a Lorentz transformation as

$$\Lambda = \exp\left(\frac{\mathrm{i}}{2}\omega_{\mu\nu}M^{\mu\nu}\right), \qquad \omega_{\mu\nu} = -\omega_{\nu\mu},$$

where

$$(M^{\mu\nu})^{\rho}{}_{\sigma} = -\mathrm{i}\,(\eta^{\mu\rho}\delta^{\nu}_{\sigma} - \eta^{\nu\rho}\delta^{\mu}_{\sigma})\,.$$

Then, for infinitesimal ω,

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu},$$

and

$$[M^{\mu\nu}, M^{\rho\sigma}] = -\mathrm{i}\,(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - (\rho \leftrightarrow \sigma))\,.$$

Let $J_{\mu\nu}$ denote the corresponding Hilbert-space generators on the one-particle representation:

$$U(\Lambda) = \exp\left(\frac{\mathrm{i}}{2}\omega_{\mu\nu}J^{\mu\nu}\right), \qquad [J^{\mu\nu}, J^{\rho\sigma}] = -\mathrm{i}\,(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - (\rho \leftrightarrow \sigma))\,.$$

The three Hilbert-space generators

$$J_3 = J_{12}, \qquad N_1 = J_{01} + J_{31}, \qquad N_2 = J_{02} + J_{32}$$

generate G_q. Their commutators are

$$[N_1, N_2] = 0, \qquad [J_3, N_1] = \mathrm{i}N_2, \qquad [J_3, N_2] = -\mathrm{i}N_1.$$

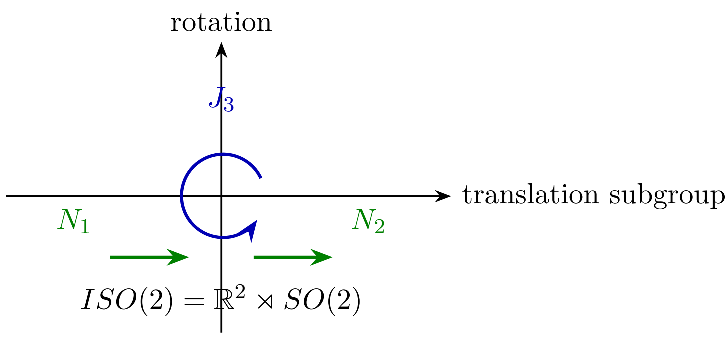
Little algebra of a future null momentum. The infinitesimal stabilizer of $q = (\kappa, 0, 0, \kappa)$ is generated by $J_3 = J_{12}$, $N_1 = J_{01} + J_{31}$, and $N_2 = J_{02} + J_{32}$, and these generators obey

$$[N_1, N_2] = 0, \qquad [J_3, N_1] = \mathrm{i}N_2, \qquad [J_3, N_2] = -\mathrm{i}N_1.$$

Hence the connected little group is the double cover of $\mathbb{R}^2 \rtimes SO(2)$. In the four-vector representation the matrices M_{12}, $M_{01} + M_{31}$, and $M_{02} + M_{32}$ annihilate q. Conversely, if an infinitesimal Lorentz transformation $\omega^{\mu}{}_{\nu}$ stabilizes q, then $\omega q = 0$. With $q^0 = q^3 = \kappa$, this condition and the η-antisymmetry of ω leave precisely one rotation parameter in the $1,2$-plane and two null-translation parameters, represented by the three generators displayed above.

The commutators follow by substituting these linear combinations into the Lorentz algebra. The translation commutator cancels between the $(01,02)$ and $(31,32)$ contributions, while J_{12} rotates the pair $(J_{01}+J_{31}, J_{02}+J_{32})$ as a two-vector. This is the Lie algebra of the Euclidean group of the transverse plane.

This is the Lie algebra of the Euclidean group $ISO(2)$: J_3 generates rotations in a two-dimensional transverse plane, while N_1, N_2 generate commuting translations in that plane. The double cover is used on Hilbert space.



21.2 Little-Group Matrices

The matrix representatives make the $ISO(2)$ structure explicit. The rotation around the x^3-axis is generated by

$$(M_{12})^{\mu}{}_{\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\mathrm{i} & 0 \\ 0 & \mathrm{i} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so

$$R_3(\theta) = \mathrm{e}^{\mathrm{i}\theta M_{12}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The two null-translation generators are

$$A = M_{01} + M_{31} = -\mathrm{i}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

and

$$B = M_{02} + M_{32} = -\mathrm{i}\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Since $A^3 = B^3 = A^2B = AB^2 = 0$ in this representation and $[A, B] = 0$, their exponential is the finite polynomial

$$T(\alpha, \beta) = \mathrm{e}^{\mathrm{i}\alpha A + \mathrm{i}\beta B} = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}, \qquad \zeta = \frac{\alpha^2 + \beta^2}{2}.$$

Every displayed matrix fixes q, and the little-group element with parameters (α, β, θ) is

$$W(\alpha, \beta, \theta) = T(\alpha, \beta)R_3(\theta).$$

The matrix $T(\alpha, \beta)$ satisfies

$$T(\alpha, \beta)^T \eta\, T(\alpha, \beta) = \eta, \qquad T(\alpha, \beta)q = q,$$

and the product law

$$T(\alpha, \beta)T(\alpha', \beta') = T(\alpha + \alpha', \beta + \beta').$$

The rotation $R_3(\theta)$ acts on the pair (α, β) by the ordinary rotation of the transverse plane. The displayed $T(\alpha, \beta)$ is the exponential of $\mathrm{i}\alpha A + \mathrm{i}\beta B$. Since A and B are Lorentz Lie-algebra matrices, the exponential preserves η. Multiplication of the displayed matrix by $(\kappa, 0, 0, \kappa)^T$ gives $(\kappa, 0, 0, \kappa)^T$, so $T(\alpha, \beta)$ fixes q. Because A and B commute, their exponentials multiply by adding parameters. The commutators $[M_{12}, A] = \mathrm{i}B$ and $[M_{12}, B] = -\mathrm{i}A$ give the stated rotational action on the null-translation parameters.

The vector-space commutators

$$[M_{12}, A] = \mathrm{i}B, \qquad [M_{12}, B] = -\mathrm{i}A$$

are the matrix form of the semidirect product $\mathbb{R}^2 \rtimes SO(2)$.

21.3 Helicity Representations

The Hilbert-space counterparts of A and B are N_1 and N_2. Since $[N_1, N_2] = 0$, a fixed-q generalized basis, in the sense of Section 5.2, may be chosen so that

$$N_1|q, a, b; \lambda\rangle = a|q, a, b; \lambda\rangle, \qquad N_2|q, a, b; \lambda\rangle = b|q, a, b; \lambda\rangle.$$

Here λ denotes any remaining degeneracy label. The commutators with J_3 imply that, on these labels,

$$J_3 = \mathrm{i}b\frac{\partial}{\partial a} - \mathrm{i}a\frac{\partial}{\partial b} + J_3^{(0)},$$

where $J_3^{(0)}$ acts on the degeneracy space at fixed (a, b). Consequently

$$\mathrm{e}^{\mathrm{i}\theta J_3}|q, a, b; \lambda\rangle = |q, a\cos\theta - b\sin\theta, a\sin\theta + b\cos\theta; \lambda_\theta\rangle$$

up to the action of $J_3^{(0)}$. If $a^2 + b^2 > 0$, the one-particle labels at fixed momentum include the full $SO(2)$-orbit of (a, b). Those irreducible representations are continuous-spin representations. The helicity sectors are the massless one-particle irreducible representations for which

$$N_1|q, h\rangle = 0, \qquad N_2|q, h\rangle = 0.$$

The remaining little-group generator acts by

$$J_3|q,h\rangle = h|q,h\rangle.$$

Equivalently, for

$$W(\alpha,\beta,\theta) = \exp(\mathrm{i}\alpha N_1 + \mathrm{i}\beta N_2)\exp(\mathrm{i}\theta J_3),$$

the state transforms as

$$U(W(\alpha,\beta,\theta))|q,h\rangle = \mathrm{e}^{\mathrm{i}h\theta}|q,h\rangle.$$

The double-cover rotation condition gives

$$h \in \frac{1}{2}\mathbb{Z}.$$

Definition 21.2 (Massless helicity representation)**.** A massless one-particle helicity representation is the induced positive-energy Poincare representation built from the little-group character

$$\chi_h(W(\alpha,\beta,\theta)) = \mathrm{e}^{\mathrm{i}h\theta}, \qquad h \in \frac{1}{2}\mathbb{Z},$$

in which the translation subgroup $\mathbb{R}^2 \subset ISO(2)$ acts trivially. Equivalently, at the reference momentum q, $N_1 = N_2 = 0$ and J_3 acts by the scalar h.

Continuous-spin alternative. The commuting self-adjoint generators N_1, N_2 have a joint spectral measure. The little-algebra commutators above imply

$$\mathrm{e}^{\mathrm{i}\theta J_3}(N_1,N_2)\mathrm{e}^{-\mathrm{i}\theta J_3} = (N_1\cos\theta - N_2\sin\theta,\ N_1\sin\theta + N_2\cos\theta).$$

Consequently the support of the joint spectral measure is invariant under rotations of the (a,b)-plane. If it contains a point with $a^2+b^2=\rho^2>0$, the little-group orbit contains the full circle

$$(a,b) \mapsto (a\cos\theta - b\sin\theta,\ a\sin\theta + b\cos\theta).$$

On this circle the rotation generator acts by the regular $SO(2)$ action, or by its double-cover lift, and the angular Fourier modes form an integer-spaced infinite tower. This is the representation-theoretic continuous-spin alternative. Finite-helicity representations are the $\rho = 0$ case in which the translation subgroup acts trivially and only the character of the rotation subgroup remains.

Remark 21.3 (Continuous-spin sectors)*.* The alternative $a^2 + b^2 = \rho^2 > 0$ gives the massless continuous-spin, or infinite-spin, Wigner representations. They are legitimate irreducible positive-energy representations of the Poincare group, so the restriction to $\rho = 0$ is an additional physical and structural choice. At fixed momentum the little-group orbit is a circle, and the Fourier modes on that circle form an infinite tower of helicity labels mixed by boosts. A gas of such particles therefore has infinitely many internal polarizations at each momentum; ordinary finite-helicity thermodynamic counting is lost unless further structure controls the occupation of those modes.

There is also a sharper locality obstruction. Under the usual assumptions of positive energy, covariance, local commutativity, and pointlike covariant local fields, the zero-mass infinite-spin representations cannot be interpolated by a local field with a nonzero vacuum-to-one-particle matrix element. Equivalently, their natural localization in algebraic constructions is spacelike-cone or string localization rather than point localization in bounded regions. Such sectors are therefore possible representation-theoretic options, but they do not enter the point-local or gauge-redundant helicity field constructions used for photons, gravitons, and the gauge theories developed in this volume.

There are two useful normalization conventions. The covariant convention used below is

$$\langle k', h' | k, h \rangle = (2\pi)^3 \, 2k^0 \, \delta^{(3)}(\vec{k}' - \vec{k}) \delta_{h'h}.$$

In the delta-normalized convention

$$_\delta\langle \vec{k}', h' | \vec{k}, h \rangle_\delta = \delta^{(3)}(\vec{k}' - \vec{k}) \delta_{h'h},$$

the corresponding state is

$$|\vec{k}, h\rangle_\delta = \sqrt{\frac{q^0}{k^0}} \, U(L(k)) |\vec{q}, h\rangle_\delta.$$

The factor is fixed by the Lorentz transformation of the three-dimensional delta function on the mass shell. It gives

$$U(\Lambda)|\vec{k}, h\rangle_\delta = \sqrt{\frac{(\Lambda k)^0}{k^0}} \, e^{ih\theta(\Lambda,k)} |\overrightarrow{\Lambda k}, h\rangle_\delta.$$

For general $k \in \Sigma_0^+$, define generalized one-particle states by

$$|k, h\rangle = U(L(k))|q, h\rangle,$$

with the covariant normalization displayed above. For a Lorentz transformation Λ, the Wigner little-group element is

$$W(\Lambda, k) = L(\Lambda k)^{-1} \Lambda L(k),$$

and the helicity state transforms by

$$U(\Lambda)|k, h\rangle = e^{ih\theta(\Lambda,k)} |\Lambda k, h\rangle,$$

where $\theta(\Lambda, k)$ is the $SO(2)$ rotation angle of $W(\Lambda, k)$ in the helicity representation. Different choices of $L(k)$ change the phase convention for $|k, h\rangle$ but leave invariant matrix elements unchanged.

21.4 Helicity Frames

For a fixed helicity h, let $V_h \simeq \mathbb{C}$ be the one-dimensional little-group representation with character

$$\chi_h(W(\alpha, \beta, \theta)) = e^{ih\theta}.$$

The translation parameters α, β do not appear in χ_h because this is the helicity sector. With

$$d\mu_0(k) = \frac{d^3k}{(2\pi)^3 \, 2|\vec{k}|},$$

the one-particle Hilbert space of a single helicity is represented, after choosing $L(k)$, as

$$\mathcal{H}_{1,h} \simeq L^2(\Sigma_0^+, d\mu_0; V_h).$$

The distributional frame $F_L(k) : V_h \to \mathcal{H}_{1,h}$ satisfies

$$U(\Lambda) F_L(k) = F_L(\Lambda k) \chi_h(W(\Lambda, k)).$$

The little-group element again obeys

$$W(\Lambda_2 \Lambda_1, k) = W(\Lambda_2, \Lambda_1 k) W(\Lambda_1, k),$$

so the helicity phase is a one-dimensional cocycle.

Wigner cocycle and helicity-frame change. For a standard section $L(k)$, the little-group element $W(\Lambda, k) = L(\Lambda k)^{-1}\Lambda L(k)$ obeys

$$W(\Lambda_2\Lambda_1, k) = W(\Lambda_2, \Lambda_1 k)W(\Lambda_1, k).$$

If $L'(k) = L(k)Q(k)$, then

$$W'(\Lambda, k) = Q(\Lambda k)^{-1}W(\Lambda, k)Q(k),$$

and the corresponding helicity-frame wavefunctions obey

$$\psi_{L'}(k) = \chi_h(Q(k))^{-1}\psi_L(k).$$

The cocycle identity is the associative identity

$$L(\Lambda_2\Lambda_1 k)^{-1}\Lambda_2\Lambda_1 L(k) = \big(L(\Lambda_2\Lambda_1 k)^{-1}\Lambda_2 L(\Lambda_1 k)\big)\big(L(\Lambda_1 k)^{-1}\Lambda_1 L(k)\big).$$

For $L'(k) = L(k)Q(k)$, substitution gives

$$L'(\Lambda k)^{-1}\Lambda L'(k) = Q(\Lambda k)^{-1}L(\Lambda k)^{-1}\Lambda L(k)Q(k),$$

which is the displayed conjugation law. The distributional frame changes by right action of the one-dimensional representation,

$$F_{L'}(k) = F_L(k)\chi_h(Q(k)).$$

Since a vector is written as

$$\int_{\Sigma_0^+} F_L(k)\psi_L(k)\, d\mu_0(k),$$

invariance of this vector under the frame change forces the inverse character on the wavefunction component.

21.5 Photon Intertwiners and Vector Representatives

Consider helicity $h = \pm 1$. A vector polarization representative at the reference momentum is

$$\epsilon^+_\mu(q) = \frac{1}{\sqrt{2}}(0, 1, -\mathrm{i}, 0), \qquad \epsilon^-_\mu(q) = \frac{1}{\sqrt{2}}(0, 1, \mathrm{i}, 0).$$

It obeys

$$q^\mu\epsilon^\pm_\mu(q) = 0, \qquad \epsilon^\pm_\mu(q)\epsilon^{\pm *\,\mu}(q) = 1, \qquad \epsilon^\pm_\mu(q)\epsilon^{\mp *\,\mu}(q) = 0.$$

The rotation generated by J_3 gives

$$W(0, 0, \theta)^\nu{}_\mu\epsilon^\pm_\nu(q) = \mathrm{e}^{\pm\mathrm{i}\theta}\epsilon^\pm_\mu(q),$$

which is the vector-representative phase entering the covariance equation; the one-particle state contributes the inverse little-group phase. The sign in this equation follows from the lower-index action. With the rotation matrix displayed above, restricted to the transverse $1, 2$ components,

$$\begin{aligned} R_3(\theta)^\nu{}_\mu\epsilon^+_\nu(q) &= \frac{1}{\sqrt{2}}\big(0, \cos\theta + \mathrm{i}\sin\theta,\, \sin\theta - \mathrm{i}\cos\theta, 0\big) \\ &= \mathrm{e}^{\mathrm{i}\theta}\epsilon^+_\mu(q), \end{aligned}$$

and the complex conjugate vector $\epsilon^-_\mu(q)$ gives the phase $\mathrm{e}^{-\mathrm{i}\theta}$.

Let $\widehat{\Phi}_\mu$ be a covariant vector local field, in a positive-metric physical Hilbert space, with a nonzero one-photon matrix element. In the delta-normalized convention write

$$_\delta\langle \vec{k}, h|\widehat{\Phi}_\mu(0)|\Omega\rangle = C_h u^h_\mu(\vec{k}).$$

If

$$U(\Lambda)\widehat{\Phi}_\mu(x)U(\Lambda)^{-1} = \Lambda^\nu{}_\mu\,\widehat{\Phi}_\nu(\Lambda x),$$

then the transformation law of the delta-normalized states gives

$$u^h_\mu(\vec{k}) = \sqrt{\frac{(\Lambda k)^0}{k^0}}\,\mathrm{e}^{-\mathrm{i}h\theta(\Lambda,k)}\Lambda^\nu{}_\mu u^h_\nu(\Lambda\vec{k}).$$

At $k = q$ and $\Lambda = W(\alpha, \beta, \theta)$, this specializes to

$$u^h_\mu(\vec{q}) = \mathrm{e}^{-\mathrm{i}h\theta}W^\nu{}_\mu u^h_\nu(\vec{q}).$$

The $SO(2)$ part of this equation fixes the transverse representatives above. The null-translation part acts by a shift along q_μ. For

$$W(\alpha,\beta,0) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}, \qquad \zeta = \frac{\alpha^2+\beta^2}{2},$$

one finds

$$W(\alpha,\beta,0)^\nu{}_\mu\epsilon^\pm_\nu(q) = \frac{1}{\sqrt{2}}\begin{pmatrix} \alpha \mp \mathrm{i}\beta \\ 1 \\ \mp\mathrm{i} \\ -\alpha \pm \mathrm{i}\beta \end{pmatrix} = \epsilon^\pm_\mu(q) - \frac{\alpha\mp\mathrm{i}\beta}{\sqrt{2}\,\kappa}\,q_\mu,$$

where $q_\mu = (-\kappa, 0, 0, \kappa)$. Hence the vector representative is defined modulo the null direction:

$$\epsilon^\pm_\mu(k) \sim \epsilon^\pm_\mu(k) + c\,k_\mu.$$

For helicity $h = \pm1$, the null-translation subgroup of the massless little group acts on the covariant vector representative by the displayed q_μ-shift, and therefore the vector representative of a helicity-one state is naturally an equivalence class modulo k_μ-shifts. Multiplying $T(\alpha, \beta)$ by $\epsilon^\pm_\nu(q)$ gives the four-component column displayed above. Since $q_\mu = (-\kappa, 0, 0, \kappa)$, the difference between that column and $\epsilon^\pm_\mu(q)$ is $-(\alpha\mp\mathrm{i}\beta)q_\mu/(\sqrt{2}\,\kappa)$. For a general null momentum $k = L(k)q$, applying $L(k)$ transports the same statement to a shift proportional to k_μ.

For general k, choose

$$\epsilon^\pm_\mu(k) = L(k)_\mu{}^\nu\epsilon^\pm_\nu(q).$$

If $L'(k) = L(k)Q(k)$ and $Q(k) = W(\alpha(k), \beta(k), \theta(k))$, then

$$\epsilon^{\pm\prime}_\mu(k) = \mathrm{e}^{\pm\mathrm{i}\theta(k)}\epsilon^\pm_\mu(k) + c_\pm(k)k_\mu.$$

The phase is the vector-representative helicity phase, and the term proportional to k_μ is the representative change generated by the null-translation part of the little group.

21.6 Field Strength Matrix Elements

A vector potential A_μ is a representative variable whose polarization matrix element uses $\epsilon^h_\mu(k)$. Its representative freedom is

$$A_\mu(x) \sim A_\mu(x) + \partial_\mu \xi(x),$$

where ξ is a scalar gauge parameter. The gauge-invariant local tensor constructed from A_μ is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

For a one-photon state of helicity $h = \pm 1$, the field-strength matrix element has the form

$$\langle k, h|\, \widehat{F}_{\mu\nu}(x)\, |\Omega\rangle = C_h \mathrm{e}^{-\mathrm{i}k\cdot x} \left(-\mathrm{i}k_\mu \epsilon^{h*}_\nu(k) + \mathrm{i}k_\nu \epsilon^{h*}_\mu(k) \right),$$

where C_h is a normalization constant fixed by the field normalization. The shift

$$\epsilon^h_\mu(k) \mapsto \epsilon^h_\mu(k) + c\, k_\mu$$

leaves this expression unchanged because

$$k_\mu k_\nu - k_\nu k_\mu = 0.$$

Thus the antisymmetric field strength is well-defined on the quotient of vector representatives by the null direction. Equivalently, the linear map

$$\epsilon_\mu \longmapsto k_\mu \epsilon_\nu - k_\nu \epsilon_\mu$$

has the null line $\mathbb{C}k_\mu$ in its kernel and therefore descends to the quotient polarization line used for helicity-one states. The x-dependence and the normalization C_h are unaffected because the representative change acts only on $\epsilon^h_\mu(k)$. The local photon observable at this stage is therefore the field strength, while the potential is a representative used to write local equations, actions, and perturbative Green functions.

Remark 21.4. This statement is made in the physical positive-metric Hilbert space and its local observable algebra. Gauge-fixed quantization later introduces A_μ on an enlarged space with constraints, ghosts, or BRST cohomology. In that formulation the vector potential is a calculational representative, and $F_{\mu\nu}$ descends to the physical local algebra.

Remark 21.5 (Two external photons). For an amplitude with an incoming photon of momentum k and an outgoing photon of momentum k', write the reduced tensor before choosing polarization representatives as $\mathcal{M}^{\mu\nu}$. Descent to the two helicity lines requires two Ward identities,

$$k_\mu \mathcal{M}^{\mu\nu} = 0, \qquad k'_\nu \mathcal{M}^{\mu\nu} = 0.$$

In tree-level spinor Compton scattering these identities are obtained by summing the two electron-exchange diagrams. Let

$$N(q) = -\mathrm{i}\; \not{q} + m, \qquad D(q) = q^2 + m^2 - \mathrm{i}0,$$

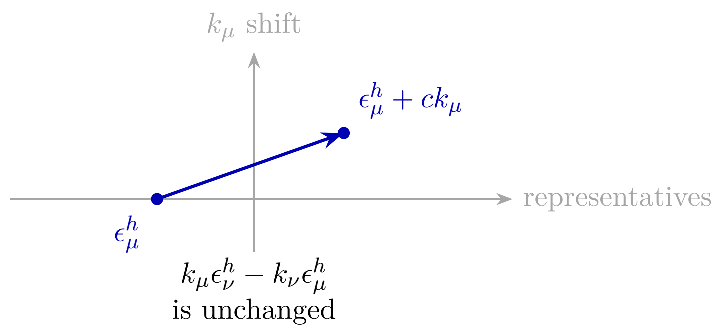
with $/q = \gamma^\rho q_\rho$, and impose the external on-shell Dirac equations. Replacing the incoming polarization by k_μ gives, up to the common charge and normalization factor, the two terms

$$\bar{u}(p') \;\not{\epsilon}'^*(k')\, \frac{N(p+k)\; \not{k}}{D(p+k)}\, u(p) = -\mathrm{i}\, \bar{u}(p') \;\not{\epsilon}'^*(k') u(p),$$

and

$$\bar{u}(p')\ \not{k}\,\frac{N(p-k')}{D(p-k')}\ \not{\epsilon}'^{*}(k')\,u(p) = +\mathrm{i}\,\bar{u}(p')\ \not{\epsilon}'^{*}(k')u(p).$$

The contact terms produced by the Ward contraction of the s- and u-channel diagrams therefore cancel. Replacing the outgoing polarization by k'_ν gives the same cancellation with k and k' interchanged.



21.7 Polarization Sums

For calculations with external helicity states it is useful to choose representatives by imposing an auxiliary linear condition. Let n^μ be a fixed vector satisfying

$$n \cdot k \neq 0.$$

Choose $\epsilon^h_\mu(k;n)$ so that

$$k^\mu\epsilon^h_\mu(k;n) = 0, \qquad n^\mu\epsilon^h_\mu(k;n) = 0, \qquad \epsilon^h(k;n)\cdot\epsilon^{h'*}(k;n) = \delta_{hh'}.$$

The sum over the two physical helicities is

$$\Pi_{\mu\nu}(k;n) = \sum_{h=\pm1}\epsilon^h_\mu(k;n)\epsilon^{h*}_\nu(k;n),$$

with

$$\Pi_{\mu\nu}(k;n) = \eta_{\mu\nu} - \frac{k_\mu n_\nu + n_\mu k_\nu}{k\cdot n} + \frac{n^2 k_\mu k_\nu}{(k\cdot n)^2}.$$

It satisfies

$$k^\mu\Pi_{\mu\nu}(k;n) = 0, \qquad n^\mu\Pi_{\mu\nu}(k;n) = 0.$$

Changing n changes the representative polarization vectors by terms proportional to k_μ. Therefore, if a quantity $M^\mu(k)$ obeys

$$k_\mu M^\mu(k) = 0,$$

then

$$M^\mu(k)\Pi_{\mu\nu}(k;n)M^{\nu*}(k)$$

is independent of the representative choice. This is the form in which polarization sums will enter gauge-invariant scattering calculations.

Let $k^2 = 0$ and $n \cdot k \neq 0$. The tensor

$$\Pi_{\mu\nu}(k;n) = \eta_{\mu\nu} - \frac{k_\mu n_\nu + n_\mu k_\nu}{k \cdot n} + \frac{n^2 k_\mu k_\nu}{(k \cdot n)^2}$$

annihilates k^μ and n^μ. If $M^\mu k_\mu = 0$, then the contraction $M^\mu \Pi_{\mu\nu}(k;n) M^{\nu *}$ is unchanged under a change of polarization representatives by k_μ-shifts. Contracting with k^μ gives

$$k_\nu - \frac{k^2 n_\nu + (k \cdot n) k_\nu}{k \cdot n} + \frac{n^2 k^2 k_\nu}{(k \cdot n)^2} = 0,$$

using $k^2 = 0$. Contracting with n^μ gives

$$n_\nu - \frac{(n \cdot k) n_\nu + n^2 k_\nu}{k \cdot n} + \frac{n^2 (n \cdot k) k_\nu}{(k \cdot n)^2} = 0.$$

If the representative basis is changed, the projector changes only by terms with at least one factor k_μ or k_ν. Such terms vanish in $M^\mu \Pi_{\mu\nu} M^{\nu *}$ when $M^\mu k_\mu = 0$.

21.8 Four-Dimensional Spinor-Helicity Variables

The preceding sections described helicity as a line bundle over the massless orbit. In four dimensions this bundle has a concrete spinorial trivialization. The statements in this and the next two sections concern tree-level perturbative massless amplitudes after the scattering operator and LSZ extraction have already been defined; they are not a definition of the S-operator.

For a real four-vector p^μ, define the Hermitian 2×2 matrix

$$p_{a\dot{a}} = p^0 \delta_{a\dot{a}} + p^i (\sigma_i)_{a\dot{a}}, \qquad a, \dot{a} = 1, 2. \tag{21.1}$$

With the mostly-plus metric used in this monograph,

$$\det(p_{a\dot{a}}) = (p^0)^2 - \vec{p}^{\,2} = -p^2.$$

Thus a complexified null momentum is equivalently a rank-one matrix, and may be written

$$p_{a\dot{a}} = \lambda_a \widetilde{\lambda}_{\dot{a}}. \tag{21.2}$$

For a real future-directed null momentum one may choose $\widetilde{\lambda}_{\dot{a}} = \overline{\lambda_a}$, up to a phase. In analytic amplitude formulae λ and $\widetilde{\lambda}$ are treated as independent complex spinors; the real physical region is a boundary value. The factorization (21.2) is unchanged by the little-group scaling

$$\lambda_a \longmapsto t\, \lambda_a, \qquad \widetilde{\lambda}_{\dot{a}} \longmapsto t^{-1} \widetilde{\lambda}_{\dot{a}}, \qquad t \in \mathbb{C}^\times. \tag{21.3}$$

This is the complexified version of the helicity-frame freedom.

Definition 21.6 (Spinor-helicity frame)**.** A spinor-helicity frame for a complex null momentum p is a factorization $p_{a\dot{a}} = \lambda_a \widetilde{\lambda}_{\dot{a}}$, with the equivalence relation

$$(\lambda, \widetilde{\lambda}) \sim (t\lambda, t^{-1}\widetilde{\lambda}), \qquad t \in \mathbb{C}^\times.$$

A helicity-h amplitude component is a section of the corresponding helicity line and therefore has weight t^{-2h} under this scaling.

Spinor contractions are

$$\langle ij\rangle = \epsilon^{ab}\lambda_{i,a}\lambda_{j,b}, \qquad [ij] = \epsilon^{\dot a\dot b}\widetilde{\lambda}_{i,\dot a}\widetilde{\lambda}_{j,\dot b}, \tag{21.4}$$

with the same $\epsilon^{12} = 1$ convention as Section 40.16. They are antisymmetric: $\langle ij\rangle = -\langle ji\rangle$ and $[ij] = -[ji]$. A direct determinant calculation gives the mostly-plus invariant

$$\langle ij\rangle[ji] = 2p_i \cdot p_j. \tag{21.5}$$

For a sum $P^\mu = \sum_{r\in I} p_r^\mu$, use the abbreviation

$$\langle i|P|j] := \lambda_i^a P_{a\dot a}\widetilde{\lambda}_j^{\dot a}.$$

Momentum conservation for an all-outgoing amplitude is

$$\sum_{i=1}^{n} \lambda_{i,a}\widetilde{\lambda}_{i,\dot a} = 0. \tag{21.6}$$

An n-point helicity amplitude with external helicities h_i is a section of the tensor product of the helicity lines. In a spinor-helicity frame this means

$$A_n(\ldots, t_i\lambda_i, t_i^{-1}\widetilde{\lambda}_i, \ldots) = t_i^{-2h_i} A_n(\ldots, \lambda_i, \widetilde{\lambda}_i, \ldots). \tag{21.7}$$

This equation is not a mnemonic: it is the helicity character χ_h of the little group, written in the coordinates (21.2).

Null factorization and mostly-plus spinor invariant. The matrix $p_{a\dot a}$ has determinant $-p^2$. Hence $p^2 = 0$ is equivalent, over the complex numbers, to the existence of a rank-one factorization $p_{a\dot a} = \lambda_a\widetilde{\lambda}_{\dot a}$. For two null momenta $p_i = \lambda_i\widetilde{\lambda}_i$ and $p_j = \lambda_j\widetilde{\lambda}_j$,

$$\langle ij\rangle[ji] = 2p_i \cdot p_j$$

in the mostly-plus convention of this monograph.

The determinant of $p^0\mathbf{1} + p^i\sigma_i$ is $(p^0)^2 - \vec{p}^{\,2} = -p^2$. A 2×2 complex matrix has determinant zero iff it has rank at most one; the zero matrix corresponds to $p = 0$, and a nonzero rank-one matrix factors as a column spinor times a row spinor. For two rank-one matrices, the antisymmetric contraction identity for 2×2 matrices gives

$$\det(p_i + p_j) = \epsilon^{ab}\lambda_{i,a}\lambda_{j,b}\,\epsilon^{\dot a\dot b}\widetilde{\lambda}_{j,\dot a}\widetilde{\lambda}_{i,\dot b} = -\langle ij\rangle[ji].$$

Since $p_i^2 = p_j^2 = 0$, the determinant side is $-(p_i + p_j)^2 = -2p_i \cdot p_j$, yielding $\langle ij\rangle[ji] = 2p_i \cdot p_j$.

The vector polarization representatives can also be written spinorially. Let $r_a, \widetilde{r}_{\dot a}$ be reference spinors with $\langle r\,\lambda\rangle \neq 0$ and $[\widetilde{\lambda}\,\widetilde{r}] \neq 0$. Define

$$\varepsilon^+_{a\dot a}(k; r) = \sqrt{2}\,\frac{r_a\widetilde{\lambda}_{\dot a}}{\langle r\,\lambda\rangle}, \qquad \varepsilon^-_{a\dot a}(k; r) = \sqrt{2}\,\frac{\lambda_a\widetilde{r}_{\dot a}}{[\widetilde{\lambda}\,\widetilde{r}]}. \tag{21.8}$$

They obey $k^{a\dot a}\varepsilon^{\pm}_{a\dot a} = 0$. Changing the reference spinors changes $\varepsilon^{\pm}$ by a term proportional to $k_{a\dot a} = \lambda_a\widetilde{\lambda}_{\dot a}$, hence by exactly the representative shift discussed above. Gauge-invariant amplitudes are independent of this reference choice.

21.9 Color-Ordered Yang–Mills Tree Amplitudes

Now specialize to the perturbative tree amplitudes of four-dimensional Yang–Mills theory, with all external momenta outgoing and the overall momentum-conservation delta function removed. The complete color-dressed amplitude is a sum over trace structures and permutations. The partial amplitude

$$A_n(1^{h_1}, 2^{h_2}, \dots, n^{h_n})$$

denotes the coefficient of a fixed cyclic ordering after stripping the common coupling and color factor. The cyclic ordering is part of the datum; poles of A_n occur only in sums of adjacent momenta in that cyclic order.

At three points, real massless momenta with momentum conservation are collinear, but complexified on-shell kinematics has two branches. On the holomorphic branch all square brackets vanish and angle brackets need not; on the antiholomorphic branch all angle brackets vanish and square brackets need not. Locality, mass dimension, and the little-group weights (21.7) fix the nonzero color-ordered three-gluon amplitudes, in the normalization used here, to be

$$A_3(1^-, 2^-, 3^+) = \frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \qquad A_3(1^+, 2^+, 3^-) = \frac{[12]^3}{[23][31]}. \tag{21.9}$$

For example, the first formula transforms as

$$A_3(1^-, 2^-, 3^+) \longmapsto t_1^2 t_2^2 t_3^{-2}\, A_3(1^-, 2^-, 3^+),$$

which is exactly the helicity weight for $(-1, -1, +1)$. Its mass dimension is that of a tree-level three-gluon partial amplitude after the coupling has been stripped.

The maximally helicity-violating, or MHV, tree amplitudes are the amplitudes with exactly two negative-helicity gluons. With negative-helicity labels r and s, the Parke–Taylor formula is

$$A_n(1^+, \dots, r^-, \dots, s^-, \dots, n^+) = \frac{\langle rs\rangle^4}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle}. \tag{21.10}$$

The expression is cyclic, has the correct little-group weight on every leg, and has only adjacent collinear and multi-particle singularities. The recursion theorem below turns these consistency checks into a construction from factorization and the three-point amplitudes.

Theorem 21.7 (Adjacent-negative Parke–Taylor tree amplitude)**.** *In color-ordered tree-level Yang–Mills theory, assuming the three-point amplitudes* (21.9)*, tree factorization with the all-outgoing internal-line convention displayed below, and a BCFW shift for which the boundary term at infinity vanishes, the n-gluon amplitude with negative-helicity gluons at adjacent labels* 1 *and* 2 *is*

$$A_n(1^-, 2^-, 3^+, \dots, n^+) = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle}.$$

Proof. The formula has the correct helicity weight because legs 1 and 2 receive t_1^2 and t_2^2, while every positive-helicity leg receives t_i^{-2}, exactly as required by (21.7). The base case is the three-point amplitude displayed in (21.9), interpreted on the holomorphic complex branch.

For the induction step use the $(-, +)$ BCFW shift involving one negative-helicity leg and one positive-helicity leg:

$$\widehat{\widetilde{\lambda}}_1 = \widetilde{\lambda}_1 + z\widetilde{\lambda}_n, \qquad \widehat{\lambda}_n = \lambda_n - z\lambda_1.$$

The boundary term in the BCFW residue formula below vanishes by hypothesis. The helicity selection rules of the three-point amplitudes imply that all recursive channels vanish except the channel

$$A_{n-1}(\widehat{1}^-, 2^-, 3^+, \dots, (n-2)^+, -\widehat{P}^+)\frac{1}{P^2_{1\cdots n-2}}A_3(\widehat{P}^-, (n-1)^+, \widehat{n}^+).$$

At the pole,

$$0 = (p_{n-1} + \widehat{p}_n)^2 = \langle n-1, \widehat{n}\rangle[\widehat{n}, n-1],$$

and the anti-MHV three-point factor is supported on the branch $\langle n-1, \widehat{n}\rangle = 0$. Hence

$$z_* = \frac{\langle n-1, n\rangle}{\langle n-1, 1\rangle}, \qquad \widehat{\lambda}_n(z_*) = \frac{\langle n1\rangle}{\langle n-1, 1\rangle}\lambda_{n-1},$$

where the second identity is Schouten's identity applied to $(\lambda_{n-1}, \lambda_n, \lambda_1)$. Use the internal spinor frame for the left momentum

$$-\widehat{P} = \lambda_{n-1}\left(\widetilde{\lambda}_{n-1} + \frac{\langle n1\rangle}{\langle n-1, 1\rangle}\widetilde{\lambda}_n\right).$$

The induction hypothesis gives

$$A_{n-1} = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\cdots\langle n-2, n-1\rangle\langle n-1, 1\rangle}.$$

With the all-outgoing internal-line convention in the factorization residue, the three-point amplitude and the propagator combine to

$$\frac{1}{P^2_{1\cdots n-2}}A_3(\widehat{P}^-, (n-1)^+, \widehat{n}^+) = \frac{\langle n-1, 1\rangle}{\langle n-1, n\rangle\langle n1\rangle}.$$

Indeed $P^2_{1\cdots n-2} = \langle n-1, n\rangle[n, n-1]$, while the three-point formula evaluates to the displayed numerator factor times $[n, n-1]$; the relative sign is fixed by assigning the two internal momenta $\widehat{P}$ and $-\widehat{P}$ the opposite all-outgoing orientations in the factorization formula. Multiplying the last two displayed equations cancels $\langle n-1, 1\rangle$ and gives the claimed cyclic denominator with the same numerator $\langle 12\rangle^4$. $\square$

21.10 BCFW Recursion

Choose two external labels i and j. The BCFW deformation

$$\widehat{\widetilde{\lambda}}_i(z) = \widetilde{\lambda}_i + z\widetilde{\lambda}_j, \qquad \widehat{\lambda}_j(z) = \lambda_j - z\lambda_i, \tag{21.11}$$

leaves all other spinors fixed. It preserves $\widehat{p}_i(z)^2 = \widehat{p}_j(z)^2 = 0$ and total momentum:

$$\widehat{p}_i(z) + \widehat{p}_j(z) = p_i + p_j.$$

Let $q_{a\dot{a}} = \lambda_{i,a}\widetilde{\lambda}_{j,\dot{a}}$. Then $\widehat{p}_i = p_i + zq$, $\widehat{p}_j = p_j - zq$, and $q^2 = q\cdot p_i = q\cdot p_j = 0$.

The shifted momenta are

$$\widehat{p}_i(z) = \lambda_i(\widetilde{\lambda}_i + z\widetilde{\lambda}_j) = p_i + zq, \qquad \widehat{p}_j(z) = (\lambda_j - z\lambda_i)\widetilde{\lambda}_j = p_j - zq.$$

Each remains rank one, hence null. Their sum is unchanged. The vector $q = \lambda_i \widetilde{\lambda}_j$ is rank one and therefore null; moreover $\langle ii \rangle = 0$ and $[jj] = 0$, so $2q \cdot p_i = 2q \cdot p_j = 0$, with the mostly-plus sign understood through the spinor-invariant calculation above.

At tree level a color-ordered amplitude $A_n(z)$ is a rational function of z. Its finite poles occur when a cyclically adjacent momentum sum

$$P_I = \sum_{\ell \in I} p_\ell$$

contains exactly one of the shifted labels and becomes null:

$$\widehat{P}_I(z_I)^2 = 0. \tag{21.12}$$

Near such a pole, tree-level unitarity and locality give the factorization

$$A_n(z) \sim \sum_h A_L(z_I) \, \frac{1}{\widehat{P}_I(z)^2} \, A_R(z_I), \tag{21.13}$$

where the two lower-point amplitudes have an internal on-shell gluon of helicity h and $-h$, respectively. Applying Cauchy's theorem to $A_n(z)/z$ gives

$$A_n(0) = \sum_{I,h} A_L(z_I) \, \frac{1}{P_I^2} \, A_R(z_I) + B_\infty. \tag{21.14}$$

The boundary term B_∞ is the residue at $z = \infty$. For Yang–Mills tree amplitudes it vanishes for the standard good shifts used in the MHV construction, including the $(-,+)$ shift below. When a shift has nonzero boundary term, equation (21.14) remains true but is not a closed recursion until B_∞ is determined by other input.

BCFW residue formula with boundary term. Let $A_n(z)$ be a tree-level color-ordered Yang–Mills partial amplitude under a BCFW shift. Suppose its only finite singularities are simple factorization poles where an adjacent shifted momentum sum becomes null, and suppose the residues are the tree factorization products

$$\operatorname*{Res}_{z=z_I} A_n(z) = \sum_h A_L(z_I) \, \operatorname*{Res}_{z=z_I} \left(\frac{1}{\widehat{P}_I(z)^2} \right) A_R(z_I).$$

Then

$$A_n(0) = \sum_{I,h} A_L(z_I) \frac{1}{P_I^2} A_R(z_I) + B_\infty,$$

where B_∞ is minus the residue of $A_n(z)/z$ at infinity. If $A_n(z) \to 0$ as $z \to \infty$, then $B_\infty = 0$. The meromorphic one-form $A_n(z)\, dz/z$ on the Riemann sphere has total residue zero. Its residue at $z = 0$ is $A_n(0)$. A finite pole occurs when $\widehat{P}_I(z)^2 = P_I^2 + z\, 2P_I \cdot q$ vanishes, so

$$z_I = -\frac{P_I^2}{2P_I \cdot q}.$$

Near this point,

$$\frac{1}{\widehat{P}_I(z)^2} = \frac{1}{(z - z_I)\, 2P_I \cdot q} = \frac{z_I}{(z - z_I)(-P_I^2)}.$$

Therefore the contribution of this finite pole to the residue of $A_n(z)/z$ is $-A_L(z_I)A_R(z_I)/P_I^2$, summed over internal helicities. Moving all finite-pole residues and the residue at infinity to the other side gives the displayed recursion formula.

Figure 21.1 records the finite-pole factorization geometry used in the recursion formula: the shifted amplitude splits into left and right on-shell subamplitudes joined by the channel $\widehat{P}_I^2 = 0$.

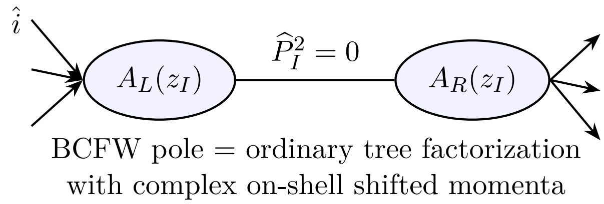


Figure 21.1: A pole in the complex BCFW parameter occurs when an adjacent momentum sum becomes null. The residue is the product of two lower-point on-shell amplitudes and the intermediate propagator.

21.11 Parke–Taylor from Recursion

Consider

$$A_n(1^-, 2^-, 3^+, \ldots, n^+).$$

Use the shift

$$\widehat{\widetilde{\lambda}}_1 = \widetilde{\lambda}_1 + z\widetilde{\lambda}_n, \qquad \widehat{\lambda}_n = \lambda_n - z\lambda_1. \tag{21.15}$$

This is a good $(-,+)$ Yang–Mills shift, so $B_\infty = 0$. The BCFW terms in which one side has fewer than two negative helicities vanish, except for the three-point anti-MHV factor allowed by complex kinematics. Inductively one obtains a single nonzero channel, consisting of an $(n-1)$-point MHV amplitude and a three-point amplitude. Substitution of (21.9) and the induction hypothesis gives

$$A_n(1^-, 2^-, 3^+, \ldots, n^+) = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle}.$$

This proves (21.10) for this placement of the two negative helicities. The nonadjacent (r, s) formula follows from the same BCFW recursion together with Schouten identities, but more than one channel can contribute; the adjacent proof above is the single-channel case needed for the worked example below.

For the first genuinely five-leg case,

$$A_5(1^-, 2^-, 3^+, 4^+, 5^+),$$

the shift (21.15) gives one nonzero recursive contribution:

$$A_5 = A_4(\widehat{1}^-, 2^-, 3^+, -\widehat{P}^+)\,\frac{1}{P_{123}^2}\,A_3(\widehat{P}^-, 4^+, \widehat{5}^+), \tag{21.16}$$

where $P_{123} = p_1 + p_2 + p_3$ and $\widehat{P} = \widehat{p}_1(z_*) + p_2 + p_3$ is null. The pole is determined by

$$\widehat{P}(z_*)^2 = 0.$$

Using the four-point Parke–Taylor amplitude on the left and the three-point amplitude (21.9) on the right, all dependence on the internal spinor representation of $\widehat{P}$ cancels. The result is

$$A_5(1^-, 2^-, 3^+, 4^+, 5^+) = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 45\rangle\langle 51\rangle}. \tag{21.17}$$

The calculation illustrates the conceptual point of BCFW recursion: tree amplitudes are determined by their on-shell factorization residues, helicity weights, and large-complex-momentum behavior, without introducing off-shell gauge-dependent subamplitudes.

The next chapter supplies the dynamics whose one-particle sector realizes these helicity states: Maxwell theory, its constraints, and the controlled introduction of gauge fixing.

Chapter 22

Haag–Ruelle Theory and Mathematical Scattering

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The source trace for the massive scattering construction is Haag's asymptotic condition and Ruelle's proof of scattering states [4, 5]. The LSZ interface later in the chapter is traced to [6]; the text keeps the Hilbert-space Haag–Ruelle construction logically prior to LSZ residues.

22.1 Vacuum Net and Isolated Mass Shell

Let $M = \mathbb{M}^D$ be Minkowski spacetime, and let

$$\mathcal{O} \longmapsto \mathcal{R}(\mathcal{O}) \subset \mathcal{B}(\mathcal{H})$$

be a Poincare-covariant vacuum net of von Neumann algebras. The standing data are a Hilbert space $\mathcal{H}$, a unit vector Ω, and a strongly continuous unitary representation U of the proper orthochronous Poincare group such that

$$U(a, \Lambda)\mathcal{R}(\mathcal{O})U(a, \Lambda)^{-1} = \mathcal{R}(\Lambda\mathcal{O} + a), \qquad U(a, \Lambda)\Omega = \Omega.$$

For translations alone we write

$$U(a) = U(a, \mathbf{1}) = \exp(\mathrm{i}a_\mu P^\mu),$$

where P^μ are the commuting self-adjoint energy-momentum generators. The scalar product in Fourier transforms is the same contraction $q\cdot x = q_\mu x^\mu$. This sign convention is part of the standing data: with the opposite convention for $U(a)$, every spectral-transfer formula below would contain $-\operatorname{supp}\widehat{f}$ instead of $\operatorname{supp}\widehat{f}$. The translation generators obey the spectrum condition

$$\mathrm{Spec}(P) \subset \overline{V}_+.$$

Let $E(\Delta)$ denote the joint spectral projection of P^μ for a Borel set $\Delta \subset \mathbb{R}^D$.

Fix $m > 0$. The positive mass shell is

$$\Sigma_m^+ = \{p \in \mathbb{R}^D : p^0 = \omega_{\vec{p}},\ p^2 = -m^2\}, \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}.$$

The one-particle subspace for this stable species is

$$\mathcal{H}_1 = E(\Sigma_m^+)\mathcal{H}.$$

This notation carries an existence hypothesis: $\mathcal{H}_1 \neq \{0\}$. The projection onto the Borel set Σ_m^+ may be zero in a perfectly consistent Hilbert-space theory, or it may fail to be isolated from continuous spectral support. In either case the ordinary massive Haag–Ruelle theorem does not produce scattering states for that putative species. The relation between a one-particle shell and an atom in a Källén–Lehmann measure was spelled out in Remark 14.17. The Haag–Ruelle construction assumes that the shell is isolated in the energy-momentum spectrum: there is an open neighborhood N of Σ_m^+ such that

$$\mathrm{Spec}(P) \cap N = \Sigma_m^+.$$

This isolation is the spectral form of stability. The formulas below are written for a single bosonic species. If several stable species, spin multiplets, or fermionic particles are present, $\mathcal{H}_1$ is replaced by the appropriate direct sum of isolated one-particle representations and $\mathcal{F}_s(\mathcal{H}_1)$ by the corresponding graded asymptotic Fock space.

For a scalar multiplicity-one species, the one-particle Hilbert space is realized by the mass-shell spectral representation

$$W_1 : L^2(\Sigma_m^+, \mathrm{d}\mu_m) \xrightarrow{\sim} \mathcal{H}_1, \qquad \mathrm{d}\mu_m(p) = \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1} 2\omega_{\vec{p}}},$$

with

$$W_1 f = \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, f(p) |p\rangle, \qquad U(a, \mathbf{1}) W_1 f = W_1\big(p \mapsto \mathrm{e}^{\mathrm{i} a_\mu p^\mu} f(p)\big).$$

The bosonic asymptotic Fock space is then the Hilbert direct sum

$$\mathcal{F}_s(\mathcal{H}_1) \simeq \mathbb{C}\Omega_F \oplus \bigoplus_{n\geq 1} L^2_{\mathrm{sym}}\big((\Sigma_m^+)^n, \mathrm{d}\mu_m^{\otimes n}\big).$$

Thus the wave operators below start from a constructed free many-particle Hilbert space whose momentum labels are coordinates in the spectral model.

22.2 Almost-Local Operators and Spectral Transfer

Almost-locality is the norm-approximation condition of Definition 17.3: an operator $B \in \mathcal{B}(\mathcal{H})$ is approximated by bounded local operators B_R localized in double cones of radius R such that, for every $N \geq 1$, some $C_N < \infty$ obeys

$$\|B - B_R\| \leq C_N R^{-N}.$$

Almost locality is the operator-algebraic substitute for using smeared local fields whose support is not strictly compact after energy-momentum filtering.

For a bounded local operator A, set

$$A(x) = U(x) A U(x)^{-1}, \qquad x \in \mathbb{R}^D.$$

For $f \in \mathcal{S}(\mathbb{R}^D)$, define

$$A(f) = \int \mathrm{d}^D x\, f(x) A(x), \qquad \widehat{f}(q) = \int \mathrm{d}^D x\, f(x) \mathrm{e}^{\mathrm{i} q \cdot x}.$$

With this convention, the energy-momentum transfer of $A(f)$ is contained in $\operatorname{supp} \widehat{f}$:

$$E(\Delta_1) A(f) E(\Delta_2) = 0 \quad \text{if} \quad \Delta_1 \cap \left(\Delta_2 + \operatorname{supp} \widehat{f}\right) = \emptyset. \tag{22.1}$$

Indeed, by the joint spectral theorem,

$$E(\mathrm{d}p)\, A(f)\, E(\mathrm{d}r) = \left(\int \mathrm{d}^D x\, f(x) \mathrm{e}^{\mathrm{i} x_\mu (p^\mu - r^\mu)} \right) E(\mathrm{d}p)\, A\, E(\mathrm{d}r) = \widehat{f}(p - r)\, E(\mathrm{d}p)\, A\, E(\mathrm{d}r).$$

After integrating $p \in \Delta_1$ and $r \in \Delta_2$, the operator can be nonzero only if $p - r \in \operatorname{supp} \widehat{f}$, equivalently $\Delta_1 \cap (\Delta_2 + \operatorname{supp} \widehat{f}) \neq \emptyset$. Equation (22.1) is the precise meaning of selecting a particle shell by smearing in spacetime.

Choose a compact set $K \subset \Sigma_m^+$. Let $\chi \in \mathcal{S}(\mathbb{R}^D)$ have $\operatorname{supp} \widehat{\chi} \subset N$, with $\widehat{\chi} = 1$ on a smaller neighborhood of K. For a local A, set

$$B = A(\chi).$$

Then B is almost local. Since the vacuum has momentum 0,

$$B\Omega = E(\operatorname{supp} \widehat{\chi}) B\Omega = P_1 B\Omega, \qquad P_1 = E(\Sigma_m^+),$$

provided the support of $\widehat{\chi}$ meets the spectrum only on the isolated shell. If $-\operatorname{supp} \widehat{\chi}$ is disjoint from the forward spectrum and $\operatorname{supp} \widehat{\chi}$ avoids 0, then

$$B^* \Omega = 0.$$

An almost-local operator satisfying these spectral support properties and $B\Omega \neq 0$ is a regular Haag–Ruelle creator for the shell.

Figure 22.1 displays the spectral-transfer condition for such a creator: the Fourier support meets the spectrum only on the isolated positive mass shell and avoids the vacuum and negative-energy regions.

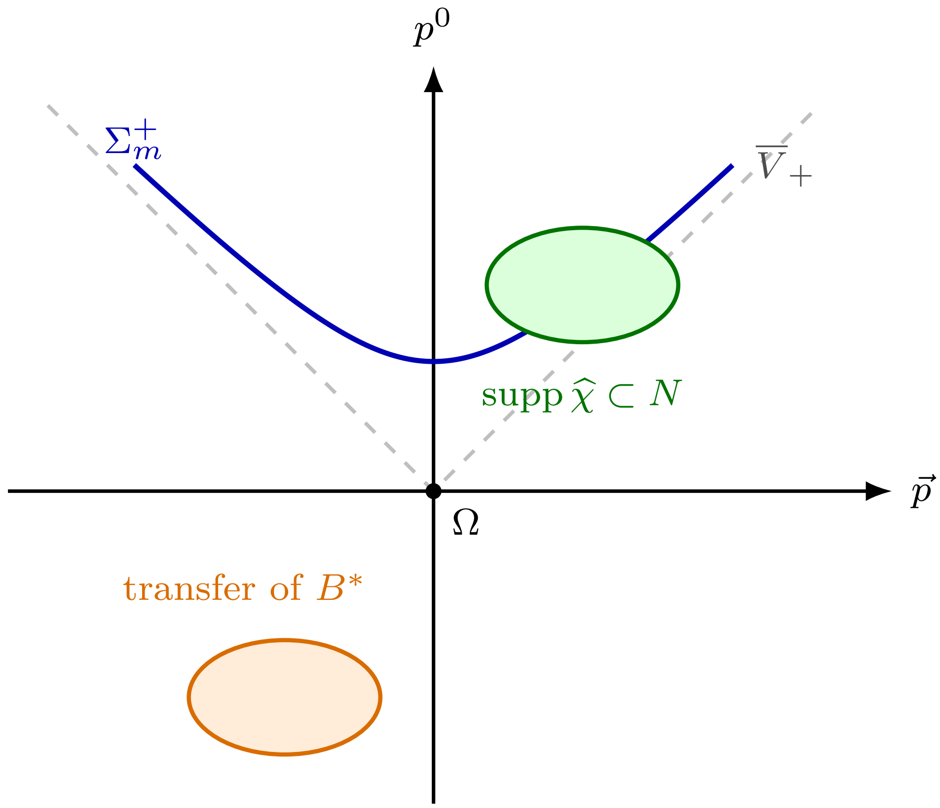


Figure 22.1: A spacetime smearing with χ restricts energy-momentum transfer to $\operatorname{supp}\widehat{\chi}$. The support is chosen inside the isolating neighborhood of the positive mass shell, while the reflected support controls the adjoint creator.

22.3 Wave Packets and Velocity Localization

Let B be a regular creator. For

$$h \in C_c^\infty(\mathbb{R}^{D-1}),$$

define the positive-energy Klein–Gordon packet

$$h_t(\vec{x}) = \int \frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}}\, h(\vec{p})\, \mathrm{e}^{\mathrm{i}\omega_{\vec{p}}t - \mathrm{i}\vec{p}\cdot\vec{x}}$$

and the time-dependent operator

$$B_t(h) = \int \mathrm{d}^{D-1}\vec{x}\, h_t(\vec{x}) B(t, \vec{x}).$$

The one-particle vector

$$\psi_h = P_1 B_t(h)\Omega$$

is independent of t; the phase in h_t cancels the free time evolution on the mass shell.

The velocity support of h is

$$V(h) = \left\{ \frac{\vec{p}}{\omega_{\vec{p}}} : \vec{p} \in \operatorname{supp} h \right\}.$$

Stationary phase says that $h_t(\vec{x})$ is rapidly decreasing away from the equal-time tube

$$\vec{x} \in t\,V(h) + O(|t|^0).$$

Thus packets with disjoint velocity supports become spacelike separated at large $|t|$. Combining this geometric separation with almost locality gives, for every $N \geq 1$,

$$\|[B_{i,t}(h_i), B_{j,t}(h_j)]\| \leq C_N(1+|t|)^{-N} \tag{22.2}$$

whenever $V(h_i) \cap V(h_j) = \emptyset$.

22.4 Haag–Ruelle Limits

Let $B_1, \ldots, B_n$ be regular creators and let

$$h_1, \ldots, h_n \in C_c^\infty(\mathbb{R}^{D-1})$$

have pairwise disjoint velocity supports. Define

$$\Psi_t = B_{1,t}(h_1) \cdots B_{n,t}(h_n)\Omega.$$

Theorem 22.1 (Haag–Ruelle scattering for a vacuum net)**.** *Assume locality, Poincare covariance, the spectrum condition, a unique vacuum, an isolated massive one-particle shell, and enough regular creators to generate a dense subspace of $\mathcal{H}_1$. Then the limits*

$$\Psi_{\text{out}} = \lim_{t\to+\infty} \Psi_t, \qquad \Psi_{\text{in}} = \lim_{t\to-\infty} \Psi_t$$

exist in Hilbert norm. They depend only on the one-particle vectors $\psi_{h_j} = P_1 B_{j,t}(h_j)\Omega$, and they define isometric wave operators from the symmetric Fock space $\mathcal{F}_s(\mathcal{H}_1)$ into $\mathcal{H}$.

Remark 22.2 (Physical scope of the isolated-shell hypothesis)*.* The nonzero isolated-shell assumption is a spectral theorem about the particular nonperturbative model, not a generic consequence of writing a formal Lagrangian. It is satisfied by free massive fields. It is also available in the massive stable sectors of constructive low-dimensional scalar models, such as $P(\phi)_2$ and massive $\lambda\phi_3^4$, in regimes where the Wightman theory, mass gap, and one-particle spectral gap have been established. For a four-dimensional $\lambda\phi^4$ Lagrangian treated only perturbatively, an isolated real pole in the formal two-point function is evidence for the intended particle interpretation, but it is not a nonperturbative theorem constructing an isolated shell in a Hilbert-space spectrum.

A Källén–Lehmann measure with no atom at m^2, even if its continuous support starts at m^2, supplies no normalizable one-particle vector of mass m for the corresponding local interpolating field. Haag–Ruelle then has no external particle of that species to scatter. The theorem exactly distinguishes a sharp stable particle from threshold spectral weight.

The hypothesis fails, or is the wrong hypothesis, in several central physical settings. In QED with long-range electromagnetic fields, charged sectors are expected to be infraparticle sectors: the energy-momentum spectrum has continuous soft-photon dressing attached to the charged state rather than a sharp Wigner mass hyperboloid. The photon is massless and belongs to a theory without the massive spectral gap used by the theorem. In QCD, colored quark and gluon shells are not expected to occur in the gauge-invariant observable Hilbert space; stable hadrons, when present below strong-decay thresholds, are the candidate isolated shells. Unstable particles and resonances do not give external Haag–Ruelle particles, because their spectral singularities are resonance poles on analytically continued sheets rather than point spectrum on the physical mass shell. Each of these cases requires a different asymptotic framework or a separate spectral theorem. The corresponding asymptotic-completeness problems for charged QED and physical QCD are stated as Open Problems 44.10 and 49.22.

22.4.1 The Cook Estimate and the Fock Inner Product

We now spell out the proof in a form that separates the analytic estimates from the algebraic conclusion. The estimates below are precisely the Haag–Ruelle estimates supplied by almost locality, spectral isolation, and stationary phase. Once they are known, the construction of scattering states and the Fock inner product is an elementary Hilbert-space argument.

Definition 22.3 (Haag–Ruelle estimate package)**.** For a finite family of regular creators B_j and smooth compactly supported packets h_j, write

$$B_{j,t} = B_{j,t}(h_j), \qquad \psi_j = B_{j,t}\Omega \in \mathcal{H}_1.$$

The family is said to satisfy the Haag–Ruelle estimate package on $|t| \geq T_0$ if the following hold.

1. The vectors $B_{j,t}\Omega = \psi_j$ are independent of t, and $B_{j,t}^{*}\Omega = 0$.
2. For every $N \geq 1$ and every pair with disjoint velocity supports,
$$\|[B_{i,t}^{\#}, B_{j,t}^{\#}]\| \leq C_N(1+|t|)^{-N}, \qquad B^{\#} \in \{B, B^*\}. \tag{22.3}$$
The same estimate holds on the finite-energy vectors obtained from the family when one entry is replaced by its strong time derivative. This derivative is the derivative of the smooth translation-smearing parameter; no norm differentiability of an unsmeared local operator is assumed.
3. For every pair of regular creators B, C with packets h, g,
$$B_t(h)^* C_t(g)\Omega = \langle B_t(h)\Omega, C_t(g)\Omega\rangle\, \Omega + r_t(B, h; C, g), \tag{22.4}$$
where for every $N \geq 1$, $\|r_t(B, h; C, g)\| \leq C_N(1+|t|)^{-N}$.
4. For two velocity-separated lists $B_1, \ldots, B_n$ and $C_1, \ldots, C_m$, the scalar products obey the recursive contraction estimate
$$\begin{aligned}&\langle B_{1,t}\cdots B_{n,t}\Omega, C_{1,t}\cdots C_{m,t}\Omega\rangle \\ &\quad = \sum_{\ell=1}^{m}\langle \psi_1, \varphi_\ell\rangle \left\langle B_{2,t}\cdots B_{n,t}\Omega, C_{1,t}\cdots \widehat{C_{\ell,t}}\cdots C_{m,t}\Omega\right\rangle + O_N((1+|t|)^{-N})\end{aligned} \tag{22.5}$$
for every $N \geq 1$, with the convention that the sum is empty when $m = 0$.
5. For each velocity-separated product $\Psi_t = B_{1,t}\cdots B_{n,t}\Omega$, there is an integrable nonnegative function $R(t)$ on $[T_0, \infty)$ and on $(-\infty, -T_0]$ such that
$$\left\|\frac{\mathrm{d}}{\mathrm{d}t}\Psi_t\right\| \leq R(t) \tag{22.6}$$
on the finite vector domain under discussion.

Proposition 22.4 (Estimate package implies Haag–Ruelle limits)**.** *Assume the estimate package Definition 22.3 for every finite velocity-separated family of regular creators. Then the limits in Theorem 22.1 exist, depend only on the one-particle vectors, and satisfy the bosonic Fock inner product formula*

$$\begin{aligned}&\big\langle \Psi^{\mathrm{out}}(\psi_1,\dots,\psi_n),\Phi^{\mathrm{out}}(\varphi_1,\dots,\varphi_m)\big\rangle\\ &\qquad=\delta_{nm}\sum_{\sigma\in S_n}\prod_{j=1}^{n}\langle\psi_j,\varphi_{\sigma(j)}\rangle_{\mathcal{H}_1}.\end{aligned}\tag{22.7}$$

The same statement holds for incoming limits. Hence the wave operators are isometries from $\mathcal{F}_s(\mathcal{H}_1)$ into $\mathcal{H}$.

Proof. We give the outgoing proof. The incoming proof is the same with $t\to-\infty$.

First, Cook's criterion gives existence. For $t'>t\geq T_0$,

$$\|\Psi_{t'}-\Psi_t\|\leq\int_t^{t'}\left\|\frac{\mathrm{d}}{\mathrm{d}s}\Psi_s\right\|\mathrm{d}s\leq\int_t^{t'}R(s)\,\mathrm{d}s.$$

Since $R\in L^1([T_0,\infty))$, the right side tends to zero as $t,t'\to+\infty$. Thus Ψ_t is Cauchy in the Hilbert norm and has an outgoing limit.

Second, compute the limiting scalar products. Let

$$\Psi_t=B_{1,t}\cdots B_{n,t}\Omega,\qquad\Phi_t=C_{1,t}\cdots C_{m,t}\Omega,$$

with both lists velocity-separated. Then

$$\langle\Psi_t,\Phi_t\rangle=\langle\Omega,B_{n,t}^*\cdots B_{1,t}^*C_{1,t}\cdots C_{m,t}\Omega\rangle.$$

The recursive contraction estimate (22.5) gives

$$\langle\Psi_t,\Phi_t\rangle=\sum_{\ell=1}^{m}\langle\psi_1,\varphi_\ell\rangle\,\langle B_{2,t}\cdots B_{n,t}\Omega,C_{1,t}\cdots\widehat{C_{\ell,t}}\cdots C_{m,t}\Omega\rangle+O_N(t^{-N}).$$

Induction on $n+m$, with the base cases following from $\langle\Omega,\Omega\rangle=1$ and $B_t^*\Omega=0$, gives (22.7). If $n\neq m$, the induction eventually leaves an annihilator acting on the vacuum, and the limit is zero.

The scalar product formula shows at once that changing an interpolating creator while keeping the one-particle vector fixed changes the limiting state by a vector of zero norm. It also identifies the norm of every finite linear combination of outgoing limits with the norm of the corresponding finite symmetric tensor in $\mathcal{F}_s(\mathcal{H}_1)$. Therefore the map from the algebraic velocity-separated Fock domain to $\mathcal{H}$ is isometric and extends uniquely to the Fock completion. Density of regular creators in $\mathcal{H}_1$ gives the stated domain of the wave operator. □

Lemma 22.5 (Stationary-phase velocity localization)**.** *Let $h\in C_c^\infty(\mathbb{R}^{D-1})$ and let*

$$h_t(\vec{x})=\int\frac{\mathrm{d}^{D-1}\vec{p}}{(2\pi)^{D-1}}\,h(\vec{p})\,\mathrm{e}^{\mathrm{i}\omega_{\vec{p}}t-\mathrm{i}\vec{p}\cdot\vec{x}}.$$

Let $V(h)=\{\vec{p}/\omega_{\vec{p}}:\vec{p}\in\operatorname{supp}h\}$. If $U\subset\mathbb{R}^{D-1}$ is an open neighborhood of $V(h)$, then for every $N\geq1$ there is C_N such that

$$|h_t(\vec{x})|\leq C_N(1+|t|+|\vec{x}|)^{-N}\qquad\textit{whenever }\vec{x}/t\notin U,\quad|t|\geq1.$$

Proof. On the compact support of h, the phase is

$$\Phi_t(\vec{p};\vec{x}) = \omega_{\vec{p}} t - \vec{p}\cdot\vec{x}.$$

Its gradient is

$$\nabla_{\vec{p}}\Phi_t = t\,\frac{\vec{p}}{\omega_{\vec{p}}} - \vec{x}.$$

If $\vec{x}/t \notin U$, compactness gives $|\nabla_{\vec{p}}\Phi_t| \ge c|t|$ on $\operatorname{supp} h$ for some $c > 0$. Apply repeatedly the integration-by-parts operator

$$L = \frac{1}{\mathrm{i}|\nabla_{\vec{p}}\Phi_t|^2}\nabla_{\vec{p}}\Phi_t\cdot\nabla_{\vec{p}}, \qquad L\mathrm{e}^{\mathrm{i}\Phi_t} = \mathrm{e}^{\mathrm{i}\Phi_t}.$$

Each integration by parts gains a factor $O(|t|^{-1})$, while derivatives of h and of the coefficients of L are bounded on the compact support. This gives $O(|t|^{-N})$. In the region $|\vec{x}| \gg |t|$, the same lower bound is $c(|t| + |\vec{x}|)$, giving the displayed estimate. □

Proposition 22.6 (Almost locality gives the commutator estimate)**.** *Let B_i and B_j be almost-local regular creators and let h_i, h_j have disjoint velocity supports. Then*

$$\|[B_{i,t}(h_i), B_{j,t}(h_j)]\| \le C_N(1+|t|)^{-N} \qquad (N \ge 1),$$

and the same estimate holds with either operator replaced by its adjoint.

Proof. Choose nested open neighborhoods

$$V(h_\ell) \subset U_\ell^0 \Subset U_\ell, \qquad \ell = i, j,$$

so that U_i and U_j have positive Euclidean distance

$$\delta = \operatorname{dist}(U_i, U_j) > 0.$$

Let χ_i^t, χ_j^t be smooth cutoffs equal to one on the smaller equal-time tubes $\vec{x}/t \in U_i^0$, $\vec{y}/t \in U_j^0$, and supported in the larger disjoint tubes $\vec{x}/t \in U_i$, $\vec{y}/t \in U_j$. Write

$$B_{i,t} = B_{i,t}^{\text{near}} + B_{i,t}^{\text{tail}}, \qquad B_{i,t}^{\text{near}} = \int \mathrm{d}^{D-1}\vec{x}\,\chi_i^t(\vec{x}) h_{i,t}(\vec{x}) B_i(t,\vec{x}),$$

and similarly for j. The stationary-phase lemma, used with an exponent larger than $N + D$, gives an L^1-tail estimate

$$\int_{\vec{x}/t\notin U_i^0} |h_{i,t}(\vec{x})|\,\mathrm{d}^{D-1}\vec{x} \le C_N(1+|t|)^{-N},$$

and the same for h_j. Since translations preserve operator norms,

$$\|B_{i,t}^{\text{tail}}\| \le \|B_i\| \int_{\vec{x}/t\notin U_i} |h_{i,t}(\vec{x})|\,\mathrm{d}^{D-1}\vec{x} = O_N(|t|^{-N}).$$

Every commutator containing at least one tail factor is therefore $O_N(|t|^{-N})$.

It remains to estimate the commutator of the near parts. For $\vec{x}/t \in U_i$ and $\vec{y}/t \in U_j$ the equal-time centers satisfy

$$|\vec{x} - \vec{y}| \ge \delta|t|.$$

Choose almost-local approximants $B_{\ell,R}$ localized in the double cone of radius R about the origin and obeying, for arbitrary K,

$$\|B_\ell - B_{\ell,R}\| \le C_K R^{-K}, \qquad \ell = i, j.$$

Set $R = \delta|t|/8$. A point in the translated localization region of $(B_{i,R})(t,\vec{x})$ has time coordinate within R of t and spatial coordinate within R of $\vec{x}$; likewise for j. Hence the absolute time separation of two points in the two translated double cones is at most $2R$, while their spatial separation is at least $\delta|t| - 2R$. With the chosen R,

$$\delta|t| - 2R = \frac{3}{4}\delta|t| > 2R,$$

so the two double cones are spacelike separated. Locality gives

$$[(B_{i,R})(t,\vec{x}), (B_{j,R})(t,\vec{y})] = 0$$

for all $\vec{x}, \vec{y}$ in the near supports, and hence the integrated commutator of the two approximated near parts vanishes.

The error made by replacing B_ℓ by $B_{\ell,R}$ is controlled in operator norm. The near-packet L^1-norm grows at most polynomially in $|t|$: inside a ball of radius $O(|t|)$ this is the trivial volume bound, while outside such a ball the same nonstationary-phase integration by parts gives rapid decay. Thus, for some fixed power M,

$$\|B_{\ell,t}^{\mathrm{near}} - (B_{\ell,R})_t^{\mathrm{near}}\| \le C(1+|t|)^M R^{-K}.$$

Since K is arbitrary in the almost-local approximation, choosing K larger than $M+N$ makes the replacement error $O_N(|t|^{-N})$. Combining the near-approximant commutator, the replacement errors, and the tail estimates proves the displayed commutator bound. The adjoint case is identical: B_R^* is localized in the same double cone as B_R, and $\|B^* - B_R^*\| = \|B - B_R\|$, so almost locality is preserved by taking adjoints. □

Proposition 22.7 (Origin of the remaining Haag–Ruelle estimates)**.** *For regular creators obtained by smooth spectral transfer inside the isolating neighborhood N, the one-particle contraction estimate* (22.4)*, the recursive contraction estimate* (22.5)*, and the Cook derivative estimate* (22.6) *follow from spectral isolation of Σ_m^+, the transfer relation* (22.1)*, and the commutator estimate of Proposition 22.6.*

Proof. The support of the spectral transfer of a regular creator meets the spectrum, when acting on the vacuum, only on Σ_m^+. Therefore $B_t(h)\Omega$ is exactly the one-particle vector determined by the mass-shell packet and is independent of t, while $B_t(h)^*\Omega = 0$ because the reflected transfer support is disjoint from $\overline{V}_+$. Acting with $B_t(h)^*C_t(g)$ on the vacuum, spectral transfer permits a vacuum component and separates all nonvacuum components from the isolated shell by a positive spectral distance fixed by the supports. Choose a smooth cutoff $\rho(P)$ that is 0 on a neighbourhood of the vacuum point and 1 on the separated nonvacuum spectral support. The oscillatory factors left after subtracting the vacuum component then have no stationary point on the relevant compact spectral support. Repeated integration by parts in the spectral variables, written equivalently as commuting powers of smooth functions of P through the translated smearing, gives the rapidly decreasing remainder in (22.4). The estimate is a strong finite-energy estimate; it is not a statement about an operator norm derivative of a pointlike field. Iterating this contraction after decomposing packet supports into finitely many small velocity cells gives (22.5); terms in which two uncontracted cells have separated velocity supports are estimated by Proposition 22.6, and the finite partition is removed by summing the finitely many cells.

For the Cook estimate, differentiate the product on the finite vector domain spanned by the vectors obtained by applying finitely many translated, spectrally smeared creators to the vacuum. The derivative of the k-th factor, after commuting it to the far right, annihilates the vacuum because the one-particle packet has been chosen to solve the free mass-shell evolution. Every commutation needed to move this derivative through later factors is controlled by the derivative version of Proposition 22.6; the derivative version is obtained by applying Lemma 22.5 to the differentiated packet. The differentiated packet only multiplies the compact momentum amplitude by a smooth polynomial expression in p, so it has the same velocity localization and rapid tube-tail estimates. The resulting bound is $C_N(1+|t|)^{-N}$ for arbitrary N, hence integrable. This is (22.6). □

Together, Propositions 22.4, 22.6, and 22.7 give the ordinary massive Haag–Ruelle theorem under the displayed vacuum-net, isolated-shell, and regular-creator hypotheses. No perturbative pole expansion or Lagrangian formalism enters this construction.

Proof of Theorem 22.1. Choose regular creators B_j whose one-particle projections are the desired wave packets and whose velocity supports are pairwise disjoint after the usual finite partition of momentum space. Spectral isolation of the mass shell gives $B_{j,t}\Omega = P_1 B_{j,t}\Omega$ independent of t, and excludes $B_{j,t}^*\Omega$ by the support condition against the forward spectrum. Lemma 22.5 and almost locality give the rapid commutator bound through Proposition 22.6. The remaining contraction and Cook bounds are supplied by Proposition 22.7; hence the family satisfies Definition 22.3. Applying Proposition 22.4 gives the outgoing and incoming Hilbert-norm limits and the Fock scalar product. The scalar product identifies vectors obtained from different interpolating creators with the same one-particle projection and extends the wave maps by completion from the algebraic symmetric Fock domain. The assumed density of regular creators in $\mathcal{H}_1$ gives the stated wave operators on $\mathcal{F}_s(\mathcal{H}_1)$. □

22.5 Gauge Charges and Nonlocal Charged Creators

The ordinary theorem uses local creators $B \in \mathcal{A}(\mathcal{O})$ whose vacuum vector $B\Omega$ has nonzero projection to an isolated one-particle shell. In a gauge theory with an unscreened Gauss-law charge, the existence of such a creator is a further dynamical and localization hypothesis. It is false for charged particles if the available local algebra is the gauge-invariant observable algebra.

Theorem 22.8 (Gauss-law obstruction to local charged creators)**.** *Let $\mathcal{A}$ be a gauge-invariant local observable net in a vacuum representation. Assume that an abelian charge Q is represented on a common invariant domain by the large-sphere Gauss-law limit*

$$Q = \lim_{R\to\infty} Q_R, \qquad Q_R = \int_{S_R} n_i E^i \, \mathrm{d}S,$$

and that locality holds between $\mathcal{A}(\mathcal{O})$ and electric-field operators localized spacelike to $\mathcal{O}$. If the vacuum is neutral, $Q\Omega = 0$, then every local gauge-invariant observable $A \in \mathcal{A}(\mathcal{O})$ creates a *neutral vector:*

$$Q\, A\Omega = 0.$$

Consequently a charged one-particle vector cannot be obtained by applying a bounded-region gauge-invariant observable to the vacuum.

Proof. The surface charge must be read as a limit of smeared operators. Choose a smooth function $\rho_R(r)$ supported in a shell $R < r < 2R$, normalized so that it represents the outward flux through a large sphere, and define the smeared Gauss operator $Q_R(\rho)$ on the common invariant domain. For R larger than the spatial diameter of $\mathcal{O}$, the support of the electric-field smearing is spacelike to $\mathcal{O}$ at the chosen time slice. Locality of the observable net with the electric-field algebra gives

$$[Q_R(\rho), A] = 0$$

on that domain. The statement is insensitive to the particular smooth representative of the large sphere: changing ρ_R inside the exterior region changes $Q_R(\rho)$ by an operator localized spacelike to $\mathcal{O}$, plus terms that vanish in the assumed Gauss-law limit.

By hypothesis the strong graph limit of $Q_R(\rho)$ on vectors generated from the vacuum by local observables exists and is the charge Q. Hence for every vector Φ in the common domain,

$$\langle \Phi, (QA - AQ)\Omega\rangle = \lim_{R\to\infty} \langle \Phi, [Q_R(\rho), A]\Omega\rangle = 0.$$

Thus $QA\Omega = AQ\Omega$ as a vector identity on the charge domain. Since the vacuum is neutral,

$$QA\Omega = AQ\Omega = 0.$$

If ψ has charge $q \neq 0$, so that $Q\psi = q\psi$, it cannot equal $A\Omega$ for such an A. □

This obstruction concerns localization and Gauss-law charge. It is distinct from the infraparticle obstruction in massless QED. The latter states that a charged sector with long-range electromagnetic flux does not contain an ordinary sharp Wigner mass-shell vector. The present theorem applies before asking whether the charged spectral distribution is an isolated shell or a soft-photon continuum: a bounded gauge-invariant observable cannot change the surface charge at infinity.

Almost-local closure. Under the hypotheses of Theorem 22.8, assume in addition that the charge operator Q is closed on a domain containing the vectors generated from Ω by local gauge-invariant observables. Let $B \in \mathcal{B}(\mathcal{H})$ be almost local with respect to the observable net: there are $B_R \in \mathcal{A}(\mathcal{O}_R)$, with $\mathcal{O}_R$ a cofinal family of bounded regions, such that

$$\|B - B_R\| \longrightarrow 0 \qquad (R \to \infty).$$

Then $B\Omega$ is neutral:

$$B\Omega \in \mathrm{Dom}(Q), \qquad QB\Omega = 0.$$

Consequently a vector with nonzero Gauss charge cannot be produced from the vacuum by an almost-local operator belonging to the gauge-invariant observable net.

For each R, Theorem 22.8 applied to the local observable B_R gives

$$B_R\Omega \in \mathrm{Dom}(Q), \qquad QB_R\Omega = 0.$$

The norm convergence $B_R \to B$ implies $B_R\Omega \to B\Omega$, while $QB_R\Omega \to 0$. Closedness of Q therefore gives

$$B\Omega \in \mathrm{Dom}(Q), \qquad QB\Omega = 0.$$

If $B\Omega$ had a nonzero component in a charge-q spectral subspace, $q \neq 0$, the self-adjoint spectral calculus for Q would give $QB\Omega \neq 0$ on that component. Hence no such component is present.

This almost-local extension is the exact point at which the ordinary Haag–Ruelle hypothesis fails. The obstruction is structural: the noncompact tail is forced by the charge superselection law. If an operator changes the Gauss-law flux at infinity, it cannot be approximated in norm, even rapidly, by bounded gauge-invariant observables localized in bounded regions.

Charged fields can be introduced in a larger field algebra, but they are not gauge-invariant local observables. A gauge-invariant charged insertion must carry flux to infinity. In an abelian convention

$$A \longmapsto A + \mathrm{d}\alpha, \qquad \psi_q(x) \longmapsto \mathrm{e}^{\mathrm{i}q\alpha(x)}\psi_q(x),$$

a path γ from x to spatial infinity gives the formal dressing

$$\Psi_{q,\gamma}(x) = \psi_q(x) \exp\left\{\mathrm{i}q \int_\gamma A\right\}. \tag{22.8}$$

For gauge functions with $\alpha(\infty) = 0$, the exponential transforms by $\mathrm{e}^{-\mathrm{i}q\alpha(x)}$, so $\Psi_{q,\gamma}(x)$ is gauge invariant. For a gauge function with nonzero boundary value, it transforms by the boundary phase and therefore carries the global charge. The price is noncompact localization: the localization region includes the curve γ reaching infinity, or a Coulombic dressing with equivalent asymptotic flux.

Hypothesis 22.9 (Nonconfining charged sector)**.** A charged sector is nonconfining, for the charged construction below, when it has the following objects and properties.

1. A vacuum observable net $\mathcal{A}$ with a positive-energy representation and a physical charged representation $(\mathcal{H}_q, U_q)$ of the observable net. The representation carries a self-adjoint boundary charge Q and, for each smooth angular test f on the sphere at spatial infinity, a Gauss-law flux operator $\Phi_\infty(f)$ obtained as the large-sphere limit of the corresponding electric-flux smearing on a common invariant domain.

2. Compactly localized gauge-invariant observables preserve the boundary charge and the angular flux sector in the large-sphere limit:

   $$[A, Q] = 0, \qquad [A, \Phi_\infty(f)] = 0$$

   on the common domain whenever A is localized in a bounded spacetime region. Thus the charged representation is labelled by data at infinity rather than by an operator in the bounded observable net.

3. The charged sector has finite-energy asymptotic content. In a massive screened or Coulomb-branch situation this means an isolated positive-energy mass hyperboloid

   $$\Sigma^+_{m,q} = \{p \in \mathbb{R}^{1,d} \mid p^2 = -m^2,\ p^0 > 0\}$$

   inside the joint spectrum of U_q, with finite-dimensional multiplicity space $E_{p,q}$ and spectral projection $P_{1,q}$. In a massless long-range abelian theory the corresponding spectral replacement is a charged threshold or infraparticle measure together with its flux-sector label; ordinary isolated-shell Haag–Ruelle estimates are then replaced by the modified asymptotic problem stated below.

4. There is a class of gauge-invariant noncompact charged creators $\Psi_{q,\Gamma}$, where Γ denotes a Wilson-line ray, Coulomb profile, or equivalent dressing geometry reaching infinity, such that their smeared vectors have a nonzero projection to the charged one-particle subspace, namely

$$P_{1,q}\Psi_{q,\Gamma}(f)\Omega \neq 0$$

in the isolated-shell case, or a nonzero threshold coefficient in the infraparticle case. The dressing geometry determines the angular flux label of the vector.

5. Truncated dressings must have a finite-energy limit after the local renormalizations belonging to the chosen charged operator. Let

$$\xi_{L,\varepsilon} = \frac{\Psi_{q,\Gamma;L,\varepsilon}(f)\Omega}{\|\Psi_{q,\Gamma;L,\varepsilon}(f)\Omega\|}$$

when the denominator is nonzero, and let $E_q(\Delta)$ be the spectral projection of the charged Hamiltonian H_q. The operational condition is spectral tightness:

$$\lim_{E\to\infty} \sup_{(L,\varepsilon)\in\mathcal{A}} \left\|(1 - E_q([0,E]))\xi_{L,\varepsilon}\right\| = 0 \tag{22.9}$$

for the admissible truncation schedules $\mathcal{A}$ used in the large-time construction. Uniform first-moment control,

$$\sup_{(L,\varepsilon)\in\mathcal{A}} \langle \xi_{L,\varepsilon}, H_q\xi_{L,\varepsilon}\rangle < \infty,$$

is a useful sufficient estimate by the spectral Markov inequality, but the scattering construction needs the tightness of the spectral measures themselves. The tightness condition is what permits extraction of finite-energy subsequential limits and makes the projection or threshold coefficient in the previous item meaningful.

The confining alternative is a spectral lower-bound statement for the declared class of gauge-invariant charged truncations. Let $\mathcal{D}_{q,L}$ be a class of normalized truncated charged vectors whose boundary flux reaches distance L. If there are $\sigma > 0$, L_0, and $c = o(L)$ such that every $\xi \in \mathcal{D}_{q,L}$, $L \geq L_0$, has spectral support above

$$E_{\text{vac}} + \sigma L - c(L), \tag{22.10}$$

or equivalently

$$\langle \xi, H_q\xi\rangle \geq E_{\text{vac}} + \sigma L - c(L),$$

then the family cannot satisfy (22.9). In that case the finite-energy charged asymptotic sector is absent and the scattering problem belongs to neutral composite sectors.

The word "nonconfining" in the rest of this section refers precisely to Hypothesis 22.9. The hypothesis does not solve the charged Haag–Ruelle problem. It states the minimum positive spectral and representation-theoretic data on which that problem can be posed. The subsequent LSZ and Dollard constructions explain what follows from these inputs and which large-time estimates still have to be proved.

Truncation topology for noncompact dressings. The symbol $\Psi_{q,\Gamma}$ in Hypothesis 22.9 is not an ordinary local operator-valued distribution with a hidden bounded localization region. For a Wilson-line ray or Coulomb dressing it must be defined as a renormalized limit of finite dressings. A precise charged construction must therefore fix, before any large-time scattering limit is taken, a family

$$\Psi_{q,\Gamma;L,\varepsilon}(f) = N_{\Gamma;L,\varepsilon}\,\psi_{q,\varepsilon}(f)\,W_{\Gamma;L,\varepsilon}$$

on a common finite-energy domain: L is the length or infrared truncation of the dressing, ε is the ultraviolet regulator at the charged endpoint and along cusps or corners, $W_{\Gamma;L,\varepsilon}$ is the finite Wilson-line or Coulombic factor, and $N_{\Gamma;L,\varepsilon}$ contains the endpoint, line, cusp, and local charged-field renormalizations belonging to the chosen scheme. The infinite dressing is a shorthand only after matrix elements against finite-energy vectors have limits, with the boundary charge and angular flux label held fixed:

$$\lim_{\varepsilon\downarrow 0}\lim_{L\to\infty}\langle \xi, \Psi_{q,\Gamma;L,\varepsilon}(f)\eta\rangle = \langle \xi, \Psi_{q,\Gamma}(f)\eta\rangle.$$

The order of limits may be replaced by a joint renormalization prescription, but the prescription is part of the operator datum.

For Haag–Ruelle purposes fixed-time convergence is not enough. The large-time approximants sample increasingly long pieces of the Gauss-law tail, so the theorem must specify admissible truncation schedules

$$L_i(t) \to \infty, \qquad \varepsilon_i(t) \downarrow 0, \qquad \sigma t \to +\infty,$$

for every charged external leg. Same-flux schedules and same-flux compact dressing changes should define the same asymptotic vector only if their modified comparison vectors obey a uniform tail condition on the scattering domain. In the notation used below this means that, for two admissible same-flux choices L and L',

$$\lim_{T\to\infty}\sup_{\sigma t\geq T}\left\|\mathcal{C}^{\sigma}_{I,L}(t) - \mathcal{C}^{\sigma}_{I,L'}(t)\right\| = 0, \tag{22.11}$$

and, whenever the norm derivative is used in Cook's argument,

$$\lim_{T\to\infty}\int_{\sigma t\geq T}\left\|\frac{\mathrm{d}}{\mathrm{d}t}\big(\mathcal{C}^{\sigma}_{I,L}(t) - \mathcal{C}^{\sigma}_{I,L'}(t)\big)\right\|\,\mathrm{d}t = 0. \tag{22.12}$$

Equations (22.11) and (22.12) are load-bearing estimates: they are the point where the formal infinite Wilson line becomes a legitimate asymptotic coordinate. A construction that only supplies $\Psi_{q,\Gamma;L,\varepsilon}(f)$ for each fixed L and then takes $L \to \infty$ at fixed time supplies a family of dressed matrix elements, while the charged Haag–Ruelle input requires control of the joint large-time/noncompact-tail limit.

Definition 22.10 (Dressed charged LSZ problem)**.** A dressed charged LSZ problem consists of:

1. a nonconfining charged sector in the sense of Hypothesis 22.9;
2. a family of gauge-invariant noncompact charged creators $\Psi_{q,\gamma}(x)$, or Coulombic variants with the same asymptotic flux;
3. a specification of the dressing geometry γ in terms of the asymptotic velocity or momentum labels;

4. a spectral statement for the vectors $\Psi_{q,\gamma}(f)\Omega$, either an isolated massive shell in an infrared-regulated or screened model, or the corresponding charged infraparticle threshold structure in massless QED;
5. a reduction prescription for Fourier-transformed dressed correlators

$$\langle\Omega|T\{\Psi_{q_1,\gamma_1}(x_1)\cdots\Psi^\dagger_{q_n,\gamma_n}(y_n)\,\mathcal{O}_1\cdots\mathcal{O}_k\}|\Omega\rangle$$

which extracts the same matrix elements as the large-time dressed wave operators.

This definition identifies the replacement for ordinary local-field LSZ in a charged gauge sector. The external residue, when an isolated charged shell is available after a regulator or screening deformation, is the residue of the dressed two-point function. It includes the short-distance renormalization of the matter field and the endpoint or cusp renormalization of the Wilson-line dressing. Different dressing geometries define different charged operators; showing that a physical on-shell answer is independent of the unobserved geometric choices is part of the reduction theorem, not an assumption inside the scalar LSZ formula.

22.5.1 Dressed Residues as Charged Coordinates

Before discussing the geometry of Wilson lines, it is useful to state the finite-dimensional algebra that any dressed LSZ theorem must implement. Fix, temporarily, an infrared regulator or screening deformation for which a charged isolated massive shell exists. Let

$$\{\Psi_a\}_{a=1}^r$$

be finitely many gauge-invariant dressed charged interpolating operators with the same charge, spin-statistics, mass, and asymptotic flux sector. Let E_p be the finite-dimensional one-particle multiplicity space over the mass shell point $p\in\Sigma_m^+$, and choose an orthonormal basis $\{|p,\lambda\rangle\}_{\lambda=1}^N$ of E_p. The one-particle matrix coefficients of the dressed coordinates are the numbers

$$z_a{}^\lambda(p)=\langle p,\lambda|\Psi_a(0)\Omega\rangle.$$

Equivalently, for a positive-frequency packet f supported in the isolating neighborhood of the shell,

$$P_1\Psi_a(f)\Omega=\int_{\Sigma_m^+}\mathrm{d}\mu_m(p)\,\widehat{f}(p)\sum_{\lambda=1}^N z_a{}^\lambda(p)|p,\lambda\rangle.$$

The singular part of the dressed two-point function is therefore the Gram matrix of these overlap vectors:

$$\lim_{p^2\to-m^2}(p^2+m^2-\mathrm{i}0)\widehat{G}_{ab}(p)=\mathrm{i}\sum_{\lambda=1}^N z_a{}^\lambda(p)\,\overline{z_b{}^\lambda(p)}\equiv\mathrm{i}Z_{ab}(p),\tag{22.13}$$

in the time-ordered convention used in the LSZ chapter. Thus the pole residue is not an intrinsic scalar attached to a raw noncompact operator; it is the coordinate Gram matrix of the physical one-particle shell in the chosen dressed operator coordinates.

The LSZ extraction of a normalized external charged state is possible from this family precisely when the overlap map

$$z(p):E_p\longrightarrow\mathbb{C}^r,\qquad \eta_\lambda|p,\lambda\rangle\longmapsto\left(\sum_\lambda z_a{}^\lambda(p)\eta_\lambda\right)_{a=1}^r$$

has full column rank. A choice of left inverse

$$L(p) : \mathbb{C}^r \longrightarrow E_p, \qquad L(p)z(p) = \mathbf{1}_{E_p},$$

then supplies the external charged residue coordinate. If

$$\widehat{H}_a(p;\Xi) = \frac{\mathrm{i}\sum_\lambda z_a{}^\lambda(p)\mathcal{A}_\lambda(p;\Xi)}{p^2+m^2-\mathrm{i}0} + \widehat{H}_a^{\text{less}}(p;\Xi)$$

is a Fourier-transformed dressed correlator with one external charged insertion and all other variables denoted by Ξ, then

$$\lim_{p^2\to -m^2}(p^2+m^2-\mathrm{i}0)\sum_a L^\rho{}_a(p)\,\widehat{H}_a(p;\Xi) = \mathrm{i}\,\mathcal{A}_\rho(p;\Xi). \tag{22.14}$$

The positive matrix $Z^{-1/2}$ in the finite-regulator theorem below is the special square-coordinate form of this left-inverse operation after the operator labels have been identified with an orthonormal multiplicity basis.

This algebra also gives the exact meaning of a harmless dressing change. If another family is related by a finite invertible change of dressed coordinates,

$$\Psi'_\alpha = \sum_a M_\alpha{}^a \Psi_a,$$

with the same charged shell and the same asymptotic flux sector, then

$$z' = Mz, \qquad Z' = MZM^\dagger, \qquad L' = LM^{-1}.$$

Equation (22.14) is unchanged. This is a coordinate statement on a fixed one-particle charged sector; it is not a path-independence theorem for arbitrary Wilson lines. A change of the asymptotic ray or Coulomb tail changes the Gauss-law flux at infinity in an unscreened theory and need not be represented by any finite invertible matrix on a common Hilbert-space shell. The soft-sector calculations below make this failure explicit.

22.5.2 Wilson-Line Covariance and Boundary Charge

The sign and orientation in (22.8) are fixed by gauge covariance. Let $\gamma_{x,\infty} : [0,\infty) \to M$ be a regulated curve with $\gamma(0) = x$ and endpoint tending to spatial infinity. With

$$A \mapsto A + \mathrm{d}\alpha, \qquad \psi_q(x) \mapsto \mathrm{e}^{\mathrm{i}q\alpha(x)}\psi_q(x),$$

the abelian Wilson factor

$$W_{q,\gamma}(x) = \exp\left\{\mathrm{i}q\int_{\gamma_{x,\infty}} A\right\}$$

transforms as

$$W_{q,\gamma}(x) \longmapsto \mathrm{e}^{\mathrm{i}q(\alpha(\infty)-\alpha(x))}W_{q,\gamma}(x).$$

Therefore

$$\Psi_{q,\gamma}(x) = \psi_q(x)W_{q,\gamma}(x) \longmapsto \mathrm{e}^{\mathrm{i}q\alpha(\infty)}\Psi_{q,\gamma}(x). \tag{22.15}$$

Equation (22.15) is the precise sense in which the operator is invariant under gauge transformations trivial at infinity and carries charge q under the boundary gauge transformation. The boundary phase is not a gauge redundancy; it is the global charge action in the charged sector.

For a nonabelian gauge field $A = A^a_\mu t^a_R \mathrm{d}x^\mu$ and a matter field $\psi_R(x)$ in representation R, choose the parallel transporter

$$U_R(\infty, x; \gamma) = P\exp\left\{\mathrm{i}\int_{\gamma_{x,\infty}} A^a_\mu t^a_R\, \mathrm{d}x^\mu\right\}.$$

Under a gauge transformation g,

$$U_R(\infty, x; \gamma) \longmapsto g(\infty) U_R(\infty, x; \gamma) g(x)^{-1}.$$

Thus $U_R(\infty, x; \gamma)\psi_R(x)$ transforms only by $g(\infty)$. In a sector where the boundary gauge group is treated as a global symmetry, this is a charged operator; in a confining theory with no colored asymptotic sectors, it is a noncompact string-like excitation rather than an external particle creator. This distinction is dynamical and cannot be read off from the formal gauge covariance alone.

Boundary-charge selection rule for dressed correlators. The endpoint transformation supplies the Ward identity that fixes which dressed correlators can enter an LSZ formula. Consider, with a fixed ultraviolet regulator and a fixed infrared endpoint prescription, a time-ordered vacuum distribution

$$G(x_1, \ldots, x_n; \Xi) = \langle\Omega| T\{\Psi^{\epsilon_1}_{q_1,\gamma_1}(x_1)\cdots\Psi^{\epsilon_n}_{q_n,\gamma_n}(x_n)\, \mathcal{O}(\Xi)\}|\Omega\rangle.$$

Here $\epsilon_a = +1$ means Ψ_{q_a,γ_a}, $\epsilon_a = -1$ means $\Psi^\dagger_{q_a,\gamma_a}$, and $\mathcal{O}(\Xi)$ is a product of neutral gauge-invariant local observables. Under a boundary gauge transformation with constant value α_∞, the vacuum and the neutral observables are invariant while

$$\Psi^{\epsilon_a}_{q_a,\gamma_a} \longmapsto \exp(\mathrm{i}\epsilon_a q_a \alpha_\infty)\Psi^{\epsilon_a}_{q_a,\gamma_a}.$$

Therefore, as an equality of regulated distributions,

$$G = \exp\left(\mathrm{i}\alpha_\infty \sum_{a=1}^{n} \epsilon_a q_a\right) G \qquad \text{for every } \alpha_\infty.$$

If the signed boundary charge does not vanish,

$$\sum_{a=1}^{n} \epsilon_a q_a \neq 0, \tag{22.16}$$

then $G = 0$. Equivalently, a nonzero vacuum correlator of dressed charged operators must be neutral after the charges carried by the endpoints at infinity are included. Matrix elements between charged superselection sectors obey the same rule with the right side of (22.16) replaced by the difference between the final and initial boundary charges. Thus the external charges in a dressed LSZ reduction form part of the boundary-charge bookkeeping of the correlator itself.

For a nonabelian boundary group, the same statement is representation theoretic. If the endpoints of the dressed insertions transform in boundary representations $R_1, \ldots, R_n$, then the vacuum correlator is an invariant tensor in

$$(R_1 \otimes \cdots \otimes R_n)^{G_\infty}.$$

If this invariant subspace is zero, the correlator vanishes. If it is nonzero, choosing a basis of invariant tensors is part of choosing the external charged channel. This is still separate from the dynamical question whether the corresponding charged sector has finite-energy asymptotic particles; in a confining phase the representation-theoretic singlet condition may be satisfied only by neutral composite operators, and ordinary Haag–Ruelle theory then applies to those composite particles rather than to individual colored endpoints.

22.5.3 Ray Dressings as Gauge-Slice Coordinates

The preceding covariance statement has a second, more computational, interpretation. A Wilson-line dressed charged coordinate is the ordinary charged coordinate written on a gauge slice adapted to the same ray. The statement below is elementary in abelian gauge theory, but it is a useful guardrail for perturbative calculations of dressed correlators.[1]

Let u^μ be a fixed nonzero vector and let the half-line from x be

$$\gamma_{x,u}(\tau) = x + \tau u, \qquad \tau \geq 0.$$

Work on a class of gauge potentials A and gauge functions α for which $u^\mu A_\mu(x+\tau u)$ and $u^\mu \partial_\mu \alpha(x+\tau u)$ are absolutely integrable on $[0, \infty)$ and for which

$$\alpha(x + \tau u) \longrightarrow 0, \qquad u^\mu A_\mu(x + \tau u) \longrightarrow 0 \qquad (\tau \to \infty)$$

along every ray under consideration. Define the line functional

$$\chi_u[A](x) = \int_0^\infty \mathrm{d}\tau \, u^\mu A_\mu(x + \tau u). \tag{22.17}$$

For the abelian gauge convention

$$A_\mu \longmapsto A_\mu + \partial_\mu \alpha, \qquad \psi_q \longmapsto \mathrm{e}^{\mathrm{i}q\alpha} \psi_q,$$

one has

$$\chi_u[A + \mathrm{d}\alpha](x) = \chi_u[A](x) - \alpha(x).$$

Hence

$$\Psi_{q,u}(x) = \mathrm{e}^{\mathrm{i}q\chi_u[A](x)} \psi_q(x) \tag{22.18}$$

is invariant under gauge transformations that are trivial at the ray endpoint.

[1] A recent perturbative study of finite-distance dressed observables in electrodynamics and perturbative gravity is C. Cheung, A. Sivaramakrishnan, J. Wilson-Gerow, and L. Zhou, *On Perturbatively Dressed Observables*, arXiv:2605.26077. The use made here is only the gauge-slice viewpoint; the abelian identities and denominators in this subsection are derived directly.

Ray dressing as an axial-gauge coordinate. For every A in the class above, set

$$A_\mu^{[u]}(x) = A_\mu(x) + \partial_\mu \chi_u[A](x), \qquad \psi_q^{[u]}(x) = \mathrm{e}^{\mathrm{i}q\chi_u[A](x)}\psi_q(x).$$

Then

$$u^\mu A_\mu^{[u]}(x) = 0, \qquad \psi_q^{[u]}(x) = \Psi_{q,u}(x). \tag{22.19}$$

Thus a dressed charged insertion may be computed, in any finite regularized perturbative path integral for which this gauge slice and the corresponding change of variables have been defined with their regulator and Faddeev–Popov factors, as the ordinary charged field inserted in the gauge $u^\mu A_\mu = 0$ with the same endpoint prescription. Indeed, differentiating (22.17) in the ray direction gives

$$\begin{aligned} u^\nu \partial_\nu \chi_u[A](x) &= \int_0^\infty \mathrm{d}\tau \, \frac{d}{d\tau}\big(u^\mu A_\mu(x+\tau u)\big) \\ &= \lim_{T\to\infty} u^\mu A_\mu(x+Tu) - u^\mu A_\mu(x) = -u^\mu A_\mu(x). \end{aligned}$$

Therefore

$$u^\mu A_\mu^{[u]}(x) = u^\mu A_\mu(x) + u^\mu \partial_\mu \chi_u[A](x) = 0.$$

The second identity in (22.19) is the definition of $\psi_q^{[u]}$ together with (22.18). The preceding transformation law for χ_u shows that this construction is constant on gauge orbits once the boundary gauge value has been fixed.

The perturbative singularities produced by the ray are also fixed at this elementary level. Insert an endpoint regulator

$$\chi_{u,\epsilon}[A](x) = \int_0^\infty \mathrm{d}\tau \, \mathrm{e}^{-\epsilon\tau} u^\mu A_\mu(x+\tau u), \qquad \epsilon > 0,$$

and use the Fourier convention

$$A_\mu(y) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \mathrm{e}^{-\mathrm{i}k\cdot y} \widehat{A}_\mu(k).$$

Then

$$\chi_{u,\epsilon}[A](x) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \mathrm{e}^{-\mathrm{i}k\cdot x} \frac{u^\mu \widehat{A}_\mu(k)}{k\cdot u - \mathrm{i}\epsilon}. \tag{22.20}$$

Indeed, substituting the Fourier transform gives

$$\chi_{u,\epsilon}[A](x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \mathrm{e}^{-\mathrm{i}k\cdot x} u^\mu \widehat{A}_\mu(k) \int_0^\infty \mathrm{d}\tau \, \mathrm{e}^{-(\epsilon + \mathrm{i}k\cdot u)\tau},$$

and the last integral is

$$\frac{1}{\epsilon + \mathrm{i}k\cdot u} = -\frac{\mathrm{i}}{k\cdot u - \mathrm{i}\epsilon}.$$

Expanding the exponential in (22.18) therefore inserts the distribution $(k\cdot u - \mathrm{i}0)^{-1}$ for every photon emitted from the line. These linear denominators are part of the nonlocal composite operator and of the corresponding axial-gauge coordinate system. They must be treated as additional singular hypersurfaces in loop integrals involving dressed finite-distance correlators. They do not, by themselves, supply the large-time Haag–Ruelle estimate or the infrared representation theory required for charged scattering; those are the separate problems formulated below and in Chapter 44.

22.5.4 The Missing Haag–Ruelle Estimate

Ordinary Haag–Ruelle theory uses two facts at large time: the one-particle part of $B_t(h)\Omega$ is time independent, and commutators of creators with disjoint velocity supports are rapidly decreasing because the creators are almost local. The almost-local closure of the Gauss-law obstruction above shows that, for an unscreened Gauss-law charge, a charged creator in the observable net cannot satisfy that localization hypothesis. A Wilson-line or Coulomb-dressed charged creator has noncompact localization for structural reasons, so the local commutator estimate must be replaced rather than reused.

The geometric replacement is not "large spacelike separation" of bounded double cones. Let V_h be the compact velocity support of the charged wave packet and let $\mathcal{C}_\Gamma$ denote the asymptotic cone, ray, or Coulomb coordinate carried by the dressing. A future charged theorem must separate two different pieces of the large-time operator. The compact core of $\Psi_{\Gamma,t}(h)$ is transported into the velocity tube $tV_h + O(1)$ and can be treated by the ordinary spacelike-separation argument when the velocity supports are disjoint. The Gauss-law tail attached to $\mathcal{C}_\Gamma$, however, remains noncompact and is not made harmless by separating the cores. Consequently the commutator estimate has to be a decomposition into the following three separately estimated contributions:

$$\text{core–core spacelike term} \quad + \quad \text{tail–tail flux pairing} \quad + \quad \text{integrable remainder.}$$

The second term is the part which becomes the Dollard or Faddeev–Kulish comparison phase below. Compact deformations of the dressing ray can change the operator by neutral field-strength insertions, as in (22.47), but they do not remove the asymptotic flux tail. Thus the analytic estimate to be proved is a cone/flux estimate, not a disguised application of ordinary almost locality.

The replacement has definite analytic content: a charged construction must separate the nonintegrable long-range part of the motion from the integrable remainder. For a charged packet h, write

$$\Psi_{\Gamma,t}(h)$$

for a dressed creator with datum $\Gamma = (q, \gamma, \mathcal{E}_\infty, \mathfrak{s})$: q is the boundary charge, γ is the geometric dressing or Coulomb coordinate, $\mathcal{E}_\infty$ is the limiting Gauss-law flux density, and $\mathfrak{s}$ records spin and internal multiplicity labels. A Haag–Ruelle replacement for a finite velocity-separated list

$$I = ((\Gamma_1, h_1), \ldots, (\Gamma_n, h_n))$$

must provide a real locally absolutely continuous comparison phase $\Theta_I(t)$ and vectors

$$\mathcal{C}_I(t) = \mathrm{e}^{-\mathrm{i}\Theta_I(t)}\Psi_{\Gamma_1,t}(h_1)\cdots\Psi_{\Gamma_n,t}(h_n)\Omega. \tag{22.21}$$

The desired estimates have the following precise content.

1. The one-particle or threshold coordinate of $\Psi_{\Gamma,t}(h)\Omega$ is independent of t after the chosen comparison dynamics is applied. In an infrared-regulated massive charged sector this is an ordinary isolated-shell projection. In massless QED it is instead a threshold or detector datum in a velocity/flux representation, not a Wigner one-particle vector in a single photon Fock space.

2. For two velocity-separated dressed creators, the large-time exchange relation on the finite-energy domain has the form

$$\Psi^{\#}_{\Gamma_i,t}(h_i)\Psi^{\#}_{\Gamma_j,t}(h_j) = \mathrm{e}^{\mathrm{i}\theta^{\#\#}_{ij}(t)}\Psi^{\#}_{\Gamma_j,t}(h_j)\Psi^{\#}_{\Gamma_i,t}(h_i) + R^{\#\#}_{ij}(t), \qquad \int_{T_0}^{\infty} \|R^{\#\#}_{ij}(t)\xi\|\,\mathrm{d}t < \infty, \tag{22.22}$$

for every vector ξ in the scattering domain and for each choice in which a superscript # denotes either the operator itself or its adjoint. The phase $\theta^{\#\#}_{ij}$, when present, is determined by the asymptotic flux data. In a local massive theory this phase is zero and the remainder is rapidly decreasing in operator norm. In a long-range charged theory a term with $\dot{\theta}_{ij}(t) \sim c_{ij}/t$ is not an error; it must be included in the comparison dynamics.

3. The comparison phase in (22.21) cancels all nonintegrable pair terms in the time derivative:

$$\left\|\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}_I(t)\right\| \le R_I(t), \qquad R_I \in L^1([T_0,\infty)). \tag{22.23}$$

This is the charged analogue of Cook's estimate. It is the analytic heart of the missing theorem.

4. Scalar products of the modified vectors have limits computed in the appropriate asymptotic representation. For a screened theory with isolated massive charged shells this representation may be an ordinary charged Fock space. For massless QED it must be a representation carrying the velocity/flux labels discussed below; one cannot first form a single photon Fock space and then attach the flux by a limiting unitary.

5. Dressings with the same limiting flux and a finite Hilbert-space difference act as changes of interpolating coordinates. Dressings with different limiting fluxes define different charged sectors. The distinction is quantified below by the soft-norm and Weyl-implementation calculations.

Finite-dimensional Coulomb-tail comparison. The following elementary calculation fixes the normalization of the logarithmic comparison phase, but it is not a theorem about QFT. Let $u \in \mathbb{R}^3 \setminus \{0\}$, $b \in \mathbb{R}^3$, $a > 0$, and $\kappa \in \mathbb{R}$, and define

$$V(t) = \frac{\kappa}{\sqrt{a^2 + |b+ut|^2}}, \qquad \Phi(T) = \int_{T_0}^{T} V(t)\,\mathrm{d}t.$$

Introduce

$$\sigma = \frac{b\cdot u}{|u|^2}, \qquad b_\perp = b - \sigma u, \qquad \rho^2 = a^2 + |b_\perp|^2.$$

Then

$$a^2 + |b+ut|^2 = \rho^2 + |u|^2(t+\sigma)^2,$$

and the integral has the exact primitive

$$\Phi(T) = \frac{\kappa}{|u|}\left[\operatorname{arsinh}\left(\frac{|u|(t+\sigma)}{\rho}\right)\right]_{T_0}^{T}. \tag{22.24}$$

Using $\operatorname{arsinh} x = \log(2x) + O(x^{-2})$ as $x \to +\infty$, one obtains a finite constant $C(T_0, b, u, a, \kappa)$ with

$$\Phi(T) = \frac{\kappa}{|u|}\log T + C(T_0,b,u,a,\kappa) + O(T^{-1}) \qquad (T \to +\infty). \tag{22.25}$$

The same completed-square expression gives

$$V(t) = \frac{\kappa}{|u|}\frac{1}{t} + O(t^{-2}).$$

This finite-dimensional comparison calculation does not identify the coefficient κ in a QFT. Its role is to fix the logic of the missing charged Haag–Ruelle estimate. If the asymptotic flux-sector pairing of two charged packets produces an effective long-range pair term $V_{ij}(t)$ with leading coefficient $\kappa_{ij}/|u_{ij}|$, then the corresponding contribution to $\Theta_I(t)$ must contain

$$\frac{\kappa_{ij}}{|u_{ij}|}\log t.$$

Treating this logarithmic phase as an error would make the Cook derivative nonintegrable. The open QFT problem is precisely to derive the analogue of κ_{ij}, and the L^1 remainder after subtracting the resulting phase, from the noncompact Wilson-line or Coulomb-dressed charged creators.

Coefficient and residual estimate to be proved. The symbol κ_{ij} in the preceding comparison is not a free normalization. A nonperturbative charged Haag–Ruelle theorem must construct it from the asymptotic Gauss-law tail of the chosen charged coordinates. For two velocity-separated packet data (Γ_i, h_i), (Γ_j, h_j), and for each choice of creator or adjoint signs $\#\#$, the theorem must produce a scalar pair kernel $K_{ij}^{\#\#}(t)$ and a residual operator $\mathcal{R}_{ij}^{\#\#}(t)$ on the scattering domain. Denote by $\mathcal{D}_{ij}^{\#\#}(t)$ the pair component of the large-time exchange defect or Cook derivative in the chosen finite-regulator construction, and by $\mathcal{P}_{ij}^{\#\#}(t)$ the corresponding ordered product with the other dressed factors held fixed. The required decomposition is

$$\mathcal{D}_{ij}^{\#\#}(t)\xi = \mathrm{i}\, K_{ij}^{\#\#}(t)\, \mathcal{P}_{ij}^{\#\#}(t)\xi + \mathcal{R}_{ij}^{\#\#}(t)\xi, \tag{22.26}$$

with

$$\int_{T_0}^{\infty} \|\mathcal{R}_{ij}^{\#\#}(t)\xi\|\,\mathrm{d}t < \infty \qquad \text{for every scattering-domain vector } \xi. \tag{22.27}$$

The leading part of $K_{ij}^{\#\#}$ is fixed by a flux-pairing functional on the limiting tails:

$$K_{ij}^{\#\#}(t) = \eta_i\eta_j \frac{\kappa(\Gamma_i,\Gamma_j)}{\sqrt{a_{ij}^2 + |b_{ij} + u_{ij}t|^2}} + k_{ij}^{\#\#,\mathrm{rem}}(t), \qquad k_{ij}^{\#\#,\mathrm{rem}} \in L^1([T_0,\infty)). \tag{22.28}$$

Equivalently,

$$K_{ij}^{\#\#}(t) - \eta_i\eta_j \frac{\kappa(\Gamma_i,\Gamma_j)}{|u_{ij}|\,t} \in L^1([T_0,\infty)).$$

In an elementary abelian Coulomb normalization the functional $\kappa(\Gamma_i,\Gamma_j)$ reduces to the Coulomb constant times the product of the two scalar charges, after the velocity and angular-flux profiles defining Γ_i, Γ_j have been fixed. In a more general nonconfining gauge sector it is a matrix element, or sector label, of the asymptotic flux pairing; it is part of the representation-theoretic charged sector, not a parameter chosen after the scattering estimate is known.

This distinction is load-bearing. Replacing κ by $\kappa+\delta\kappa$ leaves an uncompensated $\delta\kappa/(|u_{ij}|t)$ term in the Cook derivative, and this term is not L^1. By contrast, a compact same-flux dressing change can only add an L^1 derivative to $K_{ij}^{\#\#}$, hence changes the comparison phase by a finite limit and does not change the logarithmic coefficient. Equal velocities are excluded from this pair estimate: when $u_{ij} = 0$, the pair must be treated as a charged cluster, threshold sector, or bound-state coordinate rather than as two separated Dollard legs.

The many-body comparison datum has the same pairwise form. For a velocity-separated list I, write v_i for a velocity in the support of h_i, $u_{ij} = v_i - v_j$, and let $\eta_i = +1$ for a charged creator and $\eta_i = -1$ for its adjoint. The sign convention records the fact that conjugating a dressed charged creator reverses its boundary charge and its asymptotic flux. Suppose that the finite-regulator charged dynamics, after all short-range terms have been separated, produces pair terms of the form

$$V_{ij}(t) = \eta_i \eta_j \frac{\kappa_{ij}}{\sqrt{a_{ij}^2 + |b_{ij} + u_{ij} t|^2}} + r_{ij}(t), \qquad r_{ij} \in L^1([T_0, \infty)). \tag{22.29}$$

Here $a_{ij} > 0$ is a regulator or transverse smearing scale, b_{ij} is the impact-parameter coordinate in the finite comparison model, and κ_{ij} is the flux-pairing coefficient that the actual QFT estimate must derive from the Wilson-line or Coulomb-tail data. The corresponding comparison phase must satisfy

$$\dot{\Theta}_I(t) = \sum_{i<j} \eta_i \eta_j \frac{\kappa_{ij}}{\sqrt{a_{ij}^2 + |b_{ij} + u_{ij} t|^2}} + \dot{\Theta}_I^{\mathrm{fin}}(t), \qquad \dot{\Theta}_I^{\mathrm{fin}} \in L^1([T_0, \infty)), \tag{22.30}$$

or, equivalently at the logarithmic level,

$$\Theta_I(T) = \sum_{i<j} \eta_i \eta_j \frac{\kappa_{ij}}{|u_{ij}|} \log T + O(1) \qquad (T \to +\infty). \tag{22.31}$$

The velocity-separation hypothesis is part of the datum: if $u_{ij} = 0$ the logarithmic coefficient is not defined, and the pair must be treated as a single charged cluster or as a separate infraparticle/bound-state sector. Equations (22.29)–(22.31) are not an independent scattering construction. They are the finite Dollard bookkeeping that the nonperturbative estimate must reproduce: all t^{-1} tails must be exhausted by oriented pair fluxes, and only the residual derivative is allowed to enter the Cook L^1 remainder.

Scalar-product Cauchy criterion. The comparison phase is useful only after it produces limits of scalar products in the charged asymptotic representation. Suppose that two velocity-separated lists I, J have modified comparison vectors $\mathcal{C}_I(t), \mathcal{C}_J(t)$ on the charged scattering domain and that, for $t \geq T_0$,

$$\|\mathcal{C}_I(t)\| \leq M_I, \qquad \|\mathcal{C}_J(t)\| \leq M_J, \qquad \left\| \frac{\mathrm{d}}{\mathrm{d}t} \mathcal{C}_I(t) \right\| \leq R_I(t), \qquad \left\| \frac{\mathrm{d}}{\mathrm{d}t} \mathcal{C}_J(t) \right\| \leq R_J(t),$$

with $R_I, R_J \in L^1([T_0, \infty))$. Then for $T > S \geq T_0$,

$$|\langle \mathcal{C}_I(T), \mathcal{C}_J(T) \rangle - \langle \mathcal{C}_I(S), \mathcal{C}_J(S) \rangle| \leq M_J \int_S^T R_I(t)\, \mathrm{d}t + M_I \int_S^T R_J(t)\, \mathrm{d}t. \tag{22.32}$$

Hence the scalar product has a limit once the modified Cook estimates are proved on a common domain and the comparison vectors are uniformly bounded. If the logarithmic coefficient in Θ_I is changed by $\delta \neq 0$, the model scalar product acquires the factor $\exp(\mathrm{i}\delta \log t)$. Along dyadic times this changes by $\exp(\mathrm{i}\delta \log 2)$, and the function $t \mapsto \exp(\mathrm{i}\delta \log t)$ has no large-time limit. Thus the scalar-product limit is another place where the flux-pairing coefficient is not a harmless convention: the same L^1 residual that makes Cook's estimate finite is also what makes the asymptotic inner product well defined.

Hypothesis 22.11 (Charged Dollard–Cook estimate package)**.** Fix a nonconfining charged sector and a dense set $\mathfrak{D}_{\mathrm{ch}}$ of charged packet data (Γ, h), where $\Gamma = (q, \gamma, \mathcal{E}_\infty, \mathfrak{s})$ records boundary charge, dressing geometry, limiting flux, and spin/internal labels, and where h has compact velocity support. For every finite velocity-separated ordered list

$$I = ((\Gamma_1, h_1), \ldots, (\Gamma_n, h_n))$$

and for each time direction $\sigma \in \{+, -\}$, assume the following data and estimates.

1. There are dressed approximants $\Psi^\sigma_{\Gamma_j,t}(h_j)$ acting on a common finite-energy domain containing Ω, and the product vector is defined there for all sufficiently large $\sigma t > 0$. These approximants are limits, in the truncation topology described above, of finite-length and ultraviolet regularized dressings; the estimates below are required to be stable under admissible same-flux truncation schedules.

2. There is a real locally absolutely continuous comparison phase $\Theta^\sigma_I(t)$, unique up to a finite large-time limit, whose nonintegrable part is the oriented pairwise Dollard logarithm (22.31) whenever the long-range tail is unscreened.

3. The modified vector

$$\mathcal{C}^\sigma_I(t) = \exp[-\mathrm{i}\Theta^\sigma_I(t)]\Psi^\sigma_{\Gamma_1,t}(h_1)\cdots\Psi^\sigma_{\Gamma_n,t}(h_n)\Omega$$

   is locally absolutely continuous in Hilbert norm and satisfies an integrable Cook bound

$$\left\| \frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}^\sigma_I(t) \right\| \le R^\sigma_I(t), \qquad R^\sigma_I \in L^1(\{t : \sigma t \ge T_0\}). \tag{22.33}$$

4. For every pair of velocity-separated lists I, J, the scalar products have limits

$$\langle J, I\rangle^\sigma_{\mathrm{as}} = \lim_{\sigma t\to+\infty} \langle \mathcal{C}^\sigma_J(t), \mathcal{C}^\sigma_I(t)\rangle, \tag{22.34}$$

   and these limits define a positive semidefinite sesquilinear form on the algebraic span of charged asymptotic symbols. The form is compatible with finite refinements of velocity partitions and with finite same-flux dressing coordinate changes.

5. If spacetime covariance is to be asserted, the approximants and phases obey the corresponding covariance law modulo vectors whose norms tend to zero in the same large-time limit.

Construction 22.12 (Charged wave maps from a Dollard–Cook package)**.** Assume Hypothesis 22.11. Let $\mathcal{H}^\sigma_{\mathrm{as},0}$ be the algebraic vector space generated by velocity-separated charged asymptotic symbols e^σ_I, modulo the finite linear relations imposed by packet refinement and same-flux coordinate changes. Equip it with the positive semidefinite form (22.34), quotient by its null space, and complete. The resulting Hilbert space is denoted $\mathcal{H}^{\sigma,\mathrm{ch}}_{\mathrm{as}}$.

For a generator e^σ_I, define

$$\Omega^{\mathrm{ch}}_\sigma e^\sigma_I = \lim_{\sigma t\to+\infty} \mathcal{C}^\sigma_I(t). \tag{22.35}$$

The limit exists because, for $\sigma t' > \sigma t \geq T_0$,

$$\|\mathcal{C}_I^\sigma(t') - \mathcal{C}_I^\sigma(t)\| \leq \int_{\sigma t}^{\sigma t'} R_I^\sigma(\sigma s)\,\mathrm{d}s$$

with the integral taken along the chosen time ray. The L^1 tail in (22.33) tends to zero, so $\mathcal{C}_I^\sigma(t)$ is Cauchy in $\mathcal{H}$.

The definition is independent of null asymptotic symbols. If $\xi = \sum_\alpha c_\alpha e_{I_\alpha}^\sigma$, then the corresponding physical approximant is

$$\mathcal{C}_\xi^\sigma(t) = \sum_\alpha c_\alpha \mathcal{C}_{I_\alpha}^\sigma(t),$$

and Hypothesis 22.11 gives

$$\lim_{\sigma t \to +\infty} \|\mathcal{C}_\xi^\sigma(t)\|^2 = \langle \xi, \xi \rangle_{\mathrm{as}}^\sigma. \tag{22.36}$$

Thus a null asymptotic vector has zero physical limit. For two algebraic vectors ξ, η, the same scalar-product limit gives

$$\langle \Omega_\sigma^{\mathrm{ch}} \eta, \Omega_\sigma^{\mathrm{ch}} \xi \rangle_{\mathcal{H}} = \langle \eta, \xi \rangle_{\mathrm{as}}^\sigma.$$

Therefore $\Omega_\sigma^{\mathrm{ch}}$ descends to an isometry from $\mathcal{H}_{\mathrm{as}}^{\sigma,\mathrm{ch}}$ into the physical Hilbert space. If the covariance clause in the hypothesis is included, the usual intertwining relation follows by applying the same norm-limit argument to the covariance defect.

Construction 22.12 is deliberately abstract. It proves that no additional Hilbert-space miracle is needed after the charged estimates have been established, but it does not establish those estimates. The remaining QFT problem is to derive (22.22) and (22.33), together with the scalar-product limit (22.34), from a nonperturbative construction of Wilson-line or Coulomb-dressed charged creators. If the gauge field is screened or massive, the long-range phase may be absent and ordinary massive scattering estimates can be recovered for gauge-invariant composite charged operators. For an unscreened Gauss-law charge, the flux tail is part of the asymptotic datum, so any theorem must include it from the start.

Finite null-quotient ledger. The quotient in Construction 22.12 is not a cosmetic step. It is the finite-dimensional algebra which lets packet refinements and same-flux coordinate redundancies coexist with a genuine asymptotic Hilbert space. Suppose three asymptotic symbols e_1, e_2, e_3 have physical large-time limits

$$\xi_1 = (1, 0), \qquad \xi_2 = (0, 1), \qquad \xi_3 = (1, 1)$$

in a two-dimensional Hilbert space. Their asymptotic scalar-product matrix is

$$G_{ab} = \langle \xi_a, \xi_b \rangle = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}. \tag{22.37}$$

The matrix is positive semidefinite, not positive definite. Its null vector is

$$n = e_1 + e_2 - e_3, \qquad Gn = 0. \tag{22.38}$$

Thus e_3 is the same asymptotic vector as the refinement $e_1 + e_2$. For any coefficient vector $c = (c_1, c_2, c_3)$,

$$c^* G c = |c_1 + c_3|^2 + |c_2 + c_3|^2 = \|c_1\xi_1 + c_2\xi_2 + c_3\xi_3\|^2 .$$

Quotienting by the null relation is therefore exactly what makes the wave-map assignment $e_a \mapsto \xi_a$ well defined and isometric. If (c_1, c_2, c_3) is replaced by $(c_1 + c_3, c_2 + c_3, 0)$, the two coefficient vectors differ by a multiple of n and define the same asymptotic state. In the charged theorem this finite cell is replaced by the directed family of packet refinements and same-flux coordinate relations. The nontrivial analytic work is still the existence of the limits G_{ab} from Wilson-line or Coulomb-dressed creators; once those limits are supplied, the null quotient is ordinary Hilbert-space bookkeeping.

Proposition 22.13 (Same-flux schedule invariance of charged wave maps)**.** *Assume Hypothesis 22.11. Let $\mathcal{S}$ and $\mathcal{S}'$ be two admissible truncation schedules for the same charged packet data, with the same boundary charges and the same limiting angular fluxes. Write $\mathcal{C}^{\sigma}_{I,\mathcal{S}}(t)$ and $\mathcal{C}^{\sigma}_{I,\mathcal{S}'}(t)$ for the corresponding modified comparison vectors. Suppose that, for every generator e^{σ}_I, there is a real locally absolutely continuous function $\alpha^{\sigma}_I(t)$ with a finite limit $\alpha^{\sigma}_I(\infty)$ and*

$$d^{\sigma}_I(T) := \sup_{\sigma t \geq T} \left\| \mathcal{C}^{\sigma}_{I,\mathcal{S}}(t) - \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(t)} \mathcal{C}^{\sigma}_{I,\mathcal{S}'}(t) \right\| \longrightarrow 0 \qquad (T \to \infty). \tag{22.39}$$

Then the two schedules define the same physical charged wave map after the finite phase coordinate identification $e^{\sigma}_{I,\mathcal{S}} \sim \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(\infty)} e^{\sigma}_{I,\mathcal{S}'}$:

$$\Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}} e^{\sigma}_I = \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(\infty)} \Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}'} e^{\sigma}_I . \tag{22.40}$$

For two generators $e^{\sigma}_I, e^{\sigma}_J$, the asymptotic Gram forms obey

$$\langle J, I \rangle^{\sigma}_{\mathrm{as},\mathcal{S}} = \mathrm{e}^{\mathrm{i}\alpha^{\sigma}_J(\infty)} \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(\infty)} \langle J, I \rangle^{\sigma}_{\mathrm{as},\mathcal{S}'} . \tag{22.41}$$

Consequently same-flux schedule changes are quotient data, not new charged asymptotic sectors.

If the dressed-correlator interface of Hypothesis 22.14 is also available for the two schedules, and the external overlap maps on each finite coordinate frame are related by

$$z_{\ell,\mathcal{S}'}(p; t) = M_\ell(t) z_{\ell,\mathcal{S}}(p; t) + o(1), \qquad M_\ell(t) \longrightarrow M_\ell$$

with $M_\ell(t)$ invertible for all sufficiently large t and M_ℓ invertible on the chosen same-flux coordinate frame, then the left-inverse residue extraction gives the same coefficient $\mathcal{A}_{\boldsymbol{\lambda}}$ in the simultaneous external-shell limit.

Proof. The Cook bounds in (22.33) give Hilbert limits for both schedules and uniform large-time bounds for the comparison vectors. For σt large,

$$\begin{aligned} &\left\| \Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}} e^{\sigma}_I - \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(\infty)} \Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}'} e^{\sigma}_I \right\| \\ &\quad \leq \left\| \Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}} e^{\sigma}_I - \mathcal{C}^{\sigma}_{I,\mathcal{S}}(t) \right\| + d^{\sigma}_I(\sigma t) + \left| \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(t)} - \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_I(\infty)} \right| \left\| \mathcal{C}^{\sigma}_{I,\mathcal{S}'}(t) \right\| \\ &\qquad + \left\| \mathcal{C}^{\sigma}_{I,\mathcal{S}'}(t) - \Omega^{\mathrm{ch}}_{\sigma,\mathcal{S}'} e^{\sigma}_I \right\| . \end{aligned}$$

Every term on the right tends to zero. This proves (22.40).

The Gram statement is the same estimate before the last limit is taken. Put

$$\widetilde{\mathcal{C}}^{\sigma}_{I,\mathcal{S}'}(t) = \mathrm{e}^{-\mathrm{i}\alpha^{\sigma}_{I}(t)}\mathcal{C}^{\sigma}_{I,\mathcal{S}'}(t).$$

If M_I, M_J are large-time norm bounds for the corresponding comparison vectors, then

$$\begin{aligned}\Big|\langle \mathcal{C}^{\sigma}_{J,\mathcal{S}}(t), \mathcal{C}^{\sigma}_{I,\mathcal{S}}(t)\rangle - \langle \widetilde{\mathcal{C}}^{\sigma}_{J,\mathcal{S}'}(t), \widetilde{\mathcal{C}}^{\sigma}_{I,\mathcal{S}'}(t)\rangle\Big|& \\ \le M_I d^{\sigma}_{J}(\sigma t) + M_J d^{\sigma}_{I}(\sigma t).&\end{aligned}$$

Taking the large-time limit and then replacing $\alpha^{\sigma}_{I}(t), \alpha^{\sigma}_{J}(t)$ by their finite limits gives (22.41). The null quotient therefore identifies the two schedules by a finite coordinate phase.

For the residue statement, at finite regulator the singular tensor in (22.42) transforms on the ℓ-th external coordinate by $M_\ell(t)$, while the left inverse transforms by $L_{\ell,\mathcal{S}}(p;t)M_\ell(t)^{-1}$, for all sufficiently large t. Contracting the external coordinates gives

$$\bigotimes_{\ell}\big(L_{\ell,\mathcal{S}}M_\ell(t)^{-1}\big)\ \bigotimes_{\ell} M_\ell(t) = \bigotimes_{\ell} L_{\ell,\mathcal{S}}.$$

The $o(1)$ schedule errors are less singular in the declared shell topology and are killed by the same left-inverse contraction in the simultaneous external-shell limit. Hence the extracted coefficient is unchanged. □

22.5.5 Dressed Correlators and Wave-Map Matrix Elements

The wave-map construction above is formulated in Hilbert space. A Wilson-line LSZ theorem must also explain how it is read from time-ordered dressed correlators. This is a separate input: a pole in a formal gauge-fixed charged-field propagator is not a gauge-invariant observable, and a Wilson-line correlator is not automatically controlled by the ordinary local-field integration-by-parts proof. The object to be proved is the following interface between dressed correlators and the charged wave maps.

Hypothesis 22.14 (Dressed-correlator reduction interface)**.** Fix a nonconfining charged sector and assume the charged Dollard–Cook package of Hypothesis 22.11 for the packet data under consideration. For each external charged leg ℓ, choose a finite family of renormalized finite-truncation dressed coordinates

$$\Psi^{(\ell)}_{a_\ell;L_\ell,\varepsilon_\ell}(x), \qquad a_\ell = 1,\dots,r_\ell,$$

converging in the truncation topology described above to a noncompact dressing with fixed charge, asymptotic flux, and velocity support. In an infrared-regulated or screened setting with an isolated charged shell, let $E_\ell(p_\ell)$ be the corresponding one-particle multiplicity space, let

$$z_\ell(p_\ell) : E_\ell(p_\ell) \longrightarrow \mathbb{C}^{r_\ell}$$

be the dressed one-particle overlap map, and choose a left inverse $L_\ell(p_\ell)z_\ell(p_\ell) = \mathbf{1}_{E_\ell(p_\ell)}$. The dressed-correlator reduction interface consists of the following assertions, with estimates uniform over admissible same-flux truncation schedules.

1. The renormalized time-ordered correlators of the finite-truncation dressed coordinates have boundary values as tempered distributions in the external momenta after wave-packet smearing and after the declared endpoint, line, cusp, and local charged-field counterterms have been included.

2. Near the product of the isolated charged mass shells, their simultaneous external singular part has finite rank in the dressed coordinates:

$$\widehat{G}_{\mathbf{a}}(\mathbf{p};\Xi) = \frac{\mathrm{i}^N \sum_{\boldsymbol{\lambda}} \prod_{\ell=1}^N z_{\ell,a_\ell}{}^{\lambda_\ell}(p_\ell)\, \mathcal{A}_{\boldsymbol{\lambda}}(\mathbf{p};\Xi)}{\prod_{\ell=1}^N (p_\ell^2 + m_\ell^2 - \mathrm{i}0)} + \widehat{G}_{\mathbf{a}}^{\mathrm{less}}(\mathbf{p};\Xi). \tag{22.42}$$

   Here Ξ denotes compact gauge-invariant insertions or the remaining external variables, and "less" means that after multiplying by $\prod_\ell (p_\ell^2 + m_\ell^2 - \mathrm{i}0)$ and applying the left inverses, the term has zero simultaneous shell limit in the wave-packet topology being used. Partial shell singularities are assigned to the corresponding lower-particle disconnected or lower-channel matrix elements before the connected amplitude is read off.

3. The coefficient $\mathcal{A}_{\boldsymbol{\lambda}}$ obtained from (22.42) is the charged wave-map matrix element:

$$\mathcal{A}_{\boldsymbol{\lambda}}(\mathbf{p};\Xi) = \left\langle \Omega_+^{\mathrm{ch}} e_{\mathrm{out},\boldsymbol{\lambda}}, \mathcal{O}_\Xi\, \Omega_-^{\mathrm{ch}} e_{\mathrm{in},\boldsymbol{\lambda}} \right\rangle_{\mathrm{conn}}, \tag{22.43}$$

   with the total momentum delta distribution and the standard disconnected vacuum factors removed. The Dollard phases appearing in Hypothesis 22.11 are part of the definition of the asymptotic symbols $e_{\mathrm{in/out},\boldsymbol{\lambda}}$; without them a long-range charged correlator need not have a simple isolated-pole LSZ limit.

4. Endpoint and collision contact terms created by differentiating time-ordered Wilson-line insertions are either absorbed into the declared renormalized dressed coordinates, assigned to lower-channel contact coordinates, or shown to be less singular in the simultaneous external-shell sense above. They are not allowed to contribute an additional independent charged one-particle residue.

5. The extraction is covariant under finite same-flux coordinate changes: if z_ℓ is replaced by $M_\ell z_\ell$, with M_ℓ invertible on the chosen coordinate frame, then L_ℓ is replaced by $L_\ell M_\ell^{-1}$, the residue tensor is multiplied by M_ℓ on that external coordinate, and the coefficient $\mathcal{A}_{\boldsymbol{\lambda}}$ is unchanged.

This hypothesis is not a theorem about gauge theory. It names the analytic work that a Wilson-line LSZ theorem must do. One must construct the boundary-value distributions for the noncompact dressed insertions, prove uniformity in the truncation schedules $L_\ell(t)$ and $\varepsilon_\ell(t)$, identify the finite-rank external singular part, control contact terms from the endpoints and cusps of the lines, and match the resulting coefficient to the Dollard–Cook wave maps. In massless QED the isolated pole in (22.42) is expected to be replaced by a threshold coefficient in a velocity/flux representation; the same interface then has to be rewritten with the threshold functional in place of the mass-shell residue.

22.5.6 Finite-Regulator Dressed LSZ

There is nevertheless a precise finite-regulator statement. It is not the full massless charged scattering theorem; it is the part of the desired theory that follows from the usual LSZ argument once a dressed charged creator and an isolated charged shell have been supplied. Equivalently, it is the isolated-shell realization of Hypothesis 22.14 under the stronger regularity assumptions listed below.

Theorem 22.15 (Finite-regulator dressed LSZ)**.** *Fix a gauge theory with a regulator or screening deformation for which:*

1. *the physical Hilbert space contains an isolated massive charged one-particle shell Σ_m^+ in a charged sector;*
2. *for each external charged sector and mass there is a finite family of translation-covariant gauge-invariant dressed operators $\Psi_a(x)$ of the form (22.8), or Coulombic variants, whose smearings are defined on a common dense domain and whose one-particle overlap map has full rank on the chosen charged one-particle multiplicity space;*
3. *the dressed two-point distribution has, in a neighborhood of Σ_m^+, the pole decomposition*
$$\widehat{G}_{ab}(p) = \int \mathrm{d}^D x\, \mathrm{e}^{\mathrm{i}p\cdot x} \langle \Omega | T\{\Psi_a(x)\Psi_b^\dagger(0)\}|\Omega\rangle = \frac{\mathrm{i}Z_{ab}}{p^2+m^2-\mathrm{i}0} + \widehat{G}^{\mathrm{reg}}_{ab}(p), \tag{22.44}$$
where $Z = (Z_{ab})$ is the positive Gram matrix of those one-particle overlaps and $\widehat{G}^{\mathrm{reg}}$ is less singular at the shell;
4. *the large-time wave operators built from these dressed creators exist, and time-ordered dressed correlators have the usual boundary-value and wave-packet regularity needed for the LSZ integration by parts.*

For the displayed square-root formula choose, for each external charged one-particle multiplicity space, a square subfamily of dressed coordinates whose overlap matrix is invertible; equivalently restrict Z to the range spanned by that frame and regard it as a positive definite matrix there. If one keeps a rectangular overcomplete family, replace every $Z^{-1/2}$ below by a chosen left inverse of the overlap map as in (22.14). With the square-frame convention, let $Z^{-1/2}$ be the positive inverse square root of this positive definite Gram matrix. Then the simultaneous external residue of the dressed time-ordered $(m+n)$-point function equals the matrix element of the scattering operator between the dressed asymptotic charged states:

$$\begin{aligned}&\prod_{r=1}^{m}\big[(Z^{-1/2})_{a'_r b'_r}({p'_r}^2+m_r^2)\big]\prod_{s=1}^{n}\big[(p_s^2+m_s^2)(Z^{-1/2})_{b_s a_s}\big]\,\widehat{G}^{\,b'_1\cdots b'_m}{}_{b_1\cdots b_n}(p'_1,\ldots;p_1,\ldots)\\ &\qquad\longrightarrow \langle p'_1,a'_1;\ldots;p'_m,a'_m,\mathrm{out}|S|p_1,a_1;\ldots;p_n,a_n,\mathrm{in}\rangle_{\mathrm{conn}},\end{aligned} \tag{22.45}$$

after wave-packet smearing and removal of the total momentum-conservation delta distribution, with the same connected/disconnected convention as in the ordinary LSZ chapter.

Proof. The hypotheses put the dressed fields in exactly the analytic position in which LSZ is a residue calculation on the one-particle spectral summand. We spell out the reduction because the fields Ψ_a are not assumed to be localized in bounded spacetime regions. Choose a positive-frequency Klein–Gordon wave packet f whose Fourier transform is supported in the isolating neighborhood of the shell $p^2+m^2=0$. The pole decomposition (22.44) says that the distribution $\langle\Omega|\Psi_b^\dagger(g)\Psi_a(f)|\Omega\rangle$ has one-particle part

$$\sum_{c,d}\langle g,c|f,d\rangle\, Z_{ad}\,\delta_{cb},$$

in the mass-shell representation, with the regular part contributing a smooth function of the external invariant. Equivalently,

$$P_1\Psi_a(f)\Omega = \sum_b (Z^{1/2})_a{}^b\,|f,b\rangle, \tag{22.46}$$

after choosing the positive square root on the finite multiplicity space.

For one incoming external leg, multiplication of the Fourier transform by (p^2+m^2) is the Fourier transform of $(-\Box+m^2)$ in the corresponding coordinate, in the present mostly-plus Lorentzian convention. Pair this differentiated coordinate with the wave packet f and integrate over the time slab $[-T,T]$. On every open time ordering chamber the differentiated time-ordered correlator is the Klein–Gordon operator applied to the corresponding distributional matrix element. The spectral regularity assumption implies that the part orthogonal to the isolated one-particle summand remains regular at the shell; after the external factor (p^2+m^2) is taken to the shell, that regular part gives zero. The integrations by parts therefore leave only the slab boundaries and the standard equal-time contact terms. The latter are local in the external time variable and are again regular functions of the external invariant after wave-packet smearing, so they disappear in the simultaneous mass-shell residue.

The boundary term at $t=-T$ is, by the assumed existence of the dressed Haag–Ruelle limit, the incoming wave operator applied to the one-particle vector (22.46); the boundary term at $t=+T$ gives the outgoing wave operator. Repeating the argument for all external legs, with the usual induction over time orderings and with disconnected vacuum components subtracted after the total momentum delta distribution is removed, gives the scattering matrix element written in the dressed one-particle coordinates. Multiplication by $Z^{-1/2}$ on every external multiplicity index converts those coordinates to the orthonormal mass-shell basis, and yields (22.45). No step requires Ψ_a to be a bounded-region local observable; the required substitute for locality is exactly the assumed existence and regularity of the dressed large-time limits. $\Box$

The theorem is useful because it isolates what is known from what remains open. At finite infrared regulator or in a screened phase, the formal dressed-correlator residue has a precise Hilbert-space meaning if the listed hypotheses hold. Removing the regulator in four-dimensional QED is a different operation: the pole in (22.44) is expected to be replaced by charged threshold behavior, and the asymptotic Hilbert space must remember the angular electric-flux sector.

22.5.7 Dressing Dependence as Coordinate Dependence

Path dependence has two different meanings. A deformation that changes the field at infinity changes the charged sector. A compact deformation, or a replacement of one short-distance dressing by another with the same asymptotic flux, changes the interpolating coordinate used to reach the same charged state.

Compact dressing changes as LSZ-coordinate changes. In the finite-regulator setting of Theorem 22.15, let Ψ_a and Ψ'_a be two families of dressed charged interpolating operators with the same charge, spin, mass, and asymptotic flux. Suppose their one-particle matrix elements are related by an invertible matrix M,

$$P_1\Psi'_a(f)\Omega = M_a{}^b\,P_1\Psi_b(f)\Omega + \text{terms orthogonal to } \mathcal{H}_1$$

for all wave packets f supported near the shell. Then the LSZ residues computed with Ψ' give the same scattering matrix after the external normalization matrix $Z'^{-1/2}$ is used. The singular part of the two-point function transforms as

$$Z' = MZM^{\dagger}.$$

The amputated external coordinates obtained from Ψ' are therefore related to those obtained from Ψ by the unitary polar part of $MZ^{1/2}$ acting inside the one-particle multiplicity space. LSZ computes matrix elements between orthonormal one-particle vectors, not between the raw operator labels. After multiplying by $Z'^{-1/2}$, this coordinate change is removed on every external leg. Contributions orthogonal to $\mathcal{H}_1$ are regular at the isolated shell and vanish in the external residue.

For abelian Wilson lines this statement has a concrete finite-regulator meaning. Let γ and γ' be two oriented paths beginning at x and sharing the same asymptotic ray. Truncate both paths at a large radius and use the same endpoint cap at the truncation surface. If the difference of the truncated paths is the boundary of an oriented compact surface $S_{\gamma',\gamma}$,

$$\partial S_{\gamma',\gamma} = \gamma'_R - \gamma_R,$$

then, for a smooth regulated connection A with curvature $F = \mathrm{d}A$, Stokes' identity gives

$$\int_{\gamma'_R} A - \int_{\gamma_R} A = \int_{S_{\gamma',\gamma}} F. \tag{22.47}$$

Hence

$$\Psi_{q,\gamma'}(x) = \Psi_{q,\gamma}(x)\, \exp\left\{ \mathrm{i}q \int_{S_{\gamma',\gamma}} F \right\}$$

in the abelian regulated theory, with the same short-distance prescription at the common starting point. The extra factor is gauge invariant, neutral, and localized in the compact region swept out by the deformation. It can change the overlap vector $P_1\Psi_{q,\gamma}(f)\Omega$ by a finite amount, but it does not change the Gauss-law boundary charge or the asymptotic flux label. This is the precise setting in which the preceding LSZ coordinate calculation applies.

For a nonabelian connection the corresponding infinitesimal statement is a curvature-insertion formula rather than an ordinary scalar Stokes identity. With $U_\gamma(s_2, s_1)$ defined by

$$\partial_{s_2} U_\gamma(s_2, s_1) = -\dot{\gamma}^\mu(s_2) A_\mu(\gamma(s_2)) U_\gamma(s_2, s_1), \qquad U_\gamma(s_1, s_1) = 1,$$

and with a compact path variation $\eta^\nu(s)$ that vanishes at the endpoint and on the asymptotic tail, one has

$$\delta U_\gamma(\infty, 0) = \int_0^\infty \mathrm{d}s\, U_\gamma(\infty, s)\, F_{\mu\nu}(\gamma(s)) \dot{\gamma}^\mu(s) \eta^\nu(s)\, U_\gamma(s, 0),$$

in the same sign convention as the preceding differential equation. Thus a compact nonabelian deformation inserts covariantly transported curvature along the finite strip between the two dressings; it is not a change of the endpoint representation at infinity. None of these identities proves independence under changes of the asymptotic ray, because such changes are not compact: they modify the electric flux at infinity. In massless QED the asymptotic ray or Coulomb profile is tied to the charged velocity, and changing it changes the infrared representation rather than merely the interpolating coordinate.

22.5.8 Asymptotic Wilson Lines, Flux, and Eikonal Data

The preceding definitions identify a charged dressing as part of the asymptotic datum. We now spell out the elementary calculation behind this statement. Work in four spacetime dimensions with an abelian gauge field, and let a massive charged particle have charge q, momentum p^μ, mass $m > 0$, and four-velocity

$$u^\mu = \frac{p^\mu}{m}, \qquad u_\mu u^\mu = -1, \qquad p^0 > 0.$$

The future-directed straight dressing from x is represented, before the infrared limit is taken, by the regulated current

$$J^\mu_{q,u,\epsilon}(y;x) = q\int_0^\infty \mathrm{d}\tau\, \mathrm{e}^{-\epsilon\tau} u^\mu \delta^{(4)}(y - x - u\tau), \qquad \epsilon > 0. \tag{22.48}$$

The corresponding Wilson factor is the exponential of $\mathrm{i}\int A_\mu J^\mu$. The factor ϵ is not a physical photon mass; it is an adiabatic endpoint regulator used to define the half-line Fourier transform. The endpoint at $\tau = 0$ is the charged field insertion, and the endpoint at $\tau = \infty$ is the boundary charge recorded by (22.15).

Worldline dressing and the soft eikonal denominator. With the Fourier convention

$$\widehat{J}^\mu(k) = \int \mathrm{d}^4 y\, \mathrm{e}^{\mathrm{i}k\cdot y} J^\mu(y),$$

the regulated current (22.48) has

$$\widehat{J}^\mu_{q,u,\epsilon}(k;x) = q\, u^\mu\, \mathrm{e}^{\mathrm{i}k\cdot x} \frac{1}{\epsilon - \mathrm{i}k\cdot u}. \tag{22.49}$$

For an on-shell soft photon of momentum k and physical polarization $e_h(k)$, the homogeneous soft factor carried by the dressing is therefore

$$f^h_{p,\epsilon}(k) = q\, \frac{p\cdot e^*_h(k)}{\epsilon m - \mathrm{i}p\cdot k}\, \mathrm{e}^{\mathrm{i}k\cdot x}. \tag{22.50}$$

After the endpoint prescription is fixed, this is the same eikonal denominator that appears in the leading Faddeev–Kulish soft cloud and in the leading soft-photon theorem. Substituting (22.48) gives

$$\begin{aligned}\widehat{J}^\mu_{q,u,\epsilon}(k;x) &= qu^\mu \int_0^\infty \mathrm{d}\tau\, \mathrm{e}^{-\epsilon\tau} \mathrm{e}^{\mathrm{i}k\cdot(x+u\tau)} \\ &= qu^\mu \mathrm{e}^{\mathrm{i}k\cdot x} \int_0^\infty \mathrm{d}\tau\, \mathrm{e}^{-(\epsilon - \mathrm{i}k\cdot u)\tau} = qu^\mu \mathrm{e}^{\mathrm{i}k\cdot x} \frac{1}{\epsilon - \mathrm{i}k\cdot u}.\end{aligned}$$

Since $p^\mu = mu^\mu$, contraction with $e^*_h(k)$ gives (22.50). The sign of the infinitesimal imaginary part depends on whether the half-line is future or past directed and on the incoming/outgoing convention. The load-bearing structure is the homogeneous denominator $p\cdot k$, not the regulator sign.

The same charged sector can be described at spacelike infinity by its electric flux. If $\mathbf{v} = \mathbf{p}/p^0$ and $\mathbf{n} \in S^2$, the boosted Coulomb radial flux density is

$$\mathcal{E}_{q,\mathbf{v}}(\mathbf{n}) = \frac{q}{4\pi} \frac{1 - |\mathbf{v}|^2}{(1 - \mathbf{v}\cdot\mathbf{n})^2}. \tag{22.51}$$

It defines the value of the limiting flux functional

$$\Phi_\infty(f) = \int_{S^2} \mathrm{d}\Omega(\mathbf{n})\, f(\mathbf{n})\mathcal{E}_{q,\mathbf{v}}(\mathbf{n})$$

on a one-particle charged asymptotic datum with velocity $\mathbf{v}$.

Velocity-labelled flux sectors. For $|\mathbf{v}| < 1$,

$$\int_{S^2} \mathrm{d}\Omega(\mathbf{n})\, \mathcal{E}_{q,\mathbf{v}}(\mathbf{n}) = q. \tag{22.52}$$

If $q \neq 0$, the function $\mathcal{E}_{q,\mathbf{v}}(\mathbf{n})$ determines $\mathbf{v}$. Thus different charged velocities carry different sharp angular flux data. Rotate coordinates so that $\mathbf{v} = \beta \mathbf{e}_3$. Then

$$\begin{aligned}\int_{S^2} \mathrm{d}\Omega\, \frac{1-\beta^2}{(1-\beta\cos\theta)^2} &= 2\pi(1-\beta^2)\int_{-1}^{1} \frac{\mathrm{d}z}{(1-\beta z)^2} \\ &= 2\pi(1-\beta^2)\left[\frac{1}{\beta(1-\beta z)}\right]_{z=-1}^{z=1} = 4\pi,\end{aligned}$$

which proves (22.52). For $q \neq 0$, the maximum of $\mathcal{E}_{q,\mathbf{v}}$ occurs in the direction of $\mathbf{v}$ when $\mathbf{v} \neq 0$, while the ratio of the maximum to the minimum is

$$\left(\frac{1+|\mathbf{v}|}{1-|\mathbf{v}|}\right)^2.$$

This ratio determines $|\mathbf{v}|$, and the direction of the maximum determines the direction of $\mathbf{v}$. The case $\mathbf{v} = 0$ is the unique rotationally invariant density.

Neutral boundary charge versus angular flux sector. The boundary-charge condition (22.16) is the zero-mode of a sharper asymptotic datum. For an oriented list of charged asymptotic coordinates

$$I = ((\epsilon_1, q_1, \mathbf{v}_1), \ldots, (\epsilon_N, q_N, \mathbf{v}_N)), \qquad \epsilon_a = \pm 1,$$

define the signed angular flux profile

$$\mathcal{E}_I(\mathbf{n}) = \sum_{a=1}^{N} \epsilon_a\, \mathcal{E}_{q_a,\mathbf{v}_a}(\mathbf{n}). \tag{22.53}$$

By (22.52),

$$\int_{S^2} \mathrm{d}\Omega(\mathbf{n})\, \mathcal{E}_I(\mathbf{n}) = \sum_{a=1}^{N} \epsilon_a q_a.$$

Thus a vacuum dressed correlator can obey the ordinary boundary-charge Ward identity while still having nonzero angle-dependent flux data. For instance, two opposite charges with the same velocity have $\mathcal{E}_I(\mathbf{n}) = 0$ pointwise. Two opposite charges with distinct velocities have zero total boundary charge but, unless the velocity labels coincide, the profile $\mathcal{E}_I$ is not the zero function. The charged LSZ datum therefore contains more information than the integer or real total charge: it contains the limiting angular Gauss-law profile, or an equivalent large-gauge-charge functional. A same-flux dressing change is a finite coordinate change of the type described above; a change of $\mathcal{E}_I$ changes the asymptotic charged sector and belongs to the open Haag–Ruelle problem, not to a harmless path-independence argument.

22.5.9 Soft Coherent Norm and Velocity Superselection

The preceding flux density is visible already in the finite-cutoff soft photon Fock calculation. The calculation is not a construction of massless QED scattering; it is a precise diagnostic of why a single ordinary photon Fock representation cannot contain sharp charged particles of all velocities as normal vectors after the infrared cutoff is removed.

Fix $0 < \lambda < \Lambda < \infty$. The finite soft-photon one-particle space is

$$\mathfrak{h}_{\lambda,\Lambda} = L^2\left([\lambda,\Lambda] \times S^2, \frac{r\,\mathrm{d}r\,\mathrm{d}\Omega(\mathbf{n})}{2(2\pi)^3}; \mathbb{C}^2\right),$$

where $k = (r, r\mathbf{n})$ is null and the two complex components label a measurable orthonormal transverse polarization frame $\epsilon_h(\mathbf{n})$, $h = 1, 2$, satisfying $\mathbf{n} \cdot \epsilon_h(\mathbf{n}) = 0$. The polarization frame is a coordinate choice; the sum over h below is the transverse projector

$$P_{\mathbf{n}}^{ij} = \delta^{ij} - n^i n^j.$$

For a massive charge q with velocity $\mathbf{v}$, $|\mathbf{v}| < 1$, the homogeneous soft part of the worldline dressing (22.50) is represented by

$$F^h_{q,\mathbf{v},\lambda,\Lambda}(r,\mathbf{n}) = -q\,\frac{\mathbf{v} \cdot \epsilon_h(\mathbf{n})}{r(1 - \mathbf{v} \cdot \mathbf{n})}\mathbf{1}_{\lambda<r<\Lambda}. \tag{22.54}$$

The mass and Lorentz factor cancel because

$$\frac{p \cdot e_h(k)}{p \cdot k} = -\frac{\mathbf{v} \cdot \epsilon_h(\mathbf{n})}{r(1 - \mathbf{v} \cdot \mathbf{n})}$$

for $e^0_h = 0$, $p^\mu = m\gamma(1, \mathbf{v})$, and the mostly-plus Minkowski product.

Proposition 22.16 (Velocity separation in the soft coherent norm)**.** *For two velocities $\mathbf{v}, \mathbf{w} \in B_1(0) \subset \mathbb{R}^3$ and the same nonzero charge q,*

$$\|F_{q,\mathbf{v},\lambda,\Lambda} - F_{q,\mathbf{w},\lambda,\Lambda}\|^2_{\mathfrak{h}_{\lambda,\Lambda}} = \frac{q^2}{2(2\pi)^3}\log\frac{\Lambda}{\lambda}\,\mathcal{A}(\mathbf{v},\mathbf{w}), \tag{22.55}$$

where

$$\mathcal{A}(\mathbf{v},\mathbf{w}) = \int_{S^2} \mathrm{d}\Omega(\mathbf{n}) \left| P_{\mathbf{n}}\left(\frac{\mathbf{v}}{1 - \mathbf{v} \cdot \mathbf{n}} - \frac{\mathbf{w}}{1 - \mathbf{w} \cdot \mathbf{n}}\right)\right|^2. \tag{22.56}$$

The coefficient obeys $\mathcal{A}(\mathbf{v},\mathbf{w}) \geq 0$, and $\mathcal{A}(\mathbf{v},\mathbf{w}) = 0$ if and only if $\mathbf{v} = \mathbf{w}$.

Proof. The radial dependence of (22.54) is r^{-1}. Therefore the measure $r\,\mathrm{d}r\,\mathrm{d}\Omega/(2(2\pi)^3)$ gives the logarithm

$$\int_\lambda^\Lambda \frac{\mathrm{d}r}{r} = \log\frac{\Lambda}{\lambda}.$$

After summing over the two physical polarizations,

$$\sum_{h=1}^{2}\left|\left(\frac{\mathbf{v}}{1 - \mathbf{v} \cdot \mathbf{n}} - \frac{\mathbf{w}}{1 - \mathbf{w} \cdot \mathbf{n}}\right) \cdot \epsilon_h(\mathbf{n})\right|^2 = \left|P_{\mathbf{n}}\left(\frac{\mathbf{v}}{1 - \mathbf{v} \cdot \mathbf{n}} - \frac{\mathbf{w}}{1 - \mathbf{w} \cdot \mathbf{n}}\right)\right|^2,$$

which proves (22.55). The integrand in $\mathcal{A}(\mathbf{v},\mathbf{w})$ is the squared norm of a transverse vector for each $\mathbf{n}$, and the sphere measure is positive, so $\mathcal{A}(\mathbf{v},\mathbf{w}) \geq 0$.

It remains to characterize the zero set. If $\mathcal{A}(\mathbf{v},\mathbf{w}) = 0$, continuity implies that the displayed transverse projection vanishes for every $\mathbf{n}$. Suppose first that $\mathbf{v} \neq 0$. Put $\mathbf{n} = \mathbf{v}/|\mathbf{v}|$. The projection of $\mathbf{v}$ onto the plane orthogonal to $\mathbf{n}$ is zero, hence the projection of $\mathbf{w}$ onto that plane must also vanish; $\mathbf{w}$ is parallel to $\mathbf{v}$. Writing $\mathbf{v} = \beta\mathbf{e}_3$ and $\mathbf{w} = \alpha\mathbf{e}_3$, and choosing any $\mathbf{n}$ not parallel to $\mathbf{e}_3$, the vanishing condition becomes

$$\frac{\beta}{1-\beta n_3} = \frac{\alpha}{1-\alpha n_3}.$$

Thus $\beta = \alpha$. If $\mathbf{v} = 0$, the same condition says that the transverse projection of $\mathbf{w}$ vanishes for every $\mathbf{n}$; this forces $\mathbf{w} = 0$. Hence the coefficient vanishes exactly on the diagonal $\mathbf{v} = \mathbf{w}$. □

Let $\mathcal{E}(F)$ denote the normalized bosonic coherent vector in the finite-cutoff photon Fock space,

$$\mathcal{E}(F) = \mathrm{e}^{-\|F\|^2/2} \bigoplus_{n\geq 0} \frac{F^{\otimes_s n}}{\sqrt{n!}}.$$

Summing the Fock inner products gives

$$|\langle \mathcal{E}(F), \mathcal{E}(G)\rangle| = \exp\left\{-\frac{1}{2}\|F-G\|^2\right\}.$$

Therefore Proposition 22.16 implies that, for $\mathbf{v} \neq \mathbf{w}$,

$$|\langle \mathcal{E}(F_{q,\mathbf{v},\lambda,\Lambda}), \mathcal{E}(F_{q,\mathbf{w},\lambda,\Lambda})\rangle| = \exp\left\{-\frac{q^2}{4(2\pi)^3}\log\frac{\Lambda}{\lambda}\,\mathcal{A}(\mathbf{v},\mathbf{w})\right\} \longrightarrow 0 \qquad (\lambda \downarrow 0).$$

This is the finite-Fock-space calculation behind velocity-labelled charged sectors. It is stronger than saying that a bare electron emits many soft photons: different charged velocities determine different infrared coherent directions. A Haag–Ruelle replacement for massless charged sectors must therefore build its asymptotic representation with the charged velocity and flux label included from the start.

22.5.10 Weyl Implementation and the Failure of a Common Photon Fock Space

The preceding norm calculation has an operator-algebraic meaning. For $0 < \Lambda < \infty$, let $\mathfrak{h}_{0,\Lambda}$ be the photon one-particle space obtained by replacing the interval $[\lambda,\Lambda]$ in $\mathfrak{h}_{\lambda,\Lambda}$ by $(0,\Lambda]$. For each $\lambda > 0$, $\mathfrak{h}_{\lambda,\Lambda}$ is a closed subspace of $\mathfrak{h}_{0,\Lambda}$ by extension by zero on $(0,\lambda]$. Let

$$\mathcal{F}_{0,\Lambda} = \mathcal{F}_s(\mathfrak{h}_{0,\Lambda})$$

be the corresponding fixed photon Fock space, and let

$$W(f) = \exp\Big(a^\dagger(f) - a(f)\Big), \qquad f \in \mathfrak{h}_{0,\Lambda},$$

be the Weyl operator. With the real symplectic form

$$\sigma(f,g) = 2\,\mathrm{Im}\langle f,g\rangle_{\mathfrak{h}},$$

the Weyl relations are

$$W(f)W(g) = \exp\left(-\frac{\mathrm{i}}{2}\sigma(f,g)\right)W(f+g), \qquad W(f)^* = W(-f), \tag{22.57}$$

and the Fock vacuum state is

$$\omega_0(W(f)) = \langle \Omega, W(f)\Omega\rangle = \exp\left(-\frac{1}{2}\|f\|_{\mathfrak{h}}^2\right).$$

The coherent state with soft profile F is

$$\omega_F(A) = \langle \mathcal{E}(F), A\mathcal{E}(F)\rangle = \langle W(F)\Omega, AW(F)\Omega\rangle.$$

On Weyl generators this gives

$$\omega_F(W(f)) = \exp(\mathrm{i}\sigma(F,f))\exp\left(-\frac{1}{2}\|f\|_{\mathfrak{h}}^2\right). \tag{22.58}$$

Thus the soft cloud changes the representation by adding a linear phase on the Weyl algebra. At finite cutoff this phase is inner, implemented by the unitary $W(F)$. The infrared question is whether the implementing unitaries have a limit on one fixed Fock representation as $\lambda \downarrow 0$.

Proposition 22.17 (No common Fock implementer for distinct charged velocities)**.** *Fix $q \neq 0$, $0 < \Lambda < \infty$, and two distinct velocities $\mathbf{v}, \mathbf{w} \in B_1(0)$. Put*

$$D_{\lambda,\Lambda} = F_{q,\mathbf{v},\lambda,\Lambda} - F_{q,\mathbf{w},\lambda,\Lambda}.$$

Then $\|D_{\lambda,\Lambda}\| \to \infty$ as $\lambda \downarrow 0$. Consequently the finite-cutoff Weyl implementers $W(D_{\lambda,\Lambda})$ have no strong operator limit, and no nonzero weak operator limit, on the fixed photon Fock representation.

Proof. The divergence of $\|D_{\lambda,\Lambda}\|$ is exactly Proposition 22.16: for $\mathbf{v} \neq \mathbf{w}$,

$$\|D_{\lambda,\Lambda}\|^2 = \frac{q^2}{2(2\pi)^3}\log\frac{\Lambda}{\lambda}\,\mathcal{A}(\mathbf{v},\mathbf{w}), \qquad \mathcal{A}(\mathbf{v},\mathbf{w}) > 0.$$

It remains to translate this divergence into an operator statement. Let $\mathcal{D}_{\text{fin}}$ be the dense subspace of finite-particle vectors. Matrix elements of $W(D)$ between vectors in $\mathcal{D}_{\text{fin}}$ are finite sums of polynomials in the finitely many inner products $\langle f_i, D\rangle$ and their complex conjugates, multiplied by the vacuum Gaussian factor

$$\langle \Omega, W(D)\Omega\rangle = \exp\left(-\frac{1}{2}\|D\|^2\right).$$

This follows by commuting $a(f)$ and $a^\dagger(f)$ through $W(D)$, or equivalently by differentiating the Weyl relation (22.57) with respect to finite test directions. Polynomial growth cannot compensate the Gaussian decay. Hence for every $\xi, \eta \in \mathcal{D}_{\text{fin}}$,

$$\langle \xi, W(D_{\lambda,\Lambda})\eta\rangle \longrightarrow 0 \qquad (\lambda \downarrow 0).$$

The operators $W(D_{\lambda,\Lambda})$ are unitary and therefore uniformly bounded, so density of $\mathcal{D}_{\text{fin}}$ gives weak convergence to zero on all Fock vectors. A strong limit of unitaries, if it existed, would have the same weak limit and would therefore be the zero operator; but a strong limit of unitaries is an isometry on every vector whose norm limit exists:

$$\|U\eta\| = \lim_{\lambda\downarrow 0}\|W(D_{\lambda,\Lambda})\eta\| = \|\eta\|.$$

The zero operator is not an isometry. Thus no strong limit exists. ∎

Definition 22.18 (Hilbert-equivalent soft dressings)**.** Fix the ultraviolet cutoff Λ and use the photon one-particle space $\mathfrak{h}_{0,\Lambda}$ introduced above. Let

$$F_\lambda, G_\lambda \in \mathfrak{h}_{\lambda,\Lambda} \subset \mathfrak{h}_{0,\Lambda}, \qquad 0 < \lambda < \Lambda,$$

be two infrared-regulated soft profiles for the same charged insertion. They are called Hilbert-equivalent at the cutoff Λ if there exists a vector $\Delta \in \mathfrak{h}_{0,\Lambda}$ such that

$$\lim_{\lambda \downarrow 0} \|G_\lambda - F_\lambda - \Delta\|_{\mathfrak{h}_{0,\Lambda}} = 0. \tag{22.59}$$

The vector Δ is the finite soft-coordinate change between the two dressings. This definition is deliberately a Hilbert-space definition: it does not identify dressings whose difference changes the limiting Gauss-law flux, because such a difference is not represented by a vector of $\mathfrak{h}_{0,\Lambda}$.

Proposition 22.19 (Hilbert soft changes are inner Weyl coordinate changes)**.** *Let F_λ and G_λ be Hilbert-equivalent soft profiles in the sense of Definition 22.18, and put*

$$\Delta_\lambda = G_\lambda - F_\lambda.$$

For every finite Weyl polynomial A in the generators $W(f)$, $f \in \mathfrak{h}_{0,\Lambda}$,

$$\omega_{G_\lambda}(A) - \omega_{F_\lambda}(W(\Delta_\lambda)^* A W(\Delta_\lambda)) = 0, \tag{22.60}$$

and the implementing unitaries have the strong limit

$$W(\Delta_\lambda) \longrightarrow W(\Delta) \qquad \text{on } \mathcal{F}_{0,\Lambda}.$$

Consequently

$$|\langle \mathcal{E}(F_\lambda), \mathcal{E}(G_\lambda)\rangle| = \exp\left(-\frac{1}{2}\|\Delta_\lambda\|^2\right) \longrightarrow \exp\left(-\frac{1}{2}\|\Delta\|^2\right) > 0.$$

Thus a Hilbert-equivalent soft change is a finite reparametrization of the same infrared representation, not a change of charged sector. The velocity change in Proposition 22.17 fails this criterion because $\|F_{q,\mathbf{v},\lambda,\Lambda} - F_{q,\mathbf{w},\lambda,\Lambda}\| \to \infty$ for $\mathbf{v} \neq \mathbf{w}$.

Proof. The equality (22.60) is first checked on one Weyl generator. The Weyl relations give

$$W(\Delta_\lambda)^* W(f) W(\Delta_\lambda) = \exp(\mathrm{i}\sigma(\Delta_\lambda, f))\, W(f),$$

and hence, using (22.58),

$$\begin{aligned}\omega_{F_\lambda}(W(\Delta_\lambda)^* W(f) W(\Delta_\lambda)) &= \exp(\mathrm{i}\sigma(\Delta_\lambda, f)) \exp(\mathrm{i}\sigma(F_\lambda, f)) \exp\left(-\frac{1}{2}\|f\|^2\right) \\ &= \exp(\mathrm{i}\sigma(G_\lambda, f)) \exp\left(-\frac{1}{2}\|f\|^2\right) = \omega_{G_\lambda}(W(f)).\end{aligned}$$

Linearity and multiplication by the Weyl relations extend this identity to finite Weyl polynomials.

It remains to prove the strong convergence of the implementers. The coherent vectors

$$\mathcal{E}(K) = \mathrm{e}^{-\|K\|^2/2} \bigoplus_{n \geq 0} \frac{K^{\otimes_s n}}{\sqrt{n!}}, \qquad K \in \mathfrak{h}_{0,\Lambda},$$

span a dense subspace of $\mathcal{F}_{0,\Lambda}$. On such vectors,

$$W(H)\mathcal{E}(K) = \exp\left(-\frac{\mathrm{i}}{2}\sigma(H,K)\right)\mathcal{E}(K+H),$$

because $\mathcal{E}(K) = W(K)\Omega$ and (22.57) gives $W(H)W(K) = \exp[-\mathrm{i}\sigma(H,K)/2]W(H+K)$. If $H_n \to H$ in $\mathfrak{h}_{0,\Lambda}$, the displayed formula and the coherent-vector inner product

$$\langle \mathcal{E}(K_1), \mathcal{E}(K_2)\rangle = \exp\left(-\frac{1}{2}\|K_1\|^2 - \frac{1}{2}\|K_2\|^2 + \langle K_1, K_2\rangle\right)$$

show that $W(H_n)\mathcal{E}(K) \to W(H)\mathcal{E}(K)$ in norm. Since all $W(H_n)$ and $W(H)$ are unitary, density upgrades convergence on coherent vectors to strong convergence on the whole Fock space. Applying this to $H_n = \Delta_{\lambda_n}$ along every sequence $\lambda_n \downarrow 0$ proves

$$W(\Delta_\lambda) \to W(\Delta) \quad \text{strongly.}$$

The overlap formula follows by summing the Fock inner products of coherent vectors, and the limiting overlap is nonzero because $\Delta \in \mathfrak{h}_{0,\Lambda}$ has finite norm. The last assertion is then exactly Proposition 22.17. □

Proposition 22.17 is a precise version of the representation-theoretic obstruction. At every positive infrared cutoff, changing the soft cloud is an ordinary Weyl displacement on Fock space. Removing the cutoff separates two cases. A Hilbert-equivalent change remains an inner coordinate change by Proposition 22.19. A change of charged velocity escapes to infinity in the one-particle norm and is not implemented on the original Fock representation. Therefore a charged asymptotic Hilbert space must be organized as a direct integral or measurable field of infrared representations labelled by the limiting flux data, not as a single photon Fock space tensored with a Wigner one-particle electron space. This is still not the missing Haag–Ruelle theorem, but it is one of the structural inputs that theorem must contain.

The flux statement and the eikonal statement are two aspects of the same nonlocal charged datum. The Wilson line fixes how the charged insertion transforms under boundary gauge transformations; the boosted Coulomb field records the corresponding Gauss-law flux at spacelike infinity; the half-line Fourier transform gives the soft denominator used by perturbative dressed-state constructions. A local observable cannot change $\Phi_\infty(f)$, so a local observable cannot replace this noncompact dressing in a Haag–Ruelle construction. Conversely, specifying only the formal denominator $p \cdot k$ is not yet a scattering theorem: a full charged-sector Haag–Ruelle theory must still construct large-time limits in the representation labelled by the flux density (22.51).

22.5.11 Velocity-Fibered Soft Representations

The preceding propositions rule out one tempting but incorrect construction: one cannot keep a single photon Fock representation and implement all charged velocity changes by limiting Weyl unitaries. The minimal replacement is a fibered infrared representation. A scattering theory must still construct large-time limits, while this representation gives the correct representation-theoretic target for such a construction in a nonconfining charged sector.

This qualification is essential. The charged-sector construction discussed here is not a proposed scattering theory for colored quarks in a confining four-dimensional Yang–Mills theory. It applies only when the theory has physical charged superselection sectors with charged asymptotic data: for example electric charge in a nonconfining abelian gauge theory, or suitable unconfined charges on a Coulomb branch or in a deconfined regime after the infrared and representation-theoretic hypotheses have been stated. In a confining theory the gauge-invariant Wilson-line operator ending at infinity creates a state with a macroscopic flux string or fails to have a finite-energy charged-particle limit; the physical massive one-particle states, if present, are instead neutral hadrons or glueballs created by local gauge-invariant operators. Ordinary Haag–Ruelle theory is then the correct starting point for those neutral particles.

Fix $q \neq 0$ and $0 < \Lambda < \infty$. Let $\mathfrak{h}_{\mathrm{reg}} \subset \mathfrak{h}_{0,\Lambda}$ be a complex linear subspace of infrared-regular photon test functions such that, for every $\mathbf{v} \in B_1(0)$, the limit

$$\ell_{\mathbf{v}}(f) = \lim_{\lambda \downarrow 0} \sigma(F_{q,\mathbf{v},\lambda,\Lambda}, f) \tag{22.61}$$

exists for all $f \in \mathfrak{h}_{\mathrm{reg}}$. For example, the condition is satisfied on smooth compactly supported one-photon wave packets whose radial support is bounded away from $r = 0$. Larger infrared test spaces require an explicit condition on the radial behavior of f; the definition above records exactly the condition needed for the soft phase to be a finite functional.

Let $\mathfrak{W}_{\mathrm{reg}}$ be the Weyl $*$-algebra generated by $W(f)$, $f \in \mathfrak{h}_{\mathrm{reg}}$, with relations (22.57). The charged velocity $\mathbf{v}$ defines the state

$$\omega_{\mathbf{v}}(W(f)) = \exp(\mathrm{i}\ell_{\mathbf{v}}(f)) \exp\left(-\frac{1}{2}\|f\|^2_{\mathfrak{h}_{0,\Lambda}}\right), \qquad f \in \mathfrak{h}_{\mathrm{reg}}. \tag{22.62}$$

Let $(\mathcal{H}_{\mathbf{v}}, \pi_{\mathbf{v}}, \Omega_{\mathbf{v}})$ be the GNS representation of $\omega_{\mathbf{v}}$. If ν is a Borel measure on $B_1(0)$ for which the field $\mathbf{v} \mapsto (\mathcal{H}_{\mathbf{v}}, \pi_{\mathbf{v}})$ has been given a measurable structure, define

$$\mathcal{H}^{\mathrm{soft}}_{q,\nu} = \int^{\oplus}_{B_1(0)} \mathcal{H}_{\mathbf{v}}\, \mathrm{d}\nu(\mathbf{v}), \qquad \big(\pi_{q,\nu}(A)\Psi\big)(\mathbf{v}) = \pi_{\mathbf{v}}(A)\Psi(\mathbf{v}). \tag{22.63}$$

Remark 22.20 (Soft photon algebra preserves the velocity fibers). In the direct-integral representation (22.63), let P_E be multiplication by the characteristic function of a Borel set $E \subset B_1(0)$. Then

$$[P_E, \pi_{q,\nu}(A)] = 0 \qquad \text{for all } A \in \mathfrak{W}_{\mathrm{reg}}. \tag{22.64}$$

Consequently, if $E \cap E' = \varnothing$, then

$$\langle P_E\xi, \pi_{q,\nu}(A) P_{E'}\eta\rangle = 0 \qquad \text{for all } \xi, \eta \in \mathcal{H}^{\mathrm{soft}}_{q,\nu}, \quad A \in \mathfrak{W}_{\mathrm{reg}}. \tag{22.65}$$

Thus the soft photon Weyl algebra is diagonal in the charged velocity label. Any charged scattering operator that changes the massive particle momentum must include additional charged-sector dynamics supplied by operators beyond the soft photon algebra alone. Indeed, by the definition of the direct-integral action, for $\Psi \in \mathcal{H}^{\mathrm{soft}}_{q,\nu}$,

$$(P_E \pi_{q,\nu}(A)\Psi)(\mathbf{v}) = \mathbf{1}_E(\mathbf{v})\pi_{\mathbf{v}}(A)\Psi(\mathbf{v}) = (\pi_{q,\nu}(A) P_E \Psi)(\mathbf{v})$$

for ν-almost every $\mathbf{v}$. If $E \cap E' = \varnothing$, then $P_E P_{E'} = 0$, and therefore

$$\langle P_E\xi, \pi_{q,\nu}(A) P_{E'}\eta\rangle = \langle \xi, P_E \pi_{q,\nu}(A) P_{E'}\eta\rangle = \langle \xi, \pi_{q,\nu}(A) P_E P_{E'}\eta\rangle = 0.$$

This fibered construction is the algebraic form of the velocity superselection conclusion. It also states a precise target for the still missing charged Haag–Ruelle theorem: the theorem must construct incoming and outgoing charged wave operators whose asymptotic photon part lives in a measurable field of infrared representations labelled by the Gauss-law flux, and whose hard charged part supplies the off-diagonal momentum-changing dynamics. Treating the charged soft cloud as a removable unitary dressing on one Fock space would erase exactly the sector information proved above.

Open Problem 22.21 (Haag–Ruelle and LSZ for nonconfining dressed charged sectors). Construct a large-time scattering theory for nonconfining gauge-theory charged sectors using noncompact gauge-invariant charged creators. The construction should prove the analogue of Haag–Ruelle limits for the chosen dressing class, identify the asymptotic flux labels, and prove a dressed-correlator reduction formula whose external factors are the residues or threshold coefficients of the corresponding dressed two-point functions. The hypothesis "nonconfining" means that the charged sector is a genuine finite-energy physical sector with asymptotic charged data; it excludes colored charges in a confining phase, for which the physical scattering problem is the neutral hadron or glueball scattering problem. In massless QED this problem must be combined with the infraparticle flux-sector analysis of Chapter 44.

22.6 Wave Operators, Scattering Operator, and Completeness

Let

$$\Omega_{\mathrm{in}}, \Omega_{\mathrm{out}} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{H}$$

be the Haag–Ruelle wave operators, in the terminology introduced in Chapter 17, produced by Haag–Ruelle limits. They are isometric maps. Their construction has two equivalent topological descriptions. First, for each finite velocity-separated Fock vector

$$F = \psi_1 \odot \cdots \odot \psi_n$$

represented by regular creators B_j and packets h_j, the vector family

$$\Omega^{\mathrm{HR}}_{\mathrm{out},t} F = B_{1,t}(h_1) \cdots B_{n,t}(h_n)\Omega$$

converges in the Hilbert norm of $\mathcal{H}$. Second, after these vector limits are packaged as maps from the algebraic velocity-separated Fock domain to $\mathcal{H}$, the same assertion is strong convergence of the approximating maps:

$$\Omega^{\mathrm{HR}}_{\mathrm{out},t} F \longrightarrow \Omega_{\mathrm{out}} F \quad \text{in } \mathcal{H} \quad \text{for every fixed } F. \tag{22.66}$$

The incoming map is obtained by the same statement with $t \to -\infty$. Weak convergence of matrix elements follows from this norm convergence for each F, while weak convergence alone would not determine the isometric identification. No operator-norm limit over the unit sphere of $\mathcal{F}_s(\mathcal{H}_1)$ is part of the theorem. Their ranges

$$\mathcal{H}_{\mathrm{in}} = \operatorname{Ran} \Omega_{\mathrm{in}}, \qquad \mathcal{H}_{\mathrm{out}} = \operatorname{Ran} \Omega_{\mathrm{out}}$$

are the incoming and outgoing scattering subspaces. The comparison operator

$$S = \Omega^*_{\mathrm{out}} \Omega_{\mathrm{in}} \quad \text{on} \quad \mathcal{F}_s(\mathcal{H}_1)$$

is unitary on the sector when $\mathcal{H}_{\text{in}} = \mathcal{H}_{\text{out}}$. Its wave-packet matrix elements are

$$\begin{aligned}&\langle \varphi_1 \odot \cdots \odot \varphi_m,\, S(\psi_1 \odot \cdots \odot \psi_n)\rangle_{\mathcal{F}_s(\mathcal{H}_1)} \\ &\qquad = \langle \Omega_{\text{out}}(\varphi_1 \odot \cdots \odot \varphi_m),\, \Omega_{\text{in}}(\psi_1 \odot \cdots \odot \psi_n)\rangle_{\mathcal{H}}\,.\end{aligned}$$

This is the nonperturbative definition of the scattering matrix element in the massive isolated-shell sector.

Asymptotic completeness is a further assertion:

$$\mathcal{H}_{\text{sector}} = \mathcal{H}_{\text{in}} = \mathcal{H}_{\text{out}}.$$

It is a dynamical property of the model or sector. For nontrivial interacting massive local Wightman theories in four spacetime dimensions, full asymptotic completeness is presently an open problem rather than a theorem following from the axioms. Existing proofs are confined to restricted model classes, notably constructive or integrable models in lower spacetime dimensions and related weakened-locality settings. Bound states require additional isolated shells. Confinement can remove charged or colored one-particle shells from the observable spectrum while retaining hadronic shells. Long-range gauge forces and infraparticle spectra require modified asymptotic state spaces or detector observables; the charged-QED and physical-QCD versions are the open asymptotic-completeness problems formulated in Open Problems 44.10 and 49.22. Each case changes the large-time construction rather than the meaning of the massive theorem above.

22.7 Interface With LSZ and Perturbation Theory

The Haag–Ruelle construction defines the Hilbert-space target of LSZ. If a local field or local observable has a nonzero vacuum-to-one-particle matrix element on the isolated shell, then its two-point function contains the corresponding one-particle pole or spectral atom. LSZ extracts those external singularities from time-ordered correlation functions and thereby computes matrix elements of the scattering operator already defined by wave operators.

The logical dependencies are

$$\text{local net and spectrum} \longrightarrow \text{Haag–Ruelle wave operators} \longrightarrow S,$$

$$\text{time-ordered Green functions} \xrightarrow{\text{LSZ external residues}} \langle \text{out}|S|\text{in}\rangle,$$

$$\text{renormalized perturbation theory} \longrightarrow \text{Green-function expansion} \longrightarrow \text{expansion of } \langle \text{out}|S|\text{in}\rangle.$$

Thus Feynman graphs for Green functions acquire a scattering interpretation only after the asymptotic Hilbert-space data and the reduction theorem are in place. The same order applies whether the theory is presented by a Lagrangian, a Euclidean construction satisfying reconstruction hypotheses, a local net, or another nonperturbative definition.

Chapter 23

Local Data and Kernel Conventions

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

23.1 Purpose of This Chapter

The preceding part constructed the local Hilbert-space frameworks, the Källén–Lehmann spectral representation, Haag–Ruelle scattering states, and LSZ reduction. This chapter does not rederive those constructions. Its role is to fix the notation used in the analytic scattering chapters: Poincare-action conventions, one-particle normalizations, spectral-threshold symbols, and Green-function boundary-value conventions.

23.2 Status of Scalar Model Examples

Definition 23.1 (Three uses of scalar Lagrangian notation)**.** Scalar Lagrangian densities appear in this part at three logically different levels. First, at finite ultraviolet cutoff they define regulated Green functions and perturbative kernels. Second, in low spacetime dimensions, certain scalar models have constructive non-Gaussian continuum realizations, as catalogued in Table 11.2 and recalled in Chapter 15. Third, a four-dimensional $\lambda\phi^4$ expression may be used as a renormalized perturbative or effective-field-theory model below a cutoff, or as a conditional notation for an exact massive local QFT if such a theory has been separately constructed.

Remark 23.2 (Four-dimensional scalar status). For the standard reflection-positive lattice $\lambda\phi^4$

family in four Euclidean dimensions with $\lambda \geq 0$, the known triviality theorems apply to the usual critical or near-critical continuum scaling limits: the resulting continuum limits are Gaussian, so the interacting connected correlations vanish after the continuum field normalization. This is a theorem about that specified constructive scaling-limit problem. It is a different statement from the broader open question whether an ultraviolet-complete four-dimensional QFT could have a small-coupling expansion agreeing with formal ϕ_4^4 perturbation theory to all orders while differing nonperturbatively. The latter possibility would require its own construction and verification of locality, positivity, spectrum, and scattering hypotheses.

Consequently, when the following chapters use $D = 4$ scalar amplitudes, they are to be read either as finite-cutoff perturbative or effective descriptions, or as conditional consequences of the stated Hilbert-space and analytic hypotheses. The scattering machinery derives consequences from a massive local QFT with stable particles; it does not prove that a particular bare four-dimensional scalar polynomial supplies such a QFT after removing the cutoff. The corresponding Wilsonian cutoff-removal assertion is stated later as the finite-coordinate cutoff-removal estimate in Chapter 36. There the retained-coordinate limit is derived from explicit tuning, semigroup, and generated-integral convergence hypotheses, rather than being imported as informal renormalizability lore.

23.3 Poincare and Field Conventions

Convention 23.3 (Poincare action, metric, and field covariance)**.** The unitary representation is written

$$U(a, \Lambda), \qquad U(a, \mathbf{1}) = \exp(\mathrm{i}a_\mu P^\mu) = \exp(\mathrm{i}a^\mu P_\mu),$$

with group law

$$U(a', \Lambda')U(a, \Lambda) = U(a' + \Lambda' a, \Lambda'\Lambda).$$

The metric is mostly plus,

$$\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +), \qquad p^2 = \eta_{\mu\nu} p^\mu p^\nu,$$

and the spectrum condition is

$$\mathrm{Spec}(P) \subset \overline{V}_+ = \{p : p^0 \geq 0,\ p^2 \leq 0\}.$$

Here P in the spectrum condition denotes the contravariant energy-momentum P^μ, while $P_\mu = \eta_{\mu\nu}P^\nu$ is used in the exponent when the translation parameter is written with an upper index. The vacuum vector is denoted by Ω and satisfies

$$U(a, \Lambda)\Omega = \Omega.$$

For a local field transforming in a finite-dimensional Lorentz representation R,

$$U(a, \Lambda)^{-1}\widehat{\Phi}_\alpha(x)U(a, \Lambda) = R(\Lambda)_\alpha{}^\beta \widehat{\Phi}_\beta(\Lambda x + a).$$

Thus translations give

$$\widehat{\Phi}_\alpha(x) = \mathrm{e}^{-\mathrm{i}x^\mu P_\mu}\widehat{\Phi}_\alpha(0)\mathrm{e}^{\mathrm{i}x^\mu P_\mu}.$$

All such identities are identities of smeared operator-valued distributions on the stated common domains.

23.4 One-Particle Normalization

Definition 23.4 (One-particle normalization and spectral threshold)**.** For a stable scalar species of mass $m > 0$,

$$\Sigma_m^+ = \{(\omega_{\vec{p}}, \vec{p}) : \omega_{\vec{p}} = (\vec{p}^2 + m^2)^{1/2}\}$$

is the positive mass shell. Delta-normalized vectors and covariantly normalized vectors are related by

$$|\vec{p}\rangle = \left((2\pi)^d 2\omega_{\vec{p}}\right)^{1/2} |\vec{p}\rangle_\delta,$$

with

$${}_\delta\langle\vec{q} \mid \vec{p}\rangle_\delta = \delta^{(d)}(\vec{q} - \vec{p}), \qquad \langle\vec{q} \mid \vec{p}\rangle = (2\pi)^d 2\omega_{\vec{p}} \delta^{(d)}(\vec{q} - \vec{p}).$$

The invariant mass-shell measure is

$$\mathrm{d}\mu_m(\vec{p}) = \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d 2\omega_{\vec{p}}}.$$

If a scalar interpolating field $\widehat{\phi}$ creates the species, its one-particle residue is fixed by

$$\langle\Omega|\widehat{\phi}(0)|\vec{p}\rangle = Z_\phi^{1/2}, \qquad Z_\phi > 0.$$

This Z_ϕ is the atom of the Källén–Lehmann measure at $\mu^2 = m^2$, as derived in Chapter 14. The symbol M_{cont} denotes the lowest invariant mass in the continuous spectral support with the same quantum numbers. When the first continuum consists of two particles of the same mass m, $M_{\text{cont}} = 2m$.

23.5 Green-Function Boundary Values

Definition 23.5 (Time-ordered Green functions and boundary-value kernels)**.** For local fields $\widehat{\Phi}_{a_1}, \ldots, \widehat{\Phi}_{a_N}$, the time-ordered distribution is

$$G_{a_1 \cdots a_N}(x_1, \ldots, x_N) = \langle\Omega|\,\mathrm{T}\{\widehat{\Phi}_{a_1}(x_1) \cdots \widehat{\Phi}_{a_N}(x_N)\}|\Omega\rangle.$$

Products at coincident points require a specified time-ordering prescription or a renormalized composite-operator convention.

For a scalar two-point function, the positive spectral measure gives the boundary-value formula

$$G_2(x) = \int_0^\infty \mathrm{d}\rho_\phi(\mu^2)\, \Delta_F(x; \mu^2),$$

where

$$\Delta_F(x; \mu^2) = -\mathrm{i} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k \cdot x}}{k^2 + \mu^2 - \mathrm{i}0}.$$

The Euclidean two-point Schwinger function, when OS reconstruction hypotheses are satisfied, is

$$S_2(x_E) = \int_0^\infty \mathrm{d}\rho_\phi(\mu^2) \int \frac{\mathrm{d}^D k_E}{(2\pi)^D} \frac{\mathrm{e}^{\mathrm{i}k_E \cdot x_E}}{k_E^2 + \mu^2}.$$

The passage among Wightman, time-ordered, and Euclidean functions is governed by the tube-domain and OS hypotheses stated in the foundational chapters. In the present part these formulas serve only as notation and as inputs to LSZ, dispersion theory, and threshold analysis.

23.6 Regulated Functional Integrals as Downstream Data

Definition 23.6 (Regulated functional integral notation)**.** Regulated path integrals are used later to compute Green functions and their renormalized limits. If Λ is a regulator and J is a source, the notation

$$Z_\Lambda[J] = \int_{\mathcal{C}_\Lambda} D_\Lambda\varphi \, \exp\left(-S_{\Lambda,E}[\varphi; g_I(\Lambda)] + \int \mathrm{d}^D x_E \, J(x_E)\varphi(x_E)\right)$$

means the finite-dimensional or otherwise regulated integral specified by the chosen cutoff. For a finite-dimensional bosonic scalar regulator, $D_\Lambda\varphi$ is the reference density $D_\Lambda^{\mathrm{ref}}\varphi$ of Definition 11.3. If the quadratic part has been absorbed into a Gaussian reference measure, the same regulated integral is written with $\mathrm{d}\mu_{C_\Lambda}$ and an interaction weight. For gauge, fermionic, oscillatory, BV, or algebraic regularizations, $D_\Lambda\varphi$ names the regulated integration object supplied by that construction and must be specified before use. A continuum Schwinger function is a distributional limit of source derivatives of Z_Λ, after the cutoff dependence of the coefficients $g_I(\Lambda)$ has been chosen. Existence of such a limit requires a renormalization or construction theorem: the notation records the regulated objects and the topology of the intended limit. The systematic renormalized-source treatment begins in the renormalization and effective-action chapters.

Chapter 24

Scattering Kernels, LSZ, and Cluster Decomposition

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Definition 24.1 (Status of time-ordered correlators in the scattering chapters)**.** The time-ordered distributions used in this part have a fixed status. In an exact scattering statement, G_N^{c} denotes the connected time-ordered distribution associated with local Hilbert-space fields $\widehat{\phi}(x)$, obtained from the Wightman distributions by the specified ordering and boundary-value prescription. The input needed for LSZ is then spectral: the two-point function has an isolated one-particle pole of mass m and residue $Z_\phi > 0$, and the N-point time-ordered distribution has the corresponding external pole boundary values after smearing. No path integral is part of the definition of the scattering operator in this case; S remains $\Omega_{\mathrm{out}}^* \Omega_{\mathrm{in}}$ from Haag–Ruelle theory.

In perturbative examples, Feynman graphs, Feynman parameters, and loop integrals denote coefficientwise contributions to a regulated source functional $Z[J]$. Its connected functional is the coefficientwise logarithm of $Z[J]$, with the Lorentzian or Euclidean factor fixed by the chosen regulator convention; the full source-functional status declaration is given in Chapter 30, and the BPHZ status declaration is given in Chapter 32. These are formal or regulated calculations of the time-ordered distributions whose pole residues may then be inserted into the LSZ formula. Such graph calculations do not define S and do not assert the existence of an unregulated continuum path-integral measure.

If a Euclidean construction is used instead, the time-ordered Lorentzian distributions are obtained only after the stated Osterwalder–Schrader reconstruction or analytic-continuation hypotheses have been verified. Thus a symbol such as G^{c}_N, Z_ϕ, or a Feynman-parameter integral always carries one of these statuses: exact Hilbert-space distribution, Euclidean reconstructed distribution, or formal/regulator-dependent perturbative coefficient.

Definition 24.2 (Mostly-plus Mandelstam variables)**.** Throughout the scattering and analyticity chapters, physical $2 \to 2$ momenta are written as incoming positive-energy momenta p_1, p_2 and outgoing positive-energy momenta p_3, p_4, with

$$p_i^2 = -m_i^2, \qquad p_1 + p_2 = p_3 + p_4$$

in the mostly-plus metric. The Mandelstam variables are

$$s = -(p_1 + p_2)^2, \qquad t = -(p_1 - p_3)^2, \qquad u = -(p_1 - p_4)^2.$$

They obey

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2.$$

For equal masses $m_i = m$, in the center-of-mass frame,

$$s = E_{\mathrm{cm}}^2, \qquad t = -2|\vec{p}_*|^2(1 - \cos\theta), \qquad u = -2|\vec{p}_*|^2(1 + \cos\theta),$$

so physical s-channel scattering has $s \geq 4m^2$ and

$$-(s - 4m^2) \leq t \leq 0, \qquad -(s - 4m^2) \leq u \leq 0.$$

Positive real t or u in later fixed-t and crossing discussions is therefore a crossed-channel timelike invariant, not a physical s-channel scattering angle.

24.1 Cross-Volume Scattering Architecture

The scattering chapters use several inequivalent objects. This section is a routing convention for those objects before the detailed chapters branch apart. Its main rule is that an amplitude, an inclusive rate, a detector distribution, and a resonance pole are different outputs of different hypotheses. A formula may connect two of them only after the hypotheses on both sides have been stated.

Definition 24.3 (Status labels for scattering constructions)**.** In this part the following labels have fixed meanings.

1. A *theorem* is a statement proved from the stated Hilbert-space, locality, spectral, and domain hypotheses.
2. A *conditional theorem* is a theorem after additional analytic hypotheses have been supplied, for example a factorization, asymptotic completeness, or convergence assumption.
3. A *regulated construction* is a finite-cutoff or perturbative object with a specified regulator and subtraction prescription.
4. *Numerical evidence* is a finite-volume, lattice, bootstrap, or Monte Carlo datum that supports but does not replace the continuum construction.
5. An *open problem* is a required bridge whose hypotheses, convergence, or physical state-space construction are not supplied in this monograph.

Framework matrix.

Massive Haag–Ruelle/LSZ. The primary objects are the in/out maps, the scattering operator, and the external residues used to extract amplitudes. The input is a mass gap or isolated shell, local fields or almost-local creators, velocity separation, and distributional time ordering. Under those hypotheses this is a theorem producing wave-packet S-matrix elements and invariant amplitudes. It stops when charged sectors, confinement, or massless thresholds remove the isolated particle shell.

Charged QED and long-range sectors. The primary objects are dressed hard states, the soft photon cloud, and an inclusive detector resolution. The input is Gauss-law flux, the massless photon spectrum, an infrared cutoff or detector resolution, and a dressing or inclusive prescription. This gives conditional perturbative and algebraic constructions. The exact charged electron is not an ordinary LSZ particle of the massless theory; inclusive rates or dressed matrix elements replace bare charged S-matrix kernels.

Detector and event-shape observables. The primary observables are event-shape distributions and energy-flow measures. Light-ray detector products give one operator realization. The input is an outgoing physical state space, a positive detector measure, a contact convention, and either IRC safety or a declared nonperturbative function. The output is an observable distribution rather than an amplitude. Perturbative kernels enter only after matching, subtraction, and detector limits have been fixed.

Analytic S-matrix and resonances. The primary objects are partial waves, dispersion relations, and the sheet structure in which poles, cuts, and other singularities can be characterized. The input is a previously constructed amplitude, unitarity, crossing domain, spectral condition, and analyticity assumptions. This gives a conditional analytic continuation of amplitudes. A particular resonance pole is additional dynamical information, for example a zero of a continued inverse partial wave or a finite-volume/spectral analysis that fixes the continued singularity; it is not produced by analyticity alone. Resonance poles encode unstable states; they are not normalizable asymptotic particles.

Confining gauge theory and QCD factorization. The primary objects are hadronic cross sections, PDFs, TMDs, jets, and energy correlators. The input is color-singlet external states, operator definitions, factorization, Glauber status, power counting, and a detector or parton-density scheme. The output is a conditional factorized observable. Colored partonic amplitudes are short-distance coordinates, not physical external S-matrix elements.

Integrable two-dimensional scattering. The primary objects are factorized scattering functions, form factors, and TBA data. The input is elastic factorization, Yang–Baxter constraints, locality axioms for form factors, convergence, and completeness assumptions. The output is exact algebraic and thermodynamic data. Reconstruction of local observables and local algebras is a separate analytic bridge.

Massive route. In a massive theory with a stable isolated one-particle shell, the route is: local fields and spectrum; Haag–Ruelle wave operators $\Omega_{\text{in/out}}$; the scattering operator S; the invariant amplitude $\mathcal{M}$; and finally the cross section σ. The first arrow is Haag–Ruelle theory (Chapters 17 and 22). The second is the definition of the scattering operator. The third is LSZ residue extraction (Chapter 18 and this chapter). The fourth is phase-space integration with the flux and normalization conventions of Chapter 19. Analytic continuation, partial waves, and resonance poles may then be studied in Chapters 26 and 29, but those analytic objects start from the amplitude already obtained on the physical sheet.

Charged QED route. In massless QED the charged sector does not satisfy the massive LSZ input. Gauss-law flux and arbitrarily soft photons change the spectral support of a charged state, so the bare electron Green-function residue is not an exact one-particle S-matrix coordinate of the full theory. The route is instead: a hard charged process; soft dressing or an inclusive sum; and a detector-resolved rate. Chapter 42 fixes external-state conventions for regulated perturbative calculations. Chapter 44 then supplies the inclusive and dressed replacements. A perturbative hard amplitude remains useful, but only as an ingredient in the resolution dependent observable after the soft sector has been attached.

Detector and QCD route. For QCD event shapes and energy correlators the physical object is a color-singlet detector distribution. The route is: a current or hadron source; factorized operator data; jet, soft, or parton-density functions; and the detector measure. Chapter 49 gives the QCD operator and factorization layers, Chapter 50 develops event shapes, track functions, and hadronization coordinates, and Chapter 65 supplies the conformal light-ray and detector-algebra comparison. A partonic amplitude or splitting kernel is a short-distance coordinate inside this route. It becomes a physical prediction only after the observable, subtraction scheme, Glauber status, endpoint/contact convention, and nonperturbative inputs have been declared.

Map and non-map ledger. The following implications are used in later chapters.

1. Haag–Ruelle plus LSZ hypotheses give massive wave-packet amplitudes.
2. Massive amplitudes plus unitarity, boundedness, and analyticity hypotheses give partial-wave and dispersion-relation machinery, together with a continuation framework in which resonance poles, cuts, and other singularities can be characterized if present. The existence and location of a resonance pole require separate channel-specific dynamical or spectral evidence.
3. A charged hard process plus soft or inclusive data gives infrared-safe charged rates.
4. QCD operator definitions plus factorization data give detector and event-shape distributions.

The converse arrows are absent without extra hypotheses. A detector distribution does not determine a unique amplitude; a resonance pole does not produce an asymptotic particle vector; an inclusive QED rate does not restore a bare charged LSZ state; a finite partonic or lattice check does not supply the continuum factorization or reconstruction theorem. This ledger is the cross-volume convention used whenever the later chapters compare these languages.

24.2 Scattering Operator and Kernel Notation

This chapter fixes the distributional kernel notation used in the analytic chapters. The nonperturbative construction of S by Haag–Ruelle wave operators was given in Chapters 17 and 22; it is not being redefined here.

Work in one massive Haag–Ruelle scattering sector satisfying the hypotheses of Theorem 22.1, with the scalar point-field construction and Cook estimates reviewed in Proposition 17.8. Fix a stable scalar species of mass $m > 0$. Its positive mass shell is

$$\Sigma_m^+ = \{(\omega_{\vec{p}}, \vec{p}) : \vec{p} \in \mathbb{R}^d\}, \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}.$$

The corresponding one-particle Hilbert space is denoted by $\mathcal{H}_1$. In the scalar case it is represented as $L^2(\Sigma_m^+, \mathrm{d}\mu_m)$, where

$$\mathrm{d}\mu_m(\vec{p}) = \frac{\mathrm{d}^d\vec{p}}{(2\pi)^d 2\omega_{\vec{p}}}.$$

The symmetric Fock space over $\mathcal{H}_1$ is

$$\mathcal{F}_s(\mathcal{H}_1) = \bigoplus_{n=0}^{\infty} \mathcal{H}_1^{\odot n}.$$

The Haag–Ruelle wave operators from the preceding theorem layers are isometries

$$\Omega_{\mathrm{in/out}} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{H}.$$

On a scattering sector for which the incoming and outgoing ranges coincide, the scattering operator is

$$S = \Omega_{\mathrm{out}}^* \Omega_{\mathrm{in}} : \mathcal{F}_s(\mathcal{H}_1) \longrightarrow \mathcal{F}_s(\mathcal{H}_1).$$

This is the object whose kernels will be studied throughout this part.

Generalized momentum states on Σ_m^+ are normalized by

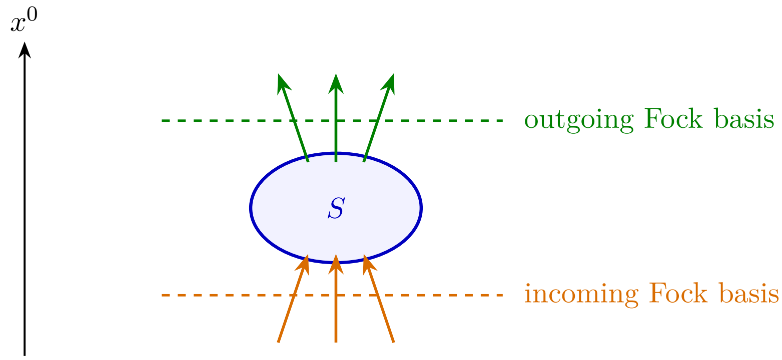
$$\langle \vec{q} | \vec{p} \rangle = (2\pi)^d 2\omega_{\vec{p}}\, \delta^{(d)}(\vec{q} - \vec{p}).$$

For outgoing momenta $q_i = (\omega_{\vec{q}_i}, \vec{q}_i)$ and incoming momenta $p_j = (\omega_{\vec{p}_j}, \vec{p}_j)$, write

$$S(q_1, \ldots, q_m | p_1, \ldots, p_n) = {}_{\mathrm{out}}\langle \vec{q}_1, \ldots, \vec{q}_m | \vec{p}_1, \ldots, \vec{p}_n \rangle_{\mathrm{in}}.$$

This notation denotes a distributional kernel. Its value on wave packets $f_i, g_j \in C_c^\infty(\mathbb{R}^d)$ is

$$\begin{aligned} &S[f_1, \ldots, f_m; g_1, \ldots, g_n] \\ &= \int \prod_{i=1}^{m} \mathrm{d}\mu_m(\vec{q}_i)\, f_i(\vec{q}_i)^* \prod_{j=1}^{n} \mathrm{d}\mu_m(\vec{p}_j)\, g_j(\vec{p}_j) S(q_1, \ldots, q_m | p_1, \ldots, p_n). \end{aligned}$$



24.3 Connected Kernels and Invariant Amplitudes

Let

$$P_{\text{out}} = \sum_{i=1}^{m} q_i, \qquad P_{\text{in}} = \sum_{j=1}^{n} p_j.$$

Translation invariance implies that the non-identity part of every connected scattering kernel is supported on $P_{\text{out}} = P_{\text{in}}$. The phase convention is fixed once and for all by distinguishing the connected S-kernel from the connected T-kernel. With

$$S = \mathbf{1} + \mathrm{i}T,$$

write

$$\begin{aligned} &S^{\mathrm{c}}(q_1, \ldots, q_m | p_1, \ldots, p_n) \\ &\qquad = I^{\mathrm{c}}(q_1, \ldots, q_m | p_1, \ldots, p_n) + \mathrm{i}\, T^{\mathrm{c}}(q_1, \ldots, q_m | p_1, \ldots, p_n), \end{aligned}$$

where I^{c} is zero except for one incoming and one outgoing particle of the same species, in which case

$$I^{\mathrm{c}}(q|p) = (2\pi)^d 2\omega_{\vec{p}} \delta^{(d)}(\vec{q} - \vec{p}).$$

The invariant amplitude $\mathcal{M}$ is the reduced connected T-kernel:

$$\begin{aligned} &T^{\mathrm{c}}(q_1, \ldots, q_m | p_1, \ldots, p_n) \\ &\qquad = (2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) \mathcal{M}(q_1, \ldots, q_m | p_1, \ldots, p_n). \end{aligned}$$

Equivalently, the transition part of the connected S-kernel is

$$S^{\mathrm{c}} - I^{\mathrm{c}} = \mathrm{i}(2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) \mathcal{M}.$$

The connected T-matrix kernel is therefore

$${}_{\text{out}}\langle q|T|p\rangle_{\mathrm{c}} = (2\pi)^D \delta^{(D)}(P_{\text{out}} - P_{\text{in}}) \mathcal{M}(q|p).$$

The word connected has a recursive distributional meaning. If the full process splits into independent subprocesses, the full kernel is the product of the kernels of those subprocesses, with separate momentum conservation in each component. The connected kernel is the part that remains after subtracting all such products.

24.4 LSZ as Pole Extraction

The pole-extraction theorem is Theorem 18.5 of Chapter 18; it is not restated here. In this chapter we use only its kernel consequence: after wave-packet smearing, a connected time-ordered distribution whose external variables have isolated one-particle poles of residue Z_ϕ gives the connected scattering kernel by multiplying each external leg by $Z_\phi^{-1/2}\mathrm{i}(k_a^2 + m^2)$, taking the simultaneous external pole boundary value, and then setting outgoing variables $k_i = q_i$ and incoming variables $k_{m+j} = -p_j$ on Σ_m^+. The resulting distribution is a matrix element of the already-defined Haag–Ruelle operator $S = \Omega_{\text{out}}^* \Omega_{\text{in}}$; in perturbative sections the same operation is applied coefficientwise to regulated time-ordered kernels.

24.5 Cluster Decomposition

Let $\alpha = \{p_1, \dots, p_n\}$ be the incoming labels and $\beta = \{q_1, \dots, q_m\}$ the outgoing labels. A process partition Π is a finite set of nonempty blocks B, where each block contains a subset $\alpha_B \subset \alpha$ and a subset $\beta_B \subset \beta$, and the blocks partition all incoming and outgoing labels. Define

$$S_B^{\mathrm{c}} = S^{\mathrm{c}}(\beta_B|\alpha_B).$$

Connected scattering kernels. For fixed incoming and outgoing labels, the full scattering kernel has a unique expansion

$$S(\beta|\alpha) = \sum_{\Pi} \prod_{B\in\Pi} S^{\mathrm{c}}(\beta_B|\alpha_B), \tag{24.1}$$

where the sum is over process partitions. Equivalently, the connected kernel is determined recursively by

$$S^{\mathrm{c}}(\beta|\alpha) = S(\beta|\alpha) - \sum_{\Pi\neq\{\alpha\cup\beta\}} \prod_{B\in\Pi} S^{\mathrm{c}}(\beta_B|\alpha_B). \tag{24.2}$$

The one-particle identity kernel I^{c} introduced above is included as the connected kernel for a block with one incoming and one outgoing label:

$$\begin{aligned} S^{\mathrm{c}}(q|p) &= I^{\mathrm{c}}(q|p) + \mathrm{i}(2\pi)^D \delta^{(D)}(q-p)\mathcal{M}(q|p) \\ &= (2\pi)^d 2\omega_{\vec{p}}\delta^{(d)}(\vec{q}-\vec{p}) + \mathrm{i}(2\pi)^D\delta^{(D)}(q-p)\mathcal{M}(q|p). \end{aligned} \tag{24.3}$$

In a translationally invariant vacuum without external backgrounds, the nontrivial one-particle amplitude vanishes for a stable isolated species, and the identity term remains.

Order the finite label sets by their total number of external labels. For one block the statement defines S^{c}. Suppose the connected kernels have been defined for every proper subcollection. Then (24.2) defines the connected kernel for the given collection by subtracting all factorizations into proper subprocesses. Substituting this definition back into the right-hand side of (24.1) gives $S(\beta|\alpha)$; uniqueness follows because every other solution would have a difference whose highest-size nonzero component appears only in the one-block term and hence must vanish. This is the moment-cumulant recursion for scattering kernels, with process partitions replacing ordinary set partitions.

This recursion is algebraic: it defines connected kernels once the full kernel is known. The physical cluster property is the large-separation theorem below. Its hypotheses are additional Hilbert-space and locality hypotheses, not consequences of the recursion.

Theorem 24.4 (Cluster decomposition for scattering kernels)**.** *Assume a Haag–Ruelle scattering theory with a unique translation-invariant vacuum, a mass gap above the vacuum, local commutativity, and the following quantitative clustering estimate for the almost-local Haag–Ruelle approximants used to construct the scattering states. If F_T and G_T are finite products of such approximants at fixed time parameter T, and if $G_T(a) = U(a)G_TU(a)^*$, then for every N there are constants C_N, M_N, R_N, depending on the chosen operator families and wave packets but independent of T and a, such that*

$$|\langle\Omega|F_TG_T(a)|\Omega\rangle - \langle\Omega|F_T|\Omega\rangle\langle\Omega|G_T|\Omega\rangle| \leq C_N(1+|T|)^{M_N}(1+\rho(a,T))^{-N}. \tag{24.4}$$

Here $\rho(a,T)$ is the spacelike separation of the two localization tubes after replacing the almost-local operators by local approximants; for velocity supports separated from the light cone of the relative translation, $\rho(a,T) \geq c|a| - R_N(1+|T|)$ for some $c > 0$. Let $\Psi_A^{\mathrm{in/out}}$ and $\Psi_B^{\mathrm{in/out}}$ be two groups of wave-packet scattering states whose velocity supports stay disjoint after a relative spacelike translation a, with $a^2 > 0$ and $|a| \to \infty$. Then the full scattering matrix element factorizes:

$$\left\langle \Psi_A^{\mathrm{out}} \otimes U(a)\Psi_B^{\mathrm{out}}, S\, \Psi_A^{\mathrm{in}} \otimes U(a)\Psi_B^{\mathrm{in}} \right\rangle \longrightarrow \left\langle \Psi_A^{\mathrm{out}}, S\Psi_A^{\mathrm{in}} \right\rangle \left\langle \Psi_B^{\mathrm{out}}, S\Psi_B^{\mathrm{in}} \right\rangle.$$

Equivalently, the connected kernel $S^{\mathrm{c}}(\beta|\alpha)$ has no distributional component carrying separate momentum conservation for a proper subprocess; it carries only the total conservation law of the connected process.

Proof. Choose Haag–Ruelle approximants

$$\Psi_{A,T}^{\mathrm{out}} = A_T^{\mathrm{out}}\Omega, \qquad \Psi_{A,T}^{\mathrm{in}} = A_{-T}^{\mathrm{in}}\Omega, \qquad \Psi_{B,T}^{\mathrm{out}} = B_T^{\mathrm{out}}\Omega, \qquad \Psi_{B,T}^{\mathrm{in}} = B_{-T}^{\mathrm{in}}\Omega$$

for the four scattering states. The Haag–Ruelle theorem gives norm convergence as $T \to +\infty$ for the outgoing approximants and as $T \to +\infty$ for the incoming approximants written at time $-T$. Translation by $U(a)$ is unitary, so these norm errors are independent of the relative translation a. Hence, for every $\varepsilon > 0$, there exists T_ε such that for $T \geq T_\varepsilon$ the exact matrix element differs by at most ε, uniformly in a, from the finite-time expression

$$M_T(a) = \langle \Omega | (A_T^{\mathrm{out}})^* (B_T^{\mathrm{out}}(a))^* A_{-T}^{\mathrm{in}} B_{-T}^{\mathrm{in}}(a) | \Omega \rangle.$$

Almost-locality permits replacement of each approximant by a local operator localized in its Haag–Ruelle tube, with an error decreasing faster than any power of the tube size. For the fixed T chosen above, large spacelike a makes the A-tube spacelike separated from the translated B-tube. Local commutativity then orders the two groups, up to the almost-local tail estimate, into

$$M_T(a) = \langle \Omega | F_T G_T(a) | \Omega \rangle + O_N((1+|a|)^{-N}), \qquad F_T = (A_T^{\mathrm{out}})^* A_{-T}^{\mathrm{in}}, \quad G_T = (B_T^{\mathrm{out}})^* B_{-T}^{\mathrm{in}}.$$

Applying (24.4) gives

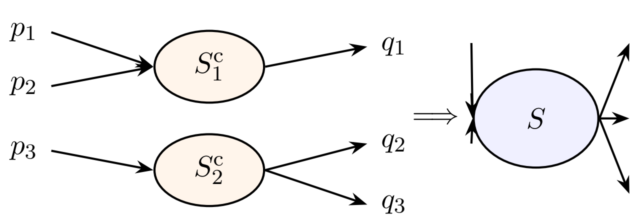
$$\lim_{|a| \to \infty} M_T(a) = \langle \Omega | F_T | \Omega \rangle \langle \Omega | G_T | \Omega \rangle.$$

Finally take $T \to \infty$ in the two factors. The same norm convergence gives

$$\langle \Omega | F_T | \Omega \rangle \longrightarrow \left\langle \Psi_A^{\mathrm{out}}, S\Psi_A^{\mathrm{in}} \right\rangle, \qquad \langle \Omega | G_T | \Omega \rangle \longrightarrow \left\langle \Psi_B^{\mathrm{out}}, S\Psi_B^{\mathrm{in}} \right\rangle.$$

Since ε was arbitrary, the scattering matrix element factorizes in the stated large-$|a|$ limit.

The kernel statement follows by testing the distributional kernel against wave packets whose velocity supports realize the two translated clusters. If the connected kernel contained a component with separate momentum conservation inside a proper subprocess, its wave-packet pairing would survive the large-separation limit as a product contribution. The recursion defining connected scattering kernels subtracts exactly those proper-subprocess products. Therefore only the total momentum-conservation delta distribution remains in S^{c}. $\square$



The same decomposition appears in connected Green functions. Since LSZ acts on external pole residues, it maps the connected part of a time-ordered Green function to the connected part of the scattering kernel with the same external labels.

24.6 Preparation for Analytic Structure

The object carried forward is the connected invariant amplitude

$$\mathcal{M}(q_1, \ldots, q_m | p_1, \ldots, p_n)$$

as a distributional boundary value on the physical mass shells. Bound states will appear as poles of this object in an appropriate channel. Resonances will appear as poles reached by analytic continuation through a continuum cut. Renormalization will be formulated as the construction of finite Green functions and finite amplitudes while preserving the local form of the counterterms.

For sectors with massless long-range forces, confinement, or infraparticle spectra, the asymptotic data require modified observables or modified state spaces. The later chapters return to those domains after the massive scattering and renormalization structures have been developed. In particular, one must not identify the charged electron labels used in perturbative QED hard kernels with exact on-shell LSZ external particles of massless QED. The ordinary LSZ formula requires an isolated charged one-particle pole; the QED infrared chapters instead construct resolution-inclusive probabilities and softly dressed charged asymptotic data.

Chapter 25

Bound States from Exchange Amplitudes

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

25.1 Exchange Model

Consider two real scalar fields in four spacetime dimensions with Lorentzian Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi_1\partial^\mu\phi_1 - \frac{1}{2}m_1^2\phi_1^2 - \frac{1}{2}\partial_\mu\phi_2\partial^\mu\phi_2 - \frac{1}{2}m_2^2\phi_2^2 - \frac{1}{2}g\phi_1^2\phi_2.$$

The masses m_1, m_2 are positive, and g is a real coupling. Use the same mostly-plus convention as before. The scalar propagators and cubic vertex are

$$\phi_i: \quad \frac{-\mathrm{i}}{k^2 + m_i^2 - \mathrm{i}0}, \qquad \phi_1\phi_1\phi_2: \quad -\mathrm{i}g.$$

This Lagrangian is used as a perturbative exchange model in the sense of Section 23.2. The pole criterion derived below is a statement about any exact massive local theory whose amplitude has the displayed analytic structure; the polynomial scalar density by itself is only the finite-cutoff or effective description from which the tree kernel is computed.

The connected $\phi_1\phi_1 \to \phi_1\phi_1$ tree amplitude receives s-, t-, and u-channel exchange contributions. The three routings and the solid/dashed line convention are fixed in Figure 25.1. With all external

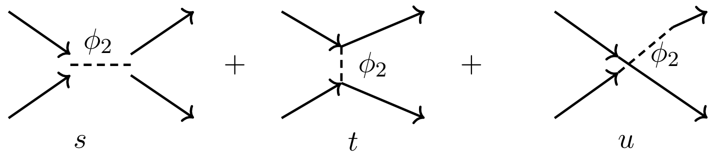


Figure 25.1: Tree exchange routings in the cubic scalar model. Solid directed lines are external ϕ_1 legs and the dashed internal line is the exchanged ϕ_2 field. The labels s, t, u specify which Mandelstam invariant flows through the internal denominator.

momenta on the ϕ_1 mass shell, define

$$s = -(k_1 + k_2)^2, \qquad t = -(k_1 - k_3)^2, \qquad u = -(k_1 - k_4)^2.$$

Here the metric is mostly plus, $k_i^2 = -m_1^2$, k_1, k_2 are incoming, and k_3, k_4 are outgoing, as in Definition 24.2. Momentum conservation gives the Mandelstam constraint

$$s + t + u = 4m_1^2.$$

The tree-level invariant amplitude is

$$\mathcal{M}_{\text{tree}}(s,t,u) = g^2 \left(\frac{1}{m_2^2 - s - \mathrm{i}0} + \frac{1}{m_2^2 - t - \mathrm{i}0} + \frac{1}{m_2^2 - u - \mathrm{i}0} \right).$$

In the center-of-mass frame for equal incoming masses m_1,

$$s = E^2, \qquad t = -\frac{E^2 - 4m_1^2}{2}(1 - z), \qquad u = -\frac{E^2 - 4m_1^2}{2}(1 + z),$$

where E is the total energy and $z = \cos\theta$. The two-ϕ_1 threshold is

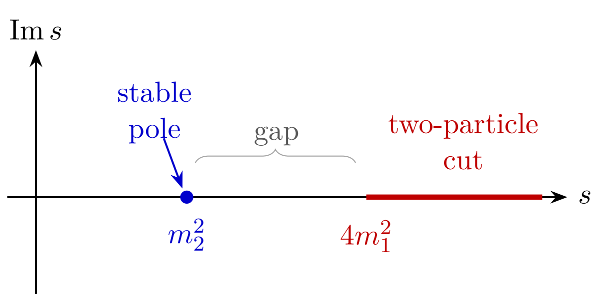
$$s = 4m_1^2.$$

25.2 Pole Below a Two-Particle Threshold

Assume

$$0 < m_2 < 2m_1.$$

Then the s-channel pole $s = m_2^2$ lies below the two-ϕ_1 threshold. It is outside the physical two-ϕ_1 scattering region $s \geq 4m_1^2$, but it is reached by analytic continuation in the invariant energy variable.



The scattering conclusion is precise: the two-ϕ_1 channel couples to a stable one-particle state of invariant mass m_2. In the present model that state is represented explicitly by the field ϕ_2. In a theory where no elementary field for that state is introduced, the same below-threshold pole is the scattering signature of a stable bound state. The pole criterion and the microscopic origin of the state are separate statements.

Definition 25.1 (Stable-channel sheets)**.** Fix a connected stable-channel scattering amplitude, with the identity term and the overall momentum-conservation delta function removed, and let s be the invariant energy variable for a two-particle channel whose threshold is s_{th}. The *first sheet*, or physical sheet, is the analytic germ reached from the Feynman boundary prescription whose physical value on the open two-particle interval is the upper-edge boundary value $\mathcal{M}^{\mathrm{I}}(s+\mathrm{i}0)$, $s > s_{\mathrm{th}}$. A real point below that threshold is on the first sheet when it is reached inside this same germ without crossing the threshold cut. Near an ordinary square-root threshold, one may use a local variable such as

$$\beta(s) = \sqrt{1 - \frac{s_{\mathrm{th}}}{s}},$$

with the branch fixed by $\beta(s+\mathrm{i}0) > 0$ on the physical interval. Continuing through the threshold cut to the adjacent germ changes the sign of this local square root. That adjacent germ is called the *second sheet* relative to the specified channel. With several open channels, a sheet is not specified by the word "second" alone: one must say which channel cuts have been crossed, or equivalently which local threshold variables have had their branches changed.

In this terminology, a stable bound state coupled to the channel appears as an isolated pole on the first sheet below the corresponding threshold. A resonance is instead a pole of an analytically continued stable-channel amplitude on an adjacent nonphysical sheet, not a pole of the physical real-axis boundary value. The resonance chapter constructs this second-sheet continuation explicitly in an exchange model, and the analyticity chapter later develops the same terminology for general $2 \to 2$ amplitudes.

Remark 25.2 (Existence is dynamical input)*.* The exchange model above contains the state at $s = m_2^2$ because the Lagrangian already contains the stable field ϕ_2 and the chosen masses place its pole below the two-ϕ_1 threshold. In a theory without such an elementary field, the existence of a below-threshold composite pole is a nonperturbative fact about the spectrum and the connected four-point function, requiring dynamical spectral input beyond locality, unitarity, and the local action. The rest of this chapter uses the conditional statement: if an isolated stable pole is present in a two-particle channel, then its position, spin, and factorized residue have the scattering consequences derived below. For a Bethe–Salpeter kernel, the corresponding converse requires the analytic Fredholm hypotheses stated later in Theorem 27.3; a below-threshold eigenvalue of an unspecified or truncated kernel is not, by itself, a theorem that the exact QFT contains a stable particle.

Remark 25.3 (Near-threshold and unitary-limit caveat)*.* The inequality $M_B < 2m_1$ is the physical location statement: the state lies below the two-particle continuum. The pole statement used in this chapter is stronger. It assumes an isolated one-particle spectral point of finite multiplicity, separated from the continuum and from other singularities by a nonzero gap, with nonzero overlap with the two-particle channel. Under those hypotheses the amplitude has a simple first-sheet pole and a factorized residue.

This hypothesis is not uniform as the binding energy tends to zero. In a two-body channel with equal masses, write the relative momentum by

$$s = 4(m_1^2 + k^2), \qquad k = \mathrm{i}\kappa, \qquad \kappa > 0$$

for a below-threshold bound state. The threshold is $k = 0$. The effective range expansion of an s-wave amplitude has the low-energy denominator

$$-a^{-1} + \frac{1}{2} r_e k^2 - \mathrm{i}k + \cdots,$$

so a shallow bound state has $\kappa \sim a^{-1}$ when the effective range is finite. As $a^{-1} \to 0^+$, the pole reaches the branch point at threshold: there is no fixed neighborhood of the pole disjoint from the two-particle cut, and the simple-pole expansion in the invariant s-plane ceases to be the right local description. Depending on the side of the continuation one may obtain a threshold bound state, a virtual state, or a resonance pole on another sheet. In three-body unitary physics, Efimov spectra give an even sharper warning: infinitely many poles may accumulate at threshold, so the isolated finite-rank pole hypothesis fails outright.

Near-threshold molecular states and unitary-limit systems therefore require a separate threshold expansion, usually in the momentum variable k and with the branch point kept explicit. The pole factorization below applies to an isolated stable bound state before this degeneration has occurred; it is not a claim that every attractive channel near threshold is described by a single simple pole in s.

Theorem 25.4 (Isolated spectral atom gives a first-sheet pole)**.** *Let $A(x)$ and $B(x)$ be local or smeared composite operators whose Fourier transforms carry the same conserved quantum numbers. Assume that, in the corresponding translation-invariant Hilbert space, the joint energy-momentum spectrum contains an isolated finite-multiplicity one-particle mass shell of mass M_B, with spectral projection Π_B, and that this shell is separated by a positive gap from the continuum in a neighborhood of total momentum P. Assume also that the matrix elements of A and B on this shell are nonzero, that their finite-rank product admits a holomorphic local off-shell extension in the complexified total momentum variables, and that the remaining spectral measure is locally finite across $P^2 = -M_B^2$. Then the Fourier transform of the connected time-ordered two-point function*

$$G_{AB}(P) = \int \mathrm{d}^4x \, \mathrm{e}^{-\mathrm{i}P\cdot x} \langle \Omega | \, \mathrm{T} \, A(x) B(0) | \Omega \rangle_{\mathrm{conn}}$$

has, in the mostly-plus Feynman convention of this volume, the local form

$$G_{AB}(P) = \frac{-\mathrm{i}\, R_{AB}(P)}{P^2 + M_B^2 - \mathrm{i}0} + G_{AB}^{\mathrm{reg}}(P), \tag{25.1}$$

where G_{AB}^{reg} is analytic in a sufficiently small first-sheet neighborhood of the pole. The numerator $R_{AB}(P)$ is any local holomorphic off-shell extension of the on-shell finite-rank matrix

$$R_{AB}(P) = \sum_{\lambda=1}^{d_B} \langle \Omega | A(0) | B, P, \lambda \rangle \langle B, P, \lambda | B(0) | \Omega \rangle. \tag{25.2}$$

defined for $P^2 = -M_B^2$, $P^0 > 0$. Here λ labels an orthonormal basis of the finite spin/internal degeneracy space of the isolated shell. The pole residue is independent of the chosen holomorphic off-shell extension; changing the extension changes only the analytic term in (25.1).

Proof. Let $U(a)$ be the translation representation and let $E(\mathrm{d}q)$ be the joint projection-valued spectral measure of its self-adjoint generators, in the mostly-plus convention $q^2 = -(q^0)^2 + |\vec{q}|^2$. Translation covariance gives

$$\langle\Omega|A(x)B(0)|\Omega\rangle = \int_{\overline{V}_+} \mathrm{e}^{\mathrm{i}q\cdot x}\,\mathrm{d}\mu_{AB}(q), \qquad \mathrm{d}\mu_{AB}(q) := \langle\Omega|A(0)E(\mathrm{d}q)B(0)|\Omega\rangle.$$

The connected subtraction removes only the vacuum atom at $q = 0$, so it does not affect a positive-mass shell with $M_B > 0$. On the isolated shell the spectral measure has the finite-rank form

$$\sum_\lambda \int \frac{\mathrm{d}^3\vec{p}}{(2\pi)^3\,2E_{\vec{p}}}\,\mathrm{e}^{\mathrm{i}p\cdot x}\langle\Omega|A(0)|B,p,\lambda\rangle\langle B,p,\lambda|B(0)|\Omega\rangle, \qquad E_{\vec{p}} = \sqrt{|\vec{p}|^2 + M_B^2}.$$

This is the restriction of $E(\mathrm{d}q)$ to the finite-dimensional range of Π_B, written in the standard covariant normalization of one-particle vectors.

The time-ordered distribution contributed by this shell is obtained by adding the reversed ordering for $x^0 < 0$. The scalar contour identity in the present convention is

$$\int \frac{\mathrm{d}P^0}{2\pi}\,\mathrm{e}^{-\mathrm{i}P^0 t}\frac{-\mathrm{i}}{P^2 + M_B^2 - \mathrm{i}0} = \frac{1}{2E_{\vec{P}}}\big[\theta(t)\mathrm{e}^{-\mathrm{i}E_{\vec{P}}t} + \theta(-t)\mathrm{e}^{\mathrm{i}E_{\vec{P}}t}\big], \qquad E_{\vec{P}} = (|\vec{P}|^2 + M_B^2)^{1/2}.$$

Indeed $P^2 + M_B^2 - \mathrm{i}0 = -((P^0)^2 - E_{\vec{P}}^2 + \mathrm{i}0)$, so the poles in the P^0-plane lie at $E_{\vec{P}} - \mathrm{i}0$ and $-E_{\vec{P}} + \mathrm{i}0$. Closing the contour below for $t > 0$ and above for $t < 0$ gives the two displayed boundary values. Fourier transformation of the shell part of the time-ordered correlator therefore gives

$$G^B_{AB}(P) = \frac{-\mathrm{i}\,R_{AB}(P)}{P^2 + M_B^2 - \mathrm{i}0}$$

in a neighborhood of the chosen shell, where the numerator is the finite sum of matrix elements displayed in (25.2). If R_{AB} is extended off the mass shell in two different holomorphic ways, their difference vanishes on the complex hypersurface $P^2 + M_B^2 = 0$. Since the differential of $P \mapsto P^2 + M_B^2$ is nonzero on the positive-energy mass shell, this function may be used as one local holomorphic coordinate. The difference of the two extensions is therefore $(P^2 + M_B^2)$ times a holomorphic function, and after division by the Feynman denominator it changes only the analytic remainder. The pole residue is the on-shell class.

It remains to check that the rest of the spectral measure cannot create a singularity at the same first-sheet point. By hypothesis there is a neighborhood $\mathcal{U}$ of the chosen real mass-shell point such that the support of $E(\mathrm{d}q)(1 - \Pi_B)$ is separated from $\{P \in \mathcal{U} : P^2 = -M_B^2,\ P^0 > 0\}$ by a positive spectral gap. For complex P in a smaller first-sheet neighborhood, the denominators arising from the contour representation of the time-ordered distribution are then bounded away from zero on this remaining support. Local finiteness of $\mathrm{d}\mu_{AB}$ permits differentiation under the spectral integral against compactly supported test functions, so the continuum contribution is holomorphic there. This holomorphic contribution, together with the off-shell-extension ambiguity just described, is G^{reg}_{AB}. □

Theorem 25.5 (LSZ factorization of an isolated bound-state pole)**.** *Consider a connected $2 \to 2$ amplitude between stable scalar external particles, and let $P = k_1 + k_2 = k_3 + k_4$ be the total momentum in the s-channel, $s = -P^2$. Assume the external one-particle LSZ hypotheses of Chapter 18, and assume that the connected four-point function has an isolated spectral atom of* *mass $M_B < 2m$ in this channel satisfying the hypotheses of Theorem 25.4. Then, in a first-sheet neighborhood of $s = M_B^2$ disjoint from the two-particle cut, the invariant amplitude has the factorized form*

$$\mathcal{M}_{fi}(s, z) = \frac{\sum_{\lambda=1}^{d_B} \Gamma_\lambda^{\text{out}}(k_3, k_4)\, \Gamma_\lambda^{\text{in}}(k_1, k_2)}{M_B^2 - s - \mathrm{i}0} + \mathcal{M}_{fi}^{\text{reg}}(s, z). \tag{25.3}$$

The functions $\Gamma_\lambda^{\text{in/out}}$ are the LSZ-amputated matrix elements coupling the two external particles to the one-particle vector $|B, P, \lambda\rangle$. They depend on the external channel, normalization of one-particle states, and finite choice of interpolating fields, but the pole position M_B^2, the finite-rank factorization, and the spin content of the residue are invariant scattering data.

Proof. Write the connected four-point function in the s-channel as a time-ordered two-point function of two composite channel operators. For test functions $f_{\text{in/out}}$ of compact support in the relative coordinates, define

$$A_{\text{out}}(X) = \int \mathrm{d}^4 r\, f_{\text{out}}(r)\, O_3\Big(X + \frac{r}{2}\Big)\, O_4\Big(X - \frac{r}{2}\Big),$$
$$A_{\text{in}}(Y) = \int \mathrm{d}^4 r\, f_{\text{in}}(r)\, O_1\Big(Y + \frac{r}{2}\Big)\, O_2\Big(Y - \frac{r}{2}\Big).$$

LSZ amputation is applied after the external one-particle Fourier variables are restored and the test functions are removed in the usual distributional order. Theorem 25.4 gives the isolated denominator $(P^2 + M_B^2 - \mathrm{i}0)^{-1}$ and a finite-rank residue equal to the product of matrix elements through the spectral projection Π_B.

The LSZ operation multiplies the four external legs by the inverse external two-point denominators and then puts each external momentum on the stable mass shell. By assumption the external LSZ residues are finite and nonzero, and M_B^2 is separated from the external one-particle poles and from the two-particle threshold. Thus LSZ changes only the numerator factors: it amputates the external legs in the two matrix elements and leaves the internal denominator $P^2 + M_B^2 - \mathrm{i}0 = M_B^2 - s - \mathrm{i}0$ untouched. The finite rank of Π_B gives the displayed sum over λ, and all remaining spectral contributions are regular in the chosen first-sheet neighborhood. □

25.3 One-Dimensional Pole Criterion as Orientation

The elementary one-dimensional Schrödinger model is useful here only as an orientation for the bound-state pole criterion. Its Hilbert-space conventions belong to the general discussion of continuous spectra in Section 5.2, and its unitary Schrödinger evolution is part of the quantum-mechanical setup of Chapter 7. The detailed one-dimensional analytic continuation, including outgoing resonance poles and the two-sheeted energy plane, is treated in Section 26.2.

For the present stable-bound-state purpose, take a short-range Hamiltonian

$$H = -\frac{1}{2\mu}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \qquad E_{\text{nr}} = \frac{k^2}{2\mu},$$

with V real. After fixing the real-k scattering normalization, an outgoing solution has the asymptotic form

$$\psi_k^{\text{out}}(x) \sim S(k)^{-1} \mathrm{e}^{-\mathrm{i}kx} + \mathrm{e}^{\mathrm{i}kx} \qquad (x \to +\infty).$$

Analytic continuation to $k = \mathrm{i}\alpha$, $\alpha > 0$, gives

$$\psi_{\mathrm{i}\alpha}^{\mathrm{out}}(x) \sim S(\mathrm{i}\alpha)^{-1}\mathrm{e}^{\alpha x} + \mathrm{e}^{-\alpha x}.$$

An L^2 bound-state wave function cannot contain the growing term $\mathrm{e}^{\alpha x}$. Thus normalizability requires

$$S(\mathrm{i}\alpha)^{-1} = 0, \qquad E_{\mathrm{nr}} = -\frac{\alpha^2}{2\mu},$$

or equivalently a pole of $S(k)$ at $k = \mathrm{i}\alpha$ on the positive imaginary k-axis. In the relativistic amplitude this statement becomes: a stable bound state below a two-particle threshold appears as a first-sheet pole below the corresponding branch cut, with the sheet convention of Definition 25.1.

25.4 Partial Waves

For an equal-mass scalar elastic channel in four dimensions, write

$$z = \cos\theta, \qquad s = E^2.$$

The underlying partial waves are generalized two-particle asymptotic states. In the center-of-mass frame, with the δ-normalized sharp-momentum basis,

$$\langle \vec{k}_1, \vec{k}_2 | E, \vec{P} = \vec{0}, \ell, m_\ell\rangle_\delta = \delta^{(3)}(\vec{k}_1 + \vec{k}_2)\,\delta(\omega_1 + \omega_2 - E)\,\mathcal{N}(E) Y_{\ell m_\ell}(\widehat{k}_1),$$

where

$$\omega_i = \sqrt{|\vec{k}_i|^2 + m^2}, \qquad |\vec{k}_1| = |\vec{k}_2| = k, \qquad E = 2\sqrt{k^2 + m^2}.$$

For identical scalar bosons,

$$\mathcal{N}(E) = \left(\frac{2E}{k\,\omega_1\omega_2}\right)^{1/2}, \qquad \ell \in 2\mathbb{Z}_{\geq 0}.$$

For a distinguishable equal-mass pair, such as a charged scalar and its antiparticle, the factor of 2 under the square root is absent and both even and odd ℓ occur. The ordered-particle invariant amplitude below uses the 16π normalization fixed in Chapter 19; for an unordered identical-boson channel, the same physical information is obtained after Bose symmetrization and restriction to even partial waves, with the 32π normalization in Equation (29.2).

Rotational invariance diagonalizes the elastic single-channel matrix element,

$$|E, \ell, m_\ell\rangle^{\mathrm{in}} = S_\ell(E)|E, \ell, m_\ell\rangle^{\mathrm{out}} \quad \text{in the elastic sector.} \tag{25.4}$$

When additional channels with the same conserved quantum numbers are open, Equation (25.4) is the elastic component of a larger unitary matrix and gives $|S_\ell(E)| \leq 1$.

In the elastic region, define partial-wave amplitudes by

$$\mathcal{M}(s, z) = 16\pi \sum_{\ell=0}^{\infty} (2\ell + 1) a_\ell(s) P_\ell(z), \tag{25.5}$$

equivalently

$$a_\ell(s) = \frac{1}{32\pi}\int_{-1}^{1} \mathrm{d}z\, P_\ell(z)\mathcal{M}(s,z). \tag{25.6}$$

This is the ordered 16π convention fixed in Chapter 19; the coefficient $a_\ell(s)$ here is the same partial-wave amplitude used there. The letter a is deliberate: M_δ remains reserved for the $\delta^{(d)}$-normalized sharp-momentum kernel of Chapter 19, whereas a_ℓ is a scalar coordinate of the invariant amplitude in a fixed angular-momentum channel. The elastic partial-wave S-matrix eigenvalue is written

$$S_\ell(s) = 1 + 2\mathrm{i}\,\beta(s)a_\ell(s), \qquad \beta(s) = \sqrt{1-\frac{4m^2}{s}},$$

with the branch chosen by $\beta(s) > 0$ for $s > 4m^2$. Elastic unitarity is

$$|S_\ell(s)| = 1$$

on the physical elastic interval.

Spin from the angular residue. Let the external particles be spinless and work in the center-of-mass frame. Suppose the isolated bound-state shell in Theorem 25.5 transforms as an irreducible spin-J representation of spatial rotations and that the channel coupling is rotationally covariant. Then the pole residue in the scalar $2 \to 2$ amplitude is proportional to $P_J(z)$, with $z = \widehat{k}_{\mathrm{out}} \cdot \widehat{k}_{\mathrm{in}}$, and only the partial wave $\ell = J$ contains the simple pole. More precisely,

$$\sum_{m=-J}^{J} \Gamma^{\mathrm{out}}_{Jm}(\widehat{k}_{\mathrm{out}})\Gamma^{\mathrm{in}}_{Jm}(\widehat{k}_{\mathrm{in}}) = g_{\mathrm{out}}g_{\mathrm{in}}\,\frac{2J+1}{4\pi}\,P_J(\widehat{k}_{\mathrm{out}}\cdot\widehat{k}_{\mathrm{in}}) \tag{25.7}$$

after the conventional normalization of spherical harmonics is absorbed into the constants $g_{\mathrm{in/out}}$. For spinless external particles, the angular dependence of a coupling to a spin-J intermediate state is an intertwiner from functions on the sphere to the spin-J representation. Rotational covariance fixes this intertwiner, up to an overall constant, to the spherical harmonics

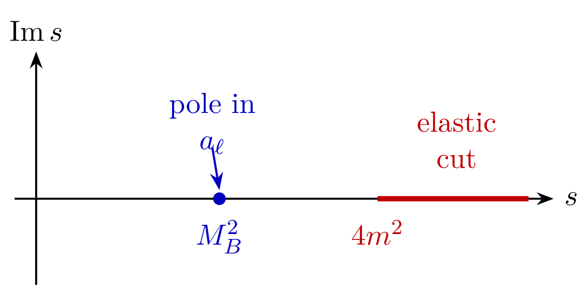
$$\Gamma^{\mathrm{in}}_{Jm}(\widehat{k}) = g_{\mathrm{in}}Y_{Jm}(\widehat{k}), \qquad \Gamma^{\mathrm{out}}_{Jm}(\widehat{k}) = g_{\mathrm{out}}\overline{Y_{Jm}(\widehat{k})}$$

with the conjugation convention determined by in/out orientation. Summing over the magnetic quantum number and using the spherical-harmonic addition theorem gives (25.7). Orthogonality of Legendre polynomials in (25.6) then projects this residue only onto the $\ell = J$ partial wave.

A stable spin-ℓ state of mass $M_B < 2m$ coupled to the two-particle channel therefore gives

$$a_\ell(s) = \frac{r_\ell}{M_B^2 - s} + \text{regular terms},$$

where the residue r_ℓ is determined by the squared coupling of the state to the two-particle channel with angular momentum ℓ.



For the exchange model, the s-channel term is independent of z. Its below-threshold pole therefore appears in the $\ell = 0$ partial wave. Higher spin bound states arise when the pole residue has the angular dependence of $P_\ell(z)$, or more generally the corresponding spin-ℓ tensor structure.

The spin carried by a channel pole can be read directly from this angular dependence. As an algebraic illustration, consider the numerator structure of scalar QED, developed later in the gauge-theory chapters; here only the Lorentz-contracted photon-exchange numerator is used, not the full gauge theory. The charged scalar-antiscalar annihilation channel has an s-channel exchange numerator proportional to

$$(p_1 - p_2)^\mu (p_3 - p_4)_\mu.$$

In the center-of-mass frame this numerator is

$$4k^2 \cos\theta = 4k^2 P_1(\cos\theta),$$

up to the overall sign fixed by momentum routing and metric conventions. The massless pole is therefore an $\ell = 1$ pole in this channel. The invariant statement is the pole position and spin-one residue in the scalar-antiscalar scattering amplitude; the microscopic description of the corresponding one-particle state is additional data of the theory.

25.5 Operator Interpretation

Let $O_B(x)$ be a local or suitably smeared composite operator with nonzero overlap with a stable state B,

$$\langle \Omega | O_B(0) | B(P) \rangle \neq 0, \qquad P^2 = -M_B^2.$$

Then the spectral representation of a two-point function of O_B contains an atom at M_B^2. The same state appears in four-point functions through a factorizing pole. For external operators O_1, O_2, O_3, O_4, and P the momentum flowing through the channel,

$$\widetilde{G}_4(P; \cdots) \sim \frac{\langle \Omega | O_3 O_4 | B(P) \rangle \langle B(P) | O_1 O_2 | \Omega \rangle}{P^2 + M_B^2 - \mathrm{i}0} + \text{regular terms}.$$

After LSZ reduction on the external stable particles, this factorization becomes the bound-state pole of the scattering amplitude.

The particle content of a theory is therefore read from spectral and asymptotic data. A field appearing in a Lagrangian can interpolate a stable state, and a composite operator can interpolate a stable state. The analytic pole and its residue state the channel coupling; additional structural data state how the corresponding state is represented inside the theory.

Chapter 26

Resonances and Dressed Propagators

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

26.1 Above-Threshold Exchange

Continue the two-scalar model of the preceding chapter,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi_1\partial^\mu\phi_1 - \frac{1}{2}m_1^2\phi_1^2 - \frac{1}{2}\partial_\mu\phi_2\partial^\mu\phi_2 - \frac{1}{2}m_2^2\phi_2^2 - \frac{1}{2}g\phi_1^2\phi_2,$$

with $m_1, m_2 > 0$ and $g \in \mathbb{R}$.

Definition 26.1 (Above-threshold scalar exchange datum)**.** The above-threshold exchange datum consists of the preceding two-scalar local Lagrangian, stable external ϕ_1 one-particle states, the mass inequality

$$m_2 > 2m_1,$$

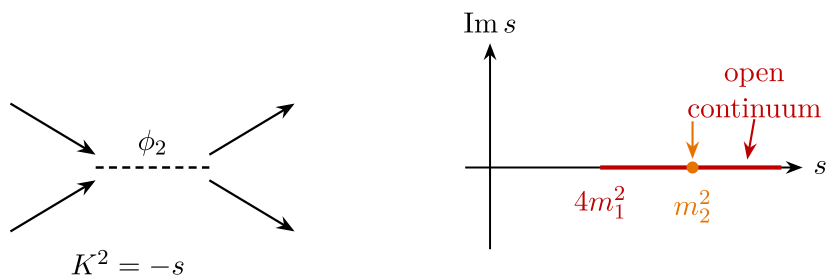
and the mostly-plus Mandelstam invariant

$$K = k_1 + k_2, \qquad s = -K^2,$$

where k_1, k_2 are the incoming ϕ_1 momenta, in the convention of Definition 24.2. The s-channel tree contribution to the connected invariant amplitude is

$$\mathcal{M}_s^{\mathrm{tree}}(s) = \frac{g^2}{m_2^2 - s - \mathrm{i}0}.$$

For the datum of Definition 26.1, the value $s = m_2^2$ lies in the two-ϕ_1 scattering region $s \geq 4m_1^2$. The relevant scattering problem is therefore a problem with stable external ϕ_1 states and an open two-particle continuum in the same channel as the exchanged ϕ_2 line. The exact physical-axis S-matrix is a bounded unitary boundary value; the pole suggested by the tree graph is the lowest-order expansion of an analytic structure reorganized by the exact two-point function of ϕ_2.



The partial-wave normalization used below is the unordered identical-boson convention for the physical $\phi_1\phi_1 \to \phi_1\phi_1$ channel:

$$\mathcal{M}(s, z) = 32\pi \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)a_\ell(s)P_\ell(z), \qquad z = \cos\theta, \tag{26.1}$$

This is the 32π, even-ℓ convention of Equation (29.2). The ordered 16π convention may still be used as an intermediate labelled-leg bookkeeping device, but the elastic S_ℓ below acts on the symmetric two-ϕ_1 Hilbert-space channel, so the physical partial-wave coordinate is a_ℓ. In an elastic one-channel interval,

$$S_\ell(s) = 1 + 2\mathrm{i}\,\rho_1(s)a_\ell(s), \qquad \rho_1(s) = \sqrt{1 - \frac{4m_1^2}{s}}. \tag{26.2}$$

The branch of ρ_1 is chosen by $\rho_1(s) > 0$ for $s > 4m_1^2$ on the physical upper boundary of the first sheet, in the terminology of Definition 25.1. Elastic unitarity gives

$$|S_\ell(s)| = 1. \tag{26.3}$$

With inelastic channels included on the same Hilbert space, the elastic matrix element instead obeys $|S_\ell(s)| \leq 1$. Thus the exact physical-axis partial wave is finite. The divergence of $\mathcal{M}_s^{\text{tree}}$ at $s = m_2^2$ records that the perturbative description has expanded around a stable reference line in a region where the state couples to an open decay channel. The exact object is a resonance: a nearby pole of the continuation of $S_\ell(s)$, or equivalently of $a_\ell(s)$, through the threshold cut. On the physical sheet the same channel is represented by the continuum part of the spectral measure; an isolated real mass atom occurs only for a stable one-particle state.

Physical-axis boundedness and resonance-sheet location. Let $I \subset (4m_1^2, \infty)$ be an elastic interval away from the threshold endpoint, and assume that the exact stable-channel partial-wave boundary value exists on I. If $S_\ell(s)$ is unitary on I, or if it is the elastic matrix element of a larger unitary channel matrix, then

$$|a_\ell(s)| \leq \frac{1}{\rho_1(s)} \qquad (s \in I).$$

Consequently a resonance associated with this channel is not a pole of the physical upper-edge boundary value $a_\ell(s+\mathrm{i}0)$. It is a pole of the analytic continuation through the channel cut.

Indeed, Equation (26.2) gives

$$a_\ell(s) = \frac{S_\ell(s)-1}{2\mathrm{i}\rho_1(s)}.$$

On I, $\rho_1(s) > 0$. If S_ℓ is elastic unitary, then $|S_\ell| = 1$, while if it is the elastic component of a larger unitary matrix, then $|S_\ell| \le 1$. In both cases

$$|a_\ell(s)| \le \frac{|S_\ell(s)|+1}{2\rho_1(s)} \le \frac{1}{\rho_1(s)}.$$

The bound is finite on compact subintervals away from threshold. Hence a simple divergence on the physical boundary is incompatible with exact stable-channel unitarity; the tree pole is a perturbative signal of a nearby singularity on an analytically continued sheet, not a physical-axis pole.

26.2 Outgoing Poles in a One-Dimensional Model

This section is the detailed one-dimensional model promised in Section 25.3. It extends the bound-state pole criterion to outgoing resonance poles; the underlying Hilbert-space interpretation of generalized scattering states is the one fixed in Section 5.2. Let

$$H = -\frac{1}{2\mu}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x),$$

where V is real and short range. For $x \to +\infty$, use the outgoing solution

$$\psi_k^{\mathrm{out}}(x,t) \sim S(k)^{-1}\mathrm{e}^{-\mathrm{i}kx-\mathrm{i}Et} + \mathrm{e}^{\mathrm{i}kx-\mathrm{i}Et}, \qquad E = \frac{k^2}{2\mu}.$$

Definition 26.2 (Outgoing one-dimensional resonance pole)**.** For the one-dimensional short-range Hamiltonian above, an outgoing resonance pole is a pole k_* of the meromorphic continuation of $S(k)$ with

$$\operatorname{Re} k_* > 0, \qquad \operatorname{Im} k_* < 0,$$

for which the corresponding continued solution satisfies the outgoing boundary condition at spatial infinity. Its complex energy is

$$E_* = \frac{k_*^2}{2\mu}.$$

If $S(k)$ has a pole at $k = k_*$, then the incoming coefficient vanishes at that analytically continued value and the asymptotic solution is purely outgoing. Write

$$E_* = \frac{k_*^2}{2\mu} = \omega - \mathrm{i}\gamma, \qquad \gamma > 0.$$

Then

$$\psi_{k_*}^{\mathrm{out}}(x,t) \sim \mathrm{e}^{\mathrm{i}k_*x-\mathrm{i}\omega t-\gamma t}.$$

The negative imaginary part of E_* is the exponential decay rate of the time dependence. The same wave grows in space along the outgoing direction. This spatial growth is forced by probability conservation for a state with only outgoing flux: a uniformly decaying packet cannot also be a normalizable stationary eigenvector of a self-adjoint Hamiltonian. The pole therefore defines a singular outgoing functional, a resonance or Gamow functional, associated with the analytically continued scattering function.

At a simple pole of $S(k)$, the coefficient $S(k)^{-1}$ of the incoming wave in the displayed outgoing solution vanishes. The continued asymptotic solution is therefore proportional to $\exp(\mathrm{i}k_*x - \mathrm{i}E_*t)$. Write

$$k_* = a - \mathrm{i}b, \qquad a > 0, \quad b > 0.$$

Then

$$E_* = \frac{a^2 - b^2}{2\mu} - \mathrm{i}\frac{ab}{\mu} = \omega - \mathrm{i}\gamma, \qquad \gamma = \frac{ab}{\mu} > 0.$$

The time factor is $\exp(-\mathrm{i}\omega t)\exp(-\gamma t)$, while the spatial factor for $x \to +\infty$ contains $\exp(bx)$. Thus the same analytic continuation that gives decay for $t > 0$ gives exponential growth in the outgoing spatial direction. Such a function is not in $L^2(\mathbb{R})$, so it is not an eigenvector of the self-adjoint Hamiltonian on the Hilbert space.

The spatial growth and temporal decay described above are summarized in Figure 26.1.

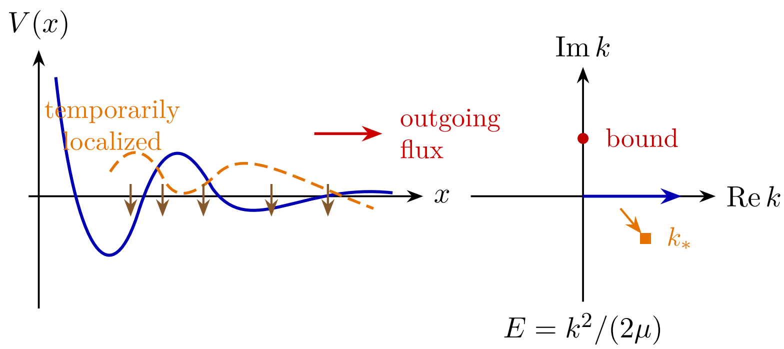


Figure 26.1: Outgoing Gamow behavior and the k-plane location of poles in the one-dimensional resonance model. The dashed packet in the left panel marks temporary localization and the arrow marks outward flux. In the right panel a circular marker denotes a bound-state pole on the positive imaginary k-axis, while a square marker denotes a resonance pole with $\mathrm{Im}\, k_* < 0$.

Because $E = k^2/(2\mu)$, the energy plane is a two-sheeted cover of the k-plane. Bound-state poles lie on the positive imaginary k-axis and map to negative real energies on the physical sheet. Resonance poles with $\mathrm{Re}\, k_* > 0$ and $\mathrm{Im}\, k_* < 0$ map to complex energies below the positive real axis on the sheet reached by continuing through the scattering cut. The continuation through the cut is shown explicitly in Figure 26.2.

26.3 The Dressed Propagator

Return to the relativistic exchange model.

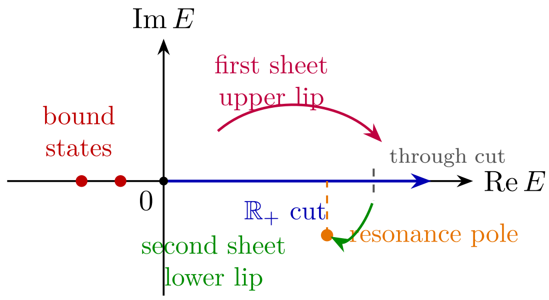


Figure 26.2: Energy-plane branch structure for a resonance pole. The first sheet is reached from the upper edge of the positive real cut; crossing the cut gives the second-sheet lower-edge branch on which the resonance pole appears.

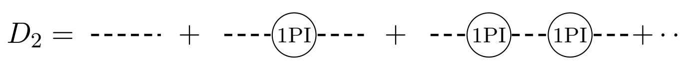
Definition 26.3 (Dressed propagator and self-energy convention)**.** The exact time-ordered two-point function of ϕ_2 in momentum space is

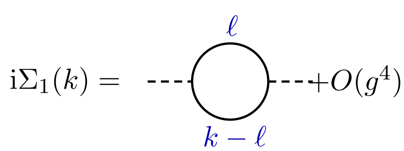
$$D_2(k) = \int \mathrm{d}^4x\, \mathrm{e}^{-\mathrm{i}k\cdot x} \langle \Omega | T\phi_2(x)\phi_2(0)|\Omega\rangle.$$

Let $\Sigma(k)$ be the sum of amputated one-particle-irreducible two-point insertions for the ϕ_2 line, with the convention

$$D_2(k) = \frac{-\mathrm{i}}{k^2 + m_{2,\Lambda}^2 - \mathrm{i}0 - \Sigma(k)}.$$

This is the geometric series obtained by inserting 1PI two-point kernels into the regulated reference propagator.





At one loop,

$$\mathrm{i}\Sigma_1(k) = \frac{(-\mathrm{i}g)^2}{2} \int \frac{\mathrm{d}^D\ell}{(2\pi)^D} \frac{-\mathrm{i}}{\ell^2 + m_1^2 - \mathrm{i}0} \frac{-\mathrm{i}}{(k-\ell)^2 + m_1^2 - \mathrm{i}0}.$$

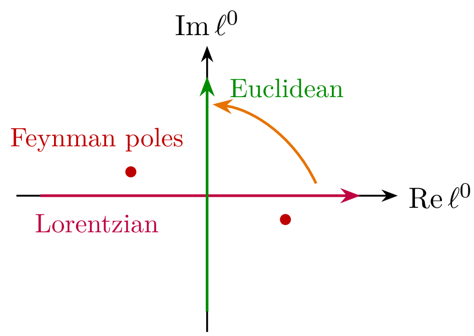
The factor $1/2$ is the symmetry factor for the two identical ϕ_1 lines in the loop. Combining denominators gives

$$\mathrm{i}\Sigma_1(k) = \frac{g^2}{2} \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D\ell}{(2\pi)^D} \frac{1}{\left[\ell^2(1-x) + (k-\ell)^2 x + m_1^2 - \mathrm{i}0\right]^2}.$$

Shift $\ell^\mu \mapsto \ell^\mu + xk^\mu$, then rotate the shifted energy contour according to the Feynman prescription. The Euclidean expression is

$$\mathrm{i}\Sigma_1(k) = \mathrm{i}\frac{g^2}{2} \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D\ell_E}{(2\pi)^D} \frac{1}{\left[\ell_E^2 + k^2x(1-x) + m_1^2\right]^2}.$$

The rotation is a contour statement, not a change of notation: for the shifted loop variable the pole displacements prescribed by $-\mathrm{i}0$ leave a sector through which the ℓ^0-contour can be moved to the Euclidean axis without crossing singularities. After the contour has been rotated, the Euclidean denominator is an ordinary real denominator. Its boundary value on a timelike cut is obtained by analytically continuing the resulting function with the original Feynman prescription; it is not encoded by leaving a residual $-\mathrm{i}0$ in the Euclidean integrand. When the answer is later written as a Lorentzian boundary value, a notation such as $k^2x(1-x)+m_1^2-\mathrm{i}0$, or $1-sx(1-x)/m_1^2-\mathrm{i}0$ with $k^2=-s$, refers to the boundary-value choice of the external invariant k^2 only. It is not a pole prescription for the already rotated Euclidean loop variable ℓ_E.



In four dimensions, with a Euclidean cutoff $|\ell_E|<\Lambda$ and with k in the domain reached by the contour rotation,

$$\Sigma_1(k)=\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\left[\log\frac{\Lambda^2}{k^2x(1-x)+m_1^2}-1+O(\Lambda^{-2})\right].$$

The divergent part is independent of k. Introduce

$$m_{2,\Lambda}^2=(m_2^R)^2+\delta m_2^2,\qquad \Sigma_1(k)=\Sigma_1^R(k)+\delta m_2^2,$$

where the counterterm cancels the cutoff-dependent constant. One convenient subtraction is

$$\Sigma_1^R(k)=-\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\log\frac{k^2x(1-x)+m_1^2}{m_1^2}.$$

This is first an analytic function on the Euclidean side of the cut. The physical boundary value is specified only after the external invariant is continued to a timelike value.

26.4 The Threshold Cut and the Cut Width

For the s-channel exchange, $k=K=k_1+k_2$ and $K^2=-s$. The dressed contribution is

$$\mathcal{M}_s(s)=\frac{g^2}{F(s)},\qquad F(s)=-s+(m_2^R)^2-\mathrm{i}0-\Sigma_1^R(K).$$

Equivalently,

$$F(s)=-s+(m_2^R)^2-\mathrm{i}0+\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\log\left[1-\frac{s\,x(1-x)}{m_1^2}-\mathrm{i}0\right].$$

Since

$$0 \le x(1-x) \le \frac{1}{4},$$

the logarithm develops a discontinuity precisely for

$$s \ge 4m_1^2.$$

For real $s > 4m_1^2$, define

$$\rho_1(s) = \sqrt{1 - \frac{4m_1^2}{s}}, \qquad x_\pm = \frac{1}{2}\left(1 \pm \rho_1(s)\right).$$

The argument of the logarithm is negative for $x \in [x_-, x_+]$. On the physical upper boundary the prescription gives

$$\log(-|a| - \mathrm{i}0) = \log|a| - \mathrm{i}\pi,$$

and therefore

$$\operatorname{Im} \Sigma_1^R(K)\big|_{s+\mathrm{i}0} = \frac{g^2}{32\pi}\rho_1(s).$$

This boundary value is fixed by the same Wick-rotation prescription that relates the Euclidean Green function to the Lorentzian time-ordered Green function. For real positive energy above the two-particle threshold, k^0 is reached by analytic continuation from the imaginary k^0-axis to the positive real axis through the upper half-plane.

Threshold discontinuity of the one-loop self-energy. For $s > 4m_1^2$, with the physical upper-edge boundary value specified above,

$$\operatorname{Im} \Sigma_1^R(K)\big|_{K^2=-s+\mathrm{i}0} = \frac{g^2}{32\pi}\rho_1(s), \qquad \rho_1(s) = \sqrt{1 - \frac{4m_1^2}{s}}.$$

Equivalently, the imaginary part of the dressed inverse denominator on the first sheet is

$$\operatorname{Im} F_{\mathrm{I}}(s + \mathrm{i}0) = -\frac{g^2}{32\pi}\rho_1(s) + O(0^+).$$

The logarithm in $F(s)$ is

$$\log\left[1 - \frac{s\,x(1-x)}{m_1^2} - \mathrm{i}0\right].$$

Its argument is negative precisely when $x \in [x_-, x_+]$, where

$$x_\pm = \frac{1}{2}(1 \pm \rho_1(s)).$$

The interval length is $x_+ - x_- = \rho_1(s)$. On the physical upper edge,

$$\log(-|a| - \mathrm{i}0) = \log|a| - \mathrm{i}\pi.$$

Since

$$\Sigma_1^R(K) = -\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\, \log\left[1 - \frac{s\,x(1-x)}{m_1^2} - \mathrm{i}0\right],$$

the imaginary part is

$$-\frac{g^2}{32\pi^2}(-\pi)\,(x_+ - x_-) = \frac{g^2}{32\pi}\rho_1(s).$$

The denominator $F_{\mathrm{I}} = m_R^2 - s - \mathrm{i}0 - \Sigma_1^R$ therefore has the opposite imaginary part, up to the infinitesimal boundary prescription.

Figure 26.3 records the two analytic continuations used in the self-energy computation: the k^0 contour deformation and the corresponding discontinuity across the s-channel cut.

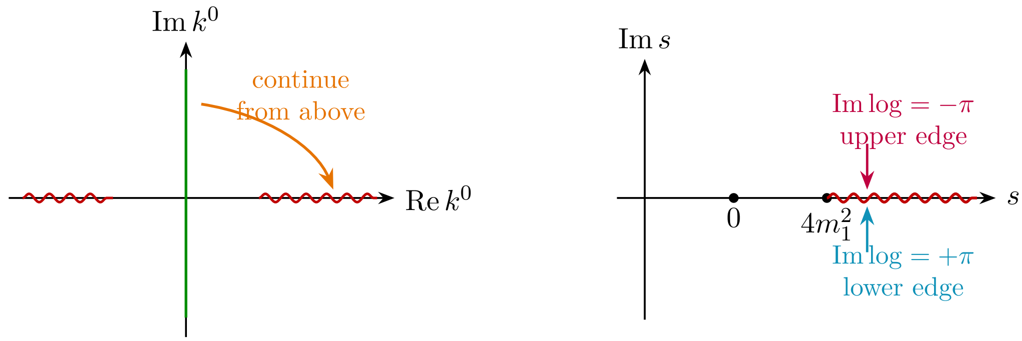


Figure 26.3: Cut continuation for the one-loop self-energy. The left panel shows the k^0 continuation from the upper edge past the on-shell singularities. The right panel records the sign convention for the logarithm on the two sides of the s-channel cut beginning at $4m_1^2$.

Choose a real resonance mass parameter M_R by the condition that the real part of the denominator vanishes at $s = M_R^2$,

$$\operatorname{Re} F(M_R^2 + \mathrm{i}0) = 0.$$

At weak coupling, and to leading order in the variation of the real part of Σ_1^R, the amplitude near this point is

$$\mathcal{M}_s(s) \simeq \frac{g^2}{M_R^2 - s - \mathrm{i}M_R\Gamma_{\mathrm{cut}}(M_R)},$$

where

$$M_R\Gamma_{\mathrm{cut}}(M_R) = \frac{g^2}{32\pi}\sqrt{1 - \frac{4m_1^2}{M_R^2}} + O(g^4).$$

The subscript “cut” is deliberate. The number above is extracted from the absorptive part of the two-point function on the physical cut. It is the quantity computed by the inclusive cut, or Fermi golden rule, calculation at the real energy M_R. The underlying discontinuity is the two-point specialization of Hilbert-space unitarity; the line-replacement calculation is its perturbative expansion, in the sense explained in Remark 28.18. It is not, by itself, a definition of every resonance width.

26.5 Width Parameters and Their Comparison

There are several operationally different quantities that are often denoted by the same letter Γ. In a narrow isolated resonance they agree to the order at which the local Breit–Wigner approximation is meaningful, but their definitions are distinct.

Definition 26.4 (Width coordinates for an isolated resonance)**.** Let a stable-channel amplitude have an isolated simple resonance pole s_* on a specified nonphysical sheet, and assume that a real energy window containing the corresponding line shape has been fixed. The following four quantities are different coordinates on different data.

1. The cut width $\Gamma_{\text{cut}}(M)$ is defined at a real energy M by the inclusive transition rate obtained from the discontinuity of the self-energy at that real energy. For a scalar resonance coupled to stable final channels X, the perturbative expression has the form

$$\Gamma_{\text{cut}}(M) = \frac{1}{2M}\sum_X \int \mathrm{d}\Phi_X(P)\ |\mathcal{A}(R(P)\to X)|^2, \qquad P^2 = -M^2, \quad P^0 > 0,$$

with the symmetry factors included in the invariant phase-space measure $\mathrm{d}\Phi_X(P)$ and in the amplitude $\mathcal{A}$. This formula should be read as a weak-coupling construction using the stable state R of the decoupled or lower-order theory as an interpolating state. Once the channel is open, the exact theory has stable asymptotic states in the final channels, not an exact external one-resonance state.

2. The line-shape width Γ_{line} is an observable extracted from a specified physical inclusive cross section or event distribution. After a specified smooth background subtraction and a specified resolution function, one may define it as the full width at half maximum in the energy variable $E = \sqrt{s}$ of the resonant contribution. This definition depends on the chosen process, on the observable used to form the inclusive distribution, and on how nonresonant background is separated from the pole term. It is therefore not an invariant analytic datum for a broad resonance.

3. The Gamow time width, or complex-energy pole width, is defined from the second-sheet pole after choosing an energy square root. If s_* is a simple resonance pole and the square-root branch is chosen so that $\mathrm{Re}\sqrt{s_*} > 0$ near the resonance, write

$$E_* = \sqrt{s_*} = M_{\text{pole}} - \frac{\mathrm{i}}{2}\Gamma_{\text{pole}}, \qquad M_{\text{pole}} > 0, \quad \Gamma_{\text{pole}} > 0.$$

Equivalently,

$$s_* = M_{\text{pole}}^2 - \mathrm{i}M_{\text{pole}}\Gamma_{\text{pole}} - \frac{1}{4}\Gamma_{\text{pole}}^2.$$

This definition is independent of a particular production process once the pole and sheet have been specified. The associated outgoing Gamow functional evolves as

$$\mathrm{e}^{-\mathrm{i}E_* t} = \mathrm{e}^{-\mathrm{i}M_{\text{pole}}t}\,\mathrm{e}^{-\Gamma_{\text{pole}}t/2},$$

so its probability norm, in the formal Gamow description, decays as $\mathrm{e}^{-\Gamma_{\text{pole}}t}$. This is the time-decay width of the pole functional. The statement concerns the pole contribution in the rigged-Hilbert analytic continuation; an exact normalizable state in a self-adjoint Hamiltonian does not have a purely exponential survival law for all times.

4. The s-plane pole-imaginary width is defined by

$$M_s = \sqrt{\mathrm{Re}\, s_*}, \qquad \Gamma_s = -\frac{\mathrm{Im}\, s_*}{M_s}.$$

This convention uses the imaginary part of the pole in the s-plane rather than the imaginary part of the energy $E = \sqrt{s}$. It agrees with Γ_{pole} only up to the kinematic conversion

$$\Gamma_s = \Gamma_{\text{pole}} \left[1 + O\left(\frac{\Gamma_{\text{pole}}^2}{M_{\text{pole}}^2} \right) \right].$$

Controlled Approximation 26.5 (Narrow-resonance comparison of widths)**.** Suppose that, in the energy window $|E - M_{\text{pole}}| = O(\Gamma_{\text{pole}})$, the continued stable-channel amplitude is dominated by a single simple pole, the residue and the nonresonant background are holomorphic and vary only on a scale of order M_{pole}, and the pole is separated from thresholds and other singularities by a squared-mass distance much larger than $M_{\text{pole}}\Gamma_{\text{pole}}$. Then the ideal resonant line shape obtained from the pole term, with the smooth residue replaced by its value at the pole and with the smooth background removed, has

$$\Gamma_{\text{line}} = \Gamma_{\text{pole}} \left[1 + O\left(\frac{\Gamma_{\text{pole}}^2}{M_{\text{pole}}^2} \right) \right],$$

as follows. Keeping only the pole term and writing $E = M_{\text{pole}} + x$,

$$|s_* - E^2|^2 = \left(2M_{\text{pole}}x + x^2 + \frac{1}{4}\Gamma_{\text{pole}}^2 \right)^2 + M_{\text{pole}}^2 \Gamma_{\text{pole}}^2.$$

The half-maximum points of the pole term, measured in the energy variable, are therefore

$$x_{\pm} = \pm\frac{1}{2}\Gamma_{\text{pole}} - \frac{1}{4M_{\text{pole}}}\Gamma_{\text{pole}}^2 + O\left(\frac{\Gamma_{\text{pole}}^3}{M_{\text{pole}}^2} \right),$$

so $x_+ - x_- = \Gamma_{\text{pole}}[1 + O(\Gamma_{\text{pole}}^2/M_{\text{pole}}^2)]$. The common shift of the two half-maximum points records that the peak in E is not centered exactly at M_{pole} once the quadratic relation $s = E^2$ is retained. If the residue, phase space, detector resolution, or subtracted background varies appreciably over the same energy interval, the measured line-shape width receives additional process-dependent corrections. That is precisely why Γ_{line} is not an invariant analytic datum for a broad resonance.

The cut calculation at the real resonance energy gives

$$M_{\text{pole}}\Gamma_{\text{pole}} = M_R \Gamma_{\text{cut}}(M_R) + O(g^4)$$

in the weak-coupling scalar model above. Indeed, on the second sheet the discontinuity computation gives

$$F_{\text{II}}(M_R^2) = -\mathrm{i} M_R \Gamma_{\text{cut}}(M_R) + O(g^4), \qquad F'_{\text{II}}(M_R^2) = -1 + O(g^2).$$

Solving $F_{\text{II}}(s_*) = 0$ by one Newton step therefore gives

$$s_* - M_R^2 = -\mathrm{i} M_R \Gamma_{\text{cut}}(M_R) + O(g^4).$$

Thus the common symbol Γ_R is justified only after these hypotheses and approximations have been stated. For a broad resonance, for an overlapping pair of poles, or near a threshold branch point, Γ_{line}, Γ_{cut}, Γ_{pole}, and Γ_s need not coincide.

26.6 Elastic Phase Motion

The preceding weak-coupling discussion defines a single-channel elastic resonance model after three choices have been made: retain the resonant s-channel exchange, discard smooth background and crossed-channel pieces, and replace the absorptive part by the width function

$$W(s) = \frac{g^2}{32\pi}\rho_1(s)$$

on the elastic real interval. These choices are the approximation. Once they have been made, the corresponding Breit–Wigner elastic S-matrix element is a definite rational function and its elastic unitarity is exact in the model.

Only the $\ell = 0$ partial wave carries the retained scalar exchange. Since the resonant s-channel term is independent of z, Equation (26.1) gives

$$a_0^{(s)}(s) = \frac{\mathcal{M}_s(s)}{32\pi}, \qquad S_0(s) - 1 = 2\mathrm{i}\rho_1(s)a_0^{(s)}(s) = \frac{\mathrm{i}\rho_1(s)}{16\pi}\mathcal{M}_s(s).$$

Equivalently,

$$\mathcal{M}_s(s) = -16\pi\mathrm{i}\sqrt{\frac{s}{s-4m_1^2}}\,\big(S_0(s) - 1\big),$$

for the angle-independent elastic partial wave in this normalization. The Breit–Wigner elastic approximant is therefore

$$S_0^{\mathrm{BW}}(s) := 1 + \frac{\mathrm{i}\rho_1(s)}{16\pi}\frac{g^2}{M_R^2 - s - \mathrm{i}W(s)} = \frac{M_R^2 - s + \mathrm{i}W(s)}{M_R^2 - s - \mathrm{i}W(s)}, \qquad W(s) = \frac{g^2}{32\pi}\rho_1(s).$$

Elastic unitarity of the Breit–Wigner factor. On a real interval where only the elastic $\phi_1\phi_1$ channel is retained and $W(s) \in \mathbb{R}$, the model S-matrix element

$$S_0^{\mathrm{BW}}(s) = \frac{M_R^2 - s + \mathrm{i}W(s)}{M_R^2 - s - \mathrm{i}W(s)}$$

obeys

$$|S_0^{\mathrm{BW}}(s)| = 1$$

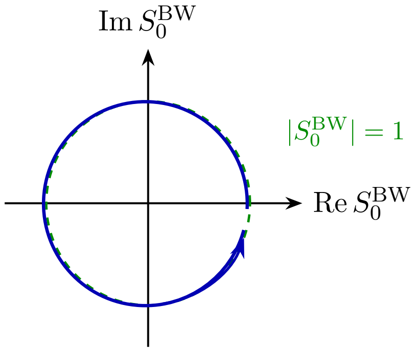
wherever the denominator is nonzero. For real s, the number $A(s) = M_R^2 - s$ is real and, by hypothesis, $W(s)$ is real. Hence $A(s)+\mathrm{i}W(s)$ is the complex conjugate of $A(s)-\mathrm{i}W(s)$. Therefore

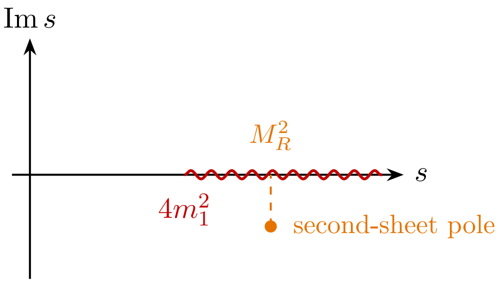
$$|S_0^{\mathrm{BW}}(s)|^2 = \frac{(A+\mathrm{i}W)(A-\mathrm{i}W)}{(A-\mathrm{i}W)(A+\mathrm{i}W)} = 1.$$

The physical partial wave differs from S_0^{BW} by the omitted background, crossed-channel, inelastic, and higher-order corrections; these are corrections to the model, not violations of the algebraic unitarity of the Breit–Wigner factor itself. The resonance is expressed as rapid phase motion. If

$$S_0^{\mathrm{BW}}(s) = \mathrm{e}^{2\mathrm{i}\delta_0^{\mathrm{BW}}(s)},$$

then the phase shift δ_0^{BW} changes rapidly through the point where $\operatorname{Re} F_{\mathrm{II}}(s)$ vanishes in the same local second-sheet approximation.





26.7 The Second Sheet

The sheet convention was fixed in Definition 25.1. Here it is made explicit for the above-threshold exchange model. The terminology applies to connected stable-channel scattering kernels.

Definition 26.6 (Adjacent second sheet in a single stable channel)**.** After the identity term and the overall momentum-conservation delta function have been removed, let

$$\mathcal{M}^{\mathrm{I}}_{ab}(s)$$

denote a connected invariant amplitude or partial-wave coordinate between stable channels a and b on the first sheet. Let $I \subset (4m_1^2, \infty)$ be an open interval on which the only cut being crossed is the $\phi_1\phi_1$ threshold cut and on which the upper and lower first-sheet boundary values exist as distributions or as ordinary functions after smearing in the remaining variables. The physical value on I is the upper-edge boundary value $\mathcal{M}^{\mathrm{I}}_{ab}(s + \mathrm{i}0)$.

The adjacent second-sheet germ relative to the $\phi_1\phi_1$ channel is the analytic germ obtained by continuing $\mathcal{M}^{\mathrm{I}}_{ab}$ from the upper edge of I through the cut once. Equivalently, in a local logarithmic coordinate for the cut, the second-sheet continuation replaces the lower first-sheet boundary value

$$\log(-r + \mathrm{i}0) = \log r + \mathrm{i}\pi, \qquad r > 0,$$

by the continued value

$$\log_{\mathrm{II}}(-r) = \log r - \mathrm{i}\pi.$$

For a square-root channel variable this is the same operation as changing the sign of the phase-space square root ρ_1 relative to the first sheet. A resonance in this channel is a pole of this adjacent germ, not a pole of the physical upper-edge boundary value and not an additional normalizable asymptotic state.

For the scalar model above the first-sheet exchange amplitude is $\mathcal{M}^{\mathrm{I}}_s(s) = g^2/F_{\mathrm{I}}(s)$, with F_{I} given by the Feynman boundary prescription in the preceding section. The first sheet contains the physical branch cut beginning at $4m_1^2$. The denominator F_{II} denotes the continuation of F_{I} through this cut in the sense of Definition 26.6. On a real point of the cut, the logarithmic discontinuity is

$$\log(-|a| - \mathrm{i}0) - \log(-|a| + \mathrm{i}0) = -2\pi\mathrm{i}.$$

Therefore the lower second-sheet boundary differs from the lower first-sheet boundary by the finite discontinuity generated by this logarithm. Near $s = M_R^2$, the branch inherited by F_{II} has

$$F_{\mathrm{II}}(s) = M_R^2 - s - \mathrm{i}W(s) + O(g^4, g^2(s - M_R^2)),$$

where $W(s) = g^2\rho_1(s)/(32\pi)$. This is the local analytic reason a pole can occur below the real axis on the adjacent sheet while no first-sheet pole appears at the same point.

Proposition 26.7 (No first-sheet resonance zero in the scalar model)**.** *Use the first-sheet branch of*

$$F_{\mathrm{I}}(s) = -s + (m_2^R)^2 + \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x \,\log\left(1 - \frac{s\,x(1-x)}{m_1^2}\right)$$

defined by continuation from the real interval $s < 4m_1^2$, with the cut placed along $s \geq 4m_1^2$. Then F_{I} has no zero in the upper half of the first sheet and no zero in the lower half of the first sheet.

Proof. Let $s = u + \mathrm{i}v$ with $v > 0$. For $0 < x < 1$, set $\alpha = x(1-x)/m_1^2 > 0$. The logarithm has argument

$$1 - \alpha s = (1 - \alpha u) - \mathrm{i}\alpha v,$$

whose imaginary part is strictly negative. On the branch reached from $s < 4m_1^2$, its argument lies in $(-\pi, 0)$; hence

$$\operatorname{Im}\log(1 - \alpha s) < 0.$$

At the endpoints $x = 0, 1$ the logarithm is zero; they do not affect the integral. Also

$$\operatorname{Im}(-s) = -v < 0.$$

Since $g^2/(32\pi^2) > 0$, every nonconstant contribution to $\operatorname{Im} F_{\mathrm{I}}(s)$ has negative sign, and the integral is strictly negative. Therefore $\operatorname{Im} F_{\mathrm{I}}(s) < 0$ in the upper half-plane, so $F_{\mathrm{I}}(s) \neq 0$ there. If $v < 0$, the argument of $1 - \alpha s = (1 - \alpha u) - \mathrm{i}\alpha v$ lies in $(0, \pi)$ on the same first-sheet branch, and $\operatorname{Im}(-s) = -v > 0$. The two terms then have positive imaginary parts, so $\operatorname{Im} F_{\mathrm{I}}(s) > 0$ in the lower half-plane. Thus the zero associated with the resonance cannot lie on either half of the first sheet. □

Proposition 26.8 (Narrow pole on the adjacent sheet)**.** *Let*

$$W_R = M_R\Gamma_{\mathrm{cut}}(M_R) = \frac{g^2}{32\pi}\sqrt{1 - \frac{4m_1^2}{M_R^2}} + O(g^4),$$

and assume $M_R^2 > 4m_1^2$ is separated from the threshold by a fixed positive amount. Suppose that in a fixed neighborhood of M_R^2 the second-sheet denominator has the weak-coupling expansion

$$F_{\mathrm{II}}(s) = M_R^2 - s - \mathrm{i}W_R + (s - M_R^2)R_1(s, g) + R_0(g), \qquad R_1 = O(g^2), \quad R_0 = O(g^4),$$

with R_1 holomorphic and uniformly bounded with its first derivative. Then, for sufficiently small g, there is a unique zero of F_{II} in a disk $|s - M_R^2| \leq Cg^2$, and it is

$$s_* = M_R^2 - \mathrm{i}M_R\Gamma_{\mathrm{cut}}(M_R) + O(g^4).$$

Proof. Write $s = M_R^2 + \delta$. The equation $F_{\mathrm{II}}(s) = 0$ is

$$\delta = -\mathrm{i}W_R + \delta R_1(M_R^2 + \delta, g) + R_0(g).$$

For $|\delta| \le Cg^2$ and C large enough, the right-hand side maps the disk to itself for all sufficiently small g, because $W_R = O(g^2)$, $\delta R_1 = O(g^4)$, and $R_0 = O(g^4)$. Its derivative with respect to δ is

$$R_1(M_R^2 + \delta, g) + \delta\, \partial_s R_1(M_R^2 + \delta, g),$$

which is $O(g^2)$ uniformly on the disk. Thus the map is a contraction, and it has a unique fixed point. The fixed-point equation immediately gives

$$\delta = -\mathrm{i}W_R + O(g^4),$$

which is the stated pole location. Equivalently, $F_{\mathrm{II}}(M_R^2) = -\mathrm{i}W_R + O(g^4)$ and $F'_{\mathrm{II}}(M_R^2) = -1 + O(g^2)$, so the same result is the controlled one-step Newton approximation. $\square$

Equivalently, the pole width defined by $E_* = \sqrt{s_*} = M_{\mathrm{pole}} - \mathrm{i}\Gamma_{\mathrm{pole}}/2$ obeys

$$M_{\mathrm{pole}}\Gamma_{\mathrm{pole}} = M_R\Gamma_{\mathrm{cut}}(M_R) + O(g^4).$$

More invariantly, s_* is defined by the denominator obtained by continuing the first-sheet F_{I} through the elastic cut:

$$F_{\mathrm{II}}(s_*) = 0.$$

The number s_* is an analytic datum of the stable-particle scattering amplitude. Its position sets the resonance location and the pole lifetime scale in the narrow-width regime; it is not the same kind of object as a physical-axis line-shape width unless the narrow-resonance hypotheses of the preceding section hold.

The same distinction is visible in the spectral representation. A stable one-particle state gives an atom in the spectral measure and supplies a Haag–Ruelle asymptotic sector. An open channel gives continuous spectral measure beginning at threshold. A narrow resonance is a pronounced analytic feature of that continuum: it can appear as a peak in the spectral density and as a pole after analytic continuation, while the external scattering states remain the stable states used in the construction of the S-matrix.

26.8 Unstable Resonances and External-Leg Reduction

The LSZ theorem applies to an external particle only when the theory supplies an isolated real mass shell in the physical Hilbert space. In the notation of Chapter 18, this means an orthogonal projection

$$P_1 = E(\Sigma_m^+)$$

onto a Wigner one-particle subspace, a positive pole residue in a local-field two-point function, and Haag–Ruelle incoming and outgoing states built from that subspace. A resonance pole $s_* = M_{\mathrm{pole}}^2 - \mathrm{i}M_{\mathrm{pole}}\Gamma_{\mathrm{pole}} - \Gamma_{\mathrm{pole}}^2/4$ has none of these properties. It is a pole of an analytically continued stable scattering amplitude on a nonphysical sheet; the associated outgoing Gamow functional is not a normalizable vector in $\mathcal{H}$, and there is no projection-valued spectral atom $E(\{p : p^2 = -s_*\})$.

Therefore a residue at $s = s_*$ is not an LSZ external-leg residue. Physical S-matrix elements with a resonance are matrix elements between stable asymptotic states. Near an isolated resonance pole the meromorphic continuation of such a stable-particle amplitude may have the local form

$$\mathcal{M}_{a\to b}(s,\alpha,\beta) = \frac{r_a(\alpha)\, r_b(\beta)}{s_* - s} + \mathcal{M}_{\mathrm{reg}}(s,\alpha,\beta),$$

where a, b label stable channels and α, β denote the remaining kinematic variables. The functions r_a, r_b are pole residues of the continued stable-channel amplitude. They can be used to describe production and decay through the resonance, but they do not define an incoming or outgoing one-resonance Hilbert-space state.

Perturbation theory often packages this pole information in a Stueckelberg–Veltman, or complex-mass-shell, prescription. One analytically continues the inverse dressed propagator to the pole,

$$F_{\mathrm{II}}(s_*) = 0, \qquad Z_*^{-1} = -F'_{\mathrm{II}}(s_*),$$

and replaces the resonant propagator near the pole by

$$\frac{g^2}{F_{\mathrm{II}}(s)} = \frac{g^2 Z_*}{s_* - s} + \text{terms holomorphic near } s_*.$$

This is a pole-scheme parametrization of the stable-channel amplitude, not a new LSZ theorem. Its usefulness requires a controlled narrow-resonance regime: the pole must be isolated from other nearby poles and branch points, the width scale $M_{\mathrm{pole}}\Gamma_{\mathrm{pole}}$ must be small compared with the squared-mass distance to thresholds or singularities whose variation is being neglected, and the nonresonant background must be slowly varying on the same scale. If the pole lies close to a threshold, or if several channels produce nearby singularities, the simple Breit–Wigner or complex-mass-shell description is replaced by the corresponding multi-channel analytic line shape. The external states remain the stable states in all cases.

26.9 Pole-Factorized Resonance Objects

The preceding discussion leaves a precise mathematical problem. Stable external particles are vectors in asymptotic Hilbert spaces. Resonances are singularities of analytic continuations of matrix elements between such vectors. A candidate "amplitude with an external resonance" must therefore be extracted from stable-channel amplitudes by pole residues.

Definition 26.9 (Local pole neighborhood)**.** Fix stable channels a, b and kinematic variables (s, α, β), where s is the invariant mass squared of a channel whose analytic continuation has a pole. A local pole neighborhood consists of a sheet $\mathfrak{s}$ of the analytically continued amplitude $\mathcal{M}^{\mathfrak{s}}_{a\to b}(s,\alpha,\beta)$, an open set $U \subset \mathbb{C}$ containing s_*, and open real or complex neighborhoods V_a, V_b of α, β, such that the only singularity in $(U \setminus \{s_*\}) \times V_a \times V_b$ is a simple pole at $s = s_*$.

Definition 26.10 (Pole-factorized resonance object)**.** In a local pole neighborhood, the pole-residue kernel is

$$R_{ab}(\alpha,\beta) = \lim_{s\to s_*} (s_* - s)\, \mathcal{M}^{\mathfrak{s}}_{a\to b}(s,\alpha,\beta). \tag{26.4}$$

If the residue kernel has rank one in the channel variables, a pole-factorized resonance object is an equivalence class of factorizations

$$R_{ab}(\alpha,\beta) = r_b(\beta)\, r_a(\alpha), \qquad (r_a, r_b) \sim (\lambda r_a, \lambda^{-1} r_b), \qquad \lambda \in \mathbb{C}^\times.$$

The functions r_a, r_b are production and decay residue functionals. The equivalence class, rather than either factor separately, is the invariant object in the one-pole problem.

For a resonance in a spin-j channel, the residue kernel carries the indices of the corresponding partial wave or covariant tensor structure. After projecting to the irreducible spin component at the pole, rank one means rank one in the remaining channel space. In gauge theory this projection must be performed on the gauge-invariant stable-channel amplitude, or equivalently on BRST cohomology representatives whose pole residue is independent of the chosen gauge-fixing representatives. Pole position and full stable-channel residues are the primary objects; a split into "production" and "decay" factors is meaningful only after the normalization equivalence in Definition 26.10 has been imposed.

The same construction extends to several resonance variables only under a strong isolation hypothesis. Suppose a stable-channel amplitude depends on complex variables $s_1, \ldots, s_N$, and the chosen sheet contains simple poles at $s_i = s_{*i}$. If there is a polydisc $\prod_i U_i$ in which the only singular divisors are these N simple pole hyperplanes, define the nested residue

$$R^{(N)} = \lim_{s_N \to s_{*N}} \cdots \lim_{s_1 \to s_{*1}} \left(\prod_{i=1}^{N} (s_{*i} - s_i) \right) \mathcal{M}^{\mathfrak{s}}(s_1, \ldots, s_N, \gamma). \tag{26.5}$$

The order of residues is then immaterial by meromorphy in the polydisc. This object is the mathematical candidate for an amplitude with N external resonance slots and any remaining stable external data γ. The definition collapses when a threshold branch point, an anomalous threshold, or another pole intersects the same neighborhood.

Remark 26.11 (Properties inherited from stable-channel scattering). Assume the stable-channel S-matrix is unitary on the physical boundary, Poincare covariant, and obtained from local fields satisfying the analyticity hypotheses used in the preceding chapters. Let R_{ab} be the pole-residue kernel in a local pole neighborhood. Then:

1. s_* and R_{ab} are invariant under changes of interpolating fields that preserve the stable-channel S-matrix.

2. covariance of the stable amplitude induces covariance of R_{ab} in the little-group or partial-wave representation labeling the pole.

3. physical unitarity gives reality and conjugation constraints relating the pole on the chosen sheet to the conjugate singularity on the reflected sheet, and it fixes the positive inclusive decay rate in the narrow isolated regime through the discontinuity of the stable-channel amplitude.

4. in a gauge theory, if the stable-channel amplitude is represented by BRST-closed external operators, the residue kernel is independent of BRST-exact changes of those representatives.

The residue kernel is obtained by applying the linear functional $\mathrm{Res}_{s=s_*}$ to a meromorphic continuation of a stable-channel matrix element. Field redefinitions that leave the stable-channel S-matrix unchanged leave this meromorphic germ unchanged, and hence leave its residue unchanged. Poincare transformations act on the stable amplitude before continuation; since analytic continuation and taking a simple residue commute with the finite-dimensional tensor or partial-wave projection, the residue transforms in the corresponding representation.

On the physical boundary, unitarity gives $S^\dagger S = 1$, or equivalently the discontinuity relation for $T = S - 1$ derived from stable-channel unitarity. Continuing this relation through the cut relates the pole on one sheet to the conjugate analytic structure on the reflected sheet. In the narrow isolated regime, projecting the discontinuity onto the pole term gives the same inclusive rate as the self-energy cut calculation of the previous section. Finally, a BRST-exact change of a stable external representative changes a physical stable-channel matrix element by zero; applying the residue functional to zero gives zero.

26.10 Gamow Functionals and the Boundary of the Construction

A rigged-Hilbert formulation packages the pole residue into generalized vectors. Choose dense test spaces

$$\Phi_- \subset \mathcal{H} \subset \Phi_-^\times, \qquad \Phi_+ \subset \mathcal{H} \subset \Phi_+^\times$$

whose energy wavefunctions have the analyticity needed to continue the incoming and outgoing scattering amplitudes to the pole. A decaying Gamow functional associated with s_* is an element of one dual space whose pairing with test vectors reproduces the pole residue. In the rest frame, with $E_* = M_{\text{pole}} - \mathrm{i}\Gamma_{\text{pole}}/2$, its time evolution is a semigroup:

$$\mathrm{e}^{-\mathrm{i}Ht}|G_*\rangle = \mathrm{e}^{-\mathrm{i}M_{\text{pole}}t}\mathrm{e}^{-\Gamma_{\text{pole}}t/2}|G_*\rangle, \qquad t \geq 0,$$

as a statement about pairings with the chosen test space. The restriction $t \geq 0$ is part of the analytic test-space construction; a self-adjoint Hamiltonian on $\mathcal{H}$ still generates a unitary group on normalizable vectors.

This rigged-Hilbert language is useful because it records the decaying pole term without pretending that the resonance is a vector in $\mathcal{H}$. It is also additional structure. Different choices of test spaces can represent the same pole residue, and a general local QFT does not come equipped with a canonical Hardy-space rigging for every multiparticle channel. For this monograph the invariant data remain the pole position, sheet, and residue kernels of stable-channel amplitudes; Gamow functionals are a representation of those data when the required analytic rigging has been supplied.

26.11 Obstructions to External-Resonance Amplitudes

The pole-residue construction has sharp failure modes.

1. **Threshold collision.** If s_* lies at a distance comparable to its width from a branch point, there is no neighborhood in which the amplitude is a single simple pole plus a holomorphic remainder. The observable line shape is controlled by the pole and the branch point together.
2. **Coupled channels.** With several open channels, the analytic continuation lives on a multi-sheeted surface. A pole may be close on one sheet and absent or shadowed on another. The residue is then a matrix in channel space, and rank-one factorization can fail.
3. **Overlapping resonances.** If two poles lie in the same local domain, the residue decomposition depends on the chosen basis in the finite-dimensional pole subspace unless the pole projectors are separated by contours.

4. **Gauge and line-shape conventions.** A complex pole of the full stable-channel amplitude is invariant, but masses, widths, branching fractions, and production factors extracted from a fitted real-axis line shape depend on the convention used to separate pole and background terms.
5. **Long-range and massless effects.** If the stable channels have infraparticle behavior or uncancelled soft radiation, the stable-channel amplitude itself must be replaced by an inclusive or dressed observable before any resonance residue is defined.

Open Problem 26.12 (External resonances). The local construction above gives a candidate object in the isolated simple-pole regime. The open problem is to promote these local residues to a general theory of amplitudes with resonance slots. Such a theory must specify the class of local QFTs and channels for which the required analytic continuations exist, define the allowed rigged-Hilbert test spaces or avoid them in favor of purely meromorphic residue data, prove compatibility with stable-channel unitarity, formulate covariance for spin and gauge theories, and give precise replacement rules when thresholds, overlapping poles, massless sectors, or coupled-channel sheets obstruct rank-one pole factorization. Perturbative complex-mass and pole schemes are calculational realizations of the isolated-pole part of this problem; they are not a nonperturbative external-state construction.

Chapter 27

Composite Bound States and Bethe–Salpeter Amplitudes

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The time-ordered four-point functions, residues, and perturbative bubble integrals in this chapter are understood with the status convention of Definition 24.1. Exact bound-state pole statements use Hilbert-space spectral and LSZ data; the bubble-chain calculation is a formal regulated graph calculation of the corresponding time-ordered kernel.

27.1 Composite Particles as Poles

Let ϕ be a real scalar field whose stable one-particle states have mass m. A stable two-particle composite species B of mass M_B is an isolated one-particle sector whose wavepacket states are normalizable and whose mass satisfies

$$0 < M_B < 2m.$$

The inequality places the state below the two-ϕ continuum threshold.

Assumption 27.1 (Bound-state pole hypothesis)**.** The existence of B is an input in this chapter. More explicitly, assume that the Hilbert space contains an isolated positive-energy one-particle subrepresentation of mass M_B, separated from the two-ϕ continuum, and that at least one operator in the two-ϕ channel has nonzero matrix element with this state. The following sections derive the consequences of that spectral hypothesis for four-point functions, scattering amplitudes, and Bethe–Salpeter residues. They do not give a general proof that an arbitrary Lagrangian with attractive interactions has such a state.

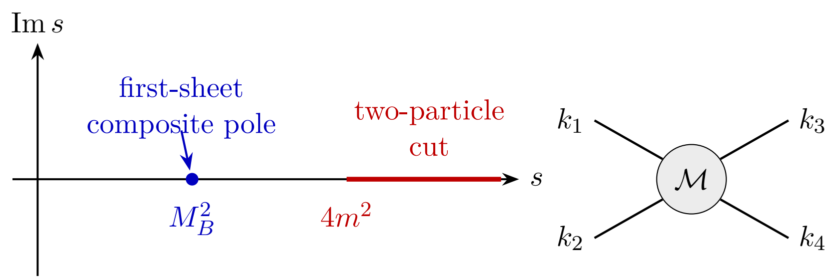
It is then represented, on the first sheet in the sense of Definition 25.1, by a pole of the appropriate four-point function at

$$s = M_B^2, \qquad s = -(k_1 + k_2)^2.$$

Equivalently, for the $2 \to 2$ invariant amplitude $\mathcal{M}(s,t)$, whose connected S-kernel transition part is $\mathrm{i}\mathcal{M}(s,t)$ after the overall momentum-conservation delta function is removed, with

$$t = -(k_1 - k_3)^2, \qquad u = -(k_1 - k_4)^2, \qquad s + t + u = 4m^2,$$

the same state appears as a pole in s, at fixed quantum numbers in the (12)-channel. After LSZ reduction of the external stable ϕ particles, the pole is present in the connected scattering amplitude; the external one-particle residues multiply the amplitude and do not determine the position of the pole. These are the mostly-plus Mandelstam variables of Definition 24.2 for on-shell external momenta: $k_i^2 = -m^2$, with k_1, k_2 incoming and k_3, k_4 outgoing.



This statement is independent of whether the Lagrangian contains a field named B. What matters is the spectral state and the residue with which operators in the two-particle channel create it.

27.2 A Bubble-Chain Example

The explicit calculation in this section is a $D = 2$ spacetime-dimensional model. The purpose is to exhibit, in a finite elementary integral, how a geometric sum of connected four-point kernels can generate a stable pole below the two-particle threshold. In $D = 4$ the same channel has a different threshold behavior and the pole condition involves the renormalized contact interaction at threshold; the numerical estimate for M_B below is therefore not a dimension-independent formula. The word generate in this paragraph refers to the specified bubble-chain approximation. A zero of the resummed denominator is a pole of that approximate kernel. Promoting it to an exact particle statement for a complete model requires additional nonperturbative control of the spectrum and of the analytic continuation of the exact four-point function.

Consider the scalar theory

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{4!}\phi^4.$$

The four-point vertex is $-\mathrm{i}g$. Before specializing the integral, the s-channel bubble-chain approximation to $2 \to 2$ scattering contains the one-loop bubble

$$\frac{(-\mathrm{i}g)^2}{2}\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\,\frac{-\mathrm{i}}{\ell^2+m^2-\mathrm{i}0}\,\frac{-\mathrm{i}}{(k_1+k_2-\ell)^2+m^2-\mathrm{i}0}.$$

The factor $1/2$ is the symmetry factor of the two identical internal lines. Feynman parametrization and Wick rotation give

$$\frac{\mathrm{i}g^2}{2}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D\ell_E}{(2\pi)^D}\,\frac{1}{\left[\ell_E^2 - sx(1-x)+m^2\right]^2}.$$

As in the self-energy calculation, the Wick-rotated denominator is real on the Euclidean side; the $-\mathrm{i}0$ prescription enters only when the resulting function is continued back to a Lorentzian boundary value.

Specializing now to $D=2$, this becomes

$$\frac{\mathrm{i}g^2}{8\pi}f(s), \qquad f(s) = \int_0^1 \mathrm{d}x\,\frac{1}{m^2 - sx(1-x)-\mathrm{i}0}.$$

With the convention of Chapter 24, the geometric bubble chain gives the connected S-kernel transition part

$$\mathrm{i}\mathcal{M}^{(s)}_{\mathrm{chain}}(s) = -\mathrm{i}g\sum_{L=0}^{\infty}\left(-\frac{g}{8\pi}f(s)\right)^L = \frac{-\mathrm{i}g}{1+\dfrac{g}{8\pi}f(s)}.$$

External wavefunction factors multiply the expression and do not move the zeros of the denominator.

For a real scalar field the external particles are identical, so the physical amplitude is Bose symmetric. The displayed chain is the contribution in the (12)-channel. The crossed chains obtained by applying the same truncated geometric construction in the (13)- and (14)-channels give functions $\mathcal{M}^{(t)}_{\mathrm{chain}}(t)$ and $\mathcal{M}^{(u)}_{\mathrm{chain}}(u)$, and the full identical-boson amplitude in this additive channel truncation is decomposed as

$$\mathcal{M}_{\mathrm{Bose}}(s,z) = \mathcal{M}^{(s)}_{\mathrm{chain}}(s) + \mathcal{M}^{(t)}_{\mathrm{chain}}(t(s,z)) + \mathcal{M}^{(u)}_{\mathrm{chain}}(u(s,z)) + \mathcal{M}_{\mathrm{rest}}(s,z),$$

where $\mathcal{M}_{\mathrm{rest}}$ denotes kernels not included in the single-channel bubble-chain truncation. In a four-dimensional partial-wave normalization the identical-boson s-wave coordinate would be

$$a_0^{\mathrm{Bose}}(s) = \frac{1}{64\pi}\int_{-1}^{1}\mathrm{d}z\,\mathcal{M}_{\mathrm{Bose}}(s,z) = \frac{\mathcal{M}^{(s)}_{\mathrm{chain}}(s)}{32\pi} + a_{0,\mathrm{cross}}(s).$$

Here, for center-of-mass momentum $p_*^2 = (s-4m^2)/4$,

$$t(s,z) = -2p_*^2(1-z), \qquad u(s,z) = -2p_*^2(1+z).$$

As s approaches the s-channel threshold $4m^2$, the crossed invariants t and u remain in a neighborhood of 0, whereas the (12)-channel invariant approaches its two-particle threshold. Therefore the crossed chains and $\mathcal{M}_{\mathrm{rest}}$, insofar as they are regular in this local neighborhood, contribute to $a_{0,\mathrm{cross}}$ as analytic background terms. They do not contain the $f(s) \sim \pi/(m\sqrt{4m^2-s})$ threshold

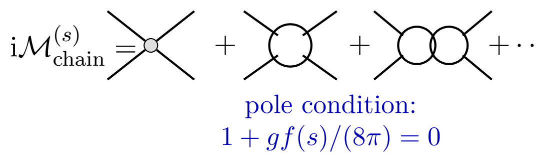


Figure 27.1: Single-channel bubble-chain contribution to the invariant amplitude. The gray vertex denotes the contact kernel in the selected channel, while the successive two-particle bubbles generate the denominator whose zero is the leading pole criterion $1 + gf(s)/(8\pi) = 0$.

enhancement that drives the denominator displayed above. Including the crossed and omitted kernels in a more complete Bethe–Salpeter kernel can change the numerical pole position; the calculation below is the leading single-channel statement. The channel separation and the pole denominator are organized in Figure 27.1.

Near the two-particle threshold from below, write

$$s = 4m^2 - \delta, \qquad \delta \to 0^+.$$

The integral is dominated by $x = 1/2$. Setting $x = 1/2 + y$, one obtains

$$m^2 - sx(1-x) = \frac{\delta}{4} + 4m^2y^2 + O(\delta y^2),$$

and hence

$$f(4m^2 - \delta) \simeq \frac{\pi}{m\sqrt{\delta}}.$$

The denominator can vanish for attractive coupling $g < 0$. At weak coupling,

$$\delta \simeq \left(\frac{g}{8m}\right)^2, \qquad M_B = \sqrt{4m^2 - \delta} \simeq 2m - \frac{g^2}{256m^3}.$$

Thus, within the $D = 2$ bubble-chain approximation and for attractive weak coupling, the resummed denominator has a stable-pole zero slightly below the two-particle threshold.

The calculation uses the local expansion around a perturbative vacuum. A complete scalar model with an attractive quartic term must specify the large-field behavior of the potential. For example,

$$V(\phi) = \frac{m^2}{\beta^2}(1 - \cos\beta\phi) = \frac{1}{2}m^2\phi^2 - \frac{m^2\beta^2}{4!}\phi^4 + O(\phi^6)$$

has an attractive quartic term in its local expansion, with local quartic coefficient $g_{\text{local}} = -m^2\beta^2$, while retaining a potential bounded from below. The bubble-chain computation should therefore be read as a statement about the local interaction coefficients around the chosen vacuum, not as a nonperturbative definition of an unstable negative-quartic polynomial theory.

27.3 Bethe–Salpeter Amplitude

Let $|B(P)\rangle$ be a stable composite one-particle state of mass M_B, with

$$P^2 = -M_B^2, \qquad P^0 > 0.$$

We use the covariant plane-wave normalization

$$\langle B(\vec{P}\,')|B(\vec{P})\rangle = (2\pi)^{D-1} 2\omega_{\vec{P}}\,\delta^{(D-1)}(\vec{P}-\vec{P}\,'), \qquad \omega_{\vec{P}} = \sqrt{\vec{P}^{\,2}+M_B^2}.$$

The momentum eigenstate is distributional; physical Hilbert-space states are obtained by smearing it with square-integrable wavepackets. The Bethe–Salpeter amplitude associated with the scalar local field $\widehat{\phi}$ is the time-ordered matrix element

$$\Phi_B(x_1,x_2;P) = \langle\Omega|T\widehat{\phi}(x_1)\widehat{\phi}(x_2)|B(P)\rangle.$$

Here Φ_B names a c-number bound-state amplitude, not a field operator. Translation covariance gives

$$\Phi_B(x_1+a,x_2+a;P) = \mathrm{e}^{\mathrm{i}P\cdot a}\Phi_B(x_1,x_2;P),$$

with the Fourier convention used below. Therefore its Fourier transform has support on total momentum P:

$$\Phi_B(x_1,x_2;P) = \int \frac{\mathrm{d}^D k_1}{(2\pi)^D}\frac{\mathrm{d}^D k_2}{(2\pi)^D}\,\mathrm{e}^{\mathrm{i}k_1\cdot x_1+\mathrm{i}k_2\cdot x_2}(2\pi)^D\delta^{(D)}(k_1+k_2-P)\widetilde{\Phi}_B(k_1,k_2;P).$$

If instead one uses unit-normalized momentum kets, a factor $\left[(2\pi)^{D-1}2\omega_{\vec{P}}\right]^{-1/2}$ appears on the right-hand side; here that factor is absorbed into the definition of $\widetilde{\Phi}_B$. The invariant normalization of $\widetilde{\Phi}_B$ is fixed by the four-point pole residue. No constituent LSZ factors are divided out in this definition. If the scalar field has one-particle residue

$$\langle\Omega|\widehat{\phi}(0)|\vec{p}\rangle = Z_\phi^{1/2},$$

then $\widetilde{\Phi}_B$ is the unamputated, field-normalized Bethe–Salpeter amplitude and contains the $Z_\phi^{1/2}$ normalization associated with each constituent field. A convention using LSZ-normalized constituent fields would instead use

$$\widetilde{\Phi}_B^{\mathrm{LSZ}} = Z_\phi^{-1}\widetilde{\Phi}_B$$

for two identical scalar constituents. In this chapter the inverse LSZ factors $Z_\phi^{-1/2}$ are not part of $\widetilde{\Phi}_B$; they are inserted only when a four-point Green function is converted to an invariant scattering amplitude.

Now consider the momentum-space time-ordered four-point function

$$\widetilde{G}_4(k_1,k_2,k_3,k_4) = \int \prod_{j=1}^{4}\mathrm{d}^D x_j\,\exp\left[-\mathrm{i}\sum_{j=1}^{4}k_j\cdot x_j\right] \times \langle\Omega|T\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)|\Omega\rangle.$$

All four momenta are taken incoming, so translation invariance supplies $(2\pi)^D\delta^{(D)}(k_1+k_2+k_3+k_4)$.

Theorem 27.2 (Bethe–Salpeter pole factorization)**.** *Assume that the* (12)*-channel contains an isolated stable scalar bound state* B *of mass* M_B*, separated from the multiparticle continuum, and that after smearing the relative coordinates of the pairs* (12) *and* (34)*, the resulting pair operators admit a channel spectral decomposition with this isolated one-particle subspace. With*

$$P = k_1+k_2, \qquad P^2 \to -M_B^2,$$

the pole part of the Fourier-transformed four-point function is

$$\widetilde{G}_4(k_1,k_2,k_3,k_4) \sim (2\pi)^D\delta^{(D)}(k_1+k_2+k_3+k_4) \times \frac{-\mathrm{i}\,\widetilde{\Phi}_B(k_1,k_2;P)\,\widetilde{\Phi}_B^*(-k_3,-k_4;P)}{P^2+M_B^2-\mathrm{i}0} + \textit{terms regular at } P^2=-M_B^2. \tag{27.1}$$

Proof. First prove the pole statement after smearing the relative coordinates. Let $\xi = x_1 - x_2$, $\eta = x_3 - x_4$, $X = (x_1+x_2)/2$, and $Y = (x_3+x_4)/2$. For test functions f, g in the relative variables set

$$\mathcal{O}_f(X) = \int \mathrm{d}^D\xi\, f(\xi)\, T\{\phi(X+\xi/2)\phi(X-\xi/2)\},$$

and define $\mathcal{O}_g(Y)$ similarly for the (34) pair. The singularity in the total momentum P is controlled by the Fourier transform in the cluster separation $R = X - Y$. Terms in which the two pairs are not separated by arbitrarily large positive or negative R^0 have compact support, or rapid decrease after the relative smearing, in the R^0 direction and therefore give functions regular at $P^2 = -M_B^2$. The pole comes from the two cluster time orderings

$$\theta(R^0)\langle\Omega|\mathcal{O}_f(X)\mathcal{O}_g(Y)|\Omega\rangle + \theta(-R^0)\langle\Omega|\mathcal{O}_g(Y)\mathcal{O}_f(X)|\Omega\rangle.$$

Insert the spectral resolution between the two pair operators. The isolated one-particle bound-state subspace contributes

$$\int \frac{\mathrm{d}^{D-1}\vec{Q}}{(2\pi)^{D-1}2\omega_{\vec{Q}}}\,|B(\vec{Q})\rangle\langle B(\vec{Q})|, \qquad \omega_{\vec{Q}} = \sqrt{\vec{Q}^{\,2}+M_B^2},$$

with the invariant one-particle normalization used throughout the scattering chapters. Translation covariance gives

$$\langle\Omega|\mathcal{O}_f(X)|B(\vec{Q})\rangle = \mathrm{e}^{\mathrm{i}Q\cdot X}F_f(\vec{Q}), \qquad \langle B(\vec{Q})|\mathcal{O}_g(Y)|\Omega\rangle = \mathrm{e}^{-\mathrm{i}Q\cdot Y}F_g(\vec{Q})^*,$$

where $Q^0 = \omega_{\vec{Q}}$, and F_f, F_g are the relative-coordinate smearings of the Bethe–Salpeter amplitudes. The bound-state part of the cluster-ordered correlator is therefore

$$\int \frac{\mathrm{d}^{D-1}\vec{Q}}{(2\pi)^{D-1}2\omega_{\vec{Q}}}\,F_f(\vec{Q})F_g(\vec{Q})^*\left[\theta(R^0)\mathrm{e}^{\mathrm{i}Q\cdot R} + \theta(-R^0)\mathrm{e}^{-\mathrm{i}Q\cdot R}\right],$$

with the second term obtained from the reversed ordering. Fourier transform the cluster separation with total momentum P. The time integral uses

$$\int_{-\infty}^{\infty}\mathrm{d}\tau\,\mathrm{e}^{\mathrm{i}E\tau}\left[\theta(\tau)\mathrm{e}^{-\mathrm{i}\omega\tau}+\theta(-\tau)\mathrm{e}^{\mathrm{i}\omega\tau}\right] = \frac{\mathrm{i}}{E-\omega+\mathrm{i}0} - \frac{\mathrm{i}}{E+\omega-\mathrm{i}0} = \frac{2\mathrm{i}\omega}{E^2-\omega^2+\mathrm{i}0}. \tag{27.2}$$

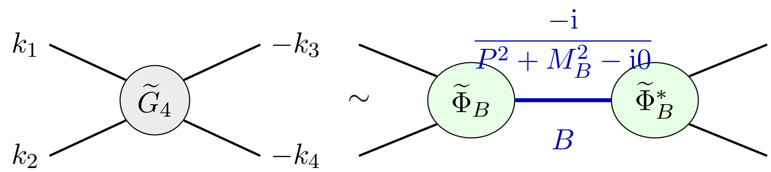
Combining this with the invariant measure $1/(2\omega_{\vec{Q}})$ gives

$$\frac{\mathrm{i}}{(P^0)^2-\omega_{\vec{P}}^2+\mathrm{i}0} = \frac{-\mathrm{i}}{P^2+M_B^2-\mathrm{i}0}$$

in the mostly-plus metric. This is the origin of the displayed Feynman prescription. The spatial Fourier transform fixes $\vec{Q}=\vec{P}$, and the remaining common translation gives $(2\pi)^D\delta^{(D)}(k_1+k_2+k_3+k_4)$.

Removing the relative-coordinate test functions gives the distributional kernel statement. The two matrix elements F_f and F_g^* unsmear to $\widetilde{\Phi}_B(k_1, k_2; P)$ and $\widetilde{\Phi}_B^*(-k_3, -k_4; P)$. The continuum spectral integral begins at invariant mass above M_B, so its denominator has no singularity at $P^2 = -M_B^2$ and contributes the terms regular at the pole. □

Equation (27.1) is the four-point analogue of the spectral pole in a two-point function. The residue is the product of the amplitudes for two fields to create and annihilate the stable composite state.



27.4 The Homogeneous Equation

Let p and q denote relative momenta at fixed total momentum P. For equal constituent masses one may write the two internal momenta as

$$\frac{P}{2} + q, \qquad \frac{P}{2} - q.$$

Let $D(k)$ be the exact time-ordered propagator of the constituent field, and define the two-particle propagator

$$\mathcal{G}_P(q) = D\left(\frac{P}{2} + q\right) D\left(\frac{P}{2} - q\right).$$

Let $\mathcal{K}_P(p, q)$ be the amputated two-particle irreducible kernel in the chosen channel: by definition, it cannot be separated into two pieces by cutting the two constituent lines carrying total momentum P. Contact vertices, crossed exchanges, and self-energy insertions on a constituent line may be part of this kernel after the constituent propagator convention has been fixed; an iteration of two kernels connected by the two constituent lines is not part of the kernel, because that iteration is precisely what the Bethe–Salpeter equation resums.

This definition is an operator statement once the relative-momentum variables are smeared. Let $\mathcal{X}$ be a chosen test-function or Hilbert space of amputated relative wavefunctions, for instance a weighted Sobolev space on a Euclidean relative-momentum contour in a massive model. The distribution $\mathcal{K}_P(p, q)$ represents a kernel only if it induces a continuous map from $\mathcal{X}$ to its dual, or, after the factors of $\mathcal{G}_P$ are included, a densely defined operator on $\mathcal{X}$. Equivalently, in a regulated theory one may define $\mathcal{K}_P$ by the exact four-point block $\mathcal{T}_P$ through

$$\mathcal{K}_P = \mathcal{T}_P(\mathbf{1} + \mathcal{G}_P\mathcal{T}_P)^{-1},$$

on any open P^2-domain where the inverse exists on the chosen space. The continuum kernel is then the limit of this regulated kernel in the same operator topology. The present chapter assumes this analytic operator realization when it writes $\mathcal{K}_P\mathcal{G}_P$ as an integral operator. Without such a realization the displayed Bethe–Salpeter equation is only distributional notation for a desired decomposition of the four-point function.

The amputated four-point scattering block $\mathcal{T}_P(p,p')$ obeys the operator equation

$$\mathcal{T}_P = \mathcal{K}_P + \mathcal{K}_P \mathcal{G}_P \mathcal{T}_P, \tag{27.3}$$

or explicitly

$$\begin{aligned} \mathcal{T}_P(p,p') &= \mathcal{K}_P(p,p') \\ &\quad + \int \frac{\mathrm{d}^D q}{(2\pi)^D}\, \mathcal{K}_P(p,q)\mathcal{G}_P(q)\mathcal{T}_P(q,p'). \end{aligned} \tag{27.4}$$

A bound-state pole at $P^2 = -M_B^2$ has the form

$$\mathcal{T}_P(p,p') \sim \frac{\Psi_B(p;P)\,\overline{\Psi}_B(p';P)}{P^2 + M_B^2 - \mathrm{i}0} + \text{regular terms}. \tag{27.5}$$

Here $\overline{\Psi}_B$ denotes the residue on the right side of the four-point function; for scalar constituents it is the complex conjugate with the momenta reversed, up to the same convention choices already present in $\widetilde{\Phi}_B^*(-k_3,-k_4;P)$. Substituting this pole form into the integral equation and equating residues gives the homogeneous Bethe–Salpeter equation

$$\Psi_B(p;P) = \int \frac{\mathrm{d}^D q}{(2\pi)^D}\, \mathcal{K}_P(p,q)\mathcal{G}_P(q)\Psi_B(q;P), \qquad P^2 = -M_B^2. \tag{27.6}$$

Equivalently, define the integral operator

$$(\widehat{V}_P F)(p) = \int \frac{\mathrm{d}^D q}{(2\pi)^D}\, \mathcal{K}_P(p,q)\mathcal{G}_P(q)F(q).$$

Then a bound state is a nonzero vector Ψ_B in the domain of this operator for which

$$(\mathbf{1} - \widehat{V}_P)\Psi_B = 0.$$

The pole condition is thus an eigenvalue-one condition for the repeated two-particle propagation at $P^2 = -M_B^2$. This statement has the logical status of a residue equation unless the operator problem is supplied with enough analytic structure. The following theorem records the standard Fredholm form of the missing structure. It is not a general existence theorem for bound states in arbitrary Lagrangians; it is the precise local theorem that converts an eigenvalue-one Bethe–Salpeter condition into a pole of the represented four-point block, and conversely, when the exact kernel has been realized as an analytic compact-operator family.

Theorem 27.3 (Analytic Fredholm Bethe–Salpeter pole criterion)**.** *Let $z = P^2$, and let $U \subset \mathbb{C}$ be a connected open neighborhood of $z_B = -M_B^2$ whose closure is separated from the two-constituent threshold and from every other singular locus of the chosen first-sheet continuation. Let $\mathcal{X}$ be the Banach space of amputated relative-momentum test functions used to realize the exact two-particle irreducible kernel. Assume that*

$$\widehat{V}(z) = \mathcal{K}_z \mathcal{G}_z : \mathcal{X} \to \mathcal{X}$$

is analytic in z as a family of compact operators, and that $\mathbf{1} - \widehat{V}(z_0)$ is invertible for at least one $z_0 \in U$. Assume also that the part of the amputated four-point block seen by the external two-field channel has an operator representation

$$\mathcal{T}_z^{\mathrm{rep}} = \mathcal{R}_z + \mathcal{F}_z\big(\mathbf{1} - \widehat{V}(z)\big)^{-1}\mathcal{E}_z, \tag{27.7}$$

where $\mathcal{E}_z : \mathcal{Y} \to \mathcal{X}$, $\mathcal{F}_z : \mathcal{X} \to \mathcal{Y}'$, and $\mathcal{R}_z : \mathcal{Y} \to \mathcal{Y}'$ are analytic bounded-operator families for the selected external test space $\mathcal{Y}$.

Then $(\mathbf{1} - \widehat{V}(z))^{-1}$, initially defined where the inverse exists, extends to a meromorphic family on U, and each coefficient in its principal part has finite rank. Therefore $\mathcal{T}_z^{\mathrm{rep}}$ is meromorphic on U. It has a pole at z_B precisely when the finite-rank principal part of $(\mathbf{1} - \widehat{V}(z))^{-1}$ at z_B is not annihilated by the channel maps $\mathcal{F}_z$ and $\mathcal{E}_z$. In particular, if $\ker(\mathbf{1} - \widehat{V}(z_B)) \neq 0$ and the corresponding root subspace has nonzero external overlap, then z_B is a pole of the represented four-point block. Conversely, any pole of $\mathcal{T}_z^{\mathrm{rep}}$ whose singular part is represented through (27.7) comes from a nonzero root vector of $\mathbf{1} - \widehat{V}(z_B)$.

If the root space is simple, so that $\ker(\mathbf{1}-\widehat{V}(z_B)) = \mathbb{C}\psi$, $\ker(\mathbf{1}-\widehat{V}(z_B))^* = \mathbb{C}\chi$, $\chi(\psi) = 1$, and the corresponding analytic eigenvalue $\lambda(z)$ of $\widehat{V}(z)$ satisfies $\lambda(z_B) = 1$, $\lambda'(z_B) \neq 0$, then the singular part is

$$\mathcal{T}_z^{\mathrm{rep}} = -\frac{(\mathcal{F}_{z_B}\psi) \otimes (\chi\mathcal{E}_{z_B})}{\lambda'(z_B)(z - z_B)} + \text{holomorphic terms}. \tag{27.8}$$

The pole is visible in the chosen channel if and only if $\mathcal{F}_{z_B}\psi \neq 0$ and $\chi\mathcal{E}_{z_B} \neq 0$. When, in addition, z_B lies on the first sheet below threshold and the exact Hilbert-space spectral representation has positive one-particle residue there, this pole is the scattering signature of a stable bound state.

Proof. The analytic Fredholm theorem applies because $\widehat{V}(z)$ is analytic and compact and because $\mathbf{1}-\widehat{V}(z_0)$ is invertible at one point of the connected domain U. Hence $(\mathbf{1}-\widehat{V}(z))^{-1}$ is meromorphic on U, with singular coefficients of finite rank supported on the finite-dimensional generalized root spaces of $\widehat{V}(z_B)$ at eigenvalue 1. Multiplication on the left and right by analytic bounded families $\mathcal{F}_z$ and $\mathcal{E}_z$, and addition of the analytic background $\mathcal{R}_z$, preserve meromorphicity and can only remove a singularity by annihilating its finite-rank principal part. This proves the pole criterion in (27.7).

For the simple-root formula, choose analytic right and left eigenvectors $\psi(z)$, $\chi(z)$ near z_B with $\chi(z)\psi(z) = 1$. On the rank-one spectral subspace, $\mathbf{1} - \widehat{V}(z)$ acts by multiplication by $1 - \lambda(z)$. Since

$$1 - \lambda(z) = -\lambda'(z_B)(z - z_B) + O\big((z - z_B)^2\big),$$

the inverse has principal part

$$-\frac{\psi \otimes \chi}{\lambda'(z_B)(z - z_B)}.$$

Substitution into (27.7) gives (27.8). The final sentence is not a Fredholm consequence: it is the separate spectral identification supplied by the exact Hilbert-space theory, namely a first-sheet isolated pole below the continuum with positive residue in a physical channel. □

Remark 27.4 (What the homogeneous equation proves). The homogeneous Bethe–Salpeter equation is therefore an existence theorem only relative to the hypotheses of Theorem 27.3: exact kernel, specified Banach space, compact analytic dependence on P^2, invertibility somewhere in the analytic neighborhood, nonzero external overlap, and spectral positivity for the first-sheet below-threshold pole. In a ladder or other truncated kernel, a nonzero solution of $(\mathbf{1} - \widehat{V}_P)\Psi = 0$ is a self-consistency condition inside that approximation. A derivation of a normalizable positive-energy bound state in the exact QFT requires control of the approximation and of the corresponding spectral subspace of the full theory.

The unamputated Bethe–Salpeter amplitude is related to the amputated vertex by

$$\widetilde{\Phi}_B(p;P) = \mathcal{G}_P(p)\Psi_B(p;P),$$

with the same field normalization used in (27.1). Here $D(k)$ is the exact propagator of the local field ϕ, including its pole residue Z_ϕ. Thus no extra Z_ϕ factor is hidden in Ψ_B. Changing to LSZ-normalized constituent fields sends

$$\widetilde{\Phi}_B \mapsto Z_\phi^{-1}\widetilde{\Phi}_B, \qquad D \mapsto Z_\phi^{-1}D, \qquad \Psi_B \mapsto Z_\phi\,\Psi_B$$

for two identical scalar constituents, so the convention can be translated without changing the pole condition. In unamputated form the equation reads

$$\widetilde{\Phi}_B(p;P) = \mathcal{G}_P(p)\int \frac{\mathrm{d}^D q}{(2\pi)^D}\,\mathcal{K}_P(p,q)\widetilde{\Phi}_B(q;P), \qquad P^2 = -M_B^2.$$

This operator formulation is the pole data used later in analytic and dispersive arguments. When a stable composite state appears in a crossed channel, crossing transports the pole $P^2 = -M_B^2$ to the corresponding Mandelstam location, while Theorem 27.2 supplies the residue as a product of Bethe–Salpeter amplitudes. In a two-constituent channel the mass of the pole is equivalently the value of P^2 for which the integral operator $\widehat{V}_P$ has eigenvalue one. Thus the bound-state pole terms added in fixed-t dispersion formulae are not independent terms in a meromorphic ansatz: their positions and residues are the spectral and Bethe–Salpeter data of the stable one-particle sector.

Figure 27.2 collects the three operator statements used in this paragraph: the definition of the two-particle irreducible kernel, its iteration into the scattering operator, and the homogeneous eigenvalue equation for a stable bound-state amplitude.

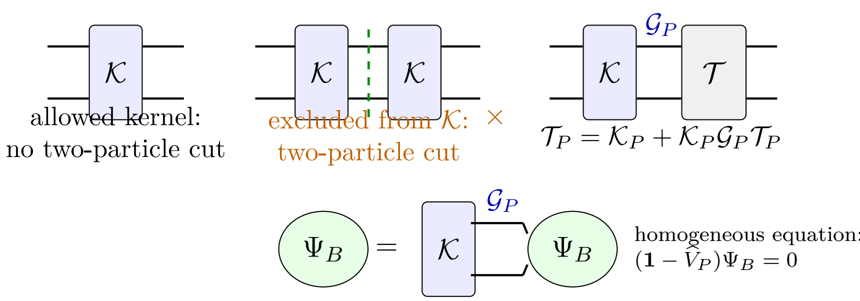


Figure 27.2: Bethe–Salpeter kernel, scattering equation, and homogeneous bound-state equation. The dashed cut marks a two-particle reducible contribution, which is excluded from $\mathcal{K}_P$ and generated instead by iterating $\mathcal{K}_P\mathcal{G}_P$.

27.5 Ladder Limit

The kernel $\mathcal{K}_P$ is exact when it contains all two-particle irreducible contributions in the chosen channel. A ladder approximation keeps only a selected low-order kernel and iterates it through $\mathcal{G}_P$. For example, a single exchanged scalar particle produces the model kernel

$$\mathcal{K}_P(p,q) \approx \frac{g_{\rm ex}^2}{(p-q)^2 + \mu^2 - \mathrm{i}0},$$

with μ the exchanged mass and with numerator factors determined by the spins and couplings of the fields.

The ladder sum has a controlled meaning only after the same operator topology has been fixed. For a selected kernel $\mathcal{K}_P^{(0)}$, set

$$\widehat{V}_P^{(0)} = \mathcal{K}_P^{(0)} \mathcal{G}_P.$$

If $\|\widehat{V}_P^{(0)}\|_{\mathcal{B}(\mathcal{X})} < 1$ in a region of complex P^2, the Neumann series

$$\mathcal{T}_P^{\mathrm{lad}} = \sum_{n=0}^{\infty} (\widehat{V}_P^{(0)})^n \mathcal{K}_P^{(0)} = (\mathbf{1} - \widehat{V}_P^{(0)})^{-1} \mathcal{K}_P^{(0)}$$

converges in operator norm there. At a bound-state pole the relevant eigenvalue reaches 1, so the Neumann series itself is no longer a convergent expansion at the pole; the pole is obtained by the meromorphic Fredholm continuation of the resolvent. Thus a ladder Bethe–Salpeter eigenvalue is a theorem inside a specified compact-operator approximation, and becomes a statement about the exact QFT only when the omitted two-particle irreducible kernels are controlled in the same topology.

In a weakly bound massive system, let

$$P^0 = 2m + E_{\mathrm{nr}}, \qquad |E_{\mathrm{nr}}| \ll m, \qquad |\vec{p}| \ll m.$$

If the kernel is instantaneous to leading order, the relative-energy integral through the two constituent propagators reduces the Bethe–Salpeter equation to a Schrödinger eigenvalue problem for the equal-time wavefunction. This nonrelativistic limit is a derived approximation to the four-point pole condition. The relativistic equation retains the full dependence on relative time, energy flow, and field-theoretic kernel structure.

Chapter 28

Analyticity, Crossing, and Landau Singularities

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The analytic amplitudes in this chapter are boundary values of exact Hilbert-space time-ordered or Wightman distributions when the stated spectral, locality, and LSZ hypotheses are being used. The Landau and Feynman-parameter discussions are instead coefficientwise perturbative graph analyses of regulated time-ordered kernels. This is the standing convention of Definition 24.1.

28.1 Physical Region and First Sheet

Definition 25.1 introduced the sheet terminology at the first point where bound-state and resonance poles required it. This section develops the same convention for the general identical massive $2 \to 2$ amplitude and fixes the physical s-channel region used in the analyticity discussion.

Consider $2 \to 2$ scattering of identical stable scalar particles of mass m. The metric is mostly plus, so a physical mass-m momentum k obeys $k^2 = -m^2$. Let the connected invariant amplitude be denoted by

$$\mathcal{M}(s, t),$$

where

$$s = -(k_1 + k_2)^2, \qquad t = -(k_1 - k_3)^2, \qquad u = -(k_1 - k_4)^2 = 4m^2 - s - t.$$

Here k_1, k_2 are incoming and k_3, k_4 are outgoing physical momenta, in the convention of Definition 24.2. Equivalently, all four external momenta may be taken incoming after replacing outgoing momenta by their negatives; the above definitions are the physical scattering convention. With $k_i^2 = -m^2$, the mostly-plus Mandelstam identity is

$$s + t + u = 4m^2.$$

In the center-of-mass frame,

$$s = E^2 = 4(k^2 + m^2), \qquad t = -2k^2(1 - \cos\theta), \qquad u = -2k^2(1 + \cos\theta).$$

Thus the physical s-channel region is

$$t \le 0, \qquad u \le 0, \qquad s \ge 4m^2 - t.$$

The last inequality is the statement $-t \le s - 4m^2$. For fixed $s \ge 4m^2$, the allowed physical interval is

$$-(s - 4m^2) \le t \le 0.$$

For fixed real $t < 0$, the physical upper edge begins at

$$s = 4m^2 - t,$$

because this is the value for which $u = 0$. The analytic function whose boundary value is the physical amplitude also has the lower s-channel threshold at $s = 4m^2$; for $4m^2 < s < 4m^2 - t$ the point is on the same right-hand cut but outside the physical s-channel scattering interval for that fixed t. Figure 28.1 fixes the corresponding s-plane branch conventions.

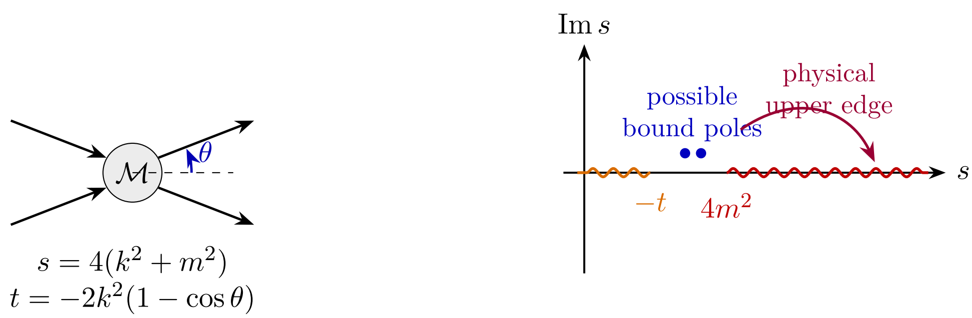


Figure 28.1: Fixed-t physical edge and cuts in the s-plane. The left panel fixes the center-of-mass kinematics and the angular variable. The right panel shows the first-sheet right-hand cut, the crossed cut, possible stable poles, and the upper-edge boundary value used for the physical amplitude.

For fixed real $t < 0$, the physical amplitude is the upper-edge boundary value

$$\lim_{\epsilon \to 0^+} \mathcal{M}(s + \mathrm{i}\epsilon, t)$$

on the first sheet. In this monograph the terms *first sheet* and *physical sheet* mean the same analytic branch: the germ reached from the Euclidean or Feynman domain by the prescribed $+\mathrm{i}0$ continuation without passing through any unitarity cut. This is a statement about a branch of the complexified invariant variables, not a synonym for the physical region of real external momenta. In the fixed-t s-plane this branch has the right-hand s-channel cut

$$C_s = [4m^2, \infty),$$

and, because $u = 4m^2 - s - t$, the crossed u-channel cut

$$C_u = (-\infty, -t].$$

A stable state of mass $M_B < 2m$ coupled to the two external scalars gives a first-sheet pole at $s = M_B^2$ in the corresponding channel. In the crossed channel this same statement appears as a pole at

$$s = 4m^2 - t - M_B^2.$$

If the stable state is composite, this pole is the same spectral pole whose four-point residue is factorized in Theorem 27.2. In a two-constituent description its mass is equivalently determined by the homogeneous Bethe–Salpeter condition (27.6), or by $(\mathbf{1} - \widehat{V}_P)\Psi_B = 0$ at $P^2 = -M_B^2$. Crossing changes the Mandelstam coordinate in which the pole is seen; it does not change the Hilbert space state or the residue matrix element from which the pole term is built. For the purposes used here, the first-sheet data consist of this cut plane together with the possible pole data and with any additional first-sheet singularities allowed by the full spectral problem.

A *second sheet* is always defined relative to a specified cut. It is the adjacent analytic germ obtained by continuing the first-sheet germ through that cut. Near a single two-particle threshold, with

$$\beta(s) = \sqrt{1 - \frac{s_{\mathrm{th}}}{s}}$$

chosen so that $\beta(s) > 0$ on the physical upper edge $s > s_{\mathrm{th}}$, the continuation through the corresponding unitarity cut sends $\beta \mapsto -\beta$. With several thresholds there is no cut-independent meaning of “the” second sheet; the sheet is labelled by the set of cuts through which the analytic germ has been continued, or equivalently by the monodromy choices for the associated channel momenta.

28.2 Real Analyticity and Crossing

For the analytic discussion in this chapter assume a mass gap, Poincare invariance, locality, unitarity, and the existence of stable one-particle external states to which LSZ applies. These assumptions specify the structural input used here, while the maximal analytic domain remains an additional problem. The continuation used in this chapter is an on-shell continuation in the invariant variables. The external legs remain on the complexified mass shell; the variables s, t, u are complex coordinates on the algebraic surface

$$\mathfrak{S}_m = \{(s, t, u) \in \mathbb{C}^3 : \ s + t + u = 4m^2\},$$

with $u = 4m^2 - s - t$. The starting object on a real physical edge is the connected LSZ amplitude as a tempered distribution in wave-packet momenta, after the overall momentum-conserving delta function and external pole residues have been extracted. Away from thresholds, stable-particle poles, and other singular loci, this distribution is represented by boundary values of holomorphic functions on tube-like domains in $\mathfrak{S}_m$.

Definition 28.1 (Tempered holomorphic amplitude germ)**.** Let $\mathcal{D} \subset \mathfrak{S}_m$ be a connected open set and let $\mathcal{D}_{\mathbb{R}}$ be a real edge of $\mathcal{D}$. A holomorphic amplitude germ on $\mathcal{D}$ is an element

$$F \in \mathcal{O}_{\text{poly}}(\mathcal{D})$$

with the following local polynomial-growth property: for every relatively compact $K \Subset \mathcal{D}$ that approaches $\mathcal{D}_{\mathbb{R}}$ inside a fixed open cone of imaginary directions, there are constants C_K, N_K such that

$$|F(z)| \le C_K(1 + |z|)^{N_K} \operatorname{dist}(z, \partial\mathcal{D})^{-N_K}.$$

Its boundary value on an open real set $U \subset \mathcal{D}_{\mathbb{R}}$ is the distribution

$$\operatorname{bv} F \in \mathcal{D}'(U)$$

defined by

$$\langle \operatorname{bv} F, \varphi\rangle = \lim_{\lambda \downarrow 0} \int_U F(x + \mathrm{i}\lambda\eta)\varphi(x)\, \mathrm{d}x, \qquad \varphi \in C_c^\infty(U),$$

whenever the limit exists and is independent of η inside the chosen cone. On noncompact real edges the same polynomial-growth condition permits Schwartz test functions and gives a tempered boundary value in $\mathcal{S}'$ in the chosen real coordinate chart. Equality of two physical amplitudes on U means equality of these boundary values in the strong dual topology of $\mathcal{D}'(U)$, or of $\mathcal{S}'$ when the noncompact tempered version is used.

28.2.1 Primitive Tube Domains from Wightman Data

We now isolate the first domain theorem used in scattering analyticity. It is not a perturbative statement. It is the theorem that converts the spectral condition and locality of Wightman fields into holomorphic domains for correlation functions. The later Lehmann–Martin and Jin–Martin arguments start from this primitive domain and add the further retarded-commutator, partial-wave, and unitarity input needed for fixed-t dispersion theory.

Let $\Phi_a(x)$, $a \in A$, be scalar Wightman fields for simplicity of notation. Spin and finite-dimensional internal labels only insert the corresponding covariance matrices and do not change the domain argument. For an ordering $\sigma \in S_n$, define the ordered n-point distribution

$$W_\sigma(x_1, \dots, x_n) = \langle \Omega, \Phi_{\sigma(1)}(x_{\sigma(1)}) \cdots \Phi_{\sigma(n)}(x_{\sigma(n)})\Omega\rangle.$$

Translation invariance allows one to remove the center-of-mass variable. For the displayed ordering, put

$$\xi_j = x_{\sigma(j)} - x_{\sigma(j+1)}, \qquad j = 1, \dots, n-1,$$

and write the reduced distribution as $w_\sigma(\xi_1, \dots, \xi_{n-1})$. Let P^μ be the self-adjoint generators of spacetime translations and let $\overline{V}_+$ denote the closed future light cone in momentum space, with the mostly-plus mass convention fixed in Volume I.

Theorem 28.2 (Primitive Wightman tube-domain theorem)**.** *Assume the Wightman hypotheses for the fields appearing above: a translation-invariant vacuum, strong continuity of translations, the spectrum condition*

$$\operatorname{Spec}(P) \subset \overline{V}_+,$$

and temperedness of all vacuum matrix elements. Then, for every ordering σ,

$$\operatorname{supp} \widehat{w}_\sigma \subset (\overline{V}_+)^{n-1} \tag{28.1}$$

in the reduced Fourier variables conjugate to $(\xi_1, \dots, \xi_{n-1})$. Consequently w_σ is the tempered boundary value of a holomorphic function

$$\mathcal{W}_\sigma(z_1, \dots, z_{n-1}), \qquad z_j = \xi_j - \mathrm{i}\eta_j,$$

on the tube

$$T_{n-1} = \{(\xi_j - \mathrm{i}\eta_j)_{j=1}^{n-1} \mid \eta_j \in V_+^\circ,\ j = 1, \dots, n-1\}, \tag{28.2}$$

with the same damping convention as the Wightman tube construction of Chapter 16: for $p, \eta \in \overline{V}_+$ the mostly-plus pairing satisfies $p \cdot \eta \leq 0$, so the factor $\exp(p \cdot \eta)$ is damping. Equivalently, it is Proposition 12.13 after the identification of its positive dual variable with $-\eta$ in the Lorentzian pairing. More precisely, for every closed subcone $K \Subset (V_+^\circ)^{n-1}$ the tested functions

$$f \longmapsto \int \mathcal{W}_\sigma(\xi - \mathrm{i}\eta) f(\xi)\, \mathrm{d}^{D(n-1)}\xi$$

are bounded by finitely many Schwartz seminorms of f, with constants polynomial in the reciprocal distance of η from the boundary of the cone, and converge in $\mathcal{S}'$ to w_σ as $\eta \to 0$ inside K.

Proof. We first prove the support statement. It is enough to test $\widehat{w}_\sigma$ against a compactly supported smooth function $\widehat{f}(p_1, \dots, p_{n-1})$ whose support is disjoint from $(\overline{V}_+)^{n-1}$. Choose Schwartz functions

$$f(\xi_1, \dots, \xi_{n-1}) = \int \mathrm{e}^{-\mathrm{i}\sum_j p_j \cdot \xi_j} \widehat{f}(p_1, \dots, p_{n-1}) \prod_j \frac{\mathrm{d}^D p_j}{(2\pi)^D}.$$

The smeared reduced Wightman distribution is obtained by inserting the joint spectral resolution of the translation representation between adjacent fields. On the common invariant domain this gives the weak identity

$$\langle \widehat{w}_\sigma, \widehat{f} \rangle = \int \widehat{f}(p_1, \dots, p_{n-1})\, \mathrm{d}\mu_\sigma(p_1, \dots, p_{n-1}), \tag{28.3}$$

where the scalar distributional measure μ_σ is built from the successive spectral projections of the translation representation applied to the vectors generated by the ordered fields. The j-th momentum p_j is the total momentum carried by the state between the j-th and $(j+1)$-st inserted fields. By the spectrum condition each such intermediate momentum lies in $\overline{V}_+$. Therefore the right hand side of (28.3) vanishes whenever $\operatorname{supp} \widehat{f} \cap (\overline{V}_+)^{n-1} = \varnothing$, which is exactly (28.1).

For completeness we spell out why the preceding spectral-resolution formula is legitimate for distributional fields. First smear each field with a Schwartz function and use the spectral theorem for the strongly continuous unitary translation group between the smeared field operators. The translation covariance relation

$$U(a)\Phi_b(h)U(a)^{-1} = \Phi_b(h_a), \qquad h_a(x) = h(x - a),$$

turns translations of adjacent insertions into the Fourier factors in (28.3). The Wightman temperedness hypothesis gives continuity in the Schwartz topology, so the identity extends from tensor-product test functions to all Schwartz tests in the reduced variables and then by density to the displayed $\widehat{f}$.

The holomorphic continuation is now the Fourier–Laplace theorem for tempered distributions carried by a closed convex cone, applied to the cone $(\overline{V}_+)^{n-1}$. The exponential damping in the imaginary directions $\eta_j \in V_+^\circ$ makes the Fourier transform into a holomorphic function of $z_j = \xi_j - \mathrm{i}\eta_j$. The finite-order Schwartz-seminorm estimate for $\widehat{w}_\sigma$ gives the displayed polynomial tube bounds after differentiating the exponential multiplier. The same estimate, with the exponential multiplier tending to 1 in the Schwartz topology after testing, gives the boundary value $\mathcal{W}_\sigma(\cdot - \mathrm{i}\eta) \to w_\sigma$ in $\mathcal{S}'$. This is precisely the construction written abstractly in Proposition 12.13. $\square$

The primitive tube is only the first part of the domain. Locality determines how different ordered tubes are glued. The relevant real edge is the Jost set. In the reduced variables of an ordering, define

$$C(\xi) = \left\{ \sum_{j=1}^{n-1} \lambda_j \xi_j \,\middle|\, \lambda_j \geq 0 \right\}.$$

A real reduced configuration ξ is a Jost configuration if every nonzero vector in $C(\xi)$ is spacelike. With the mostly-plus convention, this means $v^2 > 0$ for all $0 \neq v \in C(\xi)$. In particular, every difference $x_{\sigma(i)} - x_{\sigma(k)} = \xi_i + \cdots + \xi_{k-1}$ between two ordered insertion points is spacelike.

Theorem 28.3 (Jost-edge gluing of primitive tubes)**.** *Assume, in addition to the hypotheses of Theorem 28.2, graded local commutativity on the common Wightman domain:*

$$\Phi_a(f)\Phi_b(g) = (-1)^{|a||b|}\Phi_b(g)\Phi_a(f)$$

whenever the supports of f and g are spacelike separated. Let σ' be obtained from σ by one adjacent transposition of field labels whose two insertion points are spacelike throughout an open Jost neighbourhood U_J. Then the two boundary distributions $\operatorname{bv}\mathcal{W}_\sigma$ and $\operatorname{bv}\mathcal{W}_{\sigma'}$, expressed in the same real coordinates on U_J, agree there up to the Koszul sign of the transposition. Hence the distributional edge-of-the-wedge theorem gives a single holomorphic function on the envelope of holomorphy of the union of the two adjacent primitive tubes and the common Jost edge. Iterating adjacent transpositions gives a well-defined holomorphic germ on the primitive Bros–Epstein–Glaser envelope generated by all ordered Wightman tubes and their Jost-edge identifications.

Proof. Choose test functions supported in a coordinate neighbourhood of U_J. For the adjacent pair being transposed, every point in the product support has spacelike separation. Graded local commutativity therefore permits the two smeared fields to be exchanged on the common invariant domain, with the factor $(-1)^{|a||b|}$. Leaving all other insertions fixed gives equality of the two ordered Wightman distributions on U_J, again with the same Koszul factor. This proves equality of the real boundary values on the Jost edge.

By Theorem 28.2, each ordered distribution is the boundary value of a holomorphic function with polynomial tube bounds. The distributional edge-of-the-wedge theorem applies exactly to such a pair: holomorphic functions in two opposite or adjacent wedges with equal distributional boundary values on a common real edge extend to the envelope of holomorphy of their union. The extension is unique because a holomorphic function with zero distributional boundary value on an open edge vanishes in the connected envelope. Applying this step to a sequence of adjacent transpositions gives the asserted primitive envelope. If two different transposition sequences relate the same orderings, uniqueness on the overlaps makes the resulting germs identical; thus the iterated continuation is path independent inside the generated envelope. □

Remark 28.4 (Output of the primitive domain theorem for four-point scattering). For a massive Wightman theory with stable scalar external particles, the connected four-point Wightman distributions possess a crossing-symmetric primitive analytic germ obtained from Theorems 28.2 and 28.3. After separating the connected part and the external one-particle poles, LSZ boundary values in the physical s-, t-, and u-channel regions are boundary values of this germ on the corresponding real edges whenever those LSZ limits exist. Indeed, for $n = 4$, the preceding two theorems apply to each ordering and glue the ordered tubes along Jost edges. Passing to the connected part removes finite sums of products of lower-point distributions with the same spectral support and locality properties. The statement about scattering is not an additional analyticity theorem: it is the conditional transfer of this connected four-point germ through the already defined LSZ boundary operation, with wave-packet smearing, removal of the external one-particle pole factors, and the mass-shell limit taken only on edges where that limit exists.

28.2.2 Causal Commutators and the Dyson Representation

The next domain theorem uses the commutator rather than the ordered Wightman functions. It is the analytic reason that a coincidence region in momentum space can be converted into a representation by mass hyperboloids. This is the Jost–Lehmann–Dyson mechanism in the form needed for scattering amplitudes.

Let $C(x) \in \mathcal{S}'(\mathbb{M}^4)$ be a scalar tempered distribution whose support is contained in the closed causal cone

$$\overline{V}_{\text{caus}} = \{x \in \mathbb{M}^4 \mid x^2 \leq 0\}$$

in the mostly-plus convention. In QFT applications

$$C(x) = \langle Q_f, [J_1(x/2), J_2(-x/2)]Q_i\rangle \tag{28.4}$$

with J_i local current operators and with Q_i, Q_f finite-energy vectors. Local commutativity gives the support condition. Let

$$\widehat{C}(q) = \langle C(x), \mathrm{e}^{\mathrm{i}q\cdot x}\rangle$$

be its Fourier transform, again as a tempered distribution.

The representation also uses a momentum-space coincidence region. Let

$$R = \{q \in \mathbb{R}^4 \mid h_-(\vec{q}) < q^0 < h_+(\vec{q})\} \tag{28.5}$$

where $h_\pm : \mathbb{R}^3 \to \mathbb{R}$ are continuous functions satisfying

$$|h_\pm(\vec{q}) - h_\pm(\vec{q}\,')| \leq |\vec{q} - \vec{q}\,'|$$

and $h_-(\vec{q}) < h_+(\vec{q})$. Thus the boundary graphs are spacelike or null. Assume

$$\widehat{C}(q) = 0 \qquad q \in R \tag{28.6}$$

as a distributional restriction to R. For $u \in \mathbb{R}^4$ and $\mu \geq 0$, define the two mass-hyperboloid sheets

$$H_\pm(u, \mu) = \{q \in \mathbb{R}^4 \mid (q - u)^2 = -\mu^2,\ \pm(q^0 - u^0) > 0\}.$$

The pair (u,μ) is called R-admissible if

$$q \in H_+(u,\mu) \Rightarrow q^0 \geq h_+(\vec{q}), \qquad q \in H_-(u,\mu) \Rightarrow q^0 \leq h_-(\vec{q}). \tag{28.7}$$

Let $S_R \subset \mathbb{R}^4 \times [0,\infty)$ be the closed set of R-admissible pairs.

Lemma 28.5 (Distributional Cauchy formula for the six-dimensional wave equation)**.** *Let $Y = (\tau, \mathbf{y}) \in \mathbb{R}^{1,5}$, with $\mathbf{y} \in \mathbb{R}^5$, and let*

$$\Box_{1,5} = \partial_\tau^2 - \Delta_{\mathbf{y}}.$$

Let $F \in \mathcal{S}'(\mathbb{R}^{1,5})$ solve

$$\Box_{1,5} F = 0$$

in the distributional sense. Fix an affine spacelike hyperplane $\Sigma_{\tau_*} = \{\tau = \tau_*\}$, and suppose the Cauchy traces

$$f = F|_{\Sigma_{\tau_*}}, \qquad g = \partial_\tau F|_{\Sigma_{\tau_*}}$$

exist as tempered distributions on $\mathbb{R}^5$. Let D_6 be the odd fundamental solution of the six-dimensional wave equation, normalized by

$$\mathcal{F}_{\mathbf{y}} D_6(t,\cdot)(\mathbf{k}) = \frac{\sin(|\mathbf{k}|t)}{|\mathbf{k}|}, \qquad \partial_t D_6(0,\mathbf{y}) = \delta(\mathbf{y}), \qquad D_6(0,\mathbf{y}) = 0. \tag{28.8}$$

Then, as a tempered distribution in $Y_0 = (\tau_0, \mathbf{y}_0)$,

$$F(Y_0) = \langle f(\mathbf{y}), \partial_\tau D_6(\tau_* - \tau_0, \mathbf{y} - \mathbf{y}_0)\rangle_{\mathbf{y}} - \langle g(\mathbf{y}), D_6(\tau_* - \tau_0, \mathbf{y} - \mathbf{y}_0)\rangle_{\mathbf{y}}. \tag{28.9}$$

Equivalently, for an affine spacelike hyperplane $\Sigma \subset \mathbb{R}^{1,5}$ with future unit normal n,

$$F(Y_0) = \int_\Sigma \left(F(Y)\,\partial_{n_Y} D_6(Y - Y_0) - D_6(Y - Y_0)\,\partial_{n_Y} F(Y)\right) \mathrm{d}\Sigma(Y), \tag{28.10}$$

where the displayed integral denotes the distributional pairing of the Cauchy traces with the kernel.

Proof. It is enough to prove the formula for $\tau_* = 0$; translating τ gives the general case. Taking the Fourier transform in the five spatial variables converts the wave equation into

$$\partial_\tau^2 \widehat{F}(\tau,\mathbf{k}) + |\mathbf{k}|^2 \widehat{F}(\tau,\mathbf{k}) = 0 \tag{28.11}$$

in $\mathcal{S}'(\mathbb{R}_\tau \times \mathbb{R}^5_{\mathbf{k}})$. A distributional solution of this constant-coefficient ordinary differential equation is uniquely determined by

$$\widehat{F}(0,\mathbf{k}) = \widehat{f}(\mathbf{k}), \qquad \partial_\tau \widehat{F}(0,\mathbf{k}) = \widehat{g}(\mathbf{k}),$$

because the difference of two such solutions has zero Cauchy data and therefore pairs to zero with every compactly supported test function after solving the adjoint final-value problem. The solution is

$$\widehat{F}(\tau,\mathbf{k}) = \cos(|\mathbf{k}|\tau)\widehat{f}(\mathbf{k}) + \frac{\sin(|\mathbf{k}|\tau)}{|\mathbf{k}|}\widehat{g}(\mathbf{k}), \tag{28.12}$$

with the quotient interpreted by its smooth value τ at $\mathbf{k} = 0$. Inverse Fourier transformation and the normalization (28.8) give (28.9). The geometric form follows by applying a real Lorentz transformation carrying the chosen affine spacelike hyperplane to a constant-τ hyperplane; the wave operator, D_6, the normal derivative, and the distributional pairing are invariant under that transformation. $\square$

Lemma 28.6 (Auxiliary pushforward and the $p = 0$ momentum trace). *Let*

$$U \in \mathcal{S}'(\mathbb{R}^4_x \times \mathbb{R}^2_y)$$

and define its Fourier transform by duality from the convention

$$\Psi^\vee(x, y) = \int_{\mathbb{R}^4 \times \mathbb{R}^2} \Psi(q, p) \mathrm{e}^{\mathrm{i}q\cdot x + \mathrm{i}p\cdot y} \,\mathrm{d}^4 q \,\mathrm{d}^2 p, \qquad \langle \widehat{U}, \Psi \rangle = \langle U, \Psi^\vee \rangle. \tag{28.13}$$

Choose $\rho \in \mathcal{S}(\mathbb{R}^2_p)$ with $\int \rho(p) \,\mathrm{d}^2 p = 1$, set

$$\rho_\epsilon(p) = \epsilon^{-2} \rho(p/\epsilon), \qquad m_\epsilon(y) = \int_{\mathbb{R}^2} \rho_\epsilon(p) \mathrm{e}^{\mathrm{i}p\cdot y} \,\mathrm{d}^2 p. \tag{28.14}$$

Suppose that for every $\varphi \in \mathcal{S}(\mathbb{R}^4_x)$ the limit

$$\langle \pi_{y*} U, \varphi \rangle := \lim_{\epsilon \downarrow 0} \langle U, \varphi(x) m_\epsilon(y) \rangle \tag{28.15}$$

exists and is independent of the normalized approximate identity ρ. Then $\pi_{y*}U \in \mathcal{S}'(\mathbb{R}^4_x)$, and the trace of $\widehat{U}$ at $p = 0$, defined by

$$\left\langle \iota^* \widehat{U}, \psi \right\rangle := \lim_{\epsilon \downarrow 0} \left\langle \widehat{U}(q, p), \psi(q) \rho_\epsilon(p) \right\rangle, \qquad \iota(q) = (q, 0), \tag{28.16}$$

exists for every $\psi \in \mathcal{S}(\mathbb{R}^4_q)$ and obeys

$$\iota^* \widehat{U} = \widehat{\pi_{y*} U}. \tag{28.17}$$

Conversely, if the limits in (28.16) exist for every ψ and are continuous in the Schwartz topology, then (28.15) defines a tempered y-pushforward and (28.17) holds.

Proof. For $\psi \in \mathcal{S}(\mathbb{R}^4_q)$, define

$$\psi^\vee(x) = \int_{\mathbb{R}^4} \psi(q) \mathrm{e}^{\mathrm{i}q\cdot x} \,\mathrm{d}^4 q. \tag{28.18}$$

With the convention (28.13), the definition of the Fourier transform of a tempered distribution gives

$$\left\langle \widehat{U}(q, p), \psi(q) \rho_\epsilon(p) \right\rangle = \left\langle U(x, y), \psi^\vee(x) m_\epsilon(y) \right\rangle. \tag{28.19}$$

If the pushforward limit exists, the right side tends to

$$\langle \pi_{y*} U, \psi^\vee \rangle = \langle \widehat{\pi_{y*} U}, \psi \rangle,$$

which proves existence of the trace and (28.17). Continuity follows because the pointwise limit of the continuous functionals $\varphi \mapsto \langle U, \varphi(x) m_\epsilon(y) \rangle$ on the Schwartz space is continuous by the uniform boundedness principle for Frechet spaces; the Fourier transform is a topological automorphism of the Schwartz space.

Conversely, the map $\psi \mapsto \psi^\vee$ is a topological automorphism of $\mathcal{S}(\mathbb{R}^4)$. Given $\varphi \in \mathcal{S}(\mathbb{R}^4)$, choose ψ with $\psi^\vee = \varphi$. If the limits in (28.16) exist continuously, then (28.19) defines the limit in (28.15); the resulting functional on φ is tempered, and its Fourier transform is precisely $\iota^* \widehat{U}$. This proves the converse and the identity. $\square$

Definition 28.7 (Dyson-regular causal commutator)**.** A causal commutator distribution $C \in \mathcal{S}'(\mathbb{M}^4)$ is Dyson-regular if its Fourier transform is the restriction to the plane $p_1 = p_2 = 0$ of a rotationally invariant distribution

$$F(q, p_1, p_2) \in \mathcal{S}'(\mathbb{R}^{1,5})$$

with the following properties:

1. F solves the six-dimensional wave equation

$$\Box_{1,5} F = 0$$

   in the distributional sense.

2. On every affine spacelike hyperplane used in the Dyson argument, the traces $F|_\Sigma$ and $\partial_n F|_\Sigma$ exist as tempered distributions, and the distributional Cauchy formula (28.10) holds.

3. If U is the inverse Fourier transform of F, then U is supported on the six-dimensional light cone, the auxiliary y-pushforward $\pi_{y*}U$ exists in the sense of Lemma 28.6, and

$$\pi_{y*}U = \kappa C$$

   for a fixed nonzero normalization κ. Equivalently, after absorbing κ into the definition of the lift,

$$\iota^* F = \widehat{C}, \qquad \iota(q) = (q, 0, 0).$$

For a sufficiently regular function $C(x)$ with causal support this lift is represented by Dyson's formula

$$\widetilde{C}(x, y) = 4\pi C(x)\delta(x_D^2 - |y|^2), \qquad y \in \mathbb{R}^2,$$

in the temporary mostly-minus notation $x_D^2 = (x^0)^2 - |\vec{x}|^2$. This display is not a definition for an arbitrary Wightman distribution: multiplying an arbitrary distribution by the light-cone delta is a microlocal operation and requires a transversality or regularization argument. The definition records exactly the regularity datum needed for the Dyson proof; verifying it for a specified class of unsmeared QFT matrix coefficients is a separate theorem, not a notational convention.

Definition 28.8 (Support-pruned Dyson data in a coincidence slab)**.** Let R be the coincidence slab in (28.5), and let S_R be the corresponding closed set of admissible hyperboloids. For a distribution

$$\Psi \in \mathcal{D}'(\mathbb{R}^4_u \times [0, \infty)_\mu)$$

for which the pairing is defined, write

$$(\mathcal{H}\Psi)(q) = \big\langle \Psi(u, \mu), \varepsilon(q^0 - u^0)\delta\big((q-u)^2 + \mu^2\big)\big\rangle_{u,\mu} \tag{28.20}$$

for the Dyson hyperboloid transform. A Dyson-regular commutator has support-pruned Dyson data relative to R if the Cauchy-data distribution Ψ_Σ produced from its six-dimensional wave lift can be replaced, modulo the kernel of $\mathcal{H}$, by a distribution Ψ_R satisfying

$$\operatorname{supp} \Psi_R \subset S_R, \qquad \mathcal{H}\Psi_R = \mathcal{H}\Psi_\Sigma. \tag{28.21}$$

The condition is independent of the chosen affine Cauchy hyperplane in the following sense: changing the hyperplane changes Ψ_Σ by a distribution in the kernel of $\mathcal{H}$, because both Cauchy surfaces reconstruct the same six-dimensional wave solution by Lemma 28.5.

Proposition 28.9 (Axis trace nonuniqueness)**.** *Let the six-dimensional wave operator be written in affine coordinates*

$$Y = (\tau, \xi, p) \in \mathbb{R}_\tau \times \mathbb{R}^3_\xi \times \mathbb{R}^2_p, \qquad \Box_{1,5} = \partial_\tau^2 - \Delta_\xi - \Delta_p.$$

There exists a nonzero $O(2)$-invariant tempered smooth solution

$$F \in C^\infty(\mathbb{R}^{1,5})$$

of $\Box_{1,5} F = 0$ such that

$$F(\tau, \xi, 0) = 0 \qquad \textit{for } |\tau| < 1,\ \xi \in \mathbb{R}^3. \tag{28.22}$$

Thus vanishing of the auxiliary-momentum trace on a coincidence slab does not by itself imply that the six-dimensional Cauchy data vanish in the region of Cauchy-data space whose hyperboloids meet that slab.

Proof. Choose a nonzero function

$$a \in C_c^\infty((2,3))$$

and define radial Cauchy data on the auxiliary p-plane by

$$f(p) = a(|p|), \qquad g(p) = 0.$$

Let $F_0(\tau, p)$ be the solution of the $1+2$-dimensional wave equation

$$(\partial_\tau^2 - \Delta_p) F_0 = 0, \qquad F_0(0,p) = f(p), \qquad \partial_\tau F_0(0,p) = g(p).$$

The Cauchy problem with smooth compactly supported data has a unique smooth solution, obtained for instance by Fourier transform in p. Rotational invariance of the operator and uniqueness imply that $F_0(\tau, p)$ is $O(2)$-invariant. The finite-propagation estimate follows from the local energy identity. For a solution u of $(\partial_\tau^2 - \Delta_p)u = 0$, the stress tensor

$$e = \frac{1}{2}\big((\partial_\tau u)^2 + |\nabla_p u|^2\big), \qquad j = -\partial_\tau u \, \nabla_p u$$

satisfies $\partial_\tau e + \nabla_p \cdot j = 0$. Integrating this identity over a truncated backward cone with vertex (τ_0, p_0) gives

$$E(s_2) + \int_{\text{null side}} \frac{1}{2}\big(\partial_\tau u + \hat{n} \cdot \nabla_p u\big)^2 \, \mathrm{d}\sigma = E(s_1),$$

where $E(s)$ is the energy on the disk $|p - p_0| \le \tau_0 - s$ at time s and $\hat{n}$ is the outward radial unit vector on the side of the cone. The side flux is nonnegative, so zero Cauchy data on the base disk force u to vanish in the cone. Hence the value at (τ, p_0) depends only on the data in

$$\{p \in \mathbb{R}^2 : |p - p_0| \le |\tau|\}.$$

For $p_0 = 0$ and $|\tau| < 1$, this disk is disjoint from $\operatorname{supp} f \subset \{|p| > 2\}$ and from $\operatorname{supp} g$. Hence $F_0(\tau, 0) = 0$ for $|\tau| < 1$. On the other hand $F_0(0,p) = a(|p|)$ is not identically zero.

Define

$$F(\tau, \xi, p) = F_0(\tau, p).$$

Then F is independent of ξ, solves $\partial_\tau^2 F - \Delta_\xi F - \Delta_p F = 0$, is $O(2)$-invariant in the auxiliary variables, and satisfies (28.22). Since F_0 is smooth and has at most polynomial growth as a distribution under the Fourier-transform solution formula, its pullback constant in ξ is tempered on $\mathbb{R}^{1,5}$. This proves the claim. $\square$

Definition 28.10 (Dyson regularity modulo contact terms)**.** A causal commutator $C \in \mathcal{S}'(\mathbb{M}^4)$ is Dyson-regular modulo contact terms if there is a finite-order distribution L supported at $x = 0$ such that $C - L$ is Dyson-regular in the sense of Definition 28.7. Equivalently, $\widehat{C}$ has the Dyson hyperboloid representation of Theorem 28.15 plus a polynomial in q. This is the form actually used after source-current reduction, because applying differential wave operators to time-ordered products can produce local terms at coincident points.

Definition 28.11 (Local scaling degree)**.** Let $T \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$. For $\varphi \in C_c^\infty(\mathbb{R}^n)$, set

$$\varphi_\lambda(z) = \lambda^{-n}\varphi(z/\lambda), \qquad 0 < \lambda < 1.$$

The scaling degree of T at the origin is

$$\mathrm{sd}_0(T) = \inf\left\{\omega \in \mathbb{R} \;\middle|\; \lambda^{\omega+\epsilon}\langle T, \varphi_\lambda\rangle \to 0 \text{ for every } \epsilon > 0 \text{ and every } \varphi\right\}. \tag{28.23}$$

Finite scaling degree means $\mathrm{sd}_0(T) < \infty$. With this normalization a smooth function has scaling degree 0, while $\partial^\alpha\delta_0$ in $\mathbb{R}^n$ has scaling degree $n + |\alpha|$.

Proposition 28.12 (Microlocal status of the Dyson light-cone lift)**.** *Let*

$$X^\circ = (\mathbb{M}^4 \times \mathbb{R}^2) \setminus \{(0,0)\}, \qquad g(x,y) = x_D^2 - |y|^2, \qquad \pi(x,y) = x,$$

where $x_D^2 = (x^0)^2 - |\vec{x}|^2$*. Let*

$$\mathcal{N}_{\mathrm{lc}} = \{(x; \lambda\,\mathrm{d}_x x_D^2) \mid x \neq 0,\ x_D^2 = 0,\ \lambda \in \mathbb{R}^\times\} \subset T^*\mathbb{M}^4 \setminus 0$$

be the conormal bundle of the punctured four-dimensional light cone. If $C \in \mathcal{D}'(\mathbb{M}^4)$ *satisfies*

$$\mathrm{WF}(C) \cap \mathcal{N}_{\mathrm{lc}} = \varnothing, \tag{28.24}$$

then the product

$$U^\circ = \pi^* C \cdot \delta(g) \tag{28.25}$$

is a well-defined distribution on X°*. If* $\operatorname{supp} C \subset \overline{V}_{\mathrm{caus}}$*, then*

$$\operatorname{supp} U^\circ \subset \{(x,y) \in X^\circ \mid x \in \overline{V}_{\mathrm{caus}},\ x_D^2 = |y|^2\}.$$

Any two extensions of U° *to* $\mathbb{M}^4 \times \mathbb{R}^2$ *differ by a distribution supported at* $(0,0)$*. Hence their six-dimensional Fourier transforms differ by a polynomial in the six dual momenta, and their restrictions to* $p_1 = p_2 = 0$ *differ by a polynomial in* q*.*

Consequently, if $\mathrm{WF}(C)$ *contains a nonzero covector conormal to the punctured light cone, Dyson's displayed product*

$$C(x)\delta(x_D^2 - |y|^2)$$

is not canonically defined by the standard product of distributions. In that case a Dyson lift, if it exists, contains extra renormalized extension data; that data is invisible away from the light-cone vertex only after one proves the corresponding finite-contact ambiguity and the six-dimensional wave equation.

Proof. The projection $\pi : X^\circ \to \mathbb{M}^4$ is a submersion, so the pullback π^*C is always defined and

$$\mathrm{WF}(\pi^*C) \subset \{(x, y; \xi, 0) \mid (x; \xi) \in \mathrm{WF}(C)\}. \tag{28.26}$$

On X°, the hypersurface $g = 0$ is smooth: the only zero of $\mathrm{d}g$ is $(0, 0)$, which has been removed. Therefore

$$\mathrm{WF}(\delta(g)) = \{(x, y; \lambda\, \mathrm{d}_x g, \lambda\, \mathrm{d}_y g) \mid g(x, y) = 0,\ \lambda \in \mathbb{R}^\times\}. \tag{28.27}$$

Hormander's product criterion says that the product of two distributions is defined if their wavefront sets contain no pair of opposite covectors over the same base point. A possible obstruction to multiplying π^*C by $\delta(g)$ would therefore require

$$(\xi, 0) + (\lambda\, \mathrm{d}_x g, \lambda\, \mathrm{d}_y g) = 0.$$

If $y \neq 0$, then $\mathrm{d}_y g = -2y \cdot \mathrm{d}y \neq 0$, and the second component cannot vanish for $\lambda \neq 0$. Hence the only obstruction can occur at $y = 0$. On $g = 0$, this forces $x_D^2 = 0$, and the remaining obstruction is precisely

$$(x; \xi) = (x; -\lambda\, \mathrm{d}_x x_D^2) \in \mathcal{N}_{\mathrm{lc}}.$$

The transversality condition (28.24) excludes this case and proves that (28.25) is a well-defined distribution on X°.

The support statement follows from the elementary support rule for a defined product of distributions:

$$\mathrm{supp}(\pi^*C \cdot \delta(g)) \subset \pi^{-1}(\mathrm{supp}\, C) \cap \{g = 0\}.$$

If U_1 and U_2 are two extensions to the full six-dimensional space, then $U_1 - U_2$ vanishes on X°, so its support is contained in the single point $(0, 0)$. A distribution supported at one point is a finite linear combination of derivatives of the delta distribution at that point:

$$U_1 - U_2 = \sum_{|\alpha|+|\beta| \leq N} c_{\alpha\beta}\, \partial_x^\alpha \partial_y^\beta \delta(x)\delta(y).$$

Fourier transformation sends this finite sum to a polynomial in (q, p_1, p_2), and restriction to $p_1 = p_2 = 0$ gives a polynomial in q. This is exactly the local-contact ambiguity recorded in the Dyson representation. □

Theorem 28.13 (Punctured Dyson lift and finite scaling degree)**.** *Let $C \in \mathcal{S}'(\mathbb{M}^4)$ be supported in $\overline{V}_{\mathrm{caus}}$ and suppose that the wavefront transversality condition*

$$\mathrm{WF}(C) \cap \mathcal{N}_{\mathrm{lc}} = \varnothing$$

holds on the punctured four-dimensional light cone. Let

$$U^\circ = \kappa\, \pi^*C \cdot \delta(x_D^2 - |y|^2) \qquad \text{on } X^\circ = (\mathbb{M}^4 \times \mathbb{R}^2) \setminus \{0\}, \tag{28.28}$$

where $\kappa \neq 0$ is a fixed normalization. Assume that U° has finite scaling degree at the origin in $\mathbb{R}^6$, and that it admits a tempered $O(2)$-invariant extension

$$U \in \mathcal{S}'(\mathbb{R}^6)$$

with the following three further properties:

$$\mathrm{supp}\, U \subset \{(x, y) \mid x \in \overline{V}_{\mathrm{caus}},\ x_D^2 = |y|^2\}. \tag{28.29}$$

$$(x_D^2 - |y|^2)\, U = 0, \tag{28.30}$$

and the y-pushforward of U, with the chosen normalization κ, equals C modulo a distribution supported at $x = 0$. Then the Fourier transform

$$F(q, p_1, p_2) = \widehat{U}(q, p_1, p_2)$$

is rotationally invariant in (p_1, p_2) and solves

$$\Box_{1,5} F = 0 \tag{28.31}$$

in the distributional sense. The restriction of F to $p_1 = p_2 = 0$, if defined by the chosen extension, equals $\widehat{C}(q)$ up to a polynomial in q. Hence C is Dyson-regular modulo contact terms once the restriction and Cauchy-data requirements in Definition 28.7 are satisfied by F.

Moreover, if U_1 and U_2 are two such finite-scaling-degree extensions, then

$$U_1 - U_2 = \sum_{|\alpha|+|\beta| \le N} c_{\alpha\beta}\, \partial_x^\alpha \partial_y^\beta \delta(x)\delta(y) \tag{28.32}$$

for some N. Thus the induced ambiguity in $F(q, 0, 0)$ is a finite polynomial in q, and no cut, threshold, or hyperboloid support is changed by choosing a different extension.

Proof. The product U° on X° is well-defined by Proposition 28.12. We first recall the extension mechanism because it is the local analytic part of the argument. Let $n = 6$ and let $T^\circ \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ have finite scaling degree s at the origin. Put

$$m = \lfloor s \rfloor - n$$

when $s \ge n$, and $m = -1$ otherwise. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ equal to 1 near the origin. For a test function φ, define

$$R_m\varphi(z) = \varphi(z) - \chi(z) \sum_{|\alpha| \le m} \frac{z^\alpha}{\alpha!} \partial^\alpha \varphi(0) \tag{28.33}$$

with the sum omitted when $m = -1$. The function $R_m\varphi$ vanishes at the origin to order $m+1$. Let $\chi_\epsilon(z) = \chi(z/\epsilon)$. The finite-scaling-degree bound makes the limit

$$T_0(\varphi) = \lim_{\epsilon \downarrow 0} T^\circ\big((1 - \chi_\epsilon) R_m \varphi\big)$$

finite and continuous in the test-function topology: the Taylor remainder removes precisely the jets whose scaling would make the punctured distribution divergent at the origin. The possible values of an extension on the finite jet removed in (28.33) are exactly the constants multiplying $\partial^\alpha \delta_0$ for $|\alpha| \le m$. This gives an extension of T°, preserves the scaling degree, and shows that two extensions with the same punctured distribution differ by a finite sum of derivatives of δ_0. If T° is supported in a closed cone with vertex at the origin, the same construction applied after restricting test functions supported away from that cone gives an extension with support in the closed cone; the only new support allowed by extension is the vertex itself.

Applying this statement to $T^\circ = U^\circ$ gives the asserted local ambiguity (28.32). The additional condition (28.30) is essential: support on the cone alone would still allow derivatives of the delta distribution in the normal direction at the vertex, and multiplication by the quadratic equation need not annihilate such terms. Fourier transformation converts multiplication by $x_D^2 - |y|^2$ into the six-dimensional wave operator in momentum space, up to the sign fixed by the Fourier convention. Therefore $\square_{1,5} F = 0$. Rotational invariance in the two auxiliary momenta is one of the stated extension requirements. Equivalently, if a non-invariant extension satisfying the other conditions has been constructed, averaging it over $O(2)$ preserves the punctured value, support, (28.30), and the pushforward normalization.

It remains to identify the four-dimensional restriction. On X°, the y-pushforward of the punctured expression is the intended four-dimensional commutator with the normalization κ. The theorem therefore requires the extension to preserve this pushforward modulo a distribution supported at $x = 0$. By Lemma 28.6, Fourier transformation of this y-pushforward is precisely the $p = 0$ trace of $\widehat{U}(q, p)$, when either side exists in the stated distributional sense. Hence the distribution obtained by restricting $\widehat{U}(q, p)$ to $p = 0$ differs from $\widehat{C}(q)$ by the Fourier transform of a distribution supported at $x = 0$, namely by a polynomial in q. The same conclusion follows directly from (28.32): setting $p = 0$ turns the Fourier transform of every $\partial_x^\alpha \partial_y^\beta \delta(x)\delta(y)$ into a polynomial in q, with the p-powers evaluated at zero.

The theorem has not proved the physical source-current kernels to be Dyson-regular. It proves the exact local construction that remains once the punctured wavefront condition, finite scaling degree, cone equation, pushforward normalization, temperedness, and Cauchy-data trace hypotheses have been verified for those kernels. Under those hypotheses the difference between the constructed lift and a strict Dyson lift is only a contact term, so Definition 28.10 applies. □

Proposition 28.14 (Source-current differentials preserve the punctured input)**.** *Let Φ_b, Φ_c be local scalar fields and let Ψ_i, Ψ_f be normalizable finite-energy vectors. Define the relative-coordinate commutator matrix element*

$$C_{bc}^{\Phi}(x) = \langle \Psi_f, [\Phi_b(x/2), \Phi_c(-x/2)]\Psi_i \rangle \in \mathcal{S}'(\mathbb{M}^4). \tag{28.34}$$

Let $\mathcal{K}_b(\partial_{x_b})$ and $\mathcal{K}_c(\partial_{x_c})$ be the mostly-plus Klein–Gordon LSZ operators used to define source currents

$$J_b = \mathcal{K}_b \Phi_b, \qquad J_c = \mathcal{K}_c \Phi_c.$$

Set

$$P_{bc}(\partial_x) = \mathcal{K}_b(2\partial_x)\mathcal{K}_c(-2\partial_x), \qquad r = \operatorname{ord} P_{bc}. \tag{28.35}$$

Then

$$C_{bc}^{J}(x) = \langle \Psi_f, [J_b(x/2), J_c(-x/2)]\Psi_i \rangle = P_{bc}(\partial_x) C_{bc}^{\Phi}(x) \tag{28.36}$$

as tempered distributions. Consequently:

1. *If* $\operatorname{supp} C_{bc}^{\Phi} \subset \overline{V}_{\text{caus}}$, *then*
$$\operatorname{supp} C_{bc}^{J} \subset \overline{V}_{\text{caus}}.$$

2. *If*
$$\mathrm{WF}(C_{bc}^{\Phi}) \cap \mathcal{N}_{\text{lc}} = \varnothing,$$
then
$$\mathrm{WF}(C_{bc}^{J}) \cap \mathcal{N}_{\text{lc}} = \varnothing.$$

3. *If C_{bc}^{Φ} has finite scaling degree s at $x = 0$, then C_{bc}^{J} has finite scaling degree at most $s + r$.*

4. Under the preceding wavefront and scaling assumptions, the punctured Dyson product

$$U_{bc}^{J,\circ} = \kappa\, \pi^* C_{bc}^J \cdot \delta(x_D^2 - |y|^2) \qquad \text{on } X^\circ \tag{28.37}$$

is well-defined and has finite scaling degree at the six-dimensional origin, bounded by $s+r+2$.

Thus replacing interpolating fields by LSZ source currents supplies no new punctured light-cone obstruction beyond finite-order differentiation. The remaining, genuinely nonlocal input for Theorem 28.13 is the existence of a tempered $O(2)$-invariant extension satisfying the cone equation, pushforward normalization, and Cauchy-data trace requirements.

Proof. The coordinate $x_b = x/2$ gives

$$\partial_{x_b^\mu} = 2\partial_{x^\mu},$$

whereas $x_c = -x/2$ gives

$$\partial_{x_c^\mu} = -2\partial_{x^\mu}.$$

Since the LSZ source currents are obtained by constant-coefficient differential operators acting on the corresponding field variable, the commutator matrix element is obtained from (28.34) by the operator (28.35). This proves (28.36).

Differentiation cannot enlarge support, so causal support is preserved. For wavefront sets, every constant-coefficient differential operator P satisfies

$$\mathrm{WF}(PT) \subset \mathrm{WF}(T). \tag{28.38}$$

Indeed, after multiplying by a cutoff χ equal to one near a point, the Fourier transform of χPT is a polynomial in the Fourier variable times the Fourier transform of χT, plus lower-order terms with cutoffs supported in the same neighbourhood; polynomial multiplication cannot destroy rapid decay in a conic neighbourhood where χT already has rapid decay. This proves the wavefront assertion.

The scaling-degree estimate is equally direct. For a multi-index α,

$$\langle \partial^\alpha T, \varphi_\lambda \rangle = (-1)^{|\alpha|} \lambda^{-|\alpha|} \langle T, (\partial^\alpha \varphi)_\lambda \rangle. \tag{28.39}$$

Thus $\mathrm{sd}_0(\partial^\alpha T) \le \mathrm{sd}_0(T) + |\alpha|$. Applying this to every derivative in P_{bc} gives $\mathrm{sd}_0(C_{bc}^J) \le s + r$.

The punctured product (28.37) is defined by Proposition 28.12, because the source-current commutator has the same conormal avoidance. It remains only to record finite scaling degree in six dimensions. The pullback $\pi^* C_{bc}^J$ has the same local scaling degree as C_{bc}^J: when a six-dimensional test function is scaled, integration over the two auxiliary y-variables produces a scaled four-dimensional test function with the same power of λ relevant to the x-distribution. The distribution $\delta(x_D^2 - |y|^2)$ is homogeneous of degree -2, hence has scaling degree 2 in $\mathbb{R}^6$. The product is a continuous microlocal product on the conic wavefront neighbourhoods fixed in the previous proposition, and the defining seminorm estimates give the standard subadditivity bound

$$\mathrm{sd}_0\big(\pi^* C_{bc}^J \cdot \delta(x_D^2 - |y|^2)\big) \le \mathrm{sd}_0(C_{bc}^J) + 2 \le s + r + 2. \tag{28.40}$$

This proves the final assertion. $\square$

Theorem 28.15 (Dyson representation of a causal commutator)**.** *Let $C \in \mathcal{S}'(\mathbb{M}^4)$ be Dyson-regular in the sense of Definition 28.7. Assume* $\operatorname{supp} C \subset \overline{V}_{\text{caus}}$*. Then there is a distribution*

$$\Psi \in \mathcal{D}'(\mathbb{R}^4_u \times [0,\infty)_\mu)$$

such that, in $\mathcal{D}'(\mathbb{R}^4_q)$,

$$\widehat{C}(q) = \big\langle \Psi(u,\mu), \varepsilon(q^0 - u^0)\,\delta\big((q-u)^2 + \mu^2\big)\big\rangle_{u,\mu}. \tag{28.41}$$

Here $\varepsilon(t) = 1$ for $t > 0$ and $\varepsilon(t) = -1$ for $t < 0$. If, in addition, $\widehat{C}$ vanishes in the slab R of (28.5) and the Dyson data are support-pruned relative to R in the sense of Definition 28.8, then Ψ can be chosen with

$$\operatorname{supp}\Psi \subset S_R.$$

If a later source-current or retarded-product reduction adds local terms supported at $x = 0$, their Fourier transforms are finite polynomials in q. Such local-contact polynomials are recorded separately; they do not change the hyperboloid support or the holomorphic domain away from infinity.

Proof. The proof is a hyperbolic Cauchy problem in disguise. We give it because the support statement is the part that carries the QFT locality information into the analytic domain. The Dyson-regularity hypothesis supplies the six-dimensional wave-equation solution F; without that hypothesis, the formal product $C(x)\delta(x_D^2 - |y|^2)$ at the light cone would not be a legitimate operation for an arbitrary tempered distribution.

Temporarily write the scalar product in Dyson's mostly-minus convention

$$x_D^2 = (x^0)^2 - |\vec{x}|^2 = -x^2.$$

The causal support hypothesis becomes $C(x) = 0$ for $x_D^2 < 0$. Add two auxiliary spacelike coordinates $y = (y_1, y_2)$ and set

$$Z = (x,y) \in \mathbb{R}^{1,5}, \qquad Z_D^2 = x_D^2 - y_1^2 - y_2^2.$$

Let $R_6 = (q, p_1, p_2)$ be the six-dimensional momentum dual to Z. By Dyson regularity there is a rotationally invariant distribution $F(R_6)$ such that

$$\Box_{1,5} F = 0. \tag{28.42}$$

and the original Fourier transform is recovered as the boundary value

$$\widehat{C}(q) = F(q,0,0)$$

after absorbing the fixed normalization constant in the definition of the lift. In the regular-function case this F is the Fourier transform of $4\pi C(x)\delta(Z_D^2)$, which explains the geometry of the construction.

For solutions of (28.42) with the Dyson-regular Cauchy traces, Lemma 28.5 gives, on every affine spacelike hyperplane $\Sigma \subset \mathbb{R}^{1,5}$,

$$F(R_0) = \int_\Sigma \big(F(R)\,\partial_{n_R} D_6(R - R_0) - D_6(R - R_0)\,\partial_{n_R} F(R)\big)\,\mathrm{d}\Sigma(R), \tag{28.43}$$

where D_6 is the odd fundamental solution of the six-dimensional wave operator and n_R is the future unit normal to Σ. This is a theorem about the linear wave equation and its distributional Cauchy data; no QFT input enters it after Dyson regularity has supplied the traces.

Take $R_0 = (q, 0, 0)$. Because D_6 is supported on the six-dimensional light cone, and because F is rotation-invariant in the last two momenta, the angular integral over the (p_1, p_2)-plane rewrites (28.43) as a superposition of four-dimensional mass hyperboloids:

$$F(q, 0, 0) = \left\langle \Psi_\Sigma(u, \mu), \varepsilon(q_D^0 - u_D^0)\delta\big((q_D - u_D)^2 - \mu^2\big) \right\rangle. \tag{28.44}$$

Here u is the four-momentum part of the point $R \in \Sigma$, while $\mu = (p_1^2 + p_2^2)^{1/2}$; the distribution Ψ_Σ is the push forward of the Cauchy data $(F|_\Sigma, \partial_n F|_\Sigma)$ with the differential operator that comes from $\partial_n D_6$. Returning to the mostly-plus convention changes $(q_D - u_D)^2 - \mu^2 = 0$ into $(q-u)^2 + \mu^2 = 0$, giving the kernel in (28.41).

The support-restricted conclusion is now exactly the content of Definition 28.8. If the commutator vanishes in the slab R and its Dyson data are support-pruned relative to R, choose Ψ_R satisfying (28.21). Since $\mathcal{H}\Psi_R = \mathcal{H}\Psi_\Sigma = \widehat{C}$, the same hyperboloid representation holds with $\operatorname{supp}\Psi_R \subset S_R$.

The geometric reason for the support-pruning condition is the following. If $(u, \mu) \notin S_R$, at least one sheet of the hyperboloid $(q-u)^2 = -\mu^2$ enters the coincidence slab. Since the slab boundaries are spacelike or null graphs, a small double cone in momentum space adapted to that sheet lies inside R. The classical Jost–Lehmann–Dyson proof uses the six-dimensional wave equation and domain-of-dependence uniqueness to remove the corresponding Cauchy-data component. In this monograph that removal is recorded as the support-pruning theorem still to be proved for the LSZ source-current kernels; it is not identified with the simpler statement that the trace $F(q, 0, 0)$ vanishes in R.

The statement about finite contact polynomials follows from differentiating the commutator or from derivatives of step functions in retarded products: Fourier transformation changes local distributions supported at $x = 0$ into polynomials in q. Such terms have no cut or hyperboloid support and are separated as subtraction/contact terms in dispersion formulae. □

Proposition 28.16 (Coincidence regions from the spectral condition)**.** *Let J_b, J_c be local source-current fields whose matrix elements with the states below define tempered distributions. Let Ψ_i, Ψ_f be normalizable vectors whose joint energy-momentum spectral supports are compact sets $B_i, B_f \subset \operatorname{Spec}(P)$, and let $E(\cdot)$ be the joint spectral measure of P^μ. Define the intermediate spectral matrix distributions*

$$\begin{aligned} \mathrm{d}\nu_{bc}^+(r_f, p, r_i) &= \langle E(\mathrm{d}r_f)\Psi_f, J_b(0)E(\mathrm{d}p)J_c(0)E(\mathrm{d}r_i)\Psi_i\rangle, \\ \mathrm{d}\nu_{bc}^-(r_f, p, r_i) &= \langle E(\mathrm{d}r_f)\Psi_f, J_c(0)E(\mathrm{d}p)J_b(0)E(\mathrm{d}r_i)\Psi_i\rangle. \end{aligned} \tag{28.45}$$

Let $\mathfrak{S}_{bc}^+$ and $\mathfrak{S}_{bc}^-$ be their supports in

$$B_f \times \operatorname{Spec}(P) \times B_i.$$

With the Fourier convention

$$\widehat{f}(q) = \int \mathrm{e}^{\mathrm{i}q\cdot x} f(x)\,\mathrm{d}^4x,$$

the two ordered matrix elements

$$C_{bc}^+(x) = \langle \Psi_f, J_b(x/2)J_c(-x/2)\Psi_i\rangle, \qquad C_{bc}^-(x) = \langle \Psi_f, J_c(-x/2)J_b(x/2)\Psi_i\rangle$$

obey

$$\operatorname{supp}\widehat{C}^{+}_{bc} \subset \left\{ p - \frac{1}{2}(r_f + r_i) \;\middle|\; (r_f, p, r_i) \in \mathfrak{S}^{+}_{bc} \right\},$$
$$\operatorname{supp}\widehat{C}^{-}_{bc} \subset \left\{ \frac{1}{2}(r_f + r_i) - p \;\middle|\; (r_f, p, r_i) \in \mathfrak{S}^{-}_{bc} \right\}. \tag{28.46}$$

Hence the commutator distribution

$$C_{bc} = C^{+}_{bc} - C^{-}_{bc}$$

has Fourier transform supported in the union of the two displayed sets. Every open set R disjoint from this union is a coincidence region:

$$\widehat{C}_{bc}(q) = 0 \qquad (q \in R). \tag{28.47}$$

The coarser but often sufficient replacement

$$\mathfrak{S}^{\pm}_{bc} \subset B_f \times \overline{V}_{+} \times B_i$$

uses

$$K_{fi} = \frac{1}{2}(B_f + B_i) = \left\{ \frac{1}{2}(r_f + r_i) \;\middle|\; r_f \in B_f,\ r_i \in B_i \right\}$$

and gives the model coincidence region between the two shifted spectral cones

$$\overline{V}_{+} - K_{fi}, \qquad K_{fi} - \overline{V}_{+}.$$

For generalized sharp external momenta Q_i, Q_f, one recovers the familiar shorthand by setting $K = (Q_f + Q_i)/2$ and

$$\Sigma^{fi}_{bc} = \operatorname{supp}\langle J_b(0)^{*}\Psi_f, E(\mathrm{d}p) J_c(0)\Psi_i\rangle, \qquad \Sigma^{fi}_{cb} = \operatorname{supp}\langle J_c(0)^{*}\Psi_f, E(\mathrm{d}p) J_b(0)\Psi_i\rangle,$$

so that the two ordered supports are contained in

$$\Sigma^{fi}_{bc} - K, \qquad K - \Sigma^{fi}_{cb}.$$

Proof. Translation covariance gives

$$J_b(x/2) = \mathrm{e}^{\mathrm{i}P\cdot x/2} J_b(0) \mathrm{e}^{-\mathrm{i}P\cdot x/2}, \qquad J_c(-x/2) = \mathrm{e}^{-\mathrm{i}P\cdot x/2} J_c(0) \mathrm{e}^{\mathrm{i}P\cdot x/2}.$$

Insert the spectral resolution of P on the external vectors and between the two currents. The first ordered matrix element becomes

$$\begin{aligned} C^{+}_{bc}(x) &= \int \mathrm{e}^{\mathrm{i}(r_f + r_i)\cdot x/2} \mathrm{e}^{-\mathrm{i}p\cdot x}\, \mathrm{d}\nu^{+}_{bc}(r_f, p, r_i) \\ &= \int \mathrm{e}^{\mathrm{i}((r_f + r_i)/2 - p)\cdot x}\, \mathrm{d}\nu^{+}_{bc}(r_f, p, r_i), \end{aligned} \tag{28.48}$$

where the equality is first read after smearing all fields and then extended to the matrix-element distribution by Wightman temperedness. Pairing against a test function in x and using the Fourier convention in the statement shows that $\widehat{C}^{+}_{bc}$ is supported where

$$q + \frac{1}{2}(r_f + r_i) - p = 0.$$

Thus $q = p - \frac{1}{2}(r_f + r_i)$ with $(r_f, p, r_i) \in \mathfrak{S}^+_{bc}$. This is the first inclusion in (28.46).

The second ordered product is

$$\begin{aligned} C^-_{bc}(x) &= \int e^{-i(r_f+r_i)\cdot x/2} e^{ip\cdot x}\, d\nu^-_{bc}(r_f, p, r_i) \\ &= \int e^{i(p-(r_f+r_i)/2)\cdot x}\, d\nu^-_{bc}(r_f, p, r_i). \end{aligned} \tag{28.49}$$

Therefore its Fourier transform is supported where

$$q + p - \frac{1}{2}(r_f + r_i) = 0,$$

or $q = \frac{1}{2}(r_f + r_i) - p$, proving the second inclusion. Taking the difference of the two ordered products proves the support inclusion for the commutator. A distribution whose Fourier support is contained in a closed set vanishes on every open set disjoint from that support, which proves (28.47). □

Remark 28.17 (JLD input for the LSZ retarded commutator). Let

$$J_b = \mathcal{K}_b \Phi_b, \qquad J_c = \mathcal{K}_c \Phi_c$$

denote the source currents obtained by applying the mostly-plus Klein–Gordon operators $\mathcal{K}_b, \mathcal{K}_c$ used in the LSZ reduction formula. For finite-energy states $|Q_i\rangle, |Q_f\rangle$, assume that the distribution

$$C_{bc}(x) = \langle Q_f, [J_b(x/2), J_c(-x/2)] Q_i\rangle$$

is Dyson-regular modulo contact terms in the sense of Definition 28.10. Then the nonlocal part of C_{bc} has a Dyson representation of the form (28.41), plus a finite polynomial in q from contact terms. In a coincidence slab R where the intermediate-state spectral condition makes the nonlocal part of $\widehat{C}_{bc}$ vanish, this representation is support-restricted to S_R if and only if the corresponding Dyson data are support-pruned relative to R in the sense of Definition 28.8. The associated retarded and advanced Fourier transforms

$$C^R_{bc}(q) = \int e^{iq\cdot x} \theta(x^0) C_{bc}(x)\, d^4x, \qquad C^A_{bc}(q) = -\int e^{iq\cdot x} \theta(-x^0) C_{bc}(x)\, d^4x$$

are boundary values of the same analytic continuation across that coincidence region, up to the finite contact polynomials generated by the source-current reduction. The justification is exactly the chain of hypotheses now separated above. Locality gives causal support; Proposition 28.16 gives the spectral coincidence regions; Dyson regularity is the extra six-dimensional wave-equation input needed to use Theorem 28.15; support restriction inside a slab is a property of the Cauchy-data class modulo $\ker \mathcal{H}$, together with the separate datum that $\widehat{C}_{bc} = 0$ on the slab. Local source-current terms supported at $x = 0$ transform to finite polynomials in q. The edge-of-the-wedge continuation of retarded and advanced boundary values is therefore a conditional analytic consequence of these inputs, with theorem status supplied only after the hypotheses are verified.

Remark 28.4 is the rigorous crossing-domain input supplied directly by the Wightman locality and spectrum axioms, while Proposition 28.16 supplies the spectral coincidence regions for the LSZ source-current commutator. Theorem 28.15 then supplies the causal-commutator representation used to enlarge domains from those coincidence regions, provided the commutator is Dyson-regular modulo contact terms; the S_R-support-restricted version additionally requires support-pruned Dyson data relative to the coincidence slab. The stronger fixed-t cut-plane theorem used in dispersion relations still requires the Dyson-regularity and support-pruning verification for the relevant LSZ source-current matrix coefficients. Proposition 28.12 shows the precise microlocal obstruction to treating Dyson's light-cone lift as an ordinary product of distributions, and identifies the finite local ambiguity of any renormalized lift. Theorem 28.13 then proves the local finite-scaling-degree extension mechanism, including the cone equation, pushforward normalization, and the fact that all remaining extension ambiguity is polynomial after Fourier transformation. Proposition 28.14 checks that passing from interpolating fields to LSZ source currents preserves the punctured wavefront and finite-scaling-degree inputs by finite-order differentiation, with the correct $x/2$ and $-x/2$ factors. What remains is not the differential source-current step, but the actual QFT theorem constructing the required extension, Cauchy data, and support-pruning for the source-current kernel. After that regularity problem one still needs the Bros–Epstein–Glaser analytic-completion step and the LSZ pole-separation transfer hypotheses. The analytic LSZ transfer itself is proved in Theorem 29.17: once an off-shell fixed-t domain has normal-crossing external one-particle poles with uniform boundary growth, restricting the LSZ-amputated germ to the external mass shells preserves holomorphy and tempered discontinuities. What remains here is to prove that the primitive four-point germ and the causal-commutator Dyson representation extend far enough to give such an off-shell first-sheet s-plane domain at fixed t, with only the stated right- and left-hand cuts and with controlled boundary growth. That additional step is the Lehmann–Martin/Jin–Martin domain theorem, not a restatement of primitive tube analyticity.

The same symbol $\mathcal{M}(s,t)$ will denote the holomorphic germ $F(s,t,4m^2-s-t)$ on whichever connected domain is under discussion, and its physical value will always mean the specified tempered boundary value. Analytic continuation is continuation of this germ through overlapping domains in $\mathcal{O}_{\text{poly}}$. If two such germs have the same tempered boundary value on a nonempty real open edge, the edge-of-the-wedge theorem gives a common continuation to the envelope generated by the adjacent tubes; inside a connected holomorphic domain the ordinary identity theorem then gives uniqueness. No perturbative Borel summation procedure is part of this definition. The polynomial growth in $\mathcal{O}_{\text{poly}}$ is the local growth condition needed for distributional boundary values; polynomial boundedness in high-energy directions is a stronger global estimate and is introduced only in the dispersion-relation chapter.

Hermiticity of the theory and the Feynman boundary prescription imply real analyticity on conjugate domains:

$$\mathcal{M}(s^*,t^*) = \mathcal{M}(s,t)^*$$

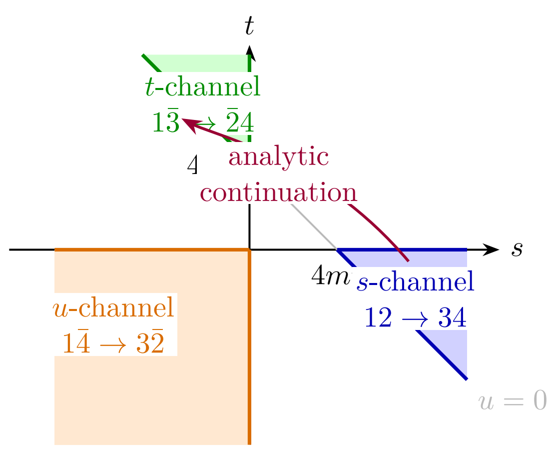
where both sides are evaluated on corresponding branches. On a real segment not lying on a cut, this says that the first-sheet value is real up to pole singularities. On a cut, it relates the lower-edge and upper-edge boundary values. In perturbation theory the same property is visible graph by graph when the couplings are real.

Crossing states that the s-, t-, and u-channel physical amplitudes are boundary values of the same analytically continued four-point function in different real regions, with the external representation conjugated when an outgoing particle is moved to the incoming side or conversely. On the real (s,t)-plane these regions are separated by thresholds and cuts. The locality-and-spectrum argument gives a common analytic germ in the Bros–Epstein–Glaser envelope generated by the relevant tube domains and spacelike edge identifications. A stronger statement, namely that full physical s- and crossed-channel regions can be joined by a path on a specified first-sheet domain, requires an analyticity domain large enough to contain that path. In four spacetime dimensions this larger Mandelstam-type domain requires input beyond the edge-of-the-wedge argument alone.

The Mandelstam representation is a stronger conjectural statement about this same four-point function. For equal external masses, with $u = 4m^2 - s - t$, it would assert that, after all first-sheet pole terms and a specified finite polynomial subtraction have been separated, the remaining amplitude admits a simultaneous two-variable dispersion representation of the form

$$\mathcal{M}_{\mathrm{reg}}(s,t) = \sum_{\chi=s,t,u} \int_{\chi_0}^{\infty} \frac{\rho_\chi(\sigma)}{\sigma - \chi}\,\mathrm{d}\sigma + \sum_{(\chi,\chi')=(s,t),(s,u),(t,u)} \int_{\chi_0}^{\infty} \mathrm{d}\sigma \int_{\chi_0'}^{\infty} \mathrm{d}\tau\, \frac{\rho_{\chi\chi'}(\sigma,\tau)}{(\sigma-\chi)(\tau-\chi')},$$

with $i0$ boundary prescriptions assigned when a physical edge of a cut is approached. Here ρ_χ and $\rho_{\chi\chi'}$ are single and double discontinuity distributions, and the support restrictions on the double spectral distributions are part of the conjectural content. This simultaneous-cut representation is stronger than the single-variable Cauchy representation at fixed t. No derivation is presently available for general interacting four-dimensional Wightman theories with a mass gap. The arguments in this monograph use fixed-t or fixed-crossed-variable dispersion relations together with crossing and angular analyticity; the Mandelstam representation is not assumed in those deductions.



28.3 Unitarity on the Physical Cut

Crossing and real analyticity constrain how different boundary values fit together. On the physical s-channel cut there is also the Hilbert-space constraint from the unitary scattering operator. Let $S_\ell(s)$ denote the angular-momentum-ℓ partial-wave scattering eigenvalue in the identical-scalar two-particle sector, with the normalization fixed earlier. The elastic $2 \to 2$ amplitude is

$$\mathcal{M}(s,t) = -16\pi \mathrm{i} \sqrt{\frac{s}{s-4m^2}} \sum_{\ell=0}^{\infty} (2\ell+1) P_\ell(\cos\theta) \big(S_\ell(s) - 1\big),$$

where

$$\cos\theta = 1 + \frac{2t}{s-4m^2}.$$

Unitarity of S implies, for $s \geq 4m^2$,

$$|S_\ell(s)| \leq 1.$$

If no inelastic channel is open, namely for

$$4m^2 \leq s \leq M_{\text{inel}}^2,$$

the partial wave is a phase and

$$|S_\ell(s)| = 1.$$

Here M_{inel} is the invariant mass threshold of the lightest state outside the two-particle elastic channel. These inequalities are boundary conditions on the upper edge of the physical cut; they are not by themselves an analytic continuation principle. They will combine with angular analyticity in the following chapter.

28.4 Perturbative Discontinuities and Cut Rules

The Hilbert-space unitarity statement determines the physical-cut discontinuity of exact on-shell amplitudes whenever the relevant upper and lower boundary values exist as distributions and the intermediate-state resolution is complete in the scattering sector under consideration. Its perturbative expansion gives the graph-level cut rules. Write

$$S = \mathbf{1} + \mathrm{i}T, \qquad \operatorname{Disc}_s \mathcal{M} = \mathcal{M}(s + \mathrm{i}0, t) - \mathcal{M}(s - \mathrm{i}0, t).$$

From $S^\dagger S = \mathbf{1}$ one obtains

$$-\mathrm{i}(T - T^\dagger) = T^\dagger T.$$

Taking connected matrix elements and removing the common momentum-conserving delta function gives, on the s-channel physical cut,

$$\operatorname{Disc}_s \mathcal{M}_{\beta\alpha} = \mathrm{i} \sum_X \int \mathrm{d}\Phi_X \, \mathcal{M}_{X\alpha} \, \mathcal{M}_{X\beta}^*. \tag{28.50}$$

Here X runs over on-shell intermediate states with total momentum P_α, and $\mathrm{d}\Phi_X$ is the Lorentz-invariant phase-space measure, including the factor $1/N!$ for N identical particles when the state sum is over unordered particles. The subscript convention is $\mathcal{M}_{\gamma\delta} = \mathcal{M}(\gamma|\delta)$, so the conjugated factor is the amplitude $\beta \to X$, complex conjugated. It is not, in a generic theory, the complex conjugate of the reversed process $X \to \beta$. Writing $\mathcal{M}_{\beta X}^*$ in this exact Hilbert-space identity would add a reciprocity assumption, such as time-reversal invariance with specified antiunitary phase conventions. The forward case $\beta = \alpha$ then reduces to the positive optical-theorem sum.

Remark 28.18 (Unitarity discontinuity and perturbative cut rules). Equation (28.50) is the non-perturbative on-shell unitarity identity: it relates boundary values of the exact amplitude to a sum over exact on-shell intermediate states and exact lower-point amplitudes. Its hypotheses are Hilbert-space unitarity, the existence of the relevant scattering states and boundary values, and the completeness of the state sum used in the channel. The Cutkosky rule is the perturbative refinement obtained after choosing a regulator and expansion, writing $T = \sum_r T^{(r)}$, representing the terms by Feynman graphs, and identifying the contribution of a specified graph cut with on-shell propagator replacements. Thus the full discontinuity formula and the graph-by-graph replacement theorem have different logical status, even though the latter is obtained by expanding the former.

Proposition 28.19 (Largest-time identity for a scalar graph)**.** *Consider a regularized scalar Feynman graph G with vertices $x_1, \dots, x_V$, local Hermitian interaction vertices, and internal lines of masses m_e. For each mass set*

$$\Delta_m^+(x) = \int \frac{\mathrm{d}^D q}{(2\pi)^D} (2\pi)\theta(q^0)\delta(q^2 + m^2)\, \mathrm{e}^{\mathrm{i}q\cdot x}, \qquad \Delta_m^-(x) = \Delta_m^+(-x),$$

and decompose the Feynman boundary value as

$$\Delta_{F,m}(x) = \theta(x^0)\Delta_m^+(x) + \theta(-x^0)\Delta_m^-(x), \qquad \Delta_{F,m}(x)^* = \theta(x^0)\Delta_m^-(x) + \theta(-x^0)\Delta_m^+(x).$$

For a circling $\sigma : \{1, \dots, V\} \to \{0, 1\}$, define G_σ by using ordinary vertex factors and Feynman propagators inside the uncircled set, complex-conjugated vertex factors and anti-Feynman propagators inside the circled set, and Wightman propagators on mixed lines. A mixed line oriented from the uncircled side to the circled side carries Δ^+; the reverse orientation carries Δ^-. Circled vertices carry the minus sign associated with complex conjugating the perturbative i-weighted vertex. Then

$$\sum_{\sigma\in\{0,1\}^V} G_\sigma(x_1, \dots, x_V) = 0. \tag{28.51}$$

Proof. First smear the vertex variables with a test function whose support avoids all equal-time hyperplanes $x_a^0 = x_b^0$. The complement is a finite union of open time-ordering chambers, so it suffices to prove the identity on one such chamber. The general smeared distribution is then obtained by a finite partition of unity and by taking the limit as the partition is refined; the equal-time hyperplanes have no independent contribution because the regularized boundary values are distributions and the displayed theta decomposition is used only after smearing.

Fix a chamber and let k be the unique vertex with largest time. Split the sum over circlings into pairs $(\sigma, \sigma^{(k)})$, where $\sigma^{(k)}$ differs from σ only at k. Lines not incident on k have identical factors in the two paired graphs. If a line joins x_k to an earlier vertex x_j, then $x_k^0 - x_j^0 > 0$. The four possible circling assignments give

circling at k, j	line factor from k to j
$0, 0$	$\Delta_F(x_k - x_j) = \Delta^+(x_k - x_j)$
$1, 1$	$\Delta_F(x_k - x_j)^* = \Delta^-(x_k - x_j)$
$0, 1$	$\Delta^+(x_k - x_j)$
$1, 0$	$\Delta^-(x_k - x_j)$.

Thus changing only the circling of the largest-time vertex leaves every incident line factor fixed: the factor is Δ^+ if the earlier vertex is uncircled and Δ^- if it is circled. The local interaction factor at x_k, however, is multiplied by -1 when x_k is circled, because complex conjugation of the perturbative i-weighted Hermitian vertex changes $i\mathcal{L}_{\text{int}}$ to $-i\mathcal{L}_{\text{int}}$. Each pair $(G_\sigma, G_{\sigma^{(k)}})$ therefore cancels. Summing the pairs over all circlings proves (28.51) on the chamber, and hence as the regularized distribution. □

Subtract from (28.51) the all-uncircled and all-circled terms. The former is the original Feynman graph. The latter is the complex-conjugated graph with the opposite boundary prescription. Every remaining term contains at least one mixed line, and each mixed line has the Fourier representation

$$\frac{\mathrm{d}^D q}{(2\pi)^D} (2\pi)\theta(q^0)\delta(q^2 + m^2)$$

after orienting q from the uncircled side to the circled side. Thus the largest-time identity expresses the difference between the two boundary values of a graph as a sum of cut graphs. A physical s-channel cut is the subsum of those circlings for which the mixed lines separate the external legs into the chosen left and right subprocesses and carry total momentum P_α across the cut. Circlings whose mixed-line energy support belongs to another channel contribute to the corresponding channel discontinuity instead.

Theorem 28.20 (Cutkosky cut rule for a perturbative physical cut). *Consider a perturbative contribution to a connected amplitude whose s-channel discontinuity is produced by putting a set C of internal lines on shell so that the graph separates into a left subamplitude and a right subamplitude. With the amplitude normalization fixed by $S = \mathbf{1} + \mathrm{i}T$, the discontinuity contribution of this cut is obtained by replacing each cut propagating line of mass m_i by the on-shell measure*

$$\frac{\mathrm{d}^D q_i}{(2\pi)^D}\,(2\pi)\theta(q_i^0)\delta(q_i^2 + m_i^2),$$

multiplying the left and right lower-order amplitudes, complex conjugating the amplitude on the lower-edge side, and integrating over the common on-shell phase space. Equivalently, the graph contribution is the corresponding term in (28.50).

Proof. There are two statements to match. The first is the Hilbert-space discontinuity. Expand $T = \sum_r T^{(r)}$ in the chosen interaction order. The order-N part of $-\mathrm{i}(T - T^\dagger) = T^\dagger T$ is

$$-\mathrm{i}\big(T^{(N)} - T^{(N)\dagger}\big) = \sum_{r=1}^{N-1} T^{(r)\dagger} T^{(N-r)}.$$

Insert the spectral resolution of the identity on the physical Hilbert space between the two factors on the right. For a sector of n stable particles of masses m_i, this resolution contains

$$\frac{1}{n!}\prod_{i=1}^{n}\left[\int \frac{\mathrm{d}^D q_i}{(2\pi)^{D-1}}\,\theta(q_i^0)\delta(q_i^2 + m_i^2)\right](2\pi)^D\delta^{(D)}\left(P_\alpha - \sum_i q_i\right).$$

After the external LSZ residues and the overall momentum-conserving delta function are removed, this is precisely the phase-space integral in (28.50).

The second statement identifies the same term inside a fixed Feynman graph. Apply Proposition 28.19 to the graph. The difference between its two physical boundary values is the sum of mixed circlings. For the chosen s-channel discontinuity, retain the circlings whose mixed lines separate the graph into the ordinary subamplitude $\alpha \to X$ and the adjoint subamplitude whose unconjugated orientation is $\beta \to X$. The mixed propagator on each separating line is exactly

$$\frac{\mathrm{d}^D q_i}{(2\pi)^D}(2\pi)\theta(q_i^0)\delta(q_i^2 + m_i^2),$$

with q_i oriented from the lower-order amplitude to its conjugate across the cut. The uncircled side contains the ordinary Feynman rules for one lower-order amplitude, while the circled side contains the complex-conjugated rules for the lower-edge amplitude. The product of the mixed-line distributions supplies the on-shell phase-space measure and the residual momentum delta function enforces $P_X = P_\alpha$. This is the graph-level line-replacement rule stated in the theorem, with the overall factor fixed by the $S = \mathbf{1} + \mathrm{i}T$ convention used in (28.50). □

For the two-particle equal-mass cut, the phase-space factor appearing in (28.50) is explicit. Let $P^2 = -s$ and

$$\mathrm{d}\Phi_2(P) = \int \frac{\mathrm{d}^4 q}{(2\pi)^3}\theta(q^0)\delta(q^2+m^2)\,\frac{\mathrm{d}^4 r}{(2\pi)^3}\theta(r^0)\delta(r^2+m^2)\,(2\pi)^4\delta^{(4)}(P-q-r).$$

In the center-of-mass frame $P = (\sqrt{s}, \vec{0})$ this becomes

$$\mathrm{d}\Phi_2(P) = \frac{\mathrm{d}^3\vec{q}}{(2\pi)^2\, 4\omega_{\vec{q}}^2}\,\delta(\sqrt{s}-2\omega_{\vec{q}}).$$

Using $\omega_{\vec{q}} = \sqrt{\vec{q}^{\,2}+m^2}$, the radial delta function fixes

$$|\vec{q}| = \frac{1}{2}\sqrt{s-4m^2},$$

and hence

$$\int \mathrm{d}\Phi_2(P) = \frac{1}{8\pi}\sqrt{1-\frac{4m^2}{s}}\,\theta(s-4m^2). \tag{28.52}$$

For identical intermediate particles this is multiplied by $1/2!$ if the state sum is over unordered two-particle states. Thus a one-loop bubble cut has discontinuity equal to i times the product of the two tree subamplitudes times (28.52), with this symmetry factor when appropriate. The square-root branch point is the same threshold that the Landau equations identify below.

28.5 Crossing From Locality and Tube Analyticity

The structural origin of crossing is the same analytic mechanism used for Wightman functions. For four scalar fields, the vacuum distribution

$$W_4(x_1,x_2,x_3,x_4) = \langle\Omega|\widehat{\phi}(x_1)\widehat{\phi}(x_2)\widehat{\phi}(x_3)\widehat{\phi}(x_4)|\Omega\rangle$$

has Fourier support fixed by the spectrum condition in relative variables. It is therefore the boundary value of a holomorphic function in a tube domain. More explicitly, after removing the center-of-mass coordinate, the Fourier transform has support in products of closed forward lightcones for appropriate partial sums of momenta. This support condition is the input which makes the imaginary spacetime variables point into forward tubes.

Local commutativity says that, at real configurations where adjacent insertions are spacelike separated,

$$W_4(\ldots,x_i,x_{i+1},\ldots) = W_4(\ldots,x_{i+1},x_i,\ldots)$$

for identical scalar fields. The edge-of-the-wedge theorem then identifies the analytic continuations attached to the two orderings in the domain generated from such spacelike configurations.

LSZ reduction is applied to the time-ordered or Wightman boundary values after isolating the one-particle poles of the external legs. Different assignments of external legs as incoming or outgoing particles correspond to different real boundary components of the same analytically continued four-point function. The s-channel process

$$1+2\to 3+4$$

and the crossed process

$$1 + \bar{3} \to \bar{2} + 4$$

are therefore boundary values of one analytic object in the domain obtained from permuted Wightman tubes by edge-of-the-wedge continuation. For a self-conjugate scalar $\bar{2} = 2$ and $\bar{3} = 3$; for charged fields the bars indicate the conjugate particle associated with the adjoint field.

The statement is local on the analytic domain. It does not erase the singularities separating physical regions. Threshold cuts, bound-state poles, and anomalous thresholds specify how the analytic continuation must be routed and which boundary value is obtained at the end. Crossing is the assertion of common analytic origin; the singularity analysis below describes the obstructions and boundary prescriptions encountered along particular continuations.

Warning 28.21 (BEG crossing versus a Mandelstam domain). The edge-of-the-wedge argument used here is the rigorous Bros–Epstein–Glaser mechanism. It proves that permuted Wightman boundary values and their LSZ residues have a common analytic continuation on the envelope generated by the corresponding tube domains. A path connecting the s-channel physical region to a crossed physical region on the same first-sheet branch, while avoiding all singularities, is additional analytic data. That physical-region-to-physical-region crossing statement is therefore a stronger analyticity assertion. In perturbation theory it is studied by tracking graph singularities and sheets; in nonperturbative four-dimensional QFT it is usually formulated as a Mandelstam-type domain hypothesis rather than as a theorem following solely from locality, spectrum, and LSZ.

28.6 Edge-of-the-Wedge Form

The analytic gluing used above is the distributional edge-of-the-wedge theorem quoted in Theorem 16.6, applied in local kinematic coordinates. Let F_+ and F_- be holomorphic functions in two domains

$$\mathcal{D}_+, \qquad \mathcal{D}_-$$

of the complexified kinematic variables. Suppose both domains have boundary values on a common real open set U, in the sense of distributional limits against test functions supported in U. Suppose these boundary values agree:

$$F_+|_U = F_-|_U.$$

The edge-of-the-wedge theorem gives a holomorphic function F on a domain containing $\mathcal{D}_+ \cup U \cup \mathcal{D}_-$ in its envelope of holomorphy, with $F = F_+$ on $\mathcal{D}_+$ and $F = F_-$ on $\mathcal{D}_-$.

In local QFT, the real set U is supplied by spacelike configurations where local commutativity equates two Wightman orderings. The domains $\mathcal{D}_\pm$ are the corresponding forward-tube domains selected by the spectrum condition. The theorem then produces a single analytic function whose different real boundary components give differently ordered correlators. LSZ reduction transfers this analytic relation from correlators to scattering amplitudes whenever the external one-particle pole hypotheses hold.

28.7 Pinches

On the first sheet one finds stable one-particle poles, ordinary thresholds, and anomalous thresholds. In perturbative graph analysis an anomalous threshold is a contour-pinch singularity on the physical sheet, caused by several internal lines going on shell in a spacetime arrangement satisfying the positive Coleman–Norton orientation condition stated in (28.53). Its invariant location is determined by the Landau configuration of those internal lines.

The singularities of perturbative amplitudes arise when an integration contour is forced against singularities of the integrand. Consider a scalar graph with internal momenta

$$q_i = q_i(\ell_1, \ldots, \ell_L; p_1, \ldots, p_n),$$

where the q_i are affine functions of the loop momenta ℓ_a and external momenta p_j. A typical denominator is

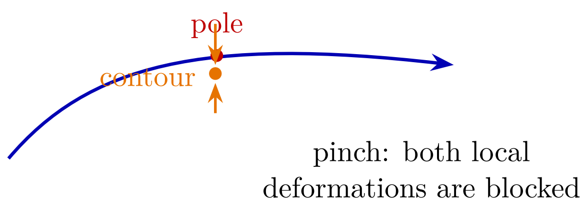
$$q_i^2 + m_i^2 - \mathrm{i}0.$$

After Feynman parametrization, a product of N denominators has the form

$$\int_{\alpha_i \geq 0} \prod_{i=1}^{N} \mathrm{d}\alpha_i\, \delta\left(\sum_{i=1}^{N} \alpha_i - 1\right) \frac{\text{numerator}}{\left[\sum_{i=1}^{N} \alpha_i (q_i^2 + m_i^2) - \mathrm{i}0\right]^N}.$$

Numerator factors and ultraviolet subtractions can change the degree or even remove a candidate singularity, but they do not change the necessary pinch conditions for the scalar denominator.

As external momenta move in the complex domain, the loop contours can be deformed as long as no singularity traps them. A necessary condition for a singularity is that the denominator vanish at a point where its first variation in every loop-momentum direction also vanishes. If the first variation were nonzero, the contour could be displaced locally away from the zero. At a stationary zero, nearby singular hypersurfaces can pinch the original real contour, and local deformations cannot remove the obstruction.



This stationary-pinch condition is the source of the Landau equations. The equations are necessary conditions for a singularity of a graph. To decide whether a singularity is actually present one must also check the sheet, the orientation of the pinch, and whether the numerator cancels the local singular term.

28.8 Landau Equations

Let

$$\mathcal{D}(\ell, \alpha; p) = \sum_{i=1}^{N} \alpha_i (q_i^2 + m_i^2).$$

A Landau singularity can occur only if there are Feynman parameters $\alpha_i \geq 0$, not all zero, such that

$$\alpha_i(q_i^2 + m_i^2) = 0 \qquad \text{for every internal line } i,$$

and, for every loop momentum ℓ_a^μ,

$$\frac{\partial \mathcal{D}}{\partial \ell_a^\mu} = 0.$$

Equivalently,

$$\sum_{i=1}^{N} \alpha_i q_{i\nu} \frac{\partial q_i^\nu}{\partial \ell_a^\mu} = 0, \qquad a = 1, \ldots, L.$$

The omitted factor of 2 is irrelevant because the right hand side is zero.

For a one-loop graph with all internal momenta oriented around the loop, this reduces to

$$\sum_i \alpha_i q_i^\mu = 0.$$

The first equation says that every line with $\alpha_i > 0$ is on shell:

$$q_i^2 = -m_i^2.$$

The second equation says that the corresponding on-shell momenta form a stationary configuration around each loop. Solutions with some $\alpha_i = 0$ describe singularities of reduced graphs in which the inactive lines have been contracted. Solutions with all relevant $\alpha_i > 0$ are the leading Landau singularities of that graph.

With the terminology fixed in Section 28.1, the physical sheet is selected by the sign of the Feynman prescription together with positive Feynman parameters. Positivity of the Feynman parameters distinguishes a first-sheet threshold from the same algebraic locus reached only after continuation through a cut.

28.9 Spacetime Interpretation of Positive Landau Solutions

The sign condition on the parameters has a spacetime meaning. Consider a Landau solution in a real physical region and keep only the active internal lines, namely those with $\alpha_i > 0$. Let the graph vertices be labelled by v, and orient each active edge i from a tail $v_-(i)$ to a head $v_+(i)$. The momentum q_i is the momentum assigned to this oriented edge by the chosen loop-momentum routing. The loop-stationarity equations say that, for every oriented cycle C of the active graph,

$$\sum_{i \in C} \sigma_{Ci} \alpha_i q_i = 0, \qquad \sigma_{Ci} = \begin{cases} +1, & \text{if the edge orientation agrees with the orientation of } C, \\ -1, & \text{if it is opposite.} \end{cases}$$

Equivalently, the vector assigned to an active edge by $\alpha_i q_i$ has zero sum around every closed path. Hence, on each connected component, there are vertex positions x_v, unique up to a common translation, such that

$$x_{v_+(i)} - x_{v_-(i)} = \lambda_i q_i, \qquad \lambda_i \geq 0,$$

with λ_i proportional to α_i. The proportionality constant depends only on the convention used for the quadratic form in the parametrized denominator.

The positive-energy condition is an additional orientation condition. Choose a time orientation in Minkowski space. A positive Coleman–Norton realization of the Landau solution consists of an orientation of every active edge in its classical direction of propagation and momenta k_i on those oriented edges such that

$$k_i^2 = -m_i^2, \qquad k_i^0 > 0, \qquad x_{v_+(i)} - x_{v_-(i)} = \lambda_i k_i, \qquad \lambda_i \geq 0. \tag{28.53}$$

If the loop-momentum routing originally assigned the opposite orientation to that edge, then $k_i = -q_i$ and the tail and head used in the spacetime equation are interchanged. Equivalently, one may reorient the active graph so that $q_i = k_i$ on every active edge. Thus the compatibility condition is precisely that each active internal momentum be a future-directed positive-energy on-shell momentum in the direction in which the corresponding edge is traversed in spacetime.

For $m_i > 0$, this is the worldline equation for a classical massive particle propagating for nonnegative proper time, with $\lambda_i m_i$ proportional to that proper time. For $m_i = 0$, it is the corresponding future-directed null geodesic equation with nonnegative affine parameter. Momentum conservation at each vertex is the graph momentum conservation for the same oriented momenta, with the external momenta assigned to that vertex. Thus a real leading Landau solution satisfying the positive Coleman–Norton orientation condition is a classical spacetime process: all active internal particles go on shell, carry positive energy in their direction of propagation, and meet at the vertices with energy and momentum conserved. This is the Coleman–Norton interpretation. It is a criterion for first-sheet physical singularities that must be combined with the pinch analysis: a formal algebraic solution contributes to the physical singularity only on the relevant sheet, with the integration contour trapped, and with a numerator that leaves a nonzero local singular term.

Warning 28.22 (Scope of the Coleman–Norton converse). The implication just proved is the geometric interpretation of a real positive Landau solution: such a solution is a classical spacetime process. The converse statement, often phrased as “every allowed classical process gives a singularity,” has a narrower status. For a fixed perturbative Feynman graph, a nondegenerate positive pinch on the relevant sheet produces a local singularity of the scalar integral; numerator factors, symmetry factors, subtractions, or sums of graphs may cancel the leading local term. Thus the perturbative converse is a theorem only after the pinch orientation, nondegeneracy, and non-cancellation hypotheses have been checked. In an exact nonperturbative QFT there is no general theorem that every such classical spacetime picture produces a singularity of the full S-matrix. There it should be regarded as a diagnostic expectation, to be justified from the spectral and locality structure of the theory in each setting.

Subleading Landau solutions have some $\alpha_i = 0$. Geometrically, the corresponding internal lines are contracted, and the classical propagation picture applies to the reduced graph. Ordinary thresholds are reduced classical processes in which a set of particles can be produced on shell with the total external energy-momentum supplied to the subgraph.

28.10 Two-Particle Threshold

For the equal-mass two-propagator loop, take internal momenta

$$q_1 = \ell, \qquad q_2 = \ell + P.$$

The Landau equations with $0 < \alpha < 1$ are

$$\ell^2 + m^2 = 0, \qquad (\ell + P)^2 + m^2 = 0,$$

and

$$(1-\alpha)\ell + \alpha(\ell + P) = 0.$$

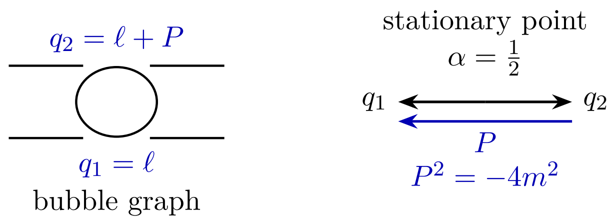
The stationary equation gives

$$\ell = -\alpha P.$$

Substitution into the two on-shell equations gives

$$\alpha = \frac{1}{2}, \qquad \ell = -\frac{P}{2}, \qquad P^2 = -4m^2.$$

Thus the ordinary two-particle threshold is a Landau singularity. In terms of the invariant $s = -P^2$, this is $s = 4m^2$.



For unequal masses, the same equations give

$$\ell = -\alpha P, \qquad \alpha^2 P^2 = -m_1^2, \qquad (1-\alpha)^2 P^2 = -m_2^2.$$

The positive-parameter solution with $0 < \alpha < 1$ is

$$\alpha = \frac{m_1}{m_1 + m_2}, \qquad P^2 = -(m_1 + m_2)^2.$$

This is the ordinary threshold: the two on-shell internal particles propagate with zero relative spatial momentum in the center-of-mass frame. The algebraic equations also contain the pseudo-threshold

$$P^2 = -(m_1 - m_2)^2,$$

but its parameter signs do not describe the same first-sheet physical production process.

28.11 Triangle Singularities

The triangle graph is the first elementary example of a first-sheet singularity produced by a genuine on-shell contour pinch rather than by a pole or endpoint threshold. Let the three external momenta be

$$P_1^2 = -M_1^2, \qquad P_2^2 = -M_2^2, \qquad P_3^2 = -M_3^2,$$

with $P_1 + P_2 + P_3 = 0$. The three internal lines are required to be on shell, and the positive-parameter stationary condition requires their momentum vectors to close consistently around the loop. For one choice of loop orientation this means

$$q_i^2 = -m_i^2, \qquad \alpha_1 q_1 + \alpha_2 q_2 + \alpha_3 q_3 = 0, \qquad \alpha_i > 0.$$

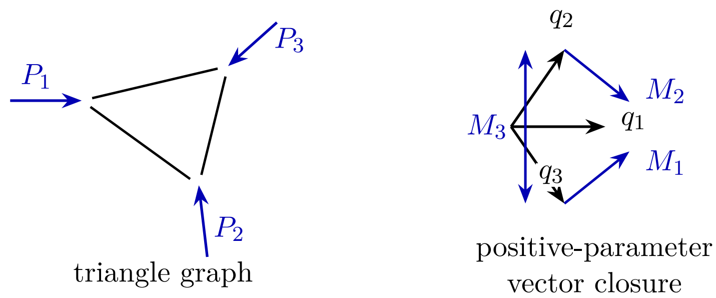


Figure 28.2: Triangle Landau data. The left panel labels the external momentum flow in the triangle graph. The right panel shows the positive-parameter closure condition $\alpha_1 q_1 + \alpha_2 q_2 + \alpha_3 q_3 = 0$ for oriented on-shell internal momenta.

Together with momentum conservation at the vertices, these equations define the candidate singular locus in the external invariants. Equivalently, the Gram matrix $G_{ij} = q_i \cdot q_j$ has a null vector $(\alpha_1, \alpha_2, \alpha_3)$ with strictly positive entries. Thus the condition is not only $\det G = 0$, but the existence of a positive null vector for the oriented on-shell momenta. The graph and vector-closure data are shown in Figure 28.2.

In the equal-internal-mass planar example, the Euclideanized internal momentum vectors in the sketch all have length m. The external invariant M_a is the length of the chord between two such vectors, so

$$M_a^2 = 2m^2(1 - \cos\varphi_a),$$

where φ_a is the angle between the corresponding internal on-shell vectors. Thus $M_a = \sqrt{2}\,m$ is the right-angle value. If $M_1, M_2 < \sqrt{2}\,m$, the relevant vectors lie in an open half-plane in the configuration shown, and no positive relation $\alpha_1 q_1 + \alpha_2 q_2 + \alpha_3 q_3 = 0$ exists. If $M_1, M_2 > \sqrt{2}\,m$, the vectors can be arranged with the origin in their convex hull; the positive-parameter Landau equation can then be satisfied. The remaining invariant is fixed by this geometry, giving a singularity at the corresponding value $P_3^2 = -M_3^2$. The convex-hull distinction is drawn in Figure 28.3.

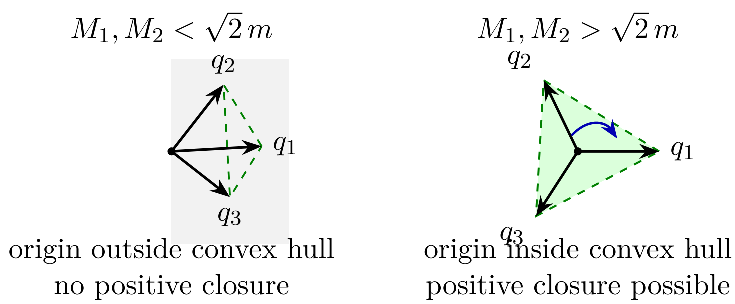


Figure 28.3: Convex-hull form of the positive-parameter Landau condition in the equal-internal-mass triangle example. When the origin lies outside the hull of the oriented on-shell vectors no positive closure exists; when it lies inside, the stationary equation can be solved with $\alpha_i > 0$.

The vector sketch represents the Coleman–Norton condition. The actual singular locus is obtained by solving the three on-shell equations together with the positive-parameter stationary equation. A solution gives a first-sheet anomalous threshold at the determined value of $P_3^2 = -M_3^2$ only when the active edges can be oriented with momenta $k_i^0 > 0$ and separations $x_{v_+(i)} - x_{v_-(i)} = \lambda_i k_i$, $\lambda_i \geq 0$. A solution requiring reversed time ordering or a negative proper-time segment lies on a continued sheet, even if it solves the same algebraic Landau equations.

Such a singularity is an anomalous threshold: a first-sheet singularity of the amplitude produced by on-shell internal propagation and contour pinching. Its local data are the masses, momentum routing, and positive Feynman parameters of the Landau solution. Ordinary endpoint thresholds are the reduced-graph case in which the on-shell process begins at the boundary of the integration domain; the triangle is the first elementary interior-pinch case. Coleman–Thun singularities in two-dimensional factorized scattering give a nonperturbative setting in which the same on-shell propagation and contour pinching mechanism appears.

28.12 Use of Landau Analysis

Landau equations are derived graph by graph. Their role here is diagnostic: they identify the local analytic mechanisms by which thresholds and more subtle first-sheet singularities arise in perturbation theory. In an exact massive theory, the singularities of amplitudes are expected to be governed by the same spectral and locality principles, with internal on-shell lines replaced by actual stable particles or by spectral continua of the theory.

Within existing axiomatic frameworks, only parts of this analytic picture are presently theorems. The terminology used here is precise, but its logical status is not uniform: stable particles appear as first-sheet poles below thresholds under the one-particle spectral hypotheses, resonances are continued-sheet pole data, ordinary thresholds are tied to multi-particle spectral continua, and anomalous thresholds are perturbative on-shell propagation patterns encoded by the Landau equations. Beyond perturbation theory the last statement is a conjectural guide unless a separate nonperturbative analytic construction is supplied. This classification will be used in the following chapter only at the level needed for angular analyticity: the nearest first-sheet crossed-channel singularity fixes the distance from the physical angular interval to the edge of the Lehmann ellipse. Landau analysis supplies a concrete perturbative mechanism for such singularities. The nonperturbative construction of the S-operator and the spectral assumptions stated earlier remain the structural input for the exact scattering problem.

Chapter 29

Partial Waves, Dispersion Relations, and High-Energy Bounds

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

29.1 Hypotheses and Angular Analyticity

Work in $D = 4$ for the detailed Lehmann-ellipse and Legendre-polynomial derivation. The higher-dimensional logarithmic-power estimate is separated near Theorem 29.25 as an additional Gegenbauer/angular-tube hypothesis. Consider scattering of the lightest stable identical scalar particle of mass m. The connected invariant amplitude is denoted by

$$\mathcal{M}(s,t).$$

The Mandelstam variables are the mostly-plus physical scattering variables

$$s = -(k_1 + k_2)^2, \qquad t = -(k_1 - k_3)^2, \qquad u = -(k_1 - k_4)^2, \qquad k_i^2 = -m^2,$$

where k_1, k_2 are incoming and k_3, k_4 are outgoing. Hence

$$s + t + u = 4m^2.$$

This is the convention of Definition 24.2; in the physical s-channel interval $t \leq 0$, while positive t below denotes the crossed-channel timelike variable that controls the nearest angular singularity.

The normalization of $\mathcal{M}$ is the identical-particle normalization fixed in Chapter 19. The forward optical theorem is rederived below in this normalization, because the unordered two-particle convention, the even-partial-wave projection, and the $1/2$ from the symmetric two-particle identity resolution must be kept together.

The hypotheses used in this chapter are the following. The theory has a mass gap above the vacuum, the external scalar is stable and is the lightest one-particle species in the sector under consideration, the S-operator on the relevant asymptotic space is unitary, and the amplitude is the boundary value of an analytic first-sheet function with the crossing properties described in Section 28.1 and Chapter 28. Whenever a displayed dispersion formula omits bound-state pole terms, those poles are assumed absent in the displayed kinematic range. If they are present, their residues are added explicitly; for a stable composite pole these residues are the Bethe–Salpeter pole-factorization data of Theorem 27.2, transported to the channel under discussion by crossing.

The results in this chapter are therefore conditional statements. They are deductions from specified analyticity, crossing, unitarity, and boundedness hypotheses, not a claim that all such hypotheses are presently derivable from one universally accepted axiom system for interacting quantum field theory. Only restricted versions of the analyticity and boundedness inputs are known as theorems under robust assumptions. This distinction matters: the mathematical content below is the implication from the hypotheses to the high-energy bound, with every hypothesis stated before it is used. The fixed-t dispersion hypotheses used here are the single-variable hypotheses described in Chapter 28; the simultaneous two-variable Mandelstam representation is not assumed. The notation $\mathcal{M}(s,t)$ is used in the function-space sense of Definition 28.1: it is a holomorphic germ with local polynomial boundary growth on the complexified on-shell invariant surface, with physical values obtained as tempered distributional boundary values. At fixed real t, the one-variable function of complex s is the corresponding restriction of that germ after possible pole loci have been separated. Cut discontinuities in the dispersion formulae below are boundary-value differences in $\mathcal{D}'$ locally, and in $\mathcal{S}'$ on noncompact real edges when the stated high-energy polynomial-growth estimates are imposed.

For a fixed physical value $s > 4m^2$, introduce

$$x = \cos\theta = 1 + \frac{2t}{s - 4m^2}.$$

The physical s-channel scattering angles give the interval

$$-1 \leq x \leq 1.$$

At the t-channel two-particle threshold $t = 4m^2$,

$$x = x_t(s) \equiv \frac{s + 4m^2}{s - 4m^2} > 1,$$

and the t-channel cut maps to $x \in [x_t, \infty)$. The u-channel threshold gives the symmetric cut $x \in (-\infty, -x_t]$. If a crossed channel contains a stable bound state of mass $M_B < 2m$, its pole appears at

$$x_B = 1 + \frac{2M_B^2}{s - 4m^2}$$

or at the corresponding point on the left. Let $t_0 > 0$ be the infimum of positive real t-values at which the crossed-channel first sheet has a singularity, including stable poles and multiparticle thresholds. If the infimum is realized by a pole, the following estimates are applied at $0 < t < t_0$ and the limit $t \to t_0^-$ is taken only at the end. For large s, the nearest right singularity is approached at

$$x_0(s) = 1 + \frac{2t_0}{s - 4m^2}.$$

In the simplest equal-mass case with no crossed-channel bound state, $t_0 = 4m^2$. If there is a stable crossed-channel bound-state pole of mass $M_B < 2m$ and no closer singularity, then $t_0 = M_B^2$. In the elementary single-pole situation this is summarized as

$$t_0 = \min\{4m^2, M_B^2\},$$

with the convention that M_B^2 is omitted when no such pole exists. When B is composite, M_B is not an additional kinematic parameter chosen independently of the dynamics. It is the invariant mass of an isolated spectral state, and in a two-constituent description it is characterized by the eigenvalue-one condition $(\mathbf{1} - \widehat{V}_P)\Psi_B = 0$ of Section 27.4. The high-energy estimate below depends on the nearest singular location t_0; the associated residue becomes relevant when the fixed-t dispersion relation is written with explicit pole terms. Figure 29.1 shows how this nearest singularity fixes the available Lehmann ellipse.

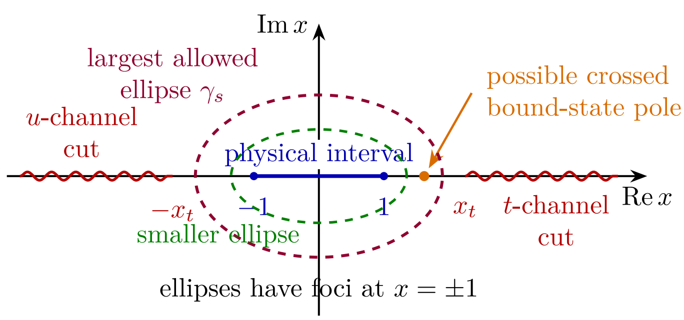


Figure 29.1: Lehmann ellipses in the angular variable $x = \cos\theta$. The physical interval has foci at $x = \pm 1$. The nearest crossed-channel singularity fixes the largest available ellipse γ_s; smaller ellipses with positive distance from the singular set are used in the estimates.

The angular analyticity hypothesis used below is this precise statement: for each such s, the boundary value $\mathcal{M}(s+\mathrm{i}0, x)$, continued in the angular variable x, is analytic in a domain containing the physical interval $[-1, 1]$. For every ellipse with foci at ± 1 whose closure lies in this domain, the amplitude is bounded on the ellipse. The largest ellipse of this type, when it exists, is called the Lehmann ellipse at energy s. The estimates use only smaller ellipses whose closures stay a positive distance from the singular set.

It is useful to parametrize these ellipses before using them. Write

$$R = \mathrm{e}^{\eta}, \qquad \eta > 0,$$

and define the ellipse E_R by $|\zeta(x)| = R$, where

$$\zeta(z) = z + \sqrt{z^2 - 1}, \qquad \zeta(z) \sim 2z \quad (z \to \infty), \tag{29.1}$$

with branch cut on $[-1, 1]$. Its right endpoint is

$$x_R = \frac{1}{2}(R + R^{-1}) = \cosh\eta.$$

Thus the point corresponding to a crossed-channel value t_* has

$$x_*(s) = 1 + \frac{2t_*}{s - 4m^2} = \cosh \eta_*(s), \qquad \eta_*(s) = 2\sqrt{\frac{t_*}{s}} + O(s^{-3/2})$$

at fixed $t_* > 0$ and large s. The singular value t_0 determines the limiting ellipse in the same way, but estimates are made at $0 < t_* < t_0$ and only then is t_* allowed to approach t_0.

29.2 Expansion Inside the Lehmann Ellipse

Let γ_s be a counterclockwise contour inside the angular analytic domain and enclosing $[-1, 1]$. For x inside γ_s,

$$\mathcal{M}(s, x) = \oint_{\gamma_s} \frac{\mathrm{d}z}{2\pi\mathrm{i}} \frac{\mathcal{M}(s, z)}{z - x}.$$

Use the branch (29.1). The level sets $|\zeta(z)| = R > 1$ are ellipses with foci at $z = \pm 1$, and the physical interval is the degenerate ellipse $R = 1$. On the level set $R = \mathrm{e}^{\eta}$ one may write

$$z = \frac{1}{2}\left(R\mathrm{e}^{\mathrm{i}\varphi} + R^{-1}\mathrm{e}^{-\mathrm{i}\varphi}\right), \qquad 0 \le \varphi < 2\pi,$$

and hence

$$\operatorname{Re} z = \frac{R + R^{-1}}{2}\cos\varphi = \cosh\eta\,\cos\varphi, \qquad \operatorname{Im} z = \frac{R - R^{-1}}{2}\sin\varphi = \sinh\eta\,\sin\varphi.$$

Thus

$$\frac{(\operatorname{Re} z)^2}{\cosh^2\eta} + \frac{(\operatorname{Im} z)^2}{\sinh^2\eta} = 1.$$

The semimajor and semiminor axes are therefore $\cosh\eta$ and $\sinh\eta$, and the focal distance is $\sqrt{\cosh^2\eta - \sinh^2\eta} = 1$, so the foci are at $z = \pm 1$.

The kernel has the Legendre expansion

$$\frac{1}{z - x} = \sum_{\ell=0}^{\infty}(2\ell + 1)P_\ell(x)Q_\ell(z),$$

where

$$Q_\ell(z) = \frac{1}{2}\int_{-1}^{1} \mathrm{d}y\, \frac{P_\ell(y)}{z - y}.$$

With the same branch choice for $\sqrt{z^2 - 1}$, the Legendre function of the second kind also has the useful representation

$$Q_\ell(z) = \int_{\zeta(z)}^{\infty} \frac{\mathrm{d}\xi}{\xi^{\ell+1}\sqrt{1 - 2z\xi + \xi^2}}, \qquad \zeta(z) = z + \sqrt{z^2 - 1},$$

where the integration path is taken in the ξ-plane with the square-root branch obtained by continuation from large positive z. This form makes the exponential large-ℓ decay transparent: the leading suppression is controlled by the lower endpoint $\zeta(z)$. This expansion converges when

$$\left|z + \sqrt{z^2 - 1}\right| > \left|x + \sqrt{x^2 - 1}\right|.$$

Indeed, at large ℓ,

$$P_\ell(x) \sim \left(x + \sqrt{x^2 - 1}\right)^\ell, \qquad Q_\ell(z) \sim \left(z + \sqrt{z^2 - 1}\right)^{-\ell},$$

where the omitted prefactors are powers of ℓ and algebraic functions of x or z that do not change the exponential rate. Thus angular analyticity implies convergence of the partial-wave expansion throughout every smaller ellipse inside the Lehmann ellipse. More quantitatively, if $\mathcal{M}(s, z)$ is bounded by $B_s(R)$ on $|\zeta(z)| = R$, and if $|\zeta(x)| \le r < R$, then the Legendre expansion inside the contour is bounded term by term by

$$C(R, r)\, B_s(R) \left(\frac{r}{R}\right)^\ell.$$

In particular, the partial-wave coefficients obtained from the contour have exponential decay $O(B_s(R) R^{-\ell})$. This is the analytic input later combined with unitarity: high angular momentum is exponentially suppressed inside an ellipse, while the physical partial-wave weights are bounded by unitarity.

29.3 Partial-Wave Normalization

Let $\mathcal{M}_{\mathrm{ord}}(s, x)$ denote the ordered two-particle amplitude with distinguishable external labels and expansion

$$\mathcal{M}_{\mathrm{ord}}(s, x) = 16\pi \sum_{\ell=0}^{\infty} (2\ell + 1) a_\ell(s) P_\ell(x).$$

The physical amplitude on the two-particle space of identical scalar bosons is Bose symmetric. With the convention fixed in Chapter 19, it is

$$\mathcal{M}(s, x) = \mathcal{M}_{\mathrm{ord}}(s, x) + \mathcal{M}_{\mathrm{ord}}(s, -x) = 32\pi \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1) a_\ell(s) P_\ell(x). \tag{29.2}$$

Equation (29.2) is the conversion between the ordered 16π convention and the unordered identical-boson convention. Only even ℓ occur, and the factor 32π is tied to the final-state factor $1/2$ in the symmetric two-particle resolution of the identity used in the identical-particle cross section. No additional symmetry factor will be inserted later.

Using

$$S_\ell(s) = 1 + 2\mathrm{i}\, \beta(s) a_\ell(s), \qquad \beta(s) = \sqrt{1 - \frac{4m^2}{s}},$$

this becomes the convention

$$\mathcal{M}(s, x) = -16\pi\mathrm{i}\, \beta(s)^{-1} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)\big(S_\ell(s) - 1\big) P_\ell(x).$$

Here $S_\ell(s)$ is the partial-wave matrix element of the full S-operator in the elastic identical-particle channel. If no inelastic channel is open in this angular momentum, it is an eigenvalue of modulus one. With inelastic channels present, it is the elastic matrix element of a unitary matrix on the channel space, hence

$$|S_\ell(s)| \le 1.$$

For real x inside the Lehmann ellipse, the partial-wave series for the physical upper-edge boundary value gives

$$\operatorname{Im} \mathcal{M}(s, x) = 16\pi\, \beta(s)^{-1} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1) P_\ell(x)\, w_\ell(s),$$

where

$$w_\ell(s) = 1 - \operatorname{Re} S_\ell(s).$$

The inequality $|S_\ell| \le 1$ implies the pointwise bound

$$0 \le w_\ell(s) \le 2.$$

The lower bound is simply $\operatorname{Re} S_\ell \le |S_\ell| \le 1$; the upper bound follows from $\operatorname{Re} S_\ell \ge -|S_\ell| \ge -1$.

We now derive the forward optical theorem in the same convention. In the center-of-mass frame put

$$p = |\vec{p}_*| = \frac{1}{2}\sqrt{s - 4m^2}, \qquad \beta(s) = \frac{2p}{\sqrt{s}}.$$

The unordered identical-boson two-particle space contains only the even orbital partial waves in this scalar channel. The same normalization that produces (29.2) gives the elastic partial-wave cross section

$$\sigma_{\text{el}} = \frac{2\pi}{p^2} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)|S_\ell(s) - 1|^2.$$

The coefficient $2\pi/p^2$ is the place where the identical final-state factor has already entered: the full labeled solid-angle integral carries the symmetric identity-resolution factor $1/2$, while the Bose-symmetrized amplitude contains only the even angular momenta. Equivalently, since $p^2 = (s - 4m^2)/4$,

$$\sigma_{\text{el}} = \frac{8\pi}{s - 4m^2} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)|S_\ell(s) - 1|^2.$$

The loss of probability from the elastic two-particle channel in the same partial wave gives

$$\sigma_{\text{inel}} = \frac{2\pi}{p^2} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)\big(1 - |S_\ell(s)|^2\big) = \frac{8\pi}{s - 4m^2} \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell + 1)\big(1 - |S_\ell(s)|^2\big).$$

Both formulas have the same prefactor because they are projections of the same unitary partial-wave channel decomposition. In the elastic term the factor

$$\frac{2\pi}{|\vec{p}_*|^2} \sum_{\ell \text{ even}} (2\ell + 1)|S_\ell - 1|^2$$

is just the identical-boson formula of Chapter 19; in the inelastic term it is the same incoming-flux normalization multiplied by the deficit $1 - |S_\ell|^2$ of elastic probability. Adding the two expressions gives

$$|S_\ell - 1|^2 + 1 - |S_\ell|^2 = 2(1 - \operatorname{Re} S_\ell) = 2w_\ell.$$

Adding them gives

$$\sigma_{\rm tot} = \frac{16\pi}{s-4m^2} \sum_{\substack{\ell=0\\ \ell \text{ even}}}^{\infty} (2\ell+1) w_\ell(s).$$

On the other hand, the forward absorptive part follows directly from the symmetrized amplitude. Setting $x = 1$, using $P_\ell(1) = 1$, and writing $S_\ell = 1 + 2\mathrm{i}\beta a_\ell$, one obtains

$$\mathcal{M}(s,1) = 32\pi \sum_{\substack{\ell=0\\ \ell \text{ even}}}^{\infty} (2\ell+1) \frac{S_\ell(s)-1}{2\mathrm{i}\beta(s)} = -16\pi\mathrm{i}\,\beta(s)^{-1} \sum_{\substack{\ell=0\\ \ell \text{ even}}}^{\infty} (2\ell+1)\big(S_\ell(s)-1\big).$$

Therefore

$$\operatorname{Im}\mathcal{M}(s,1) = 16\pi\,\beta(s)^{-1} \sum_{\substack{\ell=0\\ \ell \text{ even}}}^{\infty} (2\ell+1) w_\ell(s).$$

Since $\beta(s) = 2p/\sqrt{s} = \sqrt{s-4m^2}/\sqrt{s}$, the preceding two displays imply

$$\sigma_{\rm tot} = \frac{\beta(s)}{s-4m^2} \operatorname{Im}\mathcal{M}(s,1) = \frac{1}{\sqrt{s(s-4m^2)}} \operatorname{Im}\mathcal{M}(s,1).$$

This is the optical theorem in the present normalization. The identity-resolution factor 1/2 has already been used in the partial-wave cross-section prefactor, while the forward amplitude appearing here is the Bose-symmetrized amplitude with the 32π expansion in (29.2). Here $\sigma_{\rm tot}$ is the ideal complete-sum cross section of Chapter 19. A detector-level observable with finite angular, energy, jet, or soft-radiation resolution is obtained by replacing the outgoing identity operator in the unitarity sum by the corresponding positive detector effect. In massless theories this replacement is part of the infrared definition of the observable.

29.4 Elastic Form Factors and Watson's Theorem

The same partial-wave unitarity that fixes the phase of an elastic two-particle scattering amplitude also fixes the phase of a form factor into that two-particle channel. This statement is useful only after the channel and the boundary-value convention have been specified.

Fix a two-particle channel A. In examples A may include isospin, charge conjugation, parity, and angular momentum labels; for the identical scalar channel above it is just the even angular momentum ℓ. Suppose that on an open interval

$$s_{\rm th} < s < s_{\rm inel}$$

the only asymptotic states with the quantum numbers A and invariant mass $\sqrt{s}$ are the two particles in this channel. The elastic partial-wave S-matrix is then a scalar phase,

$$S_A(s) = 1 + 2\mathrm{i}\,\beta(s) a_A(s) = \mathrm{e}^{2\mathrm{i}\delta_A(s)}, \qquad s_{\rm th} < s < s_{\rm inel}, \tag{29.3}$$

where $a_A(s)$ is normalized by the same convention as the partial waves in Section 29.3. Equivalently,

$$a_A(s) = \frac{\mathrm{e}^{\mathrm{i}\delta_A(s)} \sin\delta_A(s)}{\beta(s)}.$$

Let $\mathcal{O}_A(x)$ be a Hermitian local operator whose vacuum quantum numbers allow it to create the channel A. Its two-particle partial-wave form factor is the boundary value of a function $\mathcal{F}_A(s)$ on the first sheet:

$$\mathcal{F}_A^+(s) = \lim_{\epsilon\downarrow 0} \mathcal{F}_A(s + \mathrm{i}\epsilon), \qquad \mathcal{F}_A^-(s) = \lim_{\epsilon\downarrow 0} \mathcal{F}_A(s - \mathrm{i}\epsilon).$$

The $+$ boundary value is the form factor with the outgoing two-particle boundary condition, while the $-$ boundary value is the corresponding incoming one. The phase convention for the one-particle states is chosen real, and Hermiticity of $\mathcal{O}_A$, together with real analyticity of the first-sheet form factor, gives

$$\mathcal{F}_A^-(s) = \mathcal{F}_A^+(s)^* \qquad (s_{\text{th}} < s < s_{\text{inel}}). \tag{29.4}$$

Figure 29.2 keeps track of the two boundary values on the elastic cut and the two operations used in Watson's theorem: real analytic conjugation and multiplication by the elastic two-body phase.

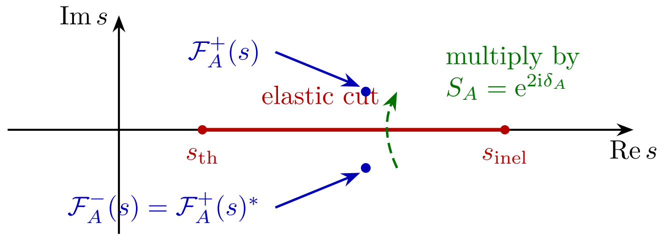


Figure 29.2: On a one-channel elastic cut, real analyticity relates the two boundary values by complex conjugation, while unitarity relates them by the dashed phase-multiplication arrow. Together they imply Watson's theorem.

Proposition 29.1 (Watson theorem in a one-channel elastic interval)**.** *Under the hypotheses above, the two boundary values of the form factor obey*

$$\mathcal{F}_A^+(s) = S_A(s)\mathcal{F}_A^-(s) = \mathrm{e}^{2\mathrm{i}\delta_A(s)}\mathcal{F}_A^-(s). \tag{29.5}$$

Consequently,

$$\mathrm{e}^{-\mathrm{i}\delta_A(s)}\mathcal{F}_A^+(s) \in \mathbb{R}. \tag{29.6}$$

At a point where $\mathcal{F}_A^+(s) \neq 0$, its phase is therefore

$$\operatorname{Arg} \mathcal{F}_A^+(s) = \delta_A(s) \mod \pi.$$

Proof. The discontinuity of the form factor across the two-particle cut is obtained by applying $S^\dagger S = 1$, or equivalently $T - T^\dagger = \mathrm{i}T^\dagger T$ for $S = 1 + \mathrm{i}T$, between the state $\mathcal{O}_A(0)|0\rangle$ and the two-particle channel A, and then projecting to the partial wave A. The LSZ reduction of this Hilbert-space identity gives, in the normalization of (29.3),

$$\mathcal{F}_A^+(s) - \mathcal{F}_A^-(s) = 2\mathrm{i}\,\beta(s)a_A(s)^*\,\mathcal{F}_A^+(s), \qquad s_{\text{th}} < s < s_{\text{inel}}. \tag{29.7}$$

There is only one intermediate state in this interval, so no channel sum is present. The coefficient is the same two-body phase-space coefficient that gives $\operatorname{disc} a_A = 2\mathrm{i}\,\beta|a_A|^2$ for elastic scattering; one side of the cut carries the complex-conjugated scattering amplitude, and the other carries the production form factor.

Using $a_A = \mathrm{e}^{\mathrm{i}\delta_A} \sin\delta_A/\beta$,

$$2\mathrm{i}\,\beta a_A^* = 2\mathrm{i}\,\mathrm{e}^{-\mathrm{i}\delta_A}\sin\delta_A = 1 - \mathrm{e}^{-2\mathrm{i}\delta_A}.$$

Thus (29.7) becomes

$$\mathcal{F}_A^+ - \mathcal{F}_A^- = (1 - \mathrm{e}^{-2\mathrm{i}\delta_A})\mathcal{F}_A^+,$$

or

$$\mathcal{F}_A^- = \mathrm{e}^{-2\mathrm{i}\delta_A}\mathcal{F}_A^+.$$

This is (29.5). Combining it with real analyticity (29.4) gives

$$\mathcal{F}_A^+(s) = \mathrm{e}^{2\mathrm{i}\delta_A(s)}\mathcal{F}_A^+(s)^*,$$

which is equivalent to (29.6). □

For the standard two-pion examples, ignore weak and electromagnetic interactions and work in the isospin limit of QCD. The isovector vector current has

$$\langle \pi^b(p_1)\pi^c(p_2), \mathrm{out}|J_a^\mu(0)|0\rangle = \mathrm{i}\epsilon^{abc}(p_1 - p_2)^\mu F_V^+(s), \qquad s = -(p_1 + p_2)^2,$$

so the two-pion state has $I = 1$ and $\ell = 1$. In the elastic interval, Watson's theorem gives

$$\operatorname{Arg} F_V^+(s) = \delta_1^1(s) \quad \mod \pi.$$

Similarly, for the isoscalar scalar source $\mathcal{S} = \widehat{m}(\bar{u}u + \bar{d}d)$,

$$\langle \pi^a(p_1)\pi^b(p_2), \mathrm{out}|\mathcal{S}(0)|0\rangle = \delta^{ab}F_S^+(s),$$

and the elastic phase is the $I = 0, \ell = 0$ pion-pion phase shift,

$$\operatorname{Arg} F_S^+(s) = \delta_0^0(s) \quad \mod \pi,$$

up to the first inelastic threshold in that channel.

The theorem has a matrix form once more than one channel is open. If $\mathbf{F}^\pm$ is the vector of form factors into all open channels with the same conserved quantum numbers, unitarity gives

$$\mathbf{F}^+(s) = S_A(s)\mathbf{F}^-(s),$$

where now $S_A(s)$ is a unitary matrix. A single form factor component then does not have a phase equal to a single elastic phase shift unless the source couples to an eigenchannel and the channel basis has been diagonalized. In general the phases are determined by the coupled-channel unitary matrix, not by one scalar function $\delta_A(s)$.

Watson's theorem is also the local input for the Omnès construction. Given a phase function on a cut and an integer N large enough for convergence, define

$$\Omega_A(s) = \exp\left\{ \frac{s^N}{\pi} \int_{s_{\mathrm{th}}}^{\infty} \frac{\delta_A(s')\,\mathrm{d}s'}{(s')^N(s' - s)} \right\}. \tag{29.8}$$

Its boundary values obey $\Omega_A^+(s) = \mathrm{e}^{2\mathrm{i}\delta_A(s)}\Omega_A^-(s)$ on the part of the cut where the phase input is valid. Hence the quotient

$$G_A(s) = \mathcal{F}_A(s)/\Omega_A(s)$$

has no discontinuity on that part of the cut. If the phase relation holds on the whole right cut, if all zeros and pole factors of $\mathcal{F}_A$ in the domain have been divided out or inserted explicitly, and if the resulting quotient is polynomially bounded on the cut plane, then the edge-of-the-wedge continuation and the usual one-variable growth theorem give

$$\mathcal{F}_A(s) = P_{N-1}(s)\,\Omega_A(s) \quad \text{up to the displayed zero and pole factors.} \tag{29.9}$$

The degree of the real polynomial P_{N-1} is fixed by the subtraction and growth hypotheses, not by Watson's theorem alone. If the phase is known only below s_{inel}, then (29.9) is not a theorem-level representation of the full form factor; the part of the cut above s_{inel}, zeros of $\mathcal{F}_A$, and possible pole factors must be included as additional input. What Watson's theorem fixes without such additional input is precisely the elastic-cut phase relation (29.5).

29.5 Eikonal Small-Angle Scattering and Regge Singularities

The Froissart–Martin argument uses partial waves to bound the total amount of probability that can be carried by large angular momenta. There is a second, more differential high-energy limit of the same partial-wave representation. It keeps the momentum transfer fixed while $s \to \infty$, so that the scattering angle tends to zero and the angular momentum becomes a transverse impact parameter. This limit is the natural setting for eikonal exponentiation and for the Regge description of fixed-t high-energy scattering.

We formulate the small-angle limit first for an ordered distinguishable amplitude. The identical-particle amplitude in (29.2) is obtained by adding the crossed backward contribution; in the forward high-energy region the ordered direct amplitude is the local object whose partial waves become impact-parameter waves. With the same $S_\ell = 1 + 2\mathrm{i}\beta a_\ell$ convention as above,

$$\mathcal{M}_{\text{ord}}(s,x) = -8\pi\mathrm{i}\,\beta(s)^{-1}\sum_{\ell=0}^{\infty}(2\ell+1)\big(S_\ell(s)-1\big)P_\ell(x).$$

Let

$$p = \frac{1}{2}\sqrt{s-4m^2}, \qquad q = \sqrt{-t} = 2p\sin\frac{\theta}{2}, \qquad b = \frac{\ell+\frac{1}{2}}{p}.$$

The high-energy small-angle scaling is

$$s \to \infty, \qquad q \text{ fixed}, \qquad \ell \to \infty, \qquad b = \frac{\ell+\frac{1}{2}}{p} \text{ fixed}.$$

In this scaling,

$$P_\ell(\cos\theta) = J_0(bq) + O(p^{-1})$$

uniformly on compact bq-intervals; this is the Mehler–Heine limit of the Legendre polynomials. If the elastic partial-wave matrix elements have a weak Riemann-sum limit

$$S_\ell(s) \longrightarrow S(s,b)$$

against smooth compactly supported test functions of b, then the partial-wave sum becomes

$$\mathcal{M}_{\text{ord}}(s,-q^2) = 2\mathrm{i}s\int_{\mathbb{R}^2}\mathrm{d}^2b\,\mathrm{e}^{\mathrm{i}q\cdot b}\,\Gamma(s,b) + o(s), \qquad \Gamma(s,b) = 1 - S(s,b). \tag{29.10}$$

Indeed,

$$\sum_{\ell}(2\ell+1)F\left(\frac{\ell+\frac{1}{2}}{p}\right) = 2p^2\int_0^\infty b\,\mathrm{d}b\,F(b) + o(p^2),$$

and

$$2\pi\int_0^\infty b\,\mathrm{d}b\,J_0(bq)F(b) = \int_{\mathbb{R}^2}\mathrm{d}^2b\,\mathrm{e}^{\mathrm{i}q\cdot b}F(b).$$

Since $4p^2 = s - 4m^2$, the coefficient in (29.10) is $2\mathrm{i}s$ at leading order. The formula is therefore not a new normalization of the S-matrix: it is the continuum large-ℓ form of the partial-wave normalization already fixed above.

At fixed b, unitarity of the partial-wave channel gives

$$|S(s,b)| \le 1$$

in the same weak limiting sense. Thus the profile $\Gamma = 1 - S$ obeys

$$2\,\mathrm{Re}\,\Gamma(s,b) = |\Gamma(s,b)|^2 + \big(1 - |S(s,b)|^2\big). \tag{29.11}$$

Consequently the high-energy limits of the total, elastic, and inelastic cross sections are

$$\sigma_{\mathrm{tot}}(s) = 2\int \mathrm{d}^2b\;\mathrm{Re}\,\Gamma(s,b), \qquad \sigma_{\mathrm{el}}(s) = \int \mathrm{d}^2b\,|\Gamma(s,b)|^2, \tag{29.12}$$

and

$$\sigma_{\mathrm{inel}}(s) = \int \mathrm{d}^2b\,\big(1 - |S(s,b)|^2\big).$$

The identity (29.11) is the continuum version of partial-wave probability conservation. The limiting profile is often parametrized by an eikonal phase

$$S(s,b) = \exp\{\mathrm{i}\chi(s,b)\}, \qquad \mathrm{Im}\,\chi(s,b) \ge 0,$$

so that

$$\mathcal{M}_{\mathrm{eik}}(s,-q^2) = 2\mathrm{i}s\int \mathrm{d}^2b\,\mathrm{e}^{\mathrm{i}q\cdot b}\left(1 - \mathrm{e}^{\mathrm{i}\chi(s,b)}\right). \tag{29.13}$$

The condition $\mathrm{Im}\,\chi \ge 0$ is precisely the statement $|S| \le 1$. A purely real χ describes elastic phase shift without absorption; a positive imaginary part describes loss of probability into inelastic channels at the same impact parameter.

The first term in the eikonal phase is fixed by the Born amplitude. Expanding (29.13) to first order gives

$$\mathcal{M}_{\mathrm{Born}}(s,-q^2) = 2s\int \mathrm{d}^2b\,\mathrm{e}^{\mathrm{i}q\cdot b}\chi_1(s,b),$$

hence

$$\chi_1(s,b) = \frac{1}{2s}\int \frac{\mathrm{d}^2Q}{(2\pi)^2}\,\mathrm{e}^{-\mathrm{i}Q\cdot b}\,\mathcal{M}_{\mathrm{Born}}(s,-Q^2). \tag{29.14}$$

For example, if a massive exchanged field of mass μ gives

$$\mathcal{M}_{\mathrm{Born}}(s,-Q^2) = \frac{g_1 g_2}{Q^2+\mu^2}$$

in the high-energy normalization used here, then

$$\chi_1(s,b) = \frac{g_1 g_2}{4\pi s} K_0(\mu b).$$

For exchange of spin J, the Born numerator may grow as s^J, so the same Fourier formula gives $\chi_1 = O(s^{J-1})$ at fixed b. The eikonal representation is then a unitarization of the repeated exchange at fixed impact parameter, not a proof that the Born approximation remains small.

Figure 29.3 identifies the impact parameter variable used in this limit and displays the exponentiation of ladder exchanges into the eikonal phase $\chi(s,b)$.

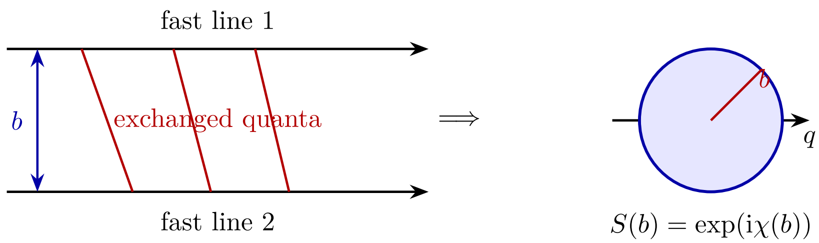


Figure 29.3: In the high-energy small-angle limit, the large angular momentum label becomes an impact parameter $b = (\ell + \frac{1}{2})/p$. The sum of ladder and crossed-ladder exchanges is encoded by the eikonal phase $\chi(s,b)$, while unitarity is the pointwise condition $|S(s,b)| \le 1$.

Controlled Approximation 29.2 (Eikonal exponentiation of repeated exchange)**.** Fix a perturbative regularization and consider two heavy or high-energy external particles whose momentum transfer remains transverse and finite as $s \to \infty$. Assume that, at each fixed order n, the hard external propagators in the ladder and crossed-ladder graphs may be replaced by their eikonal denominators with an integrable remainder. Let $\chi_1(s,b)$ be the Born phase defined by (29.14). Then the leading contribution of all n-exchange ladder and crossed-ladder graphs to the ordered amplitude is

$$\mathcal{M}^{(n)}_{\mathrm{ladd}}(s,-q^2) = 2\mathrm{i}s \int \mathrm{d}^2 b\, \mathrm{e}^{\mathrm{i}q\cdot b} \left[-\frac{(\mathrm{i}\chi_1(s,b))^n}{n!} \right] + o(s) \tag{29.15}$$

in the high-energy small-angle scaling. Summing over $n \ge 1$ gives (29.13) with $\chi = \chi_1$ at leading eikonal order.

The derivation starts from the singular part of a hard line after it absorbs momenta k_r whose transverse components are fixed and whose longitudinal components are small compared with the hard energy. For an on-shell scalar line with momentum p, the regulated propagator after a partial momentum K has flowed through the line is

$$\frac{\mathrm{i}}{(p+K)^2 + m^2 - \mathrm{i}0} = \frac{\mathrm{i}}{2p\cdot K + K^2 - \mathrm{i}0}.$$

In the eikonal scaling $K^2/(p\cdot K) \to 0$ on the leading pinch surface, so the propagator is replaced by

$$\frac{\mathrm{i}}{2p\cdot K - \mathrm{i}0}.$$

The numerator of a spinor or vector external line reduces at the same order to the matrix element of the charge or color generator transported along the classical line; spin-dependent corrections carry at least one additional power of the small momentum transfer divided by the hard energy.

The sum over all ladder and crossed-ladder orderings is exactly the perturbative expansion of two ordered exponentials along the two classical trajectories. In an abelian scalar-exchange model this statement is literal: the hard-line factor is

$$\exp\left\{ \mathrm{i}g_1 \int_{-\infty}^{\infty} \mathrm{d}\tau\, \varphi(b_1 + v_1\tau) \right\} \exp\left\{ \mathrm{i}g_2 \int_{-\infty}^{\infty} \mathrm{d}\tau\, \varphi(b_2 + v_2\tau) \right\}.$$

Contracting the exchanged field between the two lines gives the connected two-line cumulant

$$\mathrm{i}\chi_1(s,b) = -g_1 g_2 \int \mathrm{d}\tau_1 \mathrm{d}\tau_2\, D_F(b + v_1\tau_1 - v_2\tau_2),$$

with the same Feynman prescription as the original graphs. Fourier transforming the transverse separation gives (29.14). Since the exchanged field is Gaussian at this order, the exponential cumulant formula gives the coefficient $(\mathrm{i}\chi_1)^n/n!$. The connected amplitude is proportional to $1 - \exp(\mathrm{i}\chi_1)$, which supplies the sign in (29.15).

For nonabelian gauge exchange the ordered exponentials are Wilson lines, and the exponent contains color matrices ordered along the hard trajectories. Only in a color channel where the Wilson-line product is diagonal does the result reduce to a scalar phase χ. In general the impact-parameter $S(s,b)$ is a matrix on the finite-dimensional internal channel space, and (29.11) is replaced by the positive operator identity

$$(1-S) + (1-S)^\dagger = (1-S)^\dagger(1-S) + (1 - S^\dagger S).$$

Regge theory is a different organization of the same high-energy fixed-t regime. It analytically continues angular momentum rather than taking the large-ℓ continuum limit at fixed b. The continuation is not a consequence of the partial-wave expansion alone; it is an additional analyticity and growth hypothesis in the complex angular-momentum plane. The simultaneous Mandelstam representation, when available, controls the locations and discontinuities of first-sheet singularities in the invariant variables. It does not by itself assert that the partial-wave sequence has a unique meromorphic interpolation in J, nor that such an interpolation has only pole singularities. Regge poles therefore enter below only after the complex-J interpolation and its growth class have been declared.

To state the construction, use the crossed t-channel scattering angle

$$z_t = 1 + \frac{2s}{t - 4m^2}.$$

At fixed physical $t < 0$ and $s \to +\infty$, one has $-z_t \sim 2s/(4m^2 - t) > 0$. Let $a_\ell(t)$ be the t-channel partial-wave coefficient, normalized so that

$$\mathcal{M}_t(t, z_t) = 16\pi \sum_{\ell=0}^{\infty} (2\ell+1) a_\ell(t) P_\ell(z_t)$$

in the t-channel physical region. The signature components are

$$\mathcal{M}_\tau(s,t) = \frac{1}{2}\big(\mathcal{M}(s,t) + \tau\, \mathcal{M}(u,t)\big), \qquad \tau = \pm 1, \qquad u = 4m^2 - s - t.$$

Theorem 29.3 (Carlson uniqueness in the half-plane)**.** *Let F be holomorphic in $\operatorname{Re} J > 0$, continuous on $\operatorname{Re} J \geq 0$, and suppose that it has strict exponential type smaller than π in the closed right half-plane: there is a number $\kappa < \pi$ such that, for every $\delta > 0$,*

$$|F(J)| \leq C_\delta \exp\big((\kappa + \delta)|J|\big), \qquad \operatorname{Re} J \geq 0.$$

Assume also that the boundary function on the imaginary axis has strict type smaller than π: for some $\sigma < \pi$ and every $\delta > 0$,

$$|F(\mathrm{i}y)| \leq C'_\delta \exp\big((\sigma + \delta)|y|\big), \qquad y \in \mathbb{R}.$$

If

$$F(n) = 0, \qquad n = 0, 1, 2, \ldots,$$

then $F \equiv 0$ in the half-plane. Equivalently, within this growth class the values on the nonnegative integers determine the holomorphic interpolation uniquely.

Proof. The number π is the density threshold set by the model zero set $\mathbb{Z}_{\geq 0}$. The Carleman–Jensen formula in the right half-plane turns the boundary exponential type into a bound on the logarithmic density of positive real zeros. Let F be nonzero and let

$$N_F(R) = \#\{a : \ F(a) = 0, \ 0 < a < R, \ a \in \mathbb{R}\}$$

count positive real zeros with multiplicity; the possible boundary zero at $J = 0$ is irrelevant to this density estimate. Applying the half-plane Jensen formula to the half-disc

$$H_R = \{J : \operatorname{Re} J > 0, \ |J| < R\}$$

and using the growth bounds on the semicircle and on the imaginary diameter gives the zero-density estimate

$$\int_1^R \frac{N_F(r)}{r^2}\,\mathrm{d}r \leq \frac{\sigma + \delta}{\pi} \log R + O_\delta(1), \qquad R \to \infty. \tag{29.16}$$

Indeed, the zero term in Carleman's formula is $\sum_{0<a<R}(a^{-1} - aR^{-2})$, which is equivalent to $\int_1^R N_F(r) r^{-2} \mathrm{d}r$ up to $O(1)$. The logarithmic term comes from the imaginary diameter, where $\log|F(\mathrm{i}y)F(-\mathrm{i}y)| \leq 2(\sigma + \delta)|y| + O_\delta(1)$; the semicircle term is only $O_{\kappa,\delta}(1)$, since it carries the prefactor $(\pi R)^{-1} \int_{-\pi/2}^{\pi/2} \cos\theta\,\mathrm{d}\theta$. Polynomial boundary factors and small endpoint losses are removed by the standard Phragmen–Lindelof smoothing $F(J) \mapsto F(J)\mathrm{e}^{-\delta J}(1 + \epsilon J)^{-m}$, with m chosen large enough to remove polynomial boundary factors before $\epsilon \downarrow 0$.

If $F(n) = 0$ for every positive integer n, then

$$N_F(r) \geq \lfloor r \rfloor$$

and therefore

$$\int_1^R \frac{N_F(r)}{r^2}\,\mathrm{d}r \geq \log R + O(1).$$

Combining this with (29.16) gives

$$1 \leq \frac{\sigma + \delta}{\pi}$$

for every $\delta > 0$, contradicting $\sigma < \pi$. Hence a nonzero function in the stated class has too small a boundary type to vanish at every positive integer, and the theorem follows. The threshold is sharp: $\sin \pi J$ vanishes at all integers and has type exactly π on the imaginary axis. Thus the growth hypothesis is the datum that makes the complex-J interpolation unique. $\square$

Carlson uniqueness application. Fix t and a signature τ. Suppose two proposed continuations $a_{\tau,1}(J,t)$ and $a_{\tau,2}(J,t)$ of the same partial-wave sequence $a_{\tau,\ell}(t)$ are holomorphic in a right half-plane after separating the same declared Regge-pole set, have the same pole principal parts there, and obey the Carlson growth class of Theorem 29.3 on the pole-subtracted difference. Then the two continuations agree in that half-plane.

Subtract the common pole principal parts and define

$$F(J) = a_{\tau,1}(J,t) - a_{\tau,2}(J,t).$$

The hypotheses make F holomorphic in the right half-plane and put it in the Carlson growth class. Since both continuations reproduce the same integer-spin partial waves, $F(n) = 0$ for every $n \in \mathbb{Z}_{\geq 0}$. Theorem 29.3 gives $F \equiv 0$. Adding back the common pole principal parts proves the claim.

This uniqueness application has existence content only after a continuation with the stated growth has been constructed. It shows why a Regge analysis must state the vertical-strip or half-plane growth bound of the complex-J continuation: outside such a growth class, functions may agree at every integer J and differ away from the integers. The pole assumptions used below are therefore part of the analytic input rather than consequences of the physical partial-wave series alone.

Assume that, for fixed t in a domain of interest, the signature-projected partial waves admit an analytic continuation $a_\tau(J,t)$ from nonnegative integer J to a meromorphic function in a right half-plane, with sufficient decrease in vertical strips to justify the contour deformation below. Let C be a counterclockwise contour encircling the nonnegative integers and no other singularities. Then the Sommerfeld–Watson representation is

$$\mathcal{M}_\tau(s,t) = 8\pi \int_C \frac{\mathrm{d}J}{2\mathrm{i}} \, \frac{(2J+1)\big(1+\tau \mathrm{e}^{-\mathrm{i}\pi J}\big) a_\tau(J,t) P_J(-z_t)}{\sin \pi J}. \tag{29.17}$$

The normalization is fixed by residues at $J = \ell$: since

$$P_\ell(-z_t) = (-1)^\ell P_\ell(z_t), \qquad \sin \pi J = \pi(-1)^\ell (J-\ell) + O((J-\ell)^2),$$

the residue of (29.17) at $J = \ell$ is

$$16\pi \, \frac{1+\tau(-1)^\ell}{2} \, (2\ell+1) a_\ell(t) P_\ell(z_t).$$

Thus $\tau = +1$ keeps the even-spin part and $\tau = -1$ keeps the odd-spin part.

Suppose that $a_\tau(J,t)$ has a simple Regge pole

$$a_\tau(J,t) = \frac{r_\tau(t)}{J-\alpha(t)} + a_\tau^{\mathrm{reg}}(J,t), \tag{29.18}$$

and that the contour in (29.17) may be moved past this pole, with the remaining background contour lying to the left of $\operatorname{Re}\alpha(t)$. The pole contribution is then

$$\mathcal{M}_\tau^{\mathrm{pole}}(s,t) = \eta_\tau(\alpha(t)) \, \mathfrak{r}_\tau(t) \left(\frac{s}{s_0(t)}\right)^{\alpha(t)} \big(1 + O(s^{-1})\big), \tag{29.19}$$

where

$$\eta_\tau(\alpha) = \frac{1+\tau \mathrm{e}^{-\mathrm{i}\pi\alpha}}{\sin \pi\alpha}$$

is the signature factor, $s_0(t)$ is a positive kinematic scale such as $(4m^2 - t)/2$, and the Regge residue $\mathfrak{r}_\tau(t)$ includes $8\pi(2\alpha+1)r_\tau(t)$ together with the leading large-$(-z_t)$ coefficient of $P_\alpha(-z_t)$. Explicitly, for fixed complex α away from negative integers,

$$P_\alpha(Z) = \frac{\Gamma(2\alpha+1)}{\Gamma(\alpha+1)^2}\left(\frac{Z}{2}\right)^\alpha \left(1 + O(Z^{-2})\right), \qquad |Z| \to \infty$$

in any sector avoiding the negative real axis, and the branch used in (29.19) is inherited from the first-sheet continuation of z_t.

Figure 29.4 records the Sommerfeld–Watson deformation: the integer-spin contour is moved past a Regge pole, and the pole residue gives the $s^{\alpha(t)}$ fixed-t high-energy term.

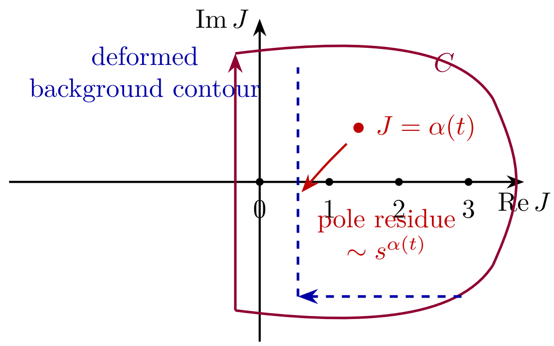


Figure 29.4: The Sommerfeld–Watson contour converts the integer-spin partial-wave sum into a complex-J integral. Deforming the contour past a Regge pole $J = \alpha(t)$ produces the fixed-t high-energy power $s^{\alpha(t)}$, while the displaced contour gives the remaining background.

The pole has a particle interpretation when its trajectory passes through an allowed integer spin. If

$$\alpha(M_n^2) = n, \qquad \tau = (-1)^n, \qquad \alpha'(M_n^2) \neq 0,$$

then the signature factor in (29.19) has the same pole as $(\alpha(t)-n)^{-1}$, and the Regge contribution contains

$$\frac{\text{const.}}{n - \alpha(t)} = \frac{\text{const.}}{-\alpha'(M_n^2)(t - M_n^2)} + \text{regular terms.}$$

This is the usual spin-n exchanged-particle pole in the crossed channel. The Regge trajectory is therefore a statement about analytic organization of a family of exchanged singularities, not an independent replacement for the spectral condition or for LSZ.

Definition 29.4 (Regge trajectory and daughter data)**.** Fix a signature τ, a t-domain, and a meromorphic continuation $a_\tau(J,t)$ satisfying the vertical-strip hypotheses needed for (29.17). A Regge trajectory system is a collection of pole curves

$$J = \alpha_r(t) - n, \qquad n \in \mathbb{Z}_{\geq 0},$$

together with residues $r_{\tau,r,n}(t)$ and a statement of the background contour. The curve $n = 0$ is called the leading trajectory and $n > 0$ curves are daughter trajectories. This terminology is only meaningful after the complex-J continuation and its growth conditions have been specified.

Daughters are not optional decorations if the amplitude is expected to encode an infinite family of crossed-channel resonances while retaining crossing symmetry and polynomial boundedness. A daughter pole contributes

$$\mathcal{M}^{\text{pole}}_{\tau,r,n}(s,t) \sim \eta_\tau(\alpha_r(t)-n)\, \mathfrak{r}_{\tau,r,n}(t) \left(\frac{s}{s_0(t)}\right)^{\alpha_r(t)-n}.$$

It is therefore suppressed at fixed t and large s relative to the leading pole, but it can carry the crossed-channel particle poles required by the finite-spin spectrum. The statement that a hadron family lies on a Regge trajectory means that the corresponding scattering amplitude has such a pole organization and that the trajectory passes through the physical pole positions. It is not a statement about a constituent potential.

Let

$$A(s,t) = g^2 \frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}, \qquad \alpha(x) = \alpha_0 + \alpha' x, \quad \alpha' > 0. \tag{29.20}$$

At $\alpha(s) = n \in \mathbb{Z}_{\geq 0}$, $A(s,t)$ has a simple pole whose residue is a polynomial of degree n in $\alpha(t)$. Explicitly,

$$A(s,t) = -\frac{g^2}{n!} \frac{\prod_{r=1}^{n}\big(\alpha(t)+r\big)}{\alpha(s)-n} + O(1). \tag{29.21}$$

Thus the n-th pole contains crossed-channel angular dependence no higher than spin n, in the same sense that a polynomial of degree n in the crossed scattering variable decomposes into partial waves with $\ell \leq n$. Near $\alpha(s) = n$, write $\epsilon = \alpha(s) - n$. Since

$$\Gamma(-n-\epsilon) = \frac{(-1)^{n+1}}{n!\,\epsilon} + O(1),$$

one has

$$A(s,t) = \frac{g^2(-1)^{n+1}}{n!\,(\alpha(s)-n)} \frac{\Gamma(-\alpha(t))}{\Gamma(-n-\alpha(t))} + O(1).$$

Using

$$\frac{\Gamma(z)}{\Gamma(z-n)} = \prod_{r=1}^{n}(z-r)$$

with $z = -\alpha(t)$ gives

$$\frac{\Gamma(-\alpha(t))}{\Gamma(-n-\alpha(t))} = (-1)^n \prod_{r=1}^{n}(\alpha(t)+r),$$

which proves (29.21).

Controlled Approximation 29.5 (Flux-tube slope and Regge phenomenology)**.** If a highly excited mesonic state is approximated by a long rotating open flux tube of tension σ with endpoint masses neglected, the classical relations

$$E = \frac{\pi\sigma}{\omega}, \qquad J = \frac{\pi\sigma}{2\omega^2}$$

give

$$J = \frac{E^2}{2\pi\sigma}. \tag{29.22}$$

This explains the frequently quoted slope $\alpha'_{\text{open}} = 1/(2\pi\sigma)$ as a long-string approximation. It is not a nonperturbative derivation of the QCD hadron spectrum. Endpoint masses, spin, string breaking, mixing with multi-hadron channels, and the fact that most physical resonances are second-sheet poles must be supplied before (29.22) becomes a quantitative QCD statement.

The leading even-signature vacuum singularity is called the Pomeron. In a pure pole model with

$$\alpha_P(0) > 1,$$

the forward optical theorem would give

$$\sigma_{\text{tot}}(s) \sim s^{\alpha_P(0)-1}$$

up to phases and logarithms. Such a single uncompensated pole cannot be the complete asymptotic answer in a massive theory satisfying the Froissart–Martin hypotheses. Eikonalization gives the mechanism by which a growing exchange may be compatible with impact-parameter unitarity. If the leading eikonal phase has large-b behavior

$$\chi_1(s,b) \sim C\, s^{\Delta} \mathrm{e}^{-\mu b}, \qquad \Delta > 0, \quad \mu > 0,$$

then the region where $|\chi_1| \gtrsim 1$ has radius

$$R(s) \sim \frac{\Delta}{\mu} \log s.$$

After the eikonal exponential has saturated $|S(b)| \le 1$ inside this region, the total cross section grows like an area,

$$\sigma_{\text{tot}}(s) \lesssim 2\pi R(s)^2 = O(\log^2 s).$$

The precise coefficient requires the same analyticity and nearest-singularity input as the Froissart–Martin theorem; the eikonal estimate explains the impact-parameter dynamics behind saturation but does not replace the axiomatic bound.

Definition 29.6 (Inclusive triple-Regge kinematics)**.** For an inclusive reaction

$$a(p_a) + b(p_b) \longrightarrow c(p_c) + X,$$

write

$$s = (p_a + p_b)^2, \qquad t = (p_a - p_c)^2, \qquad M_X^2 = (p_a + p_b - p_c)^2.$$

The triple-Regge limit is the limiting regime

$$s \to \infty, \qquad M_X^2 \to \infty, \qquad t \text{ fixed}, \qquad \xi := \frac{M_X^2}{s} \to 0. \tag{29.23}$$

The particle c is separated from the unobserved system X by the large rapidity interval $\log(s/M_X^2)$, while the internal energy available to the inclusive subsystem X grows like M_X. Choose once and for all a positive mass scale s_0 and define the dimensionless inclusive density

$$\mathcal{D}_{ab\to cX}(s, M_X^2, t) := s_0^2 \, \frac{\mathrm{d}^2\sigma(ab \to cX)}{\mathrm{d}t\, \mathrm{d}M_X^2}. \tag{29.24}$$

Hypothesis 29.7 (Triple-Regge factorization system)**.** For the inclusive density in (29.24), a triple-Regge system on a domain $t \in U$ consists of:

1. Regge trajectories $\alpha_i(t)$ for the two t-channel exchanges adjacent to the measured transition $a \to c$;
2. Regge trajectories $\alpha_k(0)$ for the forward singularities governing the inclusive b-to-X discontinuity;
3. coefficient functions $G_{ijk}(t)$, including the two external Regge residues and a local triple-Regge vertex, in a stated phase and signature convention;
4. a background term $B(s, M_X^2, t)$ whose growth in the variables s/M_X^2 and M_X^2/s_0 is bounded by exponents strictly below the retained trajectories on the chosen domain.

The system asserts the asymptotic representation

$$\mathcal{D}_{ab\to cX}(s, M_X^2, t) = \sum_{i,j,k} G_{ijk}(t) \left(\frac{s}{M_X^2}\right)^{\alpha_i(t)+\alpha_j(t)-2} \left(\frac{M_X^2}{s_0}\right)^{\alpha_k(0)-1} + B(s, M_X^2, t) \tag{29.25}$$

in the limit (29.23). Equation (29.25) is a precise factorization hypothesis about a six-point discontinuity. A derivation from local QFT requires the corresponding six-point analyticity, complex angular-momentum continuation, signature, and discontinuity hypotheses in addition to the two-body Sommerfeld–Watson representation.

Exponent bookkeeping in a triple-Regge system. Assume the system of Hypothesis 29.7 and fix a single term (i, j, k). In the variables

$$Y := \log \frac{s}{M_X^2}, \qquad Y_X := \log \frac{M_X^2}{s_0},$$

the contribution of that term to the dimensionless density is

$$G_{ijk}(t) \exp\bigl[\bigl(\alpha_i(t) + \alpha_j(t) - 2\bigr)Y + \bigl(\alpha_k(0) - 1\bigr)Y_X\bigr]. \tag{29.26}$$

For the diffractive PPP contribution, where the same even-signature vacuum trajectory appears on all three Reggeon lines,

$$\mathcal{D}^{PPP} = G_{PPP}(t) \left(\frac{s}{M_X^2}\right)^{2\alpha_P(t)-2} \left(\frac{M_X^2}{s_0}\right)^{\alpha_P(0)-1} \tag{29.27}$$

up to the declared background and lower trajectories. If $\alpha_P(t) = 1 + \Delta + \alpha_P' t + O(t^2)$, then at small fixed t the dependence on the rapidity gap is

$$\exp\bigl[2(\Delta + \alpha_P' t + O(t^2))Y\bigr].$$

Indeed, taking logarithms of the two dimensionless ratios in (29.25) gives

$$\left(\frac{s}{M_X^2}\right)^{\alpha_i(t)+\alpha_j(t)-2} \left(\frac{M_X^2}{s_0}\right)^{\alpha_k(0)-1} = \exp\bigl[\bigl(\alpha_i(t) + \alpha_j(t) - 2\bigr)Y + \bigl(\alpha_k(0) - 1\bigr)Y_X\bigr].$$

The PPP specialization is obtained by setting $i = j = k = P$. Expanding $\alpha_P(t)$ near $t = 0$ gives the final displayed expression.

Remark 29.8 (AGK rules and what the eikonal argument proves). Abramovsky–Gribov–Kancheli cutting rules are statements about discontinuities of multi-Reggeon exchange diagrams after a definite Reggeon-field-theory or eikonal representation has been specified. The minimal result that follows from the absorptive one-channel eikonal used above is the Poisson law for cut elementary exchanges at fixed impact parameter:

$$S(b) = \mathrm{e}^{-\Omega(b)/2} \quad\Longrightarrow\quad P_n(b) = \mathrm{e}^{-\Omega(b)}\,\frac{\Omega(b)^n}{n!}, \qquad \sum_{n=0}^{\infty} P_n(b) = 1. \tag{29.28}$$

This identity is a consequence of unitarity for that eikonal model. The full AGK cancellation pattern for general Reggeon diagrams requires additional analytic continuation, signature, and cutting hypotheses; in this monograph it will be used only after those hypotheses have been declared.

Definition 29.9 (Minimal Reggeon field-theory model)**.** Fix a high-energy regime in which a single even-signature vacuum Regge singularity is retained. A minimal Reggeon field-theory model consists of:

(i) a rapidity variable Y and an impact parameter $b \in \mathbb{R}^2$;

(ii) two fields $\phi(Y,b)$ and $\widetilde{\phi}(Y,b)$, where $\widetilde{\phi}$ is the response field used to write a retarded evolution problem;

(iii) a quadratic action

$$S_0[\widetilde{\phi},\phi] = \int \mathrm{d}Y\,\mathrm{d}^2b\,\widetilde{\phi}\big(\partial_Y - \alpha'_P\nabla_b^2 - \Delta_P\big)\phi, \qquad \alpha'_P > 0, \tag{29.29}$$

with retarded boundary condition in Y;

(iv) a list of interaction vertices, for example a triple-Reggeon vertex $g_{3P}\int \mathrm{d}Y\,\mathrm{d}^2b\,(\widetilde{\phi}\phi^2 - \widetilde{\phi}^2\phi)$, together with a declared cutting prescription.

This model is not a definition of QCD. It is an effective high-energy description of a selected complex-angular-momentum singularity after a Reggeon basis, signature convention, and unitarity/cutting rules have been declared.

Free Reggeon diffusion kernel. The retarded Green kernel of

$$\big(\partial_Y - \alpha'_P\nabla_b^2 - \Delta_P\big)G(Y,b) = \delta(Y)\delta^{(2)}(b) \tag{29.30}$$

is

$$G(Y,b) = \Theta(Y)\,\frac{\exp(\Delta_P Y)}{4\pi\alpha'_P Y}\,\exp\!\left(-\frac{|b|^2}{4\alpha'_P Y}\right). \tag{29.31}$$

Equivalently, in transverse momentum q,

$$\widehat{G}(Y,q) = \Theta(Y)\exp\!\big((\Delta_P - \alpha'_P q^2)Y\big). \tag{29.32}$$

Thus a pole contribution generated by the quadratic model has trajectory

$$\alpha_P(t) = 1 + \Delta_P + \alpha'_P t \tag{29.33}$$

when $t = -q^2$.

Fourier transform in b:

$$\widehat{G}(Y,q) = \int \mathrm{d}^2 b\, \mathrm{e}^{-\mathrm{i}q\cdot b} G(Y,b).$$

For $Y > 0$, (29.30) becomes

$$\partial_Y \widehat{G}(Y,q) = (\Delta_P - \alpha'_P q^2)\widehat{G}(Y,q).$$

The retarded delta-function normalization gives $\widehat{G}(0^+,q) = 1$, hence (29.32). The inverse two-dimensional Fourier transform of $\exp(-\alpha'_P q^2 Y)$ is the heat kernel $(4\pi\alpha'_P Y)^{-1}\exp[-|b|^2/(4\alpha'_P Y)]$, giving (29.31). Since $\widehat{G} \sim \exp[(\alpha_P(t)-1)Y]$ in Regge notation and $t = -q^2$, the trajectory is (29.33).

Two-Reggeon cut in the free diffusion model. In the quadratic model of Definition 29.9, the exchange of two independent Reggeons with total transverse momentum q has the rapidity kernel

$$I_2(Y,q) = \int \frac{\mathrm{d}^2 k}{(2\pi)^2}\, \widehat{G}(Y,k)\widehat{G}(Y,q-k) = \Theta(Y)\, \frac{\exp\left[\left(2\Delta_P - \frac{\alpha'_P q^2}{2}\right) Y\right]}{8\pi\alpha'_P Y}. \tag{29.34}$$

Consequently its Mellin transform in rapidity has a logarithmic branch point at

$$J_{\mathrm{cut}}(t) = 1 + 2\Delta_P + \frac{\alpha'_P}{2} t, \qquad t = -q^2, \tag{29.35}$$

rather than a simple Regge pole. The corresponding n-Reggeon free convolution has leading exponent

$$n\Delta_P - \frac{\alpha'_P q^2}{n} = n\Delta_P + \frac{\alpha'_P}{n} t \tag{29.36}$$

up to the power of Y generated by the transverse Gaussian integral.

Insert (29.32):

$$I_2(Y,q) = \Theta(Y)\mathrm{e}^{2\Delta_P Y} \int \frac{\mathrm{d}^2 k}{(2\pi)^2} \exp[-\alpha'_P Y (k^2 + (q-k)^2)].$$

Completing the square gives

$$k^2 + (q-k)^2 = 2\left(k - \frac{q}{2}\right)^2 + \frac{q^2}{2}.$$

The two-dimensional Gaussian integral

$$\int \frac{\mathrm{d}^2 \ell}{(2\pi)^2} \mathrm{e}^{-2\alpha'_P Y \ell^2} = \frac{1}{8\pi\alpha'_P Y}$$

proves (29.34). Let $\omega = J - 1$. Up to factors analytic in ω near the singularity, the rapidity Mellin transform contains

$$\int_0^\infty \frac{\mathrm{d}Y}{Y} \exp\left[-\left(\omega - 2\Delta_P + \frac{\alpha'_P q^2}{2}\right) Y\right].$$

After an infrared subtraction at small Y, this integral equals

$$-\log\left(\omega - 2\Delta_P + \frac{\alpha'_P q^2}{2}\right) + \text{analytic terms},$$

so the singularity is a branch point at $\omega = 2\Delta_P - \alpha'_P q^2/2$, which is (29.35). For n independent Reggeons with momenta k_i and $\sum_i k_i = q$, the identity

$$\sum_{i=1}^{n} k_i^2 = \sum_{i=1}^{n} \left(k_i - \frac{q}{n}\right)^2 + \frac{q^2}{n}$$

gives (29.36); the remaining constrained Gaussian integration changes only the power of Y, not the displayed exponential trajectory.

Remark 29.10 (Regge cuts as singularity data). Once more than one exchanged Reggeon is retained, complex angular momentum singularities can become branch cuts. The elementary computation above already produces a cut whose intercept and slope are determined by the two-particle diffusion saddle. Interactions, signature factors, and unitarity constraints change the singularity structure. A pole-only parametrization is therefore a restricted approximation whose scope must be declared. A QCD use of the word Pomeron must specify whether it is describing an isolated pole, a cut, or an effective fit to a restricted energy window.

Definition 29.11 (Absorptive eikonal cut generator)**.** At fixed impact parameter b, let $\Omega(s,b) \geq 0$ be the opacity of a one-channel absorptive eikonal model and write

$$S(s,b) = \exp\left[-\frac{1}{2}\Omega(s,b)\right]. \tag{29.37}$$

The cut-exchange generating function is

$$Z_b(z) = \exp[-\Omega(s,b)(1-z)]. \tag{29.38}$$

The coefficient of z^n,

$$P_n(s,b) = [z^n]Z_b(z) = \mathrm{e}^{-\Omega(s,b)} \frac{\Omega(s,b)^n}{n!}, \tag{29.39}$$

is the fixed-b topological weight for n cut elementary exchanges in this model.

Eikonal AGK cancellation as a factorial-moment identity. For the cut generator of Definition 29.11,

$$\sum_{n=0}^{\infty} P_n(s,b) = 1, \qquad \sum_{n=0}^{\infty} n\, P_n(s,b) = \Omega(s,b). \tag{29.40}$$

More generally, for the falling factorial

$$(n)_r = n(n-1)\cdots(n-r+1), \qquad r \geq 1, \tag{29.41}$$

one has

$$\sum_{n=0}^{\infty} (n)_r P_n(s,b) = \Omega(s,b)^r. \tag{29.42}$$

Equivalently, if every P_n is expanded in powers of Ω, the coefficient of Ω^m in the r-th factorial moment is

$$\sum_{n=r}^{m} \frac{(n)_r(-1)^{m-n}}{n!\,(m-n)!} = \begin{cases} 1, & m = r, \\ 0, & m \neq r. \end{cases} \tag{29.43}$$

This is the precise cancellation supplied by the one-channel absorptive eikonal. It is an AGK-type cancellation for inclusive factorial moments, not a theorem about arbitrary Reggeon-field-theory diagrams. Normalization is $Z_b(1) = 1$. Differentiating $Z_b(z)$ gives

$$\left.\frac{\partial^r Z_b}{\partial z^r}\right|_{z=1} = \Omega(s,b)^r,$$

while the same derivative of the power series

$$Z_b(z) = \sum_{n=0}^{\infty} P_n(s,b) z^n$$

is $\sum_n (n)_r P_n(s,b)$. This proves (29.42); the case $r = 1$ gives the mean.

For the coefficient form, expand

$$P_n = \sum_{q=0}^{\infty} \frac{(-1)^q}{q!\,n!} \Omega^{q+n}. \tag{29.44}$$

The coefficient of Ω^m in the factorial moment is therefore

$$\sum_{n=r}^{m} \frac{n!}{(n-r)!} \frac{(-1)^{m-n}}{(m-n)!\,n!} = \sum_{j=0}^{m-r} \frac{(-1)^{m-r-j}}{j!\,(m-r-j)!}. \tag{29.45}$$

This is the coefficient of x^{m-r} in $\mathrm{e}^x \mathrm{e}^{-x} = 1$. It is one when $m = r$ and zero otherwise.

Remark 29.12 (Soft and hard Pomeron language). The words "soft Pomeron" and "hard Pomeron" are not definitions. In this chapter a Pomeron means a declared even-signature vacuum singularity in a specified amplitude or effective high-energy datum. A phenomenological intercept fitted from hadron cross sections, a BFKL cut or pole in a perturbative color-dipole regime, and the intercept Δ_P in (29.29) are different coordinates unless a matching argument relates them. The BFKL construction in Chapter 49 is therefore a short-distance high-energy datum; it is not automatically the same object as the soft vacuum singularity used to organize low-energy hadron scattering.

29.6 The Froissart–Martin Mechanism

Define the positive total weight

$$\bar{L}^2(s) = \sum_{\substack{\ell=0 \\ \ell \text{ even}}}^{\infty} (2\ell+1) w_\ell(s) = \frac{s-4m^2}{16\pi}\,\sigma_{\mathrm{tot}}(s).$$

For real $x > 1$, the Legendre polynomial $P_\ell(x)$ is positive and strictly increases with ℓ. Consider all even-ℓ weights obeying

$$0 \le w_\ell \le 2, \qquad \sum_{\ell \text{ even}} (2\ell+1) w_\ell = \bar{L}^2.$$

Among these weights, the smallest possible value of

$$\sum_{\ell \text{ even}} (2\ell+1) P_\ell(x) w_\ell$$

is obtained by filling the lowest allowed angular momenta to height 2, with at most one partially filled endpoint. This is the elementary rearrangement inequality: if weight is moved from a larger ℓ to a smaller ℓ, the constraint is unchanged while the sum decreases. Denote this minimizing profile by b_ℓ. If L is the largest filled angular momentum, then

$$\sum_{\substack{0\le \ell \le L\\ \ell \text{ even}}} 2(2\ell+1) = L^2 + O(L), \qquad L^2 = \bar{L}^2 + O(\bar{L}).$$

The rearrangement step and the angular-analyticity domain are illustrated in Figure 29.5.

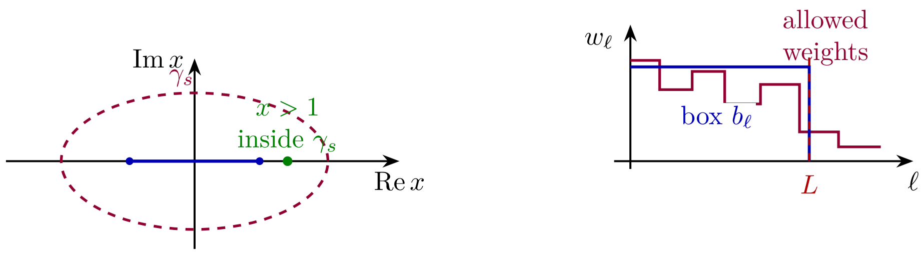


Figure 29.5: Angular point inside the Lehmann ellipse and the minimizing partial-wave weight profile. For $x > 1$ inside γ_s, positivity of $P_\ell(x)$ makes the extremal lower-bound profile a filled box b_ℓ up to the endpoint angular momentum L.

For $x > 1$, write $x = \cosh\eta$, $\eta > 0$. The integral representation

$$P_\ell(\cosh\eta) = \frac{1}{\pi}\int_0^\pi (\cosh\eta + \cos\alpha\,\sinh\eta)^\ell \, d\alpha$$

implies the following lower bound. For every fixed small $\delta > 0$ there is a constant $C_\delta > 0$, independent of s and ℓ in the large-energy regime used below, such that

$$P_\ell(\cosh\eta) \ge C_\delta \left(1 + (1-\delta)\sinh\eta\right)^\ell \equiv C_\delta y^\ell.$$

Indeed, restrict the integral to an interval of α on which $\cos\alpha \ge 1-\delta$; on that interval the integrand is bounded below by the displayed base.

The actual weighted sum is bounded below by the same sum evaluated on the minimizing box profile. Therefore, for $x > 1$ inside the angular analytic domain,

$$\operatorname{Im}\mathcal{M}(s,x) \ge C\,\beta(s)^{-1} \sum_{\substack{0\le \ell \le L\\ \ell \text{ even}}} (2\ell+1) y^\ell \ge C'\,\beta(s)^{-1} L\,\frac{y^L}{y-1}, \tag{29.46}$$

whenever $L(y-1) \ge 1$. If $L(y-1) < 1$, then $L^2 \lesssim (y-1)^{-2}$, and the relation between L^2 and σ_{tot} already gives a high-energy bound stronger than a logarithmic-square estimate. Hence the displayed lower bound covers the only case in which the Froissart estimate is nontrivial. The constants are independent of s at sufficiently large energy, for fixed δ and fixed t away from the singularity.

Now choose a number t_* with

$$0 < t_* < t_0$$

and evaluate the amplitude at

$$x_*(s) = 1 + \frac{2t_*}{s - 4m^2}.$$

This point lies strictly inside the Lehmann ellipse for sufficiently large s. At large s,

$$y - 1 = (1-\delta)\sqrt{x_*^2 - 1} + O(s^{-1}) = 2(1-\delta)\sqrt{\frac{t_*}{s}} + O(s^{-1}), \tag{29.47}$$

and

$$L \simeq \sqrt{\frac{s\,\sigma_{\rm tot}}{16\pi}}.$$

Therefore the exponential factor obeys

$$y^L \simeq \exp\left[(1-\delta)\sqrt{\frac{t_*\,\sigma_{\rm tot}}{4\pi}}\right]. \tag{29.48}$$

Definition 29.13 (Polynomial bounds used in dispersion theory)**.** For the on-shell amplitude $\mathcal{M}(s,t)$, three polynomial estimates occur in distinct domains.

1. A *physical-region bound* controls $\mathcal{M}(s+\mathrm{i}0,t)$ for real physical scattering angles $-(s-4m^2) \leq t \leq 0$.
2. A *fixed-t cut-plane bound* controls the first-sheet function of complex s at a fixed real $t < 0$, on the contours used for a fixed-t dispersion relation.
3. An *angular-tube bound* controls the analytically continued amplitude at $0 < t < t_0$, equivalently at $x = 1 + 2t/(s-4m^2) > 1$ inside a Lehmann ellipse or its higher-dimensional angular-tube replacement.

The Froissart comparison below uses the third estimate. The Jin–Martin fixed-t theorem supplies the second estimate, while the angular-tube estimate is an additional input unless it has been derived from a stronger Lehmann–Martin domain theorem in the theory under study.

Assume the angular-tube polynomial bound at the chosen nonphysical point:

$$|\mathcal{M}(s, x_*(s))| \leq C_N s^N \qquad (s \to \infty),$$

for some fixed integer N. The lower bound contains the prefactor $\beta^{-1}L/(y-1)$, which is of order $s\sqrt{\sigma_{\rm tot}}$ up to fixed constants. Hence compatibility with the polynomial upper bound implies

$$\sigma_{\rm tot}(s) \leq \frac{(N-1)^2}{(1-\delta)^2}\,\frac{4\pi}{t_*}\left(\log\frac{s}{M_0^2}\right)^2$$

for sufficiently large s. The scale M_0 changes only subleading terms in the logarithm. Taking $t_* \to t_0^-$ after the estimate gives the coefficient governed by the nearest crossed-channel singularity. This is the Froissart–Martin mechanism: angular analyticity converts polynomial boundedness into a bound on the number of strongly contributing partial waves, and unitarity bounds the weight carried by each such partial wave.

29.7 Polynomial Boundedness

The first polynomial-growth statement supplied directly by the Wightman framework is distributional. It is weaker than the pointwise high-energy estimate used at the edge of the Lehmann ellipse in the Froissart comparison, but it is the correct starting point: Wightman functions and LSZ boundary values are tempered distributions before they are functions.

Let

$$\langle s\rangle = (1+s^2)^{1/2}.$$

For $r, N \in \mathbb{Z}_{\geq 0}$, define the Schwartz seminorm

$$p_{r,N}(\varphi) = \max_{0\leq j\leq r} \sup_{s\in\mathbb{R}} \langle s\rangle^N |\partial_s^j \varphi(s)|. \tag{29.49}$$

Let $T \in \mathcal{S}'(\mathbb{R})$ be a tempered distribution. By definition, the topology on $\mathcal{S}(\mathbb{R})$ is generated by the seminorms

$$\varphi \mapsto \sup_s \langle s\rangle^a |\partial_s^b \varphi(s)|, \qquad a, b \in \mathbb{Z}_{\geq 0},$$

and continuity of T means that T is bounded by a finite linear combination of them. Increasing the two indices if necessary gives constants $C > 0$, $r \in \mathbb{Z}_{\geq 0}$, and $N \in \mathbb{Z}_{\geq 0}$ such that

$$|\langle T, \varphi\rangle| \leq C\, p_{r,N}(\varphi) \qquad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}). \tag{29.50}$$

If $\varphi \in \mathcal{S}(\mathbb{R})$ is fixed and

$$\varphi_{R,\Delta}(s) = \Delta^{-1}\varphi\left(\frac{s-R}{\Delta}\right), \qquad R \in \mathbb{R}, \quad \Delta \geq 1, \tag{29.51}$$

then

$$|\langle T, \varphi_{R,\Delta}\rangle| \leq C_\varphi\,(1+|R|+\Delta)^N. \tag{29.52}$$

The estimate follows directly from

$$\partial_s^j \varphi_{R,\Delta}(s) = \Delta^{-1-j}\varphi^{(j)}\left(\frac{s-R}{\Delta}\right).$$

Writing $s = R + \Delta y$ gives

$$\langle s\rangle^N |\partial_s^j \varphi_{R,\Delta}(s)| \leq C_N (1+|R|+\Delta)^N \Delta^{-1-j} \langle y\rangle^N |\varphi^{(j)}(y)|.$$

Since $\Delta \geq 1$, the factors Δ^{-1-j} are bounded by one; taking the supremum over y and over $0 \leq j \leq r$ gives (29.52).

Remark 29.14 (Fixed-t LSZ boundary values after smearing). Fix real $t < 0$ in a fixed-t analyticity domain and separate any first-sheet stable-particle pole terms explicitly. Suppose the connected LSZ amplitude has a boundary value

$$\mathcal{M}_t(s + \mathrm{i}0) \in \mathcal{S}'(\mathbb{R}_s)$$

on the remaining real edge. Then every family of high-energy wave-packet smearings of the form (29.51) obeys a polynomial bound in the central energy R:

$$|\langle \mathcal{M}_t, \varphi_{R,\Delta}\rangle| \leq C_{\varphi,t}\,(1+|R|+\Delta)^{N_t}. \tag{29.53}$$

This is exactly the preceding translated-packet estimate applied to the tempered distribution $\mathcal{M}_t(s+\mathrm{i}0)$. The exponent N_t and the constant $C_{\varphi,t}$ are determined by a finite list of Schwartz seminorms controlling this fixed-t boundary distribution. The constants may depend on the chosen fixed-t edge and on the pole terms that were removed before forming the cut distribution; those pole contributions are handled explicitly by their Cauchy residues in the dispersion formula.

Proposition 29.15 (Subtracted Cauchy transform of a tempered cut)**.** *Let $T \in \mathcal{S}'(\mathbb{R}_\sigma)$, and choose C_T, r_T, N_T such that*

$$|\langle T, \psi\rangle| \le C_T \max_{0\le j\le r_T} \sup_{\sigma\in\mathbb{R}} \langle\sigma\rangle^{N_T} |\partial_\sigma^j \psi(\sigma)| \tag{29.54}$$

for all $\psi \in \mathcal{S}(\mathbb{R})$. Fix $z_0 \in \mathbb{C}\setminus\mathbb{R}$ and an integer $q > N_T$. For $z \in \mathbb{C}\setminus\mathbb{R}$, set

$$K_q(\sigma; z, z_0) = \frac{(z-z_0)^q}{(\sigma-z)(\sigma-z_0)^q}, \qquad F_q(z) = \langle T, K_q(\cdot; z, z_0)\rangle. \tag{29.55}$$

Then F_q is holomorphic in the upper and lower half-planes, and for $y = |\operatorname{Im} z| > 0$,

$$|F_q(z)| \le C_{T,q,z_0}\,(1+|z|)^q\, y^{-r_T-1}. \tag{29.56}$$

Consequently, if a fixed-t amplitude with pole terms removed admits a q-subtracted Cauchy representation whose right- and left-cut discontinuities are tempered distributions of finite orders, then the amplitude is pointwise polynomially bounded on every fixed contour staying a positive distance from the cuts, up to the subtraction polynomial of degree at most $q-1$.

Proof. For real σ, the denominator $\sigma - z$ never vanishes. Since $q > N_T$, the function $K_q(\cdot; z, z_0)$ is a Schwartz function of σ: at large $|\sigma|$ it decays as $|\sigma|^{-q-1}$, and every σ-derivative decays at least as fast. Thus $F_q(z)$ is well-defined. Differentiating K_q with respect to z again gives a Schwartz function of σ, locally uniformly for z in either open half-plane, so differentiation may be passed through the continuous functional T. Hence F_q is holomorphic separately in the upper and lower half-planes.

It remains to prove the displayed bound. By Leibniz' rule, $\partial_\sigma^j K_q$ is a finite sum of terms

$$(z-z_0)^q \frac{c_{a,b}}{(\sigma-z)^{a+1}(\sigma-z_0)^{q+b}}, \qquad a+b=j, \qquad c_{a,b}\in\mathbb{Z}. \tag{29.57}$$

Because $z_0 \notin \mathbb{R}$, and because $q > N_T$,

$$\sup_{\sigma\in\mathbb{R}} \langle\sigma\rangle^{N_T} |\sigma - z_0|^{-q-b} \le C_{q,z_0,N_T}. \tag{29.58}$$

Also $|\sigma - z|^{-a-1} \le y^{-a-1} \le y^{-r_T-1}$ for $0 \le a \le j \le r_T$. Therefore

$$\max_{0\le j\le r_T} \sup_{\sigma\in\mathbb{R}} \langle\sigma\rangle^{N_T} |\partial_\sigma^j K_q(\sigma; z, z_0)| \le C_{q,z_0,N_T,r_T}\, |z-z_0|^q\, y^{-r_T-1}. \tag{29.59}$$

Combining this with (29.54) and $|z - z_0| \le C_{z_0}(1+|z|)$ gives (29.56).

For the final assertion, apply the same estimate to each cut distribution after translating the cut coordinate to σ. A q-subtracted Cauchy representation has the form

$$\mathcal{M}_{\mathrm{reg}}(z,t) = P_{q-1}(z;t) + \sum_{\mathrm{cuts}} \left\langle T_{\mathrm{cut}}(\sigma;t), \frac{(z-z_0)^q}{(\sigma-z)(\sigma-z_0)^q} \right\rangle. \tag{29.60}$$

On a contour whose distance from all cuts is bounded below by a positive number, the factor y^{-r_T-1} and its translated analogues are bounded by contour-dependent constants. The subtraction polynomial and the finite sum of Cauchy-transform bounds are therefore bounded by a power of $1+|z|$. ☐

Remark 29.16 (Where the Jin–Martin input enters). In a massive Wightman theory with stable external scalar particles, tempered LSZ boundary values and the subtracted-Cauchy Proposition 29.15 prove only the following conditional implication:

$$\begin{gathered}\text{fixed-}t\text{ first-sheet cut plane} + \text{ finite-subtraction Cauchy formula}\\ + \text{ tempered discontinuities} \Longrightarrow \text{pointwise cut-plane polynomial growth.}\end{gathered} \tag{29.61}$$

The genuinely QFT-specific theorem is the preceding line's input: locality, the spectrum condition, LSZ pole separation, and the Jost–Lehmann–Dyson/Bros–Epstein–Glaser domain analysis must supply the fixed-t first-sheet domain, the pointwise large-contour growth estimate, and the tempered cut distributions. Once those inputs are known, the finite-subtraction Cauchy formula itself follows from Theorem 29.19. This is the Jin–Martin proof layer. It is a proof obligation for the analytic-scattering part of this monograph and requires more data than Wightman temperedness alone. The implication (29.61) is the final statement of Proposition 29.15. The construction of the first-sheet fixed-t domain is an analytic-continuation theorem for Wightman functions and LSZ limits, not a consequence of the topological definition of $\mathcal{S}'$. Similarly, the pointwise large-contour bound is not a theorem about all tempered boundary values. The global Cauchy formula is proved below, but it needs those domain and growth hypotheses before it can be applied to a QFT amplitude.

Theorem 29.17 (LSZ transfer of a fixed-t holomorphic domain)**.** *Fix a real $t<0$. Let $D_t \subset \mathbb{C}_s$ be a connected one-variable first-sheet domain whose boundary contains a real open edge, and let*

$$\kappa_a = k_a^2 + m^2, \qquad a = 1, \dots, 4,$$

be local holomorphic normal coordinates to the four external mass shells in an off-shell neighbourhood of the relevant on-shell configurations. Suppose an off-shell connected four-point time-ordered germ

$$G_t(s, \kappa_1, \dots, \kappa_4)$$

is holomorphic for $s \in D_t$ and $0 < |\kappa_a| < \epsilon_a$, and has at most normal-crossing simple external one-particle poles in the following strong sense: the LSZ-amputated germ

$$H_t(s,\kappa) = \left[\prod_{a=1}^{4} Z_a^{-1/2}\, \mathrm{i}\, \kappa_a\right] G_t(s,\kappa) \tag{29.62}$$

extends holomorphically to the full polydisc

$$D_t \times \{|\kappa_a| < \epsilon_a,\ a = 1, \dots, 4\}.$$

Assume also that H_t has local polynomial boundary growth in s, uniformly for κ on compact subpolydiscs. Define

$$\mathcal{M}_t(s) = H_t(s,0,0,0,0). \tag{29.63}$$

Then $\mathcal{M}_t$ is holomorphic on D_t, has local polynomial boundary growth on every real edge inherited from H_t, and its distributional discontinuities across cuts are the external-mass-shell restrictions *of the discontinuities of H_t. In particular, if the off-shell discontinuity on a noncompact real edge is tempered in s, with the seminorm orders uniform for $|\kappa_a| \le \epsilon'_a < \epsilon_a$, then the on-shell discontinuity of $\mathcal{M}_t$ is tempered in s.*

Proof. The first assertion is the restriction theorem for holomorphic functions: because H_t is holomorphic on a neighbourhood of the complex submanifold $\kappa = 0$, its restriction $\mathcal{M}_t(s) = H_t(s, 0)$ is holomorphic in $s \in D_t$. The Cauchy formula in the normal variables gives the explicit restriction map. Choose radii $0 < r_a < \epsilon_a$. Then

$$H_t(s,0) = \frac{1}{(2\pi \mathrm{i})^4} \int_{|\zeta_1|=r_1} \cdots \int_{|\zeta_4|=r_4} \frac{H_t(s,\zeta_1,\ldots,\zeta_4)}{\zeta_1\zeta_2\zeta_3\zeta_4} \,\mathrm{d}\zeta_1 \cdots \mathrm{d}\zeta_4. \tag{29.64}$$

The integrand is holomorphic in s, and the integration tori are compact, so the integral is holomorphic in s.

The same formula transfers polynomial boundary growth. Let $K \Subset D_t$ be a subdomain whose boundary approach to a real edge is controlled in the usual tempered-holomorphic sense, and assume

$$|H_t(s,\zeta)| \le C_K (1+|s|)^N d(s, \partial D_t)^{-M}$$

for $s \in K$ and $|\zeta_a| = r_a$. Taking absolute values in (29.64) gives the same type of bound for $H_t(s,0)$, with the constant multiplied by $\prod_a r_a^{-1}$. Thus the LSZ mass-shell restriction does not create a new high-energy growth mechanism; it only changes constants and keeps the finite polynomial order.

For boundary values, test against a compactly supported smooth function $\varphi(s)$ on a real edge. The polynomial-growth hypothesis permits the boundary-value map to commute with the compact Cauchy integrals:

$$\langle \mathrm{bv}\, \mathcal{M}_t, \varphi \rangle = \frac{1}{(2\pi \mathrm{i})^4} \int_{|\zeta_1|=r_1} \cdots \int_{|\zeta_4|=r_4} \langle \mathrm{bv}\, H_t(\cdot,\zeta), \varphi \rangle \frac{\mathrm{d}^4\zeta}{\zeta_1\zeta_2\zeta_3\zeta_4}. \tag{29.65}$$

Taking the difference of the upper and lower boundary values gives the same formula for the discontinuity. If the off-shell discontinuities are tempered with seminorm orders independent of ζ on the chosen tori, the right side is bounded by the same finite Schwartz seminorms of φ, again up to the compact-torus constant. This proves temperedness of the on-shell discontinuity.

Finally, the normalization in (29.62) is exactly the external one-particle pole extraction of the LSZ theorem in Chapter 18. The theorem here does not prove that a particular Wightman theory supplies the off-shell domain and uniform normal-crossing extension; it proves that once those hypotheses are supplied by the locality, spectrum, Jost–Lehmann–Dyson, and Bros–Epstein–Glaser construction, the passage to the on-shell fixed-t amplitude is a legitimate holomorphic and distributional operation. □

Definition 29.18 (Admissible fixed-t two-cut exhaustion)**.** Let $b < a$ be real numbers and set

$$C_R = [a,\infty), \qquad C_L = (-\infty, b], \qquad \Omega = \mathbb{C} \setminus (C_R \cup C_L).$$

An admissible exhaustion of Ω is a family of bounded domains $\Omega_{\mathcal{R},\epsilon} \Subset \Omega$, indexed by $\mathcal{R} \to \infty$ and $\epsilon \downarrow 0$, whose positively oriented boundaries are piecewise C^1 and have the following structure. For each fixed $\epsilon > 0$ and all sufficiently large $\mathcal{R}$, the boundary decomposes as

$$\partial\Omega_{\mathcal{R},\epsilon} = \Gamma^{\infty}_{\mathcal{R},\epsilon} \cup \Gamma^{R,+}_{\mathcal{R},\epsilon} \cup \Gamma^{R,-}_{\mathcal{R},\epsilon} \cup \Gamma^{L,+}_{\mathcal{R},\epsilon} \cup \Gamma^{L,-}_{\mathcal{R},\epsilon} \cup E_{\mathcal{R},\epsilon}. \tag{29.66}$$

The large part $\Gamma^{\infty}_{\mathcal{R},\epsilon}$ lies in the annulus $\mathcal{R} - c_\epsilon \le |w| \le \mathcal{R} + c_\epsilon$ and obeys

$$\operatorname{length} \Gamma^{\infty}_{\mathcal{R},\epsilon} \le c_\epsilon \mathcal{R}. \tag{29.67}$$

The right lips are parameterized, up to endpoint smoothings contained in $E_{\mathcal{R},\epsilon}$, by

$$w = \sigma + \mathrm{i}\epsilon, \qquad w = \sigma - \mathrm{i}\epsilon, \qquad a + \epsilon \le \sigma \le \mathcal{R}, \tag{29.68}$$

with the induced orientation upper minus lower. The left lips are similarly parameterized by

$$w = \sigma + \mathrm{i}\epsilon, \qquad w = \sigma - \mathrm{i}\epsilon, \qquad -\mathcal{R} \le \sigma \le b - \epsilon, \tag{29.69}$$

again with upper-minus-lower orientation after writing the integral in the increasing real variable σ. The connector set $E_{\mathcal{R},\epsilon}$ consists of finitely many smooth arcs joining the lips to the large part and to small circles around the finite endpoints a and b. Its finite-endpoint components have length $O_\epsilon(1)$ and shrink to the endpoints as $\epsilon \downarrow 0$; its large-end components lie in the same annulus as $\Gamma^{\infty}_{\mathcal{R},\epsilon}$ and have total length $O_\epsilon(1)$. Finally, for every compact $K \Subset \Omega$, one has

$$K \subset \Omega_{\mathcal{R},\epsilon}$$

for all sufficiently large $\mathcal{R}$ and sufficiently small ϵ.

The definition is intentionally invariant under smoothing of the corners. If two admissible exhaustions have the same parameters, the difference of their boundaries is the boundary of a compact subset of Ω, together with connector arcs whose limits are either zero by the large-$\mathcal{R}$ estimate or are included in the endpoint boundary distributions. Therefore Cauchy's theorem gives the same limiting formula for every admissible choice.

Theorem 29.19 (Finite-subtraction Cauchy formula in a fixed-t cut plane)**.** *Let $b < a$ be real numbers and set*

$$C_R = [a, \infty), \qquad C_L = (-\infty, b], \qquad \Omega = \mathbb{C} \setminus (C_R \cup C_L).$$

Let F be meromorphic in Ω, with only finitely many simple poles $p_\nu \in \Omega$, and write its local pole part as

$$F(w) = \frac{A_\nu}{p_\nu - w} + \textit{holomorphic near } p_\nu. \tag{29.70}$$

Assume that F has polynomial boundary growth at the two cuts, so that the upper and lower boundary values on each cut exist as tempered distributions, and denote the discontinuities by

$$T_R = F(\sigma + \mathrm{i}0) - F(\sigma - \mathrm{i}0) \quad (\sigma \in C_R), \qquad T_L = F(\sigma + \mathrm{i}0) - F(\sigma - \mathrm{i}0) \quad (\sigma \in C_L).$$

Choose temperedness data N_R, N_L for the weight orders of T_R and T_L, in the sense of (29.54) after translating each cut coordinate to the real line. Suppose also that, for some $A \ge 0$, the large circular parts $\Gamma^{\infty}_{\mathcal{R},\epsilon}$ of an admissible two-cut exhaustion in the sense of Definition 29.18 obey

$$\sup_{w \in \Gamma^{\infty}_{\mathcal{R},\epsilon}} |F(w)| \le C_\epsilon \mathcal{R}^A \qquad (\mathcal{R} \to \infty) \tag{29.71}$$

for every fixed slit-opening regulator $\epsilon > 0$, with constants independent of $\mathcal{R}$. Choose distinct subtraction points $s_1, \ldots, s_q \in \Omega$, away from z and from the poles p_ν, with

$$q > \max\{A, N_R, N_L\}.$$

Define

$$B_q(w,z) = \prod_{j=1}^{q} \frac{z-s_j}{w-s_j}, \qquad L_j(z) = \prod_{k\neq j} \frac{z-s_k}{s_j-s_k}. \tag{29.72}$$

Then, for $z \in \Omega$ not equal to a pole,

$$\begin{aligned} F(z) = P_{q-1}(z) &+ \sum_{\nu} \frac{A_\nu}{p_\nu - z} B_q(p_\nu, z) \\ &+ \frac{1}{2\pi \mathrm{i}} \left\langle T_R(\sigma), \frac{B_q(\sigma,z)}{\sigma - z} \right\rangle_{C_R} + \frac{1}{2\pi \mathrm{i}} \left\langle T_L(\sigma), \frac{B_q(\sigma,z)}{\sigma - z} \right\rangle_{C_L}, \end{aligned} \tag{29.73}$$

where

$$P_{q-1}(z) = \sum_{j=1}^{q} F(s_j) L_j(z) \tag{29.74}$$

is the degree at most $q-1$ interpolation polynomial at the subtraction points. Endpoint singularities at a or b, if present as boundary distributions, are included in T_R or T_L. Thus no separate "threshold circle" term is present once the discontinuities are understood distributionally.

Proof. For fixed $z \in \Omega$, consider

$$\Phi(w;z) = \frac{F(w)}{w-z} B_q(w,z). \tag{29.75}$$

Write

$$K_z(w) = \frac{B_q(w,z)}{w-z}. \tag{29.76}$$

On a large circle $|w| = \mathcal{R}$, away from the slit regulators,

$$|\Phi(w;z)| \leq C_z \mathcal{R}^A \mathcal{R}^{-1} \mathcal{R}^{-q} = C_z \mathcal{R}^{A-q-1}.$$

By (29.67), the large circular part has length $O_\epsilon(\mathcal{R})$, hence its contribution is $O_\epsilon(\mathcal{R}^{A-q})$, which tends to zero because $q > A$. The large-end connector arcs in Definition 29.18 have $O_\epsilon(1)$ total length in the same annulus, so their contribution is $O_\epsilon(\mathcal{R}^{A-q-1})$ and also tends to zero.

The residues of Φ inside the cut-plane contour are the residue at $w = z$, the residues at the subtraction points s_j, and the residues at the poles p_ν of F. They are

$$\begin{aligned} \operatorname*{Res}_{w=z} \Phi(w;z) &= F(z), \\ \operatorname*{Res}_{w=s_j} \Phi(w;z) &= -F(s_j) L_j(z), \end{aligned} \tag{29.77}$$

and, using the pole convention (29.70),

$$\operatorname*{Res}_{w=p_\nu} \Phi(w;z) = -\frac{A_\nu}{p_\nu - z} B_q(p_\nu, z). \tag{29.78}$$

The admissible boundary has the standard upper-minus-lower orientation on each cut by (29.68) and (29.69). For fixed $(\mathcal{R}, \epsilon)$, before taking boundary values, the lip part is

$$\begin{aligned} I^{\mathrm{lip}}_{\mathcal{R},\epsilon}(z) = \frac{1}{2\pi \mathrm{i}} &\int_{a+\epsilon}^{\mathcal{R}} \mathrm{d}\sigma \, \left(F(\sigma + \mathrm{i}\epsilon) K_z(\sigma + \mathrm{i}\epsilon) - F(\sigma - \mathrm{i}\epsilon) K_z(\sigma - \mathrm{i}\epsilon)\right) \\ &+ \frac{1}{2\pi \mathrm{i}} \int_{-\mathcal{R}}^{b-\epsilon} \mathrm{d}\sigma \, \left(F(\sigma + \mathrm{i}\epsilon) K_z(\sigma + \mathrm{i}\epsilon) - F(\sigma - \mathrm{i}\epsilon) K_z(\sigma - \mathrm{i}\epsilon)\right) \end{aligned} \tag{29.79}$$

up to the connector arcs already described. After the limits $\mathcal{R} \to \infty$ and $\epsilon \downarrow 0$, this lip contribution becomes

$$\frac{1}{2\pi \mathrm{i}} \left[\int_{C_R} \mathrm{d}\sigma \, \frac{F(\sigma + \mathrm{i}0) - F(\sigma - \mathrm{i}0)}{\sigma - z} B_q(\sigma, z) + \int_{C_L} \mathrm{d}\sigma \, \frac{F(\sigma + \mathrm{i}0) - F(\sigma - \mathrm{i}0)}{\sigma - z} B_q(\sigma, z) \right]. \tag{29.80}$$

The finite-endpoint connector arcs are small circles around a and b. They are not discarded pointwise; their limits are part of the distributional boundary values at threshold. Thus any threshold-supported terms produced by endpoint singularities are represented inside T_R or T_L. The residue theorem applied before the limits says that the finite lip integral plus connector and large-contour terms equals the sum of the residues just displayed. Passing to the limits removes the large terms and absorbs the finite-endpoint limits into T_R, T_L. Solving for $F(z)$ gives (29.73).

It remains to pass from lip integrals to distributional boundary values and to justify the pairings. On C_R, for large σ,

$$\frac{B_q(\sigma, z)}{\sigma - z} = O(|\sigma|^{-q-1}),$$

and every σ-derivative has at least the same leading decay improved by one further power, up to constants depending on z and the subtraction points. The same estimate holds on C_L. Because q is larger than the tempered weight orders of T_R and T_L, the two kernels define Schwartz test functions on the cut coordinates after a smooth extension across the finite endpoints. Polynomial boundary growth is exactly the hypothesis that the lip values of F converge to the tempered boundary distributions $F(\sigma \pm \mathrm{i}0)$. Testing those convergent boundary values against the displayed kernels gives the distributional limit of (29.80); endpoint contributions, if any, are included in the same distributional limit. This limit is precisely the pairing displayed in (29.73). □

The domain theorem to be proved is therefore not the elementary estimate above, but the construction of its hypotheses from the Wightman data. Its logical pieces are: the primitive Wightman tube theorem from the spectrum condition; local commutativity at Jost configurations and the edge-of-the-wedge enlargement of the primitive tubes; the Jost–Lehmann–Dyson representation for the relevant retarded commutator matrix element; the Bros–Epstein–Glaser enlargement that gives the fixed-t Lehmann–Martin domain; and the verification that this off-shell domain satisfies the normal-crossing hypotheses of Theorem 29.17. Once those hypotheses hold, that theorem carries the domain and tempered-discontinuity bounds to the on-shell amplitude after stable-particle pole terms have been separated. The first two QFT-specific steps are proved in Theorems 28.2 and 28.3; the causal-commutator spectral coincidence regions are proved in Proposition 28.16; and the Jost–Lehmann–Dyson input is proved for Dyson-regular commutator distributions in Theorem 28.15 and Remark 28.17. The microlocal meaning and obstruction of the Dyson light-cone lift are isolated in Proposition 28.12: away from the light-cone vertex the ordinary product construction is valid under an explicit wavefront transversality condition, and any extension ambiguity is a finite contact polynomial after Fourier transform. The local finite-scaling-degree extension mechanism is proved in Theorem 28.13; it states the additional cone-equation, pushforward-normalization, temperedness, and Cauchy-data conditions that must hold before the source-current commutator is Dyson-regular modulo contact terms. Proposition 28.14 proves that the LSZ source-current differentials themselves preserve causal support, the punctured conormal wavefront avoidance, and finite scaling degree, with the correct relative-coordinate factors. The remaining steps are the Dyson-regularity verification for the actual LSZ source-current matrix coefficients and the Bros–Epstein–Glaser fixed-t enlargement with the uniform normal-coordinate control required for Theorem 29.17, together with the large-contour polynomial estimate needed in Theorem 29.19. These are needed before the Jin–Martin polynomial bound and the corresponding finite-subtraction dispersion formula can be promoted from conditional inputs to derived scattering theorems. Complex-analysis results such as the edge-of-the-wedge theorem are tools in this construction; they do not replace the field-theoretic support, locality, and LSZ arguments.

Example 29.20 (Tempered distribution with an unbounded representative)**.** There is a smooth nonnegative function $f \in L^1(\mathbb{R})$, hence a tempered distribution by

$$\langle T_f, \varphi \rangle = \int_{\mathbb{R}} f(s)\varphi(s)\,\mathrm{d}s,$$

such that $f(n) \geq e^{n^2}$ for all positive integers n. Consequently, no pointwise polynomial bound for a chosen boundary-value representative can be inferred from temperedness alone.

Choose $\psi \in C_c^\infty((-1/4, 1/4))$ with $\psi \geq 0$ and $\psi(0) = 1$. Define

$$f(s) = \sum_{n=1}^{\infty} e^{n^2} \psi\Big(e^{3n^2}(s-n)\Big).$$

The supports are contained in intervals

$$\left(n - \frac{1}{4}e^{-3n^2},\, n + \frac{1}{4}e^{-3n^2}\right),$$

which are pairwise disjoint. On every compact subset of $\mathbb{R}$, only finitely many summands are present, so f is smooth. Moreover

$$\int_{\mathbb{R}} f(s)\,\mathrm{d}s = \sum_{n=1}^{\infty} e^{n^2} e^{-3n^2} \int_{\mathbb{R}} \psi(u)\,\mathrm{d}u = \left(\int_{\mathbb{R}} \psi(u)\,\mathrm{d}u\right) \sum_{n=1}^{\infty} e^{-2n^2} < \infty.$$

Thus $f \in L^1(\mathbb{R})$, and

$$|\langle T_f, \varphi \rangle| \leq \|f\|_{L^1} \sup_s |\varphi(s)| \leq \|f\|_{L^1} p_{0,0}(\varphi),$$

so $T_f \in \mathcal{S}'(\mathbb{R})$. At the center of the n-th spike,

$$f(n) \geq e^{n^2}\psi(0) = e^{n^2}.$$

This exceeds $C(1+n)^N$ for every fixed C, N. Hence distributional temperedness is compatible with super-polynomial pointwise values of a smooth representative.

Remark 29.14 is the polynomial boundedness that follows immediately from temperedness. It says that smooth high-energy packet matrix elements with fixed energy resolution cannot grow faster than a power of the central energy. It does not by itself give a pointwise bound for a chosen boundary-value representative, nor does it control the large semicircle in a complex s-plane Cauchy deformation. The obstruction is substantive: Example 29.20 shows that the distributional statement is logically too weak even for smooth functions. Those stronger statements use additional analyticity and boundary-regularity input. The Jin–Martin fixed-t theorem is precisely such an additional theorem-level upgrade in the massive Wightman setting: it turns the locality, spectral, edge-of-the-wedge, Lehmann-domain, and LSZ hypotheses into a pointwise cut-plane polynomial estimate at fixed $t < 0$. The angular-tube estimate at positive $t < t_0$, used in the Froissart comparison above, is a still different boundedness statement and must be supplied separately when it is used.

The polynomial boundedness assumption has a causal interpretation. A one-dimensional model isolates the relevant analytic structure. Let

$$f_{\mathrm{out}}(t) = \int \mathrm{d}t'\, K(t-t') f_{\mathrm{in}}(t'), \qquad \widetilde{K}(\omega) = \int \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} K(t).$$

If the kernel has retarded support,

$$K(t) = 0 \qquad (t < 0),$$

then $\widetilde{K}(\omega)$ is analytic for $\mathrm{Im}\,\omega > 0$. Moreover a growth like $\exp(c\,\mathrm{Im}\,\omega)$ in the upper half plane would correspond to support at negative time. For relativistic scattering the hypothesis is formulated directly for the invariant amplitude: locality and causal propagation are reflected in analyticity and in the absence of exponential growth in complex energy directions.

Remark 29.21 (Jin–Martin fixed-t theorem)*.* There is a theorem-level fixed-t result in the Wightman framework. The Lehmann ellipse analysis, its Bros–Epstein–Glaser extension, and the Jin–Martin theorem imply the following kind of statement for massive two-particle scattering: for each fixed real $t < 0$ in the fixed-t analyticity domain, there are constants $C_t > 0$, $R_t > 0$, and an integer N_t such that the first-sheet amplitude in the cut s-plane obeys

$$|\mathcal{M}(s,t)| \le C_t(1+|s|)^{N_t}, \qquad |s| > R_t,$$

on the contours used for the fixed-t dispersion relation. This is the derivable polynomial boundedness input: it gives a finite number of subtractions at fixed negative t. It is narrower than the additional boundedness used in the Froissart–Martin mechanism at nonphysical positive t inside the Lehmann–Martin ellipse, and narrower than a simultaneous Mandelstam-domain polynomial bound. Those stronger bounds are stated as hypotheses whenever they are used below.

For the invariant amplitude, fix real $t < 0$ and continue s to the complex plane. Theorem 29.19 is the precise one-variable Cauchy statement behind the following physics notation. The displayed dispersion formula records the cut terms in a kinematic situation where bound-state pole terms and anomalous first-sheet terms are absent; when such terms are present, their residues are added explicitly. More concretely, if the first sheet contains simple stable-particle poles with local behavior

$$\mathcal{M}(s,t) \sim \frac{R_B^{(s)}(t)}{M_B^2 - s} \qquad \text{or} \qquad \mathcal{M}(s,t) \sim \frac{R_B^{(u)}(t)}{4m^2 - t - M_B^2 - s},$$

where $R_B^{(u)}(t)$ is defined with respect to the s-plane local coordinate $4m^2 - t - M_B^2 - s$. Equivalently, it includes the sign relating this coordinate to $M_B^2 - u$. Then the subtracted dispersion relation contains the corresponding Cauchy residue terms

$$\frac{R_B^{(s)}(t)}{M_B^2 - s} \prod_{j=1}^{N} \frac{s - s_j}{M_B^2 - s_j}, \qquad \frac{R_B^{(u)}(t)}{4m^2 - t - M_B^2 - s} \prod_{j=1}^{N} \frac{s - s_j}{4m^2 - t - M_B^2 - s_j}.$$

For a composite stable state the functions $R_B^{(s)}(t)$ and $R_B^{(u)}(t)$ are the crossed forms of the four-point pole residue described in Theorem 27.2; the homogeneous Bethe–Salpeter equation fixes the corresponding pole location when the two-constituent description is valid. Assume that for some integer N,

$$\lim_{|s|\to\infty}\left|\frac{\mathcal{M}(s,t)}{s^N}\right|=0$$

in the domain used to deform the contour. Choose subtraction points

$$s_1,\ldots,s_N$$

away from the cuts and from the evaluation point s. Then

$$\frac{\mathcal{M}(s',t)}{\prod_{j=1}^{N}(s'-s_j)}$$

has sufficient decay for the large circle to be removed in Cauchy's theorem. Equivalently,

$$\mathcal{M}(s,t)=\oint_\gamma\frac{\mathrm{d}s'}{2\pi\mathrm{i}}\,\frac{\mathcal{M}(s',t)}{s'-s}\prod_{j=1}^{N}\frac{s-s_j}{s'-s_j},$$

where γ encloses s and the subtraction points before deformation.

Figure 29.6 shows the contour deformation used in the subtracted fixed-t dispersion formula.

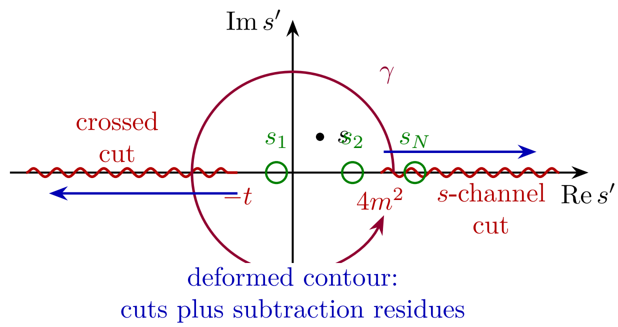


Figure 29.6: Subtracted fixed-t Cauchy contour. The original contour γ encloses the evaluation point and subtraction points; after deformation it is replaced by integrals along the right- and left-hand cuts, with the subtraction factors suppressing the large-circle contribution.

Let

$$\operatorname{disc}\mathcal{M}(s,t)=\mathcal{M}(s+\mathrm{i}0,t)-\mathcal{M}(s-\mathrm{i}0,t).$$

After deforming γ onto the right and left cuts, one obtains

$$\mathcal{M}(s,t)=P_{N-1}(s;t)+\frac{1}{2\pi\mathrm{i}}\int_{C_s\cup C_u}\mathrm{d}s'\,\operatorname{disc}\mathcal{M}(s',t)\,\frac{1}{s'-s}\prod_{j=1}^{N}\frac{s-s_j}{s'-s_j}.$$

Here $C_s=[4m^2,\infty)$ is the right cut, $C_u=(-\infty,-t]$ is the crossed cut in the s'-plane, and $P_{N-1}(s;t)$ is the polynomial of degree at most $N-1$ produced by the small circles around the subtraction points. On a real cut where real analyticity holds,

$$\operatorname{disc}\mathcal{M}(s,t)=2\mathrm{i}\,\operatorname{Im}\mathcal{M}(s+\mathrm{i}0,t).$$

The formula is an N-subtracted fixed-t dispersion relation.

Equivalently, parametrize the left cut by the positive crossed-channel variable

$$u' = 4m^2 - s' - t, \qquad u = 4m^2 - s - t, \qquad u_j = 4m^2 - s_j - t.$$

For an amplitude whose crossing convention identifies the left-cut discontinuity with the absorptive part in the u-channel, the same dispersion relation may be written as

$$\mathcal{M}(s,t) = P_{N-1}(s;t) + \int_{4m^2}^{\infty} \frac{\mathrm{d}s'}{\pi} \frac{\operatorname{Im}\mathcal{M}(s'+\mathrm{i}0,t)}{s'-s} \prod_{j=1}^{N} \frac{s-s_j}{s'-s_j}$$
$$+ \int_{4m^2}^{\infty} \frac{\mathrm{d}u'}{\pi} \frac{\operatorname{Im}\mathcal{M}_u(u'+\mathrm{i}0,t)}{u'-u} \prod_{j=1}^{N} \frac{u-u_j}{u'-u_j}.$$

Here $\mathcal{M}_u$ denotes the amplitude in the crossed channel, with external labels crossed as appropriate. For identical scalar scattering $\mathcal{M}_u$ is the same analytic function with s replaced by u; for general external species this displayed formula must be read with the corresponding crossed process.

29.8 Two Subtractions

The previous derivation used a complex-plane polynomial bound. A useful refinement is that a real-axis bound on the absorptive part controls the number of subtractions. The form needed here is the following conditional statement. Fix $t < 0$, assume a fixed-t dispersion representation with some finite number of subtractions exists, and suppose

$$|\operatorname{Im}\mathcal{M}(s,t)| \le C s^{N-\epsilon} \qquad (s \to +\infty)$$

for some $\epsilon > 0$, uniformly for t in a small real interval, with the same power bound also available at the forward endpoint $t = 0$. If a larger subtraction number $N' \ge N+1$ were genuinely required, the subtraction polynomial would contain a nonzero term

$$C_{N'-1}(t) s^{N'-1}.$$

If the coefficient $C_{N'-1}(t)$ is not identically zero, analyticity in t gives an interval of real $t < 0$ on which its magnitude is bounded below. On the physical interval for elastic scattering,

$$-(s - 4m^2) \le t \le 0,$$

write

$$k^2 = \frac{s - 4m^2}{4}, \qquad t = 2k^2(x-1).$$

Unitarity gives a positive estimate for the forward absorptive part:

$$\operatorname{Im}\mathcal{M}(s,0) = \sqrt{s(s-4m^2)}\,\sigma_{\mathrm{tot}}(s) \ge \sqrt{s(s-4m^2)}\,\sigma_{\mathrm{el}}(s).$$

The elastic term is then evaluated in the same partial-wave normalization:

$$\begin{aligned}\sqrt{s(s-4m^2)}\,\sigma_{\rm el}(s) &= 8\pi\sqrt{\frac{s}{s-4m^2}}\sum_{\substack{\ell=0\\ \ell\ \text{even}}}^{\infty}(2\ell+1)|S_\ell(s)-1|^2\\ &= \frac{1}{64\pi}\sqrt{\frac{s-4m^2}{s}}\int_{-1}^{1}\mathrm{d}x\,|\mathcal{M}(s,2k^2(x-1))|^2\\ &= \frac{1}{32\pi\sqrt{s(s-4m^2)}}\int_{-(s-4m^2)}^{0}\mathrm{d}t\,|\mathcal{M}(s,t)|^2.\end{aligned}$$

Consequently,

$$\operatorname{Im}\mathcal{M}(s,0)\geq\frac{1}{32\pi\sqrt{s(s-4m^2)}}\int_{-(s-4m^2)}^{0}\mathrm{d}t\,|\mathcal{M}(s,t)|^2.$$

The last equality in the elastic chain is the change of variables from x to t. If the nonzero polynomial coefficient $C_{N'-1}(t)$ is bounded below on a fixed real t-interval inside $(-\infty,0)$, this estimate gives, along a high-energy sequence,

$$\operatorname{Im}\mathcal{M}(s,0)\geq c\,s^{2N'-3}$$

for some $c>0$. This fixed-t version is the only estimate needed here: since $N'\geq N+1$, it contradicts the assumed bound $|\operatorname{Im}\mathcal{M}(s,0)|\leq Cs^{N-\epsilon}$ at the forward point. A stronger estimate is obtained if the leading behavior persists on a subinterval of scattering angles of nonzero length, because the corresponding t-interval has length of order s; the fixed-t estimate avoids assuming such uniformity. Therefore the real-axis absorptive bound fixes the subtraction number, apart from polynomial terms whose leading coefficients vanish identically and hence were not genuine subtractions.

The preceding Froissart mechanism, applied with any finite initial polynomial bound, gives at forward scattering

$$\operatorname{Im}\mathcal{M}(s,0)\leq C\,s(\log s)^2.$$

For fixed real $t<0$ and sufficiently large s, the corresponding angular variable satisfies $-1<x<1$. In this range $|P_\ell(x)|\leq 1$, while $P_\ell(1)=1$. Using the positive weights w_ℓ,

$$|\operatorname{Im}\mathcal{M}(s,t)|\leq\operatorname{Im}\mathcal{M}(s,0)$$

for the absorptive partial-wave sum. Thus, for every $0<\epsilon<1$,

$$|\operatorname{Im}\mathcal{M}(s,t)|\leq C_\epsilon s^{2-\epsilon}\qquad(s\to+\infty)$$

at fixed real $t<0$. Applying the preceding conditional subtraction-count argument with $N=2$ rules out genuine subtraction terms of degree two or higher. Under the finite-subtraction hypothesis stated at the beginning of this section, two subtractions are therefore sufficient in the massive scalar setting, and the twice-subtracted representation has $\mathcal{M}(s,t)/s^2\to 0$ at fixed real $t<0$ in the dispersion domain.

This conclusion is conditional in exactly the sense stated: the argument starts from the existence of a fixed-t dispersion representation with some finite subtraction number. It is not a proof, by itself, that the axioms of massive local QFT imply a two-subtracted representation without first knowing that a finite-subtracted representation exists.

Remark 29.22 (Two-subtraction theorem). There is a stronger theorem-level input in the axiomatic analyticity-unitarity program: for massive two-particle scattering satisfying the Wightman hypotheses and the usual mass-gap assumptions used in the Lehmann–Martin analysis, fixed-t dispersion theory can be brought to the two-subtraction form without assuming a finite number of subtractions as an independent premise.[1] The conditional proof above supplies a useful internal mechanism: once a finite subtraction number is available, positivity and the Froissart absorptive bound remove all genuine higher subtraction coefficients. The cited theorem supplies the additional axiomatic step. In later uses, these two inputs must not be conflated.

The extension from real $t < 0$ to complex t inside the disk set by the nearest t-plane singularity uses positivity of the Taylor coefficients of the absorptive part at $t = 0$:

$$\frac{\partial^n}{\partial t^n} \operatorname{Im} \mathcal{M}(s,t)\bigg|_{t=0} \geq 0.$$

This positivity follows from the same partial-wave expansion: derivatives of $P_\ell(1 + 2t/(s - 4m^2))$ at $t = 0$ are nonnegative, since

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} P_\ell(x)\bigg|_{x=1} = \frac{(\ell+n)!}{2^n n!(\ell-n)!} \geq 0$$

for $n \leq \ell$, and the derivative is zero for $n > \ell$. This positivity allows the Taylor expansion in t to commute with the dispersive integral inside its convergence disk.

Hypothesis 29.23 (Higher-dimensional angular tube)**.** Let $D \geq 4$ be the spacetime dimension and set

$$\lambda = \frac{D-3}{2}.$$

For scalar $2 \to 2$ scattering in the center-of-mass frame, the angular variable is a point on S^{D-2}. For an $SO(D-1)$-invariant scalar amplitude this dependence is through $x = \cos\theta$, and the partial-wave basis is the Gegenbauer basis $C_\ell^\lambda(x)$. The degeneracy of degree-ℓ spherical harmonics on S^{D-2} is

$$g_\ell^{(D)} = \frac{(2\ell + D - 3)(\ell + D - 4)!}{\ell!(D-3)!} = O(\ell^{D-3}).$$

Assume that the D-dimensional analogue of the Lehmann ellipse is available in the following form. For every $0 < t_* < t_0$ and all sufficiently large s, the amplitude has an $SO(D-1)$-invariant holomorphic continuation to a complex angular tube containing the point

$$x_*(s) = 1 + \frac{2t_*}{s - 4m^2} = \cosh \eta_*(s), \qquad \eta_*(s) = 2\sqrt{\frac{t_*}{s}} + O(s^{-3/2}),$$

and its Gegenbauer coefficients obey the Cauchy-type large-ℓ estimate

$$|a_\ell(s)| \leq A\, s^N\, \ell^q\, \mathrm{e}^{-\ell\eta_*(s)}$$

for constants A, N, q depending on the chosen tube margin but not on ℓ or large s. This hypothesis is the higher-dimensional replacement for the $D = 4$ Lehmann ellipse and Legendre-coefficient estimate; it is an additional angular-domain input beyond the displayed four-dimensional ellipse.

[1] See L. Łukaszuk and A. Martin, "Absolute upper bounds for $\pi\pi$ scattering," *Nuovo Cimento A* **52** (1967) 122–145, doi:10.1007/BF02739279; and F. J. Ynduráin, "Rigorous constraints, bounds, and relations for scattering amplitudes," *Rev. Mod. Phys.* **44** (1972) 645–667, doi:10.1103/RevModPhys.44.645.

Proposition 29.24 (Conditional angular counting in D spacetime dimensions)**.** *Assume Hypothesis 29.23, partial-wave unitarity, a two-subtraction input giving the required polynomial bound at fixed nonphysical* $0 < t_* < t_0$*, and the usual massive scalar kinematics. Then*

$$\sigma_{\text{tot}}(s) \le C_D(\log s)^{D-2}$$

for sufficiently large s*, with* C_D *depending on the normalization,* t_0*, and the angular-tube constants.*

Proof. With the normalization convention absorbed into $\mathcal{N}_D(s)$, the D-dimensional scalar partial-wave expansion has the form

$$\mathcal{M}_D(s,x) = \mathcal{N}_D(s)\sum_{\ell=0}^{\infty} g_\ell^{(D)} a_\ell(s)\frac{C_\ell^\lambda(x)}{C_\ell^\lambda(1)},$$

with only even ℓ in the identical-boson channel. Omitting the odd partial waves changes constants but not powers. Unitarity bounds the physical weights $w_\ell(s) = 1 - \operatorname{Re} S_\ell(s)$ by a constant independent of s and ℓ. The total cross section has the high-energy form

$$\sigma_{\text{tot}}(s) \le C'_D\, s^{-(D-2)/2}\sum_{\ell\ge 0} g_\ell^{(D)} w_\ell(s),$$

where C'_D records the normalization convention and the $(1+O(m^2/s))$ phase-space factor.

The angular-tube estimate gives exponential suppression of partial-wave coefficients up to powers of ℓ. More explicitly, after changing the constants to match the optical-theorem normalization, the physical absorptive weights obey a bound of the form

$$w_\ell(s) \le B\min\{1,\, s^N(1+\ell)^q \mathrm{e}^{-\ell\eta_*(s)}\}, \tag{29.81}$$

where the first entry is unitarity and the second entry is the Cauchy estimate in Hypothesis 29.23. Put

$$\eta = \eta_*(s), \qquad L(s) = \alpha\,\eta^{-1}\log s,$$

with α larger than $N + (D-2+q)/2$. Since $\eta_*(s) = 2\sqrt{t_*/s} + O(s^{-3/2})$, this is the same scale as $C\sqrt{s}\log s$. The contribution below L is controlled only by unitarity:

$$\sum_{\ell\le L(s)} g_\ell^{(D)} w_\ell(s) \le C\sum_{\ell\le L(s)}(1+\ell)^{D-3} = O(L(s)^{D-2}).$$

For the tail, set $r = D-3+q$. The same two estimates give

$$\sum_{\ell>L(s)} g_\ell^{(D)} w_\ell(s) \le C\, s^N \sum_{\ell>L(s)}(1+\ell)^r \mathrm{e}^{-\ell\eta}.$$

For large s, $L(s)\eta = \alpha\log s$ and $\eta \to 0$. Splitting the sum into unit intervals, or equivalently using a geometric comparison after observing that $(1+\ell+1)^r\mathrm{e}^{-(\ell+1)\eta}/((1+\ell)^r\mathrm{e}^{-\ell\eta}) \le \mathrm{e}^{-\eta/2}$ for $\ell \ge L(s)$, gives

$$\sum_{\ell>L(s)}(1+\ell)^r\mathrm{e}^{-\ell\eta} \le C_r\,(1+L(s))^r\,\mathrm{e}^{-L(s)\eta}\,(1-\mathrm{e}^{-\eta})^{-1}.$$

Thus

$$\sum_{\ell>L(s)} g_\ell^{(D)} w_\ell(s) \le C\, s^{N-\alpha}\,\eta^{-(r+1)}(\log s)^r = O\Big(s^{N-\alpha+(D-2+q)/2}(\log s)^{D-3+q}\Big).$$

The choice of α makes this tail $O(1)$, hence subordinate to the leading angular count. Combining the low-angular-momentum estimate with the tail estimate gives

$$\sum_{\ell \geq 0} g_\ell^{(D)} w_\ell(s) = O(L(s)^{D-2}),$$

because $g_\ell^{(D)} = O(\ell^{D-3})$. Substituting this into the preceding cross-section estimate gives

$$\sigma_{\text{tot}}(s) \leq C''_D s^{-(D-2)/2} \big(\sqrt{s} \log s\big)^{D-2} = C_D (\log s)^{D-2}.$$

□

Theorem 29.25 (Froissart–Martin high-energy bound)**.** *Consider massive $2 \to 2$ scalar scattering with mass gap and no massless exchange. Assume:*

1. *partial-wave unitarity in the physical s-channel;*
2. *analyticity in the Lehmann–Martin ellipse with nearest t-channel singularity at $t = t_0 > 0$;*
3. *crossing symmetry and a fixed-t dispersion representation;*
4. *a two-subtraction input, obtained either from the conditional finite-subtraction hypothesis together with the real-axis absorptive bound used above, or from the Łukaszuk–Martin/Ynduŕáin theorem as an additional axiomatic input;*
5. *the angular-tube polynomial bound of Definition 29.13 at $0 < t < t_0$, with the exponent $N = 2$ supplied by the two-subtraction input.*

Then, in four spacetime dimensions, for every fixed $0 < \delta < 1$ there are constants M_0 and s_0 such that

$$\sigma_{\text{tot}}(s) \leq \frac{1}{(1-\delta)^2} \frac{4\pi}{t_0} \left(\log \frac{s}{M_0^2} \right)^2, \qquad s \geq s_0. \tag{29.82}$$

In D spacetime dimensions, after replacing the $D = 4$ Lehmann-ellipse input by Hypothesis 29.23, the corresponding conditional estimate is

$$\sigma_{\text{tot}}(s) \leq C\,(\log s)^{D-2}. \tag{29.83}$$

Proof. Fix $0 < t_* < t_0$. To prove the theorem with the displayed margin δ, choose an auxiliary margin δ_0 with

$$0 < \delta_0 < \delta.$$

The Legendre lower bound below will use δ_0, leaving the remaining margin $\delta - \delta_0$ to absorb constants and lower-order logarithmic terms without changing the stated coefficient. By the Lehmann–Martin angular-domain hypothesis, the point

$$x_*(s) = 1 + \frac{2t_*}{s - 4m^2}$$

lies inside the allowed angular domain for all sufficiently large s. The angular-tube polynomial bound with exponent N gives

$$|\mathcal{M}(s, x_*(s))| \leq C_N s^N. \tag{29.84}$$

The absorptive part at this point is nonnegative in the partial-wave representation used above, so $\operatorname{Im}\mathcal{M}(s, x_*(s)) \le |\mathcal{M}(s, x_*(s))|$.

Let

$$W(s) := \sum_{\ell \text{ even}} (2\ell+1) w_\ell(s).$$

The optical-theorem normalization derived in Section 29.3 gives

$$W(s) = \frac{s-4m^2}{16\pi}\,\sigma_{\text{tot}}(s).$$

The box-profile minimization in the Froissart comparison chooses

$$L = W(s)^{1/2} + O(1) = \left(\frac{s-4m^2}{16\pi}\sigma_{\text{tot}}(s)\right)^{1/2} + O(1). \tag{29.85}$$

For $x_*(s) = \cosh\eta_*(s)$, the Legendre lower bound with margin δ_0 gives the base $y = 1 + (1-\delta_0)\sinh\eta_*(s)$. The same expansion as in (29.47), with δ replaced by δ_0, gives

$$y - 1 = 2(1-\delta_0)\sqrt{\frac{t_*}{s}} + O(s^{-1}). \tag{29.86}$$

Combining this with (29.85) yields

$$L(y-1) = (1-\delta_0)\sqrt{\frac{t_*\,\sigma_{\text{tot}}(s)}{4\pi}} + o(1). \tag{29.87}$$

The factor 4π in this expression is the coefficient produced by the partial-wave normalization $W = (s-4m^2)\sigma_{\text{tot}}/(16\pi)$ together with the angular-distance estimate $\sinh\eta_*(s) \sim 2\sqrt{t_*/s}$.

If $L(y-1) < 1$ along a high-energy subsequence, then (29.87) already gives

$$\sigma_{\text{tot}}(s) \le \frac{4\pi}{(1-\delta_0)^2 t_*} + o(1),$$

which is stronger than the logarithmic-square estimate for large s. We therefore consider the case $L(y-1) \ge 1$, where (29.46) applies. Using (29.86), the algebraic prefactor in (29.46) is bounded below by a fixed positive constant times $s\,\sigma_{\text{tot}}(s)^{1/2}$. Hence (29.84) can hold only if, for every $\varepsilon > 0$ and all sufficiently large s,

$$(1-\delta_0)\sqrt{\frac{t_*\,\sigma_{\text{tot}}(s)}{4\pi}} \le (N-1+\varepsilon)\log\frac{s}{M_0^2}. \tag{29.88}$$

Indeed, if the left side exceeded the right side along a high-energy subsequence, the exponential y^L in (29.46) would grow faster than $(s/M_0^2)^{N-1+\varepsilon}$; multiplying by the prefactor of order $s\,\sigma_{\text{tot}}^{1/2}$ would contradict (29.84).

Solving (29.88) for σ_{tot} gives

$$\sigma_{\text{tot}}(s) \le \frac{(N-1+\varepsilon)^2}{(1-\delta_0)^2}\frac{4\pi}{t_*}\left(\log\frac{s}{M_0^2}\right)^2.$$

Since $\delta_0 < \delta$, choose $\varepsilon > 0$ small enough that

$$\frac{N-1+\varepsilon}{1-\delta_0} \leq \frac{N-1}{1-\delta}.$$

This proves the same estimate with the displayed $(N-1)^2(1-\delta)^{-2}$ coefficient after increasing s_0. The two-subtraction input in the theorem supplies $N = 2$. Finally take $t_* \to t_0^-$, after the estimate has been established at fixed $t_* < t_0$, to obtain (29.82). The factor $(1-\delta)^{-2}$ is therefore the square of the final reserved margin in the Legendre lower bound, and $4\pi/t_0$ is the coefficient obtained from the optical-theorem normalization and the nearest crossed-channel angular distance.

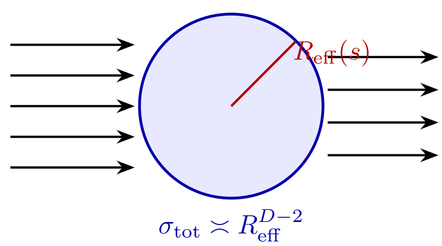
The displayed D-dimensional estimate is the separate conditional Gegenbauer/angular-counting result of Proposition 29.24. □

Equation (29.83) may be restated through an effective transverse radius whenever the higher-dimensional angular-tube hypothesis has been assumed. If one defines $R_{\text{eff}}(s)$ by the geometric scaling relation

$$\sigma_{\text{tot}}(s) \asymp R_{\text{eff}}(s)^{D-2},$$

then the massive high-energy bound implies $R_{\text{eff}}(s) = O(\log s)$. This is only a translation of the cross-section estimate, not an additional dynamical assumption.

$R_{\text{eff}}(s) = O(\log s)$ under the stated massive hypotheses



29.9 Scope of the Bound

The bound is a theorem about amplitudes satisfying the stated massive analyticity, unitarity, crossing, and boundedness hypotheses. The mass gap is part of the statement. When massless particles are present, the total cross section may diverge because arbitrarily small scattering angles or collinear emissions are experimentally unresolved. The corresponding observable is then an inclusive or angularly cut cross section, while the quantity bounded above is the total cross section of the massive problem.

For a confining gauge theory, the external particles in such a statement are gauge-invariant one-particle states in the physical Hilbert space: glueballs in pure Yang–Mills theory, and mesons or baryons in theories with matter fields when the relevant stable isolated masses exist. The scalar particle used in the preceding derivation is therefore a stand-in for a chosen stable hadron or glueball created by a local gauge-invariant operator and normalized by the Haag–Ruelle–LSZ construction. The analyticity and unitarity statements then refer to the hadronic amplitude between these physical states. In four-dimensional pure Yang–Mills theory, the existence of a positive mass gap and of the corresponding massive local quantum field theory is a separate open construction problem; the present high-energy bound uses that gap as an input. For QCD-like theories with massless pions or massless partons, the massive hypotheses must be replaced by the appropriate inclusive or gapped-channel statement before a Froissart-type conclusion is asserted.

Gravity also lies outside the same massive-QFT hypothesis. In D spacetime dimensions, a black hole of energy E has Schwarzschild radius

$$r_H \sim (G_N E)^{1/(D-3)},$$

so the effective interaction size grows as a power of energy. This is a different high-energy problem governed by long-range gravitational dynamics rather than the massive Froissart–Martin hypotheses.

Part III

Renormalization, Effective Field Theory, and Critical Phenomena

Chapter 30

Generating Functionals and the One-Particle-Irreducible Effective Action

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Notation for Z-Factors in This Part

Several unrelated objects traditionally use the letter Z. The renormalization part keeps them distinct as follows.

symbol	meaning
$Z[J]$	source or partition functional. Its logarithm generates connected correlators after a regulator and a branch or formal logarithm have been specified. This Z is not a field-strength factor.
Z_ϕ^{pole}	one-particle Källén–Lehmann/LSZ residue of a chosen interpolating field: $\mathrm{d}\rho_\phi(s) = Z_\phi^{\mathrm{pole}}\delta_{m^2}(\mathrm{d}s) + \cdots$. Earlier scattering chapters often write this residue simply as Z_ϕ, or as Z when only one scalar field is present. Once the field normalization is fixed, it is a physical pole coefficient for that interpolating field and has no renormalization-scale argument.
$Z_{\mathrm{MOM}}(\mu)$	finite momentum-subtraction field rescaling at Euclidean subtraction scale μ, for example $[\phi]_\mu = Z_{\mathrm{MOM}}(\mu)^{-1/2}\phi_R^{\mathrm{ref}}$. It is a coordinate choice in the renormalized 1PI chart, not the Källén–Lehmann pole residue. Only after analytic continuation to an isolated physical pole and the imposition of an on-shell residue condition can a finite field factor be identified with the pole-residue normalization.
$Z_\phi^{\mathrm{chart}}(\lambda(\mu), \mu, \Lambda)$	total cutoff-dependent bare-to-renormalized field factor in a chosen renormalization chart. In the Callan–Symanzik equation this factor carries the anomalous dimension through $\gamma_\phi = \frac{1}{2}\,\mathrm{d}\log Z_\phi^{\mathrm{chart}}/\mathrm{d}\log\mu$ at fixed bare data.
$Z_\phi^{\mathrm{MS}}(\lambda_{\mathrm{MS}}, \epsilon)$	minimal-subtraction field renormalization constant. It contains pole counterterms in ϵ and no finite residue normalization; it is not the LSZ residue.
$\mathcal{Z}^I{}_{A_1\cdots A_r}$	operator/source mixing distribution for composite insertions. This is a local contact-term renormalization matrix and is unrelated to either the source functional $Z[J]$ or a one-particle pole residue.

When a later formula suppresses a superscript, the surrounding chart specifies which row of this table is meant. In particular, the notation Z_ϕ in LSZ statements means Z_ϕ^{pole}, while Z_ϕ inside a Callan–Symanzik derivation means the chart field factor Z_ϕ^{chart}.

30.1 Sources and Connected Functionals

Definition 30.1 (Regulated scalar source functional)**.** Fix a quantum field theory with a specified ultraviolet regulator and a specified prescription for Lorentzian correlation functions. Let ϕ denote a real scalar field in this section; the definitions extend to several bosonic fields by adding indices, and to fermionic fields by replacing ordinary sources with Grassmann sources. Since a quantum field is a distribution, a c-number source is a real test function, for instance a smooth compactly supported function or a Schwartz function,

$$J\colon \mathbb{R}^{1,D-1} \longrightarrow \mathbb{R}$$

coupled linearly to ϕ through the pairing

$$\langle J, \phi \rangle = \int \mathrm{d}^D x \, J(x) \phi(x).$$

At finite regulator this pairing is replaced by the corresponding finite sum, finite-mode pairing, or regulated distributional pairing. The regulated source functional is

$$Z[J] = \int D_\Lambda^{\mathrm{ref}} \phi \, \exp\{\mathrm{i} S_\Lambda[\phi] + \mathrm{i} \langle J, \phi \rangle\} \, .$$

The sign in the Lorentzian source exponent must not be compared directly with a Euclidean positive-measure convention until the source coordinate has been continued. With the lower-half-plane convention of Chapter 16, $x^0 = -\mathrm{i}\tau$ and $\mathrm{d}x^0 = -\mathrm{i}\mathrm{d}\tau$, the formal chain for scalar insertions is

$$\mathrm{i} S_L[\phi] + \mathrm{i} \langle J_L, \phi \rangle_L \longmapsto -S_E[\phi_E] + \langle J_L^{\mathrm{W}}, \phi_E \rangle_E, \qquad J_L^{\mathrm{W}}(\tau, \vec{x}) := J_L(-\mathrm{i}\tau, \vec{x}).$$

Thus a Euclidean source functional written with a plus source term uses $J_E^+ = J_L^{\mathrm{W}}$, while the positive-measure/free-energy convention used later in this chapter,

$$Z_E[J_E] = \int [D\phi] \, \exp\{-S_E[\phi] - \langle J_E, \phi \rangle_E\},$$

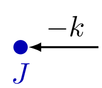
uses $J_E = -J_L^{\mathrm{W}}$. This is a change of source coordinate, not a new Green function. The derivative normalizations and the corresponding connected-log conventions are collected in Definition 30.11. With the Fourier convention

$$\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \mathrm{e}^{\mathrm{i}k \cdot x} \, \widetilde{\phi}(k),$$

the source pairing is

$$\langle J, \phi \rangle = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \widetilde{J}(k) \widetilde{\phi}(-k).$$

Perturbatively, the source is therefore a one-valent vertex with factor $\mathrm{i}\widetilde{J}(k)$:



$= \quad \mathrm{i}\widetilde{J}(k).$

Here $D_\Lambda^{\mathrm{ref}} \phi$ is the reference integration density of Definition 11.3 in a finite-dimensional bosonic scalar regulator, and S_Λ is the regulated action placed in the exponential. If a chapter instead uses a Gaussian reference measure, it writes $\mathrm{d}\mu_{C_\Lambda}$ and moves the quadratic kinetic term into that measure. Other regulators, such as fermionic, gauge, BV, Lorentzian oscillatory, or dimensional-regularization prescriptions, must specify their own regulated integration object. The notation suppresses Λ after this paragraph. All functional derivatives in this chapter are first derivatives of the regulated source functional; continuum statements are obtained only after the chosen renormalization prescription gives a distributional limit.

Definition 30.2 (Status of $\log Z$)**.** The symbol $\log Z[J]$ has one of the following meanings, fixed by the regulator and limiting prescription.

1. *Formal perturbation theory.* Let g denote the interaction couplings and work in a formal power-series ring in g, with coefficients that are regulated source distributions. If $Z[0]$ is invertible, write
$$Z[J] = Z[0]\big(1 + A[J]\big), \qquad A[0] = 0.$$
Then
$$\log Z[J] = \log Z[0] + \sum_{n\geq 1} \frac{(-1)^{n+1}}{n} A[J]^n$$
is the algebraic logarithm in that completed ring. The first term is a vacuum functional; source derivatives of positive order depend only on the second term. This is the interpretation used by dimensional regularization, minimal subtraction, and graph-by-graph perturbative constructions unless an actual positive measure or oscillatory integral has also been supplied.

2. *Positive Euclidean construction.* If $Z_E[J]$ is defined by a positive finite Euclidean measure on a real source domain and $0 < Z_E[J] < \infty$, then $\log Z_E[J]$ is the ordinary real logarithm. Convexity and Legendre–Fenchel statements in the later Euclidean section use this meaning and require the positivity and finiteness hypotheses stated there.

3. *Lorentzian oscillatory or phase construction.* If a finite Lorentzian regulator gives a complex-valued source functional $Z_L[J]$, then $\log Z_L[J]$ is defined only on a source neighborhood on which $Z_L[J] \neq 0$ and a branch has been chosen from a base source. A principal branch is a possible convention only when the image of that neighborhood avoids its branch cut. Changing the branch on a connected zero-free neighborhood changes $\log Z_L$ by an additive constant in $2\pi\mathrm{i}\mathbb{Z}$, hence changes $W_L = -\mathrm{i}\log Z_L$ by a source- independent constant and leaves connected correlators of positive order unchanged.

No statement in this chapter uses an unclassified logarithm. Every later Legendre transform or connected-kernel formula is to be read in the corresponding one of these three senses.

Definition 30.3 (Connected functional and source-dependent field)**.** For the source functional of Definition 30.1, define the Lorentzian connected functional W by
$$Z[J] = \mathrm{e}^{\mathrm{i}W[J]}, \qquad W[J] = -\mathrm{i}\log Z[J],$$
with the meaning of $\log Z$ supplied by Definition 30.2. In the Lorentzian complex-valued case this includes the choice of a zero-free source neighborhood and a logarithm branch on it; in the formal case it is the algebraic logarithm and no analytic branch is involved. The constant $W[0]$ is the vacuum functional; subtracting it removes vacuum bubbles and leaves all source derivatives of positive order unchanged. The functional $W[J]$, with this harmless constant suppressed, generates connected Green functions with the Lorentzian phase factor
$$G_{\mathrm{c}}^{(n)}(x_1, \ldots, x_n) = \mathrm{i}^{1-n} \frac{\delta^n W[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}.$$
The full Green functions are generated by $Z[J]$:
$$G^{(n)}(x_1, \ldots, x_n) = \frac{1}{Z[0]} \frac{\delta}{\mathrm{i}\delta J(x_1)} \cdots \frac{\delta}{\mathrm{i}\delta J(x_n)}\, Z[J]\bigg|_{J=0}.$$
The correlation-ordering prescription is the one built into the path integral. With the Feynman boundary prescription specified by the $+\mathrm{i}0$ deformation of quadratic denominators, these are time-ordered Lorentzian Green functions.

The source-dependent one-point function is

$$\varphi_J(x) = \frac{\delta W[J]}{\delta J(x)} = \frac{1}{Z[J]} \int D_\Lambda^{\mathrm{ref}}\phi\, \phi(x)\, \exp\{\mathrm{i}S_\Lambda[\phi] + \mathrm{i}\langle J, \phi\rangle\}\,.$$

At $J = 0$, this is the vacuum expectation value of $\phi(x)$ in the chosen state and prescription:

$$\varphi_0(x) = \langle \phi(x) \rangle.$$

Figure 30.1 records the source-to-field map used in the Legendre transform: differentiating $W_\Lambda[J]$ inserts one field, and solving for J as a function of φ defines the 1PI coordinate.

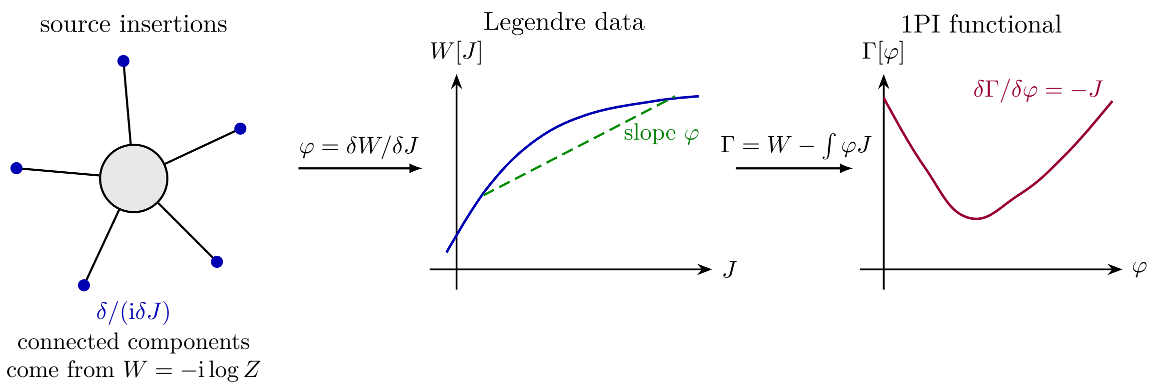


Figure 30.1: The logarithm of the source functional generates connected diagrams. The source-dependent field φ_J is the slope of $W[J]$, and the Legendre transform exchanges the source J for the field φ.

30.2 Legendre Transform

Definition 30.4 (Local one-particle-irreducible Legendre transform)**.** Assume that, in a neighborhood of the source configurations under discussion, the map

$$J \longmapsto \varphi_J = \frac{\delta W[J]}{\delta J}$$

is locally invertible. Denote the inverse by J_φ, so that

$$\varphi(x) = \left.\frac{\delta W[J]}{\delta J(x)}\right|_{J=J_\varphi}.$$

The one-particle-irreducible effective action is the Legendre transform

$$\Gamma[\varphi] = W[J_\varphi] - \langle J_\varphi, \varphi\rangle.$$

This is a local differentiable Legendre transform in the chosen source-coordinate system. It is the definition used by perturbative 1PI diagrammatics: W and Γ may be formal power series, and J_φ is obtained coefficient by coefficient by inverting $\varphi = \delta W/\delta J$. This formal operation does not require a positive measure and does not imply convexity. In particular, in dimensional regularization with minimal subtraction the regulated functional integral is not a positive finite measure on field configurations; it is a prescription for formal amplitudes. Whenever later perturbative MS or BPHZ discussions use the 1PI effective action, this local formal 1PI functional is the operative object unless a positive Euclidean regulator and the convex Legendre–Fenchel construction are explicitly invoked.

Lemma 30.5 (1PI source equation and inverse two-point kernel)**.** *For the local Legendre transform of Definition 30.4,*

$$\frac{\delta\Gamma[\varphi]}{\delta\varphi(x)} = -J_\varphi(x),$$

and the connected two-point kernel is the inverse of the 1PI two-point kernel with the Lorentzian sign

$$\int \mathrm{d}^D z\, W^{(2)}(x,z)\Gamma^{(2)}(z,y) = -\delta^{(D)}(x-y).$$

Proof. Varying the definition of Γ gives

$$\delta\Gamma[\varphi] = \int \mathrm{d}^D x\, \frac{\delta W[J_\varphi]}{\delta J_\varphi(x)}\delta J_\varphi(x) - \int \mathrm{d}^D x\, \delta\varphi(x)J_\varphi(x) \\ - \int \mathrm{d}^D x\, \varphi(x)\delta J_\varphi(x).$$

The first and third terms cancel by the defining relation $\delta W/\delta J = \varphi$. Therefore

$$\frac{\delta\Gamma[\varphi]}{\delta\varphi(x)} = -J_\varphi(x).$$

At zero source, the physical vacuum expectation value is an extremum of the effective action:

$$\left.\frac{\delta\Gamma[\varphi]}{\delta\varphi(x)}\right|_{\varphi=\langle\phi\rangle} = 0.$$

The second derivative records the inverse relation between the connected two-point function and the 1PI two-point kernel. Define

$$W^{(2)}(x,y) = \frac{\delta^2 W[J]}{\delta J(x)\delta J(y)}, \qquad \Gamma^{(2)}(x,y) = \frac{\delta^2\Gamma[\varphi]}{\delta\varphi(x)\delta\varphi(y)}.$$

Differentiating $\varphi(x) = \delta W/\delta J(x)$ with respect to $\varphi(y)$ gives

$$\delta^{(D)}(x-y) = \int \mathrm{d}^D z\, W^{(2)}(x,z)\frac{\delta J(z)}{\delta\varphi(y)}.$$

Differentiating $\delta\Gamma/\delta\varphi = -J$ gives

$$\Gamma^{(2)}(z,y) = -\frac{\delta J(z)}{\delta\varphi(y)}.$$

Combining the two identities,

$$\int \mathrm{d}^D z\, W^{(2)}(x,z)\Gamma^{(2)}(z,y) = -\delta^{(D)}(x-y).$$

The sign is a consequence of the Lorentzian convention $\Gamma = W - \int \varphi J$. In momentum space, for a translation-invariant vacuum,

$$W^{(2)}(k)\Gamma^{(2)}(k) = -1.$$

With the same Feynman $+\mathrm{i}0$ boundary prescription, $W^{(2)}$ is related to the connected propagator by the same factors of i that appear in the source-derivative definition above. □

30.3 Background Split and One-Particle-Irreducible Kernels

Shifted representation and tadpole condition. Fix a field configuration φ, set $J = J_\varphi$, and write

$$\phi = \varphi + \xi.$$

The local 1PI functional has the shifted representation

$$\begin{aligned} e^{i\Gamma[\varphi]} &= e^{iW[J]-i\langle J,\varphi\rangle} \\ &= \int [D\xi]\ \exp\{iS[\varphi+\xi] + i\langle J,\xi\rangle\}_{J=J_\varphi}. \end{aligned}$$

In this shifted functional integral,

$$\langle \xi(x)\rangle_\varphi = 0,$$

so the linear source term is chosen exactly to cancel the quantum tadpole in the background φ.

The first equality is the Legendre definition $\Gamma[\varphi] = W[J_\varphi] - \langle J_\varphi, \varphi\rangle$. In the regulated integral for $Z[J_\varphi]$, make the affine change of variables $\phi = \varphi + \xi$. The symbol $[D\xi]$ denotes the pushforward of the specified regulated integration object under this change of variables; if the chosen regulator produces a nontrivial Jacobian or density term, its logarithm is part of the shifted action. The source term $\langle J_\varphi, \phi\rangle$ becomes $\langle J_\varphi, \varphi\rangle + \langle J_\varphi, \xi\rangle$, and the first part is precisely removed by the Legendre factor. Finally,

$$\langle \xi(x)\rangle_\varphi = \langle \phi(x)\rangle_{J_\varphi} - \varphi(x) = \left.\frac{\delta W[J]}{\delta J(x)}\right|_{J=J_\varphi} - \varphi(x) = 0.$$

Expand the action around the background:

$$S[\varphi+\xi] = S[\varphi] + \int d^D x\, S^{(1)}_\varphi(x)\xi(x) + \frac{1}{2}\int d^D x\, d^D y\, \xi(x) S^{(2)}_\varphi(x,y)\xi(y) + \cdots.$$

The vertices in the shifted theory may carry two types of legs: quantum ξ-legs that are contracted in loops, and background φ-legs that remain external. The source term supplies an additional linear vertex

$$i\langle J_\varphi, \xi\rangle.$$

For example, if the microscopic action contains a cubic interaction, then the polynomial identity after the split is

$$\phi^3 = \varphi^3 + 3\varphi^2\xi + 3\varphi\xi^2 + \xi^3.$$

Thus a single local interaction produces vertices with zero, one, two, or three quantum fluctuation legs. The first term is a purely background contribution to the zero-loop effective action; the terms with ξ-legs are contracted in the shifted functional integral; and the source supplies one more one-point quantum vertex. The shifted-vertex bookkeeping is displayed in Figure 30.2. Since J_φ is fixed by $\langle\xi\rangle = 0$, all tadpole subdiagrams in the shifted theory cancel in the sum.

Remark 30.6 (One-particle-reducible graphs are Legendre trees). A connected diagram is one-particle reducible if cutting one internal ξ propagator separates it into two connected diagrams, each with at least one external background leg. In the Legendre transform, such single-propagator reducible contributions are generated by solving the stationary equation for φ_J and joining lower kernels by exact propagators. They are therefore part of the tree reconstruction of connected

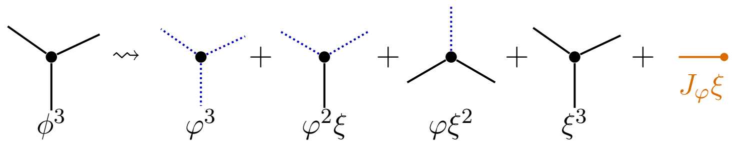


Figure 30.2: Cubic interaction after the background-field split $\phi = \varphi + \xi$. Dotted blue legs are background fields, solid legs are quantum fluctuations, and the orange one-point vertex is the source term fixed by the condition $\langle\xi\rangle_\varphi = 0$.

functions from Γ, rather than new kernels in Γ itself. The new kernels in Γ are the one-particle-irreducible ones. Equivalently, in a shifted connected vacuum diagram, cut an internal quantum propagator. Each side becomes a connected subdiagram with one open quantum leg and any number of background legs. When all graphs with the same external background data are summed, either side is precisely the source-dependent tadpole functional; it vanishes by $\langle\xi\rangle_\varphi = 0$. This is the diagrammatic content of the Legendre transform, not an additional dynamical assumption.

Figure 30.3 shows the diagrammatic content of this argument: cutting a quantum propagator in a shifted connected vacuum graph produces tadpole subgraphs, which vanish at the Legendre-transform background.

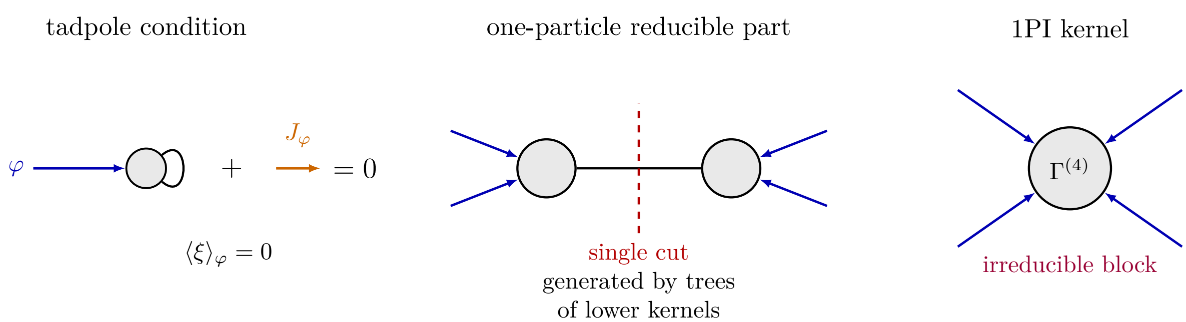


Figure 30.3: The source J_φ in the shifted functional integral is fixed by the condition that the quantum fluctuation has zero one-point function. With this condition imposed, the Legendre transform is represented by one-particle-irreducible kernels.

Perturbatively, one may write

$$\mathrm{i}\Gamma[\varphi] = \mathrm{i}S[\varphi] + \{\text{sum of connected 1PI vacuum graphs in the shifted theory}\}\,.$$

The position-space 1PI kernels are the functional derivatives

$$\Gamma^{(n)}(x_1,\ldots,x_n) = \left.\frac{\delta^n\Gamma[\varphi]}{\delta\varphi(x_1)\cdots\delta\varphi(x_n)}\right|_{\varphi=\varphi_0},$$

where φ_0 is the background about which the expansion is made. In a translation-invariant vacuum with vanishing one-point function, the expansion point is $\varphi_0 = 0$; if the one-point function is nonzero, one first shifts the field and then expands around the shifted stationary background. Equivalently, after expanding in powers of the background, $\mathrm{i}\Gamma^{(n)}$ is computed by amputated 1PI n-point diagrams whose external lines carry the background field. The momenta on these external legs range over the off-shell momentum space of the vertex function. Scattering kinematics enter only after LSZ and the corresponding on-shell limits.

30.4 Vertex Expansion

Definition 30.7 (Momentum-space 1PI vertices)**.** In a translation-invariant vacuum, the effective action has a momentum-space expansion

$$\Gamma[\varphi] = \sum_{n=0}^{\infty} \frac{1}{n!} \int \prod_{r=1}^{n} \frac{\mathrm{d}^D k_r}{(2\pi)^D} \, (2\pi)^D \delta^{(D)}\!\left(\sum_{r=1}^{n} k_r\right) \Gamma^{(n)}(k_1, \ldots, k_n) \, \widetilde{\varphi}(k_1) \cdots \widetilde{\varphi}(k_n).$$

The delta function expresses translation invariance, so only $n-1$ of the momenta are independent. With this normalization,

$$\Gamma^{(n)}(k_1, \ldots, k_n) = \left. \frac{\delta^n \Gamma}{\delta\widetilde{\varphi}(k_1) \cdots \delta\widetilde{\varphi}(k_n)} \right|_{\varphi=0}$$

after removing the overall momentum-conserving delta function.

For $n = 2$, the exact two-point 1PI kernel is the inverse propagator in the Legendre sense. For a scalar field with tree-level action

$$S_0[\phi] = -\frac{1}{2} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \, \widetilde{\phi}(k)(k^2 + m^2)\widetilde{\phi}(-k),$$

one writes

$$\mathrm{i}\Gamma^{(2)}(k) = -\mathrm{i}(k^2 + m^2) + \Pi_{\mathrm{1PI}}(k),$$

where $\Pi_{\mathrm{1PI}}(k)$ is the sum of 1PI self-energy diagrams in the same normalization. The exact connected propagator is obtained by inverting this kernel with the $i\epsilon$ prescription fixed by the underlying Lorentzian functional integral.

Definition 30.8 (Local derivative expansion of a 1PI branch)**.** When the kernels are analytic near zero momentum, their Taylor expansion gives a local derivative expansion. For a single scalar field,

$$\Gamma[\varphi] = \int \mathrm{d}^D x \left[-V_{\mathrm{eff}}(\varphi) - \frac{1}{2} Z_{\mathrm{eff}}(\varphi) \partial_\mu \varphi \, \partial^\mu \varphi + \sum_{r \geq 2} \mathcal{C}_r(\varphi, \partial\varphi, \ldots) \right],$$

where each $\mathcal{C}_r$ contains operators with higher numbers of derivatives. The coefficients are determined by derivatives of the kernels $\Gamma^{(n)}(k_1, \ldots, k_n)$ at small momenta. The expansion has to be used with its analytic domain stated: loops of massless fields commonly produce terms such as logarithms and threshold branch points at small momentum, and these terms are part of the exact 1PI action.

30.5 One-Loop Effective Potential and Dimensional Transmutation

The effective potential is the zero-derivative part of the local formal 1PI effective action. It is not, by definition, the exact convex Legendre–Fenchel effective action of a positive Euclidean measure. The latter is discussed later in this chapter. In perturbation theory one chooses a background branch, computes the local 1PI functional around that branch, and only then asks whether the branch is stable and whether the logarithms are under perturbative control.

Work in four dimensions with one real scalar field and Euclidean action

$$S_E[\phi] = \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi + V_0(\phi)\right], \qquad V_0(\phi) = \frac{\lambda}{4!}\phi^4.$$

The regulated path-integral Lagrangian contains the counterterm part needed by the chosen subtraction prescription; those terms are part of the regulated Lagrangian, not a separate physical theory. Let the background be constant, $\phi(x) = \varphi + \xi(x)$. The quadratic fluctuation operator is

$$\Delta_\varphi = -\partial^2 + M^2(\varphi), \qquad M^2(\varphi) = V_0''(\varphi) = \frac{\lambda}{2}\varphi^2.$$

In a finite box of volume Vol, the formal one-loop contribution to the 1PI effective action is

$$\Gamma_E^{(1)}[\varphi] = \frac{1}{2}\operatorname{Tr}\log\Delta_\varphi, \qquad V_1(\varphi) = \frac{1}{2\,\mathrm{Vol}}\operatorname{Tr}\log\Delta_\varphi.$$

After passing to momentum space and subtracting the φ-independent vacuum term,

$$V_1(\varphi) = \frac{1}{2}\mu^\epsilon \int \frac{\mathrm{d}^{4-\epsilon}p}{(2\pi)^{4-\epsilon}}\,\log\!\big(p^2 + M^2(\varphi)\big) + V_{\mathrm{ct}}^{(1)}(\varphi). \tag{30.1}$$

The scale μ is the dimensional-regularization scale in the selected subtraction chart. To evaluate the determinant without hiding constants, differentiate with respect to M^2:

$$\frac{\partial V_1}{\partial M^2} = \frac{1}{2}\mu^\epsilon \int \frac{\mathrm{d}^{4-\epsilon}p}{(2\pi)^{4-\epsilon}}\,\frac{1}{p^2+M^2} = \frac{1}{2}\frac{\mu^\epsilon}{(4\pi)^{2-\epsilon/2}}\Gamma\Big(-1+\frac{\epsilon}{2}\Big)\,(M^2)^{1-\epsilon/2}.$$

Integrating back in M^2 and choosing the additive constant so that $V_1(0) = 0$ gives

$$V_1^{\mathrm{reg}} = -\frac{1}{2}\frac{\mu^\epsilon}{(4\pi)^{2-\epsilon/2}}\Gamma\Big(-2+\frac{\epsilon}{2}\Big)\,(M^2)^{2-\epsilon/2} + V_{\mathrm{ct}}^{(1)}.$$

In the $\overline{\mathrm{MS}}$ chart, after subtracting the pole and the associated $\log 4\pi - \gamma_E$ convention terms, this becomes

$$V_1^{\overline{\mathrm{MS}}}(\varphi) = \frac{M^4(\varphi)}{64\pi^2}\left[\log\frac{M^2(\varphi)}{\mu^2} - \frac{3}{2}\right]. \tag{30.2}$$

Thus the one-loop formal 1PI effective potential in massless $\lambda\phi^4$ theory is

$$V_{\mathrm{eff}}^{\overline{\mathrm{MS}}}(\varphi) = \frac{\lambda(\mu)}{4!}\varphi^4 + \frac{\lambda(\mu)^2\varphi^4}{256\pi^2}\left[\log\frac{\lambda(\mu)\varphi^2}{2\mu^2} - \frac{3}{2}\right] + O(\lambda^3). \tag{30.3}$$

The logarithm is the first explicit signal that the classical scale-invariant polynomial has become a renormalized family depending on a subtraction scale. The dependence on μ is not physical by itself; it is compensated by the running coupling. To this order the field anomalous dimension starts only at higher loop order, while

$$\beta_\lambda(\lambda) = \mu\frac{\mathrm{d}\lambda}{\mathrm{d}\mu} = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3).$$

Indeed,

$$\mu\frac{\partial}{\partial\mu}\left[\frac{\lambda^2\varphi^4}{256\pi^2}\log\frac{\lambda\varphi^2}{2\mu^2}\right] = -\frac{\lambda^2\varphi^4}{128\pi^2},$$

whereas

$$\beta_\lambda\frac{\partial}{\partial\lambda}\left(\frac{\lambda\varphi^4}{24}\right) = \frac{\lambda^2\varphi^4}{128\pi^2}.$$

Thus

$$\left(\mu\frac{\partial}{\partial\mu} + \beta_\lambda\frac{\partial}{\partial\lambda} - \gamma_\phi\varphi\frac{\partial}{\partial\varphi}\right)V_{\text{eff}} = 0$$

holds through order λ^2, with $\gamma_\phi = O(\lambda^2)$ contributing only at the next order in this single-scalar example.

Coleman–Weinberg often uses a finite subtraction condition at a nonzero field value M, because the fourth derivative of $\varphi^4\log\varphi^2$ is singular at $\varphi = 0$. Define λ_{CW} by

$$\left.\frac{\mathrm{d}^4V_{\text{eff}}}{\mathrm{d}\varphi^4}\right|_{\varphi=M} = \lambda_{\text{CW}}.$$

After the corresponding finite change of quartic coordinate, the one-loop potential is

$$V_{\text{CW}}(\varphi) = \frac{\lambda_{\text{CW}}}{24}\varphi^4 + \frac{\lambda_{\text{CW}}^2}{256\pi^2}\varphi^4\left[\log\frac{\varphi^2}{M^2} - \frac{25}{6}\right] + O(\lambda_{\text{CW}}^3). \tag{30.4}$$

The constant $-25/6$ is fixed by the displayed fourth-derivative condition:

$$\frac{\mathrm{d}^4}{\mathrm{d}\varphi^4}\left[\varphi^4\log\frac{\varphi^2}{M^2}\right]\Bigg|_{\varphi=M} = 100, \qquad \frac{\mathrm{d}^4}{\mathrm{d}\varphi^4}\varphi^4 = 24.$$

The nonzero stationary points of the one-loop branch satisfy

$$0 = \frac{1}{\varphi^3}\frac{\mathrm{d}V_{\text{CW}}}{\mathrm{d}\varphi} = \frac{\lambda_{\text{CW}}}{6} + \frac{\lambda_{\text{CW}}^2}{64\pi^2}\left[\log\frac{\varphi^2}{M^2} - \frac{11}{3}\right] + O(\lambda_{\text{CW}}^3). \tag{30.5}$$

At the one-loop level this gives the formal scale

$$v = M\exp\left[\frac{11}{6} - \frac{16\pi^2}{3\lambda_{\text{CW}}(M)}\right]. \tag{30.6}$$

At such a stationary point,

$$\left.\frac{\mathrm{d}^2V_{\text{CW}}}{\mathrm{d}\varphi^2}\right|_{\varphi=v} = \frac{\lambda_{\text{CW}}^2}{32\pi^2}v^2 + O(\lambda_{\text{CW}}^3v^2).$$

Equations (30.5) and (30.6) display the algebraic mechanism called dimensional transmutation: a dimensionless coupling specified at the subtraction point is exchanged for a dimensionful scale on a chosen renormalized trajectory.

This statement has an important perturbative qualification. In the single real scalar theory, taking $M = v$ in (30.5) gives

$$\lambda_{\text{CW}}(v) = \frac{32\pi^2}{11},$$

which is not a small coupling. If instead $\lambda_{\mathrm{CW}}(M)$ is small, then $\log(v/M) = O(1/\lambda_{\mathrm{CW}})$, so higher leading logarithms are not parametrically smaller than the one-loop logarithm. The one-loop massless-scalar potential therefore demonstrates the structure of the calculation and the scale-generating relation, but it does not by itself give a controlled weak-coupling proof of a radiatively generated vacuum in pure $\lambda\phi^4$ theory.

The renormalization-group form of dimensional transmutation is independent of this particular one-loop branch. For a dimensionless coupling g with $\beta(g) \neq 0$, choosing $g(\mu)$ at a subtraction scale is equivalent, on a one-dimensional RG trajectory, to choosing the scale Λ_{g_*} at which the running coupling reaches a reference value g_*:

$$\log \frac{\Lambda_{g_*}}{\mu} = \int_{g(\mu)}^{g_*} \frac{\mathrm{d}g}{\beta(g)}. \tag{30.7}$$

For the one-loop scalar beta function $\beta_\lambda = b\lambda^2$, $b = 3/(16\pi^2)$, this gives

$$\Lambda_{\lambda_*} = \mu \exp\left[\frac{1}{b\lambda(\mu)} - \frac{1}{b\lambda_*}\right].$$

Because $b > 0$, this scale is the scale of increasing coupling toward the ultraviolet rather than an asymptotically free strong scale. In gauge theories with a negative one-loop beta function the same formula produces the familiar infrared strong scale.

A controlled Coleman–Weinberg minimum can occur in a multi-coupling theory where the tree quartic coupling is of the same order as a one-loop effect. For example, in massless scalar electrodynamics on a chosen perturbative branch and in the conventional radial-field normalization, the gauge-boson one-loop term gives

$$V_{\mathrm{eff}}(\varphi) = \frac{\lambda}{24}\varphi^4 + \frac{3e^4}{64\pi^2}\varphi^4\left[\log\frac{\varphi^2}{M^2} - \frac{25}{6}\right] + \cdots,$$

after the same type of nonzero-field quartic subtraction. Then a stationary point at $M = v$ requires $\lambda = 33e^4/(8\pi^2) + O(e^6)$, which is compatible with weak coupling when e is small. This is the controlled version of radiative breaking; the single-scalar computation above is the minimal determinant calculation and the clearest place to see how the subtraction scale enters the formal 1PI potential.

30.6 Connected Functions from Exact 1PI Vertices

The Legendre transform can be inverted. Given $\Gamma[\varphi]$, the connected functional $W[J]$ is obtained by solving

$$\frac{\delta\Gamma[\varphi]}{\delta\varphi(x)} = -J(x)$$

for $\varphi = \varphi_J$, and then setting

$$W[J] = \Gamma[\varphi_J] + \langle J, \varphi_J\rangle.$$

Equivalently, the inversion can be written as the stationary value

$$W[J] = \underset{\varphi}{\mathrm{stationary}}\left(\Gamma[\varphi] + \langle J, \varphi\rangle\right).$$

This identity is exact at the level of the Legendre transform. It also has a diagrammatic interpretation: connected Green functions are obtained as tree diagrams whose vertices are exact 1PI kernels and whose internal lines are the exact connected propagator.

Proposition 30.9 (Exact 1PI tree reconstruction of connected kernels)**.** *Work in DeWitt notation, with repeated indices summed and integrated, and set*

$$W^{i_1\cdots i_n} = \frac{\delta^n W}{\delta J_{i_1}\cdots\delta J_{i_n}}, \qquad \Gamma_{i_1\cdots i_n} = \frac{\delta^n \Gamma}{\delta\varphi^{i_1}\cdots\delta\varphi^{i_n}}$$

at the paired configurations J and φ_J. Let

$$G^{ij} = W^{ij} = \frac{\delta\varphi_J^i}{\delta J_j}$$

be the exact connected two-point kernel. Then

$$\Gamma_{ia}G^{aj} = -\delta_i^{\ j}, \tag{30.8}$$

$$W^{ijk} = G^{ia}G^{jb}G^{kc}\Gamma_{abc}, \tag{30.9}$$

and

$$\begin{aligned} W^{ijkl} &= G^{ia}G^{jb}G^{kc}G^{ld}\Gamma_{abcd} \\ &\quad + G^{ia}G^{jb}\Gamma_{abe}G^{ef}\Gamma_{fcd}G^{kc}G^{ld} \\ &\quad + G^{ia}G^{kc}\Gamma_{ace}G^{ef}\Gamma_{fbd}G^{jb}G^{ld} \\ &\quad + G^{ia}G^{ld}\Gamma_{ade}G^{ef}\Gamma_{fbc}G^{jb}G^{kc}. \end{aligned} \tag{30.10}$$

In general, every connected kernel $W^{i_1\cdots i_n}$ is obtained by a finite sum over trees whose vertices are exact 1PI kernels Γ_r with $r \geq 3$, whose internal edges are G, and whose external half-edges are attached to the external labels by G. No loop is created in this reconstruction; all loop information is already contained in the exact kernels Γ_r and the exact propagator G.

Proof. The stationarity equation is

$$\Gamma_i[\varphi_J] + J_i = 0.$$

Differentiating once with respect to J_j gives

$$\Gamma_{ia}G^{aj} + \delta_i^{\ j} = 0,$$

which proves (30.8). Differentiating this identity with respect to J_k gives

$$\Gamma_{iab}G^{bk}G^{aj} + \Gamma_{ia}W^{ajk} = 0.$$

Multiplying by $G^{\ell i}$ and using $G^{\ell i}\Gamma_{ia} = -\delta^{\ell}{}_a$, which is the same inverse relation with the two factors interchanged, gives

$$W^{\ell jk} = G^{\ell i}\Gamma_{iab}G^{aj}G^{bk}.$$

Renaming dummy indices proves (30.9).

Differentiate the preceding second-derivative identity once more:

$$\begin{aligned} 0 = {}& \Gamma_{iabc}G^{c\ell}G^{bk}G^{aj} + \Gamma_{iab}W^{bk\ell}G^{aj} + \Gamma_{iab}G^{bk}W^{aj\ell} \\ & + \Gamma_{iab}G^{b\ell}W^{ajk} + \Gamma_{ia}W^{ajk\ell}. \end{aligned}$$

Multiplication by G^{mi} solves for $W^{mjk\ell}$. The first term gives the single four-point vertex in (30.10); substituting (30.9) into the other three terms gives the three pairings of two exact three-point vertices joined by one exact propagator. This proves (30.10).

The recursive tree statement follows by induction. Suppose every source derivative of φ_J of order less than n is a sum of trees. The $(n-1)$st derivative of $\Gamma_i[\varphi_J] + J_i = 0$ is a finite Faà di Bruno sum: one differentiates some Γ_r and distributes the source derivatives among the factors $\delta^m \varphi_J / \delta J^m$. By the induction hypothesis those factors are trees. Multiplying once by G solves the remaining Γ_{ia}-factor and attaches one additional external propagator. This operation glues trees at a new exact 1PI vertex and cannot form a cycle. Therefore the result is a sum over trees only. Translating from derivatives of W to Lorentzian Green functions uses the same powers of i as in the source-derivative definitions at the beginning of the chapter. $\square$

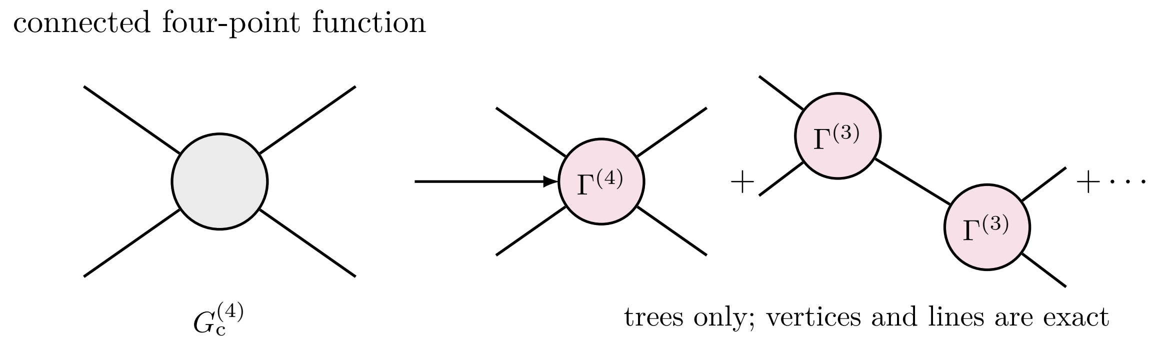


Figure 30.4: Connected Green functions are reconstructed from the exact 1PI action by forming trees. The internal lines are exact propagators, and the vertices are exact kernels $\Gamma^{(n)}$.

One can express the same statement with a bookkeeping parameter $\hbar$:

$$\mathrm{i}W[J] = \lim_{\hbar \to 0} \hbar \log \int [D\varphi]\, \exp\left\{ \frac{\mathrm{i}}{\hbar} \left(\Gamma[\varphi] + \langle J, \varphi \rangle \right) \right\}.$$

The limit keeps only tree diagrams built from Γ. Since Γ already contains the full loop information of the original theory, this tree expansion is the tree-level representation of the exact Legendre inverse.

30.7 Schwinger–Dyson Identities and the 1PI Hierarchy

The Legendre transform organizes exact kernels, but it does not by itself use the equation of motion of the integration variable. The latter information is contained in the Schwinger–Dyson identities. These identities are exact only after their hypotheses are stated: at the regulator under discussion the reference integration density must be translation invariant in the integration coordinate, or else its logarithmic derivative must be included; the contour or domain must have no boundary contribution for the variation being made; and any Jacobian from a nontrivial change of variables must be recorded as an additional insertion. Gauge theories replace the naive field translation by gauge-fixed BRST/BV identities. In this scalar section we assume the translation-invariant regulated scalar density introduced at the beginning of the chapter.

Use condensed DeWitt notation in this section. A single index i denotes both a discrete field label and a spacetime point, repeated indices are summed and integrated, and

$$J_i\phi^i = \langle J, \phi\rangle.$$

Write

$$S_i[\phi] = \frac{\delta S[\phi]}{\delta\phi^i}, \qquad S_{ij}[\phi] = \frac{\delta^2 S[\phi]}{\delta\phi^i\delta\phi^j},$$

and similarly for higher field derivatives of the regulated action.

Functional Schwinger–Dyson identity. For every regulated test functional $\mathcal{O}[\phi]$ for which the following integration by parts is legitimate,

$$\big\langle \big(S_i[\phi] + J_i\big)\mathcal{O}[\phi]\big\rangle_J = \mathrm{i}\left\langle \frac{\delta\mathcal{O}[\phi]}{\delta\phi^i}\right\rangle_J. \tag{30.11}$$

Equivalently, at the level of the source functional,

$$\left[S_i\left(\frac{1}{\mathrm{i}}\frac{\delta}{\delta J}\right) + J_i\right] Z[J] = 0. \tag{30.12}$$

At finite regulator, integrate the total derivative

$$0 = \int D_\Lambda^{\mathrm{ref}}\phi\, \frac{\delta}{\delta\phi^i}\left[\mathcal{O}[\phi]\, \exp\{\mathrm{i}S[\phi] + \mathrm{i}J_j\phi^j\}\right].$$

Expanding the derivative gives

$$0 = \left\langle \frac{\delta\mathcal{O}}{\delta\phi^i}\right\rangle_J + \mathrm{i}\,\langle \mathcal{O}(S_i + J_i)\rangle_J\,.$$

Since $-1/\mathrm{i} = \mathrm{i}$, this is (30.11). Taking $\mathcal{O} = 1$ and replacing each field insertion by $(1/\mathrm{i})\delta/\delta J$ gives (30.12). If the reference density has logarithmic derivative $R_i[\phi]$, the left side is modified by $S_i + J_i - \mathrm{i}R_i$; that is the measure contribution which, in symmetry variations, becomes the anomaly term.

The case $\mathcal{O} = 1$ gives the quantum equation of motion

$$\langle S_i[\phi]\rangle_J + J_i = 0.$$

Using $\Gamma_i[\varphi] = \delta\Gamma/\delta\varphi^i = -J_i$, this becomes

$$\Gamma_i[\varphi] = \langle S_i[\phi]\rangle_{J_\varphi}\,. \tag{30.13}$$

Thus the 1PI equation of motion is not the classical equation $S_i[\varphi] = 0$. It is the expectation value of the microscopic equation of motion in the source ensemble whose mean field is φ.

Equation (30.12) may be written directly in connected variables. Since

$$Z^{-1}\frac{1}{\mathrm{i}}\frac{\delta Z}{\delta J_i} = \frac{\delta W}{\delta J_i} = \varphi^i,$$

one has the operator identity

$$Z[J]^{-1}F\left(\frac{1}{\mathrm{i}}\frac{\delta}{\delta J}\right)Z[J] = F\left(\varphi + \frac{1}{\mathrm{i}}\frac{\delta}{\delta J}\right)1 \tag{30.14}$$

for every polynomial functional F, with the functional derivatives on the right acting on all source-dependent quantities to their right before setting them on 1. Therefore

$$S_i\left(\varphi + \frac{1}{\mathrm{i}}\frac{\delta}{\delta J}\right)1 + J_i = 0. \tag{30.15}$$

This formula is the compact exact Schwinger–Dyson equation for W. Its Legendre form is obtained by substituting $J_i = -\Gamma_i[\varphi]$ and using $W^{(2)}\Gamma^{(2)} = -1$ to convert source derivatives into field derivatives when desired.

Quartic scalar example

For a single scalar field with a quartic interaction, write the regulated action in condensed notation as

$$S[\phi] = \frac{1}{2}K_{ij}\phi^i\phi^j + \frac{\lambda}{4!}V_{ijkl}\phi^i\phi^j\phi^k\phi^l,$$

where $K_{ij} = K_{ji}$ is the regulated quadratic kernel and V_{ijkl} is totally symmetric. The sign of λV is the sign chosen in the Lorentzian action; no Euclidean positivity is being assumed in this formal identity. Since

$$S_i[\phi] = K_{ij}\phi^j + \frac{\lambda}{3!}V_{ijkl}\phi^j\phi^k\phi^l,$$

the connected Schwinger–Dyson equation becomes

$$\begin{aligned}\Gamma_i[\varphi] = K_{ij}\varphi^j + \frac{\lambda}{3!}V_{ijkl}\Big[&\varphi^j\varphi^k\varphi^l \\ &+ \frac{1}{\mathrm{i}}\big(W^{jk}\varphi^l + W^{jl}\varphi^k + W^{kl}\varphi^j\big) + \frac{1}{\mathrm{i}^2}W^{jkl}\Big]_{J=J_\varphi}.\end{aligned} \tag{30.16}$$

Here

$$W^{i_1\cdots i_n} = \frac{\delta^n W[J]}{\delta J_{i_1}\cdots\delta J_{i_n}}$$

is the connected-kernel convention already used in this chapter. The powers of 1/i in (30.16) are not optional: they come from the Lorentzian source factor $\exp(\mathrm{i}J_i\phi^i)$.

Full propagator member

Assume for this paragraph that the vacuum is translation invariant and $\phi \mapsto -\phi$ symmetric, so that $\varphi = 0$ and odd connected functions vanish at $J = 0$. Differentiate the field equation with respect to J_m and set $J = 0$. The exact two-point member of the hierarchy is

$$0 = K_{ij}W^{jm} + \frac{\lambda}{3!}V_{ijkl}\Big[\frac{1}{\mathrm{i}}\big(W^{jk}W^{lm} + W^{jl}W^{km} + W^{jm}W^{kl}\big) + \frac{1}{\mathrm{i}^2}W^{jklm}\Big] + \delta_i{}^m. \tag{30.17}$$

This is the full propagator Schwinger–Dyson equation in connected form. It is not a one-loop formula: W^{jm} is the exact connected two-point kernel and W^{jklm} is the exact connected four-point kernel. Multiplying (30.17) by the exact inverse kernel $\Gamma^{(2)}$, using

$$W^{im}\Gamma^{(2)}_{mn} = -\delta^i{}_n,$$

gives the corresponding inverse-propagator form

$$\Gamma^{(2)}_{in} = K_{in} + \frac{\lambda}{2\mathrm{i}} V_{inkl} W^{kl} - \frac{\lambda}{3!\,\mathrm{i}^2} V_{ijkl} W^{jklm} \Gamma^{(2)}_{mn}. \tag{30.18}$$

The second term is the exact tadpole contraction. The last term is the amputated contribution of the exact connected four-point kernel. A common "gap" or Hartree approximation drops the connected four-point term and then solves the remaining equation self-consistently for $W^{(2)}$. That operation is a truncation of the exact hierarchy, not a derivation from the Schwinger–Dyson identity.

Full vertex member

The next nontrivial member is obtained from (30.11) with $\mathcal{O} = \phi^a\phi^b\phi^c$, or equivalently by differentiating (30.16) three times with respect to the source. First, the insertion form gives

$$\begin{aligned} K_{ij} \left\langle \phi^j \phi^a \phi^b \phi^c \right\rangle + \frac{\lambda}{3!} V_{ijkl} \left\langle \phi^j \phi^k \phi^l \phi^a \phi^b \phi^c \right\rangle \\ = \mathrm{i}\Big(\delta_i{}^a \left\langle \phi^b \phi^c \right\rangle + \delta_i{}^b \left\langle \phi^a \phi^c \right\rangle + \delta_i{}^c \left\langle \phi^a \phi^b \right\rangle\Big). \end{aligned} \tag{30.19}$$

This equation contains disconnected pair products. They are not an ambiguity: they are removed by the lower two-point Schwinger–Dyson member. The same statement is obtained without mentioning disconnected moments by differentiating the connected equation. Let Sym_{abc} denote summation over all distinct permutations of the displayed labels a, b, c, and let $\mathrm{Sym}^{2,1}_{abc}$ denote the three splittings $(ab|c), (ac|b), (bc|a)$. In the symmetric vacuum,

$$\begin{aligned} 0 = K_{ij} W^{jabc} + \frac{\lambda}{3!} V_{ijkl} \Big[\mathrm{Sym}_{abc}\, W^{ja} W^{kb} W^{lc} \\ + \frac{3}{\mathrm{i}} \Big(W^{jk} W^{labc} + \mathrm{Sym}^{2,1}_{abc}\, W^{jkab} W^{lc} \Big) + \frac{1}{\mathrm{i}^2} W^{jklabc} \Big]. \end{aligned} \tag{30.20}$$

Equation (30.20) is often the cleanest form of the full vertex hierarchy: every term is connected, and the first term is the kinetic operator acting on the exact connected four-point kernel. The 1PI version is obtained by substituting the Legendre identities between connected kernels and exact 1PI kernels. In the $\phi \mapsto -\phi$ symmetric case,

$$W^{abcd} = W^{aa'} W^{bb'} W^{cc'} W^{dd'} \Gamma^{(4)}_{a'b'c'd'}, \tag{30.21}$$

because the exact three-point vertex vanishes. The connected six-point kernel decomposes as

$$\begin{aligned} W^{abcdef} = W^{aa'} W^{bb'} W^{cc'} W^{dd'} W^{ee'} W^{ff'} \Gamma^{(6)}_{a'b'c'd'e'f'} \\ + \sum_{\mathcal{P}_3} \left(\prod_{r \in A} W^{rr'} \right) \left(\prod_{s \in B} W^{ss'} \right) \Gamma^{(4)}_{A'u} W^{uv} \Gamma^{(4)}_{vB'}. \end{aligned} \tag{30.22}$$

Here $\mathcal{P}_3$ is the set of the ten unordered partitions $\{a, b, c, d, e, f\} = A \sqcup B$ with $|A| = |B| = 3$; A' and B' denote the primed external indices attached to the labels in the corresponding block. The signs in (30.21) and (30.22) follow from the convention $W^{im}\Gamma^{(2)}_{mn} = -\delta^i{}_n$. Substituting (30.21) and (30.22) into (30.20), and multiplying by exact inverse two-point kernels on the external labels, gives the full four-point 1PI Schwinger–Dyson equation. It relates $\Gamma^{(4)}$ to the exact propagator, the exact six-point vertex $\Gamma^{(6)}$, and the two $\Gamma^{(4)}$-vertices joined by one exact propagator in (30.22). Without the $\phi \mapsto -\phi$ symmetry, the same construction contains the additional tree terms built from exact three-point vertices described in Figure 30.4. A vertex truncation is the choice to keep only a specified subset of these terms, for example to model $\Gamma^{(4)}$ while discarding the connected six-point kernel. The exact Schwinger–Dyson hierarchy itself does not justify that closure; the closure must be supplied by a small parameter, a controlled kinematic limit, a renormalization condition plus error estimate, or an explicit nonperturbative ansatz.

30.8 Euclidean Convexity

The convexity statement is a theorem only after the Euclidean functional integral has been given positive measure-theoretic meaning. The analytic input used in the proof is isolated in the following assumption.

Assumption 30.10 (Positive finite Euclidean regulator for convexity)**.** Fix a regulator Λ. There is a measurable regulated configuration space $(\mathcal{C}_\Lambda, \mathcal{A}_\Lambda)$, a positive finite measure μ_Λ on it, and a real vector space $\mathcal{S}_\Lambda$ of regulated sources paired with the regulated field by a measurable real functional

$$(J, \phi) \longmapsto \langle J, \phi \rangle_\Lambda.$$

There is a nonempty convex source domain $\mathcal{D}_\Lambda \subset \mathcal{S}_\Lambda$ such that, for every $J \in \mathcal{D}_\Lambda$,

$$0 < Z_{E,\Lambda}[J] := \int_{\mathcal{C}_\Lambda} \exp\{-\langle J, \phi \rangle_\Lambda\} \, \mathrm{d}\mu_\Lambda(\phi) < \infty. \tag{30.23}$$

In the scalar finite-dimensional convention this positive measure may be represented as a density

$$\mathrm{d}\mu_\Lambda(\phi) = Z_\Lambda^{-1} \mathrm{e}^{-S_E[\phi]} \, D_\Lambda^{\mathrm{ref}} \phi,$$

where the normalization is optional in unnormalized source functionals. In the Gaussian-reference convention the same object is written $\mathrm{e}^{-L_\Lambda[\phi]} \mathrm{d}\mu_{C_\Lambda}(\phi)$, with the relation to $D_\Lambda^{\mathrm{ref}}\phi$ given by (11.1).

Under Assumption 30.10 define

$$Z_E[J] = Z_{E,\Lambda}[J] = \mathrm{e}^{-W_E[J]}.$$

Any continuum limit $\Lambda \to \infty$ of the statements below requires an additional existence theorem for the limiting Schwinger functional. The finite-regulator argument by itself does not prove such a limit for a proposed model.

Definition 30.11 (Lorentzian–Euclidean source-convention bridge)**.** Write the Lorentzian logarithm as

$$\mathcal{W}_L[J_L] = \log Z_L[J_L] = \mathrm{i} W_L[J_L], \qquad Z_L[J_L] = \int [D\phi] \, \exp\{\mathrm{i} S_L[\phi] + \mathrm{i}\langle J_L, \phi \rangle_L\}\,.$$

Write the Euclidean logarithm in the convention used in this and the following Euclidean chapters as

$$\mathcal{W}_E[J_E] = \log Z_E[J_E] = -W_E[J_E], \qquad Z_E[J_E] = \int [D\phi]\, \exp\{-S_E[\phi] - \langle J_E, \phi\rangle_E\}\,.$$

In this definition $[D\phi]$ is only compact continuum notation for the regulated integration object already specified earlier in the chapter. In a finite-dimensional bosonic scalar regulator, the Euclidean line is represented by $D^{\mathrm{ref}}_\Lambda \phi\, \mathrm{e}^{-S_{E,\Lambda}[\phi]}$; after splitting $S_{E,\Lambda} = S_{\mathrm{kin},\Lambda} + L_\Lambda$, the same object is $\mathrm{d}\mu_{C_\Lambda}(\phi)\, \mathrm{e}^{-L_\Lambda[\phi]}$. The Lorentzian line is the corresponding oscillatory notation and carries its own regulator and boundary-value prescription. For scalar insertions with no tensor time indices, the analytic continuation from Lorentzian time-ordered connected distributions to Euclidean connected Schwinger distributions is recorded by the derivative identities

$$G_n^{L,\mathrm{c}} = \mathrm{i}^{-n} \frac{\delta^n \mathcal{W}_L}{\delta J_L(x_1)\cdots\delta J_L(x_n)}\bigg|_{J_L=0} = \mathrm{i}^{1-n} \frac{\delta^n W_L}{\delta J_L(x_1)\cdots\delta J_L(x_n)}\bigg|_{J_L=0}, \tag{30.24}$$

and

$$G_n^{E,\mathrm{c}} = (-1)^n \frac{\delta^n \mathcal{W}_E}{\delta J_E(x_1)\cdots\delta J_E(x_n)}\bigg|_{J_E=0} = (-1)^{n+1} \frac{\delta^n W_E}{\delta J_E(x_1)\cdots\delta J_E(x_n)}\bigg|_{J_E=0}. \tag{30.25}$$

The source signs in the exponents are related as follows. Under the formal coordinate rotation $t = -\mathrm{i}\tau$, the Lorentzian pairing satisfies

$$\mathrm{i}\langle J_L, \phi\rangle_L \longmapsto +\langle J_L^{\mathrm{W}}, \phi_E\rangle_E, \qquad J_L^{\mathrm{W}}(\tau, \vec{x}) = J_L(-\mathrm{i}\tau, \vec{x}),$$

whereas the Euclidean convention above uses $-\langle J_E, \phi_E\rangle_E$. Therefore the Euclidean source variable of this chapter is

$$J_E = -J_L^{\mathrm{W}} \tag{30.26}$$

when one translates a formal source insertion from the Lorentzian convention to the Euclidean convention. Equations (30.24) and (30.25), together with the analytic continuation of the connected distributions, are the matching rule used by the later Euclidean path-integral and RG chapters. For operators with Lorentz indices, spinor indices, or time derivatives, the usual Wick-rotation matrix for those indices must also be applied.

In particular, the Euclidean free-energy convention $W_E = -\log Z_E$ is opposite to the connected-log convention $\mathcal{W}_E = \log Z_E$. The source derivative of W_E has the sign displayed in (30.25); this is the sign used in the convex Legendre transform below.

Lemma 30.12 (Euclidean convexity at finite regulator)**.** *Assume Assumption 30.10. Define*

$$K[J] = \log Z_E[J], \qquad W_E[J] = -K[J].$$

Thus W_E is the Euclidean free energy. It is the negative of the connected-log functional K. Functional derivative signs are therefore fixed by two independent choices: whether one differentiates K or $-K$, and the sign with which the source is placed in the exponent. For the present source term $-\langle J, \phi\rangle$, one has $\delta W_E/\delta J = \langle\phi\rangle_J$. For a source term $+\langle J, \mathcal{O}\rangle$, the same free-energy convention gives $\delta W/\delta J = -\langle\mathcal{O}\rangle_J$. Then K is convex on $\mathcal{D}_\Lambda$, and W_E is concave on $\mathcal{D}_\Lambda$. The finite-regulator Euclidean effective action

$$\Gamma_E[\varphi] = \sup_{J\in\mathcal{D}_\Lambda} \{W_E[J] - \langle J, \varphi\rangle\}$$

is a convex functional of φ. At a differentiability point where the supremum is attained by a locally unique source J_φ satisfying

$$\varphi(x) = \left.\frac{\delta W_E[J]}{\delta J(x)}\right|_{J=J_\varphi},$$

this definition reduces to the local Legendre transform

$$\Gamma_E[\varphi] = W_E[J_\varphi] - \langle J_\varphi, \varphi\rangle, \qquad \frac{\delta \Gamma_E[\varphi]}{\delta\varphi(x)} = -J_\varphi(x),$$

and its second variation is nonnegative on the subspace where the source-field map is nonsingular.

Proof. For $J_1, J_2 \in \mathcal{D}_\Lambda$ and $0 \le t \le 1$, Holder's inequality with respect to the positive measure μ_Λ gives

$$\begin{aligned} Z_E[tJ_1 + (1-t)J_2] &= \int_{\mathcal{C}_\Lambda} \left(\mathrm{e}^{-\langle J_1,\phi\rangle_\Lambda}\right)^t \left(\mathrm{e}^{-\langle J_2,\phi\rangle_\Lambda}\right)^{1-t} \mathrm{d}\mu_\Lambda(\phi) \\ &\le Z_E[J_1]^t Z_E[J_2]^{1-t}. \end{aligned}$$

Taking the logarithm yields

$$K[tJ_1 + (1-t)J_2] \le tK[J_1] + (1-t)K[J_2],$$

so K is convex. Equivalently,

$$W_E[tJ_1 + (1-t)J_2] \ge tW_E[J_1] + (1-t)W_E[J_2].$$

For each fixed source J, the map $\varphi \mapsto W_E[J] - \langle J, \varphi\rangle$ is affine. The supremum of affine functionals is convex, proving convexity of Γ_E. If the supremum is attained at a locally unique differentiability point, its stationarity condition is $\varphi = \delta W_E/\delta J|_{J=J_\varphi}$. Differentiating $W_E[J_\varphi] - \langle J_\varphi, \varphi\rangle$ then cancels the δJ_φ terms and gives $\delta\Gamma_E/\delta\varphi = -J_\varphi$. Convexity gives, in any nonsingular local chart,

$$\int \mathrm{d}^D x\, \mathrm{d}^D y\, f(x)\Gamma_E^{(2)}(x,y) f(y) \ge 0$$

for every real test function f in the corresponding tangent subspace. □

30.9 Legendre–Fenchel Form at Finite Regulator

This section remains in the positive-measure setting of Lemma 30.12. It supplies the positive-measure finite-regulator construction. The perturbative 1PI definition used in dimensional regularization, or in any other regulator whose "path integral" is a formal algebraic prescription, is a separate construction. The two constructions agree only in a domain where both are defined and the convex conjugate is differentiable with a locally invertible source-field map.

The local inverse J_φ used above is a convenient notation. Lemma 30.12 identifies the exact finite-regulator definition as a convex-conjugate formula. At finite regulator, the source space is a finite-dimensional vector space or a specified space of test functions, depending on the regulator. Its dual is the space in which the regulated mean field φ lives, and the pairing is

$$\langle J, \varphi\rangle = \int \mathrm{d}^D x\, J(x)\varphi(x)$$

with the corresponding finite sum on a lattice.

Let

$$K[J] = \log Z_E[J], \qquad W_E[J] = -K[J].$$

Under the positivity assumption of Lemma 30.12, K is convex. The exact Euclidean effective action is the convex conjugate with the sign convention fixed by the source term $-\langle J, \phi\rangle$:

$$\Gamma_E[\varphi] = \sup_J \{-K[J] - \langle J, \varphi\rangle\} = \sup_J \{W_E[J] - \langle J, \varphi\rangle\}.$$

This formula covers cases in which the map $J \mapsto \varphi_J$ has multiple preimages or the supremum is attained by more than one source. It is also meaningful at boundary points of the domain where the supremum is represented by a limiting sequence of sources. If K is differentiable and the supremum is attained at J_φ, the stationarity condition is

$$0 = -\frac{\delta K[J_\varphi]}{\delta J(x)} - \varphi(x), \qquad \varphi(x) = \frac{\delta W_E[J_\varphi]}{\delta J(x)}.$$

Thus the earlier Legendre transform is the differentiable local form of the Legendre–Fenchel definition.

The subgradient statement is the coordinate-free replacement for the inverse source map. A source J maximizes the displayed variational expression at φ precisely when

$$-\varphi \in \partial K[J],$$

where $\partial K[J]$ is the convex subdifferential of K at J. If K is twice differentiable and its Hessian has a kernel at a source J, the source-field map is locally degenerate there. The convex-conjugate formula remains the definition. It records the supporting hyperplanes of K: an admissible mean field φ is one for which $-\varphi$ is a subgradient of K at some source, or a limit of such subgradients at a boundary point of the domain.

A finite-dimensional model displays the same logic without functional analysis. Let $\phi \in \mathbb{R}$ and

$$Z_E(J) = \int_{\mathbb{R}} \mathrm{d}\phi\, \mathrm{e}^{-S(\phi)-J\phi} = \mathrm{e}^{-W_E(J)}$$

with $S(\phi)$ a confining function. The source-dependent mean is

$$\varphi(J) = \frac{\mathrm{d}W_E}{\mathrm{d}J}.$$

In the small-fluctuation limit, $W_E(J)$ is obtained by minimizing $S(\phi) + J\phi$ with respect to ϕ. The corresponding $\Gamma_E(\varphi)$ is the convex Legendre transform. If $S(\phi)$ has more than one local minimum, different ranges of J select different stationary branches, and the exact Legendre transform produces a convex function of φ. A perturbative expansion around a chosen branch is valid where the quadratic fluctuation operator at that branch is positive.

The last clause is an analytic condition, not a matter of terminology. Choose a point ϕ_0 on a stationary branch and choose the source

$$J_0 = -S'(\phi_0).$$

The Euclidean background representation of the effective action is

$$e^{-\Gamma_E(\phi_0)} = \int d\xi \, \exp\{-S(\phi_0+\xi) - J_0\xi\}\Big|_{\langle\xi\rangle_{\phi_0,J_0}=0}.$$

The exponent is obtained by subtracting the tangent line to S at ϕ_0. Its expansion begins

$$S(\phi_0+\xi) + J_0\xi = S(\phi_0) + \frac{1}{2}S''(\phi_0)\xi^2 + O(\xi^3).$$

If $S''(\phi_0) < 0$, the Gaussian integral around this point is not a positive quadratic approximation to the finite-dimensional Euclidean integral. The exact Γ_E is then determined by the global supporting line data encoded in the Legendre–Fenchel transform, while the coefficientwise stationary-phase expansion around that branch computes only a local branch functional. This is the finite-dimensional form of the distinction between the exact convex effective action and a perturbative effective potential computed around a specified saddle.

Figure 30.5 records the local branch condition for the finite-dimensional model: perturbation theory around a chosen saddle uses a positive quadratic part, while the exact Legendre–Fenchel transform uses global supporting lines.

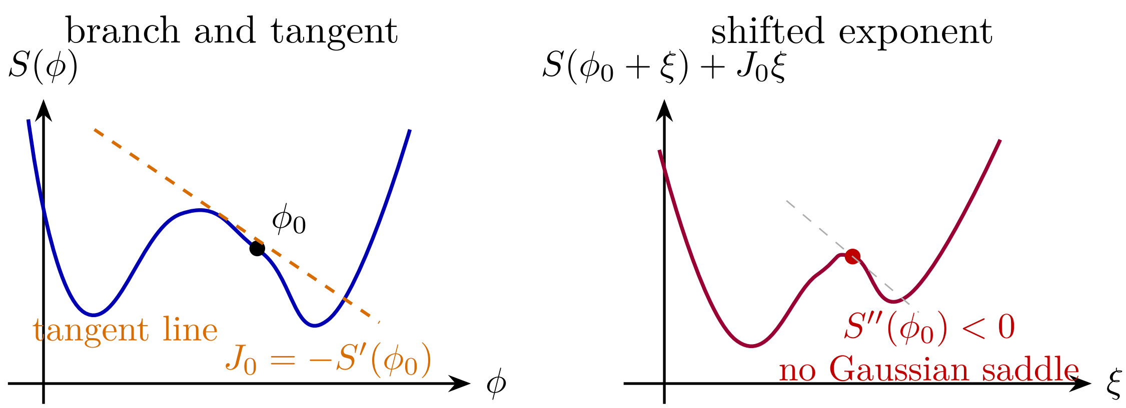


Figure 30.5: In the finite-dimensional model, the source subtracts the tangent line at the chosen mean field. Perturbation theory around that point requires the shifted exponent to have a positive quadratic part. The exact Legendre–Fenchel effective action instead uses the global supporting-line structure and is convex.

Figure 30.6 displays the convex Euclidean effective action obtained from the exact Legendre–Fenchel construction, in contrast to the nonconvex local branch just described.

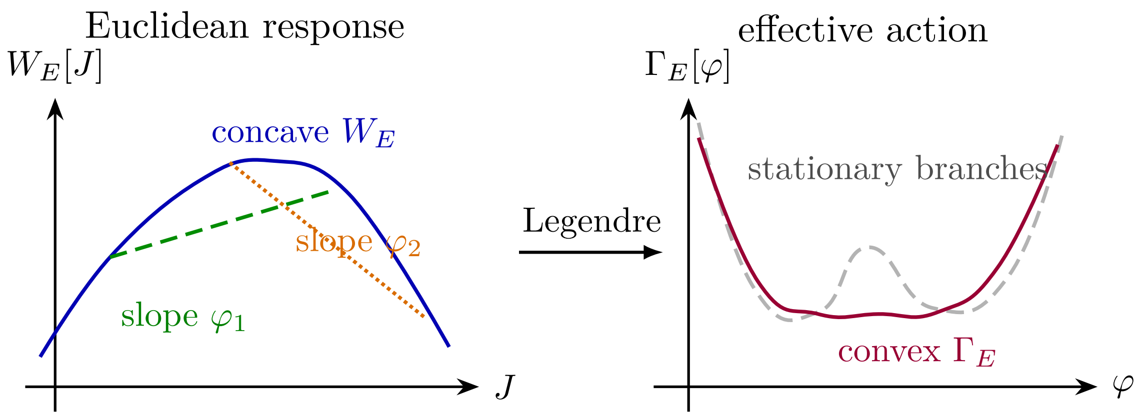


Figure 30.6: For a positive Euclidean measure and the convention $Z_E[J] = \mathrm{e}^{-W_E[J]}$, the response functional W_E is concave and the effective action Γ_E is convex. Dashed and dotted support lines distinguish two source values, while the dashed nonconvex branches are not the exact effective action. In a finite-dimensional double-well model, the exact Legendre transform connects the stable branches by the convex segment.

The Euclidean convexity theorem is a statement about the exact Legendre–Fenchel transform of a positive finite-measure source functional. A Wilsonian action at a finite cutoff scale, the regulated action that appears in the Euclidean measure, the local formal perturbative 1PI effective action, and a perturbative effective potential computed around one stationary branch are distinct functionals with distinct definitions. The next chapter uses the local formal 1PI functional as the arena in which perturbative local counterterms are identified and organized.

Chapter 31

Renormalizability and Local Counterterms

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

31.1 Regulated Euclidean Actions

Let ϕ be a real scalar field on D-dimensional Euclidean space. Fix a regulator, denoted collectively by Λ. At finite regulator the field is an element of a regulated field space $\mathcal{F}_\Lambda$: for example a finite lattice field, a finite collection of Fourier modes, or a field paired only with test functions after a smoothing kernel has been inserted. The regulated functional integral has the form

$$Z_\Lambda = \int D_\Lambda^{\mathrm{ref}}\phi \, \exp\left[-\int \mathrm{d}^D x \, \mathcal{L}_{E,\Lambda}\right],$$

where $D_\Lambda^{\mathrm{ref}}\phi$ is the finite-regulator reference density of Definition 11.3. In the alternative Gaussian-reference convention, the quadratic part of $\mathcal{L}_{E,\Lambda}$ is absorbed into $\mathrm{d}\mu_{C_\Lambda}$, and the remaining interaction is integrated with density $\exp\{-L_\Lambda[\phi]\}\mathrm{d}\mu_{C_\Lambda}(\phi)$. The Euclidean Lagrangian appearing in the path integral is written as

$$\mathcal{L}_{E,\Lambda} = \frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi + \frac{1}{2}m^2\phi^2 + \sum_I g_I\,\mathcal{O}_I(\phi).$$

Each $\mathcal{O}_I$ is a local polynomial, or more generally a local function, of $\phi(x)$ and finitely many derivatives $\partial_\mu\phi(x), \partial_\mu\partial_\nu\phi(x), \ldots$. The index I includes all discrete labels required to specify the monomial: the number of fields, the derivative multi-indices, and the internal symmetry contraction when such indices are present. Operators that differ by a total derivative or by a chosen integration-by-parts convention are represented once. The coefficient g_I is a regulated local coordinate. The collection of regulated coordinates includes the parameters chosen as finite inputs and the regulator-dependent coordinates later called counterterms.

Two regulator languages will be used repeatedly. A momentum cutoff restricts Fourier modes to

$$p^2 < \Lambda^2$$

in a Euclidean invariant notation, or to a corresponding finite Brillouin zone on a lattice. Dimensional regularization assigns analytic meaning to momentum integrals in

$$D = d - \varepsilon$$

and removes the regulator by taking $\varepsilon \to 0$ after a subtraction prescription has been specified. These two languages belong to different rows of Table 11.1. The cutoff language denotes a finite-regulator field integral only after the regulated field space and density have been supplied. Dimensional regularization is not a measure on $\mathcal{F}_\Lambda$; it is a meromorphic assignment to the formal perturbative graph distributions and their tensor algebra. In both languages the regulated coordinates are allowed to depend on the regulator or analytic continuation parameter.

Definition 31.1 (Finite-parameter renormalized family)**.** Choose an open set $U \subset \mathbb{R}^N$ with coordinates

$$\lambda = (\lambda_1, \ldots, \lambda_N),$$

and choose, for each regulator value, a map from U to the regulated coordinate space,

$$b_\Lambda \colon U \longrightarrow \{g_I\}_{I\in\mathcal{I}}, \qquad \lambda \longmapsto g_I(\lambda, \Lambda),$$

or, in dimensional regularization,

$$b_\varepsilon \colon U \longrightarrow \{g_I\}_{I\in\mathcal{I}}, \qquad \lambda \longmapsto g_I(\lambda, \varepsilon).$$

The coordinates λ_a are renormalized coordinates if they are fixed by finite conditions on regulated correlation functions or 1PI kernels and if, after inserting the corresponding map b_Λ or b_ε, the chosen class of renormalized observables has a regulator-independent limit.

Definition 31.2 (Counterterm coordinates and insertion regularization)**.** For a fixed regulator and a fixed renormalized coordinate chart $\lambda \mapsto b_\Lambda(\lambda)$, there is one regulated action in the path integral,

$$S_\Lambda[\phi;\lambda] = \int \mathrm{d}^D x\, \mathcal{L}_{E,\Lambda}\big(\phi; \{g_I(\lambda,\Lambda)\}\big).$$

A *counterterm* is not a second Lagrangian added to this action from outside. It is the regulator-dependent part of the same local coordinate map b_Λ, described relative to a chosen finite reference chart λ. Thus the split into finite reference terms and counterterm terms is a bookkeeping split of $S_\Lambda[\phi;\lambda]$.

An *insertion regularization* is separate data. Before operator mixing or continuum matching is discussed, an inserted operator is represented by a regulated functional

$$\mathcal{O}^{\mathrm{ins}}_{a,\Lambda}(x;\phi)$$

or equivalently by a derivative with respect to a regulated local source. A lattice representative, a smeared functional, a point-split expression, a heat-kernel expression, a dimensional-regularization graph rule, and a source-derivative prescription are different insertion regularizations. A renormalized operator $[O_A]_\lambda$ is the finite source coordinate or finite linear combination obtained after the regulator-dependent insertion map and the action-coordinate map have been fixed and the chosen continuum limit exists. Regularization supplies finite or meromorphic objects before the limit; renormalization is the coordinate choice that produces finite matched observables and operator insertions.

Thus a perturbatively renormalizable family consists of a finite-dimensional family of coordinate maps into regulated Lagrangians. Masses, residues, and specified 1PI vertex values at Euclidean subtraction momenta are common choices of the coordinates λ_a, but the definition only requires the chosen conditions to be finite and locally independent. Local counterterms are the regulator-dependent local coordinates in b_Λ or b_ε that make these finite conditions compatible with the existence of the continuum limit. The phrase "existence of the continuum limit" denotes a theorem or hypothesis about a specified regulator family; selected rigorous scalar and gauge examples, together with scalar triviality regimes, are catalogued in Table 11.2.

31.2 Renormalized Fields and Effective Couplings

Let $\Gamma_{E,\Lambda}$ be the regulated Euclidean 1PI effective action, defined by the Euclidean Legendre construction of the preceding chapter. A field normalization is a regulator-dependent nonzero scalar $Z_R(\lambda, \Lambda)$, or a matrix in a theory with several fields. For a single scalar we write

$$\phi_R(x) = Z_R^{-1/2}\phi(x).$$

The normalization condition can be imposed, for instance, by requiring the coefficient of p^2 in the renormalized two-point 1PI kernel at a chosen Euclidean subtraction point to equal a prescribed finite value. The precise condition is part of the coordinate choice λ.

The Green-function finiteness criterion is distributional. For every n in the chosen class and every test function $f \in C_c^\infty((\mathbb{R}^D)^n)$, require

$$\lim_{\Lambda\to\infty} \int \prod_{a=1}^{n} \mathrm{d}^D x_a\, f(x_1, \ldots, x_n)\langle \phi_R(x_1)\cdots\phi_R(x_n)\rangle_{E,\Lambda;\lambda}$$

to exist. The same statement in dimensional regularization is the existence of the $\varepsilon \to 0$ limit after the prescribed subtractions. This is a criterion for the correlation distributions themselves; it is stronger than the statement that a finite number of Taylor coefficients is finite.

On a Euclidean momentum domain where the two-point kernel is invertible and no infrared singularity is being tested, the connected Green functions are reconstructed from the exact 1PI kernels by trees. Therefore, order by order in perturbation theory, finiteness of the renormalized 1PI kernels

$$\Gamma^{(n)}_{E,R,\Lambda}(p_1, \ldots, p_n; \lambda) = Z_R^{n/2}\Gamma^{(n)}_{E,\Lambda}(p_1, \ldots, p_n; b_\Lambda(\lambda))$$

as distributions on such momentum domains is equivalent to finiteness of the connected functions on the corresponding domains. This equivalence uses the Legendre relation and the inverse two-point kernel.

For local counterterm classification one extracts only the local Taylor coefficients of the renormalized 1PI action. Define

$$\widetilde{\Gamma}_{E,\Lambda}[\phi_R] = \Gamma_{E,\Lambda}[Z_R^{1/2}\phi_R].$$

Near a subtraction point at which the relevant kernels are analytic, write the local part as

$$\widetilde{\Gamma}^{\mathrm{loc}}_{E,\Lambda}[\phi_R] = S_{\mathrm{kin}}[\phi_R] + \int \mathrm{d}^D x \sum_I g^{\mathrm{eff}}_{I,\Lambda}(\lambda)\, \mathcal{O}_I(\phi_R).$$

Here S_{kin} is the chosen finite quadratic reference action and $\widetilde{\Gamma}^{\mathrm{loc}}$ denotes the Taylor-polynomial part in external momenta. The local 1PI finiteness condition is

$$\lim_{\Lambda\to\infty} g^{\mathrm{eff}}_{I,\Lambda}(\lambda) \qquad \text{finite}$$

for every local operator coefficient included in the finite-list renormalized description. Nonlocal terms in the exact 1PI action, such as threshold logarithms, belong to the finite continuum correlation functions.

The same structure appears directly in the regulated action. Suppose the cutoff-dependent quadratic coordinate is represented by

$$S_{\mathrm{quad}}[\phi] = \frac{1}{2}\int \mathrm{d}^D x\, \phi(-\partial^2 + m_0^2)\phi.$$

After $\phi = Z_R^{1/2}\phi_R$, choose a finite mass parameter m_R and write

$$\begin{aligned} S_{\mathrm{quad}}[Z_R^{1/2}\phi_R] = \frac{1}{2}\int \mathrm{d}^D x\, \phi_R(-\partial^2 + m_R^2)\phi_R \\ + \frac{Z_R - 1}{2}\int \mathrm{d}^D x\, \phi_R(-\partial^2 + m_R^2)\phi_R + \frac{1}{2}\delta m^2 \int \mathrm{d}^D x\, \phi_R^2. \end{aligned}$$

The first line is the finite quadratic reference action. The second line is local and is treated perturbatively as the wavefunction and mass counterterm part of the interaction.

Thus perturbation theory is organized by writing

$$\widetilde{S}_\Lambda[\phi_R] = S_{\mathrm{ref}}[\phi_R] + S_{\mathrm{int}}[\phi_R;\lambda] + S_{\mathrm{ct}}[\phi_R;\lambda,\Lambda],$$

where S_{ct} is local and regulator dependent. The finite parts of the counterterms are part of the renormalization prescription: changing them changes the coordinate system λ on the same perturbative family. The displayed split is a split of the single regulated action $\widetilde{S}_\Lambda[\phi_R]$, not a division into a physical action and an external local functional.

31.3 Locality as Extension of Distributions

The locality of counterterms has an intrinsic position-space formulation. In position space, a Feynman graph with vertices

$$x_1, \ldots, x_r$$

defines a distribution on the configuration space where no ultraviolet collision has occurred. Ultraviolet renormalization extends this distribution across the collision diagonals, such as

$$x_{i_1} = \cdots = x_{i_k}.$$

The possible ambiguities in the extension are local distributions supported on those diagonals.

Definition 31.3 (Scaling degree)**.** Let $T \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$. For $\lambda > 0$, define the scaled distribution T_λ by

$$\langle T_\lambda, f \rangle = \lambda^{-n} \left\langle T, f(\lambda^{-1} \cdot) \right\rangle, \qquad f \in C_c^\infty(\mathbb{R}^n \setminus \{0\}).$$

The scaling degree of T at the origin is

$$\mathrm{sd}_0(T) = \inf \left\{ \delta \in \mathbb{R} : \lambda^\delta T_\lambda \to 0 \text{ in } \mathcal{D}'(\mathbb{R}^n \setminus \{0\}) \text{ as } \lambda \to 0^+ \right\}.$$

The extension theorem is the distribution-theoretic core of perturbative locality.

Theorem 31.4 (Extension at a diagonal)**.** *Let $T \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ have finite scaling degree $s = \mathrm{sd}_0(T)$. There exists an extension $\overline{T} \in \mathcal{D}'(\mathbb{R}^n)$ whose restriction to $\mathbb{R}^n \setminus \{0\}$ is T and whose scaling degree is still s. If $s < n$, this extension is unique. If $s \geq n$, any two extensions of scaling degree s differ by*

$$\sum_{|\alpha| \leq \lfloor s \rfloor - n} c_\alpha \, \partial^\alpha \delta^{(n)}(x),$$

with constants c_α.

Proof. We give the proof in the one-diagonal model; the general diagonal is reduced to this model by separating center-of-mass and relative coordinates. Let $m = \lfloor s \rfloor - n$. If $s < n$, choose a smooth cutoff χ that is equal to 1 near 0, and set $\chi_\epsilon(x) = \chi(x/\epsilon)$. For $f \in C_c^\infty(\mathbb{R}^n)$, define

$$\langle \overline{T}, f \rangle = \lim_{\epsilon \to 0^+} \langle T, (1 - \chi_\epsilon) f \rangle.$$

The scaling-degree hypothesis with $s < n$ implies that the contribution from the annulus $\epsilon < |x| < 2\epsilon$ is $O(\epsilon^{n-s-\eta})$ for every $\eta > 0$ small enough; hence the limit exists and is independent of the cutoff. This defines an extension with the same scaling degree. If two such extensions existed, their difference would be supported at 0. A distribution supported at 0 is a finite sum $\sum c_\alpha \partial^\alpha \delta$, and $\mathrm{sd}_0(\partial^\alpha \delta) = n + |\alpha| \geq n$. Since $s < n$, no nonzero supported-at-zero difference is allowed. This proves existence and uniqueness in the convergent case.

Assume now $s \geq n$. Choose test functions $w_\alpha \in C_c^\infty(\mathbb{R}^n)$, indexed by $|\alpha| \leq m$, such that

$$\partial^\beta w_\alpha(0) = \delta_{\alpha\beta} \qquad (|\alpha|, |\beta| \leq m).$$

Define the Taylor-subtraction projector

$$Wf = f - \sum_{|\alpha| \leq m} (\partial^\alpha f)(0) \, w_\alpha.$$

Then all derivatives of Wf at the origin through order m vanish. This vanishing improves the small-scale behavior exactly enough that the same cutoff limit

$$\langle \overline{T}_0, f \rangle = \lim_{\epsilon \to 0^+} \langle T, (1 - \chi_\epsilon) Wf \rangle$$

exists and has scaling degree s. The functional $\overline{T}_0$ agrees with T on test functions supported away from 0, because then the subtraction terms may be chosen away from the support and the cutoff is eventually zero on the support.

The choice of w_α and the values assigned to the Taylor coefficients at the origin are not canonical. Adding

$$\sum_{|\alpha|\le m} c_\alpha\, \partial^\alpha \delta^{(n)}$$

keeps the restriction to $\mathbb{R}^n \setminus \{0\}$ and has scaling degree at most $n+m \le s$. Conversely, if $\overline{T}_1$ and $\overline{T}_2$ are two extensions with scaling degree s, their difference is supported at 0, so it is a finite sum of derivatives of δ. The scaling-degree bound excludes all terms with $n+|\alpha| > s$, hence $|\alpha| \le \lfloor s \rfloor - n = m$. This is exactly the displayed ambiguity. □

Remark 31.5 (Ultraviolet content of the scaling-degree theorem). This theorem is local at the diagonal. Its hypotheses already assume that a distribution T is defined on the punctured configuration space $\mathbb{R}^n \setminus \{0\}$, and the conclusion controls only the missing ultraviolet point 0. It does not assert that a massless graph or Schwinger function exists as a global distribution before infrared renormalization, nor does it control large-distance behavior, zero-momentum limits, or exceptional momentum configurations.

For example, a massless Euclidean propagator may have finite scaling degree at the origin as a distribution on compactly supported test functions, while an integrated graph containing it may fail to have an infrared-finite meaning without an additional prescription. If an amplitude is already infrared divergent before the collision point is restored, then the assumption $T \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ has failed. In massless critical applications the extension theorem is therefore used only after an infrared-safe setup has been specified: nonexceptional external momenta, a finite-volume or small-mass limit, a Wilsonian cutoff construction, BPHZL subtractions, or an equivalent prescription. With that extra datum in place, the scaling degree classifies the local ultraviolet ambiguity, not the infrared existence problem.

For a graph subconfiguration collapsing to a diagonal, translation invariance removes the center-of-mass coordinate, so $n = D(k-1)$ for k colliding vertices. The integer

$$\omega = \lfloor \mathrm{sd}_0(T) \rfloor - n$$

is the degree of local ambiguity. A derivative $\partial^\alpha \delta^{(n)}(x)$ supported on the diagonal is, after Fourier transform, a polynomial of degree $|\alpha|$ in the momenta entering the collapsed subgraph. Thus the extension theorem gives exactly the local Taylor polynomial counterterms used in momentum-space renormalization.

Multiple collision diagonals meet along smaller diagonals. Choosing extensions compatibly over this stratified configuration space is the position-space form of the subdivergence problem. The forest formulas in the next chapter organize the same compatibility in momentum space: each divergent subgraph corresponds to one collision diagonal, and each counterterm is the allowed derivative-of-delta ambiguity on that diagonal.

31.4 First Counterterm Census in Four Dimensions

The local nature of the required terms is visible in low-order scalar examples. Vacuum-energy terms are omitted from the census because they affect only the zero-point functional. Linear tadpole terms are fixed by the chosen expansion point; in a theory without a symmetry forbidding them, their counterterms are included among the local coordinates. In the following diagrams, a shaded disk labelled 1PI denotes the full one-particle-irreducible vertex being tested after this vacuum and tadpole bookkeeping has been imposed.

First take a $D = 4$ scalar theory with a cubic vertex $\int g_3\phi^3$, expanded about a point at which the one-point function has been fixed by a tadpole condition. The one-loop two-point graph contains the integral

$$g_3^2 \int^\Lambda \frac{\mathrm{d}^4 q}{(2\pi)^4} \frac{1}{(q^2+m^2)^2} \sim g_3^2 \log \Lambda.$$

It contributes to the local operator ϕ^2, and is canceled by a mass counterterm. The corresponding one-loop triangle and box integrals are ultraviolet finite in four dimensions.

For a $D = 4$ scalar theory with a quartic vertex $\int g_4\phi^4$, the one-loop two-point 1PI graph is a tadpole. Up to the normalization chosen for the quartic vertex, it has the form

$$\Sigma^{(1)}(p) = C_4 g_4 \int^\Lambda \frac{\mathrm{d}^4 q}{(2\pi)^4} \frac{1}{q^2+m^2},$$

where C_4 is a numerical combinatorial constant. This expression is independent of the external momentum p. It therefore contributes to the local operator ϕ^2, but not to $\partial_\mu\phi\,\partial_\mu\phi$; equivalently the one-loop coefficient of the wavefunction counterterm is

$$\delta Z^{(1)}_{\phi^4} = 0.$$

The first logarithmically divergent momentum-dependent two-point contribution in ϕ^4_4 is the two-loop sunset graph, whose Taylor coefficient at order p^2 is local and is canceled by δZ. The vanishing of $\delta Z^{(1)}_{\phi^4}$ is only a one-loop census result; it is not a criterion for excluding the kinetic counterterm from the local chart. The four-point 1PI function already has a logarithmic divergence at one loop. The candidate $D = 4$, ϕ^4 local chart therefore contains

$$\delta m^2\phi^2, \qquad \delta Z\,\partial_\mu\phi\,\partial_\mu\phi, \qquad \delta g_4\phi^4.$$

The power-counting argument in the next section then proves that this candidate list is closed: higher n-point 1PI functions with $n \geq 6$ have negative superficial degree of divergence.

For $D = 4$ scalar theory with a sextic vertex $\int g_6\phi^6$, the first tests already generate both a cutoff-dependent correction to the sextic coupling and higher local terms. The six-point 1PI kernel receives a quadratically divergent local contribution of the form

$$\delta g_6\,\phi^6, \qquad \delta g_6 = O(g_6^2\Lambda^2)$$

up to logarithms and finite prescription-dependent terms. The eight-point 1PI kernel has logarithmic behavior,

$$\delta g_8\,\phi^8, \qquad \delta g_8 = O(g_6^2 \log \Lambda).$$

At the next stages, the same local-subgraph mechanism generates

$$\delta g_{10}\phi^{10}, \qquad \delta g_{12}\phi^{12}, \qquad \ldots.$$

The cutoff-removal limit then requires an unbounded sequence of independent local coefficients.

Figure 31.1 summarizes this first counterterm census: the finite local list for renormalizable scalar interactions in four dimensions and the proliferation generated by a sextic interaction.

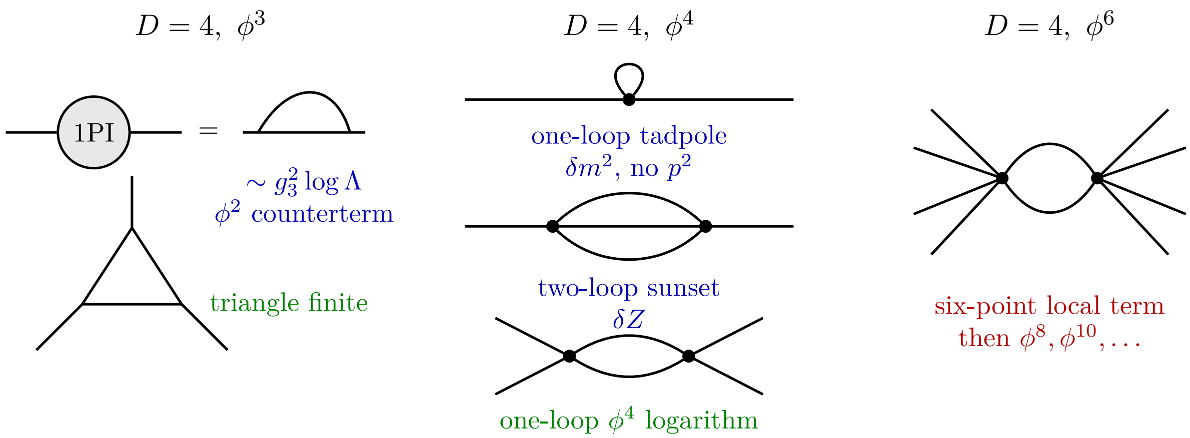


Figure 31.1: A first counterterm census in four dimensions, after vacuum-energy and tadpole bookkeeping has been fixed. Cubic and quartic scalar interactions require local terms from a finite operator list. In the quartic theory the one-loop tadpole fixes only the mass counterterm, while the field-strength counterterm first appears through the two-loop sunset. A sextic interaction generates higher local operators under loop corrections.

31.5 Engineering Dimensions and the Finite List

The preceding examples are organized by engineering dimension. Assign dimension $[x] = -1$ and $[\partial_\mu] = 1$. The Euclidean action is dimensionless. Since the kinetic term is

$$\frac{1}{2}\int \mathrm{d}^D x\, \partial_\mu\phi\, \partial_\mu\phi,$$

the field has engineering dimension

$$[\phi] = \frac{D-2}{2}.$$

Let $\mathcal{O}_I$ contain n_I fields and ℓ_I derivatives. Its engineering dimension is

$$d_I = \ell_I + n_I\frac{D-2}{2},$$

and the coupling multiplying it has dimension

$$[g_I] = D - d_I.$$

Lemma 31.6 (Power-counting finite-list criterion)**.** *Let G be a graph contributing to the coefficient of a local operator $\mathcal{O}_I$ in the 1PI effective action. Denote by $V(G)$ the multiset of interaction vertices in the graph, and let d_v be the engineering dimension of the local operator at v. After the external momentum derivatives that extract $\mathcal{O}_I$ have been taken, the superficial ultraviolet scaling has the form*

$$\left(\prod_{v\in V(G)} g_v\right)\Lambda^{(D-d_I)-\sum_{v\in V(G)}(D-d_v)}\big(\log\Lambda\big)^r, \qquad 0 \le r \le L,$$

where L is the loop number. An overall local counterterm for $\mathcal{O}_I$ can be required only when

$$D - d_I - \sum_{v \in V(G)} (D - d_v) \geq 0. \tag{31.1}$$

If all interaction vertices present in the regulated local chart have $d_v \leq D$, then every superficially required overall local counterterm has $d_I \leq D$. For scalar polynomial interactions with finitely many derivatives at fixed D, after imposing the chosen symmetries and integration-by-parts conventions, the space of such local operators is finite dimensional.

Proof. The coefficient g_I^{eff} of $\mathcal{O}_I$ in the 1PI effective action is extracted by differentiating an n_I-point 1PI vertex with respect to the appropriate external momenta and then setting those momenta to zero, or to a specified Euclidean subtraction point, provided the chosen point is infrared regular. Under the ultraviolet rescaling of all loop momenta by Λ, the coefficient of a dimension-d_I local operator must have engineering dimension $D - d_I$. The product of vertices contributes the dimension $\sum_v (D - d_v)$. Therefore the remaining cutoff dependence has the power displayed in the statement; logarithms can occur only when homogeneous powers collide or after nested integrations, and their degree is bounded at fixed loop order. This is a statement about the overall ultraviolet scaling after subdivergences have either been absent or treated recursively by the next chapter's forest formula.

If every coupling appearing in the action has $d_v \leq D$, then $D - d_v \geq 0$. Inequality (31.1) implies $d_I \leq D$. For scalar polynomial interactions with finitely many derivatives at fixed D, after imposing the chosen symmetries and integration-by-parts conventions there are only finitely many local operators with $d_I \leq D$. This is the power-counting finite-list criterion for perturbative renormalizability: every local ambiguity allowed by superficial ultraviolet scaling lies in a fixed finite-dimensional local operator space. □

If an interaction with $d_J > D$ is included and its coefficient is allowed to participate in the cutoff-removal problem on the same footing, repeated insertions can satisfy the inequality for operators of arbitrarily large dimension. Such a Lagrangian is still a valid low-energy effective description at finite cutoff. Its cutoff-removal problem, with that interaction iterated to all orders on the same footing as marginal or relevant couplings, requires an unbounded sequence of independent local coordinates by this power-counting criterion.

31.6 A Renormalizable Model in Six Dimensions

The scalar cubic interaction is marginal in $D = 6$. Use the Euclidean Lagrangian, written in terms of the renormalized field,

$$\begin{aligned}\mathcal{L}_E = \frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi + \frac{1}{2}m_R^2\phi^2 + \delta Z\left(\frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi + \frac{1}{2}m_R^2\phi^2\right) \\ + \frac{1}{2}\delta m^2\phi^2 + \frac{1}{3!}(g_R + \delta g)\phi^3.\end{aligned}$$

In dimensional regularization with $D = 6 - \varepsilon$, the last term is understood as

$$\frac{\mu^{\varepsilon/2}}{3!}(g_R + \delta g)\phi^3,$$

so that g_R is dimensionless and μ is the subtraction scale. The finite parameters are m_R, g_R, and the normalization convention for ϕ. The counterterm coordinates are

$$\delta Z, \qquad \delta m^2, \qquad \delta g.$$

They are functions of m_R, g_R, μ and of the regulator.

Lemma 31.7 (One-loop local poles in six-dimensional cubic theory)**.** *In the $D = 6 - \varepsilon$ scalar cubic model above, the order-g_R^2 self-energy pole is a local polynomial*

$$\Sigma_2(k)\big|_{\text{pole}} = -\frac{g_R^2}{6(4\pi)^3}\frac{k^2}{\varepsilon} - \frac{g_R^2}{(4\pi)^3}\frac{m_R^2}{\varepsilon} + O(\varepsilon^0).$$

It is canceled by δZ and δm^2. The order-g_R^3 one-loop three-point pole is local in the cubic vertex and is canceled by δg, while the one-loop four-point box is ultraviolet finite by power counting.

Proof. At order g_R^2, the one-loop self-energy is

$$\Sigma_2(k) = \frac{1}{2}g_R^2 \int \frac{\mathrm{d}^D\ell}{(2\pi)^D} \frac{1}{(\ell^2 + m_R^2)((k-\ell)^2 + m_R^2)}.$$

Combining denominators gives

$$\Sigma_2(k) = \frac{1}{2}g_R^2 \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D\ell}{(2\pi)^D} \frac{1}{\left(\ell^2 + x(1-x)k^2 + m_R^2\right)^2}.$$

In dimensional regularization,

$$\int \frac{\mathrm{d}^D\ell}{(2\pi)^D} \frac{1}{(\ell^2 + \Delta)^n} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n - D/2)}{\Gamma(n)} \Delta^{D/2-n}.$$

For $D = 6 - \varepsilon$, this gives

$$\Sigma_2(k) = \frac{1}{2}g_R^2 \frac{\Gamma(-1+\varepsilon/2)}{(4\pi)^{3-\varepsilon/2}} \int_0^1 \mathrm{d}x \left(m_R^2 + x(1-x)k^2\right)^{1-\varepsilon/2}.$$

The pole part is therefore a local polynomial

$$\Sigma_2(k)\big|_{\text{pole}} = A_1(\varepsilon)\, k^2 + B_1(\varepsilon)\, m_R^2,$$

where A_1 and B_1 have simple $1/\varepsilon$ poles in minimal subtraction. In the normalization of the displayed integral,

$$A_1(\varepsilon) = -\frac{g_R^2}{6(4\pi)^3}\frac{1}{\varepsilon} + O(\varepsilon^0), \qquad B_1(\varepsilon) = -\frac{g_R^2}{(4\pi)^3}\frac{1}{\varepsilon} + O(\varepsilon^0),$$

where the finite parts depend on the subtraction prescription. The finite renormalized self-energy has the large-momentum form

$$\Sigma_{2,R}(k) = (\alpha k^2 + \beta m_R^2)\log\frac{k^2}{\mu^2} + c_1 k^2 + c_0 m_R^2 + O(m_R^4/k^2) \qquad (k^2 \gg m_R^2)$$

for finite constants α, β, c_0, c_1 fixed by the chosen finite renormalization. The counterterms δZ and δm^2 are chosen so that the pole part is canceled and $\Sigma_{2,R}$ is finite at the subtraction point.

At order g_R^3, the one-loop three-point triangle has a logarithmic ultraviolet divergence. Write the corresponding scalar kernel, with the overall graph symmetry and sign convention suppressed because they do not affect locality, as

$$I_3(p_1, p_2, p_3) = \int \frac{\mathrm{d}^D \ell}{(2\pi)^D} \frac{1}{(\ell^2 + m_R^2)((\ell + p_1)^2 + m_R^2)((\ell - p_3)^2 + m_R^2)}, \qquad p_1 + p_2 + p_3 = 0. \tag{31.2}$$

Feynman parameters give

$$\begin{aligned} I_3(p_1, p_2, p_3) = 2 \int_{\Delta_2} & \mathrm{d}x_1 \mathrm{d}x_2 \mathrm{d}x_3 \, \delta(1 - x_1 - x_2 - x_3) \\ & \times \int \frac{\mathrm{d}^D L}{(2\pi)^D} \frac{1}{(L^2 + \Delta_3(x, p, m_R))^3}, \end{aligned} \tag{31.3}$$

where the shifted loop momentum L removes the linear term and Δ_3 is a polynomial in m_R^2 and the pairwise external invariants. Using the same dimensional-regularization formula as above,

$$\int \frac{\mathrm{d}^D L}{(2\pi)^D} \frac{1}{(L^2 + \Delta)^3} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(3 - D/2)}{2} \Delta^{D/2 - 3}.$$

For $D = 6 - \varepsilon$,

$$\frac{\Gamma(3 - D/2)}{2} \Delta^{D/2 - 3} = \frac{1}{\varepsilon} + O(\varepsilon^0),$$

and the simplex volume, including the prefactor 2 in (31.3), is one. Hence the triangle pole is independent of p_1, p_2, p_3:

$$I_3(p_1, p_2, p_3)\big|_{\text{pole}} = \frac{1}{(4\pi)^3} \frac{1}{\varepsilon} + O(\varepsilon^0). \tag{31.4}$$

Equivalently, write the difference from $I_3(0, 0, 0)$ as an integral of the first derivative of the integrand along the straight line from zero external momenta to (p_1, p_2, p_3). That derivative has large-ℓ behavior $O(|\ell|^{-7})$, so the six-dimensional radial integral is convergent at infinity. The pole therefore multiplies exactly the local cubic vertex and is canceled by δg. The one-loop four-point box in $D = 6$ is ultraviolet finite by power counting: four propagators give large-ℓ behavior $\mathrm{d}^6 \ell \, \ell^{-8}$. □

Figure 31.2 displays the two one-loop kernels used in this six-dimensional ϕ^3 check: the two-point kernel with mass and kinetic local parts, and the local three-point vertex subtraction.

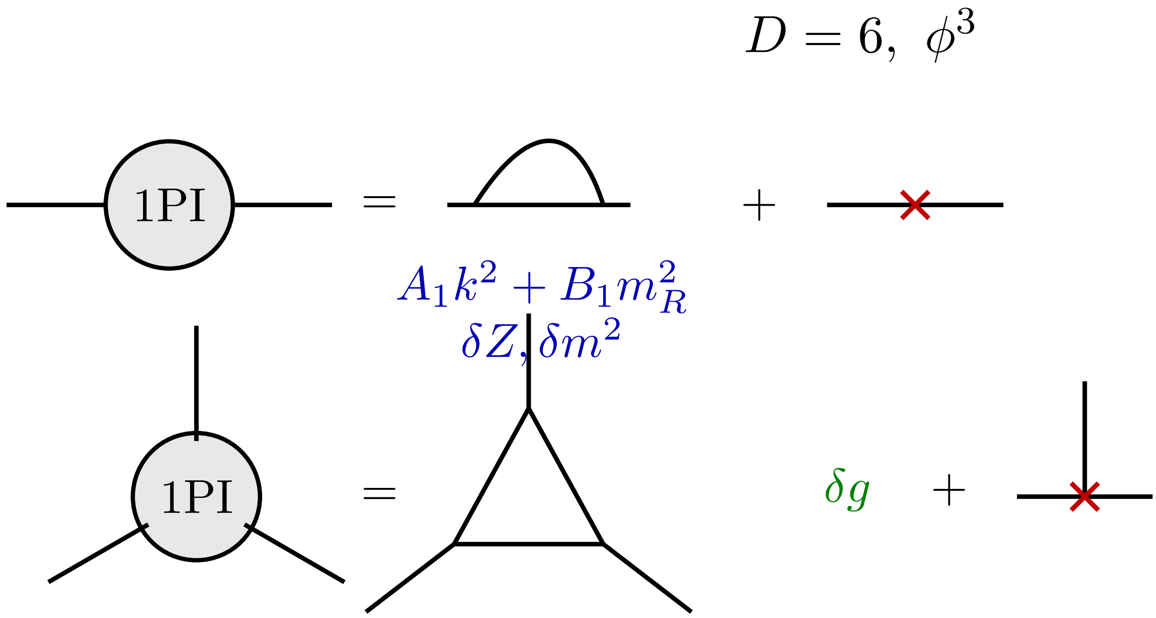


Figure 31.2: In six-dimensional ϕ^3 theory, the one-loop two-point divergence is local in k^2 and m_R^2, and the one-loop three-point divergence is local in the cubic vertex.

31.7 Locality at Higher Order

The one-loop renormalized self-energy in the six-dimensional cubic theory has large Euclidean momentum behavior

$$\Sigma_R(q) \sim (\alpha q^2 + \beta m_R^2) \log \frac{q^2}{\mu^2}, \qquad q^2 \gg m_R^2,$$

for finite constants α, β determined by the subtraction scheme. When this renormalized subgraph is inserted into a larger self-energy graph, the ultraviolet region of the remaining integral is represented by

$$\int \frac{\mathrm{d}^6 q}{(2\pi)^6} \frac{\Sigma_R(q)}{(q^2 + m_R^2)^2((k-q)^2 + m_R^2)}.$$

The external momentum dependence is isolated by expanding

$$\frac{1}{(k-q)^2 + m_R^2} = \frac{1}{q^2 + m_R^2} - \frac{k^2 - 2k \cdot q}{(q^2 + m_R^2)^2} + \frac{4(k \cdot q)^2}{(q^2 + m_R^2)^3} + R_3(q, k),$$

where Taylor's formula gives $R_3(q,k) = O(|k|^3 |q|^{-5})$ for $|q| \gg |k|, m_R$. Because

$$\frac{\Sigma_R(q)}{(q^2 + m_R^2)^2} = O\left(\frac{\log q^2}{q^2}\right), \qquad |q| \to \infty,$$

the remainder contributes an integrand

$$\mathrm{d}^6 q \; O\left(\frac{\log q^2}{|q|^7}\right),$$

which is ultraviolet convergent. The displayed Taylor terms are local in the external momentum through order k^2. After angular integration, odd powers of q vanish and $(k \cdot q)^2$ becomes proportional to $k^2 q^2$. Thus the divergent part contributes only to the ϕ^2 and $(\partial\phi)^2$ counterterms.

Dimensional regularization expresses the same fact algebraically. Integrals of the form

$$\int \frac{\mathrm{d}^D q}{(2\pi)^D} \frac{\log(q^2+\Delta)}{(q^2+\Delta)^n} = -\frac{\partial}{\partial n}\left[\frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n-D/2)}{\Gamma(n)} \Delta^{D/2-n}\right]$$

produce double and single poles in ε. For the self-energy insertion, the pole part has the local form

$$\Sigma_{\text{insertion}}(k)\big|_{\text{pole}} = A_\varepsilon k^2 + B_\varepsilon m_R^2, \qquad A_\varepsilon = \frac{a_2}{\varepsilon^2} + \frac{a_1}{\varepsilon}, \qquad B_\varepsilon = \frac{b_2}{\varepsilon^2} + \frac{b_1}{\varepsilon}.$$

The coefficients a_i, b_i are finite constants depending on the finite renormalization prescription. The pole part is polynomial in the external momentum, so it is represented by local counterterms already present in the six-dimensional cubic Lagrangian. The absence of a pole of the form

$$\frac{1}{\varepsilon} \log k^2$$

is essential: such a term would be nonlocal in position space and could not be canceled by a local Lagrangian counterterm. The reason it is absent is visible before integration over Feynman parameters. For $D = 6 - \varepsilon$ and $n = 2, 3$,

$$\frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n-D/2)}{\Gamma(n)} \Delta^{D/2-n} = \left(\frac{c_n}{\varepsilon} + c_n' \log \Delta\right) \Delta^{3-n} + O(\varepsilon^0),$$

with constants c_n, c_n'. Differentiating with respect to n, as in the logarithmic integral above, produces

$$\left(\frac{c_{n,2}}{\varepsilon^2} + \frac{c_{n,1}}{\varepsilon}\right) \Delta^{3-n} + O(\varepsilon^0),$$

because the $\varepsilon^{-1} \log \Delta$ pieces cancel between the derivative of the gamma function and the derivative of $\Delta^{D/2-n}$.

Figure 31.3 isolates the local k^0 and k^2 terms left by a renormalized self-energy insertion before the remaining large-momentum integral is made convergent.

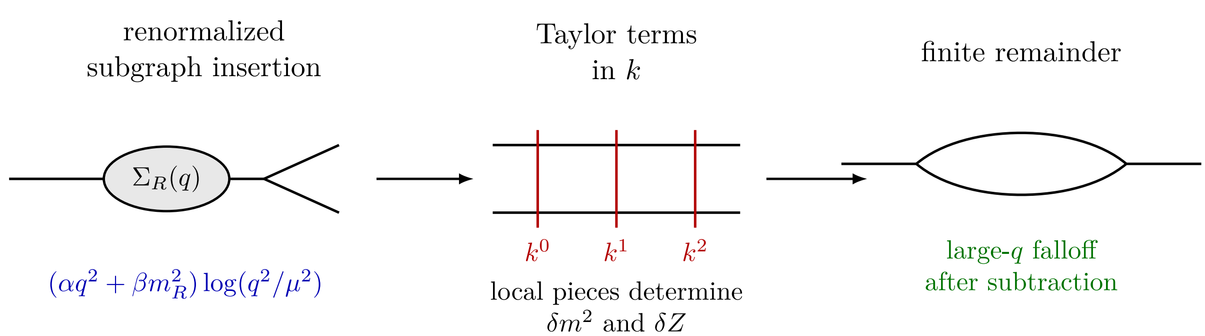


Figure 31.3: A renormalized subgraph can leave logarithmic large-momentum behavior. Expanding the remaining propagator in the external momentum separates the local k^0 and k^2 terms from a convergent remainder.

31.8 Schwinger Parameters and Subdivergent Regions

The same mechanism appears geometrically in Schwinger parameters. Consider the two-loop self-energy graph in $D = 6$ cubic theory with two loop momenta ℓ_1, ℓ_2:

$$\frac{g_R^4}{2} \int \frac{\mathrm{d}^6 \ell_1}{(2\pi)^6} \frac{\mathrm{d}^6 \ell_2}{(2\pi)^6} P(\ell_1) P(k - \ell_1) P(\ell_2) P(k - \ell_2) P(\ell_1 - \ell_2),$$

where

$$P(\ell) = \frac{1}{\ell^2 + m_R^2}.$$

Using

$$P(\ell) = \int_0^\infty \mathrm{d}\alpha \, \mathrm{e}^{-\alpha(\ell^2 + m_R^2)},$$

the Gaussian loop integrations produce an integrand of the form

$$\frac{1}{(4\pi)^6} \int_0^\infty \prod_{i=1}^5 \mathrm{d}\alpha_i \, \frac{\mathrm{e}^{-m_R^2 \sum_i \alpha_i - k^2 Q(\alpha)}}{(\det A)^3},$$

with

$$A = \begin{pmatrix} \alpha_1 + \alpha_2 + \alpha_5 & -\alpha_5 \\ -\alpha_5 & \alpha_3 + \alpha_4 + \alpha_5 \end{pmatrix}.$$

The function $Q(\alpha)$ is nonnegative. Ultraviolet regions are boundary regions in which Schwinger parameters of a subgraph go to zero. Explicitly,

$$Q(\alpha) = \alpha_2 + \alpha_4 - (\alpha_2, \alpha_4) A^{-1} \begin{pmatrix} \alpha_2 \\ \alpha_4 \end{pmatrix} \geq 0.$$

Writing

$$F(k^2; \alpha_1, \ldots, \alpha_5) = \frac{\exp[-k^2 Q(\alpha)]}{(\det A)^3},$$

one has the homogeneous scaling relation

$$\rho^6 F\left(\frac{k^2}{\rho}; \rho\alpha_1, \ldots, \rho\alpha_5\right) = F(k^2; \alpha_1, \ldots, \alpha_5).$$

There are two one-loop subdivergent regions:

$$\alpha_1, \alpha_2, \alpha_5 \to 0, \qquad \alpha_3, \alpha_4, \alpha_5 \to 0.$$

They correspond to shrinking the left or right one-loop self-energy subgraph. The counterterm insertions for the one-loop self-energy subtract the limiting functions associated with these boundary regions. At the level of the Schwinger-parameter integrand this replacement is

$$\begin{aligned} \widetilde{F} = F &- \lim_{\rho \to 0} \rho^3 F(k^2; \rho\alpha_1, \rho\alpha_2, \alpha_3, \alpha_4, \rho\alpha_5) \\ &- \lim_{\rho \to 0} \rho^3 F(k^2; \alpha_1, \alpha_2, \rho\alpha_3, \rho\alpha_4, \rho\alpha_5). \end{aligned}$$

After these subtractions, the remaining overall ultraviolet region is the simultaneous scaling

$$\alpha_1, \dots, \alpha_5 \to 0.$$

Power counting then says that its dependence on the external momentum is at most quadratic. Subtracting the k^0 and k^2 Taylor terms leaves a finite parameter integral. The overall counterterm is therefore again a linear combination of ϕ^2 and $(\partial\phi)^2$. Equivalently, the finite remainder may be represented by replacing $\widetilde{F}$ with

$$\widetilde{F}_R(k^2;\alpha) = \widetilde{F}(k^2;\alpha) - \widetilde{F}(0;\alpha) - \left.\frac{\partial \widetilde{F}}{\partial k^2}\right|_{k^2=0} k^2.$$

Figure 31.4 translates the same locality statement into Schwinger parameters: one-loop boundary faces are subtracted first, and the overall boundary face then supplies the remaining local terms.

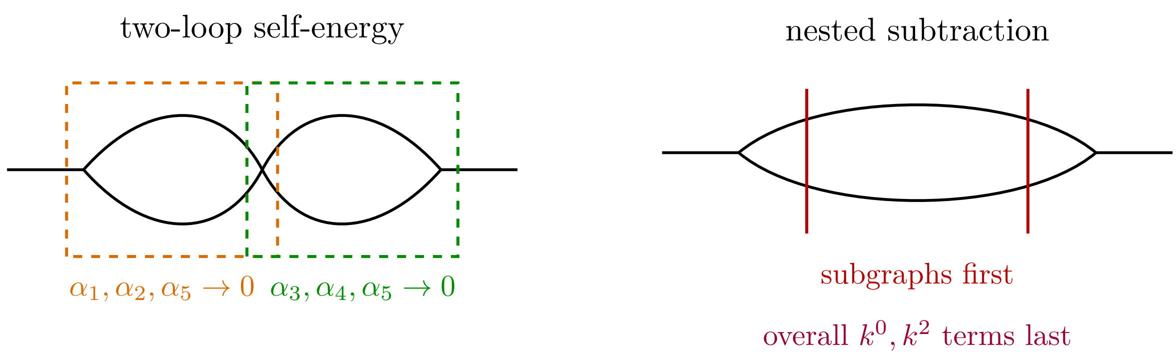


Figure 31.4: In Schwinger parameters, ultraviolet divergences are boundary regions. Subdivergent one-loop regions are subtracted before the remaining overall boundary region is expanded in local external-momentum terms.

The preceding analysis is the local core of perturbative renormalizability for power-counting finite-list theories. At all orders, the same nested structure is organized by the Bogoliubov–Parasiuk–Hepp–Zimmermann theorem: subtract local Taylor polynomials for divergent subgraphs in a compatible nested order, and the remaining integrals are finite. The next chapter develops this subtraction structure as a systematic forest formula.

Chapter 32

Subdivergences and Forest Formulas

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

32.1 Path-Integral Status: Formal Coefficientwise Renormalization

The construction in this chapter is a perturbative construction extracted from a regulated Euclidean source functional. Let $\langle - \rangle_{0,\Lambda}$ be a specified ultraviolet-regulated Gaussian expectation and let V_Λ be a finite linear combination of regulated local interaction functionals with couplings

$$g = (g_1, \ldots, g_r), \qquad V_\Lambda[\phi] = \sum_{\alpha=1}^{r} g_\alpha O_{\alpha,\Lambda}[\phi],$$

where the subscript Λ denotes the ultraviolet regulator used to define both the covariance and the inserted local functionals. The normalized source functional is

$$Z_{\Lambda,g}[J] = \frac{\big\langle \exp\big(\langle J, \phi \rangle - V_\Lambda[\phi]\big)\big\rangle_{0,\Lambda}}{\big\langle \exp\big(-V_\Lambda[\phi]\big)\big\rangle_{0,\Lambda}},$$

understood as an identity in the formal power-series ring $\mathbb{C}[[g_1, \ldots, g_r]]$. Its correlation functions are expanded as formal power series in g:

$$\big\langle \phi(x_1) \cdots \phi(x_n) \big\rangle_{\Lambda,g} = \sum_{\nu \in \mathbb{N}^r} g^\nu\, G^{(\nu)}_{\Lambda,n}(x_1, \ldots, x_n), \qquad g^\nu = \prod_{\alpha=1}^{r} g_\alpha^{\nu_\alpha}.$$

Each coefficient $G^{(\nu)}_{\Lambda,n}$ is a finite sum of graph distributions, or equivalently of momentum-space graph integrals after a choice of momentum routing and test functions. BPHZ renormalization acts on these coefficient distributions graph by graph. For a graph Γ, the forest formula constructs a renormalized coefficient $R_\Gamma I_\Gamma$ by subtracting Taylor polynomials in the external momenta of divergent subgraphs. Only after this coefficientwise operation has been performed does one assemble the formal series again.

Thus BPHZ, as used here, is not an assertion that the formal symbol $\mathcal{D}\phi$ denotes a countably additive measure on an unregulated continuum field space, and it is not a nonperturbative construction of the theory. It is a precise local subtraction scheme for the formal perturbative expansion of regulated Euclidean Green functions and 1PI kernels. Its comparison with Wilsonian effective actions is therefore an order-by-order comparison between two coordinate systems on perturbative coefficient data, not an equality of nonperturbative path integrals.

32.2 Graphs and Ultraviolet Subgraphs

Work in a massive Euclidean scalar theory in D dimensions, with local polynomial interactions and a fixed choice of momentum routing. The massive assumption isolates ultraviolet questions from infrared ones; the subtraction points will also be chosen away from exceptional momentum configurations. The graphs in this chapter represent 1PI kernels; connected Green functions are reconstructed from 1PI kernels by the Legendre tree expansion discussed earlier. A connected one-particle-irreducible Feynman graph Γ has a finite set $E(\Gamma)$ of internal lines, a finite set $V(\Gamma)$ of vertices, and external momenta

$$p = (p_1, \dots, p_N), \qquad \sum_{a=1}^{N} p_a = 0.$$

Choose independent loop momenta $\ell = (\ell_1, \dots, \ell_{L(\Gamma)})$, where

$$L(\Gamma) = |E(\Gamma)| - |V(\Gamma)| + 1.$$

With polynomial numerator $N_\Gamma(\ell, p)$ and propagator masses $m_e > 0$, the Euclidean momentum-space integrand has the form

$$I_\Gamma(\ell, p) = \frac{N_\Gamma(\ell, p)}{\prod_{e \in E(\Gamma)} \left(q_e(\ell, p)^2 + m_e^2\right)}.$$

Here $q_e(\ell, p)$ is the momentum carried by the line e, determined by the chosen momentum routing. Coupling constants, symmetry factors, and overall powers of 2π are suppressed in this chapter because they do not affect the ultraviolet degree or locality of the subtraction.

Definition 32.1 (Momentum subgraph)**.**

A momentum subgraph $\gamma \subseteq \Gamma$ is specified by a nonempty subset $E(\gamma) \subseteq E(\Gamma)$, together with all vertices incident to those internal lines. Its connected components are formed with the incidence relation inherited from Γ. The loop number of γ is

$$L(\gamma) = |E(\gamma)| - |V(\gamma)| + c(\gamma),$$

where $c(\gamma)$ is the number of connected components of γ. The external half-lines of γ are all half-lines incident to $V(\gamma)$ that are not internal lines of γ. They include original external half-lines of Γ and the half-lines obtained by cutting internal lines of $\Gamma \setminus \gamma$ at vertices of γ. The momenta on these half-lines are denoted p_γ, with momentum conservation imposed separately on each connected component. The contracted graph Γ/γ is obtained by replacing every connected component of γ by a single vertex carrying the total external momentum of that component.

The ultraviolet scaling associated with γ scales independent loop momenta internal to γ by a common parameter ρ and keeps p_γ fixed. The momenta p_γ may include loop momenta of the contracted graph; during the test of γ they are treated as external parameters. If the leading homogeneous scaling of I_γ together with the internal loop measure is

$$\mathrm{d}^{DL(\gamma)}\ell_\gamma \, I_\gamma(\rho\ell_\gamma, p_\gamma) = O(\rho^{\omega(\gamma)})\mathrm{d}^{DL(\gamma)}\ell_\gamma \qquad (\rho \to \infty),$$

then $\omega(\gamma)$ is the superficial degree of divergence. For a scalar graph with numerator degree $r(\gamma)$,

$$\omega(\gamma) = DL(\gamma) + r(\gamma) - 2|E(\gamma)|.$$

Equivalently, $\omega(\gamma)$ is the mass dimension of the local polynomial that can be produced when γ is shrunk to a point. A connected subgraph is one-particle-irreducible if it remains connected after any one of its internal lines is removed. A proper connected one-particle-irreducible subgraph with $\omega(\gamma) \geq 0$ is called a divergent subgraph; proper means $\gamma \neq \Gamma$. A disconnected subgraph is divergent when each of its connected components is divergent. In that case its local counterterm is the product of the counterterms of its connected components.

32.3 Taylor Operators and Local Vertices

Fix for each divergent subgraph γ a subtraction point p^*_γ in the space of independent external momenta of γ. The point is chosen away from infrared singularities; for massive Euclidean graphs, zero momentum is allowed when the relevant Green function is infrared regular. Let $a = (a_1, \dots, a_M)$ be affine coordinates for the independent components of $p_\gamma - p^*_\gamma$, after imposing momentum conservation. The Taylor operator of degree $\omega(\gamma)$ is

$$(t_\gamma f)(p_\gamma) = \sum_{|\alpha| \leq \omega(\gamma)} \frac{a^\alpha}{\alpha!} \, \partial_a^\alpha f(p^*_\gamma + a)\big|_{a=0}.$$

It acts on the dependence of the integrand on the external momenta of γ, while all loop momenta internal to γ remain integration variables and all momenta belonging to the contracted graph are held fixed. For a disconnected divergent subgraph, t_γ is the product of the Taylor operators for its connected components.

The image $t_\gamma I_\Gamma$ is polynomial in p_γ of degree at most $\omega(\gamma)$, with coefficients depending on the loop variables internal to γ and on the remaining variables of the contracted graph. After the internal variables of γ have been integrated or recursively prepared, this polynomial becomes a vertex factor in Γ/γ. In position space, a polynomial in the momentum entering a contracted vertex is a finite linear combination of derivatives acting at that vertex. Therefore $t_\gamma I_\Gamma$ represents a local insertion in Γ/γ. This is the graph-level form of the locality statement established in the preceding chapter.

Figure 32.1 shows this graph-level operation explicitly: Taylor expansion in the external variables of γ becomes a polynomial counterterm vertex in the contracted graph Γ/γ.

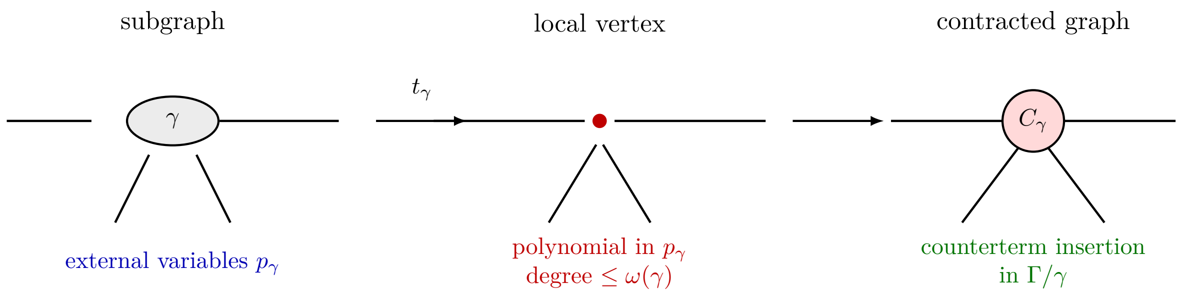


Figure 32.1: The Taylor operator t_γ extracts the polynomial part of a divergent subgraph in its external variables. The polynomial is represented in the contracted graph by a local counterterm vertex.

32.4 Spinneys, Forests, and the Recursive R-Operation

Definition 32.2 (Spinneys and forests)**.** A spinney S of Γ is a finite set of mutually disjoint connected divergent proper subgraphs of Γ. Mutually disjoint means that no internal line or vertex is shared. The empty set is allowed. A forest F of Γ is a finite set of connected divergent subgraphs such that any two elements of F are either disjoint or nested:

$$\gamma_1 \cap \gamma_2 = \emptyset, \qquad \gamma_1 \subset \gamma_2, \qquad \text{or} \qquad \gamma_2 \subset \gamma_1.$$

The full graph Γ is included among the possible forest elements when Γ is one-particle-irreducible and $\omega(\Gamma) \geq 0$. The empty forest is allowed.

Figure 32.2 fixes the compatibility relation used in the forest sum: divergent subgraphs may be disjoint or nested, whereas overlapping subgraphs appear in separate forest terms.

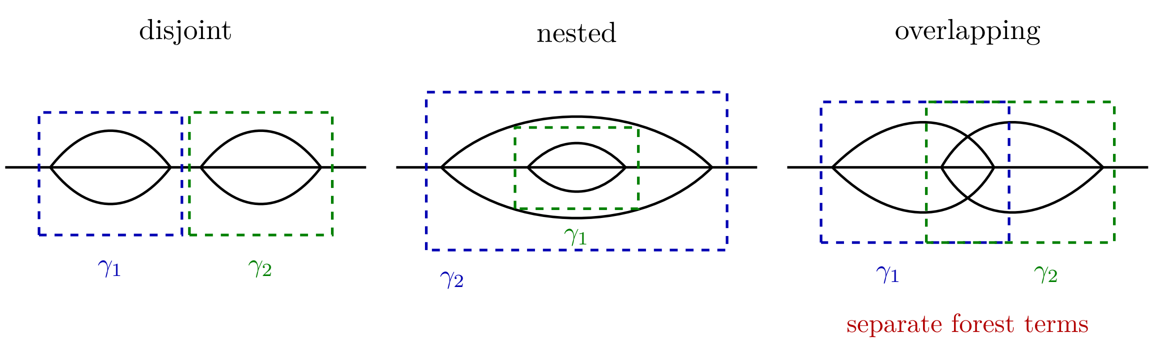


Figure 32.2: A forest contains compatible divergent subgraphs: disjoint or nested collections are compatible. Overlapping subgraphs are handled by separate forest terms.

The recursive construction is most transparent when written in two steps. First define the prepared integrand

$$\overline{R}_\Gamma I_\Gamma = I_\Gamma + \sum_{\substack{S \text{ spinney of } \Gamma \\ S \neq \emptyset}} \left(\prod_{\gamma \in S} C_\gamma \right) I_\Gamma.$$

The product of C_γ's means: for each disjoint component $\gamma \in S$, replace the prepared subgraph by its local counterterm vertex and insert the resulting product of local vertices into the contracted graph Γ/S. Since the elements of a spinney are disjoint, these replacements can be made simultaneously. The counterterm attached to a connected divergent graph is defined recursively by

$$C_\gamma I_\gamma = -\, t_\gamma \, \overline{R}_\gamma I_\gamma.$$

Thus the counterterm of γ is the negative Taylor polynomial of the subgraph after its own proper subdivergences have already been prepared. The recursion starts at divergent graphs with no proper divergent subgraphs, for which $\overline{R}_\gamma I_\gamma = I_\gamma$. For disconnected divergent subgraphs, C_γ is the product over connected components.

The full R-operation is

$$R_\Gamma I_\Gamma = \begin{cases} (1 - t_\Gamma)\, \overline{R}_\Gamma I_\Gamma, & \omega(\Gamma) \geq 0, \\ \overline{R}_\Gamma I_\Gamma, & \omega(\Gamma) < 0. \end{cases}$$

This formula enforces the order of operations: proper subdivergences are subtracted inside $\overline{R}_\Gamma$, and the overall Taylor subtraction is applied afterward when the whole graph is superficially divergent.

Proposition 32.3 (Bogoliubov preparation recursion)**.** *Fix the subtraction points and Taylor degrees described above. For every connected 1PI graph Γ, the recursive definitions of $\overline{R}_\Gamma$, C_γ, and R_Γ determine a finite linear combination of integrands obtained from Γ by replacing divergent connected subgraphs by local polynomial vertices. If γ is connected and divergent, then $C_\gamma I_\gamma$ is a polynomial in the external momenta p_γ of degree at most $\omega(\gamma)$, with coefficients that are finite linear combinations of prepared lower-loop or lower-line graph integrands.*

Proof. Order connected graphs by the number of internal lines. A proper connected subgraph of Γ has strictly fewer internal lines than Γ, so the recursion is well founded. For graphs with no proper divergent connected subgraphs, $\overline{R}_\Gamma I_\Gamma = I_\Gamma$, and, if Γ is divergent, $C_\Gamma I_\Gamma = -t_\Gamma I_\Gamma$, which is a polynomial in the external variables by the definition of t_Γ.

Assume the claim for all proper connected divergent subgraphs of Γ. A spinney is a finite set of mutually disjoint such subgraphs, so the product $\prod_{\gamma \in S} C_\gamma$ can be inserted simultaneously: each factor has already been constructed by the induction hypothesis, and disjoint replacements affect disjoint vertex and line sets. The number of spinneys is finite because Γ has finitely many internal lines. Hence $\overline{R}_\Gamma I_\Gamma$ is a finite linear combination of integrands on contracted graphs with local polynomial vertices.

If Γ itself is divergent, $C_\Gamma I_\Gamma$ is defined as $-t_\Gamma \overline{R}_\Gamma I_\Gamma$. The Taylor operator acts only on the external variables of Γ and truncates at degree $\omega(\Gamma)$; local vertices produced inside $\overline{R}_\Gamma$ are polynomial factors in slower external variables and remain local after the Taylor extraction. Thus the new counterterm is again a local polynomial vertex of the stated degree. This completes the induction. ∎

32.5 The Forest Formula

Let $\mathfrak{F}(\Gamma)$ be the set of forests whose elements are connected divergent subgraphs of Γ, including Γ itself when Γ is one-particle-irreducible and $\omega(\Gamma) \geq 0$. For $F \in \mathfrak{F}(\Gamma)$, define t_F by applying t_γ from smaller subgraphs to larger subgraphs, with each smaller subtraction regarded as a local vertex before the larger Taylor operator is applied. If two elements are disjoint, their operators commute. The forest formula is the following identity.

Theorem 32.4 (Zimmermann forest formula)**.** *For the BPHZ recursion of Proposition 32.3,*

$$R_\Gamma I_\Gamma = \sum_{F \in \mathfrak{F}(\Gamma)} (-1)^{|F|}\, t_F I_\Gamma, \tag{$*$}$$

with $t_\emptyset = 1$. The identity is an identity of formal graph integrands before loop integrations are performed. In a nested chain, Taylor operators are ordered from smaller subgraphs to larger subgraphs; for disjoint subgraphs the order is immaterial.

Proof. We prove the identity by induction on the number of internal lines. For a graph with no proper divergent connected subgraphs, the spinney sum contains only the empty spinney. If Γ is superficially convergent then $R_\Gamma I_\Gamma = I_\Gamma$, which is the forest formula with only the empty forest. If Γ is superficially divergent then

$$R_\Gamma I_\Gamma = (1 - t_\Gamma) I_\Gamma = I_\Gamma - t_\Gamma I_\Gamma,$$

which is the sum over the two forests $\emptyset$ and $\{\Gamma\}$.

Assume the statement for every proper connected divergent subgraph of Γ. Expanding the counterterm of a proper connected divergent subgraph γ gives

$$C_\gamma I_\gamma = -t_\gamma \overline{R}_\gamma I_\gamma = \sum_{\substack{F_\gamma \in \mathfrak{F}(\gamma) \\ \gamma \in F_\gamma}} (-1)^{|F_\gamma|} t_{F_\gamma} I_\gamma. \tag{32.1}$$

Indeed, by the induction hypothesis, $\overline{R}_\gamma$ is the forest sum over forests of γ not containing γ; multiplying by $-t_\gamma$ appends γ as the largest element and changes the sign.

Now expand the spinney sum in $\overline{R}_\Gamma$. A nonempty spinney S is a set of pairwise disjoint proper connected divergent subgraphs. For each $\gamma \in S$, choose a forest F_γ of γ whose largest element is γ, as in (32.1). The union

$$F = \bigcup_{\gamma \in S} F_\gamma$$

is a forest of Γ that does not contain Γ, because different members of S are disjoint and all further elements lie inside those members. Conversely, every forest of Γ not containing Γ has a unique set of maximal elements; these maximal elements form a spinney, and the portions lying below each maximal element are precisely the corresponding F_γ's. The product of the signs in the spinney expansion is $(-1)^{\sum_{\gamma \in S} |F_\gamma|} = (-1)^{|F|}$. Therefore

$$\overline{R}_\Gamma I_\Gamma = \sum_{\substack{F \in \mathfrak{F}(\Gamma) \\ \Gamma \notin F}} (-1)^{|F|} t_F I_\Gamma. \tag{32.2}$$

If Γ is superficially convergent, $R_\Gamma = \overline{R}_\Gamma$ and this is the desired formula. If Γ is superficially divergent, then

$$R_\Gamma I_\Gamma = (1 - t_\Gamma)\overline{R}_\Gamma I_\Gamma.$$

The first term is (32.2). The second term appends Γ as the unique largest element to every forest not already containing Γ, with one extra minus sign. This gives exactly the remaining forests, namely those containing Γ, and proves ($*$). $\square$

The inclusion-exclusion structure can be seen directly in a graph with one proper divergent subgraph γ and a divergent overall graph Γ. The possible forests are

$$\emptyset, \qquad \{\gamma\}, \qquad \{\Gamma\}, \qquad \{\gamma, \Gamma\}.$$

The corresponding operator is

$$R_\Gamma = 1 - t_\gamma - t_\Gamma + t_\Gamma t_\gamma = (1 - t_\Gamma)(1 - t_\gamma),$$

where t_γ acts before t_Γ. The last term restores the part counted twice at the boundary where the subgraph and the whole graph shrink in a nested way.

32.6 Order of Subtractions and Scale Hierarchies

The prescription that t_γ acts before t_Γ for $\gamma \subset \Gamma$ is determined by the geometry of ultraviolet limits. A nested forest represents a hierarchy of large internal momenta. For example, if loop momenta internal to γ are of size $\rho\sigma$ and the remaining loop momenta of Γ/γ are of size ρ, with $\sigma \to \infty$ before $\rho \to \infty$, then the innermost singular region is the subgraph γ. Its Taylor polynomial is first converted into a local vertex of the contracted graph. After that replacement, the overall Taylor operator acts on the local integrand of the larger graph.

This ordering can be expressed algebraically. If

$$C_\gamma I_\gamma = -t_\gamma \overline{R}_\gamma I_\gamma,$$

then the term $t_\Gamma t_\gamma I_\Gamma$ in a forest is interpreted as

$$t_\Gamma\big((t_\gamma I_\gamma)\, I_{\Gamma/\gamma}\big),$$

where $t_\gamma I_\gamma$ is a polynomial in the external momenta entering γ. The polynomial is part of the local vertex data of Γ/γ, so the overall subtraction is again a Taylor polynomial in the external momenta of Γ. This is the mechanism by which nested subtractions preserve locality.

Overlapping divergent subgraphs are not disjoint and neither contains the other, so they are not placed in the same forest. Their common ultraviolet region is represented by the sum over compatible forests together with the overall subtraction. The forest sum is the inclusion-exclusion expression that subtracts all ultraviolet boundary regions while assigning counterterms only to local contracted subgraphs.

32.7 The Two-Loop Self-Energy Forest

Return to the two-loop self-energy graph in the six-dimensional cubic theory. Let Γ be the full graph and let γ_L, γ_R denote the left and right one-loop self-energy subgraphs. These two subgraphs overlap: they share the central propagator. Therefore the forest set is

$$\emptyset, \quad \{\gamma_L\}, \quad \{\gamma_R\}, \quad \{\Gamma\}, \quad \{\gamma_L, \Gamma\}, \quad \{\gamma_R, \Gamma\}.$$

The overlap of γ_L and γ_R is represented by the two separate one-subgraph forests $\{\gamma_L\}$ and $\{\gamma_R\}$, together with their nested overall forests.

For this graph the R-operation is

$$\begin{aligned} R_\Gamma I_\Gamma = I_\Gamma - t_{\gamma_L} I_\Gamma - t_{\gamma_R} I_\Gamma - t_\Gamma I_\Gamma \\ + t_\Gamma t_{\gamma_L} I_\Gamma + t_\Gamma t_{\gamma_R} I_\Gamma. \end{aligned}$$

The first two subgraph subtraction terms remove the two Schwinger-parameter boundary regions

$$\alpha_1, \alpha_2, \alpha_5 \to 0, \qquad \alpha_3, \alpha_4, \alpha_5 \to 0.$$

After these subtractions, the remaining overall boundary $\alpha_1, \ldots, \alpha_5 \to 0$ is subtracted by t_Γ. Since $\omega(\Gamma) = 2$ for the self-energy in $D = 6$, t_Γ extracts the constant and quadratic terms in the external momentum. The final counterterm is therefore a linear combination of the local operators ϕ^2 and $\partial_\mu \phi \, \partial_\mu \phi$.

Figure 32.3 displays the diamond graph in which the two one-loop subdivergences overlap; the forest formula subtracts each compatible option and then the remaining overall divergence.

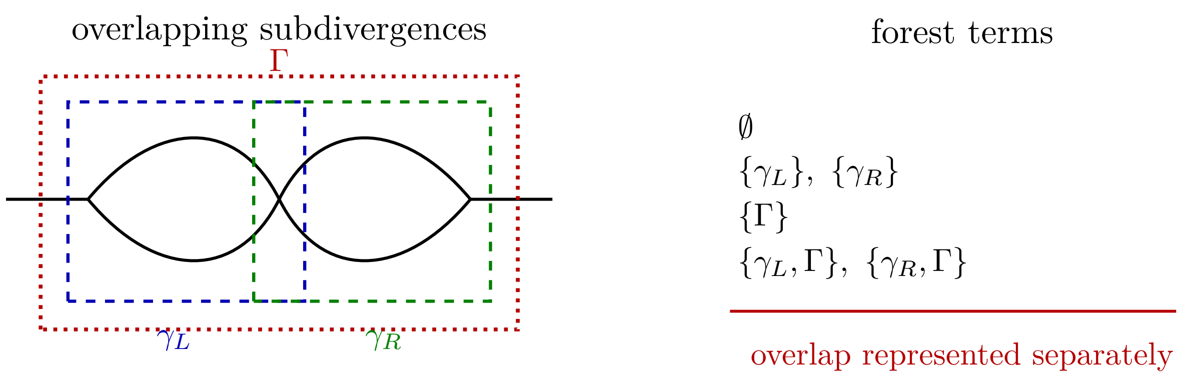


Figure 32.3: The two one-loop subdivergences of the diamond graph overlap along one internal line. The forest formula treats them in separate compatible forests, while the overall graph is subtracted after either one-loop subgraph subtraction.

32.8 Finiteness and Locality

Lemma 32.5 (Hepp-sector forest associated to an ultraviolet hierarchy)**.** *Fix a smooth Hepp-sector partition of the noncompact loop-momentum region of a graph Γ, so that in each sector the magnitudes of the line momenta are ordered up to bounded ratios. In one sector, the ultraviolet* *boundary strata determine a finite forest F_S as follows: for each threshold in the ordered list of large line momenta, form the subgraph spanned by lines above the threshold, decompose it into connected components, keep the one-particle-irreducible components that contain loops, and then discard components with negative superficial degree. The retained components are pairwise disjoint or nested. Moreover, every superficially divergent ultraviolet boundary of that sector is represented by a member of this forest, or by a quotient graph obtained after members of the forest have been contracted.*

Proof. Label the distinct momentum-size thresholds in the sector by

$$\rho_1 \gg \rho_2 \gg \cdots \gg \rho_N$$

after subdividing the partition of unity if necessary so that the ratios ρ_{j+1}/ρ_j are the sector coordinates. Let H_j be the subgraph spanned by the internal lines whose momenta have size at least ρ_j. Then

$$H_1 \subset H_2 \subset \cdots \subset H_N.$$

For two indices $i < j$, every connected component of H_i is either contained in a connected component of H_j or disjoint from it, because the edge set of H_i is a subset of the edge set of H_j.

It remains to check that passing from connected components to 1PI loop-containing components does not introduce a crossing overlap. Use the block decomposition of a connected graph into maximal one-particle-irreducible blocks, bridges, and tree attachments. If B is a 1PI block of a smaller threshold graph H_i, then any two lines of B lie on a common cycle in H_i, hence still lie on a common cycle in the larger graph H_j. Thus B is contained in a unique maximal 1PI block of the connected component of H_j that contains it. Adding further high-momentum lines can merge old 1PI blocks, but it cannot split an old 1PI block. Consequently two retained 1PI loop subgraphs obtained from different thresholds are either edge-disjoint, or one is contained in the other after the obvious inclusion of edge sets. Discarding components with negative superficial degree removes vertices from this partially ordered set and therefore preserves the forest property.

Now consider a boundary stratum of the sector at which a subset of sector ratios tends to zero, equivalently a nested collection of high-momentum scales is sent to infinity relative to the slower scales. At the first nontrivial ratio the large lines form one of the components of some H_j. If its superficial degree is negative, the Taylor-subtracted integrand has a convergent power-counting exponent at that boundary, so the boundary is not superficially divergent. If its degree is nonnegative, the corresponding 1PI loop block is one of the retained elements of F_S. For the next larger scale separation, the lines already represented by retained blocks are replaced by local vertices; the remaining divergent large-line configuration is therefore a divergent component of the quotient graph. An induction over the ordered ratios gives a nested sequence of retained subgraphs and quotient divergences. This is precisely the forest associated to the ultraviolet hierarchy in the Hepp sector. □

Theorem 32.6 (BPHZ finiteness for massive Euclidean scalar graphs)**.** *Let Γ be a connected one-particle-irreducible Euclidean Feynman graph in a massive scalar theory. Assume that all external momenta and subtraction points are chosen away from infrared singularities, that the numerator of every graph is polynomial in loop and external momenta, and that t_γ subtracts every divergent one-particle-irreducible subgraph γ through degree $\omega(\gamma)$. Then*

$$\int \prod_{i=1}^{L(\Gamma)} \mathrm{d}^D \ell_i \, R_\Gamma I_\Gamma(\ell, p)$$

is ultraviolet finite. The counterterm produced by each connected divergent subgraph is a polynomial in p_γ of degree at most $\omega(\gamma)$.

Proof. The proof is local on loop-momentum space. Choose a smooth partition of unity subordinate to one compact region and to finitely many conic Hepp sectors. Lemma 32.5 attaches to each sector S a forest F_S containing the superficially divergent boundary components of that sector. It is enough to prove absolute convergence on each sector, since the partition has finitely many sectors and the compact region is harmless for massive Euclidean propagators.

The elementary estimate is the following Taylor-remainder bound. Let γ be a connected divergent subgraph, and let $\rho \geq 1$ scale loop momenta internal to γ while the external variables p_γ of γ are kept on a slower scale. Write $p_\gamma = p_\gamma^* + a$. Taylor's formula with integral remainder gives

$$f(p_\gamma) - t_\gamma f(p_\gamma) = \sum_{|\alpha|=\omega(\gamma)+1} \frac{a^\alpha}{\alpha!} \int_0^1 (1-u)^{\omega(\gamma)} \big(\partial_a^\alpha f\big)(p_\gamma^* + ua)\, \mathrm{d}u. \tag{32.3}$$

Here f denotes the prepared integrand of γ, with all proper subdivergences already replaced by their renormalized expressions or local counterterm vertices. A derivative with respect to an external momentum of γ differentiates either a polynomial numerator or a propagator whose momentum is affine in p_γ. For a propagator,

$$\partial_{p_\mu} \frac{1}{q(\ell,p)^2 + m^2} = -\frac{2q_\nu(\ell,p)\, \partial_{p_\mu} q_\nu(\ell,p)}{(q(\ell,p)^2 + m^2)^2},$$

which has one less power under the large internal scaling than the original propagator. Derivatives of the polynomial numerator are bounded by the same rule because differentiating a polynomial lowers its degree, and derivatives falling on counterterm vertices only differentiate polynomials in slower external variables. Consequently, including the internal loop measure of γ,

$$\mathrm{d}^{DL(\gamma)}\ell_\gamma\, \partial_a^\alpha f(\rho\ell_\gamma, p_\gamma) = O\Big(\rho^{\omega(\gamma)-|\alpha|}\Big)\, \mathrm{d}^{DL(\gamma)}\ell_\gamma. \tag{32.4}$$

For $|\alpha| = \omega(\gamma)+1$, the γ-scale exponent is at most -1. In radial sector coordinates this means that the corresponding boundary integral is bounded by

$$C \int_R^\infty \rho^{-1}\, \frac{\mathrm{d}\rho}{\rho} < \infty. \tag{32.5}$$

If γ has negative superficial degree, the same estimate holds without applying $1 - t_\gamma$, because then the exponent is already at most -1.

Now fix a Hepp sector S. Let $\gamma_1, \ldots, \gamma_r$ be a chain in F_S, ordered from smaller to larger subgraph. If the forest has disjoint branches, introduce one such ordered scale chain on each branch; the sector coordinates factor into the product of these branch variables because disjoint subgraphs use disjoint sets of internal momenta. Introduce scale ratios $\rho_j \geq 1$ so that the momenta internal to γ_j but not to the previous members of the chain are of order $\rho_1 \cdots \rho_j$, while the external momenta of γ_j are on the slower scale $\rho_1 \cdots \rho_{j-1}$. The forest formula acts in precisely this order: smaller Taylor operators are applied before the larger ones, and the counterterm produced by a smaller member is viewed as a local vertex in the quotient graph. Applying (32.3) successively from the innermost members outward gives, on the sector S, a bound of the form

$$|R_\Gamma I_\Gamma(\ell,p)|\, \mathrm{d}^{DL(\Gamma)}\ell \leq C_S \prod_{j=1}^{r} \rho_j^{-1} \frac{\mathrm{d}\rho_j}{\rho_j} \prod_b \sigma_b^{-1} \frac{\mathrm{d}\sigma_b}{\sigma_b} \mathrm{d}\Omega_S. \tag{32.6}$$

The additional variables σ_b correspond to superficially convergent subgraphs or quotient subgraphs; their exponents are already negative by power counting. The angular variables Ω_S range over a compact part of the sector after the radial ratios have been extracted. Thus the right hand side is integrable in every noncompact sector variable.

It remains to explain why the estimate just obtained for a sector is the estimate of the actual forest formula, not of a sector-dependent subtraction prescription. In the sector S, the ultraviolet boundary strata are labelled by nested or disjoint components as above. Expanding the product of the corresponding Taylor remainders gives the alternating sum over all subforests of F_S. Zimmermann's forest formula is the global inclusion-exclusion sum over all compatible forests of Γ; restricted to the sector S, the terms that meet the boundary strata of S are exactly the terms in this sector product, while terms corresponding to extra subtractions are Taylor polynomials of already integrable remainders and obey the same or better estimates. Overlapping divergent subgraphs are handled by the sector decomposition: a single sector resolves the hierarchy into compatible nested or disjoint components, and the sum over sectors restores the sector-independent forest expression.

The mass and nonexceptional-momentum assumptions remove the remaining noncompact possibilities. In the compact part of loop-momentum space the integrand is smooth. Away from compact sets, the only boundaries are the ultraviolet scaling boundaries estimated above; massive denominators are bounded away from zero on the real Euclidean integration contour. Hence the renormalized integral is absolutely ultraviolet finite.

The locality assertion is algebraic and occurs before integration. By definition,

$$C_\gamma I_\gamma = -\, t_\gamma \, \overline{R}_\gamma I_\gamma,$$

and t_γ is a Taylor polynomial in the external variables p_γ of degree at most $\omega(\gamma)$. After integrating the internal variables of γ, or after inserting the recursively defined prepared counterterm, this polynomial is a local vertex in Γ/γ. Thus every connected divergent subgraph produces a counterterm polynomial of degree at most $\omega(\gamma)$. In a theory whose power-counting counterterms belong to a finite local operator space, the recursive operation remains inside that finite space, giving perturbative renormalizability in the finite-list sense used in the preceding chapter. $\square$

Remark 32.7 (Massless graphs and the BPHZL extension). The theorem just stated is a massive, nonexceptional-momentum theorem. Its proof uses the fact that the only remaining noncompact regions after subtraction are ultraviolet regions. For a massless graph, a Taylor subtraction at zero external momentum can change the small-loop-momentum behavior of the integrand. Thus the ultraviolet forest formula alone does not control the infrared boundary in which a massless subgraph momentum goes to zero.

The Lowenstein–Zimmermann extension, usually called BPHZL, augments the subtraction data so that ultraviolet and infrared degrees are controlled simultaneously.[1] In one standard formulation, each massless propagator is temporarily replaced by an s-dependent massive propagator, so that $s = 1$ gives the desired massless graph while another value of s supplies an auxiliary infrared mass.

[1] See J. H. Lowenstein and W. Zimmermann, "On the formulation of theories with zero mass propagators," *Nucl. Phys. B* **86** (1975) 77–103, doi:10.1016/0550-3213(75)90075-9; and J. H. Lowenstein and W. Zimmermann, "The power counting theorem for Feynman integrals with massless propagators," *Commun. Math. Phys.* **44** (1975) 73–86, doi:10.1007/BF01609059. For the scalar critical model see also J. H. Lowenstein and W. Zimmermann, "Infrared convergence of Feynman integrals for the massless ϕ^4-model," *Commun. Math. Phys.* **46** (1976) 105–118, doi:10.1007/BF01608491.

The subtraction operators are then Taylor operators in the external momenta together with the auxiliary parameter, with degrees chosen from both ultraviolet and infrared power counting. The resulting forest formula is local in the ultraviolet variables and has enough additional subtraction at the infrared boundary to possess a distributional $s \to 1$ massless limit.

This is the momentum-subtraction counterpart of the Wilsonian statement used in the critical chapters. The Wilson–Fisher and Ising discussions work on a massless critical surface, where zero-momentum Green functions and composite insertions must be interpreted through an infrared-safe prescription: nonexceptional external momenta, finite volume, a small mass followed by a scaling limit, BPHZL, or an equivalent Wilsonian exact-RG construction. The massive BPHZ theorem above supplies the ultraviolet locality mechanism; it is not, by itself, the full renormalization theorem for the massless critical regime.

32.9 Oversubtractions and Zimmermann Normal Products

Composite operators are renormalized by the same forest mechanism, with one distinguished local insertion. Let $O(x)$ be a local monomial in the fields and derivatives, and let

$$\Gamma_O$$

be a graph with one vertex marked as the insertion of O. Divergent subgraphs may contain the marked vertex or may be ordinary subgraphs away from it. The Taylor degree assigned to a subgraph containing the insertion is chosen at least as large as the engineering degree of O, and may be chosen larger. The resulting renormalized insertion is denoted

$$N_\delta[O](x),$$

where δ is the subtraction degree at the marked vertex.

It fixes a BPHZ subtraction scheme for insertions of O.

Increasing δ is an oversubtraction.

Proposition 32.8 (Zimmermann identity for oversubtracted normal products)**.** *Let $\delta' \geq \delta$. Fix a finite basis $\{O_A\}$ of local monomials, with the same exact global and spacetime quantum numbers as O, and with engineering degree at most δ', modulo total derivatives and field relations already chosen as part of the operator basis. For every graph with one insertion of O, the difference between the BPHZ-renormalized insertion with subtraction degree δ' and the one with subtraction degree δ is a finite sum of BPHZ-renormalized insertions of the O_A's:*

$$N_{\delta'}[O](x) = N_\delta[O](x) + \sum_A c_A\, N_{\delta'}[O_A](x). \tag{32.7}$$

The coefficients c_A are finite functions of the subtraction points, finite subtraction prescription, masses, and renormalized couplings. They are local scheme-change coordinates; they are not new nonlocal dynamical data.

Proof. Work graph by graph. For a marked subgraph γ containing the insertion vertex, write $t_\gamma^{(\delta)}$ and $t_\gamma^{(\delta')}$ for the two Taylor operators. Their difference is the finite Taylor polynomial

$$t_\gamma^{(\delta')} - t_\gamma^{(\delta)} = \sum_{\delta < |\alpha| \leq \delta'} \frac{a^\alpha}{\alpha!}\, \partial_a^\alpha|_{a=0} \tag{32.8}$$

in the external variables of the marked subgraph, with all lower subdivergences prepared in the same forest order as above. After the marked subgraph is contracted, each monomial a^α is a local derivative operator at the insertion point. By the chosen local operator basis, the resulting local insertion is a finite linear combination of the O_A's.

Unmarked divergent subgraphs are subtracted in the same way in both normal products. Hence their forest sums cancel in the difference except insofar as they prepare the coefficients multiplying the finite Taylor polynomial (32.8). The BPHZ finiteness theorem applied to graphs with the corresponding local insertion shows that these coefficients are finite after the remaining loop integrations. Regrouping the finite local insertions in the common oversubtracted scheme δ' gives (32.7). □

This is the BPHZ form of the operator mixing matrix used later in the renormalized-operator chapter.

32.10 Finite Parts and Subtraction Coordinates

The Taylor operator includes a choice of subtraction point. In addition, one may add finite local polynomial terms compatible with the same subtraction degree. Changing these choices replaces C_γ by

$$C_\gamma + \Delta C_\gamma,$$

where ΔC_γ is again a local polynomial in p_γ of degree at most $\omega(\gamma)$. Hence a change of subtraction prescription is a local redefinition of the parameters multiplying local operators. The renormalized amplitudes expressed in one set of renormalized coordinates can be re-expressed in another by a finite local change of coordinates.

The next chapter uses this freedom actively. A family of subtraction conditions indexed by a Euclidean momentum scale μ gives scale-dependent renormalized couplings. The differential comparison of neighboring values of μ is the 1PI renormalization group. Chapter 36 compares the same finite local freedom with Wilsonian local coordinates: the Taylor polynomial assigned to a divergent subgraph is a local vertex, and changing its finite part changes the corresponding coordinate on the local Wilsonian chart.

Chapter 33

The 1PI Renormalization Group

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

33.1 Scale-Dependent Coordinates on the 1PI Effective Action

The previous chapters constructed local counterterms and organized subdivergence subtractions. We now use that construction to compare renormalized 1PI data at different Euclidean momentum scales.

Work in a Euclidean perturbative quantum field theory on $\mathbb{R}^D$, with a UV regulator denoted by Λ. The symbol Λ refers to a cutoff class in Table 11.1; if the calculation is instead made in dimensional regularization, the corresponding object is a meromorphic formal graph assignment in $D = d - \varepsilon$, not a finite-regulator measure. Let $\Gamma_\Lambda[\varphi]$ be the regulated 1PI effective action, expressed as a functional of a classical field φ. Its momentum-space expansion has the form

$$\Gamma_\Lambda[\varphi] = \sum_{n\geq 0} \frac{1}{n!} \int \prod_{a=1}^{n} \frac{\mathrm{d}^D k_a}{(2\pi)^D} (2\pi)^D \delta\left(\sum_{a=1}^{n} k_a\right) \Gamma_\Lambda^{(n)}(k_1, \ldots, k_n) \widetilde{\varphi}(k_1) \cdots \widetilde{\varphi}(k_n).$$

The distribution $\Gamma_\Lambda^{(n)}$ is the regulated 1PI n-point vertex. Momentum conservation allows one momentum argument to be suppressed when convenient.

A scale-dependent 1PI coordinate is obtained by specifying:

- a finite list of local operators $\mathcal{O}_I$, each with a number n_I of fields and a derivative tensor structure;
- a Euclidean subtraction configuration $\mathcal{S}_{I,\mu}$ whose momenta have size $O(\mu)$;
- a field normalization convention at the same scale;
- a linear projector Π_I extracting the Taylor coefficient of the 1PI vertex with the tensor structure of $\mathcal{O}_I$.

The resulting coordinate is denoted $g_I(\mu)$. Its precise value depends on the chosen subtraction configuration and projector; once these choices are fixed, $g_I(\mu)$ is a finite renormalized coordinate on the family of theories under consideration.

The 1PI renormalization group is the differential comparison of these coordinates at nearby scales:

$$\frac{\mathrm{d}g_I(\mu)}{\mathrm{d}\log\mu} = \beta_I(\{g_J(\mu)\};\mu).$$

The right-hand side is the beta function in the selected coordinate system. When all dimensionful quantities have been replaced by dimensionless ratios and included among the coordinates, the vector field on this enlarged coordinate space has no separate explicit μ-argument.

The finite-cutoff Wilsonian effective action uses another construction. In that construction one defines an action at cutoff Λ by integrating over field modes between two cutoffs and studies the dependence of that action on Λ. That construction will be developed later. The present chapter uses the 1PI effective action and subtraction conditions on its vertices.

At the level of perturbative graphs, the scale comparison is a precise reuse of lower-order subgraph data. A one-loop subgraph that determines the variation of a 1PI coordinate reappears inside higher-order 1PI graphs. The renormalization-group equation records this reuse as a differential equation for the finite coordinate, after the local counterterms have removed the cutoff dependence.

Figure 33.1 records the graph-theoretic source of the perturbative RG equation: lower-order local subgraph data reappear inside higher-order 1PI graphs and therefore control the scale derivative of a finite coordinate.

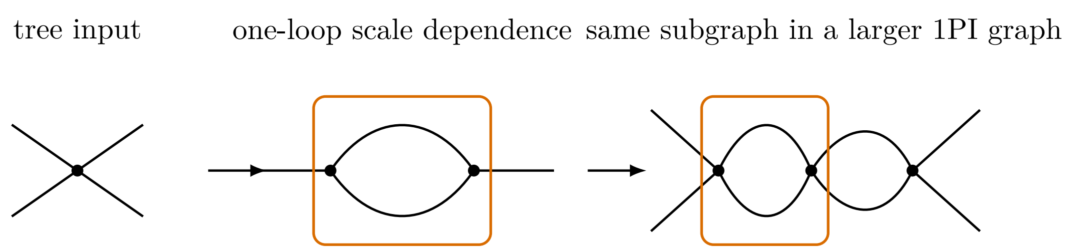


Figure 33.1: The scale derivative of a 1PI coordinate is local in perturbation theory because the scale-dependent part of a lower-order 1PI subgraph is reused inside larger graphs. The differential equation is written only after the subgraph counterterms of the previous chapters have made the coordinate comparison finite.

Figure 33.2 then identifies the coordinate comparison itself: a field normalization and a vertex subtraction condition at scale μ define the finite 1PI coordinates transported by the RG vector field.

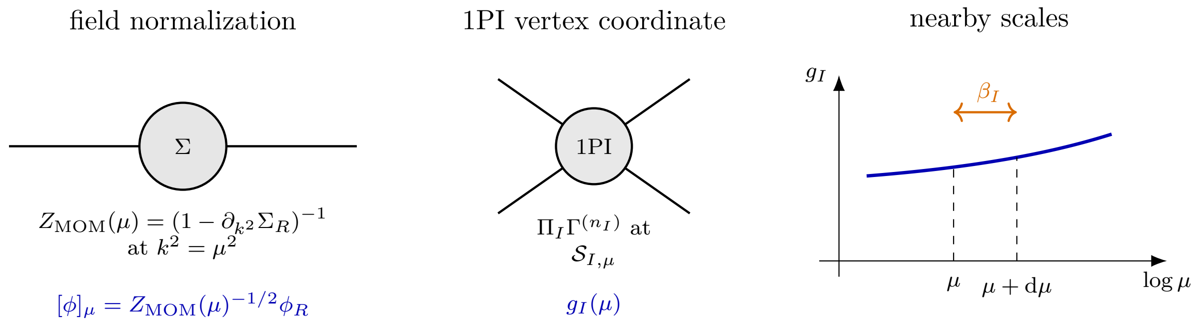


Figure 33.2: The 1PI renormalization group compares coordinate systems on renormalized 1PI vertices. A field normalization is fixed at the same scale as the vertex subtraction, and nearby scales are compared by a differential equation.

33.2 Scalar Quartic Theory in Four Euclidean Dimensions

Let $D = 4$, and consider a real scalar theory with bare Euclidean Lagrangian

$$\mathcal{L}_E = \frac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) + \frac{1}{2}m_0^2\phi^2 + \frac{1}{4!}g_0\phi^4.$$

Choose a renormalized field $\phi_R = Z_R^{-1/2}\phi$, a renormalized mass m_R, and a renormalized quartic coupling g_R. The same bare Lagrangian can be written as

$$\begin{aligned}\mathcal{L}_E = {}& \frac{1}{2}(\partial_\mu\phi_R)(\partial_\mu\phi_R) + \frac{1}{2}m_R^2\phi_R^2 \\ & + (Z_R - 1)\left[\frac{1}{2}(\partial_\mu\phi_R)(\partial_\mu\phi_R) + \frac{1}{2}m_R^2\phi_R^2\right] + \frac{1}{2}\delta m^2\phi_R^2 + \frac{1}{4!}(g_R + \delta g)\phi_R^4.\end{aligned}$$

Here Z_R, δm^2, and δg depend on the regulator and on the chosen renormalization conditions.

Let $\Gamma_R^{(4)}(k_1, k_2, k_3)$ be the renormalized 1PI four-point vertex, with $k_4 = -k_1 - k_2 - k_3$. A zero-momentum definition such as

$$g_R = -\Gamma_R^{(4)}(0, 0, 0)$$

is one possible coordinate. To compare physics at a Euclidean momentum scale μ, it is better to define a coordinate using external momenta of size μ and to normalize the external field at the same scale.

The connected two-point function is written in the form

$$G_R(k) = \frac{1}{k^2 + m_R^2 - \Sigma_R(k)},$$

which defines the self-energy $\Sigma_R(k)$ in this convention. The scale-dependent normalization is

$$Z_{\mathrm{MOM}}(\mu) = \left(1 - \frac{\partial\Sigma_R}{\partial k^2}\right)^{-1}_{k^2=\mu^2}, \qquad [\phi]_\mu = Z_{\mathrm{MOM}}(\mu)^{-1/2}\phi_R.$$

Thus the 1PI effective action, when viewed as a functional of $[\phi]_\mu$, has a canonically normalized quadratic term at momenta $k^2 = \mu^2$.

For the four-point coordinate, use the symmetric Euclidean subtraction point

$$k_i^2 = \mu^2, \qquad s = t = u = -\frac{4}{3}\mu^2,$$

where

$$s = -(k_1 + k_2)^2, \qquad t = -(k_1 + k_3)^2, \qquad u = -(k_1 + k_4)^2.$$

These are Euclidean subtraction variables obtained by analytic continuation from the mostly-plus Mandelstam convention. With all four Euclidean momenta taken incoming and $k_1+k_2+k_3+k_4 = 0$, the identity fixing the common channel value is

$$\begin{aligned} s + t + u &= -\sum_{j=2}^{4}(k_1 + k_j)^2 \\ &= -\left(3k_1^2 + \sum_{j=2}^{4} k_j^2 + 2k_1 \cdot \sum_{j=2}^{4} k_j\right) \\ &= -\sum_{i=1}^{4} k_i^2. \end{aligned}$$

At $k_i^2 = \mu^2$ this gives $s + t + u = -4\mu^2$; imposing the symmetric subtraction condition $s = t = u$ therefore gives $s = t = u = -4\mu^2/3$. After continuing to Lorentzian on-shell scattering and crossing two momenta back to the physical outgoing convention, the same algebraic relation becomes $s + t + u = \sum_i m_i^2$. Explicitly, if p_i are mostly-plus Lorentzian momenta and the Euclidean momenta are continued as $k_i = (\mathrm{i}p_i^0, \vec{p}_i)$, then

$$(k_i + k_j)^2_{\mathrm{E}} = (p_i + p_j)^2_{\mathrm{L}}, \qquad s_{\mathrm{E}} := -(k_1 + k_2)^2_{\mathrm{E}} = s_{\mathrm{L}} := -(p_1 + p_2)^2_{\mathrm{L}}.$$

The sign flip is between s_{E} and the bare mostly-plus Lorentzian quadratic form $(p_1 + p_2)^2_{\mathrm{L}}$, not between the Euclidean subtraction variable and the mostly-plus Mandelstam variable used in the scattering chapters. The running quartic coupling in this momentum-subtraction scheme is

$$g(\mu) = -Z_{\mathrm{MOM}}(\mu)^2 \Gamma_R^{(4)}(k_1, k_2, k_3)\bigg|_{k_i^2=\mu^2,\ s=t=u=-\frac{4}{3}\mu^2}.$$

Definition 33.1 (Momentum-subtraction quartic coordinate)**.** For the four-dimensional Euclidean scalar theory above, the momentum- subtraction quartic coordinate $g(\mu)$ is the negative of the four-point 1PI vertex at the symmetric Euclidean point, expressed in the field normalization $[\phi]_\mu$. The subtraction scheme consists of the renormalized field convention, the symmetric point, and the sign convention in the displayed formula.

In the 1PI vertex convention used here, the tree-level quartic vertex is $\Gamma^{(4)}_{\text{tree}} = -g_R$. The explicit minus sign in the definition of $g(\mu)$ therefore makes the coordinate agree at tree level with the positive coefficient multiplying $\phi^4/4!$ in the Euclidean action. In Section 34.9 the minimal-subtraction coordinate $\lambda(\mu)$ is written directly as the positive coefficient in

$$g^\epsilon = \mu^\epsilon \left[\lambda(\mu) + \frac{3}{16\pi^2}\frac{\lambda(\mu)^2}{\epsilon} + \cdots\right].$$

Thus the momentum-subtraction and minimal-subtraction formulas use the same positive quartic-coordinate orientation; the displayed minus sign appears only when the coordinate is extracted from the 1PI vertex.

33.3 One-Loop Running in the Quartic Example

At one loop, the four-point bubbles change the quartic vertex, while the two-point tadpole is independent of k^2. In the self-energy convention

$$G_R(k) = \frac{1}{k^2 + m_R^2 - \Sigma_R(k)},$$

the amputated one-loop tadpole is

$$\Sigma^{(1)}_{\mathrm{tad}}(k) = -\frac{g_R}{2}\int^{\Lambda} \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{1}{\ell^2 + m_R^2},$$

before the mass counterterm is chosen. The factor 1/2 is the tadpole symmetry factor. The loop momentum ℓ closes at the quartic vertex, so no external momentum flows through the loop and

$$\frac{\partial \Sigma^{(1)}_{\mathrm{tad}}}{\partial k^2} = 0.$$

The one-loop subtraction needed in the two-point function is therefore a mass subtraction, not a kinetic subtraction. Hence

$$Z_R = 1 + O(g_R^2), \qquad Z_{\mathrm{MOM}}(\mu) = 1 + O(g_R^2)$$

for the one-loop computation of $g(\mu)$.

Figure 33.3 displays the actual one-loop quartic coordinate calculation: tree vertex, three bubble channels, and the local counterterm fixed by the chosen subtraction condition.

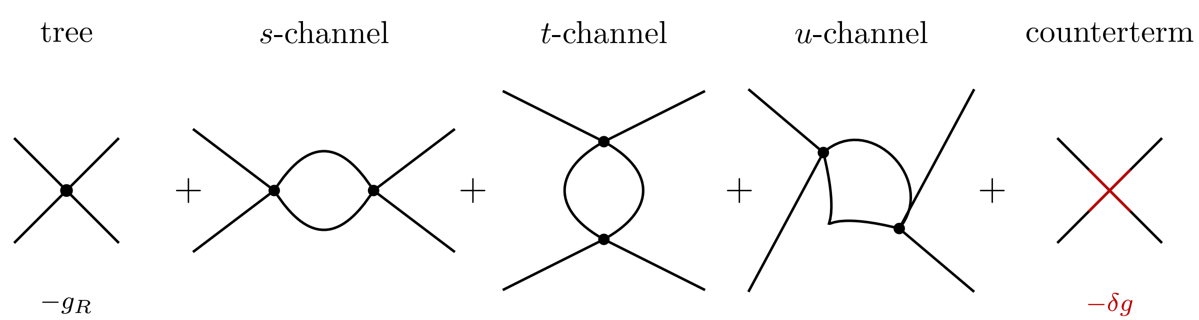


Figure 33.3: The one-loop scalar quartic coordinate receives the tree vertex, the three distinct bubble channels, and the local quartic counterterm fixed by the chosen subtraction condition.

With a momentum cutoff Λ, the one-loop relation at the symmetric subtraction point is

$$-g(\mu) = -g_R - \delta g + 3\frac{g_R^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\left[\log\frac{\Lambda^2}{m_R^2 - sx(1-x)} - 1\right]_{s=-\frac{4}{3}\mu^2} + O(g_R^3).$$

Choose the finite convention

$$g_R = g(0).$$

This condition fixes δg. Subtracting the $\mu = 0$ equation from the equation at general μ removes the cutoff and gives

$$g(\mu) = g_R - \frac{3g_R^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \log \frac{m_R^2}{m_R^2 + \frac{4}{3}\mu^2 x(1-x)} + O(g_R^3).$$

This is a finite relation between two renormalized quartic coordinates in the same theory. For $\mu \gg m_R$, the displayed correction contains a large logarithm. A comparison built from nearby scale coordinates is then better adapted to perturbation theory.

33.4 Nearby Scales and the Beta Function

Let μ' be close to μ. The same one-loop computation may be expressed as a relation between the two scale coordinates:

$$g(\mu') = g(\mu) - \frac{3}{32\pi^2} g(\mu)^2 \int_0^1 \mathrm{d}x \, \log \frac{m_R^2 + \frac{4}{3}\mu^2 x(1-x)}{m_R^2 + \frac{4}{3}\mu'^2 x(1-x)} + O(g(\mu)^3).$$

The expansion parameter is now $g(\mu)$, and the logarithm is small for μ' close to μ. Setting $\mu' = \mu + \mathrm{d}\mu$ gives

$$\frac{\mathrm{d}g(\mu)}{\mathrm{d}\log\mu} = \frac{3}{16\pi^2} g(\mu)^2 \int_0^1 \mathrm{d}x \, \frac{\frac{4}{3}\mu^2 x(1-x)}{m_R^2 + \frac{4}{3}\mu^2 x(1-x)} + O(g(\mu)^3).$$

The integral is a threshold factor. At scales much larger than the mass, $\mu \gg m_R$, it tends to 1, and

$$\frac{\mathrm{d}g}{\mathrm{d}\log\mu} = \frac{3}{16\pi^2} g^2 + O(g^3).$$

In this scheme and convention, the one-loop beta function is therefore

$$\beta(g) = \frac{3}{16\pi^2} g^2 + O(g^3) \qquad (\mu \gg m_R).$$

The one-loop differential equation can be integrated between two scales:

$$\frac{1}{g(\mu)} - \frac{1}{g(\mu')} = -\frac{3}{16\pi^2} \log \frac{\mu}{\mu'}.$$

Equivalently,

$$g(\mu) = \frac{g(\mu')}{1 - \frac{3}{16\pi^2} g(\mu') \log(\mu/\mu')}.$$

This expression contains powers of $g(\mu') \log(\mu/\mu')$ to all orders. Its perturbative domain is controlled by the running coordinate itself: $g(\widetilde{\mu})$ must remain small for the scales $\widetilde{\mu}$ used in the comparison.

33.5 The Landau Scale as a Perturbative Coordinate

The integrated one-loop equation may be written in terms of a scale μ_0 defined by

$$\mu_0 = \mu' \exp\left[\frac{16\pi^2}{3g(\mu')}\right].$$

Then

$$g(\mu) = \frac{1}{\frac{3}{16\pi^2}\log(\mu_0/\mu)}.$$

For $0 < \mu \ll \mu_0$, the denominator is large and the one-loop coordinate is small. As μ approaches μ_0, the perturbative coordinate becomes large and the one-loop approximation ceases to provide a controlled description. Within the one-loop equation, the pair $(\mu', g(\mu'))$ is equivalently represented by the single scale μ_0.

Figure 33.4 plots this one-loop coordinate flow and marks the scale μ_0 at which the perturbative coordinate leaves the controlled small-coupling region.

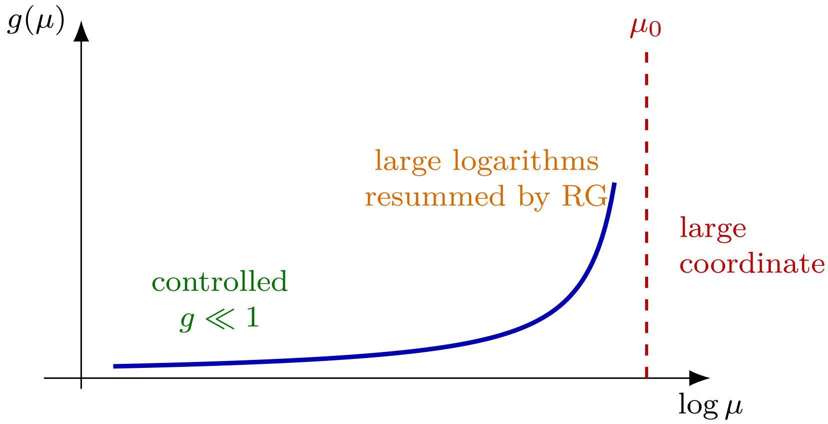


Figure 33.4: The one-loop scalar quartic flow grows with the Euclidean momentum scale. The scale μ_0 is where the one-loop coordinate diverges; the controlled perturbative domain is the region where the running coordinate remains small.

Remark 33.2 (Status of the Landau scale). The scale μ_0 is a statement about the one-loop perturbative coordinate. It gives a useful parametrization of the weak-coupling region and indicates where that coordinate loses perturbative control. A statement about the existence or nonexistence of a continuum scalar theory at arbitrarily short distances requires additional nonperturbative input.

Quoted Theorem 33.3 (Triviality of the standard scalar scaling limits)**.** *For nearest-neighbor ferromagnetic four-dimensional Ising-type spin systems at or near the critical point, and for four-dimensional $\lambda\phi^4$ fields with a lattice ultraviolet cutoff in the infinite-volume and vanishing-lattice spacing limit, the scaling limits covered by the Aizenman–Duminil-Copin theorem are Gaussian. Equivalently, after the continuum field normalization, the limiting Schwinger functions satisfy Wick's rule and the connected n-point functions with $n \geq 4$ vanish. Earlier results of Aizenman* *and Fröhlich establish complementary triviality, mean-field, and critical-approach statements in $d > 4$ and in related $d \geq 4$ scalar and Ising-type settings.* [1]

Proof mechanism and logical scope. The theorem concerns a concrete sequence of reflection-positive lattice models whose finite-cutoff observables are honest random variables. Let $a \downarrow 0$ be the lattice spacing along a critical or near-critical trajectory, let $\Phi_a(x)$ be the lattice spin or scalar field, and choose a field normalization Z_a such that the smeared variables

$$X_a(f) = Z_a\, a^4 \sum_{x \in a\mathbb{Z}^4} f(x)\Phi_a(x), \qquad f \in C_c^\infty(\mathbb{R}^4),$$

have nonzero finite two-point limits. Denote their joint cumulants by

$$\kappa_{a,n}(f_1, \ldots, f_n) = \operatorname{cum}\big(X_a(f_1), \ldots, X_a(f_n)\big).$$

Any subsequential scaling limit is Gaussian exactly when its covariance $\lim_{a\downarrow 0} \kappa_{a,2}$ exists and all limiting cumulants $\lim_{a\downarrow 0} \kappa_{a,n}$ with $n \geq 3$ vanish. This is the ordinary cumulant characterization of a Gaussian probability law: the logarithm of the characteristic function contains only its quadratic term.

For ferromagnetic even Ising and $\lambda\phi^4$ lattice measures, the $\Phi \mapsto -\Phi$ symmetry removes the odd cumulants. The load-bearing estimate is therefore the four-point truncated function

$$U_4(x_1, x_2, x_3, x_4) = \langle \Phi_{x_1}\Phi_{x_2}\Phi_{x_3}\Phi_{x_4}\rangle - \sum_{\text{pairings } P} \prod_{\{i,j\}\in P} \langle \Phi_{x_i}\Phi_{x_j}\rangle.$$

The random-current switching lemma rewrites U_4, up to a sign fixed by the ferromagnetic convention, as a two-current connection/intersection probability multiplied by two-point functions. A direct consequence is the tree-diagram comparison

$$|U_4(x_1, x_2, x_3, x_4)| \leq C \sum_{z\in\Lambda_a} \prod_{i=1}^{4} G_a(x_i, z), \qquad G_a(x,y) = \langle \Phi_x \Phi_y\rangle.$$

In dimensions $d > 4$ this bound already makes the normalized block-field fourth cumulant vanish, because the corresponding random walks are transient enough. In $d = 4$ the bound is scale-marginal. The theorem of Aizenman–Duminil-Copin supplies the missing logarithmic gain: for macroscopic boxes of lattice radius R,

$$\frac{\sum_{x_1,x_2,x_3,x_4\in B_R} |U_4(x_1,x_2,x_3,x_4)|}{\Big(\sum_{x,y\in B_R} G_a(x,y)\Big)^2} \leq \varepsilon(R), \qquad \varepsilon(R) \xrightarrow[R\to\infty]{} 0. \tag{33.1}$$

The proof of this estimate is the deep multiscale part: the random-current intersection event is decomposed by scales and is shown to carry an extra logarithmic smallness at the upper critical dimension.

[1] See M. Aizenman, "Proof of the triviality of ϕ^4_d field theory and some mean-field features of Ising models for $d > 4$," *Phys. Rev. Lett.* **47** (1981) 1–4, doi:10.1103/PhysRevLett.47.1; J. Fröhlich, "On the triviality of $\lambda\phi^4_d$ theories and the approach to the critical point in $d \geq 4$ dimensions," *Nucl. Phys. B* **200** (1982) 281–296, doi:10.1016/0550-3213(82)90088-8; M. Aizenman and H. Duminil-Copin, "Marginal triviality of the scaling limits of critical 4D Ising and ϕ^4_4 models," *Ann. of Math.* **194** (2021) 163–235, doi:10.4007/annals.2021.194.1.3, and the corrigendum in *Ann. of Math.* **199** (2024) 479, doi:10.4007/annals.2024.199.1.7.

Equation (33.1) implies, after replacing box indicators by smooth test functions and using standard approximation from above and below,

$$\kappa_{a,4}(f_1, f_2, f_3, f_4) \longrightarrow 0.$$

Ferromagnetic correlation inequalities and the random-current cumulant expansion then bound each higher even cumulant by finite sums in which at least one factor is a four-point current-intersection block and all remaining factors are two-point functions. The two-point factors have finite scaling limits by the chosen normalization, while the four-point block tends to zero by (33.1). Hence $\kappa_{a,n} \to 0$ for every $n \geq 4$. Uniform moment bounds supplied by the same current representation make the moment problem determinate for the smeared variables, so every subsequential scaling limit has characteristic function

$$\exp\left[-\frac{1}{2}\sum_{i,j} t_i t_j \lim_{a\downarrow 0} \kappa_{a,2}(f_i, f_j)\right].$$

This is precisely Wick factorization of the limiting Schwinger functions.

The theorem therefore separates two logically different statements. For the specified positive lattice scalar and nearest-neighbor ferromagnetic Ising-type scaling-limit problems, the continuum limits covered by the random-current estimates are Gaussian. The separate problem of constructing an exotic ultraviolet-complete four-dimensional local QFT whose small-coupling expansion agrees with formal ϕ_4^4 perturbation theory to all orders but differs by effects invisible to perturbation theory remains open.

33.6 General 1PI Coordinates

Definition 33.4 (Field-normalization factors)**.** Three field-normalization factors occur in this chapter, and they have different domains of definition.

symbol	meaning
Z_R	Cutoff-dependent factor used in the initial renormalization convention, for example $\phi_R = Z_R^{-1/2}\phi$ in the scalar quartic example.
$Z_{\mathrm{MOM}}(\mu)$	Finite momentum-subtraction rescaling between an already renormalized reference field and the scale-μ 1PI coordinate field: $[\phi]_\mu = Z_{\mathrm{MOM}}(\mu)^{-1/2}\phi_R^{\mathrm{ref}}$.
Z_ϕ	Total bare-to-renormalized field factor used in the Callan–Symanzik equation for the selected chart: $\phi_R^{\mathrm{chart}} = Z_\phi^{-1/2}\phi_0$.

Thus Z_R and $Z_{\mathrm{MOM}}(\mu)$ are not equal as written. If $\phi_R^{\mathrm{ref}} = (Z_\phi^{\mathrm{ref}})^{-1/2}\phi_0$, then the chart whose elementary field is $[\phi]_\mu$ has the total factor

$$Z_\phi^{\mathrm{chart}} = Z_\phi^{\mathrm{ref}}\, Z_{\mathrm{MOM}}(\mu).$$

If the reference field is itself defined by the same momentum-subtraction condition at $\mu = \mu_{\mathrm{ref}}$, then $Z_{\mathrm{MOM}}(\mu_{\mathrm{ref}}) = 1$; the cutoff dependence is in Z_ϕ^{ref}, not in the finite rescaling.

Let a regulated Euclidean action be written as

$$S_\Lambda[\phi] = S_{\mathrm{kin},\Lambda}[\phi] + \int \mathrm{d}^D x \sum_I g_I^{\mathrm{bare}}(\Lambda)\, \mathcal{O}_I(x).$$

For each label I, let $\mathcal{O}_I$ be a local operator with mass dimension d_I, field number n_I, and derivative tensor structure specified by a polynomial P_I in external momenta. The corresponding 1PI coordinate at scale μ is extracted from the n_I-point vertex by

$$g_I(\mu) = \Pi_I \left[Z_{\mathrm{MOM}}(\mu)^{n_I/2}\, \Gamma^{(n_I)}(k_1, \ldots, k_{n_I}) \right]_{\mathcal{S}_{I,\mu}}.$$

The projector Π_I includes the sign and tensor normalization conventions. In the scalar quartic example, Π_I is multiplication by -1 at the symmetric four-point configuration.

Two elementary coordinates fix the normalization conventions. First consider the two-derivative quadratic operator represented, up to total derivatives and normalization convention, by $\mathcal{O}_{2,2} = \phi\,\Box\phi$, in the same self-energy convention used above. Its 1PI Taylor coefficient at $k^2 = \mu^2$ is

$$\Gamma^{(2;2)}\big|_{k^2=\mu^2} = \left.\frac{\partial \Sigma}{\partial k^2}\right|_{k^2=\mu^2} = 1 - Z_{\mathrm{MOM}}(\mu)^{-1}.$$

With the minus sign included in the quadratic-coordinate projector, the corresponding renormalized coordinate is

$$g_{2,2}(\mu) = -Z_{\mathrm{MOM}}(\mu) \left(1 - Z_{\mathrm{MOM}}(\mu)^{-1}\right) = 1 - Z_{\mathrm{MOM}}(\mu).$$

This is the same field-normalization information expressed as a 1PI coordinate. Second, for the mass operator with bare coefficient $g_2^{\mathrm{bare}} = m_0^2$, the dimensionless coordinate may be written

$$\lambda_2(\mu) = \mu^{-2} Z_{\mathrm{MOM}}(\mu) \left[g_2^{\mathrm{bare}} - \Sigma(k)\big|_{k^2=\mu^2} \right],$$

again in the same self-energy and projector convention. These examples make explicit that the field normalization, relevant mass coordinate, and higher 1PI vertex coordinates are part of one coordinate chart.

Definition 33.5 (Mass data and mass coordinates)**.** Assume that the connected Euclidean two-point function under discussion satisfies the spectral representation of Chapter 14. If the spectral measure has an isolated atom

$$\mathrm{d}\rho_\Phi(s) = Z_\Phi\, \delta_{m_*^2} + \mathrm{d}\rho_{\Phi,\mathrm{c}}(s), \qquad Z_\Phi > 0,$$

where $\delta_{m_*^2}$ is the Dirac measure at $s = m_*^2$. If there is no support below m_*^2 in that channel, then m_* is the *spectral pole mass* seen by the interpolating operator Φ. If the corresponding state is a stable one-particle state and another local operator has nonzero overlap with the same state, it gives the same m_*; the residue changes with the operator, not the mass. In a gauge theory this statement is made for gauge-invariant operators or for physical BRST cohomology states. Pole positions of gauge-fixed elementary fields are not, by themselves, physical mass data.

By contrast, m_R is a finite 1PI subtraction coordinate, and $\lambda_2(\mu)$ is a dimensionless running coordinate for the relevant quadratic deformation. They are scheme-dependent coordinates on a family of regulated Green functions. The lattice parameter m_0^2, or its spin-model analogue such as $T - T_c$, is a regulator coordinate that must be tuned as the cutoff is removed. None of these three coordinates is a pole mass by definition.

Remark 33.6 (Relation of the mass notions). Suppose the exact renormalized two-point function in the finite 1PI chart has the form

$$G_R(k) = \frac{1}{k^2 + m_R^2 - \Sigma_R(k)}$$

in a neighborhood admitting analytic continuation to the isolated stable pole. Then the spectral pole mass satisfies

$$m_*^2 = m_R^2 - \Sigma_R(k^2 = -m_*^2), \tag{33.2}$$

with the same sign convention as the displayed Euclidean inverse propagator. An on-shell mass prescription imposes $m_R = m_*$ order by order, while a generic momentum-subtraction or minimal-subtraction prescription does not. In an MS chart, $\lambda_2(\mu)$ is related to m_* only after a finite matching condition and RG evolution have been specified. In a lattice regularization with lattice spacing a, the bare coordinate $m_0^2(a)$ is tuned to a critical surface $m_{0,c}^2(a)$ for a massless scaling limit, or away from it so that

$$m_{\text{gap}} = \lim_{a\to 0} \frac{1}{a\,\xi_{\text{lat}}(a)} \tag{33.3}$$

exists. When the lower edge of the reconstructed spectral support is an isolated one-particle atom, $m_{\text{gap}} = m_*$; otherwise m_{gap} is the threshold edge rather than a pole mass.

The first assertion is the definition of an isolated Källen–Lehmann atom: near $s = m_*^2$,

$$G_E(k) = \frac{Z_\Phi}{k^2 + m_*^2} + \text{term analytic at } k^2 = -m_*^2.$$

Thus the inverse two-point function vanishes at $k^2 = -m_*^2$. Substituting that value into the chosen 1PI inverse-propagator convention gives (33.2). A finite change of renormalization prescription changes the coordinate m_R and the finite part of Σ_R, but not the spectral singularity of the completed Green function.

Minimal subtraction fixes a coordinate by pole subtraction in dimensional regularization. It contains no condition at the physical pole, so determining m_* requires first expressing the finite Green function in that coordinate and then solving the pole equation. For the lattice statement, exponential decay of Euclidean two-point functions gives ξ_{lat}^{-1} as the lowest transfer-matrix energy in lattice units in the selected channel. Multiplication by a^{-1} converts it to continuum units. Osterwalder–Schrader reconstruction identifies the limiting transfer-matrix spectrum with the reconstructed Hamiltonian spectrum under the usual reflection-positivity and scaling-limit hypotheses. An isolated lowest eigenvalue gives the spectral atom and hence m_*; continuous spectral support beginning at the same energy gives only a gap edge.

The finite comparison of nearby scales gives

$$\frac{\mathrm{d}g_I(\mu)}{\mathrm{d}\log\mu} = \beta_I(\{g_J(\mu)\};\mu).$$

Since g_I has engineering dimension $D - d_I$, define the dimensionless coordinate

$$\lambda_I(\mu) = \mu^{d_I - D} g_I(\mu).$$

When the coordinate list contains all dimensionless ratios relevant to the chosen family of theories, the equations may be written as

$$\frac{\mathrm{d}\lambda_I}{\mathrm{d}\log\mu} = \beta_I(\{\lambda_J\}).$$

This autonomous form is a coordinate statement: the beta functions depend on the selected operator basis, subtraction configurations, field normalization, and finite part conventions.

The practical value of the 1PI RG is logarithmic organization. A fixed-order comparison between two widely separated scales can contain powers of $\log(\mu/\mu')$. The differential equation breaks that comparison into small steps and then integrates the result, thereby summing the logarithmic terms determined by the computed beta function.

33.7 Callan–Symanzik Equation from Bare-Scale Independence

The beta function is a vector field on the renormalized coordinate space. Its action on correlation functions follows from the same construction of renormalized coordinates.

There are two logically distinct field factors in the preceding discussion, after the reference renormalization convention has been chosen. The factor $Z_{\mathrm{MOM}}(\mu)$ is finite after renormalization; it relates an already renormalized reference field ϕ_R^{ref} to the momentum-subtraction field used in the scale-μ coordinate chart,

$$[\phi]_\mu = Z_{\mathrm{MOM}}(\mu)^{-1/2}\phi_R^{\mathrm{ref}}.$$

The factor denoted Z_ϕ in the Callan–Symanzik equation is instead the cutoff-dependent total factor that relates the regulated bare field to the renormalized field chosen for the chart. If

$$\phi_R^{\mathrm{ref}} = \big(Z_\phi^{\mathrm{ref}}(\lambda,\mu_{\mathrm{ref}},\Lambda)\big)^{-1/2}\phi_0,$$

then the momentum-subtraction chart whose elementary field is $[\phi]_\mu$ has

$$[\phi]_\mu = \big(Z_\phi^{\mathrm{ref}}(\lambda,\mu_{\mathrm{ref}},\Lambda)\, Z_{\mathrm{MOM}}(\mu)\big)^{-1/2}\phi_0.$$

Thus, in that chart,

$$Z_\phi^{\mathrm{chart}}(\lambda(\mu),\mu,\Lambda) = Z_\phi^{\mathrm{ref}}(\lambda,\mu_{\mathrm{ref}},\Lambda)\, Z_{\mathrm{MOM}}(\mu).$$

Below the superscript "chart" is suppressed: Z_ϕ always denotes the total bare-to-renormalized field factor for the correlation functions being studied, while $Z_{\mathrm{MOM}}(\mu)$ denotes the finite momentum-subtraction rescaling appearing in the 1PI vertex coordinates.

Let ϕ_0 be the regulated bare scalar field. Let ϕ_R now denote the elementary field of the selected chart, for instance $[\phi]_\mu$ in the momentum-subtraction chart above, and choose Z_ϕ by

$$\phi_R = Z_\phi(\lambda(\mu),\mu,\Lambda)^{-1/2}\phi_0.$$

For a connected noncoincident n-point function, write

$$G_{0,\Lambda}^{(n)}(x_1,\ldots,x_n;g_0,\Lambda) = \langle\phi_0(x_1)\cdots\phi_0(x_n)\rangle_{\mathrm{conn},\Lambda,g_0}$$

for the regulated bare function. A renormalization chart consists of functions

$$g_0^a = F^a(\lambda(\mu),\mu,\Lambda)$$

and a field factor Z_ϕ such that

$$G_R^{(n)}(x_1,\ldots,x_n;\lambda,\mu) = \lim_{\Lambda\to\infty} Z_\phi^{-n/2} G_{0,\Lambda}^{(n)}\big(x_1,\ldots,x_n;F(\lambda,\mu,\Lambda),\Lambda\big)$$

exists as a distribution away from coincident points. The finite coordinates λ_I are dimensionless coordinates; dimensionful parameters are either included as additional dimensionless ratios or are displayed separately below.

Let $t = \log \mu$. The RG vector field in this chart is

$$\beta_I(\lambda) = \left.\frac{\mathrm{d}\lambda_I}{\mathrm{d}t}\right|_{g_0,\Lambda}.$$

The anomalous dimension of the elementary field in the same chart is

$$\gamma_\phi(\lambda) = \frac{1}{2}\left.\frac{\mathrm{d}\log Z_\phi}{\mathrm{d}t}\right|_{g_0,\Lambda}.$$

The convention agrees with the normalization used earlier: $\phi_R = Z_\phi^{-1/2}\phi_0$.

Callan–Symanzik equation. With the definitions above, the renormalized connected noncoincident n-point function satisfies

$$\left(\mu\frac{\partial}{\partial\mu} + \sum_I \beta_I(\lambda)\frac{\partial}{\partial\lambda_I} + n\gamma_\phi(\lambda)\right) G_R^{(n)} = 0. \tag{33.4}$$

The derivative $\mu\partial_\mu$ is taken at fixed x_i and fixed renormalized coordinates λ_I.

The regulated bare function contains no independent dependence on the auxiliary scale μ. Therefore

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}t} G_{0,\Lambda}^{(n)}(x_1,\ldots,x_n; g_0,\Lambda)\right|_{g_0,\Lambda}.$$

Using

$$G_{0,\Lambda}^{(n)} = Z_\phi^{n/2} G_R^{(n)}$$

inside the renormalization chart and differentiating at fixed bare data gives

$$0 = Z_\phi^{n/2}\left[\left(\frac{\partial}{\partial t} + \sum_I \beta_I \frac{\partial}{\partial\lambda_I}\right) G_R^{(n)} + \frac{n}{2}\frac{\mathrm{d}\log Z_\phi}{\mathrm{d}t} G_R^{(n)}\right].$$

The cutoff limit exists by the definition of the chart, and the displayed identity becomes (33.4).

Contact terms are not part of the displayed noncoincident equation. When insertion points collide, the omitted terms are local distributions supported on the collision diagonals. Their organization requires a renormalized local operator basis and is developed in the next chapter.

33.8 Canonical Scaling and Fixed Points

The Callan–Symanzik equation records dependence on the auxiliary renormalization scale. Dimensional homogeneity is a separate statement. Suppose the elementary scalar has engineering dimension

$$d_\phi^{\mathrm{eng}} = \frac{D-2}{2}$$

and let u_A denote displayed dimensionful renormalized parameters with engineering dimensions y_A. Away from coincident points, dimensional homogeneity gives

$$\left(\sum_{i=1}^{n} x_i^\nu \frac{\partial}{\partial x_i^\nu} - \mu\frac{\partial}{\partial \mu} - \sum_A y_A u_A \frac{\partial}{\partial u_A} + n d_\phi^{\text{eng}}\right) G_R^{(n)} = 0.$$

If the u_A are displayed rather than converted into dimensionless ratios, the Callan–Symanzik equation includes the additional term $\sum_A \beta_{u_A}\partial/\partial u_A$, where

$$\beta_{u_A} = \left.\frac{\mathrm{d}u_A}{\mathrm{d}\log\mu}\right|_{g_0,\Lambda}.$$

Combining that equation with dimensional homogeneity gives

$$\left(\sum_{i=1}^{n} x_i^\nu \frac{\partial}{\partial x_i^\nu} + \sum_I \beta_I \frac{\partial}{\partial \lambda_I} + \sum_A (\beta_{u_A} - y_A u_A)\frac{\partial}{\partial u_A} + n(d_\phi^{\text{eng}} + \gamma_\phi)\right) G_R^{(n)} = 0.$$

Equivalently, one may replace u_A by dimensionless ratios u_A/μ^{y_A}; then the canonical terms are included in the beta functions of those ratios.

At a fixed point with no displayed dimensionful deformation, $\beta_I(\lambda_*) = 0$ and $\gamma_\phi(\lambda_*) = \gamma_{\phi,*}$. The noncoincident correlator then obeys

$$\left(\sum_{i=1}^{n} x_i^\nu \frac{\partial}{\partial x_i^\nu} + n\Delta_\phi\right) G_R^{(n)}(x_1,\ldots,x_n;\lambda_*) = 0, \qquad \Delta_\phi = d_\phi^{\text{eng}} + \gamma_{\phi,*}.$$

Thus the field scaling dimension at the fixed point is the engineering dimension plus the anomalous dimension defined by the field normalization in the renormalization chart. Scaling dimensions of composite operators require the mixing analysis of local operator insertions.

33.9 Logarithmic Consistency

The existence of finite beta functions imposes relations among logarithms in cutoff perturbation theory. Let

$$L = \log\frac{\Lambda}{\mu},$$

and suppose a one-coupling coordinate has the bare expansion

$$g(\mu) = g_0 - b_1 g_0^2 L - b_2 g_0^3 L - c_1 g_0^3 L^2 + O(g_0^4).$$

Since $\mathrm{d}L/\mathrm{d}\log\mu = -1$,

$$\frac{\mathrm{d}g(\mu)}{\mathrm{d}\log\mu} = b_1 g_0^2 + b_2 g_0^3 + 2c_1 g_0^3 L + O(g_0^4).$$

Inverting the first displayed relation to the order needed gives

$$g_0 = g(\mu) + b_1 g(\mu)^2 L + O(g(\mu)^3).$$

Therefore

$$\frac{\mathrm{d}g(\mu)}{\mathrm{d}\log\mu} = b_1 g(\mu)^2 + b_2 g(\mu)^3 + 2\big(c_1 + b_1^2\big) g(\mu)^3 L + O(g(\mu)^4).$$

A beta function expressed in terms of the renormalized coordinate must be finite as the cutoff is removed. Hence the explicit L-term at this order vanishes:

$$c_1 = -b_1^2.$$

Figure 33.5 shows why the nested one-loop double logarithm is constrained by the one-loop beta function: the explicit cutoff logarithm cancels only for $c_1 = -b_1^2$.

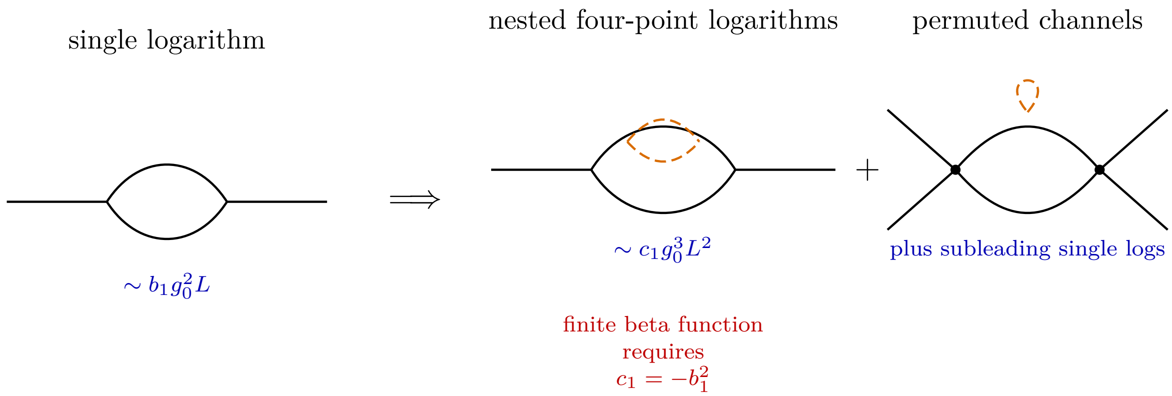


Figure 33.5: The double logarithm associated with nested one-loop subgraphs is constrained by the one-loop beta function. Dashed inner loops mark the nested subgraph whose one-loop logarithm is reused. Its coefficient is fixed by the requirement that the beta function, expressed in the renormalized coordinate, have a finite cutoff-removal limit.

This calculation is the scale-differential form of the locality statements from the previous chapters. The logarithm associated with a subgraph can be reused inside a larger graph, and the coefficient of the resulting double logarithm is determined by the lower-order beta function rather than by an independent input.

33.10 Scheme Changes

A renormalization scheme is a coordinate system on the renormalized family of theories. Let λ and $\widetilde{\lambda}$ be two coordinates related by a finite redefinition

$$\widetilde{\lambda} = f(\lambda), \qquad f(0) = 0, \qquad f'(0) = 1,$$

where the last condition fixes the leading normalization of the coordinate. If

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = \beta(\lambda), \qquad \frac{\mathrm{d}\widetilde{\lambda}}{\mathrm{d}\log\mu} = \widetilde{\beta}(\widetilde{\lambda}),$$

then the chain rule gives

$$\widetilde{\beta}(f(\lambda)) = f'(\lambda)\beta(\lambda), \qquad \beta(\lambda) = \frac{\widetilde{\beta}(f(\lambda))}{f'(\lambda)}.$$

If only the first nonlinear coefficient is displayed,

$$f(\lambda) = \lambda + \alpha\lambda^2 + O(\lambda^3),$$

then, to first order in α and without fixing the omitted cubic coefficient,

$$\beta(\lambda) = \widetilde{\beta}(\lambda) + \alpha\lambda\left(\lambda\frac{\partial}{\partial\lambda} - 2\right)\widetilde{\beta}(\lambda) + O(\alpha^2).$$

For one classically marginal coordinate,

$$\beta(\lambda) = b_1\lambda^2 + b_2\lambda^3 + b_3\lambda^4 + \cdots, \qquad \widetilde{\beta}(\lambda) = \widetilde{b}_1\lambda^2 + \widetilde{b}_2\lambda^3 + \widetilde{b}_3\lambda^4 + \cdots.$$

The relation above already implies

$$b_1 = \widetilde{b}_1, \qquad b_2 = \widetilde{b}_2.$$

To see the first scheme-dependent coefficient without hiding the role of the next finite redefinition, write

$$\widetilde{\lambda} = \lambda + \alpha\lambda^2 + \rho\lambda^3 + O(\lambda^4).$$

Using

$$\widetilde{\beta}(\widetilde{\lambda}) = f'(\lambda)\,\beta(\lambda), \qquad \widetilde{\lambda} = f(\lambda),$$

and re-expanding in powers of $\widetilde{\lambda}$ gives

$$\widetilde{b}_1 = b_1, \qquad \widetilde{b}_2 = b_2, \qquad \widetilde{b}_3 = b_3 - \alpha b_2 + (\rho - \alpha^2)b_1.$$

Thus, for a single marginal coupling and with the leading normalization fixed, the first two perturbative coefficients are invariant under finite analytic reparametrizations, while later coefficients are coordinate dependent.

For several couplings the corresponding statement is a statement about jets of a vector field, not about a list of scalar coefficients. Let a finite renormalized coordinate chart have coordinates λ^I centered at the point under discussion, and expand

$$\beta^I(\lambda) = M^I{}_J\lambda^J + \frac{1}{2}B^I{}_{JK}\lambda^J\lambda^K + O(\lambda^3), \qquad B^I{}_{JK} = B^I{}_{KJ}. \tag{33.5}$$

Let another scheme be related by the finite analytic coordinate change

$$\widetilde{\lambda}^A = R^A{}_I\lambda^I + \frac{1}{2}C^A{}_{IJ}\lambda^I\lambda^J + O(\lambda^3), \qquad C^A{}_{IJ} = C^A{}_{JI}, \tag{33.6}$$

where R is invertible. Write $S = R^{-1}$, and write $\widetilde{M} = RMS$. Then the quadratic coefficient of $\widetilde{\beta}$ is

$$\begin{aligned}\widetilde{B}^A{}_{BC} &= R^A{}_I B^I{}_{JK} S^J{}_B S^K{}_C \\ &\quad + C^A{}_{IJ}\left(S^I{}_B M^J{}_K S^K{}_C + S^I{}_C M^J{}_K S^K{}_B\right) - \widetilde{M}^A{}_D C^D{}_{IJ} S^I{}_B S^J{}_C.\end{aligned} \tag{33.7}$$

Thus the quadratic beta-function jet is not, in general, a tensorial invariant of the renormalization scheme. It is changed by the second derivative of the coordinate map whenever the linear term M is present. The invariant linear data at a fixed point are the similarity class of M, and hence its eigenvalues. The coordinate calculation is the following. The beta function transforms as a vector field:

$$\widetilde{\beta}^A(\widetilde{\lambda}) = \frac{\partial \widetilde{\lambda}^A}{\partial \lambda^I}\,\beta^I(\lambda).$$

The inverse coordinate map has the expansion

$$\lambda^I = S^I{}_A \widetilde{\lambda}^A - \frac{1}{2} S^I{}_A C^A{}_{JK} S^J{}_B S^K{}_C \widetilde{\lambda}^B \widetilde{\lambda}^C + O(\widetilde{\lambda}^3).$$

Substituting this inverse expansion and

$$\frac{\partial \widetilde{\lambda}^A}{\partial \lambda^I} = R^A{}_I + C^A{}_{IJ}\lambda^J + O(\lambda^2)$$

into the vector-field transformation law and collecting the terms quadratic in $\widetilde{\lambda}$ gives (33.7).

There is an important special case. If all displayed couplings are classically marginal at the reference point and the coordinate origin is the Gaussian fixed point, then $M = 0$. In that case (33.7) reduces to

$$\widetilde{B}^A{}_{BC} = R^A{}_I B^I{}_{JK} S^J{}_B S^K{}_C.$$

After the leading linear normalization R has been fixed, the one-loop quadratic tensor B is scheme independent. Without fixing R, the same geometric object is represented by different components in different linear bases. If $M \neq 0$, for example when relevant, irrelevant, or explicitly dimensionful coordinates are included, the quadratic coefficients are already coordinate dependent. At cubic order, even in the purely marginal case, the quadratic coordinate redefinition C combines with B; hence the single-coupling statement $b_2 = \widetilde{b}_2$ has no componentwise multi-coupling analogue without additional structure.

Remark 33.7 (Scheme coordinates for matched physical observables). Let $S_{\Lambda_0}(\theta)$ be the local action used in the regulated path integral, including all regulator-dependent local terms required by the chosen regularization procedure. Here Λ_0 denotes the ultraviolet regulator and θ denotes the full list of regulated action parameters. Let R and $\widetilde{R}$ be two renormalization schemes on the same perturbative domain. Assume the following data.

1. Each scheme assigns a finite renormalized coordinate to the same regulated action parameter θ:

   $$\lambda_\Lambda = \lambda_R(\theta, \Lambda_0), \qquad \widetilde{\lambda}_\Lambda = \lambda_{\widetilde{R}}(\theta, \Lambda_0),$$

   and the cutoff-removal limits are related by a finite analytic matching map

   $$\widetilde{\lambda} = f(\lambda), \qquad f(0) = 0, \qquad Df(0) \text{ invertible.}$$

   The same statement includes the finite field normalizations and local source coordinate maps needed to define inserted operators.

2. The observable A_Λ is defined at the regulated level before a renormalization scheme is chosen. Examples are an on-shell transition probability between normalized Haag–Ruelle particle states, a perturbative LSZ matrix element after the pole positions and residues have been matched, or an infrared-safe inclusive observable with the measurement function held fixed.

3. The ultraviolet and infrared limits required to define A exist order by order in the perturbative domain under the matching above. In massless theories this includes the infrared prescription, such as nonexceptional Euclidean momenta for Green functions or an inclusive measurement for scattering.

Let $A_R(\lambda)$ and $A_{\widetilde{R}}(\widetilde{\lambda})$ be the perturbative expansions obtained by computing the same regulated observable in the two schemes and then taking the common limit. Then, as formal perturbative series,

$$A_R(\lambda) = A_{\widetilde{R}}(f(\lambda)). \tag{33.8}$$

At finite Λ_0, A_Λ is a number, distribution, or transition probability computed in the regulated theory with action parameter θ. It does not know which finite coordinates will later be used to label the same regulated action. Thus

$$A_\Lambda(\theta) = A_{\Lambda,R}(\lambda_R(\theta, \Lambda_0)) = A_{\Lambda,\widetilde{R}}(\lambda_{\widetilde{R}}(\theta, \Lambda_0))$$

after including the field and source normalizations that belong to the two coordinate charts. The matching map is defined by eliminating the regulated action parameters between the two coordinate assignments and then taking $\Lambda_0 \to \infty$:

$$\lambda_{\widetilde{R}}(\theta, \Lambda_0) = f_{\Lambda_0}(\lambda_R(\theta, \Lambda_0)), \qquad f_{\Lambda_0} \to f.$$

If the observable has the stated perturbative limit, the finite-regulator identity passes to the limit order by order and gives (33.8).

For an LSZ amplitude, the finite field rescaling between schemes changes off-shell Green functions and amputated kernels, but the residue and one-particle-state normalizations used in the LSZ construction change by the inverse factors. The matched on-shell matrix element is therefore the same observable. For a local source insertion, a finite local source-coordinate redefinition can change the coincident-point extension of the source functional. This does not change a separated-point observable. If the observable itself probes coincident support, then its definition includes that source-response convention and the convention must be matched as part of the observable data.

Remark 33.8 (What is not scheme invariant). Equation (33.8) is a statement with a matching map. It does not say that $A_R(\lambda) = A_{\widetilde{R}}(\lambda)$ after the same coordinate values are inserted in two different schemes. It also does not make off-shell Green functions, beta-function components, anomalous-dimension matrices, or contact terms coordinate independent. Those quantities are useful precisely because they are chart-dependent representatives from which matched physical observables, fixed-point eigenvalues, anomaly classes, and other invariant data are extracted.

33.11 Perturbative Series as Asymptotic Data

The beta functions, anomalous dimensions, fixed-point coordinates, and critical exponents computed in the preceding and following chapters are usually obtained as formal expansions. The meaning of such an expansion must be specified separately from the rules used to compute its coefficients.

Definition 33.9 (Asymptotic expansion)**.** Let $A(g)$ be defined for g in a sector $S \subset \mathbb{C}$ whose closure contains $g = 0$, and let $\sum_{n\geq 0} a_n g^n$ be a formal power series. The series is an asymptotic expansion of A in S if, for every $N \geq 0$,

$$A(g) - \sum_{n=0}^{N} a_n g^n = o(g^N) \qquad (g \to 0,\ g \in S')$$

on every closed subsector S' strictly contained in S. A stronger version replaces $o(g^N)$ by $O(g^{N+1})$ with constants depending on N and S'.

An asymptotic expansion need not converge. Even when it determines every coefficient a_n, it does not by itself determine the function A: functions differing by terms such as $\exp(-c/g)$, with $c > 0$ in the sector under consideration, have the same power-series asymptotic expansion at the origin. Such nonperturbative terms are invisible to all orders in ordinary perturbation theory.

The common mechanism behind divergence is factorial growth of coefficients. In finite-dimensional integrals and in many field-theoretic expansions, the number of diagrams grows factorially. In examples where the saddle analysis of the analytically continued action is under control, the large-order coefficients have the form

$$a_n \sim C\,(-A)^{-n}\,\Gamma(n+b)$$

for constants A, C, b depending on the observable and on the expansion coordinate. Dyson's instability argument for QED is the simplest physical warning: if a perturbative series in e^2 had a nonzero radius of convergence around $e^2 = 0$, it would analytically continue to negative e^2, where like charges attract and the vacuum is not stable in the same way.[2] This argument is not a theorem about every renormalized quantity in every QFT; it is a reason not to identify perturbation theory with a convergent definition unless convergence or summability has actually been proved.

One possible reconstruction mechanism is Borel–Laplace summation. It is a theorem about a formal series together with analytic continuation and growth data for its Borel transform. The hypotheses are part of the statement.

Definition 33.10 (Gevrey-one series and directional Borel sum)**.** Let

$$\widehat{A}(g) = \sum_{n\geq 0} a_n g^n$$

be a formal power series. It is *Gevrey-one* if there are constants $C, M > 0$ such that

$$|a_n| \leq C\,M^n n! \qquad (n \geq 0).$$

Its formal Borel transform is

$$\mathcal{B}\widehat{A}(\xi) = \sum_{n\geq 0} \frac{a_n}{n!}\xi^n. \tag{33.9}$$

Fix a direction $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. The direction is admissible for $\widehat{A}$ if the series $\mathcal{B}\widehat{A}$ converges near $\xi = 0$, continues holomorphically to a sector

$$\Sigma_{\theta,\eta} = \{\xi \neq 0 : \ |\arg\xi - \theta| < \eta\}$$

for some $\eta > 0$, and obeys an exponential bound

$$|\mathcal{B}\widehat{A}(\xi)| \leq C_\eta \exp(|\xi|/R_\eta) \qquad (\xi \in \Sigma_{\theta,\eta}) \tag{33.10}$$

[2]See F. J. Dyson, "Divergence of perturbation theory in quantum electrodynamics," *Phys. Rev.* **85** (1952) 631–632, doi:10.1103/PhysRev.85.631. For explicit large-order analysis in closely related settings, see C. M. Bender and T. T. Wu, "Anharmonic oscillator," *Phys. Rev.* **184** (1969) 1231–1260, doi:10.1103/PhysRev.184.1231, and L. N. Lipatov, "Divergence of the perturbation-theory series and the quasiclassical theory," *Sov. Phys. JETP* **45** (1977) 216–223.

for some $C_\eta, R_\eta > 0$. For g satisfying

$$|\arg g - \theta| < \frac{\pi}{2} - \delta, \qquad |g| < R_\eta \cos\delta$$

with $0 < \delta < \pi/2$, the directional Borel sum is

$$\mathcal{S}_\theta \widehat{A}(g) = \frac{1}{g}\int_0^{e^{i\theta}\infty} e^{-\xi/g}\, \mathcal{B}\widehat{A}(\xi)\, d\xi, \tag{33.11}$$

where the ray is oriented from 0 to infinity.

Theorem 33.11 (Watson lemma for the Borel–Laplace transform). *Under the hypotheses of Definition 33.10, $\mathcal{S}_\theta\widehat{A}(g)$ is holomorphic in the indicated sector of the g-plane and has $\widehat{A}(g)$ as its asymptotic expansion there: for every $N \geq 0$ and every closed subsector strictly inside the domain,*

$$\mathcal{S}_\theta\widehat{A}(g) - \sum_{n=0}^{N} a_n g^n = O(g^{N+1}). \tag{33.12}$$

Proof. Rotate the integration variable so that the Borel ray is the positive real axis; the general case is the same after replacing g by $e^{-i\theta}g$. The exponential bound (33.10) gives absolute convergence of (33.11) on each closed subsector with $\Re(1/g) > 1/R_\eta + \epsilon$. Holomorphy follows by dominated convergence applied to compact subsets of that domain.

Choose $\rho > 0$ inside the convergence disk of the Borel transform. For $|\xi| \leq \rho$,

$$\mathcal{B}\widehat{A}(\xi) = \sum_{n=0}^{N} \frac{a_n}{n!}\xi^n + \xi^{N+1}R_N(\xi)$$

with R_N bounded on $|\xi| \leq \rho$. The polynomial terms give exactly

$$\frac{1}{g}\int_0^\infty e^{-\xi/g}\frac{\xi^n}{n!}\, d\xi = g^n$$

when $\Re g > 0$, and by the same rotated-ray computation in a sector. The small-ξ remainder is bounded by a constant times

$$\frac{1}{|g|}\int_0^\rho e^{-c\xi/|g|}\xi^{N+1}\, d\xi = O(|g|^{N+1})$$

on each closed subsector, where $c > 0$ is the lower bound for $\Re(\xi/g)/(|\xi|/|g|)$. The tail integral from ρ to infinity is bounded by $C\exp(-c'\rho/|g|)$ for some $c' > 0$, hence is $O(|g|^{N+1})$ for every fixed N. This proves (33.12). □

Definition 33.12 (Lateral Borel sums and Stokes discontinuity). Let $\widehat{A}(g)$ be Gevrey-one, and suppose that its Borel transform is holomorphic in a punctured sector around the ray $\arg\xi = \theta$, except for a locally finite set of singular points on the ray. If the deformed rays $\arg\xi = \theta \pm \epsilon$ have Borel–Laplace sums and these sums have limits as $\epsilon \downarrow 0$, the limits are the lateral sums

$$\mathcal{S}_{\theta^+}\widehat{A}(g), \qquad \mathcal{S}_{\theta^-}\widehat{A}(g).$$

Their difference

$$\mathrm{Disc}_\theta\, \widehat{A}(g) = \mathcal{S}_{\theta^+}\widehat{A}(g) - \mathcal{S}_{\theta^-}\widehat{A}(g) \tag{33.13}$$

is the Stokes discontinuity across the ray. This definition is a statement about boundary values of a named Borel transform; it is not a prescription unless the analytic continuation, growth bounds, and lateral limits have been established.

Simple-pole Stokes jump. Let $A > 0$, and suppose that in a neighborhood of $\xi = A$ the Borel transform has the form

$$\mathcal{B}\widehat{A}(\xi) = \frac{r}{\xi - A} + H(\xi), \tag{33.14}$$

where H is holomorphic. Assume that away from this singularity the lateral Borel integrals along the positive ray satisfy the hypotheses of Definition 33.12. Then the contribution of this pole to the upper-minus-lower discontinuity is

$$\mathrm{Disc}_0\,\widehat{A}(g) = \frac{2\pi \mathrm{i} r}{g}\,\mathrm{e}^{-A/g} + \mathrm{Disc}_0^{\mathrm{other}}\,\widehat{A}(g), \tag{33.15}$$

where the last term denotes contributions from other singularities on the same ray.

Subtract the holomorphic part H, whose upper and lower deformations across a small disk around A have the same limit. The difference between the two lateral contours for the pole term is a positively oriented small circle around A. Hence

$$\frac{1}{g}\oint_{|\xi - A| = \epsilon} \mathrm{e}^{-\xi/g}\frac{r}{\xi - A}\,\mathrm{d}\xi = \frac{2\pi \mathrm{i} r}{g}\,\mathrm{e}^{-A/g}.$$

The remaining parts of the contour cancel by deformation in the common holomorphic domain. Summing over singularities on the ray gives the stated local contribution.

Definition 33.13 (One-parameter transseries system)**.** Fix $A > 0$, $\beta \in \mathbb{C}$, and a sector in the g-plane. A one-parameter transseries system at the first exponential scale is a formal object

$$\widehat{F}(g;\sigma) = \widehat{\Phi}_0(g) + \sigma\,\mathrm{e}^{-A/g} g^{\beta}\widehat{\Phi}_1(g), \tag{33.16}$$

where $\widehat{\Phi}_0$ and $\widehat{\Phi}_1$ are formal power series and $\sigma \in \mathbb{C}$ is a parameter. A summation of (33.16) requires summation data for both formal series and a rule for how σ changes when the summation direction crosses a Stokes ray.

Remark 33.14 (Algebra of the first Stokes cancellation). Assume that the lateral sums of $\widehat{\Phi}_0$ and $\widehat{\Phi}_1$ exist on a ray and satisfy

$$\mathcal{S}_+\widehat{\Phi}_0 - \mathcal{S}_-\widehat{\Phi}_0 = C\,\mathrm{e}^{-A/g} g^{\beta}\,\mathcal{S}_-\widehat{\Phi}_1, \qquad \mathcal{S}_+\widehat{\Phi}_1 = \mathcal{S}_-\widehat{\Phi}_1, \tag{33.17}$$

for a constant C. Then the two lateral sums of $\widehat{F}(g;\sigma)$ agree if and only if

$$\sigma_- = \sigma_+ + C. \tag{33.18}$$

If the data are real on the two sides of a real Stokes ray and C is purely imaginary, the median choice

$$\sigma_+ = \sigma_{\mathrm{R}} - \frac{C}{2}, \qquad \sigma_- = \sigma_{\mathrm{R}} + \frac{C}{2}, \qquad \sigma_{\mathrm{R}} \in \mathbb{R}, \tag{33.19}$$

gives a common real value whenever the two lateral boundary values are complex conjugates.

Using (33.17),

$$\begin{aligned} \mathcal{S}_+\widehat{F}(g;\sigma_+) &= \mathcal{S}_-\widehat{\Phi}_0 + (C + \sigma_+)\mathrm{e}^{-A/g} g^{\beta}\mathcal{S}_-\widehat{\Phi}_1, \\ \mathcal{S}_-\widehat{F}(g;\sigma_-) &= \mathcal{S}_-\widehat{\Phi}_0 + \sigma_-\mathrm{e}^{-A/g} g^{\beta}\mathcal{S}_-\widehat{\Phi}_1. \end{aligned}$$

Equality is therefore equivalent to (33.18) unless the displayed one-instanton lateral sum vanishes identically, in which case the same equation is the natural continuous extension. The median choice is the symmetric solution of this affine relation. Under the stated reality hypothesis the two lateral values are conjugate, so their common symmetric value is real.

Example 33.15 (Zero-dimensional stable quartic integral)**.** The normalized integral

$$I(g) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-\frac{x^2}{2} - \frac{gx^4}{4!}\right) \mathrm{d}x \qquad (\Re g > 0) \tag{33.20}$$

is a finite-dimensional model of the distinction between coefficientwise perturbation theory and analytic reconstruction. Termwise expansion at $g = 0$ gives

$$I(g) \sim \sum_{n\geq 0} a_n g^n, \qquad a_n = (-1)^n \frac{(4n)!}{n!\,96^n (2n)!}. \tag{33.21}$$

Indeed

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^{4n} \mathrm{e}^{-x^2/2}\, \mathrm{d}x = (4n-1)!! = \frac{(4n)!}{2^{2n}(2n)!},$$

and the extra factor is $(-1)^n/(n!(4!)^n)$. Stirling's formula gives

$$\frac{a_{n+1}}{(n+1)a_n} \longrightarrow -\frac{2}{3}, \tag{33.22}$$

so the perturbative series has zero radius of convergence. Dividing by $n!$ gives the Borel series

$$\mathcal{B}I(\xi) = \sum_{n\geq 0} \frac{a_n}{n!} \xi^n = {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; -\frac{2\xi}{3}\right) \tag{33.23}$$

near the origin. The hypergeometric function has its first finite singularity at argument 1, hence at $\xi = -3/2$ in this normalization. Along the positive g-axis the integral is nevertheless well-defined and its perturbative series is a candidate for ordinary Borel summation. The opposite sign of g makes the integral unstable on the real contour; the same analytic formula must then be understood only after choosing a complex cycle.

Let $A > 0$, and suppose a Borel transform $B(\xi)$ is holomorphic on the cut plane $\mathbb{C} \setminus (-\infty, -A]$. The map

$$w(\xi) = \frac{\sqrt{1+\xi/A} - 1}{\sqrt{1+\xi/A} + 1}, \qquad \xi(w) = \frac{4Aw}{(1-w)^2}, \tag{33.24}$$

with the square root chosen positive at $\xi = 0$, maps the cut plane biholomorphically to the unit disk. It sends $\xi = 0$ to $w = 0$, the first branch point $\xi = -A$ to $w = -1$, and the positive real ξ-axis to $0 \leq w < 1$. Set $y = \sqrt{1+\xi/A}$. The cut plane is mapped by y to the right half plane $\Re y > 0$. The Cayley transform $w = (y-1)/(y+1)$ maps the right half plane to $|w| < 1$. Solving $y = (1+w)/(1-w)$ gives

$$1 + \frac{\xi}{A} = \left(\frac{1+w}{1-w}\right)^2, \qquad \xi = \frac{4Aw}{(1-w)^2}.$$

The stated images of 0, $-A$, and the positive real ray are immediate.

Remark 33.16 (Status of conformal-Borel numerics). The conformal map in (33.24) is a rigorous acceleration device only when the analytic domain of the Borel transform is known. In many applications to critical exponents one assumes, from finite-dimensional saddle analysis or from additional field-theoretic input, that the nearest Borel singularity is on the negative real axis and that no positive-ray singularity obstructs the Laplace integral. A conformal-Borel or Borel–Pade computation made under those assumptions is a controlled numerical procedure conditional on the stated singularity hypothesis; it is not a proof that the regulated QFT exists or that the perturbative expansion by itself defines it.

Definition 33.17 (Borel–Leroy and conformal-Borel approximants)**.** Let $\widehat{A}(g) = \sum_{n\geq 0} a_n g^n$ be a formal series and let $b > -1$. The Borel–Leroy transform with parameter b is

$$\mathcal{B}_b\widehat{A}(\xi) = \sum_{n\geq 0} \frac{a_n}{\Gamma(n+b+1)}\xi^n. \tag{33.25}$$

If $\mathcal{B}_b\widehat{A}$ satisfies the same analytic-continuation and growth hypotheses as in Definition 33.10, with the branch of ξ^b fixed on the chosen ray, the corresponding Borel–Leroy sum is

$$\mathcal{S}_{\theta,b}\widehat{A}(g) = \frac{1}{g^{b+1}}\int_0^{\mathrm{e}^{i\theta}\infty} \mathrm{e}^{-\xi/g}\xi^b\mathcal{B}_b\widehat{A}(\xi)\,\mathrm{d}\xi. \tag{33.26}$$

For the one-cut hypothesis of (33.24), write

$$\mathcal{B}_b\widehat{A}(\xi(w)) = \sum_{m\geq 0} c_m w^m$$

near $w = 0$, and let

$$\mathcal{B}^{\mathrm{conf}}_{b,N}(\xi) = \sum_{m=0}^{N} c_m w(\xi)^m. \tag{33.27}$$

The N-th conformal-Borel–Leroy approximant on the positive ray is

$$\mathcal{S}^{(A,b)}_N\widehat{A}(g) = \frac{1}{g^{b+1}}\int_0^{\infty} \mathrm{e}^{-\xi/g}\xi^b\mathcal{B}^{\mathrm{conf}}_{b,N}(\xi)\,\mathrm{d}\xi, \qquad g > 0, \tag{33.28}$$

when the integral converges. The data (A, b, N) are part of the approximation, not hidden conventions.

Coefficient preservation by the truncated conformal map. Assume the notation of Definition 33.17. Suppose $\mathcal{B}_b\widehat{A}$ is holomorphic at the origin, and $w(\xi)$ is the map (33.24). Then

$$\mathcal{B}^{\mathrm{conf}}_{b,N}(\xi) - \mathcal{B}_b\widehat{A}(\xi) = O(\xi^{N+1}) \qquad (\xi \to 0). \tag{33.29}$$

Consequently the formal small-g expansion of $\mathcal{S}^{(A,b)}_N\widehat{A}(g)$ reproduces the coefficients $a_0, \ldots, a_N$:

$$\mathcal{S}^{(A,b)}_N\widehat{A}(g) \sim \sum_{n=0}^{N} a_n g^n + O(g^{N+1}) \tag{33.30}$$

in the same Watson-lemma sense as Theorem 33.11, provided the positive-ray integral has a uniform exponential-growth bound. The map $w(\xi)$ is holomorphic near $\xi = 0$, satisfies $w(0) = 0$, and has nonzero derivative there:

$$w(\xi) = \frac{\xi}{4A} + O(\xi^2).$$

It therefore defines an invertible holomorphic coordinate near the origin. Truncating the Taylor expansion of $\mathcal{B}_b\widehat{A}(\xi(w))$ at order N in w changes the function by $O(w^{N+1})$, hence by $O(\xi^{N+1})$. This proves (33.29).

For each $0 \leq n \leq N$, the coefficient of ξ^n in $\mathcal{B}^{\mathrm{conf}}_{b,N}$ is therefore $a_n/\Gamma(n+b+1)$. Termwise integration of the polynomial part gives

$$\frac{1}{g^{b+1}}\int_0^{\infty} \mathrm{e}^{-\xi/g}\xi^{n+b}\,\mathrm{d}\xi = g^n\Gamma(n+b+1), \qquad b > -1, \tag{33.31}$$

and hence recovers $a_n g^n$. The remainder estimate is the same small-ray and tail estimate used in Theorem 33.11, with ξ^{N+1+b} replacing ξ^{N+1}. The hypothesis $b > -1$ is exactly the integrability condition at $\xi = 0$.

Remark 33.18 (How this applies to critical exponents). In the Wilson–Fisher ε-expansion and in fixed-dimension renormalized perturbation theory, one often applies (33.28) to a critical exponent, an anomalous dimension, or a fixed-point coordinate. The theorem-level input is the list of loop coefficients together with the renormalized chart, the Borel–Leroy parameter b, the assumed first singularity distance A, the analytic domain in the Borel plane, and a rule for estimating the truncation and parameter sensitivity. When these analytic data are not proved, the resulting number is a controlled extrapolation under declared hypotheses; it is not a proof of the existence or uniqueness of the critical QFT.

The preceding example is the finite-dimensional form of Dyson's warning: analytic continuation in a coupling can change the convergence of the integration cycle and the physical stability of the problem. The correct mathematical object is therefore not only a formal series but a pair consisting of a formal expansion and a choice of analytic sector or integration cycle. This is the point of the Lefschetz-thimble formulation.

Definition 33.19 (Holomorphic oscillatory integral and thimbles)**.** Let X be a complex n-manifold with holomorphic volume form Ω, let $S : X \to \mathbb{C}$ be a holomorphic Morse function, and let Γ be a real n-cycle on which

$$I_\Gamma(g) = \int_\Gamma \mathrm{e}^{-S(z)/g} a(z)\,\Omega \tag{33.32}$$

converges for $g = |g|\mathrm{e}^{i\theta}$. Here a is holomorphic, or more generally has controlled analytic growth compatible with the convergence of the integral. Set

$$h_\theta(z) = \mathrm{Re}\big(\mathrm{e}^{-i\theta} S(z)\big), \qquad \varphi_\theta(z) = \mathrm{Im}\big(\mathrm{e}^{-i\theta} S(z)\big).$$

For a critical point p_σ, its upward thimble $\mathcal{J}_\sigma^\theta$ is the real n-manifold obtained by flowing out of p_σ along the gradient flow of h_θ, so that $h_\theta \to +\infty$ at the ends. Its dual downward thimble $\mathcal{K}_\sigma^\theta$ is defined with the opposite flow. Along either flow,

$$\frac{\mathrm{d}h_\theta}{\mathrm{d}t} = \|\partial(\mathrm{e}^{-i\theta} S)\|^2, \qquad \frac{\mathrm{d}\varphi_\theta}{\mathrm{d}t} = 0 \tag{33.33}$$

for the upward convention. Thus the phase of the exponential is constant on each thimble.

Theorem 33.20 (Finite-dimensional thimble decomposition)**.** *Assume that h_θ is proper on the relative cycle problem defined by (33.32), that all critical values $\mathrm{e}^{-i\theta} S(p_\sigma)$ have distinct imaginary parts, and that the flows have no escape at finite flow time. Then the relative homology class of Γ decomposes as*

$$[\Gamma] = \sum_\sigma n_\sigma(\theta)[\mathcal{J}_\sigma^\theta], \qquad n_\sigma(\theta) = \langle \Gamma, \mathcal{K}_\sigma^\theta \rangle \in \mathbb{Z}, \tag{33.34}$$

and

$$I_\Gamma(g) = \sum_\sigma n_\sigma(\theta) I_\sigma(g), \qquad I_\sigma(g) = \int_{\mathcal{J}_\sigma^\theta} \mathrm{e}^{-S(z)/g} a(z)\,\Omega. \tag{33.35}$$

Near $g = 0$, each $I_\sigma(g)$ has a saddle expansion

$$I_\sigma(g) \sim \mathrm{e}^{-S(p_\sigma)/g}(2\pi g)^{n/2}\frac{a(p_\sigma)}{\sqrt{\det H_\sigma}}\left(1 + \sum_{m\geq 1} c_{\sigma,m}g^m\right), \tag{33.36}$$

after choosing the square-root phase determined by the oriented thimble; here H_σ is the Hessian of S at p_σ in holomorphic local coordinates. When two critical values have the same φ_θ, the integer basis $[\mathcal{J}_\sigma^\theta]$ can jump by a Stokes transformation. The homology class $[\Gamma]$ and the integral $I_\Gamma(g)$ are unchanged; only their expression in the thimble basis changes.

Proof. The gradient equation in Definition 33.19 gives

$$\frac{\mathrm{d}}{\mathrm{d}t}h_\theta(z(t)) = \sum_i \left|\frac{\partial}{\partial z^i}\left(\mathrm{e}^{-i\theta}S\right)\right|^2$$

in local holomorphic coordinates, while the Cauchy–Riemann equations give $\mathrm{d}\varphi_\theta/\mathrm{d}t = 0$. Thus each thimble is a constant-phase steepest-descent cycle and the exponential decays at its ends by properness of h_θ. Standard finite-dimensional Morse theory for the pair $\{h_\theta \leq H\} \subset X$, followed by $H \to +\infty$, identifies the middle-dimensional relative homology group with the free abelian group generated by the upward thimbles. The dual downward thimbles intersect them with pairing $\delta_{\sigma\tau}$, so the coefficient of $\mathcal{J}_\sigma^\theta$ in any relative cycle is $\langle \Gamma, \mathcal{K}_\sigma^\theta\rangle$. This proves (33.34), and integration of the same holomorphic form over homologous relative cycles gives (33.35).

For the expansion, apply the holomorphic Morse lemma near p_σ:

$$S(z) = S(p_\sigma) + \frac{1}{2}\sum_{i=1}^n \lambda_i w_i^2$$

in local coordinates. On the thimble the quadratic form has positive real part after multiplication by $\mathrm{e}^{-i\theta}$. Rescale $w_i = g^{1/2}y_i$, expand $a(w)$ and the higher-coordinate corrections of the volume form, and integrate term by term against the convergent Gaussian. Odd monomials vanish in the diagonal Gaussian coordinates and even monomials produce the powers in (33.36). The square-root of $\det H_\sigma$ is the Gaussian determinant with the orientation and phase chosen by the thimble.

A Stokes wall occurs precisely when two critical points can lie on a common gradient trajectory, which requires equality of the constant phase φ_θ. Crossing such a wall changes the thimble basis by adding integer multiples of thimbles connected by the flow. Since this is a change of basis in relative homology, the original cycle and integral are unchanged. □

Remark 33.21 (Resurgence as a regulated statement). In finite dimensions the Borel singularities of the saddle expansion around p_σ are controlled, under the usual analyticity and nondegeneracy hypotheses, by action differences $S(p_\tau) - S(p_\sigma)$. Lateral Borel sums across a Stokes ray differ by the contribution of thimbles whose critical values become phase-aligned with p_σ. In a QFT path integral this sentence becomes a theorem only after a regulator has turned the functional integral into a finite-dimensional or otherwise rigorously controlled analytic object, and after the regulator-removal limit has been proved in the sector of interest. Without those data, words such as "instanton resurgence" or "renormalon resurgence" name a proposed analytic mechanism, not a completed definition of the theory.

Renormalons and Borel-plane obstructions. The term *renormalon* refers to a class of large-order effects in renormalized perturbation theory whose diagnostic is a Borel singularity associated with an integration over a momentum region. It is not a theorem that every perturbative series in a QFT has a renormalon of a specified type, and it is not a replacement for a nonperturbative definition of the observable. Nor should it be stated as a property of the beta function alone. The statement must always name the observable or coefficient function, the expansion coordinate, the renormalization scheme, and the factorization prescription.

Factorial moment in a running-coupling model. For $p > 0$ and $n \geq 0$,

$$\int_0^1 x^{p-1}\big(-\log x\big)^n \,\mathrm{d}x = \frac{n!}{p^{n+1}}. \tag{33.37}$$

Consequently, if a controlled approximation to an observable contains

$$R(\alpha) = \alpha \int_0^1 \frac{x^{p-1}\,\mathrm{d}x}{1-\beta_0\alpha(-\log x)} \tag{33.38}$$

and the geometric expansion is used before the integral is performed, then

$$R(\alpha) \sim \alpha \sum_{n\geq 0} \frac{\beta_0^n n!}{p^{n+1}}\alpha^n, \qquad \mathcal{B}_R(u) = \frac{1}{p-\beta_0 u} \tag{33.39}$$

for the Borel transform of $R(\alpha)/\alpha$. The model therefore has a positive-axis Borel pole at $u = p/\beta_0$ when $\beta_0 > 0$.

Set $x = \mathrm{e}^{-t}$. Then $x^{p-1}\mathrm{d}x = \mathrm{e}^{-pt}\mathrm{d}t$ with $t \in [0,\infty)$, and

$$\int_0^1 x^{p-1}(-\log x)^n \,\mathrm{d}x = \int_0^\infty \mathrm{e}^{-pt}t^n \,\mathrm{d}t = \frac{\Gamma(n+1)}{p^{n+1}} = \frac{n!}{p^{n+1}}.$$

Expanding the denominator in (33.38) as $\sum_{n\geq 0}[\beta_0\alpha(-\log x)]^n$ gives the displayed asymptotic coefficients. Dividing the coefficient of α^{n+1} by $n!$ gives the geometric Borel series

$$\sum_{n\geq 0} \frac{\beta_0^n}{p^{n+1}} u^n = \frac{1}{p-\beta_0 u}.$$

A controlled field-theoretic model calculation has the same algebraic core. Suppose a Euclidean short-distance quantity contains, after a specified set of approximations such as insertion of a chain of one-loop self-energy bubbles, an integral over a momentum k of the form

$$\int \frac{\mathrm{d}k}{k}\, F(k/Q)\,\alpha(k),$$

where Q is the external scale, F restricts the relevant region, and $\alpha(k)$ is expanded around $\alpha(Q)$ using one-loop running. If

$$\alpha(k) = \frac{\alpha(Q)}{1+\beta_0\alpha(Q)\log(k/Q)} = \alpha(Q)\sum_{n\geq 0}\big[-\beta_0\alpha(Q)\log(k/Q)\big]^n$$

is inserted before the k-integral is performed, then integrals of powers of $\log(k/Q)$ can give $n!$-growth. In the Borel transform this factorial growth is represented, in this approximation, by singularities at finite Borel parameter.

The name of the renormalon records the momentum region. An infrared renormalon is produced by the small-k region of such an integral; in an asymptotically free theory such a singularity can lie on the positive Borel ray for particular Euclidean short-distance quantities and obstruct an unambiguous Borel sum of the perturbative coefficient function. The associated lateral-summation ambiguity then has the size of a power correction,

$$\exp[-c/\alpha(Q)] = \left(\frac{\Lambda_{\mathrm{RG}}}{Q}\right)^p$$

for suitable constants $c, p > 0$, where Λ_{RG} is the renormalization-group invariant scale. In QCD this is why infrared renormalon analysis is tied to the operator product expansion: perturbation theory for a Wilson coefficient and the definition of the corresponding long-distance matrix element are not separately physical, and their scheme/factorization ambiguities must cancel in the full observable.

Remark 33.22 (Algebraic form of OPE ambiguity cancellation). Let $Q \gg \mu$, and suppose a dimensionless Euclidean observable is represented in a specified factorization scheme by the truncated operator product expansion

$$\mathcal{X}_\pm(Q) = C_{0,\pm}(Q,\mu) + \frac{1}{Q^p} C_p(Q,\mu) M_{p,\pm}(\mu) + O(Q^{-p-\delta}) \tag{33.40}$$

with $p, \delta > 0$. Here $C_{0,\pm}$ are the two lateral Borel sums of the perturbative coefficient function for the identity operator, $M_{p,\pm}$ are the correspondingly defined matrix elements of an operator of mass dimension p, and C_p is assumed to have no lateral ambiguity at this order. If

$$C_{0,+}(Q,\mu) - C_{0,-}(Q,\mu) = -\frac{1}{Q^p} C_p(Q,\mu)\big(M_{p,+}(\mu) - M_{p,-}(\mu)\big), \tag{33.41}$$

then

$$\mathcal{X}_+(Q) - \mathcal{X}_-(Q) = O(Q^{-p-\delta}). \tag{33.42}$$

If the identity-coefficient ambiguity has the form

$$C_{0,+} - C_{0,-} = K(\mu)\left(\frac{\Lambda_{\mathrm{RG}}}{Q}\right)^p, \tag{33.43}$$

then cancellation at this order requires

$$M_{p,+} - M_{p,-} = -\frac{K(\mu)\Lambda_{\mathrm{RG}}^p}{C_p(Q,\mu)} \tag{33.44}$$

in the normalization used in (33.40).

Subtract the two lateral definitions of $\mathcal{X}$ in (33.40). The remainder contributes only $O(Q^{-p-\delta})$, and the displayed leading difference is

$$C_{0,+} - C_{0,-} + \frac{1}{Q^p} C_p(M_{p,+} - M_{p,-}).$$

This vanishes by (33.41), proving (33.42). Substituting (33.43) into (33.41) gives (33.44). The cancellation is algebraic; a QFT application still has to construct the factorization scheme, the operator M_p, and the lateral sums.

An ultraviolet renormalon is associated with the large-k region of the corresponding expansion. In theories whose perturbative coordinate grows toward the ultraviolet, such as an IR-free scalar or Abelian gauge theory, this large-momentum sensitivity is tied to the same short-distance loss of perturbative control indicated by the Landau scale. The precise location of a renormalon singularity, and whether it lies on the integration ray relevant for Borel summation, depends on the observable, the expansion variable, and the renormalization scheme. The invariant lesson is not that renormalons by themselves define the theory, but that a perturbative representation can carry ambiguities of order $\exp[-c/\alpha]$, or $\exp[-c/g^2]$ when g is the gauge coupling. Such terms are invisible to all orders in the perturbative expansion and must be fixed, canceled, or interpreted by additional nonperturbative or effective-field-theory data. [3]

Application status. The preceding definitions are analytic objects, not decorative language. A QFT application of Borel summation, transseries, or Lefschetz thimbles must name the regulated observable, the expansion coordinate, the analytic domain, and the limiting statement being asserted. The following structured input is the minimal specification used in this monograph.

Definition 33.23 (Regulated asymptotic-analysis specification for a QFT observable)**.** Let R label a regulator, let g be the perturbative expansion coordinate, and let $\mathcal{X}_R(g;\lambda)$ be a regulated observable, where λ denotes all other named parameters, including masses, subtraction scales, background sources, finite volume, and boundary conditions. A *regulated asymptotic-analysis specification* consists of:

1. a complex domain D_R in the g-plane on which $\mathcal{X}_R(g;\lambda)$ is defined as a function or distributional matrix element;
2. a formal perturbative series
$$\widehat{\mathcal{X}}_R(g;\lambda) = \sum_{n\geq 0} a_{R,n}(\lambda) g^n$$
with the coefficient-extraction operation specified;
3. a Borel transform or Borel–Leroy transform, together with its proved or hypothesized analytic continuation domain and growth bounds;
4. , when nonperturbative sectors are present, a sector set S_R, action differences $A_{R,\sigma}(\lambda)$, powers $\beta_{R,\sigma}$, and formal fluctuation series
$$\mathrm{e}^{-A_{R,\sigma}(\lambda)/g} g^{\beta_{R,\sigma}} \widehat{\Phi}_{R,\sigma}(g;\lambda);$$
5. a statement of what happens as R is removed: coefficientwise limit, Borel-plane limit, distributional limit, correlation-function limit, or no continuum-limit claim.

Remark 33.24 (Scope of a regulated asymptotic analysis). A specification in Definition 33.23 proves a Borel–Laplace reconstruction of $\mathcal{X}_R$ in a direction θ only if the transform in item 3 satisfies the hypotheses of Definition 33.10 and the resulting Borel–Laplace function is identified with $\mathcal{X}_R$ on a nonempty open subdomain of D_R. It proves a QFT statement only after the asserted regulator-removal limit in item 5 is established. Without item 5, the result is a theorem or approximation for the regulated problem, not for the continuum QFT.

[3] For a systematic review, see M. Beneke, "Renormalons," *Phys. Rept.* **317** (1999) 1–142, doi:10.1016/S0370-1573(98)00130-6, arXiv:hep-ph/9807443.

The first clause is exactly the content of the Borel–Laplace definition: Gevrey growth supplies a locally convergent Borel transform, analytic continuation and exponential bounds make the Laplace integral convergent, and Theorem 33.11 supplies the asymptotic expansion of the resulting function. The equality with the originally regulated observable is an additional identity of analytic functions or distributions. It follows from equality on a nonempty open set inside the common connected domain by the identity theorem, or by equality of distributional boundary values when the observable is defined distributionally. Removing the regulator is a separate limiting operation. Thus a statement at fixed R cannot imply a continuum-QFT statement unless item 5 is supplied.

For the zero-dimensional quartic integral of Example 33.15, the Borel transform

$$B(\xi) = {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; -\frac{2\xi}{3}\right)$$

has no singularity for $\xi \geq 0$ and is bounded on the positive real ξ-axis. Therefore

$$\frac{1}{g}\int_0^\infty \mathrm{e}^{-\xi/g} B(\xi)\,\mathrm{d}\xi$$

exists for every $g > 0$ and has the formal series $\sum_{n\geq 0} a_n g^n$ of (33.21) as its asymptotic expansion. This is a useful model calculation rather than a theorem about a continuum QFT. Euler's hypergeometric integral with

$$a = \frac{1}{4}, \qquad b = \frac{3}{4}, \qquad c = 1, \qquad z = -\frac{2\xi}{3}.$$

Since $\operatorname{Re} c > \operatorname{Re} b > 0$,

$$B(\xi) = \frac{1}{\Gamma(3/4)\Gamma(1/4)}\int_0^1 t^{-1/4}(1-t)^{-3/4}\left(1 + \frac{2\xi t}{3}\right)^{-1/4} \mathrm{d}t. \tag{33.45}$$

For $\xi \geq 0$ the last factor is real, positive, and bounded above by 1. The remaining beta integral is finite. Thus B is bounded on the positive ray and has no positive-ray singularity. The Borel–Laplace integral therefore converges for $g > 0$, and Theorem 33.11 gives the displayed asymptotic expansion because the Taylor coefficients of B are $a_n/n!$, as computed in Example 33.15.

Definition 33.25 (Quartic-oscillator spectral Borel problem)**.** Let $g \geq 0$, and let $H(g)$ be the Friedrichs realization on $L^2(\mathbb{R}, \mathrm{d}x)$ of

$$H(g) = -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}x^2 + gx^4. \tag{33.46}$$

The domain is determined by the closed quadratic form

$$q_g[\psi] = \frac{1}{2}\|\psi'\|_{L^2}^2 + \frac{1}{2}\|x\psi\|_{L^2}^2 + g\|x^2\psi\|_{L^2}^2.$$

Its spectrum is discrete and simple; write $E_n(g)$, $n = 0, 1, \ldots$, for the increasing eigenvalues. The Rayleigh–Schrödinger perturbative series at the n-th harmonic-oscillator level is

$$\widehat{E}_n(g) = \sum_{r\geq 0} E_{n,r} g^r. \tag{33.47}$$

A Borel-summability statement for (33.47) means that this particular spectral branch $E_n(g)$ is recovered by a Borel–Laplace transform satisfying Definition 33.10.

Quoted Theorem 33.26 (Quartic-oscillator Borel summability)**.** *For each fixed n, the series* (33.47) *is Borel summable on the positive real g-axis to the eigenvalue $E_n(g)$ of* (33.46)*. More precisely, the perturbative coefficients are Gevrey-one, the Borel transform has analytic continuation and growth in a sector containing the positive Borel ray, and the Borel–Laplace integral in that direction equals $E_n(g)$ for $g > 0$.*

Status of the quoted theorem. This is a theorem of Graffi–Grecchi–Simon for even polynomial anharmonic oscillators.[4] The local mechanism has three layers. First, for complex g near the positive axis one treats $H(g)$ as an analytic family of sectorial closed forms. Equivalently, after the complex dilation $x = \mathrm{e}^{\mathrm{i}\theta} y$, the quadratic form becomes

$$q_{g,\theta}[\psi] = \frac{1}{2}\mathrm{e}^{-2\mathrm{i}\theta}\|\psi'\|^2 + \frac{1}{2}\mathrm{e}^{2\mathrm{i}\theta}\|y\psi\|^2 + g\,\mathrm{e}^{4\mathrm{i}\theta}\|y^2\psi\|^2.$$

For g in a sector around the positive real axis one chooses θ so that the numerical range of this form lies in a fixed acute sector and the real part controls $\|\psi'\|^2 + \|y\psi\|^2 + |g|\|y^2\psi\|^2$. The associated operators are m-sectorial with compact resolvent, and isolated eigenvalue branches are defined by the Riesz projection

$$P_n(g) = -\frac{1}{2\pi\mathrm{i}}\oint_{\Gamma_n} (H(g) - z)^{-1}\,\mathrm{d}z,$$

where Γ_n surrounds the n-th harmonic-oscillator eigenvalue and no other one when g is small in the chosen sector. The eigenvalue

$$E_n(g) = \operatorname{tr}\big(H(g)P_n(g)\big)$$

is therefore an analytic function on this sectorial perturbation domain.

Second, the resolvent expansion around $H(0)$, combined with oscillator matrix-element estimates for x^4, gives a strong Gevrey-one remainder: for each closed subsector of the positive direction there are $C_n, A_n < \infty$ such that

$$\left| E_n(g) - \sum_{r=0}^{N-1} E_{n,r} g^r \right| \le C_n A_n^N N! |g|^N, \qquad N \ge 1. \tag{33.48}$$

The factorial comes from the repeated unbounded perturbation x^4 and the energy-denominator combinatorics; the estimate is a bound on the spectral branch $E_n(g)$ itself and therefore ties the formal Rayleigh–Schrödinger coefficients to an analytic eigenvalue function.

Third, the analytic sector together with (33.48) is a Nevanlinna–Watson Borel datum. It implies that the formal Borel transform

$$\mathcal{B}\widehat{E}_n(\xi) = \sum_{r\ge 0} \frac{E_{n,r}}{r!}\xi^r$$

continues through a sector containing $\xi \ge 0$, obeys the exponential growth needed for the Laplace integral, and satisfies

$$E_n(g) = \frac{1}{g}\int_0^\infty \mathrm{e}^{-\xi/g}\mathcal{B}\widehat{E}_n(\xi)\,\mathrm{d}\xi, \qquad g > 0.$$

[4] S. Graffi, V. Grecchi, and B. Simon, "Borel summability: Application to the anharmonic oscillator," *Phys. Lett. B* **32** (1970) 631–634, doi:10.1016/0370-2693(70)90564-2; see also B. Simon, "Coupling constant analyticity for the anharmonic oscillator," *Ann. Phys.* **58** (1970) 76–136, doi:10.1016/0003-4916(70)90240-X.

The monograph uses this theorem as a controlled comparison example: in QFT the same kind of analytic-domain, factorial-remainder, Borel-growth, and regulator-removal data must be supplied for the named regulated observable.

Remark 33.27 (Local Borel convergence from Gevrey-one growth). Suppose $\sum_{r\geq 0} a_r g^r$ satisfies $|a_r| \leq CM^r r!$. Then the formal Borel transform converges near $\xi = 0$. This local convergence is weaker than Borel summability of a specified observable.

$$\left|\frac{a_r}{r!}\xi^r\right| \leq C(M|\xi|)^r,$$

so the Borel series converges for $|\xi| < M^{-1}$. Definition 33.10 requires more: analytic continuation of the Borel transform to a sector containing the summation ray, an exponential-growth bound there, existence of the Laplace integral, and an identity between that Laplace integral and the observable on a nonempty domain. None of these follows from the local coefficient bound. Thus large-order estimates are evidence for a possible summation mechanism, but not the mechanism itself.

Definition 33.28 (Instanton-sector transseries)**.** Consider a regulated Euclidean gauge-theory path integral whose configuration space has connected components indexed by an integer topological charge $k \in \mathbb{Z}$. Assume the Euclidean action in sector k has the form

$$S_{E,R}[A,\Psi] = \frac{1}{4g^2}\int \mathrm{tr}_\Delta(F \wedge *F) - \mathrm{i}\theta k + S_{R,\mathrm{matter}}[A,\Psi], \tag{33.49}$$

where tr_Δ is the trace normalization fixed in the gauge-theory convention dictionary of this monograph. An instanton-sector transseries for a regulated observable is a formal expansion

$$\widehat{\mathcal{X}}_R(g,\theta) = \sum_{k\in\mathbb{Z}} \sum_{\alpha\in\mathcal{S}_{R,k}} \mathrm{e}^{-A_{R,k,\alpha}/g^2}\, \mathrm{e}^{\mathrm{i}k\theta}\, g^{\beta_{R,k,\alpha}}\, \widehat{\Phi}_{R,k,\alpha}(g), \tag{33.50}$$

where α labels the regulated saddle or quasi-saddle data inside the charge-k sector. The notation includes the fluctuation determinant, collective-coordinate measure, zero-mode insertions, and boundary contributions only if these have been specified by the regulator.

Algebraic constraints on an instanton-sector transseries. For a transseries of the form (33.50), the θ-dependence is 2π-periodic. If the regulated observable is real at $\theta = 0$ and the sector data obey the conjugation rule

$$A_{R,-k,\bar{\alpha}} = A_{R,k,\alpha}, \qquad \beta_{R,-k,\bar{\alpha}} = \beta_{R,k,\alpha}, \qquad \widehat{\Phi}_{R,-k,\bar{\alpha}}(g) = \overline{\widehat{\Phi}_{R,k,\alpha}(g)}$$

for $g > 0$, then the paired k and $-k$ contribution at real θ is real and equals

$$2\ \mathrm{Re}\left[\mathrm{e}^{-A_{R,k,\alpha}/g^2}\mathrm{e}^{\mathrm{i}k\theta}g^{\beta_{R,k,\alpha}}\widehat{\Phi}_{R,k,\alpha}(g)\right]. \tag{33.51}$$

Since $k \in \mathbb{Z}$,

$$\mathrm{e}^{\mathrm{i}k(\theta+2\pi)} = \mathrm{e}^{\mathrm{i}k\theta}.$$

This proves the periodicity. The stated conjugation rule makes the $-k$-sector term the complex conjugate of the k-sector term for real θ and $g > 0$. Adding the two terms gives twice the real part, which is (33.51).

BPST sector in the trace-delta convention. In the trace-delta Yang–Mills convention of this monograph,

$$S_{\mathrm{YM,E}} = \frac{1}{4g_{\mathrm{YM}}^2}\int_{\mathbb{R}^4} \mathrm{tr}_\Delta(F_{\mu\nu}F_{\mu\nu})\,\mathrm{d}^4x - \mathrm{i}\theta Q[A], \tag{33.52}$$

the $SU(2)$ BPST one-instanton sector contributes the exponential sector factor

$$\exp\left(-\frac{4\pi^2}{g_{\mathrm{YM}}^2} + \mathrm{i}\theta\right) \tag{33.53}$$

to any regulated observable whose insertions have nonzero projection onto that sector. In the common half-trace convention $\mathrm{tr}(T_aT_b) = \delta_{ab}/2$, this same factor is written $\exp[-8\pi^2/g_{\mathrm{ht}}^2 + \mathrm{i}\theta]$, with

$$g_{\mathrm{ht}} = \sqrt{2}\,g_{\mathrm{YM}}. \tag{33.54}$$

The explicit BPST calculation in Lemma 51.23 gives $Q[A] = 1$ and, in the active trace-delta convention,

$$\frac{1}{4g_{\mathrm{YM}}^2}\int_{\mathbb{R}^4} \mathrm{tr}_\Delta(F_{\mu\nu}F_{\mu\nu})\,\mathrm{d}^4x = \frac{4\pi^2}{g_{\mathrm{YM}}^2}.$$

Substitution in (33.52) gives (33.53). The half-trace basis used in much of the instanton literature differs from the trace-delta basis by $t_a = \sqrt{2}\,T_a$. Therefore the same connection is represented with $g_{\mathrm{ht}}^2 = 2g_{\mathrm{YM}}^2$, and

$$\frac{8\pi^2}{g_{\mathrm{ht}}^2} = \frac{4\pi^2}{g_{\mathrm{YM}}^2}.$$

The equality is only a convention change; it does not alter the topological charge or the physical semiclassical weight.

Controlled Approximation 33.29 (Semiclassical instanton expansion)**.** The factor (33.53) is not yet an instanton calculus formula for a continuum QFT observable. A semiclassical one-instanton approximation additionally requires a regulator, a gauge slice or BV treatment of the zero modes, the determinant ratio, the collective coordinate density on the instanton moduli space, insertion zero-mode saturation, and a prescription for boundary strata such as small instantons. Only after those data are defined and their regulator-removal behavior is proved does the sector expansion become a theorem about the observable rather than a semiclassical sector label.

Remark 33.30 (Status of instanton and renormalon applications)*.* Definition 33.28 is a bookkeeping standard, not an assertion that the instanton expansion has been justified. For a gauge theory it remains necessary to define the path integral regulator, the topological sectors, the treatment of gauge zero modes and collective coordinates, the determinant normalization, the behavior near small-size instantons or other boundary strata, and the continuum limit. Renormalon ambiguities are different objects: they arise from momentum-region singularities in perturbative coefficient functions and are tied to factorization and operator definitions. Instanton saddles and renormalon singularities can coexist in the same observable, but neither one is a substitute for the other.

In this monograph a displayed perturbative beta function or critical exponent has the following default meaning unless a stronger statement is stated: it is a coefficientwise construction in a specified regularized and renormalized chart, giving an asymptotic small-coupling expansion or an effective finite-order approximation in its stated domain. A finite truncation can be useful and sharply predictive when the running coupling and logarithms are controlled. A nonperturbative definition of the QFT, or a convergence theorem for the infinite formal series, requires additional data beyond the finite truncation.

33.12 Invariant Data at Fixed Points

The preceding one-coordinate computation is a special case of the tensorial transformation law for a vector field. Let $\mathcal{T}$ be a finite dimensional coordinate patch in the renormalized theory family, with local coordinates λ^I. The beta function is the vector field

$$\beta = \beta^I(\lambda)\frac{\partial}{\partial \lambda^I}.$$

A finite scheme change is a local diffeomorphism

$$\widetilde{\lambda}^A = f^A(\lambda), \qquad J^A{}_I(\lambda) = \frac{\partial f^A}{\partial \lambda^I}.$$

The components of the same vector field in the new coordinates are

$$\widetilde{\beta}^A(\widetilde{\lambda}) = J^A{}_I(\lambda)\, \beta^I(\lambda), \qquad \widetilde{\lambda} = f(\lambda).$$

Thus a beta function is not a coordinate-independent list of functions; the coordinate-independent object is the flow it defines on $\mathcal{T}$.

A fixed point is a zero of this vector field:

$$\beta^I(\lambda_*) = 0.$$

This statement is invariant under changes of coordinates. Let

$$M^I{}_J = \left.\frac{\partial \beta^I}{\partial \lambda^J}\right|_{\lambda_*}$$

be the linearization of the RG vector field at the fixed point. Since the extra term obtained by differentiating $J^A{}_I(\lambda)\beta^I(\lambda)$ is proportional to $\beta(\lambda_*)$, it vanishes at the fixed point. Therefore

$$\widetilde{M}^A{}_B = J^A{}_I(\lambda_*)\, M^I{}_J\, (J^{-1})^J{}_B(\lambda_*).$$

The linearized matrices in two schemes are similar. Their eigenvalues are therefore invariant, although their eigenvectors are represented by different coordinate components. These eigenvalues are the critical exponents of the linearized flow in the chosen normalization of RG time.

If M has zero eigenvalues, the linearized analysis identifies tangent directions that require higher-order study. A zero eigenvalue is the condition for linearized marginality; exact marginality is the stronger statement that the RG vector field vanishes along an actual submanifold of theory space. In local QFT this statement must also be made after quotienting redundant directions generated by field redefinitions and after imposing the symmetries used to define the coordinate patch.

Figure 33.6 illustrates this local geometry near a fixed point. The red integral curves are subsets of the same theory space; the gray and blue grids are two coordinate systems on that space. A finite scheme change alters component functions and arrow components, while preserving the fixed point, the integral curves as point sets, and the similarity class of the linearized vector field.

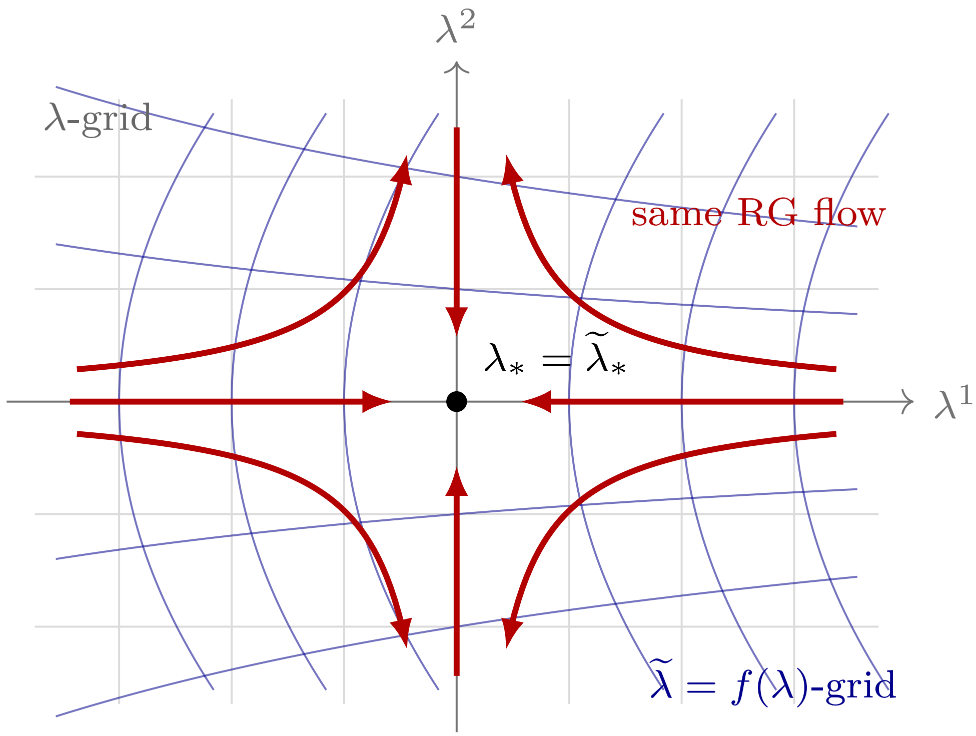


Figure 33.6: A finite scheme change is a change of local coordinates on the renormalized theory family. The gray and blue coordinate grids give different component descriptions of the same red RG vector field. The fixed point and the similarity class of the linearized flow at the fixed point are invariant geometric data.

The next chapter applies the same scale-dependent logic to local operator insertions. Infinitesimal deformations of the action define renormalized operators, and their scale dependence is governed by the linearization of the beta-function vector field.

Chapter 34

Renormalized Operators and Minimal Subtraction

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

34.1 Operators from Infinitesimal Deformations

The 1PI renormalization group compares scale-dependent coordinates $g_I(\mu)$ on a renormalized family of theories. Local operators enter the same structure through infinitesimal deformations of the action. Fix a finite or graded local operator basis $\{O_I\}_{I\in\mathcal{I}}$, chosen so that operators with the same spacetime and internal quantum numbers are included together whenever perturbation theory can mix them. To define insertions as distributions, first allow smooth compactly supported source functions $\eta_0^I(x)$ and write

$$\delta S_\eta = \int \mathrm{d}^D x \, \eta_0^I(x) \, O_I(x).$$

Constant sources $\eta_0^I(x) = \delta g_I$ give the deformation

$$\delta S = \int \mathrm{d}^D x \sum_I \delta g_I \, O_I(x).$$

Here δg_I are bare deformation coefficients. At scale μ, the same deformation can be described by renormalized source functions $\eta_\mu^J(x)$. Linearity in the infinitesimal source gives a matrix $N_{JI}(\mu)$

such that

$$\eta_0^I(x) = \sum_J \eta_\mu^J(x)\, N_{JI}(\mu).$$

For constant sources this becomes

$$\delta g_I = \sum_J \delta g_J(\mu)\, N_{JI}(\mu).$$

Substituting this into δS gives

$$\delta S = \int \mathrm{d}^D x \sum_J \delta g_J(\mu) \left(\sum_I N_{JI}(\mu)\, O_I(x) \right).$$

Definition 34.1 (Renormalized local operator)**.** In the deformation chart above, the renormalized operator associated with the source coordinate η_μ^I, or equivalently with the coupling coordinate $g_I(\mu)$ for constant sources, is

$$[O_I]_\mu = \sum_J N_{IJ}(\mu)\, O_J.$$

Equivalently,

$$\delta S = \int \mathrm{d}^D x \sum_I \delta g_I(\mu)\, [O_I]_\mu(x).$$

The matrix $N_{IJ}(\mu)$ is part of the renormalization scheme. Its rows label renormalized deformation coordinates, and its columns label the chosen bare operator basis. As an insertion in correlation functions, the equality is an equality of distributions away from coincident points; different renormalization prescriptions for contact terms differ by local terms supported on collision diagonals, in the sense of Definition 1.12.

34.2 Operators as Cotangent Data

Warning 34.2 (γ-indices). Throughout this chapter the anomalous-dimension matrix for deformation operators is indexed in the cotangent convention

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu}[O_I]_\mu = -\gamma_{IK}[O_K]_\mu, \qquad \gamma_{IK} = \frac{\partial \beta_K}{\partial g_I}.$$

Thus the first index labels the source or coupling coordinate dual to the operator being differentiated, and the second index labels the operator appearing after the cotangent action. Many column-vector conventions in the physics literature denote the same Jacobian component by

$$\gamma_{KI}^{\mathrm{col}} = \frac{\partial \beta_K}{\partial g_I} = \gamma_{IK}.$$

Comparisons of formulae must therefore transpose this pair of labels.

The preceding definition has a geometric interpretation. The infinitesimal coupling displacement

$$\delta g = \delta g_I \,\frac{\partial}{\partial g_I}$$

is a tangent vector to the renormalized family of theories. The corresponding local operator insertion is the dual object specified by the pairing

$$\langle \delta g, O\rangle = \int \mathrm{d}^D x\, \delta g_I\, [O_I]_\mu(x).$$

The deformation δS is this pairing. Therefore a change of renormalized coordinates

$$\widetilde{g}_A = f_A(g)$$

must transform the operator representatives by the inverse Jacobian:

$$[\widetilde{O}_A]_\mu = \frac{\partial g_I}{\partial \widetilde{g}_A}\, [O_I]_\mu.$$

This is the cotangent transformation law, with the Jacobian evaluated at the theory point under consideration. It is the local-operator version of the statement that beta functions are components of a vector field. The statement applies to the deformation operators dual to the selected coordinate chart; additional local operators may require enlarging the chart before their mixing can be described by the same matrix.

Definition 34.3 (Deformation-operator bundle)**.** Let U be a coordinate neighborhood in the finite-dimensional renormalized theory space with coordinates g_I. The *deformation-operator bundle* over U is the vector bundle $\mathcal{E}_{\mathrm{def}} \to U$ whose local representative column over the chart g is the ordered family of source-normalized insertion distributions

$$e_I(g, \mu) = [O_I]_\mu.$$

Its fiber at g is the span of the deformation operators dual to tangent vectors $\partial/\partial g_I$, with equality of insertions understood on noncoincident test configurations and with a separately specified contact-term extension. On an overlap with coordinates $\widetilde{g}_A = f_A(g)$, the transition functions are specified in the convention that local representative columns transform by $v_{\widetilde{g}} = M_{g\to\widetilde{g}} v_g$. The operator column therefore obeys

$$\widetilde{e}_A = M_{AI}(g)\, e_I, \qquad M_{AI}(g) = \frac{\partial g_I}{\partial \widetilde{g}_A}. \tag{34.1}$$

Thus, for deformation operators, $\mathcal{E}_{\mathrm{def}}$ is the source-defined realization of T^*U. If one includes local operators not dual to coupling coordinates, the same construction gives a larger operator bundle only after a finite-rank source-coordinate system and its transition functions have been specified.

Let $t = \log \mu$. The tangent vector δg_K evolves by the linearized RG equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta g_K = \frac{\partial \beta_K}{\partial g_I}\, \delta g_I.$$

Scale independence of the bare deformation means that the pairing $\langle \delta g, O\rangle$ is independent of t. Hence the dual operator basis evolves with the negative transpose action:

$$\frac{\mathrm{d}}{\mathrm{d}t}[O_I]_\mu = -\frac{\partial \beta_K}{\partial g_I}\, [O_K]_\mu.$$

The index placement in the anomalous-dimension matrix below is chosen to encode precisely this cotangent action.

The scale dependence of N_{IJ} follows from the scale independence of the bare deformation. Let

$$\frac{\mathrm{d}g_I(\mu)}{\mathrm{d}\log\mu} = \beta_I(g(\mu))$$

be the beta functions of the underlying finite-dimensional coordinate system. For an infinitesimal displacement $g_I(\mu) \mapsto g_I(\mu) + \delta g_I(\mu)$, the beta function changes by

$$\delta\beta_K = \sum_I \frac{\partial \beta_K}{\partial g_I}\, \delta g_I(\mu).$$

Define the anomalous-dimension matrix in this deformation convention by

$$\gamma_{IK}(g) = \frac{\partial \beta_K}{\partial g_I}(g).$$

Differentiating $\delta g_I = \sum_J \delta g_J(\mu) N_{JI}(\mu)$ with respect to $\log\mu$ and using the preceding linearized flow gives

$$\begin{aligned}
0 &= \frac{\mathrm{d}\,\delta g_I}{\mathrm{d}\log\mu} \\
&= \sum_J \left[\delta\beta_J(g(\mu))\, N_{JI}(\mu) + \delta g_J(\mu)\frac{\mathrm{d}N_{JI}(\mu)}{\mathrm{d}\log\mu}\right] \\
&= \sum_{J,K} \delta g_K(\mu)\, \frac{\partial \beta_J}{\partial g_K}(g(\mu))\, N_{JI}(\mu) + \sum_J \delta g_J(\mu)\, \frac{\mathrm{d}N_{JI}(\mu)}{\mathrm{d}\log\mu}.
\end{aligned}$$

Since the infinitesimal renormalized displacement $\delta g_J(\mu)$ is arbitrary, this is equivalent to

$$\frac{\mathrm{d}N_{IJ}}{\mathrm{d}\log\mu} = -\sum_K \gamma_{IK}(g(\mu))\, N_{KJ}(\mu).$$

Consequently the renormalized operators satisfy

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu}[O_I]_\mu = -\sum_K \gamma_{IK}(g(\mu))\, [O_K]_\mu.$$

Figure 34.1 records the two pieces of data entering this equation: operator mixing among the chosen local basis and field renormalization on the external legs of insertion kernels.

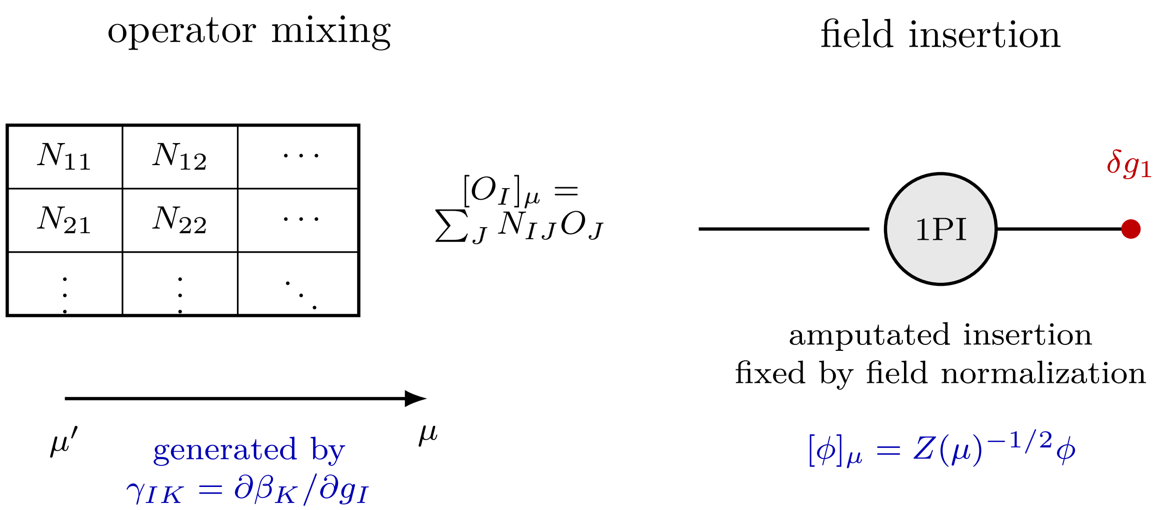


Figure 34.1: Renormalized local operators are rows of the mixing matrix N_{IJ}. The matrix evolves with the linearized beta-function vector field. For the elementary field insertion, the running is the same wavefunction normalization used in the 1PI coordinate system.

34.3 Source-Extended Wilsonian Representatives

The preceding discussion used local sources to define renormalized operators as cotangent vectors to a space of couplings. The same source coordinates also have a Wilsonian representative. The complete construction consists of three linked objects: a Wilsonian insertion functional, its connected low-field expectation, and its 1PI Legendre-transform representative.

Work at a finite ultraviolet regulator Λ_0, and let $\Lambda < \Lambda_0$ be a smooth Wilsonian cutoff. The regulated field is decomposed as a sum of an independent low field ϕ with covariance C_Λ and a shell field $\phi_{\rm sh}$ with positive covariance $\widehat{C}_{\Lambda_0,\Lambda} = C_{\Lambda_0} - C_\Lambda$, or with any equivalent smooth positive covariance decomposition. The shell kinetic term $\widehat{S}_{\rm kin}$ is the quadratic form inverse to $\widehat{C}_{\Lambda_0,\Lambda}$ on the regulated vector space. Choose a local bare operator basis $O_I(x)$ and source functions $\eta_0^I(x)$. In the Euclidean sign convention used in this volume,

$$Z_{\Lambda_0}[J,\eta_0] = \int D^{\rm ref}_{\Lambda_0}\phi \, \exp\left[-S_{\Lambda_0}[\phi] - \int \mathrm{d}^D x\, J(x)\phi(x) - \int \mathrm{d}^D x\, \eta_0^I(x) O_I(x)\right].$$

Here $D^{\rm ref}_{\Lambda_0}\phi$ is the reference density of Definition 11.3; the full quadratic kinetic term is included in S_{Λ_0}. In the Gaussian form used for the Wilsonian pushforward below, the same finite-regulator integral is expressed by placing the quadratic form in a centered Gaussian measure and retaining only the interaction/source functional in the exponential. Thus

$$-\frac{\delta W_{\Lambda_0}[J,\eta_0]}{\delta \eta_0^I(x)}\bigg|_{\eta_0=0} = \big\langle O_I(x)\big\rangle_{J,\Lambda_0}, \qquad W_{\Lambda_0} = \log Z_{\Lambda_0},$$

where the expectation value is computed with the source J still present. Higher derivatives with respect to η_0 give connected insertions, with the corresponding signs determined by the same convention.

The Wilsonian source coordinates at scale Λ are denoted $\eta_\Lambda^A(x)$. Locality permits a source-coordinate change by a local linear differential operator,

$$\eta_0^I(x) = \big(\mathcal{Z}_\Lambda\big)^I{}_A(\partial_x)\,\eta_\Lambda^A(x), \tag{34.2}$$

where

$$\big(\mathcal{Z}_\Lambda\big)^I{}_A(\partial) = \sum_{|\alpha|\geq 0} \big(\mathcal{Z}_\Lambda\big)^I{}_{A,\alpha}\,\partial^\alpha$$

as a formal derivative expansion, restricted by the spacetime and internal quantum numbers of the sources. For constant sources only the $\alpha = 0$ term contributes. The finite matrix $N_{AI}(\mu)$ introduced above is the corresponding zero-momentum source-renormalization matrix in a chosen 1PI coordinate chart.

Define the source-extended Wilsonian interaction $L_\Lambda[\phi;\eta_\Lambda]$ by the Gaussian pushforward

$$\exp\big[-L_\Lambda[\phi;\eta_\Lambda]\big] = \int \mathrm{d}\mu_{\widehat{C}_{\Lambda_0,\Lambda}}(\phi_{\mathrm{sh}})\,\exp\left[-L_{\Lambda_0}[\phi+\phi_{\mathrm{sh}};\eta_0(\eta_\Lambda)]\right], \tag{34.3}$$

where $\mathrm{d}\mu_{\widehat{C}_{\Lambda_0,\Lambda}}$ is the centered Gaussian measure whose covariance is the shell covariance between Λ_0 and Λ. Equation (34.3) is first an identity in the finite-dimensional regulated system; the continuum derivative expansion is then studied after the regulator is removed.

Definition 34.4 (Wilsonian composite insertion)**.** The Wilsonian representative of the source coordinate η_Λ^A is the local functional

$$\mathcal{O}^{\mathrm{W}}_{\Lambda,A}(x;\phi) = \left.\frac{\delta L_\Lambda[\phi;\eta_\Lambda]}{\delta\eta_\Lambda^A(x)}\right|_{\eta_\Lambda=0}.$$

It is a functional of the remaining low field ϕ. Its connected and 1PI images are obtained from the subsequent low-field integration and Legendre transform.

Let

$$W_\Lambda^{\mathrm{low}}[J,\eta_\Lambda] = \log\int \mathrm{d}\mu_{C_\Lambda}(\phi)\,\exp\left[-L_\Lambda[\phi;\eta_\Lambda] - \int J\phi\right].$$

For source functions J and η_Λ whose Fourier transforms are supported in the low-momentum region where the cutoff profile is unchanged, the Gaussian convolution identity gives

$$W_\Lambda^{\mathrm{low}}[J,\eta_\Lambda] = W_{\Lambda_0}[J,\eta_0(\eta_\Lambda)] \tag{34.4}$$

at the finite regulator. Differentiating at $\eta_\Lambda = 0$ gives

$$-\left.\frac{\delta W_\Lambda^{\mathrm{low}}[J,\eta_\Lambda]}{\delta\eta_\Lambda^A(x)}\right|_{\eta_\Lambda=0} = \big\langle\mathcal{O}^{\mathrm{W}}_{\Lambda,A}(x;\phi)\big\rangle^{\mathrm{low}}_{J,\Lambda}.$$

The equality (34.4) therefore identifies connected operator insertions computed from the bare and Wilsonian descriptions, after the source-coordinate map (34.2) has been applied.

The 1PI insertion is obtained by performing the remaining low-field integral and then taking the Legendre transform. If

$$\varphi(x) = \frac{\delta W_\Lambda^{\mathrm{low}}[J,\eta_\Lambda]}{\delta J(x)}$$

and

$$\Gamma_\Lambda[\varphi,\eta_\Lambda] = W_\Lambda^{\text{low}}[J,\eta_\Lambda] - \int \mathrm{d}^D x\, J(x)\varphi(x), \qquad J = J[\varphi,\eta_\Lambda],$$

then the cancellation of the J-variation in the Legendre transform gives

$$\left.\frac{\delta\Gamma_\Lambda[\varphi,\eta_\Lambda]}{\delta\eta_\Lambda^A(x)}\right|_\varphi = \left.\frac{\delta W_\Lambda^{\text{low}}[J,\eta_\Lambda]}{\delta\eta_\Lambda^A(x)}\right|_{J=J[\varphi,\eta_\Lambda]}.$$

Consequently the 1PI representative of the insertion is

$$\mathcal{O}^\Gamma_{\Lambda,A}(x;\varphi) = -\left.\frac{\delta\Gamma_\Lambda[\varphi,\eta_\Lambda]}{\delta\eta_\Lambda^A(x)}\right|_{\eta_\Lambda=0}.$$

Its field derivatives at $\varphi = 0$ are the amputated 1PI kernels with one operator insertion. These kernels agree with the projected 1PI insertion kernels in another subtraction scheme after the corresponding finite source-coordinate map and field normalization have been specified.

Figure 34.2 shows the nontrivial part of the source-extended comparison at two external low-field legs: differentiating the connected insertion with respect to ordinary field sources gives an insertion with exact external propagators; the 1PI representative is the amputated kernel obtained after the Legendre transform.

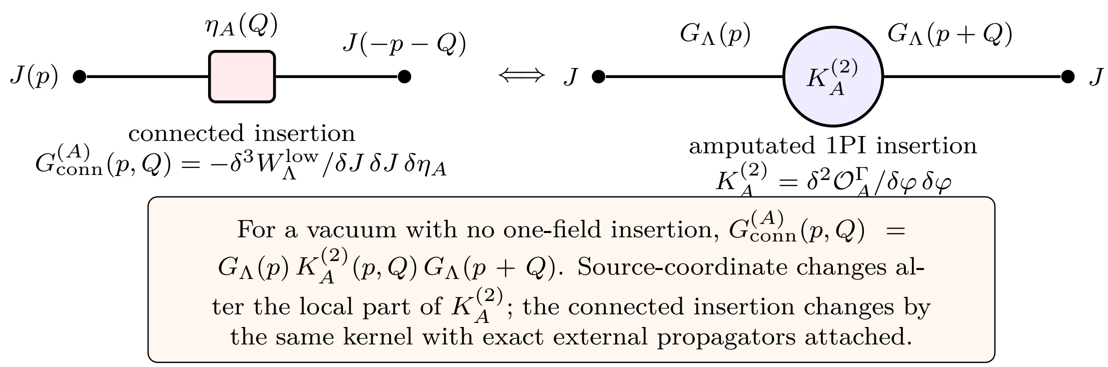


Figure 34.2: The Legendre transform changes a connected low-field insertion into an amputated 1PI insertion kernel. In the two-point sector, and when one-field insertion terms vanish by the vacuum quantum numbers, the connected insertion is obtained by attaching exact low-field propagators to the amputated kernel.

34.4 A Worked Insertion: The Mass Operator in ϕ^4

The abstract source bridge becomes concrete in the scalar quartic theory. Work at finite regulator in four Euclidean dimensions, and introduce a source η for the local operator

$$O_m(x) = \frac{1}{2}\phi(x)^2.$$

The source-dependent part of the regulated action, in the cutoff-dependent source coordinate η_0, is

$$\int \mathrm{d}^4 x\, \eta_0(x) O_m(x).$$

Let Q denote the momentum carried by the source insertion, and let

$$C_{\Lambda_0} = C_\Lambda + C_>$$

be the same covariance split used for the Wilsonian action. For covariances A, B, define the one-insertion bubble

$$\mathcal{I}_{A,B}(Q) = \int \frac{\mathrm{d}^4 q}{(2\pi)^4} A(q)\, B(q+Q).$$

Then

$$\mathcal{I}_{C_{\Lambda_0},C_{\Lambda_0}}(Q) = \mathcal{I}_{<,<}(Q) + \mathcal{I}_{>,>}(Q) + \mathcal{I}_\times(Q), \tag{34.5}$$

where

$$\mathcal{I}_{<,<} = \mathcal{I}_{C_\Lambda,C_\Lambda}, \qquad \mathcal{I}_{>,>} = \mathcal{I}_{C_>,C_>},$$

and

$$\mathcal{I}_\times = \mathcal{I}_{C_\Lambda,C_>} + \mathcal{I}_{C_>,C_\Lambda}.$$

The Wilsonian insertion functional is the source derivative of $L_\Lambda[\phi;\eta_\Lambda]$. At zeroth order in the quartic coupling, the Gaussian pushforward gives

$$\mathcal{O}^{\mathrm{W}}_{m,\Lambda}(x;\phi) = \frac{1}{2}\phi(x)^2 + \frac{1}{2}C_>(x,x) + \cdots,$$

where the second term is an identity-operator contribution. It affects the vacuum response to η, but not the two-field amputated insertion kernel. At first order in the quartic coupling g, the coefficient of $\frac{1}{2}\phi^2$ in the Wilsonian insertion receives the high-high bubble:

$$\mathcal{O}^{\mathrm{W}}_{m,\Lambda}(Q;\phi) = \left[1 - \frac{g}{2}\mathcal{I}_{>,>}(Q) + Z^{\mathrm{loc}}_{m,\Lambda}(Q)\right]\frac{1}{2}(\phi\phi)(Q) + \text{identity and higher local terms} + O(g^2). \tag{34.6}$$

Here $(\phi\phi)(Q)$ denotes the Fourier transform of $\phi(x)^2$, and $Z^{\mathrm{loc}}_{m,\Lambda}(Q)$ is the local source-coordinate counterterm and finite convention. The factor $1/2$ is the one-loop symmetry factor in the normalization $g\phi^4/4!$ and $O_m = \phi^2/2$.

Now compare with the BPHZ normal-product definition. Let $\mathcal{Q}_\mu$ be the nonexceptional source-momentum subtraction configuration used to define the two-field 1PI insertion kernel, and let

$$K^{(2)}_m(Q) = \frac{\delta^2}{\delta\varphi\,\delta\varphi}\mathcal{O}^\Gamma_m(Q;\varphi)\bigg|_{\varphi=0}$$

in the field normalization of the chosen 1PI coordinate chart. The renormalization condition

$$K^{(2)}_m(\mathcal{Q}_\mu) = 1$$

fixes the finite normalization of the renormalized operator. At one loop, the BPHZ-subtracted insertion kernel is

$$K^{(2)}_{m,\mathrm{BPHZ}}(Q) = 1 - \frac{g}{2}\left[\mathcal{I}_{C_{\Lambda_0},C_{\Lambda_0}}(Q) - \mathcal{I}_{C_{\Lambda_0},C_{\Lambda_0}}(\mathcal{Q}_\mu)\right] + O(g^2), \tag{34.7}$$

with the ultraviolet limit taken after the subtraction. The subtraction is a local normal-product convention for the operator O_m. The controlled comparison in Controlled Approximation 36.16 treats this finite normal-product choice as part of the local source-coordinate system: changing a Zimmermann normal product changes the finite local operator coordinates and therefore the source coordinate map.

Figure 34.3 displays the actual covariance assignment on the one-loop insertion graph. The loop has two internal propagator slots; inserting $C_{\Lambda_0} = C_\Lambda + C_>$ in each slot gives the low–low, high–high, and two mixed contributions in (34.5).

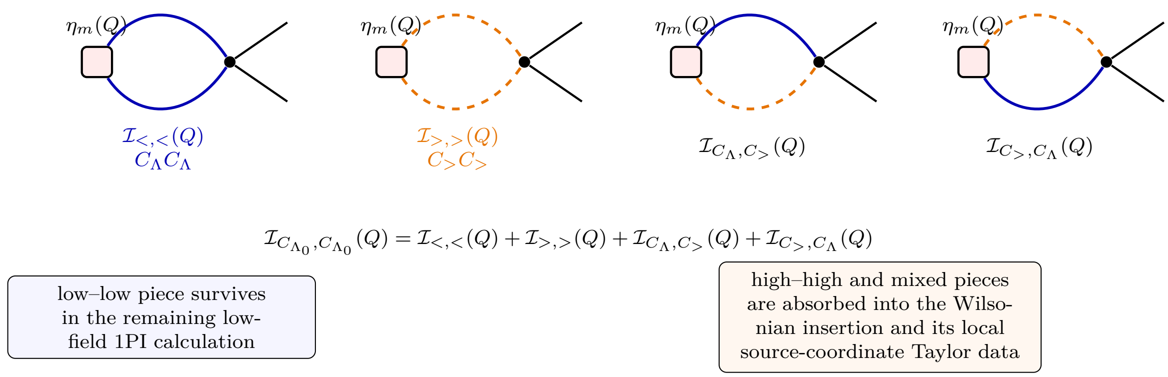


Figure 34.3: The one-loop mass-operator kernel is a single insertion graph with two internal covariance slots. Expanding $C_{\Lambda_0} = C_\Lambda + C_>$ on those slots gives the four displayed assignments; the two mixed assignments form $\mathcal{I}_\times$. This is the graph-level content behind the local Wilsonian source-coordinate matching.

The 1PI insertion computed from the Wilsonian description has the form

$$K^{(2)}_{m,\Lambda}(Q) = Z_{m,\Lambda}(Q) - \frac{g}{2}\mathcal{I}_{<,<}(Q) - \frac{g}{2}\mathcal{I}^{\text{loc}}_\times(Q) + O(g^2), \tag{34.8}$$

where $Z_{m,\Lambda}(Q)$ is the local coefficient multiplying $\frac{1}{2}\phi^2$ in the Wilsonian insertion, and $\mathcal{I}^{\text{loc}}_\times$ denotes the Taylor expansion of the mixed assignment in the chosen low-momentum source-coordinate system. Projecting at $\mathcal{Q}_\mu$ fixes the source-coordinate normalization:

$$1 = Z_{m,\Lambda}(\mathcal{Q}_\mu) - \frac{g}{2}\mathcal{I}_{<,<}(\mathcal{Q}_\mu) - \frac{g}{2}\mathcal{I}^{\text{loc}}_\times(\mathcal{Q}_\mu) + O(g^2). \tag{34.9}$$

Subtracting this condition from (34.8) gives the finite insertion kernel in the 1PI chart:

$$K^{(2)}_{m,\Lambda}(Q) = 1 - \frac{g}{2}\left[\mathcal{I}_{<,<}(Q) - \mathcal{I}_{<,<}(\mathcal{Q}_\mu)\right] + \mathcal{T}^{\text{loc}}_{\text{high+mix}}(Q;\mathcal{Q}_\mu) + O(g^2).$$

Here $\mathcal{T}^{\text{loc}}_{\text{high+mix}}(Q;\mathcal{Q}_\mu)$ is the finite local Taylor polynomial produced by the high-high and mixed assignments after the same subtraction condition has been imposed. Adding this Taylor data reconstructs the low-momentum expansion of (34.7). Thus the normal product, the Wilsonian insertion functional, and the 1PI insertion kernel are distinct representatives of the same source derivative after the finite-regulator pushforward, low-field integration, Legendre transform, and source-coordinate projection have been specified.

This construction also clarifies the transformation law of anomalous dimensions. Let $t = \log\mu$, and suppose a scale-dependent finite source-coordinate change relates two renormalized operator bases by

$$[\widetilde{O}_A]_\mu = M_{AI}(\mu)\,[O_I]_\mu.$$

If

$$\frac{\mathrm{d}}{\mathrm{d}t}[O_I]_\mu = -\gamma_{IK}[O_K]_\mu,$$

then

$$\frac{\mathrm{d}}{\mathrm{d}t}[\widetilde{O}_A]_\mu = -\widetilde{\gamma}_{AB}[\widetilde{O}_B]_\mu$$

with

$$\widetilde{\gamma} = M\gamma M^{-1} - \frac{\mathrm{d}M}{\mathrm{d}t}M^{-1}. \tag{34.10}$$

Theorem 34.5 (Anomalous dimensions as an RG connection)**.** *Let $\mathcal{E} \to U$ be a finite-rank renormalized operator bundle on a coordinate neighborhood U in theory space. Assume that the beta-function vector field $\beta = \beta_I(g)\partial/\partial g_I$ and the transition matrices $M_{\alpha\beta}(g,t)$ between source-coordinate systems are C^1 along the RG trajectories under consideration, or are formal power series for which the following differentiations are performed coefficientwise. In a local source chart, let e denote the local representative column $e_I = [O_I]_\mu$, and suppose the RG variation is*

$$\frac{\mathrm{d}}{\mathrm{d}t}e_I = -\gamma_{IK}(g(t),t)e_K. \tag{34.11}$$

Then the matrices γ define a connection on $\mathcal{E}$ along the RG vector field by

$$(D_t v)_\alpha = \frac{\mathrm{d}v_\alpha}{\mathrm{d}t} + \gamma_\alpha v_\alpha \tag{34.12}$$

for local representative columns v_α in a source-coordinate system α. Under a finite scale-dependent change of source-coordinate system with $\widetilde{e} = Me$ and $v_{\widetilde{\alpha}} = Mv_\alpha$, the connection coefficients transform as

$$\widetilde{\gamma} = M\gamma M^{-1} - \frac{\mathrm{d}M}{\mathrm{d}t}M^{-1}. \tag{34.13}$$

Conversely, any family of matrices obeying (34.13) on overlaps is precisely the local expression of a connection along the RG vector field. For the deformation-operator bundle of Definition 34.3, the connection induced by the linearized beta-function flow has

$$\gamma_{IK} = \frac{\partial \beta_K}{\partial g_I} \tag{34.14}$$

in the coordinate frame dual to g_I.

Proof. The operator frame equation (34.11) is the statement that the chosen renormalized source-representative column is parallel along the RG trajectory: $D_t e = 0$ in the local convention of (34.12). Let $\widetilde{e} = Me$. Differentiating gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{e} = \left(\frac{\mathrm{d}M}{\mathrm{d}t}M^{-1} - M\gamma M^{-1}\right)\widetilde{e}.$$

Writing the same equation in the new chart as $\mathrm{d}\widetilde{e}/\mathrm{d}t = -\widetilde{\gamma}\,\widetilde{e}$ gives (34.13). The same computation is the patching criterion for a connection along a one-dimensional direction: if local matrices obey this law, the formula $D_t v = \mathrm{d}v/\mathrm{d}t + \gamma v$ is independent of the frame because

$$\left(\frac{\mathrm{d}}{\mathrm{d}t} + \widetilde{\gamma}\right)\widetilde{v} = M\left(\frac{\mathrm{d}}{\mathrm{d}t} + \gamma\right)v \qquad (\widetilde{v} = Mv).$$

Hence the local operators D_t glue to a well-defined connection along the RG vector field.

For deformation operators, an infinitesimal tangent vector $\delta g_K \partial/\partial g_K$ obeys the variational equation obtained by linearizing the RG flow:

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta g_K = \frac{\partial \beta_K}{\partial g_I}\,\delta g_I.$$

The deformation insertion is the pairing $\int \delta g_I e_I$. Bare scale independence of this pairing gives

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\big(\delta g_I e_I\big) = \frac{\partial \beta_I}{\partial g_K}\delta g_K e_I + \delta g_I \frac{\mathrm{d}e_I}{\mathrm{d}t}.$$

Renaming dummy indices and using the arbitrariness of δg_I yields

$$\frac{\mathrm{d}e_I}{\mathrm{d}t} = -\frac{\partial \beta_K}{\partial g_I} e_K,$$

which is (34.11) with $\gamma_{IK} = \partial\beta_K/\partial g_I$. □

Thus the anomalous-dimension matrix transforms by conjugation only for scale-independent changes of operator coordinates. For scale-dependent renormalized source coordinates, (34.10) is the connection transformation law of Theorem 34.5.

34.5 Multiple Insertions and Contact-Term Coordinates

The linear source map in (34.2) is the part of the source coordinate change seen by a single insertion. Multiple insertions require the full Taylor expansion of the source-coordinate system. At finite regulator let $\mathcal{S}_\Lambda$ denote the space of regulated source functions $\eta^A_\Lambda(x)$. A local source-coordinate transformation is a smooth map

$$\mathcal{Z}_\Lambda : \mathcal{S}_\Lambda \longrightarrow \mathcal{S}_0, \qquad \eta_0 = \mathcal{Z}_\Lambda(\eta_\Lambda),$$

whose Taylor coefficients at the origin are local distributions. Explicitly,

$$\eta^I_0(z) = \sum_{r\geq 1} \frac{1}{r!} \int \prod_{a=1}^{r} \mathrm{d}^D x_a \, \mathcal{Z}^I{}_{A_1\cdots A_r}(z; x_1, \ldots, x_r)\eta^{A_1}_\Lambda(x_1)\cdots\eta^{A_r}_\Lambda(x_r). \tag{34.15}$$

The $r = 1$ coefficient is the linear differential operator in (34.2). For $r \geq 2$, locality means that $\mathcal{Z}^I{}_{A_1\cdots A_r}(z; x_1, \ldots, x_r)$ is supported on the small diagonal $z = x_1 = \cdots = x_r$ and is a finite sum of derivatives of delta functions, with coefficients depending on the chosen regulator scale and subtraction convention. These higher Taylor coefficients do not contribute for separated insertion points, but they determine the extension of inserted correlators as distributions across collision diagonals.

Let

$$G^0_{I_1\cdots I_s}(z_1, \ldots, z_s; J) = (-1)^s \frac{\delta^s W_{\Lambda_0}[J, \eta_0]}{\delta\eta_0^{I_1}(z_1)\cdots\delta\eta_0^{I_s}(z_s)}\bigg|_{\eta_0=0}$$

be the connected bare distribution with s operator insertions and arbitrary external source J. The renormalized distribution in the source coordinate η_Λ is

$$G^{\Lambda}_{A_1\cdots A_r}(x_1,\dots,x_r;J) = (-1)^r \frac{\delta^r W_{\Lambda_0}[J,\mathcal{Z}_\Lambda(\eta_\Lambda)]}{\delta\eta_\Lambda^{A_1}(x_1)\cdots\delta\eta_\Lambda^{A_r}(x_r)}\bigg|_{\eta_\Lambda=0}.$$

Set

$$H[\eta_\Lambda] = W_{\Lambda_0}[J,\mathcal{Z}_\Lambda(\eta_\Lambda)].$$

The functional Faà di Bruno formula for this composite map gives, before rewriting functional derivatives as insertion distributions,

$$\frac{\delta^r H}{\delta\eta_\Lambda^{A_1}(x_1)\cdots\delta\eta_\Lambda^{A_r}(x_r)}\bigg|_{\eta_\Lambda=0} = \sum_{\pi\in\mathrm{Part}(r)}\int\prod_{B\in\pi}\mathrm{d}^D z_B\ \frac{\delta^{|\pi|}W_{\Lambda_0}[J,\eta_0]}{\prod_{B\in\pi}\delta\eta_0^{I_B}(z_B)}\bigg|_{\eta_0=0}\prod_{B\in\pi}\mathcal{Z}^{I_B}{}_{A_B}(z_B;x_B).$$

The partition records which of the r renormalized source derivatives hit the same Taylor coefficient of the nonlinear source map $\mathcal{Z}_\Lambda$. The signs in the final partition formula are not additional signs attached to the Taylor coefficients of $\mathcal{Z}_\Lambda$. They come from rewriting the ordinary Faà di Bruno identity for the connected functional $W = \log Z$ in terms of connected insertion derivatives. Define

$$D^\Lambda_A := -\frac{\delta}{\delta\eta^A_\Lambda}, \qquad D^0_I := -\frac{\delta}{\delta\eta^I_0}.$$

Then $G^\Lambda_{A_1\cdots A_r} = D^\Lambda_{A_1}\cdots D^\Lambda_{A_r}H|_{\eta_\Lambda=0}$, while $G^0_{I_1\cdots I_s} = D^0_{I_1}\cdots D^0_{I_s}W_{\Lambda_0}|_{\eta_0=0}$. Because the displayed Faà di Bruno formula was written with ordinary derivatives $\delta/\delta\eta$, the left side obeys

$$G^{\Lambda}_{A_1\cdots A_r}(x_1,\dots,x_r;J) = (-1)^r\ \frac{\delta^r H}{\delta\eta_\Lambda^{A_1}(x_1)\cdots\delta\eta_\Lambda^{A_r}(x_r)}\bigg|_{\eta_\Lambda=0}.$$

For the bare connected cumulant attached to a partition π,

$$\frac{\delta^{|\pi|}W_{\Lambda_0}[J,\eta_0]}{\prod_{B\in\pi}\delta\eta_0^{I_B}(z_B)}\bigg|_{\eta_0=0} = (-1)^{|\pi|}G^0_{I_B\,:\,B\in\pi}(z_B\ :\ B\in\pi;J).$$

Thus a term indexed by π carries the chain-rule cumulant factor

$$(-1)^r(-1)^{|\pi|} = (-1)^{r-|\pi|}.$$

Equivalently, a block B of size $|B|$ collapses $|B|$ renormalized insertion derivatives into one bare connected cumulant and contributes the relative sign $(-1)^{|B|-1}$; multiplying over the blocks gives $(-1)^{\sum_B(|B|-1)} = (-1)^{r-|\pi|}$. The finite-regulator partition formula is therefore

$$\begin{aligned}G^{\Lambda}_{A_1\cdots A_r}(x_1,\dots,x_r;J) = \sum_{\pi\in\mathrm{Part}(r)}(-1)^{r-|\pi|}\int\prod_{B\in\pi}\mathrm{d}^D z_B\, G^0_{I_B\,:\,B\in\pi}(z_B\ :\ B\in\pi;J)\\ \times\prod_{B\in\pi}\mathcal{Z}^{I_B}{}_{A_B}(z_B;x_B).\end{aligned} \tag{34.16}$$

Here $\mathrm{Part}(r)$ is the set of partitions of $\{1,\dots,r\}$. If $B = \{b_1,\dots,b_s\}$, then

$$\mathcal{Z}^{I_B}{}_{A_B}(z_B;x_B) = \mathcal{Z}^{I_B}{}_{A_{b_1}\cdots A_{b_s}}(z_B;x_{b_1},\dots,x_{b_s}).$$

For two insertions the formula reads

$$\begin{aligned} G^{\Lambda}_{AB}(x,y;J) &= \int \mathrm{d}^D z\, \mathrm{d}^D w\, \mathcal{Z}^I{}_A(z;x)\mathcal{Z}^J{}_B(w;y) G^0_{IJ}(z,w;J) \\ &\quad - \int \mathrm{d}^D z\, \mathcal{Z}^I{}_{AB}(z;x,y) G^0_I(z;J). \end{aligned} \tag{34.17}$$

The first term is the product of the single-insertion linear maps. The second term is supported at $x = y$ and is therefore a contact term.

On the configuration space

$$\mathrm{Conf}_r = \{(x_1, \ldots, x_r) \mid x_a \neq x_b \text{ for } a \neq b\},$$

only the partition of $\{1, \ldots, r\}$ into singletons contributes to (34.16). All other partitions are local distributions supported on collision diagonals. If a block B has more than one element, the corresponding Taylor coefficient

$$\mathcal{Z}^{I_B}{}_{A_B}(z_B; x_B)$$

is supported on the diagonal where all points x_b, $b \in B$, coincide with z_B. This support does not intersect Conf_r. The only surviving term away from the collision diagonals is therefore the singleton partition, for which every Taylor coefficient is the linear source map.

The same structure appears directly in the Wilsonian representation. Define higher Wilsonian source derivatives by

$$\mathcal{O}^{\mathrm{W}}_{A_1 \cdots A_r}(x_1, \ldots, x_r; \phi) = \left. \frac{\delta^r L_\Lambda[\phi; \eta_\Lambda]}{\delta \eta_\Lambda^{A_1}(x_1) \cdots \delta \eta_\Lambda^{A_r}(x_r)} \right|_{\eta_\Lambda = 0}.$$

For two insertions, differentiating the low-field functional integral gives

$$G^{\Lambda}_{AB}(x,y;J) = \left\langle \mathcal{O}^{\mathrm{W}}_A(x) \mathcal{O}^{\mathrm{W}}_B(y) \right\rangle^{\mathrm{low}}_{\mathrm{conn},J,\Lambda} - \left\langle \mathcal{O}^{\mathrm{W}}_{AB}(x,y) \right\rangle^{\mathrm{low}}_{J,\Lambda}. \tag{34.18}$$

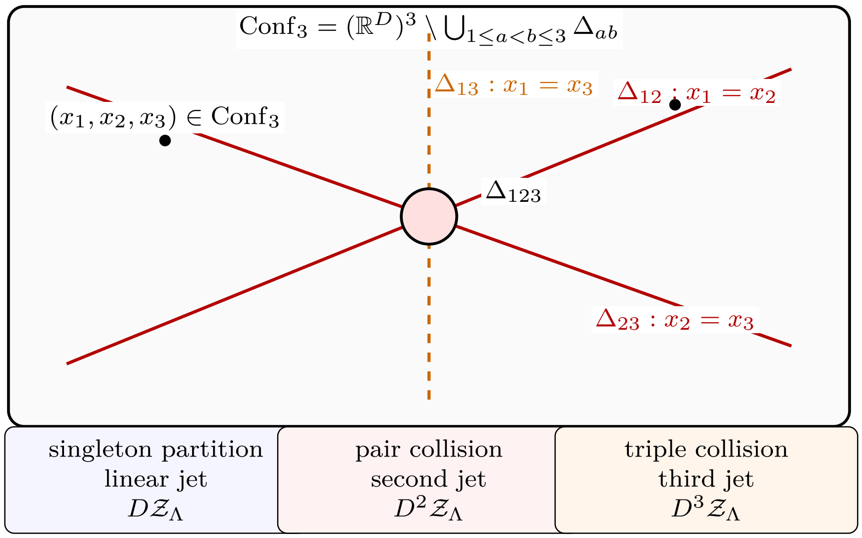
Thus the contact term is the second source derivative of the source-extended Wilsonian action in the chosen source-coordinate system.

For 1PI coordinates one must also distinguish the source Hessian of the Legendre transform from the connected two-insertion distribution. In condensed notation, with field indices i, j integrated or summed,

$$\left. \frac{\delta^2 \Gamma_\Lambda}{\delta \eta_\Lambda^A \delta \eta_\Lambda^B} \right|_{\varphi} = \left. \frac{\delta^2 W_\Lambda^{\mathrm{low}}}{\delta \eta_\Lambda^A \delta \eta_\Lambda^B} \right|_J - \frac{\delta^2 W_\Lambda^{\mathrm{low}}}{\delta \eta_\Lambda^A \delta J_i} \left(\frac{\delta^2 W_\Lambda^{\mathrm{low}}}{\delta J_i \delta J_j} \right)^{-1} \frac{\delta^2 W_\Lambda^{\mathrm{low}}}{\delta J_j \delta \eta_\Lambda^B}.$$

This is the finite-dimensional Legendre-transform identity applied to the regulated theory. The second term subtracts the part connected by one low-field two-point function. Consequently projected 1PI multi-insertion kernels have a contact-term convention inherited from both the nonlinear source-coordinate system and the Legendre-transform projection.

Figure 34.4 records the geometric support statement behind contact terms. The linear jet acts on the open configuration space of separated points. Higher source jets are local distributions supported on pairwise and higher collision diagonals, and the Legendre transform adds the separate one-low-field-reducible subtraction displayed above.



${\mathcal{Z}^I}_{AB}(z; x, y)$ is supported on $z = x = y$; ${\mathcal{Z}^I}_{ABC}(z; x, y, w)$ is supported on $z = x = y = w$.

Figure 34.4: Contact terms are support statements on configuration space. The singleton part of the source Taylor expansion acts on separated insertion points; higher Taylor coefficients are local distributions on collision strata, such as the pairwise diagonals Δ_{ab} and the triple diagonal Δ_{123} for three insertions.

34.6 Anomalous Dimensions and Scaling Operators

If a basis can be chosen along the flow in which

$$\gamma_{IK}(g(\mu)) = \delta_{IK}\gamma_I(g(\mu)),$$

then the operator equation integrates to

$$[O_I]_\mu = \exp\left[-\int_{\mu'}^{\mu} \frac{\mathrm{d}\widetilde{\mu}}{\widetilde{\mu}}\, \gamma_I(g(\widetilde{\mu}))\right] [O_I]_{\mu'}.$$

At an RG fixed point, $g(\mu) = g_*$, the entries $\gamma_I(g_*)$ are constant. The anomalous part of the scaling is then

$$[O_I]_\mu = \left(\frac{\mu}{\mu'}\right)^{-\gamma_I} [O_I]_{\mu'}.$$

An operator basis that diagonalizes γ at the fixed point is a basis of scaling operators for the anomalous part of dilatations. The full scaling dimension also includes the engineering dimension assigned before renormalization.

The elementary field is the simplest special case. Consider the deformation

$$\delta S = \int \mathrm{d}^D x\, \delta g_1\, \phi(x).$$

The amputated 1PI insertion is the external field insertion with the same normalization already used in the propagator. Hence

$$\delta g_1(\mu) = Z(\mu)^{1/2}\delta g_1, \qquad [\phi]_\mu = Z(\mu)^{-1/2}\phi.$$

The field anomalous dimension in this convention is

$$\gamma_\phi = -\frac{\mathrm{d}\log Z(\mu)^{-1/2}}{\mathrm{d}\log\mu} = \frac{1}{2}\frac{\mathrm{d}\log Z(\mu)}{\mathrm{d}\log\mu}.$$

At a fixed point, constant γ_ϕ gives

$$Z(\mu) \propto \mu^{2\gamma_\phi}.$$

34.7 Callan–Symanzik Equations With Operator Insertions

The operator mixing equation gives the insertion version of the Callan–Symanzik equation. Let

$$G^{(n)}_{0,J,\Lambda}(x;x_1,\dots,x_n;g_0,\Lambda) = \langle O_J(x)\,\phi_0(x_1)\cdots\phi_0(x_n)\rangle_{\mathrm{conn},\Lambda,g_0}$$

be the regulated bare connected correlator with one bare operator insertion. Let $G^{(n)}_I$ denote the corresponding renormalized correlator with one insertion of $[O_I]_\mu$ and n elementary renormalized fields:

$$G^{(n)}_I = \langle [O_I]_\mu(x)\,\phi_R(x_1)\cdots\phi_R(x_n)\rangle_{\mathrm{conn}}\,.$$

Away from coincident points, the renormalization chart gives

$$G^{(n)}_I = Z_\phi^{-n/2}\,N_{IJ}\,G^{(n)}_{0,J,\Lambda}$$

after the cutoff-removal limit. Repeated bare-operator labels are summed.

Insertion Callan–Symanzik equation. For noncoincident points $x, x_1, \dots, x_n$, the renormalized insertion correlators satisfy

$$\left(\mu\frac{\partial}{\partial\mu} + \beta_A(g)\frac{\partial}{\partial g_A} + n\gamma_\phi(g)\right)G^{(n)}_I + \gamma_{IK}(g)G^{(n)}_K = 0. \tag{34.19}$$

Let $t = \log\mu$. The bare correlator $G^{(n)}_{0,J,\Lambda}$ has no independent dependence on t at fixed bare couplings and cutoff. Differentiating

$$G^{(n)}_I = Z_\phi^{-n/2}N_{IJ}G^{(n)}_{0,J,\Lambda}$$

at fixed bare data gives

$$\left(\frac{\partial}{\partial t} + \beta_A\frac{\partial}{\partial g_A}\right)G^{(n)}_I = -n\gamma_\phi\,G^{(n)}_I + \frac{\mathrm{d}N_{IJ}}{\mathrm{d}t}\,Z_\phi^{-n/2}G^{(n)}_{0,J,\Lambda}.$$

Using

$$\frac{\mathrm{d}N_{IJ}}{\mathrm{d}t} = -\gamma_{IK}N_{KJ}$$

turns the last term into $-\gamma_{IK}G^{(n)}_K$. Moving both renormalization-factor contributions to the left gives (34.19).

Contact terms must be added when x collides with one of the other insertion points, or when several inserted operators are present and collide with one another. Those terms are local distributions supported on collision diagonals, as described by the source-coordinate Taylor expansion in Section 34.5. Their coefficients are governed by the same renormalized local operator basis, but they depend on the contact-term convention chosen for the source-renormalized generating functional.

At a fixed point, diagonalizing $\gamma_{IK}(g_*)$ gives scaling operators. If $[O_A]$ is an eigenoperator with anomalous dimension γ_A, then its full scaling dimension is

$$\Delta_A = d_A + \gamma_A,$$

where d_A is the engineering dimension in the fixed-point convention. Equation (34.19) is then the differential form of homogeneous scaling of correlation functions with an O_A insertion.

34.8 Dimensional Regularization and MS Coordinates

Now formulate the same 1PI RG in dimensional regularization. Let

$$D = d - \epsilon$$

and write the Euclidean action as

$$S[\phi] = S_{\mathrm{kin}}[\phi] + \int \mathrm{d}^{d-\epsilon}x \sum_I g_I^\epsilon \, O_I(x).$$

The superscript on g_I^ϵ records dependence on the regulator ϵ. The engineering dimensions are assigned as

$$[\phi] = \frac{d-\epsilon-2}{2}, \qquad [O_I] = d - \epsilon + \delta_I(\epsilon), \qquad [g_I^\epsilon] = -\delta_I(\epsilon),$$

where

$$\delta_I(\epsilon) = \delta_I^{(0)} + \epsilon\, \delta_I^{(1)}.$$

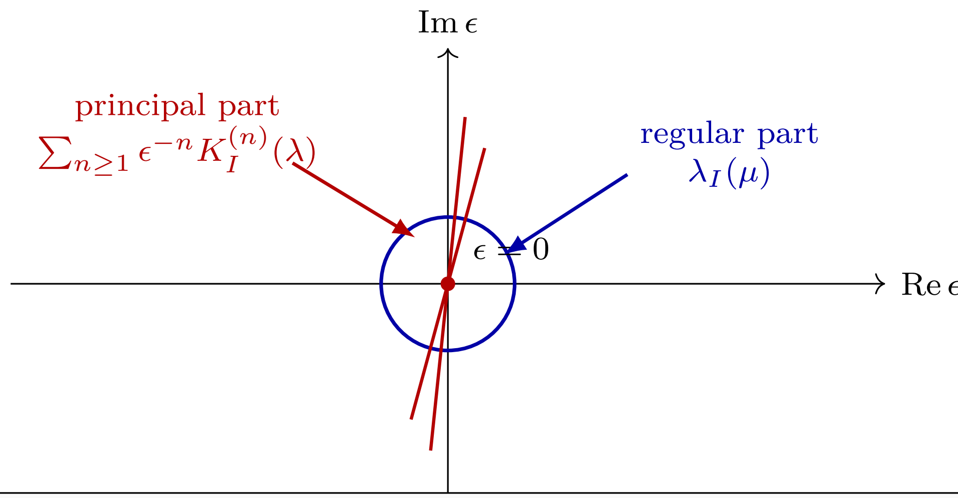
Definition 34.6 (Minimal subtraction coordinates)**.** The minimal-subtraction coordinate $\lambda_I(\mu)$ is defined by the requirement that the bare coupling have the form

$$g_I^\epsilon = \mu^{-\delta_I(\epsilon)} \left[\lambda_I(\mu) + \sum_{n\geq 1} \frac{1}{\epsilon^n} K_I^{(n)}(\lambda(\mu)) \right].$$

The functions $K_I^{(n)}$ are formal power series in the dimensionless coordinates λ_J. The bracket contains the finite coordinate λ_I and pole terms in ϵ; finite terms in the counterterm part are set to zero by the MS coordinate convention.

The MS convention fixes the finite redefinition freedom present in a general subtraction coordinate. It is especially convenient because the beta functions and anomalous dimensions are encoded in the pole coefficients.

Figure 34.5 depicts the MS coordinate as a Laurent-principal-part prescription at $\epsilon = 0$. The finite coordinate λ_I is the regular part selected after the pole part has been subtracted; the pole coefficients $K_I^{(n)}$ are the residues of the principal part in this formal meromorphic coordinate.



$$g_I^\epsilon = \mu^{-\delta_I(\epsilon)} \left[\lambda_I(\mu) + \underbrace{\sum_{n\geq 1} \epsilon^{-n} K_I^{(n)}(\lambda(\mu))}_{P_- \text{ principal part}} \right], \quad P_{\text{fin}}\big(\mu^{\delta_I(\epsilon)} g_I^\epsilon\big) = \lambda_I.$$

Figure 34.5: Minimal subtraction is a Laurent projection at the regulator point $\epsilon = 0$. The dimensionless running coordinate is the finite part after the principal pole part has been subtracted. The coefficients $K_I^{(n)}$ are then constrained by the condition $\mathrm{d}g_I^\epsilon / \mathrm{d}\log\mu = 0$.

34.9 The Scalar Quartic Pole at One Loop

For $d = 4$, consider the scalar quartic operator with dimensional regularization $D = 4 - \epsilon$. The quartic coupling g^ϵ has engineering dimension ϵ, and the interaction term is

$$\mathcal{L}_E \supset \frac{1}{4!} g^\epsilon \phi^4.$$

A general momentum-subtraction coordinate $\widetilde{\lambda}(\mu)$ can be defined at Euclidean momenta of size μ:

$$\mu^\epsilon \widetilde{\lambda}(\mu) = -Z(\mu)^2\, \Gamma^{(4)} \Big|_{k_i^2 = O(\mu^2),\, s=-\mu^2 a_s,\, t=-\mu^2 a_t,\, u=-\mu^2 a_u},$$

where $a_s, a_t, a_u > 0$ are fixed constants specifying the subtraction configuration.

At one loop,

$$\mu^\epsilon \widetilde{\lambda}(\mu) = g^\epsilon - \frac{(g^\epsilon)^2}{2} \int \frac{\mathrm{d}^{4-\epsilon} k}{(2\pi)^{4-\epsilon}} \int_0^1 \mathrm{d}x$$
$$\times \left[\frac{1}{(k^2 + \mu^2 a_s x(1-x))^2} + (a_s \to a_t) + (a_s \to a_u) \right] + O((g^\epsilon)^3).$$

Using

$$\int \frac{d^{4-\epsilon}k}{(2\pi)^{4-\epsilon}} \frac{1}{(k^2+\Delta)^2} = (4\pi)^{-2+\epsilon/2}\Gamma\left(\frac{\epsilon}{2}\right)\Delta^{-\epsilon/2},$$

and then integrating over x, this becomes

$$\mu^\epsilon \widetilde{\lambda}(\mu) = g^\epsilon - \frac{(g^\epsilon)^2}{2}\,\mu^{-\epsilon}(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)} \\ \times\left(a_s^{-\epsilon/2} + a_t^{-\epsilon/2} + a_u^{-\epsilon/2}\right) + O((g^\epsilon)^3).$$

Inverting the relation gives

$$g^\epsilon = \mu^\epsilon\left[\widetilde{\lambda}(\mu) + \widetilde{\lambda}(\mu)^2\left(\frac{3}{16\pi^2}\frac{1}{\epsilon} + O(\epsilon^0)\right) + O(\widetilde{\lambda}^3)\right].$$

The $O(\epsilon^0)$ term depends on the finite subtraction constants a_s, a_t, a_u. The MS coordinate $\lambda(\mu)$ is defined by retaining the pole coefficient and setting the finite counterterm convention to zero:

$$g^\epsilon = \mu^\epsilon\left[\lambda(\mu) + \frac{3}{16\pi^2}\frac{\lambda(\mu)^2}{\epsilon} + O(\lambda^3)\right].$$

This equation fixes the notation used later. For the Wilson–Fisher calculation, the bare quartic coefficient g_0 is this g^ϵ, and the dimensionful running quartic coefficient is

$$g_R(\mu) = \mu^\epsilon \lambda(\mu).$$

For Yang–Mills theory the coupling is placed in the inverse kinetic coefficient, so the analogous dimensionless coordinate is written

$$\lambda^2_{\mathrm{YM}}(\mu) = \mu^\epsilon g^2_{\mathrm{YM}}(\mu), \qquad \frac{1}{g^2_{\mathrm{YM}}(\mu)} = \mu^\epsilon \frac{1}{\lambda^2_{\mathrm{YM}}(\mu)}.$$

Thus the power of μ is the same engineering-dimension conversion; only the choice of coupling coordinate differs between a polynomial scalar vertex and an inverse gauge kinetic term.

Figure 34.6 displays the one-loop quartic MS structure in this convention: the logarithmic scalar bubble supplies the pole subtraction, and the engineering power of μ converts the coordinate in $4-\epsilon$ dimensions.

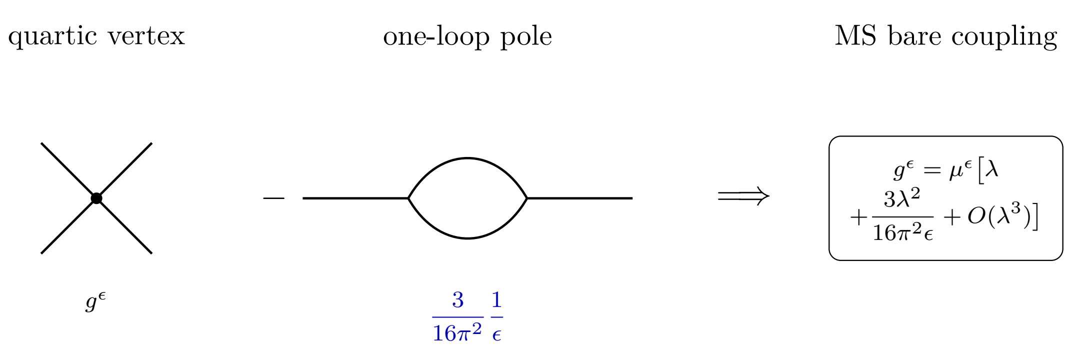


Figure 34.6: In four-dimensional scalar quartic theory, the one-loop bubble produces the pole coefficient $3/(16\pi^2\epsilon)$. The MS coordinate keeps this pole coefficient in the bare coupling and fixes the finite part by convention.

34.10 Beta Functions Away from the Target Dimension

In MS coordinates define

$$\beta_I^{\epsilon}(\lambda) = \frac{\mathrm{d}\lambda_I(\mu)}{\mathrm{d}\log\mu}.$$

The bare coupling g_I^{ϵ} is independent of μ. Differentiating the MS definition gives

$$\begin{aligned} 0 &= \frac{\mathrm{d}g_I^{\epsilon}}{\mathrm{d}\log\mu} \\ &= -\delta_I(\epsilon)\left[\lambda_I + \sum_{n\geq 1}\epsilon^{-n}K_I^{(n)}\right] + \beta_I^{\epsilon} + \sum_{n\geq 1}\epsilon^{-n}\frac{\partial K_I^{(n)}}{\partial\lambda_J}\beta_J^{\epsilon}. \end{aligned}$$

Here and below repeated J-indices are summed. Expand the $(d-\epsilon)$-dimensional beta function as

$$\beta_I^{\epsilon}(\lambda) = \sum_{m=0}^{\infty}\epsilon^m\beta_I^{(m)}(\lambda).$$

The positive powers of ϵ are constrained by the displayed equation. For example, the coefficient of ϵ^1 has the form

$$-\delta_I^{(1)}\lambda_I + \beta_I^{(1)} + \sum_{n\geq 1}\frac{\partial K_I^{(n)}}{\partial\lambda_J}\beta_J^{(n+1)} = 0,$$

and the coefficient of ϵ^2 has the form

$$\beta_I^{(2)} + \sum_{n\geq 1}\frac{\partial K_I^{(n)}}{\partial\lambda_J}\beta_J^{(n+2)} = 0.$$

The pole $K_I^{(n)}$ begins at increasingly high perturbative order as n grows. More precisely, grade the formal power series by perturbative degree and assume $K_I^{(n)} \in F^{n+1}$, where F^L denotes terms of degree at least L in the interaction coordinates. Then $\partial K_I^{(n)}/\partial\lambda_J \in F^n$. For the coefficient of ϵ^r with $r \geq 2$, the homogeneous degree-L part of

$$\beta_I^{(r)} + \sum_{n\geq 1}\frac{\partial K_I^{(n)}}{\partial\lambda_J}\beta_J^{(n+r)} = 0$$

contains $\beta_J^{(n+r)}$ only in degrees strictly smaller than L. Writing $\beta_{I,L}^{(r)}$ for the degree-L component, the lowest perturbative degree therefore gives $\beta_{I,L}^{(r)} = 0$, and induction on L gives $\beta_I^{(r)} = 0$ for every $r \geq 2$. Returning to the coefficient of ϵ^1, the remaining summation involves only $\beta_J^{(n+1)}$ with $n+1 \geq 2$, hence vanishes by the preceding induction, leaving $\beta_I^{(1)} = \delta_I^{(1)}\lambda_I$. Consequently

$$\beta_I^{(m)} = 0 \quad (m \geq 2), \qquad \beta_I^{(1)} = \delta_I^{(1)}\lambda_I.$$

Thus

$$\beta_I^{\epsilon}(\lambda) = \beta_I(\lambda) + \epsilon\,\delta_I^{(1)}\lambda_I, \qquad \beta_I(\lambda) = \beta_I^{(0)}(\lambda).$$

The function $\beta_I(\lambda)$ is the beta function of the target d-dimensional theory in the MS coordinate system.

34.11 Recursive Pole Equations

The nonpositive powers of ϵ determine the pole coefficients $K_I^{(n)}$ from $\beta_I(\lambda)$. Introduce the differential operators

$$D_\beta = \beta_J(\lambda)\frac{\partial}{\partial\lambda_J}, \qquad D_\epsilon = \delta_J^{(1)}\lambda_J\frac{\partial}{\partial\lambda_J}.$$

The coefficient of ϵ^0 gives

$$\beta_I(\lambda) - \delta_I^{(0)}\lambda_I = \big(\delta_I^{(1)} - D_\epsilon\big)K_I^{(1)}(\lambda).$$

For $n \geq 1$, the coefficient of ϵ^{-n} gives

$$\big(D_\beta - \delta_I^{(0)}\big)K_I^{(n)}(\lambda) = \big(\delta_I^{(1)} - D_\epsilon\big)K_I^{(n+1)}(\lambda).$$

The word "recursive" here has a precise algebraic meaning. Choose a perturbative filtration of the formal power-series ring in the couplings, for example by loop order or by the usual interaction degree in a finite renormalizable chart. Let $\mathcal{P}_{I,L}$ denote the finite-dimensional space of homogeneous monomials of filtration degree L that can contribute to the pole coefficient of g_I^ϵ. The operator

$$A_I := \delta_I^{(1)} - D_\epsilon$$

acts diagonally on monomials:

$$A_I(\lambda^\alpha) = \left(\delta_I^{(1)} - \sum_J \alpha_J\delta_J^{(1)}\right)\lambda^\alpha, \qquad \lambda^\alpha = \prod_J \lambda_J^{\alpha_J}.$$

Thus the only possible obstruction to solving the pole equation in a given monomial sector is a resonance

$$\delta_I^{(1)} = \sum_J \alpha_J\delta_J^{(1)}.$$

Lemma 34.7 (Existence and uniqueness of the MS pole recursion)**.** *Assume the following data.*

(i) The chosen MS coordinate chart has a finite perturbative filtration: for each I and each filtration degree L, only finitely many monomials occur in $\mathcal{P}_{I,L}$.

(ii) The pole order is bounded at fixed perturbative order: the component $K_{I,L}^{(n)}$ of $K_I^{(n)}$ in $\mathcal{P}_{I,L}$ vanishes for n larger than a finite bound depending on L.

(iii) For every sector in which a new coefficient $K_{I,L}^{(n+1)}$ is to be solved, the linear map $A_I = \delta_I^{(1)} - D_\epsilon : \mathcal{P}_{I,L} \to \mathcal{P}_{I,L}$ is invertible.

Then a formal target-dimension beta-function vector field $\beta_I(\lambda)$ determines at most one family of pole coefficients $\{K_I^{(n)}\}_{n\geq 1}$ satisfying the MS pole equations and the MS condition that no finite counterterm part is added. If the right-hand sides produced by the recursion lie in the corresponding sectors $\mathcal{P}_{I,L}$, the solution exists and is given degree by degree by

$$K_{I,L}^{(1)} = A_I^{-1}\left[\beta_I - \delta_I^{(0)}\lambda_I\right]_L,$$

and, for $n \geq 1$,

$$K_{I,L}^{(n+1)} = A_I^{-1}\left[\big(D_\beta - \delta_I^{(0)}\big)K_I^{(n)}\right]_L.$$

Here $[\cdots]_L$ denotes projection to filtration degree L.

Proof. At fixed (I, L), the sector $\mathcal{P}_{I,L}$ is finite-dimensional by hypothesis. The coefficient of ϵ^0 is the linear equation

$$A_I K^{(1)}_{I,L} = [\beta_I - \delta^{(0)}_I \lambda_I]_L.$$

Invertibility of A_I on $\mathcal{P}_{I,L}$ gives a unique solution whenever the displayed right-hand side belongs to that sector. Suppose inductively that $K^{(n)}_I$ has already been determined through the filtration degree under consideration. The coefficient of ϵ^{-n} is

$$A_I K^{(n+1)}_{I,L} = \Big[\big(D_\beta - \delta^{(0)}_I\big) K^{(n)}_I\Big]_L,$$

which is again a finite-dimensional linear equation with the same invertible left-hand side. This gives existence, when the right-hand side is in the chosen local sector, and uniqueness. The pole-order bound ensures that at a fixed perturbative degree only finitely many such equations must be solved. If two solutions existed, their difference in the first sector where they differ would lie in $\ker A_I$, contradicting the invertibility hypothesis. ☐

In ordinary renormalizable MS computations these hypotheses are supplied by the graph expansion: at fixed loop order there are finitely many local counterterm monomials in the selected chart, locality bounds the pole order, and nonresonance of A_I is checked on the monomials that actually occur. If a resonant monomial is present, the pole equations do not by themselves give a unique coefficient in that sector; one must add a normalization condition, enlarge the coordinate chart, or treat the resonance as a consistency condition rather than an automatic recursion.

In a homogeneous sector in which every coupling entering the pole coefficient for g^ϵ_I has the same ϵ-slope $\delta^{(1)}_J = \delta^{(1)}_I$, the operator D_ϵ reduces on that sector to

$$D_\epsilon = \delta^{(1)}_I \lambda_J \frac{\partial}{\partial \lambda_J}.$$

The pole equations then take the simpler form

$$\beta_I(\lambda) - \delta^{(0)}_I \lambda_I = \delta^{(1)}_I \left(1 - \lambda_J \frac{\partial}{\partial \lambda_J}\right) K^{(1)}_I(\lambda),$$

and, for $n \geq 1$,

$$\left(\beta_J \frac{\partial}{\partial \lambda_J} - \delta^{(0)}_I\right) K^{(n)}_I(\lambda) = \delta^{(1)}_I \left(1 - \lambda_J \frac{\partial}{\partial \lambda_J}\right) K^{(n+1)}_I(\lambda).$$

The single-coupling scalar quartic example belongs to this homogeneous case.

For a single marginal coupling with $\delta^{(0)} = 0$ and $\delta^{(1)} = -1$, as in $d = 4$ scalar quartic theory, the one-loop MS relation

$$g^\epsilon = \mu^\epsilon \left[\lambda + \frac{3}{16\pi^2} \frac{\lambda^2}{\epsilon} + O(\lambda^3)\right]$$

implies

$$\beta^\epsilon(\lambda) = \beta(\lambda) - \epsilon\lambda,$$

with

$$\beta(\lambda) = \frac{3}{16\pi^2} \lambda^2 + O(\lambda^3).$$

The $-\epsilon\lambda$ term is the engineering contribution in $4 - \epsilon$ dimensions; the ϵ^0 term is the four-dimensional perturbative beta function.

Figure 34.7 records the coefficient equations in the Laurent expansion. The positive powers of ϵ fix the allowed ϵ-dependence of the $d-\epsilon$ beta function, while the nonpositive powers give the displayed linear recursion for the pole coefficients.

coefficient equations in $\mathrm{d}g_I^\epsilon/\mathrm{d}\log\mu = 0$

$\epsilon^{m\geq 2}$: $\beta_I^{(m)} = 0, \qquad \epsilon^1$: $\beta_I^{(1)} = \delta_I^{(1)}\lambda_I.$

ϵ^0: $A_I K_I^{(1)} = \beta_I - \delta_I^{(0)}\lambda_I, \qquad A_I = \delta_I^{(1)} - D_\epsilon.$

$\downarrow K^{(1)} \mapsto K^{(2)}$

ϵ^{-1}: $A_I K_I^{(2)} = (D_\beta - \delta_I^{(0)})K_I^{(1)}.$

$\downarrow$ iterate

ϵ^{-n}: $A_I K_I^{(n+1)} = (D_\beta - \delta_I^{(0)})K_I^{(n)}, \quad n \geq 1.$

When A_I is invertible on the selected local monomial sector, each pole coefficient is fixed by lower data: $K_I^{(n+1)} = A_I^{-1}(D_\beta - \delta_I^{(0)})K_I^{(n)}.$

Figure 34.7: The scale independence of the regulated coupling is a Laurent identity in ϵ. Matching coefficients first fixes the positive powers in β^ϵ, then determines the pole coefficients recursively by the linear equations shown here.

The next chapter translates this scale dependence into a local spacetime statement involving the stress tensor trace. The beta-function vector field will become the coefficient of renormalized operator insertions in $T^\mu{}_\mu$.

Chapter 35

Stress Tensor Trace and Conformal Currents

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

35.1 The Stress Tensor from Local Translations

Work in flat D-dimensional Euclidean spacetime, with coordinates x^μ and metric $\delta_{\mu\nu}$. A constant translation by a vector δa^μ acts on a scalar field by

$$\phi'(x') = \phi(x), \qquad x'^\mu = x^\mu + \delta a^\mu.$$

The corresponding active variation at the same coordinate x is

$$\delta\phi(x) = \phi'(x) - \phi(x) = -\delta a^\mu\, \partial_\mu \phi(x).$$

The stress tensor is obtained by promoting δa^μ to a smooth compactly supported function $\delta a^\mu(x)$. For a multiplet of fields one is free to add to the local variation a term proportional to $\partial_\mu \delta a_\nu$:

$$\delta\phi(x) = -\delta a^\mu(x)\, \partial_\mu \phi(x) + \partial_\mu \delta a_\nu(x)\, \mathcal{D}^{\mu\nu}\phi(x).$$

Here $\mathcal{D}^{\mu\nu}$ is a fixed linear differential operator on the fields. Different choices of $\mathcal{D}^{\mu\nu}$ change the stress tensor by equation-of-motion terms and improvement terms. Once the convention is chosen, define $T^{\mu\nu}$ by

$$S[\phi+\delta\phi] = S[\phi] - \int \mathrm{d}^D x\, \partial_\mu \delta a_\nu(x)\, T^{\mu\nu}(x),$$

for compactly supported $\delta a^\mu(x)$, up to terms proportional to the equations of motion. If δa^μ is constant, the derivative vanishes, which is the Noether statement of translation invariance.

Figure 35.1 contrasts constant translations with local translations and fixes the convention that the stress tensor is the coefficient of $\partial_\mu \delta a_\nu$.

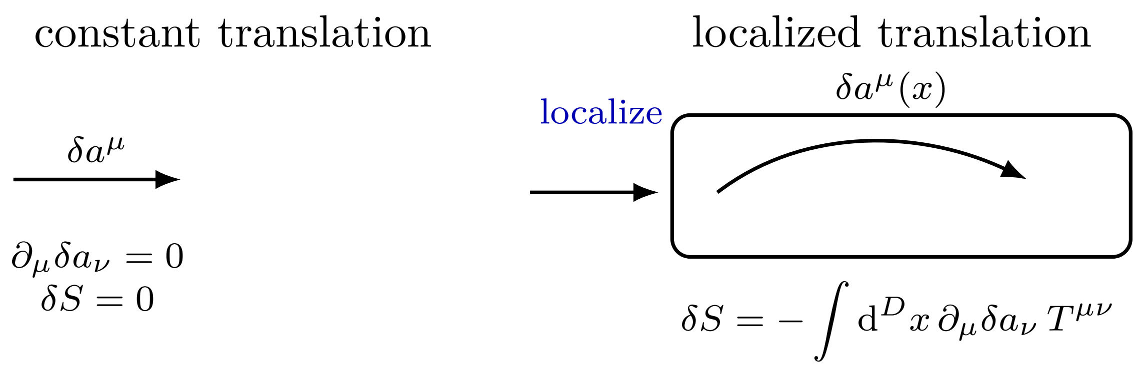


Figure 35.1: The stress tensor is the coefficient of the derivative of a local translation parameter. Constant translations are recovered by setting $\partial_\mu \delta a_\nu = 0$.

35.2 Background Metric Sources

The flat-space stress tensor above is the specialization of a source derivative with respect to a background metric. This construction requires its own data. Let $g_{\mu\nu}$ be a smooth Euclidean metric that equals $\delta_{\mu\nu}$ outside a compact set, and let $\eta^I(x)$ be smooth compactly supported sources for a chosen finite set of renormalized local operators O_I. At a regulator Λ, assume a metric-dependent regulated functional integral

$$Z_\Lambda[g,\eta] = \int \mathcal{D}^g_\Lambda \phi \, \exp\left[-S_\Lambda[g;\phi] - \int \mathrm{d}^D x\, \sqrt{g}\, \eta^I(x) O_{I,\Lambda}(x;g,\phi)\right].$$

The dependence of the cutoff, measure, and composite operators on g is part of the regulator convention. Metric-dependent local terms generated by the measure or by short-distance subtractions are absorbed into a local functional $P_\Lambda[g,\eta]$. In this chapter the source functional is the renormalized Euclidean free energy; the logarithm of the partition function is denoted K when it is needed:

$$\mathcal{W}[g,\eta] = \lim_{\Lambda\to\infty} \big(-\log Z_\Lambda[g,\eta] + P_\Lambda[g,\eta]\big).$$

If $K_\Lambda[g,\eta] = \log Z_\Lambda[g,\eta]$, then $\mathcal{W} = -K$ after the same choice of local subtraction chart. The opposite convention $K = \log Z$ is often useful for displaying connected insertions directly, but every first-derivative formula below changes sign when one replaces $\mathcal{W}$ by K. The limit is taken in the topology of distributions obtained by differentiating with respect to smooth test sources near $\eta = 0$ and $g = \delta$. It is a definition of the source-coordinate system; its existence must be established from the regulator and renormalization construction.

The stress tensor and operator insertions in this source-coordinate system are defined by

$$\langle T^{\mu\nu}(x)\rangle_{g,\eta} = -\frac{2}{\sqrt{g}(x)}\frac{\delta\mathcal{W}[g,\eta]}{\delta g_{\mu\nu}(x)}, \qquad \langle O_I(x)\rangle_{g,\eta} = \frac{1}{\sqrt{g}(x)}\frac{\delta\mathcal{W}[g,\eta]}{\delta\eta^I(x)}. \tag{35.1}$$

The stress-tensor sign follows from the metric response convention

$$\delta S[g;\phi] = -\frac{1}{2}\int \mathrm{d}^D x\,\sqrt{g}\,T^{\mu\nu}(x)\delta g_{\mu\nu}(x).$$

The operator-insertion sign follows from the Euclidean source coupling

$$S_{\mathrm{source}}[g,\eta;\phi] = \int \mathrm{d}^D x\,\sqrt{g}\,\eta^I O_I.$$

In $Z_\Lambda[g,\eta]$ this term appears in the exponential with sign $-S_{\mathrm{source}}$. The metric derivative is taken with respect to $g_{\mu\nu}$; by the chain-rule identity (58.2), this is the same stress-tensor convention as the contravariant-metric variation used in Chapter 58. At $\eta = 0$ and $g = \delta$, this metric derivative agrees with the flat-space Noether stress tensor up to contact terms, equation-of-motion operators, and improvement terms fixed by the source-coordinate system.

Assumption 35.1 (Covariant source regulator)**.** The regulated functional $Z_\Lambda[g,\eta]$, the local subtraction functional $P_\Lambda[g,\eta]$, and the limiting source-coordinate system $\mathcal{W}[g,\eta]$ are invariant under compactly supported diffeomorphisms. The metric and scalar sources transform by

$$\delta_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu, \qquad \delta_\xi\eta^I = \xi^\mu\nabla_\mu\eta^I$$

for every smooth compactly supported vector field ξ. This regulator and subtraction-scheme hypothesis requires covariance of the cutoff and counterterm prescription in addition to locality. It is available for regulator constructions and counterterm prescriptions that are covariant with respect to the background metric. A cutoff tied to a fixed coordinate lattice, a fixed momentum frame, or a noncovariant smoothing kernel requires additional improvement terms before the hypothesis can hold. Dimensional regularization on a curved background also requires a covariant continuation of the tensor algebra and a covariant subtraction prescription. In a chiral theory with a gravitational anomaly, no local counterterm choice makes $\mathcal{W}$ invariant under all such diffeomorphisms; the right-hand side of the Ward identity below is then the anomaly functional rather than zero. The derivation in this section assumes the anomaly-free case.

Under Assumption 35.1, substituting the variations above into $\delta_\xi\mathcal{W} = 0$, using (35.1), and integrating the metric term by parts gives the source Ward identity

$$\nabla_\mu\langle T^{\mu\nu}(x)\rangle_{g,\eta} = -\langle O_I(x)\rangle_{g,\eta}\nabla^\nu\eta^I(x). \tag{35.2}$$

The sign in this Ward identity is tied to the source term $-\int\sqrt{g}\,\eta^I O_I$ in $Z_\Lambda[g,\eta]$. A source term with the opposite sign gives the corresponding opposite source term in the diffeomorphism Ward identity. Spinor, vector, and tensor sources add the corresponding representation terms to $\delta_\xi\eta$.

At an RG fixed point, a source-coordinate system whose beta functions vanish and whose stress tensor has been improved as below gives a fixed-point functional

$$\mathcal{W}_*[g,\eta].$$

This is the antecedent for the fixed-point and conformal source functionals used later. The conformal chapters denote the same free-energy convention by W_* and restate the sign of the source term there. The symbol $W_*[g,\eta]$ in those chapters refers to this same renormalized metric and operator source limit, specialized to a fixed point.

35.3 Dilatations and Engineering Dimensions

A constant infinitesimal scale transformation is the special local translation

$$\delta a^\mu(x) = \sigma x^\mu,$$

where σ is a constant. The stress-tensor definition gives

$$\delta S = -\sigma \int \mathrm{d}^D x \, T^\mu{}_\mu(x).$$

Thus the trace is the local operator that measures the response to an infinitesimal dilatation in the chosen stress-tensor convention.

For a free massless scalar,

$$S_{\text{kin}}[\phi] = \frac{1}{2} \int \mathrm{d}^D x \, \partial_\mu \phi \, \partial^\mu \phi,$$

scale invariance of the kinetic term fixes the engineering field dimension to

$$\Delta_\phi^{\text{eng}} = \frac{D-2}{2}.$$

Equivalently, with $x'^\mu = (1+\sigma)x^\mu$,

$$\phi'(x') = (1+\sigma)^{-(D-2)/2} \phi(x),$$

or, at fixed x,

$$\delta\phi(x) = -\sigma x^\mu \partial_\mu \phi(x) - \sigma \frac{D-2}{2} \phi(x).$$

In the local-translation notation this means that, for the scalar field, the trace of the chosen differential operator satisfies

$$\mathcal{D}^\mu{}_\mu \phi = -\frac{D-2}{2} \phi.$$

The invariance of the kinetic term can be checked directly. Put $\Delta_\phi = (D-2)/2$. For compactly supported variations,

$$\begin{aligned}
\delta S_{\text{kin}} &= -\sigma \int \mathrm{d}^D x \, \partial_\mu \phi \, \partial^\mu \big(x^\nu \partial_\nu \phi + \Delta_\phi \phi \big) \\
&= -\sigma \int \mathrm{d}^D x \, \left[(1+\Delta_\phi) \, \partial_\mu \phi \, \partial^\mu \phi + \frac{1}{2} x^\nu \partial_\nu \big(\partial_\mu \phi \, \partial^\mu \phi \big) \right] \\
&= -\sigma \left[1 + \Delta_\phi - \frac{D}{2} \right] \int \mathrm{d}^D x \, \partial_\mu \phi \, \partial^\mu \phi = 0.
\end{aligned} \tag{35.3}$$

Let now $D = d - \epsilon$, and let O_I be a local monomial or polynomial in fields and derivatives with engineering dimension

$$d_I(\epsilon) = d - \epsilon + \delta_I(\epsilon) = D + \delta_I(\epsilon).$$

Under the same scale transformation,

$$O_I'(x') = (1+\sigma)^{-d_I(\epsilon)} O_I(x).$$

For an action of the form

$$S[\phi] = S_{\text{kin}}[\phi] + \int \mathrm{d}^D x \sum_I g_I^\epsilon \, O_I(x),$$

the interaction part transforms as

$$\int \mathrm{d}^D x' \sum_I g_I^\epsilon O_I'(x') = \int \mathrm{d}^D x \sum_I g_I^\epsilon (1+\sigma)^{-\delta_I(\epsilon)} O_I(x).$$

The total variation is the sum of the kinetic and interaction variations. Equation (35.3) shows that the kinetic contribution vanishes in this engineering convention, for compactly supported versions of the dilatation variation. Thus the nonderivative trace term comes only from the interaction coordinates. To first order in σ,

$$\delta S = -\sigma \int \mathrm{d}^D x \sum_I \delta_I(\epsilon) g_I^\epsilon O_I(x).$$

Comparing this with the stress-tensor definition gives the bare trace identity

$$T^\mu{}_\mu = \sum_I \delta_I(\epsilon) \, g_I^\epsilon \, O_I,$$

modulo total derivatives and equation-of-motion operators.

Figure 35.2 records the scaling bookkeeping used in the derivation. The kinetic term has three factors: the Jacobian of the measure, the two derivatives, and the two fields. The interaction term has only the measure factor and the engineering dimension of the local operator. Comparing the resulting first-order variation with $\delta S = -\sigma \int T^\mu{}_\mu$ fixes the sign in the bare trace identity.

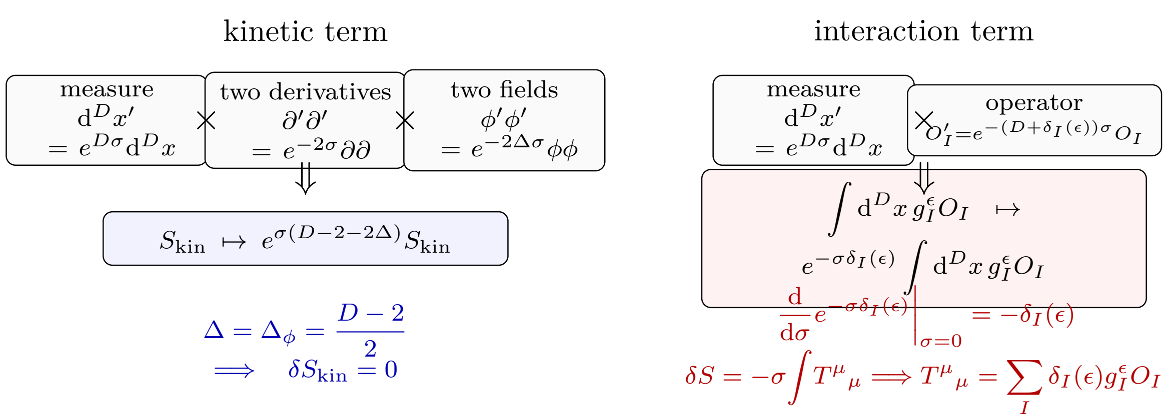


Figure 35.2: The bare trace identity follows from factor bookkeeping under a constant scale transformation. The kinetic term contributes the factor $e^{\sigma(D-2-2\Delta)}$ and is invariant at $\Delta_\phi = (D-2)/2$. An interaction operator of engineering dimension $D + \delta_I(\epsilon)$ leaves the residual factor $e^{-\sigma\delta_I(\epsilon)}$; comparison with $\delta S = -\sigma \int T^\mu{}_\mu$ gives $T^\mu{}_\mu = \sum_I \delta_I(\epsilon) g_I^\epsilon O_I$ modulo total derivatives and equation-of-motion operators.

35.4 The Renormalized Trace Identity

The preceding identity is a bare operator statement in the regulated theory. To convert it to a renormalized statement, use the same MS chart and source-renormalized operator convention as in the previous chapter. Minimal subtraction expresses each bare coupling as

$$g_I^\epsilon = \mu^{-\delta_I(\epsilon)} \left[\lambda_I(\mu) + \sum_{n \geq 1} \epsilon^{-n} K_I^{(n)}(\lambda(\mu)) \right],$$

where

$$\delta_I(\epsilon) = \delta_I^{(0)} + \epsilon \delta_I^{(1)}.$$

At finite ϵ, define the MS insertion associated with the coordinate λ_J by

$$\mathcal{O}_{J,\epsilon}^{\mathrm{MS}} = \frac{\partial}{\partial \lambda_J} \left(\sum_I g_I^\epsilon O_I \right),$$

where the derivative is taken at fixed bare operator basis O_I and fixed μ. The renormalized insertion is the finite limit

$$[O_J]_\mu = \lim_{\epsilon \to 0} \mathcal{O}_{J,\epsilon}^{\mathrm{MS}}$$

as a distribution away from coincident insertion points. Contact-term extensions of this distribution are additional scheme data.

Bare couplings are independent of μ. If

$$\beta_I^\epsilon(\lambda) = \frac{\mathrm{d}\lambda_I}{\mathrm{d}\log\mu} = \beta_I(\lambda) + \epsilon \delta_I^{(1)} \lambda_I$$

is the $D = d - \epsilon$ beta function in MS coordinates, then

$$0 = \frac{\mathrm{d}g_I^\epsilon}{\mathrm{d}\log\mu} = -\delta_I(\epsilon) g_I^\epsilon + \sum_J \beta_J^\epsilon \frac{\partial g_I^\epsilon}{\partial \lambda_J}.$$

Hence

$$\delta_I(\epsilon) g_I^\epsilon = \sum_J \big(\beta_J + \epsilon \delta_J^{(1)} \lambda_J \big) \frac{\partial g_I^\epsilon}{\partial \lambda_J}.$$

Substituting this relation into the bare trace identity gives

$$T^\mu{}_\mu = \sum_J \big(\beta_J + \epsilon \delta_J^{(1)} \lambda_J \big) \mathcal{O}_{J,\epsilon}^{\mathrm{MS}}.$$

Lemma 35.2 (Flat-space renormalized trace identity)**.** *In the MS coordinate chart above, and for separated insertions in flat space, the finite stress-tensor trace satisfies*

$$T^\mu{}_\mu = \sum_J \beta_J(\lambda) \, [O_J]_\mu. \tag{35.4}$$

Proof. Let f be a test density for the trace insertion whose support is separated from all other operator insertions. In the MS source-coordinate system the regulated operator insertion has a Laurent expansion of the form

$$\mathcal{O}^{\mathrm{MS}}_{J,\epsilon}(f) = \sum_{r\geq 1} \epsilon^{-r} C_{J,r}(f) + A_{J,0}(f) + O(\epsilon),$$

where $A_{J,0}(f) = [O_J]_\mu(f)$ and every $C_{J,r}$ is supported on a collision diagonal. For the separated test density under consideration $C_{J,r}(f) = 0$, hence

$$\operatorname*{FP}_{\epsilon=0}\Big[\big(\beta_J + \epsilon\delta^{(1)}_J\lambda_J\big)\mathcal{O}^{\mathrm{MS}}_{J,\epsilon}(f)\Big] = \beta_J [O_J]_\mu(f),$$

where $\mathrm{FP}_{\epsilon=0}$ denotes the coefficient of ϵ^0 in the Laurent expansion. Taking this finite part in the bare trace equation gives (35.4).

The order of operations in the preceding paragraph matters. For an unrestricted test density, multiplication by $\epsilon\delta^{(1)}_J\lambda_J$ sends the simple-pole contact residue $\epsilon^{-1}C_{J,1}$ to the finite local distribution $\delta^{(1)}_J\lambda_J C_{J,1}$, and it shifts the higher pole residues by one pole order. The pole equations coming from $\mathrm{d}g^\epsilon_I/\mathrm{d}\log\mu = 0$ are precisely the algebraic cancellation conditions that make the finite part of $\sum_J \beta^\epsilon_J \mathcal{O}^{\mathrm{MS}}_{J,\epsilon}$ well defined after the source counterterms have been included. Thus the separated identity above is obtained by first forming the MS-renormalized finite part of the product and then restricting away from collision diagonals. The local finite remnants for general test densities are the contact terms described after the proposition. □

At coincident insertion points the preceding proof has a narrower scope. The insertion $\mathcal{O}^{\mathrm{MS}}_{J,\epsilon}$ may have pole parts supported on collision diagonals, and multiplication by ϵ can leave a finite local distribution. Those finite remnants are the Weyl-anomaly and Ward-identity contact terms. The contact-distribution identity is formulated through the renormalized source functional of Section 35.2, together with its local source-coordinate system.

Let $\eta^A(x)$ denote all local sources in a chosen chart, including the constant coupling coordinates λ^I as the special case $\eta^I(x) = \lambda^I$. A local Weyl/RG source vector field is a first-order differential operator on source space,

$$\mathfrak{B}_\sigma = \int \mathrm{d}^d x\, \mathcal{B}^A_\sigma(x; g, \eta)\, \frac{\delta}{\delta\eta^A(x)},$$

where $\mathcal{B}^A_\sigma$ is a local polynomial in the sources, the metric, σ, and finitely many of their derivatives. Its zeroth-order part in the coupling-source directions is

$$\begin{aligned}\mathcal{B}^I_\sigma(x; g, \eta)\big|_{\eta=\lambda} = \sigma(x)\,\beta^I(\lambda)&\\ &+ \text{terms with derivatives of } \sigma\\ &+ \text{terms with positive powers of nonconstant sources.}\end{aligned}$$

The derivative-of-σ terms encode improvement and scheme-dependent local source choices; they vanish in the constant flat separated-point calculation above but contribute to contact distributions.

Proposition 35.3 (Trace identity as a contact distribution)**.** *Assume that the renormalized source functional $\mathcal{W}[g,\eta]$ exists in the sense of Section 35.2 and satisfies the local RG identity*

$$\left[\int \mathrm{d}^d x\, 2\sigma(x) g_{\mu\nu}(x)\frac{\delta}{\delta g_{\mu\nu}(x)} + \mathfrak{B}_\sigma\right]\mathcal{W}[g,\eta] = -\mathcal{A}_\sigma[g,\eta]. \tag{35.5}$$

Here σ is a compactly supported smooth Weyl parameter and $\mathcal{A}_\sigma[g,\eta]$ is a local functional, linear in σ and its derivatives. For

$$X = O_{A_1}(x_1)\cdots O_{A_n}(x_n),$$

define the full contact distribution by differentiating (35.5) with respect to the corresponding sources and then setting $g=\delta$ and $\eta=\lambda$. In insertion notation this gives

$$\int \mathrm{d}^d x\,\sigma(x)\left\langle\left(T^\mu{}_\mu - \beta^I[O_I]_\mu\right)(x)\,X\right\rangle_c = \mathcal{A}_\sigma[X] + \mathcal{C}^{\mathfrak{B}}_\sigma[X]. \tag{35.6}$$

The term $\mathcal{A}_\sigma[X]$ is the source derivative of $\mathcal{A}_\sigma[g,\eta]$. The term $\mathcal{C}^{\mathfrak{B}}_\sigma[X]$ is the finite sum obtained when one or more source derivatives hit the coefficients $\mathcal{B}^A_\sigma(x;g,\eta)$ rather than $\mathcal{W}$. Both terms are local: after differentiating with respect to sources, their kernels are supported on collision diagonals, and the remaining terms contain derivatives of the Weyl parameter associated with improvement or source-scheme choices. For constant σ and mutually distinct flat-space insertion points, these local terms vanish, and (35.6) reduces to (35.4).

Proof. The metric derivative convention (35.1) gives

$$\int \mathrm{d}^d x\,2\sigma g_{\mu\nu}\frac{\delta\mathcal{W}}{\delta g_{\mu\nu}} = -\int \mathrm{d}^d x\,\sqrt{g}\,\sigma\,\langle T^\mu{}_\mu(x)\rangle_{g,\eta}\,.$$

The zeroth-order coupling-source part of $\mathfrak{B}_\sigma$ gives

$$\int \mathrm{d}^d x\,\sigma(x)\beta^I(\lambda)\frac{\delta\mathcal{W}}{\delta\eta^I(x)} = \int \mathrm{d}^d x\,\sqrt{g}\,\sigma(x)\beta^I(\lambda)\,\langle[O_I]_\mu(x)\rangle_{g,\eta}$$

in the source-sign convention of this chapter. Substituting these two identities into (35.5) gives the one-insertion form of (35.6). Apply the functional derivative operator

$$\frac{\delta}{\delta\eta^{A_1}(x_1)}\cdots\frac{\delta}{\delta\eta^{A_n}(x_n)}$$

to (35.5). When all derivatives hit $\mathcal{W}$, the result is the connected distribution with the insertion $T^\mu{}_\mu - \beta^I[O_I]_\mu$, with the standard alternating connected-sign convention inherited from $\mathcal{W} = -\log Z$. When at least one derivative hits a coefficient $\mathcal{B}^A_\sigma$, locality of $\mathcal{B}^A_\sigma$ forces the resulting kernel to be supported where the corresponding source points collide with one another or with the point x. Those terms are, by definition, $\mathcal{C}^{\mathfrak{B}}_\sigma[X]$. The derivatives of the local functional $\mathcal{A}_\sigma[g,\eta]$ give $\mathcal{A}_\sigma[X]$, also supported on the same local loci after specializing to flat space. On the configuration space $x_i \neq x_j$ and $x \neq x_i$, these local distributions vanish, leaving the separated-point proposition. □

Because infinitesimal Weyl transformations commute, the anomaly in (35.5) must satisfy the Wess–Zumino consistency condition

$$\Delta_{\sigma_1}\mathcal{A}_{\sigma_2} - \Delta_{\sigma_2}\mathcal{A}_{\sigma_1} = 0, \qquad \Delta_\sigma = \int \mathrm{d}^d x\,2\sigma g_{\mu\nu}\frac{\delta}{\delta g_{\mu\nu}} + \mathfrak{B}_\sigma. \tag{35.7}$$

This equation is a condition on local functionals of sources and background fields. In even dimensions, curvature terms in $\mathcal{A}_\sigma[g,0]$ may produce flat-space stress-tensor contact distributions after differentiating with respect to the metric even though they vanish at separated flat points.

Thus a conformal fixed point is not specified by separated-point correlators alone; it also requires a source-coordinate system, an anomaly functional satisfying (35.7), and a stress-tensor improvement convention.

Figure 35.3 separates the two support regimes in the local RG identity. On the open configuration space of noncoincident insertions one obtains the beta-function trace identity; on the collision strata the same source-functional equation contains contact distributions and anomaly derivatives that must be retained as part of the renormalized QFT data.

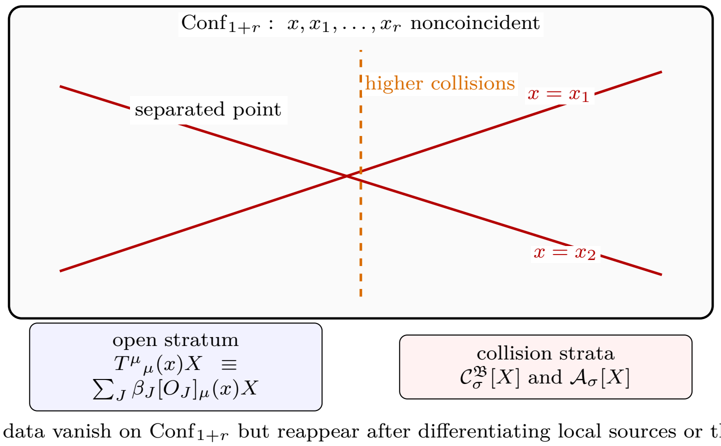


Figure 35.3: The renormalized trace identity has a separated-point part and a local contact part. Away from the collision diagonals the flat-space trace is the beta-function insertion. On the diagonals one must also keep the local distributions produced by source beta functions and by the anomaly functional; these data are invisible in ordinary separated correlators.

35.5 Fixed Points and Conformal Currents

An RG fixed point in the MS coordinate system is a point λ_* at which

$$\beta_J(\lambda_*) = 0 \qquad \text{for all } J.$$

At such a point the beta-function contribution to the trace vanishes. The remaining trace terms, if any, are total derivatives, contact terms, equation-of-motion operators, or improvement-dependent terms. Suppose that the stress tensor has been chosen symmetric, conserved, and traceless as a flat-space renormalized operator:

$$T^{\mu\nu} = T^{\nu\mu}, \qquad \partial_\mu T^{\mu\nu} = 0, \qquad T^{\mu}{}_{\mu} = 0.$$

Let $\xi^\mu(x)$ be a conformal Killing vector field in d dimensions, meaning

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \frac{2}{d}\delta_{\mu\nu}\,\partial_\rho \xi^\rho = 0.$$

Define the current

$$j^\mu_\xi(x) = T^{\mu\nu}(x)\xi_\nu(x).$$

Its divergence is

$$\partial_\mu j^\mu_\xi = T^{\mu\nu}\partial_\mu \xi_\nu = \frac{1}{2}T^{\mu\nu}\big(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu\big) = \frac{1}{d}T^\mu{}_\mu\, \partial_\rho \xi^\rho = 0.$$

Thus each conformal Killing vector gives a conserved Noether current.

The elementary conformal Killing vector fields are

$$\xi^\mu(x) = \begin{cases} a^\mu, & \text{translations,} \\ \omega^\mu{}_\nu x^\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}, & \text{rotations,} \\ c\, x^\mu, & \text{dilatations,} \\ b_\rho(2x^\rho x^\mu - x^2\delta^{\rho\mu}), & \text{special conformal transformations.} \end{cases}$$

Figure 35.4 shows the elementary Euclidean conformal Killing fields as vector-field flows in two dimensions. The algebraic conservation law $j^\mu_\xi = T^{\mu\nu}\xi_\nu$ uses the same CKV equation, and the figure places that conservation law in the geometric context of the four generating families.

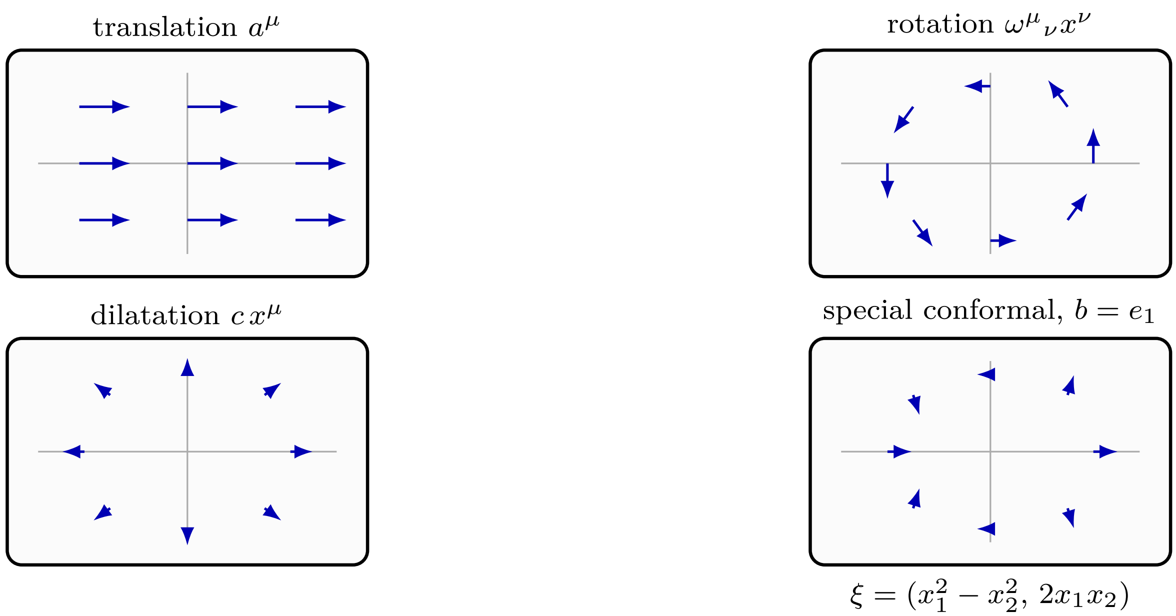


Figure 35.4: The four elementary conformal Killing vector fields. Translation, rotation, dilatation, and special conformal generators are different vector-field geometries even though the conservation proof for $j^\mu_\xi = T^{\mu\nu}\xi_\nu$ uses the same symmetric, conserved, traceless stress tensor and the same conformal Killing equation.

35.6 Virial Currents and Improvement

The fixed-point trace identity determines only the beta-function contribution to $T^\mu{}_\mu$. The stress tensor may still have a total-derivative trace

$$T^\mu{}_\mu = \partial_\mu V^\mu$$

for a local vector operator V^μ, called a virial current. In this case the dilatation current

$$D^\mu = x_\nu T^{\mu\nu} - V^\mu$$

is conserved:

$$\partial_\mu D^\mu = T^\mu{}_\mu - \partial_\mu V^\mu = 0.$$

This establishes scale invariance generated by D^μ, but it does not by itself give conserved special conformal currents of the form $T^{\mu\nu}\xi_\nu$.

If the virial current can be removed by an improvement of the stress tensor, then the same fixed point admits a traceless stress tensor and the conformal currents above. For example, if

$$V_\mu = (d-1)\partial_\mu L$$

for a local scalar operator L, then the improvement

$$T'_{\mu\nu} = T_{\mu\nu} + \big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L$$

changes the trace by

$$T'^\mu{}_\mu = T^\mu{}_\mu - (d-1)\partial^2 L,$$

and removes this virial trace.

For the free massless scalar this improvement has a definite coefficient. Keep the spacetime dimension in this paragraph denoted by D, with $D > 1$. The flat canonical stress tensor for

$$S_{\mathrm{kin}}[\phi] = \frac{1}{2}\int \mathrm{d}^D x\, \partial_\rho\phi\, \partial^\rho\phi$$

is

$$T^{\mathrm{can}}_{\mu\nu} = \partial_\mu\phi\, \partial_\nu\phi - \frac{1}{2}\delta_{\mu\nu}\partial_\rho\phi\, \partial^\rho\phi.$$

Its trace is

$$\begin{aligned} T^{\mathrm{can}\,\mu}{}_\mu &= -\frac{D-2}{2}\,\partial_\rho\phi\,\partial^\rho\phi \\ &= -\frac{D-2}{4}\,\partial^2(\phi^2) + \frac{D-2}{2}\,\phi\,\partial^2\phi. \end{aligned}$$

The second term is an equation-of-motion operator for the free action. Thus, modulo the free equation of motion and separated-insertion contact conventions, the canonical trace is the total derivative

$$T^{\mathrm{can}\,\mu}{}_\mu \equiv -\frac{D-2}{4}\,\partial^2(\phi^2).$$

Define the conformal-coupling coefficient

$$\xi_D^{\mathrm{conf}} = \frac{D-2}{4(D-1)}.$$

With the improvement convention used above,

$$L_{\mathrm{scalar}} = -\xi_D^{\mathrm{conf}}\phi^2,$$

so that

$$\begin{aligned} T^{\mathrm{conf}}_{\mu\nu} &= T^{\mathrm{can}}_{\mu\nu} + \big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L_{\mathrm{scalar}} \\ &= T^{\mathrm{can}}_{\mu\nu} + \xi_D^{\mathrm{conf}}\big(\delta_{\mu\nu}\partial^2 - \partial_\mu\partial_\nu\big)\phi^2. \end{aligned}$$

Its trace is

$$T^{\mathrm{conf}\,\mu}{}_{\mu} = \frac{D-2}{2}\,\phi\,\partial^2\phi,$$

which vanishes as an operator modulo the free equation of motion. The same ξ_D^{conf} is the coefficient of the curved source term $\frac{1}{2}\int \mathrm{d}^D x\,\sqrt{g}\,\xi_D^{\mathrm{conf}} R\phi^2$; the sign difference above is only the translation between that curvature-source coefficient and the scalar L in the flat-space convention $(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2)L$.

Stress-tensor criterion for conformal currents. Let a flat-space local QFT in dimension $d \geq 2$ have a symmetric conserved stress tensor $T_{\mu\nu}$. Suppose that scale invariance is implemented by a conserved current

$$D^\mu = x_\nu T^{\mu\nu} - V^\mu, \qquad \partial_\mu D^\mu = 0,$$

so that $T^\mu{}_\mu = \partial_\mu V^\mu$ as an operator identity, modulo equations of motion and contact terms with other insertions. If the virial current has the form

$$V_\mu = (d-1)\partial_\mu L + J_\mu, \qquad \partial_\mu J^\mu = 0,$$

for a local scalar operator L and a conserved local current J_μ, then the improved stress tensor

$$T'_{\mu\nu} = T_{\mu\nu} + \big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L$$

is symmetric, conserved, and traceless. Consequently each conformal Killing vector ξ^μ gives a conserved current

$$j_\xi^\mu = T'^{\mu\nu}\xi_\nu.$$

The improvement term is symmetric, and its divergence vanishes identically:

$$\partial^\mu\big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L = \partial_\nu\partial^2 L - \partial_\nu\partial^2 L = 0.$$

Its trace is

$$\delta^{\mu\nu}\big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L = -(d-1)\partial^2 L.$$

Using

$$T^\mu{}_\mu = \partial_\mu V^\mu = (d-1)\partial^2 L + \partial_\mu J^\mu = (d-1)\partial^2 L$$

gives $T'^\mu{}_\mu = 0$. For a conformal Killing field ξ,

$$\partial_\mu\xi_\nu + \partial_\nu\xi_\mu = \frac{2}{d}(\partial\cdot\xi)\delta_{\mu\nu}.$$

Using symmetry and conservation of T',

$$\begin{aligned}
\partial_\mu(T'^{\mu\nu}\xi_\nu) &= T'^{\mu\nu}\partial_\mu\xi_\nu \\
&= \frac{1}{2}T'^{\mu\nu}(\partial_\mu\xi_\nu + \partial_\nu\xi_\mu) \\
&= \frac{1}{d}(\partial\cdot\xi)T'^\mu{}_\mu = 0.
\end{aligned}$$

Thus every conformal Killing vector defines a conserved current.

Definition 35.4 (Virial-improvement quotient)**.** Fix the source-coordinate system, stress-tensor convention, and contact-term convention. Let $\mathfrak{V}$ be the vector space of local vector operators that are admissible as virial currents in the identity

$$T^{\mu}{}_{\mu} = \partial_\mu V^\mu,$$

and let $\mathfrak{S}$ be the vector space of local scalar operators L for which the flat-space improvement

$$T_{\mu\nu} \longmapsto T_{\mu\nu} + \big(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2\big)L$$

is an allowed change of stress-tensor representative. These vector spaces are already quotiented by equation-of-motion operators and by the contact distributions that vanish in separated insertions in the chosen source-coordinate system. Derivatives are the induced maps on these quotients. Let

$$\mathfrak{J} = \{\, J_\mu \in \mathfrak{V} : \partial_\mu J^\mu = 0 \,\}$$

be the subspace of conserved local currents. The scalar virial obstruction is the class

$$\mathrm{obs}^{\mathrm{scal}}_{\mathrm{vir}}(V) = [V] \in \mathcal{H}^{\mathrm{scal}}_{\mathrm{vir}} := \mathfrak{V}\Big/\big(\mathfrak{J} + (d-1)\partial\mathfrak{S}\big), \qquad (d-1)\partial\mathfrak{S} = \{(d-1)\partial_\mu L : L \in \mathfrak{S}\}.$$

Remark 35.5 (Scalar-improvement criterion)*.* Under the hypotheses of the stress-tensor criterion above, and restricting to the scalar improvement displayed there, the scale current can be promoted to conformal Killing currents by improving the stress tensor if and only if

$$\mathrm{obs}^{\mathrm{scal}}_{\mathrm{vir}}(V) = 0 \quad \text{in} \quad \mathcal{H}^{\mathrm{scal}}_{\mathrm{vir}}.$$

If the obstruction class vanishes, then by definition there exist $L \in \mathfrak{S}$ and $J_\mu \in \mathfrak{J}$ such that

$$V_\mu = (d-1)\partial_\mu L + J_\mu$$

in the local-operator quotient. The criterion then gives a symmetric conserved traceless improved stress tensor, hence conserved conformal Killing currents.

Conversely, suppose a scalar improvement by $L \in \mathfrak{S}$ makes the trace vanish. The improved trace is

$$0 = T^{\mu}{}_{\mu} - (d-1)\partial^2 L = \partial_\mu\big(V^\mu - (d-1)\partial^\mu L\big)$$

in the same local-operator quotient. Therefore $J_\mu := V_\mu - (d-1)\partial_\mu L$ lies in $\mathfrak{J}$, and the class of V in $\mathfrak{V}/(\mathfrak{J} + (d-1)\partial\mathfrak{S})$ is zero.

Remark 35.6 (General improvements)*.* Definition 35.4 isolates the scalar improvement used in this chapter and in the free-scalar example. If a broader stress-tensor source-coordinate system allows additional identically conserved improvement tensors, the quotient must be enlarged by the image of the corresponding trace-changing map. The precise statement is always the same: the obstruction to conformal currents is the class of the virial current in the quotient of local virial currents by conserved currents and by virial currents generated by allowed improvements.

Remark 35.7 (Scale invariance and conformal invariance: status)*.* The algebraic criterion above reduces the implication from scale invariance to conformal invariance to a concrete local-operator statement: one must prove that the virial obstruction class vanishes. The following are the status boundaries used in this monograph.

(a) In $d = 2$, the Zamolodchikov–Polchinski implication admits theorem formulations only after a definite analytic framework is specified. In the common formulation one assumes a unitary local relativistic QFT with a normalizable vacuum, a local conserved stress tensor, a conserved scale current, reflection positivity or Hilbert-space positivity, and spectral/discreteness conditions strong enough for the stress-tensor two-point functions and the c-function argument to be meaningful. Under such hypotheses one proves $\mathrm{obs}^{\mathrm{scal}}_{\mathrm{vir}}(V) = 0$, or the corresponding vanishing in the allowed improvement quotient, so Remark 35.5 applies. This is a theorem about the formulated implication under those hypotheses, not a statement about every object called a two-dimensional scale-invariant theory in the literature.

(b) In $d = 4$, the implication is not used here as an unconditional theorem. There are important perturbative checks and nonperturbative arguments based on four-dimensional anomaly monotonicity, dilaton scattering, and stress-tensor correlator constraints, but the conclusion is the same local cohomological statement: the virial class must vanish in the quotient determined by the allowed improvements. The necessary spectral, regularity, and contact-term assumptions must be stated separately. Some claims in the physics literature are better understood as conditional arguments or conjectural criteria rather than fully formulated mathematical theorems.

(c) In $d = 3$ and $d > 4$, no general theorem of the form "scale invariance implies conformal invariance" is assumed. Whenever later chapters use conformal symmetry in these dimensions, the existence of a traceless improved stress tensor, the vanishing of the virial obstruction class, or equivalent conformal Ward identities, must be part of the stated data.

[1]

Figure 35.5 displays the quotient as the actual improvement test used in the proof. A scalar improvement L changes the stress tensor and simultaneously changes the virial representative by $(d-1)\partial_\mu L$. Conserved currents do not change the trace. The class of V_μ after these two identifications is therefore precisely the remaining obstruction to a traceless representative.

[1] For theorem formulations in two dimensions and their relation to Zamolodchikov's c-function argument, see A. B. Zamolodchikov, "Irreversibility of the flux of the renormalization group in a 2D field theory," *JETP Lett.* **43** (1986) 730–732, and J. Polchinski, "Scale and conformal invariance in quantum field theory," *Nucl. Phys. B* **303** (1988) 226–236. For four-dimensional conditional arguments and limitations, see M. A. Luty, J. Polchinski, and R. Rattazzi, "The a-theorem and the asymptotics of 4D quantum field theory," *JHEP* **01** (2013) 152, arXiv:1204.5221, and A. Dymarsky, Z. Komargodski, A. Schwimmer, and S. Theisen, "On scale and conformal invariance in four dimensions," *JHEP* **10** (2015) 171, arXiv:1309.2921. See also D. Dorigoni and S. Rychkov, "Scale invariance + unitarity ⇒ conformal invariance?," arXiv:0910.1087, for perturbative checks and a discussion of why the two-dimensional proof does not directly extend to higher dimensions.

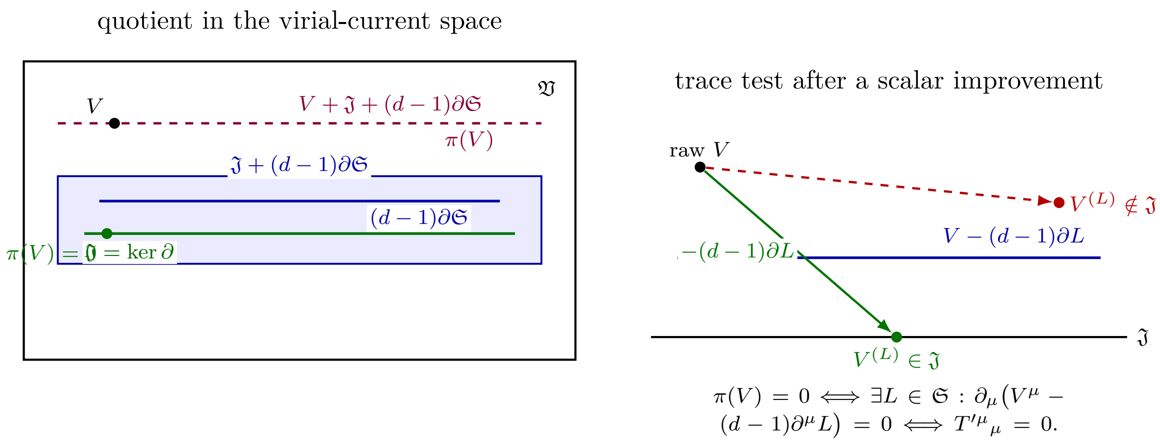


Figure 35.5: The virial obstruction as a concrete quotient test. Scalar improvements generate the subspace $(d-1)\partial\mathfrak{S} \subset \mathfrak{V}$, while conserved currents $\mathfrak{J}$ have zero divergence and hence do not affect the trace. The class $\pi(V)$ vanishes exactly when the improved virial representative is conserved, so that the improved trace is zero.

Before applying these criteria to scalar critical phenomena, we separate two renormalization constructions that will both be used below. The location of a nonzero fixed point and the anomalous dimensions of its scaling operators are computed from 1PI or minimal-subtraction coordinates. The finite-cutoff statement that irrelevant microscopic coordinates are suppressed after the relevant coordinates have been tuned is Wilsonian. The next chapter develops the Wilsonian construction and its cutoff-removal estimate; the subsequent chapters then compute the Wilson-Fisher fixed point in $4-\epsilon$ dimensions and apply the resulting data to Ising universality.

Chapter 36

Wilsonian Effective Actions and Exact Cutoff Flow

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

36.1 Smooth Cutoffs and Effective Actions

We work in Euclidean signature on $\mathbb{R}^D$, first for one real scalar field. Fourier transforms are normalized by

$$\phi(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \mathrm{e}^{\mathrm{i}k\cdot x}\widetilde{\phi}(k), \qquad \widetilde{\phi}(k) = \int \mathrm{d}^D x\, \mathrm{e}^{-\mathrm{i}k\cdot x}\phi(x).$$

For compactness, write

$$\int_k F(k) := \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, F(k), \qquad \frac{\delta\widetilde{\phi}(p)}{\delta\widetilde{\phi}(k)} = (2\pi)^D \delta^{(D)}(p-k).$$

Gauge theories require extra finite-cutoff data. A Wilsonian construction for ordinary scalar or matter variables may be written as a covariance split and a Gaussian pushforward. For gauge variables, the corresponding construction is made inside one of two gauge-compatible frameworks: a finite lattice formulation with group-valued link variables and gauge-invariant blocking maps, or a finite-cutoff BV formulation whose action, half-density, bracket, Laplacian, integration cycle, and coarse-graining map satisfy the quantum master equation. The scalar formulas in this chapter supply the ordinary field model. Their gauge-theory use is routed through the lattice chapter and the BV master-formalism chapter, where the gauge quotient, ghosts, antifields, and Ward identities are part of the finite-cutoff data.

Let $\chi : [0,\infty) \to [0,1]$ be a smooth nonincreasing cutoff profile with $\chi(0) = 1$ and rapid decay as $t \to \infty$. The generic Wilson-Polchinski algebra below only needs the covariance difference $C_\Lambda - C_{\Lambda'}$ to be a positive covariance. Exact preservation of a nonzero low-source sector by a source-independent Wilsonian action requires an additional plateau condition, stated later in (36.38); without such a plateau one must keep source-dependent Wilsonian vertices or prove an explicit leakage estimate. At cutoff scale $\Lambda > 0$, define

$$\chi_\Lambda(k) := \chi(k^2/\Lambda^2), \qquad C_\Lambda(k) := \frac{\chi_\Lambda(k)}{k^2 + m^2}.$$

Here $m^2 \geq 0$ is the mass parameter in the quadratic kernel. The Wilsonian action at scale Λ is written

$$S_\Lambda[\phi] = S_{\mathrm{kin},\Lambda}[\phi] + L_\Lambda[\phi], \tag{36.1}$$

where

$$S_{\mathrm{kin},\Lambda}[\phi] = \frac{1}{2}\int_k \widetilde{\phi}(-k)\, \frac{k^2 + m^2}{\chi_\Lambda(k)}\, \widetilde{\phi}(k). \tag{36.2}$$

Perturbation theory around this Gaussian measure uses $C_\Lambda(k)$ as propagator. The cutoff is therefore part of the definition of the covariance, so every propagator in the Gaussian expansion carries the same regulated kernel.

Figure 36.1 fixes the cutoff convention used in this chapter: the smooth profile $\chi_\Lambda(k)$ is built into the Gaussian covariance, hence every perturbative propagator carries the same ultraviolet profile.

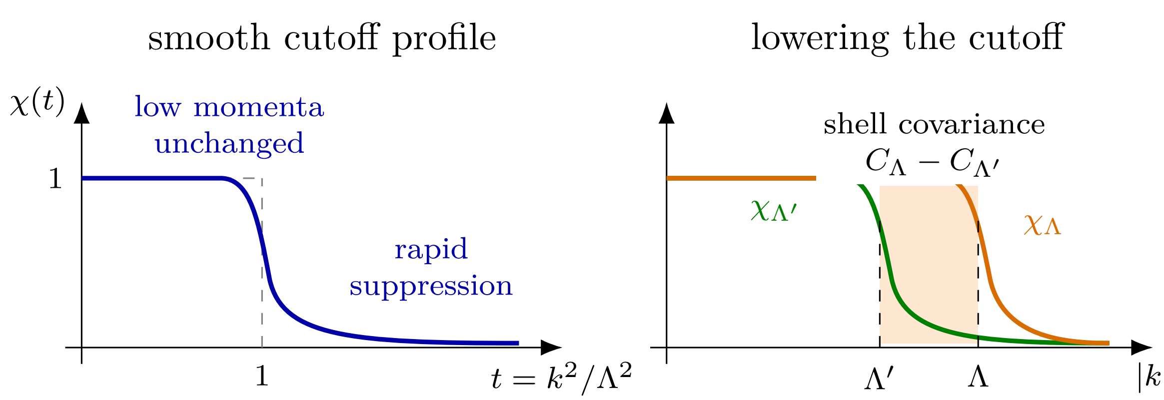


Figure 36.1: The plateau cutoff profile drawn here equals one for sufficiently small momentum and suppresses modes above Λ. When the cutoff is lowered to Λ', the difference of covariances is concentrated near the momentum shell between the two scales. Non-plateau smooth profiles do not give an exact low-source identity without adding source-dependent Wilsonian vertices.

The interaction functional L_Λ is an arbitrary translation-invariant functional compatible with the imposed symmetries. In a finite spatial volume with a finite momentum cutoff, L_Λ is an ordinary smooth or polynomial function of finitely many Fourier modes. Its vertex expansion is

$$L_\Lambda[\phi] = \sum_{n\geq 0} \frac{1}{n!} \int \prod_{i=1}^{n-1} \frac{\mathrm{d}^D k_i}{(2\pi)^D} L_\Lambda^{(n)}(k_1, \dots, k_{n-1}) \\ \times \widetilde{\phi}(k_1) \cdots \widetilde{\phi}(k_{n-1}) \widetilde{\phi}(-k_1 - \cdots - k_{n-1}). \tag{36.3}$$

The coefficient functions $L_\Lambda^{(n)}$ are symmetric in their n external momenta after the momentum-conserving argument is restored. Locality of the Wilsonian action means that, for momenta small compared with Λ, these coefficient functions admit an expansion in polynomials of the external momenta, with the nonpolynomial dependence controlled at the cutoff scale. The derivative expansion supplies an infinite set of local coordinates organized by powers of momentum over Λ.

The source-dependent generating functional at cutoff Λ is

$$Z_\Lambda[J] = \int \mathrm{d}\mu_{C_\Lambda}(\phi) \exp\left[-L_\Lambda[\phi] - \int \mathrm{d}^D x\, J(x)\phi(x)\right]. \tag{36.4}$$

The Gaussian measure $\mathrm{d}\mu_{C_\Lambda}$ carries $S_{\mathrm{kin},\Lambda}$ as in (11.1); with the reference density convention the same expression is

$$Z_{0,\Lambda}^{-1} \int D_\Lambda^{\mathrm{ref}}\phi \exp\left[-S_{\mathrm{kin},\Lambda}[\phi] - L_\Lambda[\phi] - \int \mathrm{d}^D x\, J(x)\phi(x)\right].$$

The normalization $Z_{0,\Lambda}^{-1}$ is immaterial for connected correlators with at least one source derivative. The source-independent Wilsonian condition for lowering Λ to $\Lambda' < \Lambda$ is the equality

$$Z_{\Lambda'}[J] = Z_\Lambda[J] \tag{36.5}$$

for sources J that annihilate the shell covariance, $\widehat{C}_{\Lambda,\Lambda'}J = 0$. For an orthogonal finite-mode split this means J belongs to the dual of the retained low-mode space. For a smooth profile it is an exact nonzero condition only when the profile has a plateau and the source support lies strictly inside that plateau. If the profile is non-plateau, such as a Gaussian profile, the equality in (36.5) is not an exact statement about the source-independent $L_{\Lambda'}$; one must either add source-dependent vertices to the pushforward or state and prove a profile-dependent leakage estimate. Under the annihilator condition, the scale label on S_Λ is a label on a description of the same selected source functional.

36.2 Lowering the Cutoff

The exact Wilsonian step is most transparent with a finite regulator. Let E_Λ be the finite-dimensional real vector space of Fourier modes retained by the ultraviolet regulator, and let $\mathrm{d}\mu_{C_\Lambda}$ be the centered Gaussian measure with covariance C_Λ on this space. In the notation of Definition 11.3, this chapter has absorbed the quadratic kinetic term into $\mathrm{d}\mu_{C_\Lambda}$; equivalently,

$$\mathrm{d}\mu_{C_\Lambda}(\phi) = Z_{0,\Lambda}^{-1} \exp\{-S_{\mathrm{kin},\Lambda}[\phi]\}\, D_\Lambda^{\mathrm{ref}}\phi.$$

The interaction L_Λ is integrated against this Gaussian reference measure. If

$$C_\Lambda = C_{\Lambda'} + \widehat{C}_{\Lambda,\Lambda'}$$

with both summands positive covariances on the regulated vector space, then the Gaussian convolution identity says

$$\int_{E_\Lambda} F(\phi)\,\mathrm{d}\mu_{C_\Lambda}(\phi) = \int_{E_{\Lambda'}} \int_{\widehat{E}_{\Lambda,\Lambda'}} F(\phi' + \phi_{\mathrm{sh}})\,\mathrm{d}\mu_{\widehat{C}_{\Lambda,\Lambda'}}(\phi_{\mathrm{sh}})\,\mathrm{d}\mu_{C_{\Lambda'}}(\phi') \tag{36.6}$$

for every integrable regulated function F. Equation (36.6) is a theorem about Gaussian measures on finite-dimensional vector spaces. Its proof and its continuum status are logically separate.

Finite-regulator Gaussian pushforward. Assume $E_\Lambda = E_{\Lambda'} \oplus \widehat{E}_{\Lambda,\Lambda'}$ and that $C_{\Lambda'}$ and $\widehat{C}_{\Lambda,\Lambda'}$ are positive covariance operators on the two summands. Let $C_\Lambda = C_{\Lambda'} \oplus \widehat{C}_{\Lambda,\Lambda'}$. If F is integrable with respect to the centered Gaussian measure μ_{C_Λ}, then (36.6) holds.

Let X' and X_{sh} be independent centered Gaussian random variables with values in $E_{\Lambda'}$ and $\widehat{E}_{\Lambda,\Lambda'}$ and covariances $C_{\Lambda'}$ and $\widehat{C}_{\Lambda,\Lambda'}$. For every linear functional $\ell \in E_\Lambda^*$,

$$\mathbb{E}\,\mathrm{e}^{\mathrm{i}\ell(X'+X_{\mathrm{sh}})} = \exp\Big\{-\frac{1}{2}\ell(C_{\Lambda'}\ell)\Big\}\exp\Big\{-\frac{1}{2}\ell(\widehat{C}_{\Lambda,\Lambda'}\ell)\Big\} = \exp\Big\{-\frac{1}{2}\ell(C_\Lambda\ell)\Big\}.$$

Thus $X' + X_{\mathrm{sh}}$ has the centered Gaussian law with covariance C_Λ. Equality of expectations of every integrable F gives (36.6).

Hypothesis 36.1 (Continuum interpretation of Wilsonian pushforward)**.** Choose a sequence of finite regulators $N \to \infty$, source spaces $\mathcal{S}_<$ for momenta below a fixed physical scale, and a topology $\mathcal{T}$ on Schwinger functionals or on local Wilsonian kernels. A nonperturbative continuum interpretation of the Wilsonian pushforward requires the following additional assertions.

First, the interacting finite-regulator source functionals

$$Z_{\Lambda,N}[J] = \int_{E_{\Lambda,N}} \exp\{-L_{\Lambda,N}[\phi] - \langle J,\phi\rangle\}\,\mathrm{d}\mu_{C_{\Lambda,N}}(\phi)$$

must converge in $\mathcal{T}$, after the chosen renormalization conditions are imposed, for $J \in \mathcal{S}_<$. This is an existence statement for the interacting integral, logically independent of the Gaussian convolution identity.

Second, for each pair $\Lambda' < \Lambda$, the finite-regulator pushforwards defined by (36.6) must converge in the same topology to a limiting map that preserves the limiting low-source functional. If the infinitesimal Wilson-Polchinski equation is used, the finite-regulator vector fields and their solutions must also converge in the chosen function-space topology.

The finite-regulator proposition is the theorem used in the algebra below. Hypothesis 36.1 is a separate regulator-removal problem. In a model where this hypothesis remains unproved, the continuum Wilsonian flow is a controlled formal or perturbative construction, while a nonperturbative use requires the convergence asserted in Hypothesis 36.1.

The algebraic step behind Wilsonian RG is the covariance identity

$$C_\Lambda(k) = C_{\Lambda'}(k) + \widehat{C}_{\Lambda,\Lambda'}(k), \qquad \widehat{C}_{\Lambda,\Lambda'}(k) := \frac{\chi_\Lambda(k) - \chi_{\Lambda'}(k)}{k^2+m^2}. \tag{36.7}$$

The first covariance belongs to a low field ϕ'. The second covariance belongs to an independent shell field $\phi_{\rm sh}$, with kinetic term

$$\widehat{S}_{\rm kin}[\phi_{\rm sh}] = \frac{1}{2}\int_k \widetilde{\phi}_{\rm sh}(-k)\,\frac{k^2+m^2}{\chi_\Lambda(k)-\chi_{\Lambda'}(k)}\,\widetilde{\phi}_{\rm sh}(k). \tag{36.8}$$

The shell field is denoted $\phi_{\rm sh}$. A tilde denotes Fourier transformation; thus $\widetilde{\phi}_{\rm sh}(k)$ is the shell field's Fourier mode, while $\widetilde{\phi}(k)$ elsewhere is the Fourier mode of the active field variable ϕ. The hat in $\widehat{C}_{\Lambda,\Lambda'}$ labels the covariance difference; it is not a field notation, and the shell field will not be denoted by a hatted ϕ.

Figure 36.2 records the exact propagator split used in the Wilsonian pushforward: active low modes retain $C_{\Lambda'}$, and the shell field is integrated with covariance $\widehat{C}_{\Lambda,\Lambda'}$.

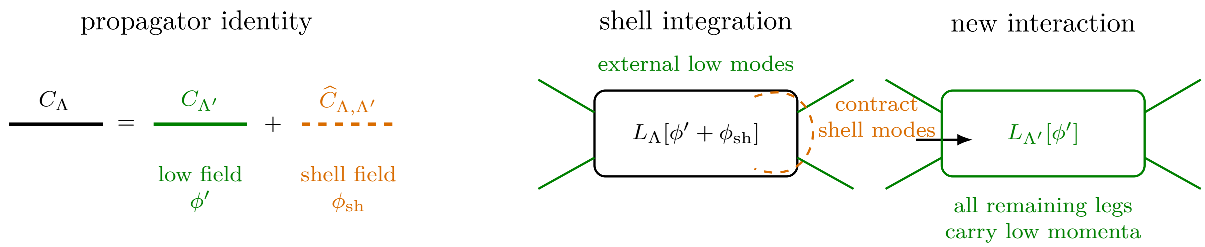


Figure 36.2: The old Gaussian covariance can be realized as the covariance of two independent fields. Integrating the shell field contracts the dashed propagators and produces a new interaction for the low field.

Before the support of the source is used, the covariance split rewrites the cutoff-Λ functional as

$$\begin{aligned} Z_\Lambda[J] = \int [D\phi']\,\exp\left[-S_{{\rm kin},\Lambda'}[\phi']\right] \int [D\phi_{\rm sh}]\,\exp\left[-\widehat{S}_{\rm kin}[\phi_{\rm sh}] - L_\Lambda[\phi'+\phi_{\rm sh}]\right] \\ \times \exp\left[-\int {\rm d}^D x\, J(x)\big(\phi'(x)+\phi_{\rm sh}(x)\big)\right]. \end{aligned} \tag{36.9}$$

The condition on J is now used in its precise regulated meaning: the source belongs to the annihilator of the shell covariance, so that the Gaussian random variable $\langle J, \phi_{\rm sh}\rangle$ has zero variance. Equivalently, for an orthogonal mode split the source belongs to the dual of the low subspace. For a smooth momentum cutoff this exact condition is available for a nonzero open low-momentum support only when the profile has a plateau; a non-plateau profile requires the source-dependent construction introduced in (36.36). Under the exact annihilator condition, the source term in (36.9) couples only to ϕ'. The shell field may then be integrated out by the defining identity

$$\exp\{-L_{\Lambda'}[\phi']\} = \int [D\phi_{\rm sh}]\,\exp\left[-\widehat{S}_{\rm kin}[\phi_{\rm sh}] - L_\Lambda[\phi'+\phi_{\rm sh}]\right], \tag{36.10}$$

up to a ϕ'-independent normalization. Combining (36.8) and (36.10) with the Gaussian measure for ϕ' gives (36.5). The definition is exact for each finite regulator by the Gaussian pushforward calculation above. Perturbation theory expands the logarithm of the shell expectation in connected shell contractions. The continuum claim is the additional Hypothesis 36.1: after the renormalization conditions are imposed, both the interacting source functional and the Wilsonian pushforward or its infinitesimal flow must have the stated regulator-removal limit.

36.3 The Wilson-Polchinski Equation

Let Λ' approach Λ. The shell covariance becomes infinitesimal, and the definition of $L_{\Lambda'}$ becomes a differential equation. With $C_\Lambda(k) = \chi_\Lambda(k)/(k^2+m^2)$ and with the functional-derivative convention already stated, the interaction functional satisfies

$$\Lambda\partial_\Lambda L_\Lambda[\phi] = \frac{1}{2}\int_k \Lambda\partial_\Lambda C_\Lambda(k)\left[\frac{\delta L_\Lambda}{\delta\widetilde{\phi}(k)}\frac{\delta L_\Lambda}{\delta\widetilde{\phi}(-k)} - \frac{\delta^2 L_\Lambda}{\delta\widetilde{\phi}(k)\delta\widetilde{\phi}(-k)}\right]. \tag{36.11}$$

In (36.11), $\widetilde{\phi}(k)$ denotes the Fourier mode of the remaining field. The notation matches the functional derivative convention; after the infinitesimal shell has been integrated, there is again only one field argument of L_Λ.

The analytic meaning of (36.11) is fixed by the following status convention.

Definition 36.2 (Function spaces for the Wilson-Polchinski equation)**.** The Wilson-Polchinski equation will be used in three nested settings.

Finite regulator. Put the theory in finite volume and impose a finite ultraviolet mode cutoff Λ_{UV}. All fields then belong to a fixed finite-dimensional real vector space $E_{\Lambda_{\mathrm{UV}}}$. In this setting L_Λ is a C^2 function on $E_{\Lambda_{\mathrm{UV}}}$, the functional derivatives in (36.11) are ordinary derivatives, and $\Lambda\partial_\Lambda L_\Lambda$ is the derivative of a C^1 curve in the topology of uniform convergence on bounded subsets, with two field derivatives continuous in the same sense. This is the status of the Gaussian convolution derivation.

Formal local expansion. Write L_Λ by the vertex expansion (36.3). For each n, each number q of momentum derivatives, and each compact set K in the momentum-conserving external-momentum domain, define the seminorm

$$p_{n,q,K}(L_\Lambda) = \max_{|\alpha|\le q}\sup_{(k_1,\ldots,k_{n-1})\in K}\left|\partial^\alpha L_\Lambda^{(n)}(k_1,\ldots,k_{n-1})\right|.$$

The Fréchet topology generated by these seminorms is the coefficientwise smooth topology on vertex kernels. A formal perturbative solution is a curve whose vertex kernels satisfy the coefficient equations obtained from (36.11) order by order in the chosen coupling, field-number, loop, and derivative expansion.

Banach RG chart. For theorem-level estimates one must choose a norm. One convenient local model fixes integers $m \ge 0$, a weight $a > 0$, and a constant $c > 1$, and takes $\mathcal{B}_\Lambda^{m,a}$ to be the space of symmetric momentum-conserving kernels for which

$$\|L_\Lambda\|_{\Lambda,m,a} = \sum_{n\ge 0} a^n \max_{|\alpha|\le m}\sup_{|k_i|\le c\Lambda}\Lambda^{d_n+|\alpha|}\left|\partial^\alpha L_\Lambda^{(n)}(k_1,\ldots,k_{n-1})\right|$$

is finite. The constants d_n encode the chosen engineering dimensions of the n-point vertices, and may be changed by a finite reparametrization of coordinates. A Banach-space solution on a scale interval I is a C^1 curve $\Lambda \mapsto L_\Lambda \in \mathcal{B}_\Lambda^{m,a}$ after identifying nearby $\mathcal{B}_\Lambda^{m,a}$ by dimensionless momenta $p_i = k_i/\Lambda$, with (36.11) holding in the Banach norm. Existence, uniqueness, and cutoff-uniform bounds are analytic assertions about this chosen normed space.

Since

$$\Lambda\partial_\Lambda C_\Lambda(k) = \frac{\Lambda\partial_\Lambda\chi(k^2/\Lambda^2)}{k^2+m^2},$$

the integration kernel is supported where the cutoff profile changes, namely near $|k| \simeq \Lambda$ for a rapidly decreasing smooth cutoff. The two terms in (36.11) have a direct graphical meaning: the product of first derivatives joins two vertices by one shell propagator, whereas the second derivative contracts two legs of a single vertex.

Figure 36.3 is the graphical form of (36.11): one term joins two Wilsonian vertices by a shell line, and the other closes two legs on one vertex.

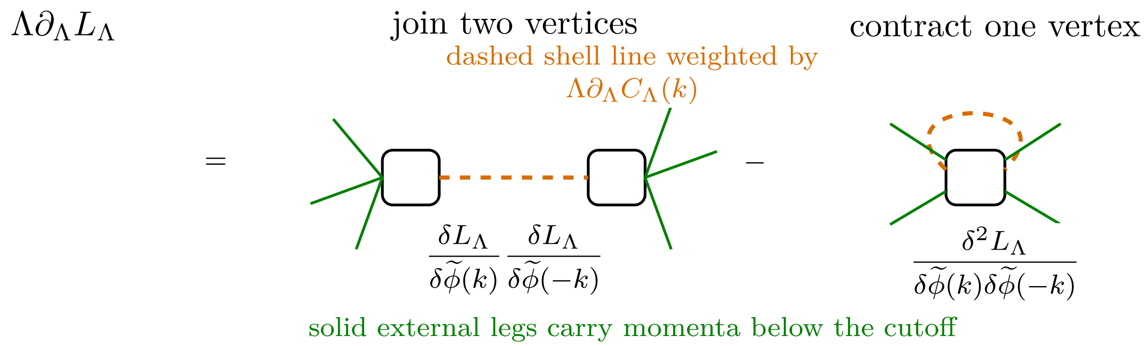


Figure 36.3: The exact cutoff flow has two local operations. The first term connects two interaction vertices with an infinitesimal shell propagator. The second term contracts two legs belonging to the same interaction vertex.

In the finite-regulator setting, equation (36.11) is a closed ordinary differential equation on L_Λ. In the formal local expansion it becomes an infinite coupled system of equations for the coefficient kernels and their derivative expansion. In a Banach RG chart it becomes a vector-field equation only after one verifies that the right-hand side maps the chosen open subset of $\mathcal{B}_\Lambda^{m,a}$ into the appropriate tangent space and satisfies the estimates needed for the claimed solution class. The use of a smooth cutoff is important: it keeps the infinitesimal shell operation compatible with a derivative expansion at external momenta small compared with Λ.

The sign in (36.11) is fixed by the direction of the cutoff derivative. For $\Lambda' = \Lambda - \delta\Lambda$, the shell covariance is

$$\delta C(k) = C_\Lambda(k) - C_{\Lambda'}(k) = \delta\Lambda\,\partial_\Lambda C_\Lambda(k) + O(\delta\Lambda^2).$$

The Gaussian convolution formula gives, to first order in $\delta\Lambda$,

$$\mathrm{e}^{-L_{\Lambda-\delta\Lambda}[\phi]} = \left[1 + \frac{1}{2}\int_k \delta\Lambda\,\partial_\Lambda C_\Lambda(k)\frac{\delta^2}{\delta\widetilde{\phi}(k)\delta\widetilde{\phi}(-k)}\right]\mathrm{e}^{-L_\Lambda[\phi]}.$$

Since

$$\frac{\delta^2 \mathrm{e}^{-L_\Lambda}}{\delta\widetilde{\phi}(k)\delta\widetilde{\phi}(-k)} = \left[\frac{\delta L_\Lambda}{\delta\widetilde{\phi}(k)}\frac{\delta L_\Lambda}{\delta\widetilde{\phi}(-k)} - \frac{\delta^2 L_\Lambda}{\delta\widetilde{\phi}(k)\delta\widetilde{\phi}(-k)}\right]\mathrm{e}^{-L_\Lambda},$$

and

$$L_{\Lambda-\delta\Lambda} = L_\Lambda - \delta\Lambda\,\partial_\Lambda L_\Lambda + O(\delta\Lambda^2),$$

the displayed Wilson-Polchinski equation follows. This derivation is an identity for the regulated finite-dimensional Gaussian integral; the continuum notation suppresses the regulator that makes the functional derivatives ordinary derivatives on a finite vector space. The ϕ-independent normalization omitted in (36.10) is the vacuum-energy coordinate. If it is retained, it replaces $L_\Lambda[\phi]$ by $L_\Lambda[\phi] + c_\Lambda$ and contributes only $\Lambda\partial_\Lambda c_\Lambda$ to the left-hand side of the flow. The functional derivatives in the right-hand side annihilate c_Λ. The displayed equation is therefore the Wilsonian interaction flow modulo this field-independent normalization, whose value may be fixed separately without changing any non-vacuum vertex or connected normalized correlator.

36.4 Effective Average Action and the Wetterich Equation

The Wilsonian interaction L_Λ is obtained by integrating a momentum shell and leaving an action for the remaining field variable. There is a second exact cutoff construction, logically different but derived from the same regulated integral, in which all modes are integrated at once while an infrared quadratic term suppresses low-momentum fluctuations. The resulting object is a cutoff-dependent 1PI functional. This section uses the name *effective average action* for that modified Legendre transform.

Because the earlier source convention in (36.4) used $-\langle J, \phi\rangle$, introduce in this section the source coordinate

$$\mathcal{J} := -J, \qquad \langle \mathcal{J}, \phi\rangle := \int \mathrm{d}^D x\, \mathcal{J}(x)\phi(x) = \int_p \widetilde{\mathcal{J}}(-p)\widetilde{\phi}(p),$$

where $\int_p$ has the same Fourier normalization as $\int_k$. This is a change of source coordinate only; no physics is attached to the sign.

Definition 36.3 (Infrared regulator and effective average action)**.** Work first at finite volume and with a finite ultraviolet mode cutoff, so the real field space is a finite-dimensional vector space E_N. Let $S_N[\phi]$ be the full regulated Euclidean action on E_N, including the quadratic part and all interaction terms allowed by the chosen regulator, and let $D_N\phi$ denote the chosen finite-dimensional reference density. An infrared regulator at scale $\kappa > 0$ is a C^1 family of symmetric nonnegative bilinear forms $R_{\kappa,N} : E_N \to E_N^*$. In a translation-invariant scalar regulator one writes

$$\Delta S_{\kappa,N}[\phi] = \frac{1}{2}\int_p \widetilde{\phi}(-p)R_\kappa(p)\widetilde{\phi}(p), \qquad R_\kappa(p) = Z_\kappa \kappa^2 r(p^2/\kappa^2), \tag{36.12}$$

with $r(s) > 0$ for s near 0 and $r(s) \to 0$ rapidly as $s \to \infty$. The displayed profile is a standard scalar model of the definition; the finite-regulator derivation below only needs a symmetric C^1 nonnegative family $R_{\kappa,N}$.

Define

$$\begin{aligned} Z_{\kappa,N}[\mathcal{J}] &:= \int_{E_N} D_N\phi\, \exp\{-S_N[\phi] - \Delta S_{\kappa,N}[\phi] + \langle \mathcal{J}, \phi\rangle\}, \\ W_{\kappa,N}[\mathcal{J}] &:= \log Z_{\kappa,N}[\mathcal{J}], \qquad \varphi_{\mathcal{J}} := \frac{\delta W_{\kappa,N}}{\delta \mathcal{J}}. \end{aligned} \tag{36.13}$$

On a source neighborhood where $\mathcal{J} \mapsto \varphi_{\mathcal{J}}$ is locally invertible, let $\mathcal{J}_{\kappa,N}[\varphi]$ denote the inverse map. The effective average action is the modified Legendre transform

$$\Gamma_{\kappa,N}[\varphi] = \langle \mathcal{J}_{\kappa,N}[\varphi], \varphi\rangle - W_{\kappa,N}[\mathcal{J}_{\kappa,N}[\varphi]] - \Delta S_{\kappa,N}[\varphi]. \tag{36.14}$$

Equivalently, $\Gamma_{\kappa,N} + \Delta S_{\kappa,N}$ is the ordinary Legendre transform of $W_{\kappa,N}$. If the integral is defined by a positive measure, $W_{\kappa,N}$ is convex in $\mathcal{J}$, and $\Gamma_{\kappa,N} + \Delta S_{\kappa,N}$ is convex in φ. The subtracted functional $\Gamma_{\kappa,N}$ need not itself be convex at positive κ.

The cutoff κ in this section is an infrared scale. It should not be identified with the Wilsonian ultraviolet cutoff Λ unless a separate matching map has been specified. At large κ, the extra quadratic term suppresses fluctuations with $p^2 \lesssim \kappa^2$. At $\kappa = 0$, if $R_\kappa \to 0$ and the limit exists, the modified Legendre transform becomes the ordinary 1PI effective action of the regulated theory.

Legendre Hessian identity. In the finite-regulator setting of Definition 36.3, assume that the source-field map is locally invertible at $\mathcal{J}$. Then

$$\frac{\delta\Gamma_{\kappa,N}}{\delta\varphi} = \mathcal{J}_{\kappa,N}[\varphi] - R_{\kappa,N}\varphi, \qquad \Gamma^{(2)}_{\kappa,N}[\varphi] + R_{\kappa,N} = \left(W^{(2)}_{\kappa,N}[\mathcal{J}]\right)^{-1}, \tag{36.15}$$

where $W^{(2)}_{\kappa,N}$ is the connected two-point covariance in the presence of the source and the inverse is the inverse linear map $E_N^* \to E_N$ on the chosen source neighborhood.

Differentiate

$$\Gamma_{\kappa,N}[\varphi] + \Delta S_{\kappa,N}[\varphi] = \langle \mathcal{J}, \varphi\rangle - W_{\kappa,N}[\mathcal{J}], \qquad \varphi = \frac{\delta W_{\kappa,N}}{\delta \mathcal{J}}.$$

The terms containing $\delta\mathcal{J}$ cancel because $\delta W_{\kappa,N} = \langle \varphi, \delta\mathcal{J}\rangle$. Hence $\delta(\Gamma_{\kappa,N} + \Delta S_{\kappa,N})/\delta\varphi = \mathcal{J}$, which gives the first identity after subtracting $\delta\Delta S_{\kappa,N}/\delta\varphi = R_{\kappa,N}\varphi$. A second variation gives

$$\Gamma^{(2)}_{\kappa,N} + R_{\kappa,N} = \frac{\delta\mathcal{J}}{\delta\varphi}.$$

On the other hand $\delta\varphi/\delta\mathcal{J} = W^{(2)}_{\kappa,N}$. Local invertibility of the source-field map says precisely that these two derivatives are inverse finite-dimensional matrices.

Finite-regulator Wetterich equation. Let $t = \log\kappa$. Suppose that $R_{\kappa,N}$ is C^1 in t, the regulated integral in (36.13) is differentiable in t, and the source-field map is locally invertible along the curve under consideration. Then, at fixed average field φ,

$$\partial_t \Gamma_{\kappa,N}[\varphi] = \frac{1}{2}\, \mathrm{Tr}_{E_N}\left[\left(\Gamma^{(2)}_{\kappa,N}[\varphi] + R_{\kappa,N}\right)^{-1} \partial_t R_{\kappa,N}\right]. \tag{36.16}$$

Differentiate the modified Legendre transform (36.14) with respect to t while holding φ fixed. As in the preceding lemma, the terms containing $\partial_t \mathcal{J}_{\kappa,N}[\varphi]$ cancel. Thus

$$\partial_t \Gamma_{\kappa,N}[\varphi] = -\partial_t W_{\kappa,N}[\mathcal{J}]\big|_{\mathcal{J}} - \partial_t \Delta S_{\kappa,N}[\varphi]. \tag{36.17}$$

The only explicit t-dependence in $Z_{\kappa,N}[\mathcal{J}]$ is the regulator, so

$$\partial_t W_{\kappa,N}[\mathcal{J}]\big|_{\mathcal{J}} = -\frac{1}{2}\,\langle \phi\,(\partial_t R_{\kappa,N})\,\phi\rangle^{\mathrm{full}}_{\mathcal{J},\kappa}.$$

Here the bracket denotes the full source-dependent expectation in the regulated integral. Decomposing the full second moment into its connected and mean parts gives

$$\langle \phi\,(\partial_t R_{\kappa,N})\,\phi\rangle^{\mathrm{full}}_{\mathcal{J},\kappa} = \langle \varphi, (\partial_t R_{\kappa,N})\varphi\rangle + \mathrm{Tr}_{E_N}\left[W^{(2)}_{\kappa,N}[\mathcal{J}]\,\partial_t R_{\kappa,N}\right].$$

Since

$$\partial_t \Delta S_{\kappa,N}[\varphi] = \frac{1}{2}\langle \varphi, (\partial_t R_{\kappa,N})\varphi\rangle,$$

the mean-field terms cancel in (36.17). The result is

$$\partial_t \Gamma_{\kappa,N}[\varphi] = \frac{1}{2}\,\mathrm{Tr}_{E_N}\left[W^{(2)}_{\kappa,N}[\mathcal{J}]\,\partial_t R_{\kappa,N}\right].$$

Substituting the Hessian identity (36.15) proves (36.16).

After suppressing the finite regulator from the notation, the formal continuum version is

$$\partial_t \Gamma_\kappa[\varphi] = \frac{1}{2}\,\mathrm{Tr}\left[\left(\Gamma^{(2)}_\kappa[\varphi] + R_\kappa\right)^{-1}\partial_t R_\kappa\right], \qquad t = \log\kappa. \tag{36.18}$$

The trace in (36.18) means the trace over spacetime, internal indices, and statistics, with a minus sign for fermionic variables when the derivation is repeated in the superalgebraic setting. For the scalar translation-invariant regulator (36.12), the factor $\partial_t R_\kappa(p)$ is concentrated at momenta $p^2 \simeq \kappa^2$ for a smooth profile. This makes the right-hand side formally both infrared and ultraviolet regulated. A theorem-level continuum use still requires more than this formal observation: one must specify the regulator-removal topology, prove convergence of $W_{\kappa,N}$ or $\Gamma_{\kappa,N}$, and prove that the finite-regulator traces converge uniformly on the field neighborhood being used.

Figure 36.4 compares the cutoff kernels rather than the names of the constructions. The Wilsonian covariance multiplies the propagator by a profile that keeps low modes and suppresses high modes. The effective-average-action regulator adds a masslike kernel to low modes and vanishes for high modes; the flow is localized by $\partial_t R_\kappa$. This is the concrete sense in which the two cutoff-dependent objects are different.

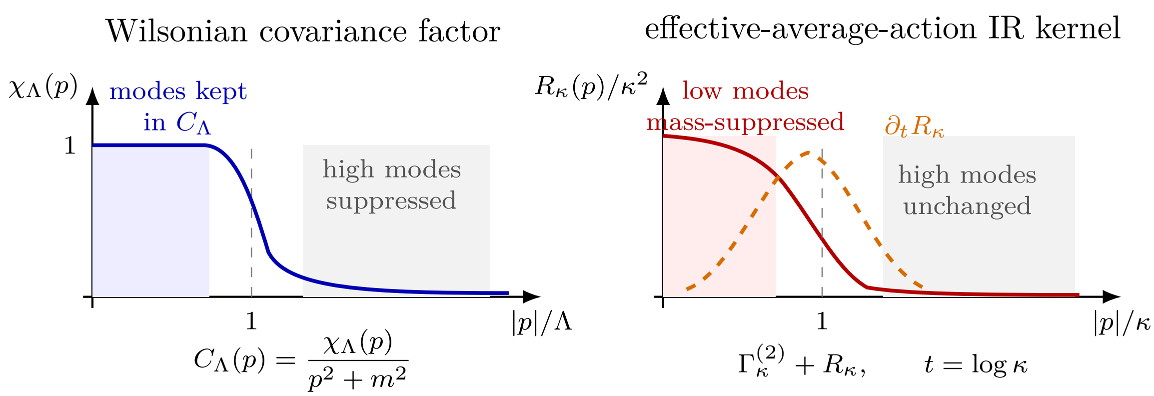


Figure 36.4: The Wilsonian and effective-average-action cutoffs act in opposite ways on momentum modes. The Wilsonian covariance keeps the low-mode propagator and removes high modes as Λ is lowered. The effective-average-action regulator adds a large quadratic penalty to modes with $|p| \lesssim \kappa$ and vanishes at high momentum; the flow kernel $\partial_t R_\kappa$ is concentrated near $|p| \simeq \kappa$. Matching the two descriptions is therefore a statement about compatible kernels and coordinates, not an identity of the cutoff-dependent functionals.

Example 36.4 (Constant-field projection and the local potential flow)**.** Assume a single scalar field and project the effective average action to the derivative expansion

$$\Gamma_\kappa[\varphi] = \int \mathrm{d}^D x\,\left\{\frac{1}{2} Z_\kappa\,\partial_\mu\varphi\,\partial_\mu\varphi + U_\kappa(\varphi)\right\} + \text{terms with more derivatives or more fields.} \tag{36.19}$$

For a constant field $\varphi(x) = v$, the Hessian in momentum space is

$$\Gamma_\kappa^{(2)}(p, q; v) = (2\pi)^D \delta^{(D)}(p + q)\big(Z_\kappa p^2 + U_\kappa''(v)\big)$$

inside this projection. Dividing (36.18) by the spacetime volume gives

$$\partial_t U_\kappa(v) = \frac{1}{2}\int_p \frac{\partial_t R_\kappa(p)}{Z_\kappa p^2 + R_\kappa(p) + U_\kappa''(v)} \tag{36.20}$$

before any expansion in powers of U_κ''. This is an exact identity only after projection to the ansatz (36.19); the omitted derivative and field operators feed back into the exact flow through $\Gamma_\kappa^{(2)}[\varphi]$.

For the illustrative limiting profile

$$R_\kappa(p) = Z_\kappa(\kappa^2 - p^2)\theta(\kappa^2 - p^2)$$

and with $\eta_\kappa := -\partial_t \log Z_\kappa$ set to zero, one obtains

$$\partial_t U_\kappa(v) = \frac{v_D\,\kappa^D}{1 + U_\kappa''(v)/(Z_\kappa \kappa^2)}, \qquad v_D := \frac{1}{D\,2^{D-1}\pi^{D/2}\Gamma(D/2)}. \tag{36.21}$$

The step-function regulator is not smooth. In a proof-oriented treatment it should be replaced by a smooth family approaching it, or else treated in a function space where the distributional boundary term is controlled. Formula (36.21) is therefore best read as an efficient projection formula, not as a substitute for the smooth-regulator finite-dimensional derivation.

Remark 36.5 (Relation to the Wilson-Polchinski equation). Both (36.11) and (36.18) are exact cutoff identities at finite regulator, but they differentiate different objects. The Wilson-Polchinski equation acts on a Wilsonian interaction before the remaining low fields have been integrated. The Wetterich equation acts on an infrared-regulated Legendre transform after the regulated integral has been performed. Therefore the vertices $L_\Lambda^{(n)}$ and $\Gamma_\kappa^{(n)}$ are not the same coordinate functions, and equality of beta functions is meaningful only after a finite matching map has been chosen.

For compatible cutoff kernels, the formal exact-RG literature constructs a Legendre-type relation between Polchinski and effective-average-action flows; one convenient account is in Rosten's review, following the earlier work of Morris and Wetterich.[1] In this monograph that relation is used only with its hypotheses displayed: a fixed finite regulator, compatible quadratic kernels, invertible Legendre map, and specified continuum or perturbative limit. It is not a license to identify arbitrary Wilsonian truncations with arbitrary effective-average-action truncations.

Remark 36.6 (Gauge theories). The scalar derivation uses an ordinary positive quadratic infrared regulator. For gauge theories, such a regulator generally breaks the gauge symmetry of the gauge-fixed functional integral unless it is embedded into a background field construction or a BV/BRST formulation with the corresponding modified Ward or Slavnov–Taylor identity. Thus a gauge-theory use of (36.18) requires extra data: the gauge fixing, ghost and antifield conventions, the regulator's transformation law, and the identity that replaces gauge invariance along the flow. These conditions are part of the definition of the regulated gauge-theory problem, not optional checks after a scalar-like flow has been written down.

[1] C. Wetterich, *Phys. Lett. B* **301** (1993) 90–94; T. R. Morris, *Int. J. Mod. Phys. A* **9** (1994) 2411–2450; O. J. Rosten, arXiv:1003.1366.

36.5 Linearized Flow and Scaling Coordinates

A fixed point of the Wilsonian flow is a scale-invariant interaction L_* after all quantities have been written in dimensionless coordinates. Let $t = \log \Lambda$. Near a fixed point write

$$L_\Lambda = L_* + \sum_I g_I(\Lambda)\, \mathcal{O}_I,$$

where $\mathcal{O}_I$ are local coordinates on theory space. The linearized flow has the form

$$\frac{\mathrm{d}g_I}{\mathrm{d}t} = \sum_J M_I{}^J g_J + O(g^2).$$

Choosing eigenvectors of M gives scaling coordinates u_α:

$$\frac{\mathrm{d}u_\alpha}{\mathrm{d}t} = -y_\alpha u_\alpha + O(u^2). \tag{36.22}$$

The number y_α is the relevance exponent.[2] In the convention $t = \log \Lambda$, decreasing the cutoff moves toward the infrared, so (36.22) implies

$$u_\alpha(\Lambda) = u_\alpha(\Lambda_0) \left(\frac{\Lambda}{\Lambda_0}\right)^{-y_\alpha} + O(u^2).$$

A coordinate with $y_\alpha > 0$ grows when Λ is lowered. It is a relevant coordinate and must be fixed by a renormalization condition, or tuned to zero when one asks for an infrared trajectory ending at the fixed point. A coordinate with $y_\alpha < 0$ is irrelevant: its value at a fixed physical scale is attracted to a function of the relevant and marginal coordinates along a renormalized trajectory, provided the nonlinear cutoff-removal estimates stated below hold.

A theorem-level codimension statement about a critical surface requires a smooth local model of theory space and a differentiable RG endpoint map. The eigenvalue calculation supplies the candidate relevant tuning directions; the local smoothness and codimension assertion used in this volume is the following.

Definition 36.7 (Banach RG chart)**.** Fix an integer $k \geq 1$. A C^k Banach RG chart near a fixed point is a triple $(\mathfrak{U}, \psi, \Phi_s)$ with the following data. The set $\mathfrak{U}$ is an open neighborhood of the fixed point in a C^k Banach manifold $\mathfrak{T}$ of regulated Wilsonian actions, modulo the redundant field redefinitions and symmetry identifications already fixed in the chart. The chart map

$$\psi : \mathfrak{U} \longrightarrow X$$

takes values in a Banach space X, and

$$X = X_{\mathrm{rel}} \oplus X_{\mathrm{rest}}$$

[2] This sign convention matches the 1PI thermal-coordinate convention used in the Wilson–Fisher fixed-point calculation. There the dimensionless temperature coordinate obeys $\mathrm{d}\tau/\mathrm{d}\log\mu = -y_t\tau + \cdots$, with $y_t = D - \Delta_{\phi^2} > 0$. Both displayed differential equations are written with respect to an increasing scale variable. The infrared-oriented variables $s_{\mathrm{W}} = \log(\Lambda_0/\Lambda)$ and $s_{1\mathrm{PI}} = \log(\mu_0/\mu)$ reverse the sign and give $\mathrm{d}u_\alpha/\mathrm{d}s_{\mathrm{W}} = y_\alpha u_\alpha + \cdots$ and $\mathrm{d}\tau/\mathrm{d}s_{1\mathrm{PI}} = y_t\tau + \cdots$. The Wilsonian cutoff flow and the 1PI subtraction-scale flow are still different vector fields until the finite matching map between Wilsonian and 1PI coordinates is specified.

is a topological direct sum with finite-dimensional relevant block $X_{\mathrm{rel}} \cong \mathbb{R}^r$. The maps Φ_s, $s \geq 0$, are the local RG maps from the ultraviolet scale to a lower scale, with $s = \log(\Lambda_0/\Lambda)$, and are C^k wherever the trajectory stays inside $\mathfrak{U}$. In coordinates write

$$\varphi_s = \psi \circ \Phi_s \circ \psi^{-1}, \qquad \pi_{\mathrm{rel}} : X_{\mathrm{rel}} \oplus X_{\mathrm{rest}} \to X_{\mathrm{rel}}$$

for the coordinate RG map and relevant projection.

Finite-reference critical surface from an implicit-function chart. Let $(\mathfrak{U}, \psi, \Phi_s)$ be a C^k Banach RG chart and fix a finite RG time $s_R = \log(\Lambda_0/\Lambda_R) > 0$. Let

$$E_{s_R}(z) = \pi_{\mathrm{rel}}\big(\varphi_{s_R}(z)\big), \qquad z = (u, w) \in X_{\mathrm{rel}} \oplus X_{\mathrm{rest}},$$

be the endpoint map that extracts the relevant coordinates at the reference scale Λ_R. Suppose E_{s_R} is defined and C^k on an open neighborhood of a point z_*, that $E_{s_R}(z_*) = 0$, and that the partial derivative

$$D_u E_{s_R}(z_*) : X_{\mathrm{rel}} \longrightarrow X_{\mathrm{rel}}$$

is an isomorphism. Then, after possibly shrinking the neighborhood of z_*, the set

$$\mathcal{C}_{s_R} = \{z : E_{s_R}(z) = 0\}$$

is a C^k embedded Banach submanifold of codimension r. In the chosen coordinates it is the graph

$$u = G_{s_R}(w)$$

for a C^k map G_{s_R} defined on an open subset of X_{rest}. This is the Banach-space implicit function theorem applied to the endpoint map E_{s_R}: the hypotheses make $E_{s_R}(u, w) = 0$ an equation in the finite-dimensional relevant block X_{rel}, with C^k dependence on the remaining Banach variable w, and the invertibility of $D_u E_{s_R}(z_*)$ supplies a unique C^k graph satisfying

$$E_{s_R}(u, w) = 0 \quad \Longleftrightarrow \quad u = G_{s_R}(w)$$

inside that neighborhood. A graph over the closed complemented subspace X_{rest} is an embedded C^k Banach submanifold, and its codimension is $\dim X_{\mathrm{rel}} = r$. The nontrivial QFT burden in any application is the construction of the chart, the differentiability of the RG endpoint map in the chosen norm, and the invertibility of the relevant block derivative; once those have been established, the graph statement is ordinary functional analysis.

The construction above is a finite-reference statement. The infinite-time critical surface consisting of trajectories that remain in the fixed-point neighborhood and converge to the fixed point as $\Lambda \to 0$ requires a stable- or center-stable-manifold theorem for the chosen Banach RG flow, with the corresponding spectral-gap, smoothness, and existence hypotheses. The monograph will not use the phrase “codimension r critical surface” as a theorem-level assertion unless the local chart, endpoint map, and invertibility or stable-manifold hypotheses have been stated.

The scaling dimension of the local operator conjugate to u_α is

$$\Delta_\alpha = D - y_\alpha$$

when the fixed point is conformal and the coupling multiplies $\int \mathrm{d}^D x \, \mathcal{O}_\alpha(x)$. Thus the Wilsonian eigenvalue problem and the conformal scaling-operator problem are the same local question seen from two sides: flow in theory space and dilation on local operators.

36.6 Redundant Directions and Field Coordinates

Theory space is coordinatized by actions, but different coordinates can describe the same correlation functions. A local change of field variable

$$\phi(x) \longmapsto \phi(x) + \epsilon F[\phi](x)$$

changes the action, to first order in ϵ, by

$$S[\phi] \longmapsto S[\phi] + \epsilon \int \mathrm{d}^D x \, \frac{\delta S}{\delta\phi(x)} F[\phi](x) - \epsilon \,\operatorname{Tr} \frac{\delta F}{\delta \phi} + O(\epsilon^2).$$

The trace term is the local Jacobian contribution in a regulated functional integral. Operators of the form

$$\int \mathrm{d}^D x \, \frac{\delta S}{\delta\phi(x)} F[\phi](x) \quad \text{together with the associated Jacobian}$$

generate redundant directions. They change the coordinate representative of the Wilsonian action and the representatives of local composite operators, but they leave separated-point correlation functions invariant after the corresponding field redefinition is applied.

The quotient by redundant directions is important in the linearized RG problem. An eigenvector of the flow that is tangent to a field redefinition is a change of local coordinates on theory space. Relevant, marginal, and irrelevant directions that classify continuum limits are therefore physical directions modulo these redundant tangent vectors, with symmetries and operator normalizations fixed as part of the coordinate choice.

36.7 Renormalized Trajectories and Universality

Let u_r denote the relevant coordinates and u_i the irrelevant coordinates near a fixed point. In a Banach RG chart where the necessary graph estimates have been proved, a renormalized trajectory is represented by a submanifold of theory space on which the irrelevant coordinates are functions of the relevant ones:

$$u_i = F_i(u_r).$$

The functions F_i are determined by the requirement that the trajectory can be extended toward the ultraviolet while remaining in the basin where the fixed-point expansion is valid. Perturbatively, they are found by solving the irrelevant beta-function equations order by order in the relevant and marginal coordinates.

Universality is the statement that many different microscopic actions flow to the same renormalized trajectory at a fixed physical scale. Suppose two bare actions at cutoff Λ_0 have the same relevant coordinates after tuning, but differ by an irrelevant coordinate $u_i(\Lambda_0)$. Linearized flow gives

$$u_i(\Lambda_R) = \left(\frac{\Lambda_R}{\Lambda_0}\right)^{-y_i} u_i(\Lambda_0) + \text{nonlinear terms}, \qquad y_i < 0,$$

for a fixed reference scale $\Lambda_R \ll \Lambda_0$. Thus the difference in any low-momentum coordinate or correlator projector whose linearization is bounded in this chart is

$$O\left(\|u_i(\Lambda_0)\| \left(\frac{\Lambda_R}{\Lambda_0}\right)^{-y_i}\right),$$

after the relevant coordinates and field normalizations have been matched. The suppression exponent is $-y_i = \Delta_i - D > 0$ for the first omitted irrelevant operator. Nonlinear versions of this statement require the Banach-norm estimates used in Controlled Approximation 36.16 and in the finite-coordinate continuum-limit theorem below.

This gives a precise meaning to effective field theory. A finite list of relevant and marginal coordinates must be measured or tuned. Irrelevant operators are still present, and their coefficients encode real short-distance physics, but their effect on low-energy observables is organized by powers of the ratio of scales and by the symmetries that determine which operators may appear.

36.8 A Quartic-Sextic Toy Flow

Consider the massless scalar theory in $D = 4$ and keep only two local coordinates in the interaction,

$$L_\Lambda[\phi] = \int \mathrm{d}^4x \left(g_4(\Lambda)\phi(x)^4 + g_6(\Lambda)\phi(x)^6\right). \tag{36.23}$$

This is a two-dimensional projection of the full Wilsonian flow. Numerical normalizations of g_4 and g_6 are included in the constants a and b below. The corresponding dimensionless coordinates are

$$\lambda_4(\Lambda) := g_4(\Lambda), \qquad \lambda_6(\Lambda) := \Lambda^2 g_6(\Lambda). \tag{36.24}$$

The canonical term in the second equation is $2\lambda_6$, because ϕ^6 has coefficient of mass dimension -2 in four dimensions. A projected Wilsonian beta function is, by definition,

$$\beta_A(\lambda) := \Lambda \frac{\mathrm{d}\lambda_A}{\mathrm{d}\Lambda}.$$

The variables λ_A are coordinates on a chosen cutoff scheme, not observables by themselves. To isolate this two-coordinate mechanism, choose local coordinates in which the small-coupling flow begins as

$$\Lambda \frac{\mathrm{d}\lambda_4}{\mathrm{d}\Lambda} = -a\lambda_6 + \cdots, \qquad \Lambda \frac{\mathrm{d}\lambda_6}{\mathrm{d}\Lambda} = 2\lambda_6 + b\lambda_4^2 + \cdots, \tag{36.25}$$

where a and b depend on the cutoff profile and on the normalization of the two operators. The ellipses denote terms of higher order in the retained local coordinates and terms involving the omitted coordinates. The coefficient 2 is the canonical irrelevant eigenvalue in the convention $t = \log \Lambda$: at fixed dimensionful g_6, the dimensionless $\lambda_6 = \Lambda^2 g_6$ decreases when the cutoff is lowered.

Assume $\lambda_6 = O(\lambda_4^2)$ along the trajectory and work to the first nontrivial order. Let

$$h(\Lambda) := \lambda_6(\Lambda) + \frac{b}{2}\lambda_4(\Lambda)^2.$$

Using (36.25),

$$\Lambda \frac{\mathrm{d}}{\mathrm{d}\Lambda}\left(\lambda_6 + \frac{b}{2}\lambda_4^2\right) = 2\left(\lambda_6 + \frac{b}{2}\lambda_4^2\right) + O(\lambda_4 h) + O(\lambda_4^3). \tag{36.26}$$

Thus, over a perturbative range in which the variation of λ_4 only affects this equation at higher order,

$$h(\Lambda) = h(\Lambda_*)\left(\frac{\Lambda}{\Lambda_*}\right)^2 + O(\lambda_4^3).$$

Along the flow toward the infrared, $\Lambda \downarrow 0$, the transverse combination h is suppressed by the canonical irrelevant power. The attracting curve is therefore

$$\lambda_6(\Lambda) = -\frac{b}{2}\lambda_4(\Lambda)^2 + O(\lambda_4^3) \tag{36.27}$$

on the attracting curve. Substitution into the λ_4 equation gives

$$\Lambda\frac{\mathrm{d}\lambda_4}{\mathrm{d}\Lambda} = \frac{ab}{2}\lambda_4^2 + O(\lambda_4^3). \tag{36.28}$$

This equation is the induced beta function for the retained marginal coordinate in the reduced coordinate system. Comparison with a 1PI renormalization convention requires an explicitly chosen finite reparametrization between Wilsonian coordinates and the renormalized coupling used to label Green functions; in the displayed coordinate system the irrelevant coordinate has been eliminated by solving its own flow equation.

Figure 36.5 displays the reduced two-coordinate flow used in the example: the irrelevant sextic coordinate is slaved to the quartic coordinate along the renormalized trajectory.

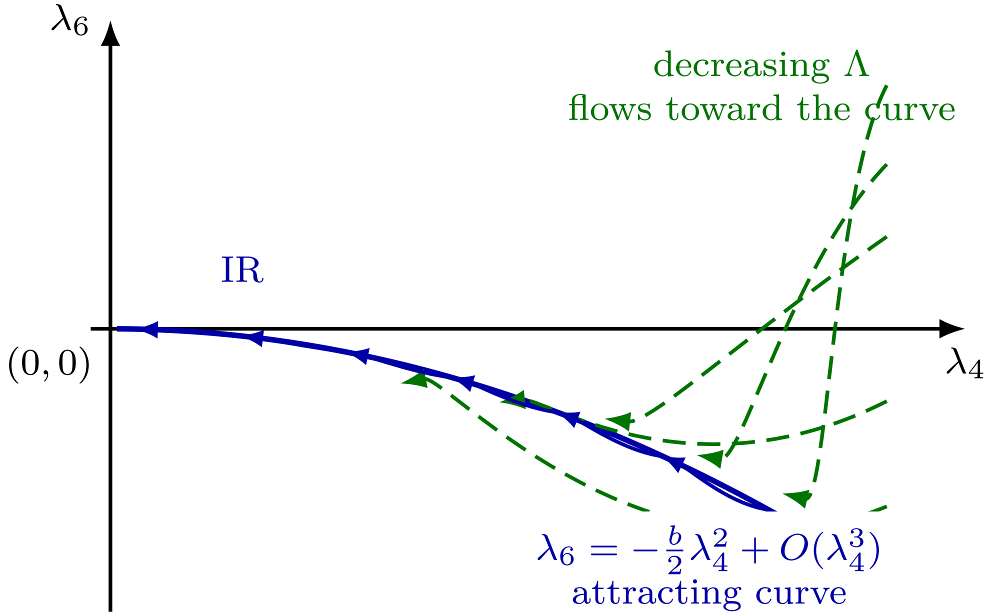


Figure 36.5: In the two-coordinate projection, the irrelevant coordinate λ_6 rapidly approaches a curve determined by λ_4. The arrows indicate the direction of decreasing cutoff. Dashed trajectories approach distinct neighborhoods of the solid attracting curve; the projection is not meant to show finite-time merging of RG characteristics. The curve is drawn in a normalization with $b > 0$; changing the sign convention for the ϕ^6 coordinate reflects the vertical axis but not the attraction statement.

This example is deliberately modest. It shows one concrete feature of the full equation: irrelevant local coordinates are generally generated even if they are absent in the chosen ultraviolet-regulated action, and near a controlled trajectory they are determined, order by order, by the coordinates that remain to be fixed at a reference scale.

36.9 Renormalizability as a Continuum Limit

The Wilsonian flow is a downward flow in cutoff. If an action S_{Λ_0} is given at a finite ultraviolet cutoff Λ_0, then (36.10) or (36.11) constructs S_Λ for $\Lambda < \Lambda_0$. The finite-cutoff construction therefore supplies a trajectory below Λ_0; a description extending to arbitrarily large Λ is a continuum-limit input.

Take a regulated path integral at cutoff Λ_0, with interaction

$$L_{\Lambda_0}[\phi] = \sum_I g_I^{\text{bare}}(\Lambda_0)\, O_I[\phi], \tag{36.29}$$

where O_I are local operators compatible with the symmetries. After flowing down to a fixed reference scale Λ_R, low-momentum Green functions can be expressed in terms of the Wilsonian coordinates $g_I(\Lambda_R)$. A continuum limit is a limit $\Lambda_0 \to \infty$ in which the bare coordinates are chosen as functions of Λ_0 so that selected physical coordinates at Λ_R are held fixed and the remaining Wilsonian coordinates at Λ_R approach limits. The finite-coordinate estimate below makes precise what is meant in a local projection. Its inputs are the endpoint tuning problem, the irrelevant semigroup estimate, a uniform bound on the retained irrelevant bare coordinates, and convergence of the generated integral. The Wilsonian mechanism is the estimate showing how these inputs produce a cutoff-independent reference-scale point.

In the quartic-sextic projection, impose the ultraviolet boundary condition

$$\lambda_4(\Lambda_0) = \lambda_4^{\text{bare}}(\Lambda_0), \qquad \lambda_6(\Lambda_0) = 0,$$

and choose $\lambda_4^{\text{bare}}(\Lambda_0)$ so that

$$\lambda_4(\Lambda_R) = \lambda_4^R$$

is fixed at the reference scale. The Wilsonian renormalizability question in this projection is whether

$$\lim_{\Lambda_0 \to \infty} \lambda_6(\Lambda_R)\,\Big|_{\lambda_4(\Lambda_R) = \lambda_4^R} \tag{36.30}$$

exists. Under perturbative control, with small couplings along the trajectory and sufficiently mild dependence of the beta functions on the irrelevant coordinate, the limit can be seen directly from the transverse coordinate h. The boundary condition $\lambda_6(\Lambda_0) = 0$ gives

$$h(\Lambda_0) = \frac{b}{2}\lambda_4(\Lambda_0)^2.$$

After tuning $\lambda_4(\Lambda_R) = \lambda_4^R$, the homogeneous memory of this ultraviolet boundary condition is

$$h(\Lambda_R) = h(\Lambda_0)\left(\frac{\Lambda_R}{\Lambda_0}\right)^2 + O\big((\lambda_4^R)^3\big)$$

within the perturbative domain. Therefore

$$\lambda_6(\Lambda_R) = -\frac{b}{2}(\lambda_4^R)^2 + O\big((\lambda_4^R)^3\big) + O\left(\left(\frac{\Lambda_R}{\Lambda_0}\right)^2\right),$$

where the coefficient of the last term is bounded by the assumed control of the bare trajectory. This displays the continuum-limit mechanism: the irrelevant coordinate at the fixed physical scale is selected by the renormalized trajectory, while the memory of the microscopic boundary condition is power-suppressed.

Figure 36.6 shows the continuum-limit mechanism in the same two-coordinate model: different microscopic boundary conditions approach the same finite-scale renormalized trajectory up to $(\Lambda_R/\Lambda_0)^2$ corrections.

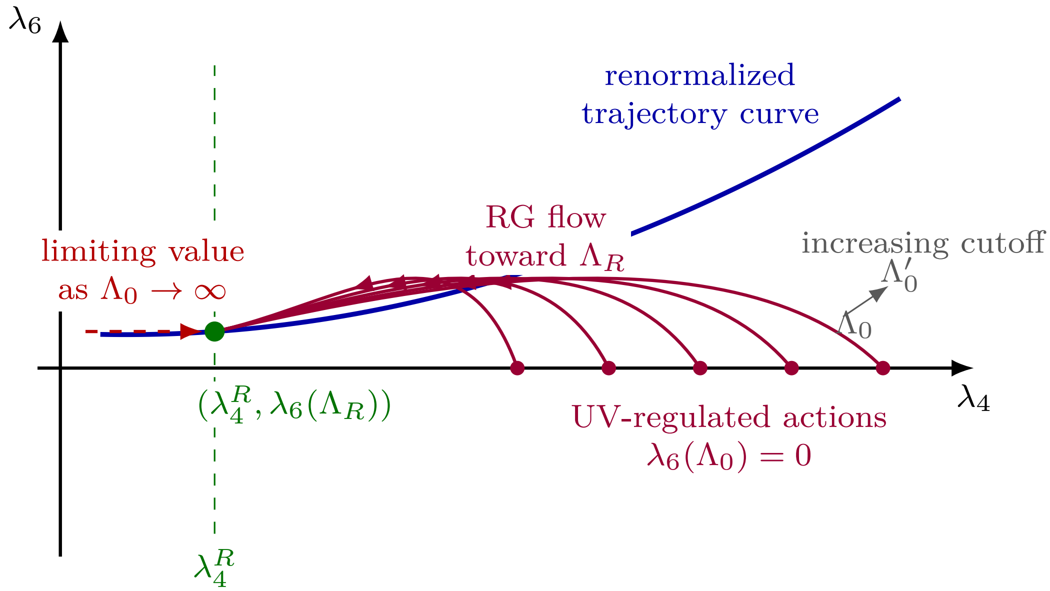


Figure 36.6: For each ultraviolet cutoff Λ_0, the bare quartic coupling is tuned so that $\lambda_4(\Lambda_R) = \lambda_4^R$. The continuum-limit statement asks whether the induced value of $\lambda_6(\Lambda_R)$ approaches a limit as Λ_0 is removed.

The same analysis allows a finite nonzero value of the dimensionless irrelevant coordinate at the ultraviolet cutoff. In the example,

$$g_6^{\mathrm{bare}}(\Lambda_0) = \Lambda_0^{-2}\lambda_6(\Lambda_0).$$

Thus a finite $\lambda_6(\Lambda_0)$ corresponds to a dimensionful bare coefficient suppressed by the cutoff. This is the form produced by lattice regularizations with lattice spacing $a = \Lambda_0^{-1}$: many local operators are present at the cutoff scale, but after tuning the relevant and marginal coordinates, irrelevant coordinates approach their continuum trajectory values at fixed physical scale. For a hypercubic lattice, the edge of the Brillouin zone is $\Lambda_{\mathrm{BZ}} = \pi/a = \pi\Lambda_0$. The estimates in this chapter use $\Lambda_0 = a^{-1}$ as the inverse-spacing cutoff convention; using Λ_{BZ} instead only rescales the constants multiplying cutoff-suppressed powers.

The perturbative Wilsonian statement has explicit hypotheses. Along the trajectory under consideration, the couplings must remain small enough for the local-coordinate expansion and the omitted-operator estimates to be valid; the derivatives such as $\partial\beta_6/\partial\lambda_6$ and $\partial\beta_4/\partial\lambda_4$ must remain controlled; and the beta function for the retained marginal coordinate must vary slowly enough along the trajectory that the irrelevant directions are genuinely suppressed. In this regime, the linearized irrelevant eigenvalues and the generated-integral estimate give the power-counting organization of local operators in perturbative renormalization.

This finite-coordinate estimate should not be mistaken for a constructive existence theorem for a non-Gaussian four-dimensional scalar continuum limit. Quoted Theorem 33.3 states that, for the positive lattice $\lambda\phi_4^4$ and nearest-neighbor ferromagnetic Ising-type scaling-limit problems covered by the rigorous triviality results cited there, the critical or near-critical continuum limits are Gaussian. The Wilsonian estimate above is a local perturbative mechanism: it explains how irrelevant coordinates lose memory of their finite-cutoff boundary values when a controlled trajectory with fixed renormalized coordinates is assumed. It does not prove that such a controlled trajectory exists to arbitrarily high cutoff at nonzero renormalized quartic coupling. For the specified standard families and covered critical regimes, the triviality result says that the limiting connected correlations beyond the two-point function vanish.

36.10 A Local Estimate for Cutoff Removal

The quartic-sextic example is the one-dimensional irrelevant-coordinate version of a general local estimate. This estimate is a finite-coordinate statement: choose a projection of the Wilsonian action onto local coordinates

$$z = (u, v) \in U \times V,$$

where $u = (u^1, \dots, u^r)$ are the coordinates fixed by renormalization conditions at the reference scale Λ_R, and $v = (v^1, \dots, v^s)$ are irrelevant coordinates retained in the projection. Let

$$t = \log \Lambda, \qquad t_R = \log \Lambda_R, \qquad t_0 = \log \Lambda_0.$$

In the coordinate neighborhood under consideration, write the projected flow as

$$\frac{\mathrm{d}u^a}{\mathrm{d}t} = B^a(u, v), \qquad \frac{\mathrm{d}v^\rho}{\mathrm{d}t} = A^\rho{}_\sigma v^\sigma + F^\rho(u, v). \tag{36.31}$$

The matrix A is the linear irrelevant block. In the convention $t = \log \Lambda$, an irrelevant coordinate has a positive eigenvalue in A; the flow is therefore contracting in that coordinate when the cutoff is lowered from t_0 to $t_R < t_0$. We assume a norm on V and constants $K_A, \omega > 0$ such that

$$\|e^{A\tau}\| \le K_A e^{\omega\tau}, \qquad \tau \le 0. \tag{36.32}$$

Thus $e^{A(t_R - t_0)}$ is bounded by

$$K_A \left(\frac{\Lambda_R}{\Lambda_0}\right)^\omega.$$

For a solution of (36.31) on $[t_R, t_0]$, variation of constants gives

$$v(t_R) = e^{A(t_R - t_0)} v(t_0) - \int_{t_R}^{t_0} e^{A(t_R - s)} F(u(s), v(s))\, \mathrm{d}s. \tag{36.33}$$

The first term is the direct memory of the irrelevant bare coordinate. The second term is generated by the flow along the trajectory. These two terms have different logical roles: suppression of the first term follows from the irrelevant eigenvalues, while existence of the cutoff-removal limit requires control of the generated integral and of the tuning used to hold $u(t_R)$ fixed.

The direct boundary-memory estimate follows by applying (36.32) with $\tau = t_R - t_0 \le 0$. If the irrelevant bare coordinate satisfies $\|v(t_0)\| \le M$ uniformly in t_0, then the direct boundary contribution to $v(t_R)$ obeys

$$\|e^{A(t_R-t_0)}v(t_0)\| \le K_A M \left(\frac{\Lambda_R}{\Lambda_0}\right)^{\omega}.$$

If the first omitted irrelevant coordinate has exponent $\omega_{\min}$, the same computation begins with the power $(\Lambda_R/\Lambda_0)^{\omega_{\min}}$. This estimate controls the homogeneous boundary-memory term in (36.33); the generated inhomogeneous term remains the analytic part of the continuum-limit problem.

The generated term in (36.33) is the part that selects the continuum graph. The precise finite-coordinate hypothesis is the existence, after tuning the retained bare coordinates, of a limit

$$V_R^{\rho}(u_R) = \lim_{\Lambda_0\to\infty}\left[-\int_{t_R}^{t_0}\left(e^{A(t_R-s)}F(u(s),v(s))\right)^{\rho}\mathrm{d}s\right], \tag{36.34}$$

where the trajectory is constrained by

$$u(t_R) = u_R.$$

When this limit exists uniformly for u_R in a small coordinate neighborhood, the finite projection of the renormalized trajectory at Λ_R is the graph

$$v^{\rho} = V_R^{\rho}(u_R).$$

For finite Λ_0, define the generated-integral remainder

$$R_F(t_0;u_R) = -\int_{t_R}^{t_0} e^{A(t_R-s)}F(u(s),v(s))\,\mathrm{d}s - V_R(u_R).$$

Finite-coordinate cutoff-removal estimate. Fix u_R and suppose that, for every sufficiently large t_0, the retained bare coordinates can be tuned so that $u(t_R) = u_R$. Assume:

(i) the solution remains in a coordinate neighborhood where (36.31) is valid;

(ii) the irrelevant semigroup satisfies (36.32);

(iii) the irrelevant bare coordinates are uniformly bounded;

(iv) the generated integral in (36.34) converges uniformly with

$$\|R_F(t_0;u_R)\| \le K_F\left(\frac{\Lambda_R}{\Lambda_0}\right)^{\omega_F}$$

for some $K_F, \omega_F > 0$.

For the tuned finite-cutoff trajectory, write

$$z_R(t_0;u_R) := \big(u(t_R;t_0,u_R),\ v(t_R;t_0,u_R)\big).$$

Then the reference-scale Wilsonian coordinates have the limit

$$\lim_{\Lambda_0\to\infty} z_R(t_0;u_R) = \big(u_R, V_R(u_R)\big) \tag{36.35}$$

in the retained coordinate space $U \times V$, and the irrelevant coordinates at the reference scale obey

$$v(t_R) = V_R(u_R) + O\left[\left(\frac{\Lambda_R}{\Lambda_0}\right)^{\min(\omega,\omega_F)}\right], \qquad \Lambda_0 \to \infty,$$

with the exponent replaced by the smallest retained irrelevant exponent when several irrelevant coordinates are present. Consequently, if Q is any continuous finite-coordinate observable, projected vertex, or low-momentum kernel functional defined in a neighborhood of $(u_R, V_R(u_R))$, then

$$\lim_{\Lambda_0 \to \infty} Q\big(z_R(t_0; u_R)\big) = Q\big(u_R, V_R(u_R)\big).$$

If Q is Lipschitz in the chosen coordinate norm, the same power gives an error bound for this convergence, with a constant multiplied by the Lipschitz constant of Q.

Indeed, equation (36.33) writes $v(t_R)$ as the sum of the boundary contribution and the generated contribution. The semigroup bound and the uniform bound on $v(t_0)$ give

$$\|e^{A(t_R - t_0)} v(t_0)\| \le K_A M \left(\frac{\Lambda_R}{\Lambda_0}\right)^{\omega}.$$

The generated contribution is $V_R(u_R) + R_F(t_0; u_R)$, with R_F bounded by the fourth hypothesis. Combining the two estimates gives the displayed asymptotic form. The relevant coordinate is fixed by the tuning condition $u(t_R; t_0, u_R) = u_R$, so the full retained coordinate vector converges to $(u_R, V_R(u_R))$. The assertion for Q follows from continuity; the quantitative version follows from applying the Lipschitz bound of Q to the coordinate difference.

The four hypotheses above are the mathematical content of a Wilsonian continuum-limit argument in a chosen local projection. The tuning condition $u(t_R) = u_R$ is a statement about the endpoint map from bare coordinates at Λ_0 to renormalized coordinates at Λ_R. In perturbation theory this endpoint map is usually solved order by order. When the linearized endpoint map in the retained coordinates is invertible, the implicit function theorem gives the local tuning functions at finite Λ_0. The remaining analytic task is to prove estimates, uniform in Λ_0, that keep the trajectory in the assumed coordinate neighborhood and make the generated integral converge.

Remark 36.8 (Status of the cutoff-removal hypotheses). The finite-coordinate estimate derives the displayed reference-scale convergence from the four listed hypotheses. Hypothesis (ii) is a backward semigroup estimate for the irrelevant part of the linearized projected RG equation in a chosen norm. It is immediate in a diagonal finite-dimensional toy model with strictly negative irrelevant exponents; in an actual Wilsonian space it is an analytic estimate about the chosen Banach or Fréchet space of functionals and the linearized cutoff-flow operator. Hypothesis (iv) is the uniform convergence, after the retained coordinates are tuned, of the generated irrelevant coordinate. This is one form of the cutoff-removal problem itself.

Several rigorous results prove analogues of these estimates only after adding substantial structure. Polchinski's smooth-cutoff argument proves perturbative renormalizability order by order for renormalizable Euclidean Lagrangians by deriving inductive bounds on cutoff-dependent vertices after the relevant parts are fixed.[3] Salmhofer's rigorous RG framework proves perturbative flow estimates, especially for fermionic systems, in specified norms and small-coupling regimes.[4]

[3]J. Polchinski, *Nucl. Phys. B* **231** (1984) 269–295.
[4]M. Salmhofer, *Renormalization: An Introduction*, Springer, 1999.

The Brydges–Kennedy tree and Mayer expansion technology supplies iterative bounds used in constructive RG, but every application still verifies its own covariance estimates, norms, tuning equations, and convergence criteria.[5]

The nonperturbative status is model-dependent. Constructive scalar field theory gives non-Gaussian ϕ_2^4 and ϕ_3^4 continuum theories under its own ultraviolet, infrared, and stability hypotheses.[6] By contrast, the standard positive four-dimensional scalar and nearest-neighbor Ising-type families covered by Quoted Theorem 33.3 have Gaussian scaling limits. Therefore later use of a Wilsonian continuum-limit estimate must identify which semigroup, tuning, and convergence estimates are available in the chosen framework.

Remark 36.9 (Renormalons and Wilsonian data). Assumption (iv) is a continuum-limit hypothesis on the Wilsonian generated integral. It is stronger than the existence of a formal perturbative expansion for the projected coordinates. A renormalon analysis is one way in which perturbation theory can diagnose the failure of a naive coefficientwise description to determine the full generated integral. In a running-coupling expansion, integration over very small or very large intermediate momenta can produce factorially growing coefficients and Borel-plane singularities. When the corresponding ambiguity has the size of a power correction, the Wilsonian description must supply the missing input through retained nonperturbative coordinates, through an operator-product-expansion matrix element, or through another matched effective description. Thus renormalons do not replace the cutoff-removal hypothesis; they identify particular ways in which that hypothesis contains information beyond formal perturbation theory.

Figure 36.7 draws the quantitative estimate used in this discussion. Increasing the ultraviolet separation $T = t_0 - t_R = \log(\Lambda_0/\Lambda_R)$ suppresses direct memory of the irrelevant boundary coordinate by $e^{-\omega T}$. After the relevant coordinates are tuned at t_R, endpoints obtained from different bounded irrelevant boundary values therefore approach the same continuum graph value up to the convergence rate of the generated integral.

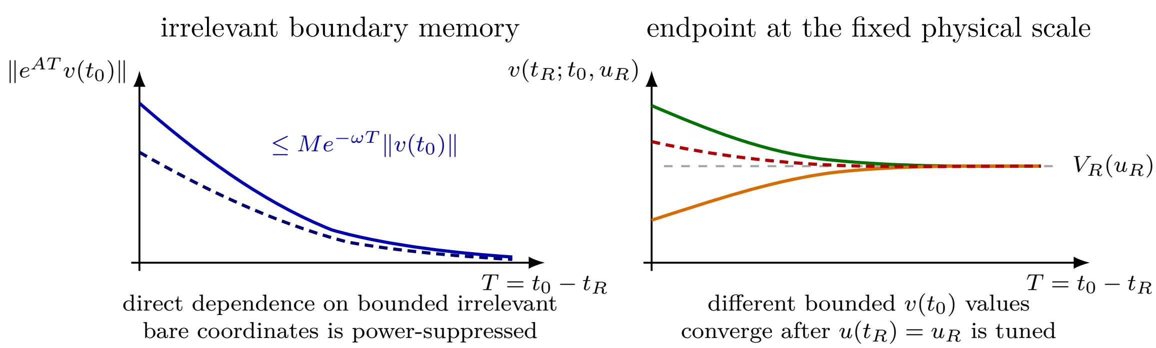


Figure 36.7: The two quantitative pieces in the finite-coordinate cutoff-removal estimate. If the irrelevant semigroup satisfies $\|e^{A(t_R-t_0)}\| \leq Me^{-\omega(t_0-t_R)}$, direct memory of a bounded irrelevant boundary coordinate is suppressed by $(\Lambda_R/\Lambda_0)^\omega$. The estimate uses uniform convergence of the generated integral; with that input, all tuned endpoints with bounded irrelevant bare coordinates converge to the same graph $v = V_R(u_R)$ at the physical scale.

[5] D. C. Brydges and T. Kennedy, *J. Stat. Phys.* **48** (1987) 19–49.

[6] See J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, second edition, Springer, 1987, for the constructive scalar-field framework and references to the original constructions.

For a lattice regulator with lattice spacing $a = \Lambda_0^{-1}$, a bounded dimensionless irrelevant bare coordinate $v^\rho(t_0)$ corresponds to a dimensionful coefficient multiplied by the appropriate positive power of a. The finite-coordinate cutoff-removal estimate explains the resulting universality statement in this local projection: after the retained coordinates are tuned at Λ_R, bounded changes of microscopic irrelevant lattice couplings alter $v(t_R)$ only by powers of Λ_R/Λ_0, while the limiting value belongs to the continuum graph selected by the flow.

36.11 The Bridge to BPHZ and 1PI Coordinates

The preceding sections define the Wilsonian action S_Λ by Gaussian pushforward. Exact preservation of a source-independent low-source functional requires the annihilator condition explained above, and for the smooth source-independent theorem below this will be implemented by an admissible plateau cutoff. Earlier chapters defined BPHZ subtractions and 1PI renormalized coordinates. These are different constructions, so their relation must be stated with the maps that connect them.

Fix a finite ultraviolet regulator Λ_0, a smooth intermediate cutoff Λ, and a Euclidean subtraction scale μ satisfying

$$\mu \ll \Lambda \ll \Lambda_0.$$

Keep distinct names for the two RG times:

$$t_{\mathrm{W}} := \log \Lambda, \qquad t_{1\mathrm{PI}} := \log \mu.$$

They parameterize different vector fields. The Wilsonian time t_{W} acts on the cutoff action before the remaining low field is integrated, while $t_{1\mathrm{PI}}$ acts on projected 1PI vertices after the connected functional has been Legendre transformed. To compare the two flows, choose a fixed matching ratio

$$\zeta := \frac{\Lambda}{\mu} \gg 1.$$

Along such a matched comparison,

$$t_{\mathrm{W}} = t_{1\mathrm{PI}} + \log \zeta, \qquad \left.\frac{\mathrm{d}}{\mathrm{d}t_{1\mathrm{PI}}}\right|_\zeta = \left.\frac{\mathrm{d}}{\mathrm{d}t_{\mathrm{W}}}\right|_\zeta.$$

If $u_A(\Lambda)$ are Wilsonian coordinates and $g_I(\mu)$ are 1PI coordinates, the bridge is a finite matching map at fixed ζ,

$$g_I(\mu) = M_I(u(\Lambda), \zeta) + O(\zeta^{-\Delta_{\mathrm{irr}}}),$$

where $\Delta_{\mathrm{irr}} > 0$ is the smallest suppression exponent among the omitted irrelevant coordinates in the chosen local expansion. Differentiating at fixed ζ gives the corresponding relation between beta functions,

$$\beta_I^{1\mathrm{PI}} = \frac{\partial M_I}{\partial u_A}\,\beta_A^{\mathrm{W}} + O(\zeta^{-\Delta_{\mathrm{irr}}}),$$

with summation over A. Thus the two uses of t in the preceding chapters have the same orientation only after this matching convention is chosen; without the matching map they refer to different mathematical constructions.

The external momenta used in the 1PI projectors are assumed to have size $O(\mu)$ and to be nonexceptional. Let

$$W_{\Lambda_0}[J] = \log Z_{\Lambda_0}[J]$$

be the connected generating functional of the theory regulated at ultraviolet scale Λ_0, with the cutoff-dependent local terms in the path-integral Lagrangian chosen so that the selected 1PI coordinates have finite limits. Let $L_\Lambda = L^{\mathrm{src}}_{\Lambda,0}$ be the source-independent Wilsonian interaction obtained from (36.10) by integrating modes between Λ_0 and Λ. For a non-plateau smooth cutoff and a nonzero open low-momentum source sector, exact source preservation requires enlarging the Wilsonian data to include source-dependent vertices. The corresponding finite-regulator object is

$$\exp[-L^{\mathrm{src}}_{\Lambda,J}[\phi_<]] := \int \mathrm{d}\mu_{C_>}(\phi_>) \, \exp\left[-L_{\Lambda_0}[\phi_< + \phi_>] - \langle J, \phi_> \rangle\right], \tag{36.36}$$

where $C_{\Lambda_0} = C_\Lambda + C_>$ and $\langle J, \phi\rangle = \int J\phi$. With

$$W^{\mathrm{src,low}}_\Lambda[J] := \log \int \mathrm{d}\mu_{C_\Lambda}(\phi_<) \, \exp\left[-L^{\mathrm{src}}_{\Lambda,J}[\phi_<] - \langle J, \phi_< \rangle\right],$$

this gives the exact identity

$$W^{\mathrm{src,low}}_\Lambda[J] = W_{\Lambda_0}[J] \tag{36.37}$$

after the remaining field is integrated. However, differentiating (36.37) with respect to J also differentiates $L^{\mathrm{src}}_{\Lambda,J}$. Therefore an exact matching implication based on this formulation must include source-vertices as part of the coordinate data. The controlled comparison below instead uses an *admissible plateau cutoff*: choose a smooth profile satisfying

$$\chi(s) = 1 \quad (0 \le s \le c_\chi^2) \tag{36.38}$$

for some $c_\chi > 0$. Let $\mathcal{E}_<$ denote the finite-dimensional low source test space chosen with

$$\operatorname{supp} \widehat{J} \subset \{|k| < c_< \Lambda\}, \qquad 0 < c_< < c_\chi. \tag{36.39}$$

Since $\Lambda < \Lambda_0$, this implies $C_>(k)\widehat{J}(k) = 0$ on the low test space. Thus the source-independent Wilsonian action L_Λ has an exact low-source identity. The connected functional computed from it is denoted

$$W^{\mathrm{low}}_\Lambda[J] = \log \int \mathrm{d}\mu_{C_\Lambda}(\phi) \, \exp\left[-L_\Lambda[\phi] - \int J\phi\right].$$

The finite-dimensional Gaussian convolution identity and (36.39) give the exact equality

$$W^{\mathrm{low}}_\Lambda[J] = W_{\Lambda_0}[J] \tag{36.40}$$

at the finite regulator. No exact source-independent statement of this form is asserted below for non-plateau profiles such as a Gaussian cutoff. Such a profile may be treated either by keeping the source-dependent vertices in (36.36), or by proving a separate profile-dependent leakage estimate; that is a different comparison problem.

Exact low-source identity for a plateau split. Assume (36.38) and (36.39). At finite regulator, (36.40) holds for every source $J \in \mathcal{E}_<$.

By Gaussian convolution,

$$\int \mathrm{d}\mu_{C_{\Lambda_0}}(\phi)\, \mathrm{e}^{-L_{\Lambda_0}[\phi]-\langle J,\phi\rangle} = \int \mathrm{d}\mu_{C_\Lambda}(\phi_<) \int \mathrm{d}\mu_{C_>}(\phi_>)\, \mathrm{e}^{-L_{\Lambda_0}[\phi_<+\phi_>]-\langle J,\phi_<+\phi_>\rangle}.$$

For $J \in \mathcal{E}_<$, the covariance of the scalar Gaussian random variable $\langle J, \phi_>\rangle$ is

$$\langle J, C_> J\rangle = \int_k |\widehat{J}(k)|^2 \frac{\chi(k^2/\Lambda_0^2) - \chi(k^2/\Lambda^2)}{k^2+m^2}.$$

On $\operatorname{supp}\widehat{J}$, both cutoff profiles equal one by (36.38) and (36.39); hence the integrand vanishes and $\langle J, \phi_>\rangle = 0$ almost surely. The previous display becomes

$$\int \mathrm{d}\mu_{C_\Lambda}(\phi_<)\, \mathrm{e}^{-\langle J,\phi_<\rangle} \int \mathrm{d}\mu_{C_>}(\phi_>)\, \mathrm{e}^{-L_{\Lambda_0}[\phi_<+\phi_>]}.$$

The inner integral is, by definition, $\exp[-L_\Lambda[\phi_<]]$ when the field-independent normalization is included as the vacuum-energy coordinate of L_Λ. With that convention this is exactly $Z_\Lambda^{\mathrm{low}}[J]$, and taking logarithms gives (36.40). If one fixes the vacuum-energy coordinate separately, the two logarithms differ only by the chosen source-independent constant and all positive source derivatives agree.

The 1PI effective action at the same finite regulator is the Legendre transform of the connected functional. If

$$\varphi(x) = \frac{\delta W[J]}{\delta J(x)},$$

then, in the Euclidean sign convention of this volume,

$$\Gamma[\varphi] = W[J] - \int \mathrm{d}^D x\, J(x)\varphi(x), \qquad J = J[\varphi].$$

Equation (36.40) therefore implies equality of the low-momentum 1PI kernels obtained from the source-independent plateau-cutoff Wilsonian description and from the original regulated description, after the same Legendre transform and the same field normalization convention are used. Here “low 1PI” means the Legendre transform of the connected functional after restriction to the finite source space $\mathcal{E}_<$. It is not being identified with the restriction of a Legendre transform already taken on the full source space; that different operation would require an additional Schur-complement statement. The Wilsonian vertices $L_\Lambda^{(n)}$ and the 1PI vertices $\Gamma^{(n)}$ are distinct vertices; their general relation includes all tree and loop diagrams built from the Wilsonian vertices and the remaining cutoff covariance C_Λ.

Figure 36.8 shows the nontrivial point in the Legendre-transform bridge. A Legendre transform performed only on the low source space inverts the low-source Hessian $W^{(2)}_{<<}$. A full Legendre transform followed by restriction to low fields instead inverts the Schur complement $W^{(2)}_{<<} - W^{(2)}_{<>}(W^{(2)}_{>>})^{-1}W^{(2)}_{><}$. The chapter uses the first operation; the second requires extra hypotheses and is not silently identified with it.

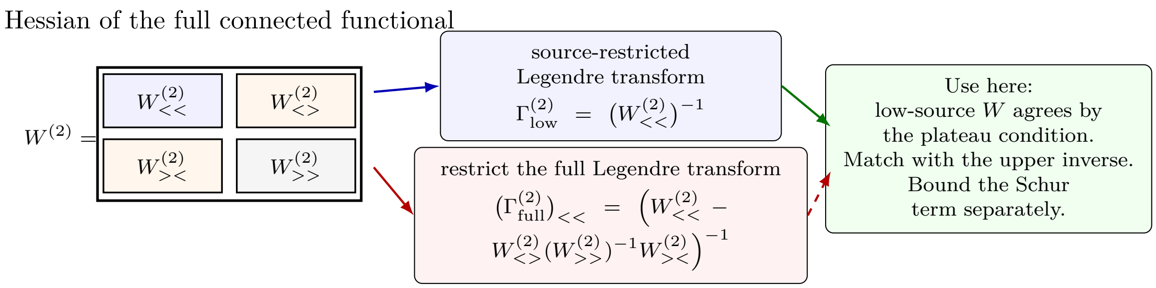


Figure 36.8: The low-source Legendre transform is not the same operation as restricting a full Legendre transform. In Hessian language, the former inverts $W^{(2)}_{<<}$, while the latter inverts the Schur complement that subtracts high-source feedback through $W^{(2)}_{<>}(W^{(2)}_{>>})^{-1}W^{(2)}_{><}$. The Wilsonian–BPHZ comparison in this section uses the source-restricted transform; identifying it with the restricted full transform would require an additional estimate.

Definition 36.10 (BPHZ–Wilsonian comparison setup)**.** A comparison setup consists of the following finite choices. First choose a BPHZ scheme $\mathfrak{b}$: a Taylor subtraction point and subtraction degree $\delta(\gamma)$ for every superficially divergent 1PI subgraph type γ, including the oversubtraction degrees used for normal products. The corresponding forest operation is

$$R^{\mathfrak{b}}_{\Gamma} I_{\Gamma} = \sum_{F\in\mathfrak{F}(\Gamma)} (-1)^{|F|} \left(\prod_{\gamma\in F} t_{\gamma}^{\delta(\gamma),\mathfrak{b}}\right) I_{\Gamma},$$

with nested products ordered from smaller subgraphs to larger ones. Second choose the Wilsonian covariance split

$$C_{\Lambda_0} = C_{\Lambda} + C_{>},$$

with an admissible plateau profile satisfying (36.38) and a low test space satisfying (36.39), the Banach chart $\mathcal{B}^{m,a}_{\Lambda}$ in which Wilsonian kernels are estimated, the finite local coordinate projectors $\rho_A(L_\Lambda) = u_A$, and the finite 1PI projectors Π_I at nonexceptional Euclidean momenta of size $O(\mu)$. Third choose a matching ratio $\zeta = \Lambda/\mu$ in a compact subinterval of $(1,\infty)$. All statements below are statements relative to this setup. A non-plateau smooth cutoff is a different setup: either source-dependent vertices must be included, or an explicit leakage estimate must be proved before the comparison theorem is invoked.

Proposition 36.11 (BPHZ local parts as Wilsonian local coordinates)**.** *Consider a power-counting finite scalar theory in a massive Euclidean momentum-subtraction setting. For every superficially divergent subgraph γ with E_γ external scalar legs, the BPHZ local part*

$$t_{\gamma}^{\delta(\gamma),\mathfrak{b}}\,\overline{R}_{\gamma} I_{\gamma}$$

is a finite sum of external-momentum monomials

$$\sum_{|\alpha|\le\delta(\gamma)} c^{\mathfrak{b}}_{\gamma,\alpha}\, p_{\gamma}^{\alpha}.$$

After Fourier transform, each term is a local vertex of the form

$$c^{\mathfrak{b}}_{\gamma,\alpha} \int \mathrm{d}^D x\, P_\alpha(\partial_1,\ldots,\partial_{E_\gamma})\phi(x_1)\cdots\phi(x_{E_\gamma})\big|_{x_i=x}.$$

For the minimal BPHZ degree $\delta(\gamma) = \omega(\gamma)$, these vertices are precisely among the relevant or marginal local Wilsonian coordinates determined by power counting. Oversubtractions add a finite list of higher-derivative local coordinates, and Zimmermann identities are the corresponding finite changes of local operator coordinates.

Proof. The prepared integrand $\overline{R}_\gamma I_\gamma$ has all proper subdivergences subtracted. The operation $t_\gamma^{\delta,\mathfrak{b}}$ is the Taylor polynomial in the independent external momenta of γ at the specified subtraction configuration, hence it has the displayed monomial form. Multiplication by p^α in momentum space is the Fourier transform of a derivative acting on the fields incident to the contracted vertex, so insertion of this Taylor polynomial into the contracted graph is the insertion of a local interaction vertex with E_γ fields and $|\alpha|$ derivatives.

For a scalar field with engineering dimension $[\phi] = (D-2)/2$, the dimension of such a local monomial in the derivative-free scalar examples of this chapter is

$$\Delta_{\gamma,\alpha} = E_\gamma \frac{D-2}{2} + |\alpha|.$$

The superficial degree $\omega(\gamma)$ is precisely the largest derivative order for which the local counterterm has scaling dimension at most D, in the power-counting convention used to define the BPHZ subtraction degree. Thus $|\alpha| \leq \omega(\gamma)$ is equivalent to $\Delta_{\gamma,\alpha} \leq D$ in this scalar setting. These are the relevant or marginal local coordinates in the Wilsonian linearized scaling convention $y_A = D - \Delta_A$. If $\delta(\gamma) > \omega(\gamma)$, the additional Taylor coefficients have $\Delta_A > D$; they are irrelevant Wilsonian coordinates retained by the chosen oversubtracted local chart. The difference between two choices of subtraction degree or finite Taylor part is therefore a finite change of coefficients multiplying local vertices, which is the Wilsonian coordinate form of the Zimmermann identity described in Chapter 32. □

Lemma 36.12 (Low-momentum Taylor remainder from a Wilsonian norm)**.** *Fix a finite loop order ℓ, a finite list of low-momentum 1PI projectors Π_I, and a comparison interval*

$$1 < \zeta_- \leq \Lambda/\mu \leq \zeta_+ < \infty.$$

Work in a Wilsonian Banach chart $\mathcal{B}_\Lambda^{m,a}$ of the form defined above, with m larger than the largest Taylor order extracted by the projectors Π_I. Let $v_{\perp,N}$ be the omitted part of L_Λ, and assume:

(i) $\|v_{\perp,N}\|_{\Lambda,m,a} \leq K_N$*;*

(ii) every Taylor monomial in $v_{\perp,N}$ that can enter one of the projectors Π_I has excess scaling exponent at least $p_N > 0$ relative to the retained projection;

(iii) the retained Wilsonian coordinates and the covariance $C_\Lambda(k)$ stay in a bounded subset of the same chart on the comparison interval.

Then, for every graph contributing through loop order ℓ to one of the projectors Π_I and containing at least one insertion of $v_{\perp,N}$, its contribution is bounded by

$$C_{I,\ell,N} K_N \left(\frac{\mu}{\Lambda}\right)^{p_N},$$

with $C_{I,\ell,N}$ depending on the bounded retained-coordinate set, on the cutoff profile, on the projectors, and polynomially on $\log(\Lambda/\mu)$ on the comparison interval.

Proof. Consider first a single omitted vertex kernel with external momenta of size $O(\mu)$. Write those momenta as $k_i = \mu\kappa_i$, where the κ_i range over a compact set fixed by the projectors. The norm $\|\cdot\|_{\Lambda,m,a}$ bounds all derivatives of the dimensionless kernel in the variables k_i/Λ up to order m. Since the Taylor monomials of excess smaller than p_N have been retained and subtracted into $L_\Lambda^{(N)}$, Taylor's theorem with integral remainder gives

$$|\Pi_I v_{\perp,N}| \le C_{I,N} K_N \left(\frac{\mu}{\Lambda}\right)^{p_N}$$

for each projected vertex contribution.

For a fixed loop order ℓ and a fixed finite family of projectors, only finitely many graph topologies and Taylor derivatives are involved. Rescale each remaining low loop momentum by $q = \Lambda x$. The smooth cutoff gives uniform bounds on the rescaled covariance and its derivatives away from the mass singularities excluded by the massive, nonexceptional hypothesis. The bounded retained vertices supply constants in the chosen chart. The Taylor-remainder factor from at least one omitted vertex is therefore carried through the whole graph. Logarithms arise only from scale integrations over the finite interval between μ and Λ, and on the fixed comparison interval they are bounded by a polynomial in $\log(\Lambda/\mu)$. Summing the finite set of graph and derivative contributions gives the stated constant $C_{I,\ell,N}$. □

Lemma 36.13 (Forest operation and the Wilsonian vertex split)**.** *Fix a graph* Γ *whose vertices are labeled by kernels appearing in the additive decomposition*

$$L_\Lambda = L_\Lambda^{(N)} + v_{\perp,N}.$$

The BPHZ forest operation is linear in every vertex kernel and depends on the graph only through its topology, momentum routing, subtraction degrees, and subtraction points. Consequently,

$$R_\Gamma^{\mathfrak{b}} I_\Gamma[L_\Lambda] = \sum_{\sigma: V(\Gamma)\to\{N,\perp\}} R_\Gamma^{\mathfrak{b}} I_{\Gamma,\sigma}[L_\Lambda^{(N)}, v_{\perp,N}], \tag{36.41}$$

where σ *records which vertices of* Γ *use retained kernels and which use* $v_{\perp,N}$*. The summands with at least one* $\perp$*-vertex are exactly the subtracted graphs to which Lemma 36.12 applies, provided the lemma's Banach norm bounds are imposed on the subtracted integrands.*

Proof. Before subtraction, the Feynman integrand is multilinear in the vertex kernels. Expanding each vertex label according to $L_\Lambda^{(N)} + v_{\perp,N}$ gives the finite sum over maps $\sigma : V(\Gamma) \to \{N, \perp\}$. For a fixed forest F, the operator

$$\prod_{\gamma\in F} t_\gamma^{\delta(\gamma),\mathfrak{b}}$$

acts by Taylor expansion in the external variables assigned to the subgraphs γ. It does not inspect whether a vertex kernel in the integrand is retained or omitted; it is a linear differential operator with coefficients fixed by the graph topology and by $\mathfrak{b}$. Therefore this Taylor operator commutes with the finite additive expansion over vertex labels. Summing over forests gives (36.41).

If a summand contains a $\perp$-vertex, every forest subtraction of that summand still contains the same $\perp$-vertex unless the Taylor contraction has replaced the whole subgraph containing it by its local Taylor polynomial. In that case the resulting local polynomial is a Taylor coefficient of a graph with a $v_{\perp,N}$-insertion. The Wilsonian Banach norm in Lemma 36.12 was imposed on the finite set of kernels and derivatives entering these Taylor coefficients, so the same $(\mu/\Lambda)^{p_N}$ bound applies to the subtracted summand. □

Definition 36.14 (Admissible finite-order BPHZ–Wilsonian matching context)**.** Consider a massive Euclidean scalar theory with local polynomial interactions, finite ultraviolet regulator Λ_0, an intermediate admissible plateau smooth cutoff Λ in the sense of (36.38), and nonexceptional external momenta of size $O(\mu)$, with $\mu < \Lambda$. Fix:

(i) a BPHZ subtraction scheme $\mathfrak{b}$ consisting of the forest formula of Chapter 32 with specified Taylor subtraction points and oversubtraction degrees;

(ii) a finite-dimensional low test space $\mathcal{E}_<$ of sources and classical fields whose Fourier transforms are supported in $|k| < c_<\Lambda$, where $0 < c_< < c_\chi$ and c_χ is the plateau constant in (36.38); this test space contains the momenta used by the selected 1PI projectors and satisfies $C_> J = 0$;

(iii) a finite list of 1PI projectors Π_I at scale μ, each extracting a Taylor coefficient of a low-momentum 1PI kernel at a prescribed nonexceptional subtraction configuration $\mathcal{S}_I$;

(iv) a finite list of Wilsonian local-coordinate projectors ρ_A at scale Λ, extracting the coefficients $u_A = \rho_A(L_\Lambda)$ of local vertex tensors P_A.

Let $L_\Lambda^{(N)}(u)$ be the local Wilsonian action reconstructed from the retained coordinates u_A, and let

$$v_{\perp,N} := L_\Lambda - L_\Lambda^{(N)}(u)$$

be the omitted part. For a fixed loop order ℓ, define the matching map

$$M_I^{(\ell,N)}(u;\Lambda,\mu,\mathfrak{b}) := \Pi_I\,\Gamma^{<,(\leq\ell)}_{\Lambda,N}[u], \tag{36.42}$$

where $\Gamma^{<,(\leq\ell)}_{\Lambda,N}[u]$ is the low-mode 1PI effective action defined as follows: compute the connected functional by integrating the full finite-regulator remaining field with covariance C_Λ and vertices $L_\Lambda^{(N)}(u)$, restrict its source argument to $\mathcal{E}_<$, then take the local Legendre transform on that finite source space with the same finite local normalization convention as $\mathfrak{b}$. The resulting formal 1PI functional is truncated after loop order ℓ.

Hypothesis 36.15 (Finite-order BPHZ–Wilsonian matching estimates)**.** In the context of Definition 36.14, assume the following estimates hold on the coordinate neighborhood under consideration:

(a) the BPHZ forest theorem applies to all graphs through loop order ℓ that contribute to the projectors Π_I;

(b) the Hessian of W_Λ^{low} on the low source space is invertible on the source neighborhood used for the Legendre transform;

(c) in a chosen Wilsonian Banach norm, $\|v_{\perp,N}\|_{\Lambda,m,a} \leq K_N$, and the first omitted local tensor that can contribute to the projectors Π_I has excess scaling exponent at least $p_N > 0$, with the retained coordinates and covariance bounded as in Lemma 36.12; the same bounds hold for the finite set of Taylor-subtracted kernels obtained by applying $R_\Gamma^{\mathfrak{b}}$ to graphs with at least one $v_{\perp,N}$-insertion. This bounded-chart hypothesis is an order-by-order perturbative hypothesis in the renormalized coupling, or a genuine nonperturbative hypothesis only for trajectories for which the stated norm remains bounded;

(d) the limit $\Lambda_0 \to \infty$, when it is taken, is taken along a Polchinski-flow counterterm trajectory: the relevant and marginal UV-cutoff coordinates in the regulated path-integral Lagrangian at

Λ_0 are tuned as functions of Λ_0 so that the selected renormalized boundary coordinates at a fixed reference scale stay fixed and the Wilsonian kernels at the intermediate scale Λ have the perturbative limits supplied by Polchinski/Salmhofer/Kopper–Müller-type cutoff estimates in the chosen norm.

Controlled Approximation 36.16 (Finite-order BPHZ–Wilsonian comparison)**.** In an admissible matching context as in Definition 36.14, and under Hypothesis 36.15, the BPHZ-renormalized 1PI coordinates through loop order ℓ obey

$$g_I^{\mathfrak{b},(\le\ell)}(\mu) = M_I^{(\ell,N)}(u(\Lambda);\Lambda,\mu,\mathfrak{b}) + R_I^{(\ell,N)}(\Lambda,\mu), \qquad |R_I^{(\ell,N)}| \le C_{I,\ell,N} K_N \left(\frac{\mu}{\Lambda}\right)^{p_N}. \tag{36.43}$$

If a sublist A_0 of Wilsonian coordinates has the same cardinality as the chosen 1PI coordinates and the Jacobian

$$J_{IA} = \frac{\partial M_I^{(\ell,N)}}{\partial u_A}\bigg|_{u=u_*}, \qquad I \in \mathcal{I},\ A \in A_0,$$

is invertible at a reference point u_*, then the finite-dimensional implicit function theorem gives a local finite change of coordinates

$$u_A = G_A\big(g^{\mathfrak{b}}, u_{\bar{A}}; \Lambda/\mu, \mathfrak{b}\big), \qquad A \in A_0,$$

with the same $O((\mu/\Lambda)^{p_N})$ truncation error.

Derivation of the comparison. The displayed comparison is an implication from the four estimates in Hypothesis 36.15; those estimates are the analytic input. The derivation records what the finite matching map becomes once they are available.

The plateau low-source identity above gives (36.40) for all sources in $\mathcal{E}_<$. This equality is an equality of functions on the same finite-dimensional source space. By Hypothesis 36.15(b), the maps

$$J \longmapsto \varphi = \frac{\delta W[J]}{\delta J}$$

are locally invertible for both sides of (36.40). Since the functions W agree, their first derivatives and local inverse maps agree. Therefore the Legendre transforms on $\mathcal{E}_<$,

$$\Gamma^<[\varphi] = W[J(\varphi)] - \langle J(\varphi), \varphi\rangle,$$

agree on the corresponding low classical-field neighborhood. This proves equality of low-mode 1PI kernels at finite regulator. The object just constructed is the 1PI functional for the low test space; it is the object used by the projectors Π_I.

Theorem 32.6 proves, by the Hepp-sector Taylor-remainder estimates in Chapter 32, that at the fixed loop order ℓ every ultraviolet subtraction in the chosen massive BPHZ scheme $\mathfrak{b}$ is a Taylor polynomial in the external variables of the subgraph and that the subtracted graph has a finite $\Lambda_0 \to \infty$ limit. Thus, after the selected subtraction points, field normalization, and oversubtraction degrees are fixed, the projected BPHZ coordinates $g_I^{\mathfrak{b},(\le\ell)}(\mu)$ are finite Taylor coefficients of the same low-mode 1PI kernels obtained from the Wilsonian action. The $\Lambda_0 \to \infty$ limit here is not taken at fixed UV-cutoff Lagrangian coordinates: by Hypothesis 36.15(d), it is taken along the Polchinski-flow counterterm trajectory that holds the chosen renormalized boundary data fixed and has a limit for the Wilsonian kernels at the intermediate scale Λ. At fixed loop order this is precisely the perturbative renormalized trajectory constructed by the Wilsonian estimates, not a nonperturbative assertion about a cutoff coupling held fixed.

Computing the projected low kernels with the retained local action $L_\Lambda^{(N)}(u)$ is exactly the definition (36.42) of $M_I^{(\ell,N)}$.

It remains only to compare the exact Wilsonian action with its finite local projection. The graph expansion is first decomposed according to $L_\Lambda = L_\Lambda^{(N)} + v_{\perp,N}$. By Lemma 36.13, the BPHZ forest operation commutes with this finite additive vertex decomposition, and the subtracted graphs with at least one $v_{\perp,N}$-insertion are still graphs to which the Wilsonian Taylor-remainder norm estimate applies. Hypothesis 36.15(c) puts those subtracted graphs in the situation of Lemma 36.12. The lemma gives the displayed power bound for their contribution to the selected Taylor coefficients of low 1PI kernels. Substituting this estimate into the equality of finite-regulator low 1PI kernels, and then taking the tuned BPHZ/Polchinski $\Lambda_0 \to \infty$ limit at fixed Λ and μ, gives (36.43).

The final assertion is finite-dimensional. If J_{IA} is invertible, the map from the selected Wilsonian coordinates u_A, $A \in A_0$, to the selected 1PI coordinates has an invertible derivative at u_*. The implicit function theorem therefore gives the displayed local coordinate map G_A, with the same remainder estimate because the previous paragraph already controls the omitted Wilsonian coordinates.

The finite change of coordinates in Controlled Approximation 36.16 can be constructed recursively in the loop expansion. Write

$$M_I^{(\ell,N)}(u) = \sum_{r=0}^{\ell} \hbar^r M_{I,r}(u), \qquad u_A = \sum_{r=0}^{\ell} \hbar^r U_{A,r}, \qquad g_I = \sum_{r=0}^{\ell} \hbar^r g_{I,r}.$$

Choose the sublist A_0 with invertible tree-level Jacobian

$$J_{IA} = \left.\frac{\partial M_{I,0}}{\partial u_A}\right|_{u=U_0}, \qquad A \in A_0.$$

At order zero, solve $g_{I,0} = M_{I,0}(U_0)$. Suppose that $U_{A,s}$ has been determined for $s < r$. The coefficient of $\hbar^r$ in $M_I(u)$ has the form

$$[\hbar^r]\, M_I(u) = J_{IA} U_{A,r} + \mathcal{F}_{I,r}(U_0, \ldots, U_{r-1}; u_{\bar{A}}),$$

where $\mathcal{F}_{I,r}$ is computed from lower-order coordinate coefficients, the explicit r-loop low-mode 1PI graphs, and the retained coordinates $u_{\bar{A}}$ outside A_0. Therefore

$$U_{A,r} = (J^{-1})_{AI} \left[g_{I,r} - \mathcal{F}_{I,r}(U_0, \ldots, U_{r-1}; u_{\bar{A}}) \right].$$

This recursion is the perturbative coordinate map

$$u_A = G_A(g, u_{\bar{A}}; \Lambda/\mu, \mathfrak{b})$$

order by order. The coefficients $\mathcal{F}_{I,r}$ are finite because BPHZ supplies finite 1PI projectors at the chosen loop order, and the Wilsonian Taylor-remainder lemma controls the omitted local coordinates by the displayed power of μ/Λ.

Remark 36.17 (Scope of the matching implication). Controlled Approximation 36.16 is a fixed-loop perturbative consequence of a specified subtraction scheme, a specified Wilsonian norm, and the admissible plateau cutoff (36.38). The plateau is load-bearing: it makes $C_{>}J = 0$ on the low test space and hence makes (36.40) exact for the source-independent Wilsonian action used in the matching map. Non-plateau smooth profiles such as Gaussian cutoffs require either the source-dependent Wilsonian action (36.36) together with source-vertex coordinates, or a separate leakage theorem; they are not covered by Controlled Approximation 36.16 as stated.

The massive BPHZ scheme used here is the forest-formula scheme of Chapter 32. Massless theories require BPHZL or another infrared-safe subtraction scheme, and dimensional regularization with minimal subtraction is a different coordinate system whose relation to the present one is another finite matching map. Since BPHZ is a perturbative forest-formula construction, there is no separate "nonperturbative BPHZ construction" being compared here. The nonperturbative question is instead whether a Wilsonian or constructive continuum theory exists and whether its correlation functions have, in the chosen local coordinates, the BPHZ-renormalized series as their perturbative asymptotic expansion. That question requires existence of the relevant measures or functionals, Legendre transforms, and cutoff-uniform Banach estimates for the model under study.

The bounded-chart hypothesis specifies the class of trajectories being compared. For scalar quartic theory in four dimensions, for example, the nonperturbative continuum limit at fixed positive bare family is obstructed in the standard reflection-positive lattice universality classes discussed earlier. The Polchinski theorem used here is an order-by-order theorem in the renormalized coupling after the counterterms have been tuned; in that perturbative coefficientwise sense the retained coordinates remain in the required neighborhood at every fixed loop order.

Remark 36.18 (Status of the comparison). The exact finite-regulator input is (36.40), valid here because the cutoff has a low-momentum plateau and the source support is strictly inside it. The Legendre-transform step is a finite-dimensional local theorem under Hessian invertibility on the chosen low test space. The BPHZ input is the order-by-order massive forest-formula theorem proved in Chapter 32, with BPHZL or another infrared prescription required for massless exceptional regions. The Wilsonian input is an estimate in the chosen normed RG chart and along the tuned Polchinski-flow counterterm trajectory. Minimal subtraction, momentum subtraction, and Wilsonian coordinates are different coordinate systems; their comparison is the finite matching problem just constructed. A theorem connecting this perturbative comparison to an actual continuum QFT requires additional existence and uniform-estimate theorems for the model under study.

Remark 36.19 (Fixed-loop comparison of renormalizability criteria). In the massive, nonexceptional, finite-projector, plateau-cutoff, and tuned Polchinski-trajectory setting of Controlled Approximation 36.16, the following two renormalizability criteria are equivalent through the fixed loop order ℓ, up to the displayed $(\mu/\Lambda)^{p_N}$ error from omitted irrelevant Wilsonian coordinates.

(i) BPHZ renormalizability: after applying the forest formula in the chosen scheme $\mathfrak{b}$, all projected 1PI coordinates $g_I^{\mathfrak{b},(\leq\ell)}(\mu)$ have finite $\Lambda_0 \to \infty$ limits, and the required subtraction freedom is a finite list of local Taylor coefficients.

(ii) Polchinski-Wilsonian renormalizability: the Wilsonian trajectory is specified at scale Λ by a finite list of relevant or marginal local coordinates u_A, together with an irrelevant remainder whose norm obeys Lemma 36.12; the selected low 1PI coordinates are then finite functions of u_A after low-mode integration.

First, BPHZ subtraction data determine Wilsonian local coordinates by Proposition 36.11. The selected 1PI coordinates are then obtained by the finite matching map $M_I^{(\ell,N)}$. Conversely, Wilsonian data satisfying the stated norm bound determines finite BPHZ projected coordinates: apply the same projectors Π_I to the low-mode 1PI functional, and assign the resulting local Taylor parts to the BPHZ subtraction coordinates. A change of BPHZ subtraction point, oversubtraction degree, or finite Taylor part changes the Wilsonian description by a finite local coordinate transformation in the retained chart.

The two directions are obtained as follows. Assume first the BPHZ criterion. Proposition 36.11 identifies every forest Taylor subtraction through the selected loop order with a local vertex in the Wilsonian chart. Minimal subtractions occupy relevant and marginal coordinates, while oversubtractions occupy the explicitly retained higher-derivative local coordinates. After these local coordinates are fixed, Controlled Approximation 36.16 gives

$$g_I^{\mathfrak{b},(\leq\ell)}(\mu) = M_I^{(\ell,N)}(u(\Lambda);\Lambda,\mu,\mathfrak{b}) + R_I^{(\ell,N)}$$

with the stated bound on $R_I^{(\ell,N)}$. Thus the BPHZ finite local subtraction freedom is represented by the finite Wilsonian coordinate list, and the omitted Wilsonian coordinates affect the selected projectors only by the Taylor-remainder estimate.

Conversely, assume the Wilsonian criterion. The finite-regulator Gaussian identity constructs the same low-source connected functional as the original regulated theory, and the Hessian hypothesis in Controlled Approximation 36.16 constructs the low-mode Legendre transform. Applying the projectors Π_I to this low-mode 1PI functional gives finite numbers through loop order ℓ. For every superficially divergent graph contributing to these projectors, its local Taylor part is one of the finitely many local monomials in the retained Wilsonian chart, again by Proposition 36.11. Assigning those Taylor coefficients as BPHZ subtraction coordinates gives a BPHZ scheme whose projected coordinates agree with the Wilsonian values up to the same remainder estimate. The construction is recursive in loop order because the forest formula subtracts proper divergent subgraphs before the overall graph, whereas the Wilsonian matching recursion displayed above solves the local coordinate map at order r using only lower-order coordinate data and the explicit r-loop low-mode graphs.

Finally, the difference between two BPHZ subtraction choices is a finite sum of Taylor polynomials in external momenta. Fourier transforming these polynomials gives a finite sum of local vertices, hence a finite change of the retained Wilsonian coordinates. Where the Jacobian in Controlled Approximation 36.16 is invertible, this finite change is a local coordinate transformation by the implicit function theorem.

Remark 36.20 (Perturbative and continuum scopes). The preceding fixed-loop comparison is a two-way implication only after the finite-loop, massive, nonexceptional, and finite-projector hypotheses, together with the plateau-cutoff and tuned counterterm-trajectory hypotheses, of Controlled Approximation 36.16 have been imposed. In that setting, BPHZ forest-formula local Taylor parts construct the Wilsonian local coordinates, and controlled Wilsonian local coordinates construct BPHZ subtraction coordinates by projecting the low-mode 1PI functional. The result is an equivalence of finite coordinate descriptions of selected perturbative 1PI data, with the omitted irrelevant coordinates contributing the stated $(\mu/\Lambda)^{p_N}$ remainder.

The corresponding continuum-QFT problem has additional objects. For the model in question one must construct the regulated and limiting functionals, choose a topology on the full Wilsonian coordinate space, prove cutoff-uniform estimates in that topology, construct the Legendre transform on the required source domain, and then prove that the BPHZ-renormalized series is the perturbative asymptotic expansion of those continuum objects. BPHZ itself supplies the perturbative forest-formula side of this comparison; the existence-and-asymptotics problem belongs to the constructive or nonperturbative Wilsonian analysis of the model.

36.12 One-Loop Matching in the Quartic Scalar Theory

The matching map in Controlled Approximation 36.16 can be displayed directly in the four-point function of the massive scalar theory in four Euclidean dimensions. Work with

$$S_{\mathrm{int}}[\phi] = \frac{g_0}{4!} \int \mathrm{d}^4x\, \phi(x)^4$$

and a finite covariance $C_{\Lambda_0}(k)$. Let

$$C_{\Lambda_0}(k) = C_\Lambda(k) + C_>(k), \qquad C_>(k) = C_{\Lambda_0}(k) - C_\Lambda(k),$$

where C_Λ is the covariance of the remaining low field and $C_>$ is the covariance integrated out in the Wilsonian step from Λ_0 to Λ. For a four-point external configuration

$$\mathcal{P} = (p_1, p_2, p_3, p_4), \qquad \sum_{i=1}^{4} p_i = 0,$$

define the channel momenta

$$P_s = p_1 + p_2, \qquad P_t = p_1 + p_3, \qquad P_u = p_1 + p_4.$$

For any two covariances A, B, define the one-loop bubble functional

$$\mathcal{B}_{A,B}(\mathcal{P}) = \sum_{c=s,t,u} \int \frac{\mathrm{d}^4q}{(2\pi)^4}\, A(q)\, B(q + P_c).$$

Thus $\mathcal{B}_{C_{\Lambda_0},C_{\Lambda_0}}$ is the regulated full one-loop bubble sum, while

$$\mathcal{B}_{<,<} := \mathcal{B}_{C_\Lambda,C_\Lambda}, \qquad \mathcal{B}_{>,>} := \mathcal{B}_{C_>,C_>}$$

record the low-low and high-high parts of the loop. The remaining mixed piece is

$$\mathcal{B}_\times = \mathcal{B}_{C_\Lambda,C_>} + \mathcal{B}_{C_>,C_\Lambda}.$$

The exact finite-regulator identity is

$$\mathcal{B}_{C_{\Lambda_0},C_{\Lambda_0}} = \mathcal{B}_{<,<} + \mathcal{B}_{>,>} + \mathcal{B}_\times. \tag{36.44}$$

In the Wilsonian action, the high-high part appears directly in the quartic vertex. Let $u_4(\Lambda;\mathcal{P})$ be the coefficient whose contribution to L_Λ has four-field kernel u_4, so that the corresponding tree 1PI vertex is $-u_4$. To second order in g_0,

$$u_4(\Lambda;\mathcal{P}) = g_0 - \frac{g_0^2}{2}\, \mathcal{B}_{>,>}(\mathcal{P}) + u_{4,\mathrm{ct}}^{\mathrm{loc}}(\Lambda;\mathcal{P}) + O(g_0^3), \tag{36.45}$$

where $u_{4,\mathrm{ct}}^{\mathrm{loc}}$ denotes the local quartic counterterm and finite local convention chosen at the ultraviolet regulator. The factor $1/2$ is the symmetry factor of the bubble in the normalization $(g_0/4!)\phi^4$.

Figure 36.9 displays the actual one-loop partition used by the matching formula: the same bubble graph is evaluated with the four possible covariance assignments on its two internal lines. This is the graph-level content behind the algebraic identity (36.44).

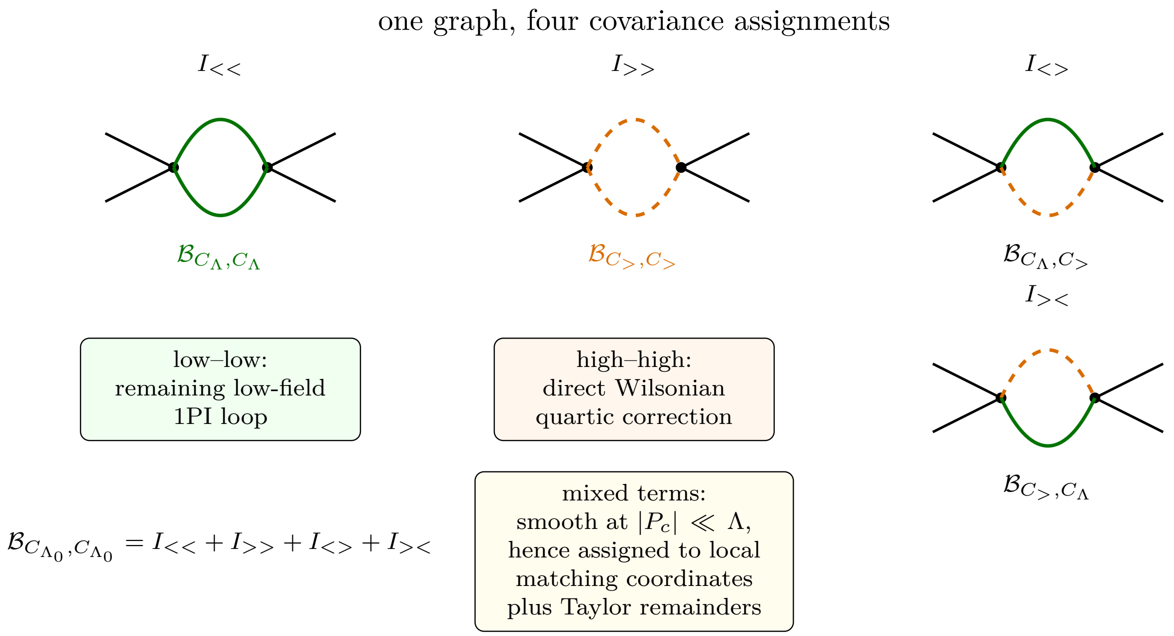


Figure 36.9: The covariance decomposition of the one-loop four-point bubble. Each panel is the same bubble graph with a different assignment of the two internal propagators to the low covariance C_Λ or the integrated covariance $C_>$. The $I_{>>}$ term is the high-momentum correction to the Wilsonian quartic vertex, $I_{<<}$ is generated by the later low-field 1PI loop, and the mixed assignments are smooth for external momenta $|P_c| \ll \Lambda$ and therefore enter the finite local matching map up to Taylor-remainder estimates.

The mixed term in (36.44) must be accounted for explicitly. In the exact Wilsonian functional it is not absent: it is produced by lower Wilsonian vertices generated by one high contraction and then used in the remaining low-field integration. If the external momenta are $O(\mu)$ with $\mu \ll \Lambda$, this contribution is smooth in the external invariants and has a Taylor expansion in powers of μ/Λ. In a finite derivative projection it is therefore part of the local matching map and of the first omitted irrelevant coordinates.

Let $\mathcal{S}_\mu$ be the symmetric Euclidean subtraction configuration used to define the 1PI quartic coordinate

$$g(\mu) = -\Gamma^{(4)}(\mathcal{S}_\mu).$$

For the logarithmically divergent one-loop four-point graph, the BPHZ overall subtraction is subtraction of the zeroth Taylor coefficient at $\mathcal{S}_\mu$. Thus, after the coupling counterterm has been fixed by the displayed condition,

$$\Gamma^{(4)}_{\mathrm{BPHZ}}(\mathcal{P}) = -g(\mu) + \frac{g(\mu)^2}{2}\left[\mathcal{B}_{C_{\Lambda_0},C_{\Lambda_0}}(\mathcal{P}) - \mathcal{B}_{C_{\Lambda_0},C_{\Lambda_0}}(\mathcal{S}_\mu)\right] + O(g^3), \tag{36.46}$$

with the limit $\Lambda_0 \to \infty$ taken after the subtraction. The difference in brackets has a finite limit at nonexceptional momenta.

The same finite-regulator 1PI kernel computed from the Wilsonian action is

$$\begin{aligned}\Gamma^{(4)}_{\Lambda,\mathrm{low}}(\mathcal{P}) = -u_4(\Lambda;\mathcal{P}) + \frac{u_4(\Lambda;\mathcal{S}_\mu)^2}{2}\mathcal{B}_{<,<}(\mathcal{P})\\ + \mathcal{M}_\times(\Lambda;\mathcal{P}) + O(u_4^3).\end{aligned} \tag{36.47}$$

Here $\mathcal{M}_\times$ is the contribution generated by the mixed covariance assignments, expressed in the low-field description. Its Taylor polynomial at $\mathcal{S}_\mu$ is local. Projecting (36.47) at $\mathcal{S}_\mu$ gives the one-loop matching map

$$g(\mu) = u_4(\Lambda;\mathcal{S}_\mu) - \frac{u_4(\Lambda;\mathcal{S}_\mu)^2}{2}\mathcal{B}_{<,<}(\mathcal{S}_\mu) - \Pi_{\mathcal{S}_\mu}\mathcal{M}_\times(\Lambda) + O(u_4^3), \tag{36.48}$$

where $\Pi_{\mathcal{S}_\mu}$ denotes evaluation of the four-point 1PI projector defining the coupling. If the Wilsonian derivative expansion is truncated after the local quartic coordinate, the finite matching map (36.42) shifts the retained coordinate by

$$-\Pi_{\mathcal{S}_\mu}\mathcal{M}_\times(\Lambda) + O(u_4^3).$$

If the first discarded Taylor coefficient in the four-point kernel has excess exponent p_N, the Taylor-remainder estimate of Lemma 36.12 gives the bound

$$C_N\|v_{\perp,N}\|_{\Lambda,m,a}\left(\frac{\mu}{\Lambda}\right)^{p_N}.$$

Subtracting the value at $\mathcal{S}_\mu$ from (36.47) and using the matching map removes the local coordinate ambiguity. The nonlocal low-momentum dependence is

$$\Gamma^{(4)}_{\Lambda,\mathrm{low}}(\mathcal{P}) - \Gamma^{(4)}_{\Lambda,\mathrm{low}}(\mathcal{S}_\mu) = \frac{g(\mu)^2}{2}\left[\mathcal{B}_{<,<}(\mathcal{P}) - \mathcal{B}_{<,<}(\mathcal{S}_\mu)\right] + \text{local Taylor terms} + O(g^3).$$

Adding the local high and mixed Taylor data reconstructs the low-momentum expansion of the BPHZ expression (36.46). This calculation displays the three roles separately: BPHZ removes the ultraviolet part of the full bubble by a local subtraction; the Wilsonian vertex stores the high-momentum part of the same graph; the 1PI coordinate is obtained only after the remaining low-field Legendre transform and the projector at $\mathcal{S}_\mu$.

The comparison proposition and the one-loop calculation also fix the role of beta functions. The BPHZ construction proves locality and finiteness of the subtracted 1PI kernels. The 1PI renormalization group differentiates finite coordinates $g_I(\mu)$ with respect to the subtraction scale μ. The Wilsonian flow differentiates the cutoff-dependent action L_Λ with respect to Λ. These two vector fields live on different coordinate spaces. They become comparable only after the matching map from L_Λ to the 1PI coordinates $g_I(\mu)$ has been specified. Under a finite change of coordinates, their components transform by the chain rule, as in the 1PI scheme-change discussion. Equality of physical low-momentum correlators is therefore a weaker statement than equality of the displayed beta-function components in two unmatched coordinate systems.

For gauge theories the same diagram of constructions must be lifted to the BRST/BV setting. The Wilsonian pushforward must preserve the quantum master equation, and the 1PI Legendre transform must satisfy the corresponding Slavnov–Taylor or BV master identity. That gauge-theoretic refinement is developed in the BV chapter; the present scalar discussion supplies the renormalization-theoretic comparison without adding gauge constraints.

Chapter 37

The Wilson-Fisher Fixed Point and Scaling Operators

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

37.1 Scalar Quartic Theory Below Four Dimensions

The trace identity makes fixed points structurally important. The next task is to construct the perturbative data of a non-Gaussian fixed point in a controlled small-ϵ calculation. Consider a single real scalar field in

$$D = 4 - \epsilon$$

Euclidean dimensions.

Definition 37.1 (Dimensional-regularized perturbative status)**.** The calculation in this chapter is a dimensionally regularized perturbative calculation. At each fixed loop order and for each fixed graph, the symbol

$$\int \frac{\mathrm{d}^{4-\epsilon}\ell}{(2\pi)^{4-\epsilon}}\, F(\ell, p, m)$$

denotes the meromorphic continuation in the complex variable $D = 4 - \epsilon$ of the corresponding rotationally invariant Euclidean loop-momentum integral, defined first in a domain where the graph integral or its Feynman-parameter representation converges. Subtractions are then applied coefficient by coefficient in the loop expansion. Thus the graph rules use analytic continuation of finite-dimensional momentum integrals; no field-configuration-space measure in noninteger dimension is being constructed.

The quantities $\beta^\epsilon(\lambda)$, $\gamma_\phi(\lambda)$, $\gamma_2(\lambda)$, $\lambda_*(\epsilon)$, and the scaling dimensions extracted from them are formal perturbative objects in a specified subtraction scheme. Before solving for the fixed point, the beta function and anomalous dimensions are formal power series in the dimensionless coupling λ, with coefficients meromorphic, and after subtraction regular, as $\epsilon \to 0$ in the displayed orders. The Wilson–Fisher fixed-point coordinate is obtained as a formal small-ϵ branch $\lambda_*(\epsilon) \in \epsilon\,\mathbb{R}[[\epsilon]]$. Substituting this branch into anomalous dimensions gives formal ϵ-expansions for Δ_ϕ, Δ_{ϕ^2}, and other scaling data. Convergence or summability of these formal series is an additional analytic question.

The local action density used to generate the perturbative graph rules is

$$\mathcal{L}_E = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \frac{1}{2}m_0^2\phi^2 + \frac{1}{4!}g_0\phi^4.$$

This action is invariant under the $\mathbb{Z}_2$ transformation $\phi \mapsto -\phi$. Unless an odd source is explicitly introduced, the renormalization-group chart used in this chapter is the $\mathbb{Z}_2$-even one. This convention matters for the count of relevant tuning conditions: inside the even chart the mass perturbation is the relevant deformation that must be tuned, while in the full local theory space an odd source for ϕ is also a relevant coordinate. The quartic coupling g_0 has mass dimension ϵ. The mass parameter m_0^2 must be tuned so that the infrared two-point function has no mass gap. This tuning is an infrared condition: it is not guaranteed by the algebra of counterterms, and it is meaningful only when the flow reaches a massless long-distance theory. The symbol m_0^2 here denotes the bare quadratic coordinate in the cutoff action, not the spectral pole mass of Definition 33.5. The massless Wilson–Fisher point is obtained by tuning this coordinate to the critical surface; massive departures from the fixed point are measured by a renormalized relevant coordinate, and only after reconstruction and spectral analysis can that coordinate be related to a physical pole or gap. The dimensionless quartic coordinate used below is therefore not the dimensionful coefficient g_0. In the notation of the minimal-subtraction chapter, g_0 is the bare $D = 4 - \epsilon$ coupling g^ϵ, and a renormalized quartic coordinate is obtained by factoring out the engineering dimension:

$$g_0 = \mu^\epsilon\left[\lambda(\mu) + \Delta_\lambda(\lambda(\mu),\epsilon)\right], \qquad g_R(\mu) := \mu^\epsilon\lambda(\mu), \qquad \Delta_\lambda = O(\lambda^2).$$

The symbol $\lambda(\mu)$ in the momentum-subtraction calculation immediately below denotes the dimensionless coordinate. The term $-\epsilon\lambda$ in the beta function is precisely the derivative of the factor $\mu^{-\epsilon}g_0$ at fixed bare coupling.

Definition 37.2 (Symmetric MOM quartic chart)**.** In the one-component $\mathbb{Z}_2$-even scalar theory, the symmetric Euclidean momentum-subtraction chart at scale $\mu > 0$ is the local renormalized coordinate system defined as follows. The dimensionless quartic coordinate is the 1PI coefficient

$$\lambda(\mu) = -\mu^{-\epsilon}Z(\mu)^2\,\Gamma^{(4)}\Big|_{k_i^2=\mu^2,\; s=t=u=-\frac{4}{3}\mu^2},$$

where

$$Z(\mu) = \left(1 - \frac{\partial\Sigma}{\partial k^2}\right)^{-1}_{k^2=\mu^2}.$$

The subtraction point is nonexceptional: no nonempty proper subset of the external momenta has vanishing total momentum. The mass coordinate of this chart is fixed separately by the endpoint condition that the infrared two-point function be massless. Thus the displayed $\lambda(\mu)$ is a coordinate on the massless endpoint slice; the mass tuning is the separate condition that places the trajectory on that slice.

If several schemes are present, the coordinates of this chart will be denoted $\lambda_{\mathrm{MOM}}(\mu)$ and $Z_{\mathrm{MOM}}(\mu)$; the lighter notation $\lambda(\mu)$ and $Z(\mu)$ is used here only while no pole-subtraction field factor is being discussed. In particular, $Z_{\mathrm{MOM}}(\mu)$ is a finite residue normalization at the subtraction momentum, whereas the minimal-subtraction field factor introduced below contains only pole counterterms in ϵ.

Lemma 37.3 (One-loop quartic vector field)**.** *In the chart of Definition 37.2, the one-loop dimensionally regularized quartic RG vector field is*

$$\beta^{\epsilon}(\lambda) = \frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = -\epsilon\lambda + \left(\frac{3}{16\pi^2} + O(\epsilon)\right)\lambda^2 + O(\lambda^3).$$

Consequently the nonzero perturbative zero in the one-coupling massless chart has the formal small-ϵ expansion

$$\lambda_*(\epsilon) = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2).$$

Proof. At one loop $Z(\mu) = 1$, and the bubble graph gives

$$\lambda(\mu) = \mu^{-\epsilon}\left[g_0 - \frac{3}{2}g_0^2(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\mu^2\right)^{-\epsilon/2} + O(g_0^3)\right].$$

Let

$$A_0(\epsilon) = -\frac{3}{2}(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\right)^{-\epsilon/2}.$$

Then the preceding equation has the form

$$\lambda(\mu) = \mu^{-\epsilon}g_0 + A_0(\epsilon)\,\mu^{-2\epsilon}g_0^2 + O(g_0^3).$$

Differentiating at fixed bare coupling gives

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = -\epsilon\,\mu^{-\epsilon}g_0 - 2\epsilon\,A_0(\epsilon)\,\mu^{-2\epsilon}g_0^2 + O(g_0^3).$$

To express the right-hand side in terms of $\lambda(\mu)$, use

$$\mu^{-\epsilon}g_0 = \lambda(\mu) - A_0(\epsilon)\lambda(\mu)^2 + O(\lambda^3), \qquad \mu^{-2\epsilon}g_0^2 = \lambda(\mu)^2 + O(\lambda^3).$$

Thus

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = -\epsilon\lambda - \epsilon\lambda^2\left[-\frac{3}{2}(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\right)^{-\epsilon/2}\right] + O(\lambda^3).$$

The pole in the bracket is multiplied by the engineering factor ϵ; the result is finite order by order in the small but nonzero parameter ϵ. Since

$$-\frac{3}{2}(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)}\left(\frac{4}{3}\right)^{-\epsilon/2} = -\frac{3}{16\pi^2}\frac{1}{\epsilon} + O(\epsilon^0),$$

the $D = 4 - \epsilon$ beta function is denoted β^ϵ:

$$\beta^\epsilon(\lambda) = \frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = -\epsilon\lambda + \left(\frac{3}{16\pi^2} + O(\epsilon)\right)\lambda^2 + O(\lambda^3).$$

This notation distinguishes the $4-\epsilon$-dimensional vector field from the strictly four-dimensional quartic beta function discussed in the 1PI RG chapter, which is obtained by setting the engineering term to zero. The zero of β^ϵ is

$$\lambda_* = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2).$$

For sufficiently small ϵ, the formal fixed-point coordinate lies in the perturbative domain in the following precise sense. The calculation constructs a formal power series

$$\lambda_*(\epsilon) = \sum_{r\geq 1} a_r\epsilon^r, \qquad a_1 = \frac{16\pi^2}{3},$$

whose coefficients are determined recursively from the beta function expanded as a formal series in λ and ϵ. At any finite loop order, one obtains a truncated polynomial vector field and a zero whose residual in the next omitted order is of higher order in ϵ. An exact RG flow or a nonperturbative continuum QFT with a fixed point at fixed positive ϵ requires additional input: for example a constructive Wilsonian or lattice continuum-limit theorem, or a summability theorem for the perturbative data together with a proof that the summed correlators satisfy the required locality, positivity, and regularity properties. In this chapter, unless a separate nonperturbative construction is explicitly stated, "the Wilson–Fisher fixed point" means this formal perturbative fixed-point data. The mass perturbation is a separate relevant direction and will be identified below from the ϕ^2 scaling operator. □

Remark 37.4 (Local uniqueness of the perturbative branch). The uniqueness statement supplied by this calculation is local and chart-dependent. Restrict to the one-component $\mathbb{Z}_2$-even scalar chart in which the mass has been tuned and the only displayed dimensionless interaction coordinate is λ. Put

$$\beta^\epsilon(\lambda) = -\epsilon\lambda + b\lambda^2 + O(\epsilon\lambda^2, \lambda^3), \qquad b = \frac{3}{16\pi^2}.$$

For a nonzero perturbative zero write $\lambda = \epsilon u$. Then

$$\epsilon^{-2}\beta^\epsilon(\epsilon u) = -u + bu^2 + O(\epsilon).$$

At $\epsilon = 0$ the nonzero root is

$$u_0 = b^{-1} = \frac{16\pi^2}{3},$$

and

$$\frac{\partial}{\partial u}(-u + bu^2)\bigg|_{u=u_0} = 1 \neq 0.$$

Equivalently, solving order by order in ϵ determines a unique formal branch $u(\epsilon) = u_0 + O(\epsilon)$ in a neighborhood of u_0. Together with the Gaussian branch $\lambda = 0$, this exhausts the small one-coupling branches visible in this local chart.

The scope is local uniqueness in the one-component $\mathbb{Z}_2$-even small-coupling scalar chart. Other perturbative fixed-point constructions belong to different charts or field contents: $O(N)$-vector

Wilson–Fisher fixed points, multicritical scalar fixed points such as the tricritical ϕ^6 branch near $D = 3 - \epsilon$, multi-coupling gauge or gauge–Yukawa fixed points of Banks–Zaks type, and decoupled product fixed points. Nonperturbative fixed points outside the small-coupling neighborhood require separate construction.

Figure 37.1 displays the local one-coordinate Wilson–Fisher construction: the symmetric Euclidean 1PI quartic vertex defines the coordinate, the beta function has the perturbative fixed point, and the critical trajectory is obtained after imposing the mass endpoint condition.

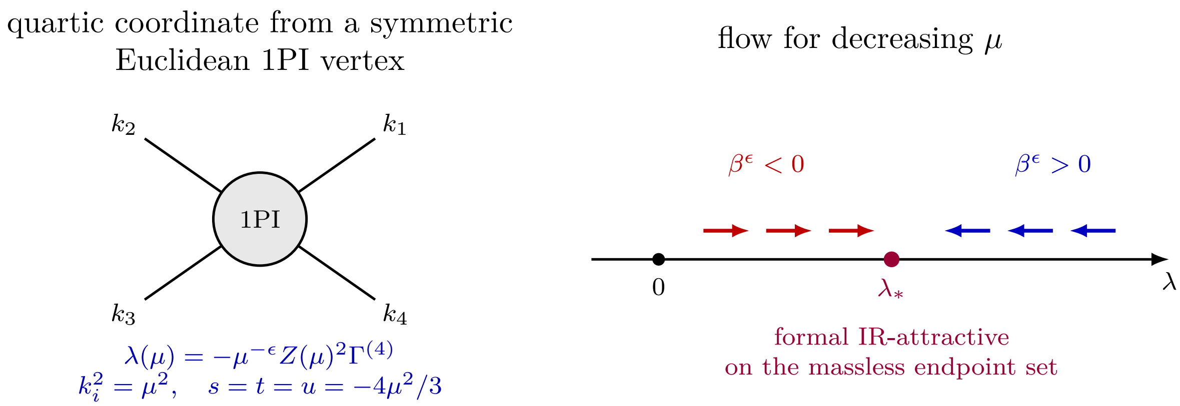


Figure 37.1: The dimensionless quartic coordinate is defined by a symmetric 1PI subtraction. In the formal $4-\epsilon$ perturbative vector field the classical term $-\epsilon\lambda$ and the one-loop term $3\lambda^2/(16\pi^2)$ balance at the nonzero zero λ_*. The displayed one-dimensional flow is the perturbative flow after the mass endpoint condition has been imposed.

The bare coupling and the renormalized coupling have different infrared behavior. The dimensionless bare coordinate $\mu^{-\epsilon}g_0$ diverges as $\mu \to 0$. In the formal perturbative RG equation, the renormalized coupling $\lambda(\mu)$, after the fluctuation correction in the 1PI coordinate definition is included, has the endpoint value $\lambda_*(\epsilon)$ along the massless critical trajectory.

Figure 37.2 separates these two coordinate behaviors: the dimensionless bare coordinate diverges under the infrared rescaling, while the renormalized coordinate approaches the Wilson–Fisher fixed-point value.

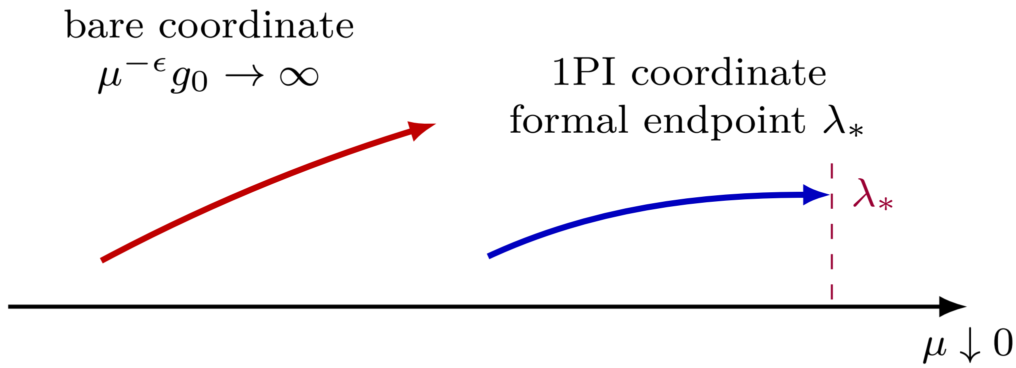


Figure 37.2: The finite infrared coupling in the formal perturbative flow is defined by the renormalized 1PI coordinate. The dimensionless bare coordinate grows without approaching a finite fixed-point value.

37.2 The First Scaling Operators

Definition 37.5 (Scaling operators in a local source-coordinate system)**.** Fix a renormalized local source-coordinate system near a fixed point. Thus one has a finite set of renormalized local fields O_A, with total derivatives, equation-of-motion fields, and the identity source either removed or placed in separate blocks, and a source deformation

$$\delta S = \sum_A \int \mathrm{d}^D x \, g^A(x) \, O_A(x).$$

For constant sources, let g^A denote the corresponding dimensionless coordinates and suppose that their RG equation has a linearization

$$\frac{\mathrm{d}g^A}{\mathrm{d}\log\mu} = -M^A{}_B \, g^B + O(g^2)$$

at the fixed point. A scaling source is an eigen-coordinate u^r of M. If

$$\frac{\mathrm{d}u^r}{\mathrm{d}\log\mu} = -y_r u^r + O(g^2),$$

then the dual scaling operator has dimension

$$\Delta_r = D - y_r.$$

Equivalently, under a change of physical scale $\mu \mapsto s\mu$ at the linearized fixed point, u^r scales as $s^{-y_r} u^r$. Relevant, marginal, and irrelevant directions are respectively those with $y_r > 0$, $y_r = 0$, and $y_r < 0$. This definition is local in the chosen source-coordinate system; if several fields with the same quantum numbers mix, the operators themselves are the eigenvectors of the mixing matrix, not the individual basis representatives.

The elementary field $O_1 = \phi$ has a fixed-point scaling dimension independent of finite analytic changes of the local field coordinate, but the anomalous-dimension function away from the fixed point depends on the renormalization chart. We compute it below in the minimal-subtraction pole chart. Let λ_{MS} be the dimensionless quartic coordinate in this chart, and let

$$\phi_0 = \left(Z_\phi^{\mathrm{MS}}\right)^{1/2} \phi_{\mathrm{MS}}, \qquad Z_\phi^{\mathrm{MS}}(\lambda_{\mathrm{MS}}, \epsilon) = 1 + \sum_{r\geq 1} \frac{Z_{\phi,r}(\lambda_{\mathrm{MS}})}{\epsilon^r},$$

where ϕ_0 is the regulated field appearing in the bare graph expansion and Z_ϕ^{MS} contains no finite part. The field anomalous dimension in this chart is

$$\gamma_\phi^{\mathrm{MS}}(\lambda_{\mathrm{MS}}) = \frac{1}{2}\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\log Z_\phi^{\mathrm{MS}}\bigg|_{g_0,\epsilon}.$$

Since $Z_\phi^{\mathrm{MS}} = 1 + O(\lambda_{\mathrm{MS}}^2)$ in scalar quartic theory, the field anomalous dimension starts only at order $\lambda_{\mathrm{MS},*}^2 = O(\epsilon^2)$. Thus

$$\Delta_\phi = \frac{D-2}{2} + \gamma_{\phi *}^{\mathrm{MS}} = 1 - \frac{\epsilon}{2} + O(\epsilon^2).$$

The conventional critical exponent η is defined by the long-distance two-point decay

$$\langle \phi(x)\phi(0)\rangle \sim |x|^{-D+2-\eta}.$$

Since a scalar primary of dimension Δ_ϕ has two-point exponent $2\Delta_\phi$, this convention gives

$$\eta = 2\Delta_\phi - (D-2) = 2\gamma_{\phi *}^{\mathrm{MS}}.$$

The first nonzero field anomalous dimension comes from the momentum-dependent part of the two-loop sunset self-energy. In the following pole computation write λ for λ_{MS}. With the interaction normalized as $\lambda\phi^4/4!$, the sunset graph has symmetry factor $1/6$, hence

$$\Sigma_{\mathrm{sun}}^{(2)}(k) = \frac{\lambda^2}{6}\,\mu^{2\epsilon} I(k), \qquad I(k) = \int_{p,q} \frac{1}{p^2 q^2 (p+q+k)^2}, \qquad \int_p := \int \frac{\mathrm{d}^{4-\epsilon}p}{(2\pi)^{4-\epsilon}}.$$

First integrate over q:

$$\int_q \frac{1}{q^2(q+p+k)^2} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)} \left((p+k)^2\right)^{-\epsilon/2}, \qquad D = 4-\epsilon.$$

The remaining massless two-point integral gives

$$\begin{aligned} I(k) &= \frac{(k^2)^{1-\epsilon}}{(4\pi)^D} \frac{\Gamma(1-\epsilon/2)^3\Gamma(-1+\epsilon)}{\Gamma(3-3\epsilon/2)} \\ &= -\frac{k^2}{2(4\pi)^4}\frac{1}{\epsilon} + \text{finite and higher-order terms in } \epsilon. \end{aligned}$$

The factor $1/2$ is the $\Gamma(3)^{-1}$ value in this last two-point integral, equivalently the corresponding Feynman-parameter normalization. Multiplying it by the sunset symmetry factor $1/6$ gives the coefficient $1/12$. Therefore the divergent part of the derivative is

$$\frac{\partial \Sigma^{(2)}(k)}{\partial k^2}\bigg|_{\mathrm{div}} = -\frac{\lambda^2}{12(16\pi^2)^2}\frac{1}{\epsilon},$$

where the self-energy sign convention is the same as in the momentum-subtraction chart, but the subtraction now retains only the pole part. Write

$$a(\lambda) = \frac{\lambda^2}{12(16\pi^2)^2}.$$

The pole part of the minimal-subtraction field factor is determined by the same residue condition:

$$Z_\phi^{\mathrm{MS}} = \Big[(1-\partial_{k^2}\Sigma)^{-1}\Big]_{\mathrm{pole}} = 1 - \frac{a(\lambda)}{\epsilon} + O(\lambda^3, \epsilon^0)$$

where $[\cdot]_{\mathrm{pole}}$ means that all finite terms are discarded. The sign in the anomalous dimension is then fixed by differentiating at fixed regulator definition. To the order needed here $\beta_{\mathrm{MS}}(\lambda) = -\epsilon\lambda + O(\lambda^2)$, so

$$\frac{1}{2}\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\log Z_\phi^{\mathrm{MS}} = \frac{1}{2}\beta_{\mathrm{MS}}(\lambda)\frac{\partial}{\partial\lambda}\left(-\frac{a(\lambda)}{\epsilon}\right) = a(\lambda) + O(\lambda^3).$$

Therefore

$$\gamma_\phi^{\mathrm{MS}}(\lambda) = \frac{\lambda^2}{12(16\pi^2)^2} + O(\lambda^3).$$

Substituting $\lambda_{\mathrm{MS},*} = (16\pi^2/3)\epsilon + O(\epsilon^2)$ gives

$$\eta = \frac{\epsilon^2}{54} + O(\epsilon^3)$$

in the interaction normalization $\lambda\phi^4/4!$. This value is not an artifact of the pole chart. If a second chart is obtained by a finite analytic coupling redefinition $\lambda' = \lambda + c_2\lambda^2 + O(\lambda^3)$ and the finite field-coordinate change $\phi' = F(\lambda)^{-1}\phi$, $F(0) = 1$, then

$$\gamma'_\phi(\lambda') = \gamma_\phi(\lambda) + \beta(\lambda)\frac{\partial \log F(\lambda)}{\partial\lambda}.$$

At an exact fixed point the second term vanishes. In the present ϵ-expansion, replacing λ_* by $\lambda'_* = \lambda_* + O(\lambda_*^2)$ changes $\gamma_\phi = A\lambda_*^2 + O(\lambda_*^3)$ only at order $O(\epsilon^3)$.

The operator $O_2 = \phi^2$ is more delicate. The constant source deformation

$$\delta S = \int \mathrm{d}^D x\, \delta g_2\, \phi^2(x)$$

corresponds to a zero-momentum insertion. In the massless theory, the one-loop graph then contains an infrared-divergent integral of the form

$$\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\frac{1}{(\ell^2)^2}.$$

The divergence is caused by the subtraction kinematics, not by the local UV definition of the operator. Equivalently, the scaling-degree extension of the coincident insertion controls the local counterterm ambiguity only after the massless infrared prescription has supplied a separated-point distribution to extend.

Use instead a spacetime-dependent source:

$$\delta S = \int \mathrm{d}^D x\, \delta g_2(x)\, O_2(x) = \int \frac{\mathrm{d}^D p}{(2\pi)^D}\, \delta\widetilde{g}_2(p)\, \widetilde{O}_2(-p).$$

Define the renormalized source by a nonzero-momentum 1PI insertion,

$$\delta\widetilde{g}_2(p;\mu) = -Z(\mu)\,(\text{amputated 1PI insertion of } O_2)\Big|_{k^2=(p+k)^2=\mu^2}.$$

The renormalized operator is determined by preserving the deformation:

$$\delta\widetilde{g}_2(p)\,\widetilde{O}_2(-p) = \delta\widetilde{g}_2(p;\mu)\,[\widetilde{O}_2(-p)]_\mu.$$

At $p^2 = \mu^2$, define

$$\gamma_2(\mu) = \frac{\mathrm{d}}{\mathrm{d}\log\mu}\log\left(\frac{\delta\widetilde{g}_2(p;\mu)}{\delta\widetilde{g}_2(p)}\right).$$

Figure 37.3 identifies the two insertion normalizations entering γ_2: the zero-momentum composite insertion and the fixed-momentum subtraction used to define the renormalized ϕ^2 coordinate.

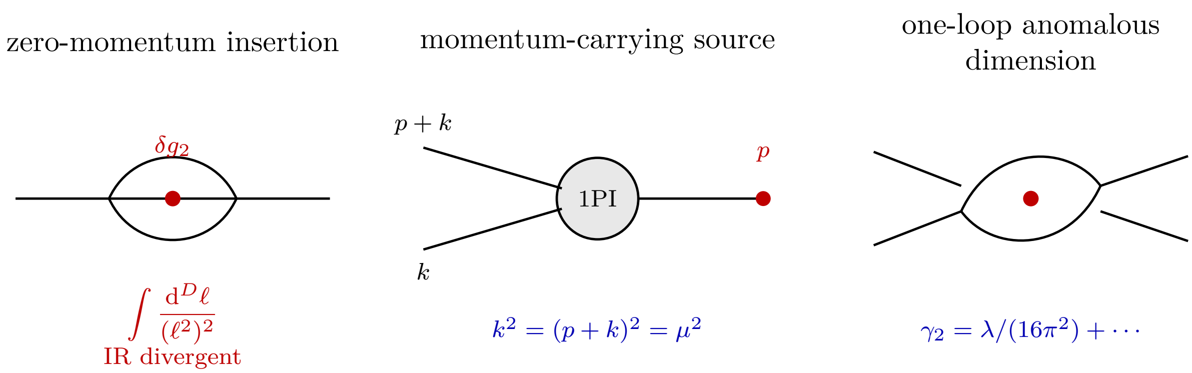


Figure 37.3: The ϕ^2 operator is renormalized by coupling it to a source with nonzero momentum. This avoids the infrared singularity of the zero-momentum subtraction and gives a finite definition of the anomalous dimension.

The one-loop source relation is

$$\delta\widetilde{g}_2(p;\mu) = \delta\widetilde{g}_2(p)\left[1 - \frac{g_4}{2}\int \frac{\mathrm{d}^{4-\epsilon}\ell}{(2\pi)^{4-\epsilon}}\frac{1}{\ell^2(p+\ell)^2} + O(g_4^2)\right],$$

or

$$\delta\widetilde{g}_2(p;\mu) = \delta\widetilde{g}_2(p)\left[1 - \frac{g_4}{2}(4\pi)^{-2+\epsilon/2}\frac{\Gamma(\epsilon/2)\Gamma(1-\epsilon/2)^2}{\Gamma(2-\epsilon)}|p|^{-\epsilon} + O(g_4^2)\right].$$

It follows that

$$\gamma_2(\mu) = \lambda(\mu)\left(\frac{1}{16\pi^2} + O(\epsilon)\right) + O(\lambda(\mu)^2).$$

At the Wilson-Fisher fixed point,

$$\gamma_{2*} = \frac{\epsilon}{3} + O(\epsilon^2), \qquad \Delta_{\phi^2} = (2-\epsilon) + \gamma_{2*} = 2 - \frac{2}{3}\epsilon + O(\epsilon^2).$$

Leading $N = 1$ scaling data. In the one-component scalar theory with interaction normalized as $\lambda\phi^4/4!$, the minimal-subtraction pole chart gives

$$\gamma_\phi(\lambda) = \frac{\lambda^2}{12(16\pi^2)^2} + O(\lambda^3), \qquad \gamma_2(\lambda) = \frac{\lambda}{16\pi^2} + O(\epsilon\lambda, \lambda^2).$$

On the nonzero small-ϵ fixed-point branch

$$\lambda_{\mathrm{MS},*} = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2),$$

the elementary field and the even quadratic scalar have dimensions

$$\Delta_\phi = 1 - \frac{\epsilon}{2} + \frac{\epsilon^2}{108} + O(\epsilon^3), \qquad \Delta_{\phi^2} = 2 - \frac{2}{3}\epsilon + O(\epsilon^2).$$

Equivalently,

$$\eta = 2\gamma_{\phi*} = \frac{\epsilon^2}{54} + O(\epsilon^3), \qquad y_t = D - \Delta_{\phi^2} = 2 - \frac{\epsilon}{3} + O(\epsilon^2).$$

These are the direct scaling consequences of the pole data already computed. The field statement follows from the pole part of the sunset self-energy:

$$\log Z_\phi^{\mathrm{MS}} = -\frac{\lambda^2}{12(16\pi^2)^2}\frac{1}{\epsilon} + O(\lambda^3).$$

Since the engineering part of the beta function is $-\epsilon\lambda + O(\lambda^2)$, differentiating $\frac{1}{2}\log Z_\phi^{\mathrm{MS}}$ at fixed regulated theory gives

$$\gamma_\phi(\lambda) = \frac{1}{2}\left(-\epsilon\lambda\right)\frac{\partial}{\partial\lambda}\left[-\frac{\lambda^2}{12(16\pi^2)^2}\frac{1}{\epsilon}\right] + O(\lambda^3) = \frac{\lambda^2}{12(16\pi^2)^2} + O(\lambda^3).$$

The quadratic-source statement follows from the nonexceptional insertion calculation: the factor 1/2 in the one-loop graph and the pole of the massless two-point integral give

$$\log Z_{2,\mathrm{src}}^{\mathrm{MS}} = -\frac{\lambda}{16\pi^2}\frac{1}{\epsilon} + O(\lambda\epsilon^0, \lambda^2),$$

and the same fixed-regulator differentiation gives $\gamma_2(\lambda) = \lambda/(16\pi^2) + O(\epsilon\lambda, \lambda^2)$. The one-loop fixed point is obtained by solving

$$-\epsilon\lambda + \frac{3\lambda^2}{16\pi^2} + O(\lambda^3) = 0$$

on the nonzero formal branch. Substitution into

$$\Delta_\phi = \frac{D-2}{2} + \gamma_{\phi*}, \qquad \Delta_{\phi^2} = D - 2 + \gamma_{2*}$$

then gives the displayed dimensions. The formula for η is the definition of the two-point exponent $2\Delta_\phi = D - 2 + \eta$, and $y_t = D - \Delta_{\phi^2}$ is Definition 37.5 applied to the source dual to $[\phi^2]$.

37.3 Two-Loop Pole Data and NLO Epsilon Expansion

The preceding calculation gives the fixed-point branch and the thermal exponent to leading order. The next coefficient of ν, and the next coefficient of the quartic irrelevant exponent, require the two-loop minimal-subtraction pole data. This section stays in the one-component $\mathbb{Z}_2$-even scalar theory. The $O(N)$ generalization is a different tensor combinatorics problem and is not folded into the notation here.

Set

$$x := \frac{\lambda_{\rm MS}}{16\pi^2}.$$

With the action normalized by $\lambda\phi^4/4!$, the local pole part of the two-loop four-point calculation is

$$\frac{g_0}{16\pi^2} = \mu^\epsilon \left[x + \frac{3x^2}{\epsilon} + x^3\left(\frac{9}{\epsilon^2} - \frac{17}{6\epsilon}\right) + O(x^4)\right]. \tag{37.1}$$

The coefficient 3 is the sum of the s, t, u one-loop bubble channels in the $1/4!$ vertex normalization. The double pole $9/\epsilon^2$ is not a new primitive datum: it is fixed by inserting the one-loop local counterterm into the one-loop four-point graph and by the forest consistency condition that the cutoff coupling be independent of μ. The new two-loop four-point information is the simple pole $-17x^3/(6\epsilon)$. In a forest subtraction this number is the sum of the primitive two-loop four-point pole and the local remnants of its subgraph subtractions, evaluated at any nonexceptional Euclidean subtraction configuration; locality of the pole part makes it independent of the chosen configuration.

For the momentum-carrying ϕ^2 source used above, let $Z_{2,\rm src}^{\rm MS}$ denote the minimal-subtraction source factor

$$Z_{2,\rm src}^{\rm MS} := \frac{\delta\widetilde{g}_2(p;\mu)}{\delta\widetilde{g}_2(p)}. \tag{37.2}$$

Its logarithm has the simple-pole coefficient

$$\log Z_{2,\rm src}^{\rm MS} = \frac{-x + \frac{5}{12}x^2}{\epsilon} + \text{double poles} + O(x^3). \tag{37.3}$$

The $-x/\epsilon$ term is the one-loop nonzero-momentum insertion computed above. The $5x^2/(12\epsilon)$ term is the forest-subtracted two-loop simple pole of the same insertion: the one-loop source counterterm removes the nested subdivergence, and the remaining local part is proportional to the same source operator. The double poles in (37.3) are determined by lower-order subdivergences and cancel out of the finite anomalous dimension after the pole equations are imposed.

For the elementary field, the two-loop sunset computation already gave

$$\log Z_\phi^{\rm MS} = -\frac{x^2}{12\epsilon} + O(x^3). \tag{37.4}$$

Only this two-loop field pole is needed for η through order ϵ^2. The next coefficient of η would require the three-loop simple pole in $Z_\phi^{\rm MS}$, so it is not part of a two-loop calculation.

Two-loop $N = 1$ Wilson–Fisher epsilon data. In the minimal-subtraction chart with $x = \lambda_{\rm MS}/(16\pi^2)$, the two-loop beta function and anomalous dimensions determined by (37.1)–(37.4) are

$$\beta_x^\epsilon(x) := \frac{\mathrm{d}x}{\mathrm{d}\log\mu} = -\epsilon x + 3x^2 - \frac{17}{3}x^3 + O(x^4), \tag{37.5}$$

$$\gamma_2(x) = x - \frac{5}{6}x^2 + O(x^3), \tag{37.6}$$

$$\gamma_\phi(x) = \frac{x^2}{12} + O(x^3). \tag{37.7}$$

The nonzero small-ϵ zero is

$$x_* = \frac{\epsilon}{3} + \frac{17}{81}\epsilon^2 + O(\epsilon^3). \tag{37.8}$$

Consequently

$$\eta = \frac{\epsilon^2}{54} + O(\epsilon^3), \tag{37.9}$$

$$\Delta_\phi = 1 - \frac{\epsilon}{2} + \frac{\epsilon^2}{108} + O(\epsilon^3), \tag{37.10}$$

$$\gamma_{2*} = \frac{\epsilon}{3} + \frac{19}{162}\epsilon^2 + O(\epsilon^3), \tag{37.11}$$

$$\Delta_{\phi^2} = 2 - \frac{2}{3}\epsilon + \frac{19}{162}\epsilon^2 + O(\epsilon^3), \tag{37.12}$$

$$y_t := D - \Delta_{\phi^2} = 2 - \frac{\epsilon}{3} - \frac{19}{162}\epsilon^2 + O(\epsilon^3), \tag{37.13}$$

$$\nu := \frac{1}{y_t} = \frac{1}{2} + \frac{\epsilon}{12} + \frac{7}{162}\epsilon^2 + O(\epsilon^3), \tag{37.14}$$

$$\omega := \left.\frac{\partial\beta_x^\epsilon}{\partial x}\right|_{x=x_*} = \epsilon - \frac{17}{27}\epsilon^2 + O(\epsilon^3). \tag{37.15}$$

The derivation is the finite algebra of minimal-subtraction pole equations and fixed-point substitution. If

$$\frac{g_0}{16\pi^2} = \mu^\epsilon\left[x + \frac{Ax^2}{\epsilon} + x^3\left(\frac{A^2}{\epsilon^2} + \frac{C}{\epsilon}\right) + O(x^4)\right],$$

then differentiating at fixed g_0 gives

$$0 = \epsilon F(x,\epsilon) + \beta_x^\epsilon(x)\,\partial_x F(x,\epsilon), \qquad F(x,\epsilon) = x + \frac{Ax^2}{\epsilon} + x^3\left(\frac{A^2}{\epsilon^2} + \frac{C}{\epsilon}\right).$$

Writing $\beta_x^\epsilon = -\epsilon x + \beta_1 x^2 + \beta_2 x^3 + O(x^4)$ and comparing powers of x, the x^2 term gives $\beta_1 = A$. The $1/\epsilon$ part of the x^3 equation cancels because the double pole is A^2, and the finite part gives $\beta_2 = 2C$. In (37.1), $A = 3$ and $C = -17/6$, proving (37.5).

For a source factor whose logarithm is

$$\log Z(x,\epsilon) = \frac{\ell_1(x)}{\epsilon} + \text{higher poles},$$

the finite anomalous dimension is obtained from $\beta_x^\epsilon \partial_x \log Z$. The pole equations make the higher-pole contribution cancel the divergent pieces; the finite part is

$$-x\,\ell_1'(x)$$

for a source renormalization and

$$-\frac{1}{2}x\,\ell_1'(x)$$

for the field factor, because $\phi_0 = (Z_\phi^{\mathrm{MS}})^{1/2}\phi_{\mathrm{MS}}$. Applying this to $\ell_1^{(2)}(x) = -x + 5x^2/12$ gives (37.6), while $\ell_1^{\phi}(x) = -x^2/12$ gives (37.7).

To solve for the fixed point, put

$$x_* = a\epsilon + b\epsilon^2 + O(\epsilon^3).$$

Substitution into (37.5) gives

$$0 = (-a + 3a^2)\epsilon^2 + \left(-b + 6ab - \frac{17}{3}a^3\right)\epsilon^3 + O(\epsilon^4).$$

The nonzero branch has $a = 1/3$, and then the ϵ^3 equation gives $b = 17/81$, proving (37.8).

Equations (37.9)–(37.12) are direct substitutions of x_* into (37.7) and (37.6). The thermal exponent is

$$y_t = (4-\epsilon) - \Delta_{\phi^2},$$

and the expansion of its reciprocal gives

$$\frac{1}{2 - \epsilon/3 - 19\epsilon^2/162} = \frac{1}{2} + \frac{\epsilon}{12} + \frac{7\epsilon^2}{162} + O(\epsilon^3).$$

Finally,

$$\partial_x\beta_x^{\epsilon} = -\epsilon + 6x - 17x^2 + O(x^3),$$

and substituting x_* gives

$$-\epsilon + 6\left(\frac{\epsilon}{3} + \frac{17}{81}\epsilon^2\right) - 17\left(\frac{\epsilon}{3}\right)^2 = \epsilon - \frac{17}{27}\epsilon^2 + O(\epsilon^3).$$

37.4 Critical Surface and the Mass Exponent

The mass perturbation is the deformation generated by the renormalized operator $[O_2]_\mu$. Let t be a renormalized coefficient for this deformation, so that near the fixed point

$$\delta S_t = \int \mathrm{d}^D x\, t\,[O_2]_\mu(x).$$

The dimension of t at the fixed point is

$$y_t = D - \Delta_{\phi^2}.$$

Using the result above,

$$y_t = (4-\epsilon) - \left(2 - \frac{2}{3}\epsilon + \frac{19}{162}\epsilon^2\right) + O(\epsilon^3) = 2 - \frac{\epsilon}{3} - \frac{19}{162}\epsilon^2 + O(\epsilon^3).$$

This identification uses the fact that the thermal source is already a linearized scaling coordinate in the $\mathbb{Z}_2$-even scalar sector at the order being computed. More explicitly, let g^A denote source coordinates for a finite list of even scalar local operators O_A, with the identity operator and equation-of-motion or total-derivative redundant directions removed. The linearized source RG at the fixed point has the form

$$\frac{\mathrm{d}g^A}{\mathrm{d}\log\mu} = -M^A{}_B\, g^B + O(g^2),$$

and the relevant exponents are the eigenvalues of $M^A{}_B$, after diagonalizing possible operator mixing. In the one-component scalar theory at this order, the even scalar sector containing the mass deformation has a single relevant nontrivial basis element, represented by $[O_2]_\mu = [\phi^2]_\mu$. The identity source is the vacuum-energy coordinate and ϕ^4 is the quartic direction, whose fixed-point eigenvalue is the positive irrelevant exponent $\omega = \epsilon - \frac{17}{27}\epsilon^2 + O(\epsilon^3)$. Hence the thermal block is one-dimensional and its eigenvalue is $D - \Delta_{\phi^2}$. In a theory with another even scalar of the same quantum numbers and degenerate dimension, t would have to be replaced by the corresponding eigen-coordinate of $M^A{}_B$, and the formula above would refer to that eigenvalue rather than to a chosen basis operator. The corresponding dimensionless coordinate

$$\tau(\mu) = \mu^{-y_t} t$$

obeys, to linear order at the fixed point,

$$\frac{\mathrm{d}\tau}{\mathrm{d}\log\mu} = -y_t\,\tau + O(\tau^2, \tau(\lambda - \lambda_*)).$$

Thus τ grows when μ is lowered. In the $\mathbb{Z}_2$-even finite chart and to linear order at the fixed point, the condition $\tau = 0$ is the codimension-one relevant-coordinate condition, with the quartic coordinate flowing to λ_* along it. The corresponding nonlinear statement that the tuned set is a C^k codimension-one submanifold is the finite-reference implicit-function construction of Chapter 36 applied to the $\mathbb{Z}_2$-even RG chart, after the endpoint map and its nondegeneracy have been established. Choosing m_0^2 so that the long-distance theory is massless is the microscopic way of imposing this endpoint condition.

If the $\mathbb{Z}_2$ symmetry is not imposed, an odd deformation

$$\delta S_h = \int \mathrm{d}^D x\, h\, [\phi]_\mu(x)$$

is present. Its fixed-point eigenvalue is

$$y_h = D - \Delta_\phi = 3 - \frac{\epsilon}{2} - \frac{\epsilon^2}{108} + O(\epsilon^3),$$

so h is also relevant. The statement that the massless critical surface has one relevant tuning condition is therefore a statement about the $\mathbb{Z}_2$-even subspace; in the local chart including h, both h and t must be set to zero to reach the symmetric Wilson-Fisher fixed point. A codimension statement again uses the Banach-chart hypotheses of the finite-reference implicit-function construction.

Let a trajectory start at a reference scale μ_0 with small nonzero $\tau_0 = \tau(\mu_0)$. Linearized running gives

$$\tau(\mu) = \tau_0 \left(\frac{\mu}{\mu_0}\right)^{-y_t}.$$

The scale at which $|\tau(\mu)|$ becomes order one sets the inverse correlation length:

$$\xi^{-1} \propto \mu_0 \, |\tau_0|^{1/y_t}.$$

Therefore the correlation-length exponent in this perturbative calculation is

$$\nu = \frac{1}{y_t} = \frac{1}{2} + \frac{\epsilon}{12} + \frac{7}{162}\epsilon^2 + O(\epsilon^3).$$

The proportionality constant in ξ^{-1} depends on the normalization of the thermal scaling field τ; the exponent does not.

Figure 37.4 records the $\mathbb{Z}_2$-even critical surface in the local coordinates used here: the thermal direction is relevant, while the critical trajectory flows into the interacting fixed point.

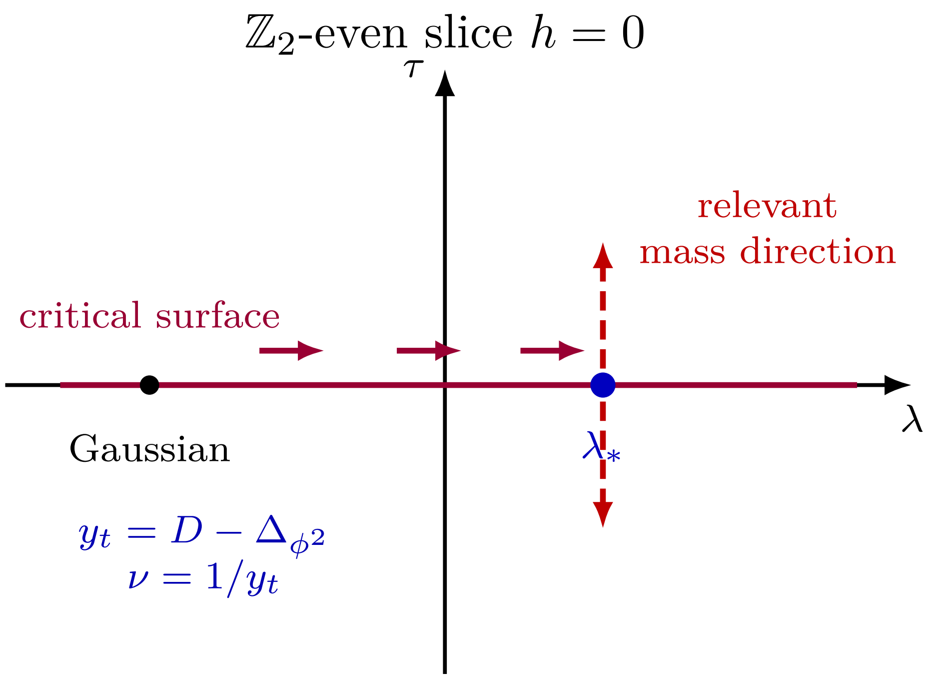


Figure 37.4: In the $\mathbb{Z}_2$-even finite chart, the massless endpoint condition removes the relevant thermal direction. The dashed transverse arrows show the mass perturbation, the direction dual to $[O_2]$, with eigenvalue $y_t = D - \Delta_{\phi^2}$.

37.5 Correlators and Local Composite Fields

At the fixed point, the running of the renormalized momentum-space operator is

$$[\widetilde{O}_2(p)]_{\mu=|p|} = \left(\frac{|p|}{\mu_0}\right)^{-\gamma_{2*}} \widetilde{O}_2(p),$$

where μ_0 is a fixed reference scale. The correlator of renormalized operators is finite and, by dimensional analysis,

$$\left\langle [\widetilde{O}_2(p)]_{\mu=|p|} [\widetilde{O}_2(q)]_{\mu=|q|} \right\rangle \propto \delta^{(4-\epsilon)}(p+q)\, |p|^{-\epsilon}.$$

Consequently

$$\mu_0^{2\gamma_{2*}} \left\langle \widetilde{O}_2(p)\widetilde{O}_2(q) \right\rangle \propto \delta^{(4-\epsilon)}(p+q)\, |p|^{-\epsilon+2\gamma_{2*}}.$$

Fourier transformation gives the position-space scaling law

$$\mu_0^{2\gamma_{2*}} \langle O_2(x)O_2(0)\rangle \propto \frac{1}{|x|^{2\Delta_{\phi^2}}}, \qquad \Delta_{\phi^2} = 2-\epsilon+\gamma_{2*}.$$

Figure 37.5 displays the invariant content of this calculation. Once a common normalization is chosen at $\mu_0|x| = 1$, the scaling dimension is the slope of the correlator on logarithmic axes. The anomalous dimension γ_{2*} changes that slope by $2\gamma_{2*}$, not by introducing a new scale.

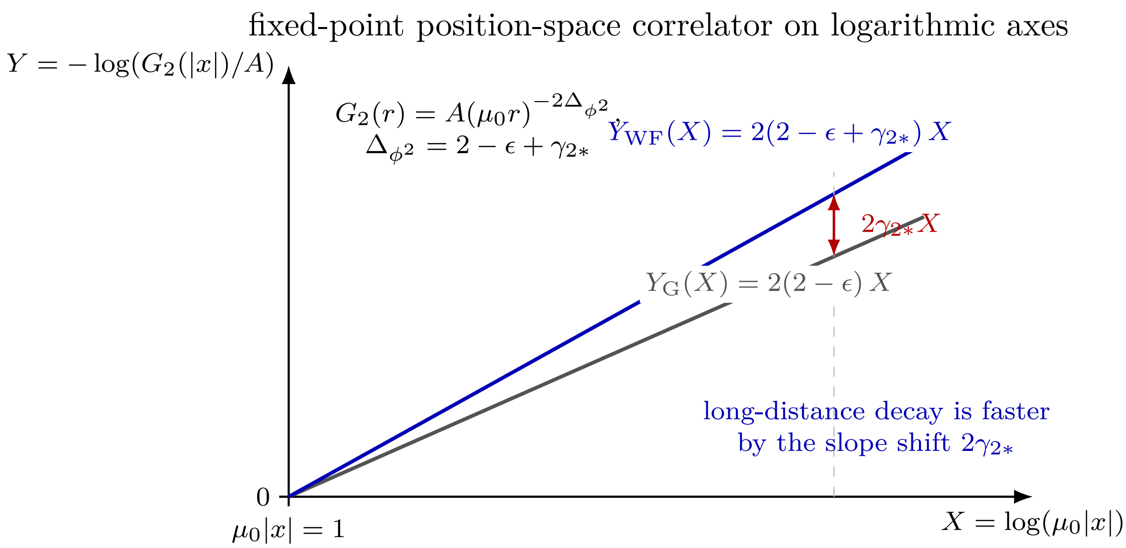


Figure 37.5: Logarithmic plot of the ϕ^2 two-point function at a fixed point, with $G_2(r) = \mu_0^{2\gamma_{2*}}\langle O_2(x)O_2(0)\rangle$, $r = |x|$, and a common normalization $G_2(1/\mu_0) = A$. The Gaussian scalar composite has slope $2(2-\epsilon)$. The Wilson–Fisher scaling operator has slope $2\Delta_{\phi^2} = 2(2-\epsilon+\gamma_{2*})$. Thus the anomalous dimension appears as a slope shift $2\gamma_{2*}$ in the log-log correlator.

There is a related locality point. The object obtained by choosing the subtraction scale separately for each Fourier mode,

$$[O_2(x)]_{\mathrm{mode}} = \int \frac{\mathrm{d}^{4-\epsilon}p}{(2\pi)^{4-\epsilon}}\, e^{ip\cdot x}\, [\widetilde{O}_2(p)]_{\mu=|p|},$$

is not a local field. By the running formula it equals

$$\int \frac{\mathrm{d}^{4-\epsilon}p}{(2\pi)^{4-\epsilon}}\, e^{ip\cdot x} \left(\frac{|p|}{\mu_0}\right)^{-\gamma_{2*}} \widetilde{O}_2(p),$$

and multiplication by the non-polynomial function $|p|^{-\gamma_{2*}}$ is nonlocal in position space. The local scaling field is obtained by fixing the reference normalization once and for all.

The notation $O_2 = \phi^2$ is a label for the renormalized local field in this operator family; it is not a definition by an unregulated product at coincident points. A point-splitting representative has the form

$$\mu_0^{\gamma_{2*}} O_2(x) \propto \lim_{x'\to x} |x'-x|^{2\gamma_{\phi*}-\gamma_{2*}} \left[\hat{\phi}(x')\hat{\phi}(x) - \left\langle \hat{\phi}(x')\hat{\phi}(x) \right\rangle\right],$$

where the subtraction removes the singular two-point function before the coincident-point limit is taken.

37.6 Descendants and Irrelevant Directions

The cubic operator is not independent at the interacting fixed point. In the renormalized operator algebra, the equation of motion gives a descendant relation with a coupling-dependent normalization. If the renormalized dimensionful quartic coupling is written as

$$g_R(\mu) = \mu^\epsilon \lambda(\mu),$$

then, away from contact terms and up to the overall sign convention for $\Box$, the renormalized equation of motion gives

$$\Box\phi(x) = \frac{g_R(\mu)}{3!}\,[\phi^3(x)]_\mu.$$

At the fixed point this is equivalently

$$[\phi^3(x)]_\mu = \frac{3!}{\mu^\epsilon \lambda_*}\,\Box\phi(x),$$

again modulo contact terms and finite normalizations of the chosen composite operator. If the proportionality coefficient is denoted by $\kappa_*(\mu)$,

$$[\phi^3(x)]_\mu = \kappa_*(\mu)\,\Box\phi(x),$$

then the action normalization above gives, at the classical level,

$$\kappa_{\rm cl}(\mu) = \frac{3!}{g_R(\mu)} = \frac{6}{g_R(\mu)}.$$

A perturbative finite normalization of the composite operator replaces this by

$$\kappa_*(\mu) = \frac{6}{g_R(\mu)}\big(1 + O(g_R(\mu))\big) = \frac{6}{g_R(\mu)} + O(1),$$

with the $O(1)$ term depending on the composite-operator convention. The statement invariant under such choices is that O_3 is an equation-of-motion descendant of $O_1 = \phi$. Its dimension is

$$\Delta_{\phi^3} = \Delta_\phi + 2.$$

The quartic operator $O_4 = \phi^4$ is instead the operator associated with motion in the quartic-coupling direction. As for O_2, one may first use a momentum-dependent source,

$$\delta S_4 = \int \mathrm{d}^D x\, \delta g_4(x)\, \phi^4(x) = \int \frac{\mathrm{d}^D p}{(2\pi)^D}\, \delta\widetilde{g}_4(p)\, \widetilde{O}_4(-p).$$

The renormalized source is extracted from a four-point 1PI insertion with external momenta of order μ and source momentum $p = O(\mu)$. In this case one may set $p = 0$: the massless infrared singularity that obstructed the zero-momentum O_2 subtraction is absent for this insertion. The momentum-dependent source definition therefore agrees with the earlier x-independent coupling-deformation definition. A small source perturbation obeys the linearized RG equation

$$\frac{\mathrm{d}\,\delta g_4(\mu)}{\mathrm{d}\log\mu} = \frac{\partial\beta^\epsilon(\lambda(\mu))}{\partial\lambda(\mu)}\,\delta g_4(\mu).$$

Therefore

$$\omega = \frac{\partial\beta^\epsilon(\lambda)}{\partial\lambda}\bigg|_{\lambda=\lambda_*} = \left(-\epsilon + \frac{3\lambda}{8\pi^2} - \frac{17\lambda^2}{(16\pi^2)^2} + O(\lambda^3)\right)\bigg|_{\lambda=16\pi^2\left(\frac{\epsilon}{3}+\frac{17}{81}\epsilon^2\right)+O(\epsilon^3)} = \epsilon - \frac{17}{27}\epsilon^2 + O(\epsilon^3),$$

which is positive. This positivity is the same statement as infrared attractiveness of the quartic direction when the flow is followed toward smaller μ. The scaling dimension is therefore

$$\Delta_{\phi^4} = D + \omega = 4 - \frac{17}{27}\epsilon^2 + O(\epsilon^3) > D.$$

Thus ϕ^4 is irrelevant at the interacting fixed point. This statement is about perturbations around the fixed point; it is compatible with the fact that the quartic interaction was needed to produce that fixed point.

Figure 37.6 summarizes the local operator-spectrum conclusion of this perturbative calculation: descendants, relevant primaries, and the quartic perturbation's irrelevant fixed-point eigenvalue occupy distinct roles.

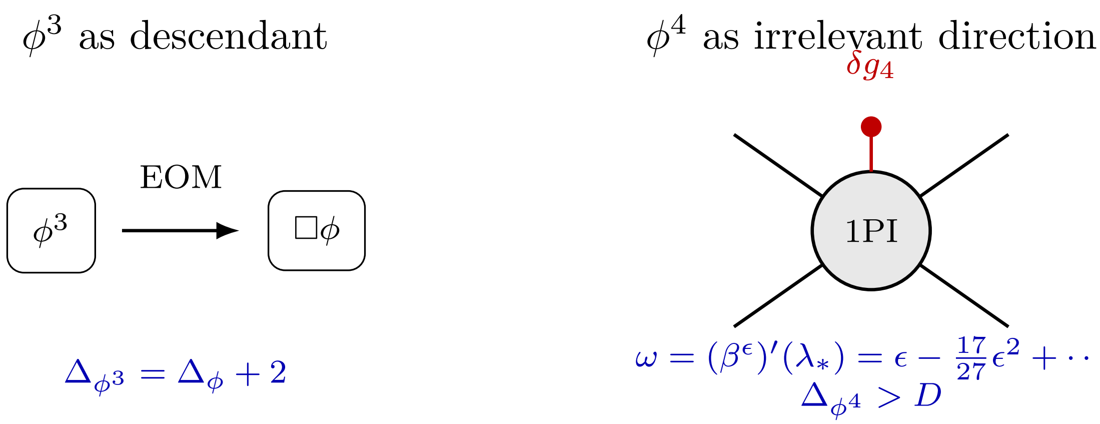


Figure 37.6: The cubic operator is an equation-of-motion descendant. The quartic operator is independent, but its dimension is larger than D at the interacting fixed point, so changing its coefficient is an irrelevant deformation.

The low-lying scalar dimensions are summarized as follows. In the third column, O_{irr} denotes the leading nonredundant $\mathbb{Z}_2$-even scalar irrelevant deformation after the identity and the energy operator have been removed. In the $4-\epsilon$ perturbative chart this direction is represented by ϕ^4; in lower-dimensional fixed-point CFTs the notation O_{irr} refers to the corresponding scaling operator, not to an unrenormalized fourth power of the spin field.

	Δ_ϕ	Δ_{ϕ^2}	$\Delta_{O_{\mathrm{irr}}}$
$d=4-\epsilon$	$1-\frac{\epsilon}{2}+\frac{\epsilon^2}{108}+O(\epsilon^3)$	$2-\frac{2}{3}\epsilon+\frac{19}{162}\epsilon^2+O(\epsilon^3)$	$4-\frac{17}{27}\epsilon^2+O(\epsilon^3)$
$d=3$	0.518148806(24)	1.41262528(29)	3.82951(61)
$d=2$	$\frac{1}{8}$	1	4

The first row is the perturbative epsilon expansion. The $d = 3$ and $d = 2$ rows are data about the corresponding fixed points themselves; they are not deduced by substituting $\epsilon = 1$ or $\epsilon = 2$ into the finite truncation displayed in the first row. In the $d = 3$ row the fixed-point fields are conventionally denoted $\sigma, \varepsilon, \varepsilon'$: ϕ, ϕ^2, ϕ^4 are the Landau representatives of the corresponding scaling directions, not definitions by unrenormalized powers at the fixed point. The displayed numerical values are conformal-bootstrap data for the three-dimensional Ising CFT. The quoted Δ_σ and Δ_ε are the mixed-correlator values reported in Chang, Dommes, Erramilli, Homrich, Kravchuk, Liu, Mitchell, Poland, and Simmons-Duffin, *Bootstrapping the 3d Ising Stress Tensor*, arXiv:2411.15300; earlier precision islands for these two dimensions were obtained by El-Showk, Paulos, Poland, Rychkov, Simmons-Duffin, and Vichi, *Phys. Rev. D* **86** (2012) 025022, arXiv:1203.6064; *J. Stat. Phys.* **157** (2014) 869–914, arXiv:1403.4545; and by Kos, Poland, Simmons-Duffin, and Vichi, *JHEP* **08** (2016) 036, arXiv:1603.04436. The value in the third column is the dimension of the leading $\mathbb{Z}_2$-even scalar irrelevant primary ε', whose Landau representative is the quartic direction; the number displayed here is the spectral estimate quoted in Simmons-Duffin, *JHEP* **03** (2017) 086, arXiv:1612.08471. The mixed-correlator crossing and positivity system underlying the $\Delta_\sigma, \Delta_\varepsilon$ island is derived from the higher-dimensional conformal-block conventions in Section 64.10; the numerical island itself remains external high-precision input rather than a theorem proved in this perturbative Wilson–Fisher chapter.

The $d = 2$ row is exact CFT data. The critical Ising CFT is the unitary minimal model $\mathcal{M}(3, 4)$ with Virasoro primaries σ and ε of weights

$$(h_\sigma, \bar{h}_\sigma) = \left(\frac{1}{16}, \frac{1}{16}\right), \qquad (h_\varepsilon, \bar{h}_\varepsilon) = \left(\frac{1}{2}, \frac{1}{2}\right),$$

so $\Delta_\sigma = h_\sigma + \bar{h}_\sigma = 1/8$ and $\Delta_\varepsilon = 1$. The scalar irrelevant entry of dimension 4 is the $T\bar{T}$ Virasoro descendant of the identity, the two-dimensional continuation of the leading even irrelevant lattice-scalar direction recorded in the third column. It is not the dimension of an unrenormalized σ^4 operator. These exact statements are the conformal-field-theoretic form of the Onsager critical Ising solution, and follow from the minimal-model classification of Belavin, Polyakov, and Zamolodchikov.

37.7 The $O(N)$ Wilson–Fisher Family

Definition 37.6 ($O(N)$ quartic scalar chart)**.** For a positive integer N, let $\phi = (\phi^1, \ldots, \phi^N)$ be an $O(N)$-vector scalar. Indices are raised and lowered with δ_{ij}, repeated $O(N)$ indices are summed, and

$$\phi^2 := \delta_{ij}\phi^i\phi^j.$$

The $O(N)$-invariant quartic theory in $D = 4 - \epsilon$ is

$$S_E[\phi] = \int \mathrm{d}^D x \left[\frac{1}{2}\partial_\mu\phi^i\partial_\mu\phi^i + \frac{1}{2}r_0\,\phi^i\phi^i + \frac{g_0}{4!}\,(\phi^i\phi^i)^2\right]. \tag{37.16}$$

The quartic vertex tensor associated with the normalization in (37.16) is

$$V_{ij;kl} = \frac{g_0}{3}\left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right). \tag{37.17}$$

For $N = 1$ the tensor in parentheses equals 3, so this is the same $g_0\phi^4/4!$ normalization used above. The singlet quadratic source is coupled to

$$S(x) := \frac{1}{2}\phi^i(x)\phi^i(x).$$

The factor $1/2$ is a source convention; it does not affect the scaling dimension.

Set again

$$x := \frac{\lambda_{\mathrm{MS}}}{16\pi^2}.$$

The one-component theory is the $N = 1$ member of this family. In the minimal-subtraction chart determined by the singlet quartic vertex (37.17), the two-loop local pole map is

$$\frac{g_0}{16\pi^2} = \mu^\epsilon\left[x + \frac{N+8}{3}\frac{x^2}{\epsilon} + x^3\left\{\frac{(N+8)^2}{9\epsilon^2} - \frac{3N+14}{6\epsilon}\right\} + O(x^4)\right]. \tag{37.18}$$

The coefficient $N + 8$ has a concrete origin. In a one-loop four-point graph, the closed internal index loop gives the N term, while the exchange contractions among the three tensor structures in (37.17) give the 8 term. The double pole in (37.18) is fixed by this one-loop tensor contraction and the forest equations. The new two-loop tensor datum is the simple-pole coefficient $-(3N+14)/6$.

The singlet source and field pole factors are

$$\log Z^{\mathrm{MS}}_{S,\mathrm{src}} = \frac{-\frac{N+2}{3}x + \frac{5(N+2)}{18}x^2}{\epsilon} + \text{double poles} + O(x^3), \tag{37.19}$$

$$\log Z^{\mathrm{MS}}_{\phi} = -\frac{N+2}{36}\frac{x^2}{\epsilon} + O(x^3). \tag{37.20}$$

The factor $N + 2$ is the index contraction of a singlet ϕ^2 insertion with the quartic tensor. The same pole-equation argument used in the one-component calculation gives the finite anomalous dimensions.

$O(N)$ **Wilson–Fisher data through two loops.** In the MS coordinate $x = \lambda_{\mathrm{MS}}/(16\pi^2)$, the singlet $O(N)$-invariant quartic chart has

$$\beta^\epsilon_x = -\epsilon x + \frac{N+8}{3}x^2 - \frac{3N+14}{3}x^3 + O(x^4), \tag{37.21}$$

$$\gamma_S(x) = \frac{N+2}{3}x - \frac{5(N+2)}{18}x^2 + O(x^3), \tag{37.22}$$

$$\gamma_\phi(x) = \frac{N+2}{36}x^2 + O(x^3). \tag{37.23}$$

The nonzero perturbative fixed-point coordinate is

$$x_* = \frac{3}{N+8}\epsilon + \frac{9(3N+14)}{(N+8)^3}\epsilon^2 + O(\epsilon^3). \tag{37.24}$$

The singlet thermal exponent, field anomalous dimension, and leading quartic irrelevant exponent are

$$\eta = \frac{N+2}{2(N+8)^2}\epsilon^2 + O(\epsilon^3), \tag{37.25}$$

$$\Delta_\phi = 1 - \frac{\epsilon}{2} + \frac{N+2}{4(N+8)^2}\epsilon^2 + O(\epsilon^3), \tag{37.26}$$

$$\Delta_S = 2 - \frac{6}{N+8}\epsilon + \frac{(N+2)(13N+44)}{2(N+8)^3}\epsilon^2 + O(\epsilon^3), \tag{37.27}$$

$$y_t := D - \Delta_S = 2 - \frac{N+2}{N+8}\epsilon - \frac{(N+2)(13N+44)}{2(N+8)^3}\epsilon^2 + O(\epsilon^3), \tag{37.28}$$

$$\nu := \frac{1}{y_t} = \frac{1}{2} + \frac{N+2}{4(N+8)}\epsilon + \frac{(N+2)(N^2+23N+60)}{8(N+8)^3}\epsilon^2 + O(\epsilon^3), \tag{37.29}$$

$$\omega := \partial_x \beta_x^\epsilon|_{x=x_*} = \epsilon - \frac{3(3N+14)}{(N+8)^2}\epsilon^2 + O(\epsilon^3). \tag{37.30}$$

This is the $O(N)$ tensor-combinatoric analogue of the previous minimal-subtraction calculation. Let

$$A = \frac{N+8}{3}, \qquad C = -\frac{3N+14}{6}.$$

Equation (37.18) has the same algebraic form as the one-component pole map:

$$\frac{g_0}{16\pi^2} = \mu^\epsilon \left[x + \frac{Ax^2}{\epsilon} + x^3\left(\frac{A^2}{\epsilon^2} + \frac{C}{\epsilon}\right) + O(x^4) \right].$$

Differentiation at fixed g_0 gives $\beta_x^\epsilon = -\epsilon x + Ax^2 + 2Cx^3 + O(x^4)$, which is (37.21). The simple-pole coefficient in (37.19) is

$$\ell_1^S(x) = -\frac{N+2}{3}x + \frac{5(N+2)}{18}x^2.$$

The finite source anomalous dimension is $-x(\ell_1^S)'(x)$, up to terms beyond two loops, giving (37.22). Similarly the field simple-pole coefficient $\ell_1^\phi(x) = -(N+2)x^2/36$ gives $-\frac{1}{2}x(\ell_1^\phi)'(x)$, namely (37.23).

Solving $\beta_x^\epsilon(x_*) = 0$ with $x_* = a\epsilon + b\epsilon^2 + O(\epsilon^3)$ gives

$$a = \frac{1}{A} = \frac{3}{N+8}, \qquad b = -\frac{2C}{A^3} = \frac{9(3N+14)}{(N+8)^3},$$

which proves (37.24). Substituting (37.24) into (37.23) gives $\eta = 2\gamma_\phi(x_*)$, hence (37.25) and (37.26). Substitution into (37.22) gives

$$\gamma_S(x_*) = \frac{N+2}{N+8}\epsilon + \frac{(N+2)(13N+44)}{2(N+8)^3}\epsilon^2 + O(\epsilon^3).$$

Since the singlet S has engineering dimension $D-2 = 2-\epsilon$, $\Delta_S = 2 - \epsilon + \gamma_S(x_*)$. Equations (37.27)–(37.29) follow by expanding $y_t = D - \Delta_S = 2 - \gamma_S(x_*)$ and then its reciprocal. Finally,

$$\partial_x \beta_x^\epsilon = -\epsilon + \frac{2(N+8)}{3}x - (3N+14)x^2 + O(x^3),$$

and substituting (37.24) gives (37.30).

For $N = 1$, the $O(N)$ two-loop data reduce to the one-component formulas above. For $N > 1$ there are additional non-singlet operator sectors. The symmetric traceless tensor

$$T^{ij} := \phi^i \phi^j - \frac{\delta^{ij}}{N} \phi^k \phi^k$$

and the vector ϕ^i have their own source-RG blocks. The formulas above concern the singlet thermal direction and the elementary vector field. A complete operator-spectrum analysis must diagonalize each $O(N)$ representation sector separately.

37.7.1 Large N, the Spherical Model, and the Sigma-Model Chart

The limit $N \to \infty$ is not obtained by holding x fixed. The fixed point in (37.24) has $x_* = O(1/N)$, while the collective coordinate

$$u := Nx$$

has a finite limit. Multiplying (37.21) by N and putting $x = u/N$ gives

$$\beta_u^{\epsilon} = -\epsilon u + \frac{1}{3} u^2 + O\left(\frac{1}{N}\right), \qquad u_* = 3\epsilon + O\left(\frac{1}{N}\right). \tag{37.31}$$

Thus the quartic coordinate of each finite-N fixed point tends to zero, but the singlet collective interaction stays finite.

The nonperturbative large-N limit is best derived before expanding around four dimensions. Work with a finite ultraviolet cutoff and an $O(N)$-invariant potential of the form

$$S[\phi] = \int \mathrm{d}^D x \left[\frac{1}{2} \partial_\mu \phi^i \partial_\mu \phi^i + N\, U(\rho)\right], \qquad \rho := \frac{\phi^i \phi^i}{N}. \tag{37.32}$$

At finite cutoff, introduce fields $\rho(x)$ and $\sigma(x)$ by a Fourier representation of the constraint $\rho(x) = \phi^i(x)\phi^i(x)/N$. The contour of σ is chosen so that the finite-dimensional regulated integral is convergent and may be deformed to the steepest-descent contour. After integrating over the N vector components, the effective action per component is

$$\frac{1}{N} S_{\text{eff}}[\rho, \sigma] = \int \mathrm{d}^D x \left[U(\rho) - \frac{1}{2} \sigma \rho\right] + \frac{1}{2} \operatorname{Tr}_\Lambda \log(-\partial^2 + \sigma). \tag{37.33}$$

For constant saddle fields, write $\sigma = m^2$. The saddle equations are

$$U'(\rho) = \frac{1}{2} m^2, \qquad \rho = \Omega_D(m), \qquad \Omega_D(m) := \int^\Lambda \frac{\mathrm{d}^D p}{(2\pi)^D} \frac{1}{p^2 + m^2}, \tag{37.34}$$

in the symmetric phase. If an ordered component $\langle \phi^1 \rangle = M$ is allowed, the second equation becomes

$$\rho = \frac{M^2}{N} + \Omega_D(m), \qquad m^2 M = 0.$$

The critical point is obtained by tuning the relevant parameter in U so that $m = 0$ is a saddle:

$$\rho_c = \Omega_D(0), \qquad U'(\rho_c) = 0.$$

For $2 < D < 4$, the small-m expansion of the regulated integral has a universal nonanalytic term

$$\Omega_D(m) - \Omega_D(0) = -K_D\, m^{D-2} + O(m^2\Lambda^{D-4}), \qquad K_D := -\frac{\Gamma(1-D/2)}{(4\pi)^{D/2}} > 0. \tag{37.35}$$

The sign is fixed by the fact that $\Omega_D(m)$ decreases with m. If τ is the linear thermal displacement from the critical saddle, then

$$\tau \propto m^{D-2}, \qquad \xi = m^{-1},$$

and therefore

$$\nu_{N=\infty} = \frac{1}{D-2}. \tag{37.36}$$

This agrees with the $N \to \infty$ limit of (37.29) after setting $D = 4 - \epsilon$:

$$\frac{1}{D-2} = \frac{1}{2-\epsilon} = \frac{1}{2} + \frac{\epsilon}{4} + \frac{\epsilon^2}{8} + O(\epsilon^3).$$

At the critical saddle the vector two-point function is $1/p^2$, so

$$\Delta_\phi = \frac{D-2}{2}, \qquad \eta_{N=\infty} = 0.$$

The singlet thermal operator is represented at large N by the auxiliary field σ, not by an unrenormalized pointwise product. Its quadratic kernel contains the bubble

$$B_D(p) := \int \frac{\mathrm{d}^D q}{(2\pi)^D}\frac{1}{q^2(q+p)^2} = b_D\,|p|^{D-4}, \qquad b_D = \frac{\Gamma(2-D/2)\Gamma(D/2-1)^2}{(4\pi)^{D/2}\Gamma(D-2)}. \tag{37.37}$$

For $2 < D < 4$, this term dominates over the local $U''(\rho_c)$ term at small momentum. Hence the σ propagator scales as

$$\langle \sigma(p)\sigma(-p)\rangle \propto |p|^{4-D}.$$

A scalar operator of dimension Δ has momentum-space two-point scaling $|p|^{2\Delta-D}$. Thus

$$\Delta_\sigma = 2, \qquad y_t = D - \Delta_\sigma = D - 2, \tag{37.38}$$

which is the same content as (37.36).

Example 37.7 (Large-N critical saddle)**.** Consider the finite-cutoff $O(N)$-invariant scalar model (37.32), with U tuned so that the translation-invariant large-N saddle has $m = 0$. Assume $2 < D < 4$, the cutoff is removed only after the saddle equations have been expressed in terms of a fixed finite critical displacement τ, and the saddle remains on a steepest-descent contour obtained from the regulated finite-dimensional integral. Then at $N = \infty$

$$\Delta_\phi = \frac{D-2}{2}, \qquad \eta = 0, \qquad \Delta_\sigma = 2, \qquad \nu = \frac{1}{D-2}.$$

At the critical saddle the inverse vector propagator is p^2, hence the vector two-point function scales as p^{-2}. Comparing with the momentum-space scaling law $|p|^{2\Delta_\phi - D}$ gives $2\Delta_\phi - D = -2$, or $\Delta_\phi = (D-2)/2$, and therefore $\eta = 0$.

The thermal displacement is measured by the saddle mass m. For $2 < D < 4$, the cutoff integral has the nonanalytic small-m expansion

$$\Omega_D(m) - \Omega_D(0) = -K_D m^{D-2} + O(m^2\Lambda^{D-4}), \qquad K_D > 0,$$

so a finite critical displacement satisfies $\tau \propto m^{D-2}$. Since the correlation length of the vector propagator is $\xi = m^{-1}$, this gives $\xi \propto |\tau|^{-1/(D-2)}$, hence $\nu = 1/(D-2)$.

Finally, the quadratic kernel of the collective field σ is dominated in the infrared by the bubble

$$B_D(p) = b_D |p|^{D-4}$$

rather than by the local term. Its propagator therefore scales as $|p|^{4-D}$. Comparing $|p|^{4-D}$ with $|p|^{2\Delta_\sigma - D}$ gives $\Delta_\sigma = 2$. The source dual to σ is the thermal source, so $y_t = D - \Delta_\sigma = D - 2$, consistently reproducing the value of ν just derived.

The first nonzero field anomalous dimension is an $O(1/N)$ effect from exchange of the collective σ field; the present saddle calculation establishes the exact $N = \infty$ dimensions and gives the expansion point for the systematic $1/N$ diagrammatics.

Equations (37.34) are the continuum field-theory form of the spherical model: the fluctuating N-component spin length is replaced, at $N = \infty$, by a saddle constraint on $\rho = \phi^2/N$. This is a statement about the regulated large-N limit and its continuum critical saddle, not an independent postulate about universality.

There is also a lower-critical-dimension chart. The $O(N)$ nonlinear sigma model with field $n^i(x)$ constrained by $n^i n^i = N$ has regulated action

$$S_\sigma[n, \alpha] = \frac{1}{2T} \int \mathrm{d}^D x \, \left[\partial_\mu n^i \partial_\mu n^i + \alpha(n^i n^i - N)\right]. \tag{37.39}$$

At $N = \infty$, integrating over $N - 1$ components gives the same saddle equations as above, with

$$T_c = \Omega_D(0)^{-1}, \qquad \Omega_D(0) - \Omega_D(m) = K_D m^{D-2} + \cdots .$$

Consequently the sigma model has the same large-N critical exponent $\nu = 1/(D-2)$ for $2 < D < 4$. Near $D = 2 + \widetilde{\epsilon}$, the usual dimensionless sigma-model coupling $\mathfrak{g}$ obeys, for $N > 2$,

$$\beta_{\mathfrak{g}} = \widetilde{\epsilon}\,\mathfrak{g} - \frac{N-2}{2\pi}\mathfrak{g}^2 + O(\mathfrak{g}^3), \qquad \mathfrak{g}_* = \frac{2\pi}{N-2}\widetilde{\epsilon} + O(\widetilde{\epsilon}^2). \tag{37.40}$$

This fixed point is the $2+\widetilde{\epsilon}$ perturbative chart for the same $O(N)$ critical family. The $N = 2$ model is exceptional: the coefficient $N-2$ vanishes, and the Berezinskii–Kosterlitz–Thouless mechanism is not captured by (37.40).

More generally, the Wilson–Fisher epsilon expansion is an asymptotic small-ϵ expansion unless an additional summability theorem is specified. Large-order analyses of related perturbative ϕ^4 series exhibit factorial coefficient growth, and no summability theorem is being assumed here for the displayed epsilon series. Consequently a finite number of terms should be read as asymptotic information about the fixed point near four dimensions. Extracting accurate $d = 3$ numbers from the epsilon series requires an extra resummation prescription, for example a Borel or conformal-Borel method together with assumptions about singularities of the Borel transform; it is not the same operation as evaluating a convergent Taylor series at $\epsilon = 1$.

37.8 Scaling Coordinates as a Local RG Chart

The preceding computations identify individual scaling operators. For the universality discussion, the corresponding coordinate statement is needed. Fix a finite projection of the local action space near a fixed point, after quotienting by total derivatives and equation-of-motion redundancies in the chosen renormalized-operator scheme. Let $g^\alpha(\mu)$ be dimensionless coordinates in this projection, with $g^\alpha = 0$ at the fixed point, and let

$$\frac{\mathrm{d}g^\alpha}{\mathrm{d}\log\mu} = \beta^\alpha(g)$$

be the projected RG vector field. The linearized matrix is

$$M^\alpha{}_\beta = \left.\frac{\partial\beta^\alpha}{\partial g^\beta}\right|_{g=0}.$$

Assume, in the finite projection under consideration, that M is diagonalizable in a basis represented by scaling operators. A relevant coordinate u^A is one whose operator has dimension $\Delta_A < D$; write

$$y_A = D - \Delta_A > 0.$$

An irrelevant coordinate v^ρ is one whose operator has dimension $\Delta_\rho > D$; write

$$\omega_\rho = \Delta_\rho - D > 0.$$

With the convention that μ is the running physical momentum scale, the linearized equations are

$$\frac{\mathrm{d}u^A}{\mathrm{d}\log\mu} = -y_A u^A + O(g^2), \qquad \frac{\mathrm{d}v^\rho}{\mathrm{d}\log\mu} = \omega_\rho v^\rho + O(g^2). \tag{37.41}$$

Thus a relevant coordinate grows toward the infrared unless it is tuned, while an irrelevant coordinate decays toward the infrared at the linearized level. This is a statement about the endpoint map of the RG flow, not about a microscopic equality of actions.

37.8.1 Dangerously Irrelevant Coordinates

Irrelevance is a statement about the RG endpoint in a specified coordinate chart. Setting the corresponding coordinate to zero inside an observable is a separate limiting operation whose validity depends on the physical limit and on the observable topology. The precise distinction is as follows.

Definition 37.8 (Dangerously irrelevant coordinate for an observable)**.** Fix a finite RG chart near a fixed point, with relevant coordinates u^A and irrelevant coordinates v^ρ as in (37.41). Let $\mathcal{O}$ be an observable whose scaling limit is represented in the same chart by an observable map

$$F_{\mathcal{O}}(u, v)$$

after the required source normalization and subtraction conventions have been specified. An irrelevant coordinate v^ρ is called dangerously irrelevant for $\mathcal{O}$, in a sector S of the relevant-coordinate space, if the following three conditions hold.

First, v^ρ is RG-irrelevant at the fixed point: $\omega_\rho = \Delta_\rho - D > 0$. Second, for flows approaching the fixed point through the sector S, the reduced map obtained by first setting $v^\rho = 0$ is not asymptotic to the full observable map. Equivalently, after choosing an infrared rescaling factor b,

$$F_{\mathcal{O}}(u, v) = b^{-\Delta_{\mathcal{O}}}\, \mathcal{F}_{\mathcal{O}}\big(b^{y_A} u^A, b^{-\omega_\rho} v^\rho, \ldots\big), \tag{37.42}$$

and the scaling function in the sector under consideration has no finite limit, or has the wrong limiting branch, as $w = b^{-\omega_\rho} v^\rho \to 0$. Third, this singular dependence changes an exponent or a leading amplitude of $\mathcal{O}$.

Thus "dangerous" is a property of a coordinate together with an observable and a sector of its scaling variables. The same coordinate remains irrelevant for the stability of the fixed point and for observables whose scaling functions have regular $w \to 0$ limits.

The canonical example is the scalar/Ising equation of state above four Euclidean dimensions. At the Gaussian fixed point in $D > 4$, the field has dimension

$$\Delta_\phi = \frac{D-2}{2}.$$

Let M denote the uniform order parameter, t the thermal coordinate, h the magnetic source, and $g > 0$ the quartic coordinate in the Landau functional

$$f_{\mathrm{L}}(M; t, h, g) = \frac{1}{2} t M^2 + \frac{g}{4!} M^4 - hM. \tag{37.43}$$

The Gaussian engineering dimensions are

$$y_t = 2, \qquad y_h = D - \Delta_\phi = \frac{D+2}{2}, \qquad y_g = 4 - D < 0.$$

The quartic coordinate is therefore irrelevant at the Gaussian fixed point. Nevertheless it is load-bearing in the ordered equation of state. Stationarity of (37.43) gives

$$h = tM + \frac{g}{6} M^3. \tag{37.44}$$

For $h = 0$ and $t < 0$,

$$M^2 = -\frac{6t}{g}. \tag{37.45}$$

Setting $g = 0$ before minimizing removes the stabilizing term and destroys the ordered saddle. The singular dependence on g is not a contradiction of RG irrelevance; it says that the ordered order-parameter observable uses a singular branch of the scaling function.

To see the exponent directly, apply the Gaussian scaling form

$$M(t, h, g) = b^{-(D-2)/2} \mathcal{M}\left(b^2 t,\ b^{(D+2)/2} h,\ b^{4-D} g\right). \tag{37.46}$$

Choose $b = |t|^{-1/2}$. In the ordered sector $t < 0$, $h = 0$, the third argument is

$$w = b^{4-D} g = g\,|t|^{(D-4)/2} \to 0^+.$$

Equation (37.45) implies

$$\mathcal{M}(-1,0,w) \sim \left(\frac{6}{w}\right)^{1/2} \qquad (w \to 0^+).$$

Substitution into (37.46) gives

$$M(t,0,g) \sim \sqrt{\frac{6}{g}}\,|t|^{1/2}, \qquad D > 4, \quad t \to 0^-. \tag{37.47}$$

The magnetic critical exponent is therefore the mean-field value $\beta_{\text{mag}} = 1/2$, not the value $(D-2)/4$ obtained by incorrectly assuming a regular $w \to 0$ limit of $\mathcal{M}(-1,0,w)$. At the critical isotherm $t = 0$, (37.44) gives

$$M(0,h,g) \sim \left(\frac{6h}{g}\right)^{1/3},$$

so $\delta = 3$. The minimized free-energy density in the ordered phase is

$$f_{\text{L,min}}(t<0, h=0, g) = -\frac{3t^2}{2g}.$$

Equivalently, the free-energy scaling function behaves as w^{-1}. This is the elementary mechanism behind the failure of naive hyperscaling above the upper critical dimension: the Gaussian value $\nu = 1/2$ and the free-energy exponent $2-\alpha = 2$ do not satisfy $2-\alpha = D\nu$ when $D > 4$.

At the upper critical dimension $D = 4$, the same coordinate is marginally irrelevant. In the one-component normalization $x = \lambda/(16\pi^2)$ used in (37.5), put

$$L = \log\frac{\mu_0}{\mu}.$$

The four-dimensional quartic flow satisfies

$$\frac{\mathrm{d}x}{\mathrm{d}L} = -3x^2 + O(x^3), \qquad x(L) = \frac{x_0}{1+3x_0L} + O(L^{-2}) \sim \frac{1}{3L}. \tag{37.48}$$

The thermal coordinate obeys, to the leading logarithmic order needed here,

$$\frac{\mathrm{d}t}{\mathrm{d}L} = \big(2 - x(L) + O(x(L)^2)\big)t. \tag{37.49}$$

Therefore

$$t(L) = t_0 e^{2L}\big(1+3x_0L\big)^{-1/3}(1+o(1)).$$

The correlation-length scale L_ξ is determined by $|t(L_\xi)| = O(1)$, and hence

$$\xi \asymp |t_0|^{-1/2}\left(\log\frac{1}{|t_0|}\right)^{1/6} \tag{37.50}$$

up to a nonuniversal multiplicative constant and lower logarithms. At that scale the ordered saddle gives $M_{\text{scaled}} \asymp x(L_\xi)^{-1/2}$, while the field rescaling contributes e^{-L_ξ} at leading logarithmic order. Thus

$$M(t_0,0,x_0) \asymp |t_0|^{1/2}\left(\log\frac{1}{|t_0|}\right)^{1/3}, \qquad t_0 \to 0^-. \tag{37.51}$$

The logarithm is the marginal version of the singular $g^{-1/2}$ dependence in (37.47). The power exponents are mean-field exponents, while the leading logarithmic corrections remember the slow RG decay of the quartic coordinate.

Definition 37.9 (Critical surface in a finite RG chart)**.** Let $u = (u^1, \ldots, u^r)$ be the relevant coordinates in a finite local RG chart at a fixed point. At a reference scale μ_R, the critical surface in that chart is the set of ultraviolet coordinates whose RG trajectory exists down to μ_R and satisfies

$$u(\mu_R) = 0.$$

If the theory is restricted by a symmetry, only relevant coordinates compatible with that symmetry are included in u.

This definition is set-theoretic. Its promotion to a C^k submanifold statement requires the finite-reference implicit-function construction: one must specify a Banach or finite-dimensional RG chart, prove that the endpoint map is C^k, and prove that its derivative in the relevant directions is invertible. The linearized eigenvalue calculation identifies the candidate relevant coordinates; it does not by itself prove smoothness or codimension of the nonlinear tuned set.

At linear order between a microscopic cutoff Λ_0 and a reference scale $\mu_R < \Lambda_0$, equation (37.41) gives

$$u^A(\mu_R) = \left(\frac{\Lambda_0}{\mu_R}\right)^{y_A} u^A(\Lambda_0), \qquad v^\rho(\mu_R) = \left(\frac{\mu_R}{\Lambda_0}\right)^{\omega_\rho} v^\rho(\Lambda_0).$$

Therefore holding $u^A(\mu_R)$ fixed as $\Lambda_0 \to \infty$ requires the ultraviolet coordinate $u^A(\Lambda_0)$ to be tuned with precision of order $(\mu_R/\Lambda_0)^{y_A}$. A bounded ultraviolet irrelevant coordinate contributes to $v^\rho(\mu_R)$ only at order $(\mu_R/\Lambda_0)^{\omega_\rho}$, apart from the generated contribution along the tuned trajectory. The Wilsonian cutoff-removal estimate of Chapter 36 gives the nonlinear finite-coordinate form of this statement.

For the Wilson-Fisher fixed point in the $\mathbb{Z}_2$-even scalar chart, the leading relevant coordinate is the thermal coordinate dual to $[O_2]$, with $y_t = D - \Delta_{\phi^2}$. In the full scalar chart the odd coordinate dual to ϕ is also relevant, with $y_h = D - \Delta_\phi$. The phrase "codimension-one critical surface" will therefore mean the $\mathbb{Z}_2$-even chart and the finite-reference submanifold theorem above, unless the magnetic coordinate has been included and separately tuned.

Chapter 38 applies this fixed-point data to the statistical Ising model. The coarse-grained Ising variables determine which local scalar deformations are compatible with spin-flip symmetry, and the Wilson-Fisher eigenvalues y_t, y_h, and ω_ρ, together with the exponents η and ν, determine the scaling of the corresponding thermal, magnetic, and irrelevant directions near criticality.

Chapter 38

The Statistical Ising Model and Universality

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

38.1 The Finite Lattice Ensemble

Fix a spatial dimension $D \geq 1$. A finite Ising system is specified by a finite subset $\Lambda \subset \mathbb{Z}^D$, a set $E(\Lambda)$ of nearest-neighbor unordered pairs, and spin variables

$$s_x \in \{-1, +1\}, \qquad x \in \Lambda.$$

The set $E(\Lambda)$ encodes the boundary convention. For example, it may come from free boundary conditions on a box, or from periodic boundary conditions after identifying opposite faces. Bulk thermodynamic quantities are defined by taking $\Lambda \nearrow \mathbb{Z}^D$ along a specified family and then checking which limits are independent of this boundary convention. For ferromagnetic coupling $J > 0$, external magnetic field h, and inverse temperature $\beta = T^{-1}$ in units with Boltzmann's constant equal to one, the energy of a spin configuration $s = \{s_x\}_{x\in\Lambda}$ is

$$H_\Lambda(s) = -J \sum_{\langle xy\rangle \in E(\Lambda)} s_x s_y - h \sum_{x\in\Lambda} s_x.$$

The finite-volume partition function and thermal expectation of an observable $F : \{-1,+1\}^\Lambda \to \mathbb{C}$ are

$$Z_\Lambda(\beta,h) = \sum_{s_x=\pm 1} \exp[-\beta H_\Lambda(s)], \qquad \langle F\rangle_{\Lambda,\beta,h} = Z_\Lambda(\beta,h)^{-1} \sum_{s_x=\pm 1} F(s)\,\mathrm{e}^{-\beta H_\Lambda(s)}.$$

This is the finite-lattice row of Table 11.1: at fixed Λ the object is an ordinary finite probability measure on a finite set of configurations. Its use as a regulator for a continuum Euclidean QFT requires thermodynamic and scaling limits with their own positivity, locality, and tightness or convergence hypotheses. At $h = 0$ the model has the global $\mathbb{Z}_2$ symmetry $s_x \mapsto -s_x$ for all x. The Wilson–Fisher chapter used the $\mathbb{Z}_2$-even RG chart unless an odd source was added. The present chapter includes that odd sector by the magnetic source: in continuum variables it is the deformation $\int h(x)\phi(x)$, and on the lattice it is represented by the $-h\sum_x s_x$ term in H_Λ. Setting $h = 0$ gives the symmetric critical theory, while differentiating with respect to h at $h = 0$ inserts the $\mathbb{Z}_2$-odd spin field σ.

The same finite probability space can be written in diagonal operator notation. Let

$$V_x = \mathrm{span}\{|+\rangle_x, |-\rangle_x\}, \qquad \mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} V_x,$$

and define commuting spin operators by $\hat{S}_x|s\rangle_\Lambda = s_x|s\rangle_\Lambda$, where $|s\rangle_\Lambda = \bigotimes_x |s_x\rangle_x$. Then

$$\hat{H}_\Lambda = -J\sum_{\langle xy\rangle} \hat{S}_x\hat{S}_y - h\sum_x \hat{S}_x, \qquad Z_\Lambda(\beta,h) = \mathrm{Tr}_{\mathcal{H}_\Lambda}\,\mathrm{e}^{-\beta\hat{H}_\Lambda}.$$

The space $\mathcal{H}_\Lambda$ in this display is a finite-dimensional representation of a classical statistical ensemble. A continuum Hilbert space, when constructed from the scaling limit of Euclidean correlators, is a further object obtained after the limit.

Figure 38.1 keeps the finite Ising ensemble data separate from the reconstructed continuum Hilbert space: spins are summed in a finite-dimensional trace before any scaling limit is taken.

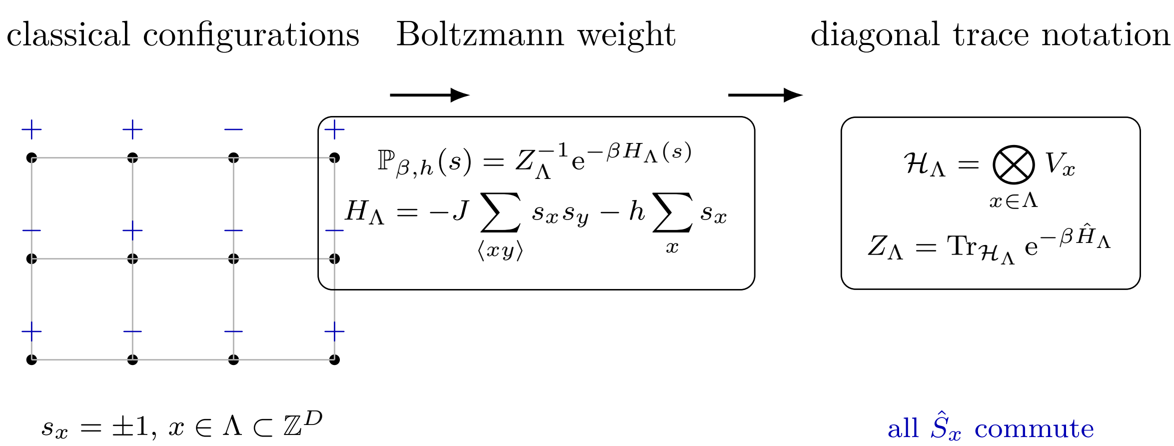


Figure 38.1: The statistical Ising model is a probability measure on lattice spin configurations. Diagonal operators provide an exact finite-volume trace notation for the same measure.

For a finite set of energy eigenstates n with energies E_n, the canonical weights

$$p_n = \frac{1}{Z}\mathrm{e}^{-\beta E_n}, \qquad Z = \sum_n \mathrm{e}^{-\beta E_n},$$

are obtained by extremizing the Shannon entropy

$$S[p] = -\sum_n p_n \log p_n$$

subject to $\sum_n p_n = 1$ and $\sum_n p_n E_n = U$. Indeed, the stationary point of

$$S[p] + \alpha\Big(\sum_n p_n - 1\Big) + \beta\Big(U - \sum_n p_n E_n\Big)$$

has $\log p_n = \alpha - 1 - \beta E_n$, hence the displayed Boltzmann form.

The entropy also has a simple monotonicity statement for symmetric probability evolution. If a finite distribution $p_n(t)$ obeys

$$\dot{p}_n = \sum_m v_{nm}(p_m - p_n), \qquad v_{nm} = v_{mn} \geq 0,$$

then $\sum_n \dot{p}_n = 0$ and

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{2}\sum_{m,n} v_{mn}(p_m - p_n)(\log p_m - \log p_n) \geq 0.$$

This is the finite-state entropy-increase mechanism underlying thermal equilibration in the idealized symmetric-rate case.

38.2 Spin Correlators and Phases

Quoted Theorem 38.1 (Ferromagnetic thermodynamic limits and phase selection)**.** *Consider the finite-volume Ising model on boxes $\Lambda \nearrow \mathbb{Z}^D$ with finite-range couplings $J_{xy} = J_{yx} \geq 0$, inverse temperature $\beta \geq 0$, and external fields h_x. Let $s_A := \prod_{x\in A} s_x$ for finite $A \subset \mathbb{Z}^D$. In finite volume the ferromagnetic measure satisfies the following correlation inequalities.*

(i) FKG positive association: if F and G are increasing functions of the spin configuration, then

$$\langle FG\rangle_{\Lambda,\beta,h} \geq \langle F\rangle_{\Lambda,\beta,h}\langle G\rangle_{\Lambda,\beta,h}.$$

(ii) The GKS inequalities, for nonnegative fields: for finite A, B,

$$\langle s_A\rangle_{\Lambda,\beta,h} \geq 0, \qquad \langle s_A s_B\rangle_{\Lambda,\beta,h} - \langle s_A\rangle_{\Lambda,\beta,h}\langle s_B\rangle_{\Lambda,\beta,h} \geq 0.$$

Equivalently, spin correlations are monotone nondecreasing under increasing ferromagnetic couplings and nonnegative external fields.

(iii) The GHS inequality, for nonnegative fields:

$$\frac{\partial^3 \log Z_\Lambda}{\partial h_x\,\partial h_y\,\partial h_z} \leq 0.$$

In particular, the finite-volume magnetization is concave as a function of a uniform positive field.

Consequently the plus and minus boundary-condition Gibbs states

$$\langle F\rangle^{\pm}_{\beta,h} := \lim_{\Lambda\nearrow\mathbb{Z}^D}\langle F\rangle^{\pm}_{\Lambda,\beta,h}$$

exist for every local observable F whenever the boxes increase through a cofinal rectangular sequence. At $h=0$, the one-sided states

$$\langle F\rangle^{\pm}_{\beta,0} = \lim_{h\downarrow 0}\langle F\rangle^{\pm}_{\beta,\pm h}$$

exist by monotonicity. If $\langle\cdot\rangle^{+}_{\beta,0} = \langle\cdot\rangle^{-}_{\beta,0}$, the thermodynamic limit is independent of the usual plus/minus/free boundary choices. If the two differ, the plus and minus limits are distinct translation-invariant Gibbs states. They are pure phases whenever they satisfy the clustering condition

$$\lim_{|a|\to\infty}\Big(\langle F\,\tau_a G\rangle^{\pm}_{\beta,0} - \langle F\rangle^{\pm}_{\beta,0}\langle G\rangle^{\pm}_{\beta,0}\Big) = 0$$

for all local F,G, where τ_a translates G. Clustering is the condition that makes the limiting Gibbs state extremal, hence pure.

Here is the finite-volume mechanism behind Quoted Theorem 38.1. Write $K_{xy} := \beta J_{xy} \geq 0$ and $H_x := \beta h_x \geq 0$. Boundary fields can be incorporated into the H_x's after adding the fixed exterior spins, so it is enough to work on a finite graph $G = (\Lambda, E)$ with weight

$$w(s) = \exp\left(\sum_{\{x,y\}\in E} K_{xy}s_xs_y + \sum_{x\in\Lambda} H_xs_x\right).$$

Order configurations by $s \leq t$ when $s_x \leq t_x$ for every x. For the coordinatewise join and meet, $s\vee t$ and $s\wedge t$, the one-edge factor obeys

$$\mathrm{e}^{K_{xy}(s\vee t)_x(s\vee t)_y}\mathrm{e}^{K_{xy}(s\wedge t)_x(s\wedge t)_y} \geq \mathrm{e}^{K_{xy}s_xs_y}\mathrm{e}^{K_{xy}t_xt_y}.$$

The single-site factors give equality in the same inequality. Hence

$$w(s\vee t)w(s\wedge t) \geq w(s)w(t), \tag{38.1}$$

which is log-supermodularity on the finite distributive lattice $\{-1,+1\}^{\Lambda}$. Positive association follows by induction over sites: conditioning on all spins except one gives a one-site measure whose conditional probability of $+1$ is increasing in the conditioned configuration by (38.1); the covariance decomposition

$$\mathrm{Cov}(F,G) = \mathbb{E}[\mathrm{Cov}(F,G\mid\mathcal{F})] + \mathrm{Cov}(\mathbb{E}[F\mid\mathcal{F}],\mathbb{E}[G\mid\mathcal{F}])$$

then propagates the one-site Chebyshev inequality to all sites. This is the finite FKG proof in the present Ising setting.

The GKS inequalities admit an equally concrete current expansion. Add a ghost vertex g and edges $\{x,g\}$ with coupling H_x; set $K_{xg} = H_x$. A current is a function $n : \bar{E} \to \mathbb{Z}_{\geq 0}$ on $\bar{E} = E \cup \{\{x,g\} : x\in\Lambda\}$, with weight

$$W(n) = \prod_{e\in\bar{E}}\frac{K_e^{n_e}}{n_e!}.$$

Let ∂n be the set of vertices incident to an odd total current. Since the ghost spin is fixed to $+1$, for $A \subset \Lambda$

$$\langle s_A \rangle_{\Lambda,K,H} = \frac{\sum_{\partial n \cap \Lambda = A} W(n)}{\sum_{\partial n \cap \Lambda = \emptyset} W(n)}. \tag{38.2}$$

Every summand is nonnegative, which gives the first GKS inequality. For the second one, the switching identity says that, for any function depending only on the sum $n_1 + n_2$,

$$\sum_{\substack{\partial n_1 \cap \Lambda = A \\ \partial n_2 \cap \Lambda = B}} W(n_1) W(n_2)\, \Phi(n_1{+}n_2) = \sum_{\substack{\partial m_1 \cap \Lambda = A \triangle B \\ \partial m_2 \cap \Lambda = \emptyset}} W(m_1) W(m_2)\, \Phi(m_1{+}m_2)\, \mathbf{1}_{\mathcal{F}_B}(m_1{+}m_2), \tag{38.3}$$

where $\mathcal{F}_B(N)$ is the event that every connected component of the support graph of N contains an even number of vertices of B. Applying (38.3) with $\Phi = 1$ gives

$$\langle s_A s_B \rangle - \langle s_A \rangle \langle s_B \rangle = \frac{\sum_{\substack{\partial m_1 \cap \Lambda = A \triangle B \\ \partial m_2 \cap \Lambda = \emptyset}} W(m_1) W(m_2) \big(1 - \mathbf{1}_{\mathcal{F}_B}(m_1 + m_2)\big)}{\big(\sum_{\partial n \cap \Lambda = \emptyset} W(n)\big)^2} \geq 0. \tag{38.4}$$

Differentiating $\langle s_A \rangle$ with respect to H_x or K_{xy} inserts respectively s_x or $s_x s_y$ and subtracts the disconnected product, so (38.4) is also the monotonicity statement in field and ferromagnetic coupling.

The GHS inequality has a finite-dimensional Gaussian proof. The third derivative

$$U_3(x,y,z) := \frac{\partial^3 \log Z_\Lambda}{\partial H_x\, \partial H_y\, \partial H_z}$$

is the third Ursell function

$$\langle s_x s_y s_z \rangle - \langle s_x \rangle \langle s_y s_z \rangle - \langle s_y \rangle \langle s_x s_z \rangle - \langle s_z \rangle \langle s_x s_y \rangle + 2 \langle s_x \rangle \langle s_y \rangle \langle s_z \rangle.$$

Choose a diagonal constant J_0 large enough that the matrix

$$V_{ij} = K_{ij} + J_0 \delta_{ij}$$

is positive definite; adding $J_0 \sum_i s_i^2/2$ changes the Hamiltonian by a constant. Since the off-diagonal entries of V are nonnegative and the diagonal entries can be chosen positive, the centered Gaussian vector X with covariance V satisfies

$$\mathbb{E} \exp\left(\sum_i X_i s_i \right) = \exp\left(\frac{1}{2} \sum_{i,j} V_{ij} s_i s_j \right).$$

Thus the finite Ising partition function is, up to a positive constant,

$$Z_\Lambda(H) = 2^{|\Lambda|} \mathbb{E}_X \prod_i \cosh(X_i + H_i). \tag{38.5}$$

To compute the third Ursell function, introduce four independent Gaussian vectors X, Y, Z, W with the same covariance and combine the five terms in U_3 into one differential expression acting on

$$G(X,Y,Z,W) = \prod_i \cosh(X_i + H_i) \cosh(Y_i + H_i) \cosh(Z_i + H_i) \cosh(W_i + H_i).$$

The orthogonal change of variables

$$\alpha = \frac{X+Y+Z+W}{2}, \qquad \beta = \frac{-X+Y-Z+W}{2},$$
$$\gamma = \frac{-X-Y+Z+W}{2}, \quad \delta = \frac{X-Y-Z+W}{2}$$

preserves the product Gaussian law and gives independent Gaussian vectors $\alpha, \beta, \gamma, \delta$, each with covariance V. In these variables the Ursell combination takes the form

$$U_3(x,y,z) = -C_{\Lambda,H}\,\mathbb{E}\left[\frac{\partial^3}{\partial\beta_x\,\partial\gamma_y\,\partial\delta_z}\prod_i Q_i(\alpha_i,\beta_i,\gamma_i,\delta_i)\right], \tag{38.6}$$

with $C_{\Lambda,H} > 0$. The single-site factor Q_i is the transformed product of the four cosh factors. Its Taylor expansion is a sum of monomials in $\alpha_i + 2H_i, \beta_i, \gamma_i, \delta_i$ with the following parity property: after differentiating once in each of $\beta_x, \gamma_y, \delta_z$, every monomial that survives the Gaussian expectation has a nonnegative coefficient and even total power in each of the four independent Gaussian families. Because V has nonnegative entries, every such Gaussian moment is nonnegative. The minus sign in (38.6) therefore gives

$$U_3(x,y,z) \le 0.$$

For a uniform field H,

$$\frac{\mathrm{d}^2}{\mathrm{d}H^2}\langle s_0\rangle = \sum_{x,y\in\Lambda}\frac{\partial^3 \log Z_\Lambda}{\partial H_0\,\partial H_x\,\partial H_y} \le 0,$$

which is the stated concavity of the finite-volume magnetization.

These finite-volume inequalities imply stochastic monotonicity in boundary condition, coupling, and field. If $\Lambda_1 \subset \Lambda_2$, the plus-boundary measure on Λ_2, restricted to Λ_1, stochastically dominates the plus-boundary measure on Λ_1; the minus-boundary expectations have the opposite monotonicity. Thus the expectations of every spin monomial form bounded monotone nets along cofinal rectangular boxes, and every local observable is a finite linear combination of spin monomials. The thermodynamic limits in the quoted theorem follow.

The one-sided $h \downarrow 0$ limits are bounded monotone limits in the field coordinate. Equality of the plus and minus limits forces every Gibbs limit squeezed between them by stochastic domination to agree with the common state, which is boundary-condition independence in the ferromagnetic class. If the limits differ, they are distinct Gibbs states selected by plus and minus boundaries, or equivalently by the sign of a symmetry-breaking field before the field is taken to zero. Finally, clustering implies extremality: if a translation-invariant Gibbs state decomposed into a nontrivial convex combination, correlations of distant translates would retain the variance of the conditional expectation onto the translation-invariant tail sigma-algebra. A clustering plus or minus state is therefore pure.

The basic observables are spin correlation functions. At $h = 0$, for translation-invariant infinite-volume limits supplied by Quoted Theorem 38.1, write

$$C_\beta(x_1,\ldots,x_n) = \langle s_{x_1}\cdots s_{x_n}\rangle_\beta.$$

The two-point function at large separation is often summarized by

$$f_\beta(r) \simeq \langle s_0 s_x\rangle_\beta, \qquad r = |x|.$$

For $D = 2, 3$ and ferromagnetic nearest-neighbor coupling, there is a critical temperature T_c. In the ordered region $T < T_c$, when the plus and minus limits are clustering and therefore pure in the sense of Quoted Theorem 38.1, boundary conditions or the one-sided field limits $h \to 0^\pm$ select states with nonzero magnetization

$$m_\pm(T) = \lim_{h\to 0^\pm} \lim_{\Lambda \nearrow \mathbb{Z}^D} \langle s_0 \rangle_{\Lambda,\beta,h}, \qquad m_+(T) = -m_-(T) \neq 0.$$

The full two-point function tends to $m_\pm(T)^2$ at large distance in such a phase. The connected correlator

$$G_\beta(x) = \langle s_0 s_x \rangle_\beta - \langle s_0 \rangle_\beta \langle s_x \rangle_\beta$$

decays at large $|x|$ away from criticality. In the disordered region $T > T_c$,

$$G_\beta(x) \sim A(T)\, \mathrm{e}^{-|x|/\xi(T)}$$

up to powers of $|x|$, where $\xi(T)$ is the correlation length. One precise definition, when the limit exists along a coordinate direction, is

$$\xi(T)^{-1} = -\lim_{r\to\infty} \frac{1}{r} \log |G_\beta(re_1)|.$$

At $T = T_c$, the correlation length diverges and the spin two-point function has power-law asymptotics

$$\langle s_0 s_x \rangle_{\beta_c} \sim \frac{A_\sigma}{|x|^{2\Delta_\sigma}}.$$

The exponent Δ_σ is the scaling dimension of the continuum spin field.

Figure 38.2 displays the three large-distance regimes encoded by the two-point function: exponential decay away from criticality, power-law behavior at criticality, and the associated divergence of the correlation length.

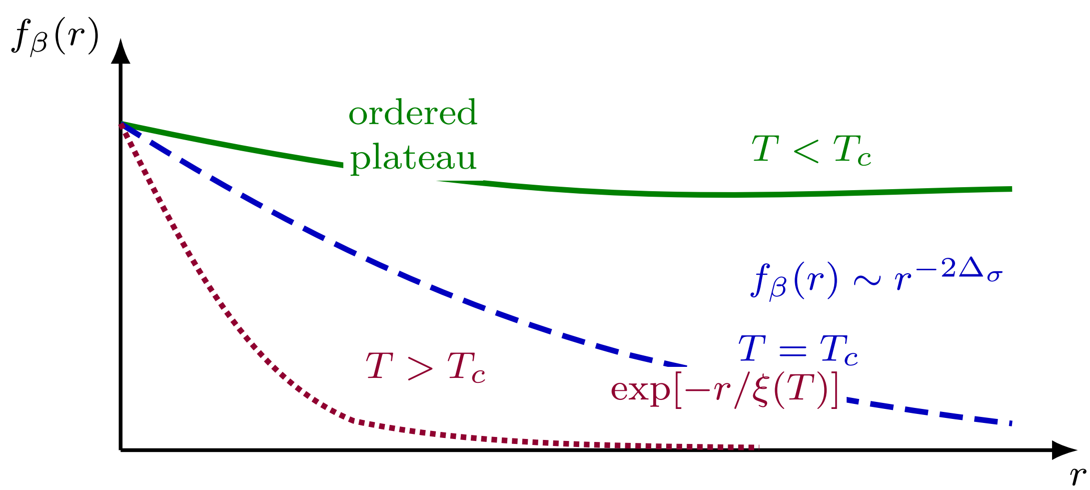


Figure 38.2: Long-distance spin correlations distinguish the ordered, critical, and disordered regimes. The ordered, critical, and disordered curves are drawn solid, dashed, and dotted respectively. In the ordered phase the full two-point function approaches the square of the magnetization; the connected correlator still decays away from the critical point.

The values relevant for the first Ising universality classes are

D	Δ_σ	ν
2	$\frac{1}{8}$	1
3	$0.518\ldots$	$0.62998\ldots$

where ν is defined by the divergence

$$\xi(T) \sim \xi_0\, |T - T_c|^{-\nu} \qquad (T \to T_c).$$

38.3 The Scaling Limit

The continuum Ising field theory is defined, when the limit exists, from large-distance lattice correlators near criticality. Let $a = L^{-1}$ be the lattice spacing in continuum units. A fixed point $x \in \mathbb{R}^D$ corresponds to a nearest lattice point $[x/a] = [Lx]$. In this subsection set the magnetic field to zero, so the $\mathbb{Z}_2$ symmetry is preserved and the thermal coordinate is the relevant coordinate being varied.

Dimension	Ising scaling-limit status	Scalar ϕ^4 comparison	Use in this chapter
$D = 2$	Critical planar Ising spin correlations and fermionic observables have rigorously constructed conformally covariant scaling limits in standard planar domains.	$P(\phi)_2$, including ϕ_2^4, is a constructive scalar QFT class after Wick ordering and the required local renormalizations.	Two-dimensional Ising may be cited as a theorem-level example of a nontrivial lattice-to-continuum critical limit, while keeping distinct the exact Ising lattice construction and the $P(\phi)_2$ constructive family.
$D = 3$	The nontrivial nearest-neighbor Ising critical scaling limit and its identification with the three-dimensional Ising CFT are not presently available as a complete theorem with OS/Wightman reconstruction data and universal correlators.	Massive ϕ_3^4 with positive quartic interaction is a constructed Glimm–Jaffe superrenormalizable scalar QFT: its Euclidean Schwinger functions satisfy the OS hypotheses, hence give a Wightman QFT by reconstruction.	Statements about the three-dimensional Ising continuum theory are therefore conditional on scaling-limit and universality hypotheses, even when supported by RG, bootstrap, Monte Carlo, and perturbative evidence. The proved massive ϕ_3^4 Wightman construction is not, by itself, a construction of the critical Wilson–Fisher/Ising fixed point.
$D = 4$	For the standard reflection-positive critical nearest-neighbor Ising scaling problem covered by the marginal triviality theorem, scaling limits are Gaussian.	The same Aizenman–Duminil-Copin theorem covers the corresponding lattice ϕ_4^4 scaling problem with positive quartic interaction.	There is no interacting four-dimensional Ising/ϕ_4^4 continuum fixed point from this standard lattice construction. This theorem is distinct from the separate logical question of nonstandard UV completions with identical formal perturbation theory.
$D > 4$	Standard critical Ising scaling limits in the theorem classes governed by mean-field/triviality results are Gaussian.	Scalar ϕ_D^4 with positive quartic coupling is trivial in the corresponding reflection-positive scaling-limit regimes.	The Ising fixed point used below is not a nontrivial $D > 4$ continuum QFT; the Wilson–Fisher deformation is meaningful below four dimensions.

Table 38.1: Dimension-dependent theorem status of Ising and scalar ϕ^4 continuum-limit claims. The table separates actual construction/triviality theorems from the conditional three-dimensional scaling-limit assertions used in the RG discussion.

References for Table 38.1. For the planar Ising entries, see Smirnov's discrete-holomorphic fermion construction for the critical random-cluster Ising model and the Chelkak–Hongler–Izyurov proof of conformal invariance of critical spin correlations. For the constructive scalar entries, see the Nelson, Simon, and Glimm–Jaffe constructions of $P(\phi)_2$ and ϕ_3^4. For the four-dimensional marginal theorem, see Aizenman–Duminil-Copin, "Marginal triviality of the scaling limits of critical 4D Ising and ϕ_4^4 models," *Ann. of Math.* **194** (2021) 163–235, together with its corrigendum. Earlier Aizenman and Fröhlich triviality theorems cover complementary above-four-dimensional scalar/Ising regimes.

The $D \geq 4$ Gaussian entries in Table 38.1 concern separated critical scaling limits and the endpoint of the tuned RG trajectory. They do not mean that the quartic coordinate may be deleted from every finite-regulator observable before the scaling limit is taken. In the terminology of Section 37.8.1, the quartic coordinate above four dimensions is dangerously irrelevant for the ordered equation of state: it flows to zero at the Gaussian fixed point, but it stabilizes the ordered saddle and appears singularly in the magnetization and free-energy amplitudes. At $D = 4$ the same coordinate is marginally irrelevant and produces the leading logarithmic corrections derived in (37.50) and (37.51). This separates the Gaussian character of the critical separated-point continuum limit from the singular quartic dependence of ordered thermodynamic observables.

Let $\xi_{\mathrm{lat}}(T)$ be the correlation length measured in lattice units, using connected correlators in a chosen pure phase when $T < T_c$. Choose a sequence of temperatures $T(L) \to T_c$ such that the signed thermal scale

$$\mathfrak{m} = \lim_{L\to\infty} \operatorname{sgn}(T(L) - T_c)\, \frac{L}{\xi_{\mathrm{lat}}(T(L))}$$

exists. The physical inverse correlation length is $|\mathfrak{m}|$. The sign records which side of the critical point is approached. More general scaling limits with a magnetic deformation require a second scaling coordinate to be held fixed. If u_h is the coefficient of the spin field, this coordinate has the form $L^{D-\Delta_\sigma} u_h$. It is set to zero here. The quantity $|\mathfrak{m}|$ is the continuum mass gap extracted from the lattice transfer-matrix or correlation-length scale. It is a spectral pole mass only under the additional condition that the reconstructed two-point spectral measure in the relevant channel has an isolated lowest one-particle atom, as in Definition 33.5; otherwise it is the lower edge of the spectrum in that channel.

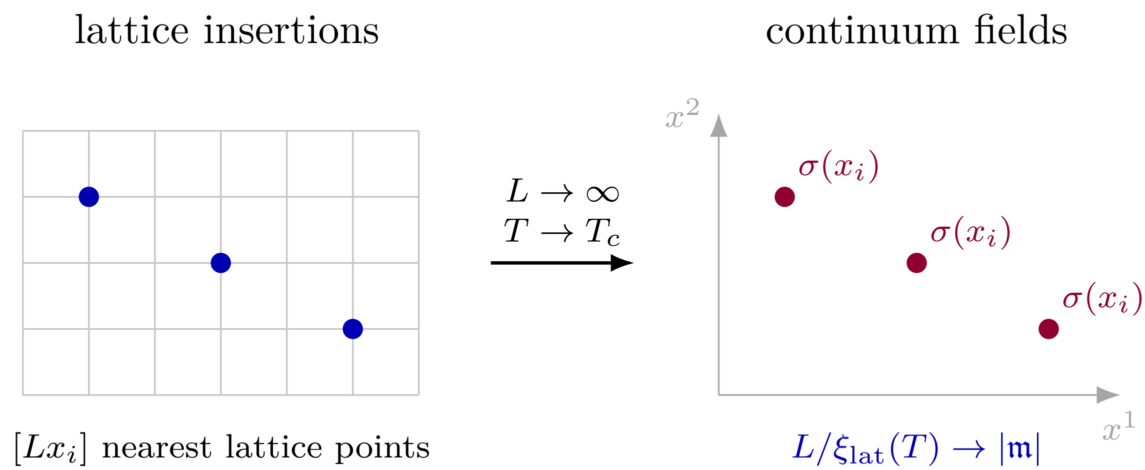
Hypothesis 38.2 (Separated spin scaling limit)**.** For the temperature trajectory above, assume that there is a spin-field renormalization factor $Z_\sigma(L,T) > 0$ such that, for every n and every configuration of pairwise distinct continuum points $x_1, \ldots, x_n$, the limit

$$\lim_{\substack{T\to T_c,\ L\to\infty \\ \operatorname{sgn}(T-T_c)L/\xi_{\mathrm{lat}}(T)\to\mathfrak{m}}} Z_\sigma(L,T)^{-n/2} \langle s_{[Lx_1]} \cdots s_{[Lx_n]} \rangle_{\beta(T)} = \langle \sigma(x_1) \cdots \sigma(x_n) \rangle_{\mathrm{Ising},\mathfrak{m}}$$

exists. The convergence is meant after the thermodynamic limit specified for the lattice model, and with the boundary/pure-phase convention fixed when $\mathfrak{m} < 0$.

When Hypothesis 38.2 holds, its right side defines the separated-point spin Schwinger functions of the continuum Ising scaling theory at thermal scale $\mathfrak{m}$. At $\mathfrak{m} = 0$, equivalently at $\xi_{\mathrm{lat}}(T)/L \to \infty$, the limiting fixed point is called the critical Ising fixed point. If this fixed point admits a local stress tensor that is conserved, symmetric, and improvable to a traceless tensor whose charges generate the conformal transformations of the separated-point Schwinger functions, then the fixed point is the Ising conformal field theory. At criticality $Z_\sigma(L, T_c)$ may be chosen proportional to $L^{-2\Delta_\sigma}$, so the rescaled spin insertions have finite correlators. The limiting objects in this display are Euclidean separated-point Schwinger functions. Reflection positivity of these limiting distributions is not a consequence of the scaling formula alone. It is a property that must either be proved from a reflection-positive regulated lattice model and a sufficiently strong mode of convergence, or imposed as a separate reconstruction hypothesis. When this positivity and the remaining reconstruction hypotheses hold, the Schwinger functions determine a Lorentzian QFT Hilbert space by the Osterwalder–Schrader reconstruction of Chapter 12. That reconstructed Hilbert space is not the finite-dimensional spin-trace space $\mathcal{H}_\Lambda$ used to define the classical thermal ensemble.

Figure 38.3 records the scaling-limit map: renormalized lattice insertions converge to separated-point Schwinger functions, and OS reconstruction is an additional step after reflection positivity and the remaining hypotheses are established.



$$\boxed{Z_\sigma^{-n/2}\,\langle s_{[Lx_1]}\cdots s_{[Lx_n]}\rangle \longrightarrow \langle \sigma(x_1)\cdots\sigma(x_n)\rangle_{\mathrm{Ising},\mathfrak{m}}}$$

Figure 38.3: The Ising field theory is the continuum scaling limit of long-distance spin correlators. The absolute value of $\mathfrak{m}$ is the inverse correlation length in the limiting continuum units, while its sign records the high- or low-temperature side.

38.4 Reflection Positivity Along the Scaling Limit

Choose one lattice coordinate and call it Euclidean time. Let θ be reflection across the hyperplane midway between the time slices $n^0 = 0$ and $n^0 = 1$:

$$\theta(n^0, n^1, \ldots, n^{D-1}) = (1 - n^0, n^1, \ldots, n^{D-1}).$$

For a finite reflection-invariant lattice region Λ, write $\Lambda_+ = \{n \in \Lambda : n^0 \geq 1\}$, and let $\mathcal{A}_+$ be the algebra of complex functions of the lattice variables with support in Λ_+. If $F \in \mathcal{A}_+$, define the anti-linear reflection

$$(\Theta F)(q) = \overline{F(q \circ \theta)},$$

where q denotes the complete lattice configuration.

Definition 38.3 (Finite-lattice reflection positivity)**.** A finite lattice probability measure μ_Λ is reflection positive with respect to the time reflection θ if

$$\int (\Theta F)(q)\, F(q)\, \mathrm{d}\mu_\Lambda(q) \geq 0 \tag{38.7}$$

for every $F \in \mathcal{A}_+$. Equivalently, the sesquilinear form $(F, G) \mapsto \int (\Theta F) G\, \mathrm{d}\mu_\Lambda$ is positive semidefinite on $\mathcal{A}_+$.

This is the finite-cutoff precursor of the Osterwalder–Schrader positivity condition. It is stronger than pointwise positivity of the Boltzmann weight: it constrains how the action couples fields on opposite sides of every reflection plane used in reconstruction.

Proposition 38.4 (Ferromagnetic nearest-neighbor models are reflection positive)**.** *Consider a reflection-invariant finite box with free boundary conditions, or a periodic torus with the reflection plane chosen so that the action decomposes into reflected halves and bonds crossing the plane. The source-free nearest-neighbor Ising measure with coupling $J \geq 0$ is reflection positive. The real scalar lattice measure*

$$\mathrm{d}\mu_\Lambda(s) = Z_\Lambda^{-1} \prod_{x\in\Lambda} \rho(\mathrm{d}s_x)\, \exp\left[\sum_{\langle xy\rangle} K\, s_x s_y\right], \qquad K \geq 0, \tag{38.8}$$

is also reflection positive whenever ρ is a positive one-site measure for which the finite-volume partition function exists. Thus the nearest-neighbor generalized Ising model with $\rho(\mathrm{d}s) = \exp[-P(s)]\mathrm{d}s$, P bounded below with sufficient growth, and $K = \beta \geq 0$, preserves reflection positivity. Convexity of P is not required; a double-well one-site potential does not by itself violate reflection positivity.

Proof. The Ising model is the special case $\rho = \frac{1}{2}(\delta_{-1} + \delta_{+1})$ and $K = \beta J$. It is therefore enough to prove the scalar statement. Decompose the action into terms strictly on the positive side, their reflected copies, and the bonds crossing the reflection plane:

$$\exp[-S_\Lambda(s)] = \exp[-S_+(s_+)]\, \exp[-S_+(\theta s_-)] \prod_{y\in\Sigma_+} \exp[K_y s_{\theta y} s_y]\,,$$

where y runs over the positive-side endpoints of the crossing bonds and $K_y \geq 0$. Expanding every crossing factor gives

$$\prod_{y\in\Sigma_+} \exp[K_y s_{\theta y} s_y] = \sum_{m:\Sigma_+\to\mathbb{Z}_{\geq 0}} \prod_{y\in\Sigma_+} \frac{K_y^{m_y}}{m_y!} \prod_{y\in\Sigma_+} s_{\theta y}^{m_y} \prod_{y\in\Sigma_+} s_y^{m_y}.$$

Substitution into the quadratic form $\int(\Theta F)F\,\mathrm{d}\mu_\Lambda$ factorizes the positive and negative half-lattice integrals. For each multi-index m the contribution is

$$\frac{1}{Z_\Lambda} \prod_{y\in\Sigma_+} \frac{K_y^{m_y}}{m_y!} \left| \int F(s_+) \prod_{y\in\Sigma_+} s_y^{m_y}\, \exp[-S_+(s_+)] \prod_{x\in\Lambda_+} \rho(\mathrm{d}s_x) \right|^2 .$$

The coefficients are nonnegative because $K_y \geq 0$. The sum of these nonnegative terms is the desired quadratic form, proving (38.7). □

The proposition identifies a sufficient condition, not a definition of the universality class. Adding next-nearest-neighbor, anisotropic, or higher-derivative lattice couplings preserves reflection positivity only when the interaction crossing each reflection plane is a positive kernel in the above sense. Negative cross-reflection couplings, or derivative discretizations whose cross-plane kernel has negative directions, generally destroy the factorization into sums of squares. Such models may still have the same formal RG relevant coordinates, but they do not automatically supply Osterwalder–Schrader positivity for a Lorentzian reconstruction.

Remark 38.5 (Closedness of reflection positivity under scaling limits). Let $a \downarrow 0$ label reflection-positive lattice measures μ_a. Let $A_a^I(f)$ be renormalized smeared lattice observables whose linear normalizations are real and whose test functions are supported in the positive Euclidean half-space. Suppose that for every finite collection of polynomial functionals $F_\alpha(A_a)$ supported in that half-space and every set of complex coefficients c_α, the reflected quadratic forms

$$Q_a(c) = \sum_{\alpha,\beta} \overline{c_\alpha} c_\beta \, \langle \Theta F_\alpha(A_a) \, F_\beta(A_a) \rangle_{\mu_a}$$

converge to the corresponding continuum quadratic form $Q_*(c)$. Then $Q_*(c) \geq 0$ for all such c_α; the limiting Schwinger distributions are reflection positive on this test-function domain. For a fixed finite list of observables and coefficients, this is simply the closedness of the cone $[0, \infty) \subset \mathbb{R}$: finite-lattice reflection positivity gives $Q_a(c) \geq 0$, and the convergence hypothesis gives $Q_*(c) = \lim_{a \downarrow 0} Q_a(c)$. The finite list and vector c are arbitrary, so every reflected finite Gram matrix built from the continuum smeared observables is positive semidefinite on the stated test domain.

The hypothesis in this remark concerns reflected quadratic forms and separated-point correlators together. Source-local subtractions used to define a distributional generating functional must either arise from genuine lattice observables whose reflected quadratic forms converge, or be separately checked to preserve the Osterwalder–Schrader quadratic inequalities. Coordinate universality and critical exponent matching do not imply this positivity statement.

Microscopic lattice observables become continuum operators through their leading terms in the scaling-field expansion. Let

$$e_x = -\sum_{\mu=1}^{D} s_x s_{x+\hat{\mu}}$$

be a local energy-density observable on the hypercubic lattice. At separated continuum points and at criticality,

$$s_{[x/a]} = C_\sigma \, a^{\Delta_\sigma} \sigma(x) + \sum_j C_j \, a^{\Delta_j} O_j(x), \qquad \Delta_j > \Delta_\sigma,$$

and

$$e_{[x/a]} - \langle e \rangle_c = C_\varepsilon \, a^{\Delta_\varepsilon} \varepsilon(x) + \sum_k C'_k \, a^{\Delta'_k} O'_k(x), \qquad \Delta'_k > \Delta_\varepsilon.$$

The constants C_σ and C_ε depend on the microscopic normalization of the lattice observables. The dimensions and operator-product coefficients of the scaling fields are fixed-point data.

In scalar continuum coordinates, the leading $\mathbb{Z}_2$-odd scalar field ϕ represents the Ising spin field σ, while the leading $\mathbb{Z}_2$-even mass operator, denoted ϕ^2 after identity mixing is subtracted, represents the energy field ε:

$$\phi \longrightarrow \sigma, \qquad [\phi^2] \longrightarrow \varepsilon.$$

If u_t denotes the thermal scaling coordinate, then

$$\Delta S_t = \int \mathrm{d}^D x \, u_t \, \varepsilon(x), \qquad [u_t] = D - \Delta_\varepsilon.$$

Since $\xi^{-1} \propto |u_t|^{1/(D-\Delta_\varepsilon)}$, comparison with $\xi \propto |T - T_c|^{-\nu}$ gives

$$\Delta_\varepsilon = D - \frac{1}{\nu}.$$

38.5 Observable Limits as Source-Dependent Distributions

The separated-point scaling formula is the restriction of a stronger distributional statement. Let a be the lattice spacing and let $\mathcal{D}(\mathbb{R}^D)$ be the space of smooth compactly supported test functions. For $f, g \in \mathcal{D}(\mathbb{R}^D)$, define the renormalized smeared lattice observables

$$\Sigma_a(f) = a^D \sum_{n\in\mathbb{Z}^D} f(an)\, C_\sigma^{-1} a^{-\Delta_\sigma} s_n,$$

and

$$E_a(g) = a^D \sum_{n\in\mathbb{Z}^D} g(an)\, C_\varepsilon^{-1} a^{-\Delta_\varepsilon} \big(e_n - \langle e\rangle_c\big),$$

where the sums are restricted to the finite lattice region before the thermodynamic limit is taken. The constants C_σ and C_ε are the same nonuniversal normalization constants appearing in the leading scaling-field expansion. The point fields $\sigma(x)$ and $\varepsilon(x)$ are the distributional kernels whose smeared limits are $\Sigma(f) = \int \mathrm{d}^D x\, f(x)\sigma(x)$ and $E(g) = \int \mathrm{d}^D x\, g(x)\varepsilon(x)$.

Definition 38.6 (Source-dependent scaling limit)**.** Fix the thermal scaling trajectory used above. A source-dependent scaling limit for a chosen collection of microscopic local observables A_a^I consists of a continuum source coordinate $\eta^A(x)$, a lattice source coordinate $\eta_a^I(n)$, and local source maps

$$\eta_a = \mathcal{Z}_a(\eta),$$

such that, after subtraction of source-local vacuum terms,

$$W_a[\mathcal{Z}_a(\eta)] - W_a[0] - P_a(\eta) \longrightarrow W_*[\eta] \tag{38.9}$$

as a functional of test sources near $\eta = 0$. Here

$$\exp W_a[\eta_a] = \left\langle \exp\left[-a^D \sum_n \eta_a^I(n) A_a^I(n)\right]\right\rangle_a,$$

and $P_a(\eta)$ is a local polynomial in the continuum sources and their derivatives. The linear part of $\mathcal{Z}_a$ fixes field normalization and operator mixing. Its higher local Taylor coefficients fix the contact extension of products of continuum fields.

The term $P_a(\eta)$ removes vacuum terms such as powers of the volume and local source self-contractions. It has no effect on separated normalized correlators, but it is part of the definition of the full distributional generating functional.

Separated kernels in a source-dependent scaling limit. Assume the source-dependent scaling limit (38.9) exists for two microscopic realizations X and Y and that their continuum source coordinates have been identified by the same fixed-point operator normalization. Then their noncoincident continuum correlators of the corresponding scaling fields agree. Their contact-term extensions may differ by local distributions supported on collision diagonals.

The point is a consequence of the source-functional definition. Functional derivatives of (38.9) at $\eta = 0$ give the scaling limits of smeared inserted correlators. Restricting the resulting distributions to the configuration space of distinct insertion points removes all terms supported on collision diagonals. By the source-coordinate partition formula of Section 34.5, only the linear source jet contributes on this configuration space. Once the linear jet has been fixed by the common operator normalization, the two realizations have the same separated-point correlators. Higher source jets and the local polynomials P_a change only the extension across collision diagonals.

Thus microscopic observables enter the continuum theory through renormalized source derivatives whose limits are operator-valued distributions. The universal data are the separated-point kernels obtained after the relevant scaling coordinates and linear operator normalizations have been fixed. These kernels are the source-coordinate input for the fixed-point operator data constructed in Chapter 56.

38.6 A Lattice Scalar Path Integral

The Ising variables may be enlarged from two values to a real local variable without changing the long-distance fixed point, provided the relevant $\mathbb{Z}_2$-even coupling is tuned. Let $P : \mathbb{R} \to \mathbb{R}$ be an even single-site potential with sufficient growth at infinity. The generalized Ising partition function is

$$\widetilde{Z}_\Lambda = \int \prod_{x\in\Lambda} \mathrm{d}s_x \, \exp\left[-\sum_{x\in\Lambda} P(s_x) + \beta \sum_{\langle xy\rangle} s_x s_y \right].$$

For P real and bounded below with sufficient growth and for $\beta \geq 0$, this nearest-neighbor scalar measure is in the reflection-positive class of Proposition 38.4. This assertion uses the positivity of the cross-reflection nearest-neighbor coupling. It would not follow for an arbitrary lattice regularization with negative next-nearest-neighbor terms or other cross-plane kernels that are not positive in the sense used in the proof of that proposition. Equivalently,

$$\widetilde{Z}_\Lambda = \int \prod_{x\in\Lambda} \mathrm{d}s_x \, \mathrm{e}^{-S_\Lambda(s)}$$

with

$$S_\Lambda(s) = \frac{\beta}{2} \sum_{\langle xy\rangle} (s_x - s_y)^2 + \sum_{x\in\Lambda} \big(P(s_x) - \beta D \, s_x^2\big),$$

where each positive lattice direction is counted once. The identity follows from

$$\sum_{\langle xy\rangle} (s_x - s_y)^2 = 2D \sum_x s_x^2 - 2 \sum_{\langle xy\rangle} s_x s_y$$

on a periodic hypercubic lattice.

For an infinite hypercubic lattice with unit spacing, introduce the Brillouin zone

$$B = [-\pi, \pi]^D$$

and write

$$s_x = \int_B \frac{\mathrm{d}^D k}{(2\pi)^D} \, \widetilde{\phi}(k) \, \mathrm{e}^{\mathrm{i}k\cdot x}, \qquad \widetilde{\phi}(k + 2\pi n) = \widetilde{\phi}(k), \quad n \in \mathbb{Z}^D.$$

The cutoff $|k^\mu| \leq \pi$ is a finite ultraviolet cutoff. The nearest neighbor kinetic term becomes

$$\sum_{\langle xy\rangle} (s_x - s_y)^2 = \int_B \frac{\mathrm{d}^D k}{(2\pi)^D} \, \widetilde{\phi}(-k)\widetilde{\phi}(k) \sum_{\mu=1}^{D} \left| \mathrm{e}^{\mathrm{i}k_\mu} - 1 \right|^2.$$

For lattice spacing a, the dimensionful momentum is $p = k/a$, and the Brillouin-zone edge is therefore

$$\Lambda_{\mathrm{BZ}} = \frac{\pi}{a}.$$

In the finite-cutoff Wilsonian estimates below, $\Lambda_0 = a^{-1}$ denotes the inverse lattice spacing rather than the Brillouin-zone edge. The two conventions are related by the fixed conversion

$$\Lambda_{\mathrm{BZ}} = \pi\Lambda_0.$$

Replacing Λ_0 by Λ_{BZ} in a cutoff-suppressed estimate only multiplies the corresponding constant by a power of π; it does not change the exponent or the continuum limit. For small momentum,

$$\sum_{\mu=1}^{D} \left| \mathrm{e}^{\mathrm{i}k_\mu} - 1 \right|^2 = k^2 - \frac{1}{12}\sum_{\mu=1}^{D} k_\mu^4 + O(k^6).$$

Thus the long-distance action has the form

$$S[\phi] = \int \mathrm{d}^D x \left[\frac{\beta}{2}\sum_{\mu=1}^{D}(\partial_\mu\phi)^2 + V_\beta(\phi) - \frac{\beta}{24}\sum_{\mu=1}^{D}(\partial_\mu^2\phi)^2 + \cdots \right],$$

with $V_\beta(\phi) = P(\phi) - \beta D\phi^2$ plus local terms generated by the field normalization convention. The higher-derivative terms reflect the lattice cutoff and the breaking of continuous rotations to the hypercubic lattice symmetry.

Figure 38.4 shows the local scalar action generated from the lattice ensemble: the single-site measure supplies $V_\beta(\phi)$, the nearest-neighbor term supplies the gradient term, and higher-derivative operators record the cutoff symmetry.

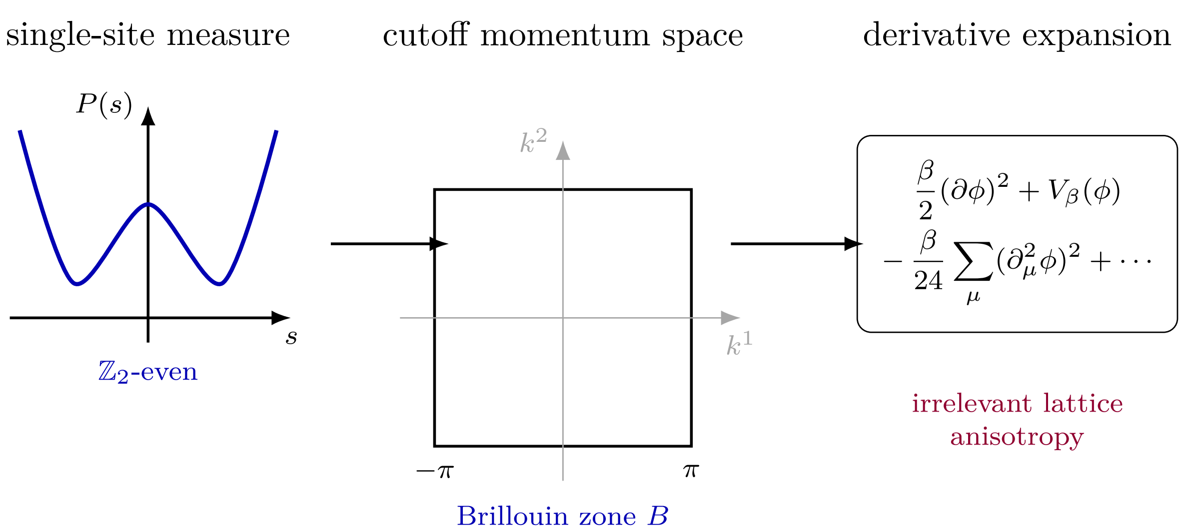


Figure 38.4: The generalized Ising model may be rewritten as a scalar lattice path integral. The single-site measure supplies the potential, the nearest neighbor interaction supplies the kinetic term, and the Brillouin zone gives a finite ultraviolet cutoff.

38.7 Universality Classes and Relevant Coordinates

The lattice scalar action just constructed and the continuum quartic scalar action differ in their short-distance coordinates. The lattice action has a finite cutoff, lattice-anisotropic derivative terms, and many allowed local couplings such as ϕ^6 and higher powers or derivatives. These differences define distinct points in the same local theory space. The universality statement is that, in $D = 2, 3$ and under the existence of the scaling limit described above, these distinct ultraviolet points have long-distance trajectories that approach the same Ising fixed point after the relevant coordinates are tuned.

Near the fixed point, write a general local deformation as

$$\Delta S = \int \mathrm{d}^D x \left(u_h\, \sigma(x) + u_t\, \varepsilon(x) + \sum_I u_I\, O_I(x) \right), \qquad \Delta_I = \dim O_I.$$

The coefficient u_I has mass dimension $D - \Delta_I$. At a physical mass scale μ, the dimensionless strength of this perturbation is

$$\lambda_I(\mu) = u_I\, \mu^{\Delta_I - D}.$$

To linear order at the fixed point,

$$\frac{\mathrm{d}\lambda_I}{\mathrm{d}\log\mu} = (\Delta_I - D)\lambda_I + O(\lambda^2).$$

If $\Delta_I > D$, the dimensionless strength tends to zero as $\mu \to 0$. These are the irrelevant perturbations. They account for microscopic differences such as the choice of square or hexagonal lattice, the details of $P(s)$, and higher-derivative cutoff terms.

The fixed-point spectral input used in the Ising universality statement is that, after omitting the identity, total derivatives, equation-of-motion descendants, and metric deformations, the only relevant scalar action deformations are the $\mathbb{Z}_2$-odd spin field σ and the $\mathbb{Z}_2$-even energy field ε:

$$\Delta_\sigma < D, \qquad \Delta_\varepsilon = D - \frac{1}{\nu} < D.$$

Their linear RG eigenvalues are

$$y_h = D - \Delta_\sigma, \qquad y_t = D - \Delta_\varepsilon = \frac{1}{\nu}.$$

The magnetic field couples to u_h. If the microscopic $\mathbb{Z}_2$ symmetry is imposed, then $u_h = 0$ and the displayed theory-space slice has only the thermal relevant direction. The temperature deviation determines the thermal scaling coordinate by

$$u_t = c_T(T - T_c) + O((T - T_c)^2), \qquad c_T \neq 0,$$

after a choice of temperature coordinate near T_c. Since u_t has dimension $D - \Delta_\varepsilon$, it generates the correlation length

$$\xi(T) \propto |u_t|^{-1/(D-\Delta_\varepsilon)} \propto |T - T_c|^{-\nu}.$$

The coefficient c_T and the amplitude of ξ are not universal; the exponent is.

Figure 38.5 gives the local RG picture of the Ising universality statement: the temperature coordinate follows the relevant direction, and microscopic data along irrelevant directions are forgotten in the scaling limit.

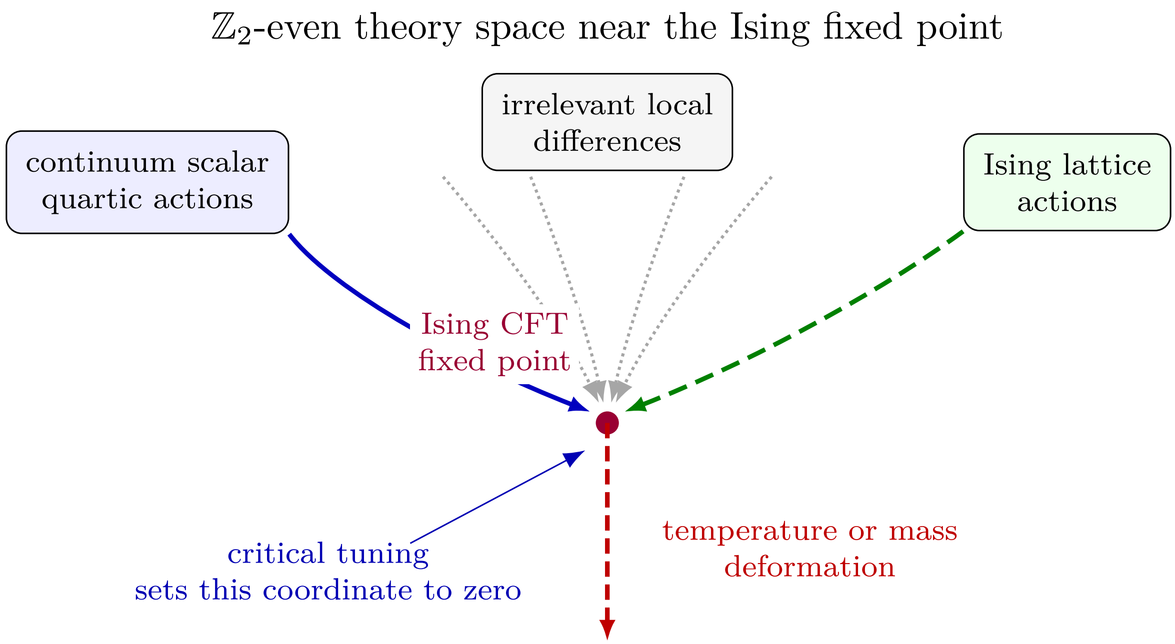


Figure 38.5: A local $\mathbb{Z}_2$-even theory-space chart for an Ising universality statement. Solid and dashed arrows distinguish continuum scalar and lattice representatives after critical tuning; dotted arrows indicate irrelevant differences whose effects must be controlled in a declared observable topology. Relevant thermal deformations move the theory away from criticality and produce massive scaling limits.

Thus the statistical Ising model supplies a constructive route from microscopic lattice ensembles to continuum Euclidean QFT. The construction is by correlators: after tuning the relevant coordinates and rescaling fields, the large-distance spin correlators define correlators of local fields at, or near, the Ising fixed point.

Definition 38.7 (Scaling-limit universality structure)**.** For the purposes of this chapter, a scaling-limit universality structure is the following collection of objects.

(i) A class $\mathfrak{U}$ of microscopic regulators X, each with a finite or Banach parameter space P_X, a cutoff a^{-1}, a reflection-positive Euclidean state or finite-volume measure, and a selected family of microscopic local observables A_I^X.

(ii) For each relevant endpoint u_R at a fixed reference scale μ_R, families of tuned microscopic parameters

$$p_X(a, u_R) \in P_X, \qquad X \in \mathfrak{U},$$

defined for all sufficiently small a.

(iii) Local normalization and source-renormalization maps

$$A_I^X \longmapsto Z_{I,a}^X A_I^X, \qquad \eta \longmapsto \mathcal{Z}_a^X(\eta),$$

together with source-local counterterms $P_a^X(\eta)$, where source-local means a polynomial functional supported on collision diagonals of the inserted sources.

(iv) A directed family of observation domains $\mathfrak{W}$. An element $W \in \mathfrak{W}$ may be a bounded set of test sources, a finite list of separated smeared correlators, or a finite observation window in the observable-germ sense of Chapter 142. Each W carries seminorms in which convergence is to be stated.

(v) A limiting source functional $W_*(\eta; u_R)$, modulo source-local polynomials, such that for every $X \in \mathfrak{U}$ and every $W \in \mathfrak{W}$,

$$W^X_{a,p_X(a,u_R)}[\mathcal{Z}_a^X(\eta)] - W^X_{a,p_X(a,u_R)}[0] - P_a^X(\eta) \longrightarrow W_*(\eta; u_R) \tag{38.10}$$

in the seminorms attached to W.

Two microscopic regulators $X, Y \in \mathfrak{U}$ are in the same scaling-limit universality class, relative to the selected observables, normalization freedom, tuning maps, and observation topology above, when they belong to a common structure with the same W_*. The relation is therefore an equivalence relation only after these normalizations, tuning maps, observation topologies, and convergence statements have been specified: equality of W_* is reflexive and symmetric, and transitivity is composition through the common limiting source functional with the permitted local source redefinitions.

The definition is deliberately source-functional. It records the object that can reconstruct separated Schwinger functions and, after positivity and regularity hypotheses, local QFT data. A Wilsonian coordinate basin supplies one possible mechanism for proving (38.10), but it is not the definition of the universality class: the definition is the common limiting observable structure with its tuning, normalization, and topology.

38.8 Finite-Cutoff Universality in a Local Chart

The preceding paragraph can be made into a local finite-cutoff statement. Let $\Lambda_0 = a^{-1}$ be the lattice cutoff and let μ_R be a fixed reference momentum scale with $\mu_R < \Lambda_0$. Choose a finite local RG chart near the Ising fixed point and write the projected Wilsonian coordinates at μ_R as

$$z(\mu_R; \Lambda_0) = (u(\mu_R; \Lambda_0), v(\mu_R; \Lambda_0)).$$

Here u denotes the relevant coordinates and v denotes the retained irrelevant coordinates. In the full scalar chart

$$u = (\lambda_h, \lambda_t),$$

where λ_h is the dimensionless coordinate coupled to the spin field σ, and λ_t is the dimensionless coordinate coupled to the energy field ε. In the $\mathbb{Z}_2$-even chart λ_h is absent. The target value

$$u_R = (\lambda_{h,R}, \lambda_{t,R})$$

specifies either the critical theory, when $u_R = 0$, or a massive scaling limit, when one of the relevant coordinates is held at a fixed nonzero value at the scale μ_R.

Let X label a microscopic realization in the same proposed universality class: a nearest-neighbor Ising model, a generalized single-site potential, or a lattice scalar action with different short-distance couplings. Let p_X denote the microscopic parameters available for tuning, such as temperature and magnetic field. The endpoint map in the finite chart is

$$E_X(\Lambda_0, p_X) = u_X(\mu_R; \Lambda_0, p_X).$$

The local tuning problem is to solve

$$E_X(\Lambda_0, p_X) = u_R.$$

In the $\mathbb{Z}_2$-even one-parameter case this is the precise version of tuning the temperature: if $p_X = T$, then the condition at criticality is $\lambda_t(\mu_R; \Lambda_0, T) = 0$.

Local RG-coordinate consequence. Fix a finite local RG chart near the Ising fixed point and a target relevant coordinate u_R. Suppose that, for every sufficiently large Λ_0 and for each microscopic realization X under consideration, the following conditions hold.

(i) The endpoint equation $E_X(\Lambda_0, p_X) = u_R$ has a solution $p_X(\Lambda_0, u_R)$, and the corresponding trajectory remains in the chosen chart between Λ_0 and μ_R.

(ii) The retained irrelevant block obeys the same backward contraction estimate used in the Wilsonian cutoff-removal theorem.

(iii) The microscopic irrelevant coordinates are uniformly bounded after expressing the ultraviolet-regulated microscopic action in the chosen local coordinates.

(iv) The generated integral along the tuned trajectory converges uniformly as $\Lambda_0 \to \infty$.

Then there is a graph $v = V_{\mu_R}(u_R)$, depending on the fixed point and on the chosen RG coordinates but not on the microscopic label X, such that

$$v_X(\mu_R; \Lambda_0, p_X(\Lambda_0, u_R)) = V_{\mu_R}(u_R) + O\left[\left(\frac{\mu_R}{\Lambda_0}\right)^{\kappa}\right]$$

for some $\kappa > 0$ determined by the irrelevant exponents and by the convergence rate of the generated integral.

This follows by applying the finite-coordinate cutoff-removal estimate of Chapter 36 to the RG trajectory whose relevant endpoint coordinate is fixed to be u_R. The irrelevant part of the bare coordinate contributes at order $(\mu_R/\Lambda_0)^{\omega}$, where ω is the smallest retained irrelevant exponent in the chart. The generated integral has the assumed cutoff limit and a convergence rate, say ω_F. Taking $\kappa = \min(\omega, \omega_F)$ gives the displayed estimate. The limiting graph is independent of X because all dependence on X has been placed either in the solved relevant-coordinate tuning or in bounded irrelevant bare coordinates whose direct memory is power-suppressed.

This coordinate form of universality separates three kinds of objects. First, the fixed point and its scaling operators determine Δ_σ, Δ_ε, y_h, y_t, and the limiting operator algebra. Second, the microscopic model determines nonuniversal coordinate maps, such as

$$\lambda_t(\Lambda_0) = A_X\,(T - T_{c,X}) + O((T - T_{c,X})^2), \qquad A_X \neq 0.$$

Third, the irrelevant microscopic coordinates determine finite-cutoff corrections. At linear order the tuning relation for a fixed thermal target $\lambda_{t,R}$ is

$$\lambda_t(\Lambda_0) = \left(\frac{\mu_R}{\Lambda_0}\right)^{y_t} \lambda_{t,R} + \text{higher-order terms},$$

so

$$T - T_{c,X} = A_X^{-1} \left(\frac{\mu_R}{\Lambda_0}\right)^{y_t} \lambda_{t,R} + \cdots .$$

The factor A_X is not fixed-point data. The exponent $y_t = D - \Delta_\varepsilon$ is.

Remark 38.8 (Coordinate universality versus scaling-limit existence). The local RG-coordinate consequence above is a statement about convergence of finitely many projected RG coordinates at the reference scale μ_R. It does not, by itself, construct a continuum probability measure, a Euclidean source functional, or a Wightman theory. To upgrade the coordinate statement to a continuum universality theorem one must additionally establish the existence of the scaling limit of the relevant Schwinger functions or, equivalently in a source-functional formulation, convergence of the renormalized lattice source functionals on a source domain large enough to determine the desired correlators. The limiting Schwinger functions must also satisfy the positivity, covariance, locality, regularity, and clustering hypotheses needed for the intended reconstruction or structural theorem.

Indeed, the local RG proposition controls the finite vector $z(\mu_R; \Lambda_0)$ after projecting the exact cutoff action to the chosen coordinate chart. A continuum measure or source functional contains infinitely many correlation data and their positivity properties. Convergence of a finite coordinate vector does not imply tightness of the underlying measures, convergence of all moments or distributions, reflection positivity in the limit, locality of the reconstructed fields, or uniqueness of the continuum state. Each of these is an additional analytic assertion about the regulated models. Hence scaling-limit existence is an independent hypothesis or theorem, not a conclusion of the coordinate estimate.

Hypothesis 38.9 (Continuum universality data)**.** Let X range over a class of reflection-positive microscopic lattice models with the same global symmetries as the Ising fixed point, and let $p_X(a, u_R)$ be microscopic parameters tuned so that the relevant endpoint coordinate at the reference scale μ_R is the same fixed value u_R. Assume the following hypotheses.

(i) The finite-coordinate RG hypotheses of the local RG-coordinate consequence above hold uniformly for the class X, so that the retained Wilsonian coordinates satisfy

$$z_X(a, u_R) \longrightarrow z_*(u_R) := (u_R, V_{\mu_R}(u_R)).$$

(ii) For the selected microscopic observables and sources, there are local source renormalization maps $\mathcal{Z}_a^X$ and source-local vacuum counterterms P_a^X on a common test-source neighborhood such that the renormalized source functionals have the representation

$$W_a^X[\mathcal{Z}_a^X(\eta)] - W_a^X[0] - P_a^X(\eta) = \mathcal{W}_X(z_X(a, u_R), \eta) + r_a^X(\eta),$$

where $r_a^X(\eta) \to 0$ as a distributional functional of η.

(iii) The continuum source coordinate has been fixed by a common normalization of the leading scaling fields, and the maps $\mathcal{W}_X(z, \eta)$ are sequentially continuous at $z = z_*(u_R)$ with the same limiting functional:

$$\lim_{z \to z_*(u_R)} \mathcal{W}_X(z, \eta) = W_*(\eta; u_R)$$

for every X in the class, modulo source-local polynomials.

(iv) The reflection-positive lattice quadratic forms for the renormalized observables converge on the positive-time test-function domain, and the limiting source functional has the Euclidean covariance, locality, regularity, and clustering properties required for the intended Osterwalder–Schrader reconstruction.

Consequence of the continuum-universality data. Under Hypothesis 38.9, the continuum separated-point Schwinger functions of the normalized scaling fields are independent of X. More precisely, functional derivatives of $W_*(\eta; u_R)$ give a common distribution on the configuration space of distinct insertion points, while the source-local polynomials in Hypothesis 38.9(iii) can change only contact terms on collision diagonals. Under the positivity and regularity hypotheses in Hypothesis 38.9(iv), the common Schwinger functions reconstruct the same Euclidean-to-Lorentzian QFT data in the observable sector generated by these fields. At $u_R = 0$, if the additional stress-tensor hypothesis stated after Hypothesis 38.2 holds, this common fixed point is the corresponding Ising CFT.

Here the substance is in the hypothesis. By Hypothesis 38.9(i), all tuned microscopic realizations approach the same retained-coordinate point $z_*(u_R)$ at the reference scale. Hypothesis 38.9(ii) separates the dependence of the renormalized source functional into a coordinate-dependent part, a vanishing remainder, and source-local vacuum terms. Hypothesis 38.9(iii) then gives

$$W_a^X[\mathcal{Z}_a^X(\eta)] - W_a^X[0] - P_a^X(\eta) \longrightarrow W_*(\eta; u_R)$$

for every X, modulo source-local polynomials. Taking functional derivatives at $\eta = 0$ gives the limiting distributions of inserted renormalized observables. Source-local polynomial differences are supported on collision diagonals, so they vanish after restriction to pairwise distinct insertion points. This proves equality of separated-point Schwinger functions. Hypothesis 38.9(iv) is precisely the additional closedness and regularity input needed to pass from equality of separated kernels to a common reconstructed Euclidean/Lorentzian theory; it is not supplied by the finite-dimensional RG coordinate estimate.

Hypothesis 38.10 (Ising lattice-to-CFT universality data)**.** Fix a spatial dimension D and a class $\mathfrak{U}$ of reflection-positive $\mathbb{Z}_2$-invariant microscopic lattice models whose thermodynamic limits exist in the pure-state convention used above. Let $A_{\sigma,a}^X$ and $A_{\varepsilon,a}^X$ be odd and even microscopic observables in a model $X \in \mathfrak{U}$, normalized by nonzero constants so that their leading scaling fields are σ and ε. Assume the following data and hypotheses.

(i) The tuned finite-coordinate Wilsonian hypotheses of the local RG-coordinate consequence above hold uniformly for $\mathfrak{U}$, with relevant endpoint coordinate $u_R = (\lambda_{h,R}, \lambda_{t,R})$. In the $\mathbb{Z}_2$-even critical slice this means $\lambda_{h,R} = \lambda_{t,R} = 0$.

(ii) The source-dependent scaling-limit hypotheses of the continuum-universality data above hold for the source coordinates coupled to $A_{\sigma,a}^X$, $A_{\varepsilon,a}^X$, and the other selected scaling observables. The reflected OS quadratic forms converge on the positive-time test-function domain.

(iii) At $u_R = 0$, the common limiting Schwinger hierarchy $W_*(\eta; 0)$ satisfies Euclidean covariance, reflection positivity, regularity, clustering, and the radial reconstruction hypotheses of Hypothesis 59.1. It has a conserved symmetric stress tensor improvable to traceless form, so the fixed point is a CFT in the sense of Chapter 56.

(iv) The fixed-point local CFT data include the local scaling-operator spectrum, two-point pairings, OPE coefficients, stress-tensor Ward data, and the OPE convergence hypotheses of Definition 64.1. For nonzero u_R, the corresponding relevant deformation of the fixed-point source functional by

$$\int \mathrm{d}^D x \left(\lambda_{h,R}\mu_R^{D-\Delta_\sigma}\sigma(x) + \lambda_{t,R}\mu_R^{D-\Delta_\varepsilon}\varepsilon(x)\right)$$

is constructed in a source-functional neighborhood large enough to determine the separated scaling functions under discussion.

Conditional Ising scaling functions. Under Hypothesis 38.10, for every $X \in \mathfrak{U}$, every list of selected microscopic observables, and every configuration of pairwise distinct continuum points $x_1, \ldots, x_n$, the normalized thermodynamic lattice correlators have the common scaling limit

$$\lim_{a \downarrow 0} \left\langle \prod_{j=1}^{n} a^{-\Delta_{I_j}} (b_{I_j}^X)^{-1} A_{I_j,a}^X(x_j) \right\rangle_{X, p_X(a,u_R)} = S_{I_1 \ldots I_n}^{\text{Ising}}(x_1, \ldots, x_n; u_R), \tag{38.11}$$

where I_j labels σ, ε, or another selected scaling operator, $b_{I_j}^X \neq 0$ is the leading microscopic normalization, and $p_X(a, u_R)$ is the tuned microscopic parameter supplied by Hypothesis 38.10(i). At $u_R = 0$, the functions on the right are the separated-point correlators of the Ising CFT. They are determined by the fixed-point CFT data in Hypothesis 38.10(iv) through radial OPE convergence and the Ward identities. At nonzero u_R inside the constructed relevant-deformation neighborhood, the same fixed-point data together with the specified relevant coordinates determine the massive or magnetized scaling functions.

The quantities $T_{c,X}$, metric factors relating $T - T_{c,X}$ and the magnetic field to $\lambda_{t,R}$ and $\lambda_{h,R}$, the constants b_I^X, finite-cutoff corrections, and contact-term extensions are not part of the universal separated-point functions (38.11).

The argument should be read as an unpacking of the hypotheses. The thermodynamic limit supplies the infinite-volume lattice state for each fixed cutoff and tuned microscopic parameter. Hypothesis 38.10(i), by the local RG-coordinate consequence above, sends every tuned microscopic realization to the same retained Wilsonian coordinate

$$z_*(u_R) = (u_R, V_{\mu_R}(u_R))$$

at the reference scale. Hypothesis 38.10(ii), by the continuum-universality consequence above, upgrades that coordinate convergence to convergence of the renormalized source functionals: after the local source maps and source-local vacuum subtractions are applied, all $X \in \mathfrak{U}$ have the same limiting functional $W_*(\eta; u_R)$ modulo source-local polynomials. Functional derivatives with respect to the normalized sources give (38.11). Source-local polynomials are supported on collision diagonals, so they vanish on the configuration space $x_i \neq x_j$.

At $u_R = 0$, Hypothesis 38.10(iii) gives the OS/radial reconstruction data and the stress-tensor condition needed to regard $W_*(\eta; 0)$ as the source functional of a unitary CFT. Hypothesis 38.10(iv) supplies the local CFT data and OPE convergence. The radial OPE theorem then reconstructs separated-point correlators from the local scaling spectrum, two-point pairings, OPE coefficients, and Ward data, with the contact chart fixed by the source functional. For $u_R \neq 0$ in the stated source-functional neighborhood, the deformation by σ and ε is part of the constructed functional data: differentiating the limiting deformed functional $W_*(\eta; u_R)$ at fixed u_R gives the corresponding separated-point scaling functions with the relevant coordinates held fixed. The last sentence lists parameters removed either by tuning, by field normalization, or by restriction to separated points; the preceding argument never identifies them with CFT data.

The same separation applies to observables. Let A^X_{spin} be a microscopic odd lattice observable and A^X_{eng} a microscopic even energy-density observable with identity part subtracted. Their scaling-field expansions have the form

$$A^X_{\mathrm{spin}} = b^X_\sigma a^{\Delta_\sigma}\sigma + \sum_J b^X_J a^{\Delta_J} O_J, \qquad A^X_{\mathrm{eng}} = b^X_\varepsilon a^{\Delta_\varepsilon}\varepsilon + \sum_K c^X_K a^{\Delta_K} O_K,$$

where the sums contain fields with larger scaling dimensions in the same symmetry sector, and $b^X_\sigma, b^X_\varepsilon$ are nonzero for the chosen observables.

Assume now, in addition to the coordinate hypotheses above, that the renormalized separated-point lattice correlators for the selected observables have scaling limits as distributions and that the coordinate-to-correlator map is continuous at the limiting graph $v = V_{\mu_R}(u_R)$. For separated continuum points x_i, the normalized lattice correlators then have the common limit

$$\lim_{a\to 0}\prod_i \left((b^X_i)^{-1}a^{-\Delta_i}\right)\left\langle \prod_i A^X_i([x_i/a])\right\rangle_{X,a,p_X(a,u_R)} = \left\langle \prod_i O_i(x_i)\right\rangle_{u_R},$$

with the same limiting continuum correlator for all X satisfying the coordinate hypotheses and the additional scaling-limit hypotheses. The constants b^X_i, the critical temperature $T_{c,X}$, and finite-cutoff corrections are microscopic. The dimensions, RG eigenvalues, and normalized continuum correlators are fixed-point data once the continuum limit has been constructed.

If the coordinate-to-correlator map is differentiable in the retained irrelevant coordinates, the same estimate gives the standard correction structure with all symbols fixed. Let

$$G^X_a(x_1, \ldots, x_n)$$

denote a normalized separated-point lattice correlator after solving the relevant endpoint equation for u_R. In the finite chart,

$$G^X_a = G_{u_R} + \sum_\rho \left(v^\rho_X(\mu_R; \Lambda_0) - V^\rho_{\mu_R}(u_R)\right) \left.\frac{\partial G}{\partial v^\rho}\right|_{v=V_{\mu_R}(u_R)} + \cdots .$$

For bounded microscopic irrelevant coordinates this gives

$$G^X_a(x_1, \ldots, x_n) = G_{u_R}(x_1, \ldots, x_n) + O((\mu_R a)^\kappa), \qquad a = \Lambda_0^{-1}.$$

When the generated-integral convergence is faster than the leading irrelevant correction, the smallest irrelevant exponent in the symmetry sector gives the leading correction-to-scaling exponent. The amplitudes multiplying these corrections depend on X; the exponents and the limiting correlator do not, under the hypotheses stated above.

Chapter 39

Effective Field Theories Without Poincare-Invariant UV Completion

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

39.1 Scaling Limits and Finite-Cutoff Effective Theories

This chapter separates two constructions that are often conflated. A lattice or many-body system may have a critical scaling limit which is a continuum QFT, or it may have only a finite-cutoff long-distance effective description. The first case can produce a UV-complete relativistic theory. The second case is an effective theory whose validity is tied to the microscopic cutoff and to a specified range of observables.

Definition 39.1 (Critical scaling limit)**.** Let $a > 0$ be a lattice spacing, let $\mu_{a,\kappa}$ be a finite- or infinite-volume Euclidean lattice probability measure with couplings κ, and let $\mathcal{O}_{a,i}(x)$ be lattice observables. A critical scaling limit consists of:

1. a tuning $\kappa(a)$ for which the correlation length in lattice units diverges;
2. normalization factors $Z_i(a)$ and continuum coordinates $x = an$;

3. limits of separated Euclidean correlators

$$\lim_{a\downarrow 0}\left\langle \prod_{j=1}^{k} Z_{i_j}(a)\mathcal{O}_{a,i_j}(n_j)\right\rangle_{\mu_{a,\kappa(a)}};$$

4. Euclidean covariance, locality, reflection positivity, clustering, and regularity estimates sufficient for OS or another stated reconstruction theorem.

When these data are supplied, the limiting continuum QFT is the scaling-limit object constructed from the lattice system.

Definition 39.2 (Finite-cutoff effective theory)**.** A finite-cutoff effective theory is a tuple

$$(\Lambda_{\mathrm{UV}}, \mathcal{A}_\Lambda, S_\Lambda, \mathcal{O}_\Lambda, \mathcal{R}, E_{\max})$$

where Λ_{UV} is a physical ultraviolet scale, $\mathcal{A}_\Lambda$ is the algebra or field-variable space retained below that scale, S_Λ is the regulated action or Hamiltonian, $\mathcal{O}_\Lambda$ is the declared operator-insertion prescription, $\mathcal{R}$ is the set of observables to be matched, and $E_{\max} \ll \Lambda_{\mathrm{UV}}$ is the energy or momentum domain in which the error estimate is claimed. It is a QFT technique only after the regulator, matching observables, and error class have been stated.

The two definitions have different logical content. A critical scaling limit aims to remove the microscopic cutoff. A finite-cutoff effective theory keeps the cutoff as part of its meaning, even when its formulas resemble a relativistic continuum Lagrangian.

Remark 39.3 (Scaling limit as continuum QFT). Suppose the limits in Definition 39.1 exist for a set of local lattice observables closed under the operator products needed to generate the limiting Schwinger functions, and suppose the limiting Schwinger functions satisfy the OS hypotheses with Euclidean covariance. Then the reconstructed theory is a continuum local QFT. If the limiting Euclidean group enhances to the full Euclidean rotation group and the OS linear-growth hypothesis needed for continuation is verified, the Lorentzian reconstruction is a Poincare-covariant Wightman theory.

The OS reconstruction theorem applies to the limiting Schwinger functions, not to the finite lattice variables themselves. Reflection positivity gives a positive Hilbert-space inner product after quotienting null vectors; Euclidean time translations give a positive Hamiltonian; Euclidean covariance and the required analytic-growth bound give the Lorentzian representation of the Poincare group; and locality follows from Euclidean symmetry and the OS permutation/reflection structure after analytic continuation. The microscopic lattice remains a regulator for the construction, while the Hilbert space, fields, and correlation functions are those of the limiting object.

Thus the two-dimensional Ising scaling limit and, conjecturally, the three-dimensional Ising critical point belong to the first category. Away from criticality, a gapped lattice system may still be described at long distances by a continuum effective action, but the continuum action is then a finite-cutoff description of selected observables, not a removed-cutoff definition of a Poincare-invariant QFT.

39.2 What Must Be Revalidated

Let $\mathcal{T}_{\rm rel}$ be a theorem whose proof in earlier volumes used Poincare covariance, the spectral condition, local commutativity, and the existence of asymptotic particle states. Applying the same theorem to a condensed-matter EFT with microscopic cutoff a^{-1} requires a replacement proof or a matching argument.

Definition 39.4 (Validation map for borrowed QFT technology)**.** A validation map from a relativistic theorem or construction $\mathcal{T}_{\rm rel}$ to a finite-cutoff effective theory consists of:

1. the precise hypotheses $H_1, \dots, H_n$ used in the relativistic proof;
2. finite-cutoff replacements H_i^Λ or a proof that H_i is emergent with a stated error bound;
3. the observable class $\mathcal{R}$ on which the conclusion is claimed;
4. an estimate comparing the microscopic observable and the effective observable in that class;
5. a statement of which parts of the relativistic conclusion are retained, modified, or unavailable.

The following table records typical outcomes.

Relativistic tool	Finite-cutoff status
Wilsonian integration	Available after the cutoff, fields, and blocking map are stated.
Power counting	Available with the correct dynamical exponent and symmetries.
OS/Wightman analyticity	Requires an independent Euclidean-to-real-time argument.
LSZ scattering	Requires stable quasiparticles and an asymptotic dynamics.
Spin-statistics and CPT	Do not transfer without Lorentz/locality hypotheses.
Wedge modular flow	Requires a relativistic wedge or a replacement modular theorem.
Anomaly matching	Transfers only for the specified symmetry and regulator response.

The entries are not restrictions on using effective field theory. They are the data needed to know what the calculation proves.

39.3 Examples

39.3.1 Fermi Liquid

A Fermi liquid is organized around a codimension-one Fermi surface Σ_F in momentum space, not around the Lorentz-invariant point $p_\mu p^\mu = m^2$. A local patch with outward normal $\hat{n}$ has linearized dispersion

$$\epsilon(k_F\hat{n} + \ell\hat{n} + q_\parallel) = v_F\ell + O(\ell^2, q_\parallel^2),$$

where $q_\parallel$ is tangent to Σ_F. The scaling that keeps the kinetic term fixed scales ℓ and frequency differently from tangential momenta. Hence relativistic spin-statistics, crossing, and LSZ are not the primitive logic. The primitive data are the microscopic Hamiltonian, the Fermi surface, quasiparticle lifetime estimates, Landau parameters, and the range of frequencies and momenta for which quasiparticles are sharp.

39.3.2 Lifshitz Fixed Points

A Lifshitz scalar with dynamical exponent z has a quadratic action of the form

$$S = \frac{1}{2}\int \mathrm{d}\tau\,\mathrm{d}^d x\,\left[(\partial_\tau\phi)^2 + \kappa^2(\Delta^{z/2}\phi)^2\right]$$

when z is even, with the obvious Fourier-symbol replacement for general integer z. The scaling is

$$x \mapsto bx, \qquad \tau \mapsto b^z\tau.$$

Power counting is meaningful, but it is not relativistic power counting. If a Lorentz-invariant relativistic action appears in the infrared, the Lorentz-violating operators must be shown to be irrelevant or tuned small in the stated scaling regime.

39.3.3 Quantum Hall Effective Theory

The integer or fractional quantum Hall response at long distances may be encoded by a Chern–Simons effective action

$$S_{\mathrm{CS}}[a, A] = \frac{K_{IJ}}{4\pi}\int a^I \mathrm{d}a^J + \frac{q_I}{2\pi}\int A\mathrm{d}a^I + \cdots .$$

This is a topological response theory for a gapped many-body phase with a microscopic non-relativistic Hamiltonian and background electromagnetic field. Its gauge invariance, boundary anomaly, and line-operator data are real low-energy statements, but they do not imply a local Poincare-invariant UV completion. The correct validation map matches ground-state response, edge anomaly, quasiparticle braiding, and finite-size gaps, not an asymptotic-particle S-matrix.

39.4 Controlled Use of Emergent Relativistic Language

Controlled Approximation 39.5 (Emergent relativistic EFT)**.** Let a microscopic system with cutoff Λ_{UV} have low-energy observables $\mathcal{R}$. A relativistic EFT with action S_{rel} is a controlled emergent description on $\mathcal{R}$ if there exists a matching map M_Λ and constants $C, \Delta > 0$ such that for every observable $O \in \mathcal{R}$ with characteristic scale $E \le E_{\max}$,

$$|\langle O\rangle_{\mathrm{micro}} - \langle M_\Lambda O\rangle_{S_{\mathrm{rel}}}| \le C_O\left(\frac{E}{\Lambda_{\mathrm{UV}}}\right)^{\Delta}$$

after the stated renormalizations. The exponent Δ is the smallest scaling dimension gap between retained operators and omitted symmetry-allowed Lorentz-violating operators in the chosen RG chart.

This formulation is deliberately observable-based. It does not require that the microscopic system possess a Lorentzian spacetime symmetry. It requires only that the selected observables are approximated by a relativistic continuum theory with a quantitative error bound. In contrast, a critical scaling limit satisfying Remark 39.3 constructs the continuum QFT as its limiting object.

Open Problem 39.6 (Validated condensed-matter EFT atlas). For each major class of condensed-matter effective theory, construct an atlas of validation maps in the sense of Definition 39.4: Fermi liquids, non-Fermi liquids, superconductors, deconfined critical points, fractional quantum Hall states, spin liquids, fracton phases, and systems with emergent gauge fields. The goal is not to force these systems into Poincare-invariant axioms, but to state exactly which QFT techniques are justified for which observables.

Part IV

Gauge Theory, Infrared Structure, and Anomalies

Chapter 40

Spinor Fields, Fermionic Statistics, and Grassmann Path Integrals

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

40.1 The Free Dirac Field

Definition 40.1 (Spinor convention and massive intertwiners)**.** Work in $D = 4$ with

$$\eta_{\mu\nu} = \mathrm{diag}(-,+,+,+), \qquad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}.$$

The convention is the one fixed globally in Section 40.16. In particular, the Dirac adjoint, chirality matrix, Majorana conjugation, and later anomaly traces are not independent sign choices. Set

$$\partial\!\!\!/ = \gamma^\mu \partial_\mu, \qquad p\!\!\!/ = p_\mu \gamma^\mu, \qquad \beta = \mathrm{i}\gamma^0.$$

For a spinor s, its Dirac adjoint is

$$\bar{s} = s^\dagger \beta.$$

Let $u^\sigma(p)$ and $v^\sigma(p)$, $\sigma = 1, 2$, be the spinor intertwiners on the positive-energy mass shell Σ_m^+ constructed in Chapter 20:

$$(p\!\!\!/ - \mathrm{i}m)u^\sigma(p) = 0, \qquad (p\!\!\!/ + \mathrm{i}m)v^\sigma(p) = 0.$$

The delta-normalized spinors used below inherit the normalization

$$\bar{u}^\sigma(p)u^\tau(p) = \frac{m}{\omega_{\vec{p}}}\delta^{\sigma\tau}, \qquad \bar{v}^\sigma(p)v^\tau(p) = -\frac{m}{\omega_{\vec{p}}}\delta^{\sigma\tau}, \qquad \bar{u}^\sigma(p)v^\tau(p) = 0,$$

where $\bar{u} = u^\dagger\beta$, $\bar{v} = v^\dagger\beta$, and $p^0 = \omega_{\vec{p}}$ on Σ_m^+. The corresponding individual completeness relations are

$$\sum_{\sigma=1}^{2} u_\alpha^\sigma(p)\bar{u}^{\sigma,\beta}(p) = \frac{(m - \mathrm{i}\, \not{p})_\alpha{}^\beta}{2\omega_{\vec{p}}}, \qquad \sum_{\sigma=1}^{2} v_\alpha^\sigma(p)\bar{v}^{\sigma,\beta}(p) = -\frac{(m + \mathrm{i}\, \not{p})_\alpha{}^\beta}{2\omega_{\vec{p}}}. \tag{40.1}$$

At the reference momentum $p_R = (m, \vec{0})$, $\not{p}_R = \mathrm{i}m\beta$, so

$$\sum_\sigma u^\sigma(p_R)\bar{u}^\sigma(p_R) = \frac{1+\beta}{2}, \qquad \sum_\sigma v^\sigma(p_R)\bar{v}^\sigma(p_R) = -\frac{1-\beta}{2}.$$

These rest-frame matrices obey

$$(\not{p}_R - \mathrm{i}m)\frac{1+\beta}{2} = 0, \qquad (\not{p}_R + \mathrm{i}m)\left[-\frac{1-\beta}{2}\right] = 0,$$

and have traces 2 and -2, reproducing the summed Dirac-adjoint normalizations at $p^0 = m$. Lorentz covariance of the intertwiners transports the identities to all $p \in \Sigma_m^+$. Let

$$\mathrm{d}\mu_m(p) = \frac{\mathrm{d}^3\vec{p}}{(2\pi)^3\, 2\omega_{\vec{p}}}, \qquad \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}, \qquad p^0 = \omega_{\vec{p}}.$$

Definition 40.2 (Free charged Dirac field and CAR)**.** Introduce two independent families of fermionic creation and annihilation operators

$$b_\sigma(\vec{p}),\ b_\sigma^\dagger(\vec{p}), \qquad d_\sigma(\vec{p}),\ d_\sigma^\dagger(\vec{p}),$$

with canonical anticommutation relations

$$\{b_\sigma(\vec{p}), b_\tau^\dagger(\vec{q})\} = (2\pi)^3\, 2\omega_{\vec{p}}\, \delta_{\sigma\tau}\delta^{(3)}(\vec{p} - \vec{q}),$$

$$\{d_\sigma(\vec{p}), d_\tau^\dagger(\vec{q})\} = (2\pi)^3\, 2\omega_{\vec{p}}\, \delta_{\sigma\tau}\delta^{(3)}(\vec{p} - \vec{q}),$$

all other anticommutators being zero. The vacuum vector Ω is annihilated by all $b_\sigma(\vec{p})$ and $d_\sigma(\vec{p})$.

The charged Dirac field is the operator-valued distribution

$$\widehat{\psi}_\alpha(x) = \sum_{\sigma=1}^{2}\int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\ \left[v_\alpha^\sigma(p)b_\sigma^\dagger(\vec{p})\mathrm{e}^{-\mathrm{i}p\cdot x} + u_\alpha^\sigma(p)d_\sigma(\vec{p})\mathrm{e}^{\mathrm{i}p\cdot x}\right].$$

Its adjoint field is

$$\widehat{\bar{\psi}}^\alpha(x) = \left(\widehat{\psi}^\dagger(x)\beta\right)^\alpha.$$

The field equation and charge convention are immediate consequences of the chosen momentum-space intertwiners and oscillator charges. Because of the two momentum-space Dirac equations above,

$$(\not{\partial} + m)\widehat{\psi}(x) = 0, \qquad \widehat{\bar{\psi}}(x)(-\overleftarrow{\not{\partial}} + m) = 0,$$

where the arrow indicates that the derivative acts on the field to its left.

The convention in this chapter assigns charge $+1$ to the one-particle states created by $b^\dagger$ and charge -1 to the states created by $d^\dagger$. If

$$U_\theta = \mathrm{e}^{\mathrm{i}\theta\widehat{Q}}, \qquad U_\theta\Omega = \Omega,$$

then

$$U_\theta\widehat{\psi}_\alpha(x)U_\theta^{-1} = \mathrm{e}^{\mathrm{i}\theta}\widehat{\psi}_\alpha(x), \qquad U_\theta\widehat{\overline{\psi}}^{\alpha}(x)U_\theta^{-1} = \mathrm{e}^{-\mathrm{i}\theta}\widehat{\overline{\psi}}^{\alpha}(x).$$

For the creation part of $\widehat{\psi}$,

$$(\not{\partial} + m)\big(v^\sigma(p)\mathrm{e}^{-\mathrm{i}p\cdot x}\big) = (-\mathrm{i}\not{p} + m)v^\sigma(p)\mathrm{e}^{-\mathrm{i}p\cdot x} = 0,$$

because $(\not{p} + \mathrm{i}m)v^\sigma(p) = 0$. For the annihilation part,

$$(\not{\partial} + m)\big(u^\sigma(p)\mathrm{e}^{\mathrm{i}p\cdot x}\big) = (\mathrm{i}\not{p} + m)u^\sigma(p)\mathrm{e}^{\mathrm{i}p\cdot x} = 0,$$

because $(\not{p} - \mathrm{i}m)u^\sigma(p) = 0$. The adjoint equation follows by taking the Dirac adjoint and using $(\gamma^\mu)^\dagger\beta = -\beta\gamma^\mu$, with the derivative acting on the field to its left.

The charge convention is fixed on the oscillator algebra by

$$[\widehat{Q}, b_\sigma^\dagger(\vec{p})] = b_\sigma^\dagger(\vec{p}), \qquad [\widehat{Q}, d_\sigma^\dagger(\vec{p})] = -d_\sigma^\dagger(\vec{p}).$$

Taking adjoints gives

$$[\widehat{Q}, b_\sigma(\vec{p})] = -b_\sigma(\vec{p}), \qquad [\widehat{Q}, d_\sigma(\vec{p})] = d_\sigma(\vec{p}).$$

Both terms in $\widehat{\psi}$ therefore have charge $+1$, while $\widehat{\overline{\psi}}$ has charge -1. Exponentiating the commutators gives the displayed U_θ-transformation laws.

40.2 The Statistics Sign from Locality

Proposition 40.3 (Free spinor locality selects the CAR sign)**.** *Once the spin-$\frac{1}{2}$ one-particle representation and the covariant spinor intertwiners have been chosen, the local free field is obtained from canonical anticommutators rather than ordinary commutators. Use the delta-normalized polarizations $\mathcal{U}, \mathcal{V}$ of Chapter 20. Let*

$$\Delta_+(z; m^2) = \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, \mathrm{e}^{\mathrm{i}p\cdot z}$$

be the positive-frequency scalar distribution, and form the trial field

$$\widehat{\Phi}_\alpha(x) = \sum_{\sigma=1}^{2} \int \frac{d^3\vec{p}}{(2\pi)^{3/2}} \Big[\mathcal{V}_\alpha^\sigma(p)a_\sigma^\dagger(\vec{p})\mathrm{e}^{-\mathrm{i}p\cdot x} + \mathcal{U}_\alpha^\sigma(p)c_\sigma(\vec{p})\mathrm{e}^{\mathrm{i}p\cdot x}\Big].$$

Suppose first that the oscillator algebra used ordinary commutators,

$$[a_\sigma(\vec{p}), a_\tau^\dagger(\vec{q})] = [c_\sigma(\vec{p}), c_\tau^\dagger(\vec{q})] = \delta_{\sigma\tau}\delta^{(3)}(\vec{p} - \vec{q}),$$

with all other commutators zero. Using (40.1) gives

$$[\widehat{\Phi}_\alpha(x), \widehat{\overline{\Phi}}^\beta(y)] = \int \frac{d^3\vec{p}}{(2\pi)^3} \left[\mathrm{e}^{-\mathrm{i}p\cdot(x-y)} \frac{m + \mathrm{i}\not{p}}{2p^0} + \mathrm{e}^{\mathrm{i}p\cdot(x-y)} \frac{m - \mathrm{i}\not{p}}{2p^0} \right]_\alpha{}^\beta.$$

The scalar part contains

$$m\big(\Delta_+(x-y;m^2) + \Delta_+(y-x;m^2)\big),$$

which has nonzero spacelike values. For example, at equal times and $\vec{x} \neq \vec{y}$ it equals the distribution

$$2m \int \frac{d^3\vec{p}}{(2\pi)^3\, 2\omega_{\vec{p}}}\, \mathrm{e}^{\mathrm{i}\vec{p}\cdot(\vec{x}-\vec{y})}.$$

Thus the field obtained with the commutator sign is nonlocal.

If instead the oscillator algebra is the canonical anticommutator algebra, (40.1) gives

$$\{\widehat{\psi}_\alpha(x), \widehat{\overline{\psi}}^\beta(y)\} = (-\not{\partial}_x + m)_\alpha{}^\beta \left[\Delta_+(x-y;m^2) - \Delta_+(y-x;m^2)\right].$$

The bracket is the Pauli–Jordan distribution. It vanishes for spacelike separation, so the spinor field is local as an odd field in a graded local algebra. This computation is the free-field derivation of the statistics sign used in the rest of the chapter.

Proof. Only the oscillator pairings $aa^\dagger$ and $cc^\dagger$, or their CAR counterparts, contribute. Inserting the mode expansion and using the spin sums (40.1) gives the two displayed kernels. For ordinary commutators the scalar part is the even combination

$$m\big(\Delta_+(x-y;m^2) + \Delta_+(y-x;m^2)\big).$$

At equal time and $\vec{x} \neq \vec{y}$ this is the Fourier transform of the positive mass-shell density $(2\omega_{\vec{p}})^{-1} d^3\vec{p}$ plus its reflected copy. It is not the zero distribution; for example smearing by a nonzero real test function supported in an equal-time ball gives a strictly positive quadratic form. Thus ordinary commutators violate locality.

For CAR, the creation and annihilation contributions enter with the relative minus sign shown above, and the scalar factor is the Pauli–Jordan distribution

$$\Delta(x-y;m^2) = \Delta_+(x-y;m^2) - \Delta_+(y-x;m^2).$$

The scalar chapter proves that $\Delta(z;m^2)$ has support in the closed light cone. Applying the local differential operator $(-\not{\partial}_x + m)_\alpha{}^\beta$ cannot enlarge support, so the anticommutator vanishes at spacelike separation. This is precisely graded locality for an odd field. □

40.3 Locality and the CAR

With the same positive-frequency scalar distribution

$$\Delta_+(x;m^2) = \int_{\Sigma_m^+} \mathrm{d}\mu_m(p)\, \mathrm{e}^{\mathrm{i}p\cdot x}.$$

With the spinor normalizations fixed by the one-particle inner product, the field anticommutator is

$$\{\widehat{\psi}_\alpha(x), \widehat{\overline{\psi}}^{\beta}(y)\} = (-\not{\partial}_x + m)_\alpha{}^\beta \left[\Delta_+(x-y;m^2) - \Delta_+(y-x;m^2)\right].$$

The bracketed distribution is the Pauli–Jordan distribution in the normalization convention already used for the scalar field. It vanishes at spacelike separation. Hence

$$\{\widehat{\psi}_\alpha(x), \widehat{\overline{\psi}}^{\beta}(y)\} = 0 \qquad \text{when } (x-y)^2 > 0.$$

The equal-time anticommutator is obtained by differentiating the Pauli–Jordan distribution:

$$\{\widehat{\psi}_\alpha(t,\vec{x}), \widehat{\overline{\psi}}^{\beta}(t,\vec{y})\} = \mathrm{i}(\gamma^0)_\alpha{}^\beta \delta^{(3)}(\vec{x}-\vec{y}),$$

and

$$\{\widehat{\psi}_\alpha(t,\vec{x}), \widehat{\psi}_\beta(t,\vec{y})\} = 0, \qquad \{\widehat{\overline{\psi}}^{\alpha}(t,\vec{x}), \widehat{\overline{\psi}}^{\beta}(t,\vec{y})\} = 0.$$

The local algebra is therefore a graded local algebra. If $|A| \in \{0,1\}$ is the fermion parity of a homogeneous operator A, the graded commutator is

$$[A,B]_{\mathrm{g}} = AB - (-1)^{|A||B|}BA.$$

Locality is the vanishing of this graded commutator for spacelike separated smeared fields. The Dirac field has parity 1, while bosonic local fields have parity 0.

Remark 40.4. The preceding computation is the free spin-$\frac{1}{2}$ instance of the spin-statistics connection. A theorem-level spin-statistics statement requires a specified Wightman framework: a common invariant dense domain for the smeared fields, a positive-energy unitary representation of the spin cover of the proper orthochronous Poincare group, covariance of the fields as operator-valued tempered distributions, locality on the common domain, positivity of the Hilbert-space inner product, and the stated adjoint compatibility between fields and their conjugates. Under those hypotheses, integer-spin fields have even local commutation and half-integer-spin fields have odd local anticommutation, except for fields that vanish in the physical Hilbert space. This chapter does not use that theorem as an input; the displayed anticommutator is the direct verification needed for the free Dirac field used below.

40.4 Dirac, Weyl, and Majorana Fields

Definition 40.5 (Dirac, Weyl, and Majorana local spinor data)**.** The Dirac field transforms in the spinor representation:

$$U(\Lambda)\widehat{\psi}_\alpha(x)U(\Lambda)^{-1} = R_\alpha{}^\beta(\Lambda)\widehat{\psi}_\beta(\Lambda x),$$

where

$$R(\Lambda) = \exp\left(-\frac{\mathrm{i}}{2}\omega_{\mu\nu}S^{\mu\nu}\right), \qquad S^{\mu\nu} = -\frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu],$$

for $\Lambda = \exp\omega$. The field $\widehat{\psi}$ has four complex components before additional Lorentz-covariant conditions are imposed.

Define

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3.$$

It obeys

$$\gamma_5^2 = 1, \qquad \{\gamma_5, \gamma^\mu\} = 0, \qquad [\gamma_5, S^{\mu\nu}] = 0.$$

Therefore

$$P_\pm = \frac{1 \pm \gamma_5}{2}$$

are Lorentz-covariant projectors. A Weyl spinor field is a Dirac spinor field subject to

$$\widehat{\chi}_\pm = P_\pm \widehat{\chi}_\pm.$$

In the explicit Clifford basis used in Chapter 20,

$$\gamma_5 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad P_+ = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_- = \begin{pmatrix} 0 & 0 \\ 0 & I_2 \end{pmatrix}.$$

Thus a Weyl condition removes two of the four complex Dirac components while preserving Lorentz covariance. The two chiralities are inequivalent complex representations of $\mathrm{Spin}^+(1,3)$. The Dirac mass term couples them because γ^μ maps one chirality to the other.

A Majorana condition is a Lorentz-covariant reality condition. Suppose a matrix B is chosen so that

$$(\gamma^\mu)^* = B\gamma^\mu B^{-1}, \qquad B^{-1}(B^{-1})^* = 1.$$

Here $*$ denotes the antilinear operation on spinor components, combined with componentwise adjunction when it is applied to operator-valued spinor fields. In the displayed gamma basis one may take $B = \gamma_2$. Since $\sigma_2^* = -\sigma_2$, this choice obeys the two identities above. Then

$$\widehat{\psi}^c = B^{-1}\widehat{\psi}^*$$

transforms in the same Lorentz representation as $\widehat{\psi}$. A Majorana field satisfies

$$\widehat{\xi} = \widehat{\xi}^c.$$

Lorentz covariance of chiral and Majorana reductions. The projectors $P_\pm = (1 \pm \gamma_5)/2$ define Lorentz-covariant Weyl subfields. If B obeys the two identities in Definition 40.5, then charge conjugation $\psi \mapsto \psi^c = B^{-1}\psi^*$ is Lorentz covariant and involutive. In four dimensions a Weyl field $\widehat{\chi}_+$ determines a Majorana field

$$\widehat{\xi} = \widehat{\chi}_+ + B^*\widehat{\chi}_+^*,$$

whose P_+-projection is $\widehat{\chi}_+$. The commutator $[\gamma_5, S^{\mu\nu}] = 0$ implies $[P_\pm, R(\Lambda)] = 0$ for the connected spin group, hence the Weyl conditions $P_\pm\chi_\pm = \chi_\pm$ are preserved by Lorentz transformations. The charge-conjugation covariance check is direct:

$$\begin{aligned} U(\Lambda)\widehat{\psi}^c(x)U(\Lambda)^{-1} &= B^{-1}\left(R(\Lambda)\widehat{\psi}(\Lambda x)\right)^* \\ &= B^{-1}BR(\Lambda)B^{-1}\widehat{\psi}^*(\Lambda x) = R(\Lambda)\widehat{\psi}^c(\Lambda x). \end{aligned}$$

The second identity for B gives $(\psi^c)^c = \psi$, so the Majorana condition is a genuine reality condition on the complex Dirac module. If $\widehat{\chi}_+ = P_+\widehat{\chi}_+$ is a Weyl field, then

$$\widehat{\xi} = \widehat{\chi}_+ + B^*\widehat{\chi}_+^*$$

is Majorana, and

$$P_+\widehat{\xi} = \widehat{\chi}_+.$$

Indeed $P_+B^* = B^*P_-^*$, so

$$P_+B^*\widehat{\chi}_+^* = B^*P_-^*\widehat{\chi}_+^* = B^*(P_-\widehat{\chi}_+)^* = 0.$$

Moreover $B^*\widehat{\chi}_+^*$ is anti-chiral:

$$\gamma_5 B^*\widehat{\chi}_+^* = -B^*\gamma_5^*\widehat{\chi}_+^* = -B^*\widehat{\chi}_+^*.$$

Thus in four dimensions a Majorana spinor is equivalent, as Lorentz representation data, to one Weyl spinor together with its conjugate. These are statements about local field representations. Particle content is determined separately by the spectral and asymptotic data of the theory.

40.5 Odd Field Variables and Configuration Superspaces

Definition 40.6 (Finite fermionic configuration superspace)**.** There are two distinct meanings of the word "fermionic field" in this chapter. The operator field $\widehat{\psi}_\alpha(x)$ is an operator-valued distribution on a $\mathbb{Z}_2$-graded Hilbert space. After smearing by test functions, it gives conventional linear operators on a common dense domain, with parity 1. Its locality property is the vanishing of the graded commutator at spacelike separation. This is a statement about the local operator algebra.

The variables $\psi_\alpha(x)$, $\bar{\psi}^\alpha(x)$ appearing in a fermionic path integral are different objects. In a finite-dimensional regularization they are odd coordinates on a supermanifold. Let V be a finite-dimensional complex vector space carrying the cutoff spinor labels, including spacetime lattice points, spinor indices, and any internal indices. The purely odd affine superspace with those labels is ΠV. As a locally super-ringed space it is

$$\Pi V = \big(\{*\}, \mathcal{O}_{\Pi V}\big), \qquad \mathcal{O}_{\Pi V} = \Lambda(V^\vee),$$

where $V^\vee$ is placed in odd degree. Equivalently, for every supercommutative test algebra $R = R_{\bar{0}} \oplus R_{\bar{1}}$, its R-points are

$$(\Pi V)(R) = R_{\bar{1}} \otimes_{\mathbb{C}} V.$$

Thus a Grassmann field variable is an odd coordinate in the structure sheaf, or equivalently an $R_{\bar{1}}$-valued component in the functor-of-points description. It is not an ordinary function on spacetime and it is not a point of an underlying set of configurations. For independent Dirac variables the finite-dimensional configuration superspace is

$$\Pi(V_\psi \oplus V_{\bar{\psi}}), \qquad \mathcal{O} = \Lambda(V_\psi^\vee \oplus V_{\bar{\psi}}^\vee),$$

with ψ and $\bar{\psi}$ treated as independent odd coordinates until a real or adjoint structure is imposed by the operator theory.

If bosonic cutoff fields are present, their configuration space is an ordinary affine or smooth space B. Fermionic directions are then encoded by an odd vector bundle $E \to B$, and the corresponding configuration supermanifold is the locally super-ringed space

$$\big(B, \mathcal{O}_B \otimes \Lambda(E^\vee)\big)$$

in a local trivialization. This description is the finite-dimensional geometric meaning of the formal phrase "integration over fermionic fields."

Purely odd Berezinian transformation. Berezin integration is the associated algebraic integration on the Berezinian line. For a purely odd coordinate system $(\eta_1, \ldots, \eta_N)$, choosing the ordered Berezinian element $\mathrm{d}\eta_N \cdots \mathrm{d}\eta_1$ defines the linear functional which extracts the coefficient of $\eta_1 \cdots \eta_N$. Under a change of odd coordinates $\eta' = A\eta$, this ordered Berezinian transforms as

$$\mathrm{d}\eta'_N \cdots \mathrm{d}\eta'_1 = (\det A)^{-1} \mathrm{d}\eta_N \cdots \mathrm{d}\eta_1.$$

This inverse determinant is the odd part of the Berezinian transformation law. The operative object is the coordinate-independent algebraic functional on Berezinian densities; positivity and countable additivity belong to Borel integration and play no role in this odd integration operation. The continuum symbol

$$\int [\mathrm{d}\bar{\psi}\, \mathrm{d}\psi]\ (\cdots)$$

used below denotes a compatible family of such finite-dimensional Berezin integrals after a regulator has supplied finitely many odd coordinates. All determinants, Pfaffians, source contractions, and fermionic signs in the formal continuum notation are inherited from these finite-dimensional Berezinian operations.

The ordered integral with density $\mathrm{d}\eta_N \cdots \mathrm{d}\eta_1$ extracts the coefficient of the ordered monomial $\eta_1 \cdots \eta_N$. If $\eta'_a = A_a{}^b \eta_b$, then antisymmetry of the odd coordinates gives

$$\eta'_1 \cdots \eta'_N = (\det A) \eta_1 \cdots \eta_N.$$

For the coefficient extracted from $f(\eta)\mathrm{d}\eta_N \cdots \mathrm{d}\eta_1$ to be coordinate independent, the ordered Berezinian element must transform by the inverse factor:

$$\mathrm{d}\eta'_N \cdots \mathrm{d}\eta'_1 = (\det A)^{-1} \mathrm{d}\eta_N \cdots \mathrm{d}\eta_1.$$

This is the odd part of the Berezinian. The construction uses only the finite-dimensional exterior algebra and its top coefficient. No underlying set of odd values, sigma-algebra, or positivity structure enters.

40.6 Classical Grassmann Mechanics and Dirac Brackets

Definition 40.7 (Finite odd first-order mechanical datum)**.** The finite-dimensional model underlying fermion path integrals is specified by its coordinate superalgebra. Let η_α, $\alpha = 1, \ldots, N$, be odd generators of a Grassmann algebra,

$$\eta_\alpha \eta_\beta = -\eta_\beta \eta_\alpha, \qquad \eta_\alpha^2 = 0.$$

Functions of the formal coordinates are elements of this algebra. Consider the first-order action

$$S[\eta] = \int \mathrm{d}t\, M_{\alpha\beta}\eta_\alpha\dot{\eta}_\beta, \qquad M_{\alpha\beta} = M_{\beta\alpha},$$

where M is nondegenerate. The symmetry of M is compatible with the odd nature of the coordinates because integration by parts gives

$$-\int \mathrm{d}t\, M_{\alpha\beta}\dot{\eta}_\alpha\eta_\beta = \int \mathrm{d}t\, M_{\alpha\beta}\eta_\beta\dot{\eta}_\alpha.$$

Using the right derivative with respect to $\dot{\eta}_\alpha$, the canonical momenta are

$$\pi_\alpha = L\frac{\overleftarrow{\partial}}{\partial\dot{\eta}_\alpha} = M_{\alpha\beta}\eta_\beta.$$

Thus the phase-space variables obey primary constraints

$$\chi_\alpha = \pi_\alpha - M_{\alpha\beta}\eta_\beta = 0.$$

For odd canonical variables use the graded Poisson brackets

$$\{\eta_\alpha, \eta_\beta\}_{\mathrm{P}} = 0, \qquad \{\pi_\alpha, \pi_\beta\}_{\mathrm{P}} = 0, \qquad \{\eta_\alpha, \pi_\beta\}_{\mathrm{P}} = \{\pi_\beta, \eta_\alpha\}_{\mathrm{P}} = \delta_{\alpha\beta}.$$

Odd second-class constraints and Dirac brackets. For the datum of Definition 40.7, the constraints $\chi_\alpha = \pi_\alpha - M_{\alpha\beta}\eta_\beta$ are second class with constraint matrix

$$K_{\alpha\beta} = \{\chi_\alpha, \chi_\beta\}_{\mathrm{P}} = -2M_{\alpha\beta},$$

and the Dirac bracket

$$\{f, g\}_{\mathrm{D}} = \{f, g\}_{\mathrm{P}} - \{f, \chi_\alpha\}_{\mathrm{P}}(K^{-1})^{\alpha\beta}\{\chi_\beta, g\}_{\mathrm{P}}.$$

annihilates the constraints:

$$\{f, \chi_\alpha\}_{\mathrm{D}} = 0.$$

For the odd coordinates,

$$\begin{aligned}\{\eta_\alpha, \eta_\beta\}_{\mathrm{D}} &= -(K^{-1})_{\alpha\beta} = \frac{1}{2}(M^{-1})_{\alpha\beta},\\ \{\eta_\alpha, \pi_\beta\}_{\mathrm{D}} &= \frac{1}{2}\delta_{\alpha\beta}.\end{aligned}$$

In the $N = 2$ model below, quantization gives a two-state Hilbert space with Hamiltonian eigenvalues $-\omega/4$ and $+\omega/4$.

Using the graded Poisson brackets in the definition,

$$\begin{aligned}\{\chi_\alpha, \chi_\beta\}_{\mathrm{P}} &= \{\pi_\alpha - M_{\alpha\gamma}\eta_\gamma, \pi_\beta - M_{\beta\delta}\eta_\delta\}_{\mathrm{P}}\\ &= -M_{\beta\alpha} - M_{\alpha\beta} = -2M_{\alpha\beta},\end{aligned}$$

because $M_{\alpha\beta} = M_{\beta\alpha}$. Nondegeneracy of M makes K invertible, hence the constraints are second class. The displayed Dirac bracket is the usual Schur-complement correction of the graded Poisson bracket by the inverse constraint matrix. Substituting one argument χ_α gives zero because $\{\chi_\rho, \chi_\alpha\}_{\mathrm{P}} = K_{\rho\alpha}$. For the coordinates, $\{\eta_\alpha, \chi_\rho\}_{\mathrm{P}} = \delta_{\alpha\rho}$, so

$$\{\eta_\alpha, \eta_\beta\}_{\mathrm{D}} = -\delta_{\alpha\rho}(K^{-1})^{\rho\sigma}\delta_{\sigma\beta} = \frac{1}{2}(M^{-1})_{\alpha\beta}.$$

The same substitution gives $\{\eta_\alpha, \pi_\beta\}_{\mathrm{D}} = \delta_{\alpha\beta}/2$.

For $N = 2$, take

$$S = \int \mathrm{d}t\, \mathrm{i}(\eta_1\dot\eta_1 + \eta_2\dot\eta_2).$$

Equivalently, with

$$\eta = \eta_1 + \mathrm{i}\eta_2, \qquad \chi = \eta_1 - \mathrm{i}\eta_2,$$

the difference between $\int \mathrm{d}t\, \mathrm{i}(\eta_1\dot\eta_1 + \eta_2\dot\eta_2)$ and $\int \mathrm{d}t\, \mathrm{i}\chi\dot\eta$ is a total derivative. The endpoint convention fixes which representative of the first-order action is used. The Dirac brackets are

$$\{\eta, \chi\}_{\mathrm{D}} = -\mathrm{i}, \qquad \{\eta, \eta\}_{\mathrm{D}} = 0 = \{\chi, \chi\}_{\mathrm{D}}.$$

Quantization replaces η_α by operators satisfying

$$\{\widehat\eta_\alpha, \widehat\eta_\beta\} = \frac{\mathrm{i}\hbar}{2}(M^{-1})_{\alpha\beta}.$$

In units $\hbar = 1$, the oscillator with

$$L = \mathrm{i}\eta_1\dot\eta_1 + \mathrm{i}\eta_2\dot\eta_2 - \mathrm{i}\omega\eta_1\eta_2$$

has Hamiltonian $H = \mathrm{i}\omega\eta_1\eta_2$. A minimal representation is

$$\widehat\eta_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \widehat\eta_2 = \frac{1}{2}\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix},$$

so that

$$\widehat H = \mathrm{i}\omega\widehat\eta_1\widehat\eta_2 = \begin{pmatrix} -\omega/4 & 0 \\ 0 & \omega/4 \end{pmatrix}.$$

The finite Grassmann system therefore quantizes to a two-state Hilbert space.

40.7 Berezin Calculus

Convention 40.8 (Berezin normalization and ordering)**.** For one odd coordinate θ, the Berezin integral is normalized by

$$\int \mathrm{d}\theta\, 1 = 0, \qquad \int \mathrm{d}\theta\, \theta = 1.$$

For ordered odd coordinates $(\theta_1, \ldots, \theta_N)$, the ordered Berezinian density $\mathrm{d}\theta_N \cdots \mathrm{d}\theta_1$ is normalized by

$$\int \mathrm{d}\theta_N \cdots \mathrm{d}\theta_1\, \theta_1\theta_2 \cdots \theta_N = 1.$$

All signs below, including the sign of a closed fermion loop and the determinant obtained from a fermionic Gaussian, use this ordering together with the Koszul rule for moving odd variables and odd differentials past one another.

Definition 40.9 (Finite Grassmann algebra and Berezin derivative)**.** Let Λ_N be the complex Grassmann algebra generated by

$$\eta_1, \ldots, \eta_N$$

with relations

$$\eta_a\eta_b = -\eta_b\eta_a, \qquad \eta_a^2 = 0.$$

A homogeneous monomial has parity equal to its degree modulo 2. If X and Y are homogeneous, then

$$XY = (-1)^{|X||Y|}YX$$

when every generator appearing in X is distinct from every generator appearing in Y. Berezin differentials are treated as odd symbols, so

$$\eta_a\, \mathrm{d}\eta_b = -\mathrm{d}\eta_b\, \eta_a.$$

The left derivative $\partial^L/\partial\eta_a$ is defined by

$$\frac{\partial^L}{\partial\eta_a}\eta_b = \delta_{ab}, \qquad \frac{\partial^L}{\partial\eta_a}(XY) = \frac{\partial^L X}{\partial\eta_a}Y + (-1)^{|X|}X\frac{\partial^L Y}{\partial\eta_a}$$

for homogeneous X. Berezin integration is the same algebraic operation, with the normalization of Convention 40.8:

$$\int \mathrm{d}\eta_a\, 1 = 0, \qquad \int \mathrm{d}\eta_a\, \eta_a = 1.$$

For N generators the ordered integral is fixed by

$$\int \mathrm{d}\eta_N \cdots \mathrm{d}\eta_1\, \eta_1\eta_2\cdots\eta_N = 1.$$

Lemma 40.10 (Finite Berezin Gaussian and contractions)**.** *For independent pairs $\eta_a, \bar\eta_a$, $a = 1, \dots, N$, and an invertible $N \times N$ complex matrix A, use the ordered measure*

$$[\mathrm{d}\bar\eta\, \mathrm{d}\eta] = \prod_{a=1}^{N} \mathrm{d}\bar\eta_a\, \mathrm{d}\eta_a.$$

With this convention,

$$\int [\mathrm{d}\bar\eta\, \mathrm{d}\eta]\, \exp\left(-\bar\eta_a A^a{}_b\eta_b\right) = \det A.$$

Adding Grassmann sources $J, \bar J$,

$$\begin{aligned}\int [\mathrm{d}\bar\eta\, \mathrm{d}\eta]\, &\exp\left(-\bar\eta_a A^a{}_b\eta_b + \bar J\eta + \bar\eta J\right) \\ &= \det A\, \exp\left(\bar J A^{-1} J\right).\end{aligned}$$

The normalized two-point contraction is

$$\langle \eta_a\bar\eta_b\rangle_A = (A^{-1})_a{}^b.$$

This identity follows from Berezin integration by parts. Since

$$0 = \int [\mathrm{d}\bar\eta\, \mathrm{d}\eta]\, \frac{\partial^L}{\partial\bar\eta_b}\left[\bar\eta_c \exp(-\bar\eta_a A^a{}_d\eta_d)\right],$$

the normalized integral obeys

$$\delta_c{}^b = A^b{}_d\langle \eta_d\bar\eta_c\rangle_A.$$

Higher Gaussian moments are obtained by repeated contractions, with the sign determined by the number of odd variables moved past one another to bring a contracted pair together.

Proof. The Gaussian identity is an identity in the finite exterior algebra generated by the $2N$ odd variables. The Berezin integral extracts one fixed top coefficient. The coefficient of the top monomial in $\exp(-\bar{\eta}_a A^a{}_b \eta_b)$ is multilinear and alternating in the columns of A, and the declared ordering normalizes the identity matrix case to 1. Hence the coefficient is $\det A$.

For the source identity, use the odd translation

$$\eta \mapsto \eta - A^{-1}J, \qquad \bar{\eta} \mapsto \bar{\eta} - \bar{J}A^{-1}.$$

Berezinian densities are invariant under translations, and the exponent becomes

$$-\bar{\eta}A\eta + \bar{J}A^{-1}J.$$

The source-free Gaussian integral then gives the factor $\det A \exp(\bar{J}A^{-1}J)$. Differentiating the source identity with the same left/right derivative convention gives $\langle \eta_a \bar{\eta}_b \rangle_A = (A^{-1})_a{}^b$. Equivalently, the integration-by-parts identity displayed above gives

$$\delta_c{}^b = A^b{}_d \langle \eta_d \bar{\eta}_c \rangle_A,$$

which is the same statement after multiplication by A^{-1}. Repeated differentiation gives Wick contractions, with Koszul signs determined by the permutation needed to place each contracted pair next to the source derivatives. □

40.8 The Fermionic Coherent-State Path Integral

The one-mode complex fermionic system has two operators

$$\hat{\eta}, \qquad \hat{\chi},$$

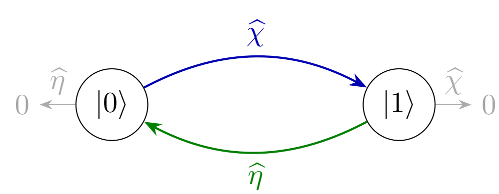
obeying

$$\{\hat{\eta}, \hat{\chi}\} = 1, \qquad \hat{\eta}^2 = \hat{\chi}^2 = 0.$$

On the two-dimensional Hilbert space with basis $|0\rangle, |1\rangle$, choose

$$\hat{\eta}|0\rangle = 0, \qquad \hat{\eta}|1\rangle = |0\rangle, \qquad \hat{\chi}|0\rangle = |1\rangle, \qquad \hat{\chi}|1\rangle = 0.$$

One-mode coherent-state resolution and time slicing.



For a Grassmann parameter η, define the right coherent ket

$$|\eta\rangle = |0\rangle + |1\rangle\eta,$$

so that

$$\widehat{\eta}|\eta\rangle = |\eta\rangle\eta.$$

Define the dual coherent bra by

$$\langle\!\langle\chi| = -\langle 1| + \chi\langle 0|.$$

It satisfies

$$\langle\!\langle\chi|\eta\rangle = \chi - \eta.$$

It also gives the left action

$$\langle\!\langle\chi|\widehat{\eta} = -\chi\,\langle\!\langle\chi|.$$

The resolution of the identity is

$$\int |\eta\rangle\,\mathrm{d}\eta\,\langle\!\langle\eta| = \mathbf{1}.$$

The sign in this formula is part of the ordered Berezin convention. Written with the differential moved to the left, it is

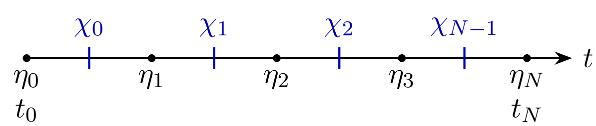
$$\begin{aligned}
\int |\eta\rangle\,\mathrm{d}\eta\,\langle\!\langle\eta| &= \int \mathrm{d}\eta\,|-\eta\rangle\,\langle\!\langle\eta| \\
&= \int \mathrm{d}\eta\,(-|1\rangle\eta + |0\rangle)(-\langle 1| + \eta\langle 0|) \\
&= |1\rangle\langle 1| + |0\rangle\langle 0|.
\end{aligned}$$

Let

$$U(T) = \mathrm{e}^{-\mathrm{i}\widehat{H}T}.$$

Insert the coherent-state resolution of identity at times

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad t_{n+1} - t_n = \varepsilon = T/N.$$



The one-step overlap identities are

$$\langle\!\langle\eta_{n+1}|\eta_n\rangle = -\int \mathrm{d}\chi_n\,\exp[-\chi_n(\eta_{n+1} - \eta_n)]\,,$$

and

$$\langle\!\langle\eta_{n+1}|\widehat{\chi}|\eta_n\rangle = -\int \mathrm{d}\chi_n\,\chi_n\,\exp[-\chi_n(\eta_{n+1} - \eta_n)]\,.$$

The endpoint convention fixes the signs in these formulas. With $\eta_0 = \eta_i$, $\eta_N = \eta_f$, and the ordered measure displayed below, the finite-N kernel is

$$\begin{aligned}
K_N(\eta_f, T; \eta_i, 0) = -\int \mathrm{d}\chi_0 \prod_{n=1}^{N-1} \mathrm{d}\eta_n\,\mathrm{d}\chi_n \\
\times \exp \sum_{n=0}^{N-1} \left[-\chi_n(\eta_{n+1} - \eta_n) - \mathrm{i}\varepsilon H(\chi_n, \eta_n)\right].
\end{aligned}$$

The kernel itself is the continuum limit of these regulated Berezin integrals,

$$\langle\!\langle \eta_f | U(T) | \eta_i \rangle = \lim_{N\to\infty} K_N(\eta_f, T; \eta_i, 0),$$

when that limit exists in the chosen topology. Equivalently, the global sign may be absorbed into the definition of the ordered finite-dimensional measure. The continuum notation for the same construction is

$$\int D\chi\, D\eta\, \exp\left\{ \int_0^T \mathrm{d}t\, \left[-\chi\dot{\eta} - \mathrm{i}H(\chi,\eta)\right] \right\}.$$

Equivalently, with the first-order action

$$S[\chi,\eta] = \int_0^T \mathrm{d}t\, \left[\mathrm{i}\chi\dot{\eta} - H(\chi,\eta)\right],$$

the continuum Grassmann path-integral notation assigns the weight $\mathrm{e}^{\mathrm{i}S}$ to the kernel. This expression denotes the limit, when it exists, of the finite time-sliced Berezin integrals displayed above; the time-sliced expression is the regulated meaning of the notation. In this first-order odd system the Hamiltonian and Lagrangian forms coincide: χ is the odd conjugate variable of η, and it remains in the first-order action rather than being removed by a nondegenerate even Gaussian integration.

The identities above are finite two-state Berezin calculations. The eigenket identity follows directly from the two-state action:

$$\widehat{\eta}(|0\rangle + |1\rangle\eta) = |0\rangle\eta = (|0\rangle + |1\rangle\eta)\eta,$$

because $\eta^2 = 0$. The dual coherent bra gives

$$\langle\!\langle \chi|\eta\rangle = (-\langle 1| + \chi\langle 0|)(|0\rangle + |1\rangle\eta) = \chi - \eta,$$

and

$$\langle\!\langle \chi|\widehat{\eta} = -\chi\langle\!\langle \chi|$$

by the same basis calculation. The displayed Berezin computation of $\int |\eta\rangle\, \mathrm{d}\eta\, \langle\!\langle \eta|$ extracts the coefficient of η and gives $|0\rangle\langle 0| + |1\rangle\langle 1|$. Inserting this identity between short-time evolution factors gives the finite-N kernel. The overlap identities follow by applying the same one-variable Berezin integral to $\langle\!\langle \eta_{n+1}|\eta_n\rangle$ and $\langle\!\langle \eta_{n+1}|\widehat{\chi}|\eta_n\rangle$.

40.9 Fermionic Traces and Thermal Boundary Conditions

Fermionic trace endpoint signs. The trace of a fermionic Hilbert-space operator carries one extra sign from the odd coherent-state resolution. For inverse temperature $\beta_{\mathrm{T}} > 0$,

$$\begin{aligned}
Z(\beta_{\mathrm{T}}) &= \mathrm{Tr}_{\mathcal{H}}\, \mathrm{e}^{-\beta_{\mathrm{T}}\widehat{H}} \\
&= \int \mathrm{Tr}_{\mathcal{H}} \left(|\eta\rangle\, \mathrm{d}\eta\, \langle\!\langle \eta|\, \mathrm{e}^{-\beta_{\mathrm{T}}\widehat{H}} \right) \\
&= \int \mathrm{d}\eta\, \langle\!\langle \eta|\mathrm{e}^{-\beta_{\mathrm{T}}\widehat{H}}|{-\eta}\rangle.
\end{aligned}$$

The state $|-\eta\rangle$ appears because moving the odd ket $|\eta\rangle = |0\rangle + |1\rangle\eta$ through the odd differential $\mathrm{d}\eta$ changes the sign of its odd component. After Euclidean time slicing this identity becomes the finite-dimensional Berezin integral with boundary condition

$$\eta(\tau + \beta_{\mathrm{T}}) = -\eta(\tau).$$

Thus ordinary thermal traces impose anti-periodic boundary conditions on fermionic variables. The supertrace

$$\mathrm{Tr}_{\mathcal{H}}\left((-1)^F \mathrm{e}^{-\beta_{\mathrm{T}}\widehat{H}}\right)$$

replaces the final $|-\eta\rangle$ by $|\eta\rangle$, and therefore imposes periodic fermionic boundary conditions. This distinction is the finite-dimensional origin of the spin-structure choice for fermionic fields on a Euclidean time circle.

For the trace sign, let A be the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

in the ordered basis $|0\rangle, |1\rangle$. Then

$$\int \mathrm{d}\eta \, \langle\!\langle \eta | A | -\eta \rangle = a + d = \mathrm{Tr}\, A,$$

whereas

$$\int \mathrm{d}\eta \, \langle\!\langle \eta | A | \eta \rangle = a - d = \mathrm{Tr}((-1)^F A).$$

Thus an ordinary trace identifies the endpoint coherent variable with a minus sign, while the insertion of $(-1)^F$ removes that sign. Time slicing carries this endpoint algebra into anti-periodic and periodic Euclidean fermion boundary conditions, respectively.

40.10 Berezin Integration as Functorial Pushforward

Definition 40.11 (Finite Berezin pushforward)**.** Let V be a finite-dimensional complex vector space and let ΠV denote the purely odd affine superspace with structure sheaf $\Lambda(V^\vee)$. A basis $e_1, \ldots, e_N$ of V determines odd coordinates $\theta_1, \ldots, \theta_N$ and the ordered Berezinian density

$$[\mathrm{d}\theta] = \mathrm{d}\theta_N \cdots \mathrm{d}\theta_1.$$

The integral of $f(\theta)[\mathrm{d}\theta]$ is the coefficient of $\theta_1 \cdots \theta_N$ in f. If $\theta'_a = A_a{}^b \theta_b$, then

$$\theta'_1 \cdots \theta'_N = (\det A)\theta_1 \cdots \theta_N, \qquad [\mathrm{d}\theta'] = (\det A)^{-1}[\mathrm{d}\theta].$$

Thus the product $f(\theta)[\mathrm{d}\theta]$ is the coordinate-independent object, and Berezin integration is the linear functional on compactly supported algebraic Berezinian densities in this finite model.

For a direct sum $V \oplus W$, choose ordered coordinates θ on ΠV and η on ΠW, with total density

$$[\mathrm{d}\eta][\mathrm{d}\theta] = \mathrm{d}\eta_M \cdots \mathrm{d}\eta_1 \, \mathrm{d}\theta_N \cdots \mathrm{d}\theta_1.$$

The projection

$$p : \Pi(V \oplus W) \longrightarrow \Pi V$$

has a pushforward

$$p_! : \Gamma(\mathrm{Ber}_{\Pi(V\oplus W)}) \longrightarrow \Gamma(\mathrm{Ber}_{\Pi V})$$

defined by

$$p_!\big(F(\theta,\eta)[\mathrm{d}\eta][\mathrm{d}\theta]\big) = \left(\int \mathrm{d}\eta_M \cdots \mathrm{d}\eta_1 \, F(\theta,\eta)\right)[\mathrm{d}\theta].$$

This finite-dimensional pushforward is the prototype of a fermionic path integral: it is a fiberwise algebraic pushforward of Berezinian densities.

For

$$\Pi(V \oplus W \oplus U) \xrightarrow{q} \Pi(V \oplus W) \xrightarrow{p} \Pi V,$$

the equality

$$(p \circ q)_! = p_! \circ q_!$$

holds when the Berezinian orders are the induced product orders. Thus a finite regulator may integrate fermionic modes in stages without changing the value, provided the induced Berezinian order is carried along. Choose ordered odd coordinates θ on V, η on W, and ζ on U, with total Berezinian density $[\mathrm{d}\zeta][\mathrm{d}\eta][\mathrm{d}\theta]$. Applying $(p\circ q)_!$ extracts first the coefficient of $\zeta_1 \cdots \zeta_{\dim U}\eta_1 \cdots \eta_{\dim W}$ from the coefficient polynomial over $\Lambda(V^\vee)$. Applying $q_!$ extracts the ζ-top coefficient and leaves a Berezinian density on $\Pi(V \oplus W)$; applying $p_!$ then extracts the η-top coefficient. Since coefficient extraction in a finite exterior algebra is associative once the ordered monomial is fixed, the two operations agree.

40.11 Supersymmetric Quantum Mechanics and the Witten Index

Definition 40.12 (One-dimensional supersymmetric quantum-mechanics datum)**.** Let $W : \mathbb{R} \to \mathbb{R}$ be a smooth function for which the operators below are essentially self-adjoint on the Schwartz domain and have discrete low-energy spectrum. The Hilbert space is

$$\mathcal{H} = L^2(\mathbb{R}) \otimes \Lambda^\bullet\mathbb{C},$$

with fermion operators $\psi, \psi^\dagger$ obeying

$$\{\psi, \psi^\dagger\} = 1, \qquad F = \psi^\dagger\psi.$$

Define

$$Q = \psi^\dagger(\partial_x + W'(x)), \qquad Q^\dagger = \psi(-\partial_x + W'(x)), \qquad H = \frac{1}{2}\{Q, Q^\dagger\}.$$

Then $Q^2 = (Q^\dagger)^2 = 0$, $[H,Q] = [H,Q^\dagger] = 0$, and

$$H = \frac{1}{2}\left[-\partial_x^2 + W'(x)^2 + W''(x)(2F-1)\right]. \tag{40.2}$$

The Witten index is

$$I_W(\beta_{\mathrm{T}}) = \mathrm{Tr}_{\mathcal{H}}\left((-1)^F \mathrm{e}^{-\beta_{\mathrm{T}} H}\right)$$

whenever this trace is defined by the finite-volume or confining regularization under discussion.

Pairing of positive-energy states. Assume $H \geq 0$, the nonzero spectrum in the regulated problem is discrete, and $Q, Q^\dagger$ preserve the domain of H. Then

$$I_W(\beta_{\mathrm{T}}) = \dim\ker H|_{F=0} - \dim\ker H|_{F=1}, \qquad \frac{d}{d\beta_{\mathrm{T}}} I_W(\beta_{\mathrm{T}}) = 0.$$

Let $E > 0$ and let $\mathcal{H}_E$ be the finite-dimensional eigenspace of energy E. Since $H = \frac{1}{2}(QQ^\dagger + Q^\dagger Q)$, a vector $\Psi \in \mathcal{H}_E$ with $E > 0$ cannot be annihilated by both Q and $Q^\dagger$. The maps Q and $Q^\dagger$ commute with H and change fermion parity. The operator

$$\frac{1}{2E}(QQ^\dagger + Q^\dagger Q)$$

is the identity on $\mathcal{H}_E$, so the even and odd parts of $\mathcal{H}_E$ have equal dimension: $Q + Q^\dagger$ gives an invertible odd map on $\mathcal{H}_E$ after division by $\sqrt{2E}$. Therefore the supertrace of $\mathrm{e}^{-\beta_{\mathrm{T}} H}$ vanishes on every positive-energy eigenspace. Only $\ker H$ remains, where $Q\Psi = Q^\dagger\Psi = 0$. The displayed formula and β_{T}-independence follow.

For the harmonic superpotential

$$W(x) = \frac{m}{2}x^2, \qquad m > 0,$$

Harmonic supersymmetric oscillator index. For $W(x) = mx^2/2$ with $m > 0$, the regulated Witten index is

$$I_W = 1.$$

The zero-mode equations are

$$(\partial_x + mx)f_0(x) = 0, \qquad (-\partial_x + mx)f_1(x) = 0.$$

Their solutions are

$$f_0(x) = C_0\mathrm{e}^{-mx^2/2}, \qquad f_1(x) = C_1\mathrm{e}^{mx^2/2}.$$

Only f_0 lies in $L^2(\mathbb{R})$. Hence

$$I_W = 1.$$

The oscillator spectrum makes the cancellation explicit. With

$$q = \mathrm{e}^{-\beta_{\mathrm{T}} m},$$

the even contribution is

$$Z_{\text{even}} = \sum_{n\geq 0} q^n = \frac{1}{1-q},$$

and the odd contribution is

$$Z_{\text{odd}} = \sum_{n\geq 0} q^{n+1} = \frac{q}{1-q}.$$

Thus

$$\operatorname{Str}\mathrm{e}^{-\beta_{\mathrm{T}} H} = Z_{\text{even}} - Z_{\text{odd}} = 1.$$

In the coherent-state path integral this is the same supertrace discussed in the previous section, so the fermionic variable is periodic on the Euclidean time circle.

40.12 Worldline Index Density from Periodic Fermions

Definition 40.13 (Twisted Dirac index and periodic worldline datum)**.** Let M be a compact oriented Riemannian spin manifold of dimension $2n$, let $E \to M$ be a Hermitian vector bundle with unitary connection, and let

$$D_E : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E)$$

be the chiral Dirac operator. The analytic index is

$$\operatorname{ind} D_E = \dim\ker D_E - \dim\ker D_E^\dagger = \operatorname{Str}_{L^2(S\otimes E)} \mathrm{e}^{-\beta_{\mathrm{T}} D_E^2}.$$

The last equality follows from the same positive-spectrum pairing as in the supersymmetric quantum-mechanical index discussion above, with $(-1)^F$ replaced by chirality.

The worldline path integral for this supertrace has periodic bosonic and fermionic loops:

$$x(\tau + \beta_{\mathrm{T}}) = x(\tau), \qquad \psi^a(\tau + \beta_{\mathrm{T}}) = \psi^a(\tau),$$

where a is an orthonormal tangent-frame index. In normal coordinates at a point $x_0 \in M$, split

$$x(\tau) = x_0 + \sqrt{\beta_{\mathrm{T}}}\, y(\tau), \qquad \psi^a(\tau) = \psi_0^a + \psi_\perp^a(\tau),$$

with y and $\psi_\perp$ having zero average. The finite-mode Gaussian calculation is performed around this zero-mode decomposition.

Example 40.14 (Finite-mode periodic-fermion index density)**.** For the preceding periodic worldline datum, the finite-mode nonzero-mode calculation and the fermion zero-mode Berezin integral give the local index density

$$\left[\widehat{A}(TM)\,\mathrm{ch}(E)\right]_{2n}.$$

When the heat-kernel proof of the McKean–Singer formula is supplied, integration over the bosonic zero mode gives

$$\operatorname{ind} D_E = \int_M \widehat{A}(TM)\,\mathrm{ch}(E).$$

The calculation is finite-dimensional after mode truncation. The nonzero-mode Gaussian ratio gives one factor for each skew-eigenvalue r_j of the curvature two-form $R/(2\pi\mathrm{i})$:

$$\prod_{j=1}^{n} \frac{r_j/2}{\sinh(r_j/2)}. \tag{40.3}$$

Indeed, in a curvature two-plane the bosonic nonzero modes carry eigenvalues $(2\pi k)^2$ shifted by the curvature matrix, while the periodic fermionic nonzero modes carry first-order eigenvalues $2\pi k$ shifted by the same skew-eigenvalue. Taking the finite-mode determinant ratio and then the symmetric product limit gives

$$\frac{z}{\sinh z} = \prod_{k=1}^{\infty} \left(1 + \frac{z^2}{\pi^2 k^2}\right)^{-1}.$$

With $z = r_j/2$, this gives the factor in (40.3). The connection on E contributes

$$\mathrm{tr}_E \exp\left(\frac{F_E}{2\pi \mathrm{i}}\right),$$

because the fermion zero modes represent exterior multiplication by tangent one-forms and the Berezin integral over $\psi_0^1, \ldots, \psi_0^{2n}$ extracts the top-degree component. Therefore the local density produced by the finite-mode worldline supertrace is

$$\left[\widehat{A}(TM)\,\mathrm{ch}(E)\right]_{2n}, \qquad \widehat{A}(TM) = \prod_{j=1}^{n} \frac{r_j/2}{\sinh(r_j/2)}, \qquad \mathrm{ch}(E) = \mathrm{tr}_E \exp\left(\frac{F_E}{2\pi \mathrm{i}}\right). \tag{40.4}$$

The equality with the Fredholm index of D_E additionally uses the McKean–Singer supertrace identity and the heat-kernel asymptotic expansion for D_E^2 on a compact manifold. With those analytic inputs established, the bosonic zero-mode integral gives

$$\mathrm{ind}\, D_E = \int_M \widehat{A}(TM)\,\mathrm{ch}(E).$$

40.13 Dirac Functional Integrals and Propagators

Definition 40.15 (Free Dirac functional-integral datum)**.** For a classical Grassmann spinor field $\psi_\alpha(x)$, define

$$\bar{\psi}^\alpha(x) = \big(\psi^\dagger(x)\beta\big)^\alpha.$$

Here $\beta = \mathrm{i}\gamma^0$, so

$$(\gamma^\mu)^\dagger \beta = -\beta\gamma^\mu.$$

This is the adjoint convention for which $\bar{\psi}(\not{\partial} + m)\psi$ is a Lorentz scalar density. The free Dirac Lagrangian density in the conventions of this volume is

$$\mathcal{L}_{\mathrm{D}} = -\bar{\psi}(\not{\partial} + m)\psi.$$

Its Euler–Lagrange equation is

$$(\not{\partial} + m)\psi = 0.$$

The momentum conjugate to ψ, using the right derivative appropriate for odd variables, is

$$\Pi_\psi{}^\alpha = \mathcal{L}_{\mathrm{D}} \frac{\overleftarrow{\partial}}{\partial(\partial_0 \psi_\alpha)} = -\bar{\psi}^\beta (\gamma^0)_\beta{}^\alpha.$$

The equal-time anticommutator computed from the mode expansion above,

$$\{\widehat{\psi}_\alpha(t,\vec{x}), \widehat{\bar{\psi}}^\beta(t,\vec{y})\} = \mathrm{i}(\gamma^0)_\alpha{}^\beta\, \delta^{(3)}(\vec{x} - \vec{y}),$$

therefore gives the canonical odd bracket

$$\{\widehat{\psi}_\alpha(t,\vec{x}), \widehat{\Pi}_\psi{}^\beta(t,\vec{y})\} = \mathrm{i}\delta_\alpha{}^\beta\, \delta^{(3)}(\vec{x} - \vec{y}).$$

The regulated Lorentzian generating functional, written in continuum shorthand, is normalized so that

$$Z[\bar{J}, J] = \int D\bar{\psi}\, D\psi\, \exp\left[\mathrm{i}\int \mathrm{d}^4x\, \mathcal{L}_{\mathrm{D}} + \mathrm{i}\int \mathrm{d}^4x\, \left(\bar{J}\psi + \bar{\psi}J\right)\right]$$
$$= Z[0,0]\, \exp\left[-\mathrm{i}\int \mathrm{d}^4x\, \mathrm{d}^4y\, \bar{J}^{\alpha}(x) S_{F,\alpha}{}^{\beta}(x-y) J_{\beta}(y)\right],$$

where J and $\bar{J}$ are Grassmann sources. The Feynman two-point function is

$$S_{F,\alpha}{}^{\beta}(x-y) = \langle\Omega|\, \mathrm{T}\{\widehat{\psi}_{\alpha}(x)\widehat{\bar{\psi}}^{\beta}(y)\}|\Omega\rangle.$$

Charge conservation gives

$$\langle\Omega|\, \mathrm{T}\{\widehat{\psi}_{\alpha}(x)\widehat{\psi}_{\beta}(y)\}|\Omega\rangle = 0, \qquad \langle\Omega|\, \mathrm{T}\{\widehat{\bar{\psi}}^{\alpha}(x)\widehat{\bar{\psi}}^{\beta}(y)\}|\Omega\rangle = 0$$

for the free charged Dirac field.

Free Dirac propagator as Feynman inverse. The free charged Dirac two-point function in Definition 40.15 is the Feynman inverse of $\not{\partial} + m$:

$$S_F(x;m) = (-\not{\partial}_x + m)\Delta_F(x;m^2).$$

Using the scalar convention

$$\Delta_F(x;m^2) = -\mathrm{i}\int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2 + m^2 - \mathrm{i}\epsilon},$$

one obtains

$$S_F(k) = \frac{-\mathrm{i}(-\mathrm{i}\not{k} + m)}{k^2 + m^2 - \mathrm{i}\epsilon}.$$

The numerator is the spinor projector algebra of Chapter 20, while the denominator is the scalar mass-shell pole with the Feynman boundary value.

Since Δ_F is the scalar Feynman inverse in the convention

$$(-\partial^2 + m^2)\Delta_F(x;m^2) = -\mathrm{i}\delta^{(4)}(x),$$

and the Clifford relation gives

$$(\not{\partial} + m)(-\not{\partial} + m) = -\not{\partial}^2 + m^2 = -\partial^2 + m^2,$$

the distribution $S_F = (-\not{\partial} + m)\Delta_F$ obeys the Dirac inverse equation with the same Feynman boundary value. Fourier transforming $\partial_\mu \mapsto \mathrm{i}k_\mu$ in the convention used in the scalar chapter gives the displayed numerator $-\mathrm{i}(-\mathrm{i}\not{k} + m)$ over $k^2 + m^2 - \mathrm{i}0$.

40.14 Spinor Spectral Poles and LSZ

Definition 40.16 (Isolated spinor one-particle pole datum)**.** Let an interacting theory have a global $U(1)$ charge $\widehat{Q}$, a vacuum satisfying $\widehat{Q}\Omega = 0$, and an isolated massive spin-$\frac{1}{2}$ one-particle sector of mass m and charge $+1$. Let $\widehat{\psi}_\alpha$ be a local spinor field of charge $+1$ with nonzero overlap with this sector:

$$[\widehat{Q}, \widehat{\psi}_\alpha] = \widehat{\psi}_\alpha, \qquad [\widehat{Q}, \widehat{\bar{\psi}}^\alpha] = -\widehat{\bar{\psi}}^\alpha.$$

In the delta-normalized one-particle convention, Lorentz covariance permits the matrix element

$$\langle \vec{p}, \sigma, +|\widehat{\psi}_\alpha(x)|\Omega\rangle = \frac{\mathrm{e}^{-\mathrm{i}p\cdot x}}{(2\pi)^{3/2}} \left[Z_v^{1/2} \mathcal{V}^\sigma_\alpha(p) + Z_u^{1/2} \mathcal{U}^\sigma_\alpha(p) \right],$$

where $\mathcal{U}, \mathcal{V}$ are the two massive spinor intertwiners constructed from the one-particle representation. A constant local redefinition inside the span of ψ and $\gamma_5\psi$, together with a phase convention for one-particle states, chooses an interpolating field whose isolated pole residue is proportional to one chosen intertwiner. After making that choice and passing to the covariant one-particle normalization used in the LSZ formulas, write

$$\langle \vec{p}, \sigma, +|\widehat{\psi}_\alpha(x)|\Omega\rangle = Z_\psi^{1/2}\, \mathrm{e}^{-\mathrm{i}p\cdot x} v^\sigma_\alpha(p), \qquad Z_\psi \geq 0.$$

Proposition 40.17 (Locality fixes conjugate spinor pole residue)**.** *The conjugate field has the corresponding overlap with the charge -1 sector,*

$$\langle \vec{p}, \sigma, -|\widehat{\bar{\psi}}^\alpha(x)|\Omega\rangle = Z_\psi^{1/2}\, \mathrm{e}^{-\mathrm{i}p\cdot x} \bar{u}^{\sigma,\alpha}(p),$$

with the same pole residue after the phase convention is fixed. The time-ordered two-point function then has the one-particle pole

$$Z_\psi(-\not{\partial}_x + m)_\alpha{}^\beta \Delta_F(x-y; m^2).$$

Proof. Let $Z_{\bar{\psi}}$ denote the conjugate-sector residue before imposing locality. The one-particle contribution to the vacuum anticommutator has the form

$$C_\alpha{}^\beta(x-y) = \sum_\sigma \int \mathrm{d}\Pi_p \left[Z_\psi\, \mathrm{e}^{-\mathrm{i}p\cdot(x-y)} v^\sigma_\alpha(p) \bar{v}^{\sigma,\beta}(p) \right. \\ \left. + Z_{\bar{\psi}}\, \mathrm{e}^{\mathrm{i}p\cdot(x-y)} u^\sigma_\alpha(p) \bar{u}^{\sigma,\beta}(p) \right].$$

The spin sums of Chapter 20 convert this to

$$C_\alpha{}^\beta(x-y) = Z_\psi(-\not{\partial}_x + m)_\alpha{}^\beta \Delta(x-y; m^2)$$

precisely when $Z_{\bar{\psi}} = Z_\psi$. Here Δ is the scalar Pauli–Jordan distribution, hence the pole contribution has support in the closed light cone. The positive- and negative-frequency terms separately do not have this support property, so the equality of residues is fixed by locality and the chosen normalization.

The time-ordered two-point function then has the one-particle pole

$$\langle \Omega|\, \mathrm{T}\{\widehat{\psi}_\alpha(x) \widehat{\bar{\psi}}^\beta(y)\}|\Omega\rangle = Z_\psi(-\not{\partial}_x + m)_\alpha{}^\beta \Delta_F(x-y; m^2) + G_\alpha{}^{\beta,\mathrm{cont}}(x-y),$$

where G^{cont} is the contribution from states outside the isolated one-particle mass shell. In momentum space the pole term is

$$Z_\psi \frac{-\mathrm{i}(-\mathrm{i}\not{k} + m)_\alpha{}^\beta}{k^2 + m^2 - \mathrm{i}\epsilon}.$$

□

Spinor mass-shell residue and external factors. The normalization content of this pole fixes the external spinor factor. Put $p = (\omega_{\vec{p}}, \vec{p})$. The delta-normalized spinor intertwiners obey

$$\sum_{\sigma=1}^{2} u^\sigma_\alpha(p)\bar{u}^{\sigma,\beta}(p) = \frac{(m - \mathrm{i}\ \not{p})_\alpha{}^\beta}{2\omega_{\vec{p}}}, \qquad \sum_{\sigma=1}^{2} v^\sigma_\alpha(p)\bar{v}^{\sigma,\beta}(p) = -\frac{(m + \mathrm{i}\ \not{p})_\alpha{}^\beta}{2\omega_{\vec{p}}}. \tag{40.5}$$

Thus the numerator of the Dirac pole contains the mass-shell Jacobian:

$$m - \mathrm{i}\ \not{p} = 2\omega_{\vec{p}} \sum_{\sigma=1}^{2} u^\sigma(p)\bar{u}^\sigma(p), \qquad m + \mathrm{i}\ \not{p} = -2\omega_{\vec{p}} \sum_{\sigma=1}^{2} v^\sigma(p)\bar{v}^\sigma(p). \tag{40.6}$$

Since

$$k^2 + m^2 - \mathrm{i}0 = -2\omega_{\vec{k}}\big(k^0 - \omega_{\vec{k}} + \mathrm{i}0\big) + O\big((k^0 - \omega_{\vec{k}})^2\big)$$

near the positive-energy pole, the pole contribution is

$$Z_\psi \frac{-\mathrm{i}(m - \mathrm{i}\ \not{k})_\alpha{}^\beta}{k^2 + m^2 - \mathrm{i}0} = \frac{\mathrm{i}Z_\psi}{k^0 - \omega_{\vec{k}} + \mathrm{i}0} \sum_{\sigma=1}^{2} u^\sigma_\alpha(k)\bar{u}^{\sigma,\beta}(k) + \text{terms regular at } k^0 = \omega_{\vec{k}}. \tag{40.7}$$

The negative-energy pole has the same form with v^σ and $k^0 + \omega_{\vec{k}} - \mathrm{i}0$. Therefore the factor $2\omega_{\vec{k}}$ from the mass-shell denominator is already accounted for by the spinor completeness relation (40.5). In the $\delta^{(3)}(\vec{p} - \vec{q})$-normalized external basis, each external wavefunction carries the Fourier factor $(2\pi)^{-3/2}$, while the spinor intertwiner carries the energy normalization in $\bar{u}^\sigma u^\tau = (m/\omega_{\vec{p}})\delta^{\sigma\tau}$ and $\bar{v}^\sigma v^\tau = -(m/\omega_{\vec{p}})\delta^{\sigma\tau}$. Hence a spinor external leg contributes

$$\frac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\, u^\sigma_\alpha(p), \qquad \frac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\, \bar{u}^{\sigma,\alpha}(p), \qquad \frac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\, v^\sigma_\alpha(p), \qquad \frac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\, \bar{v}^{\sigma,\alpha}(p), \tag{40.8}$$

with the choice among them fixed by charge and by the in/out convention. This is the spinorial normalization counterpart of the scalar one-particle normalization in Chapter 23. The spin sums (40.5) express the Dirac numerator as the mass-shell projector together with the Jacobian $2\omega_{\vec{p}}$, giving (40.6). Expanding the scalar denominator near the positive-energy shell gives the linear factor displayed before (40.7); substituting the projector identity gives the residue formula (40.7). At the negative-energy pole one expands

$$k^2 + m^2 - \mathrm{i}0 = (k^0 + \omega_{\vec{k}} - \mathrm{i}0)(k^0 - \omega_{\vec{k}} + \mathrm{i}0)$$

near $k^0 = -\omega_{\vec{k}}$, and uses the v-spin completeness relation in (40.5). The factor $(k^0 - \omega_{\vec{k}} + \mathrm{i}0)$ evaluates to $-2\omega_{\vec{k}}$ on this shell, while the v-spin sum carries the corresponding sign in the Dirac numerator. This gives the antiparticle residue with the same external normalization convention as the particle residue. Therefore no additional factor of $2\omega_{\vec{p}}$ is inserted by hand in the external spinor; it has already appeared in the projector identity. The remaining Fourier normalization of the delta-normalized one-particle wavefunction is $(2\pi)^{-3/2}$, yielding (40.8).

Definition 40.18 (Spinorial LSZ pole assignment)**.** Spinorial LSZ reduction applies the same Haag–Ruelle and pole-extraction logic as the scalar case. The difference is that the residue has spinor indices. With Fourier transform convention

$$F(k) = \int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}k\cdot x} F(x),$$

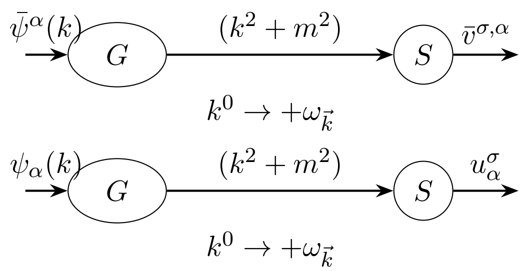
the pole assignments are:

field insertion	mass-shell limit	external one-particle state and residue
$\widehat{\bar\psi}^{\alpha}(k)$	$k^0 \to +\omega_{\vec k}$	outgoing charge $+$, $\quad Z_\psi^{1/2}\bar v^{\sigma,\alpha}(k)$
$\widehat{\bar\psi}^{\alpha}(k)$	$k^0 \to -\omega_{\vec k}$	incoming charge $-$, $\quad Z_\psi^{1/2}\bar u^{\sigma,\alpha}(k)$
$\widehat{\psi}_{\alpha}(k)$	$k^0 \to +\omega_{\vec k}$	outgoing charge $-$, $\quad Z_\psi^{1/2}u^{\sigma}_{\alpha}(k)$
$\widehat{\psi}_{\alpha}(k)$	$k^0 \to -\omega_{\vec k}$	incoming charge $+$, $\quad Z_\psi^{1/2}v^{\sigma}_{\alpha}(k)$.

Thus amputation of an external spinor leg consists of:

$$\text{multiply by } (k^2+m^2), \qquad \text{take } k^0 \to \omega_{\vec k}, \qquad \text{contract with } u^\sigma(p) \text{ or } v^\sigma(p),$$

with the choice of u or v determined by the external charge and by whether the particle is incoming or outgoing. Thus external spinor factors in perturbative scattering are the one-particle pole intertwiners of local spinor fields.



40.15 A First Fermionic Vertex

At this point the S-matrix has already been defined through asymptotic one-particle states, and LSZ has related its matrix elements to pole residues of time-ordered Green functions. Perturbation theory now enters as a method for computing those Green functions.

First four-fermion direct-minus-exchange vertex. Consider

$$\mathcal{L} = -\bar\psi(\partial\!\!\!/ + m)\psi - \frac{g}{2}(\bar\psi\psi)^2.$$

To first order in g, the connected four-point function has one local vertex. With all propagators oriented from ψ to $\bar\psi$, the spinor-index kernel of the vertex is

$$K_{\alpha\beta}{}^{\gamma\delta} = \mathrm{i}g\left(\delta_\alpha{}^\gamma\delta_\beta{}^\delta - \delta_\alpha{}^\delta\delta_\beta{}^\gamma\right).$$

The two terms are the direct and exchange contractions. Their relative sign is the sign obtained by moving odd fields through one another before applying the Gaussian contraction. Figure 40.1 draws the two index contractions; the arrows are a bookkeeping convention for the ordered contraction from ψ to $\bar\psi$, not an additional dynamical orientation.

For external spinors ζ_i selected by the LSZ table, the corresponding tree-level four-fermion amplitude has the index contraction

$$\mathcal{M} = \bar{\zeta}_3{}^{\gamma}\bar{\zeta}_4{}^{\delta}\, K_{\alpha\beta}{}^{\gamma\delta}\, \zeta_1^{\alpha}\zeta_2^{\beta}.$$

For two identical incoming and two identical outgoing charge-+ particles this reads

$$\mathcal{M} = \mathrm{i}g\left[(\bar{\zeta}_3\zeta_1)(\bar{\zeta}_4\zeta_2) - (\bar{\zeta}_4\zeta_1)(\bar{\zeta}_3\zeta_2)\right],$$

with ζ_i replaced by the charge and in/out spinors appropriate to the chosen process. This example is the first template for fermionic Feynman diagrams: every vertex has a local index tensor, every internal line is the time-ordered two-point function, and every exchange of odd fields contributes its parity sign. At first order in g, the interaction insertion is the local product $-\frac{g}{2}(\bar{\psi}\psi)(\bar{\psi}\psi)$. Contracting two incoming ψ-fields with two outgoing $\bar{\psi}$-fields gives two pairings. The first pairs (α, γ) and (β, δ) and gives $\delta_{\alpha}{}^{\gamma}\delta_{\beta}{}^{\delta}$. The second pairs (α, δ) and (β, γ). To put the odd fields in that pairing order one transposes one pair of fermionic variables, producing the minus sign. The factor $1/2$ in the Lagrangian cancels the two identical ways of choosing the two $\bar{\psi}\psi$ factors, leaving the displayed vertex normalization. The external expression is obtained by contracting this local kernel with the spinorial LSZ residues from Definition 40.18.

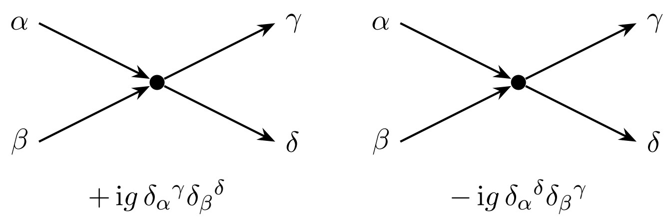


Figure 40.1: Direct and exchange contractions in the first local four-fermion vertex. The exchange contraction differs by the transposition of the two outgoing spinor indices and by the Berezin sign from moving odd variables into the contracted order.

The next chapter turns to massless particles. The logical input will again be the one-particle representation theory, now with the massless little group and helicity replacing the massive $SU(2)$ spin label.

40.16 Spinor and Gamma-Matrix Conventions

This section records the spinor conventions used throughout the monograph. The convention is the Weinberg-compatible four-dimensional gamma-matrix basis, adapted to the mostly-plus metric used in this monograph. The section fixes the data that determine signs in the Dirac adjoint, current normalization, chirality, Majorana conjugation, anomaly traces, and supersymmetry formulae. All comparisons below keep the Lorentzian metric fixed as $\eta = \mathrm{diag}(-,+,+,+)$.

40.17 Four-Dimensional Lorentzian Clifford Algebra

Definition 40.19 (Four-dimensional mostly-plus spinor convention)**.** The Lorentzian metric is

$$\eta_{\mu\nu} = \mathrm{diag}(-,+,+,+), \qquad \eta^{\mu\rho}\eta_{\rho\nu} = \delta^{\mu}{}_{\nu}.$$

Gamma matrices are endomorphisms of a complex four-dimensional vector space S and obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}_S. \tag{40.9}$$

Consequently

$$(\gamma^0)^2 = -\mathbf{1}_S, \qquad (\gamma^i)^2 = \mathbf{1}_S, \quad i = 1, 2, 3.$$

For a covector p_μ we write

$$\not{p} = p_\mu \gamma^\mu.$$

If $p^2 = \eta^{\mu\nu} p_\mu p_\nu$, then

$$\not{p}^2 = p^2 \mathbf{1}_S.$$

The explicit basis used for sign checks in the monograph is

$$\gamma^0 = -\mathrm{i}\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \gamma^j = -\mathrm{i}\begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix}, \qquad j = 1, 2, 3, \tag{40.10}$$

where

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The hermiticity identities are

$$(\gamma^0)^\dagger = -\gamma^0, \qquad (\gamma^i)^\dagger = \gamma^i.$$

Define

$$\beta = \mathrm{i}\gamma^0, \qquad \bar{\psi} = \psi^\dagger \beta. \tag{40.11}$$

Then

$$\beta^\dagger = \beta, \qquad \beta^2 = \mathbf{1}_S, \qquad (\gamma^\mu)^\dagger \beta = -\beta\gamma^\mu.$$

Thus the bilinear $\bar{\psi}\psi$ is Lorentz invariant and $\bar{\psi}\gamma^\mu\psi$ transforms as a Lorentz vector, with the Lorentz action of Definition 40.20. In these conventions the free Lorentzian Dirac density used in the monograph is

$$\mathcal{L}_{\mathrm{D}} = -\bar{\psi}(\gamma^\mu \partial_\mu + m)\psi.$$

For a real Abelian charge q, the Hermitian vector current is

$$j^\mu = \mathrm{i}q\,\bar{\psi}\gamma^\mu\psi.$$

The explicit factor of i is not optional: it follows from (40.11) and the anti-Hermitian nature of $\bar{\psi}\gamma^\mu\psi$ in the mostly-plus convention.

Basic Clifford, adjoint, and current identities. The data in Definition 40.19 obey

$$\not{p}^2 = p^2 \mathbf{1}_S, \qquad \beta^\dagger = \beta, \qquad \beta^2 = \mathbf{1}_S, \qquad (\gamma^\mu)^\dagger\beta = -\beta\gamma^\mu.$$

Consequently the pairing $s, t \mapsto s^\dagger \beta t$ is the invariant Dirac pairing used throughout the monograph, and $\mathrm{i}\bar{\psi}\gamma^\mu\psi$ is Hermitian for an even classical spinor field or for the corresponding normal-ordered quantum current. The square of $\not{p} = p_\mu\gamma^\mu$ is

$$p_\mu p_\nu \gamma^\mu \gamma^\nu = \frac{1}{2} p_\mu p_\nu \{\gamma^\mu, \gamma^\nu\} = p_\mu p_\nu \eta^{\mu\nu} \mathbf{1}_S.$$

The explicit matrices give $(\gamma^0)^\dagger = -\gamma^0$ and $(\gamma^i)^\dagger = \gamma^i$. Since $\beta = \mathrm{i}\gamma^0$ and $(\gamma^0)^2 = -1$, this gives $\beta^\dagger = \beta$ and $\beta^2 = 1$. For $\mu = 0$,

$$(\gamma^0)^\dagger\beta = -\gamma^0\mathrm{i}\gamma^0 = \mathrm{i} = -\mathrm{i}\gamma^0\gamma^0 = -\beta\gamma^0,$$

and for $i = 1, 2, 3$,

$$(\gamma^i)^\dagger\beta = \gamma^i\mathrm{i}\gamma^0 = -\mathrm{i}\gamma^0\gamma^i = -\beta\gamma^i.$$

Thus $\beta\gamma^\mu$ is anti-Hermitian. With the adjoint convention $\bar{\psi} = \psi^\dagger\beta$, the bilinear $\bar{\psi}\gamma^\mu\psi$ is anti-Hermitian in the even finite-dimensional model. Multiplication by i gives the Hermitian current normalization. The quantum statement follows after imposing the same adjoint convention on the operator-valued distribution and using normal ordering when products at the same point are represented in a regulator.

40.18 Spin Representation and Chirality

Definition 40.20 (Spin representation and chirality)**.** The spinor Lorentz generators are

$$S^{\mu\nu} = -\frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu]. \tag{40.12}$$

For $\Lambda = \exp\omega \in SO^+(1,3)$, with $\omega_{\mu\nu} = -\omega_{\nu\mu}$, the corresponding element in the spinor representation is

$$R(\Lambda) = \exp\left(-\frac{\mathrm{i}}{2}\omega_{\mu\nu}S^{\mu\nu}\right).$$

The Clifford relation gives

$$[S^{\mu\nu}, \gamma^\rho] = -\mathrm{i}\gamma^\mu\eta^{\nu\rho} + \mathrm{i}\gamma^\nu\eta^{\mu\rho},$$

and hence

$$R(\Lambda)\gamma^\mu R(\Lambda)^{-1} = \Lambda^\mu{}_\nu\gamma^\nu.$$

The β-pairing is invariant:

$$R(\Lambda)^\dagger\beta R(\Lambda) = \beta.$$

The chirality matrix is

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3. \tag{40.13}$$

It obeys

$$\gamma_5^2 = \mathbf{1}_S, \qquad \{\gamma_5, \gamma^\mu\} = 0, \qquad [\gamma_5, S^{\mu\nu}] = 0.$$

In the explicit basis (40.10),

$$\gamma_5 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad P_\pm = \frac{1 \pm \gamma_5}{2}.$$

Thus a spinor $s \in S$ decomposes as

$$s = \begin{pmatrix} \chi_a \\ \widetilde{\chi}_{\dot{a}} \end{pmatrix}, \qquad \chi \in S_+ := P_+S, \qquad \widetilde{\chi} \in S_- := P_-S, \qquad a, \dot{a} = 1, 2.$$

The matrix γ^μ maps S_+ to S_- and S_- to S_+.

With the orientation convention

$$\epsilon^{0123} = +1,$$

the four-dimensional trace convention is

$$\operatorname{tr}(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4\mathrm{i}\,\epsilon^{\mu\nu\rho\sigma}. \tag{40.14}$$

This is the convention used in the anomaly chapter.

Spin covariance and chirality trace. The generators in Definition 40.20 satisfy

$$[S^{\mu\nu}, \gamma^\rho] = -\mathrm{i}\gamma^\mu\eta^{\nu\rho} + \mathrm{i}\gamma^\nu\eta^{\mu\rho}.$$

Consequently

$$R(\Lambda)\gamma^\mu R(\Lambda)^{-1} = \Lambda^\mu{}_\nu\gamma^\nu, \qquad R(\Lambda)^\dagger\beta R(\Lambda) = \beta.$$

The chirality matrix obeys

$$\gamma_5^2 = 1, \qquad \{\gamma_5, \gamma^\mu\} = 0, \qquad [\gamma_5, S^{\mu\nu}] = 0,$$

and, with $\epsilon^{0123} = +1$,

$$\operatorname{tr}(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4\mathrm{i}\,\epsilon^{\mu\nu\rho\sigma}.$$

For any Clifford generators,

$$[[\gamma^\mu, \gamma^\nu], \gamma^\rho] = 4(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu).$$

Multiplying by $-\mathrm{i}/4$ gives the displayed commutator for $S^{\mu\nu}$. Exponentiating this infinitesimal identity gives the Clifford covariance of $R(\Lambda)\gamma^\mu R(\Lambda)^{-1}$. The infinitesimal derivative of $R^\dagger\beta R$ is zero because $(S^{\mu\nu})^\dagger\beta = \beta S^{\mu\nu}$, which follows from the adjoint identities above; the value at the identity is β.

Let $\Gamma = \gamma^0\gamma^1\gamma^2\gamma^3$. Moving each γ^μ through the other three Clifford generators shows that Γ anticommutes with every γ^μ. Since

$$\Gamma^2 = (-1)^6(\gamma^0)^2(\gamma^1)^2(\gamma^2)^2(\gamma^3)^2 = -1,$$

$\gamma_5 = -\mathrm{i}\Gamma$ squares to 1. Anticommutation with every γ^μ implies commutation with every even commutator $[\gamma^\mu, \gamma^\nu]$, hence with $S^{\mu\nu}$.

The trace with γ_5 is totally antisymmetric in the four vector indices: moving adjacent Clifford generators past one another changes the sign, while the metric term produced by the anticommutator has a repeated index and gives zero in the fully antisymmetric part. It is therefore enough to evaluate the ordered component:

$$\operatorname{tr}(\gamma_5\gamma^0\gamma^1\gamma^2\gamma^3) = -\mathrm{i}\operatorname{tr}(\Gamma^2) = 4\mathrm{i}.$$

This gives the stated $\epsilon^{0123} = +1$ normalization.

40.19 Two-Component Blocks and Index Conventions

Definition 40.21 (Two-component block and epsilon conventions)**.** Let

$$\gamma^\mu = \begin{pmatrix} 0 & \rho^\mu_{+-} \\ \rho^\mu_{-+} & 0 \end{pmatrix}$$

with respect to $S = S_+ \oplus S_-$. In the basis (40.10),

$$\rho^0_{+-} = -\mathrm{i}I_2, \qquad \rho^j_{+-} = -\mathrm{i}\sigma_j, \qquad \rho^0_{-+} = -\mathrm{i}I_2, \qquad \rho^j_{-+} = +\mathrm{i}\sigma_j.$$

Equivalently,

$$(\gamma^\mu s)_a = (\rho^\mu_{+-})_a{}^{\dot b}\widetilde{\chi}_{\dot b}, \qquad (\gamma^\mu s)_{\dot a} = (\rho^\mu_{-+})_{\dot a}{}^{b}\chi_b.$$

Spinor $SU(2)$ indices are raised and lowered by

$$\chi^a = \epsilon^{ab}\chi_b, \qquad \widetilde{\chi}^{\dot a} = \epsilon^{\dot a\dot b}\widetilde{\chi}_{\dot b}, \qquad \epsilon^{12} = \epsilon_{12} = 1, \qquad \epsilon^{\dot 1\dot 2} = \epsilon_{\dot 1\dot 2} = 1.$$

With this convention, the two chiral blocks obey the sign relation

$$(\rho^\mu_{+-})_a{}^{\dot b} = -(\rho^\mu_{-+})^{\dot b}{}_a = -\epsilon^{\dot b\dot c}(\rho^\mu_{-+})_{\dot c}{}^d\epsilon_{da}. \tag{40.15}$$

In lowered form,

$$(\rho_\mu)_{a\dot b}(\rho_\nu)^{a\dot b} = 2\eta_{\mu\nu}.$$

Complex conjugation exchanges the chiral and antichiral components:

$$\overline{\chi}^{\dot a} = (\chi_a)^*, \qquad \overline{\widetilde{\chi}}^a = (\widetilde{\chi}_{\dot a})^*.$$

The two-component sign relation is a convention check rather than an independent structural theorem. It is nevertheless worth recording explicitly because it fixes the minus sign that often changes when formulae are copied between spinor conventions. Let

$$\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The matrices in the two chiral blocks are

$$\rho^0_{+-} = -\mathrm{i}I_2, \quad \rho^j_{+-} = -\mathrm{i}\sigma_j, \qquad \rho^0_{-+} = -\mathrm{i}I_2, \quad \rho^j_{-+} = +\mathrm{i}\sigma_j.$$

Substituting the Pauli matrices gives the finite matrix identity

$$\rho^\mu_{+-} = \left(-\epsilon\,\rho^\mu_{-+}\,\epsilon\right)^T,$$

which is precisely the displayed index formula. For the contraction, lower the vector index with $\eta_{\mu\nu}$ and raise the two spinor indices with the same epsilon matrix:

$$M^{a\dot b} = \epsilon^{ac}\epsilon^{\dot b\dot d}M_{c\dot d}.$$

Direct multiplication of the four displayed 2×2 matrices gives

$$\sum_{a,\dot b}(\rho_\mu)_{a\dot b}(\rho_\nu)^{a\dot b} = -2\,\delta_{\mu 0}\delta_{\nu 0} + 2\sum_{j=1}^{3}\delta_{\mu j}\delta_{\nu j} = 2\eta_{\mu\nu}.$$

40.20 Majorana Conjugation

Definition 40.22 (Majorana conjugation in the Weinberg-compatible basis)**.** A Majorana condition is a Lorentz-covariant reality condition on the complex spinor representation. In the explicit basis (40.10), set

$$B = \gamma_2. \tag{40.16}$$

Here $\gamma_2 = \eta_{2\nu}\gamma^\nu = \gamma^2$. Directly from the Pauli matrices,

$$(\gamma^\mu)^* = B\gamma^\mu B^{-1}, \qquad B^* B = \mathbf{1}_S.$$

For a spinor s, define

$$s^c = B^{-1}s^*.$$

Then s^c transforms in the same Lorentz representation as s, and $(s^c)^c = s$. A Majorana spinor is a spinor satisfying

$$s = s^c, \qquad \text{equivalently} \qquad s^* = Bs.$$

If $\chi = P_+ s$ is the chiral component, then the Majorana spinor may be written

$$s = \chi + B^*\chi^*.$$

The charge-conjugation matrix in the convention often used for transposed spinor contractions is

$$C = B\gamma^0.$$

For a Majorana spinor s,

$$\bar{s} = s^\dagger \mathrm{i}\gamma^0 = \mathrm{i}s^T C.$$

For two Grassmann-even Majorana spinors s_1, s_2, with chiral components χ_1, χ_2, the Lorentz vector bilinear decomposes as

$$\bar{s}_1\gamma^\mu s_2 = \overline{\chi}_1\, \rho^\mu_{+-}\chi_2 + \overline{\chi}_2\, \rho^\mu_{+-}\chi_1,$$

where the precise matrix placement is the one fixed in (40.15). For Grassmann-odd spinors the same formula must be combined with the sign rules for moving odd coefficients through one another.

Lorentz covariance of Majorana conjugation. The anti-linear map $s \mapsto s^c = B^{-1}s^*$ obeys

$$(s^c)^c = s, \qquad (R(\Lambda)s)^c = R(\Lambda)s^c.$$

The Majorana condition $s = s^c$ is therefore a Lorentz-covariant real structure on the complex spinor representation. If s is Majorana, then

$$\bar{s} = \mathrm{i}s^T C, \qquad C = B\gamma^0.$$

The explicit matrices give

$$(\gamma^\mu)^* = B\gamma^\mu B^{-1}, \qquad B^* B = 1.$$

The second identity gives

$$(s^c)^c = B^{-1}(B^{-1}s^*)^* = B^{-1}(B^*)^{-1}s = s.$$

The first identity implies $(S^{\mu\nu})^* = -BS^{\mu\nu}B^{-1}$. The minus sign is exactly canceled by complex conjugating the factor $-\mathrm{i}$ in the exponential, hence $R(\Lambda)^* = BR(\Lambda)B^{-1}$. Therefore

$$(R(\Lambda)s)^c = B^{-1}R(\Lambda)^*s^* = R(\Lambda)B^{-1}s^* = R(\Lambda)s^c.$$

If $s = s^c$, then $s^* = Bs$, and

$$\bar{s} = s^\dagger \mathrm{i}\gamma^0 = (s^*)^T\mathrm{i}\gamma^0 = s^T B^T \mathrm{i}\gamma^0.$$

For the displayed basis $B^T = B$, so the last expression is $\mathrm{i}s^T B\gamma^0 = \mathrm{i}s^T C$.

40.21 Comparison with Wess-Bagger-Type Conventions

At fixed Lorentzian metric $\eta = \mathrm{diag}(-,+,+,+)$, a common chiral gamma-matrix basis, used for example in Wess-Bagger-style supersymmetry formulae, is

$$\gamma^0_{\mathrm{WB}} = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \qquad \gamma^j_{\mathrm{WB}} = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}. \tag{40.17}$$

These matrices obey the same Clifford relation

$$\{\gamma^\mu_{\mathrm{WB}}, \gamma^\nu_{\mathrm{WB}}\} = 2\eta^{\mu\nu}.$$

The Weinberg-compatible basis used in the monograph is related to (40.17) by the chiral phase change

$$U_{\mathrm{ch}} = \begin{pmatrix} I_2 & 0 \\ 0 & \mathrm{i}I_2 \end{pmatrix}, \qquad \gamma^\mu = U_{\mathrm{ch}}\gamma^\mu_{\mathrm{WB}}U_{\mathrm{ch}}^{-1}. \tag{40.18}$$

Thus if s_{WB} and s are coordinate columns for the same abstract spinor in the two bases, then

$$s = U_{\mathrm{ch}}s_{\mathrm{WB}}.$$

The chiral component is unchanged, while the antichiral component is multiplied by i. This basis change is the source of many sign differences in two-component supersymmetry formulae. In particular, with the index-raising convention used in this section, the monograph has the minus sign in (40.15); in the Wess-Bagger-type basis one often writes the corresponding relation without that minus sign, together with a different convention for upper and lower ϵ-symbols and dotted-index contractions. Formulae copied between the two systems must therefore translate both the gamma matrices and the index algebra.

The chirality matrix defined by

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3$$

is unchanged by the similarity transformation (40.18). The Dirac adjoint is also the same abstract Hermitian pairing after the basis change:

$$\bar{s} = s^\dagger(\mathrm{i}\gamma^0) = s^\dagger_{\mathrm{WB}}\left(U^\dagger_{\mathrm{ch}}\mathrm{i}\gamma^0 U_{\mathrm{ch}}\right)s_{\mathrm{WB}}.$$

The displayed matrix in the last expression is the Wess-Bagger-basis matrix for the same invariant pairing.

At the fixed metric $\eta = \mathrm{diag}(-,+,+,+)$, the Wess–Bagger-type matrices used here are related to the Weinberg-compatible matrices by a finite chiral phase change. The matrix

$$U_{\mathrm{ch}} = \begin{pmatrix} I_2 & 0 \\ 0 & \mathrm{i}I_2 \end{pmatrix}$$

relates the Wess-Bagger-type basis (40.17) to the Weinberg-compatible basis by

$$\gamma^\mu = U_{\mathrm{ch}} \gamma^\mu_{\mathrm{WB}} U_{\mathrm{ch}}^{-1}.$$

The chirality operator is unchanged as an endomorphism of the abstract spinor module, while the antichiral coordinate column is multiplied by i. Hence any formula imported from Wess-Bagger-type notation must translate both the gamma matrices and the epsilon-index convention. Since $U_{\mathrm{ch}}^{-1} = \mathrm{diag}(I_2, -\mathrm{i}I_2)$, direct block multiplication gives

$$U_{\mathrm{ch}} \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix} U_{\mathrm{ch}}^{-1} = -\mathrm{i} \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} = \gamma^0$$

and, for $j = 1, 2, 3$,

$$U_{\mathrm{ch}} \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} U_{\mathrm{ch}}^{-1} = -\mathrm{i} \begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix} = \gamma^j.$$

Both bases are block-off-diagonal with the same chiral splitting, so γ_5 remains $\mathrm{diag}(I_2, -I_2)$. The coordinate relation $s = U_{\mathrm{ch}} s_{\mathrm{WB}}$ gives $(\chi, \widetilde{\chi})^T = (\chi_{\mathrm{WB}}, \mathrm{i}\widetilde{\chi}_{\mathrm{WB}})^T$, which is the stated antichiral phase. The epsilon symbols determine how lowered and raised two-component indices are identified; therefore the matrix phase alone is insufficient for translating two-component formulae.

40.22 Dimensional Continuation and γ_5

Definition 40.23 (Dimensional-reduction chiral trace prescription)**.** Dimensional regularization and dimensional reduction are perturbative schemes; they are not definitions of a non-perturbative fermionic path integral. Whenever a chiral trace is present, the finite-dimensional four-dimensional data must be specified before continuing loop momenta. The convention used in the anomaly chapter is:

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3$$

is a strictly four-dimensional matrix. In a split $D = 4 - \varepsilon$ momentum algebra, write

$$\ell = \ell_{\parallel} + \ell_{\perp},$$

where $\ell_{\parallel}$ lies in the physical four-dimensional subspace and $\ell_{\perp}$ lies in the continued complement. Then

$$\{\gamma_5, \not{\ell}_{\parallel}\} = 0, \qquad [\gamma_5, \not{\ell}_{\perp}] = 0.$$

Together with (40.14), these rules define the chiral trace algebra used in the dimensional-reduction anomaly calculation. Changing this prescription changes intermediate contact terms and must be compensated by an explicit finite local counterterm if the same renormalized Ward identity is to be recovered.

40.23 Three-Dimensional Lorentzian Spinors

Definition 40.24 (Three-dimensional mostly-plus Majorana convention)**.** For later use in lower-dimensional QFT, take

$$\eta_{\mu\nu} = \mathrm{diag}(-,+,+), \qquad \Gamma^0 = \mathrm{i}\sigma^2, \qquad \Gamma^1 = \sigma^1, \qquad \Gamma^2 = \sigma^3.$$

Then

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu},$$

and all three matrices are real. Therefore a Majorana spinor may be taken to have real components. Raise and lower two-component indices by

$$\chi^a = \epsilon^{ab}\chi_b, \qquad \epsilon^{12} = \epsilon_{12} = 1.$$

The lowered gamma matrices

$$\Gamma^\mu_{ab} := (\Gamma^\mu)_a{}^c\epsilon_{cb}$$

are symmetric in a, b. For two Grassmann-even Majorana spinors ζ_1, ζ_2,

$$\bar{\zeta}_1\Gamma^\mu\zeta_2 = \zeta_1^T \mathrm{i}\Gamma^0\Gamma^\mu\zeta_2 = \mathrm{i}\,\zeta_1^a\Gamma^\mu_{ab}\zeta_2^b.$$

Remark 40.25 (Three-dimensional Clifford and symmetric block identities). The matrices in Definition 40.24 obey

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu},$$

and $\Gamma^\mu_{ab} = (\Gamma^\mu)_a{}^c\epsilon_{cb}$ is symmetric in a, b. This is a finite convention check: The Pauli identities $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}$ give $(\mathrm{i}\sigma^2)^2 = -1$, $(\sigma^1)^2 = (\sigma^3)^2 = 1$, and zero anticommutators between distinct displayed matrices. With $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$,

$$\Gamma^0\epsilon = -I_2, \qquad \Gamma^1\epsilon = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Gamma^2\epsilon = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which are all symmetric.

40.24 Two-Dimensional Lorentzian Spinors

Definition 40.26 (Two-dimensional mostly-plus spinor trace convention)**.** In two-dimensional Lorentzian calculations the monograph uses

$$\eta_{\mu\nu} = \mathrm{diag}(-,+), \qquad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \epsilon^{01} = +1.$$

The chirality matrix is

$$\gamma = \gamma^0\gamma^1, \qquad \gamma^2 = 1, \qquad \{\gamma, \gamma^\mu\} = 0.$$

The trace convention compatible with the anomaly calculation is

$$\mathrm{tr}(\gamma^\nu\gamma\,\gamma^\rho) = 2\epsilon^{\rho\nu}.$$

Thus

$$\mathrm{tr}(\gamma^\nu\gamma\ \not{p}) = 2\epsilon^{\rho\nu}p_\rho.$$

As in four dimensions, $\bar{\psi} = \mathrm{i}\psi^\dagger\gamma^0$ in the mostly-plus convention.

Remark 40.27 (Two-dimensional chirality trace). The convention in the preceding definition is realized by

$$\gamma^0 = \mathrm{i}\sigma^2, \qquad \gamma^1 = \sigma^1, \qquad \gamma = \gamma^0\gamma^1 = \sigma^3.$$

It satisfies

$$\mathrm{tr}(\gamma^\nu \gamma\, \gamma^\rho) = 2\epsilon^{\rho\nu}.$$

The Clifford relation follows from $(\mathrm{i}\sigma^2)^2 = -\mathbf{1}$, $(\sigma^1)^2 = \mathbf{1}$, and $\sigma^2\sigma^1 = -\sigma^1\sigma^2$, so $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ in mostly-plus signature. Also

$$\gamma^0\gamma^1 = \mathrm{i}\sigma^2\sigma^1 = \sigma^3.$$

With $\epsilon^{01} = +1$, the two nonzero traces are

$$\mathrm{tr}(\gamma^0\gamma\gamma^1) = -2, \qquad \mathrm{tr}(\gamma^1\gamma\gamma^0) = 2,$$

which equal $2\epsilon^{10}$ and $2\epsilon^{01}$, respectively; repeated indices vanish because $\gamma^\nu\gamma\gamma^\nu$ is proportional to γ, whose trace is zero.

Definition 40.28 (Two-dimensional chiral-component Majorana convention)**.** For comparison with two-dimensional CFT component conventions, it is useful to use a Majorana basis adapted to light-cone components. In flat coordinates $x^0 = t$, $x^1 = \sigma$, set

$$\widehat{\Gamma}^0 = -\mathrm{i}\sigma^1, \qquad \widehat{\Gamma}^1 = \sigma^2, \qquad \widehat{\Gamma} = \widehat{\Gamma}^0\widehat{\Gamma}^1 = \sigma^3.$$

Let $\sigma^\pm = \sigma \pm t$, so that

$$\widehat{\Gamma}^+ = \widehat{\Gamma}^1 + \widehat{\Gamma}^0, \qquad \widehat{\Gamma}^- = \widehat{\Gamma}^1 - \widehat{\Gamma}^0.$$

Spinor indices $A, B = \pm$ are raised and lowered by

$$\epsilon^{+-} = \epsilon_{+-} = 1, \qquad v^A = \epsilon^{AB} v_B.$$

The lowered light-cone gamma matrices are

$$(\widehat{\Gamma}^\pm)_{AB} := (\widehat{\Gamma}^\pm)_A{}^C \epsilon_{CB}.$$

This basis gives

$$(\widehat{\Gamma}^+)_{++} = 2\mathrm{i}, \qquad (\widehat{\Gamma}^-)_{--} = 2\mathrm{i},$$

with all other lowered light-cone components zero. It is the convention that reduces the covariant Majorana action

$$-\frac{\mathrm{i}}{4\pi} \int d^2\sigma \sqrt{-g}\, \psi \Gamma^a \nabla_a^{\mathrm{spin}} \psi$$

to

$$-\frac{1}{2\pi} \int d^2\sigma \left(\psi_+ \partial_- \psi_+ + \psi_- \partial_+ \psi_-\right)$$

in flat coordinates, with the same spinor-index contraction convention. Its role in this monograph is translational: it is the convenient bridge to the chiral free-fermion fields of two-dimensional CFT. After Wick rotation, ψ_- and ψ_+ become the holomorphic and antiholomorphic spin fields on a Riemann surface, and the global information is carried by a spin structure rather than by a preferred global spinor frame. Most two-dimensional CFT computations are therefore done directly with these components, their OPEs, and their Szego kernels. No ghost system or integration over moduli space is part of this convention; those are separate structures that are outside the fixed-surface two-dimensional QFT/CFT framework being recorded here.

Remark 40.29 (Compatibility of the two-dimensional spinor bases)*.* The Dirac-anomaly basis of Remark 40.27 and the Majorana basis of Definition 40.28 are similar representations of the same mostly-plus Clifford algebra. Explicitly, with

$$U_2 = \begin{pmatrix} -\mathrm{i} & 0 \\ 0 & 1 \end{pmatrix},$$

one has

$$\widehat{\Gamma}^0 = U_2\gamma^0 U_2^{-1}, \qquad \widehat{\Gamma}^1 = U_2\gamma^1 U_2^{-1}, \qquad \widehat{\Gamma} = U_2\gamma U_2^{-1}.$$

Consequently the two bases define the same invariant Clifford algebra and the same chirality splitting, while the free-fermion light-cone component normalization and the Dirac-anomaly trace normalization are different coordinate realizations of that algebra. The monograph uses the Dirac-anomaly basis when evaluating spinor traces and uses the chiral-component Majorana basis when comparing to two-dimensional CFT component conventions. Direct multiplication gives

$$U_2(\mathrm{i}\sigma^2)U_2^{-1} = -\mathrm{i}\sigma^1, \qquad U_2\sigma^1 U_2^{-1} = \sigma^2.$$

The chirality relation follows by multiplying the two identities. Lowering the spinor index of

$$\widehat{\Gamma}^+ = \begin{pmatrix} 0 & -2\mathrm{i} \\ 0 & 0 \end{pmatrix}, \qquad \widehat{\Gamma}^- = \begin{pmatrix} 0 & 0 \\ 2\mathrm{i} & 0 \end{pmatrix}$$

with $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ gives

$$\widehat{\Gamma}^+\epsilon = \begin{pmatrix} 2\mathrm{i} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \widehat{\Gamma}^-\epsilon = \begin{pmatrix} 0 & 0 \\ 0 & 2\mathrm{i} \end{pmatrix}.$$

Thus the light-cone components have the stated chiral-component normalization. Since U_2 is invertible, all Clifford, chirality, and trace identities are transported between the two bases after transforming the coordinate components of the spinors.

40.25 Euclidean Spinors

Definition 40.30 (Recursive Euclidean Clifford basis)**.** Euclidean gamma matrices in even dimension d are denoted $\Gamma^1, \dots, \Gamma^d$ and obey

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}.$$

A concrete recursive construction starts in $d = 2$ with

$$\Gamma^1_{(2)} = \sigma_1, \qquad \Gamma^2_{(2)} = \sigma_2,$$

and for even $d > 2$ sets

$$\Gamma^\mu_{(d)} = \Gamma^\mu_{(d-2)} \otimes (-\sigma_3), \qquad \mu = 1, \dots, d-2,$$

$$\Gamma^{d-1}_{(d)} = I_{2^{d/2-1}} \otimes \sigma_1, \qquad \Gamma^d_{(d)} = I_{2^{d/2-1}} \otimes \sigma_2.$$

The Euclidean chirality matrix is

$$\Gamma_E = \mathrm{i}^{-d/2}\Gamma^1 \cdots \Gamma^d, \qquad \Gamma_E^2 = 1.$$

One convenient charge-conjugation matrix is

$$C_E = \Gamma^1\Gamma^3\cdots\Gamma^{d-1}, \qquad d \equiv 0 \pmod 4,$$

and

$$C_E = \Gamma_E\Gamma^1\Gamma^3\cdots\Gamma^{d-1}, \qquad d \equiv 2 \pmod 4.$$

These formulae fix a basis; invariant statements should be phrased in terms of the corresponding real, complex, or quaternionic structure on the Euclidean spinor module.

The recursive construction in Definition 40.30 gives

$$\{\Gamma^i_{(d)}, \Gamma^j_{(d)}\} = 2\delta^{ij}, \qquad \Gamma_E^2 = 1$$

in every even dimension d. The $d = 2$ matrices are the Pauli matrices σ_1, σ_2, so the claim holds at the initial step. Assume it holds in dimension $d-2$. For $i, j \le d-2$, tensoring both matrices with $-\sigma_3$ preserves the anticommutator because $(-\sigma_3)^2 = 1$. The two new matrices $I \otimes \sigma_1$ and $I \otimes \sigma_2$ anticommute with each other and square to 1. They also anticommute with the old matrices because σ_1 and σ_2 anticommute with σ_3. Hence the Clifford relation holds in dimension d.

Let $\Omega_d = \Gamma^1\cdots\Gamma^d$. In Euclidean signature, $\Omega_d^2 = (-1)^{d(d-1)/2}$. For even $d = 2r$, this is $(-1)^r$. Since $\Gamma_E = \mathrm{i}^{-r}\Omega_d$,

$$\Gamma_E^2 = \mathrm{i}^{-2r}(-1)^r = (-1)^{-r}(-1)^r = 1.$$

40.26 Six- and Eight-Dimensional Euclidean Blocks

Definition 40.31 (Six- and eight-dimensional Euclidean spinor blocks)**.** For later supersymmetric and internal-symmetry applications, it is useful to record block conventions for Euclidean $SO(6)$ and $SO(8)$ spinors.

In $d = 8$, the chiral and antichiral spinor representations are both real eight-dimensional. In a basis in which the charge-conjugation matrix is the identity, write the off-diagonal gamma-matrix components as

$$(\Gamma^i)^{a\dot b} = \mathcal{C}^{ia\dot b}, \qquad (\Gamma^i)^{\dot b a} = \mathcal{C}^{i\dot b a}, \qquad \mathcal{C}^{i\dot b a} = -\mathcal{C}^{ia\dot b}.$$

They obey

$$\begin{aligned}
\mathcal{C}^{ia\dot c}\mathcal{C}^{j\dot c b} + \mathcal{C}^{ja\dot c}\mathcal{C}^{i\dot c b} &= 2\delta^{ij}\delta^{ab},\\
\mathcal{C}^{i\dot a c}\mathcal{C}^{jc\dot b} + \mathcal{C}^{j\dot a c}\mathcal{C}^{ic\dot b} &= 2\delta^{ij}\delta^{\dot a\dot b},\\
\mathcal{C}^{ia\dot c}\mathcal{C}^{i\dot d b} + \mathcal{C}^{ib\dot c}\mathcal{C}^{i\dot d a} &= 2\delta^{ab}\delta^{\dot c\dot d}.
\end{aligned}$$

The first two identities are the Clifford relations; the third records the triality symmetry that exchanges vector and spinor representations.

In $d = 6$, use $\mathrm{Spin}(6) \simeq SU(4)$. Chiral and antichiral spinors are the fundamental and antifundamental representations of $SU(4)$. Let $I, J, K, L = 1, \ldots, 4$. The gamma-matrix blocks are antisymmetric tensors

$$\mathcal{C}^i_{IJ} = -\mathcal{C}^i_{JI}, \qquad \mathcal{C}^{iIJ} = \frac{1}{2}\epsilon^{IJKL}\mathcal{C}^i_{KL} = \left(\mathcal{C}^i_{JI}\right)^*.$$

They satisfy

$$\sum_{i=1}^{6} \mathcal{C}^i_{IJ} \mathcal{C}^{iKL} = -2 \left(\delta_I{}^K \delta_J{}^L - \delta_I{}^L \delta_J{}^K \right) .$$

These identities are the ones used when four-dimensional supersymmetry or higher-dimensional reductions are translated into $SU(4)$ index notation.

Chapter 41

Maxwell Theory, Constraints, and Gauge Fixing

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

41.1 Representative Fields and Field Strength

Definition 41.1 (Local Maxwell representative datum)**.** Work on four-dimensional Minkowski space with mostly-plus metric. A Maxwell configuration in this local discussion is represented by a real one-form

$$A = A_\mu(x)\,\mathrm{d}x^\mu.$$

Two representatives related by

$$A_\mu(x) \longmapsto A_\mu(x) + \partial_\mu \xi(x)$$

describe the same electromagnetic field. Here ξ is a real scalar gauge parameter in the same boundary-condition class as A_μ. We assume throughout this chapter that fields are either smeared with compactly supported test functions or obey boundary conditions for which integrations by parts have no boundary term. Zero modes of the gauge-fixing operators are then removed separately; this is part of the definition of the local gauge slice used below.

The local gauge-invariant field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Maxwell action and Euler–Lagrange equation. The free Maxwell action is

$$S[A] = -\frac{1}{4}\int \mathrm{d}^4x\, F_{\mu\nu}F^{\mu\nu}.$$

Its Euler–Lagrange equation is

$$\partial_\mu F^{\mu\nu} = 0.$$

The variation of the field strength is $\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu$. Using antisymmetry of $F^{\mu\nu}$,

$$\delta S = -\frac{1}{2}\int \mathrm{d}^4x\, F^{\mu\nu}\delta F_{\mu\nu} = -\int \mathrm{d}^4x\, F^{\mu\nu}\partial_\mu \delta A_\nu.$$

The boundary condition convention of Definition 41.1 gives

$$\delta S = \int \mathrm{d}^4x\, (\partial_\mu F^{\mu\nu})\delta A_\nu.$$

Since δA_ν is arbitrary within the chosen test-function class, the Euler–Lagrange equation is $\partial_\mu F^{\mu\nu} = 0$. The same calculation shows gauge invariance of $S[A]$, because $\delta_\xi F_{\mu\nu} = 0$ for $\delta_\xi A_\mu = \partial_\mu \xi$.

With $x^0 = t$ and spatial indices $i, j = 1, 2, 3$, the Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{3}(\partial_0 A_i - \partial_i A_0)^2 - \frac{1}{4}\sum_{i,j=1}^{3}(\partial_i A_j - \partial_j A_i)^2.$$

The absence of $\partial_0 A_0$ is the first sign that the vector potential is a redundant representative rather than four independent propagating fields.

41.2 Canonical Constraint Analysis

Definition 41.2 (Maxwell canonical Poisson structure)**.** At fixed time let $A_\mu(t, \vec{x})$ be canonical coordinates. For functionals F, G of A_μ, Π^μ, use the equal-time Poisson bracket

$$\{F, G\}_{\mathrm{P}} = \int \mathrm{d}^3\vec{x}\, \left(\frac{\delta F}{\delta A_\mu(\vec{x})}\frac{\delta G}{\delta \Pi^\mu(\vec{x})} - \frac{\delta F}{\delta \Pi^\mu(\vec{x})}\frac{\delta G}{\delta A_\mu(\vec{x})}\right).$$

Equivalently,

$$\{A_\mu(\vec{x}), \Pi^\nu(\vec{y})\}_{\mathrm{P}} = \delta_\mu{}^\nu \delta^{(3)}(\vec{x} - \vec{y}),$$

with all A-A and Π-Π brackets zero.

Primary constraint, Gauss constraint, and canonical Hamiltonian. The canonical momenta are

$$\Pi^\mu(\vec{x}) = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu(t, \vec{x}))},$$

so

$$\Pi^0(\vec{x}) = 0, \qquad \Pi^i(\vec{x}) = \partial_0 A_i - \partial_i A_0.$$

The equation

$$\chi_1(\vec{x}) = \Pi^0(\vec{x}) = 0$$

is the primary constraint. The canonical Hamiltonian is

$$H_{\mathrm{c}} = \int \mathrm{d}^3\vec{x}\, \left[\frac{1}{2}\Pi^i\Pi^i + \frac{1}{4}F_{ij}F_{ij} - A_0\, \partial_i\Pi^i\right],$$

after integrating $\Pi^i\partial_i A_0$ by parts. Preserving the primary constraint under time evolution gives

$$\dot{\Pi}^0(\vec{x}) = \{\Pi^0(\vec{x}), H_{\mathrm{c}}\}_{\mathrm{P}} = \partial_i\Pi^i(\vec{x}),$$

hence the secondary constraint

$$\chi_2(\vec{x}) = \partial_i\Pi^i(\vec{x}) = 0.$$

This is Gauss' law in the absence of charges.

These formulae are the canonical Legendre transform. The Lagrangian density contains $\partial_0 A_i$ only through $\partial_0 A_i - \partial_i A_0$, and contains no $\partial_0 A_0$. This gives the displayed momenta and the primary constraint $\Pi^0 = 0$. Solving $\partial_0 A_i = \Pi^i + \partial_i A_0$, the Legendre transform gives

$$\Pi^i\partial_0 A_i - \mathcal{L} = \frac{1}{2}\Pi^i\Pi^i + \frac{1}{4}F_{ij}F_{ij} + \Pi^i\partial_i A_0.$$

Integrating the last term by parts gives H_{c}. Finally,

$$\{\Pi^0(\vec{x}), H_{\mathrm{c}}\}_{\mathrm{P}} = -\frac{\delta H_{\mathrm{c}}}{\delta A_0(\vec{x})} = \partial_i\Pi^i(\vec{x}),$$

so preservation of $\Pi^0 = 0$ imposes Gauss' law.

Hamiltonian generator of Maxwell gauge transformations. The constraints obey

$$\{\chi_a(\vec{x}), \chi_b(\vec{y})\}_{\mathrm{P}} = 0, \qquad a, b = 1, 2.$$

They are first-class constraints. Their relation to gauge redundancy is direct. For a spacetime function ζ, define at time t

$$G[\zeta] = \int \mathrm{d}^3\vec{y}\, \left[(\partial_0\zeta)(t,\vec{y})\, \Pi^0(\vec{y}) - \zeta(t,\vec{y})\, \partial_i\Pi^i(\vec{y})\right].$$

Then

$$\{A_0(\vec{x}), G[\zeta]\}_{\mathrm{P}} = \partial_0\zeta(t,\vec{x}), \qquad \{A_i(\vec{x}), G[\zeta]\}_{\mathrm{P}} = \partial_i\zeta(t,\vec{x}).$$

Thus the first-class constraints generate

$$\delta_\zeta A_\mu(t,\vec{x}) = \partial_\mu\zeta(t,\vec{x})$$

on the vector-potential representative. The gauge-invariant phase space is obtained by restricting to the constraint surface and quotienting by these gauge orbits, or equivalently by choosing a regular slice through the orbits.

The bracket calculation is elementary but fixes an important convention. The constraints depend only on momenta, hence their mutual Poisson brackets vanish. For the generator $G[\zeta]$,

$$\{A_0(\vec{x}), G[\zeta]\}_{\mathrm{P}} = \int \mathrm{d}^3\vec{y}\, \partial_0\zeta(t,\vec{y})\delta^{(3)}(\vec{x}-\vec{y}) = \partial_0\zeta(t,\vec{x}).$$

For A_i, integrate the derivative in the Gauss term against the delta function:

$$\{A_i(\vec{x}), G[\zeta]\}_{\mathrm{P}} = -\int \mathrm{d}^3\vec{y}\, \zeta(t,\vec{y})\partial_i^{\vec{y}}\delta^{(3)}(\vec{x}-\vec{y}) = \partial_i\zeta(t,\vec{x}).$$

The transformed field strength is unchanged because mixed partial derivatives commute. Thus the Hamiltonian first-class flow is precisely the Maxwell representative redundancy.

41.3 Axial Gauge and Reduced Phase Space

Definition 41.3 (Axial gauge local chart)**.** Axial gauge chooses the representative satisfying

$$A_3 = 0.$$

Locally along an orbit,

$$A_3 \longmapsto A_3 + \partial_3\xi,$$

so this is a regular gauge condition for modes on which ∂_3 is invertible. Modes with $\partial_3\xi = 0$ are residual gauge parameters for this slice and must be treated separately by boundary conditions or an additional gauge condition.

Axial gauge leaves two canonical degrees of freedom. There are two equivalent local descriptions. In canonical language the condition $A_3 = 0$ pairs with Gauss' law and allows one to solve

$$\partial_3\Pi^3 = -\partial_a\Pi^a, \qquad a = 1, 2,$$

for the longitudinal momentum Π^3, again modulo ∂_3 zero modes. In Lagrangian language one imposes $A_3 = 0$ before eliminating the non-dynamical field A_0. The gauge-fixed Lagrangian density is

$$\begin{aligned}\mathcal{L}_{\mathrm{ax}} = {} & \frac{1}{2}\sum_{a=1}^{2}(\partial_0 A_a - \partial_a A_0)^2 + \frac{1}{2}(\partial_3 A_0)^2 \\ & - \frac{1}{2}(\partial_1 A_2 - \partial_2 A_1)^2 - \frac{1}{2}\sum_{a=1}^{2}(\partial_3 A_a)^2.\end{aligned}$$

The A_0 equation is

$$\sum_{a=1}^{2}\partial_a(\partial_a A_0 - \partial_0 A_a) + \partial_3^2 A_0 = 0.$$

With

$$\Pi^a = \partial_0 A_a - \partial_a A_0,$$

this becomes

$$\sum_{a=1}^{2}\partial_a\Pi^a + \partial_3^2 A_0 = 0.$$

The pair

$$\Pi^0(\vec{x}) = 0, \qquad \sum_{a=1}^{2} \partial_a \Pi^a(\vec{x}) + \partial_3^2 A_0(\vec{x}) = 0$$

is second-class for nonzero k_3, since

$$\left\{ \Pi^0(\vec{x}), \sum_{a=1}^{2} \partial_a \Pi^a(\vec{y}) + \partial_3^2 A_0(\vec{y}) \right\}_{\mathrm{P}} = \partial_3^2 \delta^{(3)}(\vec{x} - \vec{y}).$$

Hence one may either introduce the corresponding Dirac bracket or solve

$$\partial_3^2 A_0 = -\sum_{a=1}^{2} \partial_a \Pi^a$$

and eliminate A_0, Π^0. For the regular modes of the axial slice the remaining canonical variables are

$$A_a(\vec{x}), \Pi^a(\vec{x}), \qquad a = 1, 2.$$

They are the two canonical degrees of freedom of the free photon. On modes where ∂_3 is invertible, Gauss' law gives $\Pi^3 = -\partial_3^{-1} \partial_a \Pi^a$, so (A_3, Π^3) is removed after imposing the axial condition and the constraint. In the Lagrangian description, A_0 has no time derivative. Its equation of motion is the displayed elliptic equation along the x^3-direction once $A_3 = 0$ is imposed. The Poisson bracket

$$\{\Pi^0(\vec{x}), \partial_3^2 A_0(\vec{y})\}_{\mathrm{P}} = \partial_3^2 \delta^{(3)}(\vec{x} - \vec{y})$$

is invertible on the same nonzero-k_3 modes, so (A_0, Π^0) is a second-class pair there and may be eliminated. The only regular canonical variables left are A_a, Π^a, $a = 1, 2$.

41.4 Gauge Slices and the Faddeev–Popov Factor

Definition 41.4 (Local Faddeev–Popov chart)**.** Before gauge fixing, the path-integral shorthand

$$Z_{\text{unfixed}} = \int DA_\mu \, \mathrm{e}^{\mathrm{i}S[A]}$$

integrates over all representatives on each gauge orbit. Let $\mathcal{G}(A)$ be a gauge condition. Locally in field space, and after removing zero modes, the Faddeev–Popov identity is

$$1 = \Delta_{\mathcal{G}}[A] \int D\xi \, \delta\big(\mathcal{G}(A + \partial\xi)\big),$$

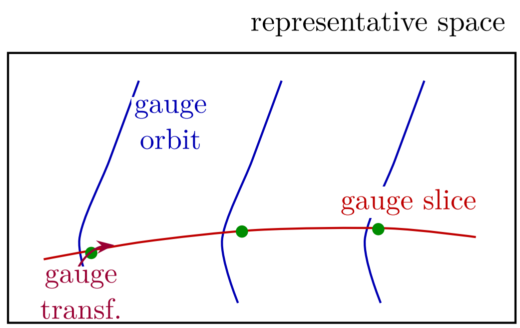
where

$$\Delta_{\mathcal{G}}[A] = \det M_{\mathcal{G}}[A], \qquad M_{\mathcal{G}}[A]\xi = \frac{\mathrm{d}}{\mathrm{d}s} \mathcal{G}(A + s\,\partial\xi) \bigg|_{s=0}.$$

Abelian linear FP determinant. For abelian Maxwell theory with a linear gauge condition, $M_{\mathcal{G}}$ is independent of A_μ. Its determinant is then an overall normalization in the free theory. This special simplification is not a definition of gauge fixing; it is a property of the abelian linear examples used in this chapter. The corresponding nonabelian construction is developed later as a local coordinate system on gauge orbits. There the analogue of $M_{\mathcal{G}}$ depends on the gauge potential, can acquire zero modes, and therefore cannot be promoted to a single global Faddeev–Popov chart without additional geometric data. If $\mathcal{G}$ is linear, then

$$\mathcal{G}(A + \partial\xi) = \mathcal{G}(A) + \mathcal{G}(\partial\xi).$$

Therefore the derivative with respect to s at $s = 0$ is simply $M_{\mathcal{G}}\xi = \mathcal{G}(\partial\xi)$, independent of A. The determinant is consequently independent of the integration variable in the free abelian representative integral and factors from normalized correlators.



Example 41.5 (Finite-dimensional gauge-slice independence model)**.** A finite-dimensional model isolates the logic. Let

$$S(x, y) = (x - y)^2$$

on $\mathbb{R}^2$, with gauge orbits

$$(x, y) \longmapsto (x + \alpha, y + \alpha).$$

The ungauged Gaussian integral diverges by the orbit volume. Choose the linear slice

$$\ell(x, y) = y + ax + b = 0, \qquad a \neq -1.$$

The Faddeev–Popov factor is the absolute value of the derivative of $\ell(x + \alpha, y + \alpha)$ with respect to α,

$$\Delta_\ell = |1 + a|.$$

For a gauge-invariant observable $O(x, y) = O(x + \alpha, y + \alpha)$,

$$\frac{\int \mathrm{d}x\, \Delta_\ell\, \mathrm{e}^{-(x+ax+b)^2} O(x, -ax - b)}{\int \mathrm{d}x\, \Delta_\ell\, \mathrm{e}^{-(x+ax+b)^2}}$$

is independent of a, b. This is the finite-dimensional form of gauge-slice independence for gauge-invariant quantities in a region where the slice is regular. On the slice $y = -ax - b$, the gauge-invariant coordinate is

$$u = x - y = (1 + a)x + b.$$

The measure changes by $\mathrm{d}u = |1 + a|\mathrm{d}x$. The factor $\Delta_\ell = |1 + a|$ therefore cancels the Jacobian in both numerator and denominator. Because O is constant on gauge orbits, $O(x, -ax - b)$ depends only on u. Both normalized integrals are thus the same Gaussian average in u, independent of a, b.

41.5 Axial-Gauge Gaussian and Propagator

Axial-gauge quadratic inverse. For axial gauge,

$$\mathcal{G}_{\mathrm{ax}}(A) = A_3, \qquad M_{\mathrm{ax}} = \partial_3.$$

After removing the ∂_3 zero modes, the determinant is independent of A_μ, and the gauge-fixed Maxwell integral may be written as

$$Z_{\mathrm{ax}} = \int DA_\mu\, \delta(A_3)\, \mathrm{e}^{\mathrm{i}S[A]}.$$

Integrating the Maxwell action by parts gives

$$S[A] = \frac{1}{2}\int \mathrm{d}^4x\, A_\mu \left(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu\right) A_\nu, \qquad \Box = \partial^\rho\partial_\rho.$$

The unfixed quadratic operator has zero directions $A_\mu = \partial_\mu\xi$. On the axial slice $A_3 = 0$, the remaining block has indices $\mu, \nu = 0, 1, 2$. With

$$A_\mu(x) = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\, \mathrm{e}^{\mathrm{i}k\cdot x}\widetilde{A}_\mu(k),$$

the quadratic form is

$$S_{\mathrm{ax}} = \frac{1}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\, \widetilde{A}_\mu(-k)\, \widetilde{D}^{\mu\nu}_{\mathrm{ax}}(k)\, \widetilde{A}_\nu(k),$$

where

$$\widetilde{D}^{\mu\nu}_{\mathrm{ax}}(k) = -\eta^{\mu\nu}k^2 + k^\mu k^\nu, \qquad \mu, \nu = 0, 1, 2.$$

For $k^2 \neq 0$ and $k_3 \neq 0$, its inverse on this 3×3 block is

$$\left(\widetilde{D}_{\mathrm{ax}}(k)^{-1}\right)_{\mu\nu} = -\frac{1}{k^2}\left(\eta_{\mu\nu} + \frac{k_\mu k_\nu}{k_3^2}\right), \qquad \mu, \nu = 0, 1, 2.$$

The abelian Faddeev–Popov operator is $M_{\mathrm{ax}}\xi = \partial_3\xi$, so its determinant is field independent after the ∂_3 zero modes are removed. Integration by parts gives the displayed quadratic operator. On the axial slice $A_3 = 0$, only the $\mu, \nu = 0, 1, 2$ block remains. Multiplying the proposed inverse by the quadratic block gives

$$\begin{aligned}
&\sum_{\rho=0}^{2}\left(-\eta^{\mu\rho}k^2 + k^\mu k^\rho\right)\left[-\frac{1}{k^2}\left(\eta_{\rho\nu} + \frac{k_\rho k_\nu}{k_3^2}\right)\right] \\
&\qquad = \delta^\mu{}_\nu - \frac{k^\mu k_\nu}{k^2} + \frac{k^\mu k_\nu}{k_3^2} - \frac{k^\mu k_\nu}{k^2 k_3^2}\sum_{\rho=0}^{2} k^\rho k_\rho = \delta^\mu{}_\nu,
\end{aligned}$$

because

$$\sum_{\rho=0}^{2} k^\rho k_\rho = k^2 - k_3^2.$$

Corollary 41.6 (Axial-gauge representative propagator and field strength)**.** *The Lorentzian Feynman boundary value therefore gives, for $\mu, \nu = 0, 1, 2$,*

$$\begin{aligned}&\langle\Omega|\,\mathrm{T}\{\widehat{A}_\mu(x)\widehat{A}_\nu(y)\}|\Omega\rangle_{\mathrm{ax}}\\ &= \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\mathrm{e}^{\mathrm{i}k\cdot(x-y)}\frac{-\mathrm{i}}{k^2-\mathrm{i}\epsilon}\left(\eta_{\mu\nu}+\frac{k_\mu k_\nu}{k_3^2}\right),\end{aligned}$$

with components involving A_3 set to zero in this gauge. The displayed kernel is singular at $k_3 = 0$, precisely where the axial slice has residual gauge directions.

The field-strength two-point function is obtained by antisymmetrizing momentum factors. Away from coincident points, equivalently modulo local contact terms fixed by the definition of time ordering, the $k_\mu k_\nu$ part of the axial propagator cancels:

$$\begin{aligned}&\langle\Omega|\,\mathrm{T}\{\widehat{F}_{\mu\nu}(x)\widehat{F}_{\rho\sigma}(y)\}|\Omega\rangle\\ &= \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\mathrm{e}^{\mathrm{i}k\cdot(x-y)}\frac{-\mathrm{i}}{k^2-\mathrm{i}\epsilon}\big(k_\mu k_\rho\eta_{\nu\sigma}-k_\nu k_\rho\eta_{\mu\sigma}-k_\mu k_\sigma\eta_{\nu\rho}+k_\nu k_\sigma\eta_{\mu\rho}\big).\end{aligned}$$

Thus the local gauge-invariant correlator has a Lorentz-covariant kernel although the axial representative propagator does not.

41.6 Covariant Gauges

Definition 41.7 (Lorenz gauge and covariant gauge parameter)**.** For perturbative calculations it is often useful to preserve manifest Lorentz covariance in the representative propagator. The Lorenz gauge condition is

$$\mathcal{G}_{\mathrm{L}}(A) = \partial_\mu A^\mu.$$

Along a gauge orbit,

$$\partial_\mu(A^\mu + \partial^\mu\xi) = \partial_\mu A^\mu + \Box\xi,$$

so the abelian Faddeev–Popov operator is $M_{\mathrm{L}} = \Box$. In the free abelian theory its determinant is field independent.

Remark 41.8 (Gaussian averaging of the Lorenz gauge condition). Instead of imposing $\partial_\mu A^\mu = 0$ sharply, introduce an auxiliary function φ and average the condition

$$\partial_\mu A^\mu = \varphi$$

with a Gaussian weight. In Euclidean signature the insertion is

$$\int D\varphi\,\exp\left[-\frac{1}{2\xi_{\mathrm{g}}}\int \mathrm{d}^4x_E\,\varphi^2\right]\delta(\partial_\mu A^\mu - \varphi), \qquad \xi_{\mathrm{g}} > 0.$$

Equivalently, writing the delta-functional with a real Lagrange multiplier λ,

$$\begin{aligned}&\int D\varphi\,D\lambda\,\exp\left[-\frac{1}{2\xi_{\mathrm{g}}}\int\varphi^2 + \mathrm{i}\int\lambda(\partial_\mu A^\mu - \varphi)\right]\\ &\propto \exp\left[-\frac{1}{2\xi_{\mathrm{g}}}\int(\partial_\mu A^\mu)^2\right].\end{aligned}$$

The Euclidean gauge-fixing term is therefore

$$\Delta S_E = \frac{1}{2\xi_{\mathrm{g}}} \int \mathrm{d}^4 x_E \, (\partial_\mu A^\mu)^2.$$

The delta-functional sets $\varphi = \partial_\mu A^\mu$. Integrating first over λ implements this constraint, and the remaining Gaussian weight in φ becomes exactly the displayed quadratic term in $\partial_\mu A^\mu$. Conversely, integrating first over φ completes the square and produces the same local quadratic functional of the Lagrange multiplier; the two orders express the same finite-mode Gaussian identity before the regulator is removed.

Covariant-gauge quadratic inverse. The corresponding Lorentzian gauge-fixed action is

$$S_{\xi_{\mathrm{g}}}[A] = \int \mathrm{d}^4 x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi_{\mathrm{g}}} (\partial_\mu A^\mu)^2 \right].$$

After integration by parts,

$$S_{\xi_{\mathrm{g}}}[A] = \frac{1}{2} \int \mathrm{d}^4 x \, A_\mu \left[\eta^{\mu\nu} \Box - \left(1 - \frac{1}{\xi_{\mathrm{g}}} \right) \partial^\mu \partial^\nu \right] A_\nu.$$

In momentum space,

$$\widetilde{D}^{\mu\nu}_{\xi_{\mathrm{g}}}(k) = -\eta^{\mu\nu} k^2 + \left(1 - \frac{1}{\xi_{\mathrm{g}}} \right) k^\mu k^\nu.$$

For $\xi_{\mathrm{g}} \neq 0$, its inverse is

$$\big(\widetilde{D}^{-1}_{\xi_{\mathrm{g}}}(k) \big)_{\mu\nu} = -\frac{\eta_{\mu\nu}}{k^2} + (1 - \xi_{\mathrm{g}}) \frac{k_\mu k_\nu}{(k^2)^2}.$$

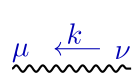
The gauge-fixed action is obtained from the Maxwell action and the Lorentzian continuation of the Euclidean term in Proposition 41.8. Integrating by parts gives the displayed quadratic operator. The inverse is verified by

$$\left[-\eta^{\mu\rho} k^2 + \left(1 - \frac{1}{\xi_{\mathrm{g}}} \right) k^\mu k^\rho \right] \left[-\frac{\eta_{\rho\nu}}{k^2} + (1 - \xi_{\mathrm{g}}) \frac{k_\rho k_\nu}{(k^2)^2} \right]$$
$$= \delta^\mu{}_\nu.$$

Corollary 41.9 (Covariant representative propagator)**.** *The Feynman two-point function of the representative field is*

$$\langle \Omega | \, \mathrm{T}\{ \widehat{A}_\mu(x) \widehat{A}_\nu(y) \} | \Omega \rangle_{\xi_{\mathrm{g}}}$$
$$= \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \mathrm{e}^{\mathrm{i}k\cdot(x-y)} \frac{-\mathrm{i}}{k^2 - \mathrm{i}\epsilon} \left[\eta_{\mu\nu} - (1 - \xi_{\mathrm{g}}) \frac{k_\mu k_\nu}{k^2 - \mathrm{i}\epsilon} \right].$$

The value $\xi_{\mathrm{g}} = 1$ is Feynman gauge:



$$= \frac{-\mathrm{i}\eta_{\mu\nu}}{k^2 - \mathrm{i}\epsilon}.$$

The value $\xi_{\mathrm{g}} \to 0$ is Landau gauge.

41.7 Gupta–Bleuler Fock Space in Lorenz Gauge

Definition 41.10 (Covariant Maxwell Krein-Fock representative)**.** The covariant representative propagator has a direct operator realization in a Krein-Fock space. The construction keeps Lorentz covariance manifest at the level of representative fields and recovers the positive physical photon space by a null quotient.

Let

$$\mathrm{d}\mu_0(k) = \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3\, 2|\vec{k}|}, \qquad k^0 = |\vec{k}|,$$

be the positive-energy massless measure. Define the one-particle covariant test space $\mathcal{K}_1^0$ to consist of smooth rapidly decreasing complex vector wavefunctions $f_\mu(k)$ on the forward light cone. It is equipped with the indefinite Hermitian form

$$[f, g]_1 = \int \mathrm{d}\mu_0(k)\, \eta^{\mu\nu}\, \overline{f_\mu(k)}\, g_\nu(k).$$

The time-like component contributes with the opposite sign because $\eta^{00} = -1$. Choosing a time-like polarization and three space-like polarizations gives a fundamental symmetry and hence a Hilbert majorant topology; the completion is a Krein space $\mathcal{K}_1$. The covariant Fock space is the symmetric Krein-Fock space

$$\mathcal{K} = \mathcal{F}_s(\mathcal{K}_1).$$

On the algebraic finite-particle domain define annihilation and creation operator-valued distributions by

$$[a_\mu(\vec{k}), a_\nu^\dagger(\vec{q})] = (2\pi)^3\, 2|\vec{k}|\, \eta_{\mu\nu}\, \delta^{(3)}(\vec{k} - \vec{q}), \qquad a_\mu(\vec{k})\Omega = 0.$$

The Lorenz-gauge representative field in Feynman gauge is

$$A_\mu(x) = \int \mathrm{d}\mu_0(k)\, \left[a_\mu(\vec{k})\mathrm{e}^{-\mathrm{i}k\cdot x} + a_\mu^\dagger(\vec{k})\mathrm{e}^{\mathrm{i}k\cdot x}\right].$$

It satisfies the free wave equation $\Box A_\mu = 0$ and has two-point function

$$\langle\Omega|\, \mathrm{T}\, A_\mu(x) A_\nu(y)|\Omega\rangle = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\, \mathrm{e}^{\mathrm{i}k\cdot(x-y)}\, \frac{-\mathrm{i}\eta_{\mu\nu}}{k^2 - \mathrm{i}0},$$

which is the $\xi_{\mathrm{g}} = 1$ propagator above. The price of this covariant representative is the indefinite form on $\mathcal{K}$.

Definition 41.11 (Gupta–Bleuler physical subspace)**.** Write

$$B(x) = \partial^\mu A_\mu(x), \qquad B^{(+)}(x) = -\mathrm{i}\int \mathrm{d}\mu_0(k)\, k^\mu a_\mu(\vec{k})\mathrm{e}^{-\mathrm{i}k\cdot x}.$$

The Gupta–Bleuler physical subspace is

$$\mathcal{K}_{\mathrm{GB}} = \{\Psi \in \mathcal{K} :\ B^{(+)}(f)\Psi = 0 \text{ for every test function } f\}.$$

Equivalently, on wavefunctions this condition is the mass-shell Lorenz condition

$$k^\mu f_\mu(k) = 0.$$

Remark 41.12 (Weak Lorenz condition). If $\Phi, \Psi \in \mathcal{K}_{\mathrm{GB}}$, then

$$\langle \Phi | B(x) | \Psi \rangle = 0.$$

Indeed $B^{(+)}(x)\Psi = 0$ by the definition of $\mathcal{K}_{\mathrm{GB}}$, while $B^{(-)}(x)$ is the Krein adjoint of $B^{(+)}(x)$, so $\langle \Phi | B^{(-)}(x) = 0$ for $\Phi \in \mathcal{K}_{\mathrm{GB}}$. Since $B = B^{(+)} + B^{(-)}$, the Lorenz condition is imposed weakly, as a condition on matrix elements between physical vectors.

Construction 41.13 (One-particle Gupta–Bleuler quotient)**.** The quotient that removes null gauge vectors can be described already at one-particle level. Let

$$\mathcal{K}_{1,\mathrm{GB}} = \{ f \in \mathcal{K}_1 : \ k^\mu f_\mu(k) = 0 \}.$$

The subspace

$$\mathcal{N}_1 = \{ f_\mu(k) = k_\mu h(k) \}$$

lies in $\mathcal{K}_{1,\mathrm{GB}}$ and is null:

$$[kh, kh']_1 = \int \mathrm{d}\mu_0(k)\, k^2\, \overline{h(k)} h'(k) = 0.$$

It is also orthogonal to every $f \in \mathcal{K}_{1,\mathrm{GB}}$, since

$$[kh, f]_1 = \int \mathrm{d}\mu_0(k)\, \overline{h(k)}\, k^\mu f_\mu(k) = 0.$$

For each null k, the quotient $k^\perp / \mathbb{C}k$ is a two-dimensional positive Hermitian space. Choosing two transverse polarization representatives $\epsilon^h_\mu(k)$, $h = \pm 1$, gives the isometry

$$[f] \longmapsto f_h(k) = \epsilon^{h\mu}(k) f_\mu(k)$$

from

$$\mathcal{H}_1^\gamma := \mathcal{K}_{1,\mathrm{GB}} / \mathcal{N}_1$$

onto the two-helicity one-photon Hilbert space

$$\bigoplus_{h=\pm 1} L^2(\mathcal{N}_+, \mathrm{d}\mu_0).$$

The positive inner product is

$$\langle [f], [g] \rangle_\gamma = \sum_{h=\pm 1} \int \mathrm{d}\mu_0(k)\, \overline{f_h(k)} g_h(k).$$

Gupta–Bleuler quotient as the two-helicity photon Fock space. With the one-particle quotient $\mathcal{H}_1^\gamma$ constructed in Construction 41.13, the many-particle physical Hilbert space is

$$\mathcal{H}_{\mathrm{GB}} = \mathcal{K}_{\mathrm{GB}} / \mathcal{N} \simeq \mathcal{F}_s(\mathcal{H}_1^\gamma),$$

where $\mathcal{N}$ is the closed subspace generated by finite-particle vectors with at least one one-particle factor in $\mathcal{N}_1$. Equivalently, on every n-particle sector,

$$\mathrm{Sym}^n(\mathcal{K}_{1,\mathrm{GB}}) / \sum_{r=1}^{n} \mathrm{Sym}^{r-1}(\mathcal{K}_{1,\mathrm{GB}}) \odot \mathcal{N}_1 \odot \mathrm{Sym}^{n-r}(\mathcal{K}_{1,\mathrm{GB}}) \cong \mathrm{Sym}^n(\mathcal{H}_1^\gamma).$$

Thus Gupta–Bleuler quantization and reduced two-helicity quantization give the same positive free-photon Hilbert space after quotienting null vectors. This is the functorial quotient calculation following the one-particle construction. Construction 41.13 identifies $\mathcal{K}_{1,\mathrm{GB}}/\mathcal{N}_1$ with the two-helicity one-photon Hilbert space. The symmetric tensor functor sends a quotient by a subspace to the quotient of each symmetric tensor power by the sum of tensors with one factor in that subspace:

$$\mathrm{Sym}^n(V/N) \cong \mathrm{Sym}^n(V)/\sum_{r=1}^{n} \mathrm{Sym}^{r-1}(V) \odot N \odot \mathrm{Sym}^{n-r}(V).$$

Taking $V = \mathcal{K}_{1,\mathrm{GB}}$ and $N = \mathcal{N}_1$ gives the displayed n-particle identification. The finite-particle direct sum therefore quotients to the symmetric Fock space over $\mathcal{H}_1^\gamma$, and completion gives $\mathcal{F}_s(\mathcal{H}_1^\gamma)$.

Proposition 41.14 (Field strengths descend to the Gupta–Bleuler quotient). *Let $u_{\mu\nu}(x)$ be a compactly supported test two-form and smear the field strength:*

$$F(u) = \int \mathrm{d}^4x\, u^{\mu\nu}(x)\big(\partial_\mu A_\nu - \partial_\nu A_\mu\big)(x).$$

Then $F(u)\Omega$ lies in $\mathcal{K}_{1,\mathrm{GB}}$, and $F(u)$ descends to a well-defined operator on $\mathcal{H}_{\mathrm{GB}} \simeq \mathcal{F}_s(\mathcal{H}_1^\gamma)$. Every free time-ordered correlator of field strengths computed in the covariant Gupta–Bleuler Fock representation agrees with the corresponding correlator computed in the reduced positive two-helicity Fock representation.

Proof. The one-particle wavefunction of $F(u)\Omega$ is proportional on the light cone to

$$f_\nu(k) = 2\mathrm{i}\, k_\mu \widetilde{u}^\mu{}_\nu(k), \qquad k^\nu f_\nu(k) = 0,$$

because $k^\mu k^\nu \widetilde{u}_{\mu\nu} = 0$. Therefore $F(u)\Omega$ lies in $\mathcal{K}_{1,\mathrm{GB}}$. If the representative field is shifted by $A_\mu \mapsto A_\mu + \partial_\mu \xi$, the field-strength smearing changes by zero; at the one-particle level this is the statement that adding $k_\mu h(k)$ changes only the null class. Consequently field-strength operators descend to well-defined operators on $\mathcal{H}_{\mathrm{GB}} \simeq \mathcal{F}_s(\mathcal{H}_1^\gamma)$.

The equality of observable correlators follows by Wick's theorem from the two-point statement. The Feynman-gauge vector propagator and the two-helicity time-ordered vector kernel differ by terms whose external vector indices are longitudinal on the light cone, together with local contact terms fixed by the time-ordering prescription. Applying the antisymmetric derivatives that form $F_{\mu\nu}$ kills the longitudinal terms. Hence every free correlator of field strengths computed in the covariant Gupta–Bleuler Fock representation agrees with the correlator computed in the reduced positive two-helicity Fock representation. The same cancellation explains why the ξ_{g}-dependent term in the covariant path-integral propagator drops out of gauge-invariant field-strength correlators. □

41.8 Field Strength and One-Photon States

Field-strength two-point function and photon helicity data. The gauge-fixed path integral defines Green functions of the representative variable A_μ. In a positive-metric physical Hilbert space the local observable obtained from the free Maxwell field is the field strength $\widehat{F}_{\mu\nu}$. The field-strength two-point function is independent of ξ_{g}, because the gauge-parameter term in the representative propagator is proportional to $k_\mu k_\nu$ and is killed by antisymmetrization.

The Wightman two-point function has the one-photon spectral form

$$\begin{aligned}&\langle\Omega|\widehat{F}_{\mu\nu}(x)\widehat{F}_{\rho\sigma}(y)|\Omega\rangle\\&=\sum_{h=\pm1}\int\frac{\mathrm{d}^3\vec{k}}{(2\pi)^3\,2|\vec{k}|}\langle\Omega|\widehat{F}_{\mu\nu}(x)|k,h\rangle\langle k,h|\widehat{F}_{\rho\sigma}(y)|\Omega\rangle,\end{aligned}$$

where the sharp states are generalized one-particle vectors in the sense of Section 5.2. With the helicity representatives of Chapter 21,

$$\langle k,h|\widehat{F}_{\mu\nu}(x)|\Omega\rangle=Z_A^{1/2}\mathrm{e}^{-\mathrm{i}k\cdot x}\left(-\mathrm{i}k_\mu\epsilon_\nu^{h*}(k)+\mathrm{i}k_\nu\epsilon_\mu^{h*}(k)\right).$$

The representative shift

$$\epsilon_\mu^h(k)\longmapsto\epsilon_\mu^h(k)+c(k)k_\mu$$

does not change this matrix element.

For a fixed null momentum k, choose a null covector $C_\mu(k)$ satisfying

$$C^2=0,\qquad k^\mu C_\mu=-1.$$

The two transverse helicity representatives may be normalized so that

$$\epsilon_\mu^h(k)\epsilon^{h'*\mu}(k)=\delta_{hh'},\qquad k^\mu\epsilon_\mu^h(k)=0,$$

and obey the completeness relation

$$\sum_{h=\pm1}\epsilon_\mu^h(k)\epsilon_\nu^{h*}(k)=\eta_{\mu\nu}+k_\mu C_\nu+k_\nu C_\mu.$$

For example, when $k^\mu=(|\vec{k}|,\vec{k})$, one may take

$$C^\mu(k)=\frac{1}{2|\vec{k}|^2}\big(|\vec{k}|,-\vec{k}\big).$$

In the field-strength spectral sum all terms involving C_μ cancel by antisymmetrization. Comparison with the explicit Wightman boundary value of the free field-strength correlator fixes the free normalization

$$Z_A=1.$$

The verification is explicit. The ξ_{g}-dependent term of the representative propagator is proportional to $k_\alpha k_\beta$. Applying the two field-strength antisymmetrizers gives the factor

$$(k_\mu\delta_\nu{}^\alpha-k_\nu\delta_\mu{}^\alpha)k_\alpha k_\beta(k_\rho\delta_\sigma{}^\beta-k_\sigma\delta_\rho{}^\beta),$$

which vanishes because $k_\mu k_\nu-k_\nu k_\mu=0$. Thus the gauge-parameter part of the representative two-point function is invisible to $F_{\mu\nu}$.

The spectral representation is the one-particle part of the Wightman two-point function for the positive photon Hilbert space constructed above by quotienting the Gupta–Bleuler space; sharp $|k,h\rangle$ are used as generalized vectors in the rigged-Hilbert-space sense. The matrix element of the field strength is obtained by applying $\partial_\mu A_\nu-\partial_\nu A_\mu$ to a one-photon vector-potential matrix element. The replacement $\epsilon^h_\mu \mapsto \epsilon^h_\mu + c\,k_\mu$ changes the parent vector potential representative, but it contributes

$$-\mathrm{i}c\,k_\mu k_\nu + \mathrm{i}c\,k_\nu k_\mu = 0$$

to the field-strength matrix element.

For the completeness relation, the tensor $\eta_{\mu\nu} + k_\mu C_\nu + k_\nu C_\mu$ annihilates k^ν, has zero components along the null complement modulo the line $\mathbb{C}k$, and restricts to the positive Euclidean metric on the two-dimensional transverse quotient $k^\perp/\mathbb{C}k$. Hence it is the sum of the two normalized helicity projectors. Substituting this completeness relation into the field-strength spectral sum, every term containing C has two symmetric factors of k entering one of the antisymmetrizers, so those terms vanish. The remaining $\eta_{\mu\nu}$ term is exactly the explicit free field-strength Wightman kernel obtained from the Feynman-gauge representative propagator. With the normalization of the photon creation and annihilation operators in Definition 41.10, this comparison gives $Z_A = 1$.

The next chapter couples this representative gauge field to a charged spinor field. The perturbative Green functions will be written using gauge-fixed representatives, while local gauge-invariant operators and external photon states are read through the field strength and helicity data constructed here.

Chapter 42

Quantum Electrodynamics and External States

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

42.1 Charged Spinors and Abelian Gauge Representatives

Definition 42.1 (QED representative datum)**.** A local representative datum for four-dimensional QED consists of a real Abelian vector-potential representative $A_\mu(x)$, a Dirac spinor representative $\psi_\alpha(x)$, its mostly-plus Dirac adjoint $\bar{\psi}^\alpha(x) = \psi^\dagger \beta$, and a real charge coordinate $g \in \mathbb{R}$. For the electron convention used below,

$$g = -e, \qquad e > 0.$$

The representative fields are therefore

$$A_\mu(x), \qquad \psi_\alpha(x), \qquad \bar{\psi}^\alpha(x),$$

with $\bar{\psi}$ defined as in Chapter 40. A real function $\xi(x)$ changes representatives by

$$A_\mu \mapsto A_\mu + \partial_\mu \xi, \qquad \psi \mapsto \mathrm{e}^{\mathrm{i}g\xi}\psi, \qquad \bar{\psi} \mapsto \bar{\psi}\,\mathrm{e}^{-\mathrm{i}g\xi}.$$

The Abelian covariant derivative is

$$D_\mu \psi = \partial_\mu \psi - \mathrm{i}g A_\mu \psi.$$

Covariance of the Abelian Dirac coupling. For the representative change of Definition 42.1, the covariant derivative transforms as

$$D_\mu(\mathrm{e}^{\mathrm{i}g\xi}\psi) = \mathrm{e}^{\mathrm{i}g\xi}D_\mu\psi.$$

Consequently the local action

$$S_{\mathrm{QED}}[A,\bar{\psi},\psi] = \int \mathrm{d}^4x \left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu D_\mu + m)\psi\right]$$

is invariant under compactly supported changes of Abelian representative, where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad \not{a} = a_\mu\gamma^\mu.$$

In the monograph's mostly-plus gamma convention the current represented by the coupling to A_μ is

$$j^\mu = \mathrm{i}g\,\bar{\psi}\gamma^\mu\psi.$$

It transforms with the same charge as ψ:

$$\begin{aligned} D_\mu(\mathrm{e}^{\mathrm{i}g\xi}\psi) &= \partial_\mu(\mathrm{e}^{\mathrm{i}g\xi}\psi) - \mathrm{i}g(A_\mu + \partial_\mu\xi)\mathrm{e}^{\mathrm{i}g\xi}\psi \\ &= \mathrm{e}^{\mathrm{i}g\xi}D_\mu\psi. \end{aligned}$$

The curvature $F_{\mu\nu}$ is unchanged because $\partial_\mu\partial_\nu\xi = \partial_\nu\partial_\mu\xi$. The Dirac term is invariant because the phase of $D_\mu\psi$ cancels the opposite phase of $\bar{\psi}$. Expanding the covariant derivative gives

$$\mathcal{L}_{\mathrm{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\not{\partial} + m)\psi + \mathrm{i}gA_\mu\bar{\psi}\gamma^\mu\psi.$$

The last term is the interaction. In these gamma-matrix conventions the electric current represented by the coupling to A_μ is

$$j^\mu = \mathrm{i}g\,\bar{\psi}\gamma^\mu\psi.$$

The leading factor of i is fixed by the mostly-plus Dirac adjoint $\bar{\psi} = \psi^\dagger\beta$, $\beta = \mathrm{i}\gamma^0$, together with $D_\mu = \partial_\mu - \mathrm{i}gA_\mu$. It is invariant under the local Abelian representative change because the endpoint phases of $\bar{\psi}$ and ψ cancel.

42.2 Local Observables and Charged Insertions

Definition 42.2 (Local Abelian charge of a representative monomial)**.** A local observable is represented by a local expression in the fields that is invariant under all compactly supported changes of gauge representative. Examples are

$$F_{\mu\nu}, \qquad \bar{\psi}\psi, \qquad \mathrm{i}g\,\bar{\psi}\gamma^\mu\psi.$$

A local monomial containing N_ψ factors of ψ and $N_{\bar{\psi}}$ factors of $\bar{\psi}$ has local charge

$$g(N_\psi - N_{\bar{\psi}}).$$

Remark 42.3 (Neutrality criterion for pointlike local monomials). Gauge invariance of a pointlike local monomial therefore requires the net local charge in Definition 42.2 to vanish. Under a compactly supported representative change the monomial is multiplied by

$$\exp\bigl(\mathrm{i}g(N_\psi - N_{\bar\psi})\xi(x)\bigr)$$

at the point of insertion. Since $\xi(x)$ can be chosen arbitrarily while remaining compactly supported, invariance for all representative changes forces $g(N_\psi - N_{\bar\psi}) = 0$. For nonzero charge coordinate g this is precisely $N_\psi = N_{\bar\psi}$.

Definition 42.4 (Wilson-line dressed charged insertion)**.** Charged insertions are represented by adding dressing data. Let C_x be an oriented curve beginning at x and ending at a prescribed asymptotic endpoint. For gauge parameters that vanish at that endpoint, set

$$\Psi_{C_x}(x) = \exp\left(\mathrm{i}g\int_{C_x} A_\mu(y)\,\mathrm{d}y^\mu\right)\psi(x).$$

Similarly, for a path $C_{y\to x}$ from y to x, set

$$\mathcal{O}_\alpha{}^\beta(x,y;C_{y\to x}) = \psi_\alpha(x)\exp\left(-\mathrm{i}g\int_{C_{y\to x}} A_\mu(z)\,\mathrm{d}z^\mu\right)\bar\psi^\beta(y).$$

The curve is part of the insertion datum; these are not pointlike local observables.

Remark 42.5 (Endpoint cancellation for Abelian Wilson dressings). The dressed operators in Definition 42.4 are invariant under compactly supported Abelian representative changes compatible with the stated endpoint convention. For the half-infinite dressing,

$$\int_{C_x}\partial_\mu\xi(y)\,\mathrm{d}y^\mu = \xi(\infty)-\xi(x) = -\xi(x),$$

so the phase acquired by the line integral cancels the phase of $\psi(x)$. For the finite bilocal dressing the Wilson-line phase changes by

$$\exp\{-\mathrm{i}g(\xi(x)-\xi(y))\}.$$

This cancels the endpoint phases of $\psi_\alpha(x)$ and $\bar\psi^\beta(y)$.

The line operator itself may require short-distance renormalization when it is defined as an operator-valued distribution. That renormalization question is separate from the finite gauge-covariance calculation displayed here.

Remark 42.6. Gauge-fixed perturbation theory uses $A_\mu, \psi, \bar\psi$ Green functions as representative correlators. After gauge fixing these correlators have propagators and vertices. Gauge-invariant local observables and physical external photon states are read from $F_{\mu\nu}$, helicity representatives, and LSZ pole data.

42.3 Gauge-Fixed Generating Functional

For perturbation theory choose the covariant gauge parameter ξ_g and add

$$\Delta\mathcal{L} = -\frac{1}{2\xi_\mathrm{g}}(\partial_\mu A^\mu)^2.$$

The Abelian Faddeev–Popov determinant for the condition $\partial_\mu A^\mu = \varphi$ is $\det\Box$, independent of A_μ; it is absorbed into the normalization of the generating functional.

Abelian Faddeev–Popov specialization. Let

$$F_\varphi[A](x) = \partial_\mu A^\mu(x) - \varphi(x)$$

be the covariant gauge condition, with φ an auxiliary scalar function in the same regulator and boundary-condition class as $\partial_\mu A^\mu$. Along the Abelian gauge orbit

$$A_\mu^\xi = A_\mu + \partial_\mu \xi, \qquad \psi^\xi = \mathrm{e}^{\mathrm{i}g\xi}\psi, \qquad \bar\psi^\xi = \bar\psi\,\mathrm{e}^{-\mathrm{i}g\xi},$$

the linearized orbit-slice map is

$$\mathcal{M}_{\mathrm{QED}}(x,y) = \left.\frac{\delta F_\varphi[A^\xi](x)}{\delta\xi(y)}\right|_{\xi=0} = \Box_x \delta^{(4)}(x-y). \tag{42.1}$$

After residual zero modes have been removed or separately fixed,

$$\Delta_{\mathrm{FP}}^{U(1)} = \det{}'\Box.$$

This determinant is independent of $A_\mu, \psi, \bar\psi$. Hence it cancels from normalized representative Green functions and may be absorbed into the normalization of Z. If it is represented by anti-commuting ghosts, the ghost action is the free decoupled quadratic action

$$S_{\mathrm{gh}}^{U(1)} = \int \mathrm{d}^4x\, \bar c\,\Box c,$$

so there is no QED ghost vertex in the perturbative rules below.

The gauge condition depends only on the vector-potential representative, so the matter-field phases do not enter the Faddeev–Popov operator. Directly,

$$F_\varphi[A^\xi](x) = \partial_\mu A^\mu(x) + \Box\xi(x) - \varphi(x).$$

Taking the functional derivative with respect to $\xi(y)$ gives (42.1). Since the result contains no field, the Jacobian is a constant determinant on the regular gauge-parameter subspace. The prime on $\det{}'\Box$ records the same zero-mode separation already needed for the gauge slice itself: constant gauge parameters, or any residual modes allowed by the chosen boundary conditions, do not change $\partial_\mu A^\mu$. For normalized correlators the same determinant multiplies numerator and denominator. The Grassmann representation follows from the finite-dimensional Berezin identity $\int D\bar c\, Dc\, \exp(\mathrm{i}\bar c M c) \propto \det M$, applied to $M = \Box$.

This is the Abelian, linear-gauge specialization of the general local orbit-measure construction in Proposition 47.2 and (47.11). In a nonabelian gauge theory the orbit map is instead

$$\delta_\zeta A_\mu^a = (D_\mu(A)\zeta)^a, \qquad \mathcal{M}^{ab}(A;x,y) = \partial_x^\mu D_\mu^{ab}(A;x)\,\delta^{(4)}(x-y)$$

for Lorenz gauge. The determinant then depends on the gauge representative and its ghost representation produces interacting ghost vertices. Moreover, outside a sufficiently small regular chart the Faddeev–Popov operator can develop zero modes and a single gauge condition can meet an orbit more than once. Those issues are not QED Feynman-rule details; they belong to the nonabelian local-coordinate and BRST analysis of Chapter 47, especially Quoted Theorem 47.5.

Definition 42.7 (Regulated gauge-fixed QED source functional)**.** Introduce an ordinary source J^μ for A_μ, and Grassmann sources $\eta_\alpha, \bar{\eta}^\alpha$ for $\bar{\psi}^\alpha, \psi_\alpha$. The regulated Lorentzian generating functional, written in continuum shorthand, is

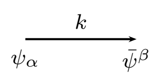
$$Z[J,\bar{\eta},\eta] = \int DA\, D\bar{\psi}\, D\psi\, \exp\Big\{ \mathrm{i} \int \mathrm{d}^4x \left(\mathcal{L}_{\mathrm{QED}} + \Delta\mathcal{L}\right) + \mathrm{i} \int \mathrm{d}^4x \left(J^\mu A_\mu + \bar{\eta}\psi + \bar{\psi}\eta\right) \Big\}.$$

As in the preceding finite-dimensional Berezin discussion, this notation is shorthand for a regulated Gaussian integral followed by perturbative expansion of the interaction term

$$\mathrm{i}g A_\mu \bar{\psi}\gamma^\mu\psi$$

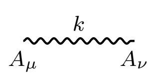
around the Gaussian Maxwell and Dirac functionals already constructed. The covariant gauge parameter ξ_{g} changes representative correlators; the local gauge-invariant correlators obtained from neutral operators and field strengths do not depend on this representative choice.

Representative QED Feynman rules. The coefficientwise expansion of Definition 42.7 gives the following momentum-space representative rules. A directed fermion line from ψ_α to $\bar{\psi}^\beta$ carries



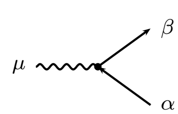
$$= \frac{-\mathrm{i}(-\mathrm{i}\,\not{k} + m)_\alpha{}^\beta}{k^2 + m^2 - \mathrm{i}\epsilon}.$$

The photon representative propagator is



$$= \frac{-\mathrm{i}}{k^2 - \mathrm{i}\epsilon}\left[\eta_{\mu\nu} - (1-\xi_{\mathrm{g}})\frac{k_\mu k_\nu}{k^2 - \mathrm{i}\epsilon}\right].$$

The cubic vertex with one photon representative and two spinor representatives is



$$= -g(\gamma^\mu)_\beta{}^\alpha.$$

For the electron convention $g = -e$, the vertex factor is $e(\gamma^\mu)_\beta{}^\alpha$. The gauge parameter ξ_{g} appears only in the representative photon propagator. Amplitudes between physical photon helicity states are obtained after LSZ projection and are independent of the longitudinal representative.

The fermion propagator is the inverse of the free Dirac kernel in the conventions of Chapter 40; the displayed numerator is the adjugate of $\not{k} + \mathrm{i}m$ in the mostly-plus spinor convention, with the Feynman boundary value fixed by $-\mathrm{i}\epsilon$. The photon representative propagator is the inverse of the quadratic Maxwell operator after adding $-(\partial_\mu A^\mu)^2/(2\xi_{\mathrm{g}})$, as derived in Chapter 41. The Abelian Faddeev–Popov determinant is field independent by the Abelian specialization above, so no ghost vertex occurs. Finally, differentiating the interaction $\mathrm{i}\int \mathrm{i}gA_\mu\bar{\psi}\gamma^\mu\psi$ with respect to one photon source and the two Grassmann sources gives the cubic representative vertex $-g\gamma^\mu$, with the displayed index placement inherited from the ordered source convention in Definition 42.7.

42.4 External Factors as Pole Data

Definition 42.8 (QED external pole-residue factors)**.** External factors in a diagrammatic amplitude are residues of poles already identified by LSZ. The particle definitions have already been supplied by the one-particle spectral data and scattering construction. With the normalizations of Chapters 18, 40, and 41, the factors are

external state	LSZ factor
$e^-_{\mathrm{out}}(\vec{p},\sigma)$	$\dfrac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\,(\bar{u}^\sigma)^\alpha(\vec{p})$
$e^+_{\mathrm{out}}(\vec{p},\sigma)$	$\dfrac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\,v^\sigma_\alpha(\vec{p})$
$\gamma_{\mathrm{out}}(\vec{k},h)$	$\dfrac{Z_A^{1/2}}{(2\pi)^{3/2}\sqrt{2\lvert\vec{k}\rvert}}\,e^{h*}_\mu(\vec{k})$
$e^-_{\mathrm{in}}(\vec{p},\sigma)$	$\dfrac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\,u^\sigma_\alpha(\vec{p})$
$e^+_{\mathrm{in}}(\vec{p},\sigma)$	$\dfrac{Z_\psi^{1/2}}{(2\pi)^{3/2}}\,(\bar{v}^\sigma)^\alpha(\vec{p})$
$\gamma_{\mathrm{in}}(\vec{k},h)$	$\dfrac{Z_A^{1/2}}{(2\pi)^{3/2}\sqrt{2\lvert\vec{k}\rvert}}\,e^{h}_\mu(\vec{k}).$

Remark 42.9 (Status of the charged entries)*.* The electron and positron rows are literal LSZ residues only in a sector whose charged spectrum contains an isolated massive one-particle shell. This is the case for an infrared-deformed theory, for instance a theory with a small gauge-invariant photon mass regulator, and it is the pole bookkeeping used in the hard part of perturbation theory. It is not an exact statement about massless four-dimensional QED after the infrared regulator is removed. Gauss' law ties nonzero electric charge to an electromagnetic field at spatial infinity, while the massless photon allows arbitrarily soft excitations with the same total charge and arbitrarily nearby energy-momentum. The charged threshold is therefore not separated from a continuum of charge plus soft photon states, and the hypothesis needed by ordinary LSZ–a delta atom or isolated pole associated with a normalizable one-electron Wigner subspace–is absent. The spinor factors below should consequently be read as hard-kernel data whose physical completion is supplied by the inclusive and dressed constructions of Chapter 44.

Proposition 42.10 (Residue origin of the external factors)**.** *Assume the spinor and photon representative two-point functions have the isolated one-particle pole data specified in Chapters 18, 40, and 41. Then the factors in Definition 42.8 are exactly the coefficients obtained by applying the LSZ boundary-value operation to the corresponding amputated kernel. A change*

$$e^h_\mu(\vec{k}) \mapsto e^h_\mu(\vec{k}) + \alpha k_\mu$$

does not change the physical photon state; it changes only the representative used before the final helicity quotient.

Proof. The spinor entries use the normalization derived in (40.8): the $2\omega_{\vec{p}}$ factor in the covariant one-particle measure is paired with the $1/(2\omega_{\vec{p}})$ in the spinor completeness relation, leaving the Fourier factor $(2\pi)^{-3/2}$ and the pole residue $Z_\psi^{1/2}$. The free spinor or photon line adjacent to the external leg is not retained: the LSZ operator removes that propagator and evaluates the residue at the one-particle pole. The displayed spinor index is contracted with the corresponding open index of the amputated Green-function kernel. In a Fourier convention in which all momenta flow into a Green function, outgoing massive particles are represented by the boundary value with the opposite energy sign; the table above is written directly in terms of the physical future-directed external momenta.

The massive spinor momenta satisfy

$$p^0 = \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2},$$

and the photon momentum satisfies

$$k^0 = |\vec{k}|, \qquad k^2 = 0.$$

The spinors obey

$$(\not{p} - \mathrm{i}m)u^\sigma(\vec{p}) = 0, \qquad (\not{p} + \mathrm{i}m)v^\sigma(\vec{p}) = 0,$$

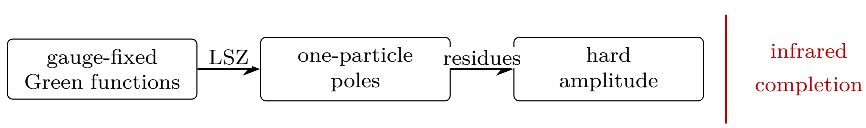
with the corresponding adjoint equations for $\bar{u}, \bar{v}$. The photon polarization representatives obey

$$k^\mu e^h_\mu(\vec{k}) = 0, \qquad e^h_\mu(\vec{k}) \sim e^h_\mu(\vec{k}) + \alpha k_\mu.$$

For any polarization representative,

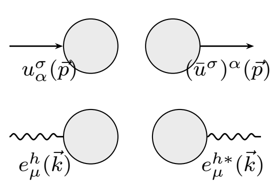
$$\not{e}^{\,h}(\vec{k}) = e^h_\mu(\vec{k})\gamma^\mu.$$

The LSZ operation removes the adjacent free propagator and leaves precisely the displayed residue vector or covector. For the photon, the quotient by longitudinal representatives is the helicity quotient constructed in Chapter 21; replacing e^h_μ by $e^h_\mu + \alpha k_\mu$ changes the representative but not the resulting one-photon helicity state. □



The vertical mark in the diagram indicates a boundary of this chapter. The hard-amplitude rules are perturbative rules for fixed external momenta and helicities. The infrared completion of charged scattering is a further construction.

Diagrammatically, the external factors attach to amputated kernels as follows:



with the common normalization factors displayed in the table.

42.5 Compton Scattering at Tree Level

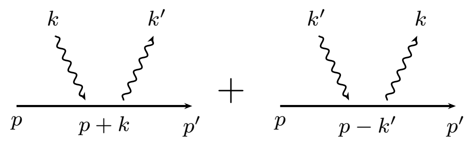
Tree-level Compton hard kernel. Consider the process

$$e^-(p,\sigma) + \gamma(k,h) \longrightarrow e^-(p',\sigma') + \gamma(k',h')$$

with

$$p + k = p' + k'.$$

At tree level two diagrams contribute:



After removing the factor

$$(2\pi)^4 \delta^{(4)}(p + k - p' - k')$$

from the scattering matrix element, the reduced amplitude is

$$\mathrm{i}\mathcal{M} = \frac{-\mathrm{i}e^2}{(2\pi)^6\sqrt{2|\vec{k}|}\sqrt{2|\vec{k}'|}}\,\bar{u}^{\sigma'}(\vec{p}')\Bigg[\not{\epsilon}^{\,h'*}(\vec{k}')\,\frac{-\mathrm{i}(\not{p}+\not{k})+m}{(p+k)^2+m^2-\mathrm{i}\epsilon}\;\not{\epsilon}^{\,h}(\vec{k})$$
$$+\;\not{\epsilon}^{\,h}(\vec{k})\,\frac{-\mathrm{i}(\not{p}-\not{k}')+m}{(p-k')^2+m^2-\mathrm{i}\epsilon}\;\not{\epsilon}^{\,h'*}(\vec{k}')\Bigg]u^{\sigma}(\vec{p}).$$

At tree level the source expansion of the representative QED Feynman rules above has exactly two connected diagrams with one internal electron line: the channel with internal momentum $p + k$ and the channel with internal momentum $p - k'$. The photon external factors are the helicity representatives in Definition 42.8; the adjacent photon propagators have already been removed by LSZ. The two cubic vertices contribute the factor e^2 in the electron convention $g = -e$, while the single internal spinor propagator contributes the displayed numerator and denominator in each channel. Removing the overall $(2\pi)^4\delta^{(4)}(p + k - p' - k')$ gives the stated reduced kernel.

In an infrared-deformed theory with ordinary charged one-particle poles, this is an LSZ-reduced expression: the external spinors and polarization vectors are the pole residues listed above. In massless QED it is the Born hard kernel. It becomes part of a physical charged-scattering prediction only after the infrared completion has been specified. At higher orders, virtual soft photons and real photons below detector resolution cannot be separated into an exact fixed-photon-number electron S-matrix element; they are combined in the inclusive or dressed constructions developed in Chapter 44.

Lemma 42.11 (Tree-level Ward identity for the Compton hard kernel)**.** *The tree-level Compton hard kernel displayed above descends from photon polarization representatives to photon helicity states. Equivalently, replacing either external photon polarization representative by that representative plus a multiple of its null momentum leaves the sum of the two tree diagrams unchanged.*

Proof. The expression descends from photon representatives to photon helicity states. Let

$$N(q) = -\mathrm{i}\,\not{q} + m, \qquad D(q) = q^2 + m^2 - \mathrm{i}\epsilon.$$

If the incoming polarization is shifted by

$$e^h_\mu(k) \mapsto e^h_\mu(k) + \alpha k_\mu,$$

then the first diagram changes by a factor proportional to

$$\frac{N(p+k)\ \not{k}}{D(p+k)} u^\sigma(p).$$

Using $p^2 = -m^2$, $k^2 = 0$, and $(\not{p} - \mathrm{i}m)u^\sigma(p) = 0$, one obtains

$$N(p+k)\ \not{k}\, u^\sigma(p) = -2\mathrm{i}(p \cdot k)u^\sigma(p), \qquad D(p+k) = 2p \cdot k - \mathrm{i}\epsilon.$$

Thus the first diagram contributes

$$-\mathrm{i}\alpha\, \bar{u}^{\sigma'}(p')\ \not{e}^{h'*}(k')u^\sigma(p)$$

up to the common prefactor. For the second diagram, momentum conservation gives $p-k' = p'-k$, and the adjoint Dirac equation gives

$$\bar{u}^{\sigma'}(p')\ \not{k}\, N(p-k') = -2\mathrm{i}(p' \cdot k)\bar{u}^{\sigma'}(p'), \qquad D(p-k') = -2p' \cdot k - \mathrm{i}\epsilon.$$

The second diagram therefore contributes

$$+\mathrm{i}\alpha\, \bar{u}^{\sigma'}(p')\ \not{e}^{h'*}(k')u^\sigma(p)$$

with the same common prefactor. The two incoming-polarization shifts cancel.

Now shift the outgoing representative by

$$e^{h'*}_\mu(k') \mapsto e^{h'*}_\mu(k') + \beta k'_\mu.$$

In the first diagram, $p + k = p' + k'$, and

$$\bar{u}^{\sigma'}(p')\ \not{k}'\, N(p+k) = -2\mathrm{i}(p' \cdot k')\bar{u}^{\sigma'}(p'), \qquad D(p+k) = 2p' \cdot k' - \mathrm{i}\epsilon.$$

It contributes

$$-\mathrm{i}\beta\, \bar{u}^{\sigma'}(p')\ \not{e}^{h}(k)u^\sigma(p)$$

up to the common prefactor. In the second diagram,

$$N(p-k')\ \not{k}'\, u^\sigma(p) = -2\mathrm{i}(p \cdot k')u^\sigma(p), \qquad D(p-k') = -2p \cdot k' - \mathrm{i}\epsilon,$$

so the contribution is

$$+\mathrm{i}\beta\, \bar{u}^{\sigma'}(p')\ \not{e}^{h}(k)u^\sigma(p).$$

The outgoing-polarization shifts cancel as well. This cancellation is the tree-level Ward identity for the Compton amplitude. □

With the cross-section normalization of Chapter 19, the total cross section for fixed incoming spin and helicity is

$$\sigma = \frac{(2\pi)^6}{v} \int \mathrm{d}^3\vec{k}'\, \mathrm{d}^3\vec{p}'\, (2\pi)^4 \delta^{(4)}(k + p - k' - p') \\ \times \sum_{\substack{h'=\pm 1 \\ \sigma'=1,2}} \left| \mathcal{M}(\vec{p}', \sigma'; \vec{k}', h' \mid \vec{p}, \sigma; \vec{k}, h) \right|^2,$$

where v is the incoming relative velocity. The spin and polarization sums are obtained from the completeness relations of Chapters 40 and 21. The reduced Compton kernel $\mathcal{M}$ is the connected on-shell matrix element after removal of the total momentum-conservation delta distribution. In the sharp-kernel event-probability formula, the final electron and photon momenta are integrated with the normalization used in the external-state chapters, while the remaining $(2\pi)^4\delta^{(4)}(k + p - k' - p')$ imposes energy-momentum conservation on the event. Summing over h' and σ' implements the inclusive measurement over final physical helicity and spin labels. The incoming spin and helicity are specified rather than averaged, so no factor averaging over initial labels appears.

42.6 Infrared Boundary of Charged Scattering

Remark 42.12 (Infrared boundary of charged QED scattering). The perturbative rules above are rules for hard Green-function residues. They are valid order by order once an infrared regulator or an inclusive prescription has been specified. The exact long-range theory has an additional constraint: Gauss' law ties electric charge to an electromagnetic field extending to arbitrarily large distances. Consequently the charged-sector asymptotic data is infrared-sensitive: its spectral support is tied to the charged mass threshold together with arbitrarily soft radiation. This lies outside the isolated massive-particle hypothesis used in the basic Haag–Ruelle construction.

This boundary is already visible in the charged dressing $\Psi_{C_x}(x)$. A dressing that retains nonzero electric charge extends to infinity. In scattering, the same long-range field is represented perturbatively by arbitrarily soft photons.

For a gauge parameter that approaches a nonzero constant on a large sphere, Gauss' law relates the total electric charge inside the sphere to the asymptotic electric flux. Thus a charged vector cannot be created from the vacuum by a compactly supported gauge-invariant operator, and a charged insertion must carry dressing data extending to the boundary as in Definition 42.4. Because the photon is massless, adding arbitrarily low-energy photons to a charged hard configuration changes the state while changing the total energy-momentum by an arbitrarily small amount. The charged mass threshold is therefore not an isolated one-particle mass shell separated by a gap from the rest of the charged spectrum. This is precisely the hypothesis ordinary massive Haag–Ruelle and the pole form of LSZ require, so the representative hard kernel must be completed by an infrared-aware construction.

The physical completions are inclusive transition probabilities, where soft photons below a resolution threshold are summed over, or dressed asymptotic charged states. Those constructions belong after renormalization and the analysis of infrared singularities. The present chapter supplies the gauge-fixed hard-amplitude rules on which that later analysis acts.

Chapter 43

QED Renormalization and Electromagnetic Form Factors

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

43.1 Regulated Variables and Ward Organization

Definition 43.1 (Regulated explicit-coupling QED coordinates)**.** Use the electron convention of Chapter 42. At a chosen perturbative regulator, use regulator-dependent representative fields and parameters

$$A^B_\mu, \qquad \psi^B, \qquad \bar{\psi}^B, \qquad m^B, \qquad e^B,$$

where the superscript B labels this cutoff-dependent coordinate system. It does not denote a second Lagrangian separate from the regulated path integral. The regulated QED Lagrangian in these coordinates is

$$\mathcal{L}_B = -\frac{1}{4} F^B_{\mu\nu} F^{\mu\nu}_B - \bar{\psi}^B \left[\gamma^\mu(\partial_\mu + \mathrm{i}e^B A^B_\mu) + m^B\right] \psi^B.$$

This is the explicit-coupling QED normalization: the photon representative has a canonical kinetic term, and the electric charge appears in the covariant derivative. It differs from the coupling-absorbed Yang–Mills normalization of Chapter 45. In the notation of (45.6), the electron convention $g = -e$ gives $A^{\mathrm{YM}}_\mu = -eA^{\mathrm{QED}}_\mu$ and $F^{\mathrm{YM}}_{\mu\nu} = -eF^{\mathrm{QED}}_{\mu\nu}$. Therefore the photon renormalization constant Z_A below is a field-residue constant in explicit-coupling coordinates; after the rescaling to Yang–Mills coordinates the same running is read from the coefficient of $(F^{\mathrm{YM}}_{\mu\nu})^2/e^2$. Introduce renormalized representative fields by

$$A^B_\mu = Z_A^{1/2} A_\mu, \qquad \psi^B = Z_\psi^{1/2}\psi, \qquad \bar\psi^B = Z_\psi^{1/2}\bar\psi.$$

The constants Z_A and Z_ψ are fixed by pole-residue normalizations in the infrared-regulated hard theory: Z_A normalizes the one-photon component of $\widehat{F}_{\mu\nu}|\Omega\rangle$, and Z_ψ normalizes the one-electron and one-positron spinor poles before the long-distance regulator is removed. In exact massless QED, Z_ψ is not a physical residue of an isolated electron Wigner particle; its infrared dependence is part of the hard-kernel bookkeeping that is combined with soft radiation in Chapter 44. Write

$$m^B = m + \delta m.$$

The renormalized charge parameter e is fixed by the electromagnetic current matrix element at zero momentum transfer. Gauge covariance of the renormalized representative derivative requires

$$e = e^B Z_A^{1/2},$$

so that

$$\partial_\mu + \mathrm{i}e^B A^B_\mu = \partial_\mu + \mathrm{i}eA_\mu.$$

Equivalently, if a separate vertex constant Z_1 is introduced by

$$e^B Z_A^{1/2} Z_\psi = eZ_1,$$

then Abelian gauge covariance gives the Ward organization

$$Z_1 = Z_\psi.$$

The same statement can be read from the renormalized representative gauge transformation,

$$A_\mu \mapsto A_\mu + \partial_\mu\xi, \qquad \psi \mapsto \mathrm{e}^{-\mathrm{i}e\xi}\psi, \qquad \bar\psi \mapsto \bar\psi\,\mathrm{e}^{\mathrm{i}e\xi}.$$

The coefficient e, not eZ_ψ, is the charge appearing in this transformation law. The common factor Z_ψ multiplies the whole spinor covariant derivative and therefore belongs to the spinor residue normalization. Thus

$$\mathcal{L}_B = -\frac{1}{4}Z_A F_{\mu\nu}F^{\mu\nu} - Z_\psi\bar\psi\left[\gamma^\mu(\partial_\mu + \mathrm{i}eA_\mu) + m + \delta m\right]\psi.$$

Equivalently, write the same regulated Lagrangian in the finite field and charge coordinates as a finite reference part plus regulator-dependent local coordinates:

$$\begin{aligned}\mathcal{L}_B = {}&\underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar\psi(\gamma^\mu\partial_\mu + m)\psi - \mathrm{i}eA_\mu\bar\psi\gamma^\mu\psi}_{\mathcal{L}_{\mathrm{ren}}}\\ &- \frac{1}{4}(Z_A - 1)F_{\mu\nu}F^{\mu\nu} - (Z_\psi - 1)\bar\psi(\gamma^\mu\partial_\mu + m)\psi - Z_\psi\delta m\,\bar\psi\psi\\ &- \mathrm{i}e(Z_\psi - 1)A_\mu\bar\psi\gamma^\mu\psi.\end{aligned}$$

The last line is the vertex counterterm tied to the spinor wavefunction counterterm by the Abelian Ward identity. The mass counterterm is independent because $m\bar\psi\psi$ is gauge-invariant. These counterterm vertices are not added to a separate bare theory; they are the regulator-dependent part of the single local Lagrangian $\mathcal{L}_B$ after it has been expressed in the renormalized coordinates $A_\mu, \psi, \bar\psi, m, e$.

Ward organization of the renormalized current. In the coordinate system of Definition 43.1, the coefficient e in the renormalized gauge transformation is the same coefficient fixed by the zero-momentum electromagnetic current matrix element. If a vertex constant Z_1 is introduced by $e^B Z_A^{1/2} Z_\psi = eZ_1$, Abelian gauge covariance gives $Z_1 = Z_\psi$, and the current normalized by the renormalized field strength is

$$\partial_\nu F^{\mu\nu} = -\mathrm{i} Z_A^{-1} e Z_\psi \,\bar{\psi}\gamma^\mu\psi \equiv j^\mu.$$

The equality $e = e^B Z_A^{1/2}$ is the condition that the covariant derivative written in regulator fields, $\partial_\mu + \mathrm{i}e^B A_\mu^B$, becomes $\partial_\mu + \mathrm{i}eA_\mu$ after the field-residue change $A_\mu^B = Z_A^{1/2} A_\mu$. Therefore the renormalized representative transformation $A_\mu \mapsto A_\mu + \partial_\mu\xi$, $\psi \mapsto \exp(-\mathrm{i}e\xi)\psi$ fixes e before the spinor residue normalization is chosen. Substituting $e = e^B Z_A^{1/2}$ into $e^B Z_A^{1/2} Z_\psi = eZ_1$ gives $Z_1 = Z_\psi$. Finally vary $\mathcal{L}_B$, expressed in the finite fields of Definition 43.1, with respect to A_μ. The Maxwell term contributes $Z_A \partial_\nu F^{\mu\nu}$ and the interaction term contributes $\mathrm{i}eZ_\psi\bar{\psi}\gamma^\mu\psi$, which gives the displayed operator identity after division by Z_A.

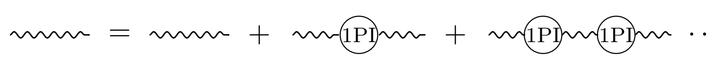
This identity is an operator identity at separated insertions in the regulated theory. In time-ordered products, differentiating the ordering symbols may also produce local contact terms; the nonlocal momentum dependence discussed below is insensitive to such polynomial contact terms.

43.2 Photon Self-Energy and Transversality

Let

$$\langle\Omega|\,\mathrm{T}\{\widehat{A}_\mu(x)\widehat{A}_\nu(0)\}|\Omega\rangle = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\mathrm{e}^{\mathrm{i}k\cdot x}\,\widetilde{G}_{\mu\nu}(k).$$

The full photon representative two-point function is obtained by inserting one-particle-irreducible photon self-energies into the free propagator:



Let $\mathrm{i}\Pi^{\mu\nu}(k)$ denote the sum of amputated one-particle irreducible photon two-point insertions.

Proposition 43.2 (Transversality of the photon self-energy)**.** *Modulo local contact terms in time-ordered products, Abelian current conservation implies*

$$k_\mu \Pi^{\mu\nu}(k) = 0, \qquad k_\nu \Pi^{\mu\nu}(k) = 0.$$

The statement is about the amputated nonlocal 1PI two-point kernel and is independent of the longitudinal representative chosen for the photon propagator.

Proof. The Maxwell equation with the renormalized current has the form

$$\partial_\nu \widehat{F}^{\mu\nu} = \widehat{j}^\mu, \qquad \partial_\mu \widehat{j}^\mu = 0$$

at separated points. Let $M^{\mu\nu}(k)$ be the noncontact coefficient in the Fourier transform of $\langle\Omega|\,\mathrm{T}\,\widehat{j}^\mu(x)\widehat{j}^\nu(0)|\Omega\rangle$. Then

$$k_\mu M^{\mu\nu}(k) = 0, \qquad k_\nu M^{\mu\nu}(k) = 0.$$

On the other hand, applying $\partial_\alpha F^{\mu\alpha}$ to the photon Dyson series acts on each external photon line by the transverse differential operator

$$k^2\eta^{\mu\rho} - k^\mu k^\rho.$$

The local contact terms are polynomials in k. The nonlocal part is therefore transverse only if the amputated 1PI insertion obeys

$$k_\mu \Pi^{\mu\nu}(k) = 0.$$

This is the two-photon consequence of a linear Ward–Takahashi identity. If $\Gamma_{\mathrm{inv}}[A,\psi,\bar\psi]$ denotes the gauge-invariant part of the QED 1PI functional, the infinitesimal transformation

$$\delta_\alpha A_\mu = \partial_\mu\alpha, \qquad \delta_\alpha\psi = -\mathrm{i}e\alpha\psi, \qquad \delta_\alpha\bar\psi = \mathrm{i}e\alpha\bar\psi$$

gives

$$\mathcal{W}_\alpha\Gamma_{\mathrm{inv}} := \int \left[\partial_\mu\alpha\,\frac{\delta\Gamma_{\mathrm{inv}}}{\delta A_\mu} - \mathrm{i}e\alpha\psi\,\frac{\delta_L\Gamma_{\mathrm{inv}}}{\delta\psi} + \mathrm{i}e\alpha\bar\psi\,\frac{\delta_R\Gamma_{\mathrm{inv}}}{\delta\bar\psi}\right]\mathrm{d}^4x = 0.$$

The operator $\mathcal{W}_\alpha$ is fixed before Γ_{inv} is known; hence the identity is linear in Γ_{inv}. Differentiating twice with respect to A and then setting the fields to zero gives transversality of the photon 1PI two-point kernel, up to the local contact terms already separated above. The nonabelian Slavnov–Taylor identity in Chapter 47 has a different mathematical form: because the BRST variations of A_μ and c are composite local fields, the corresponding 1PI identity is quadratic in Γ, as in (47.44). □

Lorentz covariance then gives the tensor decomposition

$$\Pi^{\mu\nu}(k) = (\eta^{\mu\nu}k^2 - k^\mu k^\nu)\pi(k^2)$$

for a scalar function π.

In covariant gauge, with gauge parameter ξ_{g}, the resummed propagator is

$$\widetilde{G}_{\mu\nu}(k) = \frac{-\mathrm{i}}{k^2 - \mathrm{i}\epsilon}\left[\frac{\eta_{\mu\nu}}{1-\pi(k^2)} - \frac{k_\mu k_\nu}{k^2 - \mathrm{i}\epsilon}\left(1 - \xi_{\mathrm{g}} + \frac{\pi(k^2)}{1-\pi(k^2)}\right)\right].$$

The longitudinal part depends on the representative gauge choice. The transverse pole is normalized by imposing

$$\pi(0) = 0$$

for the massive-electron theory, where the loop function is regular at the photon point $k^2 = 0$.

43.3 One-Loop Vacuum Polarization

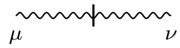
Lemma 43.3 (One-loop QED vacuum polarization in explicit-coupling coordinates)**.** *In the regulator convention of Definition 43.1, dimensional continuation to $D = 4 - \epsilon$ and the pole normalization $\pi(0) = 0$ give*

$$Z_A = 1 - \frac{e^2}{6\pi^2}\frac{1}{\epsilon} + \text{finite},$$

and the cutoff-removed one-loop transverse photon form factor is

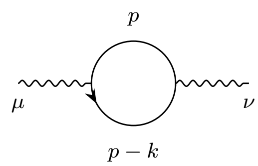
$$\pi(k^2) = \frac{e^2}{2\pi^2}\int_0^1 \mathrm{d}x\, x(1-x)\log\frac{m^2 + x(1-x)k^2}{m^2}.$$

Proof. The photon counterterm contributes



$$= -\mathrm{i}(Z_A - 1)(k^2\eta_{\mu\nu} - k_\mu k_\nu),$$

which contributes $1 - Z_A$ to $\pi(k^2)$. The one-loop electron diagram is



and equals

$$\mathrm{i}\Pi^{(1)}_{\mu\nu}(k) = -\int \frac{\mathrm{d}^4p}{(2\pi)^4}\,\mathrm{Tr}\left[e\gamma_\mu\,\frac{-\mathrm{i}(-\mathrm{i}\,\not{p} + m)}{p^2 + m^2 - \mathrm{i}\epsilon}\,e\gamma_\nu\,\frac{-\mathrm{i}(-\mathrm{i}(\not{p} - \not{k}) + m)}{(p-k)^2 + m^2 - \mathrm{i}\epsilon}\right].$$

The minus sign is the closed fermion loop. Continue the integral to $D = 4-\epsilon$ dimensions, keeping

$$\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}, \qquad \eta^{\mu\nu}\eta_{\mu\nu} = D, \qquad \mathrm{Tr}\,\mathbf{1} = 4.$$

After Wick rotation and the Feynman-parameter identity

$$\frac{1}{ab} = \int_0^1 \mathrm{d}x\,\frac{1}{\bigl(xa + (1-x)b\bigr)^2},$$

shift

$$p \mapsto p + xk.$$

The denominator becomes

$$\bigl(p^2 + x(1-x)k^2 + m^2\bigr)^2.$$

The trace identity

$$\mathrm{Tr}(\gamma_\mu\gamma_\alpha\gamma_\nu\gamma_\beta) = 4(\eta_{\mu\alpha}\eta_{\nu\beta} - \eta_{\mu\nu}\eta_{\alpha\beta} + \eta_{\mu\beta}\eta_{\nu\alpha})$$

reduces the numerator. Rotational symmetry in the continued loop momentum allows

$$p_\mu p_\nu \mapsto \frac{1}{D}\eta_{\mu\nu}p^2$$

inside the integral, and all terms linear in p vanish. With

$$M_x^2 = m^2 + x(1-x)k^2,$$

the angularly averaged numerator can be written as

$$\begin{aligned} N_{\mu\nu} = 4\Bigl[\eta_{\mu\nu}\bigl(p^2 - x(1-x)k^2 + m^2\bigr) - \frac{2}{D}\eta_{\mu\nu}p^2 \\ + 2x(1-x)k_\mu k_\nu\Bigr]. \end{aligned}$$

The two scalar integrals needed are

$$\begin{aligned} \int \frac{\mathrm{d}^Dp}{(2\pi)^D}\frac{1}{(p^2 + M_x^2)^2} &= \frac{1}{(4\pi)^{D/2}}\Gamma(2 - D/2)(M_x^2)^{D/2-2}, \\ \int \frac{\mathrm{d}^Dp}{(2\pi)^D}\frac{p^2}{(p^2 + M_x^2)^2} &= \frac{D}{2}\frac{1}{(4\pi)^{D/2}}\Gamma(1 - D/2)(M_x^2)^{D/2-1}, \end{aligned}$$

where the second formula is defined by analytic continuation in D. Substitution cancels the nontransverse local-looking pieces and leaves the transverse tensor

$$\Pi^{(1)}_{\mu\nu}(k) = -(\eta_{\mu\nu}k^2 - k_\mu k_\nu)\frac{8e^2}{(4\pi)^{D/2}}\Gamma(2-D/2)\int_0^1 \mathrm{d}x\, x(1-x)\big(m^2 + x(1-x)k^2\big)^{D/2-2}.$$

With the counterterm included,

$$\pi(k^2) = 1 - Z_A - \frac{8e^2}{(4\pi)^{D/2}}\Gamma(2-D/2)\int_0^1 \mathrm{d}x\, x(1-x)\big(m^2 + x(1-x)k^2\big)^{D/2-2}.$$

The pole normalization $\pi(0) = 0$ fixes

$$Z_A = 1 - \frac{8e^2}{(4\pi)^{D/2}}\Gamma(2-D/2)\frac{1}{6}\, m^{D-4}.$$

For $D = 4 - \epsilon$, this has the pole part

$$Z_A = 1 - \frac{e^2}{6\pi^2}\frac{1}{\epsilon} + \text{finite}.$$

Subtracting the value at $k^2 = 0$ and taking $D \to 4$ gives the finite one-loop vacuum-polarization function

$$\pi(k^2) = \frac{e^2}{2\pi^2}\int_0^1 \mathrm{d}x\, x(1-x)\log\frac{m^2 + x(1-x)k^2}{m^2}.$$

This $\pi(k^2)$ is the cutoff-removed form factor: in a cutoff realization it is the $\Lambda \to \infty$ limit of the renormalized transverse coefficient in the photon two-point matrix element, with the charge normalization $\pi(0) = 0$ held fixed. In dimensional regularization the same statement is represented by the analytic continuation to $D = 4$ after the local counterterm Z_A has subtracted the ultraviolet pole. □

43.4 The Electron Current Matrix Element

Definition 43.4 (Electron electromagnetic form factors)**.** Let $|\vec{p}, \sigma\rangle$ be a one-electron state of charge $q = -e$. Translation covariance gives

$$\langle \vec{p}\,', \sigma'|\widehat{j}^{\mu}(x)|\vec{p}, \sigma\rangle = \mathrm{e}^{\mathrm{i}(p-p')\cdot x}\langle \vec{p}\,', \sigma'|\widehat{j}^{\mu}(0)|\vec{p}, \sigma\rangle.$$

Write the matrix element as

$$\langle \vec{p}\,', \sigma'|\widehat{j}^{\mu}(0)|\vec{p}, \sigma\rangle = \frac{\mathrm{i}q}{(2\pi)^3}\bar{u}^{\sigma'}(\vec{p}\,')M^{\mu}(p', p)u^{\sigma}(\vec{p}).$$

Modulo the on-shell Dirac equations

$$(\not{p} - \mathrm{i}m)u^{\sigma}(\vec{p}) = 0, \qquad \bar{u}^{\sigma'}(\vec{p}\,')(\not{p}' - \mathrm{i}m) = 0,$$

a parity-even Lorentz vector between spinor solutions can be reduced, by the Clifford algebra, to the span of

$$\gamma^{\mu}, \qquad (p + p')^{\mu}, \qquad (p - p')^{\mu}.$$

Terms with γ_5 are excluded by parity, and terms with $[\gamma^\mu, \gamma^\nu]$ are equivalent to the displayed basis by the Gordon identity. Current conservation removes the component proportional to $(p-p')^\mu$. Therefore

$$M^\mu(p', p) = \gamma^\mu F(k^2) - \frac{\mathrm{i}}{2m}(p+p')^\mu G(k^2), \qquad k = p - p'.$$

The scalar functions F and G are the electromagnetic form factors in the present gamma-matrix convention and in the $(p+p')^\mu$ basis. For comparison with the Dirac–Pauli basis, let F_1^{DP} and F_2^{DP} be defined in this same gamma-matrix convention by

$$\bar{u}(p')M^\mu u(p) = \bar{u}(p')\left[\gamma^\mu F_1^{\mathrm{DP}}(k^2) + \left(\gamma^\mu + \frac{\mathrm{i}}{2m}(p+p')^\mu\right) F_2^{\mathrm{DP}}(k^2)\right] u(p).$$

The second structure is the Gordon-identity form of the Pauli tensor structure. Indeed, with

$$S^{\mu\nu} = -\frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu], \qquad k = p - p',$$

the on-shell identity displayed below gives

$$\bar{u}(p')\frac{1}{m}S^{\mu\nu}k_\nu u(p) = \bar{u}(p')\left(\gamma^\mu + \frac{\mathrm{i}}{2m}(p+p')^\mu\right) u(p). \tag{43.1}$$

Thus F_2^{DP} is the coefficient of the Pauli tensor $m^{-1}S^{\mu\nu}k_\nu$, not the coefficient of $(p+p')^\mu$ alone. Its matrix element vanishes at $p' = p$. Comparing coefficients with the displayed (F, G) decomposition gives

$$F_1^{\mathrm{DP}} = F + G, \qquad F_2^{\mathrm{DP}} = -G, \qquad F = F_1^{\mathrm{DP}} + F_2^{\mathrm{DP}}.$$

Thus the charge normalization below, $F(0) + G(0) = 1$, is precisely $F_1^{\mathrm{DP}}(0) = 1$, while the anomalous magnetic moment is $F_2^{\mathrm{DP}}(0) = -G(0)$.

With the plane-wave normalization and current normalization used in Definition 43.4, charge normalization fixes one combination at zero momentum transfer. The condition $\widehat{Q}|\vec{p}, \sigma\rangle = q|\vec{p}, \sigma\rangle$, where $\widehat{Q} = \int \mathrm{d}^3\vec{x}\,\widehat{j}^0(x)$, is equivalent to

$$F(0) + G(0) = 1. \tag{43.2}$$

Indeed, taking $p' \to p$, using

$$\bar{u}^{\sigma'}(\vec{p})u^\sigma(\vec{p}) = \frac{m}{p^0}\delta_{\sigma\sigma'},$$

and using the Dirac equations to rewrite

$$p^\mu \bar{u}^{\sigma'}(\vec{p})u^\sigma(\vec{p}) = \mathrm{i}m\,\bar{u}^{\sigma'}(\vec{p})\gamma^\mu u^\sigma(\vec{p}),$$

one obtains

$$\langle\vec{p}, \sigma'|\widehat{j}^\mu(0)|\vec{p}, \sigma\rangle = \frac{q}{(2\pi)^3}\frac{p^\mu}{p^0}\delta_{\sigma\sigma'}\big(F(0) + G(0)\big).$$

The charge operator

$$\widehat{Q} = \int \mathrm{d}^3\vec{x}\,\widehat{j}^0(x)$$

acts by

$$\widehat{Q}|\vec{p}, \sigma\rangle = q|\vec{p}, \sigma\rangle.$$

Thus

$$F(0) + G(0) = 1.$$

The equality is a distributional momentum-state identity: after smearing the external plane waves by a common wave packet and then removing the smearing, the spatial integral supplies the same $\delta^{(3)}(\vec{p}' - \vec{p})$ on both sides.

43.5 Magnetic Moment from the Form Factor

Proposition 43.5 (Magnetic factor in the (F, G) basis)**.** *For an electron current matrix element written in the basis of Definition 43.4, the spin-dependent coupling to a slowly varying external magnetic field is*

$$\Delta H_{\text{spin}} = -g_{\text{mag}} \frac{q}{2m} \vec{B} \cdot \frac{\vec{\sigma}}{2}, \qquad g_{\text{mag}} = 2F(0) = 2\bigl(1 - G(0)\bigr).$$

Proof. Couple the current to a weak classical background:

$$\Delta H = -\int \mathrm{d}^3\vec{x}\, A^{\text{cl}}_\mu(x) \widehat{j}^\mu(x).$$

For a slowly varying magnetic field choose

$$A^{\text{cl}}_0 = 0, \qquad \nabla \times \vec{A}^{\text{cl}} = \vec{B}.$$

The part of the form-factor matrix element proportional to $(p + p')^\mu$ gives the spin-independent Lorentz-force contribution in the nonrelativistic limit. The spin-dependent term is exposed by writing the $\gamma^\mu F$ term with the Gordon identity. Equivalently, after an integration by parts,

$$\int \mathrm{d}^3\vec{x}\, A^{\text{cl}}_i(x)\, \mathrm{e}^{\mathrm{i}(\vec{p}-\vec{p}')\cdot\vec{x}} (p' - p)_j S^{ij} = \frac{\mathrm{i}}{2} \int \mathrm{d}^3\vec{x}\, \mathrm{e}^{\mathrm{i}(\vec{p}-\vec{p}')\cdot\vec{x}} F^{\text{cl}}_{ij}(x) S^{ij},$$

up to terms higher order in the spatial momentum transfer; only the antisymmetric part contributes to the spin coupling. The spin-dependent part of $\langle \vec{p}', \sigma' | \Delta H | \vec{p}, \sigma \rangle$ is obtained from the identity

$$\begin{aligned} &\bar{u}^{\sigma'}(\vec{p}')[\gamma^\mu, \gamma^\nu](p' - p)_\nu u^\sigma(\vec{p}) \\ &\quad = -4\mathrm{i}m\, \bar{u}^{\sigma'}(\vec{p}')\gamma^\mu u^\sigma(\vec{p}) + 2(p^\mu + p'^\mu)\bar{u}^{\sigma'}(\vec{p}')u^\sigma(\vec{p}), \end{aligned}$$

with

$$S^{\mu\nu} = -\frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu].$$

In the nonrelativistic limit $|\vec{p}|, |\vec{p}'| \ll m$, the spin coupling becomes

$$\Delta H_{\text{spin}} = -\frac{qF(0)}{2m} \vec{B} \cdot \vec{\sigma}.$$

Writing this as

$$\Delta H_{\text{spin}} = -g_{\text{mag}} \frac{q}{2m} \vec{B} \cdot \frac{\vec{\sigma}}{2},$$

the magnetic factor is

$$g_{\text{mag}} = 2F(0).$$

Together with $F(0) + G(0) = 1$, this gives

$$g_{\text{mag}} = 2\bigl(1 - G(0)\bigr).$$

□

43.6 One-Loop Vertex Correction

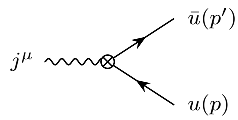
Proposition 43.6 (Schwinger term in the electron magnetic factor)**.** *In the same ultraviolet and infrared hard-kernel convention used for Lemma 43.3, the one-loop vertex correction gives*

$$G(0) = -\frac{e^2}{8\pi^2} = -\frac{\alpha}{2\pi}, \qquad g_{\mathrm{mag}} = 2 + \frac{\alpha}{\pi} + O(\alpha^2), \qquad \alpha = \frac{e^2}{4\pi}.$$

Proof. The current is represented by

$$\widehat{j}^\mu = -\mathrm{i} Z_A^{-1} e\, Z_\psi\, \bar{\psi}\gamma^\mu\psi = -\mathrm{i} e^B Z_A^{-1/2} Z_\psi\, \bar{\psi}\gamma^\mu\psi.$$

Within the same infrared-regulated hard-kernel convention, the current insertion between external electron labels is computed by reducing the external spinor legs by LSZ:



If the three-point function is written in bare fields, each external fermion leg supplies a factor $Z_\psi^{1/2}$ in the LSZ residue. If it is written in renormalized fields, those factors are absent from the external residues but reappear through the counterterm vertices. These two bookkeepings give the same matrix element. After removing the infrared regulator in massless QED, this exclusive one-electron matrix element is not an autonomous physical scattering amplitude. The hard form factor is one factor in either an inclusive probability, where unresolved photons are summed over, or a dressed-state matrix element whose charged asymptotic state already contains the soft electromagnetic field.

The tree current vertex gives

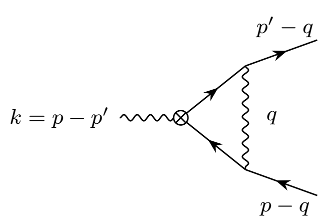
$$-\mathrm{i} Z_A^{-1} e\, \frac{Z_\psi}{(2\pi)^3} \bar{u}^{\sigma'}(\vec{p}')\gamma^\mu u^\sigma(\vec{p}).$$

The current insertion can also be followed by any number of photon self-energy insertions before the photon attaches to the electron line. The transverse photon chain contributes $Z_A/(1-\pi(k^2))$, so the whole tree-plus-vacuum-polarization chain contributes

$$-\mathrm{i} e\, \frac{Z_\psi}{(2\pi)^3} \frac{1}{1-\pi(k^2)} \bar{u}^{\sigma'}(\vec{p}')\gamma^\mu u^\sigma(\vec{p}).$$

This changes only the coefficient of γ^μ, hence only $F(k^2)$. It does not contribute to $G(k^2)$.

At order e^2, the contribution to $G(k^2)$ comes from the vertex correction



in Feynman gauge. Its integrand is

$$-\,\mathrm{i}Z_A^{-1}e\,\frac{Z_\psi}{(2\pi)^3}\int\frac{\mathrm{d}^4q}{(2\pi)^4}\frac{-\mathrm{i}}{q^2-\mathrm{i}\epsilon}\,\bar u(p')\,e\gamma^\rho\frac{-\mathrm{i}[-\mathrm{i}(\not p'-\not q)+m]}{(p'-q)^2+m^2-\mathrm{i}\epsilon}$$
$$\times\,\gamma^\mu\frac{-\mathrm{i}[-\mathrm{i}(\not p-\not q)+m]}{(p-q)^2+m^2-\mathrm{i}\epsilon}e\gamma_\rho\,u(p).$$

To isolate the term that affects the magnetic moment, Wick rotate, continue the loop integral dimensionally, and combine the three denominators

$$D_0=q^2,\qquad D_1=(p'-q)^2+m^2,\qquad D_2=(p-q)^2+m^2$$

by the simplex coordinates

$$z_0=1-x,\qquad z_1=y,\qquad z_2=x-y,\qquad 0\le y\le x\le 1.$$

On the external mass shell $p^2=p'^2=-m^2$, $k=p-p'$, one has

$$z_0D_0+z_1D_1+z_2D_2=\ell^2+m^2x^2+k^2y(x-y),\qquad q=\ell+yp'+(x-y)p.$$

Thus

$$\frac{1}{D_0D_1D_2}=2\int_0^1\mathrm{d}x\int_0^x\mathrm{d}y\,\frac{1}{\big(\ell^2+m^2x^2+k^2y(x-y)\big)^3}.$$

The external Dirac equations and the Clifford algebra reduce every parity-even vector bilinear between the on-shell spinors to the two-basis form

$$\bar u(p')\gamma^\mu u(p)\,A(k^2)+(p+p')^\mu\bar u(p')u(p)\,B(k^2).$$

The scalar $B(k^2)$ is the coefficient of the second basis vector in this on-shell quotient; for $k^2\neq 0$ it is obtained by solving the resulting two-by-two Gram system, and the $k^2=0$ value used below is the regular limit. Terms odd in ℓ vanish, and the $\ell_\alpha\ell_\beta$ terms are replaced by $D^{-1}\eta_{\alpha\beta}\ell^2$ before the scalar integrals are evaluated. The Pauli-coordinate projection of the finite part gives

$$B(k^2)=\frac{\mathrm{i}e^2m}{8\pi^2}\int_0^1\mathrm{d}x\int_0^x\mathrm{d}y\,\frac{x(1-x)}{m^2x^2+k^2y(x-y)}.$$

Since the form-factor definition is $B(k^2)=-\mathrm{i}G(k^2)/(2m)$, this is equivalent to the formula below. The scalar $A(k^2)$ contains ultraviolet and infrared singular terms whose contribution is fixed by the charge normalization and the spinor wavefunction counterterm. The scalar $B(k^2)$ is finite at this order and determines $G(k^2)$. The resulting coefficient is

$$G(k^2)=-\frac{e^2m^2}{4\pi^2}\int_0^1\mathrm{d}x\int_0^x\mathrm{d}y\,\frac{x(1-x)}{m^2x^2+k^2y(x-y)}.$$

The displayed $G(k^2)$ is again the cutoff-removed form factor: for fixed external on-shell wave packets and fixed nonzero momentum-transfer invariant away from infrared singular loci, it is the ultraviolet-regulator removal limit of the renormalized coefficient multiplying $(p+p')^\mu\bar u(p')u(p)$ in the current matrix element. The charge normalization and the wavefunction counterterm determine the accompanying γ^μ coefficient, while no additional ultraviolet subtraction is needed for this one-loop $G(k^2)$. At zero momentum transfer,

$$\int_0^1\mathrm{d}x\int_0^x\mathrm{d}y\,\frac{x(1-x)}{m^2x^2}=\frac{1}{2m^2},\qquad G(0)=-\frac{e^2}{8\pi^2}.$$

With

$$\alpha = \frac{e^2}{4\pi},$$

the one-loop magnetic factor is

$$g_{\text{mag}} = 2\big(1 - G(0)\big) = 2 + \frac{\alpha}{\pi} + O(\alpha^2).$$

$\square$

Chapter 44

Infrared Divergences and Inclusive QED

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

44.1 Soft Radiation Data

The scattering constructions developed earlier are simplest when all asymptotic particles are massive. Four-dimensional QED has a massless stable photon, and a charged particle may emit photons of arbitrarily small energy. This chapter constructs the scattering quantities adapted to that long-distance structure. The ordinary LSZ theorem for a charged external electron would require an isolated massive charged pole. Massless QED instead has charged threshold data accompanied by arbitrarily soft electromagnetic radiation. The fixed hard charged labels used below are therefore not exact fixed-photon-number Wigner-particle S-matrix labels. They are hard data in an infrared-regulated factorization problem; physical charged scattering is attached either to detector-inclusive probabilities or to asymptotic states dressed by their soft electromagnetic fields.

Throughout the chapter, spacetime is $D = 4$ Minkowski space with metric

$$\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1).$$

Consider a hard transition

$$\alpha \longrightarrow \beta$$

between massive charged particles. The hard external charged lines are labelled by n. Their momenta are p_n, their charges are g_n, and

$$p_n^2 = -m_n^2, \qquad p_n^0 > 0.$$

The sign

$$\eta_n = \begin{cases} +1, & n \in \beta \quad \text{outgoing}, \\ -1, & n \in \alpha \quad \text{incoming} \end{cases}$$

records whether the line is outgoing or incoming. Charge conservation for the hard process is

$$\sum_n \eta_n g_n = 0.$$

The charge symbols used in the QED chapters are related as follows.

context	symbol	meaning
Chapter 42	g	signed Abelian charge in $D_\mu = \partial_\mu - \mathrm{i}gA_\mu$; for an electron $g = -e$, with $e > 0$.
Chapter 43	$e,\ e^B$	positive magnitude of the electron charge and its bare counterpart; the electron covariant derivative is $\partial_\mu + \mathrm{i}eA_\mu$, and $e = e^B Z_A^{1/2}$.
this chapter	g_n	signed physical charge of the n-th hard external line; an electron has $g_n = -e$, a positron has $g_n = +e$, and an external scalar of charge g in the scalar-QED example has $g_n = g$.

Thus η_n records whether the charged line is incoming or outgoing, whereas g_n records the physical sign of the charge carried by that line.

A soft photon has momentum

$$k^\mu = (\omega, \vec{k}), \qquad \omega = |\vec{k}|,$$

and physical helicity $h = \pm$. The detector threshold is denoted by E_T: photons below this energy are included in the same observed hard event. An auxiliary infrared cutoff μ will regulate intermediate formulas, and M denotes a soft upper scale below which the eikonal approximation is valid:

$$0 < \mu < E_T < M.$$

Definition 44.1 (Regulated resolution-inclusive transition probability)**.** Let $S^{(\mu)}_{\beta;\{k_r,h_r\}|\alpha}$ denote an infrared-regulated hard kernel with hard charged configuration β, incoming hard charged configuration α, and additional photons (k_r, h_r). The superscript (μ) records the auxiliary long-distance regulator. The object $S^{(\mu)}$ may be obtained from a regulated LSZ computation, but it is not assumed to have a regulator-independent fixed-photon-number limit. The regulated resolution-inclusive probability at threshold E_T is the sum

$$P^{(\mu)}_{\beta|\alpha}(E_T) = \sum_{N=0}^{\infty} \frac{1}{N!} \sum_{h_1,\ldots,h_N} \int_{\mathcal{R}_{E_T}} \prod_{r=1}^{N} \frac{\mathrm{d}^3\vec{k}_r}{(2\pi)^3\, 2\omega_r} \left| S^{(\mu)}_{\beta;\{k_r,h_r\}|\alpha} \right|^2,$$

where $\mathcal{R}_{E_T}$ is the detector-degenerate soft region. At leading soft logarithmic order one may take $\mu < \omega_r < E_T$ independently for each unresolved photon. Refining the detector model changes finite functions of E_T/M, but not the cancellation of the regulator μ. The regulator-free inclusive probability is not the termwise limit of the summands at fixed N; it is the ordered limit constructed in Section 44.5.

44.2 External-Line Soft Factors

In scalar QED, attach an outgoing photon of momentum k and polarization vector $e_h^\mu(k)$ to an outgoing external scalar line of momentum p. The factor multiplying the hard amplitude is

$$(-\mathrm{i})\,\frac{1}{(p+k)^2+m^2-\mathrm{i}\epsilon}\,\mathrm{i}g(2p+k)^\mu e^*_{h,\mu}(k).$$

Since $p^2+m^2=0$ and $k^2=0$, the leading term as $k\to 0$ is

$$g\,\frac{p\cdot e_h^*(k)}{p\cdot k-\mathrm{i}\epsilon}.$$

The denominator is linear in the photon energy. The soft photon therefore couples to the charge and momentum of the external line, while the internal structure of the hard subdiagram is unresolved at leading order.

For the hard transition $\alpha\to\beta$, the one-soft-photon amplitude is

$$\mathcal{M}_{\beta;k,h|\alpha}=\mathcal{M}_{\beta|\alpha}\left[\sum_n \frac{g_n\,p_n\cdot e_h^*(k)}{\eta_n p_n\cdot k-\mathrm{i}\epsilon}\right]+O(k^0).$$

The same leading factor is obtained in spinor QED. The numerator of the nearly on-shell fermion propagator acts on the external spinor and reduces to the corresponding momentum factor; spin first enters in subleading soft terms. Indeed, for an outgoing fermion line the soft attachment contains

$$\left[\bar u^\sigma(p)(-g\gamma^\mu)\frac{-\mathrm{i}[-\mathrm{i}(\not p+\not k)+m]}{(p+k)^2+m^2-\mathrm{i}\epsilon}\right]e^*_{h,\mu}(k).$$

The denominator is $2p\cdot k-\mathrm{i}\epsilon+O(k^2)$. In the numerator, using the on-shell equation for the external spinor and moving γ^μ through the Dirac numerator gives

$$\bar u^\sigma(p)\gamma^\mu(-\mathrm{i}\not p+m)=\bar u^\sigma(p)(\mathrm{i}\not p+m)\gamma^\mu-2\mathrm{i}p^\mu\bar u^\sigma(p)=-2\mathrm{i}p^\mu\bar u^\sigma(p).$$

Thus the leading soft limit is

$$g\,\frac{p\cdot e_h^*(k)}{p\cdot k-\mathrm{i}\epsilon}\,\bar u^\sigma(p),$$

the same eikonal factor multiplying the original spinor amplitude.

Theorem 44.2 (Weinberg leading soft photon theorem)**.** *Fix an infrared regulator and hard external data for which the hard kernel $\mathcal{M}_{\beta|\alpha}$ is defined and smooth under the addition of one photon of momentum $k^\mu=\omega(1,\widehat{k})$ with $\omega\to 0^+$, away from nonsoft singular surfaces. Then the hard kernel with one additional outgoing physical photon of helicity h satisfies*

$$\mathcal{M}_{\beta;k,h|\alpha}=S_\gamma^{(0)}(k,h)\,\mathcal{M}_{\beta|\alpha}+O(\omega^0),\qquad S_\gamma^{(0)}(k,h)=\sum_n \eta_n g_n\,\frac{p_n\cdot e_h^*(k)}{p_n\cdot k}. \tag{44.1}$$

The sum runs over all hard charged external lines, with $\eta_n = +1$ for outgoing and $\eta_n = -1$ for incoming hard lines. The $i\epsilon$ prescription in the preceding external-line formula specifies the boundary value of the regulated kernel; the theorem displays the homogeneous leading soft factor.

Proof. The scalar and spinor calculations above show that attaching the soft photon to an external hard line produces the pole

$$\eta_n g_n \frac{p_n \cdot e_h^*(k)}{p_n \cdot k} \mathcal{M}_{\beta|\alpha}.$$

The sign η_n is the crossing sign: an incoming charged line is equivalently an outgoing line with momentum and charge flow reversed in the all-outgoing Ward identity. Attachments to internal lines are less singular in the uniform soft limit at fixed nonexceptional hard kinematics, because no internal propagator is forced onto a new mass shell by adding a momentum k of order ω. Summing the external-line poles gives (44.1).

Gauge invariance fixes the allowed leading pole. A physical photon polarization representative is defined modulo

$$e_h^\mu(k) \sim e_h^\mu(k) + c\, k^\mu.$$

Under this replacement, the leading factor shifts by

$$\delta S_\gamma^{(0)} = c^* \sum_n \eta_n g_n.$$

Thus the unphysical longitudinal polarization decouples precisely when the hard process obeys charge conservation,

$$\sum_n \eta_n g_n = 0.$$

This is the Ward identity for the global $U(1)$ charge written in the external hard basis. Conversely, a nonzero coefficient of the longitudinal soft pole would mean that the regulated hard kernel had not descended to a physical photon helicity amplitude. $\square$

Figure 44.1 records this leading-soft Ward identity: attaching a soft photon to the charged hard external lines gives the eikonal factor, and the longitudinal part cancels precisely by charge conservation.

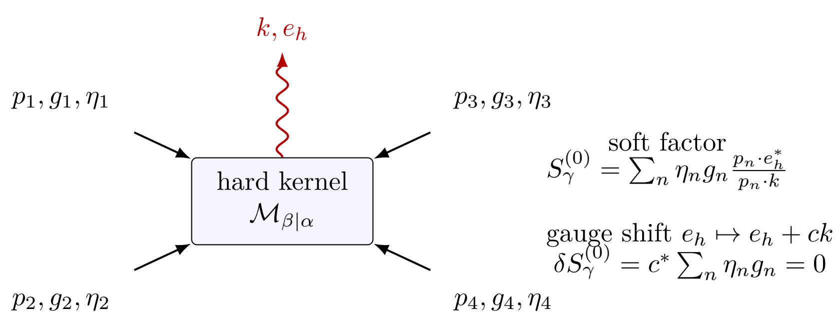


Figure 44.1: The leading soft photon factor is the sum of external-line poles. Replacing the polarization by a pure gauge vector tests the Ward identity; the pole descends to the physical photon helicity space exactly when the hard charges are conserved.

Theorem 44.3 (Weinberg leading soft graviton theorem)**.** *Consider an infrared-regulated theory with a massless spin-two gauge field whose representative is normalized by*

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}, \qquad \mathcal{L}_{\text{int}} = -\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}.$$

For one additional outgoing graviton of momentum k and physical polarization tensor $\varepsilon^{(\lambda)}_{\mu\nu}(k)$, the leading soft limit is

$$\mathcal{M}^{\text{grav}}_{\beta;k,\lambda|\alpha} = S^{(0)}_{\text{grav}}(k,\lambda)\,\mathcal{M}_{\beta|\alpha} + O(\omega^0), \qquad S^{(0)}_{\text{grav}} = \frac{\kappa}{2}\sum_n \eta_n \frac{p_n^\mu p_n^\nu \varepsilon^{(\lambda)*}_{\mu\nu}(k)}{p_n \cdot k}. \tag{44.2}$$

Proof. The stress-tensor vertex on an external massive line contains, at leading order in the soft momentum, the matrix element of the hard momentum carried by that line. For a scalar line, the one-graviton vertex contracted with a transverse traceless polarization gives

$$\kappa\, p^\mu p^\nu \varepsilon^*_{\mu\nu}(k)$$

up to terms less singular in k, while the adjacent nearly on-shell propagator gives $(2\eta_n p_n \cdot k)^{-1}$. Their product is the $(\kappa/2)\eta_n p_n^\mu p_n^\nu \varepsilon^*_{\mu\nu}/(p_n \cdot k)$ term in (44.2). Spin-dependent terms in the stress tensor contain an additional power of the soft momentum or the Lorentz generator and therefore begin at subleading order. Summing over the hard external lines gives (44.2).

The physical spin-two polarization is defined modulo the linearized diffeomorphism shift

$$\varepsilon_{\mu\nu} \sim \varepsilon_{\mu\nu} + k_\mu \xi_\nu + k_\nu \xi_\mu, \qquad k \cdot \xi = 0$$

on the null momentum k. The leading soft factor changes by

$$\delta S^{(0)}_{\text{grav}} = \kappa \xi^*_\nu \sum_n \eta_n p_n^\nu.$$

This vanishes by hard momentum conservation. If the coefficient of $h_{\mu\nu}T^{\mu\nu}$ depended on the species, the same gauge variation would contain

$$\sum_n \eta_n \kappa_n p_n^\nu,$$

which does not vanish for arbitrary allowed hard reactions unless all κ_n are the same constant. The leading soft spin-two theorem therefore packages both the external-line pole and the universality of the gravitational coupling. □

The photon theorem is the input used in the rest of this chapter. The graviton theorem is recorded here because it is the same infrared consequence of a massless gauge particle and gauge-invariant factorization. A complete quantum theory containing the spin-two field requires additional ultraviolet and nonperturbative input not used in the soft argument.

For N real soft photons the leading factors multiply:

$$\mathcal{M}_{\beta;\{k_r,h_r\}|\alpha} = \mathcal{M}_{\beta|\alpha} \prod_{r=1}^{N} \left[\sum_n \frac{g_n\, p_n \cdot e^*_{h_r}(k_r)}{\eta_n p_n \cdot k_r - \mathrm{i}\epsilon} \right] + O(k_r^0).$$

For two photons emitted from the same line, the two possible orderings of the emissions turn the nested eikonal denominators into the product displayed above. For an outgoing scalar line this is the elementary identity

$$\frac{g\,p\cdot e^*_{h_2}(k_2)}{p\cdot k_2-\mathrm{i}\epsilon}\,\frac{g\,p\cdot e^*_{h_1}(k_1)}{p\cdot(k_1+k_2)-\mathrm{i}\epsilon}+(1\leftrightarrow 2)$$
$$=\frac{g\,p\cdot e^*_{h_1}(k_1)}{p\cdot k_1-\mathrm{i}\epsilon}\,\frac{g\,p\cdot e^*_{h_2}(k_2)}{p\cdot k_2-\mathrm{i}\epsilon}.$$

Corrections from the non-eikonal parts of the propagators and vertices are less singular in the soft scaling. The same combinatorics gives the product for any finite number of soft photons.

Remark 44.4 (Subleading soft photons). The preceding product formula uses only the leading eikonal term. For a single additional photon, under the perturbative hypotheses of massive charged external particles, a smooth hard amplitude near $k=0$, and a fixed infrared regulator, the Low soft expansion has the form

$$\mathcal{M}_{\beta;k,h|\alpha}=\left(S^{(0)}_h(k)+S^{(1)}_h(k)\right)\mathcal{M}_{\beta|\alpha}+O(\omega),$$

where

$$S^{(0)}_h(k)=\sum_n\frac{g_n\,p_n\cdot e^*_h(k)}{\eta_n p_n\cdot k}$$

is the leading factor used above, and the subleading operator is

$$S^{(1)}_h(k)=-\mathrm{i}\sum_n\frac{g_n\,e^*_{h,\mu}(k)k_\nu J^{\mu\nu}_n}{\eta_n p_n\cdot k}.$$

Here $J^{\mu\nu}_n$ is the total Lorentz generator acting on the n-th hard external datum. For a scalar external line its orbital part is

$$J^{\mu\nu}_{n,\mathrm{orb}}=-\mathrm{i}\left(p^\mu_n\frac{\partial}{\partial p_{n,\nu}}-p^\nu_n\frac{\partial}{\partial p_{n,\mu}}\right),$$

and for spinning particles one adds the spin matrix $\Sigma^{\mu\nu}_n$. The formula is a distributional statement on the mass shell and momentum-conservation surface: the differential operator acts on the hard amplitude together with its conservation delta function, or equivalently on a chosen smooth local representative before restricting to the physical surface.

The subleading factor is homogeneous of degree ω^0, whereas $S^{(0)}_h$ is homogeneous of degree ω^{-1}. Consequently the real-emission integral built from $S^{(0)}_h S^{(0)*}_h$ contains $\mathrm{d}\omega/\omega$, while the interference between $S^{(0)}_h$ and $S^{(1)}_h$ contains $\mathrm{d}\omega$ and is finite at the lower endpoint in the massive case. This is why the Bloch–Nordsieck logarithm and its Poisson exponentiation are governed by $S^{(0)}$. At subleading order the soft insertion is a differential operator, and multiple soft emissions produce ordered products and contact terms. The independent one-photon multiplication factors used in the leading-log proof are a property of the leading eikonal term.

The same soft statements have an asymptotic-symmetry formulation. In four-dimensional QED, the leading soft photon theorem is equivalent, under the assumptions on null-infinity limits and charged-state sectors, to a Ward identity for large $U(1)$ gauge transformations; its classical counterpart is electromagnetic memory. In gravity, the analogous leading and subleading soft-graviton statements are tied to BMS supertranslations and to the more delicate superrotation story. These asymptotic-symmetry interpretations delimit the leading eikonal approximation used in this chapter; the inclusive cancellation above uses the exponentiation of $S^{(0)}$.

Figure 44.2 shows the leading eikonal factorization used in the inclusive calculation: soft photons attach to the external charged lines, while the hard kernel is unchanged at leading soft order.

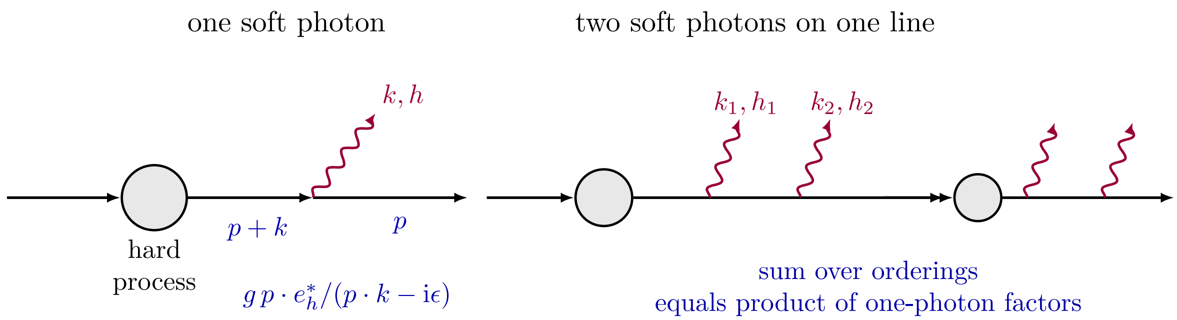


Figure 44.2: Soft factorization on external charged lines. A photon with wavelength much larger than the hard interaction region resolves only the external charge and momentum.

44.3 Real Soft Emission

Divide the real-emission contribution by the hard rate with no unresolved photons. Using the leading soft factor gives

$$R_{\text{real}}(E_T, \mu) = \sum_{N=0}^{\infty} \frac{1}{N!} \prod_{r=1}^{N} \int_{\mu<\omega_r<E_T} \frac{\mathrm{d}^3\vec{k}_r}{(2\pi)^3\, 2\omega_r} \sum_{h_r=\pm} \left| \sum_n \frac{g_n\, p_n \cdot e_{h_r}^*(k_r)}{\eta_n p_n \cdot k_r} \right|^2 .$$

For a single soft photon, define the nonnegative angular kernel

$$\mathcal{S}(k) = \sum_{h=\pm} \left| \sum_n \frac{g_n\, p_n \cdot e_h^*(k)}{\eta_n p_n \cdot k} \right|^2 .$$

A covariant physical-polarization sum may be written as

$$\sum_{h=\pm} e_{h,\mu}(k) e_{h,\nu}^*(k) = \eta_{\mu\nu} + k_\mu c_\nu + k_\nu c_\mu, \qquad k \cdot c = -1, \qquad c^2 = 0.$$

The terms involving c_μ vanish after using $\sum_n \eta_n g_n = 0$. Thus

$$\mathcal{S}(k) = \sum_{n,m} \frac{\eta_n \eta_m g_n g_m\, p_n \cdot p_m}{(p_n \cdot k)(p_m \cdot k)}.$$

Writing $k^\mu = \omega(1, \hat{k})$, the energy integral separates from the angular integral:

$$\int_\mu^{E_T} \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3\, 2\omega} \mathcal{S}(k) = \frac{1}{2(2\pi)^3} \int_\mu^{E_T} \frac{\mathrm{d}\omega}{\omega} \int \mathrm{d}^2\hat{k} \sum_{n,m} \frac{\eta_n \eta_m g_n g_m\, p_n \cdot p_m}{(p_n^0 - \vec{p}_n \cdot \hat{k})(p_m^0 - \vec{p}_m \cdot \hat{k})}.$$

For massive external particles, define the relative-velocity parameter

$$\beta_{nm} = \sqrt{1 - \frac{m_n^2 m_m^2}{(p_n \cdot p_m)^2}}.$$

The diagonal term $n = m$ is understood as the $\beta_{nm} \to 0$ limit. The angular integral may be evaluated directly by combining the two denominators:

$$\begin{aligned}
&\int \mathrm{d}^2\hat{k}\, \frac{1}{(p_n^0 - \vec{p}_n \cdot \hat{k})(p_m^0 - \vec{p}_m \cdot \hat{k})} \\
&= \int_0^1 \mathrm{d}x \int \mathrm{d}^2\hat{k}\, \frac{1}{\left[p_n^0 x + p_m^0(1-x) - (\vec{p}_n x + \vec{p}_m(1-x)) \cdot \hat{k}\right]^2}.
\end{aligned}$$

Choose the polar axis along $\vec{p}_n x + \vec{p}_m(1-x)$. The $\hat{k}$-integral becomes an elementary integral of

$$\frac{2\pi \sin\theta\, \mathrm{d}\theta}{\left[p_n^0 x + p_m^0(1-x) - |\vec{p}_n x + \vec{p}_m(1-x)| \cos\theta\right]^2}.$$

Performing the x-integral gives

$$\int \mathrm{d}^2\hat{k}\, \frac{1}{(p_n^0 - \vec{p}_n \cdot \hat{k})(p_m^0 - \vec{p}_m \cdot \hat{k})} = -\frac{2\pi}{p_n \cdot p_m} \frac{1}{\beta_{nm}} \log \frac{1+\beta_{nm}}{1-\beta_{nm}}.$$

It is useful to introduce

$$A_{\beta\alpha} = -\frac{1}{8\pi^2} \sum_{n,m} \frac{\eta_n \eta_m g_n g_m}{\beta_{nm}} \log \frac{1+\beta_{nm}}{1-\beta_{nm}}.$$

The representation by $\mathcal{S}(k)$ shows $A_{\beta\alpha} \geq 0$ for the massive charged scattering considered here. The real unresolved photons therefore give

$$R_{\text{real}}(E_T, \mu) = \sum_{N=0}^{\infty} \frac{1}{N!} \left[A_{\beta\alpha} \log \frac{E_T}{\mu}\right]^N = \left(\frac{E_T}{\mu}\right)^{A_{\beta\alpha}}$$

at leading soft logarithmic order.

44.4 Virtual Soft Photons and External Factors

A virtual soft photon exchanged between two distinct external charged lines also factorizes. For external lines n and m, set

$$J_{nm} = -\mathrm{i}\, p_n \cdot p_m \int_{\mu < |\vec{k}| < M} \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{1}{(k^2 - \mathrm{i}\epsilon)(\eta_n p_n \cdot k - \mathrm{i}\epsilon)(-\eta_m p_m \cdot k - \mathrm{i}\epsilon)}.$$

The virtual soft photons exchanged among distinct external lines exponentiate:

$$\exp\left[\frac{1}{2} \sum_{n \neq m} g_n g_m J_{nm}\right].$$

This expression omits the soft photons whose two endpoints lie on the same external charged line. Those terms are tied to the infrared part of the LSZ normalization factors. A direct way to see the accounting is to insert the electromagnetic current

$$\widehat{j}^{\mu}(x) = \partial_\nu \widehat{F}^{\mu\nu}(x)$$

between one-particle states. In scalar QED, with $k = p_2 - p_1$ and $\omega_p = \sqrt{\vec{p}^2 + m^2}$,

$$\langle \vec{p}_2 | \widehat{j}^\mu(0) | \vec{p}_1 \rangle = \frac{g}{(2\pi)^3 \sqrt{2\omega_{p_1}}\sqrt{2\omega_{p_2}}} (p_1 + p_2)^\mu F(k^2), \qquad F(0) = 1.$$

In spinor QED the corresponding decomposition is

$$\begin{aligned} &\langle \vec{p}_2, \sigma_2 | \widehat{j}^\mu(0) | \vec{p}_1, \sigma_1 \rangle \\ &\quad = \frac{\mathrm{i}g}{(2\pi)^3}\, \bar{u}^{\sigma_2}(\vec{p}_2) \left[\gamma^\mu F(k^2) - \frac{\mathrm{i}}{2m}(p_1 + p_2)^\mu G(k^2) \right] u^{\sigma_1}(\vec{p}_1), \qquad F(0) + G(0) = 1. \end{aligned}$$

In these conventions the one-loop value $G(0) = -g^2/(8\pi^2) + O(g^4)$ is a finite contribution to the magnetic form factor. The electromagnetic matrix element at $k^2 = 0$ is free of infrared divergence, although derivatives such as $\mathrm{d}F/\mathrm{d}k^2\big|_{k^2=0}$ need not be. Thus the soft virtual photons attached to the current insertion must be accompanied by the same infrared part of the LSZ factors. In the current form factor, soft photons connecting the two charged legs exponentiate as $e^{g^2 J_{12}}$. Finiteness at $k^2 = 0$ then fixes the infrared factor in Z:

$$Z_{\mathrm{IR}} = e^{-g^2 J_{12}} \Big|_{p_2 = p_1,\ \eta_2 = -\eta_1} .$$

Equivalently, compare this with the same-line expression obtained by setting the two eikonal velocities equal before interpreting the self-contraction:

$$\text{“}J_{11}\text{”} = -\mathrm{i} p_1^2 \int_{\mu < |\vec{q}| < M} \frac{\mathrm{d}^4 q}{(2\pi)^4} \frac{1}{(q^2 - \mathrm{i}\epsilon)(p_1 \cdot q - \mathrm{i}\epsilon)(-p_1 \cdot q - \mathrm{i}\epsilon)}.$$

The two matter poles in the q^0-plane contribute to the imaginary part. The real part satisfies

$$\operatorname{Re} J_{11} = -\operatorname{Re} J_{12}\big|_{p_2 = p_1,\ \eta_2 = -\eta_1}, \qquad Z_{\mathrm{IR}} = e^{g^2 \operatorname{Re} J_{11}}.$$

At one-loop order this same statement follows directly from the scalar-QED self-energy. The soft part is

$$\mathrm{i}\Sigma(p) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{-\mathrm{i}}{(p+k)^2 + m^2 - \mathrm{i}\epsilon} \frac{-\mathrm{i}}{k^2 - \mathrm{i}\epsilon} (\mathrm{i}g)^2 (2p + k)^2 + \cdots ,$$

whose infrared approximation is

$$\mathrm{i}\Sigma(p)\big|_{\mathrm{soft}} = 4g^2 p^2 \int_{|\vec{k}| < M} \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{p^2 + m^2 + 2p \cdot k - \mathrm{i}\epsilon} \frac{1}{k^2 - \mathrm{i}\epsilon}.$$

The corresponding external factor

$$Z = \left(1 - \frac{\partial \Sigma}{\partial p^2} \bigg|_{p^2 = -m^2} \right)^{-1}$$

contains the same soft logarithm as the same-line eikonal loop. In the infrared region,

$$\log Z_{\mathrm{IR}} = \mathrm{i}g^2\, 4p^2 \int_{|\vec{k}| < M} \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{(2p \cdot k - \mathrm{i}\epsilon)^2 (k^2 - \mathrm{i}\epsilon)} = g^2 \operatorname{Re} J_{11} + O(g^4).$$

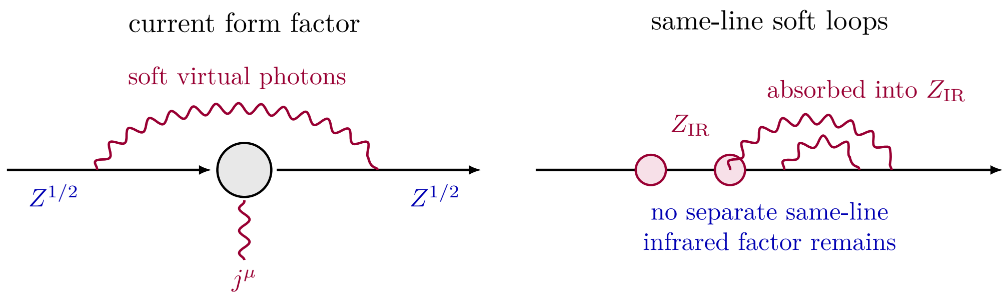


Figure 44.3: The infrared part of the external LSZ factor accounts for soft virtual photons whose two endpoints lie on the same charged external line.

Thus soft loops beginning and ending on a single external charged line are not additional uncancelled infrared factors; they are already included in the LSZ normalization of that charged external line.

Figure 44.3 identifies the same-line soft loop contribution and its placement in the external-line normalization factor rather than as a separate inclusive-rate correction.

Combining the distinct-line soft loops with the infrared parts of the $Z_n^{1/2}$ factors gives

$$\exp\left[\frac{1}{2}\sum_{n\neq m} g_n g_m J_{nm}\right]\prod_n (Z_n^{\rm IR})^{1/2} = (\text{phase})\,\exp\left[\frac{1}{2}\sum_{n,m} g_n g_m\ \mathrm{Re}\,J_{nm}\right].$$

The k^0-plane poles of J_{nm}, including only the denominators that control the contour placement, are

$$-|\vec{k}|+\mathrm{i}\epsilon,\qquad |\vec{k}|-\mathrm{i}\epsilon,\qquad \frac{\vec{p}_n\cdot\vec{k}}{p_n^0}-\mathrm{i}\eta_n\epsilon,\qquad \frac{\vec{p}_m\cdot\vec{k}}{p_m^0}+\mathrm{i}\eta_m\epsilon.$$

The photon poles give the real part of the eikonal integral, while the matter poles may contribute to its phase. Evaluating the photon-pole contribution gives

$$\mathrm{Re}\,J_{nm} = -\eta_n\eta_m\,\frac{p_n\cdot p_m}{2}\int_{\mu<|\vec{k}|<M}\frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{|\vec{k}|^3(p_n^0-\vec{p}_n\cdot\hat{k})(p_m^0-\vec{p}_m\cdot\hat{k})}.$$

Using the same angular integral as for real emission,

$$\mathrm{Re}\,J_{nm} = \eta_n\eta_m\,\frac{1}{8\pi^2\beta_{nm}}\log\frac{1+\beta_{nm}}{1-\beta_{nm}}\log\frac{M}{\mu}.$$

Therefore the virtual factor in the amplitude has modulus

$$\left(\frac{\mu}{M}\right)^{A_{\beta\alpha}/2},$$

and its contribution to a rate is

$$R_{\rm virt}(M,\mu) = \left(\frac{\mu}{M}\right)^{A_{\beta\alpha}}.$$

Figure 44.4 displays the Bloch–Nordsieck cancellation in the notation above: the virtual factor and the real-emission inclusive factor carry opposite powers of the soft regulator.

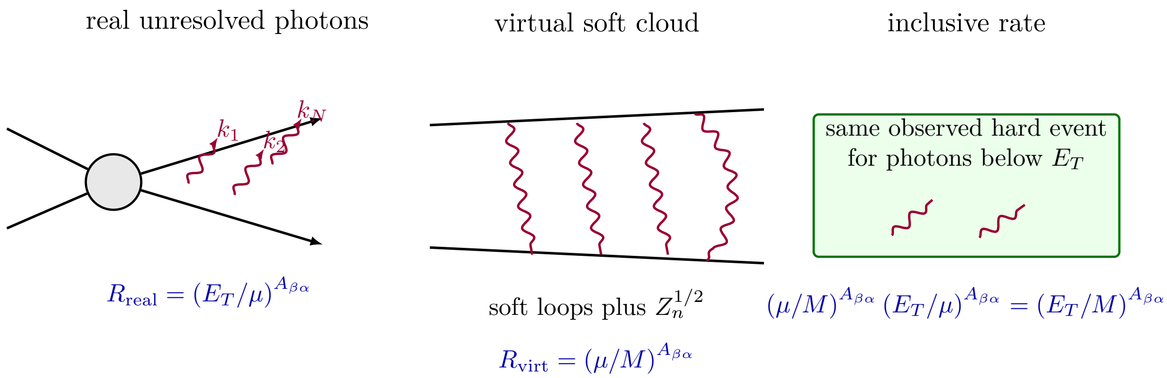


Figure 44.4: Real unresolved photons and virtual soft photons carry opposite powers of the auxiliary infrared cutoff. The product is finite for a fixed detector threshold.

44.5 Inclusive Probabilities

The cancellation of the auxiliary photon-mass regulator between real unresolved emission and virtual soft exchange is the Bloch–Nordsieck cancellation. Equivalently, real soft emission cancels virtual infrared divergences at the level of inclusive probabilities. The order of operations is part of the statement. Fix

$$0 < \mu < E_T < M$$

and compute in the regulated theory. At fixed μ, first include all detector-degenerate unresolved photons, or, coefficientwise in perturbation theory, all detector-degenerate terms contributing to the chosen coefficient. Only after that inclusive sum has been formed is the regulator $\mu \downarrow 0$ removed.

In the leading soft approximation this ordered construction is completely explicit. Put

$$\ell_T(\mu) = \log \frac{E_T}{\mu}, \qquad \ell_M(\mu) = \log \frac{M}{\mu}.$$

The real contribution with exactly N unresolved photons is

$$R_N(E_T, \mu) = \frac{1}{N!} \left[A_{\beta\alpha} \ell_T(\mu)\right]^N ,$$

and the virtual soft factor in the rate is

$$R_{\text{virt}}(M, \mu) = \exp\{-A_{\beta\alpha} \ell_M(\mu)\}.$$

Thus the leading-soft regulated inclusive factor is

$$R^{(\mu)}_{\text{incl}}(E_T, M) = R_{\text{virt}}(M, \mu) \sum_{N=0}^{\infty} R_N(E_T, \mu).$$

The infinite sum over N is taken at fixed positive μ, where it is an ordinary exponential series. It gives

$$R^{(\mu)}_{\text{incl}}(E_T, M) = \exp\{-A_{\beta\alpha} \ell_M(\mu)\} \exp\{A_{\beta\alpha} \ell_T(\mu)\} = \left(\frac{E_T}{M}\right)^{A_{\beta\alpha}},$$

which is independent of μ. Multiplying by the hard finite factor gives the leading soft result

$$P_{\beta|\alpha}(E_T) = P^{\mathrm{hard}}_{\beta|\alpha}(M) \left(\frac{E_T}{M}\right)^{A_{\beta\alpha}} \times F_{\beta\alpha}(E_T/M),$$

where $F_{\beta\alpha}$ is finite as $\mu \to 0$ and equals 1 in the leading logarithmic approximation. The scale M separates hard and soft physics; dependence on it cancels once the hard finite correction is computed to the same order.

Remark 44.5 (Coefficientwise Bloch–Nordsieck cancellation). Let $A_{\beta\alpha}$ be regarded as a formal quantity of order g^2. At every fixed power $A^L_{\beta\alpha}$, the coefficient of the inclusive leading-soft rate has a regulator-independent limit after the sum over unresolved photon numbers contributing to that coefficient is taken. More precisely,

$$\begin{aligned}
[A^L_{\beta\alpha}] & \left[\exp\{-A_{\beta\alpha}\ell_M(\mu)\} \sum_{N=0}^{\infty} \frac{[A_{\beta\alpha}\ell_T(\mu)]^N}{N!}\right] \\
&= \sum_{N=0}^{L} \frac{[-\ell_M(\mu)]^{L-N}}{(L-N)!} \frac{\ell_T(\mu)^N}{N!} = \frac{1}{L!}\left(\ell_T(\mu) - \ell_M(\mu)\right)^L \\
&= \frac{1}{L!}\left(\log\frac{E_T}{M}\right)^L.
\end{aligned}$$

Thus the cancellation is not a statement about an individual fixed photon number. It is a statement about the inclusive coefficient obtained after the degenerate real-emission terms and the virtual terms at the same perturbative order are combined.

At fixed $\mu > 0$, the product of the virtual exponential and the real emission exponential is an identity of formal power series in $A_{\beta\alpha}$. The coefficient of $A^L_{\beta\alpha}$ receives contributions from N real photons and $L - N$ powers of the virtual exponential, with $0 \le N \le L$. This gives the displayed binomial sum. Since

$$\ell_T(\mu) - \ell_M(\mu) = \log\frac{E_T}{\mu} - \log\frac{M}{\mu} = \log\frac{E_T}{M},$$

all dependence on the auxiliary regulator cancels before the limit $\mu \downarrow 0$ is taken.

For a fixed finite number N of photons, by contrast, the rate contains

$$\left(\frac{\mu}{M}\right)^{A_{\beta\alpha}} \frac{1}{N!}\left[A_{\beta\alpha}\log\frac{E_T}{\mu}\right]^N,$$

which tends to zero as $\mu \to 0$ when $A_{\beta\alpha} > 0$. The resolution-inclusive sum remains finite because arbitrarily many unresolved soft photons are included.

The coupling expansion and the infrared limit are therefore nonuniform in the exclusive fixed-N decomposition. The perturbative statement is the coefficientwise one in Remark 44.5: coefficient extraction is performed at fixed regulator, all degenerate real and virtual terms at that coefficient are added, and only then is μ removed. The exponentiated leading-soft expression is a resummation of these coefficientwise cancellations, not a license to take the regulator limit term by term in photon number.

Remark 44.6. The fixed-photon-number matrix elements are exclusive kernels for an idealized measurement that specifies the exact number of photons down to zero energy. The inclusive probability above is the transition probability associated with a finite energy resolution. In a charged sector of massless QED, the exclusive kernels should not be reinterpreted as ordinary regulator-independent LSZ S-matrix elements for bare electron states. Both the vanishing of fixed-N probabilities and the finiteness of the inclusive probability follow from the same soft factorization formulas.

The discussion so far treated massive charged particles, where the angular integral is finite. If massless charged particles occur, photon emission collinear to a charged external line produces additional singularities. The soft Bloch–Nordsieck sum is then only part of the required inclusive construction. The finite-degeneracy statement behind the Kinoshita–Lee–Nauenberg prescription is the following.

Remark 44.7 (Finite-degeneracy KLN trace identity). Let $\mathcal{H}_0$ be a Hilbert space with unperturbed Hamiltonian H_0. Let P_A and P_B be the orthogonal projectors onto two finite-dimensional degenerate subspaces of H_0, with

$$d_A = \operatorname{rank} P_A < \infty, \qquad d_B = \operatorname{rank} P_B < \infty.$$

Let $H = H_0 + V$, where V is a bounded self-adjoint perturbation, and let

$$U_T = \mathcal{T} \exp\left\{-\mathrm{i}\int_{-T}^{T} e^{\mathrm{i}H_0 t} V e^{-\mathrm{i}H_0 t}\,\mathrm{d}t\right\}$$

be the interaction-picture evolution, possibly with an adiabatic switch-on understood. Assume that, after introducing whatever ultraviolet and infrared regulators are being used, the inclusive long-time limit

$$\mathcal{P}^{(\rho)}_{B|A} = \lim_{T\to\infty} \frac{1}{d_A} \operatorname{Tr}_{\mathcal{H}_0}\left(P_B U_T P_A U_T^\dagger P_B\right)$$

exists coefficientwise in the perturbation expansion. Finally assume that the regulator-singular factors associated with exact degeneracy act by unitary rotations inside the degenerate subspaces: for the $P_A \to P_B$ block of the regulated scattering operator,

$$P_B S_\rho P_A = U_B(\rho)\, S_{\mathrm{fin}}(\rho)\, U_A(\rho)^\dagger,$$

where $U_A(\rho)$ and $U_B(\rho)$ are unitary on $P_A\mathcal{H}_0$ and $P_B\mathcal{H}_0$, respectively, and $S_{\mathrm{fin}}(\rho)$ has a finite coefficientwise limit as the regulator $\rho \downarrow 0$. Then

$$\lim_{\rho\downarrow 0} \mathcal{P}^{(\rho)}_{B|A} = \frac{1}{d_A} \operatorname{Tr}_{P_B\mathcal{H}_0}\left(S_{\mathrm{fin}} S_{\mathrm{fin}}^\dagger\right)$$

exists coefficientwise. The final-state sum removes the outgoing degenerate rotation $U_B(\rho)$, and the initial-state average removes the incoming degenerate rotation $U_A(\rho)$.

The normalized incoming density matrix for an unresolved initial degeneracy class is

$$\varrho_A = \frac{P_A}{d_A}.$$

The inclusive probability of finding the outgoing state in $P_B\mathcal{H}_0$ is

$$\operatorname{Tr}_{\mathcal{H}_0}\left(P_B S_\rho \varrho_A S_\rho^\dagger P_B\right) = \frac{1}{d_A} \operatorname{Tr}_{P_B\mathcal{H}_0}\left(P_B S_\rho P_A S_\rho^\dagger P_B\right).$$

Insert

$$P_B S_\rho P_A = U_B(\rho) S_{\mathrm{fin}}(\rho) U_A(\rho)^\dagger.$$

Because $U_A(\rho)^\dagger U_A(\rho) = P_A$ on the incoming degenerate subspace and $U_B(\rho)^\dagger U_B(\rho) = P_B$ on the outgoing degenerate subspace, cyclicity of the finite-dimensional trace gives

$$\begin{aligned}\mathcal{P}^{(\rho)}_{B|A} &= \frac{1}{d_A} \operatorname{Tr}_{P_B\mathcal{H}_0} \left[U_B(\rho) S_{\mathrm{fin}}(\rho) S_{\mathrm{fin}}(\rho)^\dagger U_B(\rho)^\dagger\right] \\ &= \frac{1}{d_A} \operatorname{Tr}_{P_B\mathcal{H}_0} \left[S_{\mathrm{fin}}(\rho) S_{\mathrm{fin}}(\rho)^\dagger\right].\end{aligned}$$

The assumed coefficientwise limit of $S_{\mathrm{fin}}(\rho)$ therefore gives the claimed coefficientwise limit of the inclusive probability.

The theorem is an abstract finite-regulator statement. In a relativistic field theory the interaction is generally an unbounded operator-valued distribution on the physical Hilbert space, and exact massless degeneracy is typically continuous rather than finite-dimensional. The theorem identifies the finite-volume, energy-resolution, and regulator structure that must be implemented before the regulator limits are taken. After those regularizations have turned detector-degenerate sectors into finite subspaces, the inclusive construction sums over final states and averages over initial states inside the same detector degeneracy class. In symbols, if $[\alpha]$ and $[\beta]$ are the incoming and outgoing equivalence classes defined by the detector resolution, one considers

$$P([\alpha] \to [\beta]) = \sum_{\alpha' \in [\alpha]} \omega_{\alpha'} \sum_{\beta' \in [\beta]} \left| S^{(\mu)}_{\beta'\alpha'} \right|^2,$$

with nonnegative weights $\omega_{\alpha'}$ normalized on the incoming degenerate class. Soft degeneracy and collinear degeneracy are therefore handled by the same operational rule: the transition probability is attached to what the detector distinguishes.

Remark 44.8 (Abelian scope of Bloch–Nordsieck cancellation). The Bloch–Nordsieck exponentiation derived in this chapter is an Abelian soft-photon statement. Its leading real-emission factor is a scalar function of the hard charges and momenta, and different soft photons multiply by commuting factors. Nonabelian gauge theory changes this algebraic input: soft gluons couple through color generators, and the soft factors are operators in color space, naturally organized by Wilson lines and color ordered products.

Consequently, the QED formula above should not be transplanted directly to partonic QCD. In perturbative QCD, final-state soft and collinear singularities cancel in sufficiently inclusive color-singlet observables or factor into universal jet and soft functions, but initial-state collinear singularities of colored partons are absorbed into the renormalization of parton distributions. The finite hadronic observable is then obtained from renormalized light-ray matrix elements and short-distance kernels, as in the DIS factorization formula (49.313) and DGLAP equation (49.314) of Chapter 49. This is the precise replacement, in QCD factorization, for an abelian fixed-charge Bloch–Nordsieck cancellation.

44.6 Soft Photon Number Distribution

The leading soft formula also determines the unresolved photon multiplicity distribution. Let

$$\overline{N}_{\beta\alpha}(\mu, E_T) = A_{\beta\alpha} \log \frac{E_T}{\mu}$$

be the mean number of photons in the energy window $\mu < \omega < E_T$ generated by the eikonal current. The probability for emitting exactly N photons in this unresolved window has the Poisson form

$$P_N(\mu, E_T) = \exp\{-\overline{N}_{\beta\alpha}\}\frac{\overline{N}_{\beta\alpha}^{\,N}}{N!}$$

at leading soft order, multiplied by the hard probability and finite corrections. The exponential factor is the virtual soft factor together with the no-resolved-emission probability. Summing over N gives unity for the soft probability distribution, while the hard probability is multiplied by the finite detector-resolution factor

$$\left(\frac{E_T}{M}\right)^{A_{\beta\alpha}}.$$

As $\mu \to 0$, the mean $\overline{N}_{\beta\alpha}$ diverges when $A_{\beta\alpha} > 0$. The inclusive probability is finite because the detector state is a distribution over arbitrarily many unresolved photons. The exclusive fixed-N probability tends to zero because it samples a set of vanishing measure inside the infrared cloud selected by the long-range electromagnetic field.

44.7 Charged Asymptotic Sectors and Infraparticles

The preceding formulas describe a scattering problem whose charged asymptotic representation is fixed by Gauss' law. Let $\mathfrak{A}_{\mathrm{loc}}$ denote the algebra of gauge-invariant observables localized in bounded spacetime regions, and let

$$E^i = F^{i0}$$

be the electric field. At a fixed time, for a smooth function f on the unit sphere, define the large-radius electric-flux observable formally by

$$\Phi_R(f) = \int_{S^2} \mathrm{d}\Omega(\mathbf{n})\, R^2 f(\mathbf{n})\, n_i E^i(0, R\mathbf{n}).$$

The constant test function gives the total electric charge in the limit

$$Q = \lim_{R\to\infty} \Phi_R(1),$$

when the limit exists in the representation under consideration. More generally, the limiting angular flux functional

$$\Phi_\infty(f) = \lim_{R\to\infty} \Phi_R(f)$$

records the electromagnetic field at spacelike infinity. Local observables cannot change this limiting flux. The precise operator statement uses a shell-smearing before the limit is taken. Choose $\chi \in C_c^\infty(\mathbb{R})$ with $\int_{\mathbb{R}} \chi(s)\mathrm{d}s = 1$, and put

$$\Phi_{R,\chi}(f) = \int_{\mathbb{R}^3} \mathrm{d}^3\mathbf{x}\, \chi(|\mathbf{x}| - R)\, f(\mathbf{x}/|\mathbf{x}|)\, \frac{x_i}{|\mathbf{x}|}\, E^i(0, \mathbf{x}).$$

This is a local field smeared in the annular shell $|\,|\mathbf{x}| - R\,| \le C_\chi$. If $A \in \mathfrak{A}_{\mathrm{loc}}$ is localized in a bounded spacetime region whose time-zero spatial projection is contained in $|\mathbf{x}| \le R_0$, then for $R > R_0 + C_\chi$ every point in the smearing shell is spacelike to that region. Locality gives

$$[\Phi_{R,\chi}(f), A] = 0$$

on the common field domain, and the same statement holds for bounded functions of $\Phi_{R,\chi}(f)$. Hence, whenever $\Phi_{R,\chi}(f)$ has a sharp limiting self-adjoint observable $\Phi_\infty(f)$ in the representation under study, the spectral projections of $\Phi_\infty(f)$ commute with every bounded local observable. The limiting angular flux is then a superselection datum for the charged sector.

The constant mode $f = 1$ is the total electric charge. The higher spherical modes are independent asymptotic data: a neutral configuration may have zero total charge while still carrying nontrivial angular flux if its charged constituents have different asymptotic velocities. Thus charge neutrality alone does not return the sector to the vacuum infrared representation; the angular Coulomb field is part of the asymptotic charged datum.

There are two logically separate consequences. First, a bounded local gauge-invariant observable cannot change the limiting surface charge: for a large sphere spacelike to its localization region it commutes with the surface-flux operator before the $R \to \infty$ limit. This is the localization obstruction to creating a charged particle by ordinary local observable Haag–Ruelle operators. Second, in massless QED the charged sector also carries velocity-dependent soft electromagnetic flux, so its spectral support is not an ordinary sharp Wigner mass shell. The next theorem concerns this second statement. The Wilson-line and Coulomb dressings needed for the first statement are formulated as a scattering-theory problem in Chapter 22.

Quoted Theorem 44.9 (Buchholz infraparticle obstruction)**.** *In a local relativistic quantum field theory satisfying Maxwell's equations and Gauss' law, a pure state carrying nonzero abelian gauge charge cannot be a Wigner one-particle state with a sharp mass shell. More precisely, charged particle weights are accompanied by asymptotic electromagnetic flux, and the representations induced by different charged velocities need not be unitarily equivalent. In particular, the charged sector is not an ordinary tensor product of a Wigner electron one-particle representation with the vacuum photon Fock representation, and Lorentz boosts of a charged particle change the flux sector rather than acting inside one fixed Fock sector.*

The electron in four-dimensional QED is therefore an infraparticle. The conclusion is an algebraic consequence of locality and Gauss' law.[1]

The mechanism behind the quoted theorem is the following. If a massive charged particle were an ordinary Wigner particle, boosts would move its one-particle vector through a single massive one-particle representation. In a theory with Gauss' law the same boost also changes the angular flux functional at spacelike infinity. Since the sharp limiting flux commutes with all bounded local observables and labels a superselection datum, the boosted charged state cannot remain a vector in one fixed photon Fock representation unless the change of flux is implemented by a square-integrable soft-photon displacement. The latter implementation fails for the Coulomb tail: the displacement has logarithmically divergent one-photon norm.

Perturbation theory displays this obstruction in a direct way. Work in a frame where $\mathbf{v} = \beta\widehat{\mathbf{z}}$. The boosted Coulomb field has radial flux density

$$\mathcal{E}_{g,\beta}(\mathbf{n}) = \frac{g}{4\pi}\frac{1-\beta^2}{(1-\beta\cos\theta)^2}, \qquad \cos\theta = \widehat{\mathbf{z}}\cdot\mathbf{n}.$$

[1] The theorem-level result used here is Buchholz's Gauss-law analysis of the infraparticle problem. The spectral analysis of charged infraparticles was developed by Fröhlich–Morchio–Strocchi, while the coherent asymptotic construction used in perturbation theory is the Faddeev–Kulish construction. The review of Buchholz–Dybalski gives a modern scattering-theoretic account of these relations.

The total flux is still g, because

$$2\pi(1-\beta^2)\int_{-1}^{1}\frac{\mathrm{d}z}{(1-\beta z)^2} = 2\pi(1-\beta^2)\left[\frac{1}{\beta(1-\beta z)}\right]_{z=-1}^{z=1} = 4\pi. \tag{44.3}$$

The angular distribution, however, determines the velocity: its value in the forward direction divided by its value in the backward direction is

$$\frac{\mathcal{E}_{g,\beta}(\widehat{\mathbf{z}})}{\mathcal{E}_{g,\beta}(-\widehat{\mathbf{z}})} = \left(\frac{1+\beta}{1-\beta}\right)^2. \tag{44.4}$$

For a charged particle of velocity

$$\mathbf{v} = \frac{\mathbf{p}}{p^0},$$

the Coulomb field boosted to that velocity has radial flux density

$$\mathcal{E}_{g,\mathbf{v}}(\mathbf{n}) = \frac{g}{4\pi}\frac{1-|\mathbf{v}|^2}{(1-\mathbf{v}\cdot\mathbf{n})^2}, \qquad \int_{S^2}\mathrm{d}\Omega(\mathbf{n})\,\mathcal{E}_{g,\mathbf{v}}(\mathbf{n}) = g. \tag{44.5}$$

Different velocities give different angular flux functionals. A multiparticle charged configuration carries the sum of the corresponding flux densities. This flux data is what the soft photon cloud implements.

The non-Fock nature of the dressing can already be read from the one-photon Hilbert norm. The physical photon one-particle space has norm

$$\|f\|_\gamma^2 = \sum_h \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3\,2\omega}\,|f^h(k)|^2.$$

For the leading charged soft profile

$$f_p^h(k) \sim g\,\frac{p\cdot e_h^*(k)}{p\cdot k}, \qquad \omega = |\mathbf{k}|,$$

one has $p\cdot k = \omega(p^0 - \mathbf{p}\cdot\mathbf{n})$ and therefore

$$\|f_p\|_\gamma^2 \sim C(p,g)\int_0^\Lambda \frac{\mathrm{d}\omega}{\omega}, \qquad C(p,g) > 0 \quad (g\neq 0).$$

Thus $f_p \notin \mathcal{H}_{1,\gamma}$ after the infrared cutoff is removed. More invariantly, choose transverse polarization vectors and write

$$V_\mathbf{v}(\mathbf{n}) = \frac{\mathbf{v}-(\mathbf{v}\cdot\mathbf{n})\mathbf{n}}{1-\mathbf{v}\cdot\mathbf{n}}.$$

This profile determines the velocity: for $\mathbf{v}\neq 0$ its zeros on the unit sphere are $\mathbf{n} = \pm\mathbf{v}/|\mathbf{v}|$, and on the great circle $\mathbf{v}\cdot\mathbf{n} = 0$ its norm is $|\mathbf{v}|$; the zero-velocity profile is identically zero. For two velocities $\mathbf{v}\neq\mathbf{w}$, the difference of the two soft profiles has norm

$$\|f_\mathbf{v} - f_\mathbf{w}\|_\gamma^2 = \frac{g^2}{2(2\pi)^3}\log\frac{\Lambda}{\mu}\int_{S^2}\mathrm{d}\Omega(\mathbf{n})\,|V_\mathbf{v}(\mathbf{n}) - V_\mathbf{w}(\mathbf{n})|^2 + O(1), \tag{44.6}$$

where μ and Λ are infrared and soft ultraviolet cutoffs. The angular integral vanishes exactly when the velocities coincide. Therefore distinct sharp charged velocities define inequivalent infrared coherent directions in the limit $\mu \to 0$. At finite cutoff the coherent operator below is a unitary Weyl operator on photon Fock space; in the limit it defines an inequivalent representation labelled by the asymptotic flux. If two dressings have different limiting flux, their difference is not square-integrable in the same norm, so they are assigned to inequivalent infrared representations rather than to two vectors in one common photon Fock representation. The inclusive construction avoids choosing a sharp charged flux sector by summing over detector-unresolved radiation; the dressed construction chooses a sector adapted to the hard charged asymptotic data.

The elementary flux-normalization, velocity-extraction, half-line Fourier, and soft-profile norm checks behind (44.3), (44.4), and (44.6) are collected in `calculation-checks/charged_flux_dressing_checks.py`. They are not a replacement for Buchholz's theorem; they are the explicit perturbative mechanism that explains why the theorem has the physical content stated above.

Open Problem 44.10 (Charged asymptotic completeness in massless QED). Give a nonperturbative Hilbert-space construction of charged scattering in four-dimensional QED with the following data and conclusions. For each charged superselection sector $\mathcal{H}^{\mathrm{ch}}_{\mathrm{phys}}$, construct an asymptotic space

$$\mathcal{H}^{\mathrm{ch}}_{\mathrm{as}}$$

whose labels include hard charged velocity and spin data, ordinary transverse photon radiation, and the limiting angular electric-flux data required by Gauss' law. Construct large-time wave operators

$$\Omega^{\mathrm{ch}}_{\mathrm{in/out}} : \mathcal{H}^{\mathrm{ch}}_{\mathrm{as}} \longrightarrow \mathcal{H}^{\mathrm{ch}}_{\mathrm{phys}}$$

which intertwine spacetime translations with the total asymptotic energy-momentum, represent the limiting electromagnetic flux consistently with Φ_∞, and are isometric after the appropriate infrared identifications of zero-resolution photon clouds. The asymptotic-completeness claim is the equality

$$\overline{\mathrm{Ran}\,\Omega^{\mathrm{ch}}_{\mathrm{in}}} = \overline{\mathrm{Ran}\,\Omega^{\mathrm{ch}}_{\mathrm{out}}} = \mathcal{H}^{\mathrm{ch}}_{\mathrm{phys}}$$

in the chosen charged sector, together with a scattering operator on $\mathcal{H}^{\mathrm{ch}}_{\mathrm{as}}$ whose detector-inclusive probabilities agree with the regulator-independent inclusive probabilities constructed above from Bloch–Nordsieck and KLN degeneracy sums.

The theorem above identifies the obstruction that fixes the form of the problem: the asymptotic space cannot be the tensor product of a Wigner one-electron Fock space with the vacuum photon Fock space. A charged creator must carry electromagnetic flux to infinity; in a gauge-invariant description this requires a noncompact dressing, for example a Dirac-Coulomb dressing or a Wilson-line dressing, and a corresponding replacement for the ordinary local-field LSZ limit. The Faddeev–Kulish construction supplies the perturbative coherent soft datum suggested by the leading eikonal factor, and the inclusive construction supplies finite detector probabilities. The open problem is to put these structures into a single nonperturbative large-time scattering theorem.

44.8 Dressed Charged States

The same eikonal data can also be incorporated into the asymptotic states. For a charged particle of momentum p, a soft coherent dressing has the leading soft form

$$\mathcal{W}_p = \exp\left[\sum_h \int_{\omega<\Lambda} \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3\,2\omega} \left(f^h_p(k)a^\dagger_h(k) - \overline{f^h_p(k)}a_h(k)\right)\right],$$

with leading coefficient

$$f_p^h(k) \sim g\,\frac{p \cdot e_h^*(k)}{p \cdot k}.$$

The operator $\mathcal{W}_p$ attaches the long-range electromagnetic field carried by the charged particle. With a positive infrared cutoff it is a well-defined Fock-space Weyl operator. Removing the cutoff is the representation-changing step described in the preceding section: the limiting state is a charged hard datum together with a coherent soft electromagnetic field in the corresponding asymptotic flux sector. This is the leading Faddeev–Kulish dressing datum. Dressed-state and inclusive descriptions use the same soft factor. The inclusive construction sums over unresolved radiation in probabilities, while the dressed construction builds part of the soft cloud into the state representation.

44.9 A Gauge-Invariant Infrared Regulator

The cutoff μ in the formulas above can be realized by deforming the theory so that the photon has a small mass, and then removing the deformation. If the same soft region is regulated dimensionally instead, the logarithms $\log(\mu/M)$ are replaced by infrared poles $1/\epsilon_{\mathrm{IR}}$ plus scheme-dependent finite constants; the coefficient multiplying the logarithm or pole is the same eikonal coefficient $A_{\beta\alpha}$, so the real–virtual cancellation is identical coefficient by coefficient. A convenient gauge-invariant deformation is an Abelian-Higgs regulator. Introduce a complex scalar φ, charged under the gauge field A_μ, with

$$D_\mu\varphi = (\partial_\mu + \mathrm{i}g_H A_\mu)\varphi$$

and Lagrangian terms

$$\mathcal{L}_{\mathrm{reg}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - |D_\mu\varphi|^2 - V(|\varphi|).$$

Choose the potential so that its vacuum satisfies

$$|\langle\varphi\rangle| = v \neq 0.$$

Expanding around this vacuum gives

$$-|D_\mu\varphi|^2 \supset -g_H^2 v^2 A_\mu A^\mu,$$

so the gauge boson acquires a mass proportional to $g_H v$. Taking v to zero after forming inclusive quantities gives a consistent version of the infrared-regulated calculation.

Figure 44.5 records the Abelian-Higgs infrared regulator used for this comparison: the gauge boson mass is generated by a regulated Higgs expectation value and is taken to zero only after inclusive quantities have been formed.

The infrared analysis will be used later whenever massless gauge fields occur. The next step in the logical development is the generating functional formalism, which packages connected Green functions and one-particle irreducible vertices in a form suitable for renormalization.

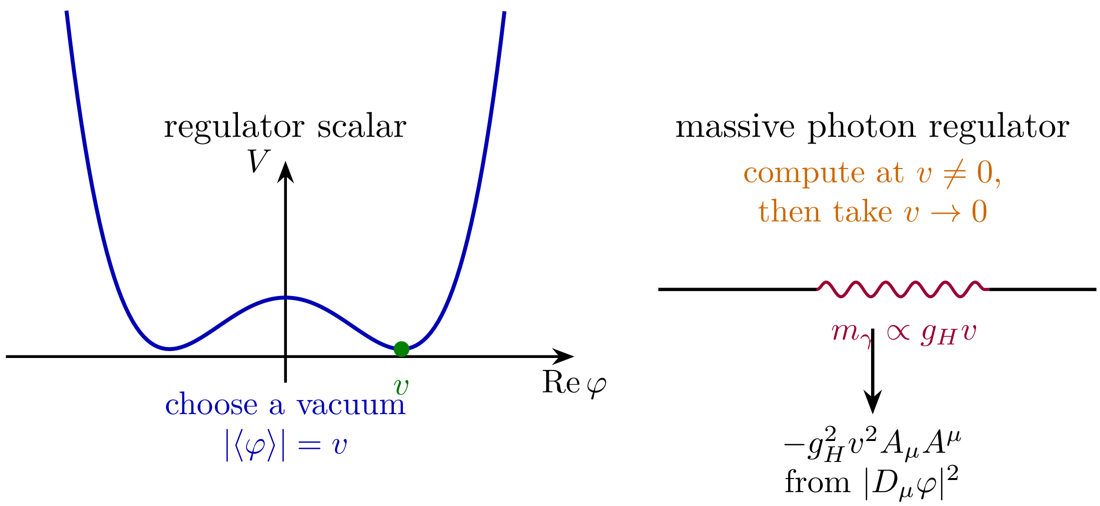


Figure 44.5: A small photon mass may be introduced through a Higgs deformation while keeping gauge invariance manifest.

Chapter 45

Classical Yang-Mills Theory and Matter Fields

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

45.1 Local Gauge Fields

This chapter formulates the local classical field theory of nonabelian gauge fields. We work first on a spacetime region over which the relevant principal G-bundle has been trivialized. In such a local trivialization, a gauge field is represented by a Hermitian matrix-valued one-form

$$A = A_\mu(x)\,\mathrm{d}x^\mu, \qquad A_\mu(x) = A^a_\mu(x) t^a.$$

The index $\mu = 0, \dots, D-1$ is a spacetime index. The index a labels a basis of the gauge algebra in the Hermitian-generator convention. If $\mathfrak{g}_{\mathrm{ah}} = \mathrm{Lie}(G)$ is realized by anti-Hermitian matrices in a unitary representation, then the Hermitian generators used here span $i\,\mathfrak{g}_{\mathrm{ah}}$. The associated anti-Hermitian connection one-form, denoted A_{ah}, is $\mathsf{A}_{\mathrm{ah}} = -\mathrm{i}A$. We write $\mathfrak{g} := i\,\mathfrak{g}_{\mathrm{ah}}$ for this Hermitian coordinate space. Throughout this chapter $[X, Y]$ denotes the ordinary matrix commutator of Hermitian representatives, while the real compact Lie bracket in Hermitian coordinates is

$$[X, Y]_{\mathrm{H}} := -\mathrm{i}[X, Y].$$

With this convention,

$$[t^a, t^b] = \mathrm{i}f^c{}_{ab} t^c. \tag{45.1}$$

The real constants ${f^c}_{ab}$ are the structure constants of $[\ ,\ ]_{\mathrm{H}}$ in this basis, with the output Lie-algebra label written first. Thus ${f^c}_{ab} = -{f^c}_{ba}$ and the Jacobi identity is

$$ {f^e}_{ab}{f^d}_{ec} + {f^e}_{bc}{f^d}_{ea} + {f^e}_{ca}{f^d}_{eb} = 0. $$

Formulas below include explicit factors of i so that A_μ, ζ, and $F_{\mu\nu}$ remain Hermitian matrix fields.

The infinitesimal local transformation with parameter $\zeta(x) = \zeta^a(x)t^a$ is

$$ \delta_\zeta A_\mu = \partial_\mu \zeta - \mathrm{i}[A_\mu, \zeta]. \tag{45.2} $$

In components,

$$ \delta_\zeta A^a_\mu = \partial_\mu \zeta^a + {f^a}_{bc} A^b_\mu \zeta^c, \tag{45.3} $$

The derivative term records the freedom to change local trivialization from point to point, while the commutator term records the adjoint action of the Lie algebra on itself.

Figure 45.1 fixes the index conventions in this gauge-potential formula: the spacetime index labels the one-form component, and the Lie-algebra index labels the chosen generator basis.

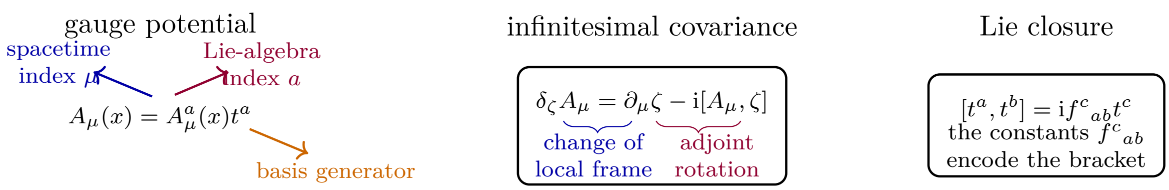


Figure 45.1: The local gauge field carries a spacetime vector index and a Lie-algebra index. The infinitesimal transformation combines the derivative of the local parameter with the adjoint action of the Lie algebra.

The finite transformation is most cleanly stated using a smooth G-valued function $g(x)$. With

$$ g(x) = \exp\{\mathrm{i}\zeta^a(x)t^a\}, $$

define

$$ A'_\mu = gA_\mu g^{-1} + \mathrm{i}g\,\partial_\mu g^{-1}. \tag{45.4} $$

The Baker-Campbell-Hausdorff formula

$$ \log(\mathrm{e}^X \mathrm{e}^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \cdots $$

shows how products of exponentials are controlled by the Lie bracket. The identity

$$ \mathrm{e}^X Y \mathrm{e}^{-X} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \cdots $$

then shows that conjugation by $g(x)$ maps $\mathfrak{g}$ to $\mathfrak{g}$. Thus (45.4) produces another local representative $A'_\mu = A'^a_\mu t^a$.

The same formula is also the transition law between local trivializations. If U_i and U_j are two coordinate patches and $h_{ji} : U_i \cap U_j \to G$ is the transition function, then their local connection representatives satisfy

$$ A^{(j)}_\mu = h_{ji}A^{(i)}_\mu h^{-1}_{ji} + \mathrm{i}h_{ji}\partial_\mu h^{-1}_{ji}. $$

Thus the local field A_μ is not itself a globally defined tensor on a nontrivial bundle. The curvature constructed below does glue covariantly:

$$F^{(j)}_{\mu\nu} = h_{ji} F^{(i)}_{\mu\nu} h_{ji}^{-1}.$$

The local formulas therefore describe a connection on a principal bundle; they do not assume that the bundle is globally trivial.

45.2 Curvature and the Yang-Mills Action

Define the covariant differential operator

$$\nabla_\mu := \partial_\mu - \mathrm{i} A_\mu. \tag{45.5}$$

The gauge coupling has been absorbed into the connection in this chapter: if g_{el} denotes the Abelian charge called g in Chapter 42, then relative to the QED convention $\partial_\mu - \mathrm{i} g_{\mathrm{el}} A_\mu^{\mathrm{QED}}$ the present connection is $A_\mu^{\mathrm{here}} = g_{\mathrm{el}} A_\mu^{\mathrm{QED}}$. With this choice the curvature and the action carry the coupling in the Yang–Mills kinetic term rather than in the covariant derivative. In the $U(1)$ specialization,

$$A_\mu^{\mathrm{here}} = g_{\mathrm{el}} A_\mu^{\mathrm{QED}}, \qquad F_{\mu\nu}^{\mathrm{here}} = \partial_\mu A_\nu^{\mathrm{here}} - \partial_\nu A_\mu^{\mathrm{here}} = g_{\mathrm{el}} F_{\mu\nu}^{\mathrm{QED}}. \tag{45.6}$$

This is a change of field normalization, not a new physical assumption; every occurrence of $F_{\mu\nu}$ below is in the coupling-absorbed Yang–Mills normalization. The finite transformation law (45.4) is equivalent to

$$\nabla'_\mu = g \nabla_\mu g^{-1}, \tag{45.7}$$

where the factors are multiplied as differential operators. The curvature is

$$F_{\mu\nu} := \mathrm{i}[\nabla_\mu, \nabla_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}[A_\mu, A_\nu] = F^a_{\mu\nu} t^a. \tag{45.8}$$

It follows from (45.7) that

$$F'_{\mu\nu} = g F_{\mu\nu} g^{-1}. \tag{45.9}$$

Connection, curvature, and Bianchi identities. In the Hermitian convention of this chapter, the finite transformation law (45.4) is equivalent to (45.7). Consequently the curvature transforms covariantly as in (45.9). Its components are

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^a{}_{bc} A^b_\mu A^c_\nu. \tag{45.10}$$

For an adjoint-valued field X, define

$$D_\mu X := \partial_\mu X - \mathrm{i}[A_\mu, X].$$

Then the curvature obeys the Bianchi identity

$$D_\lambda F_{\mu\nu} + D_\mu F_{\nu\lambda} + D_\nu F_{\lambda\mu} = 0. \tag{45.11}$$

Let s be a test section in the local trivialization. Using $\partial_\mu(g^{-1}s) = (\partial_\mu g^{-1})s + g^{-1}\partial_\mu s$,

$$\begin{aligned}(g\nabla_\mu g^{-1})s &= g\big(\partial_\mu - \mathrm{i}A_\mu\big)(g^{-1}s)\\ &= \partial_\mu s + g(\partial_\mu g^{-1})s - \mathrm{i}gA_\mu g^{-1}s.\end{aligned}$$

Since $A'_\mu = gA_\mu g^{-1} + \mathrm{i}g\partial_\mu g^{-1}$, the last line is $(\partial_\mu - \mathrm{i}A'_\mu)s$. This proves $\nabla'_\mu = g\nabla_\mu g^{-1}$. Therefore

$$[\nabla'_\mu, \nabla'_\nu] = g[\nabla_\mu, \nabla_\nu]g^{-1},$$

and multiplication by i gives (45.9).

For the component formula, substitute $A_\mu = A^a_\mu t^a$ into (45.8) and use $[t^b, t^c] = \mathrm{i}f^a{}_{bc}t^a$. The commutator term contributes $-\mathrm{i}A^b_\mu A^c_\nu[t^b, t^c] = f^a{}_{bc}A^b_\mu A^c_\nu t^a$, which gives (45.10).

Finally, the Jacobi identity for the differential operators ∇_μ is

$$[\nabla_\lambda, [\nabla_\mu, \nabla_\nu]] + [\nabla_\mu, [\nabla_\nu, \nabla_\lambda]] + [\nabla_\nu, [\nabla_\lambda, \nabla_\mu]] = 0.$$

Since $[\nabla_\mu, \nabla_\nu] = -\mathrm{i}F_{\mu\nu}$ and $[\nabla_\lambda, X] = D_\lambda X$ for multiplication by an adjoint field X, the Jacobi identity is precisely (45.11).

Figure 45.2 shows the geometric content of the commutator formula. For a small oriented rectangle based at x, with side lengths a in the x^μ direction and b in the x^ν direction, let

$$U_\mu(x, a) = \mathcal{P}\exp\left\{\mathrm{i}\int_0^a A_\mu(x + se_\mu)\,\mathrm{d}s\right\}$$

be the parallel-transport operator for $\dot{x}^\rho\nabla_\rho s = 0$ along the horizontal side. The oriented loop holonomy is

$$U_\square(x; \mu, \nu) = U_\nu(x, b)^{-1}U_\mu(x + be_\nu, a)^{-1}U_\nu(x + ae_\mu, b)U_\mu(x, a).$$

Expanding the path-ordered exponentials and using the Baker–Campbell–Hausdorff formula gives

$$U_\square(x; \mu, \nu) = 1 + \mathrm{i}F_{\mu\nu}(x)\,ab + O(a^2b + ab^2).$$

Thus $F_{\mu\nu}$ is the infinitesimal failure of parallel transports in the two coordinate directions to commute.

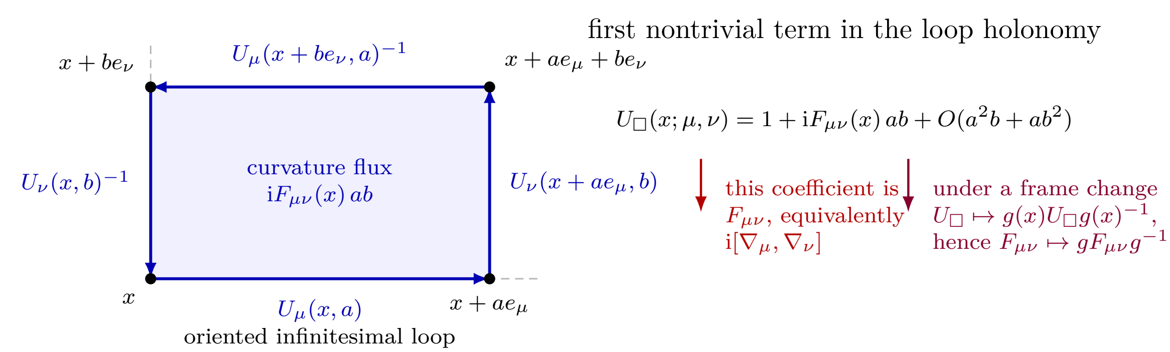


Figure 45.2: Curvature as infinitesimal loop holonomy. Parallel transport around the small rectangle returns a section to the same base point with $U_\square = 1 + \mathrm{i}F_{\mu\nu}ab + O(a^2b + ab^2)$. A change of local frame conjugates the based loop holonomy, so the curvature transforms by $F_{\mu\nu} \mapsto gF_{\mu\nu}g^{-1}$; invariant traces used in the Yang–Mills action remove this conjugation.

An invariant trace is a symmetric bilinear form on $\mathfrak{g}$, written in a chosen representation as

$$\operatorname{tr}(t^a t^b) = \kappa^{ab}. \tag{45.12}$$

Cyclicity gives

$$\operatorname{tr}([t^a, t^b]t^c) = \operatorname{tr}(t^a[t^b, t^c])$$

and, equivalently,

$$\operatorname{tr}(gXg^{-1}gYg^{-1}) = \operatorname{tr}(XY)$$

for Lie-algebra-valued X, Y. It also gives the invariant tensor

$$f_{abc} := \kappa^{ad} f^d{}_{bc} = -\mathrm{i}\ \operatorname{tr}\big(t^a[t^b, t^c]\big), \tag{45.13}$$

which is completely antisymmetric in a, b, c. Here $\kappa^{ab} := \operatorname{tr}(t^a t^b)$, while κ_{ab} denotes the inverse matrix:

$$\kappa_{ac}\kappa^{cb} = \delta_a{}^b. \tag{45.14}$$

Consequently the output-first structure constants are recovered from the trace-lowered constants by

$$f^a{}_{bc} = \kappa_{ad} f_{dbc}. \tag{45.15}$$

The trace normalization used for the Yang–Mills coupling below is $\kappa^{ab} = \delta^{ab}$ unless another normalization is explicitly declared. In that convention $f_{abc} = f^a{}_{bc}$ after the obvious identification of labels. If instead one uses the common fundamental-matrix normalization $\operatorname{tr}(t^a t^b) = \frac{1}{2}\delta^{ab}$, then $\kappa_{ab} = 2\delta_{ab}$ and

$$f^a{}_{bc} = 2\delta_{ad} f_{dbc}.$$

This factor of 2 is a normalization consequence of the chosen trace pairing, not a separate convention for the Lie bracket.

The Lorentzian Yang-Mills Lagrangian density is

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g_{\mathrm{YM}}^2} \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}). \tag{45.16}$$

With the default normalization $\operatorname{tr}(t^a t^b) = \delta^{ab}$, this is

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g_{\mathrm{YM}}^2} F^a_{\mu\nu} F^{a\mu\nu}.$$

Changing the invariant bilinear rescales the component expression and hence the coordinate called g_{YM}; the trace-form equation above is the definition of the coupling convention used in this monograph.

Warning 45.1 (Trace-form coupling convention). The Yang–Mills coupling used in this monograph is defined by the trace-form coefficient in (45.16). Changing from $\operatorname{tr}(t^a t^b) = \delta^{ab}$ to $\operatorname{tr}(t^a t^b) = \frac{1}{2}\delta^{ab}$ changes the component coordinates $F^a_{\mu\nu}$ and the component coordinate called g_{YM}. It is not a legitimate convention change to keep the same component fields, the same trace symbol, and the same coupling while inserting or removing a factor of $1/2$. The invariant datum is the matrix-valued curvature F together with the displayed trace coefficient.

Yang–Mills variation in the trace-form convention. Let

$$S_{\mathrm{YM}}[A] = -\frac{1}{4g_{\mathrm{YM}}^2}\int_M \mathrm{d}^D x\ \mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$$

on a Lorentzian spacetime region M, and assume boundary conditions for which integrations by parts have no boundary contribution. For a variation δA_μ,

$$\delta F_{\mu\nu} = D_\mu \delta A_\nu - D_\nu \delta A_\mu, \tag{45.17}$$

and

$$\delta S_{\mathrm{YM}} = \frac{1}{g_{\mathrm{YM}}^2}\int_M \mathrm{d}^D x\ \mathrm{tr}\big((D_\mu F^{\mu\nu})\delta A_\nu\big). \tag{45.18}$$

If matter coupled to the same connection has current j^ν defined by

$$\delta S_{\mathrm{matter}} = -\int_M \mathrm{d}^D x\ \mathrm{tr}(j^\nu \delta A_\nu), \tag{45.19}$$

then the classical gauge-field equation is

$$D_\mu F^{\mu\nu} = g_{\mathrm{YM}}^2 j^\nu. \tag{45.20}$$

For pure Yang–Mills, $j^\nu = 0$.

Using the definition of $F_{\mu\nu}$,

$$\begin{aligned}\delta F_{\mu\nu} &= \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu - \mathrm{i}[\delta A_\mu, A_\nu] - \mathrm{i}[A_\mu, \delta A_\nu]\\ &= D_\mu \delta A_\nu - D_\nu \delta A_\mu.\end{aligned}$$

Then

$$\delta S_{\mathrm{YM}} = -\frac{1}{2g_{\mathrm{YM}}^2}\int \mathrm{d}^D x\ \mathrm{tr}(F^{\mu\nu}\delta F_{\mu\nu}).$$

Substituting (45.17) and using antisymmetry of $F^{\mu\nu}$ gives

$$\delta S_{\mathrm{YM}} = -\frac{1}{g_{\mathrm{YM}}^2}\int \mathrm{d}^D x\ \mathrm{tr}(F^{\mu\nu} D_\mu \delta A_\nu).$$

The invariant trace satisfies $\mathrm{tr}(X D_\mu Y) = \partial_\mu \mathrm{tr}(XY) - \mathrm{tr}((D_\mu X)Y)$. After dropping the boundary term, this is (45.18). Adding (45.19) and requiring the variation to vanish for arbitrary δA_ν gives (45.20).

For a unitary Lorentzian quantum theory built from these fields, the invariant quadratic form on the gauge algebra is chosen positive definite. Equivalently, the local gauge algebra is a compact reductive Lie algebra: a direct sum of abelian $\mathbb{R}$ factors and compact simple factors. The compact simple factors are the simple Lie algebras of Cartan type

$$A,\ B,\ C,\ D,\ E,\ F,\ G.$$

For these algebras one may choose a linear basis in which

$$\kappa^{ab} = \mathrm{tr}(t^a t^b) = \delta^{ab},$$

which is the normalization used unless another normalization is explicitly declared.

Remark 45.2 (Status of four-dimensional quantum Yang–Mills). The construction in this chapter is classical and local. It supplies the connection, curvature, invariant action, and matter couplings used later in perturbation theory and in regulated path integrals. For a compact simple gauge group in four Euclidean dimensions, the nonperturbative construction of pure Yang–Mills theory on $\mathbb{R}^4$, together with a proof that the resulting quantum theory has a positive mass gap, is the Yang–Mills existence and mass-gap problem of the Clay Millennium list. A theorem at that level would construct continuum correlation functions or an equivalent operator-algebraic theory satisfying the standard structural axioms and would prove that the spectrum above the vacuum begins at a strictly positive mass.

Consequently, the local Lagrangian (45.16), its BRST gauge-fixed perturbation theory, and finite-cutoff lattice regularizations are different mathematical objects from a completed four-dimensional continuum quantum Yang–Mills theory with mass gap. Whenever later chapters use a mass gap, glueball spectrum, or confinement property in four-dimensional Yang–Mills, that input is explicitly a nonperturbative hypothesis, a lattice-supported expectation, or a statement inside a regulated approximation until the continuum existence problem has been solved. The finite-cutoff lattice formulation itself is developed in Chapter 46, where the continuum-limit hypothesis and its relation to covariant perturbation theory are stated separately.

The basic example $SU(2)$ can be written in the default trace convention as

$$t^a = \frac{\sigma^a}{\sqrt{2}}, \qquad a = 1, 2, 3,$$

where σ^a are the Pauli matrices. Then $\operatorname{tr}(t^a t^b) = \delta^{ab}$ and

$$[t^a, t^b] = \mathrm{i}\sqrt{2}\,\epsilon^c{}_{ab} t^c, \qquad f^c{}_{ab} = \sqrt{2}\,\epsilon^c{}_{ab}.$$

The frequently used basis $\sigma^a/2$ is the alternate half-trace coordinate system discussed above; it is not the default normalization of this monograph. More generally, the Hermitian coordinate space $i\,\mathfrak{su}(N)$ is represented by traceless Hermitian $N \times N$ matrices, and

$$SU(N) = \{g \in U(N) : \det g = 1\}.$$

In four spacetime dimensions there is another local gauge-invariant Lorentz scalar of power-counting dimension four, written in the trace convention just fixed.[1]

$$\Delta\mathcal{L}_\theta = \frac{\theta}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(F_{\mu\nu}F_{\rho\sigma}). \tag{45.21}$$

With $\operatorname{tr}(t^a t^b) = \delta^{ab}$, this equals

$$\frac{\theta}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}F^a_{\mu\nu}F^a_{\rho\sigma}.$$

Authors using a basis with $\operatorname{tr}(t^a t^b) = \frac{1}{2}\delta^{ab}$ write the same trace-form density with half this component coefficient, because their components of $F_{\mu\nu}$ are rescaled relative to the basis used

[1] With $F_{\mu\nu} = F^a_{\mu\nu}t^a$ and the default $\operatorname{tr}(t^a t^b) = \delta^{ab}$, $\operatorname{tr}(F_{\mu\nu}F_{\rho\sigma}) = F^a_{\mu\nu}F^a_{\rho\sigma}$. If one instead uses generators normalized by $\operatorname{tr}(t^a t^b) = \frac{1}{2}\delta^{ab}$, the components of the same matrix-valued curvature and the component form of the coupling must be rescaled accordingly. The trace-form equations are the convention-independent statements. Here and below the trace is the actual matrix trace in the chosen representation applied to the matrix-valued curvature F; it is not an independent component contraction that may be rescaled while F is held fixed.

here. The normalization is fixed by the Chern–Weil integer. For an $SU(N)$ bundle over an oriented compact four-manifold M, with

$$F = \frac{1}{2} F_{\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu,$$

the number

$$Q(P, A) := \frac{1}{8\pi^2} \int_M \mathrm{tr}(F \wedge F) = \frac{1}{32\pi^2} \int_M \mathrm{d}^4x\, \epsilon^{\mu\nu\rho\sigma}\, \mathrm{tr}(F_{\mu\nu} F_{\rho\sigma}) \in \mathbb{Z} \tag{45.22}$$

is the integral characteristic number determined by the second Chern class, with the overall sign fixed by the Hermitian curvature convention. For finite-action $SU(N)$ fields on $\mathbb{R}^4$, compactification by the point at infinity gives the same integer; equivalently, it is the degree of the limiting pure gauge map $S^3_\infty \to SU(N)$. This density is odd under parity and time reversal. In local coordinates it is a total derivative, so it does not change the local Euler-Lagrange equations under boundary conditions that keep the boundary variation fixed. Perturbation theory around the trivial gauge-field sector consequently has no local Feynman vertex associated with (45.21). Finite-action gauge fields may approach nontrivial pure gauges at spatial infinity, and then the spacetime integral of (45.21) equals $\theta Q(P, A)$. Thus θ enters through global topological sectors of the gauge field, and the Lorentzian path-integral phase is invariant under $\theta \mapsto \theta + 2\pi$. The explicit $Q = 1$ $SU(2)$ BPST representative, its action saturation, zero modes, and the resulting fermionic instanton vertex are constructed in Section 51.17.

45.3 Higher-Form Symmetry Data in Gauge Theory

Ordinary internal global symmetries act on local operators. Gauge theories also contain symmetries whose charged objects are extended operators. The basic example needed later is the center one-form symmetry of pure Yang–Mills theory. We record the definition here because it fixes the meaning of phrases such as center charge, screening, string breaking, and Wilson-loop order parameter in the QCD chapters.

Definition 45.3 (Invertible p-form symmetry, operational form)**.** Let the spacetime dimension be D, and let A be an abelian group. An invertible p-form global symmetry with group A, in the sector of the theory under discussion, consists of topological operators

$$U_a(\Sigma^{D-p-1}), \qquad a \in A,$$

assigned to closed oriented submanifolds Σ^{D-p-1} of codimension $p+1$, together with p-dimensional charged operators $\mathcal{W}_q(C^p)$. The operators satisfy

$$U_a(\Sigma) U_b(\Sigma) = U_{a+b}(\Sigma), \qquad U_0(\Sigma) = 1,$$

and are invariant under smooth deformations of Σ that do not cross charged insertions. If Σ links C, the symmetry action is

$$U_a(\Sigma)\, \mathcal{W}_q(C)\, U_a(\Sigma)^{-1} = \chi_q(a)^{\mathrm{Lk}(\Sigma, C)} \mathcal{W}_q(C), \tag{45.23}$$

where $\chi_q : A \to U(1)$ is the character carried by $\mathcal{W}_q$, and $\mathrm{Lk}(\Sigma, C)$ is the oriented linking number when the dimensions are complementary as above.

For $p = 0$, the topological operator is codimension one and separates space into two regions; moving it past a local operator implements the ordinary global symmetry action on that local operator. For $p = 1$ in four dimensions, $U_a(\Sigma)$ is a surface operator, and the charged objects are line operators. Invertibility means that the symmetry operators have inverses. More general topological defects may have non-group-like fusion; those categorical and noninvertible structures are part of the later global structure volume, after line and surface operators have been developed as QFT observables.

Consider pure $SU(N)$ Yang–Mills theory with no dynamical matter in representations of nonzero N-ality. The center is

$$Z(SU(N)) = \{\mathrm{e}^{2\pi \mathrm{i} m/N}\mathbf{1}_N : m \in \mathbb{Z}_N\}.$$

All local gauge-invariant operators built from $F_{\mu\nu}$, covariant derivatives, and adjoint fields are neutral under this center, because the adjoint representation has zero N-ality. A Wilson line in a representation R,

$$W_R(C) = \mathrm{tr}_R\, P \exp\left\{\mathrm{i}\int_C A_\mu\, \mathrm{d}x^\mu\right\},$$

has center charge $k_R \in \mathbb{Z}_N$, defined by

$$z\,\mathbf{1}_N \text{ acts on } R \text{ as } z^{k_R}.$$

The electric center one-form symmetry is the $\mathbb{Z}_N$ one-form symmetry whose charged operators are precisely these Wilson lines.

One concrete construction of the symmetry surface is as follows. Let Σ be a closed oriented two-surface in four-dimensional spacetime, and choose a three-chain V with $\partial V = \Sigma$ locally. Across V, perform a gauge transformation which is single-valued up to the center element $z_m = \mathrm{e}^{2\pi \mathrm{i} m/N}\mathbf{1}_N$. Adjoint fields and the connection in the adjoint bundle are unchanged as local gauge-invariant data; a Wilson line crossing V acquires the center phase seen by its representation. The resulting topological surface operator $U_m(\Sigma)$ obeys

$$U_m(\Sigma)W_R(C)U_m(\Sigma)^{-1} = \exp\left\{\frac{2\pi \mathrm{i}\, m k_R}{N}\,\mathrm{Lk}(\Sigma, C)\right\} W_R(C). \tag{45.24}$$

This equation is the definition of the center charge used later. The topological character of $U_m(\Sigma)$ says that deforming Σ while keeping it away from C leaves the correlation function unchanged; crossing C changes the linking number and produces exactly the displayed phase.

Dynamical matter determines the unbroken subgroup. Suppose the theory contains matter fields in representations R_α of $SU(N)$, with N-alities k_α. A Wilson line in R_α can end on a dynamical local field only when that field is present in the operator algebra. Then a center surface linked with the line endpoint fails to define a topological operation on the full set of insertions unless its phase on every dynamical endpoint is trivial. The surviving electric one-form symmetry is

$$A_{\mathrm{el}} = \{m \in \mathbb{Z}_N : \exp(2\pi \mathrm{i} m k_\alpha/N) = 1 \text{ for every dynamical } R_\alpha\}. \tag{45.25}$$

Thus pure Yang–Mills has $A_{\mathrm{el}} = \mathbb{Z}_N$, adjoint matter preserves the same electric center one-form symmetry, and dynamical fundamental quarks break it completely.

The behavior of large Wilson loops is a dynamical question, but the symmetry data determine what can serve as a sharp diagnostic. In a massive pure Yang–Mills theory, a Wilson loop with nonzero N-ality is charged under A_{el}, so its area or perimeter behavior distinguishes phases of the one-form symmetry. With dynamical fundamental quarks the same loop can be screened by pair creation; the QCD chapter therefore treats the Wilson loop as a static-source observable, while the one-form-symmetry order-parameter interpretation applies to the unbroken center subgroup.

Remark 45.4 (Higher groups, noninvertible symmetries, and placement). The definition above covers the invertible higher-form symmetry needed for the center symmetry of pure Yang–Mills. More refined global structures include mixed ordinary/higher-form symmetries organized by higher groups, background $p+1$-form gauge fields and their Postnikov constraints, and noninvertible topological defects with categorical fusion. Those structures are QFT data about extended operators and background couplings, and they are conceptually distinct from the local gauge redundancy used to coordinatize the field space. They belong to the later volume on global structure, phases, and extended operators. The present gauge-theory volume uses only the definitions required for Wilson lines, screening, confinement diagnostics, and anomaly statements.

45.4 Matter Representations

Let R be a finite-dimensional representation of G on a vector space V_R. The matter fields below use complex V_R; real representations, such as the adjoint representation, are included by complexification when they are compared with complex representations. The representation is a homomorphism

$$\rho_R : G \longrightarrow GL(V_R), \qquad \rho_R(gh) = \rho_R(g)\rho_R(h). \tag{45.26}$$

Its derived Lie-algebra representation is the linear map $\widetilde{\rho}_R : \mathfrak{g} \to \mathrm{End}(V_R)$ determined by

$$\rho_R(\mathrm{e}^{\mathrm{i}\zeta^a t^a}) = \mathrm{e}^{\mathrm{i}\zeta^a t^a_R}, \qquad t^a_R := \widetilde{\rho}_R(t^a). \tag{45.27}$$

The matrices t^a_R obey

$$[t^a_R, t^b_R] = \mathrm{i} f^c{}_{ab} t^c_R. \tag{45.28}$$

For compact G, every finite-dimensional complex representation used in a unitary theory may be equipped with a G-invariant Hermitian inner product. In such a basis the matrices t^a_R are Hermitian. This is the matter-field version of the same Hermitian convention used for the gauge potential.

For $G = SU(N)$, three representations recur throughout gauge theory:

representation	V_R	infinitesimal generators
fundamental $\square$	$\mathbb{C}^N$	$t^a_\square = t^a$
anti-fundamental $\overline{\square}$	$\mathbb{C}^N$	$t^a_{\overline{\square}} = -(t^a)^*$
adjoint	$\mathfrak{g}_\mathbb{C}$	$t^a_{\mathrm{adj}}(t^b) = \mathrm{i} f^c{}_{ab} t^c$

In the fundamental representation, $\rho_\square(g) = g$ and $t^a_\square = t^a$. In the anti-fundamental representation,

$$\rho_{\overline{\square}}(g) = g^* \qquad \text{complex conjugation, without transposition,}$$

so differentiating $\rho_{\overline{\square}}(\mathrm{e}^{\mathrm{i}\zeta^a t^a}) = \mathrm{e}^{\mathrm{i}\zeta^a t^a_{\overline{\square}}}$ gives $t^a_{\overline{\square}} = -(t^a)^*$. Here $\mathfrak{g}_\mathbb{C}$ denotes the complexification of the real Hermitian coordinate space. A real adjoint-valued field is obtained by imposing the corresponding reality condition on its components. The finite adjoint representation is conjugation on the algebra:

$$v \mapsto gvg^{-1}, \qquad \rho_{\mathrm{adj}}(g)v = gvg^{-1}.$$

Its infinitesimal form is consistent with the convention differentiating $v \mapsto e^{i\zeta^a t^a} v e^{-i\zeta^a t^a}$:

$$\delta v = i\zeta^a [t^a, v] = i\zeta^a t^a_{\mathrm{adj}} v.$$

Figure 45.3 gathers the $SU(N)$ representation data used in the matter-coupling formulas: fundamental, antifundamental, adjoint, and tensor-product conventions.

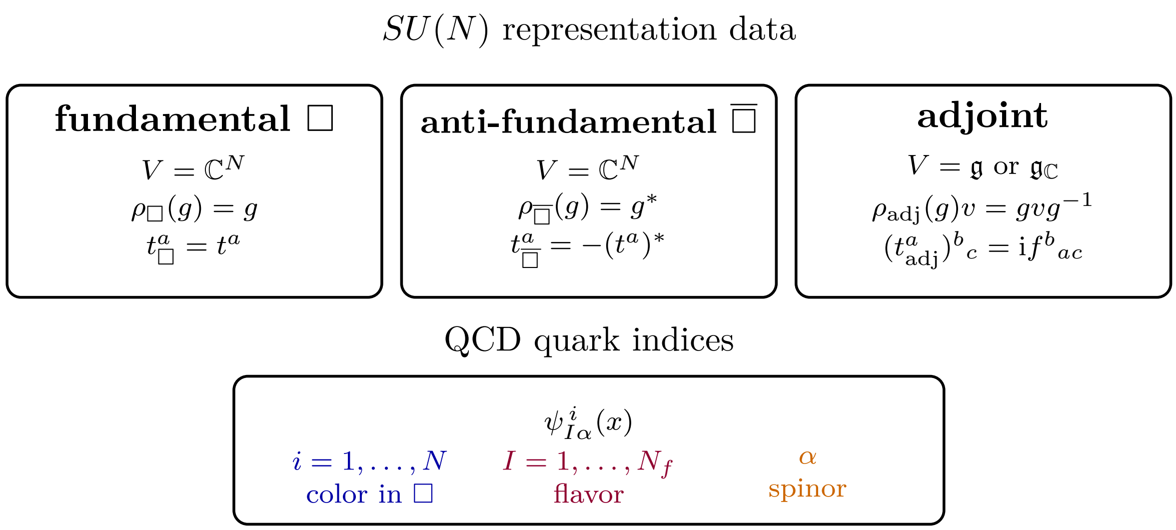


Figure 45.3: The fundamental, anti-fundamental, and adjoint representations of $SU(N)$ used in Yang-Mills theory and QCD. The anti-fundamental finite action is complex conjugation of the matrix entries, not Hermitian conjugation.

If $\psi(x) \in V_R$, its finite gauge transformation is

$$\psi'(x) = \rho_R(g(x))\psi(x). \tag{45.29}$$

The covariant derivative acting on ψ is

$$D_\mu \psi = \partial_\mu \psi - iA^a_\mu t^a_R \psi. \tag{45.30}$$

Using (45.4) and (45.26), one obtains

$$(D_\mu \psi)' = \rho_R(g) D_\mu \psi. \tag{45.31}$$

This covariance is the algebraic reason that replacing ordinary derivatives by D_μ produces gauge-invariant matter kinetic terms. For example, if $(\ , \ )_R$ is the invariant Hermitian pairing on V_R, then $(D_\mu\psi, D^\mu\psi)_R$ is gauge invariant for scalar matter. For Dirac matter, the gauge index contraction in $\bar{\psi}\gamma^\mu D_\mu \psi$ uses the pairing between R^* and R. Mass and Yukawa couplings are gauge invariant only when their flavor and representation tensors define intertwiners from the tensor product of fields to the trivial representation.

Figure 45.4 displays the geometric meaning of the same covariance statement. The matter field is a section of the associated bundle $P \times_G V_R$. Parallel transport along a curve $\gamma : x \to y$ is the representation R applied to the gauge-field parallel transport, and infinitesimal loop transport acts on matter by the represented curvature:

$$U_R(\partial\square)\psi(x) = \psi(x) + iF^a_{\mu\nu}(x) t^a_R \psi(x)\, ab + O(a^2 b + ab^2).$$

The invariant pairings used in kinetic, mass, and Yukawa terms are pairings of associated fibers.

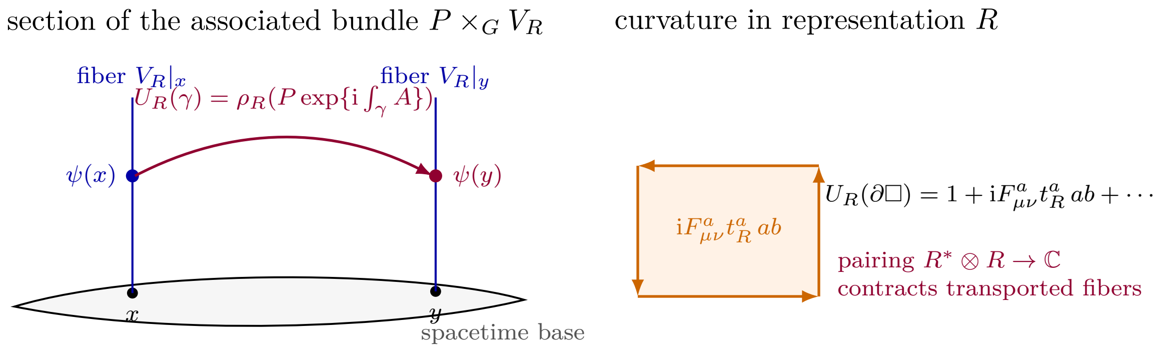


Figure 45.4: Matter covariance as associated-bundle parallel transport. A section value $\psi(x) \in V_R|_x$ is transported by applying the representation R to the gauge holonomy. Around an infinitesimal loop the first nontrivial term is the represented curvature $F^a_{\mu\nu}t^a_R$. Gauge-invariant matter contractions are pairings of associated fibers, for instance $R^* \otimes R \to \mathbb{C}$.

45.5 The Brout–Englert–Higgs Mechanism

Gauge redundancy is not a physical global symmetry acting faithfully on the Hilbert space. A gauge-theory phase with a scalar expectation value is therefore not defined by a gauge-variant order parameter. Its perturbative description is defined by choosing a coordinate chart on field space near a minimum of the gauge-invariant potential, then fixing the gauge or passing to gauge-invariant variables. The Brout–Englert–Higgs mechanism is the local quadratic statement that scalar fluctuations tangent to the gauge orbit supply the longitudinal polarizations of gauge bosons whose generators do not stabilize the chosen vacuum. The physical formulation is spectral: the BRST-invariant or gauge-invariant Hilbert space contains massive vector particles and scalar excitations, while the would-be Goldstone coordinates are not independent physical massless particles.

45.5.1 Abelian Higgs Model

Use the coupling-absorbed Abelian convention inherited from (45.6). Thus

$$D_\mu\phi = \partial_\mu\phi - \mathrm{i}A_\mu\phi,$$

$$\mathcal{L} = -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^* D^\mu\phi - \lambda\left(|\phi|^2 - \frac{v^2}{2}\right)^2. \tag{45.32}$$

The finite gauge transformation is

$$A_\mu \mapsto A_\mu + \partial_\mu\alpha, \qquad \phi \mapsto e^{\mathrm{i}\alpha}\phi.$$

Choose a local Cartesian coordinate chart near the minimum

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x) + \mathrm{i}\pi(x)\big).$$

The scalar kinetic term gives, to quadratic order,

$$(D_\mu\phi)^* D^\mu\phi = \frac{1}{2}\partial_\mu h\,\partial^\mu h + \frac{1}{2}(\partial_\mu\pi - vA_\mu)(\partial^\mu\pi - vA^\mu) + O(\text{cubic}). \tag{45.33}$$

The potential gives

$$\lambda\left(|\phi|^2 - \frac{v^2}{2}\right)^2 = \lambda v^2 h^2 + O(\text{cubic}), \qquad m_h^2 = 2\lambda v^2.$$

The gauge field A_μ is not canonically normalized; the canonical vector is $\mathcal{A}_\mu = A_\mu/e$. Therefore the vector mass is

$$m_A = ev.$$

Equation (45.33) displays the mechanism: the derivative of the orbit coordinate π appears only through the combination $\partial_\mu\pi - vA_\mu$ at quadratic order.

On the open set where $v + h \neq 0$, write polar coordinates

$$\phi(x) = \frac{v + H(x)}{\sqrt{2}}\exp\!\left(\frac{\mathrm{i}\theta(x)}{v}\right).$$

The gauge-invariant local vector coordinate

$$B_\mu := A_\mu - \frac{1}{v}\partial_\mu\theta \tag{45.34}$$

is invariant under $A_\mu \mapsto A_\mu + \partial_\mu\alpha$, $\theta \mapsto \theta + v\alpha$. In these coordinates

$$(D_\mu\phi)^* D^\mu\phi = \frac{1}{2}\partial_\mu H\,\partial^\mu H + \frac{1}{2}(v + H)^2 B_\mu B^\mu.$$

This is the unitary-gauge description: the orbit coordinate has been removed from the local chart and the vector has a mass term. It is useful for classical mode counting, but its vector propagator has poor ultraviolet power-counting behavior. Perturbative renormalizability is most transparent in R_ξ gauges.

The R_ξ gauge-fixing functional in the coupling-absorbed convention is

$$\mathcal{F}_\xi = \partial_\mu A^\mu + \xi e^2 v\pi. \tag{45.35}$$

Adding

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2e^2\xi}\mathcal{F}_\xi^2$$

cancels the quadratic mixing term $v(\partial_\mu A^\mu)\pi$ obtained from (45.33) after integrating by parts. The quadratic Lagrangian becomes

$$\mathcal{L}_2 = -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2e^2\xi}(\partial_\mu A^\mu)^2 + \frac{1}{2}v^2 A_\mu A^\mu + \frac{1}{2}(\partial h)^2 - \frac{1}{2}m_h^2 h^2 + \frac{1}{2}(\partial\pi)^2 - \frac{1}{2}\xi m_A^2\pi^2.$$

In terms of the canonical vector $\mathcal{A}_\mu = A_\mu/e$, the propagator is

$$D_{\mu\nu}(p) = \frac{-\mathrm{i}}{p^2 - m_A^2 + \mathrm{i}0}\left[\eta_{\mu\nu} - \frac{(1-\xi)p_\mu p_\nu}{p^2 - \xi m_A^2 + \mathrm{i}0}\right]. \tag{45.36}$$

It decays as p^{-2} at large momentum for fixed ξ. The field π and the ghosts have gauge-dependent mass $\sqrt{\xi}\, m_A$; they are not physical particles. The Faddeev–Popov operator follows by varying (45.35) under the infinitesimal gauge transformation. Since

$$\delta_\alpha A_\mu = \partial_\mu \alpha, \qquad \delta_\alpha \pi = (v+h)\alpha + O(\pi\alpha),$$

one obtains

$$\delta_\alpha \mathcal{F}_\xi = \left(\Box + \xi e^2 v(v+h)\right)\alpha + O(\pi\alpha).$$

Thus the ghost action contains the quadratic mass ξm_A^2 and a local ghost–ghost–h vertex. In an Abelian unbroken gauge theory ghosts decouple; in the R_ξ chart of the Abelian Higgs model this field dependence of the gauge condition makes their BRST role visible.

Figure 45.5 separates the two pieces of the Abelian BEH calculation. The angular coordinate on the circle of scalar minima is a gauge-orbit coordinate, and the physical local mode count is the same before and after rewriting the fields.

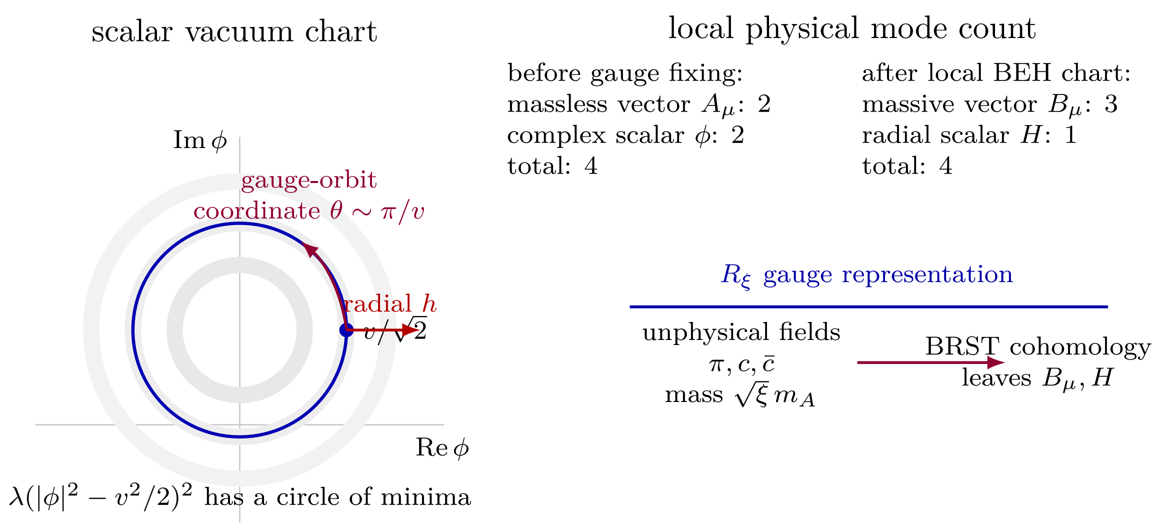


Figure 45.5: Abelian BEH geometry and mode accounting. The angular coordinate along the circle of minima is a gauge-orbit coordinate; locally it is the would-be Goldstone field. The local field variables reorganize $2+2$ degrees of freedom into a massive vector with three physical polarizations and one radial scalar. R_ξ gauges keep the orbit coordinate and ghosts as gauge-dependent representatives, but the BRST cohomology contains the same physical modes.

45.5.2 Nonabelian Mass Matrix and the $SU(2) \to U(1)$ Pattern

Let a scalar field Φ take values in a unitary representation V_R of a compact gauge group G, and let $v \in V_R$ be a minimum of a gauge-invariant potential. The stabilizer subgroup is

$$H = \{g \in G : \rho_R(g)v = v\}, \qquad \mathfrak{h} = \{X \in \mathfrak{g} : t_R(X)v = 0\}.$$

The tangent space to the gauge orbit at v is

$$T_v(G \cdot v) = \{\, \mathrm{i} t_R(X) v : X \in \mathfrak{g}\,\} \simeq \mathfrak{g}/\mathfrak{h}.$$

It is useful to isolate the finite-dimensional linear map

$$q_v : \mathfrak{g} \longrightarrow T_v V_R, \qquad q_v(X) = \mathrm{i}t_R(X)v. \tag{45.37}$$

The kernel of q_v is $\mathfrak{h}$. Thus q_v is the differential of the gauge-orbit map at the chosen vacuum, and all local mode counting in the Brout–Englert–Higgs mechanism is the mode counting of this linear map. Expanding the scalar kinetic term around $\Phi = v + \eta$ gives the gauge-boson mass bilinear

$$(D_\mu v, D^\mu v)_R = A^a_\mu A^{b\mu}\,(t^a_R v, t^b_R v)_R. \tag{45.38}$$

The Hermitian matrix

$$K_{ab} := \mathrm{Re}(t^a_R v, t^b_R v)_R \tag{45.39}$$

is positive semidefinite, and its kernel is exactly the unbroken Lie algebra:

$$K_{ab}X^b = 0 \quad \Longleftrightarrow \quad t_R(X)v = 0 \quad \Longleftrightarrow \quad X \in \mathfrak{h}.$$

Equivalently, $K = q_v^\dagger q_v$, where the adjoint is taken with respect to the invariant inner product on V_R and the invariant trace form on $\mathfrak{g}$.

Proposition 45.5 (Local BEH mode count)**.** *At a smooth vacuum v with compact stabilizer H, the scalar kinetic term gives nonzero masses to exactly* $\dim G - \dim H$ *gauge bosons. In four-dimensional Minkowski space these massive vector particles carry* $3(\dim G - \dim H)$ *physical polarizations. The scalar coordinates removed from a local unitary-gauge slice are exactly the tangent coordinates to the orbit G/H. The remaining scalar coordinates are coordinates normal to the gauge orbit, subject to the Hessian of the gauge-invariant potential on a local slice.*

Proof. Choose an H-invariant orthogonal decomposition of scalar fluctuations near v,

$$T_v V_R = \mathrm{im}\, q_v \oplus N_v,$$

where N_v is the orthogonal complement of the orbit tangent space. The existence of such a complement follows by averaging the Hermitian inner product over the compact group H. To quadratic order in fields,

$$D_\mu(v + \eta) = \partial_\mu \eta - q_v(A_\mu) + O(A_\mu \eta), \qquad A_\mu = A^a_\mu t^a.$$

If the orbit component of η is written as $q_v(\chi)$, then the quadratic expression depends on A_μ and χ through

$$\partial_\mu q_v(\chi) - q_v(A_\mu) = q_v(\partial_\mu \chi - A_\mu).$$

This identity is the precise local statement behind the phrase that the would-be Goldstone coordinate supplies the longitudinal vector polarization: the scalar orbit coordinate and the gauge potential appear in the same gauge-orbit differential. Since $\mathrm{rank}\, q_v = \dim \mathfrak{g} - \dim \mathfrak{h}$, exactly that many vector fields have nonzero mass eigenvalues. A massive vector particle in four spacetime dimensions has three physical polarizations after the BRST or gauge-invariant quotient. The normal scalar coordinates in N_v are not gauge-orbit coordinates; their quadratic masses are the eigenvalues of the potential Hessian restricted to a local slice transverse to the orbit. $\square$

After canonical normalization of the gauge kinetic term, the vector mass-squared matrix is $2g_{\mathrm{YM}}^2 K$, with the invariant trace form used to identify gauge-algebra indices. Hence precisely $\dim G - \dim H$ gauge bosons acquire mass. The same number of scalar coordinates tangent to G/H are gauge-orbit coordinates. A covariant R_ξ gauge keeps these coordinates in the field algebra but gives them gauge-dependent masses; a unitary-gauge chart removes them locally.

The nonabelian R_ξ construction is the adjoint of the same orbit differential. In a local linear chart, let $q_v^\dagger \eta \in \mathfrak{g}$ be the adjoint projection of the scalar fluctuation onto the broken gauge directions. The gauge-fixing functional

$$\mathcal{F}_\xi = \partial^\mu A_\mu + \xi g_{\mathrm{YM}}^2 q_v^\dagger \eta \tag{45.40}$$

is the nonabelian analogue of (45.35). Its quadratic cross term cancels the integration-by-parts mixing from $\|\partial_\mu \eta - q_v(A_\mu)\|^2$. In a basis of broken generators diagonalizing K, (45.40) becomes a direct sum of Abelian R_ξ gauges, with unphysical scalar and ghost masses $\sqrt{\xi}$ times the corresponding vector masses. The unbroken $\mathfrak{h}$ directions have no $q_v^\dagger \eta$ term and remain ordinary massless gauge directions at this stage.

For the adjoint $SU(2)$ model, take an adjoint scalar $\Phi = \Phi^a t^a$ and a vacuum $\Phi_0 = v t^3$. The stabilizer of Φ_0 is the $U(1)$ generated by t^3. The directions t^1, t^2 do not stabilize the vacuum, so the corresponding two gauge bosons are massive and the t^3 gauge boson remains massless. In electroweak $SU(2) \times U(1)$ with a fundamental scalar doublet, the same mass-matrix logic leaves the diagonal electromagnetic $U(1)$ unbroken and gives masses to the orthogonal charged and neutral vector combinations. The detailed electroweak parameter conventions belong to the Standard Model chapter; the structural statement is already contained in (45.38).

The perturbative renormalizability statement is also local in field space. An R_ξ gauge-fixed action for the shifted fields contains only local operators of power-counting dimension at most four when the original Yang–Mills–scalar action is renormalizable. The propagators have the large-momentum behavior required for power counting, and the BRST identities restrict counterterms to gauge-invariant deformations together with BRST-exact gauge-fixing and field-coordinate terms, as in Chapter 47. Unitary gauge is a useful reduced coordinate description of the classical spectrum, but it is not the gauge in which the renormalizability proof is manifest.

45.6 Finite-Energy Gauge-Higgs Solitons

The finite-energy configuration space of a nonabelian gauge-Higgs theory has connected components, asymptotic winding data, and nontrivial critical points of the energy functional. A classical soliton in this section is such a finite-energy critical point, together with its boundary sector and its localized energy density. Quantizing small fluctuations about such a critical point gives a semiclassical expansion with different background data from the translation-invariant vacuum expansion.

We formulate the discussion in the same Hermitian-generator convention and trace-form coupling convention used above. Let G be compact, let $\Phi(x)$ be an adjoint scalar field, and let

$$D_i \Phi := \partial_i \Phi - \mathrm{i}[A_i, \Phi], \qquad B_i := \frac{1}{2}\varepsilon_{ijk} F_{jk}$$

on a time slice $\mathbb{R}^3$. The index $i = 1, 2, 3$ is spatial and $\varepsilon_{123} = 1$. A simple gauge-Higgs energy functional is

$$E[A,\Phi] = \frac{1}{2g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\ \mathrm{tr}\big(B_iB_i + D_i\Phi D_i\Phi\big) + \frac{1}{g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\, U(\Phi), \tag{45.41}$$

where $U(\Phi) \geq 0$ is gauge invariant. For example, for $G = SU(2)$ one may take

$$U(\Phi) = \frac{\lambda}{4}\big(\mathrm{tr}\,\Phi^2 - v^2\big)^2, \qquad v > 0, \quad \lambda \geq 0.$$

The scalar Φ and the connection A_i are in the coupling-absorbed normalization of this chapter. If one instead writes the covariant derivative as $\partial_i - \mathrm{i}g_{\mathrm{phys}}A_i^{\mathrm{phys}}$, then $A_i = g_{\mathrm{phys}}A_i^{\mathrm{phys}}$; the trace-form energy (45.41) is the convention used below.

Definition 45.6 (Smooth finite-energy sector)**.** Fix a vacuum orbit

$$\mathcal{O}_v := \{\Phi \in \mathfrak{g} : U(\Phi) = 0\}.$$

In the smooth classical sector used in this section, a static configuration (A_i, Φ) is finite energy if the integral (45.41) is finite and if, after a gauge choice on the complement of a compact set,

$$U(\Phi(x)) \to 0, \qquad D_i\Phi(x) \to 0, \qquad B_i(x) \to 0 \quad \text{fast enough as } |x| \to \infty$$

to make the surface integrals below finite. The limiting Higgs field defines a map

$$\Phi_\infty : S^2_\infty \longrightarrow \mathcal{O}_v.$$

Two configurations are in the same asymptotic sector if their maps Φ_∞ are homotopic through finite-energy boundary data, modulo gauge transformations which extend over the bulk.

A more analytic treatment replaces smooth fields by Sobolev completions and states the boundary conditions as weighted norm conditions. The topological classification and the Bogomolny argument below are already visible in the smooth sector, where all integrations by parts can be justified by decay and then extended by completion when the appropriate functional-analytic hypotheses are imposed.

Let $\Phi_0 \in \mathcal{O}_v$ and let

$$H := \mathrm{Stab}_G(\Phi_0) = \{h \in G : h\Phi_0h^{-1} = \Phi_0\}.$$

For an adjoint orbit $\mathcal{O}_v \simeq G/H$, the asymptotic Higgs field is a map $S^2_\infty \to G/H$. Hence possible magnetic sectors are measured by $\pi_2(G/H)$. When G is simply connected, the long exact homotopy sequence of

$$H \longrightarrow G \longrightarrow G/H$$

gives

$$\pi_2(G/H) \simeq \pi_1(H),$$

because $\pi_2(G) = 0$ for compact connected Lie groups and $\pi_1(G) = 0$ for simply connected G. Thus the magnetic charge is the winding of the transition function of the unbroken group H seen at spatial infinity.

Example 45.7 $(SU(2) \to U(1))$**.** For $G = SU(2)$ and a nonzero adjoint vacuum value, the stabilizer is $H = U(1)$, so

$$\mathcal{O}_v \simeq SU(2)/U(1) \simeq S^2, \qquad \pi_2(\mathcal{O}_v) \simeq \mathbb{Z}.$$

The integer is the degree of the normalized asymptotic Higgs field

$$\widehat{\Phi}_\infty : S^2_\infty \to S^2, \qquad \widehat{\Phi}_\infty(\hat{x}) = \Phi_\infty(\hat{x})/v$$

when $\operatorname{tr}\Phi_\infty^2 = v^2$. This integer is invariant under gauge transformations which extend over the bulk. In a gauge where Φ_∞ is made constant separately on the northern and southern hemispheres of S^2_∞, the two gauges differ on the equator by a $U(1)$-valued transition function of the same winding.

Figure 45.6 displays the finite-energy boundary data for the monopole sector: the normalized Higgs field maps S^2_∞ to the vacuum sphere, and its equatorial transition function records the magnetic winding.

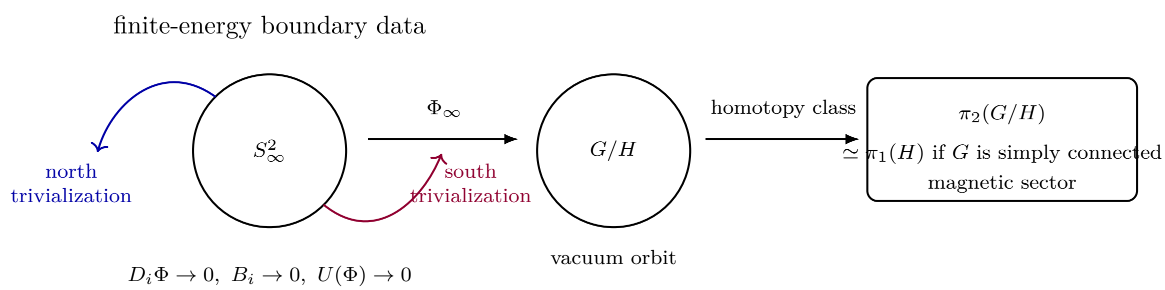


Figure 45.6: Finite energy turns spatial infinity into a two-sphere mapped to the vacuum orbit. For $SU(2) \to U(1)$, the degree of this map is the magnetic charge of the smooth monopole sector.

The Bogomolny bound is the simplest place where the topology enters the energy in addition to labeling boundary data. Set $U = 0$. This is the Prasad–Sommerfield or BPS limit of the adjoint Higgs model. Completing the square gives

$$E[A,\Phi] = \frac{1}{2g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\ \operatorname{tr}\big(B_i \mp D_i\Phi\big)^2 \pm \frac{1}{g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\ \operatorname{tr}(B_i D_i \Phi). \tag{45.42}$$

The Bianchi identity $D_iB_i = 0$ implies

$$\operatorname{tr}(B_iD_i\Phi) = \partial_i \operatorname{tr}(\Phi B_i).$$

Therefore

$$E[A,\Phi] \geq \frac{4\pi}{g_{\mathrm{YM}}^2}|\Gamma_{\mathrm{m}}(A,\Phi)|, \qquad \Gamma_{\mathrm{m}}(A,\Phi) := \frac{1}{4\pi}\int_{S^2_\infty} \mathrm{d}S_i\ \operatorname{tr}(\Phi B_i). \tag{45.43}$$

The number Γ_{m} is the magnetic central charge seen by the chosen asymptotic Higgs field. It is the trace pairing of the Higgs vacuum with the magnetic cocharacter determined by the $U(1)$ bundle at infinity. In a primitive $SU(2) \to U(1)$ sector one may write $\Gamma_{\mathrm{m}} = v_{\mathrm{m}}n$, $n \in \mathbb{Z}$, where v_{m} is the trace pairing of Φ_∞ with the primitive magnetic generator. The value of v_{m} changes if one rescales the matrix generator basis; the trace-form equation (45.43) does not. The inequality is saturated precisely when

$$B_i = \pm D_i \Phi. \tag{45.44}$$

These first-order equations imply the second-order Euler–Lagrange equations in the BPS limit: applying D_i to $B_i = \pm D_i\Phi$ gives $D_i D_i \Phi = 0$, while the variation of A_i gives $D_j F_{ji} = \mathrm{i}[\Phi, D_i\Phi]$, which follows by differentiating $B_i = \pm D_i\Phi$, using the Bianchi identity, and using $[D_i, D_j]\Phi = -\mathrm{i}[F_{ij}, \Phi]$.

Proposition 45.8 (Topological lower bound for adjoint $SU(2)$ monopoles)**.** *Consider smooth finite-energy $SU(2)$ adjoint-Higgs configurations in the BPS limit $U = 0$. Fix the asymptotic Higgs value and the primitive magnetic cocharacter so that $\Gamma_{\mathrm{m}} = v_{\mathrm{m}} n$ on the sector of degree $n \in \mathbb{Z}$. Then*

$$E \geq \frac{4\pi v_{\mathrm{m}}}{g_{\mathrm{YM}}^2}|n|.$$

Equality holds if and only if the Bogomolny equation (45.44) holds with the sign chosen to match the sign of n.

Proof. The square completion (45.42) gives, for either sign,

$$E = \frac{1}{2g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\ \mathrm{tr}(B_i \mp D_i\Phi)^2 \pm \frac{1}{g_{\mathrm{YM}}^2}\int_{\mathbb{R}^3} \mathrm{d}^3x\ \mathrm{tr}(B_i D_i \Phi).$$

The first term is nonnegative because the trace form is positive on the compact real Lie algebra. For the second term,

$$\mathrm{tr}(B_i D_i\Phi) = \partial_i\,\mathrm{tr}(B_i\Phi) - \mathrm{tr}((D_iB_i)\Phi).$$

The covariant Bianchi identity gives $D_iB_i = 0$, so the cross term is the surface integral in (45.43).

It remains to identify that surface integral with the magnetic cocharacter. Finite energy implies that, on the sphere at infinity, Φ has fixed length and $D_i\Phi \to 0$. Hence the structure group of the asymptotic bundle reduces to the stabilizer $U(1)$ of the limiting Higgs direction. Choose northern and southern patches on S^2_∞ in which Φ_∞ is gauge rotated to a fixed Cartan element H. On their equatorial overlap the transition map is

$$g_{NS}(\varphi) = \exp(\mathrm{i}n\varphi H_{\mathrm{m}})$$

for the primitive magnetic generator H_{m} and an integer winding n. In these patches the asymptotic connection is Abelian in the same Cartan direction, and Stokes' theorem gives a flux proportional to this winding:

$$\frac{1}{4\pi}\int_{S^2_\infty} \mathrm{d}S_i\ \mathrm{tr}(\Phi B_i) = \left[\frac{1}{4\pi}\int_{S^2_\infty}^{\text{primitive}} \mathrm{d}S_i\ \mathrm{tr}(\Phi B_i)\right] n = v_{\mathrm{m}} n.$$

The bracketed primitive flux is the normalization called v_{m} in the statement. This form keeps the trace-form convention explicit rather than hiding a generator normalization in H_{m}. Choosing the sign in the square completion to match the sign of n gives the lower bound

$$E \geq \frac{4\pi v_{\mathrm{m}}}{g_{\mathrm{YM}}^2}|n|.$$

Equality can hold only if the nonnegative square term vanishes almost everywhere. Smoothness makes this equivalent to the pointwise Bogomolny equation $B_i = \pm D_i\Phi$, with the same sign as the magnetic charge. $\square$

An explicit unit monopole is obtained from the hedgehog ansatz. To avoid embedding normalization factors in every radial equation, let

$$T_a := \frac{t_a}{\sqrt{2}} = \frac{\sigma_a}{2}, \qquad [T_a, T_b] = \mathrm{i}\epsilon^c{}_{ab} T_c, \qquad \mathrm{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}.$$

This T_a is only an auxiliary $SU(2)$ basis for the following explicit solution; the action and coupling convention remain the trace-form convention (45.16). With $\rho := ur$, $u > 0$, $r = |x|$, and $\hat{x}^a = x^a/r$, write

$$\Phi(x) = u\, H(\rho)\, \hat{x}^a T_a, \qquad A_i(x) = \frac{1 - K(\rho)}{r}\, \epsilon_{aij} \hat{x}^j T_a. \tag{45.45}$$

Regularity at the origin and finite energy at infinity require

$$K(0) = 1, \qquad H(0) = 0, \qquad K(\infty) = 0, \qquad H(\infty) = 1.$$

Substitution into $B_i = D_i \Phi$ gives the ordinary differential equations

$$K'(\rho) = -K(\rho)H(\rho), \qquad H'(\rho) = \frac{1 - K(\rho)^2}{\rho^2}, \tag{45.46}$$

where the prime denotes $\mathrm{d}/\mathrm{d}\rho$. The Prasad–Sommerfield solution is

$$K(\rho) = \frac{\rho}{\sinh \rho}, \qquad H(\rho) = \coth \rho - \frac{1}{\rho}. \tag{45.47}$$

Indeed,

$$\frac{\mathrm{d}}{\mathrm{d}\rho}\left(\frac{\rho}{\sinh \rho}\right) = -\frac{\rho}{\sinh \rho}\left(\coth \rho - \frac{1}{\rho}\right),$$

and

$$\frac{\mathrm{d}}{\mathrm{d}\rho}\left(\coth \rho - \frac{1}{\rho}\right) = \frac{1 - \rho^2/\sinh^2 \rho}{\rho^2}.$$

Thus (45.47) solves (45.46), is smooth at the origin, approaches the vacuum orbit at infinity, and saturates the bound (45.43) in the unit magnetic sector. Translating the parameter u to the default trace basis amounts only to replacing it by the corresponding v_{m} in the surface charge Γ_{m}.

For $\lambda > 0$ the square completion is no longer exact, but the same topological sector persists. The second-order radial equations for the profile functions have a smooth solution with a nonzero core size of order u^{-1} times a function of λ. Its mass is larger than the BPS value. The nonsingular nonabelian core replaces the singularity of a Dirac monopole: after gauge fixing Φ_∞ to a constant Cartan generator on patches, the asymptotic field is described by a $U(1)$ magnetic bundle, while the full $SU(2)$ fields remain smooth at the origin.

Figure 45.7 shows the explicit Prasad–Sommerfield profiles. This is the quantitative content hidden by the square completion: the nonabelian core is smooth, the Higgs field approaches the vacuum orbit at infinity, and the gauge profile decays to the unbroken $U(1)$ magnetic bundle.

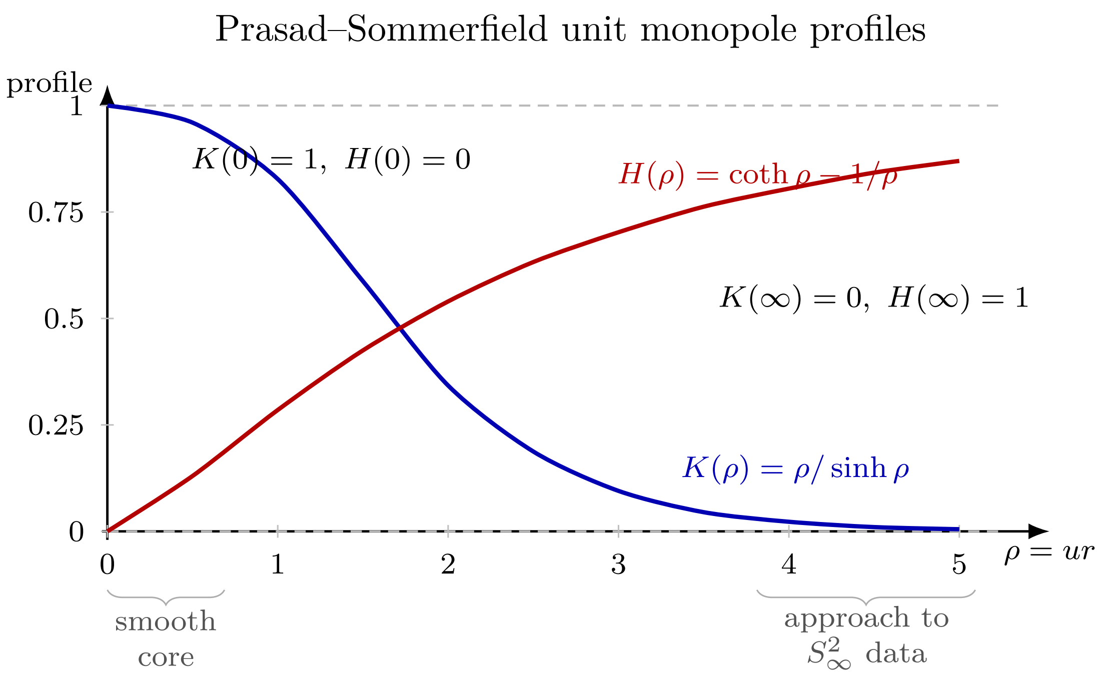


Figure 45.7: Profiles of the explicit unit BPS monopole in the auxiliary $T_a = \sigma_a/2$ basis of (45.45). The gauge profile $K(\rho) = \rho/\sinh\rho$ removes the would-be Dirac singularity at the origin and decays to zero, while $H(\rho) = \coth\rho - 1/\rho$ carries the adjoint Higgs field from the smooth core to the vacuum orbit at infinity. The first-order equations force these boundary behaviors and saturate the Bogomolny surface bound.

45.6.1 Framed Monopole Moduli Data and Low-Velocity Dynamics

The monopole also has collective coordinates, but these coordinates are not extra variables appended to the field theory. They are coordinates on the solution space of the Bogomolny equation modulo the gauge transformations declared to be redundancies. The declaration matters. To retain the electric phase whose quantization gives dyons, one divides by based gauge transformations which approach the identity at infinity; the residual constant unbroken $U(1)$ at infinity then acts on the resulting framed moduli space.

Definition 45.9 (Framed BPS monopole moduli datum)**.** Fix an $SU(2) \to U(1)$ adjoint-Higgs theory in the BPS limit and a magnetic sector n. Let $\mathcal{B}_n$ be the set of smooth finite-energy pairs (A_i, Φ) in that sector satisfying

$$B_i = D_i\Phi.$$

Let $\mathcal{G}_0$ be the group of smooth gauge transformations which tend to the identity element at spatial infinity with the decay needed to preserve the finite-energy sector. The framed monopole moduli space is the quotient

$$\mathcal{M}_n^{\mathrm{fr}} := \mathcal{B}_n/\mathcal{G}_0.$$

The unbroken asymptotic group $U(1)_\infty$ acts on $\mathcal{M}_n^{\mathrm{fr}}$. Quotienting further by this action gives the unframed moduli space. The framed version is the one used for dyonic quantization, because the primitive $U(1)_\infty$ orbit is the phase coordinate χ of Subsection 45.6.2.

Let z^a be local coordinates on a smooth chart of $\mathcal{M}_n^{\mathrm{fr}}$, and let

$$(a_{ai}, \varphi_a) = \left(\frac{\partial A_i(z)}{\partial z^a}, \frac{\partial \Phi(z)}{\partial z^a} \right)$$

be a representative tangent vector. In the Hermitian convention $D_i X = \partial_i X - \mathrm{i}[A_i, X]$, linearizing the Bogomolny equation gives

$$\varepsilon_{ijk} D_j a_{ak} - D_i \varphi_a + \mathrm{i}[a_{ai}, \Phi] = 0. \tag{45.48}$$

The infinitesimal gauge orbit through (A_i, Φ) is

$$(a_i, \varphi) = (D_i \epsilon, \mathrm{i}[\epsilon, \Phi]).$$

With respect to the kinetic inner product below, orthogonality to all such based gauge directions is exactly the background-gauge condition

$$D_i a_{ai} - \mathrm{i}[\Phi, \varphi_a] = 0. \tag{45.49}$$

Indeed, after integrating by parts and using the decay of the based parameter ϵ,

$$\int_{\mathbb{R}^3} \mathrm{tr}\big(a_i D_i \epsilon + \varphi\, \mathrm{i}[\epsilon, \Phi]\big) = \int_{\mathbb{R}^3} \mathrm{tr}\big((-D_i a_i + \mathrm{i}[\Phi, \varphi])\epsilon\big).$$

Thus a tangent vector to the quotient is represented by a pair satisfying (45.48) and (45.49), modulo based gauge parameters solving the homogeneous horizontal condition.

The metric used in the semiclassical approximation is the restriction of the real-time kinetic energy to these horizontal zero modes:

$$g_{ab}^{\mathrm{mon}}(z) = \frac{1}{g_{\mathrm{YM}}^2} \int_{\mathbb{R}^3} \mathrm{d}^3 x \; \mathrm{tr}\big(a_{ai} a_{bi} + \varphi_a \varphi_b\big). \tag{45.50}$$

The dimension assertion for $\mathcal{M}_n^{\mathrm{fr}}$ is the index statement for the elliptic complex defined by (45.48) together with (45.49). In the smooth charge $n > 0$ sector the framed dimension is $4n$: three center coordinates and one primitive phase coordinate for the total monopole, plus $4n - 4$ relative coordinates. The metric is hyperkähler in the smooth BPS sector because the linearized equations can be written as a quaternionic Dirac-type equation and the three complex structures act on the spatial one-form/scalar pair while preserving the horizontal slice. A full analytic construction of the multi-monopole moduli spaces uses the corresponding Fredholm theory and, equivalently, Nahm data; the local semiclassical formulae below require precisely the smooth chart, the horizontal tangent space, and the metric (45.50).

The low-velocity Lagrangian follows directly from Gauss' law. Let $(A_i(z(t)), \Phi(z(t)))$ be a path in a smooth moduli chart. The time-component A_0 is a Lagrange multiplier; varying it gives Gauss' law. Choosing A_0 so that the velocity is horizontal is equivalent, at linear order in $\dot{z}$, to imposing (45.49). Then

$$E_i = \dot{z}^a a_{ai} + O(\dot{z}^2), \qquad D_0 \Phi = \dot{z}^a \varphi_a + O(\dot{z}^2).$$

The magnetic and Higgs-gradient energy remains the BPS mass on the moduli space, since the topological sector and the Bogomolny equation are unchanged. The real-time kinetic terms therefore give

$$L_{\mathrm{eff}} = \frac{1}{2}\, g_{ab}^{\mathrm{mon}}(z) \dot{z}^a \dot{z}^b - M_{\mathrm{BPS}} + O(\dot{z}^3), \tag{45.51}$$

so the leading classical motion is geodesic motion for (45.50). The error term records the adiabatic expansion: massive fluctuation modes and radiation are solved for perturbatively and first affect the reduced dynamics beyond the quadratic zero-mode kinetic energy.

For the primitive unit monopole there are no relative coordinates. A framed chart is

$$(X^1, X^2, X^3, \chi) \in \mathbb{R}^3 \times S^1, \qquad \chi \sim \chi + 2\pi,$$

where X translates the core and χ is the primitive asymptotic $U(1)_\infty$ angle. In this chart the metric has the block form

$$\mathrm{d}s^2_{\mathcal{M}_1^{\mathrm{fr}}} = M_{\mathrm{BPS}}\,\delta_{ij}\mathrm{d}X^i\mathrm{d}X^j + I_\chi\,\mathrm{d}\chi^2, \qquad I_\chi > 0. \tag{45.52}$$

The translation coefficient is the soliton mass, fixed by the stress-tensor normalization of the center-of-mass kinetic energy. The phase moment of inertia I_χ is the $\chi\chi$-component of (45.50); its precise value depends on the trace and phase-coordinate normalization, while the electric-charge lattice derived below depends only on the primitive periodicity of χ.

45.6.2 Phase Coordinate, Dyonic Charge, and the Theta Angle

The unit BPS monopole has an S^1-valued phase coordinate

$$\chi \in \mathbb{R}/2\pi\mathbb{Z}.$$

The normalization used here is that $\chi \mapsto \chi + 2\pi$ is the primitive loop of the unbroken $U(1)$ gauge transformations at spatial infinity. Let $n_{\mathrm{m}} \in \mathbb{Z}$ be the magnetic charge in the same primitive cocharacter normalization. The phase coordinate is the finite-dimensional part of the gauge field which becomes electric after time dependence is allowed.

The theta density (45.21) contributes a total derivative on a path in the monopole moduli space. With the orientation convention in which a positive phase loop of a positive monopole obeys

$$\frac{1}{8\pi^2}\int_{\mathbb{R}_t\times\mathbb{R}^3} \mathrm{tr}(F\wedge F) = -\frac{n_{\mathrm{m}}}{2\pi}\int \dot\chi\,\mathrm{d}t, \tag{45.53}$$

the phase-sector effective Lagrangian has the form

$$L_\chi = \frac{1}{2}I_\chi\dot\chi^2 - \frac{\theta n_{\mathrm{m}}}{2\pi}\dot\chi, \qquad I_\chi > 0. \tag{45.54}$$

The positive constant I_χ is the $\chi\chi$-component of the monopole moduli-space metric. Its value is determined by the zero-mode inner product in (45.50); the charge-lattice shift below uses only the periodicity of χ and the topological coefficient in (45.53).

Proposition 45.10 (Dyonic tower from the monopole phase coordinate)**.** *Quantize the phase coordinate χ with Hilbert space $L^2(S^1, \mathrm{d}\chi/2\pi)$. The operator*

$$\widehat{n}_{\mathrm{e}} := -\mathrm{i}\frac{\partial}{\partial\chi}$$

has spectrum $\mathbb{Z}$. For a state $\psi_{n_{\mathrm{e}}}(\chi) = e^{\mathrm{i}n_{\mathrm{e}}\chi}$, the phase-sector Hamiltonian derived from (45.54) is

$$H_\chi(n_{\mathrm{e}}, n_{\mathrm{m}}; \theta) = \frac{1}{2I_\chi}\left(n_{\mathrm{e}} + \frac{\theta n_{\mathrm{m}}}{2\pi}\right)^2. \tag{45.55}$$

Thus the electric field carried by a charge-n_{m} monopole is graded by the shifted real coordinate

$$q_{\mathrm{E}} = n_{\mathrm{e}} + \frac{\theta n_{\mathrm{m}}}{2\pi}. \tag{45.56}$$

The periodicity $\theta \mapsto \theta + 2\pi$ preserves q_{E} after the integer electric label is relabelled by

$$n_{\mathrm{e}} \longmapsto n_{\mathrm{e}} - n_{\mathrm{m}}. \tag{45.57}$$

Proof. The canonical momentum conjugate to χ is

$$p_\chi = \frac{\partial L_\chi}{\partial \dot{\chi}} = I_\chi \dot{\chi} - \frac{\theta n_{\mathrm{m}}}{2\pi}. \tag{45.58}$$

The Legendre transform gives

$$H_\chi = p_\chi \dot{\chi} - L_\chi = \frac{1}{2I_\chi}\left(p_\chi + \frac{\theta n_{\mathrm{m}}}{2\pi}\right)^2. \tag{45.59}$$

Because χ is a coordinate on S^1, single-valued wave functions are expanded in the orthonormal Fourier basis $e^{\mathrm{i}n\chi}$, $n \in \mathbb{Z}$. Replacing p_χ by the self-adjoint generator $-\mathrm{i}\partial_\chi$ therefore gives (45.55). The combination multiplying $I_\chi^{-1/2}$ in the Hamiltonian is the electric-field coordinate (45.56). Under $\theta \mapsto \theta + 2\pi$, the relabelling (45.57) leaves this real coordinate and hence the spectrum of H_χ unchanged.

It remains to justify the topological term used in (45.54). On a slab $[t_0, t_1] \times \mathbb{R}^3$, the Chern–Weil identity gives

$$\frac{1}{8\pi^2}\int \mathrm{tr}(F \wedge F) = N_{\mathrm{CS}}[A(t_1)] - N_{\mathrm{CS}}[A(t_0)] \quad \text{plus the boundary contribution at } S^2_\infty. \tag{45.60}$$

For a path which changes only the asymptotic $U(1)$ phase, this boundary contribution is the pairing of the magnetic transition function on S^2_∞ with the $U(1)$ loop $\chi(t_0) \to \chi(t_1)$. In the primitive normalization of this subsection that pairing is

$$-\, n_{\mathrm{m}} \frac{\chi(t_1) - \chi(t_0)}{2\pi}.$$

Differentiating in time gives (45.53). The sign follows from the chosen orientation convention for a positive phase loop; reversing either the magnetic-charge orientation or the phase orientation reverses both sides and leaves (45.56) as the invariant charge-lattice statement. □

45.6.3 Collective Coordinates and the Regularized Semiclassical Measure

The word "collective coordinate" will mean a coordinate on a finite dimensional manifold of gauge-equivalence classes of classical solutions, not a new field added by hand. The measure on this manifold is not guessed from the number of zero modes. It is obtained by first defining a regulated path integral, choosing a gauge slice or working directly on the quotient, and then applying stationary phase in a neighborhood of the critical manifold. This is the point at which the distinction between a classical moduli space and the quantum weight assigned to that moduli space enters.

Definition 45.11 (Regulated collective-coordinate data)**.** Fix a finite regulator Λ. A regulated bosonic collective-coordinate data consist of the following objects.

1. A finite-dimensional smooth manifold $\mathcal{C}_\Lambda$ of regulated field variables, equipped with a smooth positive density $\mathrm{D}_\Lambda\varphi$.
2. A compact Lie group $\mathcal{G}_\Lambda$ acting smoothly on $\mathcal{C}_\Lambda$, preserving the action S_Λ and the density.
3. A gauge slice $\Sigma_\Lambda \subset \mathcal{C}_\Lambda$ meeting the gauge orbits under consideration transversely, with Faddeev–Popov Jacobian $\Delta_{\mathrm{FP},\Lambda}$. Equivalently, one may replace this item by a smooth density on the local quotient $\mathcal{C}_\Lambda/\mathcal{G}_\Lambda$.
4. A connected m-dimensional submanifold $\mathcal{M}_\Lambda \subset \Sigma_\Lambda$ of critical points of S_Λ, parametrized locally by $z = (z^1, \ldots, z^m)$, such that
$$S_\Lambda(\varphi_z) = S_0 \qquad \text{for all } z \in \mathcal{M}_\Lambda.$$
5. A regulator inner product $\langle\ ,\ \rangle_{\Lambda,z}$ on $T_{\varphi_z}\Sigma_\Lambda$. The tangent vectors
$$Z_a(z) := \frac{\partial \varphi_z}{\partial z^a}$$
define the zero-mode Gram matrix
$$G^{(\Lambda)}_{ab}(z) := \langle Z_a(z), Z_b(z)\rangle_{\Lambda,z}. \tag{45.61}$$

The data are nondegenerate if the Hessian $L_{\Lambda,z}$ of $S_\Lambda|_{\Sigma_\Lambda}$ has kernel exactly $T_{\varphi_z}\mathcal{M}_\Lambda$ and is positive on a chosen normal bundle $N_z\mathcal{M}_\Lambda$. If the residual stabilizer of φ_z in $\mathcal{G}_\Lambda$ is a compact group Γ_z, its volume is computed using the same normalization of the gauge-group density that was used to define $\Delta_{\mathrm{FP},\Lambda}$.

For a monopole, z includes center-of-mass and phase coordinates. For a BPST instanton, z includes center, size, and gauge-orientation coordinates. In both cases, the formula below is a statement at fixed regulator. The continuum semiclassical formula is obtained only after the renormalized nonzero-mode determinant and the moduli integral have been given a limit in the chosen scheme.

Theorem 45.12 (Collective-coordinate stationary phase)**.** *Let $\mathfrak{D}_\Lambda$ be a nondegenerate regulated collective-coordinate data in the sense of Definition 45.11. Denote its field space, gauge slice, critical manifold, and action by $\mathcal{C}_\Lambda$, Σ_Λ, $\mathcal{M}_\Lambda$, and S_Λ. Let $\epsilon_{\mathrm{sc}} > 0$ be the semiclassical parameter multiplying the action, and let O be a smooth gauge-invariant observable supported in a tubular neighborhood of $\mathcal{M}_\Lambda$ inside the gauge slice. Then, for each coordinate patch on $\mathcal{M}_\Lambda$,*
$$\begin{aligned}&\int_{\Sigma_\Lambda} O(\varphi)\, \exp[-S_\Lambda(\varphi)/\epsilon_{\mathrm{sc}}]\ \Delta_{\mathrm{FP},\Lambda}(\varphi)\,\mathrm{D}_\Lambda\varphi \\ &\quad = \exp(-S_0/\epsilon_{\mathrm{sc}})\,(2\pi\epsilon_{\mathrm{sc}})^{r/2} \int_{\mathcal{M}_\Lambda} \frac{O(\varphi_z)\,\Delta_{\mathrm{FP},\Lambda}(\varphi_z)}{\mathrm{Vol}(\Gamma_z)\ {\det}'_\Lambda(L_{\Lambda,z})^{1/2}}\,\mathrm{d}\nu_{\Lambda,\mathcal{M}}(z) + O\Big(\epsilon_{\mathrm{sc}}^{r/2+1}\Big),\end{aligned} \tag{45.62}$$
where $r = \dim N_z\mathcal{M}_\Lambda$ is the number of nonzero bosonic normal directions in the regulated slice, ${\det}'_\Lambda(L_{\Lambda,z})$ is the determinant of the Hessian on $N_z\mathcal{M}_\Lambda$ with respect to the regulator inner product, and
$$\mathrm{d}\nu_{\Lambda,\mathcal{M}}(z) = \sqrt{\det G^{(\Lambda)}_{ab}(z)}\,\mathrm{d}z^1\cdots\mathrm{d}z^m. \tag{45.63}$$

The product $\sqrt{\det G^{(\Lambda)}}\,\mathrm{d}^m z$ *is invariant under changes of collective-coordinate chart. If the path integral is formulated directly on the local quotient rather than on a gauge slice, the factor* $\Delta_{\mathrm{FP},\Lambda}/\operatorname{Vol}(\Gamma_z)$ *is absent and* $\mathrm{d}\nu_{\Lambda,\mathcal{M}}$ *is computed from the quotient metric.*

Proof. The proof is finite dimensional. Choose a tubular-neighborhood map

$$(z,\eta)\longmapsto\varphi(z,\eta),\qquad z\in\mathcal{M}_\Lambda,\quad \eta\in N_z\mathcal{M}_\Lambda,$$

with $\varphi(z,0)=\varphi_z$. The density in these coordinates has the form

$$\mathrm{D}_\Lambda\varphi=J_\Lambda(z,\eta)\,\mathrm{d}^m z\,\mathrm{d}^r\eta,\qquad J_\Lambda(z,0)=\sqrt{\det G^{(\Lambda)}_{ab}(z)}$$

after the normal coordinates are chosen orthonormally in $N_z\mathcal{M}_\Lambda$. This equality is precisely the definition of the Riemannian density induced on the critical manifold by the regulator inner product.

Because every φ_z is critical and the tangent directions are exactly the kernel of the Hessian,

$$S_\Lambda(\varphi(z,\eta))=S_0+\frac{1}{2}\langle\eta,L_{\Lambda,z}\eta\rangle_{\Lambda,z}+O(\|\eta\|^3).$$

By nondegeneracy, $L_{\Lambda,z}$ is positive on the normal space. For an observable with support in a fixed tubular neighborhood, the part $\|\eta\|\geq c$ is exponentially smaller than any power of ϵ_{sc}, after possibly shrinking the neighborhood. In the remaining region set $\eta=\epsilon_{\mathrm{sc}}^{1/2}\xi$. Taylor expand

$$O(\varphi(z,\epsilon_{\mathrm{sc}}^{1/2}\xi))\,\Delta_{\mathrm{FP},\Lambda}(\varphi(z,\epsilon_{\mathrm{sc}}^{1/2}\xi))\,J_\Lambda(z,\epsilon_{\mathrm{sc}}^{1/2}\xi)$$

and the cubic remainder in the action. Odd Gaussian moments vanish and the leading normal integral is

$$\int_{N_z\mathcal{M}_\Lambda}\exp\left[-\frac{1}{2}\langle\xi,L_{\Lambda,z}\xi\rangle_{\Lambda,z}\right]\mathrm{d}^r\xi=(2\pi)^{r/2}\,{\det}'_\Lambda(L_{\Lambda,z})^{-1/2}.$$

Multiplying by the Jacobian $\epsilon_{\mathrm{sc}}^{r/2}$ from $\mathrm{d}^r\eta=\epsilon_{\mathrm{sc}}^{r/2}\mathrm{d}^r\xi$ gives (45.62). The factor $\operatorname{Vol}(\Gamma_z)^{-1}$ removes the residual gauge copies left inside the slice. Under a chart change $z=z(z')$, the Gram matrix transforms as $G'=(\partial z/\partial z')^T G(\partial z/\partial z')$; hence $\sqrt{\det G'}\,\mathrm{d}^m z'=\sqrt{\det G}\,\mathrm{d}^m z$. The displayed measure is therefore intrinsic. □

For BPS monopoles, the moduli-space metric and the low-velocity geodesic Lagrangian obtained from this stationary-phase measure were derived in Subsection 45.6.1. The finite stationary-phase formula above supplies the accompanying quantum collective-coordinate density once the nonzero-mode determinant and the chosen regulator have been specified.

Construction 45.13 (Fermion zero modes and Berezin saturation)**.** At finite regulator, suppose the quadratic fermion action in a bosonic background φ_z is

$$S_F=\frac{1}{2}\Psi^T\mathcal{D}_z\Psi,$$

where Ψ is a finite list of odd variables and $\mathcal{D}_z$ is the antisymmetric matrix representing the regulated Dirac operator in the chosen pairing. Split

$$\Psi=\sum_{\alpha=1}^{n_0}\zeta^\alpha\Psi_{0,\alpha}+\Psi_\perp,\qquad \mathcal{D}_z\Psi_{0,\alpha}=0.$$

The nonzero modes give $\mathrm{Pf}'(\mathcal{D}_z)$. The zero-mode part of the Berezin integral is

$$\int \mathrm{d}\zeta^{n_0}\cdots \mathrm{d}\zeta^1\, F(\zeta^1,\ldots,\zeta^{n_0}),$$

which extracts the coefficient of $\zeta^1\cdots\zeta^{n_0}$. Therefore a correlation function in this background vanishes unless the operator insertions or mass terms supply all fermion zero-mode variables. This is the regulated algebraic origin of the 't Hooft vertex; the index theorem counts how many such variables are present, while the zero-mode wavefunctions determine the gauge, spin, and flavor tensor multiplying them.

The same finite-energy logic also produces codimension-two vortices. In the Abelian Higgs model on $\mathbb{R}^2$, with

$$D_i\phi = \partial_i\phi - \mathrm{i}A_i\phi, \qquad B = F_{12},$$

use the energy per unit length

$$T[A,\phi] = \int_{\mathbb{R}^2} \mathrm{d}^2x \left[\frac{1}{2e^2}B^2 + |D_i\phi|^2 + \frac{e^2}{2}\left(|\phi|^2 - v^2\right)^2\right]. \tag{45.64}$$

Finite energy implies $|\phi| \to v$ and $D_i\phi \to 0$ at infinity, so $\phi/|\phi| : S^1_\infty \to U(1)$ has winding n. Equivalently,

$$\int_{\mathbb{R}^2} B\,\mathrm{d}^2x = 2\pi n.$$

At the critical coupling displayed in (45.64),

$$T = \int_{\mathbb{R}^2} \mathrm{d}^2x \left[\frac{1}{2e^2}\left(B \mp e^2(v^2 - |\phi|^2)\right)^2 + |(D_1 \pm \mathrm{i}D_2)\phi|^2\right] \pm v^2 \int_{\mathbb{R}^2} B\,\mathrm{d}^2x. \tag{45.65}$$

Hence

$$T \geq 2\pi v^2 |n|,$$

with equality precisely for the Nielsen–Olesen vortex equations

$$(D_1 \pm \mathrm{i}D_2)\phi = 0, \qquad B = \pm e^2(v^2 - |\phi|^2).$$

The vortex is therefore the codimension-two analogue of the monopole: finite energy converts infinity to a sphere of one lower dimension, topological winding fixes a flux, and a first-order system appears at a special value of the scalar potential.

Finite-energy critical points also include unstable saddles. In four-dimensional $SU(2)$-Higgs theory with a fundamental Higgs doublet, the finite-energy configuration space contains a saddle point called the sphaleron. Its distinguishing data are its position between neighboring vacua of different Chern–Simons number and its single negative fluctuation mode. With the Hermitian connection convention of this chapter, the spatial Chern–Simons functional is

$$N_{\mathrm{CS}}[A] = \frac{1}{8\pi^2}\int_{\mathbb{R}^3} \mathrm{tr}\left(A\wedge \mathrm{d}A - \frac{2\mathrm{i}}{3}A\wedge A\wedge A\right), \tag{45.66}$$

defined modulo integers under large gauge transformations when the boundary conditions compactify space to S^3. The sphaleron sits, in an appropriate one-parameter minimax construction, at half-integer Chern–Simons number and has one negative fluctuation mode. It controls the height of the classical barrier between adjacent electroweak vacua; real-time transitions at finite temperature and tunneling transitions at zero temperature require additional quantum and statistical analysis. Thus the sphaleron is an unstable static saddle of the classical energy.

Solitons enter the theory as critical points or sectors of the same classical field space, selected by finite-energy boundary conditions and by the topology of gauge and Higgs data at infinity. Their quantum interpretation requires a separate construction of states, collective coordinates, fluctuation determinants, and possible fermion zero modes. In gauge theories this construction is phrased in terms of gauge-invariant observables and superselection sectors; the local gauge potential itself remains a coordinate on a redundant description.

45.7 QCD Notation and Chiral Masses

For QCD with gauge group $SU(N)$, a quark field carries three kinds of indices:

$$\psi^i_{I\alpha}(x), \qquad i = 1, \dots, N, \quad I = 1, \dots, N_f.$$

Here i is the color index in the fundamental representation, I is a flavor index, and α is a spinor index. The conjugate field $\bar{\psi}$ carries the anti-fundamental color representation.

For these quarks, the local four-dimensional terms that are Poincare invariant, gauge invariant, and power-counting renormalizable have the gauge part

$$-\frac{1}{4g^2_{\mathrm{YM}}}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{\theta}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma})$$

and the fermion part

$$\mathcal{L}_\psi = -\bar{\psi}_I\gamma^\mu D_\mu\psi_I - m_{IJ}\bar{\psi}_I\psi_J - \widetilde{m}_{IJ}\bar{\psi}_I\gamma_5\psi_J. \tag{45.67}$$

The matrices m and $\widetilde{m}$ act on flavor and are Hermitian in a Hermitian Lorentzian action.

For chiral notation, use the convention fixed in Chapter 40, $\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3$; with this sign the left-handed subspace is the $+1$ eigenspace of γ_5:

$$P_L = \frac{1+\gamma_5}{2}, \qquad P_R = \frac{1-\gamma_5}{2}, \qquad \psi_{IL} = P_L\psi_I, \quad \psi_{IR} = P_R\psi_I.$$

Then the fermion Lagrangian can be written

$$\mathcal{L}_\psi = -\bar{\psi}_{IL}\gamma^\mu D_\mu\psi_{IL} - \bar{\psi}_{IR}\gamma^\mu D_\mu\psi_{IR} - M_{IJ}\bar{\psi}_{IL}\psi_{JR} - M^*_{IJ}\bar{\psi}_{JR}\psi_{IL}. \tag{45.68}$$

The complex mass matrix M is the coefficient matrix of the left-right bilinear in (45.68); M^*_{IJ} denotes the entrywise complex conjugate of M_{IJ}. Equivalently, after substituting $\psi_{IL} = P_L\psi_I$, $\psi_{IR} = P_R\psi_I$, and the corresponding Dirac-adjoint projectors, M is defined by the identity

$$m_{IJ}\bar{\psi}_I\psi_J + \widetilde{m}_{IJ}\bar{\psi}_I\gamma_5\psi_J = M_{IJ}\bar{\psi}_{IL}\psi_{JR} + M^*_{IJ}\bar{\psi}_{JR}\psi_{IL}.$$

Thus M is the same mass data expressed in the chiral basis. The reason this package is useful is that it exposes the phase affected by anomalous axial rotations. Under a common vector rotation $\psi_L \mapsto \mathrm{e}^{\mathrm{i}\beta}\psi_L$, $\psi_R \mapsto \mathrm{e}^{\mathrm{i}\beta}\psi_R$, M is unchanged. Under a common axial rotation

$$\psi_L \mapsto \mathrm{e}^{\mathrm{i}\alpha}\psi_L, \qquad \psi_R \mapsto \mathrm{e}^{-\mathrm{i}\alpha}\psi_R,$$

the left-right bilinear transforms as $\bar{\psi}_L\psi_R \mapsto \mathrm{e}^{-2\mathrm{i}\alpha}\bar{\psi}_L\psi_R$, so the spurion transformation preserving the mass term is

$$M \mapsto \mathrm{e}^{2\mathrm{i}\alpha}M.$$

For N_f fundamental Dirac quarks the anomalous Jacobian shifts $\theta \mapsto \theta - 2N_f\alpha$ in the normalization (45.21). Hence the combination

$$\bar{\theta} := \theta + \arg\det M$$

is invariant under this common axial change of variables, provided $\det M \neq 0$. If a quark mass vanishes, $\arg\det M$ is not defined and the chiral rotation must be treated on the rank-deficient mass matrix itself.

Figure 45.8 records this invariant phase relation, which is the reason for using the complex chiral mass matrix in the QCD chiral-symmetry discussion.

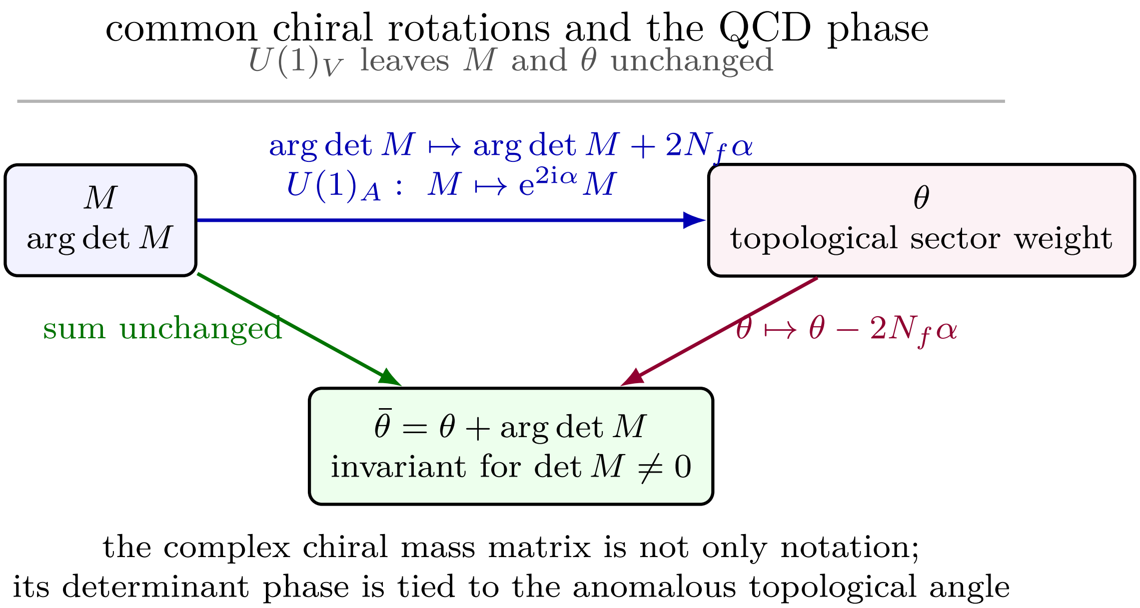


Figure 45.8: Phase information in the chiral mass matrix. With the γ_5 convention used here, a common axial field redefinition sends $M \mapsto e^{2i\alpha}M$ and shifts the topological angle by $\theta \mapsto \theta - 2N_f\alpha$. Therefore $\bar{\theta} = \theta + \arg\det M$ is the invariant phase when M is nonsingular. The vector $U(1)$ rotation leaves both quantities unchanged.

This notation prepares two later developments. Gauge fixing and BRST will quantize the gauge theory while preserving the gauge-covariant local data defined here. Anomalous chiral rotations will relate phases of the mass matrix to the θ-angle.

Chapter 46

Lattice Yang–Mills as a Nonperturbative Formulation

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter gives a regulator-level construction of pure Yang–Mills theory which does not begin by choosing a covariant gauge. The construction is a finite-dimensional integral over compact groups at finite lattice spacing and finite volume. Its continuum limit is a separate problem: one must prove, or assume with stated evidence, that a sequence of finite-cutoff theories has gauge-invariant Schwinger functions with a limit satisfying the Euclidean axioms and yielding a positive-energy Hilbert-space theory after Osterwalder–Schrader reconstruction.

46.1 Finite Lattice Gauge Fields

Fix a compact Lie group G. For most formulas below $G = SU(N)$, and tr denotes the trace in the fundamental representation with $\mathrm{tr}(t^a t^b) = \delta^{ab}$, matching the continuum Yang–Mills convention of Chapter 45. Let

$$\Lambda_{a,L} = (a\mathbb{Z}/L\mathbb{Z})^4$$

be a four-dimensional periodic hypercubic lattice, where $a > 0$ is the lattice spacing and L/a is an integer. The positively oriented links are

$$E^+(\Lambda_{a,L}) = \{(x, \mu) \mid x \in \Lambda_{a,L},\ \mu = 1, \dots, 4\}.$$

The link (x, μ) joins x to $x + a\widehat{\mu}$. A lattice gauge field is a function

$$U : E^+(\Lambda_{a,L}) \longrightarrow G, \qquad (x, \mu) \longmapsto U_\mu(x).$$

The inverse orientation is assigned the inverse group element,

$$U_{-\mu}(x + a\widehat{\mu}) = U_\mu(x)^{-1}.$$

Thus a lattice gauge field is a point of the compact manifold

$$\mathcal{A}_\Lambda = G^{E^+(\Lambda_{a,L})}.$$

The lattice gauge group is

$$\mathcal{G}_\Lambda = G^{\Lambda_{a,L}}.$$

An element $\Omega \in \mathcal{G}_\Lambda$ acts by

$$U_\mu(x) \longmapsto U_\mu^\Omega(x) = \Omega(x) U_\mu(x) \Omega(x + a\widehat{\mu})^{-1}. \tag{46.1}$$

This convention matches the Hermitian continuum convention of Chapter 45. For a smooth field with

$$U_\mu(x) = \exp\bigl(-\mathrm{i}aA_\mu(x) + O(a^2)\bigr),$$

the endpoint rule (46.1) gives

$$A_\mu^\Omega = \Omega A_\mu \Omega^{-1} + \mathrm{i}\, \Omega \partial_\mu \Omega^{-1} + O(a).$$

For $\mu < \nu$, define the oriented plaquette based at x by

$$U_{\mu\nu}(x) = U_\mu(x) U_\nu(x + a\widehat{\mu}) U_\mu(x + a\widehat{\nu})^{-1} U_\nu(x)^{-1}. \tag{46.2}$$

Under (46.1),

$$U_{\mu\nu}(x) \longmapsto \Omega(x) U_{\mu\nu}(x) \Omega(x)^{-1}. \tag{46.3}$$

Therefore every class function of $U_{\mu\nu}(x)$, and in particular $\mathrm{Re}\,\mathrm{tr}\, U_{\mu\nu}(x)$, is gauge invariant.

Figure 46.1 records the oriented plaquette convention and its based-loop gauge covariance (46.3), the input for the Wilson action.

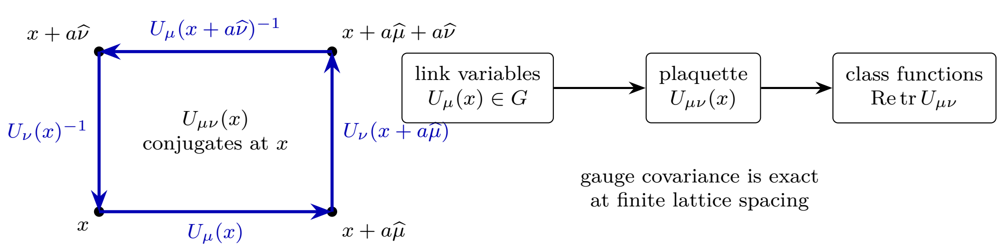


Figure 46.1: The plaquette is the elementary lattice curvature. Link variables transform at their endpoints; around a closed elementary square the endpoint factors cancel except for conjugation at the basepoint.

46.2 Wilson Action and the Finite Haar Integral

For $G = SU(N)$ the Wilson action is

$$S_{W,\Lambda}[U] = \frac{\beta}{N} \sum_{x\in\Lambda_{a,L}} \sum_{1\le\mu<\nu\le4} \operatorname{Re}\operatorname{tr}\big(\mathbf{1} - U_{\mu\nu}(x)\big), \qquad \beta = \frac{N}{g_0^2}. \tag{46.4}$$

Here g_0 is the lattice coupling parameter in the trace-form continuum normalization

$$S_{\text{cont}}[A] = \frac{1}{4g_0^2} \int \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}).$$

With the more common generator normalization $\operatorname{tr}(t^a t^b) = \frac{1}{2}\delta^{ab}$, the same Wilson action is often written with $\beta = 2N/g_0^2$. The difference is a coordinate change in the generator and coupling normalization, not a different lattice integral. The integrand $\operatorname{Re}\operatorname{tr}(\mathbf{1}-U_{\mu\nu})$ is nonnegative, because each eigenvalue of $U_{\mu\nu}$ lies on the unit circle and hence $\operatorname{Re}\operatorname{tr} U_{\mu\nu} \le N$. Gauge invariance of $S_{W,\Lambda}$ follows from (46.3).

Let $\mathrm{d}U$ be normalized Haar measure on G. At finite a and finite L, the lattice partition function is the finite-dimensional compact integral

$$Z_\Lambda(\beta) = \int_{\mathcal{A}_\Lambda} \prod_{\ell\in E^+(\Lambda_{a,L})} \mathrm{d}U_\ell \, \exp\{-S_{W,\Lambda}[U]\}. \tag{46.5}$$

Since $\mathcal{A}_\Lambda$ is compact and the integrand is continuous, $Z_\Lambda(\beta)$ is finite and strictly positive. For a bounded gauge-invariant function $O : \mathcal{A}_\Lambda \to \mathbb{C}$,

$$\langle O\rangle_{\Lambda,\beta} = Z_\Lambda(\beta)^{-1} \int_{\mathcal{A}_\Lambda} \prod_\ell \mathrm{d}U_\ell \, O[U] \, \exp\{-S_{W,\Lambda}[U]\} \tag{46.6}$$

is a completely defined number. No division by an infinite gauge-group volume and no Faddeev–Popov determinant enter this finite-cutoff definition.

The relation to the continuum Yang–Mills action is obtained by evaluating the plaquette on a smooth continuum connection. Let $x_c = x + \frac{a}{2}\widehat{\mu} + \frac{a}{2}\widehat{\nu}$ be the plaquette center. With

$$U_\mu(x) = \exp\big(-\mathrm{i}aA_\mu(x + \tfrac{a}{2}\widehat{\mu}) + O(a^3)\big),$$

the Baker–Campbell–Hausdorff expansion gives

$$U_{\mu\nu}(x) = \exp\big(-\mathrm{i}a^2 F_{\mu\nu}(x_c) + O(a^4)\big), \tag{46.7}$$

where $F_{\mu\nu}$ is the Hermitian-coordinate curvature of Chapter 45. For $SU(N)$, $\operatorname{tr} F_{\mu\nu} = 0$, and hence

$$\operatorname{Re}\operatorname{tr}(\mathbf{1} - U_{\mu\nu}) = \frac{a^4}{2} \operatorname{tr}(F_{\mu\nu}(x_c)F_{\mu\nu}(x_c)) + O(a^6).$$

Substitution in (46.4) gives

$$\sum_{\mu<\nu} \frac{a^4}{2} \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) = \frac{a^4}{4} \sum_{\mu,\nu} \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}),$$

because $F_{\nu\mu} = -F_{\mu\nu}$. Therefore

$$S_{W,\Lambda}[U(A)] = \frac{1}{4g_0^2} \int_{\mathbb{T}_L^4} \operatorname{tr}(F_{\mu\nu} F_{\mu\nu}) \, \mathrm{d}^4 x + O(a^2), \tag{46.8}$$

for smooth fields on the torus of side L. Equation (46.8) is an expansion of the classical lattice action on smooth configurations. The quantum continuum limit also requires control of the measure as $a \downarrow 0$, L is sent to the desired infrared limit, and $\beta \to \infty$ along a trajectory that keeps a physical scale fixed.

46.3 Reflection Positivity and the Transfer Matrix

The Wilson lattice theory is Euclidean from the start. To connect it to a Hilbert-space theory one uses the lattice version of reflection positivity, the discrete precursor of the Osterwalder–Schrader condition developed in Chapter 12.

Choose the reflection ϑ which sends the Euclidean time coordinate x_4 to $-x_4$ and reverses the orientation of reflected links. Let $\mathcal{B}_+$ be the algebra generated by gauge-invariant functions of link variables whose support lies in the positive-time half lattice. For $F \in \mathcal{B}_+$, set

$$\Theta F[U] = \overline{F[\vartheta U]}.$$

Theorem 46.1 (Osterwalder–Seiler reflection positivity for the Wilson action)**.** *On a reflection-invariant finite hypercubic lattice, the Wilson measure*

$$\mathrm{d}\mu_{\Lambda,\beta}(U) = Z_\Lambda(\beta)^{-1} \prod_\ell \mathrm{d}U_\ell \, \mathrm{e}^{-S_{W,\Lambda}[U]}$$

satisfies

$$\int \Theta F[U] \, F[U] \, \mathrm{d}\mu_{\Lambda,\beta}(U) \geq 0, \qquad F \in \mathcal{B}_+. \tag{46.9}$$

Proof. The proof is finite-dimensional. Haar measure is invariant under reflection, inversion, and left and right multiplication. The Wilson action splits into a positive-time part, its reflected part, and plaquette factors crossing the reflection plane. A crossing plaquette factor is a continuous central function of a group element V:

$$K_\beta(V) = \exp\left\{ \frac{\beta}{N} \operatorname{Re} \operatorname{tr} V \right\}.$$

Peter–Weyl theory expands it as

$$K_\beta(V) = \sum_{\lambda \in \widehat{G}} c_\lambda(\beta) \chi_\lambda(V), \qquad c_\lambda(\beta) = \int_G K_\beta(V) \overline{\chi_\lambda(V)} \, \mathrm{d}V. \tag{46.10}$$

For $G = SU(N)$ with the fundamental Wilson kernel this positivity is explicit. Write

$$K_\beta(V) = \exp\left[\frac{\beta}{2N} \big(\operatorname{tr} V + \operatorname{tr} V^{-1} \big) \right] = \sum_{r,s \geq 0} \frac{(\beta/2N)^{r+s}}{r! \, s!} (\operatorname{tr} V)^r (\operatorname{tr} V^{-1})^s.$$

The product $(\operatorname{tr} V)^r(\operatorname{tr} V^{-1})^s$ is the character of $(\mathbb{C}^N)^{\otimes r} \otimes (\overline{\mathbb{C}^N})^{\otimes s}$. Decomposing this finite-dimensional representation into irreducibles gives

$$(\operatorname{tr} V)^r(\operatorname{tr} V^{-1})^s = \sum_{\lambda \in \widehat{SU(N)}} m_{\lambda;r,s}\chi_\lambda(V), \qquad m_{\lambda;r,s} \in \mathbb{Z}_{\geq 0}.$$

The exponential series is absolutely and uniformly convergent on the compact group, so its character coefficients are

$$c_\lambda(\beta) = \sum_{r,s \geq 0} \frac{(\beta/2N)^{r+s}}{r!\, s!}\, m_{\lambda;r,s} \geq 0.$$

The same argument applies to a Wilson kernel built from a unitary representation R and its conjugate, with $R^{\otimes r} \otimes \overline{R}^{\otimes s}$ replacing the fundamental tensor powers. Matrix-coefficient orthogonality rewrites each crossing contribution as a finite or absolutely convergent sum of terms

$$c_\lambda(\beta)\, F_{\lambda,i}[U_+]\, \overline{F_{\lambda,i}[\vartheta U_+]}, \qquad c_\lambda(\beta) \geq 0,$$

after the links on the reflection plane have been integrated. Multiplication by the positive-time Boltzmann weight and summation over λ, i turns (46.9) into a sum of squared absolute values with nonnegative coefficients. □

Reflection positivity produces a Hilbert space at finite lattice spacing by quotienting $\mathcal{B}_+$ by the null vectors of (46.9) and completing. Translation by one lattice unit in Euclidean time gives a positive transfer matrix

$$T_a = \mathrm{e}^{-aH_a}.$$

Thus the finite lattice theory has a genuine Hilbert-space interpretation with a nonnegative lattice Hamiltonian H_a. The continuum question is whether a family of such OS-positive finite-cutoff theories has a scaling limit whose Schwinger functions satisfy the continuum hypotheses of Chapter 12.

46.4 Thermal Center Symmetry and the Polyakov Loop

The Wilson lattice also gives a precise finite-cutoff formulation of the thermal center-symmetry order parameter. This subsection is kinematic: it defines the symmetry, the charged operator, and the order of limits needed for spontaneous symmetry breaking. It does not assume a continuum proof of confinement or of a deconfinement transition.

Let the Euclidean lattice have the form

$$\Lambda_{a,N_s,N_t} = (a\mathbb{Z}/N_s a\mathbb{Z})^3 \times (a\mathbb{Z}/N_t a\mathbb{Z}), \qquad \beta_{\mathrm{T}} = aN_t.$$

The last circle is the thermal circle of inverse temperature β_{T}. To avoid confusion, the Wilson coupling parameter in (46.4) will be denoted by $\beta_{\mathrm{lat}} = N/g_0^2$ in this subsection.

For $G = SU(N)$, choose a center element

$$z_m = \mathrm{e}^{2\pi \mathrm{i} m/N}\mathbf{1}_N, \qquad m \in \mathbb{Z}_N.$$

Fix one Euclidean-time slice τ_0. The thermal center transformation acts on temporal links crossing that slice by

$$U_4(\mathbf{x},\tau_0) \longmapsto z_m\, U_4(\mathbf{x},\tau_0), \qquad U_i(\mathbf{x},\tau) \longmapsto U_i(\mathbf{x},\tau), \qquad U_4(\mathbf{x},\tau \neq \tau_0) \longmapsto U_4(\mathbf{x},\tau). \tag{46.11}$$

It is the finite thermal-lattice representative of the electric center one-form symmetry of Section 45.3 after the Euclidean time direction has been compactified.

Lemma 46.2 (Exact center symmetry of pure thermal lattice Yang–Mills)**.** *For pure $SU(N)$ Wilson lattice gauge theory on Λ_{a,N_s,N_t}, the transformation (46.11) preserves the Haar measure and the Wilson action. If dynamical matter of N-ality k_α is added, the unbroken subgroup is*

$$\{m \in \mathbb{Z}_N : \exp(2\pi \mathrm{i} m k_\alpha/N) = 1 \textit{ for every dynamical } \alpha\},$$

the same subgroup as in (45.25).

Proof. Haar measure is invariant under left multiplication by the fixed group element z_m. A spatial plaquette contains no transformed temporal link. A temporal plaquette crossing the slice contains one transformed temporal link $U_4(\mathbf{x},\tau_0)$ and one inverse transformed temporal link $U_4(\mathbf{x}+a\hat{i},\tau_0)^{-1}$. Because z_m is central, the phases cancel:

$$z_m U_4(\mathbf{x},\tau_0)\cdots\left(z_m U_4(\mathbf{x}+a\hat{i},\tau_0)\right)^{-1} = z_m U_4(\mathbf{x},\tau_0)\cdots U_4(\mathbf{x}+a\hat{i},\tau_0)^{-1} z_m^{-1}.$$

The center factors commute through the remaining links and disappear inside $\mathrm{Re}\,\mathrm{tr}$. Thus every plaquette term and the product Haar measure are invariant.

A dynamical matter hopping term in representation R_α and N-ality k_α crossing the thermal slice acquires the phase $z_m^{k_\alpha}$. Such a term is invariant for all dynamical fields precisely when the displayed phase is one for every R_α. □

The Polyakov line at spatial position $\mathbf{x}$ is the temporal holonomy around the thermal circle,

$$\mathcal{P}(\mathbf{x}) = U_4(\mathbf{x},0)U_4(\mathbf{x},a)\cdots U_4(\mathbf{x},(N_t-1)a). \tag{46.12}$$

For an irreducible representation R, define the normalized Polyakov loop

$$P_R(\mathbf{x}) = d_R^{-1}\chi_R(\mathcal{P}(\mathbf{x})). \tag{46.13}$$

Under a periodic lattice gauge transformation, $\mathcal{P}(\mathbf{x})$ is conjugated at the basepoint, so $P_R(\mathbf{x})$ is gauge invariant. Under the center transformation (46.11),

$$P_R(\mathbf{x}) \longmapsto \exp\left(\frac{2\pi \mathrm{i} m k_R}{N}\right) P_R(\mathbf{x}), \tag{46.14}$$

where k_R is the N-ality of R. The fundamental Polyakov loop is therefore a charged local operator of the three-dimensional thermal theory. Equivalently, it is a line operator of the four-dimensional Euclidean theory wrapping the thermal circle.

Figure 46.2 shows the center transformation on a thermal slice and the resulting N-ality charge of a Polyakov loop in representation R.

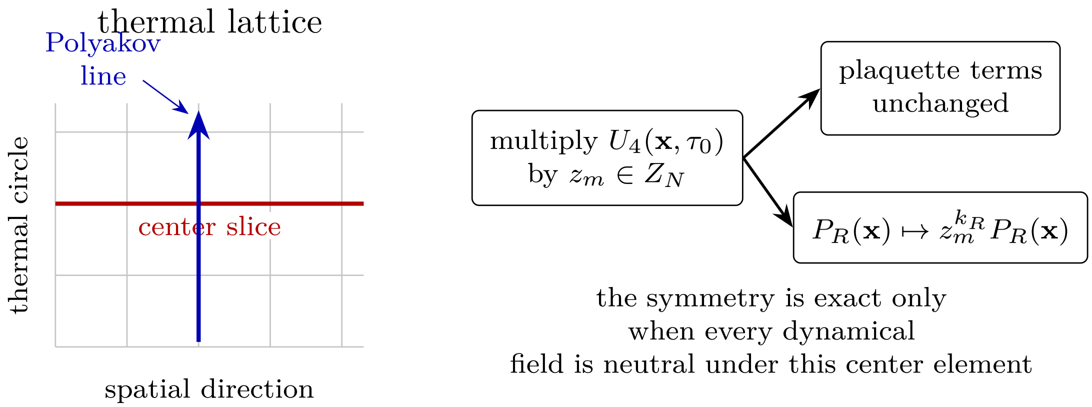


Figure 46.2: Thermal center symmetry on the lattice. Multiplying all temporal links crossing one time slice by a center element leaves plaquettes invariant, while a Polyakov loop winding the thermal circle acquires the center phase of its representation.

At finite spatial volume and with no explicit center-breaking source,

$$\langle P_\Box(\mathbf{x}) \rangle_{\Lambda, \beta_{\text{lat}}} = 0$$

in pure $SU(N)$ Yang–Mills. Indeed, averaging the expectation value over the exact $\mathbb{Z}_N$ transformations gives $\langle P_\Box \rangle = z_m \langle P_\Box \rangle$ for every m, and the only complex number with this property for a nontrivial center character is zero. A nonzero order parameter is therefore an infinite-volume spontaneous-symmetry-breaking statement, not a finite-volume expectation at zero source.

Definition 46.3 (Polyakov-loop order parameter)**.** Let

$$\overline{P}_\Box = N_s^{-3} \sum_{\mathbf{x}} P_\Box(\mathbf{x}).$$

Choose a small source h selecting one center sector, for example by adding $-hN_s^3 \operatorname{Re} \overline{P}_\Box$ to the lattice action. The fundamental Polyakov-loop order parameter is

$$\ell = \lim_{h \downarrow 0} \lim_{N_s \to \infty} \langle \overline{P}_\Box \rangle_{h, N_s, N_t, \beta_{\text{lat}}}, \tag{46.15}$$

with the continuum thermal limit, if desired, taken afterward along a scaling trajectory with $aN_t = \beta_{\text{T}}$ fixed. Center symmetry is unbroken when $\ell = 0$ and broken when $\ell \neq 0$.

The order of limits in (46.15) is part of the definition. Reversing the limits gives zero in a finite volume because the exact finite-volume measure is center invariant.

The same operator has a static-source interpretation. After a center sector has been selected, or with the small source used in (46.15), insertion of $P_R(\mathbf{x})$ at finite lattice spacing computes the partition function in the presence of a static external source in representation R, divided by the vacuum partition function:

$$\langle P_R(\mathbf{x}) \rangle_{\text{bare}} = \exp\{-\beta_{\text{T}}(F_R^{\text{bare}} - F_0)\}. \tag{46.16}$$

The adjective “bare” matters. In the continuum limit a static line has a self-energy divergence, so a renormalized Polyakov loop is obtained by a multiplicative line renormalization,

$$P_R^{\mathrm{ren}} = Z_R(a)^{N_t} P_R^{\mathrm{bare}},$$

or equivalently by subtracting the static-source self-energy from F_R^{bare}. This renormalization changes the numerical value of ℓ, but it does not change the center charge (46.14) or whether ℓ vanishes.

Similarly, the Polyakov-loop two-point function gives the free energy of a static source and anti-source pair:

$$\langle P_R(\mathbf{x}) P_{\overline{R}}(\mathbf{y})\rangle = \exp\{-\beta_{\mathrm{T}}(F_{R\overline{R}}(|\mathbf{x}-\mathbf{y}|) - F_0)\}, \tag{46.17}$$

after the same line renormalization. This correlator is center neutral even when each individual Polyakov loop is charged. If the infinite-volume thermal state satisfies clustering, then a center-broken phase has

$$\lim_{|\mathbf{x}-\mathbf{y}|\to\infty} \langle P_{\square}(\mathbf{x}) P_{\overline{\square}}(\mathbf{y})\rangle = |\ell|^2, \tag{46.18}$$

whereas a center-symmetric phase has $\ell = 0$.

Definition 46.4 (Deconfinement in the center-symmetry sense)**.** For pure $SU(N)$ Yang–Mills, a thermal phase is center confined when the thermal center symmetry is unbroken, $\ell = 0$, and center deconfined when the thermal center symmetry is spontaneously broken, $\ell \neq 0$. A deconfinement transition is a nonanalytic change, in the infinite-volume thermal free energy or its derivatives, across which this center order parameter changes its symmetry-realization pattern.

This definition is sharp for pure Yang–Mills and for theories whose dynamical matter preserves a nontrivial center subgroup. With dynamical fundamental quarks the fundamental center symmetry is explicitly broken, so the fundamental Polyakov loop is no longer a strict order parameter. It remains a useful static-source observable, and rapid finite-temperature changes in it may diagnose a crossover, but the exact symmetry statement has been lost.

The center-symmetry order parameter is logically distinct from the zero-temperature Wilson-loop area law. The latter concerns large spacelike or spacetime Wilson loops and static-source potentials at $T = 0$; the former concerns the realization of the thermal center symmetry and the free energy of a single static source at $T > 0$. In a theory where both are under control they are related by physical continuity, but neither finite-cutoff definition is a substitute for proving the other’s continuum dynamics.

46.5 Wilson Loops and the Strong-Coupling Area Law

Let C be a closed oriented lattice path. If C traverses the oriented link ℓ with sign $\epsilon_\ell = \pm 1$, set

$$U(C) = \prod_{\ell \in C} U_\ell^{\epsilon_\ell},$$

with the product ordered along the path. For an irreducible representation R of G with character χ_R and dimension d_R, define the normalized Wilson loop

$$W_R(C) = d_R^{-1} \chi_R(U(C)). \tag{46.19}$$

Gauge factors at adjacent endpoints cancel around the closed curve, so $W_R(C)$ is gauge invariant.

The strong-coupling expansion is the character expansion of each plaquette Boltzmann factor. Write

$$\exp\left\{\frac{\beta}{N}\operatorname{Re}\operatorname{tr}U_P\right\} = c_0(\beta)\left[1+\sum_{\lambda\neq 0} d_\lambda u_\lambda(\beta)\chi_\lambda(U_P)\right], \qquad u_\lambda(\beta) = O(\beta^{n_\lambda}), \tag{46.20}$$

where $n_\lambda > 0$ for every nontrivial representation. The Haar orthogonality relation

$$\int_G R^\lambda(U)_{ij} R^\mu(U^{-1})_{kl}\,\mathrm{d}U = \frac{\delta_{\lambda\mu}}{d_\lambda}\delta_{il}\delta_{jk} \tag{46.21}$$

implies a local constraint: after integrating a link, the representation matrices carried by the plaquettes and by the Wilson loop must combine to the trivial representation on that link.

For a fundamental Wilson loop in $SU(N)$, the lowest nonzero contribution is therefore obtained by filling a surface Σ with boundary C by fundamental plaquettes. If $A_{\min}(C)$ is the minimal number of plaquettes in such a surface, then

$$\langle W_\square(C)\rangle_{\Lambda,\beta} = K_C\, u_\square(\beta)^{A_{\min}(C)}\big(1+O(\beta)\big) \tag{46.22}$$

for sufficiently small β, with $K_C > 0$ determined by boundary group integrals and by the normalization of $W_\square$. More precisely, the polymer expansion obtained from (46.20) converges absolutely when β is small enough; its polymer activities obey a bound of the form

$$\sum_{\Gamma\ni P} |z(\Gamma)|\mathrm{e}^{\alpha|\Gamma|} \le \alpha$$

for a positive α, after decreasing β if necessary. This Kotecky–Preiss type estimate gives constants $C(\beta)$ and $\sigma(\beta) > 0$ such that

$$|\langle W_\square(C)\rangle_{\Lambda,\beta}| \le C(\beta)\,\exp\{-\sigma(\beta)A_{\min}(C)\} \tag{46.23}$$

uniformly in the finite volume, away from boundary effects.

For a rectangular loop with spatial length $R = na$ and Euclidean time length $T = ma$, (46.23) gives

$$\langle W_\square(C_{R,T})\rangle \sim \exp\{-\sigma_{\mathrm{lat}}(\beta)nm\} = \exp\{-[\sigma_{\mathrm{lat}}(\beta)R/a]\,T/a\}.$$

The transfer-matrix interpretation therefore gives a static-source potential which grows linearly with separation in the strong-coupling lattice theory. This is a rigorous confinement diagnostic at finite lattice cutoff for small β. The four-dimensional continuum confinement problem asks for control along a scaling trajectory with $\beta \to \infty$.

46.6 Scaling Trajectories and Dimensional Transmutation

The continuum limit is not obtained by fixing β and sending $a \downarrow 0$. One must choose a trajectory $g_0(a)$, equivalently $\beta(a) = N/g_0(a)^2$ in the trace normalization of this chapter, so that dimensionless lattice observables scale in a way compatible with finite physical observables. For example, if the continuum theory has a string tension σ_{phys}, a scaling trajectory would satisfy

$$a^2\sigma_{\mathrm{lat}}(\beta(a)) \longrightarrow a^2\sigma_{\mathrm{phys}} \quad \text{with} \quad a^2\sigma_{\mathrm{phys}} \downarrow 0.$$

Equivalently, the correlation length in lattice units must diverge.

Perturbation theory around the Gaussian ultraviolet fixed point predicts the large-β relation between a and the lattice coupling. For pure $SU(N)$, define β_0, β_1 by

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = -\beta_0 g^3 - \beta_1 g^5 + O(g^7), \qquad \beta_0 = \frac{11N}{48\pi^2}, \qquad \beta_1 = \frac{34N^2}{3(16\pi^2)^2}. \tag{46.24}$$

Then the associated lattice scale Λ_L is characterized by

$$a\Lambda_L = \exp\left\{-\frac{1}{2\beta_0 g_0^2}\right\} (\beta_0 g_0^2)^{-\beta_1/(2\beta_0^2)} \big(1 + O(g_0^2)\big), \qquad g_0^2 = \frac{2N}{\beta}. \tag{46.25}$$

The numerical value of Λ_L depends on the lattice action and on the definition of the bare coupling. Ratios between Λ_L and scales in other schemes, such as $\Lambda_{\overline{\mathrm{MS}}}$, are determined by matching a short-distance observable in the overlap of the two regularized perturbation theories.

The formula (46.25) gives the perturbative tangent to the scaling trajectory near $g_0 = 0$. A construction of four-dimensional Yang–Mills theory must also show that the gauge-invariant Schwinger functions or an equivalent algebraic object converge along such a trajectory and that the reconstructed Hilbert-space theory has the desired spectral properties.

46.7 Local Operators, Counterterms, and Lattice Symmetries

The Wilson lattice action has exact gauge invariance, lattice translations, hypercubic rotations and reflections, and charge conjugation when the θ-angle is set to zero. The continuum local terms that can be generated by the lattice regulator must respect these symmetries. This gives the Symanzik form of a local long-distance effective action:

$$S_{\mathrm{Sym}} = \int \mathrm{d}^4x \left[\mathcal{E}_0(a) + \frac{1}{4g_R(a)^2} \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) + \sum_i a^{d_i-4} c_i(a)\mathcal{O}_i(x)\right] + \mathrm{i}\theta \frac{1}{8\pi^2} \int \operatorname{tr}(F \wedge F). \tag{46.26}$$

Here $\mathcal{E}_0(a)$ is a vacuum-energy density, $d_i > 4$, and $\mathcal{O}_i$ are gauge-invariant local operators compatible with the lattice symmetries. In pure $SU(N)$ Yang–Mills, with CP imposed, the only nontrivial gauge-invariant dimension-four term is $\operatorname{tr}(F_{\mu\nu}F_{\mu\nu})$. If CP is not imposed, the topological density $\operatorname{tr}(F \wedge F)$ supplies the separate θ-parameter. Gauge invariance forbids a gluon mass term, and hypercubic symmetry makes the coefficient of $F_{\mu\nu}F_{\mu\nu}$ the same for every coordinate two-plane on an isotropic lattice.

Definition 46.5 (Tuning for a lattice continuum limit)**.** A lattice parameter is tuned when it must be chosen as a function of a in order to keep a specified renormalized observable finite and nonzero as $a \downarrow 0$. Parameters multiplying operators with dimension below four are relevant at the Gaussian fixed point; parameters multiplying dimension-four operators are marginal at tree level and must be classified by the beta function and by symmetry.

For pure CP-even $SU(N)$ Yang–Mills on an isotropic Wilson lattice, this power-counting and symmetry analysis leaves one running parameter, $g_0(a)$. The mass gap, glueball spectrum, and string tension are then outputs of the continuum theory, if the continuum limit exists. Different gauge-invariant lattice actions with the same symmetries change the coefficients of irrelevant operators in (46.26) and change the relation between their lattice scales. Symanzik improvement chooses additional irrelevant plaquette and loop terms so that the leading lattice artifacts begin at higher powers of a.

For theories whose ultraviolet flow is not asymptotically free, the preceding one-parameter mechanism may fail to produce an interacting continuum theory. Four-dimensional scalar ϕ^4 theory and four-dimensional QED are the basic cautionary examples: perturbation theory points toward a Landau-pole obstruction, and rigorous triviality results constrain large classes of lattice scalar limits. Those results do not settle every possible UV-complete theory that has the same formal perturbative expansion around the Gaussian point. The monograph therefore treats triviality and asymptotic non-freedom as theory-specific continuum-limit questions, not as consequences of formal power series alone.

46.8 Glueball Spectrum and Flux-Tube Observables

The preceding sections define finite-regulator expectation values. To discuss the pure Yang–Mills spectrum one must say which gauge-invariant Hilbert-space object is being extracted from those expectations. At finite spatial volume and finite lattice spacing, reflection positivity supplies a transfer matrix $\mathbb{T} = \exp(-aH_{\rm lat})$ on the gauge-invariant Hilbert space. A glueball is an eigenstate of $H_{\rm lat}$ in the vacuum superselection sector, created with nonzero overlap by a gauge-invariant local operator built from closed Wilson loops or lattice field-strength composites. This is a finite-regulator definition. A continuum glueball mass is a limit of energy differences along a scaling trajectory, if the limit exists.

Let $\mathcal{O}_i(t)$, $i = 1, \dots, m$, be gauge-invariant operators projected to a chosen zero-momentum lattice symmetry channel and with vacuum expectation subtracted when the channel permits it. The Euclidean correlation matrix is

$$C_{ij}(t) = \langle \mathcal{O}_i(t)\mathcal{O}_j(0)\rangle - \langle \mathcal{O}_i\rangle\langle \mathcal{O}_j\rangle. \tag{46.27}$$

If $0 = E_0 < E_1 \le E_2 \le \cdots$ are transfer-matrix energies in this channel, then the spectral theorem gives

$$C_{ij}(t) = \sum_{n\ge 1} Z_i^{(n)}\overline{Z_j^{(n)}}\,\mathrm{e}^{-aE_n t}, \qquad Z_i^{(n)} = \langle \Omega|\mathcal{O}_i|n\rangle. \tag{46.28}$$

The integer t is Euclidean time in lattice units. The connected subtraction removes the vacuum term; without it the scalar 0^{++} channel would be dominated by the identity.

Equation (46.28) is the Yang–Mills datum needed here. A finite-volume calculation must declare how the energies appearing in this spectral representation are estimated, but the estimator is not part of the definition of a glueball state. The Yang–Mills-specific content is the construction of gauge-invariant Wilson-loop source spaces, the cubic-group channel assignment, reflection positivity of the chosen lattice regulator, and the continuum/infinite-volume limit that turns finite-regulator energies into pure-glue spectral data. Correlator-matrix algebra and residual diagnostics are numerical-analysis data, not additional pure-Yang–Mills definitions.

The continuum spin labels J^{PC} are recovered only after the cubic lattice irreducible representations have been compared across lattice spacings and volumes. The following table records the operator meaning of the lowest pure-glue channels; it is not a phenomenological assignment to

QCD-with-quarks resonances.

continuum channel	typical continuum operator	cubic-lattice channel
0^{++}	$\operatorname{tr}(F_{\mu\nu}F_{\mu\nu})$	A_1^{++}
2^{++}	$\operatorname{tr}\big(F_{\mu\alpha}F_{\nu\alpha} - \frac{1}{4}\delta_{\mu\nu}F_{\rho\sigma}F_{\rho\sigma}\big)$	$E^{++} \oplus T_2^{++}$
0^{-+}	$\operatorname{tr}(F_{\mu\nu}\widetilde{F}_{\mu\nu})$	A_1^{-+}
0^{+-}	multi-loop CP-odd scalar combinations	A_1^{+-}
1^{+-}, 1^{-+}	hypercubic vector loop combinations	T_1^{+-}, T_1^{-+}

At nonzero lattice spacing the cubic irreducible representation is the exact label. A continuum spin claim is supported when the expected degeneracies among cubic irreducibles emerge under continuum extrapolation; for example a continuum $J = 2$ state splits into the E and T_2 representations of the cubic group at finite a.

Definition 46.6 (Pure-glue spectral continuum datum)**.** A pure-glue spectral continuum datum consists of:

1. a choice of gauge group and global form;
2. a reflection-positive lattice action and scaling trajectory;
3. a basis of gauge-invariant operators in each cubic PC channel, together with smearing or blocking maps stated as regulator definitions;
4. finite-volume energy estimates, with the estimator, time windows, source-basis conditioning, and stability diagnostics declared as regulator data;
5. continuum and infinite-volume extrapolations of dimensionless ratios such as $M_{J^{PC}}/\sqrt{\sigma}$ or $M_{J^{PC}}/\Lambda_{\overline{\mathrm{MS}}}$;
6. a systematic-error budget for excited-state contamination, volume effects, lattice-spacing artifacts, topological freezing, and operator-basis conditioning.

Only after these data are stated is a numerical glueball spectrum a statement about pure Yang–Mills theory rather than about a particular finite lattice calculation.

The static flux-tube sector is extracted from rectangular Wilson loops or from Polyakov-line correlators. For two static sources in a representation R, the finite-regulator static energy is

$$E_R(L) = -\lim_{T\to\infty} \frac{1}{aT} \log\langle W_R(L,T)\rangle_{\mathrm{conn}}, \tag{46.29}$$

when the limit exists. The string tension in a stable center-charge sector is

$$\sigma_R = \lim_{L\to\infty} \frac{E_R(L)}{L}.$$

Chapter 49 develops the long-string effective expansion: the Luscher term, excited open and closed flux-tube levels, k-string sectors, baryonic flux networks, and the distinction between winding closed flux tubes and contractible glueball states. The role of the present chapter is the regulator-level extraction: the operators are well-defined finite Haar-integral observables, and the continuum question is whether their dimensionless spectral data converge.

Controlled Approximation 46.7 (Large-N glueball width counting)**.** Normalize single-trace glueball operators formally by

$$\mathcal{O}_N(x) = \frac{1}{N} \operatorname{tr} \mathcal{P}(F, D)(x),$$

so that connected two-point functions are $O(N^0)$ in the confining large-N scaling regime. If the topological large-N expansion has isolated single-glueball poles with LSZ residues of order one, then connected k-glueball amplitudes scale as N^{2-k}. In particular a three-glueball coupling is $O(N^{-1})$, and a glueball decay width into two glueballs is $O(N^{-2})$, when the decay is kinematically allowed. The double-line expansion for connected correlators of k normalized single-trace insertions has leading sphere topology with k marked boundaries, hence N^{2-k}. Isolated poles with order-one residues do not change the N-power when the external legs are amputated. A decay width is quadratic in the on-shell three-point coupling and includes no compensating positive power of N in the density of a fixed number of color-singlet final states, so it is $O(N^{-2})$.

None of the preceding statements proves the four-dimensional continuum existence and mass-gap theorem for Yang–Mills theory. They give the precise finite-regulator and continuum-extrapolation language in which the expected mass gap, glueball spectrum, and confining flux-string data are formulated. The rigorous construction problem remains the one stated at the end of this chapter: produce a continuum local quantum field theory, prove positivity and locality after reconstruction, and prove a strictly positive lowest color-singlet excitation above the vacuum.

46.9 The θ-Term and Lattice Topology

For a smooth continuum $SU(N)$ connection on a closed four-manifold,

$$Q[A] = \frac{1}{8\pi^2} \int \operatorname{tr}(F \wedge F) \in \mathbb{Z}$$

is the second Chern number. The Euclidean action may include $\mathrm{i}\theta Q[A]$, and the integer-valuedness of Q gives 2π-periodicity in θ.

At finite lattice spacing, a generic lattice gauge field is only an assignment of group elements to links. There is no literal smooth bundle connection from which Q can be read without an additional construction. A plaquette discretization of the density $\operatorname{tr}(F \wedge F)$ has the correct classical limit on smooth fields, but it is not integer-valued on arbitrary lattice fields. Consequently $\exp(\mathrm{i}\theta Q_{\mathrm{plaq}})$ need not be 2π-periodic at finite a.

There are several regulator-level ways to define topological charge more carefully. A geometric definition interpolates admissible lattice fields to a bundle over the torus and evaluates the corresponding Chern number. With Ginsparg–Wilson fermions, the index of the lattice Dirac operator gives an integer

$$Q_{\mathrm{GW}} = \operatorname{Tr} \gamma_5 \left(1 - \frac{aD_{\mathrm{GW}}}{2}\right)$$

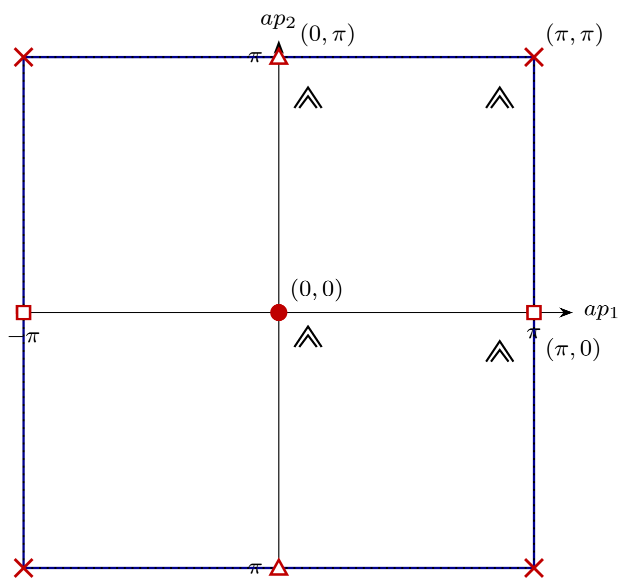
under the usual admissibility hypotheses. Gradient-flow definitions smooth the gauge field before evaluating a lattice density and are designed to have a controlled continuum limit. These choices agree in the continuum scaling regime when the required estimates hold, but they are distinct finite-cutoff definitions. Thus the θ-angle is an additional parameter whose nonperturbative lattice definition requires its own regulator convention.

46.10 Fermions: Doubling and Chiral Lattice Actions

A vectorlike gauge theory with fermions requires a lattice Dirac operator coupled to the link variables. The naive free lattice Dirac operator is

$$D_{\mathrm{n}}(p) = \frac{\mathrm{i}}{a}\sum_{\mu=1}^{4}\gamma_\mu \sin(ap_\mu), \qquad p_\mu \in [-\pi/a, \pi/a). \tag{46.30}$$

Its zeros occur whenever every ap_μ is 0 or π. In four dimensions this gives $2^4 = 16$ light species near the corners of the Brillouin zone. Figure 46.3 displays the two-dimensional Brillouin-torus version, where edge identifications turn nine drawn zeros into four inequivalent continuum cones.



edge identifications of the Brillouin torus group the nine displayed zeros into four inequivalent two-dimensional Dirac species; four dimensions have 2^4 species

Figure 46.3: The naive lattice derivative vanishes at every Brillouin-zone corner. In two dimensions the center, the two identified horizontal-edge zeros, the two identified vertical-edge zeros, and the four identified corners give four inequivalent continuum Dirac cones.

Theorem 46.8 (Degree obstruction for chiral lattice symbols)**.** *Let $f = (f_1, \dots, f_D) : \mathbb{T}^D \to \mathbb{R}^D$ be a smooth periodic map whose zeros are isolated and nondegenerate, and let*

$$D_f(p) = \mathrm{i}\sum_{\mu=1}^{D}\gamma_\mu f_\mu(p)$$

with Euclidean gamma matrices satisfying $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$. This symbol anticommutes with the fixed chirality matrix γ_{D+1}. Assign to a zero p_* the chirality sign

$$\chi(p_*) = \operatorname{sign}\det\left(\frac{\partial f_\mu}{\partial p_\nu}(p_*)\right).$$

Then

$$\sum_{f(p_*)=0}\chi(p_*) = 0.$$

Consequently a chirally symmetric periodic symbol of this form cannot contain only one nondegenerate continuum Weyl cone.

Proof. For a regular value $y \in \mathbb{R}^D$ of f, define the signed preimage count

$$I(y) = \sum_{f(p)=y} \operatorname{sign}\det\left(\frac{\partial f_\mu}{\partial p_\nu}(p)\right).$$

Sard's theorem makes the regular values dense. On a connected open set of regular values, $I(y)$ is locally constant: if $y(s)$ is a small path of regular values and $f(p_0) = y(0)$, the inverse function theorem gives a unique smooth branch $p(s)$ with $f(p(s)) = y(s)$, and the determinant sign cannot change along the branch because it never vanishes.

The image $f(\mathbb{T}^D)$ is compact, so there exists a regular value y_∞ outside the image; for it $I(y_\infty) = 0$. Join 0 to y_∞ by a smooth path and perturb it, with endpoints fixed, so that it meets the critical-value set only in isolated birth–death crossings. At such a crossing the local normal form creates or annihilates two simple preimages with opposite determinant signs. Hence the signed count is unchanged across the crossing. Therefore $I(0) = I(y_\infty) = 0$, which is the displayed chirality-sum identity.

Near a nondegenerate zero, $D_f(p)$ linearizes to $\mathrm{i}\gamma_\mu A^\mu{}_\nu(p - p_*)^\nu$. The sign of $\det A$ is exactly the orientation, hence chirality, of that continuum cone relative to the chosen gamma-matrix orientation. A single Weyl cone would have total chirality ± 1, contradicting the zero sum. □

The full Nielsen–Ninomiya theorem treats broader local, translation-invariant, chirally symmetric lattice Dirac operators by replacing $f/|f|$ with the appropriate normalized symbol on the Brillouin torus with small neighborhoods of the zeros removed. The obstruction is the same degree identity: a compact momentum torus has no boundary, so the local winding numbers of all zeros must sum to zero. Later lattice-fermion chapters spell out how Wilson and Ginsparg–Wilson constructions evade different hypotheses.

Standard lattice fermion constructions evade different hypotheses of this theorem.

Wilson fermions
: Add $ra\,\Delta/2$, giving doublers masses of order $1/a$. The Wilson term breaks chiral symmetry and produces additive mass renormalization, so the chiral limit requires tuning $m_0(a)$ to a critical value $m_c(a)$.

Staggered fermions
: Diagonalize part of the spin structure and reduce the number of tastes. The residual taste symmetry and rooting questions must be treated as finite-cutoff regularization issues.

Domain-wall fermions
: Place chiral modes on separated boundaries of an auxiliary fifth direction. Finite fifth-dimensional size produces residual chiral-symmetry breaking.

Overlap and Ginsparg–Wilson fermions
: Replace $\{\gamma_5, D\} = 0$ by $\gamma_5 D + D\gamma_5 = aD\gamma_5 D$, giving an exact modified lattice chiral symmetry and an index theorem at finite cutoff under admissibility hypotheses.

For vectorlike QCD, these formulations provide regulator choices for gauge-invariant correlation functions. Their continuum equivalence requires the same kind of scaling-limit control as in the pure gauge theory, together with the relevant mass and chiral-symmetry tunings.

46.11 Status of the Four-Dimensional Continuum Construction

Hypothesis 46.9 (Four-dimensional lattice Yang–Mills continuum limit)**.** Let $G = SU(N)$ and let the Wilson lattice measures $d\mu_{\Lambda,\beta}$ be defined by (46.5). There exists a scaling trajectory $\beta(a) \to \infty$ and an infinite-volume limit such that the gauge-invariant Schwinger functions of local and loop observables converge, after the required field and operator normalizations, to Schwinger functions satisfying the continuum Osterwalder–Schrader hypotheses. The reconstructed Hilbert-space theory has a unique vacuum and a positive mass gap.

This hypothesis is the constructive content of four-dimensional pure Yang–Mills theory. The finite lattice theory is rigorous; the hypothesis is the assertion that the rigorous finite-cutoff objects have the required continuum limit and spectral gap. Renormalization-group analyses of lattice gauge theory, including Bałaban's multiscale program, prove substantial estimates toward such a construction. Other constructive results in related models and dimensions give additional control of particular mechanisms. A complete proof of Hypothesis 46.9 for four-dimensional nonabelian Yang–Mills theory remains the Yang–Mills existence and mass-gap problem.

The strong-coupling area law is a theorem inside the small-β lattice regime. The weak-coupling continuum regime lies at $\beta \to \infty$. For four-dimensional pure $SU(2)$ and $SU(3)$ Wilson gauge theory, numerical and analytic evidence support a confining continuum limit connected to the strong-coupling phase without a physical deconfining transition at zero temperature. For other groups and lattice actions, bulk lattice-artifact transitions may occur and must be distinguished from continuum physics. The logical role of the lattice construction is therefore precise: it supplies a finite, gauge-invariant nonperturbative regulator and a clear continuum-limit problem; it does not by itself solve the mass-gap problem.

Covariant gauge fixing, Faddeev–Popov ghosts, BRST cohomology, and the BV formalism developed in the following chapters are indispensable perturbative and cohomological tools. Their nonperturbative identification with the continuum theory constructed from the lattice rests on two further inputs: first, the lattice continuum limit in Hypothesis 46.9; second, agreement of short-distance gauge-invariant observables with the covariant perturbative expansion order by order after matching the renormalized coupling and local operator normalizations.

Chapter 47

Gauge Fixing, Ghosts, and BRST Cohomology

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Remark 47.1 (Regulator-level status of covariant gauge fixing). The gauge-fixing, ghost, and BRST constructions in this chapter are local coordinate systems for perturbative gauge theory and for the cohomological classification of local operators, counterterms, and anomalies. The finite-dimensional nonperturbative regulator used in this volume for pure Yang–Mills theory is the lattice construction of Chapter 46. The identification of the covariant BRST expansion with that lattice-based continuum theory is made through gauge-invariant observables and order-by-order short-distance matching, together with the continuum-limit hypothesis stated in Hypothesis 46.9.

47.1 Gauge Orbits and Local Coordinates

Let M be the spacetime manifold and let G be the compact gauge group whose Lie algebra conventions were fixed in Chapter 45. On a fixed local trivialization, the space of gauge potentials is the affine space

$$\mathcal{A} = \Omega^1(M, \mathfrak{g}).$$

Here $\mathfrak{g} = i\,\mathfrak{g}_{\mathrm{ah}}$ is the Hermitian coordinate space associated with the anti-Hermitian Lie algebra $\mathfrak{g}_{\mathrm{ah}} = \mathrm{Lie}(G)$ in a unitary representation. If $X, Y \in \mathfrak{g}$, the compact real Lie bracket is

$$[X, Y]_{\mathrm{H}} := -\mathrm{i}[X, Y],$$

where the bracket on the right is the matrix commutator in a unitary representation. The gauge transformations form the infinite-dimensional group

$$\mathcal{G} = C^{\infty}(M, G).$$

This group must be distinguished from the finite-dimensional internal group G. In local coordinates near the identity, a gauge transformation can be written as

$$g(x) = \exp\bigl(\mathrm{i}\zeta^{a}(x)t^{a}\bigr), \qquad \xi^{\alpha} \equiv \zeta^{a}(x), \qquad \alpha = (a, x).$$

Repeated condensed indices include sums over Lie algebra labels and integrals over spacetime; for example

$$f_{\alpha}g^{\alpha} = \sum_{a} \int_{M} f_{a}(x)g^{a}(x)\,\mathrm{d}^{D}x. \tag{47.1}$$

The action of $g \in \mathcal{G}$ on $A \in \mathcal{A}$ is

$$A_{\mu} \longmapsto A_{\mu}^{g} = gA_{\mu}g^{-1} + \mathrm{i}g\partial_{\mu}g^{-1}.$$

The infinitesimal action associated with $\zeta = \zeta^{a}t^{a} \in C^{\infty}(M, \mathfrak{g})$ is

$$\delta_{\zeta}A_{\mu} = D_{\mu}(A)\zeta, \qquad D_{\mu}(A)\zeta = \partial_{\mu}\zeta + [A_{\mu}, \zeta]_{\mathrm{H}}, \qquad (D_{\mu}(A)\zeta)^{a} = \partial_{\mu}\zeta^{a} + f^{a}{}_{bc}A_{\mu}^{b}\zeta^{c}.$$

Let ϕ denote the complete list of fields: gauge potentials and, when present, matter fields. At a fixed ultraviolet and infrared regulator, before the gauge orbits are divided out, the gauge-field integral is an integral over a regulated field space $\mathcal{F}_{\Lambda}$,

$$Z_{\mathrm{unfixed},\Lambda} = \int_{\mathcal{F}_{\Lambda}} \mathrm{d}\mu_{\Lambda}(\phi)\, \exp(\mathrm{i}S_{\Lambda}[\phi]), \qquad S[\phi^{\xi}] = S[\phi], \qquad \mathrm{d}\mu_{\Lambda}(\phi^{\xi}) = \mathrm{d}\mu_{\Lambda}(\phi). \tag{47.2}$$

Here $\mathrm{d}\mu_{\Lambda}$ is the regulator measure induced by the chosen regularization of the bosonic and fermionic variables. Equation (47.2) is not a definition of a gauge theory by itself: if the gauge group acts freely, it integrates separately over orbit directions and over directions transverse to the orbits; if stabilizers or Gribov phenomena are present, the same statement holds only in sufficiently small coordinate charts. In canonical language the orbit directions are the directions generated by first-class constraints. Covariant perturbation theory therefore chooses local coordinates transverse to the orbits and keeps track of the corresponding Jacobian.

A gauge condition is a collection of functionals $F^{A}[\phi]$, with the index A carrying the same number of local degrees of freedom as the gauge parameter index α. The intended local slice is $F^{A}[\phi] = 0$. Figure 47.1 records the geometric content.

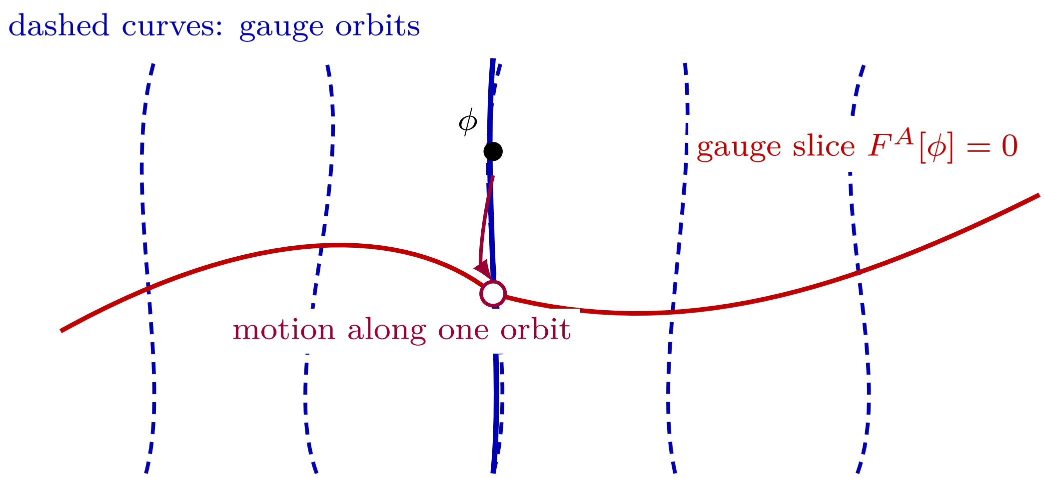


Figure 47.1: A local gauge condition is a slice through field space. The blue dashed curves are gauge orbits, the heavier red curve is the surface $F^A[\phi] = 0$, and the filled point ϕ and open point ϕ^ξ lie on the same highlighted gauge orbit. The Faddeev–Popov determinant is the Jacobian that converts integration along the orbit to integration through the slice.

The infinitesimal variation of the gauge condition along the orbit defines the Faddeev–Popov matrix

$$\mathcal{M}^A{}_\alpha[\phi] := \left.\frac{\partial F^A[\phi^\xi]}{\partial \xi^\alpha}\right|_{\xi=\mathrm{id}}. \tag{47.3}$$

For Yang–Mills gauge potentials alone this is the distributional operator

$$\mathcal{M}_F^{ab}(A; x, y) = \left.\frac{\delta F^a(A^{\exp(\mathrm{i}\zeta)})(x)}{\delta \zeta^b(y)}\right|_{\zeta=0}. \tag{47.4}$$

Proposition 47.2 (Orbit measure and the Faddeev–Popov density)**.** *Let $U \subset \mathcal{F}_\Lambda$ be a finite-regulator field-space region, or a Sobolev local Hilbert-manifold chart with determinant regularization specified, on which the gauge group $\mathcal{G}_\Lambda$ acts freely and properly. Let*

$$q : U \longrightarrow B_U := U/\mathcal{G}_\Lambda$$

be the quotient map, and let $F : U \to E$ be a gauge condition such that

$$\mathcal{S}_F = \{\phi \in U : F(\phi) = 0\}$$

is the image of a local section $s_F : B_U \to U$ of q. Equivalently, the map

$$\Theta : \mathcal{S}_F \times \mathcal{G}_\Lambda \longrightarrow U, \qquad (\sigma, \xi) \longmapsto \sigma^\xi$$

is a diffeomorphism onto U. Suppose $\mathrm{d}\mu_\Lambda$ is a $\mathcal{G}_\Lambda$-invariant density on U, and let $\mathrm{d}\xi_{\mathrm{H}}$ be a right Haar density on $\mathcal{G}_\Lambda$. For each $\phi \in U$, the orbit measure $\mathrm{d}\mu_{\mathcal{O}_\phi}$ is the pushforward of $\mathrm{d}\xi_{\mathrm{H}}$ by $\xi \mapsto \phi^\xi$. The restriction of F to the orbit has differential

$$\mathrm{d}(F|_{\mathcal{O}_\phi})_\phi = \mathcal{M}_F[\phi] : T_{\mathrm{id}}\mathcal{G}_\Lambda \longrightarrow E.$$

If the orbit intersects $\mathcal{S}_F$ once in the chart, then

$$\int_{\mathcal{O}_\phi} \mathrm{d}\mu_{\mathcal{O}_\phi}(\psi)\, \delta_E(F(\psi))\, \big|\det \mathcal{M}_F[\psi]\big| = 1. \tag{47.5}$$

Consequently the quotient measure represented in the section $\mathcal{S}_F$ is characterized by the pushed-forward density

$$(s_F)_*\mathrm{d}\nu_F = \delta_E(F(\phi))\, \big|\det \mathcal{M}_F[\phi]\big|\, \mathrm{d}\mu_\Lambda(\phi) \tag{47.6}$$

on U, as an identity of densities tested against functions supported in the coordinate chart. Equivalently, for every gauge-invariant test function H,

$$\int_{B_U} \mathrm{d}\nu_F([\,\phi\,])\, H([\,\phi\,]) = \int_U \mathrm{d}\mu_\Lambda(\phi)\, \delta_E(F(\phi))\, \big|\det \mathcal{M}_F[\phi]\big|\, H(q(\phi)). \tag{47.7}$$

The oriented perturbative Faddeev–Popov formula replaces the absolute value by $\det \mathcal{M}_F$ after choosing an orientation of the determinant line on the chart. The ghost path integral represents this oriented determinant. If $\det \mathcal{M}_F$ vanishes, or if an orbit intersects the section more than once, the single-chart quotient-measure formula has left its domain of validity.

Proof. Fix an orbit in U and write $\psi = \sigma^\xi$ with $\sigma \in \mathcal{S}_F$. The map

$$\chi_\sigma : \mathcal{G}_\Lambda \longrightarrow E, \qquad \chi_\sigma(\xi) = F(\sigma^\xi)$$

has derivative $\mathcal{M}_F[\sigma^\xi]$ in the orbit directions. Since the slice is transverse and intersects the orbit once, χ_σ is a local diffeomorphism from a neighborhood of the relevant group element to a neighborhood of $0 \in E$. The ordinary change-of-variables formula gives

$$\mathrm{d}\chi_\sigma = \big|\det \mathcal{M}_F[\sigma^\xi]\big|\, \mathrm{d}\xi_{\mathrm{H}}$$

in these coordinates, and therefore

$$\int_{\mathcal{G}_\Lambda} \mathrm{d}\xi_{\mathrm{H}}\, \delta_E(F(\sigma^\xi))\, \big|\det \mathcal{M}_F[\sigma^\xi]\big| = \int_E \mathrm{d}u\, \delta_E(u) = 1.$$

This proves (47.5). The action is free and proper on U, so $q : U \to B_U$ is a local principal bundle and the invariant density disintegrates in the product coordinates supplied by Θ into a quotient density times the Haar density on the orbit. Inserting (47.5) into the orbit factor of this disintegration gives (47.7). Multiple intersections replace the right-hand side of (47.5) by the intersection number, and zero modes of $\mathcal{M}_F$ are precisely the failure of the change of variables. □

In a field-space region where the slice is transverse to the orbit, and after separating zero modes associated with unfixed global transformations, the oriented perturbative Faddeev–Popov identity is

$$1 = \int_{\mathcal{G}} D\xi\, \delta\big(F^A[\phi^\xi]\big) \det\left(\frac{\partial F^A[\phi^\xi]}{\partial \xi^\alpha}\right). \tag{47.8}$$

Inserting (47.8) in (47.2) and changing variables $\phi \mapsto \phi^{\xi^{-1}}$ gives a factor depending only on integration over the gauge group. More explicitly, the determinant at a general point ξ on the orbit splits as

$$\det \frac{\partial F^A[\phi^{\xi'}]}{\partial \xi'^\alpha}\bigg|_{\xi'=\xi} = \det \frac{\partial(\xi'' \circ \xi)^\alpha}{\partial \xi''^\beta}\bigg|_{\xi''=\mathrm{id}} \det \frac{\partial F^A[\phi^{\xi''}]}{\partial \xi''^\alpha}\bigg|_{\xi''=\mathrm{id}}.$$

The first factor converts the coordinate measure $D\xi$ to the right Haar measure $[D\xi]_{\mathrm{H}}$, characterized formally by

$$[D(\xi \circ \xi_1)]_{\mathrm{H}} = [D\xi]_{\mathrm{H}}, \qquad \xi_1 \in \mathcal{G} \text{ fixed.} \tag{47.9}$$

Thus

$$\int_{\mathcal{G}} [D\xi]_{\mathrm{H}}$$

is the formal volume of the gauge-transformation group, and the normalized gauge-theory path integral is

$$Z = \frac{1}{\int_{\mathcal{G}} [D\xi]_{\mathrm{H}}} \int [D\phi] \exp(\mathrm{i}S[\phi]). \tag{47.10}$$

The same expression, written in local gauge-fixed coordinates, is

$$Z = \int [D\phi]\, \delta\big(F^A[\phi]\big) \Delta_{\mathrm{FP}}[\phi] \exp(\mathrm{i}S[\phi]), \qquad \Delta_{\mathrm{FP}}[\phi] = \det \mathcal{M}[\phi]. \tag{47.11}$$

Definition 47.3 (Local gauge slice and global section)**.** A local gauge slice in a region $U \subset \mathcal{F}_\Lambda$ is a subspace $\mathcal{S}_F = \{\phi \in U : F^A[\phi] = 0\}$ such that the map

$$\mathcal{S}_F \times \mathcal{G} \longrightarrow U, \qquad (\phi, \xi) \longmapsto \phi^\xi$$

is locally invertible after zero modes and stabilizers have been separated. Equivalently, the Faddeev–Popov operator ${\mathcal{M}^A}_\alpha[\phi]$ is invertible on the nonzero gauge directions. A global smooth gauge fixing would be a smooth section of the quotient map $\mathcal{F}_\Lambda \to \mathcal{F}_\Lambda/\mathcal{G}$, or of the corresponding irreducible-connection quotient after stabilizers are removed.

Proposition 47.4 (Coulomb-slice atlas on irreducible connections)**.** *Let X be a compact Euclidean spacetime, let $P \to X$ be a smooth principal G-bundle with G compact, and fix an integer $k >$* $\dim X/2 + 1$. *Write*

$$\mathcal{A}_k(P) = A_0 + L^2_k(T^*X \otimes \operatorname{ad} P), \qquad \mathcal{G}_{k+1}(P) = L^2_{k+1}\Gamma(\operatorname{Ad} P)$$

for the Sobolev-completed space of connections and gauge group. Let $A \in \mathcal{A}_k(P)$ be irreducible, and divide by a based gauge group, or equivalently remove the finite stabilizer, so that the gauge action is free at A. Denote by

$$d_A : L^2_{k+1}(\operatorname{ad} P) \longrightarrow L^2_k(T^*X \otimes \operatorname{ad} P)$$

the infinitesimal gauge action and by d_A^* its L^2-adjoint. After the stabilizer directions have been removed, the elliptic operator

$$\mathcal{M}_A = d_A^* d_A$$

has trivial kernel. Hence, for sufficiently small $\epsilon > 0$,

$$\mathcal{S}_A(\epsilon) = \{\, A + a : \ d_A^* a = 0, \ \|a\|_{L^2_k} < \epsilon \,\}$$

is a local slice: the map

$$\mathcal{S}_A(\epsilon) \times \mathcal{U}_{\mathrm{id}} \longrightarrow \mathcal{A}_k(P), \qquad (B, g) \longmapsto B^g,$$

with $\mathcal{U}_{\mathrm{id}}$ a small neighborhood of the identity in the based gauge group, is a Hilbert-manifold coordinate chart onto a neighborhood of the orbit of A. In these coordinates $\mathcal{M}_A$ is the Faddeev–Popov operator of the background Coulomb gauge condition $d_A^*(B - A) = 0$.

Proof. The derivative of the displayed map at (A, id) sends a transverse variation a and an infinitesimal gauge parameter ζ to

$$a + d_A\zeta.$$

The tangent space to the proposed slice is $\ker d_A^*$, because the Coulomb condition $d_A^*(B - A) = 0$ linearizes to $d_A^* a = 0$. Elliptic Fredholm theory gives the Hodge-type decomposition

$$L^2_k(T^*X \otimes \mathrm{ad}\, P) = \ker d_A^* \oplus \mathrm{im}\, d_A$$

after the finite-dimensional kernel of d_A has been removed. The orthogonality follows from the adjoint relation $\langle a, d_A\zeta\rangle = \langle d_A^* a, \zeta\rangle$, and closedness of $\mathrm{im}\, d_A$ follows from ellipticity of $d_A^* d_A$ on the stabilizer-free gauge algebra.

The inverse on the orbit directions is obtained explicitly. Given a one-form u, solve

$$d_A^* d_A \zeta = d_A^* u$$

for ζ; invertibility of $\mathcal{M}_A = d_A^* d_A$ is precisely the irreducibility condition with stabilizer directions removed. Then

$$u = (u - d_A\zeta) + d_A\zeta, \qquad d_A^*(u - d_A\zeta) = 0,$$

so the derivative

$$\ker d_A^* \oplus L^2_{k+1}(\mathrm{ad}\, P)_{\mathrm{free}} \longrightarrow L^2_k(T^*X \otimes \mathrm{ad}\, P)$$

is a Banach-space isomorphism. The Sobolev condition $k > \dim X/2 + 1$ makes L^2_{k+1} gauge transformations at least C^1 and makes multiplication/composition smooth in the Hilbert-manifold category. The Banach inverse function theorem therefore gives a local diffeomorphism from a neighborhood of (A, id) in $\mathcal{S}_A(\epsilon) \times \mathcal{U}_{\mathrm{id}}$ onto a neighborhood of the orbit of A. In this chart the linearization of the gauge condition along the orbit is $d_A^* d_A$, hence the background Coulomb Faddeev–Popov operator is $\mathcal{M}_A$. □

The word "local" in the Faddeev–Popov formulae below means precisely this kind of chart, or its finite-regulator analogue. Perturbation theory around a background uses one such chart and expands the resulting local Jacobian. A nonperturbative continuum gauge-fixed measure would require compatible integration data on an atlas of such charts, with boundary and overlap contributions controlled. The elementary Faddeev–Popov formula is not that global construction.

Quoted Theorem 47.5 (Singer obstruction, structural form)**.** *Let $P \to M$ be a principal bundle over a compact smooth Euclidean spacetime, let G be compact and nonabelian, let $\mathcal{A}_{\mathrm{irr}}$ be the space of irreducible connections, and divide by a based gauge group $\mathcal{G}_*$ so that the action is free. In the standard Yang–Mills cases covered by Singer's theorem, the principal $\mathcal{G}_*$-bundle*

$$\mathcal{A}_{\mathrm{irr}} \longrightarrow \mathcal{A}_{\mathrm{irr}}/\mathcal{G}_*$$

is not globally trivial. Consequently there is no smooth gauge condition that supplies one representative on every gauge orbit and is everywhere transverse to the orbit. Any Faddeev–Popov gauge condition must therefore fail globally either by encountering zero modes of $\mathcal{M}$, by intersecting some orbits more than once, or by requiring multiple coordinate charts on orbit space.

This quoted theorem is a global topological statement rather than a perturbative loop calculation.[1] The proof mechanism is the following. A smooth global gauge condition selecting exactly one point on each orbit is a smooth section

$$\sigma : \mathcal{A}_{\mathrm{irr}}/\mathcal{G}_* \longrightarrow \mathcal{A}_{\mathrm{irr}}$$

of the quotient map. Such a section is equivalent to a trivialization of the principal bundle:

$$\Theta_\sigma : (\mathcal{A}_{\mathrm{irr}}/\mathcal{G}_*) \times \mathcal{G}_* \longrightarrow \mathcal{A}_{\mathrm{irr}}, \qquad ([A], g) \longmapsto \sigma([A])^g. \tag{47.12}$$

Conversely, a trivialization gives a section by setting $g = \mathrm{id}$. Thus the existence of a global transverse gauge slice is exactly the triviality problem for the principal $\mathcal{G}_*$-bundle of connections.

In the Sobolev completions used for the local slice theorem, the full space of connections is affine and contractible. After removing stabilizers by passing to the irreducible locus and by using the based gauge group, Singer's standard Yang–Mills cases give the same universal-bundle homotopy mechanism: $\mathcal{A}_{\mathrm{irr}}$ is a free model for $E\mathcal{G}_*$ in the range relevant to the obstruction, and $\mathcal{A}_{\mathrm{irr}}/\mathcal{G}_*$ is the corresponding model for $B\mathcal{G}_*$. A principal H-bundle with contractible total space is trivial only if H is weakly contractible. Indeed, if

$$P \cong B \times H$$

and P is contractible, then H is a homotopy retract of the contractible space P by fixing a point of B, hence all homotopy groups of H vanish.

The based gauge group in nonabelian Yang–Mills theory has nonzero homotopy. For the trivial bundle over a compactified four-dimensional patch one sees the essential source already in

$$\mathcal{G}_* \simeq \mathrm{Map}_*(S^4, G), \qquad \pi_r(\mathcal{G}_*) \cong \pi_{r+4}(G), \tag{47.13}$$

with the displayed equivalence replaced by the corresponding section space for a nontrivial bundle. Compact nonabelian Lie groups have nonvanishing higher homotopy groups, for instance by Bott periodicity in the classical families. Thus $\mathcal{G}_*$ is not weakly contractible in the cases at issue, and the orbit bundle cannot be globally trivial. This is the mathematical reason why (47.11) gives a local Faddeev–Popov coordinate formula rather than a single global nonperturbative definition of the nonabelian gauge-field integral.

Definition 47.6 (Gribov region and fundamental modular domain)**.** For Landau gauge on a compact Euclidean manifold, write the gauge condition as

$$\partial^\mu A_\mu = 0$$

in a local trivialization and remove the constant gauge zero modes. The Landau Faddeev–Popov operator is

$$\mathcal{M}_L(A) = -\partial^\mu D_\mu(A).$$

The first Gribov region is

$$\Omega_{\mathrm{G}} = \{\, A : \partial^\mu A_\mu = 0,\ \mathcal{M}_L(A) \geq 0 \,\}.$$

[1]See I. M. Singer, "Some remarks on the Gribov ambiguity," *Commun. Math. Phys.* **60** (1978) 7–12, and V. N. Gribov, "Quantization of non-Abelian gauge theories," *Nucl. Phys. B* **139** (1978) 1–19.

Its boundary, where $\mathcal{M}_L(A)$ has a zero mode, is the Gribov horizon. The fundamental modular domain is the set of absolute minima along each gauge orbit of the norm functional

$$\|A\|^2 = \int_M \mathrm{d}^D x \ \mathrm{tr}(A_\mu A^\mu) : \qquad \mathfrak{F} = \{\, A : \|A\|^2 \leq \|A^g\|^2 \text{ for all } g \in \mathcal{G} \,\}.$$

The domain $\mathfrak{F}$ lies inside Ω_{G}, but boundary points of $\mathfrak{F}$ can still be gauge-equivalent. Thus even this sharper domain is not a smooth global slice without boundary identifications.

Proposition 47.7 (Where the Landau Faddeev–Popov chart is Gribov-free)**.** *On a compact flat Euclidean spacetime with constant gauge zero modes removed, the interior*

$$\Omega_{\mathrm{G}}^\circ = \{\, A : \partial^\mu A_\mu = 0, \langle \zeta, \mathcal{M}_L(A)\zeta\rangle_{L^2} > 0 \textit{ for all } 0 \neq \zeta \,\}$$

is the Landau-gauge region where the infinitesimal Faddeev–Popov chart is transverse to the gauge orbit. The Gribov horizon $\partial\Omega_{\mathrm{G}}$ is the locus where this infinitesimal chart loses invertibility. The fundamental modular domain $\mathfrak{F}$ selects absolute minima of $\|A^g\|^2$ on each orbit, but its boundary identifications mean that it is not a smooth global coordinate slice.

Proof. For a fixed connection A, define the orbit norm functional

$$N_A(g) = \|A^g\|^2 = \int_X \mathrm{d}^D x \ \mathrm{tr}(A^g_\mu A^g_\mu).$$

Let $g(t) = \exp(\mathrm{i}t\zeta)$. With the conventions of this chapter,

$$\frac{\mathrm{d}}{\mathrm{d}t} A_\mu^{g(t)}\bigg|_{t=0} = D_\mu(A)\zeta.$$

Using invariance of the trace and integrating by parts on the compact manifold gives

$$\frac{\mathrm{d}}{\mathrm{d}t} N_A(g(t))\bigg|_{t=0} = 2\int_X \mathrm{tr}(A_\mu D_\mu(A)\zeta)\,\mathrm{d}^D x = -2\int_X \mathrm{tr}((\partial^\mu A_\mu)\zeta)\,\mathrm{d}^D x.$$

Thus the stationary points of N_A on the orbit are precisely the Landau representatives. At such a stationary point,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} N_A(g(t))\bigg|_{t=0} = 2\int_X \mathrm{tr}\big(\zeta\,[-\partial^\mu D_\mu(A)]\zeta\big)\,\mathrm{d}^D x = 2\langle \zeta, \mathcal{M}_L(A)\zeta\rangle_{L^2}.$$

Therefore strict positivity of $\mathcal{M}_L(A)$ on the nonzero gauge directions is exactly the nondegeneracy condition for the Landau slice to be a local transverse chart. When a zero mode appears, the derivative of the gauge condition along the orbit is singular, and the Faddeev–Popov coordinate formula ceases to be locally invertible. Absolute minima of N_A define $\mathfrak{F}$, but two absolute minima on the boundary may be related by a gauge transformation; such boundary identifications are quotient data, not ordinary smooth coordinates. □

Figure 47.2 records the geometry behind this proof: the Landau slice, the Gribov region where the Faddeev–Popov operator is positive, the fundamental modular region of absolute minima, and the boundary identifications left by gauge equivalence.

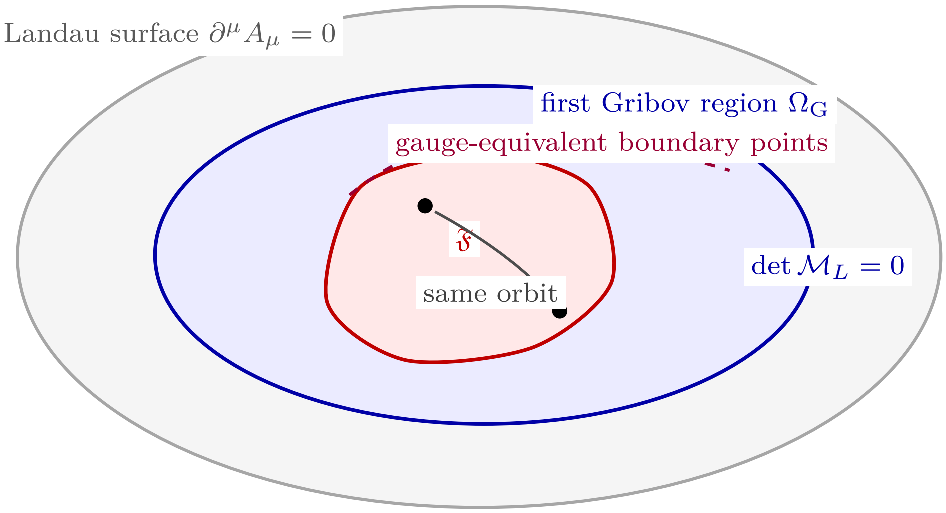


Figure 47.2: Landau gauge does not define a single global coordinate system on nonabelian orbit space. The interior of Ω_{G} is the infinitesimal Gribov-free regime where $\mathcal{M}_L$ is positive after zero modes are removed. The fundamental modular domain $\mathfrak{F}$ lies inside it but carries boundary identifications.

The perturbative Faddeev–Popov construction used in this chapter is the local construction near a chosen background, usually $A = 0$ or a fixed classical solution. Near $A = 0$, after global gauge zero modes are removed, $\mathcal{M}_L(0) = -\partial^2$ is positive on nonconstant modes, so the implicit-function theorem gives a local transverse slice. Ghost fields then represent the local Jacobian of this coordinate change. BRST symmetry, in the form used below, is the corresponding local algebraic symmetry of the gauge-fixed perturbative integrand. A global nonperturbative measure on $\mathcal{F}_\Lambda/\mathcal{G}$ requires additional data: gauge-invariant variables, patched local slices, a lattice formulation before gauge fixing, or a Gribov-region/fundamental-domain restriction with the additional boundary and BRST questions that such a restriction creates.

47.1.1 The Gribov–Zwanziger Horizon Functional

The first Gribov region gives a precise finite-regulator coordinate domain for Landau gauge, but it is still a domain with a boundary in field space. The Gribov–Zwanziger construction is a way of representing the influence of that boundary by a nonlocal functional built from the inverse Faddeev–Popov operator. The construction is used here as a controlled gauge-fixed infrared model and as a diagnostic of the failure of global covariant BRST assumptions. The gauge-invariant lattice construction remains the nonperturbative definition of Yang–Mills theory used in this volume.

Work on a compact Euclidean torus of volume V, remove the constant gauge zero modes, and impose Landau gauge. Let

$$\mathcal{M}_L(A) = -\partial^\mu D_\mu(A)$$

act on the zero-mode-removed adjoint scalar test functions. In the interior of the first Gribov region, $\mathcal{M}_L(A)$ is strictly positive and has a bounded inverse on this finite-regulator space. Define

$$B^a_{\mu,c}(x; A) := f^a{}_{bc} A^b_\mu(x), \qquad f^a{}_{cd} f^b{}_{cd} = C_A \delta^{ab}.$$

For $G = SU(N_c)$ in the trace-delta convention of this monograph, $C_A = 2N_c$. The horizon functional is

$$H(A) := \sum_{\mu,c} \left\langle B_{\mu,c}(A), \mathcal{M}_L(A)^{-1} B_{\mu,c}(A) \right\rangle_{L^2}. \tag{47.14}$$

Equivalently,

$$H(A) = \int_{M\times M} \mathrm{d}^d x\, \mathrm{d}^d y\, f^a{}_{bc} A^b_\mu(x)\, [\mathcal{M}_L(A)^{-1}]^{ad}(x,y)\, f^d{}_{ec} A^e_\mu(y),$$

where the inverse is taken after the separated zero modes have been removed. The functional is finite on $\Omega^\circ_{\mathrm{G}}$ and becomes singular along directions that approach the horizon with a nonzero projection onto a vanishing Faddeev–Popov eigenmode.

Let $A \in \Omega^\circ_{\mathrm{G}}$ at finite regulator, and let $\{e_j\}$ be an L^2-orthonormal eigenbasis of $\mathcal{M}_L(A)$ with eigenvalues $\lambda_j(A) > 0$. Then

$$H(A) = \sum_{\mu,c} \sum_j \frac{|\langle e_j, B_{\mu,c}(A)\rangle_{L^2}|^2}{\lambda_j(A)}. \tag{47.15}$$

If a path $A(s) \in \Omega^\circ_{\mathrm{G}}$ approaches a horizon point and one eigenvalue satisfies $\lambda_0(A(s)) \downarrow 0$, while

$$\liminf_s \sum_{\mu,c} |\langle e_0(s), B_{\mu,c}(A(s))\rangle|^2 > 0,$$

then $H(A(s)) \to +\infty$. Strict positivity of $\mathcal{M}_L(A)$ gives the spectral decomposition

$$\mathcal{M}_L(A)^{-1} u = \sum_j \lambda_j(A)^{-1} \langle e_j, u\rangle e_j.$$

Substituting $u = B_{\mu,c}(A)$ and summing over the displayed Lorentz and spectator color labels gives (47.15). The divergence statement follows from the single $j = 0$ term and positivity of all the other terms.

Introduce a nonnegative mass parameter γ_{G}. The nonlocal horizon-modified Landau-gauge Euclidean action is

$$S^{\mathrm{nl}}_{\mathrm{GZ}} = S^{\xi=0}_{\mathrm{FP}} + \gamma_{\mathrm{G}}^4 H(A) - \gamma_{\mathrm{G}}^4\, d\, (\dim G)\, V, \tag{47.16}$$

where $d = \dim M$, $S^{\xi=0}_{\mathrm{FP}}$ is the Landau-gauge Faddeev–Popov action, and the final term fixes the usual horizon normalization. The gap condition obtained by stationarity of the regulated vacuum functional with respect to γ_{G}^4 is

$$\langle H(A)\rangle_{\gamma_{\mathrm{G}}} = d\, (\dim G)\, V. \tag{47.17}$$

Equation (47.17) is a condition selecting the horizon parameter inside a specified finite-regulator gauge-fixed model. Its continuum and infinite-volume uses require a limiting construction for the corresponding gauge-fixed correlation functions.

The nonlocal factor $\exp[-\gamma_{\mathrm{G}}^4 H(A)]$ has a local finite-regulator representation. Introduce complex commuting fields $\varphi^{ac}_\mu, \bar\varphi^{ac}_\mu$ and Grassmann-odd fields $\omega^{ac}_\mu, \bar\omega^{ac}_\mu$, where the first adjoint index is acted on by $\mathcal{M}_L(A)$ and the second adjoint index is the spectator index c in $B_{\mu,c}$. With the contour for the bosonic fields chosen so that the Gaussian is convergent,

$$\begin{aligned}\int D\bar\varphi\, D\varphi\, D\bar\omega\, D\omega\, \exp\Big[-\bar\varphi\, \mathcal{M}_L(A)\varphi + \bar\omega\, \mathcal{M}_L(A)\omega - \mathrm{i}\gamma_{\mathrm{G}}^2 B(A)(\varphi + \bar\varphi)\Big] \\ = \mathcal{N}\, \exp[-\gamma_{\mathrm{G}}^4 H(A)].\end{aligned} \tag{47.18}$$

Here the products mean the L^2 pairing together with the sums over μ, a, c, and $\mathcal{N}$ is independent of A. The factor $\bar{\omega}\mathcal{M}_L\omega$ cancels the determinant from the commuting φ-Gaussian, leaving only the shifted source term. The factor of i records the contour convention that localizes a positive horizon functional in the Euclidean action.

The localized action at $\gamma_{\mathrm{G}} = 0$ admits the doublet assignment

$$s\varphi = \omega, \qquad s\omega = 0, \qquad s\bar{\omega} = \bar{\varphi}, \qquad s\bar{\varphi} = 0,$$

after adding the standard $s\mathcal{M}_L(A)$ term generated by applying s to $\bar{\omega}\mathcal{M}_L(A)\varphi$. Hence the auxiliary quartet does not change the local BRST cohomology when $\gamma_{\mathrm{G}} = 0$. The horizon source

$$\mathrm{i}\gamma_{\mathrm{G}}^2 B(A)(\varphi + \bar{\varphi})$$

has a nonzero s-variation for $\gamma_{\mathrm{G}} > 0$. Therefore the Gribov–Zwanziger horizon implementation and the perturbative local BRST charge are compatible only after one specifies an enlarged non-perturbative BRST or BV datum. The elementary Kugo–Ojima assumptions below, which require a global covariant BRST charge in Landau gauge, are consequently independent hypotheses rather than consequences of the horizon-restricted integral.

At the quadratic level around $A = 0$, the inverse Faddeev–Popov operator is $(-\partial^2)^{-1}$. The color contraction above gives

$$\sum_{a,c} f^a{}_{bc} f^a{}_{dc} = C_A \delta_{bd}.$$

After projecting to transverse gauge fields, the quadratic action has the form

$$S_{\mathrm{GZ},2} = \frac{1}{2} \int \frac{\mathrm{d}^d p}{(2\pi)^d}\, A_\mu^a(-p) \left[\frac{p^2}{g_{\mathrm{YM}}^2} + \frac{\lambda_{\mathrm{G}}^4}{g_{\mathrm{YM}}^2 p^2} \right] P_{\mu\nu}(p)\, A_\nu^a(p), \tag{47.19}$$

where

$$P_{\mu\nu}(p) = \delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}$$

and λ_{G}^4 is the positive horizon scale obtained from $\gamma_{\mathrm{G}}^4 C_A$ with the normalization of the quadratic action absorbed into its definition. The corresponding tree-level transverse two-point kernel is

$$\langle A_\mu^a(p) A_\nu^b(-p) \rangle_{\mathrm{GZ},2} = \delta^{ab} g_{\mathrm{YM}}^2 \frac{p^2}{p^4 + \lambda_{\mathrm{G}}^4}\, P_{\mu\nu}(p). \tag{47.20}$$

The denominator has complex p^2-plane poles at

$$p^2 = \pm \mathrm{i}\lambda_{\mathrm{G}}^2.$$

Thus the elementary tree-level horizon model already prevents the transverse gauge potential from having a positive Källén–Lehmann spectral measure. This statement concerns the gauge-fixed potential two-point function. Gauge-invariant color-singlet correlators and Wilson-loop observables remain the objects used for physical spectral questions.

Controlled Approximation 47.8 (Status of the Gribov–Zwanziger construction)**.** The Gribov–Zwanziger action is a finite-regulator Landau-gauge construction that encodes a restriction to the first Gribov region through the horizon functional (47.14). It supplies a precise infrared ansatz for gauge-fixed Green functions and a concrete test of covariant BRST assumptions. A proof that it is equivalent to the gauge-invariant Yang–Mills continuum theory requires a separate comparison of gauge-invariant observables or a construction of the gauge-fixed measure as a pushforward of the lattice continuum limit. Refined Gribov–Zwanziger models add condensates of dimension-two local fields; those refinements change the gauge-fixed infrared ansatz and require their own renormalization and BRST/BV analysis.

The determinant in (47.11) is represented by Grassmann-odd variables b_A and c^α:

$$\Delta_{\mathrm{FP}}[\phi] = \int [Db][Dc]\, \exp\left[-\mathrm{i} b_A\, \frac{\partial F^A[\phi^\xi]}{\partial \xi^\alpha}\bigg|_{\xi=\mathrm{id}} c^\alpha\right]. \tag{47.21}$$

The field c^α is the ghost associated with the gauge parameter ξ^α, and b_A is the antighost associated with the gauge condition F^A. The delta functional is represented by a commuting auxiliary field B_A:

$$\delta\big(F^A[\phi]\big) = \int [DB]\, \exp\big(\mathrm{i} B_A F^A[\phi]\big). \tag{47.22}$$

The field B_A is not an external background field or a new physical degree of freedom. It is an integrated auxiliary variable in the local gauge-fixed functional integral. Keeping it in the formula gives a polynomial local action and off-shell BRST nilpotency; carrying out the B-integration returns the delta functional imposing the sharp slice $F^A[\phi] = 0$. The gauge-fixed action in these condensed variables is therefore

$$S_{\mathrm{tot}}[\phi, B, b, c] = S[\phi] + S_{\mathrm{GF}}[\phi, B, b, c], \qquad S_{\mathrm{GF}} = B_A F^A[\phi] - b_A \mathcal{M}^A{}_\alpha[\phi] c^\alpha. \tag{47.23}$$

The term S_{GF} is not gauge invariant; its controlling symmetry is the nilpotent BRST symmetry constructed below.

47.2 Covariant Gauges and Ghost Fields

The covariant linear gauge condition is

$$F^a(A) = \partial^\mu A^a_\mu.$$

For this choice,

$$\mathcal{M}^{ab}(A) = \partial^\mu D^{ab}_\mu(A), \qquad (D_\mu c)^a = \partial_\mu c^a + f^a{}_{bc} A^b_\mu c^c. \tag{47.24}$$

For this local gauge condition the condensed labels are

$$A = (a, x), \qquad \alpha = (b, y),$$

and $\mathcal{M}^{ab}(A; x, y) = \partial^\mu D^{ab}_\mu(A)\delta^{(D)}(x - y)$. This A-dependence is the structural difference from the abelian Maxwell case discussed in Chapter 41. If the gauge group is abelian and the gauge condition is linear, $D_\mu = \partial_\mu$, the Faddeev–Popov determinant is independent of the gauge field and contributes only a field-independent factor. For a nonabelian group,

$$\partial^\mu D^{ab}_\mu(A) = \delta^{ab}\partial^2 + f^a{}_{cb}\, \partial^\mu(A^c_\mu\, \cdot),$$

so $\det \mathcal{M}(A)$ is a functional of A. The ghost fields below are therefore dynamical scalar fermions in perturbation theory; their interactions are the local representation of this A-dependent Jacobian. The b_A notation used in (47.21) is related to the conventional antighost by

$$\bar{c}_A = b_A.$$

With this convention the ghost term in the Lagrangian has the sign displayed below, while the BRST transformation of the antighost is $sb_A = B_A$. The determinant of $\mathcal{M}(A)$ is represented by Grassmann-valued scalar fields $c^a(x)$ and $\bar{c}^a(x)$ in the adjoint representation:

$$\det \mathcal{M}(A) = \int [D\bar{c}\, Dc] \exp\left\{ -\mathrm{i} \int_M \bar{c}^a \mathcal{M}^{ab}(A) c^b \,\mathrm{d}^D x \right\}. \tag{47.25}$$

This equality is understood up to a field-independent normalization and phase: in a finite-dimensional Grassmann Gaussian, changing the sign multiplying $\bar{c}\mathcal{M}c$ multiplies the integral by a power of -1 and i, but does not change its dependence on A. The sign in (47.25) is chosen so that the ghost term agrees with the BRST gauge-fixing fermion below in the Minkowski weight $\exp(\mathrm{i}S)$. The field c is called the ghost and $\bar{c}$ the antighost. They are Lorentz scalars with fermionic statistics. Their role is determined by (47.25): they encode the Jacobian of gauge fixing and produce the loop contributions required by gauge covariance of the renormalized theory.

It is useful to introduce an auxiliary real field $B^a(x)$, the Nakanishi–Lautrup field. For a real parameter ξ, and for the Yang–Mills normalization

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g_{\mathrm{YM}}^2} F^a_{\mu\nu} F^{a\mu\nu},$$

the gauge-fixing and ghost Lagrangian density can be written

$$\mathcal{L}_{\text{gf+gh}} = B^a \partial^\mu A^a_\mu + \frac{g_{\mathrm{YM}}^2 \xi}{2} B^a B^a - \bar{c}^a\, \partial^\mu D^{ab}_\mu(A) c^b. \tag{47.26}$$

At fixed regulator the covariant-gauge functional integral contains the ordinary auxiliary integration over B:

$$\int [DB] \exp\left\{ \mathrm{i} \int_M \left[B^a \partial^\mu A^a_\mu + \frac{g_{\mathrm{YM}}^2 \xi}{2} B^a B^a \right] \mathrm{d}^D x \right\} = \mathcal{N}_B(\xi) \exp\left\{ -\frac{\mathrm{i}}{2g_{\mathrm{YM}}^2 \xi} \int_M (\partial^\mu A^a_\mu)^2 \,\mathrm{d}^D x \right\}, \tag{47.27}$$

with the oscillatory Gaussian contour, or its Wick-rotated regulator-level version, fixed as part of the path integral. The factor $\mathcal{N}_B(\xi)$ is independent of A and cancels from normalized correlators. Equivalently, completing the square and using the algebraic stationary equation gives

$$B^a = -\frac{1}{g_{\mathrm{YM}}^2 \xi} \partial^\mu A^a_\mu, \qquad \mathcal{L}_{\mathrm{gf}} = -\frac{1}{2g_{\mathrm{YM}}^2 \xi} (\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu),$$

so the B-field has been Gaussian-integrated out and no longer appears among the remaining integration variables. Landau gauge is the limiting gauge condition obtained by taking $\xi \to 0$ after the regulated Gaussian B-field integral has been defined; in that limit the Gaussian becomes the Fourier representation of the delta functional imposing $\partial^\mu A^a_\mu = 0$. Setting $g_{\mathrm{YM}} = 1$, or rescaling the gauge potential to canonical quadratic normalization, gives the more common version without the displayed factor of g_{YM}^2 in the B^2 term.

47.3 The BRST Differential

The gauge-fixed field algebra is a graded-commutative algebra generated by

$$A^a_\mu, \qquad c^a, \qquad \bar{c}^a, \qquad B^a,$$

and by matter fields when present. Assign ghost numbers

$$\mathrm{gh}(A_\mu) = 0, \qquad \mathrm{gh}(B) = 0, \qquad \mathrm{gh}(c) = 1, \qquad \mathrm{gh}(\bar{c}) = -1.$$

The BRST differential s is an odd derivation of ghost number $+1$. On the Yang–Mills sector it is defined by

$$sA^a_\mu = (D_\mu c)^a, \qquad sc^a = -\frac{1}{2} f^a{}_{bc} c^b c^c, \qquad s\bar{c}^a = B^a, \qquad sB^a = 0. \tag{47.28}$$

The construction in this section uses the off-shell closure of the Yang–Mills gauge algebra. In field-space language, the infinitesimal generators S_α satisfy a Lie-bracket identity as vector fields on the space of fields,

$$[S_\beta, S_\gamma] = f^\alpha{}_{\beta\gamma} S_\alpha,$$

without imposing the Euler–Lagrange equations. Therefore $s^2 = 0$ is an identity on the algebra generated by fields and ghosts. This is a genuine hypothesis of the elementary Faddeev–Popov/BRST construction used here. For gauge systems whose commutator closes only modulo the classical equations of motion, the same formulae define at most an on-shell nilpotent operation on the original fields; the corresponding off-shell differential is obtained in the BV field-antifield formalism, where the equation-of-motion terms are recorded by higher antifield-number terms in the master action.

In condensed field-space notation this reads

$$s\phi = c^\alpha S_\alpha \phi, \qquad sB_A = 0, \qquad sb_A = B_A, \qquad sc^\alpha = -\frac{1}{2} f^\alpha{}_{\beta\gamma} c^\beta c^\gamma, \tag{47.29}$$

where

$$S_\alpha \phi = \left.\frac{\partial \phi^\xi}{\partial \xi^\alpha}\right|_{\xi = \mathrm{id}}, \qquad [S_\beta, S_\gamma] = f^\alpha{}_{\beta\gamma} S_\alpha.$$

For the gauge-transformation group of maps $M \to G$, the condensed structure constants are the distributional kernel

$$f^{(a,x)}{}_{(b,y)(c,z)} = f^a{}_{bc}\, \delta^{(D)}(x-y) \delta^{(D)}(x-z),$$

with the same output-first finite-dimensional convention as in Chapter 45. Thus the Greek-index formula is condensed notation, not a second convention for raising Lie-algebra indices. The two notations agree because $b_A = \bar{c}_A$. If ϵ is a constant Grassmann-odd parameter, the finite-parity infinitesimal variation is $\delta_B X = \epsilon\, sX$; in particular ϵc^α is even, as a gauge parameter must be. The ghost $c = c^a t^a$ is an odd field valued in the same Hermitian coordinate space as the infinitesimal gauge parameter. For an odd $\mathfrak{g}$-valued field, the expression $[c, c]_{\mathrm{H}}$ is defined by combining the Lie bracket with the Grassmann product:

$$[c, c]^a_{\mathrm{H}} = f^a{}_{bc} c^b c^c.$$

Thus $sc = -\frac{1}{2}[c, c]_{\mathrm{H}}$. The antighost $\bar{c}$ is an independent odd adjoint-valued scalar of ghost number -1, and B is an even adjoint-valued scalar of ghost number 0. For a matter field $\psi \in V_R$, with representation matrices t^a_R,

$$s\psi = \mathrm{i} c^a t^a_R \psi, \qquad s\bar{\psi} = -\mathrm{i}\bar{\psi}\, c^a t^a_R. \tag{47.30}$$

The derivation rule is graded:

$$s(XY) = (sX)Y + (-1)^{|X|}X(sY),$$

where $|X|$ is the Grassmann parity of X.

Nilpotency is an algebraic consequence of the Jacobi identity. For any original field ϕ, the condensed notation gives

$$\begin{aligned} s^2\phi &= -\frac{1}{2}f^{\alpha}{}_{\beta\gamma}c^{\beta}c^{\gamma}S_{\alpha}\phi - c^{\alpha}c^{\beta}S_{\beta}S_{\alpha}\phi \\ &= -\frac{1}{2}f^{\alpha}{}_{\beta\gamma}c^{\beta}c^{\gamma}S_{\alpha}\phi + \frac{1}{2}c^{\alpha}c^{\beta}[S_{\alpha}, S_{\beta}]\phi = 0. \end{aligned}$$

For the ghost, using the graded derivation rule gives

$$s^2c^a = \frac{1}{4}f^a{}_{bc}f^b{}_{de}c^dc^ec^c - \frac{1}{4}f^a{}_{bc}f^c{}_{de}c^bc^dc^e.$$

After antisymmetrizing over the three odd ghost factors, this is

$$s^2c^a = -\frac{1}{12}\left(f^a{}_{be}f^e{}_{cd} + f^a{}_{ce}f^e{}_{db} + f^a{}_{de}f^e{}_{bc}\right)c^bc^cc^d = 0.$$

For the gauge field, the matrix form of the same computation is

$$s^2A_{\mu} = D_{\mu}(sc) + [D_{\mu}c, c]_{\mathrm{H}} = -\frac{1}{2}D_{\mu}[c, c]_{\mathrm{H}} + [D_{\mu}c, c]_{\mathrm{H}} = 0,$$

because $D_{\mu}[c, c]_{\mathrm{H}} = 2[D_{\mu}c, c]_{\mathrm{H}}$ for odd c. For a matter field, the representation identity $[t_R^a, t_R^b] = \mathrm{i}f^c{}_{ab}t_R^c$ gives

$$s^2\psi = \mathrm{i}(sc^a)t_R^a\psi + c^ac^bt_R^at_R^b\psi = -\frac{\mathrm{i}}{2}f^a{}_{bc}c^bc^ct_R^a\psi + \frac{1}{2}c^ac^b[t_R^a, t_R^b]\psi = 0,$$

and similarly for $\bar{\psi}$. Hence

$$s^2 = 0. \tag{47.31}$$

Thus s is the infinitesimal gauge-orbit differential with the even gauge parameter replaced by the odd field c, combined with the Chevalley–Eilenberg differential of the gauge algebra and the nonminimal contractible pair $(\bar{c}, B)$.

Let

$$\Psi = \int_M \bar{c}^a\left(\partial^{\mu}A_{\mu}^a + \frac{g_{\mathrm{YM}}^2\xi}{2}B^a\right)\mathrm{d}^Dx.$$

Using (47.28),

$$s\Psi = \int_M \left[B^a\partial^{\mu}A_{\mu}^a + \frac{g_{\mathrm{YM}}^2\xi}{2}B^aB^a - \bar{c}^a\partial^{\mu}D_{\mu}^{ab}(A)c^b\right]\mathrm{d}^Dx. \tag{47.32}$$

For $\xi = 0$, this is the covariant-gauge specialization of the condensed formula

$$S_{\mathrm{GF}} = s\big(b_AF^A[\phi]\big) = B_AF^A[\phi] - b_A\mathcal{M}^A{}_{\alpha}[\phi]c^{\alpha}.$$

Equivalently, if the BRST charge Q_B is defined by the convention

$$\delta_BX = \mathrm{i}\epsilon\, Q_B \cdot X, \qquad Q_B = -\mathrm{i}s, \tag{47.33}$$

then

$$S_{\text{GF}} = \mathrm{i}Q_B \cdot \big(b_A F^A[\phi]\big), \qquad Q_B^2 = 0. \tag{47.34}$$

The ξB^2 term in (47.32) is an additional BRST-exact deformation of the same gauge-fixing fermion and does not alter the cohomology. The full gauge-fixed action is therefore

$$S_{\text{gf}} = S_{\text{YM}} + S_{\text{matter}} + s\Psi.$$

Since $S_{\text{YM}} + S_{\text{matter}}$ is gauge invariant and $s^2 = 0$, one obtains $sS_{\text{gf}} = 0$. The gauge-fixed action is therefore controlled by a nilpotent fermionic symmetry acting on the enlarged field space, including ghosts, antighosts, and the Nakanishi–Lautrup field.

47.4 Local BRST Cohomology

The BRST differential acts on local expressions before any functional integral is evaluated. Let $\mathcal{F}_{\text{loc}}$ denote the graded-commutative algebra of local functions: polynomials or formal power series in the fields

$$A_\mu, \quad c, \quad \bar{c}, \quad B, \quad \psi, \quad \bar{\psi}$$

and finitely many of their spacetime derivatives, with coefficients smooth in the spacetime point when explicit x-dependence is allowed. The differential s is extended to derivatives by

$$s(\partial_{\nu_1} \cdots \partial_{\nu_r} \Phi) = \partial_{\nu_1} \cdots \partial_{\nu_r} (s\Phi)$$

for every generator Φ. The local BRST cohomology at ghost number g is

$$H^g(s; \mathcal{F}_{\text{loc}}) = \frac{\{a \in \mathcal{F}_{\text{loc}} : \text{gh}(a) = g,\ sa = 0\}}{\{a = sb : \text{gh}(b) = g - 1\}}.$$

At ghost number zero, this cohomology is the local algebraic form of gauge-invariant composite operators, with BRST-exact representatives removed.

Integrated local functionals require a second quotient because Lagrangian densities differing by total derivatives define the same action on manifolds without boundary, or the same bulk functional after boundary data have been specified. Let Ω^p_{loc} be the space of local p-forms and let

$$d = dx^\mu \partial_\mu$$

be the spacetime exterior derivative. With the convention that s and d are odd derivations on local forms, they anticommute:

$$sd + ds = 0.$$

For a D-dimensional spacetime, an integrated local functional is represented by a local D-form a. The cohomology relevant for local functionals is the relative cohomology $H^{g,D}(s \mid d)$. Its cocycles and coboundaries are

$$sa + dm = 0, \qquad a \sim a + sb + dn,$$

where

$$a \in \Omega^D_{\text{loc}}, \quad m, n \in \Omega^{D-1}_{\text{loc}}, \quad b \in \Omega^D_{\text{loc}},$$

with ghost numbers chosen so that all displayed terms have ghost number $g+1$ in the first equation and ghost number g in the equivalence relation. Thus $H^{0,D}(s \mid d)$ classifies BRST-invariant local counterterm densities modulo field redefinitions and total derivatives, while $H^{1,D}(s \mid d)$ is the algebraic home of possible local gauge anomalies. The anomaly chapters will compute the relevant representatives in four-dimensional chiral gauge theories; the present chapter fixes the cohomological language.

The pair $(\bar{c}, B)$ belongs to the nonminimal gauge-fixing sector. Since

$$s\bar{c}^a = B^a, \qquad sB^a = 0,$$

and the BRST transformations of the minimal variables $A, c, \psi, \bar{\psi}$ do not depend on $\bar{c}$ or B, this pair is contractible in local BRST cohomology. The doublet lemma below gives the explicit homotopy. Consequently, the nonminimal variables are essential for writing local propagators and Slavnov–Taylor identities, but the cohomology classes of gauge-invariant local operators, counterterms, and anomalies may be represented in the minimal sector under the standard regularity assumptions used in perturbative Yang–Mills theory.

47.5 Krein Space Structure of Covariant Gauge Quantization

Before the BRST quotient, covariant gauge quantization uses an indefinite-metric state space with a specified topology. The functional-analytic structure used here is a Krein space.[2]

Definition 47.9 (Krein space and fundamental symmetry)**.** A Krein space is a complex vector space $\mathcal{K}$ equipped with a nondegenerate Hermitian form $[\cdot,\cdot]$ and a direct decomposition

$$\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-,$$

such that $[\cdot,\cdot]$ is positive definite on $\mathcal{K}_+$, negative definite on $\mathcal{K}_-$, and both summands are complete for the associated positive norms. The fundamental symmetry is

$$J(\psi_+ + \psi_-) = \psi_+ - \psi_-, \qquad \psi_\pm \in \mathcal{K}_\pm.$$

It defines a Hilbert inner product

$$(\psi,\chi)_J = [\psi, J\chi] = [\psi_+, \chi_+] - [\psi_-, \chi_-],$$

and the topology of $\mathcal{K}$ is the Hilbert topology determined by $(\cdot,\cdot)_J$. Different fundamental decompositions may give different fundamental symmetries; a statement about domains, closures, adjoints, or continuity is meaningful only after such a Hilbert majorant topology has been specified.

Let T be a densely defined operator on $\mathcal{K}$. Its Krein adjoint $T^\times$ is defined by

$$[T\psi, \chi] = [\psi, T^\times \chi]$$

[2] For the operator theory of Krein spaces, see J. Bognar, *Indefinite Inner Product Spaces*, Ergebnisse der Mathematik und ihrer Grenzgebiete **78**, Springer, 1974. For product-integral technology useful in controlling operator evolutions, see J. D. Dollard and C. N. Friedman, *Product Integration with Applications to Differential Equations*, Encyclopedia of Mathematics and its Applications **10**, Addison–Wesley, 1979. For the indefinite-metric framework in local covariant gauge QFT and QED, see F. Strocchi and A. S. Wightman, "Proof of the charge superselection rule in local relativistic quantum field theory," *Journal of Mathematical Physics* **15** (1974), 2198–2224, and F. Strocchi, "Local and covariant gauge quantum field theories. Cluster property, superselection rules, and the infrared problem," *Physical Review D* **17** (1978), 2010–2021.

for all $\psi \in \mathcal{D}(T)$ and all χ in the domain for which the right-hand side is represented continuously. If T_J^* denotes the Hilbert adjoint with respect to $(\cdot,\cdot)_J$, then formally

$$T^\times = JT_J^*J$$

with the corresponding transformed domain. Thus an assertion that a BRST charge is "Hermitian" in covariant gauge means Krein symmetry or Krein self-adjointness, not Hilbert self-adjointness in a positive inner product.

Hypothesis 47.10 (Krein-domain data for the BRST charge)**.** The canonical covariant-gauge construction supplies:

$$(\mathcal{K}, [\cdot,\cdot], J), \qquad N_{\mathrm{gh}}, \qquad Q : \mathcal{D}(Q) \subset \mathcal{K} \to \mathcal{K},$$

where N_{gh} is the ghost-number operator and Q is a densely defined closed odd operator of ghost number $+1$. There is a common dense invariant domain $\mathcal{D} \subset \mathcal{D}(Q)$ containing the finite particle states, stable under the fields used in the BRST construction, such that

$$Q\mathcal{D} \subset \mathcal{D}, \qquad Q^2\psi = 0 \quad (\psi \in \mathcal{D}), \qquad [Q\psi, \chi] = [\psi, Q\chi] \quad (\psi, \chi \in \mathcal{D}).$$

Equivalently, $Q \subset Q^\times$ on the stated domain; when a Krein self-adjoint closure exists, this inclusion is replaced by equality for that closure. The physical quotient below is formed first on this controlled domain and then completed only after positivity of the induced form has been proved.

The free Gupta–Bleuler and perturbative BRST Fock constructions provide the model example of this structure: longitudinal and timelike oscillator modes create the negative directions, the fundamental symmetry J records the chosen positive majorant Hilbert topology, and the BRST charge is symmetric for the indefinite form. In an interacting nonperturbative gauge theory, Hypothesis 47.10 is an additional construction problem. The path-integral Slavnov–Taylor identities derived later are formal consequences of the regularized BRST symmetry; the Krein-space operator domain is separate analytic data.

47.6 Cohomology of States and Observables

In canonical quantization of the gauge-fixed theory, the state space before the BRST quotient is the Krein space $\mathcal{K}$ described above. At a fixed time a state can be represented, in a Schrodinger notation, by a wavefunctional

$$\Psi[\phi, B, b, c]\big|_{x^0=t}.$$

The first-order functional differential operator Q_B in (47.33) is promoted to an operator $\widehat{Q}_B$ on $\mathcal{K}$:

$$\widehat{Q}_B|\Psi\rangle = |Q_B \cdot \Psi\rangle, \qquad \widehat{Q}_B^2 = 0.$$

Insertion of a functional $R[\Phi]$, where $\Phi = (\phi, B, b, c)$, corresponds to an operator $\widehat{R}$, and the insertion $Q_B \cdot R$ corresponds to the graded commutator with $\widehat{Q}_B$, with the sign fixed by the parity of R. In the rest of this section we write Q for $\widehat{Q}_B$.

A state is admissible when

$$Q|\Psi\rangle = 0. \tag{47.35}$$

This condition is selected by gauge-fixing independence. Consider the transition amplitude between two admissible boundary states,

$$\mathcal{A}_F(\Psi_f, \Psi_i) = \int [D\Phi]_{t_i}^{t_f} \, \Psi_f^*[\Phi(t_f)] \, \exp(\mathrm{i}S_F[\Phi]) \, \Psi_i[\Phi(t_i)],$$

where the subscript F records the chosen gauge condition. If $F^A[\phi]$ is deformed to $F^A[\phi]+\bar{\delta}F^A[\phi]$, then (47.34) gives

$$\bar{\delta}S_F[\Phi] = \mathrm{i}Q_B \cdot \big(b_A \bar{\delta}F^A[\phi]\big).$$

Therefore, to first order in the deformation,

$$\begin{aligned}\bar{\delta}\, \mathcal{A}_F(\Psi_f, \Psi_i) &= -\int [D\Phi] \, \Psi_f^* \, \exp(\mathrm{i}S_F) \, \Psi_i \, Q_B \cdot \big(b_A \bar{\delta}F^A\big) \\ &= -\int [D\Phi] \, Q_B \cdot \big(\Psi_f^* \, \exp(\mathrm{i}S_F) \, \Psi_i \, b_A \bar{\delta}F^A\big) = 0. \end{aligned} \tag{47.36}$$

The last equality assumes that the functional measure is BRST invariant and that the BRST integration by parts has no boundary contribution in field space. The equalities also use $Q|\Psi_i\rangle = 0$, the corresponding condition on the final bra, and $Q_B \cdot S_F = 0$.

Nilpotency implies that $Q|\chi\rangle$ is admissible for every $|\chi\rangle$. Such a state has zero matrix element against any admissible state when Q is symmetric with respect to the Krein form:

$$\langle\Psi|Q|\chi\rangle = 0, \qquad Q|\Psi\rangle = 0.$$

Thus

$$|\Psi\rangle \sim |\Psi\rangle + Q|\chi\rangle$$

is the equivalence relation on admissible states. The physical state space is the ghost-number-zero BRST cohomology

$$\mathcal{H}_{\mathrm{phys}} = H^0(Q, \mathcal{K}) = \frac{\ker(Q : \mathcal{K}^0 \to \mathcal{K}^1)}{\mathrm{im}(Q : \mathcal{K}^{-1} \to \mathcal{K}^0)}. \tag{47.37}$$

Here $\mathcal{K}^n$ is the subspace of ghost number n. Gauge-invariant local operators define classes in the ghost-number-zero cohomology of s, while BRST-exact insertions have vanishing matrix elements between physical states when the BRST symmetry is preserved by the regulator and renormalization.

The perturbative S-matrix of a gauge-fixed nonabelian theory is an operator first constructed on $\mathcal{K}$ and then descended to $\mathcal{H}_{\mathrm{phys}}$. Asymptotic states and LSZ define scattering; gauge fixing supplies local Green-function rules whose matrix elements are well defined on BRST cohomology classes.

47.7 Positivity of the BRST Quotient

The quotient in (47.37) is a Hilbert-space construction only after positivity has been checked. Assume the Krein form $[\cdot,\cdot]$ on $\mathcal{K}$ is nondegenerate and that Q is symmetric with respect to it:

$$[Q\chi, \psi] = [\chi, Q\psi].$$

If $\psi \in \ker Q$ and $\eta = Q\chi$, then

$$[\eta, \psi] = [Q\chi, \psi] = [\chi, Q\psi] = 0.$$

Thus exact states are orthogonal to all closed states, including themselves. The inner product therefore descends to the quotient $\ker Q / \operatorname{im} Q$. The remaining requirement is that the induced form on ghost number zero cohomology be positive and that zero-norm closed states represent the zero cohomology class. When this holds, completing the quotient gives the physical Hilbert space.

Positivity criterion for a BRST doublet sector. Let $Q^2 = 0$ be symmetric for the non-degenerate Krein form on the ghost-graded space $\mathcal{K}$. Assume that the ghost-number-zero closed subspace admits a direct decomposition

$$\ker(Q : \mathcal{K}^0 \to \mathcal{K}^1) = \mathcal{P} \oplus \operatorname{im}(Q : \mathcal{K}^{-1} \to \mathcal{K}^0),$$

where the restriction of the inner product to $\mathcal{P}$ is positive definite. Assume further that every closed state with zero norm lies in $\operatorname{im} Q$. Then

$$H^0(Q, \mathcal{K}) = \frac{\ker(Q : \mathcal{K}^0 \to \mathcal{K}^1)}{\operatorname{im}(Q : \mathcal{K}^{-1} \to \mathcal{K}^0)}$$

inherits a positive-definite inner product and is isometric to the positive subspace $\mathcal{P}$ by the map that sends a closed state to its $\mathcal{P}$-component. Exact states are orthogonal to closed states by the calculation above, so the inner product is independent of the representative of a cohomology class. The displayed decomposition says that each class has a unique representative in $\mathcal{P}$. The induced norm of the class is therefore the norm of this $\mathcal{P}$-representative. Positivity on $\mathcal{P}$ gives a nonnegative norm, and the additional zero-norm assumption says that a class of zero norm has an exact representative, hence is the zero class. The quotient argument itself is formal; the substantive physics and mathematics are the hypotheses. In an interacting gauge theory one must construct a common domain for Q, prove $Q^2 = 0$ after renormalization, identify the closed zero-ghost subspace, and verify the absence of nonexact zero-norm closed states. These are the Kugo–Ojima/quartet-type inputs, not consequences of nilpotency alone.

Perturbatively, the unphysical polarizations, ghosts, antighosts, and Nakanishi–Lautrup modes organize into Q-doublets. A doublet has the form

$$Qu = v, \qquad Qv = 0.$$

Any state obtained by acting with such paired excitations is either exact or paired with an exact state and does not contribute to cohomology. The cohomology is then represented by transverse physical polarizations and gauge-invariant matter states. This is the cohomological version of the statement that the physical content of the gauge-fixed construction is carried by Q-cohomology classes.

47.8 Kugo–Ojima Quartets and the Color-Confinement Criterion

The preceding doublet argument is an algebraic statement about a given BRST complex. The Kugo–Ojima mechanism is the stronger covariant-gauge scenario in which all states carrying nontrivial color belong to quartets and hence have no representatives in $H^0(Q,\mathcal{K})$.[3]

Definition 47.11 (BRST quartet)**.** Let Q be a nilpotent symmetric BRST charge on a Krein space with ghost-number operator. A BRST quartet is a four-dimensional subcomplex generated by two BRST doublets related by the conjugation structure of the indefinite metric. In a basis

$$u, \quad v, \quad \widetilde{v}, \quad \widetilde{u},$$

one has

$$Qu = v, \qquad Qv = 0, \qquad Q\widetilde{v} = \widetilde{u}, \qquad Q\widetilde{u} = 0,$$

with ghost numbers chosen so that the indefinite form pairs the two doublets nondegenerately. The quartet contributes no physical state: v and $\widetilde{u}$ are exact, while u and $\widetilde{v}$ are not closed.

For perturbative Yang–Mills theory around the trivial vacuum, the longitudinal and timelike gauge modes, together with the Faddeev–Popov ghost and antighost excitations, form such quartets mode by mode. This is the operator version of the local doublet lemma below. It proves perturbative decoupling of the unphysical gauge-fixing sector once the BRST charge is constructed on the asymptotic Fock space and the quotient is positive. It does not by itself identify which nonperturbative colored excitations, if any, exist in the state space before the quotient.

The nonperturbative Kugo–Ojima confinement criterion is most sharply stated in Landau gauge. Work in a covariantly gauge-fixed Yang–Mills theory whose BRST charge Q exists as an operator, whose global color charges exist as operators on a common invariant domain, and whose indefinite state space has a positive BRST cohomology quotient. Let

$$(X \times Y)^a = f^a{}_{bc} X^b Y^c,$$

and define, for Euclidean momentum $p \neq 0$, the Kugo–Ojima two-point function by

$$\int \mathrm{d}^D x\, \mathrm{e}^{\mathrm{i}p\cdot x} \left\langle (D_\mu c)^a(x) \left(g_{\mathrm{YM}}(A_\nu \times \bar{c})\right)^b(0) \right\rangle = \delta^{ab} P_{\mu\nu}(p)\, u(p^2) + \delta^{ab} \frac{p_\mu p_\nu}{p^2}\, w(p^2), \tag{47.38}$$

where

$$P_{\mu\nu}(p) = \delta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}$$

is the transverse projector and w records a possible longitudinal component. In strict Landau gauge the transverse part $u(p^2)$ is the quantity usually called the Kugo–Ojima function. If one instead writes

$$F_{\mathrm{KO}}(p^2) = -u(p^2),$$

then the Kugo–Ojima infrared condition is

$$\lim_{p^2 \downarrow 0} F_{\mathrm{KO}}(p^2) = 1, \qquad \text{equivalently} \qquad u(0) = -1, \tag{47.39}$$

provided the limit exists.

[3]The canonical operator formulation and confinement criterion are developed in T. Kugo and I. Ojima, "Local covariant operator formalism of non-Abelian gauge theories and quark confinement problem," *Progress of Theoretical Physics Supplement* **66** (1979), 1–130. Lattice tests and modern infrared status include H. Nakajima and S. Furui, "Test of the Kugo-Ojima confinement criterion in the lattice Landau gauge," *Nuclear Physics B Proceedings Supplements* **83–84** (2000), 521–523; K.-I. Kondo, "Infrared behavior of the ghost propagator in the Landau gauge Yang-Mills theory," *Progress of Theoretical Physics* **122** (2009), 1455–1475; and N. M. R. Brito, "Lattice computation of the Kugo-Ojima correlation function," arXiv:2310.09010.

Conjecture 47.12 (Kugo–Ojima confinement criterion)**.** *Assume that the covariant-gauge BRST construction above exists nonperturbatively in Landau gauge, that the BRST charge is unbroken, that the spacetime integral defining the global color charge has no massless boundary contribution from $\partial^\nu F^a_{\mu\nu}$, and that (47.39) holds with no compensating singular longitudinal term in $w(p^2)$. Then the global color charge is represented on physical states by a BRST-exact operator. Consequently every BRST-closed state of ghost number zero has vanishing physical matrix elements of the color charge, and colored states, if present in the indefinite space, occur in BRST quartets rather than in the physical Hilbert space.*

The relation to the ghost propagator is conditional and should be stated with the same hypotheses. If the Landau-gauge ghost two-point function is written

$$\left\langle c^a(p)\bar{c}^b(-p)\right\rangle = \delta^{ab}\frac{G_{\rm gh}(p^2)}{p^2},$$

and if the longitudinal remainder in (47.38) has a regular infrared limit in the combination entering the ghost Ward identity, then

$$G_{\rm gh}(p^2)^{-1} = 1 + u(p^2) + p^2 v(p^2) \tag{47.40}$$

for another scalar form factor v. Under this regularity assumption, $u(0) = -1$ is equivalent to an infrared-divergent ghost dressing function. This equivalence is often the form in which the criterion is compared with Dyson–Schwinger or lattice Landau-gauge data.

This is a conditional Kugo–Ojima scenario, not a theorem for four-dimensional Yang–Mills theory. It assumes a global BRST charge and a covariant-gauge state space precisely in the regime where Gribov copies and possible fundamental-domain boundaries make the nonperturbative BRST construction delicate. Moreover, large-volume lattice studies of minimal Landau gauge and Gribov–Zwanziger infrared analyses generally find a finite infrared ghost dressing function and a Kugo–Ojima function that does not approach $u(0) = -1$. These results are evidence that the original Kugo–Ojima scenario is not realized in the standard lattice Landau-gauge implementation. They are not evidence against color confinement itself, whose gauge-invariant diagnostics in this monograph are formulated through the spectrum, Wilson loops, center symmetry when available, and the nonperturbative lattice construction.

47.9 The Doublet Lemma

The decoupling of unphysical variables has a simple algebraic form. Suppose a graded commutative differential algebra contains pairs

$$u^i, \qquad v^i,$$

with

$$su^i = v^i, \qquad sv^i = 0.$$

Let N count the total number of u's and v's in a polynomial, and define a homotopy operator h by

$$hv^i = u^i, \qquad hu^i = 0,$$

extended as a graded derivation with the appropriate signs. On the subalgebra generated by the doublets one has

$$sh + hs = N.$$

If X is s-closed and homogeneous with $NX = rX$, $r > 0$, then

$$X = \frac{1}{r}(sh + hs)X = s\left(\frac{1}{r}hX\right).$$

Thus every s-closed expression with positive doublet number is s-exact. The cohomology is represented by expressions independent of the doublet variables.

In perturbative gauge theory the longitudinal gauge modes, antighosts, Nakanishi–Lautrup variables, and their paired ghost excitations are organized in precisely this way after linearization around the perturbative vacuum. The lemma explains why physical cohomology depends on gauge-invariant transverse and matter data, while the doublet variables remain necessary in intermediate local formulas for propagators, counterterms, and Slavnov–Taylor identities.

47.10 Dimensional Regularization in Perturbative Gauge Theory

Dimensional regularization is a perturbative rule for assigning meromorphic functions of a complex dimension parameter to Feynman graph distributions. It is specified graph by graph, together with algebraic rules for tensor indices, spinors, internal polarizations, and external states. The datum is this meromorphic graph-distribution assignment; the field variables remain the fields of the physical-dimensional theory whose perturbative graphs are being regularized.

Definition 47.13 (Dimensional analytic regularization of graph integrals)**.** Let G be a connected Feynman graph with loop number L, external momenta p_i obeying physical d-dimensional momentum conservation, and propagator powers ν_e. Put $D = d - \epsilon$. A dimensional regularization datum first defines the scalar integral

$$I_G(D, \nu, p, m) = \int \prod_{\ell=1}^{L} \frac{\mathrm{d}^D q_\ell}{(2\pi)^D} \prod_{e\in E(G)} \frac{1}{(Q_e(q,p)^2 + m_e^2 - \mathrm{i}0)^{\nu_e}}$$

by Schwinger or Feynman parameters in a convergence domain and by meromorphic continuation in D and ν_e. The notation $\mathrm{d}^D q$ in this formula denotes this analytic rule, not a Borel measure on a literal vector space of nonintegral dimension. Tensor integrals are then defined by adjoining a continued metric-contraction algebra. Thus the D-dimensional metric $\hat{\eta}^{\mu\nu}$ and identity $\hat{\delta}^{\mu}{}_{\nu}$ satisfy

$$\hat{\delta}^{\mu}{}_{\mu} = D, \qquad \hat{\eta}^{\mu\rho}\hat{\eta}_{\rho\nu} = \hat{\delta}^{\mu}{}_{\nu},$$

and rotational covariance in this formal algebra fixes, for example,

$$\int \mathrm{d}^D q\, q^\mu q^\nu F(q^2) = \frac{\hat{\eta}^{\mu\nu}}{D} \int \mathrm{d}^D q\, q^2 F(q^2),$$

whenever the scalar integrals on the right are defined by the same meromorphic continuation.

The continuation in D enters in three distinct places.

1. **Loop integration.** Momentum integrals, phase-space integrals in unresolved sectors, and tensor averages use the D-dimensional metric $\hat{\eta}$.
2. **Algebra.** The contractions of Lorentz indices, gauge-field polarization sums, Dirac matrices, and Levi-Civita tensors require an additional prescription. This prescription is independent data beyond the scalar analytic continuation.
3. **External kinematics.** External observed momenta and polarizations may be kept in the physical d-dimensional subspace or continued together with the loops. This choice changes finite terms and evanescent operators, hence it is part of the renormalization convention.

The principal variants used in perturbative gauge theory differ precisely in these algebraic choices.[4]

Definition 47.14 (CDR, HV, DRED, and FDH data)**.** Let $d = 4$ for definiteness.

1. Conventional dimensional regularization, CDR, continues loop momenta, unobserved momenta, gauge polarizations, and Dirac gamma matrices to D dimensions.
2. The HV prescription keeps observed external states in the physical four-dimensional subspace. Loop momenta and unresolved internal polarizations are D-dimensional.
3. Dimensional reduction, DRED, keeps a four-dimensional index algebra η_4 and equips it with formal projectors
$$\eta_4 = \hat{\eta} + \tilde{\eta}, \qquad \hat{\eta}^{\mu}{}_{\mu} = D, \qquad \tilde{\eta}^{\mu}{}_{\mu} = 4 - D,$$
where loop momenta live in the $\hat{\eta}$-sector while the gauge and spin algebra remain four-dimensional. The $\tilde{\eta}$-components of a four-dimensional gauge field behave as evanescent scalar fields and require their own local counterterm coordinates when they are generated.
4. The four-dimensional helicity prescription, FDH, is a scattering amplitude variant in which external observed helicities are four-dimensional and unresolved loop momenta are dimensionally continued. Its finite conversion to CDR or HV is a scheme transformation only after the evanescent couplings have been specified.

The distinction matters in gauge theory. In vectorlike gauge theories without chiral γ_5 insertions, CDR and HV preserve the tensor algebra needed for Ward and Slavnov–Taylor identities at the regulated level, modulo local subtractions fixed by the chosen renormalization conditions. In supersymmetric theories, DRED has the specific advantage that the spin algebra remains four-dimensional. Its consistency beyond the lowest orders requires the evanescent scalar sector and its couplings to be included as part of the local action. A calculation that drops these evanescent coordinates has changed the regularization datum.

Definition 47.15 (γ_5 and Levi-Civita data)**.** A chiral dimensional regulator must specify the algebra generated by γ_5, the physical four-dimensional Levi-Civita tensor $\epsilon^{\mu\nu\rho\sigma}$, and the split between physical and evanescent subspaces. In the Breitenlohner–Maison/'t Hooft–Veltman prescription,
$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3$$

[4] The original gauge-theory construction is G. 't Hooft and M. Veltman, "Regularization and renormalization of gauge fields," *Nuclear Physics B* **44** (1972), 189–213. For the Breitenlohner–Maison/'t Hooft–Veltman treatment of γ_5, see P. Breitenlohner and D. Maison, *Communications in Mathematical Physics* **52** (1977), 11–38, 39–54, 55–75. For dimensional reduction, see W. Siegel, "Supersymmetric dimensional regularization via dimensional reduction," *Physics Letters B* **84** (1979), 193–196, and D. Stockinger, "Regularization by dimensional reduction: consistency, quantum action principle, and supersymmetry," arXiv:hep-ph/0503129.

is four-dimensional in the mostly-plus convention fixed in Section 40.16: it anticommutes with physical gamma matrices and commutes with evanescent gamma matrices. With $\epsilon^{0123} = +1$, this is the same convention for which

$$\operatorname{tr}\left(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\right) = 4\mathrm{i}\epsilon^{\mu\nu\rho\sigma}.$$

In dimensional-reduction computations of the anomaly, γ_5 and $\epsilon^{\mu\nu\rho\sigma}$ remain four-dimensional objects while loop-momentum contractions are performed in the continued $\hat{\eta}$-sector and the evanescent $\tilde{\eta}$-sector is kept as an explicit local algebraic component.

There is no prescription in which γ_5 anticommutes with every D-dimensional gamma matrix, has the four-dimensional trace $\operatorname{tr}(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma)$, and preserves cyclic finite-dimensional trace manipulations for all divergent chiral graphs. The finite anomaly is precisely the remnant left after the regulated Ward identities, local counterterms, and evanescent algebra have been handled consistently. Chapter 51 uses this logic to keep the vector Ward identity and compute the axial anomaly.

Locality and Slavnov–Taylor control in dimensional schemes. The dimensional schemes used below are perturbative subtraction schemes for graph distributions, not measures on a noninteger-dimensional field space. In a power-counting renormalizable gauge theory whose regulated coordinate system includes every evanescent local field generated by the algebraic prescription, the forest formula subtracts Taylor polynomials in external momenta for each superficially divergent subgraph. These Taylor polynomials are local in position space, and the same locality statement applies to evanescent coordinates because they enter only through local tensor contractions in the continued algebra. The nontrivial input needed for gauge identities is the quantum action principle: at fixed loop order, a Slavnov–Taylor breaking is an integrated local functional of ghost number one. Nilpotency of the linearized Slavnov operator then makes the breaking a cocycle. An exact cocycle is removed by subtracting the corresponding local functional; a nonzero local cohomology class is an anomaly obstruction. The next theorem states the order-by-order restoration result with these hypotheses made explicit, so the cohomological mechanism is not presented as a separate proof detached from the quantum action principle.

Theorem 47.16 (All-order Slavnov–Taylor restoration)**.** *Let G be a compact semisimple gauge group and let S_0 be the gauge-fixed Yang–Mills action with matter in finite-dimensional unitary representations, Nakanishi–Lautrup fields, ghosts, antighosts, and external sources K, L for the nonlinear BRST variations. Assume that:*

1. *the classical BRST differential s_0 is off-shell nilpotent and S_0 satisfies $\mathcal{S}(S_0) = 0$;*
2. *the theory is power-counting renormalizable in dimension D, after including all evanescent local coordinates generated by the chosen dimensional scheme;*
3. *the regularization and subtraction prescription obey the quantum action principle: at a fixed loop order, the breaking of a local functional identity is an integrated local D-form whose ghost number and power-counting degree are the ones dictated by the identity;*
4. *the local gauge-anomaly class of the matter representation vanishes in*

$$H^{1,D}(s_0 \mid d),$$

with the same evanescent convention and source coordinates used to define the regularized graphs.

Then local counterterms can be chosen recursively so that the renormalized 1PI effective action

$$\Gamma^R = S_0 + \sum_{\ell \geq 1} \hbar^\ell \Gamma_\ell^R$$

satisfies the Slavnov–Taylor identity

$$\mathcal{S}(\Gamma^R) = 0$$

as an identity of formal power series in $\hbar$*. At each finite order* N*, this means*

$$\mathcal{S}\left(S_0 + \sum_{\ell=1}^{N} \hbar^\ell \Gamma_\ell^R \right) = O(\hbar^{N+1}).$$

The remaining finite freedom is the addition of ghost-number-zero local cocycles in $H^{0,D}(s_0 \mid d)$*, together with* s_0*-exact local terms that implement field, source, gauge-parameter, and coupling-coordinate redefinitions.*

Proof. Write $\mathcal{B}_X$ for the linearization of $\mathcal{S}$ at a functional X. In the variables of (47.44),

$$\mathcal{B}_X Y = \int_M \left[\frac{\delta_R X}{\delta A_\mu^a} \frac{\delta_L Y}{\delta K^{a\mu}} + \frac{\delta_R Y}{\delta A_\mu^a} \frac{\delta_L X}{\delta K^{a\mu}} \right. \\ \left. + \frac{\delta_R X}{\delta c^a} \frac{\delta_L Y}{\delta L^a} + \frac{\delta_R Y}{\delta c^a} \frac{\delta_L X}{\delta L^a} + B^a \frac{\delta_L Y}{\delta \bar{c}^a} \right] \mathrm{d}^D x.$$

For $X = S_0$, $\mathcal{B}_{S_0}$ is the integrated form of the classical BRST differential s_0. The identity $\mathcal{S}(S_0) = 0$, together with the graded Jacobi identity behind the Slavnov bracket, gives

$$\mathcal{B}_{S_0}^2 = 0$$

on integrated local functionals modulo total derivatives.

The proof is by induction on the loop order. Suppose that all counterterms through order $N - 1$ have been chosen and set

$$\Gamma_{<N} = S_0 + \sum_{\ell=1}^{N-1} \hbar^\ell \Gamma_\ell^R, \qquad \mathcal{S}(\Gamma_{<N}) = O(\hbar^N).$$

After all subdivergences have been subtracted, the quantum action principle applied to the N-loop functional gives

$$\mathcal{S}(\Gamma_{<N} + \hbar^N \Gamma_N^{\mathrm{reg}}) = \hbar^N \Delta_N + O(\hbar^{N+1}),$$

where $\Delta_N = \int_M a_N$ is an integrated local D-form of ghost number $+1$ and power-counting degree at most D. The consistency identity for the Slavnov operator,

$$\mathcal{B}_X \mathcal{S}(X) = 0,$$

is obtained by differentiating the quadratic expression $\mathcal{S}(X)$ and using the nilpotency encoded by the antibracket form. Taking the coefficient of $\hbar^N$ gives

$$\mathcal{B}_{S_0} \Delta_N = 0$$

modulo total derivatives. Thus a_N represents a class in $H^{1,D}(s_0 \mid d)$.

By the anomaly hypothesis, this class is zero for the theory under consideration. Therefore there are local forms b_N and r_N with ghost numbers 0 and 1 such that

$$a_N = s_0 b_N + dr_N.$$

Let $B_N = \int_M b_N$. Replacing the N-loop counterterm by

$$\Gamma_N^R = \Gamma_N^{\mathrm{reg}} - B_N$$

changes the order-N Slavnov breaking by

$$-\hbar^N \mathcal{B}_{S_0} B_N = -\hbar^N \Delta_N,$$

because the total derivative integrates to zero in the local functional cohomology used for counterterms. Hence

$$\mathcal{S}\big(\Gamma_{<N} + \hbar^N \Gamma_N^R\big) = O(\hbar^{N+1}).$$

This completes the induction and constructs the formal renormalized functional Γ^R.

If $C_N = \int_M c_N$ has ghost number zero and $\mathcal{B}_{S_0} C_N = 0$ modulo total derivatives, adding C_N does not spoil the identity at order N. If $C_N = \mathcal{B}_{S_0} R_N$, the change is a local canonical or field/source redefinition in the BRST variables. This identifies the finite ambiguity stated above. □

The theorem is a perturbative theorem about formal renormalized 1PI functionals. It does not construct a positive nonperturbative Hilbert space at $D \neq d$, and it does not replace a lattice or other nonperturbative regulator. Physical unitarity is recovered after renormalization and after taking the physical $D \to d$ limit of stable-channel amplitudes or infrared-safe observables. Cut identities at intermediate D are bookkeeping identities for the analytically continued integrands; the physical optical theorem is a statement about the resulting renormalized amplitudes in d dimensions.

For four-dimensional chiral Yang–Mills theories, the vanishing of the local gauge-anomaly class includes the cancellation of the cubic invariant entering the consistent nonabelian anomaly. Global gauge anomalies are separate obstructions: they are not detected by the local cohomology group $H^{1,D}(s_0 \mid d)$ and are treated in the finite-anomaly chapter.

Finally, dimensional regularization is separate from renormalization and from operator-insertion regularization. The regulator is the meromorphic assignment of graph distributions. Renormalization is the choice of local coordinate map and finite subtraction convention used to take the physical limit. A composite operator insertion requires its own source coordinate and contact-term prescription, even when the same dimensionally regularized graph integrals are used to compute its correlators.

47.11 Supersymmetric Ward Identities and Regulator Data

Supersymmetric gauge theories require one more layer of bookkeeping beyond the BRST data above. The additional symmetry generators are odd, their algebra contains translations and often gauge transformations, and in many multiplet descriptions their closure uses auxiliary fields or the equations of motion. For this reason the phrase that a regulator preserves supersymmetry has a precise meaning only after the fields, sources, evanescent variables, and operator insertions have been specified.

Definition 47.17 (Supersymmetry datum on a gauge-fixed local theory)**.** Let Φ^A denote the complete list of fields in a gauge-fixed perturbative theory, including ghosts, antighosts, Nakanishi–Lautrup fields, matter fields, and any auxiliary fields used to close the transformation law off shell. A supersymmetry datum consists of:

1. a finite-dimensional spinor space S of constant supersymmetry parameters;
2. for every $\varepsilon \in \mathsf{S}$, an odd derivation
$$q_\varepsilon : \mathcal{F}_{\text{loc}} \longrightarrow \mathcal{F}_{\text{loc}}$$
of the local jet algebra;
3. a local gauge-fixed action with external sources for the nonlinear BRST and supersymmetry variations;
4. a closure formula of the form
$$\{q_\varepsilon, q_\eta\} = a^\mu(\varepsilon, \eta)\partial_\mu + \delta_{\alpha(\varepsilon,\eta;\Phi)} + s\, r(\varepsilon, \eta; \Phi) + \mathcal{E}_i\, R^i_{\varepsilon,\eta}, \tag{47.41}$$
where $a^\mu(\varepsilon, \eta)$ is the translation parameter, δ_α is a gauge transformation with possibly field-dependent parameter, s is the BRST differential, $\mathcal{E}_i$ are the Euler–Lagrange expressions, and $R^i_{\varepsilon,\eta}$ are local differential operators.

If the last term in (47.41) is absent after auxiliary fields have been included, the supersymmetry datum is off-shell closed in the chosen field coordinates. If it is present, the closure is on shell and the correct off-shell object is the corresponding BV extended action rather than the transformation law on fields alone.

The BV formulation removes the ambiguity in the on-shell case. The Euler–Lagrange term in (47.41) is then encoded by antifield terms in the extended action. The operator combining BRST and supersymmetry is a nilpotent vector field on the field-antifield space. In algebraic renormalization this is expressed by adjoining constant ghosts: commuting spinor ghosts ϵ for the odd supersymmetry generators and an anticommuting vector ghost v^μ for translations. In a flat four-dimensional $\mathcal{N} = 1$ example the algebraic form begins as

$$s_{\text{ext}} = s_{\text{BRST}} + \epsilon^\alpha Q_\alpha + \bar{\epsilon}_{\dot\alpha}\bar{Q}^{\dot\alpha} + v^\mu\partial_\mu - 2\mathrm{i}\, \epsilon\sigma^\mu\bar{\epsilon}\, \frac{\partial}{\partial v^\mu} + \cdots, \tag{47.42}$$

where Q_α and $\bar{Q}_{\dot\alpha}$ act as the superspace derivations of Chapter 86. With the convention $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ and $[P_\mu, \Phi(x)] = \mathrm{i}\partial_\mu\Phi(x)$, the derivation represented by P_μ on coordinate-labelled fields is $\mathrm{i}\partial_\mu$. Hence
$$\{\epsilon Q, \bar{\epsilon}\bar{Q}\} = 2\mathrm{i}\, \epsilon\sigma^\mu\bar{\epsilon}\, \partial_\mu.$$

The term $-2\mathrm{i}\epsilon\sigma^\mu\bar{\epsilon}\, \partial/\partial v^\mu$ then cancels this square against the anticommutator with $v^\mu\partial_\mu$. The omitted terms are the gauge and antifield contributions required by the closure formula. The extended ghost number is the $\mathbb{Z}$-grading for which

$$\mathrm{gh}_{\text{ext}}(c) = \mathrm{gh}_{\text{ext}}(\epsilon) = \mathrm{gh}_{\text{ext}}(\bar{\epsilon}) = \mathrm{gh}_{\text{ext}}(v) = 1, \qquad \mathrm{gh}_{\text{ext}}(\text{ordinary fields and couplings}) = 0,$$

and antifields or external sources have the complementary degrees that make the extended action have total extended ghost number 0. The differential s_{ext} raises this grading by 1. The defining property is

$$s_{\text{ext}}^2 = 0 \quad \text{on integrated local functionals modulo total derivatives.}$$

This nilpotent differential, not the transformation of a single component field in isolation, is the object tested by renormalization.

Definition 47.18 (Separated regularization and scheme data)**.** A regularized supersymmetric gauge calculation contains four distinct kinds of data.

1. The *path-integral regulator* $\mathfrak{R}$: a finite cutoff, lattice, higher-derivative covariance, dimensional analytic continuation, or other rule that assigns regulated Green functions or regulated graph distributions to the field variables.
2. The *gauge-fixing and BV data*: the gauge-fixing fermion, ghosts, antifields or external sources for nonlinear variations, the extended differential s_{ext}, and the local action including all evanescent fields produced by the regulator.
3. The *operator-insertion regulator*: for every composite operator or source multiplet, a local source coordinate and contact-term prescription that define insertions before the regulator is removed.
4. The *renormalized coordinate scheme*: the finite choice of couplings, fields, gauge parameters, source coordinates, and evanescent coordinates after local subtractions have been made.

The regulator is compatible with a chosen supersymmetry datum through order N in formal perturbation theory if the source-extended renormalized effective action $\Gamma^R_{\leq N}$ satisfies

$$\mathcal{S}_{\text{ext}}(\Gamma^R_{\leq N}) = O(\hbar^{N+1}), \tag{47.43}$$

where $\mathcal{S}_{\text{ext}}$ is the Slavnov functional for the combined BRST, supersymmetry, and translation ghosts. Exact compatibility means (47.43) to all orders in the formal expansion under consideration.

This definition separates several phenomena that otherwise get conflated. A regulator may preserve the BRST identity while breaking a supersymmetry Ward identity, because the gauge orbit and the supermultiplet structure are different parts of the field-space data. Conversely, a regulator can preserve a nilpotent scalar supersymmetry of a twisted formulation while leaving the full Lorentzian supersymmetry algebra to be recovered only after taking a continuum limit and tuning the allowed local counterterms.

Proposition 47.19 (Cohomological restoration of supersymmetry Ward identities)**.** *Assume a power-counting renormalizable supersymmetric gauge theory has been written with the source and BV data of Definition 47.18. Suppose that:*

1. *the classical extended action S_{ext} satisfies the combined identity*

$$\mathcal{S}_{\text{ext}}(S_{\text{ext}}) = 0;$$

2. *the chosen regulator and subtraction procedure obey the quantum action principle for the combined identities: after lower-order subtractions, the order-N breaking of $\mathcal{S}_{\text{ext}}$ is an integrated local D-form of extended ghost number 1;*

3. *the local field-coordinate list includes every evanescent coordinate generated by the regulator, including epsilon-scalars and their independent couplings in dimensional reduction;*
4. *the class of the order-N breaking vanishes in*

$$H^{1,D}(s_{\text{ext}} \mid d)$$

computed in the same local algebra of fields, antifields, sources, constant ghosts, and evanescent variables.

Then there is a local counterterm at order N for which the renormalized effective action satisfies

$$\mathcal{S}_{\text{ext}}(\Gamma^R_{\leq N}) = O(\hbar^{N+1}).$$

If the cohomology class is nonzero, the breaking is an anomaly or a regulator obstruction in the specified symmetry problem; it cannot be removed by a local finite redefinition inside that coordinate system.

Proof. Suppose the identity has been restored through order $N-1$. The quantum action principle gives

$$\mathcal{S}_{\text{ext}}\big(\Gamma^R_{<N} + \hbar^N \Gamma^{\text{reg}}_N\big) = \hbar^N \Delta_N + O(\hbar^{N+1}), \qquad \Delta_N = \int a_N,$$

with a_N local of extended ghost number 1. Linearizing the extended Slavnov functional at the classical action gives the nilpotent operator $\mathcal{B}_{S_{\text{ext}}}$, whose action on local functionals is s_{ext} modulo total derivatives. The consistency identity

$$\mathcal{B}_X \mathcal{S}_{\text{ext}}(X) = 0$$

therefore implies

$$s_{\text{ext}} a_N + d r_N = 0$$

for a local $(D-1)$-form r_N. The vanishing of the cohomology class means

$$a_N = s_{\text{ext}} b_N + d c_N$$

with b_N local of extended ghost number 0. Adding the counterterm $-\hbar^N \int b_N$ changes the order-N breaking by $-\hbar^N \Delta_N$, while the total derivative integrates to zero in the local-functional quotient. This restores the identity through order N. The nonzero-class case follows because a local counterterm changes Δ_N only by an s_{ext}-exact term modulo d. □

Several regulators used for supersymmetric theories can now be described by their data.

1. **Dimensional reduction.** DRED keeps the four-dimensional spin and gauge index algebra used by the supersymmetry transformations while loop momenta are analytically continued in the $\hat{\eta}$-sector, where $\hat{\eta}$ is the continued $D = 4 - \epsilon$ metric in the dimensional-regularization conventions above. Its local coordinate list includes the $\tilde{\eta}$-sector epsilon-scalars, namely the evanescent $(4-D)$-dimensional components in the splitting $\eta_4 = \hat{\eta} + \tilde{\eta}$, and all couplings generated by their interactions. Within perturbation theory the question is whether the extended Slavnov identity can be restored in the local algebra containing these variables. Dropping the epsilon-scalar coordinates changes the regulator datum and invalidates the cohomology test.

2. **Conventional dimensional regularization.** CDR and HV provide BRST-compatible regulators for many vectorlike gauge calculations. For four-dimensional supersymmetry their continued tensor and spinor algebras do not carry the original component-field representation without extra evanescent data. The supersymmetry Ward identity must therefore be tested through $\mathcal{S}_{\text{ext}}$; any local breaking is either removed by the restoration criterion above or recorded as an obstruction for that scheme.

3. **Superspace and higher-derivative regulators.** In a superspace formulation, a regulator is compatible with the manifest supersymmetry when it is written as a local functional of superfields and supercovariant derivatives, with the Berezin integration and source supermultiplets specified. Gauge theories additionally require a gauge-covariant or background-covariant higher-derivative kernel and a BRST/BV treatment of the remaining gauge fixing. Residual divergences, when present, still require a local subtraction scheme whose Ward identities are checked by the same cohomological criterion.

4. **Pauli–Villars-type auxiliary determinants.** A Pauli–Villars-type regulator is specified by the auxiliary field content, statistics, mass matrix, gauge representation, and determinant prescription. Compatibility with supersymmetry requires regulator fields to be organized with the same supersymmetry datum and source identity, or else the resulting breaking must be local and cohomologically removable. The presence of heavy auxiliary fields by itself supplies no symmetry statement.

5. **Lattice and twisted regulators.** A lattice regulator gives an ordinary finite-dimensional integral before the continuum limit, but it replaces continuous translations and rotations by lattice symmetries. Full Poincare supersymmetry is therefore expressed, at finite lattice spacing, only through those subalgebras that survive the discretization. Twisted formulations may preserve a nilpotent scalar charge Q with

$$Q^2 = \delta_{\text{gauge}}(\varphi)$$

   or a lattice translation plus gauge transformation. The continuum claim is then a statement about the local counterterms allowed by the exact lattice symmetries and by the Q-Ward identity.

6. **Localization-compatible regulators.** A localization argument uses a chosen odd operator Q satisfying

$$Q^2 = \mathcal{L}_v + \delta_{\text{gauge}} + \delta_{\text{flavor}}$$

   on the fields and sources of the regulated theory. The deformation $t\,QV$ preserves a Q-closed expectation value only when the regulator, integration cycle, boundary conditions, and determinant phases are Q-compatible. The derivative of a regulated expectation value is then

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \mathcal{O}\rangle_t = -\langle Q(V\mathcal{O})\rangle_t + \text{local source/contact contributions},$$

   and it vanishes for Q-closed observables only after those contact terms and possible Q-anomalies have been controlled.

Holomorphy and nonrenormalization arguments also belong to the same scheme language. A Wilsonian superpotential statement is a statement in a renormalized superspace source-coordinate system: chiral couplings are represented by background chiral superfields, the Wilsonian action is expanded in local superspace integrals, and the allowed finite counterterms are constrained by chirality, locality, gauge invariance, global charges, and the chosen regularity domain in coupling space. The conclusion concerns the F-term coordinate in that Wilsonian chart. The Kähler potential, 1PI effective action, real D-terms, thresholds, and contact terms obey their own source identities and require separate definitions.

47.12 Slavnov–Taylor Identities and Counterterms

BRST symmetry gives functional identities for the generating functional and for the 1PI effective action. In a compact notation, introduce sources $K^{a\mu}$ and L^a coupled to sA^a_μ and sc^a. We use left and right functional derivatives defined by

$$\delta F = \int_M \delta\Phi^A \frac{\delta_L F}{\delta\Phi^A} = \int_M \frac{\delta_R F}{\delta\Phi^A}\delta\Phi^A,$$

for each homogeneous field or source Φ^A. The parities are

$$\epsilon(A_\mu) = \epsilon(B) = \epsilon(L) = 0, \qquad \epsilon(c) = \epsilon(\bar{c}) = \epsilon(K) = 1.$$

The 1PI effective action $\Gamma[A, c, \bar{c}, B; K, L]$ obeys the Slavnov–Taylor identity

$$\mathcal{S}(\Gamma) := \int_M \left[\frac{\delta_R\Gamma}{\delta A^a_\mu}\frac{\delta_L\Gamma}{\delta K^{a\mu}} + \frac{\delta_R\Gamma}{\delta c^a}\frac{\delta_L\Gamma}{\delta L^a} + B^a\frac{\delta_L\Gamma}{\delta\bar{c}^a}\right] \mathrm{d}^D x = 0. \tag{47.44}$$

There are two separate functional identities in the gauge theories developed so far. In Abelian QED, after the gauge-fixing term has been separated, the gauge-invariant part of the 1PI functional is annihilated by a fixed first-order Ward–Takahashi differential operator on field space:

$$\mathcal{W}_\alpha \Gamma_{\text{inv}} = 0.$$

The operator $\mathcal{W}_\alpha$ contains the infinitesimal field variations, but it does not contain Γ_{inv}. The equation is therefore linear in the unknown functional. Differentiating it gives the Ward–Takahashi relations, including photon self-energy transversality and the equality between the Abelian vertex and spinor residue counterterms in Chapter 43.

The nonabelian gauge-fixed identity above has a different origin and a different algebraic form. The BRST variations $sA^a_\mu = (D_\mu c)^a$ and $sc^a = -\frac{1}{2}f^a{}_{bc}c^b c^c$ are composite local fields. To express their insertions in the quantum theory one introduces the sources K, L. Let $W[J, K, L]$ be the connected generating functional with ordinary sources J_A for the fields $\Phi^A = (A, c, \bar{c}, B)$ and with K, L coupled to the nonlinear BRST variations. The BRST change of variables gives the connected identity, before Legendre transformation in the J_A variables,

$$\int \left[J_{A^a_\mu}\frac{\delta_L W}{\delta K^{a\mu}} + J_{c^a}\frac{\delta_L W}{\delta L^a} + J_{\bar{c}^a}\frac{\delta_L W}{\delta J_{B^a}}\right] \mathrm{d}^D x = 0$$

with the source signs chosen to match (47.44). Passing to the 1PI functional replaces the ordinary sources by

$$J_{\Phi^A} = -\frac{\delta_R\Gamma}{\delta\Phi^A}$$

while leaving the BRST-variation sources K, L as external coordinates. Moreover

$$\frac{\delta_L W}{\delta K^{a\mu}} = \frac{\delta_L\Gamma}{\delta K^{a\mu}}, \qquad \frac{\delta_L W}{\delta L^a} = \frac{\delta_L\Gamma}{\delta L^a}, \qquad \frac{\delta_L W}{\delta J_{B^a}} = B^a.$$

Consequently the Legendre transform converts the connected BRST identity into a product of derivatives of the same functional Γ. In the Yang–Mills coordinates displayed in (47.44), this product is precisely the quadratic Slavnov–Taylor operator $\mathcal{S}(\Gamma)$:

$$\int_M \left[\frac{\delta_R\Gamma}{\delta A^a_\mu}\frac{\delta_L\Gamma}{\delta K^{a\mu}} + \frac{\delta_R\Gamma}{\delta c^a}\frac{\delta_L\Gamma}{\delta L^a} + B^a\frac{\delta_L\Gamma}{\delta \bar{c}^a}\right] \mathrm{d}^D x.$$

The quadratic form is the 1PI shadow of the homological statement $s^2 = 0$ and is the reason antifield/BV coordinates are the natural language for renormalized nonabelian gauge theory. The sources $K^{a\mu}$ and L^a have ghost numbers -1 and -2, respectively, so that the source terms $K^{a\mu} sA^a_\mu$ and $L^a sc^a$ have ghost number zero. They are introduced because the BRST transformations of A_μ and c are nonlinear composite fields. The Batalin–Vilkovisky formulation replaces this case-by-case source assignment: for every field Φ^A, one introduces an antifield Φ^*_A of opposite Grassmann parity and ghost number

$$\mathrm{gh}(\Phi^*_A) = -1 - \mathrm{gh}(\Phi^A),$$

and encodes gauge symmetry by an odd Poisson bracket, the antibracket, and a master equation. The present Slavnov–Taylor identity is the gauge-fixed Yang–Mills specialization of that homological structure.

The signs in (47.44) follow directly from these left/right conventions. At the classical source-coupled level write

$$S_{\text{ext}} = S_{\text{gf}} + \int_M \left(K^{a\mu} sA^a_\mu + L^a sc^a\right) \mathrm{d}^D x.$$

Since K is odd and L is even,

$$\frac{\delta_L S_{\text{ext}}}{\delta K^{a\mu}} = sA^a_\mu, \qquad \frac{\delta_L S_{\text{ext}}}{\delta L^a} = sc^a.$$

Applying the odd derivation s from the left gives

$$0 = sS_{\text{ext}} = \int_M \left[(sA^a_\mu)\frac{\delta_L S_{\text{ext}}}{\delta A^a_\mu} + (sc^a)\frac{\delta_L S_{\text{ext}}}{\delta c^a} + B^a\frac{\delta_L S_{\text{ext}}}{\delta \bar{c}^a}\right] \mathrm{d}^D x.$$

For the effective action, the Legendre transform is taken in the elementary fields but not in K, L. With the derivative convention above,

$$\frac{\delta_L\Gamma}{\delta K^{a\mu}} = \left\langle sA^a_\mu\right\rangle_{\text{1PI}}, \qquad \frac{\delta_L\Gamma}{\delta L^a} = \left\langle sc^a\right\rangle_{\text{1PI}},$$

and the elementary source terms are replaced by the right derivatives $\delta_R\Gamma/\delta A^a_\mu$ and $\delta_R\Gamma/\delta c^a$. Substitution gives (47.44) with no additional hidden sign; the c-term is written with $\delta_R\Gamma/\delta c$ precisely because c is odd. Chapter 48 develops the BV formalism as the natural framework for gauge-theory 1PI effective actions, Wilsonian effective actions, and renormalization compatible with gauge symmetry.

Renormalization is compatible with gauge theory when counterterms can be chosen so that (47.44) remains valid order by order. Algebraically, the possible invariant counterterms are classified by ghost-number-zero local BRST cohomology. A gauge anomaly is an obstruction at ghost number $+1$: it is a local functional $\mathcal{A}$ satisfying the linearized consistency condition

$$\mathcal{B}_\Gamma \mathcal{A} = 0$$

that cannot be written as $\mathcal{A} = \mathcal{B}_\Gamma X$ for a local functional X. The axial anomaly and the nonabelian gauge-anomaly cancellation conditions are developed in the anomaly chapters below.

Chapter 48

The BV Master Formalism for Gauge Effective Actions

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

48.1 Why Antifields Are Introduced

The BRST chapter introduced sources $K^{a\mu}$ and L^a for the nonlinear variations sA^a_μ and sc^a. Their role is to serve as coordinates dual to the BRST variations, thereby making the Slavnov–Taylor identity a functional equation for the 1PI effective action.

The Batalin–Vilkovisky formalism is the intrinsic version of this construction. It assigns a dual coordinate, called an antifield, to every field in the gauge-fixed or ungauge-fixed complex. The antibracket pairs a field with its antifield, and the master equation is the single equation that encodes gauge invariance, BRST nilpotency, gauge fixing, and the compatible renormalization identities.

The framework in this chapter is perturbative and local. We work with a regulated field theory or, equivalently for algebraic purposes, with the graded algebra of local functionals in finitely many jets of fields. Infinite dimensional functional derivatives are therefore shorthand for the corresponding regulated or local-jet operation. When an identity involves the BV Laplacian, the regulator and measure are part of the data.

This chapter supplies the continuum gauge-theory framework for effective actions. A Wilsonian or

1PI effective action for a gauge theory is a gauge-compatible object only after the gauge symmetry has been encoded in a BRST/BV complex, or after a lattice regulator has supplied a finite gauge-invariant integral over link variables. In the BV route the finite cutoff data are the odd symplectic field-antifield space, a compatible half-density, the regulated BV Laplacian, the master action, and the coarse-graining cycle. In the lattice route the finite cutoff data are the compact link-variable space, Haar measure, gauge action, gauge-invariant observables, and blocking or scaling map. Later gauge-theory Wilsonian claims use one of these two data types and state the comparison map between them when both appear.

48.2 Fields, Antifields, and Gradings

Let $\{\Phi^A\}$ denote the full set of fields in the gauge-theory complex. The index A includes discrete labels, spacetime indices, Lie-algebra indices, and the spacetime point in condensed notation. Examples are

$$A^a_\mu, \qquad c^a, \qquad \bar{c}^a, \qquad B^a, \qquad \psi, \qquad \bar{\psi}.$$

Each field has a Grassmann parity

$$\epsilon_A \in \mathbb{Z}/2\mathbb{Z}$$

and a ghost number $\mathrm{gh}(\Phi^A)$. The antifield paired with Φ^A is denoted by Φ^*_A and is assigned

$$\epsilon(\Phi^*_A) = \epsilon_A + 1 \pmod 2, \qquad \mathrm{gh}(\Phi^*_A) = -1 - \mathrm{gh}(\Phi^A).$$

Thus antifields have opposite parity and complementary ghost number. For the Yang–Mills non-minimal sector,

Φ^A	A_μ	c	$\bar{c}$	B
$\mathrm{gh}(\Phi^A)$	0	1	-1	0
$\mathrm{gh}(\Phi^*_A)$	-1	-2	0	-1

and the parities are reversed in the antifield row.

Let $\mathcal{F}_{\mathrm{BV}}$ be the graded-commutative algebra of local functionals in Φ^A, Φ^*_A and finitely many of their spacetime derivatives, modulo integration by parts when integrated local functionals are being compared. A functional $F \in \mathcal{F}_{\mathrm{BV}}$ has parity $|F|$ and ghost number $\mathrm{gh}(F)$ when it is homogeneous. Left and right variational derivatives are defined by

$$\delta F = \int \delta\Phi^A \, \frac{\delta^L F}{\delta\Phi^A} = \int \frac{\delta^R F}{\delta\Phi^A} \, \delta\Phi^A,$$

and similarly for antifields. The distinction is necessary because odd variables do not commute with the variation symbols.

For a local functional

$$F[\Phi] = \int_M f(x, \Phi, \partial\Phi, \dots, \partial^r\Phi) \, \mathrm{d}^D x,$$

the left Euler–Lagrange derivative is the coefficient of $\delta\Phi^A$ after moving derivatives off the variation:

$$\frac{\delta^L F}{\delta\Phi^A} = \sum_{|\alpha|\le r} (-\partial)_\alpha \frac{\partial^L f}{\partial(\partial_\alpha \Phi^A)}. \tag{48.1}$$

Here α is a multi-index and $(-\partial)_\alpha = (-1)^{|\alpha|}\partial_\alpha$. Boundary conditions or compact support are part of the definition whenever the integration by parts is used. In a finite regulator $\partial_\alpha \Phi^A$ is replaced by the corresponding finite set of regulated coordinates, and (48.1) becomes the ordinary graded partial derivative. The ghost number and parity of the variational derivative are

$$\mathrm{gh}\left(\frac{\delta^L F}{\delta \Phi^A}\right) = \mathrm{gh}(F) - \mathrm{gh}(\Phi^A), \qquad \left|\frac{\delta^L F}{\delta \Phi^A}\right| = |F| + \epsilon_A.$$

These bookkeeping rules are what make the BV bracket below homogeneous of parity one and ghost number one.

48.3 The Antibracket

Definition 48.1 (BV antibracket)**.** For homogeneous functionals $F, G \in \mathcal{F}_{\mathrm{BV}}$, the BV antibracket is

$$(F, G) = \int \left[\frac{\delta^R F}{\delta \Phi^A}\frac{\delta^L G}{\delta \Phi^*_A} - \frac{\delta^R F}{\delta \Phi^*_A}\frac{\delta^L G}{\delta \Phi^A}\right] \mathrm{d}^D x, \tag{48.2}$$

with summation over all field labels A. The spacetime integral is absent when the same formula is applied to a finite-dimensional regulator.

The bracket has parity and ghost number

$$|(F, G)| = |F| + |G| + 1, \qquad \mathrm{gh}(F, G) = \mathrm{gh}(F) + \mathrm{gh}(G) + 1.$$

It satisfies

$$(F, G) = -(-1)^{(|F|+1)(|G|+1)}(G, F)$$

and the odd Jacobi identity

$$(F, (G, H)) = ((F, G), H) + (-1)^{(|F|+1)(|G|+1)}(G, (F, H)).$$

It is an odd biderivation. Explicitly,

$$(F, GH) = (F, G)H + (-1)^{(|F|+1)|G|}G(F, H),$$

and

$$(FG, H) = F(G, H) + (-1)^{|G|(|H|+1)}(F, H)G.$$

The elementary brackets are

$$(\Phi^A(x), \Phi^*_B(y)) = \delta^A{}_B\, \delta^{(D)}(x - y), \qquad (\Phi^A, \Phi^B) = 0, \qquad (\Phi^*_A, \Phi^*_B) = 0,$$

with the delta distribution understood in the same regulated or distributional sense as the fields themselves.

Lemma 48.2 (Odd symplectic origin of the antibracket)**.** *At finite regulator, let*

$$\omega = \delta \Phi^*_A\, \delta \Phi^A$$

be the odd symplectic form on the graded vector space with coordinates (Φ^A, Φ^*_A)*. For a homogeneous function F, define its Hamiltonian vector field X_F by*

$$\iota_{X_F}\omega = \delta F.$$

Then the bracket $(F, G) := X_F G$ is exactly (48.2)*. It satisfies the elementary brackets, graded antisymmetry, the odd Jacobi identity, and the two biderivation identities displayed above.*

Proof. In Darboux coordinates the equation $\iota_{X_F}\omega = \delta F$ gives

$$X_F = \frac{\delta^R F}{\delta\Phi^A}\frac{\delta^L}{\delta\Phi^*_A} - \frac{\delta^R F}{\delta\Phi^*_A}\frac{\delta^L}{\delta\Phi^A}.$$

Applying this vector field to G gives (48.2). The elementary brackets follow by applying the formula to the coordinate functions. The biderivation identities follow because X_F is a graded derivation of parity $|F| + 1$.

For the Jacobi identity, use the defining relation and the closedness of the constant Darboux form:

$$\iota_{[X_F,X_G]}\omega = \delta(X_F G) = \delta(F, G).$$

Thus $[X_F, X_G] = X_{(F,G)}$, with the graded commutator of vector fields. The graded Jacobi identity for vector-field commutators gives the displayed odd Jacobi identity for $(\,\cdot\,,\,\cdot\,)$. For continuum local functionals, first smear the fields or impose a finite regulator so that the Darboux coordinate calculation is finite-dimensional. The variational derivatives in (48.2) are then the limits of the regulated coordinate derivatives, and the bracket identities pass to the declared local-functional limit on the common domain where those variational derivatives exist. □

48.4 The Classical Master Equation

Definition 48.3 (Classical BV action)**.** A classical BV action is an even local functional $S_{\mathrm{BV}} \in \mathcal{F}_{\mathrm{BV}}$ of ghost number zero satisfying the classical master equation

$$\frac{1}{2}(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0. \tag{48.3}$$

It is required to reduce to the classical gauge-invariant action when all ghosts and antifields are set to zero.

The associated BV differential is

$$s_{\mathrm{BV}}F = (F, S_{\mathrm{BV}}).$$

If S_{BV} satisfies (48.3), then

$$s_{\mathrm{BV}}^2 = 0.$$

Using the odd Jacobi identity and the evenness of S_{BV},

$$((H, S_{\mathrm{BV}}), S_{\mathrm{BV}}) = \frac{1}{2}(H, (S_{\mathrm{BV}}, S_{\mathrm{BV}}))$$

for every functional H. The right hand side vanishes by (48.3). Hence

$$s_{\mathrm{BV}}^2 H = ((H, S_{\mathrm{BV}}), S_{\mathrm{BV}}) = 0.$$

Since this holds for every functional H in the BV algebra, the Hamiltonian vector field generated by S_{BV} is a differential. Thus the classical master equation is precisely the condition that the BRST/BV variation squares to zero off shell, before choosing a gauge-fixing Lagrangian submanifold.

This algebraic implication is the reason the master equation is stronger than a compact rewriting of a Ward identity. It constructs a differential on the space of fields, ghosts, antifields, and local functionals.

48.5 Noether Identities and the Minimal BV Complex

The master equation is constructed from the gauge identities of the classical action. Let $S_0[\phi]$ be a classical action for fields ϕ^i, and set

$$E_i(\phi) := \frac{\delta S_0}{\delta \phi^i}.$$

An infinitesimal gauge transformation is a collection of local differential operators $R^i{}_\alpha(\phi, \partial)$ acting on gauge parameters ϵ^α:

$$\delta_\epsilon \phi^i = R^i{}_\alpha \epsilon^\alpha.$$

Gauge invariance of S_0 is the Noether identity

$$E_i R^i{}_\alpha = 0 \tag{48.4}$$

after moving derivatives off ϵ^α. Condensed notation includes the spacetime integration and the signs from integration by parts.

The first BV variables are the ghosts c^α, the field antifields ϕ^*_i, and the ghost antifields c^*_α. Their ghost numbers are

$$\mathrm{gh}(\phi^i) = 0, \quad \mathrm{gh}(c^\alpha) = 1, \quad \mathrm{gh}(\phi^*_i) = -1, \quad \mathrm{gh}(c^*_\alpha) = -2.$$

The Koszul–Tate part of the BV differential resolves the equations of motion:

$$\delta_{\mathrm{KT}}\phi^i = 0, \qquad \delta_{\mathrm{KT}} c^\alpha = 0, \qquad \delta_{\mathrm{KT}}\phi^*_i = E_i, \qquad \delta_{\mathrm{KT}} c^*_\alpha = R^i{}_\alpha \phi^*_i.$$

Its square on c^*_α is

$$\delta^2_{\mathrm{KT}} c^*_\alpha = R^i{}_\alpha E_i,$$

which vanishes by (48.4). Thus the Noether identity is the first exactness condition in the antifield direction.

For an irreducible off-shell closed gauge algebra,

$$[R_\alpha, R_\beta]^i = R^i{}_\gamma C^\gamma{}_{\alpha\beta},$$

with the usual graded signs in the general case, the beginning of the minimal BV action is

$$S_{\min} = S_0 + \phi^*_i R^i{}_\alpha c^\alpha - \frac{1}{2} c^*_\gamma C^\gamma{}_{\alpha\beta} c^\alpha c^\beta + \cdots . \tag{48.5}$$

For field-independent structure constants and off-shell closure this displayed expression is complete. Field-dependent, open, or reducible gauge systems require higher antifield-number terms. Expanding $\frac{1}{2}(S_{\min}, S_{\min})$ by antifield number gives, in order, the Noether identity (48.4), the closure relation of the gauge generators, and the Jacobi relations or their higher homotopy replacements. Solving the master equation is therefore the construction of a single nilpotent differential whose antifield expansion records the complete gauge identity complex.

48.6 Minimal Yang–Mills BV Action

For a gauge theory whose BRST transformations close off shell, the minimal BV action is obtained by coupling antifields to the BRST variations. With the Hermitian Yang–Mills conventions of Chapter 45, $[t^a, t^b] = \mathrm{i}f^c{}_{ab}t^c$, so the output Lie-algebra index of $f^c{}_{ab}$ is written first. In this convention

$$sA^a_\mu = (D_\mu c)^a, \qquad sc^a = -\frac{1}{2}f^a{}_{bc}c^b c^c.$$

The minimal BV action for the gauge and ghost sector is

$$S_{\min} = S_{\mathrm{YM}}[A] + \int_M \left[A^{*\mu}_a (D_\mu c)^a - \frac{1}{2}c^*_a f^a{}_{bc}c^b c^c\right] \mathrm{d}^D x. \tag{48.6}$$

Matter fields are included by adding their classical action and antifield couplings

$$\int_M \left[\psi^* \left(\mathrm{i}c^a t^a_R \psi\right) + \left(-\mathrm{i}\bar{\psi}\, c^a t^a_R\right) \bar{\psi}^*\right] \mathrm{d}^D x,$$

with the order of odd factors fixed by the left/right derivative convention.

The master equation for (48.6) is the combined statement that S_{YM} is gauge invariant and that the BRST transformations are nilpotent. Let

$$R^a_\mu(A, c) := (D_\mu c)^a, \qquad R^a_c(c) := -\frac{1}{2}f^a{}_{bc}c^b c^c,$$

and write

$$S_{\min} = S_{\mathrm{YM}} + S_1, \qquad S_1 = \int_M \left(A^{*\mu}_a R^a_\mu + c^*_a R^a_c\right) \mathrm{d}^D x.$$

Give each antifield antifield number 1 and each field antifield number 0. Since S_{YM} is independent of antifields and S_1 is linear in antifields,

$$\frac{1}{2}(S_{\min}, S_{\min}) = (S_{\mathrm{YM}}, S_1) + \frac{1}{2}(S_1, S_1),$$

and the two displayed terms have antifield numbers 0 and 1, respectively. The antifield-number-zero component is

$$(S_{\mathrm{YM}}, S_1) = \int_M \frac{\delta^R S_{\mathrm{YM}}}{\delta A^a_\mu} R^a_\mu \, \mathrm{d}^D x = \int_M \frac{\delta S_{\mathrm{YM}}}{\delta A^a_\mu} (D_\mu c)^a \, \mathrm{d}^D x.$$

Gauge invariance of S_{YM} means that this expression vanishes when the odd ghost c is replaced by an arbitrary even infinitesimal gauge parameter ζ. Because the identity is algebraic in the parameter and its derivatives, the same coefficient identity gives the vanishing for c.

For the antifield-number-one component, the antibracket calculation gives

$$\frac{1}{2}(S_1, S_1) = \int_M \left[A^{*\mu}_a \, sR^a_\mu + c^*_a \, sR^a_c\right] \mathrm{d}^D x,$$

where s acts on functions of A, c by

$$sX = \int_M \left[R^a_\mu \frac{\delta_L X}{\delta A^a_\mu} + R^a_c \frac{\delta_L X}{\delta c^a}\right] \mathrm{d}^D x.$$

The two coefficients are explicitly

$$\begin{aligned}sR^a_\mu &= D_\mu R^a_c + f^a{}_{bc} R^b_\mu c^c \\ &= -\frac{1}{2} f^a{}_{bc} D_\mu (c^b c^c) + f^a{}_{bc} (D_\mu c)^b c^c = 0,\end{aligned}$$

where D_μ is an even derivation and the two terms in $D_\mu(c^b c^c)$ are equal after antisymmetry of $f^a{}_{bc}$ and the odd commutation of c with $D_\mu c$. Similarly,

$$\begin{aligned}sR^a_c &= -\frac{1}{2} f^a{}_{bc} \left(R^b_c c^c - c^b R^c_c \right) \\ &= -\frac{1}{12} \left(f^a{}_{be} f^e{}_{cd} + f^a{}_{ce} f^e{}_{db} + f^a{}_{de} f^e{}_{bc} \right) c^b c^c c^d = 0\end{aligned}$$

by the Jacobi identity. There are no higher antifield-number components for this off-shell closed Yang–Mills algebra. Thus $\frac{1}{2}(S_{\min}, S_{\min}) = 0$ is exactly the pair of statements $sS_{\mathrm{YM}} = 0$ and $s^2 A = s^2 c = 0$.

Remark 48.4 (Off-shell closed and open gauge algebras). The preceding construction used a special property of Yang–Mills theory. Let Φ^i denote the classical fields, let

$$E_i(\Phi) := \frac{\delta S_0}{\delta \Phi^i}$$

be the Euler–Lagrange derivative of the classical gauge-invariant action, and write an infinitesimal gauge variation as

$$\delta_\epsilon \Phi^i = R^i{}_\alpha(\Phi) \epsilon^\alpha.$$

Here i and α are condensed indices, including spacetime position and possible derivatives acting on the gauge parameter. The algebra is off-shell closed when the commutator of gauge vector fields satisfies an identity on field space,

$$R^j{}_\beta \frac{\delta R^i{}_\alpha}{\delta \Phi^j} - R^j{}_\alpha \frac{\delta R^i{}_\beta}{\delta \Phi^j} = R^i{}_\gamma C^\gamma{}_{\alpha\beta}(\Phi), \tag{48.7}$$

with no use of $E_i = 0$. Yang–Mills theory has $C^\gamma{}_{\alpha\beta} = f^\gamma{}_{\alpha\beta}$, in the local Lie-algebra notation of the BRST chapter. Then the minimal action is linear in antifields, apart from the ghost antifield term encoding (48.7), and the master equation is precisely the gauge-invariance and Jacobi-identity calculation displayed above.

An open algebra closes only modulo the equations of motion. For bosonic gauge parameters and bosonic fields, the corresponding commutator identity is

$$R^j{}_\beta \frac{\delta R^i{}_\alpha}{\delta \Phi^j} - R^j{}_\alpha \frac{\delta R^i{}_\beta}{\delta \Phi^j} = R^i{}_\gamma C^\gamma{}_{\alpha\beta}(\Phi) + M^{ij}{}_{\alpha\beta}(\Phi) E_j(\Phi), \tag{48.8}$$

with the standard Koszul signs inserted in theories with fermionic fields or fermionic gauge parameters. The coefficient $M^{ij}{}_{\alpha\beta}$ is graded antisymmetric in the pair (i, j) in the sense required by the Noether identities. If one couples antifields only linearly to the naive BRST variations, the antibracket $\frac{1}{2}(S, S)$ contains a term proportional to $\Phi^*_i M^{ij}{}_{\alpha\beta} E_j c^\alpha c^\beta$. This is zero after imposing the classical equations of motion, but it is not zero as an identity on the field algebra. The BV solution cancels it by adding higher antifield-number terms. In the same ungraded sign convention, the first such term is a quadratic antifield term of the form

$$\frac{1}{4} \int \Phi^*_i \Phi^*_j M^{ij}{}_{\alpha\beta}(\Phi) c^\alpha c^\beta,$$

with signs fixed by the parity convention. Further terms are then fixed recursively by the single equation $\frac{1}{2}(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0$. Thus BV constructs, in open algebras, an off-shell nilpotent differential on the extended field-antifield algebra.

Reducible symmetries require additional ghosts. If

$$R^i{}_\alpha Z^\alpha{}_{a_1} = 0$$

identically, or modulo the equations of motion, then the gauge parameters themselves have gauge redundancies, and one introduces ghosts-for-ghosts together with their antifields. Standard examples where the BV framework is structural rather than optional include supergravity theories without a complete off-shell auxiliary-field formulation, type-II superstring field theory, and the Poisson sigma model. The elementary Faddeev–Popov/BRST construction in the preceding chapter concerns the off-shell closed Yang–Mills case.

48.7 Nonminimal Sector and Gauge Fixing

Gauge fixing is implemented by adding a contractible nonminimal pair and then choosing a Lagrangian submanifold in the BV field-antifield space. The cohomological content is carried by the same BV differential.

For Yang–Mills theory the nonminimal pair is $(\bar{c}^a, B^a)$, with

$$s\bar{c}^a = B^a, \qquad sB^a = 0.$$

The nonminimal BV term is

$$S_{\mathrm{nm}} = \int_M \bar{c}^*_a B^a \, \mathrm{d}^D x.$$

The total classical BV action is

$$S_{\mathrm{BV}} = S_{\mathrm{min}} + S_{\mathrm{nm}}.$$

In a gauge-fixed BV functional integral the variables $\bar{c}^a$ and B^a belong to the nonminimal integration sector. The field B^a is an even auxiliary integration variable, not an external source. Its purpose is to make the nonminimal BRST pair off-shell contractible and to write the gauge-fixing term before completing the square.

Definition 48.5 (Gauge-fixing fermion)**.** A gauge-fixing fermion is an odd functional $\Psi[\Phi]$ of ghost number -1 depending only on the fields. It determines the Lagrangian submanifold

$$\Phi^*_A = \frac{\delta^L \Psi}{\delta \Phi^A}. \tag{48.9}$$

The gauge-fixed action is the restriction

$$S_\Psi[\Phi] = S_{\mathrm{BV}} \left[\Phi, \Phi^* = \frac{\delta^L \Psi}{\delta \Phi} \right].$$

For covariant Yang–Mills gauges,

$$\Psi = \int_M \bar{c}^a \left(\partial^\mu A^a_\mu + \frac{g^2_{\mathrm{YM}} \xi}{2} B^a \right) \mathrm{d}^D x.$$

Substituting (48.9) into S_{BV}, and integrating by parts with the same boundary convention as in the BRST chapter, gives

$$S_\Psi = S_{\mathrm{YM}} + \int_M \left[B^a \partial^\mu A^a_\mu + \frac{g^2_{\mathrm{YM}}\xi}{2} B^a B^a - \bar{c}^a \partial^\mu D^{ab}_\mu(A) c^b \right] \mathrm{d}^D x.$$

Thus the Faddeev–Popov action is the restriction of the BV action to a gauge-fixing Lagrangian submanifold. At fixed regulator, the subsequent integration over B is Gaussian:

$$\int [DB] \, \exp\left\{ \mathrm{i} \int_M \left[B^a \partial^\mu A^a_\mu + \frac{g^2_{\mathrm{YM}}\xi}{2} B^a B^a \right] \mathrm{d}^D x \right\} = \mathcal{N}_B(\xi) \, \exp\left\{ -\frac{\mathrm{i}}{2 g^2_{\mathrm{YM}}\xi} \int_M (\partial^\mu A^a_\mu)^2 \, \mathrm{d}^D x \right\}.$$

After this auxiliary integral is performed, B is no longer a remaining field variable in the gauge-fixed path integral; retaining it instead is an equivalent first-order presentation of the same regularized Gaussian factor. Different gauge choices correspond to different Lagrangian submanifolds. They define equivalent physical cohomology when the BV integral has no boundary contribution and the quantum master equation is preserved.

Figure 48.1 displays the geometric content of these formulae. The BV space is an odd symplectic supermanifold with Darboux coordinates (Φ^A, Φ^*_A). A gauge-fixing fermion does not add a new dynamical variable; it chooses a Lagrangian submanifold on which the antifields are functions of the fields.

BV field–antifield geometry and gauge fixing

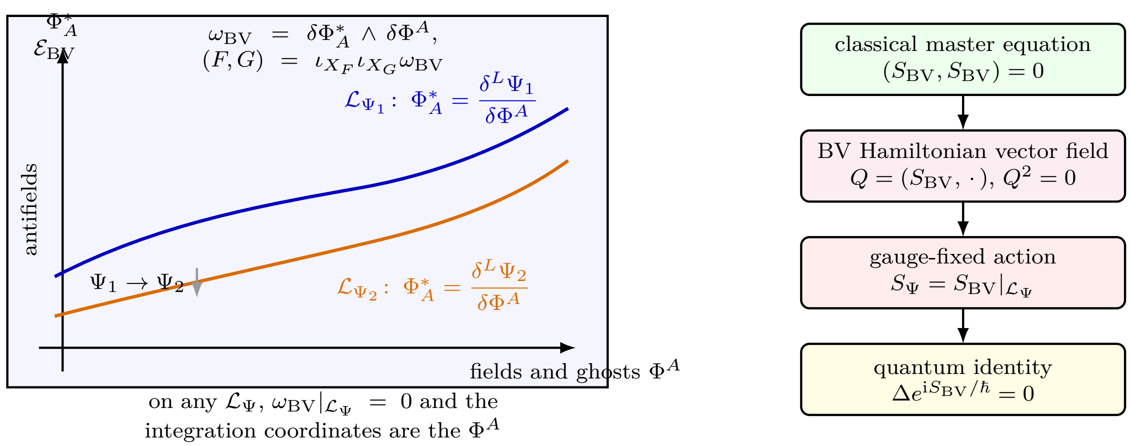


Figure 48.1: The BV antibracket is the Poisson bracket associated with the odd symplectic form on field–antifield space. A gauge-fixing fermion Ψ specifies the Lagrangian submanifold $\Phi^*_A = \delta^L \Psi / \delta \Phi^A$; restricting S_{BV} to this submanifold gives the gauge-fixed action.

48.8 Half-Densities and the BV Integration Measure

The phrase "the BV measure" has a precise finite-dimensional meaning. Let $\mathcal{E}$ be the finite-dimensional supermanifold of fields and antifields at a fixed regulator, equipped with the odd symplectic form

$$\omega = \delta\Phi^*_A \, \delta\Phi^A$$

in Darboux coordinates. A density on a supermanifold is a section of the Berezinian line $\mathrm{Ber}(\mathcal{E})$. A half-density is a section of $\mathrm{Ber}(\mathcal{E})^{1/2}$. In a coordinate system $z^I = (\Phi^A, \Phi^*_A)$, a half-density is written

$$\sigma = f(z)\,|Dz|^{1/2},$$

where f transforms by the square root of the Berezinian of the coordinate change. This transformation law is exactly what makes the restriction of σ to a Lagrangian submanifold into an ordinary Berezin density on that submanifold.

Definition 48.6 (Canonical BV operator on half-densities)**.** On an odd symplectic supermanifold $(\mathcal{E}, \omega)$, the canonical BV operator on half-densities is the odd second-order operator $\Delta_{1/2}$ defined in every Darboux chart by

$$\Delta_{1/2}\Big(f\,|D\Phi\,D\Phi^*|^{1/2}\Big) = \left[\sum_A (-1)^{\epsilon_A} \frac{\partial^R}{\partial \Phi^A} \frac{\partial^R f}{\partial \Phi^*_A}\right] |D\Phi\,D\Phi^*|^{1/2}. \tag{48.10}$$

The expression is invariant under odd-symplectic coordinate changes and $\Delta_{1/2}^2 = 0$. This is the finite-dimensional Khudaverdian semidensity operator; the displayed Darboux formula is the local expression used by perturbative regulators.

Definition 48.7 (Normal BV half-density data)**.** A finite-dimensional quantum BV regulator contains, in addition to $(\mathcal{E}, \omega)$, an even nowhere-vanishing reference half-density σ_0. It is called normal when

$$\Delta_{1/2}\sigma_0 = 0.$$

For normal half-density data, the BV Laplacian on functions is the divergence operator

$$\Delta_{\sigma_0} F := \sigma_0^{-1} \Delta_{1/2}(F\sigma_0). \tag{48.11}$$

In a Darboux chart in which $\sigma_0 = |D\Phi\,D\Phi^*|^{1/2}$, this becomes

$$\Delta_{\sigma_0} = \sum_A (-1)^{\epsilon_A} \frac{\partial^R}{\partial \Phi^A} \frac{\partial^R}{\partial \Phi^*_A}. \tag{48.12}$$

The square σ_0^2 is the corresponding Berezinian density. If a chosen regulator supplies a reference half-density with $\Delta_{1/2}\sigma_0 \neq 0$, its logarithmic defect is an additional local measure term; the symbol Δ_{BV} in this chapter denotes the function operator Δ_{σ_0} only after normal half-density data have been chosen. In Schwarz's geometric terminology, the pair consisting of the odd symplectic structure and these compatible volume/half-density data forms the finite-dimensional measure structure behind BV quantization.

Proposition 48.8 (Restriction to Lagrangians and BV Stokes)**.** *Let $L \subset \mathcal{E}$ be a Lagrangian submanifold with embedding ι_L. Every half-density σ on $\mathcal{E}$ determines a Berezin density $\iota_L^*\sigma$ on L. In a Darboux chart adapted to L, so that L is given by $\Phi^*_A = 0$,*

$$\iota_L^*\Big(f(\Phi, \Phi^*)\,|D\Phi\,D\Phi^*|^{1/2}\Big) = f(\Phi, 0)\,|D\Phi|.$$

If η is a compactly supported half-density, or if the chosen integration cycle has boundary behavior that kills total derivatives, then

$$\int_L \iota_L^*(\Delta_{1/2}\eta) = 0. \tag{48.13}$$

Proof. In an adapted Darboux chart the coordinates on the Lagrangian are the Φ^A's, and setting $\Phi^*_A = 0$ leaves the Berezin density

$$f(\Phi, 0)|D\Phi|.$$

Consider another adapted Darboux chart $(\widehat{\Phi}, \widehat{\Phi}^*)$. On L, the transformation has the block form

$$\widehat{\Phi} = \widehat{\Phi}(\Phi), \qquad \widehat{\Phi}^* = \left(\frac{\partial \Phi}{\partial \widehat{\Phi}}\right) \Phi^* + O((\Phi^*)^2),$$

with the displayed linear term forced by preservation of the odd symplectic form. Therefore the Berezinian of the full adapted transformation restricts to the square of the Berezinian of the induced transformation on L. The half-density transformation law on $\mathcal{E}$ consequently becomes exactly the Berezin-density transformation law on L. The local prescription thus glues to an intrinsic density $\iota_L^* \sigma$.

For the Stokes statement, write $\eta = h(\Phi, \Phi^*)|D\Phi\, D\Phi^*|^{1/2}$ in an adapted chart. Restricting $\Delta_{1/2}\eta$ to L gives

$$\sum_A (-1)^{\epsilon_A} \frac{\partial^R}{\partial \Phi^A} \left.\frac{\partial^R h}{\partial \Phi^*_A}\right|_{\Phi^*=0} |D\Phi|,$$

because the Darboux expression for $\Delta_{1/2}$ differentiates once in a field direction and once in its antifield partner. The displayed density is the divergence, on L, of the compactly supported Berezin vector-density whose A-component is

$$(-1)^{\epsilon_A} \left.\frac{\partial^R h}{\partial \Phi^*_A}\right|_{\Phi^*=0} |D\Phi|.$$

Its integral vanishes by the ordinary Berezin-Stokes theorem under the stated compact-support or boundary-decay hypothesis. A finite partition of unity by adapted Darboux charts gives the finite-dimensional statement for an arbitrary Lagrangian. □

48.9 The Quantum Master Equation

The classical master equation controls the algebra of gauge symmetry. The quantum functional integral also depends on the half-density data just defined. In field theory the operator Δ_{σ_0} requires a regulator because the two functional derivatives act at the same point. A quantum BV problem therefore specifies the regularized odd symplectic space, the normal reference half-density σ_0, and the action.

Definition 48.9 (Quantum master equation)**.** For Lorentzian weight $\exp(\mathrm{i}S/\hbar)$, the BV integrand is the half-density

$$\sigma_S := \exp(\mathrm{i}S/\hbar)\, \sigma_0.$$

The quantum master equation is the intrinsic condition

$$\Delta_{1/2}\sigma_S = 0. \tag{48.14}$$

When σ_0 is normal, (48.14) is equivalent to

$$\frac{1}{2}(S, S) - \mathrm{i}\hbar\, \Delta_{\sigma_0} S = 0. \tag{48.15}$$

Equivalently,

$$\Delta_{\sigma_0} \exp(\mathrm{i}S/\hbar) = 0.$$

Component form of the quantum master equation. For normal BV half-density data and an even action S,

$$\Delta_{\sigma_0} \exp(\mathrm{i}S/\hbar) = 0$$

is equivalent to

$$\frac{1}{2}(S, S) - \mathrm{i}\hbar\, \Delta_{\sigma_0} S = 0.$$

In Darboux coordinates with normal reference half-density, the BV Laplacian is an odd second-order operator. For homogeneous functions F, G, its failure to be a derivation is the antibracket:

$$\Delta_{\sigma_0}(FG) = (\Delta_{\sigma_0} F)G + (-1)^{|F|} F \Delta_{\sigma_0} G + (-1)^{|F|}(F, G). \tag{48.16}$$

This identity follows by applying (48.12) to the product FG and collecting the two terms in which one derivative hits each factor. Since S is even, induction using (48.16) gives

$$\Delta_{\sigma_0} S^n = nS^{n-1}\Delta_{\sigma_0} S + \frac{n(n-1)}{2} S^{n-2}(S, S). \tag{48.17}$$

Expanding the exponential and summing the power series yields

$$\Delta_{\sigma_0} \mathrm{e}^{\mathrm{i}S/\hbar} = \left[\frac{\mathrm{i}}{\hbar}\Delta_{\sigma_0} S - \frac{1}{2\hbar^2}(S, S)\right] \mathrm{e}^{\mathrm{i}S/\hbar}.$$

The exponential factor is invertible as a formal power series, so the vanishing of the left hand side is equivalent to the displayed component equation.

The half-density form is the coordinate-free statement. Gauge independence is then a version of Proposition 48.8: changing the gauge-fixing Lagrangian submanifold changes the integral by the integral of a $\Delta_{1/2}$-exact half-density, provided no boundary contribution is present.

Definition 48.10 (Gauge-fixed Wilsonian BV consistency)**.** At cutoff Λ, a gauge-fixed Wilsonian path integral is BV-consistent when it is obtained by restricting a BV half-density

$$\sigma_\Lambda = \exp(\mathrm{i}S_\Lambda/\hbar)\sigma_{0,\Lambda}$$

to a gauge-fixing Lagrangian submanifold $\mathcal{L}_{\Psi,\Lambda} \subset \mathcal{E}_\Lambda$, with

$$\Delta_{1/2,\Lambda}\sigma_\Lambda = 0.$$

Equivalently, for normal $\sigma_{0,\Lambda}$, the cutoff action satisfies

$$\frac{1}{2}(S_\Lambda, S_\Lambda)_\Lambda - \mathrm{i}\hbar\, \Delta_{\sigma_{0,\Lambda}} S_\Lambda = 0.$$

The gauge-fixed source functional is the restriction

$$Z_{\Lambda,\Psi}[J] = \int_{\mathcal{L}_{\Psi,\Lambda}} \iota^*_{\Psi,\Lambda} \left[\exp\left\{\frac{\mathrm{i}}{\hbar}\left(S_\Lambda + \langle J, \Phi\rangle\right)\right\}\sigma_{0,\Lambda}\right].$$

The definition identifies the master equation as the gauge-consistency condition for the gauge-fixed Wilsonian integral. The gauge condition Ψ selects a Lagrangian submanifold for computation; the equation lives on the field-antifield space before this restriction. Differentiating the restricted integral and applying BV Stokes gives the cutoff Slavnov–Taylor identities. Changing Ψ changes the representative of the same BV half-density integral when the boundary term in (48.13) vanishes. A failure of the displayed master equation is a ghost-number-one anomaly functional, and (48.19) is its Wess–Zumino consistency condition.

If the regulated theory fails to satisfy (48.15), the failure has ghost number +1. To first order in $\hbar$, write

$$\mathcal{M}(S) := \frac{1}{2}(S,S) - \mathrm{i}\hbar\,\Delta_{\sigma_0}S = \hbar\,\mathcal{A} + O(\hbar^2). \tag{48.18}$$

The BV Jacobi identity, the second-order identity for Δ_{σ_0}, and $\Delta_{\sigma_0}^2 = 0$ give the quantum consistency identity

$$(S, \mathcal{M}(S)) - \mathrm{i}\hbar\,\Delta_{\sigma_0}\mathcal{M}(S) = 0. \tag{48.19}$$

Indeed, for even S, $(S, \frac{1}{2}(S,S)) = 0$ and $\Delta_{\sigma_0}(S,S) = 2(\Delta_{\sigma_0}S, S)$, so the two order-$\hbar$ terms in (48.19) cancel. Substituting (48.18) and $S = S_0 + O(\hbar)$ into (48.19) gives

$$0 = \hbar\,(S_0, \mathcal{A}) + O(\hbar^2).$$

Equivalently, with $s_{\mathrm{BV}} := (\cdot, S_0)$,

$$s_{\mathrm{BV}}\mathcal{A} = (\mathcal{A}, S_0) = 0.$$

Changing the local counterterm by $\hbar B$ changes the anomaly by

$$\mathcal{A} \longmapsto \mathcal{A} + s_{\mathrm{BV}}B = \mathcal{A} + (B, S_0). \tag{48.20}$$

Thus the obstruction class lies in ghost-number-one local BV cohomology. This is the antifield version of the anomaly classification by $H^{1,D}(s \mid d)$. In the chiral gauge example of Chapter 51, the representative is the descent cocycle whose bottom component gives the consistent anomaly (51.30); adding a local counterterm changes the representative by an s_{BV}-exact cocycle, matching (48.20).

48.10 The 1PI Master Equation

The antifields are also the natural sources for nonlinear BRST variations in the 1PI formalism. With a gauge-fixing Lagrangian submanifold chosen for the fields, let $\iota_\Psi : \mathcal{L}_\Psi \hookrightarrow \mathcal{E}$ be its embedding and let Φ^*_{ext} denote the antifield coordinates left as external sources after the gauge-fixing canonical transformation. Define

$$Z[J, \Phi^*_{\mathrm{ext}}] = \int_{\mathcal{L}_\Psi} \iota^*_\Psi \left[\exp\left\{ \frac{\mathrm{i}}{\hbar} \left(S_{\mathrm{BV}}[\Phi, \Phi^*_{\mathrm{ext}}] + \int J_A \Phi^A \right) \right\} \sigma_0 \right].$$

In an adapted finite-regulator chart this is the compact formula with

$$D_{\sigma_0,\Psi}\Phi := \iota^*_\Psi \sigma_0$$

in place of the undefined symbol $[D\Phi]$. The external antifields are not integrated; they record the response to BRST variations. Let $W[J, \Phi^*_{\mathrm{ext}}]$ be the connected generator and let

$$\varphi^A = \frac{\delta W}{\delta J_A}, \qquad \Gamma[\varphi, \Phi^*_{\mathrm{ext}}] = W[J, \Phi^*_{\mathrm{ext}}] - \int J_A \varphi^A$$

be the 1PI effective action in the Lorentzian convention used earlier.

Assume that the regularized half-density integrand obeys the quantum master equation at $J = 0$ and that the Legendre transform is well defined on the field directions under consideration. The Ward identity obtained by applying $\Delta_{1/2}$ to the source-dependent integrand and then using BV Stokes gives

$$\int \frac{\delta^R \Gamma}{\delta \varphi^A} \frac{\delta^L \Gamma}{\delta \Phi^*_{\text{ext,A}}} \mathrm{d}^D x = 0.$$

Equivalently,

$$\frac{1}{2}(\Gamma, \Gamma) = 0, \tag{48.21}$$

where the bracket now pairs the mean field φ^A with the antifield $\Phi^*_{\text{ext,A}}$. In the Yang–Mills variables of the preceding chapter, (48.21) is the Slavnov–Taylor identity (47.44), with A^* and c^* replacing the special sources K and L, and with the nonminimal term producing $B\, \delta\Gamma/\delta\bar{c}$. The all-order restoration theorem 47.16 is therefore the gauge-fixed Yang–Mills form of the perturbative BV statement: when the local ghost-number-one obstruction class vanishes, counterterms may be chosen so that the renormalized 1PI functional satisfies (48.21) as a formal power series in $\hbar$. The ghost-number-zero BV cohomology records the remaining finite coordinate freedom.

48.11 Wilsonian BV Pushforward

The Wilsonian effective action is obtained by integrating out degrees of freedom above a scale. BV gives the condition under which this operation is compatible with gauge symmetry.

The input is more than a momentum cutoff. A gauge-compatible Wilsonian step contains a finite-dimensional or regularized BV phase space $(\mathcal{E}_\Lambda, \omega_\Lambda)$, a normal half-density $\sigma_{0,\Lambda}$, a solution S_Λ of the quantum master equation, a split into low and high BV variables, and a Lagrangian cycle for the high variables. Equivalently, in a lattice formulation it contains a finite lattice gauge integral and a gauge-invariant blocking map for observables or actions. These data are what make the phrase "integrating out gauge-field modes" a defined operation.

For a gauge-fixed Wilsonian path integral, the flow between cutoffs preserves Definition 48.10. In BV terms this means that the high-variable integration is a BV pushforward of the cutoff half-density. The resulting low-cutoff half-density again satisfies the quantum master equation, and every gauge-fixed representative obtained from it satisfies the corresponding cutoff Slavnov–Taylor identities.

Consider first a finite-dimensional odd symplectic vector space split as

$$\mathcal{E} = \mathcal{E}_{\text{low}} \oplus \mathcal{E}_{\text{high}}, \qquad \Delta_{1/2,\mathcal{E}} = \Delta_{1/2,\text{low}} + \Delta_{1/2,\text{high}}.$$

Assume the normal half-density factorizes compatibly as

$$\sigma_0 = \sigma_{0,\text{low}} \otimes \sigma_{0,\text{high}}.$$

Let $\mathcal{L}_{\text{high}} \subset \mathcal{E}_{\text{high}}$ be a Lagrangian subspace used to integrate out the high variables. Define the effective low-energy half-density by

$$\exp(\mathrm{i}S_{\text{eff}}/\hbar)\, \sigma_{0,\text{low}} = \int_{\mathcal{L}_{\text{high}}} \iota^*_{\text{high}} \left[\exp(\mathrm{i}S/\hbar)\, \sigma_0\right]. \tag{48.22}$$

Proposition 48.11 (BV pushforward)**.** *Assume*

$$\Delta_{1/2,\mathcal{E}}\left(\exp(\mathrm{i}S/\hbar)\sigma_0\right) = 0$$

and assume the integral over $\mathcal{L}_{\mathrm{high}}$ *has no boundary contribution for* $\Delta_{1/2,\mathrm{high}}$. *Then*

$$\Delta_{1/2,\mathrm{low}}\left(\exp(\mathrm{i}S_{\mathrm{eff}}/\hbar)\sigma_{0,\mathrm{low}}\right) = 0.$$

Proof. The split of the odd symplectic vector space and the compatible factorization of the reference half-density identify a full semidensity with a family of high-variable semidensities parametrized by the low fields. Since $\Delta_{1/2,\mathrm{low}}$ differentiates only the low coordinates and the fiber integral is assumed convergent, it may be passed through the high integration:

$$\begin{aligned}\Delta_{1/2,\mathrm{low}}\left(\exp(\mathrm{i}S_{\mathrm{eff}}/\hbar)\sigma_{0,\mathrm{low}}\right) &= \int_{\mathcal{L}_{\mathrm{high}}} \iota^*_{\mathrm{high}}\left[\Delta_{1/2,\mathrm{low}}\left(\exp(\mathrm{i}S/\hbar)\sigma_0\right)\right] \\ &= -\int_{\mathcal{L}_{\mathrm{high}}} \iota^*_{\mathrm{high}}\left[\Delta_{1/2,\mathrm{high}}\left(\exp(\mathrm{i}S/\hbar)\sigma_0\right)\right] = 0.\end{aligned}$$

The second equality uses the full QME and the compatible split $\Delta_{1/2,\mathcal{E}} = \Delta_{1/2,\mathrm{low}} + \Delta_{1/2,\mathrm{high}}$. The last equality is Proposition 48.8 applied to the high Lagrangian integration cycle, together with the explicit no-boundary hypothesis. Thus QME preservation is a chain-map statement for the BV pushforward; if the high cycle has a boundary contribution, the pushforward satisfies the anomalous boundary formula developed later rather than the homogeneous QME. □

This proposition is the regulated mathematical core of the Wilsonian BV identity. In a field theory with cutoff Λ, the Wilsonian action S_Λ is a scale-dependent local or quasi-local representative of the BV pushforward. Its master equation has the form

$$\frac{1}{2}(S_\Lambda, S_\Lambda)_\Lambda - \mathrm{i}\hbar\,\Delta_\Lambda S_\Lambda = 0,$$

where the bracket and Laplacian include the cutoff covariance and the remaining low-mode half-density. Exact RG flow is compatible with gauge symmetry when it preserves this equation. Changes of cutoff scheme are then BV canonical transformations together with possible local counterterm changes, with the gauge symmetry encoded by the transformed master action.

The lattice route gives a second gauge-compatible Wilsonian construction. A finite blocking map sends link variables on a fine lattice to link variables or gauge-invariant observables on a coarser lattice, and the pushforward of the compact Haar integral defines the coarse action or coarse observable law. The continuum Wilsonian statement then concerns a scaling limit of these finite lattice pushforwards. A comparison between the lattice and BV descriptions must identify the gauge-invariant observables, the local BV cohomology classes, and the limiting topology for correlation functions or effective-action coordinates.

48.12 Infinitesimal Exact RG in BV Form

The finite pushforward above is the primary definition. A differential renormalization-group equation is obtained from a differentiable family of finite pushforwards or from an infinitesimal change of regulator. The following finite-regulator statement records exactly what must be preserved.

Definition 48.12 (BV-compatible infinitesimal RG step)**.** Let $(\mathcal{E}, \omega, \sigma_0)$ be a finite-dimensional odd symplectic space with normal half-density data and let

$$\rho_t = \exp(\mathrm{i}S_t/\hbar)\sigma_0$$

be an even family of half-densities. An infinitesimal RG step is BV-compatible if there is an odd ghost-number -1 function R_t such that

$$\partial_t \rho_t = \Delta_{1/2}(R_t \rho_t). \tag{48.23}$$

The function R_t is part of the scheme data. It encodes the infinitesimal change of covariance, splitting, or gauge-fixing cycle after the regulator has made these words finite.

Infinitesimal BV-compatible RG identity. The condition in Definition 48.12 is the substantive hypothesis: it says that the velocity of the regulated half-density is $\Delta_{1/2}$-exact. Once this has been established by a finite cutoff construction, preservation of the quantum master equation is the cohomological tangent identity

$$\partial_t(\Delta_{1/2}\rho_t) = \Delta_{1/2}^2(R_t \rho_t) = 0.$$

Hence, if $\Delta_{1/2}\rho_{t_0} = 0$ at one value t_0, then $\Delta_{1/2}\rho_t = 0$ throughout the same interval, provided the same normal half-density data are used.

The action-coordinate form is a bookkeeping consequence of the same finite-regulator identity. Write $\rho_t = \mathrm{e}^{\mathrm{i}S_t/\hbar}\sigma_0$, use the function Laplacian induced by the normal half-density, and use that R_t is odd. The second-order identity gives

$$\Delta_{\sigma_0}(R_t \mathrm{e}^{\mathrm{i}S_t/\hbar}) = (\Delta_{\sigma_0} R_t)\mathrm{e}^{\mathrm{i}S_t/\hbar} - (R_t, \mathrm{e}^{\mathrm{i}S_t/\hbar}).$$

The bracket term is

$$(R_t, \mathrm{e}^{\mathrm{i}S_t/\hbar}) = \frac{\mathrm{i}}{\hbar}(R_t, S_t)\mathrm{e}^{\mathrm{i}S_t/\hbar} = -\frac{\mathrm{i}}{\hbar}(S_t, R_t)\mathrm{e}^{\mathrm{i}S_t/\hbar},$$

because S_t is even. Dividing

$$\partial_t \mathrm{e}^{\mathrm{i}S_t/\hbar} = \Delta_{\sigma_0}(R_t \mathrm{e}^{\mathrm{i}S_t/\hbar})$$

by $\mathrm{e}^{\mathrm{i}S_t/\hbar}$ and multiplying by $\hbar/\mathrm{i}$ gives

$$\partial_t S_t = (S_t, R_t) - \mathrm{i}\hbar\, \Delta_{\sigma_0} R_t. \tag{48.24}$$

The equation is a quantum canonical flow, not an arbitrary beta-function ansatz. It is the differential version of BV pushforward: the right hand side is $\Delta_{1/2}$-exact on half-densities, so it cannot create a master equation defect. A cutoff flow proposed for a gauge theory is therefore gauge-compatible only after its regulator supplies R_t, the odd symplectic structure, the half-density data, and the boundary condition making the finite pushforward or the differential equation legitimate.

48.13 Counterterms, Anomalies, and Physical Operators

Let S_{BV} be a classical solution of the master equation and let

$$s_{\mathrm{BV}} = (\cdot, S_{\mathrm{BV}}).$$

A first-order local deformation

$$S_{\mathrm{BV}} \longmapsto S_{\mathrm{BV}} + \epsilon C$$

preserves the classical master equation to order ϵ precisely when

$$s_{\mathrm{BV}} C = 0, \qquad \mathrm{gh}(C) = 0.$$

If $C = s_{\mathrm{BV}} B$, then the deformation is generated by a BV canonical transformation and does not change the physical theory. Thus invariant counterterms are classified by ghost-number-zero BV cohomology of local functionals, modulo total derivatives and canonical redefinitions.

An anomaly candidate is a ghost-number-one functional $\mathcal{A}$ satisfying

$$s_{\mathrm{BV}} \mathcal{A} = 0.$$

It is removable by a local counterterm exactly when

$$\mathcal{A} = s_{\mathrm{BV}} B.$$

This gives the precise cohomological meaning of anomaly cancellation in the perturbative BV framework: the obstruction class must vanish in the relevant local BV cohomology.

Physical local operators are treated in the same language. An insertion $\mathcal{O}$ is compatible with the gauge symmetry when

$$s_{\mathrm{BV}} \mathcal{O} = 0, \qquad \mathrm{gh}(\mathcal{O}) = 0,$$

and two insertions differing by $s_{\mathrm{BV}} \mathcal{X}$ define the same cohomology class. After gauge fixing, this statement reduces to the BRST cohomology of local observables described in the preceding chapter. Before gauge fixing, it is expressed without choosing a particular gauge slice.

48.14 BV–BRST Equivalence for Gauge-Fixed Yang–Mills

The gauge-fixed BRST complex used in the preceding chapter is obtained from the BV complex by two homological operations: a BV canonical transformation generated by the gauge-fixing fermion, and deletion of contractible nonminimal pairs. The next theorem states the equivalence in the form needed for perturbative Yang–Mills with matter.

Theorem 48.13 (Local BV–BRST comparison for Yang–Mills)**.** *Let G be compact and reductive, and let the classical Yang–Mills–matter action use the trace and coupling convention of Chapter 45,*

$$S_{\mathrm{YM}} = \frac{1}{4g_{\mathrm{YM}}^2} \int \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}) + S_{\mathrm{matter}}.$$

Assume the perturbative local jet algebra is taken near a background for which the Koszul–Tate differential is acyclic in positive antifield number for the fields under consideration, and assume boundary conditions that make local functionals modulo total derivatives the correct cochain space. Then:

1. *The canonical BV transformation generated by a gauge-fixing fermion Ψ preserves the antibracket and the classical master equation.*

2. *Restriction to the Lagrangian submanifold $\Phi^* = \delta^L\Psi/\delta\Phi$ gives the gauge-fixed BRST differential on fields, with the nonminimal pair $s\bar{c} = B$, $sB = 0$.*

3. *The inclusion of antifield-independent representatives induces an isomorphism, for nonnegative ghost number, between local BV cohomology modulo* d *and gauge-fixed BRST cohomology modulo* d, *after quotienting the contractible pair $(\bar{c}, B)$:*

$$H^{g,D}(s_{\mathrm{BV}} \mid \mathrm{d}) \cong H^{g,D}(s_\Psi \mid \mathrm{d}), \qquad g \geq 0.$$

The isomorphism identifies counterterm classes, anomaly classes, and gauge-invariant local operator classes.

Proof. The finite-regulator BV phase space has Darboux coordinates (Φ^A, Φ^*_A). The gauge-fixing fermion generates the canonical change of variables

$$\Phi^A_\Psi = \Phi^A, \qquad \Phi^*_{\Psi,A} = \Phi^*_A - \frac{\delta^L\Psi}{\delta\Phi^A}.$$

It is canonical because

$$\delta\Phi^*_{\Psi,A}\,\delta\Phi^A_\Psi = \delta\Phi^*_A\,\delta\Phi^A - \delta\left(\frac{\delta^L\Psi}{\delta\Phi^A}\right)\delta\Phi^A,$$

and the second term is the odd-symplectic differential of the exact one-form generated by Ψ; its antisymmetrized second variation vanishes. Hence the antibracket is unchanged, and $(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0$ is equivalent to the same equation in the transformed coordinates. Setting $\Phi^*_\Psi = 0$ is precisely the Lagrangian restriction $\Phi^* = \delta^L\Psi/\delta\Phi$.

On antifield-independent functions of the fields, the Hamiltonian vector field $(\,\cdot\,, S_{\mathrm{BV}})$ restricted to this Lagrangian is the gauge-fixed BRST variation. For Yang–Mills the displayed minimal action gives $sA = Dc$, $sc = -\frac{1}{2}[c, c]$, and the nonminimal term gives $s\bar{c} = B$, $sB = 0$. The proof of the classical master equation above therefore gives $s_\Psi^2 = 0$ before eliminating B.

It remains to compare cohomology. Filter the BV local functional complex by antifield number. The antifield-number -1 piece of s_{BV} is the Koszul–Tate differential δ_{KT}, with $\delta_{\mathrm{KT}}\Phi^*_i = E_i$ and $\delta_{\mathrm{KT}} c^*_\alpha = R^i{}_\alpha \Phi^*_i$. By the stated regularity hypothesis, δ_{KT} has no cohomology in positive antifield number. Therefore the spectral sequence of the antifield-number filtration collapses to representatives of antifield number zero for $g \geq 0$, modulo the gauge variation induced by the longitudinal BRST part.

The nonminimal pair is contractible. Let N_{nm} count the number of $\bar{c}$, B, and their derivatives in a local monomial, and let

$$h_{\mathrm{nm}} = \sum_{|\alpha| \geq 0} \partial_\alpha \bar{c}^a \frac{\partial^L}{\partial(\partial_\alpha B^a)}.$$

On the subalgebra generated by the nonminimal pair and its derivatives,

$$s h_{\mathrm{nm}} + h_{\mathrm{nm}} s = N_{\mathrm{nm}}.$$

If a cocycle has $N_{\mathrm{nm}} > 0$, division by this positive integer writes that component as an s-exact term. Hence the nonminimal sector contributes no cohomology. Combining this doublet contraction with the Koszul–Tate acyclicity gives the displayed isomorphism. $\square$

48.15 Local Cohomology Classification for Yang–Mills With Matter

The comparison theorem turns the abstract statement "a BV class" into a computable local object. In four-dimensional power-counting-renormalizable Yang–Mills with scalars and spinors in finite-dimensional unitary representations, the local classes used by perturbation theory have the following concrete representatives.

Proposition 48.14 (Renormalizable ghost-number-zero representatives)**.** *Under the hypotheses of Theorem 48.13, every ghost-number-zero counterterm density of engineering dimension at most four is cohomologous, modulo total derivatives and field redefinitions, to a gauge invariant density built from covariant fields:*

$$F_{\mu\nu}, \qquad D_\mu, \qquad \psi, \bar{\psi}, \qquad \phi,$$

with all gauge indices contracted by invariant tensors of the representation. For the pure gauge sector this gives

$$\frac{1}{4}\operatorname{tr}(F_{\mu\nu}F^{\mu\nu}), \qquad \frac{1}{4}\operatorname{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu})$$

in the monograph trace convention $\operatorname{tr}_\square(t^a t^b) = \delta^{ab}$*. With matter, the same list includes covariant kinetic terms, mass terms allowed by the global symmetries, Yukawa couplings, and scalar potential terms whose gauge indices are contracted invariantly. Antifield-dependent ghost-number-zero representatives encode field redefinitions, coupling-coordinate changes, and global-symmetry currents rather than new gauge-invariant interaction densities.*

Proof. Filter the local BV complex by antifield number. The Koszul–Tate differential is the first differential in the associated spectral sequence. By the acyclicity hypothesis in Theorem 48.13, positive antifield number classes in ghost number zero are either killed or represented by the standard Noether-current and field-redefinition classes. The nonminimal pair is a contractible BRST doublet, as shown in the proof of the comparison theorem, so $(\bar{c}, B)$ and their derivatives do not contribute to the cohomology. Thus the interaction-density part of a ghost-number-zero class can be represented by an antifield-independent local polynomial in

$$A_\mu, \quad \psi, \bar{\psi}, \quad \phi$$

and finitely many ordinary derivatives, modulo total derivatives.

On this antifield-independent representative the remaining differential is the longitudinal gauge differential

$$s_{\text{gauge}} A = Dc, \qquad s_{\text{gauge}} X = \mathrm{i} c^a t^a_R X$$

for every matter field X in representation R. Expanding the condition $s_{\text{gauge}} a + \mathrm{d}b = 0$ in powers of the ghost c says that the local density a is invariant under infinitesimal gauge transformations, up to the already-quotiented total derivative. Expand the local polynomial in monomials of covariant jets. Each monomial carries a finite tensor product of gauge representations, and infinitesimal invariance is exactly the statement that the coefficient tensor is fixed by the Lie-algebra action on that finite tensor product. Since G is compact reductive, Lie-algebra invariance integrates to G-invariance on these finite-dimensional representation spaces. Thus the allowed coefficients are precisely invariant tensors of the matter representations and traces of the adjoint curvature representation. Ordinary derivatives of matter fields are replaced recursively by covariant derivatives, and commutators of covariant derivatives give curvature insertions

$$[D_\mu, D_\nu]X = \mathrm{i}F^a_{\mu\nu}t^a_R X.$$

In the pure gauge sector the invariant tensors are traces in representations of G; in the monograph normalization $\mathrm{tr}_\square(t^a t^b) = \delta^{ab}$, the two independent dimension-four Lorentz-scalar densities without matter are the parity-even and parity-odd contractions

$$\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}), \qquad \mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}).$$

Finally impose the engineering-dimension bound in $D = 4$:

$$[F_{\mu\nu}] = 2, \qquad [D_\mu] = 1, \qquad [\phi] = 1, \qquad [\psi] = [\bar{\psi}] = \frac{3}{2}.$$

The possible covariant monomials of dimension at most four are therefore: gauge kinetic and topological densities; scalar kinetic terms $(D\phi)^2$; Dirac/Weyl kinetic terms; fermion and scalar mass terms allowed by gauge and global symmetries; Yukawa contractions $\phi\psi\psi$ or their complex-conjugate representation-theoretic variants; and invariant scalar polynomials of degree at most four, including possible singlet tadpoles or cubic terms when the representation permits them. All gauge indices must be contracted by invariant tensors. Total derivatives vanish in the relative cohomology $H^{0,4}(s \mid \mathrm{d})$, while BRST-exact ghost-number-zero terms are generated by BV canonical transformations and hence are precisely local field redefinitions or coupling-coordinate changes. This proves the stated list and separates genuine interaction-density coordinates from antifield representatives of redefinitions and currents. □

Proposition 48.15 (Ghost-number-one anomaly calculation)**.** *For a four-dimensional chiral Yang–Mills theory in the same perturbative local setting, a perturbative local gauge-anomaly candidate is represented in*

$$H^{1,4}(s_{\mathrm{BV}} \mid \mathrm{d})$$

by the descent of the six-form invariant polynomial

$$I_6 = \frac{\mathrm{i}}{24\pi^2}\,\mathrm{Tr}_R(\mathsf{F}^3)$$

in the anti-Hermitian background-connection convention of Chapter 51. Its component coefficient is proportional to the symmetric invariant tensor

$$d_R^{abc} := \mathrm{Tr}_R(t^{(a}t^b t^{c)}).$$

If the sum of d_R^{abc} over all left-handed Weyl fermions vanishes for every triple of gauge generators, then the perturbative local gauge anomaly class vanishes. If it does not vanish, no local counterterm restores the quantum master equation in this perturbative local cohomology.

Proof from the QME and descent. Let the loop expansion of the regulated 1PI effective action be

$$\Gamma_\Lambda = S_\Lambda + \sum_{\ell \geq 1} \hbar^\ell \Gamma_{\Lambda,\ell}.$$

Set $\Gamma_{\Lambda,0} = S_\Lambda$. The quantum master defect at the first loop order ℓ_0 where it is nonzero has the local form

$$\mathcal{B}_{\ell_0} = \frac{1}{2}\sum_{\ell_1+\ell_2=\ell_0}(\Gamma_{\Lambda,\ell_1},\Gamma_{\Lambda,\ell_2}) - \mathrm{i}\Delta_\Lambda\Gamma_{\Lambda,\ell_0-1} = \int_M a_4^{(1)},$$

where the BV antibracket has ghost number 1 and the density $a_4^{(1)}$ is a local four-form of ghost number 1. Applying the linearized classical BV differential $s_{\mathrm{BV}} = (S_\Lambda, -)$ to the QME and using the Jacobi identity gives

$$s_{\mathrm{BV}}a_4^{(1)} = \mathrm{d}a_3^{(2)}.$$

Adding a local counterterm $B_4^{(0)}$ to Γ_{Λ,ℓ_0} changes the density by

$$a_4^{(1)} \longmapsto a_4^{(1)} + s_{\mathrm{BV}}B_4^{(0)} + \mathrm{d}B_3^{(1)},$$

so the obstruction to restoring the QME at this order is a class of $H^{1,4}(s_{\mathrm{BV}} \mid \mathrm{d})$.

For the chiral Yang–Mills field content, the BV–BRST comparison map sends this class to the ordinary consistent gauge-anomaly class described in Chapter 51. In the anti-Hermitian descent convention that chapter uses,

$$I_6 = \frac{\mathrm{i}}{24\pi^2}\sum_i \mathrm{Tr}_{R_i}(\mathsf{F}^3)$$

is closed and gauge invariant, and

$$I_6 = \mathrm{d}I_5^{(0)}, \qquad sI_5^{(0)} = \mathrm{d}I_4^{(1)}$$

produces the representative $I_4^{(1)}$. Since F^a are two-forms, their products commute, and the coefficient multiplying $\mathsf{F}^a\mathsf{F}^b\mathsf{F}^c$ is exactly the symmetric invariant tensor

$$\sum_i d_{R_i}^{abc} = \sum_i \mathrm{Tr}_{R_i}(t^{(a}t^b t^{c)}).$$

The local cohomology theorem identifies these degree-three invariant polynomials with the cubic gauge part of the relative group $H^{1,4}(s_{\mathrm{BV}} \mid \mathrm{d})$. Hence a vanishing total tensor gives the zero class in this sector, while a nonzero total tensor gives a QME defect whose class is invariant under local counterterm changes.

Equation (48.18) and the consistency identity (48.19) show that the first nonzero QME defect is a ghost-number-one s_{BV}-cocycle modulo total derivatives. The BV–BRST comparison theorem identifies this group with the gauge-fixed relative BRST cohomology. The descent construction of Chapter 51 starts from the closed invariant polynomial $\mathrm{Tr}_R(\mathsf{F}^3)$, writes it locally as $\mathrm{d}I_5^{(0)}$, and then applies s to obtain

$$sI_5^{(0)} + \mathrm{d}I_4^{(1)} = 0.$$

The integrated four-form $I_4^{(1)}$ is the consistent anomaly representative. Its cubic group factor is $\mathrm{Tr}_R(t^a\{t^b,t^c\})$, equivalently the symmetric tensor d_R^{abc} after symmetrization over external gauge fields. Adding a local counterterm changes $I_4^{(1)}$ by an s-exact term plus a total derivative, as in (48.20). Therefore the class vanishes exactly when the total symmetric invariant coefficient vanishes in the chosen representation content. □

Remark 48.16 (Scope of the obstruction). The word "anomaly" in Proposition 48.15 refers to the perturbative local obstruction detected by $H^{1,4}(s_{\mathrm{BV}} \mid \mathrm{d})$. Global gauge anomalies, determinant-line topology, and finite background-anomaly phases are different objects; they are treated in Chapter 55 and in the later chapters on invertible field theories and eta invariants.

48.16 Local Quantum Observables as a BV Factorization Algebra

The BV effective-action formalism has a local-observable counterpart. This is the perturbative factorization-algebra language. It is useful here because it states locality at the level of observables rather than only at the level of a single action functional.

Definition 48.17 (Prefactorization algebra of BV observables)**.** Let M be spacetime and let $\mathrm{Open}(M)$ be the category of open subsets. A BV prefactorization algebra assigns to each open $U \subset M$ a cochain complex

$$\mathrm{Obs}_\Lambda(U)$$

of regulated quantum observables supported in U. If $U_1, \ldots, U_n \subset V$ are pairwise disjoint opens, it assigns a cochain map

$$m_{U_1,\ldots,U_n;V} : \mathrm{Obs}_\Lambda(U_1) \otimes \cdots \otimes \mathrm{Obs}_\Lambda(U_n) \longrightarrow \mathrm{Obs}_\Lambda(V),$$

compatible with permutation of the U_i's and with iterated inclusions. It is a factorization algebra when these maps satisfy descent for Weiss covers: observables on V are recovered, up to quasi-isomorphism, from observables on covers by opens that contain every finite subset of V.

For a finite BV regulator at scale Λ, the cochain differential on observables is

$$Q_\Lambda \mathcal{O} := (S_\Lambda, \mathcal{O})_\Lambda - \mathrm{i}\hbar\, \Delta_\Lambda \mathcal{O}. \tag{48.25}$$

The bracket and Laplacian are the regulated ones associated with the same half-density data used in the quantum master equation. The support condition in $\mathrm{Obs}_\Lambda(U)$ means that the kernels of $\mathcal{O}$ are supported in U, with the regulator determining what class of kernels or jets is allowed.

Proposition 48.18 (QME implies the observable differential)**.** *If S_Λ satisfies*

$$\frac{1}{2}(S_\Lambda, S_\Lambda)_\Lambda - \mathrm{i}\hbar\, \Delta_\Lambda S_\Lambda = 0,$$

then $Q_\Lambda^2 = 0$. Moreover the disjoint-support product maps in Definition 48.17 are cochain maps.

Proof. Let $\mathrm{ad}_{S_\Lambda}(\mathcal{O}) = (S_\Lambda, \mathcal{O})_\Lambda$. The regulated odd Laplacian is assumed to be square-zero on the finite regulator algebra, and it generates the regulated antibracket:

$$\Delta_\Lambda (F, G)_\Lambda = (\Delta_\Lambda F, G)_\Lambda - (-1)^{|F|}(F, \Delta_\Lambda G)_\Lambda.$$

Since S_Λ is even, this identity gives

$$\Delta_\Lambda (S_\Lambda, \mathcal{O})_\Lambda + (S_\Lambda, \Delta_\Lambda \mathcal{O})_\Lambda = (\Delta_\Lambda S_\Lambda, \mathcal{O})_\Lambda.$$

The odd Jacobi identity gives

$$\mathrm{ad}_{S_\Lambda}^2 \mathcal{O} = \frac{1}{2}\big((S_\Lambda, S_\Lambda)_\Lambda, \mathcal{O}\big)_\Lambda.$$

Therefore, using $\Delta_\Lambda^2 = 0$,

$$\begin{aligned} Q_\Lambda^2 \mathcal{O} &= \mathrm{ad}_{S_\Lambda}^2 \mathcal{O} - \mathrm{i}\hbar \left[\Delta_\Lambda (S_\Lambda, \mathcal{O})_\Lambda + (S_\Lambda, \Delta_\Lambda \mathcal{O})_\Lambda\right] \\ &= \left(\frac{1}{2}(S_\Lambda, S_\Lambda)_\Lambda - \mathrm{i}\hbar\, \Delta_\Lambda S_\Lambda, \mathcal{O}\right)_\Lambda. \end{aligned}$$

The quantum master equation sets the Hamiltonian in the final bracket to zero, so $Q_\Lambda^2 = 0$.

For disjoint supports, the regulated bracket and Laplacian have no coincident-point contraction between two different factors except the contractions included in the product map. Therefore Q_Λ obeys the graded Leibniz rule with respect to the product of disjoint observables, and the maps $m_{U_1,\dots,U_n;V}$ commute with Q_Λ. □

Changing the cutoff from Λ to Λ' gives a cochain map

$$W_{\Lambda\Lambda'} : \mathrm{Obs}_\Lambda(U) \longrightarrow \mathrm{Obs}_{\Lambda'}(U)$$

when it is produced by BV pushforward with the same boundary and locality data as in Section 48.12. The homotopy-RG content is the statement

$$Q_{\Lambda'} W_{\Lambda\Lambda'} = W_{\Lambda\Lambda'} Q_\Lambda$$

and compatibility of $W_{\Lambda\Lambda'}$ with the disjoint-support products. Thus the factorization-algebra form is the observable-side record of the same BV renormalization problem: local counterterms repair failures of the cochain and product identities precisely when the relevant local BV cohomology class is trivial.

48.17 Logical Output

The output of this chapter is the following framework.

1. A gauge theory is represented on an enlarged graded field space with fields, ghosts, nonminimal variables, and antifields.
2. The antibracket is the odd symplectic pairing between fields and antifields.
3. The classical master equation constructs a nilpotent differential.
4. Gauge fixing is restriction to a Lagrangian submanifold determined by a gauge-fixing fermion.
5. The quantum master equation is the regulated half-density statement of measure-compatible gauge symmetry.
6. The 1PI and Wilsonian effective actions in continuum gauge theory are defined as BV objects satisfying the corresponding master identities; a lattice gauge formulation supplies the alternative finite-cutoff route by compact group integrals and gauge-invariant blocking maps.
7. Counterterms, anomalies, and physical operators are local BV cohomology classes at the appropriate ghost numbers.

This is the framework used below whenever gauge-theory effective actions are discussed beyond elementary Faddeev–Popov perturbation theory.

Chapter 49

QCD Renormalization, Asymptotic Freedom, and Deep Inelastic Scattering

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

49.1 QCD as a Renormalized Gauge Theory

Quantum chromodynamics in four dimensions is Yang–Mills theory with gauge group $SU(N_c)$ coupled to N_f Dirac fermions in the fundamental representation. Write the quark fields as

$$q^i_{I\alpha}(x), \qquad i = 1, \ldots, N_c, \quad I = 1, \ldots, N_f,$$

where i is color, I is flavor, and α is a spinor index. We use the Hermitian-generator normalization fixed in Chapter 45, specifically (45.1), (45.12), and (45.16),

$$\mathrm{tr}_\square(t^a t^b) = \delta^{ab}.$$

The group-theoretic constants used in renormalization are defined by

$$f^a{}_{cd} f^b{}_{cd} = C_A \delta^{ab}, \qquad \mathrm{tr}_R(t^a_R t^b_R) = T_R \delta^{ab}, \qquad t^a_R t^a_R = C_R \mathbf{1}_R. \tag{49.1}$$

For $SU(N_c)$,

$$C_A = 2N_c, \qquad T_F = 1, \qquad C_F = \frac{N_c^2 - 1}{N_c}.$$

These are twice the values often quoted with generators T^a normalized by $\mathrm{tr}_\square(T^aT^b) = \frac{1}{2}\delta^{ab}$. The physical matrix-valued connection is unchanged; only the component coordinates and the coupling coordinate are rescaled. In the present convention the gauge action is $-\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$.

The renormalized coupling $g(\mu)$ depends on a scale μ and on the renormalization scheme. The beta function is

$$\beta(g) := \mu\frac{\mathrm{d}g}{\mathrm{d}\mu}.$$

The first coefficient is obtained below from the background-field 1PI effective action. For N_f Dirac fermions in a representation R, in a mass-independent scheme the result is

$$\beta(g) = -\frac{g^3}{16\pi^2}\left(\frac{11}{3}C_A - \frac{4}{3}T_RN_f\right) + O(g^5). \tag{49.2}$$

Equivalently,

$$\beta(g) = -\frac{g^3}{24\pi^2}\,(11N_c - 2N_f) + O(g^5) \tag{49.3}$$

for fundamental quarks in this trace normalization.

49.2 Ordinary Covariant-Gauge Green-Function Rules

The covariant gauge-fixed rules used in ordinary Yang–Mills perturbation theory are rules for time-ordered Green functions of the local variables $A^a_\mu, c^a, \bar{c}^a, q, \bar{q}$. Their physical interpretation is supplied by the BRST cohomology and by gauge-invariant operators, but the local rules themselves are needed for renormalization and for comparing different gauges.

In this section write g for g_{YM}, and use the normalization (45.16),

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g^2}F^a_{\mu\nu}F^{a\mu\nu}, \qquad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^a{}_{bc}A^b_\mu A^c_\nu.$$

This is the coupling-absorbed normalization of Chapter 45: $D_\mu = \partial_\mu - \mathrm{i}A_\mu$, and the coupling is the coefficient in front of the curvature term. In the Abelian specialization it is related to the explicit-coupling QED convention by (45.6). Thus the beta function extracted below from the coefficient of $F^a_{\mu\nu}F^{a\mu\nu}/g^2$ is written in Yang–Mills field coordinates; QED wavefunction renormalization constants in Chapter 43 use the explicit-coupling field coordinate and must first be translated by the same rescaling before being compared term by term. The Lorenz gauge condition is

$$F^a(A) = \partial^\mu A^a_\mu.$$

Under the infinitesimal gauge transformation

$$\delta_\zeta A^{a\mu} = \partial^\mu\zeta^a + f^a{}_{bc}A^{b\mu}\zeta^c,$$

the variation of $F^a(A)$ is

$$\delta_\zeta F^a(A) = \partial_\mu\left(\partial^\mu\zeta^a + f^a{}_{bc}A^{b\mu}\zeta^c\right).$$

Replacing the gauge parameter by the ghost c^a, and pairing the gauge condition with the antighost $\bar{c}^a$, gives the Faddeev–Popov action

$$S_{\mathrm{gh}} = -\int \mathrm{d}^4x\,\bar{c}^a\partial_\mu\left(\partial^\mu c^a + f^a{}_{bc}A^{b\mu}c^c\right). \tag{49.4}$$

Equivalently, after integrating by parts and discarding the boundary term,

$$S_{\text{gh}} = \int \mathrm{d}^4x\, \partial_\mu \bar{c}^a \left(\partial^\mu c^a + f^a{}_{bc} A^{b\mu} c^c \right).$$

The operator is the massless scalar kinetic operator, while the fields are Grassmann odd. With the momentum convention displayed in the figure below,

$$\langle \Omega | \, \mathrm{T}\, \widehat{\bar{c}}^a(x) \widehat{c}_b(y) | \Omega \rangle = \delta^a{}_b \int \frac{\mathrm{d}^4k}{(2\pi)^4} \mathrm{e}^{\mathrm{i}k\cdot(x-y)} \frac{-\mathrm{i}}{k^2 - \mathrm{i}\epsilon}. \tag{49.5}$$

The extra sign of a closed ghost loop is a Berezin-Gaussian fact, not an independent rule. Let K denote the free ghost kinetic operator and let V denote any matrix of background insertions carrying the ghost-gluon vertices. In a finite-dimensional regulator, with the ordered Berezin convention used for Grassmann Gaussians in the spinor path-integral chapter,

$$\frac{Z_{\text{gh}}[V]}{Z_{\text{gh}}[0]} = \frac{\int [\mathrm{d}\bar{c}\, \mathrm{d}c]\, \exp[-\bar{c}(K+V)c]}{\int [\mathrm{d}\bar{c}\, \mathrm{d}c]\, \exp[-\bar{c}Kc]} = \det(1 + K^{-1}V).$$

Hence the connected n-insertion ghost cycle is the term

$$\operatorname{Tr} \log(1 + K^{-1}V) = \sum_{n\geq 1} \frac{(-1)^{n+1}}{n} \operatorname{Tr}\big((K^{-1}V)^n\big).$$

A commuting complex scalar with the same quadratic operator would instead produce $\det(1 + K^{-1}V)^{-1}$, and therefore the opposite sign for the same closed cycle after the local vertex factors are held fixed. Equivalently, in the Wick-contraction expansion, closing a ghost cycle requires a cyclic permutation of Grassmann-odd variables before the c's and $\bar{c}$'s are paired; this cyclic parity is the same closed-loop sign as for fermions. The Feynman rules below implement this determinant sign by assigning a factor -1 to each closed ghost loop.

The ghost-gluon vertex is read from the term

$$\int \mathrm{d}^4x\, \partial_\mu \bar{c}^a\, f^a{}_{bc} A^{b\mu} c^c.$$

If the momentum k is assigned to the antighost line as in figure 49.1, the displayed local vertex factor is

$$V_{\bar{c}^a c^c A^{b\mu}}(k) = -k_\mu f^a{}_{bc}. \tag{49.6}$$

Changing the direction of the ghost arrow or the sign convention for incoming momenta changes the displayed sign in the corresponding way; the invariant object is the action (49.4). There is no local $\bar{c}\, c\, A\, A$ vertex in this Lorenz gauge, because $\partial^\mu D_\mu(A)$ is only linear in A_μ.

The B-field form of the gauge-fixing term is

$$\int \mathrm{d}^4x \left(B^a \partial^\mu A^a_\mu + \frac{g^2\xi}{2} B^a B^a \right).$$

Eliminating B^a gives

$$S_{\text{gf}} = -\frac{1}{2g^2\xi} \int \mathrm{d}^4x\, (\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu). \tag{49.7}$$

The quadratic gauge-field action is then inverted by

$$\langle\Omega|\,\mathrm{T}\;\widehat{A}_{a\mu}(x)\widehat{A}_{b\nu}(y)|\Omega\rangle = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\mathrm{e}^{\mathrm{i}k\cdot(x-y)}g^2\frac{-\mathrm{i}\delta_{ab}}{k^2-\mathrm{i}\epsilon}\left[\eta_{\mu\nu}+(\xi-1)\frac{k_\mu k_\nu}{k^2}\right]. \tag{49.8}$$

The factor g^2 is a consequence of the $1/g^2$ multiplying the Yang–Mills kinetic term in the connection normalization used here.

Expanding $F^a_{\mu\nu}F^{a\mu\nu}$ gives the cubic and quartic gluon self-interactions:

$$\begin{aligned} S_{AAA} &= -\frac{1}{2g^2}\int \mathrm{d}^4x\, f^a{}_{bc}\big(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu\big)A^{b\mu}A^{c\nu}, \\ S_{AAAA} &= -\frac{1}{4g^2}\int \mathrm{d}^4x\, f^a{}_{bc}f^a{}_{de}A^{b\mu}A^{c\nu}A^d_\mu A^e_\nu. \end{aligned} \tag{49.9}$$

These displayed action terms specify the convention before any momentum-flow choice is made for a vertex tensor. Figure 49.1 records the corresponding propagator and vertex conventions in a notation that distinguishes quark, ghost, and gauge lines, and also distinguishes present from absent local vertices; the visual distinction is used below when the ghost-loop sign and the absent four-ghost vertex are discussed.

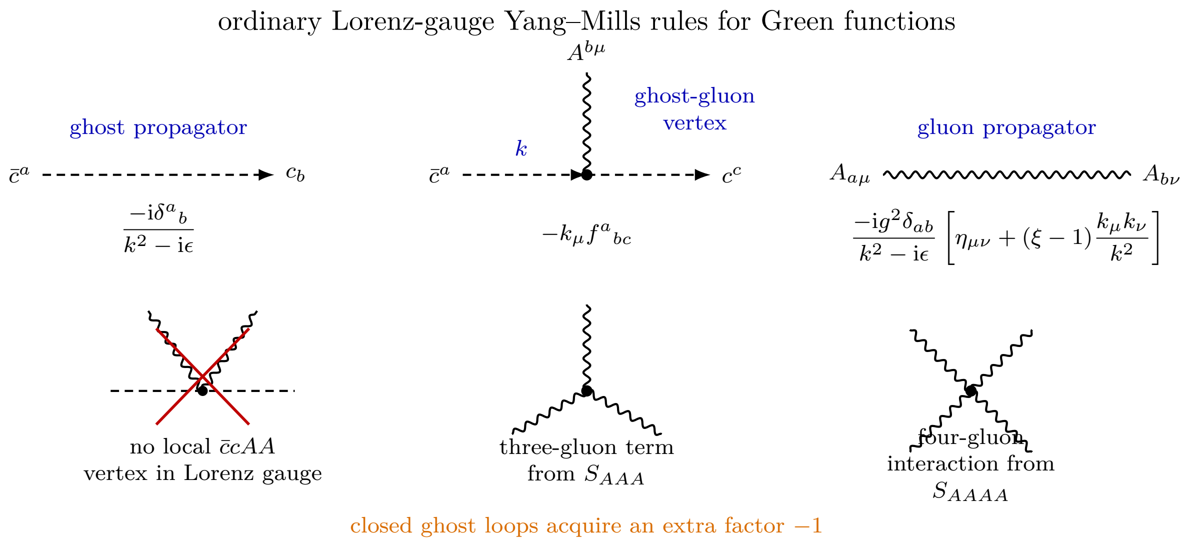


Figure 49.1: The ordinary covariant-gauge rules following from the Lorenz Faddeev–Popov action. Dashed arrows denote the ghost-number flow from antighost to ghost; wavy lines denote gauge fields. The rules organize gauge-fixed Green functions, while physical state spaces are described by BRST cohomology.

49.3 Background-Field Derivation of the One-Loop Beta Function

49.3.1 Background Gauge Datum and Effective Coupling Coordinate

The coupling in a nonabelian gauge theory is most cleanly extracted from the coefficient of the background curvature term in the 1PI effective action. Let $A_\mu = \widetilde{A}_\mu + a_\mu$, where $\widetilde{A}_\mu$ is a background connection and a_μ is the quantum fluctuation. The background covariant derivative in a representation R is denoted

$$\widetilde{D}_\mu = \partial_\mu - \mathrm{i}\widetilde{A}^a_\mu t^a_R, \qquad \widetilde{F}_{\mu\nu} = \partial_\mu \widetilde{A}_\nu - \partial_\nu \widetilde{A}_\mu - \mathrm{i}[\widetilde{A}_\mu, \widetilde{A}_\nu].$$

In the adjoint representation,

$$(\widetilde{D}_\mu X)^a = \partial_\mu X^a + f^a{}_{bc}\widetilde{A}^b_\mu X^c.$$

The structure constants and trace invariants in this calculation are those of (49.1); in particular, for fundamental quarks of $SU(N_c)$ in the monograph's trace normalization,

$$C_A = 2N_c, \qquad T_F = 1, \qquad C_F = \frac{N_c^2 - 1}{N_c}.$$

Choose the background-field gauge condition

$$\widetilde{D}^\mu a_\mu = 0.$$

Under a background gauge transformation with parameter $\zeta = \zeta^a t^a$,

$$\delta\widetilde{A}_\mu = \widetilde{D}_\mu \zeta, \qquad \delta a_\mu = -\mathrm{i}[a_\mu, \zeta],$$

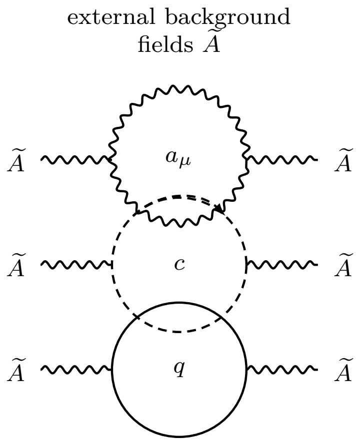
and the ghosts and quarks transform in the adjoint and R representations, respectively. Since this transformation acts linearly on the variables integrated over in the path integral and as a connection transformation on $\widetilde{A}$, the resulting 1PI functional $\Gamma[\widetilde{A}]$ is invariant under background gauge transformations. Its dimension-four term therefore has the form

$$\Gamma[\widetilde{A}] \supset -\frac{1}{4g^2(\mu)} \int \mathrm{d}^4x\, \widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu} \tag{49.10}$$

in Lorentzian signature, up to terms with more powers of $\widetilde{F}$, covariant derivatives, or quark fields. The problem is therefore reduced to computing one coefficient in a gauge-invariant functional, rather than separately renormalizing gauge-fixed vertices.

Figure 49.2 displays the actual coefficient extraction used below. The useful object is not a separate renormalization of each gauge-fixed vertex, but the determinant-weighted coefficient of $\int \widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}$ in the background effective action.

background-field one-loop coefficients



determinant weight	**coefficient in** b	**contribution to** $\beta_0 = 4b$
$-\frac{1}{2}\operatorname{Tr}\log D_A$ vector fluctuation	$+\dfrac{5}{6}C_A$	$+\dfrac{10}{3}C_A$
$+\operatorname{Tr}\log D_{\rm gh}$ Faddeev–Popov ghosts	$+\dfrac{1}{12}C_A$	$+\dfrac{1}{3}C_A$
$+N_f \operatorname{Tr}\log D_\psi$ Dirac quarks in R	$-\dfrac{1}{3}T_R N_f$	$-\dfrac{4}{3}T_R N_f$
sum	$b = \dfrac{11}{12}C_A - \dfrac{1}{3}T_R N_f$	$\beta_0 = \dfrac{11}{3}C_A - \dfrac{4}{3}T_R N_f$

$$\Gamma^{(1)}[\widetilde{A}] \supset \frac{b}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon}\int \mathrm{d}^D x\, \widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}, \text{ hence } \mu\frac{\mathrm{d}g}{\mathrm{d}\mu} = -\frac{g^3}{16\pi^2}\beta_0 + O(g^5).$$

Figure 49.2: The background-field method keeps the effective action invariant under gauge transformations of $\widetilde{A}$, so the one-loop running is read from the coefficient of $\widetilde{F}^2$. The table records the determinant weights and the coefficient b used in (49.23); the usual one-loop coefficient is $\beta_0 = 4b$.

Set the background quark and ghost fields to zero and use background Feynman gauge. The part of the action quadratic in $a_\mu, q, \bar{q}, c, \bar{c}$ has the form

$$S^{(2)} = \int \mathrm{d}^4 x \left[-\frac{1}{4g_0^2} a^\mu_a (D_A)_{a\mu,b\nu} a^\nu_b + \bar{q} D_\psi q + \bar{c}^a D^{ab}_{\rm gh} c^b \right], \tag{49.11}$$

where g_0 is the bare coupling. The operators are differential operators whose coefficients are built from $\widetilde{D}_\mu$ and $\widetilde{F}_{\mu\nu}$. In background Feynman gauge their principal pieces are

$$D_\psi = -\gamma^\mu \widetilde{D}_\mu, \qquad D_{\rm gh} = \widetilde{D}_\mu \widetilde{D}^\mu,$$

and D_A is the adjoint vector operator obtained from the quadratic part of $(\widetilde{F} + \widetilde{D}a - \widetilde{D}a + [a,a])^2$ together with the gauge-fixing term. The Gaussian integral gives

$$\mathrm{i}\Gamma^{(1)}[\widetilde{A}] = \log\left[\frac{(\det D_\psi)^{N_f}\det D_{\rm gh}}{(\det D_A)^{1/2}}\right]. \tag{49.12}$$

The numerator reflects Grassmann integration for quarks and ghosts; the denominator is the square root from the bosonic gauge-field Gaussian.

49.3.2 Constant-Background Extraction of the Logarithmic Coefficient

To extract the logarithmic coefficient, take $\widetilde{A}_\mu$ to be approximately constant inside a large box and to vanish smoothly outside. The box size supplies an infrared scale $L \sim \mu^{-1}$. In the constant region, the functional trace of a function of a differential operator reduces to an ordinary momentum integral. Figure 49.3 separates the position-space infrared regulator from the momentum-space large-$|k|$ tail whose logarithm gives the ultraviolet pole:

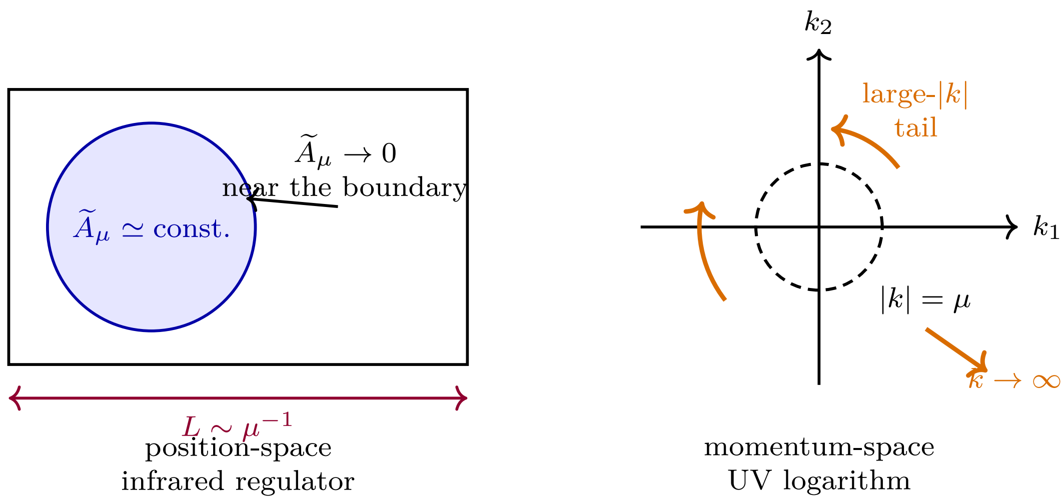


Figure 49.3: The constant-background computation is a local extraction of the logarithmic coefficient. The background is chosen nearly constant in a region of size L, and the scale $\mu \sim L^{-1}$ is used as the infrared cutoff in the momentum integral; the ultraviolet pole is the large-momentum tail of the same local trace.

$$\operatorname{Tr} f(D) = \int \mathrm{d}^D x \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, \operatorname{tr} f(M(k)). \tag{49.13}$$

Here Tr includes integration over spacetime and trace over all indices, while tr is the finite-dimensional trace over spin, Lorentz, and color indices of the symbol $M(k)$.

The coefficient can be extracted either by expanding the finite-dimensional symbols in (49.13) or, equivalently, by computing the logarithmic heat-kernel coefficient of the same local operators. The equivalence follows because both calculations isolate the parity-even curvature-squared functional that renormalizes $1/g^2$. In the symbol expansion, the only tensor average needed at fourth order is

$$\frac{k^{\nu_1}k^{\nu_2}k^{\nu_3}k^{\nu_4}}{(k^2)^4} \longrightarrow \frac{1}{(k^2)^2 D(D+2)}\left(\delta^{\nu_1\nu_2}\delta^{\nu_3\nu_4} + \delta^{\nu_1\nu_3}\delta^{\nu_2\nu_4} + \delta^{\nu_1\nu_4}\delta^{\nu_2\nu_3}\right). \tag{49.14}$$

The scalar integral carrying the logarithmic divergence is

$$I_\epsilon(\mu) := \int_{|k|>\mu} \frac{\mathrm{d}^{4-\epsilon}k}{(2\pi)^{4-\epsilon}} \frac{1}{(k^2)^2} = \frac{1}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon} + \text{finite}. \tag{49.15}$$

49.3.3 Heat-Kernel Determinant Calculation

For a self-contained evaluation, use the Euclidean logarithmic coefficient of a Laplace-type operator

$$\mathcal{L} = -\nabla^2 + E$$

on a vector bundle $\mathcal{E}$. Let $\Omega_{\mu\nu} := [\nabla_\mu, \nabla_\nu]$. In flat space,

$$\operatorname{Tr}\log\mathcal{L} \supset -I_\epsilon(\mu)\int \mathrm{d}^D x\ \operatorname{tr}_{\mathcal{E}}\left(\frac{1}{2}E^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\right). \tag{49.16}$$

This is obtained from $\operatorname{Tr}\log\mathcal{L} = -\int_0^\infty \frac{\mathrm{d}t}{t}\operatorname{Tr}\mathrm{e}^{-t\mathcal{L}}$ and the flat-space heat kernel

$$\operatorname{Tr}\mathrm{e}^{-t\mathcal{L}} \sim (4\pi t)^{-D/2}\int \mathrm{d}^D x\ \operatorname{tr}_{\mathcal{E}}\left[1 + tE + t^2\left(\frac{1}{2}E^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\right) + \cdots\right].$$

The pole in dimensional regularization is the same pole as (49.15).

Let t^a_{ad} denote the Hermitian adjoint-representation generators, defined by

$$(t^a_{\mathrm{ad}})^b{}_c = \mathrm{i}f^b{}_{ac}, \qquad \operatorname{tr}_{\mathrm{ad}}(t^a_{\mathrm{ad}}t^b_{\mathrm{ad}}) = C_A\delta^{ab}.$$

The curvature endomorphism of the adjoint covariant derivative is the anti-Hermitian operator

$$\operatorname{ad}\widetilde{F}_{\mu\nu} := -\mathrm{i}\widetilde{F}^a_{\mu\nu}t^a_{\mathrm{ad}}.$$

Thus the adjoint trace used in the heat-kernel coefficient satisfies

$$\operatorname{tr}_{\mathrm{ad}}\big(\operatorname{ad}\widetilde{F}_{\mu\nu}\ \operatorname{ad}\widetilde{F}^{\mu\nu}\big) = (-\mathrm{i})^2\widetilde{F}^a_{\mu\nu}\widetilde{F}^{b\mu\nu}\operatorname{tr}_{\mathrm{ad}}(t^a_{\mathrm{ad}}t^b_{\mathrm{ad}}) = -C_A\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}, \tag{49.17}$$

where the sign comes from the two explicit factors of $-\mathrm{i}$. For a matter representation R with Hermitian generators t^a_R,

$$\operatorname{tr}_R\big((-\mathrm{i}\widetilde{F}^a_{\mu\nu}t^a_R)(-\mathrm{i}\widetilde{F}^{b\mu\nu}t^b_R)\big) = -T_R\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}.$$

First consider one Dirac quark in R. The first-order operator is $\mathcal{D}_\psi := \gamma^\mu\widetilde{D}_\mu$, and the logarithmic part of $\operatorname{Tr}\log\mathcal{D}_\psi$ is one half of that of the Laplace-type operator $-\mathcal{D}_\psi^2$. With

$$-\mathcal{D}_\psi^2 = -\widetilde{D}^2 - \frac{1}{2}\gamma^{\mu\nu}\Omega_{\mu\nu}, \qquad \gamma^{\mu\nu} := \frac{1}{2}[\gamma^\mu, \gamma^\nu],$$

one has

$$E_\psi = -\frac{1}{2}\gamma^{\mu\nu}\Omega_{\mu\nu}.$$

Using

$$\operatorname{tr}_{\mathrm{spin}}\gamma^{\mu\nu}\gamma^{\rho\sigma} = 4\left(\delta^{\mu\sigma}\delta^{\nu\rho} - \delta^{\mu\rho}\delta^{\nu\sigma}\right),$$

the heat-kernel density is

$$\operatorname{tr}_{\mathrm{spin}\otimes\mathrm{R}}\left(\frac{1}{2}E_\psi^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\right) = \frac{2}{3}T_R\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}.$$

The extra factor $1/2$ relating $\operatorname{Tr}\log\mathcal{D}_\psi$ to $\operatorname{Tr}\log(-\mathcal{D}_\psi^2)$ gives

$$\operatorname{Tr}\log D_\psi \supset -\frac{1}{3}T_R I_\epsilon(\mu)\int \mathrm{d}^D x\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu} \tag{49.18}$$

for one Dirac flavor.

For the Faddeev–Popov ghosts in background gauge, the relevant Laplace-type operator is the scalar adjoint operator $\mathcal{L}_{\rm gh} = -\widetilde{D}^2$. Hence $E_{\rm gh} = 0$, and (49.16) gives

$$\operatorname{Tr}\log D_{\rm gh} \supset \frac{1}{12} C_A I_\epsilon(\mu) \int \mathrm{d}^D x\, \widetilde{F}^a_{\mu\nu} \widetilde{F}^{a\mu\nu}. \tag{49.19}$$

For the gauge fluctuation in background Feynman gauge, the Euclidean vector operator has the form

$$(\mathcal{L}_A)_\mu{}^\nu = -\widetilde{D}^2 \delta_\mu{}^\nu - 2\,\operatorname{ad}(\widetilde{F}_\mu{}^\nu).$$

It is a Laplace-type operator on one-forms valued in the adjoint bundle with

$$(E_A)_\mu{}^\nu = -2\,\operatorname{ad}(\widetilde{F}_\mu{}^\nu), \qquad (\Omega_{\rho\sigma})_\mu{}^\nu = \delta_\mu{}^\nu\,\operatorname{ad}(\widetilde{F}_{\rho\sigma}).$$

The finite-dimensional trace is taken in $D = 4 - \epsilon$ one-form components and in the adjoint representation. Using (49.17) and the antisymmetry of $\widetilde{F}_{\mu\nu}$,

$$\operatorname{tr}\frac{1}{2}E_A^2 = \frac{1}{2}\operatorname{tr}_{\rm ad}\left[(-2\,\operatorname{ad}\widetilde{F}_\mu{}^\nu)(-2\,\operatorname{ad}\widetilde{F}_\nu{}^\mu)\right] = 2C_A\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}, \tag{49.20}$$

$$\operatorname{tr}\frac{1}{12}\Omega_{\rho\sigma}\Omega^{\rho\sigma} = \frac{D}{12}\operatorname{tr}_{\rm ad}\big(\operatorname{ad}\widetilde{F}_{\rho\sigma}\,\operatorname{ad}\widetilde{F}^{\rho\sigma}\big) = -\frac{D}{12}C_A\,\widetilde{F}^a_{\rho\sigma}\widetilde{F}^{a\rho\sigma}. \tag{49.21}$$

The D-dependence in (49.21) has an evanescent part. Write

$$\mu^{-\epsilon} = 1 - \epsilon L_\mu + O(\epsilon^2),$$

where L_μ is the finite logarithm determined by the subtraction convention in (49.15). Then

$$(2-D)\frac{\mu^{-\epsilon}}{\epsilon} = (-2+\epsilon)\frac{\mu^{-\epsilon}}{\epsilon} = -\frac{2}{\epsilon} + 1 + 2L_\mu + O(\epsilon), \qquad D\frac{\mu^{-\epsilon}}{\epsilon} = \frac{4}{\epsilon} - 1 - 4L_\mu + O(\epsilon).$$

Consequently, replacing D by 4 is legitimate for the pole coefficient; the omitted $(D-4)I_\epsilon(\mu)$ part is a finite local contribution. Equations (49.20) and (49.21) therefore give the pole density

$$\left(2 - \frac{4}{12}\right) C_A\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu} = \frac{5}{3}C_A\,\widetilde{F}^a_{\mu\nu}\widetilde{F}^{a\mu\nu}.$$

The vector determinant contribution is therefore

$$\operatorname{Tr}\log D_A \supset -\frac{5}{3} C_A I_\epsilon(\mu) \int \mathrm{d}^D x\, \widetilde{F}^a_{\mu\nu} \widetilde{F}^{a\mu\nu}. \tag{49.22}$$

49.3.4 One-Loop Coupling Flow

Combining (49.18), (49.19), and (49.22) with the determinant weights in (49.12) gives the coefficient

$$b = -\frac{N_f}{3}T_R + \left[-\frac{1}{2}\left(-\frac{5}{3}\right) + \frac{1}{12}\right] C_A = \frac{11}{12}C_A - \frac{N_f}{3}T_R. \tag{49.23}$$

The term $5C_A/6$ comes from the vector determinant after the bosonic factor $-1/2$, the term $C_A/12$ from the Faddeev–Popov determinant, and the term $-N_f T_R/3$ from the N_f quark determinants.

Comparing the logarithmic part of $\Gamma[\widetilde{A}]$ with (49.10) gives

$$-\frac{1}{4g^2(\mu)} = -\frac{1}{4g_0^2} + \frac{b}{8\pi^2}\frac{\mu^{-\epsilon}}{\epsilon} + \text{finite}. \tag{49.24}$$

In $D = 4 - \epsilon$, write the dimensionless coupling as

$$\lambda^2(\mu) = \mu^\epsilon g^2(\mu).$$

This is the gauge-theory analogue of the scalar convention $g^\epsilon = \mu^\epsilon[\lambda + \cdots]$: the square appears because the Yang–Mills kinetic term is parameterized by $1/g^2$. Thus λ is the dimensionless D-dimensional coupling coordinate, while $g(\mu)$ retains the engineering dimension $-\epsilon/2$. Equivalently,

$$\frac{1}{g_0^2} = \mu^\epsilon \left[\frac{1}{\lambda^2(\mu)} + \frac{b}{2\pi^2}\frac{1}{\epsilon} + \text{finite} + O(\lambda^2)\right]. \tag{49.25}$$

The bare coupling g_0 is independent of the auxiliary scale μ. Differentiating (49.25) at fixed g_0 and then taking $\epsilon \to 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu}\frac{1}{\lambda^2(\mu)} = \frac{b}{2\pi^2} + O(\lambda^2), \qquad \beta(\lambda) = -\frac{b}{4\pi^2}\lambda^3 + O(\lambda^5). \tag{49.26}$$

Setting $\epsilon = 0$ and writing g for the four-dimensional dimensionless coupling,

$$\frac{b}{4\pi^2} = \frac{1}{16\pi^2}\left(\frac{11}{3}C_A - \frac{4}{3}T_R N_f\right),$$

which is (49.2). For fundamental quarks of $SU(N_c)$, $T_R = T_F = 1$ and $C_A = 2N_c$, giving (49.3).

49.4 Dimensional Transmutation

Assume

$$b_0 := 11N_c - 2N_f > 0.$$

At one loop,

$$\mu\frac{\mathrm{d}g}{\mathrm{d}\mu} = -\frac{b_0}{24\pi^2}g^3.$$

Integrating between μ_0 and μ gives

$$\frac{1}{g^2(\mu)} = \frac{1}{g^2(\mu_0)} + \frac{b_0}{12\pi^2}\log\frac{\mu}{\mu_0}. \tag{49.27}$$

Thus $g(\mu)$ decreases at large μ. This property is asymptotic freedom.

The integration constant may be traded for a scale. At one-loop accuracy,

$$\Lambda = \mu\,\exp\left\{-\frac{12\pi^2}{b_0 g^2(\mu)}\right\}. \tag{49.28}$$

Higher-loop definitions multiply the right side by powers and series in $g(\mu)$, and the numerical value depends on the renormalization scheme. The scheme-independent statement is that a dimensionless renormalized coupling has been replaced, after a renormalization condition is chosen, by a dimensionful RG-invariant scale, together with the quark masses. This is dimensional transmutation.

Perturbative predictions at external momentum scale Q are organized by the small parameter $g(Q)$ when $Q \gg \Lambda$. At scales comparable to Λ, the perturbative expansion is no longer a controlled expansion in small coupling. Statements about the spectrum, confinement, and the mass gap then require nonperturbative input.

Remark 49.1 (Mass-gap problem and the status of continuum Yang–Mills)*.* For pure compact $SU(N_c)$ Yang–Mills theory in four dimensions, the existence of the continuum quantum field theory on $\mathbb{R}^4$ and the existence of a positive mass gap form the Yang–Mills Millennium problem; see Remark 45.2. The issue is not the formal writing of the action or the perturbative construction of renormalized Green functions. It is the construction of a nonperturbative continuum measure, or equivalently a Hilbert-space or algebraic quantum theory, whose correlation functions satisfy the required structural properties and whose physical spectrum above the vacuum has a strictly positive lower bound.

The finite-lattice theory at nonzero lattice spacing is a well-defined statistical-mechanical system, and numerical as well as analytic evidence strongly supports a gapped pure Yang–Mills continuum limit. The proof of that continuum limit and its gap in $D = 4$ remains a separate mathematical problem. The QCD discussion below therefore treats confinement, flux tubes, and hadronic spectral statements as nonperturbative physical inputs or as properties of regulated/lattice formulations, not as consequences of the one-loop beta function.

49.5 Regulated Gauge-Invariant Matrix Elements

The basic nonperturbative quantities of QCD are not matrix elements of isolated colored fields. They are limits, when such limits exist, of regulated gauge-invariant matrix elements. A calculation may use gauge-fixed quark and gluon variables internally, but the object being defined must be a color-singlet operator, a line operator, or a BRST-cohomology class with a specified regulator and renormalization prescription.

Definition 49.2 (Regulated gauge-invariant QCD matrix-element specification)**.** A regulated gauge-invariant QCD matrix-element specification consists of:

1. a regulator family $\mathcal{R}_\epsilon$, such as a finite lattice, a gauge-covariant heat-kernel cutoff, or a gauge-fixed BV regulator together with its quantum master equation;
2. a regulated operator $\mathcal{O}_\epsilon$ built from local gauge-invariant fields, closed Wilson loops, or colored fields whose exposed color indices are joined by Wilson transporters along specified paths;
3. the external states or density matrix in which the operator is evaluated, for example hadron states, the vacuum, a thermal state, or a finite-volume transfer-matrix state;
4. ultraviolet, endpoint, cusp, and rapidity subtractions whenever the operator contains coincident fields, cusped Wilson lines, infinite Wilson lines, or lightlike separations;
5. a topology in which the renormalized matrix elements are required to converge as $\epsilon \to 0$, usually a distribution topology in spacetime, momentum fraction, or detector angle.

Only after these data have been supplied does a symbol such as a Wilson loop, PDF, fragmentation function, soft function, quasi-PDF, static potential, or energy-flow correlator denote a nonperturbative QCD quantity. If the continuum limit or lightcone limit is not known to exist, the same formula is a regulated definition plus a continuum-existence hypothesis, not a theorem.

The Wilson transporter is the mechanism by which separated colored insertions become gauge-covariant before their endpoint indices are contracted. For example,

$$\bar{q}(x)W_{\square}(x,y)q(y), \qquad F^a_{\mu\nu}(x)W^{ab}_{\mathrm{ad}}(x,y)F^b_{\rho\sigma}(y),$$

are gauge-invariant bilocals after the spinor and Lorentz indices are contracted as specified. Closed Wilson loops are obtained by taking the color trace of a transporter around a closed curve. Baryonic static-source operators join N_c fundamental Wilson lines at a color-antisymmetric junction. Soft functions and parton distributions use Wilson lines whose directions are fixed by the high-energy kinematics; their lightlike or cusped nature is a source of additional renormalization, not an optional convention.

This rule is not limited to perturbation theory. On a lattice, the transporter is the ordered product of link variables and gauge invariance is exact at nonzero lattice spacing. In a continuum path-integral or Hilbert-space construction, the same Wilson-line operator must be obtained as a renormalized composite. Perturbative Feynman graphs then compute short-distance coefficients, anomalous dimensions, and matching kernels for these nonperturbative matrix elements. They do not replace the matrix elements by colored asymptotic particles. Energy correlators are a useful complementary case: the detector operator is built from the gauge-invariant stress tensor, so no Wilson line is needed in the operator definition, although perturbative representatives of the same observable still involve colored internal coordinates after factorization.

49.6 Infrared Possibilities and Wilson Loop Diagnostics

The one-loop beta function determines the ultraviolet direction of the flow when $B_0 > 0$. It does not by itself determine the infrared theory. The first controlled alternative to strong infrared coupling is visible only after the two-loop term is kept. Put

$$a(\mu) := \frac{g(\mu)^2}{16\pi^2}.$$

In a mass-independent gauge-invariant renormalization scheme, the two-loop beta function has the form

$$\beta_a(a) := \mu\frac{\mathrm{d}a}{\mathrm{d}\mu} = -2B_0a^2 - 2B_1a^3 + O(a^4), \tag{49.29}$$

where the one-loop coefficient is the one derived above,

$$B_0 = \frac{11}{3}C_A - \frac{4}{3}T_RN_f,$$

and the two-loop coefficient is

$$B_1 = \frac{34}{3}C_A^2 - \left(\frac{20}{3}C_AT_R + 4C_RT_R\right)N_f. \tag{49.30}$$

The constants C_A, T_R, C_R are those of (49.1); in particular no hidden $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ convention is being used. The coefficient B_1 is the universal two-loop coefficient of the mass-independent gauge beta function. The one-loop derivation in the preceding section fixes B_0 in the present normalization; the present subsection uses the two-loop coefficient to derive the fixed point and its domain of validity.

The universality of B_0 and B_1 is elementary at the level of coupling coordinates. Let a' be another dimensionless coupling coordinate with

$$a' = a + r_1 a^2 + r_2 a^3 + O(a^4).$$

Then

$$\beta_{a'}(a') = \frac{\mathrm{d}a'}{\mathrm{d}a}\,\beta_a(a) = -2B_0 a'^2 - 2B_1 a'^3 + O(a'^4).$$

Thus a small two-loop zero is not an artifact of a quadratic coordinate change. Its precise location beyond the first nontrivial order is scheme-dependent, because $O(a^4)$ terms affect the next coefficient in the small-B_0 expansion.

For fundamental quarks in the trace convention of this monograph,

$$C_A = 2N_c, \qquad T_F = 1, \qquad C_F = \frac{N_c^2 - 1}{N_c}.$$

Hence

$$B_0 = \frac{2}{3}(11N_c - 2N_f), \qquad B_1 = \frac{136}{3}N_c^2 - \left(\frac{52}{3}N_c - \frac{4}{N_c}\right)N_f. \tag{49.31}$$

Asymptotic freedom requires $B_0 > 0$, while a perturbative infrared zero requires $B_1 < 0$. These inequalities give the weakly coupled Banks–Zaks window

$$\frac{34N_c^3}{13N_c^2 - 3} < N_f < \frac{11}{2}N_c. \tag{49.32}$$

inside the larger asymptotically free range $N_f < 11N_c/2$. More precisely, define the distance from the upper edge by

$$\epsilon_{\mathrm{BZ}} := \frac{11N_c - 2N_f}{3}, \qquad N_f = \frac{11}{2}N_c - \frac{3}{2}\epsilon_{\mathrm{BZ}}. \tag{49.33}$$

Then $B_0 = 2\epsilon_{\mathrm{BZ}}$, and

$$B_1 = -2(25N_c^2 - 11) + \left(26N_c - \frac{6}{N_c}\right)\epsilon_{\mathrm{BZ}}.$$

The two-loop zero is

$$a_*^{(2)} = -\frac{B_0}{B_1} = \frac{2\epsilon_{\mathrm{BZ}}}{2(25N_c^2 - 11) - (26N_c - 6/N_c)\epsilon_{\mathrm{BZ}}}, \tag{49.34}$$

and the fixed-point coordinate of the full perturbative beta function is

$$a_* = \frac{\epsilon_{\mathrm{BZ}}}{25N_c^2 - 11} + O(\epsilon_{\mathrm{BZ}}^2). \tag{49.35}$$

The $O(\epsilon_{\mathrm{BZ}}^2)$ coefficient is not determined by (49.29) alone, because it receives a contribution from the three-loop coefficient. The leading coefficient in (49.35), by contrast, is scheme-independent.

The linearized exponent of the coupling perturbation is obtained without an extra assumption:

$$\omega_{\mathrm{BZ}} := \left.\frac{\partial \beta_a}{\partial a}\right|_{a=a_*} = -\frac{2B_0^2}{B_1} + O(B_0^3) = \frac{4\epsilon_{\mathrm{BZ}}^2}{25N_c^2 - 11} + O(\epsilon_{\mathrm{BZ}}^3). \tag{49.36}$$

With the convention $\beta_a = \mathrm{d}a/\mathrm{d}\log\mu$, the positive sign means that the perturbation $a - a_*$ decays when the flow is followed toward the infrared. The massless theory is essential: quark mass terms are relevant deformations of the fixed point and must be tuned to zero. At fixed finite N_c, N_f is an integer, so ϵ_{BZ} is not continuously adjustable in the literal theory. The perturbative expansion may be viewed as an analytic expansion in the fermion-determinant power N_f, or more physically in the Veneziano regime with N_f/N_c just below $11/2$, where the 't Hooft coupling $N_c a_*$ is small.

The fixed points obtained in this way are the Banks–Zaks fixed points. The statement is perturbative: it constructs a weak-coupling zero of the renormalized massless beta function and gives perturbative scaling data for gauge-invariant local operators. It is not a derivation of confinement or of the low-flavor infrared behavior of QCD.

Controlled Approximation 49.3 (Banks–Zaks perturbative fixed point versus a conformal-window theorem)**.** The Banks–Zaks construction used here is the following perturbative construction, not a theorem that every theory with flavor number in (49.32) is a nonperturbative conformal field theory. One fixes a mass-independent gauge-invariant renormalization scheme, restricts to the massless theory, treats N_f near the upper edge of asymptotic freedom so that $a_* = O(\epsilon_{\mathrm{BZ}})$, and computes renormalized correlators of gauge-invariant local or light-ray operators as formal asymptotic series in the small fixed-point coordinate. The first two beta coefficients determine the leading location (49.35) and the leading irrelevant exponent (49.36); higher coefficients and finite operator normalizations are scheme coordinates. If a nonperturbative regulator and scaling limit are supplied and the RG trajectory remains in the same small-coupling neighborhood, these perturbative data are the local coordinates of the resulting infrared theory. Without that extra construction they are a controlled perturbative fixed-point expansion. The lower edge of the nonperturbative conformal window, if it exists for the chosen gauge group and matter content, is not determined by (49.29).

Figure 49.4 compares the two beta-function shapes relevant here: asymptotic freedom without a weak infrared zero and the Banks–Zaks case with a perturbative infrared fixed point.

two infrared possibilities visible in a two-loop beta function

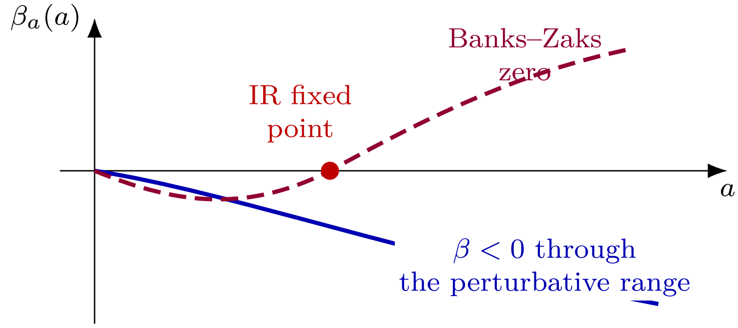


Figure 49.4: A negative one-loop coefficient implies ultraviolet asymptotic freedom. The infrared behavior depends on the subsequent terms. The solid curve has no weak infrared zero, while the dashed curve has a small two-loop zero giving a controlled perturbative fixed point; absence of such a zero leaves the flow outside perturbative control in the infrared.

For smaller N_f, perturbation theory around the ultraviolet fixed point no longer determines the infrared limit. The confining regime discussed below is therefore a conditional infrared scenario, not a conclusion derived from the one-loop beta function alone. The condition called confinement here is a spectral and observable hypothesis: isolated finite-energy asymptotic particles carry no color, and physical states are created from the vacuum by gauge-invariant operators. Deriving this spectral statement from the four-dimensional continuum QCD Lagrangian is an open nonperturbative problem. Quark and gluon fields remain part of the short-distance gauge-fixed description; the confinement statement concerns the gauge-invariant Hilbert-space spectrum and the algebra of physical observables.

The basic nonlocal observable for this confinement diagnostic is the Wilson line. For an oriented path C from x_1 to x_2, define

$$W_R(C) = P \exp\left\{ \mathrm{i} \int_C A^a_\mu(x) t^a_R \, \mathrm{d}x^\mu \right\}, \tag{49.37}$$

where R is a representation of the gauge group, t^a_R are its generators, and P denotes ordering along the path. The nonlocal operator

$$\bar{q}(x_2)\, W_R(C)\, q(x_1)$$

is gauge invariant when the endpoints transform in conjugate representations. It creates a color-neutral state containing a quark, an antiquark, and the gauge flux connecting them.

Figure 49.5 fixes the Wilson-line convention used for gauge-invariant quark bilinears and for closed Wilson-loop observables: path ordering is along the chosen curve, and endpoint representations are conjugate.

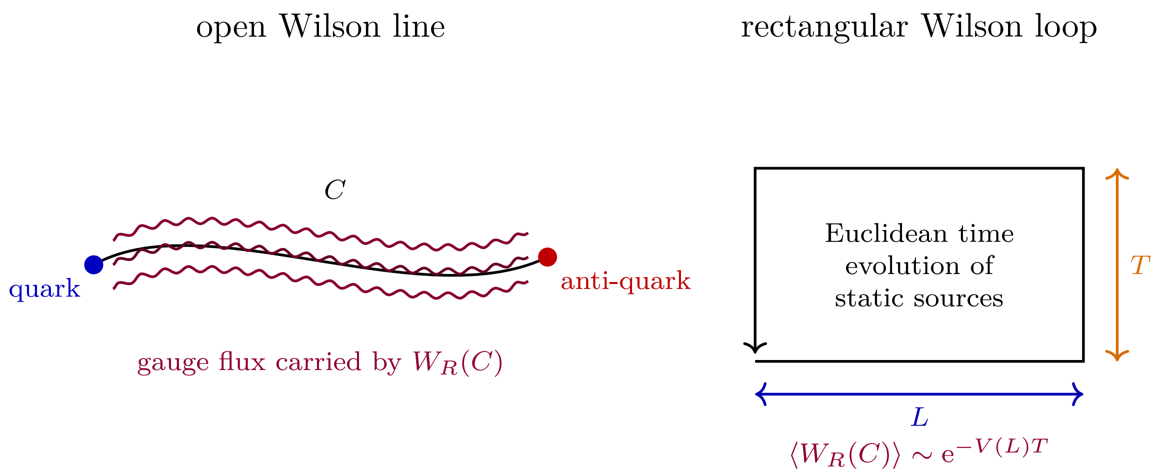


Figure 49.5: An open Wilson line attached to endpoint fields gives a gauge-invariant operator with color flux between the endpoints. A large Euclidean rectangular Wilson loop extracts the static potential between external color sources.

In pure Yang–Mills theory, where there are no dynamical fundamental quarks, a fundamental flux string cannot break by pair creation of fundamental matter. It carries a charge under the center one-form symmetry. For a large Euclidean rectangle $C_{L,T}$ of spatial width L and time height T,

$$\langle W_R(C_{L,T})\rangle \sim \mathrm{e}^{-V_R(L)T} \qquad (T\to\infty). \tag{49.38}$$

An area law

$$\langle W_R(C_{L,T})\rangle \sim \mathrm{e}^{-\gamma LT}$$

is equivalent, in this static-source setup, to a linearly growing potential

$$V_R(L) \sim \gamma L.$$

Chapter 46 gives the finite-cutoff version of this statement: at sufficiently small lattice β, the character expansion of the Wilson action converges and gives an area bound for fundamental Wilson loops. That theorem belongs to the strong-coupling lattice regime. A proof of the corresponding continuum four-dimensional pure Yang–Mills area law would require control of the $\beta\to\infty$ scaling trajectory and is part of the same nonperturbative construction problem as the mass gap. In QCD with dynamical fundamental quarks, the same Wilson loop is no longer a sharp order parameter for confinement because the flux tube can break by quark-antiquark pair creation. The physical spectral statement remains that asymptotic particles are color singlets.

49.7 The ’t Hooft Large-N_c Expansion

49.7.1 Planar Counting and Factorization

The large-N_c expansion is a statement about a sequence of gauge theories, not about a single theory at $N_c = 3$. The sequence used here is $SU(N_c)$ Yang–Mills theory with the same trace convention as above,

$$\mathrm{tr}_{\Box}(t^a t^b) = \delta^{ab}, \qquad C_A = 2N_c, \qquad T_F = 1,$$

and with

$$\lambda := g^2 N_c \tag{49.39}$$

held fixed as $N_c \to \infty$. In the common half-trace convention $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, the corresponding coupling is $g_{\mathrm{ht}}^2 = 2g^2$ and the conventional 't Hooft coupling is $\lambda_{\mathrm{ht}} = g_{\mathrm{ht}}^2 N_c = 2\lambda$. The constant factor 2 has no effect on the N_c-power counting, but it matters when comparing formulae across conventions.

For the color-counting argument it is useful to write the gauge field as a traceless matrix,

$$A_\mu = A_\mu^a t^a, \qquad (A_\mu)^i{}_i = 0,$$

and to separate the color-index contractions from the spacetime integrals, spin tensors, gauge-fixing tensors, and momentum denominators. The fundamental finite-dimensional identity in the present trace normalization is

$$\sum_{a=1}^{N_c^2-1} (t^a)^i{}_j (t^a)^k{}_\ell = \delta^i{}_\ell \delta^k{}_j - \frac{1}{N_c}\delta^i{}_j \delta^k{}_\ell. \tag{49.40}$$

Here is the derivation, since this is where convention errors most often enter. On $\operatorname{End}(\mathbb{C}^{N_c})$ use the bilinear pairing $(X, Y) = \operatorname{tr}_\square(XY)$. The matrix units have completeness kernel $\delta^i{}_\ell \delta^k{}_j$, which is the identity operator on all matrices. The one-dimensional scalar subspace is spanned by $\mathbf{1}$, whose squared norm is N_c, so the projector onto the scalar subspace has kernel $N_c^{-1}\delta^i{}_j\delta^k{}_\ell$. Since $\{t^a\}$ is an orthonormal basis of the traceless subspace, summing $(t^a)^i{}_j(t^a)^k{}_\ell$ gives the identity projector minus the scalar projector, which is exactly (49.40). The first term is the $U(N_c)$ double-line propagator: one color line carries the fundamental index and the other carries the antifundamental index. The second term subtracts the trace part and is explicitly lower order in $1/N_c$. It is therefore included in the expansion as a subleading insertion; it cannot increase the leading power of N_c. Ghost fields in covariant gauges are adjoint-valued and have the same double-line color structure, with the Grassmann sign already accounted for by the determinant argument above. Quark propagators carry one fundamental color line.

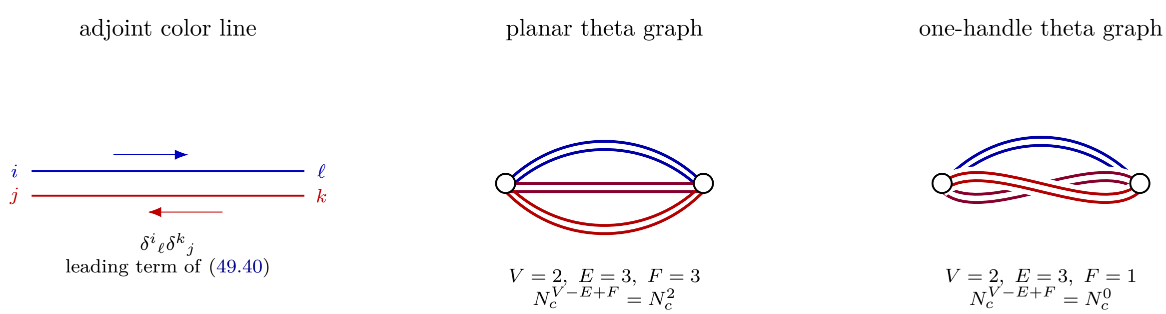


Figure 49.6: The adjoint propagator is represented by two oriented color lines. Thickening a double-line graph gives an oriented surface. The planar and one-handle theta graphs have the same number of vertices and propagators; the nonplanar routing has two fewer closed color faces and is therefore suppressed by N_c^{-2} at fixed $\lambda = g^2 N_c$.

The power counting is most transparent after the gauge-fixed adjoint quadratic form and all adjoint interaction monomials are written in the single-trace form

$$S_{\rm adj} = \frac{N_c}{\lambda} \int \mathrm{d}^4x \ \mathrm{tr}_\Box \left(\frac{1}{2}\Phi K \Phi + \sum_\alpha c_\alpha \, \Phi_{\alpha_1} \cdots \Phi_{\alpha_{r_\alpha}} \right), \tag{49.41}$$

where Φ denotes adjoint gauge, ghost, or auxiliary fields, K is the appropriate kinetic operator including spacetime and Lorentz indices, and the c_α are N_c-independent tensors once λ is held fixed. This rewriting is only a normalization of perturbative variables; it is not a new definition of QCD. It records the same fact already visible from

$$\frac{1}{4g^2}\,\mathrm{tr}_\Box\, F_{\mu\nu}F^{\mu\nu} = \frac{N_c}{4\lambda}\,\mathrm{tr}_\Box\, F_{\mu\nu}F^{\mu\nu}.$$

With the normalization (49.41), each adjoint propagator contributes a factor λ/N_c, each single-trace adjoint vertex contributes a factor N_c/λ, and each closed color face contributes a factor N_c from summing an index from 1 to N_c. Spacetime loop integrals and Lorentz numerators multiply this color factor but do not alter the topological exponent.

Let Γ be a connected adjoint ribbon graph with V single-trace vertices, E adjoint propagators, and F closed color faces. Thickening the ribbons and vertices produces a compact oriented surface Σ_Γ possibly with boundaries. If the surface has genus h and b boundaries, then

$$\chi(\Sigma_\Gamma) = V - E + F = 2 - 2h - b. \tag{49.42}$$

Therefore a fixed ribbon graph contributes

$$\mathcal{A}_\Gamma = N_c^{V-E+F}\lambda^{E-V}\,\mathcal{I}_\Gamma(\lambda,\mu,\{p\}) = N_c^{2-2h-b}\lambda^{E-V}\,\mathcal{I}_\Gamma(\lambda,\mu,\{p\}), \tag{49.43}$$

up to the explicitly subleading terms generated by the $-N_c^{-1}\delta^i{}_j\delta^k{}_\ell$ part of (49.40). The factor $\mathcal{I}_\Gamma$ contains the regulated momentum integrals, tensor contractions, signs, and scheme-dependent finite parts. Equation (49.43) is the topological content of the 't Hooft expansion: at fixed λ, changing the genus by one suppresses a connected contribution by N_c^{-2}.

The theta graph in figure 49.6 gives the elementary computation. Both routings have $V = 2$ cubic vertices and $E = 3$ propagators, hence both carry the same power $\lambda^{E-V} = \lambda$. The planar routing has $F = 3$, so

$$\mathcal{A}_{\rm planar} \sim N_c^2 \lambda\, \mathcal{I}_{\rm planar}.$$

The crossed routing thickens to a torus with $F = 1$, so

$$\mathcal{A}_{\text{one handle}} \sim N_c^0 \lambda\, \mathcal{I}_{\text{one handle}}.$$

For comparable regulated integrals at the same external data, the color topology alone gives

$$\frac{\mathcal{A}_{\text{one handle}}}{\mathcal{A}_{\rm planar}} = O(N_c^{-2}) \qquad (\lambda \text{ fixed}). \tag{49.44}$$

This is a finite color-index statement at each fixed ribbon graph. It does not by itself prove convergence of the perturbation series, nor does it define the four-dimensional continuum gauge theory nonperturbatively.

Single-trace operator insertions are counted as boundaries. For adjoint words

$$\mathcal{O}_r(x) = \mathrm{tr}_\Box\big(\Phi_{r_1}(x)\cdots\Phi_{r_{\ell_r}}(x)\big),$$

the connected genus-h contribution to an m-point function of unnormalized single traces scales as

$$\left\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_m(x_m)\right\rangle_{c,h} = O\left(N_c^{2-2h-m}\right), \tag{49.45}$$

again at fixed λ and fixed operator words. If $\widehat{\mathcal{O}}_r = N_c^{-1}\mathcal{O}_r$, then

$$\left\langle \widehat{\mathcal{O}}_1\cdots\widehat{\mathcal{O}}_m\right\rangle_{c,h} = O\left(N_c^{2-2h-2m}\right).$$

In particular,

$$\left\langle \widehat{\mathcal{O}}_1\widehat{\mathcal{O}}_2\right\rangle_c = O(N_c^{-2}), \qquad \langle\widehat{\mathcal{O}}_1\widehat{\mathcal{O}}_2\rangle = \langle\widehat{\mathcal{O}}_1\rangle\langle\widehat{\mathcal{O}}_2\rangle + O(N_c^{-2}). \tag{49.46}$$

This large-N_c factorization is often the most useful nonperturbative consequence of the counting: it says that fluctuations of normalized single-trace observables are suppressed in the planar limit, provided the large-N_c limit of the regulated theory exists.

49.7.2 Eguchi–Kawai Volume Reduction

Large-N_c factorization suggests a further possibility: in some center-symmetric large-N_c regimes, single-trace Wilson-loop expectation values on a large lattice may be obtained from a reduced matrix model with fewer sites. Double-line counting supplies factorization of normalized single traces; volume reduction additionally requires matching the finite-regulator loop equations in a center-symmetric sector.

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$ be a periodic hypercubic lattice, and let $U_\mu(x) \in SU(N_c)$ be the link from x to $x+\hat{\mu}$. The pure-gauge Wilson measure is

$$\mathrm{d}\nu_L(U) = Z_L^{-1}\exp\left\{bN_c\sum_{x\in\Lambda_L}\sum_{\mu<\nu}\operatorname{Re}\operatorname{Tr}_\Box U_{\mu\nu}(x)\right\}\prod_{x,\mu}\mathrm{d}U_\mu(x),$$

where

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x+\hat{\mu})U_\mu(x+\hat{\nu})^{-1}U_\nu(x)^{-1}.$$

The one-site Eguchi–Kawai reduced model has d matrices $V_\mu \in SU(N_c)$ and measure

$$\mathrm{d}\nu_{\mathrm{EK}}(V) = Z_{\mathrm{EK}}^{-1}\exp\left\{bN_c\sum_{\mu<\nu}\operatorname{Re}\operatorname{Tr}_\Box V_\mu V_\nu V_\mu^{-1}V_\nu^{-1}\right\}\prod_\mu \mathrm{d}V_\mu. \tag{49.47}$$

The parameter b is a lattice coupling coordinate. The present discussion uses it at fixed regulator; continuum statements require a separate scaling limit.

A path word in the reduced model is a sequence

$$p = ((\mu_1,\epsilon_1),\ldots,(\mu_\ell,\epsilon_\ell)), \qquad \epsilon_r = \pm 1,$$

with associated matrix

$$V(p) = V_{\mu_1}^{\epsilon_1}\cdots V_{\mu_\ell}^{\epsilon_\ell}$$

and displacement vector

$$n_\rho(p) = \sum_{r=1}^{\ell}\epsilon_r\,\delta_{\rho\mu_r}. \tag{49.48}$$

The normalized trace

$$W_{\mathrm{red}}(p) = \frac{1}{N_c}\,\mathrm{Tr}_{\Box}\,V(p)$$

is gauge invariant under simultaneous conjugation $V_\mu \mapsto \Omega V_\mu \Omega^{-1}$ even when $n(p) \neq 0$. This is precisely why the reduced model needs an additional center-selection rule: in the extended lattice theory, an open path is not a Wilson-loop observable, whereas its reduced trace would otherwise be an allowed matrix observable.

The reduced action is invariant under the independent center transformations

$$V_\mu \longmapsto z_\mu V_\mu, \qquad z_\mu \in Z(SU(N_c)) \simeq \mathbb{Z}_{N_c}. \tag{49.49}$$

Under this transformation,

$$W_{\mathrm{red}}(p) \longmapsto \left(\prod_{\rho=1}^{d} z_\rho^{\,n_\rho(p)}\right) W_{\mathrm{red}}(p). \tag{49.50}$$

Center selection for reduced open words. Assume the reduced finite-N_c state is invariant under (49.49). If the center charge of a word is nontrivial, namely if $n_\rho(p) \not\equiv 0 \pmod{N_c}$ for at least one ρ, then

$$\big\langle W_{\mathrm{red}}(p)\big\rangle_{\mathrm{EK}} = 0.$$

Consequently, if a charged reduced word has a nonzero expectation value in a chosen large-N_c state, that state cannot implement ordinary Eguchi–Kawai volume reduction for Wilson loops. Choose a direction ρ for which $n_\rho(p) \not\equiv 0 \pmod{N_c}$. There exists $z_\rho \in \mathbb{Z}_{N_c}$ such that $z_\rho^{\,n_\rho(p)} \neq 1$. Invariance of the state gives

$$\langle W_{\mathrm{red}}(p)\rangle_{\mathrm{EK}} = z_\rho^{n_\rho(p)} \langle W_{\mathrm{red}}(p)\rangle_{\mathrm{EK}},$$

where all other z_σ's are set to 1. The prefactor differs from one, hence the expectation value is zero. If a charged expectation is nonzero, the assumed center invariance is false. Such an expectation is an extra reduced-model observable with no counterpart as a closed Wilson loop in the unreduced lattice theory, so the reduced loop equations cannot be identified with the unreduced Wilson-loop equations without further modification.

The loop-equation form of volume reduction can now be stated cleanly.

Definition 49.4 (Volume-reduction structure for Wilson loops)**.** A volume-reduction structure for pure large-N_c lattice Yang–Mills Wilson loops consists of the following assumptions and regulator objects:

1. a sequence of finite-volume Wilson lattice measures with fixed lattice coupling coordinate b, and a sequence of reduced measures such as (49.47) or a twisted/quenched variant;

2. a class $\mathcal{C}$ of closed lattice paths whose normalized Wilson-loop expectations have large-N_c limits;

3. translation invariance of the large-volume loop expectations and large-N_c factorization of normalized single-trace loop observables;

4. unbroken reduced center symmetry strong enough to enforce the vanishing of all reduced open words that appear in the loop equations for $\mathcal{C}$;

5. uniqueness, in the selected center-symmetric sector, of the solution to the limiting Schwinger–Dyson loop equations with the specified boundary and reflection-positivity data.

When these assumptions and regulator objects are supplied, volume reduction means

$$\lim_{N_c\to\infty}\lim_{L\to\infty}\left\langle \frac{1}{N_c}\operatorname{Tr}U(C)\right\rangle_L = \lim_{N_c\to\infty}\left\langle \frac{1}{N_c}\operatorname{Tr}V(C)\right\rangle_{\text{red}} \tag{49.51}$$

for every $C \in \mathcal{C}$, where $V(C)$ is obtained from the sequence of directions in the lattice path C. The equality concerns single-trace Wilson-loop coordinates at fixed regulator. Colored local fields, baryons, finite-N_c spectra, and continuum limits require additional definitions and additional limiting statements.

The content of this structure is not decorative. In a Schwinger–Dyson equation for a Wilson loop, varying a link on the loop produces terms in which the loop splits at repeated visits to the varied link. In the unreduced theory these split loops may be based at different lattice sites; translation invariance and large-N_c factorization replace their joint expectation by the product of one-loop expectations. In the reduced model the corresponding matrix words are all based at the single site, and open words can appear algebraically. The center-selection rule above is what removes the charged open words. The uniqueness clause in Definition 49.4 is also load-bearing: two different center-symmetric solutions of the same limiting loop equations would not be distinguished by the formal loop-equation matching alone.

Controlled Approximation 49.5 (Status of Eguchi–Kawai reduction)**.** The original one-site Eguchi–Kawai model is a valid reduced candidate only in regimes where its center symmetry is not spontaneously broken in the large-N_c limit. At weak lattice coupling in dimensions larger than two, the center of the original one-site model breaks; the naive model then produces a reduced large-N_c state with charged reduced-word expectation values rather than the desired large-volume Wilson-loop theory. Quenched Eguchi–Kawai, twisted Eguchi–Kawai, partial reduction, and adjoint matter with periodic boundary conditions are different attempts to stabilize the needed center realization. Each changes the finite-regulator construction and therefore requires its own loop-equation and symmetry analysis. This monograph uses volume reduction only when the relevant center, factorization, and uniqueness hypotheses are explicitly part of the statement.

Fundamental quarks introduce boundaries rather than handles. A closed quark loop carries one fundamental color sum and one flavor sum. In the fixed-flavor 't Hooft limit N_f is held fixed, and a graph with b_q quark-loop boundaries scales as

$$N_f^{b_q} N_c^{2-2h-b_q}. \tag{49.52}$$

Thus each additional closed quark loop is suppressed by one power of N_c^{-1} relative to a gluonic planar vacuum contribution. In the Veneziano limit, by contrast, N_f/N_c is held fixed; the factor

$$N_f^{b_q} N_c^{2-2h-b_q} = \left(\frac{N_f}{N_c}\right)^{b_q} N_c^{2-2h}$$

shows that quark-loop boundaries are no longer suppressed. The Banks–Zaks discussion above naturally belongs to this Veneziano scaling when N_f/N_c is tuned close to 11/2.

The same counting explains the leading N_c-dependence of hadronic interactions in a confining large-N_c theory, subject to the separate spectral assumptions stated below. A mesonic color-singlet operator $M_{IJ} = \bar{q}_I q_J$ has a connected m-point function of order N_c when all insertions lie on one quark boundary. The canonically normalized meson source is therefore $N_c^{-1/2} M_{IJ}$, and connected m-meson amplitudes scale as $N_c^{1-m/2}$. Glueball operators normalized to have order-one connected two-point functions have connected m-point couplings of order N_c^{2-m}.

49.7.3 Large-N_c Baryons and Spin-Flavor Coordinates

Baryons require a separate order of limits and a separate normalization. In an $SU(N_c)$ theory with N_f fundamental quark flavors, a local baryon operator in a chosen spin-flavor channel has the color-antisymmetric local form

$$B_{\mathcal{A}}(x) = \epsilon_{i_1 \cdots i_{N_c}} \, \mathcal{A}^{I_1\alpha_1, \dots, I_{N_c}\alpha_{N_c}} \, q^{i_1}_{I_1\alpha_1}(x) \cdots q^{i_{N_c}}_{I_{N_c}\alpha_{N_c}}(x), \tag{49.53}$$

where i_r are color indices, I_r are flavor indices, α_r are spinor indices, and $\mathcal{A}$ is a tensor selecting the spin-flavor channel. The color tensor is totally antisymmetric. Therefore, for baryons built from quarks in the same spatial orbital, the remaining spin-flavor wavefunction is symmetric under exchange of the N_c quarks. This is a kinematic consequence of Fermi statistics and the ϵ-contraction, not a dynamical theorem about the spectrum.

Take the large-N_c limit with N_f fixed, quark masses fixed in N_c-independent units, and the spatial volume sent to infinity in the usual order needed to define particle states. If a baryon state created by (49.53) exists and has a Hartree description in which each quark moves in an N_c-independent mean field, then its leading mass is $O(N_c)$. Indeed there are N_c one-body contributions, while the two-body estimate gives

$$\binom{N_c}{2} g^2 = \frac{N_c(N_c-1)}{2} \, \frac{\lambda}{N_c} = \frac{\lambda}{2}(N_c - 1), \tag{49.54}$$

which is also $O(N_c)$ at fixed λ. The Hartree statement is a controlled large-N_c organization only after the existence of such color-singlet baryon states has been assumed or established in a regulator. It is not a derivation of baryons from the continuum Yang–Mills path integral.

Spin and flavor splittings are subleading in the same limit. If the baryon is described at low energy by a slowly rotating spin-flavor collective coordinate with moment of inertia $I_{\mathrm{rot}} = O(N_c)$, then the rotor contribution has the form

$$\Delta M_J = \frac{J(J+1)}{2 I_{\mathrm{rot}}}, \tag{49.55}$$

so fixed-spin splittings are $O(N_c^{-1})$. In theories with chiral symmetry breaking, the same scaling is compatible with a soliton description in the mesonic effective theory: $F_\pi^2 = O(N_c)$, meson interaction vertices have the large-N_c scaling $g_m = O(N_c^{1-m/2})$, and a soliton mass built from the mesonic action is $O(N_c)$. The spin-flavor multiplet structure often called contracted $SU(2N_f)$ is the algebraic form of the large-N_c consistency conditions on baryon matrix elements, for example on pion-baryon scattering. In this monograph it is used only with the displayed hypotheses: fixed N_f, a confining theory with baryon states, and a low-energy regime where the collective-coordinate description is valid. In the Veneziano limit with N_f/N_c fixed, the spin-flavor representation theory and baryon entropy grow differently and must be analyzed as a distinct limit.

For two light flavors the contracted spin-flavor algebra can be written without appealing to a quark-model Hamiltonian. It is the large-N_c limit of the exact one-body spin-flavor algebra acting on the symmetric spin-flavor part of a color-singlet baryon source.

Definition 49.6 (Two-flavor spin-flavor one-body generators)**.** On the spin-flavor space

$$\left(\mathbb{C}^2_{\text{spin}} \otimes \mathbb{C}^2_{\text{isospin}}\right)^{\otimes N_c}$$

write σ_r^i and τ_r^a for Pauli matrices acting on the spin and isospin factors of the r-th quark slot. Define

$$J^i = \sum_{r=1}^{N_c} \frac{\sigma_r^i}{2}, \qquad I^a = \sum_{r=1}^{N_c} \frac{\tau_r^a}{2}, \qquad G^{ia} = \sum_{r=1}^{N_c} \frac{\sigma_r^i \tau_r^a}{4}. \tag{49.56}$$

Here J^i generates spatial spin, I^a generates isospin, and G^{ia} is the mixed spin-isospin one-body generator. The normalization is fixed by the displayed Pauli matrices; it is independent of the gauge-group trace convention used in the Yang–Mills action.

Contraction of the two-flavor baryon spin-flavor algebra. The generators of Definition 49.6 satisfy the exact finite-N_c commutation relations

$$[J^i, J^j] = \mathrm{i}\varepsilon^{ijk} J^k, \qquad [I^a, I^b] = \mathrm{i}\varepsilon^{abc} I^c, \qquad [J^i, I^a] = 0, \tag{49.57}$$

$$[J^i, G^{ja}] = \mathrm{i}\varepsilon^{ijk} G^{ka}, \qquad [I^a, G^{ib}] = \mathrm{i}\varepsilon^{abc} G^{ic}, \tag{49.58}$$

$$[G^{ia}, G^{jb}] = \frac{\mathrm{i}}{4}\delta^{ij}\varepsilon^{abc} I^c + \frac{\mathrm{i}}{4}\delta^{ab}\varepsilon^{ijk} J^k. \tag{49.59}$$

Set $X^{ia} = G^{ia}/N_c$. On any sequence of baryon states whose matrix elements of J^i and I^a grow no faster than $O(N_c)$, the commutator of the rescaled mixed generators obeys

$$[X^{ia}, X^{jb}] = \frac{\mathrm{i}}{4N_c^2}\delta^{ij}\varepsilon^{abc} I^c + \frac{\mathrm{i}}{4N_c^2}\delta^{ab}\varepsilon^{ijk} J^k = O(N_c^{-1}) \tag{49.60}$$

in matrix elements, and it is $O(N_c^{-2})$ on fixed-spin and fixed-isospin families. The limiting algebra has X^{ia}'s transforming as vectors under spin and isospin but commuting among themselves. This is the two-flavor contracted $SU(4)$ algebra. Operators acting on different quark slots commute, so it is enough to compute the one-slot algebra. Let $s^i = \sigma^i/2$ and $t^a = \tau^a/2$. The Pauli relations give

$$[s^i, s^j] = \mathrm{i}\varepsilon^{ijk} s^k, \qquad \{s^i, s^j\} = \frac{1}{2}\delta^{ij},$$

and the same formulae with s, i, j, k replaced by t, a, b, c. Since spin and isospin matrices commute with each other,

$$[s^i t^a, s^j t^b] = \frac{1}{2}\{s^i, s^j\}[t^a, t^b] + \frac{1}{2}[s^i, s^j]\{t^a, t^b\}.$$

Substituting the Pauli relations gives

$$[s^i t^a, s^j t^b] = \frac{\mathrm{i}}{4}\delta^{ij}\varepsilon^{abc} t^c + \frac{\mathrm{i}}{4}\delta^{ab}\varepsilon^{ijk} s^k. \tag{49.61}$$

Summing (49.61) over the N_c quark slots gives (49.59). The remaining commutators follow in the same way from the facts that s^i transforms as a spin vector and t^a transforms as an isospin vector.

Dividing G^{ia} by N_c gives

$$[X^{ia}, X^{jb}] = \frac{1}{N_c^2}[G^{ia}, G^{jb}],$$

which is (49.60). If matrix elements of J and I are at most order N_c, the right-hand side is at most order N_c^{-1}. If the spin and isospin labels are held fixed as $N_c \to \infty$, it is order N_c^{-2}. The commutators with J and I in (49.58) survive the rescaling because $[J, X] = [J, G]/N_c$ and $[I, X] = [I, G]/N_c$. Thus the limiting algebra is a semidirect product of spin and isospin rotations with commuting generators X^{ia}.

The expansion also gives the perturbative origin of the string language used in the next section. Ribbon graphs are organized by the genus of an oriented surface, while the connected vacuum functional has the form

$$\log Z = \sum_{h \geq 0} N_c^{2-2h}\, F_h(\lambda)$$

as a formal expansion in $1/N_c^2$ at fixed λ. A confining flux tube, when it exists, is a different and more infrared object: it is defined by line operators and static-source sectors. The large-N_c topological expansion is therefore compatible with a genus expansion for a string description of gauge dynamics, but it is not itself a proof of the continuum QCD string or of confinement.

49.8 QCD Strings and Flux-Tube Effective Theory

The QCD string is the long-distance effective dynamics of a chromoelectric flux tube, defined by gauge-invariant line observables and by the spectrum in sectors with external color sources. In pure $SU(N_c)$ Yang–Mills theory there is a center one-form symmetry with charge in $\mathbb{Z}_{N_c}$. A Wilson line in a representation R has center charge equal to its N-ality $k_R \in \mathbb{Z}_{N_c}$. A static pair of external sources with nonzero k_R lies in a superselection sector that cannot relax to the vacuum by local gluonic excitations alone. The flux tube connecting the sources is the lowest-energy state in this sector at fixed separation.

For a Euclidean rectangle $C_{L,T}$, the transfer-matrix interpretation of $\langle W_R(C_{L,T})\rangle$ gives the spectral expansion

$$\langle W_R(C_{L,T})\rangle = \sum_\alpha |\langle \alpha; L, R|\mathcal{O}_R(L)\Omega\rangle|^2 \mathrm{e}^{-E_\alpha(L)T}, \qquad T > 0, \tag{49.62}$$

where $\mathcal{O}_R(L)$ is a gauge-invariant operator that creates the two static sources separated by L, and $|\alpha; L, R\rangle$ are eigenstates of the Hamiltonian in the corresponding static-source sector. If the lowest energy is isolated and the overlap in (49.62) is nonzero, then

$$V_R(L) = E_0(L) = -\lim_{T\to\infty} \frac{1}{T} \log\langle W_R(C_{L,T})\rangle.$$

The string tension in this sector is the asymptotic slope, when the limit exists,

$$\sigma_R := \lim_{L\to\infty} \frac{E_0(L)}{L}. \tag{49.63}$$

For representations with the same N-ality, gluon screening can change the short-distance representation carried by the flux, so the asymptotic tension depends on the center charge k_R, not on the full microscopic representation R, after all lower-energy screening channels are included.

49.8.1 Center Sectors and k-String Tensions

The preceding definition becomes sharper when the gauge theory has no dynamical fields of nonzero center charge. Let the microscopic gauge group be the simply connected group $SU(N_c)$, and let the only dynamical fields be adjoint fields. The electric one-form symmetry is then $\mathbb{Z}_{N_c}$. An external source in a representation R has center charge

$$k_R \equiv \text{number of fundamental boxes of } R \pmod{N_c}.$$

The static-source Hilbert space decomposes into sectors labelled by this charge. If dynamical matter of center charge q is present, the label is only the quotient charge left unscreened by such matter; if the global form of the gauge group is changed, the allowed Wilson lines and the one-form symmetry are changed as in the line-operator classification of Chapter 113. Thus a k-string is a spectral sector of a specified gauge theory, including its global form and its dynamical matter content, rather than an intrinsic label of a Lie algebra.

Definition 49.7 (Center-sector string tension)**.** In pure $SU(N_c)$ Yang–Mills theory, fix $k \in \{1, \dots, N_c - 1\}$. A k-string tension is the number

$$\sigma_k = \lim_{L\to\infty} \frac{E_k(L) - E_{\text{vac}}}{L}, \tag{49.64}$$

when the following objects and limits exist: a regulator with a positive transfer matrix, a static-source or winding-flux sector of electric center charge k, the lowest energy $E_k(L)$ in that sector for flux length L, the vacuum energy subtraction E_{vac} in the same finite box, and the thermodynamic and continuum limits in which the dimensionless ratio σ_k/σ_1 stabilizes. Charge conjugation identifies the sectors k and $N_c - k$, hence

$$\sigma_k = \sigma_{N_c-k}. \tag{49.65}$$

A sector is elementary at the level of tensions only if

$$\sigma_k < \min_{\substack{k_1+\cdots+k_m \equiv k \pmod{N_c} \\ m\geq 2}} (\sigma_{k_1} + \cdots + \sigma_{k_m}) \tag{49.66}$$

for all nontrivial decompositions into allowed unscreened center charges. If the inequality is saturated, the lowest state may be a multi-string threshold rather than a single stable flux tube.

The normalized ratios

$$R_k(N_c) = \frac{\sigma_k}{\sigma_1}$$

are nonperturbative observables. Two closed formulas are often useful as comparison coordinates, but neither is a theorem of nonsupersymmetric four-dimensional Yang–Mills theory:

$$R_k^{\text{Casimir}}(N_c) = \frac{k(N_c - k)}{N_c - 1}, \tag{49.67}$$

$$R_k^{\text{sine}}(N_c) = \frac{\sin(\pi k/N_c)}{\sin(\pi/N_c)}. \tag{49.68}$$

The first is the ratio of quadratic Casimirs of the rank-k antisymmetric representation and the fundamental representation. The second is the sine profile that appears in particular Abelianized supersymmetric BPS-vortex regimes, and is derived in that setting in Volume VII. In the present Yang–Mills discussion, both formulas are comparison functions for lattice or other nonperturbative determinations of σ_k.

Trace-delta Casimir comparison. Let t^a be $SU(N_c)$ fundamental generators normalized by

$$\mathrm{tr}_\square(t^a t^b) = \delta^{ab}.$$

Let $A_k = \Lambda^k \mathbb{C}^{N_c}$ be the rank-k antisymmetric representation, $1 \le k \le N_c - 1$. The comparison function (49.67) comes from the following representation-theory calculation in the same trace convention as the Yang–Mills action. The quadratic Casimir eigenvalues are

$$C_2(A_k) = \frac{k(N_c - k)(N_c + 1)}{N_c}, \qquad C_2(\square) = \frac{N_c^2 - 1}{N_c}, \tag{49.69}$$

and hence

$$\frac{C_2(A_k)}{C_2(\square)} = \frac{k(N_c - k)}{N_c - 1}.$$

The trace-delta Fierz identity in the fundamental representation is

$$\sum_a (t^a)^i{}_j (t^a)^m{}_n = \delta^i{}_n \delta^m{}_j - \frac{1}{N_c} \delta^i{}_j \delta^m{}_n. \tag{49.70}$$

Contracting $j = n$ and summing over j gives

$$\sum_a (t^a t^a)^i{}_m = \left(N_c - \frac{1}{N_c}\right) \delta^i{}_m,$$

so $C_2(\square) = (N_c^2 - 1)/N_c$.

On $V^{\otimes k}$, with $V = \mathbb{C}^{N_c}$, the total generator is

$$T^a_{\mathrm{tot}} = \sum_{r=1}^{k} t^a_{(r)},$$

where $t^a_{(r)}$ acts on the r-th tensor factor. Its quadratic Casimir is

$$\sum_a T^a_{\mathrm{tot}} T^a_{\mathrm{tot}} = \sum_{r=1}^{k} \sum_a t^a_{(r)} t^a_{(r)} + 2 \sum_{1 \le r < s \le k} \sum_a t^a_{(r)} t^a_{(s)}.$$

By (49.70), the two-site operator is

$$\sum_a t^a_{(r)} t^a_{(s)} = P_{rs} - \frac{1}{N_c},$$

where P_{rs} exchanges the two tensor factors. On $\Lambda^k V$, $P_{rs} = -1$. Therefore

$$C_2(A_k) = k\, \frac{N_c^2 - 1}{N_c} + 2\binom{k}{2}\left(-1 - \frac{1}{N_c}\right) = \frac{k(N_c - k)(N_c + 1)}{N_c}.$$

Dividing by the fundamental value gives (49.67). The factor-of-two convention in the generator trace cancels in the ratio, but it is displayed because this chapter uses the trace-delta normalization throughout.

For $1 \leq k \leq N_c - 1$, the sine comparison function (49.68) has the charge-conjugation symmetry

$$R_k^{\text{sine}}(N_c) = R_{N_c-k}^{\text{sine}}(N_c),$$

the strict subadditivity property

$$R_{k+\ell}^{\text{sine}}(N_c) < R_k^{\text{sine}}(N_c) + R_\ell^{\text{sine}}(N_c), \qquad k, \ell \geq 1, \quad k + \ell \leq N_c - 1, \tag{49.71}$$

and the fixed-k, large-N_c expansion

$$R_k^{\text{sine}}(N_c) = k - \frac{\pi^2}{6} \frac{k(k^2-1)}{N_c^2} + O(N_c^{-4}). \tag{49.72}$$

The first identity is $\sin(\pi(N_c - k)/N_c) = \sin(\pi k/N_c)$. The inequality follows from

$$\sin a + \sin b - \sin(a+b) = 4 \sin \frac{a}{2} \sin \frac{b}{2} \sin \frac{a+b}{2} > 0, \qquad a = \frac{\pi k}{N_c}, \quad b = \frac{\pi \ell}{N_c},$$

in the range $0 < a, b$ and $a + b < \pi$. The expansion is the quotient of the Taylor series for $\sin(\pi k/N_c)$ and $\sin(\pi/N_c)$. These identities are finite comparison algebra; the dynamical question is whether a continuum gauge theory has flux-sector transfer-matrix eigenvalues whose ratios approach one of these comparison functions.

The Casimir comparison function obeys the same charge-conjugation symmetry, and its fixed-k expansion begins as

$$R_k^{\text{Casimir}}(N_c) = k - \frac{k(k-1)}{N_c} + O(N_c^{-2}). \tag{49.73}$$

This is the expansion of

$$\frac{k(N_c - k)}{N_c - 1} = k \frac{1 - k/N_c}{1 - 1/N_c}.$$

The different $1/N_c$ behavior in (49.72) and (49.73) is one reason that k-strings are useful probes of large-N_c confinement dynamics. For the low ranks often used in lattice comparisons, the two functions give

N_c	k	R_k^{Casimir}	R_k^{sine}
4	2	4/3	$\sqrt{2}$
5	2	3/2	$(1+\sqrt{5})/2$
6	2	8/5	$\sqrt{3}$
6	3	9/5	2
8	2	12/7	$\sqrt{2+\sqrt{2}}$
8	3	15/7	$1+\sqrt{2}$
8	4	16/7	$\sqrt{4+2\sqrt{2}}$

Published lattice determinations for $SU(4)$, $SU(5)$, $SU(6)$, and $SU(8)$ determine the transfer-matrix quantity σ_k/σ_1 after continuum extrapolation, with operator-overlap, finite-volume, and lattice-spacing errors stated. A derivation of either comparison formula would require an independent spectral argument inside the gauge theory. The improved pure-gauge computations generally place the measured ratios between the two comparison curves; the support supplied by those measurements is for that interpolation pattern rather than for exact Casimir scaling in four-dimensional nonsupersymmetric Yang–Mills theory.

Controlled Approximation 49.8 (k-string comparison formulas)**.** When this monograph compares lattice k-string data to (49.67) or (49.68), the comparison means precisely this: the lattice calculation supplies a nonperturbative estimate of σ_k/σ_1, while the two displayed functions are finite comparison coordinates with distinct group-theoretic or supersymmetric origins. No conclusion about the microscopic mechanism of confinement follows from numerical closeness alone. A mechanism claim would require an operator-level derivation of the relevant transfer-matrix sector and its large-L spectrum.

49.8.2 Effective String Action and the Luscher Term

Figure 49.7 records the static-source flux tube as the object whose ground-state energy $E_0(L)$ defines the string tension σ_R in (49.63).

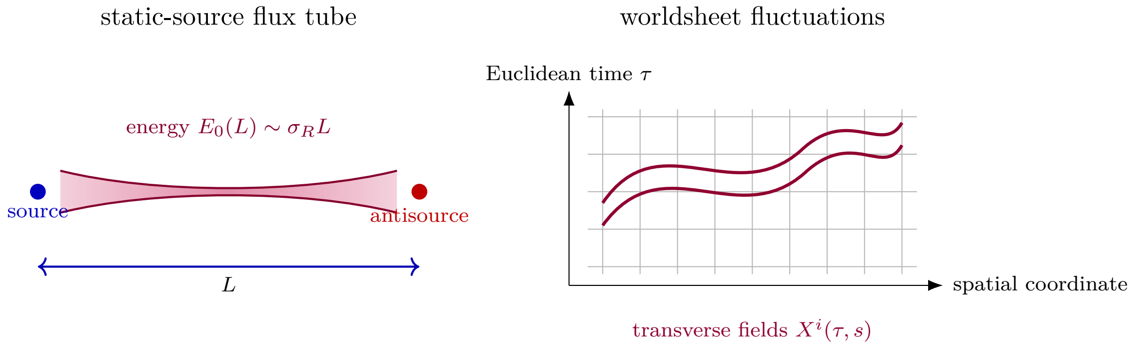


Figure 49.7: The QCD string is the long-distance flux-tube sector selected by gauge-invariant line operators. At large separation the lowest energy is extensive in the source separation, while the low-energy excitations are transverse fluctuations of the flux-tube worldsheet.

The effective string description uses QCD as its microscopic definition and expands the flux-tube energies and matrix elements at large L. Choose worldsheet coordinates (τ, s), with $0 \le s \le L$, and fix static gauge so that the embedding of the worldsheet in D-dimensional spacetime is

$$X^0(\tau, s) = \tau, \qquad X^1(\tau, s) = s, \qquad X^i(\tau, s) \;\; (i = 2, \dots, D-1)$$

for the transverse fluctuations. The leading local action compatible with worldsheet locality, Euclidean time translation, spatial rotations transverse to the string, and the nonlinear realization of target-space Poincare symmetry is the area action

$$S_{\text{eff}} = \sigma_R \int \mathrm{d}\tau\, \mathrm{d}s \sqrt{\det(\delta_{ab} + \partial_a X^i \partial_b X^i)} + \mu_R \int \mathrm{d}\tau + S_{\text{higher}}, \tag{49.74}$$

where $a, b \in \{\tau, s\}$, repeated transverse indices i are summed, and μ_R is an endpoint energy for an open string ending on external sources. The term S_{higher} contains local operators with more derivatives and possible boundary terms. Expanding the square root gives

$$S_{\text{eff}} = \sigma_R L T + \frac{\sigma_R}{2} \int \mathrm{d}\tau\, \mathrm{d}s\, \partial_a X^i \partial_a X^i + \cdots .$$

The first quantum correction to the flux-tube energy is therefore a worldsheet determinant. The determinant must be renormalized as an effective field-theory determinant: local divergent terms are absorbed into the worldsheet area coefficient, endpoint energies, and higher local operators, while the nonlocal finite term is a prediction.

Lemma 49.9 (Free transverse determinant and the Luscher term)**.** *Assume that, at distances $L\sqrt{\sigma_R} \gg 1$, the only gapless worldsheet fields of the open static-source flux tube are the $c_\perp = D-2$ transverse displacement fields X^i, and that the static sources impose Dirichlet boundary conditions on these displacements at $s = 0, L$. After subtracting local counterterms proportional to the worldsheet area and to the length of the two boundaries, the Gaussian fluctuation determinant contributes*

$$\Delta E_0^{\mathrm{open}}(L) = -\frac{\pi c_\perp}{24L}$$

to the long-time energy extracted from the rectangular Wilson loop. For a closed flux tube winding a spatial circle of length L, with periodic transverse fields and with the transverse zero mode treated as the center-of-mass coordinate rather than as an oscillator, the corresponding finite part is

$$\Delta E_0^{\mathrm{closed}}(L) = -\frac{\pi c_\perp}{6L}.$$

Proof. The coefficient σ_R in the quadratic action is removed from the oscillator frequencies by the canonical rescaling $Y^i = \sqrt{\sigma_R} X^i$. For one open transverse scalar with Dirichlet boundary conditions,

$$Y(\tau, s) = \sum_{n\geq 1} q_n(\tau) \sin\frac{\pi n s}{L}, \qquad \omega_n = \frac{\pi n}{L}.$$

The formal zero-point energy is $\frac{1}{2}\sum_{n\geq 1}\omega_n$. A spectral regulator gives

$$E_\alpha^{\mathrm{open}}(L) = \frac{1}{2}\sum_{n\geq 1}\frac{\pi n}{L}\, n^{-\alpha} = \frac{\pi}{2L}\zeta_{\mathrm{R}}(\alpha - 1).$$

The heat-kernel expansion of this regulated determinant contains local divergences proportional to L and to the endpoints. These renormalize σ_R and μ_R in (49.74). The nonlocal finite part is the analytic continuation at $\alpha = 0$:

$$E_{\mathrm{fin}}^{\mathrm{open}}(L) = \frac{\pi}{2L}\zeta_{\mathrm{R}}(-1) = -\frac{\pi}{24L}.$$

Multiplication by $c_\perp$ gives the open-string coefficient.

For one periodic transverse scalar on a circle, the nonzero modes are labelled by $n \in \mathbb{Z}\setminus\{0\}$ and have $\omega_n = 2\pi|n|/L$. Equivalently, for each $n \geq 1$ there are sine and cosine oscillators of frequency $2\pi n/L$. The zero-point contribution is therefore

$$E_\alpha^{\mathrm{closed}}(L) = \sum_{n\geq 1}\frac{2\pi n}{L}\, n^{-\alpha} = \frac{2\pi}{L}\zeta_{\mathrm{R}}(\alpha - 1),$$

whose finite part at $\alpha = 0$ is $-\pi/(6L)$. The omitted zero mode is the transverse center-of-mass coordinate of the closed string; it produces the ordinary volume normalization of the state rather than an L^{-1} Casimir energy. This proves the closed coefficient. □

Consequently, under the hypotheses of Lemma 49.9, the open-string ground-state energy has the universal large-L form

$$E_0(L) = \sigma_R L + \mu_R - \frac{\pi(D-2)}{24L} + O(L^{-3}) \tag{49.75}$$

for a long open flux tube. For a closed flux tube winding a spatial circle, the corresponding leading form is

$$E_0^{\text{closed}}(L) = \sigma_k L - \frac{\pi(D-2)}{6L} + O(L^{-3}), \tag{49.76}$$

again under the assumption that the only gapless worldsheet fields are the transverse displacement modes. The coefficients of the displayed $1/L$ terms are universal because they depend only on the central charge $D-2$ of these free transverse modes. The higher powers in $1/L$, the spectrum of massive worldsheet excitations, and the validity range of the expansion are dynamical properties of the particular gauge theory.

The Nambu–Goto square-root spectrum is a useful reference point, not the definition of the QCD string. If all higher-derivative and boundary coefficients in S_{higher} are set to their Nambu–Goto values, the open and closed ground-state energies expand as

$$\begin{aligned}
E_0^{\text{NG,open}}(L) &= \sigma_R L\sqrt{1 - \frac{\pi(D-2)}{12\sigma_R L^2}} \\
&= \sigma_R L - \frac{\pi(D-2)}{24L} - \frac{\pi^2(D-2)^2}{1152\,\sigma_R L^3} + O(L^{-5}), \qquad (49.77)\\
E_0^{\text{NG,closed}}(L) &= \sigma_R L\sqrt{1 - \frac{\pi(D-2)}{3\sigma_R L^2}} \\
&= \sigma_R L - \frac{\pi(D-2)}{6L} - \frac{\pi^2(D-2)^2}{72\,\sigma_R L^3} + O(L^{-5}). \qquad (49.78)
\end{aligned}$$

The equality of the $1/L$ term with (49.75) and (49.76) is forced by the free transverse massless fields. Agreement at later orders is a statement about the effective-string matching coefficients and the nonlinearly realized target-space Lorentz symmetry, and must be checked or imposed in the particular effective theory being used.

Controlled Approximation 49.10 (Effective string expansion of a confining flux tube)**.** An effective-string prediction for a flux-tube level consists of a static source sector or closed-string winding sector, a renormalized string tension, boundary counterterms when endpoints are present, a list of gapless and massive worldsheet fields, a local derivative expansion for S_{higher}, and a declared large-distance regime

$$L\sqrt{\sigma_R} \gg 1, \qquad E_{\text{gap}}^{\text{ws}} L \gg 1$$

for every massive worldsheet excitation not retained explicitly. The universal Luscher term is the first finite prediction under these hypotheses. Every later coefficient is either fixed by the worldsheet symmetry constraints within the chosen effective theory or is a matching parameter to be determined from the microscopic gauge theory, for example from lattice static-potential or flux-tube spectrum data.

49.8.3 Excited Flux-Tube Levels

The same quadratic worldsheet theory gives the first approximation to the excited flux-tube spectrum. For an open string with fixed endpoints, the transverse oscillator modes are labelled by

$$n \in \mathbb{Z}_{>0}, \qquad i = 1, \ldots, c_\perp, \qquad c_\perp = D-2.$$

Let $N_{n,i} \in \mathbb{Z}_{\geq 0}$ be the occupation number of the oscillator of frequency $\pi n/L$. The total oscillator level is

$$N = \sum_{i=1}^{c_\perp} \sum_{n\geq 1} n\, N_{n,i}. \tag{49.79}$$

The Gaussian effective string therefore predicts

$$E_N^{\text{open}}(L) = \sigma_R L + \mu_R + \frac{\pi}{L}\left(N - \frac{c_\perp}{24}\right) + O(L^{-3}) \tag{49.80}$$

before the L^{-3} corrections from the nonlinear action and boundary operators are included. The number of oscillator states at level N, prior to resolving the finite symmetry group of the particular flux-tube channel, is the coefficient of q^N in

$$Z_{\text{open,osc}}(q) = \prod_{n=1}^{\infty} (1-q^n)^{-c_\perp}. \tag{49.81}$$

For $D = 4$, where $c_\perp = 2$, the first coefficients are

$$1,\ 2,\ 5,\ 10, \ldots.$$

These are not yet a complete list of lattice quantum numbers. A flux tube between static sources also carries parity, charge conjugation when defined, and angular momentum around the source axis; those labels are obtained by decomposing the oscillator states under the residual spacetime symmetries of the static-source sector.

In four spacetime dimensions, place a static fundamental source and static antifundamental source on the z-axis at $z = \pm L/2$. The continuum symmetry group preserving this source pair is the usual cylindrical group $D_{\infty h}$. Its labels are:

$$\Lambda = 0, 1, 2, \ldots \quad \Longleftrightarrow \quad \Sigma, \Pi, \Delta, \Phi, \ldots,$$

where Λ is the absolute value of the angular momentum about the source axis; a sign $\eta = \pm 1$, denoted g/u, for the combined operation which exchanges the source and antisource and inverts about the midpoint; and, only for $\Lambda = 0$, a reflection sign $\epsilon = \pm$ for reflection through a plane containing the source axis. At finite lattice spacing with the axis chosen along a lattice direction, $D_{\infty h}$ is reduced to the corresponding lattice subgroup. The continuum labels are therefore continuum and large-distance assignments obtained after subduction and extrapolation, not exact labels of a single finite lattice calculation.

Open flux-tube oscillator labels. Assume the Gaussian open-string sector in $D = 4$, with circular transverse oscillators $a^\dagger_{n,+}$ and $a^\dagger_{n,-}$ of frequency $\pi n/L$, where $n \geq 1$ and the signs denote angular momentum ± 1 about the source axis. For a Fock monomial

$$\prod_{n\geq 1} \left(a^\dagger_{n,+}\right)^{N_{n,+}} \left(a^\dagger_{n,-}\right)^{N_{n,-}} \Omega_{\text{str}},$$

the oscillator level, axial angular-momentum label, and g/u sign are

$$N = \sum_{n\geq 1} n(N_{n,+} + N_{n,-}), \tag{49.82}$$

$$\Lambda = \left| \sum_{n\geq 1} (N_{n,+} - N_{n,-}) \right|, \tag{49.83}$$

$$\eta = (-1)^{\sum_{n\geq 1} n(N_{n,+}+N_{n,-})}. \tag{49.84}$$

For $\Lambda = 0$, reflection through a plane containing the source axis exchanges $+$ and $-$ oscillators, so reflection eigenstates are the even and odd combinations of each reflection orbit; if the monomial is pointwise fixed by this exchange, only the even Σ^+ state is obtained.

This is bookkeeping in the Gaussian reference Hilbert space. The level formula is the definition of the open-string oscillator level in (49.79). A circular transverse oscillator is built from

$$X^+ = X^2 + iX^3, \qquad X^- = X^2 - iX^3.$$

Under a rotation by angle α about the source axis, $X^\pm \mapsto e^{\pm i\alpha} X^\pm$. A monomial with occupations $N_{n,\pm}$ therefore has axial angular momentum $\sum_n (N_{n,+} - N_{n,-})$, and the irreducible cylindrical label records its absolute value Λ.

The operation defining g/u exchanges the endpoints and sends $(z, X^2, X^3) \mapsto (-z, -X^2, -X^3)$ in the static-source coordinate system. The mode function transforms as

$$\sin \frac{\pi n(L-s)}{L} = (-1)^{n+1} \sin \frac{\pi n s}{L},$$

while the transverse vector itself contributes the additional sign -1. Thus a single n-th transverse oscillator has eigenvalue $(-1)^n$ under the combined operation, and multiplication over all occupied oscillators gives the sign η in (49.84). Reflection through a plane containing the axis complex-conjugates X^+ and X^-, hence exchanges the two circular polarizations. For $\Lambda > 0$ this reflection relates the two helicity components of the same two-dimensional real representation. For $\Lambda = 0$ it can be diagonalized, giving $\epsilon = \pm$ when the reflection orbit has two monomials and only the even state when the monomial is fixed.

For the first oscillator levels in $D = 4$ this gives

level N	$D_{\infty h}$ content of the Gaussian reference sector	total oscillator degeneracy
0	Σ_g^+	1
1	Π_u	2
2	$\Sigma_g^+ \oplus \Pi_g \oplus \Delta_g$	5
3	$\Sigma_u^+ \oplus \Sigma_u^- \oplus 2\Pi_u \oplus \Delta_u \oplus \Phi_u$	10

At the next level the decomposition contains

$$3\Sigma_g^+ \oplus \Sigma_g^- \oplus 3\Pi_g \oplus 3\Delta_g \oplus \Phi_g \oplus (\Lambda = 4)_g,$$

with total oscillator degeneracy 20. Thus Σ_g^- is an allowed cylindrical channel, but it is not present in the first three free transverse-oscillator levels. This distinction matters when one compares the long-string reference spectrum with hybrid static-potential data.

The table is a long-string reference decomposition. A measured hybrid static potential in the same Λ_η^ϵ channel may contain massive surface excitations, endpoint-local excitations, or short-distance gluelump physics that are not present in the free transverse oscillator Hilbert space. Conversely, the free oscillator labels become useful at large L only after a transfer-matrix calculation has isolated the corresponding static sector and after the effective-string expansion is within its declared regime.

In the lattice program initiated by Juge, Kuti, and Morningstar, the objects called Π_u, Σ_g^-, $\Sigma_u^\pm$, Π_g, and so on are channel-resolved static energies extracted from correlation matrices of gauge-invariant source operators subduced to the finite lattice subgroup of $D_{\infty h}$. The label is therefore assigned before an effective-string fit is made. At small separation the same channel can be governed by a gluelump-like excitation of the gauge field bound to a compact static color-octet source pair; at large separation, if no lower massive worldsheet or endpoint excitation intervenes, the channel may approach one of the long-string oscillator towers above. The observed L-dependence of a hybrid static potential is the interpolation between these two spectral descriptions, with the Nambu–Goto reference formula supplying only the asymptotic massless-transverse-oscillator comparison. The channels Π_u and Σ_g^- are especially useful diagnostics: Π_u is the first transverse oscillator tower in the Gaussian long-string limit, whereas a low-lying Σ_g^- level indicates additional channel dynamics until the higher-level long-string asymptotics are actually reached.

For a closed flux tube winding a spatial circle, the quadratic theory has left- and right-moving oscillators. Define

$$N_L = \sum_{i=1}^{c_\perp} \sum_{n\geq 1} n\, N_{n,i}^L, \qquad N_R = \sum_{i=1}^{c_\perp} \sum_{n\geq 1} n\, N_{n,i}^R.$$

Translation along the winding circle gives a quantized longitudinal momentum

$$p_\parallel = \frac{2\pi q}{L}, \qquad q \in \mathbb{Z},$$

and the residual worldsheet translation constraint imposes the level-matching condition

$$N_L - N_R = q. \tag{49.85}$$

At Gaussian order the closed-string energy is

$$E_{N_L,N_R,q}^{\text{closed}}(L) = \sigma_k L + \frac{2\pi}{L}\left(N_L + N_R - \frac{c_\perp}{12}\right) + O(L^{-3}), \tag{49.86}$$

with the q-dependent kinetic contribution first entering at order L^{-3} in the large-L expansion. The $N_L = N_R = 0$ case recovers (49.76).

In the Nambu–Goto reference theory these oscillator levels are packaged into the square-root formulas

$$E_N^{\text{NG,open}}(L) = \sigma_R L \sqrt{1 + \frac{2\pi}{\sigma_R L^2}\left(N - \frac{c_\perp}{24}\right)}, \tag{49.87}$$

$$E_{N_L,N_R,q}^{\text{NG,closed}}(L) = \sigma_k L \sqrt{1 + \frac{4\pi}{\sigma_k L^2}\left(N_L + N_R - \frac{c_\perp}{12}\right) + \left(\frac{2\pi q}{\sigma_k L^2}\right)^2}, \tag{49.88}$$

with (49.85) imposed in the closed case. Expanding these formulas gives

$$E_N^{\mathrm{NG,open}}(L) = \sigma_R L + \frac{\pi}{L}\left(N - \frac{c_\perp}{24}\right) - \frac{\pi^2}{2\sigma_R L^3}\left(N - \frac{c_\perp}{24}\right)^2 + O(L^{-5}), \tag{49.89}$$

$$\begin{aligned} E_{N_L,N_R,q}^{\mathrm{NG,closed}}(L) = \sigma_k L &+ \frac{2\pi}{L}\left(N_L + N_R - \frac{c_\perp}{12}\right) \\ &+ \frac{2\pi^2}{\sigma_k L^3}\left[q^2 - \left(N_L + N_R - \frac{c_\perp}{12}\right)^2\right] + O(L^{-5}). \end{aligned} \tag{49.90}$$

The square-root formulas are useful because they organize a large set of levels with the same conventions, but the QCD statement is weaker and more precise: the Gaussian $1/L$ structure follows from the assumed gapless transverse modes, while the $1/L^3$ and later terms are effective-string matching data unless a universality theorem for the relevant order and boundary condition has been established.

49.8.4 Oscillator Growth and the Flux-String Hagedorn Scale

The degeneracy of the oscillator levels in (49.81) grows exponentially. This growth is a statement about the effective long-string Hilbert space. The deconfinement transition of the four-dimensional gauge theory is a property of the thermal gauge-theory transfer matrix and is discussed in Volume X. The calculation here gives the scale at which a gas of arbitrarily long effective-string states ceases to have a finite canonical partition function inside the regime where the long-string description is used.

Colored-partition growth and the string Hagedorn scale. Let $p_c(N)$ be the coefficient of q^N in

$$\prod_{n=1}^{\infty}(1-q^n)^{-c}, \qquad c \in \mathbb{Z}_{>0}.$$

Then

$$\log p_c(N) = 2\pi\sqrt{\frac{cN}{6}} + O(\log N) \qquad (N \to \infty). \tag{49.91}$$

For the Nambu–Goto reference spectrum of a long confining flux string with $c_\perp = D-2$ transverse massless fields and tension σ, the oscillator density has the exponential scale

$$\rho(E) \sim \exp(\beta_{\mathrm{H}}^{\mathrm{ws}} E), \qquad \beta_{\mathrm{H}}^{\mathrm{ws}} = \sqrt{\frac{\pi c_\perp}{3\sigma}}. \tag{49.92}$$

Set $q = \mathrm{e}^{-\varepsilon}$, with $\varepsilon > 0$. The logarithm of the oscillator generating function is

$$-c\sum_{n\geq 1}\log(1-\mathrm{e}^{-n\varepsilon}) = c\sum_{m\geq 1}\frac{1}{m}\frac{\mathrm{e}^{-m\varepsilon}}{1-\mathrm{e}^{-m\varepsilon}}.$$

As $\varepsilon \downarrow 0$, comparison with the corresponding Riemann sum, or equivalently the modular transformation of the Dedekind eta function, gives

$$\log\prod_{n=1}^{\infty}(1-\mathrm{e}^{-n\varepsilon})^{-c} = \frac{\pi^2 c}{6\varepsilon} + O(\log\varepsilon). \tag{49.93}$$

Cauchy's coefficient formula writes $p_c(N)$ as the contour integral of $q^{-N-1}\prod_n(1-q^n)^{-c}$. The exponential part of the integrand on the positive saddle is

$$N\varepsilon + \frac{\pi^2 c}{6\varepsilon}.$$

The stationary point is

$$\varepsilon_* = \pi\sqrt{\frac{c}{6N}},$$

and the saddle exponent is

$$N\varepsilon_* + \frac{\pi^2 c}{6\varepsilon_*} = 2\pi\sqrt{\frac{cN}{6}}.$$

Gaussian fluctuations and the $O(\log\varepsilon)$ term in (49.93) contribute only logarithmic corrections to $\log p_c(N)$, proving (49.91).

For an open Nambu–Goto reference string at large excitation number, (49.87) gives

$$E^2 = 2\pi\sigma N + O(N^0)$$

in the oscillator-dominated regime. Substituting $N = E^2/(2\pi\sigma) + O(1)$ into (49.91) gives

$$\log\rho(E) = 2\pi\sqrt{\frac{c_\perp}{6}\frac{E^2}{2\pi\sigma}} + O(\log E) = E\sqrt{\frac{\pi c_\perp}{3\sigma}} + O(\log E).$$

For a closed string with level matching at zero longitudinal momentum, $N_L = N_R = N$. The degeneracy is $p_{c_\perp}(N)^2$, while (49.88) gives

$$E^2 = 8\pi\sigma N + O(N^0).$$

The doubled entropy and the doubled energy coefficient combine to the same $\beta_{\mathrm{H}}^{\mathrm{ws}}$. This proves (49.92).

Controlled Approximation 49.11 (Flux-tube Hagedorn scale)**.** Equation (49.92) is a consequence of three declared inputs: a stable long flux-tube sector, $c_\perp$ massless transverse fields with the oscillator generating function above, and the Nambu–Goto high-level relation between excitation level and energy. It is a worldsheet-density statement. A four-dimensional deconfinement temperature requires the thermal gauge-theory path integral, center symmetry and matter-content data, and the infinite-volume limit. Equality between a thermal transition scale and the worldsheet Hagedorn scale is an additional dynamical statement whose proof would have to relate the thermal transfer matrix to the long-string density (49.92).

49.8.5 Static Observables, Lattice Extraction, and the Meaning of String Tension

The static potential and the effective string are nonperturbative objects. Their definition does not use perturbative gluon states. It uses the Hilbert space obtained from the gauge theory with external static sources, or equivalently the transfer matrix acting on gauge-invariant line-operator states. On a Euclidean lattice with spacing a_{lat}, a rectangular Wilson loop is the trace of the ordered product of link variables around the rectangle. In the continuum notation used in this chapter it is written

$$W_R(C) = \mathrm{tr}_R\ \mathcal{P}\exp\left(i\oint_C A_\mu\,\mathrm{d}x^\mu\right),$$

where R is a representation of the gauge group and $\mathcal{P}$ denotes ordering along the contour. The continuum expression is a compact notation for the renormalized line operator obtained from a chosen regulator; a Wilson loop with cusps or corners has additional local renormalizations at those singular points.

The area-law statement relevant for the static flux tube is therefore a statement after local perimeter and corner terms have been separated. For large rectangles $C_{L,T}$, a pure gauge theory in a confining center sector has

$$-\log\langle W_R(C_{L,T})\rangle = \sigma_k LT + \rho_R(L+T) + \kappa_R + o(LT) \tag{49.94}$$

when both L and T are large in a regime where the lowest flux-tube channel of N-ality $k = k_R$ dominates. The perimeter coefficient ρ_R and the corner coefficient κ_R are not the string tension. They depend on the definition of the line operator and on its ultraviolet renormalization. The coefficient σ_k is the asymptotic energy per unit length of the lowest flux tube in the corresponding center-charge sector.

Rectangular area coefficient and static-potential slope. The equality between the rectangular area coefficient and the large-distance static-potential slope is a transfer-matrix extraction, not an independent proof of confinement. Assume the static-source spectral expansion (49.62), assume a nonzero overlap with the lowest state for all sufficiently large L, and assume that the gap above $E_0(L)$ is bounded below by a positive function whose large-T suppression is uniform in the large-L window under consideration. If

$$E_0(L) = \sigma_k L + \mu_R + o(L) \qquad (L\to\infty),$$

then the coefficient of LT in $-\log\langle W_R(C_{L,T})\rangle$, with $T\to\infty$ taken before the large-L slope, is σ_k. Conversely, if the rectangular Wilson loop has the form (49.94) with a controlled transfer-matrix limit at each fixed L, then

$$\lim_{L\to\infty}\frac{E_0(L)}{L} = \sigma_k.$$

Indeed, writing the spectral expansion in the form

$$\langle W_R(C_{L,T})\rangle = c_0(L)\mathrm{e}^{-E_0(L)T}\left[1+\sum_{\alpha>0}\frac{c_\alpha(L)}{c_0(L)}\mathrm{e}^{-(E_\alpha(L)-E_0(L))T}\right],$$

where $c_0(L)\neq 0$ by hypothesis. The uniform gap assumption implies that the logarithm of the bracket divided by T vanishes as $T\to\infty$, so

$$-\lim_{T\to\infty}\frac{1}{T}\log\langle W_R(C_{L,T})\rangle = E_0(L).$$

If $E_0(L) = \sigma_k L + \mu_R + o(L)$, the coefficient multiplying LT in the large-rectangle logarithm is σ_k; the L-independent endpoint energy μ_R, the overlap $c_0(L)$, perimeter terms, and corner terms do not contribute to the double slope $\lim_{L\to\infty} L^{-1}\lim_{T\to\infty} T^{-1}(-\log\langle W\rangle)$. The converse is the same identity read in the opposite direction: the transfer-matrix limit extracts $E_0(L)$, and the assumed rectangular area coefficient fixes the large-L slope of this extracted energy.

On a Euclidean time circle of circumference β, the corresponding static observable is the Polyakov loop

$$P_R(\mathbf{x}) = \mathrm{tr}_R\, \mathcal{P} \exp\left(i \int_0^\beta A_0(\tau, \mathbf{x})\, \mathrm{d}\tau\right).$$

The two-source free energy is defined by

$$F_R(L; \beta) = -\frac{1}{\beta} \log\langle P_R(\mathbf{0})^\dagger P_R(\mathbf{L})\rangle_\beta.$$

When the zero-temperature transfer-matrix limit is taken by sending $\beta \to \infty$, $F_R(L; \beta)$ approaches the same static energy extracted from a long rectangular Wilson loop, up to the vacuum-energy and line-renormalization conventions used in defining the loop. At finite temperature in pure Yang–Mills theory, the expectation value of a single fundamental Polyakov loop is charged under the temporal center symmetry and is therefore an order parameter for that finite-temperature symmetry after the infinite-volume limit is specified. This is related to, but not identical with, the zero-temperature string tension σ_k. The latter is a property of the large-distance static-source spectrum.

In practical lattice calculations one does not measure σ_k by fitting a bare Wilson loop alone. One constructs a family of gauge-invariant operators in the same static-source sector, computes their Euclidean correlation matrix, and uses a variational transfer-matrix problem to isolate the lowest and excited energies. The effective string expansion then describes the large-L behavior of these extracted energies. This is the operational reason that the QCD string is a nonperturbative observable: it is a statement about the spectrum selected by gauge-invariant line operators, not about perturbative gluons propagating between color charges.

49.8.6 Baryonic Flux Networks

Baryonic confinement is not a copy of the mesonic static potential with one more endpoint. In an $SU(N_c)$ gauge theory a gauge-invariant static baryon operator is made by joining N_c fundamental Wilson lines to a common color junction and contracting their color indices with $\epsilon_{i_1\cdots i_{N_c}}$. For paths Γ_r running from a junction point y to source positions $\mathbf{x}_r$,

$$\mathcal{B}_{\Gamma_1,\ldots,\Gamma_{N_c}} = \epsilon_{i_1\cdots i_{N_c}} U(\Gamma_1)^{i_1}{}_{j_1} \cdots U(\Gamma_{N_c})^{i_{N_c}}{}_{j_{N_c}} Q^{j_1}(\mathbf{x}_1) \cdots Q^{j_{N_c}}(\mathbf{x}_{N_c}).$$

Here $Q^j(\mathbf{x}_r)$ denotes a static fundamental source insertion; it is not a dynamical asymptotic quark state. The determinant-one property of $SU(N_c)$ makes the contraction at the junction gauge invariant. For $N_c = 3$, and for source separations large compared with the inverse mass gap, the leading string model predicts an energy proportional to the length of the minimal three-pronged network, not to a unique pairwise potential. Endpoint and junction energies are local terms and must be separated from the coefficient of the large lengths:

$$V_{3Q}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \sigma\, L_Y(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) + \mu_1 + \mu_2 + \mu_3 + \mu_J + o(L_Y). \tag{49.95}$$

Proposition 49.12 (Steiner length for a three-source baryonic string)**.** *Let $a = |\mathbf{x}_2 - \mathbf{x}_3|$, $b = |\mathbf{x}_3 - \mathbf{x}_1|$, and $c = |\mathbf{x}_1 - \mathbf{x}_2|$ be the side lengths of a Euclidean triangle, and let Δ be its area. If* *every angle of the triangle is less than $2\pi/3$, the length minimizing $\sum_{r=1}^{3} |\mathbf{y} - \mathbf{x}_r|$ is realized by a point $\mathbf{y}_F$ from which the three segments meet at angles $2\pi/3$, and*

$$L_Y^2 = \frac{1}{2}\left(a^2 + b^2 + c^2 + 4\sqrt{3}\,\Delta\right). \tag{49.96}$$

If one angle is at least $2\pi/3$, the minimizing network collapses to the two sides adjacent to that angle.

Proof. For an interior minimizer $\mathbf{y}_F$, vary $\mathbf{y} \mapsto \mathbf{y} + \delta\mathbf{y}$. The first variation of $\sum_r |\mathbf{y} - \mathbf{x}_r|$ is

$$\delta\mathbf{y} \cdot \sum_{r=1}^{3} \frac{\mathbf{y}_F - \mathbf{x}_r}{|\mathbf{y}_F - \mathbf{x}_r|}.$$

It vanishes for all $\delta\mathbf{y}$ precisely when the three unit vectors from the sources to the junction sum to zero. Three unit vectors sum to zero only when their pairwise angles are $2\pi/3$.

Set $r_i = |\mathbf{y}_F - \mathbf{x}_i|$. Because the angles at the junction are $2\pi/3$,

$$a^2 = r_2^2 + r_3^2 + r_2 r_3, \qquad b^2 = r_3^2 + r_1^2 + r_3 r_1, \qquad c^2 = r_1^2 + r_2^2 + r_1 r_2.$$

Also the total area is the sum of the three triangles with common vertex $\mathbf{y}_F$:

$$\Delta = \frac{\sqrt{3}}{4}(r_1 r_2 + r_2 r_3 + r_3 r_1).$$

Therefore

$$\frac{1}{2}(a^2 + b^2 + c^2 + 4\sqrt{3}\,\Delta) = r_1^2 + r_2^2 + r_3^2 + 2(r_1 r_2 + r_2 r_3 + r_3 r_1) = (r_1 + r_2 + r_3)^2,$$

which proves (49.96). If an angle is at least $2\pi/3$, the stationarity condition cannot be realized by a point inside the triangle with all three meeting angles $2\pi/3$. The convexity of the sum of distances then places the minimum on the boundary; reducing the same one-dimensional problem along the boundary gives the vertex at the large angle, and the network is the union of the two adjacent sides. $\square$

Figure 49.8 illustrates the Steiner-network length entering the baryonic flux-string model: three strings meet at $2\pi/3$ when the Fermat point lies inside the source triangle, and otherwise the minimum lies on the boundary.

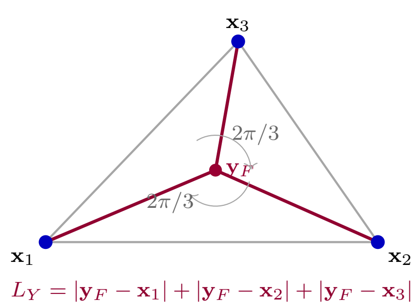


Figure 49.8: For three static fundamental sources in pure $SU(3)$ Yang–Mills theory, the long-distance baryonic string model uses a gauge-invariant color junction. When all triangle angles are below $2\pi/3$, the leading network is the Steiner tree whose three segments meet at $2\pi/3$.

For an equilateral triangle of side a, (49.96) gives

$$L_Y = \sqrt{3}\,a.$$

The frequently used pairwise comparison length

$$L_\Delta = \frac{1}{2}(a+b+c)$$

would give $L_\Delta = 3a/2$ in the same equilateral geometry. These are different large-distance hypotheses for the static baryon sector, not two forms of the same formula. The gauge-invariant object in the baryonic Wilson-line construction contains a junction, and the effective string description must decide, by matching to the gauge theory, whether the long-distance energy is governed by a junction network, by an approximately pairwise regime over the available distances, or by a crossover between the two.

49.8.7 Closed Flux Tubes, Glueball Channels, and Large-N String Coupling

A closed flux tube is cleanly defined on a spatial torus by a spatial Polyakov loop winding a noncontractible cycle,

$$\mathcal{P}_R(\tau) = \mathrm{tr}_R\, \mathcal{P} \exp\left(i \oint_{S^1_L} A_i(\tau,\mathbf{x})\,\mathrm{d}x^i\right).$$

Its Euclidean time correlator has a transfer-matrix expansion

$$\langle \mathcal{P}_R(\tau)^\dagger \mathcal{P}_R(0)\rangle = \sum_n |\langle n;k,L|\mathcal{P}_R(0)\Omega\rangle|^2 \mathrm{e}^{-E_n^{(k)}(L)\tau},$$

where k is the center charge of R. The closed-string formulas (49.76), (49.86), and (49.88) apply to this winding sector under the large-L effective-string hypotheses stated above.

This winding-sector statement is sharper than the phrase "glueballs are closed strings." A glueball is a state in the vacuum sector created by local gauge-invariant operators such as $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ and $\mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu})$. A noncontractible winding flux tube belongs to a different finite-volume sector. A contractible closed flux loop, by contrast, can shrink to a size of order $\sigma^{-1/2}$, where the long-string derivative expansion is no longer controlled. Thus the closed-string picture of glueball spectra and Regge trajectories is a dynamical large-N_c and effective-string hypothesis, not a consequence of the Gaussian Luscher calculation. What is nonperturbatively well defined are the gauge-invariant correlators and their spectral decomposition; the string interpretation is an additional infrared organization whose range must be tested or derived.

Large N_c supplies the natural weak-coupling parameter for such a string interpretation once a confining flux-tube spectrum exists. The ribbon-graph counting in the preceding section gives a connected contribution of order

$$N_c^{2-2h-b}$$

for a surface of genus h with b color-trace boundaries before any chosen normalization of the external loop operators. A handle is therefore suppressed by N_c^{-2}. Identifying the closed-string genus expansion as g_s^{2h-2} gives the parametric relation

$$g_s \sim \frac{1}{N_c}.$$

The same counting implies that splitting and joining processes of normalized closed-string states carry the corresponding factors of N_c^{-1} fixed by the number of topology-changing vertices. This statement is a consequence of color topology; it is not a derivation of the worldsheet action. The microscopic input still required is the existence of a confining large-N_c gauge theory with isolated flux-tube and glueball states in the relevant channels.

Three-dimensional confining gauge theories are especially clean testbeds for the same logic. In $D = 3$ there is a single transverse displacement field, so the universal terms become

$$\Delta E_0^{\text{open}}(L) = -\frac{\pi}{24L}, \qquad \Delta E_0^{\text{closed}}(L) = -\frac{\pi}{6L}.$$

The reduced transverse symmetry makes the lattice spectral assignments less crowded than in four dimensions, while the effective-string expansion has the same conceptual structure: line operators define the sector, the transfer matrix extracts energies, the free transverse field gives the universal Casimir term, and later coefficients encode symmetry constraints and microscopic matching.

With dynamical fundamental quarks the fundamental one-form symmetry is explicitly broken. The static-source Hilbert space then contains states made from two color-singlet static-light mesons, and the ground-state energy at very large L approaches the two-meson threshold rather than growing without bound. A flux tube can still be a well-defined long-lived state over a range of L, and its worldsheet effective theory can describe the corresponding resonant or metastable channel, but the limiting string tension (49.63) is no longer an order parameter in the same sense as in pure Yang–Mills theory. This distinction is essential when using string language in QCD: line operators, static-source sectors, and calorimetric observables are physical, and asymptotic final states are color-singlet hadrons.

49.9 Heavy-Quark Effective Theory and the Wilson-Line Limit

The static-source sector just described is also the first term in a controlled expansion for a dynamical quark whose mass is much larger than the hadronic scale. This expansion is called heavy-quark effective theory (HQET). Its mathematical content is the regulator-dependent comparison of gauge-invariant QCD matrix elements with gauge-invariant matrix elements of a local expansion in inverse powers of the heavy mass. The heavy quark is still a colored field variable, so physical matrix elements are made gauge invariant by the same Wilson-line and color-singlet constructions used elsewhere in this chapter.

Throughout this section the spacetime metric is mostly plus, $\eta_{\mu\nu} = \mathrm{diag}(-,+,+,+)$. A heavy-quark velocity is a future-directed timelike unit vector v^μ satisfying $v^2 = -1$. The orthogonal projector onto the three-plane transverse to v is

$$\Pi_v{}^\mu{}_\nu = \delta^\mu{}_\nu + v^\mu v_\nu, \qquad D_\perp^\mu = \Pi_v{}^\mu{}_\nu D^\nu. \tag{49.97}$$

The identity $v_\mu D_\perp^\mu = 0$ follows immediately from $v^2 = -1$. A near-mass-shell heavy momentum is written

$$p^\mu = m_Q v^\mu + k^\mu, \tag{49.98}$$

where m_Q is the heavy-quark mass coordinate in the chosen regulated QCD Lagrangian and the residual momentum k^μ is required to be small compared with m_Q on the states or test functions under consideration. The split is not unique: replacing v by $v + \delta v$, with $v \cdot \delta v = 0$ to first order, and replacing k by $k - m_Q \delta v$ leaves p fixed. This elementary redundancy is the kinematic origin of reparametrization constraints on the effective action.

Let $\gamma_v = \gamma_\mu v^\mu$. Since $\gamma_v^2 = -1$, the operator $\mathrm{i}\gamma_v$ has eigenvalues ± 1, and

$$P_v^\pm = \frac{1}{2}(1 \pm \mathrm{i}\gamma_v) \tag{49.99}$$

are complementary spinor projections. The sign in (49.99) is tied to the phase convention below. Define $\tau_v(x) = -v \cdot x$; in the rest frame $v = (1, 0, 0, 0)$, this is the ordinary time coordinate. A heavy Dirac field Q is decomposed at fixed velocity by

$$Q(x) = \mathrm{e}^{-\mathrm{i}m_Q \tau_v(x)}\big(h_v(x) + H_v(x)\big), \qquad P_v^+ h_v = h_v, \qquad P_v^- H_v = H_v. \tag{49.100}$$

The phase removes the rest-energy oscillation associated with $m_Q v^\mu$. The field H_v carries the complementary Dirac components and is separated from h_v by a gap of order m_Q in the free theory.

Definition 49.13 (HQET regulator specification)**.** A finite-cutoff HQET regulator specification at velocity v consists of:

1. A gauge-covariant ultraviolet regulator for QCD with the heavy field Q and mass coordinate m_Q, together with an infrared domain of states or test functions whose residual momenta obey $|k|/m_Q \le \varepsilon \ll 1$.
2. The projected field $h_v = P_v^+ h_v$, transforming in the fundamental representation of the gauge group, and a Wilsonian procedure which integrates out H_v and modes whose virtuality is of order m_Q.
3. A local gauge-covariant action

$$S^{(N)}_{\mathrm{HQET}}[h_v, A] = \int \bar{h}_v\, \mathrm{i}v \cdot D\, h_v \,\mathrm{d}^4x + \sum_{r=1}^{N} \frac{1}{(2m_Q)^r} \sum_a C_{r,a}(\mu, m_Q) \int \mathcal{O}_{r,a}(x)\,\mathrm{d}^4x, \tag{49.101}$$

   where each $\mathcal{O}_{r,a}$ is a local gauge-invariant density built from $h_v, \bar{h}_v, F_{\mu\nu}$, and transverse covariant derivatives $D_\perp$, modulo equations of motion and integration by parts in the chosen coordinate system on local functionals.
4. Renormalized composite operators, for example

$$\bar{q}\,\Gamma Q \longmapsto \sum_{r,a} \frac{1}{(2m_Q)^r} C_{\Gamma,r,a}(\mu, m_Q)\, \mathcal{J}^{\mathrm{HQET}}_{\Gamma,r,a}[q, h_v, A],$$

   with the same regulator and subtraction data. A current matching statement is a statement about matrix elements of these renormalized operators, not an identity between classical monomials.

The leading action is normalized so that the coefficient of $\bar{h}_v \mathrm{i}v \cdot D h_v$ is one. The first dimension-five operators may be represented by

$$\mathcal{O}_{\mathrm{kin}} = \bar{h}_v D_{\perp\mu} D_\perp^\mu h_v, \qquad \mathcal{O}_{\mathrm{mag}} = \bar{h}_v \sigma_{\mu\nu} F^{\mu\nu} h_v, \tag{49.102}$$

with coefficients fixed by the chosen matching convention. In a reparametrization-compatible scheme the coefficient of the kinetic operator is fixed relative to the leading term; the chromomagnetic coefficient is a genuine short-distance matching coefficient.

The leading HQET equation is the equation for parallel transport along the heavy-quark worldline. This is the precise sense in which an infinitely heavy quark becomes a Wilson line.

Static propagator as a Wilson line. Let $x(s) = x_0 + sv$, and let

$$W_v(s_2, s_1) = \mathcal{P} \exp \left\{ \mathrm{i} \int_{s_1}^{s_2} v^\mu A_\mu^a(x(s)) t_\square^a \, \mathrm{d}s \right\} \tag{49.103}$$

be the fundamental Wilson transporter along the heavy trajectory. At finite regulator and with background A_μ smooth on the scale of the regulator, the retarded Green kernel of the leading HQET operator on this trajectory is

$$G_v(s_2, s_1) = \theta(s_2 - s_1) \, W_v(s_2, s_1) \, P_v^+, \tag{49.104}$$

up to the conventional overall factor fixed by the normalization of the first-order inverse. In particular, every gauge-invariant heavy-light or heavy-heavy correlator at leading power contains the Wilson line carried by the heavy source. The calculation is the Green-kernel inversion of the first-order covariant derivative. The transporter satisfies the initial-value problem

$$\frac{\mathrm{d}}{\mathrm{d}s_2} W_v(s_2, s_1) = \mathrm{i} v^\mu A_\mu^a(x(s_2)) t_\square^a W_v(s_2, s_1), \qquad W_v(s_1, s_1) = \mathbf{1}.$$

Equivalently $v \cdot D_{x(s_2)} W_v(s_2, s_1) = 0$, using the chapter convention $D_\mu = \partial_\mu - \mathrm{i} A_\mu$. Applying the first-order operator to $\theta(s_2 - s_1) W_v(s_2, s_1) P_v^+$ gives only the derivative of the step function, because the covariant derivative annihilates the transporter for $s_2 \neq s_1$. This is the delta distribution on the oriented line. The projector P_v^+ records the spinor components retained in the heavy field. Gauge covariance follows from the same Wilson-transporter endpoint transformation law displayed in (49.292).

Consider N_h heavy flavors whose masses have been removed by separate phases and whose residual dynamics are described by the leading HQET action at the same velocity v. The leading action

$$S_0 = \sum_{A=1}^{N_h} \int \bar{h}_{v,A} \, \mathrm{i} v \cdot D \, h_{v,A} \, \mathrm{d}^4 x$$

is invariant under unitary rotations of the heavy-flavor labels and under unitary rotations of the projected spin components commuting with P_v^+. These symmetries act on the heavy variables alone and leave the gauge and light-quark fields fixed. The operator $\mathrm{i} v \cdot D$ is scalar in the heavy-flavor labels and contains no gamma matrix acting inside the projected spin space. If $h_{v,A} \mapsto U_A{}^B S h_{v,B}$, where U is a unitary matrix on flavor labels and S is unitary on the image of P_v^+, then $\bar{h}_{v,A} \mathrm{i} v \cdot D h_{v,A}$ is unchanged after summing over A. The Wilson-line form of the propagator makes the same statement in correlator language: at leading power the heavy source transports color, while its spin and heavy-flavor labels are parallel spectators.

For a free heavy particle with $p^\mu = m_Q v^\mu + k^\mu$ and $p^2 + m_Q^2 = 0$, the residual momentum obeys

$$2 m_Q \, v \cdot k + k^2 = 0. \tag{49.105}$$

In the rest frame, writing $k^\mu = (E_{\mathrm{res}}, \mathbf{k})$, this gives

$$E_{\mathrm{res}} = \sqrt{m_Q^2 + |\mathbf{k}|^2} - m_Q = \frac{|\mathbf{k}|^2}{2 m_Q} - \frac{|\mathbf{k}|^4}{8 m_Q^3} + O\left(\frac{|\mathbf{k}|^6}{m_Q^5} \right). \tag{49.106}$$

Thus the first correction to the static theory is the transverse kinetic energy. Gauge covariance promotes the transverse momentum to $D_\perp$, giving the operator $\mathcal{O}_{\rm kin}$ in (49.102). Indeed, substituting $p = m_Q v + k$ and $v^2 = -1$ into $p^2 + m_Q^2 = 0$ gives (49.105). In the rest frame, $p = (m_Q + E_{\rm res}, \mathbf{k})$, so the positive-energy solution is $m_Q + E_{\rm res} = \sqrt{m_Q^2 + |\mathbf{k}|^2}$. Expanding the square root in $|\mathbf{k}|^2/m_Q^2$ gives (49.106). The covariant operator statement follows from replacing the residual transverse momentum by the transverse covariant derivative in the local Wilsonian expansion; the coefficient is fixed by matching the free dispersion and, in an interacting reparametrization-compatible scheme, by the invariance of the momentum split (49.98).

Controlled Approximation 49.14 (Meaning of the heavy-mass expansion)**.** Fix a regulator, a velocity v, a renormalization scale $\mu \ll m_Q$, and a class of gauge-invariant external states and operator insertions whose residual momenta and invariant hadronic scales are bounded by $\Lambda_{\rm had}$. A truncation of HQET at order N asserts an estimate of the form

$$\left| \langle \mathcal{O} \rangle_{\rm QCD} - \langle \mathcal{O}^{(N)}_{\rm HQET} \rangle_{\rm HQET} \right| \le C_{\mathcal{O},N} \left(\frac{\Lambda_{\rm had}}{m_Q} \right)^{N+1}$$

for the chosen smeared matrix elements, after the QCD operator and the HQET operator have been matched in the same renormalization scheme. The constant $C_{\mathcal{O},N}$, the topology in which the estimate is meant, and the range of μ are part of the assertion. The Wilson-line limit derived above supplies the leading operator structure; the existence of a uniform heavy-mass expansion for a specified family of hadronic matrix elements is additional dynamical and renormalization information.

49.10 Heavy-Light Currents and Heavy-Quark Symmetry

The static Wilson-line limit becomes predictive only after composite currents and hadron states are matched. Let q be a light quark field and let Q be a heavy quark field. The mostly-plus spinor convention of Definition 40.19 fixes the Hermitian vector and axial currents with an explicit factor of i. Thus in this section

$$A^\mu_{qQ} = \mathrm{i}[\bar{q}\gamma^\mu\gamma_5 Q]_{\rm R}, \qquad V^\mu_{qQ} = \mathrm{i}[\bar{q}\gamma^\mu Q]_{\rm R}.$$

More generally a renormalized heavy-light QCD current is a renormalized gauge-invariant local operator

$$J^{\rm QCD}_\Gamma(x) = [\kappa_\Gamma \bar{q}(x)\Gamma Q(x)]_{\rm R},$$

where Γ is a fixed spinor matrix, $\kappa_\Gamma \in \{1, \mathrm{i}\}$ is part of the current convention, and the subscript records the regulator and subtraction convention. In HQET this current is represented by an expansion

$$J^{\rm QCD}_\Gamma = C_\Gamma(m_Q,\mu)\,[\kappa_\Gamma \bar{q}\Gamma h_v]_{\rm R} + \sum_a \frac{B_{\Gamma,a}(m_Q,\mu)}{2m_Q}\,\mathcal{J}^{(1)}_{\Gamma,a} + O(m_Q^{-2}) \tag{49.107}$$

inside the chosen class of matrix elements. The operators $\mathcal{J}^{(1)}_{\Gamma,a}$ are local gauge-invariant HQET currents with one additional transverse derivative, chromomagnetic insertion, or light-quark mass insertion, modulo the leading equations of motion in the chosen operator basis. Equation (49.107) is a matching statement for renormalized operator-valued distributions tested on low-residual-momentum states.

Assume that for a heavy flavor Q there is an isolated stable pseudoscalar meson P_Q and an isolated stable vector meson V_Q in the same light-cloud channel. Their decay constants are defined by

$$\langle 0|\, A^{\mu}_{qQ}\, |P_Q(p)\rangle = f_{P_Q}\, p^{\mu}, \tag{49.108}$$

$$\langle 0|\, V^{\mu}_{qQ}\, |V_Q(p,\varepsilon)\rangle = f_{V_Q}\, M_{V_Q}\, \varepsilon^{\mu}, \tag{49.109}$$

where $p^2 = -M^2$, $p^0 > 0$, and the vector polarization satisfies $p \cdot \varepsilon = 0$. Momentum eigenstates in QCD are normalized by

$$\langle H(p')|H(p)\rangle = 2p^0(2\pi)^3\delta^3(\mathbf{p} - \mathbf{p}').$$

The corresponding HQET residual-momentum state is normalized by

$${}_{\mathrm{HQET}}\langle H_v(k')|H_v(k)\rangle_{\mathrm{HQET}} = 2v^0(2\pi)^3\delta^3(\mathbf{k} - \mathbf{k}'),$$

with $p^{\mu} = M_H v^{\mu} + k^{\mu}$, and the leading state matching is

$$|H_Q(p = M_H v + k)\rangle_{\mathrm{QCD}} = \sqrt{M_H}\, |H_v(k)\rangle_{\mathrm{HQET}} + O(\Lambda_{\mathrm{had}}/m_Q) \tag{49.110}$$

for matrix elements in the stated residual-momentum domain.

Heavy-light decay-constant scaling. Let

$${}_{\mathrm{HQET}}\langle 0|\, \mathrm{i}[\bar{q}\gamma^{\mu}\gamma_5 h_v]_{\mathrm{R}}|P_v(0)\rangle_{\mathrm{HQET}} = F_P(\mu)\, v^{\mu}.$$

Then the leading HQET matching of the axial current gives

$$f_{P_Q}\sqrt{M_{P_Q}} = C_A(m_Q,\mu)F_P(\mu) + O(\Lambda_{\mathrm{had}}/m_Q). \tag{49.111}$$

The analogous vector-meson relation has the same static reduced matrix element up to the spin rotation dictated by the leading static symmetry described above. At $p^{\mu} = M_{P_Q} v^{\mu}$, the QCD definition (49.108) gives

$$\langle 0|A^{\mu}_{qQ}|P_Q(p)\rangle_{\mathrm{QCD}} = f_{P_Q} M_{P_Q} v^{\mu}.$$

Using the leading current matching

$$A^{\mu}_{qQ} = C_A(m_Q,\mu)\, \mathrm{i}[\bar{q}\gamma^{\mu}\gamma_5 h_v]_{\mathrm{R}} + O(m_Q^{-1})$$

and the state relation (49.110) gives

$$\langle 0|A^{\mu}_{qQ}|P_Q(p)\rangle_{\mathrm{QCD}} = C_A\sqrt{M_{P_Q}}\, F_P(\mu)v^{\mu} + O(\Lambda_{\mathrm{had}}/m_Q).$$

Equating the coefficients of v^{μ} gives (49.111). The vector statement follows because the leading static action transports color but contains no operator acting on the projected heavy spin; pseudoscalar and vector states in the same light cloud are related by the heavy-spin symmetry of the leading theory.

Heavy-to-heavy currents compare two static sectors with velocities v and v'. Define the recoil variable

$$w = -v \cdot v' \geq 1. \tag{49.112}$$

For pseudoscalar ground-state mesons in the same light-cloud channel, the leading HQET vector-current matrix element has the form

$$_{\mathrm{HQET}}\langle H_{v'}|\, \mathcal{V}^{\mu}_{Q'Q}(v', v)\, |H_v\rangle_{\mathrm{HQET}} = \xi(w)(v + v')^{\mu}, \tag{49.113}$$

where

$$\mathcal{V}^{\mu}_{Q'Q}(v', v) = \mathrm{i}[\bar{h}^{(Q')}_{v'} \gamma^{\mu} h^{(Q)}_{v}]_{\mathrm{R}}$$

with the coefficient fixed by coupling the leading HQET action to a background heavy-flavor gauge field. The scalar function ξ is the leading heavy-quark symmetry form factor. For the full QCD states this corresponds to

$$\begin{aligned} &\langle H_{Q'}(M_{Q'}v')|\, \mathrm{i}[\bar{Q}'\gamma^{\mu}Q]_{\mathrm{R}}\, |H_Q(M_Q v)\rangle \\ &\quad = \sqrt{M_{Q'}M_Q}\, C_{Q'Q}(m_{Q'}, m_Q, \mu)\, \xi(w)(v + v')^{\mu} + O(\Lambda_{\mathrm{had}}/m_Q, \Lambda_{\mathrm{had}}/m_{Q'}). \end{aligned} \tag{49.114}$$

More general spin states are obtained from the same light-cloud overlap by inserting the appropriate heavy-spin trace factors.

Lemma 49.15 (Zero-recoil normalization)**.** *At zero recoil, $v' = v$, the leading heavy-quark symmetry form factor obeys*

$$\xi(1) = 1 \tag{49.115}$$

when the current is normalized as the generator of the heavy-flavor symmetry of the leading HQET action.

Proof. At $v' = v$, the leading HQET action has a unitary flavor symmetry rotating the heavy labels Q and Q' while leaving the light fields and gauge field fixed. This statement is made inside the leading static theory, before the short-distance matching coefficient $C_{Q'Q}$ and before $1/m_Q$ corrections are restored. Couple this symmetry to a smooth background matrix-valued connection ${B_{\mu}}^{A}{}_{B}$ by replacing $D_{\mu}h_{v,A}$ with $(D_{\mu}{\delta_A}^{B} - \mathrm{i}{B_{\mu}}^{B}{}_{A})h_{v,B}$. Differentiation of the action with respect to ${B_{\mu}}^{Q}{}_{Q'}$ defines the current $\mathcal{V}^{\mu}_{Q'Q}(v, v)$, and the associated charge

$$Q_{Q'Q} = \int_{\Sigma_t} \mathrm{d}^3x\, n_{\mu}\, \mathcal{V}^{\mu}_{Q'Q}(v, v)$$

is the generator of the heavy-flavor rotation on the one-heavy-particle sector. Here n_{μ} is the future unit normal covector to the quantization slice. Choose the isolated ground states in the same light-cloud channel with the same Hilbert-space norm and with phases fixed so that the flavor rotation maps one state to the other. The Ward identity for the background flavor source then gives

$$Q_{Q'Q}|H^{(Q)}_v\rangle = |H^{(Q')}_v\rangle.$$

Translation covariance in the fixed-velocity sector and the absence of any other vector built from a single velocity imply that the local current matrix element has the form $A\, v^{\mu}$. Integrating $n_{\mu}Av^{\mu}$ over the normalization volume and using the preceding charge action fixes $A = 2$ in the velocity-state normalization used in (49.113). Equivalently,

$$_{\mathrm{HQET}}\langle H^{(Q')}_v|\mathcal{V}^{\mu}_{Q'Q}(v, v)|H^{(Q)}_v\rangle_{\mathrm{HQET}} = 2v^{\mu}.$$

Comparing with (49.113) at $v' = v$ gives $\xi(1) = 1$. ∎

Controlled Approximation 49.16 (Heavy-light current expansion)**.** A heavy-light or heavy-to-heavy form-factor prediction must specify the renormalized QCD current, the HQET current basis, the state normalization, the matching coefficients, and the spectral assumption that the meson states used in the matrix element are isolated in the relevant channel. The leading relations (49.111) and (49.113) are symmetry consequences of the static theory after these data are fixed. Corrections of order $1/m_Q$ are matrix elements of the subleading current operators and of insertions from the subleading HQET action; their absence or suppression at special kinematic points is an additional theorem only when the corresponding operator basis and symmetry argument are displayed.

49.11 Nonrelativistic Heavy Pairs and Potential Matching

Quarkonium requires a heavy-mass expansion with two heavy fields and a specified time foliation. Choose the rest frame of the heavy pair and write t for its time coordinate. A nonrelativistic heavy quark is represented by a two-component Pauli field $\psi(t, \mathbf{x})$ in the fundamental representation. The corresponding antiquark is represented, after charge conjugation, by another Pauli field $\chi(t, \mathbf{x})$ which also transforms in the fundamental representation; its adjoint $\chi^\dagger$ creates an antiquark and transforms as a row antifundamental. With

$$D_t = \partial_t - \mathrm{i}A_0, \qquad \mathbf{D} = \nabla - \mathrm{i}\mathbf{A},$$

the leading nonrelativistic action is

$$S^{(0)}_{\mathrm{NR}} = \int \mathrm{d}t\,\mathrm{d}^3\mathbf{x}\,\left[\psi^\dagger\left(\mathrm{i}D_t + \frac{\mathbf{D}^2}{2m_Q}\right)\psi + \chi^\dagger\left(\mathrm{i}D_t + \frac{\mathbf{D}^2}{2m_Q}\right)\chi\right]. \tag{49.116}$$

The sign of the spatial Laplacian in (49.116) is fixed by the convention $D_i e^{\mathrm{i}\mathbf{p}\cdot\mathbf{x}} = \mathrm{i}p_i e^{\mathrm{i}\mathbf{p}\cdot\mathbf{x}}$ in the free theory.

Definition 49.17 (NRQCD regulator and operator specification)**.** An NRQCD regulator and operator specification for a heavy flavor of mass m_Q consists of: items.

1. A gauge-covariant regulator for QCD and a matching scale below m_Q, together with a class of states whose heavy-quark three-momenta obey $|\mathbf{p}|/m_Q \ll 1$.
2. Pauli fields ψ, χ, the gauge fields, and a local action of the form

   $$S^{(N)}_{\mathrm{NRQCD}} = S^{(0)}_{\mathrm{NR}} + \sum_a \frac{c_a(\mu, m_Q)}{m_Q^{d_a-4}} \int \mathcal{O}^{\mathrm{NR}}_a(t, \mathbf{x})\,\mathrm{d}t\,\mathrm{d}^3\mathbf{x},$$

   where $\mathcal{O}^{\mathrm{NR}}_a$ ranges over gauge-invariant local operators built from $\psi, \chi, F_{\mu\nu}$, covariant derivatives, Pauli matrices, and local four-heavy-field annihilation operators, ordered by the chosen velocity and $1/m_Q$ power counting.
3. Renormalized heavy-pair composite operators. A quarkonium interpolator or current is part of the specification only after its Wilson-line dressing, regulator, and subtraction convention have been specified.

The Pauli magnetic term

$$\frac{c_F}{2m_Q}\,\psi^\dagger\sigma^i B_i\psi-\frac{c_F}{2m_Q}\,\chi^\dagger\sigma^i B_i\chi,\qquad B_i=\frac{1}{2}\epsilon_{ijk}F_{jk},$$

is the first spin-dependent single-particle correction in a parity-invariant coordinate system. Its coefficient is a matching coefficient; it is not deduced from the leading Schrödinger action alone.

Free NRQCD dispersion. The free equation following from the leading quark term in (49.116) has plane-wave solutions

$$\psi(t,\mathbf{x})=u\,\mathrm{e}^{-\mathrm{i}Et+\mathrm{i}\mathbf{p}\cdot\mathbf{x}}$$

with

$$E=\frac{|\mathbf{p}|^2}{2m_Q}.$$

Thus (49.116) realizes the leading term of the relativistic residual-energy expansion (49.106). Substitution of the plane wave gives

$$\left(E-\frac{|\mathbf{p}|^2}{2m_Q}\right)u=0,$$

because $\nabla^2\mathrm{e}^{\mathrm{i}\mathbf{p}\cdot\mathbf{x}}=-|\mathbf{p}|^2\mathrm{e}^{\mathrm{i}\mathbf{p}\cdot\mathbf{x}}$. A nonzero spinor amplitude u therefore requires the displayed dispersion relation.

Gauge-invariant quarkonium operators are bilocal at nonzero separation. Let

$$\mathbf{x}_1=\mathbf{R}+\frac{\mathbf{r}}{2},\qquad \mathbf{x}_2=\mathbf{R}-\frac{\mathbf{r}}{2},$$

and let $W(\mathbf{x}_2,\mathbf{x}_1;t)$ be the equal-time fundamental Wilson transporter from $\mathbf{x}_1$ to $\mathbf{x}_2$. Define the singlet bilocal

$$S(t,\mathbf{R},\mathbf{r})=\frac{1}{\sqrt{N_c}}\,\chi^\dagger(t,\mathbf{x}_2)W(\mathbf{x}_2,\mathbf{x}_1;t)\psi(t,\mathbf{x}_1). \tag{49.117}$$

An octet bilocal with a color label based at $\mathbf{R}$ is represented by

$$O^a(t,\mathbf{R},\mathbf{r})=\chi^\dagger(t,\mathbf{x}_2)W(\mathbf{x}_2,\mathbf{R};t)\,t^a\,W(\mathbf{R},\mathbf{x}_1;t)\psi(t,\mathbf{x}_1). \tag{49.118}$$

The octet transforms in the adjoint representation at $\mathbf{R}$; the singlet is gauge invariant.

Gauge covariance of the bilocals. Under a gauge transformation $U(t,\mathbf{x})$,

$$\psi(\mathbf{x}_1)\mapsto U(\mathbf{x}_1)\psi(\mathbf{x}_1),\qquad \chi^\dagger(\mathbf{x}_2)\mapsto\chi^\dagger(\mathbf{x}_2)U(\mathbf{x}_2)^{-1},$$

and

$$W(\mathbf{x}_2,\mathbf{x}_1)\mapsto U(\mathbf{x}_2)W(\mathbf{x}_2,\mathbf{x}_1)U(\mathbf{x}_1)^{-1}.$$

The endpoint matrices cancel in (49.117). In (49.118) the endpoint matrices again cancel, while the generator at $\mathbf{R}$ transforms by conjugation, $t^a\mapsto U(\mathbf{R})t^aU(\mathbf{R})^{-1}$, which is the adjoint action.

Potential NRQCD is obtained by integrating out the relative-momentum scale $m_Q v_{\rm rel}$ while retaining the lower energy scale $m_Q v_{\rm rel}^2$. The singlet part of the resulting bilocal action has the form

$$S^S_{\rm pNRQCD} = \int {\rm d}t\, {\rm d}^3\mathbf{R}\, {\rm d}^3\mathbf{r}\, S^\dagger \left[{\rm i}\partial_t + \frac{\nabla_{\mathbf{r}}^2}{m_Q} - V_S(r;\mu) + \cdots \right] S, \tag{49.119}$$

where $r = |\mathbf{r}|$. The potential $V_S(r;\mu)$ is a matching coefficient for the bilocal singlet operator. In a weak-coupling matching regime, the tree-level Coulomb exchange gives

$$V_S^{(0)}(r) = -\frac{g^2 C_F}{4\pi r}, \tag{49.120}$$

with $C_F = (N_c^2 - 1)/N_c$ and the same trace-delta coupling used throughout this chapter. This formula is invariant under conversion to the common half-trace convention because $g_{\rm ht}^2 = 2g^2$ and $C_F^{\rm ht} = C_F/2$.

Tree-level singlet color factor. In the trace convention of (49.1), the color factor for one-gluon exchange in a normalized quark-antiquark singlet is $-C_F$. Hence the attractive Coulomb potential is (49.120).

The normalized singlet color tensor is $\delta^i{}_j/\sqrt{N_c}$. The antiquark generator is the negative transpose of the quark generator. Acting with $t^a \otimes (-t^{aT})$ on the singlet gives

$$-\frac{1}{\sqrt{N_c}}(t^a t^a)^i{}_j = -C_F \frac{\delta^i{}_j}{\sqrt{N_c}}.$$

The spatial Fourier transform of the Coulomb kernel is

$$\int \frac{{\rm d}^3\mathbf{q}}{(2\pi)^3} \frac{{\rm e}^{{\rm i}\mathbf{q}\cdot\mathbf{r}}}{|\mathbf{q}|^2} = \frac{1}{4\pi r},$$

so the product of the exchange kernel $g^2/|\mathbf{q}|^2$ with the singlet color eigenvalue gives (49.120).

Leading Coulombic $1S$ quarkonium level. Assume a weak-coupling pNRQCD chart in which the spin-independent singlet Hamiltonian is truncated to

$$H_C = -\frac{\nabla_{\mathbf{r}}^2}{m_Q} - \frac{\alpha_C}{r}, \qquad \alpha_C := \frac{g^2 C_F}{4\pi}, \qquad r = |\mathbf{r}|. \tag{49.121}$$

Here m_Q is the heavy-quark mass coordinate in the same short-distance scheme as the potential coordinate. The normalized $1S$ eigenfunction

$$\psi_{1S}(\mathbf{r}) = \left(\frac{\kappa^3}{\pi}\right)^{1/2} {\rm e}^{-\kappa r}, \qquad \kappa := \frac{m_Q \alpha_C}{2}, \tag{49.122}$$

has energy

$$E_{1S}^{(0)} = -\frac{m_Q \alpha_C^2}{4}, \qquad |\psi_{1S}(0)|^2 = \frac{(m_Q\alpha_C)^3}{8\pi}. \tag{49.123}$$

The corresponding leading spin-averaged quarkonium mass coordinate is

$$M_{1S}^{(0)} = 2m_Q + E_{1S}^{(0)}$$

before the additive mass–potential rearrangements of Section 49.12 and before ultrasoft, relativistic, and spin-dependent corrections are included.

For $r > 0$,

$$\nabla^2 \mathrm{e}^{-\kappa r} = \left(\kappa^2 - \frac{2\kappa}{r}\right) \mathrm{e}^{-\kappa r}.$$

Thus

$$H_C \mathrm{e}^{-\kappa r} = \left[-\frac{\kappa^2}{m_Q} + \left(\frac{2\kappa}{m_Q} - \alpha_C\right)\frac{1}{r}\right] \mathrm{e}^{-\kappa r}.$$

With $\kappa = m_Q\alpha_C/2$ the $1/r$ term cancels and the eigenvalue is $-\kappa^2/m_Q = -m_Q\alpha_C^2/4$. The normalization follows from

$$4\pi \left(\frac{\kappa^3}{\pi}\right) \int_0^\infty r^2 \mathrm{e}^{-2\kappa r}\, \mathrm{d}r = 4\kappa^3 \frac{2}{(2\kappa)^3} = 1,$$

and the value at the origin is the square of the normalization constant. The mass formula is then the definition of the spin-averaged singlet energy coordinate in this leading Hamiltonian. This calculation does not define the physical charmonium or bottomonium spectrum by itself: it computes the first term in a declared pNRQCD expansion whose mass scheme, matching scale, and omitted operators must be specified.

Quarkonium fine and hyperfine spin algebra. Fix a quarkonium radial level and orbital angular momentum L in a pNRQCD specification. Let

$$\mathbf{S}_Q = \frac{1}{2}\boldsymbol{\sigma}_Q, \qquad \mathbf{S}_{\bar{Q}} = \frac{1}{2}\boldsymbol{\sigma}_{\bar{Q}}, \qquad \mathbf{S} = \mathbf{S}_Q + \mathbf{S}_{\bar{Q}}, \qquad \mathbf{J} = \mathbf{L} + \mathbf{S}.$$

Assume that, to first order in the spin-dependent part of the matched Hamiltonian, the restriction to this radial-orbital subspace has the form

$$H_{nL}^{\mathrm{sd}} = A_{nL}\, \mathbf{S}_Q \cdot \mathbf{S}_{\bar{Q}} + B_{nL}\, \mathbf{L} \cdot \mathbf{S} + C_{nL}\, \mathcal{T}_{LS}, \tag{49.124}$$

where A_{nL}, B_{nL}, C_{nL} are radial matrix elements or matching coefficients in the chosen scheme, and $\mathcal{T}_{LS}$ is the rotational scalar obtained by coupling a traceless rank-two orbital tensor to a traceless rank-two spin tensor. Then

$$\mathbf{S}_Q \cdot \mathbf{S}_{\bar{Q}} = \begin{cases} -\frac{3}{4}, & S = 0, \\ \frac{1}{4}, & S = 1, \end{cases} \tag{49.125}$$

and, on the triplet $S = 1$ states,

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}\big(J(J+1) - L(L+1) - 2\big) = \begin{cases} -(L+1), & J = L - 1, \\ -1, & J = L, \\ L, & J = L + 1. \end{cases} \tag{49.126}$$

If $M(^1L_L)$ denotes the singlet mass and $M(^3L_J)$ the triplet masses computed to this first order from a common spin-independent radial level $E_{nL}^{(0)}$, then the triplet centroid obeys

$$\overline{M}(^3L) := \frac{\sum_{J=L-1}^{L+1}(2J+1)M(^3L_J)}{3(2L+1)} = E_{nL}^{(0)} + \frac{1}{4}A_{nL}, \tag{49.127}$$

with the obvious omission of the nonexistent $J = L - 1$ term when $L = 0$. The centroid hyperfine difference is therefore

$$\overline{M}(^3L) - M(^1L_L) = A_{nL}. \tag{49.128}$$

The two heavy spins are spin-$\frac{1}{2}$ representations. Since

$$\mathbf{S}^2 = \mathbf{S}_Q^2 + \mathbf{S}_{\bar{Q}}^2 + 2\mathbf{S}_Q \cdot \mathbf{S}_{\bar{Q}}, \qquad \mathbf{S}_Q^2 = \mathbf{S}_{\bar{Q}}^2 = \frac{3}{4},$$

one has

$$\mathbf{S}_Q \cdot \mathbf{S}_{\bar{Q}} = \frac{1}{2}\left(S(S+1) - \frac{3}{2}\right),$$

which gives (49.125) for $S = 0, 1$. Similarly

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S}$$

gives the first equality in (49.126); substituting $S = 1$ and $J = L - 1, L, L + 1$ gives the displayed three values.

It remains to check the centroid. The trace of $\mathbf{L} \cdot \mathbf{S}$ on the tensor product of the orbital L space and the triplet spin space vanishes:

$$\mathrm{Tr}_{L \otimes S=1}(\mathbf{L} \cdot \mathbf{S}) = \sum_i \mathrm{Tr}_L(L_i)\, \mathrm{Tr}_{S=1}(S_i) = 0,$$

because every generator of an irreducible $SU(2)$ representation is traceless. The operator $\mathcal{T}_{LS}$ is a contraction of traceless rank-two tensors, so its trace on the same tensor product also vanishes. The weighted sum $\sum_J (2J+1) M(^3L_J)$ is precisely this trace after decomposition into irreducible J summands. Therefore only the spin-independent energy and the triplet spin-spin eigenvalue $\frac{1}{4}A_{nL}$ survive in the centroid, proving (49.127). The singlet has $\mathbf{L} \cdot \mathbf{S} = 0$, no rank-two spin tensor, and spin-spin eigenvalue $-\frac{3}{4}$; subtracting its first-order mass from the triplet centroid gives (49.128). The spin algebra above is the part of quarkonium spectroscopy that follows from angular momentum and the matched operator form alone. The numerical values of A_{nL}, B_{nL}, C_{nL} are QCD matching data. In weak-coupling pNRQCD they are computed from perturbative potentials and wavefunctions in a stated mass scheme; in strong-coupling pNRQCD they are Wilson-loop and chromofield insertion matrix elements. In either case the symbols "hyperfine splitting" and "fine structure" refer to the spectral differences defined by (49.126)–(49.128), rather than to a freestanding constituent-potential model.

Leading weak-coupling $1S$ hyperfine coordinate. In the weak-coupling $1S$ chart above, retain the leading chromomagnetic contact operator

$$H_{\mathrm{ss}}^{1S} = \frac{8\pi\alpha_C\, c_F(\mu)^2}{3m_Q^2}\, \mathbf{S}_Q \cdot \mathbf{S}_{\bar{Q}}\, \delta^{(3)}(\mathbf{r}), \tag{49.129}$$

where $c_F(\mu)$ is the chromomagnetic matching coefficient in the same short-distance mass and coupling scheme as m_Q and α_C. To first order in this operator,

$$M(^3S_1) - M(^1S_0) = \frac{8\pi\alpha_C\, c_F(\mu)^2}{3m_Q^2}|\psi_{1S}(0)|^2 = \frac{m_Q\, c_F(\mu)^2\, \alpha_C^4}{3}. \tag{49.130}$$

The displayed number is a leading coordinate in the declared weak-coupling pNRQCD scheme. It is not by itself a precision prediction for charmonium or bottomonium until the mass scheme, higher-order potentials, ultrasoft terms, nonperturbative corrections, and continuum-limit uncertainties have been specified. First-order perturbation theory in the normalized Coulombic $1S$ state gives

$$\langle H_{\rm ss}^{1S}\rangle_S = \frac{8\pi\alpha_C\, c_F(\mu)^2}{3m_Q^2}|\psi_{1S}(0)|^2\,\langle \mathbf{S}_Q\cdot\mathbf{S}_{\bar Q}\rangle_S.$$

For the triplet $S=1$ the spin expectation value is $1/4$, and for the singlet $S=0$ it is $-3/4$, as shown in (49.125). Their difference is one, so the triplet-singlet mass splitting is exactly the contact coefficient times $|\psi_{1S}(0)|^2$. Substituting (49.123),

$$|\psi_{1S}(0)|^2 = \frac{(m_Q\alpha_C)^3}{8\pi},$$

gives the second equality in (49.130). The proof uses only the declared pNRQCD contact operator and the Coulombic $1S$ wavefunction; all other terms belong to the stated remainder of the weak-coupling chart.

Controlled Approximation 49.18 (Quarkonium scale separation)**.** Let $v_{\rm rel}$ be the velocity coordinate assigned to a family of quarkonium states. The weak-coupling potential expansion assumes

$$m_Q \gg m_Q v_{\rm rel} \gg m_Q v_{\rm rel}^2, \qquad m_Q v_{\rm rel} \gg \Lambda_{\rm QCD},$$

so that the soft relative-momentum scale can be integrated out perturbatively. Strong-coupling pNRQCD instead treats the potential as a nonperturbative matching coefficient defined by static Wilson-loop and chromofield-insertion matrix elements. In both regimes a claim of accuracy requires a norm or class of smeared matrix elements and an explicit remainder estimate in powers of $v_{\rm rel}$, $1/m_Q$, and $\Lambda_{\rm QCD}/(m_Q v_{\rm rel})$ when that ratio is used.

49.12 Heavy-Quark Mass Coordinates and Static Energies

The mass m_Q appearing in HQET, NRQCD, and pNRQCD is a coordinate in a regulated Lagrangian and in its matched effective descriptions. In a confining four-dimensional QCD theory an isolated colored quark is not an asymptotic particle, so a quark pole mass is not a gauge-invariant spectral observable. Perturbation theory in a fixed gauge nevertheless defines formal mass coordinates, including an on-shell coordinate order by order. Such coordinates are useful only after their infrared sensitivity and their relation to gauge-invariant quantities have been stated.

The gauge-invariant static quark-antiquark energy is extracted from Wilson loops. With a specified line-renormalization prescription, define

$$E_{\rm stat}(r;\mu) = -\lim_{T\to\infty}\frac{1}{T}\log\langle W_\Box(r,T)\rangle_{\text{ren},\mu}, \tag{49.131}$$

where $W_\Box(r,T)$ is the rectangular fundamental Wilson loop of spatial size r and time extent T. The subscript records the ultraviolet and line-renormalization convention. Additive constants in $E_{\rm stat}$ are not absolute observables; energy differences and matched hadron masses are.

At distances r in a weak-coupling matching range, the pNRQCD singlet potential is a coordinate in the decomposition

$$E_{\mathrm{stat}}(r;\mu) = 2m_{\mathrm{sd}}(\mu_f) + V_{\mathrm{sd}}(r;\mu,\mu_f) + \Delta_{\mathrm{us}}(r;\mu,\mu_f), \tag{49.132}$$

where m_{sd} is a short-distance heavy-quark mass coordinate, V_{sd} is the corresponding singlet potential coordinate, and Δ_{us} denotes lower-energy contributions retained in pNRQCD. The auxiliary scale μ_f separates contributions assigned to the mass coordinate from contributions assigned to the potential coordinate.

Remark 49.19 (Constant mass-coordinate transformations). Let δm be a finite scheme change independent of r. The coordinate transformation

$$m'_{\mathrm{sd}} = m_{\mathrm{sd}} + \delta m, \qquad V'_{\mathrm{sd}}(r) = V_{\mathrm{sd}}(r) - 2\delta m \tag{49.133}$$

leaves the static energy (49.132) unchanged when Δ_{us} is held fixed. If a pNRQCD Hamiltonian is shifted by $V_{\mathrm{sd}} \mapsto V_{\mathrm{sd}} - 2\delta m$, every singlet eigenvalue E_n is shifted by $-2\delta m$, and $2m_{\mathrm{sd}} + E_n$ is unchanged.

Substitution into (49.132) gives

$$2(m_{\mathrm{sd}} + \delta m) + (V_{\mathrm{sd}} - 2\delta m) + \Delta_{\mathrm{us}} = 2m_{\mathrm{sd}} + V_{\mathrm{sd}} + \Delta_{\mathrm{us}}.$$

For a Hamiltonian $H_{\mathrm{sd}} = T + V_{\mathrm{sd}}$, the transformed Hamiltonian is $H'_{\mathrm{sd}} = H_{\mathrm{sd}} - 2\delta m$. If $H_{\mathrm{sd}}\psi_n = E_n\psi_n$, then $H'_{\mathrm{sd}}\psi_n = (E_n - 2\delta m)\psi_n$. Therefore

$$2(m_{\mathrm{sd}} + \delta m) + (E_n - 2\delta m) = 2m_{\mathrm{sd}} + E_n.$$

The statement is finite-dimensional spectral algebra applied to the regularized Hamiltonian, and then to any limit in which the relevant eigenvalue exists.

One useful short-distance coordinate is the potential-subtracted mass. Start from a formal perturbative pole coordinate m_{pole} and define

$$m_{\mathrm{PS}}(\mu_f) = m_{\mathrm{pole}} - \delta m_{\mathrm{PS}}(\mu_f), \qquad \delta m_{\mathrm{PS}}(\mu_f) = -\frac{1}{2}\int_{|\mathbf{q}|<\mu_f} \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\,\widetilde{V}(\mathbf{q}), \tag{49.134}$$

where $\widetilde{V}(\mathbf{q})$ is the perturbative momentum-space singlet potential in the same convention. At tree level

$$\widetilde{V}^{(0)}(\mathbf{q}) = -\frac{g^2 C_F}{|\mathbf{q}|^2}.$$

Consequently

$$\delta m^{(0)}_{\mathrm{PS}}(\mu_f) = \frac{g^2 C_F}{4\pi^2}\,\mu_f. \tag{49.135}$$

In the common half-trace convention this is the same expression because $g^2_{\mathrm{ht}} C_F^{\mathrm{ht}} = g^2 C_F$.

Using spherical coordinates in momentum space,

$$\int_{|\mathbf{q}|<\mu_f} \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\,\frac{1}{|\mathbf{q}|^2} = \frac{4\pi}{(2\pi)^3}\int_0^{\mu_f} \mathrm{d}q = \frac{\mu_f}{2\pi^2}.$$

Multiplying by $g^2 C_F/2$, as dictated by (49.134) and $\widetilde{V}^{(0)} = -g^2 C_F/|\mathbf{q}|^2$, gives (49.135).

Remark 49.20 (Renormalon language in the heavy-quark sector). The leading infrared sensitivity of the formal perturbative pole coordinate and the additive constant in the perturbative static potential are tied to the same long-distance momentum region. In perturbative large-order language this is described as a cancellation of the leading static-energy renormalon between $2m_{\text{pole}}$ and $V_{\text{pole}}(r)$. The mathematically precise statement used here is the finite coordinate transformation (49.133) and the definition of short-distance coordinates such as (49.134). A claim about a Borel singularity, its residue, or its cancellation is a claim about a specified perturbative expansion and summation prescription, not a nonperturbative definition of the quark mass.

Controlled Approximation 49.21 (Mass schemes in quarkonium predictions)**.** A quarkonium prediction based on pNRQCD must state the mass coordinate, the potential coordinate, the subtraction scale μ_f when one is used, and the class of matrix elements or energy levels being approximated. Replacing (m, V) by a short-distance pair (m', V') is exact at the level of the matched Hamiltonian if (49.133) is imposed. Truncating perturbative series introduces residual μ_f- and scheme-dependence; the size of that dependence is an error diagnostic only after the truncation order, velocity counting, and ultrasoft treatment have been fixed.

49.13 Color and Physical External States

The gauge-fixed quark and gluon fields are useful local variables in perturbation theory. Physical states are BRST cohomology classes, and local physical operators are represented by gauge-invariant or BRST-closed operators. This has an immediate consequence for scattering: LSZ reduction applied to the elementary quark or gluon fields does not by itself define a physical hadron S-matrix. Physical external states in QCD are created by color-singlet operators, such as

$$\bar{q}_I q_J, \qquad \bar{q}_I \gamma^\mu q_J, \qquad \epsilon_{i_1 \cdots i_{N_c}} q^{i_1}_{I_1} \cdots q^{i_{N_c}}_{I_{N_c}}, \qquad \operatorname{tr}(F_{\mu\nu} F^{\mu\nu}).$$

The first three are mesonic current, vector-current, and baryonic examples; the last is a gluonic example.

Confinement, in the spectral sense used here, is the assertion that isolated finite-energy asymptotic particles carry no color. In four-dimensional QCD this assertion is a nonperturbative physical input, supported by the observed hadron spectrum and by lattice evidence, rather than a theorem presently derived from the continuum Lagrangian. The monograph uses this precise spectral hypothesis when discussing QCD phenomenology: the asymptotic states are hadrons, while the short-distance operator expansion is expressed in terms of quark and gluon fields because those are the local variables of the renormalized gauge theory.

Open Problem 49.22 (Asymptotic completeness of physical QCD). Starting from a nonperturbative four-dimensional QCD Hilbert space $\mathcal{H}_{\text{phys}}$, its unitary translation representation, and its gauge-invariant local observable algebra, construct the physical incoming and outgoing scattering theory. The required asymptotic space

$$\mathcal{H}_{\text{as}}^{\text{QCD}}$$

should be built from stable color-singlet Wigner particles in the relevant superselection sectors: mesons, baryons, glueballs in pure Yang–Mills subsectors, and, in chiral limits, the massless Goldstone modes with their long-distance scattering structure. Unstable hadronic resonances should enter through the analytic structure of the color-singlet scattering operator rather than as ordinary external vectors. The wave operators

$$\Omega^{\mathrm{QCD}}_{\mathrm{in/out}} : \mathcal{H}^{\mathrm{QCD}}_{\mathrm{as}} \longrightarrow \mathcal{H}_{\mathrm{phys}}$$

should be isometric, should intertwine translations, and should have common closed range equal to the finite-energy physical sector under consideration:

$$\overline{\operatorname{Ran}\Omega^{\mathrm{QCD}}_{\mathrm{in}}} = \overline{\operatorname{Ran}\Omega^{\mathrm{QCD}}_{\mathrm{out}}} = \mathcal{H}^{\mathrm{sector}}_{\mathrm{phys}}.$$

One must also construct the detector limits, such as the energy-flow operators below, on the same scattering domain and prove that the inclusive calorimetric observables used in high-energy QCD are obtained from this Hilbert-space scattering theory.

This problem is logically stronger than asymptotic freedom, factorization, a mass gap in pure Yang–Mills theory, or the spectral confinement hypothesis used above. Asymptotic freedom controls short-distance operator products and partonic coefficient functions. Factorization gives a perturbative expansion for selected inclusive observables after the relevant operator matrix elements have been defined. Asymptotic completeness is a large-time statement about the range of physical wave operators in the gauge-invariant Hilbert space.

49.14 Hadron Spectroscopy as Ab Initio QCD

49.14.1 Spectral Sources and Finite-Volume Data

Hadron spectroscopy is not a list of model names assigned to observed peaks. It is a collection of spectral and scattering statements about the gauge-invariant Hilbert space of QCD. A spectroscopic claim must therefore state which Hilbert-space object is being computed. Stable hadrons are isolated eigenvalues of the mass operator $M^2 = P_\mu P^\mu$ in color-singlet superselection sectors. Resonances are poles of color-singlet scattering amplitudes on specified nonphysical sheets, as in Chapter 26; they are not additional vectors in the asymptotic Hilbert space. Finite-volume energy levels are regulator observables whose infinite-volume interpretation requires a quantization map, not a direct equality between a box eigenvalue and a resonance mass.

Let I, J denote flavor labels and let i, j be color labels. Typical local color-singlet sources are

$$M_{\Gamma,IJ}(x) = \bar{q}_I(x)\Gamma q_J(x), \tag{49.136}$$

$$B_{\alpha_1\cdots\alpha_{N_c};I_1\cdots I_{N_c}}(x) = \epsilon_{i_1\cdots i_{N_c}}\, q^{i_1}_{\alpha_1 I_1}(x)\cdots q^{i_{N_c}}_{\alpha_{N_c} I_{N_c}}(x), \tag{49.137}$$

$$G_{\mathcal{P}}(x) = \operatorname{tr}\mathcal{P}(F_{\mu\nu}, D_\rho)(x). \tag{49.138}$$

Here Γ is a Dirac matrix, possibly with covariant derivatives inserted and with Lorentz indices projected to a definite spin channel, and $\mathcal{P}$ is a gauge-covariant polynomial whose color indices are traced using the convention fixed in (49.1). Nonlocal sources are allowed, but their Wilson lines, smearing kernels, and endpoint renormalizations are part of the operator definition. For example, a hybrid meson source may be represented by

$$\bar{q}_I(x)\Gamma^{\mu\nu}F_{\mu\nu}(x)q_J(x),$$

or by a point-split bilinear with a Wilson-line path and a field-strength insertion on that path. These are different regulator coordinates unless a continuum matching relation has been supplied.

Definition 49.23 (Hadron spectral extraction setup)**.** A QCD hadron spectral extraction setup consists of:

1. the quark masses, gauge group, global form, spacetime volume, boundary conditions, and regulator action;
2. a list of gauge-invariant interpolating operators, including their flavor, parity, charge conjugation, spin or cubic-group channel, smearing, and renormalization prescriptions;
3. the Euclidean correlation matrix
$$C_{AB}(t) = \langle \mathcal{O}_A(t)\mathcal{O}_B^\dagger(0)\rangle - \langle \mathcal{O}_A\rangle\langle \mathcal{O}_B^\dagger\rangle;$$
4. a declared estimator of finite-volume color-singlet energies from the transfer matrix or Euclidean correlators, including its source basis, fit window, covariance treatment, and stability diagnostics;
5. a map from finite-volume energies to infinite-volume observables: isolated stable masses for one-particle states, and scattering amplitudes or pole positions for resonances;
6. continuum, chiral, heavy-quark, and infinite-volume extrapolations, with the systematics stated separately.

The setup is called *ab initio* only when the matrix elements are defined from the QCD path integral or Hilbert space itself. A constituent model may be useful as a coordinate chart or mnemonic, but it is not an ab initio definition of the spectrum.

Definition 49.24 (Operator-sector spectroscopy chart)**.** Fix a finite-volume regulator and a symmetry channel Γ. Let $\mathcal{V}_\Gamma$ be a finite-dimensional vector space of renormalized gauge-invariant interpolating operators in that channel, modulo operators whose matrix elements vanish on the finite-volume Hilbert subspace being sampled. A spectroscopy chart is a choice of ordered basis $(\mathcal{O}_1, \ldots, \mathcal{O}_m)$ of $\mathcal{V}_\Gamma$, together with any declared decomposition

$$\mathcal{V}_\Gamma = \mathcal{V}_{\mathrm{loc}} \oplus \mathcal{V}_{\text{two-hadron}} \oplus \mathcal{V}_{\mathrm{flux}} \oplus \cdots . \tag{49.139}$$

The summands in (49.139) are source subspaces, not Hilbert-space superselection sectors. For example, $\mathcal{V}_{\mathrm{loc}}$ may contain local $q\bar{q}$, baryon, or pure-glue operators, $\mathcal{V}_{\text{two-hadron}}$ may contain products of separately projected hadron operators, and $\mathcal{V}_{\mathrm{flux}}$ may contain point-split bilinears with field-strength insertions or extended Wilson-line paths. The decomposition is part of the chart; it is not an invariant classification of states unless a regulator, renormalization scheme, and continuum matching prescription have also been supplied.

For a finite spatial torus and a reflection-positive lattice regulator, the spectral decomposition has the same form as (140.66):

$$C_{AB}(t) = \sum_{n\geq 0} Z_A^{(n)}\overline{Z_B^{(n)}}\mathrm{e}^{-E_n t}, \qquad Z_A^{(n)} = \langle \Omega|\mathcal{O}_A|n, L\rangle.$$

The vectors $|n, L\rangle$ are finite-volume color-singlet eigenvectors. If an eigenvalue converges, after removing the center-of-mass motion and taking the regulator limits, to an isolated one-particle mass below all relevant thresholds, the resulting number is a stable-hadron mass. Above threshold the finite-volume levels instead sample the scattering operator. In a single elastic two-particle channel with total momentum zero, particle masses m_1, m_2, relative momentum k, and energy

$$E = \sqrt{m_1^2 + k^2} + \sqrt{m_2^2 + k^2},$$

the simplest unmixed partial-wave quantization condition is

$$\delta_\ell(k) + \phi_\ell(q) = n\pi, \qquad q = \frac{kL}{2\pi}, \tag{49.140}$$

where δ_ℓ is the infinite-volume elastic phase shift and ϕ_ℓ is a known geometric function built from the finite-volume zeta function. Equation (49.140) is not a model of a resonance. It is a finite-volume reconstruction theorem under elasticity, locality, and regulator hypotheses. In realistic QCD channels one must use the coupled-channel determinant version, because cubic symmetry mixes partial waves and inelastic thresholds are often nearby.

Source coordinates and invariant output. The QCD output of the preceding construction is not a preferred source basis. At finite regulator the invariant spectral data are the color-singlet transfer-matrix energies and their symmetry labels; after the continuum and infinite-volume limits they become stable masses, scattering amplitudes, or sheet-labeled pole data according to the channel. Source-overlap phrases such as "mostly two-hadron", "mostly hybrid", or "mostly glueball" are coordinate statements about the chosen operator family. They acquire unambiguous meaning only after the source subspaces in (49.139), their renormalization and smearing prescriptions, and the positive form used to compare overlap covectors have all been declared. Changing these source coordinates can change the quoted overlap fractions without changing the finite-volume energy or the reconstructed pole. This chapter therefore records the QCD-specific operators and the spectral or scattering quantity being extracted, while the finite-regulator estimator itself is part of the numerical methodology.

Remark 49.25 (Spectroscopic names as derived coordinates)*.* In the finite-volume QCD framework above, the invariant output of a stable sector is an energy level and its quantum numbers, while the invariant output of a resonance sector is a sheet-labeled pole and residue data after a finite-volume-to-infinite-volume reconstruction. Names such as conventional meson, glueball, hybrid, molecule, tetraquark, or flux-tube excitation are therefore derived coordinates on source overlaps, pole residues, large-N_c scaling, and deformation behavior. They are not additional definitions of QCD states.

49.14.2 Elastic Finite-Volume Reconstruction

Definition 49.26 (Periodic Green function and S-wave zeta coordinate)**.** For a cubic spatial torus of side L, let

$$\mathbf{p_n} := \frac{2\pi}{L}\mathbf{n}, \qquad \mathbf{n} \in \mathbb{Z}^3,$$

and let $k > 0$ be such that $k^2 \neq \mathbf{p_n}^2$ for all $\mathbf{n}$. The periodic Helmholtz Green function is the distribution

$$G_L(\mathbf{r}; k) := \frac{1}{L^3} \sum_{\mathbf{n} \in \mathbb{Z}^3} \frac{\mathrm{e}^{\mathrm{i}\mathbf{p_n}\cdot\mathbf{r}}}{\mathbf{p_n}^2 - k^2}, \qquad (\Delta + k^2) G_L(\mathbf{r}; k) = -\delta_L(\mathbf{r}), \tag{49.141}$$

where δ_L is the periodic delta distribution. Its singular expansion at the origin is written

$$G_L(\mathbf{r};k) = \frac{1}{4\pi|\mathbf{r}|} + c_L(k) + O(|\mathbf{r}|), \qquad \mathbf{r} \to 0. \tag{49.142}$$

Equivalently, with $q = kL/(2\pi)$, define the analytically continued Luescher zeta coordinate $\mathcal{Z}_{00}(1;q^2)$ by

$$4\pi c_L(k) = \frac{2}{\sqrt{\pi}\,L}\,\mathcal{Z}_{00}(1;q^2), \qquad \mathcal{Z}_{00}(s;q^2) := \frac{1}{\sqrt{4\pi}}\sum_{\mathbf{n}\in\mathbb{Z}^3}\frac{1}{(\mathbf{n}^2-q^2)^s} \quad (\operatorname{Re} s > \tfrac{3}{2}), \tag{49.143}$$

continued meromorphically to $s = 1$. The singular summands at $\mathbf{n}^2 = q^2$ are absent from the elastic interval under discussion; when a free two-particle level is crossed, the condition is read by the usual limiting prescription from adjacent q^2.

Proposition 49.27 (S-wave finite-volume quantization from wave matching)**.** *Consider a single elastic two-particle channel in the center-of-mass frame. Assume the two stable particles have masses m_1, m_2, the interaction range in the relative coordinate is contained in $|\mathbf{r}| < R$, the box obeys $L > 2R$, and the finite-volume corrections from virtual particles wrapping around the torus are bounded by $C\mathrm{e}^{-m_{\mathrm{gap}}L}$ for some positive mass scale m_{gap} in the omitted exchange channel. In an exact finite-range model this exponential remainder is absent. Assume also that the cubic irrep isolates the S-wave, so higher partial waves are not retained in the truncation. Let k be the relative momentum determined by*

$$E = \sqrt{m_1^2 + k^2} + \sqrt{m_2^2 + k^2}.$$

Then the finite-volume energy is allowed precisely when

$$k\cot\delta_0(k) = 4\pi c_L(k) = \frac{2}{\sqrt{\pi}\,L}\mathcal{Z}_{00}(1;q^2), \qquad q = \frac{kL}{2\pi}. \tag{49.144}$$

Equivalently,

$$q\cot\delta_0(k) = \frac{1}{\pi^{3/2}}\mathcal{Z}_{00}(1;q^2), \qquad \delta_0(k) + \phi_0(q) = n\pi, \qquad \cot\phi_0(q) = -\frac{\mathcal{Z}_{00}(1;q^2)}{\pi^{3/2}q}. \tag{49.145}$$

Proof. Outside the interaction region the relative wave function $\psi(\mathbf{r})$ satisfies the free Helmholtz equation $(\Delta + k^2)\psi = 0$ and periodic boundary conditions. In the S-wave truncation any such finite-volume solution with a source at the origin is proportional, in the exterior region $R < |\mathbf{r}| < L/2$, to the periodic Green function $G_L(\mathbf{r};k)$. The proportionality constant is irrelevant; the quantization condition is obtained by comparing the ratio of the regular and singular terms in the short-distance expansion.

The infinite-volume elastic S-wave outside the interaction range has the standard real form

$$\psi_\infty(r) = A\,\frac{\sin(kr + \delta_0(k))}{kr}, \qquad r > R.$$

Expanding this expression at small r gives

$$\psi_\infty(r) = A\left[\frac{\sin\delta_0(k)}{kr} + \cos\delta_0(k) + O(r)\right].$$

Thus the ratio of the regular coefficient to the $1/r$ coefficient is

$$\frac{A\cos\delta_0(k)}{A\sin\delta_0(k)/k} = k\cot\delta_0(k). \tag{49.146}$$

On the other hand, the finite-volume exterior solution proportional to (49.141) has expansion

$$B\,G_L(\mathbf{r};k) = B\left[\frac{1}{4\pi r} + c_L(k) + O(r)\right], \qquad r = |\mathbf{r}|.$$

Its regular-to-singular ratio is $4\pi c_L(k)$. Matching the exterior finite-volume solution to the infinite-volume scattering boundary condition therefore gives the first equality in (49.144). The second equality is the zeta coordinate definition (49.143).

Multiplying the first form by $L/(2\pi)$ and using $q = kL/(2\pi)$ gives the first equation in (49.145). The phase form follows by defining ϕ_0 through $\cot\phi_0 = -\mathcal{Z}_{00}(1;q^2)/(\pi^{3/2}q)$:

$$\cot\delta_0(k) = -\cot\phi_0(q) = \cot\bigl(n\pi - \phi_0(q)\bigr),$$

which is equivalent to $\delta_0(k) + \phi_0(q) = n\pi$ modulo π. The only physics input beyond the spectral existence of the two-particle level is the replacement of the interacting QFT outside the range by the elastic scattering boundary condition, with exponential finite-volume corrections controlled by the lightest omitted exchange. Thus the formula is a reconstruction statement with explicit hypotheses, not a model of the resonance. □

49.14.3 Single- and Coupled-Channel Pole Coordinates

For the single-channel finite-volume pole reconstruction target, assume that, on an open real energy interval I, a QCD channel is elastic, has two stable color-singlet particles of masses m_1, m_2, and has been reduced, by a stated cubic-irrep analysis, to a single partial wave ℓ. Put $J = \{E^2 : E \in I\}$ and

$$\rho_{12}(s) = \frac{2k(s)}{\sqrt{s}}, \qquad E = \sqrt{m_1^2 + k(s)^2} + \sqrt{m_2^2 + k(s)^2},$$

for $s = E^2 \in J$, and use the partial-wave normalization

$$S_\ell(s) = 1 + 2\mathrm{i}\rho_{12}(s)t_\ell^{(I)}(s) = \mathrm{e}^{2\mathrm{i}\delta_\ell(s)}.$$

Then the physical-sheet elastic amplitude is equivalently written

$$t_\ell^{(I)}(s) = \frac{1}{K_\ell^{-1}(s) - \mathrm{i}\rho_{12}(s)}, \qquad K_\ell^{-1}(s) = \rho_{12}(s)\cot\delta_\ell(s), \tag{49.147}$$

where $K_\ell^{-1}(s)$ is real on J. Each finite-volume energy $E_n \in I$ satisfying (49.140) determines the real-axis coordinate

$$K_\ell^{-1}(E_n^2) = -\rho_{12}(E_n^2)\cot\phi_\ell(q_n), \qquad q_n = \frac{k(E_n)L}{2\pi}. \tag{49.148}$$

If K_ℓ^{-1} and ρ_{12} have the chosen analytic continuations to a simply connected neighborhood of a candidate resonance pole and no additional channel threshold crosses that neighborhood, the corresponding second-sheet pole s_* is a zero of

$$K_\ell^{-1}(s_*) + \mathrm{i}\rho_{12}(s_*) = 0. \tag{49.149}$$

This is the algebraic target of a one-channel finite-volume resonance extraction, not by itself a proof that the required analytic continuation has been controlled. Elastic unitarity in the normalization above says $S_\ell^* S_\ell = 1$ on I. Hence

$$t_\ell^{(I)}(s) = \frac{\mathrm{e}^{2\mathrm{i}\delta_\ell(s)} - 1}{2\mathrm{i}\rho_{12}(s)} = \frac{\mathrm{e}^{\mathrm{i}\delta_\ell(s)}\sin\delta_\ell(s)}{\rho_{12}(s)}$$

and therefore

$$\big(t_\ell^{(I)}(s)\big)^{-1} = \rho_{12}(s)\big(\cot\delta_\ell(s) - \mathrm{i}\big).$$

This proves (49.147) and identifies the real coordinate $K_\ell^{-1} = \rho_{12}\cot\delta_\ell$. At a finite-volume level satisfying $\delta_\ell(k_n) + \phi_\ell(q_n) = n\pi$,

$$\cot\delta_\ell(k_n) = \cot\big(n\pi - \phi_\ell(q_n)\big) = -\cot\phi_\ell(q_n),$$

which gives (49.148). Across the elastic unitarity cut the chosen branch of the two-body momentum changes sign, so ρ_{12} changes sign while K_ℓ^{-1}, being the real analytic coordinate fixed on the cut, continues through the same germ. Thus the amplitude on the adjacent sheet is

$$t_\ell^{(II)}(s) = \frac{1}{K_\ell^{-1}(s) + \mathrm{i}\rho_{12}(s)},$$

and its isolated poles are precisely the zeros of the denominator, giving (49.149).

A concrete determination of s_* also contains an analytic representation or parametrization of K_ℓ^{-1}, regulator and volume limits for the input energies, and separate bounds for neglected partial waves and inelastic channels.

Lemma 49.28 (Coupled-channel K-matrix determinant and sheet continuation)**.** *Let $a, b = 1, \dots, N$ label finitely many stable two-particle color-singlet channels in a fixed set of conserved quantum numbers, and let α, β label the partial waves and cubic-irrep components retained in a finite-volume analysis. On a real interval below the next omitted threshold, let $\rho(s)$ be the positive diagonal matrix with entries $\rho_a(s) = 2k_a(s)/\sqrt{s}$, repeated on the retained angular-momentum indices. With the normalization*

$$S = 1 + 2\mathrm{i}\,\rho^{1/2} t^{(I)} \rho^{1/2},$$

elastic coupled-channel unitarity is equivalent, on the interval, to

$$t^{(I)}(s) = \big(K^{-1}(s) - \mathrm{i}\rho(s)\big)^{-1}, \qquad K^{-1}(s)^\dagger = K^{-1}(s). \tag{49.150}$$

Suppose, in addition, that the coupled-channel finite-volume reconstruction theorem has been established for this channel and written in the same normalization, so that the finite-volume energies in a cubic irrep Γ obey

$$\det\big(K^{-1}(E^2) + F^\Gamma(E, L)\big) = 0, \tag{49.151}$$

where $F^\Gamma(E, L)$ is the real geometric matrix determined by the box, total momentum, masses, and irrep projection. In the one-channel unmixed limit $F^\Gamma = \rho_{12}\cot\phi_\ell$, so (49.151) reduces to (49.148). If Σ is the diagonal matrix whose entry is $+1$ on a channel kept on the physical branch and -1 on a channel analytically continued through its unitary cut, the corresponding sheet amplitude is

$$t^{(\Sigma)}(s) = \big(K^{-1}(s) - \mathrm{i}\,\Sigma\rho(s)\big)^{-1}, \qquad \det\big(K^{-1}(s_*) - \mathrm{i}\,\Sigma\rho(s_*)\big) = 0. \tag{49.152}$$

Proof. From $S^\dagger S = 1$ and $S = 1 + 2\mathrm{i}\rho^{1/2} t^{(I)} \rho^{1/2}$, one obtains

$$t^{(I)} - t^{(I)\dagger} = 2\mathrm{i}\, t^{(I)\dagger} \rho\, t^{(I)}.$$

Multiplying by $\left(t^{(I)\dagger}\right)^{-1}$ on the left and by $\left(t^{(I)}\right)^{-1}$ on the right gives

$$\left(t^{(I)\dagger}\right)^{-1} - \left(t^{(I)}\right)^{-1} = 2\mathrm{i}\rho, \qquad \left(t^{(I)}\right)^{-1} - \left(t^{(I)\dagger}\right)^{-1} = -2\mathrm{i}\rho.$$

Thus $K^{-1} := \left(t^{(I)}\right)^{-1} + \mathrm{i}\rho$ is Hermitian on the physical interval, and (49.150) follows. The finite-volume determinant relation is the stated reconstruction input expressed in this K-matrix coordinate. In the one-channel unmixed reduction, its zero condition is

$$K_\ell^{-1}(E^2) + \rho_{12}(E^2) \cot \phi_\ell(q) = 0, \qquad K_\ell^{-1}(E^2) = -\rho_{12}(E^2) \cot \phi_\ell(q),$$

which is exactly (49.148). Finally, analytic continuation through a chosen channel cut changes the sign of that channel's two-body momentum, hence of the corresponding diagonal entry of ρ, while the K^{-1} germ is held fixed. Repeating this sign choice for all channels gives the matrix Σ, the sheet amplitude in (49.152), and the determinant pole condition. □

Definition 49.29 (QCD resonance pole data)**.** The resonance pole data in a hadronic channel consist of a channel space of stable color-singlet asymptotic states, a physical-region scattering matrix $S_{ab}(s)$, a specified analytic continuation across specified unitary cuts, and an isolated pole s_* of the continued matrix or of an equivalent partial-wave amplitude. The mass and width coordinates

$$\sqrt{s_*} = M_* - \frac{\mathrm{i}}{2}\Gamma_*$$

are pole coordinates after a branch of the square root has been declared. For a broad or overlapping resonance, line-shape parameters on the real axis are not invariant substitutes for s_*.

This definition is the correct language for the ρ, the light scalar isoscalar pole usually denoted $f_0(500)$, many excited baryons, and near-threshold exotic candidates. The notation "tetraquark", "molecule", "hybrid", or "glueball component" is then a statement about operator overlaps, pole motion under controlled deformations, large-N_c scaling, or residue factorization. It is not a separate definition of the state.

Definition 49.30 (Controlled pole-motion chart)**.** Let η be a real coordinate on a family of QCD-like theories or regulators whose action, operator definitions, stable-channel spectrum, and renormalization prescription have been specified for η in an open interval I. Examples include a quark-mass coordinate, $1/N_c$ in a large-N_c family, a finite-volume boundary condition used as an auxiliary probe before the infinite-volume limit, or a coupling to a source that is set to zero after differentiating. A pole-motion chart in a fixed set of conserved quantum numbers consists of:

1. a channel space $\mathcal{C}_\eta$ of stable color-singlet asymptotic states for each η, with an identification of these channel spaces over the interval where no threshold crossing changes the channel list;
2. an analytic family of continued partial-wave amplitudes $t_{ab}^{(\Sigma)}(s;\eta)$ on a declared sheet Σ;
3. an isolated simple pole $s_*(\eta)$ and residue vectors $g_a(\eta)$ satisfying

$$t_{ab}^{(\Sigma)}(s;\eta) = \frac{g_a(\eta) g_b(\eta)}{s_*(\eta) - s} + h_{ab}(s;\eta),$$

with h_{ab} holomorphic near $s_*(\eta)$.

The derivative $\mathrm{d}s_*/\mathrm{d}\eta$, the projective residue vector $[g_a(\eta)]$, and their behavior near thresholds are coordinates on the declared chart. They are not intrinsic labels unless the family, channel identification, and sheet have been included in the statement.

49.14.4 Near-Threshold Effective-Range Coordinates

For a shallow S-wave bound-state effective-range chart, consider one stable two-hadron channel with reduced mass μ, threshold energy set to $E = 0$, and center-of-mass momentum $k = \sqrt{2\mu E}$, where the physical branch has $k > 0$ for $E > 0$ and $\operatorname{Im} k > 0$ below threshold. Suppose that in an energy disk whose radius is large compared with the binding energy but small compared with the first omitted left-hand or inelastic singularity, the elastic S-wave amplitude is represented by the effective-range form

$$f(k) = \frac{1}{-a^{-1} + \frac{1}{2} r_e k^2 - \mathrm{i}k}, \tag{49.153}$$

where a is the scattering length and r_e is the effective range in this normalization. If f has a bound-state pole at $E = -B = -\kappa^2/(2\mu)$, with $\kappa > 0$, then

$$\frac{1}{a} = \kappa - \frac{1}{2} r_e \kappa^2. \tag{49.154}$$

Moreover, near $E = -B$,

$$f(E) = -\frac{g_{\text{eff}}^2}{E+B} + O(1), \qquad g_{\text{eff}}^2 = \frac{\kappa}{\mu(1 - r_e\kappa)}. \tag{49.155}$$

When $1 - r_e\kappa \neq 0$, the dimensionless coordinate

$$X_{\text{ERE}} := \frac{a\kappa}{2 - a\kappa} = \frac{1}{1 - r_e\kappa} \tag{49.156}$$

is determined by the low-energy pole and the first two effective-range parameters. It is a compositeness coordinate in this one-channel effective-range chart, not a regulator-independent probability attached to a QCD eigenstate without the hypotheses above.

The bound-state pole lies on the physical sheet at $k = \mathrm{i}\kappa$. The denominator of (49.153) therefore vanishes:

$$-a^{-1} + \frac{1}{2} r_e (\mathrm{i}\kappa)^2 - \mathrm{i}(\mathrm{i}\kappa) = -a^{-1} - \frac{1}{2} r_e \kappa^2 + \kappa = 0,$$

which gives (49.154).

Let

$$D(k) = -a^{-1} + \frac{1}{2} r_e k^2 - \mathrm{i}k.$$

Since $E = k^2/(2\mu)$, one has

$$\frac{\mathrm{d}D}{\mathrm{d}E} = (r_e k - \mathrm{i})\frac{\mu}{k}.$$

Evaluating at $k = \mathrm{i}\kappa$ gives

$$\left.\frac{\mathrm{d}D}{\mathrm{d}E}\right|_{E=-B} = \frac{\mu(r_e\kappa - 1)}{\kappa}.$$

Thus

$$D(E) = \frac{\mu(r_e\kappa - 1)}{\kappa}(E + B) + O((E + B)^2),$$

and

$$f(E) = \frac{\kappa}{\mu(r_e\kappa - 1)} \frac{1}{E + B} + O(1) = -\frac{\kappa}{\mu(1 - r_e\kappa)} \frac{1}{E + B} + O(1),$$

which is (49.155). Finally, (49.154) gives

$$a\kappa = \frac{1}{1 - \frac{1}{2}r_e\kappa}.$$

Substituting this expression into $a\kappa/(2 - a\kappa)$ yields

$$\frac{a\kappa}{2 - a\kappa} = \frac{1}{1 - r_e\kappa},$$

proving (49.156).

The calculation is deliberately local. It does not classify every near-threshold QCD pole. In coupled channels, or when the omitted range scale is not separated from κ^{-1}, the residue vector and pole motion of Definition 49.30 are the appropriate coordinates. A statement that a pole is molecular, compact, hybrid, or predominantly glueball-like must then be translated into source overlaps, residues, large-N_c behavior, or a specified deformation path.

The light pseudoscalar sector has additional structure because chiral symmetry and the anomaly constrain the low-energy coordinates. Chapter 55 derives the pion effective theory, the Gell-Mann–Oakes–Renner relation, and the Witten–Veneziano large-N_c formula for the singlet mass. The vector and scalar resonances require the scattering-amplitude language above.

49.14.5 Light-Meson Scattering and Roy Coordinates

Definition 49.31 ($\pi\pi$ isospin amplitudes)**.** For equal pion masses let $s + t + u = 4m_\pi^2$ and write the $SU(2)_V$-invariant four-pion amplitude as in (55.58),

$$\mathcal{A}_{ab\to cd} = \delta_{ab}\delta_{cd}A(s, t, u) + \delta_{ac}\delta_{bd}A(t, s, u) + \delta_{ad}\delta_{bc}A(u, t, s).$$

The scalar function A is symmetric in its last two arguments; this is the Bose symmetry obtained by interchanging the two pions in an equal-mass initial or final pair while keeping the tensor decomposition above fixed. The s-channel isospin amplitudes are

$$\begin{pmatrix} T^0 \\ T^1 \\ T^2 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} A(s, t, u) \\ A(t, s, u) \\ A(u, t, s) \end{pmatrix}. \tag{49.157}$$

Equivalently,

$$T^0 = 3A(s, t, u) + A(t, s, u) + A(u, t, s), \quad T^1 = A(t, s, u) - A(u, t, s), \quad T^2 = A(t, s, u) + A(u, t, s).$$

Crossing matrices in the $\pi\pi$ isospin basis. With $\mathbf{T} := (T^0, T^1, T^2)^T$, crossing $s \leftrightarrow t$ and $s \leftrightarrow u$ is

$$\mathbf{T}(t,s,u) = C_{st}\mathbf{T}(s,t,u), \qquad \mathbf{T}(u,t,s) = C_{su}\mathbf{T}(s,t,u), \tag{49.158}$$

where

$$C_{st} = \begin{pmatrix} \frac{1}{3} & 1 & \frac{5}{3} \\ \frac{1}{3} & \frac{1}{2} & -\frac{5}{6} \\ \frac{1}{3} & -\frac{1}{2} & \frac{1}{6} \end{pmatrix}, \qquad C_{su} = \begin{pmatrix} \frac{1}{3} & -1 & \frac{5}{3} \\ -\frac{1}{3} & \frac{1}{2} & \frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}. \tag{49.159}$$

The matrices in (49.159) are part of the normalization of the Roy coordinates used below. Let

$$\mathbf{A} := \begin{pmatrix} A(s,t,u) \\ A(t,s,u) \\ A(u,t,s) \end{pmatrix}, \qquad M := \begin{pmatrix} 3 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Equation (49.157) is $\mathbf{T} = M\mathbf{A}$, and

$$M^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The exchange $s \leftrightarrow t$ sends $\mathbf{A}$ to $P_{st}\mathbf{A}$, where P_{st} interchanges the first two components, and $s \leftrightarrow u$ sends $\mathbf{A}$ to $P_{su}\mathbf{A}$, where P_{su} interchanges the first and third components. Therefore $C_{st} = MP_{st}M^{-1}$ and $C_{su} = MP_{su}M^{-1}$, which gives (49.159). In particular $C_{st}^2 = C_{su}^2 = 1$, as required by applying the same crossing operation twice.

For $\pi\pi$ scattering, write the isospin partial waves as $t_\ell^I(s)$ by

$$T^I(s,t(s,z),u(s,z)) = 32\pi \sum_{\ell=0}^{\infty} (2\ell+1) P_\ell(z)\, t_\ell^I(s), \qquad t(s,z) = -\frac{s-4m_\pi^2}{2}(1-z), \tag{49.160}$$

with projection

$$t_\ell^I(s) = \frac{1}{64\pi} \int_{-1}^{1} P_\ell(z) T^I(s,t(s,z),u(s,z))\, \mathrm{d}z.$$

In this normalization physical-region elastic unitarity is

$$\operatorname{Im} t_\ell^I(s) = \rho_{\pi\pi}(s) |t_\ell^I(s)|^2, \qquad \rho_{\pi\pi}(s) = \sqrt{1 - \frac{4m_\pi^2}{s}},$$

below the first inelastic threshold. The S-wave scattering-length coordinates used in this section are

$$a_0^0 := t_0^0(4m_\pi^2), \qquad a_0^2 := t_0^2(4m_\pi^2).$$

Definition 49.32 (Roy input for $\pi\pi$ scattering)**.** Roy input on a real interval $J \subset [4m_\pi^2, s_{\max})$ consists of:

1. the pion mass, the isospin amplitudes $\mathbf{T}$, and the partial wave normalization (49.160);
2. a domain of fixed-t analyticity in which twice-subtracted Cauchy representations are valid and the boundary values on the right- and left-hand cuts exist as tempered distributions;
3. the crossing matrices (49.159);

4. subtraction constants a_0^0, a_0^2;
5. high-energy and high-partial-wave absorptive input sufficient for the two subtractions and for the convergence of the partial-wave projections.

The subtraction polynomial projected to partial waves is

$$k_0^0(s; a_0^0, a_0^2) = a_0^0 + \frac{s - 4m_\pi^2}{12m_\pi^2}\,(2a_0^0 - 5a_0^2), \tag{49.161}$$

$$k_1^1(s; a_0^0, a_0^2) = \frac{s - 4m_\pi^2}{72m_\pi^2}\,(2a_0^0 - 5a_0^2), \tag{49.162}$$

$$k_0^2(s; a_0^0, a_0^2) = a_0^2 - \frac{s - 4m_\pi^2}{24m_\pi^2}\,(2a_0^0 - 5a_0^2), \tag{49.163}$$

and $k_\ell^I = 0$ for the other partial waves in the twice-subtracted polynomial. This normalization fixes what is meant below by the Roy kernels $K_{\ell\ell'}^{II'}$.

Proposition 49.33 (Roy projection from fixed-t analyticity and crossing)**.** *Assume the Roy input of Definition 49.32. Then, for $s \in J$, the real part of each retained partial wave obeys*

$$\operatorname{Re} t_\ell^I(s) = k_\ell^I(s; a_0^0, a_0^2) + \sum_{I',\ell'} \int_{4m_\pi^2}^{\infty} \mathrm{d}s'\, K_{\ell\ell'}^{II'}(s, s')\, \operatorname{Im} t_{\ell'}^{I'}(s'). \tag{49.164}$$

More explicitly, if the twice-subtracted fixed-t representation is written after crossing the left-hand cut into the form

$$\operatorname{Re} T^I(s, t(s, z)) = P^I(s, z; a_0^0, a_0^2) + 32\pi \sum_{I',\ell'} (2\ell' + 1) \int_{4m_\pi^2}^{\infty} \mathrm{d}s'\, R_{\ell'}^{II'}(s, z; s')\, \operatorname{Im} t_{\ell'}^{I'}(s'), \tag{49.165}$$

then

$$k_\ell^I(s) = \frac{1}{64\pi} \int_{-1}^{1} P_\ell(z) P^I(s, z)\, \mathrm{d}z, \qquad K_{\ell\ell'}^{II'}(s, s') = \frac{2\ell' + 1}{2} \int_{-1}^{1} P_\ell(z) R_{\ell'}^{II'}(s, z; s')\, \mathrm{d}z. \tag{49.166}$$

The functions $R_{\ell'}^{II'}$, hence the kernels $K_{\ell\ell'}^{II'}$, are determined by the Cauchy denominators, the two subtractions, the crossing matrices C_{st}, C_{su}, and the Legendre projection of the absorptive parts.

Proof. Fix t in the declared fixed-t analyticity domain. The twice-subtracted Cauchy representation expresses the boundary value of $\mathbf{T}(s, t)$ as a subtraction polynomial plus integrals over the discontinuities on the right-hand and left-hand cuts. By crossing, the left-hand discontinuity is rewritten as a right-hand discontinuity of the crossed isospin vector, multiplied by C_{st} or C_{su} according to which crossed channel has become physical. This is the step where (49.158) enters; without it the left-hand cut is not a QCD input in the same isospin coordinates.

For $s' \geq 4m_\pi^2$, expand the absorptive part in partial waves,

$$\operatorname{Im} T^{I'}(s', t(s', z')) = 32\pi \sum_{\ell'=0}^{\infty} (2\ell' + 1) P_{\ell'}(z')\, \operatorname{Im} t_{\ell'}^{I'}(s').$$

Performing the z'-projection contained in the crossed-channel Cauchy terms gives the functions $R_{\ell'}^{II'}(s, z; s')$ in (49.165). The convergence assumptions in the Roy input justify interchanging this partial-wave expansion with the subtracted dispersion integrals on J.

Finally project the resulting fixed-t formula in the s-channel:

$$\operatorname{Re} t_\ell^I(s) = \frac{1}{64\pi} \int_{-1}^{1} P_\ell(z) \operatorname{Re} T^I(s, t(s,z))\, \mathrm{d}z.$$

Substituting (49.165) and using the orthogonality normalization already built into (49.160) gives exactly (49.164) with the projected polynomial and kernel in (49.166). The explicit polynomials (49.161)–(49.163) are the projection of the two independent isospin-even subtraction constants; the $I = 1$, $\ell = 1$ subtraction is proportional to the same combination $2a_0^0 - 5a_0^2$ because crossing fixes the first $s - 4m_\pi^2$ derivative of the odd isospin amplitude at threshold. □

Remark 49.34 (Status of a Roy pole extraction). A pole extraction from (49.164) is controlled only after the high-energy absorptive input, inelastic channels, matching point, subtraction constants, and error correlations have been specified. For the ρ, the relevant pole is obtained by continuing the $I = 1$, $\ell = 1$ partial wave through the elastic cut. For the light scalar isoscalar pole $f_0(500)$, the corresponding continuation is in the $I = 0$, $\ell = 0$ wave and is especially sensitive to the subtraction constants and low-energy crossing constraints. The same logic governs Roy–Steiner equations for πK, πN, and related channels, with unequal masses and additional crossed-channel partial waves changing the kernel construction but not the Cauchy-crossing-projection logic.

49.14.6 Light-Meson Pole Charts

Definition 49.35 (Concrete light-meson pole chart)**.** A concrete light-meson pole statement in this chapter consists of a color-singlet channel, a partial wave, a reconstruction method, and a sheet-labeled pole or stable mass. The minimal chart used for the examples below is:

(i) π, K, η, η'. The QCD channel is the pseudoscalar flavor sector. The amplitude coordinate is a stable mass, together with mixing coordinates where appropriate. The controlled route uses chiral effective theory matched to QCD, lattice masses, and the anomaly constraints of Chapter 51.

(ii) ρ. The QCD channel is $I = 1$, $\ell = 1$ $\pi\pi$. The amplitude coordinate is the second-sheet pole of $t_1^1(s)$. The controlled route is Roy input or finite-volume P-wave $\pi\pi$ quantization followed by analytic continuation.

(iii) $f_0(500)$. The QCD channel is $I = 0$, $\ell = 0$ $\pi\pi$. The amplitude coordinate is a broad second-sheet pole of $t_0^0(s)$. The controlled route is Roy input with subtraction constants and correlated high-energy input.

(iv) K^*. The QCD channel is $I = \frac{1}{2}$, P-wave πK. The amplitude coordinate is the second-sheet pole of the unequal-mass partial wave. The controlled route is Roy–Steiner input or finite-volume πK quantization.

(v) K_0^* channel. The QCD channel is $I = \frac{1}{2}$, S-wave πK. The amplitude coordinate is nearby pole data, often broad. The controlled route is Roy–Steiner input with explicit treatment of inelastic and left-hand input.

(vi) Hybrid or exotic 1^{-+} channels. The QCD channel is specified by a finite operator-source family containing $q\bar{q}F$ or Wilson-line operators. The amplitude coordinate is pole data plus source-overlap and pole-motion coordinates. The controlled route supplies color-singlet finite-volume energies from the declared source family and then performs finite-volume scattering reconstruction.

The entries are not model assignments. They are instructions for which QCD operator sector, partial wave, analytic sheet, and reconstruction theorem must be supplied before the label denotes a spectroscopic output.

Linear elastic pole coordinate near a narrow vector resonance. Consider a one-channel elastic partial wave in the normalization

$$t^{(I)}(s) = \frac{1}{K^{-1}(s) - \mathrm{i}\rho(s)}, \qquad t^{(II)}(s) = \frac{1}{K^{-1}(s) + \mathrm{i}\rho(s)}.$$

Let M_0^2 be a real point in the elastic interval such that

$$K^{-1}(M_0^2) = 0, \qquad A := \left.\frac{\mathrm{d}K^{-1}}{\mathrm{d}s}\right|_{M_0^2} > 0, \qquad \rho_0 := \rho(M_0^2) > 0,$$

and suppose that, in a neighborhood whose size is small compared with the nearest omitted singularity,

$$K^{-1}(s) = A(s - M_0^2) + O((s - M_0^2)^2), \qquad \rho(s) = \rho_0 + O(s - M_0^2).$$

Then the adjacent-sheet pole in this linearized chart is

$$s_* = M_0^2 - \mathrm{i}\,\frac{\rho_0}{A} + O\!\left(\frac{\rho_0^2}{A^2}\,\sup|(K^{-1})''| + \frac{\rho_0}{A}\sup|\rho'|\right), \tag{49.167}$$

where the suprema are taken on the same neighborhood. If the pole is parametrized by

$$\sqrt{s_*} = M_* - \frac{\mathrm{i}}{2}\Gamma_*$$

and the pole is narrow enough that $|s_* - M_0^2| \ll M_0^2$, then

$$M_* = M_0 + O(\Gamma_*^2/M_0), \qquad \Gamma_* = \frac{\rho_0}{AM_0} + O(\rho_0^2/A^2M_0^3). \tag{49.168}$$

In the exactly linear and constant approximation the second-sheet pole condition is

$$K^{-1}(s_*) + \mathrm{i}\rho(s_*) = 0 \quad\Longrightarrow\quad A(s_* - M_0^2) + \mathrm{i}\rho_0 = 0,$$

which gives $s_* = M_0^2 - \mathrm{i}\rho_0/A$. The displayed remainder follows by substituting

$$s_* = M_0^2 - \mathrm{i}\rho_0/A + \Delta$$

into the full denominator and solving once for Δ by Taylor's theorem on the neighborhood where the derivative of $K^{-1} + \mathrm{i}\rho$ remains bounded away from zero. This is an ordinary one-complex-variable implicit-function estimate; its small parameters are the quadratic variation of K^{-1} and the variation of the phase-space factor across the pole displacement.

Finally,

$$(M_* - \mathrm{i}\Gamma_*/2)^2 = M_*^2 - \frac{\Gamma_*^2}{4} - \mathrm{i}M_*\Gamma_*.$$

Comparing the imaginary part with $\operatorname{Im} s_* = -\rho_0/A + O(\rho_0^2/A^2M_0^2)$ gives (49.168); the real part then gives the stated correction to M_*. The calculation supplies a coordinate linearization of the pole equation. It is appropriate for a narrow ρ-type vector pole in an elastic window. For a broad scalar such as the $f_0(500)$, the pole coordinate itself is the invariant output and a real-axis mass-width pair is a secondary chart, if it is used at all.

Remark 49.36 (How the chart is used). For the ρ, one may use the $I = 1$, $\ell = 1$ entry of Definition 49.35 and, when the elastic window and narrowness assumptions are adequate, the linear pole coordinate above. For $f_0(500)$, the same one-channel formula is not the correct primary language: the broad pole is obtained by solving the analytically continued Roy equations for $t_0^0(s)$. For πK resonances the unequal-mass Roy–Steiner construction above fixes the kinematics and the crossing input. For near-threshold exotic candidates, the effective-range and pole-motion charts of Definition 49.30 and the shallow effective-range calculation above specify what is meant by a molecular or compact coordinate. In every case the QCD statement is a pole or a stable mass extracted from a declared operator and regulator setup.

49.14.7 Unequal-Mass and Baryon-Meson Channels

Definition 49.37 (Unequal-mass elastic kinematics). Let $A(m_1) + B(m_2) \to A(m_1) + B(m_2)$ be elastic scattering of two stable color-singlet hadrons, and set

$$s_\pm := (m_1 \pm m_2)^2, \qquad \lambda(s, m_1^2, m_2^2) := (s - s_+)(s - s_-).$$

For $s > s_+$, define the center-of-mass momentum

$$k_s(s) := \frac{\lambda(s, m_1^2, m_2^2)^{1/2}}{2\sqrt{s}}.$$

If z_s is the cosine of the s-channel scattering angle, the Mandelstam variables in the physical elastic region are

$$t(s, z_s) = -2k_s(s)^2(1 - z_s), \qquad u(s, z_s) = 2(m_1^2 + m_2^2) - s - t(s, z_s). \tag{49.169}$$

The identity

$$s + t + u = 2(m_1^2 + m_2^2) \tag{49.170}$$

is part of the kinematic convention. For πK scattering one takes $(m_1, m_2) = (m_\pi, m_K)$.

Elastic unequal-mass projection. In the kinematics of Definition 49.37, a scalar amplitude $T(s, t, u)$ with fixed internal quantum numbers has the s-channel partial-wave expansion

$$T(s, t(s, z_s), u(s, z_s)) = 16\pi \sum_{\ell=0}^{\infty} (2\ell + 1) P_\ell(z_s) f_\ell(s), \tag{49.171}$$

with inverse

$$f_\ell(s) = \frac{1}{32\pi} \int_{-1}^{1} P_\ell(z_s) T(s, t(s, z_s), u(s, z_s)) \, \mathrm{d}z_s. \tag{49.172}$$

With

$$\rho_{12}(s) := \frac{2k_s(s)}{\sqrt{s}} = \frac{\lambda(s, m_1^2, m_2^2)^{1/2}}{s},$$

elastic unitarity in this normalization is

$$\operatorname{Im} f_\ell(s) = \rho_{12}(s) |f_\ell(s)|^2 \tag{49.173}$$

below the first inelastic threshold. In the center-of-mass frame the incoming and outgoing spatial momenta have magnitude $k_s(s)$. The energy of each species is the same before and after scattering, hence the momentum transfer has zero energy component and spatial magnitude squared

$$|\mathbf{k} - \mathbf{k}'|^2 = 2k_s(s)^2(1 - z_s).$$

With the Mandelstam convention used in this chapter this gives $t = -|\mathbf{k} - \mathbf{k}'|^2$, proving (49.169). The sum rule (49.170) follows directly from the definitions of s, t, u and the four on-shell conditions.

The partial-wave inversion is the Legendre orthogonality identity

$$\int_{-1}^{1} P_\ell(z) P_{\ell'}(z)\, \mathrm{d}z = \frac{2}{2\ell + 1}\delta_{\ell\ell'}$$

applied to (49.171). Finally the two-body phase-space density in these conventions is $\rho_{12}(s) = 2k_s(s)/\sqrt{s}$. Applying $S_\ell^* S_\ell = 1$ to

$$S_\ell(s) = 1 + 2\mathrm{i}\rho_{12}(s) f_\ell(s)$$

gives (49.173).

Definition 49.38 (Roy–Steiner input)**.** Roy–Steiner input for an unequal-mass two-hadron problem consists of:

1. the masses, exact internal quantum numbers, and invariant amplitudes $\mathbf{T}(s, t, u)$ in a declared basis of isospin or charge channels;
2. the crossing matrices relating the chosen s-channel basis to the u- and t-channel bases;
3. s-channel partial waves $f_\ell^a(s)$ and crossed-channel partial waves $g_\ell^c(t)$, each with an explicit normalization such as (49.171);
4. a fixed-t, fixed-u, or hyperbolic dispersion domain in which the required subtracted Cauchy representations are valid as boundary values of holomorphic functions;
5. subtraction constants, matching points, and high-energy/high-partial wave input sufficient to make all projected integrals converge.

This input is a reconstruction coordinate for amplitudes. It includes the analyticity, crossing, subtraction, and asymptotic assumptions; QCD supplies the channel Hilbert spaces and current or hadron matrix elements to which those assumptions are applied.

Proposition 49.39 (Roy–Steiner projection construction)**.** *Assume Roy–Steiner input in the sense of Definition 49.38. Let J_s and J_t be real intervals below the first omitted inelastic thresholds in the s- and crossed t-channels. Then the retained s-channel partial waves have the form*

$$\begin{aligned}\operatorname{Re} f_\ell^a(s) = k_\ell^a(s; \mathbf{a}) &+ \sum_{b,\ell'} \int_{s_+}^{\infty} \mathrm{d}s'\, K_{\ell\ell'}^{ab}(s, s')\, \operatorname{Im} f_{\ell'}^b(s') \\ &+ \sum_{c,\ell'} \int_{t_{\mathrm{th}}}^{\infty} \mathrm{d}t'\, G_{\ell\ell'}^{ac}(s, t')\, \operatorname{Im} g_{\ell'}^c(t'), \qquad s \in J_s, \end{aligned} \tag{49.174}$$

and the crossed-channel partial waves have analogous equations

$$\operatorname{Re} g_\ell^c(t) = \widetilde{k}_\ell^c(t;\mathbf{a}) + \sum_{b,\ell'} \int_{s_+}^{\infty} \mathrm{d}s'\, \widetilde{G}_{\ell\ell'}^{cb}(t,s')\ \operatorname{Im} f_{\ell'}^b(s') + \sum_{d,\ell'} \int_{t_{\mathrm{th}}}^{\infty} \mathrm{d}t'\, \widetilde{K}_{\ell\ell'}^{cd}(t,t')\ \operatorname{Im} g_{\ell'}^d(t'), \qquad t \in J_t. \tag{49.175}$$

Here $\mathbf{a}$ *denotes the declared subtraction constants. The kernels* $K, G, \widetilde{G}, \widetilde{K}$ *are the Legendre projections of the subtracted Cauchy denominators after the crossed-channel discontinuities have been rewritten with the declared crossing matrices.*

Proof. Begin with the subtracted Cauchy representation in the chosen Roy–Steiner domain. Its discontinuity terms are located on the physical cuts of the s, u, and t variables. The right-hand s-channel cut is already expressed in the basis whose partial waves are f_ℓ^a. The u- and t-channel cuts are first moved into their physical variables and then rewritten in the chosen s-channel basis using the crossing matrices included in the input. This produces a fixed-angle representation

$$\begin{aligned}
&\operatorname{Re} T^a(s, t(s,z_s), u(s,z_s)) \\
&= P^a(s,z_s;\mathbf{a}) + \sum_{b,\ell'} \int_{s_+}^{\infty} \mathrm{d}s'\, R_{\ell'}^{ab}(s,z_s;s')\ \operatorname{Im} f_{\ell'}^b(s') \\
&\quad + \sum_{c,\ell'} \int_{t_{\mathrm{th}}}^{\infty} \mathrm{d}t'\, H_{\ell'}^{ac}(s,z_s;t')\ \operatorname{Im} g_{\ell'}^c(t').
\end{aligned}$$

The convergence part of the input justifies the interchange of the partial-wave sums, dispersion integrals, and z_s-projection on J_s. Projecting with

$$\frac{1}{32\pi} \int_{-1}^{1} P_\ell(z_s)(\,\cdot\,)\,\mathrm{d}z_s$$

defines

$$k_\ell^a(s) = \frac{1}{32\pi} \int_{-1}^{1} P_\ell(z_s) P^a(s,z_s)\,\mathrm{d}z_s,$$

and similarly defines $K_{\ell\ell'}^{ab}$ and $G_{\ell\ell'}^{ac}$ as the same projection of $R_{\ell'}^{ab}$ and $H_{\ell'}^{ac}$. This gives (49.174). For the t-channel, use the crossed-channel angle z_t, the t-channel partial-wave normalization, and the crossed basis of amplitudes before projection. The same subtracted Cauchy denominators then project to $\widetilde{G}_{\ell\ell'}^{cd}$ and $\widetilde{K}_{\ell\ell'}^{cd}$, while the polynomial part projects to $\widetilde{k}_\ell^c(t)$. This gives (49.175). No additional dynamical assumption enters at the projection stage; all dynamical input is already contained in the analytic domain, crossing matrices, absorptive parts, and subtraction constants of the input. □

49.14.8 Quarkonium, Baryon, and Hybrid Source Charts

Definition 49.40 (Quarkonium spectroscopy chart)**.** A neutral heavy quarkonium entry is a color-singlet $Q\bar{Q}$ spectral coordinate with flavor $Q = c$ or $Q = b$, radial label n, orbital angular momentum $L = 0, 1, 2, \ldots$, total heavy-quark spin $S = 0, 1$, and total angular momentum J. The spectroscopic notation is

$$n\,^{2S+1}L_J, \qquad L = S, P, D, F, \ldots \quad \text{for} \quad L = 0, 1, 2, 3, \ldots. \tag{49.176}$$

For hidden flavor, charge conjugation is an exact quantum number in the isospin-symmetric QCD-plus-QED-decoupled convention. The low-lying chart is:

state type	$n\,^{2S+1}L_J$	J^{PC}	standard entries
pseudoscalar	$n\,^1S_0$	0^{-+}	$\eta_c(nS),\ \eta_b(nS)$
vector	$n\,^3S_1$	1^{--}	$J/\psi,\ \psi(nS),\ \Upsilon(nS)$
scalar triplet	$n\,^3P_0$	0^{++}	$\chi_{c0}(nP),\ \chi_{b0}(nP)$
axial triplet	$n\,^3P_1$	1^{++}	$\chi_{c1}(nP),\ \chi_{b1}(nP)$
tensor triplet	$n\,^3P_2$	2^{++}	$\chi_{c2}(nP),\ \chi_{b2}(nP)$
singlet P-wave	$n\,^1P_1$	1^{+-}	$h_c(nP),\ h_b(nP)$

The entries in the last column are names for QCD spectral levels or sheet-labeled poles in the corresponding channels. The chart by itself is not a potential model. A prediction of the level $M_{n\,^{2S+1}L_J}$ requires a regulator or an effective Hamiltonian with a stated heavy-quark mass scheme, matching coefficients, ultrasoft treatment, and error estimate.

Quarkonium J^{PC} rule. For a neutral $Q\bar{Q}$ color-singlet state in the chart of Definition 49.40, the allowed angular momenta are

$$J \in \{|L - S|, |L - S| + 1, \ldots, L + S\},$$

and the parity and charge-conjugation quantum numbers are

$$P = (-1)^{L+1}, \qquad C = (-1)^{L+S}. \tag{49.177}$$

The angular-momentum statement is the Clebsch–Gordan decomposition of the orbital representation L with the total spin representation S. For a fermion-antifermion pair, intrinsic parities multiply to -1. The orbital wavefunction contributes $(-1)^L$ under spatial inversion, giving

$$P = (-1)(-1)^L = (-1)^{L+1}.$$

Charge conjugation exchanges the quark and antiquark. In the center-of-mass frame this exchange reverses the relative coordinate, contributing $(-1)^L$. The spin wavefunction contributes $+1$ on the spin-singlet state and -1 on the spin-triplet state after the fermion exchange convention is included; this factor is $(-1)^S$ relative to the orbital factor in the standard $Q\bar{Q}$ convention. Therefore

$$C = (-1)^L(-1)^S = (-1)^{L+S}.$$

Substitution gives the entries in Definition 49.40: for example $^1S_0$ has 0^{-+}, $^3S_1$ has 1^{--}, ^3P_J has J^{++}, and $^1P_1$ has 1^{+-}.

Heavy quarkonium uses a different controlled coordinate system. If

$$m_Q \gg m_Q v \gg m_Q v^2$$

is a useful hierarchy, the pNRQCD Hamiltonian of the preceding sections predicts the charmonium and bottomonium level spacings through a potential coordinate, spin-dependent $1/m_Q$ operators, and ultrasoft corrections. The spin-averaged levels, hyperfine splittings, and fine structure are then matrix elements in a stated effective Hamiltonian. They are not predictions of a nonrelativistic potential unless the matching coefficients, mass scheme, and ultrasoft treatment have been fixed as part of the input. The angular momentum algebra that turns these matched operators into centroid, fine, and hyperfine coordinates is the finite-dimensional content of the J^{PC} rule above and the spin algebra paragraph.

Baryon spectroscopy is likewise a QCD spectral problem, not a three-body mechanics problem imposed from outside. For $N_f = 3$ light flavors, the lowest baryons organize approximately into flavor $SU(3)$ octet and decuplet representations when the quark masses are close. To first order in an octet symmetry-breaking Hamiltonian, the octet masses may be written as

$$M(Y, I) = M_0 + aY + b\left(I(I+1) - \frac{Y^2}{4}\right), \tag{49.178}$$

where Y is hypercharge and I is isospin. This coordinate chart gives the Gell-Mann–Okubo relation

$$2(M_N + M_\Xi) = 3M_\Lambda + M_\Sigma. \tag{49.179}$$

For the decuplet, first-order octet breaking gives equal spacing along the hypercharge ladder. These relations are symmetry consequences of a stated approximation; the actual QCD masses require lattice, dispersive, and chiral extrapolation inputs.

Assume that first-order $SU(3)$-breaking baryon masses are described by (49.178) in the octet and by $M_{10}(Y) = M_{10,0} + cY$ in the decuplet. Then (49.179) holds and

$$M_{\Sigma^*} - M_\Delta = M_{\Xi^*} - M_{\Sigma^*} = M_\Omega - M_{\Xi^*}. \tag{49.180}$$

For the octet,

$$(Y, I)_N = (1, \tfrac{1}{2}), \quad (Y, I)_\Lambda = (0, 0), \quad (Y, I)_\Sigma = (0, 1), \quad (Y, I)_\Xi = (-1, \tfrac{1}{2}).$$

Substitution in (49.178) gives

$$M_N = M_0 + a + \frac{b}{2}, \quad M_\Xi = M_0 - a + \frac{b}{2}, \quad M_\Lambda = M_0, \quad M_\Sigma = M_0 + 2b,$$

which implies (49.179). For the decuplet the hypercharges are $1, 0, -1, -2$. A linear function of Y has equal successive differences, giving (49.180).

In the large-N_c limit with N_f fixed, the baryon operator (49.137) contains N_c quark fields. The pair-counting argument around (49.54) gives a mass of order N_c, while collective spin-flavor splittings are typically $O(1/N_c)$ in the rotor regime. This is a controlled limit of QCD under its own hypotheses, and it explains why baryons require a separate large-N_c analysis from mesons. It does not license importing constituent-quark spectra as QCD predictions.

Definition 49.41 (Baryon-meson resonance reconstruction data)**.** Fix conserved quantum numbers: baryon number one, total isospin I, total strangeness S, total angular momentum and parity J^P, and any other exact charges in the chosen QCD mass convention. A two-body baryon-meson resonance reconstruction data consist of:

1. a finite list of stable color-singlet two-body channels $a = (B_a, M_a)$ in the chosen energy window, with baryon mass M_{B_a}, meson mass m_{M_a}, spin s_{B_a}, meson spin s_{M_a}, and intrinsic parities P_{B_a}, P_{M_a};

2. a list of retained orbital angular momenta L_a and spin couplings producing the declared J^P;
3. the infinite-volume partial-wave matrix $t_{ab}^{J^P}(s)$, with phase-space matrix

$$\rho_a(s) = \frac{2k_a(s)}{\sqrt{s}}, \qquad \sqrt{s} = \sqrt{M_{B_a}^2 + k_a(s)^2} + \sqrt{m_{M_a}^2 + k_a(s)^2};$$

4. a reconstruction method for t^{J^P}, for example a coupled-channel finite-volume determinant or a Roy–Steiner representation;
5. a sheet label $\Sigma = \mathrm{diag}(\sigma_a)$, with $\sigma_a = \pm 1$, specifying which channel momenta have been continued through their unitarity cuts;
6. pole positions s_* and residue vectors g_a defined by

$$t_{ab}^{(\Sigma),J^P}(s) = \frac{g_a g_b}{s_* - s} + h_{ab}(s), \tag{49.181}$$

   where h_{ab} is holomorphic at s_* on the same sheet.

The invariant content is the pole position, the sheet, and the residue projective vector g_a, together with the operator and regulator data by which the amplitude was obtained. Names such as radial excitation, molecule, compact multiquark configuration, or meson-baryon bound state are interpretive coordinates on these data.

Pseudoscalar-meson–baryon partial-wave quantum numbers. Let a positive-parity spin-$\frac{1}{2}$ baryon B scatter with a pseudoscalar meson M of spin 0 and parity -1. In a two-body partial wave of orbital angular momentum L, the allowed total angular momenta and parity are

$$J \in \begin{cases} \{\frac{1}{2}\}, & L = 0, \\ \{L - \frac{1}{2}, L + \frac{1}{2}\}, & L \geq 1, \end{cases} \qquad P = (-1)^{L+1}. \tag{49.182}$$

Consequently an S-wave channel has $J^P = \frac{1}{2}^-$, while a P-wave channel has $J^P = \frac{1}{2}^+$ or $J^P = \frac{3}{2}^+$. The two-body Hilbert space at fixed center-of-mass momentum decomposes under spatial rotations as the tensor product of the orbital representation of spin L, the baryon spin representation $\frac{1}{2}$, and the meson spin representation 0. Therefore the total spin content is

$$L \otimes \frac{1}{2} = \begin{cases} \frac{1}{2}, & L = 0, \\ (L - \frac{1}{2}) \oplus (L + \frac{1}{2}), & L \geq 1, \end{cases}$$

which gives the allowed J's. Parity multiplies the intrinsic parities by the orbital factor $(-1)^L$:

$$P = P_B P_M (-1)^L = (+1)(-1)(-1)^L = (-1)^{L+1}.$$

The S- and P-wave cases are the first two values of L.

This elementary representation-theoretic step is often where baryon resonance nomenclature enters the analysis. The $\Delta(1232)$ channel is the $I = \frac{3}{2}$, $J^P = \frac{3}{2}^+$ P-wave πN amplitude. A Roper-type nucleon resonance is a pole in the $I = \frac{1}{2}$, $J^P = \frac{1}{2}^+$ channel, with residues into the retained πN, ηN, and multi-hadron channels determined by the chosen coupled-channel amplitude. The $\Lambda(1405)$-type channel has $I = 0$, $S = -1$, $J^P = \frac{1}{2}^-$, and its minimal two-body description couples the $\bar{K}N$ and $\pi\Sigma$ S-wave channels. In each case the QCD statement is the sheet-labeled pole of Definition 49.41. A finite-volume calculation supplies color-singlet energy levels through finite source-family correlators; the pole information is then reconstructed only after a coupled-channel quantization condition and analytic continuation have been specified. A dispersive calculation supplies it through the corresponding analyticity and crossing assumptions.

Proposition 49.42 (Two-channel sheet continuation and residue vector)**.** *Fix a partial wave and an energy neighborhood in which two stable color-singlet two-body channels are the only open channels retained in the analysis. Suppose that, in this neighborhood, the coupled-channel amplitude admits a K-matrix representation*

$$t^{\text{phys}}(s) = \left[K(s)^{-1} - \mathrm{i}\ \mathrm{diag}\big(\rho_1(s), \rho_2(s)\big)\right]^{-1}, \tag{49.183}$$

where $K(s)^{-1}$ is a complex-analytic symmetric matrix that is real symmetric on the physical real interval, and each $\rho_a(s)$ is the chosen square-root phase-space branch for channel a. For a sheet label $\sigma = (\sigma_1, \sigma_2)$, $\sigma_a = \pm 1$, define

$$D_\sigma(s) := K(s)^{-1} - \mathrm{i}\ \mathrm{diag}\big(\sigma_1\rho_1(s), \sigma_2\rho_2(s)\big), \qquad t^{(\sigma)}(s) := D_\sigma(s)^{-1}. \tag{49.184}$$

Then a resonance pole on this sheet is a zero of

$$\Delta_\sigma(s) := \det D_\sigma(s). \tag{49.185}$$

If s_* is a simple zero of Δ_σ and $\operatorname{rank} D_\sigma(s_*) = 1$, *then*

$$t^{(\sigma)}(s) = -\frac{\operatorname{adj} D_\sigma(s_*)}{\Delta'_\sigma(s_*)}\,\frac{1}{s_* - s} + O(1). \tag{49.186}$$

Because $D_\sigma(s_*)$ is complex symmetric, the adjugate in (49.186) *has rank one and can be written as $c\, v_a v_b$, where v spans* $\ker D_\sigma(s_*)$. *Thus the projective residue vector in Definition 49.41 is the null direction of the analytically continued denominator, with the overall complex phase fixed only after choosing a residue normalization.*

In the constant local coordinate chart

$$K^{-1} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad \rho_a(s) = \rho_a,$$

the pole equation on sheet σ is

$$(a - \mathrm{i}\sigma_1\rho_1)(c - \mathrm{i}\sigma_2\rho_2) - b^2 = 0. \tag{49.187}$$

Proof. Equation (49.183) is the representation assumed in the proposition. On the physical interval it is equivalent to the two-channel elastic-unitarity identity

$$\mathrm{Im}\big(t^{\text{phys}}(s)^{-1}\big) = -\,\mathrm{diag}(\rho_1(s), \rho_2(s))$$

on the physical interval. Analytic continuation through the unitarity cut of channel a changes the sign of the square root $k_a(s)$, hence of $\rho_a(s)$, and leaves the other channel branches unchanged. This gives (49.184). Since the matrix inverse is meromorphic wherever the entries are analytic, its poles are precisely the zeros of $\det D_\sigma$, unless the adjugate vanishes to sufficient order to cancel the zero. Under the stated rank-one and simple-zero hypotheses, the adjugate does not vanish, and

$$D_\sigma(s)^{-1} = \frac{\operatorname{adj} D_\sigma(s)}{\Delta_\sigma(s)} = \frac{\operatorname{adj} D_\sigma(s_*)}{(s-s_*)\Delta'_\sigma(s_*)} + O(1),$$

which is (49.186) after rewriting $(s-s_*)^{-1} = -(s_*-s)^{-1}$. A 2×2 matrix of rank one has rank-one adjugate; for a complex symmetric matrix the left and right null directions agree, so this rank-one matrix is proportional to vv^T. The constant formula (49.187) is the determinant of the displayed 2×2 denominator. □

This proposition is the minimal algebra behind statements about the $\Lambda(1405)$-type $\bar{K}N$-$\pi\Sigma$ sector, hidden-charm pentaquark channels, and near-threshold heavy-meson channels. The number of poles, their locations, and their residues are outputs of a finite-volume or dispersive reconstruction. The phrase "molecular" can be used only after a specific residue, pole-motion, or source-overlap coordinate has been declared; it is not an independent QCD definition.

Definition 49.43 (Light-baryon spectroscopy charts)**.** Work with baryon number $B = 1$ and three light flavors (u, d, s). Let strangeness be denoted by $S_{\rm str}$, so the hypercharge used in (49.178) is $Y = B + S_{\rm str} = 1 + S_{\rm str}$. The ground-state flavor multiplet charts used below are:

multiplet	**entry**	**I**	**Y**	$\mathbf{J}^P$
8	$N = (p, n)$	$\frac{1}{2}$	1	$\frac{1}{2}^+$
8	Λ	0	0	$\frac{1}{2}^+$
8	$\Sigma = (\Sigma^+, \Sigma^0, \Sigma^-)$	1	0	$\frac{1}{2}^+$
8	$\Xi = (\Xi^0, \Xi^-)$	$\frac{1}{2}$	-1	$\frac{1}{2}^+$
10	Δ	$\frac{3}{2}$	1	$\frac{3}{2}^+$
10	Σ^*	1	0	$\frac{3}{2}^+$
10	Ξ^*	$\frac{1}{2}$	-1	$\frac{3}{2}^+$
10	Ω^-	0	-2	$\frac{3}{2}^+$

The table is a chart of exact conserved quantum numbers and approximate flavor-$SU(3)$ organization. The masses are QCD spectral values obtained from the color-singlet Hamiltonian or from a regulator converging to it. Equation (49.179) and (49.180) are first-order coordinates on these spectral values; they are not definitions of the baryons.

Definition 49.44 (Excited light-baryon channel chart)**.** An excited light-baryon entry in this monograph is specified by:

(i) exact charges $B = 1$, I, $S_{\rm str}$, J^P, and any additional exact discrete quantum numbers of the chosen mass convention;

(ii) a finite list of two-body and effective multi-hadron channels retained in the reconstruction of Definition 49.41;

(iii) a sheet-labeled pole s_* of the corresponding color-singlet amplitude, or an isolated stable level if the state lies below every open channel in the stated volume and mass regime;

(iv) the residue projective vector into the retained channels, when the interpretation of the pole requires an overlap or composition coordinate.

The following entries are the minimal light-baryon examples that must be carried through any concrete coupled-channel analysis:

(a) $\Delta(1232)$: charges $(I, S_{\text{str}}, J^P) = (\frac{3}{2}, 0, \frac{3}{2}^+)$; retained channel πN in the P-wave; QCD output a nearby second-sheet pole of $t^{3/2^+}_{\pi N,\pi N}$.

(b) $N(1440)$-type channel: charges $(\frac{1}{2}, 0, \frac{1}{2}^+)$; retained channels $\pi N, \eta N, \pi\pi N$; QCD output a coupled-channel pole whose radial-label interpretation comes from pole motion and source overlaps.

(c) $N(1535)$-type channel: charges $(\frac{1}{2}, 0, \frac{1}{2}^-)$; retained channels $\pi N, \eta N, K\Lambda, K\Sigma$; QCD output an S-wave pole with residue data.

(d) $\Sigma(1385)$: charges $(1, -1, \frac{3}{2}^+)$; retained channels $\pi\Lambda, \pi\Sigma, \bar{K}N$; QCD output a strange P-wave pole.

(e) $\Lambda(1405)$-type channel: charges $(0, -1, \frac{1}{2}^-)$; retained channels $\bar{K}N, \pi\Sigma$; QCD output one or more sheet-labeled S-wave poles.

(f) $\Lambda(1520)$: charges $(0, -1, \frac{3}{2}^-)$; retained channels $\bar{K}N, \pi\Sigma$; QCD output a D-wave dominated pole coordinate.

The words "Roper", "$\Lambda(1405)$", and similar labels are therefore shorthand for entries in this chart. A lattice calculation supplies the finite-volume spectrum and operator overlaps; a dispersive or Hamiltonian-scattering analysis supplies an analytic continuation. Without that continuation there is no invariant resonance mass or width.

Definition 49.45 (Heavy-baryon QCD/HQET chart)**.** Let $Q = c$ or b, and consider one-heavy-quark baryons Qqq. After the heavy-quark velocity v^μ and mass scheme for m_Q are fixed, the color-singlet Hilbert-space sector admits an HQET expansion in which the light degrees of freedom carry angular momentum j_ℓ and flavor quantum numbers. The basic entries are:

family	**light flavor**	$\mathbf{j}_\ell$	J^P **at leading order**
Λ_Q, Ξ_Q	$\bar{\mathbf{3}}$ of light flavor	0	$\frac{1}{2}^+$
$\Sigma_Q, \Xi'_Q, \Omega_Q$	$\mathbf{6}$ of light flavor	1	$\frac{1}{2}^+, \frac{3}{2}^+$

The leading HQET statement is a spectral degeneracy at fixed j_ℓ and light-flavor representation, before $1/m_Q$ chromomagnetic terms are included. Doubly heavy baryons such as Ξ_{cc}, Ξ_{bc}, and Ω_{cc} are QCD sectors with two heavy fields and one light field. They may be organized by heavy-diquark variables only after a scale separation and color-$\bar{\mathbf{3}}$ matching calculation has been specified.

Definition 49.46 (Near-threshold heavy spectroscopy chart)**.** Heavy-flavor spectroscopy contains stable baryon sectors and resonance or near-threshold pole sectors. The following chart fixes the QCD object before any model language is introduced:

(a) Ξ_{cc}, Ω_{cc}: exact charges $B = 1,\ C = 2$; minimal sources ccq; QCD output isolated color-singlet masses when the levels lie below the relevant strong thresholds.

(b) $\Lambda_b, \Sigma_b, \Xi_b, \Omega_b$: exact charges $B = 1$ with bottom flavor -1; minimal sources bqq; QCD output stable or narrow levels organized by the HQET chart above.

(c) P_c-type hidden-charm baryon sector: exact charges $B = 1,\ C = 0$ with hidden $c\bar{c}$; retained channels $J/\psi\, p, \Sigma_c\bar{D}, \Sigma_c\bar{D}^*, \ldots$; QCD output a sheet-labeled baryon-meson pole.

(d) T_{cc}-type sector: exact charges $B = 0,\ C = 2$; retained channels $DD^*, D^*D^*, \ldots$; QCD output a near-threshold meson-meson pole or bound-state pole.

(e) $X(6900)$-type sector: exact charges $B = 0$, $C = 0$ with hidden $c\bar{c}c\bar{c}$; retained channels $J/\psi\, J/\psi, \eta_c\eta_c, \ldots$; QCD output a coupled heavy-quarkonium pole.

The entries with $B = 0$ are included because they use the same pole-motion and effective-range machinery as the baryon-number-one pentaquark channels. Labels such as compact tetraquark, hadronic molecule, or diquark-antidiquark are then coordinates on residues, pole motion under quark-mass variation, large-N_c behavior, and source-overlap coordinates. They are not primitive spectral definitions.

Definition 49.47 (Glueball and hybrid spectroscopy source families)**.** Fix J^{PC}, flavor quantum numbers, and a finite-volume regulator with a transfer matrix. A glueball source family is a finite-dimensional subspace $\mathcal{V}_G$ spanned by gauge-invariant pure-glue operators, such as traces of small Wilson loops or local curvature polynomials after operator renormalization. A flavor-singlet meson source family is a subspace $\mathcal{V}_M$ spanned by color-singlet quark bilinears and multi-hadron operators with the same exact quantum numbers. A hybrid source family is a subspace $\mathcal{V}_H$ spanned by gauge-invariant sources in which quark fields are connected by Wilson lines and one or more local curvature insertions, for example representative operator families of the form $\bar{q}(x)\Gamma\, U(x,y)F_{\mu\nu}(y)U(y,z)q(z)$ after the endpoint and composite-operator renormalizations are specified.

The QCD spectrum is the spectrum extracted from the full correlation matrix on the span of these source spaces and their multi-hadron completions. The labels glueball, hybrid, and flavor-singlet meson are source-family coordinates on eigenvectors and pole residues. They become invariant only after a supplementary limit or comparison is stated, for example the pure Yang–Mills limit, the large-N_c limit, or a declared quark-mass deformation path.

Controlled Approximation 49.48 (Two-state mixing coordinate)**.** A two-state mixing matrix is a coordinate on a controlled finite-dimensional projection of the transfer-matrix problem, not a definition of the QCD spectrum. It is admissible only after the source subspaces, the inner product, the fitting window, and the error made by omitting other levels have been stated. Within such a projection, suppose two normalized basis vectors $|G\rangle \in \mathcal{V}_G$ and $|M\rangle \in \mathcal{V}_M \oplus \mathcal{V}_H$ give the self-adjoint matrix

$$H_{\text{eff}}(x) = \begin{pmatrix} M_G(x) & h(x) \\ h(x) & M_M(x) \end{pmatrix}, \tag{49.188}$$

where x is a declared deformation coordinate, such as a quark mass or $1/N_c$.

Remark 49.49 (Avoided crossing in a two-state spectroscopy chart). In the controlled approximation 49.48, the two projected energies are

$$E_\pm(x) = \frac{M_G(x) + M_M(x)}{2} \pm \sqrt{\left(\frac{M_M(x) - M_G(x)}{2}\right)^2 + h(x)^2}. \tag{49.189}$$

If $M_G(x_0) = M_M(x_0)$ and $h(x_0) \neq 0$, the projected levels are separated by $2|h(x_0)|$. Away from degeneracy, a real orthogonal diagonalization angle $\theta(x)$ may be chosen so that

$$\tan 2\theta(x) = \frac{2h(x)}{M_M(x) - M_G(x)}. \tag{49.190}$$

The characteristic polynomial of (49.188) is

$$E^2 - (M_G + M_M)E + M_G M_M - h^2.$$

Solving this quadratic gives (49.189). At $M_G = M_M$, the square root is $|h|$, so the separation is $2|h|$. For the angle, write the rotated off-diagonal entry as

$$\frac{1}{2}(M_G - M_M)\sin 2\theta + h\cos 2\theta.$$

Setting it to zero gives $(M_M - M_G)\sin 2\theta = 2h\cos 2\theta$, which is (49.190) when the denominator is nonzero. At degeneracy the diagonalizing angle is $\theta = \pi/4$ modulo basis signs.

The proposition explains why an eigenstate cannot be assigned an invariant glueball or hybrid percentage from a two-by-two table alone. The invariant quantities are the QCD energies or pole positions and the declared matrix elements. The percentages are coordinates depending on the finite source space, normalization, and deformation path.

Heavy-baryon spin average removes the chromomagnetic coordinate. Fix a one-heavy-quark baryon family with $j_\ell > 0$. Suppose that, through order $1/m_Q$, the only spin-dependent term in the effective Hamiltonian on the two-state multiplet $J = j_\ell \pm \frac{1}{2}$ is

$$H_{\mathrm{mag}} = \frac{2\kappa_{j_\ell}}{m_Q}\,\mathbf{S}_Q\cdot\mathbf{j}_\ell, \tag{49.191}$$

where $\mathbf{S}_Q^2 = \frac{3}{4}$ and $\mathbf{j}_\ell^2 = j_\ell(j_\ell+1)$. Then the degeneracy-weighted spin average

$$\overline{M}_{j_\ell} := \frac{(2j_\ell)M_{j_\ell-\frac{1}{2}} + (2j_\ell+2)M_{j_\ell+\frac{1}{2}}}{4j_\ell+2} \tag{49.192}$$

is independent of κ_{j_ℓ} at this order. In particular, for the Σ_Q, Σ_Q^*-type doublet, $j_\ell = 1$ and

$$\overline{M}_{\Sigma_Q} = \frac{M_{\Sigma_Q} + 2M_{\Sigma_Q^*}}{3} \tag{49.193}$$

has no chromomagnetic $1/m_Q$ contribution in the stated scheme.

On a state with total spin J,

$$\mathbf{S}_Q\cdot\mathbf{j}_\ell = \frac{1}{2}\left[J(J+1) - j_\ell(j_\ell+1) - \frac{3}{4}\right].$$

For $J_- = j_\ell - \frac{1}{2}$, this eigenvalue is

$$\frac{1}{2}\left[(j_\ell - \tfrac{1}{2})(j_\ell + \tfrac{1}{2}) - j_\ell(j_\ell+1) - \frac{3}{4}\right] = -\frac{j_\ell+1}{2}.$$

For $J_+ = j_\ell + \frac{1}{2}$, it is

$$\frac{1}{2}\left[(j_\ell + \tfrac{1}{2})(j_\ell + \tfrac{3}{2}) - j_\ell(j_\ell+1) - \frac{3}{4}\right] = \frac{j_\ell}{2}.$$

The degeneracies are $2J_- + 1 = 2j_\ell$ and $2J_+ + 1 = 2j_\ell + 2$. Hence

$$(2j_\ell)\left(-\frac{j_\ell+1}{2}\right) + (2j_\ell+2)\left(\frac{j_\ell}{2}\right) = 0.$$

The spin-average (49.192) therefore cancels the coefficient κ_{j_ℓ} independently of its value. The remaining terms are the heavy-quark mass coordinate, the spin-independent light-sector energy, the kinetic $1/m_Q$ coordinate, and higher-order corrections.

Remark 49.50 (Large-N_c baryon catalog coordinate). The large-N_c baryon discussion preceding (49.54) gives a second controlled organization of baryon spectra. Its invariant inputs are the QCD Hamiltonian, baryon number one, fixed N_f, and the large-N_c scaling limit. Under the Hartree and rotor hypotheses stated there, baryon masses have an $O(N_c)$ common part and $O(1/N_c)$ spin-flavor splittings. The contracted $SU(2N_f)$ algebra is a leading-order relation among baryon matrix elements in this limit. It complements the finite-N_c spectral and coupled-channel charts above; it does not replace the QCD pole or eigenvalue definitions.

49.14.9 Current Matrix Elements and Forward Sum Rules

Definition 49.51 (Nucleon current matrix elements)**.** Let $|N(p,s)\rangle$ be an isolated one-nucleon state in the color-singlet QCD Hilbert space, with $p^2 = -M_N^2$, $p^0 > 0$, and spin label s. In the mostly-plus spinor convention of Section 40.16, use the same Dirac–Pauli basis as in Definition 43.4. For the Hermitian electromagnetic current J^μ_{em}, normalized by the flavor charges of the quarks, define the nucleon Dirac and Pauli form factors by

$$\langle N(p',s')|J^\mu_{\text{em}}(0)|N(p,s)\rangle = \frac{\mathrm{i}e}{(2\pi)^3}\,\bar u_{s'}(p')\left[\gamma^\mu F_1^N(k^2) + \frac{1}{M_N}S^{\mu\nu}k_\nu F_2^N(k^2)\right]u_s(p), \qquad k := p - p', \tag{49.194}$$

where $e > 0$ is the elementary charge and $S^{\mu\nu} = -\mathrm{i}[\gamma^\mu,\gamma^\nu]/4$. The charge normalization is

$$F_1^p(0) = 1, \qquad F_1^n(0) = 0.$$

The anomalous magnetic moment coordinate is $F_2^N(0)$, and the magnetic moment in nuclear magnetons is $F_1^N(0) + F_2^N(0)$ after the conventional normalization by $e/(2M_N)$.

For spacelike transfer set

$$Q^2 := -k^2 > 0, \qquad \tau := \frac{Q^2}{4M_N^2}.$$

The Sachs coordinates are the linear combinations

$$G_E^N(Q^2) = F_1^N(-Q^2) - \tau F_2^N(-Q^2), \qquad G_M^N(Q^2) = F_1^N(-Q^2) + F_2^N(-Q^2). \tag{49.195}$$

They are useful Breit-frame coordinates for elastic electron–nucleon scattering, but the nonperturbative definition is the current matrix element (49.194). A lattice, dispersive, or experimental extraction must state the current renormalization, finite-volume matrix-element method, excited-state removal, and analytic continuation or fit ansatz used to obtain the functions of Q^2.

Remark 49.52 (Sachs and Dirac–Pauli coordinates). For $\tau \neq -1$, the coordinate change (49.195) is invertible:

$$F_1^N(-Q^2) = \frac{G_E^N(Q^2) + \tau G_M^N(Q^2)}{1+\tau}, \qquad F_2^N(-Q^2) = \frac{G_M^N(Q^2) - G_E^N(Q^2)}{1+\tau}. \tag{49.196}$$

At $Q^2 = 0$,

$$G_E^N(0) = F_1^N(0), \qquad G_M^N(0) = F_1^N(0) + F_2^N(0).$$

The two equations in (49.195) form the linear system

$$\begin{pmatrix} G_E \\ G_M \end{pmatrix} = \begin{pmatrix} 1 & -\tau \\ 1 & 1 \end{pmatrix}\begin{pmatrix} F_1 \\ F_2 \end{pmatrix}.$$

The determinant is $1+\tau$. Inverting the matrix gives (49.196). Setting $Q^2=0$, hence $\tau=0$, gives the stated zero-transfer normalizations. No model of nucleon substructure enters this argument; it is only a change of coordinates on the same current matrix element.

Definition 49.53 (Isovector axial form factors)**.** Let A_a^μ be the renormalized isovector axial current, with $a=1,2,3$ and flavor generator $\tau_a/2$. Between one-nucleon states, parity, Lorentz covariance, and isospin give the decomposition

$$\langle N(p',s')|A_a^\mu(0)|N(p,s)\rangle = \frac{\mathrm{i}}{(2\pi)^3}\,\bar u_{s'}(p')\left[\gamma^\mu\gamma_5\,G_A(Q^2)+\frac{k^\mu}{2M_N}\gamma_5\,G_P(Q^2)\right]\frac{\tau_a}{2}\,u_s(p), \quad (49.197)$$

again with $Q^2=-k^2$. The number $g_A := G_A(0)$ is the axial charge in this current normalization. The induced pseudoscalar form factor G_P contains the pion-pole contribution dictated by the chiral Ward identity and the pion effective theory of Chapter 55; beyond that pole, its regular part is another QCD matrix element. Thus nucleon axial data are part of the same ab initio program as the mass spectrum: they are not defined by a quark-model wave function but by renormalized current insertions between color-singlet Hilbert-space states.

Proposition 49.54 (Stable-current three-point extraction)**.** *Let a reflection-positive finite-volume regulator of QCD on a spatial torus $\mathbb{T}_L^3$ be fixed, with Hilbert space $\mathcal{H}_L$, vacuum Ω_L, self-adjoint Hamiltonian $H_L \geq 0$, and momentum operators $\mathbf{P}_L$. Let $J_{\mathcal{R}}$ be a renormalized, gauge-invariant current in a specified operator-renormalization scheme $\mathcal{R}$, and let $J_{\mathcal{R},\mathbf{q}}$ denote its spatial Fourier component carrying momentum $\mathbf{q}$. Choose momenta $\mathbf{p},\mathbf{p}'\in(2\pi/L)\mathbb{Z}^3$ with $\mathbf{p}'=\mathbf{p}+\mathbf{q}$. Suppose that in the two relevant baryon-number-one momentum sectors there are normalized one-nucleon eigenvectors*

$$H_L|N,L;\mathbf{p},r\rangle = E_L(\mathbf{p})|N,L;\mathbf{p},r\rangle, \qquad H_L|N,L;\mathbf{p}',r'\rangle = E_L(\mathbf{p}')|N,L;\mathbf{p}',r'\rangle,$$

isolated from the rest of the spectrum in the same quantum-number channels by positive gaps $\Delta_L(\mathbf{p})$ and $\Delta_L(\mathbf{p}')$. Here r,r' label chosen rows in the relevant finite-volume symmetry irreducible representations; after the $L\to\infty$ limit these rows are matched to the usual spin labels. Let $\mathcal{O}^\dagger_{\mathbf{p},r}$ and $\mathcal{O}^\dagger_{\mathbf{p}',r'}$ be gauge-invariant baryon sources, projected to the indicated momentum and row, and set

$$\psi_{\mathbf{p},r} := \mathcal{O}^\dagger_{\mathbf{p},r}\Omega_L, \qquad \psi_{\mathbf{p}',r'} := \mathcal{O}^\dagger_{\mathbf{p}',r'}\Omega_L.$$

Assume their one-particle overlaps

$$Z_{\mathbf{p},r} := \langle N,L;\mathbf{p},r|\psi_{\mathbf{p},r}\rangle, \qquad Z_{\mathbf{p}',r'} := \langle N,L;\mathbf{p}',r'|\psi_{\mathbf{p}',r'}\rangle$$

are nonzero. At fixed regulator, assume $J_{\mathcal{R},\mathbf{q}}$ is bounded on the finite-energy subspace generated by these sources; on a lattice with compact gauge variables and local fermion bilinears this is the ordinary boundedness statement for the regulated local insertion.

Define the Euclidean two- and three-point functions

$$C_2(t;\mathbf{p},r) := \langle\psi_{\mathbf{p},r}, e^{-H_L t}\psi_{\mathbf{p},r}\rangle, \quad (49.198)$$

$$C_3^{J_{\mathcal{R}}}(t,\tau;\mathbf{p}',r';\mathbf{p},r) := \langle\psi_{\mathbf{p}',r'}, e^{-H_L(t-\tau)}J_{\mathcal{R},\mathbf{q}}e^{-H_L\tau}\psi_{\mathbf{p},r}\rangle, \qquad 0<\tau<t. \quad (49.199)$$

Then

$$C_2(t;\mathbf{p},r) = |Z_{\mathbf{p},r}|^2 e^{-E_L(\mathbf{p})t} + R_2(t;\mathbf{p},r), \qquad |R_2(t;\mathbf{p},r)| \leq \|\eta_{\mathbf{p},r}\|^2 e^{-(E_L(\mathbf{p})+\Delta_L(\mathbf{p}))t}, \quad (49.200)$$

where $\eta_{\mathbf{p},r} := (1 - P_{\mathbf{p},r})\psi_{\mathbf{p},r}$ *and* $P_{\mathbf{p},r}$ *is the spectral projection onto* $|N, L; \mathbf{p}, r\rangle$*. Moreover*

$$\begin{aligned} C_3^{J_{\mathcal{R}}}(t, \tau; \mathbf{p}', r'; \mathbf{p}, r) = \overline{Z_{\mathbf{p}',r'}} Z_{\mathbf{p},r}\, e^{-E_L(\mathbf{p}')(t-\tau)} e^{-E_L(\mathbf{p})\tau} \langle N, L; \mathbf{p}', r' | J_{\mathcal{R},\mathbf{q}} | N, L; \mathbf{p}, r\rangle \\ + R_3(t, \tau; \mathbf{p}', r'; \mathbf{p}, r), \end{aligned} \tag{49.201}$$

with the explicit bound

$$\begin{aligned} |R_3| \le \|J_{\mathcal{R},\mathbf{q}}\| \Big(&\|\eta_{\mathbf{p}',r'}\|\, |Z_{\mathbf{p},r}|\, e^{-(E_L(\mathbf{p}')+\Delta_L(\mathbf{p}'))(t-\tau)} e^{-E_L(\mathbf{p})\tau} \\ &+ |Z_{\mathbf{p}',r'}|\, \|\eta_{\mathbf{p},r}\|\, e^{-E_L(\mathbf{p}')(t-\tau)} e^{-(E_L(\mathbf{p})+\Delta_L(\mathbf{p}))\tau} \\ &+ \|\eta_{\mathbf{p}',r'}\|\, \|\eta_{\mathbf{p},r}\|\, e^{-(E_L(\mathbf{p}')+\Delta_L(\mathbf{p}'))(t-\tau)} e^{-(E_L(\mathbf{p})+\Delta_L(\mathbf{p}))\tau} \Big). \end{aligned} \tag{49.202}$$

Consequently the ratio

$$\mathcal{R}_{J_{\mathcal{R}}}(t, \tau) := \frac{C_3^{J_{\mathcal{R}}}(t, \tau; \mathbf{p}', r'; \mathbf{p}, r)}{\overline{Z_{\mathbf{p}',r'}} Z_{\mathbf{p},r}\, e^{-E_L(\mathbf{p}')(t-\tau)} e^{-E_L(\mathbf{p})\tau}} \tag{49.203}$$

satisfies

$$\mathcal{R}_{J_{\mathcal{R}}}(t, \tau) = \langle N, L; \mathbf{p}', r' | J_{\mathcal{R},\mathbf{q}} | N, L; \mathbf{p}, r\rangle + O(e^{-\Delta_L(\mathbf{p}')(t-\tau)}) + O(e^{-\Delta_L(\mathbf{p})\tau})$$

as $\tau \to \infty$ *and* $t - \tau \to \infty$*, with constants fixed by the source norms, overlaps, and regulated-current norm above.*

Proof. The proof is the spectral theorem for the finite-volume transfer-matrix Hamiltonian. We give the details because this is the mathematical content behind the phrase "extracting a form factor". Decompose the source vectors into their one-particle components and orthogonal remainders,

$$\psi_{\mathbf{p},r} = Z_{\mathbf{p},r} |N, L; \mathbf{p}, r\rangle + \eta_{\mathbf{p},r}, \qquad \psi_{\mathbf{p}',r'} = Z_{\mathbf{p}',r'} |N, L; \mathbf{p}', r'\rangle + \eta_{\mathbf{p}',r'}.$$

By the assumed spectral gaps, the spectral support of $\eta_{\mathbf{p},r}$ lies in $[E_L(\mathbf{p}) + \Delta_L(\mathbf{p}), \infty)$, and similarly for $\eta_{\mathbf{p}',r'}$. Hence

$$\|e^{-H_L s} \eta_{\mathbf{p},r}\| \le e^{-(E_L(\mathbf{p})+\Delta_L(\mathbf{p}))s} \|\eta_{\mathbf{p},r}\|, \qquad s \ge 0,$$

and the corresponding estimate holds in the primed channel. Substituting the decomposition of $\psi_{\mathbf{p},r}$ into (49.198) gives the leading term $|Z_{\mathbf{p},r}|^2 e^{-E_L(\mathbf{p})t}$. The cross terms vanish because $\eta_{\mathbf{p},r}$ is orthogonal to the one-particle eigenspace and $e^{-H_L t}$ preserves the spectral decomposition. The remaining η-term is bounded by the preceding estimate, proving (49.200).

For the three-point function, insert the same decompositions on the sink and source sides of (49.199). The term in which both spectral projections are one-particle projections is exactly the first line of (49.201). The three remaining terms contain an excited component on the sink side, on the source side, or on both sides. Applying the gap estimate to the side on which the excited component propagates, and applying Cauchy's inequality together with the regulated operator norm of $J_{\mathcal{R},\mathbf{q}}$, gives the three terms in (49.202). Division by the nonzero leading overlap factor gives (49.203) and the stated asymptotic error. Nothing in the argument assumes a constituent description of the nucleon; the only inputs are reflection positivity, spectral isolation, a renormalized gauge-invariant current insertion, and nonzero overlap of the chosen source with the state. □

The continuum nucleon form factors of Definition 49.51 are obtained from the finite-volume matrix element in Proposition 49.54 only after the current-renormalization scheme has been fixed and the regulator and volume limits have been taken with the chosen state normalization. A plateau in a ratio of Euclidean correlators is therefore evidence for the hypotheses of the proposition in a numerical calculation; it is not itself the definition of the form factor.

Definition 49.55 (Forward Compton amplitude and polarizabilities)**.** Let N be a stable proton or neutron state of mass M_N and spin $\frac{1}{2}$. Work in the nucleon rest frame and let a real photon have energy $\omega > 0$, direction $\mathbf{n} \in S^2$, and transverse polarization representatives $\boldsymbol{\epsilon}, \boldsymbol{\epsilon}'$ satisfying $\mathbf{n} \cdot \boldsymbol{\epsilon} = \mathbf{n} \cdot \boldsymbol{\epsilon}' = 0$. The forward real-Compton amplitude is the on-shell, connected, LSZ-reduced matrix element of two insertions of the renormalized electromagnetic current J^{μ}_{em}, with the photon coupling and the one-nucleon state normalization fixed by the scattering chapters. For a spin state with unit spin vector $\mathbf{s}$, define the scalar forward amplitudes $f_N(\omega)$ and $g_N(\omega)$ by

$$\mathcal{A}^{\mathrm{fwd}}_{\gamma N \to \gamma N} = \boldsymbol{\epsilon}'^{*} \cdot \boldsymbol{\epsilon}\, f_N(\omega) + \mathrm{i}\, \mathbf{s} \cdot \big(\boldsymbol{\epsilon}'^{*} \times \boldsymbol{\epsilon}\big)\, g_N(\omega). \tag{49.204}$$

This equation is a definition of the normalization of f_N, g_N. Let

$$f_N^{\mathrm{nb}}(\omega) := f_N(\omega) - f_N^{\mathrm{Born}}(\omega)$$

denote the non-Born amplitude after subtracting the one-nucleon pole contribution, whose normalization is specified in the low-energy expansion below. With this normalization the optical-theorem convention for the absorptive non-Born amplitude is

$$\operatorname{Im} f_N^{\mathrm{nb}}(\omega + \mathrm{i}0) = \frac{\omega}{4\pi}\, \sigma^{\mathrm{abs}}_{\gamma N}(\omega), \qquad \omega \geq \omega_{\mathrm{th}}, \tag{49.205}$$

where ω_{th} is the first inelastic threshold and $\sigma^{\mathrm{abs}}_{\gamma N}$ is the total real-photon absorption cross section in the same convention, not the elastic Compton cross section. The spin-dependent optical theorem similarly uses the helicity-difference absorption cross section $\Delta\sigma_{\gamma N} := \sigma_{1/2} - \sigma_{3/2}$:

$$\operatorname{Im} g_N(\omega + \mathrm{i}0) = \frac{\omega}{8\pi}\, \Delta\sigma_{\gamma N}(\omega). \tag{49.206}$$

The spin-independent electric-plus-magnetic polarizability is the coefficient of the first non-Born term in the low-energy expansion

$$f_N(\omega) = f_N^{\mathrm{Born}}(\omega) + \big(\alpha_E^N + \beta_M^N\big)\omega^2 + O(\omega^4). \tag{49.207}$$

The Born amplitude is the one-nucleon pole contribution determined by the charge $F_1^N(0)$, magnetic moment $F_1^N(0) + F_2^N(0)$, and the same Compton Ward identity used in the QED external-state chapter. Equation (49.207) defines only the sum $\alpha_E^N + \beta_M^N$. Separating α_E^N from β_M^N requires either non-forward Compton amplitudes or a background-field definition with separately specified electric and magnetic external fields.

Proposition 49.56 (Baldin sum rule as a forward-dispersion theorem)**.** *Assume that the spin-independent forward Compton amplitude $f_N^{\mathrm{nb}}(\omega)$ is an even function of ω analytic in the complex ω-plane cut along $(-\infty, -\omega_{\mathrm{th}}] \cup [\omega_{\mathrm{th}}, \infty)$, real on the real interval $|\omega| < \omega_{\mathrm{th}}$, and sufficiently bounded at infinity so that the once-subtracted Cauchy formula below has no boundary term. Then*

$$f_N^{\mathrm{nb}}(\omega) - f_N^{\mathrm{nb}}(0) = \frac{2\omega^2}{\pi} \int_{\omega_{\mathrm{th}}}^{\infty} \frac{\operatorname{Im} f_N^{\mathrm{nb}}(\omega' + \mathrm{i}0)}{\omega'(\omega'^2 - \omega^2)}\, \mathrm{d}\omega', \qquad |\omega| < \omega_{\mathrm{th}}. \tag{49.208}$$

If the small-ω expansion (49.207) exists and the integral may be expanded at $\omega = 0$, then

$$\alpha_E^N + \beta_M^N = \frac{1}{2\pi^2} \int_{\omega_{\rm th}}^{\infty} \frac{\sigma_{\gamma N}^{\rm abs}(\omega')}{\omega'^2} \, {\rm d}\omega'. \tag{49.209}$$

Proof. Set $F(\omega) := f_N^{\rm nb}(\omega) - f_N^{\rm nb}(0)$. By crossing in the forward channel F is even. Apply Cauchy's theorem to $F(z)/(z^2 - \omega^2)$ on a contour surrounding the two cuts and use the assumed large-$|z|$ bound to drop the circle at infinity. The discontinuity on the positive cut is

$$F(\omega' + {\rm i}0) - F(\omega' - {\rm i}0) = 2{\rm i} \ {\rm Im} \, f_N^{\rm nb}(\omega' + {\rm i}0).$$

The negative cut gives the same contribution after the change $\omega' \mapsto -\omega'$, because F is even. The residues at $z = \pm\omega$ then give (49.208). For $|\omega| < \omega_{\rm th}$,

$$\frac{1}{\omega'^2 - \omega^2} = \frac{1}{\omega'^2} + O(\omega^2),$$

under the stated integrability assumption. Therefore the coefficient of ω^2 in $F(\omega)$ is

$$\frac{2}{\pi} \int_{\omega_{\rm th}}^{\infty} \frac{{\rm Im} \, f_N^{\rm nb}(\omega' + {\rm i}0)}{\omega'^3} \, {\rm d}\omega'.$$

Using the optical theorem (49.205) gives (49.209). The proof shows precisely where the analytic and high-energy hypotheses enter; QCD supplies the current matrix elements, while the dispersion relation is an additional analyticity statement about the real-photon forward amplitude. □

Definition 49.57 (Forward spin polarizabilities)**.** Keep the normalization of the spin-dependent forward Compton amplitude g_N in (49.204) and the helicity difference

$$\Delta\sigma_{\gamma N}(\omega) := \sigma_{1/2}(\omega) - \sigma_{3/2}(\omega),$$

where the subscript is the total γN helicity in the rest frame of the target. Let $F_1^N(q^2)$ and $F_2^N(q^2)$ be the Dirac and Pauli form factors in the current matrix element (49.194), and set

$$Q_N := F_1^N(0), \qquad \kappa_N := F_2^N(0).$$

Thus $Q_p = 1$, $Q_n = 0$, and κ_N is the anomalous magnetic moment in units $e/(2M_N)$ in the conventions of this chapter. The spin-dependent low-energy theorem fixes the term linear in ω:

$$g_N(\omega) = -\frac{e^2\kappa_N^2}{8\pi M_N^2} \, \omega + \gamma_0^N \, \omega^3 + O(\omega^5). \tag{49.210}$$

Equation (49.210) defines the forward spin polarizability γ_0^N once the one-nucleon Born normalization has been fixed by $F_1^N(0)$ and $F_2^N(0)$. The absence of a Q_N^2 term in the coefficient of ω is part of the low-energy Ward-identity statement: the point-charge contribution is spin independent in the forward limit, while the anomalous magnetic moment controls the leading spin-dependent Born amplitude.

Proposition 49.58 (Gerasimov–Drell–Hearn and forward spin-polarizability sum rules)**.** *Assume that $g_N(\omega)$, after the Compton Ward identity has combined the s-channel, u-channel, and local Born pieces into a function regular at $\omega = 0$, is an odd function of ω analytic in the complex ω-plane cut along $(-\infty, -\omega_{\text{th}}] \cup [\omega_{\text{th}}, \infty)$, real on $|\omega| < \omega_{\text{th}}$, and sufficiently bounded at infinity that the unsubtracted odd dispersion relation below has no boundary term. Assume also that the threshold and high-energy behavior allow the first two terms of the small-ω expansion of the dispersion integral. Then, in the normalization of (49.206),*

$$g_N(\omega) = \frac{2\omega}{\pi} \int_{\omega_{\text{th}}}^{\infty} \frac{\operatorname{Im} g_N(\omega' + \mathrm{i}0)}{\omega'^2 - \omega^2} \, \mathrm{d}\omega', \qquad |\omega| < \omega_{\text{th}}, \tag{49.211}$$

and therefore

$$\int_{\omega_{\text{th}}}^{\infty} \frac{\Delta\sigma_{\gamma N}(\omega')}{\omega'} \, \mathrm{d}\omega' = -\frac{\pi e^2}{2M_N^2} \, \kappa_N^2 = -\frac{2\pi^2 \alpha_{\text{em}}}{M_N^2} \, \kappa_N^2, \tag{49.212}$$

$$\gamma_0^N = \frac{1}{4\pi^2} \int_{\omega_{\text{th}}}^{\infty} \frac{\Delta\sigma_{\gamma N}(\omega')}{\omega'^3} \, \mathrm{d}\omega'. \tag{49.213}$$

Here $\alpha_{\text{em}} := e^2/(4\pi)$. The sign in (49.212) follows from the convention $\Delta\sigma_{\gamma N} = \sigma_{1/2} - \sigma_{3/2}$.

Proof. Let ω lie in the cut plane with $|\omega| < \omega_{\text{th}}$. Apply Cauchy's theorem to $g_N(z)/(z^2 - \omega^2)$ on a contour encircling the two cuts and small circles around $z = \pm\omega$. The residue contribution is

$$\frac{g_N(\omega)}{2\omega} + \frac{g_N(-\omega)}{-2\omega} = \frac{g_N(\omega)}{\omega},$$

because g_N is odd. The large circle gives no contribution by hypothesis. On the positive cut the discontinuity is

$$g_N(\omega' + \mathrm{i}0) - g_N(\omega' - \mathrm{i}0) = 2\mathrm{i} \operatorname{Im} g_N(\omega' + \mathrm{i}0).$$

On the negative cut, the change of variable $\omega' \mapsto -\omega'$ gives the same contribution, because oddness of g_N makes $\operatorname{Im} g_N(-\omega' + \mathrm{i}0) = \operatorname{Im} g_N(\omega' + \mathrm{i}0)$ with the opposite orientation of the cut. Combining the two cuts gives (49.211).

For $|\omega| < \omega_{\text{th}}$,

$$\frac{1}{\omega'^2 - \omega^2} = \frac{1}{\omega'^2} + \frac{\omega^2}{\omega'^4} + O(\omega^4)$$

under the stated integrability assumption. Hence

$$g_N(\omega) = \omega \, \frac{2}{\pi} \int_{\omega_{\text{th}}}^{\infty} \frac{\operatorname{Im} g_N(\omega' + \mathrm{i}0)}{\omega'^2} \, \mathrm{d}\omega' + \omega^3 \, \frac{2}{\pi} \int_{\omega_{\text{th}}}^{\infty} \frac{\operatorname{Im} g_N(\omega' + \mathrm{i}0)}{\omega'^4} \, \mathrm{d}\omega' + O(\omega^5).$$

Substitution of (49.206) turns the coefficient of ω into

$$\frac{1}{4\pi^2} \int_{\omega_{\text{th}}}^{\infty} \frac{\Delta\sigma_{\gamma N}(\omega')}{\omega'} \, \mathrm{d}\omega',$$

and the coefficient of ω^3 into the right side of (49.213). Comparing the linear coefficient with the low-energy theorem (49.210) gives

$$\int_{\omega_{\text{th}}}^{\infty} \frac{\Delta\sigma_{\gamma N}(\omega')}{\omega'} \, \mathrm{d}\omega' = 4\pi^2 \left(-\frac{e^2 \kappa_N^2}{8\pi M_N^2} \right),$$

which is (49.212). Thus the sum rule is not an independent model assumption about nucleon structure; it is the equality of two definitions of the same coefficient, one from the low-energy current Ward identity and one from the absorptive spectral representation, under the stated analyticity and boundedness hypotheses. □

The practical hierarchy of spectroscopic control is therefore:

- Stable mesons and baryons are isolated color-singlet mass eigenvalues; their controlled extraction uses the finite-volume transfer matrix and continuum extrapolation.
- Elastic resonances are second-sheet poles of partial waves; their controlled extraction uses Luscher, Roy, or Roy–Steiner reconstruction.
- Heavy quarkonium is a color-singlet $Q\bar{Q}$ spectrum; its controlled extraction uses a matched NRQCD or pNRQCD Hamiltonian and lattice checks.
- Hybrids and glueballs are operator-overlap sectors and pole data; their controlled extraction uses declared color-singlet source families, finite-volume energies or scattering amplitudes, large-N_c scaling, and mixing analysis.
- Exotic near-threshold states are poles plus residues in coupled channels; their controlled extraction uses finite-volume coupled-channel amplitudes.

Every entry in this list is a statement about QCD observables. The words “quark model”, “molecule”, “diquark”, and “flux-tube excitation” may summarize patterns in a chosen coordinate system, but the monograph treats them as derived interpretations only after the corresponding spectral or pole input has been defined.

49.15 Euclidean Current Sum Rules and Borel Windows

QCD current sum rules are linear identities for gauge-invariant current correlators. Their mathematical core consists of a spectral representation, a spacelike OPE, and a chosen linear functional that suppresses subtraction constants and weights different parts of the spectrum. A finite-parameter spectral ansatz is an additional approximation layer, whose status is fixed by the error estimates stated below.

Let $J(x)$ be a renormalized local color-singlet operator with fixed quantum numbers, for example $J = \bar{q}\Gamma q$, $\operatorname{tr} F_{\mu\nu}F^{\mu\nu}$, or a baryonic color singlet. In Euclidean momentum notation $Q^2 > 0$, define the connected two-point function

$$\Pi_J(Q^2;\mu) := \int_{\mathbb{R}^4} \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}Q\cdot x} \langle J(x)J^\dagger(0)\rangle_{\mathrm{conn},\mu}, \qquad Q^2 := Q_\mu Q_\mu. \tag{49.214}$$

When J carries Lorentz indices, Π_J denotes one scalar invariant after projecting onto a specified tensor structure. Positivity of the Euclidean correlator for $J^\dagger J$ gives a positive spectral measure in channels where reflection positivity applies. With enough subtractions, the spacelike correlator has the form

$$\Pi_J(Q^2;\mu) = P_N(Q^2;\mu) + \int_{s_{\mathrm{th}}}^{\infty} \frac{\rho_J(s;\mu)}{s+Q^2}\, \mathrm{d}s, \tag{49.215}$$

where P_N is a polynomial fixed by the subtraction convention, s_{th} is the lowest physical threshold in the channel, and ρ_J is the spectral density associated with the same current normalization. Equation (49.215) is a dispersion representation of a color-singlet Wightman distribution.

At large spacelike Q^2, the current product has an operator-product expansion

$$\Pi_J(Q^2;\mu) \sim \sum_a C_{J,a}(Q^2,\mu)\,\langle \mathcal{O}_a(0;\mu)\rangle, \qquad Q^2 \to +\infty, \tag{49.216}$$

with local gauge-invariant operators $\mathcal{O}_a$ of increasing dimension and Wilson coefficients $C_{J,a}$. The symbol $\sim$ means an asymptotic expansion in a specified sector of the spacelike Q^2-plane, after the renormalization scheme and operator basis have been fixed. Vacuum expectation values such as $\langle \bar{q}q\rangle_\mu$ or $\langle \operatorname{tr} F^2\rangle_\mu$ are scheme-dependent coordinates of the vacuum state. They become physical only through the full renormalized combination in a specified observable.

Definition 49.59 (Borel–Laplace sum-rule functional)**.** For a function $F(Q^2)$ whose large-Q^2 behavior permits the limit, set

$$\mathcal{B}_{M^2}[F] := \lim_{\substack{n,Q^2\to\infty\\ Q^2/n=M^2}} \frac{(Q^2)^n}{(n-1)!}\left(-\frac{\mathrm{d}}{\mathrm{d}Q^2}\right)^n F(Q^2), \qquad M^2>0. \tag{49.217}$$

The positive parameter M^2 is the Borel or Laplace scale. The associated sum-rule coordinate is

$$\mathcal{S}_J(M^2;\mu) := \mathcal{B}_{M^2}\Pi_J = \frac{1}{M^2}\int_{s_{\text{th}}}^{\infty} \mathrm{e}^{-s/M^2}\rho_J(s;\mu)\,\mathrm{d}s, \tag{49.218}$$

provided the Borel transform can be interchanged with the spectral integral.

Borel transform of the subtracted dispersion kernel. The functional in Definition 49.59 annihilates every polynomial in Q^2. For $s \ge 0$,

$$\mathcal{B}_{M^2}\left[\frac{1}{s+Q^2}\right] = \frac{1}{M^2}\mathrm{e}^{-s/M^2}. \tag{49.219}$$

For an integer $m \ge 1$,

$$\mathcal{B}_{M^2}\left[(Q^2)^{-m}\right] = \frac{1}{(m-1)!\,(M^2)^m}. \tag{49.220}$$

If $F(Q^2)$ is a polynomial of degree d, then $(-\partial_{Q^2})^n F = 0$ for $n > d$, so $\mathcal{B}_{M^2}F = 0$. For the dispersion kernel,

$$\left(-\frac{\mathrm{d}}{\mathrm{d}Q^2}\right)^n \frac{1}{s+Q^2} = \frac{n!}{(s+Q^2)^{n+1}}.$$

Putting $Q^2 = nM^2$ gives

$$\frac{(Q^2)^n}{(n-1)!}\frac{n!}{(s+Q^2)^{n+1}} = \frac{1}{M^2}\left(1+\frac{s}{nM^2}\right)^{-n-1} \longrightarrow \frac{1}{M^2}\mathrm{e}^{-s/M^2}.$$

For $(Q^2)^{-m}$,

$$\left(-\frac{\mathrm{d}}{\mathrm{d}Q^2}\right)^n (Q^2)^{-m} = \frac{\Gamma(m+n)}{\Gamma(m)}(Q^2)^{-m-n}.$$

The Borel expression is therefore

$$\frac{\Gamma(m+n)}{\Gamma(m)\Gamma(n)}(nM^2)^{-m}.$$

Since $\Gamma(n+m)/\Gamma(n) = n^m(1+O(n^{-1}))$ for fixed integer m, the limit is (49.220).

Controlled Approximation 49.60 (QCD Borel sum-rule window)**.** A Borel sum-rule analysis in a channel J consists of:

1. a renormalized current J, tensor projection, and scheme at scale μ;
2. a truncated OPE $\Pi_J^{(D_{\max})}(Q^2;\mu)$ with a stated estimate for the omitted operator dimensions and perturbative orders after applying $\mathcal{B}_{M^2}$;
3. a spectral ansatz for ρ_J, often a low-energy part with finitely many parameters plus a perturbative continuum above a threshold s_0;
4. a nonempty interval $M^2 \in [M^2_{\min}, M^2_{\max}]$ on which the omitted-OPE estimate and the spectral-tail estimate are both controlled.

Only the equality (49.218) with the exact ρ_J is a consequence of the dispersion representation and the Borel transform. The finite-parameter extraction of a mass, residue, or coupling is a controlled approximation only after the spectral ansatz and the two error estimates above have been supplied.

For a single isolated stable state in the channel, the spectral measure has an atom $Z_R\delta(s-m_R^2)$. If the remaining spectral weight is represented by a continuum model $\rho_{\text{cont}}(s)$, then (49.218) becomes

$$\mathcal{S}_J(M^2) = \frac{Z_R}{M^2}\mathrm{e}^{-m_R^2/M^2} + \frac{1}{M^2}\int_{s_{\text{cont}}}^{\infty} \mathrm{e}^{-s/M^2}\rho_{\text{cont}}(s)\,\mathrm{d}s. \tag{49.221}$$

The common pole-plus-continuum model sets $\rho_{\text{cont}}(s)$ equal to a perturbative spectral density above a fit threshold s_0. Its validity is an empirical and analytic control question about the exact spectral measure, judged by stability under M^2, s_0, current choice, OPE truncation, and comparison with regulator observables such as lattice spectra.

For a finite retained spectral model

$$\mathcal{S}(\tau) = \sum_{r=1}^{N} Z_r e^{-\tau m_r^2}, \qquad Z_r > 0, \quad \tau := 1/M^2.$$

the logarithmic derivative used as a mass estimator is

$$-\frac{\mathrm{d}}{\mathrm{d}\tau}\log\mathcal{S}(\tau) = \frac{\sum_r Z_r m_r^2 e^{-\tau m_r^2}}{\sum_r Z_r e^{-\tau m_r^2}}. \tag{49.222}$$

because the numerator is $-\mathcal{S}'(\tau)$. The right side is a normalized average with positive weights $Z_r e^{-\tau m_r^2}$. It equals m_R^2 for a single retained atom and otherwise averages the squared masses present in the retained spectral model.

Thus a Borel plateau diagnoses the weighted average in (49.222) changes slowly over the chosen interval after the declared continuum subtraction and OPE truncation. Single-state dominance is a further spectral statement about the retained model. The object being computed remains the current correlator (49.214); the extracted spectral parameters inherit all assumptions of the Borel window.

49.16 Detector Observables and Energy Correlators

A nonperturbative formulation of high-energy QCD final states begins with detector measurements in the physical Hilbert space. The calorimetric observable is built from the renormalized, gauge-invariant stress tensor $T^{\mu\nu}$, whose normalization is fixed by the translation generators:

$$P^{\nu} = \int_{\mathbb{R}^3} \mathrm{d}^3x\, T^{0\nu}(t, \mathbf{x})$$

on the common domain of finite-energy states where the integral is defined as a quadratic form.

Construction 49.61 (Energy-flow detector from the stress tensor)**.** For a real test function $f \in C^{\infty}(S^2)$, define the formal null-infinity flux quadratic form by

$$\widehat{\mathcal{E}}(f) = \mathrm{w-}\lim_{R\to\infty} R^2 \int_{-\infty}^{\infty} \mathrm{d}t \int_{S^2} \mathrm{d}\Omega(\mathbf{n})\, f(\mathbf{n})\, n_i\, \widehat{T}^{0i}(t, R\mathbf{n}). \tag{49.223}$$

The notation means that the limit is first tested between finite-energy physical states. The operation is part of the detector definition only after the domain and closure properties below have been supplied.

Hypothesis 49.62 (Detector existence and self-adjointness)**.** There is a dense scattering domain $\mathcal{D}_{\mathrm{det}} \subset \mathcal{H}_{\mathrm{phys}}$, stable under finite products of the smeared detector forms used below, such that for every real $f \in C^{\infty}(S^2)$ the weak limit (49.223) exists on $\mathcal{D}_{\mathrm{det}}$, is symmetric there, and has a unique self-adjoint closure. For each pair $\Psi, \Phi \in \mathcal{D}_{\mathrm{det}}$, the map

$$f \longmapsto \langle \Psi | \widehat{\mathcal{E}}(f) | \Phi \rangle$$

is a distribution of finite order on S^2. The constant test function satisfies

$$\widehat{\mathcal{E}}(1) = H_{\mathrm{out}}$$

on outgoing scattering states, where H_{out} is the outgoing Hamiltonian restricted to the physical scattering Hilbert space. Finally, for every angular patch $U \subset S^2$, the quadratic form obtained by testing $\widehat{\mathcal{E}}(f)$ with $\operatorname{supp} f \subset U$ is the $R \to \infty$ limit of a family of complete null-generator energy integrals with nonnegative transverse smearings whenever $f \geq 0$. This last clause is the geometric detector/ANEC bridge; without it, ANEC and calorimetric positivity are statements about different limiting operators.

Detector positivity from ANEC. Assume Hypothesis 49.62. Assume in addition that the stress tensor satisfies the averaged null energy condition as a quadratic-form statement on the same scattering domain: for every future null direction n^{μ}, every nonnegative transverse smearing φ, and every $\Psi \in \mathcal{D}_{\mathrm{det}}$,

$$\int \mathrm{d}^2x_{\perp}\, \varphi(x_{\perp}) \int_{-\infty}^{\infty} \mathrm{d}\lambda\, \langle \Psi | T_{\mu\nu}(x_{\perp} + \lambda n) n^{\mu} n^{\nu} | \Psi \rangle \geq 0.$$

Then the self-adjoint closure of $\widehat{\mathcal{E}}(f)$ is positive for every $f \geq 0$.

Write a nonnegative smooth angular test function as a finite sum of functions supported in coordinate patches on S^2, using a partition of unity. In each patch, the large-R detector limit is the conformal or asymptotic image of a null-energy integral along the complete generator labelled by $\mathbf{n}$, with transverse smearing determined by the angular support of f. The Jacobian relating the angular measure to transverse measure is positive. The ANEC inequality therefore gives nonnegative quadratic form values for each patch. Summing the patch contributions gives

$$\langle\Psi|\widehat{\mathcal{E}}(f)|\Psi\rangle \geq 0 \qquad (\Psi \in \mathcal{D}_{\mathrm{det}},\ f \geq 0).$$

Since the operator has a unique self-adjoint closure by Hypothesis 49.62, positivity of the quadratic form on a core implies positivity of the closure by the spectral theorem.

Remark 49.63 (Status in QCD and in conformal theories). For free massless fields, the limit in (49.223) is obtained by inserting the mode expansion and applying stationary phase at null infinity. For conformal field theories satisfying the light-transform hypotheses of Chapter 65, the detector is the stress-tensor light-ray operator and positivity is supplied by the ANEC theorem stated there. In four-dimensional QCD, the detector is the expected physical calorimeter observable, but its nonperturbative construction depends on the existence of the continuum theory, the physical scattering space, and the hadronic asymptotic completeness problem stated in Open Problem 49.22. Perturbative KLN cancellation proves finiteness of regulator-expanded coefficients for IRC-continuous measurements after the relevant sums over degenerate states and the required factorization steps have been made; it does not construct the self-adjoint detector operator on the exact physical Hilbert space.

Open Problem 49.64 (Energy-flow operators in four-dimensional QCD). Construct the operators $\widehat{\mathcal{E}}(f)$ of Construction 49.61 in continuum four-dimensional QCD. The construction should start from a nonperturbative definition of the physical Hilbert space, identify a dense finite-energy scattering domain, prove the null-infinity weak limits, prove essential self-adjointness or specify the self-adjoint extension selected by detector physics, prove positivity from an ANEC theorem in the same framework, and show that $\widehat{\mathcal{E}}(1)$ agrees with the outgoing Hamiltonian on the hadronic scattering subspace.

Here $\mathbf{n} \in S^2$ is a unit spatial direction and $\mathrm{d}\Omega$ is the rotation-invariant measure on S^2. The distribution $\widehat{\mathcal{E}}(\mathbf{n})$ is defined by $\widehat{\mathcal{E}}(f) = \int_{S^2} \mathrm{d}\Omega(\mathbf{n}) f(\mathbf{n}) \widehat{\mathcal{E}}(\mathbf{n})$. On an outgoing state of stable hadrons with momenta p_r, species labels a_r, and the normalization conventions of scattering theory,

$$\widehat{\mathcal{E}}(f)|p_1, a_1; \ldots; p_N, a_N\rangle_{\mathrm{out}} = \left(\sum_{r=1}^{N} p_r^0\, f(\widehat{\mathbf{p}}_r)\right)|p_1, a_1; \ldots; p_N, a_N\rangle_{\mathrm{out}}, \qquad \widehat{\mathbf{p}}_r := \frac{\mathbf{p}_r}{|\mathbf{p}_r|}. \tag{49.224}$$

Equation (49.224) is the spectral action of the detector limit on physical hadronic asymptotic states.

Asymptotic multiplication model. The preceding formula has a precise operator-theoretic meaning once an outgoing scattering representation has been chosen. Let $\mathcal{H}_{\mathrm{out}}$ be the physical outgoing scattering subspace and write it as a direct integral over hadronic scattering configurations,

$$\mathcal{H}_{\mathrm{out}} \simeq \int_{\mathcal{X}_{\mathrm{out}}}^{\oplus} \mathcal{K}_Y \,\mathrm{d}\nu(Y).$$

Here Y specifies the number of stable hadrons, their species labels, on-shell momenta, and the discrete or continuous degeneracy labels left after the standard Bose–Fermi symmetrizations;

$\mathcal{K}_Y$ is the corresponding finite or countable multiplicity space. The outgoing Hamiltonian is multiplication by

$$E_{\mathrm{tot}}(Y) = \sum_{r\in Y} p_r^0.$$

For a bounded Borel function $f : S^2 \to \mathbb{C}$, define

$$m_f(Y) := \sum_{r\in Y} p_r^0 f(\widehat{\mathbf{p}}_r), \qquad \mathcal{E}_{\mathrm{mult}}(f) := M_{m_f}, \tag{49.225}$$

where M_{m_f} is the direct-integral multiplication operator with maximal domain

$$\mathrm{Dom}\, M_{m_f} = \left\{ \Psi : \int_{\mathcal{X}_{\mathrm{out}}} |m_f(Y)|^2 \|\Psi(Y)\|^2_{\mathcal{K}_Y}\, \mathrm{d}\nu(Y) < \infty \right\}.$$

If f is real, M_{m_f} is self-adjoint; if $f \geq 0$, it is positive; and

$$|m_f(Y)| \leq \|f\|_\infty E_{\mathrm{tot}}(Y), \qquad \mathrm{Dom}\, H_{\mathrm{out}} \subset \mathrm{Dom}\, \mathcal{E}_{\mathrm{mult}}(f). \tag{49.226}$$

Moreover the family is strongly commuting, because all members are multiplication operators on the same direct-integral space. For bounded $f_1, \dots, f_k$,

$$\mathcal{E}_{\mathrm{mult}}(f_1)\cdots\mathcal{E}_{\mathrm{mult}}(f_k) = M_{\prod_{j=1}^k m_{f_j}}, \qquad \mathrm{Dom}\, H^k_{\mathrm{out}} \subset \mathrm{Dom}\big(\mathcal{E}_{\mathrm{mult}}(f_1)\cdots\mathcal{E}_{\mathrm{mult}}(f_k)\big). \tag{49.227}$$

Indeed

$$\left|\prod_{j=1}^k m_{f_j}(Y)\right| \leq \left(\prod_{j=1}^k \|f_j\|_\infty\right) E_{\mathrm{tot}}(Y)^k,$$

which gives the displayed domain inclusion. The stress-tensor flux operator $\widehat{\mathcal{E}}(f)$ of Construction 49.61, when it exists, is required to agree with $\mathcal{E}_{\mathrm{mult}}(f)$ after the outgoing wave operator identifies finite-energy scattering states with their asymptotic configurations. This agreement is part of the nonperturbative detector construction problem; perturbation theory may test its matrix elements after the detector operator has been defined.

The corresponding eventwise calorimetric measure of the final state X is

$$\mu_X = \sum_{r\in X} p_r^0\, \delta_{\widehat{\mathbf{p}}_r}, \qquad \widehat{\mathcal{E}}_X(f) = \int_{S^2} f\, \mathrm{d}\mu_X.$$

The detector observable depends on X only through the positive measure μ_X. The multiplication model also fixes the meaning of coincident detectors. For a finite configuration X,

$$\prod_{j=1}^k \widehat{\mathcal{E}}_X(f_j) = \int_{(S^2)^k} f_1(\mathbf{n}_1)\cdots f_k(\mathbf{n}_k)\, \mathrm{d}\mu_X(\mathbf{n}_1)\cdots \mathrm{d}\mu_X(\mathbf{n}_k). \tag{49.228}$$

Thus the product measure includes the diagonal strata $\mathbf{n}_i = \mathbf{n}_j$. A separated-angle correlator is the restriction of the resulting distribution away from these diagonals; deleting the diagonal weight defines a different observable and must be declared as an additional measurement prescription.

Given a normalized incoming physical state Ψ_{in}, or a state prepared by a local color-singlet current and then evolved to future infinity, the k-point energy correlator is the distribution on $(S^2)^k$

$$\mathcal{G}_k(f_1, \dots, f_k; \Psi) = {}_{\mathrm{out}}\langle \Psi | \widehat{\mathcal{E}}(f_1)\cdots\widehat{\mathcal{E}}(f_k) | \Psi \rangle_{\mathrm{out}}. \tag{49.229}$$

After inserting a complete set of physical hadronic out-states, this becomes

$$\mathcal{G}_k(f_1,\ldots,f_k;\Psi) = \sum_X |\langle X,\mathrm{out} \mid \Psi,\mathrm{out}\rangle|^2 \prod_{\ell=1}^{k} \left(\sum_{r\in X} p_r^0\, f_\ell(\widehat{\mathbf{p}}_r) \right), \tag{49.230}$$

with the usual replacement of the displayed sum by direct integrals over continuous multiparticle phase space. This formula is nonperturbative: all states in X are physical hadronic states, and the operator being measured is built from $T^{\mu\nu}$.

Connected cumulants and contact strata. Let $\mathbb{P}_\Psi$ denote the probability measure on outgoing configurations induced by the normalized state Ψ in the multiplication model. For bounded detector tests $f_1,\ldots,f_k$, put

$$X_j(Y) := \mathcal{E}_Y(f_j) = \int_{S^2} f_j \, \mathrm{d}\mu_Y.$$

Then $\mathcal{G}_k(f_1,\ldots,f_k;\Psi)$ is the raw moment

$$\mathbb{E}_{\mathbb{P}_\Psi}[X_1\cdots X_k].$$

The connected energy correlator in this event-ensemble sense is the cumulant

$$\mathcal{G}_k^{\mathrm{conn}}(f_1,\ldots,f_k;\Psi) = \sum_{\pi\in\Pi_k} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} \mathcal{G}_{|B|}\big((f_i)_{i\in B};\Psi\big), \tag{49.231}$$

where Π_k is the set of set partitions of $\{1,\ldots,k\}$. Thus, for example,

$$\mathcal{G}_2^{\mathrm{conn}}(f,g) = \mathcal{G}_2(f,g) - \mathcal{G}_1(f)\mathcal{G}_1(g),$$

and $\mathcal{G}_3^{\mathrm{conn}}$ is obtained by subtracting the three one-block–two-block products and adding twice the product of the three one-point functions. This is a subtraction of event-to-event factorization in the probability distribution of final states.

It is a different operation from removing coincident-detector contacts. For a fixed event Y, the product measure $\mu_Y^{\otimes k}$ decomposes into strata labeled by coincidences among detector labels. Already for three detectors,

$$\begin{aligned}\mathcal{E}_Y(f)\mathcal{E}_Y(g)\mathcal{E}_Y(h) = &\sum_{\substack{r,s,t\in Y\\ r,s,t \text{ distinct}}} E_rE_sE_t f_r g_s h_t \\ &+ \sum_{\substack{r,t\in Y\\ r\neq t}} E_r^2E_t f_r g_r h_t + \sum_{\substack{r,s\in Y\\ r\neq s}} E_r^2E_s f_r g_s h_r \\ &+ \sum_{\substack{r,s\in Y\\ r\neq s}} E_rE_s^2 f_r g_s h_s + \sum_{r\in Y} E_r^3 f_r g_r h_r,\end{aligned} \tag{49.232}$$

where $E_r = p_r^0$ and $f_r = f(\widehat{\mathbf{p}}_r)$, etc. The last four terms are diagonal contact strata of the detector product; they are present before the average over events is taken. If $\mathbb{P}_\Psi$ is concentrated on a single event, all ensemble cumulants of order $k\geq 2$ vanish, but the diagonal terms in (49.232) remain in the raw detector product. Conversely, deleting diagonal strata produces a separated-detector measurement prescription, not an ensemble-connected correlator. A calculation that quotes a "connected" or "contact-subtracted" energy correlator must therefore state which of these two operations is being performed.

Soft and collinear continuity of smeared energy flow. Let $f \in C^1(S^2)$. Suppose a final-state calorimetric measure μ_X is changed either by adding one particle of energy ε at direction $\mathbf{m}$, or by replacing two particles of energies E_a, E_b and directions $\mathbf{n}_a, \mathbf{n}_b$ by one particle of energy $E_a + E_b$ in the direction

$$\mathbf{n}_{ab} = \frac{E_a \mathbf{n}_a + E_b \mathbf{n}_b}{|E_a \mathbf{n}_a + E_b \mathbf{n}_b|}.$$

The resulting detector change obeys

$$\left| \Delta \widehat{\mathcal{E}}_X(f) \right| \leq \varepsilon \|f\|_\infty$$

in the soft case, and

$$\left| \Delta \widehat{\mathcal{E}}_X(f) \right| \leq (E_a + E_b) \operatorname{Lip}(f) \, \operatorname{diam}\{\mathbf{n}_a, \mathbf{n}_b, \mathbf{n}_{ab}\}$$

in the collinear case. For a soft particle, the measure changes by $\varepsilon \delta_{\mathbf{m}}$, so the change of the detector value is $\varepsilon f(\mathbf{m})$, whose absolute value is bounded by $\varepsilon \|f\|_\infty$. For a collinear recombination, the change is

$$E_a f(\mathbf{n}_a) + E_b f(\mathbf{n}_b) - (E_a + E_b) f(\mathbf{n}_{ab}).$$

Writing this as

$$E_a \big(f(\mathbf{n}_a) - f(\mathbf{n}_{ab}) \big) + E_b \big(f(\mathbf{n}_b) - f(\mathbf{n}_{ab}) \big)$$

and using the Lipschitz constant of f on the round sphere gives

$$|\Delta \widehat{\mathcal{E}}_X(f)| \leq E_a \operatorname{Lip}(f) d(\mathbf{n}_a, \mathbf{n}_{ab}) + E_b \operatorname{Lip}(f) d(\mathbf{n}_b, \mathbf{n}_{ab}),$$

which is bounded by the displayed diameter estimate. Thus the smeared detector is continuous for the weak calorimetric topology generated by finite-energy measures tested against $C^1(S^2)$ functions: soft additions vanish with their energy, and collinear recombinations vanish uniformly on bounded-energy sectors as the angular diameter tends to zero.

The same estimate applies to products $\prod_{\ell=1}^k \widehat{\mathcal{E}}_X(f_\ell)$ on any fixed finite-energy sector by the elementary telescoping identity

$$\prod_{\ell=1}^k A_\ell - \prod_{\ell=1}^k B_\ell = \sum_{j=1}^k \left(\prod_{\ell<j} A_\ell \right) (A_j - B_j) \left(\prod_{\ell>j} B_\ell \right),$$

with $A_\ell = \widehat{\mathcal{E}}_{X'}(f_\ell)$ and $B_\ell = \widehat{\mathcal{E}}_X(f_\ell)$. Thus smeared energy correlators are continuous functions of the calorimetric measure for soft and collinear degenerations. Perturbative finiteness then has a precise theorem boundary: at each fixed order in a regulator for which the Kinoshita–Lee–Nauenberg cancellation and QCD factorization hold for the chosen inclusive channel, measurement functions with the above continuity produce finite coefficients after summing over degenerate final states and virtual corrections. The operator definition (49.229) is the nonperturbative observable; perturbation theory computes its short-distance expansion in the regime where asymptotic freedom supplies a controlled small coupling.

The unsmeared product $\widehat{\mathcal{E}}(\mathbf{n}_1) \cdots \widehat{\mathcal{E}}(\mathbf{n}_k)$ is a distribution. Coincident directions carry contact singularities because several detector insertions can act on the same hadron. The separated-angle correlator is obtained by testing against functions supported away from the coincidence diagonals in $(S^2)^k$; fully smeared correlators include the contact terms determined by the detector operator product itself.

49.16.1 Energy-Energy Correlation in Electron-Positron Annihilation

Let a color-singlet current create a state of total center-of-mass energy Q. For each physical hadronic final state X, write

$$z_r = \frac{p_r^0}{Q}, \qquad \cos\chi_{rs} = \widehat{\mathbf{p}}_r \cdot \widehat{\mathbf{p}}_s.$$

The normalized two-point energy-energy correlator is the distribution

$$\mathrm{EEC}(\chi) = \frac{1}{\sigma_{\text{tot}}} \sum_X \int \mathrm{d}\sigma_X \sum_{r,s\in X} z_r z_s\, \delta(\cos\chi - \cos\chi_{rs}), \tag{49.233}$$

where $\mathrm{d}\sigma_X$ is the differential cross-section measure for the exclusive hadronic channel X, and the sum is a direct integral over continuous phase space. The terms $r = s$ are the coincident-detector contact part at $\chi = 0$. A convention that removes this contact part defines a different distribution at $\chi = 0$ and the same separated-angle function for $0 < \chi < \pi$.

Equation (49.233) is the detector form of the Basham–Brown–Ellis–Love observable. Its perturbative expansion in asymptotically free QCD is a calculation of the current-created state and of the detector insertions in the short-distance region. The back-to-back limit $\chi \to \pi$ is controlled by soft and collinear radiation constrained by transverse momentum conservation; the small-angle limit is controlled by a collinear operator product of energy-flow insertions. Both limits require resummation because the separated-angle distribution contains large logarithms of $1 + \cos\chi$ or $1 - \cos\chi$, respectively.

EEC moment sum rules and the contact term. Let X be a physical final state in the center-of-mass frame of a color-singlet source, with total energy Q and total spatial momentum zero. Set

$$z_r = \frac{p_r^0}{Q}, \qquad \mathbf{n}_r = \frac{\mathbf{p}_r}{|\mathbf{p}_r|}, \qquad \zeta_{rs} = \mathbf{n}_r \cdot \mathbf{n}_s.$$

Define the eventwise two-point calorimetric distribution in $\zeta = \cos\chi$ by

$$E_{2,X}(\zeta) = \sum_{r,s\in X} z_r z_s\, \delta(\zeta - \zeta_{rs}). \tag{49.234}$$

Then, as distributional moments on $[-1, 1]$,

$$\int_{-1}^{1} E_{2,X}(\zeta)\, \mathrm{d}\zeta = 1, \qquad \int_{-1}^{1} \zeta\, E_{2,X}(\zeta)\, \mathrm{d}\zeta = 0. \tag{49.235}$$

The coincident-detector contact part at $\zeta = 1$ has weight

$$E_{2,X}^{\text{contact}}(\zeta) = \left(\sum_{r\in X} z_r^2\right) \delta(\zeta - 1). \tag{49.236}$$

Consequently the cross-section averaged EEC of (49.233), including the $r = s$ terms, satisfies the same zeroth and first moment sum rules after division by σ_{tot}.

The total-energy condition gives

$$\sum_{r\in X} z_r = 1.$$

Testing (49.234) against the constant function 1 gives

$$\sum_{r,s} z_r z_s = \left(\sum_r z_r\right)^2 = 1.$$

Testing against ζ gives

$$\sum_{r,s} z_r z_s \, \mathbf{n}_r \cdot \mathbf{n}_s = \left|\sum_r z_r \mathbf{n}_r\right|^2 .$$

Since the source is in its center-of-mass frame,

$$\sum_r p_r^0 \mathbf{n}_r = \sum_r \mathbf{p}_r = 0,$$

and the first moment vanishes. The contact contribution is exactly the subsum with $r = s$, for which $\zeta_{rr} = 1$. Averaging the eventwise identities with the positive cross-section measure and dividing by σ_{tot} preserves the two moment identities.

Remark 49.65 (Use of the sum rules). The eventwise sum rules above are statements about the nonperturbative detector observable, not about partons. A fixed-order calculation, a resummed endpoint formula, a lattice-inspired approximation, or a parton shower that claims to represent the normalized EEC must specify how the $\zeta = 1$ contact term is treated and must satisfy the moment identities after the same convention is applied. If the contact term is removed, the remaining separated-angle distribution no longer has unit integral by itself; the missing weight is $\langle \sum_r z_r^2 \rangle$.

Legendre moments and calorimetric multipoles. The contact-inclusive two-detector distribution has stronger eventwise constraints than the first two moments alone. For every $\ell \geq 0$ define

$$M_{\ell,X} := \int_{-1}^{1} P_\ell(\zeta) E_{2,X}(\zeta) \, \mathrm{d}\zeta = \sum_{r,s \in X} z_r z_s P_\ell(\mathbf{n}_r \cdot \mathbf{n}_s).$$

With the usual orthonormal complex spherical harmonics on S^2, the addition theorem gives

$$P_\ell(\mathbf{n} \cdot \mathbf{m}) = \frac{4\pi}{2\ell + 1} \sum_{a=-\ell}^{\ell} Y_{\ell a}(\mathbf{n}) \overline{Y_{\ell a}(\mathbf{m})}.$$

Hence

$$M_{\ell,X} = \frac{4\pi}{2\ell + 1} \sum_{a=-\ell}^{\ell} \left| \sum_{r \in X} z_r Y_{\ell a}(\mathbf{n}_r) \right|^2 \geq 0. \tag{49.237}$$

This positivity is a property of the positive calorimetric measure $\sum_r z_r \delta_{\mathbf{n}_r}$. It is not a property of the separated-angle density after the coincident-detector atom has been removed. Indeed, for $\ell = 1$,

$$M_{1,X} = \sum_{r,s} z_r z_s \, \mathbf{n}_r \cdot \mathbf{n}_s = \left|\sum_r z_r \mathbf{n}_r\right|^2 ,$$

so the color-singlet center-of-mass condition sets the full moment to zero; the separated part alone then has first Legendre moment $-\sum_r z_r^2$. The contact atom is therefore forced by positivity and momentum conservation, not by a cosmetic endpoint convention.

The next moment displays the quadrupole coordinate. Put

$$Q_{ij}(X) := \sum_{r\in X} z_r \left(n_r^i n_r^j - \frac{1}{3}\delta_{ij} \right).$$

Then

$$\operatorname{tr} Q(X)^2 = \sum_{r,s} z_r z_s \left((\mathbf{n}_r \cdot \mathbf{n}_s)^2 - \frac{1}{3} \right), \qquad M_{2,X} = \frac{3}{2} \operatorname{tr} Q(X)^2 \geq 0.$$

Cross-section averaging preserves (49.237). Thus any perturbative, resummed, shower, or finite-resolution representation of the normalized EEC must say which endpoint/contact convention realizes these multipole positivity constraints. A separated-angle approximation that fits the bulk shape but gives a negative contact-inclusive M_ℓ is not a positive detector measure.

Three-detector moment and contact ledger. The same detector definition gives a genuinely higher-point distribution before any partonic approximation is made. For the same event X, define the ordered three-detector distribution on the three angular invariants $\zeta_{12}, \zeta_{13}, \zeta_{23}$ by

$$\begin{aligned} E_{3,X}(\zeta_{12}, \zeta_{13}, \zeta_{23}) := \sum_{r,s,t\in X} & z_r z_s z_t \, \delta(\zeta_{12} - \mathbf{n}_r \cdot \mathbf{n}_s) \, \delta(\zeta_{13} - \mathbf{n}_r \cdot \mathbf{n}_t) \\ & \times \delta(\zeta_{23} - \mathbf{n}_s \cdot \mathbf{n}_t). \end{aligned} \tag{49.238}$$

Its support is not the whole cube $[-1, 1]^3$, but the Gram-positive region of triples of unit directions, plus the diagonal strata. As moments of this distribution,

$$\int E_{3,X} = 1, \qquad \int \zeta_{12} E_{3,X} = \int \zeta_{13} E_{3,X} = \int \zeta_{23} E_{3,X} = 0 \tag{49.239}$$

in a color-singlet center-of-mass event. The proof is the same detector algebra as above: the constant test gives $(\sum_r z_r)^3 = 1$, while, for example,

$$\sum_{r,s,t} z_r z_s z_t \, \mathbf{n}_r \cdot \mathbf{n}_s = \left| \sum_r z_r \mathbf{n}_r \right|^2 \left(\sum_t z_t \right) = 0.$$

The other two pair moments are identical after relabeling the ordered detectors.

The diagonal contact strata are now richer than in the EEC. The completely coincident stratum is

$$\left(\sum_{r\in X} z_r^3 \right) \delta(\zeta_{12} - 1)\delta(\zeta_{13} - 1)\delta(\zeta_{23} - 1). \tag{49.240}$$

There are also three pair-coincidence strata. For instance, the stratum where detectors 1 and 2 hit the same hadron but detector 3 hits a different hadron is

$$\sum_{\substack{r,t\in X \\ r\neq t}} z_r^2 z_t \, \delta(\zeta_{12} - 1)\delta(\zeta_{13} - \zeta_{rt})\delta(\zeta_{23} - \zeta_{rt}), \qquad \zeta_{rt} = \mathbf{n}_r \cdot \mathbf{n}_t, \tag{49.241}$$

with two analogous terms for $1 = 3$ and $2 = 3$. Removing all these diagonal strata defines a separated three-detector observable. It does not by itself obey (49.239); the missing weights are precisely the contact coordinates above. Thus higher-point energy correlators inherit the same lesson as the EEC: endpoint/contact coordinates are part of the detector observable, not optional decorations added after a partonic calculation.

49.16.2 Light-Ray Interpretation and Endpoint Regimes

The detector density $\widehat{\mathcal{E}}(\mathbf{n})$ is the QCD analogue of the stress-tensor light transform developed in Chapter 65. In a conformal theory the light transform is an operator-valued distribution controlled by Lorentzian conformal representation theory. In QCD the coupling runs, so the same formula is not a conformal operator identity. It remains an asymptotic detector definition in the physical Hilbert space, and the short-distance limits are computed by matching the detector product onto operators whose renormalization group evolution is controlled by QCD anomalous dimensions.

Proof-status architecture for the EEC program. The exact detector identities above and the endpoint formulae below live at different levels of the argument. First, the physical detector layer starts from the stress-tensor flux limit or, on the outgoing scattering representation, from multiplication by the calorimetric measure. This layer fixes positivity, total-energy moments, diagonal contact strata, multipole positivity, and the distinction between ensemble cumulants and eventwise contacts. Second, the conformal fixed-point layer is the light-ray operator framework of Chapter 65: it supplies representation-theoretic homogeneity, finite light-ray charts, diagonal-contact extensions, and endpoint pushforwards only under the Lorentzian analyticity and growth hypotheses stated there. Third, the QCD small-angle layer replaces fixed-point light-ray data by renormalized Wilson-line/light-ray operator coordinates with ultraviolet and rapidity subtractions, protected moment rows, finite chart changes, and projected curvature or omitted-channel remainders. Fourth, the perturbative endpoint layer computes coefficients, anomalous dimensions, contact counterterms, and matching functions in one declared scheme; the tree-level collinear coefficients and cusp-log flatness check are tests of this layer, not the full all-order light-ray OPE/mixing theorem. Finally, the observable prediction layer glues the separated bulk, small-angle endpoint, back-to-back endpoint, and explicit endpoint atoms so that the detector moment identities remain true.

Thus the dependency chain is

$$\begin{aligned}\text{detector measure and contacts} &\Longrightarrow \text{CFT or QCD light-ray endpoint chart}\\ &\Longrightarrow \text{renormalized mixing/factorization calculation}\\ &\Longrightarrow \text{endpoint gluing and observable prediction.}\end{aligned}$$

Skipping any arrow changes the mathematical claim. A separated-angle fixed-order distribution without contact data is not the normalized detector observable; a finite retained light-ray block without a curvature or omitted-channel estimate is not an endpoint theorem; and a resummed endpoint formula without the gluing and moment ledger is not yet a prediction for the full normalized EEC on $[-1, 1]$.

For $0 < \chi < \pi$, the EEC is a separated-angle distribution. Two endpoint regions have different expansions. In the small-angle region set

$$\theta^2 := 2(1 - \cos\chi) \ll 1.$$

The product of two nearby detector insertions is matched onto a sum of collinear light-ray operators:

$$\widehat{\mathcal{E}}(\mathbf{n}_1)\widehat{\mathcal{E}}(\mathbf{n}_2) \sim \sum_A C_A(\theta, Q, \mu)\, \mathbb{O}_A(\mathbf{n}, \mu), \qquad \mathbf{n}_1, \mathbf{n}_2 \to \mathbf{n}. \tag{49.242}$$

Here $\mathbb{O}_A$ are light-ray operators in the collinear direction, including twist-two energy-flow moments and their descendants in the appropriate channel. The symbol $\sim$ denotes an asymptotic expansion after smearing against angular test functions supported at $\theta \ll 1$. The coefficient functions C_A are perturbatively computable at large transverse scale $Q\theta$, and the evolution of $\mathbb{O}_A$ resums logarithms of θ.

In the back-to-back region set

$$q_T^2 := Q^2(1+\cos\chi) \ll Q^2.$$

Momentum conservation forces the two measured energy deposits to recoil against soft and collinear radiation of small total transverse momentum. The factorized form is the transverse-momentum resummation analogue

$$\frac{\mathrm{d}\sigma}{\mathrm{d}q_T^2} = H(Q,\mu) \int \frac{\mathrm{d}^2 b}{(2\pi)^2}\, \mathrm{e}^{\mathrm{i}b\cdot q_T}\, J_n(b,\mu,\nu)\, J_{\bar{n}}(b,\mu,\nu)\, S(b,\mu,\nu) + O(q_T/Q), \tag{49.243}$$

for a color-singlet source and in a rapidity-renormalization scheme with rapidity scale ν. The hard, jet, and soft functions have coupled μ- and ν-evolution equations; solving them resums Sudakov logarithms of q_T/Q. The power remainder and the absence or treatment of Glauber exchanges are part of the factorization hypothesis.

Here and below $\alpha_s(\mu) := g^2(\mu)/(4\pi)$ in the trace-delta gauge coordinate of this chapter. Whenever a coefficient is proportional to $\alpha_s C_F$, it is invariant under conversion to the common half-trace generator convention because $g_{\mathrm{ht}}^2 C_F^{\mathrm{ht}} = g^2 C_F$, as in (49.120).

The leading double logarithm is fixed already by the lightlike cusp anomalous dimension. Let $b = |b|$ in (49.243), set

$$b_{\mathrm{F}} := 2\mathrm{e}^{-\gamma_E}, \qquad \mu_b := \frac{b_{\mathrm{F}}}{b}, \qquad L_b := \log\frac{Q^2}{\mu_b^2} = \log\frac{Q^2 b^2}{b_{\mathrm{F}}^2}.$$

The constant b_{F} is a Fourier-convention constant, not a new physical scale. In the trace-delta convention of this chapter the one-loop quark cusp anomalous dimension is

$$\Gamma_{\mathrm{cusp}}^q(g) = \frac{g^2 C_F}{4\pi^2} + O(g^4) = \frac{\alpha_s C_F}{\pi} + O(\alpha_s^2), \tag{49.244}$$

where $C_F = (N_c^2-1)/N_c$.

Assume the back-to-back factorization datum (49.243) for a color-singlet quark-current event shape, with the hard scale Q, the impact-parameter scale $\mu_b = b_{\mathrm{F}}/b$, and a subtraction scheme in which the leading large logarithms are generated by the quark cusp anomalous dimension (49.244). If running of g and all non-cusp anomalous dimensions are omitted, the leading-logarithmic impact-parameter kernel has the form

$$W_{\mathrm{LL}}(b,Q) = W(b,\mu_b)\, \exp\left[-\frac{1}{2}\,\Gamma_{\mathrm{cusp}}^q(g)\, L_b^2\right]. \tag{49.245}$$

Equivalently, at one loop,

$$W_{\mathrm{LL}}(b,Q) = W(b,\mu_b)\, \exp\left[-\frac{\alpha_s C_F}{2\pi} L_b^2 + O(\alpha_s^2 L_b^2)\right].$$

Its content is the fixed-coupling leading-logarithmic component of the factorization formula. The factorized expression is an impact-parameter representation of the recoil constraint. The hard matching is naturally made at scale Q, while the transverse-distance functions are naturally evaluated at scale μ_b. Cusp evolution between these two scales gives the logarithmic exponent

$$S_{\mathrm{LL}}(Q,b) = \int_{\mu_b^2}^{Q^2} \frac{\mathrm{d}\kappa^2}{\kappa^2}\, \Gamma^q_{\mathrm{cusp}}(g)\, \log\frac{Q^2}{\kappa^2}$$

when g is held fixed and non-cusp anomalous dimensions are discarded. Changing variables to

$$u = \log\frac{Q^2}{\kappa^2}$$

turns the integral into

$$S_{\mathrm{LL}}(Q,b) = \Gamma^q_{\mathrm{cusp}}(g) \int_0^{L_b} u\,\mathrm{d}u = \frac{1}{2}\,\Gamma^q_{\mathrm{cusp}}(g) L_b^2.$$

The kernel in (49.243) is suppressed by the Sudakov no-resolved-recoil factor between the two scales; this fixes the minus sign in $\exp[-S_{\mathrm{LL}}]$ in the chosen evolution convention. Substituting (49.244) gives the displayed one-loop form.

Remark 49.66 (Scope of the Sudakov formula). The fixed-coupling display above supplies only the cusp-controlled double logarithm inside a full back-to-back EEC theorem. A complete perturbative prediction must specify the rapidity regulator, the hard, jet, and soft functions, their finite scheme constants, matching to the fixed separated-angle distribution, running-coupling improvement, and the leading power corrections in q_T/Q. The calculation isolates the universal cusp-controlled double logarithm that any such consistent factorization scheme must reproduce.

Lemma 49.67 (Tree-level collinear coefficient for the separated EEC)**.** *Use the standard four-dimensional, azimuth-integrated, final-state collinear splitting convention*

$$\mathrm{d}P_{a\to bc} = \frac{\alpha_s}{2\pi}\, P_{a\to bc}(x)\,\mathrm{d}x\,\frac{\mathrm{d}\theta^2}{\theta^2}, \qquad 0 < x < 1, \tag{49.246}$$

where x and $1-x$ are the daughter energy fractions and θ is their small opening angle. The real-emission contribution of this resolved splitting to the ordered separated-angle EEC is weighted by

$$w_{\mathrm{EEC}}(x) = 2x(1-x). \tag{49.247}$$

Consequently, for $\zeta = \cos\chi$ and $\theta^2 = 2(1-\zeta) + O((1-\zeta)^2)$, the leading-power separated-angle collinear density generated by a parent of type a has the form

$$\frac{\mathrm{d}P_a^{\mathrm{EEC}}}{\mathrm{d}\zeta} = \frac{\alpha_s}{2\pi}\,\frac{\Gamma_a^{\mathrm{EEC}}}{1-\zeta} + \textit{terms less singular as } \zeta\to 1, \tag{49.248}$$

with resolved real-kernel coefficients

$$\begin{aligned}
\Gamma_q^{\mathrm{EEC}} &= \int_0^1 2x(1-x)\, C_F\frac{1+x^2}{1-x}\,\mathrm{d}x = \frac{3}{2}\,C_F,\\
\Gamma_{g\to gg}^{\mathrm{EEC}} &= \int_0^1 2x(1-x)\,2C_A\left[\frac{x}{1-x} + \frac{1-x}{x} + x(1-x)\right]\mathrm{d}x = \frac{14}{5}\,C_A,\\
\Gamma_{g\to q\bar q}^{\mathrm{EEC}} &= \int_0^1 2x(1-x)\,T_F\big(x^2 + (1-x)^2\big)\mathrm{d}x = \frac{1}{5}\,T_F
\end{aligned} \tag{49.249}$$

per quark flavor in the last line. The endpoint singularities of the real splitting kernels are integrable in these coefficients because the detector weight $2x(1-x)$ vanishes when either daughter carries zero energy.

Proof. For one parent of energy E splitting into two collinear daughters of energies xE and $(1-x)E$, the ordered two-detector contribution at the nonzero opening angle consists of the two ordered pairs (b,c) and (c,b). Its energy weight divided by E^2 is therefore

$$x(1-x) + (1-x)x = 2x(1-x),$$

which proves (49.247). The coincident daughter terms have contact weight

$$x^2 + (1-x)^2,$$

and the identity

$$x^2 + (1-x)^2 + 2x(1-x) = 1$$

is the local version of the zeroth-moment sum rule in the eventwise EEC moment identities above: a collinear splitting redistributes the parent's detector weight between the contact point and the separated small angle without changing the total two-detector weight.

At leading collinear power the probability measure for the resolved splitting is (49.246). Since

$$1 - \zeta = 1 - \cos\theta = \frac{\theta^2}{2} + O(\theta^4),$$

the positive measure $\mathrm{d}\theta^2/\theta^2$, pulled back to the ζ-line near $\zeta = 1$, is

$$\frac{\mathrm{d}\theta^2}{\theta^2} = \frac{\mathrm{d}\zeta}{1-\zeta} + O(1)\,\mathrm{d}\zeta,$$

where the equality is an equality of positive densities on intervals $\zeta < 1$, not an oriented one-form identity. Multiplying the splitting probability by the ordered EEC weight and retaining only the leading distribution as $\zeta \to 1$ gives (49.248) with

$$\Gamma_a^{\mathrm{EEC}} = \sum_{a\to bc} \int_0^1 2x(1-x) P_{a\to bc}(x)\,\mathrm{d}x.$$

Substituting the real tree-level kernels displayed in (49.249) reduces the three coefficients to elementary polynomial integrals:

$$2C_F \int_0^1 x(1+x^2)\,\mathrm{d}x = \frac{3}{2} C_F,$$

$$4C_A \int_0^1 \left[x^2 + (1-x)^2 + x^2(1-x)^2\right]\mathrm{d}x = \frac{14}{5} C_A,$$

and

$$2T_F \int_0^1 x(1-x)\big(x^2 + (1-x)^2\big)\mathrm{d}x = \frac{1}{5} T_F.$$

This proves (49.249). $\square$

Remark 49.68 (What the tree coefficient does and does not prove). Lemma 49.67 is a derivation of the first resolved real-emission coefficient in the small-angle expansion under the stated collinear splitting convention. It is not the full small-angle EEC theorem. Beyond this leading coefficient, one must define the renormalized light-ray operator basis, include virtual terms and contact conventions, solve the mixing problem, and match the endpoint expansion to the separated-angle distribution in a fixed regulator and subtraction scheme. The exact observable remains the detector distribution (49.233).

Small-angle light-ray OPE structure. The symbolic matching formula (49.242) becomes a mathematical statement only after the operator coordinates, endpoint variable, regulator, and contact convention have been declared. Let

$$\rho := 1 - \zeta = 1 - \cos\chi, \qquad \rho \simeq \frac{\theta^2}{2}$$

in the small-angle region. A renormalized QCD light-ray OPE structure for this endpoint consists of the following ingredients:

$$\mathcal{D}^{\mathrm{coll}}_{\mathrm{EEC}} = \big(\mathfrak{L}_{\mathbf{n}}, \{\mathbb{O}_A\}, \mathcal{S}_{\mathrm{UV}}, \mathcal{S}_{\mathrm{rap}}, \gamma_{AB}, \eta_{AB}, C_A\big).$$

Here $\mathfrak{L}_{\mathbf{n}}$ is a linear space of gauge-invariant renormalized light-ray operator-valued distributions in the null direction specified by $\mathbf{n}$. The indexed family $\{\mathbb{O}_A(\mathbf{n};\mu,\nu)\}$ is a chosen coordinate basis of this space, including the leading twist energy-flow moments and any operators that mix with them under the declared symmetries. The symbols $\mathcal{S}_{\mathrm{UV}}$ and $\mathcal{S}_{\mathrm{rap}}$ denote, respectively, the ultraviolet subtraction scheme and, when a lightlike Wilson-line or rapidity singularity is present, the rapidity subtraction scheme. The scale μ is the ultraviolet renormalization scale and ν is the rapidity scale; if a scheme has no separate rapidity regulator, all ν-dependent objects below are omitted. The kernels γ_{AB} and η_{AB} are distributional mixing kernels in whatever momentum-fraction, Mellin, or angular variables label the basis. The coefficient $C_A(\rho, Q, \mu, \nu)$ acts on smooth endpoint test functions in ρ, with its separated-angle density, endpoint subtractions, and $\delta(\rho)$ contact coefficient all specified by the same subtraction scheme.

With column-vector operator convention, the renormalization equations are

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\mathbb{O}_A = -\sum_B \gamma_{AB}\otimes\mathbb{O}_B, \qquad \nu\frac{\mathrm{d}}{\mathrm{d}\nu}\mathbb{O}_A = -\sum_B \eta_{AB}\otimes\mathbb{O}_B, \tag{49.250}$$

where $\otimes$ denotes the convolution product in the operator labels and matrix multiplication in the discrete labels. Since the product of detectors is a physical operator-valued distribution after the detector observable is normalized, the endpoint expansion

$$\widehat{\mathcal{E}}(\mathbf{n}_1)\widehat{\mathcal{E}}(\mathbf{n}_2) \simeq \sum_A C_A\otimes\mathbb{O}_A$$

must be independent of μ and of ν. Differentiating this identity and using (49.250) gives the coefficient equations

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}C_A = \sum_B C_B\otimes\gamma_{BA}, \qquad \nu\frac{\mathrm{d}}{\mathrm{d}\nu}C_A = \sum_B C_B\otimes\eta_{BA}. \tag{49.251}$$

This sign and transpose convention fixes the cancellation between the anomalous dimension of the coefficient row and the anomalous dimension of the operator column. In a Mellin basis where the convolution diagonalizes, (49.251) reduces to ordinary matrix multiplication in the remaining operator indices.

The separated-angle real-emission coefficient in Lemma 49.67 supplies the positive density on $\rho > 0$. The renormalized endpoint coefficient also contains endpoint subtractions and coincident-detector contact data. To state this coefficient, fix a small but finite endpoint interval $0 < \rho < \rho_0$. The plus distribution on this interval is defined by

$$\int_0^{\rho_0} \left[\frac{1}{\rho}\right]_{\rho_0,+} \varphi(\rho)\,\mathrm{d}\rho := \int_0^{\rho_0} \frac{\varphi(\rho) - \varphi(0)}{\rho}\,\mathrm{d}\rho \tag{49.252}$$

for every smooth test function φ on $[0, \rho_0]$. Therefore the first singular real coefficient

$$\frac{\alpha_s}{2\pi}\,\frac{\Gamma_a^{\mathrm{EEC}}}{\rho}$$

is promoted, in a renormalized endpoint coefficient, to the distributional form

$$C_a^{(1)}(\rho) = \frac{\alpha_s}{2\pi}\,\Gamma_a^{\mathrm{EEC}}\left[\frac{1}{\rho}\right]_{\rho_0,+} + K_a^{(1)}(\rho_0, \mu, \nu)\,\delta(\rho) + C_{a,\mathrm{reg}}^{(1)}(\rho). \tag{49.253}$$

The constant $K_a^{(1)}$ and the regular distribution $C_{a,\mathrm{reg}}^{(1)}$ are scheme- and matching-dependent. The plus prescription carries the coincident-detector subtraction that makes the endpoint distribution act on test functions with finite endpoint limits. The remaining finite $\delta(\rho)$ coefficient is fixed after the virtual terms, operator renormalization, and matching convention are specified in the same scheme. Thus the real coefficient Γ_a^{EEC} is a necessary local input to the small-angle OPE; the anomalous dimensions and the full coefficient of a renormalized light-ray operator are determined by the complete scheme-dependent endpoint datum.

The dependence on the endpoint resolution boundary is also distributional, not verbal. For $0 < \rho_a < \rho_b$, define

$$D_{\rho_*}(\rho) := \left[\frac{\Theta(\rho_* - \rho)}{\rho}\right]_+$$

by

$$\langle D_{\rho_*}, \varphi\rangle = \int_0^{\rho_*} \frac{\varphi(\rho) - \varphi(0)}{\rho}\,\mathrm{d}\rho$$

for smooth endpoint tests. Then, as distributions on $[0, \rho_b]$,

$$D_{\rho_b} = D_{\rho_a} + \frac{\mathbf{1}_{\rho_a<\rho<\rho_b}}{\rho} - \log\frac{\rho_b}{\rho_a}\,\delta(\rho). \tag{49.254}$$

Testing the right-hand side against φ gives

$$\int_0^{\rho_a} \frac{\varphi - \varphi(0)}{\rho}\,\mathrm{d}\rho + \int_{\rho_a}^{\rho_b} \frac{\varphi(\rho)}{\rho}\,\mathrm{d}\rho - \log\frac{\rho_b}{\rho_a}\varphi(0) = \int_0^{\rho_b} \frac{\varphi - \varphi(0)}{\rho}\,\mathrm{d}\rho.$$

Thus enlarging the endpoint chart by the annulus $\rho_a < \rho < \rho_b$ adds the ordinary separated-angle integral over that annulus and simultaneously shifts the coincident-detector contact coordinate by the logarithmic amount required to keep constant tests unchanged. This is the finite-resolution form of the endpoint contact convention: a statement such as "drop the contact term" is incomplete until the endpoint resolution boundary and the compensating $\delta(\rho)$ coordinate have both been declared.

The eventwise moment identity (49.235) imposes a useful check on this bookkeeping. Let M_A be the energy-sum functional on the chosen light-ray operator basis, so that $M_A \mathbb{O}_A$ extracts the component that contributes to the zeroth EEC moment. Conservation of total energy says that this functional has zero anomalous dimension:

$$\sum_A M_A \, \gamma_{AB} = 0, \qquad \sum_A M_A \, \eta_{AB} = 0 \tag{49.255}$$

in any scheme that preserves the detector normalization. Combining (49.251) with (49.255) shows that the RG evolution cannot change the zeroth moment; only the declared redistribution between separated-angle weight and the $\delta(\rho)$ contact term can change the displayed representative. This is why every perturbative EEC formula must specify whether it includes the coincident-detector contact term. Without that convention, even a formally correct endpoint singularity does not define the same distribution as the nonperturbative observable (49.233).

Finite renormalized light-ray mixing chart. The preceding equations describe the continuum endpoint expansion, whose full operator basis is infinite and distributional. In an actual calculation one uses a finite coordinate chart on this space. The following finite chart is the algebraic core that every higher-order small-angle EEC calculation must instantiate before its anomalous dimensions and matching coefficients can be compared across schemes.

Fix a finite set of light-ray operator labels $\mathcal{I}_L$, for example a spin, twist, flavor, and derivative truncation, and let $\mathbb{O}_I^{\mathrm{reg}}(\Lambda, \nu_0)$, $I \in \mathcal{I}_L$, be regulated gauge-invariant light-ray composites with the endpoint and rapidity regulator specified. A finite renormalized chart consists of an invertible matrix $Z_{IA}(\Lambda, \mu, \nu; \mathcal{R})$ and renormalized operator coordinates defined by

$$\mathbb{O}_A(\mu, \nu; \mathcal{R}) = \sum_{I \in \mathcal{I}_L} (Z^{-1})_{AI}(\Lambda, \mu, \nu; \mathcal{R}) \, \mathbb{O}_I^{\mathrm{reg}}(\Lambda, \nu_0), \tag{49.256}$$

after the regulator limit is taken in matrix elements in the declared test space. If the regulated coordinates are independent of μ after the regulator and bare parameters are held fixed, then differentiating (49.256) gives

$$\gamma = Z^{-1} \mu \frac{\mathrm{d}Z}{\mathrm{d}\mu}, \qquad \mu \frac{\mathrm{d}}{\mathrm{d}\mu} \mathbb{O} = -\gamma \, \mathbb{O}, \tag{49.257}$$

with column-vector notation. A regulated coefficient row C^{reg} that pairs with the regulated operators as $C^{\mathrm{reg}} \mathbb{O}^{\mathrm{reg}}$ therefore gives the renormalized coefficient row

$$C(\rho, Q, \mu, \nu; \mathcal{R}) = C^{\mathrm{reg}}(\rho, Q, \Lambda, \nu_0) \, Z(\Lambda, \mu, \nu; \mathcal{R}), \tag{49.258}$$

and hence

$$\mu \frac{\mathrm{d}C}{\mathrm{d}\mu} = C \, \gamma. \tag{49.259}$$

This is the finite-matrix derivation of the sign and transpose convention in (49.250)–(49.251). No physical statement is attached to Z or C separately: only the paired distribution $C\mathbb{O}$, together with its endpoint contact coordinate, represents the detector product.

Finite scheme changes are coordinate changes on the same renormalized light-ray space. Let $R(\mu,\nu)$ be an invertible finite matrix and set

$$\mathbb{O}' = R^{-1}\mathbb{O}, \qquad C' = CR.$$

Then $C'\mathbb{O}' = C\mathbb{O}$, while the anomalous-dimension matrix becomes

$$\gamma' = R^{-1}\gamma R + R^{-1}\mu\frac{\mathrm{d}R}{\mathrm{d}\mu}, \tag{49.260}$$

and C' obeys $\mu \mathrm{d}C'/\mathrm{d}\mu = C'\gamma'$. Thus a quoted "light-ray anomalous dimension" is incomplete until its operator coordinate chart is specified. If $M\mathbb{O}$ is the protected total-energy moment, then $M\gamma = 0$. In the changed coordinates $M' = MR$ and

$$\mu\frac{\mathrm{d}M'}{\mathrm{d}\mu} = M'\gamma'.$$

The stronger condition that the displayed row M is unchanged is therefore not automatic; it is a restriction on admissible schemes, $MR = M$. This is the finite-coordinate version of the normalization convention behind (49.255).

Finite transport and two-scale consistency. The finite chart also makes precise what is meant by solving the light-ray mixing problem. Write

$$t = \log\mu, \qquad r = \log\nu.$$

For a single ultraviolet scale, let $U(t,t_0)$ be the matrix solution of

$$\partial_t U(t,t_0) = U(t,t_0)\gamma(t), \qquad U(t_0,t_0) = \mathbf{1}. \tag{49.261}$$

When $\gamma(t)$ at different scales does not commute, (49.261) is the definition of the path-ordered product; no choice of ordered exponential is meaningful without the side on which γ acts. With the row/column convention of (49.257)–(49.259), the transported coordinates are

$$C(t) = C(t_0)U(t,t_0), \qquad \mathbb{O}(t) = U(t,t_0)^{-1}\mathbb{O}(t_0), \tag{49.262}$$

and the paired endpoint distribution is independent of t:

$$C(t)\mathbb{O}(t) = C(t_0)\mathbb{O}(t_0).$$

Indeed, differentiating $U^{-1}U = \mathbf{1}$ gives

$$\partial_t U^{-1} = -\gamma(t)U^{-1},$$

because $\partial_t U = U\gamma$. Thus

$$\partial_t\mathbb{O}(t) = -\gamma(t)\mathbb{O}(t), \qquad \partial_t C(t) = C(t)\gamma(t),$$

and the two terms in $\partial_t(C\mathbb{O})$ cancel. If $M(t)$ is the coordinate row representing the protected total-energy moment, its transport law in a moving chart is

$$\partial_t M(t) = M(t)\gamma(t). \tag{49.263}$$

The convention $M(t) = M(t_0)$ for all t is therefore equivalent to $M\gamma(t) = 0$ in that fixed chart. This is the all-order version of the finite left-null-vector check in (49.255).

If both ultraviolet and rapidity scales are present, the operator column is declared to obey

$$\partial_t \mathbb{O} = -\gamma(t,r)\mathbb{O}, \qquad \partial_r \mathbb{O} = -\eta(t,r)\mathbb{O}.$$

The two scale evolutions define a path-independent endpoint coordinate only if the connection is flat on the chart:

$$\partial_t \eta - \partial_r \gamma + [\gamma, \eta] = 0. \tag{49.264}$$

This equation is not an optional aesthetic condition. It is the equality of mixed derivatives of the same renormalized operator coordinate. To see the sign, compute

$$\partial_r \partial_t \mathbb{O} = -(\partial_r \gamma)\mathbb{O} + \gamma\eta\,\mathbb{O}, \qquad \partial_t \partial_r \mathbb{O} = -(\partial_t \eta)\mathbb{O} + \eta\gamma\,\mathbb{O}.$$

Equating the two expressions gives (49.264). In a finite chart with nonzero curvature

$$\mathcal{F}_{tr} := \partial_t \eta - \partial_r \gamma + [\gamma, \eta],$$

transporting C and $\mathbb{O}$ along two different paths in (μ, ν)-space produces different coordinate representatives unless the curvature action is compensated by additional operators or by a prescribed scheme-change wall. A calculation that quotes separate μ- and ν-anomalous dimensions for the EEC endpoint has therefore supplied a consistent renormalized light-ray chart only after (49.264), the protected moment row, and the endpoint contact convention have been checked in the same scheme.

This paragraph gives the structural consistency condition. It is not itself a small-angle EEC loop calculation. In a concrete rapidity-factorized scheme one must display the regulated endpoint operator basis, compute the two matrices $\gamma(t,r)$ and $\eta(t,r)$ with the same subtraction and contact-term convention, and then verify $\partial_t \eta - \partial_r \gamma + [\gamma, \eta] = 0$ on that basis. If the finite basis is a truncation of an infinite operator space, the check must also state the projected curvature $P_L(\partial_t \eta - \partial_r \gamma + [\gamma, \eta])P_L$, the curvature components that leave the retained subspace, and the contact-coordinate terms that compensate endpoint scheme changes. Vanishing only after discarding an omitted operator is not flatness of the retained chart; it is an approximation statement requiring an omitted-operator remainder. The finite matrix check recorded in the companion script verifies the sign and row/column convention of the flatness equation, including the derivative terms, but it does not replace this scheme-specific QCD calculation.

Projected curvature and omitted light-ray coordinates. The most common failure mode in a finite light-ray calculation is not an incorrect commutator sign, but an omitted operator channel. Let P_L be the projection onto the retained light-ray labels and $Q_L = 1 - P_L$. In a fixed chart write

$$\gamma_{LL} = P_L \gamma P_L, \qquad \gamma_{LQ} = P_L \gamma Q_L, \qquad \gamma_{QL} = Q_L \gamma P_L,$$

and similarly for η. If the full connection is flat on the enlarged operator space and P_L is scale-independent, then the curvature computed from the retained block alone is

$$\partial_t \eta_{LL} - \partial_r \gamma_{LL} + [\gamma_{LL}, \eta_{LL}] = \eta_{LQ}\gamma_{QL} - \gamma_{LQ}\eta_{QL}. \tag{49.265}$$

Indeed, insert $P_L + Q_L = 1$ between the two factors in the commutator part of (49.264). Thus a nonzero retained curvature is not automatically an inconsistency of the full endpoint construction;

it is an explicit omitted-channel term. A finite small-angle EEC calculation that keeps only P_L must therefore either include the right-hand side of (49.265) in its remainder estimate or enlarge the light-ray/contact-coordinate chart until the flatness check is performed on the enlarged block.

This statement is coordinate covariant. For a finite change of retained coordinates $R_L(t,r)$,

$$\mathbb{O}'_L = R_L^{-1}\mathbb{O}_L, \qquad C'_L = C_L R_L,$$

one has

$$\gamma'_{LL} = R_L^{-1}\gamma_{LL}R_L + R_L^{-1}\partial_t R_L, \qquad \eta'_{LL} = R_L^{-1}\eta_{LL}R_L + R_L^{-1}\partial_r R_L,$$

and the retained curvature transforms by conjugation,

$$\mathcal{F}'_{LL} = R_L^{-1}\mathcal{F}_{LL}R_L.$$

Consequently a retained curvature cannot be removed by changing finite coordinates inside the same retained space. What can change is the split between an operator block and endpoint contact coordinates, in which case the flatness equation must be checked in the enlarged chart and the induced change of contact distribution must be recorded in the endpoint convention.

Endpoint observable transport budget. The last step in a finite endpoint calculation is not the display of γ, η, or a coefficient row. It is the statement that the retained chart gives a stable distribution when paired with endpoint detector tests. The following finite proposition is the algebraic budget behind that statement.

Proposition 49.69 (Finite endpoint observable transport budget)**.** *Fix a finite retained light-ray label set L, an endpoint test φ, and one logarithmic scale $t = \log\mu$. Suppose the retained small-angle endpoint contribution in a fixed state is represented by*

$$V_L(\varphi;t) = C_L(\varphi;t)\,O_L(t) + K_L(\varphi;t), \tag{49.266}$$

where $C_L(\varphi;t)$ is a row of coefficient distributions evaluated on φ, $O_L(t)$ is the column of retained light-ray matrix elements, and $K_L(\varphi;t)$ is the explicit endpoint contact coordinate. Let $\gamma_{LL}(t)$ be the retained anomalous-dimension matrix, and let $D_L(\varphi;t)$ be the row produced by a scale-dependent change of endpoint extension convention, such as moving an annulus into a plus-chart. Assume

$$\partial_t O_L = -\gamma_{LL}O_L + E_O, \tag{49.267}$$

$$\partial_t C_L(\varphi) = C_L(\varphi)\gamma_{LL} + D_L(\varphi) + E_C(\varphi), \tag{49.268}$$

$$\partial_t K_L(\varphi) = -D_L(\varphi)O_L + E_K(\varphi). \tag{49.269}$$

Then

$$\partial_t V_L(\varphi;t) = E_C(\varphi;t)O_L(t) + C_L(\varphi;t)E_O(t) + E_K(\varphi;t). \tag{49.270}$$

Consequently, for any interval $[t_0,t_1]$,

$$\big|V_L(\varphi;t_1) - V_L(\varphi;t_0)\big| \le \int_{t_0}^{t_1} \left(\|E_C(\varphi;t)\|\,\|O_L(t)\| + \|C_L(\varphi;t)\|\,\|E_O(t)\| + |E_K(\varphi;t)|\right)\mathrm{d}t \tag{49.271}$$

for any dual pair of finite vector norms. In particular, exact operator transport, exact coefficient transport, and the matching contact shift make the retained endpoint observable independent of t.

Proof. Differentiate (49.266) and insert (49.267)–(49.269):

$$\partial_t V_L = (C_L\gamma_{LL} + D_L + E_C)O_L + C_L(-\gamma_{LL}O_L + E_O) - D_L O_L + E_K.$$

The anomalous-dimension terms cancel because the coefficient is a row and the operator is a column. The endpoint-extension row D_L cancels only because the contact coordinate in (49.269) carries the compensating sign. This gives (49.270). Integrating in t and applying the finite norm inequality for a row-column pairing gives (49.271). □

The rows D_L are not new anomalous dimensions. They record changes of distributional representative at $\rho = 0$, such as the shift in (49.254). The residuals E_O, E_C, E_K are where an actual finite-basis endpoint calculation must place its omitted operator channels, projected curvature, approximate loop-order truncation, and endpoint matching defects. A claimed all-order renormalized light-ray OPE for the EEC must drive the right side of (49.271) to zero in the declared detector-test topology, or must state the resulting approximation error. Thus a mixing matrix and a list of endpoint plus distributions become an observable theorem only after this transport budget is supplied.

Minimal cusp-log flatness check. A useful first check is the part of a rapidity-factorized endpoint chart that is forced by a cusp logarithm. Work in a finite block in which the cusp endomorphism A is fixed and commutes with itself at different scales. Let $\Gamma_{\mathrm{cusp}}(t)$ denote the cusp coefficient in the same trace and coupling convention as the endpoint calculation, and suppose the rapidity dependence of the ultraviolet anomalous dimension contains

$$\gamma_{\mathrm{cusp}}(t, r) = \Gamma_{\mathrm{cusp}}(t)\, r\, A. \tag{49.272}$$

Then the rapidity anomalous dimension in the same chart must contain a term

$$\eta_{\mathrm{cusp}}(t, r) = G_{\mathrm{cusp}}(t)\, A, \qquad \frac{\mathrm{d}G_{\mathrm{cusp}}}{\mathrm{d}t} = \Gamma_{\mathrm{cusp}}(t), \tag{49.273}$$

up to pieces whose t-derivative is zero in this block. Indeed,

$$\partial_t \eta_{\mathrm{cusp}} = \Gamma_{\mathrm{cusp}}(t) A = \partial_r \gamma_{\mathrm{cusp}}, \qquad [\gamma_{\mathrm{cusp}}, \eta_{\mathrm{cusp}}] = 0.$$

Thus the cusp-logarithmic part of (49.264) is satisfied. With the opposite sign in (49.273), the curvature would be $-2\Gamma_{\mathrm{cusp}}(t)A$. The sign statement belongs to the declared operator convention $\partial_t \mathbb{O} = -\gamma\mathbb{O}$, $\partial_r \mathbb{O} = -\eta\mathbb{O}$. A full small-angle EEC calculation must still compute the non-cusp matrices, endpoint contact counterterms, and operator components beyond this cusp block in one declared rapidity scheme.

Controlled Approximation 49.70 (Finite light-ray mixing truncation)**.** A finite small-angle EEC calculation based on a label set $\mathcal{I}_L$ is a controlled approximation only after it supplies:

1. the regulated light-ray composites $\mathbb{O}_I^{\mathrm{reg}}$ and the subtraction matrix Z;
2. the endpoint test-function space on which the coefficient distributions $C_A(\rho)$, including plus distributions and contact coordinates, act;
3. the matrices γ and, when present, η, with the protected energy-moment row and its scheme convention;

4. the transport matrices or path-ordered products solving (49.261), and, in a two-scale scheme, the flatness condition (49.264);
5. a remainder functional $R_L(\varphi, \Psi)$ for the omitted operators, stated in the topology of the matrix elements or cross-section distributions being computed.

Without the last item, a finite light-ray basis is a useful coordinate truncation but not an approximation theorem for the EEC distribution.

Controlled Approximation 49.71 (Perturbative EEC endpoint calculations)**.** A perturbative endpoint calculation of the EEC consists of a declared endpoint variable, a factorization or light-ray OPE formula such as (49.242) or (49.243), anomalous dimensions and matching coefficients through specified orders, a prescription for matching the endpoint expansion to the separated-angle fixed-order distribution, and a power correction estimate in θ or q_T/Q. The exact observable remains the hadronic detector distribution (49.233).

Endpoint matching as distributional gluing. The endpoint formulae above live in different distribution charts. A prediction for the full normalized EEC on $[-1, 1]$ requires a gluing prescription, because the contact coefficient at $\zeta = 1$, the perturbative back-to-back delta coordinate at $\zeta = -1$, and the separated-angle density contribute to the same moment sum rules.

Let $\mathcal{R}(\zeta)$ be the distribution on the open interval $(-1, 1)$ obtained from the declared bulk expression and the overlap-subtracted small-angle and back-to-back endpoint formulae, all tested against smooth functions whose support avoids the two endpoints. Suppose its first two moments exist after the declared endpoint subtractions:

$$I_0[\mathcal{R}] = \langle \mathcal{R}, 1 \rangle, \qquad I_1[\mathcal{R}] = \langle \mathcal{R}, \zeta \rangle. \tag{49.274}$$

If the two endpoint atoms are retained as explicit scheme coordinates, write

$$\mathcal{E}_{\text{match}}(\zeta) = \mathcal{R}(\zeta) + K_+\,\delta(1-\zeta) + K_-\,\delta(1+\zeta). \tag{49.275}$$

Here K_+ is the coincident-detector contact coordinate and K_- is the back-to-back endpoint coordinate generated by the perturbative source and recoil expansion. Only K_+ has the direct eventwise interpretation (49.236). The two exact detector moment identities (49.235) impose

$$I_0[\mathcal{R}] + K_+ + K_- = 1, \qquad I_1[\mathcal{R}] + K_+ - K_- = 0. \tag{49.276}$$

When both endpoint atoms are left as matching coordinates, the solution is

$$K_+ = \frac{1 - I_0[\mathcal{R}] - I_1[\mathcal{R}]}{2}, \qquad K_- = \frac{1 - I_0[\mathcal{R}] + I_1[\mathcal{R}]}{2}. \tag{49.277}$$

If K_+ is instead fixed independently by the detector contact definition, then (49.276) becomes a consistency test on the bulk, small-angle, and back-to-back calculations in the chosen subtraction scheme. A mismatch is a statement about truncation, missing power corrections, or incompatible endpoint conventions, rather than a new observable. This is the minimal matching structure required before a perturbative endpoint formula can be compared with the nonperturbative distribution (49.233).

49.16.3 Multipoint Correlators and Energy-Flow Polynomials

For a jet or event with a chosen positive energy scale E_{ref}, define $z_i = E_i/E_{\mathrm{ref}}$ and θ_{ij} to be the angular distance between directions i and j. For $\beta > 0$, the N-point energy correlation function used in jet substructure has the form

$$e_N^{(\beta)} = \sum_{i_1 < \cdots < i_N} z_{i_1} \cdots z_{i_N} \prod_{a<b} \theta_{i_a i_b}^{\beta}. \tag{49.278}$$

The strict positivity of β makes the observable collinear continuous: when two labels become collinear, every factor distinguishing the split vanishes or recombines into the total energy weight. The soft behavior is linear in the soft energy weight. Thus (49.278) is infrared and collinear safe in the sense of the soft/collinear continuity estimate for smeared energy flow above.

More generally, a finite graph G with vertices $\{1, \ldots, N\}$ defines an energy-flow polynomial

$$\mathrm{EFP}_G^{(\beta)} = \sum_{i_1,\ldots,i_N} z_{i_1} \cdots z_{i_N} \prod_{(a,b)\in E(G)} \theta_{i_a i_b}^{\beta}. \tag{49.279}$$

This is again a polynomial functional of the calorimetric measure and angular kernel. It is a useful coordinate system on infrared-collinear safe event information because it is built from the same positive detector measure μ_X rather than from colored partonic labels. In an exact QCD calculation the final states are hadrons; in a short-distance approximation, quark and gluon variables enter through factorized coefficient functions, evolution kernels, and power corrections for this color-singlet observable.

49.16.4 Track Selectors and Charged-Energy Correlators

Collider detectors often measure a selected part of the calorimetric final state, for example the energy carried by long-lived charged hadrons. The nonperturbative definition again begins with physical hadronic final states. Let $\mathfrak{A}_{\mathrm{st}}$ be the label set of stable hadron species in the scattering representation, and fix a selector

$$\tau : \mathfrak{A}_{\mathrm{st}} \longrightarrow [0, 1].$$

For an ideal charged-track selector, $\tau(a) = 1$ for the species whose charged tracks are included in the measurement and $\tau(a) = 0$ for the species omitted by the tracking system. A detector with efficiencies or acceptance weights is represented by intermediate values of τ, once those weights have been calibrated independently of the QCD short-distance calculation.

For a hadronic final state

$$X = \{(p_r, a_r)\}_{r=1}^{N_X}, \qquad \mathbf{n}_r = \mathbf{p}_r/|\mathbf{p}_r|,$$

define the selected calorimetric measure

$$\mu_X^{\tau} = \sum_{r\in X} \tau(a_r) p_r^0 \, \delta_{\mathbf{n}_r}. \tag{49.280}$$

It satisfies $0 \le \mu_X^{\tau} \le \mu_X$ as positive measures. The smeared track detector on the outgoing multiplication representation is therefore

$$\widehat{\mathcal{E}}_X^{\tau}(f) = \int_{S^2} f \, \mathrm{d}\mu_X^{\tau} = \sum_{r\in X} \tau(a_r) p_r^0 f(\mathbf{n}_r).$$

Products of track detectors are products of this selected measure:

$$\prod_{j=1}^{k} \widehat{\mathcal{E}}_X^{\tau}(f_j) = \int_{(S^2)^k} f_1(\mathbf{n}_1)\cdots f_k(\mathbf{n}_k)\,\mathrm{d}\mu_X^{\tau}(\mathbf{n}_1)\cdots \mathrm{d}\mu_X^{\tau}(\mathbf{n}_k). \tag{49.281}$$

Thus track energy correlators are physical detector observables once the selector τ and the detector calibration have been specified.

The exact moment identities are selected-energy identities. In a center-of-mass event with $z_r = p_r^0/Q$, set

$$e_\tau(X) = \sum_r \tau(a_r) z_r, \qquad \mathbf{p}_\tau(X) = \sum_r \tau(a_r) z_r \mathbf{n}_r.$$

The eventwise two-point selected-energy distribution

$$E_{2,X}^{\tau}(\zeta) = \sum_{r,s\in X} \tau(a_r)\tau(a_s) z_r z_s\, \delta(\zeta - \mathbf{n}_r\cdot\mathbf{n}_s)$$

obeys

$$\int_{-1}^{1} E_{2,X}^{\tau}(\zeta)\,\mathrm{d}\zeta = e_\tau(X)^2, \qquad \int_{-1}^{1} \zeta E_{2,X}^{\tau}(\zeta)\,\mathrm{d}\zeta = |\mathbf{p}_\tau(X)|^2. \tag{49.282}$$

For $\tau \equiv 1$ these reduce to (49.235). For a proper track selector they are additional nonperturbative event variables: the selected charged energy and selected charged three-momentum fluctuate from event to event even when the total energy and total three-momentum are fixed.

Perturbative calculations of selected-energy correlators require nonperturbative track functions. In a factorization scheme, a track function for a parton label i is a positive distribution $T_i(x,\mu)$ on $[0,1]$, normalized by

$$\int_0^1 T_i(x,\mu)\,\mathrm{d}x = 1,$$

whose moment

$$M_i^{(n)}(\mu) = \int_0^1 x^n T_i(x,\mu)\,\mathrm{d}x$$

is the n-th moment of the selected-energy fraction carried by hadrons in a collinear jet initiated by i, in the chosen scheme. The functions T_i are nonperturbative light-ray/fragmentation data; perturbation theory controls their evolution and their convolution with hard and jet coefficients.

The local collinear bookkeeping is fixed by energy partition. If a parent parton i splits into daughters j,k with energy fractions $z, 1-z$, and the selected fractions in the daughter collinear sectors are x_j, x_k, then the selected fraction of the parent sector is

$$x = z x_j + (1-z) x_k.$$

Therefore the real-splitting contribution to the n-th selected-fraction moment has the binomial form

$$M_{i\to jk}^{(n)}(z) = \sum_{a=0}^{n} \binom{n}{a} z^a (1-z)^{n-a} M_j^{(a)} M_k^{(n-a)}. \tag{49.283}$$

In particular,

$$M_{i\to jk}^{(1)}(z) = z M_j^{(1)} + (1-z) M_k^{(1)}$$

and

$$M^{(2)}_{i\to jk}(z) = z^2 M^{(2)}_j + (1-z)^2 M^{(2)}_k + 2z(1-z) M^{(1)}_j M^{(1)}_k.$$

The last line separates the contact and resolved two-detector weights in a track EEC calculation:

$$w_{\tau,\mathrm{contact}}(z;j,k) = z^2 M^{(2)}_j + (1-z)^2 M^{(2)}_k, \qquad w_{\tau,\mathrm{sep}}(z;j,k) = 2z(1-z) M^{(1)}_j M^{(1)}_k. \quad (49.284)$$

For the full calorimeter, $M^{(n)}_i = 1$ for every parton label and every n, and the separated weight reduces to the ordinary EEC weight $2z(1-z)$ in (49.247).

A controlled track-energy calculation must specify the selector τ, the operator or fragmentation definition of the track functions T_i, the factorization scheme and scale μ, the evolution equation for T_i in that scheme, the endpoint/contact convention for the selected detector products, and the power remainder that compares the selected hadronic observable (49.281) with the finite partonic factorization formula. The selected-energy observable is defined at the hadron level; the partonic track functions are the bridge used for its short-distance expansion.

49.17 Current Correlators and the Hadronic Tensor

Deep inelastic scattering probes a hadron by a local current. The process is inclusive:

$$e^-(k,\sigma) + N(P,s) \longrightarrow e^-(k',\sigma') + X, \qquad q := k - k',$$

where N is the target hadron and X runs over all hadronic final states with total momentum $P_X = P + q$. The exchange momentum is spacelike,

$$Q^2 := -q^2 > 0, \qquad x_B := \frac{Q^2}{2P\cdot q}.$$

The variable x_B is the Bjorken scaling variable.

Figure 49.9 fixes the inclusive DIS kinematics used below: the current transfers spacelike momentum q, the hadronic final state is summed over X, and $x_B = Q^2/(2P\cdot q)$.

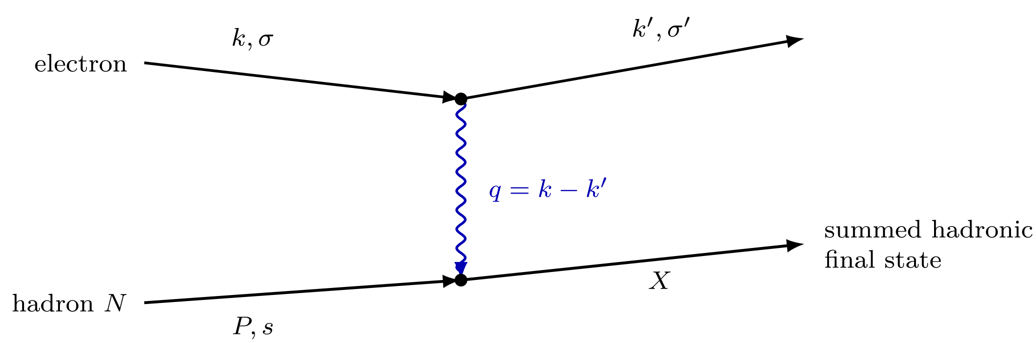


Figure 49.9: Inclusive deep inelastic scattering. The lepton line is treated perturbatively in the electromagnetic coupling, while the hadronic part is the current-current matrix element in QCD.

Let the electromagnetic current be

$$J^\mu(x) = \sum_I e_I\, \bar q_I(x)\gamma^\mu q_I(x),$$

where e_I is the electric charge of flavor I. At leading order in the electromagnetic coupling, the amplitude for a fixed hadronic final state has the form

$$\mathcal{A}_{\sigma',X;\sigma,s} = \bar{u}_{\sigma'}(k')\, e\gamma_\mu\, u_\sigma(k)\, \frac{-\mathrm{i}}{q^2 - \mathrm{i}\epsilon}\, \langle X|J^\mu(0)|P,s\rangle. \tag{49.285}$$

After summing over all hadronic final states X with the fixed transferred momentum q, the QCD part of the differential cross section is the spin-averaged positive tensor

$$W^{\mu\nu}(P,q) = \frac{1}{2}\sum_s \sum_X (2\pi)^4 \delta^{(4)}(P_X - P - q)\, \langle P,s|J^\nu(0)|X\rangle\langle X|J^\mu(0)|P,s\rangle. \tag{49.286}$$

where $\sum_X$ denotes the sum over species and the invariant phase-space integrals for a complete set of hadronic states, using the same state normalization as in the scattering chapters. Completeness and translation covariance give the equivalent form

$$W^{\mu\nu}(P,q) = \frac{1}{2}\sum_s \int \mathrm{d}^4x\, \mathrm{e}^{-\mathrm{i}q\cdot x} \langle P,s|J^\nu(x)J^\mu(0)|P,s\rangle, \tag{49.287}$$

in those conventions. Thus $W^{\mu\nu}$ is a Wightman matrix element between hadron states. Its positivity follows directly from the sum over X in (49.286).

Current conservation gives

$$q_\mu W^{\mu\nu} = q_\nu W^{\mu\nu} = 0.$$

For an unpolarized target, Lorentz covariance, current conservation, and parity give two scalar structure functions. Define

$$\Pi^\mu(P,q) := P^\mu - \frac{P\cdot q}{q^2} q^\mu.$$

Then

$$W^{\mu\nu}(P,q) = \left(\eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}\right) W_1(\nu,Q^2) + \Pi^\mu(P,q)\Pi^\nu(P,q)\, W_2(\nu,Q^2), \qquad \nu := \frac{P\cdot q}{M}, \tag{49.288}$$

where $M^2 = P^2$. Other common normalizations absorb powers of M and $P\cdot q$ into W_1, W_2.

The support condition is kinematic. If N is the lightest hadron with the chosen conserved quantum numbers, then

$$(P+q)^2 = P_X^2 \geq M^2.$$

Since $q^2 = -Q^2$, this implies

$$2P\cdot q \geq Q^2, \qquad 0 < x_B \leq 1$$

for the physical inclusive region.

The time-ordered forward Compton amplitude is

$$T^{\mu\nu}(P,q) = \frac{\mathrm{i}}{2}\sum_s \int \mathrm{d}^4x\, \mathrm{e}^{-\mathrm{i}q\cdot x} \langle P,s|\,\mathrm{T}\ J^\nu(x)J^\mu(0)|P,s\rangle. \tag{49.289}$$

Its discontinuity in the energy variable is determined by the same Wightman matrix elements:

$$\mathrm{Disc}\ T^{\mu\nu}(P,q) = 2\pi\left[W^{\mu\nu}(P,q) + W^{\nu\mu}(P,-q)\right], \tag{49.290}$$

with the second term absent in the physical region $P \cdot q > 0$ for the spacelike process. Thus the inclusive cross section is controlled by the absorptive part of a time-ordered current-current amplitude, and the observable tensor is the Wightman object (49.286).

The Bjorken limit is

$$Q^2 \to \infty, \qquad x_B \text{ fixed.}$$

The short-distance region $x \to 0$ in (49.289) is then controlled by the operator product expansion of the two currents:

$$\mathrm{T}\ J^\mu(x) J^\nu(0) \sim \sum_a C_a^{\mu\nu}(x,\mu)\, \mathcal{O}_a(0,\mu). \tag{49.291}$$

The Wilson coefficients $C_a^{\mu\nu}$ are determined by short-distance renormalization and can be computed in perturbation theory at large Q. The matrix elements $\langle P|\mathcal{O}_a(0,\mu)|P\rangle$ are long-distance hadronic data.

49.18 Twist, Parton Distributions, and Factorization

49.18.1 Twist and Light-Ray Operator Definitions

For a local symmetric traceless operator of Lorentz spin s and scaling dimension Δ, its twist is

$$\tau := \Delta - s.$$

In an interacting theory Δ is a renormalized scaling exponent of the chosen operator basis. The word "twist" in this section is therefore a statement about the renormalized leading-power expansion, not only about the engineering dimension of a classical monomial. In the Bjorken limit, the leading terms in the OPE are the smallest-twist operators compatible with the quantum numbers of the currents. A convenient way to collect the tower of leading-twist matrix elements is by light-ray operators.

Let n^μ be a future-directed lightlike vector, $n^2 = 0$, and choose the normalization $P \cdot n > 0$ for the target momentum. For a representation R of the gauge group, define the straight Wilson transporter

$$W_R(\lambda n, 0) = \mathcal{P} \exp \left\{ \mathrm{i} \int_0^\lambda n^\mu A_\mu^a(sn) t_R^a \, \mathrm{d}s \right\}.$$

Here $A_\mu = A_\mu^a t^a$ is the same connection coordinate used throughout the chapter, with $D_\mu = \partial_\mu - \mathrm{i} A_\mu$ and $-\frac{1}{4g^2} \operatorname{tr} F_{\mu\nu} F^{\mu\nu}$. The symbol $\mathcal{P}$ orders factors with larger parameter s to the left.

Gauge covariance of the open light-ray operator. Let $U(x)$ be a smooth gauge transformation in the fundamental representation, and let $U_R(x)$ be its image in a representation R. With the transformation law

$$A_\mu \longmapsto U A_\mu U^{-1} - \mathrm{i}(\partial_\mu U) U^{-1}, \qquad q(x) \longmapsto U(x) q(x),$$

the transporter satisfies

$$W_R(\lambda n, 0) \longmapsto U_R(\lambda n) W_R(\lambda n, 0) U_R(0)^{-1}. \tag{49.292}$$

Consequently

$$\bar{q}(\lambda n)W_{\Box}(\lambda n,0)\gamma\cdot n\,q(0)$$

is a color-singlet bilocal operator before renormalization, and its renormalized descendants remain gauge invariant if the regularization and subtraction scheme preserve background gauge covariance.

Let $W_R(s,0)$ denote the partial transporter from 0 to sn. It is the solution of

$$\frac{\mathrm{d}}{\mathrm{d}s}W_R(s,0)=\mathrm{i}n^\mu A^a_\mu(sn)t^a_R W_R(s,0),\qquad W_R(0,0)=\mathbf{1}.$$

After the gauge transformation, the product $U_R(sn)W_R(s,0)U_R(0)^{-1}$ obeys the same differential equation with the transformed connection and has the same initial value at $s=0$. Uniqueness of solutions to this finite-dimensional linear ordinary differential equation gives the stated transformation law. The endpoint factors cancel against the transformations of $\bar{q}(\lambda n)$ and $q(0)$, because the gamma matrix acts on spinor indices and is the identity in color space.

For an unpolarized target, write

$$\langle\mathcal{X}\rangle_{P,H}:=\frac{1}{2}\sum_s\langle P,s|\mathcal{X}|P,s\rangle,$$

with the one-particle normalization fixed in the scattering chapters. The renormalized quark light-ray operator and its parton distribution are

$$\mathbb{O}_q(\lambda n;\mu):=\left[\bar{q}(\lambda n)W_{\Box}(\lambda n,0)\gamma\cdot n\,q(0)\right]_\mu, \tag{49.293}$$

$$f_{q/H}(x,\mu):=\frac{1}{2P\cdot n}\int\frac{\mathrm{d}\lambda}{2\pi}\,\mathrm{e}^{-\mathrm{i}x\lambda P\cdot n}\langle\mathbb{O}_q(\lambda n;\mu)\rangle_{P,H}. \tag{49.294}$$

The factor $2P\cdot n$ fixes the convention so that a free quark target has $f_{q/q}(x)=\delta(1-x)$. The antiquark distribution is defined by the charge-conjugate bilocal with the fields interchanged and the transporter running between the same endpoints. The gluon distribution is defined from the adjoint light-ray operator

$$\mathbb{O}_g(\lambda n;\mu):=\Big[F^a_{n\rho}(\lambda n)W^{ab}_{\mathrm{ad}}(\lambda n,0)F^{b\rho}_n(0)\Big]_\mu,\qquad F_{n\rho}:=n^\mu F_{\mu\rho}, \tag{49.295}$$

$$f_{g/H}(x,\mu):=\frac{1}{2x(P\cdot n)^2}\int\frac{\mathrm{d}\lambda}{2\pi}\,\mathrm{e}^{-\mathrm{i}x\lambda P\cdot n}\langle\mathbb{O}_g(\lambda n;\mu)\rangle_{P,H},\qquad 0<x<1. \tag{49.296}$$

The factor $1/x$ is part of the conventional normalization of the gluon light-ray distribution; it is what makes the N-th Mellin moment correspond to the spin-N local gluon operator below. Possible singular behavior at $x=0$ belongs to the small-x problem and is not hidden inside the definition.

Definition 49.72 (Renormalized integrated PDF coordinate)**.** For a hadron species H, a renormalized integrated PDF coordinate consists of the following objects.

1. A nonperturbative QCD presentation, or a finite-regulator approximation to one, which supplies normalized color-singlet hadron states $|P,s\rangle$ with $P^2=M_H^2$ and $P\cdot n>0$.
2. A lightlike direction n^μ, an orientation of the straight segment from 0 to λn, and the representation labels of the Wilson transporters appearing in (49.293) and (49.295).

3. A finite ultraviolet regulator Λ and regulated gauge-invariant bilocal composites $\mathbb{O}_a^\Lambda(\lambda n)$, smeared in λ before any distributional limit is taken.
4. A subtraction and factorization scheme at scale μ, encoded by a convolutional coordinate map Z_{ab} from $\mathbb{O}_a^\Lambda$ to renormalized light-ray coordinates $\mathbb{O}_b(\mu)$ in the sense of (49.299).
5. A declared distribution space in the light-cone fraction variable x, including the endpoint contact convention at $x = 0, 1$, on which the Fourier transform in $\lambda P \cdot n$ is paired with test functions.

The integrated PDF $f_{a/H}(x, \mu)$ is the resulting matrix-element distribution. In the conventions of (49.294) and (49.296),

$$f_{q/H}(\,\cdot\,, \mu) = \frac{1}{2P \cdot n}\, \mathcal{F}_{\lambda P \cdot n \to x}\big[\langle \mathbb{O}_q(\lambda n; \mu)\rangle_{P,H}\big], \tag{49.297}$$

$$f_{g/H}(\,\cdot\,, \mu) = \frac{1}{2x(P \cdot n)^2}\, \mathcal{F}_{\lambda P \cdot n \to x}\big[\langle \mathbb{O}_g(\lambda n; \mu)\rangle_{P,H}\big], \tag{49.298}$$

where the second equality is an equality in the declared distribution space on which multiplication by $1/x$ has been specified. The symbol $\mathcal{F}_{\lambda P \cdot n \to x}$ denotes the Fourier transform with the phases displayed in the defining formulae above. Equations (49.297) and (49.298) define renormalized operator coordinates. Support in $0 \le x \le 1$, positivity properties, probability language, and the appearance of the same distribution in a measured structure function are additional spectral, scheme, and factorization statements; they are not part of the definition of the renormalized light-ray matrix element.

The preceding equations are nonperturbative operator definitions once the underlying quantum field theory supplies the renormalized light-ray operators $\mathbb{O}_a(\lambda n; \mu)$ as operator-valued distributions, or at least as matrix-element distributions between the hadron states under consideration. They are not definitions in terms of on-shell quarks or gluons, and they do not presuppose a colored particle interpretation. In four-dimensional continuum QCD this existence statement is tied to the still-open problem of constructing the theory and to the additional ultraviolet problem of renormalizing lightlike Wilson-line composites. A gauge-invariant regulator can define finite approximants, but the continuum lightcone limit, the operator-renormalization limit, and the factorization theorem are separate assertions. Equal-time Euclidean bilocals used in lattice quasi-PDF or pseudo-PDF methods are therefore matching observables for the light-ray definition, not the definition itself.

The subscript μ in (49.293)–(49.296) means that the singular lightlike bilocal has been defined in a specified regularization and factorization scheme. Let $\mathbb{O}_a^\Lambda(x)$ denote the Fourier-coordinate operator obtained from the same gauge-invariant light-ray composite in a finite ultraviolet regulator Λ, after the path geometry, endpoint prescription, and finite regulator have been fixed. This cutoff-regulated operator is not a new physical observable; it is the finite-regulator coordinate from which the renormalized light-ray coordinate is obtained. For integrated collinear PDFs, the coordinate change is a convolution in the light-cone momentum fraction:

$$\mathbb{O}_a^\Lambda(x) = \sum_b \int_x^1 \frac{\mathrm{d}z}{z}\, Z_{ab}\Big(\frac{x}{z}, \mu, \Lambda\Big)\, \mathbb{O}_b(z; \mu), \tag{49.299}$$

where a, b range over quarks, antiquarks, and gluons, and the convolution is understood after pairing with test functions in a distribution space whose endpoint contact conventions have been declared. This equation is a definition of the renormalized light-ray coordinates. It separates regularization, the introduction of the finite object $\mathbb{O}_a^\Lambda$, from renormalization, the choice of the finite coordinate map Z and of the operator coordinate $\mathbb{O}_b(\mu)$. For TMDs or other observables with unresolved transverse momentum, an additional rapidity renormalization is required; it is not part of the integrated PDF definition used in inclusive DIS.

The DGLAP kernel is the infinitesimal form of this operator-coordinate map, not a transition probability for physical colored asymptotic states. Let Z^{-1} be the convolution inverse in the chosen finite channel basis. Differentiating (49.299) at fixed $\mathbb{O}^\Lambda$ gives

$$0 = \sum_b (\mu\partial_\mu Z_{ab}) \otimes \mathbb{O}_b(\mu) + \sum_b Z_{ab} \otimes \mu\partial_\mu \mathbb{O}_b(\mu),$$

where $\otimes$ denotes the same light-cone fraction convolution. Hence

$$\mu\partial_\mu \mathbb{O}_c(x;\mu) = \sum_b \int_x^1 \frac{\mathrm{d}z}{z}\, P_{cb}\Big(\frac{x}{z}, g(\mu)\Big)\, \mathbb{O}_b(z;\mu), \qquad P_{cb} := -\sum_a Z_{ca}^{-1} \otimes \mu\partial_\mu Z_{ab}. \tag{49.300}$$

Taking hadron matrix elements of (49.300) gives (49.314) below. The finite perturbative calculation of P is a calculation of the ultraviolet coordinate change of the light-ray operator. The subsequent use of P in a measured DIS observable still requires the leading-power factorization statement.

Local moments of the light-ray definition. The light-ray definition contains the local twist-two tower as its Taylor coefficients at zero separation. This statement is often quoted as a formal identity between PDFs and partons; here it is only a statement about the renormalized Wilson-line operator in (49.293). Define the left covariant derivative on the transported antiquark endpoint by

$$\bar{q}\overleftarrow{D}_n := n^\mu \partial_\mu \bar{q} + \mathrm{i}\,\bar{q}\, n^\mu A_\mu, \qquad n^2 = 0.$$

The sign is fixed by the same convention $D_\mu q = (\partial_\mu - \mathrm{i}A_\mu)q$ used in the Wilson transporter above. Since

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\big(\bar{q}(\lambda n) W_\Box(\lambda n, 0)\big) = \big(\bar{q}\overleftarrow{D}_n\big)(\lambda n) W_\Box(\lambda n, 0),$$

iteration gives the short-distance expansion

$$\bar{q}(\lambda n) W_\Box(\lambda n, 0)\gamma\cdot n\, q(0) = \sum_{N\geq 0} \frac{\lambda^N}{N!}\, \bar{q}(0)(\overleftarrow{D}_n)^N \gamma\cdot n\, q(0) \tag{49.301}$$

as a formal operator identity before subtraction, and as a renormalized operator-coordinate identity after the corresponding local and bilocal subtractions have been matched. It is useful to absorb the endpoint sign into the moment operator

$$\mathcal{O}^{\leftarrow}_{q,N}(n;\mu) := \Big[\bar{q}(0)(-\mathrm{i}\overleftarrow{D}_n)^N \gamma\cdot n\, q(0)\Big]_\mu. \tag{49.302}$$

The minus sign is not optional: a free one-quark matrix element of $\bar{q}(\lambda n)\gamma\cdot n q(0)$ is proportional to $\exp(\mathrm{i}\lambda P\cdot n)$, so $-\mathrm{i}\overleftarrow{D}_n$ has eigenvalue $P\cdot n$ on the left endpoint. With the Fourier convention in (49.294), the inverse distributional relation is

$$\frac{1}{2P\cdot n}\langle \mathbb{O}_q(\lambda n;\mu)\rangle_{P,H} = \int_{\mathbb{R}} \mathrm{d}x\, \mathrm{e}^{\mathrm{i}x\lambda P\cdot n} f_{q/H}(x,\mu),$$

whenever the pairing with test functions justifies inversion. Differentiating at $\lambda = 0$ therefore gives, for $N \geq 0$,

$$\int_0^1 \mathrm{d}x\, x^N f_{q/H}(x,\mu) = \frac{\langle \mathcal{O}^{\leftarrow}_{q,N}(n;\mu)\rangle_{P,H}}{2(P\cdot n)^{N+1}}. \tag{49.303}$$

The familiar symmetric traceless tensor $\bar{q}\gamma^{(\mu_0}\mathrm{i}D^{\mu_1}\cdots\mathrm{i}D^{\mu_N)}q$ is obtained by polarizing in the null vector n and changing from the left-endpoint representative to a symmetric representative. In forward matrix elements the difference is a sum of total-derivative representatives with fixed momentum transfer zero; in off-forward matrix elements those total derivatives are precisely the terms that produce GPD polynomiality in Section 49.21. Thus (49.303) is the convention-clean moment statement for the present forward PDF definition, while the symmetric local tensor is a different coordinate on the same leading-twist operator tower.

For the gluon definition the extra factor $1/x$ in (49.296) shifts the moment index. Let $\overleftarrow{D}_{n,\mathrm{ad}}$ be the left covariant derivative acting on the adjoint field at λn, including the adjoint connection in the same Wilson-transporter convention. For $N \geq 1$, define

$$\mathcal{O}^{\leftarrow}_{g,N}(n;\mu) := \Big[F^a_{n\rho}(0)(-\mathrm{i}\overleftarrow{D}_{n,\mathrm{ad}})^{N-1}F^{a\rho}_{n}(0)\Big]_\mu, \tag{49.304}$$

with the color contraction written after transporting the left endpoint to zero. The inverse relation following from (49.296) is

$$\frac{1}{2(P\cdot n)^2}\langle \mathbb{O}_g(\lambda n;\mu)\rangle_{P,H} = \int_{\mathbb{R}} \mathrm{d}x\, \mathrm{e}^{\mathrm{i}x\lambda P\cdot n}\, x f_{g/H}(x,\mu),$$

and therefore

$$\int_0^1 \mathrm{d}x\, x^N f_{g/H}(x,\mu) = \frac{\langle \mathcal{O}^{\leftarrow}_{g,N}(n;\mu)\rangle_{P,H}}{2(P\cdot n)^{N+1}}, \qquad N \geq 1. \tag{49.305}$$

Equations (49.303) and (49.305) are moment identities inside a specified renormalized operator scheme. They do not assert positivity, support, or factorization. Positivity and support require the spectral interpretation of the hadron matrix element in the relevant scattering representation; factorization requires the separate leading-power hypothesis stated below.

49.18.2 Euclidean Spatial Bilocals and Large-Momentum Matching

Lattice access to the light-ray definitions (49.294)–(49.296) uses different operator coordinates. The basic object is a spacelike equal-time bilocal, not a lightlike bilocal. Let e_3 be a fixed unit spatial vector and let $P^\mu = (E_P, 0, 0, P_3)$, $P_3 > 0$, be a one-hadron momentum in a finite-volume regulator with a transfer matrix and a gauge-invariant Hilbert space. For a Dirac matrix Γ and a real spatial separation z, define the regulated spatial Wilson-line operator

$$\mathcal{Q}_\Gamma(z;\Lambda) := [\bar{q}(ze_3)\, W_\square(ze_3, 0)\, \Gamma\, q(0)]_\Lambda\,. \tag{49.306}$$

Here Λ denotes the complete ultraviolet and finite-volume regulator data, including the lattice spacing and box when the operator is realized on a lattice. The Wilson transporter is along the straight spatial segment. Its linear length divergence, endpoint divergences, and finite subtraction are part of the operator-renormalization datum; they are not part of the definition of the light-ray PDF.

At finite gauge-covariant regulator, the operator $\mathcal{Q}_\Gamma(z;\Lambda)$ is gauge invariant as a color-singlet bilocal. A background-gauge-covariant finite-part prescription therefore produces a gauge-invariant renormalized spatial-bilocal matrix element. This is the endpoint-cancellation argument of the Wilson-transporter covariance calculation above, with the lightlike path sn replaced by the spatial path sze_3, $0 \le s \le 1$. The parallel transporter transforms as

$$W_\square(ze_3, 0) \longmapsto U(ze_3) W_\square(ze_3, 0) U(0)^{-1}.$$

The factor $U(0)^{-1}$ cancels the transformation of $q(0)$, the factor $U(ze_3)$ cancels the transformation of $\bar{q}(ze_3)$, and Γ acts only on spinor indices. The finite-part map changes regulator coordinates and operator coordinates but carries no free color index when it is background gauge covariant.

Let

$$\mathcal{M}^{\mathcal{R}}_{\Gamma/H}(z, P_3) := \frac{1}{2P_3} \langle P, H | \mathcal{Q}^{\mathcal{R}}_\Gamma(z) | P, H \rangle, \qquad \nu := P_3 z, \tag{49.307}$$

where $\mathcal{R}$ denotes a specified finite-part prescription. A quasi-PDF coordinate is the Fourier transform

$$\widetilde{f}^{\mathcal{R}}_{\Gamma/H}(x, P_3) := P_3 \int_{\mathbb{R}} \frac{\mathrm{d}z}{2\pi} \, \mathrm{e}^{\mathrm{i}xP_3 z} \, \mathcal{M}^{\mathcal{R}}_{\Gamma/H}(z, P_3), \tag{49.308}$$

understood as a tempered distribution in x after smearing. The prefactor P_3 fixes the inverse convention

$$\mathcal{M}^{\mathcal{R}}_{\Gamma/H}(z, P_3) = \int_{\mathbb{R}} \mathrm{d}x \, \mathrm{e}^{-\mathrm{i}xP_3 z} \, \widetilde{f}^{\mathcal{R}}_{\Gamma/H}(x, P_3)$$

for distributions for which the Fourier inversion is valid. A pseudo-PDF or Ioffe-time coordinate instead regards the same matrix element as a function, or distribution, of the Ioffe time $\nu = P_3 z$ and spacelike length $z^2 > 0$ in Euclidean notation:

$$\mathfrak{M}^{\mathcal{R}}_{\Gamma/H}(\nu, z^2) := \mathcal{M}^{\mathcal{R}}_{\Gamma/H}(z, P_3). \tag{49.309}$$

When $\mathfrak{M}^{\mathcal{R}}_{\Gamma/H}(0, z^2) \neq 0$, the reduced Ioffe-time coordinate

$$\mathfrak{M}_{\mathrm{red}}(\nu, z^2) := \frac{\mathfrak{M}^{\mathcal{R}}_{\Gamma/H}(\nu, z^2)}{\mathfrak{M}^{\mathcal{R}}_{\Gamma/H}(0, z^2)}$$

cancels any multiplicative Wilson-line factor depending only on z^2 in that scheme. It does not remove matching coefficients, target-mass effects, or higher-twist matrix elements.

Controlled Approximation 49.73 (Large-momentum matching statement)**.** Fix a renormalized spatial-bilocal scheme $\mathcal{R}$, a light-ray factorization scheme at scale μ, and a test function $h \in C_c^\infty(\mathbb{R})$. A large-momentum matching statement for (49.308) consists of kernels $C^{\mathcal{R}}_{\Gamma a}$, with a running over the light-ray operator basis, such that

$$\begin{aligned}\left\langle \widetilde{f}^{\mathcal{R}}_{\Gamma/H}(\,\cdot\,, P_3), h \right\rangle = \sum_a \int_0^1 \mathrm{d}y \, f_{a/H}(y, \mu) \int_{\mathbb{R}} \mathrm{d}x \, h(x) \, \frac{1}{y} C^{\mathcal{R}}_{\Gamma a}\left(\frac{x}{y}, \frac{\mu}{P_3}\right) \\ + R^{\mathcal{R}}_{\Gamma,H}[h; P_3, \mu].\end{aligned} \tag{49.310}$$

The statement is controlled only after the topology of the remainder has been specified. A typical leading-power version requires, for fixed compact support of h,

$$|R^{\mathcal{R}}_{\Gamma,H}[h; P_3, \mu]| \le C_h \left(\frac{M_H^2}{P_3^2} + \frac{\Lambda_{\mathrm{had}}^2}{P_3^2} \right) + R_{\mathrm{reg}}[h; a, L, P_3], \tag{49.311}$$

where M_H is the hadron mass, $\Lambda_{\rm had}$ is a hadronic scale attached to higher-twist matrix elements, and $R_{\rm reg}$ records finite lattice-spacing, finite-volume, and finite-operator-length errors before the continuum and infinite-volume limits are taken. Endpoint regions, small x, and threshold $x \to 1$ require additional test-function restrictions or separate factorization data.

The kernels in Controlled Approximation 49.73 are short-distance coordinates. In perturbation theory they are determined by matching matrix elements of the spatial bilocal and the light-ray operator in the same infrared state, so the infrared dependence cancels in the difference. That perturbative statement is not, by itself, a nonperturbative theorem about the large-P_3 limit of QCD. A theorem-level version would need a renormalized construction of the spatial Wilson-line operator, a spacelike short-distance OPE with remainder estimates uniform after hadron matrix elements, and a proof that the large-momentum limit converts the spacelike operator coordinates to the light-ray coordinates in the stated distributional topology.

Scheme covariance of finite matching kernels. Let f denote the column vector of light-ray PDFs in a convolution algebra, and suppose a quasi-PDF coordinate satisfies $\widetilde{f} = C \otimes f$ at leading power. If a finite light-ray scheme change is $f' = R \otimes f$, where R has a convolution inverse, then the same quasi-PDF coordinate is written in the primed scheme as

$$\widetilde{f} = C' \otimes f', \qquad C' = C \otimes R^{-1}.$$

If a linear charge functional $N(f) = \sum_a n_a \int_0^1 f_a(y)\,\mathrm{d}y$ is preserved by R and the matching kernel satisfies

$$\sum_i m_i \int_{\mathbb{R}} C_{ia}(\xi)\,\mathrm{d}\xi = n_a$$

for the corresponding quasi-PDF charge weights m_i, then the matched quasi-PDF has the same charge N at leading power.

Associativity of convolution gives

$$(C \otimes R^{-1}) \otimes (R \otimes f) = C \otimes (R^{-1} \otimes R) \otimes f = C \otimes f.$$

For the charge statement, integrate the matching formula over x with weights m_i. The change of variables $x = y\xi$, $y > 0$, gives

$$\sum_i m_i \int_{\mathbb{R}} \mathrm{d}x\, \frac{1}{y} C_{ia}(x/y) = \sum_i m_i \int_{\mathbb{R}} \mathrm{d}\xi\, C_{ia}(\xi) = n_a.$$

The remaining y-integral is exactly $N(f)$. The assumed preservation of N by R says that the same value is obtained in the primed light-ray coordinates.

49.18.3 DIS Factorization and DGLAP Evolution

Figure 49.10 records the operator factorization used for inclusive DIS: the short-distance current product is matched to renormalized light-ray operators whose convolutional renormalization gives DGLAP evolution.

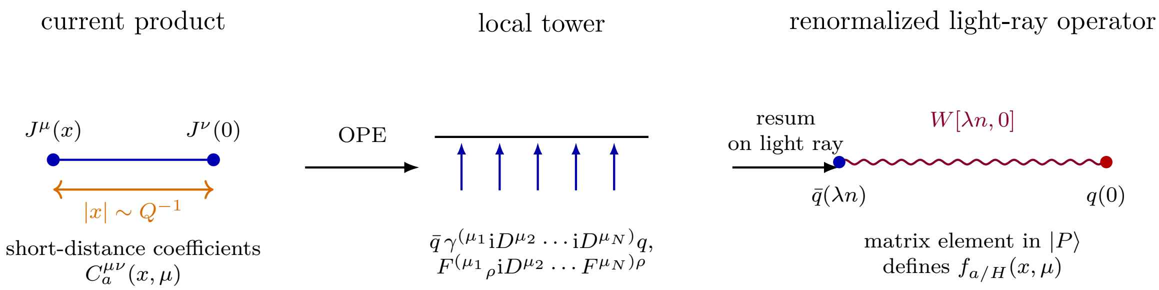


Figure 49.10: The short-distance current product is first expanded into local renormalized operators. The leading-twist tower can then be packaged into gauge-invariant light-ray operators whose hadron matrix elements define parton distributions. The Wilson segment $W[\lambda n, 0]$ is the open transporter connecting the colored fields in the bilocal operator.

The factorization assertion has two logically separate parts. The first is the exact definition of the hadronic tensor as the current-current Wightman matrix element (49.287). The second is the leading-power short-distance statement that, after the current OPE and light-ray renormalization have been performed, the same observable can be coordinatized by universal hadronic light-ray matrix elements and perturbative coefficient functions.

Common dependency ladder for QCD factorization. The local definitions above and the boundary sections below are used through a single comparison architecture. For a process family $\mathfrak{P}$, a theorem-level factorization statement has to provide:

1. the exact color-singlet observable $W_{\mathfrak{P}}$, usually a Wightman current tensor or a finite-regulator Euclidean matrix element;
2. the gauge-invariant operator coordinates, including Wilson-line geometry and the ultraviolet, cusp, endpoint, or rapidity subtractions required by that geometry;
3. the short-distance coefficient, impact-factor, or matching map in a stated distribution topology;
4. the scale-transport equations and finite-scheme covariance making the auxiliary factorization coordinates cancel in the paired observable;
5. the leading-region or boundary mechanism, including threshold, small-x, soft, rapidity, Y-term, or Glauber items whenever the process probes them;
6. a power-remainder estimate uniform on the declared test-function window;
7. the continuum, infinite-volume, large-momentum, or rapidity-regulator limit when the preceding objects are first constructed at finite regulator.

Equivalently, after pairing with a test function φ in the declared topology, the comparison between the exact observable and the factorized coordinate must be bounded by a sum of named errors,

$$
\begin{aligned}
&|\langle W_{\mathfrak{P},Q} - \mathcal{F}_{\mathfrak{P},Q}, \varphi\rangle| \le \mathcal{E}^{\mathrm{fact}}_{\mathfrak{P},Q}(\varphi),\\
&\mathcal{E}^{\mathrm{fact}}_{\mathfrak{P},Q} := \epsilon^{\mathrm{obs}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{op}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{sub}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{coeff}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{evol}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{bdry}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{rem}}_{\mathfrak{P},Q} + \epsilon^{\mathrm{lim}}_{\mathfrak{P},Q}.
\end{aligned}
\tag{49.312}
$$

Here $\mathcal{F}_{\mathfrak{P},Q}$ denotes the claimed factorized coordinate. For compact-x inclusive DIS, the boundary term records the exclusion of threshold and small-x regions. For Drell–Yan and other TMD observables it contains the rapidity subtraction, the Y-term, and the Glauber status. For small-x observables it contains the Wilson-line rapidity state, the projective generator limit, and the impact-factor map to a measured process. The finite calculations retained in this chapter check individual entries in this ladder; they are not substitutes for a proof that every required entry can be made small on a common directed window.

Hypothesis 49.74 (Leading-twist DIS factorization hypothesis)**.** Fix a factorization scheme and a compact interval $K \Subset (0,1)$ of Bjorken x. A leading-twist factorization hypothesis for an inclusive structure function F_i consists of coefficient distributions $C_{i,a}$, renormalized integrated PDFs $f_{a/H}$, and constants $C_{K,\varphi}$, Λ_{HT}, and $p > 0$, such that for every test function $\varphi \in C_c^\infty(K)$ the difference between the exact structure function and the factorized expression obeys

$$\left| \left\langle F_i - \sum_a C_{i,a} \otimes f_{a/H}, \varphi \right\rangle \right| \leq C_{K,\varphi} \left(\frac{\Lambda_{\mathrm{HT}}}{Q} \right)^p$$

for all sufficiently large Q. The convolution is the one displayed in (49.313). The compact-x restriction excludes the separate threshold $x \to 1$ and small-x limits, whose large logarithms require additional factorization data.

Under this hypothesis, factorization expresses a structure function F_i in the form

$$F_i(x_B, Q) = \sum_a \int_{x_B}^1 \frac{\mathrm{d}z}{z}\, C_{i,a}\bigg(\frac{x_B}{z}, \frac{Q}{\mu}, g(\mu) \bigg)\, f_{a/H}(z,\mu) + R_i(x_B, Q), \tag{49.313}$$

where a runs over quark, antiquark, and gluon parton species, and the remainder R_i satisfies the distributional power bound in Hypothesis 49.74. The coefficients $C_{i,a}$ are short-distance kernels; the distributions $f_{a/H}$ are universal hadronic matrix elements for the chosen factorization scheme. This paragraph is not a statement about colored asymptotic particles. The “parton” label a is a label in the light-ray operator basis and in the coefficient algebra.

The role of the parton distributions in (49.313) is also the point at which the nonabelian theory departs from the abelian Bloch–Nordsieck construction of Remark 44.8. A quark or gluon entering a short-distance hard subprocess is a colored partonic variable, not a physical asymptotic state. Its collinear gluon cloud is tied to a color generator and to the Wilson line in (49.294). Summing over unresolved final-state gluons gives the final-state part of an infrared-collinear safe observable, but the singularities associated with a massless incoming colored parton are absorbed into the renormalized light-ray operator defining $f_{a/H}$. Equivalently, the collinear subtraction that renormalizes the PDF is part of the definition of the factorized hadronic observable. The residual μ-dependence of this subtraction is the DGLAP evolution below.

The operator-coordinate RG equation (49.300) gives the evolution equation

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu} f_{a/H}(x,\mu) = \sum_b \int_x^1 \frac{\mathrm{d}z}{z}\, P_{ab}\Big(\frac{x}{z}, g(\mu) \Big)\, f_{b/H}(z,\mu), \tag{49.314}$$

where P_{ab} are splitting kernels, equivalently the anomalous-dimension kernels of the renormalized light-ray operator coordinates in the sign convention of (49.300). This is the precise sense in which short-distance quark and gluon variables control high-energy hadronic observables while physical asymptotic states remain color singlets. The factorized observable is independent of the auxiliary scale only when the coefficient functions obey the dual equation

$$\mu\partial_\mu C_{i,b} = -\sum_a C_{i,a} \otimes P_{ab}. \tag{49.315}$$

Then

$$\mu\partial_\mu \sum_a C_{i,a} \otimes f_{a/H} = \sum_b \left(\mu\partial_\mu C_{i,b} + \sum_a C_{i,a} \otimes P_{ab} \right) \otimes f_{b/H} = 0$$

inside the declared convolution algebra. Any residual scale dependence in a finite-order calculation is therefore a truncation or power-remainder effect, not a change in the nonperturbative definition of the PDF.

49.18.4 One-Loop Splitting Kernels and Sum-Rule Checks

At leading order, in the same convention as (49.314),

$$P_{ab}(x, g) = \frac{g^2}{8\pi^2} P_{ab}^{(0)}(x) + O(g^4).$$

For unpolarized distributions in the $\overline{\mathrm{MS}}$ scheme,

$$C_F = \frac{N_c^2 - 1}{N_c}, \qquad C_A = 2N_c, \qquad T_R = 1,$$

in the trace normalization of (49.1). These are twice the standard half-trace numbers, while the overall perturbative coefficient in (49.314) is written in the corresponding coupling coordinate. Let

$$D_0(x) := \left(\frac{1}{1-x} \right)_+ \tag{49.316}$$

be the plus distribution defined by

$$\int_0^1 \mathrm{d}x\, D_0(x)\, \varphi(x) = \int_0^1 \mathrm{d}x\, \frac{\varphi(x) - \varphi(1)}{1-x}$$

for every smooth test function φ. Products such as $xD_0(x)$ are defined by multiplication of this distribution by the smooth function x. With this unambiguous convention, the nonzero leading kernels are

$$\begin{aligned}
P_{q_i q_j}^{(0)}(x) &= \delta_{ij}\, C_F \left[2D_0(x) - 1 - x + \frac{3}{2}\delta(1-x) \right], \\
P_{\bar{q}_i \bar{q}_j}^{(0)}(x) &= P_{q_i q_j}^{(0)}(x), \\
P_{q_i g}^{(0)}(x) &= P_{\bar{q}_i g}^{(0)}(x) = T_R \big[x^2 + (1-x)^2 \big], \\
P_{g q_i}^{(0)}(x) &= P_{g \bar{q}_i}^{(0)}(x) = C_F\, \frac{1 + (1-x)^2}{x}, \\
P_{gg}^{(0)}(x) &= 2C_A \left[xD_0(x) + \frac{1-x}{x} + x(1-x) \right] + \frac{11C_A - 4T_R N_f}{6} \delta(1-x).
\end{aligned} \tag{49.317}$$

This notation avoids the common ambiguity between plus-prescribing the whole rational function and plus-prescribing only the pole at $x = 1$. For example, $2D_0 - 1 - x$ is the distribution often abbreviated as $(1 + x^2)/(1 - x)_+$ in particle-physics notation. Equivalently, for the flavor singlet quark distribution $\Sigma = \sum_{i=1}^{N_f}(q_i + \bar{q}_i)$,

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\begin{pmatrix}\Sigma\\ f_g\end{pmatrix} = \frac{g^2}{8\pi^2}\begin{pmatrix}P_{qq}^{(0)} & 2N_f P_{qg}^{(0)}\\ P_{gq}^{(0)} & P_{gg}^{(0)}\end{pmatrix} \otimes \begin{pmatrix}\Sigma\\ f_g\end{pmatrix} + O(g^4),$$

where $\otimes$ denotes the convolution appearing in (49.314). Nonsinglet quark combinations evolve with the same $P_{qq}^{(0)}$ and do not mix with the gluon at this order.

The kernels in (49.317) obey the following one-loop distributional sum-rule checks:

$$\int_0^1 \mathrm{d}x\, P_{qq}^{(0)}(x) = 0,$$

and the two momentum-column identities

$$\int_0^1 \mathrm{d}x\, x\big(P_{qq}^{(0)}(x) + P_{gq}^{(0)}(x)\big) = 0, \qquad \int_0^1 \mathrm{d}x\, x\big(2N_f P_{qg}^{(0)}(x) + P_{gg}^{(0)}(x)\big) = 0.$$

Thus nonsinglet quark number and the total light-cone momentum fraction are preserved by the leading-order evolution. For $m \geq 0$,

$$\int_0^1 \mathrm{d}x\, x^m D_0(x) = \int_0^1 \mathrm{d}x\, \frac{x^m - 1}{1 - x} = -H_m, \qquad H_m := \sum_{r=1}^{m} \frac{1}{r},$$

with $H_0 = 0$. The number sum rule is

$$C_F\left(2H_0 - \int_0^1 (1 + x)\,\mathrm{d}x + \frac{3}{2}\right) = 0.$$

For a quark parent,

$$\int_0^1 \mathrm{d}x\, xP_{qq}^{(0)}(x) = C_F\left(-2 - \frac{1}{2} - \frac{1}{3} + \frac{3}{2}\right) = -\frac{4}{3}C_F,$$

while

$$\int_0^1 \mathrm{d}x\, xP_{gq}^{(0)}(x) = C_F\int_0^1 \mathrm{d}x\,\left[1 + (1 - x)^2\right] = \frac{4}{3}C_F.$$

For a gluon parent,

$$\int_0^1 \mathrm{d}x\, xP_{qg}^{(0)}(x) = T_R\int_0^1 \mathrm{d}x\, x\big[x^2 + (1 - x)^2\big] = \frac{1}{3}T_R,$$

and

$$\int_0^1 \mathrm{d}x\, xP_{gg}^{(0)}(x) = 2C_A\left(-H_2 + \frac{1}{2} + \frac{1}{12}\right) + \frac{11C_A - 4T_R N_f}{6} = -\frac{2}{3}T_R N_f.$$

The two column sums therefore vanish.

49.19 Transverse-Momentum Dependent Distributions and Rapidity Renormalization

Integrated PDFs are obtained after summing over all transverse momentum in the collinear sector. Observables differential in a transverse momentum, for example color-singlet production with $q_\perp^2 \ll Q^2$, require a different operator coordinate. The relevant nonperturbative input is not the integrated light-ray operator (49.293); it is a transverse-separated light-ray matrix element with Wilson lines whose geometry creates rapidity divergences. Rapidity subtraction is therefore part of the definition, in the same sense that ultraviolet subtraction is part of the definition of a renormalized local operator.

Let $n, \bar{n}$ be future-directed lightlike vectors with $n \cdot \bar{n} = 2$, and let $b_\perp$ be transverse:

$$b_\perp \cdot n = b_\perp \cdot \bar{n} = 0.$$

For a real parameter λ, set

$$z^\mu = \lambda n^\mu + b_\perp^\mu.$$

A finite rapidity regulator may be implemented by replacing lightlike Wilson lines by Wilson lines along non-lightlike directions $v, \bar{v}$, by an analytic rapidity regulator, by a delta regulator, or by an equivalent gauge-covariant subtraction. The monograph denotes the regulator data collectively by ρ. At finite ρ and finite ultraviolet cutoff Λ, let $\mathcal{U}_v^{[\eta]}(z, 0; \rho, \Lambda)$ be a staple transporter from 0 to z, with longitudinal direction v, transverse bridge $b_\perp$, and orientation $\eta = +1$ for future-pointing and $\eta = -1$ for past-pointing staples.

Definition 49.75 (TMD operator definition)**.** A quark TMD operator definition for a hadron H with momentum P consists of:

1. the vectors $n, \bar{n}$, a transverse separation $b_\perp$, and an orientation η of the staple Wilson line;
2. a finite rapidity regulator ρ and ultraviolet regulator Λ, both chosen in a gauge-covariant way;
3. the unsubtracted beam matrix element
$$B_{q/H}^{\mathrm{unsub}}(x, b_\perp; \rho, \Lambda, \eta) = \frac{1}{2P \cdot n} \int \frac{\mathrm{d}\lambda}{2\pi}\, \mathrm{e}^{-\mathrm{i}x\lambda P\cdot n} \Big\langle \bar{q}(z)\, \mathcal{U}_v^{[\eta]}(z, 0; \rho, \Lambda)\, \gamma \cdot n\, q(0) \Big\rangle_{P,H};$$
4. the vacuum soft factor $S(b_\perp; \rho, \Lambda, \eta)$, defined by the trace of the closed Wilson geometry obtained by multiplying the corresponding v- and $\bar{v}$-staples in the vacuum;
5. a finite-part prescription, denoted $\mathrm{FP}_{\rho,\Lambda}$, which removes ultraviolet and rapidity regulator dependence and introduces the scales μ and ζ.

The renormalized TMD in this scheme is

$$F_{q/H}^{[\eta]}(x, b_\perp; \mu, \zeta) = \mathrm{FP}_{\rho,\Lambda} \left[Z_q(\mu, \zeta; \rho, \Lambda)\, \frac{B_{q/H}^{\mathrm{unsub}}(x, b_\perp; \rho, \Lambda, \eta)}{S(b_\perp; \rho, \Lambda, \eta)^{1/2}} \right]. \tag{49.318}$$

The square root is a convention for assigning half of the soft subtraction to each collinear sector. A different finite assignment is a different TMD scheme, related by a finite multiplicative transformation. The variables μ and ζ record, respectively, ultraviolet and rapidity renormalization. The orientation η is part of the operator definition; time-reversal odd TMDs may change sign between future- and past-pointing staples, as in semi-inclusive DIS versus Drell–Yan kinematics.

At finite regulator, the numerator and soft factor in (49.318) are gauge invariant after taking the color trace and endpoint fields into account. Hence a background gauge-covariant finite-part prescription produces a gauge-invariant renormalized TMD matrix element. Every Wilson segment transforms by multiplication with the gauge transformation at its endpoints, as in (49.292). Along a staple, the factors at internal junctions cancel between adjacent segments. The remaining endpoint factors at 0 and z cancel the gauge transformations of $q(0)$ and $\bar{q}(z)$. In the soft factor all Wilson lines are multiplied inside a trace, so the residual base-point factor is removed by cyclicity of the trace. The finite-part map changes only regulator and scale coordinates; if it is background gauge covariant, it cannot introduce an uncanceled color index.

Rapidity evolution is the statement that changing the auxiliary rapidity scale changes the coordinate description of the same physical transverse momentum distribution. It is most transparent in impact-parameter space, where convolutions in transverse momentum become products. Put

$$\ell_\mu := \log \mu, \qquad \ell_\zeta := \log \sqrt{\zeta}.$$

For a fixed x-test function and fixed $b_\perp$, suppose the paired TMD matrix element is nonzero and obeys

$$\frac{\partial}{\partial \ell_\zeta} \log F^{[\eta]}_{q/H}(x, b_\perp; \mu, \zeta) = \mathcal{D}_q(b_\perp; \mu), \tag{49.319}$$

$$\frac{\partial}{\partial \ell_\mu} \log F^{[\eta]}_{q/H}(x, b_\perp; \mu, \zeta) = \gamma^q_F(g(\mu), \ell_\zeta - \ell_\mu). \tag{49.320}$$

The first equation defines the Collins–Soper kernel $\mathcal{D}_q$ in the chosen convention.

Collins–Soper consistency condition. The two scale variables ℓ_μ and ℓ_ζ are coordinates on the renormalized TMD scheme, so a differentiable TMD whose mixed scale derivatives commute obeys the corresponding integrability condition. Explicitly, if the derivatives in (49.319)–(49.320) commute on the renormalized TMD distribution, then

$$\frac{\partial}{\partial \ell_\mu} \mathcal{D}_q(b_\perp; \mu) = \frac{\partial}{\partial \ell_\zeta} \gamma^q_F(g(\mu), \ell_\zeta - \ell_\mu). \tag{49.321}$$

In the convention

$$\gamma^q_F(g, \ell_\zeta - \ell_\mu) = \gamma^{q,0}_F(g) - 2\Gamma^q_{\mathrm{cusp}}(g)(\ell_\zeta - \ell_\mu),$$

this gives

$$\frac{\partial}{\partial \log \mu} \mathcal{D}_q(b_\perp; \mu) = -2\Gamma^q_{\mathrm{cusp}}(g(\mu)). \tag{49.322}$$

At small $b_\perp$, the one-loop kernel compatible with the cusp normalization of (49.369) is

$$\mathcal{D}_q(b_\perp; \mu) = -\frac{g^2(\mu) C_F}{4\pi^2} \log \frac{\mu^2 b_\perp^2 \mathrm{e}^{2\gamma_{\mathrm{E}}}}{4} + O(g^4), \tag{49.323}$$

up to a finite $b_\perp$-dependent scheme term independent of μ. The last display is a normalization check on the one-loop result: $\partial_{\log \mu} \log(\mu^2 b_\perp^2 \mathrm{e}^{2\gamma_{\mathrm{E}}}/4) = 2$, so differentiating (49.323) gives $-g^2 C_F/(2\pi^2) + O(g^4)$, which is $-2\Gamma^q_{\mathrm{cusp}}(g) + O(g^4)$ with $\Gamma^q_{\mathrm{cusp}}(g) = g^2 C_F/(4\pi^2) + O(g^4)$ in the trace-delta normalization. The substantive QCD input is the regulated Wilson-line operator definition, the existence and scheme dependence of rapidity renormalization, and the perturbative or nonperturbative determination of Γ^q_{cusp} and $\mathcal{D}_q$; the displayed consistency relation only records the integrability of the chosen scale coordinates.

Remark 49.76 (Finite TMD scheme changes). Let $R(b_\perp;\mu,\zeta)$ be a finite scalar function and set $F' = \mathrm{e}^R F$. Then

$$\mathcal{D}' = \mathcal{D} + \frac{\partial R}{\partial \ell_\zeta}, \qquad \gamma'_F = \gamma_F + \frac{\partial R}{\partial \ell_\mu}.$$

The consistency relation (49.321) is preserved. Therefore the separation of a rapidity-sensitive cross section into hard, TMD, and soft coordinates is scheme dependent, whereas the final cross section is not.

Take logarithms of $F' = \mathrm{e}^R F$ and differentiate with respect to ℓ_ζ and ℓ_μ. The displayed transformations follow. Taking ∂_{ℓ_μ} of the first transformation and ∂_{ℓ_ζ} of the second gives the same extra term $\partial_{\ell_\mu}\partial_{\ell_\zeta} R$. Hence the difference between the two sides of (49.321) is unchanged.

Hypothesis 49.77 (Small-$q_\perp$ color-singlet TMD factorization hypothesis)**.** For a color-singlet final state of invariant mass Q produced in hadron–hadron scattering, a leading-power small-$q_\perp$ TMD factorization hypothesis consists of TMD schemes for the two incoming hadrons, a hard function $H_{ij}(Q,\mu)$, rapidity scales satisfying $\zeta_A\zeta_B = Q^4$, a Y-term that matches the region $|q_\perp| \sim Q$, a statement about Glauber exchange, and a distributional remainder estimate. In the region $\Lambda_{\mathrm{QCD}} \lesssim |q_\perp| \ll Q$ the singular part has the impact-parameter form

$$\frac{\mathrm{d}\sigma_{\mathrm{sing}}}{\mathrm{d}Q^2\mathrm{d}^2q_\perp} = \sum_{i,j} H_{ij}(Q,\mu) \int \frac{\mathrm{d}^2 b_\perp}{(2\pi)^2}\, \mathrm{e}^{\mathrm{i}b_\perp\cdot q_\perp} F_{i/A}(x_A, b_\perp;\mu,\zeta_A) F_{j/B}(x_B, b_\perp;\mu,\zeta_B). \tag{49.324}$$

The formula is a theorem only when the regulator construction, cancellation or operator representation of Glauber exchange, and the topology of the power-remainder estimate have all been supplied.

49.20 Drell–Yan Factorization and Glauber Status

Drell–Yan production is the timelike current analogue of DIS:

$$h_A(P_A) + h_B(P_B) \longrightarrow \ell^+\ell^- + X, \qquad q = p_{\ell^+} + p_{\ell^-}, \qquad q^2 = Q^2 > 0.$$

The QCD object is again a Wightman current matrix element, now between a two-hadron incoming state. For unpolarized hadrons define

$$W^{\mu\nu}_{AB}(P_A,P_B,q) = \frac{1}{N_{\mathrm{spin}}} \sum_{\sigma_A,\sigma_B} \sum_X (2\pi)^4 \delta^{(4)}(P_A+P_B-q-P_X) \langle P_A,P_B|J^\mu(0)|X\rangle\langle X|J^\nu(0)|P_A,P_B\rangle. \tag{49.325}$$

Equivalently, by translation covariance and completeness,

$$W^{\mu\nu}_{AB}(P_A,P_B,q) = \frac{1}{N_{\mathrm{spin}}} \sum_{\sigma_A,\sigma_B} \int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}q\cdot x} \langle P_A,P_B|J^\mu(x)J^\nu(0)|P_A,P_B\rangle, \tag{49.326}$$

with the same state-normalization convention as in the scattering chapters. The partonic formulae below are short-distance coordinates for this tensor; they are not definitions of a cross section with colored asymptotic states.

Drell–Yan leading-power kinematics. Let

$$s := (P_A + P_B)^2, \qquad \tau := \frac{Q^2}{s},$$

and choose the hadron center-of-mass frame with the A-hadron moving in the positive longitudinal direction. At leading power in hadron masses and transverse momentum, the longitudinal momentum fractions entering the short-distance hard collision are

$$x_A = \sqrt{\tau}\, \mathrm{e}^{y}, \qquad x_B = \sqrt{\tau}\, \mathrm{e}^{-y}, \tag{49.327}$$

where

$$y := \frac{1}{2} \log \frac{q^0 + q^3}{q^0 - q^3}$$

is the lepton-pair rapidity. Hence

$$x_A x_B = \tau, \qquad y = \frac{1}{2} \log \frac{x_A}{x_B}.$$

In light-cone coordinates adapted to the beam directions, write the leading hadron momenta as

$$P_A^+ = \sqrt{s}, \quad P_A^- = 0, \qquad P_B^- = \sqrt{s}, \quad P_B^+ = 0,$$

and neglect transverse components in the leading-power kinematic identity. The active partons carry

$$q^+ = x_A P_A^+, \qquad q^- = x_B P_B^-.$$

Therefore $Q^2 = q^+ q^- + O(q_\perp^2, M_{A,B}^2) = x_A x_B s + O(q_\perp^2, M_{A,B}^2)$, which gives $x_A x_B = \tau$ at leading power. The rapidity satisfies

$$y = \frac{1}{2} \log \frac{q^+}{q^-} = \frac{1}{2} \log \frac{x_A}{x_B}.$$

Solving the two equations gives (49.327).

Integrated collinear factorization coordinate. If the lepton-pair transverse momentum is not kept differential, the appropriate leading-power coordinate is the integrated collinear factorization formula, not the TMD formula below. Fix a compact rapidity and invariant-mass window in which $x_A, x_B \in K \Subset (0,1)$ and the threshold and small-x boundaries are excluded. A leading-power integrated Drell–Yan factorization hypothesis consists of renormalized integrated PDF coordinates $f_{i/A}, f_{j/B}$, a short-distance coefficient distribution C_{ij}^{DY}, and a distributional remainder bound such that

$$\frac{\mathrm{d}\sigma_{\mathrm{DY}}}{\mathrm{d}Q^2 \mathrm{d}y} = \sum_{i,j} \int_{x_A}^1 \frac{\mathrm{d}\xi_A}{\xi_A} \int_{x_B}^1 \frac{\mathrm{d}\xi_B}{\xi_B}\, f_{i/A}(\xi_A, \mu)\, f_{j/B}(\xi_B, \mu)\, C_{ij}^{\mathrm{DY}}\left(\frac{x_A}{\xi_A}, \frac{x_B}{\xi_B}; Q, \mu\right) + R_{\mathrm{DY}}. \tag{49.328}$$

The PDFs in (49.328) are the Wilson-line light-ray matrix elements of Definition 49.72. The coefficient C_{ij}^{DY} is a hard matching distribution in the same factorization scheme; it includes the chosen flux, electroweak-current, and Jacobian normalization. The remainder R_{DY} contains the declared power corrections, threshold and small-x exclusions, perturbative truncation remainder, and the Glauber item discussed below. Thus the formula is not an identity between colored partonic cross sections. It is a comparison between the exact color-singlet Wightman tensor (49.326) and a factorized coordinate built from two incoming light-ray operator matrix elements.

The scale cancellation is the two-incoming-leg analogue of (49.315). In a Mellin or Laplace chart where the two convolutions become products, let $f_A(N_A, \mu)$ and $f_B(N_B, \mu)$ be column vectors in parton species, and write the factorized coordinate as

$$\widetilde{\Sigma}_{\mathrm{DY}}(N_A, N_B; \mu) = f_A(N_A, \mu)^T \widetilde{C}_{\mathrm{DY}}(N_A, N_B; Q, \mu) f_B(N_B, \mu). \tag{49.329}$$

If

$$\mu \frac{\mathrm{d} f_A}{\mathrm{d}\mu} = P_A f_A, \qquad \mu \frac{\mathrm{d} f_B}{\mathrm{d}\mu} = P_B f_B,$$

then auxiliary-scale independence of (49.329) requires

$$\mu \frac{\mathrm{d}\widetilde{C}_{\mathrm{DY}}}{\mathrm{d}\mu} = -P_A^T \widetilde{C}_{\mathrm{DY}} - \widetilde{C}_{\mathrm{DY}} P_B. \tag{49.330}$$

Indeed,

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu}\big(f_A^T \widetilde{C}_{\mathrm{DY}} f_B\big) = f_A^T \big(P_A^T \widetilde{C}_{\mathrm{DY}} + \mu\partial_\mu \widetilde{C}_{\mathrm{DY}} + \widetilde{C}_{\mathrm{DY}} P_B\big) f_B,$$

which vanishes by (49.330). A finite factorization-scheme change $f'_A = R_A f_A$, $f'_B = R_B f_B$ preserves the same physical coordinate only if

$$\widetilde{C}'_{\mathrm{DY}} = (R_A^{-1})^T \widetilde{C}_{\mathrm{DY}} R_B^{-1}.$$

This matrix identity is the clean bookkeeping behind statements that the two incoming collinear singularities have been absorbed into the PDFs. A factorization proof must still supply the leading-region analysis, soft cancellation or soft factor, Glauber status, and the remainder topology; the algebra above only fixes the paired RG structure once those objects exist.

Controlled Approximation 49.78 (Integrated Drell–Yan theorem boundary)**.** The exact object in integrated Drell–Yan is a tested distribution obtained from the color-singlet Wightman tensor (49.326), not the partonic coordinate (49.328). Fix a compact (Q^2, y)-window K, away from threshold and small-x boundaries, and let $\mathcal{D}_K$ be the corresponding test-function space. A leading-power integrated factorization theorem for this window must supply:

1. a finite-regulator definition of the exact tested functional $\mathfrak{T}_{\mathrm{DY}}[\varphi]$, $\varphi \in \mathcal{D}_K$, from (49.326);
2. the renormalized light-ray PDF coordinates and the hard coefficient in the same subtraction scheme;
3. a leading-region map from the exact current product to the two incoming collinear sectors, with its error measured in $\mathcal{D}_K^*$;
4. a soft-cancellation or soft-factor construction in the same regulator;
5. a Glauber statement strong enough for the chosen measurement topology, such as a commutator or inclusive-projection bound of the form described below;
6. a power-remainder topology, including hadron-mass, threshold, small-x, and perturbative-truncation qualifications;
7. a regulator-limit statement identifying which residuals vanish and which remain as declared approximation errors.

Thus the formula (49.328) becomes a theorem only after the residuals in this list are bounded for the exact tested Wightman functional. The paired RG equation (49.330) is one coordinate-transport check inside this boundary; it is not the boundary itself.

Proposition 49.79 (Tested Drell–Yan factorization residual budget)**.** *In the setting of Controlled Approximation 49.78, suppose the subtraction analysis represents the exact tested functional as*

$$\mathfrak{T}_{\rm DY}[\varphi] = f_A^T C_\varphi f_B + \Delta_{\rm lead}[\varphi] + \Delta_{\rm soft}[\varphi] + \Delta_{\rm G}[\varphi] + \Delta_{\rm pwr}[\varphi] + \Delta_{\rm reg}[\varphi], \tag{49.331}$$

where $f_A^T C_\varphi f_B$ is the tested version of (49.328). Assume that, on a window $Q \geq Q_0$,

$$|\Delta_a[\varphi]| \leq \epsilon_a(Q_0)\, \|\varphi\|_{\mathcal{D}_K}, \qquad a \in \{\mathrm{lead}, \mathrm{soft}, \mathrm{G}, \mathrm{pwr}, \mathrm{reg}\}. \tag{49.332}$$

Then

$$\left|\mathfrak{T}_{\rm DY}[\varphi] - f_A^T C_\varphi f_B\right| \leq (\epsilon_{\rm lead} + \epsilon_{\rm soft} + \epsilon_{\rm G} + \epsilon_{\rm pwr} + \epsilon_{\rm reg})\, \|\varphi\|_{\mathcal{D}_K}. \tag{49.333}$$

If the paired PDF/coefficient transport (49.330) is exact, then $f_A^T C_\varphi f_B$ is independent of the auxiliary factorization scale, but the physical mismatch in (49.333) is controlled only by the five residual norms above. Scale covariance of the factorized coordinate does not make any nonzero residual disappear.

Proof. Subtract the factorized coordinate from (49.331) and apply the triangle inequality together with (49.332). The final statement is the scale derivative calculation already displayed in (49.330): the derivative of $f_A^T C_\varphi f_B$ cancels between the two PDF legs and the dual hard coefficient. This cancellation concerns only the retained coordinate. The exact Wightman functional can agree with it in the declared topology only when the listed residual functionals are separately bounded, or vanish in the regulator and large-Q limits being claimed. □

Hypothesis 49.80 (Drell–Yan small-$q_\perp$ TMD factorization hypothesis)**.** A leading-power Drell–Yan TMD factorization hypothesis consists of:

1. the renormalized current hard function $H_{ij}(Q,\mu)$;
2. past-pointing TMD operator data $F^{[-]}_{i/A}(x_A, b_\perp; \mu, \zeta_A)$ and $F^{[-]}_{j/B}(x_B, b_\perp; \mu, \zeta_B)$, in the notation of Definition 49.75;
3. rapidity scales constrained by $\zeta_A \zeta_B = Q^4$;
4. a Y-term matching to the region $|q_\perp| \sim Q$;
5. a proof or controlled hypothesis for Glauber exchange;
6. a distributional power-remainder estimate in $q_\perp/Q$ and in the hadron-mass corrections.

The singular small-$q_\perp$ term is then represented as

$$\frac{\mathrm{d}\sigma_{\rm sing}}{\mathrm{d}Q^2 \mathrm{d}y\, \mathrm{d}^2 q_\perp} = \sum_{i,j} H_{ij}(Q,\mu) \int \frac{\mathrm{d}^2 b_\perp}{(2\pi)^2}\, \mathrm{e}^{\mathrm{i} b_\perp \cdot q_\perp} F^{[-]}_{i/A}(x_A, b_\perp; \mu, \zeta_A) F^{[-]}_{j/B}(x_B, b_\perp; \mu, \zeta_B), \tag{49.334}$$

plus the stated Y-term and power remainder. The past-pointing staple orientation is part of the operator definition. It is the operator-level source of the sign reversal of time-reversal-odd TMD coordinates between semi-inclusive DIS and Drell–Yan, once the relevant antiunitary transformation and spin conventions are specified.

Finite unitarity model for Glauber cancellation. Let $\mathcal{H}_{\rm obs}$ be a finite-dimensional observed Hilbert space, let $\mathcal{H}_{\rm un}$ be an unobserved Hilbert space, let ρ be a trace-class density matrix on $\mathcal{H}_{\rm obs} \otimes \mathcal{H}_{\rm un}$, and let M be an observable acting only on $\mathcal{H}_{\rm obs}$. If the effect of a Glauber exchange on the unobserved degrees of freedom is represented by a unitary U_G acting only on $\mathcal{H}_{\rm un}$, then

$$\mathrm{Tr}_{\rm obs,un}\left[(M \otimes \mathbf{1})(\mathbf{1} \otimes U_G)\rho(\mathbf{1} \otimes U_G^\dagger)\right] = \mathrm{Tr}_{\rm obs,un}\left[(M \otimes \mathbf{1})\rho\right]. \tag{49.335}$$

Use cyclicity of the full trace and $(\mathbf{1} \otimes U_G^\dagger)(M \otimes \mathbf{1})(\mathbf{1} \otimes U_G) = M \otimes \mathbf{1}$. The result is

$$\mathrm{Tr}\left[\rho(\mathbf{1} \otimes U_G^\dagger)(M \otimes \mathbf{1})(\mathbf{1} \otimes U_G)\right] = \mathrm{Tr}\left[\rho(M \otimes \mathbf{1})\right],$$

which is (49.335). Equivalently, the partial trace over the unobserved Hilbert space is invariant under unitary conjugation on that same unobserved factor. The finite model therefore isolates the algebraic input used in inclusive Glauber-cancellation arguments: the observed measurement must be insensitive to the unitary acting on the summed-over sector.

The same finite model also gives the residual which must be bounded when the measurement is not fully inclusive. For a general tested measurement M_φ on $\mathcal{H}_{\rm obs} \otimes \mathcal{H}_{\rm un}$, define

$$\Delta_{\rm G}^{\rm fin}[\varphi] = \mathrm{Tr}\left[\rho\big((\mathbf{1} \otimes U_G^\dagger)M_\varphi(\mathbf{1} \otimes U_G) - M_\varphi\big)\right]. \tag{49.336}$$

Then

$$\Delta_{\rm G}^{\rm fin}[\varphi] = 0 \quad \text{if} \quad [M_\varphi, \mathbf{1} \otimes U_G] = 0,$$

and in finite dimension

$$|\Delta_{\rm G}^{\rm fin}[\varphi]| \le \|\rho\|_1 \left\|(\mathbf{1} \otimes U_G^\dagger)M_\varphi(\mathbf{1} \otimes U_G) - M_\varphi\right\|_{\rm op}. \tag{49.337}$$

Thus the integrated Drell–Yan Glauber residual $\Delta_{\rm G}[\varphi]$ in (49.331) is not a ceremonial slot. It asks whether, after the leading-region and contour-deformation analysis has identified the regulated eikonal Glauber action, the tested color-singlet observable is inclusive enough for the commutator to vanish, or whether the remaining commutator is quantitatively small in the declared measurement topology.

Controlled Approximation 49.81 (What the finite Glauber calculation proves)**.** The finite unitarity calculation above is the algebraic identity that a Drell–Yan factorization proof must realize after the leading regions, color flow, rapidity regulator, and measurement topology have been fixed. It does not prove by itself that QCD Glauber exchanges act only on an unobserved tensor factor, nor that the contour deformations needed to isolate the leading regions are valid. Those statements are part of the Glauber item in Hypothesis 49.80. If the observable measures hadronic radiation in a way that resolves the spectator sector, or if a color exchange remains entangled with the measured hard sector, the finite identity above no longer supplies the required cancellation.

49.21 Off-Forward Light-Ray Matrix Elements and GPDs

Generalized parton distributions are light-ray matrix elements between different hadron momenta. They do not require a rapidity subtraction in the same way as TMDs, because the operator remains integrated over transverse momentum. Their new feature is off-forwardness.

Let p' and p be on-shell spin-$\frac{1}{2}$ hadron momenta, and define

$$P = \frac{p' + p}{2}, \qquad \Delta = p' - p, \qquad t = \Delta^2, \qquad \xi = -\frac{\Delta \cdot n}{2P \cdot n}.$$

For the symmetric light-ray separation

$$z^\mu = \lambda n^\mu,$$

define the quark vector GPDs H^q and E^q by

$$\begin{aligned} &\int \frac{\mathrm{d}\lambda}{2\pi}\, \mathrm{e}^{-\mathrm{i}x\lambda P\cdot n} \langle p', s'| \left[\bar{q}\left(\frac{\lambda n}{2}\right) W_\square\left(\frac{\lambda n}{2}, -\frac{\lambda n}{2}\right) \gamma \cdot n\, q\left(-\frac{\lambda n}{2}\right)\right]_\mu |p, s\rangle \\ &\quad = \bar{u}_{s'}(p') \left[\gamma \cdot n\, H^q(x, \xi, t; \mu) + \frac{\mathrm{i}\sigma^{n\alpha}\Delta_\alpha}{2M}\, E^q(x, \xi, t; \mu)\right] u_s(p), \end{aligned} \tag{49.338}$$

where $\sigma^{n\alpha} := n_\mu \sigma^{\mu\alpha}$, and M is the hadron mass. The support and endpoint behavior of the distributions are properties of the renormalized light-ray matrix elements and of the chosen hadronic state; they are not imposed by a parton model.

Lemma 49.82 (Polynomiality of GPD moments)**.** *Assume the renormalized twist-two local operators obtained from the light-ray operator in* (49.338) *exist as local operator-valued distributions between the hadron states under consideration. Then the N-th Mellin moments*

$$\int_{-1}^{1} \mathrm{d}x\, x^{N-1} H^q(x, \xi, t; \mu), \qquad \int_{-1}^{1} \mathrm{d}x\, x^{N-1} E^q(x, \xi, t; \mu)$$

are polynomials in ξ of degree at most N, with coefficients that are Lorentz-scalar form factors depending on t and μ.

Proof. Expanding the symmetric light-ray operator in λ gives the local twist-two tower

$$\bar{q}\, \gamma^{(\mu_1} \mathrm{i}D^{\mu_2} \cdots \mathrm{i}D^{\mu_N)} q - \text{traces},$$

contracted with $n_{\mu_1} \cdots n_{\mu_N}$. Since $n^2 = 0$, all trace terms vanish after contraction with these n's. Lorentz covariance of the off-forward matrix element gives a finite decomposition into symmetric tensors made from P, Δ, the metric, and the spinor bilinears $\bar{u}(p')\gamma^\alpha u(p)$ and $\bar{u}(p')\mathrm{i}\sigma^{\alpha\beta}\Delta_\beta u(p)$. After contraction with $n_{\mu_1} \cdots n_{\mu_N}$, each term contains

$$(P \cdot n)^{N-k} (\Delta \cdot n)^k = (P \cdot n)^N (-2\xi)^k$$

for some $0 \le k \le N$, multiplied by a scalar form factor depending only on t and μ. Dividing by the overall powers of $P \cdot n$ in the GPD decomposition therefore gives a polynomial in ξ of degree no larger than N. This argument uses only locality, Lorentz covariance, and the existence of the renormalized local twist-two tower; it does not use a probability interpretation. □

Deeply virtual Compton scattering and related exclusive processes use these GPDs in coefficient-function convolutions. As in inclusive DIS and TMD factorization, the GPD itself is the renormalized hadronic light-ray matrix element. The factorization theorem connecting it to a measured exclusive amplitude is an additional leading-power assertion with its own endpoint, Glauber, and power-remainder data.

49.22 Asymptotic Freedom and Exclusive Pion Amplitudes

49.22.1 Pion Light-Ray Data and Factorization Scheme

Inclusive DIS uses forward light-ray matrix elements. Exclusive pion amplitudes use light-ray matrix elements between the vacuum and a one-pion state. This is the part of the pion S-matrix where asymptotic freedom gives a direct large-momentum prediction: the short-distance subgraph is a perturbatively renormalized quark-gluon amplitude, while the pion appears only through a renormalized gauge-invariant light-ray matrix element. The statement is not that a pion is a weakly bound pair of on-shell quarks. The pion input is the same hadronic state defined by the nonperturbative QCD sector; only the coefficient multiplying its light-ray matrix element is perturbative.

Let P^μ be the momentum of a charged pion and let n^μ be a future-directed lightlike vector with $P \cdot n > 0$. In the isospin-symmetric two-flavor theory, define the charged-pion light-cone distribution amplitude by the renormalized bilocal matrix element

$$\langle \Omega | \left[\bar{d}(\lambda n) \, W_\square(\lambda n, 0) \, \gamma \cdot n \, \gamma_5 \, u(0) \right]_\mu | \pi^+(P) \rangle = \mathrm{i} f_\pi \, P \cdot n \int_0^1 \mathrm{d}x \, \mathrm{e}^{-\mathrm{i}x\lambda P\cdot n} \, \varphi_\pi(x, \mu). \tag{49.339}$$

The Wilson transporter is the same path-ordered fundamental transporter used in (49.293). The normalization of f_π is the axial-current normalization of Chapter 55,

$$\langle \Omega | A_a^\mu(0) | \pi^b(P) \rangle = \mathrm{i} f_\pi P^\mu \delta_a{}^b.$$

Setting $\lambda = 0$ in (49.339) therefore fixes

$$\int_0^1 \mathrm{d}x \, \varphi_\pi(x, \mu) = 1. \tag{49.340}$$

The support interval $[0, 1]$ is a property of the leading-twist light-ray operator between the vacuum and the one-pion state. It is not a probability postulate. The function φ_π is a renormalized distribution in x, depending on the factorization scheme and scale.

Definition 49.83 (Exclusive pion factorization scheme)**.** An exclusive large-Q pion factorization statement consists of:

1. a renormalized QCD matrix element whose external hadrons and currents are gauge-invariant;
2. a light-ray operator basis, here represented at leading twist by $\varphi_\pi(x, \mu)$;
3. a hard coefficient T_H, computed with external collinear partons and subtracted in the same scheme as the light-ray operator;
4. a convolution prescription in the momentum fractions;
5. a topology for the remainder, for example $O(\Lambda_{\mathrm{QCD}}^2/Q^2)$ after excluding endpoint regions by the stated hypotheses, together with the order in α_s to which T_H is known.

The scheme is part of the definition of the controlled approximation. Without it, a formula involving a pion distribution amplitude is only notation.

49.22.2 ERBL Evolution and Gegenbauer Coordinates

The one-loop renormalization of the light-ray operator in (49.339) gives the Efremov–Radyushkin–Brodsky– Lepage evolution equation

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\varphi_\pi(x,\mu)=\frac{\alpha_s(\mu)}{2\pi}\int_0^1\mathrm{d}y\,V(x,y)\varphi_\pi(y,\mu)+O(\alpha_s^2),\tag{49.341}$$

where, in the trace-delta convention,

$$V(x,y)=C_F\left[\theta(y-x)\frac{x}{y}\left(1+\frac{1}{y-x}\right)+\theta(x-y)\frac{1-x}{1-y}\left(1+\frac{1}{x-y}\right)\right]_+.\tag{49.342}$$

The plus prescription acts on the x-variable:

$$\int_0^1\mathrm{d}x\,[K(x,y)]_+f(x):=\int_0^1\mathrm{d}x\,K(x,y)\big(f(x)-f(y)\big),\tag{49.343}$$

for test functions smooth near $x=y$. Equation (49.343) makes the conservation of the axial current normalization manifest:

$$\int_0^1\mathrm{d}x\,V(x,y)=0.$$

Proposition 49.84 (Diagonalization of the leading ERBL kernel)**.** *For $n\geq 0$, let $C_n^{3/2}$ be the Gegenbauer polynomial orthogonal on $[-1,1]$ with weight $(1-z^2)$. The leading ERBL kernel preserves the basis*

$$\varphi_n(x):=6x(1-x)C_n^{3/2}(2x-1),$$

and

$$\int_0^1\mathrm{d}y\,V(x,y)\varphi_n(y)=-\frac{1}{2}\gamma_n^{(0)}\,\varphi_n(x),\tag{49.344}$$

with

$$\gamma_n^{(0)}=C_F\left[1-\frac{2}{(n+1)(n+2)}+4\sum_{k=2}^{n+1}\frac{1}{k}\right].\tag{49.345}$$

In particular $\gamma_0^{(0)}=0$ and $\gamma_n^{(0)}>0$ for every $n\geq 1$.

Proof. Write

$$P_n(x):=C_n^{3/2}(2x-1),\qquad \varphi_n(x)=6x(1-x)P_n(x).$$

The plus prescription in (49.343) subtracts the same raw kernel with the first argument integrated. Therefore, for any polynomial P, the singular action of (49.342) on $\varphi_P(y):=6y(1-y)P(y)$ is the following finite polynomial:

$$\begin{aligned}\frac{1}{C_F}\int_0^1\mathrm{d}y\,V(x,y)\varphi_P(y)&=6(1-x)\int_0^x\mathrm{d}y\,y\big(P(y)-P(x)\big)\left(1+\frac{1}{x-y}\right)\\&\quad+6x\int_x^1\mathrm{d}y\,(1-y)\big(P(y)-P(x)\big)\left(1+\frac{1}{y-x}\right).\end{aligned}\tag{49.346}$$

This identity is only the definition of the plus distribution written at fixed x: the factors $y(1-x)$ and $x(1-y)$ come from $(1-x)\varphi_P(y)/(1-y)$ and $x\varphi_P(y)/y$, while the delta subtraction supplies the terms with $P(x)$. The apparent poles are removable because $P(y)-P(x)$ is divisible by $y-x$.

Let $P(x) = \sum_{m=0}^{N} p_m x^m$, and put $H_m = \sum_{r=1}^{m} r^{-1}$, $H_0 = 0$. Evaluating the beta integrals in (49.346) gives the explicit coefficient operator

$$
\begin{aligned}
\mathcal{E}[P](x) &:= \frac{1}{C_F}\int_0^1 \mathrm{d}y\, V(x,y)\varphi_P(y) \\
&= 6(1-x)\left[\sum_{m=0}^{N} p_m \frac{x^{m+2}}{m+2} - \frac{x^2}{2}P(x) - \sum_{m=1}^{N} p_m x^{m+1}(H_{m+1}-1)\right] \\
&\quad + 6x\left[\sum_{m=0}^{N} p_m\left(\frac{1}{(m+1)(m+2)} - \frac{x^{m+1}}{m+1} + \frac{x^{m+2}}{m+2}\right) - \frac{(1-x)^2}{2}P(x)\right. \\
&\qquad\left. + \sum_{m=1}^{N} p_m \sum_{j=0}^{m-1} x^j\left(\frac{1}{(m-j)(m-j+1)} - \frac{x^{m-j}}{m-j} + \frac{x^{m-j+1}}{m-j+1}\right)\right].
\end{aligned}
\tag{49.347}
$$

For $P = P_n$, the coefficients are fixed by

$$
P_0(x) = 1, \qquad P_1(x) = 3(2x-1),
$$

and by the Gegenbauer recurrence

$$
(n+1)P_{n+1}(x) = (2n+3)(2x-1)P_n(x) - (n+2)P_{n-1}(x). \tag{49.348}
$$

Substitution of this recurrence into the coefficient operator (49.347) gives, by induction on n,

$$
\mathcal{E}[P_n](x) = -\frac{1}{2}\left[1 - \frac{2}{(n+1)(n+2)} + 4\sum_{k=2}^{n+1}\frac{1}{k}\right] 6x(1-x)P_n(x). \tag{49.349}
$$

The induction is finite algebra in the polynomial ring $\mathbb{Q}[x]$: (49.347) gives the action on each monomial, (49.348) gives the next P_n, and the new harmonic term changes by $H_{n+2} - H_{n+1} = 1/(n+2)$. The cases $n = 0, 1$ start the induction; $n = 0$ is the axial-current normalization already expressed by $\int_0^1 \mathrm{d}x\, V(x,y) = 0$. Restoring the overall factor C_F proves (49.344). The factor $1/2$ in (49.344) is tied to the convention that the evolution equation was written with $\alpha_s/(2\pi)$ in (49.341); in the common convention in which the moment equation is written with $\alpha_s/(4\pi)$, the same $\gamma_n^{(0)}$ appears without this extra displayed factor. □

The diagonal basis has a representation-theoretic origin: the collinear conformal group acting on two light-ray quark fields has $SL(2,\mathbb{R})$ quadratic Casimir diagonalized by $x(1-x)C_n^{3/2}(2x-1)$, and the one-loop kernel (49.342) is compatible with this action after the plus subtraction. The proof above keeps the normalization independent of that interpretation by reducing the statement to the finite polynomial identity (49.349).

Therefore a pion LCDA may be expanded as

$$
\varphi_\pi(x,\mu) = 6x(1-x)\left[1 + \sum_{n=1}^{\infty} a_n(\mu) C_n^{3/2}(2x-1)\right], \tag{49.350}
$$

with odd a_n vanishing in the isospin-symmetric charge-conjugation channel. At leading order

$$
\mu\frac{\mathrm{d}}{\mathrm{d}\mu}a_n(\mu) = -\frac{\alpha_s(\mu)}{4\pi}\gamma_n^{(0)} a_n(\mu) + O(\alpha_s^2), \tag{49.351}
$$

and asymptotic freedom drives all $n \geq 1$ moments to zero. The ultraviolet fixed shape is consequently

$$\varphi_\pi^{\mathrm{as}}(x) = 6x(1-x). \tag{49.352}$$

The approach to this shape is logarithmic and scheme dependent at finite scale. The statement (49.352) is therefore an asymptotic consequence of the light-ray anomalous dimensions, not an empirical model for the pion at moderate momentum transfer.

49.22.3 Hard Exclusive Form Factors and Boundary Regimes

The spacelike charged-pion electromagnetic form factor is defined by

$$\langle \pi^+(P')|J^\mu_{\mathrm{em}}(0)|\pi^+(P)\rangle = (P+P')^\mu F_\pi(Q^2), \qquad Q^2 := -(P'-P)^2 > 0. \tag{49.353}$$

At leading twist and leading order in the hard coefficient,

$$F_\pi(Q^2) = \frac{4\pi\alpha_s(\mu)C_F}{N_c}\,\frac{f_\pi^2}{Q^2}\left[\int_0^1 \mathrm{d}x\,\frac{\varphi_\pi(x,\mu)}{x}\right]\left[\int_0^1 \mathrm{d}y\,\frac{\varphi_\pi(y,\mu)}{y}\right] + O(\alpha_s^2, Q^{-4})_{\mathrm{fac}}. \tag{49.354}$$

The subscript "fac" means that the power remainder belongs to the factorization scheme of Definition 49.83; endpoint sensitivity can modify the practical convergence at accessible Q^2 unless controlled by an additional Sudakov or nonperturbative input.

Derivation of the leading hard coefficient. Project each pion onto the color-singlet leading-twist collinear pair defined by (49.339). A single hard gluon must transfer large transverse momentum between the two collinear quark lines; without this exchange the electromagnetic current cannot convert the initial collinear color-singlet into the final one at fixed large Q^2. The tree hard subgraph contains one gluon propagator of virtuality $O(Q^2)$ and one internal quark propagator of virtuality $O(Q^2)$. Contracting the spin projectors fixed by the axial matrix element gives the numerator proportional to $P \cdot P'$, and the color-singlet projection gives

$$\frac{1}{N_c}\,t^a_{ij}t^a_{ji} = \frac{C_F}{N_c}.$$

The two endpoint denominators are the light-cone momentum fractions carried by the active quarks. With the LCDA normalization (49.340), the sum of the two photon attachments to the charged pion gives (49.354). No on-shell colored external state appears in the final formula; colored partons were only projectors used to compute the short-distance coefficient in the chosen factorization scheme.

Substituting $\varphi_\pi^{\mathrm{as}}$ gives

$$Q^2F_\pi(Q^2) \sim \frac{36\pi\alpha_s(Q)C_F}{N_c}\,f_\pi^2. \tag{49.355}$$

For $N_c = 3$, this is

$$Q^2F_\pi(Q^2) \sim 16\pi\alpha_{\mathrm{ht}}(Q)f_\pi^2$$

in the common half-trace convention. Equivalently, if one uses the phenomenological convention $f_\pi^{(130)} = \sqrt{2}\,f_\pi$, the same statement is often written as $Q^2F_\pi \sim 8\pi\alpha_{\mathrm{ht}}(f_\pi^{(130)})^2$. The monograph keeps the Chapter 55 convention f_π fixed, so (49.355) is the convention-safe formula.

The neutral-pion transition form factor is defined by

$$\mathrm{i}\int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}q\cdot x}\langle \pi^0(P)|\,\mathrm{T}\; J^\mu_{\mathrm{em}}(x)J^\nu_{\mathrm{em}}(0)|\Omega\rangle = \epsilon^{\mu\nu\rho\sigma}q_\rho k_\sigma\, F_{\pi^0\gamma\gamma^*}(Q^2), \tag{49.356}$$

with $k^2 = 0$, $q^2 = -Q^2$, and $P = q + k$. The low-energy endpoint is fixed by the anomaly calculation of Chapters 51 and 55:

$$F_{\pi^0\gamma\gamma^*}(0) = \frac{1}{4\pi^2 f_\pi} \tag{49.357}$$

in the present f_π convention. At large spacelike Q^2, leading twist factorization gives

$$F_{\pi^0\gamma\gamma^*}(Q^2) = \frac{f_\pi}{3Q^2}\int_0^1 \mathrm{d}x\,\varphi_\pi(x,\mu)\left(\frac{1}{x}+\frac{1}{1-x}\right) + O(\alpha_s, Q^{-4})_{\mathrm{fac}}. \tag{49.358}$$

For the asymptotic LCDA,

$$Q^2F_{\pi^0\gamma\gamma^*}(Q^2) \longrightarrow 2f_\pi. \tag{49.359}$$

In the $f_\pi^{(130)} = \sqrt{2}f_\pi$ convention this is $\sqrt{2}f_\pi^{(130)}$. The anomaly value (49.357) and the asymptotic value (49.359) constrain two different boundary regimes of the same form factor. They do not imply a unique interpolation. Published high-Q^2 measurements are therefore tests of the approach to the factorized regime and of LCDA shape assumptions, not independent definitions of the anomaly or of asymptotic freedom.

The BaBar/Belle tension in the neutral-pion transition form factor is best formulated in exactly these terms. A reported finite-Q^2 trend can challenge a chosen LCDA model, a treatment of endpoint regions, or an estimate of power corrections. It cannot by itself refute (49.357) or (49.359), because those two equations live at opposite ends of the form factor and rest on different controlled arguments: the first on the axial anomaly and the second on leading-twist factorization plus ERBL evolution. Any quantitative comparison to data must therefore state the factorization scale, perturbative order, LCDA input, and power-correction model before it is a QCD statement.

Controlled Approximation 49.85 (Brodsky–Farrar scaling)**.** Consider a fixed-angle exclusive process

$$H_1 + H_2 \longrightarrow H_3 + H_4$$

whose leading Fock components contain n_i collinear elementary fields in the corresponding leading-twist light-ray operator, and set $n = \sum_i n_i$. If a leading-power exclusive factorization theorem holds, if endpoint configurations are power suppressed in the specified scheme, and if the hard coefficient is evaluated at momentum transfer of order $\sqrt{s}$, then

$$\mathcal{M}(s,t) \sim s^{\,2-n/2} \quad \text{at fixed } t/s, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}t} \sim s^{\,2-n}. \tag{49.360}$$

The derivation is the power counting of the hard coefficient after all external hadrons have been replaced by their normalized leading-twist distribution amplitudes. It is not a dimensional-analysis argument applied to hadron fields. For example, a pion electromagnetic form factor has $n = 4$ across the two pion light-ray operators and therefore scales as $F_\pi(Q^2) \sim Q^{-2}$, while fixed-angle $\pi\pi \to \pi\pi$ has $n = 8$ and gives $\mathrm{d}\sigma/\mathrm{d}t \sim s^{-6}$, up to logarithmic evolution and controlled power corrections.

The pion S-matrix is thus accessed by different controlled descriptions in different regions. Chapter 55 derives the chiral low-energy amplitudes and anomaly couplings; Chapter 29 develops the analytic and dispersive language needed for the resonance region; the present section gives the asymptotic exclusive regime governed by light-ray factorization and asymptotic freedom. These are not three theories. They are three coordinate systems on matrix elements of the same QCD operators, with different hypotheses controlling their errors. A complete nonperturbative construction of QCD would define the relevant matrix elements first; chiral perturbation theory, dispersive methods, lattice calculations, and hard exclusive factorization are then approximation or reconstruction methods applied to that common object.

49.23 Local Twist-Two Operators and Integer Moments

The light-ray definition packages an infinite family of local operators. For quarks, the symmetric traceless spin-N operators have the form

$$\mathcal{O}_q^{\mu_1\cdots\mu_N} = \bar{q}\,\gamma^{(\mu_1}\mathrm{i}D^{\mu_2}\cdots\mathrm{i}D^{\mu_N)}q - \text{traces}, \qquad N \geq 1,$$

with gauge-covariant derivatives in the appropriate representation. The parentheses denote symmetrization over Lorentz indices. The gluon operators are

$$\mathcal{O}_g^{\mu_1\cdots\mu_N} = F^{(\mu_1}{}_{\rho}\,\mathrm{i}D^{\mu_2}\cdots\mathrm{i}D^{\mu_{N-1}}F^{\mu_N)\rho} - \text{traces}, \qquad N \geq 2.$$

Both have twist two after subtracting traces. Their hadron matrix elements are fixed by Lorentz covariance up to scalar reduced matrix elements:

$$\langle P|\mathcal{O}_a^{\mu_1\cdots\mu_N}(\mu)|P\rangle = 2\,a_{a,N}(\mu)\big(P^{\mu_1}\cdots P^{\mu_N} - \text{traces}\big), \qquad a = q, g.$$

The normalization of $a_{a,N}$ is fixed already at tree level by replacing the hadron with a single parton state. Let $|p,i,s\rangle$ be a massless quark of flavor i and spin s, and use $\bar{u}_s(p)\gamma^\mu u_s(p) = 2p^\mu$. Since each covariant derivative in $\mathcal{O}_{q_j}$ acts on the external plane wave as multiplication by p^μ at tree level,

$$\langle p,i,s|\mathcal{O}_{q_j}^{\mu_1\cdots\mu_N}|p,i,s\rangle_{\text{tree}} = 2\delta_{ij}\big(p^{\mu_1}\cdots p^{\mu_N} - \text{traces}\big).$$

Thus a quark target has

$$a_{q_j,N}^{q_i,\text{tree}} = \delta_{ij}, \qquad a_{g,N}^{q_i,\text{tree}} = 0.$$

For a single gluon state $|p,A,\lambda\rangle$, the linearized field strength is

$$F^{\mu\nu,A}(x) = \mathrm{i}\big(p^\mu\varepsilon_\lambda^\nu - p^\nu\varepsilon_\lambda^\mu\big)\mathrm{e}^{-\mathrm{i}p\cdot x}, \qquad p\cdot\varepsilon_\lambda = 0.$$

The color- and spin-averaged partonic matrix element is

$$\frac{1}{2(N_c^2-1)}\sum_{A,\lambda}\langle p,A,\lambda|\mathcal{O}_g^{\mu_1\cdots\mu_N}|p,A,\lambda\rangle_{\text{tree}} = 2\big(p^{\mu_1}\cdots p^{\mu_N} - \text{traces}\big),$$

so

$$a_{g,N}^{g,\text{tree}} = 1, \qquad a_{q_j,N}^{g,\text{tree}} = 0.$$

Equivalently, the tree-level parton distributions are $f_{q_j/q_i}(x) = \delta_{ij}\delta(1-x)$, $f_{g/g}(x) = \delta(1-x)$, and the mixed quark-gluon distributions vanish.

Expanding the light-ray operator near zero separation gives these local operators contracted with the lightlike vector n^μ. Consequently the integer moments of the parton distributions are the same reduced matrix elements, with the precise flavor and charge-conjugation combination fixed by the current channel:

$$a_{a,N}(\mu) = \int_0^1 \mathrm{d}x\, x^{N-1} f_{a/H}(x,\mu)$$

for the corresponding parton combination. The local OPE and the nonlocal light-ray factorization are therefore two presentations of the same short-distance data.

Renormalization mixes operators with the same spin, twist, and quantum numbers. Taking integer moments of (49.314) gives

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu} a_{a,N}(\mu) = \sum_b P_{ab,N}(g(\mu))\, a_{b,N}(\mu), \qquad P_{ab,N}(g) = \int_0^1 \mathrm{d}x\, x^{N-1} P_{ab}(x,g).$$

The coefficient functions in the current OPE obey the dual evolution so that the physical structure functions are independent of the auxiliary factorization scale. This is the local-operator form of perturbative scaling violation in deep inelastic scattering.

For a spin-N leading-twist operator whose scaling dimension near the ultraviolet is

$$\Delta_N(g) = N + 2 + \tau_{1,N} g^2 + O(g^4),$$

the one-loop running

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu} \frac{1}{g^2(\mu)} = \frac{b}{2\pi^2} + O(g^2)$$

implies logarithmic scaling of its matrix element and Wilson coefficient. Equivalently, with

$$\omega := \frac{2P\cdot q}{Q^2} = \frac{1}{x_B},$$

the leading large-Q term in the N-th moment contribution has the form

$$W_N(Q,\omega) \propto \omega^N \left(\log\frac{Q}{\Lambda}\right)^{-2\pi^2\tau_{1,N}/b} \left[1 + O\!\left(\frac{1}{\log(Q/\Lambda)}\right)\right], \qquad (49.361)$$

up to the matrix of anomalous dimensions when several operators mix. The power ω^N is the Bjorken-scaling kinematic dependence of a spin-N operator; asymptotic freedom turns exact scaling into logarithmic scaling violation.

49.24 Cusped Wilson Lines, Large Spin, and the DIS Endpoint

The compact-x factorization statement in Hypothesis 49.74 deliberately does not claim uniform control as $x_B \to 1$. In that endpoint region the invariant mass of the hadronic final state,

$$W_X^2 = (P+q)^2 = M^2 + Q^2\left(\frac{1}{x_B} - 1\right),$$

is much smaller than Q^2. The short-distance coefficient functions then contain distributions supported near $x = 1$, such as $[\log^k(1-x)/(1-x)]_+$, and the perturbative expansion must be reorganized. The universal coefficient controlling the leading endpoint singularity is most cleanly defined by a cusped Wilson-line operator. This definition is an operator statement, not a parton-model picture.

Let R be a finite-dimensional unitary representation of $SU(N_c)$, with quadratic Casimir C_R in the trace-delta convention (49.1). In Euclidean signature choose two unit vectors e_1, e_2 with $e_1 \cdot e_2 = \cos\phi$, $0 < \phi < \pi$, and let $\Gamma_{\phi,L,\epsilon}$ be the two-segment path

$$x(s) = -se_1, \quad \epsilon \le s \le L, \qquad x(t) = te_2, \quad \epsilon \le t \le L,$$

joined at the origin after removing a ball of radius ϵ. The regulated cusped line is the path-ordered transporter

$$W_R(\Gamma_{\phi,L,\epsilon}) = \mathcal{P}\exp\left\{\mathrm{i}\int_{\Gamma_{\phi,L,\epsilon}} A^a_\mu(x) t^a_R \,\mathrm{d}x^\mu\right\}.$$

The straight-line endpoint and perimeter renormalizations are fixed by the same regulator applied to a smooth line. After these local smooth-line renormalizations are divided out, the remaining logarithmic singularity at the corner defines the Euclidean cusp coefficient $K_R(\phi, g)$ by

$$-\log\langle W_R(\Gamma_{\phi,L,\epsilon})\rangle_{\text{smooth}-\text{subtracted}} = K_R(\phi, g)\log\frac{L}{\epsilon} + O(1). \tag{49.362}$$

The minus sign in this definition is part of the convention: it combines with the i^2 from expanding the Wilson transporter, so that the one-loop coefficient below is read directly from the positive Euclidean propagator. For timelike Wilson lines in Lorentzian signature we instead use the positive lightlike cusp anomalous dimension

$$\Gamma^R_{\text{cusp}}(g) := \lim_{\chi\to\infty}\frac{\Gamma^R_{\text{cusp}}(\chi, g)}{\chi}, \qquad -v_1\cdot v_2 = \cosh\chi, \quad v_1^2 = v_2^2 = -1,$$

where $\Gamma^R_{\text{cusp}}(\chi, g)$ is obtained from the same local renormalization problem by the analytic continuation $\phi = \mathrm{i}\chi$. This convention is the one used in (49.244).

At one loop the angular dependence follows from a single eikonal exchange. In Feynman gauge, with the connection normalization of this chapter, the Euclidean short-distance propagator is

$$\langle A^a_\mu(x)A^b_\nu(0)\rangle = \frac{g^2\delta^{ab}\delta_{\mu\nu}}{4\pi^2 x^2} + O(g^4).$$

The exchange between the two rays contributes, up to cutoff-independent finite terms, the following scale-invariant part of $-\log\langle W_R\rangle$:

$$\frac{g^2 C_R}{4\pi^2}\int_\epsilon^L \mathrm{d}s\int_\epsilon^L \mathrm{d}t\,\frac{e_1\cdot e_2}{s^2 + t^2 + 2st\,e_1\cdot e_2}.$$

The coefficient of $\log(L/\epsilon)$ is local at the cusp, so set $t = sr$ in the scale-invariant part. The angular integral is

$$J(\phi) := \int_0^\infty \frac{\cos\phi\,\mathrm{d}r}{r^2 + 2r\cos\phi + 1} = \phi\cot\phi. \tag{49.363}$$

Indeed,

$$r^2 + 2r\cos\phi + 1 = (r + \cos\phi)^2 + \sin^2\phi,$$

and the antiderivative is

$$\cot\phi\,\arctan\frac{r+\cos\phi}{\sin\phi}.$$

Taking r from 0 to ∞ gives $\cot\phi(\pi/2-\arctan(\cot\phi))=\phi\cot\phi$, with the principal branch of arctan for $0<\phi<\pi$. A smooth straight line corresponds to $\phi=0$, for which $J(0)=1$. Therefore the corner-specific logarithm is

$$K_R^{(1)}(\phi,g)=\frac{g^2C_R}{4\pi^2}\big(\phi\cot\phi-1\big). \tag{49.364}$$

Continuing $\phi=\mathrm{i}\chi$ gives the Lorentzian timelike cusp coefficient

$$\Gamma^R_{\text{cusp}}(\chi,g)=\frac{g^2C_R}{4\pi^2}\big(\chi\coth\chi-1\big)+O(g^4),\qquad \Gamma^R_{\text{cusp}}(g)=\frac{g^2C_R}{4\pi^2}+O(g^4). \tag{49.365}$$

For $R=\Box$, this is exactly the quark coefficient in (49.244).

The same coefficient appears in the endpoint renormalization of light-ray operators. Near $x=1$, the emitted radiation is soft compared with the parent quark momentum, and the Wilson transporter in the light-ray operator is approximated by eikonal propagation along the two nearly lightlike directions selected by the incoming hadron and the hard current. The local renormalization of the resulting lightlike cusp supplies the coefficient of the plus distribution $D_0=(1-x)_+^{-1}$ in the nonsinglet kernel:

$$P_{\text{ns}}(x,g)=\Gamma^q_{\text{cusp}}(g)\,D_0(x)+\text{terms with less singular Mellin growth as } x\to 1. \tag{49.366}$$

Equation (49.366) is meant as a perturbative renormalization statement for the specified light-ray operator and subtraction scheme. It is not a nonperturbative theorem about the full endpoint region of every DIS observable; threshold factorization also requires a hard coefficient, a jet function, a soft matrix element, a rapidity prescription when appropriate, and a controlled power remainder.

The Mellin transform turns this endpoint problem into a large-spin problem. For a distribution $K(x)$, define

$$K_N:=\int_0^1 \mathrm{d}x\,x^{N-1}K(x).$$

Using the D_0 convention above, the nonsinglet one-loop kernel has the exact moment

$$P^{(0)}_{\text{ns},N}=C_F\left[-2H_{N-1}-\frac{1}{N}-\frac{1}{N+1}+\frac{3}{2}\right],\qquad N\geq 1. \tag{49.367}$$

The harmonic number appears because

$$\int_0^1\mathrm{d}x\,x^{N-1}D_0(x)=\int_0^1\mathrm{d}x\,\frac{x^{N-1}-1}{1-x}=-H_{N-1}. \tag{49.368}$$

The last equality follows by writing $(1-x^{N-1})/(1-x)=1+x+\cdots+x^{N-2}$ before integration. Thus (49.367) gives quark-number conservation at $N=1$, and at large N

$$P^{(0)}_{\text{ns},N}=-2C_F\log N+C_F\left(\frac{3}{2}-2\gamma_{\mathrm{E}}\right)+O(N^{-1}). \tag{49.369}$$

Therefore, with $P = (g^2/8\pi^2)P^{(0)} + O(g^4)$,

$$P_{\mathrm{ns},N}(g) = -\Gamma^q_{\mathrm{cusp}}(g)\log N + O(N^0), \qquad \Gamma^q_{\mathrm{cusp}}(g) = \frac{g^2 C_F}{4\pi^2} + O(g^4). \tag{49.370}$$

This is the evolution eigenvalue for matrix elements in the convention $\mu\,\mathrm{d}a_N/\mathrm{d}\mu = P_N a_N$. If instead one writes the anomalous dimension of the local twist-two operator itself by $\mu\,\mathrm{d}\mathcal{O}_N/\mathrm{d}\mu = -\gamma^{\mathcal{O}}_N\mathcal{O}_N + \cdots$, then

$$\gamma^{\mathcal{O}}_N(g) = \Gamma^q_{\mathrm{cusp}}(g)\log N + O(N^0).$$

The sign difference is bookkeeping: coefficient functions and operator matrix elements obey dual renormalization-group equations so that the physical hadronic tensor is independent of the factorization scale.

The product $g^2 C_R$ in (49.365) is invariant under generator-coordinate changes. If T^a denotes the common half-trace generator and $g_{\mathrm{ht}} = \sqrt{2}\,g$, then

$$g^2 C_F = g^2_{\mathrm{ht}} C^{\mathrm{ht}}_F.$$

The cusp coefficient is thus independent of the choice of generator coordinates once the coupling coordinate is transformed with the action.

Endpoint resummation is not a new definition of the PDF. It is an additional factorization of the coefficient function in a region where the compact-x remainder estimate is not uniform. A precise threshold factorization theorem must specify the hard current coefficient, the jet function for the nearly collinear final-state radiation, the soft Wilson-line matrix element, the contour prescription for Mellin or Laplace inversion, and a remainder estimate in the joint limit $Q \to \infty$, $N \to \infty$. The large-spin logarithm in (49.370) is the operator-renormalization side of that structure. The same Wilson-line cusp observable is the QCD analogue of the large-spin cusp scaling function studied in Chapter 98, but the logical status is different: in QCD it is a perturbative Wilson-line anomalous dimension tied to factorization hypotheses, while in planar $\mathcal{N} = 4$ super Yang–Mills the BES equation gives an integrability construction of the planar large-spin function in a special conformal theory. The small-x limit is a different boundary of the same DIS observable and is not obtained by continuing the threshold expansion.

49.25 Small-x Boundary, Dipoles, and the BFKL Kernel

The compact-x_B factorization hypothesis in Hypothesis 49.74 also does not claim uniform control as $x_B \to 0$. Put

$$Y := \log\frac{1}{x_B},$$

up to a process-dependent finite shift fixed by the chosen rapidity factorization scheme. In the high-energy boundary $Y \gg 1$, powers of $g^2 Y$ may compete with powers of g^2. The leading-logarithmic small-x expansion is therefore not a new definition of QCD and not a replacement for the hadronic tensor (49.286). It is a rapidity factorization of regulated Wilson-line matrix elements in a kinematic region where the ordinary twist-two evolution at fixed x_B is not uniform.

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^2$ denote transverse coordinates, and set

$$r_{xy} := |\mathbf{x} - \mathbf{y}|.$$

The leading transverse kernel is

$$M_{\mathbf{xyz}} := \frac{r_{xy}^2}{r_{xz}^2 r_{zy}^2}. \tag{49.371}$$

The singularities at $\mathbf{z} = \mathbf{x}$ and $\mathbf{z} = \mathbf{y}$ are not meaningful as separate real-emission probabilities. They are defined only after the virtual subtraction in the evolution equation, or after an equivalent rapidity regulator and finite-part map have been specified.

Definition 49.86 (Regulated small-x dipole setup)**.** A small-x dipole setup for a color-singlet probe consists of:

1. a QCD regulator $\mathcal{R}_\epsilon$ preserving background gauge covariance, or a gauge-fixed BV regulator with the quantum master equation;
2. two nearly lightlike Wilson-line directions, a rapidity regulator ρ, and a finite rapidity separation Y;
3. fundamental Wilson transporters $U_{\rho,Y}(\mathbf{x})$ and $U_{\rho,Y}(\mathbf{y})$, with transverse closures or boundary conditions that make the residual endpoint gauge transformations explicit;
4. the renormalized color dipole matrix element
$$S_Y(\mathbf{x},\mathbf{y}) := \frac{1}{N_c}\Big\langle \mathrm{tr}_\square\big[U_{\rho,Y}(\mathbf{x})U_{\rho,Y}(\mathbf{y})^\dagger\big]\Big\rangle^{\mathrm{ren}}_{\epsilon,\rho,Y}, \qquad N_Y(\mathbf{x},\mathbf{y}) := 1 - S_Y(\mathbf{x},\mathbf{y}); \tag{49.372}$$
5. a finite-part prescription for ultraviolet endpoint divergences, lightlike or near-lightlike rapidity divergences, and possible cusp divergences introduced by the chosen closures.

The quantity S_Y is a Wilson-line matrix element. Its connection to a measured small-x structure function is an additional high-energy factorization statement with its own impact factor, rapidity subtraction, and power-remainder data.

At finite regulator, the trace in (49.372) is gauge invariant after the endpoint closures or boundary conditions of Definition 49.86 are included. A background gauge-covariant finite-part prescription therefore gives a gauge-invariant renormalized dipole matrix element. Each Wilson transporter transforms by multiplication with the gauge transformation at its two endpoints. Along a closed contour the adjacent endpoint factors cancel and the residual base-point factor is removed by the cyclicity of $\mathrm{tr}_\square$. If the regulated Wilson lines are represented instead as open lines ending at a rapidity boundary, the boundary condition is part of the setup and supplies the same endpoint cancellation. The finite-part map changes only regulator and scale coordinates and is assumed background gauge covariant; hence it cannot introduce an uncanceled color index.

Hypothesis 49.87 (Leading-logarithmic dipole/BFKL evolution setup)**.** In a fixed-coupling leading-logarithmic rapidity scheme, the linearized color-singlet dipole amplitude obeys

$$\partial_Y N_Y(\mathbf{x},\mathbf{y}) = \frac{g^2 C_A}{8\pi^3}\int_{\mathbb{R}^2} \mathrm{d}^2\mathbf{z}\, M_{\mathbf{xyz}}\big[N_Y(\mathbf{x},\mathbf{z}) + N_Y(\mathbf{z},\mathbf{y}) - N_Y(\mathbf{x},\mathbf{y})\big], \tag{49.373}$$

where $C_A = 2N_c$ in the trace convention of (49.1). Equivalently, with

$$\bar{\alpha}_{\mathrm{s}} := \frac{g^2 C_A}{4\pi^2} = \frac{g^2 N_c}{2\pi^2},$$

the right-hand side is

$$\bar{\alpha}_{\mathrm{s}} \frac{1}{2\pi} \int \mathrm{d}^2\mathbf{z}\, M_{\mathbf{xyz}}[\cdots].$$

In the common half-trace convention, $g_{\mathrm{ht}}^2 = 2g^2$ and $C_A^{\mathrm{ht}} = N_c$, so the product $g^2 C_A = g_{\mathrm{ht}}^2 C_A^{\mathrm{ht}}$ and the kernel coefficient are unchanged.

For a large-N_c dipole-factorization truncation of the Balitsky hierarchy, the corresponding non-linear equation for $N_Y = 1 - S_Y$ has the additional term

$$-\frac{g^2 C_A}{8\pi^3} \int \mathrm{d}^2\mathbf{z}\, M_{\mathbf{xyz}}\, N_Y(\mathbf{x},\mathbf{z}) N_Y(\mathbf{z},\mathbf{y}). \tag{49.374}$$

At finite N_c, the exact Wilson-line rapidity hierarchy contains higher multi-trace operators; replacing them by the product in (49.374) is an additional closure assumption, not a theorem about continuum QCD.

49.25.1 Finite Wilson-Line Evolution and the JIMWLK Theorem Boundary

The preceding dipole equation is a closed equation only after a truncation has been chosen. The more invariant small-x object is the rapidity evolution of the whole algebra generated by Wilson lines. To state the mathematical content without importing the usual folklore terminology, first put the Wilson-line variables on a finite transverse regulator.

Definition 49.88 (Finite Wilson-line rapidity-evolution generator)**.** Let $\Lambda_\perp$ be a finite set of transverse sites and let

$$\mathcal{C}_{\Lambda_\perp} := SU(N_c)^{\Lambda_\perp} \tag{49.375}$$

with product Haar measure $\mathrm{d}h(U)$. For $\mathbf{x} \in \Lambda_\perp$ and Lie-algebra index a, let $L_{\mathbf{x}}^a$ be the left-invariant vector field

$$(L_{\mathbf{x}}^a F)(U) := \frac{\mathrm{d}}{\mathrm{d}t} F(\ldots, \mathrm{e}^{\mathrm{i}tt^a} U_{\mathbf{x}}, \ldots)\Big|_{t=0}, \tag{49.376}$$

where the generators obey the trace convention of (49.1). A finite Wilson-line rapidity-evolution structure consists of a family of smooth coefficients

$$\mathcal{A}_{\mathbf{xy}}^{ab}(U;Y) = \mathcal{A}_{\mathbf{yx}}^{ba}(U;Y)$$

whose finite matrix $(\mathcal{A}_{\mathbf{xy}}^{ab})$ is nonnegative at each (U,Y), is equivariant under residual global color conjugation, and is defined in the same rapidity and ultraviolet regulator scheme as the Wilson lines. The corresponding finite Wilson-line generator is the second-order operator

$$(\mathcal{H}_Y F)(U) := \frac{1}{2} \sum_{\mathbf{x},\mathbf{y} \in \Lambda_\perp} \sum_{a,b} L_{\mathbf{x}}^a \left[\mathcal{A}_{\mathbf{xy}}^{ab}(U;Y) L_{\mathbf{y}}^b F(U) \right]. \tag{49.377}$$

This finite generator is a finite-regulator replacement for the JIMWLK Hamiltonian only after the coefficient $\mathcal{A}$ has been derived from a specified QCD rapidity factorization scheme.

The finite object has two equivalent presentations. In the weak, or observable, presentation a rapidity-dependent state $W_Y \mathrm{d}h$ on $\mathcal{C}_{\Lambda_\perp}$ obeys

$$\frac{\mathrm{d}}{\mathrm{d}Y} \int_{\mathcal{C}_{\Lambda_\perp}} F(U)\, W_Y(U)\, \mathrm{d}h(U) = \int_{\mathcal{C}_{\Lambda_\perp}} (\mathcal{H}_Y F)(U)\, W_Y(U)\, \mathrm{d}h(U) \tag{49.378}$$

for every smooth gauge-invariant Wilson-line observable F. In the strong, or density, presentation

$$\partial_Y W_Y = \mathcal{H}_Y^* W_Y, \qquad \mathcal{H}_Y^* w = \frac{1}{2} \sum_{\mathbf{x},\mathbf{y},a,b} L_{\mathbf{y}}^b \left[\mathcal{A}_{\mathbf{xy}}^{ab}(U;Y) L_{\mathbf{x}}^a w \right], \tag{49.379}$$

where the adjoint is taken with respect to product Haar measure. The two forms are identical by integration by parts on the compact Lie group:

$$\int (L_{\mathbf{x}}^a F) G \, \mathrm{d}h = - \int F (L_{\mathbf{x}}^a G) \, \mathrm{d}h.$$

Applying this identity twice to (49.377) gives (49.379). Since $\mathcal{H}_Y 1 = 0$, the total mass $\int W_Y \mathrm{d}h$ is conserved. Since the coefficient matrix is nonnegative, the principal symbol is that of a diffusion on the compact manifold. When the closure of $\mathcal{H}_Y$ on continuous functions satisfies the positive maximum principle, the Hille–Yosida theorem gives the corresponding Markov semigroup. This semigroup assertion belongs to ordinary finite-dimensional diffusion analysis; the QCD-specific work is the derivation and limiting control of $\mathcal{A}$. These are finite statements about $\mathcal{C}_{\Lambda_\perp}$, not continuum QCD theorems.

For the dipole observable

$$F_{\mathbf{xy}}(U) := \frac{1}{N_c} \operatorname{tr}_\square (U_{\mathbf{x}} U_{\mathbf{y}}^\dagger), \tag{49.380}$$

the weak equation (49.378) generates the first equation of the Balitsky hierarchy. The right-hand side contains Wilson-line observables with more transverse labels. A closed BK equation is obtained only after large-N_c, mean-field, or other factorization input has replaced those higher observables by products of dipoles. Thus the logical chain is

$$\text{finite Wilson-line diffusion} \longrightarrow \text{Balitsky hierarchy} \longrightarrow \text{BK closure under extra hypotheses,}$$

not a direct theorem from the BFKL eigenvalue calculation.

Dipole projection and the BK closure boundary. The first arrow in the preceding display is an operator statement on the finite Wilson-line algebra; the second arrow is a closure operation on a subsystem of expectation values. It is useful to write this subsystem before passing to the continuum transverse plane. Fix a finite transverse set $\Lambda_\perp$ and nonnegative cell weights $\omega_{\mathbf{xy};\mathbf{z}}(Y)$, symmetric under $\mathbf{x} \leftrightarrow \mathbf{y}$, which represent the regulated cell-integrals of the leading dipole kernel in the same rapidity scheme as $\mathcal{H}_Y$. For distinct transverse labels set

$$S_{\mathbf{xy}}^{(1)}(Y) := \langle F_{\mathbf{xy}} \rangle_Y, \qquad S_{\mathbf{xz};\mathbf{zy}}^{(2)}(Y) := \langle F_{\mathbf{xz}} F_{\mathbf{zy}} \rangle_Y,$$

where the expectation is the weak Wilson-line state in (49.378). A finite leading-logarithmic dipole projection is the assertion that

$$\partial_Y S_{\mathbf{xy}}^{(1)} = \sum_{\mathbf{z} \in \Lambda_\perp \setminus \{\mathbf{x},\mathbf{y}\}} \omega_{\mathbf{xy};\mathbf{z}} \left(S_{\mathbf{xz};\mathbf{zy}}^{(2)} - S_{\mathbf{xy}}^{(1)} \right). \tag{49.381}$$

Equation (49.381) is not a closed equation for $S^{(1)}$. Its new variable is the double-dipole expectation $S^{(2)}$, whose rapidity derivative involves still higher Wilson-line observables. This is the finite-regulator form of the Balitsky hierarchy.

The BK reduction is a controlled approximation only if one supplies a decorrelation estimate for the connected double-dipole coordinate

$$C_{\mathbf{xz};\mathbf{zy}}(Y) := S^{(2)}_{\mathbf{xz};\mathbf{zy}}(Y) - S^{(1)}_{\mathbf{xz}}(Y)S^{(1)}_{\mathbf{zy}}(Y).$$

With

$$E^{\mathrm{BK}}_{\mathbf{xy}}(Y) := \sum_{\mathbf{z}\neq\mathbf{x},\mathbf{y}} \omega_{\mathbf{xy};\mathbf{z}}(Y)\, C_{\mathbf{xz};\mathbf{zy}}(Y), \tag{49.382}$$

the exact first dipole equation is

$$\partial_Y S^{(1)}_{\mathbf{xy}} = \sum_{\mathbf{z}\neq\mathbf{x},\mathbf{y}} \omega_{\mathbf{xy};\mathbf{z}} \left(S^{(1)}_{\mathbf{xz}}S^{(1)}_{\mathbf{zy}} - S^{(1)}_{\mathbf{xy}}\right) + E^{\mathrm{BK}}_{\mathbf{xy}}.$$

Thus the phrase "large-N_c BK equation" means that a separate argument has made E^{BK} negligible in the norm and rapidity interval under discussion. In the amplitude variable $N^{(1)}_{\mathbf{xy}} = 1 - S^{(1)}_{\mathbf{xy}}$, the closed finite BK equation obtained by setting $E^{\mathrm{BK}} = 0$ is

$$\partial_Y N_{\mathbf{xy}} = \sum_{\mathbf{z}\neq\mathbf{x},\mathbf{y}} \omega_{\mathbf{xy};\mathbf{z}} \left(N_{\mathbf{xz}} + N_{\mathbf{zy}} - N_{\mathbf{xy}} - N_{\mathbf{xz}}N_{\mathbf{zy}}\right). \tag{49.383}$$

This is the finite-cell version of (49.373) plus (49.374).

The finite error estimate is elementary but important. Suppose a real mean-field solution $\widetilde{S}_{\mathbf{xy}}$ and the exact first coordinates $S^{(1)}_{\mathbf{xy}}$ both take values in $[0,1]$ on a rapidity interval $0 \le Y \le Y_*$, share the same initial data, and have

$$L := \sup_{Y\le Y_*}\sup_{\mathbf{x}\neq\mathbf{y}} \sum_{\mathbf{z}\neq\mathbf{x},\mathbf{y}} \omega_{\mathbf{xy};\mathbf{z}}(Y) < \infty, \qquad \varepsilon(Y) := \sup_{\mathbf{x}\neq\mathbf{y}} |E^{\mathrm{BK}}_{\mathbf{xy}}(Y)|.$$

Writing

$$d(Y) := \sup_{\mathbf{x}\neq\mathbf{y}} |S^{(1)}_{\mathbf{xy}}(Y) - \widetilde{S}_{\mathbf{xy}}(Y)|,$$

one obtains

$$d(Y) \le \int_0^Y \mathrm{e}^{3L(Y-s)}\varepsilon(s)\,\mathrm{d}s. \tag{49.384}$$

Indeed, on the unit interval $|ab - a'b'| \le |a-a'| + |b-b'|$, so the product term contributes at most $2Ld(Y)$, the virtual term at most $Ld(Y)$, and the connected covariance term contributes $\varepsilon(Y)$; Gronwall gives (49.384). The closed BK vector field also points into the unit cube: at $S_{\mathbf{xy}} = 0$ its right-hand side is nonnegative, while at $S_{\mathbf{xy}} = 1$ it is nonpositive. This finite ODE fact explains the black-disk and transparent fixed points of the closed approximation. It does not replace the missing QCD estimate on E^{BK}, nor the continuum limit of the cell weights.

The continuum JIMWLK problem is the removal of the finite transverse regulator and rapidity regulator from this finite compact-manifold construction while keeping the Wilson-line observables, the coefficient kernel, and the factorization map to the measured process in a topology strong enough to pass to the limit in (49.378). The monograph therefore treats the color-glass weight functional as a renormalized Wilson-line state, not as a literal probability measure on classical gluon fields unless such a measure has been constructed in the chosen regulator.

Projective continuum limit for Wilson-line states. A continuum statement of JIMWLK must be stronger than a sequence of finite Balitsky equations. Let $\{\Lambda_\perp^{(m)}\}_{m\in I}$ be a directed family of finite transverse regulators and let $\mathcal{A}_m = C^\infty(SU(N_c)^{\Lambda_\perp^{(m)}})^{G_{\mathrm{diag}}}$ be the gauge-invariant Wilson-line algebra on the m-th regulator. A cylinder observable F depends on a fixed finite list of transverse labels; for all sufficiently fine regulators it has a representative $F_m \in \mathcal{A}_m$. A projective Wilson-line state limit on $[0, Y_*]$ consists of

$$\big(\omega_{m,Y}, \mathcal{H}_{m,Y}, F \mapsto F_m, \omega_Y, \mathcal{H}_Y^{\mathrm{cyl}}, \mathcal{I}_Y\big)$$

with the following data. First, the finite states are compatible on cylinder observables:

$$\sup_{0\le Y\le Y_*} |\omega_{m,Y}(F_m) - \omega_Y(F)| \le \delta_{m,F}, \qquad \delta_{m,F} \longrightarrow 0. \tag{49.385}$$

Second, the finite generators converge on the same cylinder test:

$$\sup_{0\le Y\le Y_*} \|\mathcal{H}_{m,Y} F_m - (\mathcal{H}_Y^{\mathrm{cyl}} F)_m\|_\infty \le \varepsilon_{m,F}, \qquad \varepsilon_{m,F} \longrightarrow 0. \tag{49.386}$$

Third, the rapidity dependence is uniformly controlled on cylinder tests, so the finite weak equations are equicontinuous in Y and limits may be passed through rapidity integrals. Fourth, positivity, normalization, and the positive-maximum-principle/Markov property hold at finite m and are closed under the cylinder limit. Fifth, $\mathcal{I}_Y$ is the process map: it turns the Wilson-line state into the measured small-x observable with the chosen impact factors, subtractions, and power-remainder topology.

Under these hypotheses the finite weak equation has a genuine cylinder limit. Indeed, for $Y_0 < Y_1$,

$$\omega_{m,Y_1}(F_m) - \omega_{m,Y_0}(F_m) = \int_{Y_0}^{Y_1} \omega_{m,s}(\mathcal{H}_{m,s} F_m)\, \mathrm{d}s.$$

After adding and subtracting the limiting terms one obtains the error ledger

$$\begin{aligned} &\left|\omega_{Y_1}(F) - \omega_{Y_0}(F) - \int_{Y_0}^{Y_1} \omega_s(\mathcal{H}_s^{\mathrm{cyl}} F)\, \mathrm{d}s\right| \\ &\qquad \le 2\delta_{m,F} + \int_{Y_0}^{Y_1} \big(\delta_{m,\mathcal{H}^{\mathrm{cyl}} F}(s) + \varepsilon_{m,F}(s)\big)\, \mathrm{d}s, \end{aligned} \tag{49.387}$$

where the right-hand side is written with the corresponding uniform or integrable bound on the state error for the generator observable when the errors depend on s. Letting $m \to \infty$ gives the continuum weak equation on the cylinder algebra,

$$\omega_{Y_1}(F) - \omega_{Y_0}(F) = \int_{Y_0}^{Y_1} \omega_s(\mathcal{H}_s^{\mathrm{cyl}} F)\, \mathrm{d}s.$$

If $Y \mapsto \omega_Y(F)$ is absolutely continuous, this is the differential weak JIMWLK equation for cylinder observables. The BK closure estimate above controls one projected dipole subsystem and one connected covariance coordinate; it does not supply (49.385), (49.386), or the process map. Thus continuum JIMWLK is a projective theorem about Wilson-line states and QCD factorization; the formal functional Hamiltonian names only the candidate cylinder generator.

Conformal covariance of the transverse kernel. The measure

$$\mathrm{d}^2\mathbf{z}\, M_{\mathbf{xyz}}$$

is invariant under translations, rotations, uniform scalings, and inversion $\mathbf{u} \mapsto \mathbf{u}/|\mathbf{u}|^2$, whenever the transformed points avoid the inversion center. Thus the leading fixed-coupling transverse kernel is covariant under the real Mobius transformations generated by these maps.

Translations and rotations preserve all distances and $\mathrm{d}^2\mathbf{z}$. Under the scaling $\mathbf{u} \mapsto \lambda\mathbf{u}$, the kernel $M_{\mathbf{xyz}}$ scales as λ^{-2}, while $\mathrm{d}^2\mathbf{z}$ scales as λ^2.

For inversion, write $\mathbf{u}' = \mathbf{u}/|\mathbf{u}|^2$. The Euclidean distance and area element transform as

$$|\mathbf{u}' - \mathbf{v}'|^2 = \frac{|\mathbf{u} - \mathbf{v}|^2}{|\mathbf{u}|^2|\mathbf{v}|^2}, \qquad \mathrm{d}^2\mathbf{z}' = \frac{\mathrm{d}^2\mathbf{z}}{|\mathbf{z}|^4}.$$

Substituting these identities gives

$$M_{\mathbf{x}'\mathbf{y}'\mathbf{z}'}\,\mathrm{d}^2\mathbf{z}' = \frac{r_{xy}^2/(|\mathbf{x}|^2|\mathbf{y}|^2)}{r_{xz}^2/(|\mathbf{x}|^2|\mathbf{z}|^2)\, r_{zy}^2/(|\mathbf{z}|^2|\mathbf{y}|^2)} \frac{\mathrm{d}^2\mathbf{z}}{|\mathbf{z}|^4} = M_{\mathbf{xyz}}\,\mathrm{d}^2\mathbf{z}.$$

This proves the stated covariance.

Proposition 49.89 (Mellin eigenvalue of the leading BFKL kernel)**.** *Let* $0 < \operatorname{Re}\gamma < 1$. *For the power*

$$f_\gamma(\mathbf{x}, \mathbf{y}) = \left(\mu_\perp^2 r_{xy}^2\right)^\gamma,$$

define the subtracted transverse integral by analytic regularization as in the proof below. Then

$$\frac{1}{2\pi}\int_{\mathbb{R}^2} \mathrm{d}^2\mathbf{z}\, M_{\mathbf{xyz}}\big[f_\gamma(\mathbf{x}, \mathbf{z}) + f_\gamma(\mathbf{z}, \mathbf{y}) - f_\gamma(\mathbf{x}, \mathbf{y})\big] = \chi(\gamma) f_\gamma(\mathbf{x}, \mathbf{y}), \tag{49.388}$$

where

$$\chi(\gamma) = 2\psi(1) - \psi(\gamma) - \psi(1-\gamma), \qquad \psi(u) := \frac{\mathrm{d}}{\mathrm{d}u}\log\Gamma(u). \tag{49.389}$$

Proof. By translation and scaling it suffices to set $\mathbf{y} = 0$, $r_{xy} = 1$, and $\mu_\perp = 1$. The singular terms in the three brackets are regulated in $d = 2 - 2\epsilon$ transverse dimensions. The Euclidean beta integral

$$\int \mathrm{d}^d\mathbf{z}\, \frac{1}{(\mathbf{z}^2)^a((\mathbf{z} - \mathbf{x})^2)^b} = \pi^{d/2} \frac{\Gamma(d/2 - a)\Gamma(d/2 - b)\Gamma(a + b - d/2)}{\Gamma(a)\Gamma(b)\Gamma(d - a - b)} (\mathbf{x}^2)^{d/2 - a - b} \tag{49.390}$$

is first valid in its convergence domain and then analytically continued. Applying (49.390) to the three pieces gives

$$J_\epsilon(\gamma) = \pi^{1-\epsilon} \frac{\Gamma(\gamma - \epsilon)\Gamma(-\epsilon)\Gamma(1 - \gamma + \epsilon)}{\Gamma(1-\gamma)\Gamma(\gamma - 2\epsilon)},$$

$$J_\epsilon(1-\gamma) = \pi^{1-\epsilon} \frac{\Gamma(1 - \gamma - \epsilon)\Gamma(-\epsilon)\Gamma(\gamma + \epsilon)}{\Gamma(\gamma)\Gamma(1 - \gamma - 2\epsilon)},$$

$$J_\epsilon(0) = \pi^{1-\epsilon} \frac{\Gamma(-\epsilon)^2\Gamma(1+\epsilon)}{\Gamma(-2\epsilon)}.$$

The regulated subtracted integral is

$$J_\epsilon(\gamma) + J_\epsilon(1-\gamma) - J_\epsilon(0).$$

Using

$$\Gamma(u+\delta) = \Gamma(u)\big(1 + \delta\psi(u) + O(\delta^2)\big)$$

and the corresponding expansion at the pole $u = 0$, the $1/\epsilon$ terms cancel and the finite part is

$$2\pi\,[2\psi(1) - \psi(\gamma) - \psi(1-\gamma)].$$

Dividing by 2π proves (49.388). □

The leading fixed-coupling kernel in (49.373) has

$$\chi\left(\frac{1}{2}\right) = 4\log 2, \qquad \chi\left(\frac{1}{2} + \mathrm{i}\nu\right) = 4\log 2 - 14\zeta(3)\nu^2 + O(\nu^4).$$

These values follow from

$$\psi\left(\frac{1}{2}\right) = -\gamma_{\mathrm{E}} - 2\log 2, \qquad \psi''\left(\frac{1}{2}\right) = -14\zeta(3),$$

inserted into (49.389); in particular $\chi'(\frac{1}{2}) = 0$ and $\chi''(\frac{1}{2}) = 28\zeta(3)$. A Mellin component near $\gamma = \frac{1}{2}$ therefore evolves as

$$\exp\big[\bar{\alpha}_{\mathrm{s}}\,4\log 2\,Y - \bar{\alpha}_{\mathrm{s}}\,14\zeta(3)\nu^2 Y + O(\bar{\alpha}_{\mathrm{s}}\nu^4 Y)\big]$$

inside the fixed-coupling leading-logarithmic setup.

Controlled Approximation 49.90 (Status of small-x resummation)**.** A small-x resummation used for a QCD observable must state: the regulated Wilson-line operator being evolved; the rapidity variable and subtraction scheme; the impact factor connecting the Wilson-line matrix element to the measured process; whether running coupling, energy-scale, collinear-improvement, saturation, or finite-N_c hierarchy effects are included; and the topology of the remainder estimate. The BFKL kernel in Proposition 49.89 is a theorem about the leading transverse integral. Its physical use in high-energy hadron scattering additionally requires the Wilson-line operator definition, impact factor, rapidity-factorization scheme, and unitarization or saturation framework appropriate to the observable.

Chapter 50

Jets, Infrared-Safe Observables, and Hadronization

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Jets enter quantum field theory as observables built from the energy and momentum deposited by physical final states. The final states in QCD are hadronic states in the physical Hilbert space; quarks and gluons are short-distance variables used inside a gauge-fixed or factorized calculation. The mathematical question is therefore not how to assign reality to colored partons, but when a measurement on physical final states admits a controlled short-distance expansion in which partonic variables are legitimate internal coordinates.

50.1 Weighted Cross Sections and Measurement Functions

Let $|i\rangle$ be an incoming scattering state and let X range over a complete family of outgoing physical states. A detector observable is a measurable function $F(X)$ of the final state. Its weighted inclusive cross section has the form

$$\sigma[F] = \frac{1}{\mathcal{N}_i} \sum_X (2\pi)^D \delta^{(D)}(P_i - P_X) |\langle X|T|i\rangle|^2 \, F(X), \tag{50.1}$$

where $\mathcal{N}_i$ is the incoming flux normalization and T is the transition operator defined by $S = \mathbf{1} + \mathrm{i}T$. Equation (50.1) is stated in terms of physical states. A perturbative partonic formula is a further expansion of the same quantity, after an infrared regulator and an ultraviolet renormalization scheme have been chosen:

$$\sigma_{\text{part}}[F] = \sum_{n \geq 0} \int \mathrm{d}\Phi_n \, |\mathcal{M}_n(p_1, \ldots, p_n)|^2 \, F_n(p_1, \ldots, p_n). \tag{50.2}$$

Here $\mathrm{d}\Phi_n$ is the n-body Lorentz-invariant phase-space measure, $\mathcal{M}_n$ is a renormalized partonic matrix element, and F_n is the partonic representative of the detector measurement. The functions F_n must satisfy compatibility conditions at the boundaries of phase space where a parton becomes soft or two partons become collinear.

Definition 50.1 (Infrared and collinear safety)**.** A family of measurement functions $F = \{F_n\}_{n \geq 0}$ is infrared and collinear safe, abbreviated IRC safe, if the following limits hold on the phase-space regions relevant to the observable:

$$\lim_{\lambda \downarrow 0} F_{n+1}(p_1, \ldots, p_n, \lambda q) = F_n(p_1, \ldots, p_n), \tag{50.3}$$

$$\lim_{\theta \downarrow 0} F_{n+1}(p_1, \ldots, zp, (1-z)p, \ldots, p_n) = F_n(p_1, \ldots, p, \ldots, p_n), \qquad 0 < z < 1. \tag{50.4}$$

The convergence is required with enough uniformity that the difference between the two sides is integrable against the soft and collinear factorization kernels of the theory at the perturbative order under consideration. Equivalently, near a single unresolved boundary and away from other unresolved boundaries, there must be constants $C, \alpha > 0$, depending on the compact set of resolved momenta but not on the unresolved scale, such that

$$|F_{n+1}(p_1, \ldots, p_n, \lambda q) - F_n(p_1, \ldots, p_n)| \leq C\lambda^\alpha, \tag{50.5}$$

$$|F_{n+1}(p_1, \ldots, zp, (1-z)p, \ldots, p_n) - F_n(p_1, \ldots, p, \ldots, p_n)| \leq C\theta^\alpha, \tag{50.6}$$

where θ is the splitting angle. On corners where several emissions are unresolved, the same estimate is required after replacing each unresolved cluster by its reduced soft or collinear configuration. These estimates make the products with $\mathrm{d}\lambda/\lambda$ and $\mathrm{d}\theta^2/\theta^2$ locally integrable after the real–virtual subtraction.

Controlled Approximation 50.2 (Fixed-order cancellation for IRC-safe measurements)**.** Consider an inclusive observable in a massless gauge theory for which the renormalized real and virtual matrix elements admit the soft and collinear boundary factorizations displayed below, with remainders locally integrable after subtracting the reduced lower-multiplicity channel. If the measurement family F is IRC safe in the sense of Definition 50.1, then real-emission singularities and virtual singularities cancel locally on the unresolved phase-space boundaries after the KLN sum over degenerate final states. With incoming hadrons, the remaining initial-state collinear singularities are absorbed into the renormalized parton distribution functions.

The derivation is a fixed-order perturbative one and uses the displayed factorization formulae as input. It is not a nonperturbative theorem about arbitrary jet measurements. In a soft region, with an extra gluon momentum $k = \lambda q$, the real matrix element factorizes as

$$|\mathcal{M}_{n+1}|^2 \, \mathrm{d}\Phi_{n+1} = |\mathcal{M}_n|^2 \, \mathrm{d}\Phi_n \left[\sum_{i,j} \mathcal{S}_{ij}(q) \right] \frac{\mathrm{d}\lambda}{\lambda} \mathrm{d}\Omega_q + R_{\text{soft}}.$$

The remainder R_{soft} is locally integrable by the hypothesis of the controlled approximation. The eikonal factor $\mathcal{S}_{ij}$ is the same factor that appears with opposite sign in the virtual soft correction to the n-parton channel. Equation (50.3) replaces the real measurement by F_n on the singular part of the integral, so the real and virtual soft integrals have the same weight and cancel.

In a final-state collinear region, two momenta approach

$$p_a = zp + r, \qquad p_b = (1-z)p - r, \qquad r_\perp^2 \to 0.$$

The real matrix element and phase space factorize as

$$|\mathcal{M}_{n+1}|^2 \, \mathrm{d}\Phi_{n+1} = |\mathcal{M}_n(\ldots, p, \ldots)|^2 \, \mathrm{d}\Phi_n \, \frac{\alpha_s}{2\pi} P_{ab\leftarrow c}(z) \frac{\mathrm{d}r_\perp^2}{r_\perp^2} \mathrm{d}z + R_{\text{coll}}.$$

The remainder R_{coll} is locally integrable by the hypothesis of the controlled approximation. Equation (50.4) gives the same measurement weight to the unresolved pair and to its parent momentum. The singular integral then cancels against the corresponding virtual collinear correction in the KLN sum. If the collinear parton is in the initial state of a hadron collision, the same factorization kernel multiplies the operator renormalization of the parton distribution function rather than canceling inside the hard partonic subprocess. The finite hard coefficient is obtained after this mass factorization step.

Examples of IRC-safe observables include the total e^+e^- hadronic cross section, thrust

$$T(X) = \max_{\widehat{n}} \frac{\sum_{h\in X} |\vec{p}_h \cdot \widehat{n}|}{\sum_{h\in X} |\vec{p}_h|},$$

the C-parameter, jet masses defined with an IRC-safe algorithm, and the energy correlators of Chapter 49. Observables that measure an identified hadron, a charged-particle multiplicity, or a single unsummed collinear fragment require additional nonperturbative input such as fragmentation functions or detector-specific response functions.

50.2 Sterman–Weinberg Jets

The original two-jet observable of Sterman and Weinberg is a clean example of the preceding definition. In e^+e^- annihilation at center-of-mass energy Q, choose two parameters

$$0 < \epsilon_{\text{out}} < 1, \qquad 0 < \delta_{\text{cone}} < \pi.$$

An event is counted as a two-jet event if there exist two cones of half-angle δ_{cone} such that the total energy outside the two cones is at most $\epsilon_{\text{out}} Q$. The corresponding measurement function is

$$F_{\text{SW}}(X) = \begin{cases} 1, & \text{if } X \text{ satisfies the two-cone energy condition,} \\ 0, & \text{otherwise.} \end{cases}$$

Soft safety follows because adding a particle of energy λQ changes the out-of-cone energy by at most λQ. Away from the boundary where the out-of-cone energy is exactly $\epsilon_{\text{out}} Q$, the classification is unchanged for sufficiently small λ. The boundary has zero measure in the phase-space integral, and the unresolved soft singularity is integrated with the same measurement weight as the event without the soft particle.

Collinear safety follows because a sufficiently collinear splitting of a momentum inside one of the cones remains inside the same cone, while a splitting outside the cones preserves the total out-of-cone energy in the collinear limit. The singular collinear phase-space integral therefore sees the parent momentum measurement.

For very small $\epsilon_{\rm out}$ and $\delta_{\rm cone}$, the fixed-order series contains logarithms such as

$$\alpha_s^m \log^r \epsilon_{\rm out} \log^s \delta_{\rm cone}, \qquad r+s \le 2m.$$

The two-jet cross section is then still an IRC-safe observable, but the useful short-distance expansion is a resummed expansion rather than a fixed-order polynomial in $\alpha_s(Q)$.

50.3 Sequential Recombination Jet Algorithms

Modern collider jet definitions are usually algorithmic. For a hadron collider, let p_{Ti} be transverse momentum, y_i rapidity, and ϕ_i azimuth. Define

$$\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2.$$

The generalized k_T family with radius R and exponent p uses

$$d_{ij} = \min(p_{Ti}^{2p}, p_{Tj}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \qquad d_{iB} = p_{Ti}^{2p}. \tag{50.7}$$

At each step the smallest number among all d_{ij} and d_{iB} is selected. If it is d_{ij}, the two objects are recombined, usually by adding their four-momenta. If it is d_{iB}, object i is declared a jet and removed from the list. The cases

$$p = 1, \qquad p = 0, \qquad p = -1$$

give the k_T, Cambridge–Aachen, and anti-k_T algorithms, respectively.

Figure 50.1 shows the geometry behind (50.7). The same rapidity–azimuth event can produce a circular hard catchment region for anti-k_T, while the k_T metric permits soft constituents to precluster and therefore gives a catchment boundary controlled by soft radiation.

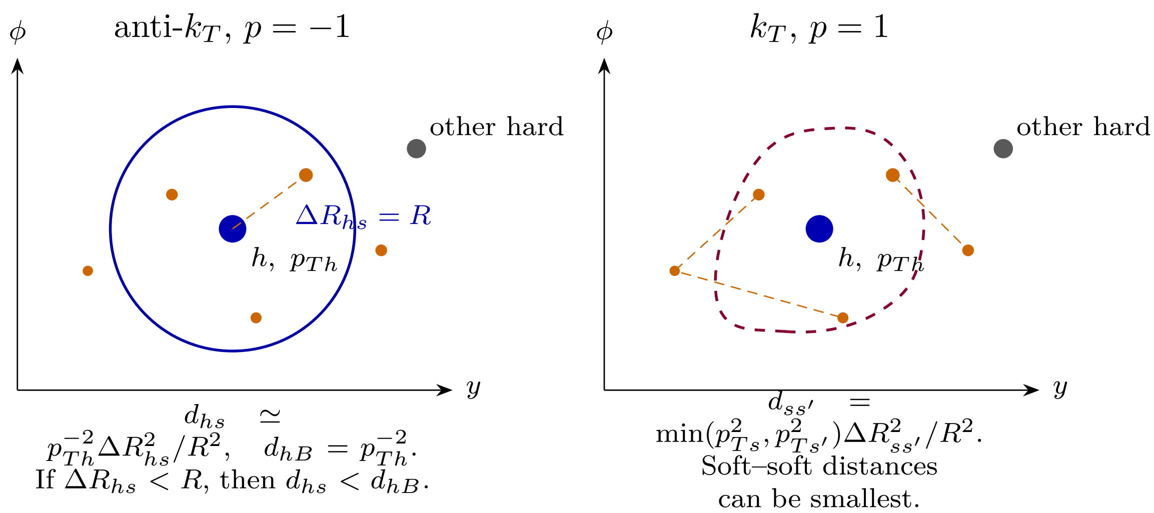


Figure 50.1: Generalized k_T clustering in the rapidity–azimuth plane. For anti-k_T, a soft constituent inside $\Delta R < R$ clusters with the hard particle before the hard particle reaches the beam distance, producing an approximately circular hard catchment region. For k_T, soft–soft distances may be smallest, so soft preclusters can control the eventual jet boundary even for the same event.

For $p = 1, 0, -1$, the generalized k_T clustering algorithm with four-momentum recombination defines IRC-safe jet momenta and IRC-safe cross sections after imposing a finite jet transverse-momentum threshold. For a collinear splitting $p \to zp, (1-z)p$, the angular distance satisfies $\Delta R_{ab} \to 0$, so $d_{ab} \to 0$. The two collinear objects are therefore recombined before they can affect finite-angle clustering, and four-momentum recombination restores the parent momentum.

For a soft momentum q, the effect of recombination with a hard object changes the hard jet momentum by $O(q)$. If the soft object is clustered among other soft objects or declared as a separate jet, a finite jet transverse-momentum threshold removes it in the soft limit. In the anti-k_T case, a soft object within radius R of a hard jet has pair distance controlled by the hard transverse momentum, so it is pulled into the hard jet without moving the hard boundary at leading power. The k_T and Cambridge–Aachen cases may cluster soft radiation differently, but the resulting change of any finite-energy jet momentum vanishes with the soft energy. Thus the measurement functions associated with the algorithm obey (50.3) and (50.4).

The anti-k_T algorithm is widely used because hard jets acquire stable, approximately circular catchment areas in the rapidity–azimuth plane while retaining IRC safety. Cone algorithms require more care. A seedless infrared-safe cone algorithm can be defined by searching over all stable cones. A seed-based cone algorithm with a transverse-momentum seed threshold can fail soft safety: adding an arbitrarily soft seed may create a stable cone that was absent before, changing the hard jets by a finite amount.

50.4 Factorization and Resummation

IRC safety makes fixed-order perturbation theory finite, but it does not make every region of the observable distribution well approximated by a few fixed orders. When an event shape $e(X)$

is small, the final state is constrained to contain narrow energetic jets plus soft radiation. A factorized description separates the scales, but the separation is not a statement that the exact QCD Hilbert space has literally split into independent colored subspaces. It is a statement about a controlled expansion of a gauge-invariant weighted cross section in which the short-distance coefficient and the long-distance matrix elements are separately defined after a regulator, an overlap-subtraction prescription, and a renormalization scheme are chosen.

For a two-jet event shape in e^+e^-, a leading singular factorized form has the convolution form

$$\frac{\mathrm{d}\sigma}{\mathrm{d}e} = H(Q,\mu)\int \mathrm{d}s_1\,\mathrm{d}s_2\,\mathrm{d}k\,J(s_1,\mu)J(s_2,\mu)S(k,\mu)\,\delta\!\left(e - \frac{s_1+s_2}{Q^2} - \frac{k}{Q}\right) + O(e^0). \tag{50.8}$$

The displayed $O(e^0)$ denotes terms that are nonsingular as distributions at the two-jet endpoint; it is not an error estimate until the observable, test function, and endpoint scaling region have been specified. The hard function H is the squared Wilson coefficient of the QCD current matched onto an effective current. The jet functions J are regulated collinear matrix elements of gauge-invariant collinear fields. The soft function S is a regulated vacuum matrix element of soft Wilson lines with the soft measurement operator inserted. Their renormalization-group equations resum logarithms of the separated scales.

Soft-collinear effective theory gives a systematic operator language for factorizations of the form (50.8). Its assumptions are visible in the operator definitions: mode scaling, Wilson-line decoupling at leading power, measurement factorization, and a power expansion in the small event-shape parameter. The same logic applies to hadron-collider observables, with the additional parton distribution functions needed for incoming hadrons and with factorization-breaking channels treated as separate hypotheses.

50.4.1 Mode Expansion and Logarithmic Accuracy

For a two-jet observable with small dimensionless event shape e, choose lightlike vectors $n, \bar{n}$ with $n \cdot \bar{n} = 2$ and write momenta as

$$p^\mu = \frac{n\cdot p}{2}\bar{n}^\mu + \frac{\bar{n}\cdot p}{2}n^\mu + p_\perp^\mu.$$

The SCET expansion parameter λ is fixed by the measured region. A mode is not an exact particle species; it is a coordinate chart on momentum regions, implemented in a regulator by smooth cutoffs or by an equivalent subtraction prescription. For thrust-like two-jet kinematics, $e \sim \lambda^2$, and the momentum scalings are

$$p_n^\mu \sim Q(\lambda^2, 1, \lambda), \qquad p_{\bar{n}}^\mu \sim Q(1, \lambda^2, \lambda), \qquad p_s^\mu \sim Q(\lambda^2, \lambda^2, \lambda^2),$$

where the three entries denote $(n\cdot p, \bar{n}\cdot p, p_\perp)$. A leading-power factorization theorem is a statement that, for $e \ll 1$ and away from additional measured hierarchies,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}e} = H \otimes J_n \otimes J_{\bar{n}} \otimes S + O(\lambda) \quad \text{or} \quad O(e^{1/2}),$$

with the precise power depending on the measurement and recoil convention. The symbol $\otimes$ denotes the convolution variables in the displayed factorization formula, not an abstract tensor product. The hard function is a Wilson coefficient of the QCD current matched onto SCET operators; the jet functions are matrix elements of collinear fields with collinear Wilson lines; the soft function is a vacuum matrix element of soft Wilson lines with the measurement operator inserted.

Definition 50.3 (SCET operator specification for a two-jet endpoint)**.** A two-jet SCET operator specification for a color-singlet source consists of:

1. two lightlike directions $n, \bar{n}$ with $n \cdot \bar{n} = 2$, a small parameter λ, and momentum regions with the scaling displayed above;
2. gauge-covariant collinear fields in the n and $\bar{n}$ sectors, with collinear Wilson lines $W_n, W_{\bar{n}}$ along the conjugate lightlike directions, and gauge-invariant building blocks such as
$$\chi_n := W_n^\dagger \xi_n, \qquad \mathcal{B}^\mu_{n\perp} := W_n^\dagger \, \mathrm{i} D^\mu_{n\perp} W_n$$
where $D_{n\perp} = \partial_{n\perp} - \mathrm{i} A_{n\perp}$ uses the coupling-absorbed Yang–Mills connection of Chapter 45. Thus the normalization of $\mathcal{B}_{n\perp}$ is fixed by the same gauge-field coordinate that appears in $-\frac{1}{4g^2_{\mathrm{YM}}} \operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$;
3. soft Wilson lines $Y_n, Y_{\bar{n}}$ in the representations carried by the energetic charges, with their future- or past-pointing orientation fixed by whether the corresponding charge is outgoing or incoming;
4. a matched current, for example
$$J_{\mathrm{QCD}}(0) = C(Q, \mu) \, \bar{\chi}_{\bar{n}}(0) \Gamma \chi_n(0) + O(\lambda),$$
before the soft Wilson-line decoupling is applied;
5. renormalized jet and soft matrix elements, with the two-jet soft matrix element having the Wilson-line form
$$S(k, \mu) = \frac{1}{N_c} \operatorname{tr} \langle \Omega | \overline{\mathrm{T}} \Big(Y_{\bar{n}}^\dagger Y_n \Big) \, \delta\Big(k - \widehat{k}_s\Big) \, \mathrm{T} \Big(Y_n^\dagger Y_{\bar{n}} \Big) \, |\Omega\rangle_\mu,$$
where $\widehat{k}_s$ is the soft measurement operator appropriate to the event shape;
6. an overlap-subtraction prescription, often called a zero-bin subtraction, preventing the same momentum region from being counted both as soft and as the soft limit of a collinear mode;
7. rapidity-regulator and rapidity-renormalization data whenever the Wilson-line geometry produces rapidity divergences not removed by ordinary ultraviolet renormalization;
8. a statement about Glauber exchange: cancellation, absorption into an additional operator, or failure of the proposed factorization.

All entries are part of the definition of the factorized approximation. The symbols H, J, S in (50.8) have no invariant meaning until these choices, including their renormalization scheme, have been specified.

Overlap subtraction as finite inclusion–exclusion. The zero-bin entry in Definition 50.3 is a mathematical subtraction, not a phrase meaning that an integral may be ignored. At finite regulator, let Λ be a finite set of momentum cells and let $\Lambda_n, \Lambda_s \subset \Lambda$ denote the cells represented by the leading n-collinear and soft approximations. Their common soft-collinear corner is

$$\Lambda_0 = \Lambda_n \cap \Lambda_s.$$

Let f_ℓ be the value of the measurement test function on cell ℓ, and let $I_n(\ell)$, $I_s(\ell)$, and $I_0(\ell)$ be the regulated leading collinear, leading soft, and common overlap densities on their domains. The finite overlap-subtracted singular coordinate is

$$I_{\mathrm{scet}}[f] = \sum_{\ell\in\Lambda_n} I_n(\ell)f_\ell + \sum_{\ell\in\Lambda_s} I_s(\ell)f_\ell - \sum_{\ell\in\Lambda_0} I_0(\ell)f_\ell. \tag{50.9}$$

If $I_n = I_s = I_0$ on Λ_0, the overlap contribution is counted once, not twice. This is the finite inclusion–exclusion content of the zero-bin subtraction. In dimensional regularization some overlap integrals are scaleless and vanish after analytic continuation; that is a feature of that regulator and subtraction scheme, not a regulator-independent proof that the overlap was absent.

A change of overlap convention is a finite scheme change only when it is paired across the collinear and soft coordinates. If δI_0 is moved from the collinear subtraction into the soft coordinate, the two displayed sums in (50.9) change separately, but their sum changes by

$$-\sum_{\ell\in\Lambda_0} \delta I_0(\ell)f_\ell + \sum_{\ell\in\Lambda_0} \delta I_0(\ell)f_\ell = 0.$$

This is the finite algebraic reason that overlap subtraction data belong in the definition of the factorized approximation. The continuum theorem still requires estimates showing that the regulated partition, the overlap approximants, and the measurement remainder converge in the topology in which the cross section is tested.

Leading soft Wilson-line decoupling identity. At finite regulator and at leading power in the n-collinear sector, the soft connection appearing through $n\cdot D_s = n\cdot\partial - \mathrm{i}n\cdot A_s$ can be removed from the collinear kinetic operator by multiplication by a soft Wilson line. More precisely, if Y_n satisfies

$$n\cdot D_s\, Y_n = 0,$$

and the original collinear field is written $\xi_n = Y_n\xi_n^{(0)}$, then

$$\mathrm{i}n\cdot D_s\, \xi_n = Y_n\, \mathrm{i}n\cdot\partial\, \xi_n^{(0)}.$$

Thus the leading soft interactions are transferred from the collinear Lagrangian to Wilson lines in the external operators and measurement matrix elements.

Using the Leibniz rule,

$$n\cdot D_s(Y_n\xi_n^{(0)}) = (n\cdot D_s Y_n)\xi_n^{(0)} + Y_n\, n\cdot\partial\, \xi_n^{(0)}.$$

The first term vanishes by the defining differential equation for Y_n. Multiplying by i gives the displayed identity. In a finite-dimensional regulated path integral, the change of variables is multiplication by an invertible color matrix at each regulated spacetime point. With a gauge-covariant regulator its Berezin or bosonic Jacobian is a local gauge-invariant factor; in the factorized construction that factor is included in the same renormalized soft and collinear operator specification rather than treated as an extra physical assumption. The identity itself is algebraic and does not assert that subleading soft interactions or Glauber exchanges decouple.

Finite Glauber unitarity diagnostic. Glauber exchange is a factorization boundary because its momentum scaling is not removed by the leading soft Wilson-line identity above. At finite regulator one useful diagnostic is purely operator-theoretic. Let $\mathcal{H}_{\text{spec}}$ be the finite Hilbert space carrying the spectator and exchanged color coordinates that are not resolved by a proposed inclusive measurement, let ρ_{spec} be the corresponding reduced density matrix, and let U_G be the finite unitary generated by the regulated Glauber operator on this space. If the measurement operator is inclusive over these coordinates, so that it is represented by $M_{\text{incl}} = \mathbf{1}_{\text{spec}}$ on this factor, then

$$\operatorname{Tr}_{\text{spec}}\Big(M_{\text{incl}} U_G \rho_{\text{spec}} U_G^{\dagger}\Big) = \operatorname{Tr}_{\text{spec}} \rho_{\text{spec}}. \tag{50.10}$$

More generally, the same finite cancellation holds whenever the measured operator M commutes with U_G:

$$\operatorname{Tr}_{\text{spec}}\Big(M U_G \rho_{\text{spec}} U_G^{\dagger}\Big) = \operatorname{Tr}_{\text{spec}}\Big(U_G^{\dagger} M U_G \rho_{\text{spec}}\Big) = \operatorname{Tr}_{\text{spec}}(M \rho_{\text{spec}}). \tag{50.11}$$

Thus an inclusive color-singlet measurement may be insensitive to a unitary Glauber rotation even though the rotation is present at amplitude level. The same algebra also shows where a factorization assertion can fail. If the measurement projects onto a spectator or color-flow coordinate that does not commute with U_G, then $U_G^{\dagger} M U_G \neq M$, and the trace need not equal the value without the Glauber exchange. In that case the exchanged Glauber coordinate must either be included as an additional operator in the factorization structure or the proposed factorization statement is false for that measurement.

There is also a quantitative finite version, useful because real factorization proofs rarely give exact commutation at an intermediate regulator. For any bounded finite measurement M, define the Glauber remainder

$$\Delta_G(M, \rho_{\text{spec}}) = \operatorname{Tr}_{\text{spec}}\Big(M U_G \rho_{\text{spec}} U_G^{\dagger}\Big) - \operatorname{Tr}_{\text{spec}}(M \rho_{\text{spec}}). \tag{50.12}$$

Cyclicity gives the exact identity

$$\Delta_G(M, \rho_{\text{spec}}) = \operatorname{Tr}_{\text{spec}}\Big((U_G^{\dagger} M U_G - M)\rho_{\text{spec}}\Big). \tag{50.13}$$

Consequently, with $\|\cdot\|$ the operator norm and $\|\cdot\|_1$ the trace norm,

$$|\Delta_G(M, \rho_{\text{spec}})| \leq \|U_G^{\dagger} M U_G - M\| \, \|\rho_{\text{spec}}\|_1. \tag{50.14}$$

If $U_G = \exp(\mathrm{i}K_G)$, the Duhamel formula refines the first factor to

$$U_G^{\dagger} M U_G - M = \mathrm{i} \int_0^1 e^{-\mathrm{i}sK_G} [M, K_G] e^{\mathrm{i}sK_G} \, \mathrm{d}s,$$

and hence

$$\|U_G^{\dagger} M U_G - M\| \leq \|[M, K_G]\|.$$

The finite diagnostic therefore has an honest error coordinate: exact factorization is recovered when the measured algebra is invariant under the Glauber unitary, and approximate factorization requires a small commutator in the measurement norm relevant to the smeared cross section. This is the operator-theoretic content behind phrases such as "Glauber cancellation"; the continuum theorem must still supply the limiting K_G, the topology, and the bound.

Equations (50.10)–(50.11) are only the finite unitarity mechanism. A continuum QCD factorization proof must also identify the regulated Glauber region, show how the eikonal approximations and contour deformations isolate it, prove the required inclusive sum or commutation property for the actual measurement, and bound the remainder in the topology of the smeared cross section. Without these extra analytic steps, saying that Glauber exchanges "cancel" is merely a claim about a missing part of the factorization structure.

Hypothesis 50.4 (Leading-power SCET factorization structure)**.** A leading-power SCET factorization statement for an observable F consists of a small parameter λ, a list of modes with momentum scaling, a mode-separated operator basis through the declared order in λ, a renormalized matrix-element definition of each hard, jet, beam, and soft function, an overlap-subtraction prescription, a measurement decomposition on those modes, a statement that Glauber or other factorization-breaking exchanges either cancel or are represented by additional operators, and a remainder estimate in powers of λ on the kinematic region where the formula is used. If the formula is asserted as a theorem, the estimate must be stated in a topology on smeared cross sections or cumulants; otherwise it is a controlled approximation hypothesis.

Controlled Approximation 50.5 (Distributional leading-power factorization estimate)**.** The content of a leading-power factorization formula is an estimate on smeared physical cross sections, not an equality of pointwise functions. Assume that, for the chosen hard scale Q, the physical QCD final-state observable e determines a positive cross-section measure $\mathrm{d}\mu_Q(e)$ on an endpoint window W_Q. Let $\mathcal{B}_{\text{end}}$ be a Banach space of endpoint test functions $\varphi\colon W_Q \to \mathbb{C}$, with norm $\|\varphi\|_{\mathcal{B}_{\text{end}}}$, strong enough to control the singular distributions under discussion. The physical smeared functional is

$$\Sigma_Q^{\text{phys}}(\varphi) = \int_{W_Q} \varphi(e)\,\mathrm{d}\mu_Q(e). \tag{50.15}$$

A two-jet operator specification $\mathcal{D}_{\text{SCET}}$ as in Definition 50.3 defines, after the specified regulator, overlap subtraction, rapidity subtraction when needed, and renormalization scheme, a singular distributional coordinate $\mathrm{d}\mu^{\text{SCET}}_{Q,\text{sing}}(e;\mathcal{D}_{\text{SCET}})$. Its smeared functional is

$$\Sigma^{\text{SCET}}_{Q,\text{sing}}(\varphi;\mathcal{D}_{\text{SCET}}) = \langle H \otimes J_n \otimes J_{\bar n} \otimes S, \varphi\rangle_{\mathcal{D}_{\text{SCET}}}\,, \tag{50.16}$$

where the pairing includes the measurement convolution, the endpoint transform convention if one is used, and the subtraction prescription.

A controlled leading-power factorization claim on W_Q is the assertion that there exist an exponent $p > 0$, a scale-dependent endpoint parameter λ_W, and a remainder bound R_Q such that

$$\left|\Sigma_Q^{\text{phys}}(\varphi) - \Sigma^{\text{SCET}}_{Q,\text{sing}}(\varphi;\mathcal{D}_{\text{SCET}})\right| \le R_Q\,\|\varphi\|_{\mathcal{B}_{\text{end}}}, \qquad R_Q \le C_Q\,\lambda_W^p \tag{50.17}$$

for every $\varphi \in \mathcal{B}_{\text{end}}$ in the declared kinematic range. If the hard, jet, soft, or anomalous-dimension entries are perturbatively truncated, the displayed R_Q must be enlarged by the declared perturbative truncation remainder in the same norm. The common phrases "leading power" and "factorization theorem" therefore have a definite meaning only after the physical functional, test space, factorization structure, endpoint parameter, and remainder topology have all been specified.

The same formulation also fixes the status of finite scheme changes. In a scalar transform-space chart, a finite multiplicative coordinate change

$$H' = HR_H, \qquad \widetilde{J}'_n = \widetilde{J}_n R_n, \qquad \widetilde{J}'_{\bar n} = \widetilde{J}_{\bar n} R_{\bar n}, \qquad \widetilde{S}' = \widetilde{S} R_S \tag{50.18}$$

represents the same singular distribution only if

$$R_H R_n R_{\bar{n}} R_S = 1 \tag{50.19}$$

as an identity in the transform variable and scale. The anomalous dimensions then transform as

$$\gamma_i' = \gamma_i + \mu \frac{\mathrm{d}}{\mathrm{d}\mu} \log R_i, \qquad i = H, J_n, J_{\bar{n}}, S, \tag{50.20}$$

and the sum of the transformed anomalous dimensions still vanishes because

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu} \log(R_H R_n R_{\bar{n}} R_S) = 0.$$

In a mixing problem the same statement is a similarity-covariant statement about matrix kernels and path-ordered products. It is part of the coordinate structure of the factorized approximation, not an optional convention added after the formula has been written.

Renormalization of the hard, jet, and soft functions gives evolution equations. The resummation structure is clearest after a Laplace transform, or any equivalent transform, has converted the measurement convolution into an ordinary product. Write the transformed singular coordinate as

$$\widetilde{C}(\nu, Q; \mu) = H(Q, \mu)\, \widetilde{J}_n(\nu, Q, \mu)\, \widetilde{J}_{\bar{n}}(\nu, Q, \mu)\, \widetilde{S}(\nu, Q, \mu),$$

where ν is conjugate to the event-shape measurement. A scalar renormalization-group transport structure consists of anomalous dimensions

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu} \log H = \gamma_H, \qquad \mu \frac{\mathrm{d}}{\mathrm{d}\mu} \log \widetilde{J}_a = \gamma_{J_a}, \qquad \mu \frac{\mathrm{d}}{\mathrm{d}\mu} \log \widetilde{S} = \gamma_S,$$

all in the same subtraction and rapidity scheme, satisfying the consistency condition

$$\gamma_H + \gamma_{J_n} + \gamma_{J_{\bar{n}}} + \gamma_S = 0. \tag{50.21}$$

In a distributional formulation the same equation is an identity of convolution kernels. If color or operator mixing is present, the scalar exponentials below are replaced by path-ordered evolution matrices and the consistency condition is the corresponding statement that the product observable has no residual common-scale dependence.

For a scalar transport structure, define

$$U_i(\mu, \mu_i) = \exp\left\{ \int_{\log \mu_i}^{\log \mu} \gamma_i(\nu, Q; \mu')\, \mathrm{d} \log \mu' \right\}, \qquad i = H, J_n, J_{\bar{n}}, S.$$

The resummed singular coordinate is then

$$\begin{aligned} \widetilde{C}_{\mathrm{sing}}(\nu, Q) = {} & H(Q, \mu_H) U_H(\mu, \mu_H)\, \widetilde{J}_n(\nu, Q, \mu_J) U_{J_n}(\mu, \mu_J) \\ & \times \widetilde{J}_{\bar{n}}(\nu, Q, \mu_J) U_{J_{\bar{n}}}(\mu, \mu_J)\, \widetilde{S}(\nu, Q, \mu_S) U_S(\mu, \mu_S). \end{aligned} \tag{50.22}$$

Differentiating $\log \widetilde{C}_{\mathrm{sing}}$ with respect to the common scale $\log \mu$ gives the left-hand side of (50.21); hence the expression is independent of μ inside the accuracy to which the anomalous dimensions and matching functions have been supplied. The natural choices $\mu_H \sim Q$, $\mu_J \sim Q\sqrt{e}$, and $\mu_S \sim Qe$, or their transform-space analogues, are not extra physics assumptions. They are coordinate choices that keep the boundary functions free of the large logarithms subsequently generated by the evolution kernels. Finite subtraction-scheme changes may redistribute terms between the anomalous dimensions and the boundary functions. Such a redistribution is a valid coordinate change only when the hard, jet, and soft entries are changed together so that (50.21) and the product coordinate $\widetilde{C}_{\rm sing}$ are preserved to the declared order.

If L denotes the large logarithm of the measured hierarchy, logarithmic accuracy is a statement about a definite truncation of (50.22). At LL one keeps the cusp anomalous dimension and beta-function data needed to generate the leading tower in the exponent, with tree-level boundary functions. NLL adds the next tower by including the next cusp data, the first non-cusp anomalous dimensions, and the corresponding running-coupling terms. NNLL adds the next tower and the matching constants required by the declared convention. Primed labels such as NLL$'$ or NNLL$'$ mean that fixed boundary functions are included one order higher than the unprimed logarithmic transport. The labels are therefore incomplete unless the observable, the factorization theorem, the transform convention, the natural scales, the anomalous-dimension orders, the matching orders, and the matching prescription to fixed order are all listed. They are formal perturbative-accuracy statements in a logarithmic hierarchy, not numerical error bounds once the hierarchy is lost, a Landau-pole region is approached, or an unproved factorization remainder dominates.

50.4.2 Massive-Vector Sudakov Boundaries

High-energy jet measurements in the Standard Model involve a second short-distance boundary in addition to the QCD soft and collinear boundaries: massive electroweak gauge bosons can be soft or collinear relative to an energetic charged line while their physical masses stop the logarithmic integration. The correct object is a massive-vector factorization coordinate for a specified observable and charge sector. The physical mass M_V and the hard scale Q are part of the electroweak perturbative datum, while the QCD hadronic measurement remains a physical final-state observable as above.

Let R be a representation of the gauge factor carried by an energetic line, and define its quadratic charge by

$$T^a_R T^a_R = C_R\,\mathbf{1}_R.$$

Here T^a_R are the canonical generators after the gauge field has been canonically normalized, so $\alpha_V = g_V^2/(4\pi)$ is the coupling that appears in the emission probability. The corresponding convention is tied to the Yang–Mills normalization of Chapter 45: the path-integral gauge field is written with kinetic term $-\frac{1}{4g^2_{\rm YM}}\operatorname{tr} F_{\mu\nu}F^{\mu\nu}$, and canonical perturbative fields carry the coupling in the vertices.

Controlled Approximation 50.6 (Single-line massive-vector Sudakov chart)**.** Fix an energetic external charge in representation R, a hard scale Q, a massive vector boson of mass M_V, and a hierarchy $Q \gg M_V$. In the strongly ordered soft-collinear chart, use the transverse momentum $k_\perp$ and energy fraction z of an unresolved virtual vector boson as coordinates. Put

$$L_V = \log\frac{Q^2}{M_V^2}, \qquad y = \log\frac{Q^2}{k_\perp^2}, \qquad x = \log\frac{1}{z}.$$

The massive vector permits the logarithmic range

$$0 < y < L_V, \qquad 0 < x < \frac{y}{2}, \tag{50.23}$$

where the second inequality is the soft-collinear energy ordering $z > k_\perp/Q$. Approximating the emission measure by $(\alpha_V C_R/\pi)\,\mathrm{d}x\,\mathrm{d}y$, the one-line virtual Sudakov coordinate is

$$\Delta_R^{(V)}(Q, M_V) = \exp\left[-\frac{\alpha_V C_R}{\pi}\, A_V(L_V)\right], \qquad A_V(L_V) = \int_0^{L_V} \mathrm{d}y \int_0^{y/2} \mathrm{d}x = \frac{1}{4} L_V^2. \tag{50.24}$$

The approximation consists precisely of the eikonal numerator, the soft-collinear phase-space chart, fixed coupling, one unresolved massive vector, and a scalar charge channel.

The derivation of (50.24) is the finite phase-space geometry. The lower transverse-momentum boundary $k_\perp = M_V$ gives $y = L_V$, the upper boundary $k_\perp = Q$ gives $y = 0$, and the condition that the emitted vector have energy larger than its transverse momentum gives $x < y/2$. The area is therefore the triangle of base L_V and height $L_V/2$. The exponent in (50.24) is the virtual no-emission coordinate in this chart. A real-emission calculation cancels the same logarithmic coordinate only for measurements that sum over the corresponding unresolved massive-vector final states and over the charge channels connected by those emissions. In electroweak applications the incoming beam and the measured final state often select weak charge. The factorized object is then a matrix in the external charge density space, with entries built from products of generators T_i^a on the participating energetic lines. A scalar exponential such as (50.24) is the one-line diagonal representative of this more general charge-transport structure.

This is the clean way to place boosted and electroweak jet statements inside the monograph. A boosted massive-particle substructure analysis is a measurement on stable hadronic and leptonic final states, together with a declared resonance-pole or narrow-width approximation when an unstable particle label such as W, Z, t, or H is used. The small parameters are then listed explicitly, for example

$$\frac{m_{\text{obj}}}{p_T}, \qquad \frac{\Gamma_{\text{obj}}}{m_{\text{obj}}}, \qquad \frac{M_W}{Q}, \qquad \lambda_{\text{jet}},$$

as applicable to the chosen observable. The perturbative electroweak Sudakov transport, the QCD jet and soft functions, the resonance approximation, and the nonperturbative hadronization or track coordinates are separate entries in the prediction. Their remainders have different meanings and must be stated separately before a numerical prediction can be assigned a claimed accuracy.

Controlled Approximation 50.7 (Perturbative prediction for an IRC-safe jet observable)**.** A perturbative prediction for a jet observable consists of: the renormalized fixed-order hard functions through a stated order in α_s; the logarithmic accuracy of any resummation, such as LL, NLL, NNLL, or beyond; the factorization theorem and its power corrections; the renormalized PDFs or fragmentation functions when they enter; and a prescription for matching resummed and fixed-order regions without double counting.

50.4.3 Global and Non-Global Soft Measurements

A factorization statement for a small event shape is much simpler when the measurement is global, meaning that every soft direction is measured with a weight compatible with the same soft function. Jet vetoes, gaps between jets, and many substructure cuts are not global. Then a soft gluon emitted into an unmeasured angular region is not vetoed, but it changes the color source that can radiate a later softer gluon into the measured region. This is the mathematical origin of non-global logarithms. The point is not the name; it is that the measurement algebra is no longer generated by a single linear Wilson-line soft function.

The following finite model records the part of the argument that can be made without invoking an unregulated continuum angular integral. Fix finitely many angular cells $\mathcal{C}$ away from the hard Wilson-line directions. Let $\mathcal{M} \subset \mathcal{C}$ be the measured, vetoed cells and $\mathcal{U} = \mathcal{C} \setminus \mathcal{M}$ the unmeasured cells. For every ordered dipole (i, j), let $K_{ij}^c \geq 0$ be the cell-integrated eikonal emission kernel into cell c. In a continuum soft-dipole approximation, one representative regulated coordinate for this rate is

$$K_{ij}^c = \int_{\Omega_c} \frac{\mathrm{d}\Omega_k}{4\pi} \frac{\alpha_s C_{\mathrm{dip}}}{\pi} \frac{1 - \widehat{n}_i \cdot \widehat{n}_j}{(1 - \widehat{n}_i \cdot \widehat{k})(1 - \widehat{k} \cdot \widehat{n}_j)}$$

after the collinear neighborhoods of i and j and the ultraviolet coupling convention have been specified. In the finite model the numbers K_{ij}^c are simply positive regulator coordinates.

Let $G_{ij}(L)$ denote the finite soft-dipole gap coordinate for the dipole (i, j), where L is the logarithm of the resolved soft-energy hierarchy, and impose the initial condition $G_{ij}(0) = 1$. The finite non-global evolution equation is

$$\frac{\mathrm{d}}{\mathrm{d}L} G_{ij}(L) = -A_{ij} G_{ij}(L) + \sum_{u \in \mathcal{U}} K_{ij}^u \big(G_{iu}(L) G_{uj}(L) - G_{ij}(L)\big), \qquad A_{ij} := \sum_{m \in \mathcal{M}} K_{ij}^m. \tag{50.25}$$

The first term is direct veto radiation into the measured cells. The second term is real–virtual balanced for radiation into an unmeasured cell: if every gap coordinate is identically one, it contributes zero. Its nonlinear product is the essential non-global structure: after an unmeasured emission at direction u, the original dipole is replaced by the two dipoles (i, u) and (u, j), each of which can subsequently radiate into $\mathcal{M}$.

Differentiating (50.25) at $L = 0$ gives a useful diagnostic. Since $G_{ij}(0) = 1$,

$$G'_{ij}(0) = -A_{ij}.$$

A second differentiation gives

$$G''_{ij}(0) = A_{ij}^2 - \sum_{u \in \mathcal{U}} K_{ij}^u \big(A_{iu} + A_{uj} - A_{ij}\big).$$

Therefore

$$G_{ij}(L) = 1 - A_{ij} L + \frac{L^2}{2} \left[A_{ij}^2 - \mathcal{N}_{ij}\right] + O(L^3), \qquad \mathcal{N}_{ij} := \sum_{u \in \mathcal{U}} K_{ij}^u \big(A_{iu} + A_{uj} - A_{ij}\big). \tag{50.26}$$

The $A_{ij}^2/2$ term is the square of the direct global Sudakov veto. The term $\mathcal{N}_{ij}$ is the first finite non-global coefficient. It vanishes when there is no unmeasured region, and it also vanishes in the special additive case $A_{iu} + A_{uj} = A_{ij}$ for every unmeasured cell u. A generic angular veto does not satisfy this identity. Thus the failure of globalness is not a cosmetic complication in logarithmic counting; it changes the evolution problem from a linear anomalous dimension to a nonlinear Wilson-line measurement problem.

Equation (50.25) is not a theorem about full QCD. It is a finite large-N_c, strongly energy-ordered, soft-dipole coordinate system for one obstruction to naive factorization. Beyond this finite model one must specify the color density matrix at finite N_c, the regulator for collinear neighborhoods of the hard lines, the matching to the hard and collinear functions, the treatment of recoil, and the status of Glauber exchanges. Super-leading logarithms, when present, are not explained by the finite non-global coefficient alone; they signal an additional factorization obstruction that must be stated in the operator structure of Hypothesis 50.4.

50.5 Jet Substructure Observables and Grooming

Jet substructure observables are measurements on the constituents of a previously identified jet. The jet definition, recombination scheme, grooming algorithm, and substructure measurement are all part of the observable. Their IRC behavior must therefore be checked as a combined map from the final-state calorimetric measure to a number or distribution.

Definition 50.8 (Soft drop)**.** Start with the constituents of a jet of radius R_0, recluster them with the Cambridge–Aachen metric, and recursively decluster the last clustering step into two branches i, j. Let

$$z_{ij} = \frac{\min(p_{Ti}, p_{Tj})}{p_{Ti} + p_{Tj}}, \qquad \theta_{ij} = \frac{\Delta R_{ij}}{R_0}.$$

For parameters $z_{\mathrm{cut}} > 0$ and $\beta_{\mathrm{SD}} \geq 0$, the soft-drop condition is

$$z_{ij} > z_{\mathrm{cut}}\, \theta_{ij}^{\beta_{\mathrm{SD}}}.$$

If the condition fails, the softer branch is removed and the recursion continues on the harder branch. If the condition holds, the remaining constituents define the groomed jet. The case $\beta_{\mathrm{SD}} = 0$ is the modified mass-drop tagger.

IRC classification of soft-drop outputs. With an IRC-safe input jet algorithm and a finite transverse-momentum threshold for the original jet, the following statements hold away from the boundary hypersurfaces of the measurement.

1. If $\beta_{\mathrm{SD}} > 0$, the soft-drop groomed four-momentum is IRC safe.
2. If $\beta_{\mathrm{SD}} = 0$, the groomed four-momentum is not collinear safe as a standalone observable.
3. For $\beta_{\mathrm{SD}} = 0$, any claimed IRC safety statement must name the measured functional of the groomed constituents, for example a groomed jet mass or an energy-correlation shape, and must check the collinear limit of that functional rather than the groomed four-vector itself.

Add a particle of transverse momentum λp_T with $\lambda \downarrow 0$. If it is tested against a hard branch at finite angle, then $z_{ij} = O(\lambda)$, while $z_{\mathrm{cut}}\theta_{ij}^{\beta_{\mathrm{SD}}}$ has a positive lower bound on compact subsets away from $\theta_{ij} = 0$. The soft branch is removed and cannot change the groomed four-momentum except on the codimension-one surface where the inequality is exactly saturated. If the soft particle is collinear to a hard branch, Cambridge–Aachen recombines or declusters it at small θ, and its contribution to the branch momentum vanishes with λ.

For a collinear splitting of a hard particle, the two branches have $\theta_{ij} \downarrow 0$. If $\beta_{\mathrm{SD}} > 0$, then

$$z_{\mathrm{cut}}\theta_{ij}^{\beta_{\mathrm{SD}}} \longrightarrow 0.$$

For every fixed splitting fraction $0 < z < 1$, the soft-drop condition is therefore eventually satisfied, and the collinear pair is retained as the parent four-momentum. This gives the collinear limit of the groomed four-vector.

If $\beta_{\rm SD} = 0$, the right-hand side of the soft-drop condition is the fixed number $z_{\rm cut}$. Consider a massless one-particle jet of four-momentum p, and replace it by a collinear pair zp and $(1-z)p$ with $0 < z < z_{\rm cut} < 1/2$. Cambridge–Aachen declustering sees the two branches, the soft-drop condition fails, and the softer branch is removed. The groomed four-vector is $(1-z)p$, not p, even in the exact collinear limit. Hence the groomed four-vector is not collinear safe for $\beta_{\rm SD} = 0$.

This counterexample does not by itself classify every functional of the groomed constituents. A groomed mass, energy-correlation ratio, or tagged substructure variable can still be IRC safe, but its safety is a statement about that functional and its normalization. The check must be made at the level of Definition 50.1.

Finite-tree formulation and decision margins. The preceding definition hides a useful finite mathematical structure: before one takes continuum, perturbative, or detector limits, soft drop is a deterministic map on a finite Cambridge–Aachen declustering tree. Let T be a rooted binary tree whose leaves are the constituents of a fixed input jet. For every internal vertex v, let v_L, v_R be the two branches obtained by undoing one Cambridge–Aachen recombination. Write

$$E_{v,a} = \sum_{\ell \in v_a} p_{T\ell}, \qquad z_v = \frac{\min(E_{v,L}, E_{v,R})}{E_{v,L} + E_{v,R}}, \qquad \theta_v = \frac{\Delta R(v_L, v_R)}{R_0},$$

where the branch direction entering $\Delta R(v_L, v_R)$ is the four-momentum recombination direction of that branch. A fixed-tree soft-drop specification consists of T, the leaf momenta, a declared tie-breaking rule when $E_{v,L} = E_{v,R}$, and the parameters $(z_{\rm cut}, \beta_{\rm SD})$. The recursion starts at the root. At a vertex v, if

$$z_v > z_{\rm cut}\theta_v^{\beta_{\rm SD}},$$

the algorithm stops and retains precisely the leaves below v. If the condition fails, it moves to the harder child and deletes the softer child.

This finite-tree specification is the object whose soft and collinear limits must be tested before it is inserted into a partonic phase-space integral. Define the soft-drop decision margin and, on failed vertices, the harder-branch margin by

$$m_v^{\rm SD} = \left| z_v - z_{\rm cut}\theta_v^{\beta_{\rm SD}} \right|, \qquad m_v^{\rm hard} = \frac{|E_{v,L} - E_{v,R}|}{E_{v,L} + E_{v,R}}. \tag{50.27}$$

For a finite event with no zero-energy branch, all quantities z_v and θ_v are continuous functions of the leaf momenta on any chart where the Cambridge–Aachen tree T itself is unchanged. Hence, if every vertex visited by the recursion has $m_v^{\rm SD} > 0$, and every failed vertex has $m_v^{\rm hard} > 0$, there is a neighborhood of the leaf momenta in which the same vertices are visited, the same branches are deleted, and the same leaf set is retained. If the tree T is generated from the momenta rather than supplied as part of the input, one must additionally require positive Cambridge–Aachen clustering margins: every pairwise angular distance comparison that determines the tree must remain separated from its nearest competitor. These margins are the precise finite-event meaning of the phrase "away from soft-drop and clustering boundaries." At such a point any groomed observable that is a smooth function of the retained momenta has the same local soft or collinear behavior as the retained-momentum map specified by the fixed recursion. The boundary hypersurfaces $m_v^{\mathrm{SD}} = 0$, $m_v^{\mathrm{hard}} = 0$, and the Cambridge–Aachen distance tie hypersurfaces require separate distributional analysis and cannot be silently discarded when a singular measurement or endpoint resummation is being defined.

Measured mMDT functionals. The $\beta_{\mathrm{SD}} = 0$ counterexample above is deliberately a counterexample to the groomed four-vector. It is not a license to classify every measured functional of the groomed set by the same sentence. A useful finite-event test is the following. For a retained finite leaf set S, define the homogeneous angular moment

$$G_\kappa(S) = \frac{1}{E_S^2 R_0^\kappa} \sum_{\ell,m\in S} E_\ell E_m \, \theta_{\ell m}^\kappa, \qquad E_S = \sum_{\ell\in S} E_\ell, \qquad \kappa > 0, \tag{50.28}$$

where $\theta_{\ell m}$ is the angular distance and the diagonal terms vanish. In the one-prong collinear split used in the mMDT counterexample, with energies $(1-z)E$ and zE, $0 < z < z_{\mathrm{cut}}$, the mMDT recursion retains only the harder daughter. Therefore

$$G_\kappa(\{(1-z)E\}) = 0 = G_\kappa(\{E\})$$

in the exact collinear limit. If $\beta_{\mathrm{SD}} > 0$, the same pair is retained for sufficiently small opening angle θ, and

$$G_\kappa(\{(1-z)E, zE\}) = 2z(1-z)\left(\frac{\theta}{R_0}\right)^\kappa,$$

which tends to zero as $\theta \downarrow 0$. Thus a positive-angular-power shape such as (50.28) passes this elementary collinear test even when the groomed four-vector does not.

By contrast, the retained-energy fraction

$$r_E(S) = \frac{E_S}{E_{\mathrm{input}}}$$

is not collinear safe for mMDT as a standalone observable: the same exact collinear split changes r_E from 1 to $1-z$. This is the measurement-level lesson. For $\beta_{\mathrm{SD}} = 0$, a groomed observable is acceptable only after its own collinear boundary behavior is checked. Angular weights with positive powers can kill the deleted one-prong collinear branch; energy-retention observables cannot. Endpoint distributions of G_κ, groomed masses, or ratios built from them still require their own factorization and resummation data, because IRC safety of the measurement is only the first finiteness condition.

Fixed-coupling soft-drop mass radiator. A useful measured-observable calibration is the groomed jet-mass cumulative distribution in the single-emission soft-collinear chart. The calculation below is the logarithmic phase-space geometry that a factorization theorem must reproduce in its leading coordinate.

Controlled Approximation 50.9 (Soft-drop mass at fixed-coupling soft-collinear accuracy)**.** Consider one energetic parton of color factor C_i and one unresolved emission with energy fraction $z \ll 1$, normalized angle $\theta = \Delta R/R_0 \ll 1$, and groomed mass coordinate

$$\rho = \frac{m_{J,\mathrm{gr}}^2}{p_T^2 R_0^2} = z\theta^2$$

in this chart. Approximate the real-emission measure by

$$\frac{\alpha_s C_i}{\pi}\frac{\mathrm{d}\theta^2}{\theta^2}\frac{\mathrm{d}z}{z},$$

hold α_s fixed, ignore finite terms in the splitting function, recoil, boundary corrections at $z \sim 1$, clustering logarithms, non-global correlations, and power corrections in ρ, z_{cut}. The soft-drop condition is $z > z_{\mathrm{cut}}\theta^{\beta_{\mathrm{SD}}}$. With

$$L_\rho = \log\frac{1}{\rho}, \qquad L_z = \log\frac{1}{z_{\mathrm{cut}}}, \qquad b = \frac{\beta_{\mathrm{SD}}}{2},$$

the leading radiator is the area

$$A_\beta(L_\rho, L_z) = \int_0^\infty \mathrm{d}t \int_0^\infty \mathrm{d}u\, \mathbf{1}_{\{u+t<L_\rho\}}\, \mathbf{1}_{\{u<L_z+bt\}}, \tag{50.29}$$

where $t = \log(1/\theta^2)$ and $u = \log(1/z)$. Hence

$$\Sigma_{\mathrm{SD}}^{\mathrm{LL}}(\rho) = \exp\left[-\frac{\alpha_s C_i}{\pi} A_\beta(L_\rho, L_z)\right] \tag{50.30}$$

inside this controlled approximation.

The area in (50.29) is elementary but instructive. If $L_\rho \le L_z$, the mass constraint is stronger than the grooming constraint throughout the triangular region $u + t < L_\rho$, so

$$A_\beta(L_\rho, L_z) = \frac{1}{2}L_\rho^2.$$

If $L_\rho > L_z$, the two boundary lines

$$u = L_\rho - t, \qquad u = L_z + bt$$

intersect at

$$t_* = \frac{L_\rho - L_z}{1+b}.$$

The area is therefore

$$A_\beta(L_\rho, L_z) = L_z t_* + \frac{b}{2}t_*^2 + \frac{1}{2}(L_\rho - t_*)^2, \qquad t_* = \frac{L_\rho - L_z}{1+b}, \qquad L_\rho > L_z. \tag{50.31}$$

For mMDT, $\beta_{\mathrm{SD}} = 0$, this reduces to

$$A_0(L_\rho, L_z) = L_\rho L_z - \frac{1}{2}L_z^2, \qquad L_\rho > L_z. \tag{50.32}$$

Thus mMDT removes the soft double logarithm below the grooming scale: the radiator grows linearly in L_ρ, with coefficient fixed by $L_z = \log(1/z_{\mathrm{cut}})$. For $\beta_{\mathrm{SD}} > 0$, the coefficient of the asymptotic L_ρ^2 term is nonzero and depends continuously on β_{SD}; the grooming boundary still removes part of the soft triangle, but no longer removes the double-logarithmic growth completely.

Equations (50.30)–(50.32) should be read as a calibration of logarithmic phase space. A physical prediction for a groomed jet-mass distribution must replace this chart by a renormalized factorization structure: the hard production process, the jet function for the initiating parton, the groomed soft or collinear-soft function, overlap subtractions, running coupling, finite splitting functions, matching to fixed order, non-global and clustering effects where present, and the hadronization coordinate appropriate to the measured mass region.

Renormalized soft-drop mass factorization structure. The preceding radiator becomes a QCD prediction only after the measured groomed-mass coordinate has been embedded in a renormalized operator definition. Let Q denote the hard energy scale of the measured jet, for instance $Q = p_T R_0$ in a small-radius hadron-collider chart, and let

$$\rho = \frac{m_{J,\mathrm{gr}}^2}{Q^2}.$$

In the soft-collinear coordinates of Controlled Approximation 50.9, the emission that lies simultaneously on the mass boundary and on the soft-drop boundary satisfies

$$z\theta^2 = \rho, \qquad z = z_{\mathrm{cut}}\theta^{\beta_{\mathrm{SD}}}.$$

Solving these two equations gives the boundary scales

$$\theta_* = \left(\frac{\rho}{z_{\mathrm{cut}}}\right)^{1/(2+\beta_{\mathrm{SD}})}, \qquad z_* = z_{\mathrm{cut}}^{2/(2+\beta_{\mathrm{SD}})}\rho^{\beta_{\mathrm{SD}}/(2+\beta_{\mathrm{SD}})}. \tag{50.33}$$

Thus $z_*\theta_*^2 = \rho$ and $z_* = z_{\mathrm{cut}}\theta_*^{\beta_{\mathrm{SD}}}$. These identities are not a factorization theorem, but they determine the scales that any such theorem must see: ordinary collinear radiation carrying the measured jet mass has virtuality of order $Q^2\rho$, while radiation near the grooming boundary has light-cone components of order

$$p_{\mathrm{cs}}^- \sim Qz_*, \qquad p_{\mathrm{cs}}^+ \sim Q\rho, \qquad |p_{\mathrm{cs}\perp}| \sim Q\sqrt{\rho z_*}.$$

Consequently its invariant transverse scale is

$$\mu_{\mathrm{cs}} \sim Q\sqrt{\rho z_*}, \qquad \mu_J \sim Q\sqrt{\rho}, \qquad \mu_G \sim Qz_{\mathrm{cut}}, \qquad \mu_H \sim Q. \tag{50.34}$$

For $\beta_{\mathrm{SD}} = 0$, $z_* = z_{\mathrm{cut}}$ and $\mu_{\mathrm{cs}} \sim Q\sqrt{\rho z_{\mathrm{cut}}}$; this is the scale separation behind the single-logarithmic mMDT radiator in (50.32).

Definition 50.10 (Soft-drop groomed-mass factorization structure)**.** Fix the jet algorithm, recoil convention, recombination scheme, hard process, partonic channel i, subtraction scheme, and soft-drop parameters $(z_{\mathrm{cut}}, \beta_{\mathrm{SD}})$. A leading singular soft-drop groomed-mass factorization structure consists of:

1. a hard coefficient $H_i(Q, \mu)$ for the short-distance production of the measured energetic parton or color representation;
2. a global-soft veto coordinate $S_i^G(z_{\mathrm{cut}}, R_0, \mu)$ describing wide-angle soft radiation that is removed by grooming but affects the probability that the event lies in the selected groomed-jet sample;
3. a renormalized collinear jet distribution $J_i(\rho_J, \mu)$ and a renormalized collinear-soft distribution $S_{i,\mathrm{SD}}^{\mathrm{cs}}(\rho_{\mathrm{cs}}; z_{\mathrm{cut}}, \beta_{\mathrm{SD}}, R_0, \mu)$, both understood as distributions paired with endpoint test functions;
4. the measurement decomposition $\rho = \rho_J + \rho_{\mathrm{cs}}$ in the leading singular chart, together with the overlap-subtraction and rapidity-renormalization data needed to avoid double counting the common soft-collinear boundary;

5. anomalous dimensions, in the same scheme, satisfying the common-scale consistency condition
$$\gamma_H + \gamma_G + \gamma_J + \gamma_{\rm cs} = 0 \tag{50.35}$$
as an identity of transform-space functions or, before transform, of convolution kernels;
6. a remainder functional $R_{\rm SD}$ on the stated test-function space, recording power corrections in $\rho, z_{\rm cut}, R_0$, finite splitting functions beyond the declared singular order, recoil and clustering effects, non-global correlations not included in S_i^G, Glauber obstructions if the measurement is not inclusive enough, and nonperturbative coordinates when one of the scales in (50.34) approaches $\Lambda_{\rm QCD}$.

The corresponding singular coordinate is
$$\begin{aligned}\left\langle \frac{{\rm d}\Sigma^{\rm SD}_{i,\rm sing}}{{\rm d}\rho}, \varphi \right\rangle &= H_i(Q,\mu)\, S_i^G(z_{\rm cut}, R_0, \mu) \int {\rm d}\rho_J\, {\rm d}\rho_{\rm cs}\, J_i(\rho_J,\mu) S^{\rm cs}_{i,\rm SD}(\rho_{\rm cs},\mu)\varphi(\rho_J+\rho_{\rm cs}) \\ &\quad + \langle R_{\rm SD}, \varphi\rangle\,,\end{aligned} \tag{50.36}$$
for every endpoint test function φ in the declared domain. The symbols S_i^G, J_i, and $S^{\rm cs}_{i,\rm SD}$ do not have standalone physical meaning outside this common structure; finite scheme changes may move terms among them only if (50.35) and the paired distribution (50.36) are preserved.

Controlled Approximation 50.11 (Perturbative soft-drop mass resummation)**.** A perturbative resummation based on Definition 50.10 is controlled on a region of endpoint tests only when the scales $\mu_H, \mu_G, \mu_J, \mu_{\rm cs}$ in (50.34) remain in the perturbative regime, the factorization remainder $R_{\rm SD}$ is bounded in the declared topology, and the logarithmic-accuracy label specifies the anomalous dimensions, boundary matching functions, fixed-order matching prescription, rapidity scheme if present, and scale-profile map used to avoid crossing nonperturbative regions. If $\mu_{\rm cs}$ is of order $\Lambda_{\rm QCD}$, the collinear-soft distribution in (50.36) must be replaced by a nonperturbative shape coordinate or by a regulator-level reconstruction observable with its own matching theorem. A Monte-Carlo tune or an empirical model may supply a useful phenomenological coordinate, but it is then an additional input, not a derivation of the soft-drop mass distribution from perturbative QCD.

Groomed momentum sharing and Sudakov-conditioned coordinates. A second soft-drop observable, often denoted z_g, records the momentum sharing at the first declustering step that passes the soft-drop condition. For a finite Cambridge–Aachen tree in the notation above, let v_* be the first vertex visited by the recursion for which
$$z_{v_*} > z_{\rm cut}\theta_{v_*}^{\beta_{\rm SD}}.$$
If such a vertex exists, define
$$z_g = z_{v_*}, \qquad \theta_g = \theta_{v_*}.$$
If no such vertex exists, the pair (z_g, θ_g) is not a number attached to the event; one must either work with the event-selection indicator or declare a separate null value outside the numerical range. This elementary point is already enough to show that z_g is not an ordinary IRC-safe measurement in the sense of Definition 50.1: in the one-particle collinear limit there is no accepted split, while a nearby two-particle configuration may have a finite accepted value. The useful short-distance statement about z_g is therefore not fixed-order IRC safety. It is an all-order, Sudakov-conditioned statement inside a specified strongly ordered soft-collinear approximation.

Controlled Approximation 50.12 (Sudakov-conditioned groomed sharing)**.** Fix an initiating parton channel i, a normalized angle $\vartheta = \theta/R_0 \in (0,1]$, a soft-collinear ordering in decreasing ϑ, and a positive symmetrized splitting kernel $\overline{P}_i(z)$ on $0 < z \le 1/2$ in the declared subtraction scheme. In the fixed-coupling, independent-emission, strongly ordered chart, the accepted-emission rate is

$$\mathrm{d}\Gamma_i = a_i\,\overline{P}_i(z)\,\frac{\mathrm{d}\vartheta}{\vartheta}\,\mathrm{d}z\,\mathbf{1}_{\{z>z_{\mathrm{cut}}\vartheta^{\beta_{\mathrm{SD}}}\}}, \qquad a_i = \frac{\alpha_s C_i}{\pi}. \tag{50.37}$$

Let

$$\mathcal{P}_i(\zeta) = \int_\zeta^{1/2} \overline{P}_i(z)\,\mathrm{d}z, \qquad \mathcal{R}_i(\vartheta) = a_i \int_\vartheta^1 \frac{\mathrm{d}\vartheta'}{\vartheta'}\,\mathcal{P}_i\Big(z_{\mathrm{cut}}\vartheta'^{\beta_{\mathrm{SD}}}\Big), \qquad \Delta_i(\vartheta) = e^{-\mathcal{R}_i(\vartheta)}. \tag{50.38}$$

Here $\Delta_i(\vartheta)$ is the probability, inside this chart, that no emission at an angle larger than ϑ has passed grooming. The joint density of the first accepted split is then

$$\mathrm{d}\mathbb{P}_i(z_g,\vartheta_g) = \Delta_i(\vartheta_g)\,a_i\,\overline{P}_i(z_g)\,\frac{\mathrm{d}\vartheta_g}{\vartheta_g}\,\mathrm{d}z_g\,\mathbf{1}_{\{z_g>z_{\mathrm{cut}}\vartheta_g^{\beta_{\mathrm{SD}}}\}}. \tag{50.39}$$

This is a controlled approximation only after the kernel, coupling prescription, ordering variable, recoil convention, endpoint regularization, and matching to the rest of the factorization structure have been specified.

The derivation of (50.39) is the stopping-time identity for the ordered emission process. The probability of no accepted split before angle ϑ_g is $\Delta_i(\vartheta_g)$. The probability of one accepted split in the infinitesimal cell $(\vartheta_g, \vartheta_g + \mathrm{d}\vartheta_g) \times (z_g, z_g + \mathrm{d}z_g)$ is the rate (50.37). Multiple emissions in the same infinitesimal ordered cell are higher order in the cell volume. Multiplying these two factors gives (50.39). This argument does not replace a QCD factorization theorem; it is the finite probabilistic core that such a theorem must reproduce in its leading logarithmic coordinate.

The special case $\beta_{\mathrm{SD}} = 0$ exposes the mechanism transparently. Then $\mathcal{P}_i(z_{\mathrm{cut}}\vartheta^{\beta_{\mathrm{SD}}}) = \mathcal{P}_i(z_{\mathrm{cut}})$ is independent of ϑ, so

$$\Delta_i(\vartheta) = \vartheta^{a_i\mathcal{P}_i(z_{\mathrm{cut}})}.$$

If the idealized chart is allowed to run to arbitrarily small angle, the probability of eventually finding an accepted split is one, and integrating (50.39) over ϑ_g gives

$$p_i^{\mathrm{mMDT}}(z_g) = \frac{\overline{P}_i(z_g)}{\displaystyle\int_{z_{\mathrm{cut}}}^{1/2} \overline{P}_i(z)\,\mathrm{d}z}\,\mathbf{1}_{\{z_{\mathrm{cut}}<z_g\le 1/2\}}. \tag{50.40}$$

Indeed,

$$a_i\overline{P}_i(z_g)\int_0^1 \vartheta^{a_i\mathcal{P}_i(z_{\mathrm{cut}})}\,\frac{\mathrm{d}\vartheta}{\vartheta} = \frac{\overline{P}_i(z_g)}{\mathcal{P}_i(z_{\mathrm{cut}})}.$$

With a finite angular cutoff $\vartheta_{\min}$, the same expression is the conditional density after dividing by the accepted-event probability $1 - \Delta_i(\vartheta_{\min})$. The cancellation of a_i in (50.40) is not a statement that the observable is nonperturbative-free; it is a consequence of conditioning on the first accepted splitting inside this leading ordered chart. Running coupling, finite splitting functions, flavor changes, hadron masses, detector thresholds, and nonperturbative collinear dynamics all belong to the factorization and remainder data, not to the stopping-time identity alone.

For $\beta_{\mathrm{SD}} > 0$, a value $z_g < z_{\mathrm{cut}}$ can occur only at an angle satisfying

$$\vartheta_g < \left(\frac{z_g}{z_{\mathrm{cut}}}\right)^{1/\beta_{\mathrm{SD}}}.$$

The fixed-order expansion of the z_g distribution is then singular at the collinear boundary, while the all-order expression (50.39) may still define a finite distribution because $\Delta_i(\vartheta)$ suppresses the small-angle region. This is the precise sense, within the stated approximation, in which the coordinate is Sudakov-conditioned. It is a useful perturbative object only together with the Sudakov factor, the conditioning convention, and the remainder estimate for the measured jet sample.

Definition 50.13 (N-subjettiness)**.** For a jet with constituents k, choose N axes $\{a_1, \ldots, a_N\}$ by a specified IRC-safe axis prescription. For $\beta_\tau > 0$,

$$\tau_N^{(\beta_\tau)} = \frac{1}{p_{TJ} R_0^{\beta_\tau}} \sum_{k \in J} p_{Tk} \min_{a=1,\ldots,N} (\Delta R_{ka})^{\beta_\tau}.$$

Ratios such as $\tau_{21} = \tau_2/\tau_1$ are substructure coordinates only on the domain where the denominator is separated from zero; near the denominator boundary their distribution requires the same factorization and resummation data as the underlying τ_N.

The axis prescription is part of the observable. One prescription with a clean nonperturbative formulation is global minimization over axes. Let (Y, d) be the compact angular patch of a jet, with $\operatorname{diam} Y \le R_0$, and let μ be the positive transverse energy measure of the jet constituents on Y. For $M = \mu(Y) > 0$, define

$$\tau_{N,\min}^{(\beta_\tau)}(\mu) = \frac{1}{M R_0^{\beta_\tau}} \inf_{(a_1,\ldots,a_N)\in Y^N} \int_Y \min_{1\le r\le N} d(\mathbf{n}, a_r)^{\beta_\tau}\, \mathrm{d}\mu(\mathbf{n}). \tag{50.41}$$

For an event consisting of finitely many hadrons this is the usual N-subjettiness value with axes chosen by global minimization. The set of minimizing axes can have several components at symmetric configurations; the observable in (50.41) is the minimized value, not a chosen point in the minimizer set.

Lemma 50.14 (Continuity of minimized N-subjettiness)**.** *Fix $N \ge 1$, $\beta_\tau > 0$, $0 < m_0 < E < \infty$, and restrict to positive measures μ on Y with $m_0 \le \mu(Y) \le E$. The functional $\tau_{N,\min}^{(\beta_\tau)}$ of (50.41) is weakly continuous. If a soft constituent of transverse energy ε is added, then*

$$\left|\tau_{N,\min}^{(\beta_\tau)}(\mu + \varepsilon\delta_{\mathbf{m}}) - \tau_{N,\min}^{(\beta_\tau)}(\mu)\right| \le \frac{2\varepsilon}{m_0}. \tag{50.42}$$

If two constituents of total transverse energy E_{ab} and angular separation θ are replaced by their transverse-energy parent direction $\mathbf{n}_{ab}$, and if

$$d(\mathbf{n}_a, \mathbf{n}_{ab}) + d(\mathbf{n}_b, \mathbf{n}_{ab}) \le c\,\theta,$$

then

$$\left|\Delta\tau_{N,\min}^{(\beta_\tau)}\right| \le \frac{E_{ab}}{m_0 R_0^{\beta_\tau}}\, C_{\beta_\tau, R_0}\,(c\theta)^{\alpha_\tau}, \qquad \alpha_\tau = \min(\beta_\tau, 1), \tag{50.43}$$

where one may take $C_{\beta,R_0} = 1$ for $0 < \beta \le 1$ and $C_{\beta,R_0} = \beta R_0^{\beta-1}$ for $\beta \ge 1$.

Proof. For $A = (a_1, \dots, a_N) \in Y^N$, set

$$\Phi_A(\mathbf{n}) = \min_{1 \le r \le N} d(\mathbf{n}, a_r)^{\beta_\tau}.$$

The space Y^N is compact, and $A \mapsto \Phi_A$ is continuous as a map from Y^N to $C(Y)$ with the uniform norm. Therefore the infimum in (50.41) is attained. The family $\{\Phi_A : A \in Y^N\}$ is compact in $C(Y)$. If $\mu_j \to \mu$ weakly with uniformly bounded total mass, then

$$\sup_{A \in Y^N} \left| \int \Phi_A \, \mathrm{d}\mu_j - \int \Phi_A \, \mathrm{d}\mu \right| \longrightarrow 0.$$

Indeed, approximate the compact family $\{\Phi_A\}$ by a finite η-net in $C(Y)$, use weak convergence on the finite net, and then let $\eta \downarrow 0$. Taking infima over A is continuous under this uniform convergence, and division by $M = \mu(Y)$, bounded away from zero, gives weak continuity of $\tau_{N,\min}^{(\beta_\tau)}$.

For a soft addition, write

$$I(\mu) = \inf_A \int \Phi_A \, \mathrm{d}\mu.$$

Since $0 \le \Phi_A \le R_0^{\beta_\tau}$, one has

$$|I(\mu + \varepsilon\delta_{\mathbf{m}}) - I(\mu)| \le \varepsilon R_0^{\beta_\tau}.$$

Also $I(\mu) \le M R_0^{\beta_\tau}$. With $M' = M + \varepsilon$,

$$\left| \frac{I(\mu + \varepsilon\delta_{\mathbf{m}})}{M' R_0^{\beta_\tau}} - \frac{I(\mu)}{M R_0^{\beta_\tau}} \right| \le \frac{\varepsilon}{M'} + \frac{\varepsilon}{M'} \le \frac{2\varepsilon}{m_0}.$$

For collinear recombination, the total mass M is unchanged. The function $t \mapsto t^{\beta_\tau}$ on $[0, R_0]$ is α_τ-Hölder with the displayed constant:

$$|u^{\beta_\tau} - v^{\beta_\tau}| \le C_{\beta_\tau, R_0} |u - v|^{\alpha_\tau}.$$

For every axis tuple A,

$$|\Phi_A(\mathbf{n}_a) - \Phi_A(\mathbf{n}_{ab})| + |\Phi_A(\mathbf{n}_b) - \Phi_A(\mathbf{n}_{ab})| \le C_{\beta_\tau, R_0} (c\theta)^{\alpha_\tau},$$

up to replacing c by a harmless larger constant if the parent-direction bound is stated separately for each daughter. Hence the change in $\int \Phi_A \, \mathrm{d}\mu$ is bounded by $E_{ab} C_{\beta_\tau, R_0} (c\theta)^{\alpha_\tau}$, uniformly in A. Taking infima before and after recombination and dividing by $M R_0^{\beta_\tau}$, with $M \ge m_0$, gives (50.43). $\square$

Thus minimizing-axis N-subjettiness is an IRC-continuous calorimetric observable as a value functional. An algorithm that returns one selected minimizing axis may jump at degenerate events, but the minimized value used in the weighted cross section is continuous in the soft and collinear senses above. Ratios such as τ_{21} inherit this control only on regions where the denominator is bounded away from zero; otherwise their endpoint distribution is a separate singular observable.

Energy-correlation functions, defined in Chapter 49, provide a detector-based substructure alternative. They depend only on the positive calorimetric measure and angular kernels, whereas N-subjettiness also depends on an axis-finding map. Both are useful, but their mathematical inputs differ.

50.6 Energy Correlators as Calorimetric Coordinates

The cleanest way to formulate jet substructure without assigning physical status to partons is to regard an event as a positive energy measure on the celestial sphere or on a jet patch. For a final state X, write

$$\mu_X = \sum_{r \in X} E_r \, \delta_{\mathbf{n}_r}, \qquad \mathbf{n}_r = \frac{\mathbf{p}_r}{|\mathbf{p}_r|},$$

or, inside a jet of reference energy $E_{\rm ref}$,

$$\bar{\mu}_X = \frac{1}{E_{\rm ref}} \mu_X.$$

The total mass of $\bar{\mu}_X$ is one for a fully normalized jet and is at most one if acceptance cuts or grooming remove energy. A calorimetric observable is a function, distribution, or positive functional of this measure. The energy-flow detector of Chapter 49 is the operator whose spectral value on a hadronic out-state is precisely μ_X.

Definition 50.15 (Smeared energy-correlator functional)**.** Let $K \in C((S^2)^k)$. The K-smeared k-point energy-correlator functional of a finite positive Borel measure μ on S^2 is

$$\mathcal{C}_K^{(k)}(\mu) = \int_{(S^2)^k} K(\mathbf{n}_1, \ldots, \mathbf{n}_k) \, \mathrm{d}\mu(\mathbf{n}_1) \cdots \mathrm{d}\mu(\mathbf{n}_k). \tag{50.44}$$

If K is supported away from coincidence diagonals, this is a separated-angle detector product. If K meets a diagonal, the functional includes the contact part in which two detector insertions act on the same hadron. Both are part of the detector observable; removing a contact term is a different distributional convention.

The usual EEC is obtained by choosing a sequence of kernels

$$K_\Delta(\mathbf{n}_1, \mathbf{n}_2; \chi) \longrightarrow \delta(\cos\chi - \mathbf{n}_1 \cdot \mathbf{n}_2)$$

as angular distributions. Energy-correlation functions used for substructure, such as (49.278), are special cases with kernels built from powers of pairwise angular distances and with diagonal terms omitted or killed by positive powers of the distance.

Lemma 50.16 (Continuity of smeared energy correlators)**.** *Let K be Lipschitz on $(S^2)^k$, and restrict to measures with total mass at most E. Then $\mathcal{C}_K^{(k)}$ is continuous in the weak topology of finite measures. Moreover, if a final state is changed by adding a soft particle of energy ε, then*

$$|\Delta \mathcal{C}_K^{(k)}| \leq k \, \|K\|_\infty \, E^{k-1} \varepsilon + O(\varepsilon^2). \tag{50.45}$$

If two particles with total energy E_{ab} and angular separation θ_{ab} are replaced by their energy-weighted parent direction, then

$$|\Delta \mathcal{C}_K^{(k)}| \leq k \; \mathrm{Lip}(K) \, E^{k-1} E_{ab} \, O(\theta_{ab}) \tag{50.46}$$

on compact regions where the parent direction is defined.

Proof. Weak continuity follows because K is continuous on the compact space $(S^2)^k$, and product measures $\mu_j^{\otimes k}$ converge weakly to $\mu^{\otimes k}$ when $\mu_j \to \mu$ weakly with uniformly bounded total mass. This can be checked first on factorized kernels $K = f_1 \otimes \cdots \otimes f_k$, where the statement is the product of $\int f_i \, \mathrm{d}\mu_j \to \int f_i \, \mathrm{d}\mu$, and then extended to all continuous K by the Stone–Weierstrass theorem on $(S^2)^k$.

For a soft addition, write $\mu' = \mu + \varepsilon\delta_{\mathbf{m}}$. Expanding $(\mu+\varepsilon\delta_{\mathbf{m}})^{\otimes k} - \mu^{\otimes k}$, the terms linear in ε insert the new point in one of k slots and integrate the other $k-1$ slots against μ. Their absolute value is bounded by $k\|K\|_\infty E^{k-1}\varepsilon$; the remaining terms have at least two factors of ε.

For a collinear recombination, the signed measure difference is

$$\delta\mu = E_a\delta_{\mathbf{n}_a} + E_b\delta_{\mathbf{n}_b} - E_{ab}\delta_{\mathbf{n}_{ab}}, \qquad E_{ab} = E_a + E_b.$$

It has total mass zero. In each tensor slot, pairing $\delta\mu$ with the Lipschitz function obtained by fixing the other $k-1$ angles gives a bound by $\mathrm{Lip}(K)E_{ab}O(\theta_{ab})$. Summing over the k possible slots and bounding the other integrations by E^{k-1} gives (50.46). In terms with two or more insertions of $\delta\mu$, choose one insertion and use its zero total mass against the Lipschitz dependence on that slot; the remaining $\delta\mu$-insertions are bounded by their total variation, at most $2E_{ab}$. Thus every such term still carries the same $O(\theta_{ab})$ factor, with a constant depending on k, E, E_{ab}, and $\|K\|_\infty + \mathrm{Lip}(K)$. This proves the displayed IRC-continuity estimates. □

Theorem 50.17 (Energy-correlator polynomials separate calorimetric observables)**.** *Let*

$$\mathcal{M}_E(S^2) = \{\mu \geq 0 : \mu(S^2) \leq E\}$$

be the compact space of finite positive Borel measures on S^2, equipped with the weak topology. Let $\mathcal{A}_E$ be the real algebra generated by the linear energy moments

$$\mu \longmapsto \int_{S^2} f(\mathbf{n}) \, \mathrm{d}\mu(\mathbf{n}), \qquad f \in C(S^2).$$

Equivalently, $\mathcal{A}_E$ is the algebra of finite linear combinations of smeared energy correlators $\mathcal{C}_K^{(k)}$ with factorized kernels $K = f_1 \otimes \cdots \otimes f_k$. Then $\mathcal{A}_E$ is uniformly dense in $C(\mathcal{M}_E(S^2))$.

Proof. The space $\mathcal{M}_E(S^2)$ is compact in the weak topology. Indeed, positive measures with total mass at most E are a weak-$*$ closed subset of the radius-E ball in the dual of $C(S^2)$, and that ball is weak-$*$ compact by Banach–Alaoglu; because $C(S^2)$ is separable, this compact topology is metrizable. The algebra $\mathcal{A}_E$ contains constants by taking the empty product, and it is closed under multiplication because products of linear moments are generated by the algebra definition.

It separates points. If $\mu \neq \nu$, the signed measure $\mu - \nu$ is a nonzero continuous linear functional on $C(S^2)$. By the Riesz representation theorem there is an $f \in C(S^2)$ such that

$$\int f \, \mathrm{d}\mu \neq \int f \, \mathrm{d}\nu.$$

Thus the linear moment $M_f(\mu) = \int f \, \mathrm{d}\mu$ distinguishes μ and ν. The real Stone–Weierstrass theorem applies to the compact Hausdorff space $\mathcal{M}_E(S^2)$: a real subalgebra of $C(\mathcal{M}_E(S^2))$ that contains constants and separates points is uniformly dense. Hence $\mathcal{A}_E$ is dense.

A monomial

$$M_{f_1}(\mu)\cdots M_{f_k}(\mu)=\int_{(S^2)^k} f_1(\mathbf{n}_1)\cdots f_k(\mathbf{n}_k)\,\mathrm{d}\mu(\mathbf{n}_1)\cdots \mathrm{d}\mu(\mathbf{n}_k)$$

is exactly $\mathcal{C}^{(k)}_K(\mu)$ for the factorized kernel $K = f_1\otimes\cdots\otimes f_k$, by the defining property of the product measure $\mu^{\otimes k}$. Finite linear combinations of such monomials are therefore exactly the finite factorized-kernel energy-correlator polynomials, so the dense algebra is an energy-correlator algebra. □

The theorem is a finite-resolution statement. It applies to continuous functions of the calorimetric measure on the compact energy ball $\mathcal{M}_E(S^2)$. Sharp step-function cuts, exact jet-counting boundaries, and distributional angular kernels are obtained as limits of continuous observables, and their boundary terms must be analyzed separately. Graph-indexed energy-flow polynomials used in jet physics choose a concrete family of angular kernels built from pairwise distances. Treating that smaller family as a practical basis requires an additional approximation statement for angular kernels on the detector resolution and symmetry class being used. The object-level reason for their robustness is the theorem above: energy correlators are coordinates on the positive energy measure itself, not on a partonic history.

50.7 Shape Functions and Track Functions

When a measured event shape is of order Λ_{QCD}/Q, the soft function in (50.8) contains nonperturbative matrix elements. The endpoint statement has two logically separate pieces. First one must define a soft operator coordinate. Then one may state a factorized approximation that convolves this coordinate with a short-distance coefficient in a specified endpoint topology.

Definition 50.18 (Leading shape-function operator definition)**.** Fix a two-jet SCET operator specification as in Definition 50.3, an event-shape soft measurement coordinate $\widehat{\ell}_e$, and a renormalization scheme $\mathcal{R}_\mu$ for the cusp and endpoint singularities of the Wilson-line operator. A leading shape-function operator definition consists of:

1. outgoing soft Wilson lines $Y_n, Y_{\bar n}$ in the representations carried by the two energetic charges;
2. for every test function h on $[0,\infty)$, a renormalized composite insertion $h_{\mathcal{R}_\mu}(\widehat{\ell}_e)$;
3. a normalization factor N_R, equal to the dimension of the color representation in the fundamental two-jet case, together with the convention that the identity insertion has unit matrix element.

The shape function $F_e(\ell,\mu;\mathcal{R})$ is the distribution on $[0,\infty)$ defined by

$$\int_0^\infty \mathrm{d}\ell\, h(\ell)F_e(\ell,\mu;\mathcal{R}) = \frac{1}{N_R}\,\mathrm{tr}\langle\Omega|\overline{\mathrm{T}}\Big(Y_{\bar n}^\dagger Y_n\Big)\, h_{\mathcal{R}_\mu}(\widehat{\ell}_e)\,\mathrm{T}\Big(Y_n^\dagger Y_{\bar n}\Big)\,|\Omega\rangle_{\mathcal{R}_\mu}. \tag{50.47}$$

At finite regulator with a positive self-adjoint measurement operator $\widehat{\ell}_e$, this pairing is the spectral measure of $\widehat{\ell}_e$ in the Wilson-line state. After renormalization it is a scheme-dependent distributional coordinate. The normalization and first moment are

$$\int_0^\infty \mathrm{d}\ell\, F_e(\ell,\mu;\mathcal{R}) = 1, \qquad \Omega_1^e(\mu;\mathcal{R}) = \int_0^\infty \mathrm{d}\ell\,\ell\, F_e(\ell,\mu;\mathcal{R}), \tag{50.48}$$

whenever the displayed moments exist in the chosen scheme.

The QFT object is the operator pairing (50.47). In a phenomenological analysis one often chooses a finite-dimensional positive family for F_e and fits its parameters to data. That fitted family is an additional statistical model for the matrix element (50.47). The lattice route to the same nonperturbative information must start from a finite gauge-field regulator as in Chapter 46 and a continuum reconstruction or matching statement as in Chapter 143. Because (50.47) contains lightlike Wilson-line geometry, a Euclidean calculation needs a separately specified quasi-, ratio-, or reconstruction observable and a matching theorem connecting it to the lightlike shape coordinate.

Controlled Approximation 50.19 (Leading endpoint shape-function convolution)**.** Let e be a dimensionless two-jet event shape and let Q be the hard scale. A leading shape-function description on an endpoint window $e \sim \Lambda_{\mathrm{QCD}}/Q$ consists of:

1. the operator definition of Definition 50.18;
2. a perturbative coefficient distribution $\mathrm{d}\sigma_{\mathrm{pert}}/\mathrm{d}e$ in the same subtraction scheme;
3. a Banach space $\mathcal{B}_{\mathrm{end}}$ of test functions supported in the endpoint window;
4. a remainder functional R_Q on $\mathcal{B}_{\mathrm{end}}$ with a stated bound, for example $\|R_Q\|_{\mathcal{B}^*_{\mathrm{end}}} \le CQ^{-2}$ in a single-soft-coordinate situation.

The approximation means that, for every $f \in \mathcal{B}_{\mathrm{end}}$,

$$\left\langle \frac{\mathrm{d}\sigma}{\mathrm{d}e}, f \right\rangle = \int_0^\infty \mathrm{d}\ell\, F_e(\ell, \mu; \mathcal{R}) \int \mathrm{d}e'\, \frac{\mathrm{d}\sigma_{\mathrm{pert}}}{\mathrm{d}e'}(e', Q, \mu; \mathcal{R})\, f\!\left(e' + \frac{\ell}{Q}\right) + R_Q(f). \tag{50.49}$$

Equivalently, when the distributions admit densities in the chosen chart,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}e} = \int_0^\infty \mathrm{d}\ell\, F_e(\ell, \mu; \mathcal{R}) \frac{\mathrm{d}\sigma_{\mathrm{pert}}}{\mathrm{d}e}\left(e - \frac{\ell}{Q}, Q, \mu; \mathcal{R}\right) + R_Q(e). \tag{50.50}$$

The elementary consequences of the convolution are useful diagnostics for normalization and scheme bookkeeping. Taking $f(e) = 1$ in (50.49) and using (50.48) gives equality of the total weighted cross section up to $R_Q(1)$. Taking $f(e) = e$ gives

$$\int \mathrm{d}e\, e\, \frac{\mathrm{d}\sigma}{\mathrm{d}e} = \int \mathrm{d}e\, e\, \frac{\mathrm{d}\sigma_{\mathrm{pert}}}{\mathrm{d}e} + \frac{\Omega_1^e(\mu; \mathcal{R})}{Q} \int \mathrm{d}e\, \frac{\mathrm{d}\sigma_{\mathrm{pert}}}{\mathrm{d}e} + R_Q(e). \tag{50.51}$$

Thus the first nonperturbative moment translates the endpoint event shape by Ω_1^e/Q in this leading single-coordinate approximation. A finite change of subtraction convention may move a constant amount $\delta\ell$ between the perturbative coefficient and the shape coordinate:

$$F_e'(\ell) = F_e(\ell - \delta\ell), \qquad \frac{\mathrm{d}\sigma_{\mathrm{pert}}'}{\mathrm{d}e}(e) = \frac{\mathrm{d}\sigma_{\mathrm{pert}}}{\mathrm{d}e}\left(e + \frac{\delta\ell}{Q}\right). \tag{50.52}$$

The convolution is invariant under the paired transformation (50.52), modulo boundary and truncation terms. Hence a numerical value quoted for Ω_1^e is meaningful only with its event-shape definition, Wilson-line orientation, regulator, and subtraction scheme.

Track-based observables measure only charged hadrons. They are generally not IRC safe as ordinary partonic measurements, because a collinear splitting can change the charged energy fraction observed by the detector. The replacement input is a track function. For a parton species i, let $T_i(x,\mu)$ be the renormalized track coordinate, in a specified operator scheme, for the fraction $x \in [0,1]$ of the parent energy carried by charged tracks. At a reference nonperturbative scale this coordinate is naturally a probability measure. After renormalization it should be viewed as a scheme-dependent distributional coordinate whose normalization is fixed by the operator definition; when it has a density we write the same symbol for that density. Perturbative coefficients are then convolved with T_i rather than evaluated with a purely partonic charged-energy measurement.

Definition 50.20 (Track-function operator definition)**.** Fix a lightlike collinear direction n, the conjugate vector $\bar{n}$ with $n \cdot \bar{n} = 2$, and the SCET operator conventions of Definition 50.3. A track-function operator definition for a parton species i consists of:

1. a gauge-invariant collinear source $\mathcal{O}_i$, for example χ_n for a quark species or $\mathcal{B}^{\mu}_{n\perp}$ for a gluon species, together with the spin, color, and polarization averaging convention used to normalize the source;
2. a detector weight w_{tr} on stable color-singlet final-state particle species, with $w_{\mathrm{tr}} = 1$ for charged tracks and $w_{\mathrm{tr}} = 0$ for neutral particles in the idealized charged-track observable;
3. a positive charged collinear momentum operator $\bar{n} \cdot \widehat{P}_{\mathrm{tr}}$ obtained by weighting the stable final-state momentum operator with w_{tr}, and the total collinear momentum operator $\bar{n} \cdot \widehat{P}_{\mathrm{col}}$;
4. the bounded measurement operator
$$\widehat{x}_{\mathrm{tr}} = \frac{\bar{n} \cdot \widehat{P}_{\mathrm{tr}}}{\bar{n} \cdot \widehat{P}_{\mathrm{col}}}$$
on the positive-energy sector created by $\mathcal{O}_i$, where $0 \le \widehat{x}_{\mathrm{tr}} \le 1$;
5. a regulator and subtraction scheme $\mathcal{R}_\mu$ for the light-ray collinear matrix element and for the endpoint singularities of the measurement insertion.

For every test function h on $[0,1]$, the track function is the renormalized distribution defined by

$$\int_0^1 \mathrm{d}x\, h(x) T_i(x,\mu;\mathcal{R}) = \frac{1}{\mathcal{N}_i(\mu;\mathcal{R})} \langle \Omega | \overline{\mathcal{O}}_i(0)\, h_{\mathcal{R}_\mu}(\widehat{x}_{\mathrm{tr}})\, \mathcal{O}_i(0) | \Omega \rangle_{\mathcal{R}_\mu}. \tag{50.53}$$

The normalization factor $\mathcal{N}_i$ is fixed by the requirement that the pairing with $h = 1$ equals 1. At finite regulator this is the spectral distribution of a bounded charged-energy fraction in the inclusive collinear state created by the gauge-invariant source. In continuum QCD it is a renormalized distributional matrix element; its value is meaningful only after the source, detector weight, regulator, and subtraction scheme have all been specified.

This definition is the nonperturbative QCD object associated with the chosen collinear source. The parton label i labels that source in the factorized approximation. Gauge invariance enters through the Wilson-line building blocks χ_n and $\mathcal{B}_{n\perp}$, while the charged-track measurement is a color-singlet final-state measurement. The renormalization of (50.53) is separate from the renormalization of the hard partonic coefficient into which the track function is inserted.

Definition 50.21 (Paired track-function evolution structure)**.** Fix a renormalization scale μ. A finite-kernel track-function evolution structure consists of:

- a finite set $\mathcal{I}$ of parton species;

- for every ordered splitting $i \to jk$, an integrable kernel $K_{i\to jk}(z,\mu)$ on $0 < z < 1$;
- the paired loss rate

$$\Gamma_i(\mu) = \sum_{j,k\in\mathcal{I}} \int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu).$$

The associated evolution equation for track coordinates is

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}T_i(x,\mu) = \sum_{j,k\in\mathcal{I}} \int_0^1 \mathrm{d}z \int_0^1 \mathrm{d}x_1\,\mathrm{d}x_2\, K_{i\to jk}(z,\mu)\, T_j(x_1,\mu)T_k(x_2,\mu)\delta(x - zx_1 - (1-z)x_2) \\ - \Gamma_i(\mu)T_i(x,\mu). \tag{50.54}$$

In perturbative QCD the kernels are renormalized splitting distributions rather than honest integrable functions before smearing; the finite-kernel definition records the algebraic real–virtual pairing that the renormalized distributional equation must preserve.

Normalization and first moment of track-function evolution. Suppose the track functions in Definition 50.21 have unit normalization,

$$\int_0^1 \mathrm{d}x\, T_i(x,\mu) = 1.$$

The paired evolution (50.54) preserves this normalization. If

$$m_i(\mu) = \int_0^1 \mathrm{d}x\, x\, T_i(x,\mu)$$

is the charged-energy first moment, then

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}m_i(\mu) = \sum_{j,k} \int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu) \left[z\, m_j(\mu) + (1-z)m_k(\mu) - m_i(\mu)\right]. \tag{50.55}$$

Integrating (50.54) over x and using

$$\int_0^1 \mathrm{d}x\, \delta(x - zx_1 - (1-z)x_2) = 1,$$

because $0 \le zx_1 + (1-z)x_2 \le 1$, gives

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\int_0^1 \mathrm{d}x\, T_i(x,\mu) = \sum_{j,k} \int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu) \left[\left(\int T_j\right)\left(\int T_k\right) - \int T_i\right].$$

If all normalizations are 1, the bracket vanishes for every splitting channel. Hence the normalization is preserved.

Multiplying by x before integrating gives

$$\int_0^1 \mathrm{d}x\, x\, \delta(x - zx_1 - (1-z)x_2) = zx_1 + (1-z)x_2.$$

The gain term therefore contributes

$$\sum_{j,k}\int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu)\left[z\, m_j(\mu)\int T_k + (1-z)\, m_k(\mu)\int T_j\right].$$

Using unit normalization and subtracting the loss term $\Gamma_i m_i$ gives (50.55).

The same finite-kernel equation controls the whole moment tower. Define

$$m_i^{(r)}(\mu) = \int_0^1 \mathrm{d}x\, x^r T_i(x,\mu), \qquad m_i^{(0)} = 1,$$

and, when the exponential test function is admissible in the chosen distributional scheme,

$$M_i(\nu,\mu) = \int_0^1 \mathrm{d}x\, e^{\nu x} T_i(x,\mu).$$

Pairing (50.54) with $e^{\nu x}$ gives

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu} M_i(\nu,\mu) = \sum_{j,k}\int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu)\left[M_j(z\nu,\mu)M_k((1-z)\nu,\mu) - M_i(\nu,\mu)\right]. \tag{50.56}$$

Expanding both sides in powers of ν yields, for every $r \geq 0$,

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu} m_i^{(r)} = \sum_{j,k}\int_0^1 \mathrm{d}z\, K_{i\to jk}(z,\mu)\left[\sum_{a=0}^{r}\binom{r}{a} z^a (1-z)^{r-a} m_j^{(a)} m_k^{(r-a)} - m_i^{(r)}\right]. \tag{50.57}$$

The $r = 0$ member is normalization preservation and the $r = 1$ member is (50.55). For $r \geq 2$ the evolution of a single moment depends on lower and equal moments of daughter species. This is why track-based jet observables require a genuine distributional nonperturbative input, or an equivalently complete moment sequence in a specified scheme; a single average charged fraction is not stable under collinear renormalization.

Track functions are therefore nonperturbative inputs with perturbative evolution, not substitutes for IRC safety. The point is sharper than a terminological distinction: a track observable is calculable only after the observable has been lifted from a partonic measurement to a convolution with the nonperturbative track coordinates and the real–virtual pairing in Definition 50.21 has been specified.

Definition 50.22 (Finite track lift of an energy-measure observable)**.** Fix a partonic short-distance configuration with labeled partons

$$a = 1,\dots,n, \qquad (i_a, E_a, y_a),$$

where i_a is the parton species, $E_a > 0$ is the energy coordinate, and y_a is the angular coordinate in the detector patch. A finite track-function lift assigns to each label a an independent random variable $X_a \in [0,1]$ with law T_{i_a} and replaces the calorimetric energy measure by the random charged-track energy measure

$$\mu_{\mathrm{tr}} = \sum_{a=1}^{n} X_a E_a\, \delta_{y_a}.$$

For a smeared k-point energy polynomial with continuous kernel f,

$$\mathcal{E}_f(\mu) = \int f(y_1, \ldots, y_k)\, \mu(dy_1) \cdots \mu(dy_k),$$

the lifted partonic coefficient is the conditional expectation

$$\mathbb{E}_T\, \mathcal{E}_f(\mu_{\mathrm{tr}}).$$

This finite definition is the algebraic core of a track-based factorization formula. A QCD prediction additionally requires the operator definition of the track functions, their subtraction scheme and RG evolution, the short-distance coefficient into which this lift is inserted, and a remainder estimate for the leading-power track-factorization approximation.

The finite lift shows why track observables cannot be reduced to replacing each energy by the average charged fraction. Let

$$m_i^{(r)} = \int_0^1 x^r\, T_i(x)\, \mathrm{d}x$$

be the r-th track moment whenever this moment is defined in the chosen scheme. For the two-point polynomial one obtains

$$\begin{aligned}\mathbb{E}_T\, \mathcal{E}_f(\mu_{\mathrm{tr}}) = &\sum_{a \neq b} E_a E_b\, f(y_a, y_b)\, m_{i_a}^{(1)} m_{i_b}^{(1)} \\ &+ \sum_a E_a^2\, f(y_a, y_a)\, m_{i_a}^{(2)}.\end{aligned} \tag{50.58}$$

The second line is a diagonal term in the energy-measure polynomial. It is controlled by the second track moment, not by the square of the first moment. For a general k-point energy polynomial the exact finite formula is the following. Let

$$\alpha : \{1, \ldots, k\} \longrightarrow \{1, \ldots, n\}$$

record which short-distance parton supplies the energy in each detector slot. Independence of the finite track variables gives

$$\mathbb{E}_T\, \mathcal{E}_f(\mu_{\mathrm{tr}}) = \sum_{\alpha:\{1,\ldots,k\}\to\{1,\ldots,n\}} \left(\prod_{r=1}^{k} E_{\alpha(r)} \right) f\big(y_{\alpha(1)}, \ldots, y_{\alpha(k)}\big) \prod_{a \in \operatorname{im} \alpha} m_{i_a}^{(|\alpha^{-1}(a)|)}. \tag{50.59}$$

Thus each block of detector slots fed by the same short-distance label uses the corresponding higher track moment. For a three-point kernel, writing $f_{abc} = f(y_a, y_b, y_c)$, the formula separates into

$$\begin{aligned}\mathbb{E}_T\, \mathcal{E}_f(\mu_{\mathrm{tr}}) = &\sum_{\substack{a,b,c \\ \text{distinct}}} E_a E_b E_c\, f_{abc}\, m_{i_a}^{(1)} m_{i_b}^{(1)} m_{i_c}^{(1)} \\ &+ \sum_{a \neq b} E_a^2 E_b \left(f_{aab} + f_{aba} + f_{baa} \right) m_{i_a}^{(2)} m_{i_b}^{(1)} \\ &+ \sum_a E_a^3\, f_{aaa}\, m_{i_a}^{(3)}.\end{aligned} \tag{50.60}$$

The diagonal and partial-diagonal pieces are not optional refinements: they are part of the definition of the lifted observable. Omitting them, or replacing every track variable by its first moment, changes the observable whose short-distance coefficient is being computed.

The same bookkeeping explains the collinear composition law behind (50.54). If a parent with energy E is resolved into daughters with energy fractions z and $1-z$, then the track fraction assigned to the unresolved pair is

$$X = zX_1 + (1-z)X_2.$$

Thus

$$\mathbb{E}[X] = z\, m_j^{(1)} + (1-z)m_k^{(1)}, \tag{50.61}$$

$$\mathbb{E}[X^2] = z^2 m_j^{(2)} + (1-z)^2 m_k^{(2)} + 2z(1-z)m_j^{(1)}m_k^{(1)}. \tag{50.62}$$

A track-based observable is therefore not IRC safe as a bare partonic function. Rather, its collinear boundary is a renormalization problem for the whole sequence of track moments, or equivalently for the track distribution. This is the precise meaning of using track functions in a jet-substructure calculation.

50.8 Parton Showers

A parton shower is a stochastic implementation of soft and collinear factorized evolution. At leading soft-collinear order, a final-state branching probability for a parton of type i is built from the Sudakov factor

$$\Delta_i(t_1, t_0) = \exp\left[-\int_{t_0}^{t_1} \frac{\mathrm{d}t}{t} \sum_{i\to jk} \int \mathrm{d}z\, \frac{\alpha_s(\kappa(t,z))}{2\pi} P_{i\to jk}(z)\right], \tag{50.63}$$

where t is the shower ordering variable, z is the splitting fraction, $P_{i\to jk}$ is an Altarelli–Parisi splitting kernel, and $\kappa(t,z)$ is the scale at which the running coupling is evaluated. The probability for the next branching between t and $t+\mathrm{d}t$ is the Sudakov no-branching probability multiplied by the differential splitting kernel.

Different showers choose different evolution variables, recoil maps, momentum conservation prescriptions, and color structures. Angular-ordered showers encode color coherence through emission angles. Dipole and antenna showers use color-connected pairs as emitters. Many practical implementations use a leading-color approximation in which the color algebra is replaced by a planar color-flow structure; subleading-color corrections require additional operator information.

Controlled Approximation 50.23 (Shower accuracy)**.** A shower calculation has a stated logarithmic accuracy only for a specified class of observables and a specified matching or merging scheme. The basic Markov shower reproduces leading soft-collinear logarithms for observables compatible with its ordering variable. NLO-matched prescriptions such as MC@NLO or POWHEG replace the first hard emission and inclusive normalization by exact NLO matrix-element information while retaining the shower resummation for subsequent unresolved radiation. Multijet merging combines samples with different resolved parton multiplicities using a merging scale whose residual dependence is part of the perturbative uncertainty estimate.

50.9 Hadronization and Fragmentation Functions

Hadronization is the nonperturbative formation of outgoing hadrons from the high-energy QCD state produced by the short-distance interaction. In the exact theory it is simply time evolution in the physical Hilbert space of QCD. In phenomenological calculations it is represented by additional nonperturbative inputs after the perturbative hard, jet, and soft evolution has been organized.

For identified hadron production, the corresponding input is a fragmentation function. Let n^μ be a lightlike vector and let $\bar{n}^\mu$ satisfy $n \cdot \bar{n} = 2$. A quark fragmentation function may be represented by a light-ray operator with Wilson lines:

$$D_{h/q}(z,\mu) = \frac{1}{2zN_c} \sum_X \int \frac{\mathrm{d}y^+}{2\pi} \mathrm{e}^{-\mathrm{i}P_h^- y^+/z} \operatorname{tr}\left[\frac{\gamma \cdot \bar{n}}{2} \langle 0|W^\dagger(y^+) q(y^+)|hX\rangle\langle hX|\bar{q}(0)W(0)|0\rangle\right]_{\mathrm{ren}}. \tag{50.64}$$

Here P_h^- is the large light-cone momentum of the observed hadron, $z = P_h^-/k^-$ is the momentum fraction relative to the parent partonic coordinate k^-, W is the appropriate lightlike Wilson line, and the subscript indicates operator renormalization. This is a nonperturbative matrix-element definition, not a probability for a physical colored quark to exist as an external state. A regulator is part of the definition of the light-ray composite, and lightlike Wilson lines may require endpoint, cusp, or rapidity subtractions depending on the observable. The scale dependence of $D_{h/q}$ is governed by DGLAP evolution, while its boundary value at a finite scale is nonperturbative data.

Lund string and cluster hadronization models are phenomenological parametrizations of this nonperturbative stage. A string model uses color flux-tube breaking with fitted parameters for string tension, flavor production, and transverse momentum. A cluster model first organizes the partonic state into color-singlet clusters and then decays those clusters into hadrons with fitted rules. Their predictions therefore carry model and tuning uncertainties in addition to perturbative uncertainties.

The lattice formulation of Chapter 46 is the regulator-level nonperturbative definition of the gauge theory used in this volume. It computes Euclidean finite-volume correlation functions directly in terms of hadronic states. Real-time fragmentation functions and fully exclusive hadronization histories involve lightlike or timelike information, so their extraction from Euclidean lattice data requires additional reconstruction, finite-volume, or large-momentum matching ideas. This is a technical obstruction in the observable, not a license to replace the nonperturbative QCD input by a perturbative calculation.

50.10 Classes of QCD Final-State Observables

The useful separation is by the mathematical input required by the observable.

Total inclusive rates and IRC-safe event shapes.
The required input is renormalized fixed-order amplitudes, KLN cancellation, and resummation when hierarchical scales are measured.

IRC-safe jet cross sections.
The required input is an IRC-safe jet algorithm, fixed-order hard functions, PDFs for incom-

ing hadrons, and logarithmic resummation where jet vetoes or small radii introduce large logarithms.

Energy correlators.
The required input is the stress-tensor energy-flow operators on the physical Hilbert space, together with angular test kernels defining continuous functionals of the calorimetric measure. The density theorem 50.17 explains why smeared energy-correlator polynomials form a natural coordinate system for finite-resolution calorimetric observables. Partonic perturbation theory is used only after the detector observable has been defined.

Identified hadron spectra.
The required input is fragmentation functions or fracture functions, their operator renormalization, and nonperturbative boundary data.

Fully exclusive hadronization patterns.
The required input is the full QCD final-state distribution or a phenomenological model with stated fit parameters and model uncertainty.

This classification keeps the physics claims separated. Perturbative QCD computes short-distance coefficients and IRC-safe partonic representatives. Fragmentation functions, PDFs, shape functions, and hadronization models are additional nonperturbative data or controlled approximations with their own definitions and uncertainties. Energy correlators are especially valuable because they are nonperturbatively defined by the stress tensor and are insensitive, after smearing, to soft emissions and collinear recombination in the precise sense used in Controlled Approximation 50.2.

Chapter 51

Chiral Axial Anomalies

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

51.1 Classical Chiral Currents

Let ψ be a Dirac fermion in four dimensions. Throughout this chapter the Lorentzian metric is mostly plus and the Dirac adjoint is

$$\bar{\psi} := \mathrm{i}\psi^\dagger \gamma^0.$$

These are the gamma-matrix conventions developed in Chapter 40 and centralized in Section 40.16. The anomaly signs below use the same orientation, chirality matrix, and dimensional-continuation prescription. The flat oriented coordinate systems used in the perturbative calculations are fixed by

$$\epsilon^{01} = +1 \quad \text{in two dimensions}, \qquad \epsilon^{0123} = +1 \quad \text{in four dimensions}.$$

The symbols $\epsilon^{\mu\nu}$ and $\epsilon^{\mu\nu\rho\sigma}$ are totally antisymmetric with these normalizations. In four dimensions the chirality convention is

$$\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3, \qquad \mathrm{tr}\big(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\big) = 4\mathrm{i}\,\epsilon^{\mu\nu\rho\sigma}.$$

Gauge generators in this chapter use the same Hermitian convention as Chapter 45: if $A_\mu = A^a_\mu t^a$, then

$$[t^a, t^b] = \mathrm{i}f^c{}_{ab}t^c, \qquad \mathrm{tr}(t^a t^b) = \delta^{ab}$$

unless a different trace is explicitly declared. The trace-form anomaly expressions are written in terms of the matrix-valued curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}[A_\mu, A_\nu],$$

so a change of generator normalization rescales the component fields and coupling coordinates but not the matrix trace formula itself. With this adjoint the Hermitian bilinear current associated with a real phase rotation contains an explicit factor of i. Couple ψ to a background or dynamical gauge field through

$$D_\mu\psi = \partial_\mu\psi - \mathrm{i}A_\mu\psi.$$

Definition 51.1 (Anomaly representative conventions)**.** An anomaly formula in this chapter means a renormalized local Ward identity for specified composite currents after four pieces of data have been fixed. First, the gauge and vector Ward identities that define the background coupling are preserved by the regulator and by the counterterm convention. Second, the remaining local term is represented in the Hermitian trace normalization stated above, namely $\mathrm{tr}(t^a t^b) = \delta^{ab}$ unless another trace is declared. Third, the orientation and chirality matrix are the displayed ones. Fourth, local counterterms have been chosen; changing them changes the representative of the anomaly but not its BRST cohomology class.

The perturbative triangle calculations below use dimensional reduction in the following restricted sense: the loop momentum is analytically continued while γ_5 and the antisymmetric tensor are kept in the physical two- or four-dimensional subspace. The heat-kernel calculation in the measure section is instead a spectral regulator for a Berezinian trace. No Pauli–Villars auxiliary field is used, and no claim is made that dimensional continuation by itself defines a nonperturbative fermionic path integral.

Warning 51.2 (Anomaly signs are tied to the full convention datum)*.* The anomaly coefficients in this chapter are not portable by changing one symbol at a time. The signs and factors use simultaneously the mostly-plus Dirac adjoint, $\gamma_5 = -\mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3$, $\epsilon^{0123} = +1$, the Hermitian covariant derivative $D_\mu = \partial_\mu - \mathrm{i}A_\mu$, the curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}[A_\mu, A_\nu]$, and the trace convention fixed in Chapter 45. Changing any one of these choices requires translating the whole convention datum, including the topological charge normalization and the local counterterm representative. Dimensional reduction below is only the perturbative regulator used to compute the displayed Ward-identity representative; it is not being used as a nonperturbative definition of the fermionic path integral.

Proposition 51.3 (Anomaly-coordinate comparison)**.** *Fix a regulated background-field groupoid $\mathcal{B}_\Lambda//\mathcal{G}_\Lambda$, with right action $A \mapsto A^g$, and a local regulator chart in which the chiral fermion functional is a scalar $W_\Lambda[A]$. Then the finite cochain*

$$C_\Lambda(g; A) = W_\Lambda[A^g] - W_\Lambda[A]$$

is the additive coordinate of the anomaly line. More precisely:

1. *functoriality of the groupoid action gives $C_\Lambda(gh; A) = C_\Lambda(h; A^g) + C_\Lambda(g; A)$;*
2. *a local counterterm $B_\Lambda[A]$ changes this coordinate by the coboundary $B_\Lambda[A^g] - B_\Lambda[A]$;*
3. *after choosing a local frame s_A for the anomaly line, the exponentiated transport cocycle is $\alpha_\Lambda(A, g) = \exp C_\Lambda(g; A)$, and the same counterterm is the inverse frame change $s_A \mapsto \exp(-B_\Lambda[A])s_A$;*

4. *the inflow bulk carries the inverse line coordinate* $\alpha_\Lambda(A,g)^{-1}$*;*
5. *differentiating* $C_\Lambda(\exp(t\zeta);A)$ *at* $t=0$*, then taking the local continuum representative, gives the consistent local anomaly density* $a_4^{(1)}(\zeta,A)$*, whose relative BRST class is the perturbative obstruction.*

Consequently local BRST cancellation removes the obstruction on the identity component of the background groupoid, while flat determinant- or Pfaffian-line holonomies on based loops remain separate global anomaly data.

Proof. The first identity is just the scalar form of functoriality:

$$W_\Lambda[A^{gh}]-W_\Lambda[A]=\big(W_\Lambda[(A^g)^h]-W_\Lambda[A^g]\big)+\big(W_\Lambda[A^g]-W_\Lambda[A]\big).$$

Replacing W_Λ by $W_\Lambda+B_\Lambda$ gives the displayed coboundary. If the line is locally framed by s_A and

$$T_g s_A=\alpha_\Lambda(A,g)s_{A^g},$$

then the frame $s'_A=\exp(-B_\Lambda[A])s_A$ gives

$$T_g s'_A=\exp(-B_\Lambda[A])\alpha_\Lambda(A,g)\exp(B_\Lambda[A^g])s'_{A^g},$$

which is the exponential of the same additive coboundary. The bulk inflow line is the inverse anomaly line, so its transport cocycle is α_Λ^{-1}, and tensoring boundary and bulk transports gives the identity transport.

For a one-parameter gauge transformation $g(t)=\exp(t\zeta)$, the derivative of the finite cocycle is

$$\left.\frac{d}{dt}\right|_{t=0}C_\Lambda(g(t);A)=\delta_\zeta W_\Lambda[A].$$

After subtracting local counterterms and passing to the continuum local functional, this derivative is represented by $\int_M a_4^{(1)}(\zeta,A)$. Differentiating the finite cocycle identity in two infinitesimal directions gives the Wess–Zumino consistency condition, and replacing the even gauge parameters by the odd ghost gives the relative BRST cocycle in $H^{1,4}(s\mid d)$, developed later in Proposition 51.9. A local counterterm changes the representative by the corresponding relative coboundary. This infinitesimal argument only sees the identity component of the background groupoid. A flat but nontrivial line holonomy along a based loop has zero local curvature and zero local descent class, but still prevents a gauge-invariant global scalar section; that global obstruction is the determinant/Pfaffian-line story developed in Chapter 153. □

For a mass m, the Dirac Lagrangian is

$$\mathcal{L}=-\bar\psi\gamma^\mu D_\mu\psi-m\bar\psi\psi.$$

When $m=0$, the classical equations of motion imply conservation of the vector and axial currents

$$j^\mu=\mathrm{i}\bar\psi\gamma^\mu\psi,\qquad j_5^\mu=\mathrm{i}\bar\psi\gamma^\mu\gamma_5\psi,\qquad \partial_\mu j^\mu=0,\qquad \partial_\mu j_5^\mu=0.$$

For $m\neq0$, the axial divergence becomes

$$\partial_\mu j_5^\mu=2\mathrm{i}m\,\bar\psi\gamma_5\psi$$

with the γ_5 convention stated above. Here $D_\mu \bar\psi = \partial_\mu \bar\psi + \mathrm{i}\bar\psi A_\mu$, and the classical equations of motion are $(\gamma^\mu D_\mu + m)\psi = 0$ and $(D_\mu \bar\psi)\gamma^\mu - m\bar\psi = 0$. Since $\{\gamma_5, \gamma^\mu\} = 0$, these equations give $\partial_\mu(\bar\psi\gamma^\mu\gamma_5\psi) = 2m\bar\psi\gamma_5\psi$; the explicit factor of i in j_5^μ gives the displayed normalization.

The corresponding infinitesimal transformations are

$$\delta_V \psi = \mathrm{i}\alpha\psi, \qquad \delta_A \psi = \mathrm{i}\beta\gamma_5\psi,$$

where α and β are constant real parameters. Classically, for massless fermions, both transformations are symmetries of the local Lagrangian. Quantum mechanically, the vector transformation is preserved when it is gauged consistently, while the axial current acquires a local divergence determined by the gauge background.

51.2 A Two-Dimensional Contact-Term Calculation

The local origin of an axial anomaly is already visible in a free two-dimensional Dirac fermion. Use the Lorentzian action

$$S_2 = -\int \mathrm{d}^2x\, \bar\psi\gamma^\mu\partial_\mu\psi, \qquad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \bar\psi = \mathrm{i}\psi^\dagger\gamma^0,$$

and define

$$\gamma := \gamma^0\gamma^1, \qquad \gamma^2 = 1, \qquad j^\mu = \mathrm{i}\bar\psi\gamma^\mu\psi, \qquad j_A^\mu = \mathrm{i}\bar\psi\gamma^\mu\gamma\psi.$$

The classical massless action is invariant under $\psi \mapsto \mathrm{e}^{\mathrm{i}\alpha}\psi$ and under $\psi \mapsto \mathrm{e}^{\mathrm{i}\beta\gamma}\psi$. The quantum question is the renormalized Ward identity obeyed by time-ordered current products. Consider

$$G_A^{\mu\nu}(x) := \langle\Omega|\,\mathrm{T}\; j_A^\mu(x) j^\nu(0)|\Omega\rangle.$$

At one loop the relevant momentum-space graph is shown in Figure 51.1; the figure fixes the loop momentum routing and distinguishes the axial-current insertion from the vector current insertion used in the Ward identity below.

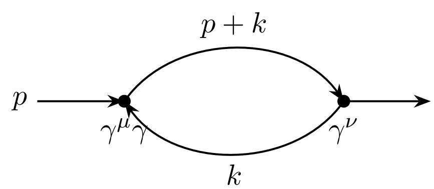


Figure 51.1: The two-dimensional current two-point function. The left insertion is the axial current and the right insertion is the vector current.

$$-\int \frac{\mathrm{d}^2k}{(2\pi)^2}\,\frac{\mathrm{tr}[\mathrm{i}\gamma^\nu(\not{p}+\not{k})\mathrm{i}\gamma^\mu\gamma\;\not{k}]}{(k^2-\mathrm{i}\epsilon)((p+k)^2-\mathrm{i}\epsilon)}.$$

Contracting with p_μ, Feynman parametrizing, and shifting $k \mapsto k - xp$ gives

$$\int \mathrm{d}^2x\, \mathrm{e}^{-\mathrm{i}p\cdot x}\partial_\mu G_A^{\mu\nu}(x) = \mathrm{i}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^2k}{(2\pi)^2}\,\frac{\mathrm{tr}(\gamma^\nu\;\not{k}\;\not{p}\gamma\;\not{k}) - x(1-x)p^2\,\mathrm{tr}(\gamma^\nu\gamma\;\not{p})}{(k^2+p^2x(1-x)-\mathrm{i}\epsilon)^2}. \tag{51.1}$$

The expression is a contact term, so its value depends on the regulator used to define the composite current. Continue the loop momentum to $D = 2-\epsilon$ dimensions while keeping the chirality matrix $\gamma = \gamma^0\gamma^1$ two-dimensional. Rotational averaging gives

$$\operatorname{tr}(\gamma^\nu \not{k} \not{p}\gamma \not{k}) \longmapsto \frac{k^2}{D}\operatorname{tr}(\gamma^\nu\gamma_\rho \not{p}\gamma\gamma^\rho) = \frac{2-D}{D}k^2 \operatorname{tr}(\gamma^\nu \not{p}\gamma).$$

The factor $2-D=\epsilon$ multiplies the logarithmic divergence of the loop integral and leaves a finite local term. With $\operatorname{tr}(\gamma^\nu\gamma \not{p}) = 2\epsilon^{\mu\nu}p_\mu$, one finds

$$\int \mathrm{d}^2x\, \mathrm{e}^{-\mathrm{i}p\cdot x}\partial_\mu G_A^{\mu\nu}(x) = -\frac{1}{\pi}\epsilon^{\mu\nu}p_\mu.$$

Equivalently,

$$\partial_\mu\langle\Omega|\,\mathrm{T}\; j_A^\mu(x)j^\nu(y)|\Omega\rangle = \frac{\mathrm{i}}{\pi}\epsilon^{\mu\nu}\partial_\mu\delta^{(2)}(x-y), \tag{51.2}$$

whereas the vector Ward identity is preserved:

$$\partial_\mu\langle\Omega|\,\mathrm{T}\; j^\mu(x)j^\nu(y)|\Omega\rangle = 0.$$

Now couple the vector current to a background $U(1)$ gauge field,

$$D_\mu = \partial_\mu - \mathrm{i}A_\mu, \qquad \Delta S = \int \mathrm{d}^2x\, j^\mu A_\mu.$$

Expanding time-ordered correlators in powers of A, the first-order term in (51.2) gives the local operator identity

$$\partial_\mu j_A^\mu(x) = \frac{1}{2\pi}\epsilon^{\mu\nu}F_{\mu\nu}(x), \qquad F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{51.3}$$

Thus the anomaly is already present before any interaction among the fermions: it is the finite remnant of a regulated contact term in a composite-current Ward identity.

Proposition 51.4 (Two-dimensional axial contact term)**.** *For the massless two-dimensional Dirac fermion with current conventions of this section, choose the renormalized current representative whose vector Ward identity is preserved. Then the axial-vector current product obeys*

$$\partial_\mu\langle\Omega|\,\mathrm{T}\; j_A^\mu(x)j^\nu(y)|\Omega\rangle = \frac{\mathrm{i}}{\pi}\epsilon^{\mu\nu}\partial_\mu\delta^{(2)}(x-y), \qquad \partial_\mu\langle\Omega|\,\mathrm{T}\; j^\mu(x)j^\nu(y)|\Omega\rangle = 0.$$

Coupling j^μ to a background Abelian gauge field with $D_\mu = \partial_\mu - \mathrm{i}A_\mu$ gives the local operator identity

$$\partial_\mu j_A^\mu = \frac{1}{2\pi}\epsilon^{\mu\nu}F_{\mu\nu}.$$

Proof. The one-loop contraction of $p_\mu\langle j_A^\mu j^\nu\rangle$ reduces, after Feynman parametrization, to (51.1). In dimensional reduction the term with two loop momenta contains the factor $(2-D)/D$, because only the physical two-dimensional gamma matrices anticommute with γ. This evanescent factor multiplies the logarithmic ultraviolet divergence of the integral and leaves the finite polynomial $-\pi^{-1}\epsilon^{\mu\nu}p_\mu$. Fourier inversion gives the displayed contact term. The vector-current representative is fixed by subtracting the local contact term that would otherwise violate the vector Ward identity.

Expanding the background-field insertion $\exp(\mathrm{i}\int j^\mu A_\mu)$ to first order and using the contact term gives

$$\partial_\mu j_A^\mu(x) = \frac{1}{\pi}\epsilon^{\mu\nu}\partial_\mu A_\nu(x) = \frac{1}{2\pi}\epsilon^{\mu\nu}F_{\mu\nu}(x),$$

where the last equality uses antisymmetry of $\epsilon^{\mu\nu}$. □

51.3 Contour Charges and Anomalous Charge Transport

The contact term has a geometric interpretation. In Euclidean two-space, a conserved current j^μ defines a charge on a closed contour C,

$$Q_C = \int_C j^\mu n_\mu \, \mathrm{d}s,$$

where n_μ is the outward unit normal to the contour. If $\partial_\mu j^\mu = 0$ away from operator insertions, the contour can be deformed through regions with no insertions. Figure 51.2 shows the allowed contour deformation and the point at which an attempted deformation would cross an operator insertion, where the contact term enters the Ward identity.

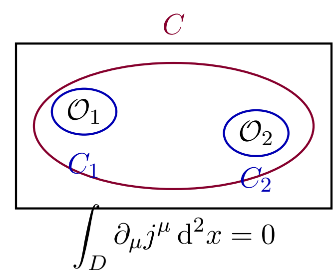


Figure 51.2: For a conserved current, the charge contour can be moved from small circles around operator insertions to a larger contour without changing the insertion.

If C encloses all insertions and can be moved to infinity, the ordinary global transformation leaves the correlation function invariant.

For the anomalous axial current the small contour still gives the local action of the axial rotation on an operator. The difference between the charge on a large contour and the sum of charges on small contours is the flux of the anomaly through the interpolating domain D, as displayed in Figure 51.3:

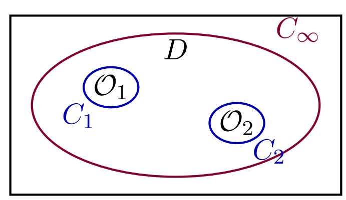


Figure 51.3: For the axial current in a background field, deforming the contour across the domain D changes the charge by the integral of the anomalous divergence.

Using (51.3),

$$\Delta Q_A = Q_{A,C_\infty} - \sum_i Q_{A,C_i} = \int_D \mathrm{d}^2 x \, \partial_\mu j^\mu_A = \frac{1}{2\pi} \int_D \mathrm{d}^2 x \, \epsilon^{\mu\nu} F_{\mu\nu}. \tag{51.4}$$

Therefore an axial rotation by α multiplies the correlation function by the background-dependent phase

$$\exp(\mathrm{i}\alpha \Delta Q_A).$$

Equivalently, in the background-field effective action the transformation shifts the action by

$$\Delta S = \frac{\alpha}{2\pi}\int d^2x\,\epsilon^{\mu\nu}F_{\mu\nu}.$$

51.4 The Four-Dimensional Triangle Contact Term

The four-dimensional axial anomaly is produced by an analogous regulated contact term with one higher short-distance singularity. For a massless Dirac fermion, study

$$\langle\Omega|\,\mathrm{T}\;j_5^\mu(x)j^\nu(y)j^\rho(z)|\Omega\rangle.$$

There are two oriented triangle graphs:

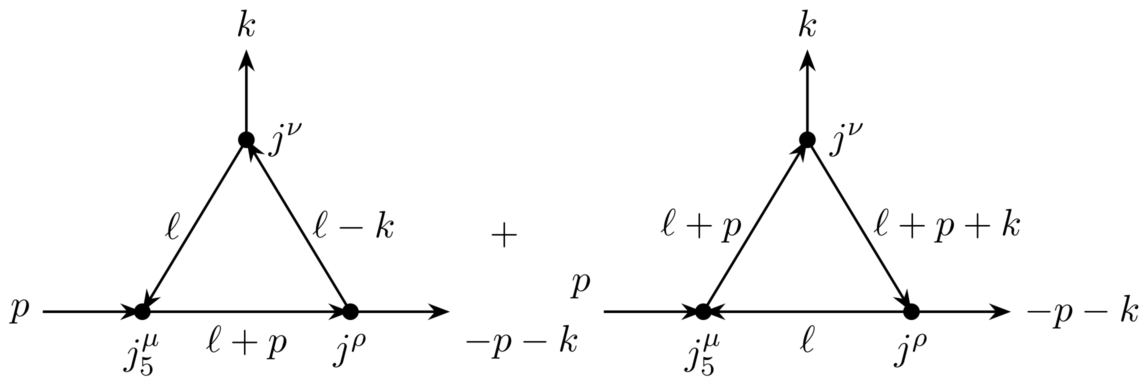


Figure 51.4: The two orientations of the one-axial, two-vector triangle. The regulator is chosen so that the vector-current Ward identities are preserved.

The divergence of the axial insertion is the Fourier transform of a local contact term. In the dimensional-reduction implementation of the perturbative regulator, set $D = 4-\epsilon$ and split the loop momentum as

$$\ell = \ell_\parallel + \ell_\perp,$$

where $\ell_\parallel$ lies in the physical four-dimensional subspace. Keep $\gamma_5 = -\mathrm{i}\gamma^{0123}$ four-dimensional, so that $\not\ell_\parallel$ anticommutes with γ_5 and $\not\ell_\perp$ commutes with γ_5. In the first triangle one uses

$$\not p\,\gamma_5 = \left[(\not\ell + \not p)\gamma_5 + \gamma_5\,\not\ell\right] - 2\,\not\ell_\perp\gamma_5.$$

The two bracketed terms are adjacent inverse-propagator numerators. In the two oriented triangle graphs they reduce one fermion propagator and become a pair of vector-current contact terms with opposite routing. Here is the cancellation explicitly for the leading routing in Figure 51.4. Set

$$D_0 = \ell^2, \qquad D_k = (\ell-k)^2, \qquad D_p = (\ell+p)^2.$$

The bracketed part of the first orientation gives

$$\mathcal{C}_1^{\nu\rho} = -\mathrm{i}\int\frac{d^D\ell}{(2\pi)^D}\left[\frac{\mathrm{tr}[\gamma_5\,\not\ell\gamma^\nu(\not\ell-\not k)\gamma^\rho]}{D_0D_k} + \frac{\mathrm{tr}[\gamma_5\gamma^\nu(\not\ell-\not k)\gamma^\rho(\not\ell+\not p)]}{D_kD_p}\right].$$

The first term came from the adjacent factor $(\not\ell + \not p)^2 = D_p$, after cyclically moving the first $(\not\ell + \not p)$ to the end of the trace; the second came from $\not\ell^2 = D_0$. The bracketed part of the opposite orientation is

$$\mathcal{C}_2^{\nu\rho} = -\mathrm{i}\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\Bigg[\frac{\mathrm{tr}[\gamma_5 \not\ell\gamma^\rho(\not\ell + \not p + \not k)\gamma^\nu]}{\ell^2(\ell+p+k)^2} + \frac{\mathrm{tr}[\gamma_5\gamma^\rho(\not\ell + \not p + \not k)\gamma^\nu(\not\ell + \not p)]}{(\ell+p+k)^2 D_p}\Bigg].$$

The dimensional-reduction regulator is defined so that the reduced two-propagator integrals are translation invariant; this is the perturbative implementation of the vector Ward identities. In the second term of $\mathcal{C}_2$, set $\ell' = \ell + p + k$ and then rename ℓ' as ℓ. It has the same denominator $D_0 D_k$ as the first term of $\mathcal{C}_1$, while

$$\begin{aligned}
&\mathrm{tr}[\gamma_5 \not\ell\gamma^\nu(\not\ell - \not k)\gamma^\rho] = 4\mathrm{i}\epsilon^{\alpha\nu\beta\rho}\ell_\alpha(\ell-k)_\beta,\\
&\mathrm{tr}[\gamma_5\gamma^\rho \not\ell\gamma^\nu(\not\ell - \not k)] = 4\mathrm{i}\epsilon^{\rho\alpha\nu\beta}\ell_\alpha(\ell-k)_\beta = -4\mathrm{i}\epsilon^{\alpha\nu\beta\rho}\ell_\alpha(\ell-k)_\beta.
\end{aligned}$$

Thus these two terms cancel. In the first term of $\mathcal{C}_2$, set $\ell' = \ell + k$. It has the same denominator $D_k D_p$ as the second term of $\mathcal{C}_1$, and

$$\begin{aligned}
\mathrm{tr}[\gamma_5\gamma^\nu(\not\ell - \not k)\gamma^\rho(\not\ell + \not p)] &= 4\mathrm{i}\epsilon^{\nu\alpha\rho\beta}(\ell-k)_\alpha(\ell+p)_\beta,\\
\mathrm{tr}[\gamma_5(\not\ell - \not k)\gamma^\rho(\not\ell + \not p)\gamma^\nu] &= 4\mathrm{i}\epsilon^{\alpha\rho\beta\nu}(\ell-k)_\alpha(\ell+p)_\beta\\
&= -4\mathrm{i}\epsilon^{\nu\alpha\rho\beta}(\ell-k)_\alpha(\ell+p)_\beta.
\end{aligned}$$

The bracketed contact terms therefore cancel pairwise once the vector Ward identities are imposed by the regulator. The entire axial contact term is carried by the evanescent $\ell_\perp$ contribution. For the first orientation this contribution is

$$2\mathrm{i}\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\frac{\mathrm{tr}[\gamma_5 \not\ell_\perp \not\ell\gamma^\nu(\not\ell - \not k)\gamma^\rho(\not\ell + \not p)]}{D_0 D_k D_p}.$$

The coefficient of the local two-vector contact term is obtained by keeping the part with two external momenta and no physical loop momentum in the γ_5-trace. Terms with a symmetric pair of physical loop momenta contracted into the antisymmetric tensor vanish after rotational averaging, and terms with an unpaired perpendicular gamma matrix or the wrong number of physical gamma matrices have zero γ_5-trace. Using

$$\not\ell_\perp \not\ell_\perp = \ell_\perp^2, \qquad \mathrm{tr}\Big(\gamma_5\gamma^\nu\gamma^\alpha\gamma^\rho\gamma^\beta\Big) = 4\mathrm{i}\epsilon^{\nu\alpha\rho\beta},$$

one obtains

$$\begin{aligned}
\mathrm{tr}[\gamma_5 \not\ell_\perp \not\ell\gamma^\nu(\not\ell - \not k)\gamma^\rho(\not\ell + \not p)]_{\mathrm{local}} &= -\ell_\perp^2 k_\alpha p_\beta\,\mathrm{tr}\Big(\gamma_5\gamma^\nu\gamma^\alpha\gamma^\rho\gamma^\beta\Big)\\
&= 4\mathrm{i}\,\ell_\perp^2\,\epsilon^{\nu\rho\alpha\beta}k_\alpha p_\beta.
\end{aligned}$$

Thus the first orientation contributes

$$-8\,\epsilon^{\nu\rho\alpha\beta}k_\alpha p_\beta\, I(k,p), \qquad I(k,p) := \int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\frac{\ell_\perp^2}{D_0 D_k D_p}.$$

The opposite orientation gives the same local tensor after $(\nu, k) \leftrightarrow (\rho, -p-k)$.

It remains to evaluate $I(k,p)$. Take the external momenta nonexceptional during the ultraviolet computation; the contact coefficient extracted below is independent of this infrared choice. Feynman parametrization gives

$$I(k,p) = 2\int_0^1 \mathrm{d}x \int_0^1 \mathrm{d}y \int_0^1 \mathrm{d}z\, \delta(1-x-y-z) \int \frac{\mathrm{d}^D r}{(2\pi)^D} \frac{r_\perp^2}{(r^2+\Delta)^3},$$
$$r = \ell - yk + zp,$$
$$\Delta = y(1-y)k^2 + z(1-z)p^2 + 2yz\, k\cdot p - \mathrm{i}0.$$

The shift does not change the perpendicular component because p and k lie in the physical four-dimensional subspace. In the continued $D = 4-\epsilon$ dimensional rotation-invariant integral,

$$\int \mathrm{d}^D r\, r_\perp^2 f(r^2) = \frac{D-4}{D} \int \mathrm{d}^D r\, r^2 f(r^2) = -\frac{\epsilon}{4-\epsilon} \int \mathrm{d}^D r\, r^2 f(r^2).$$

Therefore

$$\begin{aligned}\int \frac{\mathrm{d}^D r}{(2\pi)^D} \frac{r_\perp^2}{(r^2+\Delta)^3} &= \frac{D-4}{D} \int \frac{\mathrm{d}^D r}{(2\pi)^D} \frac{r^2}{(r^2+\Delta)^3} \\ &= \frac{D-4}{4} \frac{\Gamma(2-D/2)}{(4\pi)^{D/2}} \Delta^{D/2-2} \\ &= -\frac{1}{32\pi^2} + O(\epsilon).\end{aligned}$$

Since

$$2\int_0^1 \mathrm{d}x \int_0^1 \mathrm{d}y \int_0^1 \mathrm{d}z\, \delta(1-x-y-z) = 1$$

and $\Delta^{-\epsilon/2} = 1 + O(\epsilon)$, the required scalar integral is

$$I(k,p) = -\frac{1}{32\pi^2}.$$

With the sign convention of (51.6), this gives

$$\partial_\mu \langle \Omega |\, \mathrm{T}\, j_5^\mu(x) j^\nu(y) j^\rho(0) | \Omega \rangle = \frac{1}{2\pi^2} \epsilon^{\nu\rho\alpha\beta} \partial_\alpha \delta^{(4)}(x) \partial_\beta \delta^{(4)}(y). \tag{51.5}$$

The local coefficient is independent of the long-distance state; it is fixed by the regulated short-distance singularity of the current product.

Coupling the vector current to a background field,

$$\Delta S = \int \mathrm{d}^4 x\, A_\mu j^\mu,$$

and expanding the time-ordered exponential to second order, the contact term (51.5) yields

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$$

for a unit-charge Dirac fermion in the convention of (51.6). A fermion of electric charge q coupled with electromagnetic coupling e replaces F by eqF, giving the factor $e^2 q^2$ in the next section.

Proposition 51.5 (Four-dimensional axial contact term)**.** *For a massless four-dimensional Dirac fermion, in the anomaly representative of Definition 51.1, the regulated one-axial two-vector current product has local divergence*

$$\partial_\mu \langle \Omega | \, \mathrm{T} \; j_5^\mu(x) j^\nu(y) j^\rho(0) | \Omega \rangle = \frac{1}{2\pi^2} \epsilon^{\nu\rho\alpha\beta} \partial_\alpha \delta^{(4)}(x) \partial_\beta \delta^{(4)}(y).$$

Consequently, after coupling the vector current to a background Abelian field and preserving the vector Ward identity, a unit-charge Dirac fermion obeys

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$$

in the massless theory.

Proof. Contracting the axial vertex in the two oriented triangles writes the numerator as vector-current contact terms plus the evanescent $\ell_\perp$ term. Translation invariance of the dimensionally reduced two-propagator integrals is precisely the regulator statement that the vector Ward identities are preserved; with this prescription the vector-contact terms cancel pairwise between the two orientations. The remaining local piece is

$$-8 \, \epsilon^{\nu\rho\alpha\beta} k_\alpha p_\beta I(k,p)$$

from one orientation, with the opposite orientation contributing the corresponding exchanged tensor. The Feynman-parameter integral has the regulator-independent local limit

$$I(k,p) = \int \frac{\mathrm{d}^{4-\epsilon}\ell}{(2\pi)^{4-\epsilon}} \frac{\ell_\perp^2}{\ell^2 (\ell-k)^2 (\ell+p)^2} = -\frac{1}{32\pi^2},$$

because the factor $D-4$ from $\ell_\perp^2$ cancels the logarithmic ultraviolet pole. Substituting this value and Fourier transforming gives (51.5).

The background-field identity follows by expanding the time-ordered exponential to second order. The factor 1/2 in the exponential is cancelled by the two identical vector insertions in the contact term, and antisymmetrizing the two derivatives of A rewrites the result as $\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}/(16\pi^2)$ in the normalization of (51.6). ☐

51.5 The Abelian Axial Anomaly

For a Dirac fermion of electric charge q coupled to an electromagnetic field with coupling e, the renormalized axial current may be chosen so that

$$\partial_\mu j_5^\mu = 2im \, \bar{\psi} \gamma_5 \psi + \frac{e^2 q^2}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}. \tag{51.6}$$

The coefficient is fixed by the short-distance singularity of the triangle diagram with one axial-current insertion and two vector-current insertions. The vector Ward identity is imposed because the vector current is the current coupled to the gauge field. The remaining local term appears in the axial Ward identity.

Equation (51.6) is a statement about a renormalized composite operator identity. Each symbol in it has a prescribed meaning: $F_{\mu\nu}$ is the electromagnetic field strength, $\epsilon^{\mu\nu\rho\sigma}$ is the four-dimensional antisymmetric symbol with the orientation fixed at the start of the chapter, $\mathrm{i}m\bar{\psi}\gamma_5\psi$ is the pseudoscalar mass-breaking operator in the mostly-plus adjoint convention, and j_5^μ is the renormalized current whose vector-current Ward identities have been preserved.

The factor relating this component formula to the heat-kernel trace below is as follows. Define

$$\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \qquad F = \frac{1}{2}F_{\mu\nu}\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

Then

$$F_{\mu\nu}\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}, \qquad F \wedge F = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}\,\mathrm{d}^4x.$$

The local heat-kernel trace (51.8) gives the index density $(32\pi^2)^{-1}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}_R(F_{\mu\nu}F_{\rho\sigma})$. The Dirac axial Ward identity has twice this density because the chiral change of variables rotates ψ and $\bar{\psi}$ with the same sign, giving the prefactor $2\mathrm{i}$ in (51.7). For a single charge-one Abelian Dirac fermion, the representation trace is trivial and

$$2 \cdot \frac{1}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = \frac{1}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma},$$

which is the coefficient in (51.6); a field of charge eq contributes the additional factor e^2q^2.

51.6 Measure Derivation

The same coefficient is obtained from the Berezinian change of variables in a regulated fermion path integral. The construction uses the finite-dimensional Berezin calculus of Chapter 40; the object that transforms is an ordered Berezinian density on a finite set of odd coefficients, not a Borel measure on a set of fermion fields.

Work first on a compact Euclidean spin four-manifold, or in a finite box with elliptic boundary conditions, so that the Euclidean Dirac operator

$$\mathcal{D}_A = \gamma^\mu(\partial_\mu - \mathrm{i}A_\mu)$$

has a discrete L^2-orthonormal eigenbasis after the infrared regulator has been fixed. The local formula obtained below is independent of this infrared auxiliary choice. Let P_Λ be the spectral projection onto the span

$$V_\Lambda = \mathrm{span}\{\varphi_n :\ |\lambda_n| \le \Lambda\}, \qquad \mathcal{D}_A\varphi_n = \lambda_n\varphi_n.$$

The regulated fermion variables are odd coordinates

$$\psi_\Lambda(x) = \sum_{n\in I_\Lambda} c_n\varphi_n(x), \qquad \bar{\psi}_\Lambda(x) = \sum_{n\in I_\Lambda} \bar{c}_n\varphi_n^\dagger(x),$$

with ordered Berezinian density

$$\Omega_\Lambda = \prod_{n\in I_\Lambda} d\bar{c}_n\, dc_n.$$

An infinitesimal local chiral rotation

$$\psi(x) \mapsto \mathrm{e}^{\mathrm{i}\beta(x)\gamma_5}\psi(x), \qquad \bar{\psi}(x) \mapsto \bar{\psi}(x)\mathrm{e}^{\mathrm{i}\beta(x)\gamma_5},$$

acts on V_Λ by the finite matrix

$$U_{\Lambda,\beta} = 1 + \mathrm{i}P_\Lambda \beta\gamma_5 P_\Lambda + O(\beta^2),$$

where β denotes the multiplication operator by the smooth function $\beta(x)$. The inverse determinant in the odd Berezinian transformation law gives, with the convention that J_Λ is the factor multiplying the transformed finite-dimensional Berezin integral,

$$\log J_\Lambda = 2\mathrm{i}\ \mathrm{tr}_{V_\Lambda}(P_\Lambda\beta\gamma_5 P_\Lambda) + O(\beta^2).$$

The factor of 2 records that ψ and $\bar{\psi}$ are rotated with the same chiral sign.

The sharp projection P_Λ may be replaced, for the local asymptotic calculation, by the smooth gauge-covariant spectral regulator

$$\exp(-\mathcal{D}_A^2/M^2).$$

This is the heat-kernel regulator of the Berezinian trace. It introduces no regulator field; M is the ultraviolet scale in the heat semigroup of the elliptic operator $\mathcal{D}_A^2$. In the terminology of Table 11.1, this is a spectral regulator for an operator trace, not a Pauli–Villars auxiliary-field or determinant-ratio regulator. The regulated logarithm used below is

$$\log J_M = 2\mathrm{i} \lim_{M\to\infty} \mathrm{Tr}\left[\beta\,\gamma_5 \exp\left(-\mathcal{D}_A^2/M^2\right)\right], \tag{51.7}$$

where Tr is the trace over spacetime, spin, and internal indices. Equation (51.7) is therefore a regulated finite-mode Berezinian statement followed by a heat-kernel asymptotic limit. It is the measure-language counterpart of the dimensionally regulated triangle computation above, in the scheme where the vector gauge Ward identity is preserved and the axial Ward identity carries the local term.

The heat-kernel expansion gives the local limit

$$\lim_{M\to\infty} \mathrm{tr}_{\mathrm{spin}}\left[\gamma_5 \exp\left(-\mathcal{D}_A^2/M^2\right)\right](x,x) = \frac{1}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}_R(F_{\mu\nu}F_{\rho\sigma}), \tag{51.8}$$

with tr_R the trace in the fermion representation. The anomaly density is therefore the heat-kernel limit of the regulated Berezinian traces of γ_5 over cutoff eigenspaces. Finite-dimensional cyclic trace manipulations at fixed cutoff do not remove (51.8), because the regulator is part of the definition of the composite current.

The local heat-kernel density becomes a Fredholm index after the following global data are specified. The analytic input is the spin Dirac case of the Atiyah–Singer index theorem.[1]

Quoted Theorem 51.6 (Spin Dirac index theorem, four-dimensional form)**.** *Let M be a smooth compact oriented Riemannian spin four-manifold without boundary. Let $S = S^+ \oplus S^-$ be its*

[1]M. F. Atiyah and I. M. Singer, "The index of elliptic operators. I," Ann. of Math. (2) 87 (1968), 484–530. The boundary theorem used below is M. F. Atiyah, V. K. Patodi, and I. M. Singer, "Spectral asymmetry and Riemannian geometry. I," Math. Proc. Cambridge Philos. Soc. 77 (1975), 43–69, with Parts II and III in Math. Proc. Cambridge Philos. Soc. 78 (1975), 405–432 and 79 (1976), 71–99.

complex spinor bundle, let $E_R \to M$ be a smooth Hermitian vector bundle associated to the fermion representation R, and let ∇^E be a smooth unitary connection with curvature F_R in the same gauge convention as the local anomaly calculation. The chiral Dirac operator

$$\mathcal{D}_{A,+} : \Gamma(S^+ \otimes E_R) \longrightarrow \Gamma(S^- \otimes E_R)$$

is elliptic and hence has finite-dimensional kernel and cokernel. Its Fredholm index is

$$\operatorname{index} \mathcal{D}_{A,+} = \dim\ker \mathcal{D}_{A,+} - \dim\ker \mathcal{D}_{A,-} = \int_M \big[\widehat{A}(TM)\,\mathrm{ch}(E_R, \nabla^E)\big]_4, \tag{51.9}$$

where $[\cdots]_4$ denotes the four-form part of the Chern–Weil form, $\widehat{A}(TM)$ is the $\widehat{A}$-class of the tangent bundle, and $\mathrm{ch}(E_R, \nabla^E)$ is the Chern character form of the gauge bundle. The operator $\mathcal{D}_{A,-}$ is the opposite-chirality Dirac operator $\Gamma(S^- \otimes E_R) \to \Gamma(S^+ \otimes E_R)$.

Index mechanism and four-form normalization. The quoted theorem imports the global equality between a Fredholm index and a characteristic number. The part used by the anomaly calculation can be made more explicit. Let

$$\mathcal{D}_A = \begin{pmatrix} 0 & \mathcal{D}_{A,-} \\ \mathcal{D}_{A,+} & 0 \end{pmatrix}$$

be the self-adjoint Dirac operator on $S \otimes E_R$, and let Γ_5 denote the Euclidean chirality operator, acting by $+1$ on S^+ and by -1 on S^-. For every $t > 0$, elliptic heat-kernel theory makes $\Gamma_5 \mathrm{e}^{-t\mathcal{D}_A^2}$ trace class, and the McKean–Singer cancellation gives

$$\mathrm{Tr}\Big(\Gamma_5 \mathrm{e}^{-t\mathcal{D}_A^2}\Big) = \dim\ker \mathcal{D}_{A,+} - \dim\ker \mathcal{D}_{A,-}. \tag{51.10}$$

Indeed, every positive eigenvector of $\mathcal{D}_{A,-}\mathcal{D}_{A,+}$ is paired by $\mathcal{D}_{A,+}$ with a positive eigenvector of $\mathcal{D}_{A,+}\mathcal{D}_{A,-}$, with the same eigenvalue and opposite chirality contribution. The heat supertrace is therefore independent of t, and the index is the small-t integral of the local supertrace coefficient.

The local coefficient is the analytic part of the index theorem. In the heat-kernel proof it is obtained from the Lichnerowicz formula, Chern–Weil invariance, and Getzler rescaling of the Clifford variables. The pure gauge coefficient can be checked locally, and this check fixes the convention used in the anomaly density. Choose normal coordinates at a point x, trivialize E_R so that the connection one-form vanishes at x, and write $\Gamma^{\mu\nu} = \frac{1}{2}[\Gamma^\mu, \Gamma^\nu]$. The gauge-curvature endomorphism in the square of the Dirac operator is

$$\mathcal{E}_F(x) = \frac{1}{2}\Gamma^{\mu\nu} F_{R,\mu\nu}(x).$$

Terms with fewer than four Clifford matrices have zero Γ_5-supertrace. The first pure gauge term that survives in the diagonal heat-kernel parametrix is therefore

$$(4\pi t)^{-2}\frac{t^2}{2}\,\mathrm{tr}_{\mathrm{spin},R}\left(\Gamma_5 \mathcal{E}_F^2\right).$$

Using

$$\mathrm{tr}_{\mathrm{spin}}\left(\Gamma_5 \Gamma^{\mu\nu}\Gamma^{\rho\sigma}\right) = 4\epsilon^{\mu\nu\rho\sigma}, \qquad \epsilon^{0123} = +1,$$

one obtains

$$(4\pi)^{-2}\frac{1}{2}\frac{1}{4}\,4\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}_R(F_{R,\mu\nu}F_{R,\rho\sigma}) = \frac{1}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}_R(F_{R,\mu\nu}F_{R,\rho\sigma}).$$

The spin-curvature terms in the same Getzler expansion assemble with the pure gauge term into the Chern–Weil form $\widehat{A}(TM)\operatorname{ch}(E_R,\nabla^E)$. With the chapter's anti-Hermitian curvature convention $\mathsf{F}_R = -\mathrm{i}F_R$, the Chern character appearing in Theorem 51.6 is

$$\operatorname{ch}(E_R,\nabla^E) = \operatorname{tr}_R\exp\left(\frac{\mathrm{i}\mathsf{F}_R}{2\pi}\right) = \operatorname{tr}_R\exp\left(\frac{F_R}{2\pi}\right).$$

Consequently the gauge part of the four-form is

$$\left[\operatorname{ch}(E_R,\nabla^E)\right]_4 = \frac{1}{2(2\pi)^2}\operatorname{tr}_R(F_R\wedge F_R) = \frac{1}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}_R(F_{\mu\nu}F_{\rho\sigma})\,\mathrm{d}^4x. \tag{51.11}$$

The first equality is Chern–Weil algebra; the second uses $F_R = \frac{1}{2}F_{\mu\nu}\mathrm{d}x^\mu\wedge\mathrm{d}x^\nu$ and the orientation $\epsilon^{0123} = +1$. This explains why the local Berezinian density (51.8) and the global Fredholm index have the same coefficient once the closed spin hypotheses of the theorem are imposed.

On a flat four-torus or on $\mathbb{R}^4$ after compactification by a smooth finite-action connection for which no boundary contribution is present, the tangent-bundle factor contributes only 1, and the gauge part of (51.9) becomes the density computed in (51.8):

$$\operatorname{index}\mathcal{D}_{A,+} = n_+ - n_- = \frac{1}{32\pi^2}\int_M \epsilon^{\mu\nu\rho\sigma}\operatorname{tr}_R(F_{\mu\nu}F_{\rho\sigma})\,\mathrm{d}^4x. \tag{51.12}$$

Here $n_+ = \dim\ker\mathcal{D}_{A,+}$ and $n_- = \dim\ker\mathcal{D}_{A,-}$, equivalently the numbers of zero modes of chirality $+1$ and -1 in the convention of this chapter. The equality between the integrated heat-kernel density and this integer uses the global hypotheses in the theorem. Without those hypotheses the local Ward identity still has the density (51.8), but its integral is not by itself a Fredholm index.

If M has a boundary, assume first that the Riemannian metric and the gauge connection have product form on a collar $u\in[0,\varepsilon)$ of ∂M. In this collar the chiral Dirac operator can be written

$$\mathcal{D}_A = \gamma(n)\big(\partial_u + B_A\big),$$

where n is the outward unit normal and B_A is the self-adjoint tangential Dirac operator on ∂M. Let

$$\eta_{B_A}(s) = \sum_{\lambda\in\operatorname{Spec}(B_A)\setminus\{0\}}\operatorname{sign}(\lambda)|\lambda|^{-s}$$

for $\operatorname{Re}s$ large, continued meromorphically to $s=0$, and let $h_{B_A} = \dim\ker B_A$. With the APS boundary condition $P_{\geq 0}\psi|_{\partial M} = 0$, where $P_{\geq 0}$ is the spectral projection of B_A onto nonnegative eigenvalues, the Atiyah–Patodi–Singer theorem gives

$$\operatorname{index}\mathcal{D}^{\mathrm{APS}}_{A,+} = \int_M\left[\widehat{A}(TM)\operatorname{ch}(E_R,\nabla^E)\right]_4 - \frac{\eta_{B_A}(0)+h_{B_A}}{2}. \tag{51.13}$$

For non-product data one either deforms to product form near the boundary, or includes the corresponding local transgression term; the sum of the bulk, local boundary, and spectral terms is the invariant integer. The appearance of η_{B_A} shows that boundary or interface anomaly data include spectral terms in addition to the bulk density. Boundary extension, global-anomaly, and inflow applications use (51.13); the local Ward identity in the present chapter uses only the bulk density before this spectral correction is imposed.

51.7 Nonabelian Flavor and Gauge Anomalies

Definition 51.7 (Trace anomaly coefficient)**.** For several Weyl fermions, let T^A be a generator of a global symmetry current and let t^a, t^b be generators of gauge or background gauge fields. The Hermitian generator and curvature conventions are those stated in (45.1), (45.12), and (45.16); in particular the default basis has $\mathrm{tr}(t^a t^b) = \delta^{ab}$. The mixed anomaly coefficient is

$$\mathcal{A}^{Aab} = \frac{1}{2}\left[\mathrm{tr}_{\mathrm{left}}\big(T^A\{t^a, t^b\}\big) - \mathrm{tr}_{\mathrm{right}}\big(T^A\{t^a, t^b\}\big)\right]. \tag{51.14}$$

With background field strengths $F^a_{\mu\nu}$, the corresponding current identity has the form

$$\partial_\mu J^\mu_A = \frac{1}{32\pi^2}\mathcal{A}^{Aab}\epsilon^{\mu\nu\rho\sigma}F^a_{\mu\nu}F^b_{\rho\sigma} + \text{explicit breaking terms}. \tag{51.15}$$

The factor $1/2$ in (51.14) is part of the definition of $\mathcal{A}^{Aab}$; it compensates for the two identical vector insertions in the anticommutator. For a single Dirac fermion with an axial generator acting by $+1$ on the left-handed component and -1 on the right-handed component, and with a unit Abelian vector charge, this definition gives $\mathcal{A} = 2$, so (51.15) reduces to the coefficient $1/(16\pi^2)$ in (51.6).

Proposition 51.8 (Cubic gauge obstruction)**.** *For left-handed Weyl fermions in representations R_i of a gauge group, the cubic part of the one-loop local consistent gauge anomaly vanishes if and only if*

$$\sum_i \mathrm{tr}_{R_i}\big(t^a\{t^b, t^c\}\big) = 0 \qquad \textit{for all } a, b, c. \tag{51.16}$$

If this tensor is nonzero for a dynamical gauge symmetry, no local counterterm can make the corresponding one-loop effective action satisfy the Slavnov–Taylor identity in that background-field chart. If the tensor vanishes, this cubic local obstruction is absent; maintaining the full Slavnov–Taylor identities then reduces to the absence of the remaining BRST-anomaly classes in the chosen theory and regulator scheme.

Proof from local descent. The obstruction is computed in the perturbative local BRST bicomplex on a contractible coordinate chart of the bundle of background connections. The anti-Hermitian curvature $\mathsf{F} = \mathsf{F}^a t^a$ is an even two-form, so the exterior products $\mathsf{F}^a\mathsf{F}^b\mathsf{F}^c$ are symmetric under interchange of a, b, c. Therefore the degree-three invariant polynomial of one left-handed Weyl fermion depends on the representation only through the symmetric cubic tensor:

$$\mathrm{tr}_R(\mathsf{F}^3) = \mathrm{tr}_R(t^{(a}t^b t^{c)})\mathsf{F}^a\mathsf{F}^b\mathsf{F}^c.$$

Equivalently, after one gauge variation has been converted into a ghost and two curvatures are read as external gauge-field insertions, the same tensor is written as

$$\mathrm{tr}_R\big(t^a\{t^b, t^c\}\big)$$

up to the conventional normalization already fixed in (51.14). Right-handed Weyl fermions contribute with the opposite sign because the chirality projector entering the regulated fermion determinant is reversed.

The descent map sends the invariant polynomial to a local representative $I_4^{(1)} \in \Omega^{4,1}_{\mathrm{loc}}$:

$$\mathrm{tr}_R(\mathsf{F}^3) = \mathrm{d}I_5^{(0)}, \qquad sI_5^{(0)} = \mathrm{d}I_4^{(1)}.$$

Changing the subtraction convention for the one-loop effective action adds a local counterterm $B_4^{(0)}$, and the anomaly representative shifts by

$$I_4^{(1)} \longmapsto I_4^{(1)} + sB_4^{(0)} + \mathrm{d}B_3^{(1)}.$$

Thus counterterms change representatives inside the same class of $H^{1,4}(s \mid \mathrm{d})$. The cohomological input used here is the local BRST-classification theorem: for four-dimensional chiral Yang–Mills theories in this polynomial local setting, the semisimple cubic gauge part of $H^{1,4}(s \mid \mathrm{d})$ is generated by the descent classes of the degree-three invariant polynomials $\mathrm{tr}_R(\mathsf{F}^3)$. The vanishing of the total symmetric cubic tensor therefore removes precisely this local class, while a nonzero tensor gives a class outside the image of local counterterm variations.

The non-removability of a nonzero cubic tensor can also be seen without using the full local BRST classification. Restrict the background to a maximal torus T with Lie algebra $\mathfrak{t}$. Since an invariant polynomial on a complex semisimple Lie algebra is determined by its restriction to a Cartan subalgebra, a nonzero cubic invariant has a nonzero restriction to $\mathfrak{t}$. Indeed, regular semisimple elements are conjugate to $\mathfrak{t}$ and form a Zariski-dense subset; an invariant polynomial that vanishes on $\mathfrak{t}$ therefore vanishes on all regular semisimple elements and hence identically. Thus a nonzero semisimple cubic gauge anomaly can be detected after Abelianizing to commuting gauge fields

$$A = A^i H_i, \qquad F^i = \mathrm{d}A^i.$$

On this Abelianized chart the cubic anomaly representative has the form

$$\int_M C_{i;jk}\lambda^i F^j F^k, \qquad C_{i;jk} = C_{i;kj},$$

and its invariant cubic coordinate is the completely symmetric part of $C_{i;jk}$. The most general cubic local counterterm that can change this λFF coefficient has, modulo gauge-invariant terms and total derivatives, the form

$$B_4 = \frac{1}{2}H_{ij,k}A^i A^j F^k, \qquad H_{ij,k} = -H_{ji,k}.$$

Using $\delta_\lambda A^i = \mathrm{d}\lambda^i$, $\delta_\lambda F^i = 0$, and $\mathrm{d}F^i = 0$, its variation on a closed spacetime is

$$\delta_\lambda \int_M B_4 = -\int_M H_{ij,k}\lambda^i F^j F^k.$$

Since $F^j F^k = F^k F^j$, the effective shift of the coefficient symmetric in the two curvature labels is

$$\Delta C_{i;jk} = -\frac{1}{2}\big(H_{ij,k} + H_{ik,j}\big).$$

The complete symmetrization over i, j, k vanishes identically because H is antisymmetric in its first two labels. Therefore local counterterms cannot change the fully symmetric Cartan coefficient of the Abelianized anomaly. By the Cartan-restriction argument above, a nonzero semisimple cubic invariant tensor cannot be removed by a local counterterm. This elementary Abelianized calculation is the part of the obstruction that the anomaly-matching and WZW sections use repeatedly.

The cubic gauge part of the anomaly polynomial for a left-handed Weyl fermion in R_i is proportional to $\mathrm{tr}_{R_i}(\mathsf{F}^3)$. Expanding $\mathsf{F} = \mathsf{F}^a t^a$, the coefficient of $\mathsf{F}^a\mathsf{F}^b\mathsf{F}^c$ is the completely symmetric tensor $\mathrm{tr}_{R_i}(t^{(a}t^bt^{c)})$, equivalently $\frac{1}{6}\,\mathrm{tr}_{R_i}$ summed over all permutations. In component Ward identities the same tensor is written as $\mathrm{tr}_{R_i}(t^a\{t^b,t^c\})$, with the conventional factor absorbed into the anomaly coefficient. Summing over fermions gives (51.16).

A local counterterm changes the anomaly representative by a BRST coboundary. The Abelianized counterterm calculation above already proves that a nonzero semisimple cubic invariant tensor cannot be removed by changing local counterterms. The remaining use of local BRST cohomology is the converse classification statement: in the polynomial local perturbative Yang–Mills setting, once this invariant-polynomial coefficient vanishes, the semisimple cubic gauge-anomaly class itself is absent. Other anomaly classes, global anomaly holonomies, and regulator-specific Slavnov–Taylor restoration conditions are separate data. The last sentence is therefore not an all-purpose nonperturbative existence theorem; it is the perturbative BRST consistency reduction in a specified gauge-fixed regularized framework. □

51.8 Consistency Condition and Descent

The anomaly of a gauge symmetry is constrained before any coefficient is computed. Let $W[A]$ be the renormalized effective action for a background gauge field A. Under an infinitesimal gauge transformation with parameter $\zeta=\zeta^at^a$, write

$$\delta_\zeta W[A]=\mathcal{A}(\zeta,A).$$

Here the infinitesimal action on a functional $F[A]$ is the derivation

$$\delta_\zeta F[A]=\int_M \mathrm{d}^4x\,\frac{\delta F}{\delta A^a_\mu(x)}\,(D_\mu\zeta)^a(x),\qquad D_\mu\zeta=\partial_\mu\zeta+[A_\mu,\zeta]_{\mathrm{H}},\tag{51.17}$$

understood either after introducing a regulator that makes the functional derivatives ordinary finite-dimensional derivatives, or in the formal local functional calculus. The gauge parameters are external test functions and are not varied by δ_{ζ_i}. With these conventions the variations close on the connection:

$$[\delta_{\zeta_1},\delta_{\zeta_2}]A_\mu=D_\mu[\zeta_1,\zeta_2]_{\mathrm{H}}.\tag{51.18}$$

Indeed,

$$\delta_{\zeta_1}\delta_{\zeta_2}A_\mu-\delta_{\zeta_2}\delta_{\zeta_1}A_\mu=[D_\mu\zeta_1,\zeta_2]_{\mathrm{H}}-[D_\mu\zeta_2,\zeta_1]_{\mathrm{H}},$$

and expanding D_μ gives

$$[\partial_\mu\zeta_1,\zeta_2]_{\mathrm{H}}+[\zeta_1,\partial_\mu\zeta_2]_{\mathrm{H}}+[[A_\mu,\zeta_1]_{\mathrm{H}},\zeta_2]_{\mathrm{H}}+[\zeta_1,[A_\mu,\zeta_2]_{\mathrm{H}}]_{\mathrm{H}}=\partial_\mu[\zeta_1,\zeta_2]_{\mathrm{H}}+[A_\mu,[\zeta_1,\zeta_2]_{\mathrm{H}}]_{\mathrm{H}},$$

where the last equality is the Jacobi identity. Equation (51.18) implies, by the chain rule,

$$[\delta_{\zeta_1},\delta_{\zeta_2}]F[A]=\delta_{[\zeta_1,\zeta_2]_{\mathrm{H}}}F[A]$$

for every functional F in the same differentiability class. Applying this identity to $W[A]$, and using $\mathcal{A}(\zeta,A)=\delta_\zeta W[A]$, gives

$$\delta_{\zeta_1}\mathcal{A}(\zeta_2,A)-\delta_{\zeta_2}\mathcal{A}(\zeta_1,A)=\mathcal{A}([\zeta_1,\zeta_2]_{\mathrm{H}},A).\tag{51.19}$$

This is the Wess–Zumino consistency condition. It is an integrability condition: a proposed local expression for $\mathcal{A}$ is the gauge variation of a single effective action only if its first variations satisfy the gauge-algebra commutator above. In BRST language, replace ζ by the ghost c. The anomaly is a local ghost-number-one functional $\mathcal{A}(c, A)$ obeying

$$s\mathcal{A} = 0, \tag{51.20}$$

and adding a local counterterm $X[A]$ shifts it by

$$\mathcal{A} \longmapsto \mathcal{A} + sX.$$

Thus the possible local gauge anomalies are cohomology classes of the BRST differential at ghost number one.

The local form of this statement is the relative cohomology described in Chapter 47. Let

$$\Omega^{p,q}_{\mathrm{loc}}$$

be the vector space of local p-forms of ghost number q, built from the background connection, the ghost, and finitely many derivatives, with coefficients that are polynomial or formal power series in the fields in the chosen local coordinate chart. In the abstract cohomological notation $H(s \mid d)$, write d for the spacetime exterior derivative; in the connection formulas below the same exterior derivative is denoted d. The derivative d has bidegree $(1, 0)$,

$$d : \Omega^{p,q}_{\mathrm{loc}} \longrightarrow \Omega^{p+1,q}_{\mathrm{loc}},$$

and the BRST differential has bidegree $(0, 1)$,

$$s : \Omega^{p,q}_{\mathrm{loc}} \longrightarrow \Omega^{p,q+1}_{\mathrm{loc}}.$$

Both are odd with respect to total degree, and with the sign convention used here

$$d^2 = 0, \qquad s^2 = 0, \qquad sd + ds = 0. \tag{51.21}$$

Thus $\Omega^{\bullet,\bullet}_{\mathrm{loc}}$ is a bicomplex. For a four-dimensional integrated anomaly, choose a representative

$$a^{(1)}_4 \in \Omega^{4,1}_{\mathrm{loc}}.$$

It is a relative cocycle when there exists $a^{(2)}_3 \in \Omega^{3,2}_{\mathrm{loc}}$ such that

$$sa^{(1)}_4 - da^{(2)}_3 = 0. \tag{51.22}$$

Two representatives define the same relative class when there exist $b^{(0)}_4 \in \Omega^{4,0}_{\mathrm{loc}}$ and $b^{(1)}_3 \in \Omega^{3,1}_{\mathrm{loc}}$ such that

$$a^{(1)}_4 \sim a^{(1)}_4 + sb^{(0)}_4 + db^{(1)}_3. \tag{51.23}$$

The quotient of cocycles by this equivalence relation is

$$H^{1,4}(s \mid d).$$

Proposition 51.9 (Local anomalies as relative BRST classes)**.** *At the level of local functionals, a four-dimensional consistent gauge anomaly with local density $a^{(1)}_4$ determines a class $[a^{(1)}_4] \in H^{1,4}(s \mid d)$. Changing the renormalized effective action by a local counterterm changes $a^{(1)}_4$ by the coboundary (51.23). Conversely, a representative of $H^{1,4}(s \mid d)$ defines a local functional whose BRST variation obeys the infinitesimal Wess–Zumino consistency condition.*

Proof. Write the anomalous variation as

$$\delta_\zeta W[A] = \int_M a_4^{(1)}(\zeta, A).$$

Replacing the even parameter ζ by the odd ghost gives $\int_M a_4^{(1)}(c, A)$. The BRST differential s is the Chevalley–Eilenberg differential for infinitesimal gauge transformations, with the field-dependent term in $sc = -\frac{1}{2}[c, c]$ included. Equation (51.19) is precisely the statement that this ghost-number-one integrated functional is a Chevalley–Eilenberg cocycle, hence

$$s \int_M a_4^{(1)} = 0.$$

For a local density this means that $sa_4^{(1)}$ integrates to zero for all compactly supported field and ghost configurations. In the local variational bicomplex, the algebraic Poincare lemma identifies such local top forms with total derivatives modulo globally topological terms fixed separately by the chosen bundle sector. Thus there exists $a_3^{(2)} \in \Omega^{3,2}_{\mathrm{loc}}$ with

$$sa_4^{(1)} = da_3^{(2)},$$

which is (51.22). If $W[A]$ is changed by a local counterterm $\int_M b_4^{(0)}$, its anomalous variation changes by

$$s \int_M b_4^{(0)} = \int_M sb_4^{(0)},$$

and adding an identically integrated total derivative changes the local representative by $db_3^{(1)}$. This is exactly the equivalence (51.23); it is the statement that a finite local counterterm changes the representative but not the relative cohomology class.

Conversely, if (51.22) holds, then on a closed spacetime, or with boundary conditions for which the boundary integral is part of the specified inflow problem, the integrated BRST variation of $\int_M a_4^{(1)}$ vanishes. Expanding s in ghosts recovers the antisymmetrized commutator of two ordinary gauge variations, while $sc = -\frac{1}{2}[c, c]$ supplies the term with gauge parameter $[\zeta_1, \zeta_2]$. This is exactly the infinitesimal Wess–Zumino consistency condition (51.19). □

Finite-regulator scheme meaning of the class. The relative class above is the infinitesimal local shadow of a finite regulator statement. Let $\mathcal{B}_\Lambda$ be a regulated space of background fields, let $\mathcal{G}_\Lambda$ be the regulated background gauge group acting on the right, $A \mapsto A^g$, and suppose a local chart of the regulated determinant line has been chosen so that the fermion functional is written as a scalar $W_\Lambda[A]$. Its finite anomaly cochain is

$$C_\Lambda(g; A) = W_\Lambda[A^g] - W_\Lambda[A]. \tag{51.24}$$

At this stage the statement is an identity in the regulated chart; continuum content enters after locality, convergence, and counterterm estimates have been supplied. Functoriality of the gauge action gives the exact groupoid cocycle equation

$$C_\Lambda(gh; A) = C_\Lambda(h; A^g) + C_\Lambda(g; A), \tag{51.25}$$

with the order fixed by $(A^g)^h = A^{gh}$. Differentiating (51.25) at $g = \exp(t\zeta_1)$, $h = \exp(t\zeta_2)$, and then taking the local continuum representative gives the Wess–Zumino condition (51.19); replacing the even parameters by the ghost gives the relative BRST cocycle (51.22).

Changing the regulated Lagrangian by a finite local counterterm $B_\Lambda[A]$, or choosing a determinant-line frame whose scalar logarithm differs by $B_\Lambda[A]$, replaces W_Λ by $W_\Lambda + B_\Lambda$ and hence

$$C_\Lambda(g;A) \longmapsto C_\Lambda(g;A) + B_\Lambda[A^g] - B_\Lambda[A]. \tag{51.26}$$

The infinitesimal local density changes by $a_4^{(1)} \mapsto a_4^{(1)} + sb_4^{(0)} + db_3^{(1)}$, exactly as in (51.23). Thus regulator and subtraction choices that remain within the same local background-field scheme change the representative, not the relative cohomology class. A nonzero local class means that no local counterterm in that perturbative gauge-fixed scheme can turn the regulated determinant chart into a gauge-invariant scalar effective action. The statement is local and perturbative: vanishing of this class is the condition that removes this obstruction, while an order-by-order construction of a gauge theory must still build the counterterms and verify the Slavnov–Taylor or BV quantum master identities in the chosen regulator. Flat global holonomies of the determinant or Pfaffian line are separate finite-background data and are treated in Chapter 153.

For four-dimensional chiral fermions, this relative cohomology class is represented by the descent of a six-form anomaly polynomial. To state the normalization without hiding convention-dependent factors, introduce the anti-Hermitian connection

$$\mathsf{A} = -\mathrm{i}A, \qquad \mathsf{F} = \mathrm{d}\mathsf{A} + \mathsf{A}^2 = -\mathrm{i}F,$$

where A and F are the Hermitian gauge potential and curvature used in the Yang–Mills chapters. Let c be the corresponding odd anti-Hermitian ghost. On the descent variables,

$$s\mathsf{A} = \mathrm{d}\mathsf{c} + [\mathsf{A},\mathsf{c}], \qquad s\mathsf{c} = -\mathsf{c}^2, \qquad s\mathsf{F} = [\mathsf{F},\mathsf{c}]. \tag{51.27}$$

Products of differential forms below include both matrix multiplication and wedge product with total-degree Koszul signs. For a left-handed Weyl fermion in a representation R, define the six-form

$$I_6^R = \kappa\ \mathrm{tr}_R(\mathsf{F}^3), \qquad \kappa = \frac{\mathrm{i}}{24\pi^2}$$

in the Euclidean effective-action convention. Removing the overall Euclidean phase gives the corresponding Lorentzian Ward-identity normalization. A right-handed Weyl fermion contributes with the opposite sign. For a set of left- and right-handed Weyl fermions,

$$I_6 = \kappa\left[\sum_{i\in L}\mathrm{tr}_{R_i}(\mathsf{F}^3) - \sum_{j\in R}\mathrm{tr}_{R_j}(\mathsf{F}^3)\right].$$

The Chern–Simons five-form $I_5^{(0)}$ is any local form satisfying

$$I_6 = \mathrm{d}I_5^{(0)}.$$

For one left-handed representation R, a standard representative is

$$I_{5,R}^{(0)} = \kappa\ \mathrm{tr}_R\left(\mathsf{A}(\mathrm{d}\mathsf{A})^2 + \frac{3}{2}\mathsf{A}^3\mathrm{d}\mathsf{A} + \frac{3}{5}\mathsf{A}^5\right), \qquad \mathrm{d}I_{5,R}^{(0)} = I_6^R.$$

The numerical coefficients in this representative are fixed by the Chern–Weil homotopy, not by convention. Set

$$\mathsf{A}_t = t\mathsf{A}, \qquad \mathsf{F}_t = \mathrm{d}\mathsf{A}_t + \mathsf{A}_t^2 = t\,\mathrm{d}\mathsf{A} + t^2\mathsf{A}^2.$$

For the invariant polynomial $P(X) = \kappa\ \mathrm{tr}_R(X^3)$, the Bianchi identity for F_t and cyclicity of the trace give

$$\frac{\mathrm{d}}{\mathrm{d}t}P(\mathsf{F}_t) = 3\kappa\,\mathrm{d}\ \mathrm{tr}_R(\mathsf{A}\,\mathsf{F}_t^2).$$

Since $\mathsf{F}_0 = 0$, integration in t gives

$$I_{5,R}^{(0)} = 3\kappa\int_0^1 \mathrm{tr}_R\Big[\mathsf{A}\big(t\,\mathrm{d}\mathsf{A} + t^2\mathsf{A}^2\big)^2\Big]\,\mathrm{d}t.$$

Expanding the square and using cyclicity with the form degrees $\deg\mathsf{A} = 1$, $\deg\mathrm{d}\mathsf{A} = 2$, one obtains

$$\begin{aligned} I_{5,R}^{(0)} &= \kappa\ \mathrm{tr}_R\Big[3\int_0^1 t^2\mathrm{d}t\,\mathsf{A}(\mathrm{d}\mathsf{A})^2 \\ &\qquad + 3\int_0^1 t^3\mathrm{d}t\,\mathsf{A}\big((\mathrm{d}\mathsf{A})\mathsf{A}^2 + \mathsf{A}^2\mathrm{d}\mathsf{A}\big) + 3\int_0^1 t^4\mathrm{d}t\,\mathsf{A}^5\Big] \\ &= \kappa\ \mathrm{tr}_R\left(\mathsf{A}(\mathrm{d}\mathsf{A})^2 + \frac{3}{2}\mathsf{A}^3\mathrm{d}\mathsf{A} + \frac{3}{5}\mathsf{A}^5\right). \end{aligned}$$

Thus the factors $1, \frac{3}{2}, \frac{3}{5}$ are the explicit transgression integrals $3\int_0^1 t^2\mathrm{d}t$, $3\int_0^1 2t^3\mathrm{d}t$, and $3\int_0^1 t^4\mathrm{d}t$. They use only the fixed trace convention and the matrix-valued differential-form product. A different path of connections or a different Chern–Simons representative changes $I_5^{(0)}$ by an exact form and hence changes $I_4^{(1)}$ only by a relative coboundary of the kind already displayed in (51.23). Because $sI_6 = 0$ and $I_6 = \mathrm{d}I_5^{(0)}$,

$$\mathrm{d}(sI_5^{(0)}) = -s(\mathrm{d}I_5^{(0)}) = -sI_6 = 0.$$

The algebraic Poincare lemma on local forms then gives the first descent equation

$$sI_5^{(0)} = \mathrm{d}I_4^{(1)}(\mathsf{c}, \mathsf{A}). \tag{51.28}$$

Applying s again and using (51.21) gives a second descent equation

$$sI_4^{(1)} = \mathrm{d}I_3^{(2)} \tag{51.29}$$

for some $I_3^{(2)} \in \Omega_{\mathrm{loc}}^{3,2}$. Hence $I_4^{(1)}$ is a representative of $H^{1,4}(s \mid d)$ in the convention of (51.22). Replacing the ghost c by an even infinitesimal parameter λ gives the ordinary gauge variation

$$\delta_\lambda\mathsf{A} = \mathrm{d}\lambda + [\mathsf{A}, \lambda], \qquad \delta_\lambda I_5^{(0)} = \mathrm{d}I_4^{(1)}(\lambda, \mathsf{A}).$$

In the same left-handed convention,

$$I_{4,R}^{(1)}(\mathsf{c}, \mathsf{A}) = \kappa\ \mathrm{tr}_R\left[\mathsf{c}\,\mathrm{d}\left(\mathsf{A}\,\mathrm{d}\mathsf{A} + \frac{1}{2}\mathsf{A}^3\right)\right],$$

up to adding d-exact terms and BRST-exact local counterterm variations. The relative coefficient $\frac{1}{2}$ in this four-form is the corresponding homotopy coefficient $6\int_0^1(1-t)t^2\mathrm{d}t = \frac{1}{2}$; the coefficient of $\mathsf{A}\mathrm{d}\mathsf{A}$ is $6\int_0^1(1-t)t\mathrm{d}t = 1$. This is the same descent calculation as above, now evaluated on the vertical gauge direction $\delta_\lambda \mathsf{A} = \mathrm{d}\lambda + [\mathsf{A},\lambda]$. The displayed $\mathsf{c}\,\mathrm{d}(\cdots)$ representative is chosen because its integrated form pairs directly with the effective-action variation (51.30); the equivalent $\mathrm{d}\mathsf{c}$-representatives differ from it by exact local forms. The consistent anomaly is

$$\delta_\lambda W[\mathsf{A}] = \int_M I_4^{(1)}(\lambda,\mathsf{A}). \tag{51.30}$$

The cancellation condition (51.16) is precisely the vanishing of the six-form polynomial I_6 for the gauged symmetry, or equivalently the vanishing of the symmetric cubic tensor

$$d^{abc} = \sum_{i\in L}\mathrm{tr}_{R_i}\big(t^{(a}t^bt^{c)}\big) - \sum_{j\in R}\mathrm{tr}_{R_j}\big(t^{(a}t^bt^{c)}\big).$$

If this polynomial is nonzero only for a background field coupled to a global symmetry, it is an anomaly to be matched rather than an inconsistency.

The representative $I_4^{(1)}$ is not the invariant datum by itself. A local four-dimensional counterterm can move the anomaly among the currents, or equivalently change the Chern–Simons representative $I_5^{(0)}$ by a local transgression. In an Abelianized chart with several background one-forms A^a, the elementary Bardeen counterterm

$$B_4 = h_{ab,c}\,A^aA^bF^c, \qquad h_{ab,c} = -h_{ba,c},$$

changes a representative written as

$$I_4^{(1)} = d_{abc}\,\lambda^aF^bF^c$$

to one with

$$d'_{abc} = d_{abc} + \frac{1}{2}\big(h_{ab,c}+h_{ac,b}\big).$$

The completely symmetric part is unchanged:

$$d'_{(abc)} = d_{(abc)}.$$

This finite algebra is the local form of the statement that counterterms alter representatives but not the six-form polynomial I_6. Consequently, when the next chapter matches the pion Wess–Zumino–Witten functional to the microscopic flavor anomaly, the coefficient being compared is the symmetric cohomology coordinate carried by I_6, not an arbitrary distribution of the anomaly among particular background currents.

Abelianized descent coordinate used in WZW matching. The coefficient comparison in the Wess–Zumino–Witten application uses a particularly simple projection of the same relative class. Restrict the background connection to a commuting finite-dimensional subspace of the flavor Lie algebra, so that

$$A = A^aT_a, \qquad F^a = \mathrm{d}A^a, \qquad [T_a,T_b] = 0$$

inside the chosen test subspace. The degree-three invariant polynomial then reduces to

$$I_6 = C_{abc}\,F^aF^bF^c,$$

where $C_{abc} = C_{(abc)}$ is the symmetric coefficient obtained from the trace in the representation under discussion, multiplied by the normalization factor fixed above. On this Abelianized slice a Chern–Simons representative and its infinitesimal descent are not hidden behind non-Abelian notation:

$$I_5^{(0)} = C_{abc}\, A^a F^b F^c, \qquad \delta_\lambda I_5^{(0)} = C_{abc}\, \mathrm{d}\lambda^a F^b F^c = \mathrm{d}\Big(C_{abc}\lambda^a F^b F^c\Big), \tag{51.31}$$

using $\mathrm{d}F^a = 0$. Thus the displayed $\lambda\, \mathrm{d}A\, \mathrm{d}A$ term is not a low-order approximation to a different anomaly. It is the Abelianized coordinate of the same descent class whose full non-Abelian representatives differ by the coboundaries described in (51.23).

For a compact connected semisimple flavor group, the invariant cubic polynomial is determined by its restriction to a Cartan subalgebra; equivalently, its symmetric trilinear coefficient is recovered from the homogeneous cubic function $P(x) = C_{abc}x^a x^b x^c$ by the polarization identity

$$\begin{aligned} C(x, y, z) = \frac{1}{6}\big[&P(x + y + z) - P(x + y) - P(x + z) - P(y + z) \\ &+ P(x) + P(y) + P(z)\big]. \end{aligned} \tag{51.32}$$

Consequently matching the coefficient of $\mathrm{Tr}(\alpha\, \mathrm{d}A\, \mathrm{d}A)$ for arbitrary commuting flavor backgrounds fixes the symmetric cubic anomaly coordinate. It does not fix a preferred current representative, and it does not choose a Bardeen convention; those choices affect $I_4^{(1)}$ by local counterterm coboundaries while leaving C_{abc} unchanged.

51.9 Index-Normalized Anomaly Polynomials

The descent representative above is written in the effective-action normalization convenient for Ward identities. For global comparisons, especially when gravitational backgrounds are included, it is useful to keep a second normalization whose periods are the ones supplied by the Dirac index. Let ∇^{TM} be a Riemannian connection on the tangent bundle with anti-Hermitian curvature R, and let ∇^E be a unitary connection on the fermion bundle E_R with anti-Hermitian curvature F_R. Define the Chern–Weil forms

$$\mathrm{ch}_R(\mathsf{F}_R) = \mathrm{tr}_R \exp\bigg(\frac{\mathrm{i}\mathsf{F}_R}{2\pi}\bigg), \qquad p_1(TM) = -\frac{1}{8\pi^2}\, \mathrm{tr}_{TM}(\mathsf{R}^2), \tag{51.33}$$

where tr_{TM} is the trace in the defining real tangent representation after complexification. The six-form anomaly polynomial for a single left-handed Weyl fermion in R, in the index normalization, is

$$\mathcal{I}_6(R) = \big[\widehat{A}(TM)\, \mathrm{ch}_R(\mathsf{F}_R)\big]_6. \tag{51.34}$$

A right-handed Weyl fermion contributes the negative of (51.34); equivalently it may be rewritten as a left-handed fermion in the conjugate representation and then the same formula may be applied to the left-handed list.

Expanding

$$\widehat{A}(TM) = 1 - \frac{1}{24}p_1(TM) + O(\deg 8)$$

and keeping six-forms gives

$$\mathcal{I}_6(R) = \frac{1}{6}\, \mathrm{tr}_R \left(\frac{\mathrm{i}\mathsf{F}_R}{2\pi}\right)^3 - \frac{1}{24}\, p_1(TM)\ \mathrm{tr}_R \left(\frac{\mathrm{i}\mathsf{F}_R}{2\pi}\right). \tag{51.35}$$

The first term is the cubic gauge or flavor anomaly. The second term is the mixed gauge-gravitational anomaly. It is present only for symmetries whose generator has a nonzero trace over the chiral fermion representation. Thus it vanishes for a semisimple gauge group in any representation with traceless Lie algebra generators, but it is a genuine constraint for a gauged $U(1)$.

The relation between $\mathcal{I}_6$ and the effective-action normalization used in (51.30) is as follows. Choose a Chern–Simons form $\mathcal{I}_5^{(0)}$ with

$$d\mathcal{I}_5^{(0)} = \mathcal{I}_6, \qquad \delta_\lambda \mathcal{I}_5^{(0)} = d\mathcal{I}_4^{(1)}(\lambda).$$

Then the Euclidean anomalous variation of a left-handed determinant is

$$\delta_\lambda W_E = 2\pi\mathrm{i} \int_M \mathcal{I}_4^{(1)}(\lambda). \tag{51.36}$$

The anti-Hermitian form I_6 of the previous section is this same descent data after multiplying by $2\pi\mathrm{i}$ and rewriting the connection as $\mathsf{A} = -\mathrm{i}A$. Keeping both notations visible prevents a common coefficient error: the index-normalized polynomial is a characteristic class, whereas the Ward identity carries the additional factor $2\pi\mathrm{i}$ and the chosen Lorentzian continuation.

Example 51.10 (Four-dimensional $U(1)$ anomaly polynomial)**.** Let a left-handed Weyl fermion have integer charge q under a background $U(1)$ connection A with curvature $F = dA$. In the Hermitian curvature notation $\mathrm{i}\mathsf{F}_R = qF$, and (51.35) becomes

$$\mathcal{I}_6(q) = \frac{q^3}{6(2\pi)^3} F^3 - \frac{q}{24(2\pi)} p_1(TM) F. \tag{51.37}$$

The cubic $U(1)^3$ coefficient is $\sum_i q_i^3$ over left-handed Weyl fields. The mixed gravitational-$U(1)$ coefficient is $\sum_i q_i$. If the $U(1)$ is gauged, both sums must vanish, in addition to the mixed nonabelian-$U(1)$ coefficients

$$\sum_i q_i \, T_{G_a}(R_i) = 0$$

for each simple nonabelian factor G_a, where $\mathrm{tr}_{R_i}(t^a t^b) = T_{G_a}(R_i)\delta^{ab}$ in the chosen generator normalization.

Example 51.11 ($SU(N)$ cubic anomaly in fundamental coordinates)**.** Let $G = SU(N)$, $N \geq 3$, and choose a reference symmetric tensor

$$d_\square^{abc} = \mathrm{tr}_\square\big(t^{(a} t^b t^{c)}\big)$$

in the monograph trace convention. For an irreducible representation R define $A(R)$ by

$$\mathrm{tr}_R\big(t^{(a} t^b t^{c)}\big) = A(R) d_\square^{abc}.$$

This equation defines the cubic-anomaly coordinate; rescaling all generators rescales both sides and leaves $A(R)$ unchanged. The conjugate representation obeys $A(\overline{R}) = -A(R)$, and every real representation has $A(R) = 0$. In particular,

$$A(\square) = 1, \qquad A(\overline{\square}) = -1, \qquad A(\mathrm{adj}) = 0.$$

Therefore a chiral $SU(N)$ gauge theory containing $n_{\square}$ left-handed fundamentals, $n_{\overline{\square}}$ left-handed antifundamentals, and any number of left-handed adjoints has cubic gauge anomaly

$$d^{abc}_{\text{total}} = (n_{\square} - n_{\overline{\square}})d^{abc}_{\square}. \tag{51.38}$$

A vectorlike Dirac fundamental consists, in a left-handed description, of one $\square$ and one $\overline{\square}$; its contribution to (51.38) vanishes.

Example 51.12 (One Standard Model generation)**.** Use left-handed Weyl fields

$$Q_L : (3,2)_{1/6}, \qquad u^c : (\overline{3},1)_{-2/3}, \qquad d^c : (\overline{3},1)_{1/3},$$
$$L_L : (1,2)_{-1/2}, \qquad e^c : (1,1)_1.$$

The mixed nonabelian-$U(1)_Y$ sums, with the common quadratic index of the fundamental factored out, are

$$SU(3)^2U(1)_Y : \quad 2\left(\frac{1}{6}\right) - \frac{2}{3} + \frac{1}{3} = 0,$$

and

$$SU(2)^2U(1)_Y : \quad 3\left(\frac{1}{6}\right) - \frac{1}{2} = 0.$$

The cubic and mixed gravitational hypercharge sums are

$$\begin{aligned} \sum_{\text{Weyl}} Y^3 &= 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1 = 0, \\ \sum_{\text{Weyl}} Y &= 6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 = 0. \end{aligned} \tag{51.39}$$

The same field list contains $3 + 1 = 4$ weak doublets, so the finite $SU(2)$ anomaly discussed in Chapter 55 is absent for each generation. These cancellations are arithmetic consequences of the charge assignments; they do not follow from perturbative renormalizability alone.

Remark 51.13 (Anomaly matching as equality of anomaly lines)*.* Consider an RG flow from a ultraviolet theory $\mathcal{Q}_{\text{UV}}$ to an infrared description $\mathcal{Q}_{\text{IR}}$, and suppose a background-field groupoid $\mathcal{B}(M)$ for a global symmetry is defined along the flow. Assume that the regulator and the integrating-out procedure preserve this background coupling up to local counterterms. Then the UV and IR partition functions are sections of anomaly lines whose ratio is a local counterterm line. Consequently the anomaly classes of $\mathcal{Q}_{\text{UV}}$ and $\mathcal{Q}_{\text{IR}}$ agree in the quotient by local counterterms. At a finite cutoff, splitting the fields into high- and low-energy variables is a change of variables in the regulated path integral, together with an ordinary integration over the high-energy variables. Under a background gauge transformation, the regulated Berezinian or bosonic density changes by a local phase representing the UV anomaly. The integral over high-energy modes produces an effective functional of the low-energy variables and backgrounds. Any dependence on the arbitrary separating scale is a local functional of the backgrounds and the retained sources, because the modes integrated out have no singularities at distances larger than the cutoff scale used in the split. Thus changing the scale changes the anomaly representative by a gauge variation of a local counterterm.

Taking the continuum and infrared limits within the assumed background-field chart gives two sections over the same background groupoid. Their transformation laws differ only by the local counterterm accumulated during the flow. Passing to the anomaly class, where local counterterm variations are quotiented, gives equality. If the symmetry is spontaneously broken, the Goldstone fields and Wess–Zumino–Witten functional provide one possible infrared section of the same line; if the infrared theory is topological, its invertible response supplies the section. The statement does not assert a unique infrared realization, only equality of the background anomaly class.

51.10 Anomaly Inflow for Local Anomalies

The descent equations have a second interpretation when the four-dimensional theory is placed on the boundary of a five-dimensional invertible response. Let X be an oriented five-manifold with boundary $\partial X = M$, and let A_X be a smooth background connection on X whose restriction to M is the four-dimensional connection A. The local Chern–Simons functional

$$W_X^{\mathrm{bulk}}[\mathsf{A}_X] = -\int_X I_5^{(0)}(\mathsf{A}_X) \tag{51.40}$$

has gauge variation

$$\delta_\lambda W_X^{\mathrm{bulk}} = -\int_X \mathrm{d}I_4^{(1)}(\lambda, \mathsf{A}_X) = -\int_M I_4^{(1)}(\lambda, \mathsf{A}). \tag{51.41}$$

Writing the four-dimensional effective action in (51.30) as $W_M[\mathsf{A}]$, one obtains the bulk-boundary identity

$$\delta_\lambda \left(W_M[\mathsf{A}] + W_X^{\mathrm{bulk}}[\mathsf{A}_X] \right) = \int_M I_4^{(1)}(\lambda, \mathsf{A}) - \int_{\partial X} I_4^{(1)}(\lambda, \mathsf{A}_X) = 0, \tag{51.42}$$

where the boundary orientation on ∂X is the orientation of M. This is anomaly inflow. The boundary theory by itself transforms by the consistent anomaly; the bulk Chern–Simons response transforms by the opposite boundary term. The gauge-invariant object is the combined bulk-boundary system.

In the Callan–Harvey realization, X is a physical gapped bulk or a domain-wall geometry, and M supports chiral modes localized at the wall. Integrating out the massive bulk modes produces a Chern–Simons response whose coefficient jumps across the wall. The jump supplies precisely the term $-\int_M I_4^{(1)}$ in (51.41), while the localized chiral modes supply $+\int_M I_4^{(1)}$. The boundary chiral modes retain their consistent anomaly, and gauge invariance is a property of the combined bulk-boundary system.

The formula (51.40) is local in a chosen trivialization. On topologically nontrivial bundles the Chern–Simons form is defined patchwise, and the exponentiated bulk functional must be defined by differential-geometric data with integral periods, equivalently by an invertible five-dimensional field theory whose curvature is the anomaly polynomial I_6. The local descent computation fixes the infinitesimal variation. Global questions require the finite version: a closed five-manifold carrying a background field is assigned a phase, and cutting it open gives the anomalous transformation law of a boundary theory. That finite form is the starting point of the global-anomaly discussion in Chapter 55.

51.11 Consistent and Covariant Currents

The current obtained by differentiating the effective action,

$$J^{a\mu}_{\text{cons}}(x) = \frac{\delta W[A]}{\delta A^a_\mu(x)},$$

is the consistent current. Its anomaly is the variation of a functional and therefore obeys the Wess–Zumino consistency condition. This integrability is the reason that the consistent anomaly is the object classified by $H^{1,4}(s \mid d)$. More precisely, the anomaly class modulo local counterterms is the consistent descent class represented by $\int_M I_4^{(1)}$; covariant Ward identities are obtained after taking a local differential of this class and adding a Bardeen–Zumino polynomial.

The consistent current need not transform covariantly under background gauge transformations. There is a local polynomial current

$$J^{a\mu}_{\text{BZ}}(A)$$

called the Bardeen–Zumino current, whose addition gives a covariant current

$$J^{a\mu}_{\text{cov}} = J^{a\mu}_{\text{cons}} + J^{a\mu}_{\text{BZ}}.$$

In the anti-Hermitian form convention used in the descent formula above, the dual three-form of the Bardeen–Zumino current for one left-handed fermion in the representation R may be chosen as

$$\star J^a_{\text{BZ}} = \kappa\, \mathrm{tr}_R \left[T^a_R \left(\mathsf{AF} + \mathsf{FA} - \frac{1}{2}\mathsf{A}^3 \right) \right], \qquad \kappa = \frac{\mathrm{i}}{24\pi^2}, \tag{51.43}$$

where T^a_R is the anti-Hermitian generator used to vary A, products include wedge product and matrix multiplication, and $\star J^a_{\text{BZ}}$ denotes

$$\star J^a_{\text{BZ}} = \frac{1}{3!}\epsilon_{\mu\nu\rho\sigma} J^{a\mu}_{\text{BZ}} \mathrm{d}x^\nu \wedge \mathrm{d}x^\rho \wedge \mathrm{d}x^\sigma.$$

Equivalently, with

$$\mathsf{A} = \mathsf{A}_\mu \mathrm{d}x^\mu, \qquad \mathsf{F} = \frac{1}{2}\mathsf{F}_{\rho\sigma}\mathrm{d}x^\rho \wedge \mathrm{d}x^\sigma,$$

the component current is the local polynomial

$$J^{a\mu}_{\text{BZ}} = \frac{\kappa}{2}\,\epsilon^{\mu\nu\rho\sigma}\, \mathrm{tr}_R \left[T^a_R \left(\mathsf{A}_\nu \mathsf{F}_{\rho\sigma} + \mathsf{F}_{\rho\sigma}\mathsf{A}_\nu - \mathsf{A}_\nu \mathsf{A}_\rho \mathsf{A}_\sigma \right) \right]. \tag{51.44}$$

Expanding

$$\mathsf{F}_{\rho\sigma} = \partial_\rho \mathsf{A}_\sigma - \partial_\sigma \mathsf{A}_\rho + [\mathsf{A}_\rho, \mathsf{A}_\sigma]$$

and using antisymmetry of $\epsilon^{\mu\nu\rho\sigma}$ gives the derivative form

$$J^{a\mu}_{\text{BZ}} = \kappa\,\epsilon^{\mu\nu\rho\sigma}\, \mathrm{tr}_R \left[T^a_R \left(\mathsf{A}_\nu \partial_\rho \mathsf{A}_\sigma + \partial_\rho \mathsf{A}_\sigma\, \mathsf{A}_\nu + \frac{1}{2}\mathsf{A}_\nu \mathsf{A}_\rho \mathsf{A}_\sigma + \mathsf{A}_\rho \mathsf{A}_\sigma \mathsf{A}_\nu \right) \right]. \tag{51.45}$$

The two displayed component expressions are the same polynomial written once with F and once with $\partial\mathsf{A}$. The ordering of matrices in (51.45) is part of the convention; replacing it by a commutative shorthand would lose the distinction between consistent and covariant currents for a nonabelian background.

Abelian calibration of the Bardeen–Zumino factor. The simplest check of the normalization is the one-generator commuting subspace. Let

$$I_6 = C\,\mathsf{F}^3, \qquad \mathsf{F} = \mathrm{d}\mathsf{A},$$

where C includes the trace and the factor κ. The corresponding Chern–Simons and consistent descent forms are

$$I_5^{(0)} = C\,\mathsf{A}\mathsf{F}^2, \qquad I_4^{(1)}(\lambda, \mathsf{A}) = C\,\lambda\,\mathsf{F}^2. \tag{51.46}$$

Thus the consistent Ward identity carries the coefficient C. In the same Abelianized coordinate the Bardeen–Zumino current (51.43) reduces to

$$\star J_{\mathrm{BZ}} = 2C\,\mathsf{A}\mathsf{F}, \qquad \mathrm{d}\star J_{\mathrm{BZ}} = 2C\,\mathsf{F}^2, \tag{51.47}$$

because $\mathsf{A}^3 = 0$ for a single one-form and $\mathrm{d}\mathsf{F} = 0$. The covariant Ward representative obtained by improving the current is therefore

$$C\,\lambda\,\mathsf{F}^2 + 2C\,\lambda\,\mathsf{F}^2 = 3C\,\lambda\,\mathsf{F}^2. \tag{51.48}$$

This factor of three is a normalization check, not a new anomaly class. The first term is the descent representative of the effective-action variation; the additional two units are the local divergence of the Bardeen–Zumino polynomial. In the nonabelian case the same statement is encoded by the ordered polynomial in (51.45); the covariant current transforms covariantly, but it is not in general the differential of a local effective action. For a set of Weyl fermions, sum (51.43) over left-handed representations and subtract the corresponding right-handed representations, in parallel with the anomaly polynomial I_6. The divergence of J_{cov} is gauge covariant and is often the most compact local expression of the anomaly. However, J_{cov} is generally not the functional derivative of a local effective action. Thus it is not the current that appears directly in the integrability condition (51.19). The consistent anomaly is the cohomology class. The covariant anomaly is obtained from a particular representative by applying the local differential of that class and adding the Bardeen–Zumino polynomial; it is a covariant representative of the Ward identity, not a representative of a second obstruction class. Gauge-anomaly cancellation is therefore a statement about the consistent class; after that class vanishes, covariant currents may be used as convenient representatives of gauge-covariant Ward identities.

51.12 Local Counterterms and Preserved Currents

An anomaly representative depends on the choice of local counterterms, while its cohomology class does not. Let B denote background gauge fields for a set of currents, and let $W[B]$ be the renormalized generating functional. Adding a local functional $I_{\mathrm{ct}}[B]$ changes

$$W[B] \longmapsto W[B] + I_{\mathrm{ct}}[B].$$

If a background gauge transformation with parameter ζ has anomaly

$$\delta_\zeta W[B] = \mathcal{A}(\zeta, B),$$

then after the counterterm shift

$$\mathcal{A}(\zeta, B) \longmapsto \mathcal{A}(\zeta, B) + \delta_\zeta I_{\mathrm{ct}}[B].$$

The second term is BRST-exact in the local functional complex. It can move the displayed divergence among currents that are coupled to backgrounds, but it cannot remove a nonzero anomaly class.

This distinction is essential when one of the currents is gauged dynamically. The gauge current must satisfy the Slavnov–Taylor identity; otherwise the gauge-fixed state space cannot be quotiented to a positive physical Hilbert space. For a vector and an axial current coupled to background fields, a Bardeen counterterm can shift the local expression of the anomaly between vector and axial Ward identities. The convention used in (51.6) is the one in which the vector gauge Ward identity is preserved and the axial current carries the anomaly. For a purely global symmetry, by contrast, the anomaly is allowed and becomes an RG-invariant background response to be matched by the infrared theory.

51.13 The Strong CP Parameter

In QCD with N_f quark flavors, the Euclidean gauge action may include the topological term

$$\mathrm{i}\theta\,\frac{1}{32\pi^2}\int \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma})\,\mathrm{d}^4x.$$

The complex quark mass matrix M was introduced in (45.68). A flavor-singlet axial rotation changes the phase of $\det M$ and, by the anomaly, shifts θ. With the convention used in this chapter, the invariant CP-odd parameter is

$$\bar{\theta} = \theta + \arg\det M. \tag{51.49}$$

If the sign convention for γ_5 or for the topological density is reversed, both transformation laws change together and the invariant combination is correspondingly rewritten. The invariant statement is that only the anomaly-invariant combination of the topological angle and the quark mass phase is physically meaningful.

51.14 Topological Charge Sectors and Periodicity

For gauge group $SU(N_c)$ on a compact Euclidean four-manifold, or on $\mathbb{R}^4$ with finite-action boundary conditions so that the connection extends to a bundle over S^4, the topological charge is

$$Q[A] = \frac{1}{32\pi^2}\int \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma})\,\mathrm{d}^4x.$$

With F regarded as the matrix-valued curvature of the fundamental connection and with the matrix trace convention fixed in Chapter 45, $Q[A] \in \mathbb{Z}$. This integer statement is a Chern–Weil statement about the curvature form; rewriting it in components must use the same generator normalization as the one used to define $F = A^a t^a$. The Euclidean functional integral decomposes formally into connected components of fixed topological charge:

$$Z(\theta) = \sum_{Q\in\mathbb{Z}} \mathrm{e}^{\mathrm{i}\theta Q} Z_Q.$$

This expression makes the 2π-periodicity in θ manifest when no additional global identifications modify the allowed bundles or background fields.

The axial anomaly explains why θ cannot be assigned independently of the fermion mass phases. A flavor-singlet chiral rotation changes the fermion measure by a phase proportional to $Q[A]$ and changes $\arg\det M$ by the opposite amount in the mass term. The sector decomposition therefore gives the same invariant parameter $\bar\theta = \theta + \arg\det M$ as the local anomalous Ward identity. The local identity and the topological decomposition are two presentations of the same anomalous Jacobian in a gauge background with nonzero topological charge.

51.15 Topological Susceptibility as a Regulated Cumulant

The phrase "topological susceptibility" is often used as if it were a number attached directly to a formal continuum expression $\langle Q^2\rangle/V_4$. The object needed in QCD is more precise: it is a thermodynamic and continuum limit of a cumulant in a specified regulator.

Definition 51.14 (Finite-regulator theta data)**.** Fix a gauge-invariant Euclidean regulator on an oriented four-volume V_4. Finite-regulator theta data consist of nonnegative sector weights $Z_{\Lambda,V_4,Q}$, $Q \in \mathbb{Z}$, for the vectorlike theory at $\theta = 0$, such that

$$Z_{\Lambda,V_4}(\theta) = \sum_{Q\in\mathbb{Z}} \mathrm{e}^{\mathrm{i}\theta Q} Z_{\Lambda,V_4,Q} \tag{51.50}$$

is finite for θ in a neighborhood of zero. The finite-regulator vacuum-energy density near $\theta = 0$ is

$$E_{\Lambda,V_4}(\theta) = -\frac{1}{V_4}\log Z_{\Lambda,V_4}(\theta), \tag{51.51}$$

where the logarithm is the branch continuous from $\theta = 0$.

Remark 51.15 (Finite-volume susceptibility identity). For every finite-regulator theta data set,

$$E''_{\Lambda,V_4}(0) = \frac{1}{V_4}\left(\langle Q^2\rangle_{\Lambda,V_4,0} - \langle Q\rangle^2_{\Lambda,V_4,0}\right), \tag{51.52}$$

where the expectation is with respect to the positive weights $Z_{\Lambda,V_4,Q}/Z_{\Lambda,V_4}(0)$. If CP is preserved at $\theta = 0$, then $\langle Q\rangle_{\Lambda,V_4,0} = 0$, and the right hand side is $V_4^{-1}\langle Q^2\rangle_{\Lambda,V_4,0}$. Differentiate (51.50):

$$Z'(0) = \mathrm{i}\sum_Q Q Z_Q, \qquad Z''(0) = -\sum_Q Q^2 Z_Q,$$

where the regulator and volume labels are suppressed. Hence

$$\frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\log Z(\theta)\bigg|_{\theta=0} = \frac{Z''(0)}{Z(0)} - \left(\frac{Z'(0)}{Z(0)}\right)^2 = -\langle Q^2\rangle + \langle Q\rangle^2.$$

Multiplying by $-1/V_4$ gives (51.52). CP sends $Q \mapsto -Q$; if the $\theta = 0$ weights are CP invariant, the first moment vanishes.

Proposition 51.16 (Finite-regulator theta-cumulant hierarchy)**.** *For finite-regulator theta data, define the charge cumulants at $\theta = 0$ by*

$$\mathcal{K}_Q(t) = \log \left\langle \mathrm{e}^{tQ} \right\rangle_{\Lambda,V_4,0} = \sum_{n\geq 1} \kappa_n^{(Q)} \frac{t^n}{n!}. \tag{51.53}$$

Then, in the analytic neighborhood where Definition 51.14 applies,

$$E_{\Lambda,V_4}(\theta) - E_{\Lambda,V_4}(0) = -\frac{1}{V_4}\,\mathcal{K}_Q(\mathrm{i}\theta), \qquad \partial_\theta^n E_{\Lambda,V_4}(0) = -\frac{\mathrm{i}^n}{V_4}\,\kappa_n^{(Q)}. \tag{51.54}$$

If the $\theta = 0$ weights are CP invariant, all odd charge cumulants vanish and the finite-volume expansion can be written

$$E_{\Lambda,V_4}(\theta) - E_{\Lambda,V_4}(0) = \frac{1}{2}\chi_{\Lambda,V_4}\theta^2 \left(1 + b_{2,\Lambda,V_4}\theta^2 + b_{4,\Lambda,V_4}\theta^4 + O(\theta^6)\right), \tag{51.55}$$

with

$$\chi_{\Lambda,V_4} = \frac{\kappa_2^{(Q)}}{V_4}, \qquad b_{2,\Lambda,V_4} = -\frac{\kappa_4^{(Q)}}{12\kappa_2^{(Q)}}, \qquad b_{4,\Lambda,V_4} = \frac{\kappa_6^{(Q)}}{360\kappa_2^{(Q)}}, \tag{51.56}$$

provided $\kappa_2^{(Q)} \neq 0$.

These are finite-volume identities, not yet continuum statements. In the thermodynamic limit the derivatives in (51.54) must be taken on a selected analytic branch of the vacuum energy; branch crossings or spontaneous CP breaking can obstruct a single Taylor series. In addition, if the renormalization convention permits adding a local vacuum counterterm $f_{\mathrm{ct}}(\theta)$, then the displayed derivatives shift by $f_{\mathrm{ct}}^{(n)}(0)$. The same counterterm convention must therefore be used for χ_{top}, b_2, b_4, and any Witten–Veneziano matching data.

Proof. Divide (51.50) by $Z_{\Lambda,V_4}(0)$. The ratio is the characteristic function $\langle \mathrm{e}^{\mathrm{i}\theta Q}\rangle_{\Lambda,V_4,0}$, hence

$$\log \frac{Z_{\Lambda,V_4}(\theta)}{Z_{\Lambda,V_4}(0)} = \mathcal{K}_Q(\mathrm{i}\theta).$$

Subtracting $E_{\Lambda,V_4}(0)$ from (51.51) gives the first identity in (51.54); differentiating gives the second. CP invariance pairs the Q and $-Q$ sector weights, so $\mathcal{K}_Q(t) = \mathcal{K}_Q(-t)$ and the odd cumulants vanish. Expanding the even part gives

$$E(\theta) - E(0) = \frac{\kappa_2^{(Q)}}{2V_4}\theta^2 - \frac{\kappa_4^{(Q)}}{24V_4}\theta^4 + \frac{\kappa_6^{(Q)}}{720V_4}\theta^6 + \cdots.$$

Comparison with (51.55) gives (51.56). The last paragraph is the definition of branchwise differentiability and local counterterm covariance for the finite-volume identities just derived. □

Proposition 51.17 (Local density form and contact convention)**.** *Assume that the finite regulator is placed on a closed four-volume, such as a periodic box, or on a volume with boundary data for which the topological charge is fixed, and that it supplies a regulated local density $q_{\Lambda,V_4}(x)$ satisfying*

$$Q = \int_{V_4} q_{\Lambda,V_4}(x)\,\mathrm{d}^4 x \tag{51.57}$$

configuration by configuration. Then the finite-volume susceptibility is equivalently, with the integrals interpreted as the regulator's finite sum or integral,

$$E''_{\Lambda,V_4}(0) = \frac{1}{V_4}\int_{V_4} \mathrm{d}^4x \int_{V_4} \mathrm{d}^4y \left\langle q_{\Lambda,V_4}(x) q_{\Lambda,V_4}(y)\right\rangle^{\mathrm{c}}_{\Lambda,V_4,0}. \tag{51.58}$$

If the regulator and state are translation invariant on a periodic volume, this reduces to the zero-momentum two-point function

$$E''_{\Lambda,V_4}(0) = \int_{V_4} \mathrm{d}^4x \left\langle q_{\Lambda,V_4}(x) q_{\Lambda,V_4}(0)\right\rangle^{\mathrm{c}}_{\Lambda,V_4,0}. \tag{51.59}$$

Proof. Insert (51.57) into the finite cumulant:

$$\langle Q^2\rangle - \langle Q\rangle^2 = \int_{V_4} \mathrm{d}^4x \int_{V_4} \mathrm{d}^4y \left(\langle q_{\Lambda,V_4}(x) q_{\Lambda,V_4}(y)\rangle - \langle q_{\Lambda,V_4}(x)\rangle\langle q_{\Lambda,V_4}(y)\rangle\right).$$

Dividing by V_4 and using (51.52) gives (51.58). Translation invariance sets the connected correlator equal to a function of $x - y$; the double integral is then V_4 times the integral over one relative coordinate. □

The continuum formula $\chi_{\mathrm{top}} = \int \mathrm{d}^4x \,\langle q(x)q(0)\rangle^{\mathrm{c}}$ is therefore not a bare multiplication rule for two formal $F\widetilde{F}$ densities at coincident points. It is the limit of the regulated zero-momentum curvature above, with the contact term selected by the same local θ-counterterm convention used in Definition 51.18. Changing the convention by an additive local vacuum-energy term $\frac{1}{2}c_\theta\theta^2$ shifts both $E''(0)$ and the integrated contact part of the density two-point function by c_θ. Thus Witten–Veneziano matching uses the renormalized zero-momentum curvature of pure Yang–Mills, not an unrenormalized point-split product with its contact term silently discarded.

Definition 51.18 (Topological susceptibility)**.** The topological susceptibility of a continuum theory in a specified phase and regulator class is the limit

$$\chi_{\mathrm{top}} = \lim_{\Lambda\to\infty}\lim_{V_4\to\infty} E''_{\Lambda,V_4}(0), \tag{51.60}$$

when this limit exists and is independent of the allowed regulator choices after the local counterterm convention for the theta term has been fixed. For pure Yang–Mills we write the corresponding quantity as χ_{YM}.

This definition intentionally separates three facts which are often conflated. The finite identity (51.52) is elementary. The existence and regulator independence of (51.60) is a nonperturbative statement about the theory. The use of χ_{YM} inside a chiral effective description with dynamical quarks is a further large-N_c matching assumption, developed in the next chapter.

51.16 Theta Branches and Cluster Decomposition

The sector sum $\sum_Q \mathrm{e}^{\mathrm{i}\theta Q} Z_Q$ explains 2π-periodicity, but it does not by itself identify a pure infinite-volume vacuum. In a theory with several large-volume branches the thermodynamic limit selects the branch of smallest energy density, except at degeneracy loci where a phase-selection prescription is needed. Cluster decomposition is the local diagnostic that one has chosen a pure phase rather than a convex mixture.

Definition 51.19 (Theta-branch data)**.** Theta-branch data for a regulated Euclidean gauge theory in a volume V_4 consists of a finite or countable branch set $\mathcal{B}$, branch partition functions $Z_{\Lambda,V_4,\beta}(\theta)$, and local normalized states $\omega_{\Lambda,V_4,\beta,\theta}$ on gauge-invariant observables, such that

$$Z_{\Lambda,V_4}(\theta) = \sum_{\beta\in\mathcal{B}} Z_{\Lambda,V_4,\beta}(\theta), \qquad p_{\Lambda,V_4,\beta}(\theta) = \frac{Z_{\Lambda,V_4,\beta}(\theta)}{Z_{\Lambda,V_4}(\theta)}. \tag{51.61}$$

When the limits exist, write

$$E_\beta(\theta) = -\lim_{\Lambda\to\infty}\lim_{V_4\to\infty}\frac{1}{V_4}\log Z_{\Lambda,V_4,\beta}(\theta) \tag{51.62}$$

for the branch energy density. The data are 2π-periodic by branch relabeling when there is a bijection $T : \mathcal{B} \to \mathcal{B}$ with

$$E_{T\beta}(\theta + 2\pi) = E_\beta(\theta). \tag{51.63}$$

Remark 51.20 (Thermodynamic branch selection)*.* Assume a finite branch set and a fixed θ at which the limiting branch energies $E_\beta(\theta)$ exist. If a unique branch β_* satisfies

$$E_{\beta_*}(\theta) < E_\beta(\theta) \qquad \text{for every } \beta \neq \beta_*, \tag{51.64}$$

and if each branch partition function has the form

$$Z_{\Lambda,V_4,\beta}(\theta) = \exp\{-V_4E_\beta(\theta) + o(V_4)\} \tag{51.65}$$

uniformly over the finite branch set, then

$$\lim_{\Lambda\to\infty}\lim_{V_4\to\infty} p_{\Lambda,V_4,\beta_*}(\theta) = 1, \qquad \lim_{\Lambda\to\infty}\lim_{V_4\to\infty} p_{\Lambda,V_4,\beta}(\theta) = 0 \quad (\beta \neq \beta_*). \tag{51.66}$$

For $\beta \neq \beta_*$,

$$\frac{Z_{\Lambda,V_4,\beta}(\theta)}{Z_{\Lambda,V_4,\beta_*}(\theta)} = \exp\{-V_4(E_\beta(\theta) - E_{\beta_*}(\theta)) + o(V_4)\}.$$

The exponent tends to $-\infty$ because the energy-density gap is positive. Since the branch set is finite, the sum of all nonminimal ratios tends to zero. Dividing the branch weights by Z_{Λ,V_4,β_*} gives (51.66).

Remark 51.21 (Cluster test for mixtures of theta branches)*.* Let $\{\omega_\beta\}_{\beta\in\mathcal{B}}$ be translation-invariant pure states for gauge-invariant local observables, and assume each branch clusters: for local observables $\mathcal{O}, \mathcal{P}$,

$$\lim_{|a|\to\infty} \omega_\beta(\mathcal{O}(a)\mathcal{P}(0)) = \omega_\beta(\mathcal{O})\omega_\beta(\mathcal{P}). \tag{51.67}$$

For a convex combination

$$\omega = \sum_{\beta\in\mathcal{B}} p_\beta\omega_\beta, \qquad p_\beta \geq 0, \qquad \sum_\beta p_\beta = 1, \tag{51.68}$$

one has

$$\lim_{|a|\to\infty} [\omega(\mathcal{O}(a)\mathcal{P}(0)) - \omega(\mathcal{O})\omega(\mathcal{P})] = \mathrm{Cov}_p\big(\omega_\beta(\mathcal{O}), \omega_\beta(\mathcal{P})\big). \tag{51.69}$$

Thus the mixture clusters for all local observables only if the branch labels cannot be distinguished by local one-point functions on the support of p_β, or if the mixture is supported on a single branch. Using the branch clustering assumption,

$$\lim_{|a|\to\infty} \omega(\mathcal{O}(a)\mathcal{P}(0)) = \sum_\beta p_\beta \omega_\beta(\mathcal{O})\omega_\beta(\mathcal{P}).$$

The product of the one-point functions in the mixed state is

$$\omega(\mathcal{O})\omega(\mathcal{P}) = \left(\sum_\beta p_\beta \omega_\beta(\mathcal{O})\right)\left(\sum_\gamma p_\gamma \omega_\gamma(\mathcal{P})\right).$$

Their difference is precisely the covariance of the two branch-dependent one-point functions with respect to the probability distribution p.

Remark 51.22 (Theta vacua as theorem, hypothesis, and convention)*.* At finite regulator the sector decomposition, cumulant identity, and branch mixture algebra above are exact. A continuum statement that theta labels superselection sectors requires additional input: the existence of the continuum gauge theory, the infinite-volume limits of the local states, the cluster property inside a selected branch, and the action of large gauge transformations or theta-shift automorphisms on the physical representation. The monograph therefore uses "theta vacuum" only after these data have been specified. Otherwise it speaks of a theta parameter in the regulated action or of a theta-branch hypothesis.

51.17 The BPST Instanton and the Semiclassical Vertex

We now construct the basic finite-action representative of the sector $Q = 1$. This section is Euclidean: coordinates are x^μ, $\mu = 1, \dots, 4$, the metric is $\delta_{\mu\nu}$, and $\epsilon_{1234} = +1$. The chapter's gauge-field convention remains Hermitian, so

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}[A_\mu, A_\nu].$$

For explicit $SU(2)$ formulas it is economical to use the auxiliary basis

$$T_a = \frac{\sigma_a}{2}, \qquad [T_a, T_b] = \mathrm{i}\epsilon^c{}_{ab} T_c, \qquad \mathrm{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}.$$

This is only a coordinate choice for the displayed solution. The monograph's default trace-delta basis is $t_a = \sqrt{2}\, T_a$. If

$$\mathcal{L}_{\text{here}} = -\frac{1}{4g_{\text{YM}}^2} \mathrm{tr}_\delta(F_{\mu\nu}F^{\mu\nu})$$

is rewritten in the T_a basis, then the corresponding half-trace coupling is $g_{\text{ht}} = \sqrt{2}\, g_{\text{YM}}$. Thus the familiar instanton action $8\pi^2/g_{\text{ht}}^2$ is the same number as $4\pi^2/g_{\text{YM}}^2$ in the active trace-delta coupling coordinate.

Define the self-dual 't Hooft symbols by

$$\eta^a_{ij} = \epsilon_{aij}, \qquad \eta^a_{i4} = \delta_{ai}, \qquad \eta^a_{4i} = -\delta_{ai}, \qquad i, j, a = 1, 2, 3,$$

so that

$$\eta^a_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}\eta^a_{\rho\sigma}, \qquad \sum_{a,\mu,\nu} \eta^a_{\mu\nu}\eta^a_{\mu\nu} = 12.$$

The contraction identity used below is

$$\epsilon^a{}_{bc}\eta^b_{\mu\rho}\eta^c_{\nu\sigma}y^\rho y^\sigma = r^2\eta^a_{\mu\nu} + y_\mu\eta^a_{\nu\lambda}y^\lambda - y_\nu\eta^a_{\mu\lambda}y^\lambda. \tag{51.70}$$

It follows by checking the three possible positions of the index 4 in the defining equations for η, and then using antisymmetry in μ, ν. This identity is the finite-dimensional algebraic input that turns the apparently inhomogeneous derivative and commutator terms in the curvature into a multiple of $\eta^a_{\mu\nu}$. Let $a^\mu \in \mathbb{R}^4$ and $\rho > 0$. Put $y = x - a$ and $r^2 = y_\mu y_\mu$. The BPST one-instanton in regular gauge is

$$A_\mu(x) = \frac{2\,\eta^a_{\mu\nu}y^\nu}{r^2+\rho^2}\,T_a. \tag{51.71}$$

Write $f = r^2 + \rho^2$ and $A_\mu = A^a_\mu T_a$. Since

$$\partial_\mu A^a_\nu - \partial_\nu A^a_\mu = -\frac{4}{f}\,\eta^a_{\mu\nu} - \frac{4}{f^2}\left(y_\mu\eta^a_{\nu\lambda}y^\lambda - y_\nu\eta^a_{\mu\lambda}y^\lambda\right)$$

and

$$\epsilon^a{}_{bc}A^b_\mu A^c_\nu = \frac{4}{f^2}\left(r^2\eta^a_{\mu\nu} + y_\mu\eta^a_{\nu\lambda}y^\lambda - y_\nu\eta^a_{\mu\lambda}y^\lambda\right),$$

the curvature convention $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + \epsilon^a{}_{bc}A^b_\mu A^c_\nu$ gives

$$F_{\mu\nu}(x) = -\frac{4\rho^2}{(r^2+\rho^2)^2}\,\eta^a_{\mu\nu}T_a. \tag{51.72}$$

Hence $F_{\mu\nu} = \widetilde{F}_{\mu\nu}$, where $\widetilde{F}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F_{\rho\sigma}$. The connection is smooth at $r = 0$. At infinity it is gauge-equivalent to a pure gauge of winding one. Equivalently, applying the gauge transformation

$$g(y) = \frac{y_4\mathbf{1} + \mathrm{i}y_i\sigma_i}{r}, \qquad g : S^3_\infty \to SU(2),$$

produces the singular-gauge representative

$$A^{\text{sing}}_\mu(x) = \frac{2\rho^2\,\bar{\eta}^a_{\mu\nu}y^\nu}{r^2(r^2+\rho^2)}\,T_a,$$

where $\bar{\eta}$ is the anti-self-dual 't Hooft symbol obtained from η by reversing the signs of the components with one index equal to 4. This representative decays as r^{-3} at infinity and has a gauge-coordinate singularity at $y = 0$. The two representatives describe the same smooth connection on the one-instanton bundle over S^4.

The action saturation is an algebraic consequence of self-duality. In the half-trace component convention,

$$S^{\text{ht}}_E = \frac{1}{4g^2_{\text{ht}}}\int_{\mathbb{R}^4} \mathrm{d}^4x\, F^a_{\mu\nu}F^a_{\mu\nu}.$$

Completing the square gives, for any finite-action field in the sector of charge k,

$$S^{\text{ht}}_E = \frac{1}{8g^2_{\text{ht}}}\int (F^a_{\mu\nu} \mp \widetilde{F}^a_{\mu\nu})^2\,\mathrm{d}^4x \pm \frac{1}{4g^2_{\text{ht}}}\int F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu}\,\mathrm{d}^4x \geq \frac{8\pi^2}{g^2_{\text{ht}}}|k|. \tag{51.73}$$

Equality holds precisely for self-dual fields when $k > 0$ and anti-self-dual fields when $k < 0$. For (51.72),

$$F^a_{\mu\nu} F^a_{\mu\nu} = \frac{192\rho^4}{(r^2+\rho^2)^4}, \qquad \int_{\mathbb{R}^4} \frac{\rho^4 \, \mathrm{d}^4 x}{(r^2+\rho^2)^4} = \frac{\pi^2}{6},$$

where the last integral is

$$2\pi^2 \int_0^\infty \frac{\rho^4 r^3 \, \mathrm{d}r}{(r^2+\rho^2)^4} = \pi^2 \int_0^\infty \frac{\rho^4 u \, \mathrm{d}u}{(u+\rho^2)^4} = \frac{\pi^2}{6}.$$

Therefore

$$\int_{\mathbb{R}^4} F^a_{\mu\nu} F^a_{\mu\nu} \, \mathrm{d}^4 x = 32\pi^2, \qquad S_E^{\mathrm{ht}} = \frac{8\pi^2}{g_{\mathrm{ht}}^2}.$$

Because the field is self-dual,

$$Q[A] = \frac{1}{32\pi^2} \int F^a_{\mu\nu} \widetilde{F}^a_{\mu\nu} \, \mathrm{d}^4 x = 1.$$

In the trace-delta coupling coordinate used elsewhere in this monograph this same saddle contributes the exponential

$$\exp\left(-\frac{4\pi^2}{g_{\mathrm{YM}}^2} + \mathrm{i}\theta \right) = \exp\left(-\frac{8\pi^2}{g_{\mathrm{ht}}^2} + \mathrm{i}\theta \right). \tag{51.74}$$

Lemma 51.23 (BPST one-instanton data)**.** *With the self-dual symbols and $SU(2)$ half-trace generators $T_a = \sigma_a/2$ used in this section, the connection*

$$A_\mu(x) = \frac{2\, \eta^a_{\mu\nu} (x-a)^\nu}{(x-a)^2 + \rho^2} \, T_a, \qquad \rho > 0,$$

has curvature

$$F_{\mu\nu}(x) = -\frac{4\rho^2}{((x-a)^2+\rho^2)^2} \, \eta^a_{\mu\nu} T_a.$$

It is self-dual, has topological charge $Q = 1$, and has Euclidean action

$$S_E^{\mathrm{ht}} = \frac{8\pi^2}{g_{\mathrm{ht}}^2} = \frac{4\pi^2}{g_{\mathrm{YM}}^2}$$

when rewritten in the trace-delta coupling coordinate of the Yang–Mills chapters.

Proof. Differentiating the displayed connection gives the derivative term written above, and the commutator term is $\epsilon^a{}_{bc} A^b_\mu A^c_\nu$. The contraction identity (51.70) cancels the inhomogeneous $y_\mu \eta^a_{\nu\lambda} y^\lambda - y_\nu \eta^a_{\mu\lambda} y^\lambda$ terms and leaves (51.72). Since $\eta^a_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}\eta^a_{\rho\sigma}$, this curvature is self-dual.

The square-completion identity (51.73) gives the lower bound $8\pi^2|Q|/g_{\mathrm{ht}}^2$, saturated by a self-dual field with $Q > 0$. For the displayed curvature,

$$F^a_{\mu\nu} F^a_{\mu\nu} = \frac{192\rho^4}{(r^2+\rho^2)^4}, \qquad \int_{\mathbb{R}^4} \frac{\rho^4 \, \mathrm{d}^4 x}{(r^2+\rho^2)^4} = \frac{\pi^2}{6},$$

so $\int F^a_{\mu\nu} F^a_{\mu\nu} = 32\pi^2$. Self-duality then gives

$$Q = \frac{1}{32\pi^2} \int F^a_{\mu\nu} \widetilde{F}^a_{\mu\nu} \, \mathrm{d}^4 x = 1$$

and $S_E^{\mathrm{ht}} = 8\pi^2/g_{\mathrm{ht}}^2$. Finally, $\mathrm{tr}_\delta(t_a t_b) = \delta_{ab}$ with $t_a = \sqrt{2}\, T_a$ implies $g_{\mathrm{ht}} = \sqrt{2}\, g_{\mathrm{YM}}$, hence the trace-delta action is $4\pi^2/g_{\mathrm{YM}}^2$. □

Figure 51.5 plots the radial content of the same normalization check. In the dimensionless variable $u = r/\rho$, the normalized action-density profile and the cumulative topological charge are

$$\frac{F^a_{\mu\nu}F^a_{\mu\nu}(r)}{F^a_{\mu\nu}F^a_{\mu\nu}(0)} = \frac{1}{(1+u^2)^4}, \qquad \frac{1}{32\pi^2}\int_{|x-a|\le R} F^a_{\mu\nu}F^a_{\mu\nu}\,\mathrm{d}^4x = 1 - \frac{3}{(1+u^2)^2} + \frac{2}{(1+u^2)^3}.$$

The second formula is the radial integral in the proof of Lemma 51.23 before the limit $R/\rho \to \infty$.

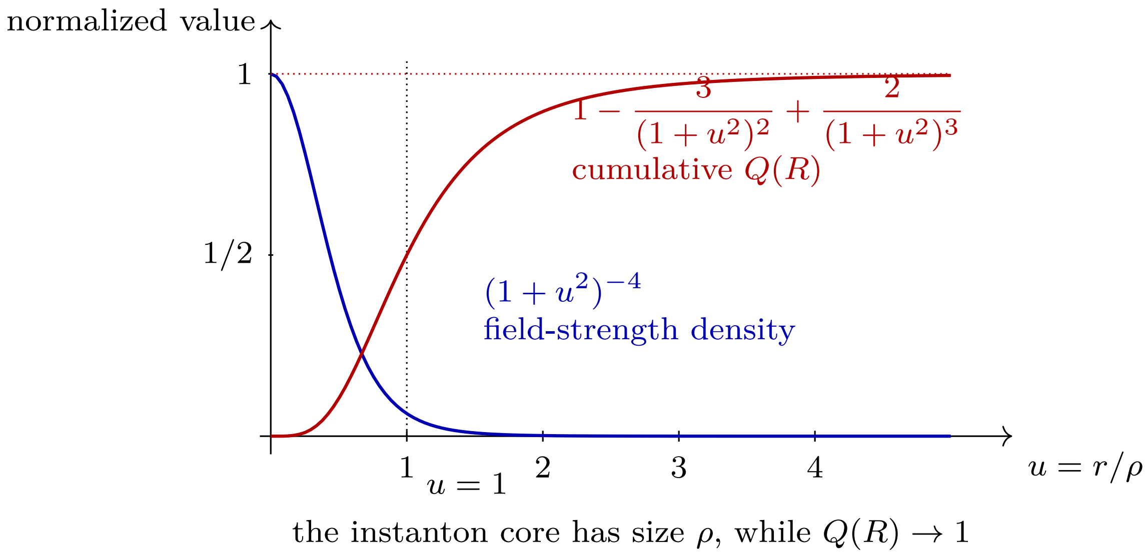


Figure 51.5: Radial BPST instanton profile in units of its size ρ. The blue curve is the normalized density $F^a_{\mu\nu}F^a_{\mu\nu}(r)/F^a_{\mu\nu}F^a_{\mu\nu}(0)$; the red curve is the cumulative topological charge inside radius $R = \rho u$. The plot makes the localized core and the normalization $Q = 1$ visible rather than merely repeating the algebraic formulas.

The bosonic zero modes are tangent vectors to the moduli of solutions modulo gauge transformations. For $SU(2)$, $k = 1$, the smooth moduli are

$$a^\mu \in \mathbb{R}^4, \qquad \rho \in \mathbb{R}_{>0}, \qquad U \in SU(2)/\mathbb{Z}_2.$$

Thus there are

$$4 + 1 + 3 = 8$$

bosonic zero modes: four translations, one dilation, and three global gauge orientations. For an $SU(N_c)$ instanton obtained by embedding the above $SU(2)$ solution, the orientation moduli are enlarged, and the one-instanton bosonic moduli space has dimension $4N_c$ before quotienting possible global identifications. The reason for the number $4N_c$ is local and elementary. The center and size contribute $4+1$ parameters. The embedded $SU(2)$ orientation is the orbit of the embedding under $SU(N_c)$, whose stabilizer is $SU(N_c-2)\times U(1)$. Hence the orientation dimension is

$$(N_c^2 - 1) - \big((N_c-2)^2 - 1\big) - 1 = 4N_c - 5,$$

and adding center and size gives $4N_c$. The precise global quotient by the center and by stabilizers affects the coordinate range, not this local dimension count.

Definition 51.24 (Framed ADHM quotient)**.** Let $W \simeq \mathbb{C}^{N_c}$ be the framing vector space and $K \simeq \mathbb{C}^k$ the instanton vector space. A framed $U(N_c)$ ADHM quadruple of charge k is

$$(B_1, B_2, I, J) \in \mathrm{End}(K)^2 \oplus \mathrm{Hom}(W, K) \oplus \mathrm{Hom}(K, W),$$

with moment maps

$$\begin{aligned} \mu_{\mathbb{C}} &:= [B_1, B_2] + IJ \in \mathrm{End}(K), \\ \mu_{\mathbb{R}} &:= [B_1, B_1^\dagger] + [B_2, B_2^\dagger] + II^\dagger - J^\dagger J \in \mathrm{Herm}(K). \end{aligned} \tag{51.75}$$

The unitary group $U(K)$ acts by

$$(B_1, B_2, I, J) \longmapsto (gB_1g^{-1}, gB_2g^{-1}, gI, Jg^{-1}). \tag{51.76}$$

The smooth framed ADHM quotient is the stable part of

$$\mathcal{M}_{k,N_c}^{\mathrm{ADHM}} := \{\, \mu_{\mathbb{C}} = 0,\ \mu_{\mathbb{R}} = 0 \,\}/U(K). \tag{51.77}$$

Here stability means that there is no proper subspace $S \subset K$ which contains $I(W)$ and is preserved by both B_1 and B_2. On this stable locus the framed gauge data have no residual endomorphism of K invisible to the framing.

The ADHM construction identifies the smooth quotient (51.77) with the framed smooth instanton moduli space on $\mathbb{R}^4$, after matching the chosen orientation convention for self-duality versus anti-self-duality. The full transform also builds the connection from the finite-dimensional linear algebra. The calculation needed for the semiclassical measure at this stage is the local quotient dimension and the boundary behavior of the quotient, which follow directly from Definition 51.24.

Proposition 51.25 (ADHM quotient dimension at smooth stable points)**.** *At a point of $\mu_{\mathbb{C}}^{-1}(0) \cap \mu_{\mathbb{R}}^{-1}(0)$ where the stable $U(K)$ action is free and the moment map is transverse to zero, the framed ADHM quotient $\mathcal{M}_{k,N_c}^{\mathrm{ADHM}}$ has local real dimension*

$$\dim_{\mathbb{R}} \mathcal{M}_{k,N_c}^{\mathrm{ADHM}} = 4kN_c.$$

The trace parts of B_1 and B_2 give the four center coordinates of the instanton configuration. The centered quotient therefore has local real dimension $4kN_c - 4$.

Proof. The real dimension of the unconstrained variables is

$$2\dim_{\mathbb{C}}\big(\mathrm{End}(K)^2\big) + 2\dim_{\mathbb{C}} \mathrm{Hom}(W, K) + 2\dim_{\mathbb{C}} \mathrm{Hom}(K, W) = 4k^2 + 4kN_c.$$

The complex equation $\mu_{\mathbb{C}} = 0$ imposes k^2 complex conditions, hence $2k^2$ real conditions. The real moment-map equation $\mu_{\mathbb{R}} = 0$ imposes k^2 real conditions because $\mathfrak{u}(K)$ has real dimension k^2. Transversality makes these conditions independent at the point under consideration.

Finally one divides by $U(K)$, whose real dimension is k^2. Freeness on the stable locus means that this quotient subtracts exactly k^2 dimensions locally. Thus

$$(4k^2 + 4kN_c) - 2k^2 - k^2 - k^2 = 4kN_c.$$

The equations contain only commutators of B_1, B_2 and the products $IJ, II^\dagger, J^\dagger J$. Shifting

$$B_\alpha \longmapsto B_\alpha + c_\alpha \mathbf{1}_K, \qquad \alpha = 1, 2,$$

leaves the moment maps unchanged. The two complex numbers $\operatorname{tr} B_1/k$ and $\operatorname{tr} B_2/k$ are therefore the four real translation coordinates in the complex structure used to write $\mathbb{R}^4 \simeq \mathbb{C}^2$. Removing them gives centered dimension $4kN_c - 4$. □

The stability condition is also where the small-instanton boundary first appears algebraically. If a stable ADHM tuple degenerates so that a nonzero subspace of K decouples from the framing, the quotient leaves the smooth instanton stratum and records instanton charge concentrated at points of $\mathbb{R}^4$. The Uhlenbeck boundary has strata of the form

$$\mathcal{M}^{\text{smooth}}_{k-\ell,N_c} \times \operatorname{Sym}^{\ell}(\mathbb{R}^4), \qquad 1 \le \ell \le k,$$

where ℓ units of charge have collapsed to pointlike ideal instantons. For a path integral this compactification carries analytic content: whether the semiclassical integral is finite depends on the determinant, operator insertions, and regulator behavior near these boundary strata.

Definition 51.26 (ADHM quotient density)**.** Equip the affine ADHM space

$$\mathbb{A}_{k,N_c} = \operatorname{End}(K)^2 \oplus \operatorname{Hom}(W,K) \oplus \operatorname{Hom}(K,W)$$

with the flat Hermitian metric

$$\|(\delta B_1, \delta B_2, \delta I, \delta J)\|^2 = \operatorname{tr}_K(\delta B_1 \delta B_1^\dagger) + \operatorname{tr}_K(\delta B_2 \delta B_2^\dagger) + \operatorname{tr}_K(\delta I \delta I^\dagger) + \operatorname{tr}_W(\delta J^\dagger \delta J).$$

On the free, stable, regular locus

$$Z^{\text{reg}}_{k,N_c} := \{\mu_{\mathbb{C}} = 0,\ \mu_{\mathbb{R}} = 0\}_{\text{stable,reg}} \subset \mathbb{A}_{k,N_c},$$

let $V_X \subset T_X Z^{\text{reg}}_{k,N_c}$ be the tangent space to the $U(K)$-orbit and let

$$H_X := V_X^{\perp} \cap T_X Z^{\text{reg}}_{k,N_c}.$$

The differential of the quotient map identifies H_X with the tangent space of the smooth quotient at $[X]$. The *ADHM quotient density* $\mathrm{d}\nu^{\text{ADHM}}_{k,N_c}$ is the Riemannian density induced by this quotient metric.

This density is a finite-dimensional object. It is not yet the complete semiclassical Yang–Mills weight: the latter also contains the nonzero-mode determinants, coupling-coordinate convention, operator renormalization, and the treatment of the boundary strata. The quotient density is the part of the measure fixed by the classical zero-mode geometry.

Proposition 51.27 (Coarea form of the ADHM quotient density)**.** *Fix an* Ad*-invariant inner product on $\mathfrak{u}(K)$ and let $\mathrm{d}\operatorname{vol}_{U(K)}$ be the associated left-invariant Riemannian density on $U(K)$. Choose an orthonormal basis $\{\xi_A\}_{A=1}^{k^2}$ of $\mathfrak{u}(K)$, and let $\xi_A^{\#}(X)$ denote the corresponding infinitesimal orbit vector at $X \in Z^{\text{reg}}_{k,N_c}$. Define the orbit Gram matrix*

$$M_{AB}(X) := \langle \xi_A^{\#}(X), \xi_B^{\#}(X) \rangle_{\mathbb{A}}.$$

If F is a compactly supported smooth $U(K)$-invariant function on Z^{reg}_{k,N_c}, then

$$\int_{\mathcal{M}^{\text{ADHM}}_{k,N_c}} F([X])\, \mathrm{d}\nu^{\text{ADHM}}_{k,N_c}([X]) = \frac{1}{\operatorname{Vol} U(K)} \int_{Z^{\text{reg}}_{k,N_c}} \frac{F(X)}{\sqrt{\det M(X)}}\, \mathrm{d}\operatorname{vol}_Z(X). \tag{51.78}$$

Equivalently, if $J_\mu(X)$ is the coarea Jacobian

$$J_\mu(X) := \det\!\big(D\mu_X D\mu_X^*\big)^{1/2}, \qquad \mu = (\operatorname{Re}\mu_{\mathbb{C}}, \operatorname{Im}\mu_{\mathbb{C}}, \mu_{\mathbb{R}}),$$

then the same density may be represented distributionally on the ambient affine space by

$$\int_{\mathcal{M}^{\mathrm{ADHM}}_{k,N_c}} F\,\mathrm{d}\nu^{\mathrm{ADHM}}_{k,N_c} = \frac{1}{\operatorname{Vol} U(K)} \int_{\mathbb{A}_{k,N_c}} F(X)\,\delta(\mu(X))\,\frac{J_\mu(X)}{\sqrt{\det M(X)}}\,\mathrm{d}X, \tag{51.79}$$

on any compact set where the regularity and freeness hypotheses hold. The delta function in (51.79) *is normalized with respect to the Euclidean density on the real target of* μ *induced by the same trace pairings.*

Proof. The restricted flat metric on Z^{reg}_{k,N_c} gives the orthogonal splitting

$$T_X Z^{\mathrm{reg}}_{k,N_c} = H_X \oplus V_X.$$

In local product coordinates (y,θ), where y is a quotient coordinate and θ^A are group coordinates near the identity, the metric is block diagonal at the chosen slice if H_X is used as the horizontal distribution. The horizontal block is the quotient metric, while the vertical block is precisely $M_{AB}(X)$. Therefore the Riemannian density on the level set factorizes as

$$\mathrm{d}\operatorname{vol}_Z = \sqrt{\det M(X)}\,\mathrm{d}\operatorname{vol}_{U(K)}\ \mathrm{d}\nu^{\mathrm{ADHM}}_{k,N_c}.$$

Because the flat metric and the moment-map equations are $U(K)$-invariant, $\det M(X)$ is constant along each orbit. Integrating an invariant function over the group coordinate gives (51.78).

For the ambient-space expression, apply the ordinary finite-dimensional coarea formula to the submersion μ:

$$\int_{\mathbb{A}_{k,N_c}} h(X)\,\delta(\mu(X))\,J_\mu(X)\,\mathrm{d}X = \int_{\mu^{-1}(0)} h(X)\,\mathrm{d}\operatorname{vol}_Z(X)$$

for compactly supported h in the regular locus. Taking $h = F/\sqrt{\det M}$ gives (51.79). $\square$

Remark 51.28 (Classical zero-mode density versus determinant weight). Equations (51.78) and (51.79) define the classical zero-mode density on the smooth ADHM quotient. They do not decide whether the instanton contribution to a continuum QFT observable is finite near a small-instanton stratum. That question depends on the nonzero-mode determinant, the operator insertions, and the renormalization scheme. This is why the Uhlenbeck boundary is part of the measure problem rather than an afterthought about topology.

Proposition 51.29 (Charge-one ADHM chart and the instanton orientation orbit)**.** *Let* $W \simeq \mathbb{C}^{N_c}$ *with its Hermitian inner product and let* $K \simeq \mathbb{C}$*. The centered charge-one framed* $U(N_c)$ *ADHM data are a pair*

$$I \in \operatorname{Hom}(W,K), \qquad J \in \operatorname{Hom}(K,W),$$

obeying

$$IJ = 0, \qquad II^\dagger - J^\dagger J = 0, \tag{51.80}$$

with $(I,J) \neq (0,0)$*, modulo the* $U(1)$ *action*

$$(I,J) \longmapsto (e^{\mathrm{i}\alpha} I, e^{-\mathrm{i}\alpha} J).$$

After identifying I *with a vector* $u \in W$ *by the Hermitian metric and writing* J *as a vector* $v \in W$*, the equations are*

$$\langle u,v\rangle = 0, \qquad \|u\| = \|v\| = \rho > 0. \tag{51.81}$$

For fixed ρ, the quotient is the homogeneous space

$$V_2(\mathbb{C}^{N_c})/U(1) \simeq U(N_c)/(U(N_c-2)\times U(1)), \tag{51.82}$$

where $V_2(\mathbb{C}^{N_c})$ is the Stiefel manifold of ordered orthonormal two-frames, $U(0)$ is interpreted as the trivial group when $N_c = 2$, and the $U(1)$ in the quotient acts by the common phase of the two frame vectors. Equivalently, for $N_c \geq 3$, after restricting to the special unitary framing, the connected orbit has the same local form

$$SU(N_c)/(SU(N_c-2)\times U(1))$$

up to a finite central quotient. For $N_c = 2$ this specialization is the usual $SU(2)/\mathbb{Z}_2$ orientation orbit. In every case the real dimension is $4N_c-5$. Adding the size ρ and the four center coordinates gives the local real dimension $4N_c$ of the charge-one instanton moduli space.

Proof. For $K \simeq \mathbb{C}$, the ADHM endomorphisms $B_1, B_2 \in \mathrm{End}(K)$ are complex numbers. They commute and therefore contribute no centered ADHM equation; their real and imaginary parts are the four center coordinates of the instanton. The remaining ADHM equations are exactly (51.80): $IJ = 0$ is the complex moment-map equation and $II^\dagger - J^\dagger J = 0$ is the real moment-map equation for the $U(1)$ acting on K.

Use the Hermitian metric to identify the row I with a vector $u \in W$ by $I(w) = \langle u, w\rangle$, and identify J with $v = J(1)$. Then $IJ = I(v) = \langle u, v\rangle$, while

$$II^\dagger = \|u\|^2\,\mathrm{id}_K, \qquad J^\dagger J = \|v\|^2\,\mathrm{id}_K\,.$$

Thus (51.80) is equivalent to (51.81). Since $(I, J) \neq (0, 0)$, the common norm is a positive number ρ. Dividing by ρ gives an ordered orthonormal two-frame $(u/\rho, v/\rho)$.

The $U(1)$ ADHM quotient sends $(u, v) \mapsto (e^{-\mathrm{i}\alpha}u, e^{-\mathrm{i}\alpha}v)$, namely the common phase of the ordered frame. The unitary group $U(N_c)$ acts transitively on ordered orthonormal two-frames. The stabilizer of the common-phase equivalence class of the reference frame (e_1, e_2) consists of an arbitrary unitary transformation on the orthogonal complement together with a common phase on the two displayed vectors, hence is $U(N_c-2)\times U(1)$. This proves (51.82). Its dimension is

$$\dim_{\mathbb{R}} U(N_c) - \dim_{\mathbb{R}} U(N_c-2) - \dim_{\mathbb{R}} U(1) = N_c^2 - (N_c-2)^2 - 1 = 4N_c - 5.$$

The size contributes one more dimension and the center contributes four. □

Cone structure and the small-instanton boundary. On the positive-size charge-one stratum, the centered ADHM quotient is the metric cone over the orientation orbit

$$\mathcal{O}_{N_c} = U(N_c)/(U(N_c-2)\times U(1)).$$

With the quotient metric induced from the flat metric $\|dI\|^2 + \|dJ\|^2$, its volume density has the local radial form

$$\rho^{4N_c-5}\,\mathrm{d}\rho\,\mathrm{d}\Omega_{N_c}, \tag{51.83}$$

for an invariant density $\mathrm{d}\Omega_{N_c}$ on $\mathcal{O}_{N_c}$. The endpoint $\rho = 0$ is the small-instanton boundary: the two-frame collapses to $(I, J) = (0, 0)$, the stabilizer jumps, and the positive-size quotient no longer supplies a smooth manifold chart. The equations (51.81) are homogeneous in (u, v). Hence every positive-size solution is uniquely written as $\rho(\hat{u},\hat{v})$, with $(\hat{u},\hat{v}) \in V_2(\mathbb{C}^{N_c})$ and $\rho > 0$, modulo the same common phase. The flat metric on (I, J) scales as

$$\|dI\|^2 + \|dJ\|^2 = c\,\mathrm{d}\rho^2 + \rho^2\,\mathrm{d}s^2_{\mathcal{O}_{N_c}}$$

for a positive constant c depending only on the normalization of ρ. A metric cone over a d-dimensional base has radial volume density proportional to $\rho^d\mathrm{d}\rho$ times the base density. Here $d = \dim_{\mathbb{R}} \mathcal{O}_{N_c} = 4N_c - 5$, giving (51.83). At $\rho = 0$ the frame vectors vanish and no two-frame remains; this is precisely the finite-dimensional ADHM model of the small-instanton degeneration.

The regulated collective-coordinate measure is the one constructed in Theorem 45.12. In a finite gauge-fixed regulator its local bosonic factor has the form

$$\mathrm{d}\mu^{\mathrm{bos}}_{k=1,\Lambda} = \frac{\Delta_{\mathrm{FP},\Lambda}(A_z)}{\mathrm{Vol}(\Gamma_z)} \frac{\sqrt{\det G^{(\Lambda)}_{ab}(z)}\,\mathrm{d}^4a\,\mathrm{d}\rho\,\mathrm{d}\Omega_{SU(N_c)/(SU(N_c-2)\times U(1))}}{\det{}'_{\Lambda}(L_{A_z})^{1/2}}, \tag{51.84}$$

multiplied by the exponential (51.74). Here z denotes the center, size, and orientation coordinates; $G^{(\Lambda)}_{ab}$ is the zero-mode Gram matrix induced by the regulated gauge-field inner product; L_{A_z} is the gauge-fixed quadratic operator on nonzero bosonic fluctuations; and Γ_z is the residual stabilizer left by the chosen slice. Equation (51.84) is not yet a continuum formula. Taking $\Lambda \to \infty$ requires combining the determinant, the coupling coordinate, and the normalization of the local operator insertions in a specified renormalization scheme. What is scheme independent at this stage is the topological sector, the zero-mode count, the finite-regulator construction of the measure, and the exponential action.

Finite-regulator nonzero-mode determinants. The symbol "determinant constant" hides several different objects unless the regulator is kept explicit. At fixed Λ, in a gauge slice near an instanton A_z, let

$$L^{\mathrm{bos}}_{\Lambda,z} : N_z\mathcal{M}_\Lambda \longrightarrow N_z\mathcal{M}_\Lambda$$

be the bosonic Hessian on the normal modes to the collective-coordinate manifold, and let

$$M^{\mathrm{gh}}_{\Lambda,z} : C^{\mathrm{gh}}_{\Lambda,z} \longrightarrow C^{\mathrm{gh}}_{\Lambda,z}$$

be the finite Faddeev–Popov ghost operator with residual stabilizer zero modes removed. For a fermion multiplet R, write

$$\mathcal{D}^R_{\Lambda,z} : \mathcal{S}^{R,+}_{\Lambda,z} \longrightarrow \mathcal{S}^{R,-}_{\Lambda,z}$$

for the regulated chiral Dirac operator. Its zero modes are not included in the nonzero-mode determinant; they are the Grassmann collective coordinates whose saturation is described below. The nonzero-mode determinant factor is

$$\mathcal{W}^{\mathrm{nz}}_\Lambda(z) := \frac{\det{}' M^{\mathrm{gh}}_{\Lambda,z}}{\det{}' M^{\mathrm{gh}}_{\Lambda,0}} \left(\frac{\det L^{\mathrm{bos}}_{\Lambda,0}}{\det{}' L^{\mathrm{bos}}_{\Lambda,z}}\right)^{1/2} \bigotimes_R \frac{\mathrm{Pf}\,{}'\mathcal{D}^R_{\Lambda,z}}{\mathrm{Pf}\,\mathcal{D}^R_{\Lambda,0}}. \tag{51.85}$$

The reference subscript 0 denotes the corresponding finite-regulator operator in the chosen topologically trivial background and in the same normalization of fields and gauge-group density. The primes mean that collective-coordinate zero modes, residual gauge zero modes, and fermion zero modes have been removed before taking the finite determinant or Pfaffian.

For a vectorlike Dirac pair, after choosing the standard conjugate pairing, the fermion factor in (51.85) is an ordinary scalar. For a genuinely chiral fermion it is a vector in the appropriate determinant or Pfaffian line over the background A_z; a scalar appears only after pairing this vector with the fermion zero-mode insertions and with any anomaly-inflow or determinant-line trivialization specified by the theory. This is the same determinant-line convention used in Chapter 153. Thus an instanton amplitude is not a number until the anomaly-line pairing and zero-mode saturation have also been specified.

With this notation the finite-regulator one-instanton contribution in a single smooth chart, for insertions whose fermion zero-mode and anomaly-line pairings have been specified, has the finite chart expression

$$\mathrm{d}\mu_{\Lambda,Q=1}(z) = \exp[-S_\Lambda(A_z)]\, \mathcal{W}_\Lambda^{\mathrm{nz}}(z)\, \mathrm{d}\nu_{\Lambda,\mathcal{M}}(z), \tag{51.86}$$

where $\mathrm{d}\nu_{\Lambda,\mathcal{M}}$ is the zero-mode density from (45.63), or equivalently the ADHM quotient density of Definition 51.26 after matching the ADHM coordinates to the regulated gauge-field slice. Formula (51.86) is a finite-dimensional statement. The continuum expression is the assertion that the product of $\mathcal{W}_\Lambda^{\mathrm{nz}}$, the coupling coordinate in S_Λ, and the renormalized operator insertions has a limit in the chosen scheme.

The finite Gaussian origin of (51.85) is as follows. A real bosonic normal integral with positive quadratic form L gives $(\det L)^{-1/2}$ up to the universal power of 2π already accounted for in the stationary-phase convention. A complex ghost Berezin integral gives $\det M^{\mathrm{gh}}$. A real fermionic quadratic form gives a Pfaffian; a complex chiral determinant is the corresponding determinant-line version of the same finite exterior-algebra construction. Removing zero modes before taking these determinants is not optional: the removed directions are precisely the collective coordinates or the fermion variables which must be integrated or saturated separately.

Finally, changing the finite-regulator Lagrangian by a local counterterm $C_\Lambda[A]$ multiplies (51.86) by $\exp[-C_\Lambda(A_z) + C_\Lambda(0)]$. If the one-instanton chart reduces this difference to

$$C_\Lambda(A_z) - C_\Lambda(0) = c_0(\Lambda) + c_1(\Lambda)\log(\mu\rho) + R_\Lambda(\rho, U),$$

then c_0 changes the scheme-dependent overall constant, c_1 changes the displayed power of $\mu\rho$, and the remainder R_Λ is part of the unresolved boundary and continuum-limit analysis. Counterterms are therefore not an after-the-fact correction to a pre-existing determinant constant; they are part of the regulated path-integral Lagrangian whose finite determinant factor is being defined.

Small-instanton boundary exponent. For the charge-one stratum the Uhlenbeck boundary is reached by $\rho \to 0$ at fixed center a and orientation U. The local boundary question is therefore a statement about the ρ-asymptotics of the finite-regulator density after the nonzero-mode determinant and the renormalized insertions have been paired. Let $\mathcal{X}$ denote the insertions under consideration, including the fermion zero-mode factors needed to obtain a scalar from the determinant line. A *one-instanton boundary exponent estimate* for $\mathcal{X}$ consists of numbers $\beta_{\mathcal{X}} \in \mathbb{R}$ and $\delta > 0$, and a locally bounded function $A_{\mathcal{X}}(a, U)$, such that in a fixed compact set of centers and in a fixed orientation chart the renormalized finite-regulator chart density has a limit of the form

$$\mathrm{d}\mu_{Q=1,\mathcal{X}} = C_{\mathcal{X}}(\mu)\, \rho^{\,b_0+\beta_{\mathcal{X}}-5}\big(A_{\mathcal{X}}(a, U) + O(\rho^\delta)\big)\, \mathrm{d}^4 a\, \mathrm{d}\rho\, \mathrm{d}\Omega_{N_c}. \tag{51.87}$$

The $b_0 - 5$ part is the universal scale and one-loop running content in (51.93). The exponent $\beta_{\mathcal{X}}$ records the net small-size contribution of the renormalized operator insertions, fermion zero-mode wavefunctions, masses or sources used to saturate zero modes, and any residual determinant factor which has a finite power-law boundary behavior. Logarithms may be included by allowing $A_{\mathcal{X}}$ to be a polynomial in $\log(\mu\rho)$; they leave the power threshold below unchanged.

On this boundary chart the local integral over $0 < \rho < \epsilon$ is controlled by

$$\int_0^{\epsilon} \rho^{\,b_0+\beta_{\mathcal{X}}-5} \,(\log(\mu\rho))^q \,\mathrm{d}\rho, \qquad q \in \mathbb{Z}_{\geq 0}.$$

It is locally finite at $\rho = 0$ precisely when

$$b_0 + \beta_{\mathcal{X}} > 4. \tag{51.88}$$

At equality the logarithmic integral diverges, and below equality the power integral diverges. Thus asymptotic freedom, $b_0 > 0$, is a statement about the running coupling at short distance, while small-instanton local integrability of a chosen observable requires the stronger observable-dependent condition (51.88). For the vacuum-density power with no extra small-ρ insertion factor $\beta_{\mathcal{X}} = 0$, the local boundary test is $b_0 > 4$. Different correlators have different $\beta_{\mathcal{X}}$, and center collisions with local operator insertions introduce additional boundary faces which must be analyzed separately from the pure $\rho \to 0$ face.

This boundary exponent estimate is weaker than a complete multi-instanton measure theorem. It isolates the analytic estimate needed near a single collapsed charge. Higher charge involves simultaneous collisions of several instantons, partial Uhlenbeck strata $\mathcal{M}_{k-\ell,N_c} \times \mathrm{Sym}^{\ell}(\mathbb{R}^4)$, and determinant behavior transverse to each stratum.

Definition 51.30 (Uhlenbeck boundary-face integrability datum)**.** Fix an instanton charge k, a gauge group $SU(N_c)$, and a scalar insertion datum $\mathcal{X}$ including fermion-zero-mode and anomaly-line pairings. A Uhlenbeck boundary-face integrability datum consists of:

(i) a bubbling charge $1 \leq \ell \leq k$, a residual smooth charge $k - \ell$, and a chart on the open Uhlenbeck stratum

$$\mathcal{M}^{\mathrm{smooth}}_{k-\ell,N_c} \times \mathrm{Conf}_{\ell}(\mathbb{R}^4)/S_{\ell},$$

with the pointlike instanton centers kept away from each other and from the local insertions;

(ii) transverse gluing scales $0 < \rho_\alpha < \epsilon$, angular variables ω_α, and boundary exponents $A_\alpha(\mathcal{X})$ for the ℓ collapsed charge units;

(iii) if some pointlike centers approach each other or an insertion support, collision distances $0 < d_\beta < \epsilon$ and collision exponents $G_\beta(\mathcal{X})$;

(iv) a finite-regulator density bound, uniform on compact subsets of the residual smooth stratum and angular variables, of the form

$$|\mathrm{d}\mu_{k,\mathcal{X}}| \leq C \,\mathrm{d}\nu^{\mathrm{sm}}_{k-\ell} \prod_{\alpha=1}^{\ell} \rho_\alpha^{A_\alpha(\mathcal{X})-1}(1+|\log\rho_\alpha|)^{q_\alpha}\,\mathrm{d}\rho_\alpha\,\mathrm{d}\omega_\alpha \prod_{\beta} d_\beta^{G_\beta(\mathcal{X})-1}(1+|\log d_\beta|)^{s_\beta}\,\mathrm{d}d_\beta\,\mathrm{d}\zeta_\beta, \tag{51.89}$$

with nonnegative integers q_α, s_β.

The exponents A_α and G_β are part of the analytic instanton-measure problem. Classical ADHM dimension counts identify the boundary coordinates; determinant, insertion, and counterterm estimates supply the powers in (51.89).

Proposition 51.31 (Uhlenbeck face dimension and local integrability budget)**.** *At a smooth residual instanton and ℓ distinct pointlike ideal instantons, the open Uhlenbeck stratum*

$$\mathcal{M}^{\mathrm{smooth}}_{k-\ell,N_c} \times \mathrm{Conf}_\ell(\mathbb{R}^4)/S_\ell$$

has local real codimension

$$4\ell(N_c - 1) \tag{51.90}$$

inside the smooth charge-k ADHM dimension count. If the ℓ pointlike charges are grouped into r collision clusters, the diagonal collision stratum contributes the additional codimension $4(\ell - r)$.

For a boundary-face datum satisfying (51.89), the local absolute integral over the displayed boundary variables is finite whenever

$$A_\alpha(\mathcal{X}) > 0 \quad \textit{for every } \alpha, \qquad G_\beta(\mathcal{X}) > 0 \quad \textit{for every } \beta. \tag{51.91}$$

If one displayed exponent is zero and the leading coefficient on that face has nonzero integral over the remaining bounded variables, the corresponding local integral has a logarithmic divergence. If one displayed exponent is negative under the same nonvanishing condition, the divergence is a power divergence.

Proof. The smooth framed charge-k dimension from Proposition 51.25 is $4kN_c$. On the open Uhlenbeck stratum with ℓ distinct ideal points, the residual smooth instanton part has dimension $4(k-\ell)N_c$, and the unordered configuration space of ℓ distinct points in $\mathbb{R}^4$ has local dimension 4ℓ. The codimension is therefore

$$4kN_c - \big(4(k-\ell)N_c + 4\ell\big) = 4\ell(N_c - 1),$$

which proves (51.90). If the ℓ ideal points lie on a diagonal with r distinct collision clusters, the point-configuration dimension drops from 4ℓ to $4r$, so the extra diagonal codimension is $4(\ell - r)$.

For the integrability statement, the compact residual and angular variables contribute only a finite multiplicative constant in the stated chart. Each scale integral is reduced to

$$\int_0^\epsilon t^{A-1}(1 + |\log t|)^q \, \mathrm{d}t.$$

With $u = -\log t$, this is bounded at $t = 0$ exactly when $A > 0$. At $A = 0$ it is the integral of a nonzero polynomial in u over an infinite ray, hence logarithmic in the cutoff scale. For $A < 0$, the antiderivative has a negative power of the cutoff scale. Applying this one-dimensional criterion to every ρ_α and d_β proves the product criterion and the two divergence alternatives. ☐

For the charge-one boundary chart, there are no point-collision variables and

$$A_1(\mathcal{X}) = b_0 + \beta_{\mathcal{X}} - 4.$$

Thus Proposition 51.31 recovers (51.88). For $k > 1$ it also records the extra data which a multi-instanton measure calculation must provide: every bubbling scale, every diagonal collision of ideal instantons, and every collision of an ideal instanton center with an inserted operator has its own exponent. A formula for the smooth ADHM quotient density and a one-loop determinant on the principal stratum become a continuum multi-instanton contribution only after these face estimates are supplied in one compatible regulator and renormalization scheme.

51.17.1 Scale and Running-Coupling Content of the One-Instanton Density

The preceding finite-regulator formula separates the zero-mode Jacobian from the determinant on nonzero fluctuations. The continuum one-instanton density contains three universal pieces which can be fixed before computing the scheme-dependent determinant constant. First, the moduli

$$a^\mu \in \mathbb{R}^4, \qquad \rho \in \mathbb{R}_{>0}$$

are the center and size of the instanton, while the remaining $4N_c - 5$ coordinates are dimensionless orientation coordinates on the embedded $SU(2)$ orbit. The translation-size part of a scale-covariant local density must therefore be

$$\frac{\mathrm{d}^4 a\, \mathrm{d}\rho}{\rho^5}.$$

Indeed a and ρ both have length dimension one, so $\mathrm{d}^4 a\, \mathrm{d}\rho/\rho^5$ is dimensionless and is invariant under the simultaneous dilation $a \mapsto \lambda a$, $\rho \mapsto \lambda\rho$. The power ρ^{-5} is the four-dimensional center plus one-dimensional scale volume element. It is the product of the zero-mode Jacobian and the nonzero-mode determinant after their regulator-dependent powers of ρ have been combined.

Second, the number of bosonic zero modes is $m = 4N_c$. In the semiclassical expansion of an action $S[A]/g_{\mathrm{ht}}^2$, replacing the zero-mode amplitudes by collective coordinates contributes one factor g_{ht}^{-1} for each bosonic zero mode. Equivalently, after absorbing normalization constants independent of g_{ht} into the orientation measure and determinant convention, the zero-mode normalization supplies

$$g_{\mathrm{ht}}^{-4N_c} = \left(\frac{8\pi^2}{g_{\mathrm{ht}}^2}\right)^{2N_c} \times \text{constant}.$$

This is the origin of the power multiplying the one-instanton density. The constant depends on the precise normalization of the path-integral density and of the invariant volume on the orientation orbit.

One-loop running factor in the one-instanton density. Use the common half-trace coupling $g_{\mathrm{ht}}(\mu)$ in the BPST subsection, and suppose the one-instanton density is renormalized in a mass-independent scheme through one loop. Let

$$\mu\frac{\mathrm{d}g_{\mathrm{ht}}}{\mathrm{d}\mu} = -\frac{b_0}{16\pi^2} g_{\mathrm{ht}}^3 + O(g_{\mathrm{ht}}^5), \qquad b_0 = \frac{11}{3}N_c - \frac{2}{3}N_f. \tag{51.92}$$

Then the part of the one-instanton measure fixed by scale covariance, zero-mode normalization, and one-loop RG invariance is

$$\mathrm{d}\mu_1^{\mathrm{univ}} = C_{\mathcal{S},N_c,N_f}\, \frac{\mathrm{d}^4 a\, \mathrm{d}\rho}{\rho^5}\, \mathrm{d}\Omega_{N_c} \left(\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)}\right)^{2N_c} (\mu\rho)^{b_0} \exp\left[-\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)} + \mathrm{i}\theta\right] \left(1 + O(g_{\mathrm{ht}}^2)\right). \tag{51.93}$$

Here $\mathrm{d}\Omega_{N_c}$ is a chosen invariant density on the local orientation orbit

$$SU(N_c)/(SU(N_c - 2) \times U(1)),$$

with the evident $SU(2)/\mathbb{Z}_2$ specialization when $N_c = 2$. The coefficient $C_{\mathcal{S},N_c,N_f}$ is a finite constant depending on the regulator, subtraction convention, and normalization of $\mathrm{d}\Omega_{N_c}$; obtaining it requires the full renormalized determinant calculation in the chosen scheme. The factors

$\mathrm{d}^4a\,\mathrm{d}\rho/\rho^5$, $\mathrm{d}\Omega_{N_c}$, and $(8\pi^2/g_{\mathrm{ht}}^2)^{2N_c}$ have just been fixed by dimension, orientation invariance, and the $4N_c$ bosonic zero modes. It remains to determine the power of $\mu\rho$ at one loop. Write the unknown power as α and apply $\mu\mathrm{d}/\mathrm{d}\mu$ to the logarithm of the density factor

$$g_{\mathrm{ht}}^{-4N_c}(\mu\rho)^{\alpha}\exp\left[-\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)}\right].$$

The derivative is

$$\alpha - 4N_c\,\frac{\beta(g_{\mathrm{ht}})}{g_{\mathrm{ht}}} + \frac{16\pi^2}{g_{\mathrm{ht}}^3}\,\beta(g_{\mathrm{ht}}).$$

Substituting (51.92) gives

$$\alpha - b_0 + O(g_{\mathrm{ht}}^2).$$

RG invariance of a physical one-instanton contribution through one loop therefore fixes $\alpha = b_0$. The derivative of the zero-mode power starts at order g_{ht}^2, so it belongs to the next perturbative order and is absorbed into the higher-loop correction multiplying (51.93). The finite constant $C_{\mathcal{S},N_c,N_f}$ is outside this argument because RG invariance fixes scale dependence, while the constant term is set by the actual regularized determinant and measure normalization.

Proposition 51.32 (Pure-gauge Pauli–Villars determinant constant)**.** *In pure $SU(N)$ Yang–Mills, $N \geq 2$, use the common half-trace coupling $g_{\mathrm{PV}}(\mu)$ and the Pauli–Villars determinant convention of the one-instanton calculation [63, 65]. For gauge-invariant insertions, integrate the embedded orientation orbit with the corresponding Haar normalization. Then the one-loop, one-instanton density is*

$$\mathrm{d}n_{I,N}^{\mathrm{PV}} = \frac{\mathrm{d}^4a\,\mathrm{d}\rho}{\rho^5}\,C_N^{\mathrm{PV}}\left(\frac{8\pi^2}{g_{\mathrm{PV}}^2(\mu)}\right)^{2N}(\mu\rho)^{11N/3}\exp\left[-\frac{8\pi^2}{g_{\mathrm{PV}}^2(\mu)} + \mathrm{i}\theta\right]\left(1 + O(g_{\mathrm{PV}}^2)\right), \tag{51.94}$$

where the finite constant is

$$C_N^{\mathrm{PV}} = \frac{2^{2-2N}}{\pi^2(N-1)!(N-2)!}\,\exp\left[-\alpha(1) - 2(N-2)\alpha\left(\frac{1}{2}\right)\right]. \tag{51.95}$$

Here

$$\alpha(1) = 0.443307\ldots, \qquad \alpha\left(\frac{1}{2}\right) = 0.145873\ldots.$$

Equivalently, in the conventional rounded form,

$$C_N^{\mathrm{PV}} = \frac{0.466\,\exp[-1.679\,N]}{(N-1)!(N-2)!} \quad \textit{to the displayed precision.} \tag{51.96}$$

For example $C_3^{\mathrm{PV}} \simeq 1.51 \times 10^{-3}$.

Proof. The orientation-integrated singular-gauge collective-coordinate calculation gives, after the Pauli–Villars zero-mode mass factors have been combined with the logarithmic determinant into $(\mu\rho)^{11N/3}$, the residual zero-mode coefficient

$$\frac{2^{4N+2}\pi^{4N-2}}{(N-1)!(N-2)!}\,g_{\mathrm{PV}}^{-4N}.$$

Rewriting

$$g_{\mathrm{PV}}^{-4N}=\left(\frac{8\pi^2}{g_{\mathrm{PV}}^2}\right)^{2N}(8\pi^2)^{-2N}$$

turns this residual coefficient into

$$\frac{2^{2-2N}}{\pi^2(N-1)!(N-2)!}\left(\frac{8\pi^2}{g_{\mathrm{PV}}^2}\right)^{2N}.$$

The finite part of the nonzero-mode determinant contributes

$$\exp\left[-\alpha(1)-2(N-2)\alpha\left(\frac{1}{2}\right)\right],$$

where $\alpha(1)$ is the $SU(2)$ vector-ghost block and the $2(N-2)$ copies of $\alpha(1/2)$ are the additional doublet blocks in the embedded $SU(2)\subset SU(N)$ decomposition. Multiplying these two factors gives (51.95); combining it with the classical action and logarithmic determinant gives (51.94). Taking logarithms of (51.95) yields

$$\log C_N^{\mathrm{PV}}=\log(0.466\ldots)-1.679\ldots\,N-\log((N-1)!)-\log((N-2)!),$$

which is the rounded form (51.96). □

Proposition 51.33 (Heat-kernel source of the instanton logarithm)**.** *In the same background-field gauge and half-trace convention, the logarithmic part of the charge-k nonzero-mode fluctuation determinant has*

$$\partial_{\log\mu}\log\mathcal{W}_{\log}^{\mathrm{nz}}=k\left[\frac{11}{3}C_2(G)-\frac{4}{3}\sum_{\text{Dirac }R}T(R)\right]. \tag{51.97}$$

For vectorlike $SU(N_c)$ QCD with N_f Dirac fundamentals this coefficient is kb_0, with b_0 as in (51.92)*. Hence, in a charge-one BPST chart,*

$$\log\mathcal{W}_{\log}^{\mathrm{nz}}(a,\rho,U;\mu)=b_0\log(\mu\rho)+\log\mathcal{W}_{\mathrm{fin}}^{\mathrm{nz}}(U;\mathcal{S}), \tag{51.98}$$

up to higher-loop and local finite-subtraction terms in the chosen scheme. Equivalently, if $X_\kappa=8\pi^2/g_{\mathrm{ht}}^2(\kappa)$ and $X_\Lambda=X_\mu+b_0\log(\Lambda/\mu)$ at one loop, then

$$\exp[-X_\Lambda+\mathrm{i}\theta]\,(\Lambda\rho)^{b_0}=\exp[-X_\mu+\mathrm{i}\theta]\,(\mu\rho)^{b_0}. \tag{51.99}$$

The factor $(\mu\rho)^{b_0}$ in (51.93) *is therefore the logarithmic shadow of the nonzero-mode fluctuation spectrum plus coupling renormalization, while the finite constant still requires the full spectral determinant calculation.*

Proof. The small-proper-time expansion of (51.131) is local in the gauge background. With background-field gauge fixing, the vector fluctuation and ghost complex contribute the coefficient $(11/3)C_2(G)$ multiplying the topological charge in the logarithmic counterterm, and a Dirac fermion in representation R contributes $-(4/3)T(R)$. Summing the fields gives (51.97). For a fundamental Dirac fermion of $SU(N_c)$, $T(\square)=1/2$, so the matter term is $-2N_f/3$. The coefficient in this convention is therefore

$$\frac{11}{3}N_c-\frac{2}{3}N_f=b_0.$$

A BPST connection with size ρ is obtained from the unit-size connection by translation, global orientation, and dilation. After the zero modes are removed, the logarithmic scale dependence of the determinant can therefore depend on the size only through the dimensionless combination $\mu\rho$, apart from finite scheme-dependent terms. This gives (51.98). Finally, the one-loop running of X_κ gives $X_\Lambda = X_\mu + b_0 \log(\Lambda/\mu)$; substituting this identity into the cutoff expression proves (51.99). □

Fermion zero modes are controlled by the index theorem already stated in (51.12). In the conventional half-trace notation, a left-handed Weyl fermion in representation R has

$$n_+ - n_- = 2T_R\, k, \qquad \mathrm{tr}_R(T_R^a T_R^b) = T_R \delta^{ab}. \tag{51.100}$$

For a fundamental Weyl fermion of $SU(N_c)$, $T_\square = 1/2$, so a $k = 1$ instanton has one normalizable zero mode of the chirality selected by the self-dual background. The anti-instanton has the opposite chirality. In the trace-delta convention the numerical value called T_R is doubled, but the topological density and the coupling coordinate are correspondingly rewritten; the integer kernel dimension in (51.100) is the invariant statement.

The zero modes determine the Grassmann selection rule. Expand a massless fermion in an instanton background as

$$\psi(x) = \sum_{\alpha=1}^{n_0} \xi_\alpha \psi_{0,\alpha}(x) + \sum_{\lambda\neq 0} \xi_\lambda \psi_\lambda(x),$$

where $\psi_{0,\alpha}$ are the zero modes and ξ_α are odd coefficients. The quadratic fermion action contains no term involving ξ_α. Therefore the Berezin integral over zero-mode variables vanishes unless operator insertions supply every missing ξ_α. This is the finite-dimensional Berezin rule $\int \mathrm{d}\xi\, 1 = 0$, $\int \mathrm{d}\xi\, \xi = 1$ applied after the spectral decomposition of the regulated fermion field.

Proposition 51.34 (Instanton sectors as physical correlator contributions)**.** *Fix a finite regulator, a $Q = 1$ instanton chart with collective coordinate $z = (a, \rho, U)$, smooth chart domain $\mathcal{U}_{Q=1}$, and renormalized insertions $\mathcal{O}_1, \ldots, \mathcal{O}_r$. After gauge fixing, collective coordinate separation, and integration of nonzero fluctuations, the instanton sector contribution has the form*

$$\big\langle \mathcal{O}_1 \cdots \mathcal{O}_r \big\rangle_{Q=1,\Lambda} = \int_{\mathcal{U}_{Q=1}} \mathrm{e}^{-S_\Lambda(A_z)+\mathrm{i}\theta}\, \mathcal{W}_\Lambda^{\mathrm{nz}}(z)\, \mathcal{Z}^0_{\mathcal{O},\Lambda}(z)\, \mathrm{d}\nu_{\Lambda,\mathcal{M}}(z), \tag{51.101}$$

provided the determinant/Pfaffian line has been paired with the fermion zero-mode and anomaly-line data to produce a scalar density. Here $\mathcal{W}_\Lambda^{\mathrm{nz}}$ is the nonzero-mode determinant datum from (51.85), while $\mathcal{Z}^0_{\mathcal{O},\Lambda}(z)$ is the finite Berezin coefficient

$$\mathcal{Z}^0_{\mathcal{O},\Lambda}(z) = \int \mathrm{d}\xi_{n_0} \cdots \mathrm{d}\xi_1\, \mathcal{O}^0_{1,\Lambda}(z,\xi) \cdots \mathcal{O}^0_{r,\Lambda}(z,\xi)\, \exp[-S^0_{\mathrm{lift},\Lambda}(z,\xi)]. \tag{51.102}$$

The superscript 0 means that each insertion and each mass, Yukawa, or source term has been restricted to the zero-mode subspace. If the integrand in (51.102) does not contain the monomial $\xi_1 \cdots \xi_{n_0}$, then $\mathcal{Z}^0_{\mathcal{O},\Lambda}(z) = 0$. If it does contain that monomial, the coefficient of that monomial is the only part of the insertions which is seen by the instanton sector.

Proof. At finite cutoff the gauge-fixed fields split into collective coordinates, normal bosonic modes, ghosts, fermion zero modes, and fermion nonzero modes. Gaussian integration over the normal bosonic and ghost variables gives the determinant factors already isolated in $\mathcal{W}_\Lambda^{\rm nz}$; Gaussian or Pfaffian integration over fermion nonzero modes gives the remaining nonzero-mode determinant-line factor. None of these operations integrates over the zero-mode variables ξ_α, because the quadratic fermion operator vanishes on them. The only finite-dimensional integral left in the fermionic zero-mode sector is (51.102). Berezin integration extracts the coefficient of the ordered top monomial $\xi_1 \cdots \xi_{n_0}$. All terms missing at least one zero-mode variable therefore vanish, while terms containing the top monomial contribute exactly their coefficient. Multiplying this coefficient by the bosonic collective-coordinate density and the nonzero-mode determinant gives (51.101). $\square$

Thus the moduli space is not the physical answer; it is the domain of a semiclassical amplitude. The physical question is which insertions, masses, or sources saturate the fermion zero modes, what nonzero-mode determinant and counterterm scheme define $\mathcal{W}_\Lambda^{\rm nz}$, and whether the resulting a, ρ, U integral is finite at its boundary faces.

From collective coordinates to an amplitude. For later use, it is helpful to compress the $Q = 1$ calculation into the ordered finite-regulator pipeline

$$K^E_{\Lambda,\mathcal{J}}(z) = \mathrm{e}^{-S_\Lambda(A_z)+\mathrm{i}\theta}\, \mathcal{W}_\Lambda^{\rm nz}(z)\, \mathcal{Z}^0_{\mathcal{J},\Lambda}(z)\, \mathrm{d}\nu_{\Lambda,\mathcal{M}}(z), \qquad G^{E,Q=1}_{\Lambda,\mathcal{J}} = \int_{\mathcal{U}_{Q=1}} K^E_{\Lambda,\mathcal{J}}(z). \tag{51.103}$$

The first factor selects the topological sector and its classical action; the last factor is the collective-coordinate density; the middle two factors are where the fluctuation and fermion-saturation calculation enters. A semiclassical local vertex is obtained only after replacing $K^E_{\Lambda,\mathcal{J}}$ on a declared window by a leading kernel and then projecting it onto an observable coordinate:

$$\mathcal{A}^{Q=1}_{\rm phys}[\mathcal{J}] = \operatorname*{Proj}_{\rm phys} \operatorname*{Cont}_{\mathcal{S}} \int_{\mathcal{U}_{\rm win}} K^E_{{\rm lead},\mathcal{J}}(z) + R_{\rm det} + R_{\rm zm} + R_{\rm src} + R_{\rm Schur} + R_\rho + R_{\rm cont} + R_{\rm spec} + R_{\rm IR} + R_{\rm cut}. \tag{51.104}$$

Here the determinant residual includes the nonzero-mode spectrum and local counterterms; $R_{\rm zm}$ records the zero-mode projection or derivative expansion of the sources; $R_{\rm src}$ records the color-singlet, hadronic, or OPE matching layer; $R_{\rm Schur}$ records finite nonzero-mode corrections to the zero-mode block; R_ρ is the size and Uhlenbeck-face endpoint remainder; and the final four terms are the continuation, spectral-projection, infrared, and unitarity-cut remnants needed before a physical amplitude is asserted.

This ordering is part of the content of the instanton calculation. The ADHM or BPST moduli space supplies the domain and the collective-coordinate Jacobian. It does not supply the nonzero-mode determinant, the fermion zero-mode Berezin coefficient, the source-to-observable matching, or the endpoint estimates. Conversely, the local 't Hooft operator is the small-size, source-adiabatic coordinate of (51.103); the four-point amplitude calculation keeps the zero-mode form factors, determinant normalization, external source residues, and size tails inside the same integral until a hard, screened, thermal, or Wilsonian window justifies separating them.

For vectorlike QCD with N_f massless Dirac fundamentals, a $Q = 1$ instanton supplies one zero mode for each flavor in one chirality sector. The semiclassical functional integral therefore generates the following leading local term in the derivative expansion around external fields whose variation scale is much larger than ρ:

$$\mathcal{L}_{\mathrm{inst}} = \int_0^\infty \frac{\mathrm{d}\rho}{\rho^5}\, C_{N_c,N_f}(\rho,\mu) \exp\!\left(-\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)} + \mathrm{i}\theta\right) \det_{1\le f,f'\le N_f} \left[\rho^3 \bar\psi_{Rf}\psi_{Lf'}\right] + \text{anti-instanton conjugate} \tag{51.105}$$

after integrating over the instanton center and color orientation and after contracting spin and color indices as dictated by the zero-mode wavefunctions and gauge invariance. The determinant is over flavor indices; it is the local tensor compatible with the nonabelian chiral flavor symmetry. For flavor diagonal external sources it reduces to the corresponding alternating sum of products of N_f bilinears. At one loop

$$C_{N_c,N_f}(\rho,\mu) = C^{(0)}_{N_c,N_f}\left(\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)}\right)^{2N_c} (\rho\mu)^{b_0}\left(1 + O(g_{\mathrm{ht}}^2(\mu))\right), \qquad b_0 = \frac{11}{3}N_c - \frac{2}{3}N_f,$$

where b_0 is defined by $\mu\, \mathrm{d}g_{\mathrm{ht}}/\mathrm{d}\mu = -b_0 g_{\mathrm{ht}}^3/(16\pi^2) + O(g_{\mathrm{ht}}^5)$. The scale and coupling dependence is the universal content of the running-factor calculation above; the constant $C^{(0)}_{N_c,N_f}$ is fixed only after choosing the determinant and orientation-volume normalization. The dimensionless factors $\rho^3\bar\psi_R\psi_L$ in (51.105) record the normalization of the external zero-mode wavefunctions in the local limit; changing the regularization scheme changes $C^{(0)}_{N_c,N_f}$ and higher-loop factors, not the zero-mode tensor or the selection rule. When the external fields vary on the scale ρ, the same zero-mode integration gives a nonlocal kernel rather than the local determinant in (51.105). Its regulator-independent selection rule is

$$\Delta Q_A = 2N_f$$

for the anomalous axial charge, in agreement with the integrated anomaly. Small quark masses may replace missing fermion insertions by mass factors, because the mass term pairs the corresponding zero-mode variables.

Proposition 51.35 (Flavor determinant from the instanton zero modes)**.** *In the $Q = 1$ sector of vectorlike $SU(N_c)$ QCD, let $\lambda_f, \widetilde\lambda_f$, $f = 1,\dots,N_f$, denote the two Grassmann zero-mode coefficients paired by the local source matrix $B = (B_{ff'})$. Choose the zero-mode orientation so that*

$$\int \prod_{f=N_f}^{1} \mathrm{d}\widetilde\lambda_f\, \mathrm{d}\lambda_f\, \exp\!\left(\sum_{f,f'=1}^{N_f} \widetilde\lambda_f B_{ff'}\lambda_{f'}\right) = \det B. \tag{51.106}$$

For slowly varying external quark fields, $B_{ff'} = \rho^3\bar\psi_{Rf}\psi_{Lf'}$, and (51.106) is precisely the flavor determinant in (51.105). If instead all zero modes are saturated by the quark mass term, the same finite-dimensional calculation replaces B by the mass-overlap matrix $B^{(m)}_{ff'}(z)$. Its phase is the phase of $\det M$*, so the one-instanton vacuum term has phase* $\exp[\mathrm{i}(\theta + \arg\det M)]$*, up to the real positive overlap and determinant normalizations.*

Proof. Expand the exponential in (51.106). Every contributing top-degree term must contain each $\widetilde\lambda_f$ once and each λ_f once. Such a term is specified by a permutation $\sigma \in S_{N_f}$:

$$\prod_{f=1}^{N_f} \widetilde\lambda_f B_{f,\sigma(f)}\lambda_{\sigma(f)}.$$

Reordering the $\lambda_{\sigma(f)}$ variables into the chosen Berezin orientation gives the sign $\operatorname{sgn}\sigma$. Berezin integration therefore extracts

$$\sum_{\sigma\in S_{N_f}} \operatorname{sgn}\sigma \prod_{f=1}^{N_f} B_{f,\sigma(f)} = \det B.$$

For $B_{ff'} = \rho^3\bar{\psi}_{Rf}\psi_{Lf'}$, this gives the local $2N_f$-fermion operator. If B is supplied by the mass term, each zero-mode overlap is real after the same regulator and zero-mode orientation have been fixed, so the only flavor phase is $\arg\det M$. □

Proposition 51.36 (Two-flavor channel content of the local instanton vertex)**.** *For $N_f = 2$, set*

$$\Phi_{ff'} = \bar{q}_{Rf}q_{Lf'}, \qquad S_{ff'} = \bar{q}_f q_{f'}, \qquad P_{ff'} = \bar{q}_f \mathrm{i}\gamma_5 q_{f'}.$$

With the chiral projectors used in this chapter,

$$\Phi = \frac{1}{2}(S + \mathrm{i}P). \tag{51.107}$$

Write the scalar and pseudoscalar flavor matrices in the Pauli basis as

$$S = s_0\mathbf{1} + \vec{s}\cdot\vec{\tau}, \qquad P = p_0\mathbf{1} + \vec{p}\cdot\vec{\tau}.$$

Then the local instanton plus anti-instanton flavor factor is

$$\begin{aligned} \mathrm{e}^{\mathrm{i}\theta}\det\Phi + \mathrm{e}^{-\mathrm{i}\theta}\det\Phi^\dagger = \frac{\cos\theta}{2}&\left(s_0^2 - \vec{s}^{\,2} - p_0^2 + \vec{p}^{\,2}\right) \\ &- \sin\theta\left(s_0p_0 - \vec{s}\cdot\vec{p}\right). \end{aligned} \tag{51.108}$$

At $\theta = 0$, the one-instanton local vertex therefore separates the isosinglet scalar, isotriplet scalar, isosinglet pseudoscalar, and isotriplet pseudoscalar channels as

$$\det\Phi + \det\Phi^\dagger = \frac{1}{2}\left(s_0^2 - \vec{s}^{\,2} - p_0^2 + \vec{p}^{\,2}\right). \tag{51.109}$$

The determinant is invariant under $\Phi \mapsto V_R^\dagger\Phi V_L$, $V_L, V_R \in SU(2)$, but under the flavor-singlet axial rotation $q \mapsto \mathrm{e}^{\mathrm{i}\alpha\gamma_5}q$ it carries phase $\det\Phi \mapsto \mathrm{e}^{-4\mathrm{i}\alpha}\det\Phi$. Hence $\mathrm{e}^{\mathrm{i}\theta}\det\Phi + \text{conjugate}$ is the local channel form of the same anomalous $U(1)_A$ selection rule as (51.105): a compensating $\theta \mapsto \theta + 4\alpha$ keeps the $Q = 1$ contribution in the same coordinate.

This channel decomposition is only the zeroth local term of the source convolution. The coefficient multiplying (51.108) is still the size/orientation/nonzero-mode determinant integral discussed below. The formula therefore explains which chiral channels the zero modes select; it is not a standalone prediction of scalar or pseudoscalar bound-state masses.

Proof. The projector identity gives

$$\bar{q}_R q_L = \bar{q}P_L q = \frac{1}{2}(\bar{q}q - \bar{q}\gamma_5 q) = \frac{1}{2}(\bar{q}q + \mathrm{i}\,\bar{q}\mathrm{i}\gamma_5 q),$$

which is (51.107). For any matrix $X = x_0\mathbf{1} + \vec{x}\cdot\vec{\tau}$, the Pauli relation $\tau_a\tau_b = \delta_{ab}\mathbf{1} + \mathrm{i}\epsilon_{abc}\tau_c$ gives $\det X = x_0^2 - \vec{x}^{\,2}$. Applying this to $X = S + \mathrm{i}P$ gives

$$\det(S + \mathrm{i}P) = s_0^2 - \vec{s}^{\,2} - p_0^2 + \vec{p}^{\,2} + 2\mathrm{i}(s_0p_0 - \vec{s}\cdot\vec{p}).$$

Since $\det\Phi = \frac{1}{4}\det(S+\mathrm{i}P)$, taking $\mathrm{e}^{\mathrm{i}\theta}\det\Phi+\mathrm{e}^{-\mathrm{i}\theta}\det\Phi^\dagger$ proves (51.108). The $\theta=0$ specialization is immediate.

Finally, an $SU(2)_L \times SU(2)_R$ rotation multiplies Φ by unit-determinant matrices on the two sides, so its determinant is unchanged. For $q \mapsto \mathrm{e}^{\mathrm{i}\alpha\gamma_5}q$, one has $\bar{q}_R q_L \mapsto \mathrm{e}^{-2\mathrm{i}\alpha}\bar{q}_R q_L$, hence $\det\Phi \mapsto \mathrm{e}^{-4\mathrm{i}\alpha}\det\Phi$. □

Controlled Approximation 51.37 (Instanton contribution to two-flavor chiral source curvatures)**.** Suppose the finite-regulator $Q=\pm1$ calculation, including the size/orientation integral, nonzero-mode determinant, source normalization, and local derivative expansion, supplies a finite coefficient $\kappa^{(2)}_{\mathrm{inst}}(\mu)$ multiplying the local $N_f=2$ source density of Proposition 51.36. Equivalently, for constant renormalized scalar and pseudoscalar source coordinates

$$S = s_0\mathbf{1}+\vec{s}\cdot\vec{\tau}, \qquad P = p_0\mathbf{1}+\vec{p}\cdot\vec{\tau},$$

the instanton part of the source generating-density coordinate $\mathcal{W}=V_4^{-1}\log Z$ has quadratic term

$$\begin{aligned}\mathcal{W}^{(2)}_{\mathrm{inst}} = \kappa^{(2)}_{\mathrm{inst}}\left[\frac{\cos\theta}{2}\left(s_0^2-\vec{s}^{\,2}-p_0^2+\vec{p}^{\,2}\right)-\sin\theta\,(s_0p_0-\vec{s}\cdot\vec{p})\right]\\ +O(J^3,\partial J),\end{aligned} \tag{51.110}$$

where J denotes the collection of source coordinates. Define the local source curvatures by differentiating $\mathcal{W}$ at zero source:

$$\chi_\sigma = \partial^2_{s_0}\mathcal{W}, \quad \chi_{\delta_a} = \partial^2_{s_a}\mathcal{W}, \quad \chi_\eta = \partial^2_{p_0}\mathcal{W}, \quad \chi_{\pi_a} = \partial^2_{p_a}\mathcal{W}, \qquad \text{no sum on } a.$$

Then the local instanton contribution is

$$\chi^{\mathrm{inst}}_\sigma = \chi^{\mathrm{inst}}_{\pi_a} = \kappa^{(2)}_{\mathrm{inst}}\cos\theta, \qquad \chi^{\mathrm{inst}}_{\delta_a} = \chi^{\mathrm{inst}}_\eta = -\kappa^{(2)}_{\mathrm{inst}}\cos\theta, \tag{51.111}$$

and the CP-odd source-mixing curvatures are

$$\partial_{s_0}\partial_{p_0}\mathcal{W}^{(2)}_{\mathrm{inst}} = -\kappa^{(2)}_{\mathrm{inst}}\sin\theta, \qquad \partial_{s_a}\partial_{p_b}\mathcal{W}^{(2)}_{\mathrm{inst}} = \kappa^{(2)}_{\mathrm{inst}}\sin\theta\,\delta_{ab}. \tag{51.112}$$

At $\theta=0$, the two $U(1)_A$-paired differences supplied by this finite-activity term are therefore

$$(\chi_{\pi_a}-\chi_{\delta_a})_{\mathrm{inst}} = (\chi_\sigma-\chi_\eta)_{\mathrm{inst}} = 2\kappa^{(2)}_{\mathrm{inst}}. \tag{51.113}$$

Equations (51.111)–(51.113) are source-curvature statements in the declared local window. The full zero-momentum QCD susceptibilities also contain connected long-distance propagation, Goldstone poles in the chiral limit, ordinary perturbative and confining contributions, and finite contact-term conventions. Thus the instanton calculation identifies the $U(1)_A$-violating local curvature it adds to the generating functional. A spectrum statement additionally requires the color-singlet spectral representation, long-distance matrix elements, and the continuum/volume limits in the chosen QCD phase.

Proposition 51.38 (Axial Ward ledger for the instanton source curvatures)**.** *The local two-flavor instanton source functional* (51.110) *obeys the finite-dimensional anomalous singlet-axial Ward identity*

$$\left[4\partial_\theta+2p_0\partial_{s_0}+2\sum_{a=1}^{3}p_a\partial_{s_a}-2s_0\partial_{p_0}-2\sum_{a=1}^{3}s_a\partial_{p_a}\right]\mathcal{W}^{(2)}_{\mathrm{inst}} = O(J^3,\partial J). \tag{51.114}$$

Here the source coordinates transform as the mass/source spurions in the finite-regulator path integral:

$$\delta_\alpha S = 2\alpha P, \qquad \delta_\alpha P = -2\alpha S, \qquad \delta_\alpha \theta = 4\alpha.$$

Thus the same anomalous Jacobian which shifts θ also rotates the local scalar and pseudoscalar source coordinates. The susceptibility splittings in (51.113) are the second-source derivatives of the same anomalous Ward identity which produces the 't Hooft vertex.

More explicitly, for each paired coordinate s, p, with $s = s_0, p = p_0$ or $s = s_a, p = p_a$, define

$$\chi_s = \partial_s^2 \mathcal{W}_{\text{inst}}^{(2)}, \qquad \chi_p = \partial_p^2 \mathcal{W}_{\text{inst}}^{(2)}, \qquad m_{sp} = \partial_s \partial_p \mathcal{W}_{\text{inst}}^{(2)}$$

at zero source. Then the differentiated Ward identity gives

$$\partial_\theta \chi_s = m_{sp}, \qquad \partial_\theta \chi_p = -m_{sp}, \qquad \chi_s - \chi_p = -2\,\partial_\theta m_{sp}. \tag{51.115}$$

For the singlet pair this is the (σ, η) relation; for a triplet pair it is the (δ_a, π_a) relation, so reversing the order of the difference gives the positive $\pi_a - \delta_a$ splitting displayed above.

Proof. Under an infinitesimal singlet axial rotation, $S + \mathrm{i}P \mapsto (1 - 2\mathrm{i}\alpha)(S + \mathrm{i}P)$, which gives $\delta S = 2\alpha P$ and $\delta P = -2\alpha S$. The anomaly shifts the source-sector theta coordinate by $\delta\theta = 4\alpha$ for $N_f = 2$. Consequently the infinitesimal Ward vector field is the operator displayed in (51.114).

Let

$$A = s_0^2 - \vec{s}^{\,2} - p_0^2 + \vec{p}^{\,2}, \qquad B = s_0 p_0 - \vec{s} \cdot \vec{p}.$$

The source rotation gives

$$\delta_\alpha A = 8\alpha B, \qquad \delta_\alpha B = -2\alpha A.$$

The theta derivative of the quadratic part of (51.110) contributes

$$4\alpha\, \kappa_{\text{inst}}^{(2)} \left[-\frac{\sin\theta}{2} A - \cos\theta\, B \right],$$

whereas the source rotation contributes

$$\kappa_{\text{inst}}^{(2)} \left[4\alpha \cos\theta\, B + 2\alpha \sin\theta\, A \right].$$

The two terms cancel, proving (51.114) to the declared local order.

Now apply ∂_s^2, ∂_p^2, and $\partial_s\partial_p$ to the one-pair form of the Ward identity $(4\partial_\theta + 2p\partial_s - 2s\partial_p)\mathcal{W} = 0$, and then set all sources to zero. The three resulting identities are

$$4\partial_\theta \chi_s - 4m_{sp} = 0, \qquad 4\partial_\theta \chi_p + 4m_{sp} = 0, \qquad 4\partial_\theta m_{sp} + 2\chi_s - 2\chi_p = 0,$$

which are exactly (51.115). □

Proposition 51.39 (Mixed mass-source expansion of the instanton determinant)**.** *In the same $Q = 1$ QCD zero-mode sector, let M be the mass-overlap matrix and let B be the matrix supplied by external bilinear sources or by a specified zero-mode-paired operator insertion. Ordinary external fermion wave packets are represented instead by the differentiated linear-source construction below. With $[N_f] = \{1, \ldots, N_f\}$, write M_{I^c, J^c} for the submatrix obtained by deleting the rows $I \subset [N_f]$*

and columns $J \subset [N_f]$, and write $B_{I,J}$ for the corresponding $I \times J$ source submatrix, both with increasing orders; empty determinants are 1*. Then*

$$\det(M+B) = \sum_{r=0}^{N_f} \sum_{\substack{I,J\subset[N_f]\\|I|=|J|=r}} (-1)^{\sum_{i\in I} i+\sum_{j\in J} j} \det M_{I^c,J^c} \det B_{I,J}. \tag{51.116}$$

Thus a correlator with r zero-mode-paired external insertions is not obtained by appending those insertions to an already-formed vacuum determinant. It is the degree-r part of the same finite Berezin determinant; the unobserved $N_f - r$ zero-mode pairs must be saturated by the complementary mass minor. Equivalently, for increasing $I = (i_1 < \cdots < i_r)$, $J = (j_1 < \cdots < j_r)$, and $\pi \in S_r$, the coefficient of $\prod_{\ell=1}^{r} B_{i_\ell, j_{\pi(\ell)}}$ in $\det(M+B)$ is

$$(-1)^{\sum I+\sum J} \operatorname{sgn}(\pi) \det M_{I^c,J^c}. \tag{51.117}$$

For $N_f = 2$,

$$\begin{aligned}\det(M+B) = \det M &+ M_{22}B_{11} - M_{21}B_{12} - M_{12}B_{21} + M_{11}B_{22}\\ &+ B_{11}B_{22} - B_{12}B_{21}.\end{aligned} \tag{51.118}$$

The last line is the massless four-fermion 't Hooft vertex. The middle line is the mixed amplitude in which one flavor is supplied by an external zero-mode pair and the other by a mass insertion. If a flavor is neither externally inserted nor present in the complementary mass minor, the instanton contribution to that correlator vanishes.

Proof. Apply the determinant expansion to $M + B$. Each contributing permutation $\sigma \in S_{N_f}$ chooses, for every row, either an M-entry or a B-entry. Let I be the set of rows where a B-entry was chosen and let $J = \sigma(I)$. Reordering the rows and columns so that the B-block appears in the I, J positions gives the cofactor sign $(-1)^{\sum I+\sum J}$. Summing over the restriction of σ to I gives $\det B_{I,J}$, while summing over the complementary restriction gives $\det M_{I^c,J^c}$. This proves (51.116). Extracting a particular source monomial from $\det B_{I,J}$ gives the $\operatorname{sgn}(\pi)$ factor in (51.117). The displayed two-flavor formula is the specialization to $N_f = 2$. □

Proposition 51.40 (Fermion determinant factor versus zero-mode determinant)**.** *Fix the same finite-regulator $Q = 1$ chart $z = (a, \rho, U)$ for vectorlike $SU(N_c)$ QCD, and add a mass matrix and an even bilinear source. Let $\mathcal{K}_{\Lambda,z}(M, B)$ be the finite Dirac bilinear between the right and left fermion coefficient spaces in this background. Decompose these finite spaces into instanton zero-mode subspaces and nonzero-mode complements. In block form,*

$$\mathcal{K}_{\Lambda,z}(M,B) = \begin{pmatrix} K_{00} & K_{0\mathrm{n}} \\ K_{\mathrm{n}0} & K_{\mathrm{nn}} \end{pmatrix}, \tag{51.119}$$

where K_{00} acts between the zero-mode coefficient spaces and K_{nn} acts on the nonzero-mode complements. If K_{nn} is invertible, then

$$\det \mathcal{K}_{\Lambda,z}(M,B) = \det K_{\mathrm{nn}} \det\!\left(K_{00} - K_{0\mathrm{n}} K_{\mathrm{nn}}^{-1} K_{\mathrm{n}0}\right). \tag{51.120}$$

Thus the finite nonzero-mode fermion determinant and the zero-mode saturation determinant are separate factors before the collective-coordinate integral is performed.

In the leading local semiclassical approximation, where the mass/source bilinear is restricted to the zero-mode subspace after the nonzero modes have supplied their determinant ratio, the Schur complement in (51.120) *reduces to*

$$K_{00}^{\mathrm{eff}}(z) = M^0(z) + B^0(z), \qquad M^0_{ff'}(z) = \rho\, M_{ff'}(\mu), \qquad B^0_{ff'}(z) = \rho^3 \bar\psi_{Rf}\psi_{Lf'} + \cdots. \tag{51.121}$$

The dots denote derivative corrections to the local source approximation and the spin-color contractions fixed by the normalized zero-mode wavefunctions. More generally, the Schur correction $-K_{0\mathrm{n}}K_{\mathrm{nn}}^{-1}K_{\mathrm{n}0}$ *is a finite nonzero-mode-propagator correction in the same regulator and scheme. Consequently the light-fermion part of a one-instanton chart has the structure*

$$\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}(z;\mu)\,\det K_{00}^{\mathrm{eff}}(z), \tag{51.122}$$

after trivial-sector normalization and local counterterms in the scheme $\mathcal{S}$*. The finite factor* $\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}$ *is the fermion nonzero-mode determinant datum. It is not determined by the Berezin determinant* $\det(M^0+B^0)$*. In particular, in the diagonal massive vacuum specialization the factor* $\prod_f(m_f\rho)$ *is the zero-mode saturation determinant; the remaining finite spectral determinant is an additional input to the physical amplitude.*

Proof. At fixed cutoff the fermion path integral is a finite Berezin Gaussian, hence equals the determinant of its finite bilinear matrix after the chosen trivial-sector normalization. Decompose the finite coefficient spaces into the zero-mode subspace and its complement. Gaussian integration over the nonzero-mode variables first gives the Schur complement of the lower-right block, which is exactly (51.120). The determinant of K_{nn} is the fermionic part of the nonzero-mode determinant factor in $\mathcal{W}^{\mathrm{nz}}_{\Lambda}$. The determinant of the Schur complement is the finite Berezin integral over the zero-mode coefficients. When the source and mass insertions are projected to the zero-mode subspace, this second determinant is the determinant of $M^0 + B^0$, whose minor expansion is Proposition 51.39. If the source or mass also couples zero and nonzero modes, the same finite Gaussian integral produces the displayed Schur correction. Since $\det K_{\mathrm{nn}}$ multiplies every zero-mode minor, no zero-mode algebra can determine its finite value. □

Proposition 51.41 (Finite source-frame covariance of the light-fermion determinant)**.** *In the setting of Proposition 51.40, write the finite zero-mode source functional as*

$$\mathcal{V}^{0,\mathcal{S}}_{\mathrm{inst}}(z) = \mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}(z;\mu)\,\det K_{00}^{\mathrm{eff}}(z). \tag{51.123}$$

Let $Z_R, Z_L \in \mathrm{GL}(N_f,\mathbb{C})$ *be a finite change of the right and left zero-mode source frames, so that the same source insertion is represented by*

$$K_{00}^{\mathrm{eff}\,\prime} = Z_R K_{00}^{\mathrm{eff}} Z_L. \tag{51.124}$$

Then the same finite-regulator amplitude coordinate is represented only if

$$\mathcal{R}^{\mathcal{S}\,\prime}_{\mathrm{f},\Lambda} = \frac{\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}}{\det Z_R\,\det Z_L}, \qquad \mathcal{R}^{\mathcal{S}\,\prime}_{\mathrm{f},\Lambda}\det K_{00}^{\mathrm{eff}\,\prime} = \mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}\det K_{00}^{\mathrm{eff}}. \tag{51.125}$$

If the finite scheme is also changed by a local fermion determinant counterterm $\Delta C_{\mathrm{f}}(z)$*, then the determinant coordinate is instead multiplied by* $\exp[-\Delta C_{\mathrm{f}}(z)]$*:*

$$\mathcal{R}^{\mathcal{S}'}_{\mathrm{f},\Lambda} = \mathrm{e}^{-\Delta C_{\mathrm{f}}(z)}\frac{\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}}{\det Z_R\,\det Z_L}. \tag{51.126}$$

Thus a numerical value for $\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}$ is not a standalone output of the zero-mode determinant. It has meaning only together with the finite source frame, local counterterm convention, and operator normalization used to extract the physical correlator.

Proof. Equation (51.124) and determinant multiplicativity give

$$\det K^{\mathrm{eff}\,\prime}_{00} = \det Z_R \,\det K^{\mathrm{eff}}_{00}\, \det Z_L.$$

The inverse transformation of $\mathcal{R}^{\mathcal{S}}_{\mathrm{f},\Lambda}$ in (51.125) is therefore exactly what preserves the finite product (51.123). A finite local counterterm changes the regulated determinant density by the exponential of minus the counterterm difference between the instanton and trivial sectors, as in (51.86); this gives (51.126). The final statement follows because source-frame changes and local counterterms act on the finite spectral determinant factor, while the Berezin determinant records only the zero-mode saturation algebra. □

Proposition 51.42 (Mass/source RG transport of the instanton vertex)**.** *Work in a mass-independent renormalization scheme in the same $Q = 1$ vectorlike QCD chart. Let $\gamma_m(g)$ be the anomalous dimension of the quark mass coordinate in the convention*

$$\mu\frac{\mathrm{d}M}{\mathrm{d}\mu} = -\gamma_m(g)M, \qquad \mu\frac{\mathrm{d}J}{\mathrm{d}\mu} = -\gamma_m(g)J, \tag{51.127}$$

where J is a renormalized source coordinate coupled to the same chiral scalar density as the mass matrix. Let $M^0(z) = \rho M$, and let $J^0(z)$ be the zero-mode projection of J, a linear function of J with the same anomalous transport as J. Let $\mathcal{R}^{\mathcal{S}}_{\mathrm{f}}(z;\mu)$ denote the finite light-fermion nonzero-mode determinant factor in the source convention of Proposition 51.40. If, at the logarithmic accuracy under consideration,

$$(\mu\partial_\mu + \beta(g)\partial_g)\log \mathcal{R}^{\mathcal{S}}_{\mathrm{f}}(z;\mu) = N_f\,\gamma_m(g), \tag{51.128}$$

then the zero-mode source functional

$$\mathcal{V}^0_{\mathrm{inst}}(z; M, J, \mu) = \mathcal{R}^{\mathcal{S}}_{\mathrm{f}}(z;\mu)\,\det\bigl(M^0(z) + J^0(z)\bigr) \tag{51.129}$$

obeys

$$\left[\mu\partial_\mu + \beta(g)\partial_g - \gamma_m(g)\sum_{f,f'}\left(M_{ff'}\frac{\partial}{\partial M_{ff'}} + J_{ff'}\frac{\partial}{\partial J_{ff'}}\right)\right]\mathcal{V}^0_{\mathrm{inst}} = 0. \tag{51.130}$$

Thus the mass/source determinant is not an RG-invariant number by itself. The source functional is covariant only after the finite nonzero-mode determinant factor and the source renormalization convention are included. The symbol J denotes the source coordinate before differentiation. The scalar-density operator obtained after differentiating the generating functional uses the corresponding operator normalization; in the local operator convention that differentiated insertion has the $B^0_{ff'} = \rho^3\bar\psi_{Rf}\psi_{Lf'} + \cdots$ normalization of (51.121).

In particular, for diagonal masses and $J = 0$, the anomalous-dimension part of the vacuum factor

$$\mathcal{R}^{\mathcal{S}}_{\mathrm{f}}(\mu)\prod_{f=1}^{N_f} m_f(\mu)$$

is scale invariant at the same logarithmic accuracy. The phase $\theta + \arg\det M$ is unaffected by the positive real mass renormalization factor; it is fixed instead by the anomalous chiral rotation described in (51.49). After differentiating (51.129) r times with respect to J and setting $J = 0$, the resulting mixed mass/source minor has the RG covariance of an r-fold insertion of the renormalized scalar density, rather than the covariance of a vacuum activity.

Proof. The matrix $M^0(z) + J^0(z)$ is homogeneous of degree one in the source coordinates M and J, while the projection factors depending on ρ, U, and the normalized zero-mode wavefunctions are collective coordinate data and are not acted on by the source anomalous dimension. The determinant is therefore homogeneous of degree N_f:

$$\sum_{f,f'} \left(M_{ff'} \frac{\partial}{\partial M_{ff'}} + J_{ff'} \frac{\partial}{\partial J_{ff'}} \right) \det(M^0 + J^0) = N_f \det(M^0 + J^0).$$

Combining this identity with (51.128) gives (51.130). Setting $J = 0$ gives the vacuum statement. Differentiating the source-functional equation r times with respect to J transfers r source anomalous-dimension terms from the transport operator to the differentiated coefficient, which is the usual Callan–Symanzik covariance of r renormalized scalar-density insertions. The phase statement follows from a separate observation. In a vectorlike mass-independent scheme, multiplicative mass renormalization changes the magnitude of M. The anomalous flavor-singlet chiral rotation changes $\arg\det M$ and θ in opposite directions. □

Proposition 51.43 (Proper-time determinant and zero-mode source functional)**.** *Fix a finite-regulator $Q = 1$ chart $z = (a, \rho, U)$, a background gauge slice, and a proper-time determinant prescription with renormalization scale μ. For each positive nonzero-mode operator $P_{\Lambda,z}$ in the instanton background, compared with the corresponding operator $P_{\Lambda,0}$ in the trivial sector, define the regulated logarithmic determinant ratio by*

$$K_\Lambda(P_z, P_0; \mu) = -\int_{\epsilon_\Lambda}^{\infty} \frac{\mathrm{d}t}{t}\, r(t\mu^2) \left(\mathrm{Tr}'\, e^{-tP_{\Lambda,z}} - \mathrm{Tr}\, e^{-tP_{\Lambda,0}} \right) + C^{\mathrm{loc}}_{\Lambda,P}(z; \mu), \tag{51.131}$$

where r is a smooth cutoff profile with $r(0) = 1$, the prime removes collective-coordinate, residual-gauge, and fermion zero modes as appropriate, and $C^{\mathrm{loc}}_{\Lambda,P}$ is the chosen local subtraction in the instanton background relative to the trivial background. The determinant density in (51.86) is then

$$\log \mathcal{W}^{\mathrm{nz}}_{\Lambda,\mathrm{pt}}(z; \mu) = K_\Lambda(M^{\mathrm{gh}}_z, M^{\mathrm{gh}}_0; \mu) - \frac{1}{2} K_\Lambda(L^{\mathrm{bos}}_z, L^{\mathrm{bos}}_0; \mu) + \sum_R K^{\mathrm{pf,R}}_\Lambda(z, 0; \mu), \tag{51.132}$$

with $K^{\mathrm{pf,R}}_\Lambda$ the Pfaffian or determinant-line version of the same nonzero-mode construction for the fermion multiplet R.

For two massless Dirac flavors there are two distinct source constructions. An even bilinear source supplies an independent finite matrix $B(z)$ in the zero-mode action $\sum_{f,f'} \widetilde{\lambda}_f B_{ff'}(z) \lambda_{f'}$; its zero-mode factor is $\det B(z)$, as in Proposition 51.35. Ordinary external fermion wave packets are instead introduced by odd generating coordinates. Let $\bar{\chi}_A, \chi_B$, $A, B = 1, 2$, be the four external source coordinates and set

$$\alpha_f(z) = \sum_{A=1}^{2} \bar{\chi}_A R_{Af}(z), \qquad \beta_f(z) = \sum_{B=1}^{2} L_{fB}(z) \chi_B, \tag{51.133}$$

where R_{Af} and L_{fB} are the zero-mode overlaps of the Ath right-handed and Bth left-handed external slots with the normalized instanton zero-mode wavefunction. With the same Berezin orientation used in (51.106), the differentiated source coefficient is

$$[\bar{\chi}_1\bar{\chi}_2\chi_1\chi_2]\prod_{f=1}^{2}\alpha_f(z)\prod_{f=1}^{2}\beta_f(z) = \det R(z)\,\det L(z). \tag{51.134}$$

Thus the one-instanton generating-functional contribution is

$$\begin{aligned}\mathcal{G}^{(4)}_{\eta,\Lambda,Q=1}[\bar{\chi},\chi] &= \int_{\mathcal{U}_{Q=1}} \exp[-S_\Lambda(A_z)+\mathrm{i}\theta]\,\mathcal{W}^{\mathrm{nz}}_{\Lambda,\mathrm{pt}}(z;\mu)\prod_{f=1}^{2}\alpha_f(z)\prod_{f=1}^{2}\beta_f(z)\,\mathrm{d}\nu_{\Lambda,\mathcal{M}}(z),\\ G^{(4)}_{\eta,\Lambda,Q=1} &= [\bar{\chi}_1\bar{\chi}_2\chi_1\chi_2]\,\mathcal{G}^{(4)}_{\eta,\Lambda,Q=1}[\bar{\chi},\chi].\end{aligned} \tag{51.135}$$

A c-number matrix formed as a single outer product $u_f v_{f'}$ is not this four-slot coefficient; for $N_f = 2$ its determinant vanishes. The small-t expansion of (51.131) supplies the local counterterms and the universal $(\mu\rho)^{b_0}$ factor, while the finite remainder fixes the scheme-dependent numerical constant in the selected source functional. Thus the physics calculation consists of the regulated fluctuation determinant, the zero-mode source differentiation, and the collective-coordinate integral as a single object.

Proof. For a finite positive operator P, the identity $\log\det P = -\int_\epsilon^\infty \mathrm{d}t\,t^{-1}\,\mathrm{Tr}\,e^{-tP}$ holds up to the subtraction convention at the lower endpoint. Applying this identity to the instanton and trivial-sector operators and adding the chosen local counterterm gives (51.131). Bosonic Gaussian integration contributes the exponent $-1/2$, ghost Berezin integration contributes $+1$, and fermionic nonzero modes contribute the Pfaffian or determinant-line factor, giving (51.132). Zero modes have already been removed from the traces and determinants, so their finite Berezin integral is exactly the calculation in Proposition 51.35 for an even bilinear source matrix. For linear external sources, expanding $\prod_f \alpha_f \prod_f \beta_f$ and extracting the $\bar{\chi}_1\bar{\chi}_2\chi_1\chi_2$ coefficient gives $\det R\,\det L$; this is the source-differentiated correlator coefficient, not the determinant of a single c-number outer product. Multiplication by the collective-coordinate density and the classical instanton exponential gives (51.135). Finally, the heat-kernel expansion of the trace difference at small t separates local counterterms, the coefficient of $\log(\mu\rho)$, and the finite determinant constant; RG invariance identifies the universal logarithmic coefficient with b_0 in the displayed one-loop density. □

Proposition 51.44 (Instanton zero-mode density form factor and local limit)**.** *In a BPST $Q = 1$ background, after the spin and color tensors of one fundamental zero-mode line are factored out, the radial scalar envelope which weights a local mass, Yukawa, or bilinear source is*

$$h_\rho(y) = \frac{2\rho^2}{\pi^2(y^2+\rho^2)^3}, \qquad \int_{\mathbb{R}^4} h_\rho(y)\,\mathrm{d}^4y = 1. \tag{51.136}$$

Thus a zero-mode-paired source matrix entry has the form

$$B^{(s)}_{ff'}(a,\rho,U) = \rho^{d_s}\int_{\mathbb{R}^4} h_\rho(y)\,\mathcal{B}_{ff'}(a+y;U)\,\mathrm{d}^4y, \tag{51.137}$$

where $d_s = 1$ for a mass source and $d_s = 3$ for the local quark-bilinear source in (51.105). Equivalently, with the Fourier convention

$$\widehat{h}_\rho(q) = \int_{\mathbb{R}^4} \mathrm{e}^{\mathrm{i}q\cdot y} h_\rho(y)\, \mathrm{d}^4 y,$$

one has the exact form factor

$$\widehat{h}_\rho(q) = zK_1(z), \qquad z = \rho|q|, \tag{51.138}$$

where K_1 is the modified Bessel function. Hence a zero-mode-paired source with Fourier momentum q is not represented by the local vertex until $z \ll 1$. The small-momentum expansion is

$$zK_1(z) = 1 + \frac{z^2}{2}\left[\log\frac{z}{2} + \gamma_{\mathrm{E}} - \frac{1}{2}\right] + O(z^4 \log z), \tag{51.139}$$

so the first nonlocal correction is $q^2\rho^2\log(|q|\rho)$, not an ordinary analytic Taylor coefficient. At $z \gg 1$,

$$zK_1(z) = \sqrt{\frac{\pi z}{2}}\,\mathrm{e}^{-z}\left(1 + \frac{3}{8z} + O(z^{-2})\right), \tag{51.140}$$

which displays the suppression of source components varying on scales shorter than the instanton size. This is the density form factor for a local bilinear or mass-source insertion. A differentiated correlator with four ordinary external fermion slots instead uses the Fourier transforms of the individual zero-mode wavefunctions in those slots; their product reduces to the local bilinear density only after the slot wavefunctions are fused into a bilinear source and the local limit is taken. In position space, for $U_R = R/\rho$,

$$\int_{|y|<R} h_\rho(y)\, \mathrm{d}^4 y = \frac{U_R^4}{(1+U_R^2)^2}, \qquad \int_{|y|>R} h_\rho(y)\, \mathrm{d}^4 y = \frac{1+2U_R^2}{(1+U_R^2)^2}. \tag{51.141}$$

The first correction to a local Taylor expansion is more delicate. The truncated second moment is

$$M_2(R) := \int_{|y|<R} |y|^2 h_\rho(y)\, \mathrm{d}^4 y = 2\rho^2\left[\log(1+U_R^2) + \frac{2}{1+U_R^2} - \frac{1}{2(1+U_R^2)^2} - \frac{3}{2}\right], \tag{51.142}$$

and rotational invariance gives

$$\int_{|y|<R} y^\mu y^\nu h_\rho(y)\, \mathrm{d}^4 y = \frac{\delta^{\mu\nu}}{4} M_2(R).$$

Consequently the leading local term in (51.137) is $\rho^{d_s}\mathcal{B}_{ff'}(a;U)$, with a tail bounded by the second quantity in (51.141). But the would-be two-derivative coefficient grows as $4\rho^2\log(R/\rho) + O(\rho^2)$ as $R/\rho \to \infty$. The local 't Hooft vertex is therefore the zeroth adiabatic term of the zero-mode source convolution; higher-derivative terms require the external wave-packet, mass, or infrared regulator that cuts off the zero-mode tail. In momentum space the same statement is the logarithm in (51.139).

Proof. The normalized radial envelope in (51.136) is the scalar part of the product of two normalized fundamental zero-mode wavefunctions in the BPST background. Using $\mathrm{d}^4 y = 2\pi^2 r^3 \mathrm{d}r$ and $u = r/\rho$,

$$\int_{\mathbb{R}^4} h_\rho(y)\, \mathrm{d}^4 y = 4\int_0^\infty \frac{u^3}{(1+u^2)^3}\, \mathrm{d}u = 1.$$

For the Fourier transform, use the Schwinger representation

$$\frac{1}{(y^2+\rho^2)^3} = \frac{1}{2}\int_0^\infty s^2 \mathrm{e}^{-s(y^2+\rho^2)}\,\mathrm{d}s.$$

Then

$$\begin{aligned}\widehat{h}_\rho(q) &= \rho^2 \int_0^\infty \exp\Big[-\rho^2 s - \frac{q^2}{4s}\Big]\,\mathrm{d}s\\ &= \rho|q|\,K_1(\rho|q|),\end{aligned}$$

where the last step is the standard integral representation of the modified Bessel function. The small-z and large-z formulas are the corresponding standard asymptotic expansions of K_1; the logarithmic small-z term is the momentum-space reflection of the logarithmically divergent second moment computed below.

The same integral cut off at U_R gives

$$4\int_0^{U_R} \frac{u^3}{(1+u^2)^3}\,\mathrm{d}u = 1 - \frac{2}{1+U_R^2} + \frac{1}{(1+U_R^2)^2} = \frac{U_R^4}{(1+U_R^2)^2},$$

and subtracting from one gives the tail formula in (51.141). For the second moment,

$$M_2(R) = 4\rho^2 \int_0^{U_R} \frac{u^5}{(1+u^2)^3}\,\mathrm{d}u.$$

With $t = 1+u^2$, this antiderivative is

$$2\rho^2\left[\log t + \frac{2}{t} - \frac{1}{2t^2} - \frac{3}{2}\right]_{t=1}^{1+U_R^2},$$

which is (51.142). Rotational invariance in four Euclidean dimensions gives the displayed $\delta^{\mu\nu}/4$ tensor moment. The Taylor expansion of $\mathcal{B}(a+y;U)$ then yields the zeroth local term and the $(M_2(R)/8)\partial^2\mathcal{B}(a;U)$ contribution inside the ball $|y|<R$, while the exterior contribution is controlled by the tail mass times the appropriate supremum norm of the source. Since $\log(1+U_R^2) = 2\log(R/\rho)+o(1)$, the full second moment is logarithmically infrared sensitive. $\square$

Proposition 51.45 (Individual BPST zero-mode slot form factor)**.** *For an ordinary external fermion slot one must use the Fourier transform of a single normalized BPST zero-mode wave-function, not the density form factor $\widehat{h}_\rho$ of Proposition 51.44. In the standard singular-gauge momentum-space convention for a charge-one instanton [63, 66], the spin-color tensor and the scalar radial function separate as*

$$\widetilde{\psi}^\alpha_{0,i}(p;\rho,U) = \sqrt{2}\,\phi'_\rho(|p|)\big(\widehat{p}\,\epsilon U\big)^\alpha_i, \qquad \phi_\rho(p) = \pi\rho^2\big[I_0(t)K_0(t) - I_1(t)K_1(t)\big], \quad t = \frac{\rho|p|}{2}. \tag{51.143}$$

Here I_ν and K_ν are modified Bessel functions, U is the color orientation, and $\widehat{p} = p/|p|$. After the adjacent free Dirac operator and fixed Fourier-normalization constants are assigned to the external-slot amputation convention, the dimensionless scalar slot form factor is

$$F_{\mathrm{zm}}(t) := -t\frac{\mathrm{d}}{\mathrm{d}t}\big[I_0(t)K_0(t) - I_1(t)K_1(t)\big] = 2t\big[I_0(t)K_1(t) - I_1(t)K_0(t)\big] - 2I_1(t)K_1(t). \tag{51.144}$$

It is normalized by

$$F_{\mathrm{zm}}(0) = 1, \tag{51.145}$$

while at large momentum

$$F_{\mathrm{zm}}(t) = \frac{3}{4t^3} + O(t^{-5}) = \frac{6}{(\rho|p|)^3} + O((\rho|p|)^{-5}). \tag{51.146}$$

Thus the individual external zero-mode slot has a power tail. The exponential large-momentum suppression in (51.140) *belongs to the fused bilinear density* h_ρ*, not to a single differentiated fermion slot.*

Proof. The first formula is the standard Fourier-Bessel transform of the normalized singular-gauge fundamental zero mode after its hedgehog spin-color tensor has been extracted. Differentiating $\Phi(t) = I_0(t)K_0(t) - I_1(t)K_1(t)$ and using

$$I_0' = I_1, \qquad I_1' = I_0 - \frac{I_1}{t}, \qquad K_0' = -K_1, \qquad K_1' = -K_0 - \frac{K_1}{t}$$

gives the second expression in (51.144). The small-t expansions $I_0(t) = 1 + O(t^2)$, $I_1(t) = t/2 + O(t^3)$, and $K_1(t) = t^{-1} + O(t \log t)$ give $F_{\mathrm{zm}}(0) = 1$. The standard large-t asymptotic expansions of I_ν and K_ν show that the constant, t^{-1}, and t^{-2} terms cancel in (51.144); the first nonzero term is $3/(4t^3)$. □

Proposition 51.46 (Amputation and source typing of the instanton kernel)**.** *Work in the finite-regulator $Q = 1$ chart of Proposition 51.43, and suppose an auxiliary infrared deformation or source scheme has been chosen in which the external quark fields used to probe the kernel have isolated nonzero external pole-residue factors. For differentiated linear source slots, let $Z_A^R(p_{R,A})$ and $Z_B^L(p_{L,B})$ be the adjacent unamputated pole residues for the two right-handed and two left-handed slots. After the spin and color external covectors are fixed, write the zero-mode overlap matrices as*

$$\begin{aligned} R_{Af}^{\mathrm{unamp}}(z) &= Z_A^R(p_{R,A})\, R_{Af}^{\mathrm{amp}}(z), \\ L_{fB}^{\mathrm{unamp}}(z) &= L_{fB}^{\mathrm{amp}}(z)\, Z_B^L(p_{L,B}), \qquad A, B, f = 1, 2. \end{aligned} \tag{51.147}$$

The four-slot coefficient from (51.134) *obeys*

$$\det R^{\mathrm{unamp}}(z) \det L^{\mathrm{unamp}}(z) = \left(\prod_{A=1}^{2} Z_A^R\right) \left(\prod_{B=1}^{2} Z_B^L\right) \det R^{\mathrm{amp}}(z) \det L^{\mathrm{amp}}(z). \tag{51.148}$$

Consequently the source-amputated differentiated four-fermion kernel is

$$\mathcal{M}_{\Lambda,Q=1}^{(4),\mathrm{amp}} = \int_{\mathcal{U}_{Q=1}} \exp[-S_\Lambda(A_z) + \mathrm{i}\theta]\, \mathcal{W}_{\Lambda,\mathrm{pt}}^{\mathrm{nz}}(z;\mu)\, \det R^{\mathrm{amp}}(z) \det L^{\mathrm{amp}}(z)\, \mathrm{d}\nu_{\Lambda,\mathcal{M}}(z). \tag{51.149}$$

Amputation removes only the adjacent external propagator residues. It does not remove the individual instanton zero-mode Fourier overlaps, the spin-color orientation tensors, the nonzero-mode determinant, or the collective-coordinate integral.

For an independent even bilinear source matrix the source type is different. If $B^{\mathrm{unamp}} = D_R B^{\mathrm{amp}} D_L$, with D_R, D_L diagonal residue matrices in flavor space, then

$$\det B^{\mathrm{unamp}} = \det D_R\, \det B^{\mathrm{amp}}\, \det D_L. \tag{51.150}$$

For a local bilinear source carrying momentum $q_{ff'}$, the BPST density form factor of Proposition 51.44 gives a representative entry

$$B^{\mathrm{amp}}_{ff'}(a,\rho,U)=\rho^3\mathrm{e}^{\mathrm{i}q_{ff'}\cdot a}\,\widehat{h}_\rho(q_{ff'})\,\mathcal{T}_{ff'}(U;q_{ff'}),\qquad q_{ff'}=p_{R,f}-p_{L,f'}. \tag{51.151}$$

This bilinear-source formula is not the determinant of a single c-number outer product of external wave packets. Ordinary external fermions use (51.147) *and the source differentiation in* (51.134).

Proof. The linear source overlaps in (51.133) attach each external slot to one normalized zero-mode wavefunction. The chosen infrared scheme supplies one adjacent pole residue per external slot; removing those residues gives (51.147). Applying the determinant to the right-slot matrix and to the left-slot matrix gives (51.148). Multiplying the unamputated differentiated Green kernel by the inverse product of the same four residues therefore gives (51.149).

For an even bilinear source, the zero-mode action already contains a matrix entry $B_{ff'}$. Row and column residue factors then multiply that matrix on the left and right, so ordinary determinant multiplicativity gives (51.150). The density form factor in (51.151) comes from the product of two zero-mode wavefunctions in a fused bilinear source. Linear external slots are differentiated before the sources are set to zero, so a single rank-one c-number matrix never appears as the four-fermion coefficient. □

Proposition 51.47 (Plane-wave assembly of the one-instanton four-fermion amplitude)**.** *Stay in the $N_f=2$ source-amputated setup of Proposition 51.46, and suppose the center coordinate factorizes from the regulated collective-coordinate density:*

$$\mathrm{d}\nu_{\Lambda,\mathcal{M}}(z)=\mathcal{J}_\Lambda(\rho,U)\,\mathrm{d}^4a\,\mathrm{d}\rho\,\mathrm{d}\Omega(U).$$

Let

$$\mathcal{P}=\sum_{A=1}^{2}p_{R,A}-\sum_{B=1}^{2}p_{L,B}.$$

For plane-wave external slots, write the individual amputated zero-mode overlaps in the form

$$R^{\mathrm{amp}}_{Af}(a,\rho,U)=\rho^{3/2}\mathrm{e}^{\mathrm{i}p_{R,A}\cdot a}\mathcal{R}_{Af}(\rho,U;p_{R,A}),\qquad L^{\mathrm{amp}}_{fB}(a,\rho,U)=\rho^{3/2}\mathrm{e}^{-\mathrm{i}p_{L,B}\cdot a}\mathcal{L}_{fB}(\rho,U;p_{L,B}),$$

where $\mathcal{R}$ and $\mathcal{L}$ are obtained from the Fourier transforms of the individual normalized zero-mode wavefunctions, with the chosen spin-color external covectors already contracted. Then the source-differentiated zero-mode coefficient in (51.149) *has the factorized form*

$$\begin{aligned}\det R^{\mathrm{amp}}(a,\rho,U)\det L^{\mathrm{amp}}(a,\rho,U)&=\rho^6\,\mathrm{e}^{\mathrm{i}\mathcal{P}\cdot a}\,\mathfrak{A}_2^{\mathrm{lin}}(\rho,U;\{p_R,p_L\}),\\ \mathfrak{A}_2^{\mathrm{lin}}&=\det\mathcal{R}(\rho,U;\{p_R\})\,\det\mathcal{L}(\rho,U;\{p_L\}).\end{aligned} \tag{51.152}$$

A separate bilinear-source specialization is obtained by inserting (51.151) *into* $\det B^{\mathrm{amp}}$. *In that case $F(z)=zK_1(z)$ appears through the density form factor and*

$$\det B^{\mathrm{amp}}=\rho^6\mathrm{e}^{\mathrm{i}\mathcal{P}\cdot a}\left[F(\rho|q_{11}|)F(\rho|q_{22}|)\mathcal{T}_{11}\mathcal{T}_{22}-F(\rho|q_{12}|)F(\rho|q_{21}|)\mathcal{T}_{12}\mathcal{T}_{21}\right],\qquad q_{ff'}=p_{R,f}-p_{L,f'},$$

provided B^{amp} is an independent bilinear source matrix. If $B^{\mathrm{amp}}_{ff'}=u_fv_{f'}$ is only a c-number outer product, the two-flavor determinant is zero. Consequently the center integral in infinite volume gives the distributional factor

$$\int_{\mathbb{R}^4}\mathrm{d}^4a\,\mathrm{e}^{\mathrm{i}\mathcal{P}\cdot a}=(2\pi)^4\delta^{(4)}(\mathcal{P}), \tag{51.153}$$

and the source-amputated one-instanton contribution may be written as

$$\begin{aligned}\mathcal{M}^{(4),\mathrm{amp}}_{\Lambda,Q=1} = \mathrm{e}^{\mathrm{i}\theta}(2\pi)^4\delta^{(4)}(\mathcal{P}) \int \exp[-S_\Lambda(A_{\rho,U})]\, \mathcal{W}^{\mathrm{nz}}_{\Lambda,\mathrm{pt}}(\rho,U;\mu) \\ \times \rho^6\, \mathfrak{A}^{\mathrm{lin}}_2(\rho,U;\{p_R,p_L\})\, \mathcal{J}_\Lambda(\rho,U)\,\mathrm{d}\rho\,\mathrm{d}\Omega(U).\end{aligned} \tag{51.154}$$

The anti-instanton term has the conjugate chirality structure and the phase $\mathrm{e}^{-\mathrm{i}\theta}$*.*

Thus the instanton calculation does not first create a local vertex and then decorate it. The finite-regulator amplitude is assembled from four load-bearing pieces: the center integral, the individual zero-mode Fourier-overlap tensors, the nonzero-mode determinant, and the size integral. The local 't Hooft operator is obtained only after the external slots have been fused into a local bilinear source, the support of the ρ*-integral lies in the small-momentum regime of the relevant zero-mode form factors, and the determinant remainder has been replaced by the finite constant of the chosen scheme.*

Proof. In $\det R^{\mathrm{amp}}$, both permutation terms contain the same phase $\mathrm{e}^{\mathrm{i}(p_{R,1}+p_{R,2})\cdot a}$. In $\det L^{\mathrm{amp}}$, both permutation terms contain the same phase $\mathrm{e}^{-\mathrm{i}(p_{L,1}+p_{L,2})\cdot a}$. The product therefore carries $\mathrm{e}^{\mathrm{i}\mathcal{P}\cdot a}$. The four individual zero-mode overlaps carry the conventional total power ρ^6, leaving $\det\mathcal{R}\,\det\mathcal{L}$ as the nontrivial form-factor and orientation coefficient. This proves (51.152). Translation invariance of the center coordinate then gives (51.153) in infinite volume, or the corresponding Kronecker delta on a finite periodic regulator. Substituting the factorized coefficient and collective-coordinate density into (51.149) gives (51.154). The bilinear-source formula follows from the same phase count applied to (51.151); the final rank-one warning is the elementary identity $\det(uv^T) = 0$ for two-component c-number vectors. □

Proposition 51.48 (Orientation projection in the instanton four-slot kernel)**.** *The global* $SU(2)$ *orientation in the embedded BPST core enters the amplitude as a projection of colored zero-mode tensors onto invariant color tensors. Normalize Haar measure by* $\int_{SU(2)}\mathrm{d}U = 1$*, and let* ${U_i}^\alpha$ *denote the fundamental matrix element carrying a core two-dimensional index* α *to a color index* i*. Then*

$$\int_{SU(2)} {U_i}^\alpha\,\mathrm{d}U = 0, \qquad \int_{SU(2)} {U_i}^\alpha\overline{{U_j}^\beta}\,\mathrm{d}U = \frac{1}{2}\delta_{ij}\delta^{\alpha\beta}, \tag{51.155}$$

and, because $\wedge^2\mathbb{C}^2$ *is the trivial representation of* $SU(2)$*,*

$$\int_{SU(2)} {U_i}^\alpha {U_j}^\beta\,\mathrm{d}U = \frac{1}{2}\varepsilon_{ij}\varepsilon^{\alpha\beta}. \tag{51.156}$$

Consequently two differentiated external zero-mode slots with core source vectors $v_A, v_B \in \mathbb{C}^2$ *have orientation average*

$$\int_{SU(2)} (Uv_A)_i(Uv_B)_j\,\mathrm{d}U = \frac{1}{2}\varepsilon_{ij}\,\varepsilon_{\alpha\beta}v_A^\alpha v_B^\beta = \frac{1}{2}\varepsilon_{ij}\det(v_A,v_B). \tag{51.157}$$

For an $N_f = 2$ *differentiated four-slot source, the four slots share the same orientation variable. One must therefore use the single four-point Haar projector, not the product of two two-point averages. When all four selected slots have been written as fundamental* $SU(2)$ *core indices, set*

$$A_{i_1i_2j_1j_2} = \varepsilon_{i_1i_2}\varepsilon_{j_1j_2}, \qquad B_{i_1i_2j_1j_2} = \varepsilon_{i_1j_1}\varepsilon_{i_2j_2},$$

and use the same symbols with upper core indices. Then

$$\begin{aligned}\int_{SU(2)} & U_{i_1}{}^{\alpha_1} U_{i_2}{}^{\alpha_2} U_{j_1}{}^{\beta_1} U_{j_2}{}^{\beta_2}\,\mathrm{d}U \\ &= \frac{1}{3} A_{i_1 i_2 j_1 j_2} A^{\alpha_1\alpha_2\beta_1\beta_2} - \frac{1}{6} A_{i_1 i_2 j_1 j_2} B^{\alpha_1\alpha_2\beta_1\beta_2} \\ &\quad - \frac{1}{6} B_{i_1 i_2 j_1 j_2} A^{\alpha_1\alpha_2\beta_1\beta_2} + \frac{1}{3} B_{i_1 i_2 j_1 j_2} B^{\alpha_1\alpha_2\beta_1\beta_2}.\end{aligned} \tag{51.158}$$

If the left slots are kept as conjugate or dual slots instead, they must first be converted to the chosen fundamental convention by the $SU(2)$ pseudoreal ε-tensor, or else the corresponding mixed $U\overline{U}$ Weingarten projector must be used. Contracting (51.158) with core source vectors $v_1^R, v_2^R, v_1^L, v_2^L$ gives

$$A_{i_1 i_2 j_1 j_2}\left[\frac{1}{3}\Delta_{12|34} - \frac{1}{6}\Delta_{13|24}\right] + B_{i_1 i_2 j_1 j_2}\left[-\frac{1}{6}\Delta_{12|34} + \frac{1}{3}\Delta_{13|24}\right], \tag{51.159}$$

where

$$\Delta_{12|34} = \varepsilon_{\alpha_1\alpha_2} v_1^{R,\alpha_1} v_2^{R,\alpha_2}\, \varepsilon_{\beta_1\beta_2} v_1^{L,\beta_1} v_2^{L,\beta_2}, \qquad \Delta_{13|24} = \varepsilon_{\alpha_1\beta_1} v_1^{R,\alpha_1} v_1^{L,\beta_1}\, \varepsilon_{\alpha_2\beta_2} v_2^{R,\alpha_2} v_2^{L,\beta_2}.$$

Thus the color-orientation collective coordinate is a genuine part of the amplitude rather than a passive coordinate volume: it enforces the full invariant four-slot projector. A product of right and left flavor determinants appears only after the Berezin source coefficient and the final color-singlet source contraction impose that additional structure. For an $SU(N_c)$ instanton embedded through an $SU(2)$ subgroup, the full orientation-orbit integral is the corresponding $SU(N_c)$-invariant projection with a convention dependent normalization; the displayed $SU(2)$ core identity is the local finite projection used by the zero-mode tensor.

Proof. There is no nonzero invariant vector in the fundamental representation, so the one-slot average vanishes. The mixed two-point average is an intertwiner $\mathbb{C}^2 \otimes (\mathbb{C}^2)^* \to \mathbb{C}^2 \otimes (\mathbb{C}^2)^*$, hence Schur orthogonality makes it a multiple of $\delta_{ij}\delta^{\alpha\beta}$. Contracting $i = j$ and $\alpha = \beta$ gives 2, so the multiple is $1/2$.

Similarly, the average of $U_i{}^{\alpha} U_j{}^{\beta}$ is an invariant tensor in $\wedge^2\mathbb{C}^2 \otimes \wedge^2(\mathbb{C}^2)^*$, and must be $c\,\varepsilon_{ij}\varepsilon^{\alpha\beta}$. Contracting with $\varepsilon^{ij}\varepsilon_{\alpha\beta}$ gives

$$4c = \int \varepsilon^{ij}\varepsilon_{\alpha\beta} U_i{}^{\alpha} U_j{}^{\beta}\,\mathrm{d}U = 2\int \det U\,\mathrm{d}U = 2,$$

so $c = 1/2$. Multiplying the averaged matrix elements by the two source vectors gives (51.157).

For four slots, the invariant subspace of $(\mathbb{C}^2)^{\otimes 4}$ is two-dimensional. Use the two spanning invariant tensors A and B above. Their Gram matrix is

$$\begin{pmatrix} \langle A, A\rangle & \langle A, B\rangle \\ \langle B, A\rangle & \langle B, B\rangle \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}, \qquad G^{-1} = \begin{pmatrix} \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{pmatrix}.$$

The Haar average is the orthogonal projection onto this invariant subspace, with one copy of G^{-1} connecting the color invariant basis to the core invariant basis. This gives (51.158). Contracting that projector with the four core source vectors gives (51.159). The calculation uses one shared Haar variable throughout; replacing it by a product of two two-slot averages would project two independent orientations and gives the wrong four-slot normalization. □

The qualification in Proposition 51.46 is essential. Colored quark legs in confining QCD do not define ordinary physical asymptotic states. The displayed object is an auxiliary source-amputated, or partonic hard-kernel, coordinate useful for matching the finite-regulator Green function to the local effective operator. Gauge-invariant amplitudes must contract the same instanton kernel with color-singlet external operators, hadronic matrix elements, or a separately specified infrared deformation. This is the amplitude-level content of the original one-instanton computation of 't Hooft [63]: collective coordinates, nonzero-mode fluctuation determinants, fermion zero-mode saturation, and external-leg normalization are one calculation.

Controlled Approximation 51.49 (Color-singlet matching of the instanton kernel)**.** Stay in the finite-regulator $Q = 1$ setup of Proposition 51.46, but replace the auxiliary colored external quark representatives by renormalized gauge-invariant source families. Local examples are chiral color-singlet bilinears $\bar{q}_{Rf}\Gamma q_{Lf'}$; nonlocal examples must include the Wilson line which makes the two quark fields a color singlet. In a selected $N_f = 2$ chirality-violating channel, suppose that the zero-mode projection of the chosen source family has the short-distance form

$$B^{\mathcal{J}}_{AB}(z) = \sum_{f,f'=1}^{2} \Phi^R_{Af}(z;\mu)\, B^{\mathrm{amp}}_{ff'}(z)\, \Phi^L_{f'B}(z;\mu) + B^{\mathrm{rem}}_{AB}(z), \tag{51.160}$$

where A, B label two independent color-singlet source coordinates, Φ^R, Φ^L are the source-to-zero-mode overlap matrices in the chosen renormalization scheme, and B^{rem} collects higher-dimension source components, soft channels not represented by the hard zero-mode kernel, and terms outside the selected chirality structure. The corresponding gauge-invariant source correlator receives the one-instanton contribution

$$G^{(4)}_{\mathcal{J},\Lambda,Q=1} = \int_{\mathcal{U}_{Q=1}} \exp[-S_\Lambda(A_z) + \mathrm{i}\theta]\, \mathcal{W}^{\mathrm{nz}}_{\Lambda,\mathrm{pt}}(z;\mu)\, \det B^{\mathcal{J}}(z)\, \mathrm{d}\nu_{\Lambda,\mathcal{M}}(z), \tag{51.161}$$

up to the same remainder specification. If the overlap matrices are square and vary only on scales which are soft compared with the hard instanton support, then pointwise in z

$$\det B^{\mathcal{J}}(z) = \det \Phi^R(z;\mu)\, \det B^{\mathrm{amp}}(z)\, \det \Phi^L(z;\mu) + \text{terms containing } B^{\mathrm{rem}}. \tag{51.162}$$

Thus the zero-mode form factors appropriate to the selected source family, the orientation tensor, the nonzero-mode determinant, and the size integral remain the short-distance instanton kernel; the color-singlet source overlaps are an additional matching layer, not a replacement for the fluctuation calculation.

For stable hadron external states, the physical amplitude is obtained only after applying the usual color-singlet spectral or LSZ extraction to the gauge-invariant correlator. For example, if interpolating sources $\mathcal{I}_H, \mathcal{I}_{H'}$ have nonzero isolated overlaps $Z_H, Z_{H'}$ with stable finite-volume states in the sense of Proposition 49.54, then a termwise semiclassical extraction requires the same pole-residue operation:

$$\mathcal{A}^{(Q=1)}_{H'\leftarrow H} = \left(\overline{Z}_{H'} Z_H\right)^{-1} \operatorname*{Res}_{H',H} G_{\mathcal{I}^\dagger_{H'}\,\mathcal{X}\,\mathcal{I}_H,\Lambda,Q=1}, \tag{51.163}$$

where $\mathcal{X}$ is the chosen gauge-invariant current or source product and the residue notation means the finite-volume large-time projection, or the corresponding infinite-volume pole extraction when it exists. The BPST calculation does not determine $Z_H, Z_{H'}$, the hadronic matrix element of the induced 't Hooft operator, or the long-distance part of $\Phi^{R,L}$. Those are nonperturbative QCD data. In a hard OPE or factorization regime, the small-ρ instanton calculation can supply a Wilson coefficient multiplying a gauge-invariant operator such as $\det[\bar{q}_R q_L]_\mu$; outside such a regime (51.161) is a source correlator, not a prediction for a hadronic amplitude by itself.

The finite-regulator bookkeeping behind (51.160) is entirely finite-dimensional. At finite regulator the fermion fields have finitely many odd coefficients, and every regulated source has a finite expansion in those coefficients. Restricting the source to the instanton zero-mode subspace produces a finite-dimensional polynomial in the zero-mode variables. The hypothesis (51.160) says that, in the chosen hard channel, the part of this polynomial which is bilinear in the right and left zero-mode variables factors through the same amputated zero-mode kernel B^{amp}, with source-dependent overlap maps on the two sides. Substituting this zero-mode source matrix into the Berezin coefficient in Proposition 51.35 gives $\det B^{\mathcal{J}}(z)$, and multiplying by the determinant density and collective-coordinate measure gives (51.161).

The determinant relation (51.162) is the finite identity

$$\det(\Phi^R B^{\mathrm{amp}} \Phi^L) = \det \Phi^R \, \det B^{\mathrm{amp}} \, \det \Phi^L,$$

applied pointwise in the collective coordinate. If the overlaps vary with z, the same identity still holds pointwise but the overlap factors must remain inside the collective-coordinate integral. Finally, because $G_{\mathcal{I}_{H'}^\dagger, \mathcal{X}\, \mathcal{I}_H, \Lambda, Q=1}$ is a correlator of gauge-invariant sources, the hadronic extraction is the standard spectral-residue operation for color-singlet correlators. Applying that operation termwise in a controlled semiclassical expansion gives (51.163); it never produces an ordinary colored-quark S-matrix element.

Controlled Approximation 51.50 (Euclidean instanton kernel to physical amplitude)**.** Let $G^{E,Q=1}_{\mathcal{J},\Lambda}$ be the finite-regulator gauge-invariant Euclidean source correlator obtained from the one-instanton chart after zero modes, nonzero-mode determinants, source projections, and collective coordinates have all been paired. Let

$$\mathcal{A}^{\mathrm{lead}}_{\mathrm{inst}} = \operatorname*{Proj}_{\mathrm{phys}} \operatorname*{Cont}_{\mathcal{S}} G^{E,Q=1}_{\mathcal{J},\mathrm{lead}}$$

denote the result of first replacing the finite-regulator density by the declared leading instanton kernel, then analytically continuing on a specified sheet $\mathcal{S}$, and finally applying the chosen physical projection: a stable-particle pole residue, an inclusive spectral weight, or a hard OPE coefficient. The corresponding physical one-instanton contribution $\mathcal{A}^{Q=1}_{\mathrm{phys}}$ differs from this coordinate by

$$\mathcal{A}^{Q=1}_{\mathrm{phys}} - \mathcal{A}^{\mathrm{lead}}_{\mathrm{inst}} = R_{\mathrm{reg}} + R_{\mathrm{cont}} + R_{\mathrm{spec}} + R_{\mathrm{IR}} + R_{\mathrm{cut}} + R_{\mathrm{match}} + R_\rho. \tag{51.164}$$

Here R_{reg} records finite-volume, cutoff, and order-of-limits errors between the regulated Euclidean sector and its continuum source correlator; R_{cont} records analytic-continuation and sheet choices, including thresholds and singularities crossed by the chosen continuation; R_{spec} records the pole, spectral, or OPE projection error; R_{IR} records confinement, massless radiation, inclusive-sum, or external Wilson-line data not contained in a colored partonic kernel; R_{cut} records unitarity-cut and conjugate-sector effects, including instanton–anti-instanton sectors needed before an S-matrix statement can obey an optical theorem; R_{match} records the matching from the chosen source family to the local operator or hadronic observable; and R_ρ is the remaining instanton-size endpoint and determinant-tail contribution after the selected hard, screened, or Wilsonian window has been declared.

If on the selected kinematic window each residual satisfies

$$|R_\kappa| \le \varepsilon_\kappa, \qquad \kappa \in \{\mathrm{reg}, \mathrm{cont}, \mathrm{spec}, \mathrm{IR}, \mathrm{cut}, \mathrm{match}, \rho\},$$

then

$$\left|\mathcal{A}^{Q=1}_{\mathrm{phys}} - \mathcal{A}^{\mathrm{lead}}_{\mathrm{inst}}\right| \le \sum_\kappa \varepsilon_\kappa. \tag{51.165}$$

The point is not to add another approximation. It is to identify where the physics lives after the moduli-space and determinant calculation has been done. In a weakly coupled Higgsed theory, a thermal screened regime, or a hard Euclidean OPE window, the physical projection and endpoint controls can sometimes be declared explicitly. In confining zero-temperature QCD, a colored four-slot kernel is only a short-distance source coordinate until the gauge-invariant spectral or OPE projection and the long-size remainder have been supplied.

Controlled Approximation 51.51 (Instanton cut pairing for inclusive rates)**.** Stay in the source channel of ControlledApproximation 51.50. After the determinant, zero-mode, orientation, and source-matching factors have been assembled, write a finite one-instanton amplitude coordinate as

$$A_I[\mathcal{J}] = \mathrm{e}^{\mathrm{i}\theta} \int_{\mathcal{U}_I} K_I(z;\mathcal{J})\, \mathrm{d}\nu_I(z), \tag{51.166}$$

where z denotes the collective coordinates and finite fluctuation/source data retained in the chosen regulator. The conjugate channel has an anti-instanton density

$$A_{\bar I}[\mathcal{J}^\dagger] = \mathrm{e}^{-\mathrm{i}\theta} \int_{\mathcal{U}_{\bar I}} K_{\bar I}(\bar z;\mathcal{J}^\dagger)\, \mathrm{d}\nu_{\bar I}(\bar z).$$

A leading inclusive spectral weight or cut rate associated to this amplitude has the paired form

$$\Gamma^{(0)}_{I\bar I}[\mathcal{J}] = \int_{\mathcal{U}_I\times\mathcal{U}_{\bar I}} K_I(z;\mathcal{J})\, K_{\bar I}(\bar z;\mathcal{J}^\dagger)\, \mathcal{C}^{(0)}_{\mathrm{cut}}(z,\bar z)\, \mathrm{d}\nu_I(z)\, \mathrm{d}\nu_{\bar I}(\bar z). \tag{51.167}$$

Here $\mathcal{C}^{(0)}_{\mathrm{cut}}$ contains the chosen leading final-state sum, on-shell or spectral projector, relative $I\bar I$ separation and orientation weight, and leading relative fluctuation determinant. At the factorized level the topological phases cancel in the cut channel,

$$\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta} = 1, \qquad \mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{\mathrm{i}\theta} = \mathrm{e}^{2\mathrm{i}\theta}.$$

Thus a same-charge pair belongs to a charge-two amplitude sector rather than to the amplitude-times-conjugate contribution to a neutral inclusive rate. The semiclassical exponential is correspondingly e^{-2S_I} at this factorized cut level.

The full cut kernel can be organized as

$$\mathcal{C}_{\mathrm{cut}} = \mathcal{C}^{(0)}_{\mathrm{cut}}(1+\delta_{\mathrm{cut}}), \qquad |\delta_{\mathrm{cut}}(z,\bar z)| \le \varepsilon_{\mathrm{cut}},$$

where δ_{cut} absorbs the difference between the declared leading cut package and the actual final-state sum, relative $I\bar I$ interaction, valley or quasi-zero-mode treatment, and nonzero-mode determinant ratio. If

$$M^{(0)}_{I\bar I} = \int |K_I(z;\mathcal{J})|\, |K_{\bar I}(\bar z;\mathcal{J}^\dagger)|\, |\mathcal{C}^{(0)}_{\mathrm{cut}}(z,\bar z)|\, \mathrm{d}\nu_I(z)\, \mathrm{d}\nu_{\bar I}(\bar z) < \infty,$$

then the exact cut contribution satisfies

$$\left|\Gamma_{I\bar{I}}[\mathcal{J}] - \Gamma^{(0)}_{I\bar{I}}[\mathcal{J}]\right| \le \varepsilon_{\text{cut}} M^{(0)}_{I\bar{I}}. \tag{51.168}$$

This is the concrete finite-regulator content of the R_{cut} entry in (51.164). The local 't Hooft vertex supplies the chirality-violating amplitude kernel. Inclusive cross sections, decay rates, and spectral weights require the conjugate anti-instanton density, final-state cut, and relative $I\bar{I}$ fluctuation determinants and interactions in addition to that kernel.

Proposition 51.52 (One-instanton amplitude assembly and error budget)**.** *Fix a finite-regulator $Q = 1$ chart and a gauge-invariant source channel. Let*

$$\mathcal{A}^{\mathcal{J}}_{\Lambda,Q=1} = \int_{\mathcal{U}_{Q=1}} \mathcal{K}_{\text{ex}}(z)\,\mathrm{d}z \tag{51.169}$$

denote the exact one-instanton contribution to the selected source correlator after gauge fixing but before replacing any finite factor by its semiclassical leading term. Choose a controlled subdomain $\mathcal{U}_{\text{hard}} \subset \mathcal{U}_{Q=1}$ and a leading density

$$\mathcal{K}_{\text{lead}}(z) = \mathrm{e}^{\mathrm{i}\theta}\mathcal{C}^{\mathcal{S}}_{\text{pg}}(z)\,\mathcal{R}^{\mathcal{S}}_{\text{f}}(z;\mu)\left(\frac{8\pi^2}{g^2_{\text{ht}}(\mu)}\right)^{2N_c}\exp\left[-\frac{8\pi^2}{g^2_{\text{ht}}(\mu)}\right](\mu\rho)^{b_0}\rho^{-5}\,\mathcal{Z}^0_{\text{lead}}(z). \tag{51.170}$$

Here $\mathcal{C}^{\mathcal{S}}_{\text{pg}}(z)$ is the pure-gauge collective-coordinate and determinant density in the chosen convention. In the Pauli-Villars convention after the embedded orientation orbit has been integrated it contains $C^{\text{PV}}_{N_c}$; in an unintegrated orientation chart it contains the corresponding orientation density instead of an already-integrated orbit volume. The factor $\mathcal{R}^{\mathcal{S}}_{\text{f}}$ is the light-fermion nonzero-mode determinant factor in the same source scheme, and $\mathcal{Z}^0_{\text{lead}}$ is the leading zero-mode source coefficient. Depending on the source type, $\mathcal{Z}^0_{\text{lead}}$ is one of $\det(M^0 + J^0)$, $\det R^{\text{amp}} \det L^{\text{amp}}$, or the color-singlet determinant $\det B^{\mathcal{J}}$ after the source matching of (51.160).

Suppose that on $\mathcal{U}_{\text{hard}}$ the exact density decomposes as

$$\mathcal{K}_{\text{ex}} = \mathcal{K}_{\text{lead}}\big(1 + \varepsilon_{\text{det}} + \varepsilon_{\text{zm}} + \varepsilon_{\text{src}}\big) + \Delta_{\text{Schur}}, \tag{51.171}$$

where ε_{det} records the higher-loop and finite determinant remainder after the chosen determinant convention, ε_{zm} records the zero-mode projection or local-source truncation error, ε_{src} records the color-singlet source-matching or hadronic/OPE matching remainder, and Δ_{Schur} records the finite nonzero-mode Schur correction from (51.120). Then

$$\mathcal{A}^{\mathcal{J}}_{\Lambda,Q=1} = \mathcal{A}^{\mathcal{J}}_{\text{lead}} + R_{\text{det}} + R_{\text{zm}} + R_{\text{src}} + R_{\text{Schur}} + R_{\text{end}}, \tag{51.172}$$

with

$$\mathcal{A}^{\mathcal{J}}_{\text{lead}} = \int_{\mathcal{U}_{\text{hard}}} \mathcal{K}_{\text{lead}}(z)\,\mathrm{d}z, \qquad R_{\text{end}} = \int_{\mathcal{U}_{Q=1}\setminus\mathcal{U}_{\text{hard}}} \mathcal{K}_{\text{ex}}(z)\,\mathrm{d}z.$$

Moreover, whenever the right-hand side is absolutely integrable,

$$\begin{aligned}\big|\mathcal{A}^{\mathcal{J}}_{\Lambda,Q=1} - \mathcal{A}^{\mathcal{J}}_{\text{lead}}\big| \le{}& \int_{\mathcal{U}_{\text{hard}}} |\mathcal{K}_{\text{lead}}|\big(|\varepsilon_{\text{det}}| + |\varepsilon_{\text{zm}}| + |\varepsilon_{\text{src}}|\big)\,\mathrm{d}z \\ &+ \int_{\mathcal{U}_{\text{hard}}} |\Delta_{\text{Schur}}|\,\mathrm{d}z + \int_{\mathcal{U}_{Q=1}\setminus\mathcal{U}_{\text{hard}}} |\mathcal{K}_{\text{ex}}|\,\mathrm{d}z.\end{aligned} \tag{51.173}$$

Thus replacing the finite-regulator instanton contribution by the local 't Hooft operator, or by a hard source-amputated kernel, is justified only after the determinant remainder, zero-mode/source projection, matching remainder, Schur correction, and endpoint integral have all been controlled in the same scheme.

Proof. Split the exact integral into the chosen hard/source domain and its complement:

$$\int_{\mathcal{U}_{Q=1}} \mathcal{K}_{\text{ex}} = \int_{\mathcal{U}_{\text{hard}}} \mathcal{K}_{\text{ex}} + \int_{\mathcal{U}_{Q=1}\setminus\mathcal{U}_{\text{hard}}} \mathcal{K}_{\text{ex}}.$$

Substituting (51.171) on $\mathcal{U}_{\text{hard}}$ gives the five remainders in (51.172). The bound (51.173) is the triangle inequality. The content is the identification of the separate physical origins of the five terms: collective-coordinate geometry supplies only $\mathcal{J}_{\text{coll}}$ or, equivalently, the collective part of $\mathcal{C}^{\mathcal{S}}_{\text{pg}}$ and the integration domain; the fluctuation determinant, light-fermion nonzero-mode factor, zero-mode source coefficient, source matching, and endpoint behavior are independent load-bearing inputs to the amplitude. □

Consequently an instanton moduli-space measure by itself is never the computed QFT amplitude. It becomes a physical contribution only after it is multiplied by the nonzero-mode fluctuation determinant, the correctly typed zero-mode source coefficient, the source or hadron matching data, and an endpoint estimate for the selected kinematic regime. The original one-instanton calculation is deep precisely because it performs this whole assembly, rather than because it merely parametrizes the BPST collective coordinates.

Controlled Approximation 51.53 (Hard-momentum size window)**.** Consider the $N_f = 2$ source-amputated four-fermion kernel (51.149) in a kinematic region where the hard arguments of the zero-mode form factors are

$$|k_\ell| = c_\ell Q, \qquad c_\ell > 0, \qquad \ell = 1, \dots, m,$$

with Q much larger than the infrared scale of the theory. For differentiated external fermion slots, $m = 4$ and the $\mathcal{F}_\ell$ below are the individual amputated zero-mode form factors of Proposition 51.45. In the singular-gauge normalization, $\mathcal{F}_\ell(c_\ell Q\rho) = F_{\text{zm}}(c_\ell Q\rho/2)$, up to the fixed spin-color tensor already included in $\mathcal{C}_{\text{hard}}$. For a fused bilinear source, $m = 2$ and $\mathcal{F}_\ell(z) = zK_1(z)$. Keep only the one-loop universal density, the total four-zero-mode power ρ^6, and the dimensionless hard form-factor product $\prod_{\ell=1}^m \mathcal{F}_\ell(c_\ell Q\rho)$. Choosing $\mu = Q$, the corresponding size factor has the form

$$\mathcal{I}_{\text{hard}}(Q) = \exp\left[-\frac{8\pi^2}{g^2_{\text{ht}}(Q)} + \mathrm{i}\theta\right] \left(\frac{8\pi^2}{g^2_{\text{ht}}(Q)}\right)^{2N_c} Q^{b_0} \int_0^\infty \rho^{\,b_0+1} \prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell Q\rho)\,\mathrm{d}\rho \times \mathcal{C}_{\text{hard}}, \tag{51.174}$$

up to finite determinant remainders, orientation tensors, source normalizations, and higher-loop matching factors. With $s = Q\rho$,

$$Q^{b_0} \int_0^\infty \rho^{\,b_0+1} \prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell Q\rho)\,\mathrm{d}\rho = Q^{-2} \int_0^\infty s^{\,b_0+1} \prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell s)\,\mathrm{d}s. \tag{51.175}$$

This Q^{-2} is the engineering dimension of the source-amputated four-fermion coefficient after the renormalization point has been set to $\mu = Q$. If the one-loop running exponential is instead traded for the RG-invariant scale

$$\Lambda_{\text{ht}}^{b_0} = Q^{b_0} \exp\left[-\frac{8\pi^2}{g^2_{\text{ht}}(Q)}\right] \quad \text{at one-loop order,}$$

then the same hard factor is

$$\mathcal{I}_{\mathrm{hard}}(Q) = \mathrm{e}^{\mathrm{i}\theta} \left(\frac{8\pi^2}{g_{\mathrm{ht}}^2(Q)} \right)^{2N_c} \Lambda_{\mathrm{ht}}^{b_0} \, Q^{-b_0-2} \, \mathcal{J}_{b_0}(\mathbf{c};\mathcal{F}) \, \mathcal{C}_{\mathrm{hard}}, \tag{51.176}$$

with

$$\mathcal{J}_{b_0}(\mathbf{c};\mathcal{F}) = \int_0^\infty s^{\,b_0+1} \prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell s) \, \mathrm{d}s.$$

Thus the semiclassical exponential and the size measure together become a power-suppressed hard coefficient. The remaining displayed $(8\pi^2/g_{\mathrm{ht}}^2(Q))^{2N_c}$ is the familiar one-loop collective-mode Jacobian prefactor; higher-loop matching changes the logarithmic factor and scheme coordinate, not the endpoint mechanism described here. The small-ρ endpoint is finite in asymptotically free QCD because the integrand behaves as ρ^{b_0+1}. The large-ρ endpoint depends on which source construction is being used. For fused bilinear density entries, $zK_1(z)$ gives the exponential cutoff

$$\prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell Q\rho) = O((Q\rho)^\gamma \exp[-d(\mathbf{c})Q\rho]) \,, \qquad d(\mathbf{c}) > 0, \qquad Q\rho \to \infty.$$

For differentiated singular-gauge fermion slots, however, (51.146) gives only the power behavior

$$\prod_{\ell=1}^{m} F_{\mathrm{zm}}(c_\ell Q\rho/2) = O\big((Q\rho)^{-3m}\big) \qquad (c_\ell > 0).$$

The large-ρ part of the size integral is then controlled only if

$$b_0 + 1 - 3m < -1. \tag{51.177}$$

For the $N_f = 2$ four-slot hard kernel, $m = 4$, so this condition is $b_0 < 10$; in $SU(3)$ QCD with two Dirac fundamentals, $b_0 = 29/3$, and the margin is $1/3$. If a determinant monomial contains a soft slot with $c_\ell = 0$, or if the power test fails, the hard source profile has not by itself produced a controlled large-size instanton amplitude. This is a physical endpoint condition in the amplitude integral, not a cutoff on the instanton moduli space. The approximation is controlled only when the dominant range also has $1/\rho$ in the weak-coupling regime.

Controlled Approximation 51.54 (Hard-size dominance versus endpoint convergence)**.** The convergence test in (51.177) is not yet a dominance statement. In the notation of (51.174), set

$$\mathcal{F}_{\mathrm{hard}}(s) = \prod_{\ell=1}^{m} \mathcal{F}_\ell(c_\ell s), \qquad a = b_0 + 1.$$

After the change of variables $s = Q\rho$, the absolute tail of the one-loop hard size integral outside the band $\rho \le R/Q$ is

$$Q^{b_0} \int_{R/Q}^{\infty} \rho^{\,b_0+1} \, |\mathcal{F}_{\mathrm{hard}}(Q\rho)| \, \mathrm{d}\rho = Q^{-2} \int_R^\infty s^{\,a} \, |\mathcal{F}_{\mathrm{hard}}(s)| \, \mathrm{d}s. \tag{51.178}$$

Thus a statement that the amplitude is dominated by instantons of size $\rho \sim Q^{-1}$ requires a bound on the right-hand tail in addition to the formal local vertex.

For example, suppose that for $s \geq s_0$

$$|\mathcal{F}_{\mathrm{hard}}(s)| \leq C_\infty s^{-\sigma}, \qquad \sigma > b_0 + 2.$$

Then for $R \geq s_0$,

$$Q^{b_0} \int_{R/Q}^{\infty} \rho^{\,b_0+1} \, |\mathcal{F}_{\mathrm{hard}}(Q\rho)| \, \mathrm{d}\rho \leq Q^{-2} \, \frac{C_\infty \, R^{\,b_0+2-\sigma}}{\sigma - b_0 - 2}. \tag{51.179}$$

If instead the fused bilinear-density source gives $|\mathcal{F}_{\mathrm{hard}}(s)| \leq C_\infty s^\gamma e^{-ds}$, $d > 0$, then the corresponding tail is bounded by

$$Q^{-2} C_\infty \int_R^\infty s^{\,b_0+1+\gamma} e^{-ds} \, \mathrm{d}s,$$

and is exponentially small once R is large. The two source constructions therefore have different endpoint control.

For the differentiated singular-gauge four-slot $SU(3)$, $N_f = 2$ hard kernel, the individual zero-mode form factors have $\sigma = 3m = 12$, while $b_0 = 29/3$. Hence

$$b_0 + 2 - \sigma = -\frac{1}{3}.$$

The contribution from $\rho > R/Q$ is only $O(R^{-1/3})$, up to the spin-color and c_ℓ-dependent constants. The size integral is convergent, but the convergence margin is narrow: claiming that the one-instanton hard kernel is numerically dominated by $\rho \sim Q^{-1}$ requires an additional estimate showing that this slowly decaying tail is small in the chosen source and kinematic window. This is a physical endpoint-control hypothesis on the amplitude, not a property of the BPST moduli coordinates alone.

The same statement is often clearer per logarithmic size shell. Define the absolute log-shell density after the hard change of variables by

$$\mathfrak{s}_{\mathrm{hard}}(s) = s^{\,b_0+2} \, |\mathcal{F}_{\mathrm{hard}}(s)|, \qquad Q^{b_0} \rho^{\,b_0+1} |\mathcal{F}_{\mathrm{hard}}(Q\rho)| \, \mathrm{d}\rho = Q^{-2} \mathfrak{s}_{\mathrm{hard}}(s) \, \mathrm{d} \log s. \tag{51.180}$$

If $|\mathcal{F}_{\mathrm{hard}}(s)| \leq C_\infty s^{-\sigma}$, the shell and the tail both decay with the exponent

$$\Delta = \sigma - b_0 - 2.$$

For the $SU(3)$, $N_f = 2$ differentiated four-slot kernel, $\Delta = 1/3$. One extra decade in the cutoff R then suppresses the tail by only $10^{-1/3}$, and a tenfold suppression of this power-tail majorant requires three decades in R. With the normalized bound used in (51.179), the tail estimate is

$$3R^{-1/3};$$

making this smaller than 0.1 requires $R > 27000$. This number is not a prediction of the amplitude. It is a warning about evidence: a formally hard one-instanton coefficient is not physically dominated by small instantons until the chosen source profile, kinematics, or matching calculation improves this log-shell bound.

Controlled Approximation 51.55 (Hard source envelope with physical screening)**.** The hard and screened size estimates above address different failures of the same amplitude integral. A hard source can suppress large instantons through external form factors, while a physical Higgs, thermal, or medium scale can screen them through the fluctuation determinant. When both mechanisms are present, the dominant size is not obtained by choosing whichever formula is more convenient. It is fixed by the combined size density.

Assume that in the selected one-instanton channel the absolute value of the fully assembled density is bounded in the retained window by

$$|\mathcal{K}(\rho)| \le K_0\, \rho^{A-1} \exp[-dQ\rho - m_{\rm scr}^2\rho^2], \qquad A > 0, \quad d > 0, \quad m_{\rm scr} > 0. \tag{51.181}$$

Here $A = b_0 + \beta_{\mathcal{X}} - 4$ contains the collective density, determinant convention, and zero-mode/source power of the chosen amplitude channel. The coefficient dQ is a proven exponential envelope for the hard form factors or source wave packets; it is absent for the differentiated individual zero-mode slots unless such a stronger source bound has actually been supplied. The scale $m_{\rm scr}$ is a physical screening mass from the same regulated problem, not a hand-inserted cutoff.

The absolute logarithmic size shell of the majorant is

$$\rho^A \exp[-dQ\rho - m_{\rm scr}^2\rho^2]\, \mathrm{d}\log\rho.$$

Its stationary point solves

$$2m_{\rm scr}^2\rho_*^2 + dQ\rho_* - A = 0, \qquad \rho_* = \frac{2A}{dQ + \sqrt{d^2Q^2 + 8Am_{\rm scr}^2}}. \tag{51.182}$$

Thus the hard-only shell $A/(dQ)$ and the screened-only shell $\sqrt{A/2}/m_{\rm scr}$ are limiting coordinates of one combined problem:

$$\frac{\rho_*}{A/(dQ)} = \frac{2}{1 + \sqrt{1 + 8Am_{\rm scr}^2/(d^2Q^2)}}.$$

The two limiting diagnostics are

$$\epsilon_{\rm hard} = \frac{Am_{\rm scr}}{dQ}, \qquad \epsilon_{\rm scr} = \frac{dQ}{m_{\rm scr}}\sqrt{\frac{A}{2}}.$$

If $\epsilon_{\rm hard} \ll 1$, the hard envelope sets the shell and screening is a perturbation. If $\epsilon_{\rm scr} \ll 1$, the physical screening scale sets the shell and the hard source is only a slowly varying factor. In the intermediate regime the two suppressions cooperate and the dominant size is smaller than either pure estimate.

This criterion is deliberately stronger than convergence. A hard instanton coefficient is a controlled short-distance object only if the combined shell and a surrounding tail bound lie inside a weak-coupling region,

$$\Lambda_{\rm QCD}\rho_* \ll 1$$

or inside the corresponding Higgs/thermal weak-coupling window. If ρ_* is instead set by a soft screening scale comparable to the confinement scale, the formula is at best an absolute majorant on a semiclassical coordinate; it is not a derivation of a physical QCD amplitude. This is exactly the point hidden by a moduli-space-only discussion: the collective coordinate supplies the integration variable, but the determinant, zero-mode form factors, physical screening, and endpoint estimate decide whether the instanton sector contributes a controlled observable coefficient.

Controlled Approximation 51.56 ($SU(3)$, two-flavor hard instanton coefficient)**.** Specialize the source-amputated four-slot construction to vectorlike $SU(3)$ QCD with $N_f = 2$ Dirac fundamentals. Use the Pauli–Villars pure-gauge convention of Proposition 51.32, choose a finite normalization of the light-fermion nonzero-mode determinant in the same scheme, and suppose all four differentiated external zero-mode slots are hard:

$$|k_\ell| = c_\ell Q, \qquad c_\ell > 0, \qquad \ell = 1, \ldots, 4.$$

Let $\mathcal{P}_{\mathrm{orient}}$ denote the shared $SU(2)$ orientation projector of (51.158), contracted with the chosen spin-color source covectors. In the leading one-instanton hard window, the short coefficient multiplying the selected four-fermion source tensor has the form

$$\begin{aligned} C^{(4)}_{\mathrm{inst}}(Q;R) = \mathrm{e}^{\mathrm{i}\theta}\, C_3^{\mathrm{PV}}\, \mathcal{R}_{\mathrm{f}}^{\mathrm{PV}}(Q) \left(\frac{8\pi^2}{g^2_{\mathrm{ht}}(Q)}\right)^6 \Lambda_{\mathrm{ht}}^{29/3}\, Q^{-35/3} \\ \times \mathcal{P}_{\mathrm{orient}} \int_0^R s^{32/3} \prod_{\ell=1}^4 F_{\mathrm{zm}}(c_\ell s/2)\,\mathrm{d}s + R_{\mathrm{det}} + R_{\mathrm{zm}} + R_{\mathrm{src}} + R_{\mathrm{Schur}} + R_{s>R}. \end{aligned} \tag{51.183}$$

Here $C_3^{\mathrm{PV}} \simeq 1.51 \times 10^{-3}$ is the orientation-integrated pure-gauge constant in (51.94), while $\mathcal{R}_{\mathrm{f}}^{\mathrm{PV}}$ is not fixed by the pure-gauge determinant calculation; it is the finite light-fermion nonzero-mode determinant factor of the same regulator and source normalization, with the finite-frame covariance fixed by Proposition 51.41. The five remainders are the determinant, zero-mode/source, source-matching, Schur, and endpoint residuals of Proposition 51.52.

The powers in (51.183) are fixed as follows. For $SU(3)$ with two Dirac fundamentals,

$$b_0 = \frac{11}{3}3 - \frac{2}{3}2 = \frac{29}{3}.$$

The four differentiated zero-mode slots supply ρ^6, so the size integrand before form factors is ρ^{b_0+1}. After $s = Q\rho$ and the one-loop trade $\exp[-8\pi^2/g^2_{\mathrm{ht}}(Q)]Q^{b_0} = \Lambda_{\mathrm{ht}}^{b_0}$, the hard coefficient carries

$$\Lambda_{\mathrm{ht}}^{29/3} Q^{-29/3-2} = \Lambda_{\mathrm{ht}}^{29/3} Q^{-35/3}.$$

Thus the dimension of the coefficient is -2, as required for a source-amputated four-fermion operator. This dimension count is not a standalone amplitude: the orientation projector, the light-fermion determinant, the zero-mode form factors, and the endpoint residual are still part of the calculation.

The endpoint term is especially restrictive. From (51.146),

$$F_{\mathrm{zm}}(c_\ell s/2) = 6 c_\ell^{-3} s^{-3} + O(s^{-5}), \qquad s \to \infty.$$

Therefore

$$\int_R^\infty s^{32/3} \prod_{\ell=1}^4 F_{\mathrm{zm}}(c_\ell s/2)\,\mathrm{d}s = 3 \cdot 6^4 \left(\prod_{\ell=1}^4 c_\ell^{-3}\right) R^{-1/3} + O(R^{-7/3}). \tag{51.184}$$

So even this hard four-slot coefficient has only a slow $R^{-1/3}$ endpoint tail unless the external source construction supplies stronger suppression. This is the amplitude-level warning in 't Hooft's calculation: the BPST moduli measure, the determinant constant, and the zero-mode tensor do not yet give a controlled physical four-point contribution until the hard-source tail and the remaining remainders have been bounded in the same scheme.

Controlled Approximation 51.57 (Instanton size-factor split)**.** Let $K_\Lambda(\rho;\mu,\mathcal{J})$ denote the fully paired finite-regulator one-instanton size integrand for a selected source family: the center phase, orientation tensor, nonzero-mode determinant, zero-mode source coefficient, and any anomaly-line pairing have already been included. Fix a finite upper regulator $\rho_{\max}$ and a factorization scale μ_I with

$$0 < \rho_I := \mu_I^{-1} < \rho_{\max}.$$

Then the exact finite-regulator contribution on this size window has the split

$$\mathcal{A}_\Lambda^{[0,\rho_{\max}]} = \mathcal{A}_\Lambda^{<}(\mu_I) + \mathcal{A}_\Lambda^{>}(\mu_I), \qquad \mathcal{A}_\Lambda^{<}(\mu_I) = \int_0^{\rho_I} K_\Lambda(\rho)\,\mathrm{d}\rho, \quad \mathcal{A}_\Lambda^{>}(\mu_I) = \int_{\rho_I}^{\rho_{\max}} K_\Lambda(\rho)\,\mathrm{d}\rho. \tag{51.185}$$

If K_Λ is continuous at ρ_I, the factorization-scale derivatives are the boundary flux through the artificial cut:

$$\partial_{\log\mu_I}\mathcal{A}_\Lambda^{<}(\mu_I) = -\rho_I K_\Lambda(\rho_I), \qquad \partial_{\log\mu_I}\mathcal{A}_\Lambda^{>}(\mu_I) = +\rho_I K_\Lambda(\rho_I). \tag{51.186}$$

Thus $\mathcal{A}_\Lambda^{[0,\rho_{\max}]}$ is independent of μ_I, while the short-size piece alone is not.

The semiclassical local 't Hooft vertex is therefore a Wilsonian object. If one replaces K_Λ by the one-loop small-size approximation $K_{\mathrm{sc}}^{<}$ only for $0 < \rho < \rho_I$, then

$$C_{\mathrm{inst}}^{<}(\mu_I;\mathcal{J}) = \int_0^{\rho_I} K_{\mathrm{sc}}^{<}(\rho;\mu,\mathcal{J})\,\mathrm{d}\rho$$

is a short-instanton coefficient. Its $-\rho_I K_{\mathrm{sc}}^{<}(\rho_I)$ scale derivative is cancelled only by the long-distance matrix element or remainder assigned to $\rho > \rho_I$, up to the matching error between K_Λ and $K_{\mathrm{sc}}^{<}$ near the cut. Discarding that long part leaves a spurious μ_I-dependence whose size is precisely the boundary density. For a hard external scale Q, choosing $\rho_I = R/Q$ identifies this boundary term with the same large-size tail diagnostic in (51.178). The computed short-instanton coefficient is useful only when this boundary flux and the remaining $\rho > \rho_I$ contribution are controlled by the physical source, screening scale, phase, or matching calculation under consideration.

Controlled Approximation 51.58 (Short-instanton OPE coefficient)**.** The short side of ControlledApproximation 51.57 becomes a local 't Hooft vertex only after it is projected onto a renormalized local operator basis. Let $\{[O_I]_\mu\}_{I\in\mathcal{B}}$ be a finite source-normalized basis in the selected quantum-number channel, containing $[\det(\bar{q}_R q_L)]_\mu$ and any operators of the same dimension and symmetry which are retained by the chosen hard OPE. Suppose the short-size part of a gauge-invariant source correlator admits the expansion

$$G_{\mathcal{J},\Lambda}^{<}(\mu_I) = \sum_{I\in\mathcal{B}} C_I^{<}(\mu_I,\mu)\ \langle \mathcal{J}\,[O_I]_\mu\rangle + R_{\mathrm{OPE}}^{<}(\mu_I,\mu). \tag{51.187}$$

The coefficient $C_I^{<}$ is obtained by integrating the small-size instanton density, the nonzero-mode determinant, the zero-mode source coefficient, and the orientation tensor against the projector onto $[O_I]_\mu$, over $0 < \rho < \rho_I = \mu_I^{-1}$. For the leading $N_f = 2$ determinant channel, this is the coefficient of $[\det(\bar{q}_R q_L)]_\mu$; derivative operators, mass insertions, and source-matching remnants belong either to the retained basis or to $R_{\mathrm{OPE}}^{<}$.

This projection has two different scale laws. The renormalization scale μ is the composite-operator scale. In the convention of Chapter 34,

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu}[O_I]_\mu = -\gamma_{IK}(\mu)[O_K]_\mu.$$

Therefore the coefficient row must obey

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu} C_K^{<}(\mu_I,\mu) = \sum_{I\in\mathcal{B}} C_I^{<}(\mu_I,\mu)\,\gamma_{IK}(\mu) \tag{51.188}$$

up to the truncation error $R_{\mathrm{OPE}}^{<}$. A finite operator-frame change $[\widetilde{O}_A]_\mu = M_{AI}(\mu)[O_I]_\mu$ changes the displayed coefficient by the inverse rule

$$\widetilde{C}_A^{<} = C_I^{<}(M^{-1})_{IA}, \qquad \sum_A \widetilde{C}_A^{<}[\widetilde{O}_A]_\mu = \sum_I C_I^{<}[O_I]_\mu. \tag{51.189}$$

Thus the local 't Hooft vertex coefficient is not a scheme-independent number; only its pairing with the renormalized operator basis and matrix elements is.

The factorization scale μ_I is different. If $K_I^{\mathrm{OPE}}(\rho;\mu)$ is the $[O_I]_\mu$-projected short-size integrand, then

$$\partial_{\log\mu_I} C_I^{<}(\mu_I,\mu) = -\rho_I K_I^{\mathrm{OPE}}(\rho_I;\mu). \tag{51.190}$$

This artificial boundary flow is not cancelled by operator anomalous dimensions. It is cancelled by the long-size matrix element or remainder, as in (51.186). Consequently a standalone local 't Hooft operator with no declared μ_I, no operator scheme, and no long-distance remainder is not a physical instanton amplitude.

This hard-size window is the amplitude counterpart of the local 't Hooft vertex. The local operator records the $Q\rho \ll 1$ tensor and selection rule, while (51.174) records how the same zero modes and fluctuation determinant enter an actual hard correlator. It also explains why the mass-saturated vacuum activity below is a different object: a vacuum theta term has no external form factor to exclude $\rho \to \infty$, so its large-size behavior is an infrared question, not a semiclassical prediction.

Thus the $Q = 1$ amplitude is the microscopic origin of the $U(1)_A$ selection rule. The determinant in (51.105) carries axial charge $2N_f$; equivalently, a mass-saturated instanton contribution depends on the CP-odd combination $\bar{\theta} = \theta + \arg\det M$ from (51.49). This gives the same anomalous Jacobian in the coordinate system where the fermion zero modes of the instanton background have been integrated.

Controlled Approximation 51.59 (Mass-saturated vacuum activity and the size integral)**.** Assume vectorlike $SU(N_c)$ QCD with N_f Dirac fundamentals, a diagonal renormalized mass matrix $M(\mu)$, and a one-instanton size window $\rho_- < \rho < \rho_+$ in which the one-loop density and the leading zero-mode mass overlaps are valid. In the same local normalization as (51.105), saturating one Dirac zero-mode pair by the mass term supplies the dimensionless factor $m_f(\mu)\rho$. Thus the mass-saturated one-instanton vacuum activity in this window has the form

$$\zeta_m^{[\rho_-,\rho_+]}(\mu) = \mathcal{C}_{\mathcal{S},N_c,N_f}\left(\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)}\right)^{2N_c} \exp\left[-\frac{8\pi^2}{g_{\mathrm{ht}}^2(\mu)}\right] |\det M(\mu)|\,\mu^{b_0} \int_{\rho_-}^{\rho_+} \rho^{\,b_0+N_f-5}\,\mathrm{d}\rho, \tag{51.191}$$

up to higher-loop/operator-renormalization factors of the chosen scheme. The constant $\mathcal{C}_{\mathcal{S},N_c,N_f}$ contains the pure-gauge orientation and determinant convention together with the light-fermion nonzero-mode factor $\mathcal{R}_{\mathrm{f}}^{\mathcal{S}}$ of Proposition 51.40, evaluated in the same small-mass window. It does not include the displayed $\prod_f(m_f\rho)$ factor: that product is the zero-mode saturation determinant. The corresponding $Q = 1$ vacuum term carries the phase

$$V_4\zeta_m^{[\rho_-,\rho_+]}\exp[\mathrm{i}(\theta + \arg\det M)]\,, \tag{51.192}$$

while the anti-instanton term carries the complex conjugate phase.

The displayed size integral isolates the two endpoint questions:

$$\int_{\rho_-}^{\rho_+} \rho^{\,b_0+N_f-5}\,\mathrm{d}\rho = \begin{cases} \dfrac{\rho_+^{\,b_0+N_f-4} - \rho_-^{\,b_0+N_f-4}}{b_0+N_f-4}, & b_0+N_f \neq 4, \\ \log(\rho_+/\rho_-), & b_0+N_f = 4. \end{cases} \tag{51.193}$$

The small-size endpoint is locally finite precisely when $b_0 + N_f > 4$. For $SU(3)$, this margin is

$$b_0 + N_f - 4 = 7 + \frac{N_f}{3},$$

so the mass-saturated one-instanton vacuum term is not obstructed at $\rho = 0$ in the asymptotically free range. The same positive power means that extending the one-loop formula to $\rho_+ = \infty$ would give an infrared power divergence. That divergence is not a new ultraviolet counterterm; it says that the zero-temperature large-size region is outside the controlled weak-coupling instanton calculation. A finite dilute activity therefore requires an additional physical infrared cutoff or mechanism, such as a Higgs scale, a thermal screening scale, or a separately justified confining/chiral-dynamics input.

If one quark is massless and no external source saturates its zero modes, $\det M = 0$ and the vacuum activity (51.191) vanishes. This is compatible with nonzero instanton contributions to fermionic correlators: those correlators use the source-differentiated zero-mode coefficient in (51.135) instead of the mass determinant in (51.191).

Proposition 51.60 (Screened one-instanton size integral)**.** *Let a selected one-instanton amplitude channel have the one-loop small-size density*

$$K_{\mathrm{sc}}(\rho) = K_0\,\rho^{A-1}, \qquad A = b_0 + \beta_{\mathcal{X}} - 4 > 0,$$

after the collective-coordinate density, nonzero-mode determinant convention, and zero-mode source coefficient have been assembled. Suppose a physical infrared mechanism in the same regulated problem contributes a positive screening factor

$$\exp[-m_{\mathrm{scr}}^2\rho^2], \qquad m_{\mathrm{scr}} > 0,$$

for example from Higgsing, thermal screening, or a declared source envelope. The screened leading contribution is then

$$\mathcal{A}_{\mathrm{scr}}^{(0)} = K_0 \int_0^\infty \rho^{A-1} \mathrm{e}^{-m_{\mathrm{scr}}^2\rho^2}\,\mathrm{d}\rho = \frac{K_0}{2}\, m_{\mathrm{scr}}^{-A}\,\Gamma\!\left(\frac{A}{2}\right). \tag{51.194}$$

In the leading screened density, even moments of the instanton size are

$$\left\langle \rho^{2n}\right\rangle_0 = m_{\mathrm{scr}}^{-2n}\frac{\Gamma(A/2+n)}{\Gamma(A/2)}, \qquad m_{\mathrm{scr}}^2\langle\rho^2\rangle_0 = \frac{A}{2}. \tag{51.195}$$

If $A > 1$, the logarithmic size-shell density $\rho^A \exp[-m_{\mathrm{scr}}^2\rho^2]$ is stationary at

$$m_{\mathrm{scr}}^2\rho_{\mathrm{shell}}^2 = \frac{A}{2}, \tag{51.196}$$

while the ordinary $\mathrm{d}\rho$*-density is stationary at*

$$m_{\mathrm{scr}}^2\rho_{\mathrm{dens}}^2 = \frac{A-1}{2}. \tag{51.197}$$

For $SU(3)$ *with* $N_f = 2$ *massive Dirac fundamentals in the mass-saturated vacuum channel,* $\beta_{\mathcal{X}} = N_f$ *and*

$$A = b_0 + N_f - 4 = \frac{23}{3}.$$

Thus the screened leading activity is proportional to

$$m_{\mathrm{scr}}^{-23/3}\Gamma(23/6), \qquad m_{\mathrm{scr}}^2\langle\rho^2\rangle_0 = \frac{23}{6}, \qquad m_{\mathrm{scr}}^2\rho_{\mathrm{shell}}^2 = \frac{23}{6}.$$

This is a controlled amplitude formula only when the same physical screening scale keeps the dominant shell inside the weak-coupling and finite-regulator window. Sending $m_{\mathrm{scr}} \to 0$ *recovers the infrared divergence of* (51.193)*; the screened formula is not a license to use the one-loop instanton density at arbitrarily large size.*

Proof. The substitution $u = m_{\mathrm{scr}}^2\rho^2$ gives

$$\int_0^\infty \rho^{A-1}\mathrm{e}^{-m_{\mathrm{scr}}^2\rho^2}\,\mathrm{d}\rho = \frac{1}{2}m_{\mathrm{scr}}^{-A}\int_0^\infty u^{A/2-1}\mathrm{e}^{-u}\,\mathrm{d}u,$$

which is (51.194). Dividing the same integral with A replaced by $A+2n$ by the $n=0$ integral gives (51.195). The stationary-shell formulas come from differentiating $A\log\rho - m_{\mathrm{scr}}^2\rho^2$ for the $\mathrm{d}\log\rho$ shell and $(A-1)\log\rho - m_{\mathrm{scr}}^2\rho^2$ for the ordinary density. The $SU(3)$, $N_f = 2$ specialization substitutes $b_0 = 29/3$ and $\beta_{\mathcal{X}} = 2$. □

Controlled Approximation 51.61 (Thermal determinant screening of the size integral)**.** Consider weak-coupling finite-temperature $SU(N_c)$ QCD with N_f light Dirac fundamentals and a one-instanton amplitude channel whose zero-temperature small-size density has the form used in Proposition 51.60. The leading high-temperature fluctuation determinant around the periodic one-instanton background supplies the Gaussian size weight

$$K_T(\rho) = K_0\,\rho^{A-1}\exp[-m_T^2\rho^2]\,\exp[R_T(\rho)], \qquad m_T^2 = \pi^2T^2\,\frac{2N_c+N_f}{3}. \tag{51.198}$$

Here R_T denotes the nonquadratic finite-temperature determinant and caloron-shape remainder in the same regulator and source convention. In the trace-delta coupling convention of Chapter 130, (130.2) gives

$$\frac{m_D^2}{g_{\mathrm{YM}}^2} = T^2\,\frac{2N_c+N_f}{3}$$

for fundamental Dirac matter, so $m_T^2 = \pi^2 m_D^2/g_{\mathrm{YM}}^2$. The thermal instanton cutoff is therefore a determinant response of the plasma, not a classical bound on the BPST moduli space.

If the chosen amplitude window controls the residual by

$$|R_T(\rho)| \le \varepsilon_T,$$

define the Gaussian approximation

$$\mathcal{A}_T^G = \frac{K_0}{2}\left(\pi^2T^2\frac{2N_c+N_f}{3}\right)^{-A/2}\Gamma\left(\frac{A}{2}\right) \tag{51.199}$$

and the absolute Gaussian majorant

$$\mathcal{M}_T^G = \frac{|K_0|}{2}\left(\pi^2 T^2 \frac{2N_c + N_f}{3}\right)^{-A/2} \Gamma\left(\frac{A}{2}\right). \tag{51.200}$$

Since $|e^{R_T(\rho)} - 1| \le e^{\varepsilon_T} - 1$, the general source-projected amplitude satisfies the manifestly positive estimate

$$\left|\mathcal{A}_T - \mathcal{A}_T^G\right| \le \left(e^{\varepsilon_T} - 1\right)\mathcal{M}_T^G \tag{51.201}$$

whenever the same size window contains the support of the amplitude. The stronger positive-activity form

$$\left|\mathcal{A}_T - \mathcal{A}_T^G\right| \le \left(e^{\varepsilon_T} - 1\right)\mathcal{A}_T^G$$

is allowed only when $K_0 \ge 0$ and the residual is real in the selected channel, so that the integrand is a positive activity density. If $K_0 = e^{i\varphi}|K_0|$ has a fixed phase and R_T is real, the same proof gives the fixed-phase estimate with $|\mathcal{A}_T^G|$ on the right-hand side. For a genuinely complex source amplitude, only (51.201) is an absolute-error bound. The leading thermal size moments are the moments in (51.195) with $m_{\rm scr} = m_T$; in a complex channel they are moments of the Gaussian majorant, not probability moments of the amplitude itself.

For the mass-saturated $SU(3)$, $N_f = 2$ vacuum channel,

$$m_T^2 = \frac{8}{3}\pi^2 T^2, \qquad A = \frac{23}{3},$$

so

$$\mathcal{A}_T^G = \frac{K_0}{2}\left(\frac{8}{3}\pi^2 T^2\right)^{-23/6} \Gamma\left(\frac{23}{6}\right), \qquad \pi^2 T^2 \langle \rho^2 \rangle_G = \pi^2 T^2 \rho_{\rm shell}^2 = \frac{23}{16}. \tag{51.202}$$

The formula is useful only in a regime where the periodic-instanton determinant, the residual R_T, and the source or volume limits are controlled. Near the thermal transition, or in zero-temperature QCD after sending $T \to 0$, it does not replace the large-size dynamics by a semiclassical prediction.

Controlled Approximation 51.62 (Dilute instanton gas as an amplitude expansion)**.** Suppose the regulated one-instanton and one-anti-instanton amplitudes obtained above admit finite, translation-invariant vacuum activities ζ_Λ per Euclidean four-volume after the size, orientation, nonzero-mode determinant, and any required zero-mode mass-overlap factors have all been included. Assume further that the multi-instanton cluster expansion has a window in which overlap corrections, instanton–anti-instanton interactions, and boundary-face remainders contribute $o(V_4)$ to $\log Z$. Then the leading dilute-gas theta dependence is not a new moduli-space statement; it is the Poisson sum of the already constructed amplitudes:

$$\frac{Z_{\rm dig}(\bar\theta)}{Z_0} = \sum_{n_+, n_- \ge 0} \frac{(V_4\zeta_\Lambda)^{n_+}}{n_+!}\frac{(V_4\zeta_\Lambda)^{n_-}}{n_-!} e^{i(n_+ - n_-)\bar\theta} = \exp\left(2V_4\zeta_\Lambda \cos\bar\theta\right). \tag{51.203}$$

Thus the vacuum-energy difference in this approximation is

$$E_{\rm dig}(\bar\theta) - E_{\rm dig}(0) = 2\zeta_\Lambda(1 - \cos\bar\theta), \tag{51.204}$$

and the topological-charge cumulants at $\bar\theta = 0$ obey

$$\kappa_{2m}(Q) = 2V_4\zeta_\Lambda, \qquad \kappa_{2m+1}(Q) = 0, \qquad \chi_{\rm top}^{\rm dig} = 2\zeta_\Lambda. \tag{51.205}$$

Equivalently, writing

$$E_{\mathrm{dig}}(\bar\theta) - E_{\mathrm{dig}}(0) = \frac{1}{2}\chi_{\mathrm{top}}^{\mathrm{dig}}\bar\theta^2\left(1 + b_2^{\mathrm{dig}}\bar\theta^2 + O(\bar\theta^4)\right)$$

gives $b_2^{\mathrm{dig}} = -1/12$.

This is a controlled semiclassical amplitude expansion only when the activity is finite and dilute in the relevant correlation volume. If a massless quark zero mode is not saturated by a mass or source, the vacuum activity for this term vanishes; instanton sectors may still contribute to fermionic correlators such as (51.135). If the ρ-integral is infrared divergent, if small-instanton boundary terms are not controlled, or if instanton interactions are not negligible, then the Poisson form (51.203) has not been derived. In particular, this dilute four-dimensional gas is not a derivation of confinement in zero-temperature Yang–Mills. Its output is the conditional theta-dependence and zero-mode-saturated amplitude data above.

Controlled Approximation 51.63 (First cluster correction to the dilute instanton gas)**.** The Poisson formula (51.203) is the zeroth term in a cluster expansion. The collective-coordinate chart supplies the integration variables, while the finite-regulator amplitude scheme must also supply finite two-body connected cluster integrals $\mathfrak{B}_{\sigma\tau}$, $\sigma,\tau \in \{+,-\}$, after all relative centers, sizes, orientations, nonzero-mode determinants, zero-mode saturations, and interaction remainders in the selected window have been included. With $z_+ = \zeta_\Lambda \mathrm{e}^{\mathrm{i}\bar\theta}$ and $z_- = \zeta_\Lambda \mathrm{e}^{-\mathrm{i}\bar\theta}$, the first Mayer correction to the logarithm has the local form

$$\frac{1}{V_4}\log\frac{Z_{\mathrm{cl}}(\bar\theta)}{Z_0} = z_+ + z_- + \frac{1}{2}\sum_{\sigma,\tau=\pm} z_\sigma z_\tau\,\mathfrak{B}_{\sigma\tau} + O(\zeta_\Lambda^3). \tag{51.206}$$

Assume CP symmetry in this window, so

$$\mathfrak{B}_{++} = \mathfrak{B}_{--} =: \mathfrak{B}_2, \qquad \mathfrak{B}_{+-} = \mathfrak{B}_{-+} =: \mathfrak{B}_0.$$

Then

$$\frac{1}{V_4}\log\frac{Z_{\mathrm{cl}}(\bar\theta)}{Z_0} = 2\zeta_\Lambda\cos\bar\theta + \zeta_\Lambda^2\big(\mathfrak{B}_2\cos(2\bar\theta) + \mathfrak{B}_0\big) + O(\zeta_\Lambda^3). \tag{51.207}$$

The neutral instanton–anti-instanton molecule $\mathfrak{B}_0$ changes the vacuum pressure and the validity of the dilute expansion, but it cancels from the energy difference $E(\bar\theta) - E(0)$ at this order. The same-charge clusters carry topological charge ± 2, and therefore generate a second theta harmonic:

$$E_{\mathrm{cl}}(\bar\theta) - E_{\mathrm{cl}}(0) = 2\zeta_\Lambda(1-\cos\bar\theta) + \zeta_\Lambda^2\mathfrak{B}_2(1-\cos(2\bar\theta)) + O(\zeta_\Lambda^3). \tag{51.208}$$

Consequently

$$\chi_{\mathrm{top}}^{\mathrm{cl}} = 2\zeta_\Lambda + 4\zeta_\Lambda^2\mathfrak{B}_2 + O(\zeta_\Lambda^3), \qquad \partial_{\bar\theta}^4 E_{\mathrm{cl}}\big|_0 = -2\zeta_\Lambda - 16\zeta_\Lambda^2\mathfrak{B}_2 + O(\zeta_\Lambda^3), \tag{51.209}$$

and, in the expansion convention of (51.204),

$$b_2^{\mathrm{cl}} = -\frac{1+8\zeta_\Lambda\mathfrak{B}_2}{12(1+2\zeta_\Lambda\mathfrak{B}_2)} + O(\zeta_\Lambda^2) = -\frac{1}{12} - \frac{1}{2}\zeta_\Lambda\mathfrak{B}_2 + O(\zeta_\Lambda^2). \tag{51.210}$$

Thus $b_2 = -1/12$ is not a universal instanton prediction. It is the Poisson limit of independent $Q = \pm 1$ amplitudes. The first same-charge connected cluster already changes the fourth theta curvature, while a neutral instanton–anti-instanton molecule first tests the cluster expansion and neutral observables rather than the theta dependence at this order.

The identities above are only as controlled as the two-body cluster integrals. They require a single regulator and source convention for the one-body amplitudes, relative collective-coordinate measure, fluctuation determinants, zero-mode overlap factors, and boundary-face estimates. They do not derive $\mathfrak{B}_{\sigma\tau}$ from ADHM data without the determinant and interaction estimates, nor do they prove that the omitted three-body and higher clusters are small.

Controlled Approximation 51.64 (Thermal dilute topological susceptibility)**.** Combine the thermal determinant-screened one-instanton activity of ControlledApproximation 51.61 with the dilute-gas hypothesis in ControlledApproximation 51.62. Write the Gaussian one-instanton activity per Euclidean four-volume in a positive mass/source-saturated channel as

$$\zeta_T^G = \frac{K_{T,0}}{2}\left(\pi^2 T^2\,\frac{2N_c+N_f}{3}\right)^{-A/2}\Gamma\!\left(\frac{A}{2}\right), \qquad A = b_0 + \beta_{\mathcal{X}} - 4, \tag{51.211}$$

where $K_{T,0}$ contains the collective-coordinate normalization, zero-temperature nonzero-mode determinant convention, mass/source zero-mode-saturation coefficient, and any finite source-frame normalization in the selected channel. Here $K_{T,0} \ge 0$ and the residual is real; otherwise one has a source amplitude controlled by the majorant (51.200), not a positive dilute-gas activity. If the finite-temperature determinant residual obeys $|R_T(\rho)| \le \varepsilon_T$ on the same size window, the exact positive activity ζ_T satisfies

$$\left|\zeta_T - \zeta_T^G\right| \le \left(\mathrm{e}^{\varepsilon_T} - 1\right)\zeta_T^G. \tag{51.212}$$

Consequently the leading dilute topological susceptibility and fourth theta curvature are

$$\chi_{\mathrm{top}}^{T,\mathrm{dig}} = 2\zeta_T, \qquad \partial_\theta^4 E_T^{\mathrm{dig}}\big|_{\theta=0} = -2\zeta_T, \qquad b_2^{T,\mathrm{dig}} = -\frac{1}{12}, \tag{51.213}$$

with Gaussian approximants obtained by replacing ζ_T by ζ_T^G. The susceptibility error is bounded by

$$\left|\chi_{\mathrm{top}}^{T,\mathrm{dig}} - \chi_{\mathrm{top}}^{T,G}\right| \le \left(\mathrm{e}^{\varepsilon_T} - 1\right)\chi_{\mathrm{top}}^{T,G}, \qquad \chi_{\mathrm{top}}^{T,G} = 2\zeta_T^G. \tag{51.214}$$

For mass-saturated $SU(3)$, $N_f = 2$ QCD,

$$A = \frac{23}{3}, \qquad \chi_{\mathrm{top}}^{T,G} = K_{T,0}\left(\frac{8}{3}\pi^2 T^2\right)^{-23/6}\Gamma\!\left(\frac{23}{6}\right).$$

After the one-loop action has been traded for $\Lambda_{\mathrm{ht}}^{29/3}$ in the hard convention used above, this is the dimension-four scaling

$$\chi_{\mathrm{top}}^{T,G} \propto |m_u m_d|\,\Lambda_{\mathrm{ht}}^{29/3}\,T^{-23/3} \tag{51.215}$$

up to the displayed finite determinant, source-frame, and running-mass coordinates. This is a statement about the theta curvature of a dilute, screened thermal instanton ensemble. It is not a proof that isolated instantons dominate the topological susceptibility near the crossover or at zero temperature.

Proposition 51.65 (Zero-mode overlap determinant in an instanton ensemble)**.** *Fix a finite-volume, finite-regulator vectorlike $SU(N_c)$ QCD background which contains n_+ instanton cores and n_- anti-instanton cores in a window where the individual zero modes are identifiable and have* *been orthonormalized. Let ψ_I^+, $I = 1, \dots, n_+$, denote the isolated zero-mode basis in the chirality sector carried by the instanton cores and let $\psi_{\bar J}^-$, $\bar J = 1, \dots, n_-$, denote the opposite-chirality basis carried by the anti-instanton cores. The leading overlap matrix is*

$$T_{I\bar J} = \langle \psi_I^+, \mathcal{D}_E[A_{\rm ens}]\psi_{\bar J}^- \rangle, \qquad T : \mathbb{C}^{n_-} \longrightarrow \mathbb{C}^{n_+}, \tag{51.216}$$

where $A_{\rm ens}$ is the chosen multi-lump gauge field. After the nonzero-mode determinant and source-convention factors have been separated as in Proposition 51.40, the one-flavor massive Dirac operator restricted to this near-zero-mode subspace is

$$\mathcal{D}_{\rm zm}(m) = \begin{pmatrix} m\,\mathbf{1}_{n_+} & T \\ -T^\dagger & m\,\mathbf{1}_{n_-} \end{pmatrix} + E_{\rm zm}, \tag{51.217}$$

where $E_{\rm zm}$ collects Gram errors, regulator chirality-breaking terms, and Schur-complement terms from the nonzero-mode sector. If $s_1, \dots, s_r$ are the singular values of T, $r = \min(n_+, n_-)$, then in the leading projected block $E_{\rm zm} = 0$,

$$\det \mathcal{D}_{\rm zm}(m) = m^{|n_+ - n_-|} \prod_{\alpha=1}^{r} \left(m^2 + s_\alpha^2\right), \tag{51.218}$$

and, for real $m > 0$,

$$\operatorname{Tr} \mathcal{D}_{\rm zm}(m)^{-1} = \frac{|n_+ - n_-|}{m} + 2m \sum_{\alpha=1}^{r} \frac{1}{m^2 + s_\alpha^2}. \tag{51.219}$$

For N_f Dirac flavors with masses m_f, the zero-mode part of the leading-block fermion determinant is the product of (51.218) over f. Hence, inside the leading projected block, a sector with $n_+ \neq n_-$ has $|n_+ - n_-|$ unpaired zero modes at $m = 0$, while a balanced instanton–anti-instanton ensemble can acquire small near-zero Dirac eigenvalues from the singular values of T.

The corresponding exact statement for the full finite-regulator Dirac operator requires stronger hypotheses. If the regulator preserves an exact chiral/index structure and the full gauge field has topological charge $Q = n_+ - n_-$, then the full massless operator has $|Q|$ index-protected zero modes of the corresponding chirality, up to the usual convention for whether one counts the kernel of $\mathcal{D}$ or $\mathcal{D}^\dagger$. A generic nonzero $E_{\rm zm}$ may lift zero modes of the leading projected block; when $E_{\rm zm}$ is small, the lifted near-zero eigenvalues are controlled by the finite-dimensional perturbation size of $E_{\rm zm}$, while the index-protected kernel survives only in an index-preserving setup.

Proof. The massless Euclidean Dirac operator anticommutes with γ_5, so in a chiral zero-mode basis it has only off-diagonal matrix elements at leading order. The choice of signs in (51.217) is the anti-Hermitian Euclidean convention; replacing it by an equivalent Hermitian convention changes phases, not the positive factors below. Take a singular-value decomposition $T = U\Sigma V^\dagger$. The unitary change of basis $\mathrm{diag}(U, V)$ preserves the determinant and decomposes the block with $E_{\rm zm} = 0$ into r two-dimensional blocks

$$\begin{pmatrix} m & s_\alpha \\ -s_\alpha & m \end{pmatrix}$$

plus $|n_+ - n_-|$ unpaired one-dimensional m-blocks. Each paired block has determinant $m^2 + s_\alpha^2$ and inverse trace $2m/(m^2 + s_\alpha^2)$; each unpaired block contributes m to the determinant and

$1/m$ to the trace. Multiplying these factors gives (51.218) and (51.219). Different flavors have independent Grassmann coefficient spaces, so their finite determinants multiply.

The exact-kernel statement is a separate index input. The displayed singular-value reduction diagonalizes the projected leading block with $E_{\mathrm{zm}} = 0$. Adding E_{zm} changes the finite matrix being diagonalized; ordinary finite-dimensional perturbation estimates bound the movement of the leading-block eigenvalues by the size of this remainder, but they do not turn the leading block count into an index theorem. Exact full-operator zero modes follow only when the regulator supplies an exact chiral index and the total charge of the full background is the stated $Q = n_+ - n_-$. $\square$

Controlled Approximation 51.66 (Instanton-liquid spectral-density criterion)**.** The determinant (51.218) is the finite linear-algebra core of the instanton-liquid mechanism [66]. It is not a statement about the instanton moduli space by itself. A physical ensemble measure must also include the bosonic action, collective-coordinate Jacobians, pure-gauge and fermionic nonzero-mode determinants, instanton–anti-instanton interactions, regulator remainders E_{zm}, and the determinant reweighting from the overlap matrix. Suppose that, after all those weights are included, the positive singular-value density

$$\rho_{\mathrm{zm},\mathcal{V}_4}(s) = \frac{1}{\mathcal{V}_4}\left\langle \sum_\alpha \delta(s - s_\alpha)\right\rangle_{\mathrm{ens}} \tag{51.220}$$

has a thermodynamic limit $\rho_{\mathrm{zm}}(s)$ continuous at $s = 0$, and that the unpaired topological zero-mode density $|n_+ - n_-|/\mathcal{V}_4$ vanishes in the same limit. Then the zero-mode-zone contribution to the scalar-source derivative has the Banks–Casher form

$$\Sigma_{\mathrm{zm}} = \lim_{m\downarrow 0}\int_0^\infty \frac{2m\,\rho_{\mathrm{zm}}(s)}{m^2+s^2}\,\mathrm{d}s = \pi\rho_{\mathrm{zm}}(0), \tag{51.221}$$

in the normalization of (135.19)–(135.20).

This is a criterion, not a derivation of the QCD chiral condensate. The hard physics is the generation and control of a finite density of small singular values of the overlap matrix after the fluctuation determinants and interactions are included. A dilute Poisson gas with finite isolated activities gives theta-cumulants as in (51.205); it does not by itself prove $\rho_{\mathrm{zm}}(0) > 0$. Conversely, if a controlled ensemble has a nonzero near-zero singular-value density, then the instanton zero modes have become physical near-zero Dirac eigenstates, and the connection to chiral symmetry breaking is the spectral statement above rather than another collective-coordinate volume formula.

Controlled Approximation 51.67 (Zero-mode-zone $U(1)_A$ susceptibility criterion)**.** The same leading zero-mode-zone block distinguishes two different physical questions. A nonzero $\rho_{\mathrm{zm}}(0)$ gives the condensate criterion (51.221); restoration of the $U(1)_A$-paired isovector scalar and pseudoscalar susceptibilities is a stronger infrared test. Use the source convention of (135.22) and the local curvature normalization of (51.110)–(51.113). For a finite leading block with positive singular values s_α, define the zero-mode-zone spectral contribution to the $U(1)_A$-odd splitting by

$$\Delta^{\mathrm{zm}}_{\pi-\delta}(m;\mathcal{V}_4) = \frac{2|n_+ - n_-|}{\mathcal{V}_4 m^2} + \frac{4m^2}{\mathcal{V}_4}\sum_\alpha \frac{1}{(m^2+s_\alpha^2)^2}. \tag{51.222}$$

The first term is the exact topological zero-mode pole in this projected coordinate; it must be removed by taking the thermodynamic limit before the chiral limit, just as in the Banks–Casher statement. The paired term follows directly from the source derivatives of a two-dimensional chiral block: for $D_{\alpha,m} = \left(\begin{smallmatrix} m & s_\alpha \\ -s_\alpha & m \end{smallmatrix}\right)$ and $\gamma_5 = \mathrm{diag}(1,-1)$,

$$\mathrm{Tr}(D_{\alpha,m}^{-1}\gamma_5 D_{\alpha,m}^{-1}\gamma_5) + \mathrm{Tr}(D_{\alpha,m}^{-1}D_{\alpha,m}^{-1}) = \frac{4m^2}{(m^2+s_\alpha^2)^2}.$$

The plus sign is the Euclidean source sign: differentiating twice with respect to the scalar source gives $-\,\mathrm{Tr}\, D^{-1}D^{-1}$, whereas the pseudoscalar source $i\gamma_5 p$ gives $+\,\mathrm{Tr}\, D^{-1}\gamma_5 D^{-1}\gamma_5$.

Assume now that the ensemble measure in (51.220) has a thermodynamic limit, the topological zero-mode density vanishes first, and the ultraviolet part of the near-zero-zone density is locally dominated so that the small-s behavior controls the chiral singularity. Then

$$\Delta^{\mathrm{zm}}_{\pi-\delta}(m) = 4m^2 \int_0^\infty \frac{\rho_{\mathrm{zm}}(s)}{(m^2+s^2)^2}\,\mathrm{d}s. \tag{51.223}$$

Consequently

$$\rho_{\mathrm{zm}}(s) = \rho_0 + o(1) \quad\Longrightarrow\quad \Delta^{\mathrm{zm}}_{\pi-\delta}(m) = \frac{\pi\rho_0}{m} + o(m^{-1}), \tag{51.224}$$

$$\rho_{\mathrm{zm}}(s) = c_1 s + o(s) \quad\Longrightarrow\quad \Delta^{\mathrm{zm}}_{\pi-\delta}(m) = 2c_1 + o(1), \tag{51.225}$$

$$\rho_{\mathrm{zm}}(s) = O(s^{1+\epsilon}), \quad \epsilon > 0 \quad\Longrightarrow\quad \Delta^{\mathrm{zm}}_{\pi-\delta}(m) \longrightarrow 0 \tag{51.226}$$

under the same local domination hypothesis. Thus an instanton ensemble that produces a Banks–Casher condensate automatically leaves a divergent $U(1)_A$-odd zero-mode-zone susceptibility. Even an ensemble with $\rho_{\mathrm{zm}}(0) = 0$ can retain a finite $U(1)_A$-odd remnant if the near-zero density is only linear. Suppression of this channel in the chiral limit requires a stronger depletion of the overlap singular values than the vanishing of the condensate alone.

This criterion is deliberately phrased in terms of the weighted ensemble spectral density, not the geometry of isolated instantons. The ADHM or collective-coordinate measure supplies only part of the input. The physical question is whether the full fluctuation determinant, fermion determinant, instanton–anti-instanton interaction, and regulator remainder produce a specific near-zero singular-value law for T; that law is what decides the π-δ channel.

For the electroweak $SU(2)_L$ gauge theory, the same zero-mode logic applies to every left-handed doublet. For one generation there are three colored quark doublets and one lepton doublet, hence four $SU(2)_L$ doublet zero modes. For N_g generations the corresponding minimal fermion operator contains

$$\mathcal{O}_{\mathrm{EW}} \propto \prod_{r=1}^{N_g} \bigl(q_{Lr} q_{Lr} q_{Lr} \ell_{Lr}\bigr), \tag{51.227}$$

with weak, color, spinor, and flavor contractions fixed by the zero-mode wavefunctions and by gauge invariance. It changes baryon and lepton numbers by

$$\Delta B = \Delta L = N_g, \qquad \Delta(B-L) = 0, \qquad \Delta(B+L) = 2N_g.$$

At zero temperature the amplitude is exponentially suppressed by the instanton action. The real-time high-temperature process is instead organized around the sphaleron barrier discussed in Section 45.6; both effects are controlled by the same anomalous relation between topological charge and fermion-number violation.

51.18 Anomalous Pion Couplings

The axial anomaly has observable low-energy consequences. In two-flavor QCD, the neutral pion couples to the electromagnetic topological density through the anomalous divergence of the axial isospin current:

$$\partial_\mu J^\mu_{5,3} = \frac{e^2}{16\pi^2}\,\mathrm{tr}\big(T_3 Q^2\big)\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} + \text{mass terms.}$$

Here Q is the quark charge matrix and T_3 is the third generator of flavor $SU(2)$. The same anomaly is reproduced in the pion effective theory by the Wess–Zumino–Witten term introduced in the next chapter.

Chapter 52

The Schwinger Model: Two-Dimensional Quantum Electrodynamics

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

52.1 Definition, Gauge Group, and Observable Sector

The Schwinger model is quantum electrodynamics in 1+1 spacetime dimensions with one massless Dirac field. We use coordinates $x^\mu = (t, x)$,

$$\eta_{\mu\nu} = \mathrm{diag}(-, +), \qquad \epsilon^{01} = +1, \qquad \gamma = \gamma^0\gamma^1.$$

The two-dimensional spinor conventions are those of Section 40.16. Let a_μ be a real Abelian gauge representative and let $e > 0$ be the charge of the dynamical Dirac field. The compact $U(1)$ connection is $e\,a_\mu$; this convention is important because external probes will be allowed to have charge Q with respect to a_μ. Thus

$$D_\mu\psi = \partial_\mu\psi - \mathrm{i}ea_\mu\psi, \qquad F_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu, \qquad E = F_{01}.$$

The local representative Lagrangian is

$$\mathcal{L}_{\mathrm{Sch}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}\gamma^\mu D_\mu\psi = \frac{1}{2}E^2 - \bar{\psi}\gamma^\mu\partial_\mu\psi + e\,a_\mu J^\mu, \tag{52.1}$$

where

$$J^\mu = \mathrm{i}\bar{\psi}\gamma^\mu\psi, \qquad J^\mu_A = \mathrm{i}\bar{\psi}\gamma^\mu\gamma\psi. \tag{52.2}$$

With these definitions eJ^μ is the dynamical electric current. Gauge representatives are related by

$$a_\mu \mapsto a_\mu + \partial_\mu\xi, \qquad \psi \mapsto \mathrm{e}^{\mathrm{i}e\xi}\psi, \qquad \bar{\psi} \mapsto \bar{\psi}\,\mathrm{e}^{-\mathrm{i}e\xi}.$$

On a spatial circle of circumference L, the gauge group includes large transformations with $\xi(x+L) - \xi(x) = 2\pi n/e$. Equivalently, the holonomy $\exp(\mathrm{i}e\oint a_x \mathrm{d}x)$ is the compact gauge-invariant zero-mode coordinate. The local computations below may be performed on the line and then lifted to the circle by including the compact zero mode and the large-gauge identifications.

A possible theta density is

$$\mathcal{L}_\theta = \frac{e\theta}{2\pi}E. \tag{52.3}$$

On a closed Euclidean two-manifold the curvature of $e\,a$ has integral periods, so $\theta \sim \theta + 2\pi$. In the massless theory an anomalous axial rotation shifts θ, and the local spectrum is independent of θ. In the massive deformation this shift is no longer a symmetry, and θ becomes a physical parameter.

Definition 52.1 (Probe-charge convention)**.** A static external probe of charge Q is defined by the conserved external current j^μ_{ext} coupled through

$$S_{\mathrm{probe}} = \int a_\mu j^\mu_{\mathrm{ext}}\,\mathrm{d}^2x.$$

A straight worldline carries the phase $\exp(\mathrm{i}Q\int a_\mu \mathrm{d}x^\mu)$. The dynamical Dirac field has $Q = e$. A probe with $Q/e \in \mathbb{Z}$ can be screened by dynamical charged particles; a probe with nonintegral Q/e is a fractional probe relative to the dynamical charge lattice.

The canonical momentum conjugate to a_x is E, while a_0 is a Lagrange multiplier. Including an external charge density $\rho_{\mathrm{ext}} = j^0_{\mathrm{ext}}$, variation of a_0 gives the operator Gauss law

$$\partial_x E + eJ^0 + \rho_{\mathrm{ext}} = 0. \tag{52.4}$$

This equation is a constraint on states and on correlation functions with line insertions. It is not optional data added after quantization; it is the local gauge-invariance condition of the theory.

Definition 52.2 (Local sector)**.** The local gauge-invariant sector of the massless Schwinger model is the local operator algebra generated, after renormalization, by $F_{\mu\nu}$, by gauge-invariant currents such as J^μ, and by neutral composite fields. The bare fermion field ψ is not a gauge-invariant local observable. Charged sectors are described by line-dressed charged insertions or by states on a spatial manifold with prescribed electric flux at infinity. The exact solution below is first a solution of the local gauge-invariant sector and of specified external line probes.

52.2 Current Algebra, Anomaly, and the Electric-Field Equation

In two dimensions the vector and axial currents are Hodge dual. With the gamma-matrix and orientation conventions fixed above,

$$J^\mu_A = -\epsilon^{\mu\nu}J_\nu. \tag{52.5}$$

The vector current is exactly conserved as a consequence of gauge symmetry:

$$\partial_\mu J^\mu = 0.$$

The axial current is not conserved. The two-dimensional anomaly formula of Chapter 51, applied to the background connection $A_\mu = ea_\mu$, gives the renormalized operator identity

$$\partial_\mu J_A^\mu = \frac{e}{2\pi}\epsilon^{\mu\nu}F_{\mu\nu} = \frac{e}{\pi}E. \tag{52.6}$$

The identity includes the finite contact term chosen in the definition of the axial-vector current correlator. It is not the Euler-Lagrange equation of the classical massless Dirac action.

Proposition 52.3 (Anomaly-induced massive electric field)**.** *In the absence of external line insertions, the gauge-invariant electric field of the massless Schwinger model obeys*

$$(\Box - m_{\mathrm{Sch}}^2)E = 0, \qquad m_{\mathrm{Sch}}^2 = \frac{e^2}{\pi}, \qquad \Box = \partial_\mu\partial^\mu. \tag{52.7}$$

Proof. The Maxwell equation obtained from (52.1) is the renormalized local operator identity

$$\partial_\nu F^{\nu\mu} + eJ^\mu = 0.$$

Here it is tested in the sector with no external line insertion. With external Wilson lines, the right side would acquire the corresponding distributional endpoint or worldline current, and the homogeneous equation below would be replaced by an inhomogeneous screened-field equation. Since $F^{10} = E$ and $F^{01} = -E$, its two components are

$$\partial_x E + eJ^0 = 0, \qquad -\partial_t E + eJ^1 = 0.$$

Hence

$$J^0 = -\frac{1}{e}\partial_x E, \qquad J^1 = \frac{1}{e}\partial_t E.$$

Using (52.5), $J_A^0 = -J^1$ and $J_A^1 = -J^0$. Therefore

$$\partial_\mu J_A^\mu = -\partial_t J^1 - \partial_x J^0 = \frac{1}{e}(-\partial_t^2 + \partial_x^2)E = \frac{1}{e}\Box E.$$

Comparing this distributional consequence of Maxwell's equation with the anomaly identity (52.6), with the same local contact-term convention for the axial current, gives $\Box E/e = eE/\pi$, which is (52.7). $\Box$

This derivation uses only gauge invariance, the exact current anomaly, and the Maxwell equation. It already proves that the local electric field creates a massive neutral particle. Bosonization gives a stronger statement: it identifies the entire current-electric-field sector as a free massive scalar theory and gives exact formulas for line-probe response.

52.3 Bosonization of the Current Sector

The conservation law for J^μ implies locally that J^μ is the Hodge dual of a derivative. The normalization is not arbitrary; it is fixed by the current algebra of one Dirac fermion.

Theorem 52.4 (Current-sector bosonization)**.** *Let the free massless Dirac field be quantized in the vacuum sector with the current normalization* (52.2)*. The gauge-invariant Abelian current subalgebra is represented by the derivative algebra of a real scalar field* φ*, normalized by*

$$\mathcal{L}_\varphi = -\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi, \qquad [\widehat{\varphi}(t,x), \partial_0\widehat{\varphi}(t,y)] = \mathrm{i}\delta(x-y),$$

through

$$J^\mu = \frac{1}{\sqrt{\pi}}\epsilon^{\mu\nu}\partial_\nu\varphi, \qquad J_A^\mu = -\frac{1}{\sqrt{\pi}}\partial^\mu\varphi. \tag{52.8}$$

Equivalently, after the same local contact-term convention is fixed on both sides, the current-generating functional satisfies

$$Z_{\mathrm{Dirac}}[b] = \int \mathcal{D}\varphi\, \exp\left\{\mathrm{i}\int \mathrm{d}^2x\left[-\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi + \frac{1}{\sqrt{\pi}}b_\mu\epsilon^{\mu\nu}\partial_\nu\varphi\right]\right\}. \tag{52.9}$$

This is the Lorentzian continuation of the Euclidean current-generating functional. When the local current algebra is extended by charge-one fermion vertex operators, the scalar is compact with period $\sqrt{\pi}$*; this compactification data is invisible to the derivative current algebra alone.*

Proof. The first step fixes the normalization independently of any path-integral argument. The equal-time commutator of the Dirac current, defined by gauge-invariant point splitting and with the vector Ward identity preserved, is

$$[\widehat{J}^0(t,x), \widehat{J}^1(t,y)] = -\frac{\mathrm{i}}{\pi}\partial_x\delta(x-y). \tag{52.10}$$

This Schwinger term follows directly from the short-distance singularity of the two chiral components of a free massless Dirac field. If (52.8) holds, then

$$J^0 = \frac{1}{\sqrt{\pi}}\partial_x\varphi, \qquad J^1 = -\frac{1}{\sqrt{\pi}}\partial_t\varphi,$$

and the canonical scalar commutator gives exactly (52.10). Hence the coefficient in (52.8) is forced.

It remains to prove that the same scalar gives all current correlators. Work first in Euclidean signature on the plane, and couple a real Schwartz one-form $b = b_\mu \mathrm{d}x^\mu$ to the conserved vector current. The Hodge decomposition on $\mathbb{R}^2$ gives

$$b_\mu = \partial_\mu\lambda + \epsilon_{\mu\nu}\partial^\nu\rho,$$

up to harmonic zero modes, which are absent for Schwartz sources. The vector Ward identity removes the exact part $\partial_\mu\lambda$, so the generating functional depends on ρ and on local contact conventions only. The transverse coupling can be written as an axial pure gauge:

$$b_\mu J^\mu = \epsilon_{\mu\nu}\partial^\nu\rho\, J^\mu = -\partial_\mu\rho\, J_A^\mu,$$

with the current convention (52.2). A local axial rotation $\psi \mapsto e^{\mathrm{i}\rho\gamma_5}\psi$ removes this coupling from the free Dirac action. The entire change of the determinant is the Fujikawa Jacobian for the same contact-term convention that gives (52.6); in the present normalization,

$$\log \frac{Z_{\mathrm{Dirac}}[b]}{Z_{\mathrm{Dirac}}[0]} = -\frac{1}{2\pi}\int_{\mathbb{R}^2} \mathrm{d}^2x\, \partial_\mu\rho\,\partial^\mu\rho.$$

Equivalently, in momentum space,

$$W_E[b] = \frac{1}{2\pi}\int \frac{\mathrm{d}^2p}{(2\pi)^2}\, b_\mu(-p)\left(\delta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right) b_\nu(p). \tag{52.11}$$

where $Z_{\mathrm{Dirac}}[b] = \exp[-W_E[b]]Z_{\mathrm{Dirac}}[0]$. This derivation is exact: after the source is removed by the axial rotation, there is no remaining interaction, and the anomaly Jacobian is quadratic in ρ.

The Euclidean scalar representative gives the same functional. With action

$$S_E[\varphi; b] = \int \mathrm{d}^2x \left[\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - \frac{\mathrm{i}}{\sqrt{\pi}} b_\mu\epsilon^{\mu\nu}\partial_\nu\varphi\right],$$

the Gaussian integral over φ, modulo the constant zero mode, gives (52.11), because

$$\epsilon^{\mu\alpha}p_\alpha\epsilon^{\nu\beta}p_\beta = p^2\delta^{\mu\nu} - p^\mu p^\nu.$$

Functional differentiation of the identical Gaussian source functional gives all Euclidean current distributions, with the already fixed local contact terms. Analytic continuation gives (52.9) and the Lorentzian current formula (52.8).

The compact period is fixed only after adjoining charged fields. A charge-one fermion operator is represented by a vertex operator whose monodromy with the current has unit charge; with the current normalization above, the fermion mass operator has the local representative $\cos(2\sqrt{\pi}\,\varphi)$. Single-valuedness of this local neutral operator gives $\varphi \sim \varphi + \sqrt{\pi}$. This compactification datum does not enter the local derivative current correlators, which are the only bosonization data used in the gauge-invariant Schwinger-model calculations below. □

52.4 Exact Bosonized Gauge Theory

Lemma 52.5 (Exact bosonized local gauge sector)**.** *In the massless Schwinger model, the gauge-invariant local current and electric-field sector is represented by the bosonic Lagrangian*

$$\mathcal{L}_{\mathrm{bos}} = -\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{e}{\sqrt{\pi}}a_\mu\epsilon^{\mu\nu}\partial_\nu\varphi + \frac{e\theta}{2\pi}E. \tag{52.12}$$

On a fixed local electric-flux branch this representative is equivalent to

$$\mathcal{L}_{\mathrm{eff}} = -\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi - \frac{1}{2}\frac{e^2}{\pi}\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right)^2. \tag{52.13}$$

The electric field is the massive scalar field

$$E = -\frac{e}{\sqrt{\pi}}\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right), \qquad (\Box - m_{\mathrm{Sch}}^2)E = 0, \qquad m_{\mathrm{Sch}}^2 = \frac{e^2}{\pi}. \tag{52.14}$$

Proof. Set $b_\mu = ea_\mu$ in the current-sector identity (52.9). Adding the Maxwell term and the theta density (52.3) gives (52.12). Since $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}E^2$ in the mostly-plus convention and

$$a_\mu \epsilon^{\mu\nu}\partial_\nu\varphi = a_0\partial_x\varphi - a_x\partial_t\varphi = E\varphi + \partial_\mu(\cdots),$$

the same local branch is represented, up to a boundary term, by

$$\mathcal{L}_{\text{bos}} = -\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi + \frac{1}{2}E^2 + E\left(\frac{e}{\sqrt{\pi}}\varphi + \frac{e\theta}{2\pi}\right). \tag{52.15}$$

In two spacetime dimensions E carries no independent transverse propagating gauge degree of freedom. Its local stationary equation is

$$E = -\frac{e}{\sqrt{\pi}}\varphi - \frac{e\theta}{2\pi}.$$

Substitution into (52.15) gives (52.13). The Euler-Lagrange equation from (52.13) is

$$\Box\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right) = \frac{e^2}{\pi}\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right),$$

and multiplying by $-e/\sqrt{\pi}$ gives the electric-field equation in (52.14). $\square$

The lemma is an exact equality for the local current and electric-field sector of the massless theory. Gauge-variant fermion fields, compact-boson vertex operators, and gauge lines belong to the charged-sector data; the local neutral algebra is the sector described above.

Remark 52.6 (Theta periodicity and branches). The local expression (52.13) is written on one electric-flux branch. In the compact theory, large gauge transformations and the compactness $\varphi \sim \varphi + \sqrt{\pi}$ identify the branch obtained by $\theta \mapsto \theta + 2\pi$ with the branch obtained by shifting φ. A formulation on a spatial circle keeps the holonomy zero mode and sums over the corresponding integer electric-flux sectors. The massless theory is independent of θ, because an axial rotation shifts θ and no fermion mass term obstructs this shift.

Figure 52.1 displays two gauge-invariant consequences of the same calculation: the local electric field has a massive pole with $m_{\text{Sch}}^2 = e^2/\pi$, and the static kernel changes from the unscreened one-dimensional Coulomb kernel to a screened Yukawa kernel.

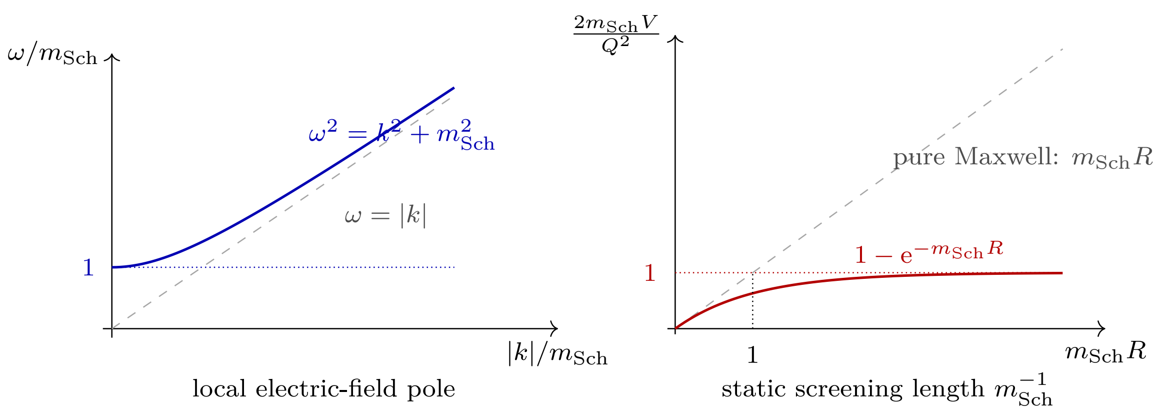


Figure 52.1: Two observable faces of the same massless Schwinger-model calculation. The anomaly-fixed current algebra gives the electric field a massive neutral pole, and the same mass changes the static probe potential from a linearly growing Maxwell kernel to the screened form derived below.

52.5 Static Probes and Screening

The response to external static charges is a line-observable question, not a statement about the local particle spectrum alone. For a neutral pair

$$\rho_{\mathrm{ext}}(x) = Q\delta(x - R/2) - Q\delta(x + R/2),$$

pure Maxwell theory in one spatial dimension gives a constant electric field between the probes. With the Gauss law $\partial_x E + \rho_{\mathrm{ext}} = 0$ in the interval between dynamical charges,

$$V_{\mathrm{pure}}(R) = \frac{Q^2}{2} R.$$

Screened static potential in the massless model. For the neutral static probe pair

$$\rho_R(x) = Q\delta(x - R/2) - Q\delta(x + R/2), \qquad R \geq 0,$$

the massless Schwinger model gives the static energy normalized by $V_{\mathrm{Sch}}(0) = 0$

$$V_{\mathrm{Sch}}(R) = Q^2 \left(G_{m_{\mathrm{Sch}}}(0) - G_{m_{\mathrm{Sch}}}(R)\right) = \frac{Q^2}{2m_{\mathrm{Sch}}} \left(1 - \mathrm{e}^{-m_{\mathrm{Sch}} R}\right), \tag{52.16}$$

where

$$\left(-\partial_x^2 + m^2\right) G_m(x) = \delta(x), \qquad G_m(x) = \frac{\mathrm{e}^{-m|x|}}{2m}.$$

Consequently the static force decays exponentially at distance m_{Sch}^{-1}, and there is no asymptotic linear electric string in the massless model.

Integrating out the massless current sector changes the static Coulomb kernel from $(-\partial_x^2)^{-1}$ to $(-\partial_x^2 + m_{\mathrm{Sch}}^2)^{-1}$. Equivalently, the temporal representative in Coulomb gauge obeys the screened static equation

$$\left(-\partial_x^2 + m_{\mathrm{Sch}}^2\right) a_0(x) = \rho_R(x).$$

The work needed to assemble the neutral pair from the coincident configuration is the positive quadratic form

$$V_{\mathrm{Sch}}(R) = \frac{1}{2} \int \frac{\mathrm{d}k}{2\pi} \frac{|\rho_R(k)|^2}{k^2 + m_{\mathrm{Sch}}^2}, \qquad \rho_R(k) = Q\big(\mathrm{e}^{-\mathrm{i}kR/2} - \mathrm{e}^{\mathrm{i}kR/2}\big). \tag{52.17}$$

Since

$$|\rho_R(k)|^2 = 2Q^2(1 - \cos kR),$$

equation (52.17) becomes

$$Q^2 \int \frac{\mathrm{d}k}{2\pi} \frac{1 - \cos kR}{k^2 + m_{\mathrm{Sch}}^2} = Q^2 \left(G_{m_{\mathrm{Sch}}}(0) - G_{m_{\mathrm{Sch}}}(R)\right).$$

The displayed Green function follows by solving the ordinary differential equation away from $x = 0$, imposing decay at infinity, and matching the jump $-G_m'(0^+) + G_m'(0^-) = 1$. This gives (52.16).

This screening statement is compatible with (52.7). The local electric field creates a massive neutral particle, while external flux is screened because the induced charge density can rearrange over the length scale m_{Sch}^{-1}. These are two different gauge-invariant observables whose common origin is the same anomaly-fixed current algebra.

52.6 Massive Fermion Variant and Fractional-Probe Confinement

Now add a Dirac mass $M\bar{\psi}\psi$. The axial rotation that removed θ in the massless theory also rotates the mass phase, so the physical parameter is the relative phase between the mass and the theta term. Choosing $M > 0$ real, bosonization gives the sine-Gordon perturbation

$$\Delta\mathcal{L}_M = \kappa M \cos(2\sqrt{\pi}\,\varphi + \theta). \tag{52.18}$$

The constant $\kappa > 0$ has mass dimension one and is fixed by the normalization of the renormalized composite mass operator. Equivalently, κ is the magnitude of the chiral condensate in the massless theory in the corresponding normal-ordering scheme. The scheme dependence of κ is exactly compensated by the scheme dependence of the renormalized mass M.

After eliminating E on a local branch, the scalar potential is

$$V_M(\varphi) = \frac{1}{2}\,\frac{e^2}{\pi}\left(\varphi + \frac{\theta}{2\sqrt{\pi}}\right)^2 - \kappa M \cos(2\sqrt{\pi}\,\varphi + \theta). \tag{52.19}$$

The quadratic term should again be understood together with the integer flux branches of the compact gauge theory. The cosine is periodic under $\varphi \mapsto \varphi + \sqrt{\pi}$; the electric term records which flux branch is being used.

Proposition 52.7 (Fractional-probe string tension at small fermion mass)**.** *Fix the normalization of the renormalized mass operator by* (52.18) *and take M small. A static probe pair with charges Q and $-Q$ shifts the effective theta angle in the interval between the probes by*

$$\theta_{\text{inside}} = \theta + \frac{2\pi Q}{e} \mod 2\pi. \tag{52.20}$$

The leading large-R energy-density shift in that interval is

$$\Delta\varepsilon_Q(\theta) = \kappa M\left[\cos\theta - \cos\!\left(\theta + \frac{2\pi Q}{e}\right)\right] + O(M^2), \tag{52.21}$$

with the physical branch chosen by minimizing over integer electric-flux sectors. At $\theta = 0$ the leading fractional-probe string tension is

$$\sigma(Q) = \kappa M\left(1 - \cos\frac{2\pi Q}{e}\right) + O(M^2). \tag{52.22}$$

It vanishes for $Q/e \in \mathbb{Z}$ and is positive, to this order, for nonintegral Q/e.

Proof. The external worldlines impose Gauss-law discontinuities. Across a probe of charge Q, the electric flux changes by Q. In the action, the theta term contributes

$$\int \mathrm{d}^2x\,\frac{e\theta}{2\pi}E.$$

Therefore a region in which the background electric flux is shifted by Q is equivalently described, in the bosonized scalar branch, by replacing θ by $\theta + 2\pi Q/e$. This proves (52.20).

For $M = 0$, the axial shift symmetry makes the vacuum energy independent of θ. To first order in M, ordinary perturbation theory around the massless Schwinger vacuum gives the vacuum-energy density

$$\varepsilon(\theta) = -\kappa M \cos\theta + O(M^2),$$

where the same κ appears in the renormalized mass perturbation (52.18). The interval between widely separated probes has energy density $\varepsilon(\theta + 2\pi Q/e)$, while the exterior vacuum has $\varepsilon(\theta)$. Their difference is (52.21). At $\theta = 0$ this is (52.22). If Q/e is an integer, the theta shift is 2π-periodic and the leading string tension vanishes; if $Q/e \notin \mathbb{Z}$, the shifted interval lies in a different point of the theta circle and has positive leading energy density at $\theta = 0$. $\square$

This is confinement of fractional test charges, not confinement of the dynamical unit-charged fermion. The latter belongs to the charge lattice that can screen integral probes.

52.7 Mechanisms Isolated by the Exact Solution

The model is useful because several mechanisms that are intertwined in higher-dimensional gauge theories become separately computable:

object	exact statement in the Schwinger model
local anomaly	$\partial_\mu J^\mu_A = (e/\pi)E$
local electric-field spectrum	E creates a scalar of $m^2_{\mathrm{Sch}} = e^2/\pi$
massless probe response	$V(R) = Q^2(1 - \mathrm{e}^{-m_{\mathrm{Sch}}R})/(2m_{\mathrm{Sch}})$
mass perturbation	$M\bar{\psi}\psi \leftrightarrow \kappa M \cos(2\sqrt{\pi}\varphi + \theta)$
fractional-probe confinement	$\sigma(Q) = \kappa M(1 - \cos(2\pi Q/e)) + O(M^2)$

Each line refers to a specified gauge-invariant local observable or to an explicit external line probe. This separation is essential: a massive electric-field two-point function, a screened Wilson-line potential, and a linear potential for fractional probes in the massive deformation are different physical statements, even though in this exactly solvable theory they are controlled by the same current algebra and anomaly.

Chapter 53

Large-N Two-Dimensional QCD and the Light-Front Bound-State Equation

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

53.1 Regulated Theory and Light-Front Coordinates

Two-dimensional QCD is a useful solvable gauge-theory laboratory precisely because the gauge field has no local propagating polarization but still imposes a nontrivial nonabelian Gauss law. The model considered in this chapter has gauge group $SU(N_c)$, N_f Dirac fermions in the fundamental representation, and the same trace convention as the four-dimensional QCD chapter:

$$\begin{aligned} \mathrm{tr}_\square(t^a t^b) &= \delta^{ab}, \\ (t^a)^i{}_j (t^a)^k{}_\ell &= \delta^i{}_\ell \delta^k{}_j - \frac{1}{N_c}\delta^i{}_j \delta^k{}_\ell. \end{aligned} \tag{53.1}$$

The second identity is the fundamental completeness relation in this normalization; it is the same color algebra used in Chapter 49. The coupling g_2 has mass dimension one, and the local action is written in the connection normalization

$$S = \int \mathrm{d}^2 x \left[-\frac{1}{4g_2^2}\,\mathrm{tr}_\square(F_{\mu\nu}F^{\mu\nu}) - \bar{q}(\gamma^\mu D_\mu + m)q \right], \qquad D_\mu = \partial_\mu - \mathrm{i}A_\mu. \tag{53.2}$$

For several flavors and masses, m is replaced by a diagonal mass matrix. The large-N_c coupling coordinate used below is

$$\lambda_2 := g_2^2 N_c. \tag{53.3}$$

It has dimensions of mass squared. Because the generator normalization is fixed by (53.1), no hidden factor of $1/2$ is allowed in this definition.

Use light-front coordinates

$$x^{\pm} = \frac{x^0 \pm x^1}{\sqrt{2}}, \qquad \mathrm{d}s^2 = -2\,\mathrm{d}x^+\mathrm{d}x^-, \tag{53.4}$$

and take x^+ as time. The momentum p^+ is conjugate to x^-, and a massive on-shell particle has

$$2p^+p^- = m^2, \qquad p^+ > 0.$$

The light-front Hamiltonian is P^-, while the invariant mass of a state with no transverse momentum in two dimensions is

$$M^2 = 2P^+P^-. \tag{53.5}$$

The general light-front regulator datum, including the mass-shell measure, the role of positive p^+, and the separate zero-mode/constraint sector, is Definition 145.16; this chapter is its large-N_c two-dimensional gauge-theory specialization.

The manipulations below are made at finite regulator before continuum limits are discussed. A convenient regulator is to put x^- on a circle of length $2\pi R$, remove or separately constrain the zero mode, impose a finite harmonic resolution $K = RP^+$, and only then study $K \to \infty$. Equivalently one may use smooth cutoffs in p^+ and a principal-value prescription for the instantaneous kernel. The zero mode of the gauge constraint is not a harmless detail: it fixes the color-neutral sector and the global electric-flux data. The bound-state equation derived below is the continuum form of the regulated color-singlet two-body problem after this constraint has been imposed.

53.2 Light-Front Gauge Constraint

Choose light-front gauge

$$A_- = 0.$$

Then $F_{+-} = -\partial_- A_+$. The A_+ field is nondynamical; it is the Lagrange-multiplier remnant of Gauss law written in a gauge where the constraint can be solved explicitly. The part of the regulated light-front action depending on A_+ has the form

$$S[A_+] = \int \mathrm{d}x^+\mathrm{d}x^- \left[\frac{1}{2g_2^2}(\partial_- A_+^a)(\partial_- A_+^a) + A_+^a J^{+a}\right], \tag{53.6}$$

where J^{+a} is the color current of the dynamical light-front fermion component. The sign convention in (53.6) is fixed by $\mathrm{d}s^2 = -2\mathrm{d}x^+\mathrm{d}x^-$ and by the action (53.2). Variation of A_+^a gives

$$-\frac{1}{g_2^2}\partial_-^2 A_+^a + J^{+a} = 0. \tag{53.7}$$

Let

$$G := (-\partial_-^2)^{-1}$$

on the zero-mode-free subspace selected by the regulator. On the infinite line, G has kernel $-|x^-|/2$ as an inverse of $-\partial_-^2$ on test functions of total integral zero; on the circle the corresponding kernel is the periodic zero-average Green function. Solving (53.7) and substituting back gives the positive instantaneous Coulomb Hamiltonian

$$P^-_{\text{inst}} = \frac{g_2^2}{2}\int \mathrm{d}x^-\mathrm{d}y^-\, J^{+a}(x^-)\, G(x^- - y^-)\, J^{+a}(y^-), \tag{53.8}$$

with the zero-mode sector understood as above.

Finite-regulator Gauss-law reduction. At finite light-front regulator, after the A_+ zero mode has been either fixed by the color-neutral Gauss-law constraint or kept as a separate compact degree of freedom, the nonzero-mode integration over A_+ is an ordinary Gaussian integration. The equivalence with the instantaneous Hamiltonian (53.8) is the following square completion. On the nonzero-mode regulated space, $-\partial_-^2$ is a positive invertible symmetric operator. Integrating (53.6) by parts in x^- gives

$$S[A_+] = \int \mathrm{d}x^+ \left[-\frac{1}{2g_2^2}\langle A_+, \partial_-^2 A_+\rangle + \langle A_+, J^+\rangle\right].$$

Completing the square with

$$A_+ = -g_2^2(-\partial_-^2)^{-1}J^+ + A'_+$$

separates a Gaussian in A'_+ from the current-current term. In the Hamiltonian P^-, the latter appears with the positive kernel G as in (53.8). The completion of the square does not determine the zero mode; that mode is exactly the global Gauss-law constraint and must be part of the regulator datum.

53.3 Planar Color-Singlet Mesons

Consider one quark flavor of mass m_1 and one antiquark flavor of mass m_2. A color-singlet light-front meson state of total momentum $P^+ > 0$ has the leading large-N_c form

$$|P^+;\phi\rangle = \frac{1}{\sqrt{N_c}}\sum_{i=1}^{N_c}\int_0^1 \mathrm{d}x\, \phi(x)\, b_i^\dagger(xP^+)d^{\dagger i}((1-x)P^+)|\Omega\rangle \tag{53.9}$$

after a choice of light-front creation-operator normalization. The function ϕ is a wavefunction on $0 < x < 1$. Its endpoint behavior is part of the domain of the mass operator. For positive renormalized endpoint masses, normalizability and finiteness of the mass expectation force sufficient vanishing at $x = 0, 1$; in the chiral limit an endpoint-constant wavefunction is allowed by the subtracted Coulomb kernel below.

The free light-front Hamiltonian contributes

$$P^-_{\text{kin}} = \frac{1}{2P^+}\left[\frac{m_1^2}{x} + \frac{m_2^2}{1-x}\right]$$

on the component with momentum fraction x. The instantaneous interaction (53.8) transfers longitudinal momentum between the quark and antiquark. The planar color contraction follows from (53.1): the first term preserves the singlet with coefficient N_c, while the second term is suppressed by N_c^{-1} in the normalized singlet matrix element. The real exchange kernel and the quark and antiquark self-energy terms combine into a subtracted principal value operator.

Definition 53.1 (Subtracted 't Hooft operator)**.** For $\gamma_2 > 0$ and endpoint masses $m_1^2, m_2^2 \geq 0$, define

$$(\mathcal{M}^2\phi)(x) = \left[\frac{m_1^2}{x} + \frac{m_2^2}{1-x}\right]\phi(x) + \gamma_2 \ \mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y. \tag{53.10}$$

The principal value is best defined by the quadratic form (53.13). For smooth test functions, it is the finite part obtained by applying the same cutoff to real exchange and self-energy terms before removing the cutoff. In the large-N_c QCD normalization of this chapter,

$$\gamma_2 = \frac{\lambda_2}{\pi} \tag{53.11}$$

for the standard continuum principal-value convention. A finite DLCQ regulator may instead take γ_2 as its displayed matrix coefficient and use (53.11) as the continuum matching condition.

Theorem 53.2 (Planar light-front meson equation)**.** *Assume the finite light-front regulator described above, take $N_c \to \infty$ with $\lambda_2 = g_2^2 N_c$ fixed and N_f fixed, and restrict to the color-singlet one-quark-one-antiquark sector with endpoint masses m_1, m_2 fixed in the chosen renormalization scheme. If the regulated planar Hamiltonians converge on this sector to the subtracted operator of Definition 53.1, then the leading large-N_c meson masses are the spectral values M^2 satisfying*

$$M^2\phi(x) = \left[\frac{m_1^2}{x} + \frac{m_2^2}{1-x}\right]\phi(x) + \frac{\lambda_2}{\pi} \ \mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y. \tag{53.12}$$

Proof. The free term follows from $M^2 = 2P^+P^-$: a quark with momentum xP^+ contributes $m_1^2/(2xP^+)$ to P^-, and the antiquark contributes $m_2^2/(2(1-x)P^+)$. Multiplication by $2P^+$ gives the endpoint mass operator.

For the interaction, write the current in momentum modes. At finite regulator, the kernel $G = (-\partial_-^2)^{-1}$ contributes a factor $(k^+)^{-2}$ for a nonzero exchanged longitudinal momentum k^+, with the zero mode excluded by Gauss law. Acting on the singlet state (53.9), the color algebra (53.1) gives a leading contraction $\delta^i{}_\ell \delta^k{}_j$ and a subleading trace subtraction. Hence the leading interaction is proportional to $g_2^2 N_c = \lambda_2$.

The real exchange term alone has a nonintegrable $1/(x-y)^2$ singularity. The self-energy terms generated by the same instantaneous Hamiltonian are the terms in which the exchanged longitudinal momentum leaves and returns to the same fermion line. With a common regulator for the real and self-energy pieces, their sum is the finite-part difference

$$\mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y.$$

The coefficient of this subtracted kernel is λ_2/π in the continuum momentum convention used for light-front creation operators. The assumed convergence of the regulated planar Hamiltonians then gives (53.12). The statement is deliberately phrased with the regulator-convergence hypothesis exposed: the algebra above derives the finite-regulator reduction and the planar kernel, while the continuum operator domain is an analytic input. □

Remark 53.3 (Relation to the unsubtracted form). A common finite-part coordinate writes the same regulated planar equation in the form

$$M^2\phi(x) = \left[\frac{\widetilde{m}_1^2}{x} + \frac{\widetilde{m}_2^2}{1-x}\right]\phi(x) - \frac{\lambda_2}{\pi}\ \mathrm{PV}\int_0^1 \frac{\phi(y)}{(x-y)^2}\,\mathrm{d}y.$$

It is a different finite-part coordinate for the same planar bound-state equation. The singular term proportional to $\phi(x)$ has been assigned to the endpoint masses $\widetilde{m}_i^2$. The subtracted form (53.12) displays the real–self-energy pairing and has a positive quadratic form, so it is the convention used in this monograph.

53.4 Quadratic Form, Positivity, and the Chiral Limit

Positive form of the subtracted kernel. For a real smooth function ϕ on $[0,1]$,

$$\int_0^1 \mathrm{d}x\,\phi(x)\,\mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y = \frac{1}{2}\int_0^1 \mathrm{d}x\int_0^1 \mathrm{d}y\,\frac{(\phi(x)-\phi(y))^2}{(x-y)^2}. \tag{53.13}$$

This identity is an elementary symmetrization of the principal-value kernel, but it is the useful way to remember the sign. Regulate the singularity by deleting $|x-y|<\epsilon$. The regulated bilinear expression is

$$I_\epsilon = \int_0^1 \mathrm{d}x\int_{|x-y|>\epsilon} \mathrm{d}y\,\phi(x)\frac{\phi(x)-\phi(y)}{(x-y)^2}.$$

Interchange x and y in the same integral. Since $(x-y)^2=(y-x)^2$, the average of the two equal expressions is

$$I_\epsilon = \frac{1}{2}\int_{|x-y|>\epsilon} \mathrm{d}x\,\mathrm{d}y\,\frac{(\phi(x)-\phi(y))^2}{(x-y)^2}.$$

For smooth ϕ, the numerator is $O((x-y)^2)$ near the diagonal, so the right hand side has a finite $\epsilon \downarrow 0$ limit. This proves (53.13). Adding the endpoint mass terms gives nonnegative contributions when $m_i^2 \geq 0$. Thus, for $m_1^2, m_2^2, \gamma_2 \geq 0$, the quadratic form of $\mathcal{M}^2$ is nonnegative on its form domain.

In the equal-mass chiral limit $m_1 = m_2 = 0$, the constant wavefunction

$$\phi_0(x) = 1$$

lies in the kernel of the subtracted Coulomb operator. Therefore

$$M_0^2 = 0$$

in the leading planar equation. This massless meson is a special feature of the large-N_c two-dimensional model and of the order of limits used here. General statements about four-dimensional QCD and the Schwinger-model mass generation of Chapter 52 require their own local gauge dynamics and anomaly structure.

53.5 Endpoint Exponents

The endpoint behavior of the wavefunction is part of the domain of the light-front mass operator. It is fixed by the cancellation of the leading singular terms in the same finite-part convention that defines (53.10).

Lemma 53.4 (Endpoint finite part)**.** *Let $0 < \beta < 1$, and define*

$$I_\beta(x) := \mathrm{PV}\int_0^1 \frac{x^\beta - y^\beta}{(x-y)^2}\,\mathrm{d}y.$$

As $x \downarrow 0$,

$$I_\beta(x) = \bigl(\pi\beta\cot(\pi\beta) - 1\bigr)x^{\beta-1} + O(1). \tag{53.14}$$

The analogous right-endpoint statement follows by replacing x with $1-x$.

Proof. Set $y = xt$. Then

$$I_\beta(x) = x^{\beta-1}\,\mathrm{PV}\int_0^{1/x} \frac{1-t^\beta}{(1-t)^2}\,\mathrm{d}t.$$

The tail from $1/x$ to ∞ contributes $O(x^{1-\beta})$ to the integral in t, hence $O(1)$ to $I_\beta(x)$. It remains to compute the finite part

$$C(\beta) := \mathrm{PV}\int_0^\infty \frac{1-t^\beta}{(1-t)^2}\,\mathrm{d}t.$$

Integrating by parts with the pole at $t = 1$ removed symmetrically gives

$$C(\beta) = -1 + \beta\,\mathrm{PV}\int_0^\infty \frac{t^{\beta-1}}{1-t}\,\mathrm{d}t.$$

The principal-value integral is

$$\mathrm{PV}\int_0^\infty \frac{t^{\beta-1}}{1-t}\,\mathrm{d}t = \int_0^1 \frac{t^{\beta-1} - t^{-\beta}}{1-t}\,\mathrm{d}t = \psi(1-\beta) - \psi(\beta) = \pi\cot(\pi\beta),$$

where $\psi = \Gamma'/\Gamma$, and the last equality is the logarithmic derivative of Euler's reflection formula $\Gamma(\beta)\Gamma(1-\beta) = \pi/\sin(\pi\beta)$. Thus $C(\beta) = \pi\beta\cot(\pi\beta) - 1$, proving (53.14). $\square$

Lemma 53.5 (Endpoint exponents)**.** *Suppose an eigenfunction of (53.10) has nonzero leading behavior*

$$\phi(x) = c_1 x^{\beta_1} + o(x^{\beta_1}), \qquad \phi(x) = c_2(1-x)^{\beta_2} + o((1-x)^{\beta_2}),$$

with $c_1, c_2 \neq 0$ and $0 \le \beta_i < 1$. For $m_i^2 > 0$, the endpoint exponents lie in $0 < \beta_i < 1$ and obey

$$\frac{m_i^2}{\gamma_2} = 1 - \pi\beta_i\cot(\pi\beta_i), \qquad i = 1, 2. \tag{53.15}$$

For $m_i^2 = 0$, the allowed leading exponent is $\beta_i = 0$ in the subtracted convention.

Proof. Near $x = 0$, the endpoint mass term in (53.10) gives

$$m_1^2 c_1 x^{\beta_1 - 1}.$$

Lemma 53.4 gives the singular part of the subtracted Coulomb term,

$$\gamma_2 c_1 \big(\pi\beta_1 \cot(\pi\beta_1) - 1\big) x^{\beta_1 - 1}.$$

The eigenvalue term $M^2\phi(x)$ is only $O(x^{\beta_1})$. Therefore the coefficient of x^{β_1-1} must vanish, giving (53.15) for $i = 1$. The right endpoint is identical after $x \mapsto 1 - x$.

The function $F(\beta) = 1 - \pi\beta\cot(\pi\beta)$ is positive and strictly increasing from 0 to $+\infty$ on $0 < \beta < 1$. Indeed

$$F'(\beta) = \pi^2\beta\csc^2(\pi\beta) - \pi\cot(\pi\beta) = \frac{\pi}{\sin^2(\pi\beta)}\big(\pi\beta - \sin(\pi\beta)\cos(\pi\beta)\big) > 0,$$

because $\pi\beta > \sin(\pi\beta)\cos(\pi\beta)$ for $0 < \beta < 1$. Hence each positive m_i^2/γ_2 determines a unique exponent. When $m_i^2 = 0$, the same equation has the endpoint solution $\beta_i = 0$, which is the constant endpoint behavior used in the massless chiral solution above. □

53.6 Closed Form Domain and Discrete Spectrum

The endpoint analysis determines the local asymptotics of eigenfunctions, but it is not by itself a definition of the spectral problem. The continuum object is the self-adjoint operator associated with the closed positive form that was already visible in (53.13). This is the coordinate in which the statement "the spectrum of the 't Hooft equation" has a precise Hilbert-space meaning.

Let

$$\mathcal{H} = L^2((0,1), \mathrm{d}x)$$

and, for $m_1^2, m_2^2, \gamma_2 \geq 0$, define

$$\begin{aligned} \mathfrak{q}_{m_1,m_2,\gamma_2}[\phi] := &\int_0^1 \left(\frac{m_1^2}{x} + \frac{m_2^2}{1-x}\right) |\phi(x)|^2 \,\mathrm{d}x \\ &+ \frac{\gamma_2}{2}\int_0^1 \mathrm{d}x \int_0^1 \mathrm{d}y\, \frac{|\phi(x) - \phi(y)|^2}{(x-y)^2}. \end{aligned} \tag{53.16}$$

The form domain is

$$\mathcal{Q}_{m_1,m_2,\gamma_2} = \left\{\phi \in L^2(0,1) \ :\ \mathfrak{q}_{m_1,m_2,\gamma_2}[\phi] < \infty\right\}. \tag{53.17}$$

When an endpoint mass is positive, the corresponding weighted L^2 term in (53.16) is part of the definition: for example $m_1^2 > 0$ requires $\int_0^1 x^{-1}|\phi(x)|^2 \mathrm{d}x < \infty$. When the endpoint mass is zero, that endpoint weight is absent. This is why the constant wavefunction is admissible in the double chiral limit but not for a positive endpoint mass.

Proposition 53.6 (Self-adjoint 't Hooft mass operator)**.** *For $m_1^2, m_2^2 \geq 0$ and $\gamma_2 > 0$, the form $\mathfrak{q}_{m_1,m_2,\gamma_2}$ on $\mathcal{H} = L^2(0,1)$ is densely defined, nonnegative, and closed. Hence there is a unique nonnegative self-adjoint operator $\mathcal{M}^2_{m_1,m_2,\gamma_2}$ such that*

$$\mathfrak{q}_{m_1,m_2,\gamma_2}[\eta, \phi] = \langle \eta, \mathcal{M}^2_{m_1,m_2,\gamma_2}\phi\rangle_{L^2}$$

for all $\eta \in \mathcal{Q}_{m_1,m_2,\gamma_2}$ and all ϕ in the operator domain. On smooth functions for which the displayed principal value exists and the endpoint terms are integrable, this operator acts by

$$(\mathcal{M}^2_{m_1,m_2,\gamma_2}\phi)(x) = \left(\frac{m_1^2}{x} + \frac{m_2^2}{1-x}\right)\phi(x) + \gamma_2\,\mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y.$$

Its resolvent is compact. Therefore its spectrum consists of eigenvalues of finite multiplicity, accumulating only at $+\infty$. If $m_1 = m_2 = 0$, the constant functions are precisely the zero eigenspace. If at least one endpoint mass is positive, zero is not an eigenvalue.

Proof. The domain is dense because $C_c^\infty(0,1)$ lies in $\mathcal{Q}_{m_1,m_2,\gamma_2}$ and is dense in $L^2(0,1)$. Positivity is immediate from (53.16). For closedness, let (ϕ_n) be Cauchy in the form norm

$$\|\phi\|^2_{\mathcal{Q}} = \|\phi\|^2_{L^2} + \mathfrak{q}_{m_1,m_2,\gamma_2}[\phi].$$

Then $\phi_n \to \phi$ in L^2 for some ϕ. Passing to a subsequence gives almost-everywhere convergence. Fatou's lemma applied to the endpoint weights and to the positive measure on $(0,1)^2 \setminus \{x=y\}$ gives

$$\mathfrak{q}_{m_1,m_2,\gamma_2}[\phi] \le \liminf_{n\to\infty} \mathfrak{q}_{m_1,m_2,\gamma_2}[\phi_n],$$

so $\phi \in \mathcal{Q}_{m_1,m_2,\gamma_2}$. Applying the same lower semicontinuity to $\phi_n - \phi_m$ and then sending $m,n \to \infty$ shows that $\phi_n \to \phi$ in the form norm. Thus the form is closed. The representation theorem for densely defined closed nonnegative forms gives the unique nonnegative self-adjoint operator associated with the form.

For a smooth ϕ and a smooth test function η, regulate the singularity by deleting $|x-y| < \epsilon$. Symmetrizing the bilinear form gives

$$\frac{\gamma_2}{2}\int_{|x-y|>\epsilon} \frac{(\overline{\eta(x)}-\overline{\eta(y)})(\phi(x)-\phi(y))}{(x-y)^2}\,\mathrm{d}x\,\mathrm{d}y = \gamma_2\int_0^1 \overline{\eta(x)}\int_{|x-y|>\epsilon}\frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y\,\mathrm{d}x.$$

The finite part as $\epsilon \downarrow 0$ is exactly the principal-value operator displayed above. The endpoint multiplication terms are ordinary weighted L^2 pairings. This proves the agreement with the differential form of the equation on the smooth core where all terms are defined.

It remains to identify the spectral type. The double integral in (53.16) is the homogeneous $H^{1/2}$-seminorm on the interval, up to the harmless normalization convention for the finite interval. More explicitly, extending ϕ to a function on a slightly larger interval by a bounded extension operator and then periodizing on a circle, the kernel form controls

$$\sum_{n\in\mathbb{Z}} |n|\,|\widehat{\phi}_n|^2$$

on the extension, while $\|\phi\|^2_{L^2}$ controls the $n=0$ coefficient. The diagonal argument for Fourier series then gives compactness of the unit ball of $H^{1/2}$ in L^2: truncate to $|n| \le N$, obtain compactness in the finite-dimensional truncated space, and make the tail uniformly small by the $H^{1/2}$ bound. The endpoint weights only shrink the form domain, so the embedding $\mathcal{Q}_{m_1,m_2,\gamma_2} \hookrightarrow L^2(0,1)$ is compact. A closed semibounded form with compact form-domain embedding has compact resolvent: $(\mathcal{M}^2_{m_1,m_2,\gamma_2}+1)^{-1}$ maps L^2 boundedly into the form domain and then compactly back to L^2. Therefore the spectrum is purely discrete with finite multiplicities and no finite accumulation point.

Finally, $\mathfrak{q}[\phi] = 0$ forces $|\phi(x) - \phi(y)|^2 = 0$ for almost every pair (x, y), hence ϕ is almost everywhere constant. If both endpoint masses vanish, these constants indeed have zero form value. If at least one endpoint mass is positive, a nonzero constant has infinite or positive endpoint-weighted form value, so only the zero vector can have zero form value. □

Remark 53.7 (Spectral coordinates). The eigenvalues can be characterized by the Rayleigh–Ritz principle applied to $\mathfrak{q}_{m_1,m_2,\gamma_2}$. This is the continuum statement behind the finite DLCQ matrices below: finite-dimensional approximations are useful only insofar as their forms converge to (53.16) in a sense strong enough to control these variational values. Changing the endpoint mass coordinate or the zero-mode prescription changes the form and hence changes the spectral problem being approximated.

53.7 Finite DLCQ Matrix and Numerical Benchmark

The finite matrix used in Chapter 145 is the DLCQ discretization of the subtracted form above. Fix $K \geq 3$, grid points

$$x_n = \frac{n}{K}, \qquad n = 1, \ldots, K-1,$$

and a coefficient $\gamma_2 > 0$. Define

$$(M_K^2)_{nm} = \delta_{nm} \left[\frac{m_1^2}{x_n} + \frac{m_2^2}{1 - x_n} + \gamma_2 K \sum_{\substack{j=1 \\ j \neq n}}^{K-1} \frac{1}{(n-j)^2} \right] - (1 - \delta_{nm}) \frac{\gamma_2 K}{(n-m)^2}. \tag{53.18}$$

Then, for $v \in \mathbb{R}^{K-1}$,

$$v^T M_K^2 v = \sum_{n=1}^{K-1} \left[\frac{m_1^2}{x_n} + \frac{m_2^2}{1 - x_n} \right] v_n^2 + \gamma_2 K \sum_{1 \leq n < m \leq K-1} \frac{(v_n - v_m)^2}{(n-m)^2}. \tag{53.19}$$

This identity is the finite-dimensional shadow of (53.13). It also fixes the sign of the off-diagonal matrix elements. In the massless case, the constant vector is an exact zero mode of the kernel part at every K, matching the continuum constant solution.

Controlled Approximation 53.8 (DLCQ use of the 't Hooft equation)**.** A DLCQ calculation of two-dimensional large-N_c QCD must state the harmonic resolution K, the treatment of zero modes, the endpoint mass coordinate, the coefficient γ_2 and its matching to λ_2/π, the normalization of finite eigenvectors, and the extrapolation model for $K \to \infty$. A finite eigenvalue of (53.18) is an eigenvalue of a regulated matrix; it becomes a continuum meson mass only after these convergence data are supplied.

The companion script

```
calculation-checks/thooft_model_checks.py
```

verifies the finite color normalization, the quadratic-form identity (53.19), positivity for positive endpoint masses, and the exact massless constant zero mode of the subtracted kernel.

53.8 Scope of the Two-Dimensional Large-N_c Construction

The model gives an exact leading-planar bound-state equation in a genuine confining gauge theory. Its content is the following precise construction.

1. The meson equation follows from a gauge constraint, the color-singlet Hilbert-space sector, and planar color algebra.
2. The absence of transverse gauge polarizations in $1 + 1$ dimensions makes the gauge field a constraint variable while preserving a nontrivial nonabelian Coulomb kernel.
3. Large-N_c factorization makes one-meson sectors stable at leading order. Meson interactions and widths are organized by subleading powers of $1/N_c$.
4. The light-front construction is a statement about a declared regulator, a zero-mode prescription, and an endpoint-mass coordinate. Changing any of these data changes the operator whose spectrum is being studied.

Thus two-dimensional large-N_c QCD is a benchmark for confinement, bound-state equations, light-front regularization, and the relation between gauge constraints and color-singlet Hilbert-space sectors. Its comparison with four-dimensional QCD is structural rather than literal: the four-dimensional theory has propagating gluons, running coupling, spin-dependent interactions, and multi-particle channels that require the separate developments of the later QCD chapters.

Chapter 54

The Standard Model as a Hybrid Quantum Field Theory

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The Standard Model used in physics is not presently a single completed constructive definition of a four-dimensional continuum QFT. It is a precise hybrid object whose components have different mathematical statuses: nonperturbative QCD, understood through its continuum Wightman target and through lattice regulators used to define hadronic matrix elements; the renormalized electroweak chiral gauge theory, defined order by order in perturbation theory in the same sense in which perturbative QED is defined; hybrid coupling data that pairs perturbative electroweak fields and sources with specified gauge-invariant local QCD operators; lower-energy weak effective theories; and matching maps between overlapping regimes. No physical ultraviolet cutoff is a parameter of a Standard Model prediction. Cutoffs and regulators enter either as auxiliary devices that are removed or absorbed into renormalized coordinates, or as extra data in a different Wilsonian or SMEFT problem. The purpose of this chapter is to write the hybrid object without hiding these domains of validity. Within each domain the standard formulae are derived as statements about explicit fields, couplings, operators, and observables. Assertions about higher energies, large field values, or physics beyond this field content require additional mathematical data; they are not supplied merely by drawing a perturbative RG curve or by writing a formal Lagrangian.

The local representation theory is developed first, because it fixes the charges, anomalies, mass coordinates, and operator coordinates that every perturbative, effective, or nonperturbative realization must reproduce. The hybrid definition of the full object is then assembled after these components have been made explicit. Until that point, the phrase “the Standard Model” in a local formula means the corresponding renormalized perturbative electroweak construction together with the stated QCD and matching data, not an assumed constructive theorem and not a finite-cutoff theory.

54.1 Gauge and Matter Data

Convention 54.1 (Electroweak charge and trace normalizations)**.** The hypercharge coordinate is normalized by

$$Q_{\mathrm{em}} = T_L^3 + Y,$$

where T_L^3 is the third generator of the weak $SU(2)_L$ doublet in the standard electroweak normalization. For transparent mass and charge formulae, this chapter temporarily uses the conventional explicit-coupling fields $\mathcal{G}_\mu, \mathcal{W}_\mu, \mathcal{B}_\mu$. The conversion to the monograph's Yang–Mills convention of Chapter 45 is

$$A_\mu^{(3)} = g_3 \mathcal{G}_\mu^A T^A, \qquad A_\mu^{(2)} = g_2 \mathcal{W}_\mu^a \tau^a, \qquad A_\mu^{(1)} = g_1 \mathcal{B}_\mu Y,$$

where T^A and $\tau^a = \sigma^a/2$ are the conventional half-trace generators, normalized by $\mathrm{tr}(\tau^a\tau^b) = \frac{1}{2}\delta^{ab}$ in the fundamental weak doublet. If $t^a = \sqrt{2}\,\tau^a$ denotes the trace-delta basis used in Chapter 45, then $g_{2,\mathrm{mono}} = g_2/\sqrt{2}$ in a matter vertex. The invariant kinetic terms are equivalently written as explicit-coupling kinetic terms for $\mathcal{G}, \mathcal{W}, \mathcal{B}$ or as $-\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ for the coupling-absorbed connections.

Definition 54.2 (One-generation left-Weyl representation)**.** For one generation, written as left-handed Weyl fields, the representation is

field	$SU(3)_c \times SU(2)_L$	Y
$Q_L = (u_L, d_L)$	$(\mathbf{3}, \mathbf{2})$	$1/6$
u^c	$(\overline{\mathbf{3}}, \mathbf{1})$	$-2/3$
d^c	$(\overline{\mathbf{3}}, \mathbf{1})$	$1/3$
$L_L = (\nu_L, e_L)$	$(\mathbf{1}, \mathbf{2})$	$-1/2$
e^c	$(\mathbf{1}, \mathbf{1})$	$1.$

Here u^c, d^c, e^c are left-handed charge-conjugate fields, not right-handed fields in the same representation. A gauge-singlet neutrino field, if included in the same left-handed convention as ν^c, has representation $(\mathbf{1}, \mathbf{1})_0$ and does not affect the local gauge anomaly sums.

Definition 54.3 (Minimal Higgs and Yukawa sector)**.** The minimal Higgs scalar is

$$H \in (\mathbf{1}, \mathbf{2})_{1/2}.$$

Set

$$\widetilde{H} := \mathrm{i}\sigma^2 H^*, \qquad \widetilde{H} \in (\mathbf{1}, \mathbf{2})_{-1/2},$$

where the second statement uses the pseudoreality of the $SU(2)$ doublet. In the left-handed Weyl convention of Definition 54.2, the renormalizable Yukawa sector is

$$\mathcal{L}_Y = -(Y_u)_{IJ}\,(Q_L^I H)\,u^{cJ} - (Y_d)_{IJ}\,(Q_L^I \widetilde{H})\,d^{cJ} - (Y_e)_{IJ}\,(L_L^I \widetilde{H})\,e^{cJ} + \text{Hermitian conjugate},$$

where $I, J = 1, 2, 3$ are generation labels and all gauge indices are contracted by the invariant tensors of $SU(3)_c$ and $SU(2)_L$. In components, $(QH)u^c$ abbreviates the contraction of the two

Weyl spinor indices by $\epsilon_{\alpha\beta}$, the two weak doublet indices by ϵ_{ij}, and the color indices by the natural $\mathbf{3} \otimes \overline{\mathbf{3}} \to \mathbf{1}$ pairing. The displayed Y_u, Y_d, Y_e are complex 3×3 matrices of local coupling coordinates.

Definition 54.4 (Local Standard Model charge and bundle data)**.** The local Standard Model charge algebra is

$$\mathfrak{g}_{\mathrm{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

The corresponding covering compact group for local representation and bundle bookkeeping is

$$\widetilde{G}_{\mathrm{SM}} = SU(3)_c \times SU(2)_L \times U(1)_y,$$

where $y := 6Y$ is the integer hypercharge coordinate for the compact $U(1)$ factor. The Lie-algebra generator appearing in the electroweak covariant derivative is still $Y = y/6$. A representation-theoretic global form is a quotient

$$G_{\mathrm{SM}} = \widetilde{G}_{\mathrm{SM}}/\Gamma, \qquad \Gamma \subset Z(\widetilde{G}_{\mathrm{SM}}),$$

for which the local fields in Definitions 54.2 and 54.3 descend to representations of G_{SM}. On a spin four-manifold M, let S_L denote the left-handed spinor bundle and let $E_R := P \times_{G_{\mathrm{SM}}} R$ be the vector bundle associated to a representation R. The local classical field data consist of P, a connection on P, left-Weyl fields in $\Gamma(S_L \otimes E_R)$ for the three generations of the representations in Definition 54.2, a Higgs field in $\Gamma(E_H)$ for $H \in (\mathbf{1}, \mathbf{2})_{1/2}$, and the local Yang–Mills, scalar, and Yukawa action. The perturbative electroweak quantum theory further requires gauge fixing or BV data, a subtraction scheme, and the renormalized construction of Green functions and operator insertions as formal power series to all orders in the electroweak couplings. A cutoff regulator may be introduced in a proof or computation, but then it is an auxiliary regularization device to be removed or matched away; it is not the definition of the electroweak sector and it is not a parameter in the Standard Model prediction.

The product $\widetilde{G}_{\mathrm{SM}}$ in Definition 54.4 is not, by itself, the gauge group of a completed quantum Standard Model. In the hybrid definition of Definition 54.37, the $SU(3)_c$ factor belongs to the nonperturbative QCD Wightman component, while $SU(2)_L \times U(1)_Y$ belongs to the all-orders renormalized perturbative electroweak component; their coupling is mediated by the operator coupling $\mathcal{C}_{\text{QCD-EW}}$. The product notation records the common local charge assignments, anomaly sums, and possible bundle/global-form bookkeeping that these components must share. The hybrid definition treats the color sector, the electroweak perturbative sector, and their coupling map as distinct pieces of structure rather than as one declared nonperturbative path integral with all gauge factors on the same footing.

The quotient Γ is invisible in the Lie algebra and in local perturbative vertices, but it changes the allowed bundles, line operators, and higher-form backgrounds in any formulation where those objects are defined. Those global questions belong to the generalized-symmetry analysis in Chapter 113. This chapter uses only the local representation-theoretic data unless a bundle or line operator is explicitly mentioned.

Remark 54.5 (Electroweak perturbation theory and optional cutoffs). The baseline electroweak component of the hybrid Standard Model is $\mathfrak{D}^{\mathrm{pert}}_{\mathrm{EW}}$: an all-orders renormalized perturbative chiral gauge theory, formulated after gauge fixing or in BV language, with a specified subtraction and composite-operator scheme. It is not a finite-cutoff theory. In a proof or computation one may introduce a cutoff-like parameter for electroweak fields, external sources, or a weak effective theory; such a parameter is then auxiliary regularization data or extra Wilsonian/EFT data for a different problem, not a hidden ultraviolet cutoff in the Standard Model prediction framework. The color sector is not thereby made into a cutoff sector of the same path integral; it is supplied through the QCD component $\mathfrak{D}^{\mathrm{W}}_{\mathrm{QCD}}$, with lattice or other nonperturbative regulators used only to approximate the required gauge-invariant QCD matrix elements.

One could instead pose a separate formal construction problem in which the QCD sector is kept as a continuum Wightman theory while an ultraviolet regulator is placed only on electroweak perturbative fields and sources. Such a construction would have to define mixed generating functionals whose coefficients are QCD Wightman distributions of the gauge-invariant operators in Definition 54.35; it would also have to specify the extension of those distributions at coincident points, the contact-term scheme, and the electroweak Ward or BV identities at finite electroweak cutoff. This is a coherent possible formalization of the hybrid viewpoint, but it is additional mathematical data beyond $\mathfrak{D}^{\mathrm{pert}}_{\mathrm{EW}}$, it is not the usual textbook path integral, and it is not ordinarily how Standard Model phenomenological computations are presented.

This is different from postulating a single ultraviolet-regulated path integral for the full local product data $SU(3)_c \times SU(2)_L \times U(1)_Y$. Such a full cutoff formulation would have to regulate the $SU(3)_c$ gauge fields and the chiral electroweak fields in one coupled object, choose the phase of the chiral determinant, provide reflection or reconstruction data, and formulate a continuum-limit problem for the coupled theory. That stronger object is the kind of data discussed later in Definition 54.40; it is not the baseline Standard Model prediction framework used in this chapter.

The kernel of the minimal local representation content is a finite center-charge calculation. Let $y := 6Y$ be the integer hypercharge coordinate, so the minimal fields have

$$y(Q_L, u^c, d^c, L_L, e^c, H) = (1, -4, 2, -3, 6, 3).$$

Let $\omega_3 = \exp(2\pi\mathrm{i}/3)$. The element

$$z = \big(\omega_3, -1, \exp(\mathrm{i}\pi/3)\big) \in SU(3)_c \times SU(2)_L \times U(1)_y \tag{54.1}$$

acts trivially on every field in Definitions 54.2 and 54.3. It generates a cyclic subgroup $\mathbb{Z}_6$ of the center, and the minimal local field content descends through any quotient by a subgroup of this $\mathbb{Z}_6$:

$$\Gamma = 1, \quad \mathbb{Z}_2, \quad \mathbb{Z}_3, \quad \mathbb{Z}_6.$$

For a representation with color triality $t \in \mathbb{Z}/3\mathbb{Z}$, weak doublet parity $s \in \mathbb{Z}/2\mathbb{Z}$, and integer hypercharge y, the element z acts by the phase

$$\exp\left[2\pi\mathrm{i}\left(\frac{t}{3} + \frac{s}{2} + \frac{y}{6}\right)\right].$$

For the six minimal representations the triples (t, s, y) are

$$\begin{array}{c|c} Q_L & (1,1,1) \\ u^c & (-1,0,-4) \\ d^c & (-1,0,2) \\ L_L & (0,1,-3) \\ e^c & (0,0,6) \\ H & (0,1,3). \end{array}$$

In each row $t/3+s/2+y/6$ is an integer. Thus z acts trivially. The element has order six, because its three center components have orders three, two, and six. Since every subgroup of a trivially acting central subgroup also acts trivially, the four displayed quotients are allowed by the minimal representation content. Distinguishing these global forms requires line-operator and bundle data, not only the local Lie algebra.

Gauge invariance of the minimal Yukawa sector. The three terms in Definition 54.3 are precisely the gauge-invariant local monomials that are bilinear in the listed left-handed Weyl fields, linear in H or $\widetilde{H}$, and of engineering dimension at most four, up to flavor tensors and Hermitian conjugation. Lorentz invariance forces the two left-handed Weyl spinor indices in each fermion bilinear to be contracted by the antisymmetric invariant tensor of $SL(2,\mathbb{C})$. Color invariance requires Q_L to pair with either u^c or d^c, because the quark singlet contraction is $\mathbf{3}\otimes\overline{\mathbf{3}}\to\mathbf{1}$; the lepton bilinear is already color neutral. Weak $SU(2)$-invariance then forces the remaining weak doublet index to be contracted with either H or $\widetilde{H}$ by the invariant tensor of the doublet representation.

The hypercharge sums determine which Higgs doublet occurs:

$$\frac{1}{6}+\frac{1}{2}-\frac{2}{3}=0,\qquad \frac{1}{6}-\frac{1}{2}+\frac{1}{3}=0,\qquad -\frac{1}{2}-\frac{1}{2}+1=0.$$

Thus Q_LHu^c, $Q_L\widetilde{H}d^c$, and $L_L\widetilde{H}e^c$ are gauge invariant. The opposite choices of H and $\widetilde{H}$ have nonzero hypercharge, hence do not define local gauge-invariant monomials in the left-handed convention. Power counting excludes additional renormalizable matter monomials with more fields or derivatives in this minimal field content. Flavor indices are spectators for the local charge algebra, so the allowed coefficients are the three arbitrary complex flavor matrices shown above.

Flavor rotations of the fermion fields remove unphysical parameters; the remaining charged-current quark mixing is encoded in the CKM matrix. If neutrino masses are included through dimension-five or singlet-neutrino operators, a separate PMNS mixing matrix appears in the lepton sector.

54.2 Flavor Coordinates and Charged Currents

The Yukawa matrices are not themselves the physical flavor parameters. A physical statement must specify which field redefinitions have already been used and which current is being observed.

Definition 54.6 (Quark mass coordinates and CKM matrix)**.** After evaluating the Higgs field at H_0, define the complex quark mass matrices

$$M_u=\frac{v}{\sqrt{2}}Y_u,\qquad M_d=\frac{v}{\sqrt{2}}Y_d.$$

Choose singular-value decompositions

$$(U_L^u)^\dagger M_uU_R^u=D_u,\qquad (U_L^d)^\dagger M_dU_R^d=D_d, \tag{54.2}$$

where $U_L^u,U_R^u,U_L^d,U_R^d\in U(3)$ and D_u,D_d are diagonal with nonnegative entries. The Cabibbo–Kobayashi–Maskawa matrix is

$$V_{\mathrm{CKM}}:=(U_L^u)^\dagger U_L^d. \tag{54.3}$$

It is the coefficient matrix in the charged weak current written in quark mass coordinates:

$$J_\mu^+ = \overline{u}_{L,a}\gamma_\mu (V_{\text{CKM}})_{ab} d_{L,b}, \qquad J_\mu^- = (J_\mu^+)^\dagger.$$

Here u_L, d_L denote mass-basis left-handed quark fields. The neutral gauge currents remain diagonal in flavor in these coordinates.

Flavor parameter counting. For n quark generations with nondegenerate masses, the pair of complex Yukawa matrices (Y_u, Y_d) contains

$$2n \quad \text{quark masses,} \qquad \frac{n(n-1)}{2} \quad \text{mixing angles,} \qquad \frac{(n-1)(n-2)}{2} \quad \text{CP-odd phases.}$$

In particular, two generations allow one mixing angle and no CKM CP phase, while three generations allow three angles and one CKM CP phase. The two complex $n \times n$ Yukawa matrices contain $4n^2$ real parameters. In the absence of Yukawa couplings the quark kinetic and gauge terms have flavor symmetry

$$U(n)_Q \times U(n)_u \times U(n)_d.$$

The Yukawa matrices break this group to the diagonal baryon-number $U(1)_B$, which does not change Y_u or Y_d. Therefore $3n^2 - 1$ real parameters are removable by flavor-coordinate choices. The physical quark-sector parameter count is

$$4n^2 - (3n^2 - 1) = n^2 + 1.$$

Among these are $2n$ nonnegative singular values, namely the quark masses. Thus the mixing matrix carries

$$n^2 + 1 - 2n = (n-1)^2$$

parameters. A unitary $n \times n$ matrix has n^2 real parameters. After the masses have been diagonalized, the $2n$ quark mass eigenfields may be rephased, but the common baryon-number rephasing leaves V_{CKM} unchanged. Hence $2n - 1$ phases are unphysical in V_{CKM}. The remaining $(n-1)^2$ parameters split into $n(n-1)/2$ real rotation angles and

$$(n-1)^2 - \frac{n(n-1)}{2} = \frac{(n-1)(n-2)}{2}$$

phases.

In the quark mass basis of Definition 54.6, the neutral electromagnetic, gluon, and Z-boson currents are flavor diagonal at tree level. Flavor change in the renormalizable gauge currents is confined to the charged current and is controlled by V_{CKM}. The gluon and photon couple to generators proportional to the identity in generation space. A unitary change of mass coordinates therefore gives

$$\overline{q}_L U_L^\dagger \gamma_\mu T U_L q_L = \overline{q}_L \gamma_\mu T q_L$$

whenever T acts only on color or electric charge and not on generation. The Z-current is a linear combination of T_L^3 and Q_{em}. Within the up sector and within the down sector these charges are again generation-independent, so the same unitarity cancellation applies. The charged current is different because it pairs the up component of Q_L with the down component before the two independent left singular-vector matrices have been chosen. Its flavor matrix is therefore $(U_L^u)^\dagger U_L^d = V_{\text{CKM}}$.

Definition 54.7 (Jarlskog invariant)**.** For three generations, a rephasing-invariant CKM CP-odd coordinate is

$$J_{\mathrm{CKM}} = \operatorname{Im}\left(V_{us}V_{cb}V_{ub}^{*}V_{cs}^{*}\right).$$

Equivalently, any expression $\operatorname{Im}(V_{ij}V_{kl}V_{il}^{*}V_{kj}^{*})$ with distinct i, k and distinct j, l is a rephasing-invariant quartet. For a three-by-three unitary matrix these quartets are related by unitarity and carry the same magnitude, with signs fixed by row and column orientation.

Rephasing invariance of J_{CKM}. Under mass-basis quark rephasings

$$V_{ij} \longmapsto \mathrm{e}^{\mathrm{i}\alpha_i}V_{ij}\mathrm{e}^{-\mathrm{i}\beta_j},$$

the quantity J_{CKM} is invariant. For $n = 2$ the corresponding quartet invariant vanishes after rephasing, so a two-generation charged current has no intrinsic CKM CP phase. The phase of the quartet

$$V_{us}V_{cb}V_{ub}^{*}V_{cs}^{*}$$

is multiplied by

$$\mathrm{e}^{\mathrm{i}\alpha_u}\mathrm{e}^{-\mathrm{i}\beta_s}\mathrm{e}^{\mathrm{i}\alpha_c}\mathrm{e}^{-\mathrm{i}\beta_b}\mathrm{e}^{-\mathrm{i}\alpha_u}\mathrm{e}^{\mathrm{i}\beta_b}\mathrm{e}^{-\mathrm{i}\alpha_c}\mathrm{e}^{\mathrm{i}\beta_s} = 1.$$

Thus its imaginary part is a coordinate-independent CP-odd function of the charged current. The equality of magnitudes of the different three-generation quartets follows directly from row and column orthogonality. For example, if $i \neq k$ and j, l, n are the three column labels, row orthogonality gives

$$V_{ij}V_{kj}^{*} + V_{il}V_{kl}^{*} + V_{in}V_{kn}^{*} = 0.$$

Multiplying by $V_{kl}V_{il}^{*}$ and taking imaginary parts gives

$$\operatorname{Im}(V_{ij}V_{kl}V_{il}^{*}V_{kj}^{*}) = -\operatorname{Im}(V_{in}V_{kl}V_{il}^{*}V_{kn}^{*}).$$

Column orthogonality gives the analogous relation for fixed column pair and varying rows. Combining the row and column relations moves any quartet with distinct row labels and distinct column labels to any other such quartet; interchanging the ordered row pair or the ordered column pair changes the sign. For two generations, the flavor-counting paragraph above shows that all phases in a 2×2 unitary charged-current matrix are removable by quark-field rephasings. A representative can therefore be chosen real, and every quartet imaginary part is zero in that coordinate.

54.3 Electroweak Breaking and Mass Coordinates

The word "breaking" in this section refers to a local field coordinate around a Higgs background and to the stabilizer of that background. Gauge redundancy is not a global symmetry acting faithfully on the physical Hilbert space, and it is not literally broken. The physical content of the Brout–Englert–Higgs phase is the spectrum, currents, and gauge-invariant operator algebra described in the renormalized perturbative construction after gauge fixing or BV treatment, or in a nonperturbative construction if one has been supplied. The formulas below are the tree-level mass and coupling coordinates of that local electroweak chart.

Definition 54.8 (Electroweak scalar chart)**.** The minimal renormalizable electroweak scalar potential is

$$V(H) = \lambda\left(H^\dagger H - \frac{v^2}{2}\right)^2, \qquad \lambda > 0.$$

In the local field-space chart used below,

$$H(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}\chi_1(x) + \mathrm{i}\chi_2(x)\\ v + h(x) + \mathrm{i}\chi_3(x)\end{pmatrix}.$$

The reference vacuum vector is

$$H_0 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}$$

in the same chart. Here $v > 0$ and $\lambda > 0$ are renormalized coordinates in the perturbative electroweak construction. The fields χ_1, χ_2, χ_3 are local coordinates tangent to the gauge orbit through H_0, while h is the radial scalar coordinate.

Proposition 54.9 (Electromagnetic stabilizer and weak-boson masses)**.** *In the convention of Convention 54.1, the stabilizer of H_0 inside the local group $SU(2)_L \times U(1)_Y$ is generated by*

$$Q_{\mathrm{em}} = T_L^3 + Y.$$

The quadratic gauge-boson mass coordinates are

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(\mathcal{W}_\mu^1 \mp \mathrm{i}\mathcal{W}_\mu^2), \qquad Z_\mu = \frac{g_2\mathcal{W}_\mu^3 - g_1\mathcal{B}_\mu}{\sqrt{g_1^2+g_2^2}}, \qquad A_\mu = \frac{g_1\mathcal{W}_\mu^3 + g_2\mathcal{B}_\mu}{\sqrt{g_1^2+g_2^2}},$$

with

$$m_W = \frac{g_2 v}{2}, \qquad m_Z = \frac{v}{2}\sqrt{g_1^2+g_2^2}, \qquad m_A = 0.$$

The coupling of the massless field is

$$e = g_2\sin\theta_W = g_1\cos\theta_W, \qquad \tan\theta_W = \frac{g_1}{g_2}.$$

Proof. The weak generator $T_L^3 = \mathrm{diag}(1/2, -1/2)$ acts on H_0 by $-H_0/2$, while the hypercharge generator acts by $H_0/2$. Hence $(T_L^3 + Y)H_0 = 0$. The generators T_L^1 and T_L^2 move H_0 into the upper component, and a general combination $aT_L^3 + bY$ annihilates H_0 only when $a = b$. Therefore the one-dimensional stabilizer Lie algebra is spanned by Q_{em}.

The covariant derivative in the explicit-coupling convention is

$$D_\mu H = \left(\partial_\mu - \mathrm{i}g_2\mathcal{W}_\mu^a\tau^a - \mathrm{i}g_1\mathcal{B}_\mu Y\right)H.$$

Using

$$\tau^1 H_0 = \frac{v}{2\sqrt{2}}\begin{pmatrix}1\\0\end{pmatrix}, \qquad \tau^2 H_0 = \frac{v}{2\sqrt{2}}\begin{pmatrix}-\mathrm{i}\\0\end{pmatrix}, \qquad \tau^3 H_0 = -\frac{1}{2}H_0, \qquad YH_0 = \frac{1}{2}H_0,$$

the quadratic term from $(D_\mu H)^\dagger(D^\mu H)$ at $H = H_0$ is

$$(D_\mu H_0)^\dagger(D^\mu H_0) = \frac{v^2}{8}\left[g_2^2(\mathcal{W}_\mu^1\mathcal{W}^{1\mu} + \mathcal{W}_\mu^2\mathcal{W}^{2\mu}) + (-g_2\mathcal{W}_\mu^3 + g_1\mathcal{B}_\mu)^2\right].$$

The charged part is $(g_2^2 v^2/4)W_\mu^+ W^{-\mu}$, so $m_W^2 = g_2^2 v^2/4$. The neutral quadratic form has rank one: the massive unit vector is Z_μ, the null vector is A_μ, and comparison with the real-vector mass term $\frac{1}{2}m_Z^2 Z_\mu Z^\mu$ gives $m_Z^2 = (g_1^2+g_2^2)v^2/4$.

Finally invert the neutral rotation:

$$\mathcal{W}_\mu^3 = \frac{g_2 Z_\mu + g_1 A_\mu}{\sqrt{g_1^2+g_2^2}}, \qquad \mathcal{B}_\mu = \frac{-g_1 Z_\mu + g_2 A_\mu}{\sqrt{g_1^2+g_2^2}}.$$

The coefficient of A_μ in $g_2 \mathcal{W}_\mu^3 T_L^3 + g_1 \mathcal{B}_\mu Y$ is

$$\frac{g_1 g_2}{\sqrt{g_1^2+g_2^2}}\,(T_L^3+Y).$$

This proves the displayed electromagnetic coupling and identifies the massless gauge field with the connection for the unbroken stabilizer. □

This is the nonabelian mass-matrix construction of Chapter 45 applied to the $(\mathbf{2})_{1/2}$ scalar representation. The three coordinates χ_1, χ_2, χ_3 are gauge-orbit coordinates in a local chart; in an R_ξ gauge they remain as unphysical fields paired by BRST, while in a unitary-gauge chart they are removed locally.

Higgs radial mass coordinate. In the scalar chart of Definition 54.8, the potential restricted to the radial coordinate is

$$V(h) = \lambda\left(vh + \frac{h^2}{2}\right)^2 = \lambda v^2 h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4.$$

Consequently the radial scalar mass coordinate is

$$m_h^2 = 2\lambda v^2.$$

The cubic and quartic self-interaction coefficients in the potential are λv and $\lambda/4$, respectively, in this field coordinate. In the unitary local chart $H = (0, (v+h)/\sqrt{2})^T$,

$$H^\dagger H - \frac{v^2}{2} = vh + \frac{h^2}{2}.$$

Substitution into $V(H)$ gives the displayed polynomial. The Lorentzian Lagrangian contains $-V$, while the standard real-scalar mass term is $-\frac{1}{2}m_h^2 h^2$. Comparing the quadratic coefficients gives $m_h^2 = 2\lambda v^2$.

Custodial $SU(2)$ and the tree-level ρ coordinate. Set

$$\Sigma := (\widetilde{H}, H).$$

The scalar kinetic term with $g_1 = 0$ and the scalar potential are invariant under

$$\Sigma \longmapsto U_L \Sigma U_R^\dagger, \qquad U_L, U_R \in SU(2),$$

and the vacuum $\Sigma_0 = (v/\sqrt{2})\mathbf{1}_2$ preserves the diagonal subgroup $SU(2)_V$. With hypercharge gauged as the T^3_R subgroup, the tree-level mass ratio

$$\rho_{\text{tree}} := \frac{m_W^2}{m_Z^2 \cos^2 \theta_W}$$

is equal to 1. The identity

$$\frac{1}{2}\,\mathrm{tr}(\Sigma^\dagger \Sigma) = H^\dagger H$$

shows that the scalar potential depends only on $\mathrm{tr}(\Sigma^\dagger\Sigma)$, hence is invariant under left and right multiplication by unitary matrices. With $g_1 = 0$, the covariant derivative acts only by left $SU(2)_L$ multiplication, so the right $SU(2)_R$ remains a global symmetry of the scalar kinetic term. The vacuum equation $U_L \Sigma_0 U_R^\dagger = \Sigma_0$ is equivalent to $U_L = U_R$, giving the diagonal stabilizer.

The hypercharge gauge field gauges only the T^3_R direction and therefore breaks the global $SU(2)_R$ explicitly, but the mass formulae of Proposition 54.9 still give

$$\cos\theta_W = \frac{g_2}{\sqrt{g_1^2 + g_2^2}}, \qquad m_W^2 = \frac{g_2^2 v^2}{4}, \qquad m_Z^2 = \frac{(g_1^2 + g_2^2)v^2}{4}.$$

Substitution gives $\rho_{\text{tree}} = 1$. Radiative corrections and higher-dimensional operators are separate coordinates of the electroweak EFT; they are not part of this tree-level identity.

54.4 Gauge-Anomaly Cancellation

Definition 54.10 (Gauge-anomaly coefficients for the local field content)**.** For a collection of left-handed Weyl fermions in representations R_i of $\mathfrak{g}_{\mathrm{SM}}$, the perturbative gauge-anomaly tensor is the symmetric cubic functional

$$\mathcal{A}(X, Y, Z) = \sum_i \mathrm{tr}_{R_i}\big(X\{Y, Z\}\big), \qquad X, Y, Z \in \mathfrak{g}_{\mathrm{SM}},$$

with spectator multiplicities included in the trace. For a mixed $G^2U(1)_Y$ component, this reduces to the sum of hypercharge times the quadratic index of the G-representation. For the mixed gravitational-$U(1)_Y$ anomaly, the coefficient is the sum of hypercharges over all left-handed Weyl components, again including spectator multiplicities.

Proposition 54.11 (SM anomaly cancellation)**.** *For the one-generation representation of Definition 54.2, all perturbative local gauge anomalies and the mixed gravitational-hypercharge anomaly vanish. The finite $SU(2)_L$ anomaly also vanishes generation by generation.*

Proof. The calculation uses only the representation table and the fact that all fermions have been written as left-handed Weyl fields. For the pure $SU(3)_c^3$ anomaly, the two weak components of Q_L contribute two fundamental color representations, while u^c and d^c contribute two antifundamentals. Since the cubic index changes sign under complex conjugation,

$$2A(\mathbf{3}) - A(\mathbf{3}) - A(\mathbf{3}) = 0.$$

The pure $SU(2)_L^3$ perturbative anomaly vanishes because the symmetric cubic invariant d^{abc} of $\mathfrak{su}(2)$ is zero. Mixed $SU(3)_c U(1)_Y^2$ and $SU(2)_L U(1)_Y^2$ components vanish because they contain $\mathrm{tr}_R(T^a)$ for a semisimple generator.

For $SU(3)_c^2 U(1)_Y$, one generation gives

$$\mathcal{A}[SU(3)^2 U(1)_Y] = 2\left(\frac{1}{6}\right) T(\mathbf{3}) + \left(-\frac{2}{3}\right) T(\overline{\mathbf{3}}) + \left(\frac{1}{3}\right) T(\overline{\mathbf{3}}) = 0.$$

Here $T(\overline{\mathbf{3}}) = T(\mathbf{3})$, and the factor 2 is the weak doublet multiplicity. For $SU(2)_L^2 U(1)_Y$,

$$\mathcal{A}[SU(2)^2 U(1)_Y] = 3\left(\frac{1}{6}\right) T(\mathbf{2}) + \left(-\frac{1}{2}\right) T(\mathbf{2}) = 0.$$

Here the factor 3 is the color multiplicity of Q_L. For the cubic hypercharge anomaly, include the full dimensions of the nonabelian representations:

$$\begin{aligned}\mathcal{A}[U(1)_Y^3] &= 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1^3 \\ &= 0.\end{aligned}$$

The mixed gravitational-hypercharge anomaly coefficient is

$$6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 = 0.$$

Finally, the finite $SU(2)$ anomaly of Chapter 55 is absent because each generation contains four left-handed $SU(2)_L$ doublets: three colored quark doublets and one lepton doublet. The number of doublets is therefore even. These equalities are finite representation-theoretic constraints on the regulated fermion determinant line; they are not consequences of perturbative renormalizability alone. □

54.5 Baryon and Lepton Number as Global Data

Definition 54.12 (Classical B, L, and $B-L$ charges)**.** In the left-handed Weyl convention, assign baryon and lepton charges

field	B	L
Q_L	$1/3$	0
u^c	$-1/3$	0
d^c	$-1/3$	0
L_L	0	1
e^c	0	-1
ν^c, if included	0	$-1.$

The signs for conjugate fields are the charges of the left-handed conjugate fields themselves. These assignments are statements about global-current coordinates in the local Lagrangian; whether a global symmetry survives in the quantum theory is determined by its anomaly polynomial and by possible operator insertions.

$B+L$ anomaly and $B-L$ mixed-gauge cancellation. For one generation of the minimal representation without ν^c, the mixed $SU(2)_L^2U(1)_B$ and $SU(2)_L^2U(1)_L$ anomaly coefficients are both nonzero:

$$\mathcal{A}[SU(2)_L^2U(1)_B] = T(\mathbf{2}), \qquad \mathcal{A}[SU(2)_L^2U(1)_L] = T(\mathbf{2}).$$

Their difference vanishes, so $B-L$ has no mixed $SU(2)_L^2$ anomaly. The mixed $SU(3)_c^2U(1)_{B-L}$ anomaly also vanishes. The gravitational and cubic anomalies of $B-L$ vanish generation by generation only after a gauge-singlet ν^c with $B-L=1$ is included. The $SU(2)^2U(1)$ coefficient is the sum of the global charge times the quadratic index of each weak doublet, with color multiplicity included. For baryon number only Q_L contributes, giving

$$3\left(\frac{1}{3}\right)T(\mathbf{2}) = T(\mathbf{2}).$$

For lepton number only L_L contributes, giving $T(\mathbf{2})$. Thus $B+L$ is anomalous and $B-L$ cancels in the $SU(2)^2$ channel. For $SU(3)^2(B-L)$, the common $T(\mathbf{3})$ factors out and the coefficient is

$$2\left(\frac{1}{3}\right) - \frac{1}{3} - \frac{1}{3} = 0,$$

where the factor 2 is the weak multiplicity of Q_L.

For the mixed gravitational anomaly of $B-L$, including multiplicities but not ν^c, one obtains

$$6\left(\frac{1}{3}\right) + 3\left(-\frac{1}{3}\right) + 3\left(-\frac{1}{3}\right) + 2(-1) + 1 = -1.$$

The cubic $U(1)_{B-L}^3$ coefficient gives the same value:

$$6\left(\frac{1}{3}\right)^3 + 3\left(-\frac{1}{3}\right)^3 + 3\left(-\frac{1}{3}\right)^3 + 2(-1)^3 + 1^3 = -1.$$

Adding a singlet ν^c with $B-L=1$ contributes $+1$ to each of these two anomaly coefficients and cancels them. This is a finite anomaly calculation, not a statement that a singlet neutrino has been observed or that one has chosen a particular neutrino-mass mechanism.

Remark 54.13 (Sphalerons as transitions between electroweak sectors). The $SU(2)_L$ anomaly means that an electroweak gauge-field history with topological charge Q_{top} changes the global charges by $\Delta B = \Delta L = N_g Q_{\text{top}}$, where N_g is the number of generations, while $\Delta(B-L)=0$. This statement assumes a field history whose boundary conditions make the $SU(2)$ second Chern number an integer; on more general backgrounds the corresponding statement must be formulated in terms of the anomaly inflow class. The sphaleron is the unstable static configuration at the top of the minimal barrier between neighboring electroweak sectors, as described in Chapter 45. Rates in a thermal state require real-time finite-temperature dynamics and are not fixed by the topological selection rule alone.

54.6 Higher-Dimensional Operator Coordinates

Definition 54.14 (SMEFT coupling chart)**.** Fix a regulator, a local operator basis, and a subtraction scheme. The corresponding Standard-Model effective coupling chart below a scale Λ is the local functional

$$S_{\mathrm{EFT}} = S_{\mathrm{SM}}^{\mathrm{ren}} + \sum_{d>4} \Lambda^{4-d} \sum_{A\in\mathcal{B}_d} C_A^{(d)}(\mu) \int_{M_4} \mathcal{O}_A^{(d)}(\mu), \tag{54.4}$$

where $\mathcal{B}_d$ is a specified basis of gauge-invariant local operators of engineering dimension d, after quotienting by total derivatives and by the chosen field-redefinition equations of motion. The symbol Λ is a reference scale used to write dimensionless Wilson coordinates in an effective theory beyond the renormalizable Standard Model. It is not a hidden cutoff parameter of Standard Model predictions. If the same expansion is interpreted as a Wilsonian theory with a physical cutoff, that interpretation is additional regulator and matching data. The coordinates $C_A^{(d)}(\mu)$ are not observables by themselves; an observable uses them together with the operator scheme, matching map, field domain, and matrix elements.

Definition 54.15 (Weinberg operator in the left-handed convention)**.** With only the minimal fields of Definitions 54.2 and 54.3, the derivative-free dimension-five lepton-number-violating operator is

$$\mathcal{O}_5^{IJ} = \epsilon^{\alpha\beta}\big(\epsilon_{ij}L^I_{\alpha i}H_j\big)\big(\epsilon_{k\ell}L^J_{\beta k}H_\ell\big), \tag{54.5}$$

with symmetric flavor coefficient $C_5^{IJ} = C_5^{JI}$. In the convention

$$\Delta\mathcal{L}_5 = -\frac{1}{2\Lambda}C_5^{IJ}\mathcal{O}_5^{IJ} + \text{Hermitian conjugate}, \tag{54.6}$$

the neutral-lepton Majorana mass matrix in the electroweak scalar chart is

$$(M_\nu)^{IJ} = \frac{v^2}{2\Lambda}C_5^{IJ}. \tag{54.7}$$

To verify (54.7), set $H = H_0$. The weak contraction $\epsilon_{ij}L^I_iH_j$ selects the neutral component of L^I and gives

$$\epsilon_{ij}L^I_{\alpha i}H_{0,j} = \frac{v}{\sqrt{2}}\,\nu^I_\alpha.$$

Substitution into (54.6) gives

$$\Delta\mathcal{L}_5 = -\frac{1}{2}\left(\frac{v^2}{2\Lambda}C_5^{IJ}\right)\epsilon^{\alpha\beta}\nu^I_\alpha\nu^J_\beta + \text{Hermitian conjugate},$$

which is the displayed Majorana mass coordinate. The symmetry of C_5^{IJ} follows because the Lorentz contraction of two left-handed Weyl fields is symmetric in the flavor labels after the anticommutation of the fields is combined with the antisymmetry of $\epsilon^{\alpha\beta}$.

Integrating out singlet neutrinos at tree level. Add gauge-singlet left-handed fields ν^{cA} with

$$\Delta\mathcal{L} = -Y_\nu^{IA}(L^IH)\nu^{cA} - \frac{1}{2}M_{AB}\nu^{cA}\nu^{cB} + \text{Hermitian conjugate},$$

where M is a nonsingular complex symmetric matrix and spinor and weak indices are contracted as in (54.5). At tree level and to leading order in external momenta over the singular values of M, the dimension-five coefficient obeys

$$\frac{C_5}{\Lambda} = Y_\nu M^{-1}Y_\nu^T$$

up to the conventional overall sign that can be changed by rephasing Majorana fields. The algebraic equation for the heavy singlet field, after suppressing derivative corrections, is

$$M_{AB}\nu^{cB} + Y_\nu^{IA}(L^I H) = 0.$$

Thus

$$\nu^{cA} = -(M^{-1})^{AB} Y_\nu^{IB}(L^I H).$$

Substituting this solution back into the quadratic expression in ν^c gives a local operator quadratic in (LH) with coefficient $\frac{1}{2}Y_\nu M^{-1} Y_\nu^T$, with the sign depending on the convention used to order the Grassmann variables in the Majorana bilinear. Comparing with Definition 54.15 gives the displayed matching coordinate. The statement is a tree-level EFT matching computation, not a claim that singlet neutrinos are part of the minimal Standard Model field content.

Definition 54.16 (Dimension-six quotient space)**.** Let $\mathcal{V}_6$ be the complex vector space of local polynomial gauge-invariant densities of engineering dimension six built from

$$G^A_{\mu\nu}, \quad W^I_{\mu\nu}, \quad B_{\mu\nu}, \quad H, \quad D_\mu, \quad q, \ell, u, d, e,$$

where q, ℓ, u, d, e denote the standard chiral Standard-Model fields in four-component notation. This notation is a bookkeeping device for this basis: q, ℓ are the left-handed fields Q_L, L_L, while u, d, e are the right-handed conjugates of the left-handed Weyl fields (u^c, d^c, e^c) used in Definition 54.2. Let $\partial\mathcal{V}_5 \subset \mathcal{V}_6$ be total derivatives and let $\mathcal{E}_6 \subset \mathcal{V}_6/\partial\mathcal{V}_5$ be the subspace generated by

$$\frac{\delta S^{\mathrm{ren}}_{\mathrm{SM}}}{\delta\Phi}\,\mathcal{F}_\Phi + \text{Hermitian conjugate,} \tag{54.8}$$

where Φ runs over the SM fields and $\mathcal{F}_\Phi$ is any gauge-covariant local polynomial of dimension $6 - \dim(\delta S^{\mathrm{ren}}_{\mathrm{SM}}/\delta\Phi)$. A dimension-six SMEFT basis is an ordered basis of the quotient

$$\mathcal{Q}_6 := \mathcal{V}_6/(\partial\mathcal{V}_5 + \mathcal{E}_6), \tag{54.9}$$

together with a Hermiticity convention and a flavor-index convention for its coefficients.

Equation-of-motion operators as coordinates. Let $S^{\mathrm{ren}}_{\mathrm{SM}}$ be the renormalized SM action in the chosen regularization and subtraction scheme. Assume the regulator treats the near-identity local field change below as an allowed coordinate change, with its Jacobian either trivial in dimensional regularization or absorbed into local counterterms in a local cutoff scheme. The infinitesimal local change of field coordinates

$$\Phi \longmapsto \Phi + \Lambda^{-2}\mathcal{F}_\Phi(\Phi) \tag{54.10}$$

changes the action, to order Λ^{-2}, by an element of $\mathcal{E}_6$. Therefore coefficients multiplying elements of $\mathcal{E}_6$ are not invariant coordinates of the effective theory; they are removed by the quotient in Definition 54.16.

Taylor expansion of the action as a functional gives

$$S^{\mathrm{ren}}_{\mathrm{SM}}[\Phi + \Lambda^{-2}\mathcal{F}_\Phi] = S^{\mathrm{ren}}_{\mathrm{SM}}[\Phi] + \Lambda^{-2}\sum_\Phi \int_{M_4} \frac{\delta S^{\mathrm{ren}}_{\mathrm{SM}}}{\delta\Phi(x)}\mathcal{F}_\Phi(x) + O(\Lambda^{-4}).$$

Boundary terms produced by moving derivatives between $\delta S/\delta\Phi$ and $\mathcal{F}_\Phi$ lie in $\partial\mathcal{V}_5$. Thus the order-Λ^{-2} shift is exactly of the form (54.8). The statement is a coordinate statement inside a specified renormalized perturbative or regulated effective path integral; it does not say that the corresponding composite operator insertion vanishes as a distribution without specifying the operator-renormalization scheme. If the chosen scheme produces a local Jacobian for the field redefinition, that Jacobian changes local coupling coordinates in the same EFT chart; it does not make the equation-of-motion operator an independent observable.

Definition 54.17 (Warsaw-type dimension-six basis)**.** For a baryon-number preserving coupling chart with no singlet neutrino fields, one convenient ordered basis of $\mathcal{Q}_6$ is the Warsaw-type basis. Write

$$\widetilde{X}_{\mu\nu} := \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}X^{\rho\sigma}, \qquad H^\dagger i\overleftrightarrow{D}_\mu H := iH^\dagger D_\mu H - i(D_\mu H)^\dagger H,$$

and

$$H^\dagger i\overleftrightarrow{D}^I_\mu H := iH^\dagger\tau^I D_\mu H - i(D_\mu H)^\dagger\tau^I H.$$

The bosonic part contains the following fifteen one-generation operator types:

$$\begin{aligned}
X^3:&\quad O_G,\ O_{\widetilde{G}},\ O_W,\ O_{\widetilde{W}},\\
H^6:&\quad O_H = (H^\dagger H)^3,\\
H^4D^2:&\quad O_{H\Box} = (H^\dagger H)\Box(H^\dagger H),\quad O_{HD} = |H^\dagger D_\mu H|^2,\\
X^2H^2:&\quad O_{HG}, O_{H\widetilde{G}}, O_{HW}, O_{H\widetilde{W}}, O_{HB}, O_{H\widetilde{B}}, O_{HWB}, O_{H\widetilde{W}B}.
\end{aligned}$$

Here

$$\begin{aligned}
O_G = f^{ABC}G^{A\nu}_\mu G^{B\rho}_\nu G^{C\mu}_\rho,&\quad O_{\widetilde{G}} = f^{ABC}\widetilde{G}^{A\nu}_\mu G^{B\rho}_\nu G^{C\mu}_\rho,\\
O_W = \epsilon^{IJK}W^{I\nu}_\mu W^{J\rho}_\nu W^{K\mu}_\rho,&\quad O_{\widetilde{W}} = \epsilon^{IJK}\widetilde{W}^{I\nu}_\mu W^{J\rho}_\nu W^{K\mu}_\rho,
\end{aligned}$$

and $O_{HX} = (H^\dagger H)X_{\mu\nu}X^{\mu\nu}$, $O_{H\widetilde{X}} = (H^\dagger H)\widetilde{X}_{\mu\nu}X^{\mu\nu}$ for $X = G, W, B$, while

$$O_{HWB} = (H^\dagger\tau^I H)W^I_{\mu\nu}B^{\mu\nu}, \qquad O_{H\widetilde{W}B} = (H^\dagger\tau^I H)\widetilde{W}^I_{\mu\nu}B^{\mu\nu}.$$

The two-fermion part contains nineteen operator types:

$$\begin{aligned}
\psi^2H^3:&\quad O_{eH}, O_{uH}, O_{dH},\\
\psi^2H^2D:&\quad O^{(1)}_{H\ell}, O^{(3)}_{H\ell}, O^{(1)}_{Hq}, O^{(3)}_{Hq}, O_{He}, O_{Hu}, O_{Hd}, O_{Hud},\\
\psi^2XH:&\quad O_{eB}, O_{eW}, O_{uG}, O_{uW}, O_{uB}, O_{dG}, O_{dW}, O_{dB}.
\end{aligned}$$

The first line means the Yukawa monomials of Definition 54.3 multiplied by $(H^\dagger H)$. The H^2D line is generated by the templates

$$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar\psi\gamma^\mu\psi), \qquad (H^\dagger i\overleftrightarrow{D}^I_\mu H)(\bar\psi\tau^I\gamma^\mu\psi),$$
$$(\widetilde{H}^\dagger iD_\mu H)(\bar u\gamma^\mu d),$$

with the displayed labels specifying the allowed SM representations. The dipole line means $(\bar\psi\sigma^{\mu\nu}\psi')HX_{\mu\nu}$, with H or $\widetilde{H}$ chosen so that the monomial has the same gauge quantum numbers as the corresponding Yukawa monomial.

The baryon-number preserving four-fermion part contains twenty-five one-generation operator types:

$$\begin{aligned}
LL:&\quad O_{\ell\ell}, O_{qq}^{(1)}, O_{qq}^{(3)}, O_{\ell q}^{(1)}, O_{\ell q}^{(3)},\\
RR:&\quad O_{ee}, O_{uu}, O_{dd}, O_{eu}, O_{ed}, O_{ud}^{(1)}, O_{ud}^{(8)},\\
LR_{\text{vector}}:&\quad O_{\ell e}, O_{\ell u}, O_{\ell d}, O_{qe}, O_{qu}^{(1)}, O_{qu}^{(8)}, O_{qd}^{(1)}, O_{qd}^{(8)},\\
LR_{\text{scalar/tensor}}:&\quad O_{\ell edq}, O_{\ell equ}^{(1)}, O_{\ell equ}^{(3)}, O_{quqd}^{(1)}, O_{quqd}^{(8)}.
\end{aligned}$$

Flavor labels I, J, K, L are part of the coefficient coordinates. When flavor labels are restored, Hermiticity and exchange symmetries must be fixed before counting real parameters. Thus the basis is an ordered coordinate system with specified flavor and reality conventions.

Definition 54.18 (Baryon-violating dimension-six classes)**.** If baryon number is not imposed as a restriction on the effective couplings, four additional gauge-invariant dimension-six classes are allowed. In standard chiral four-component notation they may be represented as

$$\begin{aligned}
O_{duq\ell} &= \epsilon_{abc}\epsilon_{ij}(d^aCu^b)(q^{ci}C\ell^j),\\
O_{qque} &= \epsilon_{abc}\epsilon_{ij}(q^{ai}Cq^{bj})(u^cCe),\\
O_{qqq\ell} &= \epsilon_{abc}\epsilon_{i\ell}\epsilon_{jk}(q^{ai}Cq^{bj})(q^{ck}C\ell^\ell),\\
O_{duue} &= \epsilon_{abc}(d^aCu^b)(u^cCe).
\end{aligned}$$

Each class carries flavor labels and changes B and L by one unit with $\Delta(B-L)=0$. In the minimal SM definition used here these coefficients are set to zero by imposing baryon number on the effective couplings; in a more general effective theory they are independent coordinates constrained by proton-decay observables.

Field-content exhaustion at dimension six. Assume the field dimensions

$$\dim H = 1, \qquad \dim D = 1, \qquad \dim X_{\mu\nu} = 2, \qquad \dim \psi = \frac{3}{2}.$$

Every Lorentz-invariant local monomial of total engineering dimension six, after integration by parts and algebraic identities but before imposing the equations of motion, has one of the following field contents:

$$X^3, \quad H^6, \quad H^4D^2, \quad X^2H^2, \quad \psi^2H^3, \quad \psi^2H^2D, \quad \psi^2XH, \quad \psi^4.$$

Consequently Definitions 54.17 and 54.18 are exhaustive as a dimension-six field-content classification. A Lorentz scalar must contain an even number of spinor fields, so the number of fermions is $0, 2$, or 4; six fermions already have dimension nine. With no fermions, nonnegative integer solutions of

$$n_H + n_D + 2n_X = 6$$

that cannot be reduced by integration by parts or by commutators of covariant derivatives are represented by H^6, H^4D^2, X^2H^2, and X^3. Terms with one field strength and only Higgs fields have uncontracted Lorentz indices unless derivatives are present; after commuting derivatives, their field-strength part is absorbed into the displayed classes or into the equation-of-motion quotient.

With two fermions, the remaining engineering dimension is three. The only possibilities are H^3, H^2D, and XH, giving ψ^2H^3, ψ^2H^2D, and ψ^2XH. With four fermions, the fermions already have dimension six, so no additional H, D, or X may appear. Gauge invariance and the finite Fierz identities then select the contractions listed in the basis definitions. The latter selection is a finite algebraic basis choice inside $\mathcal{Q}_6$, while the enumeration above establishes that no further field-content class exists.

54.7 QCD Theta Angle and Quark-Mass Phase

Proposition 54.19 (Physical strong-CP phase coordinate)**.** *In the quark sector with nonsingular mass matrices, the invariant phase coordinate involving the QCD theta angle and the quark mass matrices is*

$$\overline{\theta} = \theta_3 + \arg\det(M_u M_d) \quad \textit{mod } 2\pi. \tag{54.11}$$

Equivalently, replacing M_u, M_d by positive diagonal mass matrices by chiral flavor-coordinate changes shifts the coefficient of $\mathrm{tr}(F \wedge F)$ *by the opposite amount, leaving $\overline{\theta}$ unchanged.*

Proof. Work first with one massive Dirac quark in the fundamental color representation and write its mass as $me^{\mathrm{i}\varphi}$, $m > 0$. The field redefinition

$$q \longmapsto \exp\left(-\frac{\mathrm{i}\varphi}{2}\gamma_5\right) q$$

changes the mass term to $m\bar{q}q$. In the path integral this field redefinition is an anomalous axial rotation whose Jacobian is fixed by the chiral anomaly. With the anomaly convention fixed in Chapter 51, the above rotation shifts the coefficient of the QCD topological density by

$$\theta_3 \longmapsto \theta_3 + \varphi.$$

Thus removing the phase of the mass increases the theta coordinate by exactly that phase.

For several quarks, choose nonsingular complex mass matrices and perform singular-value decompositions

$$M_u = U_{uL}^{-1} D_u U_{uR}, \qquad M_d = U_{dL}^{-1} D_d U_{dR},$$

with D_u, D_d positive diagonal. The vector parts of these flavor coordinate changes have no QCD anomaly. The anomalous part is the product of the axial determinant phases, and the anomalous Jacobian is additive over the left-handed Weyl components in the fundamental color representation. The total phase removed from the two mass matrices is

$$\arg\det M_u + \arg\det M_d = \arg\det(M_u M_d) \qquad \text{mod } 2\pi.$$

Therefore the mass-diagonalizing coordinate change sends

$$\theta_3 \longmapsto \theta_3 + \arg\det(M_u M_d),$$

while the transformed positive diagonal mass matrices have zero determinant phase. The value assigned to the pair (θ_3, M_u, M_d) by $\theta_3 + \arg\det(M_u M_d)$ is therefore unchanged by this coordinate choice. Conversely, any further nonsingular chiral flavor coordinate change shifts θ_3 by the phase removed from the determinant of $M_u M_d$, so the same combination is the invariant coordinate on this mass chart. If one quark mass matrix has a zero singular value, the determinant phase is not a coordinate on the chart and the theta dependence is a different question; the proposition does not assume such a chart exists. □

The empirical bound on $\overline{\theta}$ is a bound on the coefficient of a specific CP-odd QCD operator in the Standard Model definition. It is not a principle for choosing parameters.

54.8 One-Loop Running in a Minimal Coupling Chart

Renormalization-group formulae in the Standard Model are statements about a renormalized coordinate system for couplings and masses. The equations below use dimensional regularization and minimal subtraction in a mass-independent unbroken-representation chart with three active generations and one Higgs doublet. Threshold matching, the broken-phase expansion, and higher-dimensional Wilson coefficients in an EFT extension are separate data. The explicit-coupling convention is the one fixed in Convention 54.1. Write

$$t := \log \mu, \qquad \beta_x := \frac{\mathrm{d}x}{\mathrm{d}t}.$$

The hypercharge coupling g_1 is the coupling to Y, not the grand-unified rescaling of it.

Lemma 54.20 (One-loop gauge beta coefficients)**.** *For three generations and one Higgs doublet, the one-loop gauge beta functions in the convention above are*

$$(4\pi)^2\beta_{g_1} = \frac{41}{6}g_1^3, \qquad (4\pi)^2\beta_{g_2} = -\frac{19}{6}g_2^3, \qquad (4\pi)^2\beta_{g_3} = -7g_3^3. \tag{54.12}$$

Proof. Use the one-loop coefficient in a gauge theory with left-handed Weyl fermions and complex scalars,

$$b_G = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_{\text{Weyl } f} T_G(R_f) + \frac{1}{3}\sum_{\text{complex } s} T_G(R_s), \qquad (4\pi)^2\beta_g = b_G g^3. \tag{54.13}$$

This is the background-field one-loop pole coefficient derived in Chapter 49; here we only evaluate the representation sums in the Standard Model coupling chart. For $U(1)_Y$ the same formula is read with $T_G(R)$ replaced by Y^2, including all nonabelian spectator multiplicities. One generation contributes

$$\begin{aligned}\sum_{\text{Weyl}} Y^2 &= 6\left(\frac{1}{6}\right)^2 + 3\left(-\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 2\left(-\frac{1}{2}\right)^2 + 1^2\\ &= \frac{10}{3}.\end{aligned}$$

Three generations give 10, and the Higgs doublet contributes

$$\sum_{\text{scalar}} Y^2 = 2\left(\frac{1}{2}\right)^2 = \frac{1}{2}.$$

Thus

$$b_1 = \frac{2}{3}\cdot 10 + \frac{1}{3}\cdot\frac{1}{2} = \frac{41}{6}.$$

For $SU(2)_L$, $C_2(SU(2)) = 2$ and $T(\mathbf{2}) = 1/2$ in the conventional half-trace normalization used for the explicit coupling g_2. One generation has three quark doublets and one lepton doublet, hence

$$\sum_{\text{Weyl}} T_{SU(2)} = 3 \cdot \frac{1}{2} + \frac{1}{2} = 2.$$

For three generations this is 6, and the Higgs doublet contributes 1/2. Therefore

$$b_2 = -\frac{11}{3} \cdot 2 + \frac{2}{3} \cdot 6 + \frac{1}{3} \cdot \frac{1}{2} = -\frac{19}{6}.$$

For $SU(3)_c$, $C_2(SU(3)) = 3$ and $T(\mathbf{3}) = T(\overline{\mathbf{3}}) = 1/2$. One generation contributes

$$2 \cdot \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2,$$

where the factor 2 is the weak multiplicity of Q_L. Three generations give 6, and there is no colored scalar in the minimal field content. Thus

$$b_3 = -\frac{11}{3} \cdot 3 + \frac{2}{3} \cdot 6 = -7.$$

□

Remark 54.21 (Hypercharge normalization). If $g_1^{\text{GUT}} := \sqrt{5/3}\, g_1$, then

$$(4\pi)^2 \beta_{g_1^{\text{GUT}}} = \frac{41}{10} (g_1^{\text{GUT}})^3.$$

This is only a coordinate change. The electroweak mass formulae in this chapter use the unrescaled g_1.

Controlled Approximation 54.22 (One-loop top-Higgs subsystem)**.** Restrict the minimal Standard Model coupling chart to the invariant subspace in which the only nonzero Yukawa singular value is the top coordinate y_t. In the same minimal-subtraction convention,

$$(4\pi)^2 \beta_{y_t} = y_t \left[\frac{9}{2} y_t^2 - 8 g_3^2 - \frac{9}{4} g_2^2 - \frac{17}{12} g_1^2 \right], \tag{54.14}$$

and

$$\begin{aligned} (4\pi)^2 \beta_\lambda = {} & 24\lambda^2 + 12\lambda y_t^2 - 6 y_t^4 - 3\lambda(3g_2^2 + g_1^2) \\ & + \frac{3}{8} \left[2g_2^4 + (g_2^2 + g_1^2)^2 \right]. \end{aligned} \tag{54.15}$$

The restriction to one Yukawa singular value is a coordinate restriction, not an estimate. In the left-handed matrix coordinates Y_u, Y_d, Y_e, the one-loop pole coefficients needed here are

$$\begin{gathered} (4\pi)^2 \beta_{Y_u} = Y_u \left[\frac{3}{2} \left(Y_u^\dagger Y_u - Y_d^\dagger Y_d \right) + T_Y - G_u \right], \\ T_Y := \operatorname{tr}\left(3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d + Y_e^\dagger Y_e \right), \qquad G_u := 8g_3^2 + \frac{9}{4} g_2^2 + \frac{17}{12} g_1^2, \end{gathered} \tag{54.16}$$

and

$$(4\pi)^2\beta_\lambda = 24\lambda^2 - 3\lambda(3g_2^2 + g_1^2) + \frac{3}{8}\big[2g_2^4 + (g_2^2 + g_1^2)^2\big] + 4\lambda T_Y - 2Q_Y, \tag{54.17}$$
$$Q_Y := \mathrm{tr}\Big(3(Y_u^\dagger Y_u)^2 + 3(Y_d^\dagger Y_d)^2 + (Y_e^\dagger Y_e)^2\Big).$$

Their scope is the one-loop minimal-subtraction pole calculation for the renormalizable electroweak coordinate chart with the active fields stated above. The primitive pole integrals are the logarithmic one-loop two-, three-, and four-point integrals already used in the renormalization chapters; the present chapter needs the group and multiplicity bookkeeping in this particular chart.

The coefficient accounting is as follows. The self-Yukawa part of (54.16) is the sum of the Yukawa vertex pole and the external-field pole terms. Gauge exchange and gauge wave-function pole terms give G_u. In the unrescaled $Q = T^3 + Y$ convention, those pole factors assemble to

$$G_u = 8g_3^2 + \frac{9}{4}g_2^2 + \frac{17}{12}g_1^2.$$

For β_λ, the scalar fish graphs give $24\lambda^2$; the gauge corrections linear in λ give $-3\lambda(3g_2^2 + g_1^2)$; the pure gauge four-Higgs pole gives the positive polynomial

$$\frac{3}{8}\left[2g_2^4 + (g_2^2 + g_1^2)^2\right];$$

the Yukawa wave-function and vertex poles give $4\lambda T_Y$; and the closed-fermion box gives $-2Q_Y$, with the minus sign from the fermion loop.

Equations (54.16)–(54.17) immediately show preservation of the one-singular-value subspace. If $Y_d = Y_e = 0$, their beta functions are proportional to Y_d and Y_e in the full matrix equations, so they remain zero. If Y_u has only one nonzero singular value, unitary flavor coordinates put it in the form

$$Y_u = \mathrm{diag}(0, 0, y_t).$$

Then $T_Y = 3y_t^2$, $Q_Y = 3y_t^4$, and (54.16) gives

$$(4\pi)^2\beta_{y_t} = y_t\left[\frac{3}{2}y_t^2 + 3y_t^2 - 8g_3^2 - \frac{9}{4}g_2^2 - \frac{17}{12}g_1^2\right],$$

which is (54.14). Substituting the same T_Y and Q_Y in (54.17) gives (54.15).

The statement is an equation for renormalized perturbative coordinates. It does not assert a non-perturbative electroweak continuum limit, and it does not turn the running of λ into a principle for choosing ultraviolet boundary data. It also is not the full physical Standard Model trajectory unless the remaining Yukawa singular values and all higher-dimensional coordinates are consistently set to zero in the chosen chart; its role here is to isolate the leading top-Higgs mechanism in the quartic running.

54.9 Higgs Quartic Running and Large-Field Claims

The running quartic $\lambda(\mu)$ is a coordinate in a chosen subtraction scheme. The sign of this coordinate is a useful perturbative fact, but it is not by itself an observable and it is not a theorem about a nonperturbatively constructed Standard Model. A statement often called "Higgs vacuum stability" therefore has to say which mathematical object it concerns. It may concern a minimal-subtraction trajectory, a perturbative effective potential, a semiclassical decay functional, or an additional Wilsonian EFT extension. These are different objects. The large-field effective-potential analysis is meaningful only after the subtraction scheme, gauge convention, field range, perturbative error estimate, omitted higher-dimensional operators if an EFT extension is being used, and the relation between a background Higgs coordinate and the observable or decay problem have been specified.

Definition 54.23 (High-scale RG continuation)**.** Fix coupling coordinates $c^A(\mu)$. In the minimal perturbative electroweak chart these include

$$g_1(\mu), \qquad g_2(\mu), \qquad g_3(\mu), \qquad y_t(\mu), \qquad \lambda(\mu), \qquad \dots .$$

Also fix a subtraction scheme, a loop order, and matching data $c^A(\mu_0)$ at a reference scale. A high-scale RG continuation to $\mu_1 > \mu_0$ is the solution of the specified ordinary differential equation

$$\frac{\mathrm{d}c^A}{\mathrm{d}\log\mu} = \beta^A(c) \tag{54.18}$$

on an interval where the perturbative chart and its error estimate are specified. This continuation is not the assertion that the Standard Model has a physical cutoff μ_1. It is a continuation of renormalized coordinates inside a perturbative scheme. To reinterpret μ_1 as a Wilsonian cutoff, one must change the problem by adding a Wilsonian QFT definition at scale μ_1, a matching map from that definition to the low-energy coordinates, and a proof or controlled hypothesis that the observables under discussion are reproduced in the overlap regime.

Remark 54.24 (Scope of minimal-subtraction running)*.* The minimal-subtraction beta functions are obtained from the pole parts of renormalized 1PI coordinate changes in dimensional regularization. Solving (54.18) transports coordinates in this renormalized coupling chart. A Wilsonian cutoff theory is different data: it requires a finite regulator or BV pushforward, a cutoff-dependent local action or half-density, and a statement that low-momentum observables are preserved by integrating out degrees of freedom. The ordinary differential equation for minimal-subtraction coordinates supplies only the coordinate transport. Treating the endpoint of that trajectory as a physical high-cutoff Wilsonian EFT is an additional assumption about the theory under discussion, and the assumption must be stated with its regulator or BV data and its matching map.

Remark 54.25 (Extensions beyond the Standard Model)*.* A proposed extension of the Standard Model is an additional mathematical hypothesis, not an explanation by itself. If it is used only through finitely many higher-dimensional operators, it is a choice of EFT coordinates together with matching assumptions. If it is claimed to be an ultraviolet completion, the claim requires a regulator or constructive framework at least as explicit as the ones demanded in Definition 54.40 and Remark 54.24. A perturbative Lagrangian, a high-scale RG trajectory, or a phenomenological motivation is not by itself such a theory.

Definition 54.26 (Quartic positivity in a specified chart)**.** Fix a subtraction scheme, a gauge for the perturbative effective potential if one is used, a loop order, matching maps from measured pole or threshold observables to coupling coordinates at μ_0, and a scale interval $I = [\mu_0, \mu_1]$. Specify whether I is only a high-scale RG continuation in the sense of Definition 54.23, or whether an additional Wilsonian or constructive definition is being attached to the endpoint. A quartic-positivity statement on I is the claim that the solution of the specified RG equations satisfies

$$\lambda(\mu) > 0, \qquad \mu \in I, \tag{54.19}$$

with an error estimate that includes threshold matching, loop truncation, input-parameter uncertainties, and the effect of higher-dimensional operators in the chosen coupling chart. It is a statement about perturbative coordinates. It is not a nonperturbative vacuum statement unless the additional regulator and observable data below make it one.

Definition 54.27 (Metastability in a specified semiclassical calculation)**.** In the same chart, a metastability claim is not the assertion $\lambda(\mu) < 0$. It is a semiclassical statement about a false-vacuum decay functional: one must specify the gauge-fixed effective action used for the bounce calculation, the fluctuation determinant convention, the renormalization scale-setting prescription, the spacetime volume or cosmological history against which the rate is compared, and the higher-dimensional operators retained in the field range of the bounce. The output is an inequality of the form

$$\Gamma_{\text{decay}}\, \mathcal{V}_4 < \varepsilon, \tag{54.20}$$

with a specified uncertainty. The sign of λ along an RG trajectory is one input to such a calculation, not the calculation itself. It becomes a prediction of a QFT only after the path integral, field range, and omitted operator remainder are controlled in the regime sampled by the bounce.

Large-field EFT expansion requires a field-domain hypothesis. Let h be the radial Higgs coordinate defined by

$$H^\dagger H = \frac{h^2}{2}.$$

In a coupling chart whose scalar potential is

$$V_{\le 6}(h) = \lambda \left(\frac{h^2 - v^2}{2}\right)^2 + \frac{C_H}{\Lambda^2}\left(\frac{h^2}{2}\right)^3 \tag{54.21}$$

the large-field quartic approximation is controlled only for h^2/Λ^2 in the specified EFT range. More precisely, for $h \gg v$,

$$V_{\le 6}(h) = \frac{\lambda}{4} h^4 \left[1 + \frac{C_H}{2\lambda}\frac{h^2}{\Lambda^2} + O\!\left(\frac{v^2}{h^2}\right)\right], \tag{54.22}$$

whenever $\lambda \neq 0$. Therefore a sign statement about the dimension-four coordinate λ alone is not a statement about the large-field potential unless the higher-dimensional coordinates and the field range have been specified. More generally, an operator

$$\frac{C_n}{\Lambda^{2n-4}}(H^\dagger H)^n, \qquad n \ge 3, \tag{54.23}$$

contributes, relative to the quartic term, by the factor

$$\frac{C_n}{\lambda\, 2^{\,n-2}}\left(\frac{h^2}{\Lambda^2}\right)^{n-2}. \tag{54.24}$$

Thus a finite truncation of the large-field potential is a controlled EFT statement only in a field domain where the omitted tower is bounded by a specified remainder.

Substituting $H^\dagger H = h^2/2$ gives

$$\lambda\left(\frac{h^2 - v^2}{2}\right)^2 = \frac{\lambda}{4}h^4 - \frac{\lambda v^2}{2}h^2 + \frac{\lambda v^4}{4}$$

and

$$\frac{C_H}{\Lambda^2}\left(\frac{h^2}{2}\right)^3 = \frac{C_H}{8\Lambda^2}h^6.$$

Factoring $\lambda h^4/4$ from the large-field expression gives (54.22). The ratio of the dimension-six term to the quartic term grows as h^2/Λ^2. Thus the EFT coupling chart, not only the dimension-four coordinate, is part of the statement.

For the general operator (54.23), substitution of $H^\dagger H = h^2/2$ gives

$$\frac{C_n}{\Lambda^{2n-4}}\frac{h^{2n}}{2^n}.$$

Dividing by $\lambda h^4/4$ gives (54.24). If h^2/Λ^2 is not small, or if the coefficients C_n are not bounded by a stated matching hypothesis, no finite truncation gives a controlled statement about the potential in that region.

Remark 54.28 (Large-field claims in the hybrid definition). The definition contains the electroweak sector as a renormalized perturbative chiral gauge theory and supplies matching maps to lower-energy EFTs and nonperturbative QCD inputs. A nonperturbative large-field stability or metastability theorem for the Higgs coordinate would require further data: a probability measure, Hilbert space, or operator algebra for the full chiral electroweak theory, or a Wilsonian/EFT construction with a declared field range and matching map. Minimal-subtraction coordinate transport has the scope described in Remark 54.24. If one changes the problem to an EFT extension with higher-dimensional operators, the large-field expansion displayed above shows that such a statement also samples the omitted operator coordinates unless the field range and matching assumptions control them. The baseline hybrid definition therefore supports conditional perturbative statements about a chosen coupling chart or decay functional; an unconditional QFT theorem about Higgs large-field stability or metastability is a separate construction problem.

Remark 54.29 (One-loop source of downward quartic running). In the one-loop top-Higgs subsystem of Controlled Approximation 54.22, the beta function at $\lambda = 0$ is

$$(4\pi)^2\beta_\lambda\big|_{\lambda=0} = -6y_t^4 + \frac{3}{8}\left[2g_2^4 + (g_2^2 + g_1^2)^2\right]. \tag{54.25}$$

Thus the top-Yukawa term supplies the negative entry in this minimal-subtraction quartic beta function. The statement belongs to the specified perturbative coupling chart.

Set $\lambda = 0$ in (54.15). The terms $24\lambda^2$, $12\lambda y_t^2$, and $-3\lambda(3g_2^2 + g_1^2)$ vanish, leaving exactly (54.25). The negative sign of the top term is the closed-fermion-loop sign recorded in the coefficient accounting of Controlled Approximation 54.22. No scheme-independent observable is obtained until the matching inputs and the observable or decay functional have been specified.

Definition 54.30 (Data required for a Higgs large-field claim)**.** A Higgs large-field claim in the hybrid Standard Model definition has a precise meaning only after specifying:

1. the input observables and matching maps used to determine $g_1, g_2, g_3, y_t, \lambda$ and the relevant higher-dimensional coefficients at the reference scale;
2. the subtraction scheme, loop order, threshold convention, and scale interval for the RG evolution;
3. whether the endpoint of the scale interval is only a formal RG continuation of renormalized coordinates or is being supplemented by an additional Wilsonian, lattice, BV, or constructive definition;
4. whether the output is a quartic-positivity statement in the sense of Definition 54.26 or a metastability statement in the sense of Definition 54.27;
5. the treatment of $O_H = (H^\dagger H)^3$ and every other retained higher-dimensional operator whose contribution is not uniformly small over the field range;
6. a remainder estimate for perturbative truncation and for omitted operators.

Without these data, a sentence about "vacuum stability" has no definite mathematical content in this chapter. Even with these data, the result is conditional unless a theorem or controlled argument relates the chosen perturbative quantity to the claimed observable or decay problem.

54.10 Precision Electroweak Self-Energy Coordinates

Precision electroweak fits use a finite-dimensional projection of the renormalized electroweak two-point functions. The projection is meaningful only after the input scheme, subtraction scheme, and class of observables have been fixed. This section defines the S, T, U coordinates as such a two-point projection.

Definition 54.31 (Transverse self-energy difference)**.** Fix a reference electroweak coupling chart with the same input values of α_{em}, G_F, and m_Z. Let

$$s_W := \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, \qquad c_W := \frac{g_2}{\sqrt{g_1^2 + g_2^2}}, \qquad e := g_2 s_W = g_1 c_W,$$

so that $m_W^2 = c_W^2 m_Z^2$ at tree level in the chart of Proposition 54.9. For $V, V' \in \{W, Z, \gamma\}$, define $\Delta\Pi_{VV'}(q^2)$ to be the difference between the renormalized transverse 1PI two-point scalar in the chart under study and the corresponding scalar in the reference chart, with identical external-current normalization. Thus

$$\Delta\Gamma^{\mu\nu}_{VV'}(q)\big|_{\mathrm{trans}} = -\mathrm{i}\left(\eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}\right)\Delta\Pi_{VV'}(q^2), \tag{54.26}$$

on conserved external currents. The electromagnetic on-shell conditions are chosen so that

$$\Delta\Pi_{\gamma\gamma}(0) = 0, \qquad \Delta\Pi_{Z\gamma}(0) = 0. \tag{54.27}$$

Assume, for the definition below, that the self-energy differences are analytic at $q^2 = 0$ on the relevant sheet and write $\Delta\Pi'_{VV'}(0) := \mathrm{d}\Delta\Pi_{VV'}(q^2)/\mathrm{d}q^2|_{q^2=0}$.

Definition 54.32 (S, T, U two-point coordinates)**.** Under the hypotheses of Definition 54.31, the Peskin–Takeuchi S, T, U coordinates are the three finite combinations

$$\begin{aligned}
\alpha_{\rm em} S &= 4s_W^2 c_W^2 \left[\Delta\Pi'_{ZZ}(0) - \frac{c_W^2 - s_W^2}{s_W c_W}\Delta\Pi'_{Z\gamma}(0) - \Delta\Pi'_{\gamma\gamma}(0)\right], \\
\alpha_{\rm em} T &= \frac{\Delta\Pi_{WW}(0)}{m_W^2} - \frac{\Delta\Pi_{ZZ}(0)}{m_Z^2}, \\
\alpha_{\rm em} U &= 4s_W^2 \left[\Delta\Pi'_{WW}(0) - c_W^2\Delta\Pi'_{ZZ}(0) - 2s_W c_W \Delta\Pi'_{Z\gamma}(0) - s_W^2\Delta\Pi'_{\gamma\gamma}(0)\right].
\end{aligned} \tag{54.28}$$

Here $\alpha_{\rm em} := e^2/(4\pi)$. These are coordinates on the two-point part of the electroweak coupling chart, not on the full space of all possible four-fermion, vertex, and detector corrections.

Remark 54.33 (What the T coordinate measures). Suppose the only low-momentum effect in the two-point chart is a shift of the charged and neutral weak mass coordinates,

$$m_W^2 \mapsto m_W^2 + \delta m_W^2, \qquad m_Z^2 \mapsto m_Z^2 + \delta m_Z^2,$$

with s_W, c_W held fixed by the chosen input convention and with $\Delta\Pi_{WW}(0) = \delta m_W^2$, $\Delta\Pi_{ZZ}(0) = \delta m_Z^2$. Then

$$\alpha_{\rm em} T = \frac{\delta m_W^2}{m_W^2} - \frac{\delta m_Z^2}{m_Z^2} \tag{54.29}$$

and the first-order shift of

$$\rho := \frac{m_W^2}{m_Z^2 c_W^2}$$

is

$$\delta\rho = \alpha_{\rm em} T + O(\delta^2). \tag{54.30}$$

The first equality is exactly the definition of T after the stated identification of the zero-momentum self-energies with mass-coordinate shifts. For the second, use the tree identity $m_W^2 = m_Z^2 c_W^2$ and expand

$$\rho_{\rm shifted} = \frac{m_W^2 + \delta m_W^2}{(m_Z^2 + \delta m_Z^2)c_W^2} = \left(1 + \frac{\delta m_W^2}{m_W^2}\right)\left(1 + \frac{\delta m_Z^2}{m_Z^2}\right)^{-1}.$$

The first-order Taylor expansion of $(1+x)^{-1}$ gives

$$\rho_{\rm shifted} = 1 + \frac{\delta m_W^2}{m_W^2} - \frac{\delta m_Z^2}{m_Z^2} + O(\delta^2),$$

which proves (54.30). This is the precise sense in which T records custodial-symmetry breaking in a two-point electroweak chart.

Controlled Approximation 54.34 (Domain of the S, T, U projection)**.** The S, T, U coordinates are a three-dimensional projection of a much larger matching problem. They are adequate only in the following restricted two-point chart: all corrections to the relevant electroweak observables enter through the analytic transverse two-point functions in Definition 54.31, and only the constant and first-derivative Taylor coefficients at $q^2 = 0$ are retained. After the input coordinates $\alpha_{\rm em}$, G_F, and m_Z are fixed, the remaining three input-independent two-point combinations can be chosen as the coordinates S, T, U in Definition 54.32.

The tree exchange amplitudes of conserved external fermion currents are obtained by inverting the quadratic form of the gauge fields and contracting with the current vertices. To first order in $\Delta\Pi$, the correction is

$$D_0(q)\,\Delta\Pi(q^2)\,D_0(q),$$

where D_0 is the tree propagator matrix in the same current basis. Under the analyticity hypothesis, the correction through order q^2 depends only on $\Delta\Pi_{VV'}(0)$ and $\Delta\Pi'_{VV'}(0)$. Electromagnetic gauge invariance and the on-shell electromagnetic normalization impose (54.27). Finite redefinitions of the three input coordinates α_{em}, G_F, and m_Z remove three further linear combinations: the electric charge normalization, the charged-current normalization used to define G_F, and the reference neutral-pole mass.

The combinations displayed in (54.28) are a convenient basis of the remaining three-dimensional quotient. The T coordinate is the residual custodial mass-ratio coordinate by Remark 54.33; the S coordinate is the neutral-current derivative combination that remains after the photon charge normalization is removed; and U is the corresponding charged-minus-neutral derivative coordinate. If nonanalytic thresholds, nonlocal detector functionals, vertex corrections, box corrections, or independent local four-fermion operators are kept at the same order, they are additional coordinates of the matched prediction and are not represented by S, T, U alone.

54.11 The Hybrid Definition

Definition 54.35 (QCD–electroweak coupling data)**.** Let $\mathfrak{D}^{\mathrm{W}}_{\mathrm{QCD}}$ be a Wightman QFT with Hilbert space $\mathcal{H}_{\mathrm{QCD}}$, vacuum Ω_{QCD}, and a net or common domain of color-gauge-invariant local operator-valued distributions. A QCD–electroweak coupling data $\mathcal{C}_{\text{QCD-EW}}$ consist of the following items.

1. A finite or countable list of renormalized color-gauge-invariant local QCD fields

   $$\mathcal{O}_A(x) \in \mathcal{O}^{\mathrm{inv}}_{\mathrm{QCD}},$$

   including the flavor vector and axial currents, charged weak currents, electromagnetic current, and scalar or pseudoscalar quark-density operators needed after an electroweak scalar chart is chosen. Each $\mathcal{O}_A$ is an operator-valued tempered distribution, with its domain, normalization, and composite-operator scheme specified.

2. The transformation laws of these operators under the flavor and electroweak source symmetries that are to be gauged perturbatively, together with their Ward identities, anomaly contact terms, and current-conservation or partial-conservation equations.

3. A local perturbative interaction functional

   $$I_{\mathrm{int}}[\varphi_{\mathrm{EW}};\mathcal{O}] = \sum_A \int_{M_4} \varphi^A_{\mathrm{EW}}(x)\,\mathcal{O}_A(x), \tag{54.31}$$

   where φ^A_{EW} denotes the corresponding perturbative electroweak field, source, or composite expression. The product in (54.31) is required to be invariant under the electroweak gauge symmetry after the perturbative gauge-fixing or BV formulation has been specified.

4. A prescription for mixed correlators as formal series in electroweak couplings whose coefficients are Wightman distributions of the QCD operators $\mathcal{O}_A$, combined with the perturbative electroweak propagators and vertices. The prescription includes the renormalization of coincident insertions and the contact-term scheme.

Remark 54.36 (Hybrid coupling is additional structure). The data $\mathcal{C}_{\text{QCD-EW}}$ in Definition 54.35 is not determined by separately giving a nonperturbative QCD Wightman theory and a perturbative electroweak Lagrangian. It is additional mathematical structure needed to define the hybrid Standard Model prediction framework.

A nonperturbative QCD Wightman theory contains physical local fields only in the color-gauge-invariant operator algebra. The colored quark fields that appear in a gauge-fixed perturbative Lagrangian are not, by themselves, Wightman fields on the physical QCD Hilbert space. Therefore an electroweak boson or Higgs fluctuation cannot be coupled to a nonperturbative QCD sector by reusing a colored perturbative quark line. The coupling has to be expressed through color-gauge-invariant QCD operator-valued distributions: for example the electromagnetic current, charged weak currents, neutral currents, and scalar density operators. Their normalizations, contact terms, Ward identities, and anomaly conventions determine the coefficients of mixed correlators with electroweak external fields.

At order n in the electroweak couplings, a mixed correlator is a finite sum of perturbative electroweak kernels paired with QCD Wightman distributions of the form

$$\big\langle \Omega_{\text{QCD}}, \mathcal{O}_{A_1}(x_1)\cdots\mathcal{O}_{A_n}(x_n)\mathcal{X}_1(y_1)\cdots\mathcal{X}_m(y_m)\Omega_{\text{QCD}}\big\rangle, \tag{54.32}$$

where the $\mathcal{X}_i$ are the QCD operators appearing in the observable. The distribution (54.32), its extension to coincident points, and its Ward identities are not supplied by the electroweak perturbative Lagrangian. Conversely, the pure QCD Wightman theory does not know which electroweak source or gauge field multiplies which operator. Thus the hybrid coupling structure is logically necessary.

Definition 54.37 (Hybrid Standard Model definition)**.** The Standard Model is used in this monograph through the following hybrid definition, not as a single completed constructive theorem for a four-dimensional continuum QFT:

$$\mathfrak{D}_{\text{SM}} = \big(\mathfrak{D}^{\text{W}}_{\text{QCD}}, \mathfrak{D}^{\text{pert}}_{\text{EW}}, \mathcal{C}_{\text{QCD-EW}}, \mathfrak{D}_{\text{weak EFT}}, \mathcal{M}\big).$$

Here $\mathfrak{D}^{\text{W}}_{\text{QCD}}$ denotes the nonperturbative continuum QCD Wightman theory selected by the $SU(3)_c$ gauge theory with quark matter; in constructive and computational practice it is approached through finite lattice regulators and a continuum-limit problem of the kind stated in Hypothesis 46.9. When a concrete lattice regulator is needed for a hadronic matrix element, it will be denoted $\mathfrak{D}^{\text{lat}}_{\text{QCD}}$; this notation refers to the regulator used to approximate the QCD component, not to a cutoff definition of the full Standard Model. $\mathfrak{D}^{\text{pert}}_{\text{EW}}$ denotes the renormalized electroweak chiral gauge theory with Higgs field, constructed as a perturbative formal power series to all orders after gauge fixing or BV treatment, anomaly cancellation, and a subtraction scheme have been fixed; $\mathcal{C}_{\text{QCD-EW}}$ is the hybrid coupling data of Definition 54.35; $\mathfrak{D}_{\text{weak EFT}}$ is the tower of lower-energy effective theories obtained by integrating out massive weak fields and heavy flavors; and $\mathcal{M}$ is the matching data fixing parameters and local operators between overlapping regimes. The symbol $\mathfrak{D}^{\text{pert}}_{\text{EW}}$ is not a finite-cutoff definition and is not a claim that a nonperturbative four-dimensional chiral electroweak continuum construction has been supplied.

Definition 54.38 (Matching data in the hybrid Standard Model)**.** The symbol $\mathcal{M}$ in Definition 54.37 consists of the following specified data for the observable class under consideration:

1. a sequence of scales or windows where adjacent descriptions overlap;
2. renormalized parameter charts for $g_3, g_2, g_1, \lambda, v$, the Yukawa matrices, theta angles, and, when an EFT extension is being used, higher-dimensional Wilson coordinates;
3. threshold matching maps when $W, Z, h, t, b, c, \tau, \ldots$ are removed from the active field content;
4. anomalous-dimension matrices for the chosen operator bases between thresholds;
5. the QCD–electroweak coupling data $\mathcal{C}_{\text{QCD-EW}}$, including the scheme for the currents, scalar densities, contact terms, and Ward identities entering the observable;
6. nonperturbative QCD matrix elements, defined by $\mathfrak{D}^{\text{lat}}_{\text{QCD}}$ or by another explicitly specified nonperturbative regulator, for hadronic operators;
7. the error model or remainder statement attached to each perturbative, lattice, or combined extrapolation.

A numerical precision claim is well defined only after these choices and their error estimates are specified.

Remark 54.39 (Logical parts of the hybrid definition). Definition 54.37 separates nonperturbative QCD input, perturbative electroweak definitions, their hybrid operator coupling, lower-energy effective theories, and observable-specific matching. These parts cannot be replaced by one another without adding an extra theorem or hypothesis.

First, the QCD sector is the nonperturbative part of the definition. It is intended as a continuum Wightman QFT with gauge-invariant local and extended observables, while compact lattice link variables, Haar measure, and finite Grassmann algebra for quarks provide the concrete nonperturbative regulators used to define approximating matrix elements. The existence of the four-dimensional continuum limit with the desired physical spectrum is still a theorem-level constructive problem, but the logical role of this component is not perturbative electroweak renormalization.

Second, the electroweak sector is a perturbatively renormalizable chiral gauge theory. Its perturbative consistency uses anomaly cancellation, BRST or BV Ward identities, and the local power-counting analysis of the BEH phase. This defines renormalized Green functions and operator insertions as formal power series to all orders in the chosen subtraction scheme. A standalone nonperturbative continuum construction is not supplied by these facts. The hypercharge $U(1)$ coupling is not asymptotically free, and the four-dimensional scalar quartic sector lies on the same marginal-triviality boundary discussed in Chapters 11 and 38. Moreover, a fully satisfactory interacting lattice construction of a four-dimensional chiral gauge theory with the Standard Model representation content is a separate problem, related to the determinant-line and chiral-lattice issues in Chapter 146.

Third, the coupling of these two parts is not automatic. It is mediated by the color-gauge-invariant QCD operator insertions specified in $\mathcal{C}_{\text{QCD-EW}}$. This is the point at which nonperturbative hadronic matrix elements enter electroweak amplitudes and where contact terms and Ward identities must be fixed.

Fourth, physical predictions are extracted in regimes where a controlled object is available. High-energy short-distance processes use perturbative renormalization and factorization with explicitly stated infrared-safe observables. Low-energy hadronic weak processes use QCD matrix elements defined by the QCD regulator together with weak effective operators matched at the electroweak scale. Precision electroweak observables use finite-order renormalized perturbation theory. A cut-off parameter is not part of the prediction; any auxiliary regulator dependence has been removed or absorbed into the stated renormalized coordinates and operator scheme.

54.12 Chiral Lattice Status of the Electroweak Sector

The statement that the electroweak sector is a chiral gauge theory has a precise regulator consequence. Vectorlike lattice fermions define a finite Grassmann integral by an ordinary determinant. A chiral gauge theory instead requires a gauge-covariant choice of a subspace of left-handed lattice fermions, and the corresponding fermion integral is a section of a determinant line over the finite gauge-field space. The phase of that section is part of the regulator, not an optional convention.

Definition 54.40 (Finite chiral-gauge regulator data)**.** A finite chiral-gauge regulator for the Standard Model representation content consists of the following data at lattice spacing a and finite volume:

1. compact lattice gauge fields for the chosen global form of the local product data $SU(3)_c \times SU(2)_L \times U(1)_Y$, with a local gauge action;
2. a local gauge-covariant lattice Dirac operator, for example an overlap/Ginsparg–Wilson operator satisfying (146.4);
3. chiral projectors assigning Q_L, L_L to left-handed subspaces and u^c, d^c, e^c to the conjugate left-handed representations of Definition 54.2;
4. a gauge-invariant, local, and reflection-compatible choice of phase of the chiral fermion determinant line over lattice gauge fields;
5. lattice Higgs and Yukawa terms whose gauge quantum numbers reduce to Definition 54.3 in the scaling window;
6. uniform locality and reflection-positivity, or a specified replacement reconstruction route, strong enough to define a candidate continuum local QFT.

Proposition 54.41 (Local anomaly cancellation as determinant-line obstruction)**.** *For finite chiral-gauge regulator data in the sense of Definition 54.40, gauge invariance of the chiral determinant line requires the perturbative gauge anomaly coefficients of Proposition 54.11 to vanish. It also requires the finite $SU(2)$ doublet-parity condition in that proposition.*

Proof. At finite lattice spacing the chiral fermion integral is not an ordinary complex-valued determinant. For each background lattice gauge field A, let $V_L(A)$ and $W_L(A)$ be the finite-dimensional left-handed domain and range selected by the chosen chiral projectors, and let

$$D_A^+ : V_L(A) \longrightarrow W_L(A)$$

be the chiral Dirac map. The Berezin integral of the left-handed fermions is an element of the determinant line

$$\mathcal{L}_A = \det W_L(A)^* \otimes \det V_L(A).$$

As A varies, these lines form a complex line bundle over the finite gauge-field space. A gauge-invariant chiral regulator must therefore supply a nonvanishing section of $\mathcal{L}$ whose pullback by every finite gauge transformation agrees with the natural action on the determinant line. This is stronger than choosing a phase at one background: the phase must be transported consistently around every gauge loop.

Choose a local determinant connection compatible with the finite regulator. For infinitesimal gauge variations generated by Lie-algebra-valued functions α, β, the curvature evaluated on the corresponding gauge-orbit tangent vectors is the obstruction to path-independent phase transport. The locality and continuum-limit requirements in Definition 54.40 force the leading local part of this curvature to be the standard descent two-form whose coefficient is the symmetric gauge-anomaly tensor of Chapter 51. Thus, in a small gauge two-cell,

$$\mathrm{hol}_{\mathcal{L}}(\alpha, \beta) = \exp\left(2\pi \mathrm{i} \int_M \omega^2_{\mathrm{anom}}(\alpha, \beta; A) + \text{higher lattice-spacing terms}\right),$$

where the local polynomial ω^2_{anom} is linear in the anomaly coefficients of the representation. If one of these coefficients is nonzero, one can choose local α, β and a smooth background approximated by the lattice fields so that the displayed local term is nonzero. The holonomy around a sufficiently small gauge two-cell is then not one, so no gauge-equivariant nonvanishing phase section can exist. Hence the perturbative local anomaly coefficients must vanish. The finite sums in Proposition 54.11 are exactly those coefficients for the Standard Model representation content.

There is also a global obstruction not detected by the local curvature. For the weak $SU(2)$ factor, $\pi_4(SU(2)) \cong \mathbb{Z}_2$. Transporting a single left-handed $SU(2)$ doublet around a representative nontrivial loop changes the sign of the corresponding Pfaffian line. With N_2 doublets, the sign is $(-1)^{N_2}$; a gauge-invariant phase is possible only if N_2 is even. One Standard Model generation contains three colored quark doublets and one lepton doublet, hence $N_2 = 4$, and the finite sign obstruction is absent generation by generation. $\square$

Remark 54.42 (Scope of the anomaly-obstruction calculation)*.* The vanishing of the local curvature obstruction and the $SU(2)$ parity obstruction is necessary regulator data. A complete finite chiral-gauge regulator still has to construct a gauge-invariant local phase functional, prove the required locality and reflection or reconstruction properties, and control the four-dimensional continuum scaling limit.

Remark 54.43 (Lattice QCD inside the hybrid Standard Model)*.* The lattice regulators used to approximate the QCD component $\mathfrak{D}^{\mathrm{W}}_{\mathrm{QCD}}$ in Definition 54.37 supply nonperturbative definitions of gauge-invariant QCD matrix elements after the electroweak matching data have selected the relevant color-singlet operators. They do so because lattice QCD uses the color gauge group and quark fields as a vectorlike fermion system for the purpose of strong-interaction matrix elements. Its finite Grassmann integral is an ordinary determinant, or a controlled product of determinants, coupled to compact $SU(3)_c$ links. The electroweak theory before matching has chiral assignments: Q_L is an $SU(2)_L$ doublet while u^c, d^c are weak singlets with different hypercharges, and L_L, e^c have the analogous chiral electroweak assignment. A finite regulator for that system must solve the determinant line and phase problem in Definition 54.40.

The hybrid construction avoids conflating these two uses. For hadronic weak observables, the weak fields are first matched onto gauge-invariant local operators such as four-quark, semileptonic, or electromagnetic-current operators. The QCD lattice regulator then computes the matrix elements of those color-singlet operators. This gives a valid matched prediction after the map $\mathcal{M}$ is specified; a nonperturbative construction of the microscopic chiral electroweak gauge theory is the distinct finite-regulator problem described above.

Definition 54.44 (Standard Model prediction)**.** A Standard Model prediction for an observable O, in the sense used in this chapter, is a number or distribution obtained only after the following data are specified:

1. the energy regime and the component of the hybrid definition used in that regime: nonperturbative QCD input, renormalized perturbative electroweak theory, a lower-energy EFT, or an additional Wilsonian/EFT extension;
2. the gauge-invariant local operator, infrared-safe inclusive observable, or detector functional representing O;
3. the matching map from the adjacent higher-energy description, including the scheme and normalization of all operator coordinates;
4. the perturbative, lattice, or combined expansion parameter and the associated remainder or systematic-error class;
5. the positivity, unitarity, gauge-invariance, and infrared-safety statements needed for the observable.

54.13 Fermi Theory as a Matched Low-Energy Layer

The tree-level Fermi coefficient is a matched low-energy coefficient, obtained by solving the massive charged-vector equation in a derivative expansion. Let $J^{\pm}_{\mu}$ be the charged left-handed weak currents normalized so that the massive charged-vector part of the electroweak EFT is

$$\mathcal{L}_{\mathrm{cc}} = -\frac{g_2}{\sqrt{2}}\big(J^{+}_{\mu}W^{+\mu} + J^{-}_{\mu}W^{-\mu}\big) + m_W^2 W^{+}_{\mu}W^{-\mu} + \cdots ,$$

where the dots include the $W^{\pm}$ kinetic term and interactions. In the low-energy EFT for external momenta $p^2 \ll m_W^2$, eliminating $W^{\pm}$ at leading order gives

$$\mathcal{L}_{\mathrm{Fermi}} = -\frac{g_2^2}{2m_W^2}J^{+}_{\mu}J^{-\mu} + O(p^2/m_W^4).$$

Therefore, with the current convention used here,

$$\mathcal{L}_{\mathrm{Fermi}} = -\frac{4G_F}{\sqrt{2}}J^{+}_{\mu}J^{-\mu} \qquad \Longleftrightarrow \qquad \frac{G_F}{\sqrt{2}} = \frac{1}{2v^2}.$$

At tree level and to leading order in derivatives, the charged vector is an auxiliary variable with algebraic Lagrangian

$$m_W^2 W^{+}_{\mu}W^{-\mu} - a(J^{+}_{\mu}W^{+\mu} + J^{-}_{\mu}W^{-\mu}), \qquad a := \frac{g_2}{\sqrt{2}}.$$

The equations obtained by varying W^+ and W^- independently are

$$m_W^2 W_\mu^- = aJ_\mu^-, \qquad m_W^2 W_\mu^+ = aJ_\mu^+.$$

Substitution gives

$$\left(\frac{a^2}{m_W^2} - 2\frac{a^2}{m_W^2}\right) J_\mu^+ J^{-\mu} = -\frac{g_2^2}{2m_W^2} J_\mu^+ J^{-\mu}.$$

Including the kinetic operator replaces m_W^{-2} by the low-momentum expansion of the massive propagator,

$$\frac{1}{m_W^2 - p^2} = \frac{1}{m_W^2} + O(p^2/m_W^4),$$

in the local derivative expansion. The mass formula from Proposition 54.9 gives

$$\frac{g_2^2}{2m_W^2} = \frac{2}{v^2}.$$

Comparing with $-4G_F J_\mu^+ J^{-\mu}/\sqrt{2}$ proves $G_F/\sqrt{2} = 1/(2v^2)$.

The equality is a tree-level matching statement in the electroweak EFT. QCD renormalization of hadronic weak operators, electromagnetic corrections, CKM factors, and matrix elements of the resulting four-fermion operators are separate parts of the low-energy matching problem.

54.14 Muon $g-2$ as a Hybrid Prediction

The anomalous magnetic moment of the muon is a useful organizing example because it is a single low-energy form-factor coordinate whose Standard Model evaluation uses all parts of $\mathfrak{D}_{\mathrm{SM}}$: perturbative QED, perturbative electroweak matching, and nonperturbative QCD current correlators.

Definition 54.45 (Muon Pauli-coordinate observable)**.** Let $\ell = \mu$ be the charged-lepton field in the low-energy EFT and let J^μ_{em} be the electromagnetic current normalized so that the muon has charge $-e$, $e > 0$. Since the muon has weak decay channels, the magnetic moment used here is the real Pauli-operator coordinate in the low-energy effective Hamiltonian for a one-muon wave packet, equivalently the real part of the Pauli form factor obtained after matching the $\mu\mu\gamma$ vertex and separating absorptive decay operators. In the zero-width projection used for this matching, define the Dirac–Pauli form factors by

$$\langle p', \sigma' | J^\mu_{\mathrm{em}}(0) | p, \sigma\rangle = -\frac{\mathrm{i}e}{(2\pi)^3}\bar{u}_{\sigma'}(p')\left[\gamma^\mu F_1^\mu(k^2) + \frac{1}{m_\mu} S^{\mu\nu} k_\nu F_2^\mu(k^2)\right] u_\sigma(p), \qquad k := p - p', \tag{54.33}$$

with $S^{\mu\nu} := -\mathrm{i}[\gamma^\mu, \gamma^\nu]/4$ in the mostly-plus spinor convention of Section 40.16. The charge normalization is $F_1^\mu(0) = 1$. The muon anomalous magnetic moment is

$$a_\mu := \frac{g_\mu - 2}{2} := F_2^\mu(0). \tag{54.34}$$

In the QED hard-kernel convention of Chapter 43,

$$a_\mu^{\mathrm{QED},1} = \frac{\alpha}{2\pi}, \qquad \alpha := \frac{e^2}{4\pi}. \tag{54.35}$$

Chapter 43 defines a form-factor basis (F, G) in which $F_2^{\mathrm{DP}}(0) = -G(0)$. The one-loop vertex calculation gives $G(0) = -\alpha/(2\pi)$. Substituting this relation into Definition 54.45 gives $a_\mu = F_2^\mu(0) = \alpha/(2\pi)$ at first order in QED. The lepton mass enters the loop integral but cancels from the zero-transfer dimensionless coefficient, as shown explicitly in Proposition 43.6.

Definition 54.46 (Matched decomposition through a specified order)**.** Fix a renormalization scheme, an input-parameter chart, and a maximum order in the electromagnetic, weak, and QCD matching expansions. The corresponding Standard Model value for a_μ is written

$$a_\mu^{\mathrm{SM}} = a_\mu^{\mathrm{QED}} + a_\mu^{\mathrm{EW}} + a_\mu^{\mathrm{HVP}} + a_\mu^{\mathrm{HLbL}} + a_\mu^{\mathrm{rem}}. \tag{54.36}$$

Here a_μ^{QED} contains the purely leptonic and photon contributions in the low-energy QED EFT; a_μ^{EW} contains diagrams or matched Pauli-operator coefficients with at least one W, Z, h, or weak Goldstone/ghost representative in a gauge-fixed computation; a_μ^{HVP} contains contributions in which the photon propagator is corrected by the gauge-invariant hadronic electromagnetic current two-point function; a_μ^{HLbL} contains the hadronic electromagnetic four-current contribution coupled to the muon line by photons; and a_μ^{rem} records the specified higher-order remainder and any separation-scheme convention.

Controlled Approximation 54.47 (Leading electroweak Pauli coefficient)**.** In the broken-phase electroweak coupling chart, and at leading order in m_μ^2/m_W^2, m_μ^2/m_Z^2, and m_μ^2/m_h^2, the one-loop weak contribution to the muon Pauli coordinate is

$$a_\mu^{\mathrm{EW},1} = \frac{G_F m_\mu^2}{8\sqrt{2}\pi^2}\left[\frac{5}{3} + \frac{1}{3}(1 - 4s_W^2)^2\right] + O\!\left(\frac{G_F m_\mu^4}{m_W^2}\right), \tag{54.37}$$

where $s_W^2 = g_1^2/(g_1^2 + g_2^2)$ in the convention $Q = T^3 + Y$.

The weak contribution is the coefficient of the dimension-five Pauli operator

$$\frac{e}{4m_\mu} a_\mu \, \bar{\mu}[\gamma^\mu, \gamma^\nu]\mu \, F_{\mu\nu} \tag{54.38}$$

in the low-energy QED EFT obtained after integrating out W, Z, h. At one loop the coefficient is obtained by matching the on-shell $\mu\mu\gamma$ vertex in the electroweak theory to the same vertex in the low-energy EFT, expanded in external momenta over m_W, m_Z, m_h. Gauge-parameter dependence cancels in the sum of vector, Goldstone, and ghost representatives because the Pauli operator is electromagnetic-gauge invariant and the matching is performed between physical on-shell matrix elements.

The charged-current W-sector diagrams give the heavy-vector coefficient

$$\frac{G_F m_\mu^2}{8\sqrt{2}\pi^2}\,\frac{5}{3}. \tag{54.39}$$

The neutral-current Z-sector contribution depends on the vector coupling of the Z to the muon. With

$$J_Z^\mu = \bar{\mu}\gamma^\mu\big(g_L P_L + g_R P_R\big)\mu, \qquad g_L = -\frac{1}{2} + s_W^2, \qquad g_R = s_W^2,$$

the vector-minus-axial combination entering the leading Pauli coefficient reduces, after using $G_F/\sqrt{2} = 1/(2v^2)$, to

$$\frac{G_F m_\mu^2}{8\sqrt{2}\pi^2}\,\frac{1}{3}(1 - 4s_W^2)^2. \tag{54.40}$$

The Higgs contribution is suppressed by an additional factor m_μ^2/m_h^2 because both the Higgs coupling and the chirality flip are proportional to the muon Yukawa coordinate. Adding the charged and neutral leading terms gives (54.37). This is a short-distance matching statement in the electroweak EFT; higher weak loops and mixed electroweak-hadronic terms are separate entries in a_μ^{rem} unless explicitly included in the specified order. A complete proof in this monograph requires the explicit one-loop Pauli projection for the charged-vector, neutral-vector, Goldstone, and ghost representatives. Until that appendix-level calculation is supplied, the displayed formula is used only as the convention-fixed leading matching coefficient, not as a theorem established here.

Definition 54.48 (Hadronic vacuum polarization from QCD current data)**.** Let

$$J^\mu_{\mathrm{had}} = \sum_q Q_q\, \bar{q}\gamma^\mu q \tag{54.41}$$

be the quark part of the electromagnetic current, with the sum taken over the active quark fields in the chosen low-energy QCD description and with the current renormalized in the specified regulator. The connected transverse current two-point function defines a scalar polarization by

$$\mathrm{i}\int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}q\cdot x}\langle\Omega|\,\mathrm{T}\, J^\mu_{\mathrm{had}}(x)J^\nu_{\mathrm{had}}(0)|\Omega\rangle_{\mathrm{conn}} = (q^\mu q^\nu - q^2\eta^{\mu\nu})\Pi_{\mathrm{had}}(q^2) + \text{local contact terms}. \tag{54.42}$$

The charge-subtracted coordinate is

$$\widehat{\Pi}_{\mathrm{had}}(q^2) := \Pi_{\mathrm{had}}(q^2) - \Pi_{\mathrm{had}}(0), \tag{54.43}$$

where the subtraction is part of the electromagnetic input normalization.

Dispersive leading-HVP formula. Assume the hadronic current two-point function has a positive spectral representation and no massless hadronic state in the electromagnetic channel. Let

$$R_{\mathrm{had}}(s) := \frac{\sigma(e^+e^- \to \text{hadrons}; s)}{4\pi\alpha^2/(3s)} \tag{54.44}$$

be the spectral density in the standard electromagnetic normalization. QED vacuum-polarization effects are removed according to the same input scheme. The leading HVP contribution is then

$$a_\mu^{\mathrm{HVP,LO}} = \frac{1}{3}\left(\frac{\alpha}{\pi}\right)^2 \int_{s_0}^\infty \frac{\mathrm{d}s}{s}\, K_{\mathrm{HVP}}(s)\, R_{\mathrm{had}}(s), \tag{54.45}$$

where s_0 is the hadronic threshold and

$$K_{\mathrm{HVP}}(s) := \int_0^1 \mathrm{d}x\, \frac{x^2(1-x)}{x^2 + (1-x)s/m_\mu^2}. \tag{54.46}$$

This is a formula derived from two inputs, not an independent theorem about QCD. First, the photon-line insertion in the one-loop Pauli form-factor calculation gives the Euclidean-momentum representation

$$a_\mu^{\mathrm{HVP,LO}} = \frac{\alpha}{\pi}\int_0^1 \mathrm{d}x\,(1-x)\big[-4\pi\alpha\,\widehat{\Pi}_{\mathrm{had}}(-Q_x^2)\big], \qquad Q_x^2 := \frac{x^2}{1-x}m_\mu^2, \tag{54.47}$$

with the same Pauli projection as in Proposition 43.6. The once-subtracted spectral representation of the hadronic current correlator is

$$-4\pi\alpha\,\widehat{\Pi}_{\rm had}(-Q^2) = \frac{\alpha}{3\pi}\int_{s_0}^{\infty} \mathrm{d}s\,\frac{Q^2}{s(s+Q^2)}R_{\rm had}(s). \tag{54.48}$$

Substituting $Q^2 = Q_x^2$ into (54.47), interchange the positive x- and s-integrals, and use

$$(1-x)\frac{Q_x^2}{s+Q_x^2} = \frac{x^2(1-x)}{x^2+(1-x)s/m_\mu^2}.$$

This gives (54.45). The formula is therefore not a perturbative expansion of QCD: all hadronic dynamics is in the spectral function $R_{\rm had}$, equivalently in the current-current correlator of Definition 54.48.

Definition 54.49 (Euclidean lattice HVP coordinate)**.** In a Euclidean nonperturbative QCD regulator, define the zero-spatial-momentum current correlator

$$G(t) := \frac{1}{3}\sum_{i=1}^{3}\int_{\mathbb{R}^3} \mathrm{d}^3\vec{x}\,\langle J^i_{\rm had}(t,\vec{x})J^i_{\rm had}(0,\vec{0})\rangle_E, \qquad t\geq 0, \tag{54.49}$$

with the conserved or renormalized lattice current specified as part of $\mathfrak{D}^{\rm lat}_{\rm QCD}$. The subtracted Euclidean polarization is then equivalently encoded by

$$\widehat{\Pi}_E(Q^2) = \int_0^{\infty} \mathrm{d}t\,G(t)\left[t^2 - \frac{4}{Q^2}\sin^2\!\left(\frac{Qt}{2}\right)\right], \tag{54.50}$$

which fixes the normalization convention for $G(t)$ relative to $\widehat{\Pi}_E$; on a finite lattice the corresponding current renormalization factor is part of the regulator data. Combining this identity with the Euclidean form of (54.47) gives

$$a_\mu^{\rm HVP,LO} = \left(\frac{\alpha}{\pi}\right)^2\int_0^{\infty} \mathrm{d}t\,\widetilde{K}_\mu(t)\,G(t), \tag{54.51}$$

where $\widetilde{K}_\mu$ is the positive QED kernel obtained by performing the Q^2- or Feynman-parameter integral after inserting (54.50). The lattice prediction additionally requires continuum, infinite-volume, physical-mass, isospin-breaking, and QED correction data; these are entries of $\mathcal{M}$, not optional refinements.

Definition 54.50 (Hadronic light-by-light coordinate)**.** The hadronic light-by-light contribution is defined from the connected Euclidean four-current correlator

$$\Pi_{\mu\nu\rho\sigma}(x,y,z) := \langle J^\mu_{\rm had}(x)J^\nu_{\rm had}(y)J^\rho_{\rm had}(z)J^\sigma_{\rm had}(0)\rangle_{E,\rm conn} \tag{54.52}$$

by convolution with the exactly known QED muon-line kernel $\mathcal{K}_{\mu\nu\rho\sigma}(x,y,z;m_\mu)$:

$$a_\mu^{\rm HLbL} = \int \mathrm{d}^4x\,\mathrm{d}^4y\,\mathrm{d}^4z\,\mathcal{K}_{\mu\nu\rho\sigma}(x,y,z;m_\mu)\,\Pi^{\mu\nu\rho\sigma}(x,y,z), \tag{54.53}$$

with the current normalization, contact-term subtraction, finite-volume treatment, and QED infrared regulator specified in $\mathcal{M}$. Dispersive descriptions of the same coordinate decompose the four-current spectral data into intermediate-channel amplitudes; lattice descriptions compute the four-current Schwinger function directly. The equality of these descriptions, when both are available, is a nontrivial matching statement about the same QCD-defined current correlator.

Remark 54.51 (Data needed for the a_μ prediction). A value assigned to a_μ^{SM} is a Standard Model prediction in the sense of Definition 54.44 only after the following data have been specified:

1. the lepton-sector QED perturbative order and charge scheme;
2. the electroweak input scheme and the order of weak matching;
3. the HVP source, either a dispersive spectral function with its radiative-correction convention or a lattice current correlator with its continuum and infinite-volume extrapolations;
4. the HLbL four-current treatment and its contact-term convention;
5. the combination rule for correlations among input parameters and systematic uncertainties.

Definition 54.45 identifies a_μ as a single Pauli-form-factor coordinate. The decomposition (54.36) is not a decomposition of the observable itself; it is a decomposition of its computation in a specified low-energy matching scheme. The QED and electroweak entries are perturbative Pauli-operator matching coefficients. The HVP and HLbL entries are functionals of gauge-invariant QCD current correlators, defined either by a spectral representation or by a finite regulator such as the lattice. Changing any item in the list changes the coordinate map from the hybrid Standard Model definition to the number called a_μ^{SM}. Thus the listed data are necessary for the five requirements in Definition 54.44.

Open Problem 54.52 (Constructive Standard Model). Construct a continuum local QFT realizing the Standard Model representation content, gauge symmetry, anomaly cancellation, BEH phase, QCD confinement data, and weak interactions with positivity, locality, covariance, and a controlled relation to the observed parameter regime. A solution may be a single nonperturbative construction or a theorem explaining why the hybrid definition above has a universal continuum domain for the specified observables. No such theorem is assumed in this monograph.

Chapter 55

Global Anomalies, Spontaneous Symmetry Breaking, and Pions

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The four-dimensional orientation convention is the one used in the local anomaly calculation of Chapter 51:

$$\epsilon^{0123} = +1, \qquad \widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$$

for any two-form F. This convention is used for the perturbative triangle contact terms, Wess–Zumino–Witten functional, and electromagnetic specialization below.

55.1 Background Fields and Anomaly Data

Let a quantum field theory have an exact internal global symmetry G_f. The symmetry acts on local fields and local operators, and on Hilbert spaces obtained after choosing a phase. To test the symmetry as a property of local correlation functions, introduce a nondynamical G_f-connection B on spacetime M. The background-field generating functional is

$$Z[B] = \int [D\Phi] \, \exp\{\mathrm{i}S[\Phi; B]\},$$

where Φ denotes the dynamical fields and B is held fixed as an external datum. Let $\mathcal{B}$ be the space of allowed background fields and $\mathcal{G}_\mathrm{f}$ the group of background gauge transformations. If $g \in \mathcal{G}_\mathrm{f}$, write B^g for the transformed connection.

Definition 55.1 (Background anomaly cocycle)**.** The background anomaly is the equivalence class, under local counterterms, of phases $\mathcal{A}_g[B]$ in the transformation law

$$Z[B^g] = \exp\{\mathrm{i}\mathcal{A}_g[B]\}\, Z[B], \tag{55.1}$$

where $g \in \mathcal{G}_\mathrm{f}$. The zero class is the class for which there exists a local counterterm $C[B]$ making the transformed functional invariant under all background gauge transformations. Multiplying $Z[B]$ by $\exp\{\mathrm{i}C[B]\}$ changes the representative by

$$\mathcal{A}_g[B] \longmapsto \mathcal{A}_g[B] + C[B^g] - C[B].$$

Thus the anomaly is represented by the equivalence class of $\mathcal{A}_g[B]$ under local-counterterm shifts.

Remark 55.2 (Finite cocycle and Wess–Zumino consistency)*.* The phases in (55.1) obey a cocycle identity. Applying h and then g gives the same transformed background as applying gh, hence

$$\mathcal{A}_{gh}[B] = \mathcal{A}_g[B^h] + \mathcal{A}_h[B] \mod 2\pi. \tag{55.2}$$

The infinitesimal part of this cocycle is the local Wess–Zumino consistency condition developed in the preceding chapter. A finite anomaly may also have a purely finite part: its differential on the identity component of $\mathcal{G}_\mathrm{f}$ vanishes, but the cocycle (55.2) is nontrivial on another component or on a noncontractible loop in background-field space.

Equation (55.1) gives

$$Z[B^{gh}] = \exp\{\mathrm{i}\mathcal{A}_{gh}[B]\}Z[B].$$

The same background is obtained by first applying h and then applying g:

$$Z[(B^h)^g] = \exp\{\mathrm{i}\mathcal{A}_g[B^h]\}Z[B^h] = \exp\{\mathrm{i}\mathcal{A}_g[B^h] + \mathrm{i}\mathcal{A}_h[B]\}Z[B].$$

The two answers differ by an integer multiple of 2π in the exponent, which is (55.2). Differentiating the finite identity at the identity element of $\mathcal{G}_\mathrm{f}$ gives the infinitesimal consistency condition for the corresponding local anomaly representative.

Geometrically, the background-field construction assigns a complex line to each gauge-equivalence class of B, and $Z[B]$ is a section of this anomaly line. Local counterterms change a trivialization of the line. When the anomaly is represented by an invertible field theory on a five-dimensional manifold Y with $\partial Y = M$, equality of boundary variations identifies the phase in (55.1). The local Chern–Simons version of this bulk-boundary identity was described in Chapter 51; below we use mapping tori to state the finite version needed for global anomalies.

Figure 55.1 records the anomaly-matching statement in background-field language: microscopic and infrared descriptions are compared as sections of the same anomaly line over background gauge fields.

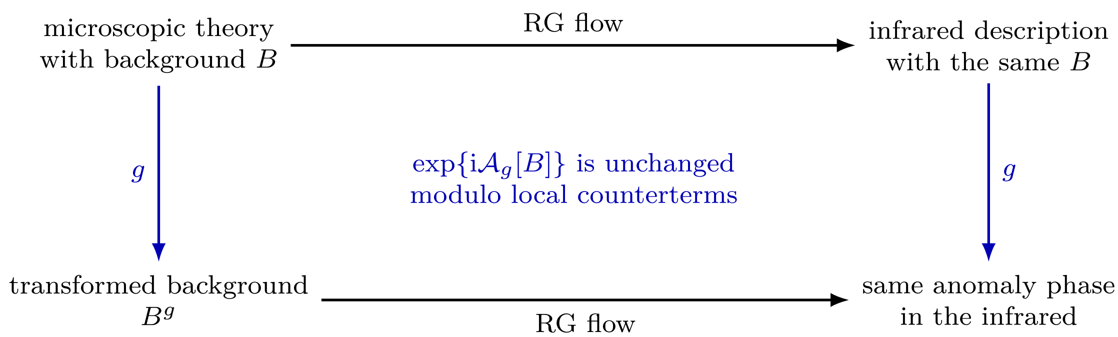


Figure 55.1: Anomaly matching is equality of background-response classes. The ultraviolet and infrared functionals may use different fields, but their transformation cocycles agree after local counterterm conventions are identified.

55.2 Anomaly Matching

Remark 55.3 (Anomaly matching as equality of background-response classes). Let a microscopic theory with exact global symmetry G_f flow to an infrared description. Couple both descriptions to the same background B. If the ultraviolet generating functional transforms as in (55.1), then the infrared functional must obey

$$Z_{\mathrm{IR}}[B^g] = \exp\{\mathrm{i}\mathcal{A}_g[B]\}\, Z_{\mathrm{IR}}[B]$$

after possible local counterterms have been matched. The equality of the equivalence class of $\mathcal{A}_g[B]$ modulo counterterms is 't Hooft anomaly matching.

The matching object is the cocycle class $[\mathcal{A}]$ defined by (55.2). Different infrared descriptions may carry the same class: massless fermions can produce it through their determinant or Pfaffian line, Goldstone fields can produce it through a Wess–Zumino–Witten functional, and a topological sector can produce it through an invertible response. The assumptions are that the symmetry remains exact along the flow, the background-field coupling is kept fixed, and any modes removed between two scales contribute only local background counterterms to the long-distance functional. Under these assumptions the cocycle class is unchanged by RG flow.

Let Λ_{UV} and Λ_{IR} be two scales in the same theory, with the same background field B coupled to the same conserved current multiplet. Integrating out modes between the two scales gives

$$Z_{\mathrm{UV}}[B] = \exp\{\mathrm{i}C_{\mathrm{mid}}[B]\}\, Z_{\mathrm{IR}}[B],$$

where $C_{\mathrm{mid}}[B]$ is local in the background at the scale where the infrared description is defined. Applying g and using Definition 55.1 gives

$$Z_{\mathrm{UV}}[B^g] = \exp\{\mathrm{i}C_{\mathrm{mid}}[B^g]\}\exp\{\mathrm{i}\mathcal{A}_g^{\mathrm{IR}}[B]\}Z_{\mathrm{IR}}[B].$$

The ultraviolet transformation law gives the same quantity as

$$\exp\{\mathrm{i}\mathcal{A}_g^{\mathrm{UV}}[B]\}\exp\{\mathrm{i}C_{\mathrm{mid}}[B]\}Z_{\mathrm{IR}}[B].$$

Therefore

$$\mathcal{A}_g^{\mathrm{UV}}[B] = \mathcal{A}_g^{\mathrm{IR}}[B] + C_{\mathrm{mid}}[B^g] - C_{\mathrm{mid}}[B] \mod 2\pi,$$

which is equality of the local-counterterm equivalence classes.

55.3 A Perturbative Example of a Global Anomaly

The local triangle computation already distinguishes two uses of a current. Consider a four-dimensional massless Dirac fermion with vector and axial currents

$$j^\mu = \mathrm{i}\bar{\psi}\gamma^\mu\psi, \qquad j_A^\mu = \mathrm{i}\bar{\psi}\gamma^\mu\gamma_5\psi.$$

The preceding chapter fixed the regularization convention in which the vector current coupled to an ordinary $U(1)$ gauge field is preserved and the axial current carries the $j_A jj$ anomaly. To test the axial $U(1)$ itself as a background symmetry, couple a background gauge field A_μ to j_A^μ,

$$\Delta S = \int \mathrm{d}^4x\, A_\mu j_A^\mu.$$

Gauge invariance of this background coupling would require the Ward identity for the vector-current divergence in the three-point function with two axial insertions:

$$\partial_\mu \langle\Omega|\, \mathrm{T}\ j^\mu(x) j_A^\nu(y) j_A^\rho(z)|\Omega\rangle.$$

The same oriented triangle graphs as in the local axial anomaly appear, with two γ_5 insertions. In the $D = 4 - \epsilon$ regulator, split $\ell = \ell_\| + \ell_\perp$ and keep γ_5 four-dimensional. The trace term that remains after imposing the two axial-vector external Ward identities is proportional to

$$\ell_\perp^2(-4\mathrm{i})\epsilon^{\nu\rho\alpha\beta}(k-2\ell)_\alpha p_\beta.$$

The loop integral is symmetric under permutations of the three external momenta $k, p, -k-p$. Its finite local part is linear in the external momenta, hence

$$\epsilon^{\nu\rho\alpha\beta}(k-2\ell)_\alpha p_\beta \longmapsto \frac{1}{3}\epsilon^{\nu\rho\alpha\beta}k_\alpha p_\beta$$

inside the regulated contact term. With the conventions of the preceding chapter this gives

$$\partial_\mu \langle\Omega|\, \mathrm{T}\ j^\mu(x) j_A^\nu(y) j_A^\rho(z)|\Omega\rangle = -\frac{1}{6\pi^2}\epsilon^{\nu\rho\alpha\beta}\partial_\alpha\delta^{(4)}(x-z)\partial_\beta\delta^{(4)}(y-z). \tag{55.3}$$

Expanding the background-field effective action

$$\mathrm{e}^{\mathrm{i}\Gamma[A]} = \int D\psi\, \exp\left\{\mathrm{i}S_0[\psi] + \mathrm{i}\int \mathrm{d}^4x\, A_\mu j_A^\mu\right\}$$

to second order in A, equation (55.3) implies

$$\Gamma[A + \mathrm{d}\zeta] - \Gamma[A] = \int \mathrm{d}^4x\, \zeta(x)\, \frac{1}{48\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{55.4}$$

Proposition 55.4 (Axial background obstruction)**.** *In the vector-current-preserving regularization convention used above, the massless Dirac fermion has a nontrivial anomaly for the axial $U(1)$ background. More precisely, no four-dimensional local counterterm depending only on the axial background A, within this convention, can make $\Gamma[A]$ invariant under all compactly supported infinitesimal background transformations $A \mapsto A + \mathrm{d}\zeta$. Therefore the axial background cannot be promoted to an ordinary dynamical gauge field unless additional matter or topological data cancels the class represented by* (55.4)*.*

Proof. Gauge invariance of an Abelian background functional would require $\delta_\zeta \Gamma[A] = 0$ for every test function ζ. Equation (55.4) gives instead

$$\delta_\zeta \Gamma[A] = \frac{1}{48\pi^2} \int_{M_4} \zeta \, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}.$$

Since $F \wedge F$ can be varied independently in a compact spacetime region, this distribution is not identically zero. A local counterterm would change the right hand side by a local Abelian Wess–Zumino coboundary. In the single-background Abelian complex there is no four-form local functional whose infinitesimal variation is $\zeta \, F \wedge F$: the only possible Chern–Simons primitive $A \wedge F$ is a three-form, while $A \wedge A \wedge F$ vanishes for an Abelian one-form. Equivalently, the class whose descent gives $\zeta F \wedge F$ is the nonzero six-form anomaly polynomial proportional to $F \wedge F \wedge F$. Thus the obstruction cannot be removed without changing the theory, changing the symmetry being gauged, or adding another sector with the opposite anomaly. $\square$

Thus the axial rotation is a symmetry of the classical massless action in flat space without background flux, but its quantum coupling to an axial background is anomalous: the partition function is naturally a section of an anomaly line rather than a function on the quotient by axial background gauge transformations.

55.4 Example: A Finite SU(2) Anomaly

Let (M, σ_M) be a compact spin four-manifold and let $G_{\mathrm{f}} = SU(2)$. The spin structure σ_M is part of the background datum: a Weyl fermion is not defined on a merely oriented four-manifold. A change of σ_M, when several spin structures exist, is a change of the fermionic QFT background and may change the Pfaffian line and its anomaly. Consider one left-handed Weyl fermion valued in the fundamental representation of $SU(2)$. For an $SU(2)$ background connection B, the Euclidean Weyl operator is

$$D^+_{B,\sigma_M} : \Gamma(S_+ \otimes E_B) \longrightarrow \Gamma(S_- \otimes E_B),$$

where $S_\pm$ are the positive and negative chirality spin bundles and E_B is the associated rank-two complex vector bundle. The fundamental representation of $SU(2)$ is pseudoreal, so the fermion functional integral defines a Pfaffian line over the space of structured backgrounds (M, σ_M, B).

Let $g : M \to SU(2)$ be a background gauge transformation, and choose a smooth path of background connections B_t, $t \in [0,1]$, from $B_0 = B$ to $B_1 = B^g$. Identifying the two ends of $M \times [0,1]$ by g gives the mapping torus

$$X_g = (M \times [0,1]) / \big((x,1) \sim (x,0) \text{ by } g\big)$$

with an induced $SU(2)$ bundle E_g. The spin structure on $M \times [0,1]$ is the product one from σ_M; gluing the two ends by the identity diffeomorphism of M and the internal gauge transformation g gives a spin structure σ_g on X_g. Thus the invariant below is a function of (X_g, σ_g, E_g), not just of the oriented manifold and the vector bundle. If the spin structure on M is changed, or if the mapping-torus spin structure is twisted by an allowed $\mathbb{Z}_2$ flat line, the mod-two index may change.

Let $\mathcal{D}_{X_g,\sigma_g,E_g}$ be the five-dimensional real Dirac operator coupled to this bundle, and define its mod-two index by

$$\zeta(X_g, \sigma_g, E_g) = \dim \ker \mathcal{D}_{X_g,\sigma_g,E_g} \mod 2.$$

The analytic framework behind this statement is the Dai–Freed formulation of fermion anomalies. For a family of chiral Dirac operators, the determinant or Pfaffian is a section of a line over background-field space, and the holonomy of this line around a closed loop of backgrounds is computed by the Atiyah–Patodi–Singer eta-invariant of the Dirac operator on the associated mapping torus. In the present pseudoreal $SU(2)$ fundamental case the real structure refines the Dai–Freed holonomy to the mod-two-index sign

$$\mathrm{Hol}_{\mathrm{DF}}(X_g, \sigma_g, E_g) = (-1)^{\zeta(X_g,\sigma_g,E_g)}. \tag{55.5}$$

Equivalently, cutting the mapping torus open gives the APS-boundary description of the same Pfaffian transport. Therefore the Pfaffian transforms by

$$\frac{\mathrm{Pf}(D^+_{B^g,\sigma_M})}{\mathrm{Pf}(D^+_{B,\sigma_M})} = (-1)^{\zeta(X_g,\sigma_g,E_g)}. \tag{55.6}$$

For $M = S^4$ and g representing the nontrivial class in $\pi_4(SU(2)) \simeq \mathbb{Z}_2$, the five-dimensional mod-two index theorem gives $\zeta(X_g, \sigma_g, E_g) = 1$ for one fundamental Weyl doublet. The following proposition is the QFT consequence of this index-theoretic input.

Proposition 55.5 (Finite $SU(2)$ anomaly criterion for doublets)**.** *Let n left-handed Weyl fermions transform as fundamental doublets of a background $SU(2)$ symmetry on a compact spin four-manifold (M, σ_M). For a closed loop of backgrounds represented by the mapping torus (X_g, σ_g, E_g), the product Pfaffian line has holonomy*

$$\mathrm{Hol}(X_g, \sigma_g, E_g) = (-1)^{n\,\zeta(X_g,\sigma_g,E_g)}.$$

For the test family $M = S^4$ with g the generator of $\pi_4(SU(2)) \simeq \mathbb{Z}_2$, one has $\zeta(X_g, \sigma_g, E_g) = 1$. Hence an odd number of doublets has a finite $SU(2)$ anomaly. This obstruction is not detected by the local cubic descent calculation, because $SU(2)$ has no independent symmetric cubic invariant.

Proof. For one pseudoreal Weyl doublet the fermionic functional integral is not an ordinary complex determinant function on the space of backgrounds; it is a section of the real Pfaffian line associated with D^+_{B,σ_M}. Equation (55.6) states that parallel transport of this line around the background loop represented by (X_g, σ_g, E_g) multiplies the Pfaffian by $(-1)^{\zeta(X_g,\sigma_g,E_g)}$. This is the Dai–Freed holonomy specialized to the pseudoreal $SU(2)$ doublet, where the eta phase reduces to the mod-two index sign.

For n independent doublets the Grassmann integral factors before the fermions are coupled by any $SU(2)$-invariant mass term, and the anomaly line is the tensor product of the n one-doublet Pfaffian lines. Holonomy is multiplicative under tensor product, so the sign is the n-th power of the one-doublet sign:

$$\mathrm{Hol}(X_g, \sigma_g, E_g) = \left((-1)^{\zeta(X_g,\sigma_g,E_g)}\right)^n = (-1)^{n\,\zeta(X_g,\sigma_g,E_g)}.$$

For the Witten mapping torus over S^4, the mod-two index input gives $\zeta = 1$. The holonomy around the nontrivial loop is therefore $(-1)^n$.

If the $SU(2)$ is gauged, the fermion measure must descend from the space of connections to the quotient by gauge transformations. A loop in the quotient whose lifted Pfaffian line has holonomy -1 prevents such a single-valued descent: after going around the loop, a putative gauge-invariant partition function would have to equal its negative. Thus odd n is inconsistent with gauging this $SU(2)$ symmetry. If the same $SU(2)$ is kept as a background global symmetry, the same sign is an anomaly class that must be reproduced by any infrared description. Since the local cubic invariant $d^{abc} = \mathrm{tr}(t^a\{t^b, t^c\})$ vanishes for $\mathfrak{su}(2)$, this anomaly is a finite global obstruction rather than an infinitesimal local descent anomaly. $\square$

55.5 Anomaly Inflow and Mapping Tori

The finite anomaly in (55.1) is represented by a five-dimensional invertible response. Its holonomy on a mapping torus gives the finite transformation phase. In the form needed here, an invertible anomaly theory assigns a phase

$$\mathcal{I}_5[Y, \sigma_Y, B_Y] \in U(1)$$

to every closed five-manifold Y equipped with a chosen spin structure σ_Y, global-symmetry background B_Y, and any additional geometric structures required by the fermions. If $(M, \sigma_M) = \partial(Y, \sigma_Y)$, the same theory assigns an anomaly line over the space of boundary backgrounds, and the four-dimensional partition function $Z[M, \sigma_M, B]$ is a vector in that line. Choosing a filling (Y, σ_Y, B_Y) trivializes the line by multiplying by the bulk phase. Changing the filling changes the trivialization by the phase of the closed structured five-manifold obtained by gluing the two fillings.

For a finite background gauge transformation g, choose a path B_t from B to B^g and identify the two ends by g. The resulting mapping torus X_g carries the induced background B_g. The inflow statement is the equality

$$\frac{Z[M, \sigma_M, B^g]}{Z[M, \sigma_M, B]} = \mathcal{I}_5[X_g, \sigma_g, B_g], \tag{55.7}$$

after a choice of local counterterm convention. On the identity component of the background gauge group, cutting the mapping torus open reduces (55.7) to the descent formula of the preceding chapter. Infinitesimally, the five-dimensional Chern–Simons response varies by the boundary term $-\int_M I_4^{(1)}$, while the boundary generating functional varies by $+\int_M I_4^{(1)}$. For a disconnected component, the mapping-torus phase can remain nontrivial even when the local polynomial anomaly vanishes.

The $SU(2)$ example above is exactly of this finite type. Its anomaly theory assigns

$$\mathcal{I}_5^{SU(2)}[X_g, \sigma_g, E_g] = (-1)^{\zeta(X_g, \sigma_g, E_g)} \tag{55.8}$$

to the mapping torus with the induced rank-two bundle. Equation (55.6) is therefore the mapping-torus inflow law (55.7). An odd number of fundamental Weyl doublets has nontrivial $\mathcal{I}_5^{SU(2)}$; an even number has the trivial product phase for this test, and the Pfaffian line has no sign holonomy around the generator detected by X_g.

This language is also the precise version of the symmetry-protected topological interpretation of anomalies. A gapped short-range-entangled five-dimensional phase with symmetry G_{f} and no intrinsic topological order has an invertible response $\mathcal{I}_5$. A four-dimensional theory with anomaly $\mathcal{I}_5$ can occur as a boundary whose anomalous transformation is cancelled by the inverse bulk response. In mathematical treatments, deformation classes of fully extended reflection-positive invertible responses with a specified tangential and G_{f}-background structure are often organized by characters of the corresponding bordism groups, together with possible free Chern–Simons or theta-type terms. This cobordism organization is a classification framework for invertible responses; each use in QFT still requires specifying the background fields, spin or pin structure, locality assumptions, and the normalization of the fermion determinant or Pfaffian line.

55.6 Low-Degree Homotopy Groups in Four-Dimensional Applications

The preceding example used $\pi_4(SU(2))$ because a gauge transformation on the compactified four-space S^4 is a map $S^4 \to G$. Other nearby homotopy groups enter different constructions. If a principal G-bundle on S^4 is described by transition data on the equator, the transition function is a map $S^3 \to G$, so the instanton number is controlled by $\pi_3(G)$ and by the normalization of the corresponding characteristic class. If a four-dimensional nonlinear sigma model has target G and its Wess–Zumino term is written by extending a field $U : M^4 \to G$ to a five-manifold, the quantization is governed by the integral cohomology class of the closed five-form used in the Wess–Zumino functional. For the standard simple-group case $G = SU(N)$, $N \geq 3$, this quantization is tested on maps $S^5 \to G$, hence by $\pi_5(G) \simeq \mathbb{Z}$. Thus π_3, π_4, π_5 play different roles and should not be conflated.

The following table records the low-degree groups most often met for compact connected classical gauge groups. The entries for $SO(N)$ with $N \geq 7$, $U(N)$ with $N \geq 3$, $SU(N)$ with $N \geq 3$, and $Sp(N)$ with $N \geq 2$ are the stable values in the displayed range, as given by Bott periodicity. For $d \geq 2$, $SO(N)$ and $Spin(N)$ have the same π_d; their π_1 differs because $Spin(N) \to SO(N)$ is the universal double cover.

G	$\pi_1(G)$	$\pi_2(G)$	$\pi_3(G)$	$\pi_4(G)$	$\pi_5(G)$
$U(1)$	$\mathbb{Z}$	0	0	0	0
$U(N),\ N \geq 3$	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$SU(2) \simeq Sp(1)$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$
$SU(N),\ N \geq 3$	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$SO(N),\ N \geq 7$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0
$Spin(N),\ N \geq 7$	0	0	$\mathbb{Z}$	0	0
$Sp(N),\ N \geq 2$	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$

The low-rank orthogonal groups are exceptional because of accidental isomorphisms:

G	$\pi_3(G)$	$\pi_4(G)$	$\pi_5(G)$
$Spin(3) \simeq SU(2)$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$
$Spin(4) \simeq SU(2) \times SU(2)$	$\mathbb{Z}^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
$Spin(5) \simeq Sp(2)$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$
$Spin(6) \simeq SU(4)$	$\mathbb{Z}$	0	$\mathbb{Z}$.

For simply connected exceptional compact groups

$$G = G_2, F_4, E_6, E_7, E_8$$

one has

$$\pi_1(G) = \pi_2(G) = 0, \qquad \pi_3(G) \simeq \mathbb{Z}, \qquad \pi_4(G) = \pi_5(G) = 0.$$

There are further low-degree torsion groups in special cases, for example $\pi_6(SU(2)) \simeq \mathbb{Z}_{12}$, $\pi_6(SU(3)) \simeq \mathbb{Z}_6$, and $\pi_6(G_2) \simeq \mathbb{Z}_3$. These become relevant in higher-dimensional global-anomaly questions.

This table is a guide to the first topological obstructions, not a replacement for the anomaly itself. A modern finite anomaly is a phase of the fermion determinant or Pfaffian line, and in its most invariant formulation it is an invertible field theory on one higher dimension. Homotopy groups detect important test families of backgrounds; the complete torsion classification can depend on spin structure, global form of the group, matter representation, and bordism data.

55.7 Vacuum Sectors and Locality

Spontaneous symmetry breaking is a statement about phases of an infinite-volume quantum system. A useful comparison is a one-dimensional quantum mechanical Hamiltonian

$$H = \frac{P^2}{2} + V(x), \qquad V(x) = V(-x),$$

where V has two degenerate minima at $x = \pm a$. Figure 55.2 uses this double-well model to fix the finite-volume comparison: the exact ground state is symmetric, while localized wave packets approximate broken-sector states only after the infinite-volume limit is taken.

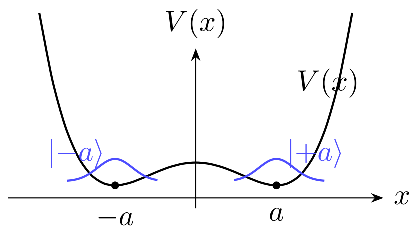


Figure 55.2: A finite-dimensional double well has localized semiclassical states near the two minima, but the exact ground state is a symmetric superposition.

The localized semiclassical states $|+a\rangle$ and $|-a\rangle$ combine into Hamiltonian eigenstates

$$|\psi_\pm\rangle = \frac{1}{\sqrt{2}}\big(|+a\rangle \pm |-a\rangle\big).$$

The splitting between them is exponentially small in the Euclidean tunneling action,

$$\Delta E \sim \mathrm{e}^{-S_0}, \qquad -\ddot{x} + V'(x) = 0, \qquad x(\pm\infty) = \pm a.$$

For a local quantum field theory with the analogous set of homogeneous minima, the tunneling action is proportional to the spatial volume. In the infinite-volume limit the transition amplitude between distinct homogeneous phases is suppressed as

$$\exp(-S_0 \ \mathrm{Vol}) \longrightarrow 0.$$

The pure vacua therefore define distinct phases rather than two nearly degenerate levels of one finite-dimensional Hamiltonian.

Locality gives the corresponding superselection statement. Let $\{|\Omega_i\rangle\}$ be translation-invariant pure vacua in infinite volume. For any local operator $\mathcal{A}(x)$, the vacuum matrix

$$A_{ij} := \langle\Omega_i|\mathcal{A}(0)|\Omega_j\rangle$$

can be chosen diagonal in the phase label. One way to see the constraint is to compare two local operators $\mathcal{A}(x)$ and $\mathcal{B}(y)$ at large spacelike separation. Insert the vacuum sector decomposition and use cluster decomposition in a pure phase:

$$\langle\Omega_i|\mathcal{A}(x)\mathcal{B}(y)|\Omega_j\rangle = \sum_k A_{ik}B_{kj} + \text{terms vanishing as } |x-y| \to \infty.$$

For spacelike separation, locality gives $[\mathcal{A}(x), \mathcal{B}(y)] = 0$, hence the vacuum matrices satisfy

$$\sum_k A_{ik}B_{kj} = \sum_k B_{ik}A_{kj}$$

for all local $\mathcal{A}, \mathcal{B}$. The commuting family of vacuum matrices is equivalently decomposed into its characters. In a character representation every element of the vacuum-sector algebra acts by a scalar. If a local observable had an off-diagonal matrix element between two distinct characters, multiplication by another element of the algebra would give different left and right scalar actions unless the two characters were equal. Thus, in the pure-sector basis, local operators do not connect distinct phases:

$$\langle\Omega_i|\mathcal{A}(x)|\Omega_j\rangle = 0 \qquad (i \neq j).$$

This is superselection by locality and clustering, not a finite-volume level splitting statement.

55.8 Spontaneous Symmetry Breaking

Definition 55.6 (Internal spontaneous symmetry breaking datum)**.** Consider a Lorentz-invariant quantum theory in infinite spatial volume with an exact continuous internal symmetry group G. Choose one pure phase and let $|\Omega\rangle$ be its vacuum vector. The unbroken subgroup is

$$H = \{h \in G : U(h)|\Omega\rangle = |\Omega\rangle\}.$$

The symmetry is spontaneously broken from G to H when H is a proper subgroup. Let $\mathfrak{g} = \mathrm{Lie}(G)$ and $\mathfrak{h} = \mathrm{Lie}(H)$. Choose an H-stable complement $\mathfrak{m}$ so that

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}, \qquad [\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}.$$

Elements $T_i \in \mathfrak{h}$ generate unbroken transformations and elements $X_a \in \mathfrak{m}$ generate broken directions.

Definition 55.7 (Goldstone effective action in CCWZ coordinates)**.** Goldstone fields are local coordinates on the coset G/H. A coset representative may be written

$$\xi(\pi(x)) = \exp\left\{\frac{\mathrm{i}}{f}\pi^a(x)X_a\right\}. \tag{55.9}$$

For $g \in G$, the defining nonlinear action is

$$g\,\xi(\pi) = \xi(\pi')\,h(g,\pi), \qquad h(g,\pi) \in H. \tag{55.10}$$

The compensator $h(g,\pi)$ is determined by the chosen coset section and by the requirement that $g\xi(\pi)$ be written again in the form $\xi(\pi')H$.

The Maurer–Cartan form decomposes as

$$\xi^{-1}\mathrm{d}\xi = \mathrm{i}e^a X_a + \mathrm{i}\omega^i T_i. \tag{55.11}$$

The one-forms $e^a = e^a_\mu \mathrm{d}x^\mu$ are the covariant vielbein on the Goldstone target G/H, while ω^i is an H-connection. If G/H carries an invariant metric G_{ab} on the broken tangent space, the leading Goldstone action is

$$S_{\text{Goldstone}} = \frac{f^2}{2}\int G_{ab}\, e^a_\mu e^{b\mu}\,\mathrm{d}^4x + \cdots\,. \tag{55.12}$$

The ellipsis denotes higher-derivative terms and explicit symmetry-breaking terms. No potential appears for exact Goldstone coordinates because constant maps into G/H label symmetry-related vacua.

55.9 Goldstone Theorem as a Spectral Statement

Proposition 55.8 (Goldstone spectral pole from a broken current)**.** *The spectral input behind* (55.12) *is as follows. Assume a conserved current j^μ_A, a local operator $\mathcal{O}$, and a nonzero order parameter*

$$\langle\Omega, [Q_A, \mathcal{O}(0)]\Omega\rangle = \langle\Omega, \delta_A\mathcal{O}(0)\Omega\rangle \neq 0, \tag{55.13}$$

where Q_A is obtained as the infinite-volume limit of smeared charges. Assume also that local contact ambiguities in the time-ordered product have been fixed so that the nonlocal part of the mixed current-operator distribution has the Lorentz-covariant scalar spectral form near $p = 0$: modulo a local polynomial in p,

$$\widehat{G}^\mu_A(p)_{\text{nonloc}} = p^\mu F_A(p^2) + \widehat{G}^\mu_{A,\perp}(p), \qquad p_\mu\widehat{G}^\mu_{A,\perp}(p) = 0,$$

where F_A is a scalar boundary-value distribution with a Källén–Lehmann representation in the broken channel. The Ward identity for the time-ordered distribution is

$$\partial_\mu\langle\Omega, T\{j^\mu_A(x)\mathcal{O}(0)\}\Omega\rangle = -\mathrm{i}\delta^{(4)}(x)\langle\Omega, \delta_A\mathcal{O}(0)\Omega\rangle. \tag{55.14}$$

Fourier transforming gives a distribution $\widehat{G}^\mu_A(p)$ satisfying

$$p_\mu\widehat{G}^\mu_A(p) = \langle\Omega, \delta_A\mathcal{O}(0)\Omega\rangle.$$

Since the right-hand side is nonzero at $p = 0$, $F_A(p^2)$ must contain a $1/(p^2 - \mathrm{i}0)$-type massless singularity. In the Källén–Lehmann language this singularity is a massless contribution to the current-operator spectral measure.

Proof. Let $c_A = \langle\Omega, \delta_A\mathcal{O}(0)\Omega\rangle \neq 0$. Integrating (55.14) against a test function equal to one near the origin gives the broken-charge matrix element (55.13); Fourier transformation converts the distributional divergence into multiplication by p_μ. After the chosen local contact polynomial has been separated, the transverse part drops out and the Ward identity reads, for the nonlocal scalar channel,

$$p^2F_A(p^2) + p_\mu P^\mu_A(p) = c_A,$$

where $P_A^\mu(p)$ is the remaining local polynomial vector contribution. The contraction $p_\mu P_A^\mu(p)$ has no constant term at $p = 0$. If F_A were regular at $p^2 = 0$, then $p^2 F_A(p^2)$ would also vanish at the origin as a boundary-value distribution modulo higher-order local polynomial terms, contradicting $c_A \neq 0$. Thus the nonlocal scalar channel contains

$$F_A(p^2) = \frac{c_A}{p^2 - \mathrm{i}0} + \text{less singular terms}$$

up to the sign fixed by the Fourier convention. In a scalar spectral representation

$$F_A(p^2) = \int_0^\infty \frac{\mathrm{d}\nu_A(\mu^2)}{p^2 + \mu^2 - \mathrm{i}0} + \text{local polynomial},$$

such an inverse p^2 boundary value is precisely an atom of $\mathrm{d}\nu_A$ at $\mu^2 = 0$. This is the massless Goldstone contribution to the spectral measure. □

Equivalently, for each broken direction there are zero-mass one-particle states $|\pi_a(p)\rangle$ created by the current:

$$\langle \Omega | j_A^\mu(0) | \pi_a(p) \rangle = \mathrm{i} F_{Aa} p^\mu, \qquad p^2 = 0. \tag{55.15}$$

If the translation generators have spectrum contained in the closed forward light cone and the vacuum is the unique translation-invariant vector in the chosen phase, the matrix F_{Aa} has rank equal to the number of broken generators. These massless states are the quanta described by the Goldstone coordinates $\pi^a(x)$.

Remark 55.9 (Goldstone theorem and gauged directions). The spectral argument above is a theorem about a global symmetry current acting on the physical positive Hilbert space. It assumes that Q_A is a charge defined on physical states and that $\delta_A \mathcal{O}$ is the variation of a gauge-invariant local observable or of a local field in an ordinary positive-metric representation.

Gauge redundancy is different. In a covariant gauge-fixed description the state space before the physical quotient is a Krein space, as in Chapter 47, and physical states are BRST cohomology classes. A scalar coordinate that moves along a gauged orbit is then a would-be Goldstone coordinate, not by itself a physical massless particle. At the level of gauge theories with scalar matter, this is the Brout–Englert–Higgs mechanism developed in Section 45.5. At quadratic order in the Abelian Higgs model, using here the conventional explicit-coupling coordinate $D_\mu = \partial_\mu - \mathrm{i} g A_\mu$, write

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x) + \mathrm{i}\pi(x)\big), \qquad m_A = gv.$$

The scalar kinetic term contains

$$|D_\mu \phi|^2 = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}(\partial_\mu \pi - m_A A_\mu)^2 + \text{higher powers of } h, \pi, A. \tag{55.16}$$

Equivalently, away from zeros of $v + h$, the gauge-invariant vector

$$\mathcal{A}_\mu = A_\mu - \frac{1}{m_A}\partial_\mu \pi$$

has the Proca mass term $\frac{1}{2} m_A^2 \mathcal{A}_\mu \mathcal{A}^\mu$ at quadratic order. In an R_ξ gauge one instead uses the gauge-fixing functional

$$F_\xi = \partial^\mu A_\mu + \xi m_A \pi,$$

which cancels the $A_\mu\partial^\mu\pi$ mixing and assigns the unphysical fields π and the Faddeev–Popov ghosts the gauge-dependent mass $\sqrt{\xi}\,m_A$. The longitudinal gauge polarization, the would-be Goldstone coordinate, and the ghost-antighost variables form BRST doublet or quartet sectors, so they have no independent positive-norm particle states in the BRST cohomology. The physical spectrum contains a massive vector particle with three polarizations and the gauge-invariant scalar excitation h, plus any Goldstone particles associated with broken global symmetries that remain after the gauge redundancy has been quotiented.

55.10 The $U(1)$ Goldstone Model

The spectral conclusion is realized explicitly by a complex scalar field with

$$\mathcal{L} = -\partial_\mu\phi^*\partial^\mu\phi - V(|\phi|).$$

Figure 55.3 displays the vacuum circle and the radial/angular coordinates used to separate the massive radial excitation from the Goldstone mode.

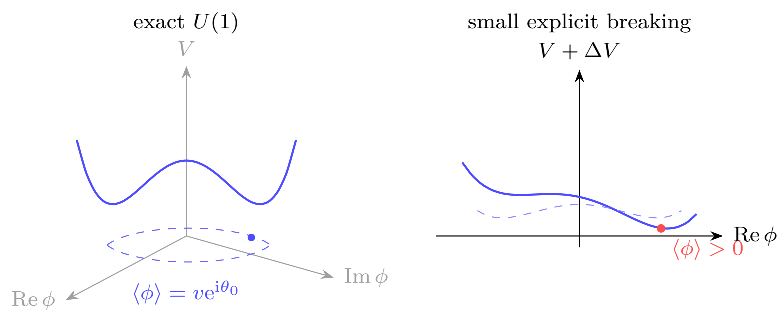


Figure 55.3: For a $U(1)$-invariant potential minimized at $|\phi| = v$, the minima form a circle of vacua. A small deformation $\Delta V = -\epsilon\,\mathrm{Re}\,\phi$ selects one point on that circle.

The $U(1)$ current is

$$j^\mu = -\mathrm{i}(\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*).$$

Write

$$\phi(x) = r(x)\mathrm{e}^{\mathrm{i}\theta(x)}.$$

Then

$$\mathcal{L} = -\partial_\mu r\,\partial^\mu r - V(r) - r^2\partial_\mu\theta\,\partial^\mu\theta, \qquad j^\mu = 2r^2\partial^\mu\theta.$$

The charge $Q = \int \mathrm{d}^3x\, j^0$ generates the shift of the angular field,

$$[Q, \theta(x)] = \mathrm{i},$$

with the sign fixed by the current convention. If $V(r)$ is minimized at $r = v > 0$, write $r = v + \rho$. The quadratic Lagrangian contains

$$-\partial_\mu\rho\,\partial^\mu\rho - \frac{1}{2}V''(v)\rho^2 - v^2\partial_\mu\theta\,\partial^\mu\theta.$$

The radial field ρ is massive when $V''(v) > 0$, while θ is a massless field whose constant shifts move along the vacuum circle.

A small explicit symmetry-breaking deformation selects a phase. For example,

$$\Delta V(\phi) = -\epsilon \ \mathrm{Re}\,\phi, \qquad \epsilon > 0,$$

chooses the vacuum with $\theta = 0$. In the angular effective theory,

$$\Delta\mathcal{L} = \epsilon v \cos\theta = \epsilon v - \frac{\epsilon v}{2}\theta^2 + O(\theta^4),$$

so the Goldstone boson becomes a pseudo-Goldstone boson with

$$m_\theta^2 \propto \frac{\epsilon}{v}.$$

This is the prototype for the pion mass generated by small quark masses.

Remark 55.10 (Masses in the $U(1)$ Goldstone model)*.* With the mostly-plus kinetic convention used in this volume, define canonical small-fluctuation fields near the vacuum by

$$h = \sqrt{2}\,\rho, \qquad \pi = \sqrt{2}\,v\,\theta.$$

If $V'(v) = 0$ and $V''(v) > 0$, then the radial excitation has

$$m_h^2 = \frac{1}{2}V''(v),$$

while the angular excitation is massless when the $U(1)$ symmetry is exact. For the explicit breaking $\Delta V = -\epsilon\,\mathrm{Re}\,\phi$, the angular excitation at $\theta = 0$ has, to first order in ϵ,

$$m_\pi^2 = \frac{\epsilon}{2v} + O(\epsilon^2).$$

The quadratic kinetic terms are

$$-\partial_\mu\rho\,\partial^\mu\rho - v^2\partial_\mu\theta\,\partial^\mu\theta = -\frac{1}{2}\partial_\mu h\,\partial^\mu h - \frac{1}{2}\partial_\mu\pi\,\partial^\mu\pi.$$

Expanding the symmetric potential gives

$$-V(v+\rho) = -V(v) - \frac{1}{2}V''(v)\rho^2 + O(\rho^3) = -V(v) - \frac{1}{4}V''(v)h^2 + O(h^3),$$

which is the canonical mass term $-\frac{1}{2}m_h^2h^2$ with $m_h^2 = V''(v)/2$. No term depending on a constant θ is allowed by the exact shift symmetry, so π has no mass term. The explicit breaking contributes

$$\epsilon v\cos\theta = \epsilon v - \frac{\epsilon v}{2}\theta^2 + O(\theta^4) = \epsilon v - \frac{\epsilon}{4v}\pi^2 + O(\pi^4),$$

and therefore $m_\pi^2 = \epsilon/(2v)$ in canonical normalization.

55.11 Chiral Symmetry of Massless QCD

Let $q_L^I = P_Lq^I$ and $q_R^I = P_Rq^I$, $I = 1, \dots, N_f$, be massless quark fields in the fundamental representation of the color gauge group. The connected continuous flavor symmetry acting faithfully on local gauge-invariant operators has Lie algebra

$$\mathfrak{su}(N_f)_L \oplus \mathfrak{su}(N_f)_R \oplus \mathfrak{u}(1)_B.$$

For local current algebra computations this Lie algebra is represented by

$$SU(N_f)_L \times SU(N_f)_R \times U(1)_B.$$

The precise global form includes finite quotients by common center actions; those quotients matter for global background fields and are developed later. The anomaly derived in the preceding chapter removes the axial $U(1)_A$ rotation from the exact continuous quantum symmetry group.

The left and right transformations act by

$$q_L \longmapsto Lq_L, \qquad q_R \longmapsto Rq_R, \qquad (L, R) \in SU(N_f)_L \times SU(N_f)_R.$$

Conjecture 55.11 (Chiral symmetry breaking in QCD)**.** *For four-dimensional $SU(N_c)$ QCD with N_f light fundamental quarks in the range where the infrared theory is confining rather than conformal, the infinite-volume vacuum selected by equal positive quark masses has a nonzero chiral order parameter in the subsequent massless limit:*

$$\lim_{m\downarrow 0}\lim_{V\to\infty} \langle \bar{q}_{R,J} q_{L,I}\rangle_{m,V} = \Sigma_0\, \delta_{IJ}, \qquad \Sigma_0 \neq 0. \tag{55.17}$$

The unbroken continuous nonabelian flavor symmetry in this phase is the diagonal vector subgroup $SU(N_f)_V$.

Conjecture 55.11 is a dynamical assertion about the QCD vacuum. It is not presently a theorem derived from the four-dimensional continuum QCD Lagrangian or from a completed continuum limit of lattice QCD. Lattice simulations and phenomenology give strong evidence for the small-N_f theory, while semiclassical instanton calculations do not supply a controlled derivation of the infrared condensate. The pion effective theory below is the low-energy theory conditional on this chiral breaking phase.

In the phase described by Conjecture 55.11, the bilinear condensate has an orientation that can be chosen as

$$\Sigma_{IJ} = \langle \bar{q}_{R,J} q_{L,I}\rangle = \Sigma_0\, \delta_{IJ},$$

so that the stabilizer is the diagonal subgroup. More explicitly, before choosing the orientation of the vacuum, write

$$A_{IJ} := \langle \bar{q}_{R,J} q_{L,I}\rangle.$$

Under $SU(N_f)_L \times SU(N_f)_R$,

$$A \longmapsto R^\dagger A L.$$

A choice of L and R can diagonalize A. If the vacuum is invariant under the non-chiral subgroup, then A is proportional to the identity. The unbroken transformations are precisely $L = R$. The breaking pattern is therefore

$$SU(N_f)_L \times SU(N_f)_R \longrightarrow SU(N_f)_V. \tag{55.18}$$

Given this dynamical input, the number of Goldstone bosons is $N_f^2 - 1$.

Definition 55.12 (QCD Goldstone target)**.** Conditional on the breaking pattern (55.18), the Goldstone target is

$$\frac{SU(N_f)_L \times SU(N_f)_R}{SU(N_f)_V} \simeq SU(N_f).$$

The coordinate $U \in SU(N_f)$ is chosen with left flavor as the row index and right flavor as the column index. With this index convention it parameterizes the condensate orientation relative to the reference condensate $U = 1$, and the left and right flavor groups act by

$$U \longmapsto LUR^{-1}.$$

This is the index convention used by the mass spurion and the pion effective action below; it is the same physical orientation as the bilinear matrix $A_{IJ} = \langle \bar{q}_{R,J} q_{L,I} \rangle$ above, written with the opposite choice of displayed row and column indices. The diagonal subgroup $L = R$ fixes the reference orientation $U = 1$ and acts on the tangent space at $U = 1$ by the adjoint representation of $\mathfrak{su}(N_f)$.

At $\theta = 0$, the non-chiral flavor symmetry is constrained by the positivity structure of the Euclidean vectorlike theory. The statement has a precise theorem boundary.

Hypothesis 55.13 (Euclidean measure needed for Vafa–Witten positivity)**.** For the vectorlike theory with equal positive quark masses $m > 0$, assume that there is a family of gauge-invariant Euclidean finite-regulator measures $d\nu_{m,\Lambda}(A)$. If a continuum statement is claimed, assume in addition that the gauge-invariant Schwinger functions and the source functionals for the order parameters used below have a continuum limit, denoted $d\nu_m(A)$ in this paragraph, with the following properties:

i. expectations of gauge-invariant polynomially bounded fermion bilinears and their source deformations are defined at the regulator level and converge in the limit in which the Vafa–Witten conclusion is asserted;

ii. the full regulator measure, including the gauge action and the vectorlike fermion discretization or determinant, is reflection positive for a Euclidean time reflection, and this positivity is preserved by the limiting Schwinger functions. Thus OS reconstruction applies to the observables used in the argument;

iii. the Dirac operator and fermion regulator have vectorlike determinant positivity in each fixed gauge-field background. In the continuum notation this is the anti-Hermiticity and eigenvalue pairing statement

$$\det(\not{D}_A + m) = \prod_{\lambda > 0} (\lambda^2 + m^2)\, m^{n_0(A)} \geq 0;$$

on a lattice it is replaced by the corresponding positivity property of the chosen vectorlike fermion action;

iv. the infinite-volume and small-mass limits used to select a phase exist for the order parameters under consideration.

At finite lattice spacing, these hypotheses are properties to verify for the chosen lattice gauge and fermion action. The Wilson plaquette gauge action is reflection positive in the sense described in Chapter 46, and standard vectorlike fermion discretizations must be checked with their own reflection-positivity and determinant-positivity conditions. These properties are not automatic for an arbitrary regulator or gauge-fixed expression. For the four-dimensional continuum gauge theory, the existence of $d\nu_m(A)$ with the required locality, reflection positivity, and scaling limits is part of the Yang–Mills/QCD construction problem; the finite-lattice positivity argument is therefore not, by itself, a construction of continuum QCD.

Under Hypothesis 55.13, the determinant factor in each fixed gauge-field background is non-negative, and for equal masses the fermion determinant and gauge measure are invariant under $SU(N_f)_V$. A vector-flavor-breaking external source may then be averaged over the compact vector flavor group without decreasing the Euclidean functional integral. In the infinite-volume limit this convexity/positivity comparison rules out a lower vacuum energy for a vector-flavor-breaking orientation. Taking the small-mass limit therefore selects, when the limit exists, a vacuum preserving $SU(N_f)_V$. This is the Vafa–Witten positivity argument in the form needed here. It does not by itself prove the existence of the chiral condensate; it constrains the orientation of the condensate once the vectorlike massive regulator and its reflection-positive measure have been constructed.

The complex phase of Σ_0 is tied to the topological angle. A flavor-singlet axial rotation would rotate Σ_0, but the same transformation is anomalous and shifts θ. Thus the invariant structure are the condensate orientation together with the anomaly-invariant combination of θ and the quark-mass phase discussed in the preceding chapter.

55.12 Topological Susceptibility and the Singlet Axial Direction

The nonabelian pion field below takes values in $SU(N_f)$, because the axial $U(1)_A$ rotation is anomalous at finite N_c. At large N_c the anomaly is suppressed in the singlet meson sector in a precise sense: the pure Yang–Mills topological susceptibility remains $O(1)$, while $f_\pi^2 = O(N_c)$. The resulting singlet mass is therefore $O(N_c^{-1})$. This paragraph is a conditional large-N_c matching statement, not a proof of the pure Yang–Mills continuum limit or of the full QCD chiral phase.

Hypothesis 55.14 (Large-N_c singlet theta matching)**.** Fix N_f while $N_c \to \infty$, and assume the following inputs.

1. Pure $SU(N_c)$ Yang–Mills has a topological susceptibility χ_{YM} in the sense of Definition 51.18, with $\chi_{\mathrm{YM}} = O(N_c^0)$.
2. In the branch near $\vartheta = 0$, the pure Yang–Mills vacuum energy has the local expansion
$$E_{\mathrm{YM}}(\vartheta) = E_{\mathrm{YM}}(0) + \frac{1}{2}\chi_{\mathrm{YM}}\vartheta^2 + O(\vartheta^4/N_c^2), \tag{55.19}$$
while 2π-periodicity is restored by the usual branch label $\vartheta \mapsto \vartheta + 2\pi k$.
3. The chiral effective field may be enlarged locally from $SU(N_f)$ to $U(N_f)$ near the identity:
$$\widehat{U} = \exp\left(\frac{2\mathrm{i}}{f_\pi}(\eta_0 T^0 + \pi^a T^a)\right), \qquad T^0 = \frac{\mathbf{1}_{N_f}}{\sqrt{2N_f}}, \qquad \mathrm{Tr}(T^A T^B) = \frac{1}{2}\delta^{AB}. \tag{55.20}$$
The leading kinetic term is
$$\frac{f_\pi^2}{4}\,\mathrm{Tr}(\partial_\mu \widehat{U}\partial^\mu \widehat{U}^\dagger), \tag{55.21}$$
so that η_0 is canonically normalized.
4. To leading order in the fixed-N_f large-N_c expansion and at vanishing quark masses, the theta dependence enters the singlet effective potential through
$$\vartheta = \theta - \mathrm{i}\log\det\widehat{U} \tag{55.22}$$

on the branch near $\widehat{U} = 1$. With the convention $U \mapsto LUR^{-1}$, a common axial change of variables sends $\theta \mapsto \theta - 2N_f\alpha$ and $\mathrm{i}\log\det\widehat{U} \mapsto \mathrm{i}\log\det\widehat{U} - 2N_f\alpha$, so this is the invariant coordinate. When the mass term aligns $\widehat{U}$ with the phase of M, it reduces to the microscopic strong-CP phase $\theta + \arg\det M$.

Witten–Veneziano mass formula from the large-N_c matching input. Under Hypothesis 55.14, the singlet field η_0 has, in the chiral limit and at leading fixed-N_f large N_c,

$$m_{\eta_0}^2 = \frac{2N_f}{f_\pi^2}\chi_{\mathrm{YM}} + O(N_c^{-2}). \tag{55.23}$$

Equivalently, because $f_\pi^2 = O(N_c)$ while $\chi_{\mathrm{YM}} = O(1)$, the singlet mass squared is $O(N_c^{-1})$.

Since the non-singlet generators are traceless,

$$-\mathrm{i}\log\det\widehat{U} = \frac{2\eta_0}{f_\pi}\operatorname{Tr}T^0 = \frac{\sqrt{2N_f}}{f_\pi}\eta_0 \tag{55.24}$$

on the branch near the identity. Substituting (55.22) into the local Yang–Mills branch (55.19) gives the singlet potential

$$V_{\mathrm{singlet}}(\eta_0;\theta) = \frac{1}{2}\chi_{\mathrm{YM}}\left(\theta + \frac{\sqrt{2N_f}}{f_\pi}\eta_0\right)^2 + O(N_c^{-2})$$

in the effective Lagrangian. At fixed $\theta = 0$, the quadratic term is

$$\frac{1}{2}\left(\frac{2N_f\chi_{\mathrm{YM}}}{f_\pi^2}\right)\eta_0^2.$$

The kinetic term (55.21) gives $\frac{1}{2}\partial_\mu\eta_0\partial^\mu\eta_0$, because $\operatorname{Tr}(T^0T^0) = 1/2$. The displayed coefficient is therefore the mass squared.

Theta-curvature matrix and the screening direction. The finite-regulator cumulant identity (51.52) fixes the sign of the pure Yang–Mills curvature:

$$\chi_{\mathrm{YM}} = E''_{\mathrm{YM}}(0) \geq 0$$

when the limit exists in a CP-invariant branch. In the local singlet chart set

$$a := \frac{\sqrt{2N_f}}{f_\pi}, \qquad V(\theta,\eta_0) = \frac{1}{2}\chi_{\mathrm{YM}}(\theta + a\eta_0)^2.$$

The Hessian in the coordinate pair (θ,η_0) is the rank-one matrix

$$H_{\theta,\eta} = \chi_{\mathrm{YM}}\begin{pmatrix}1 & a\\ a & a^2\end{pmatrix}. \tag{55.25}$$

Thus $H_{\theta,\eta}(-a,1)^T = 0$: shifting θ and η_0 so that $\theta + a\eta_0$ is fixed is the massless-quark screening direction. By contrast, the fixed-θ curvature in the η_0 direction is

$$\partial_{\eta_0}^2 V|_{\theta=0} = a^2\chi_{\mathrm{YM}} = \frac{2N_f}{f_\pi^2}\chi_{\mathrm{YM}},$$

which is exactly the mass coefficient in (55.23). The mixed derivative $\partial_\theta \partial_{\eta_0} V = a\chi_{\mathrm{YM}}$ records the sign of the anomaly-invariant coordinate $\theta - \mathrm{i}\log\det \widehat{U}$; reversing that sign would give the same positive mass but the opposite singlet screening convention.

Controlled Approximation 55.15 (Residual-controlled Witten–Veneziano matching)**.** The preceding rank-one matrix is the leading local Hessian in the chiral, fixed-N_f, large-N_c matching window. If the comparison is made in the exact massless full-QCD theory, with a continuum and regulator convention which preserves the anomalous axial Ward identity, the first residual cannot be an arbitrary symmetric 2×2 matrix. With at least one exactly massless quark the θ-angle is removable by an anomalous chiral rotation, so the full vacuum energy is independent of θ. Therefore the local residual must preserve the screening vector:

$$H^{\mathrm{loc}}_{\theta,\eta}(-a, 1)^T = 0.$$

In the two-variable local Hessian this condition forces the residual to stay in the same anomaly-invariant coordinate. Thus write

$$H^{\mathrm{loc}}_{\theta,\eta} = \chi_{\mathrm{eff}} \begin{pmatrix} 1 & a \\ a & a^2 \end{pmatrix} = (\chi_{\mathrm{YM}} + \delta\chi) \begin{pmatrix} 1 & a \\ a & a^2 \end{pmatrix}, \qquad a = \frac{\sqrt{2N_f}}{f_\pi}, \tag{55.26}$$

with $\chi_{\mathrm{eff}} > 0$ in the selected local branch. The scalar $\delta\chi$ records the declared large-N_c, branch-selection, and local counterterm uncertainty in the same renormalization convention as χ_{YM}; it shifts the fixed-θ singlet curvature and the mixed curvature together. At fixed θ, the singlet potential curvature in the leading field coordinate is

$$H^{\mathrm{loc}}_{\eta\eta} = a^2 \chi_{\mathrm{eff}} = \frac{2N_f}{f_\pi^2}(\chi_{\mathrm{YM}} + \delta\chi). \tag{55.27}$$

But minimizing the same local quadratic form over the singlet field gives

$$\chi^{m=0,\mathrm{loc}}_{\mathrm{QCD}} = \chi_{\mathrm{eff}} - \frac{(a\chi_{\mathrm{eff}})^2}{a^2 \chi_{\mathrm{eff}}} = 0. \tag{55.28}$$

The exact zero in (55.28) is part of the massless matching condition, not an optional cancellation imposed after the fact.

A physical pole mass also requires the kinetic normalization of the singlet field. At the same order at which $\delta\chi$ is kept, write the local quadratic kinetic term in the η_0 coordinate as

$$\frac{1}{2} Z_0\, \partial_\mu \eta_0 \partial^\mu \eta_0, \qquad F_0^2 := Z_0 f_\pi^2. \tag{55.29}$$

Then the mass statement is the one-field generalized eigenvalue

$$m^2_{\eta_0,\mathrm{phys}} = \frac{H^{\mathrm{loc}}_{\eta\eta}}{Z_0} = \frac{2N_f}{F_0^2}(\chi_{\mathrm{YM}} + \delta\chi), \tag{55.30}$$

while (55.27) is the corresponding potential curvature in the leading field coordinate. Thus the usable uncertainty statement is: if $|\delta\chi| \le r_\chi$, $|Z_0 - 1| \le r_Z < 1$, and the leading comparison mass is $m^2_{\mathrm{WV},0} = 2N_f \chi_{\mathrm{YM}}/f_\pi^2$, then

$$\left| m^2_{\eta_0,\mathrm{phys}} - m^2_{\mathrm{WV},0} \right| \le \frac{2N_f}{f_\pi^2} \frac{r_\chi + \chi_{\mathrm{YM}} r_Z}{1 - r_Z}, \qquad \chi^{m=0,\mathrm{loc}}_{\mathrm{QCD}} \equiv 0. \tag{55.31}$$

The first term in the numerator is the potential-curvature uncertainty; the second is the independent singlet decay-constant or wavefunction uncertainty. The anomalous Ward identity fixes the screening coordinate of the potential. It does not fix Z_0.

If one intentionally leaves this massless Ward-compatible setting, for example by keeping finite quark masses or by introducing an explicit symmetry-breaking regulator/counterterm error, one may add a separate Ward-breaking matrix $B = (b_{ij})$:

$$H^{\mathrm{break}}_{\theta,\eta} = \chi_{\mathrm{eff}} \begin{pmatrix} 1 & a \\ a & a^2 \end{pmatrix} + \begin{pmatrix} b_{\theta\theta} & b_{\theta\eta} \\ b_{\theta\eta} & b_{\eta\eta} \end{pmatrix}. \tag{55.32}$$

For $a^2\chi_{\mathrm{eff}} + b_{\eta\eta} > 0$, the Schur complement of (55.32) is

$$\chi^{\mathrm{loc}}_{\mathrm{break}} = \frac{\chi_{\mathrm{eff}} b_{\eta\eta} + a^2\chi_{\mathrm{eff}} b_{\theta\theta} + b_{\theta\theta}b_{\eta\eta} - 2a\chi_{\mathrm{eff}} b_{\theta\eta} - b^2_{\theta\eta}}{a^2\chi_{\mathrm{eff}} + b_{\eta\eta}}. \tag{55.33}$$

This last quantity is a diagnostic for the declared breaking source or for finite-mass physics. It is not the theta curvature of the exactly massless, symmetry-preserving full-QCD theory. Equations (55.26)–(55.31) are the local bookkeeping needed before comparing a measured η' mass, a lattice χ_{YM}, and a full-QCD topological susceptibility. They do not prove the Yang–Mills continuum limit, the large-N_c expansion, or the absence of branch crossings; they state how those inputs enter the observable singlet curvature once a local branch and counterterm convention have been fixed.

With nonzero quark masses the same caveat becomes a matrix statement. For the neutral fields retained in the effective theory, write

$$\mathcal{L}_2 = \frac{1}{2}K_{AB}\partial_\mu\varphi^A\partial^\mu\varphi^B - \frac{1}{2}H_{AB}\varphi^A\varphi^B + \cdots .$$

The physical pseudoscalar masses are the generalized eigenvalues

$$\det(H - m^2K) = 0, \tag{55.34}$$

so η_0-octet potential mixing, kinetic mixing, and the singlet wavefunction residual must be separated from the pure-Yang–Mills theta curvature. The scalar formula (55.30) is the chiral, unmixed, single-field specialization of (55.34).

Remark 55.16 (Why this does not contradict massless-quark theta independence). If the quark masses vanish and η_0 is included as a dynamical field, the vacuum minimizes $\theta+\sqrt{2N_f}\eta_0/f_\pi$. Thus the full massless QCD vacuum energy is locally independent of θ, and its topological susceptibility vanishes, while the curvature in the singlet field direction at fixed θ is nonzero. The susceptibility in (55.23) is the pure Yang–Mills susceptibility χ_{YM}, not the full massless-QCD susceptibility. This is the local Hessian form of the anomalous axial Ward identity and of the Leutwyler–Smilga small-mass constraint [64]: for positive small quark masses, $\chi_{\mathrm{QCD}} = \Sigma(\sum_i m_i^{-1})^{-1} + O(m^2)$, so the curvature vanishes when any one quark mass is taken to zero.

Controlled Approximation 55.17 (Instanton determinant as a chiral spurion potential)**.** Assume, in addition to the chiral phase hypothesis above, that the regulated one-instanton amplitude of Chapter 51 has been matched at a scale μ to a local color-singlet determinant source and that the instanton and anti-instanton sectors form a dilute finite-activity expansion in the sense of

(51.62). Denote the resulting renormalized activity in the chosen chiral scheme by $\zeta_\chi(\mu) > 0$. Then the zero-mode determinant and its anti-instanton conjugate induce the local singlet potential

$$\begin{aligned} V^{\chi}_{\text{inst}}(\widehat{U};\theta) = 2\zeta_\chi \left[1 - \cos\!\Big(\theta - \mathrm{i}\log\det\widehat{U}\Big)\right] \\ + \text{higher-derivative and higher-mass terms} + \text{multi-instanton terms}, \end{aligned} \tag{55.35}$$

on the same local branch as (55.22). Equivalently, in the $\widehat{U}$ convention of (55.20), the corresponding Lagrangian vertex is proportional to

$$\zeta_\chi\, \mathrm{e}^{\mathrm{i}\theta} \det\widehat{U} + \zeta_\chi\, \mathrm{e}^{-\mathrm{i}\theta} \det\widehat{U}^\dagger,$$

up to a field-independent constant in the potential. If the microscopic zero-mode determinant is written with the conjugate row/column convention, the displayed vertex is obtained by the corresponding transpose/conjugation conversion; the sign used here is fixed by the singlet coordinate (55.22).

In the η_0 chart and sign convention of (55.25), set

$$a = \frac{\sqrt{2N_f}}{f_\pi}, \qquad \theta - \mathrm{i}\log\det\widehat{U} = \theta + a\eta_0 \quad \text{on the chosen exponential branch.}$$

Expanding (55.35) gives

$$V^{\chi}_{\text{inst}}(\theta,\eta_0) = \zeta_\chi(\theta + a\eta_0)^2 + O((\theta + a\eta_0)^4), \tag{55.36}$$

so its local curvature matrix is

$$H^{\text{inst}}_{\theta,\eta} = 2\zeta_\chi \begin{pmatrix} 1 & a \\ a & a^2 \end{pmatrix}. \tag{55.37}$$

The same coefficient $2\zeta_\chi$ is the topological susceptibility of the conditional dilute gas in (51.205). Therefore a regime in which ζ_χ is actually calculable would give the singlet mass contribution

$$\Delta m^2_{\eta_0,\text{inst}} = \frac{2N_f}{f_\pi^2}\,(2\zeta_\chi) = \frac{4N_f\zeta_\chi}{f_\pi^2}. \tag{55.38}$$

This is the dilute-instanton analogue of (55.23), with χ_{YM} replaced by the finite activity susceptibility $\chi^{\text{dig}}_{\text{top}} = 2\zeta_\chi$.

The qualification is part of the statement. In zero-temperature confining QCD the large-ρ part of the one-instanton activity and the multi-instanton interaction expansion are not controlled by the BPST calculation alone. Thus (55.35) is a valid chiral coordinate for a declared finite-activity semiclassical window, such as a weakly coupled Higgsed or sufficiently screened regime, but it is not a derivation of the physical η' mass from isolated instantons in ordinary QCD. The large-N_c Witten–Veneziano input above instead uses the pure Yang–Mills susceptibility as a nonperturbative datum.

For nonzero light-quark masses the singlet field mixes with the octet pseudoscalars. This mixing is part of the susceptibility argument at the level of observables: it is the finite mass-matrix step which turns the pure Yang–Mills theta curvature into a statement about the physical pseudoscalar spectrum. For $N_f = 3$, take the isospin limit

$$M = \operatorname{diag}(m, m, m_s), \qquad m_\pi^2 = 2B_0 m, \qquad m_K^2 = B_0(m + m_s),$$

and use the local $U(3)$ coordinates of (55.20) with

$$T^3 = \frac{1}{2}\,\mathrm{diag}(1,-1,0), \qquad T^8 = \frac{1}{2\sqrt{3}}\,\mathrm{diag}(1,1,-2), \qquad T^0 = \frac{1}{\sqrt{6}}\mathbf{1}_3.$$

Expanding the leading mass spurion $\frac{f_\pi^2}{4}\,\mathrm{Tr}(\chi\widehat{U}^\dagger + \widehat{U}\chi^\dagger)$ to quadratic order gives the neutral mass matrix

$$(M^2_{\mathrm{mass}})_{AB} = 4B_0\,\frac{1}{2}\,\mathrm{Tr}\big(M\{T^A,T^B\}\big), \qquad A,B \in \{3,8,0\}.$$

The isospin generator decouples:

$$M^2_{33} = m_\pi^2, \qquad M^2_{38} = M^2_{30} = 0.$$

Adding the anomalous singlet curvature

$$m_0^2 := \frac{2N_f}{f_\pi^2}\chi_{\mathrm{YM}} \qquad (N_f = 3)$$

only in the 00 entry gives the (η_8,η_0) block

$$M^2_{\eta_8\eta_0} = \begin{pmatrix} \frac{4m_K^2 - m_\pi^2}{3} & -\frac{2\sqrt{2}}{3}(m_K^2 - m_\pi^2) \\ -\frac{2\sqrt{2}}{3}(m_K^2 - m_\pi^2) & \frac{2m_K^2 + m_\pi^2}{3} + m_0^2 \end{pmatrix}. \tag{55.39}$$

Changing the sign convention for η_8 changes the sign of the off-diagonal entry, but not the eigenvalues. If those eigenvalues are denoted m_η^2 and $m_{\eta'}^2$ at this leading order, the trace gives

$$m_{\eta'}^2 + m_\eta^2 - 2m_K^2 = m_0^2 = \frac{2N_f}{f_\pi^2}\chi_{\mathrm{YM}}. \tag{55.40}$$

The determinant records the same mixing data in the product of the two eigenvalues:

$$m_\eta^2 m_{\eta'}^2 = m_\pi^2(2m_K^2 - m_\pi^2) + \frac{m_0^2}{3}(4m_K^2 - m_\pi^2). \tag{55.41}$$

Thus the usual phenomenological form

$$m_{\eta'}^2 + m_\eta^2 - 2m_K^2 = \frac{2N_f}{f_\pi^2}\chi_{\mathrm{YM}} + \text{controlled corrections}$$

requires the quark-mass potential, the normalization of f_π, and the large-N_c and chiral remainders to be stated in the same scheme. The chiral-limit result above fixes the normalization; the finite matrix (55.39) shows how that curvature enters a physical mass spectrum once light-quark masses are restored.

Choose Hermitian flavor generators T^a with $\mathrm{Tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$. These T^a are flavor generators used to parameterize the chiral target $SU(N_f)$; they are not the color gauge generators t^a of the Yang–Mills chapters, where the default convention is $\mathrm{tr}(t^at^b) = \delta^{ab}$. The half-trace flavor convention is kept here because, together with the factor $2/f_\pi$ below, it gives the canonical leading pion coordinates and is compatible with the axial-current normalization stated below. The pion field is the $SU(N_f)$-valued matrix

$$U(x) = \exp\left\{\frac{2\mathrm{i}}{f_\pi}\pi^a(x)T^a\right\}, \qquad U(x) \in SU(N_f), \tag{55.42}$$

transforming as

$$U(x) \longmapsto L\,U(x)\,R^{-1}. \tag{55.43}$$

The axial-current normalization is

$$\langle\Omega|A^{a\mu}(0)|\pi^b(p)\rangle = \mathrm{i} f_\pi p^\mu \delta^{ab}.$$

With this convention, the leading two-derivative chiral Lagrangian is

$$\mathcal{L}_2 = -\frac{f_\pi^2}{4}\,\mathrm{Tr}\big(D_\mu U^\dagger D^\mu U\big), \tag{55.44}$$

where the covariant derivative in external left and right flavor backgrounds ℓ_μ, r_μ is

$$D_\mu U = \partial_\mu U - \mathrm{i}\ell_\mu U + \mathrm{i}U r_\mu. \tag{55.45}$$

The minus sign in (55.44) is the mostly-plus metric convention used throughout this volume. Expanding (55.42) at $\ell_\mu = r_\mu = 0$ gives

$$\mathrm{Tr}(\partial_\mu U^\dagger \partial^\mu U) = \frac{2}{f_\pi^2}\,\partial_\mu\pi^a\partial^\mu\pi^a + O(\pi^3),$$

and hence the canonical leading kinetic term

$$\mathcal{L}_2 = -\frac{1}{2}\partial_\mu\pi^a\partial^\mu\pi^a + \cdots\,.$$

The same expansion also fixes the normalization of the derivative self-interactions.

55.13 Soft-Pion Ward Identity and the Adler Zero

The axial-current normalization above fixes the normalization of the one-pion pole in current correlators. Let $\Pi^a(x)$ be a local pion interpolating field in the low-energy theory, normalized by

$$\langle\Omega|\Pi^a(0)|\pi^b(p)\rangle = Z_\pi^{1/2}\delta^{ab}, \qquad p^2 = 0,$$

and let

$$G_a^\mu(x; x_1, \ldots, x_n) = \langle\Omega|T\{A_a^\mu(x)\Pi^{a_1}(x_1)\cdots\Pi^{a_n}(x_n)\}|\Omega\rangle$$

be the time-ordered distribution with one non-singlet axial current. In the massless theory, and with the anomalous $U(1)_A$ current excluded, the local Ward identity is

$$\begin{aligned}\partial_\mu G_a^\mu(x; x_1, \ldots, x_n) = -\mathrm{i}\sum_{j=1}^{n}\delta^{(4)}(x - x_j)&\\ \times \langle\Omega|T\{\Pi^{a_1}(x_1)\cdots\delta_A^a\Pi^{a_j}(x_j)\cdots\Pi^{a_n}(x_n)\}|\Omega\rangle.&\end{aligned} \tag{55.46}$$

Here δ_A^a denotes the infinitesimal transformation generated by the broken axial charge Q_A^a. Equation (55.46) is an identity of distributions: the right side consists of contact terms because the divergence of a conserved current inside a time-ordered product differentiates the step functions.

Fourier transform in x with momentum q entering the current and in the x_j with hard pion momenta p_j. Near $q^2 = 0$, the current insertion has the pion pole fixed by

$$\langle \Omega | A_a^\mu(0) | \pi^b(q) \rangle = \mathrm{i} f_\pi q^\mu \delta^{ab}.$$

Thus its singular part has the form

$$\widetilde{G}_a^\mu(q; \{p_j, a_j\}) = \frac{-\mathrm{i} f_\pi q^\mu}{q^2 - \mathrm{i}0}\, \widetilde{G}_{n+1}^{\,a, a_1, \dots, a_n}(q, p_1, \dots, p_n) + \widetilde{R}_a^\mu(q; \{p_j, a_j\}), \tag{55.47}$$

where $\widetilde{R}_a^\mu$ is regular at the current-channel pion pole in the soft region under consideration. Let $\mathcal{L}_{\{p_j\}}$ denote the hard external-pion LSZ operation

$$\mathcal{L}_{\{p_j\}} F = \left[\prod_{j=1}^n Z_\pi^{-1/2}\, \mathrm{i} p_j^2 \right] F \quad \text{followed by the massless on-shell boundary value } p_j^2 = 0,$$

with the same residue convention as in Chapter 18. Applying $\mathcal{L}_{\{p_j\}}$ to the Fourier transform of (55.46) and using (55.47) gives

$$f_\pi\, \mathcal{M}_{n+1}^{a, a_1, \dots, a_n}(q, p_1, \dots, p_n) = \mathcal{C}_a(q; \{p_j, a_j\}) + O(q), \tag{55.48}$$

where

$$\mathcal{C}_a(q; \{p_j, a_j\}) = \mathcal{L}_{\{p_j\}} \sum_{j=1}^n \widetilde{G}_n^{a_1, \dots, \delta_A^a a_j, \dots, a_n}(p_1, \dots, p_j + q, \dots, p_n)$$

is the sum of the LSZ-reduced contact terms; the superscript means that the j-th pion operator has been replaced by its renormalized axial variation. This formula is the soft-pion theorem in its current-algebra form: the leading soft pion is determined by the action of the broken charge on the hard external states.

Theorem 55.18 (Adler zero for chiral pion amplitudes)**.** *Assume the exact chiral limit, the symmetry-breaking pattern*

$$SU(N_f)_L \times SU(N_f)_R \longrightarrow SU(N_f)_V,$$

the axial-current normalization $\langle \Omega | A_a^\mu(0) | \pi^b(p) \rangle = \mathrm{i} f_\pi p^\mu \delta^{ab}$, and the existence of the pion scattering boundary values obtained by LSZ from time-ordered Green functions. Take the soft momentum $q \to 0$ at fixed nonexceptional hard momenta, with the hard-leg soft singularities treated by the LSZ-reduced Ward identity above and no additional threshold or collinear singular surface crossed. Then every connected purely pionic amplitude obeys

$$\mathcal{M}_{n+1}^{a, a_1, \dots, a_n}(q, p_1, \dots, p_n) = O(q). \tag{55.49}$$

The vanishing of the $O(q^0)$ term is the Adler zero.

Proof. The chiral target is the symmetric space $(SU(N_f)_L \times SU(N_f)_R)/SU(N_f)_V$. In local pion coordinates normalized by the axial current, a broken axial transformation has the expansion

$$\delta_A^a \pi^b = f_\pi \delta^{ab} + O(\pi^2/f_\pi), \tag{55.50}$$

with no term linear in the pion field. For $N_f = 2$ this is visible directly from (55.54): using $\pi^a = F_{\rm st}\xi^a$ and $F_{\rm st} = 2f_\pi$, the axial variation is $\delta_A^a \pi^b = f_\pi \delta^{ab} + O(\pi^2/f_\pi)$. The absence of a linear term is the representation-theoretic statement that $[\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h}$: a broken generator translates the origin of the coset and then rotates higher-order tangent data by the unbroken group, but it does not act as a one-pion-to-one-pion matrix at nonzero momentum.

Now evaluate the contact term $\mathcal{C}_a(0; \{p_j, a_j\})$ in (55.48). The constant part $f_\pi \delta^{aa_j}$ in (55.50) replaces the j-th interpolating field by the identity operator. After the hard LSZ operation it has no one-pion pole in p_j^2, hence gives zero. The remaining terms in $\delta_A^a \pi^{a_j}$ contain at least two pion fields at the same spacetime point. They are renormalized as part of the same symmetry-preserving Ward identity, so their projection onto a single hard-pion pole is fixed to vanish; otherwise the renormalized axial transformation would have a linear pion term, contrary to (55.50). Equivalently, the broken charge has no one-pion-to-one-pion matrix element; its possible vacuum term is supported only at zero total momentum and cannot replace a hard external pion. Therefore the LSZ-reduced contact term has no $O(q^0)$ contribution:

$$\mathcal{C}_a(0; \{p_j, a_j\}) = 0.$$

Equation (55.48) then gives (55.49). □

Figure 55.4 records the Ward-identity mechanism behind the Adler zero: the axial-current insertion is reduced by LSZ, the hard contact term has no $O(q^0)$ one-pion contribution, and the soft amplitude vanishes at leading order.

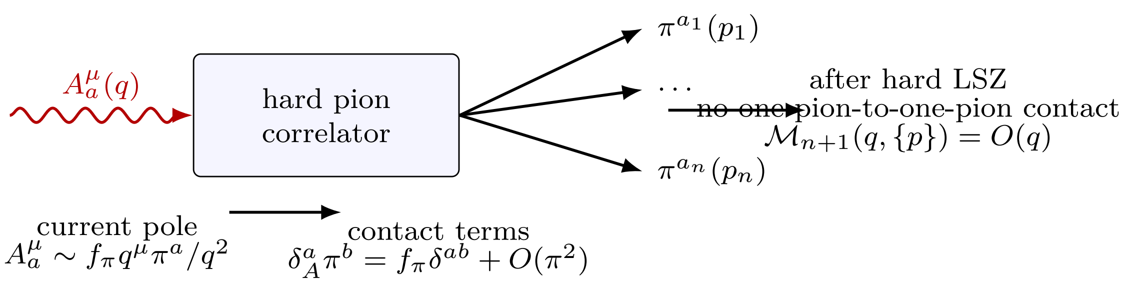


Figure 55.4: The soft-pion theorem follows by isolating the pion pole in the axial-current Ward identity. For purely pionic hard states in the chiral coset, the contact terms have no one-pion-to-one-pion component, so the leading soft term vanishes.

For external states that are not all pions, the same identity should be read before setting $\mathcal{C}_a(0) = 0$. A soft pion measures the action of the broken axial charge on the hard external sector. If the hard sector contains degenerate states connected by Q_A^a, the right side of (55.48) need not vanish. The Adler zero is the specialization to the pion S-matrix in the chiral coset.

For four pions, the theorem says that the analytic on-shell amplitude has a zero when one external pion momentum is taken soft. In the chiral limit this is the statement that the leading invariant amplitude below satisfies

$$A(s,t,u) = \frac{s}{f_\pi^2} + O(E^4/f_\pi^4), \qquad A \longrightarrow 0 \quad \text{as} \quad s,t,u \longrightarrow 0$$

in the single-soft analytic continuation. With a small common pion mass, the leading chiral formula becomes

$$A(s,t,u) = \frac{s - m_\pi^2}{f_\pi^2} + O(p^4),$$

so the zero is displaced to the corresponding Adler point in the analytically continued amplitude. The massive Adler point is not a physical threshold configuration; it is a Ward-identity zero of the analytic amplitude.

55.14 Low-Energy Pion Scattering

The two-derivative chiral Lagrangian determines the leading low-energy S-matrix of the Goldstone bosons, because every interaction is derivative and therefore ordered by powers of external momentum. For the $N_f = 2$ theory it is useful to write $SU(2) \simeq S^3$ in stereographic coordinates $\vec{\xi} = (\xi_1, \xi_2, \xi_3)$:

$$U(\xi) = \frac{1 - \vec{\xi}^{\,2} + 2\mathrm{i}\,\vec{\xi} \cdot \vec{\sigma}}{1 + \vec{\xi}^{\,2}}. \tag{55.51}$$

Let F_{st} denote the normalization used in these stereographic coordinates:

$$\mathcal{L}_2 = -\frac{F_{\mathrm{st}}^2}{16} \operatorname{Tr}(\partial_\mu U\, \partial^\mu U^\dagger) = -\frac{F_{\mathrm{st}}^2}{2} \frac{\partial_\mu \vec{\xi} \cdot \partial^\mu \vec{\xi}}{(1 + \vec{\xi}^{\,2})^2}. \tag{55.52}$$

With canonical fields

$$\vec{\pi} = F_{\mathrm{st}} \vec{\xi},$$

the expansion begins as

$$\mathcal{L}_2 = -\frac{1}{2} \partial_\mu \vec{\pi} \cdot \partial^\mu \vec{\pi} + \frac{1}{F_{\mathrm{st}}^2} \vec{\pi}^{\,2}\, \partial_\mu \vec{\pi} \cdot \partial^\mu \vec{\pi} + O(\pi^6/F_{\mathrm{st}}^4). \tag{55.53}$$

The infinitesimal $SU(2)_L \times SU(2)_R$ action in this chart is useful for organizing local terms. Write

$$L = \exp\!\left(\frac{\mathrm{i}}{2} \vec{\theta}_L \cdot \vec{\sigma}\right), \qquad R = \exp\!\left(\frac{\mathrm{i}}{2} \vec{\theta}_R \cdot \vec{\sigma}\right), \qquad \vec{\theta}_V = \frac{\vec{\theta}_L + \vec{\theta}_R}{2}, \qquad \vec{\theta}_A = \frac{\vec{\theta}_L - \vec{\theta}_R}{2}.$$

For $U \mapsto L U R^{-1}$, a direct computation from (55.51) gives

$$\delta \vec{\xi} = -\vec{\theta}_V \times \vec{\xi} + \frac{1}{2}(1 - \vec{\xi}^{\,2}) \vec{\theta}_A + \vec{\xi}(\vec{\xi} \cdot \vec{\theta}_A). \tag{55.54}$$

Thus the diagonal subgroup acts linearly on $\vec{\xi}$, while the axial directions act by a nonlinear shift plus higher powers of $\vec{\xi}$. Define

$$\vec{D}_\mu = \frac{\partial_\mu \vec{\xi}}{1 + \vec{\xi}^{\,2}}. \tag{55.55}$$

For constant transformation parameters,

$$\delta \vec{D}_\mu = -\vec{\theta}_V \times \vec{D}_\mu - (\vec{\xi} \times \vec{\theta}_A) \times \vec{D}_\mu. \tag{55.56}$$

The two independent four-derivative invariants relevant to four-pion scattering can be written as

$$(\vec{D}_\mu \cdot \vec{D}^\mu)^2, \qquad (\vec{D}_\mu \cdot \vec{D}_\nu)(\vec{D}^\mu \cdot \vec{D}^\nu),$$

up to integrations by parts and terms that first affect higher-point amplitudes. This coordinate choice is only a local chart on S^3; the scattering amplitude obtained from it is coordinate independent. In the axial-current normalization above, $T^a = \sigma^a/2$ and therefore

$$\exp\left(\frac{2\mathrm{i}}{f_\pi}\pi^a_{\mathrm{ax}}T^a\right) = 1 + \frac{\mathrm{i}}{f_\pi}\pi^a_{\mathrm{ax}}\sigma^a + O(\pi^2_{\mathrm{ax}}).$$

The stereographic chart has

$$\frac{1-\vec{\xi}^{\,2} + 2\mathrm{i}\,\vec{\xi}\cdot\vec{\sigma}}{1+\vec{\xi}^{\,2}} = 1 + 2\mathrm{i}\,\xi^a\sigma^a + O(\xi^2), \qquad \pi^a_{\mathrm{st}} = F_{\mathrm{st}}\xi^a.$$

Identifying the one-pion field defined by the axial current with the canonical stereographic field, $\pi^a_{\mathrm{ax}} = \pi^a_{\mathrm{st}}$, gives the conversion

$$F_{\mathrm{st}} = 2f_\pi. \tag{55.57}$$

A convention that rescales the stereographic coordinate rescales F_{st} and the coordinate field together; in the fixed chart (55.51) the numerical factor in (55.57) is part of the convention.

For

$$\pi^a(k_1) + \pi^b(k_2) \to \pi^c(k_3) + \pi^d(k_4),$$

use the Mandelstam variables

$$s = -(k_1+k_2)^2, \qquad t = -(k_1-k_3)^2, \qquad u = -(k_1-k_4)^2.$$

The metric is mostly plus, k_1, k_2 are incoming, and k_3, k_4 are outgoing. In the chiral limit used for the two-derivative Goldstone calculation, $k_i^2 = 0$, so

$$s + t + u = 0.$$

For equal nonzero pion masses the same convention gives $s+t+u = 4m_\pi^2$. The $SU(2)_V$-invariant amplitude has the form

$$\mathcal{A}_{ab\to cd} = \delta_{ab}\delta_{cd}A(s,t,u) + \delta_{ac}\delta_{bd}A(u,t,s) + \delta_{ad}\delta_{bc}A(t,s,u). \tag{55.58}$$

At order E^2/F^2_{st}, the four-pion vertex from (55.53) gives

$$A(s,t,u) = \frac{4s}{F^2_{\mathrm{st}}} + O(E^4/F^4_{\mathrm{st}}). \tag{55.59}$$

Equivalently, after converting to the convention in which the axial current matrix element defines f_π, this is the standard leading chiral theorem

$$A(s,t,u) = \frac{s}{f_\pi^2} + O(E^4/f_\pi^4).$$

Figure 55.5 displays the two levels of the chiral expansion used here: the leading two-derivative four-pion vertex and the next-order loop/local-counterterm structure.

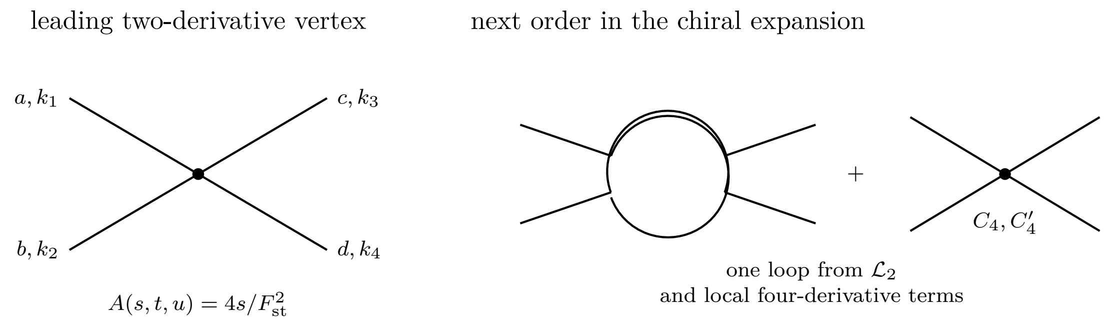


Figure 55.5: The leading pion scattering amplitude is fixed by the two-derivative nonlinear sigma model. At the next order, one-loop diagrams from $\mathcal{L}_2$ are combined with local four-derivative counterterms.

At order $E^4/F_{\rm st}^4$, the effective Lagrangian contains local four-derivative terms. In the stereographic $SU(2)$ chart these may be written, to the order needed for four-pion scattering, as

$$-\frac{C_4}{4F_{\rm st}^4}(\partial_\mu\vec{\pi}\cdot\partial^\mu\vec{\pi})^2 - \frac{C_4'}{4F_{\rm st}^4}(\partial_\mu\vec{\pi}\cdot\partial_\nu\vec{\pi})(\partial^\mu\vec{\pi}\cdot\partial^\nu\vec{\pi}).$$

The same order also receives one-loop contributions from two insertions of the two-derivative interaction. A regulator that does not preserve the nonlinearly realized symmetry may generate power terms that look like a potential on S^3; such terms must be absent in a symmetry-preserving scheme or canceled by symmetry-restoring local counterterms. After this local renormalization, the logarithmic part has the form

$$\begin{aligned} A(s,t,u) = \frac{4s}{F_{\rm st}^2} + \frac{1}{F_{\rm st}^4}\Bigg\{ \frac{1}{12\pi^2}\Bigg[& 3s^2\log\frac{\mu^2}{-s} \\ & + (u^2 - s^2 + 3t^2)\log\frac{\mu^2}{-t} \\ & + (t^2 - s^2 + 3u^2)\log\frac{\mu^2}{-u}\Bigg] \\ & - \frac{1}{2}C_4(\mu)s^2 - \frac{1}{2}C_4'(\mu)(t^2+u^2)\Bigg\} + O(E^6). \end{aligned} \tag{55.60}$$

Equivalently, in a Wilsonian cutoff notation in which the logarithms are written as $\log(\Lambda^2/-s)$, $\log(\Lambda^2/-t)$, and $\log(\Lambda^2/-u)$, the remaining cutoff dependence is absorbed by

$$C_4(\Lambda) = C_4(\mu) + \frac{2}{3\pi^2}\log\frac{\Lambda^2}{\mu^2}, \qquad C_4'(\Lambda) = C_4'(\mu) + \frac{4}{3\pi^2}\log\frac{\Lambda^2}{\mu^2}.$$

The finite constants $C_4(\mu)$ and $C_4'(\mu)$ are low-energy constants. Their values depend on the microscopic dynamics and on the subtraction convention, but the logarithmic dependence and the required μ-dependence are fixed by the low-energy effective theory.

55.15 Mass Spurions and Pseudo-Goldstone Bosons

Quark masses explicitly break the chiral symmetry. Introduce a complex mass matrix M through

$$\mathcal{L}_{\rm mass}^{\rm QCD} = -\bar{q}_L M q_R - \bar{q}_R M^\dagger q_L.$$

Treating M as a background spurion with

$$M \longmapsto LMR^{-1}$$

makes the microscopic mass term formally invariant under (L, R). The leading mass-dependent invariant in the pion Lagrangian is

$$\mathcal{L}_{\text{mass}} = \frac{f_\pi^2}{4} \operatorname{Tr}\big(\chi U^\dagger + U\chi^\dagger\big), \qquad \chi = 2B_0 M. \tag{55.61}$$

The low-energy constant B_0 is fixed by the chiral-limit condensate convention

$$\langle \bar{q}q \rangle_0 = -N_f f_\pi^2 B_0$$

at leading order. For two light flavors,

$$m_{\pi^\pm}^2 = B_0(m_u + m_d)$$

up to higher-order chiral and electromagnetic corrections. For degenerate light quarks $m_u = m_d = m_q$, this becomes $m_\pi^2 = 2B_0 m_q$. The condensate notation in this chapter is flavor-summed:

$$\Sigma_{\text{sum}} := \langle \bar{q}q \rangle_0 = \sum_{i=1}^{N_f} \langle \bar{q}_i q_i \rangle_0, \qquad \Sigma_{\text{fl}} := \frac{1}{N_f} \Sigma_{\text{sum}}.$$

The leading-order identity following from the two preceding equations is the flavor-summed form

$$m_{\pi^\pm}^2 f_\pi^2 = -(m_u + m_d) \frac{\langle \bar{q}q \rangle_0}{N_f}. \tag{55.62}$$

The conversion to the per-flavor condensate is

$$\Sigma_{\text{fl}} = \frac{\langle \bar{q}q \rangle_0}{N_f} = \langle \bar{u}u \rangle_0 = \langle \bar{d}d \rangle_0 \qquad (N_f = 2 \text{ isospin-symmetric vacuum}). \tag{55.63}$$

Using (55.63), the same leading-order statement is the per-flavor Gell-Mann–Oakes–Renner relation

$$m_\pi^2 f_\pi^2 = -(m_u + m_d)\langle \bar{u}u \rangle_0. \tag{55.64}$$

The factor of 2 is accounted for by the conversion between the flavor-summed condensate and the per-flavor condensate.

The same normalization can be seen directly in the $SU(2)$ stereographic chart. Let

$$M = \begin{pmatrix} m_u & 0 \\ 0 & m_d \end{pmatrix}.$$

In terms of the quark bilinear $\Phi^I{}_J = \bar{q}_L^I q_{R,J}$, the microscopic deformation has the form

$$\Delta\mathcal{L} = -\frac{m_u + m_d}{2} \operatorname{Tr}(\Phi + \Phi^\dagger) - \frac{m_u - m_d}{2} \operatorname{Tr}\big(\sigma_3(\Phi + \Phi^\dagger)\big).$$

Restricting the pion field to the $U(1) \subset SU(2)$ generated by σ_3,

$$U(\xi_3) = 1 + 2\mathrm{i}\xi_3\sigma_3 + \cdots = \begin{pmatrix} \mathrm{e}^{\mathrm{i}\theta} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}\theta} \end{pmatrix} + \cdots, \qquad \theta = 2\xi_3 + \cdots.$$

Since the kinetic term in (55.52) gives

$$\mathcal{L}_2 = -\frac{F_{\rm st}^2}{2}\partial_\mu\vec{\xi}\cdot\partial^\mu\vec{\xi} + \cdots\,,$$

the $U(1)$ comparison has radius $R = F_{\rm st}/2$. Expanding the mass deformation therefore gives

$$-\frac{1}{2}m_{ab}^2\pi_a\pi_b, \qquad m_{ab}^2 = \delta_{ab}\frac{4}{F_{\rm st}^2}(m_u+m_d)\left\langle -\frac{\bar{q}q}{2}\right\rangle_0,$$

which is the same leading relation as the B_0-notation after converting normalizations. Empirically,

$$m_{\pi^0} \simeq 135\,\mathrm{MeV}, \qquad m_{\pi^\pm} \simeq 140\,\mathrm{MeV},$$

with the splitting produced by isospin breaking and electromagnetism.

The chiral expansion counts $D_\mu U = O(p)$, external vector and axial backgrounds as $O(p)$, and M as $O(p^2)$. The functional obtained from $\mathcal{L}_2 + \mathcal{L}_{\rm mass}$ is therefore the leading generating functional for current Green functions at momenta small compared with the chiral-symmetry-breaking scale.

55.16 Chiral Perturbation Theory at NLO

Chiral perturbation theory is the effective expansion of the QCD current generating functional in powers of derivatives, external flavor fields, and quark masses around the chiral limit. Pion fields enter as local coordinates on the Goldstone manifold; the invariant expansion is organized by the orders of the functional's derivatives, sources, and spurion insertions. The expansion variables are assigned the orders

$$D_\mu U = O(p), \qquad \ell_\mu, r_\mu = O(p), \qquad F^L_{\mu\nu}, F^R_{\mu\nu} = O(p^2), \qquad \chi = 2B_0 M = O(p^2),$$

where the left and right field strengths in the conventions of (55.45) are

$$F^L_{\mu\nu} = \partial_\mu\ell_\nu - \partial_\nu\ell_\mu - \mathrm{i}[\ell_\mu,\ell_\nu], \qquad F^R_{\mu\nu} = \partial_\mu r_\nu - \partial_\nu r_\mu - \mathrm{i}[r_\mu, r_\nu].$$

A connected graph built from vertices of chiral order d_i and with L loops has total chiral order

$$\nu = 2 + 2L + \sum_i V_i(d_i - 2). \tag{55.65}$$

Indeed, each loop integration contributes p^4, each internal pion propagator contributes p^{-2}, and each vertex contributes p^{d_i}. For a connected graph, $L = I - \sum_i V_i + 1$, hence

$$4L - 2I + \sum_i V_i d_i = 2 + 2L + \sum_i V_i(d_i-2).$$

Thus one-loop graphs made only from $\mathcal{L}_2$ and tree graphs from the order-p^4 Lagrangian both contribute at next-to-leading order.

Controlled Approximation 55.19 (Chiral EFT truncation as a seminorm statement)**.** The power-counting identity (55.65) is only the graph algebra. To use it as a QCD statement one must also specify the observable topology in which the low-energy remainder is controlled. Fix a compact family K of external momenta and smooth flavor sources satisfying

$$|p_a| \le \epsilon\Lambda_\chi, \qquad \|\ell\|_K + \|r\|_K \le \epsilon\Lambda_\chi, \qquad \|\chi\|_K^{1/2} \le \epsilon\Lambda_\chi, \qquad 0 < \epsilon < 1,$$

together with a renormalization scheme for the local chiral coordinates. For a connected renormalized graph Γ let

$$\nu(\Gamma) = 2 + 2L(\Gamma) + \sum_d V_d(\Gamma)(d-2).$$

The useful truncation assertion is a family of seminorm estimates

$$p_K(\mathcal{A}_\Gamma) \le C_{\Gamma,K}\epsilon^{\nu(\Gamma)}$$

for the source derivatives or S-matrix coordinates under consideration, and, after retaining all graphs with $\nu \le N$,

$$p_K\left(\mathcal{A} - \sum_{\nu(\Gamma)\le N} \mathcal{A}_\Gamma\right) \le C_{K,N}\epsilon^{\nu_+(N)}, \qquad \nu_+(N) := \min\{\nu(\Gamma) : \nu(\Gamma) > N\}. \tag{55.66}$$

Equation (55.66) is not a convergence theorem for the full chiral series. It is the controlled EFT input that turns the formal graph dimension into a quantitative low-energy approximation on the declared compact source window. At $N = 4$ the retained data are precisely: tree graphs from $\mathcal{L}_2$, one-loop graphs from $\mathcal{L}_2$, and tree graphs from $\mathcal{L}_4$, including the source-contact terms. The first omitted families have $\nu = 6$: two-loop graphs from $\mathcal{L}_2$, one-loop graphs with one $\mathcal{L}_4$ vertex, and tree graphs from $\mathcal{L}_6$. Thus the $O(p^4)$ calculation below is meaningful only inside this remainder ledger; outside such a low-energy window the same local formulae are just coordinates, not controlled predictions.

For $N_f = 3$, the even-intrinsic-parity order-p^4 local functional is the Gasser–Leutwyler basis.[1] Write $\langle X\rangle_{\mathrm{f}}$ for the flavor trace $\operatorname{Tr} X$, and define

$$X_+ := \chi U^\dagger + U\chi^\dagger, \qquad X_- := \chi U^\dagger - U\chi^\dagger.$$

With the left/right convention $U \mapsto LUR^{-1}$, the local NLO Lagrangian is

$$\begin{aligned}
\mathcal{L}_4^{\mathrm{GL}} = {} & L_1\langle D_\mu U^\dagger D^\mu U\rangle_{\mathrm{f}}^2 + L_2\langle D_\mu U^\dagger D_\nu U\rangle_{\mathrm{f}}\langle D^\mu U^\dagger D^\nu U\rangle_{\mathrm{f}} \\
& + L_3\langle D_\mu U^\dagger D^\mu U D_\nu U^\dagger D^\nu U\rangle_{\mathrm{f}} + L_4\langle D_\mu U^\dagger D^\mu U\rangle_{\mathrm{f}}\langle X_+\rangle_{\mathrm{f}} \\
& + L_5\langle D_\mu U D^\mu U^\dagger X_+\rangle_{\mathrm{f}} + L_6\langle X_+\rangle_{\mathrm{f}}^2 + L_7\langle X_-\rangle_{\mathrm{f}}^2 \\
& + L_8\langle \chi U^\dagger \chi U^\dagger + U\chi^\dagger U\chi^\dagger\rangle_{\mathrm{f}} \\
& - \mathrm{i}L_9\langle F_L^{\mu\nu} D_\mu U D_\nu U^\dagger + F_R^{\mu\nu} D_\mu U^\dagger D_\nu U\rangle_{\mathrm{f}} \\
& + L_{10}\langle U^\dagger F_L^{\mu\nu} U F^R_{\mu\nu}\rangle_{\mathrm{f}} \\
& + H_1\langle F_L^{\mu\nu} F^L_{\mu\nu} + F_R^{\mu\nu} F^R_{\mu\nu}\rangle_{\mathrm{f}} + H_2\langle \chi^\dagger\chi\rangle_{\mathrm{f}}.
\end{aligned} \tag{55.67}$$

The constants $L_1, \dots, L_{10}$ multiply operators containing the pion field or its covariant derivatives. The contact constants H_1, H_2 affect only correlators of external sources and are required for

[1]The notation $L_1, \dots, L_{10}$ is the three-flavor basis introduced by Gasser and Leutwyler in the original chiral perturbation theory papers, translated here to the left/right convention $U \mapsto LUR^{-1}$.

renormalizing the source generating functional. The odd-intrinsic-parity anomaly sector is not part of $\mathcal{L}_4^{\mathrm{GL}}$; it is the Wess–Zumino–Witten functional constructed in the next section. For $N_f = 2$, Cayley–Hamilton identities reduce the number of independent even-parity p^4 invariants. The stereographic constants C_4, C_4' above are two-coordinate representatives of that reduced two-flavor sector; the L_i above are the standard three-flavor coordinates.

The L_i are renormalized coordinates on the space of local order-p^4 counterterms. In dimensional regularization one writes

$$L_i = L_i^r(\mu) + \Gamma_i \Lambda_{\mathrm{div}}, \qquad \mu \frac{\mathrm{d}L_i^r}{\mathrm{d}\mu} = -\frac{\Gamma_i}{16\pi^2},$$

where the three-flavor coefficients are

i	1	2	3	4	5	6	7	8	9	10
Γ_i	$\frac{3}{32}$	$\frac{3}{16}$	0	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{11}{144}$	0	$\frac{5}{48}$	$\frac{1}{4}$	$-\frac{1}{4}$.

The μ-dependence of $L_i^r(\mu)$ cancels the logarithmic dependence from one-loop graphs built from $\mathcal{L}_2$. This cancellation is a low-energy theorem; the finite values of $L_i^r(\mu)$ encode microscopic QCD dynamics.

The simplest chiral logarithm appears in the two-flavor pion mass. Set $m_u = m_d = m_q$ and

$$M^2 := 2B_0 m_q.$$

The one-loop tadpole integral generated by the $\mathcal{L}_2$ four-pion vertices has renormalized logarithmic part

$$A^r(M^2) = \frac{M^2}{16\pi^2} \log \frac{M^2}{\mu^2}.$$

Expanding $\mathcal{L}_2$ in normal pion coordinates gives the logarithmic on-shell self-energy contribution

$$\Sigma_{\log}(M^2) = \frac{M^2}{2f^2} A^r(M^2) = \frac{M^4}{32\pi^2 f^2} \log \frac{M^2}{\mu^2},$$

where f is the chiral-limit value of f_π. The local two-flavor NLO constant usually denoted $l_3^r(\mu)$ contributes $2l_3^r(\mu)M^4/f^2$ to the mass. Hence

$$M_{\pi,\mathrm{phys}}^2 = M^2 \left[1 + \frac{M^2}{32\pi^2 f^2} \log \frac{M^2}{\mu^2} + \frac{2l_3^r(\mu)M^2}{f^2}\right] + O(M^6). \tag{55.68}$$

This is nonanalytic in the quark mass because $M^2 \propto m_q$. The renormalization-group coefficient of the two-flavor constant is

$$\mu \frac{\mathrm{d}l_3^r}{\mathrm{d}\mu} = \frac{1}{32\pi^2},$$

so differentiating the bracket in (55.68) with respect to $\log \mu$ gives

$$-\frac{M^2}{16\pi^2 f^2} + \frac{2M^2}{f^2} \frac{1}{32\pi^2} = 0.$$

Thus the scale dependence of the logarithm is exactly canceled by the running of the local NLO coordinate. The logarithm itself cannot be removed by a local counterterm because it is the long-distance contribution of pion propagation.

55.17 The Wess–Zumino–Witten Functional

The pion effective action must reproduce the global anomalies of the microscopic quarks. For $N_f \geq 3$, let M_4 be spacetime and choose a five-manifold B_5 with $\partial B_5 = M_4$. Extend $U : M_4 \to SU(N_f)$ to $\widetilde{U} : B_5 \to SU(N_f)$. Figure 55.6 fixes this filling convention for the Wess–Zumino–Witten functional: M_4 is the boundary and $\widetilde{U}$ is an extension over B_5.

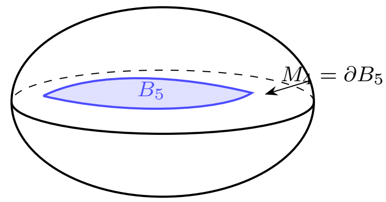


Figure 55.6: The Wess–Zumino functional is written by extending the spacetime field $U : M_4 \to SU(N_f)$ to a five-dimensional filling B_5. Changing the filling changes the integral by a quantized period.

Definition 55.20 (Ungauged Wess–Zumino functional)**.** The ungauged Wess–Zumino functional is

$$\Gamma_{\mathrm{WZ}}[U] = -\frac{\mathrm{i}}{240\pi^2} \int_{B_5} \mathrm{Tr}\big(\widetilde{U}^{-1}\mathrm{d}\widetilde{U}\big)^5. \tag{55.69}$$

The five-form is real after the factor $-\mathrm{i}$, because $\widetilde{U}^{-1}\mathrm{d}\widetilde{U}$ is anti-Hermitian. If two extensions are glued along M_4, the difference of Γ_{WZ} is the integral over a closed five-manifold. With the normalization in (55.69), this difference is 2π times an integer for maps representing $\pi_5(SU(N_f)) \simeq \mathbb{Z}$. Hence

$$\exp\{\mathrm{i}n\Gamma_{\mathrm{WZ}}[U]\}$$

is independent of the extension precisely when $n \in \mathbb{Z}$. This integer is the Wess–Zumino level.

Proposition 55.21 (Finite Wess–Zumino extension ambiguity)**.** *For $N_f \geq 3$, write*

$$\omega_5(U) = -\frac{\mathrm{i}}{240\pi^2}\,\mathrm{Tr}(U^{-1}\mathrm{d}U)^5.$$

Assume the standard generator normalization in which $\int_{S^5} g^*\omega_5 = 2\pi$ for a generator $[g] \in \pi_5(SU(N_f)) \simeq \mathbb{Z}$. If two fillings $(B_5, \widetilde{U})$ and $(B_5', \widetilde{U}')$ extend the same boundary field $U : M_4 \to SU(N_f)$, then

$$\Gamma_{\mathrm{WZ}}[B_5, \widetilde{U}] - \Gamma_{\mathrm{WZ}}[B_5', \widetilde{U}'] = 2\pi k, \qquad k \in \mathbb{Z}.$$

Consequently the exponentiated pion response $\exp\{\mathrm{i}n\Gamma_{\mathrm{WZ}}[U]\}$ *is a well-defined boundary functional for all allowed fillings exactly when $n \in \mathbb{Z}$.*

Proof. Glue B_5 to B_5' with the opposite orientation along their common boundary M_4. The result is a closed oriented five-manifold

$$X_5 = B_5 \cup_{M_4} (-B_5'),$$

and the two extensions glue to a map $\widehat{U} : X_5 \to SU(N_f)$. Additivity of the integral under gluing gives

$$\Gamma_{\mathrm{WZ}}[B_5, \widetilde{U}] - \Gamma_{\mathrm{WZ}}[B_5', \widetilde{U}'] = \int_{X_5} \widehat{U}^*\omega_5.$$

The normalized period assumption says that the right hand side lies in $2\pi\mathbb{Z}$; call it $2\pi k$. Changing the filling therefore multiplies the exponentiated level-n response by

$$\exp(2\pi i n k).$$

This factor is 1 for every possible extension ambiguity $k \in \mathbb{Z}$ if n is an integer. Conversely, the generator ambiguity $k = 1$ would give a nontrivial phase for any nonintegral level. Thus finite extension-independence is the integrality condition $n \in \mathbb{Z}$. $\square$

The finite integrality statement is separate from the infinitesimal anomaly coefficient comparison below. Local descent determines which multiple of the generator the pion functional must carry; the proposition says when that multiple exponentiates to an honest four-dimensional response rather than a choice of five-dimensional filling. In QCD the same integer is N_c, the number of color copies of a fundamental flavor Weyl fermion.

WZW level matching from the flavor descent coefficient. The equality between the QCD Wess–Zumino–Witten level and the number of colors is not an additional imported theorem. It is the coefficient matching of two representatives of the same infinitesimal background-anomaly class, after the flavor trace convention $\mathrm{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ and the unit-level normalization of Γ_{WZ} in (55.69) have been fixed. For massless QCD with N_c colors and $N_f \geq 3$ light flavors, gauge a left flavor background $A = A_\mu \mathrm{d}x^\mu$, set the right background to zero, and first retain the part quadratic in A. Matrix multiplication and wedge product are implicit in the following formulas.

The word "quadratic" here does not mean that the higher-background part of the anomaly is being discarded. The previous chapter showed that a local Bardeen counterterm can change a four-form representative $I_4^{(1)}$, but cannot change the completely symmetric degree-three coefficient of the six-form polynomial I_6. More explicitly, the Abelianized descent coordinate (51.31) proves that, on commuting flavor backgrounds, the boundary variation of the Chern–Simons representative is exactly $C_{abc}\alpha^a F^b F^c$, and the polarization identity (51.32) recovers the symmetric cubic coefficient from these commuting tests. The term $\mathrm{Tr}(\alpha\,\mathrm{d}A\,\mathrm{d}A)$ is the lowest-background coordinate of that same descent class in the convention fixed in (51.30). The $O(A^3)$ terms are the non-Abelian completion required by Wess–Zumino consistency inside the same class, up to local counterterm coboundaries. A single left-handed quark multiplet gives the consistent variation

$$\delta_\alpha \Gamma_{1\,\mathrm{color}}[A] = \frac{1}{48\pi^2}\int_{M_4} \mathrm{Tr}\big(\alpha\,\mathrm{d}A\,\mathrm{d}A\big) + O(A^3),$$

where $\alpha = \alpha^a T^a$. QCD has N_c identical color copies of this flavor representation, hence

$$\delta_\alpha \Gamma_{\mathrm{quark}}[A] = \frac{N_c}{48\pi^2}\int_{M_4} \mathrm{Tr}\big(\alpha\,\mathrm{d}A\,\mathrm{d}A\big) + O(A^3).$$

The gauged Wess–Zumino functional at level n has the corresponding Goldstone variation

$$\delta_\alpha\big(n\Gamma_{\mathrm{WZW}}[U; A, 0]\big) = \frac{n}{48\pi^2}\int_{M_4} \mathrm{Tr}\big(\alpha\,\mathrm{d}A\,\mathrm{d}A\big) + O(A^3).$$

Equality of the anomaly phases for arbitrary infinitesimal α and background A gives

$$n = N_c.$$

The comparison of the displayed quadratic term fixes the integer coefficient of the generator of the descent class. This is why the level is fixed by one coefficient rather than by a term-by-term comparison of a chosen representative. The remaining background terms are determined by the same relative BRST class after the counterterm convention has been chosen. When the flavor symmetry is coupled to background gauge fields, the Wess–Zumino functional is completed by local four-dimensional terms involving U, ℓ_μ, r_μ. The completed Wess–Zumino–Witten functional $\Gamma_{\mathrm{WZW}}[U; \ell, r]$ satisfies

$$\delta_{\alpha_L,\alpha_R}\Gamma_{\mathrm{WZW}}[U; \ell, r] = \mathcal{A}_{\mathrm{quark}}[\alpha_L, \alpha_R; \ell, r],$$

where $\mathcal{A}_{\mathrm{quark}}$ is the consistent flavor anomaly of the microscopic quarks. Thus the pion functional supplies exactly the anomaly phase required by (55.1). Equivalently, the gauged Wess–Zumino–Witten functional is a Goldstone-field realization of the same inflow class: its five-dimensional extension transgresses the microscopic anomaly polynomial, and its four-dimensional gauge variation is the boundary term that matches the quark anomaly.

55.18 ’t Hooft Anomaly Matching as an Infrared Constraint

Anomaly matching becomes a calculational constraint only after the symmetry, backgrounds, and counterterm convention have been fixed. The procedure is:

i. choose the exact global symmetry that is being tested and the class of allowed background fields;

ii. compute the ultraviolet background-gauge cocycle, or equivalently its infinitesimal Wess–Zumino representative on the identity component;

iii. write the most general proposed infrared response from massless fermions, Goldstone fields, and possible topological or invertible sectors;

iv. compare cocycle classes modulo local counterterms.

The comparison is not a comparison of Lagrangian terms or of a particular representative of a current. It is a comparison of the background response class $[\mathcal{A}]$ defined in (55.2). A candidate infrared phase with a unique gapped vacuum and no topological response has the zero class. Such a phase is ruled out when the ultraviolet class is nonzero. A candidate phase with Goldstone fields is allowed only if its Wess–Zumino–Witten term supplies the missing cocycle.

Remark 55.22 (Anomaly matching across spontaneous symmetry breaking). Suppose an exact global symmetry G_{f} is spontaneously broken to H, and the infrared fields consist of Goldstone coordinates on G/H together with possible additional massless or topological sectors. Let $[\mathcal{A}_{\mathrm{UV}}]$ be the ultraviolet background-anomaly class of Definition 55.1. The infrared description is compatible with the ultraviolet theory only if

$$[\mathcal{A}_{\mathrm{UV}}] = [\mathcal{A}_{\mathrm{Goldstone}}] + [\mathcal{A}_{\mathrm{extra}}]$$

as background-response classes modulo local counterterms. Here $[\mathcal{A}_{\mathrm{Goldstone}}]$ is produced by the gauged WZW functional whenever such a functional exists, and $[\mathcal{A}_{\mathrm{extra}}]$ is the class carried by the remaining infrared sector.

Along an RG trajectory with exact G_{f} background fields, the background-anomaly class is unchanged because changing scale changes the renormalized generating functional only by local background counterterms and by integrating out modes in a gauge-covariant manner. In the broken phase, the infrared generating functional decomposes into a Goldstone part and the functional of the remaining infrared sector, together with possible local couplings and background counterterms. Under a background gauge transformation, the phase of the product of the two nonlocal functionals is the sum of their cocycle representatives. The local couplings and local counterterms shift representatives by coboundaries. Passing to background-response classes therefore gives the displayed equality.

For massless QCD, restrict first to the connected nonabelian chiral symmetry

$$SU(N_f)_L \times SU(N_f)_R, \qquad N_f \geq 3,$$

and use the flavor generators T^a normalized by $\mathrm{Tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, as in the pion normalization above. Couple only a left background A and set the right background to zero. The ultraviolet theory contains N_c color copies of a left-handed Weyl fermion in the fundamental representation of $SU(N_f)_L$. The consistent anomaly representative is therefore

$$\delta_\alpha W_{\mathrm{UV}}[A] = \frac{N_c}{48\pi^2}\int_{M_4} \mathrm{Tr}\big(\alpha\,\mathrm{d}A\,\mathrm{d}A\big) + O(A^3), \tag{55.70}$$

where the $O(A^3)$ terms are fixed by the same descent class. A two-derivative pion sigma model without a Wess–Zumino–Witten term is invariant under background gauge transformations and would give zero on the right hand side of (55.70); it therefore cannot be the complete infrared response to the chiral backgrounds. A Goldstone theory with WZW level n gives

$$\delta_\alpha\big(n\Gamma_{\mathrm{WZW}}[U; A, 0]\big) = \frac{n}{48\pi^2}\int_{M_4} \mathrm{Tr}\big(\alpha\,\mathrm{d}A\,\mathrm{d}A\big) + O(A^3).$$

Matching (55.70) for arbitrary α and A forces

$$n = N_c. \tag{55.71}$$

This is the same equality obtained above from the Wess–Zumino functional, now read as a test of infrared phases. If the chiral symmetry were unbroken and the theory confined to a unique trivial gapped vacuum, there would be no massless determinant line, no Goldstone WZW term, and no topological response; that candidate would fail (55.70). An unbroken phase is not excluded by this argument if it contains massless composite fermions or a topological/invertible sector carrying exactly the same background cocycle. The matching condition constrains the infrared possibilities; it is not a proof of which dynamical phase QCD chooses.

The vector subgroup illustrates the same logic in a form that is often hidden by notation. Set $L = R = V$ and couple a single vector background v to both chiralities. In the anomaly coefficient (51.14), the left-handed and right-handed Dirac components have the same vector generator. Their contributions to the vector current anomaly cancel:

$$\mathcal{A}_V^{abc} = \frac{N_c}{2}\Big[\mathrm{Tr}\big(T^a\{T^b, T^c\}\big) - \mathrm{Tr}\big(T^a\{T^b, T^c\}\big)\Big] = 0. \tag{55.72}$$

Thus $SU(N_f)_V$ is an anomaly-free symmetry that can act on a purely massive color-singlet sector without requiring additional massless fields or a topological response. By contrast, a purely left chiral subgroup has the nonzero coefficient in (55.70); if it remained unbroken in a gapped infrared theory, its anomaly would have to be carried by some nontrivial infrared sector.

The positivity statement used earlier is logically separate but compatible with this anomaly test. Under Hypothesis 55.13, equal positive quark masses select a vacuum preserving the vector flavor group; taking the small-mass limit then aligns the chiral condensate with $SU(N_f)_V$ whenever the required limits exist. Anomaly matching then determines the response of the broken axial directions: the Goldstone target $SU(N_f)$ together with $N_c\Gamma_{\mathrm{WZW}}$ reproduces the ultraviolet chiral anomaly, while the unbroken vector subgroup remains nonanomalous. The two inputs have different status. The vector-alignment argument depends on Euclidean positivity and limiting assumptions; the anomaly matching equation is an exact background-response constraint once the ultraviolet symmetry and regulator convention are fixed.

For the electromagnetic specialization use the dual tensor defined at the start of the chapter and the flavor-generator normalization just stated, and fix

$$T^3 = \frac{\sigma_3}{2} = \frac{1}{2}\,\mathrm{diag}(1,-1).$$

Let q denote the dimensionless light-quark charge matrix, and embed electromagnetism by

$$\ell_\mu = r_\mu = e\,qA_\mu, \qquad q = \mathrm{diag}(q_1,\dots,q_{N_f}),$$

The relevant microscopic diagram is the quark triangle in Figure 55.7. The figure fixes the assignment of the axial flavor generator to one vertex and the two electromagnetic current insertions to the other vertices before the trace over light-quark charges is evaluated.

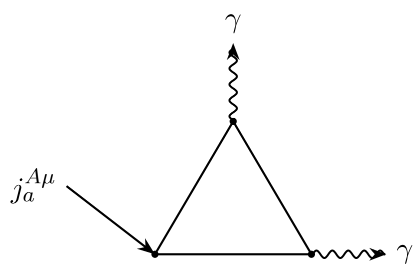


Figure 55.7: The electromagnetic background probes the axial $SU(2)$ flavor anomaly through the quark triangle with one axial-current insertion and two electromagnetic-current insertions.

Remark 55.23 (Electromagnetic specialization). The corresponding axial-flavor Ward identity is

$$\partial_\mu j_a^{A\mu} = -\frac{N_c e^2}{16\pi^2}\,\mathrm{Tr}(T^a\{q,q\})\,F_{\mu\nu}\widetilde{F}^{\mu\nu}.$$

The anticommutator records the two identical electromagnetic insertions in the triangle; since the charge matrix is diagonal,

$$\mathrm{Tr}(T^a\{q,q\}) = 2\,\mathrm{Tr}\big(T^a q^2\big)\,.$$

Equivalently,

$$\partial_\mu j_a^{A\mu} = -\frac{N_c e^2}{8\pi^2}\,\mathrm{Tr}\big(T^a q^2\big)\,F_{\mu\nu}\widetilde{F}^{\mu\nu}.$$

This is the same anomaly coefficient as the preceding chapter's Dirac-fermion formula, rewritten from $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$ to $2F_{\mu\nu}\widetilde{F}^{\mu\nu}$. For $q = \mathrm{diag}(2/3,-1/3)$, the trace is nonzero only for $a=3$, where

$$\mathrm{Tr}\big(T^3 q^2\big) = \frac{1}{6}.$$

Thus

$$\mathrm{Tr}\big(T^3\{q,q\}\big) = \frac{1}{3}.$$

Consequently the gauged WZW functional contains the neutral-pion coupling

$$\mathcal{L}_{\pi^0\gamma\gamma} = \frac{N_c e^2}{16\pi^2 f_\pi}\,\mathrm{Tr}\big(T^3\{q,q\}\big)\;\pi^0 F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{55.73}$$

namely

$$\mathcal{L}_{\pi^0\gamma\gamma} = \frac{N_c e^2}{48\pi^2 f_\pi}\pi^0 F_{\mu\nu}\widetilde{F}^{\mu\nu}.$$

For $N_c = 3$, the coefficient multiplying $\pi^0 F_{\mu\nu}\widetilde{F}^{\mu\nu}$ is $e^2/(16\pi^2 f_\pi)$. The coefficient is the low-energy coefficient required by matching the microscopic electromagnetic flavor anomaly.

Figure 55.8 records the low-energy vertex matched from that microscopic anomaly: one neutral pion couples to two photons with the coefficient fixed above.

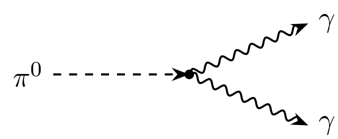


Figure 55.8: The electromagnetic flavor anomaly produces the local $\pi^0\gamma\gamma$ vertex in the pion effective action.

Writing

$$\mathcal{L}_{\pi^0\gamma\gamma} = \frac{g_{\pi\gamma\gamma}}{4}\,\pi^0 F_{\mu\nu}\widetilde{F}^{\mu\nu}$$

gives $g_{\pi\gamma\gamma} = e^2/(4\pi^2 f_\pi) = \alpha/(\pi f_\pi)$ for $N_c = 3$. The corresponding decay rate is

$$\Gamma(\pi^0 \to 2\gamma) = \frac{\alpha^2 m_{\pi^0}^3}{64\pi^3 f_\pi^2},$$

in the convention for f_π used above.

55.19 Conceptual Output of the Gauge-Dynamics Sequence

The dependencies established in this sequence are as follows. Gauge fields are local connections. Gauge fixing supplies local coordinates and ghosts. BRST cohomology defines physical states, local operators, counterterms, and anomaly candidates in the gauge-fixed formalism. QCD renormalization gives asymptotically free short-distance variables. Local anomalies are constrained by Wess–Zumino consistency and descent. Finite background anomalies are encoded by cocycles on background-field space. Exact anomaly classes are preserved along RG flow. Spontaneous symmetry breaking realizes these constraints through Goldstone fields and the Wess–Zumino–Witten functional.

Part V

Conformal Field Theory

Chapter 56

Fixed Points and Conformal Data

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

56.1 Local QFT Data Used in Fixed-Point Analysis

Let $D \geq 2$ be the spacetime dimension, let $d = D - 1$, and let $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$. The local Lorentzian data used in this volume are the following.

1. A Hilbert space $\mathcal{H}$ of physical states.
2. A strongly continuous unitary representation $U(\Lambda, a)$ of the proper orthochronous Poincare group, or of its spin cover when spinorial fields occur. The group law is
$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda' a + a').$$
3. Self-adjoint generators P_μ and $J_{\mu\nu} = -J_{\nu\mu}$, normalized near the identity by
$$U(\Lambda, a) = 1 + \mathrm{i}a^\mu P_\mu + \frac{\mathrm{i}}{2}\omega^{\mu\nu} J_{\mu\nu} + O(a^2, \omega^2, a\omega), \qquad \Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu.$$
The operators P_μ generate energy and momentum, while $J_{\mu\nu}$ generates spatial rotations and Lorentz boosts.
4. A Poincare-invariant unit vector $\Omega \in \mathcal{H}$, so $U(\Lambda, a)\Omega = \Omega$.

5. Local fields $\widehat{\Phi}_\alpha(x)$, understood as operator-valued distributions on a common invariant dense domain, with Poincare covariance

$$U(\Lambda,a)^{-1}\widehat{\Phi}_\alpha(x)U(\Lambda,a) = R(\Lambda)_\alpha{}^\beta\,\widehat{\Phi}_\beta(\Lambda x+a).$$

   Here R is the finite-dimensional Lorentz representation carried by the field component label α. Bosonic local fields commute, and fermionic local fields anticommute, after smearing with test functions whose supports are spacelike separated.

In addition, the discussion of conformal symmetry assumes a renormalized local stress tensor $\widehat{T}_{\mu\nu}(x)$. It is a symmetric tensor-valued operator distribution satisfying

$$\partial_\mu \widehat{T}^{\mu\nu} = 0, \qquad \widehat{T}^{\mu\nu} = \widehat{T}^{\nu\mu}. \tag{56.1}$$

where the conservation equation is distributional. When the spatial integrals exist with the chosen domain and infrared prescription, the translation and Lorentz generators are represented by

$$P^\mu = \int_{\Sigma_t} \mathrm{d}^d\vec{x}\,\widehat{T}^{0\mu}(t,\vec{x}). \tag{56.2}$$

and

$$J^{\mu\nu} = \int_{\Sigma_t} \mathrm{d}^d\vec{x}\,\big(x^\mu\widehat{T}^{0\nu}(t,\vec{x}) - x^\nu\widehat{T}^{0\mu}(t,\vec{x})\big). \tag{56.3}$$

with $\Sigma_t = \{x^0 = t\}$. These equations are operator equations after the same smearing, renormalization, and boundary conventions used to define $\widehat{T}_{\mu\nu}$.

Lorentz covariance of the stress tensor means

$$U(\Lambda,0)\widehat{T}_{\mu\nu}(x)U(\Lambda,0)^{-1} = \Lambda_\mu{}^\rho\Lambda_\nu{}^\sigma\widehat{T}_{\rho\sigma}(\Lambda x). \tag{56.4}$$

where indices on Λ are moved with $\eta_{\mu\nu}$. Hence $\widehat{T}^\mu{}_\mu(x)$ is a scalar local operator:

$$U(\Lambda,0)\widehat{T}^\mu{}_\mu(x)U(\Lambda,0)^{-1} = \widehat{T}^\mu{}_\mu(\Lambda x).$$

Remark 56.1 (Fixed points as supplied data, not universal ultraviolet origins). This volume analyzes the consequences of fixed-point data once those data have been constructed or assumed. A conformal perturbation problem starts with a specific fixed point $\mathcal{C}$, a specified family of scalar local operators O_A, and a prescription for the contact terms in products of $\int O_A$. The deformation coordinates g^A then describe a local chart near $\mathcal{C}$, not a general definition of QFT. In particular, the existence of an ultraviolet-complete theory does not follow from, nor is it equivalent to, being representable as a relevant deformation of an accessible CFT. Asymptotically free gauge theories are controlled in the ultraviolet by running local couplings and require a nonperturbative definition such as a lattice or other regulator framework; constructive and SPDE routes provide different fixed-point or continuum-limit data. The conformal methods developed below are used precisely where their hypotheses are present.

56.2 Renormalized Operators and the Trace Equation

Fix a renormalized coordinate chart $\{g_I(\mu)\}_{I\in\mathcal{I}}$ on a space of local scalar deformations. The index set $\mathcal{I}$ labels renormalized local insertions $[O_I]_\mu(x)$. In a Lagrangian presentation with bare scalar operators O_I^{bare}, write the interaction density as

$$\mathcal{L}_{\mathrm{int}}^{\mathrm{bare}} = \sum_I g_I^{\mathrm{bare}} O_I^{\mathrm{bare}}. \tag{56.5}$$

The renormalized insertions are defined by the response of the bare density to renormalized coordinates:

$$\delta\mathcal{L}_{\mathrm{int}}^{\mathrm{bare}} = \sum_I \delta g_I^{\mathrm{bare}} O_I^{\mathrm{bare}} = \sum_I \delta g_I(\mu)\, [O_I]_\mu. \tag{56.6}$$

This equation fixes the normalization and mixing convention for the local operators in the chosen scheme. The beta-function vector field is

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu} g_I(\mu) = \beta_I(g(\mu)). \tag{56.7}$$

Flat trace equation from a renormalized source-coordinate system. Assume that the renormalized chart $\{g_I(\mu), [O_I]_\mu\}_{I\in\mathcal{I}}$ is supplied by a local source functional whose constant Weyl variation is governed, in flat space, by the local RG vector field

$$\Delta_\sigma^{\mathrm{loc}} = \int \mathrm{d}^D x\, \sigma \left(2g_{\mu\nu}\frac{\delta}{\delta g_{\mu\nu}} + \beta_I(g)\frac{\delta}{\delta g_I}\right)$$

up to local anomaly/contact functionals and up to the divergence of a local virial current. Then, as a renormalized flat-space local-operator identity away from coincident insertions,

$$\widehat{T}^\mu{}_\mu = \sum_I \beta_I(g)\, [O_I]_\mu + \partial_\mu V^\mu. \tag{56.8}$$

The vector V^μ is a local vector operator recording the virial and improvement ambiguity in the chosen stress-tensor representative. Contact terms in (56.8) are not inferred from separated-point kernels; they are fixed together with the renormalized source Ward identities of the chart. Curvature terms appear after coupling the theory to a background metric, but vanish in the flat separated-point identity displayed here.

The source-coordinate system gives two descriptions of the same infinitesimal change of renormalization length. First, varying the metric by a constant Weyl factor $\delta_\sigma g_{\mu\nu} = 2\sigma g_{\mu\nu}$ gives, by the stress-tensor definition,

$$\delta_\sigma W = -\int \mathrm{d}^D x \sqrt{g}\, \sigma\, \langle T^\mu{}_\mu(x)\rangle\, .$$

Second, the same change expressed at fixed regulator is a displacement along the beta-function vector field in the renormalized coordinates:

$$\delta_\sigma W = -\int \mathrm{d}^D x \sqrt{g}\, \sigma\, \beta_I(g)\, \langle [O_I]_\mu(x)\rangle$$

with the sign fixed by the convention $\mu\, \mathrm{d}g_I/\mathrm{d}\mu = \beta_I$ and by (56.6). Equating the two descriptions, and restoring the possible local virial term produced by the choice of stress-tensor representative, gives (56.8) inside correlators at separated points. Local anomaly terms and derivatives of delta functions are exactly the contact terms belonging to the chosen source-coordinate system, so they do not modify the displayed separated-point flat-space operator identity.

Remark 56.2 (Scale invariance versus conformal invariance). At a fixed point the vanishing of the beta-function part of (56.8) gives

$$\widehat{T}^{\mu}{}_{\mu} = \partial_\mu V^\mu \tag{56.9}$$

in the flat-space local-operator quotient determined by the source-coordinate system. If $\widehat{T}^{\mu\nu}$ is symmetric and conserved, then

$$D^\mu = x_\nu \widehat{T}^{\mu\nu} - V^\mu \tag{56.10}$$

is a conserved dilatation current exactly when (56.9) holds. Conformal symmetry requires the stronger local statement that the virial class is removable by the allowed stress-tensor improvements. For the scalar improvement

$$\widehat{T}^{\mu\nu} \longmapsto \widehat{T}^{\mu\nu} + (\partial^\mu\partial^\nu - \eta^{\mu\nu}\partial^2)L,$$

the trace changes by $-(D-1)\partial^2 L$, so the trace can be removed when

$$V^\mu = (D-1)\partial^\mu L + J^\mu, \qquad \partial_\mu J^\mu = 0, \tag{56.11}$$

modulo the operator identities and contact-term convention of the source-coordinate system. More general improvement tensors enlarge the same quotient by the image of their trace-changing map. Thus the scale-to-conformal implication is the local problem of proving that this virial obstruction class vanishes.

In this stress-tensor framework, a scale-invariant but nonconformal local QFT is a theory for which (56.9) gives a conserved dilatation current $D^\mu = x_\nu \widehat{T}^{\mu\nu} - V^\mu$, while the class of V^μ is nonzero in the quotient by conserved currents and allowed improvement currents. Equivalently, the theory has a virial current of dimension $D-1$ whose divergence is the trace, but no admissible improvement makes the stress tensor traceless. This is the obstruction isolated by Polchinski's analysis of scale versus conformal invariance: $D^\mu = x_\nu T^{\mu\nu} - V^\mu$ may generate scale transformations even when there are no conserved special-conformal currents.[1]

The status is dimension-dependent. In $D = 2$, the Zamolodchikov–Polchinski argument proves the implication under explicit Hilbert-space positivity, locality, stress-tensor, vacuum, spectrum, and regularity hypotheses strong enough for the c-function and stress-tensor two-point analysis. In $D = 4$, the Luty–Polchinski–Rattazzi program and related anomaly/dilaton arguments give powerful conditional constraints, but this monograph does not use an unconditional theorem asserting that every unitary scale-invariant four-dimensional QFT is conformal. In $D = 3$ and $D > 4$, later uses of conformal symmetry likewise require the traceless improved stress tensor, the vanishing virial class, or the equivalent conformal Ward identities as part of the stated data. The cohomological formulation and references are given in Remark 35.7.

Virial class, improvement, and conformal currents. Let $T^{\mu\nu}$ be a symmetric conserved stress tensor in flat D-dimensional Minkowski space. Suppose that

$$T^{\mu}{}_{\mu} = \partial_\mu V^\mu$$

as a local operator identity in the chosen separated-point/contact chart. Then

$$D^\mu = x_\nu T^{\mu\nu} - V^\mu$$

[1] J. Polchinski, "Scale and conformal invariance in quantum field theory," *Nucl. Phys. B* **303** (1988) 226–236.

is a conserved dilatation current. If, for some scalar local operator L and conserved current J^μ,

$$V^\mu = (D-1)\partial^\mu L + J^\mu, \qquad \partial_\mu J^\mu = 0,$$

then the improved stress tensor

$$T^{\mu\nu}_{\rm imp} = T^{\mu\nu} + (\partial^\mu\partial^\nu - \eta^{\mu\nu}\partial^2)L$$

is symmetric, conserved, and traceless. Consequently, for every conformal Killing vector ξ^μ satisfying

$$\partial_\mu\xi_\nu + \partial_\nu\xi_\mu = \frac{2}{D}(\partial_\rho\xi^\rho)\eta_{\mu\nu},$$

the current $j^\mu_\xi = T^{\mu\nu}_{\rm imp}\xi_\nu$ is conserved. The divergence of the proposed dilatation current is

$$\partial_\mu D^\mu = T^\mu{}_\mu + x_\nu\partial_\mu T^{\mu\nu} - \partial_\mu V^\mu = 0.$$

The improvement term is symmetric. Its divergence vanishes because ordinary derivatives commute:

$$\partial_\mu(\partial^\mu\partial^\nu - \eta^{\mu\nu}\partial^2)L = \partial^\nu\partial^2 L - \partial^\nu\partial^2 L = 0.$$

Its trace is

$$\eta_{\mu\nu}(\partial^\mu\partial^\nu - \eta^{\mu\nu}\partial^2)L = -(D-1)\partial^2 L.$$

Therefore

$$T_{\rm imp}{}^\mu{}_\mu = \partial_\mu V^\mu - (D-1)\partial^2 L = \partial_\mu J^\mu = 0.$$

For a conformal Killing vector,

$$\partial_\mu j^\mu_\xi = T^{\mu\nu}_{\rm imp}\partial_\mu\xi_\nu = \frac{1}{2}T^{\mu\nu}_{\rm imp}(\partial_\mu\xi_\nu + \partial_\nu\xi_\mu) = \frac{1}{D}(\partial_\rho\xi^\rho)T_{\rm imp}{}^\mu{}_\mu = 0,$$

where conservation and symmetry of $T^{\mu\nu}_{\rm imp}$ were used. Thus the improved traceless representative supplies the currents for translations, rotations, dilatations, and special conformal transformations whenever the corresponding smeared charges are defined.

An RG fixed point in the coordinate chart is a point g_* satisfying

$$\beta_I(g_*) = 0 \qquad \text{for every } I \in \mathcal{I}. \tag{56.12}$$

At such a point the beta-function part of the trace vanishes. A flat-space conformal fixed point is obtained when the stress tensor can be chosen, by allowed improvements and contact-term conventions, so that

$$\widehat{T}^\mu{}_\mu = 0 \tag{56.13}$$

as a renormalized local operator equation in flat space. Equation (56.13), together with (56.1), is the local input from which the conformal currents and charges are derived in the next chapter.

The linearized RG flow near g_* records the scaling dimensions of deformations. Let

$$B_{IJ} = \left.\frac{\partial\beta_I}{\partial g_J}\right|_{g=g_*}.$$

Choose an eigenvector $v^{(a)}_J$ and write

$$g_I - g_{*I} = \lambda_a v^{(a)}_I + O(\lambda^2), \qquad \sum_J B_{IJ}v^{(a)}_J = -y_a v^{(a)}_I.$$

With the momentum scale μ as flow parameter,

$$\mu\frac{\mathrm{d}\lambda_a}{\mathrm{d}\mu} = -y_a\lambda_a + O(\lambda^2).$$

Equivalently, under a length rescaling $x \mapsto bx$, $b > 1$,

$$\lambda_a(b) = b^{y_a}\lambda_a + O(\lambda^2).$$

If the corresponding scalar perturbation is $\int \mathrm{d}^D x\, \lambda_a O_a(x)$, then

$$y_a = D - \Delta_a, \tag{56.14}$$

where Δ_a is the scaling dimension of O_a. Thus $y_a > 0$, $y_a = 0$, and $y_a < 0$ define relevant, marginal, and irrelevant directions, respectively, in the long-distance flow.

Linearized RG eigenvalues and operator dimensions. Assume that the beta-function vector field is differentiable at g_* and that the tangent direction $v^{(a)}$ is an ordinary eigenvector of $B_{IJ} = \partial\beta_I/\partial g_J|_{g_*}$ with eigenvalue $-y_a$. Let

$$O_a = \sum_I v_I^{(a)}[O_I]_*$$

be the corresponding scalar deformation operator in the linearized source chart. If O_a is a scaling operator at the fixed point, then its scaling dimension satisfies

$$\Delta_a = D - y_a.$$

If B has a Jordan block instead of a diagonal eigenspace, the corresponding operators form a logarithmic mixing block under scale transformations; the number $D - y_a$ is then the common diagonal scaling dimension of the block, not a complete diagonalization of the dilatation action.

Along the eigen-direction write

$$g_I(\mu) - g_{*I} = \lambda_a(\mu)v_I^{(a)} + O(\lambda_a^2).$$

The RG equation gives

$$\mu\frac{\mathrm{d}\lambda_a}{\mathrm{d}\mu} = -y_a\lambda_a + O(\lambda_a^2).$$

Changing the unit of length by $x \mapsto bx$, $b > 1$, corresponds to lowering the momentum scale by b^{-1}, hence

$$\lambda_a(b) = b^{y_a}\lambda_a + O(\lambda_a^2).$$

At the fixed point a scalar scaling operator obeys

$$O_a(x) \mapsto b^{-\Delta_a}O_a(b^{-1}x).$$

Therefore the integrated perturbation transforms as

$$\lambda_a \int \mathrm{d}^D x\, O_a(x) \mapsto \lambda_a b^{D-\Delta_a} \int \mathrm{d}^D x\, O_a(x).$$

Comparing this scaling of the coupling with $\lambda_a(b) = b^{y_a}\lambda_a$ gives $y_a = D - \Delta_a$. For a Jordan block, solving the linearized RG equation gives the same power b^{y_a} multiplied by polynomials in $\log b$, which is the RG form of logarithmic operator mixing.

56.3 Conformal Manifolds and Exactly Marginal Deformations

A marginal scalar operator at one fixed point is only a tangent candidate. A conformal manifold is a family of fixed-point data in which the tangent operators, contact terms, source normalizations, stress tensors, and redundant directions vary coherently. This section fixes the local meaning of that statement before the supersymmetric worked examples later in the monograph.

Definition 56.3 (Local conformal-family structure)**.** A local conformal-family structure of dimension r consists of the following objects.

1. A connected r-dimensional smooth or complex manifold $\mathcal{M}$ of parameters, with a base point λ_0.
2. For every $\lambda \in \mathcal{M}$, fixed-point local QFT data $\mathcal{C}_\lambda$: a Hilbert-space or Euclidean reconstruction structure, a stress-tensor source functional $W_\lambda[g, J]$, conformal Ward identities including their contact terms, and a specified local-operator space modulo null operators.
3. A vector bundle $\mathcal{V} \to \mathcal{M}$ whose fiber $\mathcal{V}_\lambda$ is the finite-dimensional space of renormalized scalar operators of dimension D that are allowed by the symmetries kept in the family, modulo total derivatives and null operators.
4. A tangent-operator map
$$\mathsf{I}_\lambda : T_\lambda\mathcal{M} \longrightarrow \mathcal{V}_\lambda, \qquad v \longmapsto O_v.$$
5. For every finite list of local operators away from collision diagonals, the separated-point distributions are differentiable in λ, and the derivative in the direction v is represented by the renormalized integrated insertion
$$\partial_v \langle X_1(x_1) \cdots X_n(x_n)\rangle_\lambda = -\left[\int_{\mathbb{R}^D} \mathrm{d}^D x \, \langle O_v(x) X_1(x_1) \cdots X_n(x_n)\rangle_\lambda\right]_{\mathrm{ren}}. \tag{56.15}$$
The brackets $[\cdots]_{\mathrm{ren}}$ include the local counterterms supported where x collides with one of the x_i and, on curved backgrounds, the corresponding source and metric contact terms.

The structure is a conformal family because every point λ already satisfies the fixed-point conditions and because (56.15) specifies how tangent vectors are represented in correlation functions.

The definition is local on $\mathcal{M}$. Global continuation of coordinates, preferred Lagrangian presentations, and discrete duality identifications require additional structure. A duality group, if present, acts on the local family after it has been defined as an isomorphism of the complete line-operator spectrum, local operator algebra, and source-contact chart.

Source-coordinate criterion for exact marginality. Let h^a, $a = 1, \ldots, m$, be local coordinates on a finite-dimensional space of candidate marginal source deformations at a fixed point $\mathcal{C}_0$. Let
$$\beta^\alpha(h), \qquad \alpha = 1, \ldots, s,$$
denote the obstruction functions in the chosen source-coordinate system: the components of the local RG vector field which must vanish for the deformed theory to remain conformal, together with any current-moment-map or virial obstruction that is part of the same Ward identity. Let

a redundancy group G_{red} act on the source coordinates by field redefinitions, flavor rotations, or other changes that do not change the fixed-point QFT structure. Its infinitesimal action is a distribution

$$\rho : \mathrm{Lie}(G_{\mathrm{red}}) \longrightarrow T_h\mathbb{R}^m.$$

The local conformal locus in this chart is

$$\mathcal{M}_{\mathrm{loc}} = \{h : \beta^\alpha(h) = 0\}/G_{\mathrm{red}}, \tag{56.16}$$

provided the quotient is taken only along orbits contained in the zero locus. If $d\beta|_{h_0}$ has constant rank r_β near the base point and the redundancy distribution has constant rank r_{red} tangent to $\beta^{-1}(0)$, then the local dimension in the source-coordinate system is

$$\dim \mathcal{M}_{\mathrm{loc}} = m - r_\beta - r_{\mathrm{red}}. \tag{56.17}$$

The substantive input behind (56.17) is the preceding source-coordinate system: all obstruction functions, all redundancies, and all contact-term conventions have to be known before the rank count is meaningful. This is the field-theoretic core of Leigh–Strassler type exact-marginality arguments in supersymmetric examples.

At second order in conformal perturbation theory, the obstruction functions are visible in the OPE. If O_a are scalar primaries of dimension D, then the collision region in

$$\int \mathrm{d}^D x\, \mathrm{d}^D y\, O_a(x) O_b(y)$$

is controlled by the $O_a(x)O_b(y)$ OPE. Terms proportional to another dimension-D scalar O_c, to a divergence of a dimension-$D-1$ vector, or to a source-contact counterterm are precisely the terms that can contribute to β^c, to a virial/current obstruction, or to a scheme change. The finite-dimensional chart above is therefore a compressed record of a distributional statement about collision singularities. A claim that a dimension-D operator is exactly marginal is justified only after these collision terms have been controlled to all orders in the proposed family, or after an independent nonperturbative construction of the family has been supplied.

Zamolodchikov metric. For a conformal-family structure, the separated two-point function of tangent operators has the form

$$\langle O_v(x) O_w(0)\rangle_\lambda = \frac{G_\lambda(v,w)}{|x|^{2D}} \qquad (x \neq 0), \tag{56.18}$$

after fixing the normalization of O_v by (56.15). Reflection positivity makes G_λ positive semidefinite on tangent candidates. Its null directions are redundant deformations: total derivatives, null operators, and infinitesimal source reparametrizations that vanish in separated-point correlators. After quotienting by these directions, G_λ is the Zamolodchikov metric on the local conformal family.

This metric is independent of the particular source coordinates in the usual tensorial sense. If $h^a = h^a(\lambda)$ and $\lambda^i = \lambda^i(h)$ are two coordinates on the same family, then

$$O_i = \frac{\partial h^a}{\partial \lambda^i} O_a \quad \text{modulo contact terms and redundant operators,}$$

so separated two-point functions give

$$G'_{ij} = \frac{\partial h^a}{\partial \lambda^i}\frac{\partial h^b}{\partial \lambda^j} G_{ab}. \tag{56.19}$$

Contact terms can change the distributional extension of (56.18) to $x = 0$, but they do not change the separated coefficient $G_\lambda(v, w)$. This distinction is essential when the same exactly marginal source also appears in local RG equations, curved-background Ward identities, or supersymmetric localization formulae.

Supersymmetric examples and global identifications. The supersymmetric volume contains worked source-coordinate examples of (56.16). In Chapter 90, the Klebanov–Witten chart has three holomorphic source coordinates and one independent beta-function constraint under the stated $\mathcal{N} = 1$, flavor-symmetry, and Wilsonian-scheme hypotheses, giving a two-complex-dimensional local conformal locus. In Chapter 97, four-dimensional $\mathcal{N} = 4$ Yang–Mills has the local coordinate

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi \mathrm{i}}{g_{\mathrm{YM}}^2}, \qquad \operatorname{Im}\tau > 0,$$

so the local family is one complex dimensional before electric-magnetic duality identifications. The quotient by a subgroup of $PSL(2, \mathbb{Z})$ is a global statement about the line-operator lattice and global form of the gauge group, separate from the local exact-marginality proof. In Chapter 95, the standard $\mathcal{N} = 6$ ABJM specification at fixed integers (N, k) has no continuous conformal coupling: the Chern–Simons level is quantized and the superpotential coefficient is fixed by the supersymmetry completion. These examples illustrate the rule: a parameter belongs to a conformal manifold only if it is represented by a genuine tangent coordinate in a fixed QFT specification. The two-dimensional CFT volume gives the complementary symmetric-product example in Construction 66.32: seed exactly marginal directions and dressed two-cycle twist fields define the local tangent data of the compact $\mathcal{N} = (4, 4)$ conformal families, while the arithmetic quotient and singular loci are global CFT data.

56.4 RG Monotones and Fixed-Point Ordering

Let $\mathfrak{T}$ be a class of unitary Lorentzian QFTs equipped with renormalized stress-tensor source functionals and let an RG trajectory in $\mathfrak{T}$ have a UV endpoint $\mathcal{C}_{\mathrm{UV}}$ and an IR endpoint $\mathcal{C}_{\mathrm{IR}}$, each a CFT in the sense of the preceding section. The length scale R increases toward the infrared. An endpoint monotone is a real number $\mathfrak{M}(\mathcal{C})$, assigned to every endpoint CFT in the class, together with an interpolating quantity $\mathfrak{M}(R)$ on the trajectory such that

$$\lim_{R\to 0} \mathfrak{M}(R) = \mathfrak{M}(\mathcal{C}_{\mathrm{UV}}), \qquad \lim_{R\to\infty} \mathfrak{M}(R) = \mathfrak{M}(\mathcal{C}_{\mathrm{IR}}), \qquad \frac{\mathrm{d}}{\mathrm{d}\log R}\mathfrak{M}(R) \le 0.$$

The endpoint consequence is $\mathfrak{M}(\mathcal{C}_{\mathrm{UV}}) \ge \mathfrak{M}(\mathcal{C}_{\mathrm{IR}})$. This definition keeps the mathematical content in the class $\mathfrak{T}$: positivity, stress-tensor existence, analyticity, infrared behavior, and contact-term conventions are hypotheses on the trajectory, not decorations on the conclusion.

In two spacetime dimensions the fixed-point number is the Virasoro central charge c_{2d}. Under the hypotheses used in the Zamolodchikov–Polchinski formulation–unitarity, Poincare invariance, a normalizable vacuum, a local stress tensor, reflection positivity or Hilbert-space positivity, and regularity of the stress-tensor two-point distributions–write the Euclidean two-point functions in complex coordinates as

$$\begin{aligned} \langle T(z,\bar z)T(0)\rangle &= \frac{F(R)}{z^4}, \\ \langle T(z,\bar z)\Theta(0)\rangle &= \frac{G(R)}{z^3\bar z}, \\ \langle \Theta(z,\bar z)\Theta(0)\rangle &= \frac{H(R)}{z^2\bar z^2}, \qquad R=|z|, \qquad \Theta = T^\mu{}_\mu. \end{aligned} \tag{56.20}$$

The powers of $z, \bar z$ isolate the tensor weights, while the functions F, G, H depend only on R. Conservation of the stress tensor gives the Zamolodchikov combination

$$C(R) = 2F(R) - G(R) - \frac{3}{8}H(R) \tag{56.21}$$

with derivative

$$\frac{\mathrm{d}C(R)}{\mathrm{d}\log R} = -\frac{3}{4}H(R) \le 0. \tag{56.22}$$

Reflection positivity gives $H(R) \ge 0$. At a conformal endpoint $\Theta = 0$ at separated flat-space points, $G = H = 0$ and $F = c_{2d}/2$ in the normalization $\langle T(z)T(0)\rangle = c_{2d}/(2z^4)$, so $C(R)$ tends to the central charge of that endpoint. Thus $c_{\mathrm{UV}} \ge c_{\mathrm{IR}}$ for trajectories satisfying the displayed hypotheses.

In three spacetime dimensions a fixed-point quantity of central importance is the finite sphere free energy

$$F_{\mathcal{C}} = -\log |Z_{\mathcal{C}}(S^3)|,$$

after the local counterterms allowed on S^3 have been fixed. This quantity is part of conformal data once a scheme for the curved-background partition function has been declared. Monotonicity statements involving $F_{\mathcal{C}}$ require additional input about the RG trajectory and its regulated observables; they are not used here as definitions of a CFT. Gauge constraints, zero modes, topological sectors, and background counterterms must be included in the definition of $Z(S^3)$.

In four spacetime dimensions the endpoint quantity is the Euler coefficient a_{W} in the Weyl anomaly,

$$\langle T^\mu{}_\mu\rangle_g = \frac{1}{16\pi^2}\left(c_{\mathrm{W}}W^2 - a_{\mathrm{W}}E_4 + b_{\mathrm{D}}\nabla^2 R\right),$$

with the notation developed in Chapter 58. The standard Komargodski–Schwimmer construction should be read here as a conditional dilaton-dispersion argument, not as a theorem-level proof from the general axioms of four-dimensional QFT. Its input is a unitary Lorentz-invariant flow between CFT endpoints, a renormalized stress-tensor source functional along the flow, a weak spectator dilaton τ coupled by $g_{\mu\nu} = \mathrm{e}^{-2\tau}\eta_{\mu\nu}$, a controlled large-$f$ decoupling limit for the spectator sector, and an infrared prescription under which the dilaton scattering amplitude exists. Under these hypotheses, Wess–Zumino consistency for the mismatch of the UV and IR Weyl anomalies fixes the flat-space four-derivative dilaton interaction whose on-shell four-dilaton amplitude is

$$\mathcal{A}(s,t,u) = \frac{a_{\mathrm{UV}} - a_{\mathrm{IR}}}{f^4}\left(s^2+t^2+u^2\right) + O(p^6), \qquad s+t+u=0. \tag{56.23}$$

In the forward limit $t = 0$, this becomes

$$\mathcal{A}(s, 0, -s) = \frac{2(a_{\mathrm{UV}} - a_{\mathrm{IR}})}{f^4} s^2 + O(s^4).$$

If, in addition, the forward amplitude obeys the analyticity, crossing, polynomial boundedness, unitarity, and infrared-regulator conditions needed for the dispersion relation, and if the large-circle and small-circle contour contributions are controlled, Cauchy's theorem applied to $\mathcal{A}(s)/s^3$ gives the formal sum rule

$$a_{\mathrm{UV}} - a_{\mathrm{IR}} = \frac{f^4}{\pi} \int_0^\infty \mathrm{d}s \, \frac{\sigma_{\tau\tau}(s)}{s^2}, \tag{56.24}$$

where the optical theorem has been used in the form $\operatorname{Im} \mathcal{A}(s, 0) = s\, \sigma_{\tau\tau}(s)$. Under the same scattering assumptions, unitarity gives $\sigma_{\tau\tau}(s) \geq 0$, hence

$$a_{\mathrm{UV}} \geq a_{\mathrm{IR}}.$$

If such a positive cross-section representation is available with a scale cutoff, the same formal sum rule defines an interpolating quantity

$$a(\mu) = a_{\mathrm{UV}} - \frac{f^4}{\pi} \int_\mu^\infty \mathrm{d}s \, \frac{\sigma_{\tau\tau}(s)}{s^2}, \qquad \frac{\mathrm{d}a(\mu)}{\mathrm{d}\mu} = \frac{f^4}{\pi} \frac{\sigma_{\tau\tau}(\mu)}{\mu^2} \geq 0. \tag{56.25}$$

Thus $a(\mu)$ increases with the energy scale μ and decreases with the length scale $R \sim \mu^{-1}$, inside the same conditional scattering framework. This $a(\mu)$ is not by itself a nonperturbative local RG c-function constructed for every four-dimensional QFT. The coefficient c_{W} does not obey an analogous general endpoint inequality; its role in fixed-point data is through stress-tensor correlators, not through this dilaton sum rule.

Remark 56.4 (Status of the monotonicity statements). The preceding paragraphs state the monotonicity statements at the level at which this volume uses them. The two-dimensional result is a theorem after the analytic assumptions entering the $C(R)$ construction have been fixed. The three-dimensional sphere quantity is recorded here as fixed-point data; the core CFT volume does not use an entropic construction as a structural input. The four-dimensional dilaton discussion is an inspired and highly constraining physical argument whose conclusion follows only after the displayed source-functional, decoupling, infrared, analyticity, crossing, boundedness, and positivity assumptions have been supplied. The monograph does not treat it as a rigorous proof of a universal a-theorem, nor as a substitute for constructing the RG trajectory or proving that its endpoints satisfy the CFT hypotheses.[2]

56.5 Conformal Field Theory as Local Fixed-Point Data

The local data of a conformal field theory in flat D-dimensional space are organized as follows.

[2] For the two-dimensional construction see A. B. Zamolodchikov, "Irreversibility of the flux of the renormalization group in a 2D field theory," *JETP Lett.* **43** (1986) 730–732, and J. Polchinski, "Scale and conformal invariance in quantum field theory," *Nucl. Phys. B* **303** (1988) 226–236. For the four-dimensional dilaton argument see Z. Komargodski and A. Schwimmer, "On renormalization group flows in four dimensions," *JHEP* **12** (2011) 099, arXiv:1107.3987. For the three-dimensional sphere partition-function quantity see D. L. Jafferis, I. R. Klebanov, S. S. Pufu, and B. R. Safdi, "Towards the F-theorem," arXiv:1103.1181.

1. An algebraic vector space $\mathcal{V}_{\rm loc}$ of renormalized local operators, graded by scaling dimension after diagonalizing the dilatation action. Its elements are denoted $\mathcal{O}_A(x)$, with A labeling a basis or a collection of multiplets. In a unitary CFT this algebraic space maps, by radial quantization, to a dense subspace of the Hilbert space on S^{D-1}; the Hilbert completion contains general sphere states and is not itself the space of local operators.
2. A stress tensor $T_{\mu\nu}$ satisfying conservation, symmetry, and the flat-space tracelessness condition
$$\partial_\mu T^{\mu\nu} = 0, \qquad T^{\mu\nu} = T^{\nu\mu}, \qquad T^\mu{}_\mu = 0.$$
3. An action of the conformal algebra on $\mathcal{V}_{\rm loc}$, generated by the stress-tensor charges associated with conformal Killing fields.
4. Euclidean separated-point correlation functions
$$\langle \mathcal{O}_{A_1}(x_1) \cdots \mathcal{O}_{A_n}(x_n) \rangle, \qquad x_i \in \mathbb{R}^D,$$
interpreted as distributions on the configuration space $x_i \neq x_j$, together with contact-term extensions specified by the source Ward identities.

For unitary Lorentzian theories these Euclidean distributions are related to Wightman distributions by analytic continuation and reflection positivity, under the same reconstruction hypotheses used for ordinary Euclidean Green functions.

56.6 Source Functionals and Ward-Identity Contact Terms

The separated-point correlators in the preceding list are restrictions of distributions obtained from a source functional. Work in Euclidean signature for this construction. The fixed-point functional used here is the renormalized metric-and-operator source limit constructed in Section 35.2, specialized to a point of the RG flow where the beta functions vanish and to a stress-tensor improvement convention appropriate to the fixed point. Thus $g_{\mu\nu}$ is a smooth background metric, $J^A(x)$ are smooth compactly supported sources for a chosen local operator basis $\mathcal{O}_A$, and local functionals of g and J specify the contact-term chart. The sign normalization of W_* is the Euclidean free-energy, or effective-action, normalization $W_* = -\log Z_*$. The connected-log functional is denoted $K_* = \log Z_*$ when it is used.

Definition 56.5 (Status of fixed-point source brackets)**.** The fixed-point partition functional
$$Z_*[g, J] := \exp\{-W_*[g, J]\}$$
is source-functional data of the fixed point. The symbol $\langle \cdots \rangle_{g,*}$ used below denotes evaluation by this curved-background source functional. A literal Borel measure or an unregulated continuum path integral would require an additional construction. If a presentation of the CFT supplies only flat separated-point correlators, then a compatible curved-background functional $Z_*[g, J]$, including its contact-term and Weyl-anomaly extension, is an additional datum or theorem to be stated before the metric derivatives below are used. The bracket notation
$$\left\langle \exp\left(\int \mathrm{d}^D x \sqrt{g}\, J^A(x) \mathcal{O}_A(x)\right)\right\rangle_{g,*} := Z_*[g, J]$$
is a shorthand for that data. Its mathematical construction depends on the framework in which the fixed point has been supplied:

(i) in a constructive Euclidean model, $Z_*[g, J]$ is obtained from the constructed measure or generalized random field after the local source counterterms have been fixed;

(ii) in a lattice or Wilsonian construction, $Z_*[g, J]$ is the continuum limit of regulated source functionals in a specified topology of distributions, after tuning and renormalizing the source-coordinate system;

(iii) in an abstract CFT presentation, $Z_*[g, J]$ is a generating functional for a compatible family of Euclidean Schwinger distributions and contact-term extensions satisfying the stated Ward identities, positivity or reflection-positivity hypotheses when unitarity is required, and the reconstruction assumptions used in this volume;

(iv) in perturbative fixed-point calculations, $Z_*[g, J]$ is a formal power series whose coefficients are renormalized source distributions.

Thus the brackets in this section are typed by the chosen fixed-point construction. They are expectation brackets when a measure construction has been provided, continuum-limit brackets when a regulated limit has been proved or assumed as part of the data, and distributional generating brackets in an axiomatic or perturbative presentation.

Hypothesis 56.6 (Existence of the fixed-point source-coordinate system)**.** All differentiations of $W_*[g, J]$ below are made in a source-coordinate system with the following properties. There is an open neighborhood $\mathcal{U}_*$ of $(g, J) = (\delta, 0)$ in the Frechet space of smooth Euclidean metrics equal to $\delta_{\mu\nu}$ outside a compact set and smooth compactly supported operator sources. The functional W_* is a real-valued functional on $\mathcal{U}_*$, defined modulo local functionals of g and J, whose Gateaux derivatives of all orders exist as distributions. More explicitly, for every $(g, J) \in \mathcal{U}_*$ and every finite list of compactly supported metric and source variations

$$u_r = (h^{(r)}_{\mu\nu}, j^{(r)A}), \qquad r = 1, \ldots, n,$$

the multilinear functional

$$D_{u_1} \cdots D_{u_n} W_*[g, J]$$

is continuous in the test-function topology and is represented by a distribution on the corresponding Cartesian power of spacetime.

If the fixed point is supplied by a regulator, this hypothesis means that there are regulated functionals $Z_\Lambda[g, J]$ and local source counterterms $P_\Lambda[g, J]$ such that

$$W^{\mathrm{ren}}_\Lambda[g, J] = -\log Z_\Lambda[g, J] + P_\Lambda[g, J]$$

has distributional limits of all such source derivatives on $\mathcal{U}_*$:

$$D_{u_1} \cdots D_{u_n} W^{\mathrm{ren}}_\Lambda[g, J] \longrightarrow D_{u_1} \cdots D_{u_n} W_*[g, J] \quad \text{in } \mathcal{D}'.$$

Two regulators or subtraction prescriptions define the same source-coordinate system only after a transition function has been specified; their difference is allowed to be a local functional of the background fields, which changes only contact terms in source derivatives. Diffeomorphism and Weyl Ward identities for W_* are additional properties of this chart, including the anomaly functional when one is present. They are not consequences of flat separated correlators alone.

The definitions and Ward identities in the rest of this section are conditional on Hypothesis 56.6. In perturbative constructions the hypothesis is a statement about formal power series whose coefficients are renormalized source distributions. In a constructive, lattice, or Wilsonian construction it is a nonperturbative source-dependent scaling-limit theorem when proved; otherwise it is part of the CFT data being assumed.

For a regulated path-integral representative this convention reads

$$\exp\{-W_*[g,J]\} = \int \mathcal{D}_g\Phi\, \exp\left\{-S_*[\Phi;g] + \int \mathrm{d}^D x\, \sqrt{g}\, J^A(x)\mathcal{O}_A(x)\right\},$$

up to the local counterterms that define the source-coordinate system. The minus sign in the operator derivative below is forced by combining $W_* = -\log Z_*$ with the source term $+\int J^A\mathcal{O}_A$ in the exponent; a source written with the opposite sign would reverse this derivative convention. In an abstract CFT the displayed integral is replaced by the corresponding renormalized source functional; the first derivatives below are the definition of the associated contact-term convention. This paragraph uses the Euclidean plus-source coordinate. Relative to the positive-measure convention $Z_E[J_E] = \int \exp\{-S_E - \langle J_E, \mathcal{O}\rangle\}$ of Chapter 30, the local operator source here is $J^A = -J_E^A$. Relative to the Lorentzian convention $\exp\{\mathrm{i}\langle J_L, \mathcal{O}\rangle_L\}$ and the lower-half-plane Wick rotation of Chapter 16, it is $J^A = J_L^{\mathrm{W},A}$, with the tensor/spin index Wick factors inserted when the operator is not a scalar. The source-coordinate sign is therefore fixed before any Ward identity is differentiated. The stress tensor and local operators are the source derivatives

$$\langle T^{\mu\nu}(x)\rangle_{g,J} = -\frac{2}{\sqrt{g}(x)}\frac{\delta W_*[g,J]}{\delta g_{\mu\nu}(x)}, \qquad \langle \mathcal{O}_A(x)\rangle_{g,J} = -\frac{1}{\sqrt{g}(x)}\frac{\delta W_*[g,J]}{\delta J^A(x)}. \tag{56.26}$$

The stress-tensor derivative is written with respect to $g_{\mu\nu}$. The same convention may be written using $g^{\mu\nu}$ by the chain-rule identity (58.2). The sign in the first formula is the one induced by the Euclidean effective action convention. With

$$\delta_g S_*[\Phi;g] = -\frac{1}{2}\int \mathrm{d}^D x\, \sqrt{g}\, T^{\mu\nu}(x)\delta g_{\mu\nu}(x),$$

one has $\delta_g W_* = \langle \delta_g S_*\rangle$, hence

$$\delta_g W_* = -\frac{1}{2}\int \mathrm{d}^D x\, \sqrt{g}\, \langle T^{\mu\nu}(x)\rangle_{g,J}\, \delta g_{\mu\nu}(x).$$

The operator-source derivative has the same overall minus sign for a different reason: $J^A\mathcal{O}_A$ appears with a plus sign in the exponent of Z_*, while $W_* = -\log Z_*$. Higher derivatives define Euclidean Schwinger distributions. The source functional is defined up to local functionals of the background sources. This is the continuum CFT version of the contact-term coordinate freedom described in Section 34.5: the restriction to the configuration space $x_i \neq x_j$ is fixed by the linear operator normalization, while the extension across collision diagonals is fixed by the chosen local source-coordinate system. The trace identity used in this system is the contact-distribution identity of Proposition 35.3; the Weyl-anomaly functional and its Wess–Zumino consistency condition are part of the same source data, not consequences of separated-point correlators alone.

The Ward identities are identities of these source-dependent distributions. For an infinitesimal diffeomorphism generated by a compactly supported vector field ξ^μ,

$$\delta_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu, \qquad \delta_\xi J^A = \xi^\mu\nabla_\mu J^A$$

for scalar sources. Invariance of $W_*[g, J]$ gives

$$\nabla_\mu \langle T^{\mu\nu}(x)\rangle_{g,J} = \langle \mathcal{O}_A(x)\rangle_{g,J} \nabla^\nu J^A(x), \tag{56.27}$$

with extra representation terms when the source carries spin indices. Differentiating this identity with respect to J and then setting $J = 0$ produces the contact terms in stress-tensor Ward identities. Equivalently, for scalar insertions

$$X = \mathcal{O}_{A_1}(x_1) \cdots \mathcal{O}_{A_n}(x_n),$$

the distributional translation Ward identity can be written in smeared form as

$$\int \mathrm{d}^D x\, \partial_\mu \xi_\nu(x)\, \langle T^{\mu\nu}(x) X\rangle = \sum_{i=1}^{n} \langle \mathcal{O}_{A_1}(x_1) \cdots (\xi^\rho \partial_\rho \mathcal{O}_{A_i})(x_i) \cdots \mathcal{O}_{A_n}(x_n)\rangle\,. \tag{56.28}$$

This equation is a compact way of recording the delta-function terms supported at $x = x_i$.

If the fixed point has no flat-space Weyl anomaly, the source-coordinate system for scalar primary operators of dimensions Δ_A may be chosen so that an infinitesimal Weyl transformation satisfies

$$\delta_\sigma g_{\mu\nu} = 2\sigma g_{\mu\nu}, \qquad \delta_\sigma J^A = (\Delta_A - D)\sigma J^A.$$

Differentiating the corresponding source Ward identity gives, for scalar primaries and in flat space,

$$\int \mathrm{d}^D x\, \sigma(x)\, \langle T^\mu{}_\mu(x) X\rangle = -\sum_{i=1}^{n} \Delta_{A_i} \sigma(x_i)\, \langle X\rangle + \mathcal{A}_\sigma[X]. \tag{56.29}$$

The term $\mathcal{A}_\sigma[X]$ is a local functional of the sources and background metric. It vanishes for flat separated-point correlators in a Weyl-invariant source scheme, but it must be specified when correlators are extended to coincident insertions or when the theory is placed on a curved background.

A scalar primary operator $\mathcal{O}$ has a scaling dimension $\Delta_{\mathcal{O}}$. For scalar primaries $\mathcal{O}_i$, separated-point scale covariance is

$$\langle \mathcal{O}_1(\lambda x_1) \cdots \mathcal{O}_n(\lambda x_n)\rangle = \lambda^{-\sum_i \Delta_i} \langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n)\rangle\,, \qquad \lambda > 0. \tag{56.30}$$

Operators with spin carry the corresponding finite-dimensional rotation or Lorentz matrices. The subsequent chapters derive these transformation laws from stress-tensor charges.

56.7 Ising Universality as a Fixed-Point Example

The statistical Ising model is a Euclidean lattice model whose microscopic state sum has a finite-dimensional diagonal trace representation. Let $\Lambda \subset \mathbb{Z}^D$ be a finite lattice. At every site $x \in \Lambda$ there is a spin variable $s_x = \pm 1$, represented in diagonal operator notation by

$$V_x = \mathrm{span}\{|\uparrow\rangle_x, |\downarrow\rangle_x\}, \qquad \widetilde{\mathcal{H}}_\Lambda = \bigotimes_{x\in\Lambda} V_x,$$

and by commuting diagonal operators $\widehat{S}_x$ with eigenvalues ± 1. With nearest-neighbor coupling normalized to one,

$$\widetilde{H} = -\sum_{\langle xx'\rangle} \widehat{S}_x \widehat{S}_{x'}.$$

The thermal expectation of an observable $\widehat{O}$ is

$$\langle \widehat{O} \rangle_{\beta_{\mathrm{T}}} = \frac{1}{Z}\,\mathrm{Tr}_{\widetilde{\mathcal{H}}_\Lambda}\left(\mathrm{e}^{-\beta_{\mathrm{T}}\widetilde{H}}\widehat{O}\right), \qquad Z = \mathrm{Tr}_{\widetilde{\mathcal{H}}_\Lambda}\,\mathrm{e}^{-\beta_{\mathrm{T}}\widetilde{H}}, \qquad \beta_{\mathrm{T}} = \frac{1}{T}. \tag{56.31}$$

The tilde distinguishes the finite-dimensional trace space of the statistical ensemble from the Hilbert space reconstructed from a continuum scaling limit.

Figure 56.1 keeps track of the finite Ising source construction used here: lattice spins, source insertions, and the thermal trace belong to the regulated statistical ensemble before the continuum CFT limit is taken.

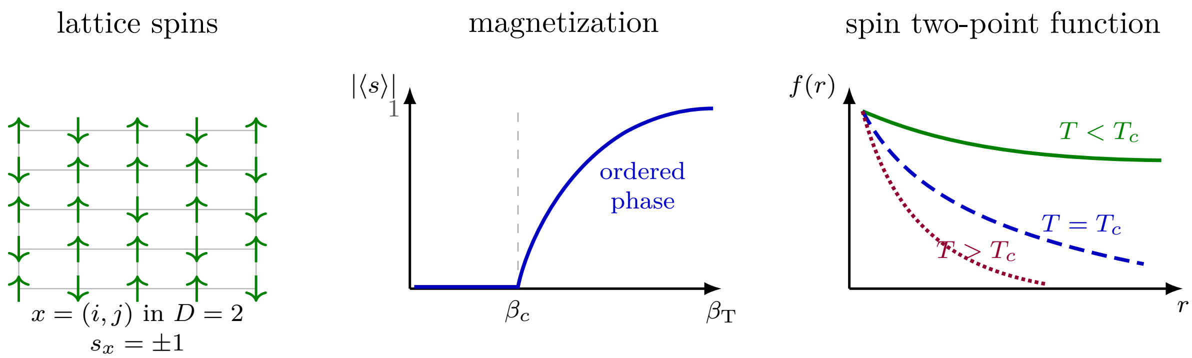


Figure 56.1: The Ising fixed point is reached by tuning the temperature. The finite lattice trace space defines the statistical ensemble, spontaneous magnetization appears for $T < T_c$, and the critical two-point function has power-law decay. In the two-point plot the ordered, critical, and disordered curves are solid, dashed, and dotted respectively.

In $D = 2$, write $S_{N,M}$ for the spin observable at $x = (N, M)$ and set

$$f_{N,M} = \langle S_{0,0} S_{N,M} \rangle_{\beta_{\mathrm{T}}} \simeq f(r), \qquad r = (N^2 + M^2)^{1/2},$$

for $r \gg 1$ and T close to T_c. The long-distance regimes are

$$f(r) \longrightarrow |\langle s \rangle|^2 > 0 \qquad (T < T_c),$$

$$f(r) \sim \mathrm{e}^{-r/\xi(T)} \qquad (T > T_c), \qquad \xi(T) \sim (T - T_c)^{-\nu},$$

and

$$f(r) \sim r^{-2\Delta_\sigma} \qquad (T = T_c). \tag{56.32}$$

The first numerical values to keep in view are

D	Δ_σ	ν
2	$\frac{1}{8}$	1
3	$0.5181488\ldots$	$0.629971\ldots$

where Δ_σ is the scaling dimension of the continuum spin field and ν is the correlation-length exponent.

Remark 56.7 (Numerical provenance in three dimensions). The $D = 3$ entries come from numerical conformal-bootstrap computations. They are obtained by applying crossing symmetry and reflection-positivity constraints to mixed four-point functions of σ, ε, and $T_{\mu\nu}$, followed by finite-dimensional semidefinite relaxations of the resulting bootstrap inequalities and numerical stability tests as the relaxation order changes. In such a computation the assumptions are the existence of a unitary three-dimensional CFT with the $\mathbb{Z}_2$ quantum numbers and operator gaps imposed in the search; the output is a small allowed region for low-lying CFT data. A constructive proof of the lattice scaling limit is a separate theorem. Chapter 64 develops the OPE, reflection-positivity, and crossing equations used by this method, but the present chapter records only the numerical CFT data and their provenance. A recent mixed-correlator computation including the stress tensor gives $\Delta_\sigma = 0.518148806(24)$ and $\Delta_\varepsilon = 1.41262528(29)$ for the leading $\mathbb{Z}_2$-odd and $\mathbb{Z}_2$-even scalar primaries of the critical Ising CFT; using $\Delta_\varepsilon = D - 1/\nu$ at $D = 3$ gives $\nu = 0.629970975\ldots$. The source is Chang, Dommes, Erramilli, Homrich, Kravchuk, Liu, Mitchell, Poland, and Simmons-Duffin, *Bootstrapping the 3d Ising Stress Tensor*, JHEP **03** (2025) 136, arXiv:2411.15300.

Definition 56.8 (Universal scaling functions in the Ising setting)**.** Fix a class $\mathfrak{R}$ of reflection-positive microscopic regulators with $\mathbb{Z}_2$ symmetry, one spinlike relevant direction, and one temperaturelike relevant direction. For each regulator $R \in \mathfrak{R}$, let a_R be the lattice spacing, let p_R denote microscopic couplings near the critical surface, and let $X_A^{(R)}(x)$ be lattice observables whose leading scaling fields are O_A. A separated n-point scaling function is a distributional limit on the configuration space $x_i \neq x_j$ of the form

$$\mathcal{F}_{A_1\ldots A_n}(x_1,\ldots,x_n;s) = \lim_{a_R\to 0}\prod_{i=1}^{n} Z_{A_i}^{(R)} \left\langle X_{A_1}^{(R)}(a_R^{-1}x_1)\cdots X_{A_n}^{(R)}(a_R^{-1}x_n)\right\rangle_{p_R(s)}^{\mathrm{conn}},$$

where $s = (s_t, s_h)$ are fixed continuum scaling coordinates for the thermal and magnetic relevant deformations, and $Z_A^{(R)}$ are multiplicative normalizations of the scaling fields. The scaling function is called universal for $\mathfrak{R}$ when any two regulators in $\mathfrak{R}$ give the same $\mathcal{F}$ after analytic changes of the relevant coordinates $(p_R - p_{R,c}) \mapsto (s_t, s_h)$, multiplicative field normalizations, and subtraction of analytic background terms in the free energy. This definition concerns separated-point kernels and their finite-size or thermodynamic limits. Critical temperatures, lattice metric factors, operator normalizations, analytic free-energy backgrounds, and contact-term extensions are regulator data rather than universal scaling functions.

Conditional consequence of complete CFT data. Assume that a regulator class $\mathfrak{R}$ has a continuum critical limit which is a unitary Euclidean CFT $\mathcal{C}_{\mathrm{Ising}}$, that the lattice spin and energy observables converge to the primary fields σ and ε after the normalizations in Definition 56.8, and that the OPE of $\mathcal{C}_{\mathrm{Ising}}$ converges as a distribution on every separated Euclidean configuration. Then the critical separated-point scaling functions are determined by the complete CFT data of $\mathcal{C}_{\mathrm{Ising}}$: the primary spectrum with spin and $\mathbb{Z}_2$ parity, two-point normalizations, and OPE coefficients.

If, in addition, the relevant deformation

$$S_{\mathcal{C}} \longmapsto S_{\mathcal{C}} + s_t \int \mathrm{d}^D x\, \varepsilon(x) + s_h \int \mathrm{d}^D x\, \sigma(x)$$

is constructed as a continuum source-dependent QFT with a specified renormalized contact-term chart, then the off-critical universal scaling functions in the variables s_t, s_h are the separated connected correlators of this deformed continuum theory. Their short-distance expansion near the critical point is obtained by conformal perturbation theory from the complete CFT data and the chosen deformation scheme. Their global massive scaling functions require the deformed continuum theory itself, because finitely many critical exponents such as Δ_σ and ν do not determine all massive correlation functions.

This paragraph records a conditional consequence of the stated convergence package. The convergence hypothesis is the substantive input: it identifies the critical scaling limit of the normalized lattice observables with the separated-point correlators of $\mathcal{C}_{\text{Ising}}$. The complete CFT data determine those correlators as follows. In a ball of configuration space where x_1 approaches x_2 with all other insertions separated, OPE convergence gives

$$O_{A_1}(x_1)O_{A_2}(x_2) = \sum_B C_{A_1A_2}{}^B(x_{12}, \partial_{x_2})\, O_B(x_2)$$

as an equality of distributions after pairing with test functions supported in that ball. The coefficient distributions $C_{A_1A_2}{}^B$ are fixed by the dimensions, spins, representation labels, two-point normalizations, and OPE coefficients. Iterating the OPE reduces an n-point function on such a configuration-space chart to a convergent sum of two- and three-point tensor structures with the same data. Compatibility of the convergent OPE expansions on overlapping charts is part of the CFT data through associativity/crossing. Hence the critical separated-point kernels are fixed by the complete CFT data.

For the deformed statement, the assumed continuum source-dependent QFT supplies a generating functional $W_{\mathcal{C}}[s_t, s_h; J]$. Its functional derivatives with respect to the sources J^A, restricted to separated points, are precisely the off-critical scaling functions in the chosen normalization and scaling-coordinate chart. Expanding $\exp[-s_t \int \varepsilon - s_h \int \sigma]$ near $s_t = s_h = 0$ expresses the short-distance coefficients as integrated CFT correlators. The specified renormalized contact-term chart is the datum that defines those integrals at coincident integration points. The full finite-s functions are therefore correlators of the relevantly deformed continuum theory.

Remark 56.9 (Status of the three-dimensional Ising scaling functions). For $D = 2$, the conformal data are exact data of the Ising minimal model, and many scaling limits of the planar lattice model are theorem-level objects. For $D = 3$, the statement that the nearest-neighbor Ising model and the scalar $\mathbb{Z}_2$-invariant ϕ^4 regulator class converge to a single unitary Ising CFT with the scaling functions of Definition 56.8 is a mathematically open continuum-limit statement. The numerical values displayed above are bootstrap estimates for the candidate CFT data. The parenthetical uncertainties attached to $\Delta_\sigma = 0.518148806(24)$ and $\Delta_\varepsilon = 1.41262528(29)$ are numerical uncertainties reported by the finite semidefinite bootstrap computation; they are neither convergence rates for the lattice spacing $a_R \to 0$ nor rigorous error bounds for a constructive lattice-to-CFT theorem. If a leading irrelevant exponent $\omega > 0$ and a regulator-dependent coefficient are supplied by a separate RG analysis, one may write corrections of the form $O(a_R^\omega)$ for specified observables, but such a rate is additional data beyond the two numbers in the table.

Remark 56.10 (Existence status in $D \geq 4$). The two-row table above records Ising fixed-point data in $D = 2, 3$. For $D \geq 4$, the existence question separates into the following cases. A free conformal field theory means a local conformal theory generated by fields obeying linear equations of motion whose vacuum correlators satisfy Wick factorization; its local operator space can be large, but its separated connected correlators are fixed by two-point data. A dynamically interacting CFT means a stress-tensor CFT whose separated correlators are not generated by Wick factorization from free conformal sectors and decoupled topological sectors.

There are explicit unitary Gaussian CFTs in $D \geq 4$: the improved massless scalar and massless Dirac field in every allowed integer dimension, and the Maxwell field in $D = 4$. These theories supply exact local conformal data, but they do not supply interacting fixed-point data. The Wilson–Fisher ϵ-expansion near $D = 4$ supplies formal perturbative conformal data at $D = 4 - \epsilon$. For nonintegral D, the usual Hilbert-space positivity condition of an integer-dimensional Lorentzian QFT is not literally available, and the analytic continuation of conformal representation theory contains negative-norm states in generic nonintegral dimension. The ϵ-expansion is therefore a powerful asymptotic computation of local CFT data, not by itself a construction of a unitary CFT at a fixed integer dimension.

For the positive-measure scalar route, four dimensions are also constrained by triviality theorems for critical Ising and ϕ^4_4-type scaling limits: within those hypotheses the continuum limits are Gaussian. This result blocks the simplest scalar construction of an interacting $D = 4$ CFT; it does not classify all possible four-dimensional fixed points. Gauge-theory mechanisms such as Banks–Zaks fixed points and later supersymmetric conformal gauge theories provide physically controlled fixed-point constructions under their own assumptions. In this core CFT volume, any such $D \geq 4$ interacting example is used only after its stress tensor, unitarity/positivity, locality, operator spectrum, and conformal Ward identities have been specified as part of the data. [3]

The same fixed-point basin can be represented by a scalar Euclidean lattice field theory, in the sense of Definition 56.8. Replace the two-valued spin by a real variable ϕ_x, use a $\mathbb{Z}_2$-even double-well single-site potential, and retain the finite Brillouin-zone cutoff. At long distance its action has the form

$$S_{\rm lat}[\phi] = \int \mathrm{d}^D x \left[\frac{1}{2} Z_\phi (\partial_\mu \phi)^2 + V(\phi) + \sum_A c_A \, \mathcal{O}_A^{\rm aniso}(\phi, \partial\phi) \right], \tag{56.33}$$

where the last sum consists of higher-derivative operators allowed by the hypercubic lattice symmetry. These terms display the finite cutoff and the breaking of $SO(D)$ rotations at the microscopic scale.

At $T = T_c$, equivalently at the tuned massless critical point, the correlation length is infinite in lattice units. The separated long-distance correlators are described by the Ising conformal field theory. For $D = 2, 3$, this is the infrared fixed point of the $\mathbb{Z}_2$-invariant massless scalar ϕ^4 fixed-point basin. The continuum fields obey the dictionary

$$\phi \longmapsto \sigma, \qquad [\phi^2] \longmapsto \varepsilon,$$

where σ is the spin field and ε is the thermal energy-density field after identity mixing has been subtracted. The thermal deformation

$$\Delta S_t = \int \mathrm{d}^D x \, u_t \, \varepsilon(x)$$

[3] For nonintegral-dimensional unitarity issues in the Wilson–Fisher expansion, see M. Hogervorst, S. Rychkov, and B. C. van Rees, "Unitarity violation at the Wilson-Fisher fixed point in $4 - \epsilon$ dimensions," arXiv:1512.00013. For four-dimensional scalar triviality, see M. Aizenman and H. Duminil-Copin, "Marginal triviality of the scaling limits of critical 4D Ising and ϕ^4_4 models," *Ann. of Math.* **194** (2021) 163–235. For the perturbative gauge-theory fixed-point mechanism, see T. Banks and A. Zaks, "On the phase structure of vector-like gauge theories with massless fermions," *Nucl. Phys. B* **196** (1982) 189–204.

has RG eigenvalue $y_t = D - \Delta_\varepsilon$. Since $\xi^{-1} \propto |u_t|^{1/y_t}$, comparison with $\xi \propto |T - T_c|^{-\nu}$ gives

$$\Delta_\varepsilon = D - \frac{1}{\nu}. \tag{56.34}$$

Figure 56.2 records the scalar cutoff presentation used to compare the Ising scaling direction with a local continuum field coordinate.

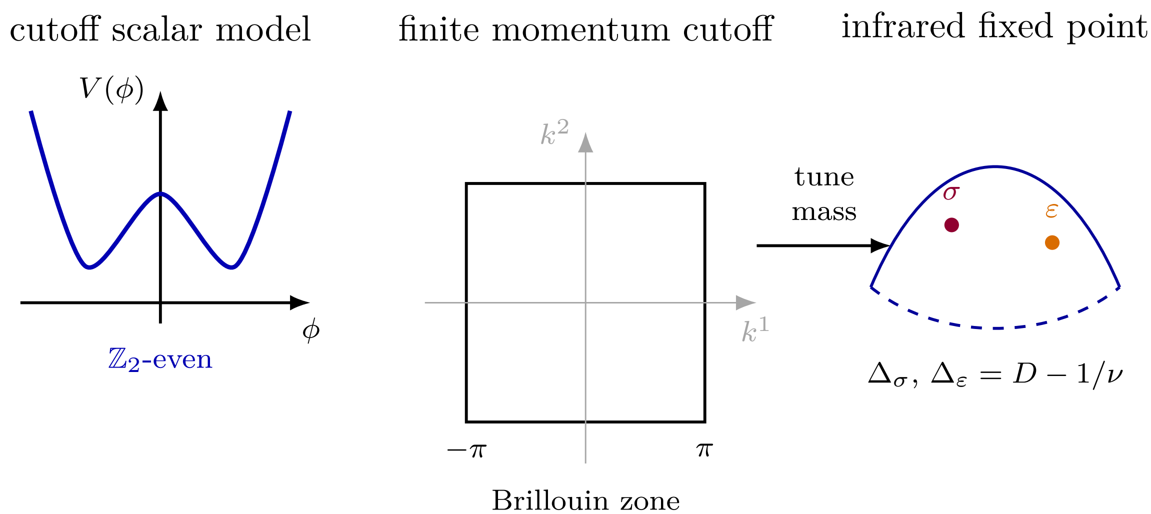


Figure 56.2: The scalar cutoff representation of the Ising fixed-point basin has a finite Brillouin-zone cutoff and lattice anisotropy. At the tuned critical point its long-distance operator data are those of the Ising conformal field theory.

Chapter 57

Conformal Killing Vectors and the Conformal Group

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- **$\hbar$.** 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

57.1 The Conformal Killing Equation

Work first in flat Euclidean space $(\mathbb{R}^D, \delta_{\mu\nu})$ with $D \geq 3$. Cartesian coordinates are denoted by x^μ, and repeated indices are contracted with $\delta_{\mu\nu}$. A smooth vector field $\epsilon^\mu(x)$ generates an infinitesimal coordinate transformation

$$x^\mu \longmapsto x^\mu + \epsilon^\mu(x).$$

Definition 57.1 (Euclidean conformal Killing vector)**.** A smooth vector field $\epsilon = \epsilon^\mu \partial_\mu$ on an open subset of $\mathbb{R}^D$ is a conformal Killing vector when the change of the metric is pointwise proportional to the metric:

$$\mathcal{L}_\epsilon \delta_{\mu\nu} = \partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = 2\sigma_\epsilon(x)\delta_{\mu\nu}$$

for some scalar function σ_ϵ. Taking the trace gives

$$\sigma_\epsilon(x) = \frac{1}{D}\partial_\rho \epsilon^\rho(x).$$

Thus the conformal Killing equation is

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu - \frac{2}{D}\delta_{\mu\nu}\partial_\rho \epsilon^\rho = 0. \tag{57.1}$$

Conformal current from a traceless stress tensor. Let $T_{\mu\nu}$ be symmetric, conserved, and traceless, and let ϵ obey (57.1). Then

$$J^\mu_\epsilon(x) = T^\mu{}_\nu(x)\epsilon^\nu(x)$$

is conserved. The charge

$$Q_\epsilon(\Sigma) = \int_\Sigma \mathrm{d}\sigma\, n_\mu J^\mu_\epsilon$$

on a closed codimension-one surface Σ depends only on the homology class of Σ relative to operator insertions. The divergence is

$$\partial_\mu J^\mu_\epsilon = (\partial_\mu T^\mu{}_\nu)\epsilon^\nu + T^{\mu\nu}\partial_\mu\epsilon_\nu = \frac{1}{2}T^{\mu\nu}(\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu) = \sigma_\epsilon T^\mu{}_\mu = 0.$$

Conservation makes the charge invariant under deformations of Σ that do not cross operator insertions. After analytic continuation to Lorentzian signature this formula is read on a spacelike Cauchy surface Σ. The measure $\mathrm{d}\sigma$ is the induced spatial volume form, n^μ is the future-directed unit timelike normal, and

$$Q_\epsilon(\Sigma) = \int_\Sigma \mathrm{d}\sigma\, n_\mu T^\mu{}_\nu\epsilon^\nu. \tag{57.2}$$

If Σ_1 and Σ_2 bound a spacetime region V containing no operator insertions, Stokes' theorem gives

$$Q_\epsilon(\Sigma_2) - Q_\epsilon(\Sigma_1) = \int_V \mathrm{d}^D x\, \partial_\mu J^\mu_\epsilon = 0.$$

This is the precise sense in which the charge depends on the homology class of the surface relative to insertions.

Figure 57.1 displays the conserved-charge surface-deformation statement: two spacelike slices homologous relative to the insertions define the same charge.

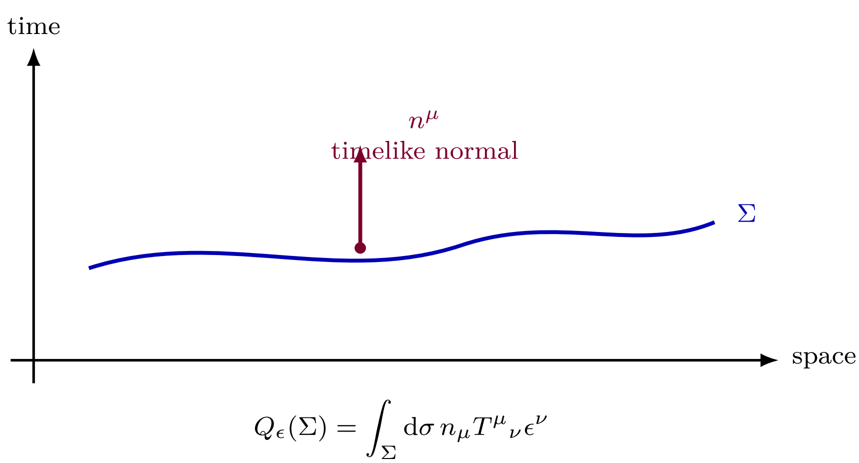


Figure 57.1: The Lorentzian conformal charge is the flux of the stress-tensor current through a spacelike slice. Current conservation and Stokes' theorem give independence under deformations of the slice through empty spacetime.

57.2 Solutions in Dimension at Least Three

Lemma 57.2 (Rigidity of the flat conformal factor)**.** *Let $D \geq 3$, and let ϵ solve the flat conformal Killing equation* (57.1) *on a connected open set. Then*

$$\partial_\mu \partial_\nu \sigma_\epsilon = 0.$$

Consequently σ_ϵ is affine-linear in x.

Proof. The conformal Killing equation is

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = 2\sigma_\epsilon \delta_{\mu\nu}.$$

Differentiate this identity and combine the three equations obtained by cyclically permuting (μ, ν, ρ). The result is

$$\partial_\mu \partial_\nu \epsilon_\rho = \delta_{\rho\nu} \partial_\mu \sigma_\epsilon + \delta_{\rho\mu} \partial_\nu \sigma_\epsilon - \delta_{\mu\nu} \partial_\rho \sigma_\epsilon. \tag{57.3}$$

Set

$$S_{\mu\nu} := \partial_\mu \partial_\nu \sigma_\epsilon.$$

Since third derivatives of ϵ_ρ commute, differentiating (57.3) with respect to x^κ and then interchanging κ and μ gives

$$\delta_{\rho\mu} S_{\kappa\nu} - \delta_{\mu\nu} S_{\kappa\rho} = \delta_{\rho\kappa} S_{\mu\nu} - \delta_{\kappa\nu} S_{\mu\rho}.$$

Contracting ρ with μ yields

$$(D-2) S_{\kappa\nu} = -\delta_{\kappa\nu} S_\lambda{}^\lambda.$$

Taking the trace gives

$$(2D-2) S_\lambda{}^\lambda = 0.$$

For $D \geq 3$, this implies $S_\lambda{}^\lambda = 0$, and then the preceding display gives $S_{\kappa\nu} = 0$. Thus all second derivatives of σ_ϵ vanish, so σ_ϵ is affine-linear on each connected component. □

Proposition 57.3 (Classification of flat conformal Killing vectors)**.** *For $D \geq 3$, every flat conformal Killing vector is at most quadratic in the Cartesian coordinates. More precisely, the general solution is*

$$\epsilon^\mu(x) = a^\mu + \omega^\mu{}_\nu x^\nu + \lambda x^\mu + c^\mu x^2 - 2(c \cdot x) x^\mu, \tag{57.4}$$

where a^μ, c^μ, and λ are constants and $\omega_{\mu\nu} = -\omega_{\nu\mu}$. These terms generate translations, rotations, dilatations, and special conformal transformations, respectively. Equivalently, the Lorentz and dilatation terms may be combined as $b^\mu{}_\nu x^\nu$, where $b_{\mu\nu}$ is either antisymmetric or proportional to $\delta_{\mu\nu}$.

Proof. By Lemma 57.2,

$$\sigma_\epsilon(x) = \lambda - 2c_\rho x^\rho$$

for constants λ and c_ρ, where the sign of c_ρ is chosen to match (57.4). Substituting this affine form in (57.3) gives

$$\partial_\mu \partial_\nu \epsilon_\rho = -2\delta_{\rho\nu} c_\mu - 2\delta_{\rho\mu} c_\nu + 2\delta_{\mu\nu} c_\rho, \tag{57.5}$$

The quadratic vector field

$$q_\rho(x) := c_\rho x^2 - 2(c \cdot x) x_\rho$$

has exactly the same Hessian,

$$\partial_\mu \partial_\nu q_\rho = -2\delta_{\rho\nu} c_\mu - 2\delta_{\rho\mu} c_\nu + 2\delta_{\mu\nu} c_\rho.$$

Hence $h_\rho := \epsilon_\rho - q_\rho$ has vanishing second derivatives. On a connected coordinate domain this means that each first derivative $\partial_\nu h_\rho$ is a constant $B_{\rho\nu}$, and integrating once more gives

$$\epsilon_\rho(x) = a_\rho + B_{\rho\nu} x^\nu + c_\rho x^2 - 2(c \cdot x) x_\rho. \tag{57.6}$$

This is the only integration step in the classification: the constant Hessian (57.5) leaves only the affine constants a_ρ and $B_{\rho\nu}$ undetermined.

It remains to impose the original first-order equation. The special conformal quadratic part satisfies

$$\partial_\mu q_\nu + \partial_\nu q_\mu = -4(c \cdot x)\delta_{\mu\nu}.$$

The constant term a_ρ contributes nothing. Therefore the symmetric part of $B_{\rho\nu}$ is fixed by the constant part of the conformal Killing equation:

$$B_{\rho\nu} + B_{\nu\rho} = 2\lambda \delta_{\rho\nu}.$$

Writing $B_{\rho\nu} = \omega_{\rho\nu} + \lambda\delta_{\rho\nu}$ with $\omega_{\rho\nu} = -\omega_{\nu\rho}$ gives (57.4).

Conversely, translations and rotations have $\sigma_\epsilon = 0$, the dilatation term λx^μ has $\sigma_\epsilon = \lambda$, and the quadratic term q^μ has $\sigma_\epsilon = -2c \cdot x$. Their sum therefore solves (57.1) with $\sigma_\epsilon = \lambda - 2c \cdot x$. The integration above shows that there are no further smooth local solutions in dimension $D \geq 3$. □

Definition 57.4 (Differential conformal generators)**.** The associated differential operators are

$$P_\mu = \partial_\mu, \qquad M_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu, \qquad D = x^\mu \partial_\mu, \tag{57.7}$$

and

$$K_\mu = x^2 \partial_\mu - 2x_\mu x^\rho \partial_\rho. \tag{57.8}$$

The signs in (57.8) are chosen so that the finite special conformal transformation below has parameter c^μ with denominator $1 + 2c \cdot x + c^2 x^2$.

Figure 57.2 makes explicit the finite-dimensional basis of conformal Killing fields in $D \geq 3$, including the dimension count that fails in the local $D = 2$ conformal problem treated later.

family	parameter	dimension	vector field $\epsilon = \epsilon^\mu \partial_\mu$	charge basis
translations	$a^\mu \in \mathbb{R}^D$	D	$\epsilon^\mu = a^\mu$	P_μ
rotations	$\omega_{\mu\nu} = -\omega_{\nu\mu}$	$D(D-1)/2$	$\epsilon^\mu = \omega^\mu{}_\nu x^\nu$	$M_{\mu\nu}$
dilatations	$\lambda \in \mathbb{R}$	1	$\epsilon^\mu = \lambda x^\mu$	D
special conformal	$c^\mu \in \mathbb{R}^D$	D	$\epsilon^\mu = c^\mu x^2 - 2(c \cdot x)x^\mu$	K_μ
total for $D \geq 3$		$(D+1)(D+2)/2$	the finite-dimensional Lie algebra generated by $P_\mu, M_{\mu\nu}, D, K_\mu$, isomorphic to $\mathfrak{so}(D+1, 1)$ in Euclidean signature	

Figure 57.2: The rigidity theorem for $D \geq 3$ leaves exactly $D + D(D-1)/2 + 1 + D = (D+1)(D+2)/2$ independent conformal Killing fields. The stress tensor pairs each vector field with the corresponding conserved charge $Q_\epsilon = \int_\Sigma \mathrm{d}\sigma\, n_\mu T^\mu{}_\nu \epsilon^\nu$. In $D = 2$, local holomorphic and antiholomorphic solutions replace this finite list, which is why the two-dimensional case is separated from the present chapter.

57.3 Charges and Vector Fields

Convention 57.5 (Charge sign convention)**.** Let $U_\epsilon(s)$ denote the one-parameter unitary family generated by Q_ϵ:

$$U_\epsilon(s) = \exp(\mathrm{i}sQ_\epsilon).$$

The normalization of Q_ϵ is fixed by translations and Lorentz transformations, whose charges are already part of the Poincare representation of the QFT. With this normalization, infinitesimal coordinate transformations compose through the ordinary Lie bracket of vector fields,

$$[\epsilon_1, \epsilon_2]^\mu = \epsilon_1^\nu \partial_\nu \epsilon_2^\mu - \epsilon_2^\nu \partial_\nu \epsilon_1^\mu. \tag{57.9}$$

Charge algebra from the vector-field bracket. Assume the charges Q_ϵ are represented on a common invariant domain and that no Schwinger contact term contributes to the finite-dimensional global conformal algebra. Then

$$[Q_{\epsilon_1}, Q_{\epsilon_2}] = -\mathrm{i}Q_{[\epsilon_1,\epsilon_2]}. \tag{57.10}$$

On local operators inserted away from the integration surfaces, the commutator of the two infinitesimal coordinate transformations is the infinitesimal transformation generated by the vector-field bracket (57.9). The sign in the charge commutator is fixed by $U_\epsilon(s) = \exp(\mathrm{i}sQ_\epsilon)$: the group commutator at order $s_1 s_2$ is generated by $-[Q_{\epsilon_1}, Q_{\epsilon_2}]$, whereas the geometric commutator is generated by $Q_{[\epsilon_1,\epsilon_2]}$. A possible discrepancy is a central local contact term; by hypothesis it is absent in the global algebra. Equation (57.10) is an operator statement on a common invariant domain, or equivalently a Ward identity for correlation functions with separated insertions. Possible central extensions or Schwinger terms are local contact terms; in dimensions $D \geq 3$ the global finite-dimensional conformal algebra has no central extension.

57.4 Finite Transformations

Definition 57.6 (Finite Euclidean conformal map)**.** A finite conformal transformation is a diffeomorphism

$$f : \mathbb{R}^D \supset U \longrightarrow f(U) \subset \mathbb{R}^D$$

whose Jacobian satisfies

$$\frac{\partial f^\rho}{\partial x^\mu}\frac{\partial f^\sigma}{\partial x^\nu}\delta_{\rho\sigma} = \Omega_f(x)^2 \delta_{\mu\nu}, \qquad \Omega_f(x) > 0. \tag{57.11}$$

For a transformation close to the identity,

$$f^\mu(x) = x^\mu + \epsilon^\mu(x) + O(\epsilon^2),$$

condition (57.11) reduces at first order to the conformal Killing equation (57.1). If f and h satisfy (57.11), then their composition satisfies it with Weyl factor

$$\Omega_{h\circ f}(x) = \Omega_h(f(x))\Omega_f(x).$$

Thus finite conformal maps form a group wherever the composed maps are defined. Expanding (57.11) to first order gives

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = 2\,\delta\Omega(x)\,\delta_{\mu\nu}.$$

Taking the trace fixes $\delta\Omega = D^{-1}\partial_\rho \epsilon^\rho$, hence (57.1). For composition, the chain rule gives the product of the two Jacobians, so the Weyl factor of $h \circ f$ is $\Omega_h(f(x))\Omega_f(x)$.

Translations, rotations, and dilatations are immediate examples. A further basic example is inversion,

$$I(x)^\mu = \frac{x^\mu}{x^2}, \qquad x^2 = \delta_{\mu\nu} x^\mu x^\nu. \tag{57.12}$$

Its Jacobian is

$$\frac{\partial I^\mu}{\partial x^\nu} = \frac{1}{x^2}\left(\delta^\mu{}_\nu - 2\frac{x^\mu x_\nu}{x^2}\right),$$

and the matrix in parentheses is orthogonal. Therefore

$$\Omega_I(x) = \frac{1}{x^2}.$$

Inversion is conformal but lies in a disconnected component of the full Euclidean conformal group: it cannot be reached by exponentiating a smooth path of conformal Killing vectors from the identity. The Jacobian is $1/x^2$ times $\delta^\mu{}_\nu - 2x^\mu x_\nu/x^2$. The latter matrix is an orthogonal reflection, so the pullback metric is $x^{-4}\delta_{\mu\nu}$, with Weyl factor $\Omega_I = 1/x^2$. Its determinant is $-|x|^{-2D}$, because the reflection has determinant -1. Thus inversion is orientation reversing on the stereographic chart, while every path starting at the identity through smooth conformal diffeomorphisms remains orientation-preserving until a singularity is reached. Hence inversion is not in the identity component generated by smooth conformal Killing vector flows.

Figure 57.3 fixes the inversion geometry: the unit sphere is mapped to itself, radial direction is reversed, and the Weyl factor is $\Omega_I = 1/x^2$.

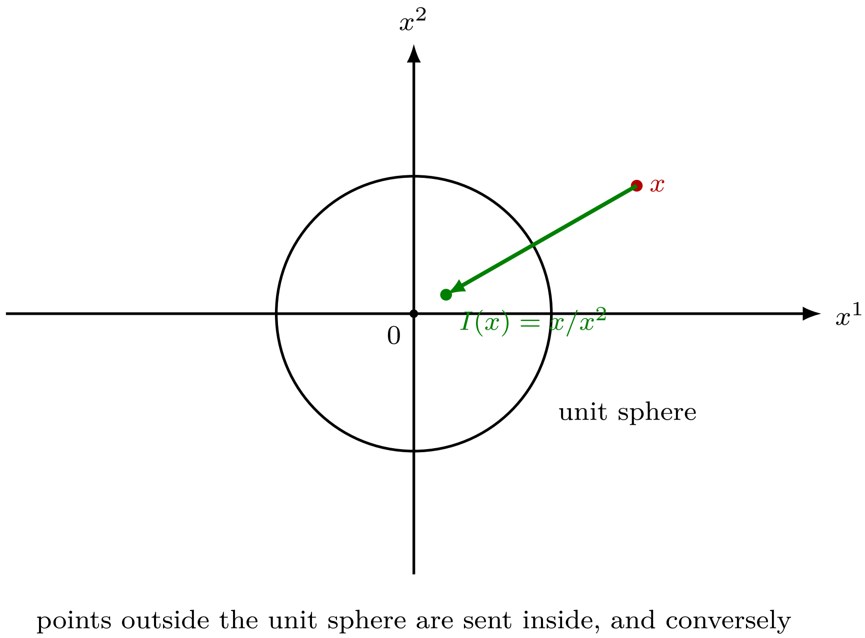


Figure 57.3: Euclidean inversion maps each ray through the origin to itself and sends the radial coordinate $r = |x|$ to $1/r$.

Special conformal transformations from inversion. The composition $I \circ T_c \circ I$, with $T_c(x) = x + c$, gives the special conformal transformation

$$x^\mu \longmapsto x'^\mu = \frac{x^\mu + c^\mu x^2}{1 + 2c \cdot x + c^2 x^2}. \tag{57.13}$$

Its infinitesimal vector field is the quadratic term in (57.4). Substituting $I(x) = x/x^2$, translating by c, and inverting once more gives

$$I(T_c(I(x)))^\mu = \frac{x^\mu + c^\mu x^2}{1 + 2c \cdot x + c^2 x^2}.$$

Expanding to first order in c gives $\delta x^\mu = c^\mu x^2 - 2(c \cdot x)x^\mu$.

The Poincare transformations, dilatations, and inversion generate all finite conformal transformations in $D > 2$, after restricting to domains on which the rational maps are defined.

57.5 Euclidean Compactification and the Radial Cylinder

Definition 57.7 (Euclidean conformal compactification)**.** The rational formulae above are local expressions for globally defined maps on the conformal compactification. The Euclidean compactification is the one-point compactification

$$\widehat{\mathbb{R}^D} = \mathbb{R}^D \cup \{\infty\} \simeq S^D.$$

In stereographic coordinates $x^\mu \in \mathbb{R}^D$, the unit sphere

$$S^D = \{N \in \mathbb{R}^{D+1} : N \cdot N = 1\}$$

is parametrized away from one pole by

$$N^\mu = \frac{2x^\mu}{1+x^2}, \qquad N^{D+1} = \frac{1-x^2}{1+x^2}, \qquad x^2 = \delta_{\mu\nu} x^\mu x^\nu. \tag{57.14}$$

The pullback of the round metric is

$$\mathrm{d}s^2_{S^D} = \frac{4\,\delta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu}{(1+x^2)^2}. \tag{57.15}$$

Thus the flat metric and the round metric are in the same conformal class on the stereographic chart. The point $x = \infty$ is the omitted pole.

Theorem 57.8 (Finite Liouville rigidity for $D \geq 3$)**.** *Let $D \geq 3$, let $U, V \subset \mathbb{R}^D$ be connected open sets, and let $f : U \to V$ be a smooth diffeomorphism satisfying*

$$f^*\delta = \Omega_f^2 \delta, \qquad \Omega_f > 0.$$

Then f is the restriction to U of a Mobius transformation. More explicitly, after writing $I_p(x) = p + (x-p)/|x-p|^2$, either

$$f(x) = b + sRx$$

or

$$f(x) = b + sRI_p(x),$$

where $b, p \in \mathbb{R}^D$, $s > 0$, and $R \in O(D)$. Consequently the smooth finite conformal transformations in dimension $D \geq 3$ are generated, on their domains of definition, by translations, orthogonal transformations, positive dilatations, and inversion.

Proof. Write $\Omega_f = e^\varphi$. Let

$$F_i{}^\alpha = \partial_i f^\alpha, \qquad g_{ij} = F_i{}^\alpha F_j{}^\alpha = e^{2\varphi}\delta_{ij}.$$

The map $f : (U, g) \to (V, \delta)$ is a local isometry. Since the Euclidean connection on V has zero Christoffel symbols in the Cartesian coordinates f^α, the coordinate functions satisfy

$$\partial_i \partial_j f^\alpha = \Gamma^k{}_{ij}(g)\,\partial_k f^\alpha, \qquad \Gamma^k{}_{ij}(g) = \delta^k{}_i \partial_j \varphi + \delta^k{}_j \partial_i \varphi - \delta_{ij}\partial^k \varphi. \tag{57.16}$$

This is just the coordinate expression for preservation of the Levi-Civita connection by a local isometry.

The pullback metric $g = f^*\delta$ is flat. Equivalently, commuting the mixed third derivatives in (57.16) gives zero curvature for the connection $\Gamma(g)$. Direct substitution of the displayed Christoffel symbols gives

$$R_{ijk\ell}(g) = e^{2\varphi}\left(\delta_{i\ell}A_{jk} + \delta_{jk}A_{i\ell} - \delta_{ik}A_{j\ell} - \delta_{j\ell}A_{ik}\right), \tag{57.17}$$

where

$$A_{ij} = \partial_i \partial_j \varphi - \partial_i \varphi\, \partial_j \varphi + \frac{1}{2}|\nabla \varphi|^2 \delta_{ij}. \tag{57.18}$$

Because $R_{ijk\ell}(g) = 0$, contraction of i with ℓ in (57.17) yields

$$(D-2)A_{jk} + \delta_{jk}A_i{}^i = 0.$$

Taking the trace gives $(2D-2)A_i{}^i = 0$. Since $D \geq 3$, it follows that $A_i{}^i = 0$, hence $A_{jk} = 0$. Thus

$$\partial_i\partial_j\varphi = \partial_i\varphi\,\partial_j\varphi - \frac{1}{2}|\nabla\varphi|^2\delta_{ij}. \tag{57.19}$$

Set $q = e^{-\varphi} = \Omega_f^{-1}$. Differentiating q twice and using (57.19) gives

$$\partial_i\partial_j q = \rho\,\delta_{ij}, \qquad \rho = \frac{1}{2}q|\nabla\varphi|^2. \tag{57.20}$$

The left side has symmetric third derivatives, so

$$\partial_k\rho\,\delta_{ij} = \partial_j\rho\,\delta_{ik}.$$

Contracting i and j gives $(D-1)\partial_k\rho = 0$, hence ρ is constant on the connected set U. Therefore

$$q(x) = \alpha|x|^2 + \beta\cdot x + \gamma \tag{57.21}$$

for constants $\alpha, \gamma \in \mathbb{R}$ and $\beta \in \mathbb{R}^D$. Substitution of (57.21) back into (57.20), now in the form

$$\partial_i\partial_j q = \frac{|\nabla q|^2}{2q}\delta_{ij},$$

gives

$$4\alpha\gamma = |\beta|^2. \tag{57.22}$$

There are two cases. If $\alpha = 0$, then (57.22) gives $\beta = 0$, so $q = \gamma > 0$ is constant. Equation (57.16) has zero right side, hence $\partial_i\partial_j f^\alpha = 0$. Thus $f(x) = b + Ax$, and the conformal condition gives $A^TA = \gamma^{-2}\mathbf{1}$. Hence $A = sR$, with $s = \gamma^{-1} > 0$ and $R \in O(D)$.

If $\alpha \neq 0$, completing the square in (57.21) and using (57.22) gives

$$q(x) = \alpha|x-p|^2, \qquad p = -\frac{\beta}{2\alpha}.$$

Since $q > 0$ on U, one has $\alpha > 0$ and $p \notin U$. The inversion $I_p(x) = p + (x-p)/|x-p|^2$ maps U diffeomorphically to a connected open set. For $h = f \circ I_p$, the Weyl factor is

$$\Omega_h(z) = \Omega_f(I_p(z))\,\Omega_{I_p}(z) = \frac{|z-p|^2}{\alpha}\frac{1}{|z-p|^2} = \alpha^{-1}.$$

The previous constant-factor case applies to h, so

$$h(z) = b + \alpha^{-1}Rz$$

for some $b \in \mathbb{R}^D$ and $R \in O(D)$. Composing again with the involution I_p gives $f(x) = b + \alpha^{-1}RI_p(x)$.

The displayed forms are precisely compositions of translations, orthogonal maps, positive dilatations, and inversion. They extend to smooth maps of the one-point compactification $S^D = \mathbb{R}^D \cup \{\infty\}$: the pole p of I_p is sent to ∞, and ∞ is sent to an ordinary point. Thus the apparent singular denominators in flat coordinates are chart singularities on the compactification. The orientation-preserving connected component is generated by translations, rotations, positive dilatations, and special conformal transformations $I \circ T_c \circ I$; in the Euclidean ambient null-cone model this is the projective action of $SO_0(D+1,1)$. □

The radial cylinder used in radial quantization is the same compactification seen in a different conformal frame. On $\mathbb{R}^D \setminus \{0\}$, write

$$x^\mu = r n^\mu, \qquad r > 0, \qquad n \in S^{D-1}, \qquad \tau = \log r.$$

Then

$$\delta_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu = \mathrm{e}^{2\tau} \left(\mathrm{d}\tau^2 + \mathrm{d}\Omega_{D-1}^2 \right). \tag{57.23}$$

After the Weyl rescaling by r^{-2}, punctured flat space becomes the Euclidean cylinder $\mathbb{R}_\tau \times S^{D-1}$. The origin and infinity are the two ends $\tau = -\infty$ and $\tau = +\infty$. Inversion acts by

$$r \mapsto r^{-1}, \qquad \tau \mapsto -\tau, \qquad n \mapsto n.$$

Consequently the cylinder appearing in radial quantization is a Weyl representative of $\widehat{\mathbb{R}^D} \setminus \{0, \infty\}$, and radial time translations are ordinary dilatations in the flat stereographic chart. Substituting $x^\mu = e^\tau n^\mu$, with $n \cdot n = 1$, into the flat metric gives (57.23). Multiplication by $r^{-2} = e^{-2\tau}$ removes the Weyl factor and leaves the product cylinder metric. The inversion $x \mapsto x/x^2$ sends r to r^{-1}, hence τ to $-\tau$, and the flat-space dilatation flow is precisely translation in τ.

57.6 The Conformal Algebra

Lemma 57.9 (Euclidean conformal charge algebra)**.** *Let the quantum charges associated with the vector fields* (57.7) *and* (57.8) *be denoted by*

$$\widehat{P}_\mu, \qquad \widehat{J}_{\mu\nu}, \qquad \widehat{D}, \qquad \widehat{K}_\mu.$$

Their commutators are the charge realization of the Lie bracket of conformal Killing vector fields. In the Euclidean convention of this chapter, indices are contracted with $\delta_{\mu\nu}$, and the algebra is

$$[\widehat{D}, \widehat{P}_\mu] = \mathrm{i}\widehat{P}_\mu, \qquad [\widehat{D}, \widehat{K}_\mu] = -\mathrm{i}\widehat{K}_\mu, \qquad [\widehat{D}, \widehat{J}_{\mu\nu}] = 0, \tag{57.24}$$

$$[\widehat{P}_\mu, \widehat{K}_\nu] = 2\mathrm{i}(\delta_{\mu\nu}\widehat{D} - \widehat{J}_{\mu\nu}), \tag{57.25}$$

together with the Poincare commutators

$$[\widehat{J}_{\mu\nu}, \widehat{P}_\rho] = \mathrm{i}(\delta_{\rho\mu}\widehat{P}_\nu - \delta_{\rho\nu}\widehat{P}_\mu), \tag{57.26}$$

$$[\widehat{J}_{\mu\nu}, \widehat{J}_{\rho\sigma}] = \mathrm{i}(\delta_{\mu\rho}\widehat{J}_{\nu\sigma} - \delta_{\nu\rho}\widehat{J}_{\mu\sigma} - \delta_{\mu\sigma}\widehat{J}_{\nu\rho} + \delta_{\nu\sigma}\widehat{J}_{\mu\rho}). \tag{57.27}$$

All unlisted commutators among the $\widehat{P}_\mu$'s and among the $\widehat{K}_\mu$'s vanish.

Proof. The vector fields in Definition 57.4 obey ordinary Lie brackets. Since $D = x^\rho \partial_\rho$, $P_\mu = \partial_\mu$, and

$$K_\nu = x^2 \partial_\nu - 2x_\nu x^\rho \partial_\rho,$$

one computes on coordinate functions, hence on all smooth functions, that

$$[D, P_\mu] = -P_\mu, \qquad [D, K_\mu] = K_\mu.$$

For the mixed bracket,

$$\begin{aligned} [P_\mu, K_\nu] &= \partial_\mu(x^2)\partial_\nu - 2\partial_\mu(x_\nu x^\rho)\partial_\rho \\ &= 2x_\mu \partial_\nu - 2\delta_{\mu\nu} x^\rho \partial_\rho - 2x_\nu \partial_\mu \\ &= 2M_{\mu\nu} - 2\delta_{\mu\nu} D. \end{aligned}$$

The rotation brackets follow from

$$M_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu$$

and the same coefficient differentiation:

$$[M_{\mu\nu}, P_\rho] = \delta_{\rho\nu} P_\mu - \delta_{\rho\mu} P_\nu,$$
$$[M_{\mu\nu}, M_{\rho\sigma}] = \delta_{\nu\rho} M_{\mu\sigma} - \delta_{\mu\rho} M_{\nu\sigma} - \delta_{\nu\sigma} M_{\mu\rho} + \delta_{\mu\sigma} M_{\nu\rho}.$$

Finally $[P_\mu, P_\nu] = 0$. The fields K_μ are obtained from translations by conjugation with inversion, or by direct coefficient differentiation, so $[K_\mu, K_\nu] = 0$. Applying (57.10) converts these vector-field brackets into the displayed charge commutators. In particular $[P_\mu, K_\nu] = 2M_{\mu\nu} - 2\delta_{\mu\nu} D$ becomes $[\,\widehat{P}_\mu, \widehat{K}_\nu\,] = 2\mathrm{i}(\delta_{\mu\nu} \widehat{D} - \widehat{J}_{\mu\nu})$. □

The Euclidean conformal algebra is isomorphic to $\mathfrak{so}(D+1,1)$. Introduce indices

$$A, B = -1, 0, 1, \ldots, D$$

and a metric

$$\eta^{(D+1,1)}_{AB} = \mathrm{diag}(-1, +1, \ldots, +1),$$

where the negative entry is the (-1)-direction. The generators $\widehat{\mathcal{J}}_{AB} = -\widehat{\mathcal{J}}_{BA}$ are related to the conformal generators by

$$\widehat{\mathcal{J}}_{\mu\nu} = \widehat{J}_{\mu\nu}, \qquad \widehat{\mathcal{J}}_{0,-1} = \widehat{D}, \qquad \widehat{\mathcal{J}}_{0,\mu} = \frac{1}{2}(\widehat{P}_\mu - \widehat{K}_\mu), \qquad \widehat{\mathcal{J}}_{-1,\mu} = \frac{1}{2}(\widehat{P}_\mu + \widehat{K}_\mu).$$

Then

$$[\widehat{\mathcal{J}}_{AB}, \widehat{\mathcal{J}}_{CD}] = \mathrm{i}(\eta^{(D+1,1)}_{AC} \widehat{\mathcal{J}}_{BD} - \eta^{(D+1,1)}_{BC} \widehat{\mathcal{J}}_{AD} - \eta^{(D+1,1)}_{AD} \widehat{\mathcal{J}}_{BC} + \eta^{(D+1,1)}_{BD} \widehat{\mathcal{J}}_{AC}).$$

The displayed change of basis is invertible. Substituting it into Lemma 57.9 gives the $\mathfrak{so}(D+1,1)$ commutator: relations with only μ, ν indices are the rotation algebra, relations involving $0, -1$ reproduce the dilatation commutators, and the mixed relations reproduce (57.25). Since the basis change is invertible, this identifies the Lie algebras. The Lorentzian conformal algebra is obtained by analytic continuation of one physical coordinate and has real form $\mathfrak{so}(D,2)$; it is not the real form used in the Euclidean derivation above.

57.7 The Lorentzian Conformal Group and Its Cover

The preceding algebra statement does not by itself identify the group that acts on a Lorentzian Hilbert space. Let

$$M = \mathbb{R}^{1,D-1}, \qquad \eta^{\mathrm{L}}_{\mu\nu} = \mathrm{diag}(-1, +1, \ldots, +1), \qquad x^2 = \eta^{\mathrm{L}}_{\mu\nu} x^\mu x^\nu.$$

Introduce an ambient vector space

$$\mathbb{R}^{D,2} = \mathbb{R}X^+ \oplus \mathbb{R}X^- \oplus \mathbb{R}^{1,D-1}$$

with bilinear form

$$X \cdot Y = \eta^{\mathrm{L}}_{\mu\nu} X^\mu Y^\nu - \frac{1}{2}(X^+ Y^- + X^- Y^+). \tag{57.28}$$

Thus $X \cdot X = \eta^{\mathrm{L}}_{\mu\nu} X^\mu X^\nu - X^+ X^-$, and the form has signature $(D, 2)$. The conformal compactification of Minkowski space is the projective null cone

$$\overline{M} = \{X \in \mathbb{R}^{D,2} \setminus \{0\} : X \cdot X = 0\}/\mathbb{R}^\times. \tag{57.29}$$

Definition 57.10 (Lorentzian conformal compactification)**.** The compactified Lorentzian spacetime $\overline{M}$ is the projective null cone (57.29) in $\mathbb{R}^{D,2}$. The Poincare patch is the affine chart $X^+ \neq 0$.

The Poincare patch $X^+ \neq 0$ is identified with M by choosing the representative

$$X^+ = 1, \qquad X^\mu = x^\mu, \qquad X^- = x^2. \tag{57.30}$$

The null condition follows immediately from (57.28). Points with $X^+ = 0$ are the conformal boundary of this patch.

The same compactification is the Einstein cylinder. Write

$$x^\mu = (t, r\omega), \qquad r \geq 0, \qquad \omega \in S^{D-2},$$

and introduce null coordinates $u = t - r$, $v = t + r$. On the Poincare patch set

$$u = \tan\frac{T-\chi}{2}, \qquad v = \tan\frac{T+\chi}{2}, \qquad 0 \leq \chi \leq \pi.$$

Then

$$\eta^{\mathrm{L}}_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = \Omega_{\mathrm{cyl}}(T,\chi)^{-2}\left(-\mathrm{d}T^2 + \mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega^2_{D-2}\right), \tag{57.31}$$

where

$$\Omega_{\mathrm{cyl}}(T,\chi) = 2\cos\frac{T-\chi}{2}\cos\frac{T+\chi}{2}. \tag{57.32}$$

The metric in parentheses is the metric on $\mathbb{R}_T \times S^{D-1}$, the universal cover of the Lorentzian conformal compactification. The projective compactification (57.29) is obtained from $S^1_T \times S^{D-1}$ by the antipodal projective identification $(T, n) \sim (T + \pi, -n)$. Thus local Lorentzian conformal transformations are restrictions to a Poincare chart of diffeomorphisms of the compactified space, and their Hilbert-space action is naturally lifted to the cylinder cover $\mathbb{R}_T \times S^{D-1}$. The coordinate substitution $u = \tan((T-\chi)/2)$ and $v = \tan((T+\chi)/2)$ converts the Minkowski line element into (57.31); the trigonometric denominator is the Weyl factor (57.32). Periodic cylinder time gives the projective compactification with the antipodal identification displayed above, while the Hilbert-space representation uses the universal cover in the T-direction.

Definition 57.11 (Projective Lorentzian conformal group)**.** The group $O(D, 2)$ acts linearly on $\mathbb{R}^{D,2}$, preserves the null cone, and therefore acts on $\overline{M}$. Scalar matrices act trivially after projectivization, so the spacetime conformal transformations in the identity component are described by the projective group

$$G^{\mathrm{Lor}}_{\mathrm{conf}} = PO_0(D, 2), \tag{57.33}$$

the identity component of the projective orthogonal group. Equivalently, one may start from $SO_0(D, 2)$ and quotient by the finite central subgroup that acts trivially on projective null lines. The Lie algebra of $G^{\mathrm{Lor}}_{\mathrm{conf}}$ is $\mathfrak{so}(D, 2)$.

The standard finite transformations on the Poincare patch are restrictions of linear ambient transformations. Lorentz transformations $\Lambda \in SO_0(1, D-1)$ act by

$$X^\mu \mapsto \Lambda^\mu{}_\nu X^\nu, \qquad X^\pm \mapsto X^\pm.$$

Translations by a^μ act by

$$X^+ \mapsto X^+, \qquad X^\mu \mapsto X^\mu + a^\mu X^+, \qquad X^- \mapsto X^- + 2a\cdot X + a^2 X^+, \tag{57.34}$$

where $a \cdot X = \eta^{\mathrm{L}}_{\mu\nu} a^\mu X^\nu$. Dilatations with parameter λ act by

$$X^+ \mapsto e^{-\lambda} X^+, \qquad X^- \mapsto e^{\lambda} X^-, \qquad X^\mu \mapsto X^\mu, \tag{57.35}$$

which becomes $x^\mu \mapsto e^\lambda x^\mu$ after rescaling back to $X^+ = 1$. Special conformal transformations with parameter c^μ act by

$$X^- \mapsto X^-, \qquad X^\mu \mapsto X^\mu + c^\mu X^-, \qquad X^+ \mapsto X^+ + 2c \cdot X + c^2 X^-. \tag{57.36}$$

On the patch $X^+ = 1$, this gives

$$x^\mu \longmapsto \frac{x^\mu + c^\mu x^2}{1 + 2c \cdot x + c^2 x^2}, \tag{57.37}$$

where the transformation is read locally whenever the denominator is nonzero, and globally as a map of the compactification (57.29). These formulae show explicitly how the infinitesimal generators integrate to the connected projective conformal group. Each displayed ambient transformation preserves the bilinear form (57.28), hence preserves the null cone and its projectivization. On the affine chart $X^+ = 1$, divide the transformed representative by its new X^+-component. This gives translations, dilatations, Lorentz rotations, and (57.37) for special conformal transformations. Differentiating at the identity recovers the Lorentzian real form of the conformal Killing generators.

Definition 57.12 (Hilbert-space conformal group)**.** For Hilbert-space representation theory, the projective spacetime group is still not the final object. Let

$$\widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}} \longrightarrow G^{\mathrm{Lor}}_{\mathrm{conf}} \tag{57.38}$$

be the universal covering group. A unitary Lorentzian CFT carries a strongly continuous unitary representation

$$U : \widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}} \longrightarrow \mathcal{U}(\mathcal{H})$$

whose derived representation gives the self-adjoint conformal charges on a common invariant finite-energy domain. The need for the cover is not a minor global convention. The maximal compact subalgebra of $\mathfrak{so}(D,2)$ is $\mathfrak{so}(D) \oplus \mathfrak{so}(2)$. In the universal cover its corresponding subgroup is covered by

$$\mathrm{Spin}(D) \times \mathbb{R}, \tag{57.39}$$

where the $\mathbb{R}$ factor is generated by Lorentzian cylinder time. Under radial quantization this generator is the nonnegative cylinder Hamiltonian D_{rad}, and

$$U_{\mathrm{cyl}}(t) = \exp(-\mathrm{i} t D_{\mathrm{rad}}). \tag{57.40}$$

If one forced the representation to descend to the finite group with an $SO(2)$ cylinder-time factor, the operator $\exp(-2\pi \mathrm{i} D_{\mathrm{rad}})$ would have to be the identity. This would impose integrality of every scaling dimension in that Hilbert-space sector. A sector containing a nonintegral scaling dimension therefore cannot descend to the $SO(2)$ quotient. Spinorial local operators similarly require the spin cover of spatial rotations. Thus later references to the Lorentzian conformal group acting on the Hilbert space mean the universal cover $\widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}}$; its projection to $G^{\mathrm{Lor}}_{\mathrm{conf}}$ is the group acting on compactified spacetime points. The $\mathbb{R}$ factor in (57.39) covers the $SO(2)$ cylinder-time circle. A representation descends to the quotient by the 2π-translation kernel precisely when $\exp(-2\pi \mathrm{i} D_{\mathrm{rad}}) = 1$. If $D_{\mathrm{rad}} \psi = \Delta \psi$, this condition gives $\exp(-2\pi \mathrm{i} \Delta) = 1$. Thus every scaling dimension in the sector must be integral. A sector with any nonintegral Δ fails this descent condition, and spinorial sectors similarly require the spin cover of rotations.

Definition 57.13 (Local primary state)**.** The vacuum vector is invariant under $\widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}}$. A local primary state is a lowest energy vector for the $\mathbb{R}$-generator in (57.39), transforms in a finite-dimensional unitary representation of $\mathrm{Spin}(D)$, and generates a positive-energy module by the action of the noncompact translation generators. This is the group-theoretic meaning of the radial primary conditions used in later chapters.

57.8 Representation-Theoretic Conventions

The same Lie algebra appears in several representation-theoretic roles, and the distinctions matter. The Euclidean conformal group

$$G_E = SO_0(D+1,1)$$

acts on the compactified Euclidean sphere S^D, or through a chosen cover when spin structures are present.

Stabilizer of a Euclidean point. Fix the point $x=0$ in the flat chart of S^D. Its stabilizer in the identity component of the Euclidean conformal group has Lie algebra

$$\mathfrak{p}_E = \mathfrak{so}(D) \oplus \mathbb{R}D \oplus \mathrm{span}\{K_\mu\}.$$

Equivalently, at the group level and in the chosen chart, it is generated by rotations, positive dilatations, and special conformal transformations. Its Levi factor is $SO(D) \times \mathbb{R}_+$, and the special conformal transformations form the nilpotent radical. Translations do not fix the origin unless their parameter is zero. A rotation and a positive dilatation fix $x=0$ and act on the tangent space by λR, with $R \in SO(D)$ and $\lambda > 0$. The finite special conformal transformation

$$x^\mu \mapsto \frac{x^\mu + c^\mu x^2}{1 + 2c\cdot x + c^2 x^2}$$

also fixes $x=0$, and its derivative at $x=0$ is the identity; hence its generator lies in the nilpotent radical of the stabilizer. The infinitesimal classification in Proposition 57.3 shows that these exhaust the conformal Killing vectors vanishing at the origin. Thus the stabilizer has the displayed Lie algebra and Levi/nilpotent decomposition.

Definition 57.14 (Euclidean primary type)**.** A Euclidean primary type at the point $x=0$ is a finite-dimensional representation of the point stabilizer P_E with the following structure. The nilpotent radical generated by K_μ acts trivially; the rotation factor acts by a finite-dimensional representation ρ of $SO(D)$, or of $\mathrm{Spin}(D)$ when spin is present; and a positive dilatation $x \mapsto \lambda x$ acts by the character $\lambda^{-\Delta}$. The real number Δ is the Euclidean scaling dimension in the covariance convention used below.

A Euclidean primary operator is encoded by a covariant section of the homogeneous vector bundle associated to this P_E-representation. In a flat chart, the transformation law has the form

Definition 57.15 (Euclidean primary covariance law)**.**

$$U_E(f)\,\mathcal{O}_i^a(x)\,U_E(f)^{-1} = \Omega_f(x)^{-\Delta_i}\,\rho_i(R_f(x))^a{}_b\,\mathcal{O}_i^b(f(x)), \tag{57.41}$$

where $\Delta_i \in \mathbb{R}$ is the scaling dimension, ρ_i is a finite-dimensional representation of $\mathrm{Spin}(D)$ or $SO(D)$, and $R_f(x)$ is the local rotation obtained from the Jacobian after extracting the positive Weyl factor $\Omega_f(x)$. The fiber representation of P_E contains a noncompact dilation character $\lambda \mapsto \lambda^{-\Delta_i}$. For the real scaling dimensions relevant to reflection-positive CFT, this fiber action is an induced covariance representation rather than the Hilbert-space unitary representation. Euclidean correlators are covariance distributions for the associated bundles, while reflection positivity is the additional input that produces the Lorentzian Hilbert-space representation.

Definition 57.16 (Positive-energy representation)**.** The Lorentzian Hilbert-space representation is a strongly continuous unitary representation of the universal cover $\widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}}$. Let

$$K_{\mathrm{cov}} = \mathrm{Spin}(D) \times \mathbb{R}$$

denote the cover of the maximal compact subgroup, with the $\mathbb{R}$ factor generated by the cylinder Hamiltonian D_{rad}. A positive-energy representation means a unitary representation U for which

$$U(r(t)) = e^{-\mathrm{i}tD_{\mathrm{rad}}}, \qquad D_{\mathrm{rad}} \geq 0,$$

with $r(t) \in \widetilde{G}^{\mathrm{Lor}}_{\mathrm{conf}}$ the lifted cylinder-time rotation and D_{rad} self-adjoint. The finite-energy subspace is

$$\mathcal{H}_{\mathrm{fin}} = \bigoplus_{\Delta<\infty} \ker(D_{\mathrm{rad}} - \Delta), \tag{57.42}$$

whenever the spectrum is discrete with finite-dimensional eigenspaces, as assumed in the radial-reconstruction hypotheses. This algebraic direct sum is a common invariant domain for the derived conformal generators. On this domain,

$$P_\mu^\dagger = K_\mu, \qquad D_{\mathrm{rad}}^\dagger = D_{\mathrm{rad}}, \qquad J_{\mu\nu}^\dagger = J_{\mu\nu},$$

with the adjoints understood in the radial Hilbert-space inner product. The closure of each one-parameter generator is the self-adjoint generator supplied by Stone's theorem from the unitary group representation.

Definition 57.17 (Generalized Verma module of a conformal primary)**.** After complexification, let

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{so}(D+2, \mathbb{C}).$$

The radial grading decomposes it as

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{n}_- \oplus (\mathfrak{so}(D, \mathbb{C}) \oplus \mathbb{C}D_{\mathrm{rad}}) \oplus \mathfrak{n}_+, \qquad \mathfrak{n}_+ = \mathrm{span}_{\mathbb{C}}\{P_\mu\}, \quad \mathfrak{n}_- = \mathrm{span}_{\mathbb{C}}\{K_\mu\}. \tag{57.43}$$

A primary vector v of dimension Δ and spin representation ρ obeys

$$D_{\mathrm{rad}}v = \Delta v, \qquad K_\mu v = 0, \qquad \mathrm{Spin}(D) \text{ acts on } \mathrm{span}\{v\} \text{ by } \rho.$$

The corresponding algebraic highest-weight, or generalized Verma, module is

$$M(\Delta, \rho) = U(\mathfrak{n}_+) \otimes V_{\Delta,\rho}, \tag{57.44}$$

where $V_{\Delta,\rho}$ is the finite-dimensional $\mathfrak{so}(D, \mathbb{C}) \oplus \mathbb{C}D_{\mathrm{rad}}$-module on which $\mathfrak{n}_-$ acts trivially. Its level-n subspace is generated by n translations P_μ.

Proposition 57.18 (Null quotient and unitarity test)**.** *The radial two-point function defines a contravariant Hermitian form on $M(\Delta,\rho)$. The null radical of this form is a submodule, and the cyclic positive-energy quotient carried by the primary datum is*

$$L(\Delta,\rho) = M(\Delta,\rho)/\operatorname{Rad} M(\Delta,\rho) \tag{57.45}$$

when the quotient form is positive and the Hilbert-space sector generated by the primary has no further invariant subspace orthogonal to the cyclic descendant space. The algebraic unitarity test on the descendant module is positivity of every finite Gram matrix. Shortening occurs when a Gram matrix develops a null vector whose conformal descendants form a nonzero submodule.

Proof. Radial reflection sends translations through the origin to special conformal transformations, so on the algebraic descendant module the two-point pairing is contravariant with respect to the anti-involution

$$P_\mu^\dagger = K_\mu, \qquad D_{\mathrm{rad}}^\dagger = D_{\mathrm{rad}}, \qquad J_{\mu\nu}^\dagger = J_{\mu\nu}.$$

Let $R \subset M(\Delta,\rho)$ be the radical of this Hermitian form. If $r \in R$, then $(r,m) = 0$ for every descendant m. For any generator X of the complexified conformal algebra,

$$(Xr, m) = (r, X^\dagger m) = 0,$$

because $X^\dagger m$ is again a descendant. Hence $Xr \in R$, and R is a submodule. Quotienting by R gives a nondegenerate cyclic descendant module generated by the primary vector. It is the irreducible primary sector precisely when the Hilbert-space sector generated by the primary contains no further invariant subspace orthogonal to this cyclic descendant space; without this extra Hilbert-space condition the algebraic quotient only describes the cyclic part generated by the chosen primary.

The level decomposition of $M(\Delta,\rho)$ is finite dimensional at each level, because each level is generated by finitely many monomials in the D translation generators acting on the finite-dimensional spin space $V_{\Delta,\rho}$. Positivity of the Hermitian form on the algebraic direct sum is therefore equivalent to positivity of every finite Gram matrix at finite total level. When such a Gram matrix has a null vector, the radical argument above shows that the conformal descendants of that vector form a submodule. This is the algebraic shortening mechanism. □

The passage from the algebraic module $L(\Delta,\rho)$ to a group representation is a globalization problem. In the unitary CFT setting the globalization is supplied by the strongly continuous representation of $\widetilde{G}_{\mathrm{conf}}^{\mathrm{Lor}}$ on the radial Hilbert space, and $L(\Delta,\rho)$ is the K_{cov}-finite subspace generated by the primary. Descent from the universal cover to a quotient group is an additional condition: every element of the covering kernel must act trivially. For example, descent to an $SO(2)$ cylinder-time factor would require $\exp(-2\pi\mathrm{i}D_{\mathrm{rad}}) = 1$, hence integral scaling dimensions on the sector under consideration; spinorial sectors similarly require keeping the spin cover rather than $SO(D)$. Throughout the conformal block chapters, statements about conformal multiplets refer to these algebraic $\mathfrak{g}_{\mathbb{C}}$-modules together with the Hilbert-space globalization supplied by radial quantization.

57.9 Two Euclidean Dimensions

For $D = 2$, the local conformal Killing equation has infinitely many solutions. In complex coordinates

$$z = x^1 + \mathrm{i}x^2, \qquad \bar{z} = x^1 - \mathrm{i}x^2,$$

the local conformal transformations are

$$z \mapsto f(z), \qquad \bar{z} \mapsto \bar{f}(\bar{z})$$

in Euclidean signature, with f holomorphic in its domain. The global conformal transformations of the Riemann sphere are the Möbius maps

$$z \mapsto \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

In two real dimensions the conformal Killing equation is equivalent to the Cauchy–Riemann equations for the complex components of the infinitesimal coordinate change. Integrating locally gives holomorphic maps $z \mapsto f(z)$ and antiholomorphic maps in the conjugate coordinate. A globally defined biholomorphism of the compact Riemann sphere is a degree-one rational map: it is meromorphic on the sphere, has a meromorphic inverse, and therefore has topological degree one. Degree-one rational maps are precisely the displayed Möbius transformations, modulo the common rescaling of (a, b, c, d).

The infinite-dimensional enhancement has its own representation theory. The global Lorentzian conformal algebra factorizes, after passing to a cover, as $\mathfrak{sl}(2, \mathbb{R}) \oplus \mathfrak{sl}(2, \mathbb{R})$, while the local chiral stress-tensor modes generate Virasoro algebras with central charge. A global primary is annihilated by the global lowering generators, whereas a Virasoro primary is annihilated by all positive chiral modes $L_n, \bar{L}_n$ with $n > 0$. The two notions coincide only after restricting attention to the global subalgebra. The present core conformal-block discussion uses the finite-dimensional global conformal group; a full two-dimensional CFT volume must keep global blocks, Virasoro blocks, and their different null-module quotients separate.

Chapter 58

Stress Tensor, Weyl Structure, and Improvement

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

58.1 The Stress Tensor from Background Metrics

Definition 58.1 (Metric-source stress tensor)**.** Let a Euclidean local theory be coupled to a smooth background metric $g_{\mu\nu}$ on a D-dimensional manifold M. The stress tensor is defined by the first variation of the action or of the renormalized generating functional with respect to the metric. This chapter uses the contravariant-metric convention

$$\delta S[\Phi; g] = \frac{1}{2} \int_M \mathrm{d}^D x \, \sqrt{g} \, T_{\mu\nu}(x) \, \delta g^{\mu\nu}(x), \tag{58.1}$$

where Φ denotes the dynamical fields and $g = \det(g_{\mu\nu})$. Since $\delta g^{\mu\nu} = -g^{\mu\rho} g^{\nu\sigma} \delta g_{\rho\sigma}$, the same convention is equivalently

$$\delta S[\Phi; g] = -\frac{1}{2} \int_M \mathrm{d}^D x \, \sqrt{g} \, T^{\mu\nu}(x) \, \delta g_{\mu\nu}(x).$$

Equivalently, at the level of functional derivatives with all other sources held fixed,

$$T^{\mu\nu}(x) = -\frac{2}{\sqrt{g}(x)} \frac{\delta}{\delta g_{\mu\nu}(x)} = \frac{2}{\sqrt{g}(x)} g^{\mu\rho}(x) g^{\nu\sigma}(x) \frac{\delta}{\delta g^{\rho\sigma}(x)}. \tag{58.2}$$

The first differential operator acts on the functional written as a function of $g_{\mu\nu}$, and the second acts on the same functional written as a function of $g^{\mu\nu}$. This identity is only the chain rule applied to $\delta g^{\mu\nu} = -g^{\mu\rho}g^{\nu\sigma}\delta g_{\rho\sigma}$. This is the convention used by the metric derivative of the Euclidean source functional. At a regulator scale for which the metric-dependent functional integral is a well-defined regulated integral, let

$$Z[g] = \int \mathcal{D}_g\Phi \, \exp\{-S[\Phi; g]\}, \qquad \mathcal{W}[g] = -\log Z[g].$$

For variations supported away from boundary components, or with boundary conditions fixed so that no boundary term is produced, differentiating the regulated integral gives

$$\delta\mathcal{W}[g] = \langle \delta S[\Phi; g] \rangle_g = -\frac{1}{2}\int_M \mathrm{d}^D x \, \sqrt{g}\, \langle T^{\mu\nu}(x) \rangle_g \, \delta g_{\mu\nu}(x).$$

Therefore

$$\langle T^{\mu\nu}(x) \rangle_g = -\frac{2}{\sqrt{g}(x)} \frac{\delta\mathcal{W}[g]}{\delta g_{\mu\nu}(x)}.$$

After subtracting local metric counterterms, the same identity defines the renormalized source-coordinate system used in (56.26). If instead one writes $K[g] = \log Z[g]$, the right-hand side carries the opposite sign; the difference is only the choice between the logarithm of the partition function and the Euclidean effective action. With this sign convention, evaluating $T_{\mu\nu}$ on the flat metric gives the translation Noether tensor after adding the divergence term that makes the Noether tensor symmetric.

Remark 58.2 (Curved extensions and flat stress tensors). A flat classical action can have several curved extensions. For example,

$$S[\phi] = \int_{\mathbb{R}^D} \mathrm{d}^D x \left(\frac{1}{2}\partial_\mu\phi \, \partial^\mu\phi + \frac{1}{2}m^2\phi^2 \right)$$

may be extended as

$$S[\phi; g] = \int_M \mathrm{d}^D x \, \sqrt{g} \left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi \, \partial_\nu\phi + \frac{1}{2}m^2\phi^2 + \text{local curvature couplings} \right).$$

Curvature couplings that vanish on flat space still affect $T_{\mu\nu}$, because $T_{\mu\nu}$ is defined by varying $g_{\mu\nu}$ before setting $g_{\mu\nu} = \delta_{\mu\nu}$.

Diffeomorphism Ward identity for metric stress tensors. If the theory is invariant under diffeomorphisms of the background, then an infinitesimal vector field ξ^μ gives

$$\delta_\xi g^{\mu\nu} = -\nabla^\mu\xi^\nu - \nabla^\nu\xi^\mu.$$

Substituting this variation into (58.1) and integrating by parts gives

$$\nabla_\mu T^{\mu\nu} = 0 \tag{58.3}$$

as an operator equation away from insertions. In flat space this becomes $\partial_\mu T^{\mu\nu} = 0$. Symmetry of $T_{\mu\nu}$ follows from variation with respect to the symmetric tensor $g_{\mu\nu}$. The calculation is the source-form of Noether's identity. Insert $\delta_\xi g^{\mu\nu} = -\nabla^\mu\xi^\nu - \nabla^\nu\xi^\mu$ into Definition 58.1. The symmetry of $T_{\mu\nu}$ gives

$$\delta_\xi S = -\int_M \mathrm{d}^D x \, \sqrt{g}\, T_{\mu\nu}\nabla^\mu\xi^\nu.$$

Integrating by parts gives

$$\delta_\xi S = \int_M \mathrm{d}^D x \,\sqrt{g}\,(\nabla_\mu T^{\mu\nu})\xi_\nu$$

up to the boundary term excluded by the support or boundary conditions. Since the background-diffeomorphism variation is zero and ξ is arbitrary, the local identity $\nabla_\mu T^{\mu\nu} = 0$ follows away from operator insertions. At coincident insertions the same source argument gives the familiar contact terms; their precise form depends on which inserted operators transform under the diffeomorphism and on the chosen contact-term extension of the correlator.

58.2 Metric Perturbations and Stress Insertions

Set $g_{\mu\nu} = \delta_{\mu\nu} + \delta g_{\mu\nu}$, with δg smooth and compactly supported. To first order, the Euclidean path integral changes by the insertion

$$\exp[-\delta_g S] = \exp\left[-\frac{1}{2}\int_{\mathbb{R}^D} \mathrm{d}^D x\,\delta g^{\mu\nu}(x) T_{\mu\nu}(x)\right], \tag{58.4}$$

where indices on $\delta g^{\mu\nu}$ are raised with the flat metric at this order. Thus, for an operator product X supported away from $\mathrm{supp}(\delta g)$,

$$\langle X\rangle_{\delta+\delta g} = \langle X\rangle_\delta - \frac{1}{2}\int \mathrm{d}^D x\,\delta g^{\mu\nu}(x)\langle T_{\mu\nu}(x) X\rangle_\delta + O(\delta g^2).$$

Higher metric variations require a prescription for coincident products of stress tensors. The short-distance singularities in $T_{\mu\nu}(x)T_{\rho\sigma}(y)$ and the corresponding contact terms are part of the renormalized definition of the theory on curved backgrounds. This is a local ultraviolet statement: in a massless theory the distributional extension of stress-tensor products is used only after the infrared state or regulator has made the separated-point correlators meaningful on the test functions under consideration. In even dimensions these contact terms include the Weyl anomaly data. Expand the metric-source action to first order in δg using Definition 58.1. Since $\delta g^{\mu\nu} = -\delta g_{\mu\nu} + O(\delta g^2)$ around the flat metric, the first-order change of the Boltzmann weight is the exponential insertion in (58.4). Expanding that exponential to first order gives the displayed variation of $\langle X\rangle$. Higher functional derivatives differentiate products of metric variations at coincident points, so their definition requires the local extension of stress-tensor products as distributions.

58.3 Weyl Transformations

Definition 58.3 (Weyl transformation and trace insertion)**.** A Weyl transformation is a local rescaling of the background metric:

$$g_{\mu\nu}(x) \mapsto \mathrm{e}^{2\omega(x)} g_{\mu\nu}(x),$$

where ω is a smooth real function. Infinitesimally,

$$\delta_\omega g_{\mu\nu} = 2\omega g_{\mu\nu}, \qquad \delta_\omega g^{\mu\nu} = -2\omega g^{\mu\nu}.$$

The corresponding metric variation gives

$$\delta_\omega S = -\int_M \mathrm{d}^D x\,\sqrt{g}\,\omega(x)\,T^\mu{}_\mu(x).$$

Flat-space trace condition from Weyl invariance. Assume that the renormalized theory is coupled to a background metric, that the stress tensor is defined by metric variation as in (58.4), and that a choice of field Weyl transformations and allowed improvements has been made for which the renormalized functional is invariant under compactly supported Weyl rescalings in flat space, apart from curvature anomaly terms and contact terms on coincident insertions. Then, at separated flat-space points,

$$T^{\mu}{}_{\mu} = 0 \tag{58.5}$$

The action may assign Weyl transformations to the dynamical fields Φ. After including these field variations and the chosen improvement terms, Weyl invariance says that the first variation of every separated correlator under $\delta_\omega g_{\mu\nu} = 2\omega g_{\mu\nu}$ vanishes, modulo the stated anomaly and contact contributions. The metric-variation formula above gives

$$\delta_\omega \langle X\rangle = -\int \mathrm{d}^D x\, \omega(x)\, \langle T^{\mu}{}_{\mu}(x) X\rangle$$

for every operator product X supported away from $\mathrm{supp}(\omega)$. Since ω is an arbitrary smooth test function in the separated region, the distributional insertion of $T^{\mu}{}_{\mu}$ vanishes there. This is the local operator equation (58.5) at separated flat-space points.

The background metric is an external source for stress-tensor insertions and Ward identities.

In even dimensions, the renormalized generating functional may transform under a Weyl rescaling by a local curvature functional. At separated flat-space points the conformal Ward identities are governed by (58.5); the anomaly contributes contact terms and curved-space response data. The present volume records this contact-term distinction; the special two-dimensional extension of conformal symmetry is deferred to a separate treatment.

58.4 Weyl Anomaly Density and Contact Ward Identities

Definition 58.4 (Weyl anomaly density)**.** Define the trace anomaly density by

$$\delta_\omega \mathcal{W}[g] = -\int_M \mathrm{d}^D x\, \sqrt{g}\, \omega(x)\, \mathcal{A}[g](x), \qquad \mathcal{A}[g](x) = \langle T^{\mu}{}_{\mu}(x)\rangle_g\,. \tag{58.6}$$

This sign follows from the convention $\langle T^{\mu\nu}\rangle_g = -2g^{-1/2}\delta\mathcal{W}/\delta g_{\mu\nu}$ and $\delta_\omega g_{\mu\nu} = 2\omega g_{\mu\nu}$. The anomaly density is a local scalar of Weyl weight D, defined modulo the Weyl variation of local metric counterterms. Its functional derivatives are contact terms in stress-tensor Ward identities. More explicitly, let

$$X_{n-1} = T^{\rho_1\sigma_1}(x_1)\cdots T^{\rho_{n-1}\sigma_{n-1}}(x_{n-1}).$$

Tracing one stress tensor in an n-point distribution gives, away from separated-point terms,

$$g_{\mu\nu}(x)\, \langle T^{\mu\nu}(x) X_{n-1}\rangle = \left.\frac{\delta^{\,n-1}\mathcal{A}[g](x)}{\prod_{j=1}^{n-1}\delta g_{\rho_j\sigma_j}(x_j)}\right|_{g=\delta} + \text{index-variation contact terms.}$$

Here the denominator denotes the iterated functional derivative with respect to the displayed metric components. Thus anomaly coefficients are part of the metric-source definition of the stress-tensor correlators, even though $T^{\mu}{}_{\mu} = 0$ at separated flat-space points.

58.5 Classification of Conformal Anomalies

Wess–Zumino consistency for Weyl anomalies. The local anomaly density is constrained because Weyl transformations form an abelian group. Let

$$\mathbf{A}_\omega[g] := \int_M \mathrm{d}^D x \sqrt{g}\, \omega(x) \mathcal{A}[g](x).$$

The identity $[\delta_{\omega_1}, \delta_{\omega_2}]\mathcal{W}[g] = 0$ gives the Wess–Zumino consistency condition

$$\delta_{\omega_1} \mathbf{A}_{\omega_2}[g] - \delta_{\omega_2} \mathbf{A}_{\omega_1}[g] = 0. \tag{58.7}$$

Changing the renormalized source functional by a local metric counterterm $\mathcal{W}[g] \mapsto \mathcal{W}[g] + B[g]$ changes $\mathbf{A}_\omega[g]$ by $-\delta_\omega B[g]$. Therefore the intrinsic trace anomaly is the cohomology of the Weyl differential on local metric functionals: solutions of (58.7) modulo Weyl variations of local counterterms. This is the source-functional form of the abelian Weyl algebra. Apply two Weyl variations to the same renormalized functional $\mathcal{W}[g]$. Since local Weyl rescalings commute, $[\delta_{\omega_1}, \delta_{\omega_2}]\mathcal{W}[g] = 0$. Using Definition 58.4, this commutator is precisely $\delta_{\omega_1}\mathbf{A}_{\omega_2} - \delta_{\omega_2}\mathbf{A}_{\omega_1}$, giving (58.7). If $\mathcal{W}$ is shifted by a local counterterm $B[g]$, its anomaly shifts by the Weyl variation of that counterterm. Therefore only the cohomology class of $\mathbf{A}_\omega$ is intrinsic.

The classification below has two separate inputs. The QFT input has just been derived from the metric source functional: the anomaly is a cocycle for the abelian Weyl differential, and local counterterms add coboundaries. The geometric input is the local invariant-theory computation of that cohomology on the jet space of smooth metrics. The quoted theorem supplies this geometric classification. A particular CFT then supplies numerical coordinates on the listed cohomology classes through its stress-tensor correlators or through a regulated path integral.

Quoted Theorem 58.5 (Parity-even local Weyl anomaly classes)**.** *In the parity-even metric sector on a closed manifold without boundary, the bulk solutions in even dimension $D = 2n$ have the form*

$$\mathcal{A}_{2n}[g] = -a_{2n} E_{2n} + \sum_\alpha c_\alpha I_\alpha + \nabla_\mu J^\mu. \tag{58.8}$$

Here

$$E_{2n} = \frac{1}{2^n} \delta^{\mu_1\nu_1\cdots\mu_n\nu_n}_{\rho_1\sigma_1\cdots\rho_n\sigma_n} R^{\rho_1\sigma_1}{}_{\mu_1\nu_1} \cdots R^{\rho_n\sigma_n}{}_{\mu_n\nu_n} \tag{58.9}$$

is the Euler density in the curvature convention used in this chapter. The term E_{2n} is the type-A anomaly. Its integral is topological: $\int_M \sqrt{g}\, E_{2n}$ is proportional to the Euler characteristic on a closed $2n$-manifold. Consequently its Weyl variation is a total derivative, so it obeys (58.7). The scalar densities I_α are type-B anomalies: $\sqrt{g}\, I_\alpha$ is locally Weyl invariant in dimension $2n$. The divergence $\nabla_\mu J^\mu$ denotes type-D, or trivial, terms; these are Weyl variations of local metric counterterms and are scheme dependent.

This classification is local. It classifies the contact terms in stress-tensor Ward identities produced by varying a smooth background metric. Global terms, boundaries, defects, and parity-odd orientation-dependent candidates require additional data and are not part of the parity-even bulk list in (58.8). In odd dimension there is no parity-even local bulk trace anomaly of this type on a closed smooth manifold, although finite nonlocal data such as the sphere free energy can be important conformal data.

Cohomological mechanism. The representative $-a_{2n}E_{2n}+\sum_\alpha c_\alpha I_\alpha$ is a coordinate choice on the quotient of local Weyl cocycles by local Weyl coboundaries. The type-B terms are cocycles for the direct reason that $\sqrt{g}\,I_\alpha$ is Weyl invariant in dimension $2n$. The type-A term is a cocycle because the Euler form is obtained by Chern–Weil transgression from an invariant polynomial: the integrated Euler density is metric-independent on a closed manifold, and its local Weyl variation is therefore an exact divergence. Type-D terms are coboundaries by definition: if $B[g]$ is a local counterterm, then $-\delta_\omega B[g]=\int\sqrt{g}\,\omega\,\nabla_\mu J^\mu$ for the corresponding representative. The nontrivial content of the quoted theorem is the converse local statement: after imposing the algebraic and differential Bianchi identities and quotienting total derivatives, every parity-even bulk solution of the Wess–Zumino condition is represented by this type-A/type-B list plus a type-D coboundary. The chapter uses that converse as geometric cohomology infrastructure, not as a substitute for computing the coefficients of a given QFT.

In two dimensions, with the convention above, write

$$\mathcal{A}_2[g]=-\frac{c_{2d}}{24\pi}R. \tag{58.10}$$

This is purely type A: $E_2=R$, and there is no independent parity-even type-B scalar in two dimensions.

Four-dimensional parity-even Weyl anomaly. In four dimensions, define

$$E_4=R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}-4R_{\mu\nu}R^{\mu\nu}+R^2,\qquad W^2=W_{\mu\nu\rho\sigma}W^{\mu\nu\rho\sigma},$$

where $W_{\mu\nu\rho\sigma}$ is the Weyl tensor. A parity-even CFT has

$$\mathcal{A}_4[g]=\frac{1}{16\pi^2}\left(c_{\mathrm{W}}W^2-a_{\mathrm{W}}E_4+b_{\mathrm{D}}\nabla^2R\right). \tag{58.11}$$

The coefficient b_{D} is changed by adding a local R^2 counterterm to $\mathcal{W}[g]$. The coefficients a_{W} and c_{W} are unchanged by such counterterms at a conformal fixed point. E_4 is the type-A term, W^2 is the unique parity-even type-B scalar, and ∇^2R is type D. More explicitly,

$$\delta_\omega\int_M \mathrm{d}^4x\,\sqrt{g}\,R^2=-12\int_M \mathrm{d}^4x\,\sqrt{g}\,\omega\,\nabla^2R$$

on a closed four-manifold, so the coefficient of ∇^2R is shifted by a choice of local R^2 counterterm. The completeness of this parity-even four-dimensional list follows from the dimension-four curvature basis $R^2_{\mu\nu\rho\sigma}$, $R^2_{\mu\nu}$, R^2, and ∇^2R: the two nontrivial Wess–Zumino classes are the Euler combination E_4 and the Weyl-invariant combination W^2, while the remaining scalar can be changed by the R^2 counterterm just displayed. The local cohomological input is the four-dimensional specialization of Quoted Theorem 58.5: among the parity-even dimension-four curvature scalars modulo total derivatives and algebraic Bianchi identities, E_4 supplies the type-A class and W^2 supplies the type-B class. The displayed Weyl variation of $\int\sqrt{g}\,R^2$ gives the explicit representative of the type-D ambiguity.

Quoted Theorem 58.6 (Six-dimensional parity-even local Weyl cohomology)**.** *In six dimensions, modulo type-D total-derivative terms, the parity-even bulk classification has one type-A coefficient and three type-B coefficients:*

$$\mathcal{A}_6[g]=\frac{1}{(4\pi)^3}\left(-a_6E_6+c_1I_1+c_2I_2+c_3I_3+\nabla_\mu J^\mu\right). \tag{58.12}$$

A convenient representative of the three type-B invariants is

$$\begin{aligned}
I_1 &= W_{\mu\rho\sigma\nu} W^{\mu\alpha\beta\nu} W_\alpha{}^{\rho\sigma}{}_\beta, \\
I_2 &= W_{\mu\nu}{}^{\rho\sigma} W_{\rho\sigma}{}^{\alpha\beta} W_{\alpha\beta}{}^{\mu\nu}, \\
I_3 &\equiv W_{\mu\nu\rho\sigma} \left(\nabla^2 \delta^\mu{}_\alpha + 4R^\mu{}_\alpha - \frac{6}{5} R\, \delta^\mu{}_\alpha \right) W^{\alpha\nu\rho\sigma} \quad \mod \nabla_\mu J^\mu.
\end{aligned} \tag{58.13}$$

The equivalence sign in the definition of I_3 is essential: different representatives differ by type-D terms, while the coefficient c_3 of the cohomology class is invariant under local counterterm changes. In dimensions higher than six the same rule applies, but the number of independent Weyl invariants grows because more contractions of Weyl tensors and conformally covariant derivatives are available. For $D = 6$, dimension counting and the algebraic Bianchi identities leave two independent cubic Weyl contractions and one independent two-Weyl-tensor invariant with two derivatives; Ricci-dependent representatives either combine with these into the conformally covariant operator in I_3, enter the Euler density E_6, or are type-D counterterm variations.

The six-dimensional statement is used here as a basis theorem for local Weyl cohomology. The derivation of the Wess–Zumino cocycle condition and the counterterm quotient is the source-functional argument above; the enumeration of the three independent type-B representatives is the quoted invariant-theory input. Consequently a later computation of a six-dimensional theory is required to determine the four numbers (a_6, c_1, c_2, c_3) in a declared regularization scheme. The quoted theorem does not supply those numbers and does not assert that a particular six-dimensional QFT exists.

58.6 Anomaly Coefficients and Stress-Tensor Correlators

Convention 58.7 ($2D$ central charge and anomaly)**.** When a two-dimensional CFT is coupled to a curved metric and its flat holomorphic stress tensor is normalized by the same metric source functional, the symbol c_{2d} denotes both the coefficient in (58.10) and the Virasoro central charge in the holomorphic OPE normalization

$$\langle T(z)T(0)\rangle = \frac{c_{2d}}{2z^4}.$$

The identification is the contact Ward identity obtained by taking two metric variations of (58.10) around a flat metric and then rewriting the result in complex coordinates. The present chapter uses this identity as a normalization convention; it is not being introduced here as a separate theorem without proof.

Convention 58.8 (Four-dimensional C_T and c_{W})**.** Use the following local normalization of the separated flat-space stress-tensor two-point function:

$$\langle T_{\mu\nu}(x)T_{\rho\sigma}(0)\rangle = \frac{C_T}{|x|^{2D}} I_{\mu\nu,\rho\sigma}(x),$$

where

$$I_{\mu\nu,\rho\sigma}(x) = \frac{1}{2}\big(I_{\mu\rho}(x)I_{\nu\sigma}(x) + I_{\mu\sigma}(x)I_{\nu\rho}(x)\big) - \frac{1}{D}\delta_{\mu\nu}\delta_{\rho\sigma}.$$

The same two-point convention is developed systematically in Chapter 63. With this normalization, the symbol c_{W} in (58.11) is normalized so that the separated stress-tensor two-point coefficient is

$$C_T = \frac{40}{\pi^4}\, c_{\mathrm{W}}. \tag{58.14}$$

The relation is a normalization statement for the metric source. Its curvature-kernel verification is obtained by taking two metric variations of the W^2 term in (58.11), Fourier transforming the resulting quadratic curvature contact term, and matching the logarithmic scale dependence of the flat-space TT correlator to the displayed two-point normalization. The Euler term begins to contribute independent information at the next metric variation.

Remark 58.9 (Four-dimensional anomaly map from TTT data). Use the local convention in which the parity-even separated TTT correlator in $D = 4$ is expanded in three Osborn–Petkou structures,

$$\langle T_{\mu\nu}(x_1)T_{\rho\sigma}(x_2)T_{\alpha\beta}(x_3)\rangle_{\text{even}} = a_{\mathrm{OP}}\mathcal{I}^{(1)} + b_{\mathrm{OP}}\mathcal{I}^{(2)} + c_{\mathrm{OP}}\mathcal{I}^{(3)}.$$

Chapter 63 constructs these three separated tensor structures explicitly from conformal covariance, conservation, and permutation symmetry. The small-sphere stress-tensor Ward identity gives, in this normalization and at $D = 4$,

$$C_T = \frac{\pi^2}{3}\left(14a_{\mathrm{OP}} - 2b_{\mathrm{OP}} - 5c_{\mathrm{OP}}\right), \tag{58.15}$$

so c_{W} is this same linear combination through (58.14). The Euler coefficient a_{W} is a second, independent linear functional of $(a_{\mathrm{OP}}, b_{\mathrm{OP}}, c_{\mathrm{OP}})$, obtained by taking three metric variations of the E_4 term in (58.11) and projecting the resulting trace Ward identity onto the three separated tensor structures. Equivalently, one may use the free-field basis of four-dimensional stress-tensor structures:

$$\langle TTT\rangle = n_s\langle TTT\rangle_{\text{real scalar}} + n_f\langle TTT\rangle_{\text{Dirac fermion}} + n_v\langle TTT\rangle_{\text{Maxwell vector}}. \tag{58.16}$$

In that basis,

$$a_{\mathrm{W}} = \frac{n_s}{360} + \frac{11n_f}{360} + \frac{31n_v}{180}, \qquad c_{\mathrm{W}} = \frac{n_s}{120} + \frac{n_f}{20} + \frac{n_v}{10}. \tag{58.17}$$

Here n_f multiplies the Dirac-fermion stress-tensor structure; using a Weyl fermion instead halves the fermion entries. The third independent linear combination of $a_{\mathrm{OP}}, b_{\mathrm{OP}}, c_{\mathrm{OP}}$ is genuine TTT data not fixed by a_{W} and c_{W}. After C_T has fixed the overall stress-tensor normalization, Chapter 65 uses the conformal-collider coordinates t_2, t_4 as detector coordinates on the remaining separated TTT data.

58.7 Improvement of the Stress Tensor

Scalar improvement of the flat stress tensor. The stress tensor has improvement freedom. If $L(x)$ is a local scalar operator, then

$$T_{\mu\nu} \longmapsto T'_{\mu\nu} = T_{\mu\nu} + \left(\delta_{\mu\nu}\partial^2 - \partial_\mu\partial_\nu\right) L \tag{58.18}$$

preserves conservation in flat space. More general improvements are possible, but (58.18) is the case needed for the free scalar and many scalar fixed points. Its trace is

$$T'^{\mu}{}_{\mu} = T^{\mu}{}_{\mu} + (D-1)\partial^2 L.$$

If $T^\mu{}_\mu$ is a total Laplacian of a local scalar, an improvement can make the trace vanish in flat space. The added tensor is identically conserved in flat space because

$$\partial^\mu(\delta_{\mu\nu}\partial^2 - \partial_\mu\partial_\nu)L = \partial_\nu\partial^2 L - \partial^2\partial_\nu L = 0.$$

Taking its trace gives $(D-1)\partial^2 L$, which is the displayed trace shift.

The improvement can be represented by adding a curvature coupling to the background-metric action. For a scalar field ϕ, the term

$$\Delta S_\xi = \frac{\xi}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,R(g)\phi^2$$

vanishes in flat space when $R(g) = 0$, but its metric variation contributes to $T_{\mu\nu}$ at flat $g_{\mu\nu}$. Thus a term absent from the flat-space action can change the flat-space stress tensor.

58.8 The Conformally Coupled Free Scalar

Proposition 58.10 (Conformally coupled free scalar)**.** *For a real massless scalar field in $D > 2$ Euclidean dimensions, the curvature coupling*

$$\xi_* = \frac{D-2}{4(D-1)}$$

is the unique scalar-curvature improvement of the form $\frac{\xi}{2}\int\sqrt{g}\,R\phi^2$ for which the flat-space stress tensor has vanishing trace modulo the scalar equation of motion. The corresponding curved action is (58.21)*.*

Proof. Consider a real massless scalar field in Euclidean signature. The minimally coupled curved-space action is

$$S_0[\phi; g] = \frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi.$$

Its metric variation is

$$\delta_g S_0 = \frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,\delta g^{\mu\nu}\left(\partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}g^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi\right).$$

On the flat background this gives

$$T^{(0)}_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\delta_{\mu\nu}(\partial\phi)^2.$$

Its trace is

$$T^{(0)\mu}{}_\mu = \left(1 - \frac{D}{2}\right)(\partial\phi)^2.$$

For $D > 2$ this trace is a nonzero local operator before improvement. Using the equation of motion $\partial^2\phi = 0$, the identity

$$\partial^2\phi^2 = 2(\partial\phi)^2 + 2\phi\,\partial^2\phi$$

shows that the trace is a Laplacian modulo the equation of motion.

Now add the curvature functional

$$S_1[\phi; g] = \frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,R(g)\phi^2.$$

The scalar curvature variation with respect to $g^{\mu\nu}$ is

$$\delta R = R_{\mu\nu}\delta g^{\mu\nu} + \nabla_\rho\left(g^{\mu\nu}\delta\Gamma^\rho_{\mu\nu} - g^{\mu\rho}\delta\Gamma^\nu_{\mu\nu}\right),$$

where $\delta\Gamma^\rho_{\mu\nu}$ is the variation of the Levi-Civita connection. Integrating the total derivative by parts and then setting $g_{\mu\nu} = \delta_{\mu\nu}$ gives

$$\delta_g S_1\big|_{g=\delta} = \frac{1}{2}\int_{\mathbb{R}^D} \mathrm{d}^D x\,\delta g^{\mu\nu}\left(-\partial_\mu\partial_\nu\phi^2 + \delta_{\mu\nu}\partial^2\phi^2\right).$$

The remaining terms in the full curved-space variation are proportional to curvature and vanish after the metric is set to $\delta_{\mu\nu}$.

For

$$S[\phi; g] = S_0[\phi; g] + aS_1[\phi; g],$$

the flat-space stress tensor is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\delta_{\mu\nu}(\partial\phi)^2 + a\left(\delta_{\mu\nu}\partial^2 - \partial_\mu\partial_\nu\right)\phi^2, \tag{58.19}$$

The total translation generators $P^\nu = \int_\Sigma \mathrm{d}^{D-1}x\,T^{0\nu}$ are unchanged by the improvement term when spatial total derivatives integrate to zero on the constant-time slice. The trace changes:

$$T^\mu{}_\mu = \left(1 - \frac{D}{2}\right)(\partial\phi)^2 + a(D-1)\partial^2\phi^2$$

and therefore

$$T^\mu{}_\mu = \left(1 - \frac{D}{2} + 2(D-1)a\right)(\partial\phi)^2 + 2(D-1)a\,\phi\,\partial^2\phi.$$

The trace vanishes modulo the scalar equation of motion if and only if

$$a = \xi_* = \frac{D-2}{4(D-1)}. \tag{58.20}$$

The corresponding curved-space action is

$$S[\phi; g] = \frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,\left[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \xi_* R(g)\phi^2\right]. \tag{58.21}$$

With the Weyl transformation

$$g_{\mu\nu} \mapsto \mathrm{e}^{2\omega}g_{\mu\nu}, \qquad \phi \mapsto \mathrm{e}^{-\frac{D-2}{2}\omega}\phi,$$

this action is Weyl invariant at the classical level by the conformal-Laplacian Weyl-covariance calculation below. In the corresponding free massless scalar QFT, the improved trace is

$$\widehat{T}^\mu{}_\mu(x) = 2(D-1)\xi_*\,\widehat{\phi}(x)\,\partial^2\widehat{\phi}(x).$$

It represents the zero class in the local-operator vector space modulo the free equation-of-motion ideal. Equivalently, for local insertions $\mathcal{O}_i(y_i)$ with $x \neq y_i$,

$$\left\langle \widehat{T}^\mu{}_\mu(x)\prod_i \mathcal{O}_i(y_i)\right\rangle = 0.$$

At coincident points the factor $\partial^2\widehat{\phi}$ generates the Schwinger–Dyson contact terms of the free Gaussian measure. $\square$

Weyl covariance of the conformal Laplacian. Let $D > 2$, set

$$\alpha = \frac{D-2}{2}, \qquad \xi_* = \frac{D-2}{4(D-1)},$$

and define the conformal Laplacian

$$P_g = -\nabla_g^2 + \xi_* R(g).$$

Under a Weyl transformation

$$g'_{\mu\nu} = \mathrm{e}^{2\omega} g_{\mu\nu}, \qquad \phi' = \mathrm{e}^{-\alpha\omega}\phi,$$

one has

$$P_{g'}\phi' = \mathrm{e}^{-\frac{D+2}{2}\omega} P_g \phi. \tag{58.22}$$

Consequently, on a closed manifold or with boundary conditions eliminating boundary terms,

$$\frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,\phi\,P_g\phi = \frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g'}\,\phi'\,P_{g'}\phi'.$$

This is precisely the Weyl invariance of (58.21). For a scalar f, the Laplacian transforms as

$$\nabla_{g'}^2 f = \mathrm{e}^{-2\omega}\left(\nabla_g^2 f + (D-2)\nabla^\mu\omega\,\nabla_\mu f\right),$$

where covariant derivatives and contractions on the right use g. Apply this to $f = \mathrm{e}^{-\alpha\omega}\phi$. Since $2\alpha = D-2$, the first derivative terms in ϕ cancel:

$$\nabla_{g'}^2(\mathrm{e}^{-\alpha\omega}\phi) = \mathrm{e}^{-(\alpha+2)\omega}\left(\nabla_g^2\phi - \alpha(\nabla_g^2\omega)\phi - \alpha^2(\nabla\omega)^2\phi\right).$$

The scalar curvature transforms by

$$R(g') = \mathrm{e}^{-2\omega}\left(R(g) - 2(D-1)\nabla_g^2\omega - (D-1)(D-2)(\nabla\omega)^2\right).$$

Therefore

$$\begin{aligned} P_{g'}(\mathrm{e}^{-\alpha\omega}\phi) = \mathrm{e}^{-(\alpha+2)\omega}\big[&-\nabla_g^2\phi + \xi_* R(g)\phi \\ &+ (\alpha - 2(D-1)\xi_*)(\nabla_g^2\omega)\phi \\ &+ (\alpha^2 - (D-1)(D-2)\xi_*)(\nabla\omega)^2\phi\big]. \end{aligned}$$

The two coefficients involving ω vanish exactly for $\xi_* = (D-2)/(4(D-1))$. Since $\alpha+2 = (D+2)/2$, this proves (58.22). Finally, $\sqrt{g'} = \mathrm{e}^{D\omega}\sqrt{g}$, and

$$D - \alpha - \frac{D+2}{2} = 0.$$

The integrated quadratic form is therefore invariant. Integrating $\phi(-\nabla_g^2)\phi$ by parts rewrites the quadratic form as (58.21).

Figure 58.1 records the dimension-dependent coefficient in the conformal Laplacian. The displayed values are a useful check on the normalization: in $D = 4$ the coefficient is $1/6$, while $D = 2$ is degenerate because the free scalar has $\Delta_\phi = 0$ and the curvature improvement coefficient vanishes.

D	$\Delta_\phi = (D-2)/2$	$\xi_* = (D-2)/[4(D-1)]$	conformal Laplacian P_g
2	0	0	$-\nabla_g^2$
3	1/2	1/8	$-\nabla_g^2 + \frac{1}{8}R(g)$
4	1	1/6	$-\nabla_g^2 + \frac{1}{6}R(g)$
5	3/2	3/16	$-\nabla_g^2 + \frac{3}{16}R(g)$
6	2	1/5	$-\nabla_g^2 + \frac{1}{5}R(g)$
10	4	2/9	$-\nabla_g^2 + \frac{2}{9}R(g)$

Figure 58.1: The conformal improvement coefficient is fixed by Weyl covariance of $P_g = -\nabla_g^2 + \xi_* R(g)$: $P_{e^{2\omega}g}(e^{-(D-2)\omega/2}\phi) = e^{-(D+2)\omega/2}P_g\phi$. Varying the metric-coupled action with this P_g, and only then setting $g_{\mu\nu} = \delta_{\mu\nu}$, gives the improved flat-space stress tensor.

58.9 Ward Identities with Stress-Tensor Insertions

Proposition 58.11 (Stress-tensor Ward identity with spinning insertions)**.** *Let*

$$X = \mathcal{O}_{A_1}(x_1)\cdots\mathcal{O}_{A_n}(x_n), \qquad X_i[B] = \mathcal{O}_{A_1}(x_1)\cdots B(x_i)\cdots\mathcal{O}_{A_n}(x_n),$$

where $B(x_i)$ replaces the i-th operator. Suppose first that each $\mathcal{O}_{A_i}$ belongs to a finite-dimensional source representation of linearized diffeomorphisms, with matrices $(\mathsf{R}^{\mu\nu}_{(i)})_{A_i}{}^{B_i}$ defined by the linear term

$$\delta_\xi\mathcal{O}_{A_i}(x_i) = \xi^\rho(x_i)\partial_\rho\mathcal{O}_{A_i}(x_i) + \partial_\mu\xi_\nu(x_i)(\mathsf{R}^{\mu\nu}_{(i)})_{A_i}{}^{B_i}\mathcal{O}_{B_i}(x_i).$$

For scalar operators the matrices $\mathsf{R}^{\mu\nu}_{(i)}$ vanish. For operators carrying only orthonormal-frame spin, the antisymmetric part of $\mathsf{R}^{\mu\nu}_{(i)}$ is the usual spin matrix $\mathsf{S}^{\mu\nu}_{(i)} = -\mathsf{S}^{\nu\mu}_{(i)}$, and the symmetric traceless part is absent. The source-functional identity (56.27), differentiated with respect to the corresponding sources and then evaluated at zero source, gives the smeared flat-space Ward identity

$$\int d^D x\, \partial_\mu\xi_\nu(x)\, \langle T^{\mu\nu}(x)X\rangle = \sum_{i=1}^n \xi^\nu(x_i)\partial_{x_i^\nu}\langle X\rangle + \sum_{i=1}^n \partial_\mu\xi_\nu(x_i)\left\langle X_i[(\mathsf{R}^{\mu\nu}_{(i)})_{A_i}{}^{B_i}\mathcal{O}_{B_i}]\right\rangle. \qquad (58.23)$$

Equivalently, as an identity of distributions in x,

$$\partial_\mu\langle T^{\mu\nu}(x)X\rangle = -\sum_{i=1}^n \delta(x-x_i)\partial_{x_i^\nu}\langle X\rangle + \sum_{i=1}^n \partial_\mu\left[\delta(x-x_i)\left\langle X_i[(\mathsf{R}^{\mu\nu}_{(i)})_{A_i}{}^{B_i}\mathcal{O}_{B_i}]\right\rangle\right]. \qquad (58.24)$$

There is no omitted spin term in (58.24). For scalar insertions $\mathsf{R}^{\mu\nu}_{(i)} = 0$, and the familiar translation Ward identity is obtained by deleting the second line. For spinning insertions that second line is part of the identity. To see its relation to the conformal Ward identities used later, let the i-th operator be a primary in an orthonormal-frame spin representation ρ_i, with infinitesimal spin matrices $S^{(i)}_{\mu\nu}$ in the convention of Chapter 61. If ϵ^μ is a conformal Killing vector and

$$J^\mu_\epsilon = T^\mu{}_\nu\epsilon^\nu, \qquad \sigma_\epsilon = \frac{1}{D}\partial_\rho\epsilon^\rho, \qquad \omega_{\epsilon,\mu\nu} = \frac{1}{2}(\partial_\nu\epsilon_\mu - \partial_\mu\epsilon_\nu),$$

then smearing (58.23) *with* $\xi = \epsilon$ *and adding the trace Ward identity gives*

$$\partial_\mu \langle J^\mu_\epsilon(x) X \rangle = \sum_{i=1}^{n} \delta^{(D)}(x - x_i) \left\langle X_i [\mathcal{D}^{(i)}_\epsilon \mathcal{O}_{A_i}] \right\rangle + \sigma_\epsilon(x) \mathcal{A}(x; X), \tag{58.25}$$

where

$$\mathcal{D}^{(i)}_\epsilon \mathcal{O}_{i,a_i}(x_i) = -\epsilon^\rho(x_i) \partial_{x_i^\rho} \mathcal{O}_{i,a_i}(x_i) - \Delta_i \sigma_\epsilon(x_i) \mathcal{O}_{i,a_i}(x_i) - \frac{\mathrm{i}}{2} \omega^{\mu\nu}_\epsilon(x_i) (S^{(i)}_{\mu\nu})_{a_i}{}^{b_i} \mathcal{O}_{i,b_i}(x_i). \tag{58.26}$$

Equations (58.25) *and* (58.26) *are the same source Ward identity as* (61.7)*; the derivative contact term in* (58.24) *is precisely what becomes the local rotation term proportional to* $\omega^{\mu\nu}_\epsilon S^{(i)}_{\mu\nu}$*. Thus one may use the scalar translation formula only after verifying that all insertions are scalars, or after the spin contact term has been carried along separately. For a basis of composite operators that mixes under the linearized diffeomorphism, the displayed representation matrix is replaced by the corresponding source-coordinate representation matrix. The trace Ward identity similarly records the response under Weyl transformations. At a flat-space conformal fixed point, the separated-point trace insertion vanishes, while contact terms encode scaling dimensions and conformal transformations of the inserted operators.*

The charge form of the Ward identity follows by integrating $T^\mu{}_\nu \epsilon^\nu$ *over a small sphere around each insertion. This is the bridge between the local stress-tensor equation and the conformal algebra acting on local operators.*

Chapter 59

Radial Quantization and the State-Operator Correspondence

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

59.1 Local Correlators as the Primary Data

Let $\widehat{\mathcal{O}}_I(x)$ denote a Lorentzian local operator-valued distribution on $\mathbb{R}^{1,D-1}$. The label I includes all internal and spin indices. Local commutativity means that for spacelike separated points

$$[\widehat{\mathcal{O}}_I(x), \widehat{\mathcal{O}}_J(y)]_\pm = 0, \qquad (x-y)^2 > 0,$$

where the graded commutator is used when fermionic operators are present. For a vacuum vector $|\Omega\rangle$, the unordered Lorentzian correlation functions are the Wightman distributions

$$W_{I_1\cdots I_n}(x_1, \dots, x_n) = \langle\Omega|\widehat{\mathcal{O}}_{I_1}(x_1)\cdots\widehat{\mathcal{O}}_{I_n}(x_n)|\Omega\rangle.$$

Time-ordered Green functions are

$$G^T_{I_1\cdots I_n}(x_1, \dots, x_n) = \langle\Omega|T\{\widehat{\mathcal{O}}_{I_1}(x_1)\cdots\widehat{\mathcal{O}}_{I_n}(x_n)\}|\Omega\rangle.$$

The corresponding Euclidean Schwinger functions are denoted by

$$S_{I_1\cdots I_n}(x_1^E, \dots, x_n^E) = \left\langle\mathcal{O}_{I_1}(x_1^E)\cdots\mathcal{O}_{I_n}(x_n^E)\right\rangle_E.$$

Hypothesis 59.1 (Radial reconstruction data)**.** The state–operator construction in this chapter is used for a Euclidean Schwinger hierarchy $S_{I_1\cdots I_n}$ satisfying the following conditions.

1. The S_n are distributional correlation functions of renormalized local operators on $\mathbb{R}^D$, with Euclidean covariance, graded permutation symmetry, and stated contact-term prescriptions at coincident points.

2. Let $\vartheta(x) = x/|x|^2$ be reflection through the unit sphere. The operator basis carries an antilinear reflection map Θ_{rad} that acts on a smeared insertion supported in the unit ball by applying ϑ, the appropriate Weyl factor, and the spin representation of the reflected orthonormal frame. For every finite linear combination F of such inside-supported insertions,
$$\langle F, F\rangle_{\mathrm{rad}} = \langle \Theta_{\mathrm{rad}} F\, F\rangle_E \geq 0.$$
The radial Hilbert space is the completion of this positive semidefinite form after quotienting by its null space.

3. The one-parameter group generated by dilatations acts on this Hilbert space by a nonnegative self-adjoint operator D_{rad}. Its spectrum on the finite-energy subspace is discrete, and for every $E < \infty$ the spectral subspace with $D_{\mathrm{rad}} \leq E$ is finite-dimensional. In particular every eigenspace has finite multiplicity, and the eigenvalue 0 subspace is spanned by the vacuum vector. Thus the conformally invariant vacuum in the radial Hilbert space is unique.

4. For each local operator $\mathcal{O}_I$ used below, the radial limit of the insertion at the origin defines a finite-energy vector. Descendant vectors are obtained by applying the translation generators, equivalently by derivatives of the insertion. The local operator basis is radially complete: after quotienting by the null radial two-point pairing, origin insertions of local operators and their descendants span precisely the finite-energy subspace
$$\mathcal{H}_{\mathrm{fin}} = \bigoplus_{\lambda<\infty} \ker(D_{\mathrm{rad}} - \lambda).$$
Equivalently, every finite-energy eigenvector is assumed to admit local radial boundary-value data of the kind specified in Definition 59.7 below. This is an existence assertion about distributional extensions at the origin; the theorem later in this chapter constructs the corresponding local-operator class from such data.

5. The Euclidean correlators are related to Lorentzian Wightman distributions by the standard tube-domain analytic continuation, with the Lorentzian distributions satisfying the spectrum condition and local commutativity. The flat-space vacuum correlators obey cluster decomposition: for two finite products of separated local insertions,
$$A(x_1, \ldots, x_m) = \mathcal{O}_{I_1}(x_1) \cdots \mathcal{O}_{I_m}(x_m), \qquad B(y_1, \ldots, y_n) = \mathcal{O}_{J_1}(y_1) \cdots \mathcal{O}_{J_n}(y_n),$$
and for a Euclidean vector a such that no coincident-point singularity is crossed along the translated configuration,
$$\lim_{\lambda\to+\infty} \langle A(x_1 + \lambda a, \ldots, x_m + \lambda a)\, B(y_1, \ldots, y_n)\rangle_E = \langle A(x_1, \ldots, x_m)\rangle_E\, \langle B(y_1, \ldots, y_n)\rangle_E.$$

Under Hypothesis 59.1, radial reflection positivity gives the inner product, D_{rad} gives the cylinder Hamiltonian, and local insertions at the origin give the complete finite-energy state space.

The completeness clause is the load-bearing part of the state–operator correspondence. Reflection positivity and the radial spectrum construct a Hilbert space and a Hamiltonian, but by themselves they only give an injection from local origin insertions into cylinder states. Surjectivity requires the additional local boundary-value assertion above: a finite-energy state must determine distributional correlators with an insertion at the origin whose contact terms, covariance, scaling law, and descendant Ward identities are those of a local operator. This condition is extra local distributional structure on top of the existence of a Hilbert-space vector.

The nonnegativity of D_{rad} in Hypothesis 59.1 is the radial spectrum condition for the vacuum representation. It is not the statement, proved later from descendant Gram positivity, that non-identity primary operators obey the unitarity lower bounds of Chapter 62. The same nonnegative self-adjoint operator D_{rad} is used in the radial OPE convergence theorem of Chapter 64.

These three versions are related by analytic continuation whenever the Lorentzian theory satisfies the spectral and locality hypotheses needed for the tube-domain construction. For an ordered Wightman product, continue

$$x_k^0 = -\mathrm{i}x_{E,k}^D$$

through a domain in which the imaginary parts remain ordered. With the convention $x^0 = -\mathrm{i}\tau_E$, the ordering

$$\operatorname{Im} x_1^0 < \operatorname{Im} x_2^0 < \cdots < \operatorname{Im} x_n^0$$

corresponds to

$$\tau_{E,1} > \tau_{E,2} > \cdots > \tau_{E,n}.$$

Here τ_E denotes Euclidean time inherited from Lorentzian analytic continuation. It is distinct from the cylinder coordinate $\tau = \log r$ introduced below. The Euclidean correlator is the boundary value obtained after this rotation. Time-ordered Green functions arise from a contour whose real-time projection places the insertions in chronological order.

Definition 59.2 (Status of the complex-time contour functional)**.** When a Lagrangian path-integral representative is available, the contour notation used for this analytic continuation is tied to the Lorentzian oscillatory convention of Definition 11.6 in Chapter 11. Choose a finite spatial regulator Λ and a finite subdivision P of the complex time contour C. Let $\mathcal{C}_{\Lambda,P}$ be the resulting finite-dimensional space of retained bosonic variables, together with the corresponding finite odd variables when fermions are present. The contour action is

$$S_{C,\Lambda,P}[\Phi] = \int_C \mathrm{d}x^0 \int \mathrm{d}^{D-1}\mathbf{x}\, \mathcal{L}_{\Lambda,P}(\Phi, \partial\Phi),$$

where the displayed integral is shorthand for the time-sliced finite regulator action along C. For $F = \prod_i \mathcal{O}_i(x_i)$, the finite-regulator contour functional is

$$I_{C,\Lambda,P}(F) = \int_{\mathcal{C}_{\Lambda,P}} d\nu_{\Lambda,P}(\Phi)\, \exp\{\mathrm{i}S_{C,\Lambda,P}[\Phi]\}\, F_{\Lambda,P}(\Phi). \tag{59.1}$$

For even variables $d\nu_{\Lambda,P}$ is an ordinary finite-dimensional reference measure. For odd variables it is a Berezin density. For gauge fields it is defined only after the finite-regulator gauge-fixing or BV data and boundary conditions have been supplied. If C contains Lorentzian segments, (59.1) is an oscillatory integral or oscillatory distribution, with Abel/Fourier boundary value or stationary-phase meaning as in Definition 11.6; Euclidean segments contribute the corresponding heat-kernel damping in the same time-sliced regulator.

The continuum symbol

$$\left\langle \prod_i \mathcal{O}_i(x_i) \right\rangle_C := \lim_{\Lambda,P} I_{C,\Lambda,P}\left(\prod_i \mathcal{O}_i(x_i)\right) \tag{59.2}$$

means the specified distributional boundary value in a construction where the limit exists. In a perturbative construction it denotes the corresponding formal asymptotic expansion obtained from the same regulator data. In an abstract CFT presentation the primary object is the Schwinger/Wightman hierarchy and its tube-domain analytic continuation; the contour functional is a representation of those boundary values.

Figure 59.1 records the boundary-value interpretation of complex-time contours used here: Euclidean, Lorentzian, and radially ordered correlators are different boundary presentations of the same analytic hierarchy once the relevant continuation exists.

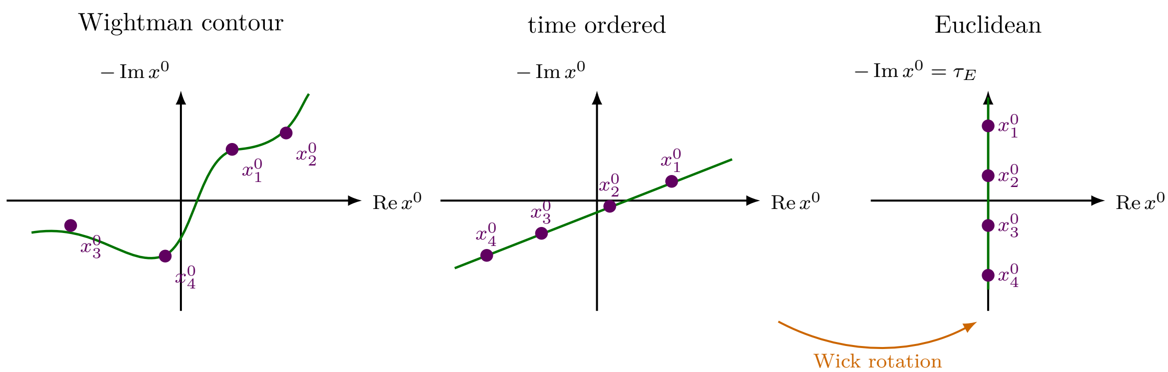


Figure 59.1: Different real-time orderings and Euclidean Schwinger functions are boundary values of the same analytic correlation functions, reached by deforming the complex-time contour without crossing singularities.

Remark 59.3 (Observables at a conformal fixed point). At a conformal fixed point, the data used in this volume are local operators, their correlation functions, and the Hilbert-space structure obtained from radial quantization. A separate scattering theory requires asymptotic particle states. Many conformal theories have no mass gap and no isolated massive particle sectors, so their intrinsic observable data are organized by local correlations rather than an S-matrix. Four-dimensional Yang–Mills theory illustrates a different situation: the classical action has no dimensionful coupling, but quantum renormalization produces Λ_{QCD} and a nonzero trace anomaly, so the quantum theory is not a conformal field theory.

59.2 The Weyl Map to the Cylinder

Let x^μ be Euclidean coordinates on $\mathbb{R}^D$, and write

$$r = |x|, \qquad x^\mu = r n^\mu, \qquad n^\mu n_\mu = 1.$$

The flat metric is

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2 \mathrm{d}\Omega_{D-1}^2,$$

where $\mathrm{d}\Omega_{D-1}^2$ is the unit metric on S^{D-1}. Define

$$\tau = \log r.$$

Then

$$\frac{\mathrm{d}s^2}{r^2} = \mathrm{d}\tau^2 + \mathrm{d}\Omega_{D-1}^2. \tag{59.3}$$

Thus the punctured space $\mathbb{R}^D \setminus \{0\}$ is Weyl equivalent to the Euclidean cylinder

$$\mathbb{R}_\tau \times S^{D-1}.$$

The origin corresponds to $\tau = -\infty$, and infinity corresponds to $\tau = +\infty$.

The use of this Weyl map assumes that the flat-space conformal theory has been extended to background metrics with a specified Weyl transformation law for renormalized local operators. In even dimensions a Weyl anomaly may add local curvature terms to the generating functional. On the punctured flat space and cylinder comparison below, such terms affect contact and vacuum normalization data; the separated-point operator map is fixed after a choice of renormalized operator basis.

Figure 59.2 displays the Weyl map from punctured flat space to the cylinder, with radial distance becoming cylinder time $\tau = \log r$.

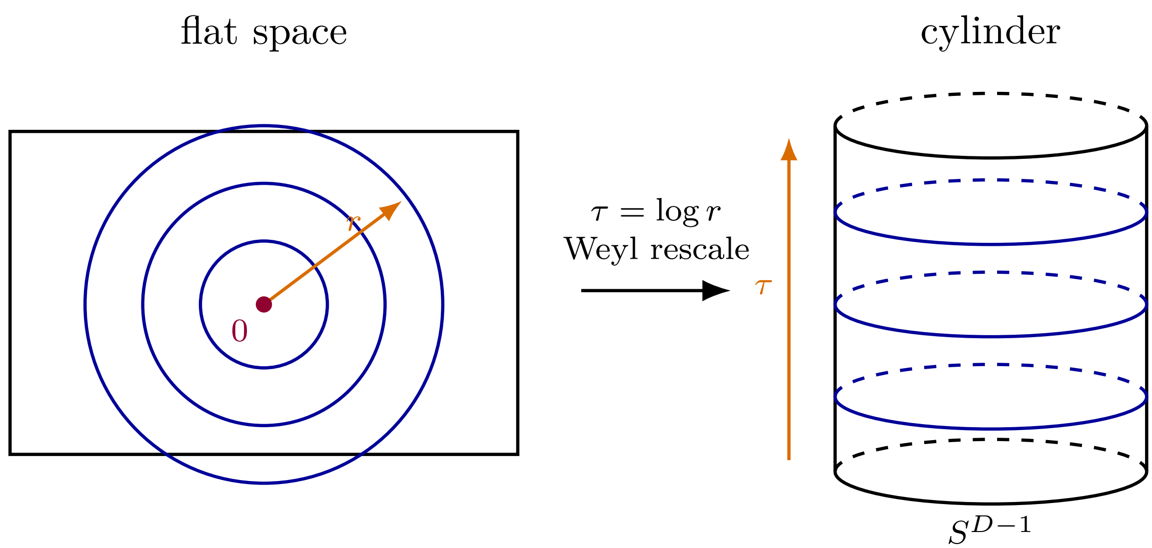


Figure 59.2: Radial quantization identifies dilatations on $\mathbb{R}^D$ with translations in cylinder time $\tau = \log r$.

59.3 Flat, Conformal, and Cylinder Vacua

Three vacuum vectors enter the radial construction, and their equality is a statement with hypotheses rather than a convention. The Lorentzian flat-space vacuum will be denoted by $|\Omega_\mathrm{P}\rangle$. It is the normalized Poincare-invariant vector in the Wightman Hilbert space on $\mathbb{R}^{1,D-1}$, with

$$P_\mu|\Omega_\mathrm{P}\rangle = 0, \qquad M_{\mu\nu}|\Omega_\mathrm{P}\rangle = 0,$$

where P_μ are the translation generators and $M_{\mu\nu}$ are the Lorentz generators. A conformal vacuum is a normalized vector $|\Omega_{\mathrm{conf}}\rangle$ invariant under the connected conformal group, or equivalently on a common invariant domain,

$$P_\mu|\Omega_{\mathrm{conf}}\rangle = M_{\mu\nu}|\Omega_{\mathrm{conf}}\rangle = D|\Omega_{\mathrm{conf}}\rangle = K_\mu|\Omega_{\mathrm{conf}}\rangle = 0.$$

The cylinder vacuum $|\Omega_0\rangle_{\mathrm{cyl}}$ is the normalized zero-eigenvalue state of the radial Hamiltonian D_{rad} in the Hilbert space on S^{D-1}:

$$D_{\mathrm{rad}}|\Omega_0\rangle_{\mathrm{cyl}} = 0.$$

It is represented by the empty Euclidean ball, or equivalently by insertion of the identity operator at the origin.

Proposition 59.4 (Identification of the flat and conformal vacua)**.** *Assume that the connected conformal group, or its universal cover, is implemented by strongly continuous unitaries on the Wightman Hilbert space on a common dense invariant domain containing $|\Omega_{\mathrm{P}}\rangle$. Assume also that the translation-invariant normalized vector is unique and that the generators obey the conformal Lie algebra on this domain. Then the flat Poincare vacuum is invariant under the connected conformal group. Thus $|\Omega_{\mathrm{conf}}\rangle = |\Omega_{\mathrm{P}}\rangle$, up to the irrelevant overall phase of a normalized vector.*

Proof. Let $U_D(s) = \exp(\mathrm{i}sD)$ be the unitary one-parameter group of dilatations. The relation

$$U_D(s)^{-1}P_\mu U_D(s) = \mathrm{e}^s P_\mu$$

implies that $U_D(s)|\Omega_{\mathrm{P}}\rangle$ is again annihilated by all translations. By uniqueness of the normalized translation-invariant vector,

$$U_D(s)|\Omega_{\mathrm{P}}\rangle = \mathrm{e}^{\mathrm{i}\alpha(s)}|\Omega_{\mathrm{P}}\rangle.$$

Strong continuity makes $\alpha(s)$ linear, so $D|\Omega_{\mathrm{P}}\rangle = d_0|\Omega_{\mathrm{P}}\rangle$ for some real number d_0. The number d_0 is forced to vanish by the conformal algebra: because $P_\nu|\Omega_{\mathrm{P}}\rangle = 0$ and P_ν is self-adjoint,

$$0 = \langle\Omega_{\mathrm{P}}|[P_\nu, K_\mu]|\Omega_{\mathrm{P}}\rangle = 2\eta_{\mu\nu}d_0.$$

Thus $D|\Omega_{\mathrm{P}}\rangle = 0$, which is the representation-theoretic normalization in which the identity operator has dimension zero.

Now use the conformal commutator

$$[P_\nu, K_\mu] = 2\eta_{\mu\nu}D + 2M_{\nu\mu}$$

with Lorentzian metric η. Since P_ν, D, and $M_{\nu\mu}$ annihilate $|\Omega_{\mathrm{P}}\rangle$,

$$P_\nu K_\mu|\Omega_{\mathrm{P}}\rangle = [P_\nu, K_\mu]|\Omega_{\mathrm{P}}\rangle = 0.$$

Therefore $K_\mu|\Omega_{\mathrm{P}}\rangle$ is translation-invariant. Uniqueness of the translation-invariant vector gives

$$K_\mu|\Omega_{\mathrm{P}}\rangle = c_\mu|\Omega_{\mathrm{P}}\rangle$$

for constants c_μ. Lorentz invariance of $|\Omega_{\mathrm{P}}\rangle$ and the vector transformation law of K_μ force c_μ to be a Lorentz invariant vector; hence $c_\mu = 0$. Thus the vacuum is annihilated by translations, Lorentz generators, dilatations, and special conformal generators, which are the infinitesimal generators of the connected conformal group. □

The relation between $|\Omega_{\mathrm{conf}}\rangle$ and $|\Omega_0\rangle_{\mathrm{cyl}}$ is supplied by the Weyl map $\mathbb{R}^D \setminus \{0\} \to \mathbb{R}_\tau \times S^{D-1}$. The Euclidean radial Hilbert space is constructed from the same flat-space vacuum correlators by cutting the Schwinger functions on spheres. Under this construction the identity operator at the origin creates the zero-energy state, so the common vacuum is identified as

$$|\Omega_{\mathrm{P}}\rangle = |\Omega_{\mathrm{conf}}\rangle \longleftrightarrow |\Omega_0\rangle_{\mathrm{cyl}}.$$

This identification fixes the notation used below: $|\Omega\rangle$ denotes this common vacuum whenever the flat or cylinder realization is clear from context.

The literal cutting-and-sewing meaning of radial quantization is a Kontsevich–Segal type Euclidean field-theoretic structure. In that language, a closed Euclidean hypersurface such as S^{D-1} is assigned a state space, a Euclidean bordism such as an annulus or punctured ball is assigned a linear amplitude between boundary state spaces, and gluing of bordisms is represented by composition of amplitudes. At this level the assigned state space should not be assumed to be a Hilbert space. In a general Euclidean field theory it is more naturally a locally convex topological vector space, often a Fréchet or nuclear space in analytic formulations, and the bordism amplitudes are continuous linear maps or distributional pairings.

Only for a unitary CFT does the radial cutting construction produce the Hilbert space used in the literal state–operator correspondence, and the extra ingredient is the inversion pairing. The map $x \mapsto x/|x|^2$ exchanges the inside and outside of a sphere, identifies the boundary sphere with its reflected copy up to the conformal Weyl factor, and turns reflection positivity of the doubled configuration into a positive semidefinite Hermitian form. Only after quotienting the null space and completing this form does one obtain the radial Hilbert space used below. Thus the hypotheses in this chapter use the part of the Kontsevich–Segal cutting/gluing structure needed for a CFT together with the inversion pairing: the state space on S^{D-1}, the positive radial inner product, the annular semigroup generated by D_{rad}, and the sewing identities for cylinder slabs. The complete Kontsevich–Segal axiom system and its construction status are developed in Chapter 13. The use here is the CFT specialization: literal radial quantization is first a hypersurface-cutting and bordism-gluing construction, and in a CFT the inversion pairing upgrades it to a Hilbert-space construction.

There remains a normalization convention. If a cylinder Hamiltonian is defined by integrating a chosen renormalized stress tensor on the curved cylinder, Weyl anomaly terms and local curvature counterterms can add a scalar Casimir energy to that operator. The radial Hamiltonian D_{rad} used in this chapter is the shifted generator whose identity state has eigenvalue zero. Equivalently the vacuum functional is normalized so that $\langle 1\rangle = 1$ on flat space and on the cylinder, with anomaly-dependent local curvature terms fixed once and for all. Separated-point operator correlators are then related by (59.9).

59.4 Operators Under the Cylinder Map

For a scalar primary $\mathcal{O}(x)$ of dimension Δ, define the cylinder operator

$$\widetilde{\mathcal{O}}(\tau, n) = \mathrm{e}^{\Delta\tau}\mathcal{O}(\mathrm{e}^{\tau} n). \tag{59.4}$$

This is the ordinary Weyl transformation of a scalar primary under $\mathrm{d}s^2 \mapsto r^{-2}\mathrm{d}s^2$.

The spin factor is the local orthonormal-frame rotation induced by the same map. Work on a coordinate patch $U \subset S^{D-1}$ with a smooth oriented orthonormal tangent frame

$$E_A^\mu(n), \qquad A = 1, \ldots, D-1, \qquad n_\mu E_A^\mu(n) = 0, \qquad E_A^\mu(n) E_{B\mu}(n) = \delta_{AB}.$$

The cylinder metric is $\widehat{g} = r^{-2}\delta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$. An orthonormal coframe for $\widehat{g}$, pulled back to the punctured flat space, is

$$\widehat{e}^{\hat{\tau}}{}_\mu(x) = r^{-1} n_\mu, \qquad \widehat{e}^{\hat{A}}{}_\mu(x) = r^{-1} E_{A\mu}(n). \tag{59.5}$$

Removing the common Weyl scale gives the $SO(D)$ rotation from the flat Cartesian orthonormal frame to the cylinder orthonormal frame,

$$R^{\hat{a}}{}_\mu(n) := r\,\widehat{e}^{\hat{a}}{}_\mu(x), \qquad R^{\hat{\tau}}{}_\mu(n) = n_\mu, \qquad R^{\hat{A}}{}_\mu(n) = E_{A\mu}(n), \tag{59.6}$$

so $R^{\hat{a}}{}_\mu R^{\hat{b}\mu} = \delta^{\hat{a}\hat{b}}$. On overlaps of two patches the frames E_A differ by an $SO(D-1)$ rotation; the formulas below transform by the corresponding tensor or spin representation.

For a covariant vector primary $V_\mu(x)$ of dimension Δ, the cylinder orthonormal-frame components are

$$\widetilde{V}_{\hat{a}}(\tau, n) = \mathrm{e}^{\Delta\tau} R_{\hat{a}}{}^\mu(n)\, V_\mu(\mathrm{e}^\tau n). \tag{59.7}$$

Equivalently,

$$V_\mu(\mathrm{e}^\tau n) = \mathrm{e}^{-\Delta\tau} R_\mu{}^{\hat{a}}(n)\, \widetilde{V}_{\hat{a}}(\tau, n).$$

The associated spin-one state map is obtained by inserting the flat operator at the origin:

$$|V_\mu\rangle := V_\mu(0)|\Omega\rangle, \qquad \lim_{\tau\to-\infty} \mathrm{e}^{-\Delta\tau}\, \widetilde{V}_{\hat{a}}(\tau, n)|\Omega\rangle = R_{\hat{a}}{}^\mu(n)|V_\mu\rangle. \tag{59.8}$$

The direction dependence in the last expression is exactly the rotation between the local cylinder frame at n and the fixed flat spin index at the origin. For a primary in a representation ρ of $SO(D)$, replace $R_{\hat{a}}{}^\mu$ in (59.7) by $\rho(R(n))$.

For a general renormalized local operator basis, the map can mix operators with the same quantum numbers when local counterterms are changed. The statement used in radial quantization is that, after fixing this basis, the correlation functions on the two Weyl-related backgrounds agree:

$$\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_{\mathbb{R}^D} = \left\langle \widetilde{\mathcal{O}}_1(\tau_1, n_1) \cdots \widetilde{\mathcal{O}}_n(\tau_n, n_n) \right\rangle_{\mathbb{R}\times S^{D-1}}, \tag{59.9}$$

for separated points $x_i = \mathrm{e}^{\tau_i} n_i$, up to the specified local curvature and contact terms.

59.5 States from Local Operators

Let

$$B_R = \{x \in \mathbb{R}^D : |x| < R\}, \qquad \Sigma_R = \partial B_R = S_R^{D-1}.$$

The state prepared by a local operator at the origin is defined, in the radial reconstruction used in this chapter, by its radial pairings with the dense states prepared by insertions inside B_R. If A is a finite linear combination of renormalized local insertions supported in $B_R \setminus \{0\}$, and Θ_R denotes reflection through Σ_R,

$$\Theta_R(x) = \frac{R^2 x}{|x|^2},$$

then the vector $|\mathcal{O};R\rangle$ is characterized on the dense finite-energy domain by

$$\langle A,\mathcal{O};R\rangle_{\mathrm{rad}}=\langle\Theta_R A\,\mathcal{O}(0)\rangle_E\,. \tag{59.10}$$

Here $\Theta_R A$ includes the Weyl factors and spin-frame rotations of the reflected insertions. Equation (59.10) is the definition used for an abstract CFT supplied by Schwinger distributions. It uses the radial reflection-positive pairing and the null quotient of Hypothesis 59.1; it does not require a continuum field-configuration measure.

Definition 59.5 (Regulated ball preparation)**.** When a Lagrangian representative is used as a mnemonic or as an actual regularized construction, the ball preparation is the following finite regulator family. Map $B_R\setminus B_\rho$ to the cylinder slab $[\log\rho,\log R]\times S^{D-1}$. Choose a radial time spacing $a>0$, a finite angular spectral cutoff L for the eigenmodes of the relevant elliptic kinetic operator on S^{D-1}, and a renormalized operator representative $\mathcal{O}_{\rho,a,L}$ supported in the inner cap $|x|\le\rho$. The regulator parameter is

$$\varepsilon=(\rho,a,L),\qquad \rho\downarrow 0,\quad a\downarrow 0,\quad L\uparrow\infty. \tag{59.11}$$

At fixed ε, the boundary configuration space $\mathcal{B}_{\varepsilon,R}$ consists of the retained boundary modes on Σ_R. The finite-dimensional trace map

$$\mathrm{Tr}_{\varepsilon,R}:\mathcal{F}_{\varepsilon,R}\longrightarrow\mathcal{B}_{\varepsilon,R}$$

restricts a regulated bulk field configuration to those boundary modes. The finite-regulator measure or density on $\mathcal{F}_{\varepsilon,R}$ factors as

$$d\nu_{\varepsilon,R}(\Phi)=d\nu^{\mathrm{bulk}}_{\varepsilon,R}(\Phi\mid\varphi_\varepsilon)\,d\lambda^{\partial}_{\varepsilon,R}(\varphi_\varepsilon),\qquad \varphi_\varepsilon=\mathrm{Tr}_{\varepsilon,R}\,\Phi. \tag{59.12}$$

For even bosonic modes $d\lambda^{\partial}_{\varepsilon,R}$ is an ordinary finite-dimensional boundary measure. For odd variables it is the corresponding Berezin density, and in gauge theories it is supplied only after the finite-regulator gauge-fixing or BV data have been specified. The conditioned ball wavefunctional is

$$\Psi_{\mathcal{O},R,\varepsilon}(\varphi_\varepsilon)=\int_{\mathrm{Tr}_{\varepsilon,R}\,\Phi=\varphi_\varepsilon}d\nu^{\mathrm{bulk}}_{\varepsilon,R}(\Phi\mid\varphi_\varepsilon)\,\exp\{-S_{\varepsilon,R}[\Phi]\}\,\mathcal{O}_{\rho,a,L}. \tag{59.13}$$

The displayed conditioning is a finite-dimensional conditional integral over interior regulated variables. Gluing two regulated regions along Σ_R means pairing their boundary wavefunctionals by integration against $d\lambda^{\partial}_{\varepsilon,R}$, with the Berezin or gauge-fixed/BV analogue in the corresponding cases.

The continuum state represented by the regulated ball is the limit

$$|\mathcal{O};R\rangle=\lim_{\varepsilon\to 0}\mathfrak{R}_{\varepsilon,R}\Psi_{\mathcal{O},R,\varepsilon}, \tag{59.14}$$

whenever this limit exists in the radial Hilbert norm, or in the stated distribution topology before Hilbert completion. The map $\mathfrak{R}_{\varepsilon,R}$ embeds or identifies the finite-regulator boundary state with the regulated radial state space. In constructive positive-measure examples, (59.14) is compatible with disintegration of the constructed Euclidean field over the chosen boundary sigma-algebra or boundary Sobolev trace space. In the general CFT setting of this chapter, the mathematical state is the vector determined by (59.10), and (59.13) is a regulator-dependent representative of that vector when such a representative has been supplied.

After the Weyl map to the cylinder, changing R translates the state in τ. The state associated with $\mathcal{O}$, with the radius dependence absorbed into cylinder evolution, is denoted

$$|\mathcal{O}\rangle.$$

Definition 59.6 (Scaling operator classes and algebraic radial states)**.** A renormalized local operator class at the origin has definite scaling dimension Δ if, in all separated-point Euclidean correlators and with the stated contact-term convention, its infinitesimal dilatation Ward identity is

$$D_{\rm op}\mathcal{O}(0) = \Delta\,\mathcal{O}(0).$$

Equivalently, after the Weyl map to the cylinder, the corresponding origin-created state is an eigenvector of $D_{\rm rad}$ with eigenvalue Δ. Let $\mathcal{V}_{\rm loc}^{(\Delta)}$ be the quotient vector space of such local scaling operators, including all spin components and descendants of dimension Δ, modulo operators whose radial two-point pairing with every local operator vanishes.

The algebraic local-operator space is the direct sum over scaling dimensions,

$$\mathcal{V}_{\rm loc}^{\rm alg} = \bigoplus_{\Delta\in\mathrm{Spec}_{\rm pp}(D_{\rm rad})} \mathcal{V}_{\rm loc}^{(\Delta)}, \tag{59.15}$$

where the symbol $\oplus$ denotes finite linear combinations. The matching algebraic finite-energy subspace of the radial Hilbert space is

$$\mathcal{H}_{\rm alg} = \bigoplus_{\Delta\in\mathrm{Spec}_{\rm pp}(D_{\rm rad})} \ker(D_{\rm rad}-\Delta) \subset \mathcal{H}_{S^{D-1}}. \tag{59.16}$$

Here $\mathrm{Spec}_{\rm pp}$ denotes the point spectrum. Under Hypothesis 59.1 the eigenspaces appearing in (59.16) have finite multiplicity and $\mathcal{H}_{\rm alg}$ is dense in $\mathcal{H}_{S^{D-1}}$.

Definition 59.7 (Local radial boundary value of a state)**.** Let ψ_a, $a = 1,\dots,m$, be a finite rotation multiplet of vectors in $\ker(D_{\rm rad}-\Delta)$. The multiplet has local radial boundary-value data if the following distributions are specified.

For every ordered list of renormalized local operators $\mathcal{O}_{I_1},\dots,\mathcal{O}_{I_n}$, define on the configuration space $x_j \neq 0$, $x_i \neq x_j$, the punctured correlator

$$W^{\circ}_{\psi_a;I_1\dots I_n}(x_1,\dots,x_n) = \langle\Omega|\widetilde{\mathcal{O}}_{I_1}(\tau_1,n_1)\cdots\widetilde{\mathcal{O}}_{I_n}(\tau_n,n_n)|\psi_a\rangle_{\rm cyl}, \qquad x_j = \mathrm{e}^{\tau_j}n_j, \tag{59.17}$$

first in every radial ordering chamber, and in other chambers by the Euclidean permutation and contact prescriptions of the Schwinger hierarchy. The required data are extensions

$$W_{\psi_a;I_1\dots I_n} \in \mathcal{D}'\big((\mathbb{R}^D)^n\big) \tag{59.18}$$

of these punctured distributions such that:

1. their singular support is contained in the union of the ordinary partial diagonals $x_i = x_j$ and the origin diagonals $x_i = 0$;
2. their Euclidean covariance, spin transformation, and internal symmetry transformation laws are those of an additional insertion at the origin transforming in the same rotation and internal representation as the multiplet ψ_a;

3. their dilatation Ward identity has weight Δ at the origin, namely scaling all x_j by $\lambda > 0$ multiplies the origin insertion by $\lambda^{-\Delta}$, with the known anomalous-dimension matrix already diagonalized inside the finite-dimensional eigenspace;

4. translation of the origin insertion is represented by the descendant multiplets $P_{\mu_1} \cdots P_{\mu_k} \psi_a$, so differentiation with respect to the origin agrees, distributionally, with minus the sum of derivatives with respect to the exterior points;

5. for every radius R and every finite inside-supported insertion state $A \subset B_R \setminus \{0\}$, the exterior pairing obtained by reflecting A through Σ_R agrees with the Hilbert-space pairing,

$$W_{\psi_a;\Theta_R A} = \langle A, \psi_a \rangle_{\mathrm{rad}}. \tag{59.19}$$

The topology of the extension in (59.18) is the ordinary distribution topology on compact subsets after choosing the local contact-term convention.

Theorem 59.8 (Operator from a definite-energy radial state)**.** *Assume Hypothesis 59.1. Let $\psi_a \in \ker(D_{\mathrm{rad}} - \Delta)$ be a finite rotation multiplet with local radial boundary-value data in the sense of Definition 59.7. Then there is a unique class*

$$[\mathcal{O}_{\psi_a}] \in \mathcal{V}_{\mathrm{loc}}^{(\Delta)}$$

such that, for every exterior product of local operators, its Euclidean correlators are the distributions (59.18). Its radial state is precisely ψ_a:

$$|\mathcal{O}_{\psi_a}\rangle = \psi_a. \tag{59.20}$$

Two representatives that give the same radial state differ by a null local operator, namely by an element whose radial two-point pairing with every local operator vanishes.

Proof. The extensions (59.18) define the correlators of an additional distributional insertion supported at the origin. The singular support condition and the stated contact prescriptions give locality at all separated points and a prescribed extension across origin diagonals. The covariance and scaling Ward identities place this insertion in the Δ-eigenspace of the local dilatation action, with the specified spin and internal quantum numbers, and the translation condition identifies its descendants with the radial states $P_{\mu_1} \cdots P_{\mu_k} \psi_a$. Thus the data define a class $[\mathcal{O}_{\psi_a}] \in \mathcal{V}_{\mathrm{loc}}^{(\Delta)}$.

The radial state of this class is computed by pairing it with the dense set of inside-supported insertion states. For every such A, the defining pairing (59.10) and the boundary-value condition (59.19) give

$$\langle A, \mathcal{O}_{\psi_a}; R \rangle_{\mathrm{rad}} = \langle A, \psi_a \rangle_{\mathrm{rad}}.$$

Since these A's are dense after the radial null quotient, the two radial vectors agree. If two local insertions have the same radial state, their difference pairs to zero with every A, hence with every local operator by radial reflection and density; this is precisely the null equivalence used in the definition of $\mathcal{V}_{\mathrm{loc}}^{(\Delta)}$. □

Thus this theorem proves a conditional construction: local radial boundary-value data determine a local-operator class. The separate radial-completeness clause of Hypothesis 59.1 asserts that every finite-energy eigenspace vector has such data. Only after that clause is imposed can one conclude that the operator-from-state map is surjective on each eigenspace.

Construction 59.9 (cylinder sewing and Heisenberg operators)**.** Let

$$\Sigma_\tau = \{\tau\} \times S^{D-1}$$

be a constant-cylinder-time slice, and write

$$H_{\mathrm{cyl}} = D_{\mathrm{rad}}$$

for the nonnegative self-adjoint generator of translations in τ. For a cylinder local operator $X(n)$ placed on the slice Σ_0, its Euclidean Heisenberg translate is the operator on the common finite-energy domain

$$X(\tau, n) = \mathrm{e}^{\tau H_{\mathrm{cyl}}} X(0, n) \mathrm{e}^{-\tau H_{\mathrm{cyl}}}. \tag{59.21}$$

For separated insertions

$$\tau_+ > \tau_1 > \tau_2 > \cdots > \tau_n > \tau_-,$$

and boundary states v_- on Σ_{τ_-} and v_+ on Σ_{τ_+}, the cylinder amplitude obtained by cutting along all intermediate slices is

$$\begin{aligned}
&\mathcal{A}_{\tau_+,\tau_-}(v_+; X_1, \ldots, X_n; v_-) \\
&\quad = \left\langle v_+, \mathrm{e}^{-(\tau_+-\tau_1)H_{\mathrm{cyl}}} X_1(0, n_1) \mathrm{e}^{-(\tau_1-\tau_2)H_{\mathrm{cyl}}} \cdots X_n(0, n_n) \mathrm{e}^{-(\tau_n-\tau_-)H_{\mathrm{cyl}}} v_- \right\rangle.
\end{aligned} \tag{59.22}$$

Equation (59.22) is the Hilbert-space form of Euclidean sewing. In a functional-integral presentation it is obtained by integrating over the boundary field data on each cut Σ_{τ_j}; in the radial reconstruction presentation it is the composition law for the positive-time semigroup on the null quotient. The two descriptions give the same sesquilinear form because both are defined by the same Schwinger correlation hierarchy.

Taking $v_+ = \Omega$ and using

$$\langle \Omega | \mathrm{e}^{-sH_{\mathrm{cyl}}} = \langle \Omega |, \qquad s \geq 0,$$

turns (59.22) into the Euclidean Heisenberg matrix element

$$\langle \Omega | X_1(\tau_1, n_1) \cdots X_n(\tau_n, n_n) | v \rangle = \mathcal{A}_{\tau_+,\tau_-}\big(\Omega; X_1, \ldots, X_n; \mathrm{e}^{-\tau_- H_{\mathrm{cyl}}} v\big), \tag{59.23}$$

for every finite-energy vector v and every lower cut $\tau_- < \tau_n$. The right side is independent of the auxiliary cuts after the boundary state is transported as displayed. Thus the cylinder correlation function with a state prepared in the far past is computed by an ordered product of Heisenberg cylinder operators.

Theorem 59.10 (State–operator correspondence)**.** *Assume the radial reconstruction data of Hypothesis 59.1. For every $\Delta \in \mathrm{Spec}_{\mathrm{pp}}(D_{\mathrm{rad}})$, insertion at the origin defines an isometric linear isomorphism between local scaling operators of dimension Δ and cylinder energy eigenstates of energy Δ:*

$$\mathsf{S}_\Delta : \mathcal{V}_{\mathrm{loc}}^{(\Delta)} \xrightarrow{\sim} \ker(D_{\mathrm{rad}} - \Delta), \qquad \mathcal{O} \longmapsto |\mathcal{O}\rangle. \tag{59.24}$$

Consequently finite linear combinations of definite-dimension local operators give an isometric algebraic isomorphism

$$\mathsf{S}_{\mathrm{alg}} : \mathcal{V}_{\mathrm{loc}}^{\mathrm{alg}} \xrightarrow{\sim} \mathcal{H}_{\mathrm{alg}}.$$

The BPZ, or radial, completion of $\mathcal{V}_{\mathrm{loc}}^{\mathrm{alg}}$ is identified with $\mathcal{H}_{S^{D-1}}$, because $\mathcal{H}_{\mathrm{alg}}$ is dense. A generic vector in this completed Hilbert space is a state on the sphere; it is represented by a local *operator precisely when it lies in the algebraic finite-energy subspace with the local boundary-value property above. For a definite-dimension operator and for insertions ordered by cylinder time $\tau_1 > \tau_2 > \cdots > \tau_n$,*

$$\langle \mathcal{O}(0)\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle_{\mathbb{R}^D} = \langle\Omega|\widetilde{\mathcal{O}}_1(\tau_1,n_1)\cdots\widetilde{\mathcal{O}}_n(\tau_n,n_n)|\mathcal{O}\rangle_{\mathrm{cyl}}. \tag{59.25}$$

Proof. Radial reflection positivity supplies the positive semidefinite inner product on local insertions inside a sphere. Quotienting its null space makes the map $\mathcal{O} \mapsto |\mathcal{O}\rangle$ injective, and the defining pairing (59.10) identifies its norm with the radial two-point pairing; hence the map is isometric. In a regulated Lagrangian representative this equality is the gluing identity obtained from (59.12), followed by the continuum limit (59.14). If $\mathcal{O} \in \mathcal{V}_{\mathrm{loc}}^{(\Delta)}$, the dilatation Ward identity for $\mathcal{O}(0)$ becomes the cylinder Hamiltonian equation

$$D_{\mathrm{rad}}|\mathcal{O}\rangle = \Delta|\mathcal{O}\rangle.$$

Thus S_Δ maps $\mathcal{V}_{\mathrm{loc}}^{(\Delta)}$ isometrically into $\ker(D_{\mathrm{rad}} - \Delta)$. Theorem 59.8, together with the radial-completeness clause of Hypothesis 59.1, gives the reverse inclusion: each vector in $\ker(D_{\mathrm{rad}} - \Delta)$ is represented by a local scaling operator of the same dimension, unique modulo null operators. This proves (59.24). Taking finite direct sums gives the algebraic isomorphism. Since D_{rad} is nonnegative self-adjoint, the spectral projections $P_{[0,E]}(D_{\mathrm{rad}})$ converge strongly to the identity as $E \to \infty$. Hence $\mathcal{H}_{\mathrm{alg}}$ is dense in $\mathcal{H}_{S^{D-1}}$, and the isometry extends by completion to the radial Hilbert space. This completion is a completion of states, not an assertion that every Hilbert-space vector is itself a local operator.

It remains to derive the correlation formula. Put

$$X_j(0,n_j) = \widetilde{\mathcal{O}}_j(0,n_j), \qquad X_j(\tau_j,n_j) = \widetilde{\mathcal{O}}_j(\tau_j,n_j),$$

where the second equality is the Heisenberg translation (59.21). Choose cuts $\tau_+ > \tau_1$ and $\tau_- < \tau_n$. By (59.23) with $v = |\mathcal{O}\rangle_{\mathrm{cyl}}$,

$$\langle\Omega|\widetilde{\mathcal{O}}_1(\tau_1,n_1)\cdots\widetilde{\mathcal{O}}_n(\tau_n,n_n)|\mathcal{O}\rangle_{\mathrm{cyl}}$$

is the cylinder Schwinger functional on the punctured cylinder with:

- the incoming boundary state prepared by the local insertion $\mathcal{O}(0)$, transported to the lower cut;
- insertions $X_n, \ldots, X_1$ placed in the slabs ordered by increasing cylinder time; and
- the outgoing boundary projected to the vacuum by the limit $\tau_+ \to +\infty$.

The vacuum projection exists because $H_{\mathrm{cyl}} \geq 0$ and the zero-energy subspace is $\mathbb{C}\Omega$. The sewing identity is unchanged if the auxiliary cuts are moved without crossing an insertion.

Finally apply the Weyl map $x_j = \mathrm{e}^{\tau_j} n_j$. The operator definition (59.4), with the corresponding spin-frame factor for spinning operators, identifies the cylinder Schwinger functional at separated points with the flat-space radially ordered correlator. Since $\tau_1 > \cdots > \tau_n$, radial ordering places $\mathcal{O}_1(x_1), \ldots, \mathcal{O}_n(x_n)$ outside the origin insertion in the displayed order. This gives (59.25).

□

The state $|\Omega\rangle$ is the ground state on S^{D-1}.

The vacuum state corresponds to the identity operator:

$$\mathbf{1} \longleftrightarrow |\Omega\rangle.$$

The algebraic local-operator space is dense in the BPZ/radial Hilbert norm but is not closed. Accordingly, the product of two separated local insertions inside a sphere defines a state on that sphere, not a new local operator at the origin. The operator product expansion is the statement that this state is the BPZ/radial-norm limit of finite sums of local scaling operators at the origin, organized by D_{rad}-eigenspaces. The density of $\mathcal{V}_{\mathrm{loc}}^{\mathrm{alg}}$ in this norm is the functional-analytic input that permits separated products to be represented by convergent local operator partial sums after radial ordering.

Figure 59.3 records the precise state-from-operator construction: inserting a local scaling operator at the origin gives an energy eigenstate on the sphere, and separated products give states approximated by local-operator partial sums.

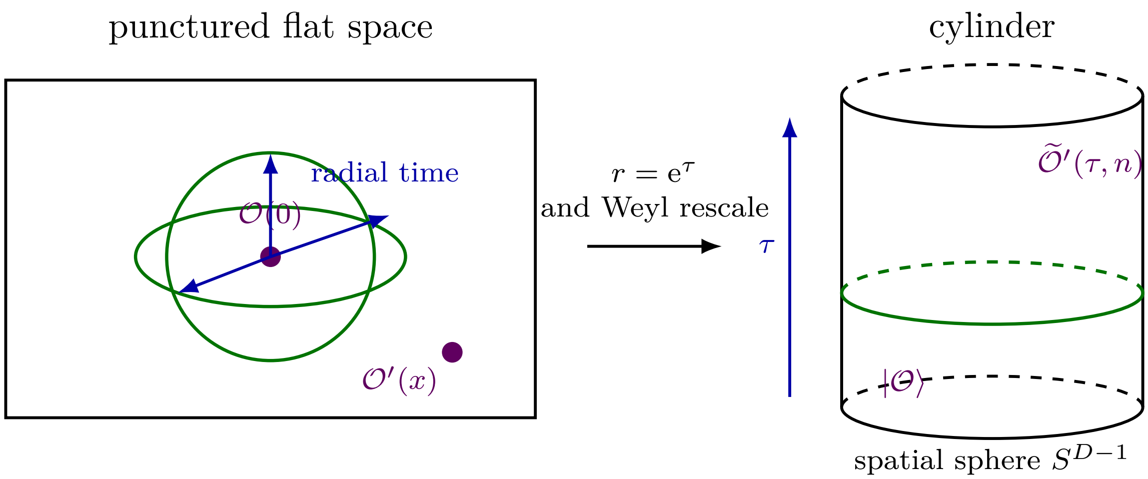


Figure 59.3: The singular endpoint $r = 0$ of the Weyl map becomes $\tau = -\infty$ on the cylinder. A local insertion at the origin therefore prepares a state on the spatial sphere.

59.6 Dilatation as the Cylinder Hamiltonian

The dilatation vector field on $\mathbb{R}^D$ is

$$x^\mu \partial_\mu = r\partial_r = \partial_\tau.$$

Therefore the dilatation charge maps to the Hamiltonian generating translations in cylinder time. Let $\widehat{H}_{\mathrm{cyl}}$ be defined by Euclidean evolution

$$|\psi(\tau + a)\rangle = \mathrm{e}^{-a\widehat{H}_{\mathrm{cyl}}}|\psi(\tau)\rangle.$$

In radial-quantization conventions the factors of i in the Lorentzian Hermitian conformal charges are absorbed, and

$$\widehat{H}_{\mathrm{cyl}} = \widehat{D}_{\mathrm{rad}}. \tag{59.26}$$

The eigenvalue of $\widehat{H}_{\rm cyl}$ is the scaling dimension.

The relation between the Hermitian Lorentzian convention and the radial Euclidean convention must be tracked at the level of the complexified conformal algebra. Let $\widehat{X}_{\rm L}$ denote the Hermitian Lorentzian charges in the convention of Chapter 57, so for example

$$[\widehat{D}_{\rm L},\widehat{P}^{\rm L}_\mu]={\rm i}\widehat{P}^{\rm L}_\mu,\qquad [\widehat{P}^{\rm L}_\mu,\widehat{K}^{\rm L}_\nu]=2{\rm i}(\delta_{\mu\nu}\widehat{D}_{\rm L}-\widehat{J}^{\rm L}_{\mu\nu}).$$

The radial descendant algebra is the real form adapted to the cylinder Hamiltonian and to the inversion pairing. With the subscript "rad" used only in this paragraph, the conversion is

$$\widehat{D}_{\rm rad}=-{\rm i}\widehat{D}_{\rm L},\qquad \widehat{J}^{\rm rad}_{\mu\nu}=-{\rm i}\widehat{J}^{\rm L}_{\mu\nu},\qquad \widehat{P}^{\rm rad}_\mu={\rm i}\widehat{P}^{\rm L}_\mu,\qquad \widehat{K}^{\rm rad}_\mu=-{\rm i}\widehat{K}^{\rm L}_\mu. \tag{59.27}$$

The opposite signs for $\widehat{P}_\mu$ and $\widehat{K}_\mu$ reflect their radial roles: translations create descendants and raise cylinder energy, while special conformal transformations annihilate primaries and lower cylinder energy. Applying (59.27) gives

$$[\widehat{D}_{\rm rad},\widehat{P}^{\rm rad}_\mu]=\widehat{P}^{\rm rad}_\mu,\qquad [\widehat{D}_{\rm rad},\widehat{K}^{\rm rad}_\mu]=-\widehat{K}^{\rm rad}_\mu,$$

and

$$[\widehat{P}^{\rm rad}_\mu,\widehat{K}^{\rm rad}_\nu]=-2(\delta_{\mu\nu}\widehat{D}_{\rm rad}-\widehat{J}^{\rm rad}_{\mu\nu}),\qquad [\widehat{K}^{\rm rad}_\mu,\widehat{P}^{\rm rad}_\nu]=2\delta_{\mu\nu}\widehat{D}_{\rm rad}-2\widehat{J}^{\rm rad}_{\nu\mu}. \tag{59.28}$$

The radial chapters suppress the subscript after this point. Thus the transition from Hermitian Lorentzian charges to the radial algebra is not a rule that deletes every visible factor of i; it is the complex change of real form displayed in (59.27). The sign in (59.28) is fixed against the conformal-Killing-vector realization: with the Hermitian charge convention $[Q_X,Q_Y]=-{\rm i}Q_{[X,Y]}$, the same vector-field brackets give the Lorentzian charge algebra, the Euclidean $SO(D+1,1)$ change of basis, and the radial real form above. The companion check `calculation-checks/conformal_algebra_sign_checks.py` verifies these three sign conversions from the vector fields rather than from mutually dependent displayed formulae. Positivity statements then use the Hilbert-space adjoint $\widehat{P}^\dagger_\mu=\widehat{K}_\mu$ supplied by radial reflection.

If $\mathcal{O}$ has scaling dimension Δ, then

$$\widehat{H}_{\rm cyl}|\mathcal{O}\rangle=\Delta|\mathcal{O}\rangle.$$

Thus the spectrum of scaling dimensions in flat space is the energy spectrum of the theory on the spatial sphere. Unitarity implies $\Delta\geq 0$ for states above the vacuum in a unitary CFT.

59.7 Primary and Descendant States

A local operator at the origin is called a conformal primary if the associated state is annihilated by all special conformal generators:

$$\widehat{K}_\mu|\mathcal{O}\rangle=0. \tag{59.29}$$

If $\mathcal{O}_\alpha$ transforms in a finite-dimensional spin representation with Hermitian matrices $(S_{\mu\nu})_\alpha{}^\beta$, so that a finite rotation near the identity is represented by $\exp(-\frac{\rm i}{2}\omega^{\mu\nu}S_{\mu\nu})$, then a primary state obeys

$$\widehat{D}_{\rm rad}|\mathcal{O}_\alpha\rangle=\Delta|\mathcal{O}_\alpha\rangle,\qquad \widehat{J}_{\mu\nu}|\mathcal{O}_\alpha\rangle=-(S_{\mu\nu})_\alpha{}^\beta|\mathcal{O}_\beta\rangle,\qquad \widehat{K}_\mu|\mathcal{O}_\alpha\rangle=0.$$

Translations generate descendants:

$$\widehat{P}_{\mu_1}\cdots\widehat{P}_{\mu_n}|\mathcal{O}\rangle\longleftrightarrow\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}(0).$$

The commutator

$$[\widehat{D}_{\mathrm{rad}}, \widehat{P}_\mu] = \widehat{P}_\mu$$

shows that each derivative raises the scaling dimension by one. Before quotienting null relations, these descendant states form the conformal generalized Verma module generated by the primary datum (Δ, ρ); the algebraic definition is given in Definition 61.3. The Hilbert-space primary multiplet is the quotient by the radical of the radial inner product, and a shortening equation is precisely the vanishing of a descendant in this quotient.

Figure 59.4 displays the conformal multiplet structure used in this quotient: a primary state generates descendants by successive actions of $\widehat{P}_\mu$, with null descendants removed in the Hilbert completion.

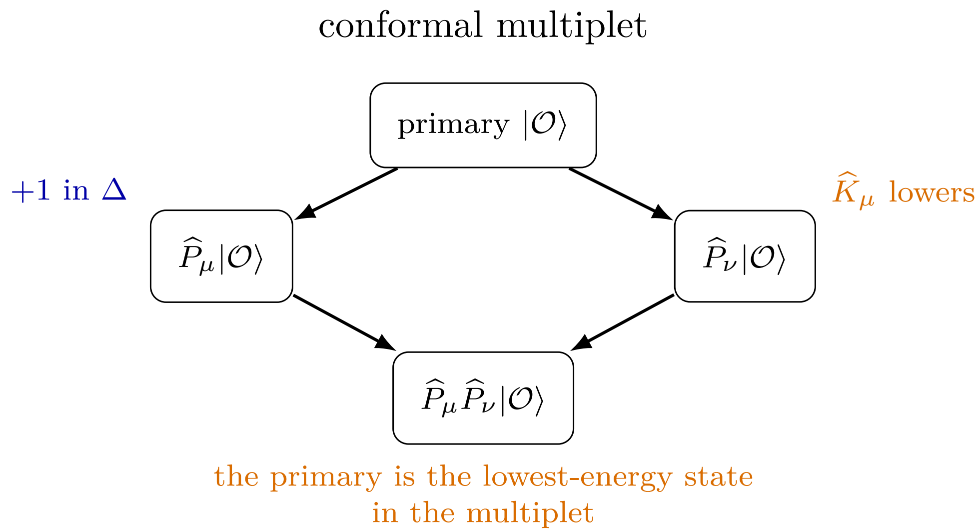


Figure 59.4: A conformal multiplet is generated from a primary by translations. On the cylinder, descendants have energies shifted upward by integers.

59.8 Reflection Positivity and Conjugation

Radial quantization gives a natural conjugation. In Euclidean flat space, reflection through the unit sphere is inversion:

$$x^\mu \mapsto \frac{x^\mu}{x^2}.$$

On the cylinder this is time reflection $\tau \mapsto -\tau$. For a scalar operator of dimension Δ, the conjugate insertion at infinity is

$$\mathcal{O}^\dagger(\infty) = \lim_{r\to\infty} r^{2\Delta}\mathcal{O}(rn)$$

when the limit is independent of n. The two-point function becomes the inner product

$$\langle \mathcal{O} \mid \mathcal{O}\rangle = \left\langle \mathcal{O}^\dagger(\infty)\mathcal{O}(0)\right\rangle.$$

For unitary theories this inner product is positive semidefinite and becomes positive definite after null states are quotiented. This positivity is the source of unitarity bounds, developed after primary transformation laws and two-point functions have been fixed.

59.9 Adjoints of the Conformal Generators

In radial quantization, Hermitian conjugation is a Hilbert-space statement in the reflection-positive completion of Hypothesis 59.1. Let $\mathcal{D}_{\text{loc}} \subset \mathcal{H}_{S^{D-1}}$ be the dense domain of finite linear combinations of local finite-energy states. For a scalar primary, the BPZ bra associated with the ket

$$|\mathcal{O}\rangle = \lim_{\tau\to-\infty} \mathrm{e}^{-\Delta\tau}\widetilde{\mathcal{O}}(\tau,n)|\Omega\rangle$$

is

$$\langle\!\langle\mathcal{O}| = \lim_{\tau\to+\infty} \mathrm{e}^{\Delta\tau}\langle\Omega|\widetilde{\mathcal{O}}^{\,\Theta_{\text{int}}}(\tau,n), \tag{59.30}$$

where Θ_{int} denotes the ordinary internal adjunction; for a spinning primary one also includes the spin-frame rotation specified by the radial reflection map in Hypothesis 59.1. Equivalently, in flat coordinates the bra is represented by the insertion obtained from $x \mapsto x/x^2$, together with the Weyl factor and spin transport. Formula (59.30) is the Hilbert-space version of the cylinder limit used later in (63.8).

The adjoint of an unbounded conformal generator is first defined on $\mathcal{D}_{\text{loc}}$. Let T_a be the finite translation of inside insertions by a small vector a, and let

$$C_a := \vartheta \circ T_a \circ \vartheta, \qquad \vartheta(x) = \frac{x}{|x|^2}, \tag{59.31}$$

be the corresponding finite special conformal transformation of the reflected insertions. The equality of Schwinger functions after applying the same conformal transformation to all separated insertions gives, for $A, B \in \mathcal{D}_{\text{loc}}$ and a small enough that no support crosses the reflection sphere,

$$\langle A, T_a B\rangle_{\text{rad}} = \langle C_a A, B\rangle_{\text{rad}}. \tag{59.32}$$

This is the step supplied by the BPZ bra construction: the transformation of the ket insertion is moved to the bra by radial reflection, and inversion turns the translated reflected insertion into a special-conformal one. With the radial generator convention in which translations create descendants,

$$\frac{\partial}{\partial a^\mu} T_a B\bigg|_{a=0} = \widehat{P}_\mu B, \qquad \frac{\partial}{\partial a^\mu} C_a A\bigg|_{a=0} = \widehat{K}_\mu A.$$

Differentiating (59.32) at $a = 0$ gives

$$\langle A, \widehat{P}_\mu B\rangle_{\text{rad}} = \langle \widehat{K}_\mu A, B\rangle_{\text{rad}}, \qquad A, B \in \mathcal{D}_{\text{loc}}. \tag{59.33}$$

Thus $\widehat{K}_\mu$ is the Hilbert-space adjoint of $\widehat{P}_\mu$ on the local finite-energy domain; after quotienting null states and taking the closure this is the adjoint relation used in the descendant Gram matrices.

Dilatations are translations in cylinder time, so

$$\widehat{D}^\dagger = \widehat{D}$$

in the cylinder Hilbert-space convention. Rotations are unitary symmetries of the spatial sphere:

$$\widehat{J}^\dagger_{\mu\nu} = \widehat{J}_{\mu\nu}.$$

Equation (59.33) is therefore summarized as

$$\widehat{P}^\dagger_\mu = \widehat{K}_\mu, \qquad \widehat{K}^\dagger_\mu = \widehat{P}_\mu. \tag{59.34}$$

Equation (59.34) is the algebraic input behind unitarity bounds. If $|\mathcal{O}\rangle$ is a primary state, then the norm of a first descendant is

$$\|\widehat{P}_\mu|\mathcal{O}\rangle\|^2 = \langle\mathcal{O}|\widehat{K}_\mu\widehat{P}_\mu|\mathcal{O}\rangle,$$

with summation over μ when a rotational scalar norm is taken. Commuting $\widehat{K}_\mu$ through $\widehat{P}_\nu$ and using $\widehat{K}_\mu|\mathcal{O}\rangle = 0$ expresses the result in terms of Δ and the spin representation of the primary. Positivity of all descendant norms is therefore a representation-theoretic constraint on allowed primary data.

59.10 The Free Scalar Check

Let ϕ be a free massless scalar primary with

$$\Delta_\phi = \frac{D-2}{2}.$$

The flat action is the $R = 0$ representative of the Weyl-covariant scalar action

$$S[\phi; g] = \frac{1}{2}\int \mathrm{d}^D x\,\sqrt{g}\,\left(g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi + \xi_* R[g]\phi^2\right), \qquad \xi_* = \frac{D-2}{4(D-1)}. \tag{59.35}$$

Thus "massless in flat space" means that the curvature scalar vanishes in the flat representative; it does not mean that the Weyl-covariant representative lacks the curvature coupling. By the Weyl transformation already described in Chapter 58, the same theory on the cylinder has $R_{\mathrm{cyl}} > 0$, so the $\xi_* R\phi^2$ term is activated and becomes the cylinder mass term. If

$$\widetilde{\phi}(\tau, n) = \mathrm{e}^{\Delta_\phi \tau}\phi(\mathrm{e}^\tau n),$$

then

$$S_{\mathrm{cyl}}[\widetilde{\phi}] = \frac{1}{2}\int_{\mathbb{R}\times S^{D-1}} \mathrm{d}\tau\,\mathrm{d}\Omega_{D-1}\left[(\partial_\tau\widetilde{\phi})^2 + \nabla_{S^{D-1}}\widetilde{\phi}\cdot\nabla_{S^{D-1}}\widetilde{\phi} + \xi_* R_{\mathrm{cyl}}\widetilde{\phi}^2\right]. \tag{59.36}$$

The cylinder metric is

$$\mathrm{d}s^2_{\mathrm{cyl}} = \mathrm{d}\tau^2 + \mathrm{d}\Omega^2_{D-1},$$

where $\mathrm{d}\Omega^2_{D-1} = \gamma_{ab}\mathrm{d}\theta^a\mathrm{d}\theta^b$ is the unit round metric on S^{D-1}. The Levi-Civita connection of this product metric splits. Thus

$$R^{\mathrm{cyl}}_{\tau\tau} = 0, \qquad R^{\mathrm{cyl}}_{ab} = R^{S^{D-1}}_{ab} = (D-2)\gamma_{ab},$$

and contraction with $g^{\tau\tau}_{\mathrm{cyl}} = 1$ and $g^{ab}_{\mathrm{cyl}} = \gamma^{ab}$ gives

$$R_{\mathrm{cyl}} = g^{MN}_{\mathrm{cyl}} R^{\mathrm{cyl}}_{MN} = \gamma^{ab}(D-2)\gamma_{ab} = (D-1)(D-2).$$

The conformal mass term is

$$m^2_{\mathrm{cyl}} = \xi_* R_{\mathrm{cyl}} = \frac{(D-2)^2}{4}.$$

For a one-particle mode on S^{D-1} with angular momentum ℓ,

$$-\nabla^2_{S^{D-1}} Y_\ell = \ell(\ell + D - 2)Y_\ell.$$

The cylinder energy is

$$E_\ell = \sqrt{\ell(\ell + D - 2) + \frac{(D-2)^2}{4}} = \ell + \frac{D-2}{2}.$$

The $\ell = 0$ mode corresponds to the primary field ϕ, whose scaling dimension is

$$\Delta_\phi = \frac{D-2}{2}.$$

Modes with $\ell > 0$ correspond to derivative descendants of ϕ. This explicit spectrum realizes the state-operator correspondence in a theory where all local operators can be built from normal-ordered products of ϕ and its derivatives, modulo the equation of motion.

59.11 The Operator Partition Function of the Free Scalar

At a fixed point, it is often useful to count linearly independent local operators by their scaling dimension. This is a graded trace on the algebraic local-operator space, not a trace over the Hilbert completion. Let

$$Z_{\text{op}}(q) = \mathrm{Tr}_{\mathcal{V}_{\text{loc}}^{\text{alg}}} q^{\widehat{D}_{\text{rad}}}$$

be the formal character of the local-operator space. In the free scalar theory, the independent single-letter operators are the symmetric traceless derivatives

$$\partial_{(\mu_1} \cdots \partial_{\mu_\ell)}\phi \quad \text{with traces removed}, \qquad \ell = 0, 1, 2, \ldots.$$

The trace removal is the equation of motion $\partial^2\phi = 0$ and all of its derivatives. Counting all symmetric derivatives and subtracting the traces gives the single-letter character

$$f_\phi(q) = q^{\frac{D-2}{2}} \frac{1-q^2}{(1-q)^D}. \tag{59.37}$$

Proposition 59.11 (Free scalar letter character as a short-module character)**.** *The factor $1-q^2$ in (59.37) is the character subtraction of the null submodule generated by the level-two scalar descendant $\widehat{P}^2|\phi\rangle$, where $\widehat{P}^2 = \delta^{\mu\nu}\widehat{P}_\mu\widehat{P}_\nu$. Equivalently,*

$$f_\phi(q) = \chi\left(M\left(\frac{D-2}{2}, \mathbf{1}\right)\right) - q^2\,\chi\left(M\left(\frac{D-2}{2}, \mathbf{1}\right)\right),$$

where $M(\Delta, \mathbf{1})$ is the scalar conformal generalized Verma module defined in Chapter 61 and $\mathbf{1}$ is the trivial $SO(D)$ representation. This is the same null relation as the scalar unitarity-bound shortening proved in Theorem 62.1.

Proof. Before quotienting null states, the scalar generalized Verma module is freely generated by the commuting translations $\widehat{P}_\mu$. Hence its graded character is

$$\chi(M(\Delta, \mathbf{1})) = q^\Delta \sum_{n=0}^{\infty} \dim \mathrm{Sym}^n(\mathbb{C}^D)\, q^n = \frac{q^\Delta}{(1-q)^D}.$$

At $\Delta_\phi = (D-2)/2$, the level-two trace descendant is null by Theorem 62.1. It is also a singular vector. Indeed, using

$$[\widehat{K}_\nu, \widehat{P}_\mu] = 2\delta_{\nu\mu}\widehat{D}_{\text{rad}} - 2\widehat{J}_{\mu\nu}, \qquad [\widehat{J}_{\mu\nu}, \widehat{P}_\rho] = \delta_{\mu\rho}\widehat{P}_\nu - \delta_{\nu\rho}\widehat{P}_\mu,$$

and $\widehat{J}_{\mu\nu}|\phi\rangle = 0$, one obtains

$$\begin{aligned}
\widehat{K}_\nu \widehat{P}^2|\phi\rangle &= \left([\widehat{K}_\nu, \widehat{P}_\mu]\widehat{P}_\mu + \widehat{P}_\mu[\widehat{K}_\nu, \widehat{P}_\mu]\right)|\phi\rangle \\
&= \left(2(\Delta_\phi + 1)\widehat{P}_\nu - 2(D-1)\widehat{P}_\nu + 2\Delta_\phi \widehat{P}_\nu\right)|\phi\rangle \\
&= 4\left(\Delta_\phi + 1 - \frac{D}{2}\right)\widehat{P}_\nu|\phi\rangle = 0.
\end{aligned}$$

Therefore the null vector generates a conformal submodule. Its translation descendants have character

$$\frac{q^{\Delta_\phi+2}}{(1-q)^D}.$$

In polynomial notation $\widehat{P}_\mu \leftrightarrow p_\mu$, this quotient is the quotient of $\mathbb{C}[p_1, \ldots, p_D]$ by the ideal $p^2\mathbb{C}[p_1, \ldots, p_D]$, where $p^2 = \delta^{\mu\nu}p_\mu p_\nu$. At homogeneous degree ℓ, the Fischer decomposition gives

$$\mathrm{Sym}^\ell(\mathbb{C}^D) = \mathcal{H}_\ell \oplus p^2\,\mathrm{Sym}^{\ell-2}(\mathbb{C}^D),$$

where $\mathcal{H}_\ell$ is the space of harmonic homogeneous polynomials, equivalently the symmetric traceless rank-ℓ representation of $SO(D)$. Thus quotienting by the null submodule leaves exactly the symmetric traceless derivatives of ϕ, and the character is

$$\frac{q^{\Delta_\phi}}{(1-q)^D} - \frac{q^{\Delta_\phi+2}}{(1-q)^D} = q^{\frac{D-2}{2}}\frac{1-q^2}{(1-q)^D}.$$

□

Equivalently,

$$f_\phi(q) = q^{\frac{D-2}{2}}\sum_{\ell=0}^{\infty} d_{\ell,D}q^\ell, \qquad d_{\ell,D} = \binom{D+\ell-1}{\ell} - \binom{D+\ell-3}{\ell-2},$$

with the second binomial interpreted as zero for $\ell < 2$. These $d_{\ell,D}$ are the dimensions of scalar spherical harmonics of degree ℓ on S^{D-1}.

Normal-ordered products of the letters form the bosonic symmetric algebra of single-particle letters. If

$$f_\phi(q) = \sum_\Delta a_\Delta q^\Delta,$$

then the free scalar operator partition function is

$$Z^\phi_{\text{op}}(q) = \prod_\Delta \frac{1}{(1-q^\Delta)^{a_\Delta}} = \exp\left(\sum_{m=1}^{\infty}\frac{1}{m}f_\phi(q^m)\right). \tag{59.38}$$

For $D = 3$, $d_{\ell,3} = 2\ell + 1$, and therefore

$$f_\phi(q) = \sum_{\ell=0}^{\infty} (2\ell + 1) q^{\ell+\frac{1}{2}} = q^{1/2} \frac{1 - q^2}{(1 - q)^3}.$$

This is the same counting obtained from the one-particle spectrum of $\widehat{H}_{\text{cyl}} = \sqrt{-\nabla^2_{S^2} + 1/4}$ on S^2.

59.12 The Free Fermion Operator Partition Function

Let S be a complex spinor representation of $\mathrm{Spin}(D)$ with dimension $s_S = \dim S$. A free spinor primary ψ_α, $\alpha = 1, \ldots, s_S$, has scaling dimension

$$\Delta_\psi = \frac{D - 1}{2}.$$

The flat equation of motion is the Dirac equation

$$\gamma^\mu{}_\beta{}^\alpha \partial_\mu \psi_\alpha = 0, \tag{59.39}$$

where $\gamma^\mu : V \otimes S \to S'$ is Clifford multiplication from the vector representation $V \simeq \mathbb{C}^D$ and the chosen spinor module S to the target spinor module S'. In even D, for a chiral Weyl field, S' is the opposite chirality spinor; for a Dirac spinor, $S' = S$ after including both chiralities. The unrefined character only uses $\dim S' = \dim S = s_S$.

Before imposing (59.39), the translation descendants form

$$S \otimes \mathbb{C}[p_1, \ldots, p_D], \qquad p_\mu \leftrightarrow \widehat{P}_\mu,$$

with character $s_S q^{\Delta_\psi}/(1 - q)^D$. The Dirac equation is a level-one singular vector: the null descendant $\gamma^\mu \widehat{P}_\mu |\psi\rangle$ generates a submodule with character $s_S q^{\Delta_\psi+1}/(1 - q)^D$. Therefore the fermion single-letter character is

$$f_\psi(q) = s_S q^{\frac{D-1}{2}} \frac{1 - q}{(1 - q)^D} = \frac{s_S q^{\frac{D-1}{2}}}{(1 - q)^{D-1}}. \tag{59.40}$$

Free fermion letter character as a short-module character. The character (59.40) is the character of the free spinor conformal module obtained by quotienting the spinor generalized Verma module by the submodule generated by the Dirac descendant $\gamma^\mu \widehat{P}_\mu |\psi\rangle$. The generalized Verma module before shortening is freely generated by the commuting translations:

$$M(\Delta_\psi, S) = \mathbb{C}[\widehat{P}_1, \ldots, \widehat{P}_D] \otimes S, \qquad \chi(M(\Delta_\psi, S)) = \frac{s_S q^{\Delta_\psi}}{(1 - q)^D}.$$

At $\Delta_\psi = (D - 1)/2$, the Dirac descendant is primary. This follows from the radial commutator

$$[\widehat{K}_\nu, \widehat{P}_\mu] = 2\delta_{\nu\mu} \widehat{D}_{\text{rad}} - 2\widehat{J}_{\mu\nu}$$

and the spinor action

$$\widehat{J}_{\mu\nu}|\psi\rangle = \Sigma_{\mu\nu}|\psi\rangle, \qquad \Sigma_{\mu\nu} = \frac{1}{4}[\gamma_\mu, \gamma_\nu].$$

Using

$$\gamma^\mu \Sigma_{\mu\nu} = \frac{D-1}{2}\gamma_\nu$$

with the displayed $\Sigma_{\mu\nu}$ convention gives

$$\widehat{K}_\nu \gamma^\mu \widehat{P}_\mu |\psi\rangle = 2\left(\Delta_\psi - \frac{D-1}{2}\right)\gamma_\nu|\psi\rangle = 0.$$

Hence the Dirac descendant generates a conformal submodule. The submodule is again freely generated by translations, now starting one level higher and with spinor multiplicity s_S. Subtracting its character gives (59.40).

If $f_\psi(q) = \sum_\Delta a_\Delta q^\Delta$, the local-operator algebra generated by one free fermion field uses the exterior algebra of the odd letters:

$$Z^\psi_{\mathrm{op}}(q) = \prod_\Delta (1+q^\Delta)^{a_\Delta} = \exp\left(\sum_{m=1}^\infty \frac{(-1)^{m+1}}{m} f_\psi(q^m)\right). \tag{59.41}$$

This is the ordinary positive-counting trace over the fermionic Fock space; the insertion of $(-1)^F$ would replace $1+q^\Delta$ by $1-q^\Delta$.

In four dimensions, a Weyl fermion has $s_S = 2$ and

$$f_{\mathrm{Weyl}}(q) = \frac{2q^{3/2}}{(1-q)^3}.$$

A Dirac fermion, or a Weyl fermion together with its conjugate, has

$$f_{\mathrm{Dirac}}(q) = \frac{4q^{3/2}}{(1-q)^3}.$$

The coefficient of $q^{\ell+3/2}$ for a Weyl fermion is $(\ell+1)(\ell+2)$, the degeneracy of one chiral tower of spinor harmonics on S^3.

59.13 The Free Maxwell Operator Partition Function

The Maxwell field is conformal in four spacetime dimensions. The local gauge-invariant letters are the field-strength components $F_{\mu\nu}$. Work in Euclidean $D = 4$ with vector representation $V \simeq \mathbb{C}^4$. The primary $F_{\mu\nu} \in \wedge^2 V$ has $\Delta_F = 2$ and $\dim \wedge^2 V = 6$. Before imposing the equations, its translation descendants have character

$$\frac{6q^2}{(1-q)^4}.$$

The Bianchi identity and Maxwell equation are

$$\partial_{[\lambda}F_{\mu\nu]} = 0, \qquad \partial^\mu F_{\mu\nu} = 0. \tag{59.42}$$

At the level of polynomial jets, these are respectively a three-form constraint and a one-form constraint. Each has dimension 4, so together they subtract $8q^3/(1-q)^4$. The two constraints have one scalar reducibility identity each:

$$\partial_{[\rho}\partial_\lambda F_{\mu\nu]} = 0, \qquad \partial^\nu\partial^\mu F_{\mu\nu} = 0.$$

The first is $d^2F = 0$; the second follows from the antisymmetry of $F_{\mu\nu}$. They restore $2q^4/(1-q)^4$. Thus the Maxwell single-letter character is

$$f_{\mathrm{Max}}(q) = \frac{q^2(6-8q+2q^2)}{(1-q)^4} = \frac{6q^2-2q^3}{(1-q)^3}. \tag{59.43}$$

Equivalently, decompose

$$F = F^+ + F^-, \qquad F^+ \in (1,0), \qquad F^- \in (0,1)$$

under $\mathrm{Spin}(4) \simeq SU(2)_L \times SU(2)_R$. Each chiral field-strength module has character

$$f_{F^\pm}(q) = \frac{q^2(3-4q+q^2)}{(1-q)^4}, \qquad f_{\mathrm{Max}}(q) = f_{F^+}(q) + f_{F^-}(q).$$

This chiral expression makes the conformal shortening transparent: the level-one divergence descendant is removed, and the level-two scalar reducibility identity restores the over-subtraction.

If

$$f_{\mathrm{Max}}(q) = \sum_\Delta b_\Delta q^\Delta,$$

then the abelian Maxwell operator partition function generated by normal-ordered products of field-strength letters is the bosonic symmetric algebra

$$Z_{\mathrm{op}}^{\mathrm{Max}}(q) = \prod_\Delta \frac{1}{(1-q^\Delta)^{b_\Delta}} = \exp\left(\sum_{m=1}^\infty \frac{1}{m} f_{\mathrm{Max}}(q^m)\right). \tag{59.44}$$

Potential variables A_μ, gauge parameters, and ghosts belong to a gauge-fixed complex for path-integral determinants. The character (59.43) counts the BRST-invariant local field-strength cohomology in the free abelian theory.

59.14 High-Temperature Operator Entropy and the Cardy Limit

For a CFT satisfying Hypothesis 59.1, assume in addition that the spectral projection of D_{rad} onto $[0,\Delta]$ has finite trace for every finite Δ. The cylinder thermal trace is then

$$Z_{\mathrm{cyl}}(\beta) = \mathrm{Tr}_{\mathcal{H}_{S^{D-1}}}\, \mathrm{e}^{-\beta D_{\mathrm{rad}}} = \int_0^\infty \mathrm{e}^{-\beta E}\, dN(E), \qquad \beta > 0, \tag{59.45}$$

where

$$N(\Delta) = \mathrm{Tr}_{\mathcal{H}_{S^{D-1}}}\, \mathbf{1}_{[0,\Delta]}(D_{\mathrm{rad}}).$$

If the spectrum is pure point with finite degeneracies, this integral is $\sum_\Delta d(\Delta)\mathrm{e}^{-\beta\Delta}$, where $d(\Delta) = \dim\ker(D_{\mathrm{rad}} - \Delta)$. The operator partition function $Z_{\mathrm{op}}(q)$ is this same trace with $q = \mathrm{e}^{-\beta}$ when the local operator basis is complete and the trace converges. The large-Δ density should therefore be understood as the integrated spectral counting function $N(\Delta)$, or as a smoothed density obtained from it, rather than as an everywhere-defined ordinary function.

Assume that the high-temperature cylinder free energy has the leading form

$$\log Z_{\mathrm{cyl}}(\beta) = A\,\beta^{-(D-1)} + o(\beta^{-(D-1)}), \qquad A > 0. \tag{59.46}$$

This hypothesis is a thermodynamic input about the CFT on S^{D-1}: in $D > 2$ the constant A is not fixed by the stress-tensor two-point coefficient alone. It is the finite-volume version of the flat thermal pressure coefficient. The precise positivity input is the following Hardy–Littlewood Tauberian implication: if dN is a positive spectral measure on $[0, \infty)$ and $\log \int_0^\infty \mathrm{e}^{-\beta E} dN(E) = A\beta^{-\alpha} + o(\beta^{-\alpha})$ with $A, \alpha > 0$, then

$$\log N(E) = (\alpha+1)A^{1/(\alpha+1)} \left(\frac{E}{\alpha}\right)^{\alpha/(\alpha+1)} + o(E^{\alpha/(\alpha+1)}).$$

Applying this with $\alpha = D - 1$ gives the leading microcanonical entropy

$$\log N(\Delta) = D\,A^{1/D} \left(\frac{\Delta}{D-1}\right)^{\frac{D-1}{D}} + o(\Delta^{\frac{D-1}{D}}). \tag{59.47}$$

The saddle calculation is explicit. Extremize

$$\Phi(\beta) = \beta\Delta + A\beta^{-(D-1)}.$$

The stationary point satisfies

$$\Delta = (D-1)A\beta_*^{-D}, \qquad \beta_* = \left(\frac{(D-1)A}{\Delta}\right)^{1/D}.$$

Substitution gives (59.47). Thus the power $\Delta^{(D-1)/D}$ is fixed by extensivity of high-temperature thermal physics in $D-1$ spatial dimensions, while the coefficient records the specific CFT thermal free energy.

In $D = 2$ the torus supplies a stronger statement because modular invariance relates the high-temperature trace to the low-temperature vacuum. Let the spatial circle have circumference 2π, and let

$$H_{\mathrm{cyl}} = L_0 + \bar{L}_0 - \frac{c_L + c_R}{24}, \qquad \Delta = h + \bar{h}, \qquad J = h - \bar{h}.$$

For a compact unitary two-dimensional CFT with discrete finite-multiplicity spectrum, unique vacuum, and modular-invariant torus partition function, the thermal trace

$$Z(\beta) = \mathrm{Tr}\exp\left[-\beta\left(L_0 + \bar{L}_0 - \frac{c_L + c_R}{24}\right)\right]$$

obeys $Z(\beta) = Z(4\pi^2/\beta)$ for the rectangular torus. As $\beta \to 0^+$, the transformed trace is dominated by the vacuum, and

$$\log Z(\beta) = \frac{\pi^2(c_L + c_R)}{6\beta} + o(\beta^{-1}). \tag{59.48}$$

If $c_L = c_R = c$, then $A = \pi^2 c/3$ in (59.46). Equation (59.47) gives

$$\log N(\Delta) = 2\pi\sqrt{\frac{c\,\Delta}{3}} + o(\sqrt{\Delta}). \tag{59.49}$$

This is the Cardy entropy-dimension relation for total scaling dimension. Keeping left and right energies separately gives the spin-resolved form

$$\log \rho(h, \bar{h}) = 2\pi\sqrt{\frac{c_L h}{6}} + 2\pi\sqrt{\frac{c_R \bar{h}}{6}} + o(\sqrt{h} + \sqrt{\bar{h}}), \tag{59.50}$$

for $h, \bar{h} \to \infty$ with $h/\bar{h}$ remaining in a compact subinterval of $(0, \infty)$. If the lowest left or right conformal weight is not the vacuum value, the same derivation replaces c_L, c_R by the corresponding effective central charges $c_L - 24h_{\min}$ and $c_R - 24\bar{h}_{\min}$.

Chapter 60

Conformal Charges and Ward Identities

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

60.1 Charges as Surface Operators

Definition 60.1 (Conformal surface charge)**.** Let $\epsilon^\mu(x)\partial_\mu$ be a conformal Killing vector on flat Euclidean space, and let

$$J^\mu_\epsilon(x) = T^\mu{}_\nu(x)\epsilon^\nu(x)$$

be the corresponding conserved current at separated points. If $\Sigma \subset \mathbb{R}^D$ is an oriented smooth codimension-one surface with unit normal n_μ, define the charge supported on Σ by

$$Q_\epsilon(\Sigma) = \int_\Sigma \mathrm{d}\sigma\, n_\mu T^\mu{}_\nu(x)\epsilon^\nu(x). \tag{60.1}$$

Here $\mathrm{d}\sigma$ is the induced volume element.

Surface deformation of a conserved charge. Let Σ_0 and Σ_1 be homologous oriented smooth codimension-one surfaces in a region $U \subset \mathbb{R}^D$ that contains no operator insertions and no boundary. If $\partial_\mu J^\mu_\epsilon = 0$ as a separated-point distribution on U, then $Q_\epsilon(\Sigma_0) = Q_\epsilon(\Sigma_1)$ inside every correlation function whose other insertions lie outside the deformation region.

Let $V \subset U$ be the oriented region swept out by the deformation, with $\partial V = \Sigma_1 - \Sigma_0$. Smearing the distributional conservation law with a test function that is one on V and whose support avoids all insertions gives

$$0 = \int_V \mathrm{d}^D x \, \partial_\mu J^\mu_\epsilon(x) = \int_{\Sigma_1} \mathrm{d}\sigma \, n_\mu J^\mu_\epsilon - \int_{\Sigma_0} \mathrm{d}\sigma \, n_\mu J^\mu_\epsilon.$$

The last equality is Stokes' theorem for the smoothed distributional pairing; shrinking the transition region of the test function to zero gives the surface statement. No contact term appears because the support of the deformation avoids the insertion diagonals.

This deformation property is the local content of a conserved charge. A charge may be evaluated on a large sphere surrounding several insertions, or on a union of small spheres surrounding them separately. The equality of these two evaluations is the Ward identity.

Definition 60.2 (Lorentzian charge commutator across an insertion)**.** The Lorentzian operator meaning of the same construction is the following. Let $\widehat{Q}_\epsilon(\Sigma)$ be the flux of the conserved current through a spacelike Cauchy surface Σ, with future-directed unit normal. If a local operator $\widehat{\mathcal{O}}(0)$ is inserted between two nearby Cauchy surfaces Σ_- and Σ_+, then

$$[\widehat{Q}_\epsilon, \widehat{\mathcal{O}}(0)] = \widehat{Q}_\epsilon(\Sigma_+)\widehat{\mathcal{O}}(0) - \widehat{\mathcal{O}}(0)\widehat{Q}_\epsilon(\Sigma_-). \tag{60.2}$$

Here conservation identifies the charge on $\Sigma_\pm$ with the same abstract symmetry generator when no insertion lies between the two surfaces. Equation (60.2) is the definition of the commutator after the charge has been represented on the Hilbert spaces immediately to the future and past of the insertion. After Euclidean continuation, the two surfaces close into a small hypersurface surrounding the insertion, and the commutator becomes the small-surface action used below.

Figure 60.1 records this Lorentzian-to-Euclidean translation of a charge commutator: the difference between future and past charge slices becomes a closed surface surrounding the insertion.

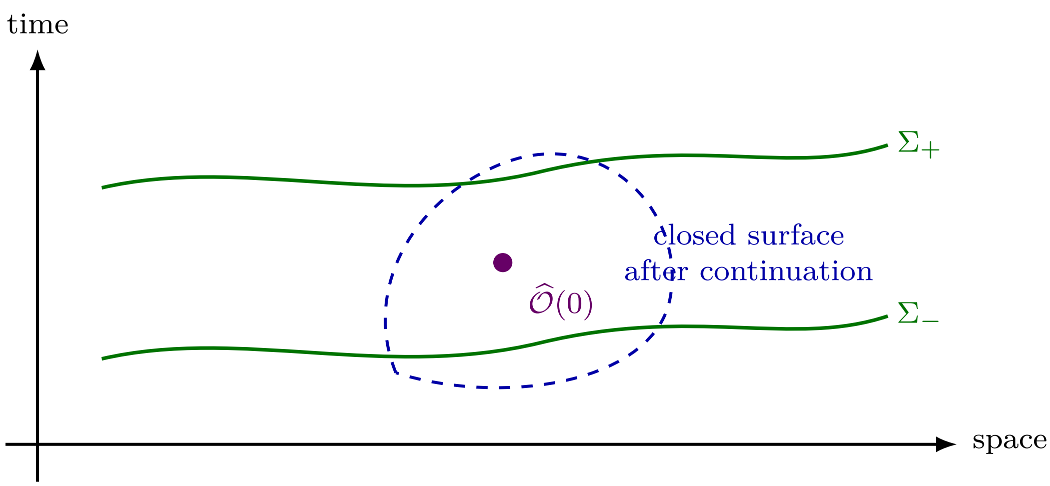


Figure 60.1: A Lorentzian commutator with a conserved charge is represented by the difference between charge insertions on slices immediately after and before the local operator. In Euclidean signature this difference is encoded by a closed surface enclosing the insertion.

Figure 60.2 displays the corresponding Euclidean Ward-identity deformation: a closed charge surface can be moved through regions free of insertions, and crossing an insertion gives the local symmetry action.

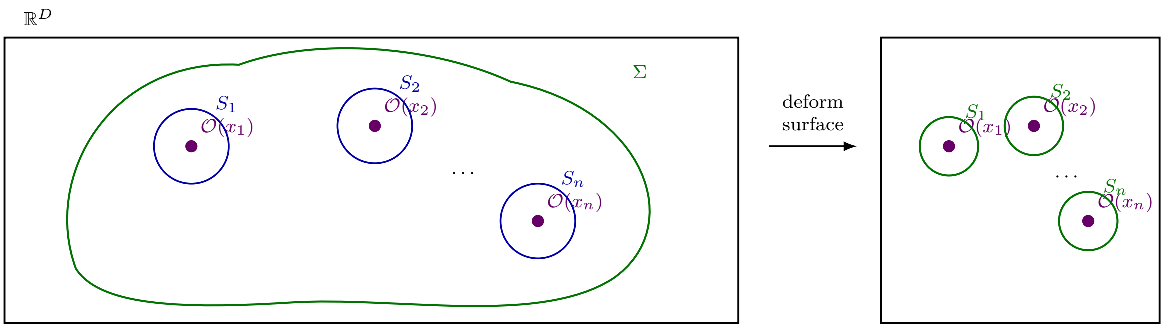


Figure 60.2: A conserved conformal charge can be moved from a large enclosing surface to small enclosing surfaces around the insertions. The Ward identity is this equality inside correlation functions.

60.2 Operator Variations from Small Spheres

Definition 60.3 (Small-surface action on a local operator)**.** Let S_x be a small sphere enclosing a single insertion at x, with orientation inherited from the boundary of a small ball removed from the correlation-function domain. The infinitesimal action of the symmetry on the operator $\mathcal{O}(x)$ is defined by

$$\delta_\epsilon \mathcal{O}(x) = \mathrm{i}\, Q_\epsilon(S_x)\mathcal{O}(x) \tag{60.3}$$

in Euclidean radial-ordering conventions. In Lorentzian operator language, this convention corresponds to

$$\delta_\epsilon \widehat{\mathcal{O}} = \mathrm{i}[\widehat{Q}_\epsilon, \widehat{\mathcal{O}}].$$

Changing the orientation convention changes both sides of (60.3) by the same sign; the Ward identity below is then unchanged.

Hypothesis 60.4 (Vacuum and large-boundary behavior)**.** For the conformal Killing vector $\epsilon^\mu \partial_\mu$ and the correlators under consideration, assume the following two properties.

(i) The vacuum is invariant under the corresponding charge. In Lorentzian notation this means

$$\widehat{Q}_\epsilon|\Omega\rangle = 0, \qquad \langle\Omega|\widehat{Q}_\epsilon = 0$$

on the common invariant domain on which the correlators are defined.

(ii) If Σ_R is a family of smooth enclosing surfaces whose distance from the insertions tends to infinity, then clustering makes the large boundary flux vanish:

$$\lim_{R\to\infty} \left\langle Q_\epsilon(\Sigma_R) \prod_{i=1}^{n} \mathcal{O}_i(x_i) \right\rangle = 0.$$

More explicitly, with

$$Q_\epsilon(\Sigma_R) = \int_{\Sigma_R} \mathrm{d}S_\mu\, \epsilon_\nu(x) T^{\mu\nu}(x),$$

the connected correlator

$$\left\langle T^{\mu\nu}(x) \prod_{i=1}^{n} \mathcal{O}_i(x_i) \right\rangle_c$$

must decay at large $|x|$ fast enough to overcome the area growth of Σ_R and the polynomial growth of the displayed conformal Killing vectors. This is the cluster-decomposition input in the deformation of the large surface to the small spheres S_i.

Global conformal Ward identity from surface deformation. Under Hypothesis 60.4, for separated insertions,

$$0 = \left\langle Q_\epsilon(\Sigma) \prod_{i=1}^{n} \mathcal{O}_i(x_i) \right\rangle = \sum_{i=1}^{n} \langle \mathcal{O}_1(x_1) \cdots Q_\epsilon(S_i)\mathcal{O}_i(x_i) \cdots \mathcal{O}_n(x_n) \rangle , \tag{60.4}$$

where Σ encloses all insertions and S_i is a small sphere enclosing x_i. Using (60.3), this becomes

$$\sum_{i=1}^{n} \langle \mathcal{O}_1(x_1) \cdots \delta_\epsilon \mathcal{O}_i(x_i) \cdots \mathcal{O}_n(x_n) \rangle = 0. \tag{60.5}$$

This is the global Ward identity associated with the conformal Killing vector ϵ^μ. It is a statement about distributions on the configuration space of distinct points, completed by contact terms on diagonals when operator products collide.

By vacuum invariance, the charge on a sufficiently large surface enclosing all insertions has zero expectation value after the large-boundary limit in Hypothesis 60.4. By the surface-deformation argument at the beginning of the chapter, that large surface can be deformed through the punctured configuration space to the union of small surfaces S_i surrounding the insertions. The induced orientation on the small boundaries is the one used in Definition 60.3, and hence each small surface gives

$$Q_\epsilon(S_i)\mathcal{O}_i(x_i) = -\mathrm{i}\, \delta_\epsilon \mathcal{O}_i(x_i).$$

Multiplying the equality of surface insertions by i gives (60.5). This derivation is carried out on the configuration space of distinct points; the extension to coincident configurations is a further source-convention statement about contact terms.

60.3 Source-Derived Local Ward Identities

The surface identity is the integral form of the source Ward identities introduced in Section 56.6. Let

$$X = \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n), \qquad x_i \neq x_j,$$

where the $\mathcal{O}_i$ are scalar local operators. In the source-coordinate system used for the stress tensor and the operators, the following distributional identities are the local form of the conformal Ward identity.

Proposition 60.5 (Local conformal-current Ward identity)**.** *Assume that the stress tensor is symmetric and that scalar primary sources are defined in the same source-coordinate system as the metric-source stress tensor. Let $\mathcal{O}_i$ have scaling dimensions Δ_i, and set*

$$\sigma_\epsilon = \frac{1}{D}\partial_\rho \epsilon^\rho, \qquad J^\mu_\epsilon = T^\mu{}_\nu \epsilon^\nu.$$

The translation Ward identity is

$$\partial_\mu \langle T^{\mu\nu}(x)X\rangle = -\sum_{i=1}^{n} \delta^{(D)}(x-x_i)\,\langle \mathcal{O}_1(x_1)\cdots \partial_i^\nu \mathcal{O}_i(x_i)\cdots \mathcal{O}_n(x_n)\rangle\,. \tag{60.6}$$

Here ∂_i^ν differentiates the ith insertion coordinate. A spin-carrying insertion adds the local representation term dictated by the diffeomorphism action on its source. Equation (60.6) is equivalent to the smeared identity (56.28); the displayed delta functions record the extension of the separated-point distribution across the diagonals $x = x_i$.

The flat-space trace Ward identity in the same source-coordinate system is

$$\langle T^\mu{}_\mu(x)X\rangle = -\sum_{i=1}^{n} \Delta_i\,\delta^{(D)}(x-x_i)\,\langle X\rangle + \mathcal{A}(x;X). \tag{60.7}$$

The distribution $\mathcal{A}(x;X)$ is the local anomaly or contact term obtained by differentiating the local source functional $\mathcal{A}_\sigma[X]$ in (56.29) with respect to the Weyl parameter $\sigma(x)$. It vanishes on the configuration space of distinct flat-space points; on collision diagonals or curved backgrounds it is part of the specified source convention.

If ϵ^μ is a conformal Killing vector, then

$$\partial_\mu \langle J^\mu_\epsilon(x)X\rangle = \sum_{i=1}^{n} \delta^{(D)}(x-x_i)\,\langle \mathcal{O}_1(x_1)\cdots \delta_\epsilon \mathcal{O}_i(x_i)\cdots \mathcal{O}_n(x_n)\rangle + \sigma_\epsilon(x)\mathcal{A}(x;X), \tag{60.8}$$

where for a scalar primary

$$\delta_\epsilon \mathcal{O}_i(x_i) = -\epsilon^\nu(x_i)\partial_{i,\nu}\mathcal{O}_i(x_i) - \Delta_i\sigma_\epsilon(x_i)\mathcal{O}_i(x_i).$$

For spin ρ_i, $\delta_\epsilon\mathcal{O}_i$ includes the local rotation term displayed later in (61.9).

Proof. The translation identity is the functional derivative of the metric-source diffeomorphism Ward identity with respect to the scalar sources, followed by setting all sources to their flat background values. The sign is fixed by the active variation $\delta_\epsilon\mathcal{O}_i = -\epsilon^\nu\partial_{i,\nu}\mathcal{O}_i$ for translations. The trace identity is the corresponding functional derivative of the Weyl Ward identity in the same source-coordinate system; the displayed Δ_i-term comes from the Weyl weight of the scalar source, while $\mathcal{A}(x;X)$ is the differentiated local Weyl anomaly functional.

For a conformal Killing vector,

$$\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = 2\sigma_\epsilon\delta_{\mu\nu}.$$

Using symmetry of $T^{\mu\nu}$,

$$\partial_\mu\langle J^\mu_\epsilon(x)X\rangle = \epsilon_\nu(x)\partial_\mu\langle T^{\mu\nu}(x)X\rangle + \sigma_\epsilon(x)\langle T^\mu{}_\mu(x)X\rangle.$$

Substituting (60.6) and (60.7) gives (60.8), with the delta functions evaluating ϵ^μ and σ_ϵ at the insertion points. □

Integrating (60.8) over a region whose boundary is $\Sigma \cup_i (-S_i)$ gives the surface decomposition (60.4); if the large boundary contribution vanishes and the anomaly term integrates to zero in the chosen background, the global Ward identity (60.5) follows.

60.4 Conformal Charges on the Cylinder

Cylinder conformal charges and radial adjointness. Under the Weyl map

$$\mathbb{R}^D \setminus \{0\} \longrightarrow \mathbb{R}_\tau \times S^{D-1}, \qquad x^\mu = \mathrm{e}^\tau n^\mu, \qquad n^\mu n_\mu = 1,$$

closed spheres surrounding the origin become constant-τ slices. The dilatation charge becomes the cylinder Hamiltonian, rotations become the rotations of S^{D-1}, and the translation and special conformal charges become mutually adjoint raising and lowering operators:

$$\widehat{P}_\mu^\dagger = \widehat{K}_\mu. \tag{60.9}$$

This adjointness is a consequence of reflection positivity under $\tau \mapsto -\tau$, which is inversion in flat space. It is the positivity input used below to derive unitarity bounds. The map $x^\mu = \mathrm{e}^\tau n^\mu$ sends the flat metric on $\mathbb{R}^D \setminus \{0\}$ to $\mathrm{d}s^2 = \mathrm{e}^{2\tau}(\mathrm{d}\tau^2 + \mathrm{d}\Omega_{D-1}^2)$; after the Weyl rescaling by $\mathrm{e}^{-2\tau}$, spheres $|x| = \mathrm{e}^\tau$ are the constant cylinder-time slices. The conformal Killing vector $x^\mu \partial_\mu$ therefore becomes ∂_τ, so its charge is the generator of Euclidean cylinder translations. Reflection positivity pairs an insertion at τ with its reflected insertion at $-\tau$, and reflection in flat coordinates is inversion $x^\mu \mapsto x^\mu/x^2$. Inversion conjugates translations to special conformal transformations; hence the positive radial inner product identifies the adjoint of the translation charge with the special conformal charge, giving (60.9). Rotations commute with the Weyl factor and act as the ordinary isometries of S^{D-1}.

More explicitly, let

$$\Pi_{\mu\nu}(n) = \delta_{\mu\nu} - n_\mu n_\nu$$

be the projector onto the tangent space of S^{D-1} at n. Acting on functions on $\mathbb{R}_\tau \times S^{D-1}$, the flat-space conformal Killing vectors become

$$D_{\mathrm{rad}} = \partial_\tau, \tag{60.10}$$

$$P_\mu = \mathrm{e}^{-\tau}\left(n_\mu \partial_\tau + \Pi_{\mu\nu}(n)\frac{\partial}{\partial n_\nu}\right), \tag{60.11}$$

$$K_\mu = \mathrm{e}^{\tau}\left(-n_\mu \partial_\tau + \Pi_{\mu\nu}(n)\frac{\partial}{\partial n_\nu}\right), \tag{60.12}$$

with rotations acting as vector fields tangent to the sphere. Thus inversion $x^\mu \mapsto x^\mu/x^2$, which is $\tau \mapsto -\tau$, exchanges P_μ and K_μ. As vector fields their weights under $D_{\mathrm{rad}}^{\mathrm{vec}} = \partial_\tau$ are

$$[\partial_\tau, P_\mu^{\mathrm{vec}}] = -P_\mu^{\mathrm{vec}}, \qquad [\partial_\tau, K_\mu^{\mathrm{vec}}] = K_\mu^{\mathrm{vec}}.$$

The Hilbert-space grading has the opposite sign because the induced infinitesimal action on local operators contains the differential operator $-\mathcal{L}_\epsilon$, as in (60.8). In the radial Hilbert-space representation recalled in Chapter 59,

$$[\widehat{D}_{\mathrm{rad}}, \widehat{P}_\mu] = \widehat{P}_\mu.$$

Since $\widehat{D}_{\mathrm{rad}}$ is self-adjoint and $\widehat{P}_\mu^\dagger = \widehat{K}_\mu$ by (60.9), the adjoint commutator is $[\widehat{D}_{\mathrm{rad}}, \widehat{K}_\mu] = -\widehat{K}_\mu$. Consequently $\widehat{P}_\mu$ raises and $\widehat{K}_\mu$ lowers the cylinder energy by one unit.

The Hamiltonian identification is also visible directly from the stress tensor. On the sphere S_r^{D-1} of radius r, with $x^\mu = rn^\mu$,

$$\widehat{D}_{\mathrm{rad}} = \int_{S_r^{D-1}} \mathrm{d}\sigma\, n^\mu \widehat{T}_{\mu\nu}(x) x^\nu = r^D \int_{S^{D-1}} \mathrm{d}\Omega\, n^\mu n^\nu \widehat{T}_{\mu\nu}(rn). \tag{60.13}$$

Under the Weyl map this is the cylinder Hamiltonian

$$\widehat{H}_{\mathrm{cyl}} = \int_{S^{D-1}} \mathrm{d}\Omega\, \widehat{T}^{\mathrm{cyl}}_{\tau\tau}.$$

In Lorentzian cylinder time $t = -\mathrm{i}\tau$, the corresponding Hermitian generator differs from the Euclidean radial generator by the standard Wick-rotation factor described in Chapter 59. The monograph uses $\widehat{D}_{\mathrm{rad}}$ for the positive Euclidean cylinder Hamiltonian that grades local operators by their scaling dimensions; whenever Hermitian Lorentzian charges appear in Ward identities, the conversion to radial commutators is made by the same analytic continuation of the group parameter.

60.5 Two-Dimensional Chiral Stress-Tensor Charges

Two Euclidean dimensions require a refinement of the preceding finite conformal-charge construction. Let U be a coordinate chart on an oriented Euclidean surface with complex coordinate z. In a flat chart away from operator insertions, assume the separated-point stress-tensor Ward identities take the chiral form

$$\partial_{\bar z} T(z) = 0, \qquad \partial_z \bar{T}(\bar z) = 0 \tag{60.14}$$

inside correlation functions. Equation (60.14) has the separated distributional meaning that, for any product X of local insertions whose supports avoid z,

$$\partial_{\bar z}\langle T(z)X\rangle = 0, \qquad \partial_z\langle \bar{T}(\bar z)X\rangle = 0.$$

Contact terms at the insertions are part of the source convention discussed in Section 56.6.

Definition 60.6 (Chiral stress-tensor contour charge)**.** The chiral contour charge associated with a holomorphic vector field $v(z)\partial_z$ on an annulus A is the oriented contour operator

$$Q_v(C) = -\oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, v(z)T(z), \tag{60.15}$$

where $C \subset A$ is positively oriented around the insertion on which the charge acts. The sign in (60.15) is the active-operator sign: the infinitesimal change generated by $z \mapsto z + \varepsilon v(z)$ is

$$\delta_v \mathcal{O} = \varepsilon Q_v(C)\mathcal{O}.$$

Suppose $\mathcal{O}(0)$ has holomorphic weight h, so that its response to translations and linear rescalings is finite and diagonal. Then the singular part of the stress-tensor OPE begins as

$$T(z)\mathcal{O}(0) = \cdots + \frac{h}{z^2}\mathcal{O}(0) + \frac{1}{z}\partial\mathcal{O}(0) + O(z^0), \tag{60.16}$$

where the omitted terms to the left of the displayed terms are possible poles of order z^{-3} and higher. These higher poles are the additional transformation data under nonlinear chiral vector fields.

Definition 60.7 (Virasoro primary and contour modes)**.** A Virasoro primary is a local operator ϕ whose stress-tensor singular part is exactly

$$T(z)\phi(0) = \frac{h}{z^2}\phi(0) + \frac{1}{z}\partial\phi(0) + O(z^0), \qquad \bar{T}(\bar{z})\phi(0) = \frac{\bar{h}}{\bar{z}^2}\phi(0) + \frac{1}{\bar{z}}\bar{\partial}\phi(0) + O(\bar{z}^0). \tag{60.17}$$

This is stronger than global conformal primarity: it is a statement about the whole local chiral stress-tensor representation, including the $\mathrm{PSL}(2,\mathbb{C})$ subalgebra.

On a punctured disc, the holomorphic modes are

$$L_n = \oint_0 \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, z^{n+1} T(z), \qquad n \in \mathbb{Z}, \tag{60.18}$$

acting first on the common radial domain generated by finitely many local insertions and then, when reflection positivity and finite-energy decomposition are available, on the corresponding dense domain in the radial Hilbert space. With this convention,

$$L_0|\phi\rangle = h|\phi\rangle, \qquad L_{-1}|\phi\rangle = |\partial\phi\rangle$$

for a Virasoro primary ϕ. The antiholomorphic modes $\bar{L}_n$ are defined analogously from $\bar{T}(\bar{z})$.

Lemma 60.8 (Stress-tensor OPE and the Virasoro algebra)**.** *Assume that the singular part of the holomorphic stress-tensor OPE is*

$$T(z)T(w) \sim \frac{c}{2(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{60.19}$$

Then the contour modes (60.18) *obey, on the radial finite-insertion domain,*

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{c}{12}(n^3-n)\delta_{n+m,0}. \tag{60.20}$$

Proof. Let C_2 be a positively oriented contour around the origin and let C_1 be a positively oriented contour outside C_2. The ordered product L_nL_m is represented by

$$\oint_{C_1} \frac{\mathrm{d}z}{2\pi\mathrm{i}} \oint_{C_2} \frac{\mathrm{d}w}{2\pi\mathrm{i}}\, z^{n+1} w^{m+1} T(z)T(w).$$

The product L_mL_n is the same expression with the z-contour moved to a contour C_1' inside C_2. Hence the commutator is obtained by integrating z over $C_1 - C_1'$. Since $C_1 - C_1'$ encloses the point w but not the origin, it can be shrunk to a small positive contour around w. Only the singular part of (60.19) contributes:

$$[L_n, L_m] = \oint_{C_2} \frac{\mathrm{d}w}{2\pi\mathrm{i}} w^{m+1} \operatorname*{Res}_{z=w}\left[z^{n+1}\left(\frac{c}{2(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}\right)\right].$$

The residues are

$$\operatorname*{Res}_{z=w} \frac{z^{n+1}}{(z-w)^4} = \frac{(n+1)n(n-1)}{6}\, w^{n-2}, \qquad \operatorname*{Res}_{z=w} \frac{z^{n+1}}{(z-w)^2} = (n+1)w^n.$$

Thus

$$[L_n, L_m] = \oint_{C_2} \frac{\mathrm{d}w}{2\pi\mathrm{i}} \left[\frac{c}{12}(n^3-n)w^{m+n-1} + 2(n+1)w^{m+n+1}T(w) + w^{m+n+2}\partial T(w)\right].$$

The first term contributes precisely when $m+n=0$. For the last term, integrating by parts on the closed contour gives

$$\oint \frac{\mathrm{d}w}{2\pi\mathrm{i}}\, w^{m+n+2}\partial T(w) = -(m+n+2)\oint \frac{\mathrm{d}w}{2\pi\mathrm{i}}\, w^{m+n+1}T(w).$$

The coefficient of the remaining stress-tensor mode is therefore $2(n+1)-(m+n+2)=n-m$, giving (60.20). □

Remark 60.9 (Antiholomorphic sector and gravitational anomaly). The antiholomorphic modes satisfy the same algebra with c replaced by $\bar{c}$, and separated holomorphic and antiholomorphic contours commute up to possible contact terms that would be a gravitational anomaly. A reflection-positive parity-invariant two-dimensional CFT has $c=\bar{c}$. In a theory with a gravitational anomaly, the two chiral central charges are different and the spatial momentum on the cylinder acquires the corresponding vacuum shift.

Infinitesimal stress-tensor transformation. With the contour-charge convention (60.15) and the stress-tensor OPE (60.19), a holomorphic vector field $v(z)\partial_z$ acts on $T(w)$ by

$$\delta_v T(w) = -v(w)\partial T(w) - 2v'(w)T(w) - \frac{c}{12}v'''(w). \tag{60.21}$$

This is an equality of insertions on any annulus on which the contour for Q_v can be chosen without crossing other operators. The formula is therefore a separated-point Ward identity; contact extensions at collisions are fixed by the source convention for the stress tensor.

The contour calculation is local around w. By definition,

$$\delta_v T(w) = Q_v(C)T(w) = -\oint_C \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, v(z)T(z)T(w),$$

where C is a small positive contour around w. Insert (60.19). The three required residues are

$$\operatorname*{Res}_{z=w}\frac{v(z)}{z-w} = v(w), \qquad \operatorname*{Res}_{z=w}\frac{v(z)}{(z-w)^2} = v'(w), \qquad \operatorname*{Res}_{z=w}\frac{v(z)}{(z-w)^4} = \frac{v'''(w)}{6}.$$

The overall minus sign in Q_v gives

$$-\left(v(w)\partial T(w) + 2v'(w)T(w) + \frac{c}{2}\,\frac{v'''(w)}{6}\right),$$

which is (60.21).

Schwarzian cocycle. For locally univalent holomorphic maps f and g, the Schwarzian derivative

$$\{f,z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2$$

satisfies

$$\{g\circ f, z\} = (f'(z))^2\{g, f(z)\} + \{f,z\}. \tag{60.22}$$

Moreover, if $f_s(z) = z + sv(z) + O(s^2)$, then

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \{f_s, z\} = v'''(z). \tag{60.23}$$

Set

$$A_f(z) = \frac{f''(z)}{f'(z)} = \partial_z \log f'(z).$$

Then

$$\{f, z\} = A_f'(z) - \frac{1}{2}A_f(z)^2.$$

For a composition,

$$A_{g\circ f}(z) = A_f(z) + f'(z)A_g(f(z)).$$

Differentiate and subtract one half of the square:

$$\begin{aligned} A_{g\circ f}' - \frac{1}{2}A_{g\circ f}^2 = A_f' - \frac{1}{2}A_f^2 + (f')^2\left(A_g' - \frac{1}{2}A_g^2\right)\circ f \\ + f''A_g \circ f - A_f f' A_g \circ f. \end{aligned}$$

Since $A_f = f''/f'$, the last two terms cancel. This proves (60.22). Finally, for $f_s(z) = z + sv(z) + O(s^2)$,

$$f_s' = 1 + sv' + O(s^2), \qquad f_s'' = sv'' + O(s^2), \qquad f_s''' = sv''' + O(s^2).$$

Substitution into the definition of the Schwarzian gives $\{f_s, z\} = sv'''(z) + O(s^2)$, proving (60.23).

Proposition 60.10 (Schwarzian transformation and plane-cylinder shift)**.** *The stress tensor itself carries an affine chiral transformation law fixed by* (60.19)*. If $z' = f(z)$ is a holomorphic coordinate change, then*

$$T'(z') = \left(\frac{\mathrm{d}f}{\mathrm{d}z}\right)^{-2}\left[T(z) - \frac{c}{12}\{f, z\}\right], \qquad \{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2. \tag{60.24}$$

The Schwarzian derivative satisfies the cocycle identity

$$\{g \circ f, z\} = (f'(z))^2\{g, f(z)\} + \{f, z\},$$

which is exactly the condition that successive coordinate changes act associatively on T.

For the cylinder coordinate $w = \sigma + \mathrm{i}\tau$, $\sigma \sim \sigma + 2\pi$, we use

$$z = \mathrm{e}^{-\mathrm{i}w}, \qquad w = \mathrm{i}\log z.$$

Applying (60.24) *to the coordinate change $z \mapsto w(z)$, for which $\{w, z\} = 1/(2z^2)$, gives*

$$T^{(w)}(w) = \left(\frac{\mathrm{d}z}{\mathrm{d}w}\right)^2 T^{(z)}(z) + \frac{c}{24} = -z^2 T^{(z)}(z) + \frac{c}{24}. \tag{60.25}$$

If

$$T^{(z)}(z) = \sum_{n\in\mathbb{Z}} \frac{L_n}{z^{n+2}}, \qquad \bar{T}^{(\bar{z})}(\bar{z}) = \sum_{n\in\mathbb{Z}} \frac{\bar{L}_n}{\bar{z}^{n+2}},$$

then

$$T^{(w)}(w) = -\sum_{n\in\mathbb{Z}} \mathrm{e}^{\mathrm{i}nw} L_n + \frac{c}{24}, \qquad \bar{T}^{(\bar{w})}(\bar{w}) = -\sum_{n\in\mathbb{Z}} \mathrm{e}^{-\mathrm{i}n\bar{w}} \bar{L}_n + \frac{\bar{c}}{24}.$$

In the cylinder normalization used in Chapter 59, the Hamiltonian and spatial momentum on the unit circle are therefore

$$H_{\mathrm{cyl}} = L_0 + \bar{L}_0 - \frac{c+\bar{c}}{24}, \qquad P_{\mathrm{cyl}} = L_0 - \bar{L}_0 - \frac{c-\bar{c}}{24}. \tag{60.26}$$

Thus the radial Hamiltonian differs from the naive dilatation eigenvalue by the Casimir term forced by the stress-tensor anomaly under the plane-cylinder coordinate change.

Proof. First prove the finite formula. For a local coordinate change f, define the right side of (60.24) by

$$\mathcal{T}_f(f(z)) = (f'(z))^{-2}\left[T(z) - \frac{c}{12}\{f,z\}\right].$$

If $f_s(z) = z + sv(z) + O(s^2)$, then

$$(f_s')^{-2} = 1 - 2sv' + O(s^2), \qquad T(z) = T(z_s - sv(z_s)) + O(s^2)$$

at the fixed final point $z_s = f_s(z)$. Together with (60.23),

$$\mathcal{T}_{f_s}(z_s) = T(z_s) - s\left(v\partial T + 2v'T + \frac{c}{12}v'''\right)(z_s) + O(s^2),$$

which is exactly the infinitesimal Ward action (60.21).

It remains to check compatibility with finite composition. Applying first f and then g gives

$$(g'(f)f')^{-2}\left[T - \frac{c}{12}\{f,z\} - \frac{c}{12}(f')^2\{g,f(z)\}\right],$$

which equals the transformation for $g\circ f$ precisely by (60.22). Thus the displayed formula is a projective cocycle for the connected local conformal group whose tangent action is the stress-tensor Ward action. On a simply connected neighborhood of the identity this characterizes the integrated action uniquely; analytic continuation along paths in the chosen coordinate-change group gives the stated finite law, with possible global monodromy recorded separately by the gravitational-anomaly data.

For $w = \mathrm{i}\log z$, direct differentiation gives

$$w' = \frac{\mathrm{i}}{z}, \qquad w'' = -\frac{\mathrm{i}}{z^2}, \qquad w''' = \frac{2\mathrm{i}}{z^3}, \qquad \{w,z\} = \frac{1}{2z^2}.$$

Using $z = \mathrm{e}^{-\mathrm{i}w}$ and $\mathrm{d}z/\mathrm{d}w = -\mathrm{i}z$ in (60.24) yields (60.25). Substituting the Laurent expansions of $T^{(z)}$ and $\bar{T}^{(\bar{z})}$ and taking the zero modes of cylinder translations and rotations gives (60.26). □

Remark 60.11 (Local operators and the radial Hilbert completion in $D=2$). The Virasoro mode construction refines the domain statement of Chapter 59. The local operators at the origin, including their Virasoro descendants, span an algebraic finite-energy subspace. After quotienting null vectors and completing in the BPZ/radial norm, this subspace is dense in the Hilbert space on the circle under the hypotheses of Hypothesis 59.1. The precise state–operator statement is therefore the correspondence between local scaling fields of definite $L_0, \bar{L}_0$ eigenvalues and the finite-energy radial eigenstates in this algebraic domain. The completed Hilbert space is the BPZ/radial completion of that dense domain. This density is the analytic input behind the radial OPE: the product of two nearby local insertions is represented by a convergent expansion in local scaling fields in the radial Hilbert norm.

Chapter 61

Primary Operators and Finite Conformal Transformations

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

61.1 Spin Representations and Primary Data

Definition 61.1 (Spin primary datum)**.** Let ρ be a finite-dimensional representation of $SO(D)$ on a vector space V_ρ. A local operator valued in this representation is written

$$\mathcal{O}_a(x), \qquad a = 1, \dots, \dim V_\rho.$$

The infinitesimal spin matrices are denoted

$$(S_{\mu\nu})_a{}^b = -(S_{\nu\mu})_a{}^b.$$

They obey the $SO(D)$ Lie algebra in the representation ρ, in the convention in which a finite rotation R near the identity is represented as $\rho(R) = \exp(-\frac{\mathrm{i}}{2}\omega^{\mu\nu}S_{\mu\nu})$. Thus the $S_{\mu\nu}$ are Hermitian spin matrices in a unitary representation.

A conformal primary operator of spin ρ and scaling dimension Δ is an operator whose state at the origin obeys

$$\widehat{K}_\mu|\mathcal{O}_a\rangle = 0, \qquad \widehat{D}_{\mathrm{rad}}|\mathcal{O}_a\rangle = \Delta|\mathcal{O}_a\rangle, \qquad \widehat{J}_{\mu\nu}|\mathcal{O}_a\rangle = -(S_{\mu\nu})_a{}^b|\mathcal{O}_b\rangle. \tag{61.1}$$

Remark 61.2 (Global primaries, quasi-primaries, and Virasoro primaries). In $D \geq 3$ the conformal Killing equation on flat space has only the finite-dimensional conformal algebra. Therefore the state condition (61.1) is the intrinsic notion of a conformal primary used throughout this volume. In $D = 2$ there is an additional local conformal enhancement: on the plane the globally defined conformal group is $\mathrm{PSL}(2, \mathbb{C})$, generated by L_{-1}, L_0, L_1 and $\bar{L}_{-1}, \bar{L}_0, \bar{L}_1$, while the local holomorphic and antiholomorphic conformal transformations are represented quantum mechanically by the Virasoro algebras. A local operator whose state is annihilated by L_1 and $\bar{L}_1$ is often called a *quasi-primary*; it transforms covariantly under the global conformal group. A Virasoro primary is a stronger object: its state is annihilated by all $L_n, \bar{L}_n$ with $n > 0$, after choosing the stress-tensor and Virasoro representation. Thus, whenever this core volume uses the word primary in $D = 2$ without further qualification, it means global primary in the preceding sense, i.e. quasi-primary in much of the two-dimensional CFT literature. A later treatment that uses full Virasoro symmetry must state the stronger Virasoro-primary condition separately.

The translation generators create the descendant states:

$$\widehat{P}_{\mu_1} \cdots \widehat{P}_{\mu_k} |\mathcal{O}_a\rangle \longleftrightarrow \partial_{\mu_1} \cdots \partial_{\mu_k} \mathcal{O}_a(0).$$

Definition 61.3 (Conformal generalized Verma module)**.** Let

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{so}(D + 2, \mathbb{C})$$

be the complexified conformal Lie algebra, written in the radial triangular decomposition

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{n}_+ \oplus \mathfrak{l} \oplus \mathfrak{n}_-, \qquad \mathfrak{n}_+ = \mathrm{span}_{\mathbb{C}}\{\widehat{P}_\mu\}, \qquad \mathfrak{n}_- = \mathrm{span}_{\mathbb{C}}\{\widehat{K}_\mu\},$$

$$\mathfrak{l} = \mathfrak{so}(D, \mathbb{C}) \oplus \mathbb{C}\, \widehat{D}_{\mathrm{rad}}.$$

Let $V_{\Delta,\rho}$ be the finite-dimensional $\mathfrak{l} \oplus \mathfrak{n}_-$-module on which $\widehat{D}_{\mathrm{rad}}$ acts by Δ, $\mathfrak{so}(D)$ acts by ρ, and every $\widehat{K}_\mu$ acts by zero. The conformal generalized Verma module generated by this primary datum is the induced module

$$M(\Delta, \rho) = U(\mathfrak{g}_{\mathbb{C}}) \otimes_{U(\mathfrak{l} \oplus \mathfrak{n}_-)} V_{\Delta,\rho}. \tag{61.2}$$

By the Poincare–Birkhoff–Witt theorem and the commutativity of the translations, its level-n vector space is

$$M(\Delta, \rho)_n \simeq \mathrm{Sym}^n(\mathbb{C}^D) \otimes V_\rho, \qquad n \geq 0, \tag{61.3}$$

before equations of motion, conservation laws, and null relations are imposed.

Conformal multiplet generated by a primary datum. Under the state–operator correspondence, a positive-energy conformal module generated from a lowest-energy state is obtained by repeated action of $\widehat{P}_\mu$; algebraically this is the generalized Verma module $M(\Delta, \rho)$. A descendant vector annihilated by all $\widehat{K}_\mu$ is a singular vector and generates a submodule. In a reflection-positive theory the radial inner product has a radical consisting of null states, invariant under the conformal algebra because $\widehat{P}_\mu^\dagger = \widehat{K}_\mu$. The cyclic physical multiplet generated by the primary is obtained by quotienting $M(\Delta, \rho)$ by this null radical. When the radical is the maximal proper submodule, this quotient is the irreducible highest-weight module in the algebraic category.

The radial decomposition of the conformal algebra has $[\widehat{D}_{\mathrm{rad}}, \widehat{P}_\mu] = \widehat{P}_\mu$ and $[\widehat{D}_{\mathrm{rad}}, \widehat{K}_\mu] = -\widehat{K}_\mu$. A primary state is therefore a lowest-energy vector, and every algebraic descendant obtained from it by the enveloping algebra may be represented, by the Poincare–Birkhoff–Witt theorem, with all $\widehat{P}_\mu$'s to the left of the primary vector. Since the translations commute, the level-n space is the symmetric tensor space in (61.3). A descendant annihilated by all $\widehat{K}_\mu$ satisfies the same lowest-energy condition as a primary and therefore generates an invariant submodule. In a reflection-positive theory $\widehat{P}_\mu^\dagger = \widehat{K}_\mu$; hence the radical of the radial Gram form is invariant under both raising and lowering operators and may be quotiented. The quotient is the cyclic physical descendant module. It is irreducible once the radical has been identified with the maximal proper submodule, equivalently once no further invariant descendant subspace remains.

The determinant of the radial Gram form restricted to $M(\Delta, \rho)_n$, after decomposing into irreducible $SO(D)$ summands, is the finite conformal-algebra analogue of the Kac determinant. Its zeros as a function of Δ locate possible singular descendants. BGG embeddings and resolutions organize the pattern of nested singular submodules. Chapter 62 computes the low-level Gram factors needed for the unitarity bounds directly from the commutation relations, without invoking the full determinant formula.

Translation from the origin and descendant fields. An operator at a general point is obtained by translating the insertion from the origin, equivalently by its Taylor expansion in descendants:

$$\mathcal{O}_a(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^{\mu_1} \cdots x^{\mu_k} \partial_{\mu_1} \cdots \partial_{\mu_k} \mathcal{O}_a(0). \tag{61.4}$$

In the Hermitian translation convention $[\widehat{P}_\mu, \mathcal{O}(x)] = \mathrm{i}\partial_\mu \mathcal{O}(x)$, the corresponding state notation is

$$|\mathcal{O}(x)\rangle = \mathrm{e}^{-\mathrm{i}x^\mu \widehat{P}_\mu} |\mathcal{O}\rangle.$$

This is a convention-level consequence of the translation Ward identity. The local field is an operator-valued distribution, so the displayed Taylor series is a shorthand for equality after smearing with test functions in a neighborhood on which the expansion converges as a radial state. The translation commutator gives

$$\frac{\mathrm{d}}{\mathrm{d}s} \left(\mathrm{e}^{-\mathrm{i}sx\cdot \widehat{P}} \mathcal{O}(0) \mathrm{e}^{\mathrm{i}sx\cdot \widehat{P}} \right) = x^\mu \partial_\mu \mathcal{O}(sx), \qquad \mathcal{O}(0) \text{ at } s = 0.$$

Integrating this first-order equation gives the Taylor expansion (61.4). Passing to the radial state created by the insertion gives the state formula with the stated Hermitian translation convention.

61.2 Spin Sources and Contact Terms

Definition 61.4 (Spin source convention and contact operator)**.** The primary module data must agree with the source-defined Ward identities. Introduce a source $\eta^a(x)$ valued in the dual representation $V_\rho^\vee$ and couple it by

$$\int \mathrm{d}^D x\, \eta^a(x) \mathcal{O}_a(x).$$

For a primary of dimension Δ, the flat-space source convention associated with a conformal Killing vector ϵ^μ is

$$\delta_\epsilon^{\mathrm{src}}\eta^a = \epsilon^\mu\partial_\mu\eta^a + (\Delta - D)\sigma_\epsilon\eta^a + \frac{\mathrm{i}}{2}\omega_\epsilon^{\mu\nu}\eta^b(S_{\mu\nu})_b{}^a, \tag{61.5}$$

where

$$\sigma_\epsilon = \frac{1}{D}\partial_\mu\epsilon^\mu, \qquad \omega_{\epsilon,\mu\nu} = \frac{1}{2}(\partial_\nu\epsilon_\mu - \partial_\mu\epsilon_\nu).$$

The last term is the dual spin action. The sign convention in (61.5) is the source convention of Section 56.6; operator insertions then acquire the contact operator

$$\mathcal{D}_\epsilon\mathcal{O}_a(x) = -\epsilon^\mu(x)\partial_\mu\mathcal{O}_a(x) - \Delta\sigma_\epsilon(x)\mathcal{O}_a(x) - \frac{\mathrm{i}}{2}\omega_\epsilon^{\mu\nu}(x)(S_{\mu\nu})_a{}^b\mathcal{O}_b(x). \tag{61.6}$$

Spinful local conformal-current Ward identity. Differentiating the source Ward identity with respect to the sources and then setting the sources to zero gives, for

$$X = \mathcal{O}_{1,a_1}(x_1)\cdots\mathcal{O}_{n,a_n}(x_n), \qquad x_i \neq x_j,$$

the local current Ward identity

$$\partial_\mu\left\langle J_\epsilon^\mu(x)X\right\rangle = \sum_{i=1}^n \delta^{(D)}(x-x_i)\left\langle\mathcal{O}_{1,a_1}(x_1)\cdots\mathcal{D}_\epsilon^{(i)}\mathcal{O}_{i,a_i}(x_i)\cdots\mathcal{O}_{n,a_n}(x_n)\right\rangle + \sigma_\epsilon(x)\mathcal{A}(x;X). \tag{61.7}$$

Here $\mathcal{D}_\epsilon^{(i)}$ uses the spin representation and scaling dimension of the ith insertion. Equation (61.7) is the spinful version of (60.8). With the small-sphere orientation used in (60.3),

$$\mathrm{i}\,Q_\epsilon(S_x)\mathcal{O}_a(x) = \mathcal{D}_\epsilon\mathcal{O}_a(x). \tag{61.8}$$

To fix the contact signs, write the source coupling in the local chart as

$$\int \mathrm{d}^D x\,\eta^a(x)\mathcal{O}_a(x).$$

Under an infinitesimal conformal transformation generated by ϵ^μ, the integration measure changes by $(\partial_\mu\epsilon^\mu)\mathrm{d}^Dx = D\sigma_\epsilon\mathrm{d}^Dx$. The transport part of $\delta_\epsilon^{\mathrm{src}}\eta^a$ in (61.5) is $\epsilon^\mu\partial_\mu\eta^a$. Integrating it by parts in the source term, and keeping the measure variation, produces the contact action $-\epsilon^\mu\partial_\mu\mathcal{O}_a$ on the operator. The scalar source weight $(\Delta - D)\sigma_\epsilon\eta^a$ then combines with the measure variation $D\sigma_\epsilon$ to give $-\Delta\sigma_\epsilon\mathcal{O}_a$. Finally the source carries the dual spin representation, so moving the spin variation from the source to the operator gives the last term in (61.6). This proves the local insertion operator $\mathcal{D}_\epsilon$.

The source Ward identity has the distributional form that the divergence of the Noether current insertion is compensated by the variation of the source coupling, plus the possible Weyl anomaly. Differentiating it once with respect to each source that creates the insertions $\mathcal{O}_{i,a_i}(x_i)$, and then setting all sources to zero, produces a delta distribution whenever the source variation acts on the ith source factor. Because the insertion points are pairwise distinct, no additional coincident-operator contact terms occur in the displayed separated-point formula. This gives (61.7). Integrating the current identity over a small punctured ball around x, then shrinking the ball while keeping the orientation convention of Definition 60.3, leaves only the delta distribution at the insertion and gives (61.8).

61.3 Infinitesimal Action at a General Point

Infinitesimal conformal action on a spin primary. Let

$$\epsilon^\mu(x) = a^\mu + \omega^\mu{}_\nu x^\nu + \lambda x^\mu + c^\mu x^2 - 2(c\cdot x)x^\mu$$

be a conformal Killing vector in $D \geq 3$. Define

$$\sigma_\epsilon(x) = \frac{1}{D}\partial_\mu\epsilon^\mu(x), \qquad \omega_{\epsilon,\mu\nu}(x) = \frac{1}{2}(\partial_\nu\epsilon_\mu - \partial_\mu\epsilon_\nu).$$

For a primary operator, the infinitesimal variation generated by the charge is the contact operator $\mathcal{D}_\epsilon$:

$$\delta_\epsilon\mathcal{O}_a(x) = -\epsilon^\mu(x)\partial_\mu\mathcal{O}_a(x) - \Delta\sigma_\epsilon(x)\mathcal{O}_a(x) - \frac{\mathrm{i}}{2}\omega_\epsilon^{\mu\nu}(x)(S_{\mu\nu})_a{}^b\mathcal{O}_b(x). \tag{61.9}$$

The first term moves the insertion point. The second term is the local scale weight. The third term rotates the spin indices by the orthogonal part of the Jacobian. The formula is (61.6) written for a general conformal Killing vector. The special-conformal part is the only nonconstant point dependence not visible from translations, rotations, and dilatations, and it is checked explicitly below.

Elementary generator actions. For the four basic generators this gives

$$\delta_a\mathcal{O}(x) = -a^\mu\partial_\mu\mathcal{O}(x), \tag{61.10}$$
$$\delta_\lambda\mathcal{O}(x) = -\lambda(x^\mu\partial_\mu + \Delta)\mathcal{O}(x), \tag{61.11}$$

and

$$\delta_\omega\mathcal{O}_a(x) = -\omega^\mu{}_\nu x^\nu\partial_\mu\mathcal{O}_a(x) - \frac{\mathrm{i}}{2}\omega^{\mu\nu}(S_{\mu\nu})_a{}^b\mathcal{O}_b(x), \tag{61.12}$$
$$\begin{aligned}\delta_c\mathcal{O}_a(x) = &-\big(c^\mu x^2 - 2(c\cdot x)x^\mu\big)\partial_\mu\mathcal{O}_a(x) + 2\Delta(c\cdot x)\mathcal{O}_a(x)\\ &- 2\mathrm{i}c^\mu x^\nu(S_{\mu\nu})_a{}^b\mathcal{O}_b(x).\end{aligned} \tag{61.13}$$

The special-conformal line is obtained directly from (61.9). For

$$\epsilon_c^\mu(x) = c^\mu x^2 - 2(c\cdot x)x^\mu$$

one has

$$\partial_\mu\epsilon_c^\mu = -2D(c\cdot x), \qquad \sigma_{\epsilon_c} = -2(c\cdot x),$$

and

$$\omega_{\epsilon_c,\mu\nu} = \frac{1}{2}(\partial_\nu\epsilon_{c,\mu} - \partial_\mu\epsilon_{c,\nu}) = 2(c_\mu x_\nu - c_\nu x_\mu).$$

Substitution gives

$$-\Delta\sigma_{\epsilon_c}\mathcal{O}_a = 2\Delta(c\cdot x)\mathcal{O}_a,$$

and, using $S_{\mu\nu} = -S_{\nu\mu}$,

$$-\frac{\mathrm{i}}{2}\omega_{\epsilon_c}^{\mu\nu}(S_{\mu\nu})_a{}^b\mathcal{O}_b = -2\mathrm{i}c^\mu x^\nu(S_{\mu\nu})_a{}^b\mathcal{O}_b.$$

This proves (61.13) from the source-Ward contact operator. At $x = 0$, all three terms vanish for ϵ_c, so the same formula is consistent with $\widehat{K}_\mu|\mathcal{O}_a\rangle = 0$ in the state–operator correspondence.

61.4 Finite Transformation Law

Definition 61.5 (Conformal factor and local rotation)**.** Let

$$f : U \longrightarrow f(U)$$

be an orientation-preserving conformal map between open subsets of $\mathbb{R}^D$. Its Jacobian is

$$M_f(x)^\mu{}_\nu = \frac{\partial f^\mu}{\partial x^\nu}.$$

The conformal condition says that there is a positive function $\Omega_f(x)$ such that

$$M_f(x)^T M_f(x) = \Omega_f(x)^2 \mathbf{1}. \tag{61.14}$$

The orthogonal matrix

$$R_f(x) = \Omega_f(x)^{-1} M_f(x)$$

is the local rotation. Since f is orientation preserving, $\Omega_f(x)^D = \det M_f(x)$. The factors obey the cocycle identities

$$\Omega_{f\circ h}(x) = \Omega_f(h(x))\Omega_h(x), \qquad R_{f\circ h}(x) = R_f(h(x))R_h(x).$$

Theorem 61.6 (Finite transformation law for a primary operator)**.** *The finite primary transformation law is the integrated form of* (61.9) *on the connected conformal group:*

$$\mathcal{O}'_a(f(x)) = \Omega_f(x)^{-\Delta} \rho(R_f(x))_a{}^b \mathcal{O}_b(x). \tag{61.15}$$

Proof. Let f_s be the one-parameter flow of a conformal Killing vector,

$$\frac{\mathrm{d}}{\mathrm{d}s} f_s^\mu(x) = \epsilon^\mu(f_s(x)), \qquad f_0(x) = x,$$

and write $M_s = \partial f_s/\partial x = \Omega_s R_s$. At $s = 0$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\Omega_s(x)\right|_{s=0} = \sigma_\epsilon(x), \qquad \left.\frac{\mathrm{d}}{\mathrm{d}s}R_s(x)_\mu{}^\nu\right|_{s=0} = \omega_{\epsilon,\mu}{}^\nu(x), \tag{61.16}$$

because the conformal Killing equation decomposes $\partial_\nu\epsilon_\mu$ into its trace $\sigma_\epsilon\delta_{\mu\nu}$ and antisymmetric part $\omega_{\epsilon,\mu\nu}$. The transformed field at a fixed final point y is obtained from $x = f_s^{-1}(y)$:

$$\mathcal{O}'_{s,a}(y) = \Omega_s(f_s^{-1}y)^{-\Delta} \rho(R_s(f_s^{-1}y))_a{}^b \mathcal{O}_b(f_s^{-1}y).$$

Since $\frac{\mathrm{d}}{\mathrm{d}s} f_s^{-1}(y)\big|_{s=0} = -\epsilon(y)$ and

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\rho(R_s)_a{}^b\right|_{s=0} = -\frac{\mathrm{i}}{2}\omega_\epsilon^{\mu\nu}(S_{\mu\nu})_a{}^b,$$

differentiation at $s = 0$ gives exactly (61.9). Conversely, the vector fields close under the Lie bracket and the factors Ω_f, R_f obey the cocycle identities above; hence the finite law is the integrated representation of the infinitesimal contact action on every connected patch where the conformal map is defined. □

Elementary finite conformal transformations. The elementary generators make the signs transparent. A translation $f_a(x) = x + a$ has $\Omega = 1$, $R = 1$, and gives $\delta_a \mathcal{O} = -a^\mu \partial_\mu \mathcal{O}$ at fixed final coordinate. A dilatation $f_\lambda(x) = \mathrm{e}^\lambda x$ has $\Omega = \mathrm{e}^\lambda$, $R = 1$, and therefore

$$\mathcal{O}'(\mathrm{e}^\lambda x) = \mathrm{e}^{-\Delta\lambda}\mathcal{O}(x), \qquad \mathcal{O}'(y) = \mathrm{e}^{-\Delta\lambda}\mathcal{O}(\mathrm{e}^{-\lambda} y),$$

whose derivative is $-\lambda(y^\mu \partial_\mu + \Delta)\mathcal{O}(y)$. A rotation has $\Omega = 1$ and R equal to its orthogonal matrix, so exponentiating $\rho(R) = \exp[-\frac{\mathrm{i}}{2}\omega^{\mu\nu} S_{\mu\nu}]$ gives the spin term in (61.9). Finally the finite special conformal transformation is

$$f_c = I \circ T_c \circ I, \qquad f_c(x) = \frac{x + c\,x^2}{1 + 2c \cdot x + c^2 x^2};$$

expanding at $c = 0$ gives $\epsilon_c^\mu = c^\mu x^2 - 2(c \cdot x) x^\mu$, so the differentiated finite law reproduces (61.13).

Remark 61.7 (Tensor primary convention). For tensor representations this formula is the tensor transformation law with the Jacobian decomposed into the scale factor $\Omega_f(x)$ and the orthogonal rotation $R_f(x)$. At coincident points the corresponding statement is made at the level of the transformed source functional; the displayed formula is its restriction to noncoincident insertions.

Finite transformation of tensor primaries. For a covariant rank-s primary tensor $\mathcal{O}_{\mu_1 \cdots \mu_s}$, the same formula can be written without abstract representation notation as

$$\mathcal{O}'_{\mu_1 \cdots \mu_s}(f(x)) = \left(\det \frac{\partial f}{\partial x}\right)^{-(\Delta - s)/D} \frac{\partial x^{\nu_1}}{\partial f^{\mu_1}} \cdots \frac{\partial x^{\nu_s}}{\partial f^{\mu_s}}\, \mathcal{O}_{\nu_1 \cdots \nu_s}(x). \tag{61.17}$$

For a contravariant rank-s primary tensor,

$$\mathcal{O}'^{\mu_1 \cdots \mu_s}(f(x)) = \left(\det \frac{\partial f}{\partial x}\right)^{-(\Delta + s)/D} \frac{\partial f^{\mu_1}}{\partial x^{\nu_1}} \cdots \frac{\partial f^{\mu_s}}{\partial x^{\nu_s}}\, \mathcal{O}^{\nu_1 \cdots \nu_s}(x). \tag{61.18}$$

The different determinant powers in (61.17) and (61.18) follow from $M_f = \Omega_f R_f$. Each covariant index contributes one factor Ω_f^{-1} from M_f^{-1}, each contravariant index contributes one factor Ω_f from M_f, and the remaining determinant factor supplies the primary weight $\Omega_f^{-\Delta}$. For a conformal diffeomorphism f, the Jacobian matrix

$$M^\mu{}_\nu(x) = \frac{\partial f^\mu}{\partial x^\nu}$$

satisfies $M^T M = \Omega_f(x)^2 \mathbf{1}$. Hence

$$R^\mu{}_\nu(x) := \Omega_f(x)^{-1} M^\mu{}_\nu(x)$$

is orthogonal on the Euclidean tangent space, and

$$\det M = \Omega_f^D \det R.$$

On the connected orientation-preserving component, $\det R = 1$, so $\Omega_f = (\det M)^{1/D}$; on an orientation-reversing component the orientation sign is part of the $O(D)$ tensor representation and the positive scale is $\Omega_f = |\det M|^{1/D}$. The formulas above are written for the orientation-preserving component, which is the component used in the local conformal Ward identities.

The abstract primary law proved in Theorem 61.6 is

$$\mathcal{O}'_a(f(x)) = \Omega_f(x)^{-\Delta}\rho(R_f(x))_a{}^b\mathcal{O}_b(x).$$

For a covariant vector index, the orthogonal representation acts by

$$(R^{-1})^\nu{}_\mu = \Omega_f\frac{\partial x^\nu}{\partial f^\mu},$$

because $M^{-1} = \partial x/\partial f$. For a contravariant vector index it acts by

$$R^\mu{}_\nu = \Omega_f^{-1}\frac{\partial f^\mu}{\partial x^\nu}.$$

Tensor-product representations multiply these factors index by index. Thus a covariant rank-s tensor acquires

$$\Omega_f^{-\Delta}\Omega_f^s\frac{\partial x^{\nu_1}}{\partial f^{\mu_1}}\cdots\frac{\partial x^{\nu_s}}{\partial f^{\mu_s}} = (\det M)^{-(\Delta-s)/D}\frac{\partial x^{\nu_1}}{\partial f^{\mu_1}}\cdots\frac{\partial x^{\nu_s}}{\partial f^{\mu_s}},$$

which is (61.17). Similarly a contravariant rank-s tensor acquires

$$\Omega_f^{-\Delta}\Omega_f^{-s}\frac{\partial f^{\mu_1}}{\partial x^{\nu_1}}\cdots\frac{\partial f^{\mu_s}}{\partial x^{\nu_s}} = (\det M)^{-(\Delta+s)/D}\frac{\partial f^{\mu_1}}{\partial x^{\nu_1}}\cdots\frac{\partial f^{\mu_s}}{\partial x^{\nu_s}},$$

which is (61.18).

Remark 61.8 (Finite covariance of primary correlators). If the vacuum is invariant under the conformal transformation, correlation functions of primaries obey

$$\left\langle\prod_{i=1}^n\mathcal{O}_{i,a_i}(f(x_i))\right\rangle = \prod_{i=1}^n\Omega_f(x_i)^{-\Delta_i}\rho_i(R_f(x_i))_{a_i}{}^{b_i}\left\langle\prod_{i=1}^n\mathcal{O}_{i,b_i}(x_i)\right\rangle. \tag{61.19}$$

Equation (61.19) is the finite form of the global Ward identities on the configuration space of distinct insertion points, with contact extensions fixed by the same source convention as in (61.7). Let U_f denote the Hilbert-space or Euclidean functional action of the connected conformal transformation on the common domain of separated insertions. Vacuum invariance means that inserting $U_f^{-1}U_f$ in the expectation value does not change it:

$$\left\langle\prod_i\mathcal{O}_{i,a_i}(f(x_i))\right\rangle = \left\langle U_f^{-1}\left(\prod_i\mathcal{O}_{i,a_i}(f(x_i))\right)U_f\right\rangle.$$

On the configuration space $x_i \neq x_j$, the conjugated product is the ordinary product of the individually transformed fields, because no two insertions collide while the transformation is applied. Substituting (61.15) for each primary therefore produces one factor $\Omega_f(x_i)^{-\Delta_i}\rho_i(R_f(x_i))$ for each insertion and gives (61.19). Extensions to collision diagonals are not determined by this separated-point argument; they are fixed by the source Ward identity (61.7).

61.5 Inversion

Inversion of scalar and spin primaries. The inversion map

$$I(x)^\mu = \frac{x^\mu}{x^2}$$

has

$$\Omega_I(x) = \frac{1}{x^2}, \qquad R_I(x)^\mu{}_\nu = \delta^\mu{}_\nu - 2\frac{x^\mu x_\nu}{x^2}.$$

For a scalar primary,

$$\mathcal{O}'(I(x)) = |x|^{2\Delta}\mathcal{O}(x). \tag{61.20}$$

This formula, together with translations, rotations, and dilatations, determines the separated-point covariance constraints used below. The matrix $R_I(x)$ lies in $O(D)$ and has determinant -1. For operators with spin, inversion therefore requires a specified extension of the spin action from $SO(D)$ to the relevant disconnected orthogonal or pin action; once that choice is made, $R_I(x)$ acts on each spin index through that representation.

Differentiating $I^\mu = x^\mu/x^2$ gives

$$\frac{\partial I^\mu}{\partial x^\nu} = \frac{1}{x^2}\left(\delta^\mu{}_\nu - 2\frac{x^\mu x_\nu}{x^2}\right).$$

The matrix in parentheses squares to the identity and has determinant -1, since it reflects the component parallel to x^μ and fixes the orthogonal hyperplane. Thus $\Omega_I = 1/x^2$ and $R_I = \delta - 2xx/x^2$. Substituting these factors into (61.15) gives (61.20) for scalar primaries and the stated orthogonal or pin action for spin primaries.

61.6 Descendants

Descendant transformation laws from primary data. The descendants of a primary are obtained by applying derivatives. Their transformation laws are obtained by differentiating (61.15) and by decomposing the result into irreducible $SO(D)$ representations. Once the primary dimension, spin representation, and normalization are fixed, the action of the whole conformal algebra on the generated multiplet is fixed. Concretely, every descendant at level n is represented algebraically by an element of $\mathrm{Sym}^n(\mathbb{C}^D)\otimes V_\rho$ in (61.3), modulo any null or equation-of-motion relations. The conformal generators act on this space by the commutation relations of $\mathfrak{so}(D+2,\mathbb{C})$. Equivalently, at separated points, differentiating (61.15) with respect to the insertion coordinate gives the finite transformation of derivatives of $\mathcal{O}$. Decomposing the resulting tensor products into irreducible $SO(D)$ summands changes the basis of descendants while preserving the representation generated by the primary datum.

Chapter 62

Unitarity Bounds and Short Multiplets

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

62.1 Positivity in Radial Quantization

Assume the conformal theory is unitary after analytic continuation to Lorentzian signature and satisfies reflection positivity in Euclidean radial quantization. Then the Hilbert space on S^{D-1} has a positive semidefinite inner product, and null states are quotiented to obtain the physical Hilbert space. The adjoint relation

$$\widehat{P}_\mu^\dagger = \widehat{K}_\mu$$

implies that every Gram matrix of descendant states must be positive semidefinite. This adjoint relation is the BPZ Hilbert-space adjoint on the dense local finite-energy domain, derived in (59.33); it is not a separate formal rule about flat-space vector fields. For example, for every complex vector a^μ and every state created by a local operator $\mathcal{O}(0)$,

$$\left\| a^\mu \widehat{P}_\mu |\mathcal{O}\rangle \right\|^2 = \overline{a}^\mu a^\nu \langle \mathcal{O}| \widehat{K}_\mu \widehat{P}_\nu |\mathcal{O}\rangle \geq 0. \tag{62.1}$$

Thus $\langle \mathcal{O}|\widehat{K}_\mu \widehat{P}_\nu|\mathcal{O}\rangle$ is a positive semidefinite Hermitian matrix in the vector indices and in any additional spin or degeneracy labels. The unitarity bounds are obtained by computing such matrices from the conformal algebra.

In this chapter the generators are the radial-quantization generators, with the conventional factors of i absorbed relative to Lorentzian Hermitian charges. The commutator used below is

$$[\widehat{K}_\mu, \widehat{P}_\nu] = 2\delta_{\mu\nu}\widehat{D}_{\mathrm{rad}} - 2\widehat{J}_{\nu\mu}, \tag{62.2}$$

where $\widehat{J}_{\mu\nu}$ acts on the spin indices of the operator multiplet. This convention makes the descendant Gram matrices real Euclidean inner products. Its sign is the radial sign derived from the Hermitian Lorentzian charge algebra in (59.27)–(59.28); it is not obtained by simply deleting the factors of i in the Lorentzian commutators.

Let $|\mathcal{O}_a\rangle$ be a primary of dimension Δ and spin representation ρ, normalized so that

$$\widehat{D}_{\mathrm{rad}}|\mathcal{O}_a\rangle = \Delta|\mathcal{O}_a\rangle, \qquad \widehat{K}_\mu|\mathcal{O}_a\rangle = 0.$$

Level-one descendant Gram calculation. Assume reflection positivity in radial quantization, the adjoint relation $\widehat{P}_\mu^\dagger = \widehat{K}_\mu$, and a primary multiplet $V_\rho = \mathrm{span}\{|\mathcal{O}_a\rangle\}$ of dimension Δ. The level-one descendant Gram matrix

$$G^{(1)}_{\mu a,\nu b} = \langle\mathcal{O}_a|\widehat{K}_\mu\widehat{P}_\nu|\mathcal{O}_b\rangle. \tag{62.3}$$

is positive semidefinite and is given by

$$G^{(1)}_{\mu a,\nu b} = 2\langle\mathcal{O}_a|(\delta_{\mu\nu}\widehat{D}_{\mathrm{rad}} - \widehat{J}_{\nu\mu})|\mathcal{O}_b\rangle. \tag{62.4}$$

For any finite array $c^{\mu a}$,

$$\bar{c}^{\mu a}G^{(1)}_{\mu a,\nu b}c^{\nu b} = \left\| c^{\nu b}\widehat{P}_\nu|\mathcal{O}_b\rangle \right\|^2 \geq 0,$$

which is the positivity assertion. Since $\widehat{K}_\mu|\mathcal{O}_b\rangle = 0$, the Gram matrix is

$$\langle\mathcal{O}_a|[\widehat{K}_\mu, \widehat{P}_\nu]|\mathcal{O}_b\rangle.$$

Substituting (62.2) gives the displayed formula.

This equation is a representation-theoretic constraint. Positivity of $G^{(1)}$ gives lower bounds on Δ once the $SO(D)$ representation ρ is specified.

The Hilbert-space convention is the following. A primary multiplet is a finite-dimensional subspace $V_\rho \subset \mathcal{H}_{\mathrm{rad}}$ on which $\widehat{D}_{\mathrm{rad}}$ acts by the scalar Δ, and the rotation generators act as a unitary $SO(D)$ representation. Descendants are obtained by applying the closed operators $\widehat{P}_\mu$ on their common finite-energy invariant domain. Positivity is imposed on every finite Gram matrix of descendants in this domain; null vectors are then quotiented. Thus each shortening equation below is an operator equation in the quotient of the local-operator space by null operators.

Equivalently, the calculation is the restriction of the Shapovalov form on the conformal generalized Verma module $M(\Delta, \rho)$ of Definition 61.3 to finite descendant levels, with the real radial adjoint $P_\mu^\dagger = K_\mu$. At each level the determinant of this form factorizes into irreducible $SO(D)$ channels; this finite conformal-algebra determinant is the Kac-determinant framework used to locate singular vectors. The proofs below compute the channels needed here explicitly, so every stated bound follows from the displayed Gram matrix rather than from an unstated determinant formula.

62.2 Scalar Primaries

For a scalar primary ϕ, the spin matrices vanish. The level-one Gram matrix reduces to

$$\langle\phi|\widehat{K}_\mu \widehat{P}_\nu|\phi\rangle = 2\Delta_\phi \delta_{\mu\nu} \langle\phi \mid \phi\rangle.$$

Thus

$$\Delta_\phi \geq 0.$$

If $\Delta_\phi = 0$, all first descendants have zero norm. After quotienting null states, this gives

$$\widehat{P}_\mu|\phi\rangle = 0.$$

Since $\widehat{P}_\mu$ represents the derivative of the local insertion at the origin, the corresponding local field obeys $\partial_\mu\phi(x) = 0$. If the theory has a unique conformally invariant vacuum and cluster decomposition for local correlators, as specified in Hypothesis 59.1, a spacetime-constant scalar local field is proportional to the identity operator, so the primary belongs to the identity representation. These hypotheses are stated because a direct sum with a decoupled topological or superselection sector can otherwise contain additional zero-energy central operators.

The sharper scalar bound comes from level two. Define

$$M^{(2)}_{\alpha\beta,\mu\nu} = \langle\phi|\widehat{K}_\alpha \widehat{K}_\beta \widehat{P}_\mu \widehat{P}_\nu|\phi\rangle.$$

Repeated use of the conformal commutation relations gives, for a scalar primary,

$$M^{(2)}_{\alpha\beta,\mu\nu} = 4\left[(\delta_{\alpha\mu}\delta_{\beta\nu} + \delta_{\alpha\nu}\delta_{\beta\mu})\Delta_\phi(\Delta_\phi + 1) - \delta_{\alpha\beta}\delta_{\mu\nu}\Delta_\phi\right]\langle\phi \mid \phi\rangle. \tag{62.5}$$

This formula is obtained by commuting the two $\widehat{K}$'s to the right until they annihilate the primary. No dynamical equation has been assumed; the only inputs are (62.2), the scalar action of $\widehat{D}_{\mathrm{rad}}$, and the vanishing spin action on $|\phi\rangle$. Contracting with $\delta^{\alpha\beta}\delta^{\mu\nu}$ computes the norm of $\widehat{P}^2|\phi\rangle$:

$$\delta^{\alpha\beta}\delta^{\mu\nu} M^{(2)}_{\alpha\beta,\mu\nu} = \left\|\widehat{P}^2|\phi\rangle\right\|^2 = 4\left[2D\,\Delta_\phi(\Delta_\phi + 1) - D^2\Delta_\phi\right]\langle\phi \mid \phi\rangle = 4D\,\Delta_\phi\big(2\Delta_\phi + 2 - D\big)\langle\phi \mid \phi\rangle. \tag{62.6}$$

Theorem 62.1 (Scalar unitarity bound and scalar shortening)**.** *Assume a positive-norm scalar primary ϕ in a reflection-positive radial-quantized CFT in $D > 2$. Then either $\Delta_\phi = 0$ and $\widehat{P}_\mu|\phi\rangle = 0$, or*

$$\Delta_\phi \geq \frac{D-2}{2}. \tag{62.7}$$

If $\Delta_\phi = (D-2)/2$, the scalar trace descendant is null and the quotient local-operator space obeys

$$\partial^2\phi(x) = 0. \tag{62.8}$$

Proof. In radial quantization $\widehat{K}_\mu = \widehat{P}^\dagger_\mu$. The level-one descendant Gram matrix is therefore

$$\langle\phi|\widehat{K}_\mu \widehat{P}_\nu|\phi\rangle = 2\Delta_\phi \delta_{\mu\nu}\langle\phi \mid \phi\rangle,$$

using the scalar-primary condition $\widehat{K}_\mu|\phi\rangle = 0$ and the commutator (62.2). Positivity gives $\Delta_\phi \geq 0$. If $\Delta_\phi = 0$, every first descendant has zero norm. After quotienting the radical of the reflection-positive form this means $\widehat{P}_\mu|\phi\rangle = 0$, so the state is translation-invariant and belongs to the identity branch under the vacuum assumptions stated in the preceding section.

Assume now $\Delta_\phi > 0$. The scalar trace descendant $\widehat{P}^2|\phi\rangle = \delta^{\mu\nu}\widehat{P}_\mu\widehat{P}_\nu|\phi\rangle$ is a vector in the level-two descendant space, hence its norm is nonnegative. The explicit Gram computation in (62.6) gives

$$0 \le 4D\,\Delta_\phi(2\Delta_\phi + 2 - D)\langle\phi \mid \phi\rangle.$$

Because $D > 2$, $\Delta_\phi > 0$, and $\langle\phi \mid \phi\rangle > 0$, this is equivalent to $2\Delta_\phi + 2 - D \ge 0$, namely (62.7). At equality the trace descendant is a null vector. In the Hilbert-space quotient by null descendants, the local state-field map sends that null relation to the operator equation $\partial^2\phi(x) = 0$ inside separated-point correlators; contact terms at coincident insertions are separate local-extension data. □

The $\Delta_\phi = 0$ branch was identified above as the identity representation under the stated vacuum assumptions. The second branch is the scalar unitarity bound for non-identity scalar primaries. At the level of conformal representation theory, the resulting shortened multiplet is the massless scalar multiplet.

62.3 Level-One Decomposition for Spin

Let $\mathcal{H}_\ell$ denote the $SO(D)$ representation of rank-ℓ symmetric traceless tensors, with $D \ge 3$. For a primary in $\mathcal{H}_\ell$, the level-one descendants transform in

$$\mathcal{H}_1 \otimes \mathcal{H}_\ell.$$

The $SO(D)$-orthogonal decomposition is

$$\mathcal{H}_1 \otimes \mathcal{H}_\ell = \mathcal{H}_{\ell+1} \oplus \mathcal{H}_{\ell,1} \oplus \mathcal{H}_{\ell-1},$$

where $\mathcal{H}_{\ell,1}$ is the traceless mixed-symmetry representation with two-row Young shape $(\ell, 1)$, and absent summands are omitted. For $\ell = 0$ the decomposition reduces to the vector descendant of a scalar. For $\ell \ge 1$, the last summand is the divergence channel.

The level-one Gram operator is $SO(D)$-equivariant. Because the above decomposition is multiplicity free, Schur's lemma implies that the Gram operator is scalar on each summand. The scalar is computed from quadratic Casimirs. Write

$$C_\ell = \ell(\ell + D - 2), \qquad C_1 = D - 1$$

for the $SO(D)$ Casimirs of $\mathcal{H}_\ell$ and the vector representation. In the descendant space, the spin-coupling operator appearing in (62.3) has eigenvalue

$$L_Q = -\frac{1}{2}\left(C_Q - C_1 - C_\ell\right) \tag{62.9}$$

on an irreducible summand $Q \subset \mathcal{H}_1 \otimes \mathcal{H}_\ell$. The minus sign is fixed by the radial commutator (62.2). The relevant Casimirs are

$$C_{\ell+1} = (\ell + 1)(\ell + D - 1), \tag{62.10}$$
$$C_{\ell,1} = \ell(\ell + D - 2) + D - 3, \tag{62.11}$$
$$C_{\ell-1} = (\ell - 1)(\ell + D - 3). \tag{62.12}$$

Substitution in (62.9) gives

$$L_{\ell+1} = -\ell, \qquad L_{\ell,1} = 1, \qquad L_{\ell-1} = \ell + D - 2.$$

Since $G^{(1)} = 2(\Delta - L_Q)$ on the summand Q, positivity gives

$$\Delta + \ell \geq 0, \qquad \Delta - 1 \geq 0, \qquad \Delta - \ell - D + 2 \geq 0. \tag{62.13}$$

For $\ell \geq 1$, the last inequality is the strongest one. Its null channel is the $\mathcal{H}_{\ell-1}$ projection of $P_\mu\mathcal{O}$, namely the divergence of the symmetric traceless tensor. Thus conservation at the unitarity bound is the null-state equation forced by positivity of the conformal representation.

The same first-level test applies to any irreducible $SO(D)$ representation ρ. If Q runs over the irreducible summands in $\mathcal{H}_1 \otimes \rho$, then positivity of the level-one Gram form requires

$$\Delta + \frac{1}{2}\left(C_Q - C_1 - C_\rho\right) \geq 0 \qquad \text{for every } Q \subset \mathcal{H}_1 \otimes \rho. \tag{62.14}$$

When equality holds for one summand, the corresponding projected first descendant is a null operator. Higher-level Gram matrices can impose further conditions in special cases, as the scalar trace descendant did above.

62.4 Two-Row Mixed-Symmetry Tensor Primaries

The first non-symmetric tensor case already displays the mechanism for mixed-symmetry shortening. Let $\mathcal{H}_{a,b}$ be the traceless $SO(D)$ tensor representation with Young row lengths (a, b), where $a \geq b \geq 1$. Work first in the stable tensor range in which the Young diagrams displayed below are distinct irreducible tensor representations; in lower dimensions Hodge duality may split or identify summands, and the same Gram-matrix test is then applied to the resulting irreducible $\mathrm{Spin}(D)$ summands.

With the normalization $C_1 = D - 1$ for the vector representation, the quadratic Casimir of a traceless tensor representation with row lengths $\lambda = (\lambda_1, \lambda_2, \ldots)$ is

$$C_\lambda = \sum_{i\geq 1} \lambda_i(\lambda_i + D - 2i). \tag{62.15}$$

To derive this formula, let $r = \lfloor D/2 \rfloor$ and choose the orthonormal basis e_i for the weight space of $\mathfrak{so}(D)_{\mathbb{C}}$ in which the vector representation has highest weight e_1. In the stable tensor range the highest weight of the tensor representation is $\Lambda = \sum_i \lambda_i e_i$, while the Weyl vector ϱ obeys $2\varrho_i = D - 2i$. The quadratic Casimir eigenvalue is $(\Lambda, \Lambda + 2\varrho)$, normalized so that the vector highest weight e_1 has $C_1 = (e_1, e_1 + 2\varrho) = D - 1$, and this gives (62.15). Thus

$$C_{a,b} = a(a + D - 2) + b(b + D - 4).$$

The level-one descendant space decomposes by adding or removing one box from the Young diagram. The derivative vector index may be projected into an added first, second, or third row, or it may be contracted with an index in the first or second row and then projected back to the traceless Young symmetry. This gives

$$\mathcal{H}_1 \otimes \mathcal{H}_{a,b} = \mathcal{H}_{a+1,b} \oplus \mathcal{H}_{a,b+1} \oplus \mathcal{H}_{a,b,1} \oplus \mathcal{H}_{a-1,b} \oplus \mathcal{H}_{a,b-1}. \tag{62.16}$$

Invalid summands are omitted: for example $\mathcal{H}_{a,b+1}$ and $\mathcal{H}_{a-1,b}$ are absent when $a = b$, and $\mathcal{H}_{a,b-1}$ becomes $\mathcal{H}_a$ when $b = 1$. The $\mathcal{H}_{a,b,1}$ channel is present only when the corresponding three-row tensor representation is not removed by a low-dimensional duality. Each displayed summand has multiplicity one in the stable range, so the level-one Gram operator is diagonal on these channels.

Substituting the Casimirs in (62.14) gives the following table:

$$\begin{array}{c|c|c} Q & \frac{1}{2}(C_Q - C_1 - C_{a,b}) & \text{level-one positivity condition} \\ \hline \mathcal{H}_{a+1,b} & a & \Delta + a \geq 0 \\ \mathcal{H}_{a,b+1} & b-1 & \Delta + b - 1 \geq 0 \\ \mathcal{H}_{a,b,1} & -2 & \Delta - 2 \geq 0 \\ \mathcal{H}_{a-1,b} & -(a+D-2) & \Delta - a - D + 2 \geq 0 \\ \mathcal{H}_{a,b-1} & -(b+D-3) & \Delta - b - D + 3 \geq 0. \end{array} \tag{62.17}$$

For instance,

$$\frac{1}{2}(C_{a-1,b} - C_1 - C_{a,b}) = -a - D + 2,$$

because replacing a by $a-1$ changes the first-row contribution to the Casimir by $-2a-D+3$, and the vector Casimir subtracts another $D-1$. The other entries follow from the same row-by-row subtraction in (62.15).

Proposition 62.2 (Two-row tensor unitarity bound at first level)**.** *Let $\mathcal{O}$ be a primary whose rotation representation is $\mathcal{H}_{a,b}$, $a \geq b \geq 1$, in the stable tensor range described above. If $a > b$, positivity of the first descendant Gram matrix implies*

$$\Delta_{\mathcal{O}} \geq a + D - 2. \tag{62.18}$$

At equality the $\mathcal{H}_{a-1,b}$ projection of $P_\mu|\mathcal{O}\rangle$ is null. In local-operator notation this is

$$\Pi_{a-1,b}\, \partial\mathcal{O} = 0, \tag{62.19}$$

where $\Pi_{a-1,b} : \mathcal{H}_1 \otimes \mathcal{H}_{a,b} \to \mathcal{H}_{a-1,b}$ is the $SO(D)$-equivariant traceless Young projector. If $a = b$, the channel $\mathcal{H}_{a-1,b}$ is not an allowed Young shape, and the leading first-level condition is instead

$$\Delta_{\mathcal{O}} \geq a + D - 3. \tag{62.20}$$

At equality the $\mathcal{H}_{a,a-1}$ projection of the first descendant is null:

$$\Pi_{a,a-1}\, \partial\mathcal{O} = 0. \tag{62.21}$$

Proof. The inequalities in (62.17) are precisely the eigenvalue inequalities for the positive semidefinite level-one Gram form. When $a > b$, the removal channel $\mathcal{H}_{a-1,b}$ is present, and its condition $\Delta \geq a + D - 2$ is stronger than $\Delta \geq b + D - 3$ because $a - b + 1 > 0$; it is also stronger than the add-a-box conditions for a positive-energy representation. If equality holds, the Gram norm on that irreducible summand is zero, hence the corresponding state is in the null subspace and becomes the operator equation (62.19) in the quotient by null operators.

When $a = b$, the Young diagram $(a-1, a)$ is not allowed, so the $\mathcal{H}_{a-1,b}$ row in the table is absent. The strongest remaining condition is $\Delta \geq a + D - 3$, coming from $\mathcal{H}_{a,a-1}$. Equality makes the $\mathcal{H}_{a,a-1}$ component of $P_\mu|\mathcal{O}\rangle$ null, giving (62.21). The special case $(a, b) = (1, 1)$ is the antisymmetric two-tensor primary: the shortening equation is the projected divergence $\partial^\mu J_{\mu\nu} = 0$, after traces are removed according to the representation. □

62.5 Spinor Primaries

For spinor primaries the rotation representation is a representation of $\mathrm{Spin}(D)$, the double cover of $SO(D)$. Let $D \geq 3$, and let S be an irreducible complex spinor representation. In even D, S may be one of the two chiral spinor representations; Clifford multiplication by a vector then maps S to the opposite chirality. Denote the spinor representation reached by Clifford multiplication by S_γ. Thus $S_\gamma \simeq S$ in odd dimension, while S_γ is the opposite chiral spinor in even dimension. Choose Euclidean Clifford maps

$$\Gamma_\mu : S \longrightarrow S_\gamma, \qquad \Gamma_\mu^\dagger : S_\gamma \longrightarrow S, \qquad \Gamma_\mu^\dagger \Gamma_\nu + \Gamma_\nu^\dagger \Gamma_\mu = 2\delta_{\mu\nu} \mathrm{id}_S.$$

These maps are intertwiners for the $\mathrm{Spin}(D)$ action and therefore identify the gamma-trace summand in the level-one descendant space.

Theorem 62.3 (Spinor unitarity bound and Dirac shortening)**.** *Let ψ be a spinor primary of scaling dimension Δ_ψ whose radial-quantization primary space carries the unitary spin representation S. Then*

$$\Delta_\psi \geq \frac{D-1}{2}. \tag{62.22}$$

If equality holds, the gamma-trace first descendant is null. In local-operator notation the shortening equation is

$$\Gamma^\mu \partial_\mu \psi(x) = 0. \tag{62.23}$$

Proof. The relevant $SO(D)$ Casimir eigenvalues, in the normalization used in (62.9), are

$$C_S = C_{S_\gamma} = \frac{D(D-1)}{8}, \qquad C_1 = D - 1.$$

The tensor product of the vector representation with S decomposes as

$$\mathcal{H}_1 \otimes S = S_\gamma \oplus S_{3/2},$$

where $S_{3/2}$ is the gamma-traceless vector-spinor representation. The orthogonal projection onto the first summand is the gamma-trace projection: for $A_\mu \in \mathcal{H}_1 \otimes S$,

$$(\Pi_\gamma A)_\mu = \frac{1}{D} \Gamma_\mu^\dagger \Gamma^\nu A_\nu, \qquad \Gamma^\mu (A_\mu - \Pi_\gamma A_\mu) = 0.$$

Equivalently, after normalizing the primary state so that $\| |\psi(s)\rangle \|^2 = \|s\|_S^2$ for $s \in S$, the gamma-trace descendant has Gram norm

$$\left\| \frac{1}{\sqrt{D}} \Gamma^\mu \widehat{P}_\mu |\psi(s)\rangle \right\|^2 = 2 \left(\Delta_\psi - \frac{D-1}{2} \right) \|s\|_S^2. \tag{62.24}$$

This is precisely (62.3) restricted to the $Q = S_\gamma$ irreducible summand, because

$$\Delta_\psi + \frac{1}{2}(C_{S_\gamma} - C_1 - C_S) = \Delta_\psi - \frac{D-1}{2}.$$

Reflection positivity of radial quantization requires the left-hand side of (62.24) to be nonnegative for every s, giving (62.22). If equality holds, the gamma-trace descendant is a null state and therefore vanishes in the quotient local-operator space, which gives (62.23).

For completeness, the complementary vector-spinor summand has

$$C_{S_{3/2}} = C_S + D.$$

Its level-one inequality is

$$\Delta_\psi + \frac{1}{2}(C_{S_{3/2}} - C_1 - C_S) = \Delta_\psi + \frac{1}{2} \geq 0,$$

so it gives no stronger lower bound for a positive-energy spinor primary. □

62.6 Symmetric Traceless Tensor Primaries

Let

$$\mathcal{O}_{\mu_1\cdots\mu_\ell}(x)$$

be a primary transforming as a rank-ℓ symmetric traceless tensor of $SO(D)$, with $\ell \geq 1$. Thus the indices are symmetric and

$$\delta^{\mu_1\mu_2}\mathcal{O}_{\mu_1\mu_2\cdots\mu_\ell} = 0.$$

Theorem 62.4 (Symmetric-traceless unitarity bound)**.** *Assume a positive-norm primary $\mathcal{O}_{\mu_1\cdots\mu_\ell}$, $\ell \geq 1$, transforming in the rank-ℓ symmetric traceless representation of $SO(D)$, with $D \geq 3$. Then*

$$\Delta_{\mathcal{O}} \geq \ell + D - 2. \tag{62.25}$$

At equality, the divergence descendant is null and therefore

$$\partial^{\mu_1}\mathcal{O}_{\mu_1\mu_2\cdots\mu_\ell}(x) = 0. \tag{62.26}$$

Proof. The level-one descendant space decomposes as

$$\mathcal{H}_1 \otimes \mathcal{H}_\ell = \mathcal{H}_{\ell+1} \oplus \mathcal{H}_{\ell,1} \oplus \mathcal{H}_{\ell-1}.$$

This is an orthogonal decomposition for the $SO(D)$-invariant radial Gram form, because the Gram operator commutes with rotations. On each irreducible summand, Schur's lemma makes the level-one Gram operator a scalar. Formula (62.3) identifies that scalar as

$$2\left[\Delta_{\mathcal{O}} + \frac{1}{2}(C_Q - C_1 - C_\ell)\right],$$

where Q is the corresponding irreducible summand. The Casimir computation above gives the three inequalities in (62.13); for $\ell \geq 1$, the strongest is

$$\Delta_{\mathcal{O}} - \ell - D + 2 \geq 0.$$

The representation that carries this eigenvalue is $\mathcal{H}_{\ell-1}$. At the level of tensor fields this projection is the trace-free part of the divergence $\partial^{\mu_1}\mathcal{O}_{\mu_1\mu_2\cdots\mu_\ell}$; because the original operator is symmetric traceless, the displayed divergence is already the relevant $\mathcal{H}_{\ell-1}$ descendant. At saturation the entire $\mathcal{H}_{\ell-1}$ descendant subspace is null. Passing to the quotient by the null radical gives the separated-point local-operator equation (62.26); contact terms in Ward identities are additional coincident-point extension data. □

For $\ell = 1$ this is current conservation and $\Delta_J = D - 1$. For $\ell = 2$ it is stress-tensor conservation and $\Delta_T = D$.

62.7 Short Multiplets as Operator Equations

The scalar, spinor, and symmetric-traceless calculations above give the following collected reference list. The proof of each entry is the corresponding descendant Gram-matrix computation: Theorem 62.1 for scalars, Theorem 62.3 for spinors, and Theorem 62.4 for rank-ℓ symmetric traceless tensors.

1. A scalar primary ϕ either lies in the $\Delta_\phi = 0$ identity branch described in Theorem 62.1, or obeys
$$\Delta_\phi \geq \frac{D-2}{2}.$$
At $\Delta_\phi = (D-2)/2$, the level-two trace descendant is null and
$$\partial^2\phi = 0$$
in the quotient local-operator space.
2. A spinor primary ψ obeys
$$\Delta_\psi \geq \frac{D-1}{2}.$$
At equality, the gamma-trace first descendant is null and
$$\Gamma^\mu \partial_\mu \psi = 0.$$
3. A rank-ℓ symmetric traceless tensor primary $\mathcal{O}_{\mu_1\cdots\mu_\ell}$, $\ell \geq 1$, obeys
$$\Delta_{\mathcal{O}} \geq \ell + D - 2.$$
At equality, the divergence first descendant is null and
$$\partial^{\mu_1}\mathcal{O}_{\mu_1\mu_2\cdots\mu_\ell} = 0.$$

A short conformal multiplet is a primary multiplet for which a descendant has zero norm and is set to zero in the physical Hilbert space. In local-operator language, the null descendant is an operator equation. The equations

$$\partial^2\phi = 0, \qquad \partial^\mu J_\mu = 0, \qquad \partial^\mu T_{\mu\nu} = 0$$

are therefore representation-theoretic statements in a unitary conformal theory. They also carry dynamical information: a conserved current generates a global symmetry, the stress tensor generates spacetime symmetries, and a scalar saturating the scalar bound obeys the scalar Laplace shortening equation.

The unitarity bounds thus convert positivity of the Hilbert-space inner product into exact restrictions on the allowed local operator spectrum.

Chapter 63

Correlation Functions and Conformal Frames

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

63.1 Ward Identities for Correlators

Throughout this chapter, correlators are Euclidean vacuum correlators at separated insertion points. The symbol $|\Omega\rangle$ denotes the common vacuum identified in Section 59.3: the Poincare vacuum of the flat Wightman theory, the conformally invariant vacuum under the hypotheses of Proposition 59.4, and the zero-energy cylinder ground state after the Weyl-map normalization of radial quantization. With this identification, and with the special-conformal vacuum invariance derived in Lemma 63.2, the global Ward identity of Chapter 60 applies. If $\epsilon^\mu(x)\partial_\mu$ is a conformal Killing vector and $\delta_\epsilon \mathcal{O}_i$ is the corresponding infinitesimal variation of the ith local operator, then

$$\sum_{i=1}^{n} \langle \mathcal{O}_1(x_1) \cdots \delta_\epsilon \mathcal{O}_i(x_i) \cdots \mathcal{O}_n(x_n)\rangle = 0 \tag{63.1}$$

as a distribution on the configuration space $x_i \neq x_j$. Contact terms on coincident-point diagonals are part of the local Ward identities and are not used in the kinematic classification below.

Definition 63.1 (Separated Euclidean CFT correlator)**.** For an ordered n-tuple of local operators $\mathcal{O}_1, \ldots, \mathcal{O}_n$, the separated configuration space is

$$\mathrm{Conf}_n(\mathbb{R}^D) = \{(x_1, \ldots, x_n) \in (\mathbb{R}^D)^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

The separated correlator is the restriction of the corresponding Schwinger distribution to this open set,

$$G_{\mathcal{O}_1\cdots\mathcal{O}_n}(x_1,\dots,x_n)=\langle\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle\,,$$

with values in the dual tensor product of the spin representation spaces when the insertions carry indices. A scalar primary tuple $(\mathcal{O}_i,\Delta_i)$ is separated conformally covariant if, for every finite conformal transformation f whose image avoids coincidences,

$$G_{\mathcal{O}_1\cdots\mathcal{O}_n}(f(x_1),\dots,f(x_n))=\prod_{i=1}^{n}\Omega_f(x_i)^{-\Delta_i}\,G_{\mathcal{O}_1\cdots\mathcal{O}_n}(x_1,\dots,x_n),\tag{63.2}$$

where $f^*\delta=\Omega_f^2\delta$. This definition contains no extension across the collision diagonals. Such extensions are local renormalization data and can differ by contact terms supported on partial diagonals.

Lemma 63.2 (Special-conformal invariance from radial spectrum positivity)**.** *Let $|\Omega\rangle$ be the common vacuum of Section 59.3. Assume that $\widehat{D}_{\mathrm{rad}}$ is a nonnegative self-adjoint generator of cylinder time translations, that*

$$\widehat{D}_{\mathrm{rad}}|\Omega\rangle=0,$$

and that the special conformal generators are defined on the vacuum and obey, on the relevant common domain,

$$[\widehat{D}_{\mathrm{rad}},\widehat{K}_\mu]=-\widehat{K}_\mu.\tag{63.3}$$

Then

$$\widehat{K}_\mu|\Omega\rangle=0.$$

Thus the special-conformal part of the vacuum Ward identity follows from radial spectral positivity and the conformal commutation relation; it is not an extra invariance assumption.

Proof. Let $\psi_\mu=\widehat{K}_\mu|\Omega\rangle$. The domain hypothesis is included so that both $\widehat{K}_\mu|\Omega\rangle$ and $\widehat{D}_{\mathrm{rad}}\widehat{K}_\mu|\Omega\rangle$ are honest Hilbert-space vectors. On this common domain, (63.3) gives

$$\widehat{D}_{\mathrm{rad}}\psi_\mu=\widehat{D}_{\mathrm{rad}}\widehat{K}_\mu|\Omega\rangle=\widehat{K}_\mu\widehat{D}_{\mathrm{rad}}|\Omega\rangle-\widehat{K}_\mu|\Omega\rangle=-\psi_\mu.$$

Thus, if $\psi_\mu\neq 0$, the spectral measure of ψ_μ for the nonnegative self-adjoint operator $\widehat{D}_{\mathrm{rad}}$ would be supported at the single point -1. The spectral theorem gives $\langle\psi_\mu,\widehat{D}_{\mathrm{rad}}\psi_\mu\rangle\geq 0$ for every vector in the quadratic-form domain, while the displayed eigenvalue equation gives the same expectation value as $-\|\psi_\mu\|^2$. Hence $\|\psi_\mu\|=0$, and $\widehat{K}_\mu|\Omega\rangle=0$. □

Equivalently, for a finite conformal transformation $f:x\mapsto\widetilde{x}$ implemented by a unitary operator U_f, invariance of the vacuum gives

$$\langle\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle=\left\langle(U_f\mathcal{O}_1U_f^{-1})(x_1)\cdots(U_f\mathcal{O}_nU_f^{-1})(x_n)\right\rangle.$$

The operators on the right are written at the same coordinate labels as before; their transformation laws supply the Jacobian and spin factors. For a scalar primary ϕ of dimension Δ_ϕ,

$$\widetilde{\phi}(\widetilde{x})=\left|\det\frac{\partial\widetilde{x}}{\partial x}\right|^{-\Delta_\phi/D}\phi(x).\tag{63.4}$$

63.2 One- and Two-Point Functions

Let $\mathcal{O}_i$ be scalar primary operators of dimensions Δ_i. If the separated Ward identities and the finite scalar primary covariance law hold, then

$$\langle \mathcal{O}_i(x) \rangle = 0 \qquad \text{unless} \qquad \Delta_i = 0.$$

In a reflection-positive CFT whose $\Delta = 0$ scalar primary sector is the vacuum representation, the only nonzero scalar-primary one-point function is the identity expectation value, after choosing the identity normalization. Translation invariance makes a one-point function constant:

$$\langle \mathcal{O}_i(x) \rangle = c_i.$$

Scale covariance gives

$$c_i = \lambda^{-\Delta_i} c_i \qquad \text{for all } \lambda > 0.$$

If $\Delta_i \neq 0$, choosing any $\lambda \neq 1$ forces $c_i = 0$. The final statement uses the $\Delta = 0$ branch of the scalar unitarity analysis in Chapter 62: the local field has vanishing translation descendants and hence acts as a multiple of the identity on the cyclic vacuum representation.

Scalar two-point selection rule. Assume the separated-point global conformal Ward identities (63.1) and scalar primary transformation law (63.4). If $\mathcal{O}_i, \mathcal{O}_j$ are scalar primaries of dimensions Δ_i, Δ_j, then

$$\langle \mathcal{O}_i(x)\mathcal{O}_j(y) \rangle = \begin{cases} \dfrac{C_{ij}}{|x-y|^{2\Delta_i}}, & \Delta_i = \Delta_j, \\ 0, & \Delta_i \neq \Delta_j. \end{cases} \tag{63.5}$$

Translation and rotation invariance imply

$$\langle \mathcal{O}_i(x)\mathcal{O}_j(y) \rangle = F_{ij}(|x-y|).$$

Scale covariance gives

$$F_{ij}(r) = \frac{C_{ij}}{r^{\Delta_i+\Delta_j}}.$$

The condition $\Delta_i = \Delta_j$ for a nonzero coefficient follows from a conformal transformation that fixes both points but has different local scale factors at the two points. Put the two points at 0 and e_1, where e_1 is a fixed unit vector, and define

$$f_\lambda(x)^\mu = \lambda \frac{x^\mu + a^\mu x^2}{1 + 2a\cdot x + a^2 x^2}, \qquad a^\mu = (\lambda - 1)e_1^\mu, \qquad \lambda > 0.$$

Then $f_\lambda(0) = 0$, $f_\lambda(e_1) = e_1$,

$$\left.\frac{\partial f_\lambda^\mu}{\partial x^\nu}\right|_{x=0} = \lambda\, \delta^\mu{}_\nu, \qquad \left.\frac{\partial f_\lambda^\mu}{\partial x^\nu}\right|_{x=e_1} = \lambda^{-1} \delta^\mu{}_\nu.$$

Applying (63.4) to $\langle \mathcal{O}_i(e_1)\mathcal{O}_j(0) \rangle$ gives

$$C_{ij} = \lambda^{\Delta_i - \Delta_j} C_{ij} \qquad \text{for all } \lambda > 0,$$

so $C_{ij} = 0$ unless $\Delta_i = \Delta_j$. Substituting this condition in the scale-covariant form gives (63.5). Within a finite-dimensional space of primaries with the same dimension and the same spin, one may choose a basis that diagonalizes the Hermitian matrix C_{ij}.

63.3 Radial Inner Product and BPZ Conjugation

Let $x = rn$, $r > 0$, $n \in S^{D-1}$, and set $\tau = \log r$. The Weyl map from $\mathbb{R}^D \setminus \{0\}$ to $\mathbb{R}_\tau \times S^{D-1}$ relates a scalar primary on flat space to the cylinder operator by

$$\mathcal{O}(x) = r^{-\Delta}\widetilde{\mathcal{O}}(\tau, n). \tag{63.6}$$

The state created by insertion at the origin is

$$|\mathcal{O}\rangle = \lim_{\tau \to -\infty} \mathrm{e}^{-\Delta\tau}\widetilde{\mathcal{O}}(\tau, n)|\Omega\rangle, \tag{63.7}$$

where $|\Omega\rangle$ is the common vacuum specified above. For a primary the limit is independent of n. The conjugate bra is obtained by the opposite cylinder-time limit,

$$\langle\!\langle \mathcal{O}| = \lim_{\tau \to +\infty} \mathrm{e}^{\Delta\tau}\langle\Omega|\widetilde{\mathcal{O}}(\tau, n). \tag{63.8}$$

This radial conjugation is often called BPZ conjugation. In a reflection-positive theory, after including the ordinary internal adjoint of the local operator, it is the Hilbert-space adjoint of the state. Consequently the two-point coefficient has the inner-product interpretation

$$\langle\!\langle \mathcal{O}_i|\mathcal{O}_j\rangle = \lim_{\tau_i \to +\infty,\ \tau_j \to -\infty} \mathrm{e}^{\Delta_i\tau_i - \Delta_j\tau_j}\left\langle \widetilde{\mathcal{O}}_i(\tau_i, n_i)\widetilde{\mathcal{O}}_j(\tau_j, n_j)\right\rangle = C_{ij}, \tag{63.9}$$

for primaries with equal dimension. Positivity of this matrix is the same positivity used in the unitarity-bound derivation.

Figure 63.1 records the two-point cylinder inner product used in radial quantization: one insertion is reflected to the opposite cylinder end, and the coefficient matrix C_{ij} is the BPZ inner product for equal dimensions.

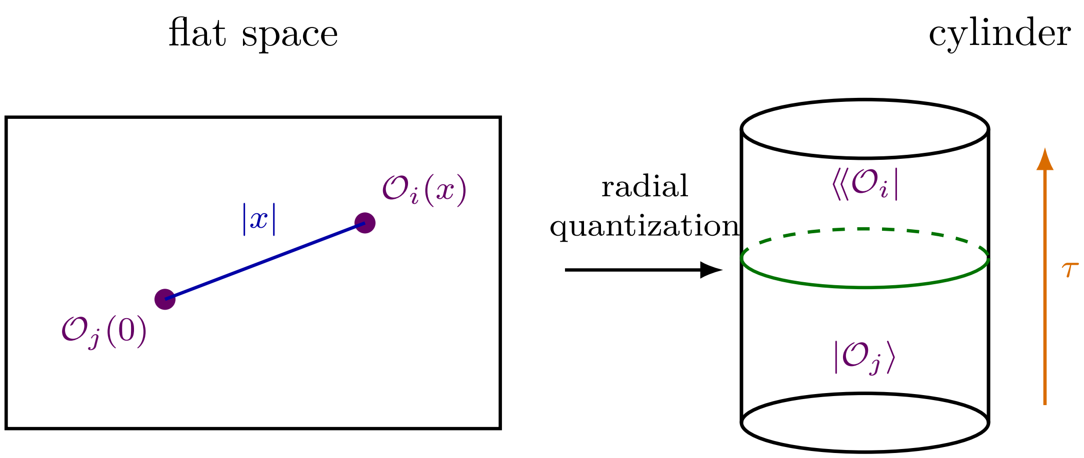


Figure 63.1: The scalar two-point coefficient is the radial-quantization inner product between the state created at the origin and the BPZ-conjugate insertion at future cylinder time.

63.4 Spin Structures

Definition 63.3 (Inversion tensor and symmetric-traceless transport)**.** The inversion tensor

$$I_{\mu\nu}(x) = \delta_{\mu\nu} - 2\frac{x_\mu x_\nu}{x^2} \tag{63.10}$$

is the orthogonal part of the inversion Jacobian. It is the basic building block for two-point functions with vector or tensor indices. Its action on the line spanned by x^μ is multiplication by -1, while its action on the orthogonal complement of x^μ is the identity. Hence $\det I(x) = -1$: inversion lies in $O(D)\backslash SO(D)$. Every use of $I_{\mu\nu}(x)$ in a spin tensor structure therefore presupposes the extension of the $SO(D)$ spin action to the disconnected orthogonal, or pin, action specified in Chapter 61.

For a rank-s symmetric traceless tensor representation, define

$$\mathcal{I}^{(s)}_{\mu_1\cdots\mu_s,\nu_1\cdots\nu_s}(x) = \Pi_{\mathrm{STT}}\left[I_{\mu_1\nu_1}(x)\cdots I_{\mu_s\nu_s}(x)\right],$$

where Π_{STT} symmetrizes separately in the μ-indices and the ν-indices and removes all traces with respect to $\delta_{\mu\nu}$.

Spinning two-point structures. Let V_μ be a vector primary of dimension Δ_V. After diagonalizing degeneracy labels within the same primary type,

$$\langle V_\mu(x)V_\nu(0)\rangle = \frac{C_V}{|x|^{2\Delta_V}} I_{\mu\nu}(x). \tag{63.11}$$

More generally, if $\mathcal{O}_{\mu_1\cdots\mu_s}$ is a rank-s symmetric traceless tensor primary of dimension $\Delta_{\mathcal{O}}$, then

$$\langle \mathcal{O}_{\mu_1\cdots\mu_s}(x)\mathcal{O}_{\nu_1\cdots\nu_s}(0)\rangle = \frac{C_{\mathcal{O}}}{|x|^{2\Delta_{\mathcal{O}}}}\mathcal{I}^{(s)}_{\mu_1\cdots\mu_s,\nu_1\cdots\nu_s}(x), \tag{63.12}$$

up to the same positive Hermitian coefficient matrix in possible degeneracy labels. Two-point functions between inequivalent $SO(D)$ representations vanish by rotational covariance. Translation covariance puts one insertion at the origin, and dilatation covariance gives the overall power $|x|^{-2\Delta}$ once the two dimensions have been matched by the scalar selection rule generalized to the chosen spin type. The remaining tensor must intertwine the spin representation at the origin with the spin representation transported to the point x. Applying inversion sends x to x/x^2, fixes the origin as the point at infinity in radial quantization, and contributes the orthogonal Jacobian $I_{\mu\nu}(x)$ on vector indices. Hence the unique vector intertwiner is $I_{\mu\nu}(x)$. For symmetric traceless rank s, the tensor product of s vector intertwiners must be projected to the symmetric-traceless irreducible component at both ends, giving $\mathcal{I}^{(s)}$. Schur's lemma on each irreducible $SO(D)$ type gives vanishing between inequivalent representations and a Hermitian degeneracy matrix within equivalent representations. Reflection positivity makes that matrix positive semidefinite before quotienting null states and positive definite on the Hilbert quotient.

For the stress tensor, $s = 2$, $\Delta_T = D$, and conservation gives a canonical normalization convention through the Ward identity. Its two-point function is

$$\langle T_{\mu\nu}(x)T_{\rho\sigma}(0)\rangle = \frac{C_T}{|x|^{2D}} I_{\mu\nu,\rho\sigma}(x), \tag{63.13}$$

where

$$I_{\mu\nu,\rho\sigma}(x) = \frac{1}{2}\big(I_{\mu\rho}(x)I_{\nu\sigma}(x) + I_{\mu\sigma}(x)I_{\nu\rho}(x)\big) - \frac{1}{D}\delta_{\mu\nu}\delta_{\rho\sigma}.$$

The coefficient C_T is part of the intrinsic CFT data because the normalization of $T_{\mu\nu}$ is fixed by the translation Ward identity.

63.5 Three-Point Functions

Lemma 63.4 (Scalar three-point kinematics)**.** *Let $\mathcal{O}_i$ be scalar primaries of dimensions Δ_i. On the configuration space of three distinct points, separated conformal covariance fixes the three-point function to the form*

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{C_{123}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta_3}|x_{23}|^{\Delta_2+\Delta_3-\Delta_1}|x_{13}|^{\Delta_1+\Delta_3-\Delta_2}}, \tag{63.14}$$

where

$$x_{ij} = x_i - x_j.$$

Proof. The connected Euclidean conformal group acts transitively on ordered triples of distinct points. Indeed, translate x_3 to the origin, use an inversion composed with a translation to send one of the remaining points to infinity, and then rotate and dilate the last finite point to a fixed unit vector. Thus there is no conformal cross-ratio for three separated points; the correlator is determined by one coefficient once its local weights at the three insertions are fixed.

It remains to determine the powers of the pairwise distances. Write the unique possible separated scalar form as

$$C_{123}|x_{12}|^{-\alpha_{12}}|x_{23}|^{-\alpha_{23}}|x_{13}|^{-\alpha_{13}}.$$

Under inversion $x \mapsto x/x^2$,

$$|x_{ij}| \longmapsto \frac{|x_{ij}|}{|x_i|\,|x_j|}, \qquad \Omega(x_i) = |x_i|^{-2}.$$

Comparison with (63.2) gives the three local weight equations

$$\alpha_{12}+\alpha_{13} = 2\Delta_1, \qquad \alpha_{12}+\alpha_{23} = 2\Delta_2, \qquad \alpha_{13}+\alpha_{23} = 2\Delta_3.$$

Solving this linear system gives

$$\alpha_{12} = \Delta_1+\Delta_2-\Delta_3, \quad \alpha_{23} = \Delta_2+\Delta_3-\Delta_1, \quad \alpha_{13} = \Delta_1+\Delta_3-\Delta_2,$$

which is precisely (63.14). Uniform dilatation covariance is then automatic because the sum of the three powers is $\Delta_1+\Delta_2+\Delta_3$. □

The constants C_{123}, after fixing the two-point normalization of primaries, are the scalar three-point OPE coefficients. If the primaries carry spin, let a_i denote indices in the corresponding finite-dimensional $SO(D)$ representations. Conformal covariance says that

$$\langle \mathcal{O}_{1,a_1}(x_1)\mathcal{O}_{2,a_2}(x_2)\mathcal{O}_{3,a_3}(x_3)\rangle = \sum_{r=1}^{n_{123}} C_{123}^{(r)}\mathcal{I}^{(r)}_{a_1a_2a_3}(x_{12},x_{13}), \tag{63.15}$$

where the tensors $\mathcal{I}^{(r)}$ form a basis of conformally covariant three-point tensor structures, including the required homogeneity in the separations. The finite integer n_{123} is representation-theoretic; the numbers $C_{123}^{(r)}$ are dynamical data after a convention for two-point normalization has been fixed.

The stress tensor illustrates why some three-point coefficients are constrained but not removed from the data. In a parity-even CFT in $D \geq 4$, the separated three-point function of a symmetric conserved traceless stress tensor is classified by the following Osborn–Petkou basis.[1] Set

$$X \equiv X_{12,3} = \frac{x_{13}}{x_{13}^2} - \frac{x_{23}}{x_{23}^2}, \qquad \widehat{X}_\mu = \frac{X_\mu}{(X^2)^{1/2}},$$

and, for a dimensionless tensor $t_{\mu\nu,\rho\sigma,\alpha\beta}(\widehat{X})$, define

$$\mathfrak{C}[t]_{\mu\nu,\rho\sigma,\alpha\beta} = \frac{I_{\mu\nu,\mu'\nu'}(x_{13}) I_{\rho\sigma,\rho'\sigma'}(x_{23}) t_{\mu'\nu',\rho'\sigma',\alpha\beta}(\widehat{X})}{|x_{12}|^D |x_{13}|^D |x_{23}|^D}. \tag{63.16}$$

Let $\mathrm{Sym}_{\mu\nu}$ denote the unnormalized symmetrization obtained by adding the expression with μ and ν interchanged. The following symmetric-traceless tensors are

$$\begin{aligned}
h^1_{\mu\nu} &= \widehat{X}_\mu \widehat{X}_\nu - \frac{1}{D}\delta_{\mu\nu}, \\
h^2_{\mu\nu,\rho\sigma} &= \widehat{X}_\mu \widehat{X}_\rho \delta_{\nu\sigma} + \widehat{X}_\nu \widehat{X}_\rho \delta_{\mu\sigma} + \widehat{X}_\mu \widehat{X}_\sigma \delta_{\nu\rho} + \widehat{X}_\nu \widehat{X}_\sigma \delta_{\mu\rho} \\
&\quad - \frac{4}{D}\left(\widehat{X}_\mu \widehat{X}_\nu \delta_{\rho\sigma} + \widehat{X}_\rho \widehat{X}_\sigma \delta_{\mu\nu}\right) + \frac{4}{D^2}\delta_{\mu\nu}\delta_{\rho\sigma}, \\
h^3_{\mu\nu,\rho\sigma} &= \delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} - \frac{2}{D}\delta_{\mu\nu}\delta_{\rho\sigma}, \\
h^4_{\mu\nu,\rho\sigma,\alpha\beta} &= \mathrm{Sym}_{\rho\sigma}\,\mathrm{Sym}_{\alpha\beta}\left[h^3_{\mu\nu,\rho\alpha}\widehat{X}_\sigma \widehat{X}_\beta\right] - \frac{2}{D}\delta_{\rho\sigma} h^2_{\mu\nu,\alpha\beta} \\
&\quad - \frac{2}{D}\delta_{\alpha\beta} h^2_{\mu\nu,\rho\sigma} - \frac{8}{D^2}\delta_{\rho\sigma}\delta_{\alpha\beta} h^1_{\mu\nu}, \\
h^5_{\mu\nu,\rho\sigma,\alpha\beta} &= \mathrm{Sym}_{\mu\nu}\,\mathrm{Sym}_{\rho\sigma}\,\mathrm{Sym}_{\alpha\beta}\left[\delta_{\mu\rho}\delta_{\nu\alpha}\delta_{\sigma\beta}\right] \\
&\quad - \frac{4}{D}\left(\delta_{\mu\nu} h^3_{\rho\sigma,\alpha\beta} + \delta_{\rho\sigma} h^3_{\mu\nu,\alpha\beta} + \delta_{\alpha\beta} h^3_{\mu\nu,\rho\sigma}\right) - \frac{8}{D^2}\delta_{\mu\nu}\delta_{\rho\sigma}\delta_{\alpha\beta}.
\end{aligned} \tag{63.17}$$

Each h^r is symmetric and traceless in every displayed index pair. The most general parity-even separated-point structure satisfying stress-tensor conservation is the three-parameter tensor

$$\begin{aligned}
t[a,b,c]_{\mu\nu,\rho\sigma,\alpha\beta} &= a\, h^5_{\mu\nu,\rho\sigma,\alpha\beta} + b\, h^4_{\alpha\beta,\mu\nu,\rho\sigma} - (2a+b)\left(h^4_{\mu\nu,\rho\sigma,\alpha\beta} + h^4_{\rho\sigma,\mu\nu,\alpha\beta}\right) \\
&\quad + c\, h^3_{\mu\nu,\rho\sigma} h^1_{\alpha\beta} + c\left(h^3_{\rho\sigma,\alpha\beta} h^1_{\mu\nu} + h^3_{\mu\nu,\alpha\beta} h^1_{\rho\sigma}\right) \\
&\quad + e\, h^2_{\mu\nu,\rho\sigma} h^1_{\alpha\beta} + e'\left(h^2_{\rho\sigma,\alpha\beta} h^1_{\mu\nu} + h^2_{\mu\nu,\alpha\beta} h^1_{\rho\sigma}\right) \\
&\quad + f\, h^1_{\mu\nu} h^1_{\rho\sigma} h^1_{\alpha\beta},
\end{aligned} \tag{63.18}$$

where conservation fixes the auxiliary coefficients to be

$$\begin{aligned}
e = (D+2)(Da+b-c), \qquad e' &= -(D+4)(D-2)a - (D-2)b + Dc, \\
f &= (D+4)(D-2)(4a+2b-c).
\end{aligned} \tag{63.19}$$

The three tensor structures in (63.20) are therefore

$$\mathcal{I}^{(1)} = \mathfrak{C}[t[1,0,0]], \qquad \mathcal{I}^{(2)} = \mathfrak{C}[t[0,1,0]], \qquad \mathcal{I}^{(3)} = \mathfrak{C}[t[0,0,1]].$$

[1] This is the position-space basis of H. Osborn and A. C. Petkou, *Implications of Conformal Invariance in Field Theories for General Dimensions*, Ann. Phys. 231 (1994), 311–362, arXiv:hep-th/9307010, specialized here to the separated-point, parity-even stress-tensor three-point function and rewritten in the conventions of this chapter.

Equivalently,

$$\langle T_{\mu\nu}(x_1)T_{\rho\sigma}(x_2)T_{\alpha\beta}(x_3)\rangle = a\,\mathcal{I}^{(1)}_{\mu\nu,\rho\sigma,\alpha\beta} + b\,\mathcal{I}^{(2)}_{\mu\nu,\rho\sigma,\alpha\beta} + c\,\mathcal{I}^{(3)}_{\mu\nu,\rho\sigma,\alpha\beta} = \mathfrak{C}[t[a,b,c]]_{\mu\nu,\rho\sigma,\alpha\beta}, \tag{63.20}$$

with three independent coefficients a, b, c when $D \geq 4$. The independence statement is not a counting by inspection. Before imposing conservation, conformal covariance, tracelessness, and the exchange symmetries of the three stress tensors reduce the parity-even separated-point ansatz to the five coefficients a, b, c, e, f; the coefficient e' appearing in (63.18) is then fixed by

$$e + e' = 8a + 4b - 2c.$$

The conservation equation

$$\left(\partial_\mu - D\frac{X_\mu}{X^2}\right) t_{\mu\nu,\rho\sigma,\alpha\beta}(X) = 0$$

imposes two independent linear equations. In the parametrization above these equations are exactly the formulae (63.19), which determine e, e', and f in terms of a, b, c. To see that no further relation is hidden for $D \geq 4$, put the three insertion points in a collinear frame by a conformal transformation. The residual group is $O(D-1)$. The completely transverse part of the conserved tensor is then parametrized by three $O(D-1)$-invariant tensors with coefficients r, s, u, and the Osborn–Petkou comparison gives

$$D^2 r = 16a - 2Db + (D+4)c, \qquad Ds = -4a - c, \qquad u = a. \tag{63.21}$$

For $D \geq 4$, the transverse space has dimension at least three, and these three invariant tensors are linearly independent. Hence $(a, b, c) \mapsto (r, s, u)$ has no universal null direction in the space of separated-point parity-even TTT structures.

For $D = 3$, the transverse space in the collinear frame is two-dimensional. [2] The three transverse invariant tensors obey one Schouten identity, and the collinear tensor depends only on

$$r - 4u, \qquad s + 2u. \tag{63.22}$$

Using (63.21) at $D = 3$, the null direction is

$$(a, b, c) = (1, -1, 2).$$

Thus, as separated parity-even three-dimensional structures,

$$\mathcal{I}^{(1)}_{D=3} - \mathcal{I}^{(2)}_{D=3} + 2\mathcal{I}^{(3)}_{D=3} = 0, \qquad \mathcal{I}^{(3)}_{D=3} = \frac{1}{2}\left(\mathcal{I}^{(2)}_{D=3} - \mathcal{I}^{(1)}_{D=3}\right). \tag{63.23}$$

Consequently a parity-even three-dimensional stress-tensor three-point function has two separated structures, for instance

$$\langle TTT\rangle^{\text{even}}_{D=3} = A\,\mathcal{I}^{(1)}_{D=3} + B\,\mathcal{I}^{(2)}_{D=3}. \tag{63.24}$$

If parity is not imposed in $D = 3$, there is a further parity-odd separated structure; it is not part of the parity-even Osborn–Petkou basis displayed here.

[2] For the three-dimensional stress-tensor/current structure count see J. Maldacena and A. Zhiboedov, *A Note on CFT Correlators in Three Dimensions*, arXiv:1104.4317. For the general spinning-correlator counting technology and its later conformal-block implementation, see M. S. Costa, J. Penedones, D. Poland, and S. Rychkov, *Spinning Conformal Correlators*, arXiv:1107.3554, and M. S. Costa, T. Hansen, J. Penedones, and E. Trevisani, *Radial Expansion for Spinning Conformal Blocks*, arXiv:1603.05552.

The Ward identity fixes one linear combination of these coefficients in terms of the two-point coefficient C_T. Let

$$S_D = \frac{2\pi^{D/2}}{\Gamma(D/2)}$$

be the area of the unit $(D-1)$-sphere. The small-sphere Ward identity uses the angular averages

$$\int_{S^{D-1}} \mathrm{d}\Omega\, \widehat{n}_\mu \widehat{n}_\nu = \frac{S_D}{D}\delta_{\mu\nu}, \qquad \int_{S^{D-1}} \mathrm{d}\Omega\, \widehat{n}_\mu \widehat{n}_\nu \widehat{n}_\rho \widehat{n}_\sigma = \frac{S_D}{D(D+2)}\left(\delta_{\mu\nu}\delta_{\rho\sigma} + \delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho}\right).$$

Substituting (63.18) into the flux $\int_{\partial B_\epsilon(x_3)} \mathrm{d}S\, n_\lambda T^{\lambda\tau}\epsilon_\tau$, taking $\epsilon \to 0$, and comparing with the conformal variation of the two remaining stress tensors in (63.13) gives

$$4S_D\, \frac{(D-2)(D+3)a - 2b - (D+1)c}{D(D+2)} = C_T. \tag{63.25}$$

The remaining two independent combinations are genuine CFT data. In lower dimensions or in parity-violating theories the list of admissible structures changes; the statement above is tied to the stated assumptions.

Figure 63.2 shows the surface Ward-identity argument fixing one linear combination of the three stress-tensor three-point structures in terms of C_T.

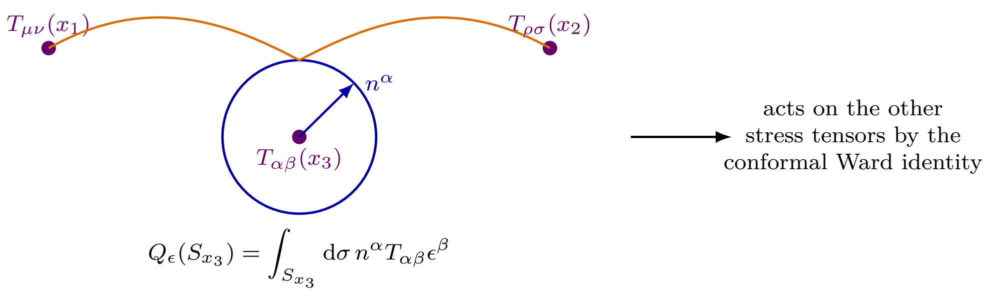


Figure 63.2: The Ward identity relates a small-sphere stress-tensor flux inside $\langle TTT\rangle$ to the conformal variation of the remaining insertions. This fixes the Ward-normalized part of the TTT data in terms of C_T.

As an important mixed example, the Ward identity fixes the coefficient of the stress-tensor three-point function with two identical scalar primaries. For $\Delta_1 = \Delta_2 = \Delta$,

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)T_{\mu\nu}(x_3)\rangle = \frac{C_{\mathcal{O}\mathcal{O}T}}{|x_{13}|^D |x_{23}|^D |x_{12}|^{2\Delta - D}}\left(\frac{X_\mu X_\nu}{X^2} - \frac{1}{D}\delta_{\mu\nu}\right), \tag{63.26}$$

with

$$X^\mu = \frac{x_{13}^\mu}{x_{13}^2} - \frac{x_{23}^\mu}{x_{23}^2}.$$

More generally, for two scalar primaries $\mathcal{O}_1, \mathcal{O}_2$ of the same dimension Δ with

$$\langle \mathcal{O}_1(x)\mathcal{O}_2(0)\rangle = \frac{N_{12}}{|x|^{2\Delta}},$$

the same tensor structure appears with coefficient

$$C_{12T} = -\frac{D\Delta}{(D-1)A_{D-1}} N_{12}, \qquad A_{D-1} = \mathrm{vol}(S^{D-1}) = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \tag{63.27}$$

Here the sign is the sign of the stress-tensor convention in (60.1) and (60.3). The derivation is a small-sphere Ward identity for the dilatation vector $\epsilon^\nu = x^\nu$. Put $x_1 = x$, $x_2 = 0$, $x_3 = y = rn$, where $|n| = 1$, and send $r \to 0$ with x fixed. Then

$$X^\mu = \frac{(x-y)^\mu}{|x-y|^2} - \frac{(-y)^\mu}{|y|^2} = \frac{n^\mu}{r} + \frac{x^\mu}{|x|^2} + O\left(\frac{r}{|x|^2}\right),$$

and therefore

$$\frac{X_\mu X_\nu}{X^2} - \frac{1}{D}\delta_{\mu\nu} = n_\mu n_\nu - \frac{1}{D}\delta_{\mu\nu} + O\left(\frac{r}{|x|}\right), \qquad \frac{1}{|x_{13}|^D |x_{23}|^D |x_{12}|^{2\Delta-D}} = \frac{r^{-D}}{|x|^{2\Delta}}\left(1 + O\left(\frac{r}{|x|}\right)\right).$$

The sphere S_2 is oriented as the boundary of the punctured integration domain, hence $\mathrm{d}S_\mu = -r^{D-1} n_\mu \mathrm{d}\Omega$, while $\epsilon_\nu(y) = rn_\nu$. The leading flux is consequently

$$\langle \mathcal{O}_1(x) Q_\epsilon(S_2) \mathcal{O}_2(0)\rangle = -\frac{C_{12T}}{|x|^{2\Delta}} \int_{S^{D-1}} \mathrm{d}\Omega\, n^\mu n^\nu \left(n_\mu n_\nu - \frac{1}{D}\delta_{\mu\nu}\right),$$

up to terms that vanish as $r \to 0$. The angular projection is

$$\begin{aligned}\int_{S^{D-1}} \mathrm{d}\Omega\, n^\mu n^\nu \left(n_\mu n_\nu - \frac{1}{D}\delta_{\mu\nu}\right) &= \int_{S^{D-1}} \mathrm{d}\Omega \left[(n^\mu n_\mu)(n^\nu n_\nu) - \frac{1}{D} n^\mu n_\mu\right] \\ &= \int_{S^{D-1}} \mathrm{d}\Omega \left[1 - \frac{1}{D}\right] = A_{D-1}\left(1 - \frac{1}{D}\right).\end{aligned} \tag{63.28}$$

In the second line we used $n^\mu n_\mu = 1$ pointwise on the unit sphere; this is the entire reduction of the fourth-order angular expression in this particular contraction. so the flux equals

$$-\frac{D-1}{D} A_{D-1} \frac{C_{12T}}{|x|^{2\Delta}}.$$

The same orientation convention converts the local dilatation Ward identity $\delta_\epsilon \mathcal{O}_2(0) = -\Delta \mathcal{O}_2(0)$ into

$$\langle \mathcal{O}_1(x) Q_\epsilon(S_2) \mathcal{O}_2(0)\rangle = \frac{\Delta N_{12}}{|x|^{2\Delta}}.$$

Matching the two expressions gives (63.27). If $\Delta_1 \neq \Delta_2$, the two-point coefficient N_{12} vanishes by dilatation covariance, and the corresponding stress-tensor three-point function vanishes.

63.6 Four-Point Functions and Cross Ratios

Definition 63.5 (Ordered four-point conformal frame)**.** For $D \geq 2$, let

$$x = (x_1, x_2, x_3, x_4) \in \mathrm{Conf}_4(\mathbb{R}^D).$$

A four-point conformal frame for x is a conformal transformation, possibly followed by the one-point compactification of $\mathbb{R}^D$, that sends the quadruple to

$$z_1 = 0, \qquad z_2 = z, \qquad z_3 = 1, \qquad z_4 = \infty$$

inside an oriented two-plane. The associated scalar cross ratios are

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \qquad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}. \tag{63.29}$$

In the frame they obey

$$z = \frac{z_{12} z_{34}}{z_{13} z_{24}}, \qquad u = z\bar{z}, \qquad v = (1-z)(1-\bar{z}).$$

Proposition 63.6 (Completeness of four-point cross ratios)**.** *For a generic ordered quadruple of distinct Euclidean points in $D \geq 2$, an oriented four-point frame is labelled by z. The pair (u, v) is equivalent to z modulo complex conjugation. For $D \geq 3$, this pair separates connected conformal orbits. For $D = 2$, it separates orbits only after adjoining the orientation-reversing conformal transformation $z \mapsto \bar{z}$; the connected orientation-preserving group distinguishes z from $\bar{z}$. Thus scalar four-point correlators invariant under the relevant Euclidean parity operation can have nontrivial position dependence only through two independent real conformal invariants.*

Proof. We give the explicit frame construction. Translate so that $x_1 = 0$. The special conformal map

$$y \longmapsto \frac{y - b\,y^2}{1 - 2b \cdot y + b^2 y^2}, \qquad b = \frac{x_4}{x_4^2},$$

fixes the origin and sends x_4 to the point at infinity, because the denominator vanishes at $y = x_4$. For a generic quadruple, the image of x_3 is finite and nonzero; a rotation and a dilatation send it to the unit vector e_1. The stabilizer of 0, e_1, and ∞ is the orthogonal group acting on the $(D-1)$-dimensional subspace orthogonal to e_1. This stabilizer rotates the image of x_2 into the (e_1, e_2)-plane, leaving a complex coordinate z. In this frame the Euclidean distances give exactly

$$u = z\bar{z}, \qquad v = (1-z)(1-\bar{z}).$$

Conversely, two framed quadruples with the same z are equal, and replacing the orientation of the two-plane sends z to $\bar{z}$ without changing (u, v). When $D \geq 3$, this orientation reversal of the chosen two-plane is implemented by a rotation in the orthogonal complement, for instance by flipping one in-plane basis vector and one transverse basis vector. When $D = 2$, it is not in the connected orientation-preserving conformal group and must be added as a separate parity operation. The dimension count is consistent:

$$4D - \frac{(D+1)(D+2)}{2} + \frac{(D-2)(D-3)}{2} = 2,$$

where the last term is the dimension of the residual rotation group fixing a generic two-plane frame. $\square$

Figure 63.3 displays the conformal frame used for four scalar insertions: the four positions are reduced to the two cross ratios (u, v), with a residual rotation group fixing the chosen two-plane.

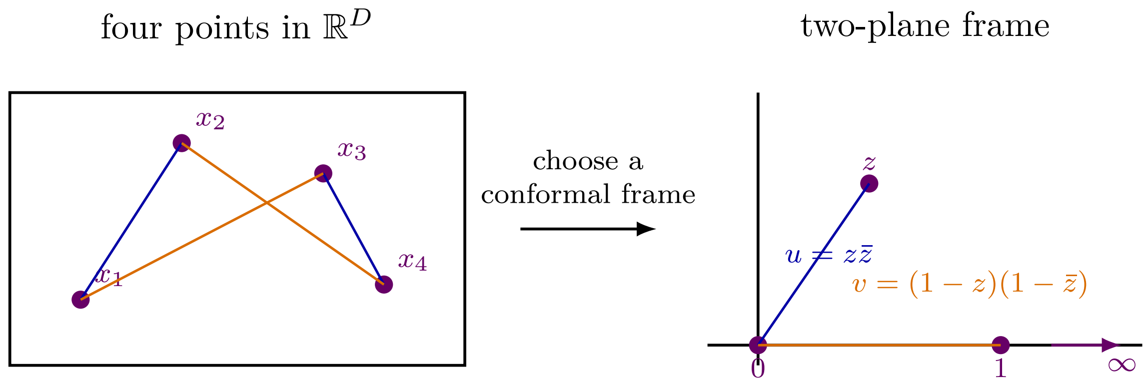


Figure 63.3: A scalar four-point function has two conformal invariants. A conformal frame places the points at $0, z, 1, \infty$ in a two-plane.

Scalar four-point prefactor covariance. Let $\mathcal{O}_i$ be scalar primaries of dimensions Δ_i. Every separated conformally covariant four-point function can be written uniquely in the form

$$\begin{aligned}&\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\mathcal{O}_4(x_4)\rangle \\ &\quad = \frac{\mathcal{G}(u,v)}{|x_{12}|^{\Delta_1+\Delta_2}|x_{34}|^{\Delta_3+\Delta_4}} \left(\frac{|x_{24}|}{|x_{14}|}\right)^{\Delta_1-\Delta_2} \left(\frac{|x_{14}|}{|x_{13}|}\right)^{\Delta_3-\Delta_4}.\end{aligned} \tag{63.30}$$

For identical scalar primaries of dimension Δ, this simplifies to

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta}|x_{34}|^{2\Delta}}. \tag{63.31}$$

The quotient of a scalar four-point function by the displayed prefactor is invariant under translations and rotations because it is built from distances. Under a uniform dilatation, the prefactor has degree $-(\Delta_1+\Delta_2+\Delta_3+\Delta_4)$, the required total degree. It remains to check the local weights under inversion. Since

$$|x_{ij}| \longmapsto \frac{|x_{ij}|}{|x_i|\,|x_j|},$$

write the prefactor as $\prod_{i<j} |x_{ij}|^{a_{ij}}$, with

$$\begin{gathered} a_{12} = -(\Delta_1+\Delta_2), \qquad a_{34} = -(\Delta_3+\Delta_4), \qquad a_{24} = \Delta_1-\Delta_2, \\ a_{14} = -(\Delta_1-\Delta_2)+(\Delta_3-\Delta_4), \qquad a_{13} = -(\Delta_3-\Delta_4), \qquad a_{23} = 0. \end{gathered}$$

The inversion power at x_i is $-\sum_{j\neq i} a_{ij}$. Direct substitution gives $2\Delta_i$ for each $i = 1, \dots, 4$. This is exactly the inversion weight of four scalar primaries because $\Omega_I(x_i) = |x_i|^{-2}$. Therefore the remaining factor is a dimensionless conformal invariant and hence a function of u, v on the generic configuration space by Proposition 63.6. Uniqueness follows because the prefactor is nonzero on $\mathrm{Conf}_4(\mathbb{R}^D)$. The identical-scalar formula is the specialization $\Delta_1 = \cdots = \Delta_4 = \Delta$.

The function $\mathcal{G}(u,v)$ is dynamical CFT data. The next stage is to determine how the operator product expansion encodes local multiplication of operators and how associativity constrains the allowed spectrum and OPE coefficients. Detailed computational machinery for solving crossing equations is deferred from the core monograph until conformal theory is rebuilt as a separate systematic development.

Chapter 64

The Operator Product Expansion

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

64.1 Local Expansion in Radial Quantization

Let $\mathcal{O}_i(x)$ and $\mathcal{O}_j(0)$ be local operators inserted inside a sphere of radius R, and let all other insertions lie outside that sphere. Radial quantization turns the separated pair

$$\Psi_{ij}(x) := \mathcal{O}_i(x)\mathcal{O}_j(0)|\Omega\rangle, \qquad |x| < R,$$

into a state on S_R^{D-1}. This state is not, as an object, a local operator at the origin, and there is no operation in the local-operator space which sends the ordered pair $(\mathcal{O}_i(x), \mathcal{O}_j(0))$, with $x \neq 0$, to a new local operator at 0. The local-operator states of definite scaling dimension form an algebraic dense subspace of the BPZ/radial Hilbert space, and the finite-energy spectral projections of the separated-pair state are finite linear combinations of those local-operator states. If $P_{\leq E}$ denotes the spectral projection of D_{rad} onto scaling dimensions at most E, the OPE is the Hilbert-norm convergence

$$\lim_{E\to\infty} P_{\leq E}\Psi_{ij}(x) = \Psi_{ij}(x).$$

Written in local-operator coordinates on the dense finite-energy subspace, the same statement is conventionally abbreviated as

$$\Psi_{ij}(x) \sim_{\mathrm{BPZ}} \sum_{\mathcal{O}} C_{ij\mathcal{O}}(x, \partial)\, \mathcal{O}(0). \tag{64.1}$$

The sum runs over primary operators $\mathcal{O}$ and the differential operator $C_{ij\mathcal{O}}(x,\partial)$ generates the descendants of $\mathcal{O}$. The symbol $\sim_{\mathrm{BPZ}}$ does not denote equality in a pointwise algebra of local operators. It denotes that the finite partial sums over dimensions $\Delta \le \Delta_*$ define local-operator states at the origin whose sequence converges to $\Psi_{ij}(x)$ in the BPZ/radial Hilbert norm on the separating sphere, and hence inside correlation functions with all other insertions farther from the origin than $|x|$.

Equivalently, write $x = \mathrm{e}^{\tau} n_x$ under the Weyl map $\mathbb{R}^D \setminus \{0\} \simeq \mathbb{R}_\tau \times S^{D-1}$. For scalar primaries, $\mathcal{O}_i(x) = |x|^{-\Delta_i} \widetilde{\mathcal{O}}_i(\tau, n_x)$, so the product creates the cylinder state

$$\widetilde{\mathcal{O}}_i(\tau, n_x)|\mathcal{O}_j\rangle \in \mathcal{H}_{S^{D-1}}.$$

The OPE is the statement that this state is represented, in every exterior correlation function, by a convergent limit of local operators at the origin, organized into primary states and their descendants. Thus the expansion is not a formal Taylor series in elementary fields and not a multiplication law closing the local-operator space. It is the spectral expansion of the Hilbert-space state created by separated local insertions by a sequence in the dense algebraic subspace of local-operator states.

The convergence parameter is radial separation. If the nearest outside insertion is at radial distance R, the expansion is organized by powers of $|x|/R$, with larger scaling dimensions contributing at higher radial energy. The OPE is therefore a convergent spectral expansion controlled by the local operator spectrum and the conformal representation theory of descendants.

Definition 64.1 (Radial OPE inputs from radial reconstruction)**.** The Euclidean radial OPE theorem below is stated for a Schwinger hierarchy satisfying the radial reconstruction data of Hypothesis 59.1. The following list is not an independent set of assumptions; it records which clauses of that single radial-reconstruction hypothesis enter the Euclidean convergence proof:

1. distributional Euclidean correlators of renormalized local operators, with covariance, graded permutation symmetry, and contact-term prescriptions;
2. radial reflection positivity and the corresponding null quotient and Hilbert completion;
3. a nonnegative self-adjoint radial Hamiltonian D_{rad} with discrete point spectrum and finite-dimensional spectral subspaces below every finite energy;
4. radial local completeness, so the algebraic direct sum of local scaling-operator states is dense and its homogeneous components identify with the D_{rad}-eigenspaces as in Theorem 59.10.

The Lorentzian continuation statements in this chapter additionally use the tube-domain analytic continuation, spectrum condition, and local commutativity part of the same hypothesis. Thus the phrase "radial OPE convergence under reflection positivity" denotes this listed Hilbert-space and spectral package.

Theorem 64.2 (Euclidean radial OPE convergence)**.** *Assume Hypothesis 59.1, including its spectral clause that D_{rad} is a nonnegative self-adjoint operator with discrete point spectrum, finite-dimensional spectral subspaces $\mathbf{1}_{[0,E]}(D_{\mathrm{rad}})\mathcal{H}_{\mathrm{rad}}$ for every $E < \infty$, and dense algebraic span by local scaling-operator states and their descendants. Let $\mathcal{O}_i(x_E)\mathcal{O}_j(0)\Omega$ be inserted at Euclidean points inside a sphere of radius r in $\mathbb{R}^D_E$, and let all remaining Euclidean insertions lie outside radius $R > r$. Then the expansion of this state in eigenspaces of D_{rad} converges in Hilbert norm for $r < R$, and the resulting OPE converges inside the corresponding Euclidean correlation functions.*

If the expansion is truncated to states of scaling dimension at most Δ_*, the remainder is bounded by the spectral tail above Δ_*, with the damping factor determined by $(r/R)^{D_{\mathrm{rad}}}$.

Proof. The proof is the spectral theorem applied to Euclidean radial quantization. Radial time is $\tau = \log|x_E|$, and the dilatation operator is the nonnegative self-adjoint radial Hamiltonian D_{rad} of Hypothesis 59.1. A separating sphere at radius R inserts the damping factor

$$\exp[-(\log(R/r))D_{\mathrm{rad}}].$$

Set $q = r/R \in (0,1)$, and write $\Psi_{ij}(x_E) = \mathcal{O}_i(x_E)\mathcal{O}_j(0)\Omega$ for the radial Hilbert-space vector defined by the inside insertions. For a spectral cutoff Δ_*, let

$$P_{\le\Delta_*} := \mathbf{1}_{[0,\Delta_*]}(D_{\mathrm{rad}}), \qquad P_{>\Delta_*} := 1 - P_{\le\Delta_*}.$$

The finite-dimensionality of $\mathbf{1}_{[0,\Delta_*]}(D_{\mathrm{rad}})\mathcal{H}_{\mathrm{rad}}$ gives an orthonormal basis $\{|A\rangle\}$ of local scaling-operator states in this subspace, and hence

$$P_{\le\Delta_*}\Psi_{ij}(x_E) = \sum_{\Delta_A\le\Delta_*} |A\rangle\,\langle A, \Psi_{ij}(x_E)\rangle_{\mathrm{rad}}.$$

This is the finite radial OPE partial sum, with the coefficients read as the radial matrix elements of the product insertion. Strong convergence of the spectral projections gives

$$\lim_{\Delta_*\to\infty} \|P_{>\Delta_*}\Psi_{ij}(x_E)\|_{\mathrm{rad}} = 0,$$

so the partial sums converge in Hilbert norm.

Pairing the inside vector with an outside vector Ξ whose support begins at radius R inserts $q^{D_{\mathrm{rad}}}$. Therefore

$$\left|\langle \Xi, q^{D_{\mathrm{rad}}} P_{>\Delta_*}\Psi_{ij}(x_E)\rangle_{\mathrm{rad}}\right| \le q^{\Delta_*}\,\|\Xi\|_{\mathrm{rad}}\,\|P_{>\Delta_*}\Psi_{ij}(x_E)\|_{\mathrm{rad}},$$

and sharper bounds follow by replacing q^{Δ_*} with the exact spectral tail of $q^{D_{\mathrm{rad}}}$ on the omitted dimensions. Since local operator states are dense in the finite-energy subspace, the Hilbert-space partial sums are precisely finite sums of local scaling operators and their descendants. For a radial Hamiltonian with continuous spectrum, the same spectral theorem produces a spectral integral; the displayed discrete OPE requires the finite-multiplicity point-spectrum hypothesis stated in the theorem. □

The preceding theorem is a statement in the Euclidean radial Hilbert space. It does not by itself assert pointwise convergence at arbitrary real Lorentzian configurations. Lorentzian OPE convergence is obtained by applying the Euclidean expansion inside a common holomorphic domain and then taking the appropriate Wightman boundary value.

Proposition 64.3 (Lorentzian continuation of the radial OPE)**.** *Assume the tube-domain part of Hypothesis 59.1. Work with a fixed Wightman ordering π and write $V_+ \subset \mathbb{R}^{1,D-1}$ for the open future light cone in the mostly-plus metric. Let $\mathcal{W}_\pi(z_1,\ldots,z_n)$ be the holomorphic continuation in the tube component*

$$z_{\pi(a)} - z_{\pi(a+1)} = \xi_a - \mathrm{i}\eta_a, \qquad \eta_a \in V_+.$$

For a chosen OPE channel, let $\rho, \bar\rho$ denote the radial cross-ratios of that channel. On every compact subset of the tube component on which

$$|\rho| \le q, \qquad |\bar\rho| \le q, \qquad q < 1, \tag{64.2}$$

the channel OPE converges locally uniformly as a holomorphic function. Its boundary value is the Lorentzian Wightman OPE in that ordering. Strictly timelike configurations in which the fused pair lies in each other's future or past light cone are included when the specified $i\epsilon$ approach remains in such a compact radial domain. Null separation, coincident points, and crossings of external light cones are boundary strata and require the distributional boundary-value interpretation.

Proof. The Euclidean radial theorem gives a convergent expansion in the open polydisc of radial variables. On a compact set satisfying (64.2), the same spectral estimate supplies a summable majorant: each level-Δ descendant contribution is multiplied by a factor bounded by q^{Δ}, while finite multiplicity below each level and radial Hilbert convergence control the corresponding coefficient norm. Therefore the OPE partial sums converge uniformly on compact subsets of the radial polydisc after analytic continuation of $\rho, \bar{\rho}$.

Each finite partial sum is a finite linear combination of Wightman matrix elements of local operators at the origin with the remaining insertions. By the spectrum condition, these matrix elements are boundary values of functions holomorphic in the tube component displayed above. Local uniform convergence of holomorphic partial sums gives a holomorphic limit by the Weierstrass theorem. On the Euclidean subdomain this limit equals $\mathcal{W}_\pi$, because both sides are the Euclidean radial OPE. The identity theorem then identifies the holomorphic limit with $\mathcal{W}_\pi$ throughout the connected tube component reached without crossing singular hypersurfaces. Taking ordered $i\epsilon$ boundary values gives the stated Lorentzian distributional convergence. ☐

Remark 64.4 (Status of Lorentzian OPE convergence)*.* The criterion above separates the Euclidean Hilbert-space theorem from its Lorentzian continuation. The tube condition fixes the operator ordering, and the radial inequality $|\rho|, |\bar{\rho}| < 1$ fixes the OPE channel. For scalar four-point functions in unitary Euclidean CFT, Kravchuk, Qiao, and Rychkov prove the needed forward-tube radial bound and distributional Minkowski OPE convergence; Qiao gives a channel-by-channel classification of real Lorentzian four-point configurations.[1]

Figure 64.1 records the radial OPE geometry: the smaller sphere encloses the approaching operator pair, and convergence is controlled by the ratio of radii in the radial Hilbert norm.

[1]See G. Mack, "Convergence of operator product expansions on the vacuum in conformal invariant quantum field theory," Commun. Math. Phys. **53** (1977) 155–184; P. Kravchuk, J. Qiao, and S. Rychkov, "Distributions in CFT II. Minkowski Space," JHEP **08** (2021) 094, arXiv:2104.02090; and J. Qiao, "Classification of Convergent OPE Channels for Lorentzian CFT Four-Point Functions," SciPost Phys. **13** (2022) 093, arXiv:2005.09105.

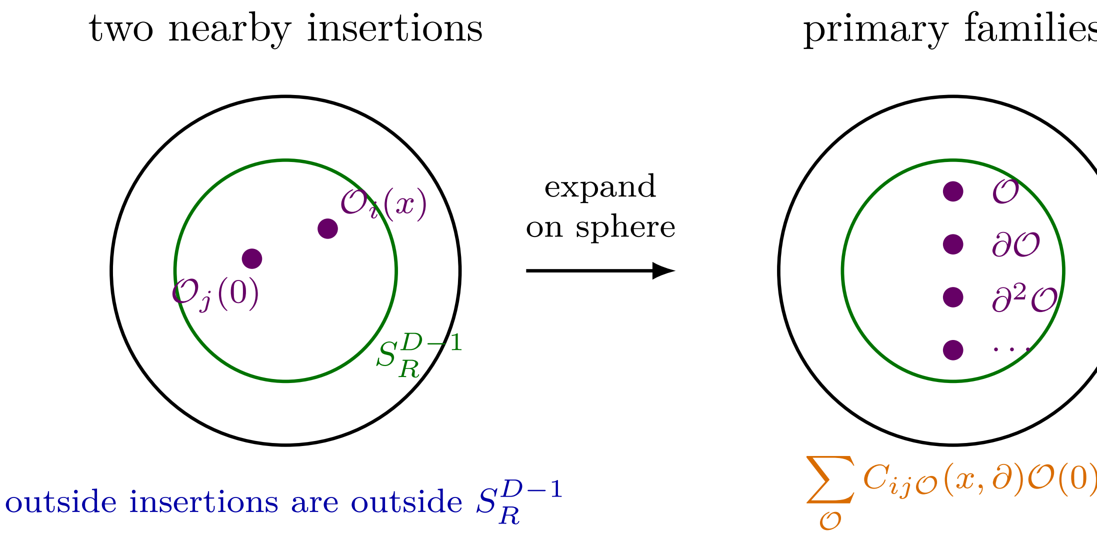


Figure 64.1: The OPE is the expansion of the state created by nearby insertions in the basis of primary conformal multiplets on a separating sphere.

64.2 Scalar-Scalar OPE

Let ϕ_1 and ϕ_2 be scalar primaries of dimensions Δ_1 and Δ_2. Let $\mathcal{O}_{\mu_1\cdots\mu_\ell}$ be a symmetric traceless primary of spin ℓ and dimension Δ. Its leading contribution to the $\phi_1 \times \phi_2$ OPE has the following state-valued asymptotic coordinate in the BPZ/radial expansion:

$$\phi_1(x)\phi_2(0) \supset \lambda_{12\mathcal{O}} \frac{x^{\mu_1}\cdots x^{\mu_\ell}}{|x|^{\Delta_1+\Delta_2-\Delta+\ell}} \mathcal{O}_{\mu_1\cdots\mu_\ell}(0) + \text{descendants}. \tag{64.3}$$

The tensor $x^{\mu_1}\cdots x^{\mu_\ell}$ selects the symmetric traceless part of the spin-ℓ representation. Antisymmetric components do not appear in the product of powers of a single vector, and trace parts are absorbed into descendants or lower-spin structures.

For an exchanged scalar primary $\mathcal{O}$, the first descendant term is fixed by conformal symmetry:

$$\phi_1(x)\phi_2(0) \supset \lambda_{12\mathcal{O}}|x|^{\Delta-\Delta_1-\Delta_2}\left(1 + \frac{\Delta+\Delta_1-\Delta_2}{2\Delta} x^\mu \partial_\mu + \cdots\right)\mathcal{O}(0). \tag{64.4}$$

The coefficient of the derivative is determined by requiring the OPE to reproduce the conformally fixed three-point function $\langle \phi_1\phi_2\mathcal{O}\rangle$. To see the coefficient, place the third operator at z with $|x| \ll |z|$ and normalize $\langle \mathcal{O}(y)\mathcal{O}(z)\rangle = |y-z|^{-2\Delta}$. The three-point function gives

$$\langle \phi_1(x)\phi_2(0)\mathcal{O}(z)\rangle = \lambda_{12\mathcal{O}}|x|^{\Delta-\Delta_1-\Delta_2}|z|^{-2\Delta}\left[1 + (\Delta+\Delta_1-\Delta_2)\frac{x\cdot z}{z^2} + O(x^2)\right].$$

The OPE ansatz

$$\lambda_{12\mathcal{O}}|x|^{\Delta-\Delta_1-\Delta_2}\big(1 + a\, x^\mu \partial_\mu + \cdots\big)\mathcal{O}(0)$$

gives instead

$$\lambda_{12\mathcal{O}}|x|^{\Delta-\Delta_1-\Delta_2}|z|^{-2\Delta}\left[1 + 2a\Delta\frac{x\cdot z}{z^2} + O(x^2)\right],$$

because $\partial_{y^\mu}|y-z|^{-2\Delta}|_{y=0} = 2\Delta z_\mu |z|^{-2\Delta-2}$. Matching the two expansions gives

$$a = \frac{\Delta + \Delta_1 - \Delta_2}{2\Delta}.$$

Higher descendants are fixed in the same way by matching higher orders, or equivalently by the conformal algebra.

For identical scalar primaries this reduces to the especially important normalization

$$\phi(x)\phi(0) \supset \lambda_{\phi\phi\mathcal{O}} |x|^{\Delta_{\mathcal{O}}-2\Delta_\phi} \left(1 + \frac{1}{2}x^\mu \partial_\mu + \cdots\right) \mathcal{O}(0) \tag{64.5}$$

for scalar exchange. The coefficient 1/2 is not an extra convention; it is the specialization of the conformal Ward constraints to two equal external dimensions.

64.3 OPE Data

Choose a basis of primaries with diagonal two-point functions. The local CFT data are then

$$\left\{\Delta_{\mathcal{O}},\ \rho_{\mathcal{O}},\ \lambda^{(a)}_{ij\mathcal{O}}\right\}. \tag{64.6}$$

Here $\rho_{\mathcal{O}}$ is the spin representation, and the label a enumerates independent three-point tensor structures when more than one structure exists. Once this data is specified, all descendant coefficients in the OPE are representation-theoretic consequences of the conformal algebra.

In representation language, if V_i is the conformal multiplet generated by a primary $\mathcal{O}_i$, the OPE map is a conformally covariant state-valued intertwining map from the tensor product of two local-state multiplets, together with a nonzero radial separation, to the Hilbert space on the separating sphere. On the decomposition into irreducible conformal multiplets its finite-energy projections have the form

$$V_i \otimes V_j \longrightarrow \bigoplus_{\mathcal{O}} m^{\mathcal{O}}_{ij} V_{\mathcal{O}} \subset \mathcal{H}_{S^{D-1}}.$$

The multiplicity $m^{\mathcal{O}}_{ij}$ equals the number of independent three-point structures connecting i, j, and $\mathcal{O}$. When $\mathcal{O}_i$ and $\mathcal{O}_j$ are both scalars, rotational covariance permits only symmetric traceless tensor primaries in the leading term, and the multiplicity for each such representation is one. For spinning external operators, several independent tensor structures may couple the same three representations, and the OPE coefficients must then carry the corresponding structure label.

Repeated OPEs determine every separated Euclidean correlation function from the primary spectrum, the two-point pairing, and the three-point coefficients, provided the radial convergence assumptions stated above hold. The data are therefore local in a strong sense: higher-point functions do not bring in independent local constants. Their nontrivial content is the compatibility of the same OPE data in different channels.

64.4 Reconstruction from OPE Data

The preceding paragraph has a converse direction only after the word "data" is given a precise mathematical meaning. A complete primary OPE table $\{\Delta_{\mathcal{O}}, \rho_{\mathcal{O}}, \lambda^{(a)}_{ij\mathcal{O}}\}$, with all primary labels, degeneracies, tensor structures, and operator-basis conventions resolved, is the table of radial expansion coefficients for separated-pair states in the dense local-operator subspace. It is therefore a candidate for an inverse determination of the theory, in the same sense that a sufficiently rich exact set of low-energy observables can in principle determine the parameters and operator content of an effective theory. If a smaller set of local operators generates the finite-energy local-operator subspace under iterated radial OPE limits, the complete OPE spectrum and coefficients visible in suitable four-point functions of those generators may likewise determine the rest of the local theory. Turning such rigidity into a reconstruction statement requires the Hilbert topology, positivity, radial convergence, channel compatibility, and distributional interpretation of separated correlators. The following formulation starts after those inputs have been supplied: it is a radial construction from a fully specified OPE system, rather than a derivation of that system from finite lists of four-point data.

Definition 64.5 (Abstract radial OPE system)**.** An abstract radial OPE system in dimension $D \geq 2$ consists of the following objects.

(i) A countable set $\mathcal{P}$ of primary labels, including the identity label $\mathbf{1}$. Each $a \in \mathcal{P}$ carries a scaling dimension $\Delta_a \in \mathbb{R}_{\geq 0}$ and a finite-dimensional unitary representation ρ_a of $\mathrm{Spin}(D)$. For every $E < \infty$, only finitely many descendant states in the modules below have dimension at most E.

(ii) For each a, a positive-energy conformal module V_a generated by a primary vector $|a\rangle$, with null descendants quotiented. The direct sum

$$\mathcal{H}_{\mathrm{fin}}^{\mathrm{abs}} = \bigoplus_{a \in \mathcal{P}} V_a$$

is equipped with a positive Hermitian inner product for which the conformal generators obey the radial adjointness relations $D^\dagger = D$, $M_{\mu\nu}^\dagger = -M_{\mu\nu}$, and $P_\mu^\dagger = K_\mu$.

(iii) A conjugation $a \mapsto a^\dagger$ and a nondegenerate two-point pairing between V_a and $V_{a^\dagger}$, normalized so that $V_{\mathbf{1}}$ is the one-dimensional vacuum module.

(iv) For every ordered pair a, b and every $c \in \mathcal{P}$, a finite list of conformally covariant three-point tensor structures $\mathcal{T}_{ab}^{c,\alpha}$, $\alpha = 1, \ldots, m_{ab}^c$, and complex coefficients $\lambda_{ab}^{c,\alpha}$. The descendant OPE coefficients are the coefficients obtained by applying the conformal algebra and the quotient Gram matrices inside V_c.

(v) For every finite ordered list of primary or descendant labels and every rooted binary OPE tree T, an iterated OPE series

$$S_T(x_1, \ldots, x_n)$$

on the radial configuration domain of T. This series is required to converge in the radial Hilbert norm for state-valued partial products and locally uniformly, after pairing with exterior finite-energy states, on compact subsets of the separated-point domain of T.

(vi) Channel compatibility: if two trees T and T' have overlapping radial domains, the functions S_T and $S_{T'}$ agree on the common domain after the canonical permutation and tensor-index identifications. For four-point functions this condition is crossing, namely equality of two binary radial-expansion limits tested between exterior states. For higher-point functions it is the equality of all iterated OPE parenthesizations on their common radial domains. A separate coherence theorem may allow the higher-point identities to be derived from a smaller list of associativity identities; absent such a theorem, the all-tree compatibility is part of the system.

(vii) Euclidean covariance under the conformal group, graded permutation symmetry at separated points, and radial reflection positivity for every finite linear combination of inside-supported products.

(viii) A prescription for extending the separated-point functions to distributions with contact terms on diagonals, whenever coincident-point Ward identities or deformations are part of the theory.

The finite-multiplicity condition in item (i) is part of the system because the radial OPE theorem uses spectral projections onto finite-dimensional energy windows. The compatibility condition in item (vi) is also part of the system. A single four-point crossing equation is one instance of this condition and is normally far from determining even that four-point function uniquely. Complete all-primary four-point associativity constraints are a different object: once the OPE table is known, they constrain the whole system of binary radial-expansion limits. The construction below therefore does not hide this analytic and higher-coherence substance in a theorem statement. The hard input is Definition 64.5; the construction records how such input is assembled into a hierarchy.

Construction 64.6 (Separated correlators from an abstract radial OPE system)**.** Let $\mathfrak{D}$ be an abstract radial OPE system in the sense of Definition 64.5. It determines a separated-point Euclidean conformal correlation hierarchy

$$S^{\mathfrak{D}}_{I_1\cdots I_n}(x_1,\ldots,x_n), \qquad x_i \neq x_j,$$

unique up to unitary equivalence of the radial Hilbert space and invertible changes of local operator basis preserving the two-point pairing. This hierarchy satisfies Euclidean conformal covariance, graded permutation symmetry, radial reflection positivity, the spectral discreteness and finite-multiplicity conditions of radial quantization, and the OPE with the prescribed primary dimensions, spin representations, and three-point coefficients. If the system also includes tube-domain analytic continuation, the spectrum condition, and local commutativity for the boundary values, then the reconstructed Euclidean hierarchy gives a Lorentzian Wightman CFT satisfying the radial reconstruction hypotheses used in this volume.

Verification of the construction. First construct the dense finite-energy vector space $\mathcal{H}^{\mathrm{abs}}_{\mathrm{fin}} = \bigoplus_a V_a$ with the positive inner product specified in Definition 64.5. Its Hilbert completion is the radial Hilbert space. The conformal generators act on each V_a, and the adjointness relations make the representation unitary on the finite-energy domain. The identity module supplies the vacuum vector.

For a separated Euclidean configuration, choose a rooted binary OPE tree whose radial inequalities are satisfied by the insertion points. Apply the intertwining OPE maps at the internal vertices of the tree and sum over intermediate primary modules and descendants. Item (v) gives a convergent state-valued expansion at each internal radial step and locally uniform convergence after pairing with the remaining exterior states. Hence the tree defines a correlation function $S_T(x_1,\ldots,x_n)$ on its radial domain.

If two admissible trees describe the same configuration on an overlap, item (vi) identifies the two sums there. The separated Euclidean configuration space is covered by such radial domains. Patching the locally defined functions through their overlaps gives a single separated-point hierarchy $S^{\mathfrak{D}}$. Conformal covariance and graded permutation symmetry are inherited from the covariance of the three-point tensor structures and from item (vii). Reflection positivity follows from the assumed positivity of all inside-supported products; quotienting by zero-norm vectors gives the physical radial Hilbert space.

The OPE of the reconstructed fields is built into the construction. When a subset of insertions is separated from the rest by a sphere, using an OPE tree that first fuses that subset gives precisely the convergent expansion with coefficients $\lambda_{ab}^{c;\alpha}$ and conformally determined descendant coefficients. Since compatible trees give the same patched correlator, the same OPE holds independently of the auxiliary tree used to compute the correlator. The stated uniqueness follows because any Euclidean hierarchy with the same system has the same convergent OPE sums on every radial tree domain, and those domains cover separated configurations.

Finally, if tube-domain analytic continuation, the Lorentzian spectrum condition, and local commutativity are included, the usual Wightman boundary value construction applied to the reconstructed Euclidean hierarchy gives the corresponding Lorentzian distributions. These are precisely the additional analytic hypotheses listed in Hypothesis 59.1.

Open Problem 64.7 (Completeness of four-point OPE data). The useful reconstruction problem is not the claim that a bare crossing equation for a single correlator produces a CFT, or even uniquely determines that correlator. The crossing equation is a functional constraint on an unknown function or distribution with unknown spectral decomposition; without additional input it has many formal solutions and many possible completions. A single correlator probes only the sector generated by its external operators, and even there one must resolve degeneracies, tensor structures, normalizations, and null states. The natural problem is sharper. Suppose one is given a complete all-primary OPE table

$$\{\Delta_a, \rho_a, \lambda_{ab}^{c;\alpha}\}$$

or, in a generated theory, the complete OPE data appearing in the four-point functions of a set of generators, with enough information to recover every primary and every independent three-point tensor structure. Suppose further that this table obeys the unitary representation-theoretic bounds, reflection-positivity inequalities, and the four-point associativity equations for all relevant choices of external operators. This is an exact infinite-data problem, analogous to determining a theory from a sufficiently complete set of precision observables; it is not the problem of solving a single bootstrap equation. Under what additional analytic hypotheses does it extend to an abstract radial OPE system satisfying Definition 64.5?

In dimension $D > 2$, complete such data are expected to determine at most one CFT up to unitary equivalence and changes of operator basis, once the generator or all-primary hypothesis is made. The existence part still requires theorem-level control of radial convergence, spectral finiteness, coherence of higher iterated OPEs, distributional extensions and contact-term prescriptions, and Lorentzian boundary-value properties. Those requirements are precisely why Construction 64.6 is formulated from an abstract radial OPE system rather than from a single informal crossing statement.

64.5 Conformal Blocks as Multiplet Contributions

The representation-theoretic object underlying a conformal block is a channel projector, or more generally a pair of intertwiners through a multiplicity space. Work in a radial Hilbert space satisfying Hypothesis 59.1, and use the positive-energy $\widetilde{G}_{\mathrm{conf}}^{\mathrm{Lor}}$-module conventions of Section 57.8.

For a primary label $\mathfrak{m} = (\Delta, \rho)$, let

$$L_{\mathfrak{m}} = M(\Delta, \rho)/\operatorname{Rad} M(\Delta, \rho)$$

be the irreducible null-quotient conformal multiplet, with its radial Hilbert globalization. In a fixed OPE channel, the finite-energy part of the separated-pair state generated by $\mathcal{O}_i$ and $\mathcal{O}_j$ has a spectral decomposition of the form

$$\mathcal{H}_{ij}^{\mathrm{ch}} \simeq \widehat{\bigoplus_{\mathfrak{m}}} \mathcal{M}_{ij}^{\mathfrak{m}} \otimes L_{\mathfrak{m}}, \tag{64.7}$$

where $\mathcal{M}_{ij}^{\mathfrak{m}}$ is a finite-dimensional multiplicity space when the finite-multiplicity radial hypotheses hold. The hat records Hilbert completion after the algebraic finite-energy direct sum is formed. For finite energy cutoffs this is an ordinary finite direct sum.

Let

$$\iota_{ij}^{\mathfrak{m},a} : L_{\mathfrak{m}} \longrightarrow \mathcal{H}_{ij}^{\mathrm{ch}}, \qquad a = 1, \ldots, \dim \mathcal{M}_{ij}^{\mathfrak{m}},$$

be a basis of conformal intertwiners for the ij channel, and let $\iota_{kl}^{\mathfrak{m},b}$ be the corresponding basis for the opposite pairing. The block contribution of $L_{\mathfrak{m}}$ is the matrix element of the finite-energy channel projector

$$\Pi_{\mathfrak{m}} : \mathcal{H}_{\mathrm{rad}} \longrightarrow \mathcal{M}^{\mathfrak{m}} \otimes L_{\mathfrak{m}}$$

between the two separated-pair states. Equivalently, after choosing the two-point pairing on $L_{\mathfrak{m}}$ and the basis of multiplicity spaces,

$$\mathcal{B}_{\mathfrak{m}}^{ab}(x_i) = \left\langle \Psi_{34}^{b}(x_3, x_4), \Pi_{\mathfrak{m}} \Psi_{12}^{a}(x_1, x_2) \right\rangle_{\mathrm{rad}}. \tag{64.8}$$

This equation is the definition of the conformal block as a Hilbert-space object. The differential-equation, shadow-integral, Mellin, and radial-series formulae below are coordinate descriptions of (64.8) under additional analytic hypotheses. If $L_{\mathfrak{m}}$ is short, the projector is formed after quotienting the generalized Verma module by its null radical; Gram matrices are inverted only on the quotient spaces.

For a scalar four-point function, this channel-projector definition separates the contribution of each primary conformal multiplet. For the general scalar four-point function (63.30), the $12 \to 34$ OPE gives

$$\mathcal{G}(u, v) = \sum_{\mathcal{O},a,b} \lambda_{12\mathcal{O}}^{(a)} N_{\mathcal{O}}^{ab} \lambda_{34\mathcal{O}}^{(b)} G_{\Delta,\ell}^{(a,b;12|34)}(u, v). \tag{64.9}$$

Here $\mathcal{O}$ runs over primary multiplets of dimension Δ and spin ℓ, a, b label possible tensor structures in the $\mathcal{O}_1\mathcal{O}_2\mathcal{O}$ and $\mathcal{O}_3\mathcal{O}_4\mathcal{O}$ three-point functions, and $N_{\mathcal{O}}^{ab}$ is the inverse two-point pairing on the corresponding degeneracy space. In an orthonormal basis and for scalar external operators with a single structure, this reduces to

$$\mathcal{G}(u, v) = \sum_{\mathcal{O}} \lambda_{12\mathcal{O}} \lambda_{34\mathcal{O}}\, G_{\Delta,\ell}^{(12|34)}(u, v).$$

The function $G_{\Delta,\ell}^{(12|34)}$ is the conformal block: it is the sum of the primary $\mathcal{O}$ and all of its descendants, with descendant coefficients fixed by the conformal algebra.

Figure 64.2 displays the two OPE association patterns for a four-point function; equality of the corresponding expansions is the crossing equation in the common convergence domain.

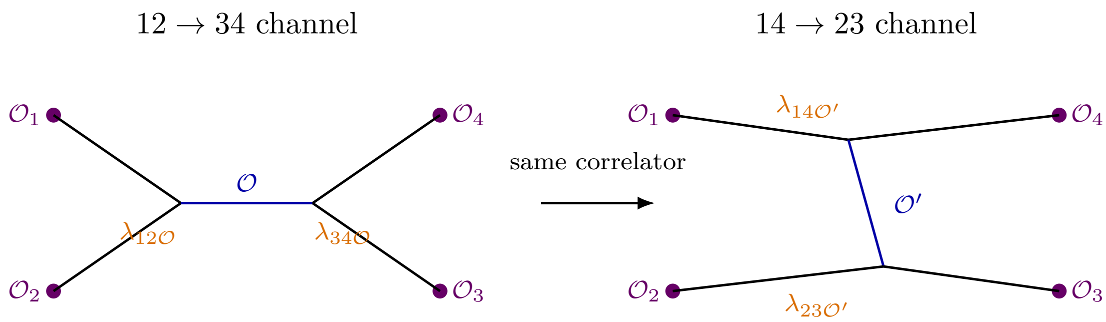


Figure 64.2: A four-point function may be expanded in different OPE trees. The equality of these decompositions is the four-point form of OPE associativity.

The block is characterized representation-theoretically by the quadratic Casimir, together with an OPE boundary condition. Put

$$u = z\bar{z}, \qquad v = (1-z)(1-\bar{z}), \qquad a = \frac{\Delta_2 - \Delta_1}{2}, \qquad b = \frac{\Delta_3 - \Delta_4}{2}, \qquad \varepsilon_D = \frac{D}{2} - 1.$$

After the external scalar prefactor in (63.30) has been removed, the 12-channel quadratic Casimir acts on the reduced function by

$$\mathcal{C}_{12}^{(u,v)} = 2\mathcal{D}_{a,b}, \tag{64.10}$$

where

$$\begin{aligned} \mathcal{D}_{a,b} = {} & u^2(1-u+v)\partial_u^2 - 2uv(1+u-v)\partial_u\partial_v + v\big((1-v)^2 - u(1+v)\big)\partial_v^2 \\ & - u\big((a+b+1)(1+u-v) + 2\varepsilon_D\big)\partial_u + (a+b+1)\big((1-v)^2 - u(1+v)\big)\partial_v \\ & - ab(1+u-v). \end{aligned} \tag{64.11}$$

Equivalently, in the algebraic coordinates $z, \bar{z}$,

$$\begin{aligned} \mathcal{D}_{a,b} &= D_z + D_{\bar{z}} + 2\varepsilon_D \frac{z\bar{z}}{z-\bar{z}}\big((1-z)\partial_z - (1-\bar{z})\partial_{\bar{z}}\big), \\ D_z &= z^2(1-z)\partial_z^2 - (a+b+1)z^2\partial_z - abz, \qquad D_{\bar{z}} = D_z\big|_{z\mapsto\bar{z}}. \end{aligned} \tag{64.12}$$

The conversion from (64.12) to (64.11) is the chain rule together with

$$z + \bar{z} = 1 + u - v, \qquad \partial_z = \bar{z}\,\partial_u - (1-\bar{z})\partial_v, \qquad \partial_{\bar{z}} = z\,\partial_u - (1-z)\partial_v.$$

The apparently singular term in (64.12) is regular on symmetric functions of $z, \bar{z}$, since

$$(1-z)\partial_z - (1-\bar{z})\partial_{\bar{z}} = (\bar{z} - z)\partial_u.$$

The factor 2 in (64.10) fixes the normalization of the conformal-algebra quadratic Casimir used in this chapter:

$$\mathcal{C}_{12}^{(u,v)} G_{\Delta,\ell}^{(12|34)}(u,v) = C_{\Delta,\ell} G_{\Delta,\ell}^{(12|34)}(u,v), \qquad C_{\Delta,\ell} = \Delta(\Delta - D) + \ell(\ell + D - 2). \tag{64.13}$$

The equality $C_{\Delta,\ell} = C_{D-\Delta,\ell}$ already shows that the differential equation alone does not select an OPE block. In the radial frame of (64.33), the two local Frobenius branches at the 12 OPE boundary are

$$G^{(12|34)}_{\Delta,\ell}(u,v) = r^{\Delta} C^{(D/2-1)}_{\ell}(\cos\alpha) + O(r^{\Delta+1}), \tag{64.14}$$

$$G^{(12|34)}_{D-\Delta,\ell}(u,v) = r^{D-\Delta} C^{(D/2-1)}_{\ell}(\cos\alpha) + O(r^{D-\Delta+1}). \tag{64.15}$$

The conformal block in the OPE sum is the solution with the first leading coefficient normalized as in (64.14) and with zero coefficient multiplying the shadow branch (64.15).

Short multiplets require a levelwise quotient, not a single informal deletion of dependent vectors. Section 62.7 explains how saturation of a unitarity bound produces a null descendant at a definite level: for example the scalar Laplace descendant $\widehat{P}^2|\phi\rangle$ at level 2, the divergence of a conserved symmetric-traceless tensor at level 1, or the spinor gamma-trace descendant at level 1. If N_α denotes such a singular null descendant at level r_α, then every

$$\widehat{P}_{\mu_1} \cdots \widehat{P}_{\mu_k} N_\alpha \qquad (k \geq 0)$$

is null at level $r_\alpha + k$. The descendant space used in the block at level n is therefore the raw level-n descendant space modulo the span of all null descendants whose total level is n. This is the same null submodule subtraction that produces the short-module characters in Chapter 59; it is performed before any Gram matrix is inverted.

Theorem 64.8 (Local existence and uniqueness of the OPE block)**.** *Assume the radial reconstruction hypotheses of Hypothesis 59.1, finite multiplicity of descendant states at each level, and the levelwise null-submodule quotient just described. For scalar external operators and one exchanged symmetric-traceless primary of dimension Δ and spin ℓ, there is a unique* 12*-channel block with the OPE normalization* (64.14)*. It is a convergent radial series for $r < 1$, satisfies the Casimir equation* (64.13)*, and contains no shadow branch.*

Proof. Let $V_{\Delta,\ell}$ be the conformal multiplet generated from the exchanged primary. Let

$$W_n \simeq \mathrm{Sym}^n(\mathbb{C}^D) \otimes V_{\Delta,\ell}$$

be the raw level-n descendant space before equations of motion, conservation laws, or other shortening relations are imposed. Its null radical for the radial Gram form is

$$\mathcal{N}_n = \{v \in W_n : \langle v, w\rangle = 0 \text{ for all } w \in W_n\}.$$

When the exchanged multiplet is short, $\mathcal{N}_n$ is generated level-by-level by the singular null descendants fixed by the saturated unitarity bounds and by all of their translation descendants at total level n. Reflection positivity implies that every vector in $\mathcal{N}_n$ has zero matrix element against every physical exterior state, so replacing W_n by

$$Q_n = W_n / \mathcal{N}_n$$

does not change any physical four-point function. Choose a basis $\{|A,n\rangle\}_A$ of Q_n and let

$$M^{(n)}_{AB} = \langle A,n|B,n\rangle$$

be the quotient Gram matrix. The quotient has no null radical, hence $M^{(n)}$ is invertible on this finite-dimensional space. The contribution of this level to the four-point function is

$$r^{\Delta+n} \sum_{A,B} \langle \mathcal{O}_4\mathcal{O}_3|A,n\rangle \, (M^{(n)})^{-1}_{AB} \, \langle B,n|\mathcal{O}_1\mathcal{O}_2\rangle, \tag{64.16}$$

with the angular dependence carried by the cylinder wavefunctions of the descendants. The three-point Ward identities fix the two vectors of matrix elements in (64.16) recursively from the primary three-point structures, because moving a conformal generator from a descendant state onto the external primaries is governed by the conformal algebra and by the scalar transformation law. Summing over all $n \geq 0$ therefore constructs a radial series whose leading term is the primary matrix element

$$r^{\Delta} C_{\ell}^{(D/2-1)}(\cos\alpha).$$

The convergence for $r < 1$ is the radial OPE convergence theorem applied after projecting the inside state to the multiplet $V_{\Delta,\ell}$.

The same construction gives the Casimir equation. The operator $\mathcal{C}_{12}$ is the square of the total conformal generator acting on the pair $(1, 2)$. On the projected intermediate multiplet it acts by the scalar $C_{\Delta,\ell}$, while the external scalar prefactor converts this action on the four-point function into the differential operator $\mathcal{C}_{12}^{(u,v)}$ displayed in (64.11). Hence the constructed radial series satisfies (64.13).

For uniqueness, let $H(r, \alpha)$ be the difference of two local solutions with the OPE normalization (64.14) and no shadow branch. Its first nonzero radial coefficient, if present, has some level $n > 0$ and lies in the finite descendant space at that level. The Casimir equation and the Ward identities impose the same finite linear system as in (64.16). Since $M^{(n)}$ is invertible on the quotient by null states and all lower-level coefficients of H vanish, that coefficient must vanish. This contradicts the choice of the first nonzero level. Therefore $H = 0$, proving uniqueness. □

Thus the spectrum and OPE coefficients are the dynamical data, while the blocks are kinematic functions determined by conformal representation theory and by the OPE branch condition.

64.5.1 Blocks, Shadows, and Partial Waves

The object just defined is an OPE-channel conformal block. Its normalization in this chapter is the OPE normalization: in the $12 \to 34$ channel the leading term as $x_1 \to x_2$ is the contribution of a primary of dimension Δ and spin ℓ, with no contribution from the shadow dimension. This is the normalization used in the discrete OPE sum (64.9).

A *conformal partial wave* is a different, although closely related, object. It is the harmonic-analysis kernel obtained from the conformal projector onto an irreducible principal-series representation. Let $\widetilde{\Delta} = D - \Delta$ be the shadow dimension of Δ. For scalar external operators and a single exchanged symmetric-traceless tensor structure, the shadow-projector normalization may be represented, with the symmetric-traceless spin-index contraction suppressed from the notation, by

$$\Psi_{\Delta,\ell}^{(12|34)}(x_i) = \frac{1}{\mathcal{N}_{\Delta,\ell}} \int_{\mathbb{R}^D} \mathrm{d}^D x_0 \, \big\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_{\Delta,\ell}(x_0)\big\rangle \big\langle \widetilde{\mathcal{O}}_{\widetilde{\Delta},\ell}(x_0)\mathcal{O}_3(x_3)\mathcal{O}_4(x_4)\big\rangle. \tag{64.17}$$

The spin indices of $\mathcal{O}_{\Delta,\ell}$ and $\widetilde{\mathcal{O}}_{\widetilde{\Delta},\ell}$ are contracted with the invariant two-point pairing. The constant $\mathcal{N}_{\Delta,\ell}$ is fixed once the two-point functions and the shadow transform are normalized; the convention is that the resulting projector is idempotent on the irreducible representation and that the Plancherel formula below has no additional scalar normalization. With spinning external operators or multiple three-point tensor structures, $\Psi_{\Delta,\ell}$ becomes a matrix in the corresponding structure labels.

After the same external kinematic prefactor has been factored out as in the four-point decomposition, the partial wave gives a function $\Psi^{(12|34)}_{\Delta,\ell}(u,v)$. It satisfies the same quadratic Casimir equation as the block with eigenvalue $C_{\Delta,\ell}$, but it has different analytic boundary data. In the 12 OPE limit it is a fixed linear combination

$$\Psi^{(12|34)}_{\Delta,\ell}(u,v) = A^{(12|34)}_{\Delta,\ell} G^{(12|34)}_{\Delta,\ell}(u,v) + A^{(12|34)}_{\widetilde{\Delta},\ell} G^{(12|34)}_{\widetilde{\Delta},\ell}(u,v), \qquad \widetilde{\Delta} = D - \Delta, \tag{64.18}$$

where the coefficients $A_{\Delta,\ell}$ are determined by the chosen shadow-transform normalization. Thus a partial wave is not the same object as the conformal block appearing in a physical Euclidean OPE sum. The block selects one OPE boundary condition; the partial wave is a single-valued Euclidean conformal harmonic, containing both the block and the shadow block.

For functions of cross-ratios satisfying the hypotheses of conformal harmonic analysis, the partial waves on the principal series $\Delta = D/2 + \mathrm{i}\nu$, $\nu \in \mathbb{R}$, form the continuous basis

$$\mathcal{F}(u,v) = \sum_{\ell=0}^{\infty} \int_{-\infty}^{\infty} \frac{\mathrm{d}\nu}{2\pi}\, \mu(\nu,\ell)\, a(\nu,\ell)\, \Psi_{D/2+\mathrm{i}\nu,\ell}(u,v). \tag{64.19}$$

Here $\mu(\nu,\ell)$ is the Plancherel density associated with the normalization in (64.17), and $a(\nu,\ell)$ is the partial-wave transform of $\mathcal{F}$. Physical OPE data are recovered, when the required analyticity and growth hypotheses hold, by moving the ν-contour and taking residues at poles in Δ. This is why one must distinguish the two objects: conformal blocks organize the discrete OPE in a channel, whereas conformal partial waves provide the representation-theoretic resolution of identity used in integral inversion formulae.

64.5.2 Mellin Representation and OPE Poles

Mellin space is a transform representation of scalar separated-point correlators. Its use requires an existence and growth hypothesis parallel to the partial-wave transform above: a meromorphic Mellin amplitude belongs to a specific Mellin-Barnes representation of the correlator. When the representation exists with suitable vertical-contour decay, it turns the 12-channel OPE expansion into a residue expansion in a complex variable conjugate to the cross-ratio u.

Let $X_{ij} = x_{ij}^2$ for Euclidean separated points and let $\mathcal{O}_i$ be scalar primaries of dimensions Δ_i. Define the affine Mellin constraint space

$$\mathfrak{A}_\Delta = \left\{ \delta_{ij} = \delta_{ji} \in \mathbb{C}, \quad \delta_{ii} = 0, \quad \sum_{j\neq i} \delta_{ij} = \Delta_i \text{ for } i = 1,\ldots,n \right\}. \tag{64.20}$$

The constraints are exactly the conditions that make $\prod_{i<j} X_{ij}^{-\delta_{ij}}$ transform like an n-point function of scalar primaries under independent local scale factors at the insertion points.

Definition 64.9 (Scalar Mellin representation)**.** An n-point scalar Euclidean correlator is Mellin representable on a configuration-space domain U if there are vertical cycles $\mathcal{C} \subset \mathfrak{A}_\Delta$, a meromorphic or distributional function $M(\delta_{ij})$, and a choice of logarithm for each X_{ij} on U, such that

$$\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle = \int_{\mathcal{C}} [\mathrm{d}\delta]\, M(\delta_{ij}) \prod_{1\le i<j\le n} \Gamma(\delta_{ij})\, X_{ij}^{-\delta_{ij}}. \tag{64.21}$$

The measure $[\mathrm{d}\delta]$ is the product of $\mathrm{d}\delta/(2\pi\mathrm{i})$ over a basis of independent affine coordinates on $\mathfrak{A}_\Delta$. The contour is chosen so that the pole families whose residues give the desired OPE expansions lie on the side prescribed by the corresponding radial domain. The gamma factors in (64.21) are part of the normalization of M; moving any universal gamma factor between M and the kernel gives a different Mellin amplitude representing the same correlator.

For four points it is useful to replace the constrained variables by three Mellin-channel variables

$$\mathfrak{s}+\mathfrak{t}+\mathfrak{u}=\Delta_\Sigma, \qquad \Delta_\Sigma=\Delta_1+\Delta_2+\Delta_3+\Delta_4,$$

where

$$\begin{aligned} \delta_{12} &= \frac{\Delta_1+\Delta_2-\mathfrak{s}}{2}, & \delta_{34} &= \frac{\Delta_3+\Delta_4-\mathfrak{s}}{2}, \\ \delta_{14} &= \frac{\Delta_1+\Delta_4-\mathfrak{t}}{2}, & \delta_{23} &= \frac{\Delta_2+\Delta_3-\mathfrak{t}}{2}, \\ \delta_{13} &= \frac{\Delta_1+\Delta_3-\mathfrak{u}}{2}, & \delta_{24} &= \frac{\Delta_2+\Delta_4-\mathfrak{u}}{2}. \end{aligned} \tag{64.22}$$

The letters $\mathfrak{s},\mathfrak{t},\mathfrak{u}$ are Mellin variables; they are not scattering Mandelstam invariants. They are named by the OPE channel whose twist poles they expose.

With the four-point prefactor of (63.30), Definition 64.9 becomes

$$\mathcal{G}(u,v)=\int_{\mathcal{C}_{\mathfrak{s},\mathfrak{t}}} \frac{\mathrm{d}\mathfrak{s}\,\mathrm{d}\mathfrak{t}}{(4\pi\mathrm{i})^2}\, u^{\mathfrak{s}/2}v^{(\mathfrak{t}-\Delta_2-\Delta_3)/2}\Gamma_\Delta(\mathfrak{s},\mathfrak{t})\, M(\mathfrak{s},\mathfrak{t}), \tag{64.23}$$

where $\mathfrak{u}=\Delta_\Sigma-\mathfrak{s}-\mathfrak{t}$ and

$$\begin{aligned} \Gamma_\Delta(\mathfrak{s},\mathfrak{t}) = {}& \Gamma\!\left(\frac{\Delta_1+\Delta_2-\mathfrak{s}}{2}\right)\Gamma\!\left(\frac{\Delta_3+\Delta_4-\mathfrak{s}}{2}\right)\Gamma\!\left(\frac{\Delta_1+\Delta_4-\mathfrak{t}}{2}\right)\Gamma\!\left(\frac{\Delta_2+\Delta_3-\mathfrak{t}}{2}\right) \\ & \times\Gamma\!\left(\frac{\Delta_1+\Delta_3-\mathfrak{u}}{2}\right)\Gamma\!\left(\frac{\Delta_2+\Delta_4-\mathfrak{u}}{2}\right). \end{aligned} \tag{64.24}$$

The normalization $(4\pi\mathrm{i})^{-2}$ comes from $\mathrm{d}\delta_{12}\mathrm{d}\delta_{14}=\mathrm{d}\mathfrak{s}\,\mathrm{d}\mathfrak{t}/4$.

Compatibility with the four-point prefactor. The representation (64.23) is exactly the $n=4$ constrained Mellin representation (64.21) after factoring out the kinematic prefactor in (63.30). The Mellin product has exponent $-\delta_{ij}$ on X_{ij}. Subtract the exponents of the prefactor in (63.30). For X_{12} and X_{34} the remaining exponent is

$$-\delta_{12}+\frac{\Delta_1+\Delta_2}{2}=-\delta_{34}+\frac{\Delta_3+\Delta_4}{2}=\frac{\mathfrak{s}}{2}.$$

For X_{14} and X_{23} the remaining exponent is

$$\frac{\mathfrak{t}-\Delta_2-\Delta_3}{2}.$$

For X_{13} and X_{24} the remaining exponent is the negative of the sum of these two numbers, because $\mathfrak{u}=\Delta_\Sigma-\mathfrak{s}-\mathfrak{t}$. These are precisely the exponents of

$$u^{\mathfrak{s}/2}v^{(\mathfrak{t}-\Delta_2-\Delta_3)/2}=\left(\frac{X_{12}X_{34}}{X_{13}X_{24}}\right)^{\mathfrak{s}/2}\left(\frac{X_{14}X_{23}}{X_{13}X_{24}}\right)^{(\mathfrak{t}-\Delta_2-\Delta_3)/2}.$$

The six gamma factors are those in (64.21) with (64.22) substituted.

The pole interpretation is a contour-shift statement. Write

$$\mathcal{I}(\mathfrak{s}, \mathfrak{t}) = \Gamma_{\Delta}(\mathfrak{s}, \mathfrak{t}) M(\mathfrak{s}, \mathfrak{t})$$

for the full Mellin integrand without the cross-ratio powers. Suppose the $\mathfrak{s}$-contour is a vertical line, $0 < u < 1$, the integrand is meromorphic in the strip swept by moving the contour to the right, and the horizontal contour arcs vanish in that strip. If $\mathcal{I}$ has a simple pole at $\mathfrak{s} = \sigma$ with residue $R_\sigma(\mathfrak{t})$, then that pole contributes

$$u^{\sigma/2} \int_{\mathcal{C}_{\mathfrak{t}}} \frac{\mathrm{d}\mathfrak{t}}{2\pi \mathrm{i}}\, v^{(\mathfrak{t}-\Delta_2-\Delta_3)/2} R_\sigma(\mathfrak{t}) \tag{64.25}$$

to $\mathcal{G}(u, v)$. A pole of order $r+1$ gives a polynomial of degree r in $\log u$ multiplying $u^{\sigma/2}$. Conversely, under the same vertical-growth hypothesis, a term $u^\alpha H(v)$ in the asymptotic expansion at fixed v is represented in the Mellin transform by a pole at $\mathfrak{s} = 2\alpha$, with residue equal to the one-dimensional Mellin transform of $H(v)$ in the $\mathfrak{t}$ variable.

Now apply this to the radial OPE. An exchanged primary of dimension Δ and spin ℓ has twist

$$\tau = \Delta - \ell.$$

In the $12 \to 34$ OPE channel, the corresponding conformal block has a radial expansion of the form

$$G^{(12|34)}_{\Delta,\ell}(u, v) = u^{\tau/2} \sum_{m=0}^{\infty} u^m\, H_m^{(\Delta,\ell)}(v), \tag{64.26}$$

inside the radial convergence domain, after the spin-dependent angular dependence is expressed through v in the chosen conformal frame. Therefore the full Mellin integrand has poles at

$$\mathfrak{s} = \tau + 2m, \qquad m = 0, 1, 2, \ldots. \tag{64.27}$$

If the gamma-stripped amplitude M is meromorphic and the pole does not coincide with a gamma-kernel pole, the same family appears in M. With the OPE normalization of the block fixed in (64.14), its local form is

$$M(\mathfrak{s}, \mathfrak{t}) = \sum_{m=0}^{\infty} \frac{\lambda_{12\mathcal{O}} \lambda_{34\mathcal{O}}\, Q^{(12|34)}_{m,\ell}(\mathfrak{t})}{\mathfrak{s} - \tau - 2m} + M_{\mathrm{hol}}(\mathfrak{s}, \mathfrak{t}), \tag{64.28}$$

in a neighborhood of the displayed pole family. The residue $Q^{(12|34)}_{m,\ell}$ is the Mack polynomial in this normalization. It is characterized intrinsically as the unique polynomial residue which, after substitution in (64.23), produces the level-m term of the Casimir eigenfunction with OPE boundary condition (64.14). Its degree in $\mathfrak{t}$ is ℓ; the descendant label m changes the coefficients and the gamma kernel, not the spin degree.

Proposition 64.10 (OPE poles in Mellin space). *Assume the radial OPE theorem of Theorem 64.2, the four-point Mellin representation* (64.23)*, meromorphic continuation of M in the relevant $\mathfrak{s}$-strip, and the vertical growth needed for the contour shift above. Then every exchanged symmetric-traceless primary $\mathcal{O}$ in the $12 \to 34$ scalar OPE whose twist-pole family does not coincide with a gamma-kernel pole contributes the pole family* (64.28)*. The residue is proportional to the product $\lambda_{12\mathcal{O}}\lambda_{34\mathcal{O}}$, and the proportionality function is fixed by conformal representation theory.*

Proof. The radial OPE theorem gives an absolutely convergent sum of conformal families in the channel domain. Project to a single exchanged conformal multiplet and use the block construction in Theorem 64.8. Its radial series has the form (64.26): the primary supplies the exponent $\tau/2$, and the descendant contribution can be grouped into integer powers of u after the remaining angular dependence is kept in the functions $H_m(v)$. The coefficient multiplying the block in the four-point function is $\lambda_{12\mathcal{O}}\lambda_{34\mathcal{O}}$ in the orthonormal single-structure case, or the corresponding contraction with the inverse two-point pairing in the notation of (64.9).

The contour-shift identity (64.25) identifies every term $u^{\tau/2+m}H_m(v)$ with a pole at $\mathfrak{s} = \tau + 2m$ in the full Mellin integrand. When this pole is not a pole already forced by Γ_Δ, it is represented by a pole of M. The v-dependence of the residue is converted to a polynomial in $\mathfrak{t}$ by the one-dimensional Mellin transform with the fixed gamma kernel in (64.24). The Casimir equation (64.13) and the OPE boundary condition (64.14) fix this residue uniquely. The highest power of $\mathfrak{t}$ is fixed by the spin-ℓ leading angular structure of the primary three-point functions, hence the residue has degree ℓ. This proves the stated pole family and normalization. □

Crossing acts linearly on Mellin space. A permutation $\sigma \in S_4$ sends δ_{ij} to $\delta_{\sigma(i)\sigma(j)}$, and the induced action on $(\mathfrak{s}, \mathfrak{t}, \mathfrak{u})$ permutes the channel variables with $\mathfrak{s} + \mathfrak{t} + \mathfrak{u} = \Delta_\Sigma$ preserved. For identical scalar primaries, crossing of the separated-point correlator is equivalent, in a common Mellin domain, to the corresponding invariance of the Mellin integrand up to contour deformation across the pole families. Thus the Mellin form expresses the same OPE-associativity constraints as the block expansion, but in a language where exchanged twists are pole locations and spins are polynomial degrees of residues.

64.5.3 Lorentzian Inversion as a Boundary Theorem

The Euclidean OPE sum determines correlators by summing over the discrete operator spectrum. Lorentzian inversion goes in the opposite direction: under additional Lorentzian analyticity and boundedness hypotheses, it reconstructs the spin-analytic OPE coefficient function from a double discontinuity of the four-point function. For identical scalar primaries, define the t-channel double discontinuity across the branch cut $\bar{z} \in [1, \infty)$ by

$$\mathrm{dDisc}_t[\mathcal{G}](z, \bar{z}) = \mathcal{G}(z, \bar{z}) - \frac{1}{2}\,\mathcal{G}^{\circlearrowleft}(z, \bar{z}) - \frac{1}{2}\,\mathcal{G}^{\circlearrowright}(z, \bar{z}), \tag{64.29}$$

where the arrows denote the two analytic continuations of $\bar{z}$ around $\bar{z} = 1$. For unequal external dimensions the two continued terms are multiplied by the corresponding monodromy phases fixed by the external weights.

In the scalar normalization compatible with the partial-wave convention above, the t-channel contribution to the Caron-Huot Lorentzian inversion formula has the form

$$c^t(\Delta, J) = \frac{\kappa_{\Delta+J}}{4} \int_0^1 \mathrm{d}z \int_0^1 \mathrm{d}\bar{z}\, \frac{|z-\bar{z}|^{D-2}}{(z\bar{z})^D}\, G^{(12|34)}_{J+D-1,\,\Delta-D+1}(z, \bar{z})\, \mathrm{dDisc}_t[\mathcal{G}](z, \bar{z}), \tag{64.30}$$

with

$$\kappa_\beta = \frac{\Gamma(\beta/2)^4}{2\pi^2\,\Gamma(\beta-1)\Gamma(\beta)} \qquad \text{for identical external scalars.} \tag{64.31}$$

The full identical-scalar coefficient includes the crossed u-channel contribution

$$c(\Delta, J) = c^t(\Delta, J) + (-1)^J c^u(\Delta, J),$$

with the same kernel after the corresponding interchange of points. The block appearing in the kernel has analytically continued labels: its first label is $J+D-1$, and its second label is $\Delta-D+1$. This is the same block/partial-wave duality already visible in (64.18); the inversion integral pairs the Lorentzian double discontinuity with the shadow-side harmonic kernel.

The coefficient function $c(\Delta, J)$ is meromorphic in Δ for the spins covered by the Regge boundedness hypothesis. If a primary of spin J and dimension $\Delta_{\mathcal{O}}$ appears in the $\phi \times \phi$ OPE, then near the pole

$$c(\Delta, J) = \frac{\lambda^2_{\phi\phi\mathcal{O}}}{\Delta - \Delta_{\mathcal{O}}} + \text{holomorphic near } \Delta_{\mathcal{O}}. \tag{64.32}$$

Thus the formula extracts OPE data from Lorentzian branch-cut data rather than from Euclidean projection onto a discrete radial spectrum.

Remark 64.11 (Hypotheses and scope). Equation (64.30) is not an additional axiom of CFT data. It is a theorem under hypotheses: unitarity or reflection positivity sufficient for the Lorentzian continuation, Euclidean OPE convergence in the required domains, analyticity in the relevant tube domains, and Regge boundedness strong enough to suppress the arcs used in the contour deformation. The spin range begins above the Regge intercept; in many unitary CFT applications this gives $J > 1$, with possible lower-spin terms requiring separate subtractions. Later analytic-bootstrap and light-ray discussions use the inversion formula only at this theorem boundary.[2]

64.5.4 Radial Expansion of a Block

For identical Hermitian scalar primaries ϕ, there is a particularly transparent block expansion in the Hogervorst–Rychkov radial variable ρ. Put the four points in a plane at

$$x_1 = \rho, \qquad x_2 = -\rho, \qquad x_3 = -1, \qquad x_4 = 1, \qquad \rho = r e^{i\alpha}, \quad 0 \le r < 1.$$

With the cross-ratio convention of (63.29), this frame gives

$$z = \frac{4\rho}{(1+\rho)^2}, \qquad \rho = \frac{z}{(1+\sqrt{1-z})^2}. \tag{64.33}$$

The Weyl map to the cylinder places the two inner insertions at radial time $\tau = \log r$ and angular positions separated by π, while the two outer insertions lie at $\tau = 0$. The map (64.33) is chosen so that $|\rho| < 1$ is precisely the radial ordering domain for the $12 \to 34$ OPE channel; the descendant expansion below is the radial-coordinate expansion introduced by Hogervorst and Rychkov.[3]

Theorem 64.12 (Hogervorst–Rychkov radial block convergence)**.** *Assume the radial reconstruction hypotheses of Hypothesis 59.1, finite multiplicity of radial energy eigenspaces, and reflection positivity in the channel under consideration. For each exchanged conformal family in a scalar four-point function, the radial block expansion constructed below converges absolutely, with convergence uniform on every closed subdisk $|\rho| \le q < 1$, in the Euclidean radial configuration where*

[2]See S. Caron-Huot, "Analyticity in spin in conformal theories," JHEP **09** (2017) 078, arXiv:1703.00278; D. Simmons-Duffin, D. Stanford, and E. Witten, "A spacetime derivation of the Lorentzian OPE inversion formula," JHEP **07** (2018) 085, arXiv:1711.03816; and P. Kravchuk and D. Simmons-Duffin, "Light-ray operators in conformal field theory," JHEP **11** (2018) 102, arXiv:1805.00098.

[3]See M. Hogervorst and S. Rychkov, "Radial Coordinates for Conformal Blocks," Phys. Rev. D **87** (2013) 106004, arXiv:1303.1111.

$\bar\rho = \rho^*$. Equivalently, after analytic continuation of z with the square-root branch $\operatorname{Re}\sqrt{1-z} > 0$,

$$\rho(z) = \frac{z}{(1+\sqrt{1-z})^2} = \frac{1-\sqrt{1-z}}{1+\sqrt{1-z}}$$

maps the cut z-plane $\mathbb{C} \setminus [1,\infty)$ to the unit disk, and every compact subset of the cut plane maps into $|\rho| \le q$ for some $q < 1$. The radial series therefore gives the analytic continuation of the block throughout that regularity domain.

Proof. The geometric part is elementary. Put $w = \sqrt{1-z}$ with $\operatorname{Re} w > 0$. Then

$$\rho = \frac{1-w}{1+w}.$$

The Cayley transform sends the right half-plane to the unit disk because

$$|\rho|^2 = \frac{|1-w|^2}{|1+w|^2} = 1 - \frac{4\operatorname{Re} w}{|1+w|^2} < 1.$$

Its inverse is $z = 4\rho/(1+\rho)^2$. Hence the unit disk in ρ is not only the old $|z| < 1$ power-series domain in disguise; it is the uniformizing radial disk for the whole z-plane cut along the crossed channel singular locus.

The analytic convergence is the radial Hilbert-space argument in the symmetric cylinder frame. After the standard external kinematic factors have been removed, the contribution of one exchanged conformal family is obtained by inserting the orthogonal projection to that family between the two states created by the two operator pairs. Denote these states by A and B. Let P_E be the spectral projection of the cylinder Hamiltonian D_{rad} onto the finite-dimensional eigenspace of radial energy E inside the chosen family. Rotations of S^{D-1} commute with D_{rad}, so the level expansion is a matrix element of

$$r^{D_{\mathrm{rad}}} U(\alpha), \qquad \rho = r e^{\mathrm{i}\alpha},$$

where $U(\alpha)$ is the unitary rotation relating the angular positions of the two pairs. For $r \le q < 1$, Cauchy–Schwarz gives

$$\sum_E r^E \,|\langle A, P_E U(\alpha) B\rangle| \le \left(\sum_E q^E \|P_E A\|^2\right)^{1/2} \left(\sum_E q^E \|P_E B\|^2\right)^{1/2}.$$

The two sums on the right are

$$\|q^{D_{\mathrm{rad}}/2} A\|^2, \qquad \|q^{D_{\mathrm{rad}}/2} B\|^2.$$

By reflection positivity these norms are Euclidean correlation functions in which the two operator pairs are separated by an annulus of radial modulus q. The radial reconstruction hypothesis asserts exactly that such separated Euclidean configurations define finite Hilbert-space vectors. Hence the displayed bound is finite and independent of the partial-sum cutoff and of α. This proves absolute convergence uniformly for $r \le q$.

Finite multiplicity makes each energy-level contribution a finite matrix element in the corresponding space of spherical harmonics; for scalar external operators it is a finite linear combination of

Gegenbauer polynomials in the angular variable. The same majorant therefore controls every compact subdisk $|\rho| \le q$, and the Weierstrass test gives local uniform convergence of the radial series. The identity with the conformal block follows on the real radial domain from the complete-set construction; the locally uniform limit is analytic in the radial variable, so the identity theorem continues the equality to the ρ-disk and, through the inverse map $z = 4\rho/(1+\rho)^2$, to the cut z-plane. $\square$

At the crossing-symmetric Euclidean point $z = \bar{z} = 1/2$,

$$\rho(1/2) = \frac{1-\sqrt{1/2}}{1+\sqrt{1/2}} = 3 - 2\sqrt{2} \approx 0.1716. \tag{64.34}$$

This small value, and for identical external scalars the additional even-level selection discussed below, is the convergence mechanism behind efficient radial-block evaluation near the bootstrap crossing point.

Figure 64.3 shows the ρ-coordinate cylinder frame used for radial-block convergence, including the small value $\rho(1/2) = 3 - 2\sqrt{2}$ at the crossing-symmetric Euclidean point.

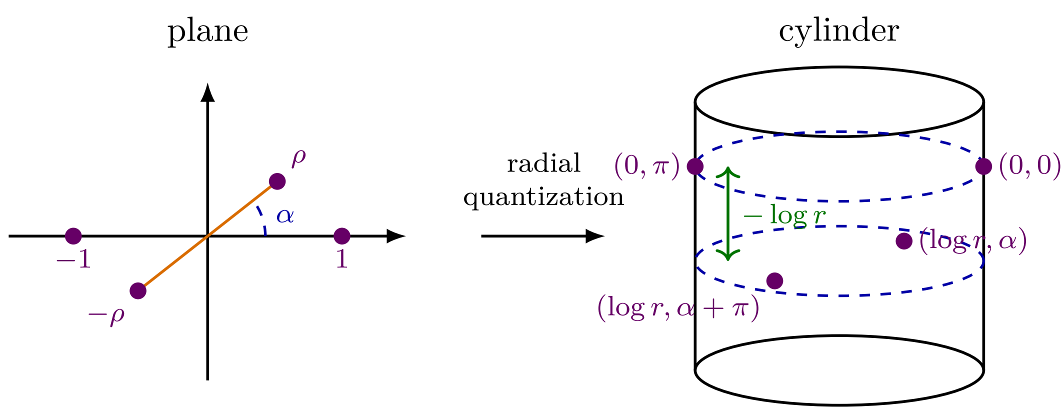


Figure 64.3: The radial variable $\rho = r\mathrm{e}^{\mathrm{i}\alpha}$ puts the $12 \to 34$ OPE channel into a cylinder frame. Inserting a complete set of cylinder energy eigenstates gives a convergent expansion in powers of r.

Assume ϕ is normalized by $\langle\langle\phi|\phi\rangle\rangle = 1$. In the cylinder frame, inserting a complete basis of states gives

$$\begin{aligned} 2^{-4\Delta_\phi}\mathcal{G}(u,v) &= \langle\Omega|\widetilde{\phi}(0,0)\widetilde{\phi}(0,\pi)\widetilde{\phi}(\log r,\alpha)\widetilde{\phi}(\log r,\alpha+\pi)|\Omega\rangle \\ &= \sum_n r^{E_n}\langle\Omega|\widetilde{\phi}(0,0)\widetilde{\phi}(0,\pi)|n\rangle\langle n|\widetilde{\phi}(0,\alpha)\widetilde{\phi}(0,\alpha+\pi)|\Omega\rangle. \end{aligned} \tag{64.35}$$

For a primary Φ_i of spin s_i and dimension Δ_i, its level-n descendants have energy $\Delta_i + n$. If $a = (\beta; N_1, \ldots, N_D)$, with $\sum_\mu N_\mu = n$, labels a raw descendant

$$|\Phi_i, a\rangle = \prod_{\mu=1}^{D} \widehat{P}_\mu^{N_\mu}|\Phi_i,\beta\rangle,$$

let

$$G^{\mathrm{raw}}_{ab} = \langle \Phi_i, a \,|\, \Phi_i, b\rangle$$

be the level-n Gram form. This raw Gram form is finite-dimensional at fixed n, but it can be singular. Its null subspace is

$$\mathcal{N}_{i,n} = \{v \in W_{i,n} : \langle v|w\rangle = 0 \text{ for every } w \in W_{i,n}\},$$

where $W_{i,n}$ is the raw level-n descendant space. Reflection positivity implies that every $v \in \mathcal{N}_{i,n}$ has zero matrix element against every physical state, by the Cauchy–Schwarz inequality. Hence null descendants are first quotiented:

$$Q_{i,n} = W_{i,n}/\mathcal{N}_{i,n}.$$

Choose any basis $\{|e^{(n)}_A\rangle\}_A$ of $Q_{i,n}$, represented by raw descendants, and define the nonsingular quotient Gram matrix

$$M^{(n)}_{AB} = \langle e^{(n)}_A|e^{(n)}_B\rangle.$$

The level-n projector onto the exchanged conformal family is

$$\Pi_{i,n} = \sum_{A,B} |e^{(n)}_A\rangle \left(M^{(n)}\right)^{-1}_{AB} \langle e^{(n)}_B|. \tag{64.36}$$

This expression is independent of the chosen quotient basis: under $|e_A\rangle \mapsto S_A{}^B|e_B\rangle$, the Gram matrix and its inverse transform contragrediently. The complete-set term at level n is therefore

$$r^{\Delta_i+n} \sum_{A,B} \langle\Omega|\widetilde{\phi}(0,0)\widetilde{\phi}(0,\pi)|e^{(n)}_A\rangle \left(M^{(n)}\right)^{-1}_{AB} \langle e^{(n)}_B|\widetilde{\phi}(0,\alpha)\widetilde{\phi}(0,\alpha+\pi)|\Omega\rangle. \tag{64.37}$$

The block is obtained by applying (64.37) to each finite level and then summing over levels. Thus no inverse of an infinite descendant Gram matrix is being assumed; the inverse is taken after quotienting null states at each energy level. Conservation laws and other shortening conditions enter exactly through the null subspaces $\mathcal{N}_{i,n}$. Consequently a block has a radial expansion

$$2^{-4\Delta_\phi} G_{\Delta_i,s_i}(u,v) = \sum_{n=0}^{\infty} A_{\Delta_i,s_i,n}(\alpha)\, r^{\Delta_i+n}, \tag{64.38}$$

where each coefficient is fixed by conformal representation theory.

The leading coefficient is especially simple. At level zero the only angular dependence is the contraction of two symmetric traceless spin-s tensors made from the two unit vectors separated by angle α. Hence, with a standard normalization of the block,

$$G_{\Delta,s}(u,v) = r^{\Delta} C^{(D/2-1)}_s(\cos\alpha) + O(r^{\Delta+1}), \tag{64.39}$$

where $C^{(\lambda)}_s$ is the standard Gegenbauer polynomial. Thus $C^{(1/2)}_s(\cos\alpha) = P_s(\cos\alpha)$ in $D = 3$, while

$$C^{(1)}_s(\cos\alpha) = \frac{\sin((s+1)\alpha)}{\sin\alpha}$$

in $D = 4$. The $D = 2$ angular functions are the separate limiting harmonic $\cos(s\alpha)$, since the Gegenbauer parameter $\lambda = D/2 - 1$ vanishes.

64.6 Reflection Positivity and OPE Coefficients

Reflection positivity constrains the numerical OPE data. Choose Hermitian scalar primaries with two-point normalization

$$\langle \mathcal{O}_A(x)\mathcal{O}_B(0)\rangle = \frac{\delta_{AB}}{|x|^{2\Delta_A}}.$$

With this normalization, phases of nondegenerate Hermitian primaries can be chosen so that three-point coefficients of Hermitian scalar operators are real. If ϕ is a Hermitian scalar primary, the contribution of a primary family $\mathcal{O}$ to the $\phi\phi\phi\phi$ four-point function has coefficient

$$\lambda^2_{\phi\phi\mathcal{O}} \geq 0$$

in the channel where the first pair of operators is radially inside the second pair. This nonnegative number is the squared norm of the projection of the state $\phi(x)\phi(0)\Omega$ onto the conformal multiplet of $\mathcal{O}$.

When several operators with the same quantum numbers mix, the statement is matrix-valued. Let P_Δ be the spectral projection onto a finite degenerate subspace of fixed spin and scaling dimension. For any coefficients α_i,

$$\left\| P_\Delta \sum_i \alpha_i \, \mathcal{O}_i(x_i)\Omega \right\|^2 \geq 0.$$

The corresponding OPE coefficients form a positive semidefinite matrix in that degenerate subspace. Positivity of OPE data is therefore a direct Hilbert-space consequence of radial quantization, not a separate dynamical assumption.

64.7 Associativity

Consider a Euclidean correlation function with three operators inside a sphere that separates them from all other insertions. If the insertion points admit nested separating spheres, there are two ways to apply the OPE:

$$(\mathcal{O}_1\mathcal{O}_2)\mathcal{O}_3, \qquad \mathcal{O}_1(\mathcal{O}_2\mathcal{O}_3).$$

Let S_{out} be the outer separating sphere. The product of the three separated insertions defines a vector in the radial Hilbert space on S_{out}, not a local operator. The two nested OPE procedures are two ways of inserting complete spectral resolutions of the same vector. On a common nested radial domain the two partial sums converge in Hilbert norm to that vector, hence have the same matrix elements against exterior states. Away from the common domain, the statement becomes an analytic-continuation statement: each nested expansion defines a holomorphic function in its own channel domain, and both are restrictions of the same separated-point correlation function on the connected component of complexified configuration space reached without crossing coincident-point or light-cone singular hypersurfaces.

For identical scalar primaries ϕ of dimension Δ_ϕ, write the four-point function in the form

$$\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle = \frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta_\phi}|x_{34}|^{2\Delta_\phi}},$$

where $x_{ij} = x_i - x_j$ and the conformal cross-ratios are

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \qquad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}.$$

Introduce $z, \bar{z}$ by $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$. The $12 \to 34$ channel is controlled by

$$\rho_s = \frac{z}{(1+\sqrt{1-z})^2}, \qquad \bar{\rho}_s = \frac{\bar{z}}{(1+\sqrt{1-\bar{z}})^2},$$

and its Euclidean radial block expansion converges on compact subsets of

$$\mathcal{D}_s = \{(z, \bar{z}) : |\rho_s| < 1,\ |\bar{\rho}_s| < 1\}. \tag{64.40}$$

The $14 \to 23$ channel uses the crossed variables

$$\rho_t = \frac{1-z}{(1+\sqrt{z})^2}, \qquad \bar{\rho}_t = \frac{1-\bar{z}}{(1+\sqrt{\bar{z}})^2},$$

and converges on compact subsets of

$$\mathcal{D}_t = \{(z, \bar{z}) : |\rho_t| < 1,\ |\bar{\rho}_t| < 1\}. \tag{64.41}$$

These domains are channel domains, not identical power-series neighborhoods: $\mathcal{D}_s$ contains the $u \to 0$ OPE boundary, while $\mathcal{D}_t$ contains the $v \to 0$ OPE boundary. Their Euclidean intersection is nonempty; for example $0 < z = \bar{z} < 1$ lies in both domains. On that intersection the two convergent Hilbert-space expansions compute the same four-point function. The identity theorem then identifies their holomorphic continuations on every connected component where both continuations are defined.

The crossing equation itself follows from the permutation symmetry of the identical Euclidean scalar correlator. Exchanging x_2 and x_4 sends (u, v) to (v, u). The same separated-point function may therefore be written as

$$\frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta_\phi} |x_{34}|^{2\Delta_\phi}} = \frac{\mathcal{G}(v,u)}{|x_{14}|^{2\Delta_\phi} |x_{23}|^{2\Delta_\phi}}.$$

Since

$$\frac{u}{v} = \frac{x_{12}^2 x_{34}^2}{x_{14}^2 x_{23}^2},$$

this equality is equivalent to

$$v^{\Delta_\phi} \mathcal{G}(u,v) = u^{\Delta_\phi} \mathcal{G}(v,u). \tag{64.42}$$

If ϕ is Hermitian and the exchanged primaries are orthonormal, the $12 \to 34$ expansion takes the form

$$\mathcal{G}(u,v) = 1 + \sum_{\mathcal{O} \neq \mathbf{1}} \lambda_{\phi\phi\mathcal{O}}^2 G_{\Delta_\mathcal{O}, s_\mathcal{O}}(u,v), \qquad \lambda_{\phi\phi\mathcal{O}}^2 \geq 0. \tag{64.43}$$

The identity block gives the constant 1. The $14 \to 23$ expansion gives the same formula with $(u, v) \mapsto (v, u)$, valid in the channel domain $\mathcal{D}_t$. Equation (64.42) is therefore the equality of two OPE-defined holomorphic functions after both have been identified with the analytic continuation of one Euclidean separated-point correlator. It is not obtained by comparing two unrelated power series at their distinct expansion corners.

64.8 Worked Example: Generalized-Free Scalar Crossing

The preceding discussion becomes completely concrete for a scalar generalized-free datum. Assume $D > 2$ and $\Delta_\phi \geq (D-2)/2$, or $D = 2$ and $\Delta_\phi \geq 0$ for a spin-zero global conformal scalar. Normalize

$$\langle \phi(x)\phi(y)\rangle_{\mathrm{GFF}} = |x-y|^{-2\Delta_\phi}.$$

The separated-point $2m$-point functions are defined by Wick pairings,

$$\langle \phi(x_1)\cdots\phi(x_{2m})\rangle_{\mathrm{GFF}} = \sum_{\pi\in\mathrm{Pair}(2m)} \prod_{\{i,j\}\in\pi} |x_i - x_j|^{-2\Delta_\phi}, \qquad \langle \phi(x_1)\cdots\phi(x_{2m+1})\rangle_{\mathrm{GFF}} = 0. \tag{64.44}$$

Here $\mathrm{Pair}(2m)$ denotes the set of partitions of $\{1,\ldots,2m\}$ into unordered two-element subsets. In the special case $D > 2$ and $\Delta_\phi = (D-2)/2$, the quotient by the null descendant $\partial^2\phi$ gives the ordinary massless free scalar CFT. For larger Δ_ϕ the same formula defines a generalized-free scalar sector. The statements below use the scalar module generated by ϕ, the Wick pairing rule, and radial reflection positivity.

In radial quantization, $\phi(0)|\Omega\rangle$ generates the scalar lowest-weight conformal module V_{Δ_ϕ}, quotienting null vectors when the unitarity bound is saturated. Equation (64.44) makes the m-particle Gram matrix the permanent Gram matrix on $\mathrm{Sym}^m V_{\Delta_\phi}$, meaning that its entries are sums over pairings with no sign character. Thus the reconstructed radial Hilbert space of this sector is the symmetric Fock space over V_{Δ_ϕ}, before Hilbert completion and after the same null-vector quotient. This finite Gram-matrix statement is the reason the OPE below has nonnegative squared coefficients.

For four points Wick factorization gives the three pairings

$$\begin{aligned}\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle_{\mathrm{GFF}} = \frac{1}{|x_{12}|^{2\Delta_\phi}|x_{34}|^{2\Delta_\phi}} + \frac{1}{|x_{13}|^{2\Delta_\phi}|x_{24}|^{2\Delta_\phi}} \\ + \frac{1}{|x_{14}|^{2\Delta_\phi}|x_{23}|^{2\Delta_\phi}}.\end{aligned} \tag{64.45}$$

With the four-point prefactor and cross-ratios used in (64.42), the reduced correlator is therefore

$$\mathcal{G}_{\mathrm{GFF}}(u,v) = 1 + u^{\Delta_\phi} + \left(\frac{u}{v}\right)^{\Delta_\phi}. \tag{64.46}$$

The crossing relation is now an algebraic identity:

$$v^{\Delta_\phi}\mathcal{G}_{\mathrm{GFF}}(u,v) = v^{\Delta_\phi} + (uv)^{\Delta_\phi} + u^{\Delta_\phi} = u^{\Delta_\phi} + (uv)^{\Delta_\phi} + v^{\Delta_\phi} = u^{\Delta_\phi}\mathcal{G}_{\mathrm{GFF}}(v,u). \tag{64.47}$$

The $12 \to 34$ OPE contains the identity and bilinear conformal primaries. Let z^μ be a null auxiliary vector, $z^2 = 0$, used to encode symmetric traceless spin. For $n \in \mathbb{Z}_{\geq 0}$ and $\ell \in 2\mathbb{Z}_{\geq 0}$, define

$$\mathcal{O}_{n,\ell}(x,z) = \mathcal{N}_{n,\ell}\,\Pi_{\mathrm{prim}}\Big[:\phi(x_1)\big(z\cdot(\partial_{x_1}-\partial_{x_2})\big)^\ell\big((\partial_{x_1}-\partial_{x_2})^2\big)^n\phi(x_2):\Big]_{x_1=x_2=x}. \tag{64.48}$$

The colons denote Wick subtraction with respect to (64.44). The projection Π_{prim} is defined as follows. At fixed total relative derivative degree $N = 2n+\ell$, the rotational spin-ℓ bilinears form a finite-dimensional inner-product space. Total derivatives are P_μ-descendants of lower energy states. Since radial reflection makes $K_\mu = P_\mu^\dagger$, the primary subspace is the orthogonal complement of the descendant subspace, equivalently the kernel of every K_μ. The finite orthogonal projection onto this kernel is Π_{prim}. Homogeneous polynomials in the relative derivative decompose as

$$\mathrm{Sym}^N(\mathbb{R}^D) = \bigoplus_{j=0}^{\lfloor N/2 \rfloor} (q^2)^j \, \mathcal{H}_{N-2j},$$

where $\mathcal{H}_m$ is the space of degree-m harmonic polynomials. For $N = 2n + \ell$, the quotient by total derivatives contains one copy of the spin-ℓ harmonic representation. Its projection has nonzero Wick norm, so it gives one orthonormal primary family for each (n, ℓ) that survives the null quotient. Exchange of the two identical fields sends the relative derivative to its negative, so only even ℓ survives in the symmetric product.

The normalization $\mathcal{N}_{n,\ell}$ is chosen so that

$$\langle \mathcal{O}_{n,\ell}(x, z) \mathcal{O}_{n',\ell'}(0, z') \rangle = 0 \quad \text{unless} \quad (n, \ell) = (n', \ell'),$$

and the nonzero two-point function has unit coefficient in the spin-ℓ conformal normalization used for the blocks in this chapter. The corresponding scaling dimension and twist are

$$\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell, \qquad \tau_{n,\ell} = \Delta_{n,\ell} - \ell = 2\Delta_\phi + 2n. \tag{64.49}$$

The dimension formula follows directly from radial energy: two scalar one-particle states contribute $2\Delta_\phi$, each derivative contributes one unit, and the primary projection only subtracts states with the same total energy.

With orthonormal primaries the $12 \to 34$ channel expansion is

$$\mathcal{G}_{\text{GFF}}(u, v) = 1 + \sum_{n=0}^{\infty} \sum_{\ell \in 2\mathbb{Z}_{\geq 0}} a_{n,\ell}^{\text{GFF}} \, G_{2\Delta_\phi + 2n + \ell, \ell}(u, v), \qquad a_{n,\ell}^{\text{GFF}} = \lambda_{\phi\phi\mathcal{O}_{n,\ell}}^2 \geq 0, \tag{64.50}$$

valid in the s-channel radial domain $\mathcal{D}_s$ of (64.40). This formula is the radial spectral resolution of the vector $\phi(x_1)\phi(x_2)|\Omega\rangle$. Its identity component is the contraction $|x_{12}|^{-2\Delta_\phi}|\Omega\rangle$; its normal-ordered component lies in $\mathrm{Sym}^2 V_{\Delta_\phi}$, whose primary decomposition was constructed in the preceding paragraph. Pairing this convergent vector expansion with the exterior state $\phi(x_3)\phi(x_4)|\Omega\rangle$ gives (64.50). Since the same matrix element was already computed by the finite Wick sum (64.45), uniqueness of the orthogonal spectral resolution identifies the block sum with (64.46) throughout $\mathcal{D}_s$.

Equivalently, with spin indices encoded by the auxiliary null vector or displayed explicitly at level zero, the OPE has the radial Hilbert-space form

$$\phi(x)\phi(0)|\Omega\rangle = |x|^{-2\Delta_\phi}|\Omega\rangle + \sum_{n=0}^{\infty} \sum_{\ell \in 2\mathbb{Z}_{\geq 0}} \lambda_{\phi\phi\mathcal{O}_{n,\ell}} \sum_{r=0}^{\infty} \mathcal{C}_{n,\ell}^{(r)}(x, P) \, \mathcal{O}_{n,\ell}(0)|\Omega\rangle, \tag{64.51}$$

where P_μ acts on the primary state to generate descendants. The level-r coefficient is a homogeneous polynomial-valued differential operator

$$\mathcal{C}_{n,\ell}^{(r)}(x, P) = \sum_{|\alpha| = r} c_\alpha^{(n,\ell)}(x) \, P^\alpha,$$

obtained by applying the finite projector (64.36) at descendant level r. Its primary-level term is

$$\mathcal{C}^{(0)}_{n,\ell}(x,P)\,\mathcal{O}_{n,\ell}(0) = |x|^{2n} x_{\mu_1}\cdots x_{\mu_\ell}\mathcal{O}^{\mu_1\cdots\mu_\ell}_{n,\ell}(0).$$

The power of $|x|$ is fixed by dimension: $\Delta_{n,\ell} - 2\Delta_\phi - \ell = 2n$.

The first nontrivial scalar primary illustrates the normalization. Put

$$\mathcal{O}_{0,0}(x) = \frac{1}{\sqrt{2}} : \phi(x)^2 : .$$

Wick contraction gives

$$\langle : \phi(x)^2 :: \phi(0)^2 :\rangle = \frac{2}{|x|^{4\Delta_\phi}}, \qquad \langle \phi(x_1)\phi(x_2) : \phi(x_3)^2 :\rangle = \frac{2}{|x_{13}|^{2\Delta_\phi}|x_{23}|^{2\Delta_\phi}}.$$

Hence

$$\lambda_{\phi\phi\mathcal{O}_{0,0}} = \sqrt{2}, \qquad a^{\mathrm{GFF}}_{0,0} = 2. \tag{64.52}$$

The first two terms of (64.51) are therefore

$$\phi(x)\phi(0)|\Omega\rangle = |x|^{-2\Delta_\phi}|\Omega\rangle + \sqrt{2}\,\mathcal{O}_{0,0}(0)|\Omega\rangle + \cdots,$$

where the omitted terms are precisely the positive-level terms in (64.51). The remaining terms in the two functions u^{Δ_ϕ} and $(u/v)^{\Delta_\phi}$, expanded near the s-channel OPE boundary, are reproduced by the descendants of $\mathcal{O}_{0,0}$ together with the higher-spin and higher-n bilinear primaries in (64.50).

The $14 \to 23$ OPE is the same representation-theoretic decomposition in the crossed channel. Written with the $12 \to 34$ prefactor, it reads

$$\mathcal{G}_{\mathrm{GFF}}(u,v) = \left(\frac{u}{v}\right)^{\Delta_\phi}\left[1 + \sum_{n=0}^{\infty}\sum_{\ell\in 2\mathbb{Z}_{\geq 0}} a^{\mathrm{GFF}}_{n,\ell}\, G_{2\Delta_\phi+2n+\ell,\ell}(v,u)\right], \tag{64.53}$$

valid in $\mathcal{D}_t$. The identity operator in the crossed channel is responsible for the term $(u/v)^{\Delta_\phi}$ in (64.46); the crossed bilinear tower sums to $(u/v)^{\Delta_\phi}(v^{\Delta_\phi} + (v/u)^{\Delta_\phi}) = u^{\Delta_\phi} + 1$. Thus the two OPE channels are two convergent radial expansions of the same Wick-pairing function, and (64.47) is the prefactor form of that equality.

64.9 A First Consequence of Crossing and Positivity

The simplest use of (64.42) is to see why a unitary CFT cannot have an arbitrarily large gap above the identity in the $\phi \times \phi$ OPE while keeping Δ_ϕ fixed. This subsection is deliberately a controlled approximation, not a sharp bootstrap theorem: we keep the leading radial behavior of heavy exchanged blocks near the crossing-symmetric point and state the assumptions used.

Take $z = \bar{z} \in (0,1)$, so that $u = z^2$ and $v = (1-z)^2$. The crossing equation becomes

$$(1-z)^{2\Delta_\phi}\mathcal{G}(z) = z^{2\Delta_\phi}\mathcal{G}(1-z), \tag{64.54}$$

where $\mathcal{G}(z)$ abbreviates $\mathcal{G}(z^2,(1-z)^2)$. Using (64.43),

$$\left[(1-z)^{2\Delta_\phi} - z^{2\Delta_\phi}\right] + \sum_{\mathcal{O}\neq\mathbf{1}} \lambda^2_{\mathcal{O}}\left[(1-z)^{2\Delta_\phi}G_{\mathcal{O}}(z) - z^{2\Delta_\phi}G_{\mathcal{O}}(1-z)\right] = 0. \tag{64.55}$$

For $0 < z < 1$, define

$$r(z) = \frac{z}{(1+\sqrt{1-z})^2}.$$

If the exchanged dimensions contributing above the identity are large, and if we keep only the leading radial behavior

$$G_{\mathcal{O}}(z) \simeq r(z)^{\Delta_{\mathcal{O}}}$$

on this line, then (64.55) is approximated by

$$\left[(1-z)^{2\Delta_\phi} - z^{2\Delta_\phi}\right] + \sum_{\mathcal{O}\neq \mathbf{1}} \lambda_{\mathcal{O}}^2 \left[(1-z)^{2\Delta_\phi} r(z)^{\Delta_{\mathcal{O}}} - z^{2\Delta_\phi} r(1-z)^{\Delta_{\mathcal{O}}}\right] = 0. \tag{64.56}$$

The crossing-symmetric point is $z = 1/2$, where

$$r(1/2) = 3 - 2\sqrt{2}.$$

Set $z = 1/2 + x$ and expand (64.56) in x. The identity contribution is

$$(1-z)^{2\Delta_\phi} - z^{2\Delta_\phi} = -\Delta_\phi 2^{3-2\Delta_\phi}\left[x + \frac{4}{3}(\Delta_\phi - 1)(2\Delta_\phi - 1)x^3 + \cdots\right].$$

The coefficient follows directly from $(1/2-x)^{2\Delta_\phi} - (1/2+x)^{2\Delta_\phi} = 2^{-2\Delta_\phi}[(1-2x)^{2\Delta_\phi} - (1+2x)^{2\Delta_\phi}]$; the odd cubic term is $-2^{4-2\Delta_\phi}\binom{2\Delta_\phi}{3}x^3$, which becomes the displayed $\frac{4}{3}(\Delta_\phi - 1)(2\Delta_\phi - 1)$ after the common factor $-\Delta_\phi 2^{3-2\Delta_\phi}$ is extracted. For a heavy exchanged primary, the leading odd part of the approximate block contribution has the form

$$\widetilde{\lambda}_{\mathcal{O}}^2\left[x + \frac{4}{3}\Delta_{\mathcal{O}}^2 x^3 + \cdots\right], \qquad \widetilde{\lambda}_{\mathcal{O}}^2 \geq 0,$$

where the positive factor relating $\widetilde{\lambda}_{\mathcal{O}}^2$ to $\lambda_{\mathcal{O}}^2$ includes $(3-2\sqrt{2})^{\Delta_{\mathcal{O}}}$ and other positive constants. Matching the coefficients of x and x^3 gives, in this approximation,

$$\sum_{\mathcal{O}\neq \mathbf{1}} \widetilde{\lambda}_{\mathcal{O}}^2\left[\Delta_{\mathcal{O}}^2 - (\Delta_\phi - 1)(2\Delta_\phi - 1)\right] = 0.$$

Since every coefficient in the sum is nonnegative, not all exchanged dimensions can be much larger than

$$\sqrt{(\Delta_\phi - 1)(2\Delta_\phi - 1)}.$$

The robust conclusion is qualitative: crossing plus reflection positivity forces nontrivial operator content in the $\phi \times \phi$ OPE at dimensions controlled by Δ_ϕ. A rigorous numerical or analytic bound requires the full conformal blocks and lies beyond this first approximation.

64.10 Mixed Correlators and the Three-Dimensional Ising Bootstrap System

The preceding argument used one four-point function. The first sharp bootstrap examples in $D > 2$ require several four-point functions at once, because positivity is then matrix-valued rather than a list of nonnegative numbers. This section records the exact crossing system for the scalar pair used in the three-dimensional Ising bootstrap. It is not a numerical solution of that system. It is the CFT information to which finite-dimensional semidefinite programs are applied.

Definition 64.13 ($\mathbb{Z}_2$-graded scalar bootstrap system)**.** A $\mathbb{Z}_2$-graded scalar bootstrap system in a unitary CFT in dimension $D > 2$ consists of:

1. two Hermitian scalar primaries σ and ε, with $\mathbb{Z}_2$-charges odd and even and dimensions $\Delta_\sigma, \Delta_\varepsilon$;
2. two-point normalization
$$\langle\sigma(x)\sigma(0)\rangle = |x|^{-2\Delta_\sigma}, \qquad \langle\varepsilon(x)\varepsilon(0)\rangle = |x|^{-2\Delta_\varepsilon};$$
3. real OPE coefficients in the three scalar OPEs
$$\sigma\times\sigma \subset \mathcal{S}_+, \qquad \varepsilon\times\varepsilon \subset \mathcal{S}_+, \qquad \sigma\times\varepsilon \subset \mathcal{S}_-,$$
where $\mathcal{S}_+$ and $\mathcal{S}_-$ are respectively the $\mathbb{Z}_2$-even and odd primary spectra;
4. the spin selection rule that $\sigma\times\sigma$ and $\varepsilon\times\varepsilon$ contain only even spins, while $\sigma\times\varepsilon$ may contain any integer spin.

The identity operator belongs to $\mathcal{S}_+$, with $\lambda_{\sigma\sigma\mathbf{1}} = \lambda_{\varepsilon\varepsilon\mathbf{1}} = 1$.

The sign in the last OPE coefficient is convention-sensitive. For a symmetric-traceless spin-ℓ primary $\mathcal{O}_{\mu_1\cdots\mu_\ell}$, the scalar-scalar-spin three-point tensor contains the homogeneous factor $(Z\cdot x_{12})^\ell$ in an index-free polarization Z. Therefore exchanging the two scalar insertions gives

$$\lambda_{\varepsilon\sigma\mathcal{O}} = (-1)^\ell \lambda_{\sigma\varepsilon\mathcal{O}}. \tag{64.57}$$

For scalar external labels i, j, k, l, define the crossing combinations

$$\mathsf{F}^{ij,kl}_{\pm,\Delta,\ell}(u,v) := v^{(\Delta_j+\Delta_k)/2} G^{ij|kl}_{\Delta,\ell}(u,v) \pm u^{(\Delta_j+\Delta_k)/2} G^{ij|kl}_{\Delta,\ell}(v,u), \tag{64.58}$$

where $G^{ij|kl}_{\Delta,\ell}$ is the $12 \to 34$ OPE-normalized scalar block in the convention of (64.9). The superscript $ij|kl$ records the external dimension differences used in the Casimir equation. When $i = j = k = l = \phi$, F_- is exactly the identical scalar crossing kernel

$$v^{\Delta_\phi} G_{\Delta,\ell}(u,v) - u^{\Delta_\phi} G_{\Delta,\ell}(v,u).$$

Proposition 64.14 (Five crossing equations for the σ, ε system)**.** *Let Definition 64.13 hold. The three four-point functions*

$$\langle\sigma\sigma\sigma\sigma\rangle, \qquad \langle\sigma\sigma\varepsilon\varepsilon\rangle, \qquad \langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$$

imply the following five crossing equations:

$$0 = \sum_{\mathcal{O}\in\mathcal{S}_+} \lambda^2_{\sigma\sigma\mathcal{O}}\, \mathsf{F}^{\sigma\sigma,\sigma\sigma}_{-,\Delta,\ell}(u,v), \tag{64.59}$$

$$0 = \sum_{\mathcal{O}\in\mathcal{S}_+} \lambda^2_{\varepsilon\varepsilon\mathcal{O}}\, \mathsf{F}^{\varepsilon\varepsilon,\varepsilon\varepsilon}_{-,\Delta,\ell}(u,v), \tag{64.60}$$

$$0 = \sum_{\mathcal{O}\in\mathcal{S}_-} \lambda^2_{\sigma\varepsilon\mathcal{O}}\, \mathsf{F}^{\sigma\varepsilon,\sigma\varepsilon}_{-,\Delta,\ell}(u,v), \tag{64.61}$$

$$0 = \sum_{\mathcal{O}\in\mathcal{S}_+} \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}\, \mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{-,\Delta,\ell}(u,v) + \sum_{\mathcal{O}\in\mathcal{S}_-} (-1)^\ell \lambda^2_{\sigma\varepsilon\mathcal{O}}\, \mathsf{F}^{\varepsilon\sigma,\sigma\varepsilon}_{-,\Delta,\ell}(u,v), \tag{64.62}$$

$$0 = \sum_{\mathcal{O}\in\mathcal{S}_+} \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}\, \mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{+,\Delta,\ell}(u,v) - \sum_{\mathcal{O}\in\mathcal{S}_-} (-1)^\ell \lambda^2_{\sigma\varepsilon\mathcal{O}}\, \mathsf{F}^{\varepsilon\sigma,\sigma\varepsilon}_{+,\Delta,\ell}(u,v). \tag{64.63}$$

The sums include the identity in $\mathcal{S}_+$; the last three equations contain no identity contribution from $\mathcal{S}_-$.

Proof. Equations (64.59) and (64.60) are the identical-scalar crossing equation (64.42) applied to σ and ε. Equation (64.61) is the same statement applied to the ordered correlator $\langle\sigma\varepsilon\varepsilon\sigma\rangle$; the exchanged primaries in the 12 channel are $\mathbb{Z}_2$-odd.

For the mixed correlator, expand $\langle\sigma_1\sigma_2\varepsilon_3\varepsilon_4\rangle$ in the $12 \to 34$ channel:

$$\mathcal{G}_{\sigma\sigma\varepsilon\varepsilon}(u,v) = \sum_{\mathcal{O}\in\mathcal{S}_+} \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}\, G^{\sigma\sigma|\varepsilon\varepsilon}_{\Delta,\ell}(u,v).$$

Only $\mathbb{Z}_2$-even primaries occur, because both pair OPEs are even. Now exchange the first and third insertions and use the same scalar four-point prefactor as in (63.30). A direct comparison of the prefactors gives

$$\mathcal{G}_{\sigma\sigma\varepsilon\varepsilon}(u,v) = \frac{u^{\Delta_\sigma}}{v^{(\Delta_\sigma+\Delta_\varepsilon)/2}}\, \mathcal{G}_{\varepsilon\sigma\sigma\varepsilon}(v,u). \tag{64.64}$$

The right-hand side has a $\sigma\times\varepsilon$ OPE and therefore contains only $\mathbb{Z}_2$-odd primaries. With $\lambda_{\varepsilon\sigma\mathcal{O}} = (-1)^\ell\lambda_{\sigma\varepsilon\mathcal{O}}$, this crossing relation is equivalent to

$$\sum_{\mathcal{O}\in\mathcal{S}_+} \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}\, v^{(\Delta_\sigma+\Delta_\varepsilon)/2} G^{\sigma\sigma|\varepsilon\varepsilon}_{\Delta,\ell}(u,v) - \sum_{\mathcal{O}\in\mathcal{S}_-} (-1)^\ell \lambda^2_{\sigma\varepsilon\mathcal{O}}\, u^{\Delta_\sigma} G^{\varepsilon\sigma|\sigma\varepsilon}_{\Delta,\ell}(v,u) = 0. \tag{64.65}$$

Repeating the same prefactor comparison for the exchanged ordering $\langle\varepsilon_1\sigma_2\sigma_3\varepsilon_4\rangle$, and then writing the result in the same block notation after the simultaneous exchange of the two scalar pairs, gives the companion equation

$$\sum_{\mathcal{O}\in\mathcal{S}_+} \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}\, u^{(\Delta_\sigma+\Delta_\varepsilon)/2} G^{\sigma\sigma|\varepsilon\varepsilon}_{\Delta,\ell}(v,u) - \sum_{\mathcal{O}\in\mathcal{S}_-} (-1)^\ell \lambda^2_{\sigma\varepsilon\mathcal{O}}\, v^{\Delta_\sigma} G^{\varepsilon\sigma|\sigma\varepsilon}_{\Delta,\ell}(u,v) = 0. \tag{64.66}$$

Subtracting (64.66) from (64.65), and then adding the same two equations, gives respectively (64.62) and (64.63). This proves all five equations. $\square$

It is useful to combine these equations into the form used by semidefinite programming. Define a five-vector $\vec{\mathsf{V}}_{-,\Delta,\ell}$ and a five-vector $\vec{\mathsf{V}}_{+,\Delta,\ell}$ whose entries are 2×2 matrices by

$$\vec{\mathsf{V}}_{-,\Delta,\ell} := \begin{pmatrix} 0 \\ 0 \\ \mathsf{F}^{\sigma\varepsilon,\sigma\varepsilon}_{-,\Delta,\ell} \\ (-1)^{\ell}\mathsf{F}^{\varepsilon\sigma,\sigma\varepsilon}_{-,\Delta,\ell} \\ -(-1)^{\ell}\mathsf{F}^{\varepsilon\sigma,\sigma\varepsilon}_{+,\Delta,\ell} \end{pmatrix}, \tag{64.67}$$

$$\vec{\mathsf{V}}_{+,\Delta,\ell} := \begin{pmatrix} \begin{pmatrix} \mathsf{F}^{\sigma\sigma,\sigma\sigma}_{-,\Delta,\ell} & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & \mathsf{F}^{\varepsilon\varepsilon,\varepsilon\varepsilon}_{-,\Delta,\ell} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & \frac{1}{2}\mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{-,\Delta,\ell} \\ \frac{1}{2}\mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{-,\Delta,\ell} & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & \frac{1}{2}\mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{+,\Delta,\ell} \\ \frac{1}{2}\mathsf{F}^{\sigma\sigma,\varepsilon\varepsilon}_{+,\Delta,\ell} & 0 \end{pmatrix} \end{pmatrix}. \tag{64.68}$$

Then Proposition 64.14 is equivalent to

$$\sum_{\mathcal{O}\in\mathcal{S}_+} \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}} & \lambda_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix} \vec{\mathsf{V}}_{+,\Delta,\ell} \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}} \\ \lambda_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix} + \sum_{\mathcal{O}\in\mathcal{S}_-} \lambda^2_{\sigma\varepsilon\mathcal{O}} \vec{\mathsf{V}}_{-,\Delta,\ell} = 0. \tag{64.69}$$

This equation is a vector equation in the space of functions of u, v. The $\mathbb{Z}_2$-even coefficients are not individually positive in the mixed entries, but they form positive semidefinite matrices. More precisely, for a fixed even primary family, or for a finite degenerate even subspace,

$$\Lambda^+_{\Delta,\ell} = \sum_a \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}_a} \\ \lambda_{\varepsilon\varepsilon\mathcal{O}_a} \end{pmatrix} \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}_a} & \lambda_{\varepsilon\varepsilon\mathcal{O}_a} \end{pmatrix} \succeq 0. \tag{64.70}$$

The odd-sector coefficients are ordinary nonnegative numbers $\lambda^2_{\sigma\varepsilon\mathcal{O}}$.

Finite functional exclusion proof object. Fix a candidate $\mathbb{Z}_2$-graded scalar spectrum satisfying the unitarity bounds and any additional gap assumptions. The crossing equation (64.69) cannot be solved if there is a linear functional separating the identity vector from the cone generated by the allowed nonidentity blocks. Concretely, let $\vec{\alpha} = (\alpha_1, \ldots, \alpha_5)$ be linear functionals on the five components of (64.69). Suppose that:

1. the identity contribution is strictly positive,

$$\begin{pmatrix} 1 & 1 \end{pmatrix} \vec{\alpha} \cdot \vec{\mathsf{V}}_{+,0,0} \begin{pmatrix} 1 \\ 1 \end{pmatrix} > 0;$$

2. for every allowed even nonidentity primary, $\vec{\alpha} \cdot \vec{\mathsf{V}}_{+,\Delta,\ell}$ is a positive semidefinite 2×2 matrix;
3. for every allowed odd primary, $\vec{\alpha} \cdot \vec{\mathsf{V}}_{-,\Delta,\ell} \geq 0$.

Then no unitary CFT with that spectrum and those gaps realizes the candidate spectrum. To see this, apply $\vec{\alpha}$ to (64.69) and separate the identity contribution. Every even nonidentity term is a quadratic form of a positive semidefinite matrix on the vector $(\lambda_{\sigma\sigma\mathcal{O}}, \lambda_{\varepsilon\varepsilon\mathcal{O}})$. Every odd term is a nonnegative number multiplying $\lambda^2_{\sigma\varepsilon\mathcal{O}}$. Thus the whole left-hand side is strictly positive, contradicting the crossing equation. The displayed contradiction is elementary once such a functional is known. The nontrivial mathematical work in a rigorous bootstrap exclusion is the construction and verification of $\vec{\alpha}$: one must specify the allowed spectral set, control the conformal blocks and their derivatives on that set, and prove the matrix and scalar positivity conditions without relying on uncontrolled floating-point output.

The three-dimensional Ising numerical bootstrap studies the case $D = 3$ with σ the lowest $\mathbb{Z}_2$-odd scalar and ε the lowest nonidentity $\mathbb{Z}_2$-even scalar. The hypotheses that create the small Ising island are spectral hypotheses, not consequences of the five crossing equations alone: one assumes, for example, that no other $\mathbb{Z}_2$-odd or nonidentity even scalar is relevant,

$$\Delta_{\sigma'} \geq 3, \qquad \Delta_{\varepsilon'} \geq 3, \tag{64.71}$$

together with the $D = 3$ unitarity bounds

$$\Delta_{\text{scalar}} \geq \frac{1}{2}, \qquad \Delta_\ell \geq \ell + 1 \quad (\ell \geq 1).$$

Derivative functionals at $z = \bar{z} = 1/2$ then turn the finite-functional criterion above into a finite semidefinite search after one chooses a block approximation and proves or verifies its positivity on the relevant ranges of Δ. Floating-point SDP output is numerical evidence until it is accompanied by a rigorous proof object, for example interval bounds for the block derivatives together with positive-polynomial or sum-of-squares witnesses. This is the precise status boundary between the mathematical CFT crossing system (64.69) and the high-precision numerical islands quoted in the Wilson–Fisher comparison table.

Open Problem 64.15 (Interval-verified mixed-correlator Ising island). Give a fully verified exclusion proof, with interval-controlled conformal block derivatives and exact semidefinite proof objects, that isolates the three-dimensional Ising point inside a specified rectangle in $(\Delta_\sigma, \Delta_\varepsilon)$ under stated gap assumptions such as (64.71). The monograph uses the numerical island as external high-precision data until such a verified proof object is supplied.

Chapter 65

Light-Ray Operators and Energy Correlators

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Energy correlators provide a common observable language for conformal field theory and high-energy gauge theory. The primitive object is a detector at future null infinity, represented in local QFT by an asymptotic flux of the stress tensor. The same object can also be described, after conformal compactification, as a null-integrated light-ray operator.

65.1 Detector States and Energy Flux

Let the Lorentzian CFT live on $\mathbb{M}^D$, $D \geq 3$, with Hilbert space $\mathcal{H}$, vacuum vector Ω, translation generators P^μ, and a conserved symmetric stress tensor $T^{\mu\nu}$. A normalizable collider state is obtained by smearing a local operator with timelike momentum support. For a local operator $\mathcal{O}_A$, where A denotes all tensor and internal indices, choose a smooth compactly supported wavepacket $h(q)$ on the open future cone and a polarization tensor ε^A, and set

$$|\Psi_{\mathcal{O},h,\varepsilon}\rangle = \int_{V_+} \mathrm{d}\mu(q)\, h(q)\, \varepsilon^A\, \widetilde{\mathcal{O}}_A(q)|\Omega\rangle.$$

Here $\mathrm{d}\mu(q)$ is a positive measure on the chosen momentum slice. The Fourier-transformed operator is an operator-valued distribution applied to the test function h. Sharp momentum states used in collider formulas are distributional limits of these wavepackets.

Hypothesis 65.1 (CFT energy detector)**.** For every real $f \in C^\infty(S^{D-2})$, the limit

$$\mathcal{E}(f) = \mathrm{w}-\lim_{r\to\infty} r^{D-2} \int_{-\infty}^{\infty} \mathrm{d}u \int_{S^{D-2}} \mathrm{d}\Omega(\mathbf{n})\, f(\mathbf{n})\, n_i T^{0i}(u+r, r\mathbf{n}) \tag{65.1}$$

exists as a quadratic form on the collider domain generated by such states. If $f \geq 0$, the resulting quadratic form is positive. The constant function gives the total outgoing energy,

$$\mathcal{E}(1) = P^0$$

on the same domain.

The distributional detector density is denoted by

$$\mathcal{E}(f) = \int_{S^{D-2}} \mathrm{d}\Omega(\mathbf{n})\, f(\mathbf{n})\mathcal{E}(\mathbf{n}).$$

For k test functions $f_1, \ldots, f_k$, the k-point energy correlator in a normalized state Ψ is

$$\mathcal{G}_k(f_1, \ldots, f_k; \Psi) = \langle \Psi | \mathcal{E}(f_1) \cdots \mathcal{E}(f_k) | \Psi \rangle. \tag{65.2}$$

The product is defined first for separated angular supports and then extended, when possible, as a distribution on $(S^{D-2})^k$. Coincident detector directions carry contact terms determined by the distributional product of light-ray operators.

65.1.1 Finite Angular Partitions and Positivity

Before multiplying detector operators, it is useful to extract the one-detector positivity consequence of Hypothesis 65.1. Let Ψ be a finite-energy collider state in the detector domain and define the state functional

$$\ell_\Psi(f) = \langle \Psi | \mathcal{E}(f) | \Psi \rangle, \qquad f \in C^\infty(S^{D-2}), \quad f \text{ real.}$$

If $f \geq 0$, then $\ell_\Psi(f) \geq 0$. Moreover $-\|f\|_\infty \leq f \leq \|f\|_\infty$ implies, by positivity and $\mathcal{E}(1) = P^0$,

$$|\ell_\Psi(f)| \leq \|f\|_\infty \, \langle \Psi | P^0 | \Psi \rangle. \tag{65.3}$$

Thus ℓ_Ψ extends uniquely from smooth tests to a positive bounded linear functional on $C(S^{D-2})$. By the Riesz representation theorem there is a finite positive regular Borel measure μ_Ψ on the detector sphere such that

$$\langle \Psi | \mathcal{E}(f) | \Psi \rangle = \int_{S^{D-2}} f(\mathbf{n})\, \mathrm{d}\mu_\Psi(\mathbf{n}), \qquad \mu_\Psi(S^{D-2}) = \langle \Psi | P^0 | \Psi \rangle. \tag{65.4}$$

This is a statewise measure statement. Promoting it to an operator-valued measure requires compatible polarized matrix elements and a common domain; those domain questions are additional parts of the detector-existence hypothesis beyond the scalar Riesz representation step.

For a finite angular partition $\mathcal{P} = \{B_1, \ldots, B_m\}$ of the sphere by Borel sets, the numbers

$$\mu_a(\Psi) = \mu_\Psi(B_a) \geq 0, \qquad \sum_{a=1}^{m} \mu_a(\Psi) = \langle \Psi | P^0 | \Psi \rangle$$

are the finite-resolution calorimeter readout in the state Ψ. If $v, w \in \mathbb{C}^m$ are bin weights, the statewise one-detector pairing

$$(v, w)_\Psi = \sum_{a=1}^{m} \mu_a(\Psi)\, \overline{v_a} w_a$$

is positive semidefinite and obeys

$$|(v, w)_\Psi|^2 \le (v, v)_\Psi\, (w, w)_\Psi. \tag{65.5}$$

These inequalities are consequences of positivity of a single detector measure. Positivity statements for products $\mathcal{E}(f_1) \cdots \mathcal{E}(f_k)$ require the diagonal-contact extension discussed next.

Finite-resolution Lipschitz estimates. The finite partition readout gives the operational topology in which an ideal detector is approached by finite angular bins. Let d be the round metric on S^{D-2}. For a finite Borel partition

$$\mathcal{P} = \{B_1, \ldots, B_m\}, \qquad \operatorname{diam}(\mathcal{P}) = \max_a \sup_{\mathbf{n}, \mathbf{n}' \in B_a} d(\mathbf{n}, \mathbf{n}'),$$

choose a representative point $\mathbf{n}_a \in B_a$ for each nonempty bin and replace the statewise detector measure by the binned measure

$$\mu_{\Psi,\mathcal{P}} = \sum_a \mu_\Psi(B_a)\, \delta_{\mathbf{n}_a}.$$

If $f : S^{D-2} \to \mathbb{R}$ is L_f-Lipschitz, then

$$\left| \int f \, \mathrm{d}\mu_\Psi - \int f \, \mathrm{d}\mu_{\Psi,\mathcal{P}} \right| \le L_f \, \operatorname{diam}(\mathcal{P})\, \mu_\Psi(S^{D-2}). \tag{65.6}$$

Indeed, on each bin,

$$|f(\mathbf{n}) - f(\mathbf{n}_a)| \le L_f \, \operatorname{diam}(\mathcal{P}),$$

and summing the resulting measure estimate over bins gives (65.6). Thus convergence of finite bin masses along partitions with $\operatorname{diam}(\mathcal{P}) \to 0$, together with the uniform total energy bound $\mu_\Psi(S^{D-2}) = \langle P^0 \rangle_\Psi$, implies weak convergence of the statewise detector measures against continuous tests by uniform approximation.

The same estimate controls finite-resolution k-detector products after a positive joint detector measure has been constructed. Let $\mu_\Psi^{(k)}$ be such a measure on $(S^{D-2})^k$, let $\mu_{\Psi,\mathcal{P}}^{(k)}$ be the pushforward to the representative points, and let F be L_F-Lipschitz with respect to the product metric

$$d_k((\mathbf{n}_1, \ldots, \mathbf{n}_k), (\mathbf{n}'_1, \ldots, \mathbf{n}'_k)) = \sum_{r=1}^{k} d(\mathbf{n}_r, \mathbf{n}'_r).$$

Then

$$\left| \int F \, \mathrm{d}\mu_\Psi^{(k)} - \int F \, \mathrm{d}\mu_{\Psi,\mathcal{P}}^{(k)} \right| \le k \, L_F \, \operatorname{diam}(\mathcal{P})\, \mu_\Psi^{(k)}((S^{D-2})^k). \tag{65.7}$$

For the product measure of a finite-energy state, the total mass is $\langle (P^0)^k \rangle_\Psi$ when the diagonal-contact extension has been included. Equations (65.6) and (65.7) are therefore the finite-resolution Lipschitz estimates behind the statement that energy correlators are detector observables on the celestial sphere. The construction of detector products remains the separate domain question; once a product measure is available, these estimates identify the finite calorimeter data it supplies.

65.1.2 Angular Moment Data and Measure Reconstruction

The statewise measure μ_Ψ turns angular detector data into a compact moment problem. Let $K = S^{D-2}$, and let $\mathcal{A}_{\mathrm{ang}} \subset C(K;\mathbb{R})$ be a unital real subalgebra that separates points and is uniformly dense in $C(K;\mathbb{R})$. A concrete choice is the algebra generated by the restrictions of the ambient coordinate functions n^i to S^{D-2}; equivalently, one may use any real spherical-harmonic basis closed under finite multiplication. The angular moments of a finite detector measure are the numbers

$$m_\Psi(a) = \int_K a(\mathbf{n})\,\mathrm{d}\mu_\Psi(\mathbf{n}), \qquad a \in \mathcal{A}_{\mathrm{ang}}. \tag{65.8}$$

For $a = 1$ this moment is the total energy $\langle\Psi|P^0|\Psi\rangle$.

Compact angular moment reconstruction. Let $M : \mathcal{A}_{\mathrm{ang}} \to \mathbb{R}$ be linear and assume

$$M(a) \geq 0 \quad \text{whenever} \quad a(\mathbf{n}) \geq 0 \text{ for all } \mathbf{n} \in K, \qquad M(1) = E < \infty.$$

Then there is a unique finite positive regular Borel measure μ_M on K satisfying

$$M(a) = \int_K a\,\mathrm{d}\mu_M, \qquad a \in \mathcal{A}_{\mathrm{ang}}.$$

Consequently the complete angular moment functional $a \mapsto m_\Psi(a)$ determines the statewise detector measure μ_Ψ uniquely.

Argument. For real $a \in \mathcal{A}_{\mathrm{ang}}$, the inequalities $-\|a\|_\infty \leq a \leq \|a\|_\infty$ on K imply

$$0 \leq M(\|a\|_\infty 1 \pm a) = \|a\|_\infty E \pm M(a).$$

Hence $|M(a)| \leq E\|a\|_\infty$. Thus M is uniformly bounded on the dense subalgebra $\mathcal{A}_{\mathrm{ang}}$ and extends uniquely by continuity to a bounded positive linear functional $\overline{M}$ on $C(K;\mathbb{R})$. Positivity of the extension follows by approximating any nonnegative $f \in C(K;\mathbb{R})$ uniformly by $a_j \in \mathcal{A}_{\mathrm{ang}}$ and applying the bound to $a_j - f$ together with the positivity assumption to the functions $a_j + \epsilon_j 1$, where $\epsilon_j = \|a_j - f\|_\infty$.

The Riesz–Markov representation theorem gives a finite positive regular Borel measure μ_M with $\overline{M}(f) = \int_K f\,\mathrm{d}\mu_M$ for all continuous real f. The restriction to $\mathcal{A}_{\mathrm{ang}}$ gives the displayed moment identities. If two finite positive measures have the same moments on $\mathcal{A}_{\mathrm{ang}}$, their integrals agree on a uniformly dense subset of $C(K)$. The bound by total mass extends this equality to all continuous functions, and the uniqueness clause in the Riesz–Markov theorem makes the two measures equal.

This criterion records the information content of complete angular detector moments. A finite list of moments is a finite-resolution projection of μ_Ψ. It gives positivity constraints and linear coordinates for a specified detector resolution, while many distinct positive measures may share the same truncated list. For example on any detector great circle, the measures $\frac{1}{2}\delta_{\mathbf{n}} + \frac{1}{2}\delta_{-\mathbf{n}}$ and $\delta_{\mathbf{n}_\perp}$, with $\mathbf{n}_\perp \cdot \mathbf{n} = 0$, have the same total energy and the same dipole moment but different quadrupole moments. The complete moment functional separates them because $\mathcal{A}_{\mathrm{ang}}$ separates points.

The same compact moment logic applies to detector products after the product has been extended to a finite positive joint detector measure $\mu_\Psi^{(k)}$ on K^k. Then moments of $\mu_\Psi^{(k)}$ against the algebraic tensor products of $\mathcal{A}_{\mathrm{ang}}$ determine that k-detector measure uniquely. The substantive QFT problem is the construction of the product measure and its diagonal-contact coordinates; once that finite measure is supplied, compact moment reconstruction is ordinary functional analysis. The QCD calorimetric-coordinate density theorem, Theorem 50.17, is the corresponding statement on the compact ball of finite positive energy measures.

65.1.3 Detector Products and Diagonal Contacts

The contact terms in detector products have a concrete measure-theoretic origin. A finite-resolution calorimetric configuration X with detector directions $\mathbf{n}_a \in S^{D-2}$ and positive energies $E_a > 0$ determines the positive finite measure

$$\varepsilon_X = \sum_{a=1}^{N} E_a\, \delta_{\mathbf{n}_a}. \tag{65.9}$$

The corresponding detector functional is

$$E_X(f) = \int f\, \mathrm{d}\varepsilon_X = \sum_a E_a f(\mathbf{n}_a).$$

For two detector tests f, g, the product decomposes exactly as

$$\begin{aligned} E_X(f)E_X(g) &= \sum_{a,b} E_a E_b f(\mathbf{n}_a) g(\mathbf{n}_b) \\ &= \sum_{a\neq b} E_a E_b f(\mathbf{n}_a) g(\mathbf{n}_b) + \sum_a E_a^2 f(\mathbf{n}_a) g(\mathbf{n}_a). \end{aligned} \tag{65.10}$$

The first term is the restriction of the product measure $\varepsilon_X \otimes \varepsilon_X$ to the complement of the diagonal in $S^{D-2} \times S^{D-2}$. The second term is supported on the diagonal $\mathbf{n}_1 = \mathbf{n}_2$. Thus a separated-angle two-detector distribution and a full two-detector distribution differ by a specified diagonal distribution; deleting or retaining that diagonal part is a choice of observable coordinate, not a change of the underlying detector.

For k detector insertions, the same bookkeeping is indexed by partitions. Let P_k be the set of set partitions of $\{1, \ldots, k\}$. If $\pi \in \mathsf{P}_k$ and $B \in \pi$ is a block, write a_B for the detector atom assigned to that block. Grouping equal labels in the product $\prod_i \sum_{a_i} E_{a_i} f_i(\mathbf{n}_{a_i})$ gives

$$\prod_{i=1}^{k} E_X(f_i) = \sum_{\pi\in\mathsf{P}_k} \sum_{\substack{a_B,\ B\in\pi \\ a_B\neq a_{B'} \text{ for } B\neq B'}} \prod_{B\in\pi} E_{a_B}^{|B|} \prod_{i\in B} f_i(\mathbf{n}_{a_B}). \tag{65.11}$$

The discrete partition into singletons is the fully separated detector support. Every coarser partition is a diagonal contact stratum in $(S^{D-2})^k$. Formula (65.11) is the finite positive measure model for the contact-term structure of (65.2).

In the operator language of the CFT, Hypothesis 65.1 supplies the smeared detector as a quadratic form. Products with mutually disjoint angular supports are defined by the off-diagonal part of the operator-valued measure. Extending the product to overlapping supports requires a distributional extension across the diagonals, and the extension must preserve the stress-tensor Ward identity

$$\mathcal{G}_2(1, 1; \Psi) = \langle \Psi | (P^0)^2 | \Psi \rangle. \tag{65.12}$$

For a state with sharp total energy Q, the normalized two-detector correlator therefore has unit zeroth moment when the diagonal contact coordinate is included. A convention that studies only separated detector directions must display the removed diagonal contribution separately. This is the CFT counterpart of the endpoint-contact bookkeeping for the QCD energy-energy correlator in Section 50.6.

65.2 Relation to Null-Integrated Operators

Let $n^\mu = (1, \mathbf{n})$ be a future null vector. For a point $x_\perp$ in a spacelike plane transverse to n^μ, the averaged null energy operator on the null line through $x_\perp$ is the quadratic form

$$\mathcal{A}_n(x_\perp) = \int_{-\infty}^{\infty} \mathrm{d}\lambda\, T_{\mu\nu}(x_\perp + \lambda n) n^\mu n^\nu. \tag{65.13}$$

The conformal map from a null line to future null infinity identifies $\mathcal{E}(\mathbf{n})$, up to the fixed conformal Jacobian determined by the chosen compactification, with the light-ray transform (65.13) of the stress tensor. Thus the detector operator is a nonlocal operator obtained by integrating a local field along a complete null generator.

This formulation separates three pieces of data:

$$\begin{gathered}\text{local stress tensor} \longrightarrow \text{null-integrated light-ray operator}\\ \longrightarrow \text{calorimetric distribution on } S^{D-2}.\end{gathered}$$

The first belongs to the local operator algebra. The second requires control of a null integral of an operator-valued distribution. The third is a detector observable in states with finite total energy.

For $f \geq 0$, positivity of $\mathcal{E}(f)$ is the detector version of the averaged null energy theorem stated below. Hypothesis 65.1 packages two analytic inputs: the existence of the null-infinity detector as a quadratic form and the positivity supplied by the theorem for nonnegative angular smearings.

65.3 The Light Transform of a Local Primary

The stress-tensor detector is one member of a broader class of light-ray operators. We spell out the conformal bookkeeping because the same representation-theoretic data are used later in the light-ray OPE, and because it prevents a common ambiguity: a light-ray operator is not an ordinary local operator placed at a point of spacetime. It is an integral transform of a local operator-valued distribution, with its own domain and growth hypotheses.

Use the standard embedding-space notation for a symmetric traceless primary $\mathcal{O}_{\Delta,J}$. A physical point is represented by a null ray $P \in \mathbb{R}^{D,2}$, and tensor indices are packaged by a null auxiliary vector Z satisfying

$$P^2 = Z^2 = P \cdot Z = 0, \qquad Z \sim Z + \beta P.$$

The encoded primary is homogeneous as

$$\mathcal{O}_{\Delta,J}(\lambda P, \alpha Z) = \lambda^{-\Delta} \alpha^J\, \mathcal{O}_{\Delta,J}(P, Z), \qquad \lambda, \alpha > 0. \tag{65.14}$$

This is a compact way to say that Δ is the scaling dimension and J is the spin. Gauge invariance under $Z \to Z + \beta P$ encodes tracelessness and transversality.

For a primary satisfying the Lorentzian growth hypotheses needed to make the integral a distribution, its light transform is

$$\mathbb{L}[\mathcal{O}_{\Delta,J}](P,Z) = \int_{-\infty}^{\infty} \mathrm{d}\alpha\, \mathcal{O}_{\Delta,J}(Z - \alpha P, -P). \tag{65.15}$$

The point $Z - \alpha P$ runs along a null generator, while the new auxiliary polarization is $-P$. The formula is first a pairing with test functions in the transverse and ray variables; convergence or analytic continuation is part of the light-transform datum, not an automatic consequence of local operator covariance.

The homogeneity of $\mathbb{L}[\mathcal{O}_{\Delta,J}]$ follows directly from (65.14). Scaling P and changing variables $\alpha' = \lambda\alpha$ gives

$$\mathbb{L}[\mathcal{O}](\lambda P, Z) = \int \frac{\mathrm{d}\alpha'}{\lambda}\, \mathcal{O}(Z - \alpha' P, -\lambda P) = \lambda^{J-1}\mathbb{L}[\mathcal{O}](P,Z). \tag{65.16}$$

Scaling Z and changing variables $\alpha = \beta\alpha'$ gives

$$\mathbb{L}[\mathcal{O}](P, \beta Z) = \int \beta\, \mathrm{d}\alpha'\, \mathcal{O}(\beta(Z - \alpha' P), -P) = \beta^{1-\Delta}\mathbb{L}[\mathcal{O}](P,Z). \tag{65.17}$$

Thus the light transform has conformal weights

$$(\Delta, J) \longmapsto (1 - J, 1 - \Delta), \tag{65.18}$$

where the second entry is understood as the analytically continued spin coordinate of the nonlocal light-ray representation. Equations (65.16)–(65.18) are representation-theoretic bookkeeping, not an existence theorem. The existence theorem requires the Lorentzian analyticity and growth assumptions appearing in Quoted Theorem 65.5. The signs in (65.18) are tied to the conformal charge convention of Lemma 57.9, not to an independent convention for detector operators. In embedding space the conformal generators act as linear first-order vector fields on the pair (P, Z). Acting on $\mathcal{O}_{\Delta,J}(Z - \alpha P, -P)$ therefore gives the same generator acting on the transformed line, plus a total derivative in the line coordinate α. Under the growth hypotheses that make the light transform a distribution, that total derivative has no endpoint contribution after pairing with line test functions. The transform consequently intertwines the complexified conformal algebra, while its projective weights change only by the two Jacobian factors displayed above: P-rescaling contributes the $\mathrm{d}\alpha/\lambda$ factor and Z-rescaling contributes the $\beta\,\mathrm{d}\alpha'$ factor. The radial real form used for descendant Gram matrices is a separate convention fixed in (59.27)–(59.28); it is obtained by the displayed change of real form before Gram matrices are computed.

For the stress tensor, $(\Delta, J) = (D, 2)$. The light transform therefore has formal embedding weights $(-1, 1 - D)$. After conformal compactification and restriction to future null infinity, this object is represented by the positive energy detector density $\mathcal{E}(\mathbf{n})$, a measure-valued operator on S^{D-2} rather than a local primary on Minkowski space. This is the reason the detector normalization is fixed by $\mathcal{E}(1) = P^0$ and by the ANEC theorem, not by treating $\mathcal{E}(\mathbf{n})$ as an ordinary local operator.

65.4 ANEC and Conformal Collider Bounds

The averaged null energy condition used in conformal collider physics is a quadratic-form statement about a null light transform of the stress tensor. It is stronger than positivity of the

Hamiltonian, because it localizes the energy measurement on one complete null generator rather than integrating over a spacelike Cauchy surface. The statement requires a domain on which the null integral of the stress-tensor distribution exists.

Let $N_n = n^{\perp}/\mathbb{R}n$ be the $(D-2)$-dimensional transverse quotient to the null direction n^{μ}. For a nonnegative test function $\varphi \in C_c^{\infty}(N_n)$, define the smeared null-energy quadratic form

$$\mathcal{A}_n(\varphi) = \int_{N_n} \mathrm{d}^{D-2}x_{\perp}\, \varphi(x_{\perp}) \int_{-\infty}^{\infty} \mathrm{d}\lambda\, T_{\mu\nu}(x_{\perp} + \lambda n) n^{\mu} n^{\nu}. \tag{65.19}$$

The transverse smearing is part of the definition. The unsmeared $\mathcal{A}_n(x_{\perp})$ in (65.13) is the corresponding distributional kernel.

Quoted Theorem 65.2 (Averaged null energy theorem for Lorentzian CFT states)**.** *Consider a unitary Lorentzian CFT on $\mathbb{M}^D$, $D \geq 3$, with a unique translation-invariant vacuum Ω, spectrum condition, local stress tensor $T_{\mu\nu}$ as a Wightman distribution, and the Lorentzian analyticity and operator-growth bounds needed for the null light transform of $T_{\mu\nu}$. Let $\mathcal{D}_{\mathrm{ANEC}} \subset \mathcal{H}$ be a dense finite-energy domain that contains the wavepacket states generated by local operators and is stable under the smearings used in (65.19). Then, for every future null n^{μ}, every nonnegative $\varphi \in C_c^{\infty}(N_n)$, and every $\Psi \in \mathcal{D}_{\mathrm{ANEC}}$,*

$$\langle \Psi | \mathcal{A}_n(\varphi) | \Psi \rangle \geq 0. \tag{65.20}$$

The theorem is used here as an external theorem boundary with its hypotheses visible.[1] The two proof mechanisms have the same local output: they identify $\mathcal{A}_n(\varphi)$ as the operator whose expectation measures the monotone or causal response of a state to a null displacement.

We spell out the modular route because it fixes the sign conventions. Choose light-cone coordinates

$$x^{\pm} = x^0 \pm x^1, \qquad y = (x^2, \ldots, x^{D-1}), \qquad \mathrm{d}s^2 = -\mathrm{d}x^+ \mathrm{d}x^- + \mathrm{d}y^2,$$

and take $n = \partial_+$. For a smooth cut $x^+ = f(y)$ of the null plane $x^- = 0$, let R_f be the domain of dependence of the portion $x^- = 0$, $x^+ \geq f(y)$. Denote by $K_f = -\log \rho^{\Omega}_{R_f}$ the vacuum modular Hamiltonian on R_f, with the additive scalar fixed by $\mathrm{Tr}_{R_f} \rho^{\Omega}_{R_f} = 1$. The null-cut modular Hamiltonian theorem gives, as an equality of quadratic forms on finite-energy states after transverse smearing,

$$K_f = 2\pi \int \mathrm{d}^{D-2}y \int_{f(y)}^{\infty} \mathrm{d}x^+ \, \big(x^+ - f(y)\big) T_{++}(x^+, x^- = 0, y). \tag{65.21}$$

This formula is a local statement on the null surface. For $f_s(y) = s\varphi(y)$, differentiation at $s = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}s} \Delta_{\Psi} \langle K_{f_s} \rangle \bigg|_{s=0} = -2\pi \int \mathrm{d}^{D-2}y\, \varphi(y) \int_0^{\infty} \mathrm{d}x^+ \, \Delta_{\Psi} \langle T_{++}(x^+, 0, y) \rangle, \tag{65.22}$$

where

$$\Delta_{\Psi}\langle X \rangle = \langle \Psi | X | \Psi \rangle - \langle \Omega | X | \Omega \rangle.$$

[1]For the modular-Hamiltonian proof, see T. Faulkner, R. G. Leigh, O. Parrikar, and H. Wang, "Modular Hamiltonians for Deformed Half-Spaces and the Averaged Null Energy Condition," JHEP **09** (2016) 038, arXiv:1605.08072. For the Lorentzian causality proof in CFT, see T. Hartman, S. Kundu, and A. Tajdini, "Averaged null energy condition from causality," JHEP **07** (2017) 066, arXiv:1610.05308.

The sign in the last step is a small but important piece of modular bookkeeping. Write

$$S_{\rm rel}(\rho_R^\Psi\|\rho_R^\Omega)=\Delta_\Psi\langle K_R\rangle-\Delta_\Psi S_R,\qquad R_1\subset R_2\Rightarrow S_{\rm rel}(R_1)\le S_{\rm rel}(R_2),$$

for Araki relative entropy. If $s>0$ and $\varphi\ge 0$, the region $R_{s\varphi}$ is contained in R_0. Hence the right derivative at the origin obeys

$$\frac{\mathrm{d}}{\mathrm{d}s}S_{\rm rel}(R_{s\varphi})\Big|_{0^+}\le 0. \tag{65.23}$$

Using (65.22), this gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\Delta_\Psi S_{R_{s\varphi}}\Big|_{0^+}\ge -2\pi\int \mathrm{d}^{D-2}y\,\varphi(y)\int_0^\infty \mathrm{d}x^+\,\Delta_\Psi\langle T_{++}(x^+,0,y)\rangle. \tag{65.24}$$

For the complementary null cut $R^c_{s\varphi}$, the inclusion is reversed: $R^c_0\subset R^c_{s\varphi}$. Its modular Hamiltonian has the same local null-cut form on the left part of the generator,

$$K^c_{s\varphi}=2\pi\int \mathrm{d}^{D-2}y\int_{-\infty}^{s\varphi(y)}\mathrm{d}x^+\,\big(s\varphi(y)-x^+\big)T_{++}(x^+,0,y),$$

so

$$\frac{\mathrm{d}}{\mathrm{d}s}\,\Delta_\Psi\langle K^c_{s\varphi}\rangle\Big|_{0^+}=2\pi\int \mathrm{d}^{D-2}y\,\varphi(y)\int_{-\infty}^0 \mathrm{d}x^+\,\Delta_\Psi\langle T_{++}(x^+,0,y)\rangle. \tag{65.25}$$

Monotonicity for the complementary inclusion gives a nonnegative right derivative of $S_{\rm rel}(R^c_{s\varphi})$. For a vector state on the global Hilbert space, the entropy of a region and of its complement agree in a split-regulated exhaustion and hence have the same first variation at the common cut. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}s}\Delta_\Psi S_{R_{s\varphi}}\Big|_{0^+}\le 2\pi\int \mathrm{d}^{D-2}y\,\varphi(y)\int_{-\infty}^0 \mathrm{d}x^+\,\Delta_\Psi\langle T_{++}(x^+,0,y)\rangle. \tag{65.26}$$

The same entropy derivative is squeezed between the right-hand sides of (65.24) and (65.26). Compatibility of the two inequalities is exactly

$$2\pi\int \mathrm{d}^{D-2}y\,\varphi(y)\int_{-\infty}^\infty \mathrm{d}x^+\,\Delta_\Psi\langle T_{++}(x^+,0,y)\rangle\ge 0. \tag{65.27}$$

The vacuum term vanishes by Poincare invariance and the stress-tensor normalization, so this is (65.20) in these coordinates. The distributional statement follows by testing this expression against $\varphi\in C_c^\infty(N_n)$; the detector statement follows after the conformal map from the complete null generator to future null infinity.

ANEC-to-detector bridge. The last sentence contains a real limiting operation. Quoted Theorem 65.2 proves positivity for smooth compact transverse tests on a null plane. The angular detector $\mathcal{E}(f)$, by contrast, is a null-infinity object tested by a smooth function $f\in C^\infty(S^{D-2})$, and finite calorimeter bins are obtained by approximating indicator functions on the compact detector sphere. To pass from ANEC to the statewise detector measure μ_Ψ of (65.4), one needs the following additional bridge data:

$$f_j\ge 0,\qquad f_j\to f\text{ in }C(S^{D-2}),\qquad \langle\Psi|\mathcal{E}(f_j)|\Psi\rangle\longrightarrow\langle\Psi|\mathcal{E}(f)|\Psi\rangle,$$

with a bound controlled by the total energy of the state, so that the positive quadratic form on smooth angular tests extends to a bounded positive functional on $C(S^{D-2})$. Riesz–Markov then gives the finite positive measure μ_Ψ. Without this bounded-extension step, ANEC positivity for each smooth transverse test is not yet a statement about Borel detector bins, nor about products of detectors at separated or coincident angles. Detector products require their own domain and contact-extension data, as in the partition discussion following (65.11). This is why the chapter keeps the one-detector positivity theorem, the statewise measure construction, and the k-detector contact algebra as distinct pieces of the detector framework.

The causality route packages the same sign in Lorentzian analyticity. Let $\mathcal{O}$ be a scalar primary used only as a probe, and let $\mathcal{X}$ be a local operator whose wavepacket creates the finite-energy state $\Psi_{\mathcal{X}}$. In the Regge light-cone regime of the normalized four-point function

$$G = \frac{\langle\Omega|\mathcal{O}(x_1)\,\mathcal{X}^\dagger(x_3)\mathcal{X}(x_4)\,\mathcal{O}(x_2)|\Omega\rangle}{\langle\Omega|\mathcal{O}(x_1)\mathcal{O}(x_2)|\Omega\rangle\langle\Omega|\mathcal{X}^\dagger(x_3)\mathcal{X}(x_4)|\Omega\rangle},$$

the stress-tensor contribution to the first nontrivial discontinuity is proportional to the ANEC matrix element

$$\langle\Psi_{\mathcal{X}}|\mathcal{A}_n(\varphi)|\Psi_{\mathcal{X}}\rangle, \qquad \varphi \geq 0,$$

where φ is determined by the probe wavepacket. Reflection positivity gives a positive norm on the Euclidean ordering, while Lorentzian causality gives analyticity and a bounded phase in the continuation around the light cone. The maximum-modulus step in that analytic domain fixes the sign of the stress-tensor discontinuity, hence gives (65.20). The derivation is conditional on the same analyticity and growth assumptions stated in Quoted Theorem 65.2.

For conformal collider physics, the consequence is immediate. If a detector state Ψ has total energy $Q = \langle\Psi|P^0|\Psi\rangle/\langle\Psi \mid \Psi\rangle$ and $\mathcal{E}(\mathbf{n})$ is the null-infinity representative of the light transform, then

$$\frac{\langle\Psi|\mathcal{E}(f)|\Psi\rangle}{\langle\Psi \mid \Psi\rangle} \geq 0 \qquad (f \geq 0), \qquad \int_{S^{D-2}} \mathrm{d}\Omega(\mathbf{n})\,\frac{\langle\Psi|\mathcal{E}(\mathbf{n})|\Psi\rangle}{\langle\Psi \mid \Psi\rangle} = Q.$$

Thus the angular energy profile in any fixed family of collider states is a positive distribution on the celestial sphere with fixed total mass Q. The Hofman–Maldacena inequalities are the finite-dimensional specialization of this positivity to stress-tensor states in four-dimensional CFTs.

65.5 Conformal Collider One-Point Functions

Consider a four-dimensional parity-even CFT and a stress-tensor state in the rest frame $q^\mu = (Q, \mathbf{0})$. Let ε_{ij} be a complex symmetric traceless spatial polarization, $i, j = 1, 2, 3$, and write

$$|\Psi_{\varepsilon,q}\rangle = \varepsilon_{ij}\widetilde{T}^{ij}(q)|\Omega\rangle$$

with wavepacket regularization understood. Rotational covariance around $\mathbf{q} = \mathbf{0}$, stress-tensor conservation, and the Ward identity normalization of $T_{\mu\nu}$ constrain the separated-angle one-detector expectation to a two-parameter tensor structure.

Lemma 65.3 (Rotational tensor basis for a stress-tensor collider state)**.** *Let $E_\varepsilon(\mathbf{n})$ be a real quadratic form in a complex symmetric traceless tensor ε_{ij}, depending on $\mathbf{n} \in S^2$, and covariant under simultaneous spatial rotations of ε and $\mathbf{n}$. Assume that its spherical-harmonic content is obtained from the tensor product of two spin-two polarizations, as happens for the separated-angle* *detector expectation in a stress-tensor state. Then after fixing the total-energy normalization, $E_\varepsilon(\mathbf{n})$ is a linear combination of*

$$1, \qquad I_2(\varepsilon, \mathbf{n}) = \frac{\varepsilon^*_{ij}\varepsilon_{ik}n^j n^k}{\varepsilon^*_{ij}\varepsilon_{ij}}, \qquad I_4(\varepsilon, \mathbf{n}) = \frac{|\varepsilon_{ij}n^i n^j|^2}{\varepsilon^*_{ij}\varepsilon_{ij}}.$$

Their angular averages are

$$\frac{1}{4\pi}\int_{S^2} \mathrm{d}\Omega\, I_2 = \frac{1}{3}, \qquad \frac{1}{4\pi}\int_{S^2} \mathrm{d}\Omega\, I_4 = \frac{2}{15}. \tag{65.28}$$

Proof. The bilinear expression in the polarization transforms in $(2) \otimes (2)$ under $SO(3)$, hence decomposes into angular momenta $\ell = 0, 1, 2, 3, 4$. Parity-even detector one-point functions and the symmetry of the stress tensor keep only the even scalar structures. The $\ell = 0$ structure is the norm $\varepsilon^*_{ij}\varepsilon_{ij}$. Contracting one pair of polarization indices with $n^j n^k$ gives the independent $\ell = 2$ structure I_2. Contracting both pairs with $n^i n^j n^k n^\ell$ gives the $\ell = 4$ structure I_4 together with lower harmonics. Thus $1, I_2, I_4$ span the parity-even rotational tensor basis.

The averages follow from the unique rotationally invariant tensors

$$\frac{1}{4\pi}\int_{S^2} \mathrm{d}\Omega\, n^i n^j = \frac{\delta^{ij}}{3},$$

and

$$\frac{1}{4\pi}\int_{S^2} \mathrm{d}\Omega\, n^i n^j n^k n^\ell = \frac{1}{15}\left(\delta^{ij}\delta^{k\ell} + \delta^{ik}\delta^{j\ell} + \delta^{i\ell}\delta^{jk}\right). \tag{65.29}$$

The first identity gives

$$\langle I_2 \rangle = \frac{\varepsilon^*_{ij}\varepsilon_{ik}\delta^{jk}}{3\,\varepsilon^*_{ab}\varepsilon_{ab}} = \frac{1}{3}.$$

For I_4, the term $\delta^{ij}\delta^{k\ell}$ in (65.29) vanishes because $\varepsilon_{ij}\delta^{ij} = 0$, while the remaining two contractions each give $\varepsilon^*_{ij}\varepsilon_{ij}$. Hence $\langle I_4 \rangle = 2/15$. □

Consequently the separated-angle one-detector expectation has the form

$$\frac{\langle \Psi_{\varepsilon,q}|\mathcal{E}(\mathbf{n})|\Psi_{\varepsilon,q}\rangle}{\langle \Psi_{\varepsilon,q} \mid \Psi_{\varepsilon,q}\rangle} = \frac{Q}{4\pi}\left[1 + t_2\left(\frac{\varepsilon^*_{ij}\varepsilon_{ik}n^j n^k}{\varepsilon^*_{ij}\varepsilon_{ij}} - \frac{1}{3}\right) + t_4\left(\frac{|\varepsilon_{ij}n^i n^j|^2}{\varepsilon^*_{ij}\varepsilon_{ij}} - \frac{2}{15}\right)\right]. \tag{65.30}$$

The real constants t_2, t_4 are coordinates on the parity-even stress-tensor three-point function after C_T and Ward-identity contact terms have been fixed. The subtractions $1/3$ and $2/15$ are precisely the averages in (65.28), so integration over S^2 gives the total energy Q for every polarization.

Lemma 65.4 (Spectral diagonalization of the stress-tensor collider form)**.** *Fix the detector direction $\mathbf{n} = e_3 = (0, 0, 1)$, and let $\mathcal{V}$ be the complex vector space of symmetric traceless tensors ε_{ij}, equipped with $\|\varepsilon\|^2 = \varepsilon^*_{ij}\varepsilon_{ij}$. Decompose $\mathcal{V}$ under rotations around e_3 as*

$$\mathcal{V} = \mathcal{V}_2 \oplus \mathcal{V}_1 \oplus \mathcal{V}_0,$$

where

$$\begin{aligned}
\mathcal{V}_2 &= \{\varepsilon : \varepsilon_{13} = \varepsilon_{23} = \varepsilon_{33} = 0\},\\
\mathcal{V}_1 &= \{\varepsilon : \varepsilon_{11} = \varepsilon_{22} = \varepsilon_{12} = \varepsilon_{33} = 0\},\\
\mathcal{V}_0 &= \mathbb{C}\ \mathrm{diag}(1, 1, -2).
\end{aligned}$$

For the normal form (65.30)*, define the dimensionless quadratic form*

$$B_{t_2,t_4}(\varepsilon) = \|\varepsilon\|^2 \left[1 + t_2\left(I_2(\varepsilon, e_3) - \frac{1}{3}\right) + t_4\left(I_4(\varepsilon, e_3) - \frac{2}{15}\right)\right].$$

If $\varepsilon = \varepsilon_2 + \varepsilon_1 + \varepsilon_0$ is the above orthogonal decomposition, then

$$B_{t_2,t_4}(\varepsilon) = \lambda_2\|\varepsilon_2\|^2 + \lambda_1\|\varepsilon_1\|^2 + \lambda_0\|\varepsilon_0\|^2, \tag{65.31}$$

with

$$\lambda_2 = 1 - \frac{1}{3}t_2 - \frac{2}{15}t_4, \qquad \lambda_1 = 1 + \frac{1}{6}t_2 - \frac{2}{15}t_4, \qquad \lambda_0 = 1 + \frac{1}{3}t_2 + \frac{8}{15}t_4. \tag{65.32}$$

Proof. The three displayed subspaces are pairwise orthogonal for $\varepsilon_{ij}^*\eta_{ij}$. They exhaust $\mathcal{V}$: a symmetric traceless 3×3 tensor is determined by the two transverse traceless entries $(\varepsilon_{11} - \varepsilon_{22}, \varepsilon_{12})$, the two mixed entries $(\varepsilon_{13}, \varepsilon_{23})$, and the scalar trace-free diagonal direction proportional to $\mathrm{diag}(1, 1, -2)$.

For $\varepsilon_2 \in \mathcal{V}_2$, the vector $\varepsilon_2 e_3$ and the scalar $e_3^i\varepsilon_{2,ij}e_3^j$ both vanish, so $I_2 = I_4 = 0$. For $\varepsilon_1 \in \mathcal{V}_1$, writing $\varepsilon_{13} = a$ and $\varepsilon_{23} = b$, one has

$$\|\varepsilon_1\|^2 = 2(|a|^2 + |b|^2), \qquad \|\varepsilon_1 e_3\|^2 = |a|^2 + |b|^2, \qquad e_3^i\varepsilon_{1,ij}e_3^j = 0,$$

so $I_2 = 1/2$ and $I_4 = 0$. For $\varepsilon_0 = c\ \mathrm{diag}(1, 1, -2)$, one has

$$\|\varepsilon_0\|^2 = 6|c|^2, \qquad \|\varepsilon_0 e_3\|^2 = 4|c|^2, \qquad |e_3^i\varepsilon_{0,ij}e_3^j|^2 = 4|c|^2,$$

so $I_2 = I_4 = 2/3$. These three evaluations give the eigenvalues (65.32). The mixed terms vanish because the subspaces are orthogonal and because the numerators $\|\varepsilon e_3\|^2$ and $|e_3^i\varepsilon_{ij}e_3^j|^2$ separate into the same $\mathcal{V}_2, \mathcal{V}_1, \mathcal{V}_0$ components. This proves (65.31). □

Four-dimensional conformal-collider inequalities. The theorem-level content in this finite step is Lemma 65.4: it identifies the three irreducible $SO(2)$ polarization sectors and computes the corresponding eigenvalues of the detector quadratic form. Once that diagonalization is in hand, positivity has no further hidden analytic content. The normal form (65.30) is positive for every stress-tensor polarization and every detector direction $\mathbf{n}$ exactly when

$$1 - \frac{1}{3}t_2 - \frac{2}{15}t_4 \geq 0, \qquad 1 + \frac{1}{6}t_2 - \frac{2}{15}t_4 \geq 0, \qquad 1 + \frac{1}{3}t_2 + \frac{8}{15}t_4 \geq 0.$$

Indeed, by rotational covariance one may put $\mathbf{n} = e_3$; the lemma then writes the form as $\lambda_2\|\varepsilon_2\|^2 + \lambda_1\|\varepsilon_1\|^2 + \lambda_0\|\varepsilon_0\|^2$, and rotation transports the same three eigenvalues to any direction.

The derivation of the normal form (65.30) is a calculation of a Wightman $\langle TTT\rangle$ distribution integrated against the detector kernel. Contact terms in the local $\langle TTT\rangle$ Ward identities fix the normalization of the stress tensor and do not change separated-angle detector values after the state wavepacket and detector smearing have been specified.

65.6 Energy-Energy Correlators in a CFT

For a state of sharp total energy Q, the normalized energy-energy correlator is the distribution

$$\mathrm{EEC}_\Psi(\chi) = \frac{1}{Q^2\langle\Psi \mid \Psi\rangle}\langle\Psi| \int_{S^{D-2}} \mathrm{d}\Omega_1 \mathrm{d}\Omega_2\, \delta(\cos\chi - \mathbf{n}_1\cdot\mathbf{n}_2)\mathcal{E}(\mathbf{n}_1)\mathcal{E}(\mathbf{n}_2)|\Psi\rangle. \tag{65.33}$$

For a wavepacket in Q, replace Q^2 by the corresponding expectation of $(P^0)^2$, or keep the unnormalized numerator and denominator separately. The separated-angle CFT observable (65.33) is the same type of detector distribution as the QCD observable (49.233). The difference is dynamical: in a CFT there is no mass gap and no hadronic scattering basis, while the detector remains defined by the stress tensor and null infinity.

When two detector directions approach one another, the product of light-ray operators has a short-distance expansion on the celestial sphere. The following theorem is used as an external theorem boundary for the modern light-ray formalism.[2]

Quoted Theorem 65.5 (Light-ray OPE under Lorentzian CFT hypotheses)**.** *In a unitary CFT satisfying the Lorentzian analyticity, spectrum, and operator-growth hypotheses required for the light-ray OPE, products of null-integrated operators on the same null plane admit a convergent expansion in light-ray operators. For energy detectors, the separated-angle product has the form, as a distribution tested near* $\mathbf{n}_1 = \mathbf{n}_2$*,*

$$\mathcal{E}(\mathbf{n}_1)\mathcal{E}(\mathbf{n}_2) = \sum_\alpha C_\alpha(\mathbf{n}_1, \mathbf{n}_2)\, \mathbb{L}_\alpha(\mathbf{n}_2),$$

where $\mathbb{L}_\alpha$ *are light-ray transforms of local CFT primaries and their descendants, analytically continued in spin as specified by the Lorentzian inversion formula.*

The word "convergent" in Quoted Theorem 65.5 is a distributional statement, not a pointwise multiplication rule for two unbounded operators. On a small celestial patch write a detector direction as a transverse coordinate $y \in \mathbb{R}^{D-2}$. A separated detector product is first paired with a test function

$$\Phi(y_1, y_2) \in C_c^\infty\big((U \times U) \setminus \{y_1 = y_2\}\big).$$

The diagonal limit is probed by families whose support approaches $|y_{12}| = |y_1 - y_2| = 0$. The theorem supplies coefficient distributions

$$C_{\alpha,m}(y_{12})$$

and light-ray operators $\mathbb{L}_\alpha(y)$, together with a topology of test functions and state wavepackets in which

$$\mathcal{E}(y_1)\mathcal{E}(y_2) \sim \sum_{\alpha,m} C_{\alpha,m}(y_{12})\, \partial_y^m \mathbb{L}_\alpha(y_2) \tag{65.34}$$

after pairing with such tests. The multi-index m labels transverse descendants and tensor structures. The coefficients are homogeneous or polyhomogeneous distributions away from $y_{12} = 0$,

[2] See M. Kologlu, P. Kravchuk, D. Simmons-Duffin, and A. Zhiboedov, "The light-ray OPE and conformal colliders," arXiv:1905.01311. For the conformal-collider formulation, see D. M. Hofman and J. Maldacena, "Conformal collider physics: energy and charge correlations," JHEP 05 (2008) 012, arXiv:0803.1467.

with logarithmic terms when anomalous dimensions or degeneracies are present. Their normalization is fixed by three-point data involving the two detector light transforms and $\mathbb{L}_\alpha$, or equivalently by the Lorentzian inversion transform in the conformal frame used to define the light transform.

The proof mechanism has three distinct analytic steps. First, the product of the local stress tensors is not expanded by the Euclidean local OPE at ordinary spacelike separation. The two null integrations and the detector limit force one to control Lorentzian configurations in which the integration contours approach lightlike singularities. Lorentzian analyticity and the spectrum condition provide domains in complexified configuration space in which the contours can be displaced and the relevant discontinuities can be defined. Second, Regge or operator-growth bounds control the arcs at infinity and justify exchanging the null-line integrals with the partial-wave or inversion expansion. Without these bounds the formal operation "integrate the local OPE along a null line" is not a proof. Third, the resulting expansion is completed in a transverse distribution topology. A usable version states that, after retaining all light-ray primaries in a set $\mathcal{A}_N$, the remainder R_N obeys an estimate of the form

$$\left|\langle\Psi|R_N(\Phi)|\Psi\rangle\right| \le C_N(\Psi)\,\|\Phi\|_{\mathcal{T}_N}\,\epsilon^{\sigma_N} \tag{65.35}$$

for tests supported in $|y_{12}| < \epsilon$, with a declared test-function seminorm $\|\cdot\|_{\mathcal{T}_N}$ and an exponent $\sigma_N > 0$ that improves as the retained operator set is enlarged. Equation (65.35) is not an additional theorem proved here; it is the kind of bound that makes (65.34) a mathematical statement rather than a formal series. Whenever the monograph uses the light-ray OPE below, the use is restricted to consequences that follow from such Lorentzian analyticity, growth, and transverse-distribution convergence hypotheses.

Finite light-ray OPE charts. For calculations and for comparison with detector data, one never manipulates the entire light-ray OPE at once. One chooses a finite chart and states its remainder. Work in a small angular patch with transverse coordinates (y_1, y_2), relative coordinate $\xi = y_1 - y_2$, and center coordinate $y = y_2$. Let $\mathfrak{A}_N$ be a finite set of retained light-ray primary and descendant labels (α, m). A finite CFT light-ray OPE chart for the two-detector product in a state domain $\mathcal{D}_N$ consists of:

1. coefficient distributions $C_{\alpha,m}$ in the relative variable ξ, including their specified extension across $\xi = 0$ whenever contact terms are retained;
2. light-ray quadratic forms $\mathbb{L}_\alpha(\psi)$ on $\mathcal{D}_N$, where ψ is a smooth compactly supported center-coordinate test;
3. for each test $\Phi \in C_c^\infty(U \times U)$, the induced center tests
$$\psi_{\alpha,m,\Phi}(y) = \left\langle C_{\alpha,m}(\xi), \partial_y^m \Phi(y+\xi, y)\right\rangle_\xi; \tag{65.36}$$
4. a seminorm $\|\cdot\|_{\mathcal{T}_N}$, a neighborhood condition on the support of Φ, and a remainder functional $R_N(\Phi;\Psi)$ such that
$$|R_N(\Phi;\Psi)| \le C_N(\Psi)\,\|\Phi\|_{\mathcal{T}_N}\,\epsilon^{\sigma_N}, \qquad \sigma_N > 0, \tag{65.37}$$
for tests supported in $|\xi| < \epsilon$.

The finite chart represents the detector product by

$$\langle\Psi|\mathcal{E}\mathcal{E}(\Phi)|\Psi\rangle = \sum_{(\alpha,m)\in\mathfrak{A}_N} \langle\Psi|\mathbb{L}_\alpha(\psi_{\alpha,m,\Phi})|\Psi\rangle + R_N(\Phi;\Psi). \tag{65.38}$$

Here $\mathcal{E}\mathcal{E}(\Phi)$ denotes the distributional pairing of the two-detector product with Φ, including the declared contact convention.

This finite chart is the CFT analogue of the QCD finite renormalized light-ray mixing chart in Chapter 49, but the coordinates are fixed-point conformal data rather than scale-dependent renormalized QCD operators. The coefficient extension across $\xi = 0$ is part of the chart. A separated-angle chart that only tests Φ supported away from the diagonal does not determine the coincident detector contact coordinate, hence does not determine the full moment $\mathcal{G}_2(1,1;\Psi) = \langle\Psi|(P^0)^2|\Psi\rangle$.

For a fixed state Ψ, suppose the retained light-ray forms and coefficient maps obey

$$|\langle\Psi|\mathbb{L}_\alpha(\psi)|\Psi\rangle| \le L_\alpha(\Psi)\, p_\alpha(\psi), \qquad p_\alpha(\psi_{\alpha,m,\Phi}) \le K_{\alpha,m}\, \|\Phi\|_{\mathcal{T}_N}. \tag{65.39}$$

Then (65.38) gives the finite detector-functional bound

$$|\langle\Psi|\mathcal{E}\mathcal{E}(\Phi)|\Psi\rangle| \le \left(\sum_{(\alpha,m)\in\mathfrak{A}_N} L_\alpha(\Psi)K_{\alpha,m} + C_N(\Psi)\epsilon^{\sigma_N} \right) \|\Phi\|_{\mathcal{T}_N}. \tag{65.40}$$

Thus the finite light-ray chart consists of coefficient distributions, their diagonal extensions, continuity estimates for the light-ray forms, and a remainder bound in the detector-test topology. These data are what turn a truncated representation-theoretic expansion into a controlled approximation to the energy correlator.

Finite chart changes and contact coordinates. The retained light-ray basis and the diagonal extension convention are part of the finite chart. Write the retained contribution to (65.38) in finite-dimensional notation as

$$V_\Psi(\Phi) = c_\Phi\, \ell_\Psi + k_\Phi + R_N(\Phi;\Psi).$$

Here $\ell_\Psi \in \mathbb{C}^r$ is the column vector of retained light-ray matrix elements, $c_\Phi \in (\mathbb{C}^r)^*$ is the row vector obtained by pairing the coefficient distributions with Φ, and k_Φ denotes the explicit diagonal-contact coordinate in the chosen extension. If $B \in GL(r,\mathbb{C})$ changes the retained light-ray basis, the same detector functional is represented by

$$\ell'_\Psi = B^{-1}\ell_\Psi, \qquad c'_\Phi = c_\Phi B, \qquad c'_\Phi\ell'_\Psi = c_\Phi\ell_\Psi. \tag{65.41}$$

This is ordinary coordinate covariance of the retained finite vector space; it is not a dynamical statement about the exact OPE.

Changing the extension of the relative-coordinate coefficient distributions across the diagonal is different. Let $d_\Phi \in (\mathbb{C}^r)^*$ be the row vector produced by adding diagonal distributions to the retained coefficient maps. The full detector distribution is unchanged only if the explicit contact coordinate is shifted simultaneously:

$$c_\Phi^{\mathrm{new}} = c_\Phi + d_\Phi, \qquad k_\Phi^{\mathrm{new}} = k_\Phi - d_\Phi\ell_\Psi. \tag{65.42}$$

Thus a separated-angle calculation, for which $d_\Phi = 0$ on all tests used, does not determine k_Φ. The Ward identity $\mathcal{G}_2(1,1;\Psi) = \langle\Psi|(P^0)^2|\Psi\rangle$ fixes a condition on the sum $c_\Phi\ell_\Psi + k_\Phi$ for the constant detector test; it does not fix a separated-angle coefficient convention by itself. A finite light-ray OPE chart used for a full energy correlator must therefore state both its retained-basis convention and its diagonal-contact convention.

One-parameter finite chart transport. The same bookkeeping can be differentiated. This is useful because it separates a harmless change of finite coordinates from a genuine change of the detector distribution. Let s parametrize a family of retained-basis changes $B_s \in GL(r,\mathbb{C})$ and diagonal-extension shifts $d_{s,\Phi} \in (\mathbb{C}^r)^*$, both stated in the original retained coordinates. For

$$V_\Psi(\Phi) = c_\Phi\ell_\Psi + k_\Phi + R_N(\Phi;\Psi)$$

define

$$\ell_{s,\Psi} = B_s^{-1}\ell_\Psi, \qquad c_{s,\Phi} = (c_\Phi + d_{s,\Phi})B_s, \qquad k_{s,\Phi} = k_\Phi - d_{s,\Phi}\ell_\Psi. \tag{65.43}$$

Then

$$c_{s,\Phi}\ell_{s,\Psi} + k_{s,\Phi} = c_\Phi\ell_\Psi + k_\Phi.$$

Put $A_s = B_s^{-1}\partial_s B_s$. Differentiating (65.43) gives

$$\partial_s\ell_{s,\Psi} = -A_s\ell_{s,\Psi}, \qquad \partial_s c_{s,\Phi} = c_{s,\Phi}A_s + (\partial_s d_{s,\Phi})B_s, \tag{65.44}$$

$$\partial_s k_{s,\Phi} = -(\partial_s d_{s,\Phi})B_s\ell_{s,\Psi}. \tag{65.45}$$

The $c_{s,\Phi}A_s$ and $-A_s\ell_{s,\Psi}$ terms cancel in $\partial_s(c_s\ell_s + k_s)$; the remaining $(\partial_s d_s)B_s\ell_s$ term cancels only against the contact derivative (65.45). With the opposite contact sign the constant detector test would change by twice this endpoint contribution. Equations (65.44)–(65.45) are finite-chart identities at a conformal fixed point, not anomalous dimensions. Their QCD analogue is the renormalized light-ray mixing chart of Chapter 49, where s is replaced by physical renormalization or rapidity scales after the regulator, subtraction scheme, and endpoint contact convention have been fixed.

Small-angle pushforward to the EEC variable. The finite light-ray chart lives first as a distribution in the two detector directions. The one-variable EEC is obtained by pushing that distribution forward by the opening angle. This geometric step contributes a Jacobian that must be included together with the light-ray OPE coefficient. Put

$$d_\perp = D - 2$$

for the dimension of the detector sphere, fix the second detector direction, and write θ for the geodesic distance of the first detector from it. In geodesic polar coordinates on $S^{d_\perp}$,

$$\mathrm{d}\Omega_1 = (\sin\theta)^{d_\perp - 1}\mathrm{d}\theta\,\mathrm{d}\omega, \qquad \omega \in S^{d_\perp - 1}.$$

The EEC kernel in (65.33) is $\delta(\cos\chi - \cos\theta)$, hence for $0 < \chi < \pi$

$$\int_{S^{d_\perp}} \mathrm{d}\Omega_1\,\delta(\cos\chi - \mathbf{n}_1\cdot\mathbf{n}_2)\,F(\theta,\omega) = (\sin\chi)^{d_\perp - 2}\int_{S^{d_\perp - 1}} \mathrm{d}\omega\,F(\chi,\omega). \tag{65.46}$$

Thus a retained coefficient whose relative-coordinate leading term is homogeneous of degree $-\lambda$,

$$C(\xi) = |\xi|^{-\lambda} C_0(\widehat{\xi}) + \text{higher radial orders}, \qquad \xi = \theta\,\omega,$$

contributes to the small-angle EEC density with leading power

$$\mathrm{EEC}_C(\chi) \sim \chi^{d_\perp - 2 - \lambda} \int_{S^{d_\perp - 1}} \mathrm{d}\omega\, C_0(\omega)\, \langle \Psi | \mathbb{L}(\psi_\omega) | \Psi \rangle, \qquad \chi \downarrow 0, \tag{65.47}$$

up to the center-coordinate smearing and the declared finite-chart remainder. Polyhomogeneous terms $|\xi|^{-\lambda_a} (\log|\xi|)^s C_{a,s}(\widehat{\xi})$ produce the corresponding powers $\chi^{d_\perp - 2 - \lambda_a} (\log \chi)^s$, with curvature corrections from $\sin\chi/\chi$ entering at even higher powers. If one changes from χ to $z = (1 - \cos\chi)/2$, the leading power is divided by two because $z = \chi^2/4 + O(\chi^4)$. Therefore the exponent quoted for a small-angle EEC singularity is a statement about the light-ray coefficient homogeneity, the detector-sphere dimension, and the chosen one-variable kernel.

This formulation also explains the relation with the QCD small-angle EEC. In a CFT the expansion coefficients are fixed-point data of light-ray representations. In asymptotically free QCD, the corresponding small-angle statement is a renormalized collinear light-ray expansion with a scale μ, an anomalous-dimension matrix, endpoint contact coordinates, and a mass-gap-sensitive long-distance remainder. The QCD object is therefore not obtained by simply copying the CFT theorem; it is a separate factorization datum whose detector side is the same stress-tensor observable.

Endpoint distribution charts in the EEC variable. The pushforward to one angular variable is still not the final endpoint distribution. A small-angle light-ray coefficient gives a density on $z > 0$, where

$$z = \frac{1 - \cos\chi}{2}, \qquad z = 0$$

is the coincident-detector endpoint. To compare with the full detector distribution, the density must be extended across $z = 0$ together with an explicit contact coordinate. This is the same logical distinction as in the two-detector chart above: separated-angle data determine a distribution on $(0, z_0)$, while the full detector product also contains diagonal contact strata.

Fix an endpoint interval $0 < z < z_0$. Let $f(z)$ be one singular density appearing in a finite light-ray chart, and suppose that subtracting the Taylor jet of order J of the test function makes the integral finite. The subtracted endpoint extension is the distribution

$$\langle [f]_{J,z_0,+}, \varphi \rangle := \int_0^{z_0} f(z) \left(\varphi(z) - \sum_{j=0}^{J} \frac{z^j}{j!} \varphi^{(j)}(0) \right) \mathrm{d}z. \tag{65.48}$$

For the logarithmic $1/z$ endpoint singularity this reduces to the familiar plus prescription

$$\left\langle \left[\frac{\Theta(z_0 - z)}{z} \right]_+, \varphi \right\rangle = \int_0^{z_0} \frac{\varphi(z) - \varphi(0)}{z}\, \mathrm{d}z.$$

Terms more singular than this are not hidden in the word “plus”: they require the corresponding higher Taylor subtraction and explicit $\delta^{(j)}(z)$ contact coordinates, with $\langle \delta^{(j)}, \varphi \rangle = (-1)^j \varphi^{(j)}(0)$.

A finite endpoint chart for the CFT EEC therefore has the form

$$\mathrm{EEC}_{\Psi}^{(N)} = \mathcal{R}_{z_0,\Psi}^{(N)} + \sum_{a\in A_N} c_a(\Psi)\,[f_a]_{J_a,z_0,+} + \sum_{j=0}^{J_N} K_j^{(N)}(\Psi)\,\delta^{(j)}(z) + R_{z_0,\Psi}^{(N)}. \tag{65.49}$$

Here $\mathcal{R}_{z_0,\Psi}^{(N)}$ is the ordinary distributional bulk representative away from $z=0$, the $c_a(\Psi)$ are the retained light-ray matrix elements and coefficient normalizations after the angular pushforward, and $R_{z_0,\Psi}^{(N)}$ is the declared finite-chart remainder. The coefficients $K_j^{(N)}$ are not determined by knowing the separated-angle asymptotic densities f_a alone. They are fixed only after the diagonal detector extension, Ward identities, and endpoint matching convention have all been specified. In particular, the constant-detector moment fixes a linear combination of the separated-angle integral and the contact coordinates; it does not by itself choose the representative in (65.49).

The dependence on z_0 is a useful sign check. For the basic plus distribution

$$D_{z_*} := \left[\frac{\Theta(z_*-z)}{z}\right]_+$$

one has, as distributions on $[0, z_b]$,

$$D_{z_b} = D_{z_a} + \frac{\mathbf{1}_{z_a<z<z_b}}{z} - \log\frac{z_b}{z_a}\,\delta(z), \qquad 0<z_a<z_b. \tag{65.50}$$

Testing (65.50) against φ gives exactly

$$\int_0^{z_a} \frac{\varphi-\varphi(0)}{z}\mathrm{d}z + \int_{z_a}^{z_b} \frac{\varphi(z)}{z}\mathrm{d}z - \log\frac{z_b}{z_a}\varphi(0) = \int_0^{z_b} \frac{\varphi-\varphi(0)}{z}\mathrm{d}z.$$

Equivalently, if the ordinary annulus $z_a<z<z_b$ is moved from the bulk representative into the endpoint plus-chart, the explicit $\delta(z)$ coordinate must shift by the annulus constant $\log(z_b/z_a)$ times the coefficient of $1/z$. With the opposite sign, the constant test changes, so the total-energy Ward identity would fail. The QCD endpoint formula (49.254) is the running-coupling, scheme-dependent analogue of this fixed-point distributional identity.

This endpoint chart gives a nonperturbative meaning to the collinear expansion of energy correlators in a CFT. It is the conformal counterpart of the QCD statement that small-angle energy correlators are governed by collinear operator data and anomalous dimensions. In QCD the running coupling and the mass gap alter the long-distance behavior; in a CFT the light-ray OPE is an operator expansion intrinsic to the fixed point, supplemented by the distributional contact data needed to recover the full detector product.

65.7 Comparison with QCD Detector Observables

The QCD and CFT constructions share the same operator-theoretic skeleton:

$$T^{\mu\nu} \quad\rightsquigarrow\quad \mathcal{E}(\mathbf{n}) \quad\rightsquigarrow\quad \mathcal{G}_k(\mathbf{n}_1,\dots,\mathbf{n}_k).$$

In QCD, the physical final-state resolution is a hadronic spectral sum and perturbation theory enters through asymptotically free short-distance coefficient functions. In a CFT, the state can be created directly by local operators and the angular dependence is constrained by conformal representation theory. The common lesson is mathematical rather than phenomenological: a detector observable is defined by the stress tensor in the physical Hilbert space, and partonic or conformal expansions are calculational representations of that observable under stated hypotheses.

Chapter 66

Two-Dimensional Sigma Models, Orbifolds, and Twist Fields

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The two-dimensional construction block beginning here uses several coordinates on CFT data, and they have different proof burdens. This chapter starts with regulated sigma-model couplings, exact toroidal and WZW/coset laboratories, finite orbifolds, and twist-field deformations. Chapter 67 then treats chiral algebra, modular data, and fixed-channel sewing estimates. Chapter 68 uses Liouville theory as the nonrational test case where probabilistic construction, bootstrap coordinates, and direct integrals must be compared. Chapter 69 adds boundary conditions and bordered-surface sewing. Finally Chapter 70 records the spin and superconformal chiral-algebra constraints that any full spin CFT or supersymmetric RG construction must satisfy.

The status convention is the same throughout this block. A local calculation such as a one-loop beta tensor, KZ flatness identity, finite orbifold sector count, Cardy matrix entry, or spectral-flow coefficient proves only the local algebra it displays. A fixed-channel analytic statement additionally needs convergence, topological vector spaces, positive pairings, or common operator domains. A full CFT construction requires Hilbert spaces or state spaces, local fields as distributions, reflection positivity when appropriate, modular covariance, anomaly-line bookkeeping, and sewing maps compatible across decompositions. Comparisons such as sigma-model exactness, Liouville probabilistic/bootstrap equivalence, rational versus nonrational sewing, and GLSM mirror-flow identifications remain theorem targets unless those data have been supplied in one common normalization.

66.1 Sigma-Model Data as Two-Dimensional QFT Data

Let Σ be an oriented Euclidean surface with metric γ, and let M be a smooth target manifold. A bosonic nonlinear sigma-model regularized datum consists of:

$$X : \Sigma \longrightarrow M, \qquad G \in \Gamma(S^2T^*M), \qquad \mathcal{B}, \qquad \Phi \in C^\infty(M),$$

where G is a Riemannian metric, $\mathcal{B}$ is a $U(1)$-gerbe connection with curvature $H \in \Omega^3_{\mathrm{cl}}(M)$, and Φ is the dilaton coupling to the scalar curvature of Σ. In a local trivialization of the gerbe, with two-form potential B, the Euclidean action representative is

$$\begin{aligned} S[X;\gamma] = \frac{1}{4\pi\alpha'} \int_\Sigma \mathrm{d}^2\sigma \sqrt{\gamma}\, \gamma^{ab} G_{ij}(X) \partial_a X^i \partial_b X^j \\ + \frac{\mathrm{i}}{4\pi\alpha'} \int_\Sigma \mathrm{d}^2\sigma\, \epsilon^{ab} B_{ij}(X) \partial_a X^i \partial_b X^j + \frac{1}{4\pi} \int_\Sigma \mathrm{d}^2\sigma \sqrt{\gamma}\, R_\gamma \Phi(X). \end{aligned} \tag{66.1}$$

The invariant meaning of the B-term is the gerbe holonomy of $X^*\mathcal{B}$. When the gerbe is trivial this holonomy is represented by the integral of a globally defined two-form B. The consistency condition is that $[H/(2\pi\alpha')]$ has integral periods in the normalization of (66.1). On a surface with boundary one must also specify brane data that trivialize the pulled-back gerbe on the boundary; this chapter restricts to closed Σ.

Definition 66.1 (Conformal sigma-model point)**.** A conformal sigma-model point is a continuum two-dimensional QFT obtained from a specified sigma-model regulator and coupling coordinates for which the renormalized stress tensor has vanishing trace on flat separated-point correlators and has only the stated Weyl-anomaly contact terms on curved source surfaces. Equivalently, in a coordinate chart on the space of sigma-model couplings, the local Weyl-anomaly coefficients

$$\widehat{\beta}^G_{ij}, \qquad \widehat{\beta}^B_{ij}, \qquad \widehat{\beta}^\Phi$$

vanish up to target-space diffeomorphisms and B-field gauge transformations, with $\widehat{\beta}^\Phi$ allowing a constant term that records the central charge. The exact QFT datum is the continuum hierarchy of local correlators and defect correlators. A finite-order loop approximation to the tensors above is a controlled large-radius criterion inside a specified asymptotic expansion.

The diffeomorphism ambiguity in Definition 66.1 is physical: if $\xi \in \Gamma(TM)$, the field redefinition $X^i \mapsto X^i + \xi^i(X)$ changes the metric and two-form couplings by

$$\delta_\xi G_{ij} = (\mathcal{L}_\xi G)_{ij}, \qquad \delta_\xi B_{ij} = (\mathcal{L}_\xi B)_{ij},$$

and therefore changes beta functions by a vector field on coupling space. Similarly $B \mapsto B + \mathrm{d}\Lambda$ leaves the closed-surface path integral unchanged except for the corresponding gerbe-gauge convention. The scheme-independent trace identity is consequently formulated modulo these redundant coordinate directions.

66.2 The One-Loop Metric Beta Tensor

The first local test of conformality is the one-loop Weyl-anomaly coefficient. We record the derivation with its regulator and expansion hypotheses because the Ricci tensor is a statement about the local logarithmic divergence of the two-dimensional QFT.

66.2.1 Covariant Background Fields and the Mean-Zero Split

The symbol $X = x + \xi$ is not meaningful on a curved target. The background field method used here is instead the following local construction. Fix a background map $x : \Sigma \to M$. A fluctuation is a section $\xi \in \Gamma(x^*TM)$, and the full map is

$$X(\sigma) = \exp_{x(\sigma)} \xi(\sigma).$$

This is defined whenever $\xi(\sigma)$ lies in a normal neighborhood of $x(\sigma)$. Equivalently, $X(\sigma) = \lambda_\sigma(1)$, where $\lambda_\sigma(t)$ is the target geodesic satisfying

$$\lambda_\sigma(0) = x(\sigma), \qquad \dot{\lambda}_\sigma(0) = \xi(\sigma).$$

In local coordinates this begins as

$$X^i = x^i + \xi^i - \frac{1}{2}\Gamma^i{}_{jk}(x)\xi^j\xi^k + O(\xi^3). \tag{66.2}$$

The fluctuation field ξ is therefore a target vector at the background point, not a coordinate difference between two points of M.

The effective action is also defined with a mean-field convention. Introduce an auxiliary covector source $J_i(\sigma)$ along x, and define the source-dependent local functional integral by

$$\exp(-\Gamma[x]) = \int [D\xi] \, \exp\left(-S[\exp_x \xi] + \int_\Sigma \mathrm{d}^d\sigma\sqrt{\gamma}\, J_i\xi^i\right)\Bigg|_{\langle\xi^i(\sigma)\rangle_J=0}. \tag{66.3}$$

The source J is not an additional sigma-model coupling. It is the Legendre-transform device that enforces that the chosen background map x, rather than $x + \langle\xi\rangle$, is the mean field.

Mean-zero source at one loop. Assume that, in a chosen coordinate patch on field space, the action expanded around x has the form

$$S[\exp_x \xi] = S[x] + \int \sqrt{\gamma}\, L_i\xi^i + \frac{1}{2}\int \sqrt{\gamma}\, \xi^i\mathcal{A}_{ij}\xi^j + O(\xi^3), \tag{66.4}$$

where $\mathcal{A}$ is the quadratic fluctuation operator and L_i is the first variation density. Ignoring zero modes and using a regulator for the determinant, the Gaussian part of (66.3) has

$$\langle\xi\rangle_J = \mathcal{A}^{-1}(J - L). \tag{66.5}$$

Thus the condition $\langle\xi\rangle_J = 0$ fixes $J = L$, and the one-loop effective action is

$$\Gamma[x] = S[x] + \frac{1}{2}\log\det{}'\mathcal{A} + \text{local measure terms} + \text{higher-loop terms}. \tag{66.6}$$

At Gaussian order the exponent in (66.3) is

$$-S[x] - \frac{1}{2}\langle\xi, \mathcal{A}\xi\rangle + \langle J - L, \xi\rangle,$$

where the brackets include the $\sqrt{\gamma}$ density and target indices. Completing the square gives

$$-S[x] - \frac{1}{2}\langle\xi - \mathcal{A}^{-1}(J-L), \mathcal{A}(\xi - \mathcal{A}^{-1}(J-L))\rangle + \frac{1}{2}\langle J - L, \mathcal{A}^{-1}(J-L)\rangle.$$

The Gaussian mean is therefore $\mathcal{A}^{-1}(J-L)$, which proves (66.5). Setting $J=L$ removes the linear term and leaves the regulated determinant of $\mathcal{A}$, giving (66.6). If $\mathcal{A}$ has zero modes, they are separated as collective coordinates or fixed by an additional gauge slice; this affects the global measure but not the local ultraviolet pole that determines the beta tensor.

For the pure metric action, the first variation density is

$$L_i = -\frac{1}{2\pi\alpha'} G_{ij}(x)\nabla^a \partial_a x^j, \tag{66.7}$$

on a closed source surface. Hence $J=0$ is allowed only when the background map is a classical harmonic map. The effective-action definition above permits one to compute the local counterterms for an arbitrary slowly varying background x; this is why the beta tensor is an off-shell local tensor on coupling space, rather than only an equation evaluated on classical solutions.

Proposition 66.2 (Second variation and the curvature vertex)**.** *Let $B=\Phi=0$, assume Σ is closed, and set*

$$S_G[X] = \frac{1}{4\pi\alpha'}\int_\Sigma \mathrm{d}^d\sigma\sqrt{\gamma}\,\gamma^{ab}G_{ij}(X)\partial_a X^i\partial_b X^j.$$

For the geodesic variation $X_s=\exp_x(s\xi)$, the expansion through second order is

$$\begin{aligned} S_G[X_s] = S_G[x] &- \frac{s}{2\pi\alpha'}\int_\Sigma \sqrt{\gamma}\,G_{ij}(x)\xi^i\nabla^a\partial_a x^j \\ &+ \frac{s^2}{4\pi\alpha'}\int_\Sigma \sqrt{\gamma}\,\gamma^{ab}\left[G_{ij}(x)D_a\xi^i D_b\xi^j - R_{ikj\ell}(x)\partial_a x^i\partial_b x^j\xi^k\xi^\ell\right] + O(s^3). \end{aligned} \tag{66.8}$$

Here

$$D_a\xi^i = \partial_a\xi^i + \Gamma^i{}_{jk}(x)\partial_a x^j\xi^k$$

and the curvature convention is

$$R(U,V)W = \nabla_U\nabla_V W - \nabla_V\nabla_U W - \nabla_{[U,V]}W. \tag{66.9}$$

The displayed component ordering is defined by

$$R_{ikj\ell}\partial_a x^i\partial_b x^j\xi^k\xi^\ell = \langle R(\xi,\partial_a x)\partial_b x,\xi\rangle.$$

Consequently the quadratic fluctuation action used below is precisely the s^2-coefficient in (66.8) at $s=1$.

Proof. Let $V=\partial_s X_s$ and $W_a=\partial_a X_s$. Since the target connection is torsion free,

$$\nabla_s W_a = \nabla_a V,$$

and since $s\mapsto X_s(\sigma)$ is a geodesic for every σ,

$$\nabla_s V = 0.$$

Differentiate the Lagrangian density $\gamma^{ab}\langle W_a,W_b\rangle$. At $s=0$,

$$\frac{\mathrm{d}}{\mathrm{d}s}\gamma^{ab}\langle W_a,W_b\rangle = 2\gamma^{ab}\langle D_a\xi,\partial_b x\rangle.$$

Integrating by parts on the closed source surface gives the first-order term in (66.8). For the second derivative,

$$\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}s^2}\gamma^{ab}\langle W_a, W_b\rangle = \gamma^{ab}\left[\langle \nabla_s\nabla_a V, W_b\rangle + \langle \nabla_a V, \nabla_s W_b\rangle\right].$$

The commutator identity and $\nabla_s V = 0$ give

$$\nabla_s\nabla_a V = R(V, W_a)V.$$

Metric compatibility and the skew-adjointness of $R(V, W_a)$ in its last two entries imply

$$\langle R(V, W_a)V, W_b\rangle = -\langle R(V, W_a)W_b, V\rangle.$$

Evaluating at $s = 0$, where $V = \xi$ and $W_a = \partial_a x$, therefore gives

$$\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}s^2}\gamma^{ab}\langle W_a, W_b\rangle\Big|_{s=0} = \gamma^{ab}\left[\langle D_a\xi, D_b\xi\rangle - \langle R(\xi, \partial_a x)\partial_b x, \xi\rangle\right],$$

which is (66.8) in components. □

Lemma 66.3 (One-loop metric divergence)**.** *Let M be Riemannian, assume Σ is closed, and set $B = \Phi = 0$. Use dimensional regularization around $d = 2 - \epsilon$ two source dimensions and the covariant background-field expansion*

$$X(\sigma) = \exp_{x(\sigma)}\xi(\sigma),$$

where $\xi(\sigma) \in T_{x(\sigma)}M$. With the action normalization (66.1), the logarithmic one-loop divergence in the background effective action is

$$\Gamma^{(1)}_{\mathrm{div}}[x] = -\frac{1}{4\pi\epsilon}\int_\Sigma \mathrm{d}^2\sigma\sqrt{\gamma}\,\gamma^{ab}R_{ij}(x)\partial_a x^i\partial_b x^j. \tag{66.10}$$

Consequently the metric beta tensor begins as

$$\mu\frac{\mathrm{d}G_{ij}}{\mathrm{d}\mu} = \alpha' R_{ij} + O(\alpha'^2) \tag{66.11}$$

in this coupling coordinate.

Proof. By Proposition 66.2, the quadratic action for ξ is

$$S_2 = \frac{1}{4\pi\alpha'}\int_\Sigma \mathrm{d}^d\sigma\sqrt{\gamma}\left[\gamma^{ab}G_{ij}(x)D_a\xi^i D_b\xi^j - \gamma^{ab}R_{ikjl}(x)\partial_a x^i\partial_b x^j\xi^k\xi^\ell\right], \qquad d = 2 - \epsilon.$$

The covariant derivative $D_a\xi^i = \partial_a\xi^i + \Gamma^i{}_{jk}(x)\partial_a x^j\xi^k$ makes target diffeomorphism covariance manifest. To extract a local pole, freeze the background at a point σ_0, choose geodesic normal coordinates on Σ at σ_0 and an orthonormal frame of x^*TM parallel at σ_0, and keep only terms with two derivatives of the slowly varying background map x. Derivatives of the connection and intrinsic curvature of Σ either contribute higher-derivative terms or the universal two-dimensional metric anomaly, not the metric counterterm being computed here.

The free kinetic action in this frame is

$$S_{\text{kin}} = \frac{1}{4\pi\alpha'} \int \mathrm{d}^d\sigma\, \delta_{ij}\partial_a\xi^i\partial^a\xi^j = \frac{1}{2}\int \mathrm{d}^d\sigma\, \xi^i \left(\frac{-\partial^2}{2\pi\alpha'}\right)\xi^i.$$

Thus the free covariance is $2\pi\alpha'(-\partial^2)^{-1}$. Introduce an auxiliary mass m only to separate the ultraviolet pole from the infrared zero mode. The coincident scalar Green kernel has the local heat-kernel representation

$$\begin{aligned}(-\partial^2 + m^2)^{-1}(\sigma_0, \sigma_0) &= \int_0^\infty \mathrm{d}s\,(4\pi s)^{-d/2}\mathrm{e}^{-m^2 s} \\ &= (4\pi)^{-d/2}\Gamma(1-d/2)(m^2)^{d/2-1} = \frac{1}{2\pi\epsilon} + O(\epsilon^0).\end{aligned} \tag{66.12}$$

Restoring covariance gives the singular part

$$\langle\xi^k(\sigma)\xi^\ell(\sigma)\rangle_{\text{div}} = 2\pi\alpha'\,\frac{1}{2\pi\epsilon}\,G^{k\ell}(x(\sigma)) = \frac{\alpha'}{\epsilon}G^{k\ell}(x(\sigma)). \tag{66.13}$$

The curvature interaction in S_2 is

$$S_{\text{curv}} = -\frac{1}{4\pi\alpha'}\int \sqrt{\gamma}\,\gamma^{ab}R_{ikj\ell}(x)\partial_a x^i\partial_b x^j\xi^k\xi^\ell.$$

Since

$$\Gamma[x] = S[x] - \log\langle \mathrm{e}^{-S_{\text{curv}}}\rangle_{\text{kin}} = S[x] + \langle S_{\text{curv}}\rangle_{\text{kin}} + O((\partial x)^4),$$

the logarithmic pole is

$$\begin{aligned}\Gamma^{(1)}_{\text{div}}[x] &= -\frac{1}{4\pi\alpha'}\int\sqrt{\gamma}\,\gamma^{ab}R_{ikj\ell}\partial_a x^i\partial_b x^j\langle\xi^k\xi^\ell\rangle_{\text{div}} \\ &= -\frac{1}{4\pi\epsilon}\int\sqrt{\gamma}\,\gamma^{ab}R_{ij}\partial_a x^i\partial_b x^j,\end{aligned}$$

where $R_{ikj\ell}G^{k\ell} = R_{ij}$ in the component ordering fixed in Proposition 66.2. Since a local metric term in the action is multiplied by $1/(4\pi\alpha')$, cancelling (66.10) requires $\delta G_{ij} = (\alpha'/\epsilon)R_{ij}$ in minimal subtraction. Requiring the bare coupling representative to be independent of μ then gives (66.11). □

With $H = \mathrm{d}B$ and dilaton included, the one-loop Weyl-anomaly coefficients in the same normalization have the form

$$\widehat{\beta}^G_{ij} = \alpha'\left(R_{ij} - \frac{1}{4}H_i{}^{k\ell}H_{jk\ell} + 2\nabla_i\nabla_j\Phi\right) + O(\alpha'^2), \tag{66.14}$$

$$\widehat{\beta}^B_{ij} = \alpha'\left(-\frac{1}{2}\nabla^k H_{kij} + \nabla^k\Phi\, H_{kij}\right) + O(\alpha'^2). \tag{66.15}$$

The tensor $\widehat{\beta}^\Phi$ fixes the scalar Weyl anomaly and the central charge. Its nonconstant part is scheme dependent under dilaton and metric redefinitions, while its constant part is the physical c-number in the two-dimensional stress-tensor anomaly.

66.3 One-Loop Weyl Anomaly and the Target-Space Variational Package

The formulas above are often quoted as target-space equations of motion. The precise statement is more structured. After a regulator and a local renormalization scheme have been chosen, ordinary coordinate beta functions are not themselves the local Weyl-anomaly coefficients. The local Weyl coefficients differ from them by redundant directions: target diffeomorphisms, B-field gauge transformations, and the dilaton-gradient terms displayed below. The hypotheses in this section are therefore: Σ is closed, the target fields are smooth on the coordinate patch under consideration, dimensional regularization is used near two source dimensions, and all formulas are first-order statements in the large-radius α'-expansion.

Lemma 66.4 (One-loop string-frame functional)**.** *Let $H = \mathrm{d}B$. Up to the constant central-charge term in $\widehat{\beta}^{\Phi}$, the one-loop Weyl-anomaly equations for (G, B, Φ) are generated by the local target-space functional*

$$\mathcal{I}_0[G, B, \Phi] = \int_M \mathrm{d}^D x \sqrt{G}\, \mathrm{e}^{-2\Phi} \left(R(G) + 4|\nabla\Phi|^2 - \frac{1}{12} H_{ijk} H^{ijk} \right). \tag{66.16}$$

Assume either that M is compact or that all variations have compact support, so that target-space boundary terms can be discarded. Define

$$E_{ij} = R_{ij} + 2\nabla_i \nabla_j \Phi - \frac{1}{4} H_i{}^{k\ell} H_{jk\ell}, \tag{66.17}$$

$$S_\Phi = R + 4\nabla^2\Phi - 4|\nabla\Phi|^2 - \frac{1}{12} H^2. \tag{66.18}$$

Then the fixed-Φ metric variation is

$$\delta_G \mathcal{I}_0 = \int_M \mathrm{d}^D x \sqrt{G}\, \mathrm{e}^{-2\Phi} \left(E_{ij} - \frac{1}{2} G_{ij} S_\Phi \right) \delta G^{ij}, \tag{66.19}$$

and the dilaton variation is

$$\delta_\Phi \mathcal{I}_0 = -2 \int_M \mathrm{d}^D x \sqrt{G}\, \mathrm{e}^{-2\Phi} S_\Phi \, \delta\Phi. \tag{66.20}$$

Variation with respect to B gives

$$\nabla^k H_{kij} - 2\nabla^k \Phi \, H_{kij} = 0, \tag{66.21}$$

and variation with respect to G, after the scalar equation is used to choose the trace representative, gives

$$R_{ij} - \frac{1}{4} H_i{}^{k\ell} H_{jk\ell} + 2\nabla_i \nabla_j \Phi = 0. \tag{66.22}$$

The scalar Weyl-anomaly coefficient may be represented as

$$\widehat{\beta}^{\Phi} = \frac{D - 26}{6} + \alpha' \left(|\nabla\Phi|^2 - \frac{1}{2}\nabla^2\Phi - \frac{1}{24} H_{ijk} H^{ijk} \right) + O(\alpha'^2). \tag{66.23}$$

With the reparametrization ghosts included, the conformality condition is the vanishing of the matter-plus-ghost scalar anomaly.

Proof. For the B-variation, use

$$\delta H_{ijk} = \nabla_i \delta B_{jk} + \nabla_j \delta B_{ki} + \nabla_k \delta B_{ij}. \tag{66.24}$$

Because H is antisymmetric,

$$\delta(H_{ijk}H^{ijk}) = 2H^{ijk}\delta H_{ijk} = 6H^{ijk}\nabla_i \delta B_{jk}. \tag{66.25}$$

Thus

$$\delta_B \mathcal{I}_0 = -\frac{1}{2}\int \sqrt{G}\,\mathrm{e}^{-2\Phi} H^{ijk}\nabla_i \delta B_{jk}.$$

Integrating by parts gives

$$\delta_B \mathcal{I}_0 = \frac{1}{2}\int \sqrt{G}\,\nabla_i(\mathrm{e}^{-2\Phi} H^{ijk})\delta B_{jk},$$

which is equivalent to (66.21). Multiplying the left side of (66.21) by $-\alpha'/2$ gives exactly (66.15).

For the metric variation, the Palatini identity gives

$$\begin{aligned}\delta(\sqrt{G}\,\mathrm{e}^{-2\Phi}R) = \sqrt{G}\,\mathrm{e}^{-2\Phi}\Big[R_{ij} - \frac{1}{2}G_{ij}R + 2\nabla_i\nabla_j\Phi \\ - 4\nabla_i\Phi\nabla_j\Phi + G_{ij}(4|\nabla\Phi|^2 - 2\nabla^2\Phi)\Big]\delta G^{ij} + \text{boundary terms.}\end{aligned}$$

The variation of $4|\nabla\Phi|^2 = 4G^{ij}\partial_i\Phi\partial_j\Phi$, including the variation of $\sqrt{G}$, is

$$\sqrt{G}\,\mathrm{e}^{-2\Phi}\left(4\nabla_i\Phi\nabla_j\Phi - 2G_{ij}|\nabla\Phi|^2\right)\delta G^{ij}.$$

The nontrace $\nabla_i\Phi\nabla_j\Phi$ terms therefore cancel. The coefficient of the H-term is fixed directly: $H_{abc}H^{abc}$ contains three inverse metrics, while the variation of $\sqrt{G}$ contributes the trace term. Thus

$$\delta_G\left(-\frac{1}{12}\sqrt{G}\,\mathrm{e}^{-2\Phi}H^2\right) = \sqrt{G}\,\mathrm{e}^{-2\Phi}\left(-\frac{1}{4}H_i{}^{k\ell}H_{jk\ell} + \frac{1}{24}G_{ij}H^2\right)\delta G^{ij}.$$

Combining the three fixed-Φ variations gives (66.19), because the trace coefficient is

$$-\frac{1}{2}R + 2|\nabla\Phi|^2 - 2\nabla^2\Phi + \frac{1}{24}H^2 = -\frac{1}{2}S_\Phi.$$

The scalar equation $S_\Phi = 0$ therefore removes exactly the trace part of the metric variation and leaves the chosen representative (66.22), which is the vanishing of (66.14).

Finally, varying Φ gives

$$-2\left(R + 4|\nabla\Phi|^2 - \frac{1}{12}H^2\right)\delta\Phi + 8\nabla^i\Phi\,\nabla_i\delta\Phi.$$

After integration by parts with the $\mathrm{e}^{-2\Phi}$ density, this becomes (66.20); equivalently the scalar combination is

$$R - \frac{1}{12}H^2 + 4\nabla^2\Phi - 4|\nabla\Phi|^2$$

before the universal free-field determinant is added. That determinant supplies the constant $(D-26)/6$ after the bosonic ghosts are included. In the flat linear-dilaton test $G_{ij} = \delta_{ij}$, $H = 0$, and $\Phi = V_iX^i + \Phi_0$, equation (66.23) becomes

$$\frac{D-26}{6} + \alpha' V^2 = 0,$$

equivalently $D + 6\alpha' V^2 = 26$, matching the central-charge convention used below in Remark 66.9. □

Lemma 66.5 (Local one-loop origin of the H-coefficients)**.** *Work in the same dimensional-regularization and minimal-subtraction scheme as Lemma 66.3. Freeze the background map at a point, choose a local orthonormal target frame parallel at that point, and keep only logarithmic poles with two derivatives of the slowly varying background $x : \Sigma \to M$. The B-field interaction has the local covariant expansion*

$$
\begin{aligned}
S_B[x,\xi] &= \frac{\mathrm{i}}{4\pi\alpha'}\int \epsilon^{ab}B_{ij}(x)\partial_a x^i\partial_b x^j \\
&\quad + \frac{\mathrm{i}}{4\pi\alpha'}\int \epsilon^{ab}H_{k\ell i}(x)\xi^k D_a\xi^\ell\partial_b x^i \\
&\quad + \frac{\mathrm{i}}{8\pi\alpha'}\int \epsilon^{ab}\nabla_\ell H_{kij}(x)\xi^\ell\xi^k\partial_a x^i\partial_b x^j + O(\xi^3,\partial^3 x).
\end{aligned}
\tag{66.26}
$$

With the dimensional-continuation convention

$$
\epsilon^{ab}\epsilon^{cd}\gamma_{ac} = \gamma^{bd} + O(\epsilon) \quad \textit{inside logarithmic poles,} \tag{66.27}
$$

the one-loop pole from the last line of (66.26) is

$$
\Gamma^{(1)}_{B,\mathrm{tad,div}} = \frac{\mathrm{i}}{8\pi\epsilon}\int \epsilon^{ab}\nabla^k H_{kij}(x)\partial_a x^i\partial_b x^j, \tag{66.28}
$$

and the connected bubble from the middle line is

$$
\Gamma^{(1)}_{B,\mathrm{bub,div}} = \frac{1}{16\pi\epsilon}\int \sqrt{\gamma}\,\gamma^{ab}H_i{}^{k\ell}H_{jk\ell}(x)\partial_a x^i\partial_b x^j. \tag{66.29}
$$

Consequently the minimal-subtraction counterterms contain

$$
\delta B_{ij} = -\frac{\alpha'}{2\epsilon}\nabla^k H_{kij}, \qquad \delta G_{ij} = -\frac{\alpha'}{4\epsilon}H_i{}^{k\ell}H_{jk\ell}, \tag{66.30}
$$

which are precisely the H-dependent non-dilaton coefficients in (66.14) and (66.15).

Proof. The expansion (66.26) is most cleanly obtained by differentiating the pulled-back gerbe representative along the geodesic family $X_s = \exp_x(s\xi)$. The linear term in s is $\iota_\xi H$ up to an exact two-form on Σ, which is a B-field gauge variation on closed Σ. Covariantizing the Σ-derivative of ξ gives the middle vertex. Differentiating once more gives the ∇H vertex, and the Taylor factor $1/2$ gives the coefficient in the third line.

The tadpole uses only the coincident pole (66.13):

$$
\left\langle \xi^\ell(\sigma)\xi^k(\sigma)\right\rangle_{\mathrm{div}} = \frac{\alpha'}{\epsilon}G^{\ell k}.
$$

Substituting this in the third line of (66.26) gives (66.28). Since the local B-coupling in the action has prefactor $\mathrm{i}/(4\pi\alpha')$, cancellation of this pole requires the first counterterm in (66.30).

For the bubble, let $\Delta = (-\partial^2 + m^2)^{-1}$ be the scalar local Green kernel used in (66.12). The only local logarithmic contraction needed, after freezing x, is

$$
\int \mathrm{d}^d\sigma'\,\partial_a\Delta(\sigma-\sigma')\,\partial_c\Delta(\sigma-\sigma') = \frac{\gamma_{ac}}{4\pi\epsilon} + O(\epsilon^0). \tag{66.31}
$$

In momentum space this is the rotational identity

$$\int \frac{\mathrm{d}^d p}{(2\pi)^d} \frac{p_a p_c}{(p^2+m^2)^2} = \frac{\gamma_{ac}}{d} \int \frac{\mathrm{d}^d p}{(2\pi)^d} \frac{p^2}{(p^2+m^2)^2} = \frac{\gamma_{ac}}{4\pi\epsilon} + O(\epsilon^0),$$

using $d = 2 - \epsilon$; terms proportional to $m^2/(p^2+m^2)^2$ are finite in the logarithmic pole part. The free covariance contributes $(2\pi\alpha')^2$, the two Wick pairings give the same tensor after using antisymmetry of $H_{k\ell i}$, and the connected effective action contains

$$-\frac{1}{2}\left(\frac{\mathrm{i}}{4\pi\alpha'}\right)^2 \left\langle \left(\int \epsilon^{ab} H_{k\ell i} \xi^k D_a \xi^\ell \partial_b x^i \right)^2 \right\rangle_c .$$

Combining the factors gives

$$-\frac{1}{2}\left(\frac{\mathrm{i}}{4\pi\alpha'}\right)^2 \cdot 2 \cdot \frac{\pi(\alpha')^2}{\epsilon} = \frac{1}{16\pi\epsilon}.$$

The target-index contractions give $H_{k\ell i} H_{mnj} G^{km} G^{\ell n} = H_i{}^{k\ell} H_{jk\ell}$, and (66.27) converts the two antisymmetric two-dimensional tensors into γ^{bd}. This proves (66.29). Comparing with the metric action prefactor $1/(4\pi\alpha')$ gives the second counterterm in (66.30).

The two counterterms are local and gauge invariant after being expressed in terms of H. Adding the dilaton-gradient redundant direction below converts these coordinate beta representatives into the hatted Weyl-anomaly coefficients. □

66.4 Minimal Subtraction as Geometry on Coupling Space

The sigma-model coupling is not a finite list of numbers. It is the point (G, B, Φ) in an infinite-dimensional space of target-space fields, quotiented by target diffeomorphisms and gerbe gauge transformations. A minimal-subtraction counterterm is therefore a local tensorial functional of (G, H, Φ), and a change of renormalization scheme is a local change of coordinates on this space.

Definition 66.6 (Local sigma-model scheme)**.** A local perturbative sigma-model renormalization scheme assigns bare fields (G^0, B^0, Φ^0) to renormalized fields (G, B, Φ) by expansions

$$G^0_{ij} = \mu^{-\epsilon} \left[G_{ij} + \sum_{\ell \geq 1} \sum_{r=1}^{\ell} \frac{(\alpha')^\ell}{\epsilon^r} C^{(\ell,r)}_{ij}(G, H, \Phi) \right], \tag{66.32}$$

$$B^0_{ij} = \mu^{-\epsilon} \left[B_{ij} + \sum_{\ell \geq 1} \sum_{r=1}^{\ell} \frac{(\alpha')^\ell}{\epsilon^r} D^{(\ell,r)}_{ij}(G, H, \Phi) \right], \tag{66.33}$$

where the coefficients are local covariant expressions built from finitely many target derivatives. Minimal subtraction keeps only pole terms in ϵ. Two such schemes are equivalent if their renormalized fields are related by local target diffeomorphisms, B-field gauge transformations, and finite local tensor redefinitions.

Finite redefinitions and beta-vector fields. Use condensed notation g^A for the collection of local couplings (G_{ij}, B_{ij}, Φ), including the target point and tensor indices in the label A. Let a finite local scheme change be

$$g'^A = g^A + F^A(g), \qquad F^A = O((\alpha')^r), \tag{66.34}$$

where F is a covariant local tensor functional modulo the redundant diffeomorphism and B-gauge directions. If $\beta = \beta^A(g)\partial_A$ is the beta-vector field in the original coordinates, then, to first order in F,

$$\beta'^A(g') = \beta^A(g') + [\beta, F]^A(g') + O(F^2), \qquad [\beta, F]^A = \beta^B \partial_B F^A - F^B \partial_B \beta^A. \tag{66.35}$$

Since the leading sigma-model beta field starts at order α', a finite counterterm of order $(\alpha')^r$ first affects the displayed representative one order later, at order $(\alpha')^{r+1}$.

Differentiate (66.34) with respect to $\log \mu$ at fixed bare fields:

$$\beta'^A(g') = \frac{\mathrm{d}g'^A}{\mathrm{d}\log\mu} = \beta^A(g) + \partial_B F^A(g)\, \beta^B(g) + O(F^2).$$

To express the result in the new coordinate g', invert (66.34) to first order,

$$g^A = g'^A - F^A(g') + O(F^2),$$

and expand

$$\beta^A(g) = \beta^A(g') - F^B(g')\partial_B \beta^A(g') + O(F^2).$$

Combining the two displayed equations gives (66.35). The same computation is valid for local tensor fields once the index A is interpreted as a DeWitt index and ∂_A as the corresponding functional derivative.

Hatted beta functions and redundant directions. Let β^G, β^B be coordinate beta functions in a chosen local scheme. A target vector field W and one-form Λ change the representatives by

$$\widehat{\beta}^G_{ij} = \beta^G_{ij} + (\mathcal{L}_W G)_{ij}, \tag{66.36}$$

$$\widehat{\beta}^B_{ij} = \beta^B_{ij} + (\mathcal{L}_W B)_{ij} + (\mathrm{d}\Lambda)_{ij}. \tag{66.37}$$

In particular, the dilaton-gradient terms in (66.14)–(66.15) are the one-loop representative obtained from

$$W^i = \alpha' \nabla^i \Phi + O(\alpha'^2), \qquad \Lambda_i = O(\alpha'^2). \tag{66.38}$$

A target diffeomorphism generated by W changes the local action by replacing G and B with their Lie derivatives. Since the trace of the stress tensor measures the response of the renormalized action to a Weyl change at fixed bare fields, adding a redundant flow direction changes the coordinate beta functions by the same Lie derivatives. A gerbe gauge change adds $\mathrm{d}\Lambda$ to the two-form representative and leaves $H = \mathrm{d}B$ fixed on closed source surfaces. Finally, $(\mathcal{L}_{\nabla\Phi} G)_{ij} = 2\nabla_i \nabla_j \Phi$, and

$$(\mathcal{L}_{\nabla\Phi} B)_{ij} = \nabla^k \Phi\, H_{kij} + \nabla_i(\nabla^k \Phi B_{kj}) - \nabla_j(\nabla^k \Phi B_{ki}),$$

so the non-gauge part is the $\nabla\Phi \cdot H$ term in (66.15).

Lemma 66.7 (Torsionful one-loop package)**.** *Let $H = \mathrm{d}B$, and define the metric connections*

$$\nabla_i^{\pm} v^k = \nabla_i v^k \pm \frac{1}{2} H^k{}_{i\ell} v^\ell.$$

Write R^-_{ij} for the Ricci tensor of ∇^-. Through one loop, and in the same local scheme as (66.14)–(66.15), the symmetric and antisymmetric Weyl-anomaly coefficients are the two parts of the nonsymmetric tensor

$$\alpha'^{-1}\left(\widehat{\beta}^G_{ij} + \widehat{\beta}^B_{ij}\right) = R^-_{ij} + 2\nabla^-_i \nabla_j \Phi. \tag{66.39}$$

Equivalently,

$$(R^-_{ij})_{(ij)} = R_{ij} - \frac{1}{4} H_i{}^{k\ell} H_{jk\ell}, \tag{66.40}$$

$$(R^-_{ij})_{[ij]} = -\frac{1}{2} \nabla^k H_{kij}. \tag{66.41}$$

The exterior derivative of the antisymmetric beta representative is a three-form beta tensor for H, and it automatically obeys $\mathrm{d}\beta^H = 0$. This is the one-loop seed of the Curci–Paffuti consistency condition: higher-loop representatives must be chosen so that target diffeomorphism invariance, B-field gauge invariance, and the Bianchi identity $\mathrm{d}H = 0$ are preserved simultaneously.

Proof. Let $K^k{}_{ij} = -H^k{}_{ij}/2$, so that $\Gamma^{-\,k}_{ij} = \Gamma^k_{ij} + K^k{}_{ij}$. Since $H^k{}_{ik} = 0$, the Ricci tensor of ∇^- is

$$R^-_{ij} = R_{ij} + \nabla_k K^k{}_{ij} + K^k{}_{k\ell} K^\ell{}_{ij} - K^k{}_{j\ell} K^\ell{}_{ik}. \tag{66.42}$$

The trace term vanishes and the derivative term is $-\frac{1}{2}\nabla^k H_{kij}$. The quadratic term is

$$-K^k{}_{j\ell} K^\ell{}_{ik} = -\frac{1}{4} H^k{}_{j\ell} H^\ell{}_{ik} = -\frac{1}{4} H_i{}^{k\ell} H_{jk\ell},$$

after lowering indices and using total antisymmetry of H. This proves (66.40) and (66.41).

For the dilaton term, $\nabla^-_i \nabla_j \Phi$ means applying ∇^- to the covector $\nabla_j \Phi$:

$$\nabla^-_i \nabla_j \Phi = \nabla_i \nabla_j \Phi + \frac{1}{2} H^k{}_{ij} \nabla_k \Phi.$$

Therefore $2\nabla^-_i \nabla_j \Phi$ contributes $2\nabla_i \nabla_j \Phi$ to the symmetric part and $\nabla^k \Phi\, H_{kij}$ to the antisymmetric part. Combining these terms with the displayed Ricci tensor gives exactly (66.14) and (66.15). Finally $\beta^H = \mathrm{d}\widehat{\beta}^B$ is exact as a three-form representative, so $\mathrm{d}\beta^H = \mathrm{d}^2 \widehat{\beta}^B = 0$; possible global gerbe data affect large gauge sectors, not this local identity. □

Controlled Approximation 66.8 (Two-loop metric beta tensor)**.** For $B = \Phi = 0$, dimensional regularization and minimal subtraction give, in the standard Riemann-normal-coordinate scheme,

$$\mu \frac{\mathrm{d}G_{ij}}{\mathrm{d}\mu} = \alpha' R_{ij} + \frac{(\alpha')^2}{2} R_{ik\ell m} R_j{}^{k\ell m} + O((\alpha')^3). \tag{66.43}$$

The displayed coefficient is a controlled perturbative statement in the large-radius expansion. Starting at two loops, finite local redefinitions of G change the displayed tensor by Lie derivatives and by terms proportional to lower-order equations of motion. Thus the tensor (66.43) is a scheme representative, while vanishing of the Weyl anomaly modulo redundant directions is the invariant statement.

RG time versus Hamilton Ricci flow. Let $B = \Phi = 0$, and restrict to the one-loop metric flow

$$\frac{\partial G_{ij}}{\partial \tau} = \alpha' R_{ij}(G), \qquad \tau = \log \mu. \tag{66.44}$$

If

$$t = -\frac{\alpha'}{2}\tau, \tag{66.45}$$

then $G(t)$ satisfies Hamilton's Ricci-flow equation

$$\frac{\partial G_{ij}}{\partial t} = -2R_{ij}(G). \tag{66.46}$$

Thus the ultraviolet sigma-model flow, in the convention $\tau = \log \mu$, is backward relative to the usual increasing Ricci-flow time t. This is a statement about the one-loop local beta-vector field on the space of target metrics. It does not assert that the nonlinear sigma model has been constructed nonperturbatively at every point of the flow, nor that a finite-time singularity of Ricci flow is by itself a theorem about the continuum QFT.

By the chain rule,

$$\frac{\partial G_{ij}}{\partial t} = \frac{\partial \tau}{\partial t}\frac{\partial G_{ij}}{\partial \tau} = \left(-\frac{2}{\alpha'}\right)\alpha' R_{ij} = -2R_{ij}.$$

The remaining assertions are hypotheses bookkeeping. Equation (66.44) is a perturbative beta-function equation in a specified coupling coordinate and regulator scheme, while Ricci flow is a geometric partial differential equation for smooth metrics. The two equations agree at the displayed order after the time reparametrization (66.45); this agreement does not supply the operator algebra, Hilbert space, stress tensor, or sewing data of a continuum CFT.

Constant-curvature radius flow. Let M have dimension $n \geq 2$, and let γ be a metric of constant sectional curvature σ, where $\sigma = +1$ for the unit sphere and $\sigma = -1$ for the unit hyperbolic metric. Put

$$G_{ij} = r^2\gamma_{ij}, \qquad \lambda = \frac{\alpha'}{r^2}, \qquad B = \Phi = 0. \tag{66.47}$$

In the pure-metric minimal-subtraction representative (66.43), the radius parameter obeys

$$\frac{\mathrm{d}r^2}{\mathrm{d}\log \mu} = (n-1)\alpha'\left(\sigma + \frac{\alpha'}{r^2}\right) + O\!\left(\frac{(\alpha')^3}{r^4}\right), \tag{66.48}$$

and the dimensionless large-radius coupling obeys

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log \mu} = -(n-1)\lambda^2(\sigma + \lambda) + O(\lambda^4). \tag{66.49}$$

In particular, for the sphere,

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log \mu} = -(n-1)\lambda^2 - (n-1)\lambda^3 + O(\lambda^4), \tag{66.50}$$

which is the asymptotic-freedom sign in the convention $\lambda = \alpha'/r^2$. For the hyperbolic sign,

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log \mu} = (n-1)\lambda^2 - (n-1)\lambda^3 + O(\lambda^4). \tag{66.51}$$

The hyperbolic formula is a local curvature check of signs and scaling; it is not a claim that an arbitrary noncompact hyperbolic target, without additional infrared and spectral data, defines a full CFT.

For the scaled constant-curvature metric $G = r^2\gamma$, the sectional curvature is $K = \sigma/r^2$. Hence

$$R_{ij} = (n-1)K\, G_{ij} = (n-1)\sigma\, \gamma_{ij}, \tag{66.52}$$

and the standard constant-curvature contraction gives

$$R_{ik\ell m}R_j{}^{k\ell m} = 2(n-1)K^2 G_{ij} = \frac{2(n-1)}{r^2}\gamma_{ij}. \tag{66.53}$$

Substituting these identities into (66.43) gives

$$\frac{\partial G_{ij}}{\partial \log \mu} = \left[(n-1)\sigma\alpha' + (n-1)\frac{(\alpha')^2}{r^2}\right]\gamma_{ij} + O\!\left(\frac{(\alpha')^3}{r^4}\right)\gamma_{ij}.$$

Because $G_{ij} = r^2\gamma_{ij}$, this is exactly (66.48). Finally, $\lambda = \alpha'/r^2$ and α' is held fixed, so

$$\frac{\mathrm{d}\lambda}{\mathrm{d}\log\mu} = -\frac{\alpha'}{r^4}\frac{\mathrm{d}r^2}{\mathrm{d}\log\mu} = -(n-1)\lambda^2(\sigma+\lambda) + O(\lambda^4),$$

which proves (66.49).

Remark 66.9 (Flat target, linear dilaton, and the critical constant). For flat D-dimensional target and linear dilaton $\Phi(X) = V_i X^i + \Phi_0$, the nonconstant tensor beta functions vanish, and the scalar Weyl anomaly is the central-charge condition

$$c_{\text{matter}} = D + 6\alpha' V^2.$$

Coupling to the bosonic reparametrization ghosts imposes $c_{\text{matter}} = 26$. This is a convention-sensitive statement about the constant part of $\widehat{\beta}^{\Phi}$; it is not the same as the vanishing of the nonconstant metric and B-field tensors.

66.5 Supersymmetric Sigma-Model CFTs

The supersymmetric cases are not separate from CFT; they are additional structure on the same two-dimensional local QFT. A $\mathcal{N} = (1,1)$ sigma model is obtained by replacing X by a map of the source supermanifold into M. Its component fields include X^i and two-dimensional Majorana fermions valued in X^*TM. The target metric connection is replaced by the torsionful connections

$$\nabla^{(\pm)} = \nabla^G \pm \frac{1}{2}G^{-1}H$$

in the left- and right-moving fermion covariant derivatives.

The heterotic $\mathcal{N} = (0,1)$ sigma model adds two-dimensional fermions valued in a vector bundle $E \to M$ with connection A. Its assumptions are stronger than those of the bosonic model: E must carry a compatible fiber metric, the spin structure on Σ must be specified, and gauge and gravitational anomalies must be cancelled by the B-field transformation law. In a convention where Tr denotes the gauge trace and tr the Lorentz trace, the Green–Schwarz-improved three-form is

$$H = \mathrm{d}B + \frac{\alpha'}{4}\big(\omega_3(\nabla^+) - \omega_3(A)\big), \qquad \mathrm{d}H = \frac{\alpha'}{4}\big(\mathrm{tr}\, R(\nabla^+)\wedge R(\nabla^+) - \mathrm{Tr}\, F\wedge F\big). \tag{66.54}$$

Here ∇^+ is the torsionful target connection chosen in this representative. Replacing ∇^+ by another connection changes (66.54) by a local counterterm and hence by a scheme change, provided the Chern–Simons convention is changed consistently.

Controlled Approximation 66.10 (Heterotic one-loop representative)**.** In the large-radius and weak-bundle-curvature expansion, the heterotic one-loop Weyl-anomaly representative contains the gauge equation

$$\widehat{\beta}^A_i = -\frac{\alpha'}{2}\left(D^j F_{ji} + \frac{1}{2}H_i{}^{jk}F_{jk} - 2\nabla^j\Phi\, F_{ji}\right) + O(\alpha'^2), \tag{66.55}$$

where D is induced by A and by the Levi-Civita connection on target indices. The metric and B-field equations are modified at the same order by gauge-curvature terms and by the Bianchi identity (66.54). The invariant statement is not the vanishing of the displayed coordinate tensor alone, but the vanishing of the local Weyl anomaly modulo target diffeomorphisms, bundle gauge transformations, and B-field gauge transformations.

Derivation of the displayed gauge equation. The pullback connection couples the bundle fermions through $A_i(X)\partial X^i$. In a covariant background-field expansion, the linear fluctuation of this coupling is $F_{ij}\xi^i\partial X^j$, and the quadratic fluctuation contains $D_iF_{jk}\xi^i\xi^j\partial X^k$. The one-loop tadpole therefore gives a pole proportional to D^jF_{ji}. The mixed contraction with the torsion vertex in the B-coupling contributes $\frac{1}{2}H_i{}^{jk}F_{jk}$. These two terms are the ordinary coordinate beta function for A in this scheme.

The dilaton-gradient term is a redundant direction. The vector field $W^i = \alpha'\nabla^i\Phi$ acts on the pulled-back connection by

$$(\mathcal{L}_W A)_i = W^j F_{ji} + D_i(W^j A_j).$$

The second term is a bundle gauge transformation, so the gauge-invariant part of the Weyl-anomaly representative receives only W^jF_{ji}. This is the $-2\nabla^j\Phi\, F_{ji}$ term inside the parentheses in (66.55).

For $\mathcal{N} = (2,2)$, the target must carry compatible complex structures. In the basic Kähler case, M is a complex manifold with Kähler metric

$$G_{I\bar{J}} = \partial_I\partial_{\bar{J}}K$$

in local coordinates. The classical $U(1)_V \times U(1)_A$ currents are the candidates for the R-symmetry currents of the infrared superconformal algebra. Their anomaly is controlled by $c_1(TM)$ in the large-radius presentation. Thus $c_1(TM) = 0$ is the first topological condition for an unbroken axial R-symmetry, and Ricci-flatness of a Kähler metric is the one-loop version of (66.11). In a compact Kähler class with $c_1(TM) = 0$, Yau's theorem supplies a Ricci-flat representative; the QFT statement that the corresponding sigma model is a conformal field theory is the existence of the continuum SCFT obtained after all higher-order local terms and nonperturbative sectors of maps $\Sigma \to M$ are included.

$\mathcal{N} = (2,2)$ **one-loop Kähler beta tensor.** Assume the regulator preserves $\mathcal{N} = (2,2)$ supersymmetry and that the target is Kähler in the coordinate patch under consideration. In a local superspace scheme whose counterterms are Kähler-potential counterterms, the one-loop metric beta tensor is

$$\beta_{I\bar{J}} = \alpha' R_{I\bar{J}} = -\alpha' \partial_I \partial_{\bar{J}} \log\det(G_{K\bar{L}}), \tag{66.56}$$

and the II and $\bar{I}\bar{J}$ components vanish in holomorphic coordinates. A one-loop conformal representative therefore requires the Ricci form

$$\rho = -\mathrm{i}\partial\bar{\partial} \log\det(G_{K\bar{L}})$$

to vanish in its Kähler class, modulo Kähler-potential redefinitions and target diffeomorphisms.

For a Kähler metric, the only nonzero Christoffel symbols are $\Gamma^I{}_{JK} = G^{I\bar{L}}\partial_J G_{K\bar{L}}$ and their complex conjugates. Hence

$$R_{I\bar{J}} = -\partial_{\bar{J}}\Gamma^K{}_{IK} = -\partial_I\partial_{\bar{J}} \log\det(G_{K\bar{L}}),$$

where the last equality is the matrix identity $\partial_I \log\det G = \mathrm{tr}(G^{-1}\partial_I G)$. The background-field one-loop divergence is the restriction of Lemma 66.3 to complex coordinates. Supersymmetry rules out a local metric counterterm that is not of type $(1,1)$ and, in the stated superspace scheme, writes the counterterm as a local correction to K. This gives (66.56).

Remark 66.11 (Higher-loop supersymmetric sigma-model status). For two-dimensional supersymmetric nonlinear sigma models, higher-loop beta functions can be computed in superspace schemes that preserve the manifest two-dimensional supersymmetry. In $\mathcal{N} = (2,2)$ Kähler models, the classic four-loop calculations exhibit the first scheme-invariant higher-curvature obstruction in the corresponding supersymmetric scheme, while many lower-loop terms can be moved by local Kähler-potential and target-coordinate redefinitions. This statement is used here only as a perturbative large-radius status boundary: it records what the local superspace calculation controls, but it does not by itself prove nonperturbative existence of a compact Calabi–Yau sigma-model SCFT, nor does it specify the global CFT Hilbert space.

The local operators corresponding to Kähler and complex-structure deformations are represented at large radius by

$$\delta G_{I\bar{J}}(X)\psi^I\bar{\psi}^{\bar{J}}, \qquad \delta J_{\bar{I}}{}^J(X)G_{J\bar{K}}\psi^{\bar{I}}\bar{\psi}^{\bar{K}},$$

with the precise operator representatives depending on the regularization and on the choice of local coordinates on the conformal manifold. The cohomological labels $H^{1,1}(M)$ and $H^{d-1,1}(M)$ are the large-radius tangent-space labels; the corresponding QFT objects are the renormalized local operators together with their contact-term convention.

66.6 Toroidal Sigma Models and Narain Lattice CFTs

Flat torus sigma models are the first place where the target-space description and the exact operator description can both be written without an asymptotic expansion. The exact description is a lattice CFT. We record it before discussing orbifolds because most elementary orbifolds act on a specified lattice theory, and because locality, spin, and modular covariance are already visible in the parent theory.

Let
$$\mathbb{R}^{n_L,n_R} = \mathbb{R}^{n_L} \oplus \mathbb{R}^{n_R}$$
with bilinear form
$$k \circ \ell = k_L \cdot \ell_L - k_R \cdot \ell_R.$$
For a lattice $\Gamma \subset \mathbb{R}^{n_L,n_R}$, define the dual lattice
$$\Gamma^\vee = \{x \in \Gamma \otimes_{\mathbb{Z}} \mathbb{R} \mid x \circ y \in \mathbb{Z} \text{ for all } y \in \Gamma\}.$$
The lattice is integral if $\Gamma \subset \Gamma^\vee$, even if $k \circ k \in 2\mathbb{Z}$ for every $k \in \Gamma$, and unimodular if $\Gamma = \Gamma^\vee$.

The chiral bosons are normalized by
$$X_L^i(z)X_L^j(0) \sim -\delta^{ij}\log z, \qquad X_R^{i'}(\bar z)X_R^{j'}(0) \sim -\delta^{i'j'}\log \bar z.$$
For $k = (k_L, k_R) \in \Gamma$, the exponential field is
$$V_k(z,\bar z) = \varepsilon_k : \exp(\mathrm{i}k_L \cdot X_L(z) + \mathrm{i}k_R \cdot X_R(\bar z)) :, \tag{66.57}$$
where the symbols ε_k are cocycle operators. They multiply by
$$\varepsilon_k\varepsilon_\ell = \epsilon(k,\ell)\varepsilon_{k+\ell}, \qquad \epsilon(k,\ell) \in \{\pm 1\}, \tag{66.58}$$
after a choice of basis and cocycle representative. Oscillator descendants are obtained by applying $\partial^r X_L^i$ and $\bar\partial^s X_R^{i'}$. The conformal weights of a descendant at oscillator levels $N, \bar N \in \mathbb{Z}_{\geq 0}$ are
$$h = \frac{1}{2}k_L^2 + N, \qquad \bar h = \frac{1}{2}k_R^2 + \bar N. \tag{66.59}$$

Proposition 66.12 (Locality, spin, and cocycles)**.** *The exponential fields (66.57) are mutually local bosonic fields only if Γ is integral and even. Given an even integral lattice, the cocycle can be chosen so that*
$$\epsilon(k,\ell+m)\epsilon(\ell,m) = \epsilon(k,\ell)\epsilon(k+\ell,m), \tag{66.60}$$
$$\epsilon(\ell,k) = (-1)^{k\circ\ell}\epsilon(k,\ell). \tag{66.61}$$
With such a choice the OPE algebra of the exponential fields is associative and has the required exchange law.

Proof. The free-field contraction gives
$$V_k(z,\bar z)V_\ell(0) \sim \epsilon(k,\ell)\, z^{k_L\cdot\ell_L}\,\bar z^{k_R\cdot\ell_R}V_{k+\ell}(0) + \cdots . \tag{66.62}$$
Taking z once counterclockwise around the origin sends $\bar z$ once clockwise around the origin and therefore multiplies the leading term by
$$\exp(2\pi\mathrm{i}(k_L \cdot \ell_L - k_R \cdot \ell_R)) = \exp(2\pi\mathrm{i}k \circ \ell).$$
Single-valued separated-point correlators of the exponential fields require $k \circ \ell \in \mathbb{Z}$ for all $k, \ell \in \Gamma$, which is integrality. The spin of V_k is
$$h - \bar h = \frac{1}{2}(k_L^2 - k_R^2) = \frac{1}{2}k \circ k,$$

modulo integers from oscillators, so integer spin requires evenness.

Associativity of the product of three cocycle operators is exactly (66.60). The exchange of two radially ordered exponentials changes $z-\zeta$ to $\zeta - z$ and $\bar z - \bar\zeta$ to $\bar\zeta - \bar z$, producing the phase $(-1)^{k\circ\ell}$. Hence the cocycle must obey (66.61). To see existence concretely, choose an integral basis e_i of Γ, write $k=\sum_i m^i e_i$, $\ell = \sum_i n^i e_i$, and put

$$A_{ij} = e_i \circ e_j, \qquad \epsilon(k,\ell) = (-1)^{\sum_{i>j} A_{ij} m^i n^j}. \tag{66.63}$$

The exponent is bilinear, so (66.60) follows by expanding both sides. Since A_{ii} is even and $A_{ij} = A_{ji}$,

$$\sum_{i>j} A_{ij} n^i m^j - \sum_{i>j} A_{ij} m^i n^j \equiv \sum_{i,j} A_{ij} m^i n^j = k\circ\ell \pmod 2,$$

which gives (66.61). □

The torus amplitude of the oscillator Fock spaces and lattice zero modes is

$$Z_\Gamma(\tau,\bar\tau) = \frac{1}{\eta(\tau)^{n_L}\overline{\eta(\tau)}^{n_R}} \sum_{k\in\Gamma} q^{k_L^2/2} \bar q^{k_R^2/2}, \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau}. \tag{66.64}$$

This formula is a trace over the Hilbert space on the circle. If $n_L \neq n_R$, the theory has chiral gravitational anomaly $c_L - c_R = n_L - n_R$; then the torus amplitude is naturally a section of the corresponding determinant line rather than an invariant scalar function.

Lemma 66.13 (Modular behavior of the lattice trace)**.** *For an even integral Γ, the lattice sum in (66.64) has trivial $T:\tau\mapsto\tau+1$ phase before the oscillator determinant is included. The full trace obeys*

$$Z_\Gamma(\tau+1,\bar\tau+1) = \exp\left(-\frac{2\pi\mathrm{i}}{24}(n_L - n_R)\right) Z_\Gamma(\tau,\bar\tau). \tag{66.65}$$

Under $S:\tau\mapsto -1/\tau$, Poisson summation sends the lattice Γ to $\Gamma^\vee$. Therefore an ordinary scalar modular invariant lattice partition function requires unimodularity together with the cancellation of the gravitational-anomaly phase. In particular, an even unimodular lattice of signature (n_L,n_R) can exist only when $n_L - n_R \equiv 0 \pmod 8$, whereas a standalone scalar torus partition function further requires $n_L - n_R \equiv 0 \pmod{24}$. For an ordinary toroidal sigma model $n_L = n_R$, the anomaly condition is automatic.

Proof. For the T-transformation, each lattice term is multiplied by

$$\exp\bigl(\pi\mathrm{i}(k_L^2 - k_R^2)\bigr) = \exp(\pi\mathrm{i}k\circ k),$$

which is 1 precisely for an even lattice. The Dedekind eta function obeys $\eta(\tau+1) = \mathrm{e}^{2\pi\mathrm{i}/24}\eta(\tau)$, and its antiholomorphic factor carries the conjugate phase. This gives (66.65).

For S, write the theta sum as a Gaussian sum over Γ with positive real part controlled by $\operatorname{Im}\tau > 0$. Poisson summation replaces the sum over Γ by a sum over $\Gamma^\vee$ and produces the Gaussian determinant factors $(-\mathrm{i}\tau)^{n_L/2}(\mathrm{i}\bar\tau)^{n_R/2}$. These are exactly the factors by which $\eta(\tau)^{n_L}\overline{\eta(\tau)}^{n_R}$ transforms, after the same branch convention is used for the determinant line. Thus S closes on a single lattice trace only when $\Gamma = \Gamma^\vee$; otherwise it mixes the trace with the discriminant-group components. The classification theorem for even unimodular lattices gives the congruence $n_L - n_R \equiv 0 \pmod 8$. The remaining scalar-invariance condition is the absence, or cancellation by another sector, of the chiral gravitational anomaly, which is the T-phase in (66.65). □

We now relate this exact operator construction to the Lagrangian torus sigma model. Let the target be

$$T^d = \mathbb{R}^d/2\pi\mathbb{Z}^d$$

with constant metric G and constant antisymmetric B-field. Choose a vielbein E with $G = E^T E$. A circle state is labeled by momentum $m \in (\mathbb{Z}^d)^\vee$ and winding $w \in \mathbb{Z}^d$. With the action normalization (66.1), the left and right momenta are

$$k_L(m,w) = \frac{1}{\sqrt{2\alpha'}}\left(E^{-T}(\alpha' m - Bw) + Ew\right), \tag{66.66}$$

$$k_R(m,w) = \frac{1}{\sqrt{2\alpha'}}\left(E^{-T}(\alpha' m - Bw) - Ew\right). \tag{66.67}$$

The set of all $(k_L(m,w), k_R(m,w))$ is a lattice $\Gamma(G,B) \subset \mathbb{R}^{d,d}$, and

$$k(m,w) \circ k(m',w') = m(w') + m'(w). \tag{66.68}$$

Thus $k(m,w) \circ k(m,w) = 2m(w)$. The B-dependence cancels from the integral pairing; it changes the embedding of the fixed abstract lattice $(\mathbb{Z}^d)^\vee \oplus \mathbb{Z}^d$ into $\mathbb{R}^{d,d}$, not its bilinear form.

The moduli of such toroidal CFTs are consequently better described as positive d-planes in $\mathbb{R}^{d,d}$ modulo automorphisms of the integral lattice:

$$\mathcal{M}_{\text{torus},d} = O(d,d;\mathbb{Z})\backslash O(d,d;\mathbb{R})/(O(d)\times O(d)), \tag{66.69}$$

up to the separate constant-dilaton coupling to the Euler characteristic and the usual fixed-point qualifications of an orbifold quotient. A T-duality is an isomorphism of the exact CFT data: it is an automorphism of the even unimodular lattice together with the induced map of cocycles, vertex operators, stress tensors, and correlation functions. A Buscher path-integral derivation is a derivation of this equality in a Lagrangian coordinate patch; the exact statement is the lattice-theoretic isomorphism of the local QFT.

66.7 Buscher Duality from a Gauged Isometry

We now record the local path-integral derivation of the Buscher rules. The derivation is formal unless a regulator preserving the gauge symmetry and the measure transformation is supplied, but it is the correct perturbative calculation behind the toroidal lattice isomorphism above.

Let X^0 be a coordinate along a circle isometry, and assume

$$\partial_0 G_{\mu\nu} = \partial_0 B_{\mu\nu} = \partial_0 \Phi = 0.$$

Write i,j for the remaining target coordinates and

$$E_{\mu\nu} = G_{\mu\nu} + B_{\mu\nu}.$$

In local light-cone coordinates $\sigma^\pm$ on Σ, the isometry is gauged by replacing $\partial_\pm X^0$ with gauge fields $A_\pm$ and adding a Lagrange multiplier $\widetilde{X}^0$ for flatness:

$$\begin{aligned} S_{\text{gauged}} = \frac{1}{4\pi\alpha'}\int &\Big[E_{00}A_+A_- + E_{0i}A_+\partial_- X^i + E_{i0}\partial_+X^i A_- + E_{ij}\partial_+X^i\partial_-X^j\Big] \\ &+ \frac{1}{4\pi\alpha'}\int \widetilde{X}^0(\partial_+A_- - \partial_-A_+). \end{aligned} \tag{66.70}$$

Integrating out $\widetilde{X}^0$ forces $A_\pm = \partial_\pm X^0$ locally and recovers the original model, modulo the usual global sum over circle bundles. Integrating out $A_\pm$ instead gives the dual model.

Proposition 66.14 (Buscher rules)**.** *The Gaussian integration over $A_\pm$ in (66.70) gives the dual background*

$$E'_{00} = \frac{1}{E_{00}}, \qquad E'_{0i} = \frac{E_{0i}}{E_{00}}, \qquad E'_{i0} = -\frac{E_{i0}}{E_{00}}, \tag{66.71}$$

$$E'_{ij} = E_{ij} - \frac{E_{i0}E_{0j}}{E_{00}}. \tag{66.72}$$

Equivalently,

$$G'_{00} = \frac{1}{G_{00}}, \qquad G'_{0i} = \frac{B_{0i}}{G_{00}}, \qquad B'_{0i} = \frac{G_{0i}}{G_{00}}, \tag{66.73}$$

$$G'_{ij} = G_{ij} - \frac{G_{0i}G_{0j} - B_{0i}B_{0j}}{G_{00}}, \tag{66.74}$$

$$B'_{ij} = B_{ij} - \frac{G_{0i}B_{0j} - B_{0i}G_{0j}}{G_{00}}. \tag{66.75}$$

The one-loop determinant of the $A_\pm$ Gaussian changes the dilaton by

$$\Phi' = \Phi - \frac{1}{2}\log G_{00}. \tag{66.76}$$

Proof. Integrate the Lagrange-multiplier term by parts and collect the terms linear in $A_\pm$:

$$S_{\text{gauged}} = \frac{1}{4\pi\alpha'}\int \left[E_{00}A_+A_- + A_+J_- + J_+A_- + E_{ij}\partial_+X^i\partial_-X^j\right],$$

where

$$J_- = E_{0i}\partial_-X^i + \partial_-\widetilde{X}^0, \qquad J_+ = E_{i0}\partial_+X^i - \partial_+\widetilde{X}^0.$$

The equations of motion for $A_\pm$ are

$$E_{00}A_- + J_- = 0, \qquad E_{00}A_+ + J_+ = 0.$$

Substituting the solution back completes the square and gives the effective term

$$-\frac{1}{E_{00}}J_+J_-.$$

Reading off the coefficients of $\partial_+\widetilde{X}^0\partial_-\widetilde{X}^0$, $\partial_+\widetilde{X}^0\partial_-X^i$, $\partial_+X^i\partial_-\widetilde{X}^0$, and $\partial_+X^i\partial_-X^j$ gives (66.71)–(66.72). Splitting $E' = G' + B'$ into symmetric and antisymmetric parts gives (66.73)–(66.75).

The functional determinant from the Gaussian integral over $A_\pm$ is the only remaining convention-sensitive point. A gauge-invariant cell regularization makes its coefficient transparent. First take G_{00} to be constant and choose a finite cell decomposition of the closed source surface with N_0 vertices, N_1 edges, and N_2 faces. Put the gauged field A on edges, the gauge parameter on vertices, and the dual scalar $\widetilde{X}^0$ on faces. Normalize the scalar and gauge-parameter measures by the L^2 norm induced by G_{00}, and normalize the dual scalar measure by the dual metric $G'_{00} = G_{00}^{-1}$.

Ignoring only zero modes and holonomy sums, which do not change the local coefficient of $\log G_{00}$, the G_{00}-dependent factors in the finite-dimensional Gaussian are:

$$\underbrace{(G_{00})^{-N_1/2}}_{A\text{-edge Gaussian}} \quad \underbrace{(G_{00})^{N_0/2}}_{\text{gauge normalization}} \quad \underbrace{(G_{00})^{N_2/2}}_{\text{dual scalar measure}} .$$

The product is

$$(G_{00})^{(N_0-N_1+N_2)/2} = (G_{00})^{\chi(\Sigma)/2}. \tag{66.77}$$

This is the finite-dimensional origin of the half coefficient; it is not a choice that can be read from the classical saddle alone.

For slowly varying $G_{00}(X^i)$, locality of the ultraviolet regulator promotes the constant factor (66.77) to the local curvature counterterm

$$\exp\left[\frac{1}{8\pi}\int_\Sigma \sqrt{\gamma}\, R_\gamma \, \log G_{00}(X)\right],$$

up to terms with derivatives of G_{00} that belong to the same higher-derivative α'-scheme choice as the other Buscher covariance corrections. Since the dilaton coupling in (66.1) contributes $\exp[-(4\pi)^{-1}\int \sqrt{\gamma}\, R_\gamma \Phi]$ to the path integral, this local determinant factor is exactly equivalent to (66.76). This last step is where the path-integral measure, rather than only the classical action, enters the Buscher rules. □

Remark 66.15 (Involutivity). Applying the transformation (66.73)–(66.75) twice returns (G, B). The dilaton also returns to its original value after the second shift. In E-variables the rules are

$$E'_{00} = E_{00}^{-1}, \qquad E'_{0i} = E_{0i}E_{00}^{-1}, \qquad E'_{i0} = -E_{i0}E_{00}^{-1}, \qquad E'_{ij} = E_{ij} - E_{i0}E_{0j}E_{00}^{-1}.$$

A direct substitution gives $E''_{00} = E_{00}$, $E''_{0i} = E_{0i}$, $E''_{i0} = E_{i0}$, and $E''_{ij} = E_{ij}$. For the dilaton,

$$\Phi'' = \Phi - \frac{1}{2}\log G_{00} - \frac{1}{2}\log G'_{00} = \Phi - \frac{1}{2}\log G_{00} + \frac{1}{2}\log G_{00} = \Phi.$$

Controlled Approximation 66.16 (Beta functions and Buscher covariance)**.** Order by order in α', Buscher duality maps Weyl-anomaly coefficients of the original background to those of the dual background after the corresponding local field redefinitions have been chosen. This is a perturbative covariance statement about the sigma-model renormalization scheme. Nonperturbative existence or isomorphism of two arbitrary curved-target continuum CFTs requires separate construction.

66.8 WZW Models, Affine Currents, and Cosets

Wess–Zumino–Witten models are sigma models whose exact conformal structure is fixed by an affine current algebra. We give the construction in a normalization independent of the four-dimensional gauge-theory trace normalization used elsewhere in this monograph.

Let G be a compact connected simply connected simple Lie group, let $\mathfrak{g}$ be its Lie algebra, and let $(\cdot,\cdot)$ be the basic invariant inner product, normalized so that long roots have squared length 2. Choose an orthonormal basis T_a of $\mathfrak{g}$, with

$$[T_a, T_b] = f_{ab}{}^c T_c, \qquad f_{abc} = f_{ab}{}^d \delta_{dc}.$$

The dual Coxeter number $h^\vee$ is defined by

$$f_{acd}f_{bcd} = 2h^\vee \delta_{ab}. \tag{66.78}$$

This convention makes the adjoint quadratic Casimir equal to $2h^\vee$.

Let $g : \Sigma \to G$. The topological term is defined using the integral class

$$[\Omega_G] \in H^3(G, \mathbb{Z}),$$

chosen as the generator compatible with the basic inner product. If $\Sigma = \partial B$ and $\widetilde{g} : B \to G$ extends g, the WZW action at level $k \in \mathbb{Z}$ is

$$S_k[g] = \frac{k}{4\pi}\int_\Sigma (g^{-1}\mathrm{d}g) \wedge *(g^{-1}\mathrm{d}g) + 2\pi \mathrm{i} k \int_B \widetilde{g}^*\Omega_G. \tag{66.79}$$

The first term is the sigma-model kinetic term on G with bi-invariant metric $k(\,\cdot\,,\,\cdot\,)$. The second term is most naturally the holonomy of a gerbe with curvature $2\pi k\,\Omega_G$; local two-form potentials represent it only after choosing a trivialization on coordinate patches.

Quantization of the WZW level. The exponent $\exp(-S_k[g])$ is independent of the filling B for every closed two-manifold Σ if $k \in \mathbb{Z}$. Conversely, for simply connected simple G, filling independence for all maps requires k to be integral in this normalization.

Let $(B_1, \widetilde{g}_1)$ and $(B_2, \widetilde{g}_2)$ be two fillings of the same boundary map. Gluing B_1 to $-B_2$ gives a closed oriented three-manifold C and a map $\widehat{g} : C \to G$. The change of the topological part of the action is

$$2\pi \mathrm{i} k \int_C \widehat{g}^*\Omega_G.$$

Since Ω_G represents an integral cohomology class, the integral is an integer. Thus the exponential is unchanged when $k \in \mathbb{Z}$. For compact simply connected simple G, $H_3(G, \mathbb{Z}) \simeq \mathbb{Z}$, and there is a map $C \to G$ pairing to 1 with $[\Omega_G]$. Applying filling independence to that map forces $\exp(-2\pi \mathrm{i} k) = 1$, hence $k \in \mathbb{Z}$.

In a local complex coordinate z, the equations of motion can be written as chiral conservation laws

$$\bar{\partial} J = 0, \qquad \partial \bar{J} = 0,$$

where $J = J^a T_a$ and $\bar{J} = \bar{J}^a T_a$ are the left and right currents. Their separated-point OPEs are the exact affine current algebra

$$J^a(z)J^b(0) \sim \frac{k\delta^{ab}}{z^2} + \frac{f_{ab}{}^c J^c(0)}{z}, \qquad \bar{J}^a(\bar{z})\bar{J}^b(0) \sim \frac{k\delta^{ab}}{\bar{z}^2} + \frac{f_{ab}{}^c \bar{J}^c(0)}{\bar{z}}. \tag{66.80}$$

The mixed $J\bar{J}$ OPE is regular at separated points. Equivalently, with

$$J^a(z) = \sum_{n\in\mathbb{Z}} J^a_n z^{-n-1},$$

the modes obey

$$[J^a_m, J^b_n] = f_{ab}{}^c J^c_{m+n} + k\, m\, \delta^{ab}\delta_{m+n,0}. \tag{66.81}$$

The central term is the quantum remnant of the Wess–Zumino term. The integer k is therefore simultaneously the gerbe level and the affine level.

Proposition 66.17 (Sugawara stress tensor)**.** *For $k + h^\vee \neq 0$, define*

$$T_G(z) = \frac{1}{2(k + h^\vee)} \sum_a : J^a J^a : (z). \tag{66.82}$$

Then J^a has conformal weight 1, and T_G has central charge

$$c(G, k) = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \tag{66.83}$$

Proof. The coefficient in (66.82) is fixed by the requirement that the $T_G J^b$ OPE have leading singular terms

$$T_G(z) J^b(0) \sim \frac{J^b(0)}{z^2} + \frac{\partial J^b(0)}{z}.$$

Contract one current in $: J^a J^a :$ with J^b. The central term in (66.80) gives $2kJ^b/z^2$ before the overall coefficient. The simple-pole contraction gives the adjoint action twice. Normal ordering the resulting single current past the remaining current adds the adjoint Casimir contribution $2h^\vee J^b/z^2$, using (66.78). Thus the numerator is $2(k+h^\vee)J^b/z^2$, which fixes the factor $[2(k+h^\vee)]^{-1}$; the first-order pole is fixed by translation covariance of the normal-ordered product.

For the $T_G T_G$ OPE, the fourth-order pole comes from double contractions of the two quadratic normal-ordered products. The central-central contractions give $2k^2 \dim \mathfrak{g}$. The current-algebra contractions with two structure constants give the quantum correction $2kh^\vee \dim \mathfrak{g}$. Dividing by $[2(k + h^\vee)]^2$ and comparing with

$$T_G(z) T_G(0) \sim \frac{c/2}{z^4} + \frac{2T_G(0)}{z^2} + \frac{\partial T_G(0)}{z}$$

gives (66.83). The remaining second- and first-order poles reproduce the Virasoro action because the same coefficient already makes each current a primary field of weight 1. □

66.8.1 Knizhnik–Zamolodchikov Equations from Current Ward Identities

The affine current algebra does more than determine the stress tensor. It turns chiral correlation functions of WZW primary fields into flat sections of a logarithmic connection on the configuration space of marked points. We give the derivation because the equation is often quoted as representation theory data, while in the CFT it is a direct consequence of the local current OPE and the Sugawara expression for translations.

Let V_λ be the finite-dimensional irreducible $\mathfrak{g}$-module of highest weight $\lambda \in P_k^+$, and let

$$t^a_\lambda : V_\lambda \to V_\lambda$$

be the action of the basis element T_a. Thus $[t^a_\lambda, t^b_\lambda] = f_{ab}{}^c t^c_\lambda$, and the quadratic Casimir acts by

$$\sum_a t^a_\lambda t^a_\lambda = (\lambda, \lambda + 2\rho)\, 1_{V_\lambda}$$

in the normalization of (66.78). A chiral WZW primary field with horizontal type V_λ is an intertwining field $\Phi_\lambda(v, z)$, $v \in V_\lambda$, whose current OPE is

$$J^a(\zeta)\Phi_\lambda(v, z) \sim \frac{\Phi_\lambda(t^a_\lambda v, z)}{\zeta - z} + \text{regular}. \tag{66.84}$$

Equivalently, $J^a_n \Phi_\lambda(v,0)|0\rangle = 0$ for $n > 0$, while J^a_0 acts as t^a_λ.

For distinct points $z_1, \dots, z_n \in \mathbb{C}$, choose primary labels $\lambda_1, \dots, \lambda_n$. A genus-zero chiral block is a multilinear function

$$\mathcal{F}(z_1, \dots, z_n) : V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_n} \to \mathbb{C}$$

obtained by inserting the corresponding chiral primary fields and projecting to the vacuum channel at infinity. We write t^a_i for $t^a_{\lambda_i}$ acting on the i-th tensor factor, and

$$\Omega_{ij} = \sum_a t^a_i t^a_j \qquad (i \neq j). \tag{66.85}$$

The global current Ward identity is

$$\mathcal{F} \circ \left(\sum_{i=1}^n t^a_i \right) = 0, \tag{66.86}$$

which is the statement that the contour integral of J^a around all finite insertions can be moved to infinity and annihilates the vacuum.

Theorem 66.18 (Knizhnik–Zamolodchikov equation)**.** *For every $i = 1, \dots, n$, the chiral block satisfies*

$$(k + h^\vee)\, \partial_{z_i} \mathcal{F} = \mathcal{F} \circ \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}. \tag{66.87}$$

With invariant Hermitian forms used to identify V_{λ_i} with its dual, the same equation may be written for vector-valued blocks with Ω_{ij} acting on the left.

Proof. The translation Ward identity gives

$$\partial_{z_i} \Phi_{\lambda_i}(v_i, z_i) = \Phi_{\lambda_i}(L_{-1} v_i, z_i).$$

Apply the Sugawara formula (66.82) to the affine primary state v_i. In the mode expression

$$L_{-1} = \frac{1}{2(k + h^\vee)} \sum_{a, m \in \mathbb{Z}} : J^a_{-1-m} J^a_m :,$$

all terms vanish on v_i except the two terms with $m = 0$ and $m = -1$. Their sum is

$$\frac{1}{2(k + h^\vee)} \sum_a \left(J^a_{-1} J^a_0 + J^a_0 J^a_{-1} \right) v_i = \frac{1}{k + h^\vee} \sum_a J^a_{-1} t^a_i v_i, \tag{66.88}$$

because $\sum_a [J^a_0, J^a_{-1}] = \sum_{a,c} f_{aa}{}^c J^c_{-1} = 0$.

It remains to evaluate a block with J^a_{-1} acting on the i-th primary. By definition of modes around z_i,

$$J^a_{-1} \Phi_{\lambda_i}(v_i, z_i) = \oint_{z_i} \frac{\mathrm{d}\zeta}{2\pi \mathrm{i}} \frac{J^a(\zeta) \Phi_{\lambda_i}(v_i, z_i)}{\zeta - z_i}.$$

Insert this contour integral in the chiral block. Deforming the contour away from z_i gives the negative of the sum of small contours around the other insertions; the integrand has residue $\Phi_{\lambda_j}(t^a_j v_j, z_j)/(z_j - z_i)$ at z_j. Therefore

$$\mathcal{F}(\dots, J^a_{-1} t^a_i v_i, \dots) = \sum_{j \neq i} \frac{1}{z_i - z_j} \mathcal{F}(\dots, t^a_i v_i, \dots, t^a_j v_j, \dots).$$

Substituting this identity into (66.88) gives (66.87). □

Lemma 66.19 (Flatness and four-point reduction)**.** *The operators*

$$\nabla_i = (k + h^\vee)\partial_{z_i} - \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} \tag{66.89}$$

define a flat logarithmic connection on the configuration space $\mathrm{Conf}_n(\mathbb{C})$*, after restricting to the invariant subspace* $(V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_n})^G$ *or, equivalently, imposing* (66.86)*. After fixing three points by a Mobius transformation, a four-point block with insertions at* $0, x, 1, \infty$ *obeys*

$$(k + h^\vee)\frac{\mathrm{d}}{\mathrm{d}x}\mathcal{F}(x) = \left(\frac{\Omega_{21}}{x} + \frac{\Omega_{23}}{x-1}\right)\mathcal{F}(x), \tag{66.90}$$

with Ω_{2j} *acting on the remaining invariant tensor space.*

Proof. Only commutators of Ω's sharing an index can be nonzero. The quadratic tensor

$$\Omega = \sum_a T_a \otimes T_a \in \mathfrak{g} \otimes \mathfrak{g}$$

is invariant under the diagonal adjoint action, hence

$$[\Omega_{ij}, t_i^b + t_j^b] = 0$$

on $V_{\lambda_i} \otimes V_{\lambda_j}$. Multiplying by t_k^b on a third tensor factor and summing over b gives, for distinct i, j, k,

$$[\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0. \tag{66.91}$$

These are the infinitesimal braid relations. Expanding $[\nabla_i, \nabla_j]$, the derivative terms cancel the double-pole derivatives of $(z_i - z_j)^{-1}$, while the remaining triple-index commutators are precisely the combinations in (66.91). Thus the curvature vanishes.

For $n = 4$, use conformal covariance to place the points at $(z_1, z_2, z_3, z_4) = (0, x, 1, \infty)$. The equation with $i = 2$ gives (66.90); terms involving the insertion at infinity are absorbed by the global Ward identity, which identifies the fourth action with minus the sum of the first three finite actions on invariant tensors. □

$SU(2)$ **four-fundamental reduction.** The abstract connection becomes a concrete finite system already for four spin-$\frac{1}{2}$ insertions in $SU(2)_k$. Let $V = \mathbb{C}^2$ with indices $\alpha, \beta \in \{0, 1\}$, let $\epsilon_{01} = 1 = -\epsilon_{10}$, and choose the generators normalized by

$$\sum_{a=1}^{3} (t^a)_\alpha{}^\gamma (t^a)_\beta{}^\delta = \delta_\alpha{}^\delta \delta_\beta{}^\gamma - \frac{1}{2}\delta_\alpha{}^\gamma \delta_\beta{}^\delta. \tag{66.92}$$

This is the $SU(2)$ specialization of the WZW convention above: the fundamental quadratic Casimir is $3/2$, while $h^\vee = 2$. The invariant space in $V^{\otimes 4}$ is two-dimensional. A convenient spanning pair is

$$\mathcal{I}_{s\alpha_1\alpha_2\alpha_3\alpha_4} = \epsilon_{\alpha_1\alpha_2}\epsilon_{\alpha_3\alpha_4}, \qquad \mathcal{I}_{t\alpha_1\alpha_2\alpha_3\alpha_4} = \epsilon_{\alpha_1\alpha_3}\epsilon_{\alpha_2\alpha_4},$$

with the third pairing $\mathcal{I}_u = \epsilon_{\alpha_1\alpha_4}\epsilon_{\alpha_2\alpha_3}$ obeying

$$\mathcal{I}_s - \mathcal{I}_t + \mathcal{I}_u = 0. \tag{66.93}$$

Using (66.92), equivalently $\Omega_{ij} = P_{ij} - \frac{1}{2}$ on two fundamental factors, one obtains

$$\Omega_{12}\mathcal{I}_s = -\frac{3}{2}\,\mathcal{I}_s, \qquad\qquad \Omega_{12}\mathcal{I}_t = -\mathcal{I}_s + \frac{1}{2}\,\mathcal{I}_t, \tag{66.94}$$

$$\Omega_{23}\mathcal{I}_s = -\frac{1}{2}\,\mathcal{I}_s + \mathcal{I}_t, \qquad\qquad \Omega_{23}\mathcal{I}_t = \mathcal{I}_s - \frac{1}{2}\,\mathcal{I}_t. \tag{66.95}$$

Thus, for a vector-valued block

$$\Psi(x) = F_s(x)\mathcal{I}_s + F_t(x)\mathcal{I}_t$$

with insertions at $0, x, 1, \infty$, the KZ equation is the explicit rank-two Fuchsian system

$$(k+2)\frac{\mathrm{d}}{\mathrm{d}x}\begin{pmatrix} F_s \\ F_t \end{pmatrix} = \left[\frac{1}{x}\begin{pmatrix} -\frac{3}{2} & -1 \\ 0 & \frac{1}{2} \end{pmatrix} + \frac{1}{x-1}\begin{pmatrix} -\frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}\right]\begin{pmatrix} F_s \\ F_t \end{pmatrix}. \tag{66.96}$$

Near $x = 0$, the residue matrix is triangular in this basis. Its eigenvalues $-3/2$ and $1/2$ are exactly the singlet and triplet eigenvalues

$$\frac{1}{2}(C_\mu - C_{\text{fund}} - C_{\text{fund}}), \qquad \mu = 0, 2,$$

where $C_0 = 0$, $C_2 = 4$, and $C_{\text{fund}} = 3/2$ in the present normalization. Hence the local monodromy exponents of the chiral invariant tensor before multiplying by external scalar prefactors are

$$-\frac{3}{2(k+2)}, \qquad \frac{1}{2(k+2)}.$$

This finite reduction is the elementary model for the general statement that KZ monodromy represents the braid action on WZW conformal blocks; the identification with the full unitary CFT still requires the sewing and positivity data of the completed WZW model.

The highest-weight representations that occur in the compact unitary WZW model at positive integral k are the integrable highest weights

$$P_k^+ = \{\lambda \in P^+ \mid (\lambda, \theta) \le k\}, \tag{66.97}$$

where P^+ is the set of dominant integral weights and θ is the highest root. A primary field Φ_λ has holomorphic conformal weight

$$h_\lambda = \frac{(\lambda, \lambda + 2\rho)}{2(k + h^\vee)}, \tag{66.98}$$

where ρ is the Weyl vector. The full diagonal WZW model has Hilbert space

$$\mathcal{H}_{G,k}^{\text{diag}} = \bigoplus_{\lambda \in P_k^+} \mathcal{H}_\lambda \otimes \overline{\mathcal{H}_\lambda}. \tag{66.99}$$

Other modular invariants correspond to different pairings of left and right integrable modules and, in general, to additional global data of the full CFT rather than to a different local affine OPE.

Now let $H \subset G$ be a compact connected subgroup with Lie algebra $\mathfrak{h}$. Let the embedding index x_e be defined by

$$(X, Y)_{\mathfrak{g}} = x_e (X, Y)_{\mathfrak{h}} \qquad X, Y \in \mathfrak{h},$$

where both inner products are basic. The $\mathfrak{h}$-currents inside the G_k current algebra have level

$$k_H = kx_e.$$

The coset stress tensor is

$$T_{G/H} = T_G - T_H, \tag{66.100}$$

where T_H is the Sugawara tensor built from the embedded $\mathfrak{h}$-currents at level k_H. The corresponding central charge is

$$c(G_k/H_{k_H}) = \frac{k \dim \mathfrak{g}}{k + h^\vee_{\mathfrak{g}}} - \frac{k_H \dim \mathfrak{h}}{k_H + h^\vee_{\mathfrak{h}}}. \tag{66.101}$$

Since T_H is built from a subalgebra of the G-currents, the difference (66.100) has regular OPE with every $\mathfrak{h}$-current. Thus coset local fields are represented by $\mathfrak{h}$-invariant branching components of G_k affine modules, with conformal weights given by the G weight minus the H weight modulo integers from descendants.

The same construction has a Lagrangian presentation as a gauged WZW model. The group H is gauged by coupling the H-current algebra to a two-dimensional gauge field and adding the local terms forced by the Polyakov–Wiegmann identity. Gauge consistency requires the left and right H-levels to match; otherwise the gauge variation is an anomaly. After gauge fixing, the physical local operators are the BRST cohomology classes. The equality between the gauged WZW presentation and the coset vertex algebra is therefore a statement about a gauge-fixed local QFT with its ghost system and BRST differential. A target-space quotient picture, when available, is a geometric shadow of this operator construction.

For the diagonal embedding

$$SU(2)_k \times SU(2)_1 \supset SU(2)_{k+1},$$

the coset central charge is

$$c_k = \frac{3k}{k+2} + 1 - \frac{3(k+1)}{k+3} = 1 - \frac{6}{(k+2)(k+3)}. \tag{66.102}$$

This gives the unitary Virasoro minimal-model central charges. The central charge is the first scalar invariant of the construction; the full CFT data are the coset Hilbert space, OPE coefficients, sewing constraints, and modular invariant built from the branching of affine representations.

66.8.2 Rank-One Solvable Cosets: Parafermions and the Cigar

The cosets $SU(2)_k/U(1)$ and $SL(2,\mathbb{R})_k/U(1)$ are small enough to compute explicitly, but they are also good tests of the logical order of the construction. The exact CFT is first a current-algebra coset with specified representation data. A gauged sigma-model metric and dilaton, when available, are then a useful representative in a controlled large-level regime. Reversing this order can hide selection rules, field identifications, reflection amplitudes, and normalizability conditions.

Definition 66.20 (The compact parafermion coset)**.** Let $k \in \mathbb{Z}_{>0}$. In the $SU(2)_k$ WZW model write the Cartan current as J^3, normalized so that the embedded abelian current algebra has level k and primary charges are naturally recorded by an integer m modulo $2k$. The compact parafermion coset is

$$\mathrm{PF}_k = SU(2)_k/U(1)_k, \qquad T_{\mathrm{PF}_k} = T_{SU(2)_k} - T_{U(1)_k}.$$

Its affine branching labels are pairs

$$(\ell, m), \qquad 0 \le \ell \le k, \quad m \in \mathbb{Z}/2k\mathbb{Z}, \qquad \ell + m \in 2\mathbb{Z}.$$

The selection rule means that the $U(1)$ charge must actually occur in the finite $SU(2)$ spin-$\ell/2$ multiplet before its affine descendants are included.

Equivalently, the integrable $SU(2)_k$ character decomposes as

$$\chi_\ell^{SU(2)_k}(q,z) = \sum_{\substack{m\in\mathbb{Z}/2k\mathbb{Z} \\ m\equiv\ell \pmod 2}} b^{(k)}_{\ell,m}(q)\, \Theta_{m,k}(q,z). \tag{66.103}$$

Here $\Theta_{m,k}$ is the abelian level-k theta character and $b^{(k)}_{\ell,m}$ is the parafermion branching function. The formula is a representation-theoretic decomposition of the $SU(2)_k$ affine module; it is not a pointwise quotient of a target manifold.

Parafermion central charge, weights, and identification. The compact parafermion coset has central charge

$$c(\mathrm{PF}_k) = \frac{3k}{k+2} - 1 = \frac{2(k-1)}{k+2}. \tag{66.104}$$

The holomorphic conformal weight of the coset primary labelled by (ℓ, m) is

$$h^{\mathrm{PF}}_{\ell,m} = \frac{\ell(\ell+2)}{4(k+2)} - \frac{m^2}{4k} \mod \mathbb{Z}, \tag{66.105}$$

and the branching labels are identified by

$$(\ell, m) \sim (k-\ell, m+k). \tag{66.106}$$

The central charge is the difference between the $SU(2)_k$ Sugawara central charge $3k/(k+2)$ and the abelian current central charge 1. For the weight, write the $SU(2)$ highest weight as $\lambda = \ell\omega$. Since $(\omega,\omega) = 1/2$ and $\rho = \omega$,

$$h_\ell^{SU(2)_k} = \frac{(\lambda, \lambda+2\rho)}{2(k+2)} = \frac{\ell(\ell+2)}{4(k+2)}.$$

An abelian charge m at level k, in this doubled-charge normalization, has conformal weight $m^2/(4k)$. Subtracting the abelian weight gives (66.105), with integer shifts coming from which affine descendants occur in the branching component. The identification (66.106) is generated by the simple current of $SU(2)_k$. Its compatibility with the displayed weight formula is checked directly:

$$h^{\mathrm{PF}}_{k-\ell,m+k} - h^{\mathrm{PF}}_{\ell,m} = -\frac{\ell+m}{2}.$$

This is integral precisely on the selected branching labels $\ell + m \in 2\mathbb{Z}$.

For $k = 2$ one obtains the Ising central charge $c = 1/2$: the labels $(1,1)$ and $(2,0)$ give the weights $1/16$ and $1/2$. For $k = 3$ one obtains $c = 4/5$, the three-state Potts value: the labels $(1,1)$ and $(2,0)$ give weights $1/15$ and $2/5$. The parafermion current can be represented by $(0,2)$; using (66.105) modulo integers gives

$$h^{\mathrm{PF}}_{0,2} = 1 - \frac{1}{k}.$$

Proposition 66.21 (Parafermion fusion, modular data, and diagonal invariant)**.** *Let $\mathcal{L}_k$ be the set of selected labels*

$$\mathcal{L}_k = \{(\ell, m) \mid 0 \le \ell \le k,\ m \in \mathbb{Z}/2k\mathbb{Z}, \ell + m \in 2\mathbb{Z}\},$$

and let $\mathcal{I}_k = \mathcal{L}_k/\sim$, where $(\ell, m) \sim (k-\ell, m+k)$. The simple-current action is free, so

$$|\mathcal{I}_k| = \frac{k(k+1)}{2}.$$

For orbit labels $[\ell, m] \in \mathcal{I}_k$, the fusion rule is

$$[\ell_1, m_1] \boxtimes [\ell_2, m_2] = \bigoplus_{\substack{|\ell_1-\ell_2| \le \ell_3 \le \min(\ell_1+\ell_2,\, 2k-\ell_1-\ell_2) \\ \ell_1+\ell_2+\ell_3 \in 2\mathbb{Z}}} [\ell_3, m_1 + m_2], \tag{66.107}$$

where the charge is reduced modulo $2k$ and the result is again read as an orbit. With the theta-function convention used in (66.103), the modular S-matrix on these orbits is

$$S^{\mathrm{PF}}_{[\ell,m],[\ell',m']} = \frac{2}{\sqrt{k(k+2)}} \sin\!\left(\frac{\pi(\ell+1)(\ell'+1)}{k+2}\right) \exp\!\left(-\frac{\pi i m m'}{k}\right). \tag{66.108}$$

The diagonal compact parafermion torus partition function is

$$Z^{\mathrm{diag}}_{\mathrm{PF}_k}(\tau, \bar\tau) = \sum_{[\ell,m] \in \mathcal{I}_k} b^{(k)}_{\ell,m}(\tau)\, \overline{b^{(k)}_{\ell,m}(\tau)}. \tag{66.109}$$

Moreover S^{PF} is unitary and the Verlinde formula recovers exactly the fusion rule (66.107):

$$N_{ab}{}^{c} = \sum_{x \in \mathcal{I}_k} \frac{S^{\mathrm{PF}}_{ax} S^{\mathrm{PF}}_{bx} \overline{S^{\mathrm{PF}}_{cx}}}{S^{\mathrm{PF}}_{[0,0]x}}, \qquad a, b, c \in \mathcal{I}_k, \tag{66.110}$$

where $N_{ab}{}^{c}$ is 1 when c appears in (66.107) and 0 otherwise.

Proof. The action $(\ell, m) \mapsto (k-\ell, m+k)$ has no fixed point: a fixed point would require $m \equiv m+k \pmod{2k}$, impossible for $k > 0$. Since $|\mathcal{L}_k| = k(k+1)$, the orbit count follows.

Fusion in the parent $SU(2)_k$ WZW model is the truncated tensor-product rule

$$\ell_3 = |\ell_1 - \ell_2|,\ |\ell_1 - \ell_2| + 2, \ldots, \min(\ell_1 + \ell_2, 2k - \ell_1 - \ell_2).$$

The abelian factor conserves $U(1)$ charge modulo $2k$. Taking the coset branching component therefore gives (66.107). The parity condition is automatic from the $SU(2)$ fusion parity and the two input selection rules.

The modular matrix is the product of the $SU(2)_k$ modular matrix and the complex conjugate of the $U(1)_k$ theta modular matrix, multiplied by the order 2 of the free simple-current identification. This gives (66.108). The displayed expression is well-defined on orbits because replacing (ℓ, m) by $(k-\ell, m+k)$ multiplies the sine factor by $(-1)^{\ell'}$ and the exponential by $(-1)^{m'}$, whose product is 1 on selected labels $\ell' + m' \in 2\mathbb{Z}$. The diagonal invariant is the branchwise diagonal invariant after quotienting by this free identification; it is equivalently one half of the selected-label diagonal sum before quotient.

It remains to check that no normalization has been lost in passing to orbits. Put

$$S^{su}_{\ell\ell'} = \sqrt{\frac{2}{k+2}}\,\sin\!\left(\frac{\pi(\ell+1)(\ell'+1)}{k+2}\right), \qquad S^{u(1)}_{mm'} = \frac{1}{\sqrt{2k}}\,\exp\!\left(\frac{\pi i m m'}{k}\right).$$

The finite sine and Fourier orthogonality relations give

$$\sum_{\ell'=0}^{k} S^{su}_{\ell\ell'} S^{su}_{\ell''\ell'} = \delta_{\ell,\ell''}, \qquad \sum_{m'\in\mathbb{Z}/2k\mathbb{Z}} S^{u(1)}_{mm'} \overline{S^{u(1)}_{m''m'}} = \delta_{m,m''}.$$

Before quotienting, the branching functions transform with the product kernel $\widehat{S}_{(\ell,m),(\ell',m')} = S^{su}_{\ell\ell'}\overline{S^{u(1)}_{mm'}}$. The selection rule is the zero-monodromy condition for the simple current $\mathsf{J}(\ell,m) = (k-\ell, m+k)$, and the previous paragraph shows that $\widehat{S}$ is constant on selected J-orbits. Since every orbit has two elements, the induced orbit matrix is $2\widehat{S}$, namely (66.108). Summing first over selected labels and then dividing by the orbit size gives

$$\sum_{x\in\mathcal{I}_k} S^{\mathrm{PF}}_{ax}\overline{S^{\mathrm{PF}}_{bx}} = \delta_{a,b},$$

with $\delta_{a,b}$ understood on orbit labels.

For (66.110), insert (66.108). The finite Fourier sum over m_x imposes $m_c \equiv m_a + m_b \pmod{2k}$, up to the same free field-identification orbit. The remaining sine-kernel quotient is the $SU(2)_k$ Verlinde coefficient, equivalently the truncated Clebsch–Gordan condition on ℓ_c. These two constraints are precisely (66.107). Because the simple-current action is free, no fixed-point resolution or multiplicity factor is hidden in this last step. □

For $k=2$, the three orbits may be represented by $(0,0)$, $(1,1)$, and $(2,0)$. The fusion rule gives

$$(1,1)\boxtimes(1,1) = (0,0)\oplus(2,0), \qquad (2,0)\boxtimes(2,0) = (0,0),$$

which is the Ising fusion algebra under the identifications $(1,1)=\sigma$ and $(2,0)=\varepsilon$. For general k, the field $(0,2)$ is a simple current of order k and shifts $[\ell,m]$ to $[\ell,m+2]$.

Controlled Approximation 66.22 (The bell as a gauged-WZW representative)**.** Choose Euler-angle coordinates on $SU(2)$ and gauge the Cartan circle axially. In a local complex coordinate, write $\partial = \partial_z$ and $\bar\partial = \partial_{\bar z}$. After parametrizing the group by base coordinates (θ,φ) and a fiber coordinate ψ, the gauged action has the local form

$$\begin{aligned} S_{\text{gauged}} = \frac{k}{2\pi}\int d^2z\,\Big[&\partial\theta\,\bar\partial\theta + \tan^2\theta\,\partial\varphi\,\bar\partial\varphi \\ &+ \cos^2\theta\,\big(\partial\psi + \tan^2\theta\,\partial\varphi + A\big)\big(\bar\partial\psi - \tan^2\theta\,\bar\partial\varphi + \bar A\big)\Big]. \end{aligned} \tag{66.111}$$

This formula is a coordinate representative of the gauged WZW model; the global action is still defined by the WZW gerbe and the gauged Polyakov–Wiegmann construction. Fixing the gauge $\psi=0$, the gauge fields $A,\bar A$ are algebraic. Their equations of motion are

$$\bar A = \tan^2\theta\,\bar\partial\varphi, \qquad A = -\tan^2\theta\,\partial\varphi.$$

Substitution leaves

$$\frac{k}{2\pi}\int d^2z\,\big(\partial\theta\,\bar\partial\theta+\tan^2\theta\,\partial\varphi\,\bar\partial\varphi\big).$$

The Gaussian integration over $A,\bar A$ has quadratic coefficient $k\cos^2\theta$; its determinant shifts the dilaton by $-\log(k\cos\theta)$, up to the constant convention for Φ_0. Thus the semiclassical large-k representative is

$$ds^2=k\alpha'\big(d\theta^2+\tan^2\theta\,d\varphi^2\big),\qquad \Phi=\Phi_0-\log\cos\theta. \tag{66.112}$$

after absorbing $-\log k$ into Φ_0. This "bell" geometry is a useful sigma-model shadow of the coset. The exact finite-k CFT is nevertheless the parafermion coset with the branching, selection, field-identification, and sewing data specified above; it is not defined by the unrenormalized metric alone.

Definition 66.23 (The bosonic cigar coset)**.** Let $k>2$. The bosonic cigar CFT is the axial Cartan coset

$$\mathrm{Cig}_k=SL(2,\mathbb{R})_k/U(1),\qquad T_{\mathrm{Cig}_k}=T_{SL(2,\mathbb{R})_k}-T_{U(1)},$$

with Euclidean continuation chosen so that the residual angular coordinate is compact. A primary is labelled locally by $SL(2,\mathbb{R})$ quantum numbers $j,m,\bar m$, but a full noncompact CFT datum also includes the choice of continuous and discrete representation ranges, spectral-flow sectors, reflection relation, normalizability prescription, and modular invariant pairing of left and right labels.

Cigar central charge, weights, and spin. In the above convention the bosonic cigar has

$$c(\mathrm{Cig}_k)=\frac{3k}{k-2}-1=2+\frac{6}{k-2}. \tag{66.113}$$

The holomorphic conformal weight of a coset primary with labels (j,m) is

$$h^{\mathrm{Cig}}_{j,m}=-\frac{j(j-1)}{k-2}+\frac{m^2}{k}. \tag{66.114}$$

For momentum $n\in\mathbb{Z}$ and winding $w\in\mathbb{Z}$ around the asymptotic circle, use

$$m=\frac{n+kw}{2},\qquad \bar m=\frac{-n+kw}{2}. \tag{66.115}$$

Then the spin is integral:

$$h^{\mathrm{Cig}}_{j,m}-\bar h^{\mathrm{Cig}}_{j,\bar m}=nw.$$

For $SL(2,\mathbb{R})$ the dual-Coxeter shift has the opposite sign from the compact $SU(2)$ case, giving the WZW central charge $3k/(k-2)$. Subtracting the abelian current central charge gives (66.113). The parent $SL(2,\mathbb{R})$ affine primary of spin j has Sugawara weight

$$h^{SL(2)}_j=-\frac{j(j-1)}{k-2}.$$

Writing the state as a coset branching field times the Cartan vertex, the timelike Cartan vertex has weight $-m^2/k$. Therefore

$$h^{SL(2)}_j=h^{\mathrm{Cig}}_{j,m}-\frac{m^2}{k},$$

which gives (66.114). Finally,

$$h^{\text{Cig}}_{j,m} - \bar{h}^{\text{Cig}}_{j,\bar{m}} = \frac{m^2 - \bar{m}^2}{k} = \frac{(m-\bar{m})(m+\bar{m})}{k} = nw,$$

using (66.115).

Remark 66.24 (Bosonic cigar spectrum status). For the standard axial $SL(2,\mathbb{R})_k/U(1)$ cigar CFT with $k > 2$, the continuous normalizable spectrum contains

$$j = \frac{1}{2} + \mathrm{i}s, \qquad s \in \mathbb{R}_{\geq 0}, \qquad n, w \in \mathbb{Z},$$

with $m, \bar{m}$ as in (66.115). Its weights are therefore

$$h_{s,n,w} = \frac{s^2 + \frac{1}{4}}{k-2} + \frac{(n+kw)^2}{4k}, \qquad \bar{h}_{s,n,w} = \frac{s^2 + \frac{1}{4}}{k-2} + \frac{(-n+kw)^2}{4k}.$$

The analytic continuation in j obeys a reflection relation

$$\mathcal{V}_{j,m,\bar{m}} \simeq \mathcal{R}(j;m,\bar{m})\, \mathcal{V}_{1-j,m,\bar{m}}, \qquad \mathcal{R}(j;m,\bar{m})\mathcal{R}(1-j;m,\bar{m}) = 1,$$

and the normalizable discrete states are obtained from the corresponding pole and residue prescription. The exact discrete ranges and multiplicity choices are part of the noncompact CFT and modular-invariant datum; they do not follow from the local stress tensor and metric formulas alone.

The equality $h^{\text{Cig}}_{j,m} = h^{\text{Cig}}_{1-j,m}$ is immediate from (66.114). Thus the reflection relation is compatible with Virasoro weights, but its coefficient and the allowed discrete residues are global analytic data of the noncompact theory. This is one of the points where the cigar differs sharply from the compact parafermion coset: the compact model has finitely many ordinary characters and a finite modular matrix, while the cigar has delta-normalized continuum states and a scattering/reflection problem at infinity.

Controlled Approximation 66.25 (The cigar as a large-level background)**.** In the same gauged-WZW derivation, now for $SL(2,\mathbb{R})$ with the Euclidean axial gauging, choose local coordinates (ρ, θ, t) so that t is the gauged fiber coordinate. The local action is

$$\begin{aligned} S_{\text{gauged}} = \frac{k}{2\pi} \int d^2 z \, \Big[&\partial\rho\, \bar{\partial}\rho + \tanh^2 \rho\, \partial\theta\, \bar{\partial}\theta \\ &- \cosh^2 \rho \left(\partial t + \tanh^2 \rho\, \partial\theta + A\right)\left(\bar{\partial} t - \tanh^2 \rho\, \bar{\partial}\theta + \bar{A}\right)\Big]. \end{aligned} \tag{66.116}$$

The minus sign records the noncompact Cartan signature; in the Euclidean coset path integral it is accompanied by the corresponding Wick-rotated gauge-field contour. Fixing $t = 0$, the algebraic equations of motion are

$$\bar{A} = \tanh^2 \rho\, \bar{\partial}\theta, \qquad A = -\tanh^2 \rho\, \partial\theta.$$

Substitution gives the kinetic term

$$\frac{k}{2\pi} \int d^2 z \left(\partial\rho\, \bar{\partial}\rho + \tanh^2 \rho\, \partial\theta\, \bar{\partial}\theta\right),$$

and the Gaussian determinant has coefficient $k\cosh^2\rho$, producing $-\log(k\cosh\rho)$ in the dilaton. Absorbing $-\log k$ into Φ_0 gives the large-k sigma-model representative

$$ds^2 = k\alpha'\big(d\rho^2 + \tanh^2\rho\, d\theta^2\big), \qquad \Phi = \Phi_0 - \log\cosh\rho, \qquad \theta \sim \theta + 2\pi. \tag{66.117}$$

The tip is smooth, while the end is an asymptotic cylinder with a linear dilaton. The exact central charge is reproduced by the asymptotic background-charge convention $Q = 1/\sqrt{k-2}$. At finite k, however, the exact object is the noncompact coset CFT with its representation, reflection, normalizability, and modular data; the metric and dilaton alone do not determine those analytic choices.

One-loop Weyl identities for the bell and cigar. Let

$$ds^2 = K\big(du^2 + f(u)^2 d\varphi^2\big), \qquad \Phi = \Phi(u), \qquad K > 0,$$

and set $H = 0$. The only nonzero Christoffel symbols are

$$\Gamma^u_{\varphi\varphi} = -ff', \qquad \Gamma^\varphi_{u\varphi} = \Gamma^\varphi_{\varphi u} = \frac{f'}{f},$$

and the Ricci tensor and dilaton Hessian are

$$R_{uu} = -\frac{f''}{f}, \qquad R_{\varphi\varphi} = -ff'', \qquad \nabla_u\nabla_u\Phi = \Phi'', \qquad \nabla_\varphi\nabla_\varphi\Phi = ff'\Phi'.$$

For the cigar,

$$f(\rho) = \tanh\rho, \qquad \Phi(\rho) = \Phi_0 - \log\cosh\rho,$$

and for the bell,

$$f(\theta) = \tan\theta, \qquad \Phi(\theta) = \Phi_0 - \log\cos\theta.$$

In both cases the one-loop metric Weyl-anomaly equation

$$R_{ij} + 2\nabla_i\nabla_j\Phi = 0$$

holds identically on the coordinate patch. The scalar combination entering (66.23) is constant:

$$|\nabla\Phi|^2 - \frac{1}{2}\nabla^2\Phi = \begin{cases} K^{-1}, & \text{cigar}, \\ -K^{-1}, & \text{bell}. \end{cases} \tag{66.118}$$

For a rotational metric with constant overall scale K, the connection is independent of K. The displayed Christoffel symbols follow by direct substitution in

$$\Gamma^a_{bc} = \frac{1}{2}g^{ad}\big(\partial_b g_{cd} + \partial_c g_{bd} - \partial_d g_{bc}\big).$$

The Ricci components reduce to

$$R_{uu} = -\frac{f''}{f}, \qquad R_{\varphi\varphi} = -ff'',$$

and the Hessian formulas follow from $\nabla_i\nabla_j\Phi = \partial_i\partial_j\Phi - \Gamma^a_{ij}\partial_a\Phi$.

For the cigar one has

$$f' = 1 - f^2, \qquad f'' = -2f(1 - f^2), \qquad \Phi' = -f, \qquad \Phi'' = -(1 - f^2).$$

Therefore

$$R_{uu} + 2\Phi'' = 2(1 - f^2) - 2(1 - f^2) = 0$$

and

$$R_{\varphi\varphi} + 2\nabla_\varphi \nabla_\varphi \Phi = 2f^2(1 - f^2) - 2f^2(1 - f^2) = 0.$$

Also

$$K\left(|\nabla\Phi|^2 - \frac{1}{2}\nabla^2\Phi\right) = f^2 - \frac{1}{2}\left(\Phi'' + \frac{f'}{f}\Phi'\right) = f^2 + (1 - f^2) = 1.$$

For the bell one has

$$f' = 1 + f^2, \qquad f'' = 2f(1 + f^2), \qquad \Phi' = f, \qquad \Phi'' = 1 + f^2.$$

The same substitutions give

$$R_{uu} + 2\Phi'' = -2(1 + f^2) + 2(1 + f^2) = 0$$

and

$$R_{\varphi\varphi} + 2\nabla_\varphi \nabla_\varphi \Phi = -2f^2(1 + f^2) + 2f^2(1 + f^2) = 0.$$

Finally

$$K\left(|\nabla\Phi|^2 - \frac{1}{2}\nabla^2\Phi\right) = f^2 - \frac{1}{2}\left(\Phi'' + \frac{f'}{f}\Phi'\right) = f^2 - (1 + f^2) = -1.$$

Taking $K = k\alpha'$, the scalar constants in (66.118) give the leading large-k central-charge shifts $2 + 6/k$ for the cigar and $2 - 6/k$ for the bell in the convention of (66.23). The exact finite-level current-algebra formulas replace these leading denominators by $k-2$ and $k+2$, respectively. This is precisely why the geometry check is useful but not a definition of the exact CFT.

The $\mathcal{N} = 2$ supersymmetric versions of these rank-one cosets add fermions, an R-current, spectral flow, chiral primaries, and mirror relations to $\mathcal{N} = 2$ Liouville theory. The exact chiral-algebraic interface for the compact minimal coset and the noncompact cigar coset is recorded in Section 70.7. The bosonic discussion here supplies the current-algebra and gauged-WZW baseline, while the GLSM and mirror-duality construction remains part of the supersymmetric-QFT development.

66.9 Finite Orbifolds as Gauging of Topological Lines

Let $\mathcal{C}$ be a two-dimensional CFT with a finite internal symmetry group G. In the defect-line formulation this means that, for each $g \in G$, there is an invertible topological line $\mathcal{L}_g$, with fusion

$$\mathcal{L}_g \otimes \mathcal{L}_h \simeq \mathcal{L}_{gh}.$$

The associativity of line junctions can carry an anomaly class

$$[\omega] \in H^3(G, U(1)).$$

An orbifold $\mathcal{C}/G$ requires a choice of trivialization of this class. Inequivalent trivializations differing by a two-cocycle give the discrete torsion group $H^2(G, U(1))$.

For each $g \in G$, let $\mathcal{H}_g$ be the Hilbert space on the circle with $\mathcal{L}_g$ inserted along the spatial cycle. Equivalently, in a Lagrangian representative this is the sector with boundary condition

$$\Phi(\sigma + 2\pi) = g \cdot \Phi(\sigma)$$

for every field Φ. If $hgh^{-1} = g'$, encircling the insertion by $\mathcal{L}_h$ gives an isomorphism

$$\widehat{h} : \mathcal{H}_g \longrightarrow \mathcal{H}_{g'}.$$

The orbifold Hilbert space is

$$\mathcal{H}_{\mathcal{C}/G} = \bigoplus_{[g]} \mathcal{H}_g^{C_g}, \tag{66.119}$$

where $[g]$ runs over conjugacy classes and C_g is the centralizer of g. Formula (66.119) is the operator form of gauging: the sum over g adds bundles on the spatial circle, and the C_g-invariants impose the residual gauge projection.

Torus partition function of a finite orbifold. For an anomaly-free finite symmetry, and for a fixed discrete-torsion two-cocycle $\epsilon(g, h)$ defined on commuting pairs, the orbifold torus partition function is

$$Z_{\mathcal{C}/G}(\tau, \bar{\tau}) = \frac{1}{|G|} \sum_{\substack{g,h \in G \\ gh=hg}} \epsilon(g, h)\, Z_g^h(\tau, \bar{\tau}), \qquad Z_g^h = \mathrm{Tr}_{\mathcal{H}_g} \widehat{h}\, q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}. \tag{66.120}$$

The modular generators act on the labels by

$$S : (g, h) \mapsto (h^{-1}, g), \qquad T : (g, h) \mapsto (g, gh), \tag{66.121}$$

with the cocycle ϵ supplying the corresponding discrete-torsion phases.

The trace over $\mathcal{H}_g$ is the path integral on a torus with g-holonomy around the spatial cycle and h-holonomy around the Euclidean-time cycle. A flat G-bundle on the torus is specified by a commuting pair (g, h), modulo simultaneous conjugation. Dividing by $|G|$ is the finite gauge-volume factor. Cutting the torus along the spatial cycle gives the trace in (66.120). The modular transformations change the homology basis of the torus; the standard generators send the pair of holonomies as in (66.121). If the $H^3(G, U(1))$ anomaly has been trivialized, the defect junctions are associative, so this change of cutting is an equality of amplitudes. A nontrivial two-cocycle modifies the equality by the displayed $\epsilon(g, h)$, and the cocycle equation is exactly the condition that the two generators obey the modular group relations on the summed expression.

66.10 Twist Fields and Their Weights

A g-twist field is the endpoint, or small-circle state, of the topological line $\mathcal{L}_g$. In the parent theory it is not an ordinary local operator unless $g = 1$; a local field transported around it is acted on by g. In the orbifold theory, the C_g-invariant combinations of such defect endpoints are ordinary local operators. The term "twisted sector" therefore has an operator meaning: it is the radial Hilbert space generated by local endpoints of a fixed monodromy line.

Cyclic permutation twist weight. Let $\mathcal{C}$ have central charge c_0, and consider the cyclic permutation of K copies in $\mathcal{C}^{\otimes K}$. The ground twist field for a K-cycle has

$$h_K = \frac{c_0}{24}\left(K - \frac{1}{K}\right), \qquad \bar{h}_K = \frac{\bar{c}_0}{24}\left(K - \frac{1}{K}\right). \tag{66.122}$$

Near the twist insertion, the K-sheeted base coordinate z is covered by a single coordinate t with $z = t^K$. The total base stress tensor is the sum of the K copy stress tensors. In the cover vacuum, $\langle T(t)\rangle = 0$, and the only contribution to the base one-point function is the Schwarzian term in the stress-tensor transformation. For $t = z^{1/K}$,

$$\{t, z\} = \frac{1 - K^{-2}}{2z^2}.$$

Multiplying by the K sheets and by $c_0/12$ gives

$$\langle T_{\text{base}}(z)\rangle_{\sigma_K} = \frac{c_0}{24}\left(K - \frac{1}{K}\right)\frac{1}{z^2}.$$

The Ward identity $\langle T(z)\sigma_K(0)\rangle / \langle \sigma_K(0)\rangle = h_K/z^2$ then gives (66.122).

Free-boson rotation twist weight. For a single complex free boson Z with rotation

$$Z(\mathrm{e}^{2\pi \mathrm{i}} z, \mathrm{e}^{-2\pi \mathrm{i}}\bar{z}) = \mathrm{e}^{2\pi \mathrm{i}\alpha} Z(z, \bar{z}), \qquad 0 \le \alpha < 1,$$

the holomorphic ground twist field has

$$h_\alpha = \frac{1}{2}\alpha(1-\alpha). \tag{66.123}$$

The monodromy shifts the holomorphic oscillator moding of ∂Z and $\partial \bar{Z}$ by α and $1-\alpha$. With the convention that positive oscillator frequencies contribute $\frac{1}{2}r$ to the normal-ordering constant, the twisted zero-point contribution relative to the untwisted complex boson is

$$E_0(\alpha) - E_0(0) = \frac{1}{2}\sum_{n=0}^{\infty}(n+\alpha) + \frac{1}{2}\sum_{n=0}^{\infty}(n+1-\alpha) - \sum_{n=1}^{\infty} n.$$

The sums are not ordinary convergent sums; they define the local normal-ordering constant by Hurwitz-zeta continuation,

$$\sum_{n=0}^{\infty}(n+x) := \zeta(-1, x) = -\frac{1}{2}\left(x^2 - x + \frac{1}{6}\right).$$

At the endpoint $\alpha = 0$, this expression is read as the untwisted sector value, equivalently as the continuous limit $\alpha \downarrow 0$ after the zero-frequency mode has been omitted from the oscillator energy. Therefore

$$E_0(\alpha) - E_0(0) = \frac{1}{2}\zeta(-1,\alpha) + \frac{1}{2}\zeta(-1, 1-\alpha) - \zeta(-1,1) = \frac{1}{2}\alpha(1-\alpha).$$

The twist ground state is a Virasoro primary, and its L_0 eigenvalue is this shifted zero-point energy, giving (66.123).

For one real boson with $\mathbb{Z}_2$ reflection, this is half of the complex-boson answer at $\alpha = 1/2$:

$$h = \bar{h} = \frac{1}{16}.$$

The same number follows from the two-point function with two twist insertions and two ∂X insertions: the branch condition fixes the meromorphic differential up to a polynomial, and the $T = -\frac{1}{2} : \partial X \partial X :$ coincident limit reads off the coefficient of z^{-2} in the $T\sigma$ OPE.

66.11 Symmetric-Product Orbifolds

Let $\mathcal{C}$ be a seed CFT with central charges $(c_0, \bar{c}_0)$. For a positive integer N, the symmetric-product orbifold is

$$\mathrm{Sym}^N(\mathcal{C}) = \mathcal{C}^{\otimes N}/S_N,$$

where S_N permutes the tensor factors. The central charges before gauging are additive,

$$c_N = N c_0, \qquad \bar{c}_N = N \bar{c}_0,$$

and gauging a finite anomaly-free permutation symmetry does not change the stress-tensor OPE coefficient. Thus these are also the central charges of the orbifold theory.

A conjugacy class of S_N is a partition

$$\lambda = (1^{m_1} 2^{m_2} \cdots N^{m_N}), \qquad \sum_{K=1}^{N} K\, m_K = N.$$

For a permutation with this cycle type, the centralizer is

$$C_\lambda \simeq \prod_{K=1}^{N} \left(\mathbb{Z}_K^{m_K} \rtimes S_{m_K}\right), \qquad |C_\lambda| = \prod_{K=1}^{N} K^{m_K} m_K!. \tag{66.124}$$

The K-cycle twisted Hilbert space $\mathcal{H}_{(K)}$ is the seed Hilbert space placed on a circle of length K, with the vacuum-energy shift computed in (66.122). If a seed state has weights $(h, \bar{h})$, its corresponding long-string weights are

$$h^{(K)} = \frac{h}{K} + \frac{c_0}{24}\left(K - \frac{1}{K}\right), \qquad \bar{h}^{(K)} = \frac{\bar{h}}{K} + \frac{\bar{c}_0}{24}\left(K - \frac{1}{K}\right). \tag{66.125}$$

The cyclic subgroup $\mathbb{Z}_K \subset C_\lambda$ acts as translation by one sheet on the cover, so the $\mathbb{Z}_K$-invariant subspace imposes

$$h - \bar{h} \in K\mathbb{Z}$$

on the seed state before fractional oscillator excitations are included. For the full partition λ, the sector contribution to the orbifold Hilbert space is

$$\mathcal{H}_\lambda^{\mathrm{orb}} = \left[\bigotimes_{K=1}^{N} \mathcal{H}_{(K)}^{\otimes m_K}\right]^{C_\lambda}. \tag{66.126}$$

Consequently

$$\mathcal{H}_{\mathrm{Sym}^N(\mathcal{C})} = \bigoplus_{\lambda \vdash N} \mathcal{H}_\lambda^{\mathrm{orb}}.$$

This is the symmetric-product specialization of (66.119).

Symmetric-product torus partition function. Let

$$Z_{\mathcal{C}}(\tau,\bar\tau)=\mathrm{Tr}_{\mathcal{H}_{\mathcal{C}}}\, q^{L_0-c_0/24}\bar q^{\bar L_0-\bar c_0/24}.$$

With no discrete torsion, the generating function of symmetric-product partition functions is

$$\sum_{N=0}^{\infty} p^N Z_{\mathrm{Sym}^N(\mathcal{C})}(\tau,\bar\tau)=\exp\left[\sum_{m=1}^{\infty} p^m\, T_m Z_{\mathcal{C}}(\tau,\bar\tau)\right], \tag{66.127}$$

where the nonholomorphic Hecke transform used here is

$$T_m Z(\tau,\bar\tau)=\frac{1}{m}\sum_{\substack{ad=m\\ 0\le b<d}} Z\!\left(\frac{a\tau+b}{d},\frac{a\bar\tau+b}{d}\right). \tag{66.128}$$

Apply the finite-orbifold formula (66.120) to $G=S_N$. A commuting pair (g,h) is a finite S_N-bundle on the torus, equivalently a finite cover of the torus together with a labeling by sheets. Each connected orbit of the subgroup generated by g and h is an unbranched cover of the torus of degree m. Such a connected cover is represented by integers

$$a,d>0,\qquad ad=m,\qquad b\in\mathbb{Z}/d\mathbb{Z},$$

and its modulus is $(a\tau+b)/d$. The seed partition function on that connected cover is the summand in (66.128). The factor $1/m$ is the automorphism factor of the connected cover.

A general finite cover is a disjoint union of connected covers. The exponential in (66.127) is the ordinary exponential formula for a set of connected components, with p^m recording the degree. The coefficient of p^N is therefore the finite gauge sum over all degree-N covers, which is exactly the S_N-orbifold torus partition function.

Connected torus covers. Let $E_\tau=\mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z})$, with $\mathrm{Im}\,\tau>0$. Connected unbranched degree-m covers of E_τ, with a chosen lift of the origin, are represented uniquely by integers

$$a,d>0,\qquad ad=m,\qquad 0\le b<d.$$

The covering torus has modulus

$$\tau'=\frac{a\tau+b}{d}, \tag{66.129}$$

and the number of such marked connected covers is

$$\sigma_1(m)=\sum_{d|m} d. \tag{66.130}$$

A connected unbranched cover of E_τ is equivalent to a finite-index sublattice

$$\Lambda'\subset\Lambda=\mathbb{Z}\tau+\mathbb{Z}$$

of index m. The chosen lift of the origin removes translations of the covering torus. With respect to the ordered basis $(\tau,1)$ of Λ, every index-m sublattice has a unique Hermite normal form:

$$\Lambda'=\mathbb{Z}(a\tau+b)+\mathbb{Z}d,\qquad a,d>0,\quad ad=m,\quad 0\le b<d.$$

The index is the determinant ad. Dividing the lattice by d gives an isomorphic complex torus with lattice generated by $(a\tau + b)/d$ and 1, which proves (66.129). For each divisor $d \mid m$, there are exactly d choices of b, hence (66.130).

The automorphism factor in (66.128) is also visible in this lattice description. Since Λ is abelian, every finite-index sublattice is normal, so the deck group is Λ/Λ' and has order m. Thus a connected degree-m component contributes with weight $1/m$, exactly as in the finite gauge-volume normalization of the orbifold sum.

Constant-seed covering-count test. If the seed partition function in (66.127) is replaced by the formal constant 1, then

$$T_m 1 = \frac{\sigma_1(m)}{m}$$

and

$$\exp\left[\sum_{m=1}^{\infty} p^m \frac{\sigma_1(m)}{m}\right] = \prod_{r=1}^{\infty} \frac{1}{1-p^r}. \tag{66.131}$$

Consequently the coefficient of p^N is the partition number $p_{\text{part}}(N)$, the number of conjugacy classes of S_N.

The first assertion is the connected-cover count (66.130) with the deck-automorphism factor $1/m$. For the product identity, expand the logarithm:

$$-\sum_{r=1}^{\infty} \log(1-p^r) = \sum_{r=1}^{\infty}\sum_{k=1}^{\infty} \frac{p^{rk}}{k}.$$

The coefficient of p^m in this logarithm is

$$\sum_{r|m} \frac{1}{m/r} = \frac{1}{m}\sum_{r|m} r = \frac{\sigma_1(m)}{m}.$$

Exponentiating gives (66.131). The same coefficient counts cycle types of S_N, because a cycle type is a partition $N = \sum_r r\, m_r$. This is only a finite-cover normalization test; the constant seed is not being used as a substitute for a specified unitary seed CFT.

The twist fields of the symmetric product are organized by cycle type. A bare twist field σ_λ has ground weight

$$h_\lambda = \sum_{K=1}^{N} m_K \frac{c_0}{24}\left(K - \frac{1}{K}\right), \qquad \bar{h}_\lambda = \sum_{K=1}^{N} m_K \frac{\bar{c}_0}{24}\left(K - \frac{1}{K}\right). \tag{66.132}$$

More generally, a state in the sector λ is obtained by adding seed states and fractional oscillators on each long-string component, followed by the centralizer projection in (66.126).

The operator product of twist fields is group multiplication of monodromies plus a sum over allowed long-string excitations. For representatives $\gamma, \delta \in S_N$,

$$\sigma_\gamma(z,\bar{z})\sigma_\delta(0) = \sum_{\mathcal{O}} C_{\gamma,\delta}^{\gamma\delta;\mathcal{O}}\, z^{h_{\gamma\delta,\mathcal{O}} - h_\gamma - h_\delta}\, \bar{z}^{\bar{h}_{\gamma\delta,\mathcal{O}} - \bar{h}_\gamma - \bar{h}_\delta} \sigma_{\gamma\delta,\mathcal{O}}(0) + \cdots, \tag{66.133}$$

before conjugacy-class averaging. The orbifold local operators are obtained by summing such endpoints over conjugacy classes and projecting to centralizer-invariant states. If a transposition joins two disjoint cycles of lengths K and L, the leading bare-twist exponent is controlled by

$$h_{K+L} - h_K - h_L = \frac{c_0}{24}\left[K + L - \frac{1}{K+L} - K + \frac{1}{K} - L + \frac{1}{L}\right], \tag{66.134}$$

with the analogous antiholomorphic formula. A transposition acting inside a single cycle splits it into two cycles. The covering surface of the correlator determines the numerical OPE coefficient after the seed correlator and the chosen normalization of twist fields have been specified.

Riemann–Hurwitz constraints for twist correlators. For a connected genus-zero component of a symmetric-product twist correlator, let $z_1, \ldots, z_n$ be the branch points. Choose monodromies

$$\gamma_1, \ldots, \gamma_n \in S_N$$

acting transitively on the active sheets and ordered so that

$$\gamma_1\gamma_2\cdots\gamma_n = 1. \tag{66.135}$$

Let $c(\gamma_a)$ denote the number of cycles of γ_a on those active sheets, including fixed points. If the corresponding covering surface has genus g_{cov}, then

$$g_{\text{cov}} = 1 - N + \frac{1}{2}\sum_{a=1}^{n}\bigl(N - c(\gamma_a)\bigr). \tag{66.136}$$

Thus a proposed leading twist correlator must pass three finite tests before any seed-CFT correlator is evaluated: the monodromy product is the identity, the generated subgroup is transitive on the active sheets, and the integer (66.136) is a nonnegative genus.

The monodromies specify the connected covering of the punctured sphere. A cycle of length r over z_a contributes ramification $r-1$. Summing over all cycles of γ_a gives

$$\sum_{\text{cycles } C\subset\gamma_a}(|C| - 1) = N - c(\gamma_a). \tag{66.137}$$

Riemann–Hurwitz for a degree-N map $\widehat{\Sigma} \to \mathbb{P}^1$ says

$$2 - 2g_{\text{cov}} = 2N - \sum_{a=1}^{n}\bigl(N - c(\gamma_a)\bigr).$$

Solving for g_{cov} gives (66.136). The identity product is the statement that the loop enclosing all branch points on the sphere is contractible, and transitivity is exactly connectedness of the active cover.

Example 66.26 (Two-point and primitive joining covers)**.** For a normalized two-point function of length-K twists, take active degree $N = K$ and monodromies

$$\gamma = (1\,2\,\cdots\,K), \qquad \gamma^{-1}.$$

Both have one cycle on the active sheets, so

$$g_{\mathrm{cov}} = 1 - K + \frac{1}{2}\big((K-1) + (K-1)\big) = 0. \tag{66.138}$$

For a primitive connected three-point joining component, let

$$\alpha = (1\,2\,\cdots\,K), \qquad \beta = (K\,K+1\,\cdots\,K+L-1), \qquad \gamma = (\alpha\beta)^{-1}, \tag{66.139}$$

where $K, L \geq 2$. The active degree is $N = K + L - 1$. The product is $\alpha\beta\gamma = 1$, and α, β overlap on the sheet K, so the generated subgroup is transitive. The product $\alpha\beta$ is a single cycle of length $K + L - 1$. Hence

$$c(\alpha) = L, \qquad c(\beta) = K, \qquad c(\gamma) = 1,$$

where fixed points on the active sheets are included. Formula (66.136) gives

$$g_{\mathrm{cov}} = 1 - (K + L - 1) + \frac{1}{2}\big((K-1) + (L-1) + (K+L-2)\big) = 0. \tag{66.140}$$

This genus-zero calculation fixes the covering topology. It does not determine the OPE coefficient without the seed correlator, local coordinate choices at the branch points, and the normalization of the twist fields.

Proposition 66.27 (Primitive joining cover in local coordinates)**.** *Let $K, L \geq 2$, and set $N = K + L - 1$. After moving the three branch points on the base sphere to $0, 1, \infty$, the primitive joining cover of Example 66.26 has a polynomial representative*

$$z_{K,L}(t) = B_{K,L}^{-1} \int_0^t u^{K-1}(u-1)^{L-1}\,\mathrm{d}u, \qquad B_{K,L} = (-1)^{L-1}\frac{(K-1)!(L-1)!}{N!}. \tag{66.141}$$

It is a degree-N map $\mathbb{P}^1_t \to \mathbb{P}^1_z$ with ramification indices K, L, N over $z = 0, 1, \infty$, respectively. More precisely, the local expansions are

$$z_{K,L}(t) = \binom{N}{K} t^K + O(t^{K+1}), \tag{66.142}$$

$$z_{K,L}(t) - 1 = (-1)^{L-1}\binom{N}{L}(t-1)^L + O((t-1)^{L+1}), \tag{66.143}$$

$$\frac{1}{z_{K,L}(t)} = (-1)^{L-1}\frac{(K-1)!(L-1)!}{(N-1)!}\, s^N + O(s^{N+1}), \qquad s = \frac{1}{t}. \tag{66.144}$$

Proof. The integral in (66.141) is a polynomial of degree N, because its derivative is

$$\frac{\mathrm{d}z_{K,L}}{\mathrm{d}t} = B_{K,L}^{-1} t^{K-1}(t-1)^{L-1}.$$

The beta-integral normalization gives

$$\int_0^1 u^{K-1}(u-1)^{L-1}\,\mathrm{d}u = (-1)^{L-1}\frac{(K-1)!(L-1)!}{(K+L-1)!} = B_{K,L},$$

so $z_{K,L}(0) = 0$ and $z_{K,L}(1) = 1$. The derivative factorization shows that the only finite critical points are $t = 0$ with multiplicity $K-1$ and $t = 1$ with multiplicity $L-1$. Since $z_{K,L}$ is a degree-N

polynomial, the point $t = \infty$ maps to $z = \infty$ with ramification index N. There are no further branch points, because the total finite critical multiplicity is $(K-1)+(L-1) = N-1$, the finite critical multiplicity of a degree-N polynomial with a single point above infinity.

The local coefficient at $t = 0$ is obtained by replacing $(u-1)^{L-1}$ by its constant term $(-1)^{L-1}$:

$$B_{K,L}^{-1}\frac{(-1)^{L-1}}{K} = \frac{N!}{K!(L-1)!} = \binom{N}{K}.$$

Near $t = 1$, write $u = 1 + v$. The first nonzero term is

$$B_{K,L}^{-1}\frac{(t-1)^L}{L} = (-1)^{L-1}\frac{N!}{(K-1)!L!}(t-1)^L = (-1)^{L-1}\binom{N}{L}(t-1)^L.$$

Finally, the leading term at infinity is

$$z_{K,L}(t) = \frac{B_{K,L}^{-1}}{N}t^N + O(t^{N-1}).$$

Inverting this leading term and using $s = 1/t$ gives (66.144). These local coefficients are the Jacobian data that enter a covering-space path-integral calculation. A numerical OPE coefficient still requires the seed correlator, the twist-field normalization, and a regulator convention for the Liouville or Schwarzian anomaly associated with the chosen local coordinates. □

Schwarzian and OPE-exponent calculation for primitive joining. Assume the seed theory is in the cover vacuum near the branch points and that the bare twist fields are normalized as Virasoro primaries with weights

$$h_r = \frac{c_0}{24}\left(r - \frac{1}{r}\right), \qquad \bar{h}_r = \frac{\bar{c}_0}{24}\left(r - \frac{1}{r}\right).$$

Let $z - z_a = aw^r + O(w^{r+1})$, with $a \neq 0$, be a local ramified cover coordinate of index r. Then the Schwarzian contribution to the base stress-tensor one-point function has double-pole coefficient h_r:

$$\langle T_{\text{base}}(z)\rangle = \frac{h_r}{(z-z_a)^2} + O((z-z_a)^{-1}). \tag{66.145}$$

For the primitive joining branch data of Proposition 66.27, with $N = K + L - 1$, the holomorphic OPE power in the channel

$$(K)\cdot(L) \longrightarrow (N)$$

is therefore

$$h_N - h_K - h_L = \frac{c_0}{24}\left[N - \frac{1}{N} - K + \frac{1}{K} - L + \frac{1}{L}\right], \tag{66.146}$$

with the analogous antiholomorphic formula using $\bar{c}_0$. Equivalently, before conjugacy-class averaging and before inserting excited seed states,

$$\sigma_K(z,\bar{z})\sigma_L(0) = C_{K,L}^N\, z^{h_N-h_K-h_L}\,\bar{z}^{\bar{h}_N-\bar{h}_K-\bar{h}_L}\sigma_N(0) + \cdots, \tag{66.147}$$

where $C_{K,L}^N$ is not fixed by the covering and Schwarzian calculation alone.

Verification. The inverse branches near z_a have the form

$$w_j(z) = \omega_r^j a^{-1/r}(z - z_a)^{1/r} + O((z - z_a)^{2/r}), \qquad j = 0, \dots, r-1.$$

The leading Schwarzian of any such branch is independent of a and of the root label:

$$\{w_j, z\} = \frac{1 - r^{-2}}{2(z - z_a)^2} + O((z - z_a)^{-1}).$$

Using the stress-tensor transformation law (60.24) in the inverse direction, and using the cover-vacuum assumption, the ramified point contributes

$$\frac{c_0}{12} \sum_{j=0}^{r-1} \frac{1 - r^{-2}}{2(z - z_a)^2} = \frac{c_0}{24} \left(r - \frac{1}{r} \right) \frac{1}{(z - z_a)^2}.$$

Unramified preimages over the same base point have regular inverse branches and do not contribute to the double pole. This proves (66.145).

For three primary insertions at z_1, z_2, z_3, global conformal covariance fixes the holomorphic coordinate dependence to

$$C\, z_{12}^{h_3 - h_1 - h_2} z_{13}^{h_2 - h_1 - h_3} z_{23}^{h_1 - h_2 - h_3}.$$

Applying this with branch indices K, L, N and then taking the OPE limit of the K- and L-twists gives (66.146) and (66.147). The constant $C_{K,L}^N$ also contains the cover anomaly functional, the local coordinate coefficients in (66.142)–(66.144), the chosen twist-field normalization, and any nontrivial seed correlator. Those data are not determined by monodromy, Riemann–Hurwitz, or conformal covariance alone.

Lemma 66.28 (Conjugacy-class normalization for primitive joining)**.** *Let M be the total number of copies in $\mathrm{Sym}^M(\mathcal{C})$. For $r \le M$, let $[r] \subset S_M$ denote the conjugacy class consisting of one r-cycle and $M - r$ fixed points, and assume the labelled bare twists are normalized by*

$$\langle \sigma_\gamma(\infty) \sigma_\delta(0) \rangle = \delta_{\delta, \gamma^{-1}}.$$

The normalized single-cycle class operator is then

$$\Sigma_r^{(M)} = |[r]|^{-1/2} \sum_{\gamma \in [r]} \sigma_\gamma, \qquad |[r]| = \frac{M!}{r(M - r)!}. \tag{66.148}$$

Let $K, L \ge 2$, set $R = K + L - 1$, and assume $M \ge R$. If $C_{K,L}^R$ denotes the labelled primitive joining coefficient in (66.147), with the same local coordinate and seed-correlator convention for all conjugates, then the class-normalized leading primitive channel has the form

$$\Sigma_K^{(M)}(z, \bar z) \Sigma_L^{(M)}(0) \supset \mathfrak{g}_{K,L}^{(M)}\, C_{K,L}^R\, z^{h_R - h_K - h_L} \bar z^{\bar h_R - \bar h_K - \bar h_L} \Sigma_R^{(M)}(0), \tag{66.149}$$

where no additional symmetry factor is inserted in the operator product, and

$$\left(\mathfrak{g}_{K,L}^{(M)} \right)^2 = KLR\, \frac{(M - K)!(M - L)!}{M!(M - R)!}. \tag{66.150}$$

Equivalently, $\mathfrak{g}_{K,L}^{(M)} = R\sqrt{|[R]|/(|[K]||[L]|)}$. Thus the class-sum normalization supplies a universal finite-group factor, while $C_{K,L}^R$ remains the analytic covering-space coefficient discussed in the Schwarzian/OPE exponent calculation above.

Proof. The class-size formula in (66.148) follows by choosing the r-element support and then choosing a cyclic ordering:

$$\binom{M}{r}(r-1)! = \frac{M!}{r(M-r)!}.$$

Since inversion preserves the conjugacy class $[r]$, the labelled diagonal two-point normalization gives

$$\left\langle \Sigma_r^{(M)}(\infty)\Sigma_r^{(M)}(0)\right\rangle = |[r]|^{-1}\sum_{\gamma,\delta\in[r]}\delta_{\delta,\gamma^{-1}} = 1.$$

It remains to count primitive factorizations of a fixed R-cycle η. Write the support of η in cyclic order as

$$x_0 \longmapsto x_1 \longmapsto \cdots \longmapsto x_{R-1} \longmapsto x_0.$$

For each choice of the unique overlap sheet x_j, indices understood modulo R, define

$$\alpha_j = (x_{j-K+1}\, x_{j-K+2}\, \cdots\, x_j), \qquad \beta_j = (x_j\, x_{j+1}\, \cdots\, x_{j+L-1}).$$

With the convention that products act from right to left, one checks directly that $\alpha_j\beta_j = \eta$. These are R distinct factorizations. Conversely, if α is a K-cycle and β is an L-cycle with $\alpha\beta = \eta$, then the support of η has size $K+L-1$. If the supports of α and β were disjoint, the product would have two nontrivial cycles rather than one. If their intersection had more than one element, the union of the supports would have size strictly smaller than $K+L-1$, too small to contain the support of η. Thus the supports meet in exactly one sheet and their union is the support of η. Removing the overlap sheet cuts the two cycles into two directed strings, and their concatenation is the displayed cyclic order of η. The overlap sheet therefore reconstructs α and β uniquely, so the number of primitive factorizations of a fixed $\eta \in [R]$ is R.

When the two normalized class sums are multiplied, each fixed $\eta \in [R]$ therefore receives R labelled primitive contributions. Projecting the resulting sum

$$\frac{R\, C_{K,L}^{R}}{\sqrt{|[K]||[L]|}}\sum_{\eta\in[R]}\sigma_\eta$$

onto $\Sigma_R^{(M)} = |[R]|^{-1/2}\sum_{\eta\in[R]}\sigma_\eta$ gives

$$\mathfrak{g}_{K,L}^{(M)} = R\sqrt{\frac{|[R]|}{|[K]||[L]|}}.$$

Substituting the class-size formula and using $R = K+L-1$ gives (66.150). □

Lemma 66.29 (Transposition class algebra: join and split factors)**.** *Let $\Sigma_2^{(M)}$ be the normalized transposition class operator from* (66.148)*. Assume the same labelled twist normalization as in Lemma 66.28. For $K \geq 2$ and $M \geq K+1$, the class-normalized channel in which a transposition joins a fixed point to a K-cycle is*

$$\Sigma_2^{(M)}(z,\bar z)\Sigma_K^{(M)}(0) \supset \mathfrak{j}_K^{(M)}\, C_{2,K}^{K+1}\, z^{h_{K+1}-h_2-h_K}\,\bar z^{\bar h_{K+1}-\bar h_2-\bar h_K}\,\Sigma_{K+1}^{(M)}(0), \tag{66.151}$$

with

$$\left(\mathfrak{j}_K^{(M)}\right)^2 = 2K(K+1)\frac{M-K}{M(M-1)}. \tag{66.152}$$

For a split channel, let $K \geq 4$, let $a, b \geq 2$, and let $a + b = K$. Let $[a, b] \subset S_M$ be the conjugacy class with one a-cycle, one b-cycle, and $M - K$ fixed points, and define

$$\Sigma_{a,b}^{(M)} = |[a, b]|^{-1/2} \sum_{\eta \in [a,b]} \sigma_\eta.$$

Then

$$\Sigma_2^{(M)}(z, \bar{z})\Sigma_K^{(M)}(0) \supset \mathfrak{s}_{a,b}^{(M)}\, C_{2,K}^{a,b}\, z^{h_a+h_b-h_2-h_K}\, \bar{z}^{\bar{h}_a+\bar{h}_b-\bar{h}_2-\bar{h}_K} \Sigma_{a,b}^{(M)}(0), \tag{66.153}$$

where $C_{2,K}^{a,b}$ is the labelled split-channel coefficient and

$$\left(\mathfrak{s}_{a,b}^{(M)}\right)^2 = \frac{2Kab}{(1 + \delta_{a,b})M(M - 1)}. \tag{66.154}$$

If one piece is a fixed point, say $K = b + 1$ with $b \geq 2$, the output class is the single-cycle class $[b]$ and the endpoint split factor is

$$\left(\mathfrak{s}_{1,b}^{(M)}\right)^2 = 2(b + 1)b\, \frac{M - b}{M(M - 1)}. \tag{66.155}$$

Proof. The join channel is the special case $L = 2$ of Lemma 66.28. Substituting $R = K + 1$ into (66.150) gives (66.152).

For the split channel, fix a permutation $\eta = \alpha\beta$, where α and β are disjoint cycles of lengths a and b. If p is in the support of α and q is in the support of β, then $\tau = (p\, q)$ cuts and reglues the two directed cycles into a single K-cycle

$$\gamma = \tau\eta.$$

Since $\tau^2 = 1$, this is equivalent to $\tau\gamma = \eta$. Conversely, if $\tau\gamma = \eta$ with τ a transposition and γ a K-cycle, then τ cannot involve a fixed point of η, because then $\gamma = \tau\eta$ would move a sheet outside the support of η. It also cannot exchange two points in the same component of η: the other component would remain invariant under γ, contradicting the assumption that γ is a single K-cycle. Hence τ must connect one point of the a-cycle to one point of the b-cycle, so a fixed output $\eta \in [a, b]$ has exactly

$$ab$$

labelled split factorizations.

The class size is

$$|[a, b]| = \frac{M!}{ab(1 + \delta_{a,b})(M - K)!},$$

where the factor $1 + \delta_{a,b}$ is the permutation symmetry of the two equal-length cycles when $a = b$. Projecting

$$\frac{ab\, C_{2,K}^{a,b}}{\sqrt{|[2]||[K]|}} \sum_{\eta \in [a,b]} \sigma_\eta$$

onto $\Sigma_{a,b}^{(M)}$ gives

$$\left(\mathfrak{s}_{a,b}^{(M)}\right)^2 = a^2b^2 \frac{|[a, b]|}{|[2]||[K]|}.$$

Using $|[2]| = M!/(2(M-2)!)$ and $|[K]| = M!/(K(M-K)!)$ gives (66.154).

If one component has length 1, fix the nontrivial output b-cycle β. The split-off fixed point may be any of the $M-b$ sheets outside the support of β, and the cut point on β may be chosen in b ways. Thus a fixed output $\beta \in [b]$ receives $b(M-b)$ labelled factorizations. Projecting to the single-cycle class $\Sigma_b^{(M)}$ and using $K = b+1$ gives (66.155). This is the same finite factor as the join coefficient $\mathrm{j}_b^{(M)}$, as required by the diagonal two-point normalization. □

Discrete torsion is incorporated by replacing the centralizer projection by a projective centralizer action determined by a class in

$$H^2(S_N, U(1)).$$

For S_N, this group is trivial for $N \le 3$ and is $\mathbb{Z}_2$ for $N \ge 4$. The choice changes the phases in the commuting-pair sum and the projective representation carried by twisted-sector ground states. It is part of the orbifold definition.

66.12 Twist-Field Deformations

Let $\{\sigma_A\}$ be spinless orbifold local fields built from twisted sectors, with dimensions

$$\Delta_A = h_A + \bar{h}_A, \qquad s_A = h_A - \bar{h}_A \in \mathbb{Z}.$$

A deformation by twist fields is specified by the conformal-perturbation series

$$\langle X \rangle_\lambda = \left\langle \exp\left(-\sum_A \lambda^A \int_\Sigma \mathrm{d}^2 z\, \sigma_A(z, \bar{z}) \right) X \right\rangle_{\mathcal{C}/G} \tag{66.156}$$

with a regulator for collision points and with local counterterms determined by the OPEs of the σ_A. The coupling dimension is

$$[\lambda^A] = 2 - \Delta_A.$$

Separate this engineering dimension from the operator-theoretic obstruction as follows. Choose a local flat coordinate patch and a short-distance cutoff a. Write dimensionless couplings $g^A(a)$ by

$$S_{\text{int}}(a) = \sum_A g^A(a)\, a^{\Delta_A - 2} \int \mathrm{d}^2 z\, \sigma_A(z, \bar{z}), \qquad y_A = 2 - \Delta_A.$$

Proposition 66.30 (Cutoff OPE beta function)**.** *Assume that the spinless primary fields σ_A have diagonal two-point normalization and short-distance OPE*

$$\sigma_B(z, \bar{z}) \sigma_C(0) \sim \sum_A C_{BC}{}^A |z|^{\Delta_A - \Delta_B - \Delta_C} \sigma_A(0) + \textit{descendants and other spins.} \tag{66.157}$$

In the minimal annular cutoff scheme that discards derivative descendants and projects to the listed spinless primaries, the length-scale Wilsonian beta function

$$\beta_\ell^A = \frac{\mathrm{d}g^A}{\mathrm{d}\ell}, \qquad \ell = \log a,$$

is

$$\beta_\ell^A = y_A g^A - \pi C_{BC}{}^A g^B g^C + O(g^3), \tag{66.158}$$

with ordered repeated indices. Equivalently, for the energy scale $\mu = a^{-1}$*,*

$$\mu \frac{\mathrm{d}g^A}{\mathrm{d}\mu} = -y_A g^A + \pi C_{BC}{}^A g^B g^C + O(g^3). \tag{66.159}$$

Proof. Expand $\exp(-S_{\text{int}})$ to second order and isolate pairs with relative coordinate $r = z - w$ in the annulus $a < |r| < ae^{\mathrm{d}\ell}$. The term in the exponential is

$$\frac{1}{2} g^B g^C a^{\Delta_B+\Delta_C-4} \int \mathrm{d}^2 w \int_{a<|r|<ae^{\mathrm{d}\ell}} \mathrm{d}^2 r\, \sigma_B(w+r)\sigma_C(w).$$

Using (66.157), the contribution to the $\sigma_A(w)$ channel is

$$\frac{1}{2} g^B g^C C_{BC}{}^A a^{\Delta_B+\Delta_C-4} \int \mathrm{d}^2 w\, \sigma_A(w) \int_a^{ae^{\mathrm{d}\ell}} 2\pi\rho\, \mathrm{d}\rho\, \rho^{\Delta_A-\Delta_B-\Delta_C}.$$

To first order in $\mathrm{d}\ell$, the radial integral is

$$2\pi\, \mathrm{d}\ell\, a^{2+\Delta_A-\Delta_B-\Delta_C},$$

so the local term is

$$\pi\, g^B g^C C_{BC}{}^A\, \mathrm{d}\ell\, a^{\Delta_A-2} \int \mathrm{d}^2 w\, \sigma_A(w).$$

Because this term appears with a plus sign in the expansion of $\exp(-S_{\text{int}})$, it decreases the local coefficient in S_{int} by the same amount. The classical rescaling of the dimensionless coupling contributes $y_A g^A \mathrm{d}\ell$. Combining the two effects gives (66.158), and (66.159) follows from $\log\mu = -\ell$. □

Remark 66.31 (Contact terms and coupling coordinates). A local contact convention is a choice of finite counterterms supported at operator collisions. To quadratic order it is equivalently a coupling coordinate change

$$g'^A = g^A + M^A{}_{BC} g^B g^C + O(g^3).$$

If

$$\beta_\ell^A = y_A g^A + b^A{}_{BC} g^B g^C + O(g^3),$$

then

$$b'^A{}_{BC} = b^A{}_{BC} + (y_B + y_C - y_A) M^A{}_{BC}. \tag{66.160}$$

Indeed,

$$\frac{\partial g'^A}{\partial g^D} \beta_\ell^D = y_A g^A + \left(b^A{}_{BC} + (y_B + y_C) M^A{}_{BC}\right) g^B g^C + O(g^3),$$

and substituting $g^A = g'^A - M^A{}_{BC} g'^B g'^C + O(g'^3)$ gives (66.160). Thus a quadratic term is scheme-invariant only in resonant channels $y_A = y_B + y_C$, including the marginal-to-marginal channel $y_A = y_B = y_C = 0$. Exact marginality of a spinless dimension-two twist deformation therefore starts with the condition that the projection of $C_{BC}{}^A$ to dimension-two nonredundant primaries vanish in the chosen operator algebra; higher orders require the full contact-term calculus, not only the bare group-theoretic monodromy.

The relevance classification is the linear part of (66.158): if $\Delta_A < 2$, then $y_A > 0$ and the coupling grows toward longer distances; if $\Delta_A = 2$, the first nontrivial obstruction is the dimension-two projection of the quadratic OPE coefficient in (66.159).

In the symmetric-product case of Section 66.11, a deformation by a single seed operator has the untwisted form

$$\mathcal{O}_{\mathrm{seed},N} = \sum_{i=1}^{N} \mathcal{O}_i. \tag{66.161}$$

It changes the seed CFT in every copy. A deformation that joins and splits cycles is built instead from conjugacy-class sums of twist fields, for example

$$\mathcal{O}_{[2]} = \frac{1}{\sqrt{N(N-1)/2}} \sum_{1\le i<j\le N} \mathcal{O}_{(ij)}, \tag{66.162}$$

where $\mathcal{O}_{(ij)}$ is a specified dressed transposition-twist operator. The normalization gives an order-one two-point function when the individual twist operators are orthonormal in the orbifold two-point normalization.

In supersymmetric symmetric products the undressed twist-sector ground field is one ingredient in the physical marginal operator. Fermion zero modes, R-symmetry spin fields, or supercurrent descendants supply the additional quantum numbers so that the final operator is invariant, spinless, and has $(h, \bar{h}) = (1, 1)$. The following construction is the local CFT datum behind the example often called the D1–D5 symmetric-product point.

Construction 66.32 ($\mathcal{N} = (4, 4)$ two-cycle marginal tangent)**.** Let $\mathcal{C}_0$ be a compact unitary seed CFT with small $\mathcal{N} = (4, 4)$ symmetry and

$$c_0 = \bar{c}_0 = 6.$$

Let $\mathrm{Sym}^N(\mathcal{C}_0)$ be the finite orbifold defined in Section 66.11. For a transposition (ij), the bare twist ground field has

$$h_{\mathrm{bare}} = \bar{h}_{\mathrm{bare}} = \frac{6}{24}\left(2 - \frac{1}{2}\right) = \frac{3}{8}. \tag{66.163}$$

Assume that the transposition sector contains an S_N-covariant spin-field dressing $S^A_{(ij)}\widetilde{S}^B_{(ij)}$ of weights $(1/8, 1/8)$ and R-symmetry quantum numbers such that

$$\chi^{AB}_{(ij)} = \sigma_{(ij)} S^A_{(ij)} \widetilde{S}^B_{(ij)} \tag{66.164}$$

is an $\mathcal{N} = (4, 4)$ chiral primary with

$$h_\chi = \bar{h}_\chi = \frac{1}{2}.$$

Let $G^a_{-1/2}$ and $\widetilde{G}^{\dot{a}}_{-1/2}$ be left and right supercharge modes in a convention where each mode raises h or $\bar{h}$ by $1/2$. Contract the R-symmetry indices to a singlet in one of the allowed short multiplets and define

$$\mathcal{O}^{\mathfrak{a}}_{\mathrm{blow}} = \frac{1}{\sqrt{N(N-1)/2}} \sum_{1\le i<j\le N} \Pi^{\mathfrak{a}}_{a\dot{a},AB}\, G^a_{-1/2} \widetilde{G}^{\dot{a}}_{-1/2} \chi^{AB}_{(ij)}. \tag{66.165}$$

Here $\Pi^{\mathfrak{a}}$ is a fixed invariant tensor selecting one real component of the short multiplet. The normalization is the same conjugacy-class normalization as (66.162). The operator $\mathcal{O}^{\mathfrak{a}}_{\mathrm{blow}}$ is spinless and has

$$(h, \bar{h}) = (1, 1).$$

It is the local two-cycle blow-up tangent operator. Its integration defines an exactly marginal coordinate precisely when the $\mathcal{N} = (4,4)$-preserving contact-term and current-moment-map obstructions in the source-coordinate criterion of Section 56.3 vanish or, more strongly, when the surrounding local conformal-family datum is constructed.

The construction separates three different facts that are often compressed together. First, the equality $(h, \bar{h}) = (1, 1)$ is a representation theory statement:

$$\frac{3}{8} + \frac{1}{8} + \frac{1}{2} = 1$$

on the left and similarly on the right. Second, the S_N-invariant class sum is a finite-orbifold projection, not a geometric quotient operation on operators. Third, exact marginality is an integrability statement in renormalized conformal perturbation theory. It requires absence of nonredundant dimension-two obstructions in the OPE of the top components and compatibility with the $\mathcal{N} = (4,4)$ Ward identities.

The local tangent-space count for the standard compact examples is then a protected multiplet count, not a claim extracted from a low-energy geometric picture. Let d_{seed} be the real dimension of the seed $\mathcal{N} = (4,4)$ conformal-family tangent space. The untwisted seed directions give the operators

$$\sum_{i=1}^{N} \mathcal{O}_i^{\alpha}, \qquad \alpha = 1, \ldots, d_{\mathrm{seed}},$$

and the transposition short multiplet contributes four real blow-up directions in (66.165). Thus the local protected tangent count is

$$d_{\mathrm{local}} = d_{\mathrm{seed}} + 4. \tag{66.166}$$

For a toroidal seed with Narain moduli

$$SO(4,4)/(SO(4) \times SO(4)),$$

one has $d_{\mathrm{seed}} = 16$, hence $d_{\mathrm{local}} = 20$. This is the local CFT count underlying the tangent space of

$$SO(5,4)/(SO(5) \times SO(4)).$$

For a K3 seed, the seed sigma-model tangent space has real dimension 80, and the symmetric-product blow-up multiplet gives 84, matching the local dimension of

$$SO(4,21)/(SO(4) \times SO(21)).$$

The arithmetic quotients and singular loci of these spaces require the integral charge lattice and the global CFT identification group; they are global data beyond the local tangent construction above.

A current-current expression such as $J\bar{J}$ defines an exactly marginal deformation only after the current algebra, OPE coefficient, and contact-term scheme have been specified; in a symmetric product one must additionally specify whether the operator is the untwisted seed sum (66.161), a single-cycle/twist-sector operator such as (66.162), or a multi-cycle product.

66.13 Development Tasks for the CFT Volume

The material in this chapter is the entry point for a larger two-dimensional CFT development. Subsequent passes must expand the following topics in the same formalism:

higher-genus sewing of Narain and toroidal CFT correlators,
WZW and gauged WZW/coset sigma-model CFTs,
Buscher duality as an equality of regulated path integrals,
orbifold defects, discrete torsion, and anomaly obstructions,
higher-point Hurwitz sums and normalized twist OPE coefficients,
supersymmetric twist fields and exactly marginal blow-up modes.

Each item requires explicit operator definitions, modular and sewing compatibility, and examples with computable correlators. In particular, twist-field deformations are deformations of a fully specified QFT; geometric resolution language is secondary and must be backed by operator construction.

Chapter 67

Vertex Operator Algebras, Modular Sewing, and Logarithmic CFT

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter treats chiral algebra as the algebraic coordinate system for two-dimensional CFT, but it keeps that coordinate system distinct from the analytic construction of a full QFT on surfaces. The first layer is formal: VOA locality, modes, Zhu's algebra, minimal-model characters, Coulomb-gas screenings, and logarithmic pseudo-traces are algebraic or finite-level statements once the relevant modules are specified. The second layer is analytic within a fixed channel: chiral blocks, Gram/Shapovalov projectors, trace-class propagation factors, and direct-integral OPE decompositions require convergence, topological vector spaces, and positive pairings. The third layer is global sewing: local plumbing estimates have to be continued across moduli space and compared between pants decompositions before they become a modular functor or a full CFT. The fourth layer is operator algebraic: conformal nets and strongly local VOAs replace formal fields by von Neumann algebras and make the domain-to-locality passage a theorem boundary rather than a slogan.

The chapter's quoted-theorem boundaries are therefore not all of the same kind. Zhu classification and rational character modularity import finite cofiniteness and trace-recursion theorems beyond the local top-level calculations. Complete rationality of conformal nets imports subfactor/DHR machinery after the localized-endomorphism mechanism has been displayed. Strong locality of VOAs imports analytic estimates turning smeared fields on a common smooth domain into local von Neumann algebras. Rational sewing imports analytic continuation and mapping-class compatibility beyond fixed plumbing charts. Nonrational and logarithmic CFT require still different replacements for finite semisimple sums: direct integrals, measurable fusing kernels, projective pseudo-traces, modified traces, and nonsemisimple tensor-category data. The local calculations below are included to expose these mechanisms; they are not meant to hide the remaining analytic existence problem.

67.1 Chiral Algebra Data

Two-dimensional conformal field theory has an additional algebraic structure: local holomorphic fields may be multiplied by operator products that depend only on one complex coordinate. A vertex operator algebra is the precise algebraic encoding of this chiral operator product.

Definition 67.1 (Vertex operator algebra)**.** A vertex operator algebra consists of the following data.

1. A complex vector space
$$V = \bigoplus_{n\in\mathbb{Z}} V_n$$
with finite-dimensional graded pieces and $V_n = 0$ for n sufficiently negative.
2. A vacuum vector $\mathbf{1} \in V_0$.
3. A state-field map
$$Y(\,\cdot\,, z) : V \longrightarrow \mathrm{End}(V)[[z, z^{-1}]], \qquad a \longmapsto Y(a, z) = \sum_{m\in\mathbb{Z}} a_m z^{-m-1}.$$
4. A conformal vector $\omega \in V_2$, whose modes
$$Y(\omega, z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}$$
obey the Virasoro algebra
$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

These data satisfy:
$$Y(\mathbf{1}, z) = \mathrm{id}_V, \qquad Y(a, z)\mathbf{1} \in a + zV[[z]],$$
$$[L_{-1}, Y(a, z)] = \partial_z Y(a, z), \qquad L_0 a = na \quad \text{for } a \in V_n,$$
and locality: for every $a, b \in V$, there is $N \geq 0$ such that
$$(z-w)^N [Y(a, z), Y(b, w)] = 0$$
as a formal distribution on V.

The modes a_m are operators on V. If a has conformal weight Δ_a, then a_m has degree $\Delta_a - m - 1$. The vacuum axiom says that a is recovered from the field $Y(a, z)$ by acting on the vacuum and then taking the value at $z = 0$:
$$a = Y(a, z)\mathbf{1}\big|_{z=0}.$$

The locality axiom is the algebraic form of the chiral OPE. It implies the Jacobi identity for modes, and conversely the Jacobi identity packages locality, associativity, and commutativity of OPE expansions.

Proposition 67.2 (OPE from VOA locality). *For $a, b \in V$, the product $Y(a,z)Y(b,w)$ has an expansion in the region $|z| > |w|$ of the form*

$$Y(a,z)Y(b,w) = \sum_{m\in\mathbb{Z}} Y(a_m b, w)(z-w)^{-m-1},$$

where for each matrix element only finitely many positive powers of $(z-w)^{-1}$ occur.

Proof. Fix $a, b \in V$. Lower truncation says that $a_m b = 0$ for all sufficiently large m; hence the singular part of the candidate expansion has only finitely many negative powers of $z - w$ in every matrix element. Locality gives an integer N such that

$$(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z)$$

as formal distributions. The coefficient of $(z-w)^{-m-1}$ in the singular part is determined by the formal residue

$$a_m b = \mathrm{Res}_{z-w}(z-w)^m Y(a,z)b.$$

For $m < N$, these residues specify the complete principal part at $z = w$. Subtract the finite sum of principal parts from $Y(a,z)Y(b,w)$. After multiplication by $(z-w)^N$, locality makes the difference regular at $z = w$, so the remaining term is the Taylor expansion in nonnegative powers of $z - w$ in the region $|z| > |w|$. Equivalently, all matrix elements against graded-dual vectors agree as Laurent series in that region. This is the displayed OPE, and lower truncation gives the finiteness of its singular part. □

67.2 Modules, Characters, and Chiral Blocks

A V-module M is a graded vector space with fields

$$Y_M(a,z) = \sum_{m\in\mathbb{Z}} a_m^M z^{-m-1}$$

obeying the same vacuum, translation, and locality identities with one entry acting on M. The conformal vector gives an operator L_0 on M. For an ordinary positive-energy module,

$$M = \bigoplus_{\lambda\in\mathbb{C}} M_\lambda, \qquad \dim M_\lambda < \infty,$$

and the set of λ's is bounded below in every coset modulo $\mathbb{Z}$. The character is

$$\chi_M(\tau) = \mathrm{Tr}_M\, q^{L_0 - c/24}, \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau}, \qquad \mathrm{Im}\,\tau > 0. \tag{67.1}$$

The shift by $c/24$ is the cylinder vacuum energy fixed by the conformal map from the plane to the circle.

Let C be a compact Riemann surface with marked points p_i, local coordinates z_i at the marked points, and V-modules M_i. A chiral conformal block is a linear functional

$$\mathcal{F} : \bigotimes_i M_i \longrightarrow \mathbb{C}$$

that obeys the Ward identities for meromorphic chiral fields on C. For the stress tensor, the identity says that inserting a meromorphic vector field regular away from the p_i's acts by the sum of the corresponding Virasoro modes at the marked points and gives zero. For a general chiral state $a \in V$, the meromorphic one-form valued in the field a gives the same condition with modes of $Y_{M_i}(a, z_i)$.

On the sphere with three marked points, conformal blocks are intertwiners. If M_i, M_j, M_k are simple modules, the fusion coefficient is

$$N_{ij}{}^{k} := \dim \mathrm{CB}_{0,3}(M_i, M_j, M_k^\vee),$$

where $M_k^\vee$ is the contragredient module. This definition is intrinsic: it counts chiral three-point structures, not a particular coordinate presentation of a three-point function.

67.3 Zhu Algebra and Top Levels

The character and sewing constructions use the whole graded module. The classification of ordinary modules begins with a smaller algebra obtained from genus-one coordinate changes. This algebra is Zhu's algebra. Its role in the monograph is structural: it converts the lowest-energy part of a positive-energy module into a finite algebraic object, while leaving the reconstruction of the full module to the VOA theorem boundary.

Let $a \in V$ be homogeneous of conformal weight Δ_a, and write

$$Y(a, z) = \sum_{n \in \mathbb{Z}} a_n z^{-n-1}.$$

Define bilinear operations on V by

$$a * b := \mathrm{Res}_{z=0} \frac{(1+z)^{\Delta_a}}{z} Y(a, z) b = \sum_{j \geq 0} \binom{\Delta_a}{j} a_{j-1} b, \tag{67.2}$$

$$a \circ b := \mathrm{Res}_{z=0} \frac{(1+z)^{\Delta_a}}{z^2} Y(a, z) b = \sum_{j \geq 0} \binom{\Delta_a}{j} a_{j-2} b. \tag{67.3}$$

The sums are finite on each pair a, b by lower truncation. Let

$$O(V) = \mathrm{span}_{\mathbb{C}}\{a \circ b \mid a, b \in V \text{ homogeneous}\}.$$

Definition 67.3 (Zhu algebra)**.** The Zhu algebra of a VOA V is the quotient vector space

$$A(V) = V/O(V) \tag{67.4}$$

with product

$$[a]\,[b] := [a * b]. \tag{67.5}$$

The formal identities below imply that this product is independent of the chosen representatives and is associative; they are the same component-Jacobi residue identities used in Proposition 67.4. We denote the class of $a \in V$ by $[a] \in A(V)$.

The exponent $(1+z)^{\Delta_a}$ is not decorative. It is the coordinate factor relating the punctured plane coordinate to the genus-one coordinate used in torus one-point functions. The quotient by $O(V)$ removes the descendant terms that vanish on lowest-energy states.

Proposition 67.4 (Top-level action)**.** *Let M be an ordinary positive-energy V-module with lowest conformal weight h_M, so that*

$$M = \bigoplus_{n\geq 0} M_{h_M+n}.$$

For a homogeneous $a \in V$, set

$$o_M(a) := a_{\Delta_a-1}\big|_{M_{h_M}}. \tag{67.6}$$

Then $o_M(a \circ b) = 0$ on M_{h_M}, and

$$o_M(a * b) = o_M(a)o_M(b). \tag{67.7}$$

Thus M_{h_M} is an $A(V)$-module.

Proof. The mode a_n sends an L_0-eigenvector of weight λ to weight

$$\lambda + \Delta_a - n - 1. \tag{67.8}$$

Thus every lowest-weight vector $v \in M_{h_M}$ is killed by u_n whenever u is homogeneous and $n > \Delta_u - 1$. Equivalently,

$$M_{h_M} \subset \Omega(M) := \{v \in M \mid u_n v = 0 \text{ for all homogeneous } u \in V,\ n > \Delta_u - 1\}. \tag{67.9}$$

The remaining step is a finite component calculation from the module Jacobi identity. In mode form, that identity says

$$\sum_{i\geq 0} \binom{m}{i} (a_{n+i}b)_{m+k-i} = \sum_{i\geq 0} (-1)^i \binom{n}{i} \big(a_{m+n-i}b_{k+i} - (-1)^n b_{n+k-i}a_{m+i}\big). \tag{67.10}$$

All sums are finite on a fixed vector by lower truncation. Substitute $n = j-2$ and sum with the coefficients $\binom{\Delta_a}{j}$ from (67.3). The resulting finite sum is reorganized by the binomial identity $(1+z)^{\Delta_a} = (1+z)(1+z)^{\Delta_a-1}$; every summand in $o_M(a\circ b)v$ contains a factor $u_r v$ with $r > \Delta_u - 1$, hence vanishes for $v \in M_{h_M}$. Substituting $n = j-1$ and summing with the coefficients in (67.2) leaves exactly the term $a_{\Delta_a-1}b_{\Delta_b-1}v$; the remaining terms again contain a mode killing $\Omega(M)$. Therefore

$$o_M(a \circ b)v = 0, \qquad o_M(a * b)v = o_M(a)o_M(b)v \tag{67.11}$$

for every $v \in M_{h_M}$. This proves that $O(V)$ is in the kernel of the top-level action and that the induced action is multiplicative. □

Quoted Theorem 67.5 (Zhu classification theorem)**.** *Let V be a grading-restricted VOA of CFT type:*

$$V_n = 0 \quad (n < 0), \qquad V_0 = \mathbb{C}\mathbf{1}, \qquad \dim V_n < \infty.$$

For every simple ordinary positive-energy V-module M, the lowest space M_{h_M} is a simple $A(V)$-module. Conversely, every simple $A(V)$-module U has a universal $\mathbb{N}$-gradable weak V-module generated from U as its top level; when the irreducible quotient is ordinary, this construction is *inverse to $M \mapsto M_{h_M}$. In particular, if every simple $A(V)$-module induces an ordinary simple module, then simple ordinary positive-energy V-modules are classified by simple $A(V)$-modules.*

If V is rational in the sense that the ordinary module category is semisimple with finitely many simple objects and finite-dimensional graded pieces, then $A(V)$ is finite-dimensional and semisimple. If, in addition,

$$C_2(V) := \mathrm{span}_{\mathbb{C}}\{a_{-2}b \mid a, b \in V\} \quad \textit{has finite codimension in } V,$$

then the same finite label set appears in the vector of characters $(\chi_i(\tau))_{i\in I}$ entering the modular-invariance theorem. This is the Zhu module-classification and character-finiteness theorem in the form used here [37].

Induction mechanism behind Zhu's theorem. The quoted theorem hides a concrete algebraic construction, so we record the mechanism used later when examples are checked. Let $\mathfrak{g}(V)$ be the Lie algebra generated by symbols $a[m]$, for homogeneous $a \in V$ and $m \in \mathbb{Z}$, modulo the translation relation

$$(L_{-1}a)[m] = -m\, a[m-1],$$

with bracket

$$[a[m], b[n]] = \sum_{j\geq 0} \binom{m}{j} (a_j b)[m+n-j]. \tag{67.12}$$

The degree of $a[m]$ is $\Delta_a - m - 1$. Thus $\mathfrak{g}(V)$ has a triangular decomposition into positive, zero, and negative degree modes. On the top level of an ordinary module, positive degree modes vanish and zero degree modes act by $o(a) = a_{\Delta_a - 1}$. Proposition 67.4 identifies this zero-degree action with the Zhu algebra action.

Conversely, let U be a simple $A(V)$-module. Give U an action of the nonnegative part of $\mathfrak{g}(V)$ by letting positive degree modes act by zero and letting degree-zero modes act through $A(V)$. The induced graded vector space

$$\mathcal{M}_{\mathrm{ind}}(U) = U(\mathfrak{g}(V)) \otimes_{U(\mathfrak{g}(V)_{\geq 0})} U \tag{67.13}$$

is generated by U under negative degree modes. The VOA Jacobi identity imposes additional relations beyond the Lie brackets (67.12); quotienting by those relations gives the universal N-gradable weak module generated from U. Its maximal proper graded submodule with zero intersection with U, when it exists, gives the irreducible quotient. Every simple ordinary module is generated from its top level by the negative modes, so it is obtained from this construction. The analytic and finiteness content of Quoted Theorem 67.5 is exactly the assertion that, under the stated hypotheses, the quotient produced in this way is ordinary and that the process exhausts the ordinary simple modules.

Example 67.6 (Ising Zhu algebra as a top-weight test)**.** For the Ising VOA $L(1/2, 0)$, the Zhu algebra is generated by the class $[\omega]$ of the conformal vector. The three simple modules have top weights

$$0, \qquad \frac{1}{16}, \qquad \frac{1}{2}.$$

Equivalently, the semisimple Zhu algebra is the quotient

$$A(L(1/2,0)) \cong \mathbb{C}[x] \Big/ \left(x \left(x - \frac{1}{16} \right) \left(x - \frac{1}{2} \right) \right), \tag{67.14}$$

where x acts as L_0 on the top level. The idempotents of this three-dimensional algebra are the Lagrange polynomials

$$e_0 = 32\left(x - \frac{1}{16}\right)\left(x - \frac{1}{2}\right), \tag{67.15}$$

$$e_{1/16} = -\frac{256}{7}x\left(x - \frac{1}{2}\right), \tag{67.16}$$

$$e_{1/2} = \frac{32}{7}x\left(x - \frac{1}{16}\right). \tag{67.17}$$

They obey $e_h(h') = \delta_{h,h'}$, $e_h^2 = e_h$ in the quotient (67.14), and $e_0 + e_{1/16} + e_{1/2} = 1$. Thus they project to the vacuum, spin, and energy top levels, respectively.

67.4 Unitary Virasoro Minimal Models

The preceding VOA definitions are deliberately general. The smallest nontrivial unitary chiral CFTs arise when the chiral algebra is just the Virasoro algebra and the representation category is finite. This finiteness is not automatic: it is a consequence of null vectors in highest-weight Virasoro modules together with positivity.

Let $M(c, h)$ be the Verma module generated by a highest-weight vector $|h\rangle$ satisfying

$$L_0|h\rangle = h|h\rangle, \qquad L_n|h\rangle = 0 \quad (n > 0),$$

for the Virasoro algebra of central charge c. In radial quantization of a unitary chiral theory,

$$L_n^\dagger = L_{-n}.$$

Thus each finite descendant level carries a Hermitian Gram form. The first two levels already show the mechanism. At level one,

$$\|L_{-1}|h\rangle\|^2 = \langle h|L_1L_{-1}|h\rangle = 2h,$$

so a unitary module has $h \geq 0$, and $h = 0$ makes $L_{-1}|0\rangle$ a null translation descendant of the vacuum. At level two, in the ordered basis $(L_{-2}|h\rangle, L_{-1}^2|h\rangle)$, the Gram matrix is

$$M^{(2)}(c, h) = \begin{pmatrix} 4h + \frac{c}{2} & 6h \\ 6h & 4h(2h+1) \end{pmatrix}. \tag{67.18}$$

Indeed,

$$\langle h|L_2L_{-2}|h\rangle = \langle h|(4L_0 + c/2)|h\rangle,$$
$$\langle h|L_2L_{-1}^2|h\rangle = 6h,$$

and

$$\langle h|L_1^2L_{-1}^2|h\rangle = 4h(2h+1),$$

obtained by commuting positive modes to the right until they annihilate $|h\rangle$. If $\det M^{(2)}(c, h) = 0$ and $h \neq 0, -1/2$, the null vector is

$$\left(L_{-2} - \frac{3}{2(2h+1)}L_{-1}^2\right)|h\rangle. \tag{67.19}$$

Null descendants are not optional in a unitary CFT: they have zero inner product with every state and must be quotiented to obtain the physical irreducible module.

Quoted Theorem 67.7 (Unitary Virasoro highest-weight classification)**.** *Let $L(c,h)$ be an irreducible positive-energy highest-weight module for the Virasoro algebra with positive-definite Hermitian form and $L_n^\dagger = L_{-n}$. Then either*

$$c \geq 1, \qquad h \geq 0,$$

or there is an integer $m \geq 3$ such that

$$c = c_m := 1 - \frac{6}{m(m+1)} \tag{67.20}$$

and

$$h = h_{r,s}^{(m)} := \frac{\big((m+1)r - ms\big)^2 - 1}{4m(m+1)} \tag{67.21}$$

for integers

$$1 \leq r \leq m-1, \qquad 1 \leq s \leq m.$$

The identification

$$h_{r,s}^{(m)} = h_{m-r,m+1-s}^{(m)} \tag{67.22}$$

leaves $m(m-1)/2$ distinct irreducible modules. The rational unitary Virasoro minimal model $\mathcal{M}(m, m+1)$ has precisely these simple chiral modules. This is the Friedan–Qiu–Shenker necessity theorem together with the Goddard–Kent–Olive coset sufficiency theorem, stated here only at the representation-classification boundary [39, 40].

The theorem is the deep unitarity input. The monograph uses it as a classification boundary, not as a substitute for the local representation calculations above. The mechanism begins with the Kac determinant. Introduce a real parameter $b > 0$ by

$$c = 1 - 6(b^{-1} - b)^2, \qquad h_{r,s}(b) = \frac{(rb^{-1} - sb)^2 - (b^{-1} - b)^2}{4}.$$

At descendant level N, the determinant of the Virasoro Verma Gram form is, up to a positive normalization depending on the chosen ordered monomial basis,

$$\det M^{(N)}(c,h) = C_N \prod_{\substack{r,s \geq 1 \\ rs \leq N}} \big(h - h_{r,s}(b)\big)^{p(N-rs)}, \qquad C_N > 0, \tag{67.23}$$

where $p(k)$ is the number of integer partitions of k. Formula (67.23) is algebraic: it records exactly where a singular vector can occur in a Verma module. Unitarity then imposes an inequality on every principal minor of every level Gram matrix. For $c < 1$, these inequalities are so restrictive that the zero hyperplanes $h = h_{r,s}(b)$ must occur in an interlacing pattern compatible with quotienting null submodules. The full Friedan–Qiu–Shenker/Goddard–Kent–Olive argument proves that this happens precisely when

$$b^2 = \frac{m}{m+1}, \qquad m = 3, 4, \ldots,$$

and the surviving irreducible highest weights are the Kac-table weights in (67.21). Thus the quoted theorem supplies the all-level part of the classification: positivity of the null quotients at every descendant level and exhaustion of the $0 < c < 1$ positive-energy highest-weight modules.

A convenient nonredundant list of representatives is the triangular set

$$1 \leq s \leq r \leq m-1.$$

Rocha–Caridi characters. The next step uses one further representation-theoretic input: the Rocha–Caridi alternating character formula for the irreducible quotients in the unitary Kac table. Its content is the Euler-character statement of the singular-vector resolution of the irreducible Virasoro module. Once this character formula is accepted, the modular S-matrix below is a finite theta-function calculation.

Put $K = m(m+1)$, $q = e^{2\pi i\tau}$, and

$$\Theta_{\lambda,K}(\tau) = \sum_{n\in\mathbb{Z}} q^{(2Kn+\lambda)^2/(4K)}, \qquad \lambda \in \mathbb{Z}/2K\mathbb{Z}.$$

For $1 \le r \le m-1$, $1 \le s \le m$, define

$$\lambda^-_{r,s} = (m+1)r - ms, \qquad \lambda^+_{r,s} = (m+1)r + ms.$$

The irreducible minimal-model character is

$$\chi^{(m)}_{r,s}(\tau) = \frac{\Theta_{\lambda^-_{r,s},K}(\tau) - \Theta_{\lambda^+_{r,s},K}(\tau)}{\eta(\tau)}. \tag{67.24}$$

Proposition 67.8 (Unitary minimal-model modular data from characters)**.** *Let $\mathcal{M}(m, m+1)$ be the diagonal unitary Virasoro minimal model with $m \ge 3$, and assume that its irreducible chiral characters are the Rocha–Caridi characters (67.24). Its chiral character vector is indexed by Kac-table orbits*

$$[r, s] = [m-r, m+1-s], \qquad 1 \le r \le m-1, \quad 1 \le s \le m,$$

or equivalently by the triangular representatives $1 \le s \le r \le m-1$. The modular transformation $\tau \mapsto -1/\tau$ is represented by

$$S_{(r,s),(r',s')} = 2\sqrt{\frac{2}{m(m+1)}}\,(-1)^{1+sr'+rs'} \sin\left(\frac{\pi(m+1)rr'}{m}\right) \sin\left(\frac{\pi m s s'}{m+1}\right), \tag{67.25}$$

and $\tau \mapsto \tau+1$ is represented by the diagonal matrix

$$T_{(r,s),(r,s)} = \exp\left(2\pi i \left(h^{(m)}_{r,s} - \frac{c_m}{24}\right)\right). \tag{67.26}$$

The formula for S is well-defined on Kac-table orbits, is unitary on the orbit basis, and together with T gives the genus-one modular representation on characters.

Proof. The representation-theoretic content imported into the proposition is exactly the character formula (67.24), whose source is the Virasoro minimal-model character theory following the Friedan–Qiu–Shenker and Goddard–Kent–Olive classification [39, 40]. The modular matrix itself is then computed as follows. Poisson summation gives

$$\Theta_{\lambda,K}(-1/\tau) = \sqrt{\frac{-i\tau}{2K}} \sum_{\mu\in\mathbb{Z}/2K\mathbb{Z}} e^{-\pi i\lambda\mu/K}\Theta_{\mu,K}(\tau), \qquad \eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau). \tag{67.27}$$

Therefore

$$\chi^{(m)}_{r,s}(-1/\tau) = \frac{1}{\sqrt{2K}} \sum_{\mu\in\mathbb{Z}/2K\mathbb{Z}} \left(e^{-\pi i\lambda^-_{r,s}\mu/K} - e^{-\pi i\lambda^+_{r,s}\mu/K}\right) \frac{\Theta_{\mu,K}(\tau)}{\eta(\tau)}.$$

Because $\Theta_{\mu,K} = \Theta_{-\mu,K}$, the coefficient of $\Theta_{\lambda^-_{r',s'},K}/\eta$ after grouping the pair $\pm\lambda^-_{r',s'}$ is

$$\begin{aligned}\frac{2}{\sqrt{2K}}&\left[\cos\left(\frac{\pi\lambda^-_{r,s}\lambda^-_{r',s'}}{K}\right)-\cos\left(\frac{\pi\lambda^+_{r,s}\lambda^-_{r',s'}}{K}\right)\right]\\&=2\sqrt{\frac{2}{m(m+1)}}\,(-1)^{1+sr'+rs'}\sin\left(\frac{\pi(m+1)rr'}{m}\right)\sin\left(\frac{\pi mss'}{m+1}\right).\end{aligned}\tag{67.28}$$

The same calculation gives the negative of (67.28) for $\Theta_{\lambda^+_{r',s'},K}/\eta$, so the coefficient of the character $\chi^{(m)}_{r',s'}$ is exactly (67.25). The T-matrix follows from $\Theta_{\lambda,K}(\tau+1) = e^{\pi\mathrm{i}\lambda^2/(2K)}\Theta_{\lambda,K}(\tau)$, $\eta(\tau+1) = e^{\pi\mathrm{i}/12}\eta(\tau)$, and $(\lambda^+_{r,s})^2 - (\lambda^-_{r,s})^2 = 4Krs$, which makes the two theta terms in (67.24) acquire the same phase $e^{2\pi\mathrm{i}(h^{(m)}_{r,s}-c_m/24)}$.

The label identification is a quotient, not a mnemonic. The involution

$$J(r,s) = (m-r, m+1-s)$$

has no fixed points: a fixed point would require $2r = m$ and $2s = m+1$, which cannot hold simultaneously in integers. Thus every orbit has two members, and the factor $2\sqrt{2/(m(m+1))}$ in (67.25) is the quotient normalization. If one starts from the product of the two sine matrices for $SU(2)_{m-2}$ and $SU(2)_{m-1}$, restriction to the J-orbits multiplies the parent normalization by $\sqrt{2}$. This is the finite normalization step that is often hidden in physics derivations. □

It is useful to state the resulting fusion rule without appealing to any picture of fields. Let $n^{(k)c}_{ab}$ be the $SU(2)_k$ fusion coefficient for $0 \le a, b, c \le k$:

$$n^{(k)c}_{ab} = \begin{cases}1, & |a-b| \le c \le \min(a+b, 2k-a-b) \text{ and } a+b+c \in 2\mathbb{Z},\\ 0, & \text{otherwise.}\end{cases}$$

For triangular representatives $a = (r,s)$, $b = (r',s')$, and $c = (r'',s'')$, the unitary minimal-model Verlinde coefficient is

$$\begin{aligned}N_{ab}{}^{c} &= n^{(m-2)\quad r''-1}_{r-1,r'-1}\, n^{(m-1)\quad s''-1}_{s-1,s'-1}\\&\quad + n^{(m-2)\quad m-r''-1}_{r-1,r'-1}\, n^{(m-1)\quad m-s''}_{s-1,s'-1}.\end{aligned}\tag{67.29}$$

The first term fuses to the displayed representative of the target orbit; the second fuses to its J-image.

Proposition 67.9 (Minimal-model fusion from the finite quotient)**.** *Assume the modular data of Proposition 67.8. The Verlinde formula, derived later in this chapter from modular diagonalization and displayed in* (67.77)*, gives exactly* (67.29)*. In particular the coefficients are nonnegative integers.*

Proof. Let

$$S^{(k)}_{ab} = \sqrt{\frac{2}{k+2}}\sin\left(\frac{\pi(a+1)(b+1)}{k+2}\right)$$

be the $SU(2)_k$ sine matrix. The finite sine orthogonality relation

$$\sum_{x=0}^{k} S^{(k)}_{ax} S^{(k)}_{bx} = \delta_{ab}$$

is the ordinary orthogonality of sine functions on the finite interval. Put

$$\theta_x = \frac{\pi(x+1)}{k+2}, \qquad \chi_a(\theta) = \frac{\sin((a+1)\theta)}{\sin\theta}.$$

Then $S^{(k)}_{ax} = S^{(k)}_{0x}\chi_a(\theta_x)$, and the Verlinde coefficient is the finite inner product

$$n^{(k)c}_{ab} = \sum_{x=0}^{k} \big(S^{(k)}_{0x}\big)^2 \chi_a(\theta_x)\chi_b(\theta_x)\chi_c(\theta_x). \tag{67.30}$$

The ordinary $SU(2)$ character identity gives

$$\chi_a\chi_b = \sum_{\substack{|a-b|\le d\le a+b \\ a+b+d \text{ even}}} \chi_d. \tag{67.31}$$

On the finite grid θ_x there are the affine-wall relations

$$\chi_{k+1}(\theta_x) = 0, \qquad \chi_{k+1+r}(\theta_x) = -\chi_{k+1-r}(\theta_x) \quad (r \ge 1), \tag{67.32}$$

which follow from $(k+2)\theta_x = \pi(x+1)$. Substituting (67.31) in (67.30), reducing all terms by (67.32), and using finite orthogonality gives

$$n^{(k)c}_{ab} = \begin{cases} 1, & |a-b| \le c \le \min(a+b, 2k-a-b), \quad a+b+c \in 2\mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases} \tag{67.33}$$

Indeed, if $a+b \le k$ no wall term occurs and this is the ordinary $SU(2)$ tensor-product interval. If $a+b > k$, the reflected terms across the wall $k+1$ cancel exactly the labels $c > 2k-a-b$, leaving the truncated alcove interval displayed above.

For the minimal model, the matrix (67.25) is the signed product of the $SU(2)_{m-2}$ and $SU(2)_{m-1}$ sine matrices, followed by the fixed-point-free quotient by J. The sign is the simple-current selection rule in the Verlinde sum. Therefore the quotient coefficient is obtained from the parent product by summing over the two representatives c and Jc of the target orbit. Written in triangular labels, this gives exactly (67.29). Each summand is a product of 0-1 $SU(2)_k$ fusion coefficients, so the result is a nonnegative integer. □

67.4.1 Coulomb-Gas Blocks and Screening Charges

There is a second, complementary description of Virasoro minimal-model chiral blocks: a free chiral scalar with background charge, together with contour integrals of screening fields. This construction is useful only after its domain of validity is stated carefully. It constructs chiral blocks and intertwining maps in a free-field resolution of Virasoro modules; it is not by itself a definition of the full nonchiral CFT, nor does it replace the Hilbert space positivity and sewing data discussed above.

Let $\varphi(z)$ be a chiral scalar normalized by

$$\varphi(z)\varphi(w) \sim -\log(z-w), \qquad \partial\varphi(z)\partial\varphi(w) \sim -\frac{1}{(z-w)^2}. \tag{67.34}$$

For a complex background-charge parameter α_0, define

$$T_{\alpha_0}(z) = -\frac{1}{2} :\partial\varphi\,\partial\varphi:(z) + \mathrm{i}\alpha_0\partial^2\varphi(z). \tag{67.35}$$

For $V_\alpha(z) =: e^{\mathrm{i}\alpha\varphi(z)}:$, Wick contraction gives

$$T_{\alpha_0}(z)V_\alpha(w) \sim \frac{h(\alpha)V_\alpha(w)}{(z-w)^2} + \frac{\partial V_\alpha(w)}{z-w}, \qquad h(\alpha) = \frac{1}{2}\alpha(\alpha - 2\alpha_0), \tag{67.36}$$

and

$$c = 1 - 12\alpha_0^2. \tag{67.37}$$

The sign of the improvement term in (67.35) is fixed by the convention (67.34); changing the sign of φ changes α and α_0 together and leaves (67.36) invariant.

For the unitary model $\mathcal{M}(m, m+1)$, set

$$\alpha_+^2 = \frac{2(m+1)}{m}, \qquad \alpha_-^2 = \frac{2m}{m+1}, \qquad \alpha_+\alpha_- = -2, \qquad \alpha_0 = \frac{\alpha_+ + \alpha_-}{2}. \tag{67.38}$$

Thus $m\alpha_+ + (m+1)\alpha_- = 0$. The charge associated with the Kac label (r,s) is

$$\alpha_{r,s} = \frac{1-r}{2}\alpha_+ + \frac{1-s}{2}\alpha_-. \tag{67.39}$$

Substitution into (67.37) and (67.36) gives

$$c = 1 - \frac{6}{m(m+1)} = c_m, \qquad h(\alpha_{r,s}) = \frac{\big((m+1)r - ms\big)^2 - 1}{4m(m+1)} = h_{r,s}^{(m)}. \tag{67.40}$$

The Kac reflection is charge conjugation in the background-charge scalar:

$$\alpha_{m-r,m+1-s} = 2\alpha_0 - \alpha_{r,s}, \qquad h(2\alpha_0 - \alpha) = h(\alpha).$$

Definition 67.10 (Screening currents and Coulomb-gas block datum)**.** The screening currents are

$$S_\pm(u) = V_{\alpha_\pm}(u). \tag{67.41}$$

Because $\alpha_+\alpha_- = -2$, (67.36) gives $h(\alpha_+) = h(\alpha_-) = 1$. Given external charges β_i, insertion points z_i, and nonnegative integers $N_\pm$, remove from $\mathbb{C}^{N_+ + N_-}$ all diagonals $u_a = u_{a'}$, $v_b = v_{b'}$, $u_a = v_b$, and all collision hyperplanes with the external points z_i. The multivalued factor in (67.43) determines a rank-one local system $\mathcal{L}$ on this complement; a twisted cycle Γ means a singular chain with coefficients in $\mathcal{L}^\vee$ whose twisted boundary vanishes. Endpoint prescriptions in a compactification, when used, are part of the choice of this relative twisted homology class. A genus-zero Coulomb-gas block datum consists of such a twisted cycle Γ and the neutrality condition

$$\sum_i \beta_i + N_+\alpha_+ + N_-\alpha_- = 2\alpha_0. \tag{67.42}$$

The corresponding chiral integral is

$$\mathcal{F}_\Gamma(z_i) = \int_\Gamma \prod_{a=1}^{N_+} \mathrm{d}u_a \prod_{b=1}^{N_-} \mathrm{d}v_b \prod_{i<j} (z_i - z_j)^{\beta_i\beta_j} \\ \times \prod_{a,i} (u_a - z_i)^{\alpha_+\beta_i} \prod_{b,i} (v_b - z_i)^{\alpha_-\beta_i} \prod_{a<a'} (u_a - u_{a'})^{\alpha_+^2} \prod_{b<b'} (v_b - v_{b'})^{\alpha_-^2} \prod_{a,b} (u_a - v_b)^{-2}. \tag{67.43}$$

Lemma 67.11 (Screenings produce Virasoro intertwiners)**.** *For any closed contour or twisted cycle Γ on which boundary terms vanish, insertion of the screening charges*

$$Q_\pm(\Gamma) = \int_\Gamma S_\pm(u)\,\mathrm{d}u$$

commutes with the Virasoro action generated by T_{α_0}. Consequently (67.43), when convergent or defined by analytic continuation from a convergent chamber, satisfies the Virasoro Ward identities for a chiral block with external weights $h(\beta_i)$.

Proof. First suppose that Γ is represented by a compact twisted chain in a convergence chamber, so that all contour deformations and integrations by parts below are ordinary absolutely convergent operations after choosing local branches. Since $S_\pm$ have weight 1,

$$T(z)S_\pm(u) \sim \frac{S_\pm(u)}{(z-u)^2} + \frac{\partial_u S_\pm(u)}{z-u}.$$

The commutator of $L_n = \oint z^{n+1} T(z)\mathrm{d}z/(2\pi\mathrm{i})$ with the local field $S_\pm(u)$ is therefore

$$[L_n, S_\pm(u)] = \partial_u\big(u^{n+1} S_\pm(u)\big)\,.$$

Thus

$$[L_n, Q_\pm(\Gamma)] = \int_\Gamma d_u\big(u^{n+1} S_\pm(u)\big)\,,$$

where the differential is read as the flat covariant derivative on the rank-one local system carried by the multivalued integrand. Twisted Stokes gives

$$\int_\Gamma \nabla_{\mathcal{L}}\eta = \int_{\partial_{\mathcal{L}}\Gamma} \eta.$$

The right hand side vanishes because $\partial_{\mathcal{L}}\Gamma = 0$. In the single-valued special case this is the usual statement that the integral of a total derivative around a closed contour is zero; in the multivalued case the coefficients of the twisted chain are exactly what cancels the monodromy terms on adjacent boundary faces. For several screenings the same argument is applied to each screening coordinate, and the commutator with the product of screening integrations is the sum of these twisted exact terms. Hence the integrated screening charges commute with all Virasoro modes in the sense of chiral blocks.

It remains to identify the Ward identity obeyed by the unintegrated free field correlator. Wick contraction with (67.35) gives the primary terms (67.36) at each finite exponential insertion. The contour of $\partial\varphi$ around all finite exponential insertions measures the total finite charge $\sum_i \beta_i + N_+\alpha_+ + N_-\alpha_-$. The improvement term in T_{α_0} is equivalently a background charge $2\alpha_0$ at infinity on the sphere. Therefore the stress-tensor contour at infinity has the vacuum behavior needed for the genus-zero Virasoro Ward identity exactly when (67.42) holds. The screening integrations preserve that identity because their Virasoro commutators are the twisted exact terms just shown to integrate to zero.

When the displayed integral is defined by meromorphic continuation, the Ward identity is first proved in a nonempty convergence chamber with the same twisted homology class. Both sides are meromorphic functions of the charges and insertion positions away from collision and resonance divisors, so the identity extends to the continued block on the common domain of definition. □

One-screening Dotsenko–Fateev beta integral. Let $A, B \in \mathbb{C}$ with initially $\operatorname{Re} A > -1$, $\operatorname{Re} B > -1$, and choose the principal branch on $(0,1)$. Then the one-screening integral

$$I(A,B) = \int_0^1 u^A (1-u)^B \, \mathrm{d}u \tag{67.44}$$

is

$$I(A,B) = \frac{\Gamma(A+1)\Gamma(B+1)}{\Gamma(A+B+2)}. \tag{67.45}$$

Both sides define the meromorphic continuation of the same chiral block coordinate in A, B, with poles exactly where endpoint subtractions or resonant descendants have to be specified.

In the convergence chamber this is Euler's beta integral: write

$$\Gamma(A+1)\Gamma(B+1) = \int_0^\infty \int_0^\infty e^{-x-y} x^A y^B \, \mathrm{d}x \, \mathrm{d}y.$$

Use the change of variables $t = x+y$, $u = x/(x+y)$. The Jacobian is t, so the double integral factors as

$$\left(\int_0^\infty e^{-t} t^{A+B+1} \mathrm{d}t \right) \left(\int_0^1 u^A (1-u)^B \mathrm{d}u \right) = \Gamma(A+B+2) I(A,B).$$

Division by $\Gamma(A+B+2)$ gives (67.45). Meromorphic continuation follows because the gamma-function expression is meromorphic and agrees with the integral on a connected open convergence chamber.

Two-screening Selberg chamber. The first place where the Dotsenko–Fateev integrals are not just repeated Euler beta integrals is the two-screening factor. In a real convergence chamber, suppressing the common external-position prefactors, put

$$I_2(A,B,C) = \int_0^1 \int_0^1 u^A (1-u)^B v^A (1-v)^B |u-v|^{2C} \, \mathrm{d}u \, \mathrm{d}v. \tag{67.46}$$

For $\operatorname{Re} A, \operatorname{Re} B > -1$ and

$$\operatorname{Re} C > -\min\{1/2, \operatorname{Re}(A+1), \operatorname{Re}(B+1)\},$$

Selberg's $n = 2$ formula gives

$$I_2(A,B,C) = \frac{\Gamma(A+1)\Gamma(B+1)\Gamma(A+C+1)\Gamma(B+C+1)\Gamma(2C+1)}{\Gamma(A+B+C+2)\Gamma(A+B+2C+2)\Gamma(C+1)}. \tag{67.47}$$

For nonnegative integer A, B, C the same identity is already visible by elementary finite algebra. By symmetry,

$$I_2(A,B,C) = 2\int_{0<v<u<1} u^A(1-u)^B v^A(1-v)^B(u-v)^{2C}\,\mathrm{d}v\,\mathrm{d}u.$$

Expand the three polynomials

$$(1-u)^B, \qquad (1-v)^B, \qquad (u-v)^{2C}.$$

The term with powers i, j, k contributes

$$2(-1)^{i+j+k}\binom{B}{i}\binom{B}{j}\binom{2C}{k}\frac{1}{A+j+k+1}\,\frac{1}{2A+i+j+2C+2}. \tag{67.48}$$

Summing (67.48) over $0 \le i, j \le B$ and $0 \le k \le 2C$ gives exactly (67.47) with gamma functions replaced by factorials. Thus the new datum in the two-screening block is not an additional copy of the one-screening endpoint powers; it is the screening–screening exponent $2C$, which is $\alpha_\pm^2$ in the Coulomb-gas block and is the source of the Selberg/Vandermonde factors in multi-screening Dotsenko–Fateev formulas.

Screening-to-BPZ construction budget. The preceding formulas are coordinates on a block space; they are not yet a proof that a chosen contour family is the correct minimal-model correlator. On a finite jet test space in the cross-ratio, one should separate the following finite vectors:

$$\mathcal{C}_{\mathrm{DF}}, \quad \mathcal{C}_{\mathrm{Ward}}, \quad \mathcal{C}_{\mathrm{tw}}, \quad \mathcal{C}_{\mathrm{BPZ}}, \quad \mathcal{C}_{\mathrm{cont}}, \quad \mathcal{C}_{\mathrm{full}}.$$

They denote, respectively, the raw Dotsenko–Fateev screening integrals, the same coordinates after the Virasoro Ward/intertwiner check, the projection to closed twisted cycles, the comparison with a BPZ solution basis, the analytically continued block coordinates in a second channel, and the nonchiral sewing pairing. If $\mathsf{M}_{\mathrm{cont}}$ is the finite connection matrix between the chosen BPZ bases, define

$$E_{\mathrm{scr}} = \mathcal{C}_{\mathrm{Ward}} - \mathcal{C}_{\mathrm{DF}}, \qquad E_{\mathrm{tw}} = \mathcal{C}_{\mathrm{tw}} - \mathcal{C}_{\mathrm{Ward}}, \tag{67.49}$$

$$E_{\mathrm{BPZ}} = \mathcal{C}_{\mathrm{BPZ}} - \mathcal{C}_{\mathrm{tw}}, \qquad E_{\mathrm{cont}} = \mathcal{C}_{\mathrm{cont}} - \mathsf{M}_{\mathrm{cont}}\mathcal{C}_{\mathrm{BPZ}}, \tag{67.50}$$

$$E_{\mathrm{pair}} = \mathcal{C}_{\mathrm{full}} - \mathcal{C}_{\mathrm{cont}}. \tag{67.51}$$

Then the finite comparison identity is

$$\mathcal{C}_{\mathrm{full}} - \mathsf{M}_{\mathrm{cont}}\mathcal{C}_{\mathrm{DF}} = \mathsf{M}_{\mathrm{cont}}(E_{\mathrm{scr}} + E_{\mathrm{tw}} + E_{\mathrm{BPZ}}) + E_{\mathrm{cont}} + E_{\mathrm{pair}}. \tag{67.52}$$

Indeed, add and subtract the intermediate coordinates before applying the connection matrix. If the norm of $\mathsf{M}_{\mathrm{cont}}$ on this finite test space is at most κ, and if the five residuals are bounded by $\varepsilon_{\mathrm{scr}}, \varepsilon_{\mathrm{tw}}, \varepsilon_{\mathrm{BPZ}}, \varepsilon_{\mathrm{cont}}, \varepsilon_{\mathrm{pair}}$, then

$$\|\mathcal{C}_{\mathrm{full}} - \mathsf{M}_{\mathrm{cont}}\mathcal{C}_{\mathrm{DF}}\| \le \kappa(\varepsilon_{\mathrm{scr}} + \varepsilon_{\mathrm{tw}} + \varepsilon_{\mathrm{BPZ}}) + \varepsilon_{\mathrm{cont}} + \varepsilon_{\mathrm{pair}}. \tag{67.53}$$

The residuals have distinct meanings:

- E_{scr} measures failure of the screening insertion to satisfy the Ward/intertwiner relation in the declared regularization.

- E_{tw} measures failure of the contour to be a closed twisted cycle, including endpoint and relative-boundary prescriptions.
- E_{BPZ} measures the finite BPZ differential-equation residual after quotienting by null vectors.
- E_{cont} measures analytic continuation and connection-matrix error between the two chosen channel bases.
- E_{pair} measures the nonchiral reflection-positive pairing and sewing residual.

Thus a Dotsenko–Fateev formula becomes a theorem-level minimal-model block construction only when these five residuals vanish compatibly as the finite jet space is enlarged.

Remark 67.12 (What the Dotsenko–Fateev formulas do and do not prove). Multiple screening integrals are Selberg-type generalizations of (67.44). Their contour choices span spaces of solutions to the BPZ differential equations and are a powerful way to compute chiral-block connection matrices and selected structure constants. The load-bearing data are the charges, the neutrality equation, the twisted cycle, and the analytic continuation prescription. A single displayed integral, without those data and without the nonchiral sewing pairing, is not a construction of the full minimal model. Conversely, once the Hilbert space, null quotient, modular data, and sewing constraints have been fixed, the Coulomb-gas integrals give concrete coordinates on the same chiral block spaces and make many BPZ solutions explicitly computable.

Consequently the diagonal A-series full CFT has genus-one partition function

$$Z_m(\tau, \bar{\tau}) = \sum_{1 \le s \le r \le m-1} \left|\chi_{r,s}(\tau)\right|^2,$$

which is modular invariant because the same unitary S and diagonal T act on the holomorphic and antiholomorphic character vectors.

For $m = 3$, one obtains

$$c_3 = \frac{1}{2}, \qquad h_{1,1} = 0, \qquad h_{2,1} = \frac{1}{2}, \qquad h_{2,2} = \frac{1}{16}.$$

The same spin field may also be represented as $h_{1,2} = 1/16$, which is the label making its level-two null vector manifest. For $c = 1/2$ and $h = 1/16$, the matrix (67.18) has determinant zero and (67.19) becomes

$$\left(L_{-2} - \frac{4}{3}L_{-1}^2\right)|\sigma\rangle = 0. \tag{67.54}$$

Proposition 67.13 (Ising spin four-point function from the BPZ equation)**.** *Normalize the Ising spin field by $\langle\sigma(z, \bar{z})\sigma(0)\rangle = |z|^{-1/4}$. Let*

$$\langle\sigma(z_1, \bar{z}_1)\cdots\sigma(z_4, \bar{z}_4)\rangle = |z_{12}|^{-1/4}|z_{34}|^{-1/4} f(z, \bar{z})$$

in the usual four-point cross-ratio frame. The null vector (67.54) implies the holomorphic BPZ equation

$$\left[\partial_z^2 + \frac{2-5z}{4z(1-z)}\partial_z - \frac{3}{64(1-z)^2}\right] f(z, \bar{z}) = 0, \tag{67.55}$$

and the analogous antiholomorphic equation. A basis of local holomorphic solutions is

$$f_\pm(z) = (1-z)^{-1/8}\sqrt{1 \pm \sqrt{z}}.$$

If

$$\mathcal{F}_0(z) = \frac{f_+(z) + f_-(z)}{2}, \qquad \mathcal{F}_\varepsilon(z) = f_+(z) - f_-(z),$$

then single-valuedness, normalization, and crossing fix

$$f(z, \bar{z}) = |\mathcal{F}_0(z)|^2 + \frac{1}{4}|\mathcal{F}_\varepsilon(z)|^2, \qquad C_{\sigma\sigma\varepsilon} = \frac{1}{2} \tag{67.56}$$

up to the sign convention for ε.

Proof. The BPZ equation follows from the null vector as follows. Insert (67.54) at one point in the four-point function. The L_{-1}^2 term is the second derivative with respect to that point. The L_{-2} term is computed by writing L_{-2} as a small contour integral of $T(\zeta)$, expanding the contour to surround the other three insertions, and using the primary Ward identity

$$T(\zeta)\sigma(z_i) \sim \frac{(1/16)\sigma(z_i)}{(\zeta - z_i)^2} + \frac{\partial_{z_i}\sigma(z_i)}{\zeta - z_i}.$$

Fixing three points by $PSL_2(\mathbb{C})$ covariance reduces the resulting Ward identity to (67.55). Direct substitution checks that f_+ and f_- solve the second-order equation.

The two solutions are multivalued, but the full correlator must be single-valued. Around $z = 0$, the combinations $\mathcal{F}_0$ and $\mathcal{F}_\varepsilon$ have the OPE behavior of the identity and energy channels in $\sigma \times \sigma = \mathbf{1} \oplus \varepsilon$. Therefore the most general diagonal single-valued expression in this basis is

$$f(z, \bar{z}) = |\mathcal{F}_0(z)|^2 + C^2|\mathcal{F}_\varepsilon(z)|^2$$

with $C^2 \geq 0$ by reflection positivity. The holomorphic solutions obey the reduced crossing transformation

$$f_+(1-z) = \left(\frac{1-z}{z}\right)^{1/8} \frac{f_+(z) + f_-(z)}{\sqrt{2}}, \qquad f_-(1-z) = \left(\frac{1-z}{z}\right)^{1/8} \frac{f_+(z) - f_-(z)}{\sqrt{2}}.$$

The scalar prefactor is not part of the fusion matrix: it is the conformal weight factor coming from the external spin fields. In the cross-ratio frame $G(z, \bar{z}) = |z|^{-1/4} f(z, \bar{z})$, crossing says $|1 - z|^{-1/4} f(1-z, 1-\bar{z}) = |z|^{-1/4} f(z, \bar{z})$, and this exactly cancels the modulus square of $((1-z)/z)^{1/8}$. Thus the matrix part is the invariant datum. Equivalently,

$$\mathcal{F}_0(1-z) = \left(\frac{1-z}{z}\right)^{1/8} \left(\frac{1}{\sqrt{2}}\mathcal{F}_0(z) + \frac{1}{2\sqrt{2}}\mathcal{F}_\varepsilon(z)\right),$$

$$\mathcal{F}_\varepsilon(1-z) = \left(\frac{1-z}{z}\right)^{1/8} \left(\sqrt{2}\,\mathcal{F}_0(z) - \frac{1}{\sqrt{2}}\mathcal{F}_\varepsilon(z)\right).$$

The remaining crossing condition for the matrix part then forces the coefficient of the mixed term $\mathcal{F}_0\overline{\mathcal{F}_\varepsilon} + \text{c.c.}$ to vanish:

$$\frac{1}{2} - 2C^2 = 0.$$

Thus $C^2 = 1/4$. The coefficients of $|\mathcal{F}_0|^2$ and $|\mathcal{F}_\varepsilon|^2$ then also match, giving (67.56). $\square$

67.5 Sewing and Higher-Genus Blocks

Sewing is the operation that constructs higher-genus surfaces and higher-point blocks from lower-genus pieces. Let C_1 and C_2 have marked points with coordinates z and w. Removing disks and imposing

$$zw = q, \qquad |q| < 1,$$

glues the two punctures into an annulus. If the two glued punctures carry a module M_a and its dual $M_a^\vee$, a sewn chiral block has the form

$$\mathcal{F}_{\mathrm{sewn}}(q) = \sum_\alpha q^{h_a+n_\alpha-c/24}\mathcal{F}_1(\cdots, e_\alpha)\mathcal{F}_2(e^\alpha, \cdots). \tag{67.57}$$

Here $\{e_\alpha\}$ is a homogeneous basis of M_a, e^α is the dual basis with respect to the invariant pairing, and $L_0 e_\alpha = (h_a + n_\alpha)e_\alpha$. The exponent records propagation along the sewn annulus.

The formula is meaningful only after specifying a topology on the module and a convergence domain for the sum. For a unitary positive-energy chiral module M_a write

$$M_a = \widehat{\bigoplus_{n\geq 0}} M_{a,h_a+n}, \qquad d_a(n) = \dim M_{a,h_a+n},$$

where L_0 has eigenvalue $h_a + n$ on M_{a,h_a+n}. For $0 < |q| < 1$ define the annular propagation operator

$$P_a(q) = q^{L_0-c/24}, \qquad P_a(q)e = q^{h_a+n-c/24}e \quad (e \in M_{a,h_a+n}), \tag{67.58}$$

with a fixed branch of $\log q$ on the sewing coordinate chart. Its trace norm is

$$\|P_a(q)\|_1 = |q|^{h_a-c/24}\sum_{n\geq 0} d_a(n)|q|^n = |\chi_a(q)|, \tag{67.59}$$

up to the phase carried by $q^{L_0-c/24}$. Thus the first analytic condition behind sewing is ordinary character convergence at the modulus $|q|$.

Definition 67.14 (Hilbert sewing datum)**.** Let M_a be a unitary chiral module with nondegenerate invariant Hermitian pairing. A one-channel Hilbert sewing datum consists of:

1. two continuous linear functionals ℓ_1, ℓ_2 on the Hilbert completion of M_a, representing the two lower-genus chiral blocks with one M_a-leg and one $M_a^\vee$-leg;
2. a sewing parameter q for which the propagator $P_a(q)$ is trace class;
3. a choice of dual homogeneous bases $\{e_\alpha\}$, $\{e^\alpha\}$ satisfying $\langle e^\alpha, e_\beta\rangle = \delta^\alpha{}_\beta$.

The sewn block is the scalar

$$\mathrm{Sew}_{a,q}(\ell_1, \ell_2) = \sum_\alpha \ell_1(e_\alpha)\, q^{L_0(e_\alpha)-c/24}\, \ell_2(e^\alpha), \tag{67.60}$$

provided the series is absolutely convergent.

Lemma 67.15 (Trace-class estimate for one-channel sewing)**.** *Assume $P_a(q)$ is trace class and that $\ell_i(v) = \langle u_i, v\rangle$ for vectors u_i in the Hilbert completion of M_a. Then the series (67.60) is absolutely convergent and satisfies*

$$|\mathrm{Sew}_{a,q}(\ell_1, \ell_2)| \leq \|u_1\|\, \|u_2\|\, \|P_a(q)\|_1. \tag{67.61}$$

In particular the sewn block is independent of the chosen orthonormal homogeneous basis.

Proof. Choose the homogeneous basis orthonormal. With $\lambda_\alpha = q^{L_0(e_\alpha)-c/24}$,

$$\mathrm{Sew}_{a,q}(\ell_1,\ell_2) = \sum_\alpha \langle u_1, e_\alpha\rangle \lambda_\alpha \langle e_\alpha, u_2\rangle = \langle u_1, P_a(q)u_2\rangle$$

when the finite-level partial sums are taken first. Trace class implies compactness and boundedness, with $\|P_a(q)\| \le \|P_a(q)\|_1$. Hence

$$|\langle u_1, P_a(q)u_2\rangle| \le \|u_1\|\,\|P_a(q)\|_1\,\|u_2\|.$$

Absolute convergence of the displayed basis expansion follows directly from

$$\sum_\alpha |\langle u_1, e_\alpha\rangle|\,|\lambda_\alpha|\,|\langle e_\alpha, u_2\rangle| \le \|u_1\|\,\|u_2\| \sup_{\|v\|=\|w\|=1} \sum_\alpha |\lambda_\alpha||\langle v, e_\alpha\rangle|\,|\langle e_\alpha, w\rangle| \le \|u_1\|\,\|u_2\| \sum_\alpha |\lambda_\alpha|.$$

The last sum is the trace norm in this diagonal homogeneous basis. If $P_a(q)$ is not diagonal in the chosen orthonormal basis, apply the same argument to its singular-value decomposition

$$P_a(q) = \sum_r s_r\,|v_r\rangle\langle w_r|, \qquad s_r \ge 0, \qquad \sum_r s_r = \|P_a(q)\|_1,$$

which gives the same estimate by Cauchy–Schwarz in the r-sum. The finite-level partial sums converge in trace norm to $P_a(q)$, so their matrix elements converge to $\langle u_1, P_a(q)u_2\rangle$. Since this inner product is an operator-theoretic scalar, not a coordinate expression, changing the orthonormal homogeneous basis does not change the sewn block. □

Remark 67.16 (Torus trace from self-sewing). Let M_a be a unitary positive-energy module and let v be a chiral state whose vertex operator $Y(v,1)$ preserves the common domain of $P_a(q)$. Sewing the two punctures of the sphere two-point block $\langle e^\alpha, Y(v,1)e_\alpha\rangle$ gives

$$Z_a(v;q) = \mathrm{Tr}_{M_a}\left(Y(v,1)q^{L_0-c/24}\right). \tag{67.62}$$

For $v = \mathbf{1}$, this is the character

$$Z_a(\mathbf{1};q) = \chi_a(q) = \mathrm{Tr}_{M_a}\, q^{L_0-c/24}. \tag{67.63}$$

Indeed, if $\{e_\alpha\}$ is a homogeneous orthonormal basis, the self-sewn block is

$$\sum_\alpha q^{L_0(e_\alpha)-c/24}\langle e_\alpha, Y(v,1)e_\alpha\rangle,$$

which is exactly the trace of $Y(v,1)q^{L_0-c/24}$ on finite-level truncations and hence, whenever the trace is convergent, the operator trace. Setting $v = \mathbf{1}$ gives $Y(\mathbf{1},1) = \mathrm{id}_{M_a}$.

Definition 67.17 (Sewing graph for a chiral block)**.** A chiral sewing graph Γ consists of:

1. vertices v, each labelled by a genus g_v, external insertion labels, and a finite-dimensional chiral block functional $\mathcal{F}_v$;
2. internal edges e, each joining two marked local coordinates or two marked coordinates on the same vertex surface;
3. for every internal edge e, a channel label a_e, a pair of dual chiral modules M_{a_e}, $M_{a_e}^\vee$, and a sewing parameter q_e with $|q_e| < 1$;

4. an orientation convention specifying which end of each edge carries M_{a_e} and which carries $M_{a_e}^{\vee}$.

Its geometric genus is

$$g(\Gamma) = \sum_v g_v + b_1(\Gamma), \tag{67.64}$$

where $b_1(\Gamma)$ is the first Betti number of the graph.

Given Γ, the formal sewn block is obtained by inserting a dual basis on every internal edge and multiplying the propagator factors:

$$\mathcal{F}_\Gamma(\{q_e\}) = \sum_{\{\alpha_e\}} \left(\prod_e q_e^{L_0(e_{\alpha_e}) - c/24} \right) \prod_v \mathcal{F}_v(\ldots, e_{\alpha_e}, e^{\alpha_e}, \ldots). \tag{67.65}$$

The genus formula (67.64) follows because sewing two connected components adds genera, while sewing two punctures on the same connected component adds one independent cycle.

Finite-channel sewing estimate. Assume a rational unitary chiral theory with finitely many channel labels, and fix a sewing graph Γ. Suppose that for each edge e, $P_{a_e}(q_e)$ is trace class, and that every vertex block functional $\mathcal{F}_v$ extends continuously to the Hilbert tensor product of its incident channel modules. Then the edge-basis sum (67.65) converges absolutely for the chosen $\{q_e\}$. Its value is independent of the orthonormal homogeneous bases used on internal edges.

Insert finite-level projectors $E_{a_e,\le N}$ on every internal edge. For fixed cutoffs the sum is finite. Choose orthonormal homogeneous bases on the cutoff subspaces and write

$$\lambda_{e,\alpha} = q_e^{L_0(e_\alpha) - c/24}$$

for the diagonal matrix coefficient of the annular propagator on edge e. If $\mathbf{e}_v(\boldsymbol{\alpha})$ denotes the tensor product of the unit basis vectors incident on v, then continuity of the vertex block functional gives the pointwise bound

$$|\mathcal{F}_v(\mathbf{e}_v(\boldsymbol{\alpha}))| \le \|\mathcal{F}_v\|.$$

Therefore the absolute value of the finite cutoff contraction is bounded by

$$\begin{aligned} \sum_{\{\alpha_e\}} \left(\prod_e |\lambda_{e,\alpha_e}| \right) \left(\prod_v |\mathcal{F}_v(\mathbf{e}_v(\boldsymbol{\alpha}))| \right) &\le \left(\prod_v \|\mathcal{F}_v\| \right) \prod_e \left(\sum_{\alpha_e} |\lambda_{e,\alpha_e}| \right) \\ &\le \left(\prod_v \|\mathcal{F}_v\| \right) \left(\prod_e \|P_{a_e}(q_e)\|_1 \right). \end{aligned} \tag{67.66}$$

The first inequality uses only that the same edge index is seen by the two incident vertices; after taking absolute values the indices separate into a product of one-edge sums. The second inequality is equality in a homogeneous diagonal basis and is bounded by the trace norm in any orthonormal basis. Since the right hand side of (67.66) is finite and independent of the cutoffs, the finite absolute sums are dominated by the full product of the edge ℓ^1 sums. Enlarging the cutoff on one or more edges adds a tail whose absolute value is bounded by the same product with at least one edge sum replaced by its trace-norm tail; these tails tend to zero. Hence the cutoff net is absolutely Cauchy. Equivalently, if one replaces $P_{a_e}(q_e)$ by its singular-value truncations, the tensor product of the edge operators converges in trace norm because

$$\left\| \bigotimes_e P_{a_e}(q_e) \right\|_1 = \prod_e \|P_{a_e}(q_e)\|_1$$

for finite tensor products of trace-class operators, and the vertex functionals are bounded on the Hilbert tensor products. This operator description also proves basis independence: two homogeneous orthonormal bases give the same trace-class tensor contraction after the cutoff limit.

The estimate is intentionally local. It controls a plumbing chart with fixed channel labels and $|q_e| < 1$. It does not prove that the resulting functions analytically continue between different pants decompositions, nor that two such continuations agree on the overlap of coordinate charts. Those are the analytic sewing and Moore–Seiberg compatibility statements isolated next.

Hypothesis 67.18 (Analytic sewing theorem for rational chiral data)**.** For the rational chiral theories treated as fully constructed in this chapter, the absolutely convergent local sewing series (67.65) analytically continue over the appropriate cover of the moduli space of pointed curves, assemble into finite-rank vector bundles of chiral blocks, and are compatible with changes of pants decomposition by the fusing, braiding, and modular moves. The compatibility maps obey the pentagon, hexagon, and genus-one modular identities.

This hypothesis is not a cosmetic assumption. The trace-class estimates above control a local plumbing coordinate chart near $q_e = 0$. They do not by themselves prove global analytic continuation, mapping-class-group covariance, or equality of two different decompositions of the same Riemann surface. Those assertions are precisely the higher-genus sewing theorem.

Definition 67.19 (Sewing-compatible rational chiral theory)**.** For this chapter, a rational chiral theory is a VOA V with finitely many simple ordinary modules $\{M_i\}_{i \in I}$, semisimple module category, finite-dimensional conformal-block spaces, nondegenerate two-point pairings, and convergent sewing expansions (67.57) in a neighborhood of $q = 0$, satisfying Hypothesis 67.18. The analytic continuations of these blocks along the moduli space of pointed curves give a projectively flat vector bundle of conformal blocks.

The mapping class group acts on this bundle. In genus zero the elementary moves are associativity and braiding of OPE channels. In genus one the generators are

$$S : \tau \mapsto -1/\tau, \qquad T : \tau \mapsto \tau + 1.$$

The compatibility of sewing with these moves is the Moore–Seiberg system: pentagon, hexagon, and modular identities for associativity, braiding, and twist data. The modern algebraic home of this system is a modular tensor category when the module category is semisimple.

67.5.1 Moore–Seiberg Data as Defect-Junction Consistency

The Moore–Seiberg equations should be read as equalities of conformal-block transport maps, or equivalently as equalities of topological-line junction networks after the physical null quotient of Definition 121.3. This is the same physical-defect category used in Chapter 121; here its objects are the topological lines labelled by chiral sectors.

Let $\mathcal{C}$ be a semisimple chiral sector category with simple objects $\{M_i\}_{i\in I}$. For simple labels define the fusion spaces

$$V_{ij}^k := \mathrm{Hom}_{\mathcal{C}}(M_i \boxtimes M_j, M_k), \qquad N_{ij}{}^k = \dim V_{ij}^k. \tag{67.67}$$

A vector in V_{ij}^k is a trivalent junction at which the two lines i, j fuse to the line k. For four external labels i, j, k, ℓ, the two standard pants decompositions have block spaces

$$\bigoplus_m V_{ij}^m \otimes V_{mk}^\ell, \qquad \bigoplus_n V_{jk}^n \otimes V_{in}^\ell. \tag{67.68}$$

The fusing matrix is the change of basis

$$F_{ijk}^\ell : \bigoplus_m V_{ij}^m \otimes V_{mk}^\ell \longrightarrow \bigoplus_n V_{jk}^n \otimes V_{in}^\ell. \tag{67.69}$$

The braiding maps are linear isomorphisms

$$R_{ij}^k : V_{ij}^k \longrightarrow V_{ji}^k, \tag{67.70}$$

and the twist is a scalar θ_i on each simple line in the semisimple case. They are the local coordinates of a topological defect category: F changes the parenthesization of three fusions, R exchanges two adjacent lines, and θ records a full rotation of one line.

The pentagon is the assertion that the two composites from $(((a \boxtimes b) \boxtimes c) \boxtimes d)$ to $a \boxtimes (b \boxtimes (c \boxtimes d))$ agree:

$$\alpha_{a,b,c\boxtimes d} \circ \alpha_{a\boxtimes b,c,d} = (\mathbf{1}_a \boxtimes \alpha_{b,c,d}) \circ \alpha_{a,b\boxtimes c,d} \circ (\alpha_{a,b,c} \boxtimes \mathbf{1}_d). \tag{67.71}$$

Here $\alpha_{a,b,c} : (a \boxtimes b) \boxtimes c \to a \boxtimes (b \boxtimes c)$ is represented, in the bases (67.68), by the corresponding F-matrix. In multiplicity sectors the pentagon is an equality of linear maps between direct sums of junction spaces. Physically it says that the correlator obtained by fusing four nearby topological lines is independent of the order in which the intermediate collars are shrunk.

The two hexagons say that moving one line past a fused pair is the same as moving it past the two constituents one at a time, with the associators included. With braiding $c_{a,b} : a \boxtimes b \to b \boxtimes a$, one of the two hexagons is

$$c_{a,b\boxtimes c} = \alpha_{b,c,a}^{-1} \circ (\mathbf{1}_b \boxtimes c_{a,c}) \circ \alpha_{b,a,c} \circ (c_{a,b} \boxtimes \mathbf{1}_c) \circ \alpha_{a,b,c}^{-1}, \tag{67.72}$$

as a map $a \boxtimes (b \boxtimes c) \to (b \boxtimes c) \boxtimes a$. The other hexagon is the parallel identity for $c_{a\boxtimes b,c}$. The balancing identity

$$\theta_{a\boxtimes b} = c_{b,a} \circ c_{a,b} \circ (\theta_a \boxtimes \theta_b)$$

then fixes how a Dehn twist acts on a fused pair. In a rational unitary CFT, these equalities are the local algebraic part of sewing: they identify different cuts of the same punctured surface after analytic continuation around the moduli space.

This formulation also fixes the status of the algebraic classification problem. A solution of the pentagon, hexagon, and modular equations is a candidate topological-line and conformal-block datum. To obtain a QFT one must still construct the Hilbert space of states, local fields or local algebras, positive pairings, convergent correlation functions, and sewing maps whose analytic continuations realize the displayed F, R, and θ. Conversely, once those analytic and positivity structures are constructed, the Moore–Seiberg equations follow from equality of two correlation-function limits in the defect category. The monograph therefore uses the equations as structural constraints and as coordinates on known classes of rational CFT data; the definition of the CFT itself includes the analytic and Hilbert-space structures just listed.

67.6 Modular Tensor Category and Verlinde Formula

Let $\mathcal{C}$ be the semisimple ribbon category of modules of a rational chiral theory. Its simple objects are M_i, with tensor product

$$M_i \boxtimes M_j \cong \bigoplus_k N_{ij}{}^k M_k.$$

The matrices $(N_i)_j{}^k = N_{ij}{}^k$ are the fusion matrices. The category has a braiding, duals, and twists $\theta_i = \mathrm{e}^{2\pi \mathrm{i} h_i}$. Its modular S-matrix is the Hopf-link pairing normalized so that the vacuum label 0 corresponds to V. In a unitary theory S is unitary and

$$d_i := \frac{S_{i0}}{S_{00}}$$

is the quantum dimension of M_i.

Quoted Theorem 67.20 (Modularity of rational chiral characters)**.** *For a rational, C_2-cofinite, positive-energy VOA satisfying the analytic finiteness hypotheses needed for sewing, the characters $\chi_i(\tau)$ of simple modules span a finite-dimensional representation of $SL(2,\mathbb{Z})$:*

$$\chi_i(-1/\tau) = \sum_j S_{ij}\chi_j(\tau), \qquad \chi_i(\tau+1) = \sum_j T_{ij}\chi_j(\tau).$$

The representation agrees with the modular data of the semisimple module category. This is Zhu's modular-invariance theorem, with the tensor-category identification supplied by the Huang–Lepowsky theory in the rational setting [37, 38].

Finite trace-recursion mechanism. The proof boundary in Quoted Theorem 67.20 is analytic, but the algebraic source of finite modular closure is explicit. Use Zhu's square-bracket coordinate

$$Y[a,z] := Y(\mathrm{e}^{zL_0}a, \mathrm{e}^z - 1) = \sum_{n\in\mathbb{Z}} a[n]z^{-n-1}, \tag{67.73}$$

and write

$$Z_i(v,\tau) = \mathrm{Tr}_{M_i}\, o(v)\, q^{L_0 - c/24}.$$

Let

$$\wp(z,\tau) = z^{-2} + \sum_{\ell\geq 1}(2\ell+1)G_{2\ell+2}(\tau)z^{2\ell}, \qquad G_{2k}(\tau) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{0\}} (m\tau+n)^{-2k}. \tag{67.74}$$

The genus-one Ward identity obtained by moving an a-contour around the torus gives, for homogeneous $a, b \in V$,

$$Z_i(a[-2]b,\tau) + \sum_{\ell\geq 1}(2\ell+1)G_{2\ell+2}(\tau)Z_i(a[2\ell]b,\tau) = 0. \tag{67.75}$$

Equation (67.75) is the torus analogue of the sphere residue calculation in Proposition 67.2: the pole z^{-2} records the $a[-2]b$ descendant, while the holomorphic terms of $\wp$ record the allowed windings of the contour around the elliptic curve.

Now C_2-cofiniteness supplies a finite spanning set for the square-bracket descendant quotient. Repeated use of (67.75), together with the corresponding $L[-2]$-insertion identity that gives the modular covariant τ-derivative, reduces all trace functions to a finite system of linear differential equations with modular-form coefficients. The quoted theorem adds the hard analytic conclusions: convergence of the trace functions on the upper half-plane, continuation to a finite-dimensional space of holomorphic functions, and invariance of that space under the generators $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. The Verlinde calculation below uses only the finite modular representation after these analytic inputs have been secured.

Verlinde formula from modular diagonalization. Assume the fusion matrices are simultaneously diagonalized by a unitary matrix S with $S_{0m} \neq 0$:

$$(N_i S)_{jm} = \frac{S_{im}}{S_{0m}} S_{jm}. \tag{67.76}$$

Then

$$N_{ij}{}^k = \sum_m \frac{S_{im} S_{jm} \overline{S_{km}}}{S_{0m}}. \tag{67.77}$$

The coefficient $N_{ij}{}^k$ is the j, k matrix element of N_i:

$$N_{ij}{}^k = (N_i)_j{}^k.$$

Equation (67.76) says

$$N_i = S D_i S^{-1}, \qquad (D_i)_{mm} = \frac{S_{im}}{S_{0m}}.$$

Since S is unitary, $S^{-1}_{mk} = \overline{S_{km}}$. Therefore

$$(N_i)_j{}^k = \sum_m S_{jm} \frac{S_{im}}{S_{0m}} \overline{S_{km}},$$

which is (67.77).

This calculation separates the deep input from the algebra. The deep input is that the same S-matrix obtained by the modular transformation of torus characters diagonalizes the fusion ring. Once this is granted, the formula is linear algebra.

67.7 The Ising Chiral Theory

The Ising minimal model has central charge

$$c = \frac{1}{2}$$

and three simple modules:

$$\mathbf{1}, \qquad \sigma, \qquad \varepsilon,$$

with conformal weights

$$h_{\mathbf{1}} = 0, \qquad h_\sigma = \frac{1}{16}, \qquad h_\varepsilon = \frac{1}{2}.$$

In the ordered basis $(\mathbf{1}, \sigma, \varepsilon)$, the modular S-matrix is

$$S_{\text{Ising}} = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}. \tag{67.78}$$

The corresponding T-matrix is diagonal:

$$T_{\text{Ising}} = \operatorname{diag}\left(\mathrm{e}^{-\pi\mathrm{i}/24}, \mathrm{e}^{\pi\mathrm{i}/12}, \mathrm{e}^{23\pi\mathrm{i}/24}\right), \tag{67.79}$$

because $T_{ii} = \mathrm{e}^{2\pi\mathrm{i}(h_i - c/24)}$ and $c = 1/2$. It is real, unitary, and obeys $S^2 = 1$, because every simple object is self-dual. The quantum dimensions are

$$d_{\mathbf{1}} = 1, \qquad d_\sigma = \sqrt{2}, \qquad d_\varepsilon = 1, \qquad \mathcal{D}^2 = \sum_i d_i^2 = 4.$$

Applying (67.77) gives the fusion rules

$$\varepsilon \boxtimes \varepsilon = \mathbf{1}, \qquad \varepsilon \boxtimes \sigma = \sigma, \qquad \sigma \boxtimes \sigma = \mathbf{1} \oplus \varepsilon.$$

The verification is pure Verlinde algebra. The identity $S^2 = 1$ is a direct matrix multiplication using $(\sqrt{2})^2 = 2$. The vacuum row gives the quantum dimensions by $d_i = S_{i0}/S_{00}$. For example,

$$N_{\sigma\sigma}{}^{\mathbf{1}} = \sum_m \frac{S_{\sigma m} S_{\sigma m} S_{\mathbf{1} m}}{S_{\mathbf{1} m}} = S_{\sigma\mathbf{1}}^2 + S_{\sigma\sigma}^2 + S_{\sigma\varepsilon}^2 = \frac{1}{2} + 0 + \frac{1}{2} = 1,$$

and

$$N_{\sigma\sigma}{}^{\sigma} = \sum_m \frac{S_{\sigma m} S_{\sigma m} S_{\sigma m}}{S_{\mathbf{1} m}} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

The same substitution gives $N_{\sigma\sigma}{}^{\varepsilon} = 1$, $N_{\varepsilon\varepsilon}{}^{\mathbf{1}} = 1$, and $N_{\varepsilon\sigma}{}^{\sigma} = 1$, with all other coefficients fixed by commutativity and the unit object.

67.8 Operator-Algebraic Conformal Nets

The VOA formalism starts from chiral fields as formal distributions and then extracts algebraic consequences. The operator-algebraic formalism starts from bounded observables localized in intervals on the circle. This difference has operator-algebraic content. Smeared chiral fields are usually unbounded operators; commutation on a common invariant domain does not by itself imply commutation of the generated von Neumann algebras. A conformal net records the bounded local algebras directly, and treats fields as affiliated coordinates when such coordinates exist.

Definition 67.21 (Chiral conformal net)**.** Let $\mathcal{I}(S^1)$ be the set of connected, nonempty, nondense open intervals $I \subset S^1$. A Möbius covariant chiral local net consists of a Hilbert space $\mathcal{H}$, a unit vector $\Omega \in \mathcal{H}$, a positive energy unitary representation U of the Möbius group on $\mathcal{H}$, and von Neumann algebras

$$\mathcal{A}(I) \subset B(\mathcal{H}), \qquad I \in \mathcal{I}(S^1),$$

such that:

1. isotony: $I_1 \subset I_2$ implies $\mathcal{A}(I_1) \subset \mathcal{A}(I_2)$;
2. locality: $I_1 \cap I_2 = \varnothing$ implies $[\mathcal{A}(I_1), \mathcal{A}(I_2)] = 0$;
3. covariance: $U(\gamma)\mathcal{A}(I)U(\gamma)^* = \mathcal{A}(\gamma I)$;
4. positive energy: the generator of rotations has spectrum bounded below by 0;
5. vacuum: $U(\gamma)\Omega = \Omega$, and Ω is cyclic for $\bigvee_{I\in\mathcal{I}(S^1)} \mathcal{A}(I)$.

The net is irreducible if $\bigvee_I \mathcal{A}(I) = B(\mathcal{H})$. It is a conformal net, or diffeomorphism-covariant net, if U extends to a strongly continuous projective positive-energy representation of $\mathrm{Diff}^+(S^1)$ satisfying the same covariance and the local implementer condition: if a diffeomorphism is the identity on I, its implementer acts trivially on $\mathcal{A}(I)$.

The complement I' of an interval is the interior of $S^1 \setminus I$. Haag duality is the equality

$$\mathcal{A}(I)' = \mathcal{A}(I').$$

For irreducible Möbius covariant nets on the circle it follows under the standard vacuum-net hypotheses; in higher-dimensional AQFT it is a separate condition, as in Chapter 4.

Definition 67.22 (Split property, strong additivity, and complete rationality)**.** Let $\mathcal{A}$ be an irreducible Möbius covariant net.

1. The split property means that whenever $\overline{I}_1 \subset I_2$ there is a type I factor N with
$$\mathcal{A}(I_1) \subset N \subset \mathcal{A}(I_2).$$
2. Strong additivity means that if deleting one point from an interval I gives the two intervals I_1, I_2, then
$$\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2).$$
3. Assume the split property. Choose four intervals I_1, I_2, I_3, I_4 in cyclic order and set $E = I_1 \cup I_3$. The μ-index is the Jones index
$$\mu_{\mathcal{A}} = [\mathcal{A}(E')' : \mathcal{A}(E)]$$
when this index is finite and independent of the chosen four intervals.

The net is completely rational if it has the split property, is strongly additive, and has finite $\mu_{\mathcal{A}}$.

Quoted Theorem 67.23 (Representation category of a completely rational net)**.** *For a completely rational chiral conformal net, the finite-index localized representations form a unitary modular tensor category. The categorical global dimension agrees with the μ-index:*

$$\mu_{\mathcal{A}} = \sum_{\rho\in\mathrm{Irr}(\mathcal{A})} d_\rho^2,$$

where d_ρ is the statistical dimension of the sector ρ. This is the Kawahigashi–Longo–Müger conformal-net rationality theorem. The proof uses subfactor index theory and DHR localization, and is not reproduced here [41].

DHR mechanism behind the conformal-net theorem. The quoted theorem is not only a finiteness statement; it identifies the operator-algebraic origin of the braiding and of the modular S-matrix. Fix an interval I_0. A DHR sector localized in I_0 is represented, after Haag duality, by a normal unital endomorphism ρ of the quasilocal algebra such that

$$\rho(A) = A, \qquad A \in \mathcal{A}(I_0'), \tag{67.80}$$

where I_0' is the interior of the complement. If ρ, σ are two such endomorphisms, the morphism space is

$$\mathrm{Hom}(\rho, \sigma) = \{T : T\rho(A) = \sigma(A)T \text{ for all local } A\}. \tag{67.81}$$

The tensor product is composition of endomorphisms, $\rho \otimes \sigma := \rho \circ \sigma$, with tensoring of intertwiners defined by $S \otimes T = S\rho(T)$ for $S \in \mathrm{Hom}(\rho, \rho')$ and $T \in \mathrm{Hom}(\sigma, \sigma')$. Finite Jones index of the subfactor $\rho(\mathcal{A}(I)) \subset \mathcal{A}(I)$, for an interval $I \supset I_0$, gives the statistical dimension

$$d_\rho = [\mathcal{A}(I) : \rho(\mathcal{A}(I))]^{1/2}, \tag{67.82}$$

independent of I in the finite-index sector. Conjugate sectors are endomorphisms $\bar{\rho}$ equipped with intertwiners

$$R \in \mathrm{Hom}(\mathrm{id}, \bar{\rho}\rho), \qquad \bar{R} \in \mathrm{Hom}(\mathrm{id}, \rho\bar{\rho}),$$

obeying the conjugate equations

$$\bar{R}^* \rho(R) = \mathrm{id}_\rho, \qquad R^* \bar{\rho}(\bar{R}) = \mathrm{id}_{\bar{\rho}}, \tag{67.83}$$

after the conventional normalization that minimizes $\|R\|\,\|\bar{R}\|$. This is the C^*-tensor-categorical form of particle-antiparticle pairing in the chiral net.

The braiding is constructed from charge transporters. Move ρ and σ, by unitary intertwiners, to disjoint intervals with a specified cyclic order on S^1. Locality says that the transported endomorphisms commute on all local algebras. Transporting back gives a unitary intertwiner

$$\varepsilon_{\rho,\sigma} \in \mathrm{Hom}(\rho\sigma, \sigma\rho). \tag{67.84}$$

Changing the transporters changes the intermediate representative but not the resulting natural braiding morphism. Naturality and the hexagon identities are therefore consequences of isotony, locality, and the composition law for charge transporters. The two possible cyclic orders on the circle give the braiding and its opposite.

Complete rationality supplies the missing finiteness and nondegeneracy. For two disjoint intervals $E = I_1 \cup I_3$, the finite index

$$\mu_{\mathcal{A}} = [\mathcal{A}(E')' : \mathcal{A}(E)]$$

is the two-interval index measuring the failure of additivity across the two gaps. The subfactor analysis shows that only finitely many irreducible finite-index DHR sectors occur and that

$$\mu_{\mathcal{A}} = \sum_{\rho \in \mathrm{Irr}(\mathcal{A})} d_\rho^2. \tag{67.85}$$

The categorical S-matrix is the normalized trace of monodromy:

$$S_{\rho\sigma} = \frac{1}{\mathcal{D}}\, \mathrm{Tr}_{\rho\sigma}\big(\varepsilon_{\sigma,\rho}\varepsilon_{\rho,\sigma}\big), \qquad \mathcal{D} = \sqrt{\sum_\lambda d_\lambda^2}. \tag{67.86}$$

Here $\mathrm{Tr}_{\rho\sigma}$ is the standard categorical trace coming from the conjugate equations. The vacuum row is $S_{\mathrm{id},\rho} = d_\rho/\mathcal{D}$. Nondegeneracy means that the only sector whose double braiding with every sector is the identity is the vacuum sector; equivalently the matrix $(S_{\rho\sigma})$ is invertible. This is the precise operator-algebraic content of the word "modular" in the theorem. The full proof that the two-interval index forces this nondegeneracy remains the subfactor-theoretic theorem boundary, but the objects entering the statement are the concrete endomorphisms, intertwiners, dimensions, and charge-transporter braidings defined above.

The comparison with VOAs is therefore a comparison of two different coordinates on chiral locality. A VOA gives formal fields and mode algebra; a net gives bounded interval algebras. Passing from one to the other is an analytic theorem, not a definition.

Quoted Theorem 67.24 (Strongly local VOAs and conformal nets)**.** *Let V be a simple unitary VOA whose vertex operators obey polynomial energy bounds. If the smeared fields $Y(a,f)$, $a \in V$, $f \in C^\infty(S^1)$, generate mutually local von Neumann algebras*

$$\mathcal{A}_V(I) = W^*\{\overline{Y(a,f)} : \operatorname{supp} f \subset I,\ a \in V\}, \qquad I \in \mathcal{I}(S^1),$$

then V is called strongly local and $I \mapsto \mathcal{A}_V(I)$ is an irreducible conformal net. The net is independent, up to isomorphism, of the normalized invariant scalar product on V.

Conversely, the Fredenhagen–Jörß reconstruction applied to $\mathcal{A}_V$ recovers the smeared vertex operators of V. More generally, if an irreducible conformal net is generated by quasi-primary reconstructed fields, those fields satisfy polynomial energy bounds, and the L_0-eigenspaces are finite-dimensional, then the finite-energy subspace $\mathcal{H}^{\mathrm{fin}}$ carries a simple unitary strongly local VOA structure V with $\mathcal{A} \simeq \mathcal{A}_V$.

This is the Carpi–Kawahigashi–Longo–Weiner VOA-to-net theorem and its Fredenhagen–Jörß converse under the stated analytic hypotheses [42].

Mechanism of the VOA-to-net passage. Let $\mathcal{H}$ be the Hilbert completion of the unitary VOA V, let L_0 be the positive rotation generator, and set

$$\mathcal{H}^\infty := \bigcap_{k \geq 0} \mathrm{Dom}(1+L_0)^k.$$

A polynomial energy bound for $a \in V$ means that there exist constants C, r, s, depending on a, such that for every mode a_n and every $\psi \in \mathcal{H}^\infty$,

$$\|a_n\psi\| \leq C(1+|n|)^r \|(1+L_0)^s\psi\|. \tag{67.87}$$

For a smooth test function

$$f(\mathrm{e}^{\mathrm{i}\theta}) = \sum_{n\in\mathbb{Z}} \hat{f}_n \mathrm{e}^{-\mathrm{i}n\theta}$$

the rapid decay of $\hat{f}_n$ implies

$$Y(a,f)\psi := \sum_{n\in\mathbb{Z}} \hat{f}_n a_n \psi \tag{67.88}$$

converges in $\mathcal{H}$, because $\sum_n |\hat{f}_n|(1+|n|)^r < \infty$. Unitarity supplies the adjoint formula for vertex operators, so $Y(a,f)$ is closable on $\mathcal{H}^\infty$. The local von Neumann algebra in the theorem is therefore not obtained by treating $Y(a,z)$ as a pointwise operator; it is generated from closed smeared operators with supports contained in I:

$$\mathcal{A}_V(I) = W^*\{\overline{Y(a,f)} : \operatorname{supp} f \subset I,\ a \in V\}.$$

The delicate step is locality. VOA locality gives, for each pair a, b, an integer N such that

$$(z-w)^N [Y(a,z), Y(b,w)] = 0$$

as a formal distribution identity. After smearing with disjointly supported smooth functions, this gives

$$[Y(a,f), Y(b,g)]\psi = 0, \qquad \psi \in \mathcal{H}^\infty.$$

This common-domain commutator identity is not yet locality of the net: the latter requires the von Neumann algebras generated by the closures to commute. The analytic content of strong locality is precisely the upgrade from finite-energy-domain locality of unbounded fields to strong commutativity of the closed smeared fields and their bounded functional calculi. Polynomial energy bounds, positivity of L_0, and the adjoint formula give the operator control used in that upgrade.

Covariance is more formal once the smeared fields exist. The projective unitary Möbius and diffeomorphism action on the Hilbert completion sends a smeared field to a finite linear combination of smeared fields whose test functions are pulled back with the appropriate conformal-weight factors. Thus

$$U(\gamma)\mathcal{A}_V(I)U(\gamma)^* = \mathcal{A}_V(\gamma I)$$

provided the implementers preserve the common smooth domain and the energy bounds. Vacuum cyclicity follows from the state-field correspondence and the density of VOA descendants in the Hilbert completion. The theorem boundary is the simultaneous proof that these analytic requirements are stable for the whole field family, not only for a chosen finite generating set.

Mechanism of the net-to-VOA reconstruction. The converse starts with bounded interval algebras rather than formal fields. For a finite-energy vector $\psi \in \mathcal{H}^{\mathrm{fin}}$, the Fredenhagen–Jörß construction seeks a closed operator-valued distribution Φ_ψ affiliated with the local algebras and normalized by

$$\Phi_\psi(f)\Omega = \int_{S^1} f(z)\, U(z)\psi\, dz$$

in the distributional sense, where $U(z)$ denotes rotation of the state. Locality of the net gives commutation of $\Phi_\psi(f)$ with $\mathcal{A}(I')$ when $\operatorname{supp} f \subset I$. Möbius covariance decomposes $\mathcal{H}^{\mathrm{fin}}$ into finite-dimensional L_0-eigenspaces and fixes the mode degrees of the reconstructed fields. If the reconstructed quasi-primary fields obey polynomial energy bounds, their smeared products have vacuum matrix elements with the same analytic-continuation and locality properties as VOA correlation functions. The modes of these fields then define a state-field map on $\mathcal{H}^{\mathrm{fin}}$, with the vacuum, translation, locality, and Virasoro covariance axioms recovered from the net axioms plus the Fredenhagen–Jörß field-state normalization.

The converse theorem boundary is again analytic rather than formal. One must prove that enough finite-energy vectors admit such reconstructed fields, that the fields satisfy the required energy bounds, and that the mode products close on $\mathcal{H}^{\mathrm{fin}}$. Without those hypotheses a conformal net need not be identified with a VOA merely because both describe chiral locality.

Ising net μ-index and DHR fusion. Assume the unitary Ising VOA is taken in its strongly local realization and that its associated conformal net is completely rational with sectors $\mathbf{1}, \sigma, \varepsilon$. Then

$$\mu_{\mathcal{A}_{\mathrm{Ising}}} = 4,$$

and the DHR fusion ring of the net agrees with the Ising Verlinde fusion ring displayed above.

From the explicit Ising S-matrix (67.78), the quantum dimensions are

$$d_{\mathbf{1}} = 1, \qquad d_\sigma = \sqrt{2}, \qquad d_\varepsilon = 1.$$

The quoted complete-rationality theorem gives

$$\mu_{\mathcal{A}_{\mathrm{Ising}}} = d_{\mathbf{1}}^2 + d_\sigma^2 + d_\varepsilon^2 = 1 + 2 + 1 = 4.$$

The same theorem identifies finite-index DHR sectors with the modular tensor category of the net. Under the VOA-to-net comparison theorem, the strongly local Ising VOA and the Ising net have the same chiral sectors, so the DHR fusion coefficients are the Verlinde coefficients already computed.

Open Problem 67.25 (Unqualified VOA/net equivalence)*.* Remove or sharply weaken the analytic hypotheses in Theorem 67.24. In particular, determine whether every simple unitary positive-energy VOA satisfying the specified finiteness conditions is strongly local, and whether every irreducible conformal net of finite chiral type is recovered from a strongly local VOA. The monograph uses the proved strongly local correspondence as a theorem boundary and does not assume this unqualified equivalence.

67.9 Full CFT and the Modular Bootstrap

A full local CFT is not a chiral theory alone. In a rational diagonal case, one constructs a torus partition function from left and right characters:

$$Z(\tau, \bar{\tau}) = \sum_{i,j} M_{ij} \chi_i(\tau) \overline{\chi_j(\tau)}. \tag{67.89}$$

The matrix M must have nonnegative integer entries, $M_{00} = 1$, and must commute with the modular representation:

$$MS = SM, \qquad MT = TM. \tag{67.90}$$

These are genus-one sewing constraints. Higher-genus sewing imposes further compatibility between M, chiral blocks, duality moves, and the pairing of left and right chiral sectors.

Ising genus-one invariant. For the Ising chiral data of Section 67.7, restrict to full CFT torus partition functions whose left and right chiral labels both lie in $(\mathbf{1}, \sigma, \varepsilon)$. Let

$$M = (M_{ij})_{i,j \in \{\mathbf{1}, \sigma, \varepsilon\}} \in \mathrm{Mat}_3(\mathbb{Z}_{\geq 0})$$

with $M_{\mathbf{11}} = 1$. If M satisfies (67.90) with (67.78) and (67.79), then $M = I_3$. Thus the only genus-one modular invariant with one vacuum in this chiral Ising label set is the diagonal invariant

$$Z_{\mathrm{Ising}}(\tau, \bar{\tau}) = |\chi_{\mathbf{1}}|^2 + |\chi_\sigma|^2 + |\chi_\varepsilon|^2. \tag{67.91}$$

The T-constraint is already a spin-selection rule. Since T is diagonal,

$$(MT)_{ij} - (TM)_{ij} = M_{ij}\left(\mathrm{e}^{2\pi\mathrm{i}(h_j - c/24)} - \mathrm{e}^{2\pi\mathrm{i}(h_i - c/24)}\right).$$

Therefore $M_{ij} \neq 0$ implies $h_i - h_j \in \mathbb{Z}$. For the Ising weights

$$0, \qquad \frac{1}{16}, \qquad \frac{1}{2},$$

no two distinct weights differ by an integer. Hence M is diagonal:

$$M = \mathrm{diag}(1, a, b), \qquad a, b \in \mathbb{Z}_{\geq 0}.$$

The S-constraint now gives, entry by entry,

$$(MS)_{\mathbf{1}\sigma} = S_{\mathbf{1}\sigma}, \qquad (SM)_{\mathbf{1}\sigma} = aS_{\mathbf{1}\sigma}.$$

Since $S_{\mathbf{1}\sigma} = 1/\sqrt{2} \neq 0$, one gets $a = 1$. Similarly

$$(MS)_{\mathbf{1}\varepsilon} = S_{\mathbf{1}\varepsilon}, \qquad (SM)_{\mathbf{1}\varepsilon} = bS_{\mathbf{1}\varepsilon},$$

and $S_{\mathbf{1}\varepsilon} = 1/2 \neq 0$, so $b = 1$. Thus $M = I_3$, and (67.91) follows from (67.89).

67.9.1 Torus One-Point Blocks and Verlinde Defect Lines

The partition function is the genus-one zero-point function. A complete genus-one discussion also needs one-point blocks and topological defect lines. Both are often drawn pictorially; here we spell out the algebraic content because the assumptions are different from those used in the scalar partition function.

Let V be a rational chiral theory satisfying Definition 67.19, and let $v \in V$ be homogeneous of conformal weight h_v. The zero mode

$$o(v) := v_{h_v - 1}$$

has L_0-degree zero. The chiral torus one-point block on the simple module M_i is

$$F_i(v;\tau) := \mathrm{Tr}_{M_i}\, o(v)\, q^{L_0 - c/24}. \tag{67.92}$$

Only the zero mode can contribute to a trace over L_0-eigenspaces: a mode of nonzero L_0-degree maps each finite-dimensional eigenspace to a different eigenspace and therefore has zero diagonal trace. For $v = \mathbf{1}$, $o(v) = 1$ and $F_i(\mathbf{1};\tau) = \chi_i(\tau)$.

Quoted Theorem 67.26 (Zhu modular covariance of torus one-point blocks)**.** *Assume V is rational, C_2-cofinite, of CFT type, and has ordinary simple modules M_i. For each homogeneous $v \in V$, the finite vector $(F_i(v;\tau))_i$ spans a representation of $SL_2(\mathbb{Z})$. If v is Virasoro primary of weight h_v, then for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$*

$$F_i\left(v; \frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{h_v} \sum_j \rho_v(\gamma)_{ij} F_j(v;\tau). \tag{67.93}$$

For $v = \mathbf{1}$, ρ_v is the character modular representation used above. This is the one-point-function part of Zhu's genus-one modular-covariance theorem [37].

The factor $(c\tau+d)^{h_v}$ is the conformal weight factor for the change of torus coordinate. The theorem input is the finite-dimensionality and holomorphic modular closure of the trace functions.

Mechanism behind torus one-point covariance. The one-point theorem is the genus-one version of transporting a conformal block on a once-punctured torus. The point of the zero mode $o(v)$ is that it preserves L_0-eigenspaces; hence the trace

$$F_i(v;\tau) = \mathrm{Tr}_{M_i} o(v) q^{L_0 - c/24}$$

is a genuine finite-energy trace after summing over each L_0-eigenspace. In Zhu's square-bracket coordinate, the inserted state is transported by the coordinate change at the puncture. For a Virasoro primary v, this transport is scalar:

$$v \longmapsto (c\tau + d)^{h_v} v$$

under the modular coordinate change

$$(z,\tau) \longmapsto \left(\frac{z}{c\tau + d}, \frac{a\tau + b}{c\tau + d} \right).$$

For descendants, the same coordinate change also produces lower-descendant terms involving the Schwarzian and the square-bracket Virasoro modes; those terms are the source of the quasimodular G_2-mixing in genus-one Ward recursions. The primary hypothesis in Theorem 67.26 is exactly the condition that this coordinate transport has a single leading weight factor.

The finite-dimensional closure is the same C_2-cofinite trace-recursion mechanism used for characters, but with v kept as a marked insertion. Moving a square-bracket field around the torus expresses traces with $[-2]$-descendants in terms of traces with finitely many lower square-bracket descendants and modular forms. Rationality identifies the simple ordinary modules as a finite set of sectors, while C_2-cofiniteness keeps the marked-state quotient finite. Therefore the traces

$$\{F_i(w;\tau) :\ i \in I,\ w \in W_v\}$$

for the finite descendant span W_v determined by v satisfy a finite modular differential system. Zhu's analytic theorem supplies convergence on $\mathbb{H}$, continuation of this finite solution space, and its invariance under S and T. Formula (67.93) is the primary, one-marked-state component of that finite system.

A full-CFT torus one-point function of a bulk primary Φ of weights $(h, \bar{h})$ is a sesquilinear pairing of the corresponding chiral one-point-block vectors. Modular covariance therefore requires an invariant coefficient tensor for $\rho_\Phi(\gamma) \otimes \overline{\rho_\Phi(\gamma)}$, together with the weight factor $(c\tau + d)^h (c\bar{\tau} + d)^{\bar{h}}$. Thus torus one-point functions constrain bulk OPE data beyond the scalar partition function.

Topological defect lines give a second genus-one probe. In a diagonal rational full CFT built from a unitary modular tensor category $\mathcal{C}$, the Verlinde defects are labelled by simple objects $a \in I$. Let

$$\mathcal{H}_{\mathrm{diag}} = \bigoplus_{i \in I} M_i \otimes \overline{M_i^\vee}$$

be the diagonal Hilbert space. Define D_a on the summand labelled by i by

$$D_a\big|_{M_i \otimes \overline{M_i^\vee}} = \lambda_a(i)\,\mathrm{id}, \qquad \lambda_a(i) := \frac{S_{ai}}{S_{0i}}. \tag{67.94}$$

The denominator is nonzero in a unitary modular tensor category because $S_{0i} = d_i S_{00}$ and $d_i > 0$.

Verlinde defect fusion and modular covariance. For the diagonal rational full CFT above,

$$D_a D_b = \sum_c N_{ab}{}^c D_c. \tag{67.95}$$

Moreover, the torus trace with D_a wrapping the Euclidean-time cycle is

$$Z_a^{\mathrm{time}}(\tau,\bar\tau) = \sum_i \lambda_a(i)\chi_i(\tau)\overline{\chi_i(\tau)}. \tag{67.96}$$

After the modular S-move it becomes the spatial-defect trace

$$Z_a^{\mathrm{time}}(-1/\tau,-1/\bar\tau) = \sum_{p,q} N_{ap}{}^q\,\chi_p(\tau)\overline{\chi_q(\tau)}. \tag{67.97}$$

The Verlinde formula says that the matrices N_a are simultaneously diagonalized by S, with eigenvalue S_{ai}/S_{0i} on the i-th character. Hence for every i,

$$\lambda_a(i)\lambda_b(i) = \sum_c N_{ab}{}^c \lambda_c(i),$$

which proves (67.95) on each diagonal summand of $\mathcal{H}_{\mathrm{diag}}$.

For the modular transform, insert the character transformation law into (67.96). The coefficient of $\chi_p\overline{\chi_q}$ is

$$\sum_i \frac{S_{ai}}{S_{0i}}\,S_{ip}\,\overline{S_{iq}} = \sum_i \frac{S_{ai}S_{pi}S_{q^\vee i}}{S_{0i}} = N_{ap}{}^q,$$

where unitarity gives $\overline{S_{iq}} = S_{q^\vee i}$, and the last equality is Verlinde. This gives (67.97). The right side is a Hilbert-space trace over

$$\mathcal{H}_a = \bigoplus_{p,q} N_{ap}{}^q\, M_p \otimes \overline{M_q^\vee},$$

so the modular move has turned a temporal defect operator into a spatial defect sector.

For the Ising labels $(\mathbf{1},\sigma,\varepsilon)$, the eigenvalue vectors $(\lambda_a(\mathbf{1}),\lambda_a(\sigma),\lambda_a(\varepsilon))$ are

$$\lambda_{\mathbf{1}} = (1,1,1), \qquad \lambda_\sigma = (\sqrt{2},0,-\sqrt{2}), \qquad \lambda_\varepsilon = (1,-1,1).$$

Thus

$$D_\varepsilon^2 = D_{\mathbf{1}}, \qquad D_\varepsilon D_\sigma = D_\sigma, \qquad D_\sigma^2 = D_{\mathbf{1}} + D_\varepsilon,$$

so D_ε is the invertible spin-flip line and D_σ is the noninvertible Kramers–Wannier line. Their modular covariance is completely finite. For example,

$$Z_\varepsilon^{\mathrm{time}} = |\chi_{\mathbf{1}}|^2 - |\chi_\sigma|^2 + |\chi_\varepsilon|^2$$

is sent by S to

$$\chi_{\mathbf{1}}\overline{\chi_\varepsilon} + \chi_\varepsilon\overline{\chi_{\mathbf{1}}} + |\chi_\sigma|^2,$$

the Hilbert space twisted by the spin-flip defect. Similarly

$$Z_\sigma^{\mathrm{time}} = \sqrt{2}\left(|\chi_{\mathbf{1}}|^2 - |\chi_\varepsilon|^2\right)$$

is sent by S to

$$\chi_{\mathbf{1}}\overline{\chi_\sigma} + \chi_\varepsilon\overline{\chi_\sigma} + \chi_\sigma\overline{\chi_{\mathbf{1}}} + \chi_\sigma\overline{\chi_\varepsilon}.$$

The same line-sliding argument gives selection rules for torus one-point functions. Since D_ε acts by -1 on the Ising spin primary, the torus one-point function of σ equals its negative and therefore vanishes. This topological-defect Ward identity is independent of any particular lattice realization.

For a unitary compact two-dimensional CFT with discrete spectrum on the circle, the rectangular torus partition function

$$Z(\beta) = \mathrm{Tr}_{\mathcal{H}(S^1)} \exp\{-\beta(H - c_{\mathrm{tot}}/24)\}, \qquad c_{\mathrm{tot}} = c + \bar{c},$$

obeys modular invariance

$$Z(\beta) = Z(4\pi^2/\beta)$$

when the spatial circle has circumference 2π. If the low-temperature trace is dominated by a unique vacuum, then as $\beta \to 0^+$,

$$\log Z(\beta) = \frac{\pi^2 c_{\mathrm{tot}}}{6\beta} + o(\beta^{-1}). \tag{67.98}$$

The derivation is only the modular transformation: low temperature at $4\pi^2/\beta$ gives

$$Z(4\pi^2/\beta) = \exp\left\{\frac{4\pi^2}{\beta}\frac{c_{\mathrm{tot}}}{24}\right\}(1 + o(1)),$$

which is (67.98). Tauberian hypotheses convert this high-temperature partition-function statement into an asymptotic density of states. Those hypotheses are analytic input; modular invariance alone is not a pointwise formula for individual degeneracies.

Quoted Theorem 67.27 (Exponential Tauberian theorem)**.** *Let $N(E)$ be a nondecreasing right-continuous function on $[0,\infty)$, and suppose its Laplace transform*

$$Z(\beta) = \int_{[0,\infty)} \mathrm{e}^{-\beta E}\,\mathrm{d}N(E)$$

is finite for every $\beta > 0$. If, for some $A > 0$,

$$\log Z(\beta) = \frac{A}{\beta} + o(\beta^{-1}) \qquad (\beta \downarrow 0),$$

then

$$\log N(E) = 2\sqrt{AE} + o(\sqrt{E}) \qquad (E \to \infty).$$

This is the exponential Tauberian input used in Cardy asymptotics; the regular-variation reference is a source trace, while the chapter below spells out the one-sided Laplace estimate and isolates the inverse Tauberian step [43].

The theorem contains an elementary one-sided estimate and a genuinely Tauberian inverse estimate. Let μ_N be the positive Stieltjes measure with $\mu_N([0,E]) = N(E)$. For every $\beta > 0$,

$$Z(\beta) \geq \int_{[0,E]} \mathrm{e}^{-\beta E}\,\mathrm{d}\mu_N(E') = \mathrm{e}^{-\beta E} N(E),$$

and hence

$$\log N(E) \le \inf_{\beta>0}\{\beta E + \log Z(\beta)\}. \tag{67.99}$$

If $\log Z(\beta) = A/\beta + o(\beta^{-1})$, choosing $\beta = \sqrt{A/E}$ in (67.99) gives the upper half

$$\limsup_{E\to\infty} \frac{\log N(E)}{\sqrt{E}} \le 2\sqrt{A}.$$

The inverse half is the nontrivial content: positivity and monotonicity of the counting measure force enough spectral mass in the saddle window $E' \sim A/\beta^2$ to produce the same leading exponent from below. Without that positivity, cancellations in a signed trace can obey the same modular transformation law while carrying no comparable counting interpretation.

The coefficient $2\sqrt{A}$ is fixed by the Legendre saddle. If a candidate cumulative growth has the leading form $\log N(E) \sim B\sqrt{E}$, the formal Laplace exponent is

$$\Phi_\beta(E') = B\sqrt{E'} - \beta E', \qquad E'_\beta = \frac{B^2}{4\beta^2},$$

and

$$\sup_{E'\ge 0} \Phi_\beta(E') = \frac{B^2}{4\beta}.$$

Matching this with A/β gives $B = 2\sqrt{A}$. The quoted theorem is precisely the analytic statement that this saddle reasoning is legitimate for a positive cumulative spectral measure at the exponential scale.

This theorem is the precise analytic bridge between the modular high-temperature statement and the density of states. Its positivity hypothesis is essential: $N(E)$ is a cumulative spectral counting function, not a signed distribution obtained by subtracting sectors.

CFT-internal Cardy growth. Let $\mathcal{C}$ be a unitary compact two-dimensional CFT on a circle of circumference 2π. Assume the Hilbert-space spectrum of

$$E = L_0 + \bar{L}_0$$

is discrete with finite multiplicities, $E \ge 0$, and has a unique vacuum at $E = 0$. Assume also that the torus partition function is modular invariant and has no continuous spectral contribution. Let

$$N(E) = \dim\{\psi \in \mathcal{H}(S^1) \mid (L_0 + \bar{L}_0)\psi \le E\}$$

be the cumulative degeneracy. Then

$$\log N(E) = 2\pi\sqrt{\frac{c_{\mathrm{tot}}E}{6}} + o(\sqrt{E}), \qquad E \to \infty. \tag{67.100}$$

This is a direct consequence of the quoted Tauberian theorem after identifying the positive counting measure. The shifted cylinder partition function used above is

$$Z(\beta) = \operatorname{Tr}\exp\{-\beta(L_0 + \bar{L}_0 - c_{\mathrm{tot}}/24)\}.$$

Multiplying by $\exp\{-\beta c_{\rm tot}/24\}$ gives the ordinary Laplace transform of the positive counting measure $\mathrm{d}N(E)$:

$$\widetilde{Z}(\beta) = \operatorname{Tr}\exp\{-\beta(L_0+\bar{L}_0)\} = \int_{[0,\infty)} \mathrm{e}^{-\beta E}\,\mathrm{d}N(E).$$

Since $\beta c_{\rm tot}/24 = o(\beta^{-1})$, the high-temperature asymptotic (67.98) gives

$$\log \widetilde{Z}(\beta) = \frac{\pi^2 c_{\rm tot}}{6\beta} + o(\beta^{-1}).$$

Apply Theorem 67.27 with $A = \pi^2 c_{\rm tot}/6$.

A saddle calculation checks the coefficient but does not replace the Tauberian theorem or the positivity hypothesis above. With $A = \pi^2 c_{\rm tot}/6$, if a smooth density had entropy

$$S(E) = 2\sqrt{AE},$$

then the Laplace exponent $S(E) - \beta E$ is stationary at

$$E_* = \frac{A}{\beta^2},$$

and the stationary value is A/β. Thus the density-growth coefficient in (67.100) is exactly the Legendre dual of the modular coefficient in (67.98); the theorem above is what justifies this conclusion for a discrete positive spectrum.

The modular bootstrap is the study of the constraints (67.90), reflection positivity, spin integrality, and higher-genus sewing on the spectrum and OPE data. In rational theories it is finite-dimensional algebra. In nonrational compact theories it is an infinite-dimensional positivity and growth problem for torus and higher-genus partition functions.

67.10 Direct-Integral Data for Nonrational CFT

The rational theory discussed above has two simplifying properties: the chiral sector category has finitely many simple objects, and every sewing formula reduces to a finite sum. Neither property is part of the local definition of a two-dimensional conformal field theory. A compact CFT may have a countably infinite discrete spectrum, as in an irrational compact boson. A noncompact CFT may have continuous normalizable spectrum, as in Liouville theory. In both cases the finite sum over intermediate primaries must be replaced by a spectral resolution of Hilbert-space vectors.

Definition 67.28 (Spectral radial product datum)**.** Fix a unitary two-dimensional CFT on the round circle with Hilbert space $\mathcal{H}$, vacuum Ω, self-adjoint nonnegative operators $L_0, \bar{L}_0$, and dense local domain $\mathcal{D}_{\rm loc}$. A spectral radial product datum for two local fields A, B consists of the following data.

1. For every $0 < r < 1$ and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, the radially ordered product vector

$$\Phi_{BA}(r,\theta) = B(re^{\mathrm{i}\theta})A(0)\Omega$$

is a vector in $\mathcal{H}$.

2. Let $\mathcal{H}_{BA} \subset \mathcal{H}$ be the closed subspace generated by all $\Phi_{BA}(r,\theta)$ and by the action of the two Virasoro algebras. The representation of the commuting spectral observables used to label intermediate sectors is represented as a direct integral

$$\mathcal{H}_{BA} \simeq \int_{X_{BA}}^{\oplus} \mathcal{K}_x \, \mathrm{d}\mu_{BA}(x), \tag{67.101}$$

where X_{BA} is a standard Borel space, μ_{BA} is a positive measure, and $x \mapsto \mathcal{K}_x$ is a measurable field of Hilbert spaces carrying the descendant and multiplicity degrees of freedom.

3. In this representation the product vector is a square-integrable measurable vector field

$$\widehat{\Phi}_{BA}(x;r,\theta) \in \mathcal{K}_x, \qquad \int_{X_{BA}} \|\widehat{\Phi}_{BA}(x;r,\theta)\|^2_{\mathcal{K}_x} \mathrm{d}\mu_{BA}(x) < \infty.$$

The variable x is an intermediate spectral label, not necessarily the label of a normalizable primary vector. When x lies in a continuous spectrum, the formal object called a primary of label x is a generalized eigenvector; the Hilbert-space object is the square-integrable field $\widehat{\Phi}_{BA}$.

This definition is deliberately phrased before choosing a basis of chiral blocks. The direct integral is supplied by the spectral theorem for the commuting self-adjoint operators and by the decomposition of the chiral representation on the closed cyclic subspace. A conformal-block expansion is obtained only after choosing measurable bases of intertwiners in the fibers $\mathcal{K}_x$. The Plancherel measure and the multiplicity fields are therefore part of the CFT data.

Remark 67.29 (Direct-integral OPE identity). Let $A, B, C, D \in \mathcal{D}_{\mathrm{loc}}$ and assume that the radial products $\Phi_{BA}(r,\theta)$ and $\Phi_{CD}(\rho,\varphi)$ have spectral radial product data using a common direct-integral decomposition

$$\mathcal{H}_{ABCD} \simeq \int_X^{\oplus} \mathcal{K}_x \, \mathrm{d}\mu(x)$$

of the closed subspace generated by the two product families. Then the four-point function in the radial domain $0 < r < \rho < 1$ is

$$\langle C(\rho e^{\mathrm{i}\varphi})D(0)\Omega,\, B(re^{\mathrm{i}\theta})A(0)\Omega\rangle_{\mathcal{H}} = \int_X \Big\langle \widehat{\Phi}_{CD}(x;\rho,\varphi),\, \widehat{\Phi}_{BA}(x;r,\theta) \Big\rangle_{\mathcal{K}_x} \mathrm{d}\mu(x). \tag{67.102}$$

If the measure is counting measure on a finite set this is the rational finite sum over intermediate states. If the measure has a continuous part, the same formula is an L^2-inner product, not a pointwise sum over delta-normalized states. Indeed, choose the unitary map

$$U : \mathcal{H}_{ABCD} \longrightarrow \int_X^{\oplus} \mathcal{K}_x \, \mathrm{d}\mu(x)$$

implementing the direct integral. By definition, $U\Phi_{BA} = \widehat{\Phi}_{BA}$ and $U\Phi_{CD} = \widehat{\Phi}_{CD}$. Unitarity of U gives the Plancherel identity

$$\langle \Phi_{CD}, \Phi_{BA}\rangle_{\mathcal{H}} = \int_X \langle (U\Phi_{CD})(x), (U\Phi_{BA})(x)\rangle_{\mathcal{K}_x} \mathrm{d}\mu(x),$$

which is (67.102). No assumption about a discrete basis of normalizable primaries has entered the argument.

The usual OPE notation is a compressed way of writing (67.102). For a continuous spectrum it should be read as an identity of matrix elements after smearing the intermediate label against the Plancherel measure and after specifying the domain of radial convergence.

67.10.1 Worked Example: The Noncompact Free Boson

The simplest continuous-spectrum example is a single real free boson with target $\mathbb{R}$. We use the same chiral normalization as in Section 66.6,

$$X_L(z)X_L(0) \sim -\log z, \qquad X_R(\bar{z})X_R(0) \sim -\log \bar{z}.$$

The zero-mode Hilbert space is $L^2(\mathbb{R}, \mathrm{d}x)$. After Fourier transform it is $L^2(\mathbb{R}, \mathrm{d}p)$, and the full Hilbert space is the direct integral

$$\mathcal{H}_{\mathbb{R}} = \int_{\mathbb{R}}^{\oplus} \mathcal{F}_p \,\mathrm{d}p, \qquad \mathcal{F}_p \simeq \mathcal{F}^L_{\mathrm{osc}} \otimes \mathcal{F}^R_{\mathrm{osc}}. \tag{67.103}$$

The fiber ground state at momentum p has

$$h_p = \bar{h}_p = \frac{p^2}{2}. \tag{67.104}$$

Thus p is a spectral label in the direct integral. The plane wave $\mathrm{e}^{\mathrm{i}px}$ is a generalized eigenvector; an actual Hilbert vector is a square-integrable wave packet $f(p) \in L^2(\mathbb{R}, \mathrm{d}p)$, together with oscillator excitations.

For $a \in \mathbb{R}$, let $V_a(z, \bar{z})$ denote the diagonal exponential field

$$V_a(z, \bar{z}) =: \exp(\mathrm{i}aX_L(z) + \mathrm{i}aX_R(\bar{z})) : .$$

As a zero-mode operator it translates the momentum fiber,

$$V_a : \mathcal{F}_p \longrightarrow \mathcal{F}_{p+a}.$$

The field at a fixed real label a is distributional in the spectral label. For $f \in C_c^\infty(\mathbb{R})$, the smeared exponential

$$V_f(z, \bar{z}) = \int_{\mathbb{R}} f(a)V_a(z, \bar{z})\,\mathrm{d}a \tag{67.105}$$

has finite matrix elements between finite-energy wave packets in every closed annulus avoiding coincident points.

The leading product of two labelled exponentials is

$$V_a(z, \bar{z})V_b(0) \sim |z|^{2ab}V_{a+b}(0) + \text{oscillator descendants}. \tag{67.106}$$

Consequently the smeared leading OPE coefficient is the weighted convolution

$$V_f(z, \bar{z})V_g(0) \sim \int_{\mathbb{R}} (f \star_z g)(c)\, V_c(0)\,\mathrm{d}c + \cdots, \tag{67.107}$$

where

$$(f \star_z g)(c) = \int_{\mathbb{R}} f(a)g(c-a)|z|^{2a(c-a)}\,\mathrm{d}a. \tag{67.108}$$

This is the continuous analogue of the lattice charge-conservation rule in (66.62). Associativity of the leading charge factor is the elementary identity

$$ab + (a+b)c = bc + a(b+c), \tag{67.109}$$

which says that either radial grouping of $V_aV_bV_c$ gives the same charge $a+b+c$ and the same power of pairwise distances after the standard coordinate factors are restored. The full diagonal field is separated-point single-valued because the left and right monodromy phases cancel.

The genus-one zero-point function is a trace density. If one puts the target coordinate in a box of length L, the ordinary trace is proportional to L. Dividing by L and letting $L \to \infty$ gives the density

$$Z_{\mathbb{R}}^{\mathrm{dens}}(\tau,\bar{\tau}) = \frac{1}{|\eta(\tau)|^2}\int_{\mathbb{R}} \exp(-2\pi\tau_2 p^2)\,\mathrm{d}p = \frac{1}{\sqrt{2\tau_2}\,|\eta(\tau)|^2}, \qquad \tau_2 = \operatorname{Im}\tau. \tag{67.110}$$

The equality uses the ordinary Gaussian integral in the chosen Plancherel normalization. The density is modular invariant: under $S:\tau\mapsto -1/\tau$, $\tau_2\mapsto \tau_2/|\tau|^2$ while $|\eta(-1/\tau)|^2 = |\tau|\,|\eta(\tau)|^2$, so the factor $\tau_2^{-1/2}|\eta(\tau)|^{-2}$ is unchanged. This is a statement about the semifinite trace per unit target volume; it is the correct replacement for a finite Hilbert-space trace in a translation-invariant noncompact target.

The same density is obtained from the compact circle by decompactification. For a circle of radius R, the momentum spacing is $1/R$. At fixed $\tau_2 > 0$, the winding sectors with nonzero winding have weights exponentially small as $R\to\infty$, and

$$\frac{1}{R}\sum_{m\in\mathbb{Z}} \exp\!\left(-2\pi\tau_2\frac{m^2}{R^2}\right) \longrightarrow \int_{\mathbb{R}} \exp(-2\pi\tau_2 p^2)\,\mathrm{d}p.$$

Thus compact-boson theta functions converge, after division by the target volume normalization, to the Plancherel density (67.110).

This example isolates the continuous-spectrum mechanics from the additional analytic content of Liouville theory. In the free noncompact boson the Plancherel measure is Lebesgue measure, the reflection relation is absent, and the fusing transform is the Fourier/Plancherel transform attached to linear momentum conservation. In Liouville theory the same direct-integral language remains, but the spectrum is $\alpha = Q/2 + \mathrm{i}P$ modulo reflection, the three-point coefficient is DOZZ, and the fusing transform is the Virasoro noncompact fusion kernel. The difference is data, not notation.

Definition 67.30 (Measurable fusing transform)**.** For fixed ordered external fields A, B, C, D, let

$$\int_{X_s}^{\oplus} \mathcal{K}_x^{(s)}\,\mathrm{d}\mu_s(x), \qquad \int_{X_t}^{\oplus} \mathcal{K}_y^{(t)}\,\mathrm{d}\mu_t(y)$$

be the direct-integral Hilbert spaces obtained from the s-channel radial grouping $(BA)(DC)$ and the t-channel grouping $(CB)(DA)$. A measurable fusing transform is a unitary operator

$$\mathbb{F}_{st} : \int_{X_s}^{\oplus} \mathcal{K}_x^{(s)}\,\mathrm{d}\mu_s(x) \longrightarrow \int_{X_t}^{\oplus} \mathcal{K}_y^{(t)}\,\mathrm{d}\mu_t(y) \tag{67.111}$$

which intertwines the two analytic continuations of the same chiral four-point block from the s- and t-channel radial domains. When the fibers are finite-dimensional and a measurable basis has been chosen, this operator is represented by a kernel

$$(\mathbb{F}_{st} f)(y) = \int_{X_s} F_{st}(y,x) f(x)\,\mathrm{d}\mu_s(x), \tag{67.112}$$

understood as an equality in the direct-integral Hilbert space, with the unitarity condition

$$\int_{X_t} F_{st}(y,x)^{\dagger} F_{st}(y,x')\,\mathrm{d}\mu_t(y) = \delta_{\mu_s}(x,x')\,\mathbf{1} \tag{67.113}$$

in distributional notation.

Plancherel form of crossing. Assume the hypotheses of Definition 67.30. Let v_s, w_s be the two s-channel square-integrable coefficient fields whose inner product gives the four-point function in an s-channel radial domain, and let v_t, w_t be the corresponding t-channel coefficient fields. If

$$v_t = \mathbb{F}_{st} v_s, \qquad w_t = \mathbb{F}_{st} w_s, \tag{67.114}$$

then unitarity of the fusing transform gives

$$\int_{X_s} \langle w_s(x), v_s(x)\rangle_{\mathcal{K}_x^{(s)}}\,\mathrm{d}\mu_s(x) = \int_{X_t} \langle w_t(y), v_t(y)\rangle_{\mathcal{K}_y^{(t)}}\,\mathrm{d}\mu_t(y). \tag{67.115}$$

In this formulation the displayed Hilbert-space identity is the final Plancherel step. The mathematical content is the construction of the direct-integral decompositions, the proof that the conformal blocks admit the unitary transform $\mathbb{F}_{st}$, and the verification of (67.114) for the actual OPE kernels. In a rational CFT the same statement reduces to a finite matrix identity. In Liouville theory the label spaces are continuous and the identity becomes the Ponsot–Teschner Virasoro fusion-kernel identity, together with the DOZZ coefficient relation.

Definition 67.31 (Nonrational genus-zero bootstrap problem)**.** A nonrational genus-zero bootstrap problem consists of:

1. a class of local fields and their dense radial-product domain;
2. for every ordered pair of fields, direct-integral intermediate spectra with Plancherel measures and multiplicity fibers;
3. measurable three-point/OPE coefficient fields;
4. chiral conformal blocks with specified radial convergence domains and analytic continuations;
5. measurable fusing and braiding transforms satisfying unitarity, pentagon, and hexagon identities as identities of direct-integral operators;
6. a reflection-positive pairing of left and right chiral data producing single-valued Euclidean correlation distributions.

Solving this problem means proving existence, uniqueness up to unitary equivalence of the resulting correlation theory, reflection positivity, locality, and compatibility with sewing in the domains under consideration. Crossing equations for individual four-point functions are necessary consequences of such a solution, not a replacement for the direct-integral and sewing data.

Example 67.32 (Two spectral types)**.** For an irrational compact free boson the intermediate label space is countable, the measure is counting measure, and modular covariance is governed by theta functions and Poisson summation rather than by a finite modular matrix. For Liouville theory the normalizable bulk spectrum is

$$\alpha = \frac{Q}{2} + \mathrm{i}P, \qquad P \in \mathbb{R}_{\geq 0},$$

with Plancherel measure fixed by the two-point normalization and reflection amplitude. The four-point function is the direct integral displayed in Definition 68.8. These two examples have different spectral measures, but both are governed by the same Hilbert-space identity (67.102).

67.11 Logarithmic Theories

A logarithmic CFT is a conformal field theory whose chiral module category is not semisimple and in which the energy operator L_0 need not be diagonalizable. Diagonal L_0 is what made the rational characters honest sums of powers of q over simple modules; a nilpotent part of L_0 changes both local conformal covariance and the genus-one trace problem.

Definition 67.33 (Rank-two logarithmic Virasoro pair)**.** A rank-two logarithmic pair of chiral fields of weight h is a pair $(\mathcal{C}, \mathcal{D})$ whose states C, D satisfy

$$L_0 C = hC, \qquad L_0 D = hD + C, \qquad L_n C = L_n D = 0 \quad (n > 0).$$

Equivalently, the singular part of the stress-tensor OPE is

$$T(\zeta)\mathcal{C}(z) \sim \frac{h\mathcal{C}(z)}{(\zeta - z)^2} + \frac{\partial \mathcal{C}(z)}{\zeta - z},$$

$$T(\zeta)\mathcal{D}(z) \sim \frac{h\mathcal{D}(z) + \mathcal{C}(z)}{(\zeta - z)^2} + \frac{\partial \mathcal{D}(z)}{\zeta - z}.$$

Thus $\mathcal{C}$ is primary, while $\mathcal{D}$ is its logarithmic partner. With $ND = C$ and $NC = 0$, the two-dimensional span of (C, D) carries $L_0 = h\,\mathrm{id} + N$.

The finite consequences are immediate and worth recording because this is where signs and 2πi's enter. On states,

$$q^{L_0 - c/24} C = q^{h - c/24} C, \qquad q^{L_0 - c/24} D = q^{h-c/24}\big(D + 2\pi\mathrm{i}\tau\, C\big), \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau}.$$

On fields, the Virasoro commutators are, for $n = -1, 0, 1$,

$$[L_n, \mathcal{C}(z)] = \big(z^{n+1}\partial_z + (n+1)hz^n\big)\mathcal{C}(z),$$

$$[L_n, \mathcal{D}(z)] = \big(z^{n+1}\partial_z + (n+1)hz^n\big)\mathcal{D}(z) + (n+1)z^n\mathcal{C}(z).$$

In particular,

$$\mathrm{e}^{sL_0}\mathcal{C}(z)\mathrm{e}^{-sL_0} = \mathrm{e}^{sh}\mathcal{C}(\mathrm{e}^s z),$$

$$\mathrm{e}^{sL_0}\mathcal{D}(z)\mathrm{e}^{-sL_0} = \mathrm{e}^{sh}\big(\mathcal{D}(\mathrm{e}^s z) + s\mathcal{C}(\mathrm{e}^s z)\big).$$

The additive $s = \log\lambda$ term is the local origin of logarithms in correlators.

Proposition 67.34 (Two-point functions of a logarithmic pair)**.** *Assume the vacuum is invariant under L_{-1}, L_0, L_1, the two fields in Definition 67.33 are mutually local, and their two-point functions are single-valued on a chosen simply connected punctured coordinate chart after a branch of* $\log z$ *is fixed. Then translation, scaling, and special-conformal Ward identities force*

$$\langle \mathcal{C}(z)\mathcal{C}(0)\rangle = 0,$$

$$\langle \mathcal{C}(z)\mathcal{D}(0)\rangle = b\, z^{-2h},$$
$$\langle \mathcal{D}(z)\mathcal{D}(0)\rangle = (d - 2b\log z)z^{-2h},$$

for constants $b, d \in \mathbb{C}$. The constant d depends on the choice of logarithmic partner D, while b is invariant under $D \mapsto D + \lambda C$.

Proof. Translation invariance makes the correlators functions of $z = z_1 - z_2$. Write

$$G_{CC}(z) = \langle \mathcal{C}(z)\mathcal{C}(0)\rangle, \qquad G_{CD}(z) = \langle \mathcal{C}(z)\mathcal{D}(0)\rangle,$$
$$G_{DD}(z) = \langle \mathcal{D}(z)\mathcal{D}(0)\rangle.$$

The L_0 Ward identity gives

$$(z\partial_z + 2h)G_{CC} = 0,$$
$$(z\partial_z + 2h)G_{CD} + G_{CC} = 0,$$
$$(z\partial_z + 2h)G_{DD} + 2G_{CD} = 0.$$

Hence, before imposing L_1,

$$G_{CC} = Az^{-2h}, \qquad G_{CD} = (B - A\log z)z^{-2h},$$
$$G_{DD} = (D_0 - 2B\log z + A(\log z)^2)z^{-2h}.$$

This calculation is the scale-covariant source of logarithms.

Now impose the L_1 Ward identity on $\langle \mathcal{C}(z_1)\mathcal{D}(z_2)\rangle$. The ordinary primary part of the differential operator is

$$(z_1 + z_2)(z\partial_z + 2h)G_{CD}(z), \qquad z = z_1 - z_2,$$

and the nilpotent term from the $\mathcal{D}$-insertion is

$$2z_2 G_{CC}(z).$$

Using the L_0 equation, the total L_1 variation is

$$-(z_1 + z_2)G_{CC}(z) + 2z_2 G_{CC}(z) = -z\, G_{CC}(z).$$

It must vanish for $z \neq 0$, so $G_{CC} = 0$ and therefore $A = 0$. The remaining L_0 equations give the displayed forms with $b = B$ and $d = D_0$. The L_1 equation for G_{DD} is then automatic: the primary differential part gives $-2b(z_1 + z_2)z^{-2h}$, while the two nilpotent insertions contribute $2b(z_1 + z_2)z^{-2h}$.

Finally replace D by $D + \lambda C$. Since $G_{CC} = 0$,

$$G_{CD} \mapsto G_{CD}, \qquad G_{DD} \mapsto G_{DD} + 2\lambda G_{CD}.$$

Thus b is unchanged and d is shifted to $d + 2\lambda b$. $\square$

The ordinary character sees only the trace of $q^{L_0-c/24}$. On the span of C, D,

$$\mathrm{Tr}\big(q^{L_0-c/24}\big) = 2q^{h-c/24},$$

because the nilpotent coefficient $2\pi i\tau N$ has trace zero. Thus an ordinary character records the generalized eigenvalue and multiplicity but not the extension class producing the logarithmic partner. In a finite-length C_2-cofinite logarithmic theory, the genus-one space closed under modular transformations is therefore larger than the span of ordinary characters: it includes generalized characters and pseudo-trace functions on projective modules. A pseudo-trace is the replacement for the usual matrix trace when the endomorphism algebra of a projective module is not semisimple; it is a linear functional with the cyclicity property needed for torus sewing, and it can couple to the nilpotent part invisible to the ordinary trace.

Finite projective pseudo-trace model. The finite algebraic mechanism is already visible without invoking the full logarithmic VOA modular theorem. Let A be a finite-dimensional associative algebra over $\mathbb{C}$, let P be a finitely generated projective left A-module, and let

$$\varphi : A \longrightarrow \mathbb{C}$$

be a symmetric linear functional, meaning

$$\varphi(ab) = \varphi(ba), \qquad a, b \in A.$$

Choose a projective dual basis

$$p_r \in P, \qquad f^r \in \mathrm{Hom}_A(P, A), \qquad x = \sum_r p_r f^r(x) \quad (x \in P).$$

For $T \in \mathrm{End}_A(P)$, define the φ-pseudo-trace

$$\mathrm{ptr}_P^{\varphi}(T) := \sum_r \varphi\big(f^r(Tp_r)\big). \tag{67.116}$$

The expression is independent of the chosen dual basis. This is the Hattori–Stallings trace paired with φ: changing dual bases changes $\sum_r f^r(Tp_r)$ by a commutator in A, and the symmetric property of φ kills commutators. It follows in particular that

$$\mathrm{ptr}_P^{\varphi}(ST) = \mathrm{ptr}_P^{\varphi}(TS), \qquad S, T \in \mathrm{End}_A(P),$$

which is the finite cyclicity property needed when the torus sewing trace is rotated around a nonsemisimple channel.

The smallest example is the dual-number algebra

$$A = \mathbb{C}[\epsilon]/(\epsilon^2), \qquad P = A.$$

Let $\varphi_{\mathrm{res}}(a + b\epsilon) = b$. Multiplication by $a + b\epsilon$ on the vector space with basis $(1, \epsilon)$ has ordinary trace $2a$, so the ordinary trace kills the nilpotent coefficient b. The projective pseudo-trace with $p_1 = 1$ and $f^1(x) = x$, however, gives

$$\mathrm{ptr}_A^{\varphi_{\mathrm{res}}}(R_{a+b\epsilon}) = \varphi_{\mathrm{res}}(a + b\epsilon) = b,$$

where $R_{a+b\epsilon}$ denotes right multiplication, an A-module endomorphism of the regular left module. If the Jordan cell of L_0 is modeled by $L_0 = h + \epsilon$, then

$$q^{L_0 - c/24} = q^{h-c/24}\big(1 + 2\pi \mathrm{i}\tau\, \epsilon\big),$$

and the ordinary trace gives $2q^{h-c/24}$, whereas the pseudo-trace gives

$$\mathrm{ptr}_A^{\varphi_{\mathrm{res}}}\big(q^{L_0-c/24}\big) = 2\pi\mathrm{i}\tau\, q^{h-c/24}.$$

Thus the polynomial τ-dependence that ordinary characters miss is not a decorative correction; it is the projective trace of the nilpotent extension data. In logarithmic VOA applications the algebra A is replaced by endomorphism algebras of projective covers or by the appropriate finite quotients controlling torus one-point functions, and (67.116) is the finite model for the pseudo-trace functions that enter modular closure.

The semisimple Verlinde formula (67.77) is consequently replaced by categorical statements about the Grothendieck ring, projective modules, modified traces, and the factorizable braided finite tensor category attached to the logarithmic chiral theory. The right question is not whether the same matrix formula survives unchanged, but which categorical pairing diagonalizes which quotient of the tensor product. This is one of the places where the algebraic formulation is essential: it names the objects whose semisimplicity has failed and identifies the extra data needed to recover modular sewing.

67.12 Output

The output of this chapter is a hierarchy of definitions and hypotheses: The construction has the following order. A VOA defines modules and chiral blocks. Chiral blocks sew over the moduli space of pointed curves. In the semisimple finite case, sewing data assemble into a modular tensor category. The full CFT is then obtained by choosing a compatible pairing of left and right chiral theories, equivalently by imposing the modular-invariance and factorization constraints on torus and higher-genus amplitudes. The Ising model gives the finite rational example. Logarithmic theories show which parts of the rational construction depend on semisimplicity and which parts persist after replacing simple-module linear algebra by finite tensor category data.

Chapter 68

Liouville Conformal Field Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Liouville theory is used here as a test case for noncompact, nonrational CFT: the same theory has a probabilistic construction, a Virasoro bootstrap coordinate system, continuous spectral decompositions, and boundary states with distributional wavefunctions. These presentations are not interchangeable until comparison maps and sewing estimates have been specified. The chapter therefore separates four layers. The local field layer fixes the background-charge stress tensor, conformal weights, BPZ null-vector equations, and finite Coulomb-gas residues. The probabilistic layer constructs correlation distributions from a GFF plus Gaussian multiplicative chaos inside the Seiberg domain and then meromorphically continues them. The bootstrap layer uses DOZZ, reflection, elliptic recursion, FZZT/ZZ wavefunctions, and finite-difference constraints as coordinates on conformal blocks. The functorial sewing layer asks whether the probabilistic distributions and the bootstrap kernels define the same positive, continuous gluing theory on all closed and bordered surfaces.

This organization is deliberately stricter than listing Liouville formulae. The zero-mode gamma integral, BPZ hypergeometric connection calculation, DOZZ shift equations, and FZZT finite-difference identities are local mechanisms that can be checked in the chapter's conventions. They do not by themselves prove the probabilistic construction, the DOZZ theorem, the boundary bootstrap, positivity of annulus spectral measures, or all-surface sewing. Those theorem boundaries are named at their points of use so that Liouville serves as a genuine comparison problem between probabilistic QFT, bootstrap coordinates, and nonrational CFT sewing rather than as a collection of famous closed forms.

68.1 Classical Datum and Stress Tensor

Liouville theory is the noncompact two-dimensional conformal theory whose interaction is an exponential of a scalar field. This chapter treats it as a two-dimensional QFT construction problem whose data are a background surface, a scalar distribution, a curvature coupling, exponential insertions, Virasoro Ward identities, degenerate fields, and sewing constraints. The development below fixes the normalizations, derives the local representation-theoretic identities, and states the present theorem boundaries for the probabilistic and bootstrap constructions.

Definition 68.1 (Classical Liouville datum)**.** A classical Liouville datum consists of an oriented smooth closed surface Σ, a Riemannian metric g on Σ, a real scalar field $\phi \in C^\infty(\Sigma, \mathbb{R})$, parameters

$$b > 0, \qquad Q \in \mathbb{R}, \qquad \mu > 0,$$

and the Euclidean action

$$S_L[\phi; g] = \frac{1}{4\pi} \int_\Sigma \mathrm{d}^2x \sqrt{g} \left(g^{ab} \partial_a \phi \, \partial_b \phi + Q R_g \phi + 4\pi\mu \, \mathrm{e}^{2b\phi} \right). \tag{68.1}$$

Here R_g is the scalar curvature of g. The exponential term is a function of the smooth classical field in this definition; in the quantum theory it is a renormalized random measure or a normal-ordered local field, depending on the construction framework.

Varying a compactly supported smooth $\delta\phi$ gives

$$\delta S_L = \frac{1}{4\pi} \int_\Sigma \sqrt{g} \left(-2\Delta_g \phi + Q R_g + 8\pi b \mu \mathrm{e}^{2b\phi} \right) \delta\phi.$$

Thus the classical Euler–Lagrange equation is

$$-\Delta_g \phi + \frac{Q}{2} R_g + 4\pi b \mu \mathrm{e}^{2b\phi} = 0. \tag{68.2}$$

On a flat complex coordinate patch z, the holomorphic stress tensor representative is

$$T(z) = - :(\partial\phi)^2:(z) + Q \, \partial^2 \phi(z). \tag{68.3}$$

The normal ordering in (68.3) is a free-field short-distance prescription. It is used here as a perturbative coordinate on the Virasoro representation data; the nonperturbative probabilistic construction is stated later as a separate object.

Convention 68.2 (Free-field normalization)**.** In this chapter the local free field has two-point singularity

$$\phi(z)\phi(w) \sim -\frac{1}{2} \log |z - w|^2, \qquad \partial\phi(z)\partial\phi(w) \sim -\frac{1}{2(z-w)^2}.$$

The holomorphic exponential field is denoted

$$V_\alpha(z, \bar{z}) = :\mathrm{e}^{2\alpha\phi(z,\bar{z})}:.$$

With this convention the standard Liouville relation is

$$Q = b + b^{-1}. \tag{68.4}$$

Improved central charge and Liouville weights. Assume the free-field OPE convention 68.2. The stress tensor (68.3) has Virasoro central charge

$$c_L = 1 + 6Q^2. \tag{68.5}$$

The exponential V_α is Virasoro primary in the free-field coordinate with conformal weights

$$h_\alpha = \bar{h}_\alpha = \alpha(Q - \alpha). \tag{68.6}$$

This is an OPE computation. The stress tensor is $T = T_0 + T_Q$, where

$$T_0 = - :(\partial\phi)^2:, \qquad T_Q = Q\partial^2\phi.$$

The double contraction in $T_0(z)T_0(w)$ gives the central term $\frac{1}{2}(z-w)^{-4}$, hence $c = 1$ for $Q = 0$. The mixed terms have at most third-order poles and contribute to the transformation of T, not to the central term. Since

$$\partial^2\phi(z)\partial^2\phi(w) = \partial_z^2\partial_w^2\left(-\frac{1}{2}\log(z-w)\right) \sim \frac{3}{(z-w)^4},$$

the improvement contributes $3Q^2(z-w)^{-4}$. Comparing

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$$

gives $c = 1 + 6Q^2$.

For the exponential,

$$\partial\phi(z)V_\alpha(w,\bar{w}) \sim -\frac{\alpha}{z-w}V_\alpha(w,\bar{w}), \qquad \partial^2\phi(z)V_\alpha(w,\bar{w}) \sim \frac{\alpha}{(z-w)^2}V_\alpha(w,\bar{w}).$$

The double contraction of $- :(\partial\phi)^2:$ contributes $-\alpha^2(z-w)^{-2}V_\alpha$, and the improvement contributes $\alpha Q(z-w)^{-2}V_\alpha$. The simple-pole term is $(z-w)^{-1}\partial_w V_\alpha$. Therefore V_α has the weight $\alpha(Q-\alpha)$.

The interaction $\mu e^{2b\phi}$ is marginal precisely when $b(Q-b) = 1$, which gives $Q = b + b^{-1}$. Thus (68.4) is an operational normalization condition: it is the condition that the exponential potential have weight $(1,1)$ in the background-charge representation.

68.2 Exponential Operators and the Probabilistic Object

Liouville correlation functions are infinite-dimensional Euclidean QFT objects. The rigorous probabilistic construction separates the constant mode from the mean-zero Gaussian free field and interprets the exponential interaction as Gaussian multiplicative chaos.

Definition 68.3 (Seiberg domain in physics normalization)**.** For insertions $V_{\alpha_1}(x_1), \ldots, V_{\alpha_n}(x_n)$ on a closed surface Σ with Euler characteristic $\chi(\Sigma)$, the Seiberg domain is the set of momenta satisfying

$$\alpha_i < \frac{Q}{2} \quad \text{for every } i, \qquad 2\sum_{i=1}^{n}\alpha_i > Q\,\chi(\Sigma). \tag{68.7}$$

The first inequalities are local integrability conditions near the marked points. The second inequality is the negative-constant-mode integrability condition. On the sphere it reads $\sum_i \alpha_i > Q$.

The zero-mode check is elementary and fixes the sign conventions. Write $\phi = c + \phi_0$, with $\int_\Sigma \phi_0 \operatorname{vol}_g = 0$. The curvature term contributes $\exp[-Q\chi(\Sigma)c]$, while the insertions contribute $\exp[2c\sum_i \alpha_i]$. As $c \to -\infty$, integrability requires $2\sum_i \alpha_i - Q\chi(\Sigma) > 0$. The positive c direction is controlled by the Liouville potential when its multiplicative-chaos measure is nonzero.

Equivalently, with

$$\varepsilon_\Sigma(\boldsymbol{\alpha}) := \sum_i \alpha_i - \frac{Q}{2}\chi(\Sigma),$$

the negative-constant-mode tail has the model form

$$\int_{-\infty}^{c_0} \mathrm{e}^{2\varepsilon_\Sigma c}\, \mathrm{d}c.$$

The second Seiberg-domain inequality is precisely $\operatorname{Re}\varepsilon_\Sigma > 0$. The pole calculation in Paragraph 68.5 is the same zero-mode mechanism evaluated on the boundary of this convergence region; outside the Seiberg domain, the correlator is represented by the meromorphic continuation whose boundary residues are described in Subsection 68.7.1.

Definition 68.4 (Gaussian multiplicative chaos input)**.** Let X_g be a mean-zero Gaussian free field on (Σ, g), normalized by a Green kernel with logarithmic singularity $-\log d_g(x,y)$. For $\gamma \in (0,2)$, let $X_{g,\varepsilon}$ be a smooth mollification at scale ε. The subcritical Gaussian multiplicative-chaos measure is the probability-limit object

$$M_{\gamma,g}(\mathrm{d}x) = \lim_{\varepsilon \downarrow 0} \varepsilon^{\gamma^2/2} \mathrm{e}^{\gamma X_{g,\varepsilon}(x)} \operatorname{vol}_g(\mathrm{d}x) \tag{68.8}$$

in the topology of weak convergence in probability against continuous test functions. In Liouville theory $\gamma = 2b$ after translating from the physics field ϕ of Convention 68.2 to the probabilistic GFF normalization.

Quoted Theorem 68.5 (Probabilistic Liouville correlation functions)**.** *The probabilistic Liouville constructions of David–Kupiainen–Rhodes–Vargas and Guillarmou–Kupiainen–Rhodes–Vargas [44, 46] define Liouville correlation functions by integrating the constant mode against the Gaussian multiplicative-chaos functional, with insertions in the Seiberg domain* (68.7)*. In the regimes covered by these theorems the construction gives finite correlation functions, conformal covariance with central charge $1 + 6Q^2$, and meromorphic continuation in the insertion momenta after the initial probabilistic domain is established. These theorems supply the measure-theoretic construction of the closed-surface Liouville path integral; the full Segal sewing theory for higher-genus boundary operations is recorded below as a separate boundary problem.*

Comparison datum for the closed theory. The probabilistic theorem produces meromorphic families of closed-surface correlation distributions. The bootstrap formulas used later in this chapter use a different coordinate system: the Hilbert space is decomposed into Virasoro representation fibers, and correlators are expanded by cutting the surface into pairs of pants. Comparing the two descriptions requires the following maps.

First, for each closed marked surface $(\Sigma; x_1, \ldots, x_n)$, the probabilistic construction supplies a distribution

$$\mathcal{L}_{\Sigma,n}(\alpha_1, \ldots, \alpha_n) \in \mathcal{D}'(\operatorname{Conf}_n(\Sigma))$$

in the insertion positions, initially for $\boldsymbol{\alpha}$ in the Seiberg domain and then by meromorphic continuation in the momenta. Second, the cylinder state space must be identified with the direct integral

$$\mathcal{H}^{\mathrm{L}}_{S^1} = \int_0^{\infty}{}^{\oplus} \mathcal{V}_{Q/2+\mathrm{i}P} \otimes \overline{\mathcal{V}_{Q/2+\mathrm{i}P}}\, \frac{\mathrm{d}P}{\pi}, \tag{68.9}$$

with the reflection convention and two-point normalization fixed by Definition 68.7. Third, three-punctured spheres must be represented by distributional trilinear maps on the fibers of (68.9); in the normalization of this chapter their scalar kernel is the DOZZ constant (68.39). Fourth, cutting a closed surface along a separating circle should express the probabilistic distribution as a Plancherel integral over the intermediate momentum:

$$\mathcal{L}_{\Sigma,n} = \int_0^{\infty} \mathcal{L}_{\Sigma_L,\{P\}}\, \mathcal{L}_{\Sigma_R,\{P\}}\, \frac{\mathrm{d}P}{\pi}, \tag{68.10}$$

in a topology strong enough to justify the gluing limit. The notation in (68.10) suppresses the left/right descendant pairings, conformal-block coordinates, and possible discrete residue terms that appear after meromorphic continuation of external momenta through poles. These four items are the comparison datum; the probabilistic construction, the DOZZ theorem, and the conformal-block calculus verify different parts of it.

Constant-mode reduction of the probabilistic construction. The theorem just quoted has a concrete core. Let

$$\gamma = 2b, \qquad s(\boldsymbol{\alpha},\Sigma) = 2\sum_{i=1}^{n} \alpha_i - Q\chi(\Sigma).$$

After separating the constant mode, write the mean-zero field in the probabilistic normalization as X_g, with chaos parameter γ. The local covariance used in Convention 68.2 is $-\frac{1}{2}\log|z-w|^2 = -\log|z-w|$, so it has the same logarithmic coefficient as the probabilistic GFF. The only notation change is that a physics vertex $V_{\alpha_i} = {:}\mathrm{e}^{2\alpha_i\phi}{:}$ is the probabilistic insertion with exponent

$$\alpha_i^{\mathrm{prob}} = 2\alpha_i.$$

The c-dependent part of an n-point function has the form

$$\int_{-\infty}^{\infty} \exp(s(\boldsymbol{\alpha},\Sigma)c - \mu\, \mathrm{e}^{\gamma c} Z_g(X_g;\boldsymbol{x},\boldsymbol{\alpha}))\, \mathrm{d}c,$$

where the random functional multiplying the exponential potential is

$$Z_g(X_g;\boldsymbol{x},\boldsymbol{\alpha}) = \int_{\Sigma} \prod_{i=1}^{n} \exp\bigl(\gamma\alpha_i^{\mathrm{prob}}\, G_g(y,x_i)\bigr)\, M_{\gamma,g}(\mathrm{d}y). \tag{68.11}$$

The Green kernel G_g records the singularity of the GFF conditioned by the vertex insertions; all coincident-point self-contractions are absorbed into the renormalized exponential fields. For $s>0$, the elementary change of variable $t = \mu Z_g \mathrm{e}^{\gamma c}$ gives

$$\int_{\mathbb{R}} \mathrm{e}^{sc-\mu Z_g \mathrm{e}^{\gamma c}}\, \mathrm{d}c = \frac{1}{\gamma}\, \mu^{-s/\gamma}\, \Gamma(s/\gamma)\, Z_g^{-s/\gamma}. \tag{68.12}$$

This identity is the source of the global Seiberg inequality $s(\boldsymbol{\alpha}, \Sigma) > 0$. The local inequalities are a separate Gaussian-multiplicative-chaos input. Near a marked point x_i, the potential mass contains the singular GMC integral

$$I_i = \int_{U_i} d_g(y, x_i)^{-\gamma\alpha_i^{\mathrm{prob}}} M_{\gamma,g}(\mathrm{d}y), \qquad \alpha_i^{\mathrm{prob}} = 2\alpha_i.$$

The first moment of I_i is finite only for $\alpha_i^{\mathrm{prob}} < 2/\gamma$, because $\mathbb{E}[M_{\gamma,g}]$ is the background volume measure. The Liouville construction criterion is the larger subcritical threshold supplied by the GMC thick-point theorem. That theorem controls rare thick points of the log-correlated field and gives the almost-sure local integrability and the negative-moment estimates needed for (68.12) precisely under

$$\alpha_i^{\mathrm{prob}} < Q_\gamma, \qquad Q_\gamma := \frac{2}{\gamma} + \frac{\gamma}{2}.$$

Since $\gamma = 2b$, $Q_\gamma = b + b^{-1} = Q$, so this is the physics condition $\alpha_i < Q/2$. Thus the probabilistic correlation function in the initial domain is the expectation of the explicit negative moment

$$\mathbb{E}\Big[Z_g(X_g; \boldsymbol{x}, \boldsymbol{\alpha})^{-s/\gamma}\Big],$$

multiplied by the deterministic Green-function and gamma-function factors fixed above. The analytic theorem boundary supplies existence of these negative moments, Weyl covariance of the renormalized expression, and meromorphic continuation away from the initial domain.

Definition 68.6 (Reflection relation)**.** The Liouville spectrum is parametrized by

$$\alpha = \frac{Q}{2} + \mathrm{i}P, \qquad P \in \mathbb{R}_{\geq 0},$$

with conformal weight $Q^2/4 + P^2$. The reflection relation is the identification, at the level of meromorphic correlation functions,

$$V_\alpha = R_b(\alpha)\, V_{Q-\alpha}, \tag{68.13}$$

where $R_b(\alpha)$ is the meromorphic reflection amplitude fixed by the two-point normalization and equivalently by the DOZZ three-point function. Thus a single bulk primary label is represented by the reflection orbit $\{\alpha, Q - \alpha\}$, with the normalization encoded in $R_b(\alpha)$.

Definition 68.7 (Asymptotic scattering basis)**.** Let ϕ_0 be the constant mode of ϕ on a long cylinder. In the weak-coupling end $\phi_0 \to -\infty$, the Liouville potential is negligible and the Hilbert space is described by the background-charge free field with oscillators. A momentum-P incoming state has asymptotic wavefunction

$$\Psi_P^{\mathrm{in}}(\phi_0, \{\phi_n\}) \sim \left(e^{2\mathrm{i}P\phi_0} + S(P)e^{-2\mathrm{i}P\phi_0}\right) \exp\left(-\sum_{n\geq 1} n\phi_n\phi_{-n}\right), \qquad P \geq 0, \tag{68.14}$$

where $S(P)$ is the unitary reflection phase in the scattering normalization. The corresponding incoming primary is represented asymptotically by

$$V_P^{\mathrm{in}} \sim e^{(Q+2\mathrm{i}P)\phi} + S(P)e^{(Q-2\mathrm{i}P)\phi}, \qquad h_P = \frac{Q^2}{4} + P^2. \tag{68.15}$$

The delta-normalized primary used in the continuous spectrum is $V_P = S(P)^{-1/2} V_P^{\mathrm{in}}$, with

$$\langle V_P(1) V_{P'}(0) \rangle = \pi \delta(P - P')$$

after fixing the conventional measure. This is the canonical-quantization version of the reflection orbit $\alpha = Q/2 + \mathrm{i}P \sim Q/2 - \mathrm{i}P$.

Definition 68.8 (Continuous conformal-block decomposition)**.** For four normalizable external Liouville primaries $\alpha_j = Q/2 + \mathrm{i}P_j$, the bootstrap ansatz on the sphere is a direct integral over the same spectrum:

$$\langle V_{P_1}(\infty) V_{P_2}(1) V_{P_3}(z) V_{P_4}(0) \rangle = \int_0^\infty C(P_1, P_2, P) C(P_3, P_4, P) \left| \mathcal{F}_P \begin{bmatrix} P_1 & P_2 \\ P_4 & P_3 \end{bmatrix} (z) \right|^2 dP. \tag{68.16}$$

Here $\mathcal{F}_P$ is the Virasoro conformal block with internal weight $Q^2/4 + P^2$. Equation (68.16) is a direct-integral decomposition. Its analytic content includes the choice of Plancherel measure, the location of possible discrete residues after analytic continuation, and the crossing kernel for the Virasoro blocks.

68.3 Virasoro Blocks from Gram Matrices

The symbol $\mathcal{F}_P$ in (68.16) hides a representation theoretic construction. The construction is local in the moduli variable z and does not require the DOZZ formula. It uses only Virasoro Ward identities, the contravariant form on a Verma module, and the three-point matrix elements of descendants.

Definition 68.9 (Four-point Virasoro block coefficients)**.** Fix a central charge c, an internal highest weight h, and four external weights h_1, h_2, h_3, h_4. Consider the chiral four-point geometry

$$V_1(\infty) V_2(1) V_3(z) V_4(0),$$

with the intermediate channel separating $(1, 2)$ from $(3, 4)$. Assume that the contravariant Gram matrix on the Verma module $\mathcal{V}_{c,h}$ is nondegenerate through the level under discussion. Through level 3 the normalized Virasoro block is written

$$\mathcal{F}_h^{(c)} \begin{bmatrix} h_1 & h_2 \\ h_4 & h_3 \end{bmatrix} (z) = z^{h - h_3 - h_4} \left(1 + f_1 z + f_2 z^2 + f_3 z^3 + O(z^4) \right).$$

The assumption of nondegeneracy excludes $h = 0$ at level 1 and excludes the zeros of the relevant Kac determinants at higher level. At those parameters one must quotient by null states, take a meromorphic limit, or work with logarithmic extensions; the inverse Gram matrices below are not literal inverses on the unreduced Verma module.

Lemma 68.10 (Level-one and level-two block coefficients)**.** *In the notation of Definition 68.9, set*

$$a_1 = h + h_3 - h_4, \qquad b_1 = h + h_2 - h_1,$$
$$a_2 = h + 2h_3 - h_4, \qquad b_2 = h + 2h_2 - h_1,$$

and

$$a_{11} = a_1(a_1 + 1), \qquad b_{11} = b_1(b_1 + 1).$$

Then

$$f_1 = \frac{b_1 a_1}{2h}. \tag{68.17}$$

Let

$$G_2 = \begin{pmatrix} 4h + c/2 & 6h \\ 6h & 4h(2h+1) \end{pmatrix}, \qquad \Delta_2 = \det G_2 = 2h\{16h^2 + 2(c-5)h + c\}.$$

With the ordered level-two basis $(L_{-2}|h\rangle, L_{-1}^2|h\rangle)$*, the second coefficient is*

$$\begin{aligned} f_2 = \frac{1}{\Delta_2}\big(&4h(2h+1)b_2 a_2 - 6h(b_2 a_{11} + b_{11} a_2) \\ &+ (4h + c/2) b_{11} a_{11}\big). \end{aligned} \tag{68.18}$$

For Liouville blocks one substitutes

$$c = 1 + 6Q^2, \qquad h = \frac{Q^2}{4} + P^2, \qquad h_i = \alpha_i(Q - \alpha_i).$$

For generic real P *the denominators above are nonzero; the degenerate values are precisely the places where a BPZ quotient changes the block recursion.*

Proof. Let $\rho_{12}(Y)$ be the normalized three-point matrix element with the descendant $L_{-Y}|h\rangle$ in $\langle V_1(\infty)V_2(1)\cdot\rangle$, and let $\rho_{34}(Y)$ be the corresponding matrix element for the OPE of $V_3(z)V_4(0)$. The Ward identity gives, for $n = 1, 2$,

$$\rho_{12}(L_{-n}) = h + nh_2 - h_1, \qquad \rho_{34}(L_{-n}) = h + nh_3 - h_4.$$

For the repeated L_{-1} descendant one differentiates the three-point function

$$\langle V_1(\infty)V_2(1)V_h(x)\rangle = (1-x)^{-h_2 - h + h_1}$$

and its right-channel analogue. This gives

$$\rho_{12}(L_{-1}^2) = b_1(b_1 + 1), \qquad \rho_{34}(L_{-1}^2) = a_1(a_1 + 1).$$

The level-one Gram matrix is $(2h)$, so the projector onto the intermediate module gives (68.17). At level 2, the Virasoro commutation relation gives

$$\langle L_{-2}h, L_{-2}h\rangle = 4h + c/2, \qquad \langle L_{-2}h, L_{-1}^2 h\rangle = 6h,$$
$$\langle L_{-1}^2 h, L_{-1}^2 h\rangle = 4h(2h+1).$$

Therefore the level-two contribution is

$$f_2 = \begin{pmatrix} b_2 & b_{11} \end{pmatrix} G_2^{-1} \begin{pmatrix} a_2 \\ a_{11} \end{pmatrix}.$$

Inverting the displayed 2×2 matrix gives

$$G_2^{-1} = \frac{1}{\Delta_2}\begin{pmatrix} 4h(2h+1) & -6h \\ -6h & 4h + c/2 \end{pmatrix},$$

which is exactly (68.18). $\square$

Lemma 68.11 (Level-three block coefficient and determinant)**.** *In the same four-point geometry, assume that the level-three Gram matrix is nondegenerate. Put*

$$a_3 = h + 3h_3 - h_4, \qquad b_3 = h + 3h_2 - h_1,$$

$$a_{21} = a_2(a_1+2), \quad b_{21} = b_2(b_1+2), \qquad a_{111} = a_1(a_1+1)(a_1+2), \quad b_{111} = b_1(b_1+1)(b_1+2).$$

With ordered basis

$$\big(L_{-3}|h\rangle,\, L_{-2}L_{-1}|h\rangle,\, L_{-1}^3|h\rangle\big),$$

the level-three Gram matrix is

$$G_3 = \begin{pmatrix} 6h+2c & 10h & 24h \\ 10h & h(c+8h+8) & 12h(3h+1) \\ 24h & 12h(3h+1) & 24h(h+1)(2h+1) \end{pmatrix}. \tag{68.19}$$

Its determinant factors as

$$\det G_3 = 48h^2\big(16h^2 + 2(c-5)h + c\big)\big(3h^2 + (c-7)h + c + 2\big). \tag{68.20}$$

Let

$$A_3 = \begin{pmatrix} a_3 \\ a_{21} \\ a_{111} \end{pmatrix}, \qquad B_3 = \begin{pmatrix} b_3 \\ b_{21} \\ b_{111} \end{pmatrix}.$$

Then the third block coefficient is the finite projector expression

$$f_3 = B_3^{\mathsf{T}} G_3^{-1} A_3. \tag{68.21}$$

Moreover, with h, h_i fixed and $c \to \infty$,

$$f_3 = \frac{b_{111}a_{111}}{24h(h+1)(2h+1)} + O(c^{-1}), \tag{68.22}$$

which is the level-three coefficient of the global SL_2 block.

Proof. The same Ward identities used at level two give the level-three descendant matrix elements. For the single mode, $\rho_{34}(L_{-3}) = a_3$ and $\rho_{12}(L_{-3}) = b_3$. For the mixed descendant, the adjoint modes act in the order L_1L_2. Thus L_2 first sends the right three-point function to $a_2 z^{h-h_3-h_4+2}$, and then L_1 gives $a_2(a_1+2)$. The left vector is identical with (h_3, h_4) replaced by (h_2, h_1), giving $b_2(b_1+2)$. Three derivatives of the three-point function give the L_{-1}^3 entries a_{111} and b_{111}.

The entries of G_3 follow by commuting the positive modes in

$$\langle h|L_Y L_{-Y'}|h\rangle$$

to the right using

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}, \qquad L_n|h\rangle = 0 \quad (n>0), \quad L_0|h\rangle = h|h\rangle.$$

For example, $\langle L_{-3}h, L_{-2}L_{-1}h\rangle = \langle h|L_3L_{-2}L_{-1}|h\rangle = 10h$, while $\langle L_{-2}L_{-1}h, L_{-1}^3 h\rangle = 12h(3h+1)$. Explicitly, the six independent entries are

$$\begin{aligned}
\langle h|L_3L_{-3}|h\rangle &= 6h + 2c, \\
\langle h|L_3L_{-2}L_{-1}|h\rangle &= 10h, \\
\langle h|L_3L_{-1}^3|h\rangle &= 24h, \\
\langle h|L_1L_2L_{-2}L_{-1}|h\rangle &= h(c + 8h + 8), \\
\langle h|L_1L_2L_{-1}^3|h\rangle &= 12h(3h+1), \\
\langle h|L_1^3L_{-1}^3|h\rangle &= 24h(h+1)(2h+1).
\end{aligned}$$

These identities give (68.19). Taking the determinant of this displayed matrix gives (68.20); the first nontrivial factor is the level-two Kac factor and the second is the new level-three factor.

The projector onto level 3 in the intermediate Verma module is G_3^{-1} in this basis, so contraction with the left and right three-point descendant vectors gives (68.21). Finally, in (68.19) the first two basis vectors have norms growing with c, while the L_{-1}^3 norm is independent of c. Equivalently, the $(3,3)$ entry of G_3^{-1} tends to $[24h(h+1)(2h+1)]^{-1}$ and all inverse entries involving L_{-3} or $L_{-2}L_{-1}$ are $O(c^{-1})$. This proves (68.22). □

68.4 Elliptic Nome and Recursion Coordinates

The z-series in Definition 68.9 is the local OPE expansion near $z = 0$. For crossing and numerical bootstrap questions it is better to use the elliptic nome of the four-punctured sphere. This change of coordinate is not a cosmetic acceleration trick: it separates the universal geometry of the double cover from the representation-theoretic recursion in the intermediate Virasoro module.

Definition 68.12 (Elliptic nome of the four-punctured sphere)**.** Let

$$K(z) = \frac{\pi}{2}\, {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right)$$

be the complete elliptic integral on the principal branch. For $z \in \mathbb{C} \setminus ((-\infty, 0] \cup [1, \infty))$, define

$$q(z) = \exp\left(-\pi\, \frac{K(1-z)}{K(z)}\right). \tag{68.23}$$

Equivalently, if $|q|$ is small,

$$z = \lambda(q) = \frac{\vartheta_2(q)^4}{\vartheta_3(q)^4}, \tag{68.24}$$

where

$$\vartheta_2(q) = 2q^{1/4} \sum_{n \geq 0} q^{n(n+1)}, \qquad \vartheta_3(q) = 1 + 2\sum_{n \geq 1} q^{n^2}.$$

The first terms are

$$\lambda(q) = 16q - 128q^2 + 704q^3 + O(q^4). \tag{68.25}$$

Remark 68.13 (First conversion from z-coefficients to q-coefficients)*.* Let the normalized block be as in Definition 68.9, and set

$$a = h - h_3 - h_4.$$

Write its elliptic-coordinate expansion before the universal theta-function factorization as

$$\mathcal{F}_h^{(c)}\begin{bmatrix} h_1 & h_2 \\ h_4 & h_3 \end{bmatrix}(\lambda(q)) = (16q)^a \left(1 + g_1 q + g_2 q^2 + O(q^3)\right). \tag{68.26}$$

Then

$$g_1 = 16f_1 - 8a, \tag{68.27}$$
$$g_2 = 256f_2 - 128(1+a)f_1 + 32a^2 + 12a. \tag{68.28}$$

Equation (68.25) gives

$$\lambda(q) = 16q(1 - 8q + 44q^2 + O(q^3)).$$

Therefore

$$\lambda(q)^a = (16q)^a \left(1 - 8aq + (32a^2 + 12a)q^2 + O(q^3)\right).$$

The nonleading part of the z-block contributes

$$1 + f_1\lambda(q) + f_2\lambda(q)^2 + O(q^3) = 1 + 16f_1 q + (-128f_1 + 256f_2)q^2 + O(q^3).$$

Multiplying these two displayed series gives (68.27)–(68.28).

Quoted Theorem 68.14 (Zamolodchikov elliptic recursion)**.** *Zamolodchikov's recursion formula [47, 48] has the following form in the normalizations used here. Fix a central charge $c = 1 + 6Q^2$, external weights h_i, and a nondegenerate internal weight h. After multiplying $\mathcal{F}_h^{(c)}(z)$ by the universal powers of z, $1-z$, $16q$, and $\vartheta_3(q)$ determined by (68.24), one obtains an elliptic block*

$$H(c, h_i, h; q) = 1 + O(q)$$

which is meromorphic as a function of h and satisfies the recursive expansion

$$H(c, h_i, h; q) = 1 + \sum_{m,n\geq 1} \frac{(16q)^{mn} R_{m,n}(c, h_i)}{h - h_{m,n}(c)} H(c, h_i, h_{m,n}(c) + mn; q). \tag{68.29}$$

Here

$$h_{m,n}(c) = \frac{Q^2 - (mb + nb^{-1})^2}{4}$$

when $Q = b + b^{-1}$, and the residues $R_{m,n}$ are explicit products of the inverse null-vector norm residue and the two chiral fusion polynomials for the pairs (h_1, h_2) and (h_3, h_4).

The theorem boundary in Theorem 68.14 is deliberate. The monograph has derived the local z-coefficients and the first coordinate-conversion coefficients above. The full recursion also requires normalizing null vectors and chiral vertex-operator matrix elements at every degenerate level and proving the analytic meromorphy/convergence statements that make the infinite recursive expression meaningful. The residue products must be inserted with the same external-leg and block normalizations as the DOZZ constants; otherwise the recursion is a formal mnemonic rather than a defined object.

Residue mechanism behind the elliptic recursion. The recursion is a compact way of organizing a Verma-module spectral calculation. Let $M(c,h)$ be the Virasoro Verma module generated by $|h\rangle$, and let $G_N(c,h)$ be the Shapovalov matrix on the level N descendant space. In the normalization used above, the coefficient of the ordinary four-point block at level N is

$$F_N(c,h_i,h) = \sum_{|I|=|J|=N} \rho_R(L_{-I}|h\rangle)\left(G_N(c,h)^{-1}\right)_{IJ}\rho_L(L_{-J}|h\rangle), \tag{68.30}$$

where ρ_L and ρ_R are the two chiral three-point functionals fixed by the external fields. Poles in the internal weight h can occur only where the Shapovalov form degenerates. For generic c, the Kac determinant has a simple zero at

$$h = h_{m,n}(c) \tag{68.31}$$

coming from a singular vector

$$\chi_{m,n} \in M(c,h_{m,n})_{mn}. \tag{68.32}$$

The submodule generated by $\chi_{m,n}$ is itself a highest-weight Verma module with highest weight $h_{m,n}+mn$, because

$$L_0\chi_{m,n} = (h_{m,n}+mn)\chi_{m,n}, \qquad L_k\chi_{m,n} = 0 \quad (k>0). \tag{68.33}$$

Near $h = h_{m,n}$, the inverse Shapovalov matrix has a rank-one singular part on this null direction. At every level $N \geq mn$, this singular part projects the descendant space of $M(c,h)$ onto the descendants of $\chi_{m,n}$:

$$G_N(c,h)^{-1} = \frac{1}{h-h_{m,n}}\,\frac{\Pi_{m,n;N}}{\partial_h\langle\chi_{m,n}(h),\chi_{m,n}(h)\rangle|_{h=h_{m,n}}} + O(1). \tag{68.34}$$

Here $\Pi_{m,n;N}$ denotes the rank-one null projection tensored with the inverse Shapovalov form of the quotient descendant module at level $N-mn$. The two chiral functionals evaluated on the singular vector are the fusion polynomials:

$$\rho_L(\chi_{m,n})\,\rho_R(\chi_{m,n}) = \mathcal{P}_{m,n}(h_1,h_2)\,\mathcal{P}_{m,n}(h_3,h_4) \tag{68.35}$$

up to the same singular-vector normalization that appears in the denominator of (68.34). Therefore the residue of the block at $h = h_{m,n}$ is the block of the embedded highest-weight module:

$$\mathrm{Res}_{h=h_{m,n}}\mathcal{F}^{(c)}_h(z) = z^{mn}\,R_{m,n}(c,h_i)\,\mathcal{F}^{(c)}_{h_{m,n}+mn}(z) \tag{68.36}$$

before the elliptic universal prefactors are stripped. In the q-block normalization used in (68.29), the leading sewing coordinate is $z = 16q + O(q^2)$. This converts the level shift z^{mn} into the factor $(16q)^{mn}$, while the remaining prefactors are analytic and are absorbed into the displayed residue $R_{m,n}(c,h_i)$.

After all these simple-pole residues are subtracted, the elliptic block is holomorphic in h and tends to 1 as $h\to\infty$ in the standard large-internal-weight domain. The recursion is then the corresponding Mittag–Leffler reconstruction, iterated with the shifted internal weights $h_{m,n}+mn$. The proof debt in Theorem 68.14 is the analytic part of this last sentence: one must control the large-h bound, the convergence of the iterated residue expansion, and the compatibility of the singular vector normalization with the DOZZ normalization used in the nonchiral Liouville four-point integral.

68.5 The DOZZ Formula and Shift Checks

The exact three-point function is the Dorn–Otto–Zamolodchikov–Zamolodchikov structure constant. It is theorem-level input for this chapter, but the special functions and convention checks are part of the manuscript.

Definition 68.15 (The Υ_b function)**.** For $b > 0$ and $Q = b + b^{-1}$, $\Upsilon_b(x)$ is the entire function characterized by self-duality $\Upsilon_b(x) = \Upsilon_{1/b}(x)$, reflection $\Upsilon_b(x) = \Upsilon_b(Q - x)$, zeros at

$$x = -mb - nb^{-1} \quad \text{and} \quad x = Q + mb + nb^{-1}, \qquad m, n \in \mathbb{Z}_{\geq 0},$$

and shift relations

$$\Upsilon_b(x + b) = \gamma(bx)\, b^{1-2bx}\, \Upsilon_b(x), \tag{68.37}$$

$$\Upsilon_b(x + b^{-1}) = \gamma(x/b)\, b^{2x/b-1}\, \Upsilon_b(x), \tag{68.38}$$

where $\gamma(x) = \Gamma(x)/\Gamma(1 - x)$. These equations fix the normalization up to the conventional value of $\Upsilon_b'(0)$, which is kept explicit in the three-point constant.

Quoted Theorem 68.16 (DOZZ three-point function)**.** *In the normalization of Convention 68.2, the sphere three-point function has the form*

$$\langle V_{\alpha_1}(z_1) V_{\alpha_2}(z_2) V_{\alpha_3}(z_3)\rangle_{\mathbb{P}^1} = \frac{C_b(\alpha_1, \alpha_2, \alpha_3)}{|z_{12}|^{2(h_1+h_2-h_3)} |z_{13}|^{2(h_1+h_3-h_2)} |z_{23}|^{2(h_2+h_3-h_1)}},$$

with $h_i = \alpha_i(Q - \alpha_i)$ and

$$C_b(\alpha_1, \alpha_2, \alpha_3) = \left[\pi\mu\, \gamma(b^2)\, b^{2-2b^2}\right]^{(Q-\alpha_1-\alpha_2-\alpha_3)/b} \times \frac{\Upsilon_b'(0) \prod_{i=1}^{3} \Upsilon_b(2\alpha_i)}{\Upsilon_b(\alpha_1 + \alpha_2 + \alpha_3 - Q)\Upsilon_b(\alpha_1 + \alpha_2 - \alpha_3)\Upsilon_b(\alpha_1 + \alpha_3 - \alpha_2)\Upsilon_b(\alpha_2 + \alpha_3 - \alpha_1)}. \tag{68.39}$$

The formula was proposed by Dorn–Otto and by the Zamolodchikovs and was proved in the probabilistic construction by Kupiainen–Rhodes–Vargas [49, 50, 45]. Its role here is to fix the normalization whose reflection and shift consequences are checked in the representation-theoretic calculations below.

Shift-equation uniqueness mechanism. The DOZZ theorem boundary combines a construction theorem with a bootstrap uniqueness statement. The local part of the uniqueness argument is visible from the two degenerate fields developed later in this chapter. Fusing $V_{-b/2}$ with a generic V_α gives the two channels $\alpha - b/2$ and $\alpha + b/2$; the BPZ equation and hypergeometric connection matrix give the b-shift ratio

$$\frac{C_b(\alpha_1 + b/2, \alpha_2, \alpha_3)}{C_b(\alpha_1 - b/2, \alpha_2, \alpha_3)} = \mathcal{R}_b(\alpha_1; \alpha_2, \alpha_3)$$

displayed explicitly in (68.74). The dual degenerate insertion $V_{-1/(2b)}$ gives the same statement with b replaced by b^{-1}. Thus any meromorphic three-point constant in the same normalization obeys two independent finite-difference equations:

$$C(\alpha_1 + b, \alpha_2, \alpha_3) = A_b(\alpha_1; \alpha_2, \alpha_3)\, C(\alpha_1, \alpha_2, \alpha_3),$$

$$C(\alpha_1 + b^{-1}, \alpha_2, \alpha_3) = A_{1/b}(\alpha_1; \alpha_2, \alpha_3)\, C(\alpha_1, \alpha_2, \alpha_3),$$

with A_b and $A_{1/b}$ fixed by the degenerate OPE coefficients and connection matrices. The ratio of two solutions is therefore invariant under translations by b and b^{-1} in each momentum.

The analytic step hidden in this sentence is the following. Let C and C^{DOZZ} be two meromorphic three-point constants obeying the same two shift equations in a connected meromorphic domain $U \subset \mathbb{C}^3$, after the same reflection and leg-normalization conventions have been chosen. On a component of U with the common pole and zero divisors removed, put

$$F(\alpha_1, \alpha_2, \alpha_3) = \frac{C(\alpha_1, \alpha_2, \alpha_3)}{C^{\mathrm{DOZZ}}(\alpha_1, \alpha_2, \alpha_3)}.$$

Then F is holomorphic and satisfies

$$F(\alpha_1 + b, \alpha_2, \alpha_3) = F(\alpha_1 + b^{-1}, \alpha_2, \alpha_3) = F(\alpha_1, \alpha_2, \alpha_3)$$

wherever the two sides are defined, and likewise in the other momentum variables. If $b^2 \notin \mathbb{Q}$, the additive subgroup $\{mb + nb^{-1} : m, n \in \mathbb{Z}\} \subset \mathbb{R}$ is dense. Fix the other variables and a horizontal line segment in the first variable avoiding the removed divisor. Periodicity on the dense subgroup and continuity imply that F is constant along that real segment. Hence $\partial_{\mathrm{Re}\,\alpha_1} F = 0$ there, and the Cauchy–Riemann equations give $\partial_{\mathrm{Im}\,\alpha_1} F = 0$. Analytic continuation then makes F independent of α_1 on the connected component. Repeating the argument in α_2 and α_3 makes F a constant. The same reasoning also explains the divisor hypothesis: a nonzero meromorphic divisor invariant under a dense real period group would have non-discrete accumulation in one complex variable, which is incompatible with meromorphicity unless the divisor is common to the two compared solutions and has already been removed.

Thus, for irrational b^2, the shift equations determine the meromorphic three-point constant up to an overall constant once the analytic domain, divisor structure, and moderate-growth assumptions have been fixed. The reflection relation and the residue normalization at the screening poles fix that constant. Rational b^2 is not obtained from the dense-period argument itself: there the two shifts generate a lattice, so additional elliptic-periodic factors are not excluded by finite-difference equations alone. The DOZZ statement at rational b^2 is reached by meromorphic continuation in b from the irrational set, together with the absence of extra singularities not already visible in the special-function formula. The probabilistic theorem supplies an independently constructed solution satisfying these properties, and the special-function formula (68.39) is the explicit representative.

The shift relations (68.37)–(68.38) are the convention-sensitive part of (68.39). They are what make the degenerate-field shift equations compatible with the normalization of the exponential interaction. The companion script `calculation-checks/liouville_bpz_checks.py` verifies the level-two null-vector arithmetic used below.

Scattering-normalized DOZZ representative. Set $\alpha_j = Q/2 + \mathrm{i}P_j$. After absorbing the cosmological-constant power and the two-point reflection factors into the delta-normalized fields of Definition 68.7, a standard scattering-normalized three-point representative is

$$\begin{aligned} C(P_1, P_2, P_3) = {} & \frac{\Upsilon_b'(0)}{\Upsilon_b(Q/2 + \mathrm{i}(P_1 + P_2 + P_3))} \\ & \times \prod_{j=1}^{3} \frac{(\Upsilon_b(2\mathrm{i}P_j)\Upsilon_b(-2\mathrm{i}P_j))^{1/2}}{\Upsilon_b(Q/2 + \mathrm{i}(P_1 + P_2 + P_3) - 2\mathrm{i}P_j)}. \end{aligned} \tag{68.40}$$

Different square-root and reflection-amplitude conventions multiply (68.40) by phase factors depending separately on the external legs; the four-point integrand (68.16) is invariant after the same convention is used for both three-point constants and two-point normalization.

Substitute $\alpha_j = Q/2 + \mathrm{i}P_j$ into (68.39). The four denominator arguments become

$$\frac{Q}{2} + \mathrm{i}(P_1 + P_2 + P_3), \qquad \frac{Q}{2} + \mathrm{i}(P_1 + P_2 + P_3) - 2\mathrm{i}P_j, \quad j = 1, 2, 3.$$

The numerator factors are $\Upsilon_b(Q+2\mathrm{i}P_j) = \Upsilon_b(-2\mathrm{i}P_j)$ by the reflection identity $\Upsilon_b(x) = \Upsilon_b(Q-x)$. Passing from V_{α_j} to the delta-normalized scattering fields contributes one square root of the two-point reflection normalization per external leg, replacing the numerator by $(\Upsilon_b(2\mathrm{i}P_j)\Upsilon_b(-2\mathrm{i}P_j))^{1/2}$ up to the leg phases just mentioned. The remaining μ-dependent power is a zero-mode convention for the origin of ϕ_0, so it is absent from this representative.

Pole normalization and the reflection phase. The scattering normalization also determines the phase $S(P)$ in Definition 68.7. Let

$$\epsilon = \frac{Q}{2} + \mathrm{i}(P_1 + P_2 + P_3).$$

When $\epsilon \to 0$, the pole of $\Upsilon_b(\epsilon)^{-1}$ in (68.40) gives

$$C(P_1, P_2, P_3) \sim \frac{1}{\epsilon} \prod_{j=1}^{3} \left(\frac{\Upsilon_b(2\mathrm{i}P_j)}{\Upsilon_b(-2\mathrm{i}P_j)} \right)^{1/2}. \tag{68.41}$$

The same pole can be read in the weak-coupling region $\phi_0 \ll 0$ from the linear-dilaton zero-mode integral. Introduce a reference cutoff by integrating the zero mode over $\phi < \phi_{\mathrm{ref}} - \frac{1}{2b}\log\mu$. Since the background charge contributes $\mathrm{e}^{-2Q\phi}$ in the three-point sphere computation, the asymptotic integral is

$$\int^{\phi_{\mathrm{ref}} - \frac{1}{2b}\log\mu} \mathrm{d}\phi\, \mathrm{e}^{-2Q\phi} \prod_{j=1}^{3} S(P_j)^{-1/2} \mathrm{e}^{(Q+2\mathrm{i}P_j)\phi} \sim \frac{1}{\epsilon} \left(\mu \mathrm{e}^{2b\phi_{\mathrm{ref}}} \right)^{-\epsilon/b} \prod_{j=1}^{3} S(P_j)^{-1/2}.$$

Comparison with (68.41), for arbitrary triples approaching the pole hyperplane, gives

$$S(P) = \left(\mu \mathrm{e}^{2b\phi_{\mathrm{ref}}} \right)^{-2\mathrm{i}P/b} \frac{\Upsilon_b(-2\mathrm{i}P)}{\Upsilon_b(2\mathrm{i}P)}. \tag{68.42}$$

Using the two Υ_b shift relations reduces this to the gamma-ratio form

$$S(P) = \left(\mu \mathrm{e}^{2b\phi_{\mathrm{ref}}} b^{2b^2-2} \right)^{-2\mathrm{i}P/b} \frac{\gamma(2\mathrm{i}P/b)}{b^2\, \gamma(-2\mathrm{i}bP)}. \tag{68.43}$$

The factor containing ϕ_{ref} is the origin convention for the zero mode; changing it multiplies V_P^{in} by a momentum-dependent phase and leaves the direct-integral four-point function unchanged when the two-point normalization is transformed simultaneously.

68.6 Boundary Liouville States

Boundary Liouville theory is the first place in this volume where the Cardy-state intuition from rational CFT has to be rebuilt rather than merely generalized by notation. The closed spectrum is the continuous direct integral over $P \geq 0$, the boundary states are distributional wavefunctions on that spectrum, and the annulus kernels are Fourier or hyperbolic kernels rather than entries of a finite modular S-matrix. This section fixes the normalization used later in the BCFT chapter; it does not assert a complete Segal-style sewing construction.

Definition 68.17 (Boundary Liouville datum)**.** Let (Σ, g) be an oriented smooth Riemannian surface with nonempty smooth boundary. Let n be the outward unit normal, K_g the geodesic curvature of $\partial\Sigma$ in the convention paired with n, and $\mathrm{d}s_g$ the induced line element. A boundary Liouville datum consists of $\phi \in C^\infty(\Sigma, \mathbb{R})$, $b > 0$, $Q \in \mathbb{R}$, $\mu > 0$, and a boundary cosmological constant $\mu_B \in \mathbb{R}$, with action

$$\begin{aligned} S_L^\partial[\phi; g, \mu_B] = \frac{1}{4\pi} \int_\Sigma \mathrm{d}^2x\, \sqrt{g}\, \left(g^{ab} \partial_a\phi\, \partial_b\phi + Q R_g \phi + 4\pi\mu\, \mathrm{e}^{2b\phi} \right) \\ + \int_{\partial\Sigma} \mathrm{d}s_g\, \left(\frac{Q}{2\pi} K_g \phi + \mu_B \mathrm{e}^{b\phi} \right). \end{aligned} \tag{68.44}$$

The factor $QK_g/(2\pi)$ is not optional: it is the boundary partner of the bulk curvature coupling and is needed for conformal covariance under Weyl rescaling on surfaces with boundary.

Boundary Euler equation. For fixed metric g, the first variation of (68.44) with arbitrary smooth $\delta\phi$ is stationary exactly when the bulk equation (68.2) holds and the boundary field satisfies

$$\nabla_n \phi + QK_g + 2\pi b \mu_B \mathrm{e}^{b\phi} = 0 \qquad \text{on } \partial\Sigma. \tag{68.45}$$

On the upper half-plane with the Euclidean metric, $K_g = 0$ and the outward normal is $n = -\partial_y$, so $-\partial_y\phi + 2\pi b\mu_B \mathrm{e}^{b\phi} = 0$. The bulk variation is the one already displayed in (68.2), except that integration by parts now also gives the boundary term

$$\frac{1}{2\pi} \int_{\partial\Sigma} \mathrm{d}s_g\, (\nabla_n\phi)\, \delta\phi.$$

The variation of the boundary curvature coupling and of the boundary exponential, with g fixed, is

$$\int_{\partial\Sigma} \mathrm{d}s_g\, \left(\frac{Q}{2\pi} K_g + b\mu_B \mathrm{e}^{b\phi} \right) \delta\phi.$$

Since the boundary value of $\delta\phi$ is arbitrary, the coefficient of $\delta\phi$ on $\partial\Sigma$ must vanish. Multiplying that boundary coefficient by 2π gives (68.45).

In the quantum conformal normalization $Q = b + b^{-1}$, the boundary exponential

$$B_\beta(x) = {:}\mathrm{e}^{\beta\phi(x)}{:}$$

has boundary scaling dimension

$$h_\beta^\partial = \beta(Q - \beta) \tag{68.46}$$

in the doubled free-field coordinate. Thus B_b has dimension 1. This is the perturbative Ward-identity reason why the coupling μ_B is a conformal boundary coupling. Existence of the non-compact interacting boundary measure is a separate problem; the formulas below are bootstrap boundary data whose sewing status is recorded explicitly.

Definition 68.18 (FZZT parameter and wavefunction)**.** Assume $b > 0$, $Q = b + b^{-1}$, $\mu > 0$, and write

$$\gamma(x) = \frac{\Gamma(x)}{\Gamma(1-x)}.$$

The FZZT boundary parameter $s \in \mathbb{C}$ is related to the boundary cosmological constant by

$$\cosh^2(\pi b s) = \frac{\mu_B^2}{\mu}\sin(\pi b^2), \tag{68.47}$$

with the equality interpreted by analytic continuation when the displayed right hand side is outside the real positive range. For physical Liouville momentum $P \in \mathbb{R}$, set

$$u_b(P) = \frac{1}{\mathrm{i}P}\left(\pi\mu\,\gamma(b^2)\right)^{-\mathrm{i}P/b}\Gamma(1+2\mathrm{i}bP)\Gamma(1+2\mathrm{i}P/b) \tag{68.48}$$

as a meromorphic distribution in P, and

$$U_s(P) = u_b(P)\cos(2\pi sP). \tag{68.49}$$

With the Ishibashi states $\langle\!\langle P|$ normalized compatibly with (69.15), the formal FZZT boundary state is

$$\langle B_s| = \int_0^\infty \frac{\mathrm{d}P}{\pi}\,U_s(P)\,\langle\!\langle P|. \tag{68.50}$$

Quoted Theorem 68.19 (FZZT one-point function)**.** *The FZZT boundary formula [51, 52, 53] gives the following disk one-point representative in the conventions of this chapter. For generic b and generic boundary parameter s, the disk one-point function of the bulk primary $V_\alpha = {:}\mathrm{e}^{2\alpha\phi}{:}$ on the upper half-plane has the form*

$$\langle V_\alpha(z,\bar z)\rangle_s = \frac{U_s(\alpha)}{|z-\bar z|^{2h_\alpha}}, \qquad h_\alpha = \alpha(Q-\alpha), \tag{68.51}$$

where

$$U_s(\alpha) = \frac{2}{b}\left(\pi\mu\,\gamma(b^2)\right)^{(Q-2\alpha)/(2b)}\Gamma(2b\alpha-b^2)\Gamma\!\left(\frac{2\alpha}{b}-\frac{1}{b^2}-1\right)\cosh((2\alpha-Q)\pi s)\,. \tag{68.52}$$

The theorem boundary is the boundary bootstrap: the degenerate $V_{-b/2}$ and $V_{-1/(2b)}$ shift equations, the reflection relation, and the annulus consistency checks determine the meromorphic solution under the standard nonresonance and analytic-continuation hypotheses.

The conversion from (68.52) to the momentum-space wavefunction in Definition 68.18 is elementary but convention sensitive. If $\alpha = Q/2 + \mathrm{i}P$, then

$$2b\alpha - b^2 = 1 + 2\mathrm{i}bP, \qquad \frac{2\alpha}{b} - \frac{1}{b^2} - 1 = \frac{2\mathrm{i}P}{b}.$$

Using $\Gamma(1+x) = x\Gamma(x)$, the prefactor $2/b$ changes $\Gamma(2\mathrm{i}P/b)$ into $\Gamma(1+2\mathrm{i}P/b)/(\mathrm{i}P)$, while $\cosh(2\pi\mathrm{i}sP) = \cos(2\pi sP)$. Hence (68.52) gives exactly (68.49).

Degenerate boundary finite-difference mechanism. The boundary parameter s enters the one-point function through the hyperbolic factor

$$H_s(\alpha) = \cosh((2\alpha - Q)\pi s)\,.$$

The two bulk degenerate fields generate finite-difference operators in the bulk momentum:

$$(\mathcal{D}_b H_s)(\alpha) := H_s(\alpha + b/2) + H_s(\alpha - b/2), \qquad (\mathcal{D}_{1/b} H_s)(\alpha) := H_s(\alpha + 1/(2b)) + H_s(\alpha - 1/(2b)).$$

The elementary identities

$$(\mathcal{D}_b H_s)(\alpha) = 2\cosh(\pi b s)\, H_s(\alpha), \tag{68.53}$$

$$(\mathcal{D}_{1/b} H_s)(\alpha) = 2\cosh(\pi s/b)\, H_s(\alpha) \tag{68.54}$$

are the algebraic core of the FZZT boundary parameter. The first eigenvalue is the square-root coordinate for the boundary cosmological constant in (68.47); the second is its $b \leftrightarrow b^{-1}$ dual coordinate.

Let $G_b(\alpha)$ denote the gamma-function and μ-dependent prefactor in (68.52), so that $U_s(\alpha) = G_b(\alpha) H_s(\alpha)$. Equations (68.53)–(68.54) are equivalently

$$\frac{U_s(\alpha + b/2)}{G_b(\alpha + b/2)} + \frac{U_s(\alpha - b/2)}{G_b(\alpha - b/2)} = 2\cosh(\pi b s)\frac{U_s(\alpha)}{G_b(\alpha)}, \tag{68.55}$$

$$\frac{U_s(\alpha + 1/(2b))}{G_b(\alpha + 1/(2b))} + \frac{U_s(\alpha - 1/(2b))}{G_b(\alpha - 1/(2b))} = 2\cosh(\pi s/b)\frac{U_s(\alpha)}{G_b(\alpha)}. \tag{68.56}$$

These identities also explain the analytic uniqueness class behind the boundary parameter. Write

$$W_s(\alpha) := \frac{U_s(\alpha)}{G_b(\alpha)}.$$

If W_s is holomorphic in a strip and obeys the two constant-coefficient relations

$$W_s(\alpha + \delta/2) + W_s(\alpha - \delta/2) = 2\cosh(\pi s \delta)\, W_s(\alpha), \qquad \delta = b, b^{-1},$$

then each pure exponential $W_s(\alpha) = \mathrm{e}^{\lambda(2\alpha - Q)}$ must satisfy

$$2\cosh(\lambda\delta) = 2\cosh(\pi s \delta), \qquad \delta = b, b^{-1}.$$

For irrational b^2, and with the moderate-growth condition excluding extra functions periodic under both shifts, the simultaneous characteristic equations force $\lambda = \pi s$ or $\lambda = -\pi s$. Thus the local solution space in this analytic class is spanned by $\mathrm{e}^{\pi s(2\alpha - Q)}$ and $\mathrm{e}^{-\pi s(2\alpha - Q)}$. The Liouville reflection relation $\alpha \leftrightarrow Q - \alpha$, in the one-point normalization used here, selects the even combination

$$W_s(\alpha) \propto \cosh((2\alpha - Q)\pi s)\,.$$

For rational b^2, or without a growth condition, finite-difference equations alone permit additional periodic factors. The boundary-bootstrap theorem boundary is precisely where those analytic restrictions, resonance prescriptions, and normalization conditions are supplied. The full boundary-bootstrap theorem supplies the remaining analytic and operator-theoretic inputs: the disk two-point function with one degenerate bulk insertion, the connection matrix relating the bulk OPE channel to the bulk-to-boundary channel, the normalization of the boundary identity operator, the treatment of resonant momenta, and the growth control needed for meromorphic uniqueness. The displayed finite-difference equations are the part of that mechanism visible directly in the FZZT representative.

Finite-difference checks of the FZZT representative. The displayed one-point function represents the meromorphic solution selected by the boundary-bootstrap theorem. The local constraints used later are the two shift equations coming from the bulk degenerate insertions $V_{-b/2}$ and $V_{-1/(2b)}$ in the presence of the boundary. The complete derivation of the boundary connection coefficients belongs to the nonrational boundary-sewing theorem boundary. Once those coefficients are fixed, the proposed one-point function can be checked directly.

Put

$$K_b := \pi\mu\,\gamma(b^2), \qquad \gamma(x) = \frac{\Gamma(x)}{\Gamma(1-x)}.$$

For nonresonant α, meaning that no gamma factor below is evaluated at a pole and the denominator one-point functions are nonzero, the b-shift ratio obtained from (68.52) is

$$\begin{aligned}\frac{U_s(\alpha+b/2)}{U_s(\alpha-b/2)} = K_b^{-1}&\frac{\Gamma(2b\alpha)}{\Gamma(2b\alpha-2b^2)}\frac{\Gamma(2\alpha/b-b^{-2})}{\Gamma(2\alpha/b-b^{-2}-2)}\\ &\times\frac{\cosh\big(\pi s(2\alpha+b-Q)\big)}{\cosh\big(\pi s(2\alpha-b-Q)\big)}.\end{aligned} \tag{68.57}$$

The dual degenerate insertion gives the independent b^{-1}-shift check,

$$\begin{aligned}\frac{U_s(\alpha+1/(2b))}{U_s(\alpha-1/(2b))} = K_b^{-1/b^2}&\frac{\Gamma(2b\alpha+1-b^2)}{\Gamma(2b\alpha-1-b^2)}\frac{\Gamma(2\alpha/b-1)}{\Gamma(2\alpha/b-2b^{-2}-1)}\\ &\times\frac{\cosh\big(\pi s(2\alpha+b^{-1}-Q)\big)}{\cosh\big(\pi s(2\alpha-b^{-1}-Q)\big)}.\end{aligned} \tag{68.58}$$

Equations (68.57) and (68.58) are the component form of the two degenerate boundary shift constraints in the normalization of (68.52). The theorem boundary supplies these shift constraints, the reflection relation, and the analytic nonresonance conditions with enough growth control to fix the meromorphic solution. The companion script checks the gamma-function and hyperbolic bookkeeping in both ratios against the displayed $U_s(\alpha)$.

Definition 68.20 (ZZ boundary states as FZZT differences)**.** For positive integers m, n, define the two imaginary FZZT parameters

$$s^+_{m,n} = \mathrm{i}\left(\frac{m}{b}+nb\right), \qquad s^-_{m,n} = \mathrm{i}\left(\frac{m}{b}-nb\right).$$

In the FZZT normalization above, the formal ZZ boundary state is the finite difference

$$\langle ZZ_{m,n}| = \langle B_{s^+_{m,n}}| - \langle B_{s^-_{m,n}}|. \tag{68.59}$$

Remark 68.21 (ZZ wavefunction in FZZT normalization). The boundary state $\langle ZZ_{m,n}|$ has the momentum-space expansion

$$\langle ZZ_{m,n}| = \int_0^\infty \frac{\mathrm{d}P}{\pi}\,U^{ZZ}_{m,n}(P)\,\langle\!\langle P|, \tag{68.60}$$

where

$$U^{ZZ}_{m,n}(P) = 2u_b(P)\,\sinh\!\left(\frac{2\pi mP}{b}\right)\sinh(2\pi nbP). \tag{68.61}$$

Insert $s^{\pm}_{m,n}$ into $\cos(2\pi sP)$. Since $\cos(\mathrm{i}X) = \cosh X$,

$$U_{s^{+}_{m,n}}(P) - U_{s^{-}_{m,n}}(P) = u_b(P)\left[\cosh\!\left(2\pi P\left(\frac{m}{b} + nb\right)\right) - \cosh\!\left(2\pi P\left(\frac{m}{b} - nb\right)\right)\right]$$
$$= 2u_b(P)\,\sinh\!\left(\frac{2\pi mP}{b}\right)\sinh(2\pi nbP),$$

using $\cosh(X+Y) - \cosh(X-Y) = 2\sinh X \sinh Y$. The same identity is verified as an exact Laurent-polynomial equality in `calculation-checks/bcft_cardy_checks.py`.

The discrete labels (m, n) are therefore not new rational-Cardy labels. They are residues or finite differences inside a continuous boundary family. Their annulus interpretation uses the same open–closed equality as Chapter 69, but with distributional wavefunctions and continuous characters. In annulus-normalized conventions one often rewrites the degenerate (m, n) wavefunction by dividing by the basic $(1, 1)$ hyperbolic factors,

$$\frac{\sinh(2\pi mP/b)\sinh(2\pi nbP)}{\sinh(2\pi P/b)\sinh(2\pi bP)},$$

which makes the finite degenerate fusion sums visible. The algebraic hyperbolic identities are elementary; the analytic sewing theorem for the resulting nonrational BCFT is not elementary and remains part of the open problem below.

68.7 Degenerate Insertions and BPZ Equations

68.7.1 Pole-Residue Structure of Liouville Correlators

The Coulomb-gas computations below are residue computations in the meromorphic Liouville correlator. This is the point that the Coulomb-gas notation tends to obscure: expansion by screening number is Laurent-coefficient bookkeeping at pole hyperplanes, not an approximation to the Liouville correlator at small μ. The constant-mode calculation following Definition 68.3 gives the organizing coordinate. On a closed genus g surface,

$$\varepsilon_\Sigma(\boldsymbol{\alpha}) = \sum_i \alpha_i - Q(1-g) = \sum_i \alpha_i - \frac{Q}{2}\chi(\Sigma),$$

and the weak-coupling tail $\phi_0 \to -\infty$ has the form

$$\int_{-\infty}^{c_0} \mathrm{e}^{2\varepsilon_\Sigma c}\,\mathrm{d}c.$$

The Seiberg domain is the open convergence region $\operatorname{Re}\varepsilon_\Sigma > 0$, together with the local integrability inequalities $\alpha_i < Q/2$. The meromorphic continuation of the correlator has boundary poles when the exponent can be neutralized by screening insertions. Expanding the interaction only as a device for extracting residues,

$$\exp\!\left[-\mu\int_\Sigma V_b\right] = \sum_{N\geq 0}\frac{(-\mu)^N}{N!}\left(\int_\Sigma V_b\right)^N,$$

the N-screening term changes the zero-mode exponent from ε_Σ to $\varepsilon_\Sigma + Nb$. Hence it contributes to the residue on the pole hyperplane

$$\varepsilon_\Sigma(\boldsymbol{\alpha}) = -Nb, \qquad N \in \mathbb{Z}_{\geq 0}. \tag{68.62}$$

At this hyperplane the remaining integral over the mean-zero field and the screening positions is the Coulomb-gas integral for the residue. Terms with a different number of screenings compute residues at different hyperplanes; each pole hyperplane has its own neutralizing screening number.

The $b \leftrightarrow b^{-1}$ duality of the DOZZ meromorphic function gives the parallel family

$$\varepsilon_\Sigma(\boldsymbol{\alpha}) = -\frac{\widehat{N}}{b}, \qquad \widehat{N} \in \mathbb{Z}_{\geq 0}, \tag{68.63}$$

whose residues are represented by $\widehat{N}$ insertions of the dual screening $V_{1/b}$ with a normalization $\widetilde{\mu}$. The dual screening is therefore a coordinate on the dual residue family of the meromorphic correlator. The classical action remains (68.1); the relation between $\widetilde{\mu}$ and μ is part of the global DOZZ normalization.

This is the logical bridge between the probabilistic convergence statement, the scattering pole calculation, and the degenerate BPZ shift derivation. The pole at $\epsilon = 0$ used in Paragraph 68.5 is the zero-screening member of the same volume-divergence mechanism. The one-screening coefficient in Lemma 68.23 is the residue at the $N = 1$ pole in (68.62). The dual coefficient in Lemma 68.24 is the $\widehat{N} = 1$ residue in (68.63). The full DOZZ three-point function is the meromorphic solution reconstructed from these residue families, the reflection relation, the two degenerate BPZ shift equations, and the $b \leftrightarrow b^{-1}$ symmetry, subject to the analytic theorem boundary stated in Quoted Theorem 68.16.

The simplest degenerate Liouville momentum is $\alpha = -b/2$. Its conformal weight is

$$h_{-b/2} = -\frac{1}{2} - \frac{3b^2}{4}.$$

Level-two Liouville null vector. Let $c = 1 + 6Q^2$, $Q = b + b^{-1}$, and

$$h = -\frac{1}{2} - \frac{3b^2}{4}.$$

In a highest-weight Virasoro module with highest vector $|h\rangle$, the level-two vector

$$\left(L_{-1}^2 + b^2 L_{-2}\right)|h\rangle \tag{68.64}$$

is annihilated by L_1 and L_2. Hence it is a null vector whenever the module is quotiented to the degenerate representation carried by $V_{-b/2}$. For a candidate $(L_{-1}^2 + \kappa L_{-2})|h\rangle$, the Virasoro commutators give

$$L_1 L_{-1}^2|h\rangle = (4h+2)L_{-1}|h\rangle, \qquad L_1 L_{-2}|h\rangle = 3L_{-1}|h\rangle,$$

and

$$L_2 L_{-1}^2|h\rangle = 6h|h\rangle, \qquad L_2 L_{-2}|h\rangle = (4h + c/2)|h\rangle.$$

The L_1 equation requires $\kappa = -(4h+2)/3 = b^2$. Substituting $h = -1/2 - 3b^2/4$ and $c = 1 + 6(b + b^{-1})^2$ into the L_2 equation gives

$$6h + b^2(4h + c/2) = 0.$$

Thus both positive modes annihilate (68.64).

Dual level-two null vector and BPZ equation. Let $c = 1 + 6Q^2$, $Q = b + b^{-1}$, and

$$h^\vee = -\frac{1}{2} - \frac{3}{4b^2}.$$

In a highest-weight Virasoro module with highest vector $|h^\vee\rangle$, the level-two vector

$$\left(L_{-1}^2 + b^{-2}L_{-2}\right)|h^\vee\rangle \tag{68.65}$$

is annihilated by L_1 and L_2. Therefore the degenerate insertion $V_{-1/(2b)}$ satisfies the dual BPZ equation

$$\left[\partial_z^2 + b^{-2}\sum_{i=1}^{n}\left(\frac{h_i}{(z-z_i)^2} + \frac{1}{z-z_i}\partial_{z_i}\right)\right]\left\langle V_{-1/(2b)}(z)\prod_{i=1}^{n}V_{\alpha_i}(z_i)\right\rangle = 0, \tag{68.66}$$

for separated-point correlators in any domain where the Ward identity and the degenerate quotient are valid. The Virasoro commutator calculation just displayed applies to every candidate $(L_{-1}^2 + \kappa L_{-2})|h\rangle$. Substituting $h = h^\vee$ gives from the L_1 equation

$$\kappa = -\frac{4h^\vee + 2}{3} = b^{-2}.$$

The remaining L_2 coefficient is

$$6h^\vee + b^{-2}(4h^\vee + c/2).$$

With $c = 1 + 6(b + b^{-1})^2$ and $h^\vee = -1/2 - 3/(4b^2)$, this coefficient is zero. Hence (68.65) is a null vector in the degenerate quotient. Inserting $(L_{-1}^2 + b^{-2}L_{-2})V_{-1/(2b)} = 0$ into the same stress-tensor Ward identity used below gives (68.66). The restriction to separated points is the same restriction as for (68.67); collision limits require OPE renormalization and meromorphic continuation.

Proposition 68.22 (BPZ differential equation)**.** *Let*

$$F(z; z_1, \ldots, z_n) = \left\langle V_{-b/2}(z)\prod_{i=1}^{n}V_{\alpha_i}(z_i)\right\rangle$$

be a separated-point correlation function in a domain where the Virasoro Ward identity and the degenerate null-vector quotient are valid. Then F satisfies

$$\left[\partial_z^2 + b^2\sum_{i=1}^{n}\left(\frac{h_i}{(z-z_i)^2} + \frac{1}{z-z_i}\partial_{z_i}\right)\right]F = 0, \qquad h_i = \alpha_i(Q - \alpha_i). \tag{68.67}$$

Proof. The proof uses only two inputs: the degenerate quotient in the Verma module of $V_{-b/2}$, and the separated-point Virasoro Ward identity. The state-field relation gives

$$\left\langle (L_{-1}^2 V_{-b/2})(z)\prod_i V_{\alpha_i}(z_i)\right\rangle = \partial_z^2 F.$$

For the L_{-2} insertion, write

$$(L_{-2}V_{-b/2})(z) = \oint_z \frac{\mathrm{d}w}{2\pi\mathrm{i}}\,\frac{T(w)V_{-b/2}(z)}{w - z}.$$

The contour may be moved, inside the separated-point domain, to the contours around the other insertions and the contour at infinity. The latter vanishes for the sphere correlator after the usual Ward identity falloff. Around z_i,

$$T(w)V_{\alpha_i}(z_i) \sim \frac{h_i}{(w-z_i)^2}V_{\alpha_i}(z_i) + \frac{1}{w-z_i}\partial_{z_i}V_{\alpha_i}(z_i),$$

and expanding $1/(w-z)$ at $w=z_i$ gives

$$\left\langle (L_{-2}V_{-b/2})(z)\prod_i V_{\alpha_i}(z_i)\right\rangle = \sum_i \left(\frac{h_i}{(z-z_i)^2} + \frac{1}{z-z_i}\partial_{z_i}\right)F.$$

The null-vector equation in the degenerate quotient,

$$(L_{-1}^2 + b^2L_{-2})V_{-b/2} = 0,$$

can now be inserted into the correlator. Combining the two displayed insertions gives (68.67). This is an identity for separated-point correlators in the specified Ward-identity domain; coincident limits require the OPE and contact-term renormalization rather than evaluation of the differential equation on the collision diagonals. □

The BPZ equation constrains correlators recursively because fusing $V_{-b/2}$ with a generic V_α has only two allowed Liouville momenta,

$$V_{-b/2}V_\alpha \sim C_+(\alpha)V_{\alpha-b/2} + C_-(\alpha)V_{\alpha+b/2}.$$

The dual degenerate field $V_{-1/(2b)}$ similarly shifts a generic momentum to $\alpha \pm 1/(2b)$ and supplies the b^{-1}-shift equation. The two degenerate equations are logically separate: the first uses the null vector (68.64), while the second uses (68.65). Treating the second as an unstated symmetry argument would hide one of the hypotheses needed in the standard bootstrap determination of the DOZZ constant. Solving (68.67) near one puncture and matching it to the same equation near another puncture gives shift equations for the three-point structure constants. The DOZZ constant is the meromorphic solution with the reflection, shift, and analyticity properties stated above.

Lemma 68.23 (Screening computation of the degenerate OPE coefficient)**.** *In the free-field normalization of Convention 68.2, and with the interaction normalization in (68.1), the degenerate OPE is normalized so that $C_+(\alpha)=1$. The residue at the single V_b-screening pole in the second channel is represented by the meromorphic coefficient*

$$\begin{aligned} C_-(\alpha) &= -\pi\mu\,\gamma(1-2b\alpha)\,\gamma(1+b^2)\,\gamma(2b\alpha-1-b^2) \\ &= -\pi\mu\,\frac{\gamma(2b\alpha-1-b^2)}{\gamma(2b\alpha)\gamma(-b^2)}, \end{aligned} \tag{68.68}$$

where $\gamma(x)=\Gamma(x)/\Gamma(1-x)$. Equivalently,

$$C_-(\alpha) = \pi\mu\, b^4\gamma(b^2)\,\frac{\gamma(2b\alpha-1-b^2)}{\gamma(2b\alpha)}.$$

The equalities are identities of meromorphic functions of b and α; the intermediate Coulomb-gas integral is defined by analytic continuation.

Proof. The coefficient C_+ is fixed by the no-screening normal-ordered product: the charges add as $\alpha - b/2$, and the free contraction gives the expected OPE power

$$V_{-b/2}(z)V_\alpha(0) = |z|^{2b\alpha}V_{\alpha-b/2}(0) + \cdots .$$

For the second channel, the relevant pole of the meromorphic correlator is the hyperplane on which the zero-mode deficit is neutralized by exactly one V_b screening, as in (68.62). The charge of $V_b(w)V_{-b/2}(z)V_\alpha(0)$ is $\alpha + b/2$, so the single-screening configuration computes the residue multiplying the channel $V_{\alpha+b/2}$. Contributions with two, three, or more V_b insertions have zero-mode exponents matched at $\varepsilon_\Sigma = -2b, -3b, \ldots$; they are residues at different poles of the same meromorphic object. The free contractions for the single-screening residue give

$$-\mu \int_{\mathbb{C}} \mathrm{d}^2 w \, |z|^{2b\alpha} \, |w-z|^{2b^2} \, |w|^{-4b\alpha} \, V_{\alpha+b/2}(0).$$

Set $w = zu$. The z-dependent factor becomes

$$|z|^{2+2b^2-2b\alpha} = |z|^{2(h_{\alpha+b/2}-h_\alpha-h_{-b/2})},$$

which is exactly the required OPE power in the second channel. Thus

$$C_-(\alpha) = -\mu \int_{\mathbb{C}} \mathrm{d}^2 u \, |u|^{-4b\alpha}|1-u|^{2b^2}.$$

This integral is the local Coulomb-gas representative of the residue at the single-screening pole. Its dependence on μ records the normalization of that residue in the classical action convention. The full Liouville correlator is the meromorphic object reconstructed from all residue data and the reflection and shift constraints. For complex exponents a, c in a convergence domain, the Dotsenko–Fateev beta integral is

$$\int_{\mathbb{C}} \mathrm{d}^2 u \, |u|^{2a}|1-u|^{2c} = \pi \, \gamma(1+a) \, \gamma(1+c) \, \gamma(-1-a-c). \tag{68.69}$$

Both sides are meromorphic in a, c, so (68.69) defines the continued value when the physical exponents do not lie in an absolute-convergence strip. Substituting $a = -2b\alpha$ and $c = b^2$ gives the first line of (68.68). The remaining forms follow from $\gamma(1-x) = \gamma(x)^{-1}$, $\gamma(1+b^2) = \gamma(-b^2)^{-1}$, and $\gamma(1+b^2) = -b^4\gamma(b^2)$. □

Lemma 68.24 (Dual screening computation)**.** *The $b \leftrightarrow b^{-1}$ dual residue family of the DOZZ meromorphic correlator is represented locally by the dual screening $V_{1/b}$ with coefficient $\widetilde{\mu}$, as in* (68.63)*. This coefficient fixes the normalization of the dual residues. The classical action remains* (68.1)*. Normalize the no-dual-screening channel in*

$$V_{-1/(2b)}V_\alpha \sim \widetilde{C}_+(\alpha)V_{\alpha-1/(2b)} + \widetilde{C}_-(\alpha)V_{\alpha+1/(2b)}$$

by $\widetilde{C}_+(\alpha) = 1$. Then the one-dual-screening coefficient is the meromorphic function

$$\begin{aligned} \widetilde{C}_-(\alpha) &= -\pi\widetilde{\mu}\,\gamma\!\left(1-\frac{2\alpha}{b}\right)\gamma\!\left(1+\frac{1}{b^2}\right)\gamma\!\left(\frac{2\alpha}{b}-1-\frac{1}{b^2}\right) \\ &= \pi\widetilde{\mu}\, b^{-4}\gamma(b^{-2})\,\frac{\gamma\!\left(\frac{2\alpha}{b}-1-\frac{1}{b^2}\right)}{\gamma(2\alpha/b)}. \end{aligned} \tag{68.70}$$

Proof. The no-dual-screening product has charge $\alpha - 1/(2b)$ and fixes $\widetilde{C}_+(\alpha) = 1$. The channel $\alpha + 1/(2b)$ lies on the pole whose zero-mode deficit is neutralized by one insertion of $V_{1/b}(w)$. This single dual screening computes the $\widehat{N} = 1$ residue in (68.63); further dual screenings compute the residues at $\varepsilon_\Sigma = -2/b, -3/b, \ldots$. The free contractions for the single dual residue give

$$-\widetilde{\mu} \int_{\mathbb{C}} \mathrm{d}^2 w \, |z|^{2\alpha/b} \, |w - z|^{2/b^2} \, |w|^{-4\alpha/b} \, V_{\alpha+1/(2b)}(0).$$

After setting $w = zu$, the z-dependent factor is

$$|z|^{2+2/b^2-2\alpha/b} = |z|^{2(h_{\alpha+1/(2b)} - h_\alpha - h_{-1/(2b)})},$$

so the scaling power matches the dual OPE channel. The remaining Dotsenko–Fateev integral is

$$-\widetilde{\mu} \int_{\mathbb{C}} \mathrm{d}^2 u \, |u|^{-4\alpha/b} |1 - u|^{2/b^2}.$$

Applying (68.69) with $a = -2\alpha/b$ and $c = b^{-2}$ gives the first line of (68.70). The second line follows from $\gamma(1-x) = \gamma(x)^{-1}$ and $\gamma(1+b^{-2}) = -b^{-4}\gamma(b^{-2})$. The formula is therefore the precise $b \leftrightarrow b^{-1}$ counterpart of Lemma 68.23, with $\widetilde{\mu}$ kept explicit because the relation between μ and $\widetilde{\mu}$ is part of the full meromorphic DOZZ normalization; this local residue integral fixes only the displayed coefficient. $\square$

Proposition 68.25 (Degenerate BPZ blocks and the DOZZ b-shift)**.** *Fix three generic momenta $\alpha_1, \alpha_2, \alpha_3$, put the degenerate field $V_{-b/2}$ at z, and use global conformal covariance to set the other insertions at $0, 1, \infty$. After extracting the elementary powers fixed by the OPE with $V_{\alpha_1}(0)$, the two local solutions of (68.67) near $z = 0$ are Gauss hypergeometric functions. In a normalization where the two fusion channels are $\alpha_1 - b/2$ and $\alpha_1 + b/2$, one may write*

$$\mathcal{F}_-(z) = z^{b\alpha_1}(1 - z)^{b\alpha_2} {}_2F_1(A, B; C; z), \tag{68.71}$$
$$\mathcal{F}_+(z) = z^{1+b^2-b\alpha_1}(1 - z)^{b\alpha_2} {}_2F_1(A - C + 1, B - C + 1; 2 - C; z), \tag{68.72}$$

with

$$A = b(\alpha_1 + \alpha_2 + \alpha_3 - Q - b/2), \quad B = b(\alpha_1 + \alpha_2 - \alpha_3 - b/2), \quad C = 1 + b(2\alpha_1 - Q).$$

Analytic continuation from the $z = 0$ channel to the $z = 1$ channel gives the corresponding gamma-function connection coefficients, hence the degenerate shift equation

$$\frac{C_b(\alpha_1 + b/2, \alpha_2, \alpha_3)}{C_b(\alpha_1 - b/2, \alpha_2, \alpha_3)} = \mathcal{R}_b(\alpha_1; \alpha_2, \alpha_3), \tag{68.73}$$

where, in the DOZZ normalization of (68.39),

$$\begin{aligned}\mathcal{R}_b(\alpha_1; \alpha_2, \alpha_3) = &\frac{\gamma(2b\alpha_1)\,\gamma(2b\alpha_1 - b^2)\,\gamma(b(\alpha_2 + \alpha_3 - \alpha_1 - b/2))}{\pi\mu\,\gamma(b^2)\,b^4} \\ &\times \frac{1}{\gamma(b(\alpha_1 + \alpha_2 + \alpha_3 - Q - b/2))} \\ &\times \frac{1}{\gamma(b(\alpha_1 + \alpha_2 - \alpha_3 - b/2))} \frac{1}{\gamma(b(\alpha_1 + \alpha_3 - \alpha_2 - b/2))}.\end{aligned} \tag{68.74}$$

The same function is obtained from the hypergeometric connection matrix and the OPE normalization

$$V_{-b/2}V_{\alpha_1} \sim C_+(\alpha_1)V_{\alpha_1-b/2} + C_-(\alpha_1)V_{\alpha_1+b/2}.$$

The connection matrix used here is the standard one for the Gauss equation:

$$\begin{aligned} {}_2F_1(A,B;C;z) = \frac{\Gamma(C)\Gamma(C-A-B)}{\Gamma(C-A)\Gamma(C-B)}{}_2F_1(A,B;A+B-C+1;1-z) \\ + (1-z)^{C-A-B}\frac{\Gamma(C)\Gamma(A+B-C)}{\Gamma(A)\Gamma(B)}{}_2F_1(C-A,C-B;C-A-B+1;1-z). \end{aligned} \tag{68.75}$$

For generic momenta, none of the denominator gamma functions in (68.75) *is at a pole. When a parameter hits a pole, the equation is understood by meromorphic continuation, and logarithmic resonances must be treated separately.*

Proof. With insertions fixed at $0, 1, \infty$, equation (68.67) has only regular singular points $\{0, 1, \infty\}$. The indicial equation at 0 has exponents equal to the OPE weight differences

$$h_{\alpha_1-b/2} - h_{\alpha_1} - h_{-b/2} = b\alpha_1, \qquad h_{\alpha_1+b/2} - h_{\alpha_1} - h_{-b/2} = 1 + b^2 - b\alpha_1.$$

After factoring these powers and the analogous elementary factor at 1, the remaining second-order Fuchsian equation is the hypergeometric equation with parameters A, B, C above. The standard connection formula for Gauss hypergeometric functions, (68.75), expresses the same two-dimensional solution space in the $z = 1$ fusion basis. The exponents at $z = 1$ are

$$b\alpha_2, \qquad b\alpha_2 + C - A - B = 1 + b^2 - b\alpha_2,$$

which are the two OPE exponents for fusing $V_{-b/2}$ with $V_{\alpha_2}(1)$. Crossing symmetry of the four-point function with one degenerate insertion identifies the products of OPE coefficients in the two bases, giving (68.73).

Here is the coefficient extraction explicitly. Let $\mathcal{H}_-$ and $\mathcal{H}_+$ be the two local blocks in the $z = 1$ channel, normalized with exponents $b\alpha_2$ and $1 + b^2 - b\alpha_2$. The connection formula gives

$$\mathcal{F}_i = \sum_{j=\pm} M_{ij}\mathcal{H}_j, \qquad i = \pm,$$

with

$$\begin{aligned} M_{--} &= \frac{\Gamma(C)\Gamma(C-A-B)}{\Gamma(C-A)\Gamma(C-B)}, & M_{-+} &= \frac{\Gamma(C)\Gamma(A+B-C)}{\Gamma(A)\Gamma(B)}, \\ M_{+-} &= \frac{\Gamma(2-C)\Gamma(C-A-B)}{\Gamma(1-A)\Gamma(1-B)}, & M_{++} &= \frac{\Gamma(2-C)\Gamma(A+B-C)}{\Gamma(A-C+1)\Gamma(B-C+1)}. \end{aligned}$$

In the $z = 0$ channel the diagonal Liouville four-point function has the form

$$\mathcal{G}(z,\bar{z}) = A_s\,|\mathcal{F}_-|^2 + B_s\,|\mathcal{F}_+|^2,$$

where

$$A_s = C_b(\alpha_1 - b/2, \alpha_2, \alpha_3), \qquad B_s = C_-(\alpha_1)C_b(\alpha_1 + b/2, \alpha_2, \alpha_3).$$

The absence of $\mathcal{F}_-\overline{\mathcal{F}_+}$ terms is not a normalization convention: for generic momenta the two intermediate primaries have nonintegrally separated conformal weights, so single-valuedness around $z = 0$ and the diagonal spectrum force the hermitian form on the local solution space to be diagonal. The same statement in the $z = 1$ OPE basis requires the coefficient of $\mathcal{H}_-\overline{\mathcal{H}_+}$ to vanish. In a real nonresonant chamber the connection coefficients are real, and this condition is

$$A_s M_{--}M_{-+} + B_s M_{+-}M_{++} = 0.$$

Both sides are meromorphic in the momenta, so the resulting identity extends away from the polar and logarithmic resonance divisors. Hence

$$\frac{C_b(\alpha_1 + b/2, \alpha_2, \alpha_3)}{C_b(\alpha_1 - b/2, \alpha_2, \alpha_3)} = -\frac{M_{--}M_{-+}}{C_-(\alpha_1)M_{+-}M_{++}}. \tag{68.76}$$

Substituting A, B, C and the one-screening coefficient (68.68) into (68.76), then using $\Gamma(x)\Gamma(1 - x) = \pi/\sin(\pi x)$ only through the ratio $\gamma(x) = \Gamma(x)/\Gamma(1-x)$, gives precisely (68.74). The formula therefore follows from a local BPZ connection problem plus the explicitly normalized degenerate OPE; it is not an independent assumption about the DOZZ function.

It remains to compute the right hand side in the DOZZ normalization. Let $K = \pi\mu\,\gamma(b^2)b^{2-2b^2}$. In the ratio

$$\frac{C_b(\alpha_1 + b/2, \alpha_2, \alpha_3)}{C_b(\alpha_1 - b/2, \alpha_2, \alpha_3)},$$

the cosmological-constant factor contributes K^{-1}. The numerator $\Upsilon_b(2\alpha_1)$ contributes

$$\frac{\Upsilon_b(2\alpha_1 + b)}{\Upsilon_b(2\alpha_1 - b)} = \gamma(2b\alpha_1)\,\gamma(2b\alpha_1 - b^2)\,b^{2+2b^2-8b\alpha_1}.$$

For the denominator factors one uses $\Upsilon_b(x)/\Upsilon_b(x + b) = \gamma(bx)^{-1}b^{-1+2bx}$ for the first three terms and $\Upsilon_b(y + b)/\Upsilon_b(y) = \gamma(by)b^{1-2by}$ for the last term, with

$$x_1 = \alpha_1 + \alpha_2 + \alpha_3 - Q - b/2, \qquad x_2 = \alpha_1 + \alpha_2 - \alpha_3 - b/2,$$
$$x_3 = \alpha_1 + \alpha_3 - \alpha_2 - b/2, \qquad y = \alpha_2 + \alpha_3 - \alpha_1 - b/2.$$

The powers of b from these four denominator ratios and the numerator combine to b^{-2b^2-2}; multiplying by K^{-1} leaves the total factor $1/(\pi\mu\,\gamma(b^2)b^4)$. The remaining gamma functions are exactly (68.74). □

68.8 Sewing Status and Open Boundary

The closed-surface probabilistic theory, the bootstrap formula, and the Segal/Kontsevich–Segal sewing formulation require the comparison datum described after Quoted Theorem 68.5.

Open Problem 68.26 (Liouville sewing and functorial closure). Construct, in a single framework, the Hilbert or topological vector spaces assigned to circles, the multilinear maps assigned to bordered Riemann surfaces, the boundary state spaces, and the sewing maps whose closed-surface matrix elements reproduce the probabilistic Liouville correlators and the DOZZ/bootstrap conformal blocks. In the notation of (68.9) and (68.10), the construction must specify the topology on continuous-spectrum direct integrals, the domain of unbounded vertex operators, the descendant pairing used on the two sides of a cut, the convergence of the gluing integrals over $\alpha = Q/2 + \mathrm{i}P$, and the residue prescription at meromorphic continuation walls. This is the Volume V form of the Liouville and nonrational-CFT open problem in the Kontsevich–Segal functorial-QFT chapter.

Thus the status is deliberately layered. The classical action and background-charge Ward algebra are derived in the chapter. The measure-theoretic path integral and the DOZZ formula are quoted theorem inputs with named proof boundaries. Full functorial sewing remains an open problem for the monograph's foundational program.

Chapter 69

Boundary Conformal Field Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

69.1 Boundary CFT Data

Boundary conformal field theory is not just a conformal field theory placed near a wall. The boundary condition is additional local QFT data: it specifies boundary operators, bulk-to-boundary operator expansions, and sewing maps for bordered surfaces. In two dimensions this datum admits a particularly sharp operator formulation because the conformal boundary condition identifies the holomorphic and antiholomorphic stress tensors on the boundary.

Definition 69.1 (Oriented bosonic BCFT datum)**.** Fix a closed oriented two-dimensional CFT $\mathcal{C}$. An oriented bosonic boundary conformal field theory based on $\mathcal{C}$ consists of the following data.

1. A set B of boundary conditions.
2. For each ordered pair $a, b \in \mathsf{B}$, a positive-energy Hilbert space $\mathcal{H}_{ab}$ assigned to an interval whose left boundary condition is a and whose right boundary condition is b.
3. For each $a \in \mathsf{B}$, a boundary operator algebra $\mathcal{A}_a$ acting on boundary segments with condition a, and more generally boundary-condition-changing operators in $\mathrm{Hom}(\mathcal{H}_{ab}, \mathcal{H}_{ac})$.
4. Bulk-to-boundary expansion maps expressing a bulk local field $\mathcal{O}(z, \bar{z})$, as z approaches a

boundary segment with condition a, in the form

$$\mathcal{O}(x+\mathrm{i}y, x-\mathrm{i}y) \sim \sum_{\beta} B_{\mathcal{O}}^{a,\beta}\,(2y)^{h_\beta-h-\bar{h}}\widehat{\mathcal{O}}_\beta^{a}(x), \qquad y>0, \tag{69.1}$$

where the boundary fields $\widehat{\mathcal{O}}_\beta^a$ have one-dimensional conformal weights h_β.

5. Correlation functions on bordered Riemann surfaces with boundary segments labelled by elements of B, satisfying locality, reflection positivity in the unitary case, conformal covariance, and sewing compatibility for gluing boundary and bulk circles.

The same closed CFT can admit many inequivalent choices of B and sewing data.

How the sewing data are organized in this chapter. The definition above is deliberately broader than any one construction. The chapter develops its consequences in four layers. First come the kinematic Virasoro and annulus constraints: stress-tensor gluing gives one preserved chiral algebra, Ishibashi states solve the closed-channel linear gluing equations, and Cardy's diagonal solution is the first complete rational test. Second come the genus-zero sewing equations: boundary OPE associativity, disk bulk-boundary classifying coordinates, direct-sum boundary conditions, and the finite Ising sewing cell show exactly which finite algebra is being checked in the diagonal case. Third come the two non-diagonal directions. For rational chiral data the Frobenius-algebra/module formalism supplies the local algebraic mechanism behind annuli, boundary OPEs, classifying projectors, and two-point pairings; the pointed G/H example is one coherent finite model of those four shadows. For nonrational data the Liouville/FZZT discussion replaces finite sums by direct-integral wavefunctions, contour prescriptions, residues, and continuous sewing maps. Fourth, the quoted theorem and open problem at the end record the remaining all-surface analytic burden: constructing positive Hilbert spaces, common domains for boundary fields, determinant-line compatibility, and sewing maps that make the finite rational and nonrational formulas part of one BCFT datum.

All-surface sewing as a generated compatibility problem. The point of the local calculations below is not that a disk, an annulus, or a single four-boundary-field correlator is a BCFT by itself. A bordered surface is cut into disks, strips, annuli, and pairs of pants; two such cuts are related by a finite sequence of elementary moves. In the rational semisimple case these are the annulus open/closed move (69.8), the boundary-associativity move (69.25), the disk classifying move (69.31), the bulk-boundary identity normalization (69.35), and the Frobenius/module cutting moves (69.66) and (69.69). In the nonrational examples the same list is enlarged by contour choices, residue jumps such as (69.61), direct-integral convergence, and anomaly-line trivializations. The finite cells in this chapter verify pieces of these moves; the theorem boundary is the claim that these moves generate all changes of decomposition and act on a space where every amplitude is defined.

Proposition 69.2 (Finite move-defect budget for bordered sewing)**.** *Let $\mathcal{G}$ be a finite graph whose vertices are choices of local bordered-surface decompositions and whose directed edges are elementary sewing moves. Assign to each vertex v a vector A_v in a normed finite dimensional space. For an edge $e: u \to v$, let T_e be the linear transport associated with the sewing move, with $\|T_e\| \le L_e$, and define the local defect*

$$\delta_e = A_v - T_e A_u.$$

For a path $p = e_N \cdots e_1$ from v_0 to v_N,

$$\|A_{v_N} - T_{e_N} \cdots T_{e_1} A_{v_0}\| \leq \sum_{r=1}^{N} \left(\prod_{s=r+1}^{N} L_{e_s} \right) \|\delta_{e_r}\|. \tag{69.2}$$

In particular, if every local defect on a connected move graph vanishes and the transports agree on closed loops, the assigned amplitude is independent of the chosen decomposition.

Proof. Write $T_{r:s} = T_{e_r} \cdots T_{e_s}$ for $r \geq s$. The identity

$$A_{v_N} - T_{N:1} A_{v_0} = \sum_{r=1}^{N} T_{N:r+1} \big(A_{v_r} - T_{e_r} A_{v_{r-1}} \big)$$

is a telescoping sum; the factor $T_{N:r+1}$ is absent for $r = N$. Taking norms and using submultiplicativity gives (69.2). If all defects vanish, transport along any path carries the initial amplitude to the terminal one. The closed-loop condition then removes dependence on the path inside the connected move graph. $\square$

This proposition is the finite algebraic skeleton behind the Cardy–Lewellen sewing problem. The diagonal Cardy and pointed G/H checks later in the chapter are exact zero-defect tests for selected rational moves. The Liouville contour and Plancherel checks are local models for nonrational closed/open moves. None of them supplies the missing analytic input that the move graph is generated on a complete topological state space with convergent multilinear sewing maps; that is why the all-surface construction remains a quoted theorem or open problem rather than a consequence of the examples.

Definition 69.3 (Finite sewing anomaly cocycle)**.** In the setting of Proposition 69.2, suppose the amplitudes are not scalar vectors but vectors in one-dimensional anomaly lines $\mathcal{L}_v$ over the vertices of the move graph. After choosing local trivializations of these lines, an elementary sewing move $e : u \to v$ is represented by a scalar $\lambda_e \in \mathbb{C}^\times$ times the ordinary transport T_e. For a path $p = e_N \cdots e_1$, set

$$\lambda_p = \lambda_{e_N} \cdots \lambda_{e_1}.$$

The scalar λ_ℓ on a closed loop ℓ is the scalar part of the finite projective sewing anomaly. The actual closed-loop action on an amplitude is the total operator $\lambda_\ell T_\ell$, where $T_\ell = T_{e_N} \cdots T_{e_1}$ is the ordinary vector-space transport. The anomaly cocycle is trivial on a connected move graph if there are nonzero vertex scalars η_v such that

$$\lambda_e = \eta_v \eta_u^{-1} \qquad (e : u \to v). \tag{69.3}$$

Proposition 69.4 (Trivializing a finite sewing anomaly)**.** *Let $\mathcal{G}$ be a connected finite move graph with projective sewing scalars λ_e. Suppose the local projective defects vanish:*

$$A_v = \lambda_e T_e A_u \qquad (e : u \to v).$$

If λ_e has the coboundary form (69.3), then the rescaled amplitudes

$$\widetilde{A}_v = \eta_v^{-1} A_v$$

obey ordinary zero-defect sewing,

$$\widetilde{A}_v = T_e\widetilde{A}_u.$$

Conversely, if a closed loop has $\lambda_\ell \neq 1$, no such vertex trivialization exists on the component containing that loop. This is an obstruction to removing the scalar cocycle by changing local line frames.

Scalar decomposition independence after forgetting the anomaly line is a separate condition. For a closed loop ℓ based at v, a scalar amplitude A_v is loop-independent only if

$$\lambda_\ell T_\ell A_v = A_v. \tag{69.4}$$

Therefore $\lambda_\ell \neq 1$ obstructs a nonzero scalar amplitude only after the ordinary closed-loop transport has been trivialized on that amplitude, or more generally when 1 is not an eigenvalue of $\lambda_\ell T_\ell$ on the relevant amplitude subspace.

Proof. Substituting $\lambda_e = \eta_v\eta_u^{-1}$ gives

$$\widetilde{A}_v = \eta_v^{-1}A_v = \eta_v^{-1}\lambda_e T_e A_u = T_e\eta_u^{-1}A_u = T_e\widetilde{A}_u.$$

For the converse, multiply (69.3) around a closed loop $\ell = e_N\cdots e_1$. The vertex factors telescope, so every coboundary has $\lambda_\ell = 1$. A loop with $\lambda_\ell \neq 1$ therefore cannot come from vertex rescalings. The local projective sewing relation transported around the loop gives $A_v = \lambda_\ell T_\ell A_v$, equivalently (69.4). If $T_\ell A_v = A_v$, this reduces to $(\lambda_\ell - 1)A_v = 0$, so a nonzero scalar amplitude is impossible when $\lambda_\ell \neq 1$. Without that ordinary-transport hypothesis, λ_ℓ alone is not the scalar path-independence obstruction: ordinary holonomy may cancel the scalar phase on a particular subspace. The invariant statement is the total projective loop condition $\lambda_\ell T_\ell A_v = A_v$. □

This is the finite model for determinant-line bookkeeping in bordered CFT. The analytic sewing theorem must construct the relevant anomaly line over the moduli/decomposition groupoid, identify the line transports produced by cutting and gluing, and either trivialize their cocycle or keep amplitudes as line-valued sections. Local Cardy–Lewellen algebra controls the vector part of the move; it does not by itself remove the line cocycle. Conversely, a scalar monodromy test must include the ordinary vector transport as well as the line factor, because the physical closed-loop comparison is $\lambda_\ell T_\ell$, not λ_ℓ alone.

For a boundary equal to the real axis in the upper half-plane $\mathbb{H} = \{z = x + \mathrm{i}y \mid y \geq 0\}$, the stress tensor has components

$$T(z) = T_{zz}(z), \qquad \bar{T}(\bar{z}) = T_{\bar{z}\bar{z}}(\bar{z}).$$

The component of energy flow through the boundary is proportional to $T(z) - \bar{T}(\bar{z})$ at $z = \bar{z} = x$. A conformal boundary condition sets this flux to zero:

$$T(x) = \bar{T}(x), \qquad x \in \mathbb{R}, \tag{69.5}$$

as an identity inside boundary correlation functions away from boundary operator insertions.

Preserved Virasoro algebra. Suppose a boundary condition on the upper half-plane obeys (69.5). The conformal transformations preserving the boundary are then generated by a single copy of the Virasoro algebra. In the closed-channel boundary-state convention the modes are

$$L_n^{\partial} = L_n - \bar{L}_{-n} \tag{69.6}$$

or equivalently by the diagonal combination of holomorphic and antiholomorphic conformal vector fields on the half-plane.

Let $v(z)\partial_z + \bar{v}(\bar{z})\partial_{\bar{z}}$ be a conformal vector field on the upper half-plane. It preserves the real boundary precisely when $v(x) = \bar{v}(x)$ for real x. The Ward identity for an infinitesimal transformation is the contour integral of

$$v(z)T(z)\mathrm{d}z + \bar{v}(\bar{z})\bar{T}(\bar{z})\mathrm{d}\bar{z}.$$

Moving the contour to the boundary produces a term proportional to

$$v(x)\big(T(x) - \bar{T}(x)\big)\mathrm{d}x,$$

which vanishes by (69.5). On the cylinder, the closed-channel boundary is a state at Euclidean time 0. Reversing the orientation of the antiholomorphic contour sends $\bar{L}_n$ to $\bar{L}_{-n}$, so the annihilating generator is $L_n - \bar{L}_{-n}$. These generators obey a single Virasoro algebra because the two central terms cancel in the difference.

69.2 Boundary States and Ishibashi States

The same annulus can be quantized in two ways. In the open channel, time runs along the strip and the Hilbert space is $\mathcal{H}_{ab}$. In the closed channel, time runs across the strip and the two boundary components are represented by states in the closed-string Hilbert space. The equality of these two descriptions is the annulus sewing constraint.

Let the closed Hilbert space of a diagonal rational CFT be

$$\mathcal{H}_{\mathrm{cl}} = \bigoplus_{i \in I} \mathcal{H}_i \otimes \overline{\mathcal{H}_i}, \tag{69.7}$$

where $\mathcal{H}_i$ are irreducible representations of the chiral algebra. For an interval of length L and Euclidean time β, define

$$q = \exp(-\pi\beta/L), \qquad \tilde{q} = \exp(-4\pi L/\beta).$$

The same annulus amplitude must have the form

$$Z_{ab}^{\mathrm{open}}(q) = \mathrm{Tr}_{\mathcal{H}_{ab}}\, q^{L_0 - c/24} = \langle b|\tilde{q}^{(L_0 + \bar{L}_0 - c/12)/2}|a\rangle = Z_{ab}^{\mathrm{closed}}(\tilde{q}). \tag{69.8}$$

The factor $1/2$ in the closed Hamiltonian is a convention fixed by the modular parameter of the annulus; changing cylinder coordinates changes both sides by the same reparametrization. The order $\langle b| \cdots |a\rangle$ is the orientation convention used below: $\mathcal{H}_{ab}$ is the open Hilbert space of fields that begin at boundary condition a and end at boundary condition b.

Definition 69.5 (Ishibashi state)**.** Assume $\mathcal{H}_i$ is paired with its conjugate in (69.7). Choose a homogeneous orthonormal basis $\{e_\alpha\}$ of $\mathcal{H}_i$ and an antiunitary map $U_i : \mathcal{H}_i \to \overline{\mathcal{H}_i}$ intertwining the chiral algebra with the antiholomorphic copy. The convention is

$$U_i L_m U_i^{-1} = \bar{L}_m \tag{69.9}$$

on the finite-energy domain. For an additional chiral generator W_m^a of spin s_a, the preserved boundary condition is encoded by the corresponding antiunitary adjoint convention

$$U_i (W_m^a)^\dagger U_i^{-1} = (-1)^{s_a} \Omega(\bar{W}_{-m}^a), \tag{69.10}$$

where Ω is the chosen automorphism of the chiral algebra. The Ishibashi state associated to i is the formal state

$$|i\rangle\rangle = \sum_\alpha e_\alpha \otimes U_i e_\alpha. \tag{69.11}$$

It is not usually a normalizable vector in $\mathcal{H}_{\mathrm{cl}}$; it is a distributional state whose overlaps are defined after inserting $\tilde{q}^{(L_0+\bar{L}_0-c/12)/2}$.

Proposition 69.6 (Ishibashi gluing)**.** *The state $|i\rangle\rangle$ satisfies*

$$(L_n - \bar{L}_{-n})|i\rangle\rangle = 0 \tag{69.12}$$

for all $n \in \mathbb{Z}$. More generally, if W_m^a is a mode of a chiral current of spin s_a, the symmetry-preserving Ishibashi state satisfies

$$\left(W_m^a - (-1)^{s_a} \Omega(\bar{W}_{-m}^a)\right) |i\rangle\rangle = 0 \tag{69.13}$$

for the chosen automorphism Ω of the chiral algebra.

Proof. Let

$$C_i = \sum_\alpha e_\alpha \otimes U_i e_\alpha$$

denote the formal coevaluation vector. We first record the finite-level identity which controls the antiunitary bookkeeping. If A is an operator on the finite-energy domain of $\mathcal{H}_i$ with adjoint $A^\dagger$, then

$$(A \otimes 1) C_i = (1 \otimes U_i A^\dagger U_i^{-1}) C_i \tag{69.14}$$

as a distributional vector, meaning after pairing against any finite-level vector $u \otimes U_i v$. Indeed, using the Hermitian inner product on $\mathcal{H}_i$, written antilinear in the first argument, the antiunitarity of U_i, and the finite-level resolution of the identity $\sum_\alpha |e_\alpha\rangle\langle e_\alpha| = 1$, the two pairings are

$$\sum_\alpha \langle u, A e_\alpha\rangle \langle e_\alpha, v\rangle = \langle u, Av\rangle = \langle A^\dagger u, v\rangle$$

and

$$\sum_\alpha \langle u, e_\alpha\rangle \langle A^\dagger e_\alpha, v\rangle = \langle A^\dagger u, v\rangle.$$

Since finite-level vectors separate the graded dual completion in which $|i\rangle\rangle$ lives, (69.14) follows. Taking $A = L_n$ and using positive-energy unitarity $L_n^\dagger = L_{-n}$, together with (69.9), gives

$$(L_n \otimes 1)|i\rangle\rangle = (1 \otimes \bar{L}_{-n})|i\rangle\rangle,$$

which is (69.12). For a spin-s_a current the finite-level identity (69.14), applied to $A = W^a_m$, gives the adjoint mode in the reflected copy. The boundary reflection reverses the chiral coordinate and contributes the spin phase $(-1)^{s_a}$; the chosen preserved diagonal current is then allowed to act through the chiral-algebra automorphism Ω. These three inputs give (69.13). □

The overlap of two Ishibashi states is diagonal:

$$\langle\langle i|\tilde{q}^{(L_0+\bar{L}_0-c/12)/2}|j\rangle\rangle = \delta_{ij}\chi_i(\tilde{q}), \tag{69.15}$$

where χ_i is the chiral character. Thus a boundary state preserving the chiral algebra has the expansion

$$|a\rangle = \sum_i B_a{}^i\,|i\rangle\rangle \tag{69.16}$$

with coefficients constrained by the open-channel interpretation.

69.3 Cardy Consistency

Assume the rational diagonal setting of (69.7). The open-channel Hilbert space between two elementary boundary conditions a, b must decompose as

$$\mathcal{H}_{ab} = \bigoplus_k n_{ab}{}^k\,\mathcal{H}_k, \qquad n_{ab}{}^k \in \mathbb{Z}_{\geq 0}. \tag{69.17}$$

Using $\chi_i(\tilde{q}) = \sum_k S_{ik}\chi_k(q)$, the annulus equality gives

$$n_{ab}{}^k = \sum_i \overline{B_b{}^i}\,B_a{}^i S_{ik} \tag{69.18}$$

for unitary boundary coefficients in this orientation convention. If the modular data are real and all labels are self-conjugate, this reduces to the familiar symmetric shorthand. In general the conjugation records the direction of the boundary-changing open sector. Cardy consistency is the demand that these numbers be nonnegative integers and that the resulting open operator algebras sew consistently.

Proposition 69.7 (Oriented Cardy annulus for the diagonal invariant)**.** *Let a unitary rational CFT have diagonal closed Hilbert space* (69.7)*, unitary modular matrix S, and fusion coefficients $N_{ab}{}^k$ obeying the Verlinde formula. Then the boundary states*

$$|a\rangle = \sum_i \frac{S_{ai}}{\sqrt{S_{0i}}}|i\rangle\rangle, \qquad a \in I, \tag{69.19}$$

satisfy the annulus integrality condition with

$$n_{ab}{}^k = N_{ka}{}^b. \tag{69.20}$$

Proof. Substitute $B_a{}^i = S_{ai}/\sqrt{S_{0i}}$ into (69.18). The coefficient of the open-channel character $\chi_k(q)$ is

$$n_{ab}{}^k = \sum_i \frac{\overline{S_{bi}}\,S_{ai}S_{ik}}{S_{0i}}.$$

By unitarity and charge conjugation, $\overline{S_{bi}} = S_{\bar{b}i}$, and the Verlinde formula gives

$$N_{ka}{}^{b} = \sum_i \frac{S_{ki} S_{ai} \overline{S_{bi}}}{S_{0i}}.$$

Since S is symmetric, these expressions agree. Fusion coefficients are dimensions of three-point chiral block spaces, hence are nonnegative integers. For self-conjugate labels and real S-matrix entries this orientation-aware statement collapses to the common symmetric shorthand after the corresponding relabelling of boundary orientation, but the displayed formula is the one used by boundary-changing fields in non-self-conjugate examples. □

Proposition 69.8 (Annulus nimrep spectral resolution)**.** *Let I be the set of simple chiral labels of a unitary rational theory, with fusion coefficients $N_{ij}{}^{k}$ and modular matrix S. Let $\mathcal{B}$ be a finite set of elementary boundary labels, and suppose the open annulus multiplicities form matrices*

$$(n_i)_a{}^b = n_{ab}{}^i, \qquad a, b \in \mathcal{B},$$

such that

$$n_0 = \mathbf{1}, \qquad n_i n_j = \sum_k N_{ij}{}^k n_k. \tag{69.21}$$

Assume also that these matrices are normal and mutually commuting in the unitary boundary two-point pairing. Then there is a unitary matrix Ψ_{ae}, whose columns are indexed by a finite multiset $\mathcal{E}$ of exponents e, and a chiral label $\alpha(e) \in I$ for each exponent, such that

$$(n_i)_a{}^b = \sum_{e \in \mathcal{E}} \Psi_{ae} \frac{S_{i,\alpha(e)}}{S_{0,\alpha(e)}} \overline{\Psi_{be}}. \tag{69.22}$$

The number of copies of a fixed chiral label in $\mathcal{E}$ is the dimension of the corresponding common eigenspace of the annulus matrices. If the closed channel contains one Ishibashi copy $|e\rangle\rangle$ for each exponent copy and the boundary-state coefficients are

$$B_a{}^e = \frac{\Psi_{ae}}{\sqrt{S_{0,\alpha(e)}}}, \tag{69.23}$$

then the closed-channel annulus formula reproduces precisely the open-channel matrices n_i.

Proof. The matrices n_i are simultaneously unitarily diagonalizable because they are normal and mutually commute. On a common eigenline e, the eigenvalues $\lambda_i(e)$ obey

$$\lambda_i(e)\lambda_j(e) = \sum_k N_{ij}{}^k \lambda_k(e), \qquad \lambda_0(e) = 1,$$

so $i \mapsto \lambda_i(e)$ is a one-dimensional representation of the Verlinde fusion algebra. Since S diagonalizes this semisimple algebra, every such character has the form

$$\lambda_i(e) = \frac{S_{i,\alpha(e)}}{S_{0,\alpha(e)}}$$

for a chiral label $\alpha(e)$, with repetitions allowed when the common eigenspace has multiplicity. Writing the spectral projectors in an orthonormal eigenbasis gives (69.22). Finally, (69.18) with (69.23) gives

$$\sum_{e \in \mathcal{E}} \overline{B_b{}^e}\, B_a{}^e\, S_{\alpha(e)i} = \sum_{e \in \mathcal{E}} \Psi_{ae} \frac{S_{i,\alpha(e)}}{S_{0,\alpha(e)}} \overline{\Psi_{be}},$$

using the symmetry of S, which is the displayed open annulus matrix. □

This proposition is the annulus part of the open–closed sewing problem, not a replacement for the Cardy–Lewellen equations. It says that a non-diagonal rational boundary theory is not specified by arbitrary nonnegative integer open spectra: the annulus matrices must form a fusion-ring representation, and their common eigenvectors are the closed-channel reflection coefficients of the boundary states. The later boundary OPE, disk classifying algebra, and two-point pairing data still have to explain how these reflection coefficients participate in local operator products and all bordered-surface sewings. In the diagonal Cardy solution this spectral resolution is the regular representation diagonalized by S; in non-diagonal examples the exponent multiset records which Ishibashi sectors, and with what multiplicities, are visible to the chosen boundary theory.

69.4 Boundary OPE and Cardy–Lewellen Equations

The annulus only detects the graded dimensions of open-channel Hilbert spaces. A BCFT also has boundary operator products. If the boundary condition is a to the left of an insertion and b to the right, write

$$\psi_i^{ab,\alpha}(x)$$

for a boundary-condition-changing primary transforming in the chiral representation i. The multiplicity label α runs through a basis of the corresponding space of boundary fields.

Hypothesis 69.9 (Rational sewing hypotheses for this subsection)**.** The explicit sewing equations in this subsection assume a unitary rational chiral algebra with finitely many simple representations, semisimple positive-energy modules, convergent genus-zero chiral conformal blocks, nondegenerate boundary two-point pairings, and Moore–Seiberg fusing isomorphisms satisfying the pentagon identity. The diagonal Cardy example also assumes the closed Hilbert space $\oplus_i \mathcal{H}_i \otimes \overline{\mathcal{H}_i}$. Outside these hypotheses the formulas remain useful local models, but they are not a proof of analytic sewing.

Under Hypothesis 69.9, for ordered boundary conditions a, b, c, the boundary OPE has the form

$$\psi_i^{ab,\alpha}(x)\psi_j^{bc,\beta}(0) \sim \sum_{k,\gamma} C\begin{bmatrix} a & b & c \\ i & j & k \end{bmatrix}_{\alpha\beta}^{\gamma} x^{h_k-h_i-h_j}\psi_k^{ac,\gamma}(0) + \cdots . \tag{69.24}$$

The coefficient is not just a number attached to three labels; it depends on normalizations of boundary two-point functions and on a choice of basis in each chiral block space. A change of those bases conjugates the coefficient tensors, while the sewing equations below are invariant statements.

For four boundary fields on a disk, with boundary conditions

$$a \longrightarrow b \longrightarrow c \longrightarrow d \longrightarrow a,$$

there are two ways to perform the OPE. In a rational theory the two chiral decompositions are related by the fusing matrix F. The resulting Cardy–Lewellen boundary sewing equation, with dual-basis pairings of chiral blocks suppressed from the notation, is

$$\sum_{p,\mu} C\begin{bmatrix} a & b & c \\ i & j & p \end{bmatrix}^{\mu}_{\alpha\beta} C\begin{bmatrix} a & c & d \\ p & k & \ell \end{bmatrix}^{\delta}_{\mu\gamma} F_{pq}\begin{bmatrix} i & j \\ k & \ell \end{bmatrix}$$
$$= \sum_{\eta} C\begin{bmatrix} b & c & d \\ j & k & q \end{bmatrix}^{\eta}_{\beta\gamma} C\begin{bmatrix} a & b & d \\ i & q & \ell \end{bmatrix}^{\delta}_{\alpha\eta}. \tag{69.25}$$

The displayed formula suppresses the standard dual-basis pairings of chiral blocks. Its invariant content is that boundary OPE coefficients form an associative multiplication object in the chiral tensor category, with the associator supplied by the Moore–Seiberg fusing isomorphism.

Cardy boundary fields from fusion. In the diagonal Cardy solution, the multiplicity of boundary primaries from a to b is

$$\dim \operatorname{span}\{\psi_i^{ab}\} = N_{ia}{}^{b}. \tag{69.26}$$

If the boundary three-point coefficients are chosen to be the normalized fusing matrices of the chiral theory, then (69.25) is precisely the pentagon identity for the fusing matrices.

The open spectrum for Cardy boundary labels is $\mathcal{H}_{ab} \simeq \oplus_i N_{ia}{}^{b}\mathcal{H}_i$, matching (69.20) and (69.26). Boundary three-point functions are chiral three-point blocks on the doubled disk. The operation of composing two boundary OPEs is therefore the operation of changing parenthesization in a four-point chiral block. The Moore–Seiberg fusing matrix is exactly the transition matrix between the two parenthesizations. The pentagon identity says that all ways of changing parenthesization in a five-point block agree. Specializing the pentagon to the four-boundary-field factorization diagram gives (69.25).

The disk one-point coefficient of a spinless bulk primary is another piece of sewing data. In the diagonal Cardy case the coefficient of the identity boundary field is proportional to

$$B_a{}^{i} = \frac{S_{ai}}{\sqrt{S_{0i}}}.$$

The Verlinde eigenvalue

$$\lambda_a(i) = \frac{S_{ia}}{S_{0a}} \tag{69.27}$$

is a one-dimensional character of the fusion ring:

$$\lambda_a(i)\lambda_a(j) = \sum_k N_{ij}{}^{k}\, \lambda_a(k). \tag{69.28}$$

Indeed, the Verlinde formula says that the matrices N_i are diagonalized by S, with eigenvalues S_{ia}/S_{0a}. The Cardy disk coefficient is the same eigenvalue with the normalization required by the bulk two-point function:

$$B_a{}^{i} = \frac{S_{0a}}{\sqrt{S_{0i}}}\, \lambda_a(i).$$

Thus the disk one-point functions solve the genus-zero bulk-boundary sewing equation only after this normalization has been separated from the physical two-point convention. The useful coordinate for the disk sewing equation is

$$\widehat{B}_a(i) := \frac{\sqrt{S_{0i}}}{S_{0a}}\, B_a{}^{i} = \lambda_a(i). \tag{69.29}$$

In the diagonal Cardy theory the bulk fields which can appear in the boundary identity channel form the fusion algebra object with basis $\{e_i\}$ and product

$$e_i e_j = \sum_k N_{ij}{}^k e_k. \tag{69.30}$$

The boundary condition a defines the linear functional $\chi_a(e_i) = \widehat{B}_a(i)$. The disk sewing constraint obtained by comparing the two orders of operation, first fusing two bulk fields and then taking the boundary identity channel or first taking the two identity bulk-boundary channels and multiplying the results, is the multiplicativity condition

$$\chi_a(e_i e_j) = \chi_a(e_i)\chi_a(e_j). \tag{69.31}$$

Substituting (69.30) and $\chi_a(e_i) = \lambda_a(i)$ gives precisely (69.28). This is a statement about the normalized classifying coordinate. It is not the assertion that the raw bulk OPE constants are fusion coefficients, nor that all metric factors have disappeared; those factors have been absorbed into the chosen basis of the classifying algebra and into the normalization of the identity boundary field. In a non-diagonal rational CFT the same low-genus sewing constraint is formulated in the classifying algebra associated to the chosen Frobenius algebra object; in a nonrational theory there need not be a finite-dimensional classifying algebra at all, and the finite character calculation must be replaced by an analytic statement about the relevant continuous spectrum.

69.5 Bulk-Boundary Normalization and Direct Sums

The boundary state coefficient, the disk one-point function, and the identity term in the bulk-boundary OPE are often denoted by the same symbol. That is harmless only after the two-point metric and the identity normalization have been fixed. We record the convention because otherwise Cardy formulas can appear to disagree by factors of S_{0i}.

Assume that bulk spinless primaries ϕ_i are normalized on the plane by

$$\langle \phi_i(z,\bar{z})\phi_j(0,0)\rangle = \frac{D_{ij}}{|z|^{4h_i}}, \qquad D_{ij} = 0 \text{ unless } h_i = h_j, \tag{69.32}$$

with inverse matrix D^{ij}. For a boundary condition a, write the upper-half-plane one-point function as

$$\langle \phi_i(x+\mathrm{i}y, x-\mathrm{i}y)\rangle_a = \frac{U_i^a}{(2y)^{2h_i}}, \qquad y > 0. \tag{69.33}$$

If the boundary identity $\mathbf{1}_a$ has expectation value 1, the identity term in the bulk-boundary OPE is

$$\phi_i(x+\mathrm{i}y, x-\mathrm{i}y) = R^a_{i0}(2y)^{-2h_i}\mathbf{1}_a + \cdots, \qquad R^a_{i0} = U^a_i. \tag{69.34}$$

Equivalently, if one raises the bulk label by defining

$$U_a{}^j = D^{ji}U^a_i,$$

then the same coefficient is

$$R^a_{i0} = D_{ij}U_a{}^j. \tag{69.35}$$

Thus (69.19) determines a physical one-point function only after the Ishibashi overlap normalization and the bulk two-point metric have both been declared.

Remark 69.10 (Direct sums and Chan–Paton matrix units)*.* Let a, b be boundary conditions in a unitary BCFT, and suppose finite direct sums of boundary conditions are allowed. For

$$A = a^{\oplus n}, \qquad B = b^{\oplus m},$$

with no additional boundary fields beyond the direct-sum completion, the open Hilbert space and boundary fields are

$$\mathcal{H}_{AB} \simeq \mathcal{H}_{ab} \otimes \mathrm{Mat}_{n\times m}(\mathbb{C}). \tag{69.36}$$

If E_{rs} denotes the standard matrix unit, boundary OPEs obey

$$(\psi_\alpha \otimes E_{rs})(x)(\psi_\beta \otimes E_{tu})(0) = \delta_{st}\big(\psi_\alpha(x)\psi_\beta(0)\big) \otimes E_{ru}. \tag{69.37}$$

Consequently

$$Z^{\text{open}}_{AB}(q) = nm\, Z^{\text{open}}_{ab}(q), \qquad g_{a^{\oplus n}} = n\, g_a. \tag{69.38}$$

Indeed, a direct sum boundary condition is a finite biproduct in the boundary category. An interval with left boundary $a^{\oplus n}$ and right boundary $b^{\oplus m}$ therefore has one copy of $\mathcal{H}_{ab}$ for each ordered pair of summands. Choosing the standard bases of the multiplicity spaces identifies this direct sum with (69.36). Locality on the boundary forces composition to match endpoint labels: an operator carrying the right endpoint index s can compose with a second operator whose left endpoint is t only when $s = t$. This is precisely the multiplication rule for matrix units, giving (69.37). Taking the graded trace over $\mathcal{H}_{ab} \otimes \mathrm{Mat}_{n\times m}(\mathbb{C})$ gives the factor nm in the open annulus. In the closed channel, the boundary state of $a^{\oplus n}$ is the sum of n identical boundary states, so its vacuum overlap is ng_a.

69.6 Compact Free Boson Boundaries

Let $X \sim X + 2\pi R$ be a compact free boson with

$$X_L(z)X_L(0) \sim -\log z, \qquad X_R(\bar z)X_R(0) \sim -\log \bar z.$$

Circle states are labelled by momentum $m \in \mathbb{Z}$ and winding $w \in \mathbb{Z}$, with

$$k_L = \frac{m}{R} + wR, \qquad k_R = \frac{m}{R} - wR, \qquad h = \frac{1}{2}k_L^2 + N, \quad \bar h = \frac{1}{2}k_R^2 + \bar N. \tag{69.39}$$

The oscillator modes obey $[a_m, a_n] = m\delta_{m+n,0}$ and similarly for $\bar a_n$.

On the upper half-plane, Neumann and Dirichlet boundary conditions are

$$\partial_y X|_{y=0} = 0, \qquad X|_{y=0} = x_0.$$

In closed-channel boundary-state language they become

$$(a_n + \bar a_{-n})|N, \theta\rangle = 0, \tag{69.40}$$
$$(a_n - \bar a_{-n})|D, x_0\rangle = 0, \tag{69.41}$$

for all $n \in \mathbb{Z}$. The zero-mode parts are the important point:

$$\text{Neumann: } k_L + k_R = 0 \iff m = 0, \qquad \text{Dirichlet: } k_L - k_R = 0 \iff w = 0.$$

Thus Neumann boundary states sum over winding and carry a Wilson-line phase θ, while Dirichlet boundary states sum over momentum and carry a position phase x_0:

$$|N,\theta\rangle = \mathcal{N}_N \sum_{w\in\mathbb{Z}} e^{\mathrm{i}w\theta} \exp\left(-\sum_{n\geq 1}\frac{1}{n}a_{-n}\bar{a}_{-n}\right)|0,w\rangle, \tag{69.42}$$

$$|D,x_0\rangle = \mathcal{N}_D \sum_{m\in\mathbb{Z}} e^{\mathrm{i}mx_0/R} \exp\left(\sum_{n\geq 1}\frac{1}{n}a_{-n}\bar{a}_{-n}\right)|m,0\rangle. \tag{69.43}$$

The constants $\mathcal{N}_N, \mathcal{N}_D$ are fixed by the annulus normalization. With the oscillator trace normalized by $1/\eta(q)$, one may choose $\mathcal{N}_N = \sqrt{R}$ and $\mathcal{N}_D = 1/\sqrt{2R}$ in the present momentum convention; changing the normalization of the circle volume rescales both boundary states and the open-string ground-state density by the same factor.

The radius-duality map

$$R \longmapsto R^{-1}, \qquad m \longleftrightarrow w, \qquad X_R \longmapsto -X_R$$

exchanges the gluing equations (69.40) and (69.41). It sends a Neumann brane with Wilson line θ to a Dirichlet brane at the dual position θ, and sends a Dirichlet position x_0 to a Wilson line on the dual circle. The transformation $X_R \mapsto -X_R$ sends $\bar{a}_n \mapsto -\bar{a}_n$, while leaving a_n unchanged. Therefore $a_n + \bar{a}_{-n}$ becomes $a_n - \bar{a}_{-n}$. On zero modes, $(m,w) \mapsto (w,m)$ exchanges the condition $m = 0$ with $w = 0$. The phases in (69.42) and (69.43) then exchange Wilson-line and position data.

69.7 Majorana Fermion and Ising Boundary States

The chiral Majorana field has modes

$$\psi(z) = \sum_r \psi_r z^{-r-1/2}, \qquad \{\psi_r, \psi_s\} = \delta_{r+s,0}.$$

In the Neveu–Schwarz sector $r \in \mathbb{Z}+1/2$, while in the Ramond sector $r \in \mathbb{Z}$. A conformal fermion boundary condition is

$$\left(\psi_r + \mathrm{i}\eta\,\bar{\psi}_{-r}\right)|B,\eta\rangle = 0, \qquad \eta = \pm 1. \tag{69.44}$$

The stress-tensor gluing follows because the fermion stress tensor is quadratic in ψ, but the converse is weaker: a Virasoro boundary condition need not preserve the fermion field itself.

For the bosonic Ising model, the three Virasoro primaries are

$$\mathbf{1}, \qquad \epsilon, \qquad \sigma$$

with weights $0, 1/2, 1/16$. In the ordered basis $(\mathbf{1}, \epsilon, \sigma)$, the modular matrix is

$$S_{\text{Ising}} = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}. \tag{69.45}$$

Cardy's formula gives the three elementary boundary states

$$|+\rangle = \frac{1}{\sqrt{2}}|\mathbf{1}\rangle\rangle + \frac{1}{\sqrt{2}}|\epsilon\rangle\rangle + 2^{-1/4}|\sigma\rangle\rangle, \tag{69.46}$$

$$|-\rangle = \frac{1}{\sqrt{2}}|\mathbf{1}\rangle\rangle + \frac{1}{\sqrt{2}}|\epsilon\rangle\rangle - 2^{-1/4}|\sigma\rangle\rangle, \tag{69.47}$$

$$|f\rangle = |\mathbf{1}\rangle\rangle - |\epsilon\rangle\rangle. \tag{69.48}$$

The labels $+$, $-$, and f denote the two fixed spin boundaries and the free boundary. The annulus spectra are the Ising fusion rules:

$$\mathcal{H}_{++} \simeq \mathcal{H}_{\mathbf{1}}, \qquad \mathcal{H}_{+-} \simeq \mathcal{H}_{\epsilon}, \qquad \mathcal{H}_{ff} \simeq \mathcal{H}_{\mathbf{1}} \oplus \mathcal{H}_{\epsilon},$$

and

$$\mathcal{H}_{+f} \simeq \mathcal{H}_{\sigma}, \qquad \mathcal{H}_{-f} \simeq \mathcal{H}_{\sigma}.$$

The NS/R bookkeeping is visible in the fermionic presentation: the $\mathbf{1}$ and ϵ Virasoro modules come from NS fermion parity sectors, while σ is the Ramond spin field. A spin CFT keeps the spin-structure dependence explicitly; the bosonic Ising CFT is obtained after the corresponding fermion-parity gauging.

Ising boundary-changing OPE constants. Identify the Cardy labels

$$+ = \mathbf{1}, \qquad - = \epsilon, \qquad f = \sigma.$$

Let ψ_i^{ab} denote the boundary primary of chiral type i between boundary labels a and b, when the fusion multiplicity ${N_{ia}}^b$ is one. Use the Cardy boundary-field basis determined by the Ising fusing symbols, and fix the residual scalar normalizations of one-dimensional boundary-field spaces so that the one-channel products displayed below have coefficient 1. The only products below with two allowed final channels are the products in which two σ-type boundary-changing fields leave and return to the free boundary through a fixed boundary. For $x > 0$,

$$\psi_\sigma^{f+}(x)\psi_\sigma^{+f}(0) \sim 2^{-1/2}x^{-1/8}\mathbf{1}_f + 2^{-1/2}x^{3/8}\psi_\epsilon^{ff}(0) + \cdots, \tag{69.49}$$

$$\psi_\sigma^{f-}(x)\psi_\sigma^{-f}(0) \sim 2^{-1/2}x^{-1/8}\mathbf{1}_f - 2^{-1/2}x^{3/8}\psi_\epsilon^{ff}(0) + \cdots. \tag{69.50}$$

The opposite orientation has no channel multiplicity:

$$\psi_\sigma^{+f}(x)\psi_\sigma^{f+}(0) \sim x^{-1/8}\mathbf{1}_+ + \cdots, \tag{69.51}$$

$$\psi_\sigma^{-f}(x)\psi_\sigma^{f-}(0) \sim x^{-1/8}\mathbf{1}_- + \cdots, \tag{69.52}$$

$$\psi_\sigma^{+f}(x)\psi_\sigma^{f-}(0) \sim x^{3/8}\psi_\epsilon^{+-}(0) + \cdots, \tag{69.53}$$

$$\psi_\sigma^{-f}(x)\psi_\sigma^{f+}(0) \sim x^{3/8}\psi_\epsilon^{-+}(0) + \cdots. \tag{69.54}$$

The multiplicities follow from (69.26) and the Ising fusion rules. Thus the nontrivial boundary-changing fields are

$$\psi_\epsilon^{+-}, \quad \psi_\epsilon^{-+}, \quad \psi_\epsilon^{ff}, \quad \psi_\sigma^{+f}, \quad \psi_\sigma^{f+}, \quad \psi_\sigma^{-f}, \quad \psi_\sigma^{f-}.$$

In the diagonal Cardy case the boundary OPE constants are the fusing coefficients of the chiral theory in the chosen basis. The only nontrivial Ising fusing matrix needed here is the recoupling of three σ lines with total charge σ:

$$F_\sigma^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \tag{69.55}$$

where the row records the intermediate fixed boundary $\mathbf{1}$ or ϵ, and the column records the final channel $\mathbf{1}_f$ or ψ_ϵ^{ff} on the free boundary. Equations (69.49)–(69.50) are precisely the two rows of (69.55). The powers of x are the boundary OPE powers $h_k - h_i - h_j$:

$$0 - \frac{1}{16} - \frac{1}{16} = -\frac{1}{8}, \qquad \frac{1}{2} - \frac{1}{16} - \frac{1}{16} = \frac{3}{8}.$$

All remaining displayed OPEs have one-dimensional fusion spaces. Their coefficients have been fixed to be 1 by the residual scalar normalizations specified in the statement; changing those normalizations rescales the corresponding raw three-point constants but not the sewing class represented by (69.55). Thus rescaling the fields ψ_i^{ab} changes the displayed constants by the corresponding two-point normalization factors. The invariant content is the Ising fusing matrix and, in particular, the relative sign of the ψ_ϵ^{ff} coefficient for the two fixed boundaries.

A finite Ising sewing cell. The preceding constants give a completely explicit instance of the Cardy–Lewellen equation. Let $r, s \in \{+, -\}$, and consider four boundary fields arranged around the disk as

$$f \xrightarrow{\psi_\sigma^{fr}} r \xrightarrow{\psi_\sigma^{rf}} f \xrightarrow{\psi_\sigma^{fs}} s \xrightarrow{\psi_\sigma^{sf}} f.$$

Write

$$\psi_\sigma^{fr}(x)\psi_\sigma^{rf}(0) \sim v_r(\mathbf{1})\, x^{-1/8}\mathbf{1}_f + v_r(\epsilon)\, x^{3/8}\psi_\epsilon^{ff}(0) + \cdots .$$

Equations (69.49) and (69.50) give

$$v_+ = 2^{-1/2}(1,1), \qquad v_- = 2^{-1/2}(1,-1),$$

where the two entries are the $\mathbf{1}$ and ϵ channels on the free boundary. If the two adjacent pairs are fused first through the free boundary, the coefficient multiplying the common free-boundary intermediate channel is

$$\sum_{p\in\{\mathbf{1},\epsilon\}} v_r(p)v_s(p).$$

If instead one first fuses across the fixed-boundary side, the corresponding open-sector morphism space is $\mathrm{Hom}(U_r, U_s)$, which has dimension δ_{rs}. The sewing equality for this cell is therefore

$$\sum_{p\in\{\mathbf{1},\epsilon\}} v_r(p)v_s(p) = \delta_{rs}. \tag{69.56}$$

This is the matrix identity

$$F_\sigma^{\sigma\sigma\sigma}\,(F_\sigma^{\sigma\sigma\sigma})^T = \mathbf{1}$$

for the nontrivial Ising fusing matrix (69.55). In this example, the abstract module associativity statement in (69.25) reduces to a visible orthogonality relation between the two fixed-boundary rows. The calculation does not prove analytic sewing for a general BCFT; it identifies exactly which finite algebraic identity the sewing theorem uses in the diagonal Ising cell.

69.8 Nonrational Example: Liouville Boundary States

Chapter 68, Section 68.6, constructs the Liouville boundary states used as the basic nonrational test case for this chapter. The closed state space is no longer a finite diagonal sum; in the scattering normalization used here it is a direct integral

$$\mathcal{H}_{\mathrm{cl}}^{\mathrm{L}} = \int_0^\infty \mathcal{V}_{Q/2+\mathrm{i}P} \otimes \overline{\mathcal{V}}_{Q/2+\mathrm{i}P}\, \mathrm{d}P,$$

and the Ishibashi label i is replaced by the continuous momentum P. A boundary state is therefore a distributional wavefunction $\Psi_a(P)$ multiplying $|P\rangle\!\rangle$, not a finite vector of Cardy coefficients.

For the FZZT family, the wavefunction is

$$\Psi_s(P) = \frac{1}{\mathrm{i}P} \left(\pi\mu\,\gamma(b^2)\right)^{-\mathrm{i}P/b} \Gamma(1+2\mathrm{i}bP)\Gamma(1+2\mathrm{i}P/b)\cos(2\pi sP),$$

where $\gamma(x) = \Gamma(x)/\Gamma(1-x)$, up to the common normalization and measure convention fixed in (68.50). The discrete ZZ states are finite differences at imaginary FZZT parameters, so their wavefunctions contain

$$2\sinh(2\pi mP/b)\sinh(2\pi nbP).$$

This is the nonrational replacement for the finite modular S-matrix entry in Cardy's rational formula: annulus consistency is expressed by integral kernels and hyperbolic identities, followed by an analytic statement about the resulting spectral measure.

Hyperbolic kernels behind Liouville annuli. For formal variables X, Y and for every $n \in \mathbb{Z}_{>0}$,

$$\cosh(X+Y) - \cosh(X-Y) = 2\sinh X \sinh Y, \tag{69.57}$$

$$\frac{\sinh(nX)}{\sinh X} = \sum_{j=0}^{n-1} \exp((n-1-2j)X)\,. \tag{69.58}$$

The first identity is the exact algebra turning an imaginary FZZT difference into a ZZ wavefunction. The second identity is the exact algebra turning degenerate Liouville annulus kernels into a finite sum of shifted nondegenerate characters. The verification is elementary algebra: $\cosh Z = (\mathrm{e}^Z + \mathrm{e}^{-Z})/2$, and multiplying the finite sum in (69.58) by $2\sinh X = \mathrm{e}^X - \mathrm{e}^{-X}$ leaves only the telescoping endpoints $\mathrm{e}^{nX} - \mathrm{e}^{-nX}$. The identities are deliberately only the algebraic part of the story. A complete Liouville BCFT construction must still specify the direct-integral topology, the domains of boundary-condition-changing fields, the treatment of poles encountered under contour motion, and the positivity or distribution interpretation of the annulus spectral density. Those are analytic sewing requirements, not consequences of the finite hyperbolic identities alone.

Continuous annulus Plancherel cell. It is useful to isolate one piece of the nonrational annulus which is not a finite Cardy sum in disguise. Strip off the common Liouville gamma prefactor and the Virasoro character, and keep only the even FZZT momentum wave

$$C_s(P) := 2\cos(2\pi sP), \qquad P \geq 0.$$

With a heat regulator $t > 0$, the closed-channel momentum pairing is

$$\begin{aligned} K_t(s, s') &:= \int_0^\infty \mathrm{e}^{-tP^2} C_s(P) C_{s'}(P)\,\mathrm{d}P \\ &= G_t(s - s') + G_t(s + s'), \qquad G_t(x) := \sqrt{\frac{\pi}{t}}\,\exp\!\left(-\frac{\pi^2 x^2}{t}\right). \end{aligned} \tag{69.59}$$

Indeed,

$$4\cos(2\pi s P)\cos(2\pi s' P) = 2\cos(2\pi(s - s')P) + 2\cos(2\pi(s + s')P),$$

and

$$\int_0^\infty \mathrm{e}^{-tP^2} \cos(aP)\,\mathrm{d}P = \frac{1}{2}\sqrt{\frac{\pi}{t}}\,\exp\!\left(-\frac{a^2}{4t}\right).$$

The kernel G_t has total mass one on the real line and tends to $\delta(x)$ as $t \downarrow 0$. Thus (69.59) tends distributionally to

$$\delta(s - s') + \delta(s + s').$$

The second term is not an extra open-sector multiplicity. It records that the wavefunction is even in the FZZT parameter: $C_s = C_{-s}$. Hence a nonrational open-channel label is a point of the quotient $s \sim -s$, and the open annulus is a spectral measure on this quotient rather than an integer matrix ${n_{ab}}^k$. In a finite cyclic regulator this same bookkeeping is exact: for odd N, nonzero orbits $s, s' \in \mathbb{Z}_N/\{\pm 1\}$, and

$$c_s(p) := \mathrm{e}^{2\pi \mathrm{i} sp/N} + \mathrm{e}^{-2\pi \mathrm{i} sp/N},$$

one has

$$\frac{1}{2N} \sum_{p \in \mathbb{Z}_N} c_s(p) c_{s'}(p) = \delta_{s,s'}. \tag{69.60}$$

If s and $-s$ are kept as separate labels, the Gram matrix has off-diagonal unit entries between them; the quotient is part of the boundary data, not a later convention. The full FZZT annulus restores the Liouville prefactors, modular kernels, contour prescriptions, and pole contributions. Equation (69.59) proves only the Plancherel/reflection coordinate of that analytic sewing problem.

Pole crossing as a discrete sewing summand. The contour prescription mentioned above has a precise local algebraic content. Let $\phi(P)$ be a holomorphic test function in a strip around the real P-contour, with rapid decay on horizontal translates, and suppose a boundary kernel has near a moving pole $P = \alpha$ the form

$$K_\alpha(P) = \frac{\rho(P)}{P - \alpha} + K_\alpha^{\mathrm{reg}}(P),$$

where ρ is holomorphic. If the pole is moved through the contour and the amplitude is defined by deforming the contour rather than by simply discarding the crossing, Cauchy's theorem gives

$$\int_{\mathcal{C}_+} \phi(P) K_\alpha(P)\,\mathrm{d}P = \int_{\mathcal{C}_-} \phi(P) K_\alpha(P)\,\mathrm{d}P + 2\pi \mathrm{i}\,\phi(\alpha)\rho(\alpha), \tag{69.61}$$

with the sign fixed by the orientation of the small circle left behind by the deformation. Thus a crossed pole contributes a rank-one evaluation functional on the closed-channel test space. In open-channel language this is exactly the local model for a discrete summand in the spectral measure ν_{ab}: the continuous contour integral and the residue term are two pieces of one sewing map. Omitting the residue in one factorization channel but not in another changes the amplitude by $2\pi\mathrm{i}\,\phi(\alpha)\rho(\alpha)$. Therefore a nonrational Cardy–Lewellen datum must specify one global contour/residue prescription before comparing disk, annulus, and higher bordered-surface sewings. The finite checkable shadow is the polynomial identity

$$\operatorname*{Res}_{P=\alpha} \frac{\rho(P)\sum_{m=0}^{M} c_m P^m}{P-\alpha}\,\mathrm{d}P = \rho(\alpha)\sum_{m=0}^{M} c_m \alpha^m,$$

and multiple simple crossings commute because these evaluation functionals add linearly.

Analytic sewing datum for nonrational boundaries. The preceding wavefunctions become a boundary conformal field theory only after they are placed in a functional-analytic sewing datum. One convenient way to state the target is the following. First choose, inside the closed Liouville direct integral, a dense nuclear test space $\mathcal{S}_{\mathrm{cl}}$ whose vectors have rapidly decreasing smooth dependence on P and lie levelwise in the smooth Virasoro-module vectors. Boundary states in this setting are continuous linear functionals

$$|a\rangle \in \mathcal{S}_{\mathrm{cl}}'.$$

The FZZT and ZZ formulas above specify candidate distributional coordinates of such functionals. The regulated closed-channel annulus is then the distributional pairing

$$Z^{\mathrm{cl}}_{ab}(\tilde{q}) = \big\langle a \;\big|\; \tilde{q}^{(L_0+\bar{L}_0-c/12)/2} \;\big|\; b\big\rangle, \qquad 0<|\tilde{q}|<1, \tag{69.62}$$

initially defined by testing the two distributions against the trace-class heat kernel on each positive-energy Virasoro module, with the continuous P-integration still treated distributionally, and then by analytic continuation in the boundary parameters when poles are crossed.

Second, for every ordered pair of boundary conditions a, b, one must construct an open-channel positive-energy Hilbert space $\mathcal{H}_{ab}$ and a dense common smooth domain $\mathcal{D}_{ab} \subset \mathcal{H}_{ab}$. Annulus positivity is the existence of positive spectral measures ν_{ab} and multiplicity fields $\mathcal{M}_{ab}(h)$ such that

$$Z^{\mathrm{op}}_{ab}(q) = \int_{\mathbb{R}_{\geq 0}} \mathrm{Tr}_{\mathcal{M}_{ab}(h)}\, q^{h-c/24}\,\mathrm{d}\nu_{ab}(h) \tag{69.63}$$

agrees with (69.62) in the common domain of convergence, with the usual matrix-positivity condition for finite linear combinations of boundary labels. In degenerate cases the measure may have a discrete part. The ZZ finite differences are meaningful precisely because the contour prescription turns the corresponding closed-channel functional into such a positive open-channel spectral datum; the hyperbolic telescoping identity (69.58) is its finite algebraic shadow.

Third, boundary-condition-changing fields must be defined as operator-valued distributions on these domains. A primary changing a to b, inserted on a boundary segment, should give multilinear matrix elements

$$\big\langle v_0,\, \psi^{ab}_{\lambda_1}(x_1)\cdots\psi^{cd}_{\lambda_n}(x_n)\, v_n\big\rangle, \qquad x_1 > \cdots > x_n,$$

with v_0, v_n in the appropriate open-channel smooth domains or their duals. The labels λ_i may be continuous, discrete, or resonant. The OPE coefficients are kernels between spectral measures, so their associativity is an equality of distributions after smearing in the spectral variables. This is the nonrational replacement for finite sums over boundary primaries in (69.24).

Fourth, the analytic continuation in FZZT parameters and the passage to ZZ states must include a contour and residue prescription. When the boundary parameter is moved to an imaginary degenerate value, the P-contour in closed-channel pairings can encounter poles of the wavefunctions and of bulk-boundary or boundary-boundary structure kernels. A sewing datum must state whether the amplitude is defined by contour deformation, by principal value plus residues, or by an equivalent distributional boundary value, and must verify that the same prescription is used in all channels. A channel-dependent prescription defines incompatible disk or annulus amplitudes even when each individual channel satisfies the displayed finite-difference algebra.

Finally, bordered-surface sewing requires continuous multilinear maps on the chosen test spaces. Closed-circle gluing is integration over

$$\int_0^\infty \mathrm{d}P \, \mu_{\mathrm{L}}(P)$$

with the Liouville Plancherel measure and descendant two-point pairing; open-interval gluing is integration or summation over the open spectral measure ν_{ab}. In a plumbing chart the resulting series or integral must converge for $|q_e| < 1$ and must be compatible with changing the order of gluing. Anomaly determinant lines, if present, must be tensored into these sewing maps. This paragraph is therefore a definition of the analytic burden carried by a nonrational BCFT construction. The Liouville FZZT/ZZ formulas supply strong candidate coordinates and many exact residue identities; proving the full bordered-surface functor requires the analytic data above.

69.9 Sewing, Boundary Entropy, and Defect Interfaces

Section 69.4 displayed the boundary four-point sewing equation and the fusion-ring character equation for disk one-point functions in the diagonal Cardy case. A full BCFT also has bulk-boundary OPE coefficients with nontrivial boundary fields, bulk three-point coefficients, and compatibility with defect endpoints. These data obey additional compatibility equations from sewing a disk with two bulk fields and one boundary field in different orders, and from comparing bulk, boundary, and defect factorizations.

Quoted Theorem 69.11 (Rational Cardy–Lewellen construction boundary)**.** *For rational chiral data satisfying the analytic sewing hypotheses of a modular functor, a special symmetric Frobenius algebra object A in the chiral modular tensor category produces a full rational CFT with boundaries; boundary conditions are left A-modules, and the genus-zero Cardy–Lewellen equations follow from the Frobenius and module identities. Converses and classification statements require additional equivalence data and are not used below. This chapter uses only the local finite-categorical consequences developed in the following paragraphs, not a proof of nonrational or continuous-spectrum BCFT sewing.*

Algebraic core of the rational boundary construction. The quoted theorem has a concrete finite-categorical mechanism. Let C be the semisimple ribbon tensor category of chiral representations, with tensor product $\otimes$, unit object $\mathbf{1}$, associator α, braiding c, and representatives $\{U_i\}_{i\in I}$ of simple objects. A rational full CFT with a chosen bulk modular invariant and a chosen family of topological boundary conditions is encoded algebraically by an object $A \in \mathsf{C}$ equipped with maps

$$m : A \otimes A \to A, \qquad \eta : \mathbf{1} \to A, \qquad \Delta : A \to A \otimes A, \qquad \varepsilon : A \to \mathbf{1}.$$

The multiplication and unit make A an algebra object:

$$m(m\otimes \mathrm{id}_A)=m(\mathrm{id}_A\otimes m)\alpha_{A,A,A},\qquad m(\eta\otimes \mathrm{id}_A)=\mathrm{id}_A=m(\mathrm{id}_A\otimes \eta).$$

The comultiplication and counit obey the dual coalgebra identities, and the Frobenius compatibility is

$$(m\otimes \mathrm{id}_A)(\mathrm{id}_A\otimes \Delta)=\Delta m=(\mathrm{id}_A\otimes m)(\Delta\otimes \mathrm{id}_A), \tag{69.64}$$

with associators inserted in the unique type-correct positions. The word special means that $m\Delta$ is a nonzero scalar multiple of id_A, so cutting out a short A-ribbon segment does not change amplitudes except for a fixed normalization. The word symmetric means that the two natural maps $A\to A^\vee$ obtained from εm agree; this is the categorical form of a nondegenerate two-point pairing compatible with orientation reversal. For a local oriented bosonic full CFT one also imposes the commutativity condition

$$m\circ c_{A,A}=m$$

when the algebra describes a local extension of the chiral theory. Boundary conditions for this full CFT are left A-modules

$$\rho_M: A\otimes M\to M,\qquad \rho_M(m\otimes \mathrm{id}_M)=\rho_M(\mathrm{id}_A\otimes \rho_M)\alpha_{A,A,M},\qquad \rho_M(\eta\otimes \mathrm{id}_M)=\mathrm{id}_M\,.$$

The Frobenius identity is the algebraic form of the elementary cutting move, not an abstract decoration. In the ordinary finite model $A=\mathrm{Mat}_d(\mathbb{C})$, take matrix units E_{ij}, ordinary multiplication m, and trace counit $\varepsilon(X)=\operatorname{Tr}X$. The coproduct dual to multiplication with respect to $\varepsilon(XY)$ is

$$\Delta(E_{ij})=\sum_{r=1}^{d}E_{ir}\otimes E_{rj}. \tag{69.65}$$

Then, on $E_{ij}\otimes E_{k\ell}$,

$$(m\otimes \mathrm{id})(\mathrm{id}\otimes\Delta)=\Delta m=(\mathrm{id}\otimes m)(\Delta\otimes \mathrm{id})=\delta_{jk}\sum_{r=1}^{d}E_{ir}\otimes E_{r\ell}, \tag{69.66}$$

and

$$m\Delta(E_{ij})=d\,E_{ij}. \tag{69.67}$$

Thus sliding an A-line through a multiplication junction is exactly the Frobenius identity, while a contractible A-loop contributes the fixed specialness scalar. The categorical ribbon construction replaces these matrix units by dual bases of morphism spaces and inserts associators, braidings, and twists; the local algebraic reason for genus-zero boundary sewing is already visible in (69.66).

This language converts the open sector into a finite-dimensional statement. For two boundary conditions M,N, the multiplicity of chiral type i in the open Hilbert space is

$${n_{MN}}^{i}=\dim\mathrm{Hom}_A(M\otimes U_i,N),\qquad \mathcal{H}_{MN}=\bigoplus_{i\in I}\mathrm{Hom}_A(M\otimes U_i,N)\otimes\mathcal{H}_i. \tag{69.68}$$

Here $M\otimes U_i$ is given the left A-action $\rho_M\otimes \mathrm{id}_{U_i}$, with the associator understood. Thus the annulus multiplicities are dimensions of spaces of A-linear morphisms, and integrality is no longer a separate miracle.

Boundary OPE associativity is likewise a module-theoretic composition law. Let

$$f \in \mathrm{Hom}_A(M \otimes U_i, N), \qquad g \in \mathrm{Hom}_A(N \otimes U_j, P).$$

Then

$$g \circ (f \otimes \mathrm{id}_{U_j}) \circ \alpha_{M,U_i,U_j} \in \mathrm{Hom}_A(M \otimes (U_i \otimes U_j), P). \tag{69.69}$$

Choose a basis of chiral fusion morphisms

$$e_{ij}^{k,\mu} : U_i \otimes U_j \to U_k, \qquad \bar{e}_{k,\mu}^{ij} : U_k \to U_i \otimes U_j, \qquad e_{ij}^{k,\mu} \bar{e}_{\ell,\nu}^{ij} = \delta_{k\ell}\delta_{\mu\nu}\, \mathrm{id}_{U_k}.$$

Inserting the resolution

$$\mathrm{id}_{U_i \otimes U_j} = \sum_{k,\mu} \bar{e}_{k,\mu}^{ij} e_{ij}^{k,\mu}$$

in (69.69) gives the boundary OPE coefficients in (69.24). Associativity of the boundary OPE is the statement that the two ways to compose three such A-linear morphisms agree after changing the parenthesization of $U_i \otimes U_j \otimes U_k$. The only nontrivial transformation between the two bases is the chiral fusing isomorphism, so the module associativity identity above becomes the Cardy–Lewellen equation (69.25). The pentagon for α is the algebraic reason that the five-boundary-field sewing diagrams agree.

The diagonal Cardy solution is the special case $A = \mathbf{1}$. A left $\mathbf{1}$-module is just an object of C, and the elementary modules are $M = U_a$. Equation (69.68) reduces to

$${n_{ab}}^i = \dim \mathrm{Hom}(U_a \otimes U_i, U_b) = {N_{ai}}^b,$$

which is the boundary-field multiplicity (69.26) after the same orientation convention used in the annulus. The fusing-matrix expression for the Cardy boundary three-point constants is precisely the expansion of (69.69) in the basis $\{e_{ij}^{k,\mu}\}$.

The annulus coefficients in a non-diagonal rational boundary theory carry a stronger constraint than integrality. For simple A-modules M, N, form the matrices

$${(n_i)_M}^N = \dim \mathrm{Hom}_A(M \otimes U_i, N).$$

Taking the graded trace of a boundary interval after inserting two successive chiral labels i, j can be computed either by first summing over an intermediate boundary module N, or by fusing $U_i \otimes U_j$ in the chiral category. Module associativity therefore gives the annulus representation identity

$$\sum_N {(n_i)_M}^N {(n_j)_N}^P = \sum_k {N_{ij}}^k {(n_k)_M}^P. \tag{69.70}$$

This is the finite algebra behind the usual statement that annulus multiplicity matrices form a nonnegative-integer matrix representation of the fusion ring. It is not a separate classification theorem: it is the decategorified shadow of the same A-module composition law that produces (69.69).

Bulk fields in the same algebraic construction are bimodule morphisms. If $U_i^+ \otimes A \otimes U_j^-$ denotes the A-bimodule obtained by using the braiding for the left chiral label and the inverse braiding for the right chiral label, the closed bulk multiplicity is

$$Z_{ij}(A) = \dim \mathrm{Hom}_{A|A}(U_i^+ \otimes A \otimes U_j^-, A). \tag{69.71}$$

The diagonal invariant is again recovered from $A = \mathbf{1}$, for which the bimodule morphism space is one-dimensional only when $j = i^\vee$. The bulk-boundary OPE is obtained by evaluating such a bimodule morphism on the boundary module M and then projecting to $\mathrm{End}_A(M)$. This is the categorical form of the classifying algebra mentioned above: disk one-point functions are simultaneous eigenvalues of the commutative algebra of bulk fields acting on elementary boundary modules. Formula (69.28) is the $A = \mathbf{1}$ version of this statement.

Finite algebra shadow of the classifying algebra. The preceding categorical sentence has a simple finite algebra model which keeps the logic visible in non-diagonal examples. Let

$$A_{\mathrm{fin}} = \bigoplus_{r=1}^{R} \mathrm{Mat}_{d_r}(\mathbb{C})$$

be an ordinary finite-dimensional semisimple algebra, regarded as the decategorified shadow of a special symmetric Frobenius algebra object and its module category. Its center is

$$Z(A_{\mathrm{fin}}) = \bigoplus_{r=1}^{R} \mathbb{C}\, e_r, \qquad e_r e_s = \delta_{rs} e_r, \qquad \sum_{r=1}^{R} e_r = 1.$$

The simple left A_{fin}-modules are the column modules $M_r = \mathbb{C}^{d_r}$. A central element $z = \sum_r z_r e_r$ acts on M_r by the scalar z_r, and therefore defines a character

$$\chi_r : Z(A_{\mathrm{fin}}) \to \mathbb{C}, \qquad \chi_r(z) = z_r. \tag{69.72}$$

The disk identity-channel sewing equation is, in this finite model, exactly the multiplicativity of this central character:

$$\chi_r(zz') = \chi_r(z)\chi_r(z'), \qquad z, z' \in Z(A_{\mathrm{fin}}). \tag{69.73}$$

This is the non-diagonal analogue of (69.31): the bulk field that survives in the boundary identity channel is not a raw element of the fusion ring but its image in the commutative classifying algebra acting on the elementary boundary module.

Two cautions are part of the statement. First, a noncentral element of A_{fin} does not act by a scalar on M_r, so it cannot be read as a disk one-point coefficient of the identity boundary field. Its boundary image is an endomorphism-valued insertion. Second, if one replaces the elementary boundary condition M_r by a direct sum $M = \oplus_r M_r^{\oplus n_r}$, the boundary identity algebra becomes $\mathrm{End}_{A_{\mathrm{fin}}}(M) \simeq \oplus_r \mathrm{Mat}_{n_r}(\mathbb{C})$. The scalar character χ_r is replaced by a matrix-valued action, or by a chosen trace after adding Chan–Paton data. Such a trace is additive over summands but is not a multiplicative character unless the boundary module is simple. Thus Cardy–Lewellen disk sewing distinguishes elementary boundary conditions from finite direct sums even before any analytic issue about convergence of bordered-surface correlators enters.

A small pointed module example shows how this differs from the diagonal Cardy case. The next group of formulas should be read as one sewing laboratory, not as independent finite identities. The annulus action records how chiral labels move boundary conditions; its Fourier diagonalization records the classifying data visible to annuli; the boundary-field composition records Cardy–Lewellen associativity; the stabilizer idempotents record the bulk-boundary classifying sectors invisible to the annulus quotient; and the Frobenius pairing records the two-point normalization that detects those sectors. A full rational construction must realize all five pieces by the same A-module category and the same analytic sewing maps.

Let the chiral fusion skeleton have labels $G = \mathbb{Z}_2 \times \mathbb{Z}_2$, with $U_g \otimes U_h = U_{g+h}$, and take the subgroup $H = \{(0,0),(1,0)\}$. Boundary labels are the two cosets G/H. The open multiplicity matrix attached to $g = (g_1, g_2)$ is

$$(n_g)_{xH}{}^{yH} = \begin{cases} 1, & yH = (x+g)+H, \\ 0, & \text{otherwise.} \end{cases} \tag{69.74}$$

Thus $n_{(0,0)} = n_{(1,0)} = \mathbf{1}$, while $n_{(0,1)} = n_{(1,1)}$ swaps the two boundary labels. Equation (69.70) is then just the coset action law $n_g n_h = n_{g+h}$. The example is finite and algebraic; realizing it as a unitary local rational BCFT requires the corresponding modular, Frobenius, and analytic sewing data in addition to the displayed matrices.

The annulus matrices in this cell also display the modular diagonalization that replaces the diagonal Cardy formula. Put $Q = G/H \cong \mathbb{Z}_2$. For a character $\eta \in \widehat{Q}$, define the Fourier projector

$$P_\eta = \frac{1}{|Q|} \sum_{\bar{g} \in Q} \eta(\bar{g})^{-1} n_{\bar{g}}. \tag{69.75}$$

These projectors satisfy

$$P_\eta P_{\eta'} = \delta_{\eta,\eta'} P_\eta, \qquad \sum_{\eta \in \widehat{Q}} P_\eta = \mathbf{1},$$

and the spectral resolution of the annulus action is

$$n_g = \sum_{\eta \in \widehat{Q}} \eta(g+H)\, P_\eta. \tag{69.76}$$

Equivalently, with the normalized boundary Fourier matrix

$$\Psi_{xH,\eta} = |Q|^{-1/2} \eta(xH),$$

one has

$$(n_g)_{xH}{}^{yH} = \sum_{\eta \in \widehat{Q}} \Psi_{xH,\eta}\, \eta(g+H)\, \overline{\Psi_{yH,\eta}}. \tag{69.77}$$

This is the finite version of the rational BCFT statement that annulus nimreps are diagonalized by boundary classifying data, with eigenvalues given by fusion-ring characters. It also records a limitation of annulus spectra: if $h \in H$, then $h + H = H$, so n_h has the same eigenvalues and the same matrix as the vacuum. The stabilizer label is nevertheless visible in boundary endomorphism fields and in the mixed projector identity below. Thus annulus modular diagonalization sees the quotient action G/H, while full Cardy–Lewellen sewing still needs the stabilizer endomorphism and classifying-idempotent data.

The same example also shows the boundary-field composition behind the annulus identity. Let $x \in G/H$, and let $\psi_{x,g}$ denote the one-dimensional generator of the open sector with chiral label g from the boundary xH to the boundary $(x+g)H$. For $g, h \in G$ the pointed fusion space is one-dimensional, so the module-composition coordinate can be normalized as

$$\psi_{x+g,h} \circ \psi_{x,g} = \psi_{x,g+h}. \tag{69.78}$$

Then

$$(\psi_{x+g+h,k} \circ \psi_{x+g,h}) \circ \psi_{x,g} = \psi_{x,g+h+k} = \psi_{x+g,h+k} \circ (\psi_{x+g,h} \circ \psi_{x,g}),$$

which is the Cardy–Lewellen boundary associativity equation in this finite module cell. The stabilizer H is still visible: both $\psi_{x,(0,0)}$ and $\psi_{x,(1,0)}$ are boundary endomorphism fields of the same boundary label, but they carry different chiral labels. Thus the non-diagonal boundary theory is not obtained by merely identifying chiral labels that act on the same coset; the open sector remembers the chiral fusion label even when the boundary endpoint is unchanged.

The mixed bulk-boundary sewing shadow is visible in the same finite cell. At a fixed boundary xH, the stabilizer fields $\{\psi_{x,h}\}_{h\in H}$ form the group algebra of H. For a character $\chi \in \widehat{H}$, define the primitive stabilizer idempotent

$$e_{x,\chi} = \frac{1}{|H|} \sum_{h\in H} \chi(h)^{-1} \psi_{x,h}. \tag{69.79}$$

Then

$$e_{x,\chi} e_{x,\chi'} = \delta_{\chi,\chi'} e_{x,\chi}, \qquad \sum_{\chi\in\widehat{H}} e_{x,\chi} = \psi_{x,(0,0)}. \tag{69.80}$$

The compatibility of this classifying projector with a boundary-changing field is the slide identity

$$\psi_{x,g} \circ e_{x,\chi} = e_{x+g,\chi} \circ \psi_{x,g}. \tag{69.81}$$

Using (69.78), both sides are

$$\frac{1}{|H|} \sum_{h\in H} \chi(h)^{-1} \psi_{x,g+h},$$

where G is abelian. This is a finite mixed Cardy–Lewellen sewing cell: projecting to a stabilizer classifying sector and then changing the boundary condition gives the same amplitude as first changing the boundary condition and then using the transported projector at the new endpoint. It is still only the algebraic shadow of bulk-boundary sewing; the analytic rational BCFT construction must supply the full center object, correlators, and mapping-class-compatible sewing maps.

The same finite cell also displays the boundary two-point normalization that is lost if the stabilizer labels are collapsed. Equip the stabilizer group algebra at the boundary xH with the Frobenius trace

$$\varepsilon_x(\psi_{x,h}) = \delta_{h,0}.$$

Then

$$\varepsilon_x(e_{x,\chi} e_{x,\chi'}) = \frac{\delta_{\chi,\chi'}}{|H|}. \tag{69.82}$$

Thus $|H|^{1/2} e_{x,\chi}$ are orthonormal classifying sectors for this finite pairing. For a boundary-changing label g, define the projected sector field

$$\psi_{x,g;\chi} := \psi_{x,g} e_{x,\chi} = e_{x+g,\chi} \psi_{x,g}.$$

Since every element of $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ is its own inverse,

$$\psi_{x+g,g;\chi'} \circ \psi_{x,g;\chi} = \delta_{\chi,\chi'} e_{x,\chi}. \tag{69.83}$$

Applying ε_x gives the two-point pairing

$$\varepsilon_x(\psi_{x+g,g;\chi'}\psi_{x,g;\chi}) = \frac{\delta_{\chi,\chi'}}{|H|}.$$

The stabilizer character is therefore a genuine finite sewing coordinate: it is invisible to the annulus quotient G/H, but it is detected by boundary endomorphism pairings and by inverse boundary-changing fields.

The point of the finite laboratory is not the separate existence of the annulus matrices, Fourier projectors, boundary products, stabilizer idempotents, and two-point pairings. They are five shadows of one module datum. For each $g \in G$,

$$(n_g)_{xH,yH} = \dim \operatorname{span}\{\psi_{x,g} \mid (x+g)H = yH\}. \tag{69.84}$$

Thus the open annulus records the graded dimension of exactly the boundary-changing sector whose composition law is (69.78). The quotient Fourier projectors diagonalize only the G/H-motion of the endpoints, whereas the stabilizer Fourier inversion

$$\psi_{x,h} = \sum_{\chi \in \widehat{H}} \chi(h)\, e_{x,\chi}, \qquad h \in H, \tag{69.85}$$

recovers the endpoint-preserving boundary endomorphism fields from the classifying idempotents. Equations (69.84) and (69.85) are the finite end-to-end check that the annulus, disk classifying sector, boundary OPE, and two-point pairing are compatible projections of one rational boundary coordinate. A full rational BCFT construction replaces this finite group laboratory by an A-module category in a chiral tensor category and then adds the analytic sewing maps; the finite formulas above should be read as one dependency diagram, not as independent evidence for the all-surface theorem.

Annulus shadows versus sewing data. In the pointed G/H laboratory, the endpoint annulus matrices do not determine the endpoint-preserving boundary algebra, its classifying idempotents, or its two-point sewing pairing. More precisely, the annulus shadow

$$n_{(0,0)} = n_{(1,0)} = \mathbf{1}$$

is compatible with the semisimple stabilizer algebra

$$E_{\mathrm{ss}} = \mathbb{C}\{1, \tau\}, \qquad \tau^2 = 1,$$

which has primitive idempotents

$$e_\pm = \frac{1}{2}(1 \pm \tau),$$

but the same endpoint-dimension shadow is also compatible, as a graded vector space, with the nilpotent algebra

$$E_{\mathrm{nil}} = \mathbb{C}\{1, \epsilon\}, \qquad \epsilon^2 = 0.$$

The latter has no nontrivial idempotent decomposition and the coefficient-of-1 two-point pairing is degenerate on the ϵ sector. Hence the annulus endpoint spectrum by itself is not enough data for Cardy–Lewellen sewing; the boundary OPE multiplication, the classifying idempotents, and the Frobenius two-point functional are additional structure.

The finite verification is direct. Both algebras have a one-dimensional vacuum-labeled endomorphism sector and a one-dimensional stabilizer-labeled endomorphism sector at the same boundary endpoint. Therefore the annulus matrix which records only endpoint multiplicity is the identity matrix for both labels 1 and the nontrivial stabilizer label.

In $E_{\rm ss}$,

$$e_\pm^2 = e_\pm, \qquad e_+e_- = 0, \qquad e_+ + e_- = 1.$$

With $\varepsilon_{\rm ss}(1) = 1$ and $\varepsilon_{\rm ss}(\tau) = 0$, the pairing $(x, y) \mapsto \varepsilon_{\rm ss}(xy)$ satisfies

$$\varepsilon_{\rm ss}(e_\pm e_\pm) = \frac{1}{2}, \qquad \varepsilon_{\rm ss}(e_+e_-) = 0.$$

This is the finite classifying-sector splitting used above.

In $E_{\rm nil}$, an idempotent $p = a + b\epsilon$ satisfies

$$a^2 = a, \qquad 2ab = b.$$

The first equation gives $a = 0$ or $a = 1$, and then the second gives $b = 0$. Thus the only idempotents are 0 and 1. With the same annulus-normalized coefficient-of-1 functional $\varepsilon_{\rm nil}(1) = 1$, $\varepsilon_{\rm nil}(\epsilon) = 0$, the pairing obeys

$$\varepsilon_{\rm nil}(\epsilon\epsilon) = 0,$$

so the stabilizer sector has no inverse two-point partner in that pairing. The two finite systems therefore have the same endpoint annulus shadow but different boundary OPE and classifying-sector data. The semisimple rational construction excludes the nilpotent alternative by its Frobenius-module identities, while positivity supplies the corresponding two-point nondegeneracy. The example shows why these inputs are not recoverable from annulus spectra alone.

The disk partition function defines the Affleck–Ludwig boundary entropy. For a Cardy boundary state in a diagonal rational CFT,

$$g_a = \langle 0|a\rangle = \frac{S_{a0}}{\sqrt{S_{00}}}, \tag{69.86}$$

with the vacuum Ishibashi state normalized as in (69.15).

Hypothesis 69.12 (Boundary entropy gradient input)**.** Let a unitary two-dimensional QFT on a half-cylinder have a fixed conformal bulk and a renormalized finite-dimensional chart of boundary couplings λ^a. Let $B^a(\lambda)$ be the boundary RG vector field and $\phi_a(\tau)$ the renormalized boundary fields conjugate to λ^a. Assume that the boundary trace of the stress tensor has the renormalized form

$$\theta(\tau) = B^a(\lambda)\phi_a(\tau) + \partial_\tau j(\tau) \tag{69.87}$$

and quotient the coupling chart by redundant directions generated by total boundary derivatives. Let $z(\lambda, L)$ be the boundary contribution to the cylinder partition function after subtracting the bulk extensive term and the boundary ground-state energy, and define

$$s(\lambda, L) = \left(1 - L\frac{\partial}{\partial L}\right) \log z(\lambda, L). \tag{69.88}$$

Assume the local contact counterterms can be fixed compatibly with boundary translation invariance, reflection positivity, and the subtraction in (69.88), so that the finite Ward susceptibility identity

$$\partial_a s(\lambda, L) = -\frac{1}{2}\int_0^L \mathrm{d}\tau_1 \int_0^L \mathrm{d}\tau_2\, K_L(\tau_1 - \tau_2)\big\langle \phi_a(\tau_1)\theta(\tau_2)\big\rangle_c^{\rm ren}, \qquad K_L(\tau) = 1 - \cos\frac{2\pi\tau}{L} \tag{69.89}$$

holds on the quotient chart. Assume also that the connected boundary two-point functions define the positive symmetric bilinear form

$$G_{ab}(\lambda) = \frac{1}{2}\int_0^L \mathrm{d}\tau_1 \int_0^L \mathrm{d}\tau_2 \left[1 - \cos\!\left(\frac{2\pi(\tau_1 - \tau_2)}{L}\right)\right] \left\langle \phi_a(\tau_1)\phi_b(\tau_2)\right\rangle_c^{\mathrm{ren}} \tag{69.90}$$

on the quotient by redundant total-derivative directions.

The positivity in (69.90) comes from the spectral representation of the boundary two-point function. The finite-regulator mechanism is as follows. Let $\Phi = u^a\phi_a$ be a real boundary perturbing field after subtracting its one-point function, and first place the boundary theory on a finite half-cylinder with discrete positive-energy boundary Hamiltonian. Grouping matrix elements by a positive energy gap $\Delta_r > 0$, reflection positivity and the KMS trace give the connected Euclidean boundary two-point function in the form

$$\langle \Phi(\tau)\Phi(0)\rangle_c = \sum_r A_r \frac{\mathrm{e}^{-\Delta_r\tau} + \mathrm{e}^{-\Delta_r(L-\tau)}}{1 - \mathrm{e}^{-L\Delta_r}}, \qquad A_r \geq 0, \qquad 0 < \tau < L, \tag{69.91}$$

where the connected trace has removed the zero-gap diagonal part. Put $\kappa = 2\pi/L$. For a single gap,

$$\int_0^L \mathrm{d}\tau \left(1 - \cos\kappa\tau\right) \frac{\mathrm{e}^{-\Delta\tau} + \mathrm{e}^{-\Delta(L-\tau)}}{1 - \mathrm{e}^{-L\Delta}} = \frac{2\kappa^2}{\Delta(\Delta^2 + \kappa^2)}. \tag{69.92}$$

Thus the quadratic form obtained from (69.90) is a sum of nonnegative matrix-element weights

$$L\sum_r A_r \frac{2\kappa^2}{\Delta_r(\Delta_r^2 + \kappa^2)}.$$

This calculation explains the shape of the trigonometric kernel in the gradient formula: after the connected trace removes the zero-gap diagonal part, the kernel assigns a positive finite weight to each nonzero boundary excitation. In the continuum theorem, the hard analytic work is to renormalize the coincident boundary singularities, preserve the boundary Ward identity (69.87), and quotient total derivative directions so that this finite spectral positivity survives in the renormalized coupling chart.

Mechanism of the gradient formula. Hypothesis 69.12 is the analytic input behind the finite proposition. It identifies a particular renormalized susceptibility with the differential of the finite boundary entropy. We spell out the mechanism because the contact-term prescription, the quotient by redundant boundary directions, and reflection positivity are part of the mathematical content of the statement.

Use the sign convention in which the renormalized boundary action is deformed by

$$S_\partial(\lambda) = S_\partial(0) + \lambda^a \int_0^L \phi_a(\tau)\,\mathrm{d}\tau.$$

Then differentiation of the renormalized boundary partition function gives

$$\partial_a \log z(\lambda, L) = -\int_0^L \left\langle \phi_a(\tau)\right\rangle_{\lambda,L}^{\mathrm{ren}}\,\mathrm{d}\tau. \tag{69.93}$$

The scale Ward identity on the half-cylinder states that, after the bulk extensive term and the boundary ground-state energy have been subtracted, the operation $L\partial_L$ is represented by the renormalized insertion of the boundary trace

$$\Theta = \int_0^L \theta(\tau)\, \mathrm{d}\tau$$

together with local contact counterterms supported at coincident boundary points. Thus $(1 - L\partial_L)\partial_a \log z$ contains the integrated two-point insertion $\int\int \langle \phi_a(\tau_1)\theta(\tau_2)\rangle_c$ together with local counterterms. Those contact terms must be fixed by a renormalization prescription compatible with boundary translation invariance, reflection positivity, and the subtraction defining $s = (1 - L\partial_L) \log z$. Equation (69.89) is the real analytic step in the gradient formula: the kernel K_L is the circle kernel, in this normalization, that implements the finite entropy subtraction and removes the ambiguous zero-mode and total-derivative pieces after passing to the quotient of boundary coupling space. With weaker analytic control the displayed susceptibility records the expected Ward-identity structure rather than a proved gradient formula.

Proposition 69.13 (Boundary entropy monotonicity from the gradient input)**.** *Under Hypothesis 69.12,*

$$\partial_a s(\lambda, L) = -\, G_{ab}(\lambda)B^b(\lambda). \tag{69.94}$$

Consequently, along an RG trajectory with $t = \log L$ and $\dot\lambda^a = B^a(\lambda)$,

$$\frac{d}{dt} s(\lambda(t), L) = B^a\partial_a s = -\, B^a G_{ab} B^b \leq 0. \tag{69.95}$$

If the trajectory has conformal endpoints and the finite boundary entropy has endpoint limits, then $g_{\mathrm{UV}} \geq g_{\mathrm{IR}}$, where $g = \exp s$ at a boundary fixed point.

Proof. Substitute the trace identity (69.87) into the Ward susceptibility (69.89). At fixed λ,

$$\big\langle \phi_a(\tau_1)\theta(\tau_2)\big\rangle_c^{\mathrm{ren}} = B^b \big\langle \phi_a(\tau_1)\phi_b(\tau_2)\big\rangle_c^{\mathrm{ren}} + \partial_{\tau_2} \big\langle \phi_a(\tau_1) j(\tau_2)\big\rangle_c^{\mathrm{ren}}.$$

The total-derivative term is a redundant direction; on the quotient in Hypothesis 69.12 it vanishes after integration against K_L and after the prescribed contact subtractions. The remaining term is exactly $-G_{ab}B^b$ by (69.90), proving (69.94). Contracting with the RG vector field gives (69.95), and positivity of G gives the inequality. At conformal endpoints $B = 0$ and $g = \exp s$; integrating the monotone inequality from the ultraviolet endpoint to the infrared endpoint gives $g_{\mathrm{UV}} \geq g_{\mathrm{IR}}$. □

This conclusion gives an ordering constraint on boundary fixed points reached by a given unitary boundary RG flow. It does not determine which conformal boundary condition is reached, nor does it supply the annulus or Cardy–Lewellen sewing data.

Chapter-scale proof-status ledger. The examples in this chapter should now be read as a dependency chain, not as a collection of unrelated BCFT cells. A genuine construction of a boundary conformal field theory has to pass through the following layers.

1. The closed chiral input must be fixed first: either a rational semisimple chiral theory with modular functor and fusing data, or a nonrational direct-integral theory with declared test spaces and Plancherel measure.

2. Boundary labels must then be promoted to open-channel Hilbert spaces or spectral measures, with common domains for boundary-condition-changing fields and a reflection-positive two-point pairing.

3. Local coordinates must be compatible: annulus multiplicities or spectral measures, disk classifying coordinates, boundary OPE products, bulk-boundary maps, and two-point pairings must be projections of the same boundary structure.

4. Local moves must be checked in the same normalization. In the rational case these include open/closed annulus exchange, boundary OPE associativity, disk classifying multiplication, and Frobenius/module cutting moves. In the nonrational case they also include contour choices, residue summands, direct-integral convergence, and determinant-line bookkeeping.

5. All-surface sewing is the final analytic step: the local moves must generate the changes of bordered-surface decomposition, their defects must telescope as in Proposition 69.2, and closed-loop transports must either be trivialized as in Proposition 69.4 or kept as honest anomaly-line transports.

The diagonal Ising example closes the first few rational finite checks for a specific modular tensor category and boundary basis. The pointed G/H laboratory goes further by showing that annulus matrices, quotient Fourier data, stabilizer idempotents, boundary OPEs, and two-point pairings are compatible shadows of one module-category mechanism. The Liouville block closes only local nonrational coordinates: hyperbolic FZZT/ZZ identities, the even-parameter Plancherel quotient, and pole-crossing residue bookkeeping. None of these examples proves the final generated all-surface sewing theorem. Conversely, the boundary-entropy gradient result uses renormalized boundary RG and positivity data once a unitary boundary QFT exists; it is not a replacement for constructing the BCFT sewing maps or for identifying the infrared boundary condition.

Finally, two-dimensional BCFT fits into the broader boundary/defect language of Chapter 120. A topological line defect can end on a boundary only when its endpoint is a specified boundary condition-changing operator. Noninvertible defects, such as the Ising Kramers–Wannier line, act on the set of Cardy boundary conditions by fusion when the required endpoint operators exist. Boundary anomalies are likewise not optional decorations: a boundary partition function may be a section of an anomaly line, and anomaly cancellation is part of the boundary datum. The present chapter supplies the Virasoro and modular-sewing machinery special to two-dimensional conformal boundaries; the global categorical defect viewpoint is developed in Volume IX.

Open Problem 69.14 (BCFT sewing beyond rational diagonal examples). Construct a single analytic framework that includes rational Cardy boundaries, compact-boson continuous families, the Liouville FZZT and ZZ states of Section 69.8, boundary-condition-changing operators, and defect endpoints, with convergent sewing maps for bordered Riemann surfaces. The framework must realize the test-space, open spectral-measure, operator-domain, pole-prescription, and determinant-line requirements stated above, and must prove that the rational finite-sum construction and the Liouville direct-integral construction are two limits of the same sewing notion rather than two unrelated calculational recipes. In particular, closed-loop sewing phases must be identified as a trivializable determinant-line cocycle, or the resulting amplitudes must be kept as line-valued rather than scalar correlators.

Chapter 70

Two-Dimensional Superconformal Algebras

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$, with radial quantization invoked where stated.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the radial-quantization vacuum when a Hilbert space is used.
- **Generator convention.** represented symmetry transformations use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$; differential conformal generators are defined with the sign displayed in the relevant formula.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

70.1 Spin and Sector Data

The superconformal algebras in two dimensions are chiral symmetry algebras. They are not, by themselves, definitions of a continuum quantum field theory. The underlying QFT must supply local operator domains, correlation functions, positivity when the theory is unitary, and sewing data. This chapter isolates the chiral algebraic layer because it is shared infrastructure for two-dimensional CFT. The dynamical construction of supersymmetric field theories, including Landau–Ginzburg and gauged linear sigma models, belongs to the supersymmetric-QFT development in Volume VII. The statements below are therefore algebraic consequences and consistency tests for any such model, not a replacement for the supersymmetric RG analysis.

Definition 70.1 (Spin two-dimensional CFT structure)**.** A spin two-dimensional CFT structure consists of the following items.

1. An oriented two-dimensional Euclidean CFT $\mathcal{C}$, with holomorphic stress tensor $T(z)$, antiholomorphic stress tensor $\bar{T}(\bar{z})$, and central charges $c, \bar{c}$.
2. For each spin surface $(\Sigma, \mathfrak{s})$, correlation functions of local fields which may be even or odd under the fermion-parity operator $(-1)^F$.

3. Chiral state spaces on the circle decomposed into Neveu–Schwarz and Ramond sectors,

$$\mathcal{H}^{\mathrm{ch}} = \mathcal{H}^{\mathrm{ch}}_{\mathrm{NS}} \oplus \mathcal{H}^{\mathrm{ch}}_{\mathrm{R}},$$

   where a chiral fermionic field has antiperiodic boundary condition in the NS sector and periodic boundary condition in the R sector.

4. A locality rule: even fields have ordinary single-valued OPEs with all local fields, while odd chiral fields are mutually local only after the spin structure and fermion-parity signs have been specified.

The same chiral algebra may occur in several full spin CFTs, because the full theory also requires an antiholomorphic sector, a choice of spin-structure sum or projection, and modular/sewing data.

Let $a(z)$ be a chiral field of holomorphic weight h_a. In a sector where a has monodromy $e^{-2\pi i \nu_a}$ around the spatial circle, we write

$$a(z) = \sum_{r \in \mathbb{Z}+\nu_a} a_r z^{-r-h_a}. \tag{70.1}$$

For the supercurrent of an $\mathcal{N}=1$ or $\mathcal{N}=2$ superconformal algebra, $h_a = 3/2$, $\nu_a = 1/2$ in the NS sector, and $\nu_a = 0$ in the R sector. The Ramond sector therefore contains supercharge zero modes. This is the operator form of the difference between antiperiodic and periodic spin structures on the cylinder.

70.2 $\mathcal{N}=1$ Superconformal Algebra

Definition 70.2 ($\mathcal{N}=1$ chiral superconformal algebra)**.** An $\mathcal{N}=1$ chiral superconformal algebra of central charge c consists of a chiral stress tensor $T(z)$, an odd chiral field $G(z)$ of weight $3/2$, and a sector choice

$$r \in \mathbb{Z} + \frac{1}{2} \quad (\mathrm{NS}), \qquad r \in \mathbb{Z} \quad (\mathrm{R})$$

for the modes

$$T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}, \qquad G(z) = \sum_r G_r z^{-r-3/2}.$$

The singular OPEs are

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \tag{70.2}$$

$$T(z)G(w) \sim \frac{(3/2)G(w)}{(z-w)^2} + \frac{\partial G(w)}{z-w}, \tag{70.3}$$

$$G(z)G(w) \sim \frac{2c/3}{(z-w)^3} + \frac{2T(w)}{z-w}. \tag{70.4}$$

Lemma 70.3 (Mode algebra from the $\mathcal{N}=1$ OPEs)**.** *The modes in Definition 70.2 obey*

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3-m)\delta_{m+n,0}, \tag{70.5}$$

$$[L_m, G_r] = \left(\frac{m}{2} - r\right) G_{m+r}, \tag{70.6}$$

$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}. \tag{70.7}$$

Proof. The Virasoro bracket is the stress-tensor calculation of Lemma 60.8. For the second formula, write

$$[L_m, G_r] = \oint_0 \frac{\mathrm{d}w}{2\pi\mathrm{i}} w^{r+1/2} \oint_w \frac{\mathrm{d}z}{2\pi\mathrm{i}} z^{m+1} T(z)G(w).$$

Using (70.3), the inner residue is

$$(m+1)w^m \frac{3}{2} G(w) + w^{m+1}\partial G(w).$$

The outer residue of the first term is $\frac{3}{2}(m+1)G_{m+r}$. For the second term, integration by parts gives

$$\oint_0 \frac{\mathrm{d}w}{2\pi\mathrm{i}} w^{m+r+3/2}\partial G(w) = -\left(m+r+\frac{3}{2}\right)G_{m+r}.$$

The coefficient is therefore $m/2 - r$.

For the anticommutator, the odd fields require the supercommutator contour. The singular part (70.4) gives

$$\{G_r, G_s\} = \oint_0 \frac{\mathrm{d}w}{2\pi\mathrm{i}} w^{s+1/2} \operatorname{Res}_{z=w} z^{r+1/2} \left(\frac{2c/3}{(z-w)^3} + \frac{2T(w)}{z-w}\right).$$

The T-term gives $2L_{r+s}$. The cubic pole gives

$$\frac{2c}{3}\cdot\frac{1}{2}\left(r+\frac{1}{2}\right)\left(r-\frac{1}{2}\right)\delta_{r+s,0} = \frac{c}{3}\left(r^2-\frac{1}{4}\right)\delta_{r+s,0}.$$

□

Ramond zero-mode bound. In a unitary Ramond representation of the $\mathcal{N}=1$ algebra,

$$L_0 - \frac{c}{24} \geq 0. \tag{70.8}$$

A Ramond ground state is a state annihilated by G_0, equivalently a state with $L_0 = c/24$. Set $r = s = 0$ in (70.7). Then

$$\{G_0, G_0\} = 2L_0 - \frac{c}{12} = 2\left(L_0 - \frac{c}{24}\right).$$

If the representation is unitary, $G_0^\dagger = G_0$, so the expectation value of $\{G_0, G_0\} = 2G_0^2$ on any vector is nonnegative. Equality holds precisely on the kernel of G_0, because $\langle v, G_0^2 v\rangle = \|G_0 v\|^2$. Hence the bottom of a unitary Ramond module is bounded by $c/24$, and states at the bound are exactly the Ramond states annihilated by the zero mode.

70.3 $\mathcal{N}=2$ Superconformal Algebra

The $\mathcal{N}=2$ algebra adds a holomorphic $U(1)$ current. This extra current is the reason that spectral flow, chiral-primary bounds, and topological twists are algebraic operations rather than model-dependent accidents.

Definition 70.4 ($\mathcal{N}=2$ chiral superconformal algebra)**.** An $\mathcal{N}=2$ chiral superconformal algebra of central charge c consists of a chiral stress tensor $T(z)$, a weight-one even current $J(z)$, and two odd weight-3/2 chiral fields $G^+(z)$ and $G^-(z)$. Their modes are

$$J(z)=\sum_{n\in\mathbb{Z}} J_n z^{-n-1}, \qquad G^{\pm}(z)=\sum_r G_r^{\pm} z^{-r-3/2},$$

with $r\in\mathbb{Z}+1/2$ in the NS sector and $r\in\mathbb{Z}$ in the R sector. The singular OPEs, in addition to the Virasoro OPE for T, are

$$T(z)J(w)\sim\frac{J(w)}{(z-w)^2}+\frac{\partial J(w)}{z-w}, \tag{70.9}$$

$$T(z)G^{\pm}(w)\sim\frac{(3/2)G^{\pm}(w)}{(z-w)^2}+\frac{\partial G^{\pm}(w)}{z-w}, \tag{70.10}$$

$$J(z)J(w)\sim\frac{c/3}{(z-w)^2}, \tag{70.11}$$

$$J(z)G^{\pm}(w)\sim\pm\frac{G^{\pm}(w)}{z-w}, \tag{70.12}$$

$$G^+(z)G^-(w)\sim\frac{2c/3}{(z-w)^3}+\frac{2J(w)}{(z-w)^2}+\frac{2T(w)+\partial J(w)}{z-w}, \tag{70.13}$$

$$G^{\pm}(z)G^{\pm}(w)\sim 0. \tag{70.14}$$

Mode algebra for $\mathcal{N}=2$. The modes in Definition 70.4 obey

$$[L_m,J_n]=-nJ_{m+n}, \tag{70.15}$$

$$[J_m,J_n]=\frac{c}{3}m\delta_{m+n,0}, \tag{70.16}$$

$$[L_m,G_r^{\pm}]=\left(\frac{m}{2}-r\right)G^{\pm}_{m+r}, \tag{70.17}$$

$$[J_m,G_r^{\pm}]=\pm G^{\pm}_{m+r}, \tag{70.18}$$

$$\{G_r^+,G_s^-\}=2L_{r+s}+(r-s)J_{r+s}+\frac{c}{3}\left(r^2-\frac{1}{4}\right)\delta_{r+s,0}, \tag{70.19}$$

$$\{G_r^{\pm},G_s^{\pm}\}=0. \tag{70.20}$$

The formulas involving L_m are the primary-field residue calculation used in Lemma 70.3, with weights 1 and 3/2. The current commutator follows from the double-pole residue in (70.11):

$$[J_m,J_n]=\oint_0\frac{\mathrm{d}w}{2\pi\mathrm{i}}w^n\,\mathrm{Res}_{z=w}\,z^m\frac{c/3}{(z-w)^2}=\frac{c}{3}m\delta_{m+n,0}.$$

Equation (70.18) is the simple-pole residue of (70.12). For the mixed supercurrent anticommutator, the triple pole in (70.13) gives $\frac{c}{3}(r^2-1/4)\delta_{r+s,0}$, exactly as in the $\mathcal{N}=1$ calculation. The simple pole gives $2L_{r+s}$. The double pole and the derivative term combine as

$$2\left(r+\frac{1}{2}\right)J_{r+s}-(r+s+1)J_{r+s}=(r-s)J_{r+s}.$$

Spectral-flow algebra check. For any $\eta \in \frac{1}{2}\mathbb{Z}$, the assignment

$$L_n^{(\eta)} = L_n + \eta J_n + \frac{c}{6}\eta^2\delta_{n,0}, \tag{70.21}$$

$$J_n^{(\eta)} = J_n + \frac{c}{3}\eta\delta_{n,0}, \tag{70.22}$$

$$(G_r^{\pm})^{(\eta)} = G_{r\pm\eta}^{\pm} \tag{70.23}$$

preserves the $\mathcal{N} = 2$ mode algebra, with the same central charge c. Integer η preserves NS and R sectors separately; half-integer η exchanges them.

This is an automorphism of the displayed abstract mode algebra. Checking it is a finite residue calculation in (70.15)–(70.20). For example,

$$[J_m^{(\eta)}, J_n^{(\eta)}] = [J_m, J_n] = \frac{c}{3}m\delta_{m+n,0},$$

which is the same formula in the flowed generators because the central shifts commute. Also

$$[J_m^{(\eta)}, (G_r^{\pm})^{(\eta)}] = [J_m, G_{r\pm\eta}^{\pm}] = \pm G_{m+r\pm\eta}^{\pm} = \pm(G_{m+r}^{\pm})^{(\eta)}.$$

For the mixed supercurrent relation,

$$\{G_{r+\eta}^{+}, G_{s-\eta}^{-}\} = 2L_{r+s} + (r - s + 2\eta)J_{r+s} + \frac{c}{3}\left((r+\eta)^2 - \frac{1}{4}\right)\delta_{r+s,0}.$$

Substituting

$$L_{r+s} = L_{r+s}^{(\eta)} - \eta J_{r+s} - \frac{c}{6}\eta^2\delta_{r+s,0}, \qquad J_{r+s} = J_{r+s}^{(\eta)} - \frac{c}{3}\eta\delta_{r+s,0},$$

the J-coefficient becomes $r - s$, and the central coefficient reduces to $\frac{c}{3}(r^2 - 1/4)$. The remaining brackets are checked in the same residue algebra.

If a simultaneous L_0, J_0 eigenstate has eigenvalues h, q, then spectral flow sends them to

$$h^{(\eta)} = h + \eta q + \frac{c}{6}\eta^2, \qquad q^{(\eta)} = q + \frac{c}{3}\eta. \tag{70.24}$$

This formula is often the cleanest way to see why NS chiral primaries and Ramond ground states are the same representation-theoretic object viewed in different spin sectors.

Proposition 70.5 (NS unitarity bound and chiral primaries)**.** *Let v be a simultaneous L_0, J_0 eigenvector in the common domain of the four NS modes $G_{\pm 1/2}^{\pm}$ in a unitary NS representation, with*

$$L_0 v = hv, \qquad J_0 v = qv.$$

Then

$$h \geq \frac{|q|}{2}. \tag{70.25}$$

If $q \geq 0$, equality $h = q/2$ holds precisely when

$$G_{-1/2}^{+} v = 0, \qquad G_{1/2}^{-} v = 0. \tag{70.26}$$

If $q \leq 0$, equality $h = -q/2$ holds precisely when

$$G_{-1/2}^{-} v = 0, \qquad G_{1/2}^{+} v = 0. \tag{70.27}$$

In particular, if v is an NS superconformal primary, so all positive supercurrent modes annihilate it, the first equality condition reduces to $G_{-1/2}^{+} v = 0$ and the second to $G_{-1/2}^{-} v = 0$.

Proof. Unitarity gives $(G_r^{\pm})^{\dagger} = G_{-r}^{\mp}$ on the common invariant domain. With $r = 1/2$ and $s = -1/2$, (70.19) gives

$$\{G_{1/2}^{-}, G_{-1/2}^{+}\} = 2L_0 - J_0,$$

because the central term vanishes at $r = \pm 1/2$. Taking the expectation value in v gives the positive-operator identity

$$(2h - q)\|v\|^2 = \|G_{-1/2}^{+}v\|^2 + \|G_{1/2}^{-}v\|^2.$$

Similarly,

$$\{G_{1/2}^{+}, G_{-1/2}^{-}\} = 2L_0 + J_0$$

and hence

$$(2h + q)\|v\|^2 = \|G_{-1/2}^{-}v\|^2 + \|G_{1/2}^{+}v\|^2.$$

Both right-hand sides are nonnegative, so $2h \geq q$ and $2h \geq -q$. If $q \geq 0$, saturation of $h = q/2$ is equivalent to the vanishing of both summands in the first displayed identity. If $q \leq 0$, saturation of $h = -q/2$ is equivalent to the vanishing of both summands in the second identity. The primary-state reduction follows by imposing the positive-mode annihilation conditions $G_{1/2}^{\pm}v = 0$. □

NS–R relation for chiral primaries. If v is an NS chiral primary with $h = q/2$, then spectral flow by $\eta = -1/2$ maps it to a Ramond state with

$$h^{(-1/2)} = \frac{c}{24}. \tag{70.28}$$

Thus it saturates the Ramond ground-state energy. This is obtained by inserting $h = q/2$ and $\eta = -1/2$ into (70.24):

$$h^{(-1/2)} = \frac{q}{2} - \frac{q}{2} + \frac{c}{24} = \frac{c}{24}.$$

70.4 Extended $\mathcal{N} = 2$ Spectral-Flow Operators

The spectral-flow algebra check above is a statement about the abstract mode algebra. It does not say that spectral flow is implemented by a local operator in a particular CFT. That extra assertion needs a charge lattice and locality hypothesis. This distinction is small in notation and large in logic: the extended $\mathcal{N} = 2$ algebra is an additional chiral-algebra structure requiring more than the singular OPEs in Definition 70.4.

Put

$$\kappa = \frac{c}{3}.$$

In this section assume $\kappa > 0$ and assume that the $U(1)_R$ current exponentiates to a rank-one Heisenberg/lattice chiral algebra. Concretely, choose a chiral boson φ with

$$\varphi(z)\varphi(0) \sim -\log z, \qquad J(z) = \mathrm{i}\sqrt{\kappa}\,\partial\varphi(z). \tag{70.29}$$

Then $J(z)J(0) \sim \kappa z^{-2}$, matching (70.11). A field of R-charge q has a Heisenberg factor

$$\exp\left(\mathrm{i}\frac{q}{\sqrt{\kappa}}\varphi\right)$$

up to a neutral factor. The charge lattice of the actual CFT specifies which such exponentials exist and which are mutually local.

Definition 70.6 (Spectral-flow vertex)**.** Let $\eta \in \frac{1}{2}\mathbb{Z}$. A spectral-flow vertex of parameter η is a chiral field

$$X_\eta(z) =: \exp\left(\mathrm{i}\eta\sqrt{\kappa}\,\varphi(z)\right) : \tag{70.30}$$

provided the momentum $\eta\sqrt{\kappa}$ belongs to the chosen Heisenberg/lattice extension. It is a local chiral operator only when its monodromy with the fields of the chiral algebra is trivial. If its monodromy is spin-dependent, it is instead an operator between spin sectors.

Operator implementation of spectral flow. Assume the factorization above, and let

$$\mathcal{O}(z) =: \exp\left(\mathrm{i}\frac{q}{\sqrt{\kappa}}\varphi(z)\right) : \mathcal{O}_0(z)$$

where $\mathcal{O}_0$ is neutral under J. If X_η exists, then

$$X_\eta(z)\mathcal{O}(0) \sim z^{\eta q}\,\mathcal{O}^{(\eta)}(0) + \cdots \tag{70.31}$$

where $\mathcal{O}^{(\eta)}$ has

$$q^{(\eta)} = q + \eta\kappa, \qquad h^{(\eta)} = h + \eta q + \frac{\eta^2\kappa}{2}. \tag{70.32}$$

Since $\kappa = c/3$, this is exactly (70.24). The branch factor $z^{\eta q}$ is the locality test: integer powers give ordinary chiral locality, while half-integer powers signal the NS/R spin-sector change.

The OPE of bosonic exponentials gives

$$: e^{\mathrm{i}a\varphi(z)} :: e^{\mathrm{i}b\varphi(0)} :\sim z^{ab} : e^{\mathrm{i}(a+b)\varphi(0)} : + \cdots .$$

With $a = \eta\sqrt{\kappa}$ and $b = q/\sqrt{\kappa}$, the branch factor is $z^{\eta q}$ and the new charge is

$$\sqrt{\kappa}\left(\frac{q}{\sqrt{\kappa}} + \eta\sqrt{\kappa}\right) = q + \eta\kappa.$$

The Heisenberg part of the conformal weight changes by

$$\frac{(q+\eta\kappa)^2 - q^2}{2\kappa} = \eta q + \frac{\eta^2\kappa}{2}.$$

The neutral factor $\mathcal{O}_0$ is unchanged, so the same difference is the total weight shift.

For $\eta = \pm 1$, write

$$X^{\pm} = X_{\pm 1}.$$

These fields, when present as local chiral fields, are the integer spectral-flow generators of the extended $\mathcal{N} = 2$ algebra. They are obtained from the vacuum by spectral flow and are NS highest-weight states, hence

$$h(X^{\pm}) = \frac{c}{6}, \qquad q(X^{\pm}) = \pm\frac{c}{3}. \tag{70.33}$$

In a unitary NS representation X^+ is chiral and X^- is antichiral:

$$G^+_{-1/2}X^+ = 0, \qquad G^-_{-1/2}X^- = 0, \tag{70.34}$$

because (70.33) saturates the bound of Proposition 70.5. Define the opposite superdescendants

$$Y^+ = G^-_{-1/2}X^+, \qquad Y^- = G^+_{-1/2}X^-. \tag{70.35}$$

Then

$$h(Y^{\pm}) = \frac{c}{6} + \frac{1}{2}, \qquad q(Y^{\pm}) = \pm\left(\frac{c}{3} - 1\right). \tag{70.36}$$

These four fields $X^{\pm}, Y^{\pm}$, together with $T, J, G^{\pm}$, are the finite generator set usually meant by the extended $\mathcal{N} = 2$ chiral algebra.

The singular part of the spectral-flow-generator OPE is also fixed by the Heisenberg current. From (70.30),

$$X_{\eta}(z)X_{-\eta}(0) \sim z^{-\eta^2\kappa}\left[1 + \eta z J(0) + z^2\left(\frac{\eta}{2}\partial J(0) + \frac{\eta^2}{2} : JJ : (0)\right) + O(z^3)\right]. \tag{70.37}$$

The pole order, the coefficient of J, and the two second-order coefficients are therefore convention-sensitive consequences of (70.29), not independent structure constants.

Example 70.7 (Calabi–Yau-type charge lattice)**.** If $c = 3n$, then $\kappa = n$. The integer spectral-flow fields have

$$h(X^{\pm}) = \frac{n}{2}, \qquad q(X^{\pm}) = \pm n,$$

and the half-flow fields $X_{\pm 1/2}$, when present on the spin cover, have

$$h(X_{\pm 1/2}) = \frac{n}{8}, \qquad q(X_{\pm 1/2}) = \pm\frac{n}{2}. \tag{70.38}$$

In a large-radius Calabi–Yau sigma-model representative these are realized by the holomorphic top-form fermion product and the Ramond spin fields, respectively. The present chapter uses only the algebraic implication: their existence is a charge-lattice and spin-sector input whose weights and charges must be exactly those displayed above.

70.5 Elliptic Genus and Spectral-Flow Covariance

The elliptic genus is the protected genus-one observable that links the chiral algebraic data above to the supersymmetric models of Volume VII. Its definition requires a full spin CFT, not just a chiral algebra. We therefore state the hypotheses before using the trace.

Definition 70.8 (Elliptic genus setup)**.** Let $\mathcal{C}$ be a compact unitary $\mathcal{N} = (2, 2)$ spin CFT with discrete Ramond–Ramond Hilbert space

$$\mathcal{H}_{\mathrm{RR}} = \bigoplus_{h,\bar{h},q} \mathcal{H}_{h,\bar{h},q}$$

of finite multiplicities. Assume $c = \bar{c}$, a left $U(1)_R$ current J, a fermion parity $(-1)^F$, and convergence of the Ramond trace for $\operatorname{Im}\tau > 0$. The elliptic genus is

$$\varphi_{\mathcal{C}}(\tau, z) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{RR}}}(-1)^F y^{J_0} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - c/24}, \qquad y = \mathrm{e}^{2\pi \mathrm{i} z}, \quad q = \mathrm{e}^{2\pi\mathrm{i}\tau}, \quad \bar{q} = \mathrm{e}^{-2\pi\mathrm{i}\bar{\tau}}. \tag{70.39}$$

Proposition 70.9 (Right-moving localization)**.** *Under Definition 70.8, and assuming the usual trace-class cyclicity for the Ramond supertrace, the elliptic genus is independent of $\bar{\tau}$ and receives contributions only from right-moving Ramond ground states:*

$$\bar{L}_0 = \frac{c}{24}. \tag{70.40}$$

Proof. In the right Ramond sector the zero-mode relation is

$$\{\bar G_0^+, \bar G_0^-\} = 2\left(\bar L_0 - \frac{c}{24}\right). \tag{70.41}$$

Differentiating (70.39) with respect to $\bar\tau$ inserts the operator $\bar L_0 - c/24$, up to the scalar factor $-2\pi\mathrm{i}$. By (70.41) this insertion is one half of the anticommutator $\{\bar G_0^+, \bar G_0^-\}$. The zero modes are odd and commute with the left-moving factors $y^{J_0}q^{L_0-c/24}$. The trace-class cyclicity hypothesis therefore gives

$$\mathrm{Str}_{\mathcal{H}_{\mathrm{RR}}}\, y^{J_0}q^{L_0-c/24}\bar q^{\bar L_0-c/24}\{\bar G_0^+, \bar G_0^-\} = 0,$$

because it is the supertrace of a supercommutator. Hence $\partial_{\bar\tau}\varphi_{\mathcal{C}} = 0$.

The same identity gives the state-by-state localization. On an eigenspace with $\bar E := \bar L_0 - c/24 > 0$, the odd operator

$$\bar Q_{\mathrm{R}} = \frac{\bar G_0^+ + \bar G_0^-}{\sqrt{2\bar E}}$$

is an involution after restricting to that finite-multiplicity eigenspace, because $\bar Q_{\mathrm{R}}^2 = 1$. It reverses right-moving fermion parity and preserves the left-moving quantum numbers appearing in the trace. Thus the positive-$\bar E$ eigenspaces cancel in the supertrace. The only unpaired states are the kernel of $\bar L_0 - c/24$, namely the right Ramond ground states. □

The elliptic variable z remembers left R-charge, so spectral flow imposes a finite functional equation. Put

$$m_{\mathrm{ell}} := \frac{c}{6}. \tag{70.42}$$

Assume that integer left spectral flow is implemented on the Ramond Hilbert space and that the chosen left R-charge lattice has a multiplier $\zeta(\mu) \in U(1)$ under $z \mapsto z+\mu$, meaning that $\mathrm{e}^{2\pi\mathrm{i}\mu J_0}$ acts by this scalar on the Ramond states contributing to the trace. Then

$$\varphi_{\mathcal{C}}(\tau, z+\lambda\tau+\mu) = \exp\left[-2\pi\mathrm{i} m_{\mathrm{ell}}\left(\lambda^2\tau + 2\lambda z\right)\right]\zeta(\mu)\varphi_{\mathcal{C}}(\tau, z), \tag{70.43}$$

for $\lambda, \mu \in \mathbb{Z}$. If all Ramond left R-charges in the trace are integral, the last multiplier is 1.

Here is the trace-level derivation of the λ-shift. Replacing z by $z+\lambda\tau$ multiplies each state by $q^{\lambda q_L}$. Under left spectral flow by λ,

$$h' = h + \lambda q_L + \frac{c}{6}\lambda^2, \qquad q_L' = q_L + \frac{c}{3}\lambda. \tag{70.44}$$

Thus

$$q^{h-c/24+\lambda q_L}y^{q_L} = q^{-c\lambda^2/6}y^{-c\lambda/3}q^{h'-c/24}y^{q_L'}. \tag{70.45}$$

Reindexing the trace by the spectral-flow bijection gives (70.43), since $m_{\mathrm{ell}} = c/6$. The μ-shift is exactly the assumed charge-lattice scalar $\zeta(\mu)$.

Hypothesis 70.10 (Spin modular covariance of the elliptic-genus trace)**.** In addition to Definition 70.8, assume that the Ramond trace with a flat left $U(1)_R$ background is supplied by the spin modular functor of the full CFT, and that the chosen R-symmetry background has no anomaly except the level-$c/3$ current-algebra determinant line displayed below. In the trace trivialization, assume that the charge-lattice multiplier is the scalar $\zeta(\mu)$ specified above and that, for

$$\gamma = \begin{pmatrix} a & b \\ c_\gamma & d \end{pmatrix} \in SL_2(\mathbb{Z}), \tag{70.46}$$

the modular change of spin elliptic curve and flat R-background acts by

$$\varphi_{\mathcal{C}}\left(\frac{a\tau+b}{c_\gamma\tau+d}, \frac{z}{c_\gamma\tau+d}\right) = \exp\left(2\pi i m_{\mathrm{ell}} \frac{c_\gamma z^2}{c_\gamma\tau+d}\right) \varphi_{\mathcal{C}}(\tau, z) \tag{70.47}$$

up to the corresponding spin and charge-lattice multiplier. This is the full-CFT sewing input. The left-moving mode algebra supplies the spectral-flow law (70.43), while modular transport of the spin theory on the torus is part of the full spin modular functor datum.

Under Hypothesis 70.10, the holomorphic function $\varphi_{\mathcal{C}}(\tau, z)$, together with (70.43) and (70.47), is a weak Jacobi form of weight 0 and index $m_{\mathrm{ell}} = c/6$, with the charge-lattice multiplier when the index is half-integral. This sentence unpacks the definition of a weak Jacobi form in the present normalization; the nontrivial CFT input is the spin modular covariance hypothesis, and the local algebra is the spectral-flow and determinant-line computation recorded here.

The mechanism behind the exponential factor in (70.47) is local. The global sewing input is the spin modular covariance of Hypothesis 70.10. The trace is a partition function on the elliptic curve

$$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$$

with a flat background for the left $U(1)_R$ current. In the trace trivialization the holonomy parameter is z, and a change of homology basis by

$$\gamma = \begin{pmatrix} a & b \\ c_\gamma & d \end{pmatrix} \in SL_2(\mathbb{Z})$$

sends the same flat background to the coordinate

$$z_\gamma = \frac{z}{c_\gamma\tau+d}.$$

The current algebra OPE fixes the anomaly coefficient:

$$J(z)J(0) \sim \frac{c/3}{z^2}, \qquad m_{\mathrm{ell}} = \frac{1}{2}\frac{c}{3} = \frac{c}{6}.$$

Thus the Ramond trace is most precisely a section of an index-m_{ell} anomaly determinant line over the moduli stack of elliptic curves with flat R-backgrounds. In the trace trivialization, the transition function of this line is precisely the quadratic factor

$$\exp\left(2\pi i m_{\mathrm{ell}} \frac{c_\gamma z^2}{c_\gamma\tau+d}\right).$$

The elliptic law (70.43) and the modular law (70.47) therefore form one Jacobi-group automorphy system. Its cocycle identity is an elementary consequence of matrix multiplication in $SL_2(\mathbb{Z})$. The quadratic transition function is the local current-algebra part of the calculation. The theorem boundary is the existence of the spin modular functor and anomaly-cancelled R-background for the full interacting CFT.

The adjective "weak" refers to Fourier support. Under the compact Ramond-trace hypotheses and the discrete-spectrum hypothesis in Definition 70.8, the supertrace converges absolutely on $\operatorname{Im}\tau > 0$ and has an expansion of the form

$$\varphi_{\mathcal{C}}(\tau, z) = \sum_{n\geq 0}\sum_{r\in\Lambda_R} c(n, r)\, q^n y^r,$$

after the conventional Ramond-ground-state normalization has been absorbed into n. The elliptic law then gives the coefficient orbit relation

$$c(n,r) = c\big(n + \lambda r + m_{\mathrm{ell}}\lambda^2, r + 2m_{\mathrm{ell}}\lambda\big), \qquad \lambda \in \mathbb{Z}, \tag{70.48}$$

whenever both sides lie in the same charge-lattice multiplier sector. The combination

$$4m_{\mathrm{ell}}n - r^2$$

is invariant along this spectral-flow orbit. Thus the Jacobi symmetry identifies Fourier coefficients along fixed-discriminant charge orbits. The weakness condition is the lower bound $n \geq 0$ supplied by the discrete Ramond spectrum after right-moving ground-state localization; it is a statement about the controlled Ramond trace defined above.

There is a finite Landau–Ginzburg shadow of the elliptic genus which can be checked before any infrared-flow theorem is invoked. Suppose a quasihomogeneous isolated polynomial W of Definition 70.11 is realized by a compact unitary $\mathcal{N} = 2$ fixed point, and suppose the Jacobi ring classes are represented by NS chiral primaries with charges

$$Q_R(X^a) = \sum_i a_i q_i.$$

Spectral flow by $\eta = -1/2$ gives the corresponding left Ramond charges

$$Q_R^{\mathrm{R}}(X^a) = Q_R(X^a) - \frac{c_{\mathrm{LG}}}{6}. \tag{70.49}$$

Therefore the q^0 protected charge polynomial has the form

$$\chi_y^{\mathrm{LG}} = \sum_{[X^a]\in \mathrm{Jac}(W)} y^{Q_R(X^a) - c_{\mathrm{LG}}/6}. \tag{70.50}$$

For $W = X^{k+2}$, this becomes

$$\chi_y^{A_k} = \sum_{\ell=0}^{k} y^{\frac{\ell}{k+2} - \frac{k}{2(k+2)}}. \tag{70.51}$$

Setting $y = 1$ gives the Witten index $k + 1 = \dim \mathrm{Jac}(W)$. The equality between this protected finite polynomial and the infrared elliptic genus is precisely the kind of LG/GLSM construction statement assigned to Volume VII.

70.6 Protected Algebraic Tests for Supersymmetric Dynamics

The full construction of two-dimensional $\mathcal{N} = (2,2)$ Landau–Ginzburg and gauged linear sigma-model flows belongs to the supersymmetric-QFT chapters. The part needed in the CFT volume is narrower: the chiral-ring and central-charge data that any proposed $\mathcal{N} = 2$ superconformal fixed point must reproduce. We state this as an algebraic interface between the CFT and supersymmetric-QFT volumes, not as a proof of an infrared fixed point.

Definition 70.11 (Quasihomogeneous Landau–Ginzburg chiral polynomial)**.** Let $X_1, \dots, X_N$ be complex variables with rational weights $q_i \in (0, 1/2]$. A quasihomogeneous Landau–Ginzburg chiral polynomial consists of a polynomial

$$W \in \mathbb{C}[X_1, \dots, X_N]$$

such that

$$W(\lambda^{q_1} X_1, \dots, \lambda^{q_N} X_N) = \lambda W(X_1, \dots, X_N) \qquad \lambda \in \mathbb{C}^\times, \tag{70.52}$$

and such that the critical locus

$$\partial_1 W = \cdots = \partial_N W = 0$$

is isolated at the origin. Its Jacobi ring is

$$\mathrm{Jac}(W) = \mathbb{C}[X_1, \dots, X_N]/(\partial_1 W, \dots, \partial_N W). \tag{70.53}$$

The $U(1)$ charge of a monomial

$$X^a = X_1^{a_1} \cdots X_N^{a_N}$$

is

$$Q_R(X^a) = \sum_{i=1}^{N} a_i q_i. \tag{70.54}$$

If the polynomial is realized by a unitary compact $\mathcal{N} = 2$ superconformal fixed point with these fields as chiral primaries, the chiral-primary relation gives

$$h(X^a) = \frac{1}{2} Q_R(X^a). \tag{70.55}$$

The central charge predicted by the $U(1)_R$ anomaly is

$$c_{\mathrm{LG}} = 3 \sum_{i=1}^{N} (1 - 2q_i). \tag{70.56}$$

In this chapter (70.56) is used only as a protected chiral-algebra consistency check. The assertion that a given ultraviolet Landau–Ginzburg functional flows to a particular continuum SCFT requires the regulator and RG construction requested in the Volume VII supersymmetric-QFT program.

Example 70.12 (A-series algebra)**.** For

$$W = X^{k+2}, \qquad k \in \mathbb{Z}_{\geq 0},$$

the quasihomogeneous weight is $q = 1/(k+2)$. The Jacobi ring is

$$\mathrm{Jac}(W) = \mathbb{C}[X]/(X^{k+1}), \tag{70.57}$$

with basis $1, X, \dots, X^k$. The central charge predicted by (70.56) is

$$c = \frac{3k}{k+2}, \tag{70.58}$$

and the basis element X^ℓ, $0 \leq \ell \leq k$, has

$$q_\ell = \frac{\ell}{k+2}, \qquad h_\ell = \frac{\ell}{2(k+2)} \tag{70.59}$$

when represented by a chiral primary.

Example 70.13 (Quintic algebraic check)**.** For

$$W = X_1^5 + \cdots + X_5^5$$

with all weights $q_i = 1/5$, the anomaly formula gives

$$c = 3 \cdot 5 \left(1 - \frac{2}{5}\right) = 9. \tag{70.60}$$

This is an algebraic $U(1)_R$-anomaly calculation. Interpreting it as a Calabi–Yau sigma-model phase is not part of the present chapter.

70.7 Supersymmetric Rank-One Coset Interfaces

The rank-one cosets of Section 66.8.2 have supersymmetric refinements which are too important to leave as informal shorthand. This section records the exact chiral-algebraic interface needed by the CFT volume. It does not assert a construction of a UV Landau–Ginzburg or GLSM flow. The latter requires the regulator, supersymmetric Ward identities, and infrared-comparison topology of Volume VII. The point here is narrower and sharper: once the supersymmetric WZW coset exists as an $\mathcal{N} = 2$ chiral algebra, its central charge, primary weights, R-charges, spectral flow, and basic field-identification arithmetic are fixed.

We use the following conventions. A supersymmetric $SU(2)_k$ WZW model, with $k \in \mathbb{Z}_{\geq 2}$, is represented by a bosonic $\widehat{\mathfrak{su}}(2)_{k-2}$ current algebra with currents $k^a(z)$, $a = 1, 2, 3$, together with three free Majorana fermions $\chi^a(z)$. The total Cartan current is denoted K^3. The supersymmetric $SL(2, \mathbb{R})_k$ model, with $k > 0$, is represented by a bosonic $\widehat{\mathfrak{sl}}(2, \mathbb{R})_{k+2}$ current algebra with currents $j^a(z)$, three free fermions $\psi^a(z)$ with the noncompact-signature bilinear form, and total Cartan current J^3. In both cases the coset removes the $\mathcal{N} = 1$ abelian sector generated by the Cartan fermion and total Cartan current. This is an operator statement: the displayed coset fields are required to have regular OPE with the removed $\mathcal{N} = 1$ $U(1)$ sector.

Definition 70.14 (Compact supersymmetric rank-one coset)**.** For $k \in \mathbb{Z}_{\geq 2}$, the compact supersymmetric rank-one coset is the chiral algebra obtained from the supersymmetric $SU(2)_k$ model by decoupling the $\mathcal{N} = 1$ $U(1)$ sector generated by χ^3 and K^3. Put

$$k^{\pm} = k^1 \pm \mathrm{i}k^2, \qquad \chi^{\pm} = \frac{\chi^1 \pm \mathrm{i}\chi^2}{\sqrt{2}}.$$

In the standard free-fermion realization the $\mathcal{N} = 2$ generators of the coset are

$$G^{\pm} = \sqrt{\frac{2}{k}}\, k^{\mp}\chi^{\pm}, \qquad J_R = -\frac{2}{k}k^3 + \frac{k-2}{k}\chi^+\chi^- = -\frac{2}{k}K^3 + \chi^+\chi^-. \tag{70.61}$$

Compact coset central charge and primary data. The compact coset of Definition 70.14 has

$$c_{\mathrm{comp}}(k) = \frac{3(k-2)}{k}. \tag{70.62}$$

Let $V^{su}_{j,m,\bar{m}}$ be a diagonal bosonic $\widehat{\mathfrak{su}}(2)_{k-2}$ primary, where

$$j \in \left\{0, \frac{1}{2}, \ldots, \frac{k-2}{2}\right\}, \qquad m, \bar{m} \in \{-j, -j+1, \ldots, j\}.$$

Bosonize the total Cartan current by

$$K^3 \simeq \mathrm{i}\sqrt{\frac{k}{2}}\,\partial Y$$

and factor the Cartan vertex by

$$V^{su}_{j,m,\bar m} \simeq \exp\left[\mathrm{i}\sqrt{\frac{2}{k}}\,(mY + \bar m\,\bar Y)\right]\Psi^{su}_{j,m,\bar m}. \tag{70.63}$$

Then the coset primary $\Psi^{su}_{j,m,\bar m}$ has

$$h = \frac{j(j+1) - m^2}{k}, \qquad q = -\frac{2m}{k}, \qquad \bar h = \frac{j(j+1) - \bar m^2}{k}, \qquad \bar q = -\frac{2\bar m}{k}. \tag{70.64}$$

The central-charge calculation is a useful place to keep the level shifts visible. The bosonic $SU(2)_{k-2}$ algebra has central charge

$$\frac{3(k-2)}{(k-2)+2} = \frac{3(k-2)}{k}.$$

The three free fermions contribute $3/2$. The removed abelian $\mathcal{N} = 1$ sector contributes $1 + 1/2 = 3/2$. Hence the coset central charge is $\frac{3(k-2)}{k} + 3/2 - 3/2$, which is (70.62).

For the weights, the bosonic $SU(2)_{k-2}$ primary of spin j has Sugawara weight $j(j+1)/k$, while the Cartan factor in (70.63) has holomorphic weight m^2/k. Subtracting the Cartan part gives the displayed h, and the antiholomorphic sector gives $\bar h = (j(j+1) - \bar m^2)/k$ with $\bar m$ in place of m. The R-charge follows by taking the simple pole of J_R with the coset factor after the K^3-vertex has been removed. With the normalization above, this pole is $-2m/k$ times the field; the antiholomorphic formula is identical.

The compact coset is the $\mathcal{N} = 2$ minimal model whose usual minimal-model level is

$$K = k - 2.$$

Thus $c_{\mathrm{comp}} = 3K/(K+2)$. The chiral-primary representatives in the convention (70.64) are obtained by setting $m = -j$. Then

$$h = \frac{j}{k}, \qquad q = \frac{2j}{k}, \qquad h = \frac{q}{2}.$$

Writing $\ell = 2j$, with $0 \le \ell \le K$, gives

$$q_\ell = \frac{\ell}{K+2}, \qquad h_\ell = \frac{\ell}{2(K+2)}.$$

These are exactly the A-series algebraic values in Example 70.12, after identifying $W = X^{K+2}$. The equality of protected chiral data is a consistency interface. A proof that a chosen UV Landau–Ginzburg regulator flows to this coset requires control of the corresponding RG trajectory.

Proposition 70.15 (Compact coset chiral ring)**.** *Let $K = k - 2$. For $0 \le \ell \le K$, put*

$$\Phi_\ell = \Psi^{su}_{\ell/2,-\ell/2}$$

for the holomorphic compact-coset chiral primary in the NS sector. In the protected chiral-primary fusion ring,

$$\Phi_a\Phi_b = \begin{cases} \Phi_{a+b}, & a+b \le K, \\ 0, & a+b > K, \end{cases} \qquad 0 \le a, b \le K, \tag{70.65}$$

after normalizing $\Phi_0 = 1$ and the generator Φ_1. Hence

$$\mathcal{R}_{\mathrm{ch}}\big(SU(2)_k/U(1)\big) \cong \mathbb{C}[x]/(x^{K+1}), \qquad x^\ell \longleftrightarrow \Phi_\ell. \tag{70.66}$$

Here the $K = 0$ endpoint means $\mathbb{C}[x]/(x) \cong \mathbb{C}$, and the normalization of Φ_1 applies only when $K \ge 1$.

Proof. The fields Φ_ℓ are chiral because

$$h_\ell = \frac{\ell}{2k}, \qquad q_\ell = \frac{\ell}{k}, \qquad h_\ell = q_\ell/2.$$

The OPE of two NS chiral primaries has no negative powers in the chiral channel. Indeed any term of R-charge $q_a + q_b$ has, by the unitarity bound, weight at least $(q_a + q_b)/2 = h_a + h_b$. Thus the chiral-primary projection of the product has additive R-charge.

When $a + b \le K$, the parent $\widehat{\mathfrak{su}}(2)_{k-2}$ fusion of spins $a/2$ and $b/2$ contains the top spin $(a+b)/2$ with multiplicity one. The Cartan charges add from $-a/2$ and $-b/2$ to $-(a+b)/2$, so the coset branching contains precisely the chiral primary Φ_{a+b} in the BPS channel. The one-dimensionality of this channel allows the displayed normalization.

If $a + b > K$, then the additive charge would be $(a+b)/k > K/k = c/3$. The compact coset has no chiral primary with that charge: the complete chiral list is $0 \le \ell \le K$. Equivalently, the affine fusion truncation removes the would-be top spin before it can appear as a chiral coset branch. The BPS projection of the product is therefore zero. The resulting associative algebra is the truncated polynomial algebra $\mathbb{C}[x]/(x^{K+1})$. This proves the equality of the compact-coset protected chiral ring with the A-series Jacobi algebra as a chiral-data statement, while leaving the dynamical LG-to-coset flow theorem to Volume VII. □

The compact coset has independent holomorphic and antiholomorphic spectral flow operations. Applying (70.24) with (70.62) to (70.64) gives the flowed labels

$$\begin{aligned} h^{(\eta)} &= \frac{j(j+1) - (m+\eta)^2}{k} + \frac{\eta^2}{2}, & q^{(\eta)} &= -\frac{2(m+\eta)}{k} + \eta, \\ \bar{h}^{(\bar\eta)} &= \frac{j(j+1) - (\bar m+\bar\eta)^2}{k} + \frac{\bar\eta^2}{2}, & \bar{q}^{(\bar\eta)} &= -\frac{2(\bar m+\bar\eta)}{k} + \bar\eta, \end{aligned} \tag{70.67}$$

where $\eta, \bar\eta \in \frac{1}{2}\mathbb{Z}$. The compact simple-current identification becomes, for the corresponding spectral-flowed fields,

$$\Psi^{su,(\eta,\bar\eta)}_{j,m,\bar m} \simeq \Psi^{su,(\eta-1,\bar\eta-1)}_{\frac{k-2}{2}-j,\, m-\frac{k-2}{2},\, \bar m-\frac{k-2}{2}} \simeq \Psi^{su,(\eta-1,\bar\eta+1)}_{\frac{k-2}{2}-j,\, m-\frac{k-2}{2},\, \bar m+\frac{k-2}{2}}. \tag{70.68}$$

The full diagonal compact theory is obtained only after the left and right representations are paired. In the conventional diagonal pairing one imposes

$$m + \eta - \bar m - \bar\eta \in k\mathbb{Z}, \qquad \eta - \bar\eta \in \mathbb{Z}. \tag{70.69}$$

The remaining angular symmetry of the disc sigma-model representative is a $\mathbb{Z}_k$ symmetry acting by

$$\Psi^{su,(\eta,\bar\eta)}_{j,m,\bar m} \longmapsto \exp\left[\frac{2\pi\mathrm{i}}{k}(m+\eta+\bar m+\bar\eta)\right]\Psi^{su,(\eta,\bar\eta)}_{j,m,\bar m}. \tag{70.70}$$

This last statement is a warning against a common shortcut: the disc metric is not the exact CFT data. The finite-level CFT remembers the simple-current identification and the $\mathbb{Z}_k$ selection rule.

Definition 70.16 (Noncompact supersymmetric rank-one coset)**.** For $k>0$, the noncompact supersymmetric rank-one coset is obtained from the supersymmetric $SL(2,\mathbb{R})_k$ model by decoupling the $\mathcal{N}=1$ abelian sector generated by ψ^3 and J^3. Put

$$j^\pm = j^1 \pm \mathrm{i}j^2, \qquad \psi^\pm = \frac{\psi^1 \pm \mathrm{i}\psi^2}{\sqrt{2}}.$$

In the standard realization its $\mathcal{N}=2$ generators are

$$G^\pm = \sqrt{\frac{2}{k}}\,j^\mp\psi^\pm, \qquad J_R = \frac{2}{k}j^3 + \frac{k+2}{k}\psi^+\psi^- = \frac{2}{k}J^3 + \psi^+\psi^-. \tag{70.71}$$

Cigar central charge and primary data. The noncompact coset of Definition 70.16 has

$$c_{\mathrm{cig}}(k) = \frac{3(k+2)}{k}. \tag{70.72}$$

Let $V^{sl}_{j,m,\bar m}$ be a bosonic $\widehat{\mathfrak{sl}}(2,\mathbb{R})_{k+2}$ primary, with j the standard $SL(2,\mathbb{R})$ spin parameter and $m,\bar m$ the total Cartan eigenvalues. Bosonize the total Cartan current by

$$J^3 \simeq -\sqrt{\frac{k}{2}}\,\partial X$$

and factor

$$V^{sl}_{j,m,\bar m} \simeq \exp\left[\sqrt{\frac{2}{k}}\,(mX+\bar m\,\bar X)\right]\Psi^{sl}_{j,m,\bar m}. \tag{70.73}$$

Then

$$h = \frac{-j(j-1)+m^2}{k}, \qquad q = \frac{2m}{k}, \qquad \bar h = \frac{-j(j-1)+\bar m^2}{k}, \qquad \bar q = \frac{2\bar m}{k}. \tag{70.74}$$

The bosonic $SL(2,\mathbb{R})_{k+2}$ current algebra has central charge $3(k+2)/k$ in the convention of Section 66.8.2. Adding three fermions and removing the abelian $\mathcal{N}=1$ Cartan sector changes the central charge by $+3/2-3/2$, so the coset central charge is (70.72). The bosonic primary has Sugawara weight $-j(j-1)/k$. The Cartan boson in (70.73) is timelike with respect to the noncompact bilinear form, so removing the Cartan factor adds m^2/k to the coset weight. The R-charge is the simple-pole coefficient of the J_R OPE after the J^3-vertex has been factored out, giving $2m/k$. The antiholomorphic sector uses $\bar m$ in the same factorization and gives $\bar h = (-j(j-1)+\bar m^2)/k$ and $\bar q = 2\bar m/k$.

The noncompact spectrum is not determined by (70.74) alone. In the conventional cigar theory the continuous series has

$$j = \frac{1}{2}+\mathrm{i}s, \qquad s\in\mathbb{R}, \qquad m-\bar m\in\mathbb{Z},$$

while discrete normalizable states are residues in an allowed strip such as

$$\frac{1}{2} < j < \frac{k+1}{2}, \qquad m, \bar{m} \in j + \mathbb{Z}_{\geq 0} \quad \text{or} \quad m, \bar{m} \in -j - \mathbb{Z}_{\geq 0},$$

together with the winding condition

$$m + \bar{m} \in k\mathbb{Z}. \tag{70.75}$$

The exact reflection coefficient, its poles, and the spectral density are analytic CFT data. They should not be replaced by the cigar metric.

Spectral flow gives

$$\begin{aligned} h^{(\eta)} &= \frac{-j(j-1) + (m+\eta)^2}{k} + \frac{\eta^2}{2}, & q^{(\eta)} &= \frac{2(m+\eta)}{k} + \eta, \\ \bar{h}^{(\bar{\eta})} &= \frac{-j(j-1) + (\bar{m}+\bar{\eta})^2}{k} + \frac{\bar{\eta}^2}{2}, & \bar{q}^{(\bar{\eta})} &= \frac{2(\bar{m}+\bar{\eta})}{k} + \bar{\eta}. \end{aligned} \tag{70.76}$$

The momentum and winding labels in the asymptotic cigar circle are shifted to

$$n = m + \eta - \bar{m} - \bar{\eta}, \qquad w = \frac{m + \eta + \bar{m} + \bar{\eta}}{k}, \tag{70.77}$$

so the winding condition becomes

$$m + \eta + \bar{m} + \bar{\eta} \in k\mathbb{Z}. \tag{70.78}$$

For discrete states one extends the range by the identification

$$\Psi^{sl,(\eta,\bar{\eta})}_{j,m,\bar{m}} \simeq \Psi^{sl,(\eta+1,\bar{\eta}+1)}_{\frac{k+2}{2}-j,\, m-\frac{k+2}{2},\, \bar{m}-\frac{k+2}{2}}. \tag{70.79}$$

There is no compact-coset analogue shifting η and $\bar{\eta}$ with opposite signs. The reason is structural: the cigar has an exact circle translation quantum number n, whereas the compact disc representative has only the finite $\mathbb{Z}_k$ remnant in (70.70).

Remark 70.17 (Spectral-flow derivation of the rank-one formulas). Equations (70.67) and (70.76) follow from the general $\mathcal{N} = 2$ spectral-flow law (70.24) and the unflowed data (70.64), (70.74). For the compact coset,

$$h + \eta q + \frac{c_{\text{comp}}\eta^2}{6} = \frac{j(j+1) - m^2}{k} - \frac{2\eta m}{k} + \frac{k-2}{2k}\eta^2.$$

Since

$$-\frac{(m+\eta)^2}{k} + \frac{\eta^2}{2} = -\frac{m^2 + 2m\eta}{k} + \frac{k-2}{2k}\eta^2,$$

this is the first formula in (70.67). The charge identity is

$$q + \frac{c_{\text{comp}}\eta}{3} = -\frac{2m}{k} + \frac{k-2}{k}\eta = -\frac{2(m+\eta)}{k} + \eta.$$

The barred compact formulas are identical. For the noncompact coset,

$$h + \eta q + \frac{c_{\text{cig}}\eta^2}{6} = \frac{-j(j-1) + m^2}{k} + \frac{2\eta m}{k} + \frac{k+2}{2k}\eta^2,$$

and

$$\frac{(m+\eta)^2}{k}+\frac{\eta^2}{2}=\frac{m^2+2m\eta}{k}+\frac{k+2}{2k}\eta^2.$$

The charge formula follows from

$$q+\frac{c_{\rm cig}\eta}{3}=\frac{2m}{k}+\frac{k+2}{k}\eta=\frac{2(m+\eta)}{k}+\eta.$$

The mirror relation between the noncompact $SL(2,\mathbb{R})_k/U(1)$ coset and $\mathcal{N}=2$ Liouville theory belongs at the interface between this chiral algebra, the cigar sigma-model representative in Section 66.8.2, and the Volume VII GLSM duality construction. In this volume it is safest to treat mirror symmetry as a target theorem for the supersymmetric-QFT development: the present section supplies the exact coset side, including the R-charges and spectral-flow labels that any such theorem must match.

70.8 Small $\mathcal{N}=4$ Boundary

For completeness, we record the boundary between what is used in this monograph and the full small $\mathcal{N}=4$ representation theory. The small $\mathcal{N}=4$ algebra contains an affine $\mathfrak{su}(2)$ current algebra and four supercurrents. It includes $\mathcal{N}=2$ subalgebras after choosing a Cartan $U(1)\subset SU(2)_R$, but the representation theory involves the full $SU(2)_R$ spin.

Definition 70.18 (Small $\mathcal{N}=4$ chiral algebra)**.** A small $\mathcal{N}=4$ chiral algebra at level $k\in\mathbb{Z}_{\geq 0}$ consists of:

1. a stress tensor $T(z)$ of central charge $c=6k$;
2. affine $\mathfrak{su}(2)$ currents $J^a(z)$, $a=1,2,3$, with level k;
3. four weight-3/2 supercurrents transforming as two doublets under $\mathfrak{su}(2)$;
4. OPEs such that the anticommutator of opposite supercurrents closes on T, J^a, and the central term, while the current OPEs act on the supercurrents by the doublet representation.

Definition 70.18 records only the part of the small-$\mathcal{N}=4$ algebra used in this volume. The present monograph uses two consequences: the central charge is tied to the affine level by $c=6k$, and every Cartan choice supplies an $\mathcal{N}=2$ subalgebra whose spectral-flow operation is inherited from the affine $SU(2)_R$ structure. A later chapter that needs small-$\mathcal{N}=4$ multiplet classification must state the full OPE tensor and derive the corresponding unitarity bounds.

Open Problem 70.19 (Intrinsic two-dimensional supersymmetric RG construction)*.* Coordinate the CFT-volume algebraic layer with the Volume VII construction of two-dimensional $\mathcal{N}=(2,2)$ Landau–Ginzburg models, gauged linear sigma models, and their proposed infrared SCFTs as intrinsic two-dimensional QFTs with a regulator, supersymmetric Ward identities, renormalized $U(1)_R$ current, and a topology in which the infrared chiral ring and central charge are obtained. The algebraic checks in Definition 70.11 and Examples 70.12–70.13 are necessary consistency tests, not a construction of the continuum fixed point.

Part VI

Integrable Quantum Field Theory

Chapter 71

Factorized Scattering and Integrability

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This volume begins with massive two-dimensional QFT because that is the place where integrability has a sharp scattering-theoretic meaning. Let the theory have a vacuum representation on $\mathbb{M}^2$, a mass gap above the vacuum, and isolated one-particle species labelled by $a, b, \dots$, with masses $m_a > 0$. A one-particle momentum is parametrized by rapidity

$$p_a^\mu(\theta) = (m_a \cosh\theta,\ m_a \sinh\theta), \qquad \theta \in \mathbb{R}.$$

The Lorentz group acts by translating θ. Hence a two-particle invariant depends on the rapidity difference $\theta_{12} = \theta_1 - \theta_2$.

Definition 71.1 (Rapidity chart and ordered chambers)**.** For a species a, the rapidity chart is the map

$$\theta \longmapsto p_a(\theta) = m_a(\cosh\theta, \sinh\theta)$$

from $\mathbb{R}$ to the positive-energy mass shell $\{p \in \mathbb{R}^{1,1} : p^2 = -m_a^2,\ p^0 > 0\}$. The positive two-particle Mandelstam invariant is

$$s_{ab}(\theta_1 - \theta_2) := -\big(p_a(\theta_1) + p_b(\theta_2)\big)^2.$$

For N particles, an ordered rapidity chamber is a connected component of

$$\{(\theta_1, \dots, \theta_N) \in \mathbb{R}^N : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

The fundamental chamber used below is $\theta_1 > \theta_2 > \cdots > \theta_N$. Other chambers are reached by permuting rapidities and then relating the corresponding scattering bases by two-body scattering operators.

Remark 71.2 (Two-particle kinematics in rapidity variables). In the rapidity chart of Definition 71.1,

$$s_{ab}(\theta) = m_a^2 + m_b^2 + 2m_a m_b \cosh\theta, \qquad \theta = \theta_1 - \theta_2.$$

The Lorentz-invariant two-body kinematics of massive particles in $1+1$ dimensions is therefore encoded by the rapidity difference θ, together with the species labels.

With mostly-plus metric,

$$p_a(\theta_1)\cdot p_b(\theta_2) = -m_a m_b \cosh\theta_1 \cosh\theta_2 + m_a m_b \sinh\theta_1 \sinh\theta_2 = -m_a m_b \cosh(\theta_1 - \theta_2).$$

Since $p_a^2 = -m_a^2$ and $p_b^2 = -m_b^2$,

$$-\left(p_a + p_b\right)^2 = m_a^2 + m_b^2 + 2m_a m_b \cosh(\theta_1 - \theta_2).$$

A proper orthochronous Lorentz boost of rapidity λ sends $\theta_i \mapsto \theta_i + \lambda$, so $\theta_1 - \theta_2$ is invariant. Conversely, once the masses and the total invariant mass are fixed, $\cosh(\theta_1 - \theta_2)$ is fixed; the sign of $\theta_1 - \theta_2$ records the ordering of the two momenta.

71.1 Reconstruction Spine and Status Labels

The exact data in this volume do not all have the same logical status. A two-body S-matrix satisfying unitarity, crossing, and Yang–Baxter is on-shell scattering data. A TBA equation is a thermodynamic or mirror-channel functional equation after Bethe–Yang quantization has been accepted. A form-factor solution is a family of candidate local-operator matrix elements. None of these, by itself, is a constructed local quantum field theory. The volume therefore uses a common dependency spine with two intertwined chains. The Hilbert-space chain runs from a local massive QFT with higher-spin charges to elastic factorized scattering, then to S-Fock/ZF data, wedge-local fields, local algebras, and finally local fields and Wightman functions. The thermodynamic chain runs from Bethe–Yang finite-volume quantization to TBA pseudoenergies, mirror finite-size data, and GHD currents. The arrows in these chains are not equivalent. Some are theorems under standard Hilbert-space hypotheses; some are conditional theorems after analytic and nuclearity assumptions are imposed; some are formal constructions whose output still lacks positivity or locality; some are numerical or finite-regulator evidence; and some are open problems.

Definition 71.3 (Status labels for integrable-QFT bridges)**.** For the rest of this volume, a bridge between two layers of exact data carries one of the following labels.

(i) *Theorem*: the stated hypotheses construct the target object in a Hilbert space or algebra of local observables and prove the claimed property.

(ii) *Conditional theorem*: the proof is valid once explicitly stated analytic, domain, nuclearity, thermodynamic-limit, or state-space hypotheses are supplied.

(iii) *Formal construction*: the formula produces algebraic or distributional candidates, but positivity, locality, convergence, or domain control is not yet part of the statement.

(iv) *Numerical or finite-regulator evidence*: a calculation check verifies a finite identity, normalization, sign, or approximation window. It is not evidence for continuum reconstruction unless the limiting theorem is separately stated.

(v) *Open problem*: a named missing implication whose hypotheses and target object are known well enough to be stated.

The main bridges in this volume should be read with the following status map. The entries are deliberately conservative; upgrading a row requires proving the missing analytic statement, not adding another finite identity.

Bridge	**Status**	**Load-bearing evidence**
local QFT to factorized scattering	conditional theorem	Haag–Ruelle scattering, mass gap, separating higher-spin charges, regular distribution kernels, and control of threshold and bound-state singularities.
factorized S to S-Fock/ZF algebra	theorem at the algebraic Hilbert-space level	Unitarity gives positive chamber inner products; Yang–Baxter gives path-independent chamber continuation and associative exchange.
S-Fock data to wedge-local fields	conditional theorem	Regularity, crossing, and analytic bounds strong enough to define polarization-free generators and wedge-local operator domains.
wedge-local fields to local algebras	conditional theorem	Nontriviality of double-cone intersections, usually via modular nuclearity or related phase-space bounds.
form-factor equations to local fields	formal construction unless convergence and locality are proved	Watson, cyclicity, and residue equations plus growth bounds; the missing analytic work is convergence of spectral sums, positivity, domains, and locality of reconstructed fields.
Bethe–Yang data to ground-state TBA	conditional theorem	Finite-volume quantization with controlled errors, density convergence, convex free-energy functional, and fixed normalization of the mirror channel.
ground-state TBA to excited-state or mirror data	conditional or formal, model by model	Specified singularity motion, contour deformation, source terms, and a check that no untracked pole crosses the continuation path.
TBA densities to GHD	conditional hydrodynamic limit	Local generalized Gibbs ensembles, dressing invertibility, Euler-scale limit, clustering/mixing, and absence or controlled size of integrability breaking.

Two models serve as reference points for this status discipline. In the massive Ising model, the scattering datum is the scalar $S = -1$, the S-Fock space is the antisymmetric free-fermion Fock space, and the local energy field is a Wick polynomial of the Majorana field. The local QFT and its Wightman functions are therefore constructed by the free-field theorem; the form-factor expansion for separated Euclidean correlators is a theorem after the usual Fock-space domain and convergence estimates. The spin and disorder fields require the additional order/disorder semi-locality structure developed later in the volume, but their status is not inferred merely from the equation $S = -1$.

For a regular scalar factorizing model such as sinh-Gordon, the exact two-body scattering function gives an S-symmetric Fock space and wedge-local polarization-free generators under the standard analyticity and crossing bounds. The existence of nontrivial double-cone algebras is then a conditional theorem once the relevant modular nuclearity estimate is proved. Pointlike local fields obtained from form-factor series, and completeness of a declared family of such fields, remain separate analytic tasks. TBA predictions for the same model are therefore thermodynamic consequences of the scattering specification under Bethe–Yang and mirror-channel hypotheses; they are not, by themselves, a proof that the local double-cone net has been constructed.

This status map is also how calculation companions are used. A finite Yang–Baxter, residue, TBA-kernel, or form-factor check verifies the stated finite algebra in its row of the map. It does not upgrade a formal form-factor solution to a local field, nor a TBA equation to a theorem about finite-size spectra, unless the missing Hilbert-space, convergence, and thermodynamic-limit assumptions have been supplied at the point of use.

71.2 Factorized Scattering Datum

Definition 71.4 (Elastic factorized scattering system)**.** An elastic factorized scattering system consists of a finite or countable set of massive species, a positive mass m_a for each species, and meromorphic functions

$$S_{ab}^{cd}(\theta), \qquad a, b, c, d \in I,$$

defined on a common rapidity-domain cover, satisfying the following algebraic and analytic conditions.

First, two-particle unitarity is

$$\sum_{e,f} S_{ab}^{ef}(\theta)\, S_{ef}^{cd}(-\theta) = \delta_a^c \delta_b^d.$$

Second, Hermitian analyticity is

$$S_{ab}^{cd}(\theta)^* = S_{cd}^{ab}(-\theta^*)$$

on the boundary where both sides are defined. Third, if $\bar{a}$ denotes the charge-conjugate species, crossing has the form

$$S_{ab}^{cd}(\theta) = C_{a\bar{a}'} C^{c\bar{c}'}\, S_{\bar{c}'b}^{\bar{a}'d}(\mathrm{i}\pi - \theta),$$

with the charge-conjugation matrix C fixed as part of the particle-state normalization. Fourth, three-particle consistency is the Yang–Baxter equation

$$S_{12}(\theta_{12})S_{13}(\theta_{13})S_{23}(\theta_{23}) = S_{23}(\theta_{23})S_{13}(\theta_{13})S_{12}(\theta_{12})$$

as an identity on the tensor product of three one-particle internal spaces.

The notation S_{ij} means that the two-particle matrix S acts on the i-th and j-th tensor factors and as the identity on the remaining factor. The Yang–Baxter equation is the associativity condition for reordering three rapidities by adjacent transpositions. It is a structural part of factorized scattering: it identifies the pairwise decompositions of a single ordered three-particle process.

Definition 71.5 (Separating higher-spin family)**.** Let $\{Q_{s,\lambda}\}_{(s,\lambda)\in\mathcal{A}}$ be commuting conserved charges whose one-particle eigenvalues have the form

$$q_{s,\lambda,a}(\theta) = \kappa_{s,\lambda,a}\, \mathrm{e}^{s\theta}, \qquad s > 1.$$

For fixed particle number N, set $z_i = \mathrm{e}^{\theta_i}$. The family is N-separating on a rapidity domain U_N if the moment map

$$\Phi_N\big(\{(a_i, z_i)\}_{i=1}^N\big) = \left(\sum_{i=1}^N \kappa_{s,\lambda,a_i} z_i^s\right)_{(s,\lambda)\in\mathcal{A}}$$

is injective on $U_N/\mathfrak{S}_N$, outside a specified closed exceptional set consisting of coincident rapidities and declared threshold or bound-state singular loci. The quotient means that only the unordered multiset of species and rapidities is to be recovered.

A concrete sufficient condition, for a finite species set, is the following. For each species a and for each r needed up to the maximal particle number under consideration, finite linear combinations of the charges produce the species-resolved power sums

$$p_{r,a} = \sum_{i\colon a_i=a} z_i^r.$$

Then Newton identities recover the monic polynomial

$$\prod_{i\colon a_i=a} (t - z_i)$$

for each species a, and hence recover the unordered species-resolved rapidity multiset wherever the roots are distinct. The definition above is the hypothesis used below; this sufficient condition is only a convenient checkable way to obtain it.

Support test by multiplication. Let $U \subset \mathbb{R}^n$ be open, let $T \in \mathcal{D}'(U)$, and let $f \in C^\infty(U)$. If $fT = 0$ and f has no zero on an open set $V \subset U$, then T vanishes on V.

For any test function $\varphi \in C_c^\infty(V)$, the quotient φ/f is smooth with compact support in V, because f is nowhere zero on V. Since $fT = 0$,

$$\langle T, \varphi\rangle = \left\langle T, f\frac{\varphi}{f}\right\rangle = \left\langle fT, \frac{\varphi}{f}\right\rangle = 0.$$

Thus T pairs to zero with every test function supported in V, which means $T|_V = 0$.

Proposition 71.6 (Factorization from conserved higher-spin charges)**.** *Assume a massive* $1+1$*-dimensional QFT has Haag–Ruelle asymptotic particle states, a family of conserved charges* $\{Q_{s,\lambda}\}_{(s,\lambda)\in\mathcal{A}}$ *as in Definition 71.5, and scattering kernels which, after smearing by compactly supported wave packets in rapidity chambers, depend continuously on the wave packets away from threshold and bound-state singular loci. Equivalently, the unsmeared kernels are distributional boundary values whose singular support in the rapidity variables is contained in those declared loci.*

If the scattering operator commutes with every $Q_{s,\lambda}$*, and if the charge family is* N*-separating on the incoming and outgoing* N*-particle domains under consideration, then the* $N \to N$ *scattering kernel is supported, away from the declared singular loci, on the union of permutation graphs*

$$\{(\theta_1', b_1; \ldots; \theta_N', b_N) = (\theta_{\sigma(1)}, a_{\sigma(1)}; \ldots; \theta_{\sigma(N)}, a_{\sigma(N)})\}.$$

If the corresponding moment maps for $M \neq N$ *have disjoint generic images, then generic* $N \to M$ *particle production is absent.*

Proof. Let $K_{N\to M}$ be the distribution kernel of the scattering operator between ordered rapidity chambers, away from the declared singular loci. Commutation with $Q_{s,\lambda}$ gives, in the sense of distributions,

$$\left(\sum_{i=1}^{N}\kappa_{s,\lambda,a_i}\mathrm{e}^{s\theta_i}-\sum_{j=1}^{M}\kappa_{s,\lambda,b_j}\mathrm{e}^{s\theta'_j}\right)K_{N\to M}=0$$

for every (s,λ). The following support property of distributions is being used: the support test above implies that the kernel vanishes on any open set where one of the displayed charge differences is nonzero. Therefore the support of the kernel is contained in the common zero locus of all differences of charge eigenvalues.

For $M=N$, the N-separation hypothesis says that this common zero locus, outside the exceptional set, is precisely the union of permutation graphs displayed in the statement. Thus, after smearing with wave packets supported in a connected component of the regular rapidity chamber, the scattering operator cannot have support away from elastic permutations. For $M\neq N$, the assumed disjointness of the generic moment-map images leaves no regular common zero locus, so the production kernel vanishes on the corresponding open domain. Threshold and bound-state loci are excluded from the statement because the distribution kernels may have additional singular components there; those components must be described after the relevant channel is resolved. □

71.3 Ordered Bases and the Scattering Groupoid

Factorized scattering is most precise when formulated as a comparison of ordered bases for the same asymptotic Hilbert-space sector. Fix a regular rapidity point with distinct rapidities. For each chamber $\theta_{\sigma(1)}>\cdots>\theta_{\sigma(N)}$, choose the ordered asymptotic basis in which the creation operators are written in that order. Crossing a codimension-one wall $\theta_i=\theta_{i+1}$ is represented by the two-body operator

$$\mathsf{T}_i(\theta_i-\theta_{i+1})=S_{i,i+1}(\theta_i-\theta_{i+1})$$

acting on the adjacent species spaces and reversing the adjacent rapidity order. Thus the elementary maps T_i are morphisms in the groupoid whose objects are rapidity chambers and whose arrows are paths crossing walls. A consistent factorized scattering theory requires this groupoid representation to be path independent on the complement of the declared threshold and bound-state singular loci.

Proposition 71.7 (Scattering-chamber consistency)**.** *Let $S_{ab}^{cd}(\theta)$ be an elastic factorized scattering system. Away from poles and coincident rapidities, the adjacent chamber maps T_i define a path-independent representation of the chamber groupoid if and only if the two-particle unitarity identities and the Yang–Baxter equation hold.*

Proof. The chamber groupoid for ordered points on the line is generated by adjacent wall crossings. Its local relations are:

$$\mathsf{T}_i^2=1,\qquad \mathsf{T}_i\mathsf{T}_j=\mathsf{T}_j\mathsf{T}_i\quad(|i-j|>1),\qquad \mathsf{T}_i\mathsf{T}_{i+1}\mathsf{T}_i=\mathsf{T}_{i+1}\mathsf{T}_i\mathsf{T}_{i+1}.$$

The first relation is exactly two-particle unitarity,

$$S_{ab}(\theta)S_{ba}(-\theta)=\mathrm{id},$$

written in components as the identity in Definition 71.4. The second relation holds because disjoint adjacent crossings act on disjoint tensor factors. For three adjacent rapidities $\theta_1 > \theta_2 > \theta_3$, the two reduced paths from the chamber (123) to the chamber (321) give

$$S_{12}(\theta_{12})S_{13}(\theta_{13})S_{23}(\theta_{23}) \quad \text{and} \quad S_{23}(\theta_{23})S_{13}(\theta_{13})S_{12}(\theta_{12}),$$

respectively. Equality is precisely the Yang–Baxter equation. These local relations present the chamber groupoid away from the excluded diagonals and singular loci, so unitarity plus Yang–Baxter gives path independence. Conversely, path independence around the two local loops just described gives unitarity and the Yang–Baxter equation. □

71.4 Zamolodchikov–Faddeev Operators

The factorized scattering system becomes an operator algebra on asymptotic wavefunctions. For each species a introduce distributional creation operators $Z_a^\dagger(\theta)$ obeying

$$Z_a^\dagger(\theta_1)Z_b^\dagger(\theta_2) = S_{ab}^{cd}(\theta_1 - \theta_2)\, Z_d^\dagger(\theta_2)Z_c^\dagger(\theta_1), \qquad \theta_1 > \theta_2. \tag{71.1}$$

Repeated species labels are summed. These operator-valued distributions are a compact encoding of ordered asymptotic creation operators and their chamber-transition functions. Local fields enter through form factors and operator reconstruction below.

Associativity of the ZF creation algebra. Assume two-particle unitarity. The exchange rule (71.1) gives an associative algebra of ordered creation symbols, independent of the sequence of adjacent re-orderings, if and only if the Yang–Baxter equation in Definition 71.4 holds.

It is enough to compare the two ways of reordering three strictly ordered rapidities $\theta_1 > \theta_2 > \theta_3$. Starting from

$$Z_a^\dagger(\theta_1)Z_b^\dagger(\theta_2)Z_c^\dagger(\theta_3),$$

the reduced word which first moves the first rapidity past the second and then through the third gives the coefficient

$$S_{ab}^{a'b'}(\theta_{12})S_{a'c}^{a''c'}(\theta_{13})S_{b'c'}^{b''c''}(\theta_{23})$$

multiplying the final ordered product

$$Z_{c''}^\dagger(\theta_3)Z_{b''}^\dagger(\theta_2)Z_{a''}^\dagger(\theta_1).$$

The other reduced word gives the same final ordered product with coefficient

$$S_{bc}^{b'c'}(\theta_{23})S_{ac'}^{a'c''}(\theta_{13})S_{a'b'}^{a''b''}(\theta_{12}).$$

Equality of these coefficients for all external species is exactly the component form of the Yang–Baxter equation. The inverse adjacent move is well defined by two-particle unitarity, and disjoint moves commute because they involve disjoint creation symbols. Hence the braid relations are equivalent to associativity and sequence independence of the exchange algebra.

Definition 71.8 (S-symmetric finite-particle wavefunctions)**.** An N-particle S-symmetric wavefunction is a collection of functions

$$\psi_{a_1 \ldots a_N}(\theta_1, \ldots, \theta_N)$$

on rapidity chambers, square-integrable with respect to $\mathrm{d}\theta_1 \cdots \mathrm{d}\theta_N$ on compactly supported wave packets, whose boundary values across an adjacent chamber wall obey

$$\psi_{\ldots a_i a_{i+1} \ldots}(\ldots, \theta_i, \theta_{i+1}, \ldots) = S^{bc}_{a_i a_{i+1}}(\theta_i - \theta_{i+1})\, \psi_{\ldots cb \ldots}(\ldots, \theta_{i+1}, \theta_i, \ldots).$$

The finite-particle S-Fock space is obtained by imposing the L^2 positive inner product on compactly supported wavefunctions in one fundamental chamber, extending them to all chambers by the displayed relations, and completing the resulting pre-Hilbert space.

Well-defined chamber extension. If the two-body scattering system satisfies unitarity and the Yang–Baxter equation, then the chamber relations in Definition 71.8 determine the values of an S-symmetric wavefunction in every chamber from its value in one fundamental chamber, independently of the chosen adjacent-transposition path.

Crossing a wall that exchanges adjacent rapidities applies the adjoint two-body scattering matrix to the corresponding tensor factors. Two-particle unitarity makes this wall-crossing map invertible and implements $\mathsf{T}_i^2 = 1$. Wall crossings involving disjoint adjacent pairs act on disjoint tensor factors, hence commute. The only remaining ambiguity in a path through adjacent chambers is the braid move $\mathsf{T}_i \mathsf{T}_{i+1} \mathsf{T}_i = \mathsf{T}_{i+1} \mathsf{T}_i \mathsf{T}_{i+1}$, and this is exactly the Yang–Baxter equation for the three involved tensor factors. Therefore every word representing the same permutation gives the same continuation from the fundamental chamber.

71.5 Local Operators and Form Factors

Let $\mathcal{O}(x)$ be a local operator whose matrix elements between the vacuum and ordered out-states exist as distributions. Its n-particle form factor is

$$F^{\mathcal{O}}_{a_1 \ldots a_n}(\theta_1, \ldots, \theta_n) = \langle \Omega | \mathcal{O}(0) Z^\dagger_{a_1}(\theta_1) \cdots Z^\dagger_{a_n}(\theta_n) | \Omega \rangle, \qquad \theta_1 > \cdots > \theta_n.$$

Changing the ordered rapidity chamber changes the scattering basis by the two-body S-matrix.

Remark 71.9 (Watson exchange from ordered asymptotic bases)*.* For adjacent rapidities in a regular chamber, the form-factor boundary values obey

$$F^{\mathcal{O}}_{\ldots a_i a_{i+1} \ldots}(\ldots, \theta_i, \theta_{i+1}, \ldots) = S^{bc}_{a_i a_{i+1}}(\theta_i - \theta_{i+1})\, F^{\mathcal{O}}_{\ldots cb \ldots}(\ldots, \theta_{i+1}, \theta_i, \ldots),$$

with summation over b, c. This statement uses only the factorized scattering normalization of ordered asymptotic bases. The role of locality of $\mathcal{O}$ begins with the cyclicity and semi-locality conditions developed in Chapter 76.

Take the defining matrix element in the chamber $\theta_1 > \cdots > \theta_i > \theta_{i+1} > \cdots > \theta_n$. Applying the ZF exchange relation (71.1) to the adjacent pair gives

$$Z^\dagger_{a_i}(\theta_i) Z^\dagger_{a_{i+1}}(\theta_{i+1}) = S^{bc}_{a_i a_{i+1}}(\theta_i - \theta_{i+1})\, Z^\dagger_c(\theta_{i+1}) Z^\dagger_b(\theta_i).$$

Insert this identity between $\langle\Omega|\mathcal{O}(0)$ and $|\Omega\rangle$, leaving all other ordered creation operators fixed. The resulting matrix element on the right is exactly the boundary value in the neighboring chamber with the adjacent rapidities interchanged. This gives the displayed Watson equation. No commutation of $\mathcal{O}$ through the ZF operator has been used.

Together with Watson exchange, form-factor bootstrap uses crossing, cyclicity, and residue equations at bound-state poles. The exchange equation is a statement about boundary values in neighboring rapidity chambers and the normalization of ordered scattering states; locality of $\mathcal{O}$ enters only in those additional cyclicity and semi-locality conditions. The operator reconstruction problem is to determine which solutions of these functional equations come from genuine local operators in a Hilbert-space QFT. The theorem-boundary data include the local operator domain, positivity, convergence of form-factor expansions, and clustering.

Open Problem 71.10 (Completeness of form-factor reconstruction)*.* Give intrinsic hypotheses on a factorized scattering system and a family of form-factor solutions that imply the existence of a local QFT with those asymptotic particles, those local operators, and Wightman functions obtained by convergent form-factor sums. Existing constructive approaches prove this for important classes of two-dimensional models; the monograph will develop the exact hypotheses and examples in later chapters of this volume.

Chapter 72

Two-Dimensional Scattering Analyticity and Bootstrap Data

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The factorized scattering chapter fixed the algebraic data of an integrable massive theory. This chapter adds the analytic data in the rapidity plane. The object studied here is an on-shell two-body scattering function together with analytic hypotheses. A local QFT realizing that function is a further object; its construction requires Hilbert-space, locality, and domain data.

72.1 Rapidity Plane and Physical Strip

For equal masses m, the two-particle Mandelstam invariant is

$$s(\theta) = 2m^2(1 + \cosh\theta) = 4m^2 \cosh^2\left(\frac{\theta}{2}\right).$$

The physical two-particle scattering line is $\theta \in \mathbb{R}$. The crossed channel is reached by $\theta \mapsto \mathrm{i}\pi - \theta$. Hence the natural analytic domain for a two-body amplitude is the physical strip

$$\mathcal{S}_{\mathrm{phys}} = \{\theta \in \mathbb{C} \mid 0 < \mathrm{Im}\,\theta < \pi\}.$$

For unequal masses m_a, m_b,

$$s_{ab}(\theta) = m_a^2 + m_b^2 + 2m_a m_b \cosh\theta,$$

and the same strip organizes the direct and crossed channels after the species labels are crossed.

Definition 72.1 (Factorized scattering bootstrap data)**.** Factorized scattering bootstrap data are an elastic factorized scattering system of Definition 71.4 together with the following extra data and hypotheses:

(i) meromorphic continuation of each component $S_{ab}^{cd}(\theta)$ to the physical strip $\mathcal{S}_{\mathrm{phys}}$;

(ii) a specified finite or locally finite list of simple-pole locations $\theta = \mathrm{i}u_{ab}^{e}$, $0 < u_{ab}^{e} < \pi$, labelled by proposed direct-channel bound-state species e;

(iii) finite-dimensional multiplicity spaces for these bound-state channels and intertwiners $\Gamma_{ab}^{e,\alpha}$ specifying residues;

(iv) horizontal-strip growth bounds stated as follows: for every closed substrip $K \subset \mathcal{S}_{\mathrm{phys}}$ whose distance from the declared pole set is positive, there are constants C_K, N_K such that

$$\|S_{ab}^{cd}(\theta)\| \leq C_K(1 + |\operatorname{Re}\theta|)^{N_K}, \qquad \theta \in K.$$

Near each declared simple pole the corresponding principal part is subtracted before applying this estimate. When a thermodynamic Bethe-ansatz kernel $\phi_{ab}(\theta) = -\mathrm{i}\partial_\theta \log S_{ab}(\theta)$ is used later, the datum must also specify the branch and the additional integrability hypothesis on ϕ_{ab}, usually $L^1(\mathbb{R})$ after removal of explicitly treated CDD or bound-state factors.

The growth condition in part (iv) is part of the datum, not a placeholder for later arguments. Stronger estimates may be imposed in form-factor or TBA applications, but any such strengthening is an additional hypothesis on top of the polynomial substrip bound displayed above.

The mass of a direct-channel bound state at $\theta = \mathrm{i}u_{ab}^{e}$ is fixed by the pole location:

$$m_e^2 = m_a^2 + m_b^2 + 2m_a m_b \cos u_{ab}^{e}.$$

Indeed, write the incoming momenta as

$$p_a(\theta_a) = m_a(\cosh\theta_a, \sinh\theta_a), \qquad p_b(\theta_b) = m_b(\cosh\theta_b, \sinh\theta_b),$$

with $\theta = \theta_a - \theta_b$. Then

$$-\big(p_a + p_b\big)^2 = m_a^2 + m_b^2 + 2m_a m_b \cosh(\theta_a - \theta_b) = m_a^2 + m_b^2 + 2m_a m_b \cosh\theta.$$

Analytic continuation to the pole $\theta = \mathrm{i}u_{ab}^{e}$ gives $\cosh(\mathrm{i}u_{ab}^{e}) = \cos u_{ab}^{e}$, hence the displayed formula. A one-particle vector of mass m_e in a Hilbert-space QFT is additional data to be constructed or assumed. The pole and residue record the on-shell criterion that such a vector would have to satisfy; a physical-strip pole with the wrong residue sign, or a pole related by crossing rather than the direct-channel residue, is not by itself a new stable direct-channel particle.

72.2 Residues and Fusion

Let V_a denote the internal one-particle space of species a. A simple pole at $\theta = \mathrm{i}u_{ab}^{e}$ is interpreted as a direct-channel bound-state pole only after one specifies a finite-dimensional multiplicity space M_{ab}^{e} and maps

$$\Gamma_{e,\alpha}^{ab} \colon V_e \longrightarrow V_a \otimes V_b, \qquad \Gamma_{ab}^{e,\alpha} \colon V_a \otimes V_b \longrightarrow V_e, \qquad \alpha \in M_{ab}^{e},$$

normalized by

$$\Gamma^{e,\alpha}_{ab}\Gamma^{ab}_{e,\beta} = \delta^\alpha{}_\beta\, \mathbf{1}_{V_e}.$$

With this convention the residue condition is the operator identity

$$-\mathrm{i} \operatorname*{Res}_{\theta=\mathrm{i}u^e_{ab}} S_{ab}(\theta) = \sum_{\alpha\in M^e_{ab}} \Gamma^{ab}_{e,\alpha}\Gamma^{e,\alpha}_{ab} \quad \text{on } V_a\otimes V_b. \tag{72.1}$$

In components this is the factorized residue formula with one map carrying the incoming ab-indices to the e-channel and the other carrying the e-channel back to outgoing ab-indices. Positivity of the Hilbert-space norm fixes the sign in (72.1) after the one-particle normalization has been chosen.

The fusing angles are not independent symbols. They are defined by the complex on-shell momentum identity

$$p_a(\theta+\mathrm{i}\bar u^b_{ae}) + p_b(\theta-\mathrm{i}\bar u^a_{be}) = p_e(\theta), \qquad \bar u^b_{ae}+\bar u^a_{be} = u^e_{ab}.$$

Equivalently,

$$m_a\sin\bar u^b_{ae} = m_b\sin\bar u^a_{be}, \qquad m_e = m_a\cos\bar u^b_{ae} + m_b\cos\bar u^a_{be}.$$

These equations are obtained by expanding the complex boosts:

$$\begin{aligned} p_a(\theta+\mathrm{i}\alpha)+p_b(\theta-\mathrm{i}\beta) = {}&\big(m_a\cos\alpha+m_b\cos\beta\big)(\cosh\theta,\sinh\theta)\\ &+\mathrm{i}\big(m_a\sin\alpha-m_b\sin\beta\big)(\sinh\theta,\cosh\theta), \end{aligned}$$

with $\alpha=\bar u^b_{ae}$ and $\beta=\bar u^a_{be}$. Equality to the real on-shell vector $p_e(\theta)$ forces the imaginary coefficient to vanish and the real coefficient to be m_e. Squaring this real coefficient then recovers the pole-location mass formula because

$$\begin{aligned} m_e^2 &= \big(m_a\cos\alpha+m_b\cos\beta\big)^2\\ &= m_a^2+m_b^2+2m_am_b\cos(\alpha+\beta), \end{aligned}$$

where the last step uses $m_a\sin\alpha=m_b\sin\beta$ to replace $m_a^2\sin^2\alpha+m_b^2\sin^2\beta$ by $2m_am_b\sin\alpha\sin\beta$. Thus the fusing-angle data are not extra kinematics beyond the pole location and the on-shell momentum identity; they are the component form of that identity.

Fusion is the compatibility condition between the residue projection (72.1) and scattering with an additional particle k. With the written composition convention that the rightmost map acts first, the bootstrap equation is the meromorphic identity

$$S_{ek}(\eta) = \sum_{\alpha\in M^e_{ab}} \Gamma^{e,\alpha}_{ab}\, S_{ak}(\eta+\mathrm{i}\bar u^b_{ae})\, S_{bk}(\eta-\mathrm{i}\bar u^a_{be})\, \Gamma^{ab}_{e,\alpha}, \qquad \eta=\theta-\theta_k.$$

The Yang–Baxter equation is the statement that fusing before or after an allowed adjacent reordering of three particles gives the same meromorphic operator.

Proposition 72.2 (Fusion identity as a residue identity)**.** *Assume that the three-particle scattering amplitude has meromorphic factorization on the rapidity chamber under consideration and that the pair (a,b) has a simple direct-channel pole at $\theta_a-\theta_b=\mathrm{i}u^e_{ab}$ with residue (72.1). Assume also that, on the small contour used below, no pole of the ak or bk two-body factors is met. Choose the bound-state rapidity θ so that the two constituent rapidities are*

$$\theta_a=\theta+\mathrm{i}\bar u^b_{ae}, \qquad \theta_b=\theta-\mathrm{i}\bar u^a_{be}, \qquad \bar u^b_{ae}+\bar u^a_{be}=u^e_{ab}.$$

Then the residue of the factorized three-particle amplitude at the (a,b)*-pole gives*

$$S_{ek}(\theta-\theta_k)=\sum_{\alpha\in M^e_{ab}}\Gamma^{e,\alpha}_{ab}\,S_{ak}(\theta-\theta_k+\mathrm{i}\bar u^b_{ae})\,S_{bk}(\theta-\theta_k-\mathrm{i}\bar u^a_{be})\,\Gamma^{ab}_{e,\alpha}$$

as an identity on the bound-state sector, after the one-particle normalization used in the residue has been fixed. If the multiplicity space has dimension greater than one, the displayed expression is the trace over the chosen residue basis; equivalently one may keep the corresponding matrix-valued $S^{\alpha\beta}_{ek}$*.*

Proof. Put

$$\eta=\theta-\theta_k,\qquad \zeta=\theta_a-\theta_b,$$

and, near the pole $\zeta=\mathrm{i}u^e_{ab}$, parametrize the two constituent rapidities by

$$\theta_a(\zeta)=\theta+\mathrm{i}\bar u^b_{ae}+\frac{\zeta-\mathrm{i}u^e_{ab}}{2},\qquad \theta_b(\zeta)=\theta-\mathrm{i}\bar u^a_{be}-\frac{\zeta-\mathrm{i}u^e_{ab}}{2}.$$

Thus $\theta_a(\zeta)-\theta_b(\zeta)=\zeta$, while at the pole the two continued momenta add to $p_e(\theta)$. Choose a small positively oriented circle C_e around $\mathrm{i}u^e_{ab}$ on which no singularity of the factors $S_{ak}(\theta_a(\zeta)-\theta_k)$ or $S_{bk}(\theta_b(\zeta)-\theta_k)$ is encountered. The pole extraction is the contour operation

$$-\mathrm{i}\operatorname*{Res}_{\zeta=\mathrm{i}u^e_{ab}}A(\zeta)=-\frac{\mathrm{i}}{2\pi\mathrm{i}}\oint_{C_e}A(\zeta)\,d\zeta.$$

In the unresolved three-particle channel, factorization gives a meromorphic amplitude of the form

$$A_{ab|k}(\zeta)=S_{ak}(\theta_a(\zeta)-\theta_k)\,S_{bk}(\theta_b(\zeta)-\theta_k)\,S_{ab}(\zeta),$$

up to the common convention of writing operators in the order in which they act. The first two factors are holomorphic inside C_e. Hence Cauchy's residue theorem and (72.1) give

$$-\mathrm{i}\operatorname*{Res}_{\zeta=\mathrm{i}u^e_{ab}}A_{ab|k}(\zeta)=S_{ak}(\eta+\mathrm{i}\bar u^b_{ae})\,S_{bk}(\eta-\mathrm{i}\bar u^a_{be})\sum_\alpha\Gamma^{ab}_{e,\alpha}\Gamma^{e,\alpha}_{ab}.$$

Project this residue onto a fixed bound-state multiplicity component by applying $\Gamma^{e,\alpha}_{ab}$ on the outgoing ab-pair and $\Gamma^{ab}_{e,\alpha}$ on the incoming e-channel. Using $\Gamma^{e,\alpha}_{ab}\Gamma^{ab}_{e,\beta}=\delta^\alpha{}_\beta\mathbf{1}_{V_e}$, the unresolved residue becomes

$$\Gamma^{e,\alpha}_{ab}\,S_{ak}(\eta+\mathrm{i}\bar u^b_{ae})\,S_{bk}(\eta-\mathrm{i}\bar u^a_{be})\,\Gamma^{ab}_{e,\alpha}.$$

The same contour, evaluated after resolving the pole as a one-particle intermediate state, is by definition the scattering amplitude $S^{(\alpha)}_{ek}(\eta)$ of that bound-state component with k. Since both calculations are residues of the same meromorphic three-particle amplitude and no other singularity crossed the contour, the two expressions are equal. Summing over the multiplicity basis gives the displayed fusion identity. The proof uses only meromorphic factorization, the residue normalization, and the existence of the one-particle e-channel; it does not use an off-shell Lagrangian. □

The noncommuting algebra in this residue step is finite-dimensional. If $P_e=\sum_\alpha\Gamma^{ab}_{e,\alpha}\Gamma^{e,\alpha}_{ab}$ and

$$S_{ab}(\zeta)=\frac{\mathrm{i}P_e}{\zeta-\mathrm{i}u^e_{ab}}+S^{\mathrm{reg}}_{ab}(\zeta),\qquad B(\zeta)=S_{ak}(\theta_a(\zeta)-\theta_k)S_{bk}(\theta_b(\zeta)-\theta_k),$$

with B holomorphic on C_e, then

$$-\mathrm{i} \operatorname*{Res}_{\zeta=\mathrm{i}u_{ab}^e} B(\zeta)S_{ab}(\zeta) = B(\mathrm{i}u_{ab}^e)P_e.$$

Projecting to the resolved bound-state coordinate gives

$$\Gamma_{ab}^{e,\alpha}B(\mathrm{i}u_{ab}^e)P_e\Gamma_{e,\beta}^{ab} = \Gamma_{ab}^{e,\alpha}B(\mathrm{i}u_{ab}^e)\Gamma_{e,\beta}^{ab}.$$

Thus derivatives or regular coefficients of the ak and bk factors do not enter this simple-pole residue, while the sign in (72.1) fixes the sign of the fused amplitude. The companion script `calculation-checks/integrable_scattering_bootstrap_checks.py` checks this point in an exact rational finite-rank model with a nonorthogonal projection and noncommuting holomorphic scattering factor.

72.3 Minimal Solutions and CDD Factors

For diagonal scattering, where $S_{ab}^{cd}(\theta) = \delta_a^c\delta_b^d S_{ab}(\theta)$, unitarity and crossing become

$$S_{ab}(\theta)S_{ab}(-\theta) = 1, \qquad S_{ab}(\theta) = S_{\bar{a}b}(\mathrm{i}\pi - \theta).$$

A scalar meromorphic factor $C(\theta)$ satisfying the same two equations and having no assigned particle pole is a CDD factor. Thus the analytic constraints and the bound-state list determine the amplitude only after an additional growth or minimality condition is specified.

Definition 72.3 (Elementary scalar block)**.** For $x \in \mathbb{C}$, the elementary scalar block is

$$[x]_\theta = \frac{\sinh((\theta + \mathrm{i}\pi x)/2)}{\sinh((\theta - \mathrm{i}\pi x)/2)},$$

as a meromorphic function of θ.

This block is the basic unitary zero-pole pair used in diagonal bootstrap formulae. Products of such blocks place zeros and poles at prescribed locations. The bootstrap construction problem is to choose the product and possible CDD factors so that unitarity, crossing, physical-strip pole assignments, residue signs, and asymptotic bounds hold simultaneously.

Remark 72.4 (Elementary block algebra). The elementary scalar block satisfies

$$[x]_\theta[x]_{-\theta} = 1.$$

Its poles are

$$\theta = \mathrm{i}\pi x + 2\pi\mathrm{i}n, \qquad n \in \mathbb{Z},$$

and its zeros are

$$\theta = -\mathrm{i}\pi x + 2\pi\mathrm{i}n, \qquad n \in \mathbb{Z}.$$

For real $0 < x < 1$, the pole at $\theta = \mathrm{i}\pi x$ lies in the physical strip.

The displayed properties follow immediately from the zero set of $\sinh z$ and from $\sinh(-z) = -\sinh z$. The point worth keeping in the main text is the residue sign. At the physical-strip pole $\theta = \mathrm{i}\pi x$, $0 < x < 1$,

$$\operatorname*{Res}_{\theta=\mathrm{i}\pi x}[x]_\theta = 2\mathrm{i}\sin(\pi x), \qquad -\mathrm{i}\operatorname*{Res}_{\theta=\mathrm{i}\pi x}[x]_\theta = 2\sin(\pi x) > 0.$$

Thus a single block has the residue sign compatible with a direct-channel scalar bound-state pole in the convention (72.1). Crossing is a separate constraint. The elementary relation

$$[x]_{\mathrm{i}\pi-\theta} = -[1-x]_\theta$$

shows that the crossing-symmetric scalar pair

$$\mathcal{C}_x(\theta) := [x]_\theta[1-x]_\theta$$

is unitary and crossing invariant. For $0 < x < 1$ with $x \neq 1/2$, it has physical-strip poles at $\theta = \mathrm{i}\pi x$ and $\theta = \mathrm{i}\pi(1-x)$, but their residue signs are opposite:

$$-\mathrm{i}\operatorname*{Res}_{\theta=\mathrm{i}\pi x}\mathcal{C}_x(\theta) = -2\tan(\pi x), \qquad -\mathrm{i}\operatorname*{Res}_{\theta=\mathrm{i}\pi(1-x)}\mathcal{C}_x(\theta) = 2\tan(\pi x). \tag{72.2}$$

Equation (72.2) is a useful diagnostic: a CDD zero-pole pair can satisfy scalar unitarity and crossing while still failing the positivity sign for one of its physical-strip poles. The particle interpretation of a pole therefore uses the full residue datum, not only its position in the strip. The companion check `calculation-checks/integrable_scattering_bootstrap_checks.py` verifies the scalar-block identities and the two residue signs without printing the elementary hyperbolic cancellation in the chapter.

72.4 Hilbert-Space Boundary

The bootstrap data are on-shell information. A local QFT realization also requires a Hilbert space, Wightman distributions, local operator domains, and cluster properties. In constructive treatments of factorizing models, wedge- localized fields and modular localization provide one route from an admissible S-matrix to local algebras. The theorem boundary is precise: analyticity and pole data enter as hypotheses for constructing wedge operators and then proving that double-cone intersections are nontrivial.

Open Problem 72.5 (Bootstrap completeness for local algebras). Characterize the bootstrap data for which the resulting wedge-local construction has nontrivial local algebras, a unique vacuum, asymptotic completeness, and local fields whose form factors solve the equations of Chapter 71. This problem is the operator-algebraic completion of the on-shell bootstrap.

Chapter 73

Yang–Baxter Consistency and Internal Symmetry

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The two-body bootstrap data of Chapter 72 becomes a many-particle scattering theory only if different decompositions of the same multi-particle process agree. This chapter formulates that requirement as an identity of meromorphic maps on species spaces and explains how internal symmetry reduces the identity to scalar functions on irreducible channels.

Definition 73.1 (Internal two-body R-matrix specification)**.** An internal two-body R-matrix specification is a tuple

$$\mathfrak{R} = \big(\mathcal{I}, \{m_a, V_a\}_{a\in\mathcal{I}}, \{S_{ab}\}_{a,b\in\mathcal{I}}, \iota, \{C_a\}_{a\in\mathcal{I}}, \mathcal{A}, \{\rho_a\}_{a\in\mathcal{I}}\big)$$

with the following entries.

(i) $\mathcal{I}$ is the set of stable particle species, $m_a > 0$ is the mass of species a, and V_a is the finite-dimensional complex internal space carried by a one-particle state of species a.

(ii) $S_{ab}(\theta) : V_a \otimes V_b \to V_b \otimes V_a$ is a meromorphic two-body scattering map in the rapidity difference $\theta = \theta_a - \theta_b$. Its R-matrix representative is $R_{ab}(\theta) = P_{ab} S_{ab}(\theta)$, where $P_{ab} : V_b \otimes V_a \to V_a \otimes V_b$ is the ordinary flip.

(iii) $\iota : \mathcal{I} \to \mathcal{I}$ is the antiparticle involution, written $a \mapsto \bar{a}$, and $C_a : V_a^\vee \to V_{\bar{a}}$ fixes the crossing identification used to compare physical-strip boundary values.

(iv) $\mathcal{A}$ is an optional internal symmetry algebra or group, and ρ_a is its representation on V_a. The specification is $\mathcal{A}$-invariant when each $R_{ab}(\theta)$ belongs to $\mathrm{End}_{\mathcal{A}}(V_a \otimes V_b)$.

The specification is called a regular factorized scattering specification when, after undoing the flip in item (ii), it obeys the physical unitarity, real analyticity, crossing, residue, and growth requirements of Definition 72.1, and when the Yang–Baxter identity of (73.1) below holds as an identity of meromorphic maps on every triple species space. The terminology is deliberately on-shell: this specification is not a construction of local fields or local algebras.

73.1 Species Spaces and Two-Body Operators

Let $\mathfrak{R}$ be an internal two-body specification in the sense of Definition 73.1. The one-particle momentum variable of a particle of mass m_a is the rapidity θ, with

$$p_a(\theta) = m_a(\cosh\theta, \sinh\theta).$$

For ordered species a, b, a two-body scattering operator is a meromorphic map

$$S_{ab}(\theta) : V_a \otimes V_b \longrightarrow V_b \otimes V_a, \qquad \theta = \theta_a - \theta_b.$$

The codomain order is part of the definition: after scattering in one spatial dimension the outgoing ordered particles have exchanged their positions.

It is often convenient to compose S_{ab} with the ordinary flip $P_{ab} : V_b \otimes V_a \to V_a \otimes V_b$ and define

$$R_{ab}(\theta) = P_{ab}S_{ab}(\theta) : V_a \otimes V_b \longrightarrow V_a \otimes V_b.$$

The S-operator records physical ordering, while R records an endomorphism on a fixed tensor product. All equations below are stated in the R-matrix convention because the domains and codomains are then identical.

Regularity includes the inverse relation

$$R_{ab}(\theta)R_{ab}(-\theta) = \mathrm{id}_{V_a \otimes V_b}$$

on every common domain of holomorphy. This is the fixed-order version of two-body unitarity; the remaining physical constraints are the crossing and residue conditions imposed on S_{ab} before applying the flip.

73.2 Derivation of the Yang–Baxter Equation

Consider three particles with rapidities $\theta_1 > \theta_2 > \theta_3$ and species spaces $V_{a_1}, V_{a_2}, V_{a_3}$. Factorization asserts that the full three-particle scattering is a product of two-body scatterings. There are two reduced decompositions of the permutation reversing the order of the three particles:

$$(12)(23)(12) = (23)(12)(23).$$

The equality of the resulting linear maps on

$$V_{a_1} \otimes V_{a_2} \otimes V_{a_3}$$

is therefore the condition that factorized scattering be independent of the chosen reduced decomposition.

Write $\theta_{ij} = \theta_i - \theta_j$. Let R_{ij} denote the endomorphism R acting on the i-th and j-th tensor factors and as the identity on the remaining factor. Equality of the two decompositions is

$$R_{12}(\theta_{12})R_{13}(\theta_{13})R_{23}(\theta_{23}) = R_{23}(\theta_{23})R_{13}(\theta_{13})R_{12}(\theta_{12}).$$

Since $\theta_{13} = \theta_{12} + \theta_{23}$, this is the additive spectral-parameter Yang–Baxter equation

$$R_{12}(\theta)R_{13}(\theta + \phi)R_{23}(\phi) = R_{23}(\phi)R_{13}(\theta + \phi)R_{12}(\theta). \tag{73.1}$$

Three-particle consistency. For a regular two-body R-system, the factorized three-particle scattering operator is independent of the reduced decomposition of the three-particle permutation if and only if the Yang–Baxter equation holds as a meromorphic identity on every triple species space.

Each adjacent transposition is represented by the corresponding two-body operator with the rapidity difference of the two lines at the moment they cross. In the R-matrix convention this representation acts on a fixed tensor product. The two reduced decompositions of the longest element of the three-particle permutation group give exactly the two sides of the displayed Yang–Baxter equation. Thus independence of the decomposition gives the equation. Conversely, if the equation holds, the two possible three-particle factorizations define the same meromorphic map. The remaining relations of the permutation group are the inverse relation, supplied by unitarity, and commutation of disjoint transpositions, which is automatic because the operators act on disjoint tensor factors.

Many-particle consistency. For n ordered particles the longest permutation, and more generally every permutation, has many reduced words in the adjacent transpositions s_i. The only local moves relating reduced words in the symmetric group are

$$s_i s_j = s_j s_i \quad (|i - j| > 1), \qquad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}.$$

The first move is represented by commuting operators on disjoint tensor factors. The second move is represented by (73.1) with the appropriate rapidity differences. Thus the Yang–Baxter identity is the local compatibility condition whose repeated use makes the full factorized n-body scattering map independent of the chosen reduced word. This argument uses only the Coxeter presentation of the symmetric group and the meromorphic identity (73.1); it does not use perturbation theory.

73.3 Spectral-Parameter Conventions and Classical Limit

The rapidity variable gives the additive convention: two-body amplitudes depend on $\theta_i - \theta_j$, and the middle argument in (73.1) is the sum $\theta + \phi$. In lattice-model and quantum-group conventions one often uses a multiplicative parameter

$$x = e^{\theta},$$

so that the same identity reads

$$R_{12}(x)R_{13}(xy)R_{23}(y) = R_{23}(y)R_{13}(xy)R_{12}(x).$$

The two forms are equivalent after choosing a logarithm on a simply connected domain avoiding the singularities of the meromorphic R-matrix. The choice matters in crossing formulas, because crossing shifts rapidities by imaginary half-periods; it does not change the algebraic content of the Yang–Baxter identity itself.

There is also a braid convention. If

$$\check{R}_{ab}(\theta) = S_{ab}(\theta) \quad \text{as a map } V_a \otimes V_b \to V_b \otimes V_a,$$

then the same consistency statement is a braid relation for maps whose codomains are reordered after each crossing. The $R = P\check{R}$ convention used in this chapter trades that geometric reordering for endomorphisms on a fixed tensor product. Confusing these two conventions is a common source of incorrect sign and argument order in component calculations.

Classical expansion. Suppose a family of solutions has a formal parameter $\hbar$ and admits an expansion

$$R_{ab}(\theta;\hbar) = \mathrm{id} + \hbar\, r_{ab}(\theta) + \hbar^2 q_{ab}(\theta) + O(\hbar^3)$$

as a meromorphic endomorphism of $V_a \otimes V_b$. Substitution into (73.1) gives no condition at order $\hbar$: both sides contain

$$r_{12}(\theta) + r_{13}(\theta+\phi) + r_{23}(\phi).$$

At order $\hbar^2$, the q-terms also cancel between the two sides, while the ordered products of the first-order terms give

$$[r_{12}(\theta), r_{13}(\theta+\phi)] + [r_{12}(\theta), r_{23}(\phi)] + [r_{13}(\theta+\phi), r_{23}(\phi)] = 0. \tag{73.2}$$

This is the spectral-parameter classical Yang–Baxter equation. It is a necessary condition for a classical r-matrix to be quantized into a factorized scattering R-matrix with the stated expansion. It is not, by itself, a sufficiency theorem: quantization also requires the higher $\hbar$-orders, analytic constraints, and the physical unitarity/crossing conditions of the scattering problem.

73.4 Internal Symmetry and Projector Decomposition

Let G be a group or Lie algebra acting on the species spaces by representations $\rho_a : G \to GL(V_a)$. Internal-symmetry invariance of the two-body scattering system means

$$R_{ab}(\theta)(\rho_a(g) \otimes \rho_b(g)) = (\rho_a(g) \otimes \rho_b(g))R_{ab}(\theta), \qquad g \in G.$$

Equivalently, $R_{ab}(\theta)$ is an element of

$$\mathrm{End}_G(V_a \otimes V_b),$$

the algebra of G-intertwiners. If

$$V_a \otimes V_b \simeq \bigoplus_\lambda M_{ab}^\lambda \otimes W_\lambda$$

is the decomposition into irreducible G-modules W_λ with multiplicity spaces M_{ab}^λ, then Schur's lemma gives

$$\mathrm{End}_G(V_a \otimes V_b) \simeq \bigoplus_\lambda \mathrm{End}(M_{ab}^\lambda).$$

If every multiplicity space is one-dimensional, the most general invariant R-matrix has the form

$$R_{ab}(\theta) = \sum_\lambda r_{ab}^\lambda(\theta) P_\lambda,$$

where P_λ is the projector onto the irreducible channel λ, and r_{ab}^λ is a scalar meromorphic function.

Example 73.2 (Vector representation of $O(N)$)**.** Let $V = \mathbb{C}^N$ be the vector representation of $O(N)$. The tensor product decomposes as

$$V \otimes V = \mathbf{1} \oplus \Lambda^2 V \oplus \mathrm{Sym}_0^2 V.$$

An $O(N)$-invariant two-body R-matrix without multiplicity mixing is therefore

$$R(\theta) = r_0(\theta)P_0 + r_A(\theta)P_A + r_T(\theta)P_T,$$

with projectors onto the singlet, antisymmetric, and traceless-symmetric channels. The Yang–Baxter equation becomes a coupled system of functional equations for r_0, r_A, r_T, together with the analytic constraints of the bootstrap data.

73.5 Algebraic Origin of the Matrix Identity

The Yang–Baxter equation can be produced from an algebraic object before any scattering interpretation is imposed. Let $\mathcal{A}$ be an associative algebra with coproduct

$$\Delta : \mathcal{A} \longrightarrow \mathcal{A} \otimes \mathcal{A}$$

and opposite coproduct $\Delta^{\mathrm{op}} = \tau \circ \Delta$, where τ is the tensor flip. A universal R-element $\mathcal{R}$ is an invertible element of a suitable completion of $\mathcal{A} \otimes \mathcal{A}$ satisfying

$$\mathcal{R}\,\Delta(x) = \Delta^{\mathrm{op}}(x)\,\mathcal{R}, \qquad x \in \mathcal{A},$$

and

$$(\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \qquad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

Then $\mathcal{R}$ obeys the algebraic Yang–Baxter identity

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}. \tag{73.3}$$

Indeed,

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{12}(\Delta \otimes \mathrm{id})(\mathcal{R}) = (\Delta^{\mathrm{op}} \otimes \mathrm{id})(\mathcal{R})\mathcal{R}_{12} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}.$$

The middle equality is the intertwining relation applied in the first two tensor factors to each tensor component of $\mathcal{R}$. Applying representations $\rho_a \otimes \rho_b \otimes \rho_c$ gives an ordinary matrix solution on $V_a \otimes V_b \otimes V_c$.

For scattering, this algebraic mechanism is only a source of candidate matrix parts. One must still provide the rapidity dependence, scalar CDD factors, physical-strip pole interpretation, crossing identification, and Hilbert-space realization. The algebraic equation (73.3) explains why families such as rational, trigonometric, and elliptic R-matrices appear repeatedly; it does not replace the QFT construction problem.

73.6 Boundaries and Reflection Consistency

On a half-line or in the presence of a reflecting boundary, a one-particle reflection matrix $K_a(\theta) : V_a \to V_a$ must be compatible with the same bulk R-matrix. For two particles with rapidities θ_1, θ_2, there are two ways to move both particles to the boundary, reflect them, and restore the order. Equality of the two maps on $V_{a_1} \otimes V_{a_2}$ gives the boundary Yang–Baxter equation

$$R_{12}(\theta_1 - \theta_2)K_1(\theta_1)R_{21}(\theta_1 + \theta_2)K_2(\theta_2) = K_2(\theta_2)R_{12}(\theta_1 + \theta_2)K_1(\theta_1)R_{21}(\theta_1 - \theta_2). \tag{73.4}$$

Here $R_{21} = P_{12}R_{12}P_{12}$, and K_i acts on the i-th tensor factor. The argument $\theta_1 + \theta_2$ occurs because after one reflection the second collision is between rapidities of opposite sign. Boundary integrability therefore requires more than choosing a bulk factorized R-matrix; it requires reflection data satisfying (73.4), boundary unitarity, boundary crossing, and a boundary pole interpretation. Those additional analytic requirements are the boundary analogue of the bulk bootstrap constraints.

73.7 What the Equation Enforces

The Yang–Baxter equation is the compatibility condition for factorized multi-particle on-shell scattering. It does not by itself construct local operator algebras, form factors, or a Hilbert-space QFT. The next chapter adds the form-factor problem, where local operators are represented by meromorphic functions satisfying exchange, crossing, residue, and growth conditions.

Open Problem 73.3 (From R-matrices to local fields)*.* Given a meromorphic R-system satisfying unitarity, crossing, the Yang–Baxter equation, residue positivity, and stated growth bounds, construct or obstruct a local QFT realization. The construction must identify a Hilbert space, wedge-local or local algebras, form factors of a dense set of local operators, and the topology in which locality and clustering are verified.

Chapter 74

Algebraic Bethe Ansatz and Transfer Matrices

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Chapter 73 derived the Yang–Baxter equation as the condition that factorized three-particle scattering be independent of the chosen reduced decomposition. This chapter develops the finite-dimensional algebraic consequence needed for non-diagonal finite-volume spectra. The input is an R-matrix satisfying the Yang–Baxter equation. The output is a commuting family of transfer matrices and, in the rational $SU(2)$ case, explicit Bethe equations for their simultaneous eigenvectors.

74.1 Auxiliary Space, Quantum Space, and Monodromy

Let $V \simeq \mathbb{C}^d$ be a finite-dimensional complex vector space. In spin chains V is the one-site Hilbert space. In finite-volume integrable QFT it is the internal species space transported through a sequence of two-body scatterings. Let

$$R(u) \in \mathrm{End}(V \otimes V), \qquad u \in \mathbb{C},$$

be a meromorphic endomorphism. The parameter u is called the spectral parameter. For three tensor factors, $R_{ij}(u)$ denotes $R(u)$ acting on the i-th and j-th factors and the identity on the remaining factor. The Yang–Baxter equation is

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{74.1}$$

Fix a chain length L. The quantum space is

$$\mathcal{H}_L = V_1 \otimes \cdots \otimes V_L, \qquad V_j \simeq V.$$

An additional copy $V_0 \simeq V$ is called the auxiliary space. For complex inhomogeneities $\xi_1, \ldots, \xi_L$, define the monodromy matrix

$$T_0(u) = R_{0L}(u - \xi_L) \cdots R_{02}(u - \xi_2) R_{01}(u - \xi_1) \in \mathrm{End}(V_0 \otimes \mathcal{H}_L). \tag{74.2}$$

The order is part of the convention. A different order conjugates or relabels later formulae but does not change the transfer-matrix commutativity argument.

Proposition 74.1 (RTT relation)**.** *Assume $R(u)$ satisfies (74.1). Let V_0 and $V_{0'}$ be two auxiliary copies of V. Then*

$$R_{00'}(u - v) T_0(u) T_{0'}(v) = T_{0'}(v) T_0(u) R_{00'}(u - v) \tag{74.3}$$

as meromorphic endomorphisms of $V_0 \otimes V_{0'} \otimes \mathcal{H}_L$.

Proof. Write

$$T_0(u) T_{0'}(v) = \prod_{j=L}^{1} R_{0j}(u - \xi_j) \prod_{j=L}^{1} R_{0'j}(v - \xi_j),$$

where products decrease in j. The Yang–Baxter equation applied to $V_0 \otimes V_{0'} \otimes V_j$ gives

$$R_{00'}(u - v) R_{0j}(u - \xi_j) R_{0'j}(v - \xi_j) = R_{0'j}(v - \xi_j) R_{0j}(u - \xi_j) R_{00'}(u - v).$$

Operators with distinct quantum labels commute unless they share an auxiliary factor. Move $R_{00'}(u - v)$ successively through the pair $R_{0j}(u - \xi_j) R_{0'j}(v - \xi_j)$ for $j = L, L - 1, \ldots, 1$. After the last move all $R_{0'j}$ factors stand to the left of all R_{0j} factors in the order defining $T_{0'}(v) T_0(u)$, and the auxiliary $R_{00'}(u - v)$ stands at the right. This is (74.3). □

74.2 Commuting Transfer Matrices

The transfer matrix is the auxiliary trace of the monodromy:

$$t(u) = \mathrm{tr}_{V_0} T_0(u) \in \mathrm{End}(\mathcal{H}_L). \tag{74.4}$$

The trace is the ordinary finite-dimensional trace on V_0. No trace is taken over $\mathcal{H}_L$.

Proposition 74.2 (Transfer-matrix commutativity)**.** *Assume $R(u)$ satisfies the Yang–Baxter equation and is invertible for generic u. Then, for all u, v in the common domain of meromorphy,*

$$[t(u), t(v)] = 0.$$

Proof. The trace in (74.4) is a partial trace. If

$$X \in \mathrm{End}(V_0 \otimes V_{0'} \otimes \mathcal{H}_L),$$

then $\mathrm{tr}_{0,0'} X$ is the unique endomorphism of $\mathcal{H}_L$ satisfying

$$\mathrm{tr}_{\mathcal{H}_L}\big(Y \ \mathrm{tr}_{0,0'} X\big) = \mathrm{tr}_{0,0',\mathcal{H}_L}\big((1_{0,0'} \otimes Y) X\big)$$

for finite-rank test operators Y on $\mathcal{H}_L$. In components this definition preserves the order of the quantum-space endomorphisms, so

$$t(u)t(v) = \mathrm{tr}_{0,0'}(T_0(u)T_{0'}(v)), \qquad t(v)t(u) = \mathrm{tr}_{0,0'}(T_{0'}(v)T_0(u)).$$

The proof therefore uses cyclicity only in the finite auxiliary trace, never cyclicity in the quantum space.

On the open set where $R_{00'}(u-v)$ is invertible, the RTT relation gives

$$T_0(u)T_{0'}(v) = R_{00'}(u-v)^{-1}T_{0'}(v)T_0(u)R_{00'}(u-v).$$

Since $R_{00'}(u-v)$ acts trivially on $\mathcal{H}_L$, it may be moved cyclically inside the finite trace over $V_0 \otimes V_{0'}$:

$$\begin{aligned} t(u)t(v) &= \mathrm{tr}_{0,0'}(T_0(u)T_{0'}(v)) \\ &= \mathrm{tr}_{0,0'}\big(R_{00'}(u-v)^{-1}T_{0'}(v)T_0(u)R_{00'}(u-v)\big) \\ &= \mathrm{tr}_{0,0'}(T_{0'}(v)T_0(u)) = t(v)t(u). \end{aligned}$$

Both sides are meromorphic $\mathrm{End}(\mathcal{H}_L)$-valued functions of u, v. The complement of the auxiliary invertibility locus is a proper analytic subset, so equality on the nonempty open invertibility domain extends as a meromorphic identity on the common domain of meromorphy. □

The proposition is the algebraic source of the conservation laws used in integrable finite-volume problems. Expanding $t(u)$ or $\log t(u)$ around a regular point gives commuting operators. Which one is the Hamiltonian is a choice of representation and normalization, not part of the Yang–Baxter equation alone.

74.3 The Rational SU(2) R-Matrix

Let $V = \mathbb{C}^2$ with basis $|\uparrow\rangle, |\downarrow\rangle$, and let $P \in \mathrm{End}(V \otimes V)$ be the flip

$$P(x \otimes y) = y \otimes x.$$

The rational $SU(2)$ R-matrix in the normalization used below is

$$R(u) = u\,\mathrm{id}_{V\otimes V} + \mathrm{i}P.$$

It is $SU(2)$-invariant and satisfies the Yang–Baxter equation. One direct verification uses only

$$P_{ij}^2 = 1, \qquad P_{12}P_{23}P_{12} = P_{23}P_{12}P_{23},$$

and the commutativity of disjoint flips; expanding both sides of (74.1) gives identical words in the symmetric group algebra of S_3.

For the homogeneous spin chain it is convenient to use the shifted local operator

$$L_{0j}(u) = R_{0j}\left(u - \frac{\mathrm{i}}{2}\right) = \left(u - \frac{\mathrm{i}}{2}\right)\mathrm{id} + \mathrm{i}P_{0j}.$$

The monodromy is

$$T_0(u) = L_{0L}(u)\cdots L_{01}(u).$$

Relative to the auxiliary basis, write

$$T_0(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix},$$

where A, B, C, D are operators on $\mathcal{H}_L$. The ferromagnetic reference vector is

$$\Omega_L = |\uparrow\rangle^{\otimes L}.$$

The local triangular form of $L_{0j}(u)$ on $|\uparrow\rangle_j$ gives

$$C(u)\Omega_L = 0, \qquad A(u)\Omega_L = \left(u + \frac{\mathrm{i}}{2}\right)^L \Omega_L, \qquad D(u)\Omega_L = \left(u - \frac{\mathrm{i}}{2}\right)^L \Omega_L.$$

74.4 Creation Operators and Bethe Equations

The RTT relation gives commutation relations among A, B, C, D. The relations needed to compute $t(u) = A(u) + D(u)$ on a Bethe vector are

$$B(u)B(v) = B(v)B(u), \tag{74.5}$$

$$A(u)B(v) = \frac{u - v - \mathrm{i}}{u - v} B(v)A(u) + \frac{\mathrm{i}}{u - v} B(u)A(v), \tag{74.6}$$

$$D(u)B(v) = \frac{u - v + \mathrm{i}}{u - v} B(v)D(u) - \frac{\mathrm{i}}{u - v} B(u)D(v). \tag{74.7}$$

The signs in (74.6)–(74.7) are tied to the convention $T_0(u) = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, $R(u) = u\,1 + \mathrm{i}P$, and $L(u) = R(u - \mathrm{i}/2)$. In this convention B lowers one spin relative to the ferromagnetic vector, $A(u)\Omega_L = (u + \mathrm{i}/2)^L \Omega_L$, and $D(u)\Omega_L = (u - \mathrm{i}/2)^L \Omega_L$. For a single site the relations reduce to

$$A(u)B = (u - \mathrm{i}/2)B, \qquad BA(u) = (u + \mathrm{i}/2)B,$$

and similarly $D(u)B = (u + \mathrm{i}/2)B$, $BD(u) = (u - \mathrm{i}/2)B$; this one-site check already fixes the displayed signs. For general L the same relations are the component form of RTT, obtained by inserting auxiliary matrix units $E^{(0)}_{ab} E^{(0')}_{cd}$ in (74.3) and comparing the corresponding $\mathrm{End}(\mathcal{H}_L)$-valued coefficients. The apparent poles at $u = v$ are not singularities of the operator identity after multiplication by $u - v$; they encode the rational R-matrix normalization. For complex numbers $u_1, \ldots, u_M$, define

$$|u_1, \ldots, u_M\rangle = B(u_1) \cdots B(u_M)\Omega_L.$$

Theorem 74.3 (Algebraic Bethe ansatz for the $XXX_{1/2}$ chain)**.** *Suppose the numbers $u_1, \ldots, u_M$ are distinct and satisfy*

$$\left(\frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2}\right)^L = \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k + \mathrm{i}}{u_j - u_k - \mathrm{i}}, \qquad j = 1, \ldots, M. \tag{74.8}$$

Then $|u_1, \ldots, u_M\rangle$, if nonzero, is an eigenvector of $t(u)$ with eigenvalue

$$\Lambda(u) = \left(u + \frac{\mathrm{i}}{2}\right)^L \prod_{j=1}^{M} \frac{u - u_j - \mathrm{i}}{u - u_j} + \left(u - \frac{\mathrm{i}}{2}\right)^L \prod_{j=1}^{M} \frac{u - u_j + \mathrm{i}}{u - u_j}. \tag{74.9}$$

Proof. Let

$$a(u) = \left(u + \frac{\mathrm{i}}{2}\right)^L, \qquad d(u) = \left(u - \frac{\mathrm{i}}{2}\right)^L.$$

The exact normal-ordering identities needed below are

$$\begin{aligned} A(u) \prod_{k=1}^{M} B(u_k)\Omega_L &= a(u) \prod_{k=1}^{M} \frac{u - u_k - \mathrm{i}}{u - u_k} \prod_{k=1}^{M} B(u_k)\Omega_L \\ &\quad + \sum_{j=1}^{M} \frac{\mathrm{i}\, a(u_j)}{u - u_j} \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k - \mathrm{i}}{u_j - u_k} \left(B(u) \prod_{\substack{k=1 \\ k \neq j}}^{M} B(u_k)\Omega_L \right). \end{aligned} \tag{74.10}$$

$$\begin{aligned} D(u) \prod_{k=1}^{M} B(u_k)\Omega_L &= d(u) \prod_{k=1}^{M} \frac{u - u_k + \mathrm{i}}{u - u_k} \prod_{k=1}^{M} B(u_k)\Omega_L \\ &\quad - \sum_{j=1}^{M} \frac{\mathrm{i}\, d(u_j)}{u - u_j} \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k + \mathrm{i}}{u_j - u_k} \left(B(u) \prod_{\substack{k=1 \\ k \neq j}}^{M} B(u_k)\Omega_L \right). \end{aligned} \tag{74.11}$$

For $M = 1$ these are exactly (74.6) and (74.7) applied to Ω_L. The induction step moves $A(u)$, respectively $D(u)$, through the leftmost B-factor. The wanted part gives the first displayed product factor. The unwanted part changes the moving diagonal entry to $A(u_j)$, respectively $D(u_j)$, and the remaining B-factors are then passed only by wanted moves; if two unwanted moves are chosen in opposite orders, the two normally ordered terms cancel by the rational identity obtained from the same RTT component relation that gives $B(u)B(v) = B(v)B(u)$. Thus the induction produces exactly one replaced rapidity in the final normal form and no terms with two replaced rapidities.

Adding (74.10) and (74.11), the wanted terms give the eigenvalue in (74.9). The coefficient of

$$B(u) \prod_{\substack{k=1 \\ k \neq j}}^{M} B(u_k)\Omega_L$$

is

$$\frac{\mathrm{i}}{u - u_j} \left[\left(u_j + \frac{\mathrm{i}}{2}\right)^L \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k - \mathrm{i}}{u_j - u_k} - \left(u_j - \frac{\mathrm{i}}{2}\right)^L \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k + \mathrm{i}}{u_j - u_k} \right].$$

Vanishing of this coefficient is precisely (74.8). After these unwanted terms vanish, the displayed wanted term is the full action of $t(u)$. □

The Bethe equations also have a pole-cancellation interpretation. The right-hand side of (74.9) is written with apparent simple poles at $u = u_j$. The residue at $u = u_j$ vanishes exactly when (74.8) holds. Thus the Bethe equations make $\Lambda(u)$ a polynomial of degree L, as it must be because $t(u)$ is a polynomial operator.

74.5 Hamiltonian Normalization and Momentum

The periodic $XXX_{1/2}$ Hamiltonian in the normalization used by the calculation checks is

$$H_{XXX} = \sum_{j=1}^{L} (1 - P_{j,j+1}), \qquad P_{L,L+1} = P_{L,1}.$$

It is obtained from the logarithmic derivative of the transfer matrix at the regular point, up to an additive constant fixed by $H_{XXX}\Omega_L = 0$. For a Bethe vector satisfying (74.8),

$$E = \sum_{j=1}^{M} \frac{1}{u_j^2 + 1/4}, \qquad \mathrm{e}^{\mathrm{i}P_{\mathrm{tot}}} = \prod_{j=1}^{M} \frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2}. \tag{74.12}$$

For $M = 1$, set

$$\mathrm{e}^{\mathrm{i}p} = \frac{u + \mathrm{i}/2}{u - \mathrm{i}/2}.$$

Then $u = (1/2)\cot(p/2)$, the Bethe equation gives $\mathrm{e}^{\mathrm{i}pL} = 1$, and

$$E = 4 \sin^2 \frac{p}{2}.$$

74.6 Use in Relativistic Integrable QFT

In a relativistic two-dimensional integrable QFT with non-diagonal internal scattering, the R-matrix in the monodromy is the matrix part of the two-body S-matrix. A particle transported once around a circle scatters through every other particle. The resulting internal monodromy is a transfer matrix. Its eigenvalues replace the scalar product of phase shifts in the diagonal Bethe–Yang equation. The next chapter makes this statement explicit and then develops the nested equations that diagonalize the internal transfer matrix for $SU(N)$-type scattering.

Chapter 75

Nested Bethe Ansatz and Matrix Bethe–Yang Equations

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The algebraic Bethe ansatz of Chapter 74 diagonalizes a transfer matrix by turning the Yang–Baxter equation into commuting finite-dimensional operators. This chapter develops the higher-rank version needed for non-diagonal finite-volume scattering. The physical rapidities are fixed by Bethe–Yang quantization, while auxiliary Bethe roots diagonalize the internal transfer matrix. The auxiliary roots are not extra particles: they are spectral coordinates for the internal wavefunction of a multiparticle state.

75.1 Matrix Bethe–Yang Quantization

Let a massive integrable QFT have n stable particles with rapidities $\theta_1, \ldots, \theta_n$, masses m_{a_j}, and internal species spaces V_{a_j}. Write

$$\mathcal{V}_{\boldsymbol{a}} = V_{a_1} \otimes \cdots \otimes V_{a_n}.$$

For a chosen particle j, transporting it once around a spatial circle of circumference L produces the phase $\exp(\mathrm{i}m_{a_j} L \sinh\theta_j)$ and the ordered product of two-body scatterings through the other particles. In the R-matrix convention, this ordered product is an internal transfer matrix

$$\mathcal{T}_j(\theta_j \mid \theta_1, \ldots, \widehat{\theta}_j, \ldots, \theta_n) \in \mathrm{End}(\mathcal{V}_{\boldsymbol{a}}).$$

The hat means that θ_j is omitted from the list of spectator rapidities. The Yang–Baxter equation implies that the different $\mathcal{T}_j$'s commute on the common meromorphic domain after the ordering convention for identical particles has been fixed.

Definition 75.1 (Matrix Bethe–Yang state)**.** A finite-volume n-particle matrix Bethe–Yang state is a vector $\Psi \in \mathcal{V}_{\boldsymbol{a}}$ and rapidities $\theta_1, \ldots, \theta_n$ such that Ψ is a simultaneous eigenvector of all internal transfer matrices,

$$\mathcal{T}_j \Psi = \tau_j \Psi,$$

and

$$\exp\bigl(\mathrm{i} m_{a_j} L \sinh\theta_j\bigr)\, \tau_j = 1, \qquad j = 1, \ldots, n. \tag{75.1}$$

When the scattering is diagonal, τ_j is the product of scalar two-particle phases and (75.1) reduces to the Bethe–Yang equation in Chapter 77. In the non-diagonal case the eigenvalue τ_j is obtained by an auxiliary Bethe ansatz.

75.2 RTT Algebra and Yangian Evaluation Data

Let $V = \mathbb{C}^N$ and let $E_{ab} \in \mathrm{End}(V)$ be the elementary matrix with $E_{ab} e_c = \delta_{bc} e_a$. For the rational $SU(N)$ chain we use

$$R(u) = u\,\mathrm{id} + \mathrm{i}P, \qquad P = \sum_{a,b=1}^{N} E_{ab} \otimes E_{ba}.$$

The monodromy

$$T_0(u) = R_{0L}(u - \xi_L) \cdots R_{01}(u - \xi_1)$$

is written as

$$T_0(u) = \sum_{a,b=1}^{N} E^{(0)}_{ab} \otimes T_{ab}(u), \qquad T_{ab}(u) \in \mathrm{End}(\mathcal{H}_L),$$

where $\mathcal{H}_L = (\mathbb{C}^N)^{\otimes L}$. The RTT relation

$$R_{00'}(u - v) T_0(u) T_{0'}(v) = T_{0'}(v) T_0(u) R_{00'}(u - v)$$

is the defining finite-dimensional representation of the rational Yangian RTT algebra. Equivalently, the matrix entries obey the component relation

$$(u - v)[T_{ab}(u), T_{cd}(v)] = \mathrm{i}\bigl(T_{cb}(v) T_{ad}(u) - T_{cb}(u) T_{ad}(v)\bigr). \tag{75.2}$$

For $R(u) = u + \mathrm{i}P$, the operator identity RTT is equivalent to (75.2) for all $a, b, c, d = 1, \ldots, N$. Insert $P = \sum_{p,q} E^{(0)}_{pq} E^{(0')}_{qp}$ and $T_0(u) = \sum E^{(0)}_{ab} T_{ab}(u)$, $T_{0'}(v) = \sum E^{(0')}_{cd} T_{cd}(v)$ into RTT. The coefficient of $E^{(0)}_{ab} E^{(0')}_{cd}$ in the term proportional to $u - v$ is $(u - v)(T_{ab}(u) T_{cd}(v) - T_{cd}(v) T_{ab}(u))$. The coefficient in the permutation terms is $\mathrm{i}(T_{cb}(v) T_{ad}(u) - T_{cb}(u) T_{ad}(v))$. Equality of the coefficients is exactly (75.2), and the converse is obtained by summing these coefficient identities.

Expanding $T_{ab}(u)$ for large u gives level-zero and higher Yangian charges. For a finite chain the representation is not an additional postulate: it is the evaluation representation produced by the concrete monodromy above. The Yangian language is nevertheless useful because the nested roots are organized by the simple roots of $\mathfrak{sl}_N$, and the recursive auxiliary monodromy below is the same RTT construction applied to smaller residual color spaces.

75.3 Recursive Nested Monodromy

Choose the reference vector

$$\Omega_L = e_1^{\otimes L}.$$

With the shifted L-operator convention of Chapter 74,

$$T_{11}(u)\Omega_L = \left(u + \frac{\mathrm{i}}{2}\right)^L \Omega_L, \qquad T_{aa}(u)\Omega_L = \left(u - \frac{\mathrm{i}}{2}\right)^L \Omega_L \quad (a = 2, \ldots, N),$$

and the lower-left entries annihilate Ω_L. The first-level creation operators are the entries $B_\alpha(u) = T_{1,\alpha+1}(u)$, $\alpha = 1, \ldots, N-1$. A first-level Bethe vector has the form

$$\Psi^{(1)} = \sum_{\alpha_1,\ldots,\alpha_{M_1}=1}^{N-1} F^{\alpha_1\cdots\alpha_{M_1}}\, B_{\alpha_1}(u_1^{(1)}) \cdots B_{\alpha_{M_1}}(u_{M_1}^{(1)})\Omega_L.$$

The coefficient tensor F is not arbitrary. RTT relations among the B_α's show that exchanging two first-level roots acts on their color indices by the rational R-matrix on $\mathbb{C}^{N-1}$. Therefore the unwanted terms in the action of the full transfer matrix are controlled by an auxiliary $SU(N-1)$ transfer matrix on $(\mathbb{C}^{N-1})^{\otimes M_1}$. This is the origin of nesting.

Construction 75.2 (Nested auxiliary monodromy)**.** Let $V^{(r)} \simeq \mathbb{C}^{N-r}$ be the residual color space at nesting level r, with $r = 1, \ldots, N-1$. Put $M_0 = L$, and let $\{u_j^{(0)}\}$ denote the physical inhomogeneities in the spin-chain problem or the physical rapidities after conversion to the rational auxiliary variable. Given roots $\{u_j^{(r-1)}\}_{j=1}^{M_{r-1}}$, define the level-$r$ auxiliary monodromy by

$$T_0^{(r)}(u) = R_{0M_{r-1}}^{(N-r+1)}(u - u_{M_{r-1}}^{(r-1)}) \cdots R_{01}^{(N-r+1)}(u - u_1^{(r-1)}),$$

where $R^{(m)}(u) = u + \mathrm{i}P$ acts on $\mathbb{C}^m \otimes \mathbb{C}^m$. The level-$r$ transfer matrix is $t^{(r)}(u) = \operatorname{tr}_0 T_0^{(r)}(u)$. The roots $\{u_j^{(r)}\}_{j=1}^{M_r}$ are the Bethe roots used to diagonalize $t^{(r)}$ by the same pseudovacuum construction, now with residual rank smaller by one.

Theorem 75.3 (Recursive origin of the nested equations)**.** *For the rational fundamental $SU(N)$ chain with generic inhomogeneities and generic roots, the repeated construction 75.2 gives an eigenvector of the transfer matrix whenever the roots satisfy the pole-cancellation equations*

$$\left(\frac{u_j^{(r)} + \frac{\mathrm{i}}{2}\delta_{r1}}{u_j^{(r)} - \frac{\mathrm{i}}{2}\delta_{r1}}\right)^L = \prod_{s=1}^{N-1} \prod_{\substack{k=1 \\ (s,k)\neq(r,j)}}^{M_s} \frac{u_j^{(r)} - u_k^{(s)} + \frac{\mathrm{i}}{2}A_{rs}}{u_j^{(r)} - u_k^{(s)} - \frac{\mathrm{i}}{2}A_{rs}}, \tag{75.3}$$

where $A_{rs} = 2\delta_{rs} - \delta_{r,s+1} - \delta_{r,s-1}$ is the A_{N-1} Cartan matrix. The energy in the nearest-neighbor rational Hamiltonian normalization depends only on first-level roots:

$$E = \sum_{j=1}^{M_1} \frac{1}{(u_j^{(1)})^2 + 1/4}. \tag{75.4}$$

Proof. The proof is an induction on N. The case $N = 2$ is precisely Theorem 74.3: the transfer eigenvalue is meromorphic with possible simple poles at first-level roots, and the residue at $u = u_j^{(1)}$ vanishes exactly when the $SU(2)$ Bethe equation holds.

Assume the statement for residual rank $N-r$. At level r, the RTT commutation relations move the diagonal entries of $T^{(r)}(u)$ through the creation operators $B_\alpha^{(r)}(u_j^{(r)})$. The wanted term is the one in which no spectral parameter is exchanged. The unwanted terms replace one creation operator by $B_\alpha^{(r)}(u)$ and carry coefficients that are matrix elements of the next auxiliary transfer matrix $t^{(r+1)}$ on $(V^{(r+1)})^{\otimes M_r}$. By the induction hypothesis those matrix coefficients are diagonal exactly when the roots at levels $r+1,\dots,N-1$ obey their own pole-cancellation equations.

Only three kinds of factors can appear in the residue at a root $u_j^{(r)}$. First, exchanging $u_j^{(r)}$ with another root at the same level uses the two-body rational R-matrix on the same residual color space and contributes the shifts $\pm\mathrm{i}$, which are the entries $\pm\mathrm{i}A_{rr}/2$. Second, the next auxiliary transfer matrix contributes only adjacent Dynkin-node factors, because after one nesting step the root $u^{(r)}$ is an inhomogeneity for the $SU(N-r)$ residual problem; this gives the shifts $\mp\mathrm{i}/2$ for $s=r\pm1$, namely $\pm\mathrm{i}A_{rs}/2$. Third, the vacuum diagonal eigenvalues of the original monodromy enter only at $r=1$, giving the source factor $((u+\mathrm{i}/2)/(u-\mathrm{i}/2))^L$. Non-adjacent nodes correspond to commuting residual color sectors and give $A_{rs}=0$, hence the factor 1. Setting every residue to zero is exactly (75.3). The Hamiltonian is the logarithmic derivative of the first-level transfer matrix at the regular point, so the same calculation as in the $SU(2)$ case gives (75.4); auxiliary levels select the internal eigenvector but carry no direct nearest-neighbor energy. □

Expanding (75.3) gives the familiar displayed form

$$\begin{aligned}
\left(\frac{u_j^{(1)}+\mathrm{i}/2}{u_j^{(1)}-\mathrm{i}/2}\right)^L &= \prod_{\substack{k=1\\k\neq j}}^{M_1}\frac{u_j^{(1)}-u_k^{(1)}+\mathrm{i}}{u_j^{(1)}-u_k^{(1)}-\mathrm{i}}\prod_{k=1}^{M_2}\frac{u_j^{(1)}-u_k^{(2)}-\mathrm{i}/2}{u_j^{(1)}-u_k^{(2)}+\mathrm{i}/2},\\
1 &= \prod_{\substack{k=1\\k\neq j}}^{M_r}\frac{u_j^{(r)}-u_k^{(r)}+\mathrm{i}}{u_j^{(r)}-u_k^{(r)}-\mathrm{i}}\prod_{\sigma=\pm1}\prod_{k=1}^{M_{r+\sigma}}\frac{u_j^{(r)}-u_k^{(r+\sigma)}-\mathrm{i}/2}{u_j^{(r)}-u_k^{(r+\sigma)}+\mathrm{i}/2},\\
&\qquad\qquad r=2,\dots,N-2,\\
1 &= \prod_{\substack{k=1\\k\neq j}}^{M_{N-1}}\frac{u_j^{(N-1)}-u_k^{(N-1)}+\mathrm{i}}{u_j^{(N-1)}-u_k^{(N-1)}-\mathrm{i}}\prod_{k=1}^{M_{N-2}}\frac{u_j^{(N-1)}-u_k^{(N-2)}-\mathrm{i}/2}{u_j^{(N-1)}-u_k^{(N-2)}+\mathrm{i}/2}.
\end{aligned}\tag{75.5}$$

Products over an empty set are 1.

75.4 Dressed-Vacuum Transfer Eigenvalue

The nested equations are most useful when one keeps track of the transfer eigenvalue that produced them. We record the expression in a gauge in which the pole-cancellation calculation is transparent. Let

$$Q_r(u)=\prod_{j=1}^{M_r}(u-u_j^{(r)}),\qquad Q_0(u)=Q_N(u)=1,$$

and use the half-shift notation

$$f^{[k]}(u)=f\left(u+\frac{\mathrm{i}k}{2}\right).$$

For the homogeneous fundamental chain define

$$\phi_1(u) = (u + \mathrm{i})^L, \qquad \phi_a(u) = u^L \quad (a = 2, \ldots, N).$$

With a diagonal twist $g = \mathrm{diag}(g_1, \ldots, g_N)$ the fundamental transfer eigenvalue can be written, after an overall scalar normalization and a harmless shift of the spectral coordinate, as

$$\Lambda(u) = \sum_{a=1}^{N} g_a\, \phi_a(u)\, \frac{Q_{a-1}^{[a+1]}(u)}{Q_{a-1}^{[a-1]}(u)} \frac{Q_a^{[a-2]}(u)}{Q_a^{[a]}(u)}. \tag{75.6}$$

This is the dressed-vacuum form of the eigenvalue. The adjective means only that the vacuum diagonal eigenvalues ϕ_a are multiplied by rational "dressing" factors built from the Q-functions; it is not an additional physical vacuum.

Pole-cancellation derivation of the twisted nested equations. Assume the roots are generic and the twist eigenvalue ratios are finite. The meromorphic function (75.6) has no pole at the would-be pole associated with $u_j^{(r)}$ precisely when

$$\frac{g_r}{g_{r+1}} \frac{\phi_r(u_j^{(r)} - \frac{\mathrm{i}r}{2})}{\phi_{r+1}(u_j^{(r)} - \frac{\mathrm{i}r}{2})} = \prod_{s=1}^{N-1} \prod_{\substack{k=1 \\ (s,k)\neq(r,j)}}^{M_s} \frac{u_j^{(r)} - u_k^{(s)} + \frac{\mathrm{i}}{2} A_{rs}}{u_j^{(r)} - u_k^{(s)} - \frac{\mathrm{i}}{2} A_{rs}}. \tag{75.7}$$

For the untwisted homogeneous chain this reduces to (75.3). This is a residue calculation inside the dressed-vacuum formula, not an independent theorem: it is the local algebraic step that converts polynomiality of the transfer eigenvalue into the nested Bethe equations.

The root $u_j^{(r)}$ appears as a pole of $\Lambda(u)$ at the common point $u = u_j^{(r)} - \mathrm{i}r/2$ in precisely two neighboring summands, namely the $a = r$ and $a = r + 1$ summands. All other summands are regular there because their denominator Q-functions have different node labels. Multiplying the residue equation by the nonzero common factors gives

$$\begin{aligned} 0 = g_r \phi_r \frac{Q_{r-1}^{[r+1]}}{Q_{r-1}^{[r-1]}} Q_r^{[r-2]} \\ + g_{r+1} \phi_{r+1} Q_r^{[r+2]} \frac{Q_{r+1}^{[r-1]}}{Q_{r+1}^{[r+1]}} \end{aligned}$$

evaluated at $u = u_j^{(r)} - \mathrm{i}r/2$; the omitted common denominator is the same derivative $Q_r'(u_j^{(r)})$. Solving for the ratio of the two vacuum terms gives

$$\frac{g_r \phi_r}{g_{r+1} \phi_{r+1}} = -\frac{Q_r(u_j^{(r)} + \mathrm{i})}{Q_r(u_j^{(r)} - \mathrm{i})} \frac{Q_{r-1}(u_j^{(r)} - \mathrm{i}/2)}{Q_{r-1}(u_j^{(r)} + \mathrm{i}/2)} \frac{Q_{r+1}(u_j^{(r)} - \mathrm{i}/2)}{Q_{r+1}(u_j^{(r)} + \mathrm{i}/2)}.$$

The factor involving Q_r contains the root $u_j^{(r)}$ itself:

$$-\frac{Q_r(u_j^{(r)} + \mathrm{i})}{Q_r(u_j^{(r)} - \mathrm{i})} = \prod_{\substack{k=1 \\ k\neq j}}^{M_r} \frac{u_j^{(r)} - u_k^{(r)} + \mathrm{i}}{u_j^{(r)} - u_k^{(r)} - \mathrm{i}}.$$

Combining this same-node factor with the two adjacent-node factors is exactly the Cartan-matrix product in (75.7). For the homogeneous untwisted chain,

$$\frac{\phi_1(u_j^{(1)}-\mathrm{i}/2)}{\phi_2(u_j^{(1)}-\mathrm{i}/2)}=\left(\frac{u_j^{(1)}+\mathrm{i}/2}{u_j^{(1)}-\mathrm{i}/2}\right)^L,\qquad \frac{\phi_r}{\phi_{r+1}}=1\quad(r\geq 2),$$

which gives (75.3).

Formula (75.6) is often the cleanest way to compare the algebraic Bethe ansatz, functional TQ/QQ systems, and separation of variables. The finite-dimensional transfer matrix is polynomial after all pole residues vanish; polynomiality, not a separate appeal to a formal Bethe equation, is the algebraic reason the spectrum is well-defined.

75.5 Worked SU(3) Configuration

For $SU(3)$ there are first-level roots u_j and second-level roots v_α. The equations are

$$\left(\frac{u_j+\mathrm{i}/2}{u_j-\mathrm{i}/2}\right)^L=\prod_{\substack{k=1\\k\neq j}}^{M_1}\frac{u_j-u_k+\mathrm{i}}{u_j-u_k-\mathrm{i}}\prod_{\alpha=1}^{M_2}\frac{u_j-v_\alpha-\mathrm{i}/2}{u_j-v_\alpha+\mathrm{i}/2}, \tag{75.8}$$

$$1=\prod_{\substack{\beta=1\\\beta\neq\alpha}}^{M_2}\frac{v_\alpha-v_\beta+\mathrm{i}}{v_\alpha-v_\beta-\mathrm{i}}\prod_{j=1}^{M_1}\frac{v_\alpha-u_j-\mathrm{i}/2}{v_\alpha-u_j+\mathrm{i}/2}. \tag{75.9}$$

Set $L=6$, $M_1=2$, $M_2=1$,

$$u_1=-\frac{\sqrt{3}}{2},\qquad u_2=\frac{\sqrt{3}}{2},\qquad v_1=0.$$

Let

$$z=\frac{u_2+\mathrm{i}/2}{u_2-\mathrm{i}/2}=\mathrm{e}^{\mathrm{i}\pi/3}.$$

Then $z^6=1$. The second-level equation is

$$\frac{\sqrt{3}/2-\mathrm{i}/2}{\sqrt{3}/2+\mathrm{i}/2}\frac{-\sqrt{3}/2-\mathrm{i}/2}{-\sqrt{3}/2+\mathrm{i}/2}=z^{-1}z=1.$$

For the first root u_2, the right side of (75.8) is

$$\frac{2u_2+\mathrm{i}}{2u_2-\mathrm{i}}\frac{u_2-\mathrm{i}/2}{u_2+\mathrm{i}/2}=z\,z^{-1}=1,$$

and the left side is $z^6=1$. The equation for u_1 is the inverse of the same equality. The energy is $E=2$. This example is deliberately small: every phase can be checked exactly, and the second-level root is essential even though it carries no direct energy.

75.6 Trigonometric Nesting

The relativistic sine-Gordon and affine-Toda families use trigonometric matrix parts rather than the rational $R(u) = u + \mathrm{i}P$. A convenient $\mathfrak{sl}_N$ trigonometric normalization is

$$R_\eta(u) = \sinh(u+\eta)\sum_a E_{aa}\otimes E_{aa} + \sinh u \sum_{a\neq b} E_{aa}\otimes E_{bb} \\ + \sinh\eta \sum_{a\neq b} \mathrm{e}^{\mathrm{sgn}(a-b)u} E_{ab}\otimes E_{ba}.$$

It solves the Yang–Baxter equation and degenerates to the rational matrix after $u, \eta \to 0$ with u/η fixed, up to scalar normalization. The nested pole-cancellation equations become

$$\left(\frac{\sinh(u_j^{(r)} + \frac{\eta}{2}\delta_{r1})}{\sinh(u_j^{(r)} - \frac{\eta}{2}\delta_{r1})}\right)^L = \prod_{s=1}^{N-1} \prod_{\substack{k=1 \\ (s,k)\neq(r,j)}}^{M_s} \frac{\sinh(u_j^{(r)} - u_k^{(s)} + \frac{\eta}{2}A_{rs})}{\sinh(u_j^{(r)} - u_k^{(s)} - \frac{\eta}{2}A_{rs})}. \tag{75.10}$$

The proof is the same recursive RTT proof as Theorem 75.3, with rational functions replaced by hyperbolic functions. The scalar factor multiplying R_η is not fixed by the Yang–Baxter equation; in a relativistic QFT it is fixed separately by unitarity, crossing, bound-state residues, and normalization of asymptotic states.

75.7 Relativistic SU(N)-Symmetric Scattering

For a relativistic integrable QFT whose particles transform in a fundamental $SU(N)$ multiplet, the two-body S-matrix often factorizes as

$$S(\theta) = S_0(\theta)\,\widehat{R}(\theta),$$

where S_0 is a scalar meromorphic factor and $\widehat{R}$ is an $SU(N)$-invariant rational or trigonometric matrix normalized to satisfy unitarity and crossing in the physical rapidity variable θ. The finite-volume quantization separates into a scalar phase and a nested transfer-matrix eigenvalue:

$$mL\sinh\theta_j - \mathrm{i}\sum_{k\neq j}\log S_0(\theta_j - \theta_k) - \mathrm{i}\log\tau_j(\theta_1,\ldots,\theta_n;\{u^{(r)}\}) = 2\pi I_j. \tag{75.11}$$

The auxiliary roots $\{u^{(r)}\}$ obey rational or trigonometric equations of the form (75.3) or (75.10), with the map from physical rapidity θ to auxiliary spectral parameter determined by $\widehat{R}$. Changing this map by a meromorphic reparametrization changes the displayed coordinate form but not the transfer-matrix eigenvalue.

Controlled Approximation 75.4 (Large-volume Bethe–Yang regime)**.** Equation (75.11) is a large-L quantization condition for stable particles whose separations are much larger than the interaction range set by the mass gap. Corrections are exponentially small in mL for a massive theory without massless channels, provided no bound-state pole is pinched by the finite-volume contour. Power corrections and wrapping corrections require the TBA or mirror-channel formalism of later chapters.

75.8 Nested Strings and Thermodynamic Input

In a thermodynamic limit the roots of nested equations often organize into strings. In the rational normalization, an m-string at node r with real center x has the asymptotic form

$$u_\ell^{(r,m)} = x + \frac{\mathrm{i}}{2}(m + 1 - 2\ell) + \varepsilon_\ell(L), \qquad \ell = 1, \ldots, m, \tag{75.12}$$

with $\varepsilon_\ell(L) \to 0$ in the regime where the string hypothesis is valid. The string center becomes the variable in the thermodynamic density equations; the string length m and Dynkin node r become species labels.

Controlled Approximation 75.5 (Nested string hypothesis)**.** The replacement of finite-volume roots by exact string species is not a formal consequence of the Bethe equations alone. It is a model-dependent thermodynamic hypothesis requiring control of singular strings, deviations $\varepsilon_\ell(L)$, possible holes, and the order of limits. When used below, the input is: for the thermodynamic states under consideration, all roots not on real physical rapidity lines can be partitioned into families of the form (75.12), with deviations whose contribution to intensive free energy vanishes in the limit. This is the precise assumption that converts the finite nested Bethe equations into the multi-species TBA equations of Chapter 78.

75.9 Principal Chiral Model and Gross–Neveu Families

The $SU(N)$ principal chiral model has global $SU(N)_L \times SU(N)_R$ symmetry. Its massive particles carry left and right internal indices, so the matrix part of its scattering problem contains two nested $SU(N)$ systems, one for each chiral factor. Gross–Neveu families have a similar nested structure, with the details controlled by the vector, spinor, or fundamental representation carried by the particles. The Bethe equations therefore contain physical rapidities together with auxiliary roots attached to the Dynkin diagram of the relevant symmetry algebra.

The important conceptual point is that auxiliary roots are not additional particles in the Hilbert space. They parametrize the internal wavefunction of a multiparticle state. The energy and momentum are functions of the physical rapidities; the auxiliary roots enter by selecting the transfer-matrix eigenvalue that appears in the quantization condition.

75.10 Completeness and Its Precise Scope

Quoted Theorem 75.6 (Generic finite-chain completeness)**.** Proof references.[1] *For the inhomogeneous rational $SU(N)$ fundamental chain with a generic diagonal twist whose eigenvalue ratios are not roots of unity, and with generic inhomogeneities, the nested Bethe equations have a set of admissible solutions whose Bethe vectors form a basis of $(\mathbb{C}^N)^{\otimes L}$. The transfer-matrix spectrum is simple on a Zariski-open set of the twist and inhomogeneity parameters.*

[1] The algebraic nested Bethe vectors and higher transfer-matrix eigenvalues for rational $\mathfrak{gl}_N$ XXX-type models are developed in Mukhin–Tarasov–Varchenko [73]. Simple-spectrum and completeness results for XXX Bethe algebras are proved in the Mukhin–Tarasov–Varchenko and Tarasov works cited in [74, 75]. The formulation used here is the generic finite-chain theorem-boundary form: generic twist and inhomogeneity are hypotheses, not decorative parameters.

Mechanism and singular-limit boundary. The quoted theorem is a finite-dimensional algebraic theorem about the commutative Bethe algebra generated by transfer matrices. The generic parameters serve two purposes. First, they separate transfer-matrix eigenvalues: on the Zariski-open parameter set the joint spectrum is simple, so an eigenvector is determined by its eigenvalue data. Second, the inhomogeneities and twist remove the singular Bethe-root configurations that make the coordinate Bethe equations incomplete at special points. The proof strategy can be summarized as follows. The transfer matrices generate a commutative algebra whose joint eigenvalues are encoded by a rational difference operator or, equivalently, by Wronskian/Casoratian polynomial data. For generic parameters the relevant algebraic variety is reduced and has the dimension predicted by the representation space. Admissible Bethe solutions map to this variety, the Bethe vectors attached to distinct solutions have distinct joint eigenvalues, and dimension counting gives a basis. None of these steps is a thermodynamic statement; every step takes place at fixed finite chain length L.

The theorem is a finite-dimensional algebraic statement. It does not by itself prove completeness for the periodic homogeneous chain, because taking the twist to the identity and the inhomogeneities to equal values can create singular roots and degeneracies. It also does not prove that a continuum QFT finite-volume Hamiltonian is approximated by matrix Bethe–Yang equations. For continuum use we therefore separate three assertions:

algebraic completeness	basis theorem for a finite transfer matrix,
Bethe–Yang control	large-volume approximation to QFT levels,
thermodynamic control	existence of the density/string limit.

Each assertion has different hypotheses. The monograph uses the generic finite-chain theorem as an algebraic guide, but it states explicitly whenever periodic homogeneous completeness, continuum spectral control, or the string hypothesis is being assumed rather than proved.

75.11 Interface with Later Volumes

The relativistic integrable-QFT use of nested TBA differs from planar gauge-theory integrability. In the present volume, physical particles have relativistic dispersion $E = m\cosh\theta$ and finite-size effects are organized by the mirror channel of a two-dimensional QFT. In the planar $\mathcal{N} = 4$ SYM volume, magnons have a non-relativistic dispersion on a spin chain, wrapping effects arise from trace cyclicity and finite operator length, and the quantum spectral curve replaces the finite-rank Baxter polynomial by a system of analytic Q-functions. The algebraic bridge is the RTT, nested monodromy, TQ, QQ, and Hirota logic developed here and in Chapter 78; the physical interpretation is different.

Chapter 76

Form-Factor Bootstrap and Local Operators

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The scattering specification of Chapters 72 and 73 is on-shell. Local QFT contains more structure: local operator domains, Wightman distributions, and matrix elements of local fields between finite-particle states. In a massive two-dimensional theory with factorized scattering, the first operator-level object is the family of form factors.

Definition 76.1 (Local-operator form-factor structure)**.** Let $\mathfrak{R}$ be a regular factorized scattering datum in the sense of Definition 73.1. A local-operator form-factor structure over $\mathfrak{R}$ is a tuple

$$\mathfrak{F}_{\mathcal{O}} = \big(\widehat{\mathcal{O}}, \{F^{\mathcal{O}}_{a_1\ldots a_n}\}_{n\geq 0,\ a_i\in\mathcal{I}}, \{\omega_{\mathcal{O},a}\}_{a\in\mathcal{I}}, \mathcal{P}_{\mathrm{kin}}, \mathcal{P}_{\mathrm{bs}}, \mathcal{G}_{\mathcal{O}}\big)$$

with the following entries.

(i) $\widehat{\mathcal{O}}(f)$ is a scalar local or semi-local operator-valued distribution on a common dense scattering domain $\mathcal{D}_{\mathrm{scat}} \subset \mathcal{H}$, with vacuum $\Omega \in \mathcal{D}_{\mathrm{scat}}$. Semi-locality means that the exchange of $\widehat{\mathcal{O}}$ with a charged particle line may carry the monodromy specified in item (iii); ordinary locality is the special case with trivial monodromy for every species.

(ii) For every ordered species word $\mathbf{a} = (a_1, \ldots, a_n)$, $F^{\mathcal{O}}_{\mathbf{a}}$ is a meromorphic covector-valued function on the rapidity variables,

$$F^{\mathcal{O}}_{\mathbf{a}}(\theta_1, \ldots, \theta_n) \in \big(V_{a_1} \otimes \cdots \otimes V_{a_n}\big)^\vee,$$

whose physical real boundary value in the chamber $\theta_1 > \cdots > \theta_n$ equals the vacuum-to-in-state matrix element of $\widehat{\mathcal{O}}(0)$. The word $\mathbf{a}$ is part of the argument; when adjacent rapidities are analytically continued around one another, the corresponding species spaces are reordered by the scattering maps of $\mathfrak{R}$.

(iii) $\omega_{\mathcal{O},a} \in \mathbb{C}^\times$ is the semi-locality monodromy acquired when a particle of species a is analytically continued once around the insertion of $\mathcal{O}$. For a strictly local bosonic scalar operator all $\omega_{\mathcal{O},a}$ are 1; for order/disorder fields in two-dimensional examples the same symbol records the controlled failure of single-valuedness.

(iv) $\mathcal{P}_{\mathrm{kin}}$ consists of the annihilation-pole residue tensors determined by the antiparticle and crossing data of $\mathfrak{R}$. These tensors specify the residues at $\theta_i - \theta_j = \mathrm{i}\pi$ after the chosen tensor-ordering convention is fixed.

(v) $\mathcal{P}_{\mathrm{bs}}$ consists of the bound-state residue tensors inherited from the physical-strip poles of the two-body scattering maps. If no stable bound-state channel is present, this entry is empty.

(vi) $\mathcal{G}_{\mathcal{O}}$ is a stated family of growth and boundary-value bounds. It must be strong enough for each spectral series later claimed in the chapter to define the announced distribution or separated Euclidean function. A form-factor structure without such bounds is only algebraic bootstrap input, not a reconstruction of a local field.

The exchange, cyclicity, kinematic-pole, bound-state-pole, covariance, and growth equations are developed below as constraints on $\mathfrak{F}_{\mathcal{O}}$. The definition records the object to which those constraints apply; it does not assert that a solution of the algebraic equations has already constructed a Wightman QFT.

76.1 Finite-Particle Matrix Elements

Let $\mathfrak{F}_{\mathcal{O}}$ be a local-operator form-factor structure over $\mathfrak{R}$. Thus $\mathcal{I}$ is the set of stable particle species, V_a their internal species spaces, and $S_{ab}(\theta) : V_a \otimes V_b \to V_b \otimes V_a$ the two-body scattering maps. Ordered incoming states are denoted

$$|\theta_1, a_1; \ldots; \theta_n, a_n\rangle_{\mathrm{in}}, \qquad \theta_1 > \cdots > \theta_n.$$

For a local scalar operator $\widehat{\mathcal{O}}(x)$, the n-particle form factor is the distributional boundary value

$$F^{\mathcal{O}}_{a_1 \ldots a_n}(\theta_1, \ldots, \theta_n) = \langle \Omega | \widehat{\mathcal{O}}(0) | \theta_1, a_1; \ldots; \theta_n, a_n\rangle_{\mathrm{in}}.$$

If the species spaces are not one-dimensional, this is a covector on $V_{a_1} \otimes \cdots \otimes V_{a_n}$. Translation covariance gives

$$\langle \Omega | \widehat{\mathcal{O}}(x) | \theta_1, a_1; \ldots; \theta_n, a_n\rangle_{\mathrm{in}} = \exp\left[-\mathrm{i} x_\mu \sum_{j=1}^{n} p^\mu_{a_j}(\theta_j)\right] F^{\mathcal{O}}_{a_1 \ldots a_n}(\theta_1, \ldots, \theta_n).$$

Thus the nontrivial information is at $x = 0$, while the distributional character of $\widehat{\mathcal{O}}$ enters through growth and boundary-value conditions on $F^{\mathcal{O}}$.

76.2 Exchange Equation

Analytic continuation in rapidities extends the ordered form factor to domains in $\mathbb{C}^n$. Fix an adjacent pair and write

$$\mathcal{C}_+ = \{\theta_i > \theta_{i+1}\}, \qquad \mathcal{C}_- = \{\theta_{i+1} > \theta_i\}.$$

The form factor has two distributional boundary values at the wall $\theta_i = \theta_{i+1}$, obtained by approaching the wall from the two ordered chambers, or equivalently by the two standard $i0$-prescriptions for the rapidity difference

$$F_+ = \lim_{\epsilon \downarrow 0} F(\dots, \theta_i + \mathrm{i}\epsilon, \theta_{i+1} - \mathrm{i}\epsilon, \dots), \qquad F_- = \lim_{\epsilon \downarrow 0} F(\dots, \theta_i - \mathrm{i}\epsilon, \theta_{i+1} + \mathrm{i}\epsilon, \dots).$$

The ordered scattering-state convention says that the corresponding finite-particle boundary vectors are related by the adjacent two-body scattering operator,

$$|\dots; \theta_i, a_i; \theta_{i+1}, a_{i+1}; \dots\rangle_+ = S^{bc}_{a_i a_{i+1}}(\theta_i - \theta_{i+1}) |\dots; \theta_{i+1}, c; \theta_i, b; \dots\rangle_-. \tag{76.1}$$

This relation is the rapidity-space form of the scattering theorem for ordered Haag–Ruelle states in a factorizing theory; locality is used upstream in constructing the asymptotic fields and their scattering relations, not as a one-line replacement for the boundary-value statement. Pairing (76.1) with the local-operator covector gives Watson's exchange equation

$$F^{\mathcal{O}}_{\dots a_i a_{i+1} \dots}(\dots, \theta_i, \theta_{i+1}, \dots) = F^{\mathcal{O}}_{\dots a_{i+1} a_i \dots}(\dots, \theta_{i+1}, \theta_i, \dots) S_{a_i a_{i+1}}(\theta_i - \theta_{i+1}),$$

where the S-matrix acts on the two corresponding species factors before the covector is evaluated.

Proposition 76.2 (Exchange from factorized scattering)**.** *Assume that the finite-particle scattering states satisfy asymptotic completeness in the massive sector and that the ordered in-state convention is related across adjacent rapidity chambers by the factorized S-matrix. Then every local operator whose finite-particle matrix elements admit rapidity boundary values satisfies the exchange equation above.*

Proof. Let g_+ and g_- be compactly supported test functions in the two neighboring rapidity chambers whose supports approach the same compact set on the wall but do not meet any bound-state or threshold singularity. Smearing (76.1) with these test functions gives an identity of Hilbert-space vectors in the finite-particle scattering domain. The identity is first proved for separated velocity supports by the Haag–Ruelle construction of ordered in-states; analytic continuation in the rapidity difference then supplies the displayed boundary values. The factorizing hypothesis identifies the change of ordered basis with the adjacent two-body S-matrix and leaves all other tensor factors unchanged.

Applying the continuous distributional covector $\langle\Omega|\widehat{\mathcal{O}}(0)$ to both sides gives the equality of smeared form-factor boundary values. Since this holds for all such test functions, the equality holds as a distributional boundary-value identity in the rapidity variables. No off-shell Lagrangian is used. The hypotheses are the existence of the finite-particle scattering states, their boundary-value analyticity in rapidity chambers, and the factorized two-body change of ordered basis. $\square$

76.3 Cyclicity, Locality, and Semi-Locality

The cyclicity equation is a crossing-locality statement. It is not obtained from the two-body S-matrix alone. The analytic input is the same as in the Wightman reconstruction chapters: the spectrum condition gives forward-tube analyticity of vacuum matrix elements, and locality identifies boundary values after permuting spacelike separated insertions. In rapidity variables the complex Lorentz transformation

$$\theta \longmapsto \theta + \mathrm{i}\pi$$

changes $p_a(\theta)$ to $-p_a(\theta)$, and hence crosses an incoming particle to the outgoing antiparticle channel. A second shift by $\mathrm{i}\pi$ crosses it back to the incoming side after it has passed through the local operator.

Proposition 76.3 (Cyclicity from crossing and locality)**.** *Assume that the finite-particle matrix elements of a scalar local operator $\widehat{\mathcal{O}}$ are boundary values of Wightman functions satisfying the spectrum condition, locality, and the one-particle crossing relation obtained by analytic continuation through the physical rapidity strip. Assume that the path $\theta_1 \mapsto \theta_1 + 2\pi\mathrm{i}$ crosses no singularity other than the standard crossing boundary and no bound-state pole. Then*

$$F^{\mathcal{O}}_{a_1 a_2 \ldots a_n}(\theta_1 + 2\pi\mathrm{i}, \theta_2, \ldots, \theta_n) = F^{\mathcal{O}}_{a_2 \ldots a_n a_1}(\theta_2, \ldots, \theta_n, \theta_1).$$

If $\widehat{\mathcal{O}}$ is semi-local with respect to species a_1, the right side is multiplied by the declared monodromy endomorphism $\omega_{\mathcal{O},a_1}$.

Proof. For separated velocity supports, represent the incoming particle of species a_1 by a Haag–Ruelle wave packet and write the matrix element as a boundary value of a Wightman distribution with one interpolating field for that particle. The joint spectrum condition implies that this distribution extends holomorphically when the imaginary parts of successive spacetime differences lie in the future cone. Fourier transforming the first wave packet and using $p_a(\theta + \mathrm{i}\pi) = -p_a(\theta)$, continuation $\theta_1 \mapsto \theta_1 + \mathrm{i}\pi$ converts the incoming a_1-particle into an outgoing $\bar{a}_1$-particle. In bracket notation this is the crossing boundary value

$$F^{\mathcal{O}}_{a_1 a_2 \ldots a_n}(\theta_1 + \mathrm{i}\pi, \theta_2, \ldots, \theta_n) = {}_{\mathrm{out}}\langle \theta_1, \bar{a}_1 | \widehat{\mathcal{O}}(0) | \theta_2, a_2; \ldots; \theta_n, a_n \rangle_{\mathrm{in}},$$

with the charge-conjugation map suppressed in the notation.

Continue once more by $\mathrm{i}\pi$. At the boundary of the tube one may choose the spacetime supports of the crossed particle and of $\widehat{\mathcal{O}}$ to be spacelike separated before the wave-packet limit is taken. Local commutativity then permits the crossed particle interpolating field to be moved past $\widehat{\mathcal{O}}$. For a scalar local operator this gives no extra factor; for a semi-local operator it gives the declared monodromy $\omega_{\mathcal{O},a_1}$. After the second crossing the particle is again incoming, and the ordered-state convention places its rapidity at the right end of the ordered list. Thus the boundary value is the displayed cyclic form factor. The equality is distributional because it was derived after smearing in rapidity and then unsmearing by uniqueness of boundary values. The excluded pole crossings are precisely the cases where additional residue terms must be added. □

76.4 Kinematic Pole Equation

Let $\bar{a}$ denote the antiparticle species of a. The pair $(\bar{a}, a)$ at rapidity difference $\mathrm{i}\pi$ has total momentum zero and can annihilate into the vacuum. Consequently a local form factor may have a kinematic pole at

$$\theta_{\bar a} = \theta_a + \mathrm{i}\pi.$$

For a scalar local operator the residue has the structural form

$$-\mathrm{i} \operatorname*{Res}_{\theta'=\theta+\mathrm{i}\pi} F^{\mathcal{O}}_{\bar a\, a\, a_1\ldots a_n}(\theta',\theta,\theta_1,\ldots,\theta_n) = \left(1 - \prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j)\right) F^{\mathcal{O}}_{a_1\ldots a_n}(\theta_1,\ldots,\theta_n), \tag{76.2}$$

with tensor contractions and charge-conjugation maps inserted according to the species spaces. The product is ordered by moving the particle a through the n remaining particles before annihilation.

Proposition 76.4 (Kinematic annihilation pole)**.** *Assume the hypotheses of Proposition 76.3, asymptotic completeness in the finite-particle sector under consideration, factorized scattering for ordered states, and the standard rapidity normalization of one-particle states. Assume further that the only singularity encountered at $\theta' = \theta + \mathrm{i}\pi$ is the vacuum annihilation of the adjacent pair $(\bar a, a)$. Then the singular part of the form factor is*

$$F^{\mathcal{O}}_{\bar a\, a\, a_1\ldots a_n}(\theta',\theta,\theta_1,\ldots,\theta_n) = \frac{\mathrm{i}}{\theta'-\theta-\mathrm{i}\pi}\left(1 - \prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j)\right) F^{\mathcal{O}}_{a_1\ldots a_n} + \textit{holomorphic},$$

with the same tensor-ordering convention as in (76.2). Equivalently, (76.2) holds.

Proof. Set $z = \theta' - \theta - \mathrm{i}\pi$. The coefficient of the crossed one-particle contraction is fixed by matching its two boundary values to the rapidity delta function, not by an independent sign convention. In the rapidity normalization, valid for any stable species b,

$$\langle \theta', b | \theta, b\rangle = 2\pi\delta(\theta'-\theta),$$

the contraction term must have boundary-value jump $2\pi\delta(x)$ when $x = \operatorname{Re} z$. The distributional identity

$$\frac{1}{x+\mathrm{i}0} - \frac{1}{x-\mathrm{i}0} = -2\pi\mathrm{i}\,\delta(x)$$

therefore selects

$$\frac{\mathrm{i}}{z}.$$

Indeed, $\mathrm{i}(x+\mathrm{i}0)^{-1} - \mathrm{i}(x-\mathrm{i}0)^{-1} = 2\pi\delta(x)$, exactly the one-particle contraction coefficient. This is the rapidity form of the mass-shell phase-space pole: near $z=0$, the continued total momentum $p_{\bar a}(\theta+\mathrm{i}\pi+z) + p_a(\theta)$ vanishes linearly in z, and the chosen positive-energy boundary value is the one just normalized. Thus direct annihilation of the adjacent $(\bar a, a)$-pair into the vacuum contributes

$$\frac{\mathrm{i}}{z} F^{\mathcal{O}}_{a_1\ldots a_n}(\theta_1,\ldots,\theta_n). \tag{direct}$$

There is a second boundary value of the same meromorphic matrix element. Use Watson exchange to move the particle a through the spectator particles $a_1,\ldots,a_n$ before performing the annihilation. This ordered-basis change contributes the ordered product

$$\prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j).$$

The annihilation contour now approaches the same Cauchy pole from the opposite side of the rapidity ordering. Equivalently, after applying cyclicity, the crossed particle encircles the local operator and returns to the incoming side with the opposite orientation for the small circle around $z = 0$. In local coordinates this is the statement that the reverse ordering uses

$$z_{\mathrm{rev}} = \theta - \theta' + \mathrm{i}\pi = -z,$$

so a positively oriented small circle in z_{rev} is a negatively oriented small circle in z. Therefore the residue of the same annihilation pole enters with the opposite sign. The second singular contribution is

$$-\frac{\mathrm{i}}{z}\left(\prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j)\right) F^{\mathcal{O}}_{a_1\ldots a_n}(\theta_1,\ldots,\theta_n). \qquad \text{(scattered)}$$

For a scalar local operator there is no extra semi-locality phase; a semi-local operator would multiply this second contribution by its monodromy factor. Adding the two boundary contributions gives the displayed singular part. Taking the residue at $z = 0$ gives

$$-\mathrm{i}\operatorname*{Res}_{z=0}\frac{\mathrm{i}}{z}\left(1-\prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j)\right) F^{\mathcal{O}}_{n} = \left(1-\prod_{j=1}^{n} S_{aa_j}(\theta-\theta_j)\right) F^{\mathcal{O}}_{n}.$$

The proof is local in the rapidity variables; additional bound-state poles or threshold singularities on the deformation path would add further residue terms and are excluded by hypothesis. □

76.5 Free Majorana Examples

The form-factor axioms become concrete already in the massive Ising field theory. In the fermionic particle basis there is one self-conjugate particle of mass m, and the diagonal two-body scattering function is the constant

$$S(\theta) = -1.$$

The constant phase is physically important: the ordered rapidity variables are still bosonic coordinates on asymptotic particle states, but exchanging two particles multiplies the finite-particle basis by -1. Thus a scalar form factor must be antisymmetric in rapidity variables when it has an even number of Majorana particles.

Example 76.5 (Energy-density form factor)**.** Fix a scalar normalization constant κ_ε. The massive Majorana energy-density operator ε may be normalized by

$$F^{\varepsilon}_0 = 0, \qquad F^{\varepsilon}_2(\theta_1,\theta_2) = \kappa_\varepsilon \sinh\!\left(\frac{\theta_1-\theta_2}{2}\right), \qquad F^{\varepsilon}_n = 0 \quad (n \neq 2).$$

The exchange equation follows because

$$\sinh\!\left(\frac{\theta_2-\theta_1}{2}\right)(-1) = \sinh\!\left(\frac{\theta_1-\theta_2}{2}\right).$$

Cyclicity follows from

$$\sinh\!\left(\frac{\theta_1+2\pi\mathrm{i}-\theta_2}{2}\right) = -\sinh\!\left(\frac{\theta_1-\theta_2}{2}\right) = \sinh\!\left(\frac{\theta_2-\theta_1}{2}\right).$$

The kinematic-pole equation is also satisfied: the displayed form factors are entire, and the right side vanishes because $F_0^\varepsilon = 0$ for no spectators and because $1 - S^2 = 0$ for two spectators. This example is finite rather than recursive; it checks that the sign conventions of Watson exchange and $2\pi\mathrm{i}$ cyclicity are compatible with a familiar local quadratic field.

Proposition 76.6 (Two-particle reconstruction for the Majorana energy operator)**.** *Use the one-particle normalization*

$$\langle \theta' \mid \theta \rangle = 2\pi\delta(\theta' - \theta), \qquad p^\mu(\theta) = m(\cosh\theta, \sinh\theta), \qquad p(\theta)^2 = -m^2.$$

Let $W_\varepsilon(x) = \langle \Omega|\varepsilon(x)\varepsilon(0)|\Omega\rangle$ at noncoincident points, and define the Fourier transform by

$$\widetilde{W}_\varepsilon(P) = \int_{\mathbb{R}^{1,1}} \mathrm{d}^2x\, e^{\mathrm{i}P\cdot x} W_\varepsilon(x), \qquad W_\varepsilon(x) = \int \frac{\mathrm{d}^2P}{(2\pi)^2}\, e^{-\mathrm{i}P\cdot x} \widetilde{W}_\varepsilon(P).$$

With $s = -P^2$, the two-particle form factor in Example 76.5 gives

$$\widetilde{W}_\varepsilon(P) = \theta(P^0)\theta(s - 4m^2)\, \frac{|\kappa_\varepsilon|^2}{2m^2} \sqrt{1 - \frac{4m^2}{s}} \tag{76.3}$$

in this Fourier convention. Equivalently, for Euclidean time separation $r > 0$,

$$G_\varepsilon(r) = \langle \varepsilon(r)\varepsilon(0)\rangle_E = \frac{|\kappa_\varepsilon|^2}{2\pi^2} \int_0^\infty \mathrm{d}\alpha\, \sinh^2\!\Big(\frac{\alpha}{2}\Big)\, K_0\Big(2mr\cosh\Big(\frac{\alpha}{2}\Big)\Big), \tag{76.4}$$

where, for $z > 0$,

$$K_0(z) = \int_0^\infty e^{-z\cosh u}\, \mathrm{d}u.$$

Proof. The finite-particle resolution of the identity in the one-species sector is

$$\mathbf{1}_{\mathrm{fin}} = \sum_{n\geq 0} \frac{1}{n!} \int_{\mathbb{R}^n} \prod_{j=1}^n \frac{\mathrm{d}\theta_j}{2\pi}\, |\theta_1, \dots, \theta_n\rangle_{\mathrm{in}\,\mathrm{in}}\langle\theta_1, \dots, \theta_n|.$$

For the free Majorana energy operator only the two-particle term is present. Therefore, at non-coincident Lorentzian separation,

$$W_\varepsilon(x) = \frac{1}{2} \int \frac{\mathrm{d}\theta_1}{2\pi}\frac{\mathrm{d}\theta_2}{2\pi}\, |F_2^\varepsilon(\theta_1, \theta_2)|^2\, e^{-\mathrm{i}x\cdot(p(\theta_1)+p(\theta_2))}.$$

The factor $1/2$ is not optional: the rapidity integration is over all ordered labels, while the two Majorana particles are identical. Fourier transformation gives

$$\widetilde{W}_\varepsilon(P) = \frac{1}{2} \int \frac{\mathrm{d}\theta_1}{2\pi}\frac{\mathrm{d}\theta_2}{2\pi}\, (2\pi)^2\delta^{(2)}(P - p(\theta_1) - p(\theta_2))\, |F_2^\varepsilon(\theta_1, \theta_2)|^2.$$

The support condition is already visible: P must be future directed and $s = -P^2 \geq 4m^2$.

Because the integrand is Lorentz invariant except for the explicit vector P, evaluate it in the frame $P = (M, 0)$, $M = \sqrt{s}$. Put

$$\Theta = \frac{\theta_1 + \theta_2}{2}, \qquad \alpha = \theta_1 - \theta_2.$$

Then

$$p(\theta_1) + p(\theta_2) = 2m \cosh\left(\frac{\alpha}{2}\right) (\cosh\Theta, \sinh\Theta),$$

and the equations imposed by the delta function have the two roots

$$\Theta = 0, \qquad \alpha = \pm\alpha_s, \qquad \cosh\left(\frac{\alpha_s}{2}\right) = \frac{\sqrt{s}}{2m}.$$

The absolute value of the Jacobian of $(\Theta, \alpha) \mapsto (p_1^0 + p_2^0, p_1^1 + p_2^1)$ at either root is

$$\left|\det\begin{pmatrix} \partial_\Theta P^0 & \partial_\alpha P^0 \\ \partial_\Theta P^1 & \partial_\alpha P^1 \end{pmatrix}\right| = \sqrt{s}\, m \sinh\left(\frac{\alpha_s}{2}\right).$$

The two roots in α cancel the identical-particle factor $1/2$. Using

$$|F_2^\varepsilon|^2 = |\kappa_\varepsilon|^2 \sinh^2\left(\frac{\alpha_s}{2}\right), \qquad \sinh^2\left(\frac{\alpha_s}{2}\right) = \frac{s - 4m^2}{4m^2},$$

one obtains (76.3). The threshold behavior is therefore $\sqrt{s - 4m^2}$, not a pole: the two-body phase-space square-root singularity is softened by the antisymmetric energy-density form factor, which vanishes at equal rapidities.

For the Euclidean formula, take a positive Euclidean time separation r. The positive-energy spectral term is the analytic continuation of the same finite-particle contribution:

$$G_\varepsilon(r) = \frac{1}{2} \int \frac{\mathrm{d}\theta_1}{2\pi} \frac{\mathrm{d}\theta_2}{2\pi} |\kappa_\varepsilon|^2 \sinh^2\left(\frac{\theta_1 - \theta_2}{2}\right) e^{-mr(\cosh\theta_1 + \cosh\theta_2)}.$$

Changing variables to (Θ, α) has unit Jacobian and gives

$$G_\varepsilon(r) = \frac{|\kappa_\varepsilon|^2}{8\pi^2} \int_{\mathbb{R}} \mathrm{d}\Theta \int_{\mathbb{R}} \mathrm{d}\alpha \, \sinh^2\left(\frac{\alpha}{2}\right) e^{-2mr\cosh(\alpha/2)\cosh\Theta}.$$

By the integral definition of K_0,

$$\int_{-\infty}^{\infty} e^{-2mr\cosh(\alpha/2)\cosh\Theta} \, \mathrm{d}\Theta = 2K_0\left(2mr\cosh\left(\frac{\alpha}{2}\right)\right).$$

The remaining integrand is even in α, giving (76.4). All equalities in this proof are separated-point equalities. Possible contact terms at $x = 0$ are not part of the finite-particle spectral calculation. □

The spin and disorder fields of the Ising theory are more instructive because their form factors form infinite families and because their locality relative to the Majorana particle is a branch-field structure with an explicit monodromy. Let

$$P_N(\theta_1, \ldots, \theta_N) = \prod_{1 \le i < j \le N} \tanh\left(\frac{\theta_i - \theta_j}{2}\right), \qquad P_0 = P_1 = 1.$$

The function P_N is antisymmetric and satisfies

$$P_N(\theta_1 + 2\pi \mathrm{i}, \theta_2, \ldots, \theta_N) = P_N(\theta_1, \ldots, \theta_N), \qquad P_N(\theta_2, \ldots, \theta_N, \theta_1) = (-1)^{N-1} P_N(\theta_1, \ldots, \theta_N).$$

Thus a nonzero product family with even N solves cyclicity with semi-locality phase -1, while a nonzero product family with odd N solves cyclicity with phase $+1$. This phase is part of the operator sector. It is not determined by the diagonal scattering function alone.

In the thermally perturbed Ising field theory, the sign of the Majorana mass selects which of the lattice order/disorder scaling fields has a nonzero vacuum expectation value. We write σ_+ for the spin field in the ordered convention in which

$$\bar{\sigma} = \langle \Omega | \sigma_+(0) | \Omega \rangle \neq 0.$$

With the conformal normalization $\langle \sigma(x)\sigma(0)\rangle \sim |x|^{-1/4}$ at short distance, the standard infinite-plane normalization is

$$\bar{\sigma} = 2^{1/12} e^{-1/8} A_{\mathrm{G}}^{3/2} |m|^{1/8},$$

where A_{G} is the Glaisher constant. The derivation below uses only the symbol $\bar{\sigma}$, because the form-factor equations determine the rapidity dependence and relative phases, not the ultraviolet normalization of the spin field.

Even spin-field form factors in the free massive fermion. Let the Majorana particle have $S = -1$, and let σ_+ be a scalar branch field with semi-locality phase -1 relative to the Majorana particle. The rapidity dependence selected by the form-factor equations is the even family

$$F_{2k}^{\sigma_+}(\theta_1, \ldots, \theta_{2k}) = \bar{\sigma}\, \mathrm{i}^k P_{2k}(\theta_1, \ldots, \theta_{2k}), \qquad F_{2k+1}^{\sigma_+} = 0, \tag{76.5}$$

It satisfies Watson exchange, semi-local cyclicity

$$F_{2k}^{\sigma_+}(\theta_1 + 2\pi\mathrm{i}, \theta_2, \ldots, \theta_{2k}) = -F_{2k}^{\sigma_+}(\theta_2, \ldots, \theta_{2k}, \theta_1),$$

and the semi-local annihilation-pole equation

$$-\mathrm{i} \operatorname*{Res}_{\theta' = \theta + \mathrm{i}\pi} F_{n+2}^{\sigma_+}(\theta', \theta, \eta_1, \ldots, \eta_n) = (1 + (-1)^n)\, F_n^{\sigma_+}(\eta_1, \ldots, \eta_n).$$

In particular,

$$F_0^{\sigma_+} = \bar{\sigma}, \qquad F_2^{\sigma_+}(\theta_1, \theta_2) = \mathrm{i}\bar{\sigma} \tanh\!\left(\frac{\theta_1 - \theta_2}{2}\right).$$

Away from coincident bra and ket rapidities, crossing K outgoing particles gives the connected analytic part of the mixed matrix element

$$\begin{aligned}
&{}_{\mathrm{out}}\langle \beta_1, \ldots, \beta_K | \sigma_+(0) | \beta'_1, \ldots, \beta'_N \rangle_{\mathrm{in}}^{\mathrm{conn}} \\
&\quad = \bar{\sigma}\, \mathrm{i}^{(K+N)/2} \prod_{1 \le i < j \le K} \tanh\!\left(\frac{\beta_i - \beta_j}{2}\right) \prod_{1 \le p < q \le N} \tanh\!\left(\frac{\beta'_p - \beta'_q}{2}\right) \prod_{\substack{1 \le s \le K \\ 1 \le t \le N}} \coth\!\left(\frac{\beta_s - \beta'_t}{2}\right),
\end{aligned} \tag{76.6}$$

for $K + N$ even; it vanishes for $K + N$ odd. The superscript "conn" reminds us that the full distributional matrix element also has disconnected delta-function contractions when some bra rapidity coincides with a ket rapidity.

The verification is the following explicit product calculation. Watson exchange follows because P_N is antisymmetric and $S = -1$. For $N = 2k$, the two displayed identities for P_N give

$$P_N(\theta_1 + 2\pi\mathrm{i}, \theta_2, \ldots, \theta_N) = -P_N(\theta_2, \ldots, \theta_N, \theta_1),$$

which is exactly cyclicity with semi-locality phase -1.

For the pole equation, set $\theta' = \theta + \mathrm{i}\pi + \epsilon$. The annihilating pair contributes

$$\tanh\left(\frac{\theta' - \theta}{2}\right) = \coth\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} + O(\epsilon),$$

while each spectator pair gives

$$\tanh\left(\frac{\theta' - \eta_j}{2}\right)\tanh\left(\frac{\theta - \eta_j}{2}\right) \longrightarrow 1.$$

Hence

$$\operatorname*{Res}_{\theta'=\theta+\mathrm{i}\pi} P_{n+2}(\theta', \theta, \eta_1, \ldots, \eta_n) = 2P_n(\eta_1, \ldots, \eta_n).$$

When n is even, $F_n^{\sigma+} = \bar{\sigma}\mathrm{i}^{n/2}P_n$ and $F_{n+2}^{\sigma+} = \bar{\sigma}\mathrm{i}^{n/2+1}P_{n+2}$, so

$$-\mathrm{i}\ \mathrm{Res}\, F_{n+2}^{\sigma+} = 2F_n^{\sigma+}.$$

This is the displayed equation because $1 + (-1)^n = 2$. When n is odd, both $F_n^{\sigma+}$ and $F_{n+2}^{\sigma+}$ vanish.

The crossed formula is a direct consequence of the same even family. Move K outgoing particles to the incoming side by the boundary-value continuation $\beta_s \mapsto \beta_s + \mathrm{i}\pi$. Products among two crossed rapidities and among two uncrossed rapidities remain tanh-products, because $\tanh(z + \pi\mathrm{i}) = \tanh z$. A crossed-uncrossed pair gives

$$\tanh\left(\frac{\beta_s + \mathrm{i}\pi - \beta_t'}{2}\right) = \coth\left(\frac{\beta_s - \beta_t'}{2}\right).$$

Substitution in $F_{K+N}^{\sigma+}$ gives (76.6). The exclusion of coincident rapidities is necessary because crossing a particle onto an identical particle in the other state also produces the ordinary Fock-space contraction terms; those delta terms are separate from the meromorphic connected form factor derived here.

The case $K = N = 1$ gives the convention-sensitive spin-field matrix element used in the Fonseca–Zamolodchikov treatment:

$${}_{\mathrm{out}}\langle\beta_1|\sigma_+(0)|\beta_2\rangle_{\mathrm{in}}^{\mathrm{conn}} = \mathrm{i}\bar{\sigma}\coth\left(\frac{\beta_1 - \beta_2}{2}\right).$$

In the present chapter this formula is the crossed boundary value of the two-particle spin form factor above.[1]

The dual odd product family is obtained by the same algebra with the opposite cyclicity phase. In a normalization $v \in \mathbb{C}$ it is

$$F_{2k+1}^{\Sigma}(\theta_1, \ldots, \theta_{2k+1}) = v\,\mathrm{i}^k \prod_{1\le i<j\le 2k+1} \tanh\left(\frac{\theta_i - \theta_j}{2}\right), \qquad F_{2k}^{\Sigma} = 0. \tag{76.7}$$

The constant $v = F_1^{\Sigma}$ is the one-particle normalization in this sector. Which product family is called order or disorder after matching to the lattice scaling limit depends on the sign convention for the Majorana mass and on the Kramers–Wannier identification. The mathematical point here is independent of that naming convention: the parity of the nonzero form-factor family is fixed by the declared cyclicity phase.

[1] A useful comparison point is P. Fonseca and A. Zamolodchikov, "Ising field theory in a magnetic field: analytic properties of the free energy", arXiv:hep-th/0112167, and "Ward identities and integrable differential equations in the Ising field theory", arXiv:hep-th/0309228. The formulas here are rederived in the rapidity and crossing conventions of this chapter.

Odd-family bootstrap verification. For the odd product family (76.7), the three bootstrap checks are elementary but convention-sensitive. The family satisfies Watson exchange, the local cyclicity equation for odd particle number, and the kinematic-pole equation with $S=-1$:

$$-\mathrm{i}\operatorname*{Res}_{\theta'=\theta+\mathrm{i}\pi} F^{\Sigma}_{n+2}(\theta',\theta,\eta_1,\dots,\eta_n)=(1-(-1)^n)\,F^{\Sigma}_{n}(\eta_1,\dots,\eta_n)$$

for every odd n. For even n, both sides vanish because $F^{\Sigma}_{n}=F^{\Sigma}_{n+2}=0$.

Interchanging adjacent variables changes exactly one factor by a sign and permutes the rest, so P_N is antisymmetric. Since $S=-1$, this is precisely Watson's equation.

For N odd,

$$P_N(\theta_1+2\pi\mathrm{i},\theta_2,\dots,\theta_N)=P_N(\theta_1,\theta_2,\dots,\theta_N),$$

because $\tanh(z+\pi\mathrm{i})=\tanh z$. Moving θ_1 from the first slot to the last reverses its order relative to $N-1$ variables and hence multiplies the antisymmetric product by $(-1)^{N-1}=1$. This proves cyclicity for the odd family.

It remains to compute the annihilation residue. Put

$$\theta'=\theta+\mathrm{i}\pi+\epsilon.$$

The pair factor is

$$\tanh\!\left(\frac{\theta'-\theta}{2}\right)=\tanh\!\left(\frac{\mathrm{i}\pi+\epsilon}{2}\right)=\coth\!\left(\frac{\epsilon}{2}\right)=\frac{2}{\epsilon}+O(\epsilon).$$

For each spectator rapidity η_j,

$$\tanh\!\left(\frac{\theta'-\eta_j}{2}\right)\tanh\!\left(\frac{\theta-\eta_j}{2}\right)\longrightarrow\coth\!\left(\frac{\theta-\eta_j}{2}\right)\tanh\!\left(\frac{\theta-\eta_j}{2}\right)=1.$$

Therefore

$$\operatorname*{Res}_{\theta'=\theta+\mathrm{i}\pi} P_{n+2}(\theta',\theta,\eta_1,\dots,\eta_n)=2P_n(\eta_1,\dots,\eta_n).$$

If $n=2k+1$, then $F^{\Sigma}_{n}=v\mathrm{i}^kP_n$, while $F^{\Sigma}_{n+2}=v\mathrm{i}^{k+1}P_{n+2}$. Hence

$$-\mathrm{i}\,\operatorname{Res}F^{\Sigma}_{n+2}=-\mathrm{i}\,2v\mathrm{i}^{k+1}P_n=2v\mathrm{i}^kP_n=2F^{\Sigma}_{n}.$$

Since n is odd, $1-(-1)^n=2$, giving the claimed kinematic-pole equation.

Proposition 76.7 (Separated Euclidean spectral series for the Ising branch fields)**.** *Use the free Majorana one-particle normalization of Proposition 76.6, and consider Euclidean time separation $r>0$. Define*

$$I_m(r)=\int_{\mathbb{R}}\frac{\mathrm{d}\theta}{2\pi}\,e^{-mr\cosh\theta}=\frac{1}{\pi}K_0(mr). \tag{76.8}$$

The even spin-field form factors (76.5) determine an absolutely convergent separated-point Euclidean spectral series

$$G_{\sigma_+}(r)=\sum_{k=0}^{\infty}\frac{1}{(2k)!}\int_{\mathbb{R}^{2k}}\prod_{j=1}^{2k}\frac{\mathrm{d}\theta_j}{2\pi}\,\left|F^{\sigma_+}_{2k}(\theta_1,\dots,\theta_{2k})\right|^2e^{-mr\sum_{j=1}^{2k}\cosh\theta_j}. \tag{76.9}$$

For every $r > 0$,

$$|G_{\sigma_+}(r)| \le |\bar{\sigma}|^2 \cosh I_m(r). \tag{76.10}$$

After removing the $k = 0$ term, the connected series is bounded by $|\bar{\sigma}|^2(\cosh I_m(r) - 1)$.

The odd product family (76.7) similarly gives

$$G_\Sigma(r) = \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} \int_{\mathbb{R}^{2k+1}} \prod_{j=1}^{2k+1} \frac{\mathrm{d}\theta_j}{2\pi} \left|F^{\Sigma}_{2k+1}(\theta_1, \ldots, \theta_{2k+1})\right|^2 e^{-mr\sum_{j=1}^{2k+1}\cosh\theta_j}, \tag{76.11}$$

with

$$|G_\Sigma(r)| \le |v|^2 \sinh I_m(r). \tag{76.12}$$

Both series converge uniformly with all r-derivatives on compact subsets of $(0, \infty)$. Hence they define smooth separated Euclidean two-point functions on the temporal ray; Euclidean covariance then gives the corresponding separated radial functions. This proposition is a separated-point reconstruction statement for this free model. The full local-reconstruction problem for a general factorizing S-matrix is the open problem stated below.

Proof. The finite-particle identity resolution gives (76.9) and (76.11) exactly as in the energy-density derivation, with the factorial compensating for integration over labelled rapidity variables of identical Majorana particles. For real rapidities,

$$\left|\tanh\left(\frac{\theta_i - \theta_j}{2}\right)\right| \le 1.$$

Therefore

$$\left|F^{\sigma_+}_{2k}(\theta_1, \ldots, \theta_{2k})\right|^2 \le |\bar{\sigma}|^2, \qquad \left|F^{\Sigma}_{2k+1}(\theta_1, \ldots, \theta_{2k+1})\right|^2 \le |v|^2. \tag{76.13}$$

The factorized exponential then gives the termwise bounds

$$\frac{1}{(2k)!} \int_{\mathbb{R}^{2k}} \prod_{j=1}^{2k} \frac{\mathrm{d}\theta_j}{2\pi} |\bar{\sigma}|^2 e^{-mr\sum_j \cosh\theta_j} = |\bar{\sigma}|^2 \frac{I_m(r)^{2k}}{(2k)!} \tag{76.14}$$

and

$$\frac{1}{(2k+1)!} \int_{\mathbb{R}^{2k+1}} \prod_{j=1}^{2k+1} \frac{\mathrm{d}\theta_j}{2\pi} |v|^2 e^{-mr\sum_j \cosh\theta_j} = |v|^2 \frac{I_m(r)^{2k+1}}{(2k+1)!}. \tag{76.15}$$

Summing the even and odd majorants gives (76.10) and (76.12). The identity (76.8) follows from the integral definition of K_0 and evenness in θ.

Let $0 < r_0 < r_1 < \infty$. A derivative of order q with respect to r inserts a finite sum of factors bounded by

$$m^q \left(\sum_j \cosh\theta_j\right)^q e^{-mr\sum_j \cosh\theta_j}.$$

For $r \in [r_0, r_1]$, split

$$e^{-mr\sum_j \cosh\theta_j} = e^{-(mr_0/2)\sum_j \cosh\theta_j} e^{-m(r-r_0/2)\sum_j \cosh\theta_j}.$$

The polynomial factor in $\cosh\theta_j$ is dominated by a constant depending on q, r_0, m times the first exponential, and the remaining integrals are bounded by the same exponential series with $r_0/2$ in place of r. The Weierstrass test therefore applies to every derivative order on $[r_0, r_1]$. The resulting functions are smooth for $r > 0$, and the spectral construction gives the separated Euclidean two-point functions on the temporal ray. Contact terms at coincident points are outside this separated-point estimate and belong to the extension problem on the collision diagonal. $\square$

76.6 Bound-State Poles and Operator Couplings

Suppose species a and b have a bound state c. If the two-body S-matrix has a physical-strip pole at $\theta = \mathrm{i}u_{ab}^c$ with residue tensor $\Gamma_{ab}^c : V_c \to V_a \otimes V_b$, then the form factor has the compatible pole

$$-\mathrm{i} \operatorname*{Res}_{\theta_a - \theta_b = \mathrm{i}u_{ab}^c} F_{aba_1\ldots a_n}^{\mathcal{O}} = F_{ca_1\ldots a_n}^{\mathcal{O}} \Gamma_{ab}^c,$$

again with the stated tensor-ordering convention. This equation is the operator-level statement that a local field couples to the bound state through the same one-particle channel that appears in the scattering residue.

76.7 Form-Factor Reconstruction Boundary

The structure of Definition 76.1 supplies the bootstrap input. Its growth entry is essential: it controls whether the spectral series for Wightman functions,

$$\langle\Omega|\widehat{\mathcal{O}}(x)\widehat{\mathcal{O}}(0)|\Omega\rangle = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{a_1,\ldots,a_n} \int d\mu_n \, e^{-\mathrm{i}x\cdot\sum_j p_j} F_{a_1\ldots a_n}^{\mathcal{O}}(\theta) \overline{F_{a_1\ldots a_n}^{\mathcal{O}}(\theta)}$$

defines a distribution. Here $d\mu_n$ is the Lorentz-invariant n-particle rapidity measure with the normalization fixed by the one-particle Hilbert-space convention.

Status checkpoint: from form factors to fields. The preceding equations are necessary matrix-element identities for a local or semi-local field in a factorizing theory. They become a local-field construction only after five analytic steps have been supplied in the same normalization. First, the finite-particle vectors must sit in a positive Hilbert space with a common invariant domain on which the smeared operators are defined and closable. Second, the spectral sums must converge as tempered Wightman distributions, with separated Euclidean functions for $r > 0$ serving only as one part of that construction; this includes control of the extension to collision diagonals and of contact terms. Third, the resulting distributions must satisfy covariance, the spectrum condition, and positivity. Fourth, the commutator or semi-locality relation must vanish on spacelike test-function supports as an operator statement on the common domain. Fifth, the finite-particle states used in the series must be complete in the sector claimed. The Ising Majorana examples above meet these requirements because the underlying free field already supplies the Hilbert space and local fields, and the product bounds prove separated Euclidean convergence for the displayed branch families. For an interacting scalar factorizing model, the same list is a set of additional analytic hypotheses beyond Watson, cyclicity, and residue equations.

Open Problem 76.8 (Local reconstruction from form factors). Given a factorized S-matrix and a family of form-factor data satisfying the displayed equations and explicit growth bounds, construct the Wightman domain, prove convergence of the spectral series as distributions, verify local commutativity, and identify a dense set of local operators. This is the operator-level completion of the scattering bootstrap.

Chapter 77

Thermodynamic Bethe Ansatz

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The thermodynamic Bethe ansatz turns the exact two-body scattering data of a massive integrable theory into finite-temperature or finite-size thermodynamics. The construction in this chapter is stated first for a diagonal elastic S-matrix. Matrix-valued scattering, bound-state strings, and nested Bethe ansatz data require additional species and auxiliary-root bookkeeping but obey the same variational logic once the allowed densities are specified.

Definition 77.1 (Diagonal massive TBA specification)**.** A diagonal massive thermodynamic Bethe ansatz specification consists of the following objects.

(i) A finite species set A, with a mass $m_a > 0$ for each $a \in \mathsf{A}$.

(ii) Diagonal two-body scattering amplitudes $S_{ab}(\theta) \in U(1)$ for real rapidity difference θ, with a declared real-axis phase branch

$$S_{ab}(\theta) = \exp\bigl(\mathrm{i}\delta_{ab}(\theta)\bigr)$$

on each rapidity interval used in the thermodynamic calculation. The logarithmic derivative kernel is

$$\varphi_{ab}(\theta) = \frac{1}{2\pi}\delta'_{ab}(\theta) = \frac{1}{2\pi\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\theta}\log S_{ab}(\theta)$$

as a locally integrable function or distribution. The branch convention is part of the specification whenever S_{ab} has constant signs, poles, or CDD factors crossing the chosen real-axis chart.

(iii) Circle quantization shifts $\nu_a \in \mathbb{R}/\mathbb{Z}$, encoding the statistics and boundary convention for each species.

(iv) A thermodynamic root-density statement: for circle circumference $L \to \infty$, the Bethe–Yang solutions whose rapidities stay in a fixed compact interval have empirical measures converging to particle densities $\rho_a(\theta)\mathrm{d}\theta$ and hole densities $\rho_a^h(\theta)\mathrm{d}\theta$, and the differentiated Bethe–Yang map gives the density constraint

$$\rho_a + \rho_a^h = \frac{m_a}{2\pi}\cosh\theta + \sum_{b\in\mathsf{A}} \varphi_{ab} * \rho_b.$$

(v) The exclusion-statistics entropy convention

$$s[\rho,\rho^h] = \sum_{a\in\mathsf{A}} \int d\theta \left[\rho_a^t \log\rho_a^t - \rho_a\log\rho_a - \rho_a^h\log\rho_a^h\right], \qquad \rho_a^t = \rho_a + \rho_a^h.$$

This convention is the one matched to one occupied or empty Bethe quantum number in each microscopic rapidity cell. Other generalized exclusion-statistics systems require the corresponding entropy and density matrix in place of this item.

(vi) A thermal or mirror-channel normalization. At inverse temperature $R > 0$, the energy density coordinate is $\sum_a \int d\theta\, m_a \cosh\theta\, \rho_a(\theta)$. When the same specification is used for a finite-size ground-state energy, the chapter assumes the mirror-channel identification and the normalization of $E_0(R)$ stated in Section 77.5.

77.1 Bethe–Yang Quantization

For the diagonal specification of Definition 77.1, place the theory on a spatial circle of circumference L. For n particles of masses m_a, rapidities θ_j, and diagonal scattering phases

$$S_{ab}(\theta) = \exp\bigl(\mathrm{i}\delta_{ab}(\theta)\bigr),$$

periodicity of the wavefunction of particle j gives the Bethe–Yang equation

$$m_{a_j} L\sinh\theta_j + \sum_{k\neq j} \delta_{a_j a_k}(\theta_j - \theta_k) = 2\pi I_j, \qquad I_j \in \mathbb{Z} + \nu_j.$$

The shift ν_j depends on statistics and boundary convention. The one-particle energy and momentum are

$$E_a(\theta) = m_a\cosh\theta, \qquad p_a(\theta) = m_a\sinh\theta.$$

77.2 Density Equations

In the thermodynamic limit $L \to \infty$, introduce particle and hole densities $\rho_a(\theta)$ and $\rho_a^h(\theta)$, normalized so that

$$L\rho_a(\theta)d\theta$$

is the number of occupied rapidities of species a in $d\theta$. Define the scattering kernel

$$\varphi_{ab}(\theta) = \frac{1}{2\pi}\frac{d}{d\theta}\delta_{ab}(\theta) = \frac{1}{2\pi \mathrm{i}}\frac{d}{d\theta}\log S_{ab}(\theta).$$

Differentiating the Bethe–Yang equations gives the density constraint

$$\rho_a(\theta) + \rho_a^h(\theta) = \frac{m_a}{2\pi}\cosh\theta + \sum_b \int_{\mathbb{R}} d\theta'\, \varphi_{ab}(\theta-\theta')\rho_b(\theta').$$

This equation is purely kinematic: it counts available rapidity levels after the scattering phase shifts have shifted the quantization conditions.

77.3 Entropy and Free Energy

For fixed total density $\rho_a^t = \rho_a + \rho_a^h$, the number of microscopic occupancy choices in an interval is

$$\binom{L\rho_a^t\, d\theta}{L\rho_a\, d\theta}.$$

Stirling's formula gives the entropy density functional

$$s[\rho,\rho^h] = \sum_a \int d\theta\, \left[\rho_a^t \log\rho_a^t - \rho_a\log\rho_a - \rho_a^h\log\rho_a^h\right].$$

At inverse temperature R, the free energy per length is

$$f[\rho,\rho^h] = \sum_a \int d\theta\, m_a\cosh\theta\,\rho_a(\theta) - \frac{1}{R}s[\rho,\rho^h],$$

subject to the density constraint.

77.4 Pseudoenergy Equation

Define the pseudoenergy by

$$\epsilon_a(\theta) = \log\frac{\rho_a^h(\theta)}{\rho_a(\theta)}.$$

Varying the constrained free energy with respect to ρ_a gives

$$\epsilon_a(\theta) = Rm_a\cosh\theta - \sum_b\int_{\mathbb{R}} d\theta'\,\varphi_{ab}(\theta-\theta')\log\Big(1+e^{-\epsilon_b(\theta')}\Big).$$

This is the thermodynamic Bethe ansatz equation. The equilibrium free energy is

$$f(R) = -\frac{1}{2\pi R}\sum_a m_a\int_{\mathbb{R}} d\theta\,\cosh\theta\,\log\Big(1+e^{-\epsilon_a(\theta)}\Big).$$

Proposition 77.2 (Variational derivation of TBA)**.** *For diagonal elastic scattering with smooth phase kernels and a thermodynamic limit in which the Bethe–Yang root densities exist, the stationary point of the constrained free-energy functional is exactly the pseudoenergy equation above.*

Proof. Work on the open cone where $\rho_a > 0$ and $\rho_a^h > 0$, and take compactly supported smooth variations of the particle densities. The density constraint then determines the variation of the total density:

$$\delta\rho_c^t(\theta) = \sum_a \int d\theta' \, \varphi_{ca}(\theta - \theta')\delta\rho_a(\theta').$$

Since $\rho_a^h = \rho_a^t - \rho_a$, the corresponding hole-density variation is $\delta\rho_a^h = \delta\rho_a^t - \delta\rho_a$. The entropy variation is

$$\delta s = \sum_a \int d\theta \left[\log \frac{\rho_a^h}{\rho_a} \, \delta\rho_a + \log \frac{\rho_a^t}{\rho_a^h} \, \delta\rho_a^t\right].$$

Let

$$L_a(\theta) = \log \frac{\rho_a^t(\theta)}{\rho_a^h(\theta)} = \log(1 + e^{-\epsilon_a(\theta)}).$$

Substituting the constraint variation into $\delta f = \delta e - R^{-1}\delta s$ and collecting the coefficient of $\delta\rho_a(\theta)$ gives

$$0 = m_a \cosh\theta - \frac{1}{R}\epsilon_a(\theta) - \frac{1}{R}\sum_c \int d\theta' \, \varphi_{ca}(\theta' - \theta) L_c(\theta').$$

With the diagonal-scattering convention $\varphi_{ca}(\theta' - \theta) = \varphi_{ac}(\theta - \theta')$, this is exactly the displayed TBA equation.

It remains to derive the compact expression for the stationary free energy. The entropy density identity

$$\rho_a^t \log\rho_a^t - \rho_a \log\rho_a - \rho_a^h \log\rho_a^h = \rho_a \, \epsilon_a + \rho_a^t L_a$$

gives

$$Rf_* = \sum_a \int d\theta \, \rho_a(Rm_a \cosh\theta - \epsilon_a) - \sum_a \int d\theta \, \rho_a^t L_a.$$

Use the stationary equation for the first parenthesis and the density constraint for ρ_a^t. The convolution terms cancel after interchanging species labels and integration variables, leaving only the one-particle state-density term

$$Rf_* = -\sum_a \int d\theta \, \frac{m_a}{2\pi} \cosh\theta \, L_a(\theta),$$

which is the stated formula for $f(R)$. □

77.5 Finite-Size Ground-State Energy

Interchanging space and Euclidean time maps the thermal free energy on an infinite line at inverse temperature R to the ground-state energy on a circle of circumference R, after the mirror-channel scattering data have been identified. In a relativistic theory the mirror dispersion is obtained by a rapidity shift by $\mathrm{i}\pi/2$, so the same TBA equation gives

$$E_0(R) = -\frac{1}{2\pi}\sum_a m_a \int_{\mathbb{R}} d\theta \, \cosh\theta \, \log\Big(1 + e^{-\epsilon_a(\theta)}\Big).$$

The effective central charge extracted from the ultraviolet limit is

$$c_{\text{eff}}(R) = -\frac{6R}{\pi} E_0(R), \qquad R \to 0,$$

provided the limit exists and the infrared bulk energy has been subtracted in the declared normalization.

77.6 Free Majorana Example and the Ultraviolet Central Charge

The massive Ising field theory in its fermionic phase gives the minimal diagonal example. It has one neutral particle of mass m. The two-particle scattering phase is the constant

$$S(\theta) = -1.$$

The derivative kernel is therefore zero:

$$\varphi(\theta) = \frac{1}{2\pi \mathrm{i}} \frac{\mathrm{d}}{\mathrm{d}\theta} \log S(\theta) = 0,$$

after the constant branch of $\log(-1) = \mathrm{i}\pi$ has been fixed. The TBA equation reduces to

$$\epsilon(\theta) = r \cosh\theta, \qquad r = mR.$$

The finite-size ground-state energy in the normalization of the preceding section is

$$E_0(R) = -\frac{m}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}\theta \, \cosh\theta \, \log\Big(1 + \mathrm{e}^{-r\cosh\theta}\Big). \tag{77.1}$$

The ultraviolet central charge follows from a direct scaling limit. Put

$$x = \sinh\theta, \qquad \mathrm{d}x = \cosh\theta \, \mathrm{d}\theta.$$

Then

$$I(r) := \int_{-\infty}^{\infty} \mathrm{d}\theta \, \cosh\theta \, \log(1 + \mathrm{e}^{-r\cosh\theta}) = \int_{-\infty}^{\infty} \mathrm{d}x \, \log(1 + \mathrm{e}^{-r\sqrt{1+x^2}}).$$

Rescale $y = rx$. This gives

$$rI(r) = \int_{-\infty}^{\infty} \mathrm{d}y \, \log(1 + \mathrm{e}^{-\sqrt{r^2+y^2}}).$$

For $0 < r \le 1$ the integrand is bounded by $\log(1+\mathrm{e}^{-|y|})$, which is integrable on $\mathbb{R}$, and it converges pointwise to $\log(1 + \mathrm{e}^{-|y|})$. Dominated convergence gives

$$\lim_{r \downarrow 0} rI(r) = 2 \int_0^{\infty} \mathrm{d}y \, \log(1 + \mathrm{e}^{-y}).$$

Expanding $\log(1 + \mathrm{e}^{-y})$ for $y > 0$ and integrating termwise gives

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2} = \int_0^{\infty} \mathrm{d}y \, \log(1 + \mathrm{e}^{-y}) = \frac{\pi^2}{12},$$

Equation (77.1) gives

$$E_0(R) = -\frac{\pi}{12R} + o(R^{-1}).$$

Thus

$$c_{\text{eff}} = \lim_{R\downarrow 0}\left(-\frac{6R}{\pi}E_0(R)\right) = \frac{1}{2}.$$

The result matches the central charge of the ultraviolet Majorana conformal field theory, but the computation above uses only the declared finite-size TBA specification and the normalization of $E_0(R)$.

77.7 Scaling Lee–Yang Example and the Effective Central Charge

The first interacting one-particle TBA example is the massive scaling Lee–Yang theory. It is not a unitary Hilbert-space example; it is included here because it is the minimal nontrivial diagonal bootstrap specification for which the TBA ultraviolet limit can be carried to an exact number. The ultraviolet conformal theory is the nonunitary minimal model $M(2,5)$, whose central charge is $c = -22/5$. Its lowest conformal weight is $-1/5$, so the cylinder ground-state energy is governed by the effective central charge

$$c_{\text{eff}} = c - 24h_{\text{min}} = \frac{2}{5}.$$

The TBA computation below derives the same number from the scattering specification.

The diagonal one-particle amplitude is

$$S_{\text{LY}}(\theta) = \frac{\sinh\theta + \mathrm{i}\sin(\pi/3)}{\sinh\theta - \mathrm{i}\sin(\pi/3)}.$$

With the kernel convention of this chapter,

$$\varphi_{\text{LY}}(\theta) = \frac{1}{2\pi\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\theta}\log S_{\text{LY}}(\theta) = -\frac{\sin(\pi/3)\cosh\theta}{\pi\left(\sinh^2\theta + \sin^2(\pi/3)\right)} = -\frac{2\sqrt{3}\cosh\theta}{\pi(2\cosh 2\theta + 1)}. \tag{77.2}$$

It has total integral

$$N_{\text{LY}} = \int_{\mathbb{R}} \mathrm{d}\theta\, \varphi_{\text{LY}}(\theta) = -1. \tag{77.3}$$

Indeed, putting $x = \sinh\theta$ and $s = \sin(\pi/3)$ gives

$$N_{\text{LY}} = -\frac{s}{\pi}\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{x^2+s^2} = -1.$$

The TBA equation is therefore

$$\epsilon(\theta) = r\cosh\theta - \int_{\mathbb{R}} \mathrm{d}\theta'\, \varphi_{\text{LY}}(\theta-\theta')\log(1+\mathrm{e}^{-\epsilon(\theta')}). \tag{77.4}$$

Controlled Approximation 77.3 (Lee–Yang ultraviolet plateau)**.** Assume the standard one-plateau ultraviolet limit of (77.4): as $r\downarrow 0$, there is a rapidity region whose width tends to infinity in which $\epsilon(\theta)$ approaches a constant ϵ_0, while the kernel is replaced by its integral N_{LY}. Then

$$Y_0 := \mathrm{e}^{\epsilon_0} = \phi_{\mathrm{g}} = \frac{1+\sqrt{5}}{2}, \qquad c_{\text{eff}} = \frac{2}{5}.$$

The plateau equation is obtained from (77.4) by dropping the vanishing driving term and replacing the convolution by N_{LY}:

$$\epsilon_0 = -N_{\mathrm{LY}} \log(1 + \mathrm{e}^{-\epsilon_0}) = \log(1 + \mathrm{e}^{-\epsilon_0}).$$

Writing $Y_0 = \mathrm{e}^{\epsilon_0}$ gives

$$Y_0 = 1 + Y_0^{-1}, \qquad Y_0^2 = Y_0 + 1.$$

The positive solution is the golden ratio ϕ_{g}.

The ultraviolet TBA central charge is the one-species Rogers-dilogarithm value

$$c_{\mathrm{eff}} = \frac{6}{\pi^2} L\left(\frac{1}{1 + Y_0}\right) = \frac{6}{\pi^2} L(\phi_{\mathrm{g}}^{-2}).$$

For $0 < x < 1$, use

$$L(x) = \mathrm{Li}_2(x) + \frac{1}{2} \log x \log(1 - x).$$

The two identities needed here are

$$L(x) + L(1 - x) = \frac{\pi^2}{6}, \qquad L(x^2) = 2L(x) - 2L\left(\frac{x}{1 + x}\right). \tag{77.5}$$

The first follows from Euler's dilogarithm identity. The second follows by differentiating the difference of the two sides, using $\mathrm{Li}_2'(x) = -\log(1 - x)/x$, and evaluating at $x = 0$. Set $x = \phi_{\mathrm{g}}^{-1}$. Then

$$1 - x = x^2, \qquad \frac{x}{1 + x} = x^2.$$

The identities become

$$L(x) + L(x^2) = \frac{\pi^2}{6}, \qquad L(x^2) = 2L(x) - 2L(x^2).$$

Thus $3L(x^2) = 2L(x)$. Solving the two linear equations gives

$$L(\phi_{\mathrm{g}}^{-2}) = L(x^2) = \frac{\pi^2}{15}.$$

Therefore $c_{\mathrm{eff}} = 6/15 = 2/5$.

Remark 77.4 (What the example proves)*.* The calculation proves an internal consistency statement for the declared TBA datum and its ultraviolet plateau asymptotics. It is not by itself a construction of the Lee–Yang local QFT, nor a derivation that the finite-volume Hamiltonian spectrum of such a QFT is governed by (77.4). Those statements require the Hilbert-space and locality input separated in the theorem boundary below.

Status checkpoint: from scattering data to TBA. The TBA pseudoenergy equation has a different logical status from the factorized scattering equations. The scattering input supplies the kernel $\varphi = (2\pi \mathrm{i})^{-1} \partial_\theta \log S$. To turn that kernel into a thermodynamic statement about a local QFT one must also prove the finite-volume Bethe–Yang description with controlled errors, convergence of Bethe counting measures to root and hole densities, the microscopic entropy of the allowed Bethe cells, existence and uniqueness of the free-energy minimizer, and the channel identification that converts mirror free energy into direct finite-size ground-state energy. The free Majorana and Lee–Yang computations above verify the internal finite algebra and ultraviolet asymptotics of specific TBA data. They do not by themselves prove that an arbitrary local Hamiltonian with the same proposed scattering phase has that finite-volume spectrum.

77.8 Euler-Scale Generalized Hydrodynamics

The equilibrium TBA datum also supplies the kinematic variables used in generalized hydrodynamics. In this chapter it is treated as a conditional Euler-scale closure for inhomogeneous states whose local cells are assumed to be described by thermodynamic Bethe root densities.

Let $\rho_a(x,t,\theta)$ and $\rho_a^h(x,t,\theta)$ be slowly varying particle and hole densities, and set

$$\rho_a^t = \rho_a + \rho_a^h, \qquad n_a = \frac{\rho_a}{\rho_a^t}.$$

The local density constraint is the pointwise version of the TBA counting equation:

$$\rho_a^t(x,t,\theta) = \frac{p_a'(\theta)}{2\pi} + \sum_b \int d\alpha\, \varphi_{ab}(\theta-\alpha)\rho_b(x,t,\alpha), \qquad p_a'(\theta) = m_a \cosh\theta. \tag{77.6}$$

For any vector of one-particle functions $h_a(\theta)$, define its dressing by the linear equation

$$h_a^{\mathrm{dr}}(\theta) = h_a(\theta) + \sum_b \int d\alpha\, \varphi_{ab}(\theta-\alpha) n_b(\alpha) h_b^{\mathrm{dr}}(\alpha). \tag{77.7}$$

The inverse of $1-\varphi n$ is part of the hydrodynamic chart; if the operator has a zero mode, this chart has left its domain of validity. Applying (77.7) to p_a' gives

$$(p_a')^{\mathrm{dr}} = 2\pi\rho_a^t,$$

which is precisely (77.6). Applying it to

$$E_a'(\theta) = m_a \sinh\theta$$

defines the Euler-scale effective velocity

$$v_a^{\mathrm{eff}}(\theta) = \frac{(E_a')^{\mathrm{dr}}(\theta)}{(p_a')^{\mathrm{dr}}(\theta)}. \tag{77.8}$$

Hypothesis 77.5 (Euler-scale GHD closure)**.** Fix a diagonal massive TBA specification and a family of inhomogeneous finite-volume states whose mesoscopic cells have thermodynamic root densities $\rho_a(x,t,\theta)$. The Euler-scale generalized-hydrodynamic closure is the assertion that, for every compact rapidity interval and every test function in x,θ, the empirical quasiparticle measures converge in the scaling limit to densities obeying

$$\partial_t\rho_a(x,t,\theta) + \partial_x\left(v_a^{\mathrm{eff}}(x,t,\theta)\rho_a(x,t,\theta)\right) = 0, \tag{77.9}$$

with v_a^{eff} defined from the local dressing equations (77.7)–(77.8).

This hypothesis has a direct conservation check. For a local conserved charge whose one-particle eigenvalue is $h_a(\theta)$, define the Euler-cell density and current by

$$q_h(x,t) = \sum_a \int d\theta\, h_a(\theta)\rho_a(x,t,\theta), \qquad j_h(x,t) = \sum_a \int d\theta\, h_a(\theta) v_a^{\mathrm{eff}}(x,t,\theta)\rho_a(x,t,\theta). \tag{77.10}$$

Multiplying (77.9) by h_a, summing over species, and integrating in rapidity gives

$$\partial_t q_h + \partial_x j_h = 0$$

whenever boundary terms vanish. The closure is therefore compatible with the infinite family of Bethe charges by construction. The difficult assertion is not this algebraic conservation check, but the scaling theorem that the microscopic inhomogeneous QFT actually converges to (77.9).

The formula for v^{eff} is also the linearized counting formula for a dressed wave packet. Insert a single quasiparticle into a thermodynamic cell and perturb the Bethe counting function. The shift of available rapidity levels is governed by the same kernel φ that appears in (77.6). The dressed momentum and dressed energy gradients of the excitation are therefore $(p')^{\text{dr}}$ and $(E')^{\text{dr}}$, and the group velocity of the dressed packet is their ratio. This is the kinematic content of (77.8); it does not prove equilibration to a locally stationary Bethe state.

Hard-rod calibration. A useful finite algebraic laboratory is the classical hard-rod gas, which is not a relativistic local QFT but tests the collision-shift logic. Let rods have length $a > 0$, velocity labels v_i, and local velocity densities ρ_i with $a\rho_{\text{tot}} < 1$,

$$\rho_{\text{tot}} = \sum_i \rho_i.$$

The exact collision shift gives the effective-velocity equations

$$v_i^{\text{eff}} = v_i + a \sum_j \rho_j \left(v_i^{\text{eff}} - v_j^{\text{eff}} \right). \tag{77.11}$$

Writing $J = \sum_j \rho_j v_j^{\text{eff}}$, this becomes

$$(1 - a\rho_{\text{tot}}) v_i^{\text{eff}} = v_i - aJ.$$

Multiplying by ρ_i and summing over i gives

$$J = \sum_i \rho_i v_i.$$

Hence

$$v_i^{\text{eff}} = \frac{v_i - a\sum_j \rho_j v_j}{1 - a\rho_{\text{tot}}}. \tag{77.12}$$

The formula displays the same structure as (77.8): bare velocities are corrected by the finite density of collision shifts, and the correction depends on the state. In the quantum integrable case the scalar shift a is replaced by the rapidity-dependent scattering kernel and the algebraic solution by the dressing equation (77.7).

Open Problem 77.6 (GHD from local integrable QFT). Starting from a constructed local integrable QFT with factorized scattering, derive the Euler-scale generalized-hydrodynamic closure (77.9) for inhomogeneous finite-energy states. The proof must identify the local thermodynamic Bethe state, control finite-volume and mesoscopic errors, justify the dressing inverse, and show that nonintegrable perturbations or inelastic channels are absent on the Euler time scale under consideration.

77.9 Weak Integrability Breaking And Kinetic Cells

Perturbing an integrable QFT changes the role of the Bethe charges. The Hamiltonian still has exact energy, momentum, and internal-symmetry charges declared by the perturbation, but the higher commuting charges of the integrable point are generally lost. A controlled kinetic description must therefore specify which finite-cell transitions are allowed, which charges they preserve exactly, and what entropy functional is monotone under the coarse-grained collision dynamics. These are finite algebraic requirements before any limiting theorem from a local QFT is attempted.

Definition 77.7 (Finite collision-cell Markov model)**.** A finite collision-cell Markov model consists of the following objects.

1. A finite set Ω of mesoscopic cell states.
2. A strictly positive stationary weight π_ω on Ω, normalized by $\sum_{\omega\in\Omega}\pi_\omega = 1$.
3. Transition rates $k_{\omega\omega'} \geq 0$, for $\omega \neq \omega'$, satisfying detailed balance
$$\pi_\omega k_{\omega\omega'} = \pi_{\omega'} k_{\omega'\omega}. \tag{77.13}$$
4. A finite family of exactly conserved cell observables $Q^A : \Omega \to \mathbb{R}$, meaning that $k_{\omega\omega'} \neq 0$ implies $Q^A(\omega) = Q^A(\omega')$ for every A.
5. Occupation variables $N_i : \Omega \to \mathbb{Z}_{\geq 0}$, where i labels a finite rapidity/species bin.

The cell probability $P_\omega(T)$ evolves by
$$\frac{dP_\omega}{dT} = \sum_{\omega'\neq\omega} \left(P_{\omega'} k_{\omega'\omega} - P_\omega k_{\omega\omega'}\right). \tag{77.14}$$

The projected occupation collision functional is
$$C_i[P] = \sum_{\omega,\omega'} P_\omega k_{\omega\omega'} \left(N_i(\omega') - N_i(\omega)\right). \tag{77.15}$$

If a conserved observable has the occupation form
$$Q^A(\omega) = Q^A_0 + \sum_i h^A_i N_i(\omega),$$
then the projected functional obeys
$$\sum_i h^A_i C_i[P] = 0. \tag{77.16}$$

Lemma 77.8 (Finite collision-cell entropy identity)**.** *For a finite collision-cell Markov model, set*
$$r_\omega = \frac{P_\omega}{\pi_\omega}, \qquad \mathcal{H}(P|\pi) = \sum_{\omega\in\Omega} P_\omega \log\frac{P_\omega}{\pi_\omega}.$$

Along (77.14),

$$\frac{d}{dT}\mathcal{H}(P|\pi) = -\frac{1}{2}\sum_{\omega,\omega'} \pi_\omega k_{\omega\omega'} \left(r_\omega - r_{\omega'}\right) \left(\log r_\omega - \log r_{\omega'}\right) \leq 0. \tag{77.17}$$

Moreover

$$\frac{d}{dT}\sum_{\omega} P_\omega Q^A(\omega) = 0$$

for every exactly conserved cell observable Q^A.

Proof. The master equation conserves total probability, so the derivative of the relative entropy is

$$\frac{d}{dT}\mathcal{H}(P|\pi) = \sum_{\omega} \dot{P}_\omega \log r_\omega.$$

Insert (77.14) and exchange the dummy indices in the gain term:

$$\frac{d}{dT}\mathcal{H}(P|\pi) = \sum_{\omega,\omega'} P_\omega k_{\omega\omega'} \left(\log r_{\omega'} - \log r_\omega\right).$$

Using $P_\omega = \pi_\omega r_\omega$ and detailed balance, the same sum with ω and ω' exchanged is

$$\sum_{\omega,\omega'} \pi_\omega k_{\omega\omega'} r_{\omega'} \left(\log r_\omega - \log r_{\omega'}\right).$$

Averaging the two expressions gives (77.17). The final sign follows from monotonicity of the logarithm:

$$(x-y)(\log x - \log y) \geq 0 \qquad x, y > 0.$$

For an exactly conserved cell observable, multiply the master equation by $Q^A(\omega)$, sum over ω, and symmetrize the result as

$$\frac{d}{dT}\sum_{\omega} P_\omega Q^A(\omega) = \sum_{\omega,\omega'} P_\omega k_{\omega\omega'} \left(Q^A(\omega') - Q^A(\omega)\right).$$

Every nonzero rate has $Q^A(\omega') = Q^A(\omega)$, hence the derivative vanishes. If Q^A is linear in the occupation variables, the same identity is exactly (77.16). $\square$

Controlled Approximation 77.9 (Weak-breaking kinetic scaling)**.** Let H_0 be an integrable Hamiltonian whose thermodynamic cells are described by the TBA variables of Section 77.8, and perturb it by a local operator,

$$H_\varepsilon = H_0 + \varepsilon V, \qquad V = \int dx\, \mathcal{O}(x),$$

with $0 < \varepsilon \ll 1$. A weak-breaking kinetic description is the following scaling ansatz. Choose a mesoscopic cell length ℓ and a macroscopic length X with microscopic scale $\ll \ell \ll X$, pass to the kinetic time $T = \varepsilon^2 t$, and assume that the local finite-cell transition rates are obtained from the on-shell matrix elements of V after the finite-volume limit, resonance averaging, and subtraction of disconnected pieces have all been performed in a specified scheme. The hydrodynamic equation is then replaced by

$$\partial_t \rho_i(x,t) + \partial_x \Big(v_i^{\text{eff}}[\rho](x,t)\rho_i(x,t) \Big) = \varepsilon^2 C_i[P_{x,t}] + R_i^{(\varepsilon,\ell,X)}(x,t), \tag{77.18}$$

where $i = (a, \theta)$ denotes a species/rapidity bin after discretization, $P_{x,t}$ is the local cell probability whose occupation means give $\rho_i(x,t)$, and C_i is the projected collision functional (77.15). The remainder records gradient corrections, memory effects, finite-cell correlations, and errors in the kinetic limit. The approximation is controlled only after one proves a norm in which $R_i^{(\varepsilon,\ell,X)} \to 0$ under a declared joint limit of ε, ℓ, X.

The finite-cell lemma identifies the part of the weak-breaking story that can be checked before a local-QFT limit theorem has been proved. If the perturbation preserves energy and momentum, the collision graph must be restricted to transitions with the same energy and momentum labels; (77.16) then gives the exact conservation laws of the kinetic equation. If a higher Bethe charge Q_h is not preserved by the perturbation, the same finite formula displays its decay:

$$\frac{d}{dT}\langle Q_h \rangle = \sum_{\omega,\omega'} P_\omega k_{\omega\omega'} \left(Q_h(\omega') - Q_h(\omega) \right).$$

Thus weak integrability breaking is not a modification of the dressing formula alone. It is a change in the collision graph of the mesoscopic cell, with the remaining exact charges and the entropy production fixed before any model-dependent scattering calculation is inserted.

Open Problem 77.10 (Kinetic limit of weakly broken integrable QFT). Derive (77.18) from a local integrable QFT perturbed by a specified local operator. The proof must construct the finite-volume transition rates from regulated matrix elements, prove the ε^{-2} kinetic time scaling, control secular terms and near-degenerate resonances, identify the local stationary weights π_ω, and show convergence of the remainder $R_i^{(\varepsilon,\ell,X)}$ in a norm strong enough to imply convergence of the declared local observables.

77.10 Theorem Boundary

The derivation above assumes existence of thermodynamic root densities, validity of Bethe–Yang quantization up to errors negligible at large L, and uniqueness of the free-energy minimizer. In concrete models these are mathematical hypotheses to be proved or checked. Bound states, non-diagonal scattering, zero modes, massless flows, and weak integrability-breaking collisions change the density system and must be included explicitly.

Open Problem 77.11 (TBA from local QFT). Derive the TBA equations directly from a constructed local QFT with factorized scattering: prove Bethe–Yang quantization with controlled finite-volume errors, establish thermodynamic root-density convergence, and identify the finite-size ground-state energy with the mirror free energy.

Chapter 78

Nested TBA, Baxter Relations, and Separation Variables

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Diagonal TBA converts scalar Bethe–Yang equations into density and pseudoenergy equations. Non-diagonal scattering adds auxiliary roots. The thermodynamic limit then contains physical densities and magnonic auxiliary densities. This chapter develops the finite-dimensional functional relations that control those auxiliary roots: Baxter TQ equations, QQ-systems, Hirota T-systems, and separation of variables. These are algebraic objects first; their use in a continuum QFT requires the separate large-volume and thermodynamic hypotheses stated below.

78.1 Nested Densities

Start from logarithmic nested equations of the form

$$Lp_a(\theta_j) + \sum_{b,k} \delta_{ab}(\theta_j - \theta_k) + \sum_{\alpha,r} \chi_{ar}(\theta_j - u_\alpha^{(r)}) = 2\pi I_j,$$

together with auxiliary equations

$$\sum_{b,k} \chi_{rb}(u_\alpha^{(r)} - \theta_k) + \sum_{s,\beta} \psi_{rs}(u_\alpha^{(r)} - u_\beta^{(s)}) = 2\pi J_\alpha^{(r)}.$$

The labels a, b enumerate physical particle species; r, s enumerate auxiliary Dynkin nodes or nesting levels. The functions δ, χ, ψ are chosen branches of logarithms of the scalar and matrix-scattering eigenvalue factors.

In the thermodynamic limit introduce physical densities ρ_a, ρ_a^h and auxiliary densities σ_r, σ_r^h. Differentiating the logarithmic equations gives

$$\rho_a + \rho_a^h = \frac{1}{2\pi}\frac{\mathrm{d}p_a}{\mathrm{d}\theta} + \sum_b K_{ab} * \rho_b + \sum_r K_{ar} * \sigma_r, \tag{78.1}$$

$$\sigma_r + \sigma_r^h = \sum_a K_{ra} * \rho_a + \sum_s K_{rs} * \sigma_s. \tag{78.2}$$

Here

$$(K * f)(x) = \int_{\mathbb{R}} K(x-y) f(y)\, \mathrm{d}y,$$

and each kernel is $1/(2\pi)$ times the derivative of the corresponding phase. Auxiliary roots carry no direct one-particle energy in the examples under discussion; they affect the free energy through the constraints.

Proposition 78.1 (Nested TBA variational equations)**.** *Assume the density equations* (78.1)–(78.2) *have a thermodynamic limit with positive total densities, and use the fermionic occupancy entropy for each physical and auxiliary root species. Define pseudoenergies*

$$\epsilon_a = \log\frac{\rho_a^h}{\rho_a}, \qquad \eta_r = \log\frac{\sigma_r^h}{\sigma_r}.$$

Then stationarity of the constrained free energy at inverse temperature R gives

$$\epsilon_a = RE_a - \sum_b K_{ba} * \log(1 + \mathrm{e}^{-\epsilon_b}) - \sum_r K_{ra} * \log(1 + \mathrm{e}^{-\eta_r}), \tag{78.3}$$

$$\eta_r = -\sum_a K_{ar} * \log(1 + \mathrm{e}^{-\epsilon_a}) - \sum_s K_{sr} * \log(1 + \mathrm{e}^{-\eta_s}). \tag{78.4}$$

Proof. Let A denote either a physical label a or an auxiliary label r. Write n_A for the occupied density, n_A^h for the hole density, and $n_A^t = n_A + n_A^h$ for the total density. The density equations have the uniform form

$$n_A^t = d_A + \sum_B K_{AB} * n_B, \qquad d_a = \frac{1}{2\pi}\frac{\mathrm{d}p_a}{\mathrm{d}\theta}, \qquad d_r = 0, \tag{78.5}$$

where the energy is E_a on physical labels and 0 on auxiliary labels. The entropy density of one fermionic occupancy species is

$$s_A = n_A^t \log n_A^t - n_A \log n_A - n_A^h \log n_A^h,$$

so, using $n_A^h = n_A^t - n_A$,

$$\delta s = \sum_A \int \left[\log\frac{n_A^h}{n_A}\, \delta n_A + \log\frac{n_A^t}{n_A^h}\, \delta n_A^t\right].$$

The dimensionless free-energy functional is

$$\mathcal{F}_R[n] = \sum_a \int RE_a\, \rho_a \, \mathrm{d}\theta - \sum_A \int s_A \, \mathrm{d}\theta.$$

Varying (78.5) gives

$$\delta n_A^t = \sum_B K_{AB} * \delta n_B.$$

For test variations of compact support the convolution pairing satisfies

$$\int L_A(\theta)(K_{AB} * \delta n_B)(\theta)\, \mathrm{d}\theta = \int (K_{AB}^\dagger * L_A)(u)\, \delta n_B(u)\, \mathrm{d}u, \qquad K_{AB}^\dagger(x) = K_{AB}(-x).$$

With

$$\epsilon_A = \log \frac{n_A^h}{n_A}, \qquad L_A = \log \frac{n_A^t}{n_A^h} = \log(1 + \mathrm{e}^{-\epsilon_A}),$$

stationarity of $\mathcal{F}_R$ therefore gives, for every label B,

$$0 = RE_B - \epsilon_B - \sum_A K_{AB}^\dagger * L_A, \qquad E_r = 0. \tag{78.6}$$

Writing the adjoint-transposed kernels as K_{BA}, separating physical and auxiliary labels, and substituting the definitions of ϵ_a, η_r gives exactly (78.3)–(78.4). $\square$

78.2 Constant A2 Magnonic Example

The ultraviolet limit of many nested systems reduces to constant equations for the plateau values $Y_A = \mathrm{e}^{\epsilon_A}$. A minimal A_2 magnonic example has two auxiliary nodes and incidence matrix

$$I_{rs} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The constant Y-system is

$$Y_1^2 = 1 + Y_2, \qquad Y_2^2 = 1 + Y_1.$$

The symmetric solution has $Y_1 = Y_2 = \phi_{\mathrm{g}}$, where

$$\phi_{\mathrm{g}} = \frac{1 + \sqrt{5}}{2}, \qquad \phi_{\mathrm{g}}^2 = 1 + \phi_{\mathrm{g}}.$$

This finite calculation is the algebraic core behind Rogers-dilogarithm central-charge evaluations. To turn it into a central-charge theorem for a specific QFT one must also justify the scaling limit, the kernel reduction to the constant Y-system, and the absence or subtraction of bulk-energy terms.

78.3 An SU(3) Nested Density System

The finite $SU(3)$ equations (75.8)–(75.9) give a useful nontrivial density example. Define

$$\varphi_\alpha(u) = \mathrm{i}\log\frac{u+\mathrm{i}\alpha}{u-\mathrm{i}\alpha}, \qquad a_\alpha(u) = \frac{1}{2\pi}\varphi'_\alpha(u) = \frac{1}{2\pi}\frac{2\alpha}{u^2+\alpha^2}.$$

The logarithmic branches are fixed continuously along the real root lines. For first-level roots u_j and second-level roots v_α, the counting functions may be chosen as

$$Z_u(u) = L\varphi_{1/2}(u) - \sum_k \varphi_1(u-u_k) + \sum_\beta \varphi_{1/2}(u-v_\beta),$$
$$Z_v(v) = -\sum_\beta \varphi_1(v-v_\beta) + \sum_j \varphi_{1/2}(v-u_j).$$

Differentiating and passing to densities gives

$$\rho + \rho^h = a_{1/2} - a_1 * \rho + a_{1/2} * \sigma, \tag{78.7}$$
$$\sigma + \sigma^h = a_{1/2} * \rho - a_1 * \sigma. \tag{78.8}$$

The signs are not arbitrary: they are the logarithmic derivatives of the same-level scattering factors and adjacent-node inverse factors in (75.8)–(75.9). If $\widehat{f}(\omega) = \int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}\omega u} f(u)\,\mathrm{d}u$, then

$$\widehat{a}_\alpha(\omega) = \mathrm{e}^{-\alpha|\omega|},$$

so the coupled linear density constraints are diagonalized in Fourier space. This elementary calculation is the finite-rank shadow of the nested TBA kernel manipulations used in principal chiral and Gross–Neveu models: the Cartan matrix controls which auxiliary species couple, while the analytic input lies in the physical scalar factors and in the thermodynamic string hypothesis.

78.4 Baxter TQ Relation

For the $XXX_{1/2}$ transfer eigenvalue (74.9), define the Baxter polynomial

$$Q(u) = \prod_{j=1}^{M}(u-u_j).$$

Then the eigenvalue equation can be rewritten as the finite-difference relation

$$\Lambda(u)Q(u) = \left(u+\frac{\mathrm{i}}{2}\right)^L Q(u-\mathrm{i}) + \left(u-\frac{\mathrm{i}}{2}\right)^L Q(u+\mathrm{i}). \tag{78.9}$$

Let $Q(u) = \prod_j (u-u_j)$ have simple roots. Equation (78.9) is regular at the zeros of Q precisely when the roots u_j obey the $XXX_{1/2}$ Bethe equations (74.8). Evaluating the numerator at $u = u_j$ gives

$$\left(u_j+\frac{\mathrm{i}}{2}\right)^L Q(u_j-\mathrm{i}) + \left(u_j-\frac{\mathrm{i}}{2}\right)^L Q(u_j+\mathrm{i}) = 0.$$

Since

$$\frac{Q(u_j+\mathrm{i})}{Q(u_j-\mathrm{i})} = -\prod_{\substack{k=1\\k\neq j}}^{M} \frac{u_j-u_k+\mathrm{i}}{u_j-u_k-\mathrm{i}},$$

the displayed equality is precisely (74.8). Conversely, if the Bethe equations hold, the numerator on the right side of (78.9) vanishes at every root of Q, so division by Q produces a polynomial Λ.

78.5 Baxter Casoratian and the Second Solution

The scalar TQ relation is a second-order finite-difference equation. Its two independent solutions carry the same information as a rank-two Wronskian system. This elementary point becomes structurally important in higher rank and in the quantum spectral curve: a Q-system is not just a list of Bethe polynomials, but a family of finite-difference solutions tied together by Casoratian identities and boundary conditions.

Write the rank-one Baxter equation in the form

$$\Lambda(u)Q(u) = a(u)Q(u-\mathrm{i}) + d(u)Q(u+\mathrm{i}), \tag{78.10}$$

where a, d, Λ are meromorphic functions on a common shift domain. For two solutions Q_1, Q_2, define the half-shift Casoratian

$$\mathcal{W}[Q_1,Q_2](u) = Q_1\left(u+\frac{\mathrm{i}}{2}\right)Q_2\left(u-\frac{\mathrm{i}}{2}\right) - Q_1\left(u-\frac{\mathrm{i}}{2}\right)Q_2\left(u+\frac{\mathrm{i}}{2}\right). \tag{78.11}$$

Lemma 78.2 (Baxter Casoratian transport)**.** *If Q_1 and Q_2 solve (78.10) with the same a, d, Λ, then*

$$d(u)\,\mathcal{W}[Q_1,Q_2]\left(u+\frac{\mathrm{i}}{2}\right) = a(u)\,\mathcal{W}[Q_1,Q_2]\left(u-\frac{\mathrm{i}}{2}\right). \tag{78.12}$$

At a point where $a(u)d(u) \neq 0$, vanishing of the Casoratian on one adjacent half-shift is therefore transported to the other.

Proof. Multiply (78.10) for Q_1 by $Q_2(u)$, multiply the same equation for Q_2 by $Q_1(u)$, and subtract. The terms containing $\Lambda(u)Q_1(u)Q_2(u)$ cancel, leaving

$$a(u)\left(Q_1(u-\mathrm{i})Q_2(u) - Q_2(u-\mathrm{i})Q_1(u)\right) + d(u)\left(Q_1(u+\mathrm{i})Q_2(u) - Q_2(u+\mathrm{i})Q_1(u)\right) = 0.$$

The second bracket is

$$\mathcal{W}[Q_1,Q_2]\left(u+\frac{\mathrm{i}}{2}\right),$$

whereas the first bracket is

$$-\mathcal{W}[Q_1,Q_2]\left(u-\frac{\mathrm{i}}{2}\right).$$

Substitution gives (78.12). If $a(u)d(u) \neq 0$, the displayed equation is a nonzero scalar multiple relation between the two adjacent Casoratians, so either one vanishes exactly when the other does. □

For the homogeneous $XXX_{1/2}$ normalization of (78.9),

$$a(u) = \left(u + \frac{\mathrm{i}}{2}\right)^L, \qquad d(u) = \left(u - \frac{\mathrm{i}}{2}\right)^L,$$

so the Casoratian obeys

$$\frac{\mathcal{W}(u + \mathrm{i}/2)}{\mathcal{W}(u - \mathrm{i}/2)} = \left(\frac{u + \mathrm{i}/2}{u - \mathrm{i}/2}\right)^L. \tag{78.13}$$

The polynomial Bethe solution is only one member of the two-dimensional local solution space of the difference equation. The companion solution is usually meromorphic rather than polynomial; its poles and asymptotics encode the boundary condition that selected the Bethe state. Higher-rank Wronskian Q_A's generalize this rank-one Casoratian: the determinant formula in the next section is the multi-solution version of (78.11), and the QQ-relations are the corresponding Pluecker identities.

78.6 QQ-System and Bäcklund Flow

For higher-rank systems it is more efficient to work with a family of Q-functions indexed by subsets. Let $I = \{1, \ldots, N\}$. For every subset $A \subset I$ introduce a meromorphic function $Q_A(u)$, with $Q_\varnothing(u) = 1$. Write

$$f^\pm(u) = f\left(u \pm \frac{\mathrm{i}}{2}\right).$$

The rational $\mathfrak{gl}_N$ QQ-system is

$$Q_{Aab}(u)Q_A(u) = Q^+_{Aa}(u)Q^-_{Ab}(u) - Q^-_{Aa}(u)Q^+_{Ab}(u), \tag{78.14}$$

for $a, b \notin A$, $a \neq b$. Here $Aa = A \cup \{a\}$ and $Aab = A \cup \{a, b\}$. Repeated use of (78.14) reconstructs all multi-index Q-functions from a flag of one-index functions together with normalization data. Moving from Q_A to Q_{Aa} is the functional form of a Bäcklund step: one removes one simple color direction from the auxiliary problem, exactly as the nested monodromy removes one residual color space at each level.

Let $q_1(u), \ldots, q_N(u)$ be functions for which the determinants below are defined. For an ordered subset $A = \{a_1 < \cdots < a_m\}$, set

$$Q_A(u) = \det_{1 \le r,s \le m} q_{a_r}^{[m+1-2s]}(u), \qquad f^{[k]}(u) = f\left(u + \frac{\mathrm{i}k}{2}\right),$$

and $Q_\varnothing = 1$. Then the functions Q_A obey (78.14). Fix A of size m and two indices $a, b \notin A$. Form the $(m+2) \times (m+2)$ matrix whose columns are the shifted columns corresponding to A, a, b. Apply the Desnanot–Jacobi identity to the minor obtained by deleting the two columns associated with the outer shifts. With the shifts ordered as $m+2, m, \ldots, -m, -m-2$, the four minors are precisely $Q_{Aab}Q_A$, $Q^+_{Aa}Q^-_{Ab}$, and $Q^-_{Aa}Q^+_{Ab}$, with the relative sign fixed by the column order. This gives (78.14). The identity is purely determinant algebra; no analytic assumption on the q_i's is used beyond existence of the displayed values.

78.7 Bäcklund Elimination as Q-System Restriction

The Bäcklund language used in nested Bethe ansatz should be understood as a recursive statement about Q-system data, not as a separate dynamical principle. Let $I = \{1, \ldots, N\}$ and let $\{Q_A\}_{A \subset I}$ obey (78.14). Choose a color $c \in I$, and define an unnormalized restricted family on $I \setminus \{c\}$ by

$$Q_A^{(c)}(u) = Q_{A \cup \{c\}}(u), \qquad A \subset I \setminus \{c\}. \tag{78.15}$$

Then $Q^{(c)}$ obeys the same local QQ-relation on the smaller index set:

$$Q_{Aab}^{(c)}(u) Q_A^{(c)}(u) = \big(Q_{Aa}^{(c)}\big)^+(u)\big(Q_{Ab}^{(c)}\big)^-(u) - \big(Q_{Aa}^{(c)}\big)^-(u)\big(Q_{Ab}^{(c)}\big)^+(u), \tag{78.16}$$

for $a, b \notin A \cup \{c\}$, $a \neq b$. This is simply (78.14) applied to the subset $A \cup \{c\}$ and the two remaining indices a, b. The point of recording it is not the algebraic verification; the point is the interpretation. A Bäcklund step removes one auxiliary color direction by passing from the N-color Q-system to the subsystem of minors that contain the eliminated color. Repeating the operation along a flag

$$I = I_N \supset I_{N-1} \supset \cdots \supset I_1 \supset I_0 = \varnothing, \qquad |I_r| = r,$$

produces the chain of rank-reduced Q-systems that mirrors the recursive nested monodromy of Chapter 75.

The adjective "unnormalized" in (78.15) matters. The restricted empty function is $Q_\varnothing^{(c)} = Q_c$, not the constant 1. In a finite spin chain one often chooses a Q-gauge in which the new empty function is normalized away on a domain where Q_c has no zeros. In an analytic T-system or quantum spectral-curve problem this normalization is part of the analytic datum: zeros, cuts, asymptotics, and large-u powers cannot be divided out without recording what has happened to them. Thus Bäcklund elimination is a local algebraic operation on Q-systems, while the normalized physical presentation of the reduced problem requires the gauge and analytic data of Definition 78.4.

For $N = 3$, the first elimination already displays the nesting pattern. If $c = 3$, the restricted rank-two data are

$$Q_\varnothing^{(3)} = Q_3, \qquad Q_1^{(3)} = Q_{13}, \qquad Q_2^{(3)} = Q_{23}, \qquad Q_{12}^{(3)} = Q_{123}.$$

The smaller QQ-relation is

$$Q_{123} Q_3 = Q_{13}^+ Q_{23}^- - Q_{13}^- Q_{23}^+,$$

which is the original QQ-relation with base subset $\{3\}$. In nested Bethe ansatz this is the algebraic shadow of passing from the original three-color auxiliary problem to the residual two-color one; the Bethe roots of the reduced problem are zeros of the Q-functions that survive the restriction, together with the gauge convention used to interpret them as polynomials.

78.8 Fusion and the Hirota T-System

The transfer matrix with a fundamental auxiliary space is only the first member of a fused family. In a rational $SU(N)$ chain, one obtains commuting transfer matrices $T_{a,s}(u)$ by taking auxiliary spaces whose highest weights are rectangular Young diagrams: a rows and s columns, with $1 \leq$

$a \leq N-1$ and $s \geq 0$. Fusion is implemented by projecting products of fundamental monodromies at shifted spectral parameters onto the corresponding irreducible auxiliary component. The Yang–Baxter equation ensures that the projected transfer matrices commute.

The local functional identity is the Hirota equation

$$T^+_{a,s}(u)T^-_{a,s}(u) = T_{a+1,s}(u)T_{a-1,s}(u) + T_{a,s+1}(u)T_{a,s-1}(u), \tag{78.17}$$

with boundary data $T_{0,s} = T_{a,0} = 1$ in the simplest normalization and $T_{N,s}$ fixed by the quantum determinant. The precise boundary factors depend on twist, inhomogeneity, and representation; the local bilinear identity is the invariant content.

Suppose the fused eigenvalues $T_{a,s}$ are written in a Wronskian gauge as minors of a matrix of shifted Q-functions. Then (78.17) follows from the Pluecker relation among adjacent minors. In the Wronskian gauge, $T_{a,s}$ is an $a \times a$ minor whose columns are shifts separated by i. The shifts of $T^+_{a,s}$ and $T^-_{a,s}$ differ by moving the left and right boundary columns of this minor outward. The Desnanot–Jacobi identity for the corresponding $(a+1)\times(a+1)$ matrix states that the product of the two adjacent $a \times a$ minors equals the product of the inner and outer minors plus the product of the two side minors. These four minors are, respectively, $T^+_{a,s}T^-_{a,s}$, $T_{a+1,s}T_{a-1,s}$, and $T_{a,s+1}T_{a,s-1}$. This is exactly (78.17).

The associated Y-functions are gauge-invariant ratios of neighboring T-functions:

$$Y_{a,s}(u) = \frac{T_{a,s+1}(u)T_{a,s-1}(u)}{T_{a+1,s}(u)T_{a-1,s}(u)}. \tag{78.18}$$

Using Hirota gives the local Y-system identity

$$Y^+_{a,s}(u)Y^-_{a,s}(u) = \frac{(1+Y_{a,s+1})(1+Y_{a,s-1})}{(1+Y^{-1}_{a+1,s})(1+Y^{-1}_{a-1,s})}, \tag{78.19}$$

where all functions are evaluated at u unless shifted. This is an algebraic identity once the same Hirota gauge is used at the four adjacent nodes; analytic continuation, strip domains, source terms, and asymptotics are additional model-specific data.

78.9 T-Gauge and Analytic T-System Specifications

The Hirota equation is not by itself a spectral problem. It is a local bilinear relation on a two-dimensional lattice of functions. A transfer matrix, a relativistic TBA system, or a quantum spectral curve is obtained only after one adds the boundary conditions and analytic constraints that select a particular solution. The first such constraint is the gauge freedom of the T-system.

Definition 78.3 (T-gauge transformation)**.** Let g_1, g_2, g_3, g_4 be nowhere-vanishing meromorphic functions on a shift domain containing the arguments used below. For a family $T_{a,s}(u)$, set

$$\mathcal{G}_{a,s}(u) = g_1^{[a+s]}(u)\, g_2^{[a-s]}(u)\, g_3^{[-a+s]}(u)\, g_4^{[-a-s]}(u), \qquad \widetilde{T}_{a,s}(u) = \mathcal{G}_{a,s}(u)T_{a,s}(u). \tag{78.20}$$

This transformation is called a T-gauge transformation on the chosen (a,s)-domain.

The four factors in $\mathcal{G}_{a,s}$ are arranged so that every monomial in the Hirota equation receives the same multiplier:

$$\mathcal{G}_{a,s}^{+}\mathcal{G}_{a,s}^{-} = \mathcal{G}_{a+1,s}\mathcal{G}_{a-1,s} = \mathcal{G}_{a,s+1}\mathcal{G}_{a,s-1}. \tag{78.21}$$

For example, the g_1 arguments in the first product are $[a+s+1]$ and $[a+s-1]$; the g_1 arguments in the second and third products are the same two shifts in the opposite lattice directions. The same check applies to g_2, g_3, g_4. Thus a T-gauge transformation maps a solution of (78.17) to another solution. It also leaves the Y-functions (78.18) invariant, because the four g_i-shift pairs in the numerator and denominator cancel separately. Consequently the Y-system is a gauge-invariant shadow of the T-system, while the transfer matrices themselves retain gauge-dependent boundary factors.

This distinction matters in applications. The local Hirota relation says that adjacent rectangular representations, or adjacent Wronskian minors, fit together consistently. It does not specify:

1. the allowed (a, s)-domain;
2. the boundary factors $T_{0,s}$, $T_{N,s}$, $T_{a,0}$, and their zeros and poles;
3. the strip of meromorphy or cut structure in the spectral parameter;
4. asymptotic behavior at infinity or at distinguished singular points;
5. reality, crossing, twist, and quantum-determinant constraints;
6. source terms introduced by physical particles, holes, defects, or wrapping corrections.

These items are not decorative. They are the analytic and boundary conditions that turn the Hirota identity into a spectral problem with a discrete set, or a controlled continuum family, of solutions. Two T-systems related by (78.20) describe the same Y-system, but they may place boundary divisors in different T-functions. A statement about polynomiality, zero locations, or large-u normalization is therefore meaningful only after the gauge has been fixed.

Definition 78.4 (Analytic T-system specification)**.** An analytic T-system specification on a lattice domain $\mathfrak{D} \subset \mathbb{Z}^2$ consists of:

1. meromorphic functions $T_{a,s}$ on a specified shift-stable spectral domain, for $(a, s) \in \mathfrak{D}$, satisfying Hirota at interior lattice points;
2. a T-gauge equivalence class, together with a chosen gauge when polynomiality or asymptotics are asserted;
3. boundary functions and quantum-determinant factors on $\partial\mathfrak{D}$;
4. the allowed zero, pole, branch-cut, and asymptotic conditions;
5. the map from this analytic specification to physical charges or transfer eigenvalues.

In finite spin chains the analytic specification is often polynomial: the inhomogeneities, twist, representation labels, and quantum determinant fix the boundary factors, and Bethe equations enforce cancellation of unwanted poles. In relativistic integrable QFT, the same local Hirota/Y-system algebra must be supplemented by strip analyticity, source terms, and large-rapidity asymptotics before it becomes a TBA problem. In planar gauge-theory integrability, the quantum spectral curve replaces the finite polynomial condition by a Riemann–Hilbert type problem: sheets, cuts, asymptotics, gluing conditions, and quantization of charges. The determinant identities in this chapter supply the local algebraic skeleton; the analytic T-system specification turns that skeleton into a physical spectral problem.

78.10 Q-Operators

A Q-operator is an operator-valued function $Q(u)$ commuting with the transfer matrix and satisfying an operator TQ relation whose eigenvalue version is (78.9). Operator construction requires an auxiliary representation in which the trace, or a regulated trace, is defined. In rank one this can be described as follows.

Construction 78.5 (Rank-one Baxter operator construction)**.** Let $\mathcal{A}$ be an auxiliary representation of the RTT algebra for the same R-matrix, not necessarily finite dimensional. Let $\mathcal{L}_{\mathcal{A},j}(u) \in \mathrm{End}(\mathcal{A} \otimes \mathbb{C}^2_j)$ satisfy the mixed RTT relation with the fundamental L-operator. If the regularized auxiliary trace

$$Q(u) = \mathrm{Tr}^{\mathrm{reg}}_{\mathcal{A}}\big(\mathcal{L}_{\mathcal{A},L}(u) \cdots \mathcal{L}_{\mathcal{A},1}(u)\big)$$

exists and is cyclic on the products used in the RTT proof, then $[Q(u), t(v)] = 0$. If the tensor product of the fundamental auxiliary space with $\mathcal{A}$ has a short exact sequence whose quotient and submodule shift the spectral parameter by $\pm\mathrm{i}$, the transfer-matrix trace over that sequence gives the operator TQ relation.

Exact-sequence derivation of an operator TQ relation. Assume the data of Construction 78.5. Assume further that, for every spectral parameter u, there is a short exact sequence of auxiliary RTT modules

$$0 \longrightarrow \mathcal{A}(u - \mathrm{i}) \xrightarrow{\iota} \mathbb{C}^2(u) \otimes \mathcal{A}(u) \xrightarrow{\pi} \mathcal{A}(u + \mathrm{i}) \longrightarrow 0, \tag{78.22}$$

and that the corresponding monodromy on $\mathbb{C}^2(u) \otimes \mathcal{A}(u) \otimes \mathcal{H}_L$ is block triangular in a basis adapted to ι, π, with diagonal blocks $a(u)\, M_{\mathcal{A}}(u - \mathrm{i})$ and $d(u)\, M_{\mathcal{A}}(u + \mathrm{i})$. Then

$$t(u)\, Q(u) = a(u)Q(u - \mathrm{i}) + d(u)Q(u + \mathrm{i}) \tag{78.23}$$

as operators on the quantum chain.

The product $t(u)Q(u)$ is the regularized trace over $\mathbb{C}^2(u) \otimes \mathcal{A}(u)$ of the product monodromy, because the mixed RTT relation permits the fundamental and auxiliary monodromies to be ordered inside one common auxiliary trace. Adapt the auxiliary vector space to the short exact sequence (78.22). The monodromy matrix is block triangular, hence its trace is the sum of the traces of the two diagonal blocks. The scalar diagonal factors are exactly the vacuum propagation factors $a(u)$ and $d(u)$, while the remaining block traces are the regularized traces defining $Q(u-\mathrm{i})$ and $Q(u+\mathrm{i})$. Cyclicity of the regularized trace is the only analytic input in this finite-chain argument. This gives (78.23).

The construction displays the logical ingredients rather than hiding them in the phrase "Baxter operator": a mixed RTT relation, a trace functional with cyclicity, and an exact sequence of auxiliary representations. In rational finite chains one may realize $\mathcal{A}$ as a limiting highest-weight module. In trigonometric rank-one chains one uses a q-oscillator Fock module. For $q \in \mathbb{C}^\times$ not a root of unity, one convenient q-oscillator algebra is generated by $D, \mathsf{a}, \mathsf{a}^\dagger$ with

$$q^D \mathsf{a} q^{-D} = q^{-1}\mathsf{a}, \qquad q^D \mathsf{a}^\dagger q^{-D} = q\mathsf{a}^\dagger, \qquad \mathsf{a}^\dagger \mathsf{a} = 1 - q^{2D}, \qquad \mathsf{a}\mathsf{a}^\dagger = 1 - q^{2D+2}.$$

On the Fock module $\mathcal{F} = \bigoplus_{n\ge 0} \mathbb{C}|n\rangle$,

$$D|n\rangle = n|n\rangle, \qquad \mathsf{a}|n\rangle = (1-q^{2n})|n-1\rangle, \qquad \mathsf{a}^\dagger|n\rangle = |n+1\rangle.$$

This convention is related to the Borel q-oscillator generators $\mathcal{H}, \mathcal{E}_\pm$ by

$$\mathcal{H} = 2D, \qquad \mathcal{E}_+ = \frac{\mathsf{a}^\dagger}{q-q^{-1}}, \qquad \mathcal{E}_- = \frac{\mathsf{a}}{q-q^{-1}}.$$

Indeed

$$[\mathcal{H}, \mathcal{E}_\pm] = \pm 2\mathcal{E}_\pm, \qquad q\mathcal{E}_+\mathcal{E}_- - q^{-1}\mathcal{E}_-\mathcal{E}_+ = \frac{1}{q-q^{-1}}, \tag{78.24}$$

where the second identity follows from

$$q\mathsf{a}^\dagger\mathsf{a} - q^{-1}\mathsf{a}\mathsf{a}^\dagger = q(1-q^{2D}) - q^{-1}(1-q^{2D+2}) = q - q^{-1}.$$

The local trigonometric q-oscillator L-operator. Let $V = \mathbb{C}e_+ \oplus \mathbb{C}e_-$. In the six-vertex normalization used in this paragraph, $R(x) \in \mathrm{End}(V \otimes V)$ has non-zero entries

$$\begin{aligned} &R^{++}_{++}(x) = R^{--}_{--}(x) = q + q^{-1}x, && R^{+-}_{+-}(x) = R^{-+}_{-+}(x) = 1 + x, \\ &R^{+-}_{-+}(x) = q - q^{-1}, && R^{-+}_{+-}(x) = -(q-q^{-1})x. \end{aligned} \tag{78.25}$$

The q-oscillator auxiliary L-operator is the following element of $\mathrm{End}(V \otimes \mathcal{F})$, written as a 2×2 matrix in the ordered basis e_+, e_-:

$$L^{(+)}_{\mathcal{F}}(x) = \begin{pmatrix} q^D & \mathsf{a} \\ -x\,\mathsf{a}^\dagger & q^{-D} + xq^{D+1} \end{pmatrix}. \tag{78.26}$$

Equivalently, before choosing the Fock realization,

$$\begin{aligned} L^{(+)}_{\mathcal{F}}(x) = q^{\mathcal{H}\otimes\sigma^z/2} &+ (q-q^{-1})(\mathcal{E}_- \otimes \sigma^+ - x\mathcal{E}_+ \otimes \sigma^-) \\ &+ x(q-q^{-1})q^{-\mathcal{H}/2}(\mathcal{E}_+\mathcal{E}_- - \mathcal{E}_-\mathcal{E}_+) \otimes \sigma^-\sigma^+, \end{aligned} \tag{78.27}$$

with $\sigma^z e_\pm = \pm e_\pm$, $\sigma^+ e_- = e_+$, $\sigma^- e_+ = e_-$. Substituting (78.24) gives (78.26); the bottom-right entry uses

$$\mathcal{E}_+\mathcal{E}_- - \mathcal{E}_-\mathcal{E}_+ = \frac{q^{2D+1}}{q-q^{-1}}.$$

Proposition 78.6 (Local q-oscillator RLL identity)**.** *For $q \in \mathbb{C}^\times$ not a root of unity, the operators* (78.25) *and* (78.26) *satisfy*

$$R_{12}(-x/y)\, L^{(+)}_{\mathcal{F},13}(x)\, L^{(+)}_{\mathcal{F},23}(y) = L^{(+)}_{\mathcal{F},23}(y)\, L^{(+)}_{\mathcal{F},13}(x)\, R_{12}(-x/y) \tag{78.28}$$

as an identity in $\mathrm{End}(V \otimes V \otimes \mathcal{F})$.

Proof. Write $N = q^D$. The q-oscillator relations are

$$N\mathsf{a} = q^{-1}\mathsf{a}N, \qquad N\mathsf{a}^\dagger = q\mathsf{a}^\dagger N, \qquad \mathsf{a}^\dagger\mathsf{a} = 1 - N^2, \qquad \mathsf{a}\mathsf{a}^\dagger = 1 - q^2N^2. \tag{78.29}$$

Both sides of (78.28) are finite sums of monomials in $N^{\pm1}, \mathsf{a}, \mathsf{a}^\dagger$. Moving every $N^{\pm1}$ to the middle and reducing every adjacent $\mathsf{a}^\dagger\mathsf{a}$ and $\mathsf{a}\mathsf{a}^\dagger$ by (78.29) gives a normal form with at most one creation or annihilation operator on either side of a Laurent polynomial in N. Therefore it is enough to compare the four spin-input sectors $e_{\epsilon_1} \otimes e_{\epsilon_2} \otimes |n\rangle$, $n \geq 0$.

Let

$$\begin{aligned} N_n &= q^n, & \alpha_n &= 1 - q^{2n}, & A_n &= N_n^2 - 1, & \Delta &= x - q^2 y, \\ B_x &= N_n^2 qx + 1, & B_y &= N_n^2 qy + 1, & G_y &= N_n^2 y + q. \end{aligned}$$

For the mixed sector below also write

$$\begin{aligned} C_{+-} &= N_n^2 q^2 y^2 - N_n^2 xy - q^2 xy - qx + qy + xy, \\ D_{+-} &= N_n^2 q^3 y - N_n^2 qx + q^2 + qx - qy - 1. \end{aligned}$$

In the sector $e_+ \otimes e_+$, both sides of (78.28) act as

$$\begin{aligned} e_+ \otimes e_+ \otimes |n\rangle \mapsto \; & -\frac{\Delta}{qy} N_n^2 \, e_+ \otimes e_+ \otimes |n\rangle + \frac{\Delta}{q} N_n \, e_+ \otimes e_- \otimes |n+1\rangle \\ & + \frac{x\Delta}{y} N_n \, e_- \otimes e_+ \otimes |n+1\rangle - \frac{x\Delta}{q} \, e_- \otimes e_- \otimes |n+2\rangle. \end{aligned} \tag{78.30}$$

In the sector $e_- \otimes e_-$, applying the same reductions gives

$$\begin{aligned} e_- \otimes e_- \otimes |n\rangle \mapsto \; & \frac{(q - N_n)(N_n + q) A_n \Delta}{q^3 y} \, e_+ \otimes e_+ \otimes |n-2\rangle + \frac{A_n G_y \Delta}{q N_n y} \, e_+ \otimes e_- \otimes |n-1\rangle \\ & + \frac{A_n \Delta B_x}{q N_n y} \, e_- \otimes e_+ \otimes |n-1\rangle - \frac{\Delta B_x B_y}{q N_n^2 y} \, e_- \otimes e_- \otimes |n\rangle. \end{aligned} \tag{78.31}$$

The terms with $|n-1\rangle$ or $|n-2\rangle$ vanish automatically when the displayed ket is outside the Fock module, because $\alpha_0 = 0$. The mixed sectors are the only places where the off-diagonal six-vertex weights enter. For $e_+ \otimes e_-$, both sides give

$$\begin{aligned} & -\frac{\Delta \alpha_n N_n}{q^2 y} \, e_+ \otimes e_+ \otimes |n-1\rangle + \frac{C_{+-}}{qy} \, e_+ \otimes e_- \otimes |n\rangle \\ & + \frac{x D_{+-}}{qy} \, e_- \otimes e_+ \otimes |n\rangle + \frac{x \Delta B_y}{q N_n y} \, e_- \otimes e_- \otimes |n+1\rangle. \end{aligned} \tag{78.32}$$

The $e_- \otimes e_+$ formula is obtained from (78.32) by interchanging $(x, e_1) \leftrightarrow (y, e_2)$ together with the asymmetric normalization of R_{+-}^{-+} in (78.25); the resulting coefficients are exactly those obtained by expanding the two sides. These four sector computations exhaust the basis of $V \otimes V \otimes \mathcal{F}$, hence (78.28) follows. $\square$

The regularized trace is first defined by

$$\operatorname{Tr}_{\mathcal{F}}^{\mathrm{reg}}(X) = \underset{z=1}{\text{mer. cont.}} \operatorname{Tr}_{\mathcal{F}}(z^D X), \qquad |z| < 1,$$

whenever the meromorphic continuation exists and is finite after the scalar normalization of the Q-operator has been fixed. Proposition 78.6 supplies the local algebraic input for the trigonometric rank-one Q-operator. The remaining analytic inputs are separate: cyclicity of the regularized trace on the monodromy products and the short exact sequence (78.22). Once these are checked, the operator identity (78.23) is a consequence of block-triangular algebra plus the stated trace regularization. The eigenvalue statement used in this chapter is the consequence of this construction, not its replacement.

78.11 Separation of Variables

Separation of variables uses a different complete set of commuting objects. For the 2×2 monodromy

$$T(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix},$$

RTT implies $[B(u), B(v)] = 0$. Suppose we are in a finite-dimensional representation for which the commuting coefficients of $B(u)$ are simultaneously diagonalizable with simple joint spectrum. On a joint right or left eigenline the B-eigenvalue has the polynomial form

$$B(u) = B_0 \prod_{\alpha=1}^{L-1} (u - x_\alpha).$$

The numbers x_α are separated coordinates.

Local RTT shift algebra. The separated-coordinate shift is an operator statement before it is a wavefunction equation. In the rational RTT normalization compatible with (78.9), the components involving A, D, and B are understood as identities after multiplication by $u - v$:

$$A(u)B(v) = \frac{u - v - \mathrm{i}}{u - v} B(v)A(u) + \frac{\mathrm{i}}{u - v} B(u)A(v), \tag{78.33}$$

$$D(u)B(v) = \frac{u - v + \mathrm{i}}{u - v} B(v)D(u) - \frac{\mathrm{i}}{u - v} B(u)D(v). \tag{78.34}$$

Let $\langle \boldsymbol{x}|$ be a simultaneous left eigenvector of the commuting coefficients of B:

$$\langle \boldsymbol{x}| B(v) = B_0 \prod_{\beta=1}^{L-1} (v - x_\beta) \langle \boldsymbol{x}|.$$

Set $u = x_\alpha$ in (78.33)–(78.34). The terms proportional to $B(x_\alpha)$ vanish on the left eigenvector, so

$$\langle \boldsymbol{x}| A(x_\alpha) B(v) = B_0 (v - x_\alpha + \mathrm{i}) \prod_{\beta \neq \alpha} (v - x_\beta) \langle \boldsymbol{x}| A(x_\alpha), \tag{78.35}$$

$$\langle \boldsymbol{x}| D(x_\alpha) B(v) = B_0 (v - x_\alpha - \mathrm{i}) \prod_{\beta \neq \alpha} (v - x_\beta) \langle \boldsymbol{x}| D(x_\alpha). \tag{78.36}$$

Thus, whenever the resulting covector is nonzero and the joint spectrum is simple, $\langle \boldsymbol{x}|A(x_\alpha)$ lies on the eigenline with the single coordinate x_α replaced by $x_\alpha - \mathrm{i}$, while $\langle \boldsymbol{x}|D(x_\alpha)$ lies on the eigenline with x_α replaced by $x_\alpha + \mathrm{i}$. A choice of SoV covector normalization fixes the scalar coefficients. In the normalization used in the scalar Baxter equation,

$$\langle \boldsymbol{x}| A(x_\alpha) = a(x_\alpha) \langle \boldsymbol{x} - \mathrm{i} e_\alpha|, \qquad \langle \boldsymbol{x}| D(x_\alpha) = d(x_\alpha) \langle \boldsymbol{x} + \mathrm{i} e_\alpha|. \tag{78.37}$$

Equations (78.35)–(78.37) are the finite-chain construction behind the separated Baxter equation below. The shift is not an analogy with classical separated coordinates; it is the residue of the RTT algebra at a zero of B.

Finite-chain SoV reduction under simple-spectrum hypotheses. Assume the coefficients of $B(u)$ have a simple joint eigenbasis $\{\langle \boldsymbol{x}|\}$ normalized as in (78.37). Then a transfer-matrix eigenvector with separated wavefunction $\Psi(\boldsymbol{x}) = \langle \boldsymbol{x}|\Psi\rangle$ satisfies, for each $\alpha = 1, \dots, L-1$, a one-variable Baxter equation

$$\Lambda(x_\alpha)\Psi(\boldsymbol{x}) = a(x_\alpha)\Psi(\boldsymbol{x} - \mathrm{i}e_\alpha) + d(x_\alpha)\Psi(\boldsymbol{x} + \mathrm{i}e_\alpha), \tag{78.38}$$

where $a(u)$ and $d(u)$ are the vacuum diagonal eigenvalue functions of $A(u)$ and $D(u)$, and e_α shifts only the coordinate x_α.

Act with $t(u) = A(u)+D(u)$ on a transfer eigenvector Ψ and evaluate the left covector equation at $u = x_\alpha$. The two summands are evaluated by (78.37), giving exactly (78.38). The simple-spectrum hypothesis ensures that the shifted covectors are identified by their separated coordinates rather than by a nontrivial Jordan block or multiplicity space.

When both the algebraic Bethe ansatz and SoV are valid, the Bethe polynomial $Q(u)$ is the one-variable factor whose values solve (78.38) at every separated coordinate. The pseudovacuum construction builds eigenvectors from zeros of Q; SoV builds the same spectral problem by diagonalizing $B(u)$. Their equality is not a slogan about roots: it is a comparison statement between two bases of the same finite-dimensional representation.

Finite-chain ABA–Q–SoV comparison under explicit hypotheses. For a finite rational $SU(2)$ chain with generic diagonal twist and generic inhomogeneities, assume the following finite-dimensional properties:

1. the transfer matrix has simple spectrum;
2. the algebraic Bethe ansatz produces one nonzero Bethe vector for every transfer eigenvalue;
3. a Baxter Q-operator satisfying the operator TQ relation exists;
4. the coefficients of $B(u)$ have a simple SoV spectrum.

Then the ABA Bethe roots, the zeros of the Q-operator eigenvalue, and the one-variable SoV Baxter wavefunction determine the same transfer-matrix eigenline.

The justification uses only these finite-chain inputs. By the operator TQ relation, the eigenvalue $Q_\Psi(u)$ of $Q(u)$ on a transfer eigenvector Ψ satisfies (78.9) with the same $\Lambda(u)$. Regularity of the scalar TQ relation at the simple zeros of Q_Ψ gives the ABA Bethe equations. Assumption 2 then gives an ABA vector with the same transfer eigenvalue Λ. Simplicity of the transfer spectrum identifies the two vectors up to scalar. The SoV wavefunction satisfies (78.38); the polynomial Q_Ψ gives a solution of each one-coordinate equation by evaluation at x_α. Simplicity of the SoV joint spectrum identifies the resulting wavefunction with the same eigenline. Thus all three constructions label the same line, under the stated hypotheses.

78.12 The SoV Measure Is Part of the Construction

The word "measure" in finite-chain SoV does not mean a Borel measure on a classical configuration space. It is the weight in the resolution of the identity for a finite-dimensional joint spectral problem. Let $\mathcal{X}$ be the finite joint spectrum of the commuting coefficients of $B(u)$, and let $|\boldsymbol{x}\rangle$, $\langle\widetilde{\boldsymbol{x}}|$ be right and left eigenvectors normalized so that

$$\langle\widetilde{\boldsymbol{x}} \mid \boldsymbol{y}\rangle = 0 \qquad (\boldsymbol{x} \neq \boldsymbol{y}).$$

If the joint spectrum is simple and complete, define

$$\mu(\boldsymbol{x}) = \langle\widetilde{\boldsymbol{x}} \mid \boldsymbol{x}\rangle^{-1}.$$

Then

$$\mathrm{id}_{\mathcal{H}} = \sum_{\boldsymbol{x}\in\mathcal{X}} \mu(\boldsymbol{x})|\boldsymbol{x}\rangle\langle\widetilde{\boldsymbol{x}}|. \tag{78.39}$$

For two vectors Φ, Ψ, their scalar product is therefore

$$\langle\Phi \mid \Psi\rangle = \sum_{\boldsymbol{x}\in\mathcal{X}} \mu(\boldsymbol{x})\,\langle\Phi \mid \boldsymbol{x}\rangle\langle\widetilde{\boldsymbol{x}} \mid \Psi\rangle,$$

with the obvious conjugations when the left basis is chosen as the Hermitian dual of the right basis in a unitary representation. In concrete inhomogeneous spin chains this weight can be written as a determinant or Vandermonde-type product. That explicit formula is representation-dependent, but (78.39) is the invariant finite-dimensional statement.

Under the simple-spectrum SoV hypotheses above, suppose a function $Q(u)$ solves the one-variable Baxter equation at every separated coordinate value appearing in $\mathcal{X}$. Then

$$\Psi_Q(\boldsymbol{x}) = \prod_{\alpha=1}^{L-1} Q(x_\alpha)$$

solves the separated transfer-matrix eigenvalue equations on $\mathcal{X}$. Conversely, if the separated spectrum is a Cartesian product of one-coordinate spectra and the one-coordinate Baxter equation has a one-dimensional solution space after the boundary conditions are imposed, then every separated eigenfunction factorizes in this form. Equation (78.38) shifts only one coordinate. If $\Psi_Q = \prod_\beta Q(x_\beta)$, the common factors $\prod_{\beta\neq\alpha} Q(x_\beta)$ cancel from the α-th separated equation, leaving exactly the one-variable Baxter equation for Q at x_α. Conversely, on a Cartesian separated spectrum the equations for different coordinates commute and act on distinct variables. The stated one-dimensionality of the one-coordinate solution space fixes each factor up to a scalar; the product of those scalars is one overall normalization of the eigenvector.

78.13 From SoV Data to Spectral Traces

The preceding finite-chain results identify transfer eigenlines. A local observable in a quantum field theory asks for more data: a Hamiltonian or charge extraction from the transfer matrix, a representative O_L of the local operator, matrix elements of O_L, the thermal or finite-volume state, and the limiting map from the finite chain to the QFT. The distinction is already visible before any limit is taken.

Proposition 78.7 (Finite-chain SoV spectral trace)**.** *Let $\mathcal{H}_L$ be a finite-dimensional transfer-matrix representation for which the ABA–Q–SoV comparison hypotheses above hold. Let H_L be* *a self-adjoint Hamiltonian obtained from the commuting transfer family, and let* $O_L \in \mathrm{End}(\mathcal{H}_L)$*. If* $\{|\lambda\rangle\}_{\lambda\in\mathcal{S}_L}$ *is the resulting complete orthonormal eigenbasis of* H_L*, then, for* $0 \le \tau \le R$*,*

$$G_L^O(\tau, R) = \frac{1}{Z_L}\,\mathrm{Tr}_{\mathcal{H}_L}\left(\mathrm{e}^{-(R-\tau)H_L} O_L \mathrm{e}^{-\tau H_L} O_L^\dagger\right) = \frac{1}{Z_L}\sum_{\lambda,\lambda'\in\mathcal{S}_L} \mathrm{e}^{-(R-\tau)E_\lambda - \tau E_{\lambda'}}\left|\langle\lambda|O_L|\lambda'\rangle\right|^2, \tag{78.40}$$

where $Z_L = \sum_\lambda \mathrm{e}^{-RE_\lambda}$*. The SoV resolution (78.39) computes the matrix elements in (78.40) by finite sums over separated coordinates and their measure.*

Proof. The commuting transfer family and the comparison hypotheses identify the same eigenlines whether they are labeled by Bethe roots, a Q-operator eigenvalue, or separated wavefunctions. Since H_L is extracted from that commuting family, these eigenlines diagonalize H_L. Insert the identity $\sum_\lambda |\lambda\rangle\langle\lambda| = \mathrm{id}_{\mathcal{H}_L}$ on both sides of O_L inside the trace. Cyclicity of the finite trace gives the displayed Boltzmann factors and the squared matrix element. Finally insert (78.39) between O_L and the transfer eigenvectors to express each matrix element as a finite SoV sum. In a non-unitary or twisted biorthogonal presentation, the same proof uses the left and right eigenvectors supplied by the SoV resolution, with the two matrix elements replacing the absolute square. □

Controlled Approximation 78.8 (SoV-to-observable reconstruction budget)**.** Fix a test norm for Euclidean correlators on a compact window in (τ, R). Let $G^{\mathcal{O}}_{\mathrm{phys}}$ be the target local-QFT correlator and let $G^{\mathrm{SoV}}_{L,\mathcal{S}}$ be the correlator obtained from a finite chain, a subset $\mathcal{S} \subset \mathcal{S}_L$, a chosen SoV measure, and chosen matrix-element formulae for an operator O_L. Insert the intermediate exact finite-chain trace G_L^O, the complete SoV-labeled finite trace $G^{\mathrm{SoV},*}_{L,\mathrm{all}}$, and the exact retained trace $G^{\mathrm{SoV},*}_{L,\mathcal{S}}$. Let $\widehat{G}^{\mathrm{SoV}}_{L,\mathcal{S}}$ be the retained SoV expression after the chosen operator matrix-element formulae have been used but before a TBA, mirror, generalized Gibbs, or hydrodynamic state replaces the finite trace. The difference decomposes as

$$\begin{aligned} G^{\mathcal{O}}_{\mathrm{phys}} - G^{\mathrm{SoV}}_{L,\mathcal{S}} &= R_{\mathrm{reg}} + R_{\mathrm{comp}} + R_{\mathrm{tail}} + R_O + R_{\mathrm{state}}, \\ R_{\mathrm{reg}} &= G^{\mathcal{O}}_{\mathrm{phys}} - G_L^O, \\ R_{\mathrm{comp}} &= G_L^O - G^{\mathrm{SoV},*}_{L,\mathrm{all}}, \\ R_{\mathrm{tail}} &= G^{\mathrm{SoV},*}_{L,\mathrm{all}} - G^{\mathrm{SoV},*}_{L,\mathcal{S}}, \\ R_O &= G^{\mathrm{SoV},*}_{L,\mathcal{S}} - \widehat{G}^{\mathrm{SoV}}_{L,\mathcal{S}}, \\ R_{\mathrm{state}} &= \widehat{G}^{\mathrm{SoV}}_{L,\mathcal{S}} - G^{\mathrm{SoV}}_{L,\mathcal{S}}. \end{aligned} \tag{78.41}$$

If these five residuals obey

$$\|R_{\mathrm{reg}}\| \le \varepsilon_{\mathrm{reg}}, \quad \|R_{\mathrm{comp}}\| \le \varepsilon_{\mathrm{comp}}, \quad \|R_{\mathrm{tail}}\| \le \varepsilon_{\mathrm{tail}}, \quad \|R_O\| \le \varepsilon_O, \quad \|R_{\mathrm{state}}\| \le \varepsilon_{\mathrm{state}},$$

then

$$\left\|G^{\mathcal{O}}_{\mathrm{phys}} - G^{\mathrm{SoV}}_{L,\mathcal{S}}\right\| \le \varepsilon_{\mathrm{reg}} + \varepsilon_{\mathrm{comp}} + \varepsilon_{\mathrm{tail}} + \varepsilon_O + \varepsilon_{\mathrm{state}}. \tag{78.42}$$

The status of the five terms is different. Under Proposition 78.7 and the SoV measure identity, $R_{\mathrm{comp}} = 0$ is a finite-dimensional theorem. The term R_{tail} is a spectral truncation estimate. The term R_O is the operator problem: eigenvalue data from Q-functions must be supplemented by form factors, determinant formulae, or direct SoV matrix elements for the specific local field being probed. The terms R_{reg} and R_{state} are the physical bridge from the regulated chain to the continuum QFT state and local operator. Thus the nested TQ, QQ, and SoV machinery supplies a complete finite spectral coordinate system only after its hypotheses are proved; local QFT amplitudes additionally require the regulator, tail, operator, and state residuals in (78.41).

78.14 Continuum Status

The finite-chain spectral package above isolates the remaining continuum QFT problem. For a relativistic integrable QFT one must additionally prove that finite-volume energy levels are approximated by matrix Bethe–Yang equations in the large-L regime, that the nested string hypothesis used in Chapter 75 is valid for the thermodynamic states under consideration, that local operators have controlled finite-volume matrix elements, and that the integral operators in the TBA equations have the required analytic and Fredholm properties. These are model-dependent mathematical statements. They are separate from the finite RTT algebra, and the residual budget (78.41) records where each one enters a physical correlator.

78.15 Interface with Planar Gauge-Theory Integrability

In the planar $\mathcal{N} = 4$ SYM volume, magnons have a non-relativistic dispersion on a spin chain, wrapping effects arise from trace cyclicity and finite operator length, and the quantum spectral curve replaces the finite-rank Baxter polynomial by a system of analytic Q-functions. The algebraic bridge is now explicit: the Q-system (78.14), the Hirota equation (78.17), and the TQ/SoV comparison theorems of this chapter are the finite-rank prototypes. Planar gauge theory changes the analytic domain, asymptotics, grading, and boundary conditions; it does not change the local determinant identities.

Chapter 79

Integrable RG Flows and Perturbed Two-Dimensional CFT

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter studies two-dimensional quantum field theories specified near a conformal fixed point by a relevant scalar local operator and studied away from the fixed point through factorized scattering, form factors, and finite-size spectra. The word flow will always mean a comparison between renormalized local QFT data at different length scales. An identification of such a flow includes a declared local-operator chart, infinite-volume scattering data, and a finite-volume Hilbert-space construction whose ground-state energy is computed by the stated TBA system.

79.1 Perturbed-CFT Data

Let $\mathcal{C}$ be a Euclidean two-dimensional CFT on the plane. A scalar primary operator $\mathcal{O}$ has holomorphic and antiholomorphic weights $(h, \bar{h})$ with $h = \bar{h}$, and scaling dimension

$$\Delta_{\mathcal{O}} = h + \bar{h} = 2h.$$

It is relevant if

$$y_{\mathcal{O}} := 2 - \Delta_{\mathcal{O}} > 0.$$

The number $y_{\mathcal{O}}$ is the engineering dimension of the coupling that multiplies the integrated insertion $\int d^2x\, \mathcal{O}(x)$. This dimension statement is independent of perturbation theory: under the

Euclidean dilation $x \mapsto \lambda x$, the measure contributes λ^2 and the primary insertion contributes $\lambda^{-\Delta_{\mathcal{O}}}$.

Definition 79.1 (Renormalized perturbed-CFT chart)**.** A renormalized perturbation chart for $(\mathcal{C}, \mathcal{O})$ consists of the following data.

1. A normalization of $\mathcal{O}$, for example

$$\langle \mathcal{O}(x)\mathcal{O}(0)\rangle_{\mathcal{C}} = \frac{1}{|x|^{2\Delta_{\mathcal{O}}}} \quad (x \neq 0),$$

   or another explicitly stated nonzero constant in the numerator.

2. A local source coordinate g with mass dimension $[g] = y_{\mathcal{O}}$.

3. For each collection X of separated spectator insertions and each integer $n \geq 0$, an extension prescription

$$\mathcal{R}_n[\langle X\, \mathcal{O}(x_1) \cdots \mathcal{O}(x_n)\rangle_{\mathcal{C}}]$$

   from the configuration space where all points are distinct to a distribution on the full product space. The extension prescription must state its contact terms on diagonals; different choices are different renormalized coordinates.

The formal perturbation series in this chart is

$$\langle X\rangle_g = \frac{\sum_{n\geq 0} \frac{(-g)^n}{n!} \int \prod_{j=1}^n d^2x_j\, \mathcal{R}_n[\langle X\, \mathcal{O}(x_1)\cdots \mathcal{O}(x_n)\rangle_{\mathcal{C}}]}{\sum_{n\geq 0} \frac{(-g)^n}{n!} \int \prod_{j=1}^n d^2x_j\, \mathcal{R}_n[\langle \mathcal{O}(x_1)\cdots \mathcal{O}(x_n)\rangle_{\mathcal{C}}]}.$$

The expression is a formal or regulated object until the volume, infrared boundary condition, and convergence or summability statement have been specified.

The extension maps in Definition 79.1 are part of the definition of the perturbed theory. They are where contact terms enter a perturbed CFT. In a distributional formulation the product of local fields is first defined away from diagonals. A renormalized perturbation chooses extensions to the diagonals subject to locality, covariance, and the declared source normalization. Thus the coupling g is a coordinate in the Lagrangian or source functional of a specified regularized and renormalized path integral.

Remark 79.2 (First-order scaling of a relevant source)*.* Let $g(\mu)$ denote the dimensionful source in the chart above, and let

$$\lambda(\mu) = \mu^{-y_{\mathcal{O}}} g(\mu)$$

be the corresponding dimensionless coordinate at renormalization scale μ. If contact-term extensions are not changed while comparing two nearby scales and nonlinear OPE contributions are ignored, then

$$\mu \frac{d\lambda}{d\mu} = -y_{\mathcal{O}}\lambda.$$

Equivalently, increasing the length scale $\ell = \log(\mu_0/\mu)$ gives

$$\frac{d\lambda}{d\ell} = y_{\mathcal{O}}\lambda.$$

The dimensionful source g has mass dimension $y_{\mathcal{O}}$. Holding the source functional fixed while changing only the unit of mass gives $\lambda(\mu) = \mu^{-y_{\mathcal{O}}} g$. Differentiating this identity at fixed g proves the first displayed equation. The second follows from $\ell = -\log \mu + \text{constant}$. Corrections beyond this statement come from the diagonal extension of products of $\mathcal{O}$, equivalently from the singular part of the $\mathcal{O} \times \mathcal{O}$ OPE in the chosen renormalized coordinate.

If the perturbation generates a massive theory with mass gap $M > 0$, the dimensionless finite-size variable on a circle of circumference R is

$$r = MR.$$

Dimensional analysis fixes only the power relating M and $|g|^{1/y_{\mathcal{O}}}$. Once the normalization of $\mathcal{O}$, the source coordinate, and the scattering mass convention have been declared, an integrable model may determine a numerical mass-coupling constant

$$M = \kappa_{\mathcal{O}} \, |g|^{1/y_{\mathcal{O}}}.$$

The number $\kappa_{\mathcal{O}}$ is part of the matching data after the coordinates are fixed.

79.2 Polynomial Scalar Landau–Ginzburg Interface

This section records the ordinary two-dimensional scalar Landau–Ginzburg route as a renormalized perturbed-QFT interface. It is a useful comparison point for the minimal-model flow material below, but its logical status is different from an integrable scattering construction.

Let φ be a real scalar field in two Euclidean dimensions with a Gaussian ultraviolet reference measure whose covariance at separated points has the logarithmic singularity

$$C(x) = -\kappa \log(\mu|x|) + C_{\text{smooth}}(x), \qquad \kappa > 0. \tag{79.1}$$

Wick products $:\!\varphi^{\ell}\!:$ are normal ordered with respect to this reference covariance and the scale μ. A polynomial Landau–Ginzburg source-coordinate system is

$$V(\varphi) = \sum_{\ell=0}^{K} \frac{g_\ell}{\ell!} :\!\varphi^{\ell}\!: . \tag{79.2}$$

The finite-volume, infrared-regulated perturbation of separated correlation functions is represented formally by

$$\langle X \rangle_V = \frac{\big\langle X \exp\big(-\int d^2x \, V(\varphi(x))\big)\big\rangle_0}{\big\langle \exp\big(-\int d^2x \, V(\varphi(x))\big)\big\rangle_0}, \tag{79.3}$$

where X is a product of spectator fields supported away from the integration points before diagonal extensions are chosen.

The local ultraviolet statement behind this chart is elementary but important. Products of Wick polynomials have only powers of logarithms as their short-distance singularities:

$$:\!\varphi^a(x)\!::\!\varphi^b(y)\!:= \sum_{j=0}^{\min(a,b)} j! \binom{a}{j} \binom{b}{j} C(x-y)^j \, :\!\varphi^{a+b-2j}(y)\!: + \text{Taylor terms}.$$

Since

$$\int_0^{\varepsilon} r\,|\log r|^p\,dr < \infty \qquad (p \in \mathbb{Z}_{\geq 0}),$$

these logarithmic singularities are locally integrable at a single collision. Iterating the same estimate in relative coordinates around every partial diagonal gives local integrability for every finite order in the expansion of (79.3). The remaining choices are the finite contact-term extensions on diagonals, the infrared boundary condition, and convergence or summability of the perturbation series.

Equation of motion and the finite order-field quotient. The normal-ordered chart gives a precise distributional Schwinger–Dyson identity. Acting on the covariance with $\partial\bar{\partial}$ gives

$$\partial\bar{\partial}C(x) = -\pi\kappa\,\delta^{(2)}(x)$$

in the convention of (79.1). Gaussian integration by parts applied to (79.3) therefore gives, in correlators away from spectator insertions,

$$\partial\bar{\partial}\varphi = \pi\kappa\,V'(\varphi), \tag{79.4}$$

where

$$V'(\varphi) = \sum_{\ell=1}^{K} \frac{g_\ell}{(\ell-1)!}\,{:}\varphi^{\ell-1}{:}$$

in the same normal-ordering scheme. Contact terms at spectator insertions are part of the operator-source extension data in Definition 79.1.

For the even multicritical family

$$V(\varphi) = \sum_{j=1}^{m-1} \frac{g_{2j}}{(2j)!}\,{:}\varphi^{2j}{:}, \qquad m \geq 3, \tag{79.5}$$

the highest degree is $K = 2m-2$. In the idealized scaling chart where the critical potential is represented by its top monomial, (79.4) relates ${:}\varphi^{2m-3}{:}$ to the conformal descendant $\partial\bar{\partial}\varphi$. Thus the polynomial order-field sector has representatives

$$1,\ \varphi,\ {:}\varphi^2{:}, \ldots,\ {:}\varphi^{2m-4}{:}, \tag{79.6}$$

before diagonalizing the scaling fields. Its dimension is $2m-3$. The even source-coordinate system contains $m-1$ couplings $g_2, g_4, \ldots, g_{2m-2}$; after quotienting by the overall mass scale, there are $m-2$ dimensionless ratios to tune in the $\mathbb{Z}_2$-even family.

The expected continuum endpoint of the m-th multicritical even scalar theory is the unitary minimal model $\mathcal{M}(m, m+1)$ with

$$c_m = 1 - \frac{6}{m(m+1)}.$$

In this monograph this sentence is an RG-identification problem, not an input theorem. A proof must construct the continuum limit of the scalar polynomial theory at the tuned point, identify the stress tensor and local operator algebra, and explain how the order-field sector (79.6) embeds into the full minimal-model operator content. For example, when $m = 3$ the three order-field representatives match the Ising fields $1, \sigma, \epsilon$. For larger m, the full diagonal minimal model contains sectors not exhausted by the polynomial order-field quotient; those sectors must be supplied by the continuum construction rather than by the finite polynomial algebra alone.

79.3 Unitary Minimal Models and $\phi_{1,3}$

The A-series unitary Virasoro minimal model is denoted $\mathcal{M}(m, m+1)$, with $m \geq 3$. Its central charge is

$$c_m = 1 - \frac{6}{m(m+1)},$$

and its chiral Kac weights are

$$h_{r,s}^{(m)} = \frac{\big((m+1)r - ms\big)^2 - 1}{4m(m+1)}, \qquad 1 \leq r \leq m-1, \quad 1 \leq s \leq m,$$

with the identification $(r, s) \sim (m-r, m+1-s)$. The diagonal full CFT has scalar primaries when the same chiral module is paired with its antiholomorphic copy.

Remark 79.3 (Elementary data of the $\phi_{1,3}$ perturbation). In $\mathcal{M}(m, m+1)$, the diagonal scalar primary $\phi_{1,3}$ has

$$h_{1,3}^{(m)} = \frac{m-1}{m+1}, \qquad \Delta_{1,3}^{(m)} = \frac{2(m-1)}{m+1}, \qquad y_{1,3}^{(m)} = \frac{4}{m+1}.$$

Moreover

$$c_m - c_{m-1} = \frac{12}{m(m^2-1)} \qquad (m \geq 4).$$

For $(r, s) = (1, 3)$,

$$(m+1)r - ms = m + 1 - 3m = 1 - 2m.$$

Therefore

$$h_{1,3}^{(m)} = \frac{(2m-1)^2 - 1}{4m(m+1)} = \frac{4m(m-1)}{4m(m+1)} = \frac{m-1}{m+1}.$$

The full diagonal scalar has scaling dimension $2h_{1,3}^{(m)}$, giving the displayed $\Delta_{1,3}^{(m)}$. Subtracting from 2 gives

$$y_{1,3}^{(m)} = 2 - \frac{2(m-1)}{m+1} = \frac{4}{m+1}.$$

Finally,

$$c_m - c_{m-1} = \left(1 - \frac{6}{m(m+1)}\right) - \left(1 - \frac{6}{(m-1)m}\right) = \frac{6}{m}\left(\frac{1}{m-1} - \frac{1}{m+1}\right),$$

which is $12/[m(m^2-1)]$.

The standard integrable identification asserts that, for the massless branch of the $\phi_{1,3}$ perturbation and a definite sign convention for the source g,

$$\mathcal{M}(m, m+1) + g\int d^2x\, \phi_{1,3}(x) \quad \longrightarrow \quad \mathcal{M}(m-1, m) \qquad (m \geq 4).$$

In this monograph the theorem boundary is kept explicit. The exact arithmetic in Remark 79.3 proves only that the proposed infrared central charge is smaller by the displayed positive amount and that the perturbing operator is relevant. A theorem-level identification of the flow would require a construction of the perturbed local QFT, a proof of its infrared limit, and a matching of its local operator algebra to that of $\mathcal{M}(m-1, m)$.

Example 79.4 (First minimal-model cases)**.** For $m = 3$, $\mathcal{M}(3,4)$ is the Ising CFT. The label $(1,3)$ is identified with $(2,1)$, and

$$h^{(3)}_{1,3} = \frac{1}{2}, \qquad \Delta^{(3)}_{1,3} = 1.$$

This is the thermal perturbation, whose massive scaling limit is the free massive Majorana theory discussed in Chapter 77. For $m = 4$,

$$c_4 = \frac{7}{10}, \qquad c_3 = \frac{1}{2}, \qquad c_4 - c_3 = \frac{1}{5},$$

which is the central-charge drop of the tricritical-Ising to Ising massless branch. These statements fix the conformal arithmetic; the sign of the coupling and the operator normalization still have to be declared before comparing with scattering or TBA conventions.

79.4 Finite-Size Scaling Function

Let $E_0(R)$ be the ground-state energy on a spatial circle of circumference R. Define

$$c_{\text{eff}}(R) = -\frac{6R}{\pi} E_0(R).$$

For a CFT with periodic boundary conditions and cylinder Hamiltonian

$$H_R = \frac{2\pi}{R}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right),$$

the vacuum energy is

$$E_0(R) = -\frac{\pi c}{6R},$$

so $c_{\text{eff}}(R) = c$. In a nonunitary CFT the cylinder ground state may be a primary of negative weight; then the same formula gives

$$c_{\text{eff}} = c - 24h_{\min},$$

where $h_{\min}$ is the lowest diagonal conformal weight. This is the convention used for the Lee–Yang example in Chapter 77.

For a massive infrared theory with a unique vacuum,

$$E_0(R) = \mathcal{E}_{\text{bulk}} R + o(R^0) \qquad (R \to \infty).$$

The bulk energy density $\mathcal{E}_{\text{bulk}}$ is a local extensive coordinate. Hence comparisons of ultraviolet and infrared central charges must say whether the extensive term has been subtracted. The subtracted quantity is

$$E_0^{\text{sub}}(R) = E_0(R) - \mathcal{E}_{\text{bulk}} R, \qquad c_{\text{sub}}(R) = -\frac{6R}{\pi} E_0^{\text{sub}}(R).$$

Definition 79.5 (Finite-size identification data)**.** A finite-size identification between a perturbed CFT and a factorized scattering model consists of:

1. a Hilbert space on the circle and a self-adjoint Hamiltonian H_R for each R in the scaling regime;
2. a normalization of the vacuum energy, including the value of $\mathcal{E}_{\text{bulk}}$;
3. a mass convention for the infinite-volume particles;
4. a proof, or an explicitly stated hypothesis, that the large-volume mirror-channel thermodynamics computes $E_0(R)$.

With these data declared, the equality between a TBA scaling function and a QFT finite-size energy is a precise mathematical assertion.

79.5 Trace Sum Rule as a Local Test of the Flow

Finite-size central charges are necessary but not sufficient data for identifying a local RG trajectory. A sharper test uses the trace of the stress tensor. In this section $\Theta = T^\mu{}_\mu$ denotes the flat-space Cartesian trace in the stress-tensor normalization for which the Zamolodchikov sum rule below has coefficient $3/(4\pi)$. If a different normalization of the stress tensor or the perturbing source is used, the same statements hold after replacing Θ by the correspondingly normalized trace operator.

Definition 79.6 (Zamolodchikov sum-rule hypotheses)**.** A two-dimensional Euclidean RG trajectory satisfies the Zamolodchikov sum-rule hypotheses if the following data are specified.

1. A reflection-positive local QFT on $\mathbb{R}^2$ with a symmetric conserved stress tensor $T_{\mu\nu}$, trace $\Theta = T^\mu{}_\mu$, and connected trace two-point distribution
$$G_\Theta(x) = \langle \Theta(x)\Theta(0)\rangle_{\text{c}}.$$
2. A distributional extension of G_Θ to $x = 0$ whose contact terms are fixed by the stress-tensor source-coordinate system. The weighted pairing
$$\int_{\mathbb{R}^2} d^2x\, |x|^2 G_\Theta(x)$$
is required to be finite and independent of the remaining allowed point-supported contact terms. This condition is automatic for contact terms annihilated by the test function $|x|^2$, but it must be checked when derivatives of delta functions are permitted by the source scheme.
3. A continuous function $C(R)$, $R > 0$, whose endpoint limits exist,
$$\lim_{R\downarrow 0} C(R) = c_{\text{UV}}, \qquad \lim_{R\to\infty} C(R) = c_{\text{IR}},$$
and whose radial derivative is represented by
$$\frac{dC(R)}{d\log R} = -\frac{3}{4\pi}\int_0^{2\pi} d\alpha\, R^4\, G_\Theta(Re^{\mathrm{i}\alpha}) \tag{79.7}$$
in the distributional sense after smearing in R.

The last item is the local Ward-identity content of the two-dimensional c-theorem in the present normalization. It is not a definition of G_Θ; it is a theorem-level property to be established from stress tensor conservation, Euclidean covariance, positivity, and the regularity assumptions of the chosen QFT framework.

Trace sum rule. For a trajectory satisfying the Zamolodchikov sum-rule hypotheses,

$$c_{\mathrm{UV}} - c_{\mathrm{IR}} = \frac{3}{4\pi}\int_{\mathbb{R}^2} d^2x\,|x|^2\langle\Theta(x)\Theta(0)\rangle_c. \tag{79.8}$$

If reflection positivity makes $G_\Theta(x)$ a nonnegative radial spectral density at separated points, then $c_{\mathrm{UV}} \ge c_{\mathrm{IR}}$.

Integrating (79.7) from $R=\epsilon$ to $R=L$ gives

$$C(\epsilon) - C(L) = \frac{3}{4\pi}\int_\epsilon^L \frac{dR}{R}\int_0^{2\pi} d\alpha\, R^4 G_\Theta(Re^{\mathrm{i}\alpha}).$$

Since $d^2x = R\,dR\,d\alpha$, the right side is

$$\frac{3}{4\pi}\int_{\epsilon<|x|<L} d^2x\,|x|^2 G_\Theta(x).$$

The integrability hypothesis in Definition 79.6 permits the limits $\epsilon \downarrow 0$ and $L\to\infty$, while the endpoint hypothesis identifies the left side with $c_{\mathrm{UV}} - c_{\mathrm{IR}}$. Positivity of the weighted trace two-point distribution gives the inequality.

Form-factor version of the trace sum rule. Consider a massive integrable trajectory with particle species a, masses $m_a > 0$, and trace-operator form factors

$$F^\Theta_{a_1\cdots a_n}(\theta_1,\ldots,\theta_n) = \langle 0|\Theta(0)|\theta_1,a_1;\ldots;\theta_n,a_n\rangle_{\mathrm{in}}.$$

Assume that the connected Euclidean two-point function has the absolutely convergent spectral expansion

$$G_\Theta(x) = \sum_{n\ge1}\frac{1}{n!}\sum_{a_1,\ldots,a_n}\int_{\mathbb{R}^n}\prod_{j=1}^n\frac{d\theta_j}{2\pi}\,\left|F^\Theta_{a_1\cdots a_n}(\theta_1,\ldots,\theta_n)\right|^2 e^{-|x|\sum_j m_{a_j}\cosh\theta_j} \tag{79.9}$$

and that Fubini's theorem applies to the $|x|^2$-weighted integral. Then

$$c_{\mathrm{UV}} - c_{\mathrm{IR}} = 9\sum_{n\ge1}\frac{1}{n!}\sum_{a_1,\ldots,a_n}\int_{\mathbb{R}^n}\prod_{j=1}^n\frac{d\theta_j}{2\pi}\,\frac{\left|F^\Theta_{a_1\cdots a_n}(\theta_1,\ldots,\theta_n)\right|^2}{\left(\sum_j m_{a_j}\cosh\theta_j\right)^4}. \tag{79.10}$$

Insert (79.9) into (79.8). Absolute convergence permits the exchange of the spacetime integral with the rapidity integrals and species sums. For fixed rapidities set

$$E(\theta_1,\ldots,\theta_n) = \sum_j m_{a_j}\cosh\theta_j > 0.$$

The radial integral is elementary:

$$\frac{3}{4\pi}\int_{\mathbb{R}^2} d^2x\,|x|^2 e^{-E|x|} = \frac{3}{4\pi}(2\pi)\int_0^\infty d\rho\,\rho^3 e^{-E\rho} = \frac{3}{2}\frac{3!}{E^4} = \frac{9}{E^4}.$$

Substitution gives the displayed formula.

Example 79.7 ($\phi_{1,3}$ minimal-model flow target)**.** For the massless branch expected to connect $\mathcal{M}(m, m+1)$ to $\mathcal{M}(m-1, m)$, any proposed trace form factors and normalization must satisfy the numerical target

$$9 \sum_{n \geq 1} \frac{1}{n!} \sum_{a_1, \ldots, a_n} \int \prod_{j=1}^{n} \frac{d\theta_j}{2\pi} \frac{|F^{\Theta}_{a_1 \cdots a_n}(\theta_1, \ldots, \theta_n)|^2}{(\sum_j m_{a_j} \cosh \theta_j)^4} = \frac{12}{m(m^2-1)}.$$

The right side is the central-charge drop computed in Remark 79.3. This equality is stronger than matching the endpoint values of a TBA scaling function: it tests the local normalization of Θ, the form-factor solution, and the short-distance-to-long-distance interpolation of the trajectory.

79.6 TBA Scaling Function

For a diagonal factorized massive theory, use the kernel convention of Chapter 77,

$$\varphi_{ab}(\theta) = \frac{1}{2\pi \mathrm{i}} \frac{d}{d\theta} \log S_{ab}(\theta).$$

Let $m_a = \mu_a M$, where M is the reference mass and $\mu_a > 0$. The pseudoenergies are solutions of

$$\epsilon_a(\theta; r) = r\, \mu_a \cosh \theta - \sum_b \int_{\mathbb{R}} d\theta'\, \varphi_{ab}(\theta - \theta') \log\Big(1 + e^{-\epsilon_b(\theta'; r)}\Big), \qquad r = MR.$$

With the bulk term separated, the mirror-channel formula gives

$$E_0^{\mathrm{sub}}(R) = -\frac{M}{2\pi} \sum_a \mu_a \int_{\mathbb{R}} d\theta\, \cosh\theta\, \log\Big(1 + e^{-\epsilon_a(\theta; r)}\Big).$$

Consequently

$$c_{\mathrm{TBA}}(r) = \frac{3r}{\pi^2} \sum_a \mu_a \int_{\mathbb{R}} d\theta\, \cosh\theta\, \log\Big(1 + e^{-\epsilon_a(\theta; r)}\Big).$$

This formula is a theorem about the variational TBA functional under the hypotheses of Chapter 77; its identification with the finite-size vacuum energy of a local QFT requires the additional input in Definition 79.5.

Controlled Approximation 79.8 (Plateau reduction)**.** Assume that as $r \downarrow 0$ the solution has a plateau interval whose width tends to infinity, that the boundary layers at the ends of the plateau contribute only the standard Rogers-dilogarithm endpoints, and that the convolution kernels may be replaced on the plateau by their total integrals

$$N_{ab} = \int_{\mathbb{R}} d\theta\, \varphi_{ab}(\theta).$$

The plateau values $Y_a = \exp(\epsilon_a)$ then obey

$$\log Y_a = -\sum_b N_{ab} \log(1 + Y_b^{-1}).$$

The corresponding ultraviolet contribution to the effective central charge is

$$c_{\mathrm{UV}} = \frac{6}{\pi^2} \sum_a L\left(\frac{1}{1+Y_a}\right),$$

where

$$L(x) = -\frac{1}{2}\int_0^x dt\,\left(\frac{\log(1-t)}{t} + \frac{\log t}{1-t}\right) = \mathrm{Li}_2(x) + \frac{1}{2}\log x \log(1-x)$$

for $0 < x < 1$.

The displayed algebraic equation is obtained by dropping the driving term on the plateau and replacing each convolution by the total kernel integral. To derive the dilogarithm contribution, differentiate the TBA equation with respect to rapidity in a boundary layer where the pseudoenergy changes from a constant plateau value to its asymptotic value. The same integration by parts used in the Lee–Yang calculation of Chapter 77 reduces the scaling integral for species a to

$$\frac{6}{\pi^2}\int d\epsilon_a\,\log(1+e^{-\epsilon_a})\frac{d}{d\epsilon_a}\log(1+e^{\epsilon_a})^{-1}.$$

With $x = (1+e^{\epsilon_a})^{-1}$, this is the Rogers dilogarithm evaluated between the endpoint values. Under the present plateau hypothesis the nontrivial endpoint is $x_a = (1+Y_a)^{-1}$, giving the stated sum. The argument is a controlled asymptotic reduction. A complete application to a particular model includes a proof of the stated plateau hypothesis for that model.

79.7 Massless Flows

An integrable massless flow contains right- and left-moving asymptotic particles in the infrared scaling description. A right mover is parametrized by

$$E = p = \frac{M}{2}e^{\theta},$$

and a left mover by

$$E = -p = \frac{M}{2}e^{-\theta}.$$

The parameter M is the crossover scale. The TBA now has right-right, left-left, and right-left kernels. The right-left kernel records the interaction between the two chiral sectors along the flow; its normalization controls the interpolation between the ultraviolet and infrared plateaux.

Definition 79.9 (Massless factorized-flow data)**.** Diagonal massless factorized-flow data consist of:

1. right species $a \in A_R$ and left species $\alpha \in A_L$;
2. analytic two-body amplitudes

   $$S^{RR}_{ab}(\theta), \qquad S^{LL}_{\alpha\beta}(\theta), \qquad S^{RL}_{a\alpha}(\theta)$$

   obeying the declared unitarity, crossing, and bootstrap equations in their analytic strips;
3. a TBA system with driving terms

   $$r\,\mu_a e^{\theta}/2, \qquad r\,\nu_\alpha e^{-\theta}/2,$$

   and kernels obtained from the logarithmic derivatives of the amplitudes in the same sign convention;

4. ultraviolet and infrared local-operator charts to which the finite-size scaling function and form factors are to be matched.

For the minimal-model $\phi_{1,3}$ massless branch, the expected endpoint central charges satisfy

$$\lim_{r\downarrow 0} c_{\mathrm{sub}}(r) = c_m, \qquad \lim_{r\to\infty} c_{\mathrm{sub}}(r) = c_{m-1}.$$

The value of $c_m - c_{m-1}$ is fixed by Remark 79.3. The exact TBA system must reproduce these two limits after the bulk term and mass scale have been matched. This central-charge comparison is one component of the identification; the full identification also includes the local operator algebras and their OPE data in the ultraviolet and infrared.

79.8 Operator Data Along the Flow

The local-correlator part of the flow is supplied by form factors and their short-distance normalization. For a local operator $\widehat{\mathcal{A}}$, define its infinite-volume form factors by

$$F^{\mathcal{A}}_{a_1\cdots a_n}(\theta_1,\dots,\theta_n) = \langle 0|\widehat{\mathcal{A}}(0)|\theta_1,a_1;\dots;\theta_n,a_n\rangle_{\mathrm{in}},$$

where the ordering convention and particle-state normalization are those of Chapter 76. The ultraviolet scaling dimension is tested by the short-distance behavior

$$\langle\widehat{\mathcal{A}}(x)\widehat{\mathcal{A}}(0)\rangle_{\mathrm{c}} \sim \frac{C_{\mathcal{A}}}{|x|^{2\Delta_{\mathcal{A}}}} \qquad (M|x|\to 0),$$

with $C_{\mathcal{A}}$ fixed by a source convention. The perturbing operator itself must also satisfy the trace relation in the chosen normalization:

$$\Theta(x) = \kappa_\Theta\, y_{\mathcal{O}}\, g\,\mathcal{O}(x) + O(g^2)$$

when the Euclidean action is perturbed as $S_{\mathcal{C}} + g\int d^2x\,\mathcal{O}(x)$ and the stress tensor is normalized by $T^{\mu}{}_{\mu} = \Theta$. The constant κ_Θ is fixed by the Weyl-variation convention for the stress tensor and by whether a factor such as $1/(2\pi)$ is included in the source term. A chapter using this relation states κ_Θ before using numerical trace-normalization formulas.

Remark 79.10 (Necessary matching conditions for an integrable flow). Suppose a factorized scattering model is identified with a relevant perturbation of $\mathcal{C}$ by $\mathcal{O}$ in the sense of Definition 79.1. Then the following data must be mutually compatible:

1. the particle masses and scattering amplitudes determine a TBA scaling function whose ultraviolet limit equals the effective central charge of $\mathcal{C}$;

2. if the infrared limit is conformal, the large-R subtracted finite-size energy gives the infrared central charge;

3. the form-factor spectral expansion of the operator corresponding to $\mathcal{O}$ has the ultraviolet two-point normalization fixed in the perturbed-CFT chart;

4. the trace operator extracted from the finite-size response to a change of R agrees, to first order in g, with the perturbing operator through the displayed trace relation.

The first two assertions are consequences of the definition of $c_{\mathrm{sub}}(R)$ and of the CFT cylinder Hamiltonian. A mismatch of central charge would contradict equality of the finite-size vacuum energies in either scaling limit. The third assertion follows because the source coordinate in Definition 79.1 fixes the normalization of $\mathcal{O}$; a different form-factor normalization would compute a different source derivative of the Schwinger functional. For the fourth assertion, vary the perturbed action under a Weyl rescaling. The integrated insertion has scaling defect $y_{\mathcal{O}}$, and the declared stress-tensor normalization converts the Weyl variation of the action into the trace insertion. Higher powers of g are contact-term and operator mixing data of the same renormalized chart.

Open Problem 79.11 (Rigorous integrable-flow reconstruction). Construct a theorem that starts from factorized scattering data, form-factor solutions satisfying locality and convergence estimates, and TBA finite-size data, and reconstructs a local two-dimensional QFT whose ultraviolet limit is a specified CFT perturbation. The theorem should identify the operator topology in which the ultraviolet limit is taken, prove the finite-size Hamiltonian/TBA equality, and show that the OPE coefficients obtained from short-distance limits agree with the declared CFT data.

Chapter 80

Mirror-Channel TBA and Finite-Size Effects

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter refines the finite-volume interpretation of the thermodynamic Bethe ansatz. The primitive object is a Euclidean two-torus partition function with two Hamiltonian decompositions. The mirror-channel decomposition turns the circumference of the direct theory into the inverse temperature of a gas in the rotated theory. Finite-size corrections are then organized by the number of mirror particles winding around the compact cycle.

Hypothesis 80.1 (Mirror finite-size TBA structure)**.** A mirror finite-size TBA structure is a tuple

$$\mathfrak{M} = \left(Z, \{H_R, \mathcal{H}_R\}_{R>0}, \{\widetilde{H}_L, \widetilde{\mathcal{H}}_L\}_{L>0}, \mathsf{A}, \{m_a, S_{ab}, \varphi_{ab}\}_{a,b\in\mathsf{A}}, \mathcal{E}_{\mathrm{bulk}}, \mathcal{S}_{\mathrm{mir}}, \mathcal{P}_{\mathrm{mir}}\right)$$

with the following properties.

(i) $Z(R, L)$ is the regulated Euclidean torus partition function for a massive two-dimensional local QFT on a flat torus whose oriented cycles have lengths R and L. The same regulated functional admits the two Hamiltonian decompositions

$$Z(R, L) = \mathrm{Tr}_{\mathcal{H}_R}\, e^{-LH_R} = \mathrm{Tr}_{\widetilde{\mathcal{H}}_L}\, e^{-R\widetilde{H}_L}.$$

(ii) H_R is self-adjoint and bounded below on the direct-channel Hilbert space $\mathcal{H}_R$. The large-L limit $-\lim_{L\to\infty} L^{-1} \log Z(R, L) = E_0(R)$ exists for the range of R under discussion.

(iii) The mirror channel has a thermodynamic limit at inverse temperature R. Its stable mirror species form the finite or locally finite set A, with masses $m_a > 0$ and one-particle dispersion $\widetilde{E}_a(\theta) = m_a \cosh\theta$, $\widetilde{p}_a(\theta) = m_a \sinh\theta$. When A is locally finite rather than finite, every displayed sum over species is part of the specification as an absolutely convergent sum in the norm used for that formula.

(iv) In the diagonal sector treated in this chapter, $S_{ab}(\theta) \in \mathbb{C}^\times$ is the mirror two-body scattering phase on the real rapidity line, with a declared logarithm branch and

$$\varphi_{ab}(\theta) = \frac{1}{2\pi\mathrm{i}} \frac{d}{d\theta} \log S_{ab}(\theta).$$

This is the diagonal mirror structure. Non-diagonal mirror scattering uses the nested generalization of the density system.

(v) $\mathcal{S}_{\mathrm{mir}}$ consists of the mirror thermodynamic density constraint and entropy convention used below:

$$\rho_a + \rho_a^h = \frac{m_a}{2\pi} \cosh\theta + \sum_b \varphi_{ab} * \rho_b, \qquad s_a = (\rho_a + \rho_a^h) \log(\rho_a + \rho_a^h) - \rho_a \log\rho_a - \rho_a^h \log\rho_a^h.$$

(vi) $\mathcal{E}_{\mathrm{bulk}}$ is the local extensive vacuum-energy coordinate fixed by the same renormalization convention used in the direct finite-volume theory.

(vii) $\mathcal{P}_{\mathrm{mir}}$ records the analytic strips, pole sets, and contour-orientation conventions used when mirror rapidities are crossed through direct-channel particles or operator insertions. This analytic entry is part of the data needed to derive excited-state F-terms and wrapping residues from the torus trace.

All finite-size formulae in this chapter are conditional on $\mathfrak{M}$ and on the extra bounded-kernel or pole-avoidance hypotheses stated at the point where estimates or contour deformations are used.

80.1 Two Hamiltonian Decompositions

Let $\mathfrak{M}$ be a mirror finite-size TBA structure. Quantization along the L-direction gives

$$Z(R, L) = \mathrm{Tr}_{\mathcal{H}_R} \exp[-LH_R],$$

where H_R is the Hamiltonian on a spatial circle of circumference R. Assume that H_R is self-adjoint, bounded below, and has a unique ground state separated by a gap from the rest of the spectrum. Then

$$-\lim_{L\to\infty} \frac{1}{L} \log Z(R, L) = E_0(R),$$

where $E_0(R)$ is the direct-channel vacuum energy.

Quantization after exchanging the two cycles gives

$$Z(R, L) = \mathrm{Tr}_{\widetilde{\mathcal{H}}_L} \exp[-R\widetilde{H}_L].$$

The Hilbert space $\widetilde{\mathcal{H}}_L$ is the mirror-channel Hilbert space. For a relativistic theory obtained by Euclidean rotation, the one-particle dispersion in the infinite-volume mirror channel is

$$\widetilde{E}_a(\theta) = m_a \cosh\theta, \qquad \widetilde{p}_a(\theta) = m_a \sinh\theta.$$

The same function $m_a \cosh\theta$ now appears in the Boltzmann weight $\exp[-R\widetilde{E}_a(\theta)]$.

Remark 80.2 (Vacuum energy from mirror pressure). Assume that both torus decompositions exist at a common regulator, that the large-L limit of $L^{-1}\log Z(R,L)$ exists in the direct channel, and that the mirror channel has a thermodynamic limit at inverse temperature R. Let

$$f_{\mathrm{mir}}(R) = -\lim_{L\to\infty}\frac{1}{RL}\log \mathrm{Tr}_{\widetilde{\mathcal{H}}_L}\exp[-R\widetilde{H}_L]$$

be the mirror free-energy density. Then

$$E_0(R) = R\, f_{\mathrm{mir}}(R).$$

The two traces are the same torus functional written in two Hamiltonian channels. Dividing the equality of traces by L after taking the logarithm gives

$$\frac{1}{L}\log Z(R,L) = \frac{1}{L}\log \mathrm{Tr}_{\widetilde{\mathcal{H}}_L}\exp[-R\widetilde{H}_L].$$

The direct-channel assumption gives the left-hand limit $-E_0(R)$. The mirror thermodynamic definition gives the right-hand limit $-Rf_{\mathrm{mir}}(R)$. Equality of the two limits proves the formula.

80.2 Mirror TBA Functional

Use the kernel convention of Chapter 77:

$$\varphi_{ab}(\theta) = \frac{1}{2\pi \mathrm{i}}\frac{d}{d\theta}\log S_{ab}(\theta).$$

Here $S_{ab}(\theta)$ is the diagonal two-particle scattering phase in the mirror channel, and the branch of the logarithm is fixed continuously on the real rapidity line after the particle labels and statistics convention have been declared.

Let $\rho_a(\theta)$ and $\rho_a^h(\theta)$ be particle and hole densities per unit mirror length, so that

$$L\rho_a(\theta)d\theta$$

is the number of occupied rapidities of species a in the interval $(\theta, \theta + d\theta)$. Differentiating the Bethe–Yang equations gives

$$\rho_a(\theta) + \rho_a^h(\theta) = \frac{m_a}{2\pi}\cosh\theta + \sum_b \int_{\mathbb{R}} d\theta'\, \varphi_{ab}(\theta-\theta')\rho_b(\theta').$$

The entropy density of a Bethe cell with exclusion occupancy is

$$s_a = (\rho_a + \rho_a^h)\log(\rho_a+\rho_a^h) - \rho_a\log\rho_a - \rho_a^h\log\rho_a^h.$$

Define the pseudoenergy by

$$\epsilon_a(\theta) = \log\frac{\rho_a^h(\theta)}{\rho_a(\theta)}.$$

Mirror TBA variational equation. For diagonal factorized scattering with the density equation above, the stationary point of the mirror free-energy functional at inverse temperature R satisfies

$$\epsilon_a(\theta) = Rm_a \cosh\theta - \sum_b \int_{\mathbb{R}} d\theta'\, \varphi_{ab}(\theta-\theta') \log\Big(1+e^{-\epsilon_b(\theta')}\Big).$$

At this stationary point,

$$E_0(R) = \mathcal{E}_{\mathrm{bulk}} R - \sum_a \int_{\mathbb{R}} \frac{d\theta}{2\pi}\, m_a \cosh\theta\, \log\Big(1+e^{-\epsilon_a(\theta)}\Big).$$

The variation is the same variational calculation as in Proposition 77.2, with R now interpreted as inverse temperature of the mirror gas. A variation $\delta\rho_a$ induces

$$\delta\rho_c^t(\theta) = \sum_a \int d\theta'\, \varphi_{ca}(\theta-\theta')\delta\rho_a(\theta'), \qquad \rho_c^t = \rho_c + \rho_c^h.$$

The entropy variation is

$$\delta s = \sum_a \int d\theta\, \left[\log\frac{\rho_a^h}{\rho_a}\,\delta\rho_a + \log\frac{\rho_a^t}{\rho_a^h}\,\delta\rho_a^t\right].$$

Substituting the density-constraint variation and using $\log(\rho_a^t/\rho_a^h) = \log(1+e^{-\epsilon_a})$ gives the displayed pseudoenergy equation. Inserting the stationary relation back into the mirror free-energy density gives

$$f_{\mathrm{mir}}(R) = -\frac{1}{2\pi R}\sum_a m_a \int_{\mathbb{R}} d\theta\, \cosh\theta \log(1+e^{-\epsilon_a(\theta)}).$$

Remark 80.2 then gives the subtracted direct-channel vacuum energy. The additive $\mathcal{E}_{\mathrm{bulk}}R$ is the local extensive coordinate fixed in Chapter 79.

80.3 Large-Circumference Expansion

Let

$$d_a(\theta) = Rm_a\cosh\theta, \qquad q_a(\theta) = e^{-d_a(\theta)}, \qquad L_a(\theta) = \log(1+e^{-\epsilon_a(\theta)}).$$

The large-R expansion is an expansion in mirror occupation factors q_a, equivalently in windings around the direct spatial circle. The first winding is universal; later windings also contain the scattering kernels.

Two-winding expansion of the vacuum energy. Assume that the kernel convolution operator is bounded in a weighted L^∞ norm for which all q_a are small, so that the TBA equation is solved by iteration. The TBA equation is

$$\epsilon_a = d_a - \sum_b \varphi_{ab} * L_b.$$

If $L_b = q_b + O(q^2)$, then

$$e^{-\epsilon_a} = e^{-d_a}\exp\left(\sum_b \varphi_{ab} * L_b\right) = q_a\left(1+\sum_b \varphi_{ab} * q_b\right) + O(q^3).$$

Using $\log(1+x) = x - x^2/2 + O(x^3)$ gives

$$L_a(\theta) = q_a(\theta) + q_a(\theta) \sum_b \int_{\mathbb{R}} d\theta' \, \varphi_{ab}(\theta - \theta') q_b(\theta') - \frac{1}{2} q_a(\theta)^2 + O(q^3).$$

Consequently

$$\begin{aligned} E_0(R) - \mathcal{E}_{\text{bulk}} R = &- \sum_a \int_{\mathbb{R}} \frac{d\theta}{2\pi} \, m_a \cosh\theta \, q_a(\theta) \\ &- \sum_{a,b} \int_{\mathbb{R}} \frac{d\theta}{2\pi} \int_{\mathbb{R}} d\theta' \, m_a \cosh\theta \, q_a(\theta) \varphi_{ab}(\theta - \theta') q_b(\theta') \\ &+ \frac{1}{2} \sum_a \int_{\mathbb{R}} \frac{d\theta}{2\pi} \, m_a \cosh\theta \, q_a(\theta)^2 + O(q^3). \end{aligned}$$

Proposition 80.3 (Vacuum Luescher term with a remainder bound)**.** *Let the set* A *of mirror particle species be finite, and let*

$$m_* = \min_{a \in \mathsf{A}} m_a > 0.$$

Assume that the diagonal mirror TBA equations written above have the exclusion form and that their kernels obey

$$\Phi = \sup_{a \in \mathsf{A}} \sum_{b \in \mathsf{A}} \int_{\mathbb{R}} du \, |\varphi_{ab}(u)| < \infty.$$

Then there are constants R_0 *and* C*, depending on the masses and on* Φ*, such that for* $R \geq R_0$ *the solution in the small-occupation branch satisfies*

$$E_0(R) - \mathcal{E}_{\text{bulk}} R = -\sum_{a \in \mathsf{A}} \frac{m_a}{\pi} K_1(m_a R) + \mathcal{R}_2(R), \qquad |\mathcal{R}_2(R)| \leq C R^{-1/2} e^{-2m_* R}.$$

Here K_1 *is the modified Bessel function normalized by*

$$K_1(r) = \int_0^\infty d\theta \, e^{-r\cosh\theta} \cosh\theta.$$

Proof. Write the TBA fixed-point equation directly for

$$L_a(\theta) = \log(1 + e^{-\epsilon_a(\theta)}).$$

With

$$(KL)_a(\theta) = \sum_{b \in \mathsf{A}} \int_{\mathbb{R}} d\theta' \, \varphi_{ab}(\theta - \theta') L_b(\theta'),$$

the equation is

$$L_a(\theta) = \log\Big(1 + q_a(\theta) e^{(KL)_a(\theta)}\Big), \qquad q_a(\theta) = e^{-R m_a \cosh\theta}.$$

The integral bound gives

$$\|KL\|_\infty \leq \Phi \|L\|_\infty.$$

For R large enough the map

$$T_R(L)_a(\theta) = \log\Big(1 + q_a(\theta) e^{(KL)_a(\theta)}\Big)$$

maps the closed ball $\|L\|_\infty \le 2e^{-Rm_*}$ into itself. Indeed

$$|T_R(L)_a(\theta)| \le q_a(\theta)e^{\Phi\|L\|_\infty} \le e^{-Rm_*}e^{2\Phi e^{-Rm_*}} \le 2e^{-Rm_*}$$

after increasing R_0. The mean-value theorem and $|\log(1+x)-\log(1+y)| \le |x-y|$ for $x, y \ge 0$ give

$$\|T_R(L) - T_R(M)\|_\infty \le C_\Phi e^{-Rm_*}\|L-M\|_\infty,$$

with C_Φ independent of R. Thus T_R is a contraction for $R \ge R_0$, so the small-occupation solution exists uniquely in this ball.

For that solution,

$$|(KL)_a(\theta)| \le 2\Phi e^{-Rm_*}.$$

Using $e^x = 1 + O(x)$ uniformly for $|x| \le 2\Phi e^{-Rm_*}$ and $\log(1+y) = y + O(y^2)$ uniformly for $0 \le y \le 2e^{-Rm_*}$, one obtains the pointwise estimate

$$L_a(\theta) = q_a(\theta) + \delta L_a(\theta), \qquad |\delta L_a(\theta)| \le Cq_a(\theta)e^{-Rm_*} + Cq_a(\theta)^2.$$

Substitution in the vacuum-energy formula gives

$$E_0(R) - \mathcal{E}_{\text{bulk}}R = -\sum_a \int_{\mathbb{R}} \frac{d\theta}{2\pi}\, m_a \cosh\theta\, q_a(\theta) + \mathcal{R}_2(R),$$

where

$$|\mathcal{R}_2(R)| \le C\sum_a \int_{\mathbb{R}} \frac{d\theta}{2\pi}\, m_a \cosh\theta \left(q_a(\theta)e^{-Rm_*} + q_a(\theta)^2\right).$$

The first integral is

$$\int_{\mathbb{R}} d\theta\, \cosh\theta\, e^{-Rm_a\cosh\theta} = 2K_1(m_aR),$$

so its contribution to the energy is $-m_aK_1(m_aR)/\pi$. It remains only to record the uniform large-R bound. For $\lambda \ge 1$,

$$K_1(\lambda) = \int_0^\infty d\theta\, \cosh\theta\, e^{-\lambda\cosh\theta} \le C\lambda^{-1/2}e^{-\lambda}.$$

This elementary estimate follows by splitting the integral at $\theta = 1$: on $0 \le \theta \le 1$, $\cosh\theta \ge 1 + \theta^2/3$, while on $\theta \ge 1$ the stronger decay of $e^{-\lambda\cosh\theta}$ gives a term bounded by $Ce^{-\lambda\cosh 1}$. Applying this estimate to $\lambda = m_aR$ and to $\lambda = 2m_aR$ gives

$$|\mathcal{R}_2(R)| \le CR^{-1/2}e^{-2m_*R}$$

after another harmless enlargement of C. □

Example 80.4 (Free Majorana leading winding)**.** For the massive Ising/Majorana theory the derivative kernel vanishes. The leading winding contribution is

$$E_0(R) - \mathcal{E}_{\text{bulk}}R = -\frac{m}{2\pi}\int_{-\infty}^{\infty} d\theta\, \cosh\theta\, e^{-mR\cosh\theta} + O(e^{-2mR}).$$

The modified Bessel function has the integral representation

$$K_\nu(r) = \int_0^\infty d\theta\, e^{-r\cosh\theta}\cosh(\nu\theta), \qquad r > 0.$$

Therefore

$$\int_{-\infty}^{\infty} d\theta\, \cosh\theta\, e^{-r\cosh\theta} = 2K_1(r),$$

and the first winding gives

$$E_0(R) - \mathcal{E}_{\text{bulk}}R = -\frac{m}{\pi}K_1(mR) + O(e^{-2mR}).$$

Using the standard steepest-descent expansion at the saddle $\theta = 0$,

$$K_1(r) \sim \sqrt{\frac{\pi}{2r}}\, e^{-r}\left(1 + \frac{3}{8r} - \frac{15}{128r^2} + \cdots\right),$$

so the leading finite-size vacuum correction is exponentially small with prefactor $R^{-1/2}e^{-mR}$.

80.4 Excited-State F-Term

A direct-channel finite-volume excited state is described, at leading large R, by physical rapidities θ_j and species a_j satisfying Bethe–Yang equations. A single mirror particle of species b and rapidity θ winding once around the direct spatial circle scatters through the physical particles. Euclidean rotation converts this scattering into the crossed amplitudes

$$S_{ba_j}\left(\theta - \theta_j + \frac{\mathrm{i}\pi}{2}\right).$$

Definition 80.5 (Analytic strip for the F-term)**.** For a state $\Psi = (a_1, \theta_1; \ldots; a_n, \theta_n)$, an F-term strip is a connected domain containing the real θ-axis on which every function

$$S_{ba_j}\left(\theta - \theta_j + \frac{\mathrm{i}\pi}{2}\right)$$

is meromorphic, the chosen integration contour can be kept away from poles, and the product of amplitudes has sufficient decay for the contour integral below.

Controlled Approximation 80.6 (One-winding excited-state F-term)**.** Assume diagonal scattering, a declared F-term strip with no pole crossed by the real mirror contour, and a large-R cluster expansion of the torus trace in mirror occupation factors. The one-mirror-particle correction to the energy of the direct-channel state Ψ, after subtracting the vacuum mirror loop, is

$$\Delta E_\Psi^F(R) = -\sum_b \int_{\mathbb{R}} \frac{d\theta}{2\pi}\, m_b \cosh\theta\, e^{-Rm_b\cosh\theta}\left(\prod_{j=1}^{n} S_{ba_j}\left(\theta - \theta_j + \frac{\mathrm{i}\pi}{2}\right) - 1\right).$$

In the mirror channel, the direct excited state is represented by operator insertions that a mirror particle crosses while winding around the cycle. At one winding, the trace contribution of one mirror particle of species b and rapidity θ has Boltzmann weight $\exp[-Rm_b\cosh\theta]$ and density of states $d\theta/(2\pi)$. Crossing the direct-channel physical particles multiplies this contribution by

$$\prod_{j=1}^{n} S_{ba_j}\left(\theta - \theta_j + \frac{\mathrm{i}\pi}{2}\right).$$

The logarithm of the trace converts this one-particle contribution into an energy shift with the overall factor $-m_b \cosh\theta$; the identical loop with no physical particles is the vacuum contribution and gives the subtraction by 1. Summing over mirror species gives the displayed formula.

If a pole is crossed while moving the contour to the declared F-term strip, the residue is a separate contribution. A convenient residue formula is

$$\Delta E^{\mu}_{\Psi}(R) = -\sum_{\theta_* \in \mathcal{P}} \sigma_* \,\mathrm{i} \operatorname*{Res}_{\theta=\theta_*} \left[\frac{m_b \cosh\theta}{2\pi} e^{-R m_b \cosh\theta} \left(\prod_j S_{ba_j}\left(\theta - \theta_j + \frac{\mathrm{i}\pi}{2}\right) - 1 \right) \right],$$

where $\mathcal{P}$ is the finite set of crossed poles and $\sigma_* = \pm 1$ records the orientation of the contour deformation. In a bound-state theory those residues are controlled by the same pole data that enter the factorized scattering bootstrap. The formula becomes a theorem only after the analytic strip, contour orientation, and bound-state pole normalization have been specified.

Status checkpoint: excited-state continuation. An excited-state finite-size formula is not obtained by substituting Bethe–Yang rapidities into the ground-state TBA functional. The direct state changes the mirror trace by insertions, and analytic continuation of the mirror rapidity contour can encounter poles, zeros of $1 + \mathrm{e}^{-\epsilon}$, and branch changes of the logarithm. A theorem-level excited-state TBA must therefore specify a continuation path in R and in rapidity space, record the source terms created by crossed singularities, prove that no omitted singularity crosses the contour, and compare the resulting expression with the direct-channel spectral trace. The one-winding F-term and the μ-term residue above are the first controlled pieces of this continuation ledger.

Controlled Approximation 80.7 (Excited-state continuation residual budget)**.** Let

$$\mathcal{E}^{\mathrm{dir}}_{\Psi}(R) = E_{\Psi}(R) - E_0(R)$$

be the direct-channel excitation energy above the vacuum. Let $\mathcal{E}^{\mathrm{ins}}_{\Psi}$ be the same quantity extracted from the mirror trace with the direct-state insertions kept at the regulator level, $\mathcal{E}^{\mathrm{cont}}_{\Psi}$ the expression after the declared contour and branch continuation, and $\mathcal{E}^{\mathrm{src}}_{\Psi}$ the expression after all crossed singularities on that declared path have been converted into source or residue terms. Finally set

$$\mathcal{E}^{\mathrm{1w}}_{\Psi}(R) = E^{\mathrm{BY}}_{\Psi}(R) + \Delta E^{F}_{\Psi}(R) + \Delta E^{\mu}_{\Psi}(R), \tag{80.1}$$

where E^{BY}_{Ψ} is the Bethe–Yang energy in the chosen finite-volume normalization and the last two terms are the one-winding F-term and the crossed-pole μ-term retained above. The exact difference between the direct spectral coordinate and the one-winding coordinate decomposes as

$$\mathcal{E}^{\mathrm{dir}}_{\Psi}(R) - \mathcal{E}^{\mathrm{1w}}_{\Psi}(R) = R_{\mathrm{trace}} + R_{\mathrm{cont}} + R_{\mathrm{branch}} + R_{\mathrm{pole}} + R_{\mathrm{multi}} + R_{\mathrm{dens}} + R_{\mathrm{norm}}. \tag{80.2}$$

Here R_{trace} compares the direct spectral trace with the mirror insertion trace; R_{cont} is the error in the contour deformation inside the declared analytic strip; R_{branch} records logarithm and source-branch changes; R_{pole} is the net contribution of crossed singularities not included in ΔE^{μ}_{Ψ}; R_{multi} is the omitted two- and higher-mirror-particle sector; R_{dens} records the finite-volume density and Bethe-rapidity normalization mismatch; and R_{norm} records local operator, state, and bulk-energy normalization differences between the direct and mirror coordinates. Consequently, if on a window $R \in I$ these residuals obey

$$|R_{\kappa}(R)| \le \varepsilon_{\kappa}(R), \qquad \kappa \in \{\mathrm{trace}, \mathrm{cont}, \mathrm{branch}, \mathrm{pole}, \mathrm{multi}, \mathrm{dens}, \mathrm{norm}\},$$

then

$$\left|\mathcal{E}_\Psi^{\mathrm{dir}}(R) - \mathcal{E}_\Psi^{1\mathrm{w}}(R)\right| \leq \sum_\kappa \varepsilon_\kappa(R). \tag{80.3}$$

The budget is only an identity plus declared estimates, but it is the right identity to audit. A successful excited-state TBA theorem proves that the residuals vanish in the intended limit or remain within a stated approximation window. A one-winding formula without these seven controls is a semiclassical finite-size coordinate, not a reconstruction of the direct-channel spectral trace.

80.5 Wrapping Effects for Local Operators

A wrapping contribution is a finite-volume term represented in the mirror channel by virtual propagation around a compact cycle. Its order is measured by the number and species of mirror occupation factors $q_a(\theta)$. This ordering is independent of the loop order of any Lagrangian perturbation series.

Let $\widehat{\mathcal{O}}$ be a local operator with a declared infinite-volume normalization. A diagonal finite-volume matrix element has an expansion of the form

$$\langle \Psi, R|\widehat{\mathcal{O}}|\Psi, R\rangle = \langle \Psi|\widehat{\mathcal{O}}|\Psi\rangle_\infty + \sum_{k\geq 1} \Delta_{\mathcal{O}}^{(k)}(R),$$

where $\Delta_{\mathcal{O}}^{(k)}(R)$ is assembled from k-mirror-particle form factors, crossed scattering factors, and finite-volume density determinants. The first term in this series is determined by the connected one-mirror-particle form factor in the crossed channel, after subtracting the vacuum wrapping term and the disconnected pieces already accounted for by the finite-volume normalization of $|\Psi, R\rangle$.

Definition 80.8 (Operator wrapping specification)**.** An operator wrapping specification consists of:

1. the infinite-volume form factors of $\widehat{\mathcal{O}}$ in the state normalization of Chapter 76;
2. the analytic strip and crossing prescription for moving mirror rapidities through the operator insertion and through the physical particles;
3. the finite-volume density determinant for the physical Bethe–Yang state;
4. a subtraction prescription for vacuum terms and disconnected diagonal pieces.

With this specification fixed, each wrapping correction is a contour integral whose integrand is a declared meromorphic function. Residues from crossed poles are then part of the same contour calculation rather than extra rules.

80.6 Relation To Planar Wrapping

In large-N gauge-theory spectral problems the word wrapping often denotes corrections whose spin-chain interaction range reaches the length of a single-trace operator. The relativistic finite-size mechanism developed here has a literal spacetime compact cycle and a mirror thermal trace.

Later planar-integrability chapters build the bridge by identifying the spectral problem, the auxiliary mirror channel in that spectral problem, and the analytic continuation that maps finite-size corrections between the two settings.

Open Problem 80.9 (Excited-state mirror construction). Give a theorem-level construction of excited-state TBA starting from factorized scattering, a declared analytic strip, pole data, and finite-volume Bethe states, including a proof that the contour deformations and residue terms reproduce the spectral trace on the torus.

Chapter 81

Sine-Gordon, Massive Thirring, And Affine Toda Theories

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter introduces exact relativistic model families in which the factorized-scattering, form-factor, and TBA structures of the preceding chapters can be tested against Lagrangian and ultraviolet data. Every normalization below is part of the definition of the model being discussed.

81.1 Model-Family Datum

An integrable model family is specified by:

$$(\mathcal{A}_{\mathrm{UV}}, \mathcal{O}_{\mathrm{pert}}, \lambda, \mathcal{Q}, \mathcal{S}, \mathcal{F}),$$

where $\mathcal{A}_{\mathrm{UV}}$ is a short-distance operator algebra or regulated Lagrangian chart, $\mathcal{O}_{\mathrm{pert}}$ is the relevant perturbing coordinate, λ is its coupling, $\mathcal{Q}$ is the commutative algebra of conserved charges, $\mathcal{S}$ is the factorized scattering system, and $\mathcal{F}$ is the form-factor system for local operators. The model family is under control only when these entries are matched in common conventions.

81.2 Sine-Gordon Normalization

Let φ be a real scalar field with Lorentzian action

$$S_{\mathrm{SG}} = \int d^2x \left[-\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi + \frac{\mu}{\beta^2}\big(\cos(\beta\varphi) - 1\big) \right],$$

where $0 < \beta^2 < 8\pi$ in the short-distance free-boson normalization used for the displayed formula. The potential is periodic under

$$\varphi \longmapsto \varphi + \frac{2\pi}{\beta}.$$

Finite-energy configurations on the line therefore have topological charge

$$Q_{\mathrm{top}} = \frac{\beta}{2\pi}\int_{-\infty}^{\infty} dx\,\partial_x\varphi \in \mathbb{Z}.$$

The charge $Q_{\mathrm{top}} = 1$ particle is the soliton, and $Q_{\mathrm{top}} = -1$ is the antisoliton.

81.3 Sine-Gordon Spectrum

Introduce

$$\xi = \frac{\beta^2}{8\pi - \beta^2}.$$

Equivalently $\lambda = \xi^{-1} = 8\pi/\beta^2 - 1$. The attractive regime is $0 < \xi < 1$, or $\lambda > 1$. In this regime the soliton-antisoliton S-matrix has bound-state poles corresponding to neutral breathers

$$B_n, \qquad n = 1, \dots, N_B, \qquad N_B = \max\{n \in \mathbb{Z}_{>0} \mid n\xi < 1\},$$

with masses

$$m_n = 2M_s \sin\left(\frac{\pi n\xi}{2}\right),$$

where M_s is the soliton mass. The formula follows from relativistic two-body kinematics: a pole at rapidity difference $\theta = \mathrm{i}u$ in a two-particle channel of equal masses M_s produces a bound state of mass

$$m^2 = 2M_s^2(1 + \cos u) = 4M_s^2\cos^2\left(\frac{u}{2}\right).$$

For the breather poles $u = \pi(1 - n\xi)$, this gives the displayed value.

81.4 Mass Scale And The UV Perturbing Coordinate

The symbol M_s denotes the physical soliton mass, defined by the isolated mass hyperboloid in the soliton superselection sector. The coefficient μ in a Lagrangian expression is not itself an invariant until the short-distance normal-ordering convention for the vertex operator has been specified. To make the comparison unambiguous, introduce the Euclidean free-boson normalization

$$A_0[\phi] = \frac{1}{16\pi}\int_{\mathbb{R}^2} \mathrm{d}^2x\,\partial_\alpha\phi\,\partial_\alpha\phi, \qquad \langle\phi(x)\phi(0)\rangle_0 = -2\log|x| + O(1),$$

and define the perturbation by

$$A_{\mathrm{SG}}^{\mathrm{UV}} = A_0[\phi] - 2\mu_{\mathrm{UV}} \int_{\mathbb{R}^2} \mathrm{d}^2x \ :\cos(\beta_{\mathrm{UV}}\phi(x)):_0 . \tag{81.1}$$

Then the scaling dimension of $:\exp(\mathrm{i}\beta_{\mathrm{UV}}\phi):_0$ is $2\beta_{\mathrm{UV}}^2$, the coupling μ_{UV} has mass dimension $2-2\beta_{\mathrm{UV}}^2$, and

$$\xi = \frac{\beta_{\mathrm{UV}}^2}{1-\beta_{\mathrm{UV}}^2}.$$

Equivalently, in the canonical kinetic normalization of the preceding section, $\phi = \sqrt{8\pi}\,\varphi$ and $\beta_{\mathrm{UV}} = \beta/\sqrt{8\pi}$. The exact mass-coupling relation, in the UV convention (81.1), is

$$M_s = \frac{2\Gamma(\xi/2)}{\sqrt{\pi}\,\Gamma((1+\xi)/2)} \left[\frac{\pi\mu_{\mathrm{UV}}\Gamma(1-\beta_{\mathrm{UV}}^2)}{\Gamma(\beta_{\mathrm{UV}}^2)}\right]^{(1+\xi)/2} . \tag{81.2}$$

The power is forced by dimensions because

$$\frac{1+\xi}{2} = \frac{1}{2(1-\beta_{\mathrm{UV}}^2)}.$$

The nontrivial content of (81.2) is therefore the dimensionless gamma-function coefficient. In this chapter the formula is used as a coordinate change between the UV perturbing coordinate and the physical mass coordinate; the derivation from finite-size thermodynamic Bethe ansatz data and the vacuum energy normalization belongs to the later chapter on exact RG flows.

81.5 Exact Soliton-Antisoliton Scattering Datum

The one-soliton internal space is

$$V = \mathbb{C}|+\rangle \oplus \mathbb{C}|-\rangle, \qquad Q_{\mathrm{top}}|\pm\rangle = \pm|\pm\rangle.$$

Charge conservation forces the two-particle scattering operator on $V\otimes V$ to have the block form

$$\mathsf{S}_{\mathrm{sol}}(\theta) = S_{\min}(\theta) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b(\theta) & c(\theta) & 0 \\ 0 & c(\theta) & b(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{|++\rangle,|+-\rangle,|-+\rangle,|--\rangle} , \tag{81.3}$$

where

$$b(\theta) = \frac{\sinh(\lambda\theta)}{\sinh(\lambda(\mathrm{i}\pi-\theta))}, \qquad c(\theta) = \frac{\mathrm{i}\sin(\pi\lambda)}{\sinh(\lambda(\mathrm{i}\pi-\theta))}, \qquad \lambda = \xi^{-1}.$$

One standard minimal scalar normalization is the boundary value, for real θ, of

$$S_{\min}(\theta) = -\exp\left[-\mathrm{i}\int_0^\infty \frac{\mathrm{d}t}{t}\,\frac{\sinh((1-\xi)t/2)}{\sinh(\xi t/2)\cosh(t/2)} \sin\left(\frac{\theta t}{\pi}\right)\right], \tag{81.4}$$

continued meromorphically from the physical line with the prescribed physical-strip pole data. The displayed system is on-shell bootstrap data: the construction of a local Hilbert-space QFT whose soliton sectors realize it is a separate existence problem, recorded at the end of the chapter.

Matrix unitarity and the breather pole locations. For real $\lambda > 0$, the matrix factor in (81.3) obeys

$$R_\lambda(\theta)R_\lambda(-\theta) = \mathbf{1}$$

as a meromorphic 4×4 matrix. In the attractive regime $\lambda > 1$, the denominators of $b(\theta)$ and $c(\theta)$ have physical-strip zeros at

$$\theta = \mathrm{i}u_n, \qquad u_n = \pi(1 - n\xi), \qquad n \in \mathbb{Z}_{>0}, \quad n\xi < 1, \tag{81.5}$$

and these are precisely the rapidity locations that give the breather masses

$$m_n = 2M_s \sin\left(\frac{\pi n \xi}{2}\right).$$

Only the charge-zero block is nontrivial. Set

$$X = \sinh(\lambda\theta), \qquad s = \sin(\pi\lambda), \qquad D_\pm = \sinh(\lambda(\mathrm{i}\pi \mp \theta)).$$

Using

$$D_+D_- = \sinh(\mathrm{i}\pi\lambda - \lambda\theta)\sinh(\mathrm{i}\pi\lambda + \lambda\theta) = -\big(X^2 + s^2\big),$$

one obtains

$$b(\theta)b(-\theta) + c(\theta)c(-\theta) = \frac{-X^2 - s^2}{D_+D_-} = 1,$$

and

$$b(\theta)c(-\theta) + c(\theta)b(-\theta) = \frac{X\,\mathrm{i}s + \mathrm{i}s(-X)}{D_+D_-} = 0.$$

The opposite charge-zero ordering has the same diagonal coefficient and the off-diagonal coefficient with the two terms interchanged, so its product gives the identical scalar equations. On $|{+}{+}\rangle$ and $|{-}{-}\rangle$ the matrix block is scalar, and unitarity reduces to multiplication of the corresponding scalar entries at θ and $-\theta$.

The denominator vanishes exactly when

$$\lambda(\mathrm{i}\pi - \theta) = \mathrm{i}\pi n, \qquad n \in \mathbb{Z},$$

so $\theta = \mathrm{i}\pi(1 - n/\lambda) = \mathrm{i}\pi(1 - n\xi)$. This point lies in the physical strip $0 < \operatorname{Im}\theta < \pi$ exactly when $n\xi < 1$. Substituting $u_n = \pi(1 - n\xi)$ into the equal-mass bound-state formula gives

$$m_n = 2M_s \cos\left(\frac{u_n}{2}\right) = 2M_s \sin\left(\frac{\pi n\xi}{2}\right),$$

as claimed.

Lemma 81.1 (Yang-Baxter equation for the soliton matrix part)**.** *Let $R_\lambda(\theta)$ be the 4×4 matrix factor in (81.3), without the scalar multiplier $S_{\min}$. On $V^{\otimes 3}$ it obeys*

$$R_{12}(\theta_{12})R_{13}(\theta_{13})R_{23}(\theta_{23}) = R_{23}(\theta_{23})R_{13}(\theta_{13})R_{12}(\theta_{12}), \qquad \theta_{ij} = \theta_i - \theta_j.$$

Proof. Only the charge-one and charge-two subspaces of $V^{\otimes 3}$ are nontrivial; charge zero and charge three are scalar. Write

$$B(u) = \frac{\sinh(\lambda u)}{\sinh(\mathrm{i}\pi\lambda - \lambda u)}, \qquad C(u) = \frac{\mathrm{i}\sin(\pi\lambda)}{\sinh(\mathrm{i}\pi\lambda - \lambda u)}, \qquad A = \mathrm{i}\pi\lambda.$$

Direct multiplication of the two 8×8 products reduces all component equalities to the following three scalar identities, with $w = u + v$:

$$\begin{aligned} -B(u)C(v) + B(w)C(v) - B(v)C(u)C(w) &= 0, \\ -B(u)B(v)C(w) - C(u)C(v) + C(w) &= 0, \\ -B(u)C(w)C(v) + B(w)C(u) - B(v)C(u) &= 0. \end{aligned}$$

After clearing denominators these identities are consequences of

$$\sinh(A-u)\sinh(A-v) - \sinh u\, \sinh v = \sinh A\, \sinh(A-u-v),$$

and the two identities obtained from it by replacing one of the factors with the addition formula

$$\sinh(u+v) = \sinh u \cosh v + \cosh u \sinh v.$$

Since $\sinh A = \mathrm{i}\sin(\pi\lambda)$, these are exactly the three displayed component identities. The scalar factor $S_{\min}$ cancels from the Yang-Baxter equation, because it contributes the same product of three scalar functions on both sides. $\square$

Remark 81.2 (Free-fermion point). At $\beta^2 = 4\pi$, one has $\xi = 1$ and $\lambda = 1$. Then $\sin(\pi\lambda) = 0$, so the reflection amplitude $c(\theta)$ vanishes, while $b(\theta) = 1$. The remaining scalar factor is $S_{\min}(\theta) = -1$, the fermionic exchange phase of the massive Thirring field at $g_{\mathrm{T}} = 0$. This is a useful convention check: the soliton and antisoliton then form the free massive Dirac particle and antiparticle.

81.6 Soliton-Breather Scattering

The neutral breather B_n is obtained from the residue of the soliton-antisoliton amplitude at

$$\theta = \mathrm{i}\pi(1 - n\xi).$$

Applying the fusion identity of Proposition 72.2 to the soliton matrix (81.3) gives a diagonal scalar amplitude for a soliton or antisoliton scattering through B_n:

$$S_{sB_n}(\theta) = \frac{\sinh\theta + \mathrm{i}\cos(\pi n\xi/2)}{\sinh\theta - \mathrm{i}\cos(\pi n\xi/2)} \prod_{\ell=1}^{n-1} \frac{\sin^2\left(\frac{\pi\xi}{4}(n-2\ell) - \frac{\pi}{4} + \frac{\mathrm{i}\theta}{2}\right)}{\sin^2\left(\frac{\pi\xi}{4}(n-2\ell) - \frac{\pi}{4} - \frac{\mathrm{i}\theta}{2}\right)}. \tag{81.6}$$

The same scalar function appears for $s = +$ and $s = -$, because B_n is neutral and the fused residue is a one-dimensional charge-preserving channel. The formula is minimal in the same sense as before: it contains no additional CDD factor that is analytic, nonzero, and unitary in the physical strip.

Soliton–breather unitarity, crossing, and pole locations. For $n\xi < 1$, the amplitude (81.6) obeys

$$S_{sB_n}(\theta)S_{sB_n}(-\theta) = 1, \qquad S_{sB_n}(\mathrm{i}\pi - \theta) = S_{sB_n}(\theta).$$

Its first scalar factor has physical-strip poles at

$$\theta = \mathrm{i}\left(\frac{\pi}{2} + \frac{\pi n\xi}{2}\right), \qquad \theta = \mathrm{i}\left(\frac{\pi}{2} - \frac{\pi n\xi}{2}\right).$$

The first is the direct-channel soliton pole in

$$s + B_n \longrightarrow s,$$

and the second is its crossed-channel partner. The product factors have additional double poles at

$$\theta_\ell = \mathrm{i}\left(\frac{\pi}{2} + \frac{\pi\xi}{2}(2\ell - n)\right), \qquad \ell = 1, \dots, n-1,$$

which are internal Coleman-Thun-type singularities of the fused scattering input rather than new one-particle states.

Unitarity follows factor by factor, because replacing θ by $-\theta$ in every displayed numerator gives the corresponding denominator. For crossing, use

$$\sinh(\mathrm{i}\pi - \theta) = \sinh\theta$$

for the first factor, and

$$a_\ell = \frac{\pi\xi}{4}(n - 2\ell) - \frac{\pi}{4}$$

for the product factors. Under $\theta \mapsto \mathrm{i}\pi - \theta$, the ℓ-th numerator becomes

$$\sin^2\left(a_\ell - \frac{\pi}{2} - \frac{\mathrm{i}\theta}{2}\right) = \cos^2\left(a_\ell - \frac{\mathrm{i}\theta}{2}\right).$$

Because

$$a_{n-\ell} = -\frac{\pi\xi}{4}(n - 2\ell) - \frac{\pi}{4}, \qquad \cos\left(a_\ell - \frac{\mathrm{i}\theta}{2}\right) = \sin\left(a_{n-\ell} + \frac{\mathrm{i}\theta}{2} + \pi\right),$$

the crossed numerator of the ℓ-th factor equals the uncrossed numerator of the $(n-\ell)$-th factor after squaring. The denominators are permuted in the same way. Hence the full product, not each individually labelled factor, is crossing invariant.

The first denominator vanishes when

$$\sinh\theta = \mathrm{i}\cos(\pi n\xi/2).$$

Inside the physical strip this gives the two displayed locations. Let

$$\alpha = \frac{\pi n\xi}{2}, \qquad m_n = 2M_s \sin\alpha.$$

At the direct pole $u = \pi/2 + \alpha$, the invariant mass in the soliton-breather channel is

$$M_s^2 + m_n^2 + 2M_s m_n \cos u = M_s^2 + 4M_s^2 \sin^2\alpha - 4M_s^2 \sin^2\alpha = M_s^2.$$

Thus the direct pole has precisely the mass of the external soliton. The second pole is obtained by crossing. Finally, the denominator of the ℓ-th product factor vanishes when

$$\frac{\pi\xi}{4}(n-2\ell) - \frac{\pi}{4} - \frac{\mathrm{i}\theta}{2} \in \pi\mathbb{Z},$$

which gives the displayed θ_ℓ in the physical strip. The square on the sine gives a double pole. These singularities arise because the breather was represented as a fused soliton-antisoliton pair and the external soliton can scatter through either constituent; their invariant masses do not match a new stable particle below threshold.

81.7 Lightest Breather Scattering

The lightest breather B_1 is a neutral scalar particle in the attractive regime $0 < \xi < 1$. Its diagonal two-body amplitude is

$$S_{11}(\theta) = \frac{\sinh\theta + \mathrm{i}\sin(\pi\xi)}{\sinh\theta - \mathrm{i}\sin(\pi\xi)}. \tag{81.7}$$

This formula is a useful test case because each analytic requirement can be checked directly. Unitarity follows by multiplication:

$$S_{11}(\theta)S_{11}(-\theta) = \frac{\sinh\theta + \mathrm{i}s_\xi}{\sinh\theta - \mathrm{i}s_\xi}\frac{-\sinh\theta + \mathrm{i}s_\xi}{-\sinh\theta - \mathrm{i}s_\xi} = 1, \qquad s_\xi = \sin(\pi\xi).$$

Hermitian analyticity follows from

$$S_{11}(\theta)^* = S_{11}(-\theta^*)$$

for the boundary values obtained from real rapidity. Crossing for an self-conjugate scalar requires $S_{11}(\theta) = S_{11}(\mathrm{i}\pi - \theta)$. Since $\sinh(\mathrm{i}\pi - \theta) = \sinh\theta$, the displayed amplitude satisfies this identity.

The denominator vanishes when

$$\sinh\theta = \mathrm{i}\sin(\pi\xi).$$

Inside the physical strip $0 < \operatorname{Im}\theta < \pi$, this equation has two solutions,

$$\theta = \mathrm{i}\pi\xi, \qquad \theta = \mathrm{i}\pi(1-\xi).$$

They are exchanged by crossing. Only the first one is the direct $B_1B_1 \to B_2$ pole when $2\xi < 1$; the second is its crossed-channel image and must not be counted as a separate direct-channel particle. For two incoming particles of mass m_1 and rapidities $\theta/2$ and $-\theta/2$, the squared center-of-mass energy is

$$s(\theta) = \big(p(\theta/2) + p(-\theta/2)\big)^2 = 4m_1^2\cosh^2(\theta/2).$$

At $\theta = \mathrm{i}\pi\xi$, this is

$$s_* = 4m_1^2\cos^2\left(\frac{\pi\xi}{2}\right).$$

Therefore the pole has the mass required for the second breather:

$$m_2 = 2m_1\cos\left(\frac{\pi\xi}{2}\right).$$

Using the soliton-breather mass formula

$$m_n = 2M_s \sin\left(\frac{\pi n \xi}{2}\right)$$

for $n = 1, 2$, the same relation becomes

$$2m_1 \cos\left(\frac{\pi\xi}{2}\right) = 4M_s \sin\left(\frac{\pi\xi}{2}\right)\cos\left(\frac{\pi\xi}{2}\right) = 2M_s \sin(\pi\xi) = m_2.$$

Thus the elementary analytic pole of (81.7) encodes the fusion process

$$B_1 + B_1 \longrightarrow B_2$$

when B_2 lies below the two-soliton threshold as a stable breather. The condition is $2\xi < 1$. At the endpoint $2\xi = 1$ the candidate sits at threshold, and for $2\xi > 1$ this pole does not define an additional stable breather in the Hilbert-space particle spectrum.

81.8 Neutral Breather Blocks and Minimal Fusion

It is useful to isolate the scalar factor

$$F_a(\theta) = \frac{\sinh\theta + \mathrm{i}\sin(\pi a)}{\sinh\theta - \mathrm{i}\sin(\pi a)}, \qquad a \in \mathbb{R}. \tag{81.8}$$

Then $S_{11}(\theta) = F_\xi(\theta)$. The block records both the direct pole and its crossed image in a way that makes the residue sign visible.

Neutral scalar block. For real $0 < a < 1$,

$$F_a(\theta)F_a(-\theta) = 1, \qquad F_a(\mathrm{i}\pi - \theta) = F_a(\theta).$$

The physical strip contains the two simple poles

$$\theta = \mathrm{i}\pi a, \qquad \theta = \mathrm{i}\pi(1-a).$$

The corresponding residues obey

$$-\mathrm{i}\operatorname*{Res}_{\theta=\mathrm{i}\pi a} F_a(\theta) = 2\tan(\pi a), \qquad -\mathrm{i}\operatorname*{Res}_{\theta=\mathrm{i}\pi(1-a)} F_a(\theta) = -2\tan(\pi a).$$

Thus, for $0 < a < 1/2$, the pole at $\theta = \mathrm{i}\pi a$ has the positive direct-channel residue sign, while the pole at $\theta = \mathrm{i}\pi(1-a)$ is its crossed-channel partner.

Unitarity follows by replacing $\sinh\theta$ by $-\sinh\theta$, and crossing follows from $\sinh(\mathrm{i}\pi - \theta) = \sinh\theta$. The denominator vanishes at $\theta = \mathrm{i}\pi a$ and $\theta = \mathrm{i}\pi(1-a)$. At $\theta_0 = \mathrm{i}\pi a$, the numerator is $2\mathrm{i}\sin(\pi a)$, while the derivative of the denominator is $\cosh\theta_0 = \cos(\pi a)$. Therefore

$$\operatorname*{Res}_{\theta=\mathrm{i}\pi a} F_a(\theta) = 2\mathrm{i}\tan(\pi a).$$

At $\theta_1 = \mathrm{i}\pi(1-a)$, the numerator is again $2\mathrm{i}\sin(\pi a)$, but the derivative is $\cos(\pi(1-a)) = -\cos(\pi a)$, giving the second residue formula.

The minimal diagonal scattering amplitude of breathers B_m and B_n, when both labels and every displayed fused label lie in the stable range $k\xi < 1$, is

$$S_{mn}(\theta) = F_{\frac{(m+n)\xi}{2}}(\theta)\, F_{\frac{|m-n|\xi}{2}}(\theta) \prod_{\ell=1}^{\min(m,n)-1} F_{\frac{(|m-n|+2\ell)\xi}{2}}(\theta)^2, \tag{81.9}$$

with the convention $F_0(\theta) = 1$. The formula is the minimal solution of the diagonal bootstrap equations with the breather spectrum fixed above: each F_a supplies one direct pole and its crossing partner, and the double factors are exactly the poles encountered twice when the two fused constituent rapidity strings scatter through each other. More explicitly, start from $S_{11} = F_\xi$. The residue identity of Proposition 72.2 constructs B_{r+s} from the direct pole of B_rB_s; because the neutral breather amplitudes are diagonal scalar functions, the fused scattering with a third breather is the ordinary product of the constituent scalar factors evaluated at the shifted rapidities. Iterating this statement produces the two endpoint factors $(m + n)\xi/2$ and $|m - n|\xi/2$, while every intermediate constituent difference $(|m-n|+2\ell)\xi/2$ occurs once from each side of the fused rapidity string and therefore appears squared. The word minimal means that no further CDD factor, analytic and pole-free in the physical strip, has been multiplied into the result.

Direct fusion pole in the breather amplitudes. Assume $m, n \geq 1$ and $(m + n)\xi < 1$, so that B_{m+n} is a stable breather. The amplitude $S_{mn}(\theta)$ in (81.9) has a direct-channel pole at

$$\theta = \mathrm{i}u_{mn}^{m+n}, \qquad u_{mn}^{m+n} = \frac{\pi(m+n)\xi}{2},$$

and the bound-state kinematics at this pole gives the mass m_{m+n}.

The pole is supplied by the first factor $F_{(m+n)\xi/2}(\theta)$. Since $(m+n)\xi < 1$, its parameter $(m+n)\xi/2$ lies in $(0, 1/2)$, so the neutral-block residue computation above gives the positive direct-channel residue sign at the displayed rapidity. Let

$$\alpha = \frac{\pi\xi}{2}, \qquad m_k = 2M_s \sin(k\alpha).$$

The pole angle is $u = (m + n)\alpha$. The rapidity kinematics gives

$$m_m^2 + m_n^2 + 2m_m m_n \cos u = 4M_s^2 \left[\sin^2(m\alpha) + \sin^2(n\alpha) + 2\sin(m\alpha)\sin(n\alpha)\cos((m+n)\alpha)\right].$$

Using

$$\sin((m+n)\alpha) = \sin(m\alpha)\cos(n\alpha) + \cos(m\alpha)\sin(n\alpha),$$

one obtains the elementary identity

$$\sin^2((m+n)\alpha) = \sin^2(m\alpha) + \sin^2(n\alpha) + 2\sin(m\alpha)\sin(n\alpha)\cos((m+n)\alpha).$$

Thus the bracket in the preceding display is $\sin^2((m+n)\alpha)$, and the invariant mass at the pole is

$$2M_s \sin((m+n)\alpha) = m_{m+n},$$

which proves the claim.

81.9 Massive Thirring Equivalence

We use the Lorentzian mostly-plus metric in this section, set $\epsilon^{01} = +1$, and write

$$j^\mu = \bar{\psi}\gamma^\mu\psi.$$

The massive Thirring model has Dirac field ψ and action

$$S_{\mathrm{MT}} = \int d^2x \left[\mathrm{i}\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi - \frac{g_{\mathrm{T}}}{2}(\bar{\psi}\gamma^\mu\psi)(\bar{\psi}\gamma_\mu\psi)\right].$$

The parameter g_{T} is the coefficient of the displayed renormalized current square in the vector-current point-splitting scheme used below. Changing the current contact term changes this coordinate; the relation to the sine-Gordon coupling is therefore a relation between declared renormalization charts with specified current contact terms and vertex normal-ordering schemes.

81.9.1 Free-Boson Vertex Chart

Let φ be a Euclidean free scalar with canonical action

$$A_{\mathrm{can}}[\varphi] = \frac{1}{2}\int_{\mathbb{R}^2} \mathrm{d}^2x\, \partial_\alpha\varphi\, \partial_\alpha\varphi, \qquad \langle\varphi(x)\varphi(0)\rangle_{\mathrm{can}} = -\frac{1}{2\pi}\log(\mu_{\mathrm{R}}|x|) + C,$$

where μ_{R} is the normal-ordering scale and C is removed by normal ordering. Define

$$V_\alpha(x) =:\exp(\mathrm{i}\alpha\varphi(x)): .$$

In the above canonical scalar chart,

$$V_\alpha(x)V_{-\alpha}(y) = (\mu_{\mathrm{R}}|x-y|)^{-\alpha^2/(2\pi)} :\exp(\mathrm{i}\alpha(\varphi(x)-\varphi(y))): \tag{81.10}$$

for $x \neq y$, with the right side understood as an asymptotic expansion in local derivatives at y. The scaling dimension of V_α is

$$\Delta_\alpha = \frac{\alpha^2}{4\pi}, \qquad h_\alpha = \bar{h}_\alpha = \frac{\alpha^2}{8\pi}. \tag{81.11}$$

Normal ordering removes all self-contractions, so Wick's theorem gives the single cross-contraction factor

$$\exp[(\mathrm{i}\alpha)(-\mathrm{i}\alpha)\langle\varphi(x)\varphi(y)\rangle_{\mathrm{can}}] = (\mu_{\mathrm{R}}|x-y|)^{-\alpha^2/(2\pi)}.$$

The remaining normal-ordered exponential is the displayed operator. A scalar primary of dimension Δ has two-point singularity $|x-y|^{-2\Delta}$, hence $2\Delta_\alpha = \alpha^2/(2\pi)$. Rotational invariance gives $h_\alpha = \bar{h}_\alpha = \Delta_\alpha/2$.

In this canonical chart the sine-Gordon perturbation

$$\int \mathrm{d}^2x\ :\cos(\beta\varphi):$$

has scaling dimension

$$\Delta_{\cos} = \frac{\beta^2}{4\pi}.$$

It is relevant in the ultraviolet Gaussian theory exactly for $\beta^2 < 8\pi$, marginal at $\beta^2 = 8\pi$, and irrelevant for $\beta^2 > 8\pi$.

81.9.2 Current Algebra and the Coleman Coupling Map

Let ϕ_0 denote the canonical scalar that bosonizes the current sector of a free massless Dirac fermion. With the vector-current point-splitting contact term fixed by the Schwinger term, the current map is

$$j^\mu = \frac{1}{\sqrt{\pi}}\epsilon^{\mu\nu}\partial_\nu\phi_0. \tag{81.12}$$

Equivalently, in Euclidean signature the source functional for the free Dirac vector current is the Gaussian scalar functional obtained from

$$A_E[\phi_0, b] = \frac{1}{2}\int \mathrm{d}^2x\,(\partial\phi_0)^2 - \frac{\mathrm{i}}{\sqrt{\pi}}\int \mathrm{d}^2x\, b_\mu\epsilon^{\mu\nu}\partial_\nu\phi_0,$$

up to the local contact term fixed by current conservation. This is the same normalization used in the Schwinger-model derivation in Chapter 52.

Coleman relation from the current-sector Gaussian. In the Euclidean massless Thirring current sector with $K > 0$,

$$K = 1 + \frac{g_{\mathrm{T}}}{\pi},$$

the scalar kinetic term is

$$A^E_{\mathrm{MT},m=0} = \frac{K}{2}\int \mathrm{d}^2x\,(\partial\phi_0)^2.$$

After the canonical rescaling

$$\varphi = \sqrt{K}\,\phi_0,$$

the fermion mass operator is represented by

$$\bar\psi\psi \longleftrightarrow C_m :\cos(\beta\varphi):, \qquad \beta^2 = \frac{4\pi}{K},$$

where C_m is the finite normalization of the renormalized mass operator. Therefore

$$\frac{4\pi}{\beta^2} = 1 + \frac{g_{\mathrm{T}}}{\pi}. \tag{81.13}$$

This relation is a normalization consequence of the current-sector Gaussian, not an independent theorem. For one free Dirac fermion, the current map (81.12) gives in Euclidean signature

$$j_\mu j_\mu = \frac{1}{\pi}\,\partial_\mu\phi_0\,\partial_\mu\phi_0.$$

The Euclidean Thirring weight corresponding to the Lorentzian interaction displayed at the beginning of this section therefore contributes

$$\frac{g_{\mathrm{T}}}{2}\int \mathrm{d}^2x\, j_\mu j_\mu = \frac{g_{\mathrm{T}}}{2\pi}\int \mathrm{d}^2x\,(\partial\phi_0)^2.$$

Adding the free scalar action gives the kinetic coefficient K. The condition $K > 0$ is the positivity condition for the Euclidean Gaussian current functional in this chart.

At $g_{\mathrm{T}} = 0$, chiral bosonization gives

$$\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L \longleftrightarrow C_m :\cos(\sqrt{4\pi}\,\phi_0): .$$

The current-current deformation changes only the kinetic coefficient of the current scalar. Rewriting the same operator in the canonical field $\varphi = \sqrt{K}\,\phi_0$ gives

$$\sqrt{4\pi}\,\phi_0 = \frac{\sqrt{4\pi}}{\sqrt{K}}\,\varphi.$$

Thus the sine-Gordon vertex charge is $\beta = \sqrt{4\pi/K}$, which is equivalent to (81.13).

Combining (81.12) with the same rescaling gives the current dictionary in the sine-Gordon canonical field,

$$j^\mu = \frac{\beta}{2\pi}\epsilon^{\mu\nu}\partial_\nu\varphi. \tag{81.14}$$

The integrated charge is

$$\int dx\, j^0 = \frac{\beta}{2\pi}\int dx\,\partial_x\varphi = Q_{\mathrm{top}},$$

so fermion number equals the sine-Gordon topological charge.

81.9.3 Mandelstam Operator and Exchange Phase

The fermion field is a semi-local soliton vertex operator. In the equal-time canonical chart

$$[\varphi(t,x),\Pi(t,y)] = \mathrm{i}\delta(x-y), \qquad \Pi = \partial_t\varphi,$$

Mandelstam's soliton-creating fields have the normal-ordered form

$$\psi_\pm(t,x) = Z_\psi^{1/2}\,\mathcal{K}_\pm\ :\exp\left[-\frac{2\pi\mathrm{i}}{\beta}\int_{-\infty}^{x}\mathrm{d}y\,\Pi(t,y) \mp \frac{\mathrm{i}\beta}{2}\varphi(t,x)\right]:, \tag{81.15}$$

where Z_ψ is a finite field normalization fixed by the one-particle residue, $\mathcal{K}_\pm$ are Klein factors implementing the two chiral Clifford generators, and the displayed vertex is normal ordered in the same UV scheme as the sine-Gordon perturbation.

Fermi exchange from the Mandelstam line. For

$$I(t,x) = \int_{-\infty}^{x}\mathrm{d}y\,\Pi(t,y), \qquad M_{a,b}(t,x) =:\exp(\mathrm{i}a\varphi(t,x) + \mathrm{i}bI(t,x)):,$$

the equal-time product at $x \neq y$ obeys

$$M_{a,b}(t,x)M_{a,b}(t,y) = \exp\big(\mathrm{i}ab\ \mathrm{sgn}(x-y)\big)M_{a,b}(t,y)M_{a,b}(t,x).$$

The exponential part of $\psi_\pm$ in (81.15) has $ab = \pm\pi$. Hence two identical Mandelstam fields anticommute at separated equal-time points; the Klein factors supply the anticommutation between the two chiral labels.

The canonical commutator gives

$$[\varphi(t,x), I(t,y)] = \mathrm{i}\Theta(y-x), \qquad [I(t,x),\varphi(t,y)] = -\mathrm{i}\Theta(x-y),$$

where Θ is the step function. Thus the exponent

$$A(x) = \mathrm{i}a\varphi(t,x) + \mathrm{i}bI(t,x)$$

satisfies

$$[A(x), A(y)] = \mathrm{i}ab\big(\Theta(x-y) - \Theta(y-x)\big) = \mathrm{i}ab\ \mathrm{sgn}(x-y).$$

This commutator is a scalar, so the Baker-Campbell-Hausdorff identity gives the product formula. In (81.15),

$$b = -\frac{2\pi}{\beta}, \qquad a = \mp\frac{\beta}{2},$$

and therefore $ab = \pm\pi$. Since $\exp(\pm\mathrm{i}\pi\ \mathrm{sgn}(x-y)) = -1$, the separated equal-time exchange is fermionic.

The scaling dimension of the Mandelstam exponential follows from the local field and dual-field two-point functions:

$$\Delta_\psi(\beta) = \frac{1}{4\pi}\left(\frac{\beta^2}{4} + \frac{4\pi^2}{\beta^2}\right) = \frac{\beta^2}{16\pi} + \frac{\pi}{\beta^2}. \tag{81.16}$$

At $\beta^2 = 4\pi$ this gives $\Delta_\psi = 1/2$, the free Dirac value.

81.9.4 Operator Dictionary and Correlation Checks

With the conventions fixed above, the local operator dictionary begins as

$$\begin{aligned}
j^\mu = \bar\psi\gamma^\mu\psi \qquad &\longleftrightarrow \quad \frac{\beta}{2\pi}\epsilon^{\mu\nu}\partial_\nu\varphi,\\
\bar\psi\psi = \bar\psi_L\psi_R + \bar\psi_R\psi_L \qquad &\longleftrightarrow \quad C_m :\!\cos(\beta\varphi)\!:,\\
j_\mu j_\mu \text{ in Euclidean signature} \qquad &\longleftrightarrow \quad \left(\frac{\beta}{2\pi}\right)^2 \partial_\mu\varphi\,\partial_\mu\varphi,\\
:\!\psi_R^\dagger\partial_-^s\psi_R\!: \qquad &\longleftrightarrow \quad \mathcal{P}_{s+1}^{(R)}[\varphi_R].
\end{aligned} \tag{81.17}$$

The last line is fixed by the chiral generating identity

$$:\!\psi_R^\dagger(x+\ell)\psi_R(x-\ell)\!:= \frac{1}{2\pi\mathrm{i}\,\ell}\left[:\!\exp\Big(-\mathrm{i}\sqrt{4\pi}\left(\varphi_R(x+\ell) - \varphi_R(x-\ell)\right)\Big)\!: -1\right], \tag{81.18}$$

expanded in ℓ after subtracting the singular free-fermion contraction. The left side generates chiral bilinear currents; the right side generates the normal-ordered differential polynomials $\mathcal{P}_{s+1}^{(R)}[\varphi_R]$ in $\partial_-\varphi_R, \ldots, \partial_-^{s+1}\varphi_R$.

At the free-fermion point $\beta^2 = 4\pi$, hence $K = 1$ and $g_\mathrm{T} = 0$, the chiral vertex

$$\psi_R(z) = \frac{\mathcal{K}_R}{\sqrt{2\pi a}} :\!\exp(\mathrm{i}\sqrt{4\pi}\,\varphi_R(z))\!:$$

has

$$\langle\psi_R(z)\psi_R^\dagger(0)\rangle = \frac{1}{2\pi z}$$

after the short-distance length a is identified with the chosen normal-ordering scale. This follows from the chiral version of the vertex-OPE exponent in (81.10), whose exponent is $\alpha^2/(4\pi) = 1$ for $\alpha = \sqrt{4\pi}$. The full mass operator $\bar{\psi}\psi$ has dimension 1, matching $:\cos(\sqrt{4\pi}\,\varphi):$. At the marginal sine-Gordon endpoint $\beta^2 = 8\pi$, the mass operator has dimension 2, while (81.13) gives

$$g_{\mathrm{T}} = -\frac{\pi}{2}.$$

Thus the cosine perturbation is relevant precisely for

$$\beta^2 < 8\pi \quad \Longleftrightarrow \quad 1 + \frac{g_{\mathrm{T}}}{\pi} > \frac{1}{2} \quad \Longleftrightarrow \quad g_{\mathrm{T}} > -\frac{\pi}{2}.$$

The current-current coordinate g_{T} itself is an exactly marginal current-algebra coordinate in the massless Abelian theory; it changes the compact-boson radius and thereby changes the dimension of the mass operator.

81.9.5 Chiral, Nonabelian, and Defect Extensions

The Abelian discussion has a chiral refinement. A right-moving complex fermion current

$$J_R =:\psi_R^\dagger \psi_R:$$

is represented by a chiral scalar φ_R through

$$J_R = \frac{1}{\sqrt{\pi}}\partial_- \varphi_R, \qquad \psi_R \propto \mathcal{K}_R :\exp(\mathrm{i}\sqrt{4\pi}\,\varphi_R):.$$

The chiral current OPE fixes the coefficient exactly as in the nonchiral current-sector proof: both sides have the same Schwinger term.

For N massless complex fermions, decompose the currents into the total Abelian current and the traceless $\mathfrak{su}(N)$ currents

$$J_R^a =:\psi_R^\dagger t^a \psi_R:.$$

The nonabelian current OPE is the affine $\widehat{\mathfrak{su}}(N)_1$ current algebra in the generator normalization used in the volume containing nonabelian gauge theory. The nonabelian bosonic presentation is the $SU(N)_1$ WZW model, with the Sugawara stress tensor and primary field normalizations reviewed in Section 66.8. The Abelian sine-Gordon equivalence is the $U(1)$ factor of this broader bosonization mechanism; interactions that preserve only a subgroup require a separate operator dictionary in the corresponding current algebra.

There is also a defect formulation at compact-boson radii where the topological defect category is defined. A duality defect is an object D whose fusion with its orientation reverse decomposes into a sum of symmetry defects rather than the identity alone. At rational compactification radii and after specifying spin structure, charge lattice, and boundary-condition sector, the bosonization map can be represented by such defects between the fermionic and bosonic descriptions. For the generic massive sine-Gordon and massive Thirring continuum theories, the primary statement used in this chapter remains the operator and scattering-data equivalence fixed by (81.13)–(81.17). The categorical defect formulation supplies an additional finite-symmetry organization only after those extra compactification data have been declared.

At the free-fermion point $\beta^2 = 4\pi$, hence $g_{\mathrm{T}} = 0$, the soliton and antisoliton form the massive Dirac particle and antiparticle. Away from this point the equivalence is a statement about renormalized local operator algebras and scattering data in the declared convention.

81.10 Affine Toda Data

Let $\mathfrak{g}$ be a simple Lie algebra of rank r, with simple roots α_i and highest root

$$\theta = \sum_{i=1}^{r} n_i \alpha_i.$$

Set $\alpha_0 = -\theta$ and $n_0 = 1$. The affine Toda action for a real Cartan-valued field $\phi \in \mathfrak{h}_{\mathbb{R}}$ is

$$S_{\mathrm{AT}} = \int d^2x \left[-\frac{1}{2}(\partial_\mu \phi, \partial^\mu \phi) - \frac{m^2}{\beta^2} \sum_{i=0}^{r} n_i \left(e^{\beta(\alpha_i, \phi)} - 1 \right) \right].$$

The classical equations admit a Lax representation with spectral parameter. Consequently the classical theory has commuting local conserved charges. In the quantum theory, the factorized S-matrix is organized by the root system, particle species are labelled by nodes of the Dynkin diagram in the simply-laced cases, and the pole structure encodes fusing rules compatible with the affine root data.

81.11 Mass Matrix In Affine Toda Theory

Expanding the potential at $\phi = 0$ gives

$$V(\phi) = \frac{m^2}{2} \sum_{i=0}^{r} n_i (\alpha_i, \phi)^2 + O(\phi^3).$$

Thus the classical mass matrix is

$$M_{\mathrm{cl}}^2 = m^2 \sum_{i=0}^{r} n_i \, \alpha_i \otimes \alpha_i.$$

The following explicit $A_r^{(1)}$ calculation shows what is meant by the Coxeter mass data at the classical level, without turning the linear-algebra diagonalization itself into a theorem about the quantum model.

$A_r^{(1)}$ mass matrix. Let $\mathfrak{g} = A_r$, realize the roots in

$$H = \left\{ (x_1, \dots, x_{r+1}) \in \mathbb{R}^{r+1} \;\middle|\; \sum_{k=1}^{r+1} x_k = 0 \right\}$$

with orthonormal ambient basis e_k, and take

$$\alpha_i = e_i - e_{i+1}, \qquad i = 1, \dots, r, \qquad \alpha_0 = e_{r+1} - e_1.$$

All affine Dynkin labels are $n_i = 1$, so the quadratic affine Toda mass operator is

$$M_{\mathrm{cl}}^2 = m^2 \sum_{i=0}^{r} \alpha_i \otimes \alpha_i.$$

To diagonalize it, take $v = (v_1, \dots, v_{r+1}) \in H$. Then

$$\frac{1}{m^2} M_{\mathrm{cl}}^2 v = \sum_{i=1}^{r} (v_i - v_{i+1})(e_i - e_{i+1}) + (v_{r+1} - v_1)(e_{r+1} - e_1),$$

so the k-th component is

$$2v_k - v_{k-1} - v_{k+1},$$

with indices read cyclically modulo $r+1$. Thus M_{cl}^2/m^2 is the graph Laplacian of the cycle with $r+1$ vertices, restricted to the hyperplane H. The Fourier modes

$$v_k^{(a)} = \exp\left(\frac{2\pi \mathrm{i} a k}{r+1}\right), \qquad a = 0, 1, \dots, r,$$

diagonalize this cycle Laplacian. The $a = 0$ mode is the constant vector and does not lie in H. For $a = 1, \dots, r$,

$$2 - e^{-2\pi \mathrm{i} a/(r+1)} - e^{2\pi \mathrm{i} a/(r+1)} = 2 - 2\cos\left(\frac{2\pi a}{r+1}\right) = 4\sin^2\left(\frac{\pi a}{r+1}\right),$$

and therefore the nonzero classical mass eigenvalues are

$$M_a^2 = 4m^2 \sin^2\left(\frac{\pi a}{r+1}\right), \qquad a = 1, \dots, r. \tag{81.19}$$

Real sine and cosine eigenvectors are obtained by taking real and imaginary parts of these Fourier modes. Equivalently, with Coxeter number $h = r+1$, the classical mass ratios are

$$M_a : M_1 = \sin\left(\frac{\pi a}{h}\right) : \sin\left(\frac{\pi}{h}\right), \qquad a = 1, \dots, r.$$

The same numbers also make the Coxeter/Perron–Frobenius presentation visible. Let I be the adjacency matrix of the finite A_r Dynkin chain, so $(I\mathbf{M})_a = M_{a-1} + M_{a+1}$ with boundary values $M_0 = M_{r+1} = 0$. For

$$M_a = 2m \sin\left(\frac{\pi a}{r+1}\right)$$

the sine addition formula gives

$$M_{a-1} + M_{a+1} = 2\cos\left(\frac{\pi}{r+1}\right) M_a.$$

Hence $\mathbf{M} = (M_1, \dots, M_r)$ is the positive Perron–Frobenius eigenvector of I, with eigenvalue

$$2\cos\left(\frac{\pi}{h}\right), \qquad h = r+1.$$

This calculation is the finite A_r algebra behind the Coxeter mass language used in the simply-laced affine Toda bootstrap. For D and E types, the analogous mass vector is specified by the positive Perron–Frobenius eigenvector of the finite Dynkin adjacency matrix with eigenvalue $2\cos(\pi/h)$. The first non-A check is small enough to write out. For D_4, use the node order in which node 2 is the central trivalent node and nodes $1, 3, 4$ are the leaves. The Coxeter number is $h = 6$, so the Perron–Frobenius eigenvalue is

$$2\cos\left(\frac{\pi}{6}\right) = \sqrt{3}.$$

Writing the mass vector as

$$\mathbf{M} = (x, y, x, x)$$

the adjacency eigenvalue equations are

$$y = \sqrt{3}\,x, \qquad 3x = \sqrt{3}\,y.$$

The two equations are compatible and, after the normalization $x = 1$, give

$$\mathbf{M}_{D_4} = (1, \sqrt{3}, 1, 1).$$

Equivalently, the finite Cartan matrix $C = 2\mathbf{1} - I$ sends this vector to $(2-\sqrt{3})\mathbf{M}_{D_4}$. This finite D_4 cell is the algebraic content behind the Coxeter-mass phrase: the trivalent node is heavier by the factor $\sqrt{3}$, while the three outer particles are degenerate. Matching that algebraic mass datum to an exact quantum S-matrix and to a continuum local QFT is part of the model comparison problem stated below; it is not supplied by the Hessian calculation alone.

81.12 Model-Comparison Summary

The exact-solution claim for each model has three independent tests:

UV test	operator dimensions and perturbing-coordinate normalization,
IR test	particle masses, charges, and factorized S-matrix poles,
finite-size test	TBA ground-state energy and local operator finite-volume data.

Agreement of one row does not determine the other rows. The model becomes a QFT example for this monograph only after all rows are expressed in the same regularization and source convention.

Open Problem 81.3 (Constructive integrable model comparison). Construct sine-Gordon and affine Toda theories as continuum local QFTs with operators, soliton sectors, factorized scattering, TBA finite-size energies, and form factors derived from one regulated framework rather than matched only through separate exact-solution data.

Chapter 82

O(N), Gross–Neveu, And Sigma-Model Families

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter introduces asymptotically free two-dimensional model families whose exact scattering data carry nonabelian internal symmetry. The aim is to make explicit which pieces of the exact description are consequences of symmetry, analyticity, and factorization, and which pieces require a nonperturbative construction of the underlying local QFT.

82.1 Internal-Symmetry Scattering Data

Let G be a compact Lie group and let V be a finite-dimensional unitary representation. A massive particle multiplet with rapidity θ has one-particle space

$$\mathcal{H}_1 = L^2(\mathbb{R}, \mathrm{d}\theta) \otimes V.$$

A factorized two-particle scattering operator is a meromorphic family

$$S_{VV}(\theta) : V \otimes V \longrightarrow V \otimes V,$$

where $\theta = \theta_1 - \theta_2$. G-invariance means

$$S_{VV}(\theta)\, (\rho(g) \otimes \rho(g)) = (\rho(g) \otimes \rho(g))\, S_{VV}(\theta)$$

for every $g \in G$. If

$$V \otimes V \simeq \bigoplus_\lambda m_\lambda V_\lambda,$$

then Schur decomposition gives

$$S_{VV}(\theta) = \bigoplus_\lambda S_\lambda(\theta) \otimes I_{V_\lambda}$$

whenever the multiplicities m_λ are one. With multiplicities, the coefficient $S_\lambda(\theta)$ is an endomorphism of the multiplicity space. Thus symmetry reduces the scattering problem to scalar or finite matrix meromorphic functions, but unitarity, crossing, pole residues, and the Yang–Baxter equation still constrain them.

82.2 O(N)-Invariant Two-Particle Form

For the vector representation $V = \mathbb{R}^N$ of $O(N)$, an invariant two-particle amplitude has the index form

$$S_{ij}^{kl}(\theta) = \sigma_1(\theta)\delta_{ij}\delta^{kl} + \sigma_2(\theta)\delta_i^k\delta_j^l + \sigma_3(\theta)\delta_i^l\delta_j^k.$$

Equivalently, decompose

$$V \otimes V = \mathbf{1} \oplus \mathrm{Sym}_0^2(V) \oplus \wedge^2 V.$$

Let P_0, P_S, P_A be the corresponding projectors. Then

$$S(\theta) = S_0(\theta)P_0 + S_S(\theta)P_S + S_A(\theta)P_A.$$

The two descriptions are related by applying the projectors to the index formula. For example,

$$P_{0\,ij}^{\,kl} = \frac{1}{N}\delta_{ij}\delta^{kl}, \qquad P_{A\,ij}^{\,kl} = \frac{1}{2}(\delta_i^k\delta_j^l - \delta_i^l\delta_j^k).$$

The amplitude is unitary on the real rapidity axis when

$$S_R(\theta)S_R(-\theta) = 1, \qquad R \in \{0, S, A\},$$

after diagonalization in these channels. Crossing identifies the particle with its conjugate and relates $S_{ij}^{kl}(\theta)$ to the amplitude at $\mathrm{i}\pi - \theta$ with one external index crossed.

82.3 Exact O(N) Sigma-Model Vector S-Matrix

For $N > 2$, set

$$\Delta = \frac{1}{N-2}, \qquad z = -\frac{\mathrm{i}\theta}{2\pi}.$$

The minimal $O(N)$-invariant vector S-matrix of the massive nonlinear sigma-model family is

$$S_{ij}^{kl}(\theta) = \sigma_1(\theta)\delta_{ij}\delta^{kl} + \sigma_2(\theta)\delta_i^k\delta_j^l + \sigma_3(\theta)\delta_i^l\delta_j^k, \tag{82.1}$$

with

$$\sigma_2(\theta) = \frac{\Gamma(\Delta + z)\Gamma(\frac{1}{2} + z)\Gamma(\frac{1}{2} + \Delta - z)\Gamma(1 - z)}{\Gamma(\frac{1}{2} + \Delta + z)\Gamma(z)\Gamma(1 + \Delta - z)\Gamma(\frac{1}{2} - z)}, \tag{82.2}$$

and

$$\sigma_1(\theta) = -\frac{2\pi \mathrm{i}}{N-2}\frac{\sigma_2(\theta)}{\mathrm{i}\pi-\theta}, \qquad \sigma_3(\theta) = \sigma_1(\mathrm{i}\pi-\theta) = -\frac{2\pi \mathrm{i}}{N-2}\frac{\sigma_2(\theta)}{\theta}. \tag{82.3}$$

The word minimal means that no CDD factor, unitary and crossing-symmetric and analytic without zeros or poles in the physical strip, has been multiplied into the displayed solution. The formula is a scattering input. Identifying it with a continuum local sigma-model QFT additionally requires the nonperturbative construction and matching problem stated at the end of the chapter.

The tensor in (82.1) is written in the Zamolodchikov–Faddeev component convention: it obeys

$$S_{12}(\theta)S_{13}(\theta+\phi)S_{23}(\phi) = S_{23}(\phi)S_{13}(\theta+\phi)S_{12}(\theta)$$

as a meromorphic identity on $V^{\otimes 3}$. The common scalar factor $\sigma_2(\theta)\sigma_2(\theta+\phi)\sigma_2(\phi)$ cancels from this identity, leaving the rational $O(N)$-invariant tensor

$$I - \frac{\mathrm{i}a}{\theta}P - \frac{\mathrm{i}a}{\mathrm{i}\pi-\theta}K, \qquad a = \frac{2\pi}{N-2},$$

where $P_{ij}^{kl} = \delta_i^l\delta_j^k$ and $K_{ij}^{kl} = \delta_{ij}\delta^{kl}$. Direct expansion on the basis $e_i \otimes e_j \otimes e_k$ reduces the component equality to rational identities in θ, ϕ and the contraction relation $K^2 = NK$; the accompanying calculation check enumerates finite-dimensional instances of this tensor identity together with the scalar unitarity and crossing checks.

Channel unitarity and crossing checks. Let

$$S_0 = N\sigma_1 + \sigma_2 + \sigma_3, \qquad S_S = \sigma_2 + \sigma_3, \qquad S_A = \sigma_2 - \sigma_3$$

be the singlet, symmetric-traceless, and antisymmetric channel eigenvalues of (82.1). For real rapidity $\theta \neq 0$,

$$S_R(\theta)S_R(-\theta) = 1, \qquad R \in \{0, S, A\}.$$

The coefficient functions satisfy the crossing relations

$$\sigma_2(\theta) = \sigma_2(\mathrm{i}\pi-\theta), \qquad \sigma_1(\theta) = \sigma_3(\mathrm{i}\pi-\theta), \qquad \sigma_3(\theta) = \sigma_1(\mathrm{i}\pi-\theta).$$

The role of this paragraph is the normalization check that the displayed meromorphic tensor has the required unitarity and crossing properties in the three $O(N)$ channels.

The projector decomposition gives the displayed channel eigenvalues because

$$Q_{ij}^{kl} = \delta_{ij}\delta^{kl}, \qquad I_{ij}^{kl} = \delta_i^k\delta_j^l, \qquad F_{ij}^{kl} = \delta_i^l\delta_j^k$$

act on $V \otimes V$ as $Q = NP_0$, $I = P_0 + P_S + P_A$, and $F = P_0 + P_S - P_A$. Thus

$$\sigma_1 Q + \sigma_2 I + \sigma_3 F = S_0P_0 + S_SP_S + S_AP_A.$$

For real θ, the gamma-function expression obeys

$$\sigma_2(\theta)\sigma_2(-\theta) = \frac{\theta^2}{\theta^2 + \left(\frac{2\pi}{N-2}\right)^2}. \tag{82.4}$$

This follows by multiplying (82.2) at z and $-z$ and cancelling adjacent gamma factors with $\Gamma(1+z) = z\Gamma(z)$ and $\Gamma(1-z) = (-z)\Gamma(-z)$. Set

$$a = \frac{2\pi}{N-2}.$$

The symmetric-traceless and antisymmetric rational factors are

$$\frac{S_S(\theta)}{\sigma_2(\theta)} = 1 - \frac{\mathrm{i}a}{\theta}, \qquad \frac{S_A(\theta)}{\sigma_2(\theta)} = 1 + \frac{\mathrm{i}a}{\theta}.$$

Multiplying these factors at θ and $-\theta$ gives

$$1 + \frac{a^2}{\theta^2},$$

which cancels the denominator in (82.4). For the singlet channel,

$$\frac{S_0(\theta)}{\sigma_2(\theta)} = 1 - \frac{\mathrm{i}a}{\theta} - \frac{\mathrm{i}Na}{\mathrm{i}\pi - \theta}.$$

The elementary identity

$$\left(1 - \frac{\mathrm{i}a}{\theta} - \frac{\mathrm{i}Na}{\mathrm{i}\pi - \theta}\right)\left(1 + \frac{\mathrm{i}a}{\theta} - \frac{\mathrm{i}Na}{\mathrm{i}\pi + \theta}\right) = 1 + \frac{a^2}{\theta^2}$$

again cancels (82.4). This gives channel unitarity.

For crossing of σ_2, set

$$-\frac{\mathrm{i}(\mathrm{i}\pi - \theta)}{2\pi} = \frac{1}{2} - z,$$

in the gamma-function expression (82.2). This substitution sends each numerator gamma factor at z to the corresponding numerator factor at $\frac{1}{2} - z$, and likewise for the denominators, so the ratio is unchanged in precisely the displayed way. The relations between σ_1 and σ_3 then follow from their definitions in (82.3).

82.4 Sigma-Model Ultraviolet Datum

The Euclidean $O(N)$ nonlinear sigma model is defined with field

$$n : \Sigma \longrightarrow S^{N-1} \subset \mathbb{R}^N, \qquad n \cdot n = 1,$$

and action

$$S_\sigma[n] = \frac{1}{2g^2} \int_\Sigma \mathrm{d}^2x\, \partial_\mu n \cdot \partial_\mu n.$$

The parameter g is the coefficient of the kinetic term in the displayed regularized path integral. Perturbation theory around a local coordinate chart on S^{N-1} gives the one-loop running

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = -\frac{N-2}{2\pi} g^3 + O(g^5).$$

The sign follows from the Ricci tensor of the round target:

$$R_{ab} = (N-2)h_{ab}.$$

The corresponding sigma-model beta tensor has leading term

$$\mu\frac{\mathrm{d}h_{ab}}{\mathrm{d}\mu} = \frac{1}{2\pi}R_{ab} + O(R^2),$$

and $h_{ab} = g^{-2}h_{ab}^{\mathrm{round}}$ yields the displayed scalar beta function. A nonperturbative massive QFT construction must then provide the Hilbert space, local fields or local algebras, and scattering states whose two-particle data match the factorized $O(N)$-invariant amplitude.

82.5 Large-N Sigma-Model Gap Equation

The large-N saddle gives a direct derivation of dimensional transmutation. Write the constraint with a Lagrange multiplier M^2 and integrate the N components of n at a translation-invariant saddle. With Euclidean momentum cutoff Λ, and with the 't Hooft coupling

$$\lambda = Ng^2$$

held fixed, the saddle equation is

$$\frac{1}{\lambda} = \int^{\Lambda} \frac{\mathrm{d}^2p}{(2\pi)^2}\frac{1}{p^2+M^2} = \frac{1}{4\pi}\log\left(\frac{\Lambda^2+M^2}{M^2}\right).$$

Solving gives

$$M = \frac{\Lambda}{\sqrt{\exp(4\pi/\lambda)-1}} = \Lambda\exp\left(-\frac{2\pi}{\lambda}\right)\left(1+O(e^{-4\pi/\lambda})\right)$$

in the weak-coupling scaling regime. If the physical mass M is held fixed while the subtraction scale μ is changed, the running coupling

$$\lambda(\mu) = \frac{2\pi}{\log(\mu/M)}$$

satisfies

$$\mu\frac{\mathrm{d}\lambda}{\mathrm{d}\mu} = -\frac{\lambda^2}{2\pi} + O(1/N).$$

This is the large-N form of asymptotic freedom. The finite-N one-loop coefficient replaces N by $N-2$ in the beta function for the unscaled coupling, as in the Ricci-tensor calculation above.

82.6 Projective Sigma Models

The next basic asymptotically free family is the projective model with target $\mathbb{CP}^{N-1}$. A convenient ultraviolet coordinate uses a complex N-component field z, a nondynamical $U(1)$ connection A_μ, and the local redundancy

$$z(x) \mapsto \mathrm{e}^{\mathrm{i}\alpha(x)}z(x), \qquad A_\mu(x) \mapsto A_\mu(x) + \partial_\mu\alpha(x).$$

The constraint and covariant derivative are

$$z^\dagger z = 1, \qquad D_\mu z = (\partial_\mu - \mathrm{i}A_\mu)z,$$

and the Euclidean action is

$$S_{\mathbb{CP}^{N-1}}[z, A] = \frac{1}{g^2}\int_\Sigma \mathrm{d}^2x\,(D_\mu z)^\dagger D_\mu z + \mathrm{i}\frac{\theta}{2\pi}\int_\Sigma F_A. \tag{82.5}$$

The $U(1)$ field is part of the quotient description, not an additional microscopic photon. On a closed Euclidean surface, for a smooth line bundle,

$$\frac{1}{2\pi}\int_\Sigma F_A \in \mathbb{Z},$$

so the coefficient θ is 2π-periodic in the continuum topological sectors. A lattice or other ultraviolet definition must specify how this integer is represented before the θ-dependent path integral is a mathematical object.

The projector

$$P = zz^\dagger, \qquad P^2 = P, \qquad \operatorname{tr} P = 1,$$

is gauge invariant and gives an intrinsic coordinate on $\mathbb{CP}^{N-1}$. The $U(1)$ connection in (82.5) can be eliminated at the classical level. Varying with respect to A_μ, while imposing $z^\dagger z = 1$, gives

$$A_\mu = -\mathrm{i}z^\dagger\partial_\mu z, \qquad D_\mu z = (1 - P)\partial_\mu z. \tag{82.6}$$

This formula is invariant under $z \mapsto \mathrm{e}^{\mathrm{i}\alpha}z$ together with $A \mapsto A + \mathrm{d}\alpha$, as required by the quotient description.

Projector form of the kinetic term and charge. For a smooth representative z of a map into $\mathbb{CP}^{N-1}$, with $A = -\mathrm{i}z^\dagger\mathrm{d}z$ and $P = zz^\dagger$,

$$\operatorname{tr}(\partial_\mu P\,\partial_\mu P) = 2(D_\mu z)^\dagger D_\mu z, \tag{82.7}$$
$$F_A = -\mathrm{i}\,\operatorname{tr}(P\,\mathrm{d}P \wedge \mathrm{d}P). \tag{82.8}$$

Consequently the action in (82.5) is equivalently

$$S_{\mathbb{CP}^{N-1}}[P] = \frac{1}{2g^2}\int_\Sigma \mathrm{d}^2x\,\operatorname{tr}(\partial_\mu P\,\partial_\mu P) + \mathrm{i}\theta Q[P], \qquad Q[P] = -\frac{\mathrm{i}}{2\pi}\int_\Sigma \operatorname{tr}(P\,\mathrm{d}P \wedge \mathrm{d}P).$$

The stationarity equation for A_μ follows from

$$(D_\mu z)^\dagger D_\mu z = \partial_\mu z^\dagger\partial_\mu z + A_\mu^2 + 2\mathrm{i}A_\mu z^\dagger\partial_\mu z,$$

where $\partial_\mu z^\dagger z = -z^\dagger\partial_\mu z$. Hence $A_\mu = -\mathrm{i}z^\dagger\partial_\mu z$, and then

$$D_\mu z = \partial_\mu z - z(z^\dagger\partial_\mu z) = (1 - P)\partial_\mu z.$$

Differentiating $P = zz^\dagger$ and using $z^\dagger D_\mu z = 0$ gives

$$\partial_\mu P = (D_\mu z)z^\dagger + z(D_\mu z)^\dagger.$$

Squaring and tracing gives
$$\operatorname{tr}(\partial_\mu P\,\partial_\mu P) = 2(D_\mu z)^\dagger D_\mu z,$$
because $z^\dagger D_\mu z = 0$ and $z^\dagger z = 1$. Finally
$$F_A = \mathrm{d}A = -\mathrm{i}\,\mathrm{d}z^\dagger \wedge \mathrm{d}z.$$
The same differentiated-projector identity gives
$$\operatorname{tr}(P\,\mathrm{d}P \wedge \mathrm{d}P) = \mathrm{d}z^\dagger \wedge \mathrm{d}z,$$
which proves (82.8). If the map is a section of a nontrivial line bundle, the computation is local and the two-form patches globally; its integral is the first Chern number of that line bundle.

On an oriented Euclidean surface with local coordinates x^1, x^2, write $F_{12}\mathrm{d}x^1 \wedge \mathrm{d}x^2 = F_A$. The algebraic identity
$$\left|D_1 z \pm \mathrm{i}D_2 z\right|^2 = |D_1 z|^2 + |D_2 z|^2 \mp F_{12}$$
gives the instanton bound, in the normalization of (82.5),
$$S_{\mathbb{CP}^{N-1},\theta=0} \geq \frac{2\pi}{g^2}|Q[P]|. \tag{82.9}$$
For $Q > 0$, equality is attained by $D_1 z + \mathrm{i}D_2 z = 0$, equivalently by a holomorphic map to projective space in the induced complex structure. For $Q < 0$, equality is attained by the anti-holomorphic equation. This is a finite-action Euclidean saddle statement in the continuum sigma-model functional. A real-time local QFT construction additionally supplies the Hilbert space, local observables, reflection-positive Euclidean data, and analytic continuation needed to interpret these sectors dynamically.

The one-loop beta tensor is again the Ricci tensor of the target; with the Fubini–Study metric normalized so that
$$\mathrm{Ric} = N\,h_{\mathrm{FS}},$$
the model is asymptotically free. If $t = g^2$ denotes the scalar coupling in (82.5), the leading running has the form
$$\mu\frac{\mathrm{d}t}{\mathrm{d}\mu} = -\frac{N}{2\pi}t^2 + \text{higher powers of } t, \tag{82.10}$$
in this normalization.

At large N with $\lambda = Ng^2$ fixed, introduce a Lagrange multiplier enforcing $z^\dagger z = 1$. A translation-invariant saddle gives
$$\frac{1}{\lambda} = \int^\Lambda \frac{\mathrm{d}^2 p}{(2\pi)^2}\frac{1}{p^2+m^2} = \frac{1}{4\pi}\log\left(\frac{\Lambda^2+m^2}{m^2}\right), \tag{82.11}$$
up to the convention-dependent factor already fixed in the action. Thus the projective family has the same dimensional-transmutation structure as the round $O(N)$ model. What differs is the global topology, the $U(1)$ quotient, the θ-angle, and the particle multiplet structure.

Proposition 82.1 (Large-N induced gauge-field kernel)**.** *Work at $\theta = 0$, in the translation-invariant large-N saddle of* (82.11)*, and expand the effective action around the constant mass*

$m > 0$ and $A_\mu = 0$. The one-loop quadratic term for the auxiliary $U(1)$ field has the transverse form

$$\Gamma^{(2)}[A] = \frac{N}{2}\int \frac{\mathrm{d}^2p}{(2\pi)^2}\, A_\mu(-p)\left[(p^2\delta_{\mu\nu} - p_\mu p_\nu)\,\Pi(p^2)\right] A_\nu(p), \tag{82.12}$$

where

$$\Pi(p^2) = \frac{1}{4\pi}\int_0^1 \mathrm{d}x\, \frac{(1-2x)^2}{m^2 + x(1-x)p^2}. \tag{82.13}$$

For $|p| \ll m$,

$$\Pi(p^2) = \frac{1}{12\pi m^2} - \frac{p^2}{120\pi m^4} + O(p^4/m^6). \tag{82.14}$$

Equivalently, the leading local term in the low-momentum effective action is

$$\Gamma_{\mathrm{loc}}[A] = \frac{N}{48\pi m^2}\int \mathrm{d}^2x\, F_{\mu\nu}F_{\mu\nu} = \frac{N}{24\pi m^2}\int \mathrm{d}^2x\, F_{12}^2 + O(\partial^4/m^4). \tag{82.15}$$

Proof. At fixed A, the z-integral is a Gaussian over N complex fields. The quadratic term in the logarithm of the covariant operator $-D_\mu D_\mu + m^2$, including the seagull term from $A_\mu A_\mu z^\dagger z$, gives

$$\Pi_{\mu\nu}(p) = N\int \frac{\mathrm{d}^2k}{(2\pi)^2}\left[\frac{(2k+p)_\mu(2k+p)_\nu}{(k^2+m^2)((k+p)^2+m^2)} - \frac{2\delta_{\mu\nu}}{k^2+m^2}\right],$$

after the same ultraviolet regulator and subtraction convention are used in both terms. Introduce a Feynman parameter, shift $q = k + xp$, and set

$$M_x^2 = m^2 + x(1-x)p^2.$$

The odd terms in q vanish. Rotational invariance gives

$$\int \frac{\mathrm{d}^2q}{(2\pi)^2}\frac{q_\mu q_\nu}{(q^2+M_x^2)^2} = \frac{\delta_{\mu\nu}}{2}\int \frac{\mathrm{d}^2q}{(2\pi)^2}\frac{q^2}{(q^2+M_x^2)^2}.$$

The logarithmic divergence in this expression cancels exactly against the seagull term after writing the latter with the same Feynman parameter. The remaining finite tensor is transverse:

$$\Pi_{\mu\nu}(p) = N(p^2\delta_{\mu\nu} - p_\mu p_\nu)\frac{1}{4\pi}\int_0^1 \mathrm{d}x\, \frac{(1-2x)^2}{m^2 + x(1-x)p^2}.$$

This is (82.12) and (82.13). Expanding the denominator gives

$$\int_0^1 (1-2x)^2 \mathrm{d}x = \frac{1}{3}, \qquad \int_0^1 (1-2x)^2 x(1-x)\mathrm{d}x = \frac{1}{30},$$

which proves (82.14). Finally, in two Euclidean dimensions,

$$\frac{1}{2}A_\mu(p)(p^2\delta_{\mu\nu} - p_\mu p_\nu)A_\nu(-p) = \frac{1}{4}F_{\mu\nu}(p)F_{\mu\nu}(-p),$$

and $F_{\mu\nu}F_{\mu\nu} = 2F_{12}^2$, giving (82.15). □

Linear potential in the induced two-dimensional gauge sector. Keep only the leading local Maxwell term

$$\frac{1}{4e_{\text{eff}}^2}\int F_{\mu\nu}F_{\mu\nu}, \qquad e_{\text{eff}}^2 = \frac{12\pi m^2}{N}, \tag{82.16}$$

obtained from (82.15). In Lorentzian signature on the line, two static external $U(1)$ sources of charges $+Q$ and $-Q$, separated by distance R, have electric-field energy

$$V(R) = \frac{e_{\text{eff}}^2 Q^2}{2}R = \frac{6\pi m^2 Q^2}{N}R. \tag{82.17}$$

Choose $A_1 = 0$. Gauss' law for the Hamiltonian associated with (82.16) is

$$\partial_x E(x) = e_{\text{eff}}^2 Q\big(\delta(x) - \delta(x-R)\big).$$

With vanishing field outside the interval between the charges, the solution is

$$E(x) = e_{\text{eff}}^2 Q, \qquad 0 < x < R,$$

and $E(x) = 0$ outside. The Hamiltonian density is $\frac{1}{2e_{\text{eff}}^2}E^2$, hence

$$V(R) = \int_0^R \mathrm{d}x\, \frac{(e_{\text{eff}}^2 Q)^2}{2e_{\text{eff}}^2} = \frac{e_{\text{eff}}^2 Q^2}{2}R.$$

Substitution of (82.16) gives (82.17).

Controlled Approximation 82.2 (Meaning of the induced-gauge large-N result)**.** Propositions 82.1 and the linear-potential computation above are statements about the large-N, low-momentum expansion around the massive saddle of the regulated projective path integral. They explain why the quotient $U(1)$ coordinate cannot be discarded after quantization, and why the charged homogeneous variables z are not gauge-invariant particle states of the projective model. They do not prove factorized scattering for the full $\mathbb{CP}^{N-1}$ local QFT, and they do not decide the $\theta = \pi$ massless endpoint.

82.6.1 The Projective Line and the Exact-Scattering Status

The case $N = 2$ is a useful normalization check because $\mathbb{CP}^1$ is the two-sphere. Let σ_a, $a = 1, 2, 3$, be Pauli matrices with

$$\operatorname{tr}(\sigma_a\sigma_b) = 2\delta_{ab}, \qquad \operatorname{tr}(\sigma_a\sigma_b\sigma_c) = 2\mathrm{i}\epsilon_{abc}.$$

For a unit vector $n : \Sigma \to S^2 \subset \mathbb{R}^3$, set

$$P_n = \frac{1}{2}(I + n_a\sigma_a). \tag{82.18}$$

$\mathbb{CP}^1$ **as the** $O(3)$ **sigma model.** The map $n \mapsto P_n$ identifies S^2 with the rank-one Hermitian projectors in $\mathbb{C}^2$ and obeys

$$\operatorname{tr}(\partial_\mu P_n\, \partial_\mu P_n) = \frac{1}{2}\,\partial_\mu n \cdot \partial_\mu n, \tag{82.19}$$

$$-\mathrm{i}\,\operatorname{tr}(P_n\,\mathrm{d}P_n \wedge \mathrm{d}P_n) = \frac{1}{2}\,\epsilon_{abc}n_a\,\mathrm{d}n_b \wedge \mathrm{d}n_c. \tag{82.20}$$

Consequently the $\mathbb{CP}^1$ action

$$\frac{1}{2g^2}\int_\Sigma \operatorname{tr}(\partial_\mu P\,\partial_\mu P) + \mathrm{i}\theta Q[P]$$

is the $O(3)$ action

$$\frac{1}{2g^2_{O(3)}}\int_\Sigma \partial_\mu n\cdot\partial_\mu n + \mathrm{i}\theta\,\deg(n), \qquad g^2_{O(3)} = 2g^2, \tag{82.21}$$

where

$$\deg(n) = \frac{1}{8\pi}\int_\Sigma \epsilon_{abc} n_a\,\mathrm{d}n_b\wedge\mathrm{d}n_c.$$

The matrix P_n is Hermitian and

$$P_n^2 = \frac{1}{4}(I + 2n_a\sigma_a + n_a n_b\sigma_a\sigma_b) = \frac{1}{2}(I + n_a\sigma_a) = P_n,$$

because $n_a n_b \epsilon_{abc} = 0$ and $n_a n_a = 1$. Its trace is one. Conversely every rank-one Hermitian projector in $\mathbb{C}^2$ has the form (82.18), since Hermitian 2×2 matrices are spanned by $I, \sigma_1, \sigma_2, \sigma_3$, the trace-one condition fixes the coefficient of I, and $P^2 = P$ fixes $n\cdot n = 1$.

Differentiating gives $\partial_\mu P_n = \frac{1}{2}(\partial_\mu n_a)\sigma_a$. Thus

$$\operatorname{tr}(\partial_\mu P_n\,\partial_\mu P_n) = \frac{1}{4}\,\partial_\mu n_a\partial_\mu n_b\operatorname{tr}(\sigma_a\sigma_b) = \frac{1}{2}\,\partial_\mu n\cdot\partial_\mu n.$$

For the curvature form,

$$\operatorname{tr}(P_n\,\mathrm{d}P_n\wedge\mathrm{d}P_n) = \frac{1}{8} n_a\,\mathrm{d}n_b\wedge\mathrm{d}n_c\ \operatorname{tr}(\sigma_a\sigma_b\sigma_c),$$

because the term with $\operatorname{tr}(\sigma_b\sigma_c)$ is symmetric in (b, c) and hence vanishes after wedging. The Pauli trace identity gives (82.20). Substituting (82.19) into the projector action gives the coupling match, and substituting (82.20) into $Q[P]$ gives the degree formula.

For $N = 2$, the exact massive scattering problem is therefore the $O(3)$ case of the vector sigma-model scattering problem described earlier in this chapter. For $N > 2$, one must keep a sharper distinction. The classical $\mathbb{CP}^{N-1}$ target is a compact symmetric space and has a classical zero-curvature description. A quantum factorization theorem for the bosonic projective model requires additional data: quantum conserved charges or an equivalent factorizing construction, the particle spectrum, the scalar normalization, the crossed particle-antiparticle amplitudes, and the matching map from the regulator-defined model (82.5) to the bootstrap mass coordinate.

The $N = (2, 2)$ supersymmetric $\mathbb{CP}^{N-1}$ model is a different local QFT. Its scattering proposal factorizes an $SU(N)$ chiral Gross–Neveu tensor part with an additional supersymmetric minimal factor and has an antisymmetric bound-state tower. Those statements use the extra fermionic fields and supercharges of the supersymmetric theory; the bosonic projective action (82.5) requires its own factorization and matching data.

82.6.2 Projective-Model Crossing Data

The minimal scattering language for the massive projective family contains particles in the defining representation $V = \mathbb{C}^N$ of $SU(N)$ and antiparticles in the dual representation $V^\vee$. Fix a basis $\{e_i\}_{i=1}^N$ of V and the dual basis $\{e^i\}_{i=1}^N$ of $V^\vee$. On $V\otimes V$, define

$$I_{VV}(e_i\otimes e_j) = e_i\otimes e_j, \qquad P_{VV}(e_i\otimes e_j) = e_j\otimes e_i.$$

On $V \otimes V^\vee$, define

$$I_{V\bar{V}}(e_i \otimes e^j) = e_i \otimes e^j, \qquad E_{V\bar{V}}(e_i \otimes e^j) = \delta_i^j \Omega, \qquad \Omega = \sum_{m=1}^{N} e_m \otimes e^m. \tag{82.22}$$

The operator $E_{V\bar{V}}$ is the invariant cap-cup tensor; it obeys

$$E_{V\bar{V}}^2 = N E_{V\bar{V}}. \tag{82.23}$$

Thus $N^{-1}E_{V\bar{V}}$ projects onto the singlet line $\mathbb{C}\Omega$, while

$$\Pi_{\mathrm{adj}} = I_{V\bar{V}} - \frac{1}{N} E_{V\bar{V}} \tag{82.24}$$

projects onto the adjoint summand in $V \otimes V^\vee$.

Definition 82.3 (Projective crossing system)**.** A rank-one projective bootstrap system consists of meromorphic functions $A(\theta)$, $B(\theta)$, $C(\theta)$, and $D(\theta)$, together with the operators

$$S_{VV}(\theta) = A(\theta) I_{VV} + B(\theta) P_{VV}, \tag{82.25}$$

$$S_{V\bar{V}}(\theta) = C(\theta) I_{V\bar{V}} + D(\theta) E_{V\bar{V}}. \tag{82.26}$$

The chosen charge-conjugation pairing identifies $V^\vee$ with the crossed outgoing V-leg. In this convention, crossing in the second leg is the meromorphic relation

$$C(\theta) = A(\mathrm{i}\pi - \theta), \qquad D(\theta) = B(\mathrm{i}\pi - \theta). \tag{82.27}$$

Crossing of the invariant tensors. With the convention of Definition 82.3, the crossed tensor of S_{VV} is exactly $S_{V\bar{V}}$ with coefficients (82.27).

Write matrix elements of S_{VV} as

$$(S_{VV})_{ij}^{k\ell}(\theta) = A(\theta)\delta_i^k \delta_j^\ell + B(\theta)\delta_i^\ell \delta_j^k. \tag{82.28}$$

Crossing the second outgoing leg to an incoming antiparticle replaces the pair (j, ℓ) by (ℓ, j) and evaluates the rapidity at $\mathrm{i}\pi - \theta$:

$$(S_{V\bar{V}})_{i\bar{j}}^{k\bar{\ell}}(\theta) = (S_{VV})_{i\ell}^{kj}(\mathrm{i}\pi - \theta). \tag{82.29}$$

Substitution into (82.28) gives

$$A(\mathrm{i}\pi - \theta)\delta_i^k \delta_\ell^j + B(\mathrm{i}\pi - \theta)\delta_i^j \delta_\ell^k.$$

The first tensor is $I_{V\bar{V}}$, and the second tensor is $E_{V\bar{V}}$, with the component convention in (82.22). This proves (82.27).

Let

$$S_\pm(\theta) = A(\theta) \pm B(\theta), \qquad S_{\mathrm{adj}}(\theta) = C(\theta), \qquad S_{\mathrm{sing}}(\theta) = C(\theta) + ND(\theta). \tag{82.30}$$

Then braiding unitarity on $V \otimes V$ is equivalent to

$$S_+(\theta)S_+(-\theta) = 1, \qquad S_-(\theta)S_-(-\theta) = 1, \tag{82.31}$$

and braiding unitarity on $V \otimes V^\vee$ is equivalent to

$$S_{\mathrm{adj}}(\theta)S_{\mathrm{adj}}(-\theta) = 1, \qquad S_{\mathrm{sing}}(\theta)S_{\mathrm{sing}}(-\theta) = 1. \tag{82.32}$$

The flip P_{VV} has eigenvalue $+1$ on $\mathrm{Sym}^2 V$ and eigenvalue -1 on $\wedge^2 V$. Therefore S_{VV} has eigenvalues S_+ and S_- on the two irreducible summands, and unitarity reduces to (82.31). For $V \otimes V^\vee$, (82.23) gives eigenvalue N on $\mathbb{C}\Omega$ and eigenvalue 0 on the adjoint summand. The two eigenvalues of $S_{V\bar{V}}$ are therefore S_{sing} and S_{adj}, giving (82.32).

Remark 82.4 (Pole assignment in the projective rank-one block). Suppose the VV tensor part is the rational block $\mathcal{R}_N(\theta)$ of (82.57), multiplied by a scalar factor $X_{\mathbb{CP}}(\theta)$. If the projective bootstrap system assigns no stable direct-channel particle in $\wedge^2 V$, then

$$X_{\mathbb{CP}}(\theta)$$

has a zero at $\theta = 2\pi\mathrm{i}/N$ of at least the order of the pole of $\mathcal{R}_N$ there. If instead that pole is assigned to a listed stable bound state, the residue must be projected onto $\wedge^2 V$ and must obey the positive-residue convention of the Hilbert-space normalization.

The symmetric and antisymmetric eigenvalues of $\mathcal{R}_N$ are

$$1, \qquad \frac{\theta + \frac{2\pi\mathrm{i}}{N}}{\theta - \frac{2\pi\mathrm{i}}{N}}.$$

The second eigenvalue has a simple pole at $\theta = 2\pi\mathrm{i}/N$, inside the physical strip for $N > 2$, and the corresponding eigenspace is $\wedge^2 V$. A meromorphic scattering amplitude can contain such a pole in the physical strip only as declared scattering data. If the spectrum assigns no stable direct-channel state to that tensor channel, the pole of the matrix block must be canceled by a zero of the scalar factor. If the pole is kept, its residue is precisely the projection onto the stated bound-state channel, with its normalization fixed by the Hilbert-space inner product.

Controlled Approximation 82.5 (Projective scattering status)**.** The preceding tensor constructions give finite-dimensional bootstrap infrastructure for the projective family. They specify the invariant channels, their crossing relation, and the pole-assignment rule before any scalar factor is chosen. A completed projective-model S-matrix further specifies the scalar functions, their physical-strip singularities, the CDD normalization, and the map from the regulator-defined model (82.5) to the bootstrap mass coordinate. The $\theta = \pi$ infrared behavior is a separate dynamical question; for some families it is expected to land on a level-one WZW fixed point, but that statement belongs to a specified continuum limit and cannot be read off from the perturbative beta function alone.

82.7 Principal Chiral Models and WZW Endpoints

Let G be a compact simple Lie group with invariant positive form κ on its Lie algebra $\mathfrak{g}$. The principal chiral model has field $U : \Sigma \to G$, current $j_\mu = U^{-1}\partial_\mu U$, and action

$$S_{\mathrm{PCM}}[U] = \frac{1}{2g^2}\int_\Sigma \mathrm{d}^2x\, \kappa(j_\mu, j_\mu). \tag{82.33}$$

The global symmetry is $G_L \times G_R$, acting by

$$U \mapsto g_L\, U\, g_R^{-1}.$$

The equations of motion and Maurer–Cartan identity in light-cone coordinates are

$$\partial_+ j_- + \partial_- j_+ = 0, \qquad \partial_+ j_- - \partial_- j_+ + [j_+, j_-] = 0. \tag{82.34}$$

Define

$$L_+(\zeta) = \frac{j_+}{1-\zeta}, \qquad L_-(\zeta) = \frac{j_-}{1+\zeta}. \tag{82.35}$$

Then the flatness condition

$$\partial_+ L_-(\zeta) - \partial_- L_+(\zeta) + [L_+(\zeta), L_-(\zeta)] = 0$$

for all complex ζ away from ± 1 is equivalent to the two relations in (82.34). Multiplying the curvature by $1-\zeta^2$ gives

$$(1-\zeta)\partial_+ j_- - (1+\zeta)\partial_- j_+ + [j_+, j_-] = \big(\partial_+ j_- - \partial_- j_+ + [j_+, j_-]\big) - \zeta(\partial_+ j_- + \partial_- j_+).$$

Vanishing for all ζ is therefore equivalent to the vanishing of the constant and linear coefficients, which are precisely the Maurer–Cartan identity and the equation of motion. Conversely, if those two relations hold, both coefficients vanish and the curvature of $L_\pm(\zeta)$ vanishes for all $\zeta \neq \pm 1$. The poles at $\zeta = \pm 1$ are part of the auxiliary spectral-parameter family and do not add additional equations of motion.

The Wess–Zumino deformation is defined by extending U to a three-manifold B with $\partial B = \Sigma$ and setting

$$\Gamma[U] = \frac{1}{24\pi} \int_B \kappa(U^{-1}\mathrm{d}U, [U^{-1}\mathrm{d}U, U^{-1}\mathrm{d}U]). \tag{82.36}$$

The quantum functional

$$S_{k,\lambda}[U] = \frac{1}{2\lambda^2} \int_\Sigma \kappa(j_\mu, j_\mu) + \mathrm{i}k\,\Gamma[U] \tag{82.37}$$

is single-valued only when the normalization of κ and the integer level k make the change of $k\Gamma$ under a different extension a multiple of 2π.

Polyakov–Wiegmann identity in the chapter normalization. Let $U, V : \Sigma \to G$ be smooth maps that admit extensions to a common three-manifold B with $\partial B = \Sigma$. Put

$$\Theta_U = U^{-1}\mathrm{d}U, \qquad \overline{\Theta}_V = \mathrm{d}V\, V^{-1}.$$

With Γ normalized as in (82.36),

$$\Gamma[UV] = \Gamma[U] + \Gamma[V] - \frac{1}{4\pi} \int_\Sigma \kappa(\Theta_U, \overline{\Theta}_V), \tag{82.38}$$

up to the integral periods already fixed by the level quantization of the WZ term. Here $\kappa(\Theta_U, \overline{\Theta}_V)$ denotes the scalar two-form obtained by wedging the one-forms and pairing their Lie-algebra values by κ. Let

$$H(g) = \frac{1}{24\pi} \kappa(g^{-1}\mathrm{d}g, [g^{-1}\mathrm{d}g, g^{-1}\mathrm{d}g])$$

be the closed three-form integrated in $\Gamma[g]$. The bracket of Lie-algebra-valued one-forms is the wedge bracket, so that the displayed normalization agrees with (82.36). Since

$$(UV)^{-1}\mathrm{d}(UV) = V^{-1}\Theta_U V + V^{-1}\mathrm{d}V,$$

invariance of κ and the Maurer–Cartan equations

$$\mathrm{d}\Theta_U + \frac{1}{2}[\Theta_U, \Theta_U] = 0, \qquad \mathrm{d}\overline{\Theta}_V - \frac{1}{2}[\overline{\Theta}_V, \overline{\Theta}_V] = 0$$

give the differential identity

$$H(UV) = H(U) + H(V) - \frac{1}{4\pi}\mathrm{d}\,\kappa(\Theta_U, \overline{\Theta}_V). \tag{82.39}$$

To see the coefficient, expand the cubic expression for $H(UV)$: the pure Θ_U and pure $V^{-1}\mathrm{d}V$ terms give $H(U)$ and $H(V)$, and the three mixed cyclic placements combine with the two Maurer–Cartan equations into the exterior derivative in (82.39). Integrating this identity over B and applying Stokes' theorem gives (82.38). Changing the extensions shifts the three bulk terms by integral periods; the quantization condition on k is exactly the condition that $\exp(-S_{k,\lambda})$ be independent of that choice.

82.7.1 Current Algebra at the WZW Endpoint

The conformal endpoint is specified by its local current algebra, positive energy representation space, stress tensor, and Ward identities. We record the construction here in the sigma-model normalization used in this chapter; Volume V develops the same object from the broader two-dimensional CFT point of view.

Let $(\,\cdot\,,\,\cdot\,)$ be the basic invariant form on $\mathfrak{g}$, normalized so that long roots have squared length 2. Choose an orthonormal basis T_a, write

$$[T_a, T_b] = f_{ab}{}^{c}T_c, \qquad f_{abc} = f_{ab}{}^{d}\delta_{dc},$$

and define the dual Coxeter number by

$$f_{acd}f_{bcd} = 2h^\vee \delta_{ab}. \tag{82.40}$$

With this convention the adjoint quadratic Casimir is $2h^\vee$. At a positive integer level k, the holomorphic and antiholomorphic currents have separated-point OPEs

$$J^a(z)J^b(0) \sim \frac{k\delta^{ab}}{z^2} + \frac{f_{ab}{}^{c}J^c(0)}{z}, \qquad \bar{J}^a(\bar{z})\bar{J}^b(0) \sim \frac{k\delta^{ab}}{\bar{z}^2} + \frac{f_{ab}{}^{c}\bar{J}^c(0)}{\bar{z}}. \tag{82.41}$$

The mixed $J\bar{J}$ OPE is regular at separated points. If

$$J^a(z) = \sum_{n\in\mathbb{Z}} J^a_n z^{-n-1},$$

then (82.41) is equivalent to

$$[J^a_m, J^b_n] = f_{ab}{}^{c}J^c_{m+n} + km\,\delta^{ab}\delta_{m+n,0}. \tag{82.42}$$

The integer k is therefore both the Wess–Zumino level and the central extension of the loop algebra. This equality is a theorem of the canonical quantization of the WZW functional or, equivalently, of the current Ward identity derived from the Polyakov–Wiegmann identity; it is not a separate postulate appended to the sigma model.

Lemma 82.6 (Sugawara tensor at the WZW endpoint)**.** *Assume $k + h^\vee \neq 0$. The holomorphic stress tensor is*

$$T_G(z) = \frac{1}{2(k + h^\vee)} \sum_a : J^a J^a : (z), \tag{82.43}$$

and has central charge

$$c(G, k) = \frac{k \, \dim \mathfrak{g}}{k + h^\vee}. \tag{82.44}$$

Each current J^a is a primary field of weight 1.

Proof. The coefficient in (82.43) is forced by the OPE $T_G(z)J^b(0)$. Contracting one of the two currents in $: J^a J^a :$ with J^b, the central term in (82.41) contributes $2kJ^b/z^2$ before the overall coefficient. The simple-pole contraction gives two adjoint actions. When the resulting current is normal ordered past the remaining current, the adjoint Casimir contributes $2h^\vee J^b/z^2$, by (82.40). Hence the numerator of the second-order pole is $2(k + h^\vee)J^b$, and the unique coefficient making the current have weight 1 is $1/[2(k + h^\vee)]$. The first-order pole is then $\partial J^b/z$, because the zero mode of T_G implements translations of the current field.

For the $T_G T_G$ OPE, the fourth-order pole is obtained from double contractions of the two quadratic normal-ordered products. The central central contractions give $2k^2 \dim \mathfrak{g}$, and the contractions containing two structure constants give $2kh^\vee \dim \mathfrak{g}$. After division by $[2(k + h^\vee)]^2$, comparison with

$$T_G(z)T_G(0) \sim \frac{c/2}{z^4} + \frac{2T_G(0)}{z^2} + \frac{\partial T_G(0)}{z}$$

gives (82.44). The lower-order poles agree with the Virasoro action because the coefficient was already fixed by the current-primary condition. □

Let P^+ be the dominant integral weights of G, let θ be the highest root, and let ρ be the Weyl vector. The compact unitary WZW endpoint at positive integral level k uses the integrable highest weights

$$P_k^+ = \{\lambda \in P^+ \mid (\lambda, \theta) \leq k\}. \tag{82.45}$$

The corresponding affine highest-weight Hilbert spaces are denoted $\mathcal{H}_\lambda$. A primary with horizontal highest weight λ has conformal weight

$$h_\lambda = \frac{(\lambda, \lambda + 2\rho)}{2(k + h^\vee)}. \tag{82.46}$$

The diagonal compact theory has Hilbert space

$$\mathcal{H}_{G,k}^{\mathrm{diag}} = \bigoplus_{\lambda \in P_k^+} \mathcal{H}_\lambda \otimes \overline{\mathcal{H}_\lambda}. \tag{82.47}$$

Other modular invariants change the pairing of left and right integrable modules and are therefore extra global data of the full CFT. They do not alter the local affine OPE (82.41).

Proposition 82.7 (KZ equations from WZW Ward identities)**.** *Let V_{λ_i} be finite-dimensional $\mathfrak{g}$-modules with $\lambda_i \in P_k^+$, and let $\mathcal{F}(z_1, \ldots, z_n)$ be a genus-zero chiral block with primary insertions of horizontal types V_{λ_i}. Write t_i^a for the action of T_a on the i-th tensor factor and*

$$\Omega_{ij} = \sum_a t_i^a t_j^a.$$

Then

$$(k+h^\vee)\,\partial_{z_i}\mathcal{F} = \mathcal{F}\circ\sum_{j\neq i}\frac{\Omega_{ij}}{z_i-z_j}, \qquad i=1,\dots,n. \tag{82.48}$$

Proof. For a primary field $\Phi_{\lambda_i}(v_i,z_i)$, the current OPE is

$$J^a(\zeta)\Phi_{\lambda_i}(v_i,z_i)\sim\frac{\Phi_{\lambda_i}(t_i^a v_i,z_i)}{\zeta-z_i}.$$

The translation Ward identity gives

$$\partial_{z_i}\Phi_{\lambda_i}(v_i,z_i)=\Phi_{\lambda_i}(L_{-1}v_i,z_i).$$

Applying (82.43) to an affine-primary state, all modes in

$$L_{-1}=\frac{1}{2(k+h^\vee)}\sum_{a,m}:J^a_{-1-m}J^a_m:$$

annihilate the state except the two terms with $m=0$ and $m=-1$. Their sum is

$$L_{-1}v_i=\frac{1}{k+h^\vee}\sum_a J^a_{-1}t_i^a v_i,$$

because $\sum_a[J_0^a,J_{-1}^a]=0$. The insertion of J^a_{-1} is the contour integral

$$\oint_{z_i}\frac{\mathrm{d}\zeta}{2\pi\mathrm{i}}\,\frac{J^a(\zeta)}{\zeta-z_i}.$$

Deforming this contour away from z_i gives the sum of the residues at the other insertions. The residue at z_j is $(z_i-z_j)^{-1}t_j^a$. Substitution gives (82.48). No crossing or modular-invariance input enters this derivation; it is a consequence of the local current OPE, primary-field OPE, and Sugawara translation operator. □

Let $H\subset G$ be a compact connected subgroup with Lie algebra $\mathfrak{h}$. If the basic forms satisfy

$$(X,Y)_{\mathfrak{g}}=x_e(X,Y)_{\mathfrak{h}}, \qquad X,Y\in\mathfrak{h},$$

then the induced H-current algebra inside G_k has level $k_H=kx_e$. The coset stress tensor

$$T_{G/H}=T_G-T_H \tag{82.49}$$

has central charge

$$c(G_k/H_{k_H})=\frac{k\dim\mathfrak{g}}{k+h^\vee_{\mathfrak{g}}}-\frac{k_H\dim\mathfrak{h}}{k_H+h^\vee_{\mathfrak{h}}}. \tag{82.50}$$

It has regular OPE with the embedded $\mathfrak{h}$-currents. The level rescales by the embedding index because the central term in the current OPE is the invariant form evaluated on the embedded Lie algebra. Both T_G and T_H make the same embedded H-currents primary fields of weight 1. Their difference therefore has no singular OPE with those currents. The central charge is the difference of the two Sugawara central charges in Lemma 82.6, with the H-level set to k_H.

Free-field realizations such as Wakimoto charts are useful computational coordinates on particular affine modules. They are not part of the definition of the sigma-model endpoint, and they require a separate development of bosonic first-order systems and screening charges. In this volume we use only the current algebra, integrable modules, KZ connection, and coset stress tensor, which are the endpoint structures needed for the principal chiral and current-current perturbations below.

82.8 Nonabelian Bosonization and Chiral Gross–Neveu Bridges

The relation between sigma models, WZW models, and fermionic Gross–Neveu models should be stated at the level of current algebras, not as a slogan about replacing fermions by bosons. Let

$$\psi_{L\,a}^{\,i}(x^+), \qquad \psi_{R\,a}^{\,i}(x^-), \qquad a = 1, \dots, N_c, \quad i = 1, \dots, N_f,$$

be free complex chiral fermions. The index a is acted on by $SU(N_c)$, the index i by $SU(N_f)$. The left-moving bilinears

$$J_c^A =: \psi_{L\,a}^{\,i\dagger}(T_c^A)^a{}_b \psi_{L\,b}^{\,i} :, \qquad J_f^B =: \psi_{L\,a}^{\,i\dagger}(T_f^B)^i{}_j \psi_{L\,a}^{\,j} :, \qquad J_u =: \psi_{L\,a}^{\,i\dagger}\psi_{L\,a}^{\,i} :$$

generate affine

$$\widehat{\mathfrak{su}}(N_c)_{N_f} \oplus \widehat{\mathfrak{su}}(N_f)_{N_c} \oplus \widehat{\mathfrak{u}}(1)$$

inside the free-fermion chiral algebra. The level is the number of spectator fermion flavors in the trace of the short-distance contraction: for example, the color-current two-point singularity receives N_f identical flavor contractions and hence has level N_f.

Remark 82.8 (Central-charge check for nonabelian bosonization). With the current normalization in which

$$c\big(\widehat{\mathfrak{su}}(N)_k\big) = \frac{k(N^2-1)}{k+N},$$

the chiral algebra generated by the bilinears above has the same central charge as N_cN_f free complex left-moving fermions:

$$\frac{N_f(N_c^2-1)}{N_f+N_c} + \frac{N_c(N_f^2-1)}{N_c+N_f} + 1 = N_cN_f. \tag{82.51}$$

Bring the two nonabelian terms to the common denominator $N_c + N_f$:

$$\frac{N_fN_c^2 - N_f + N_cN_f^2 - N_c}{N_c+N_f} = \frac{N_cN_f(N_c+N_f) - (N_c+N_f)}{N_c+N_f} = N_cN_f - 1.$$

Adding the $\widehat{\mathfrak{u}}(1)$ boson gives N_cN_f. The identity is only the central-charge compatibility check; the operator equivalence also requires equality of the affine OPEs, the stress tensor, and the compact $U(1)$ radius.

For $N_f = 1$ the flavor affine factor is absent and a set of N_c complex fermions decomposes into $SU(N_c)_1$ WZW data plus a compact $U(1)$ boson. Perturbing the WZW model by the marginal current-current operator

$$\delta S = \lambda \int \mathrm{d}^2x \sum_A J_L^A(x^+) J_R^A(x^-) \tag{82.52}$$

is the bosonic current-algebra presentation of the chiral Gross–Neveu family, once the $U(1)$ sector and discrete quotient data have been fixed. The exact massive $SU(N)$ chiral Gross–Neveu scattering theory is then the rapidity-space description of the current-current perturbation, while the WZW model is the ultraviolet fixed point described by the affine algebra. The statement is not that every fermionic model is identical to a principal chiral model; the dictionary is a statement about local current algebras and perturbations with specified global quotients. In the abelian rank-one subcase this reduces to the sine-Gordon/massive-Thirring bridge of Chapter 81; in the nonabelian case the replacement of the compact boson by a WZW field is the essential new structure.

82.9 SU(N) Bootstrap Spectrum and Matrix Blocks

The exact scattering descriptions of the $SU(N)$ chiral Gross–Neveu and principal-chiral families contain a common representation-theoretic skeleton. Let $V = \mathbb{C}^N$ be the defining representation of $SU(N)$, and write

$$V_a = \wedge^a V, \qquad a = 1, \dots, N-1.$$

The conjugate of V_a is V_{N-a}. A bootstrap spectrum of sine-law type assigns masses

$$m_a = M\,\frac{\sin(\pi a/N)}{\sin(\pi/N)}, \qquad a = 1, \dots, N-1, \tag{82.53}$$

for a single positive mass scale $M = m_1$. In the chiral Gross–Neveu family these multiplets are the antisymmetric tensor particles. In the principal chiral model the corresponding particles carry left and right indices, with one common convention taking the rank-a multiplet to transform in $V_a \otimes V_a^*$ under $SU(N)_L \times SU(N)_R$. The choice of the global form of the symmetry and the quotient by common centers is part of the family specification; it affects which line operators and twisted sectors exist, although not the local sine-mass formula itself.

Remark 82.9 (Fusing-angle consistency of the sine spectrum). For $1 \le a, b$ and $a + b < N$, the mass formula (82.53) is compatible with a direct-channel bound state $V_a \otimes V_b \to V_{a+b}$ at rapidity

$$\theta = \mathrm{i}u_{ab}^{a+b}, \qquad u_{ab}^{a+b} = \frac{\pi(a+b)}{N}. \tag{82.54}$$

Indeed,

$$m_{a+b}^2 = m_a^2 + m_b^2 + 2m_a m_b \cos u_{ab}^{a+b}. \tag{82.55}$$

Set $A = \pi a/N$, $B = \pi b/N$, and suppress the common positive factor $M/\sin(\pi/N)$. The required identity is

$$\sin^2(A+B) = \sin^2 A + \sin^2 B + 2\sin A \sin B \cos(A+B). \tag{82.56}$$

Using $\cos(A+B) = \cos A \cos B - \sin A \sin B$, the right-hand side becomes

$$\sin^2 A + \sin^2 B + 2\sin A \sin B \cos A \cos B - 2\sin^2 A \sin^2 B.$$

Since

$$\sin^2(A+B) = \sin^2 A \cos^2 B + \cos^2 A \sin^2 B + 2\sin A \sin B \cos A \cos B,$$

and $\cos^2 A = 1 - \sin^2 A$, $\cos^2 B = 1 - \sin^2 B$, the two expressions agree. The restriction $a+b < N$ selects the antisymmetric tensor before conjugation; fusion channels with $a + b \ge N$ are equivalently described after replacing V_c^* by V_{N-c} and keeping track of crossed-channel conventions.

The fundamental rational $SU(N)$ matrix block acts on $V \otimes V$. Let

$$P(v \otimes w) = w \otimes v, \qquad \lambda_N = \frac{2\pi}{N},$$

and define

$$\mathcal{R}_N(\theta) = \frac{\theta\, I - \mathrm{i}\lambda_N P}{\theta - \mathrm{i}\lambda_N}. \tag{82.57}$$

This normalization makes the symmetric-channel eigenvalue equal to 1, and the antisymmetric-channel eigenvalue equal to

$$\frac{\theta + \mathrm{i}\lambda_N}{\theta - \mathrm{i}\lambda_N}.$$

Thus the pole of the matrix block at $\theta = \mathrm{i}\lambda_N$ lies in the antisymmetric channel, matching the rank-two entry of (82.53). The scalar factor multiplying $\mathcal{R}_N$ is not optional: it fixes crossing, removes or adds CDD ambiguities, and supplies the complete pole-residue normalization. The matrix block records the tensor part of the exact S-matrix; the scalar minimality and the regulator-to-bootstrap matching are additional data.

Rational-block unitarity and Yang–Baxter equation. The block $\mathcal{R}_N$ obeys

$$\mathcal{R}_N(\theta)\mathcal{R}_N(-\theta) = I$$

as an operator on $V \otimes V$. Its unnormalized numerator

$$R_N(\theta) = \theta I - \mathrm{i}\lambda_N P$$

obeys the Yang–Baxter equation

$$R_{12}(\theta)R_{13}(\theta+\phi)R_{23}(\phi) = R_{23}(\phi)R_{13}(\theta+\phi)R_{12}(\theta) \tag{82.58}$$

on $V^{\otimes 3}$.

The permutation P satisfies $P^2 = I$, so the numerator product is

$$(\theta I - \mathrm{i}\lambda_N P)(-\theta I - \mathrm{i}\lambda_N P) = -(\theta^2 + \lambda_N^2)I = (\theta - \mathrm{i}\lambda_N)(-\theta - \mathrm{i}\lambda_N)I.$$

For the Yang–Baxter equation, expand both sides of (82.58) as polynomials in θ, ϕ and λ_N. The constant, linear, and quadratic terms reduce to the relations of the symmetric group acting on tensor factors:

$$P_{12}^2 = P_{23}^2 = I, \qquad P_{12}P_{23}P_{12} = P_{23}P_{12}P_{23}.$$

The mixed P_{13} terms are written as $P_{13} = P_{12}P_{23}P_{12} = P_{23}P_{12}P_{23}$. After this substitution the two expanded polynomials coincide coefficient by coefficient. The calculation companion checks this equality by finite-dimensional matrix multiplication for representative N and rapidities.

82.10 SU(N) Scalar Factors and CDD Conventions

The rational matrix block is only the tensor part of the scattering problem. The scalar factor fixes the analytic normalization. Let

$$\mathcal{P} = \{\theta \in \mathbb{C} : 0 < \operatorname{Im}\theta < \pi\}$$

be the physical rapidity strip for equal-mass relativistic scattering. In a scalar channel, unitarity is the meromorphic identity

$$X(\theta)X(-\theta) = 1,$$

and self-conjugate crossing is the identity $X(\theta) = X(\mathrm{i}\pi - \theta)$. For non-self-conjugate multiplets, crossing relates a particle-particle amplitude to a particle-antiparticle amplitude; the scalar factor must be specified together with the charge-conjugation convention.

Definition 82.10 (CDD factor in one scalar channel)**.** A scalar CDD factor is a meromorphic function $C(\theta)$, nonzero on the real rapidity axis, such that

$$C(\theta)C(-\theta) = 1$$

and such that the appropriate crossing equation of the channel is preserved. It is an allowed ambiguity only after its zeros and poles in $\mathcal{P}$ have been declared either absent or assigned to specified bound states, Coleman–Thun singularities, or other explicitly stated on-shell mechanisms. A scalar factor is called minimal relative to a listed bootstrap spectrum when it has no additional zeros or poles in $\mathcal{P}$ beyond those listed.

The elementary self-conjugate block

$$C_\alpha(\theta) = \frac{\sinh\theta + \mathrm{i}\sin\alpha}{\sinh\theta - \mathrm{i}\sin\alpha} \tag{82.59}$$

obeys $C_\alpha(\theta)C_\alpha(-\theta) = 1$ and $C_\alpha(\mathrm{i}\pi - \theta) = C_\alpha(\theta)$. Its denominator vanishes at $\theta = \mathrm{i}u$ when $\sin u = \sin\alpha$. Thus, for $0 < \alpha < \pi$, it has physical-strip singularities at $u = \alpha$ and $u = \pi - \alpha$, with coincidences at $\alpha = \pi/2$. Such a block is therefore not a harmless convention unless those singularities have been assigned. This is the reason that "multiply by a CDD factor" is not a mathematical conclusion; it is new scattering data.

For the $SU(N)$ chiral Gross–Neveu family, $N \geq 3$, put

$$x = \frac{\theta}{2\pi\mathrm{i}}, \qquad \lambda_N = \frac{2\pi}{N},$$

and define the rank-one scalar transmission eigenvalue

$$A_N(\theta) = -\frac{\Gamma(1-x)\Gamma(1-\frac{1}{N}+x)}{\Gamma(1+x)\Gamma(1-\frac{1}{N}-x)}. \tag{82.60}$$

In the Zamolodchikov–Faddeev component convention with outgoing order reversed,

$$S^{\delta\gamma}_{\alpha\beta}(\theta) = \delta^\gamma_\alpha\delta^\delta_\beta\, b_N(\theta) + \delta^\delta_\alpha\delta^\gamma_\beta\, c_N(\theta), \qquad c_N(\theta) = -\frac{\mathrm{i}\lambda_N}{\theta} b_N(\theta), \tag{82.61}$$

with

$$A_N(\theta) = b_N(\theta) + c_N(\theta), \qquad b_N(\theta) = \frac{A_N(\theta)}{1 - \mathrm{i}\lambda_N/\theta}.$$

Equivalently, if the outgoing permutation is pulled out explicitly, this is

$$S^{(1,1)}_{\mathrm{CGN}}(\theta) = A_N(\theta)\, P\, \mathcal{R}_N(\theta), \tag{82.62}$$

with $\mathcal{R}_N$ defined in (82.57). The matrix pole at $\theta = \mathrm{i}\lambda_N$ is then the first antisymmetric bound-state pole, while A_N itself supplies the minimal gamma-function normalization in this convention.

Gross–Neveu scalar unitarity identity. As meromorphic identities in θ,

$$A_N(\theta)A_N(-\theta) = 1,$$

and the component matrix (82.61) obeys

$$S(\theta)S(-\theta) = I$$

on $V \otimes V$, away from its poles.

In (82.60), replacing x by $-x$ interchanges the two gamma-function products: the factors $\Gamma(1-x)\Gamma(1/N+x)$ in $A_N(\theta)$ become the denominator factors of $A_N(-\theta)$, and the denominator factors $\Gamma(1+x)\Gamma(1/N-x)$ become the numerator factors. The two explicit minus signs multiply to 1. Thus $A_N(\theta)A_N(-\theta) = 1$. For the component matrix, write $S = bP + cI$ with $c = -(\mathrm{i}\lambda_N/\theta)b$. At $-\theta$,

$$c(-\theta) = \frac{\mathrm{i}\lambda_N}{\theta} b(-\theta).$$

Since $A = b(1 - \mathrm{i}\lambda_N/\theta)$ and

$$A(-\theta) = b(-\theta)(1 + \mathrm{i}\lambda_N/\theta),$$

the scalar identity gives

$$b(\theta)b(-\theta)\left(1 + \frac{\lambda_N^2}{\theta^2}\right) = 1.$$

Using $P^2 = I$,

$$(bP + cI)(b_-P + c_-I) = (bb_- + cc_-)I + (bc_- + cb_-)P,$$

where $b_- = b(-\theta)$ and $c_- = c(-\theta)$. The P-coefficient vanishes, and the I-coefficient is the scalar displayed above. This proves matrix unitarity.

For the $SU(N)$ principal chiral model, the fundamental particle carries left and right indices. It is convenient to distinguish the normalized block $\mathcal{R}_N$ from the unnormalized rational block

$$\widetilde{\mathcal{R}}_N(\theta) = I - \frac{\mathrm{i}\lambda_N}{\theta} P = \frac{\theta - \mathrm{i}\lambda_N}{\theta}\, \mathcal{R}_N(\theta).$$

One standard particle-particle bootstrap convention is

$$S^{(1,1)}_{\mathrm{PCM}}(\theta) = W_N(\theta)\, \widetilde{\mathcal{R}}_{N,L}(\theta) \otimes \widetilde{\mathcal{R}}_{N,R}(\theta), \tag{82.63}$$

where

$$W_N(\theta) = \frac{\sinh(\theta/2 - \pi\mathrm{i}/N)}{\sinh(\theta/2 + \pi\mathrm{i}/N)} \left[\frac{\Gamma(1-x)\Gamma(x-\frac{1}{N})}{\Gamma(1-x-\frac{1}{N})\Gamma(x)}\right]^2, \qquad x = \frac{\theta}{2\pi\mathrm{i}}. \tag{82.64}$$

In the normalized block convention used in (82.66) below, the scalar factor is

$$X_{\mathrm{PCM}}(\theta) = W_N(\theta)\left(\frac{\theta - \mathrm{i}\lambda_N}{\theta}\right)^2. \tag{82.65}$$

Thus

$$S^{(1,1)}_{\mathrm{PCM}}(\theta) = X_{\mathrm{PCM}}(\theta)\, \mathcal{R}_{N,L}(\theta) \otimes \mathcal{R}_{N,R}(\theta). \tag{82.66}$$

The sine-law tower (82.53) is generated by fusion at (82.54); higher-rank scattering is obtained by projecting tensor products onto the corresponding antisymmetric representations and taking residues at the listed poles. The CDD convention is part of the statement: multiplying W_N or X_{PCM} by any additional factor satisfying Definition 82.10 changes the bootstrap theory unless the extra singularity data are also specified.

Principal-chiral scalar unitarity identity. The scalar factor (82.64) obeys

$$W_N(\theta)W_N(-\theta) = \left(1 + \frac{\lambda_N^2}{\theta^2}\right)^{-2}. \tag{82.67}$$

Consequently $X_{\mathrm{PCM}}(\theta)X_{\mathrm{PCM}}(-\theta) = 1$, and (82.66) is unitary on the real rapidity axis away from poles.

Let $a = 1/N$ and $x = \theta/(2\pi\mathrm{i})$. The hyperbolic prefactor in (82.64) has product 1 under $\theta \mapsto -\theta$, because sinh is odd. For the gamma ratio

$$G(x) = \frac{\Gamma(1-x)\Gamma(x-a)}{\Gamma(1-x-a)\Gamma(x)},$$

use $\Gamma(1+x) = x\Gamma(x)$, $\Gamma(1-x) = -x\Gamma(-x)$, $\Gamma(1-x-a) = (-x-a)\Gamma(-x-a)$, and $\Gamma(1+x-a) = (x-a)\Gamma(x-a)$. These identities give

$$G(x)G(-x) = \frac{x^2}{(x-a)(x+a)}.$$

Since $\lambda_N = 2\pi a$ and $\theta = 2\pi\mathrm{i}x$,

$$1 + \frac{\lambda_N^2}{\theta^2} = 1 - \frac{a^2}{x^2} = \frac{(x-a)(x+a)}{x^2}.$$

Squaring the gamma-ratio product proves (82.67). Multiplying by

$$\left(\frac{\theta - \mathrm{i}\lambda_N}{\theta}\right)^2 \left(\frac{-\theta - \mathrm{i}\lambda_N}{-\theta}\right)^2 = \left(1 + \frac{\lambda_N^2}{\theta^2}\right)^2$$

then gives $X_{\mathrm{PCM}}(\theta)X_{\mathrm{PCM}}(-\theta) = 1$. The normalized matrix blocks are unitary by the rational-block calculation above, so the full particle-particle PCM matrix is unitary.

Equations (82.62) and (82.66) are exact bootstrap data with the displayed scalar convention. They become statements about the continuum sigma models only after the ultraviolet definitions of the preceding sections are matched to the factorizing QFT: one must construct the Hilbert space, local or graded-local fields, conserved currents, Haag–Ruelle scattering states, and the map between the regulator mass scale and the bootstrap mass coordinate.

82.11 Nested String Data for Gross–Neveu and PCM

The nested Bethe ansatz of Chapter 78 becomes concrete only after the physical representation carried by each particle and the number of independent internal symmetry nests are specified. For the $SU(N)$ chiral Gross–Neveu family there is one A_{N-1} nest. For the $SU(N)$ principal chiral model there are two nests, one for $SU(N)_L$ and one for $SU(N)_R$. This distinction is not a cosmetic doubling: it is the finite-volume origin of the two rational blocks in (82.66).

Let α_r, $r = 1, \ldots, N-1$, be the simple roots of A_{N-1}, and let ω_r be the fundamental weights, normalized by $\langle \alpha_r^\vee, \omega_s \rangle = \delta_{rs}$. Since A_{N-1} is simply laced,

$$\alpha_r = \sum_{s=1}^{N-1} C_{rs}\omega_s, \qquad C_{rs} = 2\delta_{rs} - \delta_{r,s+1} - \delta_{r,s-1}. \tag{82.68}$$

The inverse Cartan matrix is

$$(C^{-1})_{rs} = \frac{\min(r,s)\big(N - \max(r,s)\big)}{N}, \qquad 1 \le r, s \le N-1. \tag{82.69}$$

The quotient of the weight lattice by the root lattice is

$$P_{\mathrm{wt}}/Q_{\mathrm{rt}} \cong \mathbb{Z}_N, \qquad \omega_a \longmapsto a \pmod N.$$

This is the center charge, or N-ality, of the corresponding antisymmetric-tensor representation $V_a = \wedge^a \mathbb{C}^N$.

Nested root-count constraints. Consider a finite-volume chiral Gross–Neveu multiparticle state containing M_a particles in V_a, $a = 1, \ldots, N-1$. In the algebraic nested Bethe ansatz, an auxiliary root of Dynkin color r lowers the total highest weight by α_r. If m_r is the total number of auxiliary roots of color r, counted with string length, then the Dynkin-label vector after nesting is

$$\ell_s = M_s - \sum_{r=1}^{N-1} C_{rs} m_r, \qquad s = 1, \ldots, N-1. \tag{82.70}$$

Consequently a state in the trivial $SU(N)$ representation must satisfy

$$m_r = \sum_{s=1}^{N-1} (C^{-1})_{rs} M_s. \tag{82.71}$$

The vector on the right-hand side is integral if and only if

$$\sum_{a=1}^{N-1} a\, M_a \equiv 0 \pmod N. \tag{82.72}$$

Verification. The reference tensor product has highest weight $\Lambda_{\mathrm{ref}} = \sum_a M_a \omega_a$. The nested creation operator of color r has Cartan weight $-\alpha_r$: in the RTT presentation this follows from commuting the Cartan part of the monodromy through the off-diagonal entry that creates the root; in the representation-theoretic language it is exactly the elementary lowering by the simple root associated to the adjacent pair of colors in the A_{N-1} Dynkin diagram. After m_r such roots, the total weight is

$$\Lambda_{\mathrm{ref}} - \sum_r m_r \alpha_r = \sum_s \left(M_s - \sum_r m_r C_{rs} \right) \omega_s,$$

which is (82.70). Trivial representation forces all Dynkin labels to vanish, hence $M = Cm$ and (82.71).

It remains to identify when $C^{-1}M$ is integral. The equality $\sum_s M_s \omega_s = \sum_r m_r \alpha_r$ with $m_r \in \mathbb{Z}$ is precisely the condition that the reference weight lies in the root lattice Q_{rt}. For $SU(N)$, the map $P_{\mathrm{wt}} \to P_{\mathrm{wt}}/Q_{\mathrm{rt}} \cong \mathbb{Z}_N$ sends ω_a to a. Therefore $\sum_s M_s \omega_s \in Q_{\mathrm{rt}}$ exactly when (82.72) holds. Since the simple roots form an integral basis of Q_{rt}, the coordinates m_r in the simple-root basis are then integers; conversely, integral m_r makes the weight a root lattice element and hence gives zero N-ality.

This is only weight-space bookkeeping. It does not replace the Bethe equations, the string hypothesis, or a proof that a particular solution exists. It does, however, fix the source vector and the number of nests before any TBA kernel is written. If $M^{\rm aux}_{r,k}$ is the number of auxiliary strings of Dynkin color r and length k, then

$$m_r = \sum_{k\geq 1} k\, M^{\rm aux}_{r,k}. \tag{82.73}$$

Thus the thermodynamic limit replaces M_a/L by physical particle densities and $M^{\rm aux}_{r,k}/L$ by auxiliary string densities, while preserving the Cartan constraint

$$\frac{M_s}{L} - \sum_{r,k} k\, C_{rs}\frac{M^{\rm aux}_{r,k}}{L} \longrightarrow \ell_s/L.$$

The energy and momentum driving terms come only from the physical particles:

$$E = \sum_{a=1}^{N-1}\sum_{j=1}^{M_a} m_a \cosh\theta_{a,j}, \qquad P = \sum_{a=1}^{N-1}\sum_{j=1}^{M_a} m_a \sinh\theta_{a,j}, \tag{82.74}$$

with m_a given in (82.53). Auxiliary strings carry no independent relativistic energy; they encode internal polarization and enter the TBA through the logarithmic derivatives of the rational matrix blocks and their fused descendants.

Worked chiral Gross–Neveu root counts. For $SU(3)$,

$$C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1\\ 1 & 2\end{pmatrix}.$$

One V_1 particle and one V_2 particle have $M=(1,1)$, zero N-ality, and require $m=(1,1)$. Three fundamental particles have $M=(3,0)$ and require $m=(2,1)$. A single fundamental particle has $M=(1,0)$, nonzero N-ality, and $C^{-1}M=(2/3,1/3)$, so it cannot be a singlet-sector Bethe state. For $SU(4)$, four fundamentals have $m=(3,2,1)$, while two rank-two particles have $m=(1,2,1)$. These numbers are the total color-root counts; for example, $m=(3,2,1)$ may be refined into three length-one strings at node 1, two at node 2, and one at node 3, or into any string distribution whose length-weighted totals are the displayed integers and which solves the finite-volume Bethe equations.

Principal-chiral doubling. A rank-a principal-chiral particle transforms as $V_a\otimes V_a^*\cong V_a\otimes V_{N-a}$ under $SU(N)_L\times SU(N)_R$. For a state with physical counts M_a, the two source vectors are therefore

$$M^L_s = M_s, \qquad M^R_s = M_{N-s}, \qquad s=1,\dots,N-1, \tag{82.75}$$

and a two-sided singlet constraint has

$$m^L = C^{-1}M^L, \qquad m^R = C^{-1}M^R. \tag{82.76}$$

The two integrality conditions are equivalent because

$$\sum_s sM^R_s = \sum_s sM_{N-s} \equiv -\sum_a aM_a \pmod N.$$

Thus the PCM has the same center obstruction as the chiral Gross–Neveu family, but the auxiliary-string sector is doubled. This is the finite-volume reason that the principal-chiral TBA has left and right magnonic towers attached to the same physical sine-law particles, whereas the chiral Gross–Neveu TBA has only one such tower. The calculation companion verifies (82.69)–(82.76) exactly for representative N, including the $SU(3)$ and $SU(4)$ root counts above.

82.12 Symmetric-Space and Deformed-Sphere Families

Many integrable sigma models are organized by a homogeneous target G/H. Let $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ be a reductive split and write

$$g^{-1}\mathrm{d}g = A + P, \qquad A \in \mathfrak{h}, \quad P \in \mathfrak{m}.$$

The bosonic coset action is

$$S_{G/H}[g] = \frac{1}{2g^2}\int_\Sigma \kappa(P_\mu, P_\mu), \tag{82.77}$$

with local H redundancy $g(x) \mapsto g(x)h(x)$. If G/H is a symmetric space,

$$[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}, \qquad [\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}, \qquad [\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h},$$

the equations admit a Lax representation that is slightly different from the principal-chiral one because the H gauge connection must remain in the auxiliary connection. Write $A_\pm$ and $P_\pm$ for the light-cone components of A and P, and let

$$D_\pm X = \partial_\pm X + [A_\pm, X]$$

be the H-covariant derivative on $\mathfrak{m}$-valued fields.

Lemma 82.11 (Symmetric-space Lax equivalence)**.** *For the bosonic symmetric-space action (82.77), the Euler–Lagrange equation together with the two $\mathfrak{h}$- and $\mathfrak{m}$-components of the Maurer–Cartan identity is equivalent to flatness, for every $\zeta \in \mathbb{C}^\times$, of the connection*

$$\mathcal{L}_+(\zeta) = A_+ + \zeta P_+, \qquad \mathcal{L}_-(\zeta) = A_- + \zeta^{-1} P_-. \tag{82.78}$$

Proof. The Maurer–Cartan equation for $g^{-1}\mathrm{d}g = A+P$ splits, using the symmetric-space brackets, into

$$F_{+-}(A) + [P_+, P_-] = 0, \tag{82.79}$$
$$D_+P_- - D_-P_+ = 0. \tag{82.80}$$

Varying g by a compactly supported $\mathfrak{m}$-valued field gives the field equation

$$D_+P_- + D_-P_+ = 0. \tag{82.81}$$

Conversely, (82.80) and (82.81) are equivalent to

$$D_+P_- = 0, \qquad D_-P_+ = 0.$$

The curvature of (82.78) is

$$\partial_+\mathcal{L}_- - \partial_-\mathcal{L}_+ + [\mathcal{L}_+, \mathcal{L}_-] = F_{+-}(A) + [P_+, P_-] + \zeta^{-1}D_+P_- - \zeta D_-P_+.$$

If (82.79), (82.80), and (82.81) hold, this curvature vanishes for all ζ. If the curvature vanishes for all $\zeta \in \mathbb{C}^\times$, the coefficients of ζ^{-1}, ζ^0, and ζ give $D_+P_- = 0$, $F_{+-}(A) + [P_+, P_-] = 0$, and $D_-P_+ = 0$, which imply the Maurer–Cartan identities and the equation of motion. This proves the equivalence. □

Projective space is the example

$$\mathbb{CP}^{N-1} = SU(N)/S(U(N-1) \times U(1)).$$

82.13 Supergroup and Supersphere Sigma-Model Families

Supergroup sigma models require the same locally super-ringed-space discipline as fermionic path integrals. A supertarget is not a set of ordinary maps. Locally it is described by even coordinate functions and odd coordinate generators in a structure sheaf; the odd coordinates are integrated by Berezin rules, not by a Borel measure. This distinction matters in sigma-model examples because cancellations in beta functions are supertrace cancellations, not cancellations between two positive measures.

Let $E = E_{\bar{0}} \oplus E_{\bar{1}}$ be a finite-dimensional real super vector space, with

$$\operatorname{sdim} E = \dim E_{\bar{0}} - \dim E_{\bar{1}}.$$

For $m, n \geq 0$, the orthosymplectic supersphere

$$S^{m-1|2n} = OSp(m|2n)/OSp(m-1|2n)$$

is the supermanifold cut out in $\mathbb{R}^{m|2n}$ by

$$X \cdot X = 1.$$

Here $X \cdot X$ uses a positive quadratic form on the even coordinates and a nondegenerate symplectic pairing on the $2n$ odd coordinates. The Euclidean sigma-model functional is

$$S_{S^{m-1|2n}}[X] = \frac{1}{2g^2} \int_\Sigma \mathrm{d}^2x\, \partial_\mu X \cdot \partial_\mu X, \qquad X \cdot X = 1, \tag{82.82}$$

understood as a local expression in the supercoordinate sheaf. The ordinary round sphere is recovered at $n = 0$, $m = N$.

Supersphere Ricci coefficient. In the standard constant-curvature normalization of (82.82), the Ricci tensor entering the one-loop sigma-model beta tensor is

$$\mathrm{Ric}_{ab} = (m - 2n - 2)h_{ab}. \tag{82.83}$$

Thus the scalar coupling has leading perturbative running

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = -\frac{m - 2n - 2}{2\pi} g^3 + O(g^5). \tag{82.84}$$

The family $S^{2n+1|2n} = OSp(2n+2|2n)/OSp(2n+1|2n)$ has vanishing one-loop coefficient.

For a bosonic round sphere S^{m-1}, the constant-curvature tensor gives $\mathrm{Ric} = (m-2)h$: the tangent space has $m-1$ dimensions and the Ricci contraction counts all tangent directions except the distinguished one in the two-plane used by the curvature operator. In the supersphere, the same normal-coordinate computation uses the graded contraction

$$\mathrm{Ric}_{ab} = (-1)^{|c|} R^{c}{}_{acb}, \tag{82.85}$$

where $|c| = 0$ on even tangent directions and $|c| = 1$ on odd tangent directions. The tangent superdimension is

$$\operatorname{sdim} TS^{m-1|2n} = m - 1 - 2n.$$

The same constant-curvature contraction therefore replaces the bosonic count $(m-1)-1$ by the graded count

$$(m - 1 - 2n) - 1 = m - 2n - 2,$$

which gives (82.83). Substitution into the one-loop sigma-model beta tensor

$$\mu \frac{\mathrm{d}h_{ab}}{\mathrm{d}\mu} = \frac{1}{2\pi} \mathrm{Ric}_{ab} + O(R^2)$$

and $h_{ab} = g^{-2} h^{\mathrm{unit}}_{ab}$ gives (82.84). Setting $m = 2n + 2$ gives zero.

The projective supermodel is the quotient analogue. Let

$$Z = (z^1, \dots, z^{p+1}; \eta^1, \dots, \eta^q)$$

be a complex supervector, with z^i even and η^α odd. The superprojective target

$$\mathbb{CP}^{p|q}$$

is obtained by imposing a nonzero homogeneous coordinate and quotienting by the even $\mathbb{C}^\times$ rescaling. In the unitary coordinate used for the sigma model, one writes a $U(1)$ redundancy

$$Z(x) \mapsto \mathrm{e}^{\mathrm{i}\alpha(x)} Z(x), \qquad A_\mu \mapsto A_\mu + \partial_\mu \alpha,$$

and

$$S_{\mathbb{CP}^{p|q}}[Z, A] = \frac{1}{g^2} \int_\Sigma \mathrm{d}^2x \, (D_\mu Z)^\dagger D_\mu Z + \mathrm{i}\frac{\theta}{2\pi} \int_\Sigma F_A, \qquad D_\mu Z = (\partial_\mu - \mathrm{i}A_\mu) Z, \tag{82.86}$$

with the constraint $Z^\dagger Z = 1$ interpreted in the superalgebra of local functions. The gauge-invariant superprojector is

$$\mathcal{P} = ZZ^\dagger, \qquad \mathcal{P}^2 = \mathcal{P}, \qquad \operatorname{str} \mathcal{P} = 1,$$

where str is the supertrace.

Projective-supermodel one-loop identity. For the Fubini–Study supermetric on $\mathbb{CP}^{p|q}$, the one-loop Ricci coefficient is

$$\mathrm{Ric} = (p + 1 - q) \, h_{\mathrm{FS}}. \tag{82.87}$$

Consequently $\mathbb{CP}^{N-1|N}$ has zero one-loop beta coefficient in this normalization, while the ordinary $\mathbb{CP}^{N-1}$ coefficient is N.

In a holomorphic coordinate patch, the Fubini–Study Kähler potential is the even function

$$K = \log(1 + w^\dagger w + \eta^\dagger \eta),$$

where w has p even components and η has q odd components. The Ricci form of a Kähler supermetric is computed by the Berezinian:

$$\rho = -\mathrm{i}\,\partial\bar{\partial} \log \mathrm{Ber}(g_{A\bar{B}}). \tag{82.88}$$

For the Fubini–Study metric, the Berezinian is proportional to

$$(1 + w^\dagger w + \eta^\dagger \eta)^{-(p+1-q)}.$$

The exponent is the ordinary projective homogeneous-coordinate count $p+1$ minus the number q of odd homogeneous coordinates, because the odd block enters the Berezinian inversely. Applying $-\mathrm{i}\partial\bar{\partial}\log$ gives (82.87). The two special cases follow by setting $q = 0$, $p = N-1$, and by setting $p = N-1$, $q = N$.

Finally, let G be a Lie supergroup whose Lie superalgebra $\mathfrak{g}$ carries a nondegenerate even invariant bilinear form κ. The principal supergroup sigma-model action is

$$S_{\mathrm{PSG}}[U] = \frac{1}{2g^2} \int_\Sigma \mathrm{d}^2 x\, \kappa(U^{-1}\partial_\mu U, U^{-1}\partial_\mu U). \tag{82.89}$$

The one-loop counterterm is proportional to the Killing form

$$K(X, Y) = \mathrm{str}_{\mathfrak{g}}(\mathrm{ad}_X\, \mathrm{ad}_Y). \tag{82.90}$$

Thus supergroups with $K = 0$, such as the projective special linear family $\mathfrak{psl}(N|N)$ and the dual-Coxeter-zero orthosymplectic family $\mathfrak{osp}(2n+2|2n)$, are distinguished perturbatively. This is a statement about the first local counterterm in a regularized superfield path integral. Construction of the Hilbert space, locality, and exact factorized scattering theory requires additional data.

Controlled Approximation 82.12 (Supercoset status)**.** The supersphere actions, projective supermodels, and principal supergroup actions above are ultraviolet family data. Their perturbative cancellations explain why $OSp(2n+2|2n)$, $\mathbb{CP}^{N-1|N}$, and $\mathfrak{psl}(N|N)$-type examples recur in the integrable and logarithmic two-dimensional literature. A theorem-level identification with a local QFT requires the same ingredients demanded in Open Problem 82.26: a regulator, Hilbert space or graded state space, local observables, conserved currents, and matching between ultraviolet and bootstrap coordinates. For supertargets, one must additionally specify the real form, the positivity or graded replacement of positivity, and the interpretation of logarithmic sectors if Jordan blocks appear.

82.14 Deformed-Sphere and Sausage Family

The sausage family should be treated as a controlled deformation of the round S^2 sigma model with a stated metric and RG trajectory before any comparison to cigar or parafermion cosets is made. The sign of the deformation parameter is part of the specification: the elongated physical sausage is the negative branch of the real parameter used below.

Definition 82.13 (Sausage metric)**.** The two-dimensional deformed-sphere, or sausage, target used in the semiclassical description is the $U(1)$-invariant Riemannian metric

$$\mathrm{d}s^2_{h,q} = h\left[\frac{\mathrm{d}r^2}{(1-r^2)(1+qr^2)} + \frac{1-r^2}{1+qr^2}\,\mathrm{d}\phi^2\right], \qquad -1 < r < 1, \quad \phi \sim \phi + 2\pi, \tag{82.91}$$

with $h > 0$ and $q > -1$. The undeformed value $q = 0$ is the round sphere of radius $\sqrt{h}$: setting $r = \cos\vartheta$ gives

$$\mathrm{d}s^2_{h,0} = h(\mathrm{d}\vartheta^2 + \sin^2\vartheta\,\mathrm{d}\phi^2).$$

For $q \neq 0$ the metric preserves only the axial $U(1)$ isometry. The pair (h,q) is therefore a metric coordinate for the sigma model; an exact quantum S-matrix proposal for the sausage family is a separate rapidity-space object that must be matched to this coordinate choice.

Curvature of the sausage target. For the metric (82.91), the scalar curvature is

$$R_{h,q}(r) = \frac{2(1+q)(1-qr^2)}{h(1+qr^2)}. \tag{82.92}$$

In particular $R_{h,0} = 2/h$, as required for the round two-sphere of radius $\sqrt{h}$.

Write the metric as $E(r)\mathrm{d}r^2 + G(r)\mathrm{d}\phi^2$, where

$$E(r) = \frac{h}{(1-r^2)(1+qr^2)}, \qquad G(r) = h\frac{1-r^2}{1+qr^2}.$$

For any diagonal two-dimensional metric of this form, direct computation from the Christoffel symbols gives

$$R = -\frac{G''}{EG} + \frac{(G')^2}{2EG^2} + \frac{E'G'}{2E^2G}. \tag{82.93}$$

Substitution of E and G into (82.93) gives (82.92). At $q = 0$, the formula becomes the constant $2/h$, agreeing with the standard scalar curvature of the round sphere.

Closure under one-loop Ricci flow. Let $g_{h,q}$ denote the metric (82.91). Under the geometric Ricci flow

$$\partial_\tau g_{ij} = -\,\mathrm{Ric}_{ij}, \tag{82.94}$$

the family $g_{h,q}$ is closed. The induced parameter equations are

$$\frac{\mathrm{d}h}{\mathrm{d}\tau} = -(1+q), \qquad \frac{\mathrm{d}q}{\mathrm{d}\tau} = -\frac{2q(1+q)}{h}. \tag{82.95}$$

Equivalently, for the physical sausage branch $q = -\kappa^2$, $0 \le \kappa < 1$,

$$\frac{\mathrm{d}h}{\mathrm{d}\tau} = -(1-\kappa^2), \qquad \frac{\mathrm{d}\kappa}{\mathrm{d}\tau} = -\frac{\kappa(1-\kappa^2)}{h}. \tag{82.96}$$

The ratio q/h^2 is constant along (82.95).

In two dimensions $\mathrm{Ric}_{ij} = \frac{1}{2}Rg_{ij}$. Write

$$g_{rr} = \frac{h}{(1-r^2)(1+qr^2)}, \qquad g_{\phi\phi} = h\frac{1-r^2}{1+qr^2}.$$

The logarithmic variation of either component under a parameter change is

$$\frac{\partial_\tau g_{ij}}{g_{ij}} = \frac{\dot h}{h} - \frac{\dot q\, r^2}{1+qr^2}. \tag{82.97}$$

The right side of (82.94), using (82.92), gives the common logarithmic factor

$$-\frac{1}{2}R_{h,q}(r) = -\frac{(1+q)(1-qr^2)}{h(1+qr^2)}. \tag{82.98}$$

Equating (82.97) and (82.98) as rational functions of r^2 gives the constant term

$$\dot h = -(1+q)$$

and the coefficient of r^2,

$$q\frac{\dot h}{h} - \dot q = \frac{q(1+q)}{h}.$$

Substituting $\dot h = -(1+q)$ gives $\dot q = -2q(1+q)/h$. The κ-form follows from $q = -\kappa^2$. Finally,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{q}{h^2}\right) = \frac{\dot q}{h^2} - \frac{2q\dot h}{h^3} = 0.$$

Integrated sausage Ricci trajectory. Fix a solution of (82.95) with initial value (h_0, q_0) at $\tau = \tau_0$. If $q_0 = 0$, then

$$h(\tau) = h_0 - (\tau - \tau_0), \qquad q(\tau) = 0. \tag{82.99}$$

On the physical sausage branch $q < 0$, write the invariant as

$$\frac{q}{h^2} = -\alpha^2, \qquad \alpha > 0, \tag{82.100}$$

so that $\kappa(\tau) = \alpha h(\tau)$. While $0 < \alpha h(\tau) < 1$,

$$h(\tau) = \frac{1}{\alpha}\tanh(\operatorname{arctanh}(\alpha h_0) - \alpha(\tau - \tau_0)), \qquad q(\tau) = -\alpha^2 h(\tau)^2. \tag{82.101}$$

On the positive branch $q > 0$, write $q/h^2 = \beta^2$, $\beta > 0$. Then

$$h(\tau) = \frac{1}{\beta}\tan(\arctan(\beta h_0) - \beta(\tau - \tau_0)), \qquad q(\tau) = \beta^2 h(\tau)^2, \tag{82.102}$$

up to the first zero or pole of the tangent chart.

The preceding Ricci-flow calculation gives the invariant q/h^2. For $q_0 = 0$, the equation for h reduces to $\dot h = -1$, which gives (82.99).

On the negative branch, substitute $q = -\alpha^2 h^2$ into (82.95). The scalar equation is

$$\dot h = -(1 - \alpha^2 h^2). \tag{82.103}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}h}\operatorname{arctanh}(\alpha h) = \frac{\alpha}{1-\alpha^2 h^2},$$

(82.103) is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\operatorname{arctanh}(\alpha h(\tau)) = -\alpha.$$

Integrating from τ_0 to τ gives (82.101). The relation $q = -\alpha^2 h^2$ then supplies the second coordinate. The condition $\alpha h < 1$ is exactly $q > -1$, the Riemannian domain of Definition 82.13.

On the positive branch, substitute $q = \beta^2 h^2$. Then

$$\dot{h} = -(1 + \beta^2 h^2),$$

and

$$\frac{\mathrm{d}}{\mathrm{d}h}\arctan(\beta h) = \frac{\beta}{1 + \beta^2 h^2}.$$

Thus $\frac{\mathrm{d}}{\mathrm{d}\tau}\arctan(\beta h(\tau)) = -\beta$, and integration gives (82.102).

Remark 82.14 (Round and cylindrical endpoints of the metric family). The branch $q = -\kappa^2$, $0 \leq \kappa < 1$, interpolates at the metric level between the round two-sphere and a flat cylinder. More precisely, $\kappa = 0$ gives S^2 of radius $\sqrt{h}$, while the limiting metric as $\kappa \to 1$ is

$$\mathrm{d}s^2_{h,-1} = h(\mathrm{d}\rho^2 + \mathrm{d}\phi^2), \qquad r = \tanh\rho, \quad \rho \in \mathbb{R}. \tag{82.104}$$

The case $\kappa = 0$ is the calculation already displayed in Definition 82.13. For $\kappa \to 1$, set $q = -1$ in (82.91). Then

$$\frac{\mathrm{d}r^2}{(1-r^2)^2} + \mathrm{d}\phi^2$$

is obtained. The change of variable $r = \tanh\rho$ gives $\mathrm{d}r = (1 - \tanh^2\rho)\mathrm{d}\rho = (1 - r^2)\mathrm{d}\rho$, hence $\mathrm{d}r^2/(1-r^2)^2 = \mathrm{d}\rho^2$. This proves (82.104).

82.14.1 Repulsive-Domain Fateev–Onofri–Zamolodchikov Sausage Scattering

The metric/RG data above are not themselves a factorized scattering theory. The Fateev–Onofri–Zamolodchikov massive sausage scattering proposal is a one-parameter $U(1)$-invariant deformation of the $O(3)$ triplet scattering theory. We record the repulsive-domain version because it is the exact rapidity-space object against which any regulator-defined sausage sigma model must be matched.

Definition 82.15 (Repulsive massive sausage scattering proposal)**.** Let $0 \leq \lambda < \frac{1}{2}$. The one-particle labels are

$$a \in \{+, 0, -\}, \qquad Q(+) = 1, \quad Q(0) = 0, \quad Q(-) = -1,$$

all with the same mass $m > 0$. The two-particle scattering matrix preserves the total $U(1)$ charge. In the Zamolodchikov–Faddeev convention used in Section 82.1, the nonzero entries generated from charge conjugation and exchange by the following formulae are

$$S_{++}^{++}(\theta) = S_{+-}^{+-}(\mathrm{i}\pi - \theta) = \frac{\sinh\lambda(\theta - \mathrm{i}\pi)}{\sinh\lambda(\theta + \mathrm{i}\pi)}, \tag{82.105}$$

$$S_{+0}^{0+}(\theta) = S_{+-}^{00}(\mathrm{i}\pi - \theta) = \frac{-\mathrm{i}\sin(2\pi\lambda)}{\sinh\lambda(\theta - 2\mathrm{i}\pi)} S_{++}^{++}(\theta), \tag{82.106}$$

$$S_{+0}^{+0}(\theta) = \frac{\sinh(\lambda\theta)}{\sinh\lambda(\theta - 2\mathrm{i}\pi)} S_{++}^{++}(\theta), \tag{82.107}$$

$$S_{+-}^{-+}(\theta) = -\frac{\sin(\pi\lambda)\sin(2\pi\lambda)}{\sinh\lambda(\theta - 2\mathrm{i}\pi)\,\sinh\lambda(\theta + \mathrm{i}\pi)}, \tag{82.108}$$

$$S_{00}^{00}(\theta) = S_{+0}^{+0}(\theta) + S_{-+}^{+-}(\theta). \tag{82.109}$$

Here $S_{-+}^{+-} = S_{+-}^{-+}$ and the remaining charge-conjugate entries are fixed by the simultaneous replacement $+ \leftrightarrow -$. No other matrix element is nonzero in this scattering proposal. At $\lambda = 0$ the formula is understood as the meromorphic limit. The parameter λ is a bootstrap coordinate. A comparison with a metric-coordinate deformation such as (h, q) in Definition 82.13 requires a separate matching map; the semiclassical and TBA/NLIE analyses suggest using a coordinate ν satisfying

$$\frac{\nu}{4\pi} = \frac{\lambda}{1 - 2\lambda}, \tag{82.110}$$

but in this monograph (82.110) is a matching coordinate relation, not a theorem identifying a constructed local QFT.

Repulsive-sausage strip analyticity and block unitarity checks. For $0 \le \lambda < \frac{1}{2}$, the displayed denominator factors in (82.105)–(82.108) have no zeros in the open physical strip

$$\mathcal{S}_{\mathrm{phys}} = \{\theta \in \mathbb{C} : 0 < \operatorname{Im}\theta < \pi\}.$$

Moreover, the charge $+2$, charge $+1$, and charge 0 blocks obey two-particle unitarity on the real rapidity axis. Explicitly, for real $\theta \neq 0$,

$$S_{++}^{++}(\theta) S_{++}^{++}(-\theta) = 1,$$

and, in the ordered charge-$+1$ basis $(|+0\rangle, |0+\rangle)$, the matrix

$$S_{+1}(\theta) = \begin{pmatrix} A(\theta) & B(\theta) \\ B(\theta) & A(\theta) \end{pmatrix},$$

where

$$A(\theta) = S_{+0}^{+0}(\theta), \qquad B(\theta) = S_{+0}^{0+}(\theta),$$

satisfies

$$S_{+1}(\theta) S_{+1}(-\theta) = I_2.$$

In the ordered charge-zero basis $(|+-\rangle, |00\rangle, |-+\rangle)$, write

$$S_{Q=0}(\theta) = \begin{pmatrix} E(\theta) & F(\theta) & G(\theta) \\ F(\theta) & H(\theta) & F(\theta) \\ G(\theta) & F(\theta) & E(\theta) \end{pmatrix},$$

where

$$E = S_{+-}^{+-}, \qquad F = S_{+-}^{00}, \qquad G = S_{+-}^{-+}, \qquad H = S_{00}^{00}.$$

Then

$$S_{Q=0}(\theta)S_{Q=0}(-\theta) = I_3.$$

These are explicit bootstrap consistency checks for the written repulsive-sausage scattering proposal. The constructive question of whether this proposal is realized by a regulator-defined local sigma model is stated separately in Controlled Approximation 82.21.

The possible zeros of $\sinh \lambda(\theta + \mathrm{i}\pi)$ occur at

$$\theta = \mathrm{i}\pi \left(\frac{n}{\lambda} - 1\right), \qquad n \in \mathbb{Z},$$

when $\lambda > 0$. For $n \leq 0$ the imaginary part is at most $-\pi$; for $n \geq 1$ it is at least $\pi(1/\lambda - 1) > \pi$. Hence there is no zero in the open physical strip. The possible zeros of $\sinh \lambda(\theta - 2\mathrm{i}\pi)$ occur at

$$\theta = \mathrm{i}\pi \left(\frac{n}{\lambda} + 2\right).$$

For $n \geq 0$ the imaginary part is at least 2π; for $n \leq -1$ it is at most $2\pi - \pi/\lambda \leq 0$. Thus these denominators also have no physical-strip zeros. At $\lambda = 0$ the statement follows from the meromorphic limiting rational functions.

Set $x = \lambda\theta$ and $a = \pi\lambda$. The highest-charge block is

$$D(x) = \frac{\sinh(x - \mathrm{i}a)}{\sinh(x + \mathrm{i}a)}.$$

Since $D(-x) = D(x)^{-1}$, the charge $+2$ unitarity relation follows. For the charge $+1$ block,

$$A(x) = \frac{\sinh x}{\sinh(x - 2\mathrm{i}a)} D(x), \qquad B(x) = \frac{-\mathrm{i}\sin(2a)}{\sinh(x - 2\mathrm{i}a)} D(x).$$

Using $D(-x) = D(x)^{-1}$,

$$A(-x) = \frac{\sinh x}{\sinh(x + 2\mathrm{i}a)} D(x)^{-1}, \qquad B(-x) = \frac{\mathrm{i}\sin(2a)}{\sinh(x + 2\mathrm{i}a)} D(x)^{-1}.$$

Therefore

$$A(x)A(-x) + B(x)B(-x) = \frac{\sinh^2 x + \sin^2(2a)}{\sinh(x - 2\mathrm{i}a)\sinh(x + 2\mathrm{i}a)} = 1,$$

because $\sinh(x - 2\mathrm{i}a)\sinh(x + 2\mathrm{i}a) = \sinh^2 x + \sin^2(2a)$. Similarly

$$A(x)B(-x) + B(x)A(-x) = 0.$$

These two scalar identities are exactly $S_{+1}(\theta)S_{+1}(-\theta) = I_2$.

It remains to check the charge-zero block. With the same notation $x = \lambda\theta$, $a = \pi\lambda$, set

$$s_k = \sinh(x + \mathrm{i}ka), \qquad u = \sin a, \qquad v = \sin(2a).$$

The entries of $S_{Q=0}$ are

$$E = \frac{s_0}{s_{-2}}, \qquad F = \frac{\mathrm{i}vs_0}{s_{-2}s_1}, \qquad G = -\frac{uv}{s_{-2}s_1}, \qquad H = \frac{s_0 s_{-1} - uv}{s_{-2}s_1}. \tag{82.111}$$

At rapidity $-\theta$, the replacement is $s_k \mapsto -s_{-k}$. The antisymmetric vector $(1,0,-1)$ has eigenvalue $E-G$, while the symmetric subspace spanned by $(1,0,1)$ and $(0,1,0)$ has matrix

$$\begin{pmatrix} E+G & \sqrt{2}F \\ \sqrt{2}F & H \end{pmatrix}.$$

Thus $S_{Q=0}(\theta)S_{Q=0}(-\theta) = I_3$ is equivalent to the four scalar identities

$$(E-G)(E_- - G_-) = 1, \tag{82.112}$$

$$(E+G)(E_- + G_-) + 2FF_- = 1, \tag{82.113}$$

$$(E+G)F_- + FH_- = 0, \tag{82.114}$$

$$2FF_- + HH_- = 1, \tag{82.115}$$

where $E_- = E(-\theta)$, and similarly for F, G, H. Substituting (82.111), using $s_k(-x) = -s_{-k}(x)$, and repeatedly applying

$$\sinh(y + \mathrm{i}b)\sinh(y - \mathrm{i}b) = \sinh^2 y + \sin^2 b$$

reduces the left sides of (82.112)–(82.115) exactly to $1, 1, 0, 1$, respectively. This gives the charge-zero unitarity block.

Definition 82.16 (Full sausage triplet matrix)**.** Let V_s be the three-dimensional vector space with ordered basis (e_+, e_0, e_-) and charges $Q(e_\pm) = \pm 1$, $Q(e_0) = 0$. The entries in Definition 82.15 define a meromorphic endomorphism

$$S_\lambda(\theta) : V_\mathrm{s} \otimes V_\mathrm{s} \longrightarrow V_\mathrm{s} \otimes V_\mathrm{s}$$

by declaring all charge-nonconserving matrix elements to vanish and by using charge conjugation $+ \leftrightarrow -$ for the negative-charge blocks. In the ordered two-particle bases

$$Q = 2: \ (++), \qquad Q = 1: \ (+0, 0+), \qquad Q = 0: \ (+-, 00, -+),$$

the nontrivial blocks are respectively

$$(S^{++}_{++}), \qquad \begin{pmatrix} S^{+0}_{+0} & S^{0+}_{+0} \\ S^{0+}_{+0} & S^{+0}_{+0} \end{pmatrix}, \qquad \begin{pmatrix} S^{+-}_{+-} & S^{00}_{+-} & S^{-+}_{+-} \\ S^{00}_{+-} & S^{00}_{00} & S^{00}_{+-} \\ S^{-+}_{+-} & S^{00}_{+-} & S^{+-}_{+-} \end{pmatrix}.$$

The $Q = -1, -2$ blocks are obtained by simultaneous charge conjugation.

Example 82.17 (Exact Yang–Baxter verification for the displayed sausage matrix)**.** For $0 \le \lambda < \frac{1}{2}$, the meromorphic matrix $S_\lambda(\theta)$ of Definition 82.16 satisfies the Zamolodchikov–Faddeev Yang–Baxter equation

$$S_{\lambda,12}(\theta)S_{\lambda,13}(\theta+\phi)S_{\lambda,23}(\phi) = S_{\lambda,23}(\phi)S_{\lambda,13}(\theta+\phi)S_{\lambda,12}(\theta) \tag{82.116}$$

as a meromorphic identity on $V_\mathrm{s}^{\otimes 3}$.

This is a finite algebraic identity. We give the reduction because a sampled numerical Yang–Baxter check would not be evidence for the meromorphic statement. Put

$$q = e^{\mathrm{i}\pi\lambda}, \qquad u = e^{\lambda\theta}, \qquad v = e^{\lambda\phi},$$

and, for an integer k, abbreviate

$$[z]_k = \frac{1}{2}\left(zq^k - z^{-1}q^{-k}\right) = \sinh(\lambda\vartheta + \mathrm{i}k\pi\lambda) \quad \text{when } z = e^{\lambda\vartheta}.$$

In these variables the highest-charge entry is the rational function

$$D(z) = \frac{[z]_{-1}}{[z]_1},$$

the charge-one entries are

$$A(z) = \frac{[z]_0}{[z]_{-2}} D(z), \qquad B(z) = -\frac{q^2 - q^{-2}}{2[z]_{-2}} D(z),$$

and the charge-zero exchange entry is

$$G(z) = \frac{(q - q^{-1})(q^2 - q^{-2})}{4[z]_{-2}[z]_1}.$$

The crossed charge-zero entries are obtained by replacing z with q/z:

$$E(z) = D(q/z), \qquad F(z) = B(q/z), \qquad H(z) = A(z) + G(z).$$

Thus every entry of $S_\lambda(\theta)$, $S_\lambda(\phi)$, and $S_\lambda(\theta + \phi)$ is an element of the rational function field $\mathbb{Q}(q, u, v)$, with $z = u, v, uv$, respectively. After substituting these entries into (82.116), charge conservation leaves 141 possibly nonzero scalar component equations. Clearing the common denominators in each component gives 141 Laurent polynomial identities in q, u, v. Multiplication by a sufficiently large monomial $q^a u^b v^c$ turns each into an ordinary polynomial identity over $\mathbb{Q}$. Direct expansion and collection gives the zero polynomial in every component.

The companion file `calculation-checks/sigma_model_family_checks.py` implements exactly this reduction in sparse tensor notation. It constructs the 19 nonzero two-particle components, forms the two sides of (82.116) on $V_{\mathrm{s}}^{\otimes 3}$, and verifies the 141 numerator polynomials symbolically in $\mathbb{Q}(q, u, v)$, with no floating-point specialization of q, u, v. Since the two sides are meromorphic functions, equality of all cleared numerators proves equality away from the union of their pole divisors, and therefore proves the meromorphic identity.

Example 82.17 is an algebraic bootstrap identity for the displayed meromorphic matrix. It is not a construction of the local sausage sigma model, nor a proof that the regulator-defined metric family has scattering matrix S_λ.

Remark 82.18 (Rational $O(3)$ limit of the repulsive sausage matrix)*.* At fixed rapidity away from the displayed poles, the limit $\lambda \to 0$ of the repulsive sausage matrix exists. The highest-charge and charge-$+1$ entries become

$$S_{++}^{++}(\theta) \longrightarrow \frac{\theta - \mathrm{i}\pi}{\theta + \mathrm{i}\pi}, \tag{82.117}$$

$$S_{+0}^{+0}(\theta) \longrightarrow \frac{\theta}{\theta - 2\mathrm{i}\pi}\frac{\theta - \mathrm{i}\pi}{\theta + \mathrm{i}\pi}, \tag{82.118}$$

$$S_{+0}^{0+}(\theta) \longrightarrow \frac{-2\mathrm{i}\pi}{\theta - 2\mathrm{i}\pi}\frac{\theta - \mathrm{i}\pi}{\theta + \mathrm{i}\pi}. \tag{82.119}$$

The charge-zero block has the corresponding charge-conjugate rational limit obtained by the crossing relations in Definition 82.15. Thus the repulsive deformation is normalized so that its $\lambda = 0$ endpoint is the same rational $O(3)$ triplet block used in the round sigma-model bootstrap coordinate.

Use

$$\sinh(\lambda z) = \lambda z + O(\lambda^3) \quad \text{and} \quad \sin(2\pi\lambda) = 2\pi\lambda + O(\lambda^3).$$

Substitution in (82.105) gives (82.117). Substitution in (82.107) and (82.106), using the already computed highest-charge limit, gives (82.118) and (82.119). The charge-zero entries are generated by the same meromorphic crossing prescriptions that define them at nonzero λ, so their limits follow by applying those prescriptions to the three displayed rational functions.

Definition 82.19 (Massless $\theta = \pi$ endpoint problem)**.** A massless endpoint statement for the S^2 or sausage family at topological angle $\theta_{\text{top}} = \pi$ is a statement about a scaling limit of regulator-defined local QFTs, not a statement about the massive repulsive matrix S_λ by itself. A precise endpoint claim consists of:

1. a sequence of regularized sigma models with target metric g_ϵ, topological term $\mathrm{i}\theta_{\text{top},\epsilon}Q$, and local operator algebra $\mathcal{A}_\epsilon$;
2. a correlation-length scale $\xi_\epsilon \to \infty$ and a rule for holding finite renormalized distances as $\epsilon \to 0$;
3. candidate limiting stress tensor and current operators, obtained as renormalized limits of local fields in $\mathcal{A}_\epsilon$;
4. convergence of finite-volume energies or Euclidean correlation functions to the $SU(2)_1$ WZW conformal data, including $c = 1$, affine currents of level 1, and the allowed primary dimensions;
5. a statement of which global form of the $SO(3)$ or $SU(2)$ symmetry acts faithfully on the limiting Hilbert space.

The Haldane–Affleck picture supplies a powerful physical route to this endpoint, but the endpoint assertion is not identical to the massive factorized scattering proposal in Definition 82.15.

Open Problem 82.20 (Constructive massless endpoint for the sausage family)*.* Give a regulator-level proof of the massless $\theta_{\text{top}} = \pi$ endpoint for a declared branch of the round or deformed S^2 sigma model, including reflection-positive Euclidean correlators, the limiting stress tensor, the $SU(2)_1$ current algebra, and the convergence of finite-volume spectra. A satisfactory proof would also identify how the massive repulsive bootstrap coordinate λ, if used as an analytic continuation device, is related to the endpoint scaling coordinates.

Controlled Approximation 82.21 (Sausage bootstrap matching)**.** Equations (82.91), (82.95), and (82.104) are metric and one-loop RG data, while Definition 82.15 is a rapidity-space scattering proposal satisfying the bootstrap consistency tests above. A theorem identifying the two requires a regulator-defined local sigma model, conserved charges or an equivalent factorization construction, Haag–Ruelle scattering states, and a matching map from the regulator-defined couplings to the bootstrap coordinate λ. This matching is especially delicate near the cylindrical endpoint, where the target develops an asymptotic circle and the relation to compact parafermion or noncompact cigar descriptions depends on the precise scaling limit. The elementary strip-analyticity and block-unitarity checks above are necessary bootstrap consistency tests, and Example 82.17 is the factorization identity for the meromorphic matrix; neither replaces the missing constructive identification. In attractive analytic continuations, physical-strip poles must be accompanied by declared bound-state particles and residue normalizations before one can speak of a complete spectrum. The repulsive domain treated here has no such denominator poles by the strip-analyticity check above. A massless $\theta_{\text{top}} = \pi$ endpoint statement requires the separate scaling data of Definition 82.19.

Definition 82.22 (Sigma-model family specification)**.** A sigma-model family specification consists of:

1. a target, quotient, or supertarget X, including its global and local redundancy groups;
2. a regulator definition of maps $\Sigma \to X$ and of the functional measure or Berezin functional;
3. coupling coordinates in the metric, B-field, Wess–Zumino, and θ-angle sectors;
4. classical integrability structures, such as a Lax connection, when present;
5. ultraviolet beta functions or a nonperturbative mass-generation calculation;
6. a proposed bootstrap spectrum and S-matrix, with CDD choices and bound state poles specified;
7. a matching statement between the regulator construction and bootstrap specification.

The families discussed in this chapter are the round $O(N)$ model, $\mathbb{CP}^{N-1}$, principal chiral models, WZW endpoints, symmetric-space cosets, supergroup variants, and deformed-sphere trajectories.

82.15 Gross–Neveu Ultraviolet Action

Let ψ^i, $i = 1, \dots, N$, be Majorana fermions in two Lorentzian dimensions. The $O(N)$ Gross–Neveu action is

$$S_{\text{GN}} = \int \mathrm{d}^2x \left[\frac{\mathrm{i}}{2}\psi^i\gamma^\mu\partial_\mu\psi^i + \frac{g^2}{8}(\bar{\psi}^i\psi^i)^2\right].$$

Introducing an auxiliary scalar σ gives the equivalent finite-cutoff functional

$$S_{\text{GN}}[\psi,\sigma] = \int \mathrm{d}^2x \left[\frac{\mathrm{i}}{2}\psi^i\gamma^\mu\partial_\mu\psi^i - \frac{1}{2}\sigma\,\bar{\psi}^i\psi^i - \frac{1}{2g^2}\sigma^2\right],$$

because the Gaussian equation of motion is

$$\sigma = -\frac{g^2}{2}\bar{\psi}^i\psi^i.$$

Substitution returns the quartic interaction. The auxiliary-field form makes the large-N mass gap equation explicit: for a constant saddle $\sigma = m$,

$$\frac{m}{g^2} = Nm\int^\Lambda \frac{\mathrm{d}^2p_E}{(2\pi)^2}\frac{1}{p_E^2 + m^2}.$$

For $m \neq 0$,

$$\frac{1}{Ng^2} = \frac{1}{4\pi}\log\left(\frac{\Lambda^2 + m^2}{m^2}\right).$$

Thus a fixed physical mass m requires logarithmic dependence of g on the regulator scale. Equivalently, with $\lambda_{\mathrm{GN}} = Ng^2$,

$$m = \frac{\Lambda}{\sqrt{\exp(4\pi/\lambda_{\mathrm{GN}}) - 1}} \sim \Lambda \exp\left(-\frac{2\pi}{\lambda_{\mathrm{GN}}}\right).$$

This derivation is a saddle calculation in a specified regularized functional integral; the exact finite-N factorized QFT is a separate construction.

82.16 Regulator-to-Bootstrap Matching

The preceding sections deliberately separated three kinds of statements. First, a sigma model can be specified by ultraviolet local structures: a target or quotient, a regularized functional integral, coupling coordinates, global symmetries, and local composite fields. Second, integrability can be stated by classical Lax equations or by finite-dimensional Yang–Baxter identities for rapidity-space matrices. Third, an exact scattering proposal becomes a statement about a local QFT only after the Hilbert space, one-particle subspaces, scattering states, and local observables have been constructed from the same regulator.

Definition 82.23 (Bootstrap specification for a massive factorizing family)**.** A massive factorizing bootstrap specification consists of:

1. a finite list of one-particle species a, with masses $m_a > 0$, internal Hilbert spaces V_a, charge conjugates $\bar{a}$, and a positive inner product on each V_a;
2. meromorphic two-particle matrices

$$S_{ab}(\theta) : V_a \otimes V_b \longrightarrow V_b \otimes V_a$$

satisfying real-rapidity unitarity, the appropriate crossing equations, and the Yang–Baxter equation as identities of meromorphic operators;
3. a physical-strip pole assignment that assigns each pole either to a stable bound-state multiplet with a positive residue in the Hilbert-space convention, to a specified on-shell composite singularity, or to a zero of the scalar factor that cancels an unphysical matrix pole;
4. CDD declaration, namely a statement that no additional physical-strip zeros or poles are present except those just assigned;
5. form-factor residues for the local currents and stress tensor, or an equivalent construction of the local operator algebra;
6. a finite-volume spectral prescription, such as a Bethe–Yang or TBA system, with kernels defined from the channel eigenvalues by

$$K_R(\theta) = \frac{1}{2\pi \mathrm{i}} \partial_\theta \log S_R(\theta)$$

wherever a diagonal channel R is being used.

The word "exact" refers to this whole specification, not only to the displayed matrix part of S_{ab}.

Definition 82.24 (Regulator realization of a bootstrap specification)**.** Let $\mathcal{B}$ be a bootstrap specification in the sense of Definition 82.23. A regulator realization of $\mathcal{B}$ is a sequence of regularized local theories $\mathfrak{R}_\epsilon$, $\epsilon \downarrow 0$, together with a choice of renormalized coupling coordinates, such that:

1. the Euclidean Schwinger functions or Lorentzian Wightman functions of local observables converge, after renormalization, to distributions obeying the positivity, covariance, spectral, and locality properties needed for Haag–Ruelle scattering in the neutral massive sector under discussion;
2. the limiting mass operator has isolated one-particle subspaces $\mathcal{H}_{1,a} \simeq V_a \otimes L^2(H^+_{m_a}, \mathrm{d}\mu_{m_a})$, with the same masses and symmetry representations as $\mathcal{B}$;
3. the local currents, stress tensor, and any topological or Noether charges obtained from the regulator act on these one-particle subspaces in the representations assigned by $\mathcal{B}$;
4. Haag–Ruelle wave operators exist for the stable species and the two-particle scattering kernel in rapidity variables is the boundary value of a meromorphic function on the domain used by the bootstrap specification;
5. this meromorphic scattering kernel, including scalar factors and residue normalizations, equals the S_{ab} of $\mathcal{B}$;
6. finite-volume energies and local form factors computed from the regulator converge to the Bethe–Yang/TBA and form-factor structures specified in $\mathcal{B}$, with the same mass coordinate.

The matching map includes the relation between the regulator mass scale, renormalized couplings, θ-angle or WZ level, and the rapidity-space bootstrap parameter. Without this map, a bootstrap matrix and a sigma-model action are two compatible coordinate systems, not yet one constructed QFT.

Analytic uniqueness after regulator matching. Assume a regulator realization supplies Haag–Ruelle two-particle scattering for species a, b, and that the resulting two-particle kernel $S^{\mathrm{HR}}_{ab}(\theta)$ and a bootstrap matrix $S^{\mathrm{boot}}_{ab}(\theta)$ are meromorphic operator-valued functions on a connected domain $\Omega \subset \mathbb{C}$ containing a nonempty real rapidity interval I in its boundary-value set. Suppose their boundary values on I exist as tempered distribution kernels and agree there as operators

$$V_a \otimes V_b \longrightarrow V_b \otimes V_a.$$

Assume also that the difference carries no boundary-supported contact term in the two-particle rapidity variable. These hypotheses imply the meromorphic identity

$$S^{\mathrm{HR}}_{ab}(\theta) = S^{\mathrm{boot}}_{ab}(\theta), \qquad \theta \in \Omega.$$

Here is the complete analytic step. Choose bases of V_a and V_b. For each matrix entry set

$$F(\theta) = (S^{\mathrm{HR}}_{ab})^{\alpha'\beta'}_{\beta\alpha}(\theta) - (S^{\mathrm{boot}}_{ab})^{\alpha'\beta'}_{\beta\alpha}(\theta).$$

On any relatively compact subdomain avoiding accumulation of pole divisors, multiply by the finite product of local factors that cancel the poles met in that subdomain. The result is a holomorphic function $\widetilde{F}$ on a connected open set whose boundary still contains a real interval $I_0 \subset I$. The equality of distributional boundary values on I_0, together with the stated exclusion of a boundary-supported term, says that the boundary distribution of $\widetilde{F}$ vanishes on I_0. Fix

$x_0 \in I_0$ and a disc D centered at x_0 whose upper half lies in the holomorphy domain. Define $G = \widetilde{F}$ on the upper half-disc and $G = 0$ on the lower half-disc. For a triangle $T \subset D$ crossing the diameter, replace T by $T \cap \{\Im z > \varepsilon\}$ and let $\varepsilon \downarrow 0$. The holomorphic integral over the truncated triangle is zero, and the only limiting contribution from the diameter is the vanishing boundary distribution paired with the test density induced by the edge of T. Hence $\int_{\partial T} G \, \mathrm{d}z = 0$ for every such triangle; Morera's theorem makes G holomorphic across the interval. Since G vanishes on the lower half-disc, the identity theorem gives $\widetilde{F} = 0$ in the upper half-disc. Analytic continuation then carries the vanishing through the connected component. Dividing by the pole-cancelling factors gives $F = 0$ away from the removed divisors, and meromorphic continuation extends the equality across them.

The mathematical burden in a sigma-model bootstrap comparison is therefore the construction of the continuum regulator realization: existence of the neutral Haag–Ruelle scattering theory, control of the meromorphic boundary values, identification of poles and residues, and matching of the finite-volume and form-factor structures in Definition 82.23. The paragraph above is only the final uniqueness step once those structures have already been obtained.

Definition 82.25 (Family status summary for this chapter)**.** The families developed in this chapter have the following status in the terminology above.

1. The $O(N)$ sigma model has a regulator action, asymptotic freedom, large-N mass generation, and a vector bootstrap matrix with the standard tensor decomposition. The chapter records the bootstrap consistency data; a fully constructive realization remains an open problem.

2. The $O(N)$ Gross–Neveu model has the auxiliary-field ultraviolet action, the large-N gap equation, and the same factorizing-scattering logic in the fermionic representation. The bosonic WZW/current-current bridge is the nonabelian-bosonization coordinate on the same local current algebra, after the $U(1)$ and quotient data are fixed.

3. The $\mathbb{CP}^{N-1}$ family is specified by the constrained complex field, auxiliary $U(1)$ quotient connection, projector action, topological charge, instanton bound, large-N gap equation, induced Maxwell kernel, and projective tensor/crossing data. For $N = 2$, the projector identifies the model with the $O(3)$ target with the coupling conversion derived above. For $N > 2$, the chapter treats exact projective scattering as a bootstrap and construction problem rather than importing a scalar factor without a regulator-level proof.

4. Principal chiral models are developed through their $G_L \times G_R$ action, Lax equivalence, Wess–Zumino deformation, Polyakov–Wiegmann identity, WZW endpoint current algebra, KZ equations, GKO cosets, and the $SU(N)$ sine-law/bootstrap matrix and scalar-factor conventions. The matrix and scalar unitarity identities are proved in the displayed convention; the constructive matching to a regulator remains a separate construction problem.

5. Symmetric-space sigma models are developed through the reductive quotient G/H, local H redundancy, and the Lax equivalence for symmetric spaces. This proves the classical integrability statement used by the family classification.

6. Supersphere, projective-super, and principal-supergroup examples are defined as locally superringed target theories. Their one-loop cancellations are supertrace/Berezinian statements, not statements about positive Borel measures. Positivity, logarithmic sectors, and exact scattering require separate constructions.

7. The deformed-sphere sausage family is specified by its metric, scalar curvature, one-loop

Ricci trajectory, round and cylindrical metric limits, the repulsive-domain triplet scattering proposal, real-rapidity unitarity, and the algebraic Yang–Baxter identity for the displayed meromorphic matrix. The massless $\theta_{\mathrm{top}} = \pi$ endpoint and the regulator-to-λ matching are separate scaling problems.

Open Problem 82.26 (Constructive factorizing sigma-model families). Construct the massive factorizing sigma-model families discussed in this chapter as local QFTs with Hilbert spaces or graded state spaces, local algebras, conserved currents, mass gaps, Haag–Ruelle scattering states, and exact S-matrices in one framework where the ultraviolet regulator definitions, bootstrap specifications, form factors, and finite-volume spectra are proved to match. At present the chapter supplies the regulator definitions, classical integrability identities, large-N controls, algebraic bootstrap checks, and scalar-factor consistency checks that such a theorem must connect; the constructive identification remains a genuine theorem-level problem.

Chapter 83

Bridges To Nonintegrable Two-Dimensional QFT And CFT

Conventions in this chapter.

- **Signature.** Lorentzian local-operator data and Euclidean radial or conformal geometry where stated.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for the Lorentzian or radial-quantization vacuum.
- **Generator convention.** Poincare translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$; represented conformal charges use $\exp(\mathrm{i}\epsilon^A \widehat{Q}_A)$ when a unitary Hilbert-space action is present.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Exact integrability supplies unusually explicit coordinates on a local QFT: asymptotic particles, factorized scattering, form factors, finite-volume Bethe–Yang spectra, and TBA vacuum energies. A nearby nonintegrable theory is studied by deforming these data through a declared regulator and local operator chart. The bridge developed in this chapter is local and finite-volume first; infinite-volume resonance poles and decay widths are defined as limits of finite-volume matrix elements.

83.1 Integrability-Breaking Datum

Let $\mathcal{Q}_0$ be a massive integrable two-dimensional QFT with Hilbert space $\mathcal{H}_0$, Hamiltonian H_0, local operator algebra $\mathcal{A}_0$, and commuting conserved charges

$$Q_s, \qquad s \in \mathcal{S} \subset \mathbb{Z}.$$

The charge Q_s acts diagonally on an asymptotic multiparticle state

$$\alpha = (a_1, \theta_1; \ldots; a_n, \theta_n)$$

with eigenvalue

$$q_s(\alpha) = \sum_{j=1}^{n} q_s^{(a_j)}(\theta_j),$$

where $q_s^{(a)}(\theta)$ is the one-particle charge eigenvalue. For ordinary energy and momentum,

$$q_1^{(a)}(\theta) = m_a \cosh\theta, \qquad \widetilde{q}_1^{(a)}(\theta) = m_a \sinh\theta.$$

Definition 83.1 (Regulated integrability-breaking deformation)**.** A regulated deformation of $\mathcal{Q}_0$ is a family of finite-volume Hamiltonians

$$H(\epsilon, L, \Lambda) = H_0(L, \Lambda) + \epsilon V_b(L, \Lambda) + \sum_A c_A(\epsilon, \Lambda)\, V_A(L, \Lambda),$$

where

$$V_b(L, \Lambda) = \int_0^L dx\, \mathcal{O}_b(x)_\Lambda, \qquad V_A(L, \Lambda) = \int_0^L dx\, \mathcal{O}_A(x)_\Lambda.$$

Here L is the spatial circumference, Λ is a UV cutoff, $\mathcal{O}_b$ is the declared perturbing local operator, and the finite set of local coordinates c_A is fixed by the chosen regularization and renormalization condition. The continuum deformation is the limit of this family in matrix elements or Schwinger functions after these local coordinates have been specified.

The deformation parameter ϵ has engineering dimension

$$[\epsilon] = 2 - \Delta_b$$

when $\mathcal{O}_b$ has scaling dimension Δ_b at a conformal short-distance fixed point. Away from a CFT, the same symbol denotes the coordinate multiplying the integrated local operator in the regulated Hamiltonian.

83.2 Broken Conservation Laws

Let $V_b = \int dx\, \mathcal{O}_b(x)$ in infinite volume, with a finite-volume regulator understood. At $\epsilon = 0$,

$$[H_0, Q_s] = 0.$$

The deformed time derivative of Q_s is

$$\frac{dQ_s}{dt} = \mathrm{i}[H(\epsilon), Q_s] = \mathrm{i}\epsilon[V_b, Q_s] + O(\epsilon^2).$$

If $Q_s = \int dx\, j_s^0(x)$, this is the integrated form of

$$\partial_\mu j_s^\mu(x) = \epsilon\, \mathcal{B}_s(x) + O(\epsilon^2),$$

where $\mathcal{B}_s$ is the local operator representing the charge-breaking density in the chosen operator chart.

Broken-charge matrix element identity. Let α and β be finite-particle scattering states of the integrable theory on which Q_s has eigenvalues $q_s(\alpha)$ and $q_s(\beta)$. Then the first-order broken-charge matrix element obeys

$$\langle\beta|\mathrm{i}[V_b, Q_s]|\alpha\rangle = \mathrm{i}\big(q_s(\alpha) - q_s(\beta)\big)\langle\beta|V_b|\alpha\rangle.$$

Since $Q_s|\alpha\rangle = q_s(\alpha)|\alpha\rangle$ and $\langle\beta|Q_s = q_s(\beta)\langle\beta|$,

$$\langle\beta|V_b Q_s|\alpha\rangle = q_s(\alpha)\langle\beta|V_b|\alpha\rangle, \qquad \langle\beta|Q_s V_b|\alpha\rangle = q_s(\beta)\langle\beta|V_b|\alpha\rangle.$$

Subtracting the two identities and multiplying by i gives the formula.

The equation is useful because the integrable charges give many independent kinematic tests. A matrix element that changes any higher-spin charge eigenvalue is a direct channel through which the deformation breaks factorized scattering. Energy and momentum remain conserved when the deformed Hamiltonian is translation invariant; higher charges typically do not.

83.3 Form-Factor Perturbation Theory

Let $|A_a(\theta)\rangle$ denote a one-particle state of species a and rapidity θ in the integrable theory. The Lorentzian normalization used in this volume is

$$\langle A_a(\theta')|A_b(\theta)\rangle = 2\pi\,\delta_{ab}\delta(\theta' - \theta).$$

For a scalar perturbing operator $\mathcal{O}_b$, define the connected diagonal crossed form factor by

$$F^{b,\mathrm{conn}}_{a\bar{a}}(\theta + \mathrm{i}\pi, \theta) = {}_{\mathrm{out}}\langle A_a(\theta)|\mathcal{O}_b(0)|A_a(\theta)\rangle_{\mathrm{in,conn}}.$$

Lorentz invariance makes this connected diagonal matrix element independent of θ for a scalar operator; denote it by F^b_a.

Controlled Approximation 83.2 (First-order mass shift from a diagonal form factor)**.** Assume that the one-particle state remains an isolated stable particle under the deformation, and that the diagonal connected form factor F^b_a is finite in the declared crossing prescription. For

$$H = H_0 + \epsilon \int dx\, \mathcal{O}_b(x) + O(\epsilon^2),$$

the first-order shift of the squared mass is

$$\delta m_a^2 = 2\epsilon F^b_a.$$

Place the theory on a circle of circumference L. The one-particle Bethe–Yang density is

$$\rho_1(\theta) = m_a L \cosh\theta.$$

The finite-volume diagonal form-factor formula of Chapter 84 gives, after the vacuum term is separated into the bulk energy coordinate,

$${}_L\langle A_a(\theta)|\mathcal{O}_b(0)|A_a(\theta)\rangle^{\mathrm{conn}}_L = \frac{F^b_a}{m_a L\cosh\theta} + O(e^{-\mu L}).$$

Thus the connected first-order energy shift is

$$\delta E_a(\theta) = \epsilon L_L \langle A_a(\theta)|\mathcal{O}_b(0)|A_a(\theta)\rangle_L^{\mathrm{conn}} = \epsilon \frac{F_a^b}{m_a \cosh\theta} + O(e^{-\mu L}).$$

If the shifted dispersion remains relativistic,

$$E_a(\theta)^2 - p_a(\theta)^2 = m_a^2 + \delta m_a^2 + O(\epsilon^2),$$

then at fixed momentum

$$\delta E_a(\theta) = \frac{\delta m_a^2}{2E_a(\theta)} = \frac{\delta m_a^2}{2m_a \cosh\theta}.$$

Comparing the two expressions gives $\delta m_a^2 = 2\epsilon F_a^b$.

Definition 83.3 (Form-factor perturbation data)**.** First-order form-factor perturbation data consist of:

1. the perturbing operator normalization in the local operator chart;
2. the crossing prescription for diagonal form factors and kinematic-pole subtractions;
3. the vacuum-energy subtraction convention;
4. the finite-volume normalization used to take the infinite-volume limit;
5. the stability hypothesis specifying which isolated particles remain asymptotic states.

The stability hypothesis is part of the data because an integrable particle may become a resonance or may be confined after the deformation. The finite diagonal form factor then supplies a mass counterterm or a pole displacement only after the relevant analytic continuation has been declared.

83.4 Semi-Locality and Confinement Diagnostics

Let $\mathcal{O}_b$ be semi-local with respect to particle species a. The semi-locality phase $\omega_{b,a} \in U(1)$ is the monodromy acquired when the particle-creating field for a is moved once around $\mathcal{O}_b$. The kinematic pole equation of the form-factor chapter then gives the two-particle diagonal pole

$$-\mathrm{i} \operatorname*{Res}_{\theta'=\theta+\mathrm{i}\pi} F^b_{\bar{a}a}(\theta', \theta) = (1-\omega_{b,a})\,\langle \mathcal{O}_b \rangle.$$

Remark 83.4 (Semi-local pole obstruction to a particle mass shift)*.* If $\omega_{b,a} \neq 1$ and $\langle \mathcal{O}_b \rangle \neq 0$, the first-order diagonal form factor entering Controlled Approximation 83.2 has a kinematic pole at the required crossing point. The perturbative one-particle mass shift is therefore replaced by a confinement or resonance problem in the deformed theory.

The displayed residue is nonzero under the stated hypotheses. Hence the crossed two-particle form factor has a pole at $\theta' = \theta + \mathrm{i}\pi$, which is exactly the diagonal point used to define the connected one-particle matrix element. The finite value F_a^b in Controlled Approximation 83.2 is absent. Thus the assumption that the original one-particle state admits an ordinary first-order mass shift fails. The deformed spectral problem must be formulated with the appropriate long-distance force, resonance pole, or finite-volume mixing included.

Example 83.5 (Ising magnetic perturbation and kink confinement)**.** In the ordered phase of the thermally perturbed Ising model there are two vacua $|+\rangle$ and $|-\rangle$ with

$$\langle +|\sigma|+\rangle = v, \qquad \langle -|\sigma|-\rangle = -v.$$

Perturb the Hamiltonian by

$$\delta H = -h \int dx\, \sigma(x).$$

The two vacuum energy densities shift by $-hv$ and $+hv$. A kink–antikink pair separated by distance R encloses a segment of the higher-energy vacuum. The leading long-distance potential is therefore

$$V_{\mathrm{conf}}(R) = 2|hv|\, R.$$

For two heavy kinks of mass m, the leading nonrelativistic meson equation for the relative coordinate is

$$\left[-\frac{1}{m}\frac{d^2}{dR^2} + 2|hv|\,|R|\right]\psi(R) = (M-2m)\psi(R).$$

This equation is an infrared consequence of the vacuum energy splitting. The operator-level warning sign is the semi-local kinematic pole of the spin-field form factor with respect to the kink.

83.5 Finite-Volume Transition Rates

Let

$$V_b(L) = \int_0^L dx\, \mathcal{O}_b(x)$$

and let $|A, L\rangle$ be an eigenstate of $H_0(L)$. Let $|n, L\rangle$ be finite-volume eigenstates with energies $E_n(L)$. First order time-dependent perturbation theory gives

$$P_{A\to n}(T, L) = \epsilon^2 \left| \int_0^T dt\, e^{\mathrm{i}(E_n - E_A)t} \langle n, L|V_b(L)|A, L\rangle \right|^2 + O(\epsilon^3).$$

The elementary distributional limit is

$$\frac{1}{T}\left|\int_0^T dt\, e^{\mathrm{i}\Delta E t}\right|^2 = \frac{4\sin^2(\Delta E T/2)}{T(\Delta E)^2} \longrightarrow 2\pi\delta(\Delta E).$$

This last arrow is not a pointwise limit at finite volume. If L is fixed, the spectrum is discrete: away from exact degeneracies the expression oscillates with total weight $O(T^{-1})$, while exact degeneracies give terms growing linearly in T. The delta function notation means weak convergence of the kernels

$$K_T(\omega) = \frac{4\sin^2(\omega T/2)}{T\omega^2}$$

against smooth spectral test functions:

$$\lim_{T\to\infty} \int_{\mathbb{R}} d\omega\, K_T(\omega) f(\omega) = 2\pi f(0), \qquad f \in C_c^\infty(\mathbb{R}).$$

For a finite-volume regulator this weak limit must be paired with the $L \to \infty$ limit that turns the discrete Bethe–Yang spectrum into an absolutely continuous rapidity measure.

Finite-volume golden-rule limit. For each L, let $\nu_{A,L}$ be the finite positive spectral measure obtained from the unperturbed finite-volume states by

$$\int F(\omega,p)\,d\nu_{A,L}(\omega,p)=\sum_n |\langle n,L|V_b(L)|A,L\rangle|^2 F(E_n(L)-E_A(L),P_n(L)-P_A(L)).$$

Assume that, after inserting the Bethe–Yang state-counting measure and the finite-volume form-factor normalization, $\nu_{A,L}$ converges weakly on smooth compactly supported test functions in the channel under consideration to the absolutely continuous rapidity measure determined by the connected crossed form factors. Assume also that the crossed form factors are regular on the joint energy-momentum conserving contour. For a channel

$$\beta=(b_1,\theta_1;\ldots;b_n,\theta_n)$$

use the momentum-conserving rapidity measure

$$d\Phi_\beta=2\pi\delta\left(P_A-\sum_{j=1}^n m_{b_j}\sinh\theta_j\right)\prod_{j=1}^n\frac{d\theta_j}{2\pi}.$$

Then the leading inclusive decay rate out of A is

$$\Gamma_A=2\pi\epsilon^2\sum_\beta\int d\Phi_\beta\,\delta\left(E_A-\sum_{j=1}^n m_{b_j}\cosh\theta_j\right)|\langle\beta|\mathcal{O}_b(0)|A\rangle_{\rm conn}|^2\,,$$

At finite L, the transition probability per time is the finite sum over normalizable states

$$\frac{P_A(T,L)}{T}=\epsilon^2\sum_n\frac{4\sin^2((E_n-E_A)T/2)}{T(E_n-E_A)^2}|\langle n,L|V_b(L)|A,L\rangle|^2+O(\epsilon^3).$$

The continuum golden-rule rate is the iterated weak limit

$$\Gamma_A=\epsilon^2\lim_{T\to\infty}\lim_{L\to\infty}\int K_T(\omega)\,d\nu_{A,L}(\omega,p),$$

with the momentum component restricted by the spatial integral in $V_b(L)$. Equivalently, one may use a joint large-L, large-T limit in which the time window is long compared with microscopic scales but does not resolve the finite-volume recurrence structure before the continuum spectral measure has formed. The spatial integral in $V_b(L)$ gives finite-volume momentum conservation; in the infinite-volume limit this becomes the delta function contained in $d\Phi_\beta$. The finite-volume matrix-element normalization converges to the connected crossed form factor by the same Gaudin-density cancellation used in Chapter 84. Applying the weak kernel limit $K_T\to2\pi\delta$ to the limiting absolutely continuous spectral measure gives the displayed inclusive rate.

83.6 Two-Particle Phase Space In Two Dimensions

The finite-volume formula becomes especially sharp for the first allowed decay channel. Let the unstable particle A have mass M_A, and study the contribution of two stable particles r,s of masses m_r,m_s. In the rest frame of A, energy-momentum conservation has a real solution precisely when

$$M_A>m_r+m_s.$$

Define the Kallen polynomial

$$\lambda(x,y,z)=x^2+y^2+z^2-2xy-2xz-2yz$$

and the relative momentum

$$p_*=\frac{\sqrt{\lambda(M_A^2,m_r^2,m_s^2)}}{2M_A}.$$

The two on-shell energies at the decay point are

$$E_r^*=\frac{M_A^2+m_r^2-m_s^2}{2M_A},\qquad E_s^*=\frac{M_A^2+m_s^2-m_r^2}{2M_A}.$$

For each orientation $\sigma=\pm1$, let

$$m_r\sinh\theta_r^\sigma=\sigma p_*,\qquad m_s\sinh\theta_s^\sigma=-\sigma p_*.$$

Two-particle decay width in $1+1$ dimensions. Use the state normalization and golden-rule convention above. Suppose A is at rest, $M_A>m_r+m_s$, and the crossed connected form factor

$$\mathcal{F}_{rs;A}(\theta_r,\theta_s)=\langle A_r(\theta_r)A_s(\theta_s)|\mathcal{O}_b(0)|A\rangle_{\text{conn}}$$

has a finite boundary value at the two energy-conserving rapidity pairs $(\theta_r^\sigma,\theta_s^\sigma)$. If the channel sum is over labelled ordered particles, the rs contribution to the leading width is

$$\Gamma_{A\to rs}=\frac{\epsilon^2}{M_Ap_*}\sum_{\sigma=\pm1}|\mathcal{F}_{rs;A}(\theta_r^\sigma,\theta_s^\sigma)|^2\,.$$

If instead the channel is counted as an unordered physical channel, the right-hand side must be divided by 2 when $r=s$, and is unchanged when $r\neq s$.

Start from the finite-volume golden-rule formula with $P_A=0$ and with a two-particle labelled channel:

$$\Gamma_{A\to rs}=2\pi\epsilon^2\int\frac{d\theta_r}{2\pi}\frac{d\theta_s}{2\pi}2\pi\delta(p_r+p_s)\delta(M_A-E_r-E_s)|\mathcal{F}_{rs;A}(\theta_r,\theta_s)|^2,$$

where

$$p_i=m_i\sinh\theta_i,\qquad E_i=m_i\cosh\theta_i.$$

Change variables from rapidities to momenta. Since $d\theta_i=dp_i/E_i$, the phase-space integral without the exterior factor $2\pi\epsilon^2$ is

$$\int_{\mathbb{R}^2}\frac{dp_r\,dp_s}{2\pi E_rE_s}\,\delta(p_r+p_s)\,\delta(M_A-E_r-E_s)\,|\mathcal{F}_{rs;A}|^2.$$

The momentum delta function sets $p_r=p$ and $p_s=-p$. Therefore

$$E_r(p)=\sqrt{m_r^2+p^2},\qquad E_s(p)=\sqrt{m_s^2+p^2}.$$

The equation $E_r(p)+E_s(p)=M_A$ has the two roots $p=\sigma p_*$, $\sigma=\pm1$, because solving it gives

$$E_r^*=\frac{M_A^2+m_r^2-m_s^2}{2M_A},\qquad (E_r^*)^2-m_r^2=\frac{\lambda(M_A^2,m_r^2,m_s^2)}{4M_A^2}.$$

At either root,

$$\left|\frac{d}{dp}(E_r(p)+E_s(p))\right| = |p_*|\left(\frac{1}{E_r^*}+\frac{1}{E_s^*}\right) = \frac{|p_*|M_A}{E_r^*E_s^*}.$$

Thus each root contributes

$$\frac{1}{2\pi E_r^*E_s^*}\,\frac{E_r^*E_s^*}{|p_*|M_A}|\mathcal{F}_{rs;A}(\theta_r^\sigma,\theta_s^\sigma)|^2.$$

Multiplication by the exterior $2\pi\epsilon^2$ gives the displayed formula. If $r=s$ and the physical channel is unlabelled, the two labels of the identical particles describe the same basis of final states twice; dividing by 2 implements the chosen unordered channel convention.

The factor $1/p_*$ is the one-dimensional threshold singularity of the two-particle density of states. It is not by itself a physical divergence in an observable statement: at threshold the perturbative decay problem must be examined with the form-factor boundary value, finite-volume level spacing, and possible near-threshold bound or virtual-state poles kept in the same scaling limit.

83.7 CFT-Deformation Regulator Interface

Let $\mathcal{C}$ be a two-dimensional CFT on a circle of circumference L. The CFT Hamiltonian is

$$H_{\mathrm{CFT}} = \frac{2\pi}{L}\left(L_0+\bar{L}_0-\frac{c}{12}\right).$$

Perturbing by relevant scalar operators $\mathcal{O}_i$ gives

$$H = H_{\mathrm{CFT}} + \sum_i \lambda_i\left(\frac{2\pi}{L}\right)^{\Delta_i-1}\int_0^{2\pi} d\theta\,\mathcal{O}_i(\theta).$$

Equivalently,

$$\frac{L}{2\pi}H = L_0+\bar{L}_0-\frac{c}{12}+\sum_i g_i(L)\int_0^{2\pi} d\theta\,\mathcal{O}_i(\theta), \qquad g_i(L)=\lambda_i\left(\frac{L}{2\pi}\right)^{2-\Delta_i}.$$

The displayed Hamiltonian is a coordinate chart for relevant deformations, not a numerical calculation. A truncated conformal-space regulator may be used to approximate such a deformation only after one has chosen a CFT sector, cutoff subspaces, perturbing-operator matrix elements, counterterm coordinates, and an order of limits. Those choices, together with the Rayleigh–Ritz and finite-matrix error controls, are developed in Chapter 145. This chapter does not expose the TCSA machinery further, because no concrete nonintegrable finite-volume spectrum is extracted here. The information needed for the present bridge is only that a CFT-deformation regulator must match the same local operator normalizations and counterterm coordinates that enter the form-factor perturbation theory around an integrable massive point.

83.8 Comparison Objects

A bridge between integrable and nonintegrable two-dimensional QFT compares declared observables:

finite-volume spectra, local-operator matrix elements,
resonance pole positions and residues, thermal or hydrodynamic late-time data.

An integrable point controls nearby data through form-factor perturbation theory in ϵ. A CFT point controls nearby data through relevant deformations and Hamiltonian truncation. The two coordinate systems meet when the finite-volume regulator, local counterterms, operator normalization, and infinite-volume continuation are matched.

Open Problem 83.6 (Nonintegrable deformation). Construct a nonintegrable two-dimensional QFT near an integrable point with a finite-volume regulator, local counterterm coordinates, form-factor perturbation theory, resonance pole analysis, and a proof of convergence to a local continuum QFT in a specified scaling domain.

Chapter 84

Finite-Volume Form Factors And Spectral Expansions

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Exact scattering data become QFT data only after they are connected to local operator matrix elements and to correlation functions. A spatial circle of circumference L is the cleanest intermediate regulator in massive two-dimensional integrable theories: rapidities become discrete, scattering states become normalizable vectors, matrix elements have honest Hilbert-space normalizations, and the infinite-volume spectral expansion is recovered as a controlled limit. This chapter develops that bridge for diagonal scattering sectors. Non-diagonal scattering requires replacing scalar phase functions by eigenvalues of transfer matrices and is treated in the nested Bethe-ansatz chapters.

84.1 Finite-Volume Hilbert Space And Bethe–Yang Quantization

Let the infinite-volume theory have stable particle species

$$a \in \mathcal{I}, \qquad p_a^\mu(\theta) = m_a(\cosh\theta, \sinh\theta),$$

with diagonal two-body scattering eigenvalues

$$S_{ab}(\theta) = \exp(\mathrm{i}\delta_{ab}(\theta)).$$

The phase convention is fixed by unitarity and real analyticity on the real rapidity axis:

$$S_{ab}(\theta)S_{ba}(-\theta) = 1, \qquad \delta_{ab}(\theta) + \delta_{ba}(-\theta) \in 2\pi\mathbb{Z}.$$

We choose a continuous branch on the rapidity interval under consideration and define

$$\varphi_{ab}(\theta) = \partial_\theta \delta_{ab}(\theta).$$

The branch of δ changes the Bethe integers, while φ and the Gaudin density below are branch-independent.

For a multiparticle state with ordered rapidities

$$\theta_1, \ldots, \theta_n$$

and species $a_1, \ldots, a_n$, the Bethe–Yang equations are

$$Q_j(\theta) = m_{a_j} L \sinh\theta_j + \sum_{k\neq j} \delta_{a_j a_k}(\theta_j - \theta_k) = 2\pi I_j, \qquad j = 1, \ldots, n. \tag{84.1}$$

The I_j are integers or half-integers depending on particle statistics, twists, and branch choices. The finite-volume energy and momentum at Bethe–Yang order are

$$E_L(\theta) = \sum_{j=1}^{n} m_{a_j} \cosh\theta_j, \qquad P_L(\theta) = \sum_{j=1}^{n} m_{a_j} \sinh\theta_j.$$

The neglected effects are exponentially small in $m_* L$, where

$$m_* = \min_{a\in\mathcal{I}} m_a,$$

provided no massless channel, threshold singularity, or nearby bound-state pole invalidates the contour deformation used to wrap virtual particles around the circle.

Construction 84.1 (Bethe–Yang phase from diagonal finite-volume monodromy)**.** Assume the n-particle wave function is described away from collision hyperplanes by ordered plane waves, and that crossing a hyperplane $x_j = x_k$ multiplies the ordered amplitude by the diagonal scattering phase $S_{a_j a_k}(\theta_j - \theta_k)$. Periodicity of particle j around the circle gives (84.1).

Indeed, moving particle j once around the spatial circle while keeping the other particles fixed contributes the free propagation phase

$$\exp(\mathrm{i} p_j L), \qquad p_j = m_{a_j} \sinh\theta_j.$$

During the circuit it crosses each other particle once, so in the diagonal sector the ordered amplitude is multiplied by

$$\prod_{k\neq j} S_{a_j a_k}(\theta_j - \theta_k) = \exp\left(\mathrm{i} \sum_{k\neq j} \delta_{a_j a_k}(\theta_j - \theta_k)\right).$$

Single-valuedness of the wave function demands that the total phase be one. Taking the logarithm on the chosen branch gives

$$m_{a_j} L \sinh\theta_j + \sum_{k\neq j} \delta_{a_j a_k}(\theta_j - \theta_k) = 2\pi I_j.$$

84.2 Gaudin Density And Normalization

The density of Bethe–Yang states is the Jacobian of the map

$$\theta \longmapsto Q(\theta).$$

Define the Gaudin matrix

$$G_{ij}(\theta) = \frac{\partial Q_i}{\partial \theta_j}, \qquad \rho_n(\theta) = \det G(\theta). \tag{84.2}$$

Explicitly,

$$G_{ii} = m_{a_i} L \cosh\theta_i + \sum_{k\neq i} \varphi_{a_i a_k}(\theta_i - \theta_k), \qquad G_{ij} = -\varphi_{a_i a_j}(\theta_i - \theta_j) \quad (i \neq j).$$

For one particle,

$$\rho_1(\theta) = m_a L \cosh\theta.$$

For two particles, using $\varphi_{a_2 a_1}(\theta_2 - \theta_1) = \varphi_{a_1 a_2}(\theta_1 - \theta_2)$, one obtains

$$\rho_2(\theta_1, \theta_2) = m_{a_1} m_{a_2} L^2 \cosh\theta_1 \cosh\theta_2 + \varphi_{a_1 a_2}(\theta_{12}) L\big(m_{a_1}\cosh\theta_1 + m_{a_2}\cosh\theta_2\big), \tag{84.3}$$

where $\theta_{12} = \theta_1 - \theta_2$.

Proposition 84.2 (State-counting measure)**.** *Let $U \subset \mathbb{R}^n$ be a bounded rapidity region with piecewise C^1 boundary, and suppose that on a neighborhood of $\overline{U}$ the Bethe–Yang map Q_L is a C^2 diffeomorphism onto its image and $\rho_{n,L} = \det(\partial Q_{L,i}/\partial\theta_j) > 0$. Assume also that, for $r = 0, 1, 2$,*

$$\|\partial^r Q_L\|_{C^0(\overline{U})} \le C_r L, \qquad \|(\partial Q_L)^{-1}\|_{C^0(\overline{U})} \le C_{-1} L^{-1},$$

with constants independent of L. For every $f \in C_c^1(U)$,

$$\sum_{I:\,\theta(I)\in U} f(\theta(I)) = \int_U \frac{\mathrm{d}^n\theta}{(2\pi)^n}\, \rho_{n,L}(\theta) f(\theta) + O(L^{n-1})$$

as $L \to \infty$. The implicit constant depends on U, the displayed uniform bounds for Q_L, and $\|f\|_{C^1}$, but not on L.

Proof. The Bethe integers form a translate of the lattice $(2\pi)\mathbb{Z}^n$ in Q-space. Since Q_L is one-to-one on U, summing over Bethe solutions in U is summing the function

$$h_L(Q) = f(\theta_L(Q)), \qquad \theta_L = Q_L^{-1},$$

over lattice points in $Q_L(U)$. Cover $Q_L(U)$ by closed lattice cubes of side 2π. For a cube C contained in $Q_L(U)$ with lattice corner Q_C,

$$\left| (2\pi)^n h_L(Q_C) - \int_C h_L(Q)\, \mathrm{d}^n Q \right| \le C_n \sup_C |\nabla_Q h_L|.$$

By the chain rule and the inverse-derivative bound,

$$|\nabla_Q h_L(Q)| \le \|(\partial Q_L)^{-1}\|\, |\nabla_\theta f(\theta_L(Q))| \le C L^{-1} \|f\|_{C^1}.$$

The number of cubes meeting $Q_L(\operatorname{supp} f)$ is $O(L^n)$, because Q_L has $O(L)$ diameter on the fixed region U. Hence the interior Riemann-sum error is $O(L^{n-1})$. Cubes crossing the boundary of $Q_L(U)$ also contribute $O(L^{n-1})$: the piecewise C^1 boundary and the C^1 bound on Q_L give $(n-1)$-dimensional boundary measure $O(L^{n-1})$, hence only $O(L^{n-1})$ fixed-size cubes meet the boundary. Therefore

$$\sum_I f(\theta(I)) = \int_{Q_L(U)} \frac{\mathrm{d}^n Q}{(2\pi)^n}\, h_L(Q) + O(L^{n-1}).$$

Changing variables from Q back to θ gives

$$\mathrm{d}^n Q = \det\left(\frac{\partial Q_i}{\partial \theta_j}\right) \mathrm{d}^n\theta = \rho_{n,L}(\theta)\mathrm{d}^n\theta,$$

which proves the formula. □

The same determinant normalizes finite-volume states. If

$$|\theta_1, a_1; \ldots; \theta_n, a_n\rangle_\infty$$

is an infinite-volume rapidity state normalized by delta functions, then the corresponding unit finite-volume vector has leading normalization

$$|I_1, \ldots, I_n\rangle_L = \rho_n(\theta)^{-1/2} |\theta_1, a_1; \ldots; \theta_n, a_n\rangle_\infty + O(\mathrm{e}^{-\mu L})$$

in matrix elements of local operators. The statement compares normalized matrix elements through the finite-volume regulator; the infinite-volume delta-normalized state and the finite-volume unit vector live in different Hilbert-space normalizations.

84.3 Off-Diagonal Matrix Elements

Let $\widehat{\mathcal{O}}$ be a local scalar operator. The infinite-volume crossed form factor for an outgoing m-particle state and incoming n-particle state is written

$$F^{\mathcal{O}}_{b_1 \ldots b_m, a_1 \ldots a_n}(\theta'_1 + \mathrm{i}\pi, \ldots, \theta'_m + \mathrm{i}\pi, \theta_1, \ldots, \theta_n),$$

with the branch specified by the rapidity ordering and by the crossing prescription of Chapter 76.

Controlled Approximation 84.3 (Off-diagonal finite-volume form-factor formula)**.** Assume the incoming and outgoing Bethe–Yang rapidity sets have no coincident rapidities after crossing, and assume that the crossed form factor is regular at the corresponding point. Then

$$_L\langle J_1, \ldots, J_m | \widehat{\mathcal{O}}(0) | I_1, \ldots, I_n\rangle_L = \frac{F^{\mathcal{O}}_{b_1 \ldots b_m, a_1 \ldots a_n}(\theta'_1 + \mathrm{i}\pi, \ldots, \theta'_m + \mathrm{i}\pi, \theta_1, \ldots, \theta_n)}{\sqrt{\rho_m(\theta')\rho_n(\theta)}} + O(\mathrm{e}^{-\mu L}). \quad (84.4)$$

The derivation is a matching argument, not a theorem independent of the finite-volume construction. Smear the finite-volume Bethe states by packets whose rapidity widths are large compared with the level spacing L^{-1} and small compared with the distance to any kinematic or bound-state pole. In that window the finite-volume packet is approximated by the infinite-volume packet with the state-counting measure of Proposition 84.2. The unit normalization of the finite-volume packet therefore supplies one factor $\rho_n^{-1/2}$ for the incoming state and one factor $\rho_m^{-1/2}$ for the outgoing state. The local operator matrix element between the corresponding infinite-volume packets is the crossed form factor by the crossing prescription. Since the form factor is regular at the chosen point, shrinking the packets to the Bethe rapidities gives (84.4). Virtual particles wrapping around the circle give the exponentially small correction when the corresponding contour deformation is valid with a positive mass scale μ.

Remark 84.4 (Cancellation in the infinite-volume two-point sum). For vacuum-to-n-particle matrix elements of a scalar operator,

$$\sum_I \left| {}_L\langle \Omega_L|\widehat{\mathcal{O}}(0)|I\rangle_L \right|^2 f(\theta(I)) \longrightarrow \int \frac{\mathrm{d}^n\theta}{(2\pi)^n} |F_n^{\mathcal{O}}(\theta)|^2 f(\theta)$$

on rapidity regions with no singularities, after the usual $1/n!$ quotient for identical unordered particles.

By Controlled Approximation 84.3, the squared matrix element contributes

$$|F_n^{\mathcal{O}}(\theta(I))|^2/\rho_n(\theta(I))$$

up to exponentially small corrections. Summing over Bethe integers and using Proposition 84.2 supplies the measure

$$\rho_n(\theta)\mathrm{d}^n\theta/(2\pi)^n.$$

The Gaudin determinant cancels. If the rapidity ordering was used to label states, the ordered chamber gives one copy; if integration is over all $\mathbb{R}^n$ for identical particles, the $n!$ permutations must be divided out.

84.4 Diagonal Matrix Elements

Diagonal matrix elements require subtracting kinematic poles. Let

$$F_{2n,c}^{\mathcal{O}}(\theta_1, \ldots, \theta_n)$$

denote the connected diagonal form factor obtained as the finite part of

$$F_{\bar{a}_n\ldots\bar{a}_1a_1\ldots a_n}^{\mathcal{O}}(\theta_n + \mathrm{i}\pi + \epsilon_n, \ldots, \theta_1 + \mathrm{i}\pi + \epsilon_1, \theta_1, \ldots, \theta_n)$$

after removing all terms singular in the ϵ_j as $\epsilon_j \to 0$. The order of the crossed rapidities is part of the definition. In diagonal theories the connected finite part is symmetric in the rapidities after the same branch convention is used throughout.

Definition 84.5 (Principal minors of the Gaudin matrix)**.** For $A \subset \{1, \ldots, n\}$, let G_A be the principal submatrix of the Gaudin matrix with rows and columns in A, and set

$$\rho_A(\theta) = \det G_A, \qquad \rho_\emptyset = 1.$$

When the complement of A is denoted $\bar{A}$, the symbol $\rho_{\bar{A}}$ means the determinant of the principal minor on that complement, not the Gaudin density of an isolated smaller-volume theory.

Controlled Approximation 84.6 (Diagonal finite-volume matrix element)**.** Under the same massive diagonal assumptions and away from accidental degeneracies,

$$ {}_L\langle I_1, \ldots, I_n | \widehat{\mathcal{O}}(0) | I_1, \ldots, I_n \rangle_L = \frac{1}{\rho_n(\theta)} \sum_{A \subset \{1,\ldots,n\}} F^{\mathcal{O}}_{2|A|,c}(\theta_A)\, \rho_{\bar{A}}(\theta) + O(\mathrm{e}^{-\mu L}). \tag{84.5} $$

Here $F^{\mathcal{O}}_{0,c}$ is the infinite-volume vacuum expectation value $\langle \mathcal{O} \rangle$.

The formula is obtained by a finite-volume subtraction of the kinematic poles. Start from the off-diagonal formula with outgoing rapidities θ'_j distinct from the incoming rapidities, then approach $\theta'_j = \theta_j$. The crossed form factor develops kinematic poles. Each singular contraction identifies one crossed particle with one incoming particle and produces the finite-volume delta contribution associated with the corresponding Bethe quantum number. In the finite-volume regularization these delta contributions are replaced by derivatives of the quantization conditions, namely entries and minors of the Gaudin matrix.

After all possible disconnected contractions are grouped by the subset $\bar{A}$ of particles that contract through the finite-volume normalization, the remaining particles A contribute the connected diagonal finite part $F^{\mathcal{O}}_{2|A|,c}(\theta_A)$, while the contracted particles contribute the principal minor $\rho_{\bar{A}}$. The overall normalization of the bra and ket contributes $1/\rho_n$. Summing over all subsets gives (84.5). The only analytic input beyond the off-diagonal case is the kinematic-pole residue equation of Chapter 76, which ensures that the disconnected pieces are exactly the contractions enumerated by the subsets.

For one particle this gives

$$ {}_L\langle I | \mathcal{O} | I \rangle_L = \langle \mathcal{O} \rangle + \frac{F^{\mathcal{O}}_{2,c}(\theta)}{\rho_1(\theta)} + O(\mathrm{e}^{-\mu L}). $$

For two particles,

$$ \begin{aligned} {}_L\langle I_1, I_2 | \mathcal{O} | I_1, I_2 \rangle_L = \frac{1}{\rho_{12}} \Big[&\langle \mathcal{O} \rangle \rho_{12} + F^{\mathcal{O}}_{2,c}(\theta_1) \rho_{\{2\}} + F^{\mathcal{O}}_{2,c}(\theta_2) \rho_{\{1\}} \\ &+ F^{\mathcal{O}}_{4,c}(\theta_1, \theta_2) \Big] + O(\mathrm{e}^{-\mu L}). \end{aligned} $$

This explicit formula is often the first place where the distinction between connected diagonal form factors and symmetric finite parts matters.

84.5 Zero-Temperature Spectral Expansion

For a finite-volume two-point function

$$ C_L(x,t) = \langle \Omega_L | \widehat{\mathcal{O}}(x,t) \widehat{\mathcal{O}}(0,0) | \Omega_L \rangle, $$

insert the finite-volume identity operator:

$$ \mathbf{1}_L = \sum_\alpha |\alpha, L\rangle \langle \alpha, L|. $$

Translation covariance gives

$$C_L(x,t) = \sum_{\alpha} \big|\langle \Omega_L|\widehat{\mathcal{O}}(0)|\alpha, L\rangle\big|^2 \exp(-\mathrm{i}E_\alpha t + \mathrm{i}P_\alpha x).$$

Using the cancellation mechanism of Remark 84.4 in each particle-number sector gives the infinite-volume Wightman expansion

$$C(x,t) = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{a_1,\dots,a_n} \int_{\mathbb{R}^n} \prod_{j=1}^{n} \frac{\mathrm{d}\theta_j}{2\pi} \big|F^{\mathcal{O}}_{a_1\dots a_n}(\theta_1,\dots,\theta_n)\big|^2 e^{-\mathrm{i}\sum_j m_{a_j} t\cosh\theta_j + \mathrm{i}\sum_j m_{a_j} x \sinh\theta_j}. \quad (84.6)$$

The Euclidean Schwinger function at separation

$$r = \sqrt{\tau^2 + x^2} > 0$$

is obtained by analytic continuation $t = -\mathrm{i}\tau$. Lorentz invariance allows the exponent to be written, in a frame where the separation is purely Euclidean time, as

$$\exp\left[-r\sum_{j=1}^{n} m_{a_j}\cosh\theta_j\right].$$

The following comparison estimate is a practical convergence criterion rather than a theorem about any particular integrable model. Fix $r > 0$. If there are constants $A, B, c \geq 0$, with $c < m_* r$, such that the absolute n-particle term satisfies

$$\frac{1}{n!} \sum_{a_1,\dots,a_n} \int_{\mathbb{R}^n} \prod_{j=1}^{n} \frac{\mathrm{d}\theta_j}{2\pi} |F^{\mathcal{O}}_{a_1\dots a_n}(\theta)|^2 e^{-r\sum_j m_{a_j}\cosh\theta_j} \leq \frac{AB^n}{n!} e^{cn}.$$

for every n, then the sum of absolute values is bounded by

$$\sum_{n=0}^{\infty} \frac{A(Be^c)^n}{n!} = A\exp(Be^c).$$

This proves absolute convergence by comparison. The analytic work in a specific model lies in estimating the joint large-rapidity and large-particle-number behavior of the form factors strongly enough to obtain such a bound, or a sharper model-specific replacement for it.

84.6 Example: Free Majorana Energy Density

In the massive Ising field theory without magnetic perturbation, the free Majorana scattering phase is

$$S = -1.$$

For the energy-density operator ε, the only nonzero vacuum-to-particle form factor in the even sector needed for the first spectral term is the two-particle form factor

$$F_2^{\varepsilon}(\theta_1,\theta_2) = \kappa \sinh\left(\frac{\theta_1-\theta_2}{2}\right),$$

with normalization κ fixed by the convention for ε. The two-particle contribution to the Euclidean two-point function is

$$C_2^{\varepsilon}(r) = \frac{1}{2}\int_{\mathbb{R}^2} \frac{\mathrm{d}\theta_1 \mathrm{d}\theta_2}{(2\pi)^2} \left|\kappa \sinh \frac{\theta_1 - \theta_2}{2}\right|^2 e^{-mr(\cosh\theta_1 + \cosh\theta_2)}. \tag{84.7}$$

Introduce center and relative rapidities

$$\Theta = \frac{\theta_1 + \theta_2}{2}, \qquad \alpha = \theta_1 - \theta_2.$$

Then

$$\cosh\theta_1 + \cosh\theta_2 = 2\cosh\Theta \cosh\frac{\alpha}{2}, \qquad \mathrm{d}\theta_1 \mathrm{d}\theta_2 = \mathrm{d}\Theta\, \mathrm{d}\alpha.$$

Using

$$\int_{-\infty}^{\infty} e^{-z\cosh\Theta}\, \mathrm{d}\Theta = 2K_0(z),$$

(84.7) becomes

$$C_2^{\varepsilon}(r) = \frac{|\kappa|^2}{(2\pi)^2}\int_{-\infty}^{\infty} \mathrm{d}\alpha\, \sinh^2\frac{\alpha}{2}\, K_0\Big(2mr\cosh\frac{\alpha}{2}\Big).$$

Since the integrand is even,

$$C_2^{\varepsilon}(r) = \frac{2|\kappa|^2}{(2\pi)^2}\int_{0}^{\infty} \mathrm{d}\alpha\, \sinh^2\frac{\alpha}{2}\, K_0\Big(2mr\cosh\frac{\alpha}{2}\Big).$$

This example displays the finite-volume logic in its simplest form: the finite-volume matrix element contains the inverse Gaudin determinant, the sum over Bethe integers contains the Gaudin determinant, and the infinite-volume answer contains only the invariant rapidity measure and the form factor.

84.7 Thermal Traces And Connected Diagonal Series

At finite temperature, the Gibbs trace is organized by the thermodynamic limit at fixed rapidity density. The pseudoenergy $\varepsilon_a(\theta)$ is determined by the TBA equations in the same scattering convention used to define the form factors. For fermionic Bethe quantization conventions, the occupation factor is

$$n_a(\theta) = \frac{1}{1 + e^{\varepsilon_a(\theta)}}.$$

The Leclair–Mussardo type one-point series has the form

$$\langle \mathcal{O}\rangle_R = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{a_1,\dots,a_n} \int \prod_{j=1}^{n} \frac{\mathrm{d}\theta_j}{2\pi} \prod_{j=1}^{n} n_{a_j}(\theta_j)\, F^{\mathcal{O}}_{2n,c}(\theta_1,\dots,\theta_n), \tag{84.8}$$

where R is inverse temperature. Other quantization conventions move statistical signs between the occupation factor and the connected diagonal form factor. The formula is meaningful only with a fixed prescription for the TBA kernel, the connected diagonal finite part, and the convergence domain.

84.8 Reconstruction Boundary

The finite-volume form-factor formalism supplies a computable chain

$$S\text{-matrix} \longrightarrow F_n^{\mathcal{O}} \longrightarrow {}_L\langle\beta|\mathcal{O}|\alpha\rangle_L \longrightarrow \text{correlation functions.}$$

Each arrow has its own status. The first arrow is on-shell scattering data. The second is the operator bootstrap and crossing prescription. The third is a finite-volume normalization statement with normalizable Bethe states and Gaudin densities. The last arrow is a reconstruction step: it asks that the regulated matrix-element sums converge to distributions with positivity, covariance, locality or wedge-locality with a local subalgebra, clustering, and compatibility with the scattering theory used at the beginning. Thus the finite-volume regulator is valuable precisely because each pre-limit step has a declared Hilbert space and normalizable vectors, but removing the regulator is still an analytic theorem, not a bookkeeping identity. The free Majorana energy-density example realizes the chain inside an already constructed local theory. For an interacting regular factorizing model, the same chain is a conditional reconstruction program until convergence, domains, and locality have been proved.

Controlled Approximation 84.7 (Separated Euclidean reconstruction package)**.** Fix a separated Euclidean test space $\mathcal{T}_{r_0,r_1}$ whose tests are supported at distances $r_0 \le r \le r_1$, with $0 < r_0 < r_1 < \infty$. A form-factor reconstruction claim for a local scalar operator $\mathcal{O}$ on this window consists of the following data:

1. a finite-volume approximation $G^{\mathcal{O}}_{L,N}[\psi]$, $\psi \in \mathcal{T}_{r_0,r_1}$, built from normalizable Bethe states and Gaudin-normalized matrix elements through particle number N;
2. a finite-volume-to-rapidity-integral error for taking $L \to \infty$ on the retained sectors;
3. a particle-number tail estimate for the omitted sectors $n > N$;
4. a collision/contact extension controlling coincident rapidities, diagonal finite parts, and possible $r = 0$ contact coordinates;
5. a common-domain and positivity statement strong enough for the tested Schwinger functions to define a Hilbert-space operator or local algebra;
6. a locality or wedge-locality-with-local-subalgebra statement;
7. a sector-completeness statement comparing the scattering states used in the expansion with the physical Hilbert-space sectors under study.

The separated window removes the ultraviolet contact point from the tested observable, but it does not remove the domain, positivity, locality, or completeness obligations. Those are additional reconstruction inputs, not consequences of Bethe–Yang normalization.

Proposition 84.8 (Finite form-factor reconstruction residual budget)**.** *In the setting of Controlled Approximation 84.7, let $G^{\mathcal{O}}_{\mathrm{loc}}[\psi]$ denote the target separated Euclidean functional of a local QFT, and suppose that, for a fixed L, N,*

$$G^{\mathcal{O}}_{\mathrm{loc}}[\psi] = G^{\mathcal{O}}_{L,N}[\psi] + \Delta_{\mathrm{FV}}[\psi] + \Delta_{\mathrm{tail}}[\psi] + \Delta_{\mathrm{diag}}[\psi] + \Delta_{\mathrm{dom}}[\psi] + \Delta_{\mathrm{loc}}[\psi] + \Delta_{\mathrm{comp}}[\psi]. \tag{84.9}$$

Assume each residual is bounded in the dual test norm,

$$|\Delta_a[\psi]| \le \epsilon_a(L, N; r_0, r_1)\, \|\psi\|_{\mathcal{T}}, \qquad a \in \{\mathrm{FV}, \mathrm{tail}, \mathrm{diag}, \mathrm{dom}, \mathrm{loc}, \mathrm{comp}\}. \tag{84.10}$$

Then

$$\left|G_{\mathrm{loc}}^{\mathcal{O}}[\psi] - G_{L,N}^{\mathcal{O}}[\psi]\right| \le (\epsilon_{\mathrm{FV}} + \epsilon_{\mathrm{tail}} + \epsilon_{\mathrm{diag}} + \epsilon_{\mathrm{dom}} + \epsilon_{\mathrm{loc}} + \epsilon_{\mathrm{comp}})\, \|\psi\|_{\mathcal{T}}. \tag{84.11}$$

In particular, exact Gaudin-density cancellation and exact finite-volume matrix-element normalization control only the retained pre-limit coordinate $G_{L,N}^{\mathcal{O}}$. They do not prove local-QFT reconstruction unless the six residual budgets are driven to zero, or are retained as declared approximation errors, in the stated test topology.

Proof. Subtract $G_{L,N}^{\mathcal{O}}[\psi]$ from (84.9) and apply the triangle inequality together with (84.10). The final statement follows because the Gaudin cancellation and finite-volume normalization enter only in the definition of the retained approximant $G_{L,N}^{\mathcal{O}}$. The other terms measure analytic and operator properties that are invisible to finite state-counting algebra. □

Open Problem 84.9 (Form-factor construction)*.* Construct an interacting factorizing QFT from bootstrap scattering and form-factor data by proving convergence of the spectral expansions, locality of reconstructed fields or algebras, and agreement of finite-volume matrix elements with the Hilbert-space scattering limit.

Part VII

Supersymmetric Quantum Field Theory

Chapter 85

Supersymmetry Algebras and Representation Data

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Supersymmetry is an extension of spacetime symmetry by odd generators. The word "odd" refers to a $\mathbb{Z}_2$-graded algebra: even generators commute by ordinary commutators, odd generators anticommute with odd generators, and mixed brackets are ordinary commutators. This chapter fixes the representation-theoretic data before introducing superfields or Lagrangians.

85.1 Particle Multiplets and Field-Variable Multiplets

Definition 85.1 (Particle supersymmetry representation)**.** A particle representation of supersymmetry is a positive-energy unitary representation of the real super-Poincare algebra on a $\mathbb{Z}_2$-graded Hilbert space, or on a one-particle spectral subspace of such a Hilbert space. The odd generators are closed operators whose anticommutators are understood on a common invariant domain. A particle supermultiplet is an irreducible representation after the mass orbit, spin or helicity data, internal charges, and central charges have been fixed.

Definition 85.2 (Field-variable supersymmetry representation)**.** A field-variable representation of supersymmetry is an action of the super-Poincare algebra, or of its supertranslation subalgebra, by derivations on a space of classical field variables, superfields, antifields, or local functionals. It is off shell when the algebra closes without imposing Euler–Lagrange equations. It is on shell when closure holds only modulo equations of motion, gauge transformations, or both. Auxiliary fields are field variables added to obtain off-shell closure; they are not particle states by that fact alone.

The two notions of multiplet have different ambient categories. Particle multiplets live in Hilbert-space representation theory and are constrained by positivity of the supercharge anticommutator. Off-shell field multiplets live in a coordinate description of local dynamics. A relation between them is a derived statement: one must choose an action or equations of motion, impose constraints, quotient gauge redundancy if present, quantize or reconstruct the state space, and then identify the resulting one-particle spectral subspaces. Counting component fields in an off-shell superfield is therefore not a proof of the number or spin content of physical particles.

On-shell particle multiplets. Let $\mathcal{V}$ be a finite-dimensional linear space of component field variables on which the supertranslation algebra acts by linear differential operators in spacetime, and let S_{cl} be a quadratic supersymmetric action. The physical one-particle multiplet is determined by the positive-frequency solution space of the Euler–Lagrange equations modulo linearized gauge transformations and null vectors of the symplectic form; it is not determined by $\mathcal{V}$ alone. In particular, an auxiliary field whose equation of motion is algebraic contributes no independent one-particle oscillator in the free theory. The free one-particle space is obtained from the space of classical solutions by the standard linear quantization construction: one restricts to solutions of the linear equations, quotients pure-gauge directions if present, equips the quotient with the conserved Hermitian form coming from the quadratic action, and completes the positive-frequency part. None of these operations is encoded in the representation of the supertranslation algebra on the ambient coordinate space $\mathcal{V}$ alone.

If a component $F \in \mathcal{V}$ is auxiliary, the quadratic action contains no time derivative of F and the Euler–Lagrange equation has the form

$$AF + B(\Phi) = 0,$$

where A is an invertible algebraic operator and Φ denotes the other linearized fields. Thus $F = -A^{-1}B(\Phi)$ on the solution space. The initial data of F are fixed by the initial data of Φ, and the symplectic form has no independent F-oscillator. Consequently the auxiliary component may be necessary for off-shell closure, but it is absent from the particle Hilbert-space multiplet.

85.2 Graded Lie Algebra

Definition 85.3 (Four-dimensional $\mathcal{N} = 1$ supertranslation algebra)**.** Let P_μ be the translation generators in the convention $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$. The complexified $\mathcal{N} = 1$ supertranslation algebra is generated by P_μ, odd Weyl spinors Q_α, $\bar{Q}_{\dot\alpha}$, and the identity, with

$$[P_\mu, P_\nu] = [P_\mu, Q_\alpha] = [P_\mu, \bar{Q}_{\dot\alpha}] = 0,$$

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\} = 0, \qquad \{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2\sigma^\mu_{\alpha\dot\beta} P_\mu.$$

The matrices $\sigma^\mu_{\alpha\dot\beta}$ and the raising/lowering conventions are those of Section 40.16. A real form is obtained by a choice of adjoint operation for which P_μ is Hermitian and $(Q_\alpha)^\dagger = \bar{Q}_{\dot\alpha}$ after the stated spinor identifications.

The Poincare generators $M_{\mu\nu}$ act on Q and $\bar{Q}$ by the two Weyl spinor representations. This action is part of the full super-Poincare algebra and fixes the Lorentz covariance of the anticommutator above.

85.3 Haag–Łopuszański–Sohnius Classification

The super-Poincare algebra is not introduced as an arbitrary formal gadget. Its distinguished status in four-dimensional particle physics comes from a classification theorem about possible symmetries of a massive scattering theory. The theorem is conditional. It assumes the same kind of scattering framework as the Coleman–Mandula theorem: a positive-energy Hilbert space, an S-matrix on an asymptotic particle space, analyticity of matrix elements in the relevant variables, a finite number of one-particle species below each mass, and nontrivial scattering. These assumptions exclude massless conformal theories without an ordinary particle S-matrix, integrable two-dimensional theories with infinitely many higher-spin charges, and gauge-theory charged sectors whose asymptotic states require nonlocal dressings. They are therefore hypotheses of a classification result, not axioms of QFT.

Definition 85.4 (Finite graded scattering symmetry)**.** In the setting of a massive Haag–Ruelle scattering theory, a finite graded scattering symmetry is a finite-dimensional real Lie superalgebra

$$\mathfrak{g} = \mathfrak{g}_{\bar{0}} \oplus \mathfrak{g}_{\bar{1}}$$

represented by densely defined operators on a common invariant domain in the asymptotic Hilbert space, with the following properties.

1. The even subalgebra contains the Poincare algebra generated by P_μ and $M_{\mu\nu}$.
2. The even generators preserve fermion parity and the odd generators reverse it.
3. The generators commute with the scattering operator S on the common domain, after acting by the corresponding graded Leibniz rule on tensor-product asymptotic states.
4. The action on one-particle states has finite multiplicity at fixed mass and spin, and no generator creates a continuum of new particle species from one species.

The word "finite" refers to the symmetry algebra and to particle-species multiplicity; it does not assert that the Hilbert space is finite-dimensional.

Quoted Theorem 85.5 (Haag–Łopuszański–Sohnius form in four dimensions)**.** *Under the usual analyticity, finite-species, positive-energy, and nontrivial massive scattering hypotheses of the Coleman–Mandula and Haag–Łopuszański–Sohnius theorems, any finite graded scattering symmetry of a four-dimensional theory is, after quotienting possible ideals acting trivially on the S-matrix, a super-Poincare algebra with compact internal even symmetries. More explicitly, the even generators are*

$$P_\mu, \qquad M_{\mu\nu}, \qquad B_A,$$

where B_A generate a compact internal Lie algebra commuting with P_μ and $M_{\mu\nu}$. The odd generators are Weyl spinors

$$Q^I_\alpha, \qquad \bar{Q}_{\dot{\alpha} I}, \qquad I = 1, \ldots, \mathcal{N}, \quad \alpha, \dot{\alpha} = 1, 2,$$

possibly transforming in a unitary representation of the internal algebra. In a basis compatible with the positive inner product, the nonzero brackets may be written

$$[M_{\mu\nu}, Q^I_\alpha] = (\sigma_{\mu\nu})_\alpha{}^\beta Q^I_\beta, \qquad [M_{\mu\nu}, \bar{Q}_{\dot\alpha I}] = (\bar\sigma_{\mu\nu})_{\dot\alpha}{}^{\dot\beta}\bar{Q}_{\dot\beta I}, \tag{85.1}$$

$$[B_A, Q^I_\alpha] = (T_A)^I{}_J Q^J_\alpha, \qquad [B_A, \bar{Q}_{\dot\alpha I}] = -\bar{Q}_{\dot\alpha J}(T_A)^J{}_I, \tag{85.2}$$

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2\delta^I{}_J \sigma^\mu_{\alpha\dot\beta} P_\mu, \tag{85.3}$$

$$\{Q^I_\alpha, Q^J_\beta\} = 2\epsilon_{\alpha\beta} Z^{IJ}, \qquad \{\bar{Q}_{\dot\alpha I}, \bar{Q}_{\dot\beta J}\} = 2\epsilon_{\dot\alpha\dot\beta}\bar{Z}_{IJ}. \tag{85.4}$$

Here $Z^{IJ} = -Z^{JI}$ are central charges invariant under whatever internal symmetry is kept as a simultaneous symmetry of the fixed algebra:

$$(T_A)^I{}_K Z^{KJ} + (T_A)^J{}_K Z^{IK} = 0. \tag{85.5}$$

No odd Lorentz tensors of spin $3/2$ or higher, and no additional even spacetime-tensor charges, occur under these hypotheses.

Remark 85.6 (Status of the theorem in this monograph). The quoted theorem is a classification theorem about symmetries of a class of massive scattering theories. It is not a construction theorem for supersymmetric QFTs. It also does not apply without modification to superconformal theories, theories without an ordinary particle S-matrix, two-dimensional factorized theories with infinitely many higher-spin charges, or gauge-theory charged sectors requiring Wilson-line dressings. The algebraic consequences used below, however, are finite-dimensional representation-theoretic statements and are proved directly.

Lorentz covariance of the mixed supercharge anticommutator. Assume the odd generators transform as left- and right-handed Weyl spinors and commute with translations. Under the HLS scattering hypotheses excluding new independent even spacetime-vector charges, Lorentz covariance and positivity put the mixed anticommutator in the form

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2\delta^I{}_J \sigma^\mu_{\alpha\dot\beta} P_\mu$$

after a nonsingular change of basis in the index I. The tensor product of the left Weyl representation $(\frac{1}{2}, 0)$ with the right Weyl representation $(0, \frac{1}{2})$ is the vector representation $(\frac{1}{2}, \frac{1}{2})$. Therefore Lorentz covariance requires

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = A^I{}_J \sigma^\mu_{\alpha\dot\beta} V_\mu$$

for some even vector generator V_μ, up to possible sums of such vector generators. The Coleman–Mandula part of the HLS hypotheses rules out a new independent even vector symmetry acting nontrivially on the S-matrix; the only available spacetime vector generator is P_μ. Thus

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = A^I{}_J \sigma^\mu_{\alpha\dot\beta} P_\mu.$$

Hermitian conjugation gives $A = A^\dagger$. In a timelike positive-energy state with momentum p, the expectation value of $\{u^\alpha_I Q^I_\alpha, (u^\beta_J Q^J_\beta)^\dagger\}$ is a norm and is nonnegative for every u. Hence A is positive semidefinite on the subspace of nontrivially acting supercharges. Quotienting zero-norm supercharges and choosing an orthonormal basis in the remaining internal index diagonalizes A to a positive diagonal matrix. Rescaling each nonzero Q^I fixes the diagonal entries to 2, giving the displayed normalization.

Central-charge tensor structure. Under the same HLS hypotheses, the anticommutator of two left-handed supercharges has only the Lorentz-scalar central part

$$\{Q^I_\alpha, Q^J_\beta\} = 2\epsilon_{\alpha\beta} Z^{IJ}, \qquad Z^{IJ} = -Z^{JI},$$

with the conjugate formula for two right-handed supercharges. If an internal generator B_A is retained as an actual symmetry of the same algebra, then Z obeys the invariance condition (85.5).

The product of two left Weyl representations decomposes as

$$\left(\frac{1}{2}, 0\right) \otimes \left(\frac{1}{2}, 0\right) = (0, 0) \oplus (1, 0).$$

The scalar summand is obtained by contracting the spinor indices with $\epsilon_{\alpha\beta}$, while the $(1, 0)$ summand is a self-dual two-form spin representation. An even self-dual tensor charge would be an additional spacetime-tensor symmetry; the HLS hypotheses exclude such charges in the particle S-matrix setting. The remaining scalar term is therefore $2\epsilon_{\alpha\beta} Z^{IJ}$. The factor 2 is part of the central charge convention used below: it makes the rest-frame positivity block have eigenvalues $2(m \pm z_s)$ when z_s is a singular value of Z.

The graded anticommutator is symmetric under exchange of the combined labels $(I, \alpha) \leftrightarrow (J, \beta)$. Since $\epsilon_{\beta\alpha} = -\epsilon_{\alpha\beta}$, this symmetry requires

$$2\epsilon_{\alpha\beta} Z^{IJ} = 2\epsilon_{\beta\alpha} Z^{JI} = -2\epsilon_{\alpha\beta} Z^{JI},$$

and hence $Z^{IJ} = -Z^{JI}$. Finally, apply the graded Jacobi identity to $(B_A, Q^I_\alpha, Q^J_\beta)$. Using $[B_A, Q^I_\alpha] = (T_A)^I{}_K Q^K_\alpha$, the Jacobi identity gives

$$[B_A, Z^{IJ}] = (T_A)^I{}_K Z^{KJ} + (T_A)^J{}_K Z^{IK}.$$

For Z^{IJ} to be a numerical central element in the algebra, the left side vanishes; this is exactly (85.5). If this condition fails for a proposed internal generator, that generator is an automorphism of a family of algebras or is broken in the sector with the fixed central charge, not a simultaneous internal symmetry of that fixed algebra.

85.4 Positivity and Multiplet Size

For a one-particle state of mass $m > 0$, go to its rest frame $P^\mu = (m, 0, 0, 0)$. The supercharge anticommutator becomes

$$\{Q_\alpha, Q^\dagger_\beta\} = 2m\, \delta_{\alpha\beta}$$

after choosing the rest-frame spinor basis. Thus

$$a_\alpha = \frac{1}{\sqrt{2m}} Q_\alpha, \qquad a^\dagger_\alpha = \frac{1}{\sqrt{2m}} Q^\dagger_\alpha$$

generate a two-mode fermionic oscillator algebra.

Massive $\mathcal{N} = 1$ oscillator module. Fix a massive one-particle orbit of mass $m > 0$, and fix an irreducible spin-j representation V_j of the massive little group $SU(2)$. The corresponding irreducible rest-frame supertranslation module is, as a $\mathbb{Z}_2$-graded vector space,

$$V_j \otimes \Lambda^\bullet \mathbb{C}^2.$$

It has total dimension $4(2j+1)$, bosonic dimension $2(2j+1)$, and fermionic dimension $2(2j+1)$. Under the diagonal little-group action the fermionic one-oscillator sector carries

$$V_j \otimes \mathbb{C}^2 \cong V_{j+1/2} \oplus V_{j-1/2}, \tag{85.6}$$

with the $V_{j-1/2}$ summand omitted when $j = 0$, while the zero- and two-oscillator sectors are two copies of V_j with opposite fermion number parity from the one-oscillator sector. The normalized operators $a_\alpha, a_\alpha^\dagger$ satisfy

$$\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}, \qquad \{a_\alpha, a_\beta\} = \{a_\alpha^\dagger, a_\beta^\dagger\} = 0.$$

The irreducible representation of this Clifford algebra with a cyclic vector annihilated by the a_α is the exterior algebra $\Lambda^\bullet \mathbb{C}^2$. Tensoring with the chosen little-group representation gives the stated vector space and dimension. The two creation operators transform as a spin-1/2 doublet, so the one-oscillator sector is $V_j \otimes V_{1/2}$, which decomposes by the Clebsch–Gordan rule into (85.6). The zero-oscillator sector $\Lambda^0\mathbb{C}^2$ and the two-oscillator sector $\Lambda^2\mathbb{C}^2$ are both little-group scalars, giving two copies of V_j. Fermion parity is exterior degree modulo 2, hence the even and odd dimensions are both $2(2j+1)$.

Remark 85.7 (Equal bosonic and fermionic dimensions). Let $F = (-1)^{N_F}$ be the fermion-parity operator of the massive rest-frame $\mathcal{N} = 1$ oscillator algebra. For a two-mode fermionic Fock space,

$$\operatorname{Tr} F = (1-1)^2 = 0.$$

Tensoring with a little-group representation multiplies the trace by its dimension, so the trace remains zero. Since F has eigenvalue $+1$ on bosonic states and -1 on fermionic states, $\operatorname{Tr} F = 0$ is exactly equality of the two dimensions.

85.5 Massless Representations

For a massless momentum $p^\mu = (E, 0, 0, E)$, the matrix $\sigma^\mu p_\mu$ has rank one. Only one complex linear combination of the supercharges acts with nonzero norm on a positive-energy representation; the null combination is quotiented. The remaining creation operator changes helicity by 1/2. Hence a massless irreducible representation is built from a Clifford algebra with one fermionic oscillator, and its helicities come in adjacent pairs $(h, h+1/2)$ before CPT completion.

Massless $\mathcal{N} = 1$ helicity pair. Let $E > 0$ and $p^\mu = (E, 0, 0, E)$. In a spinor basis where $\sigma^\mu p_\mu$ is diagonal, positivity sets one complex supercharge combination to zero on the Hilbert space and leaves one normalized fermionic oscillator. An irreducible massless $\mathcal{N} = 1$ representation generated from a helicity-h state therefore has two helicity states,

$$h, \qquad h + \frac{1}{2},$$

before imposing CPT. For the chosen momentum, $\sigma^\mu p_\mu$ is a positive semidefinite 2×2 Hermitian matrix of rank one. A unitary change of spinor basis therefore puts the supercharge norm matrix into the form

$$\{Q_a, Q_b^\dagger\} = \begin{pmatrix} cE & 0 \\ 0 & 0 \end{pmatrix}_{ab}, \qquad c > 0.$$

If ψ is any vector in the common invariant domain, then

$$0 = \langle \psi, \{Q_2, Q_2^\dagger\}\psi\rangle = \|Q_2\psi\|^2 + \|Q_2^\dagger\psi\|^2$$

because the second diagonal entry is zero. Positivity of the inner product therefore forces $Q_2\psi = Q_2^\dagger\psi = 0$ for all vectors in the domain, equivalently the null supercharge combination acts trivially after quotienting zero-norm vectors. The remaining normalized operator

$$b = (cE)^{-1/2}Q_1, \qquad b^\dagger = (cE)^{-1/2}Q_1^\dagger$$

satisfies $\{b, b^\dagger\} = 1$. Because a Weyl supercharge has helicity $\pm 1/2$, the nontrivial creation operator shifts helicity by one half in the chosen convention. Thus the irreducible Clifford module has precisely the two displayed helicities.

85.6 Central Charges and BPS Bounds

For extended supersymmetry, odd generators carry an index $I = 1, \ldots, N$. The algebra may contain antisymmetric central charges,

$$\{Q_\alpha^I, Q_\beta^J\} = 2\epsilon_{\alpha\beta}Z^{IJ}, \qquad Z^{IJ} = -Z^{JI}.$$

In a rest frame, positivity of the matrix of anticommutators gives lower bounds on the mass in terms of the singular values of Z. A state saturating such a bound is annihilated by a nonzero linear combination of supercharges; its multiplet is shorter because the corresponding fermionic oscillator is absent in the quotient representation.

Proposition 85.8 (Rest-frame BPS bound from singular values)**.** *In a massive rest frame for extended supersymmetry, suppose the central-charge matrix has singular values $z_s \geq 0$ after skew-diagonalization of the antisymmetric matrix Z^{IJ}. Positivity of the Hilbert-space representation implies*

$$m \geq z_s \qquad \text{for every singular value } z_s.$$

If $m = z_s$ for some s, then a nonzero linear combination of supercharges has zero norm and annihilates the shortened representation.

Proof. The linear-algebra input is the skew Takagi factorization for a complex antisymmetric matrix: there is a unitary matrix U such that

$$(UZU^T)^{2s-1,2s} = z_s, \qquad (UZU^T)^{2s,2s-1} = -z_s, \qquad z_s \geq 0,$$

with all other entries inside the displayed 2×2 block zero, and with additional zero blocks if the number of supercharges is odd. Replacing Q_α^I by $U^I{}_J Q_\alpha^J$ preserves the mixed anticommutator

$$\{Q_\alpha^I, (Q_\beta^J)^\dagger\} = 2m\,\delta_{\alpha\beta}\delta^{IJ}$$

in the rest frame and puts the central term into these blocks.

Fix one nonzero block and write it as $I = 1, 2$, $Z^{12} = z$. Since $(Q^I_\alpha)^\dagger = \bar{Q}_{\dot{\alpha} I}$, the two-dimensional subspace spanned by the operators Q^1_1 and $\bar{Q}_{\dot{2}\,2}$ has Hermitian anticommutator matrix

$$\begin{pmatrix} \{Q^1_1, \bar{Q}_{\dot{1}\,1}\} & \{Q^1_1, Q^2_2\} \\ \{\bar{Q}_{\dot{2}\,2}, \bar{Q}_{\dot{1}\,1}\} & \{\bar{Q}_{\dot{2}\,2}, Q^2_2\} \end{pmatrix} = 2 \begin{pmatrix} m & z \\ z & m \end{pmatrix},$$

after the phase of the block has been absorbed into the unitary Takagi basis. The companion pair $Q^1_2, -\bar{Q}_{\dot{1}\,2}$ gives the same matrix. Thus the normalized combinations

$$R_\pm = \frac{1}{\sqrt{2}} \left(Q^1_1 \pm \bar{Q}_{\dot{2}\,2} \right)$$

have anticommutators

$$\{R_\pm, R^\dagger_\pm\} = 2(m \pm z).$$

The same computation applies independently to every skew block and to each multiplicity space for a repeated singular value.

In a Hilbert-space representation, $\langle \psi, \{R, R^\dagger\} \psi \rangle = \|R^\dagger \psi\|^2 + \|R\psi\|^2$ is nonnegative for every vector in the common invariant domain of the supercharges. Therefore every eigenvalue of each block must be nonnegative, and in particular $2(m - z_s) \geq 0$. This gives $m \geq z_s$ for every singular value. If $m = z_s$, the operator R_- for that block has zero norm on every vector; after quotienting zero-norm vectors and closing the operator, R_- acts trivially. The associated fermionic oscillator is absent, which is the representation-theoretic shortening mechanism. This proves only the algebraic BPS inequality and shortening criterion; whether a QFT contains states saturating it is separate dynamical information. □

Open Problem 85.9 (Nonperturbative realization of supersymmetric QFT data). For a proposed supersymmetric QFT, one must construct the Hilbert space or Euclidean/functorial object, the supercharges as operators or Ward identities, the stress tensor and supercurrent, and the required positivity structure. Perturbative superfield expressions and protected-index calculations are powerful coordinates on such data, but a nonperturbative construction requires separate control of the regulator, local operator domains, anomalies, and possible supersymmetry-breaking effects.

Chapter 86

Superspace, Superfields, and Local Actions

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Superspace is a supermanifold, understood as a locally super-ringed space. Its odd coordinates are algebraic generators of local structure sheaves, not points of an ordinary set. This distinction is essential for a precise relation between fermionic path-integral variables, component fields, and operator-valued fermion fields.

Two-component spinor conventions are those of Section 40.16. In this chapter the supersymmetry variation is written with odd constant parameters $\epsilon^\alpha, \bar{\epsilon}_{\dot{\alpha}}$. Thus $\delta_\epsilon = \epsilon^\alpha Q_\alpha + \bar{\epsilon}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}}$ is an even derivation of the supercommutative algebra of field variables. This convention is convenient for component calculations because all signs are obtained from the same Koszul rule as for the odd coordinates. If one instead uses commuting spinor parameters to label transformations of operator fields, the displayed component formulae are the same after the parameters are moved outside the graded algebra, but closure signs must be translated accordingly. We use

$$\theta^2 = \theta^\alpha \theta_\alpha, \qquad \bar{\theta}^2 = \bar{\theta}_{\dot{\alpha}} \bar{\theta}^{\dot{\alpha}}, \qquad \epsilon\psi = \epsilon^\alpha \psi_\alpha.$$

Thus, with $\epsilon^{12} = \epsilon_{12} = 1$, $\theta^2 = 2\theta^1\theta^2$ as a top monomial. The Berezin measure below is normalized against this θ^2, not against the unordered monomial $\theta^1\theta^2$.

86.1 Superfields as Field-Variable Data

Definition 86.1 (Off-shell superfield multiplet)**.** An off-shell superfield multiplet is a finite collection of superfields and constraints on them such that the supertranslation derivations act without using Euler–Lagrange equations. Its component fields are coordinates on a space of field variables. They are not, by definition, particle states in a Hilbert space.

If an action is supplied, the Euler–Lagrange equations may remove auxiliary fields or impose differential constraints. If gauge symmetry is present, the physical field configurations are further quotients or cohomology classes. Only after these operations and a Hilbert-space or Euclidean reconstruction has been specified can one compare the resulting one-particle spectrum with the particle representations of Chapter 85. This chapter therefore uses the word *multiplet* only for field-variable packages unless explicitly stated otherwise.

86.2 Affine Superspace

Four-dimensional $\mathcal{N} = 1$ affine superspace has underlying ordinary manifold $\mathbb{M}^4$ and structure sheaf

$$\mathcal{O}_{\mathbb{M}^{4|4}}(U) = C^\infty(U) \otimes \Lambda(\theta^1, \theta^2, \bar{\theta}^{\dot{1}}, \bar{\theta}^{\dot{2}})$$

for open $U \subset \mathbb{M}^4$. A superfield is a section of this sheaf or of a vector bundle tensored with it. In local coordinates,

$$F(x, \theta, \bar{\theta}) = f(x) + \theta^\alpha f_\alpha(x) + \bar{\theta}_{\dot{\alpha}} \bar{f}^{\dot{\alpha}}(x) + \cdots + \theta^2 \bar{\theta}^2 f_{\mathrm{top}}(x),$$

where the expansion terminates because the odd coordinate algebra is finite.

The supertranslation generators act as derivations of the structure sheaf:

$$Q_\alpha = \frac{\partial}{\partial\theta^\alpha} - \mathrm{i}\sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_\mu, \qquad \bar{Q}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} + \mathrm{i}\theta^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \partial_\mu.$$

The covariant derivatives are

$$D_\alpha = \frac{\partial}{\partial\theta^\alpha} + \mathrm{i}\sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_\mu, \qquad \bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - \mathrm{i}\theta^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \partial_\mu.$$

They anticommute with the supercharges and obey

$$\{D_\alpha, \bar{D}_{\dot{\beta}}\} = -2\mathrm{i}\sigma^\mu_{\alpha\dot{\beta}} \partial_\mu.$$

86.3 Chiral Superfields

Definition 86.2 (Chiral superfield)**.** A chiral superfield is a superfield Φ satisfying

$$\bar{D}_{\dot{\alpha}} \Phi = 0.$$

Equivalently, in chiral coordinates

$$y^\mu = x^\mu + \mathrm{i}\theta\sigma^\mu\bar{\theta},$$

it has the component expansion

$$\Phi(y, \theta) = \phi(y) + \sqrt{2}\,\theta^\alpha \psi_\alpha(y) + \theta^2 F(y).$$

Here ϕ is an even scalar field, ψ_α an odd Weyl spinor field, and F an even auxiliary field.

Component transformations of a chiral multiplet. With the supertranslation derivations displayed above and the odd-parameter convention fixed at the beginning of the chapter, the components of a chiral superfield transform as

$$\delta\phi = \sqrt{2}\,\epsilon\psi, \tag{86.1}$$
$$\delta\psi_\alpha = \sqrt{2}\,\epsilon_\alpha F + \mathrm{i}\sqrt{2}\,\sigma^\mu_{\alpha\dot\alpha}\bar\epsilon^{\dot\alpha}\partial_\mu\phi, \tag{86.2}$$
$$\delta F = \mathrm{i}\sqrt{2}\,\bar\epsilon_{\dot\alpha}\bar\sigma^{\mu\dot\alpha\alpha}\partial_\mu\psi_\alpha. \tag{86.3}$$

The commutator of two such even variations is the translation

$$[\delta_{\epsilon_1}, \delta_{\epsilon_2}] = a^\mu\partial_\mu, \qquad a^\mu = 2\mathrm{i}\left(\epsilon_1\sigma^\mu\bar\epsilon_2 - \epsilon_2\sigma^\mu\bar\epsilon_1\right),$$

on each of ϕ, ψ_α, F, without imposing equations of motion.

Use left derivatives with respect to odd variables. In the chiral coordinate system $(y^\mu, \theta, \bar\theta)$, the chiral condition removes the $\bar\theta$-dependence of Φ, while the supercharges restrict to

$$Q_\alpha = \frac{\partial^L}{\partial\theta^\alpha}, \qquad \bar Q_{\dot\alpha} = 2\mathrm{i}\,\theta^\beta\sigma^\mu_{\beta\dot\alpha}\partial_\mu$$

on chiral superfields. Hence

$$\delta\Phi = \epsilon^\alpha\frac{\partial^L}{\partial\theta^\alpha}\big(\phi + \sqrt{2}\,\theta^\beta\psi_\beta + \theta^2 F\big) + 2\mathrm{i}\,\bar\epsilon_{\dot\alpha}\theta^\beta\sigma_\beta^{\mu\dot\alpha}\partial_\mu\big(\phi + \sqrt{2}\,\theta^\gamma\psi_\gamma + \theta^2 F\big).$$

The coefficient of 1 is $\sqrt{2}\,\epsilon^\alpha\psi_\alpha$, giving (86.1). The coefficient of θ^α is $\sqrt{2}\,\delta\psi_\alpha$. Since $\partial^L\theta^2/\partial\theta^\alpha = 2\theta_\alpha$, the ϵQ part contributes $2\epsilon_\alpha F$ to the coefficient of θ^α. Dividing by the $\sqrt{2}$ in the component expansion gives the displayed $\sqrt{2}\,\epsilon_\alpha F$. The $\bar\epsilon\bar Q$ part contributes $\mathrm{i}\sqrt{2}\,\sigma^\mu_{\alpha\dot\alpha}\bar\epsilon^{\dot\alpha}\partial_\mu\phi$, giving (86.2).

The coefficient of θ^2 gives δF. The ϵQ part has no θ^2 contribution because $\partial^L(\theta^2 F)/\partial\theta^\alpha$ is only linear in θ. The remaining term is

$$2\mathrm{i}\sqrt{2}\,\bar\epsilon_{\dot\alpha}\theta^\beta\theta^\gamma\sigma_\beta^{\mu\dot\alpha}\partial_\mu\psi_\gamma.$$

Using $\theta^\beta\theta^\gamma = -\frac{1}{2}\epsilon^{\beta\gamma}\theta^\rho\theta_\rho$ and the spinor-index conventions of Section 40.16, this coefficient is precisely (86.3).

For closure, it is shortest to return to the supertranslation algebra. Since δ_{ϵ_i} are even derivations with odd coefficients,

$$[\delta_{\epsilon_1}, \delta_{\epsilon_2}] = \epsilon_1^\alpha\bar\epsilon_{2\dot\beta}\{Q_\alpha, \bar Q^{\dot\beta}\} - \epsilon_2^\alpha\bar\epsilon_{1\dot\beta}\{Q_\alpha, \bar Q^{\dot\beta}\},$$

and the anticommutator of the differential operators is $2\mathrm{i}\sigma_\alpha^{\mu\dot\beta}\partial_\mu$. This gives the displayed translation. The computation used only the component expansion and the auxiliary field F; no Euler–Lagrange equation was invoked.

86.4 Berezin Integration and Local Superspace Actions

Berezin integration is the projection functional onto the top odd coefficient of the exterior algebra:

$$\int d^2\theta\, \theta^2 = 1, \qquad \int d^2\bar\theta\, \bar\theta^2 = 1.$$

As a linear functional on the odd coordinate algebra, it is the integration operation intrinsic to the locally super-ringed-space description. A local superspace action for chiral fields has the form

$$S = \int d^4x\, d^4\theta\, K(\Phi^i, \bar\Phi^{\bar j}) + \left[\int d^4x\, d^2\theta\, W(\Phi^i) + \text{h.c.}\right],$$

where K is a real function of superfields and W is holomorphic in the chiral superfields. The $d^4\theta$ term is a D-term; the $d^2\theta$ term is an F-term.

Nilpotent Taylor coefficient of a chiral F-term. Let

$$\Phi^i = \phi^i + \sqrt{2}\,\theta^\alpha\psi^i_\alpha + \theta^2 F^i, \qquad i = 1, \dots, n,$$

with the θ^2 convention fixed above, and let W be holomorphic in the even variables ϕ^i. The θ^2 coefficient of $W(\Phi)$ is

$$[W(\Phi)]_{\theta^2} = F^i\,\partial_i W(\phi) - \frac{1}{2}\,\partial_i\partial_j W(\phi)\,\psi^{i\alpha}\psi^j_\alpha. \tag{86.4}$$

The repeated flavor indices are summed, and $\psi^{i\alpha}\psi^j_\alpha$ is symmetric in i, j because the spinor contraction and the odd field variables both change sign under exchange.

Set

$$\Xi^i = \sqrt{2}\,\theta^\alpha\psi^i_\alpha + \theta^2 F^i.$$

Since there are only two undotted θ-coordinates, all terms of degree higher than two in Ξ have zero θ^2 coefficient. Thus

$$W(\phi + \Xi) = W(\phi) + \partial_i W(\phi)\Xi^i + \frac{1}{2}\partial_i\partial_j W(\phi)\Xi^i\Xi^j$$

for the purpose of extracting the F-term. The linear term contributes $\theta^2 F^i \partial_i W$. In the quadratic term the only possible θ^2 contribution comes from the one-θ pieces. The monograph's spinor and odd-variable conventions give the identity

$$(\theta\psi^i)(\theta\psi^j) = -\frac{1}{2}\,\theta^2\,\psi^{i\alpha}\psi^j_\alpha. \tag{86.5}$$

Therefore

$$\frac{1}{2}\partial_i\partial_j W\big(\sqrt{2}\,\theta\psi^i\big)\big(\sqrt{2}\,\theta\psi^j\big) = -\frac{1}{2}\theta^2\partial_i\partial_j W\,\psi^{i\alpha}\psi^j_\alpha.$$

Combining the linear and quadratic contributions proves (86.4). No equation of motion or integration by parts has been used.

For one chiral field with $K = \bar\Phi\Phi$, the component expansion gives

$$\mathcal{L} = -\partial_\mu\bar\phi\,\partial^\mu\phi - \mathrm{i}\bar\psi\bar\sigma^\mu\partial_\mu\psi + \bar F F + \left[F W'(\phi) - \frac{1}{2}W''(\phi)\psi^\alpha\psi_\alpha + \text{h.c.}\right]$$

in the mostly-plus convention. Eliminating F by its algebraic Euler–Lagrange equation gives $F = -\overline{W'(\phi)}$ and potential

$$V(\phi, \bar\phi) = |W'(\phi)|^2.$$

Component form of the Wess–Zumino model. Let Φ^i, $i = 1, \ldots, n$, be chiral superfields with canonical Kahler potential $K = \sum_i \bar{\Phi}^{\bar{i}} \Phi^i$ and holomorphic superpotential $W(\Phi)$. Up to total derivatives, the superspace action

$$S_{\rm WZ} = \int d^4x\, d^4\theta\, \bar{\Phi}^{\bar{i}} \Phi^i + \left[\int d^4x\, d^2\theta\, W(\Phi) + \text{h.c.} \right]$$

has Lagrangian density

$$\begin{aligned} \mathcal{L}_{\rm WZ} = &-\partial_\mu \bar{\phi}^{\bar{i}} \partial^\mu \phi^i - \mathrm{i} \bar{\psi}^{\bar{i}} \bar{\sigma}^\mu \partial_\mu \psi^i + \bar{F}^{\bar{i}} F^i \\ &+ F^i\, \partial_i W(\phi) + \bar{F}^{\bar{i}}\, \partial_{\bar{i}} \bar{W}(\bar{\phi}) - \frac{1}{2}\, \partial_i \partial_j W(\phi)\, \psi^i \psi^j - \frac{1}{2}\, \partial_{\bar{i}} \partial_{\bar{j}} \bar{W}(\bar{\phi})\, \bar{\psi}^{\bar{i}} \bar{\psi}^{\bar{j}}. \end{aligned} \tag{86.6}$$

The auxiliary Euler–Lagrange equations are

$$F^i = -\partial_{\bar{i}} \bar{W}(\bar{\phi}), \qquad \bar{F}^{\bar{i}} = -\partial_i W(\phi),$$

and the on-shell scalar potential is

$$V(\phi, \bar{\phi}) = \sum_i |\partial_i W(\phi)|^2.$$

For the D-term, expand $\bar{\Phi}^{\bar{i}} \Phi^i$ in ordinary $(x, \theta, \bar{\theta})$ coordinates and extract the $\theta^2 \bar{\theta}^2$ coefficient. The $y = x + \mathrm{i}\theta\sigma\bar{\theta}$ dependence of Φ and the conjugate $\bar{y} = x - \mathrm{i}\theta\sigma\bar{\theta}$ dependence of $\bar{\Phi}$ give

$$[\bar{\Phi}^{\bar{i}} \Phi^i]_{\theta^2\bar{\theta}^2} = -\partial_\mu \bar{\phi}^{\bar{i}} \partial^\mu \phi^i - \mathrm{i} \bar{\psi}^{\bar{i}} \bar{\sigma}^\mu \partial_\mu \psi^i + \bar{F}^{\bar{i}} F^i + \partial_\mu(\cdots).$$

The total derivative is discarded in the action. For the F-term, apply (86.4). The θ^2 coefficient is

$$F^i \partial_i W(\phi) - \frac{1}{2} \partial_i \partial_j W(\phi) \psi^i \psi^j.$$

Adding the conjugate gives (86.6). The equations for $F^i, \bar{F}^{\bar{i}}$ are algebraic because no derivatives of auxiliary fields occur. Substitution gives

$$\bar{F}^{\bar{i}} F^i + F^i W_i + \bar{F}^{\bar{i}} \bar{W}_{\bar{i}} = -\sum_i |W_i|^2,$$

so the Lagrangian has the standard form kinetic minus potential, with $V = \sum_i |W_i|^2$.

For the elementary Wess–Zumino model one takes one chiral superfield and

$$W(\Phi) = \frac{1}{2} m \Phi^2 + \frac{1}{3} y \Phi^3.$$

The parameters m and y are coordinates in the chosen supersymmetric regularization scheme; later nonrenormalization arguments concern how these coordinates enter Wilsonian F-terms, not an undefined all-purpose "effective potential".

86.5 Real Vector Superfields and Wess–Zumino Gauge

Gauge multiplets require a separate superfield because a connection is not a gauge-invariant local field variable. A real vector superfield is a Lie-algebra-valued superfield $V = V^\dagger$. In a representation of the gauge group, a chiral gauge parameter Λ acts by

$$e^V \longmapsto e^{-\bar\Lambda} e^V e^{-\Lambda}, \qquad \bar D_{\dot\alpha}\Lambda = 0.$$

For an Abelian gauge group this reduces to

$$V \longmapsto V - \Lambda - \bar\Lambda.$$

The lowest scalar, spinor, and complex auxiliary components of a general real superfield are therefore gauge representatives, not physical fields.

Wess–Zumino gauge as a representative. Locally in field space and perturbatively in the non-Abelian case, the chiral gauge freedom can put a real vector superfield in the representative

$$V_{\mathrm{WZ}} = -\theta\sigma^\mu\bar\theta\, A_\mu + \mathrm{i}\theta^2\bar\theta_{\dot\alpha}\bar\lambda^{\dot\alpha} - \mathrm{i}\bar\theta^2\theta^\alpha\lambda_\alpha + \frac{1}{2}\theta^2\bar\theta^2 D. \tag{86.7}$$

The remaining gauge freedom is the ordinary gauge transformation of $A_\mu, \lambda, \bar\lambda, D$. A supersymmetry transformation of V_{WZ} does not preserve the representative (86.7); it must be followed by a field-dependent chiral gauge transformation. Thus Wess–Zumino gauge is a gauge-fixed coordinate choice on field variables, not a gauge-invariant realization of the supersymmetry algebra.

A general real superfield has the finite expansion

$$V = C + \theta\chi + \bar\theta\bar\chi + \theta^2 M + \bar\theta^2\bar M + \theta\sigma^\mu\bar\theta\, v_\mu + \theta^2\bar\theta\bar\rho + \bar\theta^2\theta\rho + \theta^2\bar\theta^2 E.$$

The chiral parameter has the expansion

$$\Lambda(y,\theta) = a(y) + \sqrt{2}\,\theta\zeta(y) + \theta^2 f(y).$$

In the Abelian transformation $V \mapsto V - \Lambda - \bar\Lambda$, the real part of a removes C, ζ removes χ, and f removes M. The imaginary part of a remains and acts on v_μ by the ordinary gauge transformation $v_\mu \mapsto v_\mu + \partial_\mu\alpha$ after the $y, \bar y$ expansion is written in x-coordinates. Renaming the remaining components with the normalization in (86.7) gives $(A_\mu, \lambda, \bar\lambda, D)$.

For a non-Abelian group, the same elimination is performed order by order using the Baker–Campbell–Hausdorff expansion of $e^{-\bar\Lambda}e^V e^{-\Lambda}$. Since the components removed by Λ are algebraic superfield coordinates and the identity transformation has invertible linearization on them, the formal implicit function theorem gives the local gauge representative. Supersymmetry changes the removed components because Q and $\bar Q$ do not commute with the gauge-slice condition; the compensating chiral gauge transformation is therefore necessary.

In Abelian Wess–Zumino gauge the chiral field strength

$$W_\alpha = -\frac{1}{4}\bar D^2 D_\alpha V$$

has component expansion

$$W_\alpha(y,\theta) = -\mathrm{i}\lambda_\alpha + \theta_\alpha D - \mathrm{i}(\sigma^{\mu\nu}\theta)_\alpha F_{\mu\nu} + \theta^2\sigma^\mu_{\alpha\dot\alpha}\partial_\mu\bar\lambda^{\dot\alpha}.$$

The non-Abelian formula replaces $D_\alpha V$ by $e^{-V}D_\alpha e^V$ and ordinary derivatives by covariant derivatives, as in Chapter 87.

Gauge kinetic coefficient in Wess–Zumino gauge. Let $\eta = \mathrm{diag}(-,+,+,+)$, $\epsilon^{0123} = +1$, and

$$\sigma^{\mu\nu} = \frac{1}{4}\left(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu\right)$$

with the two-component blocks fixed by Section 40.16. Interpret repeated antisymmetric pairs in the displayed W_α formula as summation over all μ, ν; equivalently, in a sum over $\mu < \nu$, replace $(\sigma^{\mu\nu}\theta)_\alpha F_{\mu\nu}$ by $2(\sigma^{\mu\nu}\theta)_\alpha F_{\mu\nu}$. Then the bosonic chiral top coefficient is

$$[W^\alpha W_\alpha]_{\theta^2,\mathrm{bos}} = D^2 - \frac{1}{2}F_{\mu\nu}F^{\mu\nu} - \frac{\mathrm{i}}{4}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{86.8}$$

Consequently, for real g,

$$\frac{1}{4g^2}\int d^2\theta\, W^\alpha W_\alpha + \mathrm{h.c.} = -\frac{1}{4g^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2g^2}D^2 + \mathrm{fermions}. \tag{86.9}$$

For a complex holomorphic gauge coupling the imaginary part of (86.8) instead contributes to the topological θ-angle term, with the normalization matched in Chapter 87.

Only the terms of W_α that are linear in θ can contribute to the bosonic θ^2 coefficient of $W^\alpha W_\alpha$. Thus set

$$L_\alpha = \theta_\alpha D - 2\mathrm{i}\sum_{\mu<\nu}(\sigma^{\mu\nu}\theta)_\alpha F_{\mu\nu}.$$

The factor 2 converts the unrestricted antisymmetric summation in the component formula to a sum over ordered pairs $\mu < \nu$. Since $\epsilon^{12} = \epsilon_{12} = 1$, lowering is $\theta_1 = \theta^2$, $\theta_2 = -\theta^1$, and the inverse operation on a lowered spinor is $\chi^\alpha = -\epsilon^{\alpha\beta}\chi_\beta$. Hence

$$\theta^\alpha\theta_\alpha = \theta^2, \qquad \theta^\alpha\theta_\beta = \frac{1}{2}\delta^\alpha{}_\beta\,\theta^2. \tag{86.10}$$

The pure D-term in $L^\alpha L_\alpha$ gives D^2. The mixed D-F term is proportional to $\mathrm{tr}\,\sigma^{\mu\nu} = 0$, and therefore vanishes. The remaining finite matrix identity, obtained by multiplying the displayed $\sigma^{\mu\nu}$ blocks and using (86.10), is

$$[(\sigma^{\mu\nu}\theta)^\alpha(\sigma^{\rho\sigma}\theta)_\alpha]_{\theta^2} = \frac{1}{4}\left(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}\right) + \frac{\mathrm{i}}{4}\epsilon^{\mu\nu\rho\sigma}, \qquad \mu<\nu,\ \rho<\sigma. \tag{86.11}$$

Substitution gives

$$[L^\alpha L_\alpha]_{\theta^2} = D^2 - 4\sum_{\mu<\nu,\rho<\sigma} F_{\mu\nu}F_{\rho\sigma}\left[(\sigma^{\mu\nu}\theta)^\alpha(\sigma^{\rho\sigma}\theta)_\alpha\right]_{\theta^2}.$$

The real part of this sum is $-\sum_{\mu<\nu}\eta^{\mu\mu}\eta^{\nu\nu}F_{\mu\nu}^2 = -\frac{1}{2}F_{\mu\nu}F^{\mu\nu}$. The oriented part uses

$$\sum_{\mu<\nu,\rho<\sigma}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = \frac{1}{4}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma},$$

where the expression on the right has unrestricted repeated indices. This proves (86.8). Adding the Hermitian conjugate with real coefficient $1/(4g^2)$ doubles the real part and cancels the imaginary topological term, giving (86.9).

The companion Python calculation check in `calculation-checks/` verifies (86.10), (86.11), and the final gauge-kinetic coefficient by exact finite exterior algebra. With the trace and gauge-coupling normalization fixed in Chapter 87, the non-Abelian superspace term

$$\frac{1}{4g^2}\int d^2\theta\ \mathrm{tr}(W^\alpha W_\alpha) + \mathrm{h.c.}$$

contains $-\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$.

86.6 Operator Fields and Grassmann Variables

Fermionic component fields in a Hilbert-space QFT are operator-valued distributions satisfying graded locality. Fermionic variables in a superspace or path-integral expression are elements of odd pieces of a Grassmann algebra. The bridge between them is the correlation functional: Grassmann sources and Berezin integration encode antisymmetric multilinear forms that represent vacuum or Euclidean correlation functions of the operator-valued fermion fields after reconstruction or perturbative definition.

86.7 Supersymmetric Wilsonian Schemes

The later dynamics of four-dimensional $\mathcal{N} = 1$ and $\mathcal{N} = 2$ gauge theories will use Wilsonian effective actions. Such statements require a scheme, not only a formal superspace expression.

Definition 86.3 (Supersymmetric Wilsonian scheme)**.** A supersymmetric Wilsonian scheme on a scale interval $\Lambda_{\mathrm{IR}} < \Lambda < \Lambda_{\mathrm{UV}}$ consists of the following data:

(i) a regulated space of field variables $\mathfrak{F}_\Lambda$, including ordinary fields, superfields, ghosts, antifields, and background sources when they are part of the construction;

(ii) an integration prescription with Lagrangian functional S_Λ: for gauge theories this means either a BV finite-cutoff integral with bracket, Laplacian, half-density, and master action specified, or a lattice construction with gauge-invariant link-variable data and blocking maps;

(iii) odd derivations or Ward operators representing the supersymmetry algebra on the regulated variables, with a stated closure relation: off shell, on shell, modulo gauge transformations, or modulo the BV differential;

(iv) a coarse-graining map from scale Λ_2 to scale Λ_1 for $\Lambda_{\mathrm{IR}} < \Lambda_1 < \Lambda_2 < \Lambda_{\mathrm{UV}}$, satisfying a composition law up to finite local coordinate changes;

(v) a coordinate chart on local and quasilocal functionals, including the separation of D-terms, F-terms, gauge kinetic terms, theta terms, and local counterterms allowed by the regulator;

(vi) a prescription for composite operator insertions and background-field variations, stated separately from the regularization of the functional integral itself.

The scheme is called supersymmetric if the regulated Ward identity is exact at finite Λ, or if the failure is a local breaking functional with a specified restoration prescription and anomaly class.

Holomorphic or nonrenormalization statements are statements about coordinates in such a scheme. They are not statements about an undefined path-integral symbol, nor are they automatically statements about the one-particle irreducible effective action. A physical statement extracted from a Wilsonian action must be invariant under the finite coordinate changes allowed by the scheme, or the coordinate dependence must be part of the statement. This distinction is essential in the exact dynamics of four-dimensional supersymmetric gauge theories.

There is a second distinction which will be used repeatedly. The BV formalism can encode gauge symmetry and supersymmetry on a space of fields, ghosts, antifields, and background superfields, and it can express a symmetry-preserving coarse-graining step as a canonical or quantum-canonical operation once a regulator has been supplied. This does not by itself produce an ultraviolet regularization. A regulated BV Laplacian, a finite field-variable space, a cutoff propagator, an integration cycle, and a proof that the regulated quantum master equation and Ward identities hold are additional data. Thus an off-shell-superfield BV description is a natural language for supersymmetric Wilsonian integration, but it is not identical to a manifestly supersymmetric regulator.

For supersymmetric gauge theories, the Wilsonian effective theory is therefore formulated through BV data unless a lattice construction is explicitly available. The BV data must include the regulated odd symplectic structure, normal half-density, BV Laplacian, integration cycle, master action, and coarse-graining map. Lattice data must include the finite gauge-field configuration space, fermion and auxiliary-field regulator, exact gauge symmetry, reflection or Hilbert-space positivity status when needed, and the blocking map for observables. Holomorphic coordinates, moduli-space coordinates, and exact superpotential arguments are verified only relative to one of these gauge-compatible finite-cutoff frameworks.

For four-dimensional supersymmetric gauge theories we shall not assume the existence of a completely satisfactory manifestly supersymmetric regularization scheme. Dimensional reduction is a perturbative prescription: loop momenta are analytically continued away from four dimensions while spinor and gauge algebra are partly kept four-dimensional. A calculation performed with this prescription is therefore verified only after the algebraic rules used at the relevant loop order have been stated and checked, including identities involving $\epsilon_{\mu\nu\rho\sigma}$, chiral matrices, and evanescent tensor structures. Higher-covariant-derivative or heat-kernel regulators supplemented by Pauli–Villars-type fields are likewise candidate schemes, not axioms: one must state the regulator fields, statistics, masses, gauge and supersymmetry transformations, BV pairing, decoupling limit, local counterterm freedom, and anomaly class before using such a construction as a proved input.

Open Problem 86.4 (Regulators preserving supersymmetry). Classify regulator schemes for supersymmetric QFT by the exact symmetries they preserve, the Ward identities they realize before removing the regulator, and the local counterterms required to restore the remaining identities. Dimensional reduction must be treated as a formal perturbative scheme whose consistency is checked at the loop order being used, rather than as a mathematical definition of a supersymmetric continuum theory. Higher-derivative heat-kernel schemes, Pauli–Villars completions, lattice constructions, and BV-compatible exact-RG schemes must be compared at the level of their regulated master equations or finite lattice Ward identities, locality of counterterms, decoupling limits, and anomaly cohomology before nonrenormalization or anomaly statements are used.

Open Problem 86.5 (Wilsonian foundations for four-dimensional supersymmetric gauge dynamics). Develop the exact dynamics of four-dimensional $\mathcal{N} = 1$ and $\mathcal{N} = 2$ gauge theories inside explicitly defined supersymmetric Wilsonian schemes. The target is to formulate holomorphic couplings, effective superpotentials, quantum moduli spaces, Seiberg-Witten effective actions, anomaly matching, decoupling, and duality tests as statements about specified Wilsonian coordinates and scheme-invariant observables, with the relation to 1PI effective actions stated separately.

Open Problem 86.6 (Supersymmetric examples across dimensions). Develop the main classes of supersymmetric QFTs in dimensions other than four with the same separation of algebra, field-variable realization, regulator, Wilsonian coordinate, and observable. In two dimensions this includes $\mathcal{N} = (2, 2)$ Landau–Ginzburg models, nonlinear sigma models with Calabi–Yau target, and gauged linear sigma models, where the relation among UV superpotential data, Kähler data, chiral rings, RG flow, and infrared CFT must be stated as a theorem, construction, controlled argument, or open problem rather than as an undeveloped phase label. In three dimensions this includes supersymmetric Chern–Simons–matter theories, where level quantization, framing, parity anomalies, boundary conditions, monopole operators, and the meaning of Wilsonian coordinates for topological terms must be part of the definition. In six dimensions this includes superconformal theories for which a conventional Lagrangian may not be part of the data; the tensor-branch effective theory, anomaly polynomial, BPS strings, compactifications, and extended operators then become diagnostic objects rather than replacements for a nonperturbative definition.

Definition 86.7 (Localization data)**.** Localization data for a supersymmetric functional integral consist of:

(i) a regulated integration space or integration cycle, including its orientation, boundary, singular strata, and gauge quotient if present;

(ii) an odd symmetry Q acting on the regulated variables, together with the precise bosonic transformation represented by Q^2;

(iii) a Q-invariant action and Q-closed observables;

(iv) a deformation $t\,QV$ whose convergence, boundary behavior, and large-t limit are controlled;

(v) the fixed locus of Q, its normal complex, zero modes, and determinant or Euler-class prescription;

(vi) a contour or residue prescription for remaining finite-dimensional integrals, including the rule used at singular hyperplane arrangements.

Finite-dimensional equivariant localization is a theorem only after the geometric hypotheses of the theorem have been stated. An infinite-dimensional supersymmetric path integral inherits no automatic theorem from this analogy. One must specify the regulator or finite-dimensional approximation, prove or assume the absence of boundary contributions, and describe singular loci. In particular, Jeffrey–Kirwan residues are contour prescriptions tied to hyperplane arrangements and chamber choices. A definition of the original path integral requires the regulator, contour, boundary behavior, and singular loci. Likewise, zero-size gauge instantons are points on a singular compactification or boundary of a moduli space; whether they are legitimate integration objects depends on the chosen compactification, measure, and local QFT interpretation.

In finite-dimensional reductions of gauge theories, a Jeffrey–Kirwan residue typically appears after abelianization or gauge fixing has replaced part of the integral by a meromorphic form on the complexified Cartan algebra with hyperplane divisors from charged fluctuations. The chamber vector encodes orientation data such as a stability parameter, Fayet–Iliopoulos parameter, or contour at infinity. A QFT localization argument must therefore derive the meromorphic form and this chamber data from the regulated integration cycle; otherwise the residue is an additional prescription selecting one boundary value of a singular integral.

Open Problem 86.8 (Rigorous supersymmetric localization)*.* Develop supersymmetric localization in QFT from regulated localization data. The target is to identify when the localized integral is well-defined, when the deformation argument has no boundary contribution, how singular instanton strata and noncompact Coulomb or Higgs directions are compactified or excluded, and how residue prescriptions such as Jeffrey–Kirwan arise from the declared integration cycle rather than from a formal rule.

Chapter 87

Supersymmetric Gauge Theory

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter introduces supersymmetric gauge theory as field-variable data. The particle spectrum, vacuum structure, Wilsonian effective action, and nonperturbative dynamics are later constructions. We keep the gauge representative, the gauge-invariant observables, and the regularization scheme separate.

87.1 Gauge and Trace Conventions

This chapter uses the Hermitian-generator convention of Chapter 45. If $\mathrm{Lie}(G)$ is realized by anti-Hermitian matrices in a unitary representation, write

$$\mathfrak{g} := \mathrm{i}\ \mathrm{Lie}(G)$$

for the corresponding real vector space of Hermitian generators. Its compact Lie bracket in Hermitian coordinates is

$$[X, Y]_{\mathrm{H}} := -\mathrm{i}[X, Y], \qquad X, Y \in \mathfrak{g},$$

where the bracket on the right is the ordinary matrix commutator. Fix an invariant nondegenerate symmetric bilinear form $\mathrm{tr} : \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{R}$. In trace-delta examples one chooses a basis t^a with $\mathrm{tr}(t^a t^b) = \delta^{ab}$.

In a local trivialization, a gauge field is a Hermitian $\mathfrak{g}$-valued one-form $A = A_\mu dx^\mu$, and the covariant derivative on a field in a unitary representation is

$$\nabla_\mu = \partial_\mu - \mathrm{i}A_\mu.$$

The curvature is

$$F_{\mu\nu} := \mathrm{i}[\nabla_\mu, \nabla_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}[A_\mu, A_\nu].$$

Under a smooth gauge transformation $u : U \to G$,

$$A^u_\mu = uA_\mu u^{-1} + \mathrm{i}u\,\partial_\mu u^{-1}.$$

Curvature covariance and gauge-invariant densities. With the definitions above,

$$F^u_{\mu\nu} = uF_{\mu\nu}u^{-1}.$$

Consequently $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ and $\mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu})$ are local gauge-invariant scalar densities.

The transformation law for A_μ is equivalent to

$$\nabla^u_\mu = u\nabla_\mu u^{-1}$$

as an equality of differential operators. Therefore

$$F^u_{\mu\nu} = \mathrm{i}[\nabla^u_\mu, \nabla^u_\nu] = \mathrm{i}u[\nabla_\mu, \nabla_\nu]u^{-1} = uF_{\mu\nu}u^{-1}.$$

Invariance of the bilinear form gives $\mathrm{tr}(uXu^{-1}uYu^{-1}) = \mathrm{tr}(XY)$. Applying this to $X = F_{\mu\nu}$ and $Y = F^{\mu\nu}$, and then to $Y = \widetilde{F}^{\mu\nu}$, proves the two density statements.

The Yang–Mills coupling convention in this monograph is

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) + \frac{\theta}{32\pi^2}\,\mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}) + \cdots,$$

where $\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ in the stated orientation convention. This convention applies uniformly to $U(1)$, $U(N)$, and simple factors after the bilinear form has been fixed.

87.2 Vector Multiplet as Field Variables

Definition 87.1 (Off-shell vector multiplet)**.** An $\mathcal{N} = 1$ off-shell vector multiplet with gauge algebra $\mathfrak{g}$ consists locally of a connection A_μ, a Weyl spinor λ_α valued in $\mathfrak{g}_{\mathbb{C}}$, its conjugate $\bar{\lambda}_{\dot{\alpha}}$, and a real auxiliary field D valued in $\mathfrak{g}$. These are gauge-representative field variables. The physical local observables are gauge-invariant cohomological or operator classes after quantization.

In Wess–Zumino gauge, the supersymmetry transformations have the form

$$\delta A_\mu = \mathrm{i}\epsilon\sigma_\mu\bar{\lambda} - \mathrm{i}\lambda\sigma_\mu\bar{\epsilon},$$

$$\delta\lambda_\alpha = (\sigma^{\mu\nu}\epsilon)_\alpha F_{\mu\nu} + \mathrm{i}\epsilon_\alpha D, \qquad \delta D = -\epsilon\sigma^\mu D_\mu\bar{\lambda} - D_\mu\lambda\sigma^\mu\bar{\epsilon}.$$

Here $D_\mu X := \partial_\mu X - \mathrm{i}[A_\mu, X]$ for adjoint-valued component fields X. The displayed transformations are not bare supertranslations restricted to the Wess–Zumino slice. They are bare supertranslations followed by the compensating chiral gauge transformation that restores the representative (86.7).

Proposition 87.2 (Closure in Wess–Zumino gauge)**.** *Let $\mathcal{V}$ be a local formal neighbourhood in the space of real vector superfields, and let $\mathcal{S} \subset \mathcal{V}$ be the Wess–Zumino slice represented by (86.7). Assume the local slice and the compensating chiral gauge parameters are chosen in this neighbourhood. For each supersymmetry parameter ϵ, write*

$$\widehat{\delta}_\epsilon = \delta^0_\epsilon + \delta^{\mathrm{g}}_{\Lambda_\epsilon}$$

for the vector field tangent to $\mathcal{S}$, where δ^0_ϵ is the supertranslation vector field on $\mathcal{V}$, and $\delta^{\mathrm{g}}_{\Lambda_\epsilon}$ is the chiral gauge vector field with parameter $\Lambda_\epsilon(V)$. Then, on $\mathcal{S}$,

$$[\widehat{\delta}_{\epsilon_1}, \widehat{\delta}_{\epsilon_2}] = a^\rho \partial_\rho + \delta^{\mathrm{g}}_{\Omega_{12}}, \qquad a^\rho = 2\mathrm{i}\left(\epsilon_1 \sigma^\rho \bar{\epsilon}_2 - \epsilon_2 \sigma^\rho \bar{\epsilon}_1\right), \tag{87.1}$$

where Ω_{12} is a real residual ordinary gauge parameter determined by the two compensators and the slice condition. Equivalently, if $\Xi_{12} := \Omega_{12} + a^\rho A_\rho$, then for every adjoint-valued component field $X \in \{\lambda, \bar{\lambda}, D\}$,

$$[\widehat{\delta}_{\epsilon_1}, \widehat{\delta}_{\epsilon_2}]X = a^\rho D_\rho X + \mathrm{i}[\Xi_{12}, X],$$

and for the connection

$$[\widehat{\delta}_{\epsilon_1}, \widehat{\delta}_{\epsilon_2}]A_\mu = a^\rho F_{\rho\mu} + D_\mu \Xi_{12}.$$

No Euler–Lagrange equation is used; the closure is off shell because the auxiliary field D is part of the coordinate system.

Proof. On the full real-vector-superfield space $\mathcal{V}$, the supertranslation vector fields obey the algebra derived in Chapter 86:

$$[\delta^0_{\epsilon_1}, \delta^0_{\epsilon_2}] = a^\rho \partial_\rho.$$

Let $\mathcal{D}_{\mathrm{g}}$ be the vertical distribution generated by chiral gauge transformations. This distribution is involutive because chiral gauge transformations form a group, and it is preserved by supertranslations: if $\bar{D}_{\dot{\alpha}}\Lambda = 0$, then the supertranslation of Λ is again a chiral gauge parameter. Field dependence of $\Lambda_\epsilon(V)$ does not change this conclusion; differentiating a chiral gauge parameter along any tangent vector in field space still gives a chiral gauge parameter. Therefore every term in the commutator

$$[\delta^0_{\epsilon_1} + \delta^{\mathrm{g}}_{\Lambda_{\epsilon_1}}, \delta^0_{\epsilon_2} + \delta^{\mathrm{g}}_{\Lambda_{\epsilon_2}}]$$

except the bare supertranslation commutator is vertical. Hence the commutator of the projected vector fields equals $a^\rho \partial_\rho$ modulo a chiral gauge vector field.

Both $\widehat{\delta}_{\epsilon_i}$ are tangent to the Wess–Zumino slice, so their commutator is tangent to that slice. A vertical vector field that is tangent to the Wess–Zumino slice is precisely the residual ordinary gauge transformation left after the components C, χ, M of a general real vector superfield have been removed. This gives (87.1). In Hermitian-generator conventions an infinitesimal residual gauge parameter Ω acts by

$$\delta^{\mathrm{g}}_\Omega A_\mu = D_\mu \Omega, \qquad \delta^{\mathrm{g}}_\Omega X = \mathrm{i}[\Omega, X]$$

on adjoint component fields. The alternative covariant form follows from the identity

$$a^\rho \partial_\rho A_\mu + D_\mu \Omega = a^\rho F_{\rho\mu} + D_\mu(\Omega + a^\rho A_\rho),$$

which uses $F_{\rho\mu} = \partial_\rho A_\mu - \partial_\mu A_\rho - \mathrm{i}[A_\rho, A_\mu]$ and $D_\mu Y = \partial_\mu Y - \mathrm{i}[A_\mu, Y]$. This proves the stated closure without a component equation of motion. □

Thus Wess–Zumino gauge is not a gauge-invariant description of supersymmetry; it is a convenient representative after ordinary gauge freedom has been partly fixed.

87.3 Superfield Strength

Let V be a real vector superfield. For matrix notation in a representation of G, a chiral gauge transformation acts by

$$e^V \longmapsto e^{-\bar{\Lambda}} e^V e^{-\Lambda}, \qquad \bar{D}_{\dot{\alpha}} \Lambda = 0.$$

The chiral field strength is

$$W_\alpha = -\frac{1}{4} \bar{D}^2 \left(e^{-V} D_\alpha e^V \right), \qquad \bar{D}_{\dot{\alpha}} W_\beta = 0.$$

It transforms covariantly,

$$W_\alpha \longmapsto e^{\Lambda} W_\alpha e^{-\Lambda}.$$

Consequently $\mathrm{tr}(W^\alpha W_\alpha)$ is a chiral gauge-invariant local superfield. With the trace convention fixed above, the gauge kinetic term is normalized so that its component expansion contains the Yang–Mills term

$$-\frac{1}{4g^2} \, \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}).$$

All later anomaly and beta-function formulae must refer back to this coupling normalization.

87.4 Matter Multiplets and the Moment Map

Let R be a finite-dimensional unitary complex representation of G. A chiral matter multiplet is a chiral superfield Φ valued in R. Gauge covariance of the kinetic term is expressed by

$$\int d^4\theta \, \bar{\Phi} e^V \Phi.$$

In components this term contains the scalar kinetic term, fermion kinetic term, gauge couplings, and the auxiliary coupling

$$\langle \mu_R(\phi, \bar{\phi}), D \rangle,$$

where $\mu_R \in \mathfrak{g}^*$ is the moment map determined by

$$\langle \mu_R(\phi, \bar{\phi}), X \rangle = \phi^\dagger X_R \phi, \qquad X \in \mathfrak{g}.$$

Here X_R is the Hermitian generator acting in R. A Fayet–Iliopoulos parameter is a linear functional $\zeta \in \mathfrak{g}^*$ satisfying

$$\zeta([X, Y]_{\mathrm{H}}) = 0, \qquad X, Y \in \mathfrak{g},$$

so it factors through the abelian quotient of $\mathfrak{g}$. In particular there is no FI parameter for a semisimple gauge algebra.

Auxiliary D-field elimination. Let $\eta = \mu_R(\phi, \bar{\phi}) - \zeta$, with $\zeta = 0$ if no FI parameter is allowed. Use the bilinear form to define $\eta^\sharp \in \mathfrak{g}$ by

$$\mathrm{tr}(\eta^\sharp X) = \langle \eta, X \rangle \quad \text{for every } X \in \mathfrak{g}.$$

For fixed scalar fields, the auxiliary part of the Lorentzian Lagrangian is

$$\mathcal{L}_D = \frac{1}{2g^2}\,\mathrm{tr}(D^2) + \langle \eta, D\rangle.$$

Eliminating D gives

$$D = -g^2\eta^\sharp, \qquad \mathcal{L}_D\big|_{\text{on shell}} = -\frac{g^2}{2}\,\mathrm{tr}(\eta^\sharp\eta^\sharp).$$

Thus the corresponding scalar potential contribution is

$$V_D = \frac{g^2}{2}\,\mathrm{tr}(\eta^\sharp\eta^\sharp)$$

in this convention.

By the definition of $\eta^\sharp$,

$$\mathcal{L}_D = \frac{1}{2g^2}\,\mathrm{tr}(D^2) + \mathrm{tr}(\eta^\sharp D).$$

Completing the square gives

$$\mathcal{L}_D = \frac{1}{2g^2}\,\mathrm{tr}\Big(D + g^2\eta^\sharp\Big)^2 - \frac{g^2}{2}\,\mathrm{tr}(\eta^\sharp\eta^\sharp).$$

The algebraic Euler–Lagrange equation is therefore $D + g^2\eta^\sharp = 0$. Substitution gives the displayed on-shell Lagrangian. Since Lorentzian scalar Lagrangians have the form kinetic minus potential, the potential is the positive final expression.

87.5 $\mathcal{N} = 2$ Gauge-Matter Coupling in $\mathcal{N} = 1$ Variables

We shall often use $\mathcal{N} = 1$ superfields to describe four-dimensional $\mathcal{N} = 2$ gauge theory. The translation is field-theoretic and does not use any string-theoretic construction. Its data are as follows. Let G be compact, let R be a finite-dimensional unitary representation, and choose the invariant trace form on $\mathfrak{g}$ used in the Yang–Mills kinetic term. An $\mathcal{N} = 2$ vector multiplet consists, in $\mathcal{N} = 1$ language, of the vector superfield V together with an adjoint chiral superfield

$$\Phi = \Phi^a T^a \in \mathfrak{g}_{\mathbb{C}}.$$

A hypermultiplet in R is represented by two chiral superfields

$$Q \in R, \qquad \widetilde{Q} \in R^\vee,$$

where $R^\vee$ is the algebraic dual representation. If $T^a_R \in \mathrm{End}(R)$ are the representation matrices, write

$$\Phi_R = \Phi^a T^a_R \in \mathrm{End}(R).$$

With canonical flat chiral-field metrics and the Yang–Mills coupling normalization already fixed in the vector kinetic term, the minimal $\mathcal{N} = 2$ gauge-matter superpotential is

$$W_{\mathcal{N}=2} = \sqrt{2}\,g_{\mathrm{YM}}\,\widetilde{Q}\,\Phi_R Q = \sqrt{2}\,g_{\mathrm{YM}}\,\Phi^a\,\widetilde{Q}\,T^a_R Q. \tag{87.2}$$

The coefficient in (87.2) is part of the normalization statement: rescaling $Q, \widetilde{Q}, \Phi$, or changing the trace convention, changes the displayed coefficient together with the kinetic terms and the gauge coupling. Additional flavor tensors or independent cubic coefficients are deformations of the $\mathcal{N} = 1$ theory; they are not the minimal $\mathcal{N} = 2$ gauge coupling.

Gauge invariance and F-term equations. Assume the preceding flat chiral-field metrics, and choose an orthonormal basis T^a for the trace form. Under a finite gauge transformation

$$Q \mapsto u_R Q, \qquad \widetilde{Q} \mapsto \widetilde{Q} u_R^{-1}, \qquad \Phi_R \mapsto u_R \Phi_R u_R^{-1},$$

the contraction $\widetilde{Q}\Phi_R Q$ is invariant. The corresponding scalar F-term potential is

$$V_F = 2g_{\mathrm{YM}}^2 \left(\|\varphi_R q\|_R^2 + \|\widetilde{q}\,\varphi_R\|_{R^\vee}^2 + \sum_a |\widetilde{q}\,T_R^a q|^2 \right),$$

where $q, \widetilde{q}, \varphi$ are the scalar components of $Q, \widetilde{Q}, \Phi$, and $\varphi_R = \varphi^a T_R^a$. Hence the F-flat equations are

$$\varphi_R q = 0, \qquad \widetilde{q}\,\varphi_R = 0, \qquad \widetilde{q}\,T_R^a q = 0 \quad \text{for every } a.$$

Gauge invariance is the finite-dimensional identity

$$(\widetilde{Q} u_R^{-1})(u_R \Phi_R u_R^{-1})(u_R Q) = \widetilde{Q}\Phi_R Q.$$

Let $c = \sqrt{2}\, g_{\mathrm{YM}}$. In coordinates,

$$W = c\, \widetilde{q}_i (\varphi_R)^i{}_j q^j = c\, \varphi^a\, \widetilde{q}_i (T_R^a)^i{}_j q^j.$$

Differentiating with respect to the three independent chiral coordinates gives

$$\frac{\partial W}{\partial \widetilde{q}_i} = c(\varphi_R q)^i, \qquad \frac{\partial W}{\partial q^j} = c(\widetilde{q}\,\varphi_R)_j, \qquad \frac{\partial W}{\partial \varphi^a} = c\, \widetilde{q}\, T_R^a q.$$

For canonical flat metrics the F-term potential is the squared norm of this holomorphic gradient. Since $|c|^2 = 2g_{\mathrm{YM}}^2$, the displayed potential follows, and its zero locus is precisely the simultaneous vanishing of the three displayed gradient components.

If R is the adjoint representation, the two chiral fields in the hypermultiplet are also adjoint-valued. Together with Φ they are the three adjoint chiral superfields of $\mathcal{N} = 4$ Yang–Mills written in $\mathcal{N} = 1$ language, and the cubic term above becomes the usual adjoint cubic after translating T_R^a into the adjoint action. The enhancement statement is a statement about the full kinetic, Yukawa, and auxiliary-field normalizations together with the cubic polynomial.

87.6 Gauge Anomalies as a Consistency Condition

The classical field-variable construction is not yet a quantum gauge theory. For a four-dimensional chiral gauge theory, gauge invariance of the quantum theory requires the perturbative gauge anomaly class to vanish. In a representation R, the cubic anomaly coefficient is the symmetric trilinear form

$$A_R(X, Y, Z) = \mathrm{Tr}_R(X_R \{Y_R, Z_R\}), \qquad X, Y, Z \in \mathfrak{g},$$

with the trace over chiral fermion variables. The local anomaly cancellation condition is

$$\sum_{\text{chiral fermions } i} A_{R_i} = 0$$

as an invariant symmetric trilinear form. Global anomalies and mixed anomalies require separate topological data and are not detected by this polynomial alone.

Conjugate-pair and real-representation cancellation. Let $R^\vee$ be the conjugate representation of a unitary representation R, in the Hermitian-generator convention. Then

$$A_{R^\vee} = -A_R.$$

Consequently a vectorlike pair $R \oplus R^\vee$ has zero perturbative local gauge anomaly. If R is equivalent to its conjugate representation, then $A_R = 0$. In particular, the adjoint gaugino in an $\mathcal{N} = 1$ vector multiplet gives no perturbative cubic gauge anomaly.

If T^a_R are Hermitian generators in R, then the conjugate representation has Hermitian generators

$$T^a_{R^\vee} = -(T^a_R)^*.$$

Therefore

$$\mathrm{Tr}_{R^\vee}\left(T^a_{R^\vee}\{T^b_{R^\vee}, T^c_{R^\vee}\}\right) = -\left[\mathrm{Tr}_R\left(T^a_R\{T^b_R, T^c_R\}\right)\right]^*.$$

The trace on the right is real because the anticommutator of Hermitian matrices is Hermitian and the invariant symmetric tensor is obtained by symmetrizing over a, b, c. Hence $A_{R^\vee} = -A_R$. The vectorlike statement follows by addition. If $R \simeq R^\vee$, anomaly coefficients are representation invariants, so $A_R = A_{R^\vee} = -A_R$, and $A_R = 0$. The adjoint representation of a compact Lie algebra is real, giving the last claim.

Open Problem 87.3 (Supersymmetric gauge theory with regulated Ward identities). Construct four-dimensional supersymmetric gauge theories with a regulator that states the gauge symmetry, supersymmetry transformations, BV master equation, local counterterm space, anomaly class, and decoupling limit at finite cutoff. This is the regulator-level input needed before exact Wilsonian statements about holomorphy, moduli spaces, and duality can be treated as quantum statements rather than coordinate calculations.

Chapter 88

Supersymmetric Wilsonian Schemes and Exact Dynamics

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Exact statements in supersymmetric QFT require a declared Wilsonian scheme. The scheme is the data that define which modes are integrated out, which local coordinates are used for the resulting action, which Ward identities hold at finite cutoff, and which anomaly classes obstruct stronger statements. Supersymmetry of the Wilsonian integration and supersymmetry of the regulator are separate requirements.

88.1 Wilsonian Data

Definition 88.1 (Supersymmetric Wilsonian datum)**.** A supersymmetric Wilsonian datum at scale Λ consists of: a space $\mathcal{E}_\Lambda$ of regulated superfield variables, a gauge group action together with either a BV gauge complex or a lattice gauge-field construction, a projection $\pi_{\Lambda\Lambda'} : \mathcal{E}_\Lambda \to \mathcal{E}_{\Lambda'}$ for $\Lambda > \Lambda'$, an integration cycle in the discarded variables, and a local action functional S_Λ satisfying the declared gauge and supersymmetry Ward identities at finite cutoff.

If the gauge symmetry is present, the natural finite-cutoff statement is a BV statement. Let $(\,\cdot\,,\,\cdot\,)_\Lambda$ be the odd BV bracket and Δ_Λ the regulated BV Laplacian. The quantum master equation is

$$\frac{1}{2}(S_\Lambda, S_\Lambda)_\Lambda - \mathrm{i}\hbar\, \Delta_\Lambda S_\Lambda = 0.$$

This equation is meaningful only after the regulator and the functional measure entering Δ_Λ have been specified. For a gauge-fixed supersymmetric Wilsonian path integral, gauge consistency is the same equation before restriction to the gauge-fixing Lagrangian submanifold. The gauge-fixed Slavnov–Taylor identities and supersymmetry Ward identities are obtained by restricting the BV half-density and applying BV Stokes. A Wilsonian coarse-graining step is gauge-consistent when it is a BV pushforward that sends a solution of the quantum master equation at Λ to a solution at Λ'.

A lattice supersymmetric gauge construction, when it exists, is the other finite-cutoff route: gauge fields are link variables, gauge invariance is exact at finite lattice spacing, and Wilsonian coarse graining is a blocking map for link variables or gauge-invariant observables. The obstruction is then shifted to the realization of supersymmetry, fermions, auxiliary variables, reflection positivity, and the continuum limit. The two routes are therefore complementary: BV is the continuum homological framework for gauge-compatible effective actions, while the lattice gives a compact gauge-invariant finite integral.

88.2 Coarse Graining as BV Pushforward

For $\Lambda > \Lambda'$, a Wilsonian step must be defined before one writes an action $S_{\Lambda'}$. The object naturally transported by BV integration is not the action itself but the BV half-density

$$\rho_\Lambda = \exp\left(\frac{\mathrm{i}}{\hbar} S_\Lambda\right) \mu_\Lambda^{1/2},$$

where $\mu_\Lambda^{1/2}$ is the chosen regulated half-density. The quantum master equation is

$$\Delta_\Lambda \rho_\Lambda = 0,$$

equivalently the displayed equation for S_Λ when Δ_Λ is written using $\mu_\Lambda^{1/2}$.

Definition 88.2 (Admissible BV Wilsonian step)**.** An admissible BV Wilsonian step from Λ to Λ' consists of the following finite-cutoff data.

(i) A split exact sequence of odd symplectic spaces

$$0 \longrightarrow K_{\Lambda\Lambda'} \longrightarrow \mathcal{E}_\Lambda \xrightarrow{p_{\Lambda\Lambda'}} \mathcal{E}_{\Lambda'} \longrightarrow 0$$

such that, after choosing the splitting,

$$\omega_\Lambda = \omega_{\Lambda'} \oplus \omega_{K_{\Lambda\Lambda'}}, \qquad \Delta_\Lambda = \Delta_{\Lambda'} + \Delta_{K_{\Lambda\Lambda'}}.$$

(ii) A Lagrangian integration cycle $\mathcal{L}_{\Lambda\Lambda'} \subset K_{\Lambda\Lambda'}$ for the odd symplectic form on the eliminated variables, together with an orientation or Berezinian convention fixing the fiber integral.

(iii) A convergence or oscillatory-integral prescription for which the fiber integral of the half-density is defined and for which differentiation by $\Delta_{\Lambda'}$ may be interchanged with the fiber integral.

(iv) A BV boundary condition:

$$\int_{\mathcal{L}_{\Lambda\Lambda'}} \Delta_{K_{\Lambda\Lambda'}} \alpha = 0 \tag{88.1}$$

for every half-density α in the class generated by the Wilsonian integrand and by the variations used in Ward identities.

(v) If supersymmetry is required at the cutoff, the regulated supersymmetry generators preserve the split BV structure and the cycle up to a deformation whose boundary term satisfies (88.1).

The adjective "admissible" is load-bearing. Item (i) supplies a genuine BV factorization of fields, ghosts, antifields, and possible auxiliary variables; a projection of bosonic gauge potentials alone is only a map of coordinates. Item (ii) chooses the integration cycle for the eliminated variables, and item (iii) supplies the analytic meaning of the fiber integral. Item (iv) is the finite-cutoff form of the statement that integrating a BV divergence over the eliminated variables produces no boundary contribution. In perturbative language this condition is where the regulator, gauge fixing, contour, and large-field behavior of the effective integration cycle enter. Supersymmetry at the cutoff is an extra compatibility datum in item (v): the regulated supercharges must act on the full BV complex and preserve the admissible cycle up to a BV-exact deformation with vanishing boundary term. The construction of such data is the substantive problem in a supersymmetric Wilsonian scheme. The local coordinate calculation behind item (iv) is the following finite-dimensional BV Stokes calculation.

BV Stokes in a finite-cutoff Darboux chart. Let K be a finite-dimensional odd symplectic space with Darboux coordinates x^i, ξ_i and BV Laplacian

$$\Delta_K = \sum_i \frac{\partial^2}{\partial x^i \partial \xi_i}$$

on functions multiplying the standard half-density. If the Lagrangian cycle is locally $\xi_i = 0$, then

$$\int_{\xi=0} \Delta_K \alpha = \int_{\partial(\xi=0)} \sum_i \left.\frac{\partial \alpha}{\partial \xi_i}\right|_{\xi=0} \iota_{\partial/\partial x^i}(dx^1 \cdots dx^n).$$

In particular, the integral vanishes for compact boundaryless cycles and for noncompact cycles on which the boundary term at infinity is zero. The same statement in another Darboux chart follows by the invariance of the BV Laplacian on half-densities under canonical coordinate changes. Indeed, on $\xi = 0$,

$$\Delta_K \alpha|_{\xi=0} = \sum_i \frac{\partial}{\partial x^i} \left.\frac{\partial \alpha}{\partial \xi_i}\right|_{\xi=0}.$$

Integration over the x-cycle converts this ordinary divergence into the displayed boundary integral. The vanishing criterion is the usual Stokes theorem applied to the integration cycle, expressed in BV half-density coordinates. Since half-density BV integration is defined invariantly, a general Lagrangian chart gives the same formula after a canonical change of Darboux coordinates. Thus the real input in an admissible Wilsonian step is the existence of a cycle and regulator for which this boundary functional vanishes.

The Wilsonian pushforward associated to these data is

$$(p_{\Lambda\Lambda'})_* \rho_\Lambda(\phi') = \int_{\mathcal{L}_{\Lambda\Lambda'}} \rho_\Lambda(\phi', \chi). \tag{88.2}$$

Only after this half-density has been computed and a nonvanishing reference half-density $\mu_{\Lambda'}^{1/2}$ has been chosen do we write

$$(p_{\Lambda\Lambda'})_*\rho_\Lambda = \exp\left(\frac{\mathrm{i}}{\hbar} S_{\Lambda'}\right)\mu_{\Lambda'}^{1/2}.$$

The logarithm is therefore a coordinate choice on half-densities; the semigroup law is first a statement about pushforwards.

Proposition 88.3 (BV pushforward preserves the quantum master equation)**.** *For an admissible BV Wilsonian step,*

$$\Delta_\Lambda\rho_\Lambda = 0 \qquad \Longrightarrow \qquad \Delta_{\Lambda'}(p_{\Lambda\Lambda'})_*\rho_\Lambda = 0.$$

Proof. Write the high variables as $K_{\Lambda\Lambda'}$, so that the admissible step supplies a factorization

$$\Delta_\Lambda = \Delta_{\Lambda'} + \Delta_{K_{\Lambda\Lambda'}}$$

on half-densities in the chosen finite-cutoff Darboux atlas. The residual operator $\Delta_{\Lambda'}$ differentiates only the unintegrated variables, so it commutes with the convergent fiber integral:

$$\Delta_{\Lambda'}(p_{\Lambda\Lambda'})_*\rho_\Lambda = \int_{\mathcal{L}_{\Lambda\Lambda'}} \Delta_{\Lambda'}\rho_\Lambda = \int_{\mathcal{L}_{\Lambda\Lambda'}} \left(\Delta_\Lambda - \Delta_{K_{\Lambda\Lambda'}}\right)\rho_\Lambda.$$

The term containing $\Delta_\Lambda\rho_\Lambda$ vanishes by the quantum master equation at scale Λ. The remaining term is a BV divergence in the integrated variables. In a Darboux chart it is exactly the boundary functional displayed above, and the coordinate-independent form of this vanishing is the admissibility condition (88.1). Hence no residual boundary semidensity is left on the low variables. The proposition proves preservation of the QME for a declared admissible step; it does not construct the regulator or the supersymmetric integration cycle. □

For gauge theories, this BV compatibility or a lattice blocking construction is part of the definition of the Wilsonian step. A projection of gauge potentials by itself does not define a gauge-theory coarse graining; the projection must be lifted to the full ghost-antifield complex with a compatible half-density and integration cycle, or replaced by a finite gauge-invariant lattice blocking map. When a gauge fixing or a supersymmetric regulator changes with Λ, the change is harmless only if it is represented by an admissible BV isotopy of cycles; otherwise the failure of (88.1) is precisely a Ward-identity anomaly of the scheme.

Definition 88.4 (Composable BV Wilsonian tower)**.** For $\Lambda > \Lambda' > \Lambda''$, an admissible tower is composable if the chosen splittings identify

$$\mathcal{E}_\Lambda \simeq \mathcal{E}_{\Lambda''} \oplus K_{\Lambda'\Lambda''} \oplus K_{\Lambda\Lambda'}$$

as odd symplectic spaces, if

$$\mathcal{L}_{\Lambda\Lambda''} = \mathcal{L}_{\Lambda'\Lambda''} \times \mathcal{L}_{\Lambda\Lambda'}$$

with the product Berezinian convention, and if the corresponding ordinary or oscillatory Fubini theorem holds for the Wilsonian half-density and its BV variations.

Semigroup property of a composable BV Wilsonian tower. For a composable BV Wilsonian tower,

$$(p_{\Lambda'\Lambda''})_*(p_{\Lambda\Lambda'})_*\rho_\Lambda = (p_{\Lambda\Lambda''})_*\rho_\Lambda.$$

Consequently, in any action coordinate for which the three half-densities are written as $\rho_\bullet = \exp(\mathrm{i}S_\bullet/\hbar)\mu_\bullet^{1/2}$, the Wilsonian maps obey

$$S_{\Lambda''} = \mathcal{R}_{\Lambda'\Lambda''}\big(\mathcal{R}_{\Lambda\Lambda'}(S_\Lambda)\big) = \mathcal{R}_{\Lambda\Lambda''}(S_\Lambda)$$

up to the additive normalization constants fixed by the chosen reference half-densities. For a test point $\phi'' \in \mathcal{E}_{\Lambda''}$, the iterated pushforward is

$$\int_{\mathcal{L}_{\Lambda'\Lambda''}}\left(\int_{\mathcal{L}_{\Lambda\Lambda'}}\rho_\Lambda(\phi'',\eta,\chi)\right),$$

where $\eta \in K_{\Lambda'\Lambda''}$ and $\chi \in K_{\Lambda\Lambda'}$. By the Fubini part of Definition 88.4, this equals

$$\int_{\mathcal{L}_{\Lambda'\Lambda''}\times\mathcal{L}_{\Lambda\Lambda'}}\rho_\Lambda(\phi'',\eta,\chi).$$

The product cycle is, by definition, the direct cycle $\mathcal{L}_{\Lambda\Lambda''}$. This is exactly the direct pushforward. The equality of actions follows only after passing from half-densities to the chosen action coordinates; any field-independent determinant factor is an additive constant in S and is fixed by the normalization convention.

Finite circle-Darboux model of BV pushforward. Let the base variables be a periodic even coordinate y and its odd BV partner η, and let the eliminated variables be a periodic even coordinate x and its odd BV partner ξ. Work on the finite algebra generated by finitely many Fourier modes

$$e^{imy}e^{inx}, \qquad m,n\in\mathbb{Z},$$

and by the odd variables η,ξ. Define

$$\Delta_B = \frac{\partial}{\partial y}\frac{\partial^L}{\partial\eta}, \qquad \Delta_K = \frac{\partial}{\partial x}\frac{\partial^L}{\partial\xi}, \qquad \Delta = \Delta_B + \Delta_K.$$

Let the fiber pushforward be the normalized circle integral over the Lagrangian cycle $\xi = 0$:

$$P(\alpha)(y,\eta) = \int_{S^1_x}\alpha(y,\eta,x,0)\,\frac{dx}{2\pi}.$$

Then

$$P(\Delta_K\alpha) = 0, \qquad P(\Delta\alpha) = \Delta_B P(\alpha). \tag{88.3}$$

Consequently, if $\Delta\alpha = 0$, then $\Delta_B P(\alpha) = 0$. If two eliminated circle-Darboux pairs are present and the product cycle is used, then the corresponding pushforwards commute and equal the direct product pushforward. It is enough to check a monomial

$$\alpha = c\,e^{imy}e^{inx}\eta^a\xi^b, \qquad a,b\in\{0,1\}.$$

If $b=0$, then $\partial^L\alpha/\partial\xi = 0$, so $P(\Delta_K\alpha)=0$. If $b=1$, then $\Delta_K\alpha$ is n times a Fourier mode e^{inx} with no ξ-factor. The normalized circle integral of a nonzero derivative mode vanishes, and the $n=0$

mode has derivative coefficient $n = 0$. Thus $P(\Delta_K \alpha) = 0$ in all cases. Since Δ_B differentiates only the base variables, it commutes with the normalized x-integral and with setting $\xi = 0$. This proves (88.3).

For two eliminated pairs, each pushforward selects the zero Fourier mode in its own even variable and the coefficient with no corresponding odd fiber variable. These two coefficient extractions act on disjoint variables, so their order is irrelevant and the result is the same as extracting both at once. This is the finite Fourier model of the Fubini and product-cycle assumption in Definition 88.4.

88.3 Local Coordinates: F-Terms and D-Terms

For four-dimensional $\mathcal{N} = 1$ theories, local supersymmetric functionals split by superspace measure:

$$\int d^4x\, d^2\theta\, \mathcal{F}(\Phi, W_\alpha, \nabla) + \int d^4x\, d^4\theta\, \mathcal{K}(\Phi, \bar{\Phi}, V, \nabla).$$

The first class is called F-term coordinates and the second class D-term coordinates. This coordinate decomposition becomes a statement about dynamics only after a Wilsonian scheme and its Ward identities have been specified. The gauge kinetic coordinate is the chiral functional

$$\frac{1}{16\pi \mathrm{i}} \int d^4x\, d^2\theta\, \tau\, \operatorname{tr}(W^\alpha W_\alpha) + \text{c.c.}, \qquad \tau = \frac{\theta}{2\pi} + \frac{4\pi \mathrm{i}}{g^2},$$

with the trace normalization fixed in Chapter 87. If a different superspace normalization for W_α is chosen, the displayed coefficient is changed by the component-matching factor; the physical convention is the component Yang–Mills term $-\frac{1}{4g^2} \operatorname{tr} F^2$.

88.4 Holomorphic Wilsonian Claims

Hypothesis 88.5 (Holomorphic Wilsonian scheme)**.** A Wilsonian scheme is holomorphic for a set of chiral couplings t^A if the regulated chiral action depends holomorphically on t^A, the coarse-graining map preserves the chiral superspace cohomology class of local F-terms, and all violations of holomorphy are represented by declared local anomaly functionals.

Under this hypothesis, nonrenormalization statements become statements about coordinates. A superpotential coordinate

$$\int d^4x\, d^2\theta\, W_{\text{tree}}(\Phi; t)$$

can receive only those Wilsonian corrections that are holomorphic functions of the chiral coordinates, respect the exact symmetries including anomalous spurion transformations, and are local at scale Λ. The selection-rule arguments are replaced here by these three stated conditions.

Selection rule for a Wilsonian F-term. Assume Hypothesis 88.5. Let $\mathcal{U}(t, \Phi)$ be a candidate correction to a chiral F-term. If $\mathcal{U}$ is not a holomorphic section of the same symmetry line bundle as the original chiral measure and operator insertion, then it is absent from the Wilsonian F-term coordinate in that scheme. The Wilsonian F-term coordinate is, by hypothesis, a holomorphic element of the chiral local cohomology with specified transformation under exact and anomalous symmetries. Local terms with different transformation law lie in a different line bundle over the coupling space. They cannot be added to the coordinate without violating the Ward identity defining the scheme. Thus only holomorphic sections with the same transformation law are admissible.

88.5 Exact Dynamics as a Scheme-Dependent Statement

The chiral ring at scale Λ is the cohomology of local chiral operators modulo $\bar{Q}$-exact operators and equations of motion in the Wilsonian scheme. A moduli space statement is an assertion about expectation values of this ring in supersymmetric states together with the effective action governing fluctuations. A duality statement is an isomorphism between two such Wilsonian data, including line operators, anomalies, contact terms, and relevant deformations.

Open Problem 88.6 (Wilsonian construction). Construct a Wilsonian scheme for four-dimensional gauge theories that preserves gauge invariance, realizes supersymmetry in BV form at finite cutoff, states the anomaly functionals, and proves the holomorphic hypotheses used in exact dynamics. This is the infrastructure needed for proofs of $\mathcal{N} = 1$ and $\mathcal{N} = 2$ gauge dynamics.

Chapter 89

Nonrenormalization and Holomorphy

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$.
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- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Nonrenormalization arguments are statements about Wilsonian coordinates in a scheme with declared supersymmetry, gauge symmetry, locality, and anomaly data. This chapter formulates the perturbative superpotential argument, the holomorphic gauge-coupling coordinate, and the passage to the NSVZ beta-function coordinate in the language of Chapter 88.

89.1 Chiral Local Coordinates

Let Φ^i be chiral superfields and t^A chiral coupling coordinates. A Wilsonian chiral coordinate is a local functional of the form

$$I_F[\Phi; t] = \int d^4x \, d^2\theta \, \mathcal{W}(\Phi; t) + \text{c.c.}$$

defined modulo $\bar{D}_{\dot{\alpha}}$-exact full-superspace terms and equations of motion appropriate to the chosen Wilsonian action. A full superspace coordinate is

$$I_D[\Phi, \bar{\Phi}; t, \bar{t}] = \int d^4x \, d^4\theta \, \mathcal{K}(\Phi, \bar{\Phi}; t, \bar{t}).$$

The distinction is cohomological: an F-term is represented by a chiral integrand and the chiral measure, while a D-term is represented by the full superspace measure.

89.2 Perturbative Superpotential Argument

Controlled Approximation 89.1 (Perturbative Wilsonian superpotential argument)**.** Assume a massive or Wilsonian infrared-regulated $\mathcal{N} = 1$ theory in a holomorphic Wilsonian scheme. Perturbative loop corrections do not generate new superpotential terms beyond those already present as holomorphic local coordinates, except through tree-level elimination of massive chiral fields and through explicitly declared anomaly-induced coordinate changes.

The mechanism is a finite statement about the Grassmann part of a regulated supergraph. Consider one connected loop supergraph after every chiral vertex has been written as a full-superspace integral with the usual covariant spinor derivatives on Grassmann delta functions. The graph has a vertex variable θ_v at each vertex. Choose a spanning tree. The delta functions on the tree identify all θ_v's and leave one integration

$$\int d^4\theta.$$

The remaining, non-tree, propagators are precisely the loop propagators. Their spinor derivatives may be integrated by parts until they act on external superfields or on the remaining Grassmann delta functions. The algebra

$$\{D_\alpha, \bar{D}_{\dot\beta}\} = -2i\sigma^\mu_{\alpha\dot\beta}\partial_\mu$$

then converts closed derivative strings into spacetime momenta and residual covariant derivatives. Thus the local Wilsonian contribution of a loop graph is represented by a full-superspace local density

$$\int d^4x\, d^4\theta\, \mathcal{K}_\Gamma(\Phi, \bar\Phi, D, \bar{D}, \partial; \Lambda).$$

The only possible escape from this conclusion is nonlocality. A nonzero chiral F-term class can be extracted from a full-superspace expression by using a chiral projector, but in flat superspace such a projector contains an inverse d'Alembertian. On the standard projector domain it is represented by

$$P_{\text{ch}} = \frac{\bar{D}^2 D^2}{16\Box}.$$

The inverse $\Box$ is exactly the signal of an infrared singularity in the external momenta. A Wilsonian effective action at scale Λ integrates out modes above Λ and expands its low-momentum kernels in nonnegative powers of p/Λ; it does not contain the $1/p^2$ pole needed to turn the full-superspace loop density into a local chiral density. The familiar exceptions in a massless 1PI effective action are therefore not exceptions to the Wilsonian statement: they use the infrared pole that the Wilsonian coordinate chart has deliberately excluded.

Consequently perturbative loop corrections in the declared Wilsonian patch are full-superspace local coordinates. Tree-level elimination of massive chiral fields is different: it substitutes the solution of a chiral equation of motion into the existing superpotential and hence stays within the chiral coordinate class. Anomaly-induced coordinate changes are also different: their chiral terms come from the regulated Berezinian of a change of variables, not from a loop correction to the superpotential. These are precisely the declared exceptions in the scheme data.

Tree-level chiral elimination. Let the chiral coordinates split as (X^a, Φ^i), where the X^a are massive in a Wilsonian patch. Suppose the holomorphic equations

$$\partial_{X^a} W(X, \Phi; t) = 0$$

have, on this patch, a unique holomorphic solution

$$X^a = X^a_*(\Phi; t),$$

and suppose the heavy-block Hessian

$$H_{ab} = \partial_{X^a}\partial_{X^b} W(X_*, \Phi; t)$$

is invertible there. Then the tree-level eliminated superpotential

$$W_{\text{eff}}(\Phi; t) = W(X_*(\Phi; t), \Phi; t)$$

is holomorphic in the remaining chiral coordinates and couplings, and its first derivatives are

$$\partial_{\Phi^i} W_{\text{eff}} = \partial_{\Phi^i} W|_{X=X_*}\,. \tag{89.1}$$

In the quadratic special case

$$W(X, \Phi; t) = W_0(\Phi; t) + J_a(\Phi; t)X^a + \frac{1}{2}M_{ab}(\Phi; t)X^a X^b, \qquad \det M \neq 0,$$

one has

$$X^a_* = -(M^{-1})^{ab} J_b, \qquad W_{\text{eff}} = W_0 - \frac{1}{2}J_a(M^{-1})^{ab}J_b. \tag{89.2}$$

The hypotheses are precisely those of the holomorphic implicit-function theorem applied to the map

$$F_a(X, \Phi; t) = \partial_{X^a} W(X, \Phi; t).$$

Invertibility of $H_{ab} = \partial F_a/\partial X^b$ at the solution gives a unique holomorphic solution $X_*(\Phi; t)$ after possibly shrinking the patch. Composing the holomorphic functions W and X_* gives a holomorphic W_{eff}. The chain rule gives

$$\partial_{\Phi^i} W_{\text{eff}} = \partial_{\Phi^i} W|_{X=X_*} + \partial_{X^a} W|_{X=X_*}\, \partial_{\Phi^i} X^a_*.$$

The second term vanishes by the heavy-field equation of motion, proving (89.1).

For the quadratic representative, the heavy-field equation is

$$J_a + M_{ab}X^b = 0,$$

so $X^a_* = -(M^{-1})^{ab}J_b$. Substitution gives

$$W_{\text{eff}} = W_0 - J_a(M^{-1})^{ab}J_b + \frac{1}{2}M_{ab}(M^{-1})^{ac}J_c(M^{-1})^{bd}J_d = W_0 - \frac{1}{2}J_a(M^{-1})^{ab}J_b,$$

where symmetry of $M_{ab} = \partial_a\partial_b W$ is used in the last step. If $\det M$ vanishes, the patch contains an additional massless chiral coordinate and the elimination is no longer a Wilsonian tree-level operation in this sense.

89.3 Spurion Symmetries

Treat a coupling t^A as a background chiral superfield. If an exact symmetry acts by

$$\Phi^i \mapsto e^{\mathrm{i}q^i\alpha}\Phi^i, \qquad t^A \mapsto e^{\mathrm{i}q^A\alpha}t^A,$$

then a Wilsonian F-term must be a holomorphic section with the same total charge as the chiral measure. This turns dimensional analysis and symmetries into a finite problem: list holomorphic monomials in fields and couplings with the required charges and locality degree. The conclusion is only as strong as the declared symmetry and anomaly accounting.

89.4 Holomorphic Gauge Coupling

For a gauge theory with chiral matter, the holomorphic gauge coordinate is

$$\frac{1}{16\pi\mathrm{i}}\int d^4x\, d^2\theta\, \tau(\Lambda)\,\mathrm{tr}(W^\alpha W_\alpha) + \text{c.c.}$$

The trace convention is the one fixed in Chapter 87. Thus

$$b_0 = 3C_2(G) - \sum_i T(R_i)$$

for chiral matter representations R_i. The indices are defined by

$$\mathrm{Tr}_{R_i}(XY) = T(R_i)\,\mathrm{tr}(XY), \qquad \mathrm{Tr}_{\mathrm{ad}}(XY) = C_2(G)\,\mathrm{tr}(XY).$$

The displayed b_0 is supplied by the regulated background-field one-loop determinant in the declared holomorphic Wilsonian coordinate. In a supersymmetric background-gauge calculation the shell contribution to the chiral gauge kinetic coordinate has the local form

$$\Delta X_h^{(1)} = \left(3C_2(G) - \sum_i T(R_i)\right)\log\frac{\Lambda}{\Lambda_0}, \tag{89.3}$$

with the same trace convention as the gauge kinetic term. The vector multiplet and its gauge-fixed ghost complex supply the $3C_2(G)$ term; a chiral matter multiplet in R_i supplies $-T(R_i)$. These are local heat-kernel or supergraph coefficients of the finite shell determinant. The holomorphy argument below uses this one-loop input to exclude further perturbative powers of the holomorphic weak-coupling coordinate.

The next proposition is conditional in an essential way. Its input is a weak-coupling Wilsonian patch with a chiral local gauge-kinetic coordinate and a spurion charge assignment in which all anomalous transformations have been accounted for. Establishing such a patch for a concrete regulator is part of the supersymmetric Wilsonian-definition problem; the holomorphy argument below then determines the perturbative scale dependence of that declared coordinate.

Proposition 89.2 (Wilsonian one-loop exactness of the holomorphic gauge coordinate)**.** *Work in a weak-coupling Wilsonian patch in which:*

(i) the gauge kinetic coefficient is a chiral local coordinate $\tau_{\mathrm{eff}}(\mu)$, holomorphic in the microscopic chiral couplings;

(ii) the theta-angle periodicity is represented by $\tau \mapsto \tau + 1$, so scale derivatives of the gauge kinetic coordinate are single-valued holomorphic functions of

$$q_h := \exp(2\pi\mathrm{i}\tau);$$

(iii) the perturbative sector is the q_h^0 coefficient in the expansion near $q_h = 0$; positive powers of q_h are instanton sectors and negative powers are excluded from the weak-coupling Wilsonian patch;

(iv) the superpotential spurion charge assignment contains no additional anomalous chiral coordinate with the transformation law of the gauge kinetic term.

Then the perturbative Wilsonian running of the holomorphic coordinate is

$$\tau(\Lambda) = \tau(\Lambda_0) - \frac{b_0}{2\pi\mathrm{i}} \log \frac{\Lambda}{\Lambda_0}, \tag{89.4}$$

up to a finite Λ-independent holomorphic redefinition of the reference coordinate. Equivalently,

$$\mu \frac{d\tau_{\mathrm{eff}}^{\mathrm{pert}}}{d\mu} = -\frac{b_0}{2\pi\mathrm{i}}.$$

Proof. The statement concerns a Wilsonian chiral coordinate rather than the canonical two-point-function coupling. By locality and chirality, the scale derivative $\mu\, d\tau_{\mathrm{eff}}/d\mu$ is itself the coefficient of the local chiral operator $\mathrm{tr}(W^\alpha W_\alpha)$. It is dimensionless, holomorphic in the chiral couplings, and invariant under $\tau \mapsto \tau + 1$. On the weak-coupling patch it therefore has an expansion

$$\mu \frac{d\tau_{\mathrm{eff}}}{d\mu} = B_0(t) + \sum_{n \geq 1} B_n(t) q_h^n, \tag{89.5}$$

after excluding negative powers by the weak-coupling regularity hypothesis. The $n \geq 1$ terms are nonperturbative instanton sectors. The perturbative projection is the constant term $B_0(t)$.

The coefficient $B_0(t)$ is the perturbative constant term selected by the holomorphic q_h-expansion. Holomorphy and periodicity identify this term as the only perturbative part in the declared weak-coupling patch. Its value is fixed by the one-loop shell determinant (89.3). With the present convention

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi\mathrm{i}}{g_h^2}, \qquad X_h := \frac{8\pi^2}{g_h^2} = 2\pi\, \mathrm{Im}\, \tau,$$

that determinant coefficient is the statement

$$\mu \frac{dX_h}{d\mu} = b_0.$$

Equivalently, $q_h = \exp(2\pi\mathrm{i}\tau)$ has

$$\mu \frac{d}{d\mu} \log q_h = -b_0,$$

so for $b_0 > 0$ the weak-coupling coordinate q_h decreases toward the ultraviolet. This is the asymptotic-freedom sign in the holomorphic coordinate. A perturbative contribution depending nontrivially on other chiral couplings would have to transform in the same affine way as τ. Hypothesis (iv) excludes such an additional coordinate. Hence

$$2\pi\mathrm{i} B_0 = -b_0,$$

or

$$B_0 = -\frac{b_0}{2\pi\mathrm{i}}.$$

Integrating in $\log\Lambda$ gives (89.4). A finite holomorphic change of the reference coordinate adds a Λ-independent function and does not change the derivative.

Canonical gauge couplings are different coordinates. Passing to them requires rescaling matter and gauge multiplets; the corresponding local Berezinian is the Konishi-type anomaly derived below. Thus the one-loop-exact statement here is compatible with higher-loop beta functions in canonical normalization. □

89.5 Konishi-Type Rescaling Anomaly

Let Φ be a chiral superfield valued in a unitary representation R of the gauge group. The representation index used in this section is fixed by

$$\mathrm{Tr}_R(XY) = T(R)\,\mathrm{tr}(XY), \qquad X, Y \in \mathfrak{g},$$

where tr is the same invariant form used in the Yang–Mills term $-\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$.

Proposition 89.3 (Konishi rescaling Jacobian)**.** *Assume a gauge-covariant supersymmetric finite-mode regulator, or a perturbative spectral regulator whose local heat-kernel coefficient agrees with the finite-mode Berezinian limit, for the chiral superfield $\Phi \in R$. Under the constant holomorphic field redefinition*

$$\Phi \mapsto e^{\alpha}\Phi$$

the gauge-field-dependent local part of the regulated super-Berezinian is equivalent to adding the chiral local functional

$$\Delta S = \frac{T(R)\alpha}{8\pi^2}\int d^4x\, d^2\theta\ \mathrm{tr}(W^{\alpha}W_{\alpha}) + c.c.$$

to the Wilsonian action, up to gauge-independent constants and local D-terms removable by the coordinate choice. For several chiral superfields the right hand side is summed over their representations and rescaling parameters.

Proof. The statement is a local Jacobian statement inside a specified regulator, not a property of a formal infinite-dimensional measure. First impose an infrared auxiliary regulator so that the kinetic operator has a discrete orthonormal basis. The odd component field ψ_α of Φ is a left Weyl fermion in R. Expanding it in a gauge-covariant finite set of modes gives odd coordinates c_n, and the ordered Berezinian density $\prod_n dc_n$ transforms by the inverse determinant of the finite matrix representing the change of variables. The scalar and auxiliary components produce ordinary determinants. Their gauge-dependent local contribution is a D-term or a gauge-independent normalization in a supersymmetric regulator; the chiral gauge-field-dependent term is fixed by the Weyl-fermion Berezinian.

The finite-mode calculation is the chiral half of the Berezinian trace in Chapter 51. Replacing the sharp finite-mode projector by the equivalent smooth gauge-covariant spectral regulator for the local coefficient gives

$$\log J^{\mathrm{loc}}_{\alpha,M} = \alpha\ \mathrm{Tr}_{R,S_+}\left[\Pi_+\ \exp(-\mathcal{D}_A^2/M^2)\right]_{\mathrm{loc}} + \text{anti-chiral part} + O(M^{-1}),$$

where S_+ is the left Weyl spinor bundle and $\mathcal{D}_A$ is the Dirac operator coupled to the same connection. The symbol $[\cdots]_{\mathrm{loc}}$ means the part depending locally on the background curvature after gauge-independent power divergences have been subtracted. The chiral asymmetry of the heat kernel is

$$\lim_{M\to\infty} \mathrm{tr}_S \left[\gamma_5 \exp(-\mathcal{D}_A^2/M^2)\right](x,x) = \frac{1}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{Tr}_R(F_{\mu\nu}F_{\rho\sigma}),$$

as in (51.8). Therefore the imaginary part of the Jacobian for the phase subcase $\alpha = \mathrm{i}\beta$ is the regulated Konishi-current anomaly of the left Weyl fermion in R. Using $\mathrm{Tr}_R(XY) = T(R)\,\mathrm{tr}(XY)$, its nontrivial curvature dependence is proportional to $T(R)\,\mathrm{tr}(F\wedge F)$.

It remains to identify the superspace representative of this local functional. The holomorphic part of the Jacobian must be a chiral, gauge-invariant, local functional, linear in α, quadratic in the background gauge field at leading order, and of engineering dimension three in the chiral integrand. In an $\mathcal{N}=1$ gauge background the unique such representative, modulo chiral total derivatives and D-terms, is

$$c\,\alpha \int d^4x\, d^2\theta\ \mathrm{tr}(W^\alpha W_\alpha).$$

The coefficient c is determined by the component anomaly just computed. With the superspace normalization of Chapter 87,

$$\frac{1}{16\pi\mathrm{i}}\tau \int d^4x\, d^2\theta\ \mathrm{tr}(W^\alpha W_\alpha) + \text{c.c.}$$

has component theta term $\frac{\theta}{32\pi^2}\,\mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu})$ when $\mathrm{Re}\,\tau = \theta/(2\pi)$. This fixes

$$c = \frac{T(R)}{8\pi^2}$$

in the action-shift convention of the proposition. The conjugate anti-chiral Berezinian gives the displayed c.c. term. Gauge-independent power-divergent constants and local full-superspace terms change the normalization of fields or the D-term coordinate and are not part of the Konishi chiral anomaly. □

Remark 89.4 (Meaning of the action-shift convention). The sign of a Jacobian may be moved between the transformed density and the action depending on whether one writes the change of variables as $\mathcal{D}\Phi = J\,\mathcal{D}\Phi'$ or $\mathcal{D}\Phi' = J^{-1}\mathcal{D}\Phi$, and on Lorentzian versus Euclidean weight conventions. Proposition 89.3 fixes ΔS by the convention used in the Wilsonian coordinate relation below: a holomorphic rescaling of matter fields shifts the holomorphic gauge coordinate by the amount encoded in the displayed $\frac{T(R)\alpha}{8\pi^2}\int W^2$ term. Ward identities obtained by varying the same change of variables carry the corresponding term on the opposite side if the Jacobian is kept in the measure instead of exponentiated into the action.

This anomaly explains why holomorphic Wilsonian couplings and canonically normalized couplings obey different coordinate beta functions. It is not a claim that a manifestly supersymmetric nonperturbative regulator has been constructed for every four-dimensional supersymmetric gauge theory; it is a perturbative local Jacobian statement in schemes where the above regulated Berezinian calculation is valid.

89.6 NSVZ Coordinate Formula

Fix the bilinear form tr of Chapter 87. Define representation indices by

$$\mathrm{Tr}_{R_i}(XY) = T(R_i)\,\mathrm{tr}(XY), \qquad \mathrm{Tr}_{\mathrm{ad}}(XY) = C_2(G)\,\mathrm{tr}(XY).$$

Let g_h be the holomorphic Wilsonian coupling and let g be the canonically normalized gauge coupling in the same momentum-subtraction scheme for the two-point function. Let $Z_i(\mu)$ be the matter wave-function coordinate and define the anomalous-dimension convention

$$\gamma_i(\mu) = \mu \frac{d}{d\mu} \log Z_i(\mu).$$

Proposition 89.5 (Holomorphic-canonical gauge-coordinate relation)**.** *Assume a perturbative supersymmetric Wilsonian scheme in which:*

(i) the holomorphic gauge coordinate

$$X_h(\mu) := \frac{8\pi^2}{g_h(\mu)^2}$$

is defined by the chiral gauge kinetic term in the convention of Chapter 87;

(ii) the canonical coordinate $g(\mu)$ is defined by normalizing the gauge-field two-point function so that the component kinetic term is $-\frac{1}{4g(\mu)^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$;

(iii) the chiral matter kinetic terms are normalized by positive wave-function coordinates $Z_i(\mu)$;

(iv) the matter rescaling Jacobian is Proposition 89.3, and the gauge-multiplet rescaling Jacobian is computed in the same background-gauge BV regulator.

Then, through the perturbative order to which the regulator and local coordinates have been constructed,

$$\frac{8\pi^2}{g_h^2} = \frac{8\pi^2}{g^2} + C_2(G)\log g^2 - \sum_i T(R_i)\log Z_i + \kappa,$$

where κ is independent of μ and records finite choices of reference normalization for the local coordinates.

Proof. Write the chiral gauge-coordinate part of the Wilsonian action as

$$S^F_{\mathrm{gauge}} = \frac{X_h}{16\pi^2}\int d^4x\, d^2\theta\ \mathrm{tr}(W^\alpha W_\alpha) + \text{theta-angle part} + \text{c.c.}$$

This equation fixes what is meant by $X_h = 8\pi^2/g_h^2$. With the same normalization, Proposition 89.3 says that a holomorphic rescaling $\Phi_i \mapsto e^{\alpha_i}\Phi_i$ shifts this real coordinate by

$$\Delta_i X_h = 2T(R_i)\alpha_i. \tag{89.6}$$

Indeed the chiral action coefficient changes by $\Delta_i X_h/(16\pi^2)$, while the Jacobian coefficient in Proposition 89.3 is $T(R_i)\alpha_i/(8\pi^2)$.

To pass from the Wilsonian fields with matter kinetic coordinate Z_i to canonical matter fields, set

$$\Phi_i^{\mathrm{hol}} = Z_i^{-1/2}\Phi_i^{\mathrm{can}}.$$

Thus $\alpha_i = -\frac{1}{2}\log Z_i$. Substitution in (89.6) gives the matter contribution

$$\Delta_{\mathrm{matter}} X_h = -\sum_i T(R_i)\log Z_i. \tag{89.7}$$

This is the part of the relation that is often called the Konishi anomaly. It is a statement about the local Berezinian of the regulated change of variables, not about an unregulated functional measure.

The vector multiplet must also be rescaled when passing from holomorphic normalization to the canonical gauge coordinate. In a background-field BV gauge fixing, the rescaling is performed on the whole quantum gauge complex: the vector superfield, ghosts, antighosts, auxiliary fields, and antifields are rescaled so that the BV pairing and gauge-fixed quadratic form are kept in canonical dual normalization. Let

$$V^{\mathrm{hol}} = g\,V^{\mathrm{can}}, \qquad \rho := \log g = \frac{1}{2}\log g^2.$$

The regulated super-Berezinian of this adjoint BV complex has the chiral local part

$$\Delta_{\mathrm{vec}} S = \frac{C_2(G)\rho}{8\pi^2}\int d^4x\, d^2\theta\ \mathrm{tr}(W^\alpha W_\alpha) + \mathrm{c.c.}$$

Equivalently,

$$\Delta_{\mathrm{vec}} X_h = 2C_2(G)\rho = C_2(G)\log g^2. \tag{89.8}$$

Here $C_2(G)$ is defined by $\mathrm{Tr}_{\mathrm{ad}}(XY) = C_2(G)\,\mathrm{tr}(XY)$. The coefficient is obtained from the same heat-kernel coefficient as in the matter calculation, but with the supertrace over the gauge-fixed adjoint BV complex. The longitudinal vector modes and chiral Faddeev–Popov/Nielsen– Kallosh ghost modes cancel except for the adjoint chiral index contribution shown above. Performing the rescaling on only A_μ and ignoring this complex would not define a gauge-consistent Jacobian.

Classically, after the vector and matter variables have been brought to the canonical coordinate chart, the real gauge kinetic coordinate is

$$X_c := \frac{8\pi^2}{g^2}.$$

Adding the two anomalous local shifts (89.7) and (89.8), and allowing a finite μ-independent normalization difference between the two declared coordinate charts, gives

$$X_h = X_c + C_2(G)\log g^2 - \sum_i T(R_i)\log Z_i + \kappa.$$

This is the displayed relation. The constant κ changes under finite renormalizations such as $Z_i \mapsto c_i Z_i$, $g \mapsto g + O(g^3)$ at a chosen reference scale, or a finite local gauge kinetic counterterm; it has zero μ-derivative in the fixed scheme and therefore drops out of the beta-function identity. $\square$

NSVZ beta function as a coordinate identity. The beta-function formula commonly called NSVZ contains no further theorem once the holomorphic-canonical coordinate relation and the one-loop holomorphic Wilsonian running have been established in the declared regulator. It is the coordinate expression of the same beta-vector field in the canonical coupling chart. Assume the hypotheses of Proposition 89.5, and assume that the holomorphic Wilsonian coordinate has the one-loop derivative

$$\mu \frac{d}{d\mu}\left(\frac{8\pi^2}{g_h^2}\right) = 3C_2(G) - \sum_i T(R_i). \tag{89.9}$$

Then, in the canonical coupling coordinate g, with

$$\gamma_i = \mu \frac{d}{d\mu} \log Z_i,$$

$$\beta_g = -\frac{g^3}{16\pi^2} \frac{3C_2(G) - \sum_i T(R_i)\big(1-\gamma_i\big)}{1 - \frac{C_2(G)g^2}{8\pi^2}}. \tag{NSVZ}$$

Here is the derivation, included to fix signs and normalizations. Differentiate the coordinate relation of Proposition 89.5. Since

$$\mu \frac{d}{d\mu}\left(\frac{8\pi^2}{g^2}\right) = -\frac{16\pi^2}{g^3}\beta_g, \qquad \mu \frac{d}{d\mu} \log g^2 = \frac{2\beta_g}{g},$$

and $\mu\, d\kappa/d\mu = 0$, one obtains

$$\left(-\frac{16\pi^2}{g^3} + \frac{2C_2(G)}{g}\right)\beta_g - \sum_i T(R_i)\gamma_i = 3C_2(G) - \sum_i T(R_i).$$

Move the anomalous-dimension term to the right:

$$\left(\frac{16\pi^2}{g^3} - \frac{2C_2(G)}{g}\right)\beta_g = -\left(3C_2(G) - \sum_i T(R_i)(1-\gamma_i)\right).$$

Factoring the left hand side as

$$\frac{16\pi^2}{g^3}\left(1 - \frac{C_2(G)g^2}{8\pi^2}\right)\beta_g$$

and dividing gives (NSVZ). Thus the displayed beta function is a coordinate statement derived from a holomorphic Wilsonian coupling, the rescaling anomaly, and a declared canonical normalization prescription. A finite redefinition of the canonical coupling changes the denominator by ordinary reparametrization of the vector field $\beta_g \partial_g$.

89.7 Status Boundary

The perturbative statements above are Wilsonian statements under explicit regularity hypotheses. Nonperturbative superpotentials, instanton effects, gaugino condensation, and strong-coupling moduli-space constraints are not contradictions: they are additional semiclassical or nonperturbative contributions to Wilsonian chiral coordinates. Each such contribution must be derived from a declared integration cycle, zero-mode measure, anomaly matching, and infrared effective description.

Open Problem 89.6 (Rigorous supersymmetric Wilsonian holomorphy). Construct a regulator and BV Wilsonian scheme in which the chiral local cohomology, supergraph measure, rescaling anomaly, and holomorphic gauge coordinate are all defined at finite cutoff, and prove the perturbative nonrenormalization statements in that scheme.

Chapter 90

Four-Dimensional N=1 Gauge Dynamics

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops the first layer of four-dimensional $\mathcal{N} = 1$ gauge dynamics in Wilsonian coordinates. The conclusions are conditional on the supersymmetric Wilsonian scheme and anomaly calculation fixed in Chapters 88 and 89.

90.1 SQCD Data

Let $G = SU(N_c)$. Supersymmetric QCD with N_f flavors has chiral superfields

$$Q^i \in \mathbf{N}_c, \qquad \widetilde{Q}_j \in \overline{\mathbf{N}}_c, \qquad i, j = 1, \ldots, N_f.$$

The vector multiplet has field strength W_α, and the Wilsonian gauge coordinate is

$$\frac{1}{16\pi\mathrm{i}} \int d^4x\, d^2\theta\, \tau\; \mathrm{tr}(W^\alpha W_\alpha) + \text{c.c.}$$

with the trace convention inherited from Chapter 87. The holomorphic scale is the invariant coordinate

$$\Lambda_h^{b_0} = \mu^{b_0} \exp(2\pi\mathrm{i}\tau(\mu)), \qquad b_0 = 3N_c - N_f.$$

This equation defines Λ_h after a branch convention has been chosen; only $\Lambda_h^{b_0}$ is invariant under the periodicity of θ.

90.2 SQCD NSVZ Coordinate Audit

The holomorphic scale is not the same coordinate as the canonical gauge coupling used in particle normalizations. This distinction is a coordinate statement, not a change of the underlying theory. Write

$$X_h(\mu) = \frac{8\pi^2}{g_h(\mu)^2}, \qquad X_c(\mu) = \frac{8\pi^2}{g(\mu)^2},$$

where g_h is the holomorphic Wilsonian coupling and g is the canonical coupling defined by the component kinetic term of Chapter 87. Let Z_Q and $Z_{\widetilde{Q}}$ be the matter wave-function coordinates, with

$$\gamma_Q(\mu) = \mu\frac{d}{d\mu}\log Z_Q(\mu), \qquad \gamma_{\widetilde{Q}}(\mu) = \mu\frac{d}{d\mu}\log Z_{\widetilde{Q}}(\mu).$$

Hypothesis 90.1 (SQCD NSVZ coordinate specialization)**.** The SQCD beta-function identity below is stated in a perturbative supersymmetric Wilsonian scheme satisfying the hypotheses of Proposition 89.5. The flavor-symmetric submanifold is imposed by taking

$$Z_Q = Z_{\widetilde{Q}}, \qquad \gamma_Q = \gamma_{\widetilde{Q}}.$$

The trace convention is the half-trace convention of this chapter:

$$C_2(SU(N_c)) = N_c, \qquad T(\mathbf{N}_c) = T(\overline{\mathbf{N}}_c) = \frac{1}{2}.$$

Finite changes of canonical coupling coordinate are allowed, but then the rational NSVZ denominator changes by the induced reparametrization. Thus the pole of the displayed denominator is a feature of this coordinate chart; the invariant tests in later sections are the numerator condition, anomaly matching, holomorphic scale matching, and protected chiral data.

SQCD specialization of the NSVZ coordinate relation. Assume Hypothesis 90.1. In the flavor-symmetric SQCD coordinate chart,

$$X_h = X_c + N_c \log g^2 - N_f \log Z_Q + \kappa, \tag{90.1}$$

where κ is independent of μ. Consequently

$$\beta_g = -\frac{g^3}{16\pi^2}\,\frac{3N_c - N_f(1-\gamma_Q)}{1 - \frac{N_c g^2}{8\pi^2}}. \tag{90.2}$$

The general relation of Proposition 89.5 gives

$$X_h = X_c + C_2(SU(N_c))\log g^2 - \sum_i T(R_i)\log Z_i + \kappa.$$

There are N_f chiral multiplets Q^i in $\mathbf{N}_c$ and N_f chiral multiplets $\widetilde{Q}_j$ in $\overline{\mathbf{N}}_c$. Hence

$$\sum_i T(R_i)\log Z_i = \frac{N_f}{2}\log Z_Q + \frac{N_f}{2}\log Z_{\widetilde{Q}}.$$

On the flavor-symmetric submanifold this is $N_f \log Z_Q$, and $C_2(SU(N_c)) = N_c$. This proves (90.1).

The holomorphic Wilsonian coordinate has one-loop derivative

$$\mu\frac{dX_h}{d\mu} = 3N_c - N_f.$$

Differentiating (90.1) therefore gives

$$\left(-\frac{16\pi^2}{g^3} + \frac{2N_c}{g}\right)\beta_g - N_f\gamma_Q = 3N_c - N_f.$$

Moving the anomalous-dimension term to the right hand side yields the NSVZ numerator $3N_c - N_f(1-\gamma_Q)$. Solving the displayed linear equation for $\beta_g = \mu\, dg/d\mu$ gives (90.2).

Equivalently, inserting

$$C_2(SU(N_c)) = N_c, \qquad T(\mathbf{N}_c) = T(\overline{\mathbf{N}}_c) = \frac{1}{2}$$

directly into the general NSVZ coordinate formula of Chapter 89 gives

$$\beta_g = -\frac{g^3}{16\pi^2}\,\frac{3C_2(SU(N_c)) - N_f\big(T(\mathbf{N}_c) + T(\overline{\mathbf{N}}_c)\big)(1-\gamma_Q)}{1 - \frac{C_2(SU(N_c))g^2}{8\pi^2}},$$

whose numerator is

$$3N_c - N_f(1-\gamma_Q).$$

The proof above displays why the same numerator arises from the holomorphic-canonical coordinate relation: the two factors of $1/2$ from the fundamental and antifundamental indices add to the single coefficient N_f, while the vector-multiplet Jacobian produces the denominator in the chosen canonical coordinate.

90.3 Gauge-Invariant Chiral Coordinates

The meson matrix is the chiral operator

$$M^i{}_j = \widetilde{Q}_j Q^i.$$

For $N_f \geq N_c$ there are baryons

$$B^{i_1\cdots i_{N_c}} = \epsilon_{a_1\cdots a_{N_c}} Q^{a_1 i_1}\cdots Q^{a_{N_c} i_{N_c}},$$

and similarly antibaryons $\widetilde{B}$. The classical chiral coordinate ring is the quotient of the polynomial ring in $M, B, \widetilde{B}$ by the relations obtained from the definitions above. Quantum dynamics changes this ring only through Wilsonian chiral coordinates compatible with holomorphy, global symmetries, and anomalies.

90.4 Anomaly-Free R-Symmetry

Assign

$$R(Q) = R(\widetilde{Q}) = 1 - \frac{N_c}{N_f}, \qquad R(W_\alpha) = 1.$$

The $SU(N_c)^2U(1)_R$ anomaly coefficient is

$$N_c + \frac{1}{2}N_f(R(Q) - 1) + \frac{1}{2}N_f(R(\widetilde{Q}) - 1) = 0$$

in the fundamental normalization where each fundamental contributes $\frac{1}{2}$. This R-symmetry is part of the chiral-coordinate selection data when $N_f > 0$. If a mass term

$$W_{\text{tree}} = m_i{}^j M^i{}_j$$

is included, then $m_i{}^j$ is treated as a background chiral coordinate with the charge needed to make the superpotential have R-charge 2.

90.5 Gaugino Condensation

For pure $SU(N_c)$ supersymmetric Yang–Mills, $b_0 = 3N_c$. Let

$$S = -\frac{1}{32\pi^2}\,\mathrm{tr}(W^\alpha W_\alpha)$$

be the chiral glueball coordinate in the same trace convention. Its lowest component is proportional to the gaugino bilinear $\mathrm{tr}\,\lambda\lambda$, so it has engineering dimension 3 and gaugino-phase charge 2.

A useful way to keep the nonperturbative status precise is to separate three statements: the anomaly fixes the possible discrete phases, the Witten index counts supersymmetric vacua under finite-volume hypotheses, and a glueball Wilsonian description produces the chiral condensate equation. A constructive mass-gap theorem would additionally require a regulator, a Hilbert-space limit, and spectral estimates for the gauge-invariant local algebra.

Pure SYM discrete chiral anomaly. In pure $SU(N_c)$ supersymmetric Yang–Mills, the classical gaugino phase rotation

$$\lambda_\alpha \longmapsto \mathrm{e}^{\mathrm{i}\alpha}\lambda_\alpha$$

is reduced by the adjoint fermion anomaly to

$$\mathbb{Z}_{2N_c}.$$

If a nonzero condensate $\langle S\rangle$ forms, this discrete symmetry is further broken to $\mathbb{Z}_2$, giving N_c possible condensate phases. The left-handed adjoint Weyl fermion has anomaly coefficient

$$2T(\text{adj}) = 2N_c$$

in the instanton normalization used throughout the chapter. Under the formal rotation by α, the functional measure shifts the Yang–Mills angle by

$$\theta \longmapsto \theta + 2N_c\alpha.$$

Since θ is 2π-periodic, the rotation is a genuine symmetry precisely when

$$2N_c\alpha \in 2\pi\mathbb{Z}, \qquad \text{that is} \qquad \alpha = \frac{\pi\ell}{N_c}, \quad \ell \in \mathbb{Z}.$$

Modulo $\alpha \sim \alpha + 2\pi$, these rotations form $\mathbb{Z}_{2N_c}$. The glueball coordinate S has charge 2 under the same rotation:

$$S \longmapsto \mathrm{e}^{2\mathrm{i}\alpha} S.$$

A nonzero value of S is fixed only when $\mathrm{e}^{2\mathrm{i}\alpha} = 1$, which is the subgroup generated by $\alpha = \pi$. Hence the unbroken subgroup is $\mathbb{Z}_2$, and the quotient has $2N_c/2 = N_c$ phases.

Hypothesis 90.2 (Pure SYM glueball F-term description)**.** The following glueball description is a Wilsonian chiral-sector hypothesis. Work in a supersymmetric holomorphic scheme for pure $SU(N_c)$ Yang–Mills with the glueball coordinate S normalized as above. On each massive chiral branch, assume that the protected F-term dependence is locally described by a holomorphic, possibly multivalued, superpotential

$$W_{\text{eff}}(S; \Lambda_h^{3N_c})$$

and that there are no further massless chiral coordinates. The source normalization of S is fixed by

$$\Lambda_h^{3N_c} \frac{\partial W_{\text{eff}}}{\partial \Lambda_h^{3N_c}} = S. \tag{90.3}$$

Finite holomorphic rescalings of S or Λ_h change the numerical constant in the final condensate but not the number of vacua or their relative phases.

VY glueball superpotential from the chiral-sector hypothesis. Assume Hypothesis 90.2. Up to the finite normalization just described, the pure-SYM glueball superpotential may be written as

$$W_{\text{VY}}(S) = S \left(\log \frac{\Lambda_h^{3N_c}}{S^{N_c}} + N_c \right). \tag{90.4}$$

It has dimension 3, satisfies the source identity (90.3), and its critical points obey

$$S^{N_c} = \Lambda_h^{3N_c}. \tag{90.5}$$

Set $L = \Lambda_h^{3N_c}$. The only dimensionless holomorphic combination of S and L is

$$u = \frac{S^{N_c}}{L}.$$

A dimension-three glueball superpotential therefore has the local form

$$W_{\text{eff}} = S f(u)$$

for a holomorphic function f on a chosen logarithmic branch. The source normalization gives

$$S = L \frac{\partial}{\partial L} \big(S f(u) \big) = -S \, u f'(u).$$

Thus

$$u f'(u) = -1,$$

so

$$f(u) = -\log u + c$$

on that branch. Hence

$$W_{\text{eff}} = S\left(\log\frac{L}{S^{N_c}} + c\right).$$

The constant c is shifted by a finite holomorphic rescaling of L or of S. The standard Veneziano–Yankielowicz representative is the normalization $c = N_c$, giving (90.4). Differentiating this representative with respect to S yields

$$\frac{\partial W_{\text{VY}}}{\partial S} = \log\frac{\Lambda_h^{3N_c}}{S^{N_c}},$$

so the $F_S = 0$ equation is (90.5). Finally, both S and Λ_h^3 have dimension 3, and the logarithm is dimensionless; therefore W_{VY} has dimension 3.

Choosing a branch of Λ_h^3, the solutions of (90.5) are

$$\langle S\rangle_k = C\,\Lambda_h^3\,\exp\left(\frac{2\pi \mathrm{i}k}{N_c}\right), \qquad k = 0,\dots,N_c-1.$$

The constant C depends on the normalization of S and on the definition of Λ_h. In the standard representative of (90.4), $C = 1$. At a critical point,

$$\frac{\partial^2 W_{\text{VY}}}{\partial S^2} = -\frac{N_c}{S},$$

so the glueball critical points are isolated whenever $S \neq 0$. The statement with invariant content is the existence of a nonzero chiral condensate with these relative phases in a scheme where the low-energy theory has a mass gap and isolated vacua.

Condensate branches and source normalization. Assume Hypothesis 90.2, and assume that the isolated critical points of (90.5) are realized as massive chiral branches of the pure $SU(N_c)$ theory. With

$$L = \Lambda_h^{3N_c}.$$

the k-th branch has

$$W_k(L) = N_c\,C\,L^{1/N_c}\mathrm{e}^{2\pi\mathrm{i}k/N_c}, \qquad \langle S\rangle_k = C\,L^{1/N_c}\mathrm{e}^{2\pi\mathrm{i}k/N_c}, \qquad k\in\mathbb{Z}/N_c\mathbb{Z}.$$

The source identity gives

$$L\frac{\partial W_k}{\partial L} = \langle S\rangle_k.$$

Analytic continuation of L once around the origin sends $k\mapsto k+1$. Thus the N_c branches form one orbit under the anomalous chiral symmetry and under a 2π theta-angle loop. The calculation is the following. At a VY critical point the logarithm in (90.4) vanishes, so the critical value of the standard representative is $W_k = N_c S_k$. The finite normalization constant C records the allowed rescaling of S and of the chosen branch of Λ_h^3. Differentiating

$$W_k(L) = N_c\,C\,L^{1/N_c}\mathrm{e}^{2\pi\mathrm{i}k/N_c}$$

gives

$$L\frac{\partial W_k}{\partial L} = C\,L^{1/N_c}\mathrm{e}^{2\pi\mathrm{i}k/N_c} = \langle S\rangle_k.$$

The remaining statement is the elementary monodromy of the N_c-th root: if L is carried around a positively oriented loop about the origin, then L^{1/N_c} is multiplied by $\mathrm{e}^{2\pi\mathrm{i}/N_c}$. This shifts the branch label by one. The same shift is produced by the generator $\alpha = \pi/N_c$ of the anomaly-preserving $\mathbb{Z}_{2N_c}$, since S has gaugino-phase charge 2.

90.6 Protected and Spectral Data in Pure SYM

The glueball coordinate S records protected chiral data. The massive particle spectrum of pure supersymmetric Yang–Mills is a different object: it is extracted from the translation representation on the Hilbert space of a chosen vacuum, or from an equivalent Euclidean transfer matrix after reconstruction. This section fixes the precise spectral problem that connects the chiral branch discussion above to the pure-Yang–Mills spectral discussion of Section 46.8.

Definition 90.3 (Vacuum spectral problem of pure SYM)**.** Fix $N_c \geq 2$ and a trace convention for the gauge algebra as in Section 51.17. A vacuum spectral problem for pure $\mathcal{N} = 1$ $SU(N_c)$ Yang–Mills consists of:

(i) a regulator or continuum construction that gives a local gauge-invariant $*$-algebra $\mathfrak{A}_{\text{loc}}$ generated by smeared components of gauge-invariant superfields;

(ii) a supersymmetric vacuum state ω_k with GNS triple $(\mathcal{H}_k, \Omega_k, \pi_k)$, where $k \in \mathbb{Z}/N_c\mathbb{Z}$ labels the chiral branch in the VY/source-normalization discussion above;

(iii) a strongly continuous translation representation $U_k(a) = \exp(\mathrm{i}a^\mu P_\mu^{(k)})$ with joint spectral measure $E_k(\cdot)$, vacuum vector $P_\mu^{(k)}\Omega_k = 0$, and mass operator $M_k^2 = -\eta^{\mu\nu}P_\mu^{(k)}P_\nu^{(k)}$;

(iv) a finite or countable source family $\mathcal{O}_i \in \mathfrak{A}_{\text{loc}}$ of gauge-invariant smeared component fields, such as the scalar, pseudoscalar, and spinor components of S and their descendants after contact terms have been fixed;

(v) the connected spectral measures

$$\rho_{ij}^{(k)}(B) = \big\langle \pi_k(\mathcal{O}_i)\Omega_k,\, E_k(B)\,\pi_k(\mathcal{O}_j)\Omega_k \big\rangle_{\mathcal{H}_k}$$

for Borel sets $B \subset \{p \in \mathbb{R}^{1,3} : p^0 \geq 0\}$, with the vacuum projection removed when $\mathcal{O}_i$ has a nonzero vacuum expectation value.

An isolated massive supermultiplet of mass $m > 0$ is represented by an atomic component of these measures on the mass shell $p^2 = -m^2$. The matrix of residues on that atom gives the overlap of the chosen sources with the one-particle states.

Chiral F-term data and quadratic masses. Assume that a neighborhood of a branch $S = S_k$ admits a single-coordinate two-derivative chiral effective action

$$\int \mathrm{d}^4\theta\, K(S,\overline{S}) + \left(\int \mathrm{d}^2\theta\, W(S) + \text{c.c.}\right), \qquad g_k = \partial_S\partial_{\overline{S}}K(S_k,\overline{S}_k) > 0,$$

and that $W'(S_k) = 0$. At quadratic order in the fluctuation $\sigma = S - S_k$, the complex scalar and its Majorana fermion partner have common mass

$$m_k = \frac{|W''(S_k)|}{g_k}. \tag{90.6}$$

For the VY representative (90.4),

$$W''(S_k) = -\frac{N_c}{S_k}. \tag{90.7}$$

Thus the protected F-term fixes the branch values and the holomorphic second derivative, while the numerical particle masses also require the Kähler metric and, in the full theory, the complete source-residue matrix of Definition 90.3. At a supersymmetric critical point $W'(S_k) = 0$, the scalar potential is

$$V(S,\overline{S}) = K^{S\overline{S}}(S,\overline{S})\,|W'(S)|^2.$$

Expanding at S_k gives

$$W'(S_k+\sigma) = W''(S_k)\sigma + O(\sigma^2), \qquad K^{S\overline{S}}(S_k,\overline{S}_k) = g_k^{-1}.$$

The quadratic scalar Lagrangian is therefore

$$g_k\,\partial_\mu\sigma\,\partial^\mu\overline{\sigma} - g_k^{-1}|W''(S_k)|^2|\sigma|^2.$$

After the canonical rescaling $\sigma_c = g_k^{1/2}\sigma$, the mass-squared coefficient is $|W''(S_k)|^2/g_k^2$. The fermion quadratic term is

$$-\frac{1}{2}W''(S_k)\psi_S\psi_S + \text{c.c.}$$

together with the kinetic coefficient g_k; the same canonical rescaling gives the fermion mass $|W''(S_k)|/g_k$. Differentiating (90.4) twice gives (90.7). In the microscopic theory, several gauge-invariant sources with the same spin, parity, charge conjugation, and fermion number can have nonzero overlap with the same atom of the spectral measure, so the physical mass extraction uses the residue matrix rather than a single holomorphic coordinate alone.

Definition 90.4 (Pure-SYM wall data)**.** For two massive supersymmetric branches $i, j \in \mathbb{Z}/N_c\mathbb{Z}$, a wall data consist of a Hilbert-space or Euclidean functional-integral construction on $\mathbb{R}^{1,2}\times\mathbb{R}$ with boundary conditions approaching the i-th vacuum as $x^3 \to -\infty$ and the j-th vacuum as $x^3 \to +\infty$, together with a conserved energy per unit $\mathbb{R}^{1,2}$-volume. In a two-derivative chiral effective description, the wall is represented by a path $S(x^3)$ in the branch coordinate space.

BPS wall tension from branch superpotentials. Assume the wall data of Definition 90.4 are governed by a two-derivative $\mathcal{N}=1$ chiral effective action with positive Kähler metric $g_{S\overline{S}}$, and assume that a wall saturates the $\mathcal{N}=1$ central-charge bound. Then its tension is

$$T_{ij} = 2\,|W_j - W_i|, \tag{90.8}$$

where W_i and W_j are the branch values of the superpotential. For the pure-SYM VY branch values

$$W_k = N_c\,C\,\Lambda_h^3\,\mathrm{e}^{2\pi \mathrm{i}k/N_c},$$

the wall between branches k and $k+\ell$ has

$$T_{k,k+\ell} = 4N_c\,|C|\,|\Lambda_h^3|\,\left|\sin\frac{\pi\ell}{N_c}\right|. \tag{90.9}$$

For a static configuration depending only on x^3, the energy density in the effective chiral model is

$$\mathcal{E} = g_{S\overline{S}}\,\partial_3 S\,\partial_3\overline{S} + g^{S\overline{S}}W_S\overline{W}_{\overline{S}}.$$

Let $\alpha \in \mathbb{R}$. Completing the square gives

$$\mathcal{E} = g_{S\overline{S}} \left| \partial_3 S - \mathrm{e}^{\mathrm{i}\alpha} g^{S\overline{S}} \overline{W}_{\overline{S}} \right|^2 + 2 \,\mathrm{Re}\big(\mathrm{e}^{-\mathrm{i}\alpha} \partial_3 W\big) .$$

After integrating over x^3, the last term is

$$2 \,\mathrm{Re}\big(\mathrm{e}^{-\mathrm{i}\alpha}(W_j - W_i)\big) .$$

Choosing $\alpha = \arg(W_j - W_i)$ gives the bound

$$T \geq 2|W_j - W_i|.$$

Saturation is equivalent to the first-order wall equation

$$\partial_3 S = \mathrm{e}^{\mathrm{i}\alpha} g^{S\overline{S}} \overline{W}_{\overline{S}}.$$

Substituting the branch values gives

$$|W_{k+\ell} - W_k| = N_c |C| \, |\Lambda_h^3| \left| \mathrm{e}^{2\pi\mathrm{i}(k+\ell)/N_c} - \mathrm{e}^{2\pi\mathrm{i}k/N_c} \right| = 2N_c |C| \, |\Lambda_h^3| \left| \sin \frac{\pi\ell}{N_c} \right| ,$$

which proves (90.9).

Thus the sine formula is a consequence of the effective chiral wall chart and the saturation hypothesis. The nontrivial QFT problem is the construction of the wall sector and the justification that this two-derivative description with the stated branch superpotential controls its central charge. The square-completion records only the effective-chart consequence of those data.

Pure-SYM confining string tensions are additional line-operator spectral data. They are defined by the area-law exponent of center-charged Wilson-loop or 't Hooft-loop correlators, or equivalently by a transfer-matrix flux-sector energy in a regulator that realizes the center sectors. The wall formula (90.9) is protected by the chiral superpotential under the stated effective-action hypotheses; the flux-string tension requires the nonchiral dynamics encoded in the spectral data.

Pure-SYM one-instanton zero-mode test. Work in the charge-one $SU(N_c)$ instanton sector in the BPST normalization of Section 51.17. Assume a semiclassical collective-coordinate separation in which the exact adjoint fermion zero modes are represented by Grassmann variables and must be saturated by separated operator insertions. Then

$$\langle S(x_1) \cdots S(x_m) \rangle_{k=1}^{\mathrm{sep}} = 0, \qquad m < N_c, \tag{90.10}$$

before contact terms or boundary contributions are considered. The first separated chiral correlator whose fermion degree can saturate the charge-one instanton measure is the N_c-point correlator

$$\langle S(x_1) \cdots S(x_{N_c}) \rangle_{k=1}^{\mathrm{sep}} . \tag{90.11}$$

Its protected charge and dimension assignment is the same assignment as the VY equation $S^{N_c} \sim \Lambda_h^{3N_c}$.

The Dirac index in the adjoint representation gives

$$n_\lambda = 2T(\mathrm{adj})\, k = 2N_c$$

left-handed gaugino zero modes for a charge-one instanton. Let $\eta_1, \ldots, \eta_{2N_c}$ denote their Grassmann coefficients. The zero-mode part of the collective-coordinate measure contains

$$d\eta_{2N_c} \cdots d\eta_1.$$

The lowest component of S is quadratic in the gaugino field. Hence a separated insertion of S is a polynomial of Grassmann degree at most two in the η_a's, and a product of m such insertions has degree at most $2m$. A Berezin integral over $2N_c$ variables extracts only the coefficient of $\eta_1 \cdots \eta_{2N_c}$. If $m < N_c$, that coefficient is absent, proving (90.10).

For $m = N_c$, the Grassmann degree can be $2N_c$. This is a necessary zero-mode condition, not yet a proof of a nonzero coefficient; the spacetime dependence, nonzero-mode determinant, instanton-size integration, and possible boundary terms still have to be controlled. The holomorphic scale factor of a charge-one pure-SYM instanton is $\Lambda_h^{3N_c}$, while S^{N_c} has engineering dimension $3N_c$ and gaugino-phase charge $2N_c$. Thus the first zero-mode-saturated chiral correlator has precisely the dimension and anomalous-charge assignment of the VY critical equation.

Saturated pure-SYM Berezin coefficient. The following exterior-algebra calculation is not a separate QFT theorem; it records the sign and factorial convention needed by the instanton zero-mode test. Keep the assumptions and notation of the one-instanton zero-mode test above. At fixed bosonic instanton collective coordinate $\mathfrak{m}$, let

$$\eta_1, \ldots, \eta_{2N_c}$$

be an oriented basis of the adjoint gaugino zero-mode Grassmann variables, with Berezin measure

$$\mathrm{d}\eta_{2N_c} \cdots \mathrm{d}\eta_1.$$

For a separated insertion point x_a, write the zero-mode part of the lowest component of $S(x_a)$ as the quadratic form

$$S(x_a)\big|_{\mathrm{zm}} = \sum_{1 \le p < q \le 2N_c} K_a^{pq}(x_a; \mathfrak{m})\, \eta_p \eta_q, \tag{90.12}$$

where the coefficients K_a^{pq} are ordinary complex numbers after the bosonic collective coordinate and the zero-mode basis have been fixed. Define the saturated Berezin coefficient

$$\mathfrak{B}_{N_c}(K_1, \ldots, K_{N_c}) = [\eta_1 \cdots \eta_{2N_c}] \prod_{a=1}^{N_c} \left(\sum_{p<q} K_a^{pq} \eta_p \eta_q \right), \tag{90.13}$$

where $[\eta_1 \cdots \eta_{2N_c}]$ denotes coefficient extraction in the ordered monomial $\eta_1 \cdots \eta_{2N_c}$. Equivalently,

$$\mathfrak{B}_{N_c} = \sum_{\substack{p_a < q_a \\ \bigsqcup_{a=1}^{N_c} \{p_a, q_a\} = \{1, \ldots, 2N_c\}}} \operatorname{sgn}(p_1, q_1, \ldots, p_{N_c}, q_{N_c}) \prod_{a=1}^{N_c} K_a^{p_a q_a}. \tag{90.14}$$

The fixed-$\mathfrak{m}$ zero-mode integral of the N_c-point insertion is

$$\int \mathrm{d}\eta_{2N_c} \cdots \mathrm{d}\eta_1 \prod_{a=1}^{N_c} S(x_a)\big|_{\mathrm{zm}} = \mathfrak{B}_{N_c}(K_1, \ldots, K_{N_c}). \tag{90.15}$$

Thus the saturated instanton calculation requires a nonzero top exterior component of the N_c two-forms K_a. In the canonical test case

$$K_a = \eta_{2a-1}\eta_{2a}, \qquad a = 1, \dots, N_c,$$

the coefficient is 1. If instead every insertion has the same canonical two-form

$$\Omega_0 = \sum_{a=1}^{N_c} \eta_{2a-1}\eta_{2a},$$

then the coefficient of $\Omega_0^{N_c}$ is $N_c!$. These factorial and sign conventions are part of the instanton calculation; they are not fixed by dimension, holomorphy, or the index.

Indeed, Berezin integration over the zero-mode variables extracts the coefficient of the oriented top monomial:

$$\int \mathrm{d}\eta_{2N_c} \cdots \mathrm{d}\eta_1 \, \eta_1 \cdots \eta_{2N_c} = 1.$$

Substituting (90.12) into the product of insertions therefore gives (90.15) directly. Expanding the product, a monomial contributes only if every variable $\eta_1, \dots, \eta_{2N_c}$ occurs exactly once. Ordering that monomial to $\eta_1 \cdots \eta_{2N_c}$ produces the sign in (90.14). If a variable is missing or repeated, the coefficient of the top monomial is zero.

For $K_a = \eta_{2a-1}\eta_{2a}$, the product is already the ordered top monomial, hence the coefficient is 1. For $\Omega_0 = \sum_a \eta_{2a-1}\eta_{2a}$, the only nonzero terms in $\Omega_0^{N_c}$ choose each pair exactly once; the $N_c!$ permutations of these commuting degree-two pair factors all contribute with positive sign. This proves the displayed coefficient and records the convention-sensitive normalization that later gets absorbed into the nonzero constant κ_{inst}.

Topological-sector selection for S-correlators. Fix an instanton number $\nu \in \mathbb{Z}_{>0}$. Work in a self-dual charge-ν sector of pure $SU(N_c)$ SYM, with the same BPST/index normalization as above. Assume that the exact adjoint gaugino zero modes are separated as Grassmann variables and that, in the separated chiral correlator under consideration, the only insertions carrying these variables are the displayed S-insertions. Then

$$\langle S(x_1) \cdots S(x_m) \rangle_\nu^{\text{sep}} = 0, \qquad m < \nu N_c, \tag{90.16}$$

at the zero-mode Berezin level. The first possible saturated chiral correlator in the charge-ν sector has

$$m = \nu N_c$$

insertions and carries the same dimension and anomalous gaugino-phase charge as $\Lambda_h^{3\nu N_c}$. Consequently the N_c-point correlator used in Hypothesis 90.5 is a charge-one test: sectors with $\nu \geq 2$ leave unsaturated adjoint zero modes in that channel. Anti-self-dual sectors have the conjugate chirality zero modes and are paired with anti-chiral $\overline{S}$-insertions rather than with the holomorphic S-correlator above.

The selection rule is zero-mode degree counting. The adjoint Dirac index gives

$$n_\lambda(\nu) = 2T(\text{adj})\, \nu = 2N_c \nu$$

left-handed gaugino zero modes in a self-dual charge-ν sector. A separated insertion of S is quadratic in the left-handed gaugino zero-mode variables, so m such insertions have Grassmann degree at most $2m$. The zero-mode Berezin integral in the charge-ν sector extracts the coefficient of a monomial of degree $2N_c\nu$. If $m < \nu N_c$, that top-degree coefficient is absent, proving (90.16).

For $m = \nu N_c$, the fermion degree can match the top degree. This is only the algebraic saturation condition; the corresponding coefficient and the bosonic moduli integral remain separate analytic inputs, as in Hypothesis 90.5. The dimension of $S^{\nu N_c}$ is $3\nu N_c$, and its gaugino-phase charge is $2\nu N_c$, matching the holomorphic scale factor $\Lambda_h^{3\nu N_c}$ and the index count of the zero modes. Setting $m = N_c$ forces $\nu = 1$ for equality. For $\nu \geq 2$, the same N_c-point insertion has degree $2N_c$, leaving $2(\nu - 1)N_c$ adjoint zero modes unsaturated.

The chirality statement is the conjugate form of the same index theorem: anti-self-dual sectors carry the opposite chiral zero modes. The holomorphic operator S is built from the chiral field strength $W_\alpha W^\alpha$, whereas the conjugate zero modes are supplied by $\overline{S}$. Thus they belong to the anti-chiral conjugate calculation, apart from contact terms, source insertions, or boundary effects outside the separated chiral selection rule.

Hypothesis 90.5 (Zero-mode-saturated pure-SYM instanton correlator)**.** The following is the additional analytic input required if one wants to use the original instanton logic as evidence for the pure-SYM condensate. In a supersymmetric holomorphic regulator, assume that the separated N_c-point chiral correlator (90.11) receives a well-defined charge-one instanton contribution

$$\langle S(x_1) \cdots S(x_{N_c}) \rangle_{k=1}^{\text{sep}} = \kappa_{\text{inst}} \Lambda_h^{3N_c}, \qquad \kappa_{\text{inst}} \neq 0, \tag{90.17}$$

for separated points in the chiral sector, in the same finite normalization of S and Λ_h used above. This hypothesis includes control of the saturated Berezin coefficient (90.13) after integration over bosonic collective coordinates, the instanton-size integral, the nonzero-mode determinant, collision singularities, and boundary strata. In un-Higgsed pure SYM these analytic controls are not supplied by the elementary zero-mode count alone.

Remark 90.6 (Cluster extraction is a separate assumption). Assume Hypothesis 90.5 and assume the infinite-volume theory has a massive pure phase ω_ℓ with cluster decomposition and one-point value

$$\omega_\ell(S) = s_\ell.$$

If the separated chiral N_c-point function is identified with the large-separation limit in that pure phase, then

$$s_\ell^{N_c} = \kappa_{\text{inst}} \Lambda_h^{3N_c}. \tag{90.18}$$

Consequently the instanton N_c-point calculation can at most supply the root equation for the condensate. The choice of a particular root and the existence of the massive pure phases require the anomaly, index, and massive-branch inputs stated in this section.

Choose separated points $x_1, \ldots, x_{N_c}$, and translate them so that their mutual distances go to infinity. Cluster decomposition in the pure phase gives

$$\lim_{\min_{a\neq b} |x_a - x_b| \to \infty} \omega_\ell(S(x_1) \cdots S(x_{N_c})) = \prod_{a=1}^{N_c} \omega_\ell(S) = s_\ell^{N_c}.$$

If this large-separation limit is the protected chiral correlator (90.17), then (90.18) follows. The equation determines only the N_c-th power. Its N_c roots are then organized by the $\mathbb{Z}_{2N_c}$ anomaly and the θ-monodromy described in the VY/source-normalization discussion above; no single-instanton Berezin integral selects one branch by itself.

The component condensate $\langle \mathrm{tr}\,\lambda\lambda\rangle_k$ is obtained from $\langle S\rangle_k$ by the fixed component convention in the definition of S. The phase information above is convention independent. In a massive infinite-volume vacuum, cluster decomposition would give

$$\lim_{|x|\to\infty} \langle S(x)S(0)\rangle_k = \langle S\rangle_k^2.$$

This is an infinite-volume statement about a chosen pure phase. A finite-volume ground-state basis can instead diagonalize the discrete symmetry, and the passage to cluster vacua is part of the massive-branch hypothesis rather than a consequence of the index alone.

90.7 Witten Index Of Pure Supersymmetric Yang–Mills

Hypothesis 90.7 (Finite-volume pure-SYM index problem)**.** Fix $N_c \geq 2$ and take the gauge group to be the simply connected group $SU(N_c)$. Put the theory on a spatial three-torus T_L^3 with periodic spin structure for the gaugino, and impose Gauss' law in the trivial discrete-flux sector. The index is computed by a regulated Hamiltonian system whose supersymmetry, gauge invariance, and fermion parity are exact at finite regulator. The physical Hilbert space $\mathcal{H}_0(T_L^3)$ has a self-adjoint Hamiltonian H, closed supercharges $Q, Q^\dagger$, and a fermion-parity operator $(-1)^F$ such that

$$H = \frac{1}{2}\{Q, Q^\dagger\}, \qquad \{(-1)^F, Q\} = \{(-1)^F, Q^\dagger\} = 0.$$

The heat operator $\mathrm{e}^{-\beta H}$ is trace class in the regulated finite-volume problem, and the zero-energy subspace is finite-dimensional. Changing the global form of the gauge group or turning on 't Hooft fluxes changes the Hilbert-space sector and is not part of the displayed $SU(N_c)$ index.

Under Hypothesis 90.7, the finite-volume index is

$$\mathcal{I}(\beta, L) = \mathrm{Tr}_{\mathcal{H}_0(T_L^3)}(-1)^F \mathrm{e}^{-\beta H}.$$

Finite-volume graded-trace identity. Assume Hypothesis 90.7. The finite-dimensional supersymmetry algebra then gives

$$\mathcal{I}(\beta, L) = \dim\ker H|_{\text{bosonic}} - \dim\ker H|_{\text{fermionic}}$$

and is independent of β. Invariance under a continuous change of coupling, volume, or regulator follows provided no zero-energy state is lost to a continuum, no state enters from a continuum, and the deformation preserves the same supercharge, spin structure, global form, and flux sector.

If $H|\psi\rangle = E|\psi\rangle$ with $E > 0$, then at least one of $Q|\psi\rangle$ and $Q^\dagger|\psi\rangle$ is nonzero; otherwise

$$E\|\psi\|^2 = \frac{1}{2}\big(\|Q|\psi\rangle\|^2 + \|Q^\dagger|\psi\rangle\|^2\big) = 0.$$

The nonzero image has the same energy and opposite fermion parity. Thus the positive-energy spectrum contributes zero to the supertrace. Only $\ker H$ remains, giving the displayed formula and β-independence.

For a continuous family of finite-volume Hamiltonians satisfying the same algebra, positive-energy levels can cross or split only in supersymmetric boson-fermion multiplets. The integer supertrace can change only if the definition of the Hilbert-space sector changes or if zero-energy states escape to, or arrive from, a continuum. This is the analytic caveat: the graded trace identity is algebraic, while deformation invariance of a continuum quantum field theory is a statement about the spectral behavior of the regulated family.

Finite-volume symmetry basis versus cluster branches. The following finite-dimensional calculation is only an algebraic model for the zero-energy sector after the massive-branch hypotheses have been imposed. Let V_0 be an N_c-dimensional zero-energy space on which all states have the same fermion parity $\varepsilon = \pm 1$. Suppose a unitary operator U, representing the generator of the quotient $\mathbb{Z}_{2N_c}/\mathbb{Z}_2$, acts on a branch basis $\{|k\rangle\}_{k\in\mathbb{Z}/N_c\mathbb{Z}}$ by

$$U|k\rangle = |k+1\rangle.$$

Suppose also that the chiral coordinate S is diagonal in this branch basis,

$$S|k\rangle = s\,\omega^k|k\rangle, \qquad \omega = \exp(2\pi\mathrm{i}/N_c), \qquad s \neq 0.$$

Then

$$\mathrm{Tr}_{V_0}(-1)^F = \varepsilon N_c, \qquad \mathrm{Tr}_{V_0}\big(U^p(-1)^F\big) = 0 \quad (p \not\equiv 0 \pmod{N_c}), \tag{90.19}$$

and the U-eigenvectors

$$|r\rangle = \frac{1}{\sqrt{N_c}}\sum_{k=0}^{N_c-1}\omega^{-rk}|k\rangle, \qquad r \in \mathbb{Z}/N_c\mathbb{Z},$$

satisfy

$$U|r\rangle = \omega^r|r\rangle, \qquad \langle r|S|r\rangle = 0, \qquad S|r\rangle = s\,|r-1\rangle. \tag{90.20}$$

Thus a finite-volume basis that diagonalizes the exact discrete symmetry need not display a nonzero one-point condensate. The nonzero values $\langle k|S|k\rangle = s\omega^k$ belong to the cluster branch basis, which is an infinite-volume pure-phase idealization requiring the additional massive-branch and cluster hypotheses, beyond the ordinary Witten index.

In the branch basis, U^p is the permutation $k \mapsto k+p$. If $p \not\equiv 0 \pmod{N_c}$, this permutation has no fixed basis vector and therefore has trace zero. If $p \equiv 0$, the trace is N_c. Since $(-1)^F = \varepsilon$ on V_0, this proves (90.19).

The displayed Fourier vectors obey

$$U|r\rangle = \frac{1}{\sqrt{N_c}}\sum_k \omega^{-rk}|k+1\rangle = \omega^r|r\rangle.$$

Their diagonal S-matrix element is

$$\langle r|S|r\rangle = \frac{s}{N_c}\sum_{k=0}^{N_c-1}\omega^k = 0,$$

because $\omega \neq 1$ and $\omega^{N_c} = 1$. Finally,

$$S|r\rangle = \frac{s}{\sqrt{N_c}}\sum_k \omega^{(1-r)k}|k\rangle = s\,|r-1\rangle.$$

The operator S therefore changes the discrete chiral charge sector rather than producing a diagonal order parameter in the symmetry-eigenstate basis. The branch basis diagonalizes S but not U, exactly as a symmetry-breaking basis should.

Hypothesis 90.8 (Small-circle affine-Toda input)**.** The following semiclassical computation is used as a controlled chiral-sector model for the same $SU(N_c)$ theory with periodic fermions on $\mathbb{R}^3 \times S^1_R$. Work at small R in a weak-coupling region with a generic Wilson line breaking

$$SU(N_c) \longrightarrow U(1)^{N_c-1}.$$

The Wilson-line scalars and dual photons are assembled into periodic chiral coordinates Y_i, $i = 1, \ldots, N_c$, with $\sum_i Y_i = 0$. In the basis

$$\alpha_i = e_i - e_{i+1}, \quad i = 1, \ldots, N_c - 1, \qquad \alpha_0 = e_{N_c} - e_1,$$

the $N_c - 1$ fundamental monopole-instantons and the affine monopole each have exactly two gaugino zero modes and contribute a chiral superpotential term. Their product has the topological charge and holomorphic weight of one four-dimensional instanton, denoted by the compactified scale coordinate η. Effects with additional fermion zero modes, nonchiral determinants, and boundary terms are assumed not to change the protected critical-point count. After the chiral variables are gapped, the remaining three-dimensional theory may be placed in finite spatial volume, and the finite-dimensional local oscillator calculation below applies to each nondegenerate massive chiral critical point.

Under Hypothesis 90.8, the monopole terms generate an affine-Toda superpotential of the form

$$W_{\text{aff}} = \sum_{i=1}^{N_c-1} \exp(Y_i - Y_{i+1}) + \eta \exp(Y_{N_c} - Y_1),$$

where η is the four-dimensional holomorphic scale coordinate in the compactified normalization. Define the nonzero variables

$$x_i = \exp(Y_i - Y_{i+1}), \quad i = 1, \ldots, N_c - 1, \qquad x_{N_c} = \eta \exp(Y_{N_c} - Y_1).$$

The constraint is

$$x_1 x_2 \cdots x_{N_c} = \eta.$$

Affine-Toda critical-point count. Assume Hypothesis 90.8 and $\eta \neq 0$. The critical equations of W_{aff} on the constraint hypersurface

$$x_1 \cdots x_{N_c} = \eta$$

have exactly N_c isolated solutions:

$$x_1 = x_2 = \cdots = x_{N_c}.$$

Equivalently, if x is the common value, then

$$x^{N_c} = \eta.$$

Write variations tangent to the constraint as

$$\delta x_i = x_i\,\delta y_i, \qquad \sum_{i=1}^{N_c} \delta y_i = 0.$$

The first variation of $W_{\mathrm{aff}} = \sum_i x_i$ is

$$\delta W_{\mathrm{aff}} = \sum_{i=1}^{N_c} x_i\,\delta y_i.$$

This vanishes for every vector (δy_i) with zero sum if and only if all x_i are equal. The constraint then gives $x^{N_c} = \eta$, which has N_c solutions when $\eta \neq 0$.

The solutions are isolated. On the tangent hyperplane $\sum_i \delta y_i = 0$, the quadratic term at a critical point is

$$\frac{x}{2}\sum_{i=1}^{N_c} (\delta y_i)^2.$$

Since $x \neq 0$, the bilinear form $\sum_i (\delta y_i)^2$ is nondegenerate on the hyperplane. Thus each critical point is Morse in the complex holomorphic sense.

Proposition 90.9 (Local index of a nondegenerate chiral critical point)**.** *Consider a finite-dimensional supersymmetric quantum mechanics obtained by keeping n complex chiral zero modes z^a, a positive Hermitian metric at a critical point p, and a holomorphic superpotential W. Assume that*

$$\partial_a W(p) = 0, \qquad H_{ab} = \partial_a \partial_b W(p)$$

is an invertible complex symmetric matrix, and that the regulator localizes the low-energy contribution to a small ball around p without introducing extra boundary zero modes. Then p contributes

$$\mathcal{I}_{\mathrm{loc}}(p) = +1$$

to the graded trace. In particular, the local contribution in the affine-Toda chamber is not the real Morse sign of $\operatorname{Re} W$*; the holomorphic chiral complex has a canonical complex orientation.*

Proof. Choose normal holomorphic coordinates for the Hermitian metric at p, and discard higher terms in W. The discarded terms give a compact perturbation of the local oscillator as long as no boundary zero mode is created, so the local index is unchanged by the finite-dimensional graded-trace identity. Since H is complex symmetric and invertible, Takagi factorization gives a unitary matrix U and nonzero singular values $\mu_a > 0$ such that

$$H = U^T \operatorname{diag}(\mu_1, \ldots, \mu_n) U.$$

After this holomorphic change of variables, the quadratic model is a tensor product of one-dimensional chiral oscillators with

$$W_{\mathrm{quad}} = \frac{1}{2}\sum_{a=1}^{n} \mu_a (z^a)^2.$$

For one factor, the supersymmetric ground-state equation is the Gaussian equation

$$\left(\frac{\partial}{\partial \bar z} + \mu z\right)\Psi_0 = 0, \qquad \Psi_0(z,\bar z) = \exp(-\mu|z|^2)$$

up to normalization and the harmless convention-dependent factor in the oscillator frequency. This state is normalizable, unique, and lies in the even fermion sector. Every excited oscillator state is paired with a state of opposite fermion parity by the local supercharge. Hence one complex factor has local supertrace $+1$, and the tensor product of n factors also has local supertrace $+1$.

The statement about signs follows from the same normal form. As a real Morse function, $\mathrm{Re}(\mu z^2/2)$ has one positive and one negative Hessian direction for each complex coordinate. That real Morse sign belongs to the de Rham Morse complex of $\mathrm{Re}\, W$, not to the holomorphic chiral oscillator used in the supersymmetric gauge-theory index. □

Hypothesis 90.10 (Index continuation and massive-branch interpretation)**.** The $SU(N_c)$ index obtained in the small-circle semiclassical chamber is identified with the four-dimensional finite-volume index only when no zero-energy state leaks through a continuum and the spin structure, global form, and flux sector are unchanged. The infinite-volume limit is assumed to have N_c massive pure phases, related by the anomalous $\mathbb{Z}_{2N_c}$ chiral symmetry and satisfying cluster decomposition. Confinement and a mass gap require the separate spectral construction described in Section 90.6.

Pure $SU(N_c)$ index and condensate relation. Assume Hypotheses 90.7, 90.8, and 90.10. Then

$$\mathcal{I}_{SU(N_c)} = N_c.$$

If, in addition, Hypothesis 90.2 realizes the massive chiral branches through the glueball coordinate S, then the N_c branches have condensates

$$\langle S\rangle_k = C\,\Lambda_h^3 \mathrm{e}^{2\pi \mathrm{i} k/N_c}, \qquad k = 0,\dots,N_c-1,$$

and the discrete chiral symmetry is broken as $\mathbb{Z}_{2N_c} \to \mathbb{Z}_2$ in each pure phase.

The affine-Toda computation gives N_c isolated supersymmetric critical points in the small-circle chamber. The finite-volume graded-trace identity turns this critical-point count into the index in that chamber, because the critical points are isolated and massive in the chiral variables of the semiclassical approximation, and Proposition 90.9 gives local contribution $+1$ at each of them. The continuation hypothesis then identifies the same integer with the four-dimensional $SU(N_c)$ finite-volume index.

The condensate statement uses an input beyond the index: the additional glueball F-term hypothesis. Under that hypothesis, the VY/source-normalization branch analysis gives precisely the displayed N_c-fold orbit of $\langle S\rangle$. Since S has gaugino-phase charge 2, a nonzero value of $\langle S\rangle_k$ leaves only the $\alpha = \pi$ subgroup unbroken, namely $\mathbb{Z}_2$.

For a simple simply connected gauge group G, the expected index is the dual Coxeter number $h_G^\vee$. The monograph does not use that general statement here; the detailed derivation above is the $SU(N_c)$ case, where $h^\vee_{SU(N_c)} = N_c$. The delicate point is still the route from the regulated finite-volume computation to the four-dimensional infinite-volume theory: the index protects a signed count only under the stated spectral and boundary-condition hypotheses, while the condensate requires the additional massive-branch and chiral-source input.

90.8 ADS Superpotential

The ADS superpotential enters here as a Wilsonian chiral quantity obtained from a controlled semiclassical seed and then continued holomorphically. Symmetry and holomorphy are powerful only after one knows which chiral quantity is being computed, on which patch it is controlled, and which boundary terms have been excluded or absorbed into the holomorphic scale. The constructive seed is the $N_f = N_c - 1$ theory on the maximal-rank Higgs patch. There the gauge group is completely broken, the instanton size integral is cut off at large size by the Higgs expectation values, and the charge-one collective-coordinate integral can be reduced to finite-dimensional bosonic and fermionic zero-mode algebra together with a regulated determinant over nonzero modes.

The logical order used below is therefore the following. First, count the BPST collective coordinates in the normalization fixed earlier and identify the exact two gaugino modes that can become the $d^2\theta$ measure. Second, use the microscopic Yukawa coupling in the Higgs background to lift all remaining matter and non-Goldstino gaugino zero modes, and compute the corresponding Berezin determinant instead of replacing it by an uncomputed "zero-mode saturation" phrase. Third, check the real instanton-size integral that shows how the Higgs vevs control the large-size endpoint. Fourth, use holomorphy, special-flavor invariance, and the scaling degree to identify the remaining chiral function as $C/\det M$. Finally, only after this semiclassical seed is in place, holomorphic mass decoupling propagates the result to $N_f < N_c - 1$.

This order matters conceptually. The explicit instanton calculation supplies the existence of a nonzero F-term, the scale exponent, the exact zero-mode selection rule, and the meson-patch denominator in a regime where the functional integral has a semiclassical meaning. Holomorphy then extends that chiral input. Confinement, a mass gap, and a complete Hilbert-space description require the Wilsonian and infrared hypotheses separated below.

Assume $N_f < N_c$ and consider the region where M has maximal rank. The holomorphic superpotential compatible with flavor symmetry, the anomaly-free R-symmetry, and the scale coordinate has the form

$$W_{\text{ADS}} = C_{N_c,N_f} \left(\frac{\Lambda_h^{3N_c-N_f}}{\det M} \right)^{\frac{1}{N_c-N_f}}.$$

Its mass dimension is 3, because

$$[\Lambda_h^{3N_c-N_f}] = 3N_c - N_f, \qquad [\det M] = 2N_f,$$

and

$$\frac{3N_c - N_f - 2N_f}{N_c - N_f} = 3.$$

Its R-charge is 2, since $R(M) = 2(1 - N_c/N_f)$, hence

$$R(\det M) = 2N_f - 2N_c, \qquad R\Big((\det M)^{-1/(N_c-N_f)}\Big) = 2.$$

The coefficient is fixed by matching to a semiclassical instanton calculation for $N_f = N_c - 1$ and by holomorphic decoupling for smaller N_f, once the instanton measure and scale convention have been specified.

Hypothesis 90.11 (SQCD one-instanton calculus input)**.** For the direct instanton derivation at $N_f = N_c - 1$, the following analytic inputs are assumed. The Euclidean functional integral is regulated in a supersymmetric holomorphic scheme preserving $SU(N_f)_L \times SU(N_f)_R \times U(1)_B$ and the anomaly-free $U(1)_R$. The vacuum is taken on the maximal-rank Higgs patch, so the gauge group is completely broken and the instanton size integral is cut off at large size by the Higgs scale. The collective-coordinate change of variables separates exact zero modes from nonzero modes, with the topological charge, θ-angle, and coupling normalization fixed by Section 51.17. Boundary terms from the small-size endpoint are absorbed into the holomorphic scale coordinate Λ_h, and no further boundary contribution with the same quantum numbers is present. Under these assumptions the one-instanton sector defines a holomorphic chiral F-term up to an overall nonzero constant depending on the normalization of Λ_h.

Higgs-patch collective-coordinate calculation. Let $N_f = N_c - 1$. On the dense maximal-rank meson patch one may choose local complex coordinates

$$Q^{ai} = v_i \delta^{ai}, \qquad \widetilde{Q}_{ia} = \widetilde{v}_i \delta_{ia}, \qquad i = 1, \ldots, N_c - 1, \tag{90.21}$$

with all $v_i \widetilde{v}_i \neq 0$, so that

$$M^i{}_j = \widetilde{v}_i v_i \, \delta^i{}_j, \qquad \det M = \prod_{i=1}^{N_c - 1} \widetilde{v}_i v_i.$$

At such a point the gauge group $SU(N_c)$ is completely Higgsed. A charge-one instanton has $4N_c$ bosonic collective coordinates, $2N_c$ adjoint gaugino zero modes, and $2N_c - 2$ matter-fermion zero modes. The Yukawa coupling to the Higgs expectation values lifts $2N_c - 2$ adjoint zero modes together with all $2N_c - 2$ matter zero modes, leaving exactly two gaugino zero modes for the $d^2\theta$ superpotential measure.

The diagonal form (90.21) is only a local coordinate choice on the open set $\det M \neq 0$; the final answer must be rewritten in flavor-covariant variables. If $g \in SU(N_c)$ preserves all nonzero columns $v_i e_i$, $i = 1, \ldots, N_c - 1$, then $g e_i = e_i$ for those i. The determinant-one condition then fixes the last basis vector as well, so the continuous stabilizer is trivial.

For $N_c \geq 3$, the charge-one $SU(N_c)$ instanton bosonic moduli may be counted as four translations, one size coordinate, and the gauge-orientation quotient

$$\frac{SU(N_c)}{SU(N_c - 2) \times U(1)}.$$

The orientation quotient has real dimension

$$(N_c^2 - 1) - \big((N_c - 2)^2 - 1\big) - 1 = 4N_c - 5.$$

For $N_c = 2$ the orientation space is the full $SU(2)$, which has the same dimension $3 = 4N_c - 5$. Thus the formula below holds uniformly. The total bosonic dimension is therefore

$$4 + 1 + (4N_c - 5) = 4N_c.$$

The fermion zero-mode counts are the index-theorem counts in the BPST normalization used earlier. For a charge-one instanton,

$$n_\lambda = 2T(\text{adj}) = 2N_c, \qquad n_{\psi_Q} = N_f, \qquad n_{\psi_{\widetilde{Q}}} = N_f,$$

because $2T(\mathbf{N}_c) = 2T(\overline{\mathbf{N}}_c) = 1$. Hence the matter zero-mode count is $2N_f = 2N_c - 2$.

The Euclidean Yukawa terms contain the finite-dimensional zero-mode couplings obtained by inserting the scalar expectation values into the gauge-matter interaction

$$\sqrt{2}\,\widetilde{Q}^\dagger \lambda \psi_{\widetilde{Q}} - \sqrt{2}\,\psi_Q \lambda Q^\dagger + \text{conjugate terms}, \tag{90.22}$$

with spinor and color contractions inherited from Chapter 87. In the diagonal patch each nonzero v_i and $\widetilde{v}_i$ supplies a nondegenerate pairing between a matter zero mode and one non-Goldstino gaugino mode. Thus the lifting matrix has rank $2N_f = 2N_c - 2$. The two gaugino modes generated by the broken supersymmetries are not lifted; they are the exact Grassmann coordinates whose integration is the $d^2\theta$ measure.

Yukawa-lifting Berezin determinant on the Higgs patch. Assume the setup of the Higgs-patch collective-coordinate calculation above. After choosing bases for the $2N_f$ matter zero modes and for the $2N_f$ non-Goldstino adjoint zero modes, the zero-mode part of (90.22) has the finite-dimensional form

$$S_{\text{lift}} = \sum_{i=1}^{N_f} \left(a_i\, \chi_i \zeta_i + b_i\, \widetilde{\chi}_i \widetilde{\zeta}_i \right), \qquad a_i = c_i\, \overline{v_i}, \quad b_i = \widetilde{c}_i\, \overline{\widetilde{v}_i}, \tag{90.23}$$

where $c_i, \widetilde{c}_i$ are nonzero constants depending only on normalizations of zero-mode wavefunctions. With the orientation convention and these nonzero normalization constants absorbed into a nonzero constant C_{or}, the lifted zero-mode integral is

$$\int \prod_{i=1}^{N_f} d\chi_i\, d\zeta_i\, d\widetilde{\chi}_i\, d\widetilde{\zeta}_i\, \exp(-S_{\text{lift}}) = C_{\text{or}} \prod_{i=1}^{N_f} \overline{v_i\, \widetilde{v}_i}. \tag{90.24}$$

Thus the Yukawa matrix has maximal rank precisely on the patch $\det M \neq 0$, saturates all lifted Grassmann coordinates, and leaves only the two exact gaugino modes. The anti-holomorphic product in (90.24) is a statement about this finite-dimensional Euclidean lifting determinant. The holomorphic ADS denominator also uses the full regulated instanton measure, the bosonic Higgs cutoff, and the hypothesis that the sector defines a holomorphic F-term; these data determine the reduced factor $\mathcal{F}(M)$ below. The diagonal Higgs background separates the flavor-color directions. The terms in (90.22) pair the Q matter zero mode in the i-th direction with one lifted adjoint mode, with coefficient proportional to $\overline{v_i}$, and pair the $\widetilde{Q}$ matter zero mode with another lifted adjoint mode, with coefficient proportional to $\overline{\widetilde{v}_i}$. Since $v_i \widetilde{v}_i \neq 0$ on the patch, each 2×2 Grassmann block has nonzero coefficient.

For a single block,

$$\exp(-a\chi\zeta) = 1 - a\chi\zeta.$$

The Berezin integral over $d\chi\, d\zeta$ extracts the coefficient of the top monomial, up to the chosen orientation sign. Hence one block contributes a nonzero multiple of a, and the tensor product of all Q and $\widetilde{Q}$ blocks contributes

$$C_{\text{or}} \prod_i a_i b_i = C'_{\text{or}} \prod_i \overline{v_i\, \widetilde{v}_i}.$$

Renaming this nonzero constant as C_{or} gives (90.24). If any lifted variable is omitted, or if one of the coefficients is zero, the top monomial in the corresponding Grassmann variables is absent and the integral vanishes. This proves both the saturation statement and the maximal-rank criterion. The two Goldstino coordinates do not appear in S_{lift}, so they remain as the $d^2\theta$ measure.

Radial Higgs cutoff in the ADS instanton patch. Keep the $N_f = N_c - 1$ Higgs-patch setup above, and write

$$H(v, \widetilde{v}) = \sum_{i=1}^{N_f} \bigl(|v_i|^2 + |\widetilde{v}_i|^2\bigr).$$

On the maximal-rank patch $\det M \neq 0$, one has $H(v, \widetilde{v}) > 0$. Suppose that a fixed angular and zero-mode term in the regulated charge-one collective-coordinate integral has radial factor

$$\rho^{2p-1} \exp(-c\,\rho^2 H(v, \widetilde{v}))\, d\rho, \qquad p \in \mathbb{Z}_{>0}, \quad c > 0, \tag{90.25}$$

where $\rho > 0$ is the instanton size. Then the radial integral is

$$I_p(H) = \int_0^\infty \rho^{2p-1} \exp(-cH\rho^2)\, d\rho = \frac{(p-1)!}{2(cH)^p}. \tag{90.26}$$

In particular, the Higgs expectation values give exponential convergence at large ρ and a real homogeneous factor of degree $-2p$ under $(v_i, \widetilde{v}_i) \mapsto t(v_i, \widetilde{v}_i)$. This real cutoff estimate supplies convergence and real homogeneity. The holomorphic ADS denominator additionally requires the Yukawa lifting rank on the boundary $\det M = 0$ and the holomorphic invariant $1/\det M$, supplied by the Yukawa-lifting calculation above and Lemma 90.12.

If $\det M \neq 0$ in the diagonal patch, then $\widetilde{v}_i v_i \neq 0$ for every i, hence each summand in H is strictly positive and $H > 0$. For $\rho \geq 1$, the radial integrand is bounded by $\rho^{2p-1} \mathrm{e}^{-(cH)\rho^2}$, and this function is integrable on $[1, \infty)$ because the exponential decay dominates every polynomial. Near $\rho = 0$, the integrand is $\rho^{2p-1}(1 + O(\rho^2))\, d\rho$, which is integrable because $p \geq 1$. Thus the model radial integral is finite.

Set $u = cH\rho^2$. Then

$$\rho^{2p-1}\, d\rho = \frac{u^{p-1}}{2(cH)^p}\, du.$$

Therefore

$$I_p(H) = \frac{1}{2(cH)^p} \int_0^\infty u^{p-1} e^{-u}\, du = \frac{\Gamma(p)}{2(cH)^p} = \frac{(p-1)!}{2(cH)^p},$$

because p is a positive integer.

If all Higgs coordinates are multiplied by t, then H is multiplied by $|t|^2$, so I_p is multiplied by $|t|^{-2p}$. The absolute values show why this radial estimate is a real analytic convergence statement, not a holomorphic transformation law. Approaching $\det M = 0$ also changes the rank of the finite-dimensional Yukawa lifting matrix, so the maximal-rank instanton calculation cannot be extended across that boundary by the radial estimate alone.

Lemma 90.12 (Maximal-rank meson invariant for the ADS patch)**.** *Let $n = N_c - 1$, and let*

$$U_n = \{M \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \det M \neq 0\}.$$

Suppose $\mathcal{F} : U_n \to \mathbb{C}$ is a single-valued holomorphic function satisfying

$$\mathcal{F}(LMR^{-1}) = \mathcal{F}(M), \qquad L, R \in \mathrm{SL}_n(\mathbb{C}), \tag{90.27}$$

and the homogeneous scaling law

$$\mathcal{F}(sM) = s^{-n}\mathcal{F}(M), \qquad s \in \mathbb{C}^\times. \tag{90.28}$$

Then

$$\mathcal{F}(M) = \frac{C}{\det M}$$

for a constant C. For the $N_f = N_c - 1$ SQCD R-charge assignment, $(\det M)^{-1}$ *has R-charge* 2.

Proof. The action in (90.27) is the complexified special-flavor action on the meson matrix. For $\delta \in \mathbb{C}^\times$, set

$$D_\delta = \mathrm{diag}(\delta, 1, \ldots, 1).$$

Every $M \in U_n$ with $\det M = \delta$ lies in the same $\mathrm{SL}_n(\mathbb{C})$ left orbit as D_δ, because

$$A = MD_\delta^{-1} \qquad \text{satisfies} \qquad \det A = (\det M)(\det D_\delta)^{-1} = 1$$

and hence $A \in \mathrm{SL}_n(\mathbb{C})$ with $M = AD_\delta$. Therefore $\mathcal{F}$ is constant on each determinant fiber. Define

$$f(\delta) = \mathcal{F}(D_\delta), \qquad \delta \in \mathbb{C}^\times.$$

Then

$$\mathcal{F}(M) = f(\det M).$$

Now scale M by $s \in \mathbb{C}^\times$. Since $\det(sM) = s^n \det M$, the homogeneity condition gives

$$f(s^n\delta) = s^{-n} f(\delta).$$

For arbitrary $u \in \mathbb{C}^\times$, choose s with $s^n = u$. The displayed identity becomes

$$f(u\delta) = u^{-1} f(\delta).$$

Taking $\delta = 1$ gives $f(u) = C/u$. Thus $\mathcal{F}(M) = C/(\det M)$.

Finally, when $N_f = N_c - 1 = n$,

$$R(M) = 2\left(1 - \frac{N_c}{N_c - 1}\right) = -\frac{2}{n}, \qquad R(\det M) = -2,$$

so $R((\det M)^{-1}) = 2$. □

One-instanton test of the ADS numerator and denominator. Assume Hypothesis 90.11. For $N_f = N_c - 1$, the one-instanton contribution to the Wilsonian chiral superpotential has the form

$$W_{k=1} = \kappa_{N_c} \frac{\Lambda_h^{2N_c+1}}{\det M}, \qquad \kappa_{N_c} \neq 0$$

in the holomorphic scale convention used in the instanton measure. This is the $N_f = N_c - 1$ specialization of W_{ADS}.

For $N_f = N_c - 1$, the Higgs-patch collective-coordinate calculation above and Hypothesis 90.11 reduce the charge-one sector to a finite-dimensional zero-mode integral of the form

$$\int d^4x_0\, d^2\eta\, \Lambda_h^{2N_c+1}\, \mathcal{F}(M), \tag{90.29}$$

where x_0 is the instanton position, η_α are the two exact gaugino modes, and $\mathcal{F}(M)$ is holomorphic on the patch $\det M \neq 0$. The one-loop holomorphic exponent is

$$b_0 = 3N_c - N_f = 2N_c + 1,$$

so a charge-one instanton carries the chiral scale factor $\Lambda_h^{2N_c+1}$. The Higgs expectation values cut off the large-size region of the instanton integral through a positive term of the form

$$\Delta S_{\rm cl} = c\,\rho^2 \sum_{i=1}^{N_c-1} \left(|v_i|^2 + |\widetilde{v}_i|^2\right), \qquad c > 0,$$

in the bosonic collective-coordinate action, while the small-size endpoint is absorbed into the holomorphic scale by the hypothesis.

The remaining dependence must be a holomorphic function of the meson matrix $M^i{}_j = \widetilde{Q}_j Q^i$, because the baryonic coordinates are absent when $N_f = N_c - 1$. The exact nonzero modes and lifted zero modes are integrated out in a regulator preserving the special flavor group, so $\mathcal{F}$ obeys the complexified $\mathrm{SL}(N_f, \mathbb{C})_L \times \mathrm{SL}(N_f, \mathbb{C})_R$ invariance used in Lemma 90.12. The reduced factor $\mathcal{F}(M)$ in (90.29) must carry engineering dimension

$$3 - (2N_c + 1) = -(2N_c - 2),$$

so under the coordinate scaling $M \mapsto sM$, where each meson entry has dimension 2, it is homogeneous of degree $-N_f$. With $N_f = N_c - 1$, Lemma 90.12 gives

$$\mathcal{F}(M) = \frac{C}{\det M}.$$

The anomaly-free R-charge gives the independent convention check

$$R(M) = 2\left(1 - \frac{N_c}{N_c - 1}\right) = -\frac{2}{N_c - 1}, \qquad R(\det M) = -2,$$

so $(\det M)^{-1}$ has R-charge 2, matching the charge of a superpotential. Thus the one-instanton sector has the displayed form, with the nonzero coefficient fixed by the chosen normalization of the regulated collective-coordinate measure.

For the smallest example, $SU(2)$ with one flavor, the calculation gives eight bosonic collective coordinates, four gaugino zero modes, and two matter zero modes. The Higgs expectation value lifts two gaugino modes together with the two matter modes, leaving $d^2\theta$. The formula becomes

$$W_{SU(2),\,N_f=1} = \kappa_2 \frac{\Lambda_h^5}{M}, \qquad M = \widetilde{Q}_a Q^a.$$

This low-rank case is often the cleanest way to audit the convention: the power 5 is $3N_c - N_f$, the denominator has dimension 2, and $R(M) = -2$, so M^{-1} supplies the required superpotential R-charge 2.

Holomorphic scale matching across a massive threshold. Give the N_f-flavor theory a mass m for the last flavor. If the low-energy $(N_f - 1)$-flavor Wilsonian theory is reached by integrating out that massive chiral pair, then the holomorphic scales obey

$$\Lambda_{N_f-1}^{3N_c-(N_f-1)} = m\,\Lambda_{N_f}^{3N_c-N_f}.$$

Match the holomorphic gauge coordinate at the threshold $\mu = m$. Above the threshold the one-loop coefficient is $3N_c - N_f$; below it is $3N_c - (N_f - 1)$. The invariant combinations

$$\mu^{3N_c - N_f} e^{2\pi \mathrm{i} \tau_{\text{above}}(\mu)} \quad \text{and} \quad \mu^{3N_c-(N_f-1)} e^{2\pi \mathrm{i} \tau_{\text{below}}(\mu)}$$

are equal at $\mu = m$ after the massive determinant is included in the Wilsonian chiral coordinate. Eliminating the common value of $\tau(m)$ gives the displayed equation.

This is a coordinate-matching calculation inside the holomorphic Wilsonian scheme. The dynamical assumption is the preceding sentence: the massive pair must be integrated out in a scheme whose chiral gauge coordinate and threshold determinant have been specified. The displayed equation is not by itself a proof that the infrared description exists.

Proposition 90.13 (ADS decoupling recursion)**.** *Fix N_c and let $1 \leq n < N_c - 1$. Suppose the $(n+1)$-flavor Wilsonian chiral theory has, on a chosen holomorphic branch of the maximal-rank meson patch, the ADS superpotential*

$$W_{n+1} = C_{n+1} \left(\frac{\Lambda_+^{3N_c-(n+1)}}{\det \mathcal{M}} \right)^{\frac{1}{N_c-n-1}},$$

with $C_{n+1} \neq 0$. Add a holomorphic mass term

$$W_{\text{mass}} = m \, \mathcal{M}^L{}_L, \qquad L = n+1,$$

and assume the Wilsonian threshold is described by the holomorphic scale matching above. On the light branch $\mathcal{M}^a{}_L = \mathcal{M}^L{}_a = 0$, $a = 1, \ldots, n$, the effective n-flavor superpotential is

$$W_n = C_n \left(\frac{\Lambda_-^{3N_c-n}}{\det M} \right)^{\frac{1}{N_c-n}}, \qquad \Lambda_-^{3N_c-n} = m \, \Lambda_+^{3N_c-(n+1)}, \tag{90.30}$$

where $M^a{}_b$ is the light meson block and, writing $d = N_c - n$,

$$C_n = d \left(\frac{C_{n+1}}{d-1} \right)^{\frac{d-1}{d}}. \tag{90.31}$$

In the holomorphic scale normalization for which the one-instanton seed has $C_{N_c-1} = 1$, the recursion gives the standard representative

$$C_n = N_c - n.$$

Proof. The statement is local on a holomorphic branch because fractional powers are involved. The branch data are the usual discrete chiral choices of the Wilsonian description and do not affect the algebraic recursion.

Set $d = N_c - n \geq 2$, split $\mathcal{M}$ into the light block M, the mixed components, and

$$X = \mathcal{M}^L{}_L.$$

On the light branch the determinant factorizes:

$$\det \mathcal{M} = X \det M.$$

Define the holomorphic quantity

$$A = \frac{\Lambda_+^{3N_c-(n+1)}}{\det M}.$$

The superpotential to be extremized with respect to the massive chiral coordinate X is therefore

$$W(X) = C_{n+1}\left(\frac{A}{X}\right)^{\frac{1}{d-1}} + mX. \tag{90.32}$$

Introduce

$$Y = \left(\frac{A}{X}\right)^{\frac{1}{d-1}}.$$

The $F_X = 0$ equation is

$$0 = -\frac{C_{n+1}}{d-1}A^{\frac{1}{d-1}}X^{-\frac{d}{d-1}} + m.$$

Multiplying by X gives the useful identity

$$mX = \frac{C_{n+1}}{d-1}Y. \tag{90.33}$$

The same equation also gives

$$Y^d = \frac{(d-1)m}{C_{n+1}}A. \tag{90.34}$$

Substituting (90.33) into (90.32) gives

$$W_{\text{eff}} = C_{n+1}Y + mX = \frac{d}{d-1}C_{n+1}Y.$$

Using (90.34), this becomes

$$W_{\text{eff}} = d\left(\frac{C_{n+1}}{d-1}\right)^{\frac{d-1}{d}}\left(\frac{m\Lambda_+^{3N_c-(n+1)}}{\det M}\right)^{\frac{1}{d}}.$$

The holomorphic scale matching above turns the numerator into $\Lambda_-^{3N_c-n}$, proving (90.30) and (90.31).

If $C_{N_c-1} = 1$, induction on $d = N_c - n$ gives $C_n = d$. Indeed, if $C_{n+1} = d-1$, then (90.31) gives

$$C_n = d\left(\frac{d-1}{d-1}\right)^{\frac{d-1}{d}} = d.$$

$\square$

Proposition 90.14 (Massive SQCD decoupling to pure SYM branches)**.** *Fix $1 \le n < N_c$, and assume the n-flavor Wilsonian chiral sector on the maximal-rank meson patch is described by the standard ADS representative*

$$W_{\text{ADS},n} = d\left(\frac{\Lambda_n^{3N_c-n}}{\det M}\right)^{1/d}, \qquad d = N_c - n, \tag{90.35}$$

on a chosen holomorphic branch. Add an invertible holomorphic mass matrix

$$W_{\text{mass}} = m_i{}^j M^i{}_j, \qquad \det m \neq 0. \tag{90.36}$$

Assume that integrating out the massive meson coordinates is legitimate in the Wilsonian chiral theory and that holomorphic threshold matching to pure SYM holds:

$$\Lambda_0^{3N_c} = (\det m)\,\Lambda_n^{3N_c-n}. \tag{90.37}$$

Then the massive SQCD branch superpotential is

$$W_{\text{eff},k} = N_c\,\Lambda_0^3 \exp\left(\frac{2\pi \mathrm{i} k}{N_c}\right), \qquad k \in \mathbb{Z}/N_c\mathbb{Z}, \tag{90.38}$$

up to the same finite normalization of Λ_0 and S used in the pure-SYM glueball description. The source identity gives

$$\Lambda_0^{3N_c} \frac{\partial W_{\text{eff},k}}{\partial \Lambda_0^{3N_c}} = \Lambda_0^3 \exp\left(\frac{2\pi \mathrm{i} k}{N_c}\right). \tag{90.39}$$

Thus massive SQCD decoupling reproduces the pure-SYM condensate root equation and its N_c branches, conditional on the ADS input and the massive threshold assumptions.

Proof. Write

$$A = \Lambda_n^{3N_c-n}, \qquad Y = \left(\frac{A}{\det M}\right)^{1/d}. \tag{90.40}$$

Then $W_{\text{ADS},n} = dY$. For a variation of an invertible meson matrix,

$$\delta \log \det M = \operatorname{tr}(M^{-1}\delta M), \qquad \delta Y = -\frac{Y}{d}\operatorname{tr}(M^{-1}\delta M).$$

Hence

$$\delta W_{\text{ADS},n} = -Y\ \operatorname{tr}(M^{-1}\delta M).$$

In indices, the $F_M = 0$ equation is therefore

$$m_i{}^j = Y\,(M^{-1})^j{}_i. \tag{90.41}$$

Multiplying by $M^i{}_j$ and summing gives

$$m_i{}^j M^i{}_j = nY, \tag{90.42}$$

because $M^i{}_j (M^{-1})^j{}_i = n$. Taking determinants in (90.41) gives

$$\det m = \frac{Y^n}{\det M}. \tag{90.43}$$

Since $Y^d = A/\det M$, equations (90.43) and (90.40) imply

$$Y^{d+n} = A\ \det m.$$

As $d + n = N_c$, this is

$$Y^{N_c} = (\det m)\Lambda_n^{3N_c-n} = \Lambda_0^{3N_c}$$

by (90.37). Thus

$$Y_k = \Lambda_0^3 \exp\left(\frac{2\pi \mathrm{i} k}{N_c}\right), \qquad k \in \mathbb{Z}/N_c\mathbb{Z},$$

after choosing a branch of Λ_0^3.

At a critical point,

$$W_{\text{eff}} = dY + m_i{}^j M^i{}_j = dY + nY = N_c Y,$$

which proves (90.38). Finally, with $L_0 = \Lambda_0^{3N_c}$, the branch value is

$$W_{\text{eff},k} = N_c\, L_0^{1/N_c} \mathrm{e}^{2\pi \mathrm{i} k/N_c}.$$

Therefore

$$L_0 \frac{\partial W_{\text{eff},k}}{\partial L_0} = L_0^{1/N_c} \mathrm{e}^{2\pi \mathrm{i} k/N_c},$$

which is (90.39). The proof is a finite-dimensional chiral-coordinate elimination. It does not by itself construct the pure-SYM Hilbert space, prove the mass gap, or justify the ADS and threshold hypotheses from a microscopic regulator. □

Mass-source and Konishi identities in massive SQCD. Assume the hypotheses and notation of Proposition 90.14. On a chosen massive branch, the holomorphic source derivative with respect to the mass matrix gives the meson expectation

$$\frac{\partial W_{\text{eff},k}}{\partial m_i{}^j} = \langle M^i{}_j \rangle_k. \tag{90.44}$$

The finite-dimensional $F_M = 0$ equation then implies the matrix identity

$$\langle M^a{}_j \rangle_k\, m_i{}^j = \langle S \rangle_k\, \delta_i{}^a, \qquad \langle S \rangle_k = \Lambda_0^3 \exp\left(\frac{2\pi \mathrm{i} k}{N_c}\right). \tag{90.45}$$

Taking the trace gives

$$m_i{}^j \langle M^i{}_j \rangle_k = n\, \langle S \rangle_k. \tag{90.46}$$

This is the massive-SQCD chiral-coordinate realization of the Konishi relation. The local operator identity with the Konishi current and $\bar{D}^2$-exact terms requires the regulator-level current statement of Chapter 89; the calculation here uses only the already stated Wilsonian source calculus.

The full chiral function before eliminating M is

$$W(M; m, L_n) = d \left(\frac{L_n}{\det M}\right)^{1/d} + m_i{}^j M^i{}_j, \qquad L_n = \Lambda_n^{3N_c - n}.$$

Let $M_k(m, L_n)$ be the critical point on the chosen branch. The envelope theorem for a holomorphic finite-dimensional critical point gives

$$\frac{\partial}{\partial m_i{}^j} W(M_k(m, L_n); m, L_n) = M_k{}^i{}_j,$$

because the terms involving $\partial M_k / \partial m_i{}^j$ are multiplied by F_M and vanish at the critical point. This proves (90.44).

Equation (90.41) says

$$m_i{}^j = Y_k (M_k^{-1})^j{}_i.$$

Multiplying by $M_k{}^a{}_j$ gives

$$M_k{}^a{}_j \, m_i{}^j = Y_k \, M_k{}^a{}_j (M_k^{-1})^j{}_i = Y_k \, \delta_i{}^a.$$

By (90.39), the same branch coordinate is the glueball source derivative:

$$\langle S \rangle_k = Y_k = \Lambda_0^3 \exp\left(\frac{2\pi \mathrm{i} k}{N_c}\right).$$

This gives (90.45). Setting $a = i$ and summing over $i = 1, \ldots, n$ gives (90.46).

90.9 Quantum Modified Constraint and Confining Decoupling

For $N_f = N_c$, the classical relation between mesons and baryons is

$$\det M - B\widetilde{B} = 0.$$

The holomorphic scale has dimension $2N_c$ and the same spurionic quantum numbers as the left side. Under the Wilsonian and infrared hypotheses made explicit below, the protected quantum chiral-coordinate ring is represented by

$$\det M - B\widetilde{B} = \Lambda_h^{2N_c}.$$

This equation is a statement about the chiral ring and its vacuum space. It does not by itself determine the full Hilbert space, massive spectrum, or metric on the moduli space.

Hypothesis 90.15 ($N_f = N_c$ and $N_f = N_c + 1$ chiral-sector descriptions)**.** The following derivations compare protected chiral-coordinate rings in a supersymmetric Wilsonian scheme. For $N_f = N_c$, the infrared chiral sector is assumed to be generated by $M, B, \widetilde{B}$, with no generated superpotential and with a possible holomorphic deformation of the classical relation. For $N_f = N_c + 1$, the infrared chiral sector is assumed to be generated by mesons $M^I{}_J$, baryons B_I, and antibaryons $\widetilde{B}^J$, $I, J = 1, \ldots, N_c + 1$, with superpotential

$$W_{N_c+1} = \frac{1}{\Lambda_+^{2N_c-1}} \left(B_I M^I{}_J \widetilde{B}^J - \det M \right). \tag{90.47}$$

Here Λ_+ is the holomorphic scale of the $N_f = N_c + 1$ theory. The statements below derive consequences of these chiral descriptions and of holomorphic decoupling; they do not construct the full Hilbert space metric or prove confinement from a microscopic regulator.

Uniqueness of the field-independent quantum modification. Assume the $N_f = N_c$ chiral-sector part of Hypothesis 90.15. If the classical constraint is deformed by a field-independent holomorphic scale term, then up to the normalization of Λ_h the only allowed deformation is

$$\det M - B\widetilde{B} = \Lambda_h^{2N_c}.$$

For $N_f = N_c$, the anomaly-free assignment gives

$$R(Q) = R(\widetilde{Q}) = 0.$$

Thus $R(M) = R(B) = R(\widetilde{B}) = 0$. The two classical terms have the same engineering dimension:

$$[\det M] = 2N_c, \qquad [B\widetilde{B}] = N_c + N_c = 2N_c,$$

and the same flavor and baryon quantum numbers. A field-independent holomorphic deformation must therefore have dimension $2N_c$, zero baryon charge, and the same spurionic flavor and R-charges. The holomorphic scale of the $N_f = N_c$ theory appears in the invariant combination

$$\Lambda_h^{b_0}, \qquad b_0 = 3N_c - N_f = 2N_c.$$

No other field-independent monomial in the holomorphic scale has the required dimension. Absorbing the nonzero numerical coefficient into the definition of $\Lambda_h^{2N_c}$ gives the displayed constraint.

Mass decoupling from $N_f = N_c + 1$ to $N_f = N_c$. Assume Hypothesis 90.15. Add a holomorphic mass term for the last flavor of the $N_f = N_c + 1$ theory,

$$W_{\text{mass}} = m\, M^L{}_L, \qquad L = N_c + 1.$$

On the branch where the mixed heavy-light chiral coordinates vanish, the F-term equations of

$$W_{N_c+1} + W_{\text{mass}}$$

imply the $N_f = N_c$ quantum-modified constraint

$$\det \widehat{M} - B\widetilde{B} = \Lambda_-^{2N_c}, \qquad \Lambda_-^{2N_c} = m\, \Lambda_+^{2N_c-1}, \tag{90.48}$$

where $\widehat{M}^a{}_b$, $a, b = 1, \ldots, N_c$, is the light meson block and $B = B_L, \widetilde{B} = \widetilde{B}^L$ are the light baryons.

Split the flavor indices as $I = (a, L)$, with $a = 1, \ldots, N_c$. On the branch used for the low-energy $N_f = N_c$ variables, set

$$M^a{}_L = M^L{}_a = 0, \qquad B_a = \widetilde{B}^a = 0,$$

and write $X = M^L{}_L$. Then

$$\det M = X \det \widehat{M}, \qquad B_I M^I{}_J \widetilde{B}^J = B_L X \widetilde{B}^L.$$

The superpotential restricted to these coordinates is

$$W = \frac{X}{\Lambda_+^{2N_c-1}} \left(B\widetilde{B} - \det \widehat{M} \right) + mX.$$

The F-term equation for X gives

$$\frac{B\widetilde{B} - \det \widehat{M}}{\Lambda_+^{2N_c-1}} + m = 0,$$

or

$$\det \widehat{M} - B\widetilde{B} = m\, \Lambda_+^{2N_c-1}.$$

The remaining F-term equations along the low-energy branch set

$$X = 0$$

unless one sits on special subloci where additional massive coordinates have not yet been eliminated. Finally, holomorphic decoupling at the mass threshold gives

$$\Lambda_-^{3N_c-N_c} = m\,\Lambda_+^{3N_c-(N_c+1)} = m\,\Lambda_+^{2N_c-1},$$

which is exactly (90.48).

Remark 90.16 (Checks on the $N_f = N_c + 1$ confining superpotential). For $N_f = N_c + 1$, the two terms in (90.47) have the same engineering dimension and R-charge, and the prefactor makes W_{N_c+1} a dimension-three, R-charge-two chiral superpotential. At $N_f = N_c + 1$,

$$R(Q) = R(\widetilde{Q}) = \frac{1}{N_c+1}.$$

Thus

$$R(M) = \frac{2}{N_c+1}, \qquad R(B_I) = R(\widetilde{B}^J) = \frac{N_c}{N_c+1}.$$

The product $B_I M^I{}_J \widetilde{B}^J$ has R-charge

$$\frac{N_c}{N_c+1} + \frac{2}{N_c+1} + \frac{N_c}{N_c+1} = 2,$$

and $\det M$, where the determinant is over the $N_f = N_c + 1$ flavor indices, has R-charge

$$(N_c+1)\frac{2}{N_c+1} = 2.$$

Classically, $\dim M = 2$ and $\dim B_I = \dim \widetilde{B}^J = N_c$. Both $B_I M^I{}_J \widetilde{B}^J$ and $\det M$ therefore have dimension

$$2N_c + 2 = 2(N_c+1).$$

Since $\Lambda_+^{2N_c-1}$ has dimension $2N_c - 1$ and zero R-charge in the anomaly-free spurion convention, the quotient has dimension 3 and R-charge 2, as required.

90.10 Superconformal SQCD Bookkeeping

The same anomaly calculation also fixes the candidate superconformal data of SQCD in the range where an interacting infrared fixed point exists. For a chiral primary operator $\mathcal{O}$ in a four-dimensional $\mathcal{N} = 1$ superconformal theory,

$$\Delta(\mathcal{O}) = \frac{3}{2}R(\mathcal{O}).$$

For an elementary chiral field Φ, write its scalar scaling dimension as

$$\Delta(\Phi) = 1 + \frac{1}{2}\gamma_\Phi.$$

Compatibility of these two conventions gives

$$\gamma_\Phi = 3R(\Phi) - 2. \tag{90.49}$$

Here γ_Φ is the chapter convention

$$\gamma_\Phi = \frac{d}{d\log\mu}\log Z_\Phi.$$

Some supersymmetric-gauge-theory references instead use $\mathcal{C}_\Phi = \frac{1}{2}\,d\log Z_\Phi/d\log\mu$. The conversion is

$$\gamma_\Phi = 2\mathcal{C}_\Phi, \qquad \Delta(\Phi) = 1 + \mathcal{C}_\Phi.$$

For a product of two chiral fields with equal anomalous dimensions, the product anomalous dimension in the $\mathcal{C}$-notation is numerically equal to the single-field γ_Φ used below:

$$\mathcal{C}_{\Phi\Psi} = \Delta(\Phi\Psi) - 2 = \mathcal{C}_\Phi + \mathcal{C}_\Psi = \gamma_\Phi \qquad (\gamma_\Phi = \gamma_\Psi).$$

This is why the conifold formulas may be written with $1 - \gamma_A$ here while the same fixed point is sometimes described by $\mathcal{C}_{AB} = -\frac{1}{2}$ for the meson A_iB_j.

For SQCD with no tree superpotential, the anomaly-free assignment of the previous section gives

$$R(Q) = R(\widetilde{Q}) = 1 - \frac{N_c}{N_f}, \qquad \gamma_Q = 1 - \frac{3N_c}{N_f}.$$

Substitution in the NSVZ numerator gives

$$3N_c - N_f(1-\gamma_Q) = 3N_c - N_f\left(\frac{3N_c}{N_f}\right) = 0.$$

Thus the anomaly-free R-charge is also the unique candidate superconformal R-charge compatible with the flavor-symmetric NSVZ coordinate.

The meson $M = \widetilde{Q}Q$ has

$$R(M) = 2\left(1 - \frac{N_c}{N_f}\right), \qquad \Delta(M) = 3\left(1 - \frac{N_c}{N_f}\right).$$

The scalar unitarity bound for a gauge-invariant chiral operator requires $\Delta(M) \geq 1$, which gives $N_f \geq \frac{3}{2}N_c$. This is the lower edge of the usual electric conformal-window test; below it the electric meson cannot be an interacting chiral primary with the displayed superconformal R-charge. The upper edge $N_f < 3N_c$ is the condition that the electric gauge coupling is asymptotically free in the holomorphic one-loop coordinate. The existence and uniqueness of the infrared fixed point in the full interval require the nonperturbative input summarized in the status boundary below.

Hypothesis 90.17 (SCFT input used in bookkeeping)**.** Whenever this chapter assigns a scaling dimension by $\Delta = \frac{3}{2}R$, the following assumptions are in force: the infrared limit exists as a unitary four-dimensional $\mathcal{N} = 1$ SCFT; the operator under discussion is a chiral primary in that SCFT; the displayed $U(1)_R$ current is the superconformal R-current after all allowed mixing with non-R flavor currents has been fixed; and accidental symmetries do not change the R-charge of the operator in question. The anomaly and NSVZ calculations below test these assumptions; they do not by themselves construct the fixed point.

SQCD central charges and free-field comparison. Assume Hypothesis 90.17 and suppose $\frac{3}{2}N_c < N_f < 3N_c$. For the candidate SQCD superconformal R-current displayed above,

$$\operatorname{Tr} R = -N_c^2 - 1, \qquad \operatorname{Tr} R^3 = N_c^2 - 1 - \frac{2N_c^4}{N_f^2}. \tag{90.50}$$

Consequently the conformal anomalies are

$$a_{\mathrm{IR}} = \frac{3}{32}\left(4N_c^2 - 2 - \frac{6N_c^4}{N_f^2}\right), \qquad c_{\mathrm{IR}} = \frac{1}{32}\left(14N_c^2 - 4 - \frac{18N_c^4}{N_f^2}\right). \tag{90.51}$$

The free ultraviolet theory has

$$a_{\mathrm{UV}} = \frac{3}{16}(N_c^2 - 1) + \frac{N_c N_f}{24}. \tag{90.52}$$

Writing $y = N_f/N_c$, the difference is

$$a_{\mathrm{UV}} - a_{\mathrm{IR}} = \frac{N_c^2}{48y^2}(3-y)^2(2y+3). \tag{90.53}$$

It is strictly positive in the interacting conformal window and vanishes at the Gaussian edge $N_f = 3N_c$. This is only an anomaly-arithmetic comparison between a free ultraviolet fixed point and the candidate infrared SCFT. It is not a proof of a monotonicity theorem or of the existence of the flow; those remain part of the independent continuum and SCFT hypotheses.

The traces are over Weyl fermions. The gaugino contributes

$$\operatorname{Tr} R_\lambda = N_c^2 - 1, \qquad \operatorname{Tr} R_\lambda^3 = N_c^2 - 1.$$

The scalar R-charge of Q and $\widetilde{Q}$ is

$$R(Q) = R(\widetilde{Q}) = 1 - \frac{N_c}{N_f},$$

so their fermions have R-charge $-N_c/N_f$. There are $2N_cN_f$ such Weyl fermions. Hence

$$\operatorname{Tr} R = (N_c^2 - 1) + 2N_cN_f\left(-\frac{N_c}{N_f}\right) = -N_c^2 - 1,$$

and

$$\operatorname{Tr} R^3 = (N_c^2 - 1) + 2N_cN_f\left(-\frac{N_c}{N_f}\right)^3 = N_c^2 - 1 - \frac{2N_c^4}{N_f^2}.$$

For a four-dimensional $\mathcal{N} = 1$ SCFT,

$$a = \frac{3}{32}(3\operatorname{Tr} R^3 - \operatorname{Tr} R), \qquad c = \frac{1}{32}(9\operatorname{Tr} R^3 - 5\operatorname{Tr} R). \tag{90.54}$$

Substituting the two traces gives (90.51).

At the ultraviolet Gaussian fixed point, a vector multiplet contributes $\frac{3}{16}\dim SU(N_c)$ to a, and a free chiral multiplet contributes $\frac{1}{48}$ per complex gauge-flavor component. Since $\dim SU(N_c) = N_c^2 - 1$ and there are $2N_cN_f$ chiral components, (90.52) follows.

Subtracting the candidate infrared value from the ultraviolet value gives

$$a_{\mathrm{UV}} - a_{\mathrm{IR}} = -\frac{3N_c^2}{16} + \frac{N_c N_f}{24} + \frac{9N_c^4}{16N_f^2}.$$

Set $y = N_f/N_c$. Then

$$a_{\mathrm{UV}} - a_{\mathrm{IR}} = N_c^2 \left(-\frac{3}{16} + \frac{y}{24} + \frac{9}{16y^2} \right) = \frac{N_c^2}{48y^2} \big(2y^3 - 9y^2 + 27 \big).$$

The polynomial factorizes as

$$2y^3 - 9y^2 + 27 = (3 - y)^2 (2y + 3).$$

For $y > 0$ this expression is nonnegative and it is zero only when $y = 3$. In the interacting window $3/2 < y < 3$, it is strictly positive. The proof has established this finite anomaly comparison; it has not established the general a-theorem.

90.11 Seiberg Duality And The SQCD Phase Structure

The preceding superconformal bookkeeping used only the electric SQCD variables. We now isolate the nonperturbative duality input that is needed whenever the chapter talks about the full SQCD phase diagram. This is a field-theoretic statement about four-dimensional $\mathcal{N} = 1$ gauge dynamics; no string construction is part of the definition or the argument.

Write the flavor symmetry as

$$SU(N_f)_L \times SU(N_f)_R \times U(1)_B \times U(1)_R.$$

The electric fields transform as

	$SU(N_c)$	$SU(N_f)_L$	$SU(N_f)_R$	$U(1)_B$	$U(1)_R$
Q^i	$\mathbf{N}_c$	$\mathbf{N}_f$	$\mathbf{1}$	$+1$	$1 - \frac{N_c}{N_f}$
$\widetilde{Q}_j$	$\overline{\mathbf{N}}_c$	$\mathbf{1}$	$\overline{\mathbf{N}}_f$	-1	$1 - \frac{N_c}{N_f}$

and the meson $M^i{}_j = \widetilde{Q}_j Q^i$ transforms as $(\mathbf{N}_f, \overline{\mathbf{N}}_f)$. The normalization of $U(1)_B$ is chosen so that an electric baryon Q^{N_c} has charge N_c. This differs by a harmless overall factor from conventions in which Q has baryon charge $1/N_c$.

For $N_f \geq N_c + 2$, set

$$\widetilde{N}_c = N_f - N_c.$$

The magnetic SQCD variables consist of an $SU(\widetilde{N}_c)$ vector multiplet, magnetic quarks $q_i, \widetilde{q}^j$, and a gauge-singlet meson coordinate $M^i{}_j$ with the global charges

	$SU(\widetilde{N}_c)$	$SU(N_f)_L$	$SU(N_f)_R$	$U(1)_B$	$U(1)_R$
q_i	fund	$\overline{\mathbf{N}}_f$	$\mathbf{1}$	$\frac{N_c}{\widetilde{N}_c}$	$\frac{N_c}{N_f}$
$\widetilde{q}^j$	$\overline{\text{fund}}$	$\mathbf{1}$	$\mathbf{N}_f$	$-\frac{N_c}{\widetilde{N}_c}$	$\frac{N_c}{N_f}$
$M^i{}_j$	$\mathbf{1}$	$\mathbf{N}_f$	$\overline{\mathbf{N}}_f$	0	$2\left(1 - \frac{N_c}{N_f}\right)$.

The magnetic superpotential is

$$W_{\text{mag}} = \frac{1}{\mu_*} M^i{}_j q_i \widetilde{q}^j. \tag{90.55}$$

Here $M^i{}_j$ is normalized as the electric composite $\widetilde{Q}_j Q^i$, so it has engineering dimension two in the microscopic matching convention; μ_* is a nonzero matching scale of dimension one. Equivalently one may use the canonically normalized magnetic singlet $M_{\text{can}} = M/\mu_*$ and absorb μ_* into the Yukawa coupling. The displayed form keeps the operator map visible.

Hypothesis 90.18 (SQCD infrared-comparison hypotheses)**.** Whenever this section compares the electric and magnetic SQCD descriptions, the following hypotheses are part of the statement.

1. The electric and magnetic Lagrangians have continuum infrared limits in a regulator preserving the gauge Ward identities, the listed flavor symmetries, and enough supersymmetry for the Wilsonian chiral-coordinate arguments of Chapters 88 and 89.
2. The unbroken global currents are represented by the same operators in the two infrared limits. Consequently their 't Hooft anomalies are invariant data of the infrared theory and must match.
3. The displayed $U(1)_R$ current is the superconformal current in the interacting range after all allowed flavor-current mixing has been fixed. At free endpoints one must include the accidental symmetries of the free fields before interpreting $\Delta = \frac{3}{2}R$.
4. The chiral coordinates $M, B, \widetilde{B}$ and their magnetic images generate the protected chiral sector being compared, after imposing F-term equations and after excluding extra accidental chiral generators not present in the stated phase.
5. Relevant deformations by holomorphic quark masses and by generic Higgs-branch expectation values commute with the comparison: after integrating out massive fields and matching holomorphic scales, the lower-rank theories satisfy the same comparison hypotheses.

These assumptions are not justified by historical authority. Their role is visible in the calculations below. The magnetic rank and baryon charges are forced by the baryon map $Q^{N_c} \leftrightarrow q^{\widetilde{N}_c}$; the meson singlet is forced because $Q\widetilde{Q}$ is an independent electric chiral coordinate; the superpotential is the unique cubic chiral coupling with the displayed flavor charges and R-charge 2; the magnetic gauge anomaly cancellation is a necessary consistency test; and the global anomaly matching below is a regulator-independent test of the common infrared current algebra. These checks do not construct the continuum infrared fixed point, but they make every algebraic burden of the duality statement explicit.

Quoted Theorem 90.19 (Seiberg-duality input for SQCD)**.** *For four-dimensional $\mathcal{N} = 1$ SQCD with gauge group $SU(N_c)$, $N_f \geq N_c + 2$ flavors, zero tree superpotential, and the Wilsonian coordinate conventions of this chapter, and under Hypothesis 90.18, the infrared theory at the origin of the quantum moduli space has an equivalent magnetic description with gauge group $SU(N_f - N_c)$, fields and charges as displayed above, and superpotential (90.55). The equivalence identifies the common global currents, chiral ring after imposing superpotential equations, deformations by masses and Higgs expectation values, and all 't Hooft anomalies. In the interval $\frac{3}{2}N_c < N_f < 3N_c$ this is an equivalence of interacting infrared SCFTs; in the free electric and free magnetic ranges it is understood after including the appropriate free variables and accidental symmetries at the endpoint.*

This quoted theorem is not a theorem proved from the preceding Wilsonian holomorphy alone. The monograph uses it as an exact nonperturbative input and then exposes the checks that make the statement sharp: charge assignments, anomaly matching, deformation matching, and consistency with NSVZ fixed-point arithmetic.

Magnetic R-charge and NSVZ consistency. With the magnetic charges displayed above, $U(1)_R$ is free of $SU(\widetilde{N}_c)^2U(1)_R$ gauge anomaly, the magnetic superpotential has R-charge 2, and the magnetic NSVZ numerator vanishes at the same candidate superconformal point as the electric numerator.

The $SU(\widetilde{N}_c)^2U(1)_R$ anomaly receives $+\widetilde{N}_c$ from the magnetic gaugino. The two magnetic quark families have fermion R-charge

$$R(q) - 1 = R(\widetilde{q}) - 1 = \frac{N_c}{N_f} - 1 = -\frac{\widetilde{N}_c}{N_f}.$$

Using $T(\mathbf{fund}) = T(\overline{\mathbf{fund}}) = \frac{1}{2}$,

$$\widetilde{N}_c + \frac{1}{2}N_f\left(-\frac{\widetilde{N}_c}{N_f}\right) + \frac{1}{2}N_f\left(-\frac{\widetilde{N}_c}{N_f}\right) = 0.$$

The superpotential charge is

$$R(M) + R(q) + R(\widetilde{q}) = 2\left(1 - \frac{N_c}{N_f}\right) + 2\frac{N_c}{N_f} = 2.$$

At a superconformal fixed point the chapter convention gives

$$\gamma_q = 3R(q) - 2 = \frac{3N_c}{N_f} - 2.$$

The magnetic NSVZ numerator is therefore

$$3\widetilde{N}_c - N_f(1 - \gamma_q) = 3(N_f - N_c) - N_f\left(3 - \frac{3N_c}{N_f}\right) = 0.$$

The electric numerator was checked in Section 90.10; both descriptions therefore pass the same fixed-point arithmetic.

Electric-magnetic anomaly matching. Normalize cubic nonabelian anomalies so that a left-handed Weyl fermion in the fundamental of $SU(N_f)$ contributes $+1$ and an antifundamental contributes -1. Normalize mixed $SU(N_f)^2U(1)$ anomalies so that a fundamental or antifundamental contributes the same quadratic index. Then the electric and magnetic descriptions have the common anomaly coefficients

anomaly	coefficient
$SU(N_f)_L^3$	N_c
$SU(N_f)_R^3$	$-N_c$
$SU(N_f)_L^2 U(1)_R$	$-\frac{N_c^2}{N_f}$
$SU(N_f)_R^2 U(1)_R$	$-\frac{N_c^2}{N_f}$
$SU(N_f)_L^2 U(1)_B$	N_c
$SU(N_f)_R^2 U(1)_B$	$-N_c$
$U(1)_B^2 U(1)_R$	$-2N_c^2$
$U(1)_R$	$-N_c^2 - 1$
$U(1)_R^3$	$N_c^2 - 1 - \frac{2N_c^4}{N_f^2}$.

All traces are over left-handed Weyl fermions. In the electric theory the quark and antiquark fermions have R-charge $-N_c/N_f$, while the gaugino has R-charge 1. Thus, for example,

$$SU(N_f)_L^3 : \quad N_c, \qquad SU(N_f)_R^3 : \quad -N_c,$$

and

$$SU(N_f)_L^2 U(1)_R : \quad N_c \left(-\frac{N_c}{N_f}\right) = -\frac{N_c^2}{N_f}.$$

The right-flavor mixed R-anomaly is the same, while the baryonic mixed anomalies are N_c and $-N_c$ because Q and $\widetilde{Q}$ have baryon charges $+1$ and -1. The purely abelian traces are

$$U(1)_B^2 U(1)_R : \quad 2N_c N_f \left(-\frac{N_c}{N_f}\right) = -2N_c^2,$$

$$\operatorname{Tr} R = (N_c^2 - 1) + 2N_c N_f \left(-\frac{N_c}{N_f}\right) = -N_c^2 - 1,$$

and

$$\operatorname{Tr} R^3 = (N_c^2 - 1) + 2N_c N_f \left(-\frac{N_c}{N_f}\right)^3 = N_c^2 - 1 - \frac{2N_c^4}{N_f^2}.$$

For the magnetic theory set $t = \widetilde{N}_c = N_f - N_c$. The q and $\widetilde{q}$ fermions have R-charge $-t/N_f$, baryon charges $\pm N_c/t$, and tN_f components each. The mesino in $M^i{}_j$ has

$$R(M) - 1 = 2\left(1 - \frac{N_c}{N_f}\right) - 1 = \frac{N_f - 2N_c}{N_f}.$$

For the left cubic flavor anomaly, q gives $-t$ and the N_f right flavor copies of the mesino give $+N_f$, hence $-t + N_f = N_c$. For the right cubic anomaly, $\widetilde{q}$ gives $+t$ and the mesino gives $-N_f$, hence $t - N_f = -N_c$. The left mixed R-anomaly is

$$t\left(-\frac{t}{N_f}\right) + N_f\left(\frac{N_f - 2N_c}{N_f}\right) = -\frac{(N_f - N_c)^2}{N_f} + N_f - 2N_c = -\frac{N_c^2}{N_f},$$

and the right mixed R-anomaly is identical. The mixed baryonic flavor anomalies are

$$SU(N_f)_L^2 U(1)_B : \quad t\frac{N_c}{t} = N_c, \qquad SU(N_f)_R^2 U(1)_B : \quad -t\frac{N_c}{t} = -N_c.$$

The $U(1)_B^2 U(1)_R$ trace is

$$2tN_f\left(\frac{N_c}{t}\right)^2\left(-\frac{t}{N_f}\right)=-2N_c^2.$$

Finally,

$$\operatorname{Tr}R=(t^2-1)+2tN_f\left(-\frac{t}{N_f}\right)+N_f^2\left(\frac{N_f-2N_c}{N_f}\right)=-N_c^2-1,$$

and

$$\operatorname{Tr}R^3=(t^2-1)-\frac{2t^4}{N_f^2}+\frac{(N_f-2N_c)^3}{N_f}=N_c^2-1-\frac{2N_c^4}{N_f^2}.$$

This completes the displayed anomaly-matching calculation.

Mass and Higgs deformation tests for SQCD duality. Let $N_f \ge N_c+2$ and set $\widetilde{N}_c = N_f - N_c$. The magnetic field content and superpotential in (90.55) pass the following necessary deformation tests. This is a bookkeeping calculation inside the proposed dual description, not a proof of Seiberg duality.

First give the last electric flavor a holomorphic mass

$$W_{\text{el,mass}} = m\, M^L{}_L, \qquad L = N_f.$$

In the magnetic variables the F-term equation for $M^L{}_L$ is

$$\frac{1}{\mu_*} q_L^\alpha \widetilde{q}_\alpha^L + m = 0. \tag{90.56}$$

For $\widetilde{N}_c \ge 2$, a rank-one solution Higgses

$$SU(\widetilde{N}_c) \longrightarrow SU(\widetilde{N}_c - 1).$$

The light magnetic theory has $N_f - 1$ flavors, gauge rank

$$\widetilde{N}_c - 1 = (N_f - 1) - N_c,$$

and the same meson-quark superpotential form for the light flavor block. For initial $\widetilde{N}_c = 2$, the resulting $SU(1)$ gauge group is trivial; this is the boundary case leading to the $N_f = N_c + 1$ confining chiral description rather than to another nonabelian magnetic gauge theory.

Second give the electric theory a rank-r meson expectation value on the classical Higgs patch, with $1 \le r < N_c$:

$$M^A{}_B = v_A \delta^A{}_B, \qquad A, B = 1, \ldots, r, \qquad v_A \neq 0.$$

The electric gauge group is Higgsed to $SU(N_c - r)$ with $N_f - r$ light flavors. In the magnetic variables the same background gives masses

$$\frac{v_A}{\mu_*}\, q_A \widetilde{q}^A, \qquad A = 1, \ldots, r,$$

to r magnetic flavor pairs. After they are integrated out, the magnetic gauge rank remains $\widetilde{N}_c$, which is precisely the dual rank of the low-energy electric theory:

$$(N_f - r) - (N_c - r) = N_f - N_c = \widetilde{N}_c. \tag{90.57}$$

For the mass deformation, split flavor indices as $i = (a, L)$, with $a = 1, \dots, N_f - 1$. The deformed magnetic superpotential is

$$W_{\text{mag}} + mM^L{}_L = \frac{1}{\mu_*} M^i{}_j q_i^\alpha \widetilde{q}_\alpha^j + mM^L{}_L.$$

The $M^L{}_L$ equation is (90.56), while the other M-equations set the corresponding magnetic quark bilinears to zero on the light branch. Choose a local representative

$$q_L^\alpha = v\, \delta^\alpha{}_{\widetilde{N}_c}, \qquad \widetilde{q}_\alpha^L = \widetilde{v}\, \delta_{\alpha \widetilde{N}_c}, \qquad v\widetilde{v} = -\mu_* m, \tag{90.58}$$

with the usual D-flat representative chosen inside the complexified gauge orbit. The stabilizer of this nonzero fundamental-antifundamental pair is $SU(\widetilde{N}_c - 1)$. The fields carrying the broken color line are massive or gauge directions. The remaining light magnetic quarks have flavor labels $a = 1, \dots, N_f - 1$ and color labels in $SU(\widetilde{N}_c - 1)$, and the surviving singlet block is $M^a{}_b$. Restricting the superpotential to these light fields gives

$$W_{\text{light}} = \frac{1}{\mu_*} M^a{}_b q_a \widetilde{q}^b,$$

which has the same form as (90.55). The rank identity is immediate:

$$\widetilde{N}_c - 1 = N_f - N_c - 1 = (N_f - 1) - N_c.$$

For the Higgs deformation, a rank-r electric meson expectation value may be represented on the dense Higgs patch by r independent nonzero quark-antiquark pairs. Their common stabilizer in $SU(N_c)$ is $SU(N_c - r)$, and the light electric flavors are the remaining $N_f - r$. In the magnetic superpotential, substituting $M^A{}_B = v_A \delta^A{}_B$ gives the displayed mass terms $(v_A/\mu_*) q_A \widetilde{q}^A$. For generic nonzero v_A, these integrate out exactly r magnetic flavor pairs. The magnetic gauge group is not Higgsed by this operation, so its rank remains $\widetilde{N}_c = N_f - N_c$. Equation (90.57) shows that this is the dual rank of the low-energy electric $SU(N_c - r)$ theory with $N_f - r$ flavors.

These deformation tests are necessary consequences of the proposed duality, not a replacement for the quoted nonperturbative input. Their value in this chapter is that the mass, Higgs, and rank bookkeeping is completely visible: one can see which fields are eliminated, which gauge group remains, and where the boundary $N_f = N_c + 1$ description enters.

The $N_f < N_c$ entries in the phase table below have a different logical status from the duality entries. The $N_f = N_c - 1$ ADS term was derived above from an explicit one-instanton zero-mode and symmetry calculation under Hypothesis 90.11. The cases $N_f < N_c - 1$ are then reached by holomorphic decoupling using the scale matching above and Proposition 90.13. Thus the nonzero superpotential in the runaway range is not inferred from Seiberg duality; it is a chiral F-term result whose direct semiclassical anchor, finite-dimensional elimination step, and decoupling assumptions are displayed separately.

SQCD infrared range synthesis. The following table is a synthesis of results and inputs with different logical status, not a single theorem with a uniform proof. The $N_f < N_c$ range follows from the ADS seed, holomorphic decoupling, and the displayed finite-dimensional elimination arguments. The $N_f = N_c$ and $N_f = N_c + 1$ entries use the quantum-modified and confining chiral-coordinate inputs stated earlier in the section. The magnetic ranges $N_f \geq N_c + 2$ use the nonperturbative Seiberg-duality input of Quoted Theorem 90.19 under Hypothesis 90.18. The one-loop signs below are Wilsonian coordinate checks on the proposed infrared variables; by themselves they do not construct the continuum infrared fixed points or prove the electric-magnetic equivalence.

$N_f < N_c$. Status: ADS/holomorphic-decoupling result under the instanton and chiral coordinate hypotheses already stated. The ADS superpotential is generated. With no tree-level mass terms there is no supersymmetric vacuum at finite meson expectation value; the potential has a runaway direction on the maximal-rank meson patch.

$N_f = N_c$. Status: quantum-modified chiral-coordinate input checked against the decoupling from the $N_f = N_c+1$ description. There is no generated superpotential, but the chiral coordinate ring is quantum modified:

$$\det M - B\widetilde{B} = \Lambda_h^{2N_c}.$$

$N_f = N_c + 1$. Status: confining chiral-coordinate input with the mass-decoupling and anomaly checks displayed above. The infrared chiral description is in terms of mesons and baryons. If B_i and $\widetilde{B}^j$ denote the baryons with one flavor index left over, the exact superpotential is

$$W_{\mathrm{conf}} = \frac{1}{\Lambda_h^{2N_c-1}} \left(B_i M^i{}_j \widetilde{B}^j - \det M \right). \tag{90.59}$$

$N_c + 2 \leq N_f < \frac{3}{2}N_c$. Status: Seiberg-duality input plus perturbative sign check in the magnetic variables. The magnetic gauge group $SU(N_f - N_c)$ is infrared free, since its one-loop coefficient is

$$b_{0,\mathrm{mag}} = 3(N_f - N_c) - N_f = 2N_f - 3N_c < 0.$$

The appropriate low-energy variables are the magnetic quarks, magnetic gluons, and meson singlets.

$\frac{3}{2}N_c < N_f < 3N_c$. Status: Seiberg-duality input plus SCFT-current and accidental-symmetry hypotheses. Both electric and magnetic variables are asymptotically free and flow, under that input, to the same interacting nonabelian Coulomb-phase SCFT. The lower endpoint is detected by the electric meson unitarity bound, and the upper endpoint by the electric loss of asymptotic freedom.

$N_f \geq 3N_c$. Status: perturbative electric infrared-freedom statement, with the marginal endpoint interpreted in the same Wilsonian coordinate convention. The electric variables are in the free electric phase. For $N_f > 3N_c$ the electric gauge coupling is infrared free in the usual perturbative sense; at $N_f = 3N_c$ the interacting Banks–Zaks fixed point has collided with the Gaussian endpoint in this table.

90.12 Klebanov–Witten Conifold SCFT

Let $N \geq 2$. The Klebanov–Witten theory has gauge group

$$G_{\mathrm{KW}} = SU(N)_1 \times SU(N)_2$$

and four chiral superfields

$$A_i \in (\mathbf{N}, \overline{\mathbf{N}}), \qquad B_j \in (\overline{\mathbf{N}}, \mathbf{N}), \qquad i, j = 1, 2.$$

Equivalently, one may start from the unitary $U(N)_1 \times U(N)_2$ presentation. Its diagonal $U(1)$ is a free center-of-mass vector multiplet, and the interacting nonabelian sector is the $SU(N)_1 \times SU(N)_2$ theory displayed above. The relative abelian factor is useful for describing the rank-one abelian quotient below.

The index i is a doublet index for a global $SU(2)_A$, and j is a doublet index for a global $SU(2)_B$. The baryonic symmetry $U(1)_B$ is normalized here by

$$q_B(A_i) = +1, \qquad q_B(B_j) = -1.$$

The tree superpotential is

$$W_{\mathrm{KW}} = h\, \epsilon^{ij} \epsilon^{kl}\, \mathrm{tr}(A_i B_k A_j B_l), \tag{90.60}$$

where the trace is over the gauge indices in the alternating product

$$\mathbf{N}_1 \xrightarrow{A_i} \mathbf{N}_2 \xrightarrow{B_k} \mathbf{N}_1 \xrightarrow{A_j} \mathbf{N}_2 \xrightarrow{B_l} \mathbf{N}_1.$$

The manifest continuous global symmetry at the superpotential point is

$$SU(2)_A \times SU(2)_B \times U(1)_B \times U(1)_R.$$

There is also a charge-conjugation symmetry exchanging the two gauge factors and the A- and B-multiplets when the two gauge couplings are exchanged.

Hypothesis 90.20 (KW Lagrangian and branch assumptions)**.** The algebraic calculations in this section use the four-dimensional $\mathcal{N} = 1$ Lagrangian with gauge group, matter representations, trace normalization, and superpotential displayed above. Statements about the mesonic vacuum branch use the classical F- and D-flat quotient of that Lagrangian and then restrict to the commuting branch where the bifundamental matrices can be simultaneously reduced to diagonal representatives. Statements about protected dimensions use Hypothesis 90.17. Claims about the full continuum conformal manifold are isolated in Hypothesis 90.24 and in the exact-marginality rank calculation below.

KW F-term equations. The F-term equations of W_{KW} are

$$A_1 B_j A_2 = A_2 B_j A_1, \qquad j = 1, 2, \tag{90.61}$$

and

$$B_1 A_i B_2 = B_2 A_i B_1, \qquad i = 1, 2. \tag{90.62}$$

Expanding the antisymmetric tensors and using cyclicity of the trace gives

$$W_{\mathrm{KW}} = 2h\ \mathrm{tr}(A_1 B_1 A_2 B_2 - A_1 B_2 A_2 B_1).$$

Varying with respect to A_1 and moving δA_1 to the left inside the trace gives

$$\delta_{A_1} W_{\mathrm{KW}} = 2h\ \mathrm{tr}\big[\delta A_1 (B_1 A_2 B_2 - B_2 A_2 B_1)\big],$$

so $B_1 A_2 B_2 = B_2 A_2 B_1$. The variation with respect to A_2 gives the same equation with A_1 and A_2 exchanged. Varying with respect to B_1 and B_2 in the same way gives

$$A_2 B_2 A_1 = A_1 B_2 A_2, \qquad A_1 B_1 A_2 = A_2 B_1 A_1,$$

which is exactly (90.61). These are matrix equations in the appropriate bifundamental Hom-spaces.

KW anomaly and superpotential R-charge. Assume the $SU(2)_A \times SU(2)_B$ symmetry and the exchange symmetry, so that

$$R(A_i) = R(B_j) = r.$$

The simultaneous conditions that $U(1)_R$ have no $SU(N)_a^2 U(1)_R$ anomaly and that W_{KW} have R-charge 2 give

$$r = \frac{1}{2}.$$

For either gauge factor the gaugino contributes $T(\mathrm{adj}) = N$ to the $SU(N)^2 U(1)_R$ anomaly coefficient. Each of the two A-fields and each of the two B-fields is a fundamental or antifundamental for that gauge factor and has N copies from the other gauge index. With $T(\mathbf{N}) = T(\overline{\mathbf{N}}) = \frac{1}{2}$, the matter contribution is

$$4 \cdot \frac{N}{2}(r-1) = 2N(r-1).$$

The anomaly coefficient is therefore

$$N + 2N(r-1) = N(2r-1),$$

which vanishes exactly at $r = \frac{1}{2}$. The superpotential contains two A-fields and two B-fields, so its R-charge is $4r$, also equal to 2 exactly at $r = \frac{1}{2}$.

Remark 90.21 (KW NSVZ numerator). At $r = \frac{1}{2}$, as fixed by the anomaly and superpotential-charge calculation above, the chiral-multiplet anomalous dimensions in the superconformal convention are

$$\gamma_A = \gamma_B = -\frac{1}{2},$$

and the NSVZ numerator of each gauge factor vanishes. Equation (90.49) gives

$$\gamma_A = \gamma_B = 3 \cdot \frac{1}{2} - 2 = -\frac{1}{2}.$$

For either $SU(N)$ factor, the total matter index is

$$4 \cdot N \cdot \frac{1}{2} = 2N.$$

Hence the NSVZ numerator is

$$3N - 2N(1-\gamma_A) = 3N - 2N\left(1 + \frac{1}{2}\right) = 0.$$

The scalar dimension of each A_i and B_j is therefore

$$\Delta(A_i) = \Delta(B_j) = 1 + \frac{1}{2}\gamma_A = \frac{3}{4} = \frac{3}{2}R(A_i).$$

KW beta-function rank count. Work in the $SU(2)_A \times SU(2)_B \times U(1)_B$-preserving source-coordinate system near the Klebanov–Witten point, and assume the NSVZ coordinate convention of this chapter with finite denominators for both gauge couplings. Let γ_A and γ_B be the anomalous dimensions of the A- and B-doublets in the monograph convention. Then the two gauge beta-function numerators and the quartic-superpotential marginality defect are all proportional to the same expression:

$$\mathcal{B}_1 = \mathcal{B}_2 = N(1 + \gamma_A + \gamma_B), \qquad \mathcal{B}_h = 1 + \gamma_A + \gamma_B. \tag{90.63}$$

Consequently the three local vanishing conditions have rank one in this symmetry-preserving chart. At the exchange-symmetric point $\gamma_A = \gamma_B = \gamma$, the common condition is $\gamma = -\frac{1}{2}$.

For the first gauge group, each of the two A-fields is a fundamental with N copies from the second gauge index, and each of the two B-fields is an antifundamental with the same multiplicity. Hence

$$\sum_i T_1(R_i)(1-\gamma_i) = 2 \cdot \frac{N}{2}(1-\gamma_A) + 2 \cdot \frac{N}{2}(1-\gamma_B) = N(2-\gamma_A-\gamma_B).$$

The NSVZ numerator of the first gauge factor is therefore

$$\mathcal{B}_1 = 3N - N(2-\gamma_A-\gamma_B) = N(1+\gamma_A+\gamma_B).$$

The same count holds for the second gauge factor, with fundamentals and antifundamentals interchanged, so $\mathcal{B}_2 = \mathcal{B}_1$.

For the superpotential term, nonrenormalization means that the local condition for a quartic chiral coordinate to be marginal is the dimension condition on the chiral operator. The operator $\mathrm{tr}(A_i B_k A_j B_l)$ has dimension

$$2\left(1+\frac{\gamma_A}{2}\right) + 2\left(1+\frac{\gamma_B}{2}\right) = 4+\gamma_A+\gamma_B.$$

Since a four-dimensional superpotential integrand must have dimension 3, the marginality defect is

$$\mathcal{B}_h = (4+\gamma_A+\gamma_B) - 3 = 1+\gamma_A+\gamma_B.$$

Equivalently, the beta function of a dimensionless quartic coupling is proportional to this defect, up to the sign convention for whether the flow parameter is an energy scale or a length scale. Only the zero locus is used here.

Thus $\mathcal{B}_1$, $\mathcal{B}_2$, and $\mathcal{B}_h$ impose a single scalar condition in the symmetric chart. This is a necessary local beta-function count. Existence of the continuum SCFT, realization of the common zero by an RG trajectory, and nonsingularity of the local zero locus are the additional inputs isolated in Hypothesis 90.24.

The anomaly and NSVZ calculations above fix the exchange-symmetric point. To see that the same value is selected by a-maximization, allow mixing with the baryonic current:

$$R_s(A_i) = \frac{1}{2}+s, \qquad R_s(B_j) = \frac{1}{2}-s.$$

The gauge-anomaly and superpotential constraints impose only $R_s(A)+R_s(B)=1$. The trial central charge is computed from Weyl fermion R-charges. The two gaugino adjoints have fermion R-charge 1; the two A-fermions have charge $-\frac{1}{2}+s$; the two B-fermions have charge $-\frac{1}{2}-s$. Therefore

$$\mathrm{Tr}\, R_s = -2,$$

and

$$\mathrm{Tr}\, R_s^3 = 2(N^2-1) + 2N^2\left[\left(-\frac{1}{2}+s\right)^3 + \left(-\frac{1}{2}-s\right)^3\right] = \frac{3}{2}N^2 - 2 - 6N^2 s^2.$$

Thus

$$a(s) = \frac{3}{32}\left(3\,\mathrm{Tr}\, R_s^3 - \mathrm{Tr}\, R_s\right) = \frac{3}{32}\left(\frac{9}{2}N^2 - 4 - 18N^2 s^2\right),$$

which is maximized at $s = 0$. The superconformal R-charge is the symmetric one:

$$R(A_i) = R(B_j) = \frac{1}{2}.$$

Remark 90.22 (KW central charges). At the symmetric superconformal R-charge,

$$a_{\mathrm{KW}} = \frac{27}{64}N^2 - \frac{3}{8}, \qquad c_{\mathrm{KW}} = \frac{27}{64}N^2 - \frac{1}{4}.$$

For a four-dimensional $\mathcal{N} = 1$ SCFT,

$$a = \frac{3}{32}(3\,\mathrm{Tr}\,R_f^3 - \mathrm{Tr}\,R_f), \qquad c = \frac{1}{32}(9\,\mathrm{Tr}\,R_f^3 - 5\,\mathrm{Tr}\,R_f),$$

where the traces run over Weyl fermions. The fermion trace data at $s = 0$ are

$$\mathrm{Tr}\,R_f = 2(N^2 - 1) + 4N^2\left(-\frac{1}{2}\right) = -2$$

and

$$\mathrm{Tr}\,R_f^3 = 2(N^2 - 1) + 4N^2\left(-\frac{1}{8}\right) = \frac{3}{2}N^2 - 2.$$

Substitution gives the displayed values.

The basic mesonic chiral operators are

$$W_{ij} = A_i B_j.$$

For a single rank-one abelian sector in the $U(1)\times U(1)$ presentation, the diagonal $U(1)$ decouples and the complexified anti-diagonal gauge group acts with charges $+1$ on the A_i and -1 on the B_j. The four coordinates W_{ij} are invariant and obey one quadratic relation.

For the rank-one abelian quotient just described, the mesonic coordinate ring is

$$\mathbb{C}[W_{11}, W_{12}, W_{21}, W_{22}]/(W_{11}W_{22} - W_{12}W_{21}).$$

Write $W_{ij} = A_i B_j$ with commuting complex variables A_1, A_2, B_1, B_2. Then

$$W_{11}W_{22} - W_{12}W_{21} = A_1B_1A_2B_2 - A_1B_2A_2B_1 = 0.$$

Conversely, the invariant functions of a $\mathbb{C}^\times$ action with charges $+1, +1, -1, -1$ are generated by the degree-zero monomials A_iB_j. The only rank-one relation among the entries of the 2×2 matrix (W_{ij}) is its determinant. Hence the quotient is the affine conifold

$$\det(W) = 0.$$

On the mesonic commuting branch for general N, the matrices A_iB_j can be diagonalized simultaneously at generic points, and the Weyl group permutes the N eigenvalue quadruples. The corresponding branch is

$$\mathrm{Sym}^N\{W_{11}W_{22} - W_{12}W_{21} = 0\}.$$

This statement concerns the commuting mesonic branch; baryonic branches and noncommuting chiral-ring strata require additional invariant-theory data.

The protected operator dimensions follow immediately from the superconformal R-charges. The mesons W_{ij} transform as $(\mathbf{2},\mathbf{2})$ under $SU(2)_A \times SU(2)_B$ and have

$$R(W_{ij}) = 1, \qquad \Delta(W_{ij}) = \frac{3}{2}.$$

The baryons

$$\mathcal{B}_A = \det A_i, \qquad \mathcal{B}_B = \det B_j$$

have

$$R(\mathcal{B}_A) = R(\mathcal{B}_B) = \frac{N}{2}, \qquad \Delta(\mathcal{B}_A) = \Delta(\mathcal{B}_B) = \frac{3N}{4},$$

and baryonic charges $+N$ and $-N$ in the normalization above. Symmetric products

$$\mathrm{tr}(A_{i_1}B_{j_1}\cdots A_{i_\ell}B_{j_\ell})$$

have $R = \ell$ and $\Delta = \frac{3}{2}\ell$ when they represent nonzero chiral-primary classes.

Hypothesis 90.23 (Local $\mathcal{N} = 1$ exact-marginality chart)**.** Let $\mathcal{T}_0$ be a four-dimensional $\mathcal{N} = 1$ SCFT with a chosen global-symmetry subalgebra $\mathfrak{f}$ to be preserved. A local holomorphic marginal-source-coordinate system consists of complex coordinates

$$u = (u^1, \ldots, u^m)$$

for gauge and superpotential sources coupled to $\mathfrak{f}$-invariant chiral operators of superpotential dimension 3, together with a holomorphic beta map

$$\mathcal{B} : U \subset \mathbb{C}^m \longrightarrow \mathbb{C}^r$$

whose zero set is exactly the set of conformal theories in this source chart. Redundant directions from holomorphic field-coordinate changes have either been quotient out or fixed by a declared source normalization. The map $\mathcal{B}$ includes the gauge beta functions, superpotential beta functions, and any current-moment-map obstruction for a broken conserved current. If $d\mathcal{B}$ has constant rank r along the zero set, the local conformal manifold in this chart is the complex manifold $\mathcal{B}^{-1}(0)$ of dimension $m - r$.

Under this hypothesis, the dimension statement is the holomorphic implicit-function theorem applied to the finite-dimensional source-coordinate system. The QFT content is not the theorem of complex analysis; it is the hypothesis that the beta map and the quotient by redundant coordinates faithfully describe the continuum SCFT family.

Hypothesis 90.24 (KW exact-marginality input)**.** Near the Klebanov–Witten point, use local holomorphic sources

$$u = (\tau_1, \tau_2, h)$$

for the two $SU(N)$ gauge couplings and the quartic superpotential in (90.60), restricted to the $SU(2)_A \times SU(2)_B \times U(1)_B$-preserving chart. Assume the continuum SCFT exists in this neighborhood, the NSVZ Wilsonian coordinate analysis of this chapter computes the vanishing conditions for these sources, and no additional exactly marginal obstruction is present after the background contact-term and source-normalization data are fixed. Finally assume the common defect

$$E(u) = 1 + \gamma_A(u) + \gamma_B(u)$$

has nonzero differential at the symmetric point in the chosen source-coordinate system.

KW local conformal manifold from rank one. Assume Hypothesis 90.24. In the $SU(2)_A \times SU(2)_B \times U(1)_B$-preserving source-coordinate system, the Klebanov–Witten conformal locus is locally a complex surface in the source space with coordinates (τ_1, τ_2, h).

The rank-count calculation above gives, in this chart,

$$\mathcal{B}_1 = NE, \qquad \mathcal{B}_2 = NE, \qquad \mathcal{B}_h = E, \qquad E = 1 + \gamma_A + \gamma_B.$$

Thus the zero set of the three beta functions is the single equation

$$E(u) = 0.$$

By Hypothesis 90.24, $dE \neq 0$ at the Klebanov–Witten point. The holomorphic implicit-function theorem therefore identifies the local zero set with a smooth complex hypersurface in the three-dimensional source-coordinate system. Its local complex dimension is

$$3 - 1 = 2.$$

The proof uses no holographic construction. Its nontrivial QFT assumptions are exactly those isolated in Hypothesis 90.24: existence of the continuum SCFT chart, completeness of the beta-function analysis, and absence of hidden current or contact-term obstructions in the preserved-symmetry sector.

The result is a conditional field-theoretic reconstruction of the Leigh–Strassler rank count for this example. The chapter has proved the anomaly, NSVZ, a-maximization, chiral-ring, and rank-one beta-function arithmetic. It has not constructed the nonperturbative continuum SCFT from a regulator.

90.13 Klebanov–Strassler Cascade

Fix integers $M > 0$ and $N > M$. The ultraviolet theory used in the Klebanov–Strassler cascade has gauge group

$$SU(N+M)_1 \times SU(N)_2$$

with the same bifundamental matter and quartic superpotential as (90.60), now with rectangular gauge indices. Treating the anomalous dimensions of the four bifundamentals as equal to a common γ, the NSVZ numerators are

$$\mathcal{B}_1 = 3(N+M) - 2N(1-\gamma), \qquad \mathcal{B}_2 = 3N - 2(N+M)(1-\gamma). \tag{90.64}$$

At the Klebanov–Witten value $\gamma = -\frac{1}{2}$,

$$\mathcal{B}_1 = 3M, \qquad \mathcal{B}_2 = -3M.$$

Since the NSVZ beta function has the sign $\beta_g = -g^3\mathcal{B}/(16\pi^2(1 - \cdots))$, the larger gauge factor is driven to stronger coupling toward the infrared in this local coordinate, while the smaller factor is driven in the opposite direction. The cascade idealizes the strongly coupled step by applying Seiberg duality to the larger factor.

Hypothesis 90.25 (Cascade input assumptions)**.** The cascade discussion assumes the same supersymmetric Wilsonian scheme and NSVZ coordinate convention used earlier in the chapter. The displayed numerator signs are perturbative-coordinate diagnostics near the Klebanov–Witten anomalous dimension, not a proof that the flow follows a unique trajectory. Each rank-reducing step additionally assumes Seiberg duality for the strongly coupled node with the other node treated during the matching as a weakly gauged flavor symmetry, and assumes that integrating out the mesons in the magnetic description is legitimate in the Wilsonian neighborhood being used. These duality assumptions are recorded again in Quoted Theorem 90.26.

Cascade rank step. Assume the Seiberg-duality transformation for the first gauge factor while the second gauge factor is treated as a weakly gauged flavor symmetry during the dualization. Then

$$SU(N+M)\times SU(N) \quad\longmapsto\quad SU(N)\times SU(N-M),$$

up to relabeling of the two factors and matching of the quartic superpotential.

The first gauge factor $SU(N+M)$ sees two A-fields and two B-fields. Because each carries an $SU(N)$ index of the second gauge group, it has

$$N_f = 2N$$

flavors in the SQCD sense. Seiberg duality maps an electric $SU(N_c)$ theory with N_f flavors to a magnetic $SU(N_f - N_c)$ theory. Here

$$N_f - N_c = 2N - (N+M) = N - M.$$

The untouched second gauge factor is $SU(N)$. Relabeling the larger factor first gives $SU(N)\times SU(N-M)$. The electric mesons of the dualized node are composites

$$M_{ij} = A_i B_j.$$

The magnetic superpotential has the local form

$$W_{\mathrm{mag}} = h\,\epsilon^{ij}\epsilon^{kl}\,\mathrm{tr}(M_{ik}M_{jl}) + \frac{1}{\mu_*}\,\mathrm{tr}(M_{ij}a_i b_j), \tag{90.65}$$

where a_i, b_j are magnetic bifundamentals and μ_* is the matching scale used to assign dimension one to M_{ij}. The quadratic form

$$(M_{ij}) \longmapsto \epsilon^{ij}\epsilon^{kl}\,\mathrm{tr}(M_{ik}M_{jl})$$

is nondegenerate on the four mesons. Thus the F-term equations of (90.65) solve M_{ij} linearly in the bilinears $a_i b_j$. Substitution gives again a quartic superpotential proportional to

$$\epsilon^{ij}\epsilon^{kl}\,\mathrm{tr}(a_i b_k a_j b_l),$$

with a coefficient depending on the normalization of M_{ij} and μ_*.

Thus the rank difference remains M, while the common rank parameter drops by M. If

$$N = qM + r, \qquad 0 \le r < M,$$

then after q formal cascade steps the pair has reached

$$SU(M+r)\times SU(r),$$

where the second factor is absent when $r = 0$. The integer q is the number of rank-reducing duality steps, and $r = N \bmod M$ is the residual common rank parameter.

KS R-anomaly remnant. For the unequal-rank theory with the field charges

$$R(A_i) = R(B_j) = \frac{1}{2}, \qquad R(\lambda_1) = R(\lambda_2) = 1,$$

the continuous $U(1)_R$ has gauge-anomaly coefficients

$$\mathcal{A}_1 = M, \qquad \mathcal{A}_2 = -M$$

for $SU(N+M)_1^2 U(1)_R$ and $SU(N)_2^2 U(1)_R$. Consequently only the $\mathbb{Z}_{2M}$ subgroup is an honest symmetry with fixed theta angles.

For $SU(N+M)_1$, the gaugino contributes $N+M$. Each chiral fermion in an A_i or B_j multiplet has R-charge $-\frac{1}{2}$, and the total matter index seen by the first gauge group is

$$4 \cdot N \cdot \frac{1}{2} = 2N.$$

Therefore

$$\mathcal{A}_1 = (N+M) - \frac{1}{2}(2N) = M.$$

For $SU(N)_2$, the total matter index is instead

$$4 \cdot (N+M) \cdot \frac{1}{2} = 2(N+M),$$

so

$$\mathcal{A}_2 = N - \frac{1}{2}\big(2(N+M)\big) = -M.$$

In the normalization where pure $SU(K)$ supersymmetric Yang–Mills leaves $\mathbb{Z}_{2K}$, an R-rotation by angle φ shifts the two theta angles by

$$\theta_1 \mapsto \theta_1 + 2M\varphi, \qquad \theta_2 \mapsto \theta_2 - 2M\varphi.$$

Because each theta angle has period 2π, the rotation is a symmetry at fixed theta angles exactly when

$$2M\varphi \in 2\pi\mathbb{Z}.$$

Modulo the 2π-periodicity of the R-rotation parameter, this leaves $2M$ elements, namely $\mathbb{Z}_{2M}$.

Quoted Theorem 90.26 (Seiberg-duality input for the cascade)**.** *The cascade rank step describes the infrared chiral data and anomalies of the successive Wilsonian theories provided Seiberg duality holds for the strongly coupled node with the neighboring gauge group treated as a weakly gauged flavor symmetry during the matching. The matching includes the flavor symmetries, baryonic symmetry, R-symmetry, holomorphic scale coordinates, and the quartic superpotential after the massive mesons are eliminated.*

This is a status boundary, not a new proof of Seiberg duality. The monograph uses the duality as an exact field-theoretic input with visible checks: the magnetic rank is $N_f - N_c$, the rank difference is preserved, the alternating quartic superpotential is regenerated, and the global anomaly assignment is unchanged after the duality map.

For $r = 0$, the cascade ends at pure $SU(M)$ supersymmetric Yang–Mills. The anomalous chiral symmetry leaves a $\mathbb{Z}_{2M}$ gaugino phase symmetry, and gaugino condensation breaks it to $\mathbb{Z}_2$, producing M massive vacua in the finite-volume/gap sense used earlier in this chapter. For $0 < r < M$, the final strongly coupled theory contains an ADS or quantum-modified chiral coordinate, depending on the precise endpoint rank. This is only a field-theoretic endpoint summary in the present monograph; any geometric interpretation is outside the argument and is not used as evidence for the QFT dynamics.

90.14 Status Boundary

The arguments in this chapter determine holomorphic coordinates from symmetry, anomaly, decoupling, and semiclassical input. Turning them into a theorem for a four-dimensional continuum QFT requires a nonperturbative construction of the regulated supersymmetric gauge theory, a proof of the required mass gaps or moduli-space descriptions, and a Wilsonian scheme in which the chiral coordinates above are defined as continuum limits.

Open Problem 90.27 (Nonperturbative $\mathcal{N} = 1$ Wilsonian scheme)*.* Construct a four-dimensional regulator and continuum limit for $\mathcal{N} = 1$ gauge theories that preserves enough supersymmetric and gauge Ward identities to make the holomorphic Wilsonian coordinate arguments above proved statements.

Chapter 91

Four-Dimensional N=2 Gauge Dynamics and Seiberg-Witten Theory

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops the first layer of four-dimensional $\mathcal{N} = 2$ gauge dynamics. The emphasis is the Wilsonian low-energy description on the Coulomb branch, where the exact two-derivative Abelian action is encoded by a holomorphic prepotential and by periods of a family of curves.

Hypothesis 91.1 (Seiberg–Witten Coulomb-branch structure)**.** A Seiberg–Witten Coulomb-branch structure of rank r is the following collection attached to an assumed four-dimensional $\mathcal{N} = 2$ local QFT $\mathfrak{T}$.

(i) A connected complex analytic vacuum branch $\mathcal{B}$ and a discriminant locus $\Delta \subset \mathcal{B}$ such that the low-energy theory over $\mathcal{B}^\circ = \mathcal{B} \setminus \Delta$ contains r Abelian vector multiplets and no additional massless charged hypermultiplets.

(ii) A locally constant electromagnetic charge lattice $\Gamma \to \mathcal{B}^\circ$ with a nondegenerate integral antisymmetric Dirac pairing $\langle \cdot, \cdot \rangle : \Gamma \otimes \Gamma \to \mathbb{Z}$. A local electric polarization is a symplectic frame $(\alpha^I, \beta_I)_{I=1}^r$ of Γ.

(iii) A holomorphic period section

$$\Pi = (a_{D,I}, a^I)$$

in the dual local system, with monodromy in $Sp(2r, \mathbb{Z})$, such that

$$Z_\gamma = \langle \gamma, \Pi \rangle + \sum_\alpha s_\alpha m_\alpha$$

is the central charge of a state of electromagnetic charge γ and flavor charges s_α.

(iv) On each simply connected electric patch, a holomorphic prepotential $\mathcal{F}(a)$ satisfying

$$a_{D,I} = \frac{\partial \mathcal{F}}{\partial a^I}, \qquad \tau_{IJ} = \frac{\partial^2 \mathcal{F}}{\partial a^I \partial a^J}, \qquad \operatorname{Im}\tau > 0,$$

and the corresponding two-derivative Abelian Wilsonian action with the superspace normalization used in this volume.

(v) Local singularity data along Δ: the primitive charges of massless BPS states, the associated Picard–Lefschetz monodromies, and the convention for the BPS inequality $M_\gamma \geq \sqrt{2}\,|Z_\gamma|$.

(vi) When a curve realization is used, a holomorphic family of compact Riemann surfaces Σ_u, meromorphic differentials λ_{SW}, and symplectic one-cycle frames whose periods reproduce the section Π. This curve realization is auxiliary data for computing the special geometry; it is not by itself a construction of the four-dimensional QFT $\mathfrak{T}$.

All dynamical conclusions in this chapter are conditional on the existence of $\mathfrak{T}$ and of such a structure. The open problem of constructing $\mathfrak{T}$ from a regulator and deriving these structures as theorems of QFT is recorded in Open Problem 91.11.

91.1 Coulomb-Branch Coordinates

Let G be a compact connected gauge group with Lie algebra $\mathfrak{g}$. An $\mathcal{N} = 2$ vector multiplet contains a gauge field, two gauginos, and a complex scalar $\phi \in \mathfrak{g}_{\mathbb{C}}$. A Coulomb-branch point is a vacuum for which ϕ has expectation value in a Cartan subalgebra and the low-energy gauge group contains an Abelian factor $U(1)^r$, where $r = \operatorname{rank} G$. Gauge-invariant holomorphic coordinates are polynomial functions of ϕ, for example

$$u_k = \left\langle \operatorname{tr}(\phi^k) \right\rangle.$$

In the notation of Hypothesis 91.1, these u_k are local coordinates on the branch $\mathcal{B}$ only after one has chosen the microscopic gauge theory and restricted to the component where the Abelian rank is r. The trace convention is the convention fixed in the gauge chapters: the Yang–Mills kinetic term uses

$$\frac{1}{4g^2}\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$$

with the chosen invariant bilinear form.

91.2 Special Coordinates And Prepotential

On a smooth patch of $\mathcal{B}^\circ$, choose electric special coordinates

$$a^I, \qquad I = 1, \ldots, r.$$

The two-derivative Abelian Wilsonian action in Hypothesis 91.1 is determined by a holomorphic function $\mathcal{F}(a)$, the prepotential, through

$$a_{D,I} = \frac{\partial \mathcal{F}}{\partial a^I}, \qquad \tau_{IJ}(a) = \frac{\partial^2 \mathcal{F}}{\partial a^I \partial a^J}.$$

The imaginary part of τ_{IJ} is the positive gauge-coupling matrix in the region where the Abelian Wilsonian description is valid. Electric-magnetic duality acts on the column vector

$$\Pi = \begin{pmatrix} a_D \\ a \end{pmatrix}$$

by integral symplectic matrices. If $\gamma = (n_m^I, n_{e,I})$ is an electromagnetic charge vector, its central charge is

$$Z_\gamma = n_{e,I} a^I + n_m^I a_{D,I} + \sum_\alpha s_\alpha m_\alpha,$$

where m_α are background flavor masses and s_α are flavor charges. The BPS inequality has the convention

$$M_\gamma \geq \sqrt{2}\,|Z_\gamma|.$$

States saturating this bound form shortened $\mathcal{N} = 2$ multiplets.

In $\mathcal{N} = 1$ language the same local Abelian $\mathcal{N} = 2$ Wilsonian vector multiplets are represented by vector superfields V^I and chiral superfields Φ^I whose scalar components are a^I. Once the two-derivative $\mathcal{N} = 2$ action is assumed to exist, its scalar metric and gauge kinetic coefficients are not independent data. They are encoded by the same holomorphic prepotential:

$$K_{\mathcal{F}}(a, \overline{a}) = \frac{1}{2\pi} \operatorname{Im} \left(\overline{a}^I \frac{\partial \mathcal{F}}{\partial a^I} \right), \qquad \tau_{IJ}(a) = \frac{\partial^2 \mathcal{F}}{\partial a^I \partial a^J}. \tag{91.1}$$

The factor $1/(2\pi)$ is the superspace normalization used in this volume. It can be absorbed into the definition of the Abelian fields, but it must not be silently dropped when comparing instanton, period, and component-action conventions.

Rigid special-Kähler metric from the prepotential. Let $U \subset \mathbb{C}^r$ be a coordinate patch with holomorphic coordinates a^I, and let $\mathcal{F}$ be holomorphic on U. Define $K_{\mathcal{F}}$ by (91.1). Then

$$g_{I\bar{J}} := \frac{\partial^2 K_{\mathcal{F}}}{\partial a^I \partial \overline{a}^J} = \frac{1}{2\pi} \operatorname{Im} \tau_{IJ}. \tag{91.2}$$

Consequently the scalar kinetic metric is positive exactly on those Wilsonian patches where $\operatorname{Im} \tau_{IJ}$ is positive definite. Linear terms in $\mathcal{F}$ change $K_{\mathcal{F}}$ by a Kähler transformation, while real quadratic terms shift only $\operatorname{Re} \tau$ and hence only the theta-angle part of the Abelian gauge action. Write

$$K_{\mathcal{F}} = \frac{1}{4\pi \mathrm{i}} \left(\overline{a}^L \mathcal{F}_L(a) - a^L \overline{\mathcal{F}}_L(\overline{a}) \right), \qquad \mathcal{F}_L = \frac{\partial \mathcal{F}}{\partial a^L}.$$

Treating a^I and $\overline{a}^J$ as independent complex coordinates,

$$\frac{\partial K_{\mathcal{F}}}{\partial a^I} = \frac{1}{4\pi \mathrm{i}} \left(\overline{a}^L \mathcal{F}_{IL} - \overline{\mathcal{F}}_I \right).$$

Differentiating once more gives

$$\frac{\partial^2 K_{\mathcal{F}}}{\partial a^I \partial \overline{a}^J} = \frac{1}{4\pi \mathrm{i}} \left(\mathcal{F}_{IJ} - \overline{\mathcal{F}}_{IJ} \right) = \frac{1}{2\pi} \operatorname{Im} \mathcal{F}_{IJ}.$$

Since $\tau_{IJ} = \mathcal{F}_{IJ}$, this proves (91.2). If $\mathcal{F} \mapsto \mathcal{F} + c_I a^I + c_0$, then $K_{\mathcal{F}}$ changes by $(2\pi)^{-1} \operatorname{Im}(\overline{a}^I c_I)$, a sum of a holomorphic and an antiholomorphic function, so $g_{I\bar{J}}$ is unchanged. If $\mathcal{F} \mapsto \mathcal{F} + \frac{1}{2} q_{IJ} a^I a^J$ with q_{IJ} real and symmetric, then $\operatorname{Im} \tau$ is unchanged and $\operatorname{Re} \tau$ shifts by q_{IJ}.

91.3 Seiberg-Witten Period Geometry

The curve realization in Hypothesis 91.1 consists of a family of compact Riemann surfaces Σ_u, a meromorphic differential λ_{SW}, and a symplectic basis of one-cycles (A_I, B^I) such that

$$a^I(u) = \frac{1}{2\pi} \oint_{A_I} \lambda_{\mathrm{SW}}, \qquad a_{D,I}(u) = \frac{1}{2\pi} \oint_{B^I} \lambda_{\mathrm{SW}}.$$

The differential is fixed up to exact terms by requiring that $\partial \lambda_{\mathrm{SW}} / \partial u_k$ span holomorphic differentials on Σ_u and that the large-field asymptotics reproduce the one-loop Wilsonian coupling.

The period coupling is integrable on a contractible Coulomb-branch patch under the following geometric hypotheses. Let U be a patch on which the a^I, $I = 1, \ldots, r$, are holomorphic coordinates. Suppose $\pi : \Sigma \to U$ is a holomorphic family of compact Riemann surfaces with a flat symplectic homology frame (A_I, B^I), and suppose that the Seiberg-Witten differential λ_{SW} is chosen so that

$$\omega_J := \frac{\partial \lambda_{\mathrm{SW}}}{\partial a^J}$$

is a holomorphic one-form on each fiber, with no residue contribution in the Riemann bilinear relation, and is normalized by

$$\frac{1}{2\pi} \oint_{A_I} \omega_J = \delta_{IJ}.$$

The period matrix

$$\tau_{IJ} := \frac{\partial a_{D,I}}{\partial a^J} = \frac{1}{2\pi} \oint_{B^I} \omega_J$$

is symmetric. Consequently there is a holomorphic prepotential $\mathcal{F}$ on U, unique up to an additive constant, such that

$$a_{D,I} = \frac{\partial \mathcal{F}}{\partial a^I}.$$

Because the homology frame is flat, differentiating a period at fixed cycle gives the period of the differentiated differential:

$$\frac{\partial a_{D,I}}{\partial a^J} = \frac{1}{2\pi} \oint_{B^I} \omega_J.$$

Apply the Riemann bilinear relation on a fiber to the two holomorphic one-forms ω_I and ω_J:

$$0 = \sum_{K=1}^{r} \left(\oint_{A_K} \omega_I \oint_{B^K} \omega_J - \oint_{B^K} \omega_I \oint_{A_K} \omega_J \right).$$

The assumed A-period normalization gives

$$0 = 2\pi \oint_{B^I} \omega_J - 2\pi \oint_{B^J} \omega_I,$$

hence $\tau_{IJ} = \tau_{JI}$. The holomorphic one-form

$$\alpha = \sum_I a_{D,I}(a)\, \mathrm{d}a^I$$

therefore obeys

$$\mathrm{d}\alpha = \sum_{I<J} \left(\frac{\partial a_{D,J}}{\partial a^I} - \frac{\partial a_{D,I}}{\partial a^J} \right) \mathrm{d}a^I \wedge \mathrm{d}a^J = 0.$$

Since U is contractible, the ordinary Poincare lemma gives a holomorphic function

$$\mathcal{F}(a) = \int_{a_*}^{a} \alpha$$

whose value is independent of the path in U. Then $\mathrm{d}\mathcal{F} = \alpha$, which is exactly $a_{D,I} = \partial\mathcal{F}/\partial a^I$. The additive constant depends on the chosen base point a_* and has no physical content.

91.4 Argument Types And Charge-Monodromy Conventions

The Seiberg-Witten construction mixes several kinds of reasoning. In this chapter every load-bearing step is to be read with one of the following status labels.

Definition 91.2 (Argument-status labels for Seiberg-Witten theory)**.** A statement in the rank-one Seiberg-Witten construction has one of these statuses.

(i) *QFT-hypothesis*: an assumption about the existence of the four-dimensional continuum theory, its Coulomb-branch vacua, or its Wilsonian effective action.

(ii) *Symmetry-derived*: a consequence of the microscopic global symmetries, anomalies, Weyl group, and dimensional analysis.

(iii) *Holomorphy-derived*: a consequence of holomorphic dependence on Coulomb parameters and couplings once the Wilsonian $\mathcal{N} = 2$ framework has been assumed.

(iv) *Special-geometry theorem*: a statement about periods, symplectic lattices, Riemann bilinear identities, or nodal degenerations of curves.

(v) *Constraint-derived*: the minimal object satisfying the preceding symmetry, monodromy, and asymptotic constraints, with uniqueness only inside the declared ansatz.

(vi) *Consistency check*: a nontrivial compatibility test, such as a monodromy product, Picard-Fuchs equation, or instanton coefficient match.

(vii) *Heuristic/status boundary*: a physical identification not derived from a complete regulator construction in this monograph.

For the pure $\mathfrak{su}(2)$ theory below, the input that the Coulomb branch exists and is described at generic points by a two-derivative Abelian Wilsonian action is a QFT-hypothesis. The one-loop logarithm and the resulting monodromy at infinity are holomorphy-derived after the perturbative Wilsonian scheme is fixed. The residual $u \mapsto -u$ action is symmetry-derived. The Picard-Lefschetz transformation at a nodal fiber is a special-geometry theorem. The curve $y^2 = (x^2 - \Lambda^4)(x-u)$ is constraint-derived inside the minimal genus-one ansatz. The hypergeometric periods, Picard-Fuchs equation, finite-monodromy product, BPS mass vanishings, and Nekrasov coefficient are consistency checks. The statement that these data are the exact nonperturbative low-energy dynamics of the four-dimensional continuum QFT remains at the status boundary recorded at the end of the chapter.

In rank one the electromagnetic charge lattice is

$$\Gamma \simeq \mathbb{Z}^2, \qquad \gamma = (n_m, n_e),$$

with central charge $Z_\gamma = n_m a_D + n_e a$. For two charges $\delta = (m_m, m_e)$ and $\gamma = (n_m, n_e)$, use the antisymmetric pairing

$$\langle \delta, \gamma \rangle = m_m n_e - m_e n_m. \tag{91.3}$$

This is the Dirac pairing in the period basis $\Pi = (a_D, a)^T$.

The local coefficient in a hypermultiplet monodromy. The phrase "Picard–Lefschetz monodromy of a massless hypermultiplet" contains two logically separate inputs. The geometric input is the Picard–Lefschetz transvection for the vanishing one-cycle of the period family. The physical normalization of that transvection is fixed by the singular Wilsonian threshold of the light charged hypermultiplet. Let $\gamma \in \Gamma$ be primitive and choose an auxiliary charge $\eta \in \Gamma$ with $\langle \eta, \gamma \rangle = 1$. In the local electric frame

$$z = Z_\gamma, \qquad z_D = Z_\eta,$$

the additional light field is an ordinary $\mathcal{N} = 2$ hypermultiplet of electric charge one. Its singular holomorphic coupling is

$$\tau_{\text{sing}}(z) = \frac{\partial z_D}{\partial z} = -\frac{\mathrm{i}}{\pi} \log z + \text{single-valued holomorphic terms.} \tag{91.4}$$

Equivalently,

$$z_D^{\text{sing}} = -\frac{\mathrm{i}}{\pi} z \log z + \text{single-valued holomorphic linear terms.}$$

The coefficient in (91.4) is the one-loop Wilsonian threshold of the two oppositely charged chiral multiplets inside a full hypermultiplet, in the Abelian normalization of this volume. A positive small loop around $z = 0$ therefore leaves z fixed and sends

$$z_D \longmapsto z_D + 2z. \tag{91.5}$$

Reversing the loop orientation reverses the sign. The factor two is not a decorative convention: it is the local threshold coefficient of a full hypermultiplet and is the same factor that appears in the finite Seiberg–Witten monodromies below.

Lemma 91.3 (Rank-one hypermultiplet monodromy). *Suppose that a primitive charge $\gamma = (n_m, n_e)$ is represented by a single massless hypermultiplet at a nodal singular fiber, and suppose the period vector is ordered as $\Pi = (a_D, a)^T$. Then the Picard-Lefschetz monodromy acting on periods is*

$$M_\gamma = \begin{pmatrix} 1 + 2n_m n_e & 2n_e^2 \\ -2n_m^2 & 1 - 2n_m n_e \end{pmatrix}. \tag{91.6}$$

Equivalently, for every test charge δ,

$$Z_\delta \longmapsto Z_\delta + 2\langle \delta, \gamma \rangle Z_\gamma. \tag{91.7}$$

The matrix M_γ is integral symplectic and unipotent.

Proof. Choose $\eta \in \Gamma$ with $\langle \eta, \gamma \rangle = 1$. Every $\delta \in \Gamma$ can be written uniquely as

$$\delta = k\eta + \ell\gamma, \qquad k = \langle \delta, \gamma \rangle,$$

because (η, γ) is an integral symplectic basis of the rank-two lattice. The local hypermultiplet calculation (91.5) gives

$$Z_\gamma \longmapsto Z_\gamma, \qquad Z_\eta \longmapsto Z_\eta + 2Z_\gamma.$$

Therefore

$$Z_\delta = kZ_\eta + \ell Z_\gamma \longmapsto k(Z_\eta + 2Z_\gamma) + \ell Z_\gamma = Z_\delta + 2\langle \delta, \gamma \rangle Z_\gamma,$$

which proves (91.7). The geometric Picard–Lefschetz theorem supplies the existence of the transvection around the vanishing cycle; the preceding Wilsonian threshold fixes its coefficient in the physical electromagnetic lattice.

Now write $\gamma = (p, q)$ in the original basis and apply the central-charge action to the two basis charges. For $\delta = (1, 0)$, $\langle \delta, \gamma \rangle = q$, hence

$$a'_D \equiv Z'_{(1,0)} = a_D + 2q(pa_D + qa) = (1 + 2pq)a_D + 2q^2 a.$$

For $\delta = (0, 1)$, $\langle \delta, \gamma \rangle = -p$, hence

$$a' \equiv Z'_{(0,1)} = a - 2p(pa_D + qa) = -2p^2 a_D + (1 - 2pq)a.$$

These two equations are exactly (91.6). The same central-charge formula implies preservation of the antisymmetric pairing:

$$\langle \delta_1 + 2\langle \delta_1, \gamma \rangle \gamma, \delta_2 + 2\langle \delta_2, \gamma \rangle \gamma \rangle = \langle \delta_1, \delta_2 \rangle,$$

because the two cross terms cancel and $\langle \gamma, \gamma \rangle = 0$. Equivalently, $M_\gamma^T J M_\gamma = J$ for

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The determinant is one, and $M_\gamma - I$ has square zero, so the monodromy is unipotent. $\square$

91.5 Pure Gauge Theory With Gauge Algebra su two

For the pure theory with gauge algebra $\mathfrak{su}(2)$, the Coulomb coordinate is

$$u = \left\langle \operatorname{tr} \phi^2 \right\rangle .$$

The Weyl group identifies $a \sim -a$, and the large-field relation is

$$u = \frac{1}{2}a^2 + O(\Lambda^4/a^2), \qquad a(u) \sim \sqrt{2u}. \tag{91.8}$$

The perturbative Wilsonian prepotential in the pure theory has the one-loop form

$$\mathcal{F}_{\text{pert}}(a) = \frac{\mathrm{i}}{2\pi} a^2 \log \frac{a^2}{\Lambda^2} + \text{terms polynomial in } a. \tag{91.9}$$

The polynomial terms change a_D by an integral affine electric redefinition and do not change the monodromy class. Thus

$$a_D = \frac{\partial \mathcal{F}}{\partial a} = \frac{\mathrm{i}}{\pi} a \log \frac{a^2}{\Lambda^2} + O(a). \tag{91.10}$$

When u circles infinity once counterclockwise, the square root in (91.8) gives $a \mapsto -a$, while the logarithm in (91.10) shifts by $2\pi\mathrm{i}$. Therefore the period vector $\Pi = (a_D, a)^T$ has monodromy

$$M_\infty = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}. \tag{91.11}$$

The anomalous $U(1)_R$ symmetry of the microscopic theory is reduced to $\mathbb{Z}_8$. In the convention used here the vector-multiplet scalar has R-charge 2, so

$$\phi \mapsto e^{2\mathrm{i}\alpha}\phi, \qquad u = \langle \operatorname{tr} \phi^2 \rangle \mapsto e^{4\mathrm{i}\alpha} u.$$

The reduction to $\mathbb{Z}_8$ is the instanton-anomaly statement in these charges. In a one-instanton background, the index theorem gives $2C_2(SU(2)) = 4$ zero modes for each adjoint Weyl gaugino. The $\mathcal{N} = 2$ vector multiplet has two such gauginos, each of R-charge 1, so the one-instanton sector acquires the phase $\exp(8\mathrm{i}\alpha)$ under the continuous $U(1)_R$ rotation. The sign convention for assigning this phase to zero-mode coordinates or to their Berezinian does not change the invariant subgroup. Thus only $\alpha = \pi k/4$, $k \in \mathbb{Z}$, remains as an exact microscopic symmetry. A generic point $u \neq 0$ is fixed by the even values of k, namely by a $\mathbb{Z}_4$ subgroup. The quotient $\mathbb{Z}_8/\mathbb{Z}_4 \simeq \mathbb{Z}_2$ therefore acts on the Coulomb coordinate by

$$u \longmapsto -u.$$

Let $S \subset \mathbb{C}_u$ be the finite singular set of the rank-one period problem. The residual exact symmetry gives $S = -S$. A single finite nodal hypermultiplet singularity is impossible in the minimal pure theory because its Picard–Lefschetz monodromy is unipotent with trace 2, whereas the weak-coupling monodromy (91.11) has trace -2. The smallest symmetry-compatible nodal set away from the fixed point $u = 0$ is therefore

$$S = \{s, -s\}.$$

Since u has mass dimension 2, s is a nonzero dimension-two multiple of the strong scale. The normalization of the strong scale in this chapter is chosen so that $s = \Lambda^2$. Thus the minimal two-singularity period problem places the finite singularities at $u = \pm\Lambda^2$. At a nodal singular fiber, a primitive charge $\gamma = (n_m, n_e)$ becomes massless and the period vector transforms by Lemma 91.3. Choosing the singularity at $u = \Lambda^2$ to be a monopole point, $\gamma_m = (1, 0)$, gives

$$M_{\Lambda^2} = M_{(1,0)} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}.$$

The second finite monodromy is fixed by

$$M_{\Lambda^2} M_{-\Lambda^2} = M_\infty,$$

hence

$$M_{-\Lambda^2} = \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix} = M_{(1,-1)}. \tag{91.12}$$

Thus the second singularity carries a massless dyon of charge $\gamma_d = (1, -1)$ in this period basis.

Minimal rank-one curve inside the two-singularity ansatz. Assume the rank-one period problem is represented by a genus-one hyperelliptic family with two fixed finite branch points and one branch point depending holomorphically on u. Assume further that the only finite singularities are exchanged by $u \mapsto -u$, occur at $\pm\Lambda^2$, and are nodal singularities, while $u = \infty$ is the weak-coupling cusp. Then, up to an affine change of the hyperelliptic coordinate and a rescaling of Λ, the minimal family is

$$\Sigma_u : \qquad y^2 = (x^2 - \Lambda^4)(x - u). \tag{91.13}$$

An affine change of x sends the two fixed finite branch points to Λ^2 and $-\Lambda^2$. A nodal singularity occurs when the moving branch point collides with one of these fixed branch points, so the moving branch point must take the values $\pm\Lambda^2$ at $u = \pm\Lambda^2$. In the minimal degree-one ansatz it is therefore $x = u$ after the same normalization of u. The fourth branch point is at infinity, giving the weak-coupling cusp. The resulting hyperelliptic equation is (91.13). Its finite branch-point discriminant is proportional to

$$(\Lambda^2 - (-\Lambda^2))^2(\Lambda^2 - u)^2(-\Lambda^2 - u)^2 = 4\Lambda^4(u^2 - \Lambda^4)^2,$$

so the only finite singular values are exactly $u = \pm\Lambda^2$. This gives uniqueness only within the stated ansatz; the QFT assertion that this ansatz captures the exact pure theory is the Seiberg-Witten status boundary.

Its branch points are

$$x = \Lambda^2, \qquad x = -\Lambda^2, \qquad x = u, \qquad x = \infty,$$

and the finite discriminant is proportional to

$$\Delta(u) = u^2 - \Lambda^4.$$

At $u = \Lambda^2$ the branch points $x = u$ and $x = \Lambda^2$ collide; at $u = -\Lambda^2$ the branch points $x = u$ and $x = -\Lambda^2$ collide. The monodromies of the corresponding vanishing cycles are exactly (91.6) with the charges above. This derivation uses the following hypotheses: the rank-one Coulomb branch has only the two finite singularities required by the residual R-symmetry, the singular fibers are nodal hypermultiplet singularities, and the weak-coupling monodromy is exhausted by the perturbative vector multiplet.

91.6 Periods, Picard-Fuchs Equation, and BPS Masses

Choose the branch of the periods in the weak-coupling region $|u| \gg |\Lambda|^2$ by

$$a(u) \sim \sqrt{2u}.$$

With the curve (91.13), the periods obey the Picard-Fuchs equation

$$(u^2 - \Lambda^4)\frac{d^2\Pi}{du^2} + \frac{1}{4}\Pi = 0, \qquad \Pi \in \{a_D, a\}. \tag{91.14}$$

The solution normalized as the electric period at infinity can be written as

$$a(u) = \sqrt{2(u + \Lambda^2)}\, {}_2F_1\left(-\frac{1}{2}, \frac{1}{2}; 1; \frac{2\Lambda^2}{u + \Lambda^2}\right), \tag{91.15}$$

with the branch chosen by (91.8). A dual period that vanishes at the monopole point is

$$a_D^{(m)}(u) = \frac{\mathrm{i}}{2\Lambda}(u - \Lambda^2)\,{}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 2; \frac{\Lambda^2 - u}{2\Lambda^2}\right). \tag{91.16}$$

The superscript records the local branch near $u = \Lambda^2$. Analytic continuation of the pair $(a_D^{(m)}, a)$ around the singularities gives the monodromy matrices above. Near the monopole point,

$$a_D^{(m)}(u) = \frac{\mathrm{i}}{2\Lambda}(u - \Lambda^2) + O((u - \Lambda^2)^2). \tag{91.17}$$

The monopole central charge is $Z_{(1,0)} = a_D$, hence

$$M_{(1,0)}(u) = \sqrt{2}\,|a_D(u)| = \frac{|u - \Lambda^2|}{\sqrt{2}|\Lambda|} + O(|u - \Lambda^2|^2). \tag{91.18}$$

At $u = -\Lambda^2$, use the local period basis obtained by transporting to that singularity. The vanishing central charge is

$$Z_{(1,-1)} = a_D - a,$$

and the same nodal-fiber analysis gives

$$a_D(u) - a(u) = c_-\,(u + \Lambda^2) + O((u + \Lambda^2)^2), \qquad c_- \neq 0, \tag{91.19}$$

on the dyon branch. Therefore

$$M_{(1,-1)}(u) = \sqrt{2}\,|a_D(u) - a(u)| = \sqrt{2}\,|c_-|\,|u + \Lambda^2| + O(|u + \Lambda^2|^2). \tag{91.20}$$

The phase and magnitude of c_- depend on the branch and symplectic basis; the linear vanishing and the charge $(1, -1)$ are invariant data of the singular fiber.

91.7 ADHM Moduli And The Nekrasov Expansion

The Seiberg-Witten curve determines the low-energy prepotential indirectly through periods. There is also a weak-coupling construction of the same instanton expansion from the ultraviolet gauge theory. This section records the construction with all convention-dependent inputs exposed.

Work in Euclidean signature on $\mathbb{R}^4 \simeq \mathbb{C}^2$, with the orientation and instanton number convention of Section 51.17. The integer

$$k = \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{tr}(F \wedge F)$$

is positive for the orientation chosen there. The instanton-counting coordinate will be denoted by q. In the common half-trace superspace coordinate, $q = \exp(2\pi \mathrm{i}\tau_{\mathrm{ht}})$. In the trace-delta component coupling coordinate of Section 51.17, the same one-instanton exponential is

$$q = \exp\left(-\frac{4\pi^2}{g_{\mathrm{YM}}^2} + \mathrm{i}\theta\right) = \exp\left(-\frac{8\pi^2}{g_{\mathrm{ht}}^2} + \mathrm{i}\theta\right).$$

The ADHM and localization formulae below use q only as the holomorphic coordinate that grades instanton number. For the pure $SU(N)$ theory, dimensional transmutation replaces q by Λ^{2N} after the renormalization scheme has been fixed.

Definition 91.4 (Framed ADHM data for $U(N)$)**.** Let $K \simeq \mathbb{C}^k$ and $W \simeq \mathbb{C}^N$. Framed ADHM data are quadruples

$$(B_1, B_2, I, J), \qquad B_1, B_2 \in \operatorname{End}(K), \quad I \in \operatorname{Hom}(W, K), \quad J \in \operatorname{Hom}(K, W),$$

obeying

$$[B_1, B_2] + IJ = 0, \tag{91.21}$$

$$[B_1, B_1^\dagger] + [B_2, B_2^\dagger] + II^\dagger - J^\dagger J = \zeta\, \mathbf{1}_K. \tag{91.22}$$

The quotient by $U(K)$ is the hyper-Kähler quotient. The parameter $\zeta > 0$ is a stability or noncommutative resolution parameter; the ordinary framed instanton moduli space is recovered at $\zeta = 0$ away from small-instanton singular strata. The framing is a trivialization at S^3_∞.

On the stable locus where the $GL(K)$ action has no stabilizer, the framed k-instanton ADHM moduli space for $U(N)$ has complex dimension

$$2kN$$

and real dimension $4kN$. Removing the center-of-mass position in $\mathbb{R}^4$ gives the centered dimension $4kN - 4$. The complex variables B_1, B_2, I, J have complex dimension

$$2k^2 + kN + kN = 2k^2 + 2kN.$$

The complex moment-map equation (91.21) is an equation in $\operatorname{End}(K)$, hence imposes k^2 complex equations on the smooth stable locus. Quotienting by the complexified gauge group $GL(K)$ removes another k^2 complex dimensions. Thus

$$(2k^2 + 2kN) - k^2 - k^2 = 2kN.$$

The real moment-map quotient by $U(K)$ is equivalent to this stable complex quotient by the Kempf–Ness correspondence. Therefore the real dimension is $2(2kN) = 4kN$. Translation of the instanton center is the trace part of B_1, B_2, a copy of $\mathbb{C}^2 \simeq \mathbb{R}^4$; fixing it removes four real dimensions.

Remark 91.5 (Gauge group and compactification). The $U(N)$ ADHM quotient is the clean equivariant model because the framing torus has N weights $a_1, \dots, a_N$. Pure $SU(N)$ instanton calculations are obtained by imposing $\sum_{\alpha=1}^{N} a_\alpha = 0$ and by discarding the decoupled abelian center in observables. Ordinary commutative $U(1)$ gauge theory has no smooth finite-action instantons on $\mathbb{R}^4$; the $U(1)$ fixed points in the sheaf or noncommutative compactification are pointlike ideal-sheaf strata. Thus the compactification or contour prescription is part of the definition of Z_{inst}, as in Definition 101.5.

Let

$$T = (\mathbb{C}^\times)^2_{\epsilon_1,\epsilon_2} \times (\mathbb{C}^\times)^N_{a_1,\dots,a_N}$$

act by rotating $\mathbb{C}^2$ and by the framing torus. The parameters ϵ_1, ϵ_2 are the equivariant weights of the two complex planes, and a_α are Coulomb weights. The equivariant deformation is called the Ω-background. It has two jobs: it makes the instanton volume integral an equivariant integral, and it isolates fixed points for localization.

Quoted Theorem 91.6 (ADHM fixed points)**.** *For the smooth framed $U(N)$ ADHM compactification on $\mathbb{C}^2$, with generic $a_\alpha, \epsilon_1, \epsilon_2$, the T-fixed points are isolated and are labeled by N-tuples of Young diagrams*

$$\vec{Y} = (Y_1, \dots, Y_N), \qquad |\vec{Y}| := \sum_{\alpha=1}^{N} |Y_\alpha| = k.$$

The equivariant integral over the k-instanton component equals the sum of the reciprocal equivariant Euler classes of the tangent complexes at these fixed points.

The quoted statement is a theorem of equivariant algebraic geometry for the framed sheaf compactification. Its use in the physical path integral additionally assumes that the supersymmetric regulator, the Ω-background, and the contour define the same equivariant integral. This physical identification is the localization input; the dimension count alone does not imply it.

Definition 91.7 (Pure vector Nekrasov partition function)**.** For pure $U(N)$ vector multiplets, define

$$Z^{U(N)}_{\text{inst}}(a, \epsilon_1, \epsilon_2; q) = \sum_{k \geq 0} q^k Z^{U(N)}_k(a, \epsilon_1, \epsilon_2), \qquad Z^{U(N)}_0 = 1,$$

where $Z^{U(N)}_k$ is the equivariant integral over the framed k-instanton compactification. The one-instanton fixed-point formula is

$$Z^{U(N)}_1 = \frac{1}{\epsilon_1\epsilon_2} \sum_{\alpha=1}^{N} \prod_{\beta\neq\alpha} \frac{1}{(a_\alpha - a_\beta)(a_\alpha - a_\beta + \epsilon_1 + \epsilon_2)}. \tag{91.23}$$

The pure $SU(N)$ specialization imposes $\sum_\alpha a_\alpha = 0$. Matter hypermultiplets multiply each fixed-point term by the equivariant Euler class of the corresponding Dirac complex; the vector formula above is the part needed for the pure $SU(2)$ Seiberg-Witten comparison.

Pure $SU(2)$ one-instanton coefficient. Set $a_1 = a$, $a_2 = -a$, and $\epsilon = \epsilon_1 + \epsilon_2$. Then

$$Z_1^{SU(2)} = \frac{2}{\epsilon_1\epsilon_2(4a^2 - \epsilon^2)}.$$

Consequently, in the Nekrasov normalization

$$\mathcal{F}_{\text{inst}}(a; q) := \lim_{\epsilon_1,\epsilon_2\to 0} \epsilon_1\epsilon_2 \log Z_{\text{inst}}^{SU(2)}(a, \epsilon_1, \epsilon_2; q),$$

the first pure-gauge term is

$$\mathcal{F}_{\text{inst}}(a; q) = \frac{q}{2a^2} + O(q^2).$$

For the pure $SU(2)$ theory, q is identified with Λ^4 in the weak-coupling scale coordinate.

Substituting $a_1 = a$, $a_2 = -a$ in (91.23) gives two fixed-point contributions:

$$Z_1^{SU(2)} = \frac{1}{\epsilon_1\epsilon_2}\left[\frac{1}{(2a)(2a+\epsilon)} + \frac{1}{(-2a)(-2a+\epsilon)}\right].$$

The second denominator is $2a(2a - \epsilon)$. Hence

$$Z_1^{SU(2)} = \frac{1}{\epsilon_1\epsilon_2}\frac{1}{2a}\left(\frac{1}{2a+\epsilon} + \frac{1}{2a-\epsilon}\right) = \frac{2}{\epsilon_1\epsilon_2(4a^2 - \epsilon^2)}.$$

Since $Z_{\text{inst}} = 1 + qZ_1 + O(q^2)$,

$$\epsilon_1\epsilon_2 \log Z_{\text{inst}} = q\,\frac{2}{4a^2 - \epsilon^2} + O(q^2).$$

Taking $\epsilon_1, \epsilon_2 \to 0$ gives $q/(2a^2)$ at order q.

The coefficient above is the first weak-coupling instanton correction to the same holomorphic prepotential whose periods are encoded by the curve (91.13), after matching the additive quadratic ambiguity in $\mathcal{F}$ and the scale coordinate. The comparison is stringent because the Seiberg-Witten curve determines all powers of Λ^4/a^4 by a period problem, while the ADHM construction determines them by finite-dimensional equivariant integrals.

The exponential size of the k-instanton sector is q^k. In a transseries analysis of the weak-coupling expansion, these actions mark the candidate locations of Borel singularities separated from the perturbative sector by integer multiples of the one-instanton action. A theorem-level resurgence statement for four-dimensional supersymmetric gauge theory would require a regulator, a summation prescription, and uniform control of the large-order perturbative coefficients. The assertion here is only the shared grading convention: the same k, action, and θ-phase grade the ADHM integral, the Ω-background partition function, and the instanton transseries sectors.

91.8 Monodromy And Light Charges

The weak-coupling region is the exterior of the curve of marginal stability on which a_D/a is real. In the charge convention fixed above, the semiclassical BPS spectrum contains the vector multiplet with charge $(0, 1)$ and a tower of hypermultiplets with charges

$$(n_m, n_e) = (1, n), \qquad n \in \mathbb{Z}, \tag{91.24}$$

together with antiparticles. The tower is generated by monodromy at infinity from either finite singular charge. The central charges are

$$Z_{(0,1)} = a, \qquad Z_{(1,n)} = a_D + na.$$

Inside the strong-coupling chamber bounded by the marginal-stability curve, the stable hypermultiplets are the two light states associated with the two singularities, with charges

$$(1,0), \qquad (1,-1),$$

and their antiparticles. The change in the BPS index occurs when

$$\arg Z_{\gamma_1} = \arg Z_{\gamma_2}$$

for charges that can form or decay as a BPS bound state. This is the wall-crossing mechanism in its rank-one Seiberg-Witten form: the central charge function is continuous, while the protected index of one-particle BPS states is locally constant only away from marginal-stability walls.

91.9 Argyres-Douglas Scaling As A Rank-One Singularity

The pure $\mathfrak{su}(2)$ theory has two separated finite singularities, so a local duality frame exists near either one and the low-energy theory contains a single light hypermultiplet. A different phenomenon occurs when two singularities with mutually nonlocal vanishing charges collide after a mass or coupling is tuned. If $\gamma_1, \gamma_2 \in \Gamma$ vanish at the same point and

$$\langle \gamma_1, \gamma_2 \rangle \neq 0,$$

then no electric-magnetic frame makes both light states purely electric. There is therefore no local Abelian Lagrangian consisting of one vector multiplet coupled to two ordinary electric hypermultiplets. The Seiberg-Witten interpretation is an interacting rank-one superconformal fixed point, usually called an Argyres-Douglas point. The existence of the four-dimensional QFT fixed point is a status-boundary statement unless a separate construction is supplied; the scaling dimensions below are consequences of the local special-geometry model.

Definition 91.8 (Rank-one cusp model)**.** The simplest rank-one Argyres-Douglas local period model is the cusp

$$\Sigma_{u,v}: \qquad y^2 = x^3 + vx + u, \tag{91.25}$$

with local Seiberg-Witten differential

$$\lambda_{\mathrm{AD}} = \frac{u\,dx}{y}. \tag{91.26}$$

The parameter u is the Coulomb-branch deformation of the cusp and v is the relevant deformation that unfolds the collision. The normalization is chosen so that a period of λ_{AD} has mass dimension one.

Discriminant of the cusp unfolding. For the local curve

$$y^2 = x^3 + vx + u,$$

the finite discriminant is

$$\Delta_{\text{cusp}}(u, v) = -4v^3 - 27u^2. \tag{91.27}$$

Thus the two nodal singularities in the u-plane collide at $(u, v) = (0, 0)$. At $v = 0$, the singular fiber is a cusp rather than a node. For a monic cubic $f(x) = x^3 + px + q$, the discriminant is

$$\operatorname{disc}(f) = \prod_{i<j}(r_i - r_j)^2, \qquad f(x) = \prod_i (x - r_i),$$

and the elementary symmetric relations give

$$\operatorname{disc}(x^3 + px + q) = -4p^3 - 27q^2.$$

Substituting $p = v$, $q = u$ gives (91.27). For $v \neq 0$, the discriminant equation has two local solutions for u, namely two nodal degenerations after a choice of branch of $v^{3/2}$. As $v \to 0$, both solutions approach $u = 0$. At $u = v = 0$ the polynomial is x^3, so all three finite branch points coincide and the singularity is cuspidal.

Mutually nonlocal light charges and electric frames. Let Γ be a rank-two electromagnetic charge lattice with symplectic pairing $\langle\cdot,\cdot\rangle$. If two primitive charges $\gamma_1, \gamma_2 \in \Gamma$ obey

$$\langle\gamma_1, \gamma_2\rangle \neq 0, \tag{91.28}$$

then no symplectic change of basis can make both charges purely electric. Consequently a neighborhood in which both central charges vanish cannot be described as a weakly coupled Abelian vector multiplet coupled only to ordinary electric hypermultiplets for both charges. In any electric-magnetic frame, the purely electric sublattice is Lagrangian: the symplectic pairing of any two purely electric charges is zero. A symplectic change of basis preserves $\langle\cdot,\cdot\rangle$. Therefore if both γ_1 and γ_2 became purely electric in some frame, their pairing would have to vanish in that frame and hence in every frame, contradicting (91.28). The last assertion follows because an ordinary local Abelian Lagrangian with hypermultiplets of electric charge uses one electric polarization of Γ; it cannot include two hypermultiplet fields whose charges have nonzero Dirac pairing in that same polarization.

For the local model (91.25) with differential (91.26), the scaling dimensions are

$$[x] = \frac{2}{5}, \qquad [y] = \frac{3}{5}, \qquad [u] = \frac{6}{5}, \qquad [v] = \frac{4}{5}. \tag{91.29}$$

Thus the Coulomb-branch operator has dimension $6/5$, and the unfolding parameter v has dimension $4/5$. Quasi-homogeneity of $y^2 = x^3 + vx + u$ gives

$$2[y] = 3[x] = [u].$$

The period of the Seiberg-Witten differential has mass dimension one, and $\lambda_{\text{AD}} = u\,dx/y$ scales with dimension $[u] + [x] - [y]$. Therefore

$$[u] + [x] - [y] = 1.$$

Solving these equations gives $[x] = 2/5$, $[y] = 3/5$, and $[u] = 6/5$. The term vx has the same dimension as u, so $[v] + [x] = 6/5$, and therefore $[v] = 4/5$.

This calculation is intentionally local. It does not claim that every collision of singular fibers produces a continuum QFT, nor does it replace the need to specify which ultraviolet theory and deformation realize the collision. What it does fix is the special-geometry consequence of a cusp with differential λ_{AD}: periods scale with dimension one, the Coulomb operator can have fractional dimension, and mutually nonlocal light charges force the infrared description outside an ordinary weakly coupled Abelian hypermultiplet frame.

91.10 Low-Energy Abelian Theory Near A Singular Point

Near a point where the monopole period a_D vanishes, the correct low-energy variables are the dual $U(1)$ vector multiplet with scalar a_D and the light monopole hypermultiplet. A local Wilsonian superpotential has the form

$$W_{\mathrm{loc}} = \sqrt{2}\, A_D M \widetilde{M},$$

where A_D is the dual chiral multiplet and $M, \widetilde{M}$ are monopole chiral multiplets of opposite dual electric charge. Let $U(A_D)$ denote the chiral coordinate whose lowest component is the Coulomb-branch coordinate u expressed in the local dual frame. Add a small $\mathcal{N} = 1$-preserving deformation

$$W_{\mathrm{def}} = \mu\, U(A_D), \qquad \mu \in \mathbb{C}, \qquad |\mu| \ll |\Lambda|, \tag{91.30}$$

so that the local superpotential is

$$W_{\mathrm{loc},\mu} = \sqrt{2}\, A_D M \widetilde{M} + \mu\, U(A_D). \tag{91.31}$$

The smallness assumption keeps the analysis inside the Abelian neighborhood where the monopole hypermultiplet is the only additional light charged field.

Monopole condensation from the local F-terms. Assume $U'(0) \neq 0$ at the monopole point $A_D = 0$, and work on the local patch described by (91.31). The F-term equations have a branch

$$A_D = 0, \qquad M\widetilde{M} = -\frac{\mu}{\sqrt{2}}\, U'(0). \tag{91.32}$$

When the $U(1)_D$ D-term equation is imposed with vanishing real Fayet–Iliopoulos parameter,

$$|M|^2 - |\widetilde{M}|^2 = 0, \tag{91.33}$$

the dual gauge group is Higgsed for $\mu U'(0) \neq 0$.

The derivatives of (91.31) are

$$\frac{\partial W_{\mathrm{loc},\mu}}{\partial M} = \sqrt{2}\, A_D \widetilde{M}, \qquad \frac{\partial W_{\mathrm{loc},\mu}}{\partial \widetilde{M}} = \sqrt{2}\, A_D M,$$

and

$$\frac{\partial W_{\mathrm{loc},\mu}}{\partial A_D} = \sqrt{2}\, M\widetilde{M} + \mu\, U'(A_D).$$

The first two equations are solved by $A_D = 0$. Substituting this value in the third equation gives (91.32). If $U'(0)\mu \neq 0$, then $M\widetilde{M} \neq 0$. Together with (91.33), this implies $|M| = |\widetilde{M}| \neq 0$, and the oppositely charged monopole fields Higgs the dual $U(1)_D$.

In the dual Abelian description, a Wilson line of the original electric charge is magnetically charged with respect to $U(1)_D$. The Higgsed dual gauge theory therefore supports Abrikosov–Nielsen–Olesen flux tubes whose magnetic flux is the original electric flux. This gives the local confinement mechanism used in the comparison between Donaldson and Seiberg–Witten theories.

Definition 91.9 (Local SW confining-string model)**.** In the neighborhood of a monopole singularity, a local confining-string model consists of:

(i) the dual Abelian gauge field $A_{D,\mu}$ with curvature F_D and coupling e_D;

(ii) monopole fields $M, \widetilde{M}$ of dual charges $+1$ and -1;

(iii) the complex parameter
$$\xi_{\mathbb{C}} = -\frac{\mu}{\sqrt{2}}U'(0), \tag{91.34}$$
defined by the local F-term equation $M\widetilde{M} = \xi_{\mathbb{C}}$;

(iv) a flux sector $n \in \mathbb{Z}$ with
$$\frac{1}{2\pi}\int_{\mathbb{R}^2} F_D = n$$
on the plane transverse to the string;

(v) boundary conditions at transverse infinity obeying $M\widetilde{M} \to \xi_{\mathbb{C}}$ and $|M| = |\widetilde{M}|$.

The string tension is the infimum of the static energy per unit length in the fixed flux sector.

Controlled Approximation 91.10 (BPS vortex tension in the local Abelian model)**.** If the local theory is approximated by an $\mathcal{N} = 2$ Abelian Higgs model with Fayet–Iliopoulos triplet of magnitude $|\xi| = |\xi_{\mathbb{C}}|$ in the normalization
$$\mathcal{E} = \frac{1}{2e_D^2}B_D^2 + |D_1 q|^2 + |D_2 q|^2 + \frac{e_D^2}{2}(|q|^2 - |\xi|)^2, \tag{91.35}$$
then a vortex of flux $2\pi n$ obeys
$$T_n \geq 2\pi |n|\,|\xi|, \tag{91.36}$$
and saturation is equivalent to the first-order vortex equations
$$(D_1 + \mathrm{i}\,\mathrm{sgn}(n) D_2)q = 0, \qquad B_D = \mathrm{sgn}(n) e_D^2(|\xi| - |q|^2). \tag{91.37}$$

Derivation of the local vortex bound. For $n > 0$, complete the square in (91.35):
$$\mathcal{E} = |(D_1 + \mathrm{i}D_2)q|^2 + \frac{1}{2e_D^2}\big(B_D - e_D^2(|\xi| - |q|^2)\big)^2 + |\xi| B_D + \partial_i J^i.$$
The current J^i is a total derivative whose integral vanishes for the finite-energy boundary conditions in Definition 91.9. Integrating over the transverse plane gives
$$T_n \geq |\xi| \int_{\mathbb{R}^2} F_D = 2\pi n |\xi|.$$
For $n < 0$, use the opposite square completion. The two square terms vanish precisely when (91.37) holds.

91.11 Status Boundary

The period construction determines an Abelian Wilsonian effective action under explicit hypotheses: existence of the $\mathcal{N} = 2$ continuum theory, existence of the Coulomb-branch vacuum family, validity of the two-derivative Abelian description away from singular loci, and BPS saturation for the states assigned to vanishing cycles. The missing nonperturbative regulator for four-dimensional supersymmetric gauge theory remains part of the Wilsonian-scheme problem recorded in Chapter 90.

Open Problem 91.11 (Theorem-level SW construction)*.* Construct the four-dimensional $\mathcal{N} = 2$ gauge theory with a regulator and continuum limit sufficient to derive the Coulomb-branch special geometry, period data, BPS spectrum near singular fibers, and low-energy Abelian Wilsonian action as theorems of QFT.

Chapter 92

Spectral Bridges Among Supersymmetric Yang–Mills Cousins

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The preceding chapters developed three separate structures: pure $\mathcal{N} = 1$ Yang–Mills chiral dynamics, pure $\mathcal{N} = 2$ Seiberg–Witten special geometry, and planar $\mathcal{N} = 4$ spectral integrability. They are connected by relevant deformations, but the connection has several layers. Holomorphic quantities can often be matched by scale coordinates and protected Ward identities. Massive spectra, confining strings, and line-operator Hilbert spaces require additional nonperturbative data. This chapter defines the comparison problem.

92.1 Comparison Problem

Let $G = SU(N_c)$ with $N_c \geq 2$. A comparison among the $\mathcal{N} = 4, \mathcal{N} = 2, \mathcal{N} = 1$, and bosonic Yang–Mills theories is a comparison among four QFT problems, not a single family of Lagrangian symbols. The amount of supersymmetry determines which quantities are protected, which deformations preserve a chiral coordinate chart, and which spectral questions remain dynamical.

Definition 92.1 (Yang–Mills cousin comparison problem)**.** A Yang–Mills cousin comparison problem is specified by:

(i) a gauge group $G = SU(N_c)$ or a specified global form with a specified line-operator lattice;

(ii) a UV description of the $\mathcal{N}=4$ theory in the $\mathcal{N}=1$ field variables

$$V, \qquad \Phi_1, \Phi_2, \Phi_3 \in \operatorname{ad} G,$$

with holomorphic coupling

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi\mathrm{i}}{g_{\mathrm{YM}}^2}$$

in the trace convention fixed in Chapter 87;

(iii) a finite list of relevant deformation parameters, for example adjoint chiral masses m_i, an $\mathcal{N}=2 \to \mathcal{N}=1$ mass coordinate μ, and a soft gaugino-mass spurion m_λ;

(iv) for each deformation regime, a declared set of observables: gauge-invariant local operators, center-charged line operators, domain-wall boundary conditions, and defect operators on Wilson lines;

(v) a status label for each comparison: holomorphic scale matching, protected F-term matching, semiclassical Abelian approximation, lattice spectral extraction, planar-integrability reconstruction, or open nonperturbative construction.

Two rows of the comparison are compared only after the observable type and status label have been fixed.

The basic ladder is:

$$\begin{aligned}
\mathcal{N}=4 &\quad \xrightarrow{m_{\text{hyper}}} \quad \mathcal{N}=2^* \xrightarrow{E\ll|m_{\text{hyper}}|} \text{pure } \mathcal{N}=2,\\
\mathcal{N}=4 &\quad \xrightarrow{m_1,m_2,m_3} \quad \mathcal{N}=1^* \xrightarrow{E\ll\min_i|m_i|} \text{pure } \mathcal{N}=1,\\
\text{pure } \mathcal{N}=2 &\quad \xrightarrow{\mu\, u} \quad \text{softly broken } \mathcal{N}=1 \text{ with Abelian confinement},\\
\text{pure } \mathcal{N}=1 &\quad \xrightarrow{X_\lambda S+\text{c.c.}} \quad \text{bosonic Yang–Mills deformation problem.}
\end{aligned}$$

The arrows are deformation regimes. A continuum construction for the endpoint theory, or a proof that a path of deformations reaches a chosen endpoint without crossing a phase boundary in the observables of interest, is extra information.

92.2 Holomorphic Scale Matching

Holomorphic scale matching is the cleanest part of the comparison. It is a statement about chiral source coordinates and threshold matching in a holomorphic Wilsonian scheme. It does not compute the nonchiral spectrum.

Definition 92.2 (Holomorphic threshold matching)**.** Let a four-dimensional supersymmetric gauge theory with gauge algebra $\mathfrak{su}(N_c)$ have holomorphic one-loop coefficient b_{high} above a chiral mass threshold m and coefficient b_{low} below the threshold. A holomorphic threshold matching at the scale m is the identity of chiral scale coordinates

$$\Lambda_{\text{low}}^{b_{\text{low}}} = m^{\,b_{\text{low}}-b_{\text{high}}}\,\Lambda_{\text{high}}^{b_{\text{high}}}, \tag{92.1}$$

up to finite holomorphic scheme rescalings already included in the definitions of m and Λ.

Write $\mathcal{N} = 4$ Yang–Mills in $\mathcal{N} = 2$ variables as one $\mathcal{N} = 2$ vector multiplet plus one adjoint hypermultiplet. Give the hypermultiplet a holomorphic mass m_{h}. In the region

$$E \ll |m_{\mathrm{h}}|,$$

the remaining pure $\mathcal{N} = 2$ holomorphic scale satisfies

$$\Lambda_{\mathcal{N}=2}^{2N_c} = m_{\mathrm{h}}^{2N_c}\, q, \qquad q = \mathrm{e}^{2\pi \mathrm{i}\tau}. \tag{92.2}$$

The $\mathcal{N} = 4$ theory has vanishing holomorphic beta coefficient, so the high-energy scale coordinate is the dimensionless instanton factor $q = \mathrm{e}^{2\pi\mathrm{i}\tau}$. Pure $\mathcal{N} = 2$ $SU(N_c)$ Yang–Mills has holomorphic coefficient

$$b_{\mathcal{N}=2} = 2N_c.$$

Applying (92.1) with

$$b_{\mathrm{high}} = 0, \qquad b_{\mathrm{low}} = 2N_c, \qquad m = m_{\mathrm{h}}$$

gives (92.2). The dimension of the right-hand side is $2N_c$, matching the dimension of $\Lambda_{\mathcal{N}=2}^{2N_c}$.

Write $\mathcal{N} = 4$ Yang–Mills in $\mathcal{N} = 1$ variables as one vector multiplet and three adjoint chiral multiplets Φ_1, Φ_2, Φ_3. Give the three adjoint chiral multiplets holomorphic masses m_1, m_2, m_3. In the region

$$E \ll \min_i |m_i|,$$

the pure $\mathcal{N} = 1$ holomorphic scale satisfies

$$\Lambda_{\mathcal{N}=1}^{3N_c} = (m_1 m_2 m_3)^{N_c}\, q, \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau}. \tag{92.3}$$

The high-energy $\mathcal{N} = 4$ coefficient is zero. Pure $\mathcal{N} = 1$ $SU(N_c)$ Yang–Mills has coefficient

$$b_{\mathcal{N}=1} = 3N_c.$$

Each massive adjoint chiral multiplet changes the holomorphic coefficient by $T(\mathrm{ad}) = N_c$. Applying (92.1) successively at the three mass thresholds gives

$$\Lambda_{\mathcal{N}=1}^{3N_c} = m_1^{N_c} m_2^{N_c} m_3^{N_c} q,$$

which is (92.3). The mass dimension is $3N_c$, as required for the pure $\mathcal{N} = 1$ scale coordinate.

Example 92.3 ($\mathcal{N} = 1^*$ chiral critical equations)**.** In an $\mathcal{N} = 1$ presentation, the equal-mass $\mathcal{N} = 1^*$ superpotential is

$$W = \sqrt{2}\ \mathrm{tr}\, \Phi_1[\Phi_2, \Phi_3] + \frac{m}{2} \sum_{i=1}^{3} \mathrm{tr}\, \Phi_i^2. \tag{92.4}$$

The F-term equations are

$$\sqrt{2}[\Phi_2, \Phi_3] + m\Phi_1 = 0, \qquad \sqrt{2}[\Phi_3, \Phi_1] + m\Phi_2 = 0, \qquad \sqrt{2}[\Phi_1, \Phi_2] + m\Phi_3 = 0. \tag{92.5}$$

If J_i are matrices obeying

$$[J_i, J_j] = \mathrm{i}\epsilon_{ijk} J_k, \tag{92.6}$$

then

$$\Phi_i = \frac{\mathrm{i}m}{\sqrt{2}}\, J_i \tag{92.7}$$

solves (92.5). Such solutions organize chiral vacua of the massive adjoint theory by embeddings of $\mathfrak{su}(2)$ into $\mathfrak{su}(N_c)$. The full massive spectrum and confinement data also depend on the nonchiral dynamics and on the chosen line-operator lattice.

92.3 What Is Uniform Across the Ladder

The uniform part of the ladder is a map of observables, not a universal numerical formula. Four classes appear repeatedly.

Definition 92.4 (Family observable comparison table)**.** For a Yang–Mills cousin comparison problem, the repeated observable classes are the following.

observable type	**representative object**	**status**
chiral local	$S,\ u,\ \operatorname{tr}\Phi^k$	holomorphic or protected
massive local spectral	$\rho_{ij}(B)$ for gauge-invariant sources	spectral measure; lattice extraction
center-charged line	$W_R(C),\ H_m(C)$, flux-sector transfer matrix	declared line-sector problem
defect spectral	$\mathcal{H}_{\rm def},\ H_{\rm def}, \mathbb{O}_a(t)$	Wilson-line or cusp spectral problem

where S is the pure $\mathcal{N}=1$ glueball coordinate of Section 90.5, u is the rank-one $\mathcal{N}=2$ Coulomb coordinate of Chapter 91, ρ_{ij} is the spectral measure of Definition 90.3, $W_R(C)$ is a Wilson line in representation R, $H_m(C)$ is a magnetic line with magnetic charge m, and $\mathcal{H}_{\rm def}$ is the defect Hilbert space of Definition 100.17.

Controlled Approximation 92.5 (Observable-type comparison across a controlled threshold)**.** Consider a relevant mass deformation with threshold M between two supersymmetric Yang–Mills cousins. The following comparison is meaningful only after the threshold problem is supplied with:

(i) a Wilsonian description exists above and below M in a common regulator or in compatible holomorphic schemes;

(ii) the local source under consideration has external momenta $|p_a| \ll |M|$;

(iii) every massive field integrated out has a gap bounded below by $c|M|$ for some $c > 0$ in the chosen background;

(iv) the line-operator sector is kept fixed by declaring the global form and allowed electric and magnetic charges before the deformation.

Under these hypotheses, chiral local sources match by holomorphic threshold coordinates, local nonchiral correlators match after adding local higher-dimension Wilsonian operators, and center-charge labels of line operators are transported as declared labels. The spectrum is a separate dynamical problem: numerical masses, string tensions, and line-defect excitation energies are functions of the deformation parameters and require the corresponding Hilbert-space, transfer-matrix, or line-defect spectral analysis.

For chiral local sources, the supersymmetric Wilsonian action separates the holomorphic F-terms from D-terms. The threshold contribution of a massive chiral multiplet to the holomorphic gauge coupling is one-loop exact in the Wilsonian holomorphic coordinate, giving the scale matching of Definition 92.2. Thus the chiral coordinates match by finite holomorphic maps such as (92.2) and (92.3).

For nonchiral local correlators at momenta $|p_a| \ll |M|$, the massive propagators admit an expansion in powers of p/M after the regulator and subtraction scheme have been fixed. The coefficients are local functionals of the light fields and sources, hence enter the Wilsonian action as higher-dimension local operators. This is an equality of low-momentum Wilsonian coordinates at fixed order in the derivative expansion, with a remainder controlled only when the chosen regulator supplies the corresponding analytic estimates.

Line operators require the global form and charge lattice as part of the definition. A mass threshold changes the dynamics of screening and confinement, but it does not change the declared lattice of allowed line charges. Therefore the center or charge label can be transported through the comparison problem. The area-law coefficient or defect excitation energy is obtained from the Hilbert-space or transfer-matrix problem in the deformed theory, so it remains a dynamical spectral quantity.

92.4 Center Strings and k-String Tension Comparisons

For $SU(N_c)$ with the usual Wilson lines, the center charge of a representation is an element of $\mathbb{Z}_{N_c}$. A confining theory may have stable flux sectors labeled by

$$k \in \{0, 1, \ldots, N_c - 1\},$$

with k and $N_c - k$ related by charge conjugation.

Definition 92.6 (k-string tension extraction)**.** A k-string tension extraction for a confining Yang–Mills theory is specified by:

(i) a regulator or continuum Hilbert-space construction with spatial transfer matrix T_L on a box whose long direction has length L;

(ii) a center-flux sector $k \in \mathbb{Z}_{N_c}$;

(iii) the lowest energy $E_k(L)$ in that sector after the vacuum energy is subtracted;

(iv) a limit

$$T_k = \lim_{L\to\infty} \frac{E_k(L)}{L} \tag{92.8}$$

whenever the limit exists.

Equivalently, in an area-law regime for a large rectangular Wilson loop in a representation of N-ality k,

$$\langle W_k(C)\rangle = \exp\bigl(-T_k \ \mathrm{Area}(C) + o(\mathrm{Area}(C))\bigr) \tag{92.9}$$

defines the same tension if the transfer-matrix and Wilson-loop formulations have been proved equivalent for the regulator under consideration.

Two comparison formulae are useful reference functions:

$$R_k^{\mathrm{sine}}(N_c) = \frac{\sin(\pi k/N_c)}{\sin(\pi/N_c)}, \qquad R_k^{\mathrm{Casimir}}(N_c) = \frac{k(N_c - k)}{N_c - 1}. \tag{92.10}$$

The first appears in Abelianized supersymmetric regimes and in several string-inspired large-N_c calculations. The second is a quadratic color-factor comparison coordinate. In this chapter they are comparison functions until a specified QFT framework supplies a derivation of a spectral tension ratio.

For $1 \le k \le N_c - 1$,

$$R_k^{\text{sine}}(N_c) = R_{N_c-k}^{\text{sine}}(N_c), \qquad R_k^{\text{Casimir}}(N_c) = R_{N_c-k}^{\text{Casimir}}(N_c). \tag{92.11}$$

At fixed k and large N_c,

$$R_k^{\text{sine}}(N_c) = k - \frac{\pi^2}{6}\frac{k(k^2-1)}{N_c^2} + O(N_c^{-4}), \tag{92.12}$$

while

$$R_k^{\text{Casimir}}(N_c) = k - \frac{k(k-1)}{N_c} + O(N_c^{-2}). \tag{92.13}$$

These are finite comparison identities. The symmetry follows from

$$\sin\left(\frac{\pi(N_c-k)}{N_c}\right) = \sin\left(\pi - \frac{\pi k}{N_c}\right) = \sin\left(\frac{\pi k}{N_c}\right)$$

and from

$$(N_c - k)k = k(N_c - k).$$

The sine-law expansion is obtained by dividing

$$\sin\frac{\pi k}{N_c} = \frac{\pi k}{N_c} - \frac{\pi^3 k^3}{6N_c^3} + O(N_c^{-5}), \qquad \sin\frac{\pi}{N_c} = \frac{\pi}{N_c} - \frac{\pi^3}{6N_c^3} + O(N_c^{-5}).$$

This gives

$$R_k^{\text{sine}} = k\left(1 - \frac{\pi^2 k^2}{6N_c^2}\right)\left(1 + \frac{\pi^2}{6N_c^2}\right) + O(N_c^{-4}),$$

which is (92.12). For the Casimir comparison,

$$\frac{k(N_c-k)}{N_c-1} = k\frac{1-k/N_c}{1-1/N_c} = k\left(1 - \frac{k}{N_c}\right)\left(1 + \frac{1}{N_c} + O(N_c^{-2})\right),$$

which gives (92.13). The physics lies in whether a specified gauge theory and regulator produce one of these finite functions as a spectral ratio; the algebra above supplies only comparison coordinates.

92.5 Entries of the Family

92.5.1 Pure Bosonic Yang–Mills

Pure bosonic Yang–Mills is the endpoint with the fewest protected observables and the strongest need for a nonperturbative construction. Its glueball and flux-tube spectra are formulated in Section 46.8. In the family comparison, the bosonic row contains:

object	role
OS-positive lattice gauge measure	candidate regulator and transfer matrix
local operators $\operatorname{tr} F^2, \operatorname{tr} F\widetilde{F}, \ldots$	J^{PC} glueball sources
center-flux sectors	k-string tensions and excited flux tubes
large-N_c expansion	$M_{\text{glueball}} = O(1), \quad \Gamma_{\text{glueball}} = O(N_c^{-2})$

The bosonic theory supplies the target spectral questions: a mass gap, isolated glueball poles, stable flux sectors, and an effective string description in the long-string regime.

92.5.2 Pure $\mathcal{N} = 1$ Yang–Mills

Pure $\mathcal{N} = 1$ Yang–Mills adds protected chiral branch information. The local coordinate

$$S = -\frac{1}{32\pi^2}\,\mathrm{tr}\, W^\alpha W_\alpha$$

has branch values

$$\langle S\rangle_k = C\Lambda_h^3 \mathrm{e}^{2\pi \mathrm{i}k/N_c}.$$

The family comparison separates three outputs:

output	source
N_c branch phases	$\mathbb{Z}_{2N_c}$ anomaly and chiral F-term
$T_{ij} = 2\lvert W_i - W_j\rvert$	$\mathcal{N} = 1$ BPS wall algebra
glueball/gluinoball masses	spectral measure $\rho_{ij}^{(k)}$

The first two are protected after the stated hypotheses of Chapter 90. The third is a massive spectral problem involving the scalar, pseudoscalar, and fermionic components of gauge-invariant superfields. Supersymmetry organizes isolated massive states into multiplets when the vacuum preserves the supercharges, but it does not supply their numerical masses.

Definition 92.7 (Pure-SYM channel source table)**.** Fix a massive branch k of pure $\mathcal{N} = 1$ $SU(N_c)$ Yang–Mills and the vacuum spectral measure of Definition 90.3. Let

$$S = -\frac{1}{32\pi^2}\,\mathrm{tr}\, W^\alpha W_\alpha$$

be the chiral coordinate used in Section 90.6. The elementary source table attached to this superfield is:

channel	**source family**	**spectral role**
bosonic even scalar	$\mathrm{Re}\, S\rvert_{\theta=0}$, $\mathrm{Re}\, D^2S\rvert$, and same-quantum-number descendants	overlaps with 0^{++}-type atoms after operator mixing is fixed
bosonic odd scalar	$\mathrm{Im}\, S\rvert_{\theta=0}$, $\mathrm{Im}\, D^2S\rvert$, and same-quantum-number descendants	overlaps with 0^{-+}-type atoms after operator mixing is fixed
fermionic	$D_\alpha S\rvert$, $\overline{D}_{\dot\alpha}\overline{S}\rvert$, and derivative descendants	overlaps with spin-$\frac{1}{2}$ atoms in the same massive multiplets

Here the vertical bar denotes restriction to $\theta = \overline{\theta} = 0$ after the chosen superspace operator has been renormalized. The table lists source families, not particles: the spectral measure may contain several isolated atoms and continuum components in any row, and the same atom can have nonzero residue for several sources with the same exact quantum numbers. Center-charged Wilson or magnetic lines belong to a separate line-sector problem; they define flux-string sectors rather than chiral-particle masses.

Mass degeneracy inside an isolated pure-SYM multiplet. Assume the vacuum branch k carries unbroken $\mathcal{N} = 1$ supercharges $Q_\alpha, \overline{Q}_{\dot\alpha}$ on the Hilbert space of Definition 90.3. Let $E_k(\Sigma_m)$ be the spectral projection onto an isolated positive-mass shell $\Sigma_m = \{p : p^2 = -m^2,\ p^0 > 0\}$, separated from the rest of the joint translation spectrum in a neighborhood invariant under the little group. Then the range of $E_k(\Sigma_m)$ is invariant under the supercharges. In particular, whenever a bosonic or fermionic source in Definition 92.7 has nonzero overlap with this atom, the complete supermultiplet generated by the supercharges has the same mass m. The supersymmetry algebra gives

$$[P_\mu, Q_\alpha] = [P_\mu, \overline{Q}_{\dot\alpha}] = 0.$$

Therefore the mass operator

$$M_k^2 = -\eta^{\mu\nu} P_\mu^{(k)} P_\nu^{(k)}$$

commutes with the supercharges on their common invariant domain. Functional calculus then gives

$$Q_\alpha E_k(B) = E_k(B) Q_\alpha, \qquad \overline{Q}_{\dot\alpha} E_k(B) = E_k(B) \overline{Q}_{\dot\alpha}$$

for every Borel set B of the joint spectrum for which the products are defined by closure; applying this to the isolated shell gives invariance of $\operatorname{Ran} E_k(\Sigma_m)$. Acting with the supercharges changes spin and fermion number but not P^2, so all nonzero states in the generated irreducible massive supermultiplet have the same mass. The statement says nothing about the numerical value of m or about whether the atom exists; those are spectral questions in the chosen nonperturbative construction.

Definition 92.8 (Regulated pure-SYM spectral extraction)**.** A regulated extraction of the pure $\mathcal{N} = 1$ SYM spectrum consists of:

(i) a gauge-field regulator and an adjoint Majorana-fermion regulator with a specified Pfaffian or determinant prescription;

(ii) a parameter $m_{\lambda,\text{reg}}$ whose tuned value defines the massless-gluino trajectory, together with the Ward identity or equivalent renormalized condition used to tune it;

(iii) a finite-volume transfer matrix or Euclidean reflection-positive substitute sufficient to define spectral estimates;

(iv) source matrices in the exact finite-regulator symmetry channels, with the continuum J^{PC} labels assigned only after the rotation and discrete-symmetry restoration limits have been taken;

(v) a limit prescription $a \to 0$, $L \to \infty$, and $m_{\lambda,\text{reg}} \to m_{\lambda,*}$, including the order of limits and the systematic error controls used in the extrapolation.

This regulated extraction is the supersymmetric analogue of the pure-glue construction in Section 46.8. It is not replaced by the Veneziano–Yankielowicz superpotential: the latter supplies protected branch information, while the spectral extraction is what can define numerical glueball, gluinoball, and mixed supermultiplet masses.

Definition 92.9 (Channel-by-channel pure-SYM correlator matrices)**.** Fix the regulated extraction of Definition 92.8. Choose zero-spatial-momentum finite-volume sources

$$B_{+,r}, \qquad B_{-,r}, \qquad F_{\alpha,r}, \qquad r = 1, \dots, R,$$

where B_+ has the quantum numbers of the even scalar row of Definition 92.7, B_- has the odd-scalar row, and F_α has the fermionic row. With the vacuum projection removed, define the Euclidean transfer-matrix correlator matrices

$$C_+^{rs}(t) = \langle\Omega|B_{+,r}\,\mathrm{e}^{-tH_{\mathrm{reg}}}\,B_{+,s}^\dagger|\Omega\rangle_c, \tag{92.14}$$

$$C_-^{rs}(t) = \langle\Omega|B_{-,r}\,\mathrm{e}^{-tH_{\mathrm{reg}}}\,B_{-,s}^\dagger|\Omega\rangle_c, \tag{92.15}$$

$$C_F^{rs}(t) = \frac{1}{2}\,\mathrm{tr}_{\mathrm{rest}}\langle\Omega|F_r\,\mathrm{e}^{-tH_{\mathrm{reg}}}\,\overline{F}_s|\Omega\rangle_c. \tag{92.16}$$

The trace in (92.16) is the declared spin average in the rest frame; at finite lattice spacing it must be replaced by the exact spin-little-group projection of the regulator. The channel matrices are not themselves a numerical result. They specify the finite-regulator spectral data whose leading isolated pole positions may be compared across the even scalar, odd scalar, and fermionic channels. The method used to estimate those pole positions is external to the present chapter; the pure-SYM content is the choice of channels, sources, spin/rest-frame projections, gluino-mass tuning condition, and joint continuum, thermodynamic, and massless-gluino limiting procedure. A pure-SYM spectral extraction has restored supersymmetry in a one-particle window only after that limiting procedure makes the pole locations extracted from the three channels agree within a stated continuum and infinite-volume error estimate.

Operationally, the pure-SYM lattice or Hamiltonian analysis should publish, for each candidate multiplet, the three leading masses

$$m_+, \qquad m_-, \qquad m_F$$

with the regulator parameters, source bases, fit windows, finite-volume trajectory, and tuning condition for the gluino mass. The supersymmetry restoration diagnostic is not the equality of two names in a table; it is the vanishing of

$$\delta_{\mathrm{SUSY}} = \frac{\max\{|m_+ - m_-|, |m_+ - m_F|, |m_- - m_F|\}}{\frac{1}{3}(m_+ + m_- + m_F)} \tag{92.17}$$

in the joint continuum, infinite-volume, and massless-gluino limit, with the limiting procedure stated as part of Definition 92.8.

92.5.3 Pure $\mathcal{N}=2$ and Softly Broken $\mathcal{N}=2$

Pure $\mathcal{N}=2$ Yang–Mills has Coulomb-branch coordinate u, period vector (a_D, a), and BPS central charge

$$Z_\gamma = n_e a + n_m a_D.$$

At a monopole singularity, a soft $\mathcal{N}=1$ deformation

$$W_{\mathrm{loc},\mu} = \sqrt{2}\,A_D M\widetilde{M} + \mu\,U(A_D)$$

gives

$$A_D = 0, \qquad M\widetilde{M} = -\frac{\mu}{\sqrt{2}}U'(0).$$

The dual $U(1)_D$ is Higgsed, and the local Abelian model supports flux tubes carrying the original electric flux. The Seiberg–Witten row therefore contains a controlled Abelian confinement mechanism near the singularity. The detailed non-Abelian k-string spectrum away from the local Abelian regime is a separate spectral problem.

Controlled Approximation 92.10 (Radial BPS string profile near a monopole point)**.** Use the local Abelian-Higgs normalization of (91.35), take $n > 0$, and choose the orientation for which

$$(D_1 + \mathrm{i}D_2)q = 0, \qquad B_D = e_D^2(|\xi| - |q|^2). \tag{92.18}$$

With polar coordinate (r, φ) on the plane transverse to the string, set

$$q(r, \varphi) = |\xi|^{1/2} f(R)\mathrm{e}^{\mathrm{i}n\varphi}, \qquad A_D = n\, a(R)\, \mathrm{d}\varphi, \qquad R = e_D|\xi|^{1/2} r. \tag{92.19}$$

Finite energy and regularity impose

$$f(0) = 0, \qquad a(0) = 0, \qquad f(\infty) = 1, \qquad a(\infty) = 1. \tag{92.20}$$

The first-order equations reduce to

$$\frac{\mathrm{d}f}{\mathrm{d}R} = \frac{n}{R}(1 - a)f, \qquad \frac{\mathrm{d}a}{\mathrm{d}R} = \frac{R}{n}(1 - f^2). \tag{92.21}$$

The flux and BPS tension are

$$\int_{\mathbb{R}^2} F_D = 2\pi n, \qquad T_n = 2\pi n|\xi|. \tag{92.22}$$

Derivation of the radial vortex equations. For the ansatz (92.19),

$$D_r q = |\xi|^{1/2} e_D |\xi|^{1/2} f'(R)\mathrm{e}^{\mathrm{i}n\varphi}, \qquad D_\varphi q = \mathrm{i}n(1 - a)|\xi|^{1/2} f(R)\mathrm{e}^{\mathrm{i}n\varphi}.$$

Since $D_1 + \mathrm{i}D_2 = \mathrm{e}^{\mathrm{i}\varphi}(D_r + \mathrm{i}r^{-1}D_\varphi)$, the first BPS equation gives

$$e_D|\xi|^{1/2} f'(R) - \frac{n}{r}(1 - a)f(R) = 0,$$

which is the first equation in (92.21). The curvature of the one-form $A_D = n\, a(R)\mathrm{d}\varphi$ is

$$F_D = n\, a'(R)\mathrm{d}R \wedge \mathrm{d}\varphi, \qquad B_D = \frac{n}{r} e_D|\xi|^{1/2} a'(R).$$

Substituting $r = R/(e_D|\xi|^{1/2})$ into $B_D = e_D^2|\xi|(1 - f^2)$ gives the second radial equation. Finally,

$$\int_{\mathbb{R}^2} F_D = \int_0^{2\pi} \mathrm{d}\varphi \int_0^\infty n\, a'(R)\mathrm{d}R = 2\pi n\,[a(\infty) - a(0)] = 2\pi n.$$

The tension formula is then the saturated bound (91.36) with positive orientation. The derivation is local in the Seiberg–Witten patch; it does not assert that a full non-Abelian k-string spectrum is determined by the Abelian profile.

92.5.4 Abelianized k-String Comparison

The local $U(1)_D$ vortex just derived is the rank-one building block. For $SU(N_c)$ with $N_c > 2$, the softly broken $\mathcal{N} = 2$ theory has singular loci where several mutually local magnetic states become light and the low-energy magnetic gauge group is an Abelian torus. The string calculation in such a region is an Abelianized calculation. Its output is a precise low-energy spectral problem; comparison with the non-Abelian pure-Yang–Mills k-string spectrum requires an additional deformation argument.

Let K_{ij}, $1 \leq i, j \leq N_c - 1$, be the A_{N_c-1} Cartan matrix

$$K_{ij} = 2\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j}. \tag{92.23}$$

The simple magnetic fields of an Abelianized Seiberg–Witten patch are labelled by these simple roots. A soft $\mathcal{N} = 1$ deformation supplies complex Fayet–Iliopoulos coordinates ξ_i for the corresponding local Abelian Higgs factors after a magnetic duality frame has been chosen. When the phases of the ξ_i are aligned, the BPS square completion of the rank-one calculation applies independently in each diagonalized magnetic charge sector.

Definition 92.11 (Abelianized A_{N_c-1} string construction)**.** An Abelianized A_{N_c-1} string construction is specified by:

(i) a magnetic low-energy patch with gauge group $U(1)^{N_c-1}$, gauge fields $A_{D,i}$, and mutually local charged monopole hypermultiplets;

(ii) complex FI coordinates ξ_i, $i = 1, \ldots, N_c - 1$, with a declared common phase, so that a BPS bound may be written in a single real central charge direction;

(iii) flux integers $n_i \in \mathbb{Z}$ with
$$\frac{1}{2\pi} \int_{\mathbb{R}^2} F_{D,i} = n_i;$$

(iv) a map from the Abelian magnetic flux vector $n = (n_i)$ to the electric center charge $k \in \mathbb{Z}_{N_c}$ of the heavy Wilson source being confined;

(v) the domain in Coulomb-branch and deformation-parameter space in which the Abelian two-derivative description is used.

This construction defines an Abelianized string spectrum in the local effective theory. It becomes a statement about the microscopic non-Abelian theory only after the Seiberg–Witten effective action and the deformation to the chosen microscopic problem have been justified in the stated regime.

Cartan BPS bound in an aligned Abelian chart. Assume the Abelianized construction of Definition 92.11 and suppose that, in the chosen magnetic frame, the static transverse energy density is a sum of rank-one Abelian-Higgs BPS densities with aligned ξ_i. For flux vector $n = (n_1, \ldots, n_{N_c-1})$, every finite-energy string obeys

$$T(n) \geq 2\pi \left| \sum_{i=1}^{N_c-1} n_i \xi_i \right|. \tag{92.24}$$

If the first-order vortex equations are solved in every occupied sector with the common central-charge orientation, and if all nonzero $n_i \xi_i$ have the phase of $\sum_j n_j \xi_j$, the bound is saturated.

For a single diagonal magnetic factor the square completion in (91.35) gives

$$T_i \geq 2\pi |n_i|\, |\xi_i|$$

with equality when the flux orientation is chosen to have the same phase as $n_i\xi_i$. Writing all ξ_i in the common phase direction as $\xi_i = \mathrm{e}^{\mathrm{i}\alpha}|\xi_i|$, the central charge of a multi-flux configuration is

$$Z(n) = 2\pi \sum_i n_i\xi_i.$$

The triangle inequality gives

$$\sum_i 2\pi|n_i|\,|\xi_i| \geq 2\pi\left|\sum_i n_i\xi_i\right| = |Z(n)|,$$

which is (92.24). Saturation of the displayed central-charge bound requires equality in the triangle inequality, namely all nonzero $n_i\xi_i$ have the same phase, and also saturation of the rank-one square in every occupied sector. The argument is a statement in the diagonal Abelian effective theory; off-diagonal massive fields and non-Abelian core effects are outside the hypotheses.

At the Chebyshev-type maximally singular point of the pure $\mathcal{N}=2$ $SU(N_c)$ Coulomb branch, the Abelianized low-energy calculation often produces an FI profile proportional to the positive A_{N_c-1} sine vector

$$s_i = \sin\frac{\pi i}{N_c}, \qquad i = 1,\ldots,N_c-1. \tag{92.25}$$

The finite algebra behind the resulting sine ratios is elementary and is recorded here because it is the part of the comparison that is independent of any literature convention.

Example 92.12 (A_{N_c-1} sine profile and normalized tensions)**.** Let $s_i = \sin(\pi i/N_c)$. Then $s = (s_i)$ is the strictly positive eigenvector of the A_{N_c-1} Cartan matrix with eigenvalue

$$\lambda_1 = 4\sin^2\frac{\pi}{2N_c}. \tag{92.26}$$

Consequently, if an Abelianized $SU(N_c)$ string construction has aligned FI magnitudes

$$|\xi_k| = |\xi_1|\frac{\sin(\pi k/N_c)}{\sin(\pi/N_c)} \qquad (1 \leq k \leq N_c-1), \tag{92.27}$$

and if the k-labelled string is the single BPS sector associated with ξ_k, then the Abelianized normalized tensions are

$$\frac{T_k^{\mathrm{SW}}}{T_1^{\mathrm{SW}}} = \frac{\sin(\pi k/N_c)}{\sin(\pi/N_c)}. \tag{92.28}$$

They obey charge conjugation

$$T_k^{\mathrm{SW}} = T_{N_c-k}^{\mathrm{SW}} \tag{92.29}$$

and strict subadditivity

$$T_{k+\ell}^{\mathrm{SW}} < T_k^{\mathrm{SW}} + T_\ell^{\mathrm{SW}}, \qquad k,\ell \geq 1, \quad k+\ell \leq N_c-1. \tag{92.30}$$

The verification is finite-dimensional. For $1 < i < N_c-1$,

$$(Ks)_i = 2\sin\frac{\pi i}{N_c} - \sin\frac{\pi(i-1)}{N_c} - \sin\frac{\pi(i+1)}{N_c}.$$

Using

$$\sin(x-a)+\sin(x+a)=2\sin x\cos a$$

with $a=\pi/N_c$, this becomes

$$2\left(1-\cos\frac{\pi}{N_c}\right)\sin\frac{\pi i}{N_c}=4\sin^2\frac{\pi}{2N_c}\,s_i.$$

The endpoint rows obey the same identity with the absent sine terms $\sin 0$ and $\sin\pi$, both zero. The entries s_i are positive for $1\le i\le N_c-1$, so this is the Perron–Frobenius eigenvector of the Cartan matrix.

Under the FI profile (92.27), the saturated rank-one BPS tension in the k-sector is $T_k^{\mathrm{SW}}=2\pi|\xi_k|$. Division by the $k=1$ result gives (92.28). Charge conjugation follows from

$$\sin\frac{\pi(N_c-k)}{N_c}=\sin\frac{\pi k}{N_c}.$$

For subadditivity, set $a=\pi k/N_c$ and $b=\pi\ell/N_c$. Since $0<a,b$ and $a+b<\pi$,

$$\sin a+\sin b-\sin(a+b)=4\sin\frac{a}{2}\sin\frac{b}{2}\sin\frac{a+b}{2}>0.$$

After multiplying by the positive normalization $(2\pi|\xi_1|)/\sin(\pi/N_c)$, this gives (92.30).

Definition 92.13 (Seiberg–Witten to pure-Yang–Mills string comparison)**.** A comparison between the Abelianized Seiberg–Witten strings and the bosonic pure-Yang–Mills k-strings consists of:

(i) a fixed global form of the gauge group and a fixed line-operator lattice, so that the center charge k means the same thing throughout;

(ii) normalized tension functions

$$R_k(t)=\frac{T_k(t)}{T_1(t)}$$

along a declared deformation path t, where $t=0$ denotes the Abelianized Seiberg–Witten regime and $t=1$ denotes the target bosonic Yang–Mills problem;

(iii) a statement of the topology in which the flux-sector transfer matrices, their lowest eigenvalues, and the large-length limits defining $T_k(t)$ vary with t;

(iv) a list of possible obstructions: loss of the Abelian patch, string breaking by dynamical matter, collision with another flux-sector level, change of line-operator lattice, or failure of the continuum limit.

What the Abelian sine calculation transfers. Assume a comparison problem as in Definition 92.13. If, for a fixed k, the tensions $T_1(t)$ and $T_k(t)$ exist, are positive, and are continuous on $0\le t\le 1$, then the normalized ratio $R_k(t)$ is continuous. If in addition $R_k(t)$ is constant along the path, then the pure-Yang–Mills ratio equals the Abelianized value

$$R_k(1)=\frac{\sin(\pi k/N_c)}{\sin(\pi/N_c)}.$$

The Abelian Seiberg–Witten calculation proves the initial value $R_k(0)$ under its low-energy hypotheses; constancy, or any other controlled evolution law for $R_k(t)$, is an additional nonperturbative statement about the deformation path. The first assertion is the continuity of a quotient whose denominator is positive. The second follows by evaluating the constant function $R_k(t)$ at $t = 0$ and using (92.28). The last sentence records the logical dependence of the two inputs: Example 92.12 is an Abelian effective-theory statement, while constancy of $R_k(t)$ is a statement about a family of nonperturbative flux-sector transfer matrices.

92.5.5 Planar $\mathcal{N} = 4$

Planar $\mathcal{N} = 4$ Yang–Mills is conformal, so the family comparison uses scaling dimensions and defect energies instead of massive glueball poles. A local-operator row is the planar spin-chain/QSC spectral problem of Chapters 97–100. A line-defect row is the Wilson-line flux-tube spectral problem of Definition 100.17. Its vacuum energy is the cusp anomalous dimension, and its excitations are the pentagon-OPE or mirror-TBA flux-tube excitations in an integrability description.

The $\mathcal{N} = 4$ row is therefore the most exactly soluble spectral row currently available inside four-dimensional interacting QFT, while its proof boundary is also sharp: the planar integrability construction is an exact spectral framework under stated integrability hypotheses, and a derivation from a nonperturbative four-dimensional regulator remains open.

92.6 Soft Breaking Toward Bosonic Yang–Mills

The deformation from pure $\mathcal{N} = 1$ Yang–Mills toward bosonic Yang–Mills is naturally expressed by a spurion source for the chiral coordinate S:

$$\Delta\mathcal{L} = \int \mathrm{d}^2\theta\, X_\lambda(\theta) S + \text{c.c.}, \qquad X_\lambda(\theta) = \theta^2 m_\lambda. \tag{92.31}$$

The component deformation is

$$\Delta\mathcal{L}_{\text{comp}} = m_\lambda\, S|_{\theta=0} + \overline{m}_\lambda\, \overline{S}|_{\overline{\theta}=0}. \tag{92.32}$$

At small $|m_\lambda|$, the leading branch energy splitting is obtained from the chiral condensate:

$$\Delta E_k = -2\, \mathrm{Re}\Big(m_\lambda\, C\Lambda_h^3 \mathrm{e}^{2\pi \mathrm{i} k/N_c}\Big) + O(|m_\lambda|^2). \tag{92.33}$$

This formula is a first-order source-response statement in the massive $\mathcal{N} = 1$ theory. Increasing $|m_\lambda|$ until the gaugino decouples is a deformation path whose spectral continuity must be examined with nonchiral observables: glueball masses, string tensions, domain-wall fates, and line-operator sectors. The endpoint comparison with bosonic Yang–Mills therefore belongs to the spectral comparison table of Definition 92.4, not only to the chiral branch comparison.

Controlled Approximation 92.14 (Small soft mass branch selection)**.** Assume pure $\mathcal{N} = 1$ $SU(N_c)$ Yang–Mills has N_c massive branches with condensates

$$S_k = C\Lambda_h^3 \mathrm{e}^{2\pi \mathrm{i} k/N_c}$$

and that the soft source (92.31) can be treated in first-order perturbation theory in a finite volume before the thermodynamic limit is taken. Then the branch selected at first order is the branch maximizing

$$\mathrm{Re}(m_\lambda S_k).$$

First-order branch selection. The first-order shift of the vacuum energy density under the component source $m_\lambda S + \overline{m}_\lambda \overline{S}$ is minus the expectation value of the source term:

$$\Delta E_k = -\left(m_\lambda S_k + \overline{m}_\lambda\, \overline{S}_k\right) + O(|m_\lambda|^2) = -2\,\mathrm{Re}(m_\lambda S_k) + O(|m_\lambda|^2).$$

At first order, minimizing ΔE_k is the same as maximizing $\mathrm{Re}(m_\lambda S_k)$.

92.7 Status Summary

The present chapter closes the elementary comparison map. It records the following theorem boundaries:

statement	status in this monograph
$\mathcal{N}=4 \to \mathcal{N}=2, \mathcal{N}=1$ scale matching	derived holomorphic threshold matching
$\mathcal{N}=1$ branches and BPS walls	derived from chiral F-term hypotheses
$\mathcal{N}=2$ monopole condensation	derived in local Abelian patch
$\mathcal{N}=2$ vortex tension	BPS Abelian-Higgs controlled approximation
$\mathcal{N}=2$ Abelianized k-strings	A-type sine profile; comparison problem
$\mathcal{N}=4$ Wilson-line flux tube	defect spectral problem plus integrability hypotheses
bosonic glueball and k-string spectra	lattice or Hilbert-space spectral problem

The frontier problem is to construct a regulator-level family in which enough of the ladder is realized simultaneously to compare the spectral measures, line-defect Hilbert spaces, and flux-sector transfer matrices directly.

Open Problem 92.15 (Regulator-level supersymmetry-breaking spectral bridge)*.* Construct a regulator and continuum-limiting framework for the deformation from supersymmetric Yang–Mills theories to bosonic Yang–Mills that controls all of the following in the same topology: gauge-invariant local spectral measures, center-flux transfer matrices, line-operator charge lattices, domain-wall sectors, and protected chiral coordinates. Prove, disprove, or replace the commonly assumed continuity of glueball and flux-string spectra along such deformation paths by a theorem with stated hypotheses.

Chapter 93

Moduli Spaces In Supersymmetric Quantum Field Theory

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter defines supersymmetric moduli spaces as spaces of supersymmetric vacua together with their low-energy QFT structures. The set of classical solutions is only the first coordinate chart; the physical object includes quotienting, singular strata, massless fields, chiral rings, metrics, and quantum corrections.

Definition 93.1 (Quantum supersymmetric moduli-space structure)**.** Fix a supersymmetric local QFT with supercharges Q_α acting on a common invariant domain in each vacuum representation. A quantum supersymmetric moduli-space structure is a tuple

$$\mathfrak{M} = \left(\mathcal{M}, \mathcal{O}^{\mathrm{hol}}_{\mathcal{M}}, \{\mathcal{H}_p, \mathcal{A}_p, \mathcal{E}_p, \mathcal{G}_p\}_{p\in\mathcal{M}}\right)$$

with the following entries.

1. $\mathcal{M}$ is a reduced stratified analytic space whose points label equivalence classes of supersymmetric vacuum sectors. A representative vacuum sector is a translation-invariant positive-energy representation with vacuum vector or vacuum state annihilated by the preserved supercharges, modulo unitary equivalence preserving the local operator net and background couplings.

2. $\mathcal{O}^{\mathrm{hol}}_{\mathcal{M}}$ is the sheaf of holomorphic branch-coordinate functions generated, where valid, by expectation values of protected chiral operators. Nilpotent or derived structure is additional

structure; if it affects operator products or low-energy dynamics it must be retained rather than replaced by the reduced space.

3. $\mathcal{H}_p$ is the Hilbert space of fluctuations in the vacuum sector p, and $\mathcal{A}_p$ is the local operator algebra or field presentation of the low-energy QFT in that sector.
4. $\mathcal{E}_p$ records the effective branch structure: massless fields, two-derivative metric, Wess–Zumino terms, superselection sectors, and the energy range on which the branch effective theory is valid.
5. $\mathcal{G}_p$ records background responses, anomaly lines, and global-form or extended-operator structures in the vacuum sector.

On a smooth stratum, $T_p\mathcal{M}$ is represented by supersymmetric massless scalar directions in $\mathcal{H}_p$. A singular stratum is a locus where this representation, the effective theory $\mathcal{E}_p$, or the background-response structure $\mathcal{G}_p$ changes. A classical quotient of scalar zero modes is therefore a coordinate construction for one approximation to $\mathfrak{M}$, not the definition of the quantum moduli-space structure.

93.1 Vacuum Equations

For four-dimensional $\mathcal{N} = 1$ chiral multiplets Φ^i with superpotential $W(\Phi)$ and gauge group G, a classical supersymmetric vacuum obeys

$$F_i = \frac{\partial W}{\partial \Phi^i} = 0, \qquad \mu^a = 0,$$

where $\mu : \mathcal{X} \to \mathfrak{g}^*$ is the moment map for the Kähler target $\mathcal{X}$ of scalar fields. The classical vacuum space is the Kähler quotient

$$\mathcal{M}_{\text{cl}} = \{F = 0,\ \mu = 0\}/G.$$

Equivalently, under the required stability hypotheses for the complexified gauge action, it is the holomorphic quotient

$$\{F = 0\}/G_{\mathbb{C}}.$$

The second description identifies holomorphic functions on $\mathcal{M}_{\text{cl}}$ with gauge-invariant chiral polynomials modulo the ideal generated by the F-term equations.

The last sentence hides several hypotheses, so we spell out the quotient construction used in this volume. Let V be the finite-dimensional Hermitian representation of G carried by the scalar fields, let $G_{\mathbb{C}}$ be a reductive complexification, and let $W \in \mathbb{C}[V]^{G_{\mathbb{C}}}$ be a polynomial superpotential. Write

$$Z_F = \{v \in V \mid dW(v) = 0\}, \qquad I_F = (\partial_1 W, \dots, \partial_n W) \subset \mathbb{C}[V].$$

The affine chiral quotient is

$$\mathcal{M}_{\text{ch,cl}} = \operatorname{Spec}\left((\mathbb{C}[V]/I_F)^{G_{\mathbb{C}}}\right)_{\text{red}}, \tag{93.1}$$

where the subscript records that the displayed affine variety forgets nilpotents unless they are kept as derived or scheme-theoretic data. The symplectic vacuum quotient is

$$\mathcal{M}_{\mu,\text{cl}} = (Z_F \cap \mu^{-1}(0))/G. \tag{93.2}$$

Classical quotient and chiral functions. Assume the Kempf–Ness comparison for the triple $(V, G, G_{\mathbb{C}})$ on the $G_{\mathbb{C}}$-invariant closed subset Z_F: every polystable $G_{\mathbb{C}}$-orbit in Z_F meets $\mu^{-1}(0)$, and two points of $Z_F \cap \mu^{-1}(0)$ lie in the same closed $G_{\mathbb{C}}$-orbit exactly when they lie in the same G-orbit. Under this comparison $\mathcal{M}_{\mu,\mathrm{cl}}$ is naturally identified with the polystable affine quotient $\mathcal{M}_{\mathrm{ch,cl}}$, and its holomorphic coordinate ring is

$$\mathbb{C}[\mathcal{M}_{\mathrm{ch,cl}}] = \left((\mathbb{C}[V]/I_F)^{G_{\mathbb{C}}}\right)_{\mathrm{red}}.$$

Thus the notation $\{F = 0\}/G_{\mathbb{C}}$ in this monograph means the affine quotient in (93.1), together with the stated stability hypothesis, not the naive set of all complexified orbits. Because W is $G_{\mathbb{C}}$-invariant, the differential dW is equivariant and the ideal I_F is $G_{\mathbb{C}}$-stable. Hence every class in $(\mathbb{C}[V]/I_F)^{G_{\mathbb{C}}}$ is represented by a gauge-invariant holomorphic polynomial on Z_F, and restriction to $Z_F \cap \mu^{-1}(0)$ gives a holomorphic function constant on G-orbits. The assumed Kempf–Ness statement identifies G-orbits in $Z_F \cap \mu^{-1}(0)$ with closed $G_{\mathbb{C}}$-orbits in the affine quotient. Invariant polynomials separate closed orbits in an affine reductive quotient, so the descended functions are precisely the regular holomorphic functions on the reduced quotient. Nonclosed complexified orbits can have the same closure and therefore the same values of all invariant polynomials; the affine quotient collapses them to the polystable representative.

Example 93.2 (Rank-one abelian quotient)**.** Let $G = U(1)$ act on $V = \mathbb{C}^2$ by

$$(x, y) \longmapsto (\mathrm{e}^{\mathrm{i}\alpha}x, \mathrm{e}^{-\mathrm{i}\alpha}y), \qquad W = 0,$$

with moment map $\mu = |x|^2 - |y|^2$. The invariant ring of the complexified action $t\cdot(x, y) = (tx, t^{-1}y)$ is

$$\mathbb{C}[x, y]^{\mathbb{C}^*} = \mathbb{C}[u], \qquad u = xy.$$

Indeed the monomial $x^a y^b$ has $U(1)$ charge $a - b$, so invariance forces $a = b$ and the monomial is u^a. On the symplectic side $\mu = 0$ gives $|x| = |y|$. The map $(x, y) \mapsto xy$ sends $\mu^{-1}(0)/U(1)$ bijectively to $\mathbb{C}$: for $u = \rho\mathrm{e}^{\mathrm{i}\theta}$, with $\rho \geq 0$, choose

$$x = \sqrt{\rho}, \qquad y = \sqrt{\rho}\,\mathrm{e}^{\mathrm{i}\theta},$$

and changing the phase split between x and y is precisely the $U(1)$ quotient. Thus this basic vacuum quotient is the complex line with coordinate u, not a two-complex-dimensional space of unquotiented scalar expectation values.

93.2 Quantum Moduli Space

For the quantum moduli-space structure $\mathfrak{M}$ of Definition 93.1, a smooth branch is a stratum on which the massless field content, preserved supersymmetry algebra, and background-response structure are locally constant. If z^i are massless scalar coordinates on such a branch, the two-derivative effective action determines a metric

$$ds^2 = g_{i\bar{j}}(z, \bar{z})\, dz^i d\bar{z}^{\bar{j}}$$

on the branch. The metric is part of the low-energy QFT structure; the chiral coordinate ring determines only holomorphic functions.

93.3 Chiral Rings And Holomorphic Functions

Let $\mathcal{R}_{\mathrm{ch}}$ be the chiral ring of gauge-invariant local operators modulo $\bar{Q}$-exact operators. In a vacuum p, expectation values define a homomorphism

$$\mathrm{ev}_p : \mathcal{R}_{\mathrm{ch}} \longrightarrow \mathbb{C}.$$

If chiral operators separate vacua on a branch and have no nilpotent relations in the reduced branch coordinate ring, the collection of ev_p identifies the reduced holomorphic coordinate ring with a quotient of $\mathcal{R}_{\mathrm{ch}}$. Nilpotents, accidental massless fields, and singular strata must be retained when they affect operator products or low-energy dynamics.

93.4 Fixed Charge As A Probe Of A Branch

There are two distinct operator-theoretic probes of a moduli branch. The first probe is the character of the chiral ring in a vacuum, $\mathrm{ev}_p : \mathcal{R}_{\mathrm{ch}} \to \mathbb{C}$. It addresses the reduced holomorphic coordinate algebra, under the separation hypotheses stated above. The second probe uses sectors of fixed global charge. This second probe is the precise setting of the large-charge expansion: a low-energy effective theory on a smooth branch can determine the asymptotic spectrum of highly charged states, and in a conformal theory those states are equivalently highly charged local operators of definite scaling dimension.

Let F be a compact internal global symmetry of the QFT, with conserved currents j^μ_A and commuting charge operators

$$Q_A = \int_\Sigma d\Sigma_\mu \, j^\mu_A, \qquad A = 1, \ldots, r,$$

on a spatial slice Σ, after the usual choice of domain and infrared regulator. For an integral charge vector $q = (q_A)$ in the joint spectrum, write

$$\mathcal{H}_q = \{\psi \in \mathcal{H} \mid Q_A \psi = q_A \psi \text{ for all } A\}.$$

A chemical potential μ^A is the Lagrange multiplier conjugate to Q_A. In Hamiltonian language it enters the variational problem through

$$H_\mu = H - \mu^A Q_A,$$

while in Euclidean finite-volume language it is an imaginary background holonomy for the corresponding global symmetry, with the precise analytic continuation depending on the reflection-positive construction of the thermal theory.

Assume that a smooth branch $\mathcal{M}_{\mathrm{sm}} \subset \mathcal{M}$ carries an F-action generated by vector fields $K_A = K^i_A(z)\partial_i + \overline{K}^{\bar{i}}_A(\bar{z})\partial_{\bar{i}}$, and that the branch EFT with metric $g_{i\bar{j}}$ is valid below a scale $M_{\mathrm{gap}}(p)$ for the massive modes transverse to the branch. A homogeneous charged saddle on the cylinder has the form

$$\dot{z}^i = \mu^A K^i_A(z), \qquad \dot{\bar{z}}^{\bar{i}} = \mu^A \overline{K}^{\bar{i}}_A(\bar{z}),$$

up to the fermionic and auxiliary-field conditions imposed by the preserved supercharges. Its charge and energy are computed from the Noether functional of the branch action. At the two-derivative level, with $V_R = \mathrm{Vol}(S_R^{d-1})$,

$$q_A = V_R\, g_{i\bar{j}}(z,\bar{z}) \left(K_A^i \dot{\bar{z}}^{\bar{j}} + \dot{z}^i \overline{K}_A^{\bar{j}} \right) + q_A^{\mathrm{WZ}} + \cdots , \tag{93.3}$$

$$E_{\mathrm{EFT}}(q,R) = \underset{z,\mu}{\mathrm{crit}} \left[V_R g_{i\bar{j}}(z,\bar{z})\, \mu^A K_A^i\, \mu^B \overline{K}_B^{\bar{j}} + E_{\mathrm{curv}}(z,R) + E_{\mathrm{WZ}}(z,\mu,R) + \cdots \right]_{q \text{ fixed}} . \tag{93.4}$$

Here R is the radius of the spatial sphere, q_A^{WZ} and E_{WZ} denote possible Wess-Zumino or background-response contributions, and the ellipses are higher-derivative terms ordered by the small parameters set by $1/(|\mu|R)$ and $|\mu|/M_{\mathrm{gap}}(p)$. The displayed equations are a reconstruction mechanism rather than a theorem: they become a controlled expansion only after specifying the charged saddle, the positivity of transverse fluctuations, and a range in which all omitted operators are uniformly smaller than the retained terms.

In a unitary CFT, put the theory on $\mathbb{R} \times S_R^{d-1}$. The state-operator theorem of Chapter 59 identifies local operators of definite scaling dimension with the corresponding energy eigenstates in the local-operator subspace of the cylinder Hilbert space. Consequently, if the lowest-energy vector in $\mathcal{H}_q$ is represented by a primary operator $\mathcal{O}_q$, then

$$\Delta(\mathcal{O}_q) = R\, E_0(q,R).$$

The large-charge program initiated by Hellerman and collaborators is exactly this bridge in the conformal setting: fixed-charge spectral data determine the coefficients of the branch EFT that control the charged saddle, while the branch EFT predicts the asymptotic dimensions and matrix elements of large-charge operators. Its output is an asymptotic charged-sector reconstruction. Recovering the full vacuum datum $(\mathcal{M}, \mathcal{H}_p, \mathcal{A}_p, \mathcal{G}_p)$, including singular strata, uncharged directions, nilpotent scheme structure, background responses, and changes of low-energy degrees of freedom, requires additional input beyond the large-charge sector alone.

The broader reconstruction problem is more structural. Wightman-type data provide a Hilbert space, local operator-valued distributions, vacuum correlation distributions, spectrum, locality, and covariance. A Kontsevich–Segal-type theory asks instead for functorial amplitudes assigned to geometric bordisms, with boundary state spaces, sewing, reflection positivity, and background-field dependence. A supersymmetric moduli space should be visible in such KS data as a family of boundary conditions or asymptotic sectors whose sewn amplitudes produce the low-energy branch theory. Chiral-ring characters and fixed-charge spectra are therefore probes of the Wightman-to-KS reconstruction map, rather than the map itself.

93.5 N One SQCD Branches

For $SU(N_c)$ SQCD with N_f flavors, the microscopic chiral fields are quarks $Q^a{}_i$ in the fundamental and antiquarks $\widetilde{Q}_a{}^i$ in the antifundamental, with color $a = 1, \ldots, N_c$ and flavor $i = 1, \ldots, N_f$. The elementary gauge-invariant chiral coordinates are the mesons

$$M^i{}_j = \widetilde{Q}_a{}^i Q^a{}_j$$

and, when $N_f \geq N_c$, the baryons and antibaryons obtained by contracting N_c quarks or antiquarks with $\epsilon_{a_1 \cdots a_{N_c}}$. The relations among these coordinates are algebraic relations in the reduced affine quotient; they are not optional decoration, because they decide which operator expectation values can occur in a supersymmetric vacuum.

The following table is only an orientation map. It is not used as a proof, and each row requires a separate derivation or an explicitly labeled Wilsonian input:

$N_f < N_c$	ADS superpotential produces a runaway direction
$N_f = N_c$	$\det M - B\widetilde{B} = \Lambda_h^{2N_c}$
$N_f = N_c + 1$	confining chiral coordinates with superpotential constraint

Each entry is a Wilsonian chiral-coordinate statement. The metric on the branch and the massive spectrum require additional low-energy data.

The table becomes meaningful only after the classical quotient coordinates have been fixed and after the status of the quantum input has been separated from the algebra done with that input. The smallest nontrivial $N_f = N_c$ example is therefore a concrete test case for the monograph's method: construct the classical quotient from invariant theory, state the protected quantum deformation as the only dynamical input, and then derive all consequences that follow from the deformed equation by ordinary algebra.

Classical $SU(2)$, $N_f = 2$ quotient as invariant theory. Consider four-dimensional $\mathcal{N} = 1$ $SU(2)$ SQCD with two flavors and $W = 0$. Package the two quark and two antiquark doublets, using the pseudoreality of the $SU(2)$ fundamental, as four chiral doublets

$$P_a^I, \qquad a = 1, 2, \qquad I = 1, \ldots, 4.$$

The complexified gauge group $SL(2, \mathbb{C})$ acts on the gauge index a. Define the antisymmetric gauge-invariant coordinates

$$V^{IJ} = \epsilon^{ab} P_a^I P_b^J = -V^{JI}, \qquad 1 \le I, J \le 4, \tag{93.5}$$

with $\epsilon^{12} = 1$. Let

$$\mathrm{Pf}(V) = V^{12}V^{34} - V^{13}V^{24} + V^{14}V^{23}. \tag{93.6}$$

Then the reduced affine classical chiral quotient is

$$\mathcal{M}_{\mathrm{ch,cl}} \simeq \{\, V \in \bigwedge^2 \mathbb{C}^4 \mid \mathrm{Pf}(V) = 0 \,\}. \tag{93.7}$$

It has complex dimension 5. In the usual $Q, \widetilde{Q}$ flavor splitting, the six entries of V^{IJ} are the mesons and baryons, and (93.7) is the classical $\det M - B\widetilde{B} = 0$ relation in an $SU(4)$-covariant notation. The quantum deformation $\mathrm{Pf}(V) = \Lambda_h^4$ is not part of this classical quotient; it is the Wilsonian nonperturbative input discussed in Chapter 90.

Write the two rows of the 2×4 matrix P as

$$u = (P_1^1, \ldots, P_1^4), \qquad v = (P_2^1, \ldots, P_2^4).$$

Then $V^{IJ} = u_I v_J - u_J v_I$, so $V = u \wedge v$ is a decomposable two-form. Hence $V \wedge V = 0$. With the convention (93.6), the coefficient of $2e_1 \wedge e_2 \wedge e_3 \wedge e_4$ in $V \wedge V$ is $\mathrm{Pf}(V)$, proving that every gauge orbit maps into the displayed hypersurface.

Conversely, suppose $V \neq 0$ and $\mathrm{Pf}(V) = 0$. After relabeling the four flavor indices, assume $a_0 = V^{12} \neq 0$. Set

$$u = (1, 0, -V^{23}/a_0, -V^{24}/a_0), \qquad v = (0, a_0, V^{13}, V^{14}).$$

The minors $u_I v_J - u_J v_I$ reproduce V^{12}, V^{13}, V^{14}, V^{23}, and V^{24} by construction, while the remaining minor is

$$u_3 v_4 - u_4 v_3 = \frac{V^{13}V^{24} - V^{14}V^{23}}{a_0} = V^{34},$$

where the last equality is exactly $\mathrm{Pf}(V) = 0$. The point $V = 0$ is obtained from $P = 0$. Thus the invariant map has precisely the hypersurface (93.7) as its image.

It remains to justify that no additional reduced affine-coordinate data are missing. In a polynomial invariant, all $SL(2, \mathbb{C})$ gauge indices must be contracted by invariant tensors. For the fundamental two-dimensional representation, the invariant tensors are generated by the alternating tensor ϵ^{ab}; equivalently, after decomposing tensor powers into irreducible $SL(2)$ representations, the singlet summands are obtained by pairwise antisymmetric contractions. Hence the reduced invariant ring is generated by the pairings V^{IJ}. For four flavor labels the only quadratic straightening relation among these six generators is the Plucker relation

$$V^{12}V^{34} - V^{13}V^{24} + V^{14}V^{23} = 0,$$

which we already identified as the Pfaffian equation. Therefore the reduced affine quotient is the displayed hypersurface. Its dimension is $6 - 1 = 5$, agreeing with the quotient count

$$\dim_{\mathbb{C}} \mathbb{C}^8 - \dim_{\mathbb{C}} SL(2, \mathbb{C}) = 8 - 3 = 5$$

on the stable maximal-rank locus.

Algebra after the $SU(2)$, $N_f = 2$ quantum-deformation input. Assume the Wilsonian chiral-coordinate input that the quantum branch of the theory whose classical chiral quotient is (93.7) is

$$\mathrm{Pf}(V) = \Lambda_h^4, \qquad \Lambda_h \neq 0, \tag{93.8}$$

where Pf is the polynomial in (93.6). The dynamical content is the stated Wilsonian input; the following paragraphs record only the algebraic consequences of that input. The deformed branch is smooth and five-complex-dimensional. Moreover, after adding a nondegenerate antisymmetric mass source and putting it, by a complex linear change of flavor basis, in Darboux form

$$W_m = m_1 V^{12} + m_2 V^{34}, \qquad m_1 m_2 \neq 0, \tag{93.9}$$

and imposing the constraint by a chiral Lagrange multiplier X,

$$W_{\mathrm{constr}} = m_1 V^{12} + m_2 V^{34} + X\big(\mathrm{Pf}(V) - \Lambda_h^4\big), \tag{93.10}$$

the F-term equations over $\mathbb{C}$ have exactly two solutions. After choosing a square root Λ_h^2 of the deformation parameter Λ_h^4 and a square root of $m_1 m_2$, they are

$$X_\pm = \pm\frac{\sqrt{m_1 m_2}}{\Lambda_h^2}, \qquad V_\pm^{12} = -\frac{m_2}{X_\pm}, \qquad V_\pm^{34} = -\frac{m_1}{X_\pm}, \tag{93.11}$$

with all other V^{IJ} equal to zero. Their superpotential values are

$$W_{\pm} = \mp\, 2\Lambda_h^2 \sqrt{m_1 m_2}. \tag{93.12}$$

Thus the deformed branch removes the classical origin, and the simplest mass deformation has the two isolated chiral vacua expected after decoupling to pure $SU(2)$ supersymmetric Yang–Mills, with scale matching $\Lambda_0^6 = m_1 m_2 \Lambda_h^4$ in this Darboux convention. The last comparison uses matching to the massive low-energy theory; the algebraic statement above uses only (93.8). Different choices of square root or of the sign convention relating V^{IJ} to the usual meson-baryon coordinates only relabel the two vacua.

Let

$$F(V) = V^{12}V^{34} - V^{13}V^{24} + V^{14}V^{23}.$$

The gradient of F in the six affine coordinates is

$$\nabla F = (V^{34}, -V^{24}, V^{23}, -V^{13}, V^{14}, V^{12}),$$

ordered as

$$(V^{12}, V^{13}, V^{14}, V^{24}, V^{23}, V^{34}).$$

If $\nabla F = 0$, then all six entries of V vanish, so $F(V) = 0$. No point on $F(V) = \Lambda_h^4$ with $\Lambda_h \neq 0$ can therefore be singular. A smooth hypersurface in six complex affine coordinates has complex dimension 5.

For the mass deformation, a general holomorphic quadratic mass source in the four doublets is an antisymmetric tensor $m_{IJ}V^{IJ}/2$. If m is nondegenerate, elementary alternating-form linear algebra gives a basis (e_1, e_2, e_3, e_4) with

$$m = m_1\, e^1 \wedge e^2 + m_2\, e^3 \wedge e^4, \qquad m_1 m_2 \neq 0;$$

one chooses e_1, then e_2 with $m(e_1, e_2) \neq 0$, takes the orthogonal complement of their span, and repeats in the remaining two-dimensional nondegenerate alternating space. In that Darboux basis, the F-terms from (93.10) are

$$\begin{array}{lll} m_1 + XV^{34} = 0, & -XV^{24} = 0, & XV^{23} = 0, \\ -XV^{13} = 0, & XV^{14} = 0, & m_2 + XV^{12} = 0, \end{array}$$

together with $F(V) = \Lambda_h^4$. If $X = 0$, the first and last equations would force $m_1 = m_2 = 0$, contrary to the hypothesis. Hence $X \neq 0$, and the middle four equations set

$$V^{13} = V^{14} = V^{23} = V^{24} = 0.$$

The remaining two linear equations give

$$V^{12} = -\frac{m_2}{X}, \qquad V^{34} = -\frac{m_1}{X}.$$

The constraint then becomes

$$\frac{m_1 m_2}{X^2} = \Lambda_h^4,$$

so $X = X_{\pm}$ and the two displayed solutions follow. Substitution into (93.9) gives

$$W_{\pm} = -\frac{2m_1 m_2}{X_{\pm}} = \mp\, 2\Lambda_h^2 \sqrt{m_1 m_2},$$

as claimed.

Finally, the comparison with pure $SU(2)$ is the standard holomorphic threshold matching applied to two massive doublet pairs. The one-loop holomorphic exponents are $b_0 = 4$ for $SU(2)$ with two flavors and $b_0 = 6$ for pure $SU(2)$. In the Darboux mass convention the Pfaffian of the antisymmetric mass matrix is $m_1 m_2$, so decoupling gives

$$\Lambda_0^6 = m_1 m_2 \Lambda_h^4.$$

Thus the two values above may be written as $W_\pm = \mp 2\Lambda_0^3$ after choosing a cube-root branch for Λ_0^3, exactly the two-branch pure $SU(2)$ chiral-source pattern. This matching paragraph uses the Wilsonian decoupling input; the smoothness and F-term solution were already proved directly from the deformed Pfaffian equation.

93.6 N Two Coulomb, Higgs, And Mixed Branches

For $\mathcal{N} = 2$ theories, Coulomb-branch coordinates come from vector multiplet scalars and have special Kähler geometry. Higgs branches come from hypermultiplet scalars and are hyperkähler quotients:

$$\mathcal{M}_H = \mu_{\mathbb{R}}^{-1}(0) \cap \mu_{\mathbb{C}}^{-1}(0)/G.$$

Mixed branches have both vector and hypermultiplet massless fields after solving compatibility conditions between vector scalar expectation values and hypermultiplet expectation values. The Coulomb-branch metric may receive quantum corrections encoded by Seiberg-Witten periods. In many Lagrangian examples the Higgs-branch hyperkähler metric is protected, but this protection is a statement about the supersymmetric Wilsonian scheme and the branch coordinates, not a statement about arbitrary coordinate choices.

The quotient statement has concrete content already in the simplest hypermultiplet model. We spell it out because this is the local model behind many later Higgs-branch calculations.

Proposition 93.3 (Rank-one hypermultiplet quotient)**.** *Let $N \geq 2$. Let $U(1)$ act on*

$$V = \mathbb{C}_q^N \oplus \mathbb{C}_{\widetilde{q}}^N$$

by

$$q_i \longmapsto \mathrm{e}^{\mathrm{i}\alpha} q_i, \qquad \widetilde{q}_i \longmapsto \mathrm{e}^{-\mathrm{i}\alpha} \widetilde{q}_i, \qquad i = 1, \ldots, N.$$

Use the complex structure in which $q_i, \widetilde{q}_i$ are holomorphic coordinates. The complex moment map and real moment map are

$$\mu_{\mathbb{C}}(q, \widetilde{q}) = \sum_{i=1}^N q_i \widetilde{q}_i, \qquad \mu_{\mathbb{R}}(q, \widetilde{q}) = \sum_{i=1}^N |q_i|^2 - \sum_{i=1}^N |\widetilde{q}_i|^2.$$

For real FI parameter $\zeta > 0$ and complex FI parameter 0, the Higgs-branch quotient

$$\mathcal{M}_H(\zeta) = \{\, \mu_{\mathbb{C}} = 0,\ \mu_{\mathbb{R}} = \zeta \,\}/U(1) \tag{93.13}$$

is biholomorphic, in this complex structure, to $T^*\mathbb{P}^{N-1}$. Its complex dimension is $2(N-1)$, equivalently its real dimension is $4(N-1)$. On the patch $U_a = \{q_a \neq 0\}$, local holomorphic coordinates are

$$z_i^{(a)} = \frac{q_i}{q_a}, \qquad p_i^{(a)} = q_a \widetilde{q}_i, \qquad i \neq a, \tag{93.14}$$

and the complex moment-map equation determines

$$q_a\widetilde{q}_a = -\sum_{i\neq a} z_i^{(a)} p_i^{(a)}. \tag{93.15}$$

The holomorphic one-form

$$\Theta = \sum_{i=1}^{N} \widetilde{q}_i \, dq_i$$

descends to the canonical cotangent one-form

$$\Theta = \sum_{i\neq a} p_i^{(a)} \, dz_i^{(a)} \tag{93.16}$$

on U_a.

Proof. First remove the real quotient from the holomorphic bookkeeping. If $\mu_{\mathbb{R}} = \zeta > 0$, then $q \neq 0$. Conversely, fix a point $(q, \widetilde{q})$ with $q \neq 0$ and $\mu_{\mathbb{C}} = 0$. Along the positive real part of the complexified orbit,

$$q \mapsto rq, \qquad \widetilde{q} \mapsto r^{-1}\widetilde{q}, \qquad r > 0,$$

the real moment map becomes

$$f(r) = r^2\|q\|^2 - r^{-2}\|\widetilde{q}\|^2.$$

This function is continuous and strictly increasing on $r > 0$, with limit $-\infty$ as $r \to 0$ if $\widetilde{q} \neq 0$ and limit 0 if $\widetilde{q} = 0$, and with limit $+\infty$ as $r \to \infty$. Hence there is a unique positive r solving $f(r) = \zeta$. The remaining phase is precisely the compact $U(1)$ action. Thus the quotient (93.13) is identified with the complex quotient

$$\{\, \mu_{\mathbb{C}} = 0, \ q \neq 0 \,\}/\mathbb{C}^*. \tag{93.17}$$

On U_a, the coordinates in (93.14) are invariant under $\mathbb{C}^*$. They determine the orbit because q_a supplies the complexified gauge scale, the ratios determine all q_i, the $p_i^{(a)}$ determine all $\widetilde{q}_i$ with $i \neq a$, and (93.15) determines $\widetilde{q}_a$. There are $2(N-1)$ complex coordinates, so the complex dimension is $2(N-1)$.

It remains to check that the local charts glue as the cotangent bundle, not just as an abstract $2(N-1)$-dimensional quotient. It suffices to compare two patches; relabel the indices as $a = 1$ and $b = 2$. Write

$$z_i = \frac{q_i}{q_1}, \quad p_i = q_1\widetilde{q}_i \quad (i \neq 1),$$

and on the patch $q_2 \neq 0$,

$$w_1 = \frac{q_1}{q_2} = \frac{1}{z_2}, \qquad w_i = \frac{q_i}{q_2} = \frac{z_i}{z_2} \quad (i \neq 1, 2),$$

$$r_1 = q_2\widetilde{q}_1 = -z_2 \sum_{i\neq 1} z_i p_i, \qquad r_i = q_2\widetilde{q}_i = z_2 p_i \quad (i \neq 1, 2),$$

where the formula for r_1 uses $\mu_{\mathbb{C}} = 0$. A direct differentiation gives

$$r_1 \, dw_1 + \sum_{i\neq 1,2} r_i \, dw_i = \sum_{i\neq 1} p_i \, dz_i. \tag{93.18}$$

Thus the transition functions preserve the canonical holomorphic cotangent one-form.

Finally, on a single patch,

$$dq_i = z_i^{(a)}\, dq_a + q_a\, dz_i^{(a)} \qquad (i \neq a).$$

Using (93.15), the coefficient of dq_a in $\sum_i \widetilde{q}_i\, dq_i$ is

$$\widetilde{q}_a + \sum_{i\neq a} z_i^{(a)}\widetilde{q}_i = 0,$$

while the coefficient of $dz_i^{(a)}$ is $q_a\widetilde{q}_i = p_i^{(a)}$. This proves (93.16) and identifies the quotient with $T^*\mathbb{P}^{N-1}$. □

93.7 Singularities

A point $p \in \mathcal{M}$ is singular when the low-energy description changes there: additional states can become massless, the effective metric can degenerate, or the coordinate quotient can have an orbifold or more severe singularity. In the $\mathcal{N} = 2$ pure gauge-algebra $\mathfrak{su}(2)$ example, the singularities $u = \pm\Lambda^2$ are interpreted by vanishing cycles of the Seiberg-Witten curve and the appearance of light monopole or dyon hypermultiplets. The correct local model is the Abelian theory with those light fields included.

93.8 Effective Theory On A Branch

Let p lie on a smooth branch and let z^i be local coordinates. The low-energy effective action has the derivative expansion

$$S_{\text{eff}} = \int d^dx\, g_{i\bar{j}}(z,\bar{z})\partial_\mu z^i\partial^\mu \bar{z}^{\bar{j}} + \text{fermions} + \text{higher-derivative terms.}$$

The branch approximation is valid for momenta small compared with the masses of fields integrated out and away from loci where additional fields become light. A complete moduli-space statement therefore includes its domain of validity inside the full QFT.

Open Problem 93.4 (Moduli-space construction). Construct the supersymmetric moduli-space datum as part of a general passage from Wightman-type local QFT data to Kontsevich–Segal-type geometric amplitude data. Starting from local operator distributions, Hilbert spaces, vacuum sectors, and correlation distributions, the construction should produce the boundary state spaces, sewing maps, reflection pairings, and background-field dependence from which the vacuum sectors, chiral-coordinate ring, low-energy Hilbert spaces, branch metrics, singular local models, and background anomaly responses are recovered. It should also explain the precise domain of the two probes described above: chiral-ring characters as reduced holomorphic-coordinate data, and fixed-charge spectra as large-charge asymptotic data for the effective theory on charged regions of the branch.

Chapter 94

Two-Dimensional Supersymmetric Models

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter begins the lower-dimensional supersymmetric examples. The main objects are $2D$ $\mathcal{N} = (2,2)$ Landau–Ginzburg models, sigma models, and gauged linear sigma models. Each is treated as a QFT presentation with declared fields, symmetries, operator coordinates, and infrared questions.

94.1 The N=(2,2) Algebra

In Lorentzian signature let $P_\pm = P_0 \pm P_1$. The supercharges

$$Q_\pm, \quad \bar{Q}_\pm$$

obey

$$\{Q_+, \bar{Q}_+\} = 2P_+, \qquad \{Q_-, \bar{Q}_-\} = 2P_-,$$

with all other anticommutators vanishing in the absence of central charges. An R-symmetry assignment consists of vector and axial charges

$$F_V, \qquad F_A,$$

whose action on supercharges is part of the model specification. A BPS state with central charge Z obeys an inequality obtained by positivity of $(Q+\mathrm{e}^{\mathrm{i}\alpha}\bar{Q})^\dagger(Q+\mathrm{e}^{\mathrm{i}\alpha}\bar{Q})$; the saturated states are kernels of a corresponding linear combination of supercharges.

94.2 Landau–Ginzburg Presentations

Definition 94.1 (Two-dimensional Landau–Ginzburg presentation)**.** A polynomial $2D\,\mathcal{N}=(2,2)$ Landau–Ginzburg presentation in this chapter is specified by:

1. a Euclidean two-manifold Σ, initially a smooth oriented surface with a chosen spin structure when fermions are kept explicitly;
2. chiral multiplet coordinates
$$\Phi^i=(\phi^i,\psi_+^i,\psi_-^i,F^i),\qquad i=1,\dots,n,$$
where $\phi^i:\Sigma\to\mathbb{C}$ are scalar field variables, $\psi_\pm^i$ are fermionic field variables of right- and left-moving chirality, and F^i are auxiliary even field variables;
3. a Kähler potential K whose Hessian
$$g_{i\bar{j}}(\phi,\bar{\phi})=\partial_i\partial_{\bar{j}}K(\phi,\bar{\phi})$$
is positive on the field chart used in the perturbative construction;
4. a holomorphic polynomial superpotential
$$W\in\mathbb{C}[\phi^1,\dots,\phi^n].$$

This is a field-variable presentation. A continuum QFT requires a regulator, a supersymmetric Ward identity, and a Hilbert-space or Euclidean reconstruction; none of these is inferred from the polynomial W alone.

Eliminating the auxiliary fields is a finite-dimensional calculation at each point of Σ. In the normalization where the auxiliary part of the Euclidean Lagrangian is
$$g_{i\bar{j}}F^i\bar{F}^{\bar{j}}+F^i\partial_iW+\bar{F}^{\bar{j}}\partial_{\bar{j}}\overline{W},$$
the algebraic equation of motion is
$$g_{i\bar{j}}\bar{F}^{\bar{j}}+\partial_iW=0.$$
Substituting this equation gives the bosonic potential
$$V(\phi)=g^{i\bar{j}}\partial_iW(\phi)\,\overline{\partial_jW(\phi)}.$$
Thus supersymmetric classical vacua in the affine field chart solve
$$\partial_iW(\phi)=0,\qquad i=1,\dots,n.$$

Definition 94.2 (Quasihomogeneous isolated Landau–Ginzburg polynomial)**.** Let $q_i \in \mathbb{Q}_{>0}$. A polynomial

$$W \in \mathbb{C}[\phi^1, \dots, \phi^n]$$

is quasihomogeneous of degree 1 with weights q_i when

$$W(\lambda^{q_1}\phi^1, \dots, \lambda^{q_n}\phi^n) = \lambda W(\phi^1, \dots, \phi^n) \qquad \lambda \in \mathbb{C}^\times. \tag{94.1}$$

It is isolated when the common zero locus of

$$\partial_1 W, \dots, \partial_n W$$

in $\mathbb{C}^n$ is the single point 0. The algebraic chiral coordinate ring attached to the polynomial is the Jacobi algebra

$$\mathrm{Jac}(W) = \mathbb{C}[\phi^1, \dots, \phi^n]/(\partial_1 W, \dots, \partial_n W). \tag{94.2}$$

The term "algebraic chiral coordinate ring" is deliberately narrower than "the infrared chiral ring of the continuum QFT". It is the finite algebraic object forced by the F-term equations and by the protected cohomological argument below. The statement that a regulated Landau–Ginzburg model flows to a compact $\mathcal{N} = 2$ superconformal field theory with this ring as its chiral-primary ring is an additional infrared construction problem.

Jacobian quotient from the B-type differential. Let $A = \mathbb{C}[\phi^1, \dots, \phi^n]$ and let $I_W \subset A$ be the ideal generated by $\partial_i W$. Assume that the regulated B-type local operator complex contains odd local variables η_i, $i = 1, \dots, n$, and a scalar odd differential Q_B satisfying

$$Q_B \phi^i = 0, \qquad Q_B \eta_i = \partial_i W(\phi), \qquad Q_B^2 = 0 \tag{94.3}$$

on polynomial local coordinates. Then the image of Q_B inside the polynomial subalgebra A is exactly I_W. Hence the polynomial Q_B-cohomology in degree zero is canonically

$$H^0_{Q_B}\big(A \otimes \Lambda(\eta_1, \dots, \eta_n)\big) \cong \mathrm{Jac}(W).$$

Every polynomial $f(\phi) \in A$ is Q_B-closed because $Q_B \phi^i = 0$ and Q_B is a derivation. Conversely, the Q_B-exact polynomials are generated by applying Q_B to degree-one expressions

$$\sum_i h^i(\phi)\eta_i, \qquad h^i \in A.$$

Using (94.3),

$$Q_B\left(\sum_i h^i(\phi)\eta_i\right) = \sum_i h^i(\phi)\partial_i W(\phi),$$

because $Q_B h^i(\phi) = 0$. Thus $\operatorname{im} Q_B \cap A = I_W$, and quotienting the closed polynomial operators by exact polynomial operators gives (94.2).

Quasihomogeneous charge and central-charge test. Let W be a quasihomogeneous isolated polynomial in the sense of Definition 94.2. Assign the monomial

$$\phi^a = (\phi^1)^{a_1} \cdots (\phi^n)^{a_n}$$

the charge

$$Q_R(\phi^a) = \sum_{i=1}^{n} a_i q_i. \tag{94.4}$$

Then every monomial appearing in W has charge 1, while every monomial appearing in $\partial_i W$ has charge $1 - q_i$. If, in addition,

$$W = \sum_{i=1}^{n} (\phi^i)^{d_i}, \qquad d_i \geq 2, \tag{94.5}$$

then $q_i = 1/d_i$, the Jacobi algebra has basis

$$\prod_{i=1}^{n} (\phi^i)^{a_i}, \qquad 0 \leq a_i \leq d_i - 2, \tag{94.6}$$

and dimension

$$\dim_{\mathbb{C}} \mathrm{Jac}(W) = \prod_{i=1}^{n} (d_i - 1). \tag{94.7}$$

The protected $U(1)_R$-anomaly prediction for a compact unitary infrared $\mathcal{N} = 2$ superconformal fixed point is

$$c_{\mathrm{LG}} = 3 \sum_{i=1}^{n} (1 - 2q_i). \tag{94.8}$$

In this chapter (94.8) is used as a necessary protected-data test for any proposed compact unitary infrared fixed point with these chiral fields and this $U(1)_R$ current.

Equation (94.1) says precisely that every monomial in W has total charge 1. Differentiating a monomial with respect to ϕ^i lowers its charge by q_i, hence the terms in $\partial_i W$ have charge $1 - q_i$.

For the Fermat polynomial (94.5), quasihomogeneity gives $d_i q_i = 1$, so $q_i = 1/d_i$. The Jacobi ideal is

$$((\phi^1)^{d_1 - 1}, \ldots, (\phi^n)^{d_n - 1}),$$

up to multiplication by the nonzero constants d_i. Reduction modulo this monomial ideal removes every factor with exponent $a_i \geq d_i - 1$, and the remaining monomials (94.6) are linearly independent in the quotient. Counting the allowed exponents gives (94.7).

The final formula is the protected anomaly calculation used by the $\mathcal{N} = 2$ superconformal-algebra interface in Chapter 70. The present paragraph records its algebraic inputs and status: once a continuum infrared fixed point with these chiral fields and $U(1)_R$ current has been constructed, the anomaly supplies (94.8).

This is therefore an algebraic and anomaly-matching datum. The existence of the infrared fixed point and the convergence of a Landau–Ginzburg regulator to it are separate construction and RG questions.

Wilsonian superpotential coordinate for Fermat LG models. The polynomial W is treated here as part of the Wilsonian F-term coordinate of a declared regulated theory. The following finite argument is the two-dimensional version of the holomorphic-coordinate logic of Chapter 89.

Fix

$$W_{\mathrm{UV}}(\Phi; g) = \sum_{i=1}^{n} g_i (\Phi^i)^{d_i}, \qquad d_i \geq 2,$$

and assume a Wilsonian scheme at scale Λ with the following properties:

1. the chiral superspace coordinate is preserved, so corrections to the superpotential are holomorphic functions of the chiral fields and of the background chiral couplings g_i;
2. the patch is regular at $g_i = 0$, so the perturbative Wilsonian expansion contains nonnegative powers of the g_i;
3. the scheme preserves the n flavor spurion symmetries

   $$\Phi^i \mapsto \lambda_i \Phi^i, \qquad g_i \mapsto \lambda_i^{-d_i} g_i, \qquad \lambda_i \in \mathbb{C}^\times,$$

   together with the quasihomogeneous R-charge convention

   $$R(\Phi^i) = \frac{1}{d_i}, \qquad R(g_i) = 0, \qquad R(W) = 1.$$

Then every regular monomial allowed in the Wilsonian superpotential is one of the original monomials $g_i(\Phi^i)^{d_i}$, up to its coefficient. Indeed, write a candidate monomial as

$$\prod_{i=1}^{n} g_i^{m_i} (\Phi^i)^{a_i}, \qquad m_i, a_i \in \mathbb{Z}_{\geq 0}.$$

The i-th spurion symmetry requires

$$a_i - d_i m_i = 0 \qquad \text{for every } i, \tag{94.9}$$

and the chiral R-charge condition requires

$$\sum_{i=1}^{n} \frac{a_i}{d_i} = 1. \tag{94.10}$$

Substituting (94.9) into (94.10) gives

$$\sum_{i=1}^{n} m_i = 1.$$

Since the m_i are nonnegative integers, exactly one m_j is 1 and all other m_i vanish. Then $a_j = d_j$ and $a_i = 0$ for $i \neq j$. Thus the holomorphic selection rule alone permits only

$$g_j (\Phi^j)^{d_j}, \qquad j = 1, \ldots, n.$$

The scope of the argument is the Wilsonian coordinate at the declared scale: after a regulator with the stated chiral and spurion Ward identities has been supplied, perturbative local corrections preserve the Fermat superpotential coordinate up to finite redefinition of the displayed couplings. Existence of the continuum infrared fixed point, convergence of the RG flow, and control of any allowed nonperturbative sectors are separate analytic inputs. The Kähler potential and other D-term coordinates may still flow.

The same protected-data interface controls the finite χ_y shadow of the elliptic genus in Section 70.5. If the infrared fixed point exists and the Jacobi algebra classes become NS chiral primaries, then spectral flow by $\eta = -1/2$ sends a monomial of charge Q_R to a left Ramond ground-state contribution of charge

$$Q_R - \frac{c_{\mathrm{LG}}}{6}. \tag{94.11}$$

Thus the finite polynomial $\sum_{[O]\in \mathrm{Jac}(W)} y^{Q_R(O)-c_{\mathrm{LG}}/6}$ is a necessary protected check on the proposed SCFT. The full elliptic genus requires the compact Ramond trace and modular sewing hypotheses stated in the CFT volume.

Example 94.3 (*A*-series and quintic polynomial checks)**.** For $W = X^{k+2}$, $k \geq 0$, one has $q = 1/(k+2)$,

$$\mathrm{Jac}(W) = \mathbb{C}[X]/(X^{k+1}), \qquad c_{\mathrm{LG}} = 3\left(1 - \frac{2}{k+2}\right) = \frac{3k}{k+2}.$$

For

$$W = X_1^5 + \cdots + X_5^5$$

the weights are $q_i = 1/5$, the algebraic Jacobi dimension is

$$4^5 = 1024,$$

and the protected central-charge test gives

$$c_{\mathrm{LG}} = 3 \cdot 5\left(1 - \frac{2}{5}\right) = 9.$$

This is an intrinsic Landau–Ginzburg charge and Jacobi-algebra calculation.

94.3 Sigma-Model Datum

A $2D$ $\mathcal{N} = (2,2)$ nonlinear sigma model is specified by a Kähler target X, metric $g_{i\bar{j}}$, B-field B, and possibly a superpotential W. The bosonic Euclidean action is

$$S_{\mathrm{bos}} = \frac{1}{4\pi\alpha'}\int_\Sigma \mathrm{d}^2x\, g_{i\bar{j}}(\phi)\partial_\mu\phi^i\partial_\mu\bar{\phi}^{\bar{j}} + \frac{\mathrm{i}}{2\pi\alpha'}\int_\Sigma \phi^*B.$$

The large-volume perturbative beta tensor begins with the Ricci form:

$$\mu\frac{\mathrm{d}g_{i\bar{j}}}{\mathrm{d}\mu} = \frac{1}{2\pi}R_{i\bar{j}} + \cdots .$$

Thus Ricci-flatness is the first condition for a conformal infrared candidate in a large-volume chart. The full QFT claim includes instanton corrections, operator renormalization, and a continuum limit; these are additional data, separate from the local tensor equation.

94.4 A And B Twisting Data

The two cohomological twists of a $2D$ $\mathcal{N} = (2,2)$ theory are intrinsic operations on the two-dimensional QFT. They require nonanomalous R-symmetries and a regulator preserving the selected scalar supercharge. The definition uses the two-dimensional rotation symmetry, the selected R-current, and the supersymmetry algebra.

Definition 94.4 (Two-dimensional twist convention)**.** Let s denote the Lorentz spin for $U(1)_{\mathrm{rot}}$. In this chapter the supercharges have charges

$$\begin{array}{c|cccc} & Q_+ & \bar{Q}_+ & Q_- & \bar{Q}_- \\ \hline s & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ F_V & 1 & -1 & 1 & -1 \\ F_A & 1 & -1 & -1 & 1. \end{array} \tag{94.12}$$

The A twist uses

$$s_A = s + \frac{1}{2}F_V, \tag{94.13}$$

and the B twist uses

$$s_B = s + \frac{1}{2}F_A. \tag{94.14}$$

Remark 94.5 (Twist bookkeeping: scalar supercharges)*.* Assume the charge convention of Definition 94.4 and assume that there are no central charges in the local-sector supersymmetry algebra. Then:

$$Q_A = \bar{Q}_+ + Q_- \tag{94.15}$$

is scalar for the A-twisted rotation action and satisfies $Q_A^2 = 0$ on gauge-invariant local observables, while

$$Q_B = \bar{Q}_+ + \bar{Q}_- \tag{94.16}$$

is scalar for the B-twisted rotation action and satisfies $Q_B^2 = 0$ on gauge-invariant local observables.

Using (94.13), the A-twisted spins are

$$\begin{array}{c|cccc} & Q_+ & \bar{Q}_+ & Q_- & \bar{Q}_- \\ \hline s_A & 1 & 0 & 0 & -1. \end{array}$$

Thus $\bar{Q}_+$ and Q_- are scalars, while the other two supercharges are not. Using (94.14), the B-twisted spins are

$$\begin{array}{c|cccc} & Q_+ & \bar{Q}_+ & Q_- & \bar{Q}_- \\ \hline s_B & 1 & 0 & -1 & 0. \end{array}$$

Thus $\bar{Q}_+$ and $\bar{Q}_-$ are scalars for the B twist. The untwisted local algebra has only

$$\{Q_+, \bar{Q}_+\} = 2P_+, \qquad \{Q_-, \bar{Q}_-\} = 2P_-$$

among these generators. Therefore

$$Q_A^2 = \frac{1}{2}\{\bar{Q}_+ + Q_-, \bar{Q}_+ + Q_-\} = 0, \qquad Q_B^2 = \frac{1}{2}\{\bar{Q}_+ + \bar{Q}_-, \bar{Q}_+ + \bar{Q}_-\} = 0$$

in the central-charge-free local sector. If gauge transformations appear in an off-shell realization, the same statement is made after passing to gauge-invariant observables.

94.5 Gauged Linear Sigma-Model Presentations

Definition 94.6 (Abelian $2D$ $\mathcal{N} = (2,2)$ GLSM presentation)**.** An abelian GLSM presentation in this chapter is specified by:

1. gauge group $G = U(1)$;
2. chiral multiplets Φ_i, $i = 1, \dots, m$, with integer charges $Q_i \in \mathbb{Z}$;
3. a $U(1)$-invariant holomorphic superpotential $W \in \mathbb{C}[\Phi_1, \dots, \Phi_m]^{U(1)}$;
4. a complexified FI-theta coordinate
$$t = \frac{\theta}{2\pi} + \mathrm{i}r, \qquad r \in \mathbb{R}, \quad \theta \in \mathbb{R}/2\pi\mathbb{Z};$$
5. a gauge coupling $e > 0$ and a specified ultraviolet regulator for the gauge theory.

The classical scalar potential contains

$$\frac{e^2}{2}\left(\sum_{i=1}^{m} Q_i|\phi_i|^2 - r\right)^2 + \sum_i |\partial_i W|^2 + |\sigma|^2 \sum_i Q_i^2 |\phi_i|^2, \tag{94.17}$$

where σ is the complex scalar in the vector multiplet. Classical Higgs vacua with at least one charged scalar nonzero therefore solve the F-term equations, the D-term equation, and then are quotiented by $U(1)$.

The axial R-symmetry anomaly coefficient of this abelian presentation is

$$\mathcal{A}_A = \sum_{i=1}^{m} Q_i. \tag{94.18}$$

This coefficient is a finite charge sum: each charged chiral multiplet contributes its charge to the axial rotation of the fermion measure. When $\mathcal{A}_A = 0$, the one-loop axial shift of the FI coordinate vanishes. This is a necessary condition for an axial R-symmetry in an infrared superconformal candidate. A conformality claim additionally requires a construction of the infrared continuum theory and its stress tensor.

The vector and axial R-symmetries have different anomaly requirements. The A twist is available only when F_V is a genuine symmetry of the regulated theory. The B twist is available only when F_A is a genuine symmetry. In the abelian GLSM presentation of Definition 94.6, the axial anomaly obstruction is the charge sum (94.18). Therefore the hypersurface GLSM with charges $(1, \dots, 1, -d)$ has an axial B-twist candidate precisely when $N = d$.

94.6 B-Twisted Landau–Ginzburg Sector

The B-twisted Landau–Ginzburg model uses the scalar supercharge (94.16). The Jacobian-quotient calculation above gives the first protected observable: polynomial local operators map to $\mathrm{Jac}(W)$. The next finite-dimensional check is the residue pairing at nondegenerate critical points.

Morse residue pairing on the LG Jacobi algebra. Let $W \in \mathbb{C}[\phi^1, \ldots, \phi^n]$ have a finite critical set

$$\mathrm{Crit}(W) = \{p \in \mathbb{C}^n : \partial_i W(p) = 0 \text{ for all } i\}.$$

Assume every critical point is nondegenerate:

$$H_W(p) = \det(\partial_i \partial_j W)(p) \neq 0.$$

Define

$$\mathrm{Res}_W(f) = \sum_{p \in \mathrm{Crit}(W)} \frac{f(p)}{H_W(p)}, \qquad f \in \mathbb{C}[\phi^1, \ldots, \phi^n]. \tag{94.19}$$

Then Res_W kills the ideal $(\partial_1 W, \ldots, \partial_n W)$, so it descends to $\mathrm{Jac}(W)$. The bilinear form

$$\langle [f], [g] \rangle_{\mathrm{res}} = \mathrm{Res}_W(fg) \tag{94.20}$$

on $\mathrm{Jac}(W)$ is nondegenerate.

If $f = \sum_i h_i \partial_i W$, then $f(p) = 0$ at every critical point p, so $\mathrm{Res}_W(f) = 0$. Hence the functional descends to the quotient. Nondegeneracy of the Hessian means that each zero of the gradient map

$$\nabla W : \mathbb{C}^n \longrightarrow \mathbb{C}^n, \qquad \phi \longmapsto (\partial_1 W, \ldots, \partial_n W)$$

is simple. The local algebra of ∇W at such a zero is therefore $\mathbb{C}$. Since the critical set is finite, the Chinese-remainder decomposition gives

$$\mathrm{Jac}(W) \cong \prod_{p \in \mathrm{Crit}(W)} \mathbb{C},$$

where a class $[f]$ is sent to its values $f(p)$. In these coordinates (94.20) is the diagonal pairing with nonzero diagonal entries $H_W(p)^{-1}$, hence is nondegenerate.

For a compact Riemann surface of genus g, the B-twisted Landau–Ginzburg zero-mode formula used in this chapter is

$$\langle f_1 \cdots f_m \rangle_g = \sum_{p \in \mathrm{Crit}(W)} \frac{f_1(p) \cdots f_m(p)}{H_W(p)^{1-g}}. \tag{94.21}$$

In this monograph (94.21) has the following status: it is the finite-dimensional zero-mode and determinant calculation after the Q_B-preserving regulator, determinant-line orientation, and absence of boundary terms have been supplied. The continuum QFT assertion is conditional on those analytic inputs.

94.7 A-Twisted Sigma-Model Zero Modes

Let X be a smooth Kähler target with real tangent bundle TX. The A-twisted sigma model uses the scalar supercharge (94.15). The full model includes nonconstant maps $\Sigma \to X$, instanton sectors, compactification of moduli spaces, and operator contact terms. The constant-map sector, however, is a finite-dimensional calculation.

In the constant-map truncation of an A-twisted sigma model with target X, let x^a be local real coordinates on X and let χ^a be odd variables transforming as dx^a. Suppose the scalar differential acts by

$$Q_A x^a = \chi^a, \qquad Q_A \chi^a = 0. \tag{94.22}$$

Then the algebra of polynomial functions in the odd variables identifies with differential forms on X, and Q_A identifies with the de Rham differential d_X. Thus the constant-map local Q_A-cohomology is

$$H^\bullet_{Q_A} \cong H^\bullet_{\mathrm{dR}}(X;\mathbb{C}).$$

Functions of x^a and the odd variables χ^a are local functions on the parity-shifted tangent bundle ΠTX. The coordinate identification

$$\chi^{a_1}\cdots\chi^{a_k} \longleftrightarrow dx^{a_1}\wedge\cdots\wedge dx^{a_k}$$

identifies these functions with differential forms. Under this identification the derivation determined by (94.22) is

$$Q_A = \chi^a \frac{\partial}{\partial x^a},$$

which is precisely the de Rham differential in local coordinates. Its cohomology is therefore de Rham cohomology after gluing the coordinate charts.

The nonconstant A-model sectors are classified, at the classical level, by homology classes $\beta = \phi_*[\Sigma] \in H_2(X,\mathbb{Z})$. For a Kähler metric with form ω, the bosonic energy admits the pointwise Bogomolny decomposition

$$E(\phi) = \int_\Sigma |\bar\partial_J \phi|^2 \,\mathrm{vol}_\Sigma + \int_\Sigma \phi^*\omega \tag{94.23}$$

in the normalization of the sigma-model action. Thus holomorphic maps $\bar\partial_J\phi = 0$ minimize the energy in a fixed homology class. A mathematical A-model correlation function requires compactifying the holomorphic-map moduli space, orienting its virtual determinant line, and controlling collisions of marked points. These are theorem-level inputs, additional to the twist construction.

94.8 Abelian Duality And Circle T-Duality

Abelian duality is used here only as a statement about two-dimensional QFT. The local calculation is a first-order path integral for a compact scalar and, in the $\mathcal{N} = (2,2)$ extension, the corresponding Legendre transform between chiral and twisted-chiral field variables. The global QFT statement requires the compact path-integral measure, spin structures, contact terms, and topological sectors to be specified.

Hypothesis 94.7 (Compact scalar first-order duality setup)**.** Let Σ be a closed oriented Euclidean surface. A compact scalar $\vartheta : \Sigma \to \mathbb{R}/2\pi\mathbb{Z}$ of radius $R > 0$ has action

$$S_R(\vartheta) = \frac{R^2}{4\pi}\int_\Sigma d\vartheta \wedge * d\vartheta. \tag{94.24}$$

The functional integral sums over all winding sectors $[d\vartheta/2\pi] \in H^1(\Sigma;\mathbb{Z})$. The first-order presentation uses a real one-form A with the same integral-period normalization and a dual periodic scalar $\widetilde{\vartheta} \sim \widetilde{\vartheta} + 2\pi$:

$$S_{\text{fo}}(A, \widetilde{\vartheta}) = \frac{R^2}{4\pi} \int_\Sigma A \wedge *A + \frac{\mathrm{i}}{2\pi} \int_\Sigma \widetilde{\vartheta}\, dA. \tag{94.25}$$

The measure is assumed to be chosen so that integrating over $\widetilde{\vartheta}$ imposes flatness with integral periods and integrating over A is the corresponding Gaussian integral along the standard Euclidean contour. Field-independent determinant constants are absorbed in the normalization of the partition function.

Compact scalar radius inversion. Under Hypothesis 94.7, the local first-order duality maps the radius R compact scalar to a compact scalar of radius $R^\vee = R^{-1}$:

$$S_{R^\vee}(\widetilde{\vartheta}) = \frac{1}{4\pi R^2} \int_\Sigma d\widetilde{\vartheta} \wedge *d\widetilde{\vartheta}. \tag{94.26}$$

Applying the same local operation twice returns the original radius.

Integrating over $\widetilde{\vartheta}$ in (94.25) imposes $dA = 0$ together with the integral-period condition. Locally $A = d\vartheta$, and globally the integral periods reconstruct the winding sectors of (94.24).

For the other elimination, integrate the second term by parts:

$$\frac{\mathrm{i}}{2\pi} \int_\Sigma \widetilde{\vartheta}\, dA = -\frac{\mathrm{i}}{2\pi} \int_\Sigma d\widetilde{\vartheta} \wedge A.$$

The Gaussian equation for A is

$$R^2 * A = \mathrm{i}\, d\widetilde{\vartheta}, \qquad A = \frac{\mathrm{i}}{R^2} * d\widetilde{\vartheta}, \tag{94.27}$$

with the usual Euclidean contour understood. Substituting the saddle and absorbing the field-independent determinant gives (94.26). Therefore the dual radius is R^{-1}, and the second duality sends R^{-1} back to R.

The Hamiltonian spectrum supplies a regulator-independent arithmetic check of the same radius inversion. On a spatial circle, write the momentum and winding labels as

$$(n, w) \in \mathbb{Z}^2.$$

In the convention of (94.24), the left and right momenta are

$$p_L(n, w; R) = \frac{n}{R} + wR, \qquad p_R(n, w; R) = \frac{n}{R} - wR. \tag{94.28}$$

The duality map

$$R \longmapsto R^{-1}, \qquad (n, w) \longmapsto (w, n) \tag{94.29}$$

preserves $p_L^2 + p_R^2$ and flips the sign of p_R:

$$p_L(w, n; R^{-1}) = p_L(n, w; R), \qquad p_R(w, n; R^{-1}) = -p_R(n, w; R). \tag{94.30}$$

Thus the finite oscillator spectrum and the zero-mode quadratic form are compatible with the first-order path integral. A proof of equality of all correlation functions must realize this exact arithmetic inside a specified compact path-integral measure and operator-insertion prescription.

Definition 94.8 (Local $\mathcal{N} = (2,2)$ abelian Legendre duality)**.** Let $\mathcal{B}$ be a real $\mathcal{N} = (2,2)$ superfield coordinate for a shift-isometry direction, and let Y be a twisted-chiral superfield whose imaginary part is periodic. Locally, suppressing normalization constants fixed by the component convention, a first-order superspace functional has the form

$$\int d^2x\, d^4\theta\, \big(\mathcal{B}\, u - K(\mathcal{B})\big), \qquad u = Y + \bar{Y}, \tag{94.31}$$

where $K''(\mathcal{B}) > 0$. Eliminating Y enforces the local constraint $\mathcal{B} = \Phi + \bar{\Phi}$ for a chiral field Φ. Eliminating $\mathcal{B}$ instead uses

$$u = K'(\mathcal{B}) \tag{94.32}$$

and gives the Legendre-dual Kähler potential

$$\widetilde{K}(u) = \mathcal{B}(u)u - K(\mathcal{B}(u)). \tag{94.33}$$

Remark 94.9 (Hessian inversion in local abelian duality)*.* Assume Definition 94.8. Then

$$\widetilde{K}''(u) = \frac{1}{K''(\mathcal{B}(u))}. \tag{94.34}$$

In particular, the free quadratic potential

$$K(\mathcal{B}) = \frac{R^2}{2}\mathcal{B}^2$$

is dual to

$$\widetilde{K}(u) = \frac{1}{2R^2}u^2,$$

the supersymmetric version of the radius inversion $R^2 \mapsto R^{-2}$.

Differentiate (94.33). Since $u = K'(\mathcal{B}(u))$,

$$\frac{d\widetilde{K}}{du} = \mathcal{B}(u) + u\mathcal{B}'(u) - K'(\mathcal{B}(u))\mathcal{B}'(u) = \mathcal{B}(u).$$

Differentiating (94.32) gives

$$1 = K''(\mathcal{B}(u))\mathcal{B}'(u),$$

and hence $\widetilde{K}''(u) = \mathcal{B}'(u) = 1/K''(\mathcal{B}(u))$. For $K(\mathcal{B}) = R^2\mathcal{B}^2/2$, equation (94.32) gives $\mathcal{B} = u/R^2$, and (94.33) gives $\widetilde{K}(u) = u^2/(2R^2)$.

For a charged chiral field in an abelian GLSM, the same local dualization couples the dual twisted-chiral variable linearly to the abelian field strength Σ, with coefficient equal to the integer gauge charge. The perturbative local calculation is therefore compatible with the GLSM charge matrix. Nonperturbative exponential terms in the dual twisted superpotential, when present, require a separate vortex-instanton calculation: compactness of the vortex moduli space, fermion zero-mode saturation, determinant-line orientation, and boundary control must be supplied before such terms are promoted from semiclassical input to a QFT statement.

94.9 Charged-Chiral Dual Variables and Abelian Mirror Presentations

The preceding Legendre transform is the local part of the abelian GLSM mirror construction. The full statement contains one additional ingredient: a holomorphic exponential term in the twisted superpotential. The Gaussian Legendre transform supplies the dual variables and their linear couplings to field strengths. The exponential term is a nonperturbative datum, attributed in the semiclassical presentation to finite-action vortex sectors, and any regulator-level equality of QFTs must derive it from a compactness, zero-mode, determinant-line, and boundary analysis of those sectors. The algebraic consequences of this datum are completely explicit and will be used below.

Consider an abelian GLSM with gauge group $U(1)^s$, twisted chiral field strengths Σ_a, complexified FI-theta coordinates t_a, and chiral multiplets Φ_i, $i = 1, \ldots, m$, of integer charges

$$Q_i{}^a \in \mathbb{Z}, \qquad a = 1, \ldots, s.$$

For each charged chiral multiplet introduce a periodic twisted chiral variable

$$Y_i \sim Y_i + 2\pi\mathrm{i}.$$

Repeated abelian gauge indices are summed in this section.

The local first-order functional for one charged chiral variable is the superspace analogue of the compact scalar construction. Let B_i be a real superfield and set

$$U_i = Y_i + \overline{Y}_i. \tag{94.35}$$

The D-term

$$\int d^2x\, d^4\theta \left[e^{B_i + 2Q_i{}^a V_a} - \frac{1}{2} U_i B_i\right] \tag{94.36}$$

is invariant under

$$V_a \mapsto V_a + \mathrm{i}(\Lambda_a - \overline{\Lambda}_a), \qquad B_i \mapsto B_i - 2\mathrm{i}Q_i{}^a(\Lambda_a - \overline{\Lambda}_a), \qquad Y_i \mapsto Y_i.$$

Integrating over Y_i imposes the twisted linear constraint on B_i. In components of superspace this is the pair of constraints

$$\overline{D}_+ D_- B_i = 0, \qquad D_+ \overline{D}_- B_i = 0.$$

Locally this says

$$B_i = \Psi_i + \overline{\Psi}_i$$

with Ψ_i a chiral superfield. Then

$$e^{B_i + 2Q_i{}^a V_a} = \overline{\Phi}_i \exp(2Q_i{}^a V_a) \Phi_i, \qquad \Phi_i = e^{\Psi_i},$$

which recovers the original charged chiral kinetic term in that field chart.

Eliminating B_i instead gives the algebraic equation

$$e^{B_i + 2Q_i{}^a V_a} = \frac{1}{2} U_i, \qquad B_i = \log \frac{U_i}{2} - 2Q_i{}^a V_a. \tag{94.37}$$

Substitution gives a dual D-term of Legendre form, together with the linear V_aU_i term

$$Q_i{}^a \int d^2x\, d^4\theta\, V_a(Y_i + \overline{Y}_i).$$

The superspace integration-by-parts identity in the present convention is

$$\int d^4\theta\, V_a(Y_i + \overline{Y}_i) = -\int d^2\widetilde{\theta}\, \Sigma_a Y_i + \text{c.c.}, \tag{94.38}$$

where Y_i is twisted chiral and $\Sigma_a = \overline{D}_+ D_- V_a$. Thus the local dualization produces the twisted F-term

$$-\int d^2\widetilde{\theta}\, Q_i{}^a \Sigma_a Y_i + \text{c.c.}$$

before any vortex effect is invoked. This derivation fixes the charge matrix in the dual variables: the integer $Q_i{}^a$ that couples Φ_i to V_a is the same integer multiplying $\Sigma_a Y_i$.

The convention in the rest of this section is chosen so that eliminating the Y_i matches the one-loop Coulomb superpotential (94.63). Put

$$M_i(\Sigma) = \sum_{a=1}^{s} Q_i{}^a \Sigma_a. \tag{94.39}$$

The dual twisted-superpotential ansatz is

$$\widetilde{W}_{\text{dual}}(\Sigma, Y) = -\sum_{a=1}^{s} t_a \Sigma_a - \sum_{i=1}^{m} M_i(\Sigma) Y_i - \mu \sum_{i=1}^{m} c_i e^{-Y_i}, \qquad c_i \in \mathbb{C}^\times. \tag{94.40}$$

The constants c_i depend on the normalization of the dual variables and on the vortex determinant convention; rescaling e^{-Y_i} shifts the corresponding logarithmic branch and changes t_a by a finite affine scheme transformation. The intrinsic data are the charge matrix, the periodicity of the Y_i, and the fact that the exponential monomial is nonzero with the charge dictated by the dualized chiral field.

The local dualization has not derived the exponential term. It has derived the periodic variable Y_i, the linear coupling $-M_i Y_i$, and the charge normalization. The remaining input is a protected connected amplitude on the asymptotic end of the dual cylinder. Because Y_i is periodic with imaginary period $2\pi\mathrm{i}$, holomorphic local functions on that end are Laurent series in

$$X_i = e^{-Y_i}.$$

The large positive $\operatorname{Re} Y_i$ region is the weak-coupling end of the dualized chiral variable, so regularity there permits nonnegative powers of X_i. Suppose, for one dual field and on a simply connected branch where $M_i \neq 0$, that the connected protected correction is an ordinary holomorphic germ $h_i(X_i)$ with $h_i(0) = 0$, so that the relevant part of the twisted superpotential is

$$-M_i Y_i - h_i(X_i) = M_i \log X_i - h_i(X_i), \qquad X_i = e^{-Y_i}. \tag{94.41}$$

The critical equation is

$$M_i = X_i h_i'(X_i). \tag{94.42}$$

At a critical point the derivative of the eliminated superpotential is

$$\frac{\partial \widetilde{W}_{\mathrm{elim}}}{\partial M_i} = \log X_i(M_i). \tag{94.43}$$

Indeed, the coefficient of $\partial X_i/\partial M_i$ is $M_i/X_i - h_i'(X_i)$, which vanishes by (94.42). Exact agreement with the Coulomb one-loop logarithm

$$\frac{\partial}{\partial M_i}\left[M_i\left(\log\frac{M_i}{\mu c_i} - 1\right)\right] = \log\frac{M_i}{\mu c_i}$$

therefore forces $X_i(M_i) = M_i/(\mu c_i)$ on the branch. Substituting into (94.42) gives $h_i'(X_i) = \mu c_i$, hence

$$h_i(X_i) = \mu c_i X_i$$

up to an irrelevant additive constant. Thus the primitive monomial $-\mu c_i e^{-Y_i}$ is singled out by the combination of periodicity, regularity at the weak-coupling end, and exact Coulomb-branch matching. This argument is not a construction of the amplitude. It reduces the nonperturbative theorem boundary to the following sharp questions: construct the single-vortex sector in the chosen regulator, prove that its fermion zero modes leave precisely the twisted F-term measure, prove $c_i \neq 0$, and justify that possible higher connected harmonics do not contribute in the protected coordinate being matched.

Proposition 94.10 (Vortex zero-mode filter for twisted F-terms)**.** *In the finite-regulator charge-one sector used below, let $\mathcal{D}_{F,i,\Lambda}(z)$ be the fermion Hessian at a vortex configuration z. Assume its kernel has been split as*

$$\ker \mathcal{D}_{F,i,\Lambda}(z) = \Pi_{\mathrm{univ}}(z) \oplus K^{\mathrm{res}}_{i,\Lambda}(z), \qquad \dim \Pi_{\mathrm{univ}} = 2,$$

where Π_{univ} is the universal pair identified with the $d^2\widetilde{\theta}$ measure. Let

$$\nu^{\mathrm{res}}_{i,\Lambda}(z) = \dim K^{\mathrm{res}}_{i,\Lambda}(z).$$

After the universal pair has been removed, the residual fermion-zero-mode coefficient has the Berezin form

$$\mathcal{Z}^0_{i,F,\Lambda}(z;P) = \int \prod_{\alpha=1}^{\nu^{\mathrm{res}}_{i,\Lambda}(z)} \mathrm{d}\eta_\alpha\, P_{\mathrm{ins}}(z,\eta), \tag{94.44}$$

where P_{ins} denotes insertions or regulator-allowed zero-mode liftings. Hence the charge-one sector can contribute to the uninserted twisted superpotential only on the locus

$$\nu^{\mathrm{res}}_{i,\Lambda}(z) = 0.$$

If $\nu^{\mathrm{res}}_{i,\Lambda}(z) > 0$ and no insertion or lifting supplies the missing residual Grassmann degree, then $\mathcal{Z}^0_{i,F,\Lambda}(z;1) = 0$. The nonzero-mode determinant entering the coefficient is therefore the determinant on the complement of the full kernel of $\mathcal{D}_{F,i,\Lambda}(z)$, not the raw determinant before zero-mode removal.

Proof. The two universal zero modes are coordinates on the superspace measure of the twisted F-term, so they are not integrated into the scalar coefficient. All remaining fermion zero modes are ordinary Grassmann variables in the finite stationary-phase integral. Berezin integration extracts the coefficient of the top exterior monomial $\eta_1 \cdots \eta_{\nu^{\mathrm{res}}_{i,\Lambda}(z)}$. If the insertion polynomial has smaller residual Grassmann degree, that coefficient is absent and the integral vanishes. With no insertion or lifting, the polynomial is constant, so the only nonzero case is $\nu^{\mathrm{res}}_{i,\Lambda}(z) = 0$. Removing the full kernel before forming the determinant is the same collective-coordinate separation on the fermionic side as the bosonic zero-mode quotient in the vortex measure. $\square$

This is the amplitude gate hidden behind the compact mirror formula. The integer vortex number fixes the exponential e^{-Y_i}, and Coulomb matching fixes the primitive harmonic once a nonzero coefficient is present. Neither step proves that the regulated vortex sector has exactly the two universal fermion zero modes or that its determinant line has been oriented consistently.

Proposition 94.11 (Single-vortex coefficient as a protected amplitude)**.** *Fix one charged chiral multiplet Φ_i in the abelian GLSM and choose a supersymmetric finite regulator preserving the integer vortex number and the twisted chiral field strength. Let $\mathcal{U}_{i,1,\Lambda}$ be a smooth finite-dimensional chart for the charge-one vortex sector attached to Φ_i, after quotienting by gauge transformations and after choosing collective-coordinate coordinates z. Write $\mathrm{d}\nu_{i,1,\Lambda}(z)$ for the bosonic zero-mode density, $\mathcal{W}^{\mathrm{nz}}_{i,1,\Lambda}(z)$ for the nonzero-mode determinant ratio in that vortex background, and $\mathcal{Z}^0_{i,F,\Lambda}(z)$ for the Berezin coefficient left after the two universal fermion zero modes have been identified with $\mathrm{d}^2\widetilde{\theta}$, as filtered by Proposition 94.10. Then the charge-one sector contributes to the Wilsonian twisted F-term in the form*

$$\Delta \widetilde{W}^{(1)}_{i,\Lambda}(Y_i) = -\mu\, c_{i,\Lambda}\, e^{-Y_i}, \tag{94.45}$$

where

$$c_{i,\Lambda} = \mu^{-1} \int_{\mathcal{U}_{i,1,\Lambda}} e^{-S^{\mathrm{red}}_{i,1,\Lambda}(z)}\, \mathcal{W}^{\mathrm{nz}}_{i,1,\Lambda}(z)\, \mathcal{Z}^0_{i,F,\Lambda}(z)\, \mathrm{d}\nu_{i,1,\Lambda}(z). \tag{94.46}$$

Here $S^{\mathrm{red}}_{i,1,\Lambda}$ is the regulator action with the universal topological coupling to the dual variable Y_i factored out. If additional fermion zero modes remain unsaturated, then $\mathcal{Z}^0_{i,F,\Lambda} = 0$ and this vortex sector supplies no twisted-superpotential term. If the continuum limit of (94.46) *exists and is nonzero, then its limit is the coefficient c_i in* (94.40)*.*

Proof. At fixed regulator the semiclassical expansion around the charge-one vortex is a finite stationary-phase calculation. The dual periodic variable couples to vortex number, so the charge-one topological factor is e^{-Y_i}; the period $Y_i \sim Y_i + 2\pi\mathrm{i}$ fixes this monomial unambiguously. The remaining finite integral splits into the bosonic collective-coordinate density, the nonzero-mode determinant ratio, and the finite Berezin integral over fermion zero modes. Two universal fermion zero modes are precisely the superspace measure of a twisted F-term; every extra zero mode must be saturated by an insertion or lifted by an allowed deformation. This gives (94.46) and hence (94.45). Multiplying e^{-Y_i} by a nonzero finite constant rescales c_i, which is the same finite normalization freedom recorded above as an affine shift of the FI-theta coordinate. $\square$

Formula (94.46) isolates the data hidden in the coefficient c_i. The integer vortex-number coupling fixes the monomial e^{-Y_i}; the determinant ratio, the orientation of the two universal fermion zero modes, the saturation of possible additional zero modes, and the boundary behavior of $\mathcal{U}_{i,1,\Lambda}$ decide the value and possible vanishing of the coefficient. Agreement with the Coulomb logarithm is then a protected consistency test: it fixes the primitive harmonic and exposes any finite FI-coordinate shift caused by determinant normalization.

The Y_i-critical equation is

$$\partial_{Y_i}\widetilde{W}_{\text{dual}} = 0 \quad \Longleftrightarrow \quad e^{-Y_i} = \frac{M_i(\Sigma)}{\mu c_i}. \tag{94.47}$$

On a branch where $M_i(\Sigma) \neq 0$, substituting (94.47) gives

$$\widetilde{W}_{\text{dual}}\big|_{\partial_Y \widetilde{W}=0} = -\sum_{a=1}^{s} t_a \Sigma_a + \sum_{i=1}^{m} M_i(\Sigma)\left(\log\frac{M_i(\Sigma)}{\mu c_i} - 1\right). \tag{94.48}$$

For $s = 1$, $M_i = Q_i\Sigma$, and $c_i = 1$, this is precisely the charge-resolved form of (94.63), with the constants c_i and logarithm branches absorbed into the finite definition of the FI coordinate. Thus the charged-chiral dual variables give a nontrivial consistency check: after the exponential terms have been supplied, their elimination reproduces the protected Coulomb-branch determinant.

The finite shift of the FI coordinate is explicit. In the rank-one case, (94.48) is

$$-t\Sigma + \sum_i Q_i\Sigma\left(\log\frac{Q_i\Sigma}{\mu} - 1\right) - \Sigma\sum_i Q_i \log c_i.$$

The last term is affine in Σ, hence is the same as replacing

$$t \longmapsto t + \sum_i Q_i \log c_i.$$

This is why the constants c_i are convention data for the protected Coulomb matching unless one separately fixes a normalization of the vortex measure.

Proposition 94.12 (Vortex normalization and all-rank FI coordinates)**.** *For an abelian gauge group $U(1)^s$, fix nonzero vortex coefficients c_i and choices of $\log c_i$. Set*

$$\widehat{X}_i = c_i e^{-Y_i}.$$

Then the mirror superpotential has normalized primitive terms

$$-\mu\sum_i \widehat{X}_i,$$

while the logarithmic-torus constraints are written in the shifted FI coordinates

$$\prod_{i=1}^{m} \widehat{X}_i^{{Q_i}^a} = \exp\left(t_a + \sum_{i=1}^{m} {Q_i}^a \log c_i\right), \qquad a = 1,\ldots,s. \tag{94.49}$$

Equivalently, in the Coulomb-eliminated expression the constants c_i replace

$$t_a \longmapsto t_a^{\text{phys}} = t_a + \sum_{i=1}^{m} {Q_i}^a \log c_i. \tag{94.50}$$

Proof. The constraint $\sum_i {Q_i}^a Y_i = -t_a$ is equivalent, on the logarithmic torus $X_i = e^{-Y_i}$, to

$$\prod_i X_i^{{Q_i}^a} = e^{t_a}.$$

Multiplying X_i by c_i gives

$$\prod_i \widehat{X}_i^{Q_i{}^a} = \left(\prod_i X_i^{Q_i{}^a}\right)\left(\prod_i c_i^{Q_i{}^a}\right),$$

which is (94.49). The same shift appears before imposing the low-energy constraint. Since

$$\sum_i M_i(\Sigma)\log c_i = \sum_i\sum_a Q_i{}^a\Sigma_a \log c_i = \sum_a \Sigma_a \sum_i Q_i{}^a \log c_i,$$

the c_i-dependent part of (94.48) is $-\sum_a \Sigma_a \sum_i Q_i{}^a \log c_i$, precisely the affine FI-theta shift in (94.50). □

Thus the Hori–Vafa expression is not a normalization-free statement of a bare path integral. Its invariant protected input is the charge lattice, the primitive nonzero vortex monomial for each allowed charge-one sector, and the FI-theta coordinate after the finite determinant normalization has been chosen. Changing the regulator normalization rescales the mirror torus coordinates and moves the same finite data into t_a^{phys}; it does not by itself prove the existence or nonvanishing of the vortex coefficient. Changing the branch by $\log c_i \mapsto \log c_i + 2\pi \mathrm{i} n_i$ changes t_a^{phys} by

$$2\pi\mathrm{i}\sum_i Q_i{}^a n_i, \qquad n_i \in \mathbb{Z},$$

which is an integral theta-period shift. The exponentiated constraints (94.49) are therefore the branch-independent protected statement.

There is a complementary low-energy use of the same expression. In a regime where the abelian vector multiplet has no independent low-energy propagating mode, Σ_a acts as a twisted chiral Lagrange multiplier. The Σ_a-critical equations of (94.40) are

$$\sum_{i=1}^m Q_i{}^a Y_i = -t_a, \qquad a = 1, \dots, s. \tag{94.51}$$

The dual presentation is then a Landau–Ginzburg model on the logarithmic torus with coordinates Y_i, subject to the linear constraints (94.51), and with twisted superpotential

$$-\mu \sum_i c_i e^{-Y_i}$$

restricted to that subspace. This is a presentation of protected twisted-chiral data. A full mirror equivalence requires a regulator, identification of the local and extended operator algebras, control of the singular loci and topological sectors, and convergence of the two continuum limits.

Example 94.13 ($\mathbb{P}^{N-1}$ dual critical points)**.** Take one $U(1)$ gauge factor and N chiral multiplets of charge 1. The constraint (94.51) is

$$Y_1 + \cdots + Y_N = -t.$$

Set $X_i = e^{-Y_i}$. Then

$$X_1 \cdots X_N = e^t.$$

For $c_i = 1$, the constrained twisted superpotential is proportional to

$$X_1 + \cdots + X_N$$

on the algebraic torus with fixed product e^t. Eliminating

$$X_N = \frac{e^t}{X_1 \cdots X_{N-1}}$$

gives the critical equations

$$X_a = \frac{e^t}{X_1 \cdots X_{N-1}}, \qquad a = 1, \ldots, N-1.$$

Thus all X_i are equal and

$$X_i^N = e^t.$$

For $e^t \neq 0$ there are N simple critical points, in agreement with the Coulomb-branch vacuum count in Example 94.17. The agreement is algebraic; it is the protected finite calculation that a full path-integral equivalence must reproduce after its regulator, local operators, and topological sectors have been specified.

Proposition 94.14 ($\mathbb{P}^{N-1}$ mirror residue and the quantum-product test)**.** *For the charge-one $\mathbb{P}^{N-1}$ GLSM, use the normalized mirror variables of Proposition 94.12 and write*

$$q_{\text{phys}} = \exp\left(t + \sum_{i=1}^{N} \log c_i\right) \neq 0. \tag{94.52}$$

On the logarithmic torus

$$\prod_{i=1}^{N} \widehat{X}_i = q_{\text{phys}}, \qquad \widetilde{W} = -\mu \sum_{i=1}^{N} \widehat{X}_i,$$

let H denote the protected hyperplane/Coulomb class represented in the mirror Jacobi quotient by $[\widehat{X}_1]$. With the determinant-line normalization that strips the universal factor $(-\mu)^{N-1}$ from the logarithmic Hessian, the mirror residue trace is

$$\mathcal{R}_{\mathbb{P}^{N-1}}(H^k) = \sum_{x^N = q_{\text{phys}}} \frac{x^k}{N x^{N-1}} = \begin{cases} q_{\text{phys}}^d, & k = N-1+dN, \quad d \geq 0, \\ 0, & k \not\equiv N-1 \pmod{N}. \end{cases} \tag{94.53}$$

Consequently the protected Frobenius algebra seen by this trace has

$$H^N = q_{\text{phys}}, \qquad \mathcal{R}_{\mathbb{P}^{N-1}}(H^{N-1}) = 1.$$

Proof. Use logarithmic coordinates

$$u_a = \log \widehat{X}_a, \qquad a = 1, \ldots, N-1,$$

and eliminate

$$\widehat{X}_N = q_{\text{phys}} \exp\left(-\sum_{a=1}^{N-1} u_a\right).$$

For the reduced function

$$S(u) = \sum_{a=1}^{N-1} e^{u_a} + q_{\mathrm{phys}} \exp\left(-\sum_{a=1}^{N-1} u_a\right),$$

the critical equations are

$$e^{u_a} = \widehat{X}_N, \qquad a = 1, \dots, N-1.$$

Thus every critical point has

$$\widehat{X}_1 = \cdots = \widehat{X}_N = x, \qquad x^N = q_{\mathrm{phys}}.$$

At such a point the logarithmic Hessian of S is

$$x(I_{N-1} + J_{N-1}),$$

where J_{N-1} is the all-ones matrix. Since $I_{N-1} + J_{N-1}$ has eigenvalues $1, \dots, 1, N$, its determinant is Nx^{N-1}. The factor from $\widetilde{W} = -\mu S$ is the universal $(-\mu)^{N-1}$ determinant-line factor already fixed in the trace normalization, leaving the denominator in (94.53).

It remains to evaluate the finite root sum. Put

$$\ell = k - (N-1).$$

Then

$$\sum_{x^N = q_{\mathrm{phys}}} \frac{x^k}{Nx^{N-1}} = \frac{1}{N} \sum_{x^N = q_{\mathrm{phys}}} x^\ell.$$

If ℓ is not divisible by N, the sum over the N-th roots vanishes. If $\ell = dN$ with $d \geq 0$, each critical point contributes $x^{dN} = q_{\mathrm{phys}}^d$, and the prefactor $1/N$ cancels the number of roots. The same critical-point equations identify H^N and q_{phys} in the Jacobi quotient; the nondegenerate residue pairing separates the quotient classes. □

The status of Proposition 94.14 is a protected observable test of the vortex-normalized mirror, not a standalone proof of mirror equivalence. Once a compactified A-twisted GLSM or sigma-model construction identifies H with the hyperplane class and q_{phys} with the degree-one vortex amplitude, the formula becomes the primary selection rule

$$\langle H^{N-1+dN} \rangle_0 = q_{\mathrm{phys}}^d, \qquad \langle H^k \rangle_0 = 0 \quad (k \not\equiv N-1 \pmod N).$$

The mirror residue supplies the finite Hessian and root-sum side of that statement. The physical identification still depends on the same determinant, orientation, zero-mode, compactification, and contact-term data isolated in (94.46).

Proposition 94.15 (From vortex coefficient to protected observable)**.** *In the charge-one $\mathbb{P}^{N-1}$ GLSM, fix a finite supersymmetric regulator and let $c_{i,\Lambda}$ be the regulated coefficients in* (94.46)*. Put*

$$q_\Lambda = \exp(t) \prod_{i=1}^{N} c_{i,\Lambda}.$$

For $k \geq 0$, define the finite mirror-residue prediction

$$S_k(q_\Lambda) = \begin{cases} q_\Lambda^d, & k = N-1+dN, \quad d \geq 0, \\ 0, & k \not\equiv N-1 \pmod N. \end{cases}$$

Let $A^A_{\Lambda,k}$ be the corresponding regulated A-twisted correlator after the local operator assigned to H^k has been chosen. Suppose the comparison is organized through the stage residuals

$$A^A_{\Lambda,k} - S_k(q_\Lambda) = R_{\mathrm{coeff}} + R_{\mathrm{det}} + R_{\mathrm{zm}} + R_{\mathrm{cpt}} + R_{\mathrm{gl}} + R_{\mathrm{op}} + R_{\mathrm{cont}}. \tag{94.54}$$

Here R_{coeff} measures the difference between the continuum charge-one vortex amplitude and the regulated coefficient (94.46), R_{det} the nonzero-mode determinant and determinant-line normalization error, R_{zm} the residual zero-mode saturation and source-insertion error, R_{cpt} the compactification, obstruction, boundary, and contact-term error, R_{gl} the multi-vortex or degree-d gluing error when $d > 1$, R_{op} the map from the GLSM operator to the hyperplane/Coulomb class, and R_{cont} the continuum limit remainder. If

$$|R_\alpha| \leq \epsilon_\alpha \quad (\alpha = \mathrm{coeff}, \mathrm{det}, \mathrm{zm}, \mathrm{cpt}, \mathrm{gl}, \mathrm{op}, \mathrm{cont}),$$

then

$$\left|A^A_{\Lambda,k} - S_k(q_\Lambda)\right| \leq \epsilon_{\mathrm{coeff}} + \epsilon_{\mathrm{det}} + \epsilon_{\mathrm{zm}} + \epsilon_{\mathrm{cpt}} + \epsilon_{\mathrm{gl}} + \epsilon_{\mathrm{op}} + \epsilon_{\mathrm{cont}}. \tag{94.55}$$

Consequently the quantum-product statement $H^N = q_{\mathrm{phys}}$ follows from the mirror residue only after $q_\Lambda \to q_{\mathrm{phys}} \neq 0$ and all seven residuals vanish in the protected comparison. If some $c_{i,\Lambda} = 0$ because the determinant line is unoriented, an extra fermion zero mode is unsaturated, or the compactified vortex integral vanishes, then $q_\Lambda = 0$; the nonzero degree-one quantum product cannot be recovered by holomorphy or charge matching alone.

Proof. The identity (94.54) is obtained by adding and subtracting the successive finite approximants: the regulated vortex coefficient, the determinant-normalized coefficient, the zero-mode-saturated coefficient, the compactified A-model amplitude, the glued degree-d sector, the local-operator identification, and the continuum protected correlator. The estimate (94.55) is the triangle inequality. Proposition 94.14 supplies the finite residue value $S_k(q_\Lambda)$, but it contains no information about whether the regulated vortex coefficient is nonzero or whether the comparison residuals vanish. That information is precisely the physical amplitude input isolated above. □

94.10 The Cigar–Liouville Mirror Chain as a QFT Duality Problem

The same dualization mechanism gives a useful intrinsic two-dimensional description of the relation between the $\mathcal{N} = 2$ cigar coset and $\mathcal{N} = 2$ Liouville theory. This subsection records the construction as a QFT comparison problem. The names "cigar" and "Liouville" refer to two-dimensional field theories and their semiclassical target descriptions.

Start with a $U(1)$ vector multiplet V, an ordinary chiral multiplet Φ of charge 1, and a logarithmic chiral multiplet

$$P \sim P + 2\pi \mathrm{i}$$

whose imaginary part shifts under the gauge symmetry so that

$$P+\overline{P}+V$$

is gauge invariant. With $k>0$, the bosonic low-derivative part of the GLSM contains

$$|(\partial_\mu-\mathrm{i}A_\mu)\phi|^2+\frac{k}{2}(\partial_\mu\operatorname{Re}p)^2+\frac{k}{2}(\partial_\mu\operatorname{Im}p-A_\mu)^2 \tag{94.56}$$

together with the auxiliary constraint

$$|\phi|^2+\frac{k}{2}(p+\overline{p})=0$$

in the low-energy quotient. Write

$$\rho=\sqrt{2}\,|\phi|,\qquad \vartheta=\arg\phi,\qquad p=u+\mathrm{i}\chi.$$

Then

$$|(\partial_\mu-\mathrm{i}A_\mu)\phi|^2=\frac{1}{2}(\partial_\mu\rho)^2+\frac{\rho^2}{2}(\partial_\mu\vartheta-A_\mu)^2.$$

The auxiliary constraint gives $u=-\rho^2/(2k)$, hence

$$\frac{k}{2}(\partial_\mu u)^2=\frac{\rho^2}{2k}(\partial_\mu\rho)^2.$$

Fix the gauge $\chi=0$. The A_μ-dependent angular part is

$$\frac{1}{2}\rho^2(\partial_\mu\vartheta-A_\mu)^2+\frac{k}{2}A_\mu A^\mu.$$

Its algebraic equation is

$$(\rho^2+k)A_\mu=\rho^2\partial_\mu\vartheta. \tag{94.57}$$

Substituting

$$A_\mu=\frac{\rho^2}{\rho^2+k}\partial_\mu\vartheta$$

gives the angular coefficient

$$\frac{1}{2}\frac{k\rho^2}{\rho^2+k}(\partial_\mu\vartheta)^2=\frac{1}{2}\frac{\rho^2}{1+\rho^2/k}(\partial_\mu\vartheta)^2.$$

Combining the radial and angular terms, the classical two-real-dimensional quotient metric is

$$ds^2=\left(1+\frac{\rho^2}{k}\right)d\rho^2+\frac{\rho^2}{1+\rho^2/k}\,d\vartheta^2. \tag{94.58}$$

This metric is the classical quotient metric. The conformal cigar coset is an infrared fixed-point statement whose definition requires the corresponding quantum current algebra or continuum fixed-point construction, rather than the low-derivative quotient metric alone.

Dualize both logarithmic directions. Introduce twisted chiral variables

$$Y,\qquad Y_P,\qquad Y\sim Y+2\pi\mathrm{i},\quad Y_P\sim Y_P+2\pi\mathrm{i}.$$

The ordinary charged chiral multiplet supplies the exponential term μe^{-Y}. The logarithmic chiral multiplet contributes a quadratic dual Kahler term rather than a second exponential term. The absence of a P-vortex exponential has a finite topological reason in the smooth finite-action sector. The periodic field $e^{\mathrm{i}\chi}$ is a unit-norm charged field for the same $U(1)$ connection A:

$$D(e^{\mathrm{i}\chi}) = \mathrm{i}e^{\mathrm{i}\chi}(d\chi - A).$$

Thus a smooth P-configuration on a Euclidean surface Σ_E supplies a nowhere-vanishing section of the associated unit circle bundle. If the gauge bundle has vortex number

$$n = \frac{1}{2\pi}\int_{\Sigma_E} F \in \mathbb{Z},$$

such a nowhere-vanishing section trivializes the line bundle and forces $n = 0$. Equivalently, on a disk, a boundary winding of χ extends smoothly through the center only when the winding is zero; unlike the ordinary charged scalar $\phi = \rho e^{\mathrm{i}\vartheta}/\sqrt{2}$, the logarithmic multiplet has no radial modulus whose zero can absorb the winding. Allowing singular P-fields would be a different defect insertion, not the smooth finite-action vortex sector used to generate twisted F-terms.

In the convention compatible with (94.40), the low-energy twisted superpotential has the form

$$\widetilde{W}_{\mathrm{cig}}(\Sigma, Y, Y_P) = -\Sigma(Y + Y_P) - \mu e^{-Y}, \tag{94.59}$$

up to affine redefinitions of Y_P and Σ. The vector multiplet imposes

$$Y + Y_P = 0. \tag{94.60}$$

Substituting the constraint leaves one twisted chiral variable Y with twisted superpotential

$$\widetilde{W}_{\mathrm{Liouville}}(Y) = -\mu e^{-Y}. \tag{94.61}$$

The dual Kahler potential has the asymptotic large-$\mathrm{Re}\,Y$ behavior

$$K(Y, \overline{Y}) = \frac{1}{8\pi k}\overline{Y}Y + O(e^{-\mathrm{Re}\,Y}), \qquad \mathrm{Re}\,Y \to +\infty, \tag{94.62}$$

with the sign convention appropriate to twisted chiral superspace. This is the $\mathcal{N} = 2$ Liouville presentation: an asymptotic cylinder coordinate Y together with the exponential twisted superpotential (94.61).

The construction supplies a precise protected-data bridge:

$$\begin{aligned}&\text{charged chiral dualization} + \text{nonzero vortex exponential} + \text{IR vector constraint}\\ &\quad \Longrightarrow \mathcal{N} = 2 \text{ Liouville data.}\end{aligned}$$

A theorem identifying this with the cigar coset QFT requires construction of both continuum theories, equality of their Hilbert spaces and local algebras, matching of the $\mathcal{N} = 2$ superconformal generators, matching of defects and spectral-flow sectors, and control of the strong-coupling region where the semiclassical quotient metric (94.58) is only a coordinate on the asymptotic target geometry.

94.11 Abelian GLSM Coulomb-Branch One-Loop Calculation

The GLSM has a second protected local calculation beyond the classical Higgs quotient. Move to a Coulomb-branch region where $\sigma \neq 0$, so charged chiral multiplets have masses

$$m_i(\sigma) = Q_i \sigma.$$

The calculation below is a Wilsonian twisted F-term calculation whose scope is the protected Coulomb coordinate chart. The full infrared QFT requires a separate continuum construction.

Hypothesis 94.16 (Abelian Coulomb one-loop determinant)**.** Fix an abelian GLSM presentation as in Definition 94.6, assume all charges Q_i are nonzero, and restrict to a simply connected branch of the region

$$Q_i \sigma \neq 0, \qquad i = 1, \dots, m.$$

Assume a supersymmetric Wilsonian regulator preserving the twisted chiral field strength Σ, the gauge symmetry, and the integer charge normalization. The finite part of the massive chiral determinant is assumed to be represented by the local twisted superpotential

$$\widetilde{W}_{\text{eff}}(\Sigma) = -t\Sigma + \sum_{i=1}^{m} Q_i \Sigma \left(\log \frac{Q_i \Sigma}{\mu} - 1 \right) \tag{94.63}$$

up to an additive constant and an affine function of Σ. The renormalization scale $\mu > 0$ and the logarithm branches are part of the scheme choice. Boundary contributions from $\sigma = 0$, vortex sectors, and singular loci are not included in this local determinant hypothesis.

Local determinant derivation. The part of Hypothesis 94.16 that is a perturbative determinant calculation can be made explicit. Work in a background in which the twisted chiral field strength is slowly varying and has lowest component σ. For a chiral multiplet of charge Q, the Coulomb-region complex mass is

$$M = Q\sigma.$$

The local Wilsonian determinant is obtained by expanding the massive Gaussian path integral to first order in the top component of Σ, or equivalently to first order in the background $D + \mathrm{i}F_{12}$ that couples to the FI-theta coordinate. The bosonic and fermionic determinants cancel at zero background $D + \mathrm{i}F_{12}$; the uncancelled linear term is the massive propagator at coincident points. With a proper-time subtraction at the reference mass μ,

$$4\pi \int_0^\infty \frac{\mathrm{d}s}{4\pi s} \left(e^{-s|M|^2} - e^{-s\mu^2} \right) = \log \frac{\mu^2}{|M|^2}. \tag{94.64}$$

The equality follows by differentiating both sides with respect to $|M|^2$ and fixing the constant at $|M| = \mu$:

$$\frac{\partial}{\partial |M|^2} \int_0^\infty \frac{\mathrm{d}s}{s} \left(e^{-s|M|^2} - e^{-s\mu^2} \right) = -\int_0^\infty \mathrm{d}s\, e^{-s|M|^2} = -\frac{1}{|M|^2}.$$

Equation (94.64) determines the real logarithm in the finite FI shift. Its imaginary partner is fixed by the two-dimensional axial anomaly: changing the phase of M by $M \mapsto e^{\mathrm{i}\alpha} M$ is a chiral rotation of a charge-Q massive multiplet and shifts the theta coordinate by $Q\alpha$ in the same normalization in which the flux of a unit-charge $U(1)$ bundle is integral. Thus the real subtraction and the anomaly combine into the holomorphic branch

$$Q\log\frac{M}{\mu}=Q\log\frac{Q\Sigma}{\mu}$$

for the derivative of the twisted superpotential. For m massive chiral multiplets this gives, on the chosen simply connected Coulomb branch,

$$\frac{\partial\widetilde{W}_{\text{eff}}}{\partial\Sigma}=-t+\sum_{i=1}^{m}Q_i\log\frac{Q_i\Sigma}{\mu}. \tag{94.65}$$

Integrating (94.65) in the holomorphic coordinate Σ gives

$$-t\Sigma+\sum_i Q_i\Sigma\left(\log\frac{Q_i\Sigma}{\mu}-1\right),$$

up to an additive constant and an affine function of Σ. This is (94.63). The calculation is local on the Coulomb chart: it uses $Q_i\sigma\neq 0$, chooses logarithm branches, and says nothing by itself about vortex sectors, singular loci, or the existence of the infrared continuum QFT.

Coulomb critical equation. Under Hypothesis 94.16, the local Coulomb critical equation is

$$-t+\sum_{i=1}^{m}Q_i\log\frac{Q_i\sigma}{\mu}=0. \tag{94.66}$$

Equivalently, on the chosen logarithm branch,

$$\prod_{i=1}^{m}\left(\frac{Q_i\sigma}{\mu}\right)^{Q_i}=\mathrm{e}^{t}. \tag{94.67}$$

If all $Q_i>0$, then this equation has

$$Q_{\text{tot}}=\sum_{i=1}^{m}Q_i \tag{94.68}$$

simple solutions for σ whenever $\mathrm{e}^t\neq 0$. For $m=N$ fields of charge 1, the count is N.

Differentiate (94.63) with respect to Σ. For a single summand,

$$\frac{d}{d\Sigma}\left[Q_i\Sigma\left(\log\frac{Q_i\Sigma}{\mu}-1\right)\right]=Q_i\log\frac{Q_i\Sigma}{\mu}.$$

The critical equation is therefore (94.66). Exponentiating gives (94.67).

When all charges are positive, write

$$C_Q=\prod_{i=1}^{m}Q_i^{Q_i}.$$

Then (94.67) becomes

$$C_Q\,\mu^{-Q_{\text{tot}}}\sigma^{Q_{\text{tot}}}=\mathrm{e}^{t}.$$

Since $Q_{\text{tot}}>0$ and the right side is nonzero, this polynomial equation has Q_{tot} nonzero roots. Its derivative is proportional to $\sigma^{Q_{\text{tot}}-1}$, so every nonzero root is simple. For N charge-one fields, $Q_{\text{tot}}=N$.

Example 94.17 ($\mathbb{P}^{N-1}$ GLSM quantum vacua)**.** Take $m = N$, $Q_i = 1$, and $W = 0$. The local equation is

$$\left(\frac{\sigma}{\mu}\right)^N = \mathrm{e}^t.$$

Thus the Coulomb-branch one-loop calculation predicts N massive twisted chiral vacua in the chamber where this effective description is valid. This is the protected arithmetic behind the usual $\mathbb{P}^{N-1}$ vacuum count. The statement that these vacua exhaust the continuum theory requires the same regulator, compactness, and singular-locus controls stated in Hypothesis 94.16.

For the hypersurface charge vector $(1, \ldots, 1, -d)$, the exponent of σ in (94.67) is $N - d$, the same integer as the axial anomaly coefficient (94.18). In the anomaly-free case $N = d$, the Coulomb equation becomes a condition on t and the logarithm branch rather than an equation with finitely many isolated σ-solutions. This is a protected signal of a singular locus in the FI-theta parameter space. A full infrared phase-transition statement requires control of the continuum theory at the singular locus.

Classical chambers for the hypersurface GLSM. Fix integers $N \geq 2$ and $d \geq 2$. Let the GLSM have chiral fields

$$X_1, \ldots, X_N, \quad P$$

with charges

$$Q(X_i) = 1, \qquad Q(P) = -d, \tag{94.69}$$

and superpotential

$$W = P\, G_d(X_1, \ldots, X_N), \tag{94.70}$$

where G_d is homogeneous of degree d. Assume that the projective hypersurface

$$Y_G = \{[X] \in \mathbb{P}^{N-1} : G_d(X) = 0\} \tag{94.71}$$

is smooth, equivalently the equations

$$G_d(X) = \partial_1 G_d(X) = \cdots = \partial_N G_d(X) = 0$$

have no solution with $X \neq 0$. Then the classical Higgs-vacuum quotient has the following chamber description.

1. If $r > 0$, then $X \neq 0$, $P = 0$, and the quotient is $Y_G \subset \mathbb{P}^{N-1}$.
2. If $r < 0$, then $P \neq 0$, $X = 0$ on the vacuum locus, and the unbroken finite gauge group is

   $$\mu_d = \{\zeta \in U(1) : \zeta^d = 1\}.$$

   Fluctuations of the X_i are therefore organized as a Landau–Ginzburg model with polynomial $G_d(X)$, together with the residual finite μ_d gauge quotient.
3. At $r = 0$, the point $X = P = 0$ obeys the D-term equation and the gauge quotient is singular. A definition of the infrared QFT at the phase boundary requires additional continuum data beyond the chamber calculation.

The F-term equations of (94.70) are

$$G_d(X) = 0, \qquad P\,\partial_i G_d(X) = 0, \quad i = 1, \dots, N. \tag{94.72}$$

The D-term equation is

$$\sum_{i=1}^{N} |X_i|^2 - d|P|^2 = r. \tag{94.73}$$

If $r > 0$, equation (94.73) forces $X \neq 0$. On the locus $G_d(X) = 0$, smoothness of Y_G implies that the gradient $(\partial_i G_d(X))$ is not the zero vector. Hence (94.72) forces $P = 0$. Quotienting nonzero X by the charge-one $U(1)$ action gives $[X] \in \mathbb{P}^{N-1}$, and the remaining equation is $G_d(X) = 0$.

If $r < 0$, equation (94.73) forces $P \neq 0$. Then (94.72) gives $\partial_i G_d(X) = 0$ for all i. Homogeneity gives Euler's identity

$$d\,G_d(X) = \sum_{i=1}^{N} X_i \partial_i G_d(X),$$

so $G_d(X) = 0$ follows as well. By the smoothness assumption, the common zero of G_d and all $\partial_i G_d$ is $X = 0$. A gauge transformation $\zeta \in U(1)$ preserves a nonzero value of P exactly when $\zeta^{-d} = 1$, equivalently $\zeta \in \mu_d$. The transverse coordinates are the X_i, and the residual superpotential is $G_d(X)$.

At $r = 0$, substituting $X = 0$ and $P = 0$ into (94.73) gives $0 = 0$, and the F-term equations are also satisfied. The whole $U(1)$ gauge group fixes this point, so the quotient contains a positive-dimensional stabilizer stratum. That stabilizer stratum is the singular locus of the classical quotient at the phase boundary.

Adjunction, central charge, and the residual finite gauge theory. The smooth degree-d hypersurface

$$Y_G \subset \mathbb{P}^{N-1}$$

has complex dimension $N - 2$. Let

$$H = c_1(\mathcal{O}_{\mathbb{P}^{N-1}}(1)).$$

The normal bundle of Y_G in $\mathbb{P}^{N-1}$ is

$$N_{Y_G/\mathbb{P}^{N-1}} \cong \mathcal{O}(d)|_{Y_G}.$$

The exact sequence

$$0 \longrightarrow TY_G \longrightarrow T\mathbb{P}^{N-1}|_{Y_G} \longrightarrow \mathcal{O}(d)|_{Y_G} \longrightarrow 0$$

gives

$$c_1(TY_G) = (N-d)H|_{Y_G}, \qquad K_{Y_G} \cong \mathcal{O}(d-N)|_{Y_G}. \tag{94.74}$$

Thus the topological Calabi–Yau condition in this hypersurface family is

$$d = N.$$

In that case the large-volume sigma-model central-charge target is

$$c_{\text{sigma}} = 3\dim_{\mathbb{C}} Y_G = 3(N-2), \tag{94.75}$$

provided the infrared fixed point exists and realizes the standard $\mathcal{N} = (2, 2)$ superconformal stress tensor.

On the Landau–Ginzburg side, the Fermat reference polynomial

$$G_d(X) = X_1^d + \cdots + X_N^d$$

has quasihomogeneous weights $q_i = 1/d$, hence the protected Landau–Ginzburg central-charge calculation gives

$$c_{\mathrm{LG}} = 3 \sum_{i=1}^{N} \left(1 - \frac{2}{d}\right) = 3N \left(1 - \frac{2}{d}\right). \tag{94.76}$$

The equality

$$c_{\mathrm{LG}} = 3(N - 2)$$

holds precisely when $d = N$. This is a protected numerical consistency condition linking the two chambers. The equality of central charges is one piece of data in an infrared comparison; the equivalence of continuum QFTs also requires the operator algebra, stress tensor, state space, and extended sectors to be constructed and matched.

The finite group in the $r < 0$ chamber is an actual residual gauge group. Choose a gauge in which the nonzero field P is fixed to a positive real constant. A gauge transformation $\zeta \in U(1)$ preserves this gauge choice exactly when

$$\zeta^{-d} = 1.$$

Therefore $\zeta \in \mu_d$, and the remaining chiral fields transform as

$$X_i \longmapsto \zeta X_i.$$

Homogeneity gives

$$G_d(\zeta X) = \zeta^d G_d(X) = G_d(X),$$

so the Landau–Ginzburg superpotential descends to the finite gauge quotient. The finite-gauge QFT contains the invariant untwisted local operators together with twisted sectors determined by the finite gauge theory. The invariant part of $\mathrm{Jac}(G_d)$ is therefore only one protected component of the orbifold theory; a full state-space comparison must include the sectors created by the μ_d defect lines. For the Fermat Jacobi basis

$$X_1^{a_1} \cdots X_N^{a_N}, \qquad 0 \le a_i \le d - 2,$$

the untwisted invariant condition is

$$a_1 + \cdots + a_N \equiv 0 \pmod{d}.$$

Coulomb coordinate for the phase-boundary signal. For the charge vector $(1, \ldots, 1, -d)$, the one-loop Coulomb equation (94.67) becomes

$$\left(\frac{\sigma}{\mu}\right)^N \left(\frac{-d\sigma}{\mu}\right)_{\mathrm{branch}}^{-d} = \mathrm{e}^t. \tag{94.77}$$

Equivalently,

$$\left(\frac{\sigma}{\mu}\right)^{N-d} (-d)_{\text{branch}}^{-d} = \mathrm{e}^{t}. \tag{94.78}$$

The subscript records the chosen logarithm branch for the negatively charged field. In the anomaly-free case $d = N$, the σ-dependence cancels and the local Coulomb chart selects the branch-dependent condition

$$\mathrm{e}^{t} = C_{\text{branch}}, \qquad C_{\text{branch}} = (-N)_{\text{branch}}^{-N}. \tag{94.79}$$

In the additive local FI coordinate

$$T = t - \log C_{\text{branch}},$$

the condition is $T = 0$. At this value the one-loop large-σ twisted superpotential has a critical direction rather than isolated local Coulomb vacua. This is the protected Coulomb-coordinate signal associated with the chamber boundary. The continuum behavior at that coordinate is a separate infrared construction problem.

Example 94.18 (The quintic GLSM as intrinsic two-dimensional QFT data)**.** Set $N = d = 5$. The charges are

$$(1, 1, 1, 1, 1, -5),$$

so the axial anomaly coefficient (94.18) is $5 - 5 = 0$. For $r > 0$, the classical Higgs quotient is the degree-five hypersurface

$$G_5(X) = 0 \quad \text{inside} \quad \mathbb{P}^4.$$

For $r < 0$, the residual finite gauge group is μ_5, and the local Landau–Ginzburg polynomial is $G_5(X_1, \ldots, X_5)$. These are statements about chambers of a two-dimensional supersymmetric gauge theory. The associated infrared equivalence claim requires a two-dimensional continuum construction and a comparison of its protected and unprotected operator data.

94.12 Infrared Comparison Problem

For each two-dimensional supersymmetric model one must distinguish:

UV regulator, supersymmetric Wilsonian flow,
protected operator algebra, full infrared QFT.

Protected rings and indices can match before the full Hilbert space, correlators, and stress tensor are identified. A complete infrared equivalence therefore requires a map of local algebras, states, stress tensors, supercharges, and extended operators, with convergence or scaling limits stated.

Open Problem 94.19 (Two-dimensional supersymmetric RG construction)*.* Construct $2D$ $\mathcal{N} = (2, 2)$ GLSM flows to Landau–Ginzburg and hypersurface sigma-model fixed points with a regulator, supersymmetric Wilsonian maps, protected-ring comparison, stress-tensor convergence, and Hilbert-space reconstruction in one declared framework. The target-space terminology here refers only to an intrinsic two-dimensional QFT comparison between continuum limits.

Chapter 95

Three-Dimensional Chern–Simons–Matter Theories

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

In three dimensions, Chern–Simons–matter theories combine topological gauge terms, parity anomalies, monopole operators, and matter dynamics. This chapter treats them as field-theoretic data with specified definitions and duality maps. The first part fixes the local Chern–Simons and light-cone conventions, including the vector-model large-N framework in which the gauge action becomes quadratic. The second part fixes the $3D$ $\mathcal{N} = 2$ Chern–Simons–matter conventions used by the localization chapter. The final part gives the Aharony–Bergman–Jafferis–Maldacena theory as a worked example with gauge group, levels, matter, superpotential, conformal locus, moduli space, monopole enhancement boundary, and S^3 localization matrix model all stated in the same normalization.

95.1 Three-Dimensional Supersymmetry

Let $\eta_{\mu\nu} = \mathrm{diag}(-,+,+)$. A minimal Majorana spinor has two real components. The $\mathcal{N} = 2$ algebra may be written with complex supercharges $Q_\alpha, \bar{Q}_\alpha$:

$$\{Q_\alpha, \bar{Q}_\beta\} = 2(\gamma^\mu)_{\alpha\beta} P_\mu + 2\mathrm{i}\epsilon_{\alpha\beta} Z,$$

where Z is a possible central charge. Particle multiplets are representations of this Hilbert-space algebra. Superfields are off-shell coordinate packages for local variables; the two notions are connected only after dynamics and quantization have been specified.

A $3D$ $\mathcal{N}=2$ vector multiplet contains

$$(A_\mu, \sigma, \lambda, \bar{\lambda}, D),$$

where A_μ is a gauge field, σ is a real adjoint scalar, $\lambda, \bar{\lambda}$ are gauginos, and D is an auxiliary scalar. A chiral multiplet contains

$$(\phi, \psi, F)$$

in a unitary representation of the gauge group. These are field variables in a chosen off-shell presentation. A statement about BPS particles or local operators must also specify the Hamiltonian, Gauss law, and operator completion of the theory.

95.2 Chern–Simons Datum

Let G be compact and let $P \to M_3$ be a principal G-bundle. A Chern–Simons action is

$$S_{\mathrm{CS}}[A] = \frac{k}{4\pi} \int_{M_3} \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

The trace convention and the allowed levels k are tied to the lattice of large gauge transformations. Gauge invariance of $\exp(\mathrm{i}S_{\mathrm{CS}})$ requires the change of S_{CS} under a large gauge transformation to lie in $2\pi\mathbb{Z}$. For a spin Chern–Simons theory the allowed counterterms may differ from the purely bosonic theory by half-integral contact terms; this choice is part of the three-dimensional QFT specification.

Varying the action gives

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_{M_3} \mathrm{Tr}(\delta A \wedge F_A) + \frac{k}{4\pi} \int_{\partial M_3} \mathrm{Tr}(A \wedge \delta A).$$

On a closed manifold, the field equation of pure Chern–Simons theory is

$$F_A = 0.$$

On a manifold with boundary, a boundary condition or boundary degrees of freedom are part of the gauge-theory specification.

95.3 Conformal Chern–Simons–Matter Charts and Light-Cone Gauge

Chern–Simons–matter theories give a class of three-dimensional conformal fixed points whose perturbative expansion is not organized by a dimensionful Yang–Mills coupling. The level k is quantized, while the perturbative coordinate is a dimensionless parameter such as $1/k$, or in vector models the 't Hooft coordinate

$$\lambda := \frac{N}{k_{\mathrm{eff}}}, \tag{95.1}$$

where k_{eff} includes the parity-anomaly and regulator shift specified with the theory. This statement is not a claim that every Lagrangian written with massless matter is automatically a conformal theory. Relevant deformations, such as scalar and fermion masses, must be tuned to the fixed point, and scalar self-interactions must be placed at their fixed-point values when they are present. Once these data have been fixed, correlation functions can be computed in ordinary diagrammatic perturbation theory around the conformal point.

The light-cone gauge makes a special feature of Chern–Simons theory explicit. Use

$$x^{\pm} = \frac{x^0 \pm x^1}{\sqrt{2}}, \qquad x^{\perp} = x^2, \qquad ds^2 = -2\,dx^+dx^- + (dx^{\perp})^2, \tag{95.2}$$

and choose the orientation convention $\epsilon^{+-\perp} = +1$. The gauge condition is

$$A_- = 0. \tag{95.3}$$

The cubic Chern–Simons vertex then vanishes because every wedge product of three one-forms in three dimensions contains one dx^-. Thus the pure gauge part is first order and quadratic:

$$S_{\text{CS}}\big|_{A_-=0} = \frac{k}{2\pi}\int d^3x\ \text{Tr}(A_+\partial_-A_\perp)\,, \tag{95.4}$$

up to boundary terms and zero-mode data.

Component check of the light-cone Chern–Simons action. The component form of the local Chern–Simons density is

$$\frac{k}{4\pi}\epsilon^{\mu\nu\rho}\,\text{Tr}\left(A_\mu\partial_\nu A_\rho + \frac{2}{3}A_\mu A_\nu A_\rho\right).$$

The cubic term is antisymmetric in the three spacetime labels and therefore contains one A_-; it is zero after imposing $A_- = 0$. The remaining quadratic terms with no A_- are

$$\frac{k}{4\pi}\,\text{Tr}(A_+\partial_-A_\perp - A_\perp\partial_-A_+)\,,$$

because $\epsilon^{+-\perp} = +1$ and $\epsilon^{\perp-+} = -1$. Integrating the second term by parts in x^- and dropping the boundary term gives twice $\text{Tr}(A_+\partial_-A_\perp)$, giving (95.4).

Let matter in a representation R of G define the gauge current

$$J^{\mu a} := -\frac{\delta S_{\text{matter}}}{\delta A^a_\mu}$$

in the same connection normalization as above. At finite light-front regulator, remove the ∂_- zero mode or keep it as a separate Gauss-law degree of freedom. On the zero-mode-free subspace write ∂_-^{-1} for the inverse with the declared prescription.

Light-cone Chern–Simons current kernel. At finite regulator, the light-cone-gauge integration over A_+ and $A_\perp$ is an ordinary Gaussian integration on the nonzero-mode subspace. With the source convention

$$S[A_+, A_\perp; J] = \int d^3x \left[\frac{k}{2\pi} A_+^a \partial_- A_\perp^a - A_+^a J^{+a} - A_\perp^a J^{\perp a} \right], \tag{95.5}$$

substitution of the classical solution gives the effective current interaction

$$S_{\text{eff}}[J] = -\frac{2\pi}{k} \int d^3x \, J^{+a} \, \partial_-^{-1} J^{\perp a}, \tag{95.6}$$

with the ordering and zero-mode prescription displayed.

Set $a = k/(2\pi)$ and $D = \partial_-$. The equations following from (95.5) are

$$aDA_\perp^a = J^{+a}, \qquad aDA_+^a = -J^{\perp a},$$

where the second equation uses integration by parts in x^-. Hence

$$A_\perp^a = a^{-1} D^{-1} J^{+a}, \qquad A_+^a = -a^{-1} D^{-1} J^{\perp a}.$$

Substituting into the action, the bilinear term cancels the $A_+ J^+$ source term and leaves

$$-A_\perp^a J^{\perp a} = -a^{-1} (D^{-1} J^{+a}) J^{\perp a}.$$

This is (95.6). The calculation is a finite-dimensional completion of the square after the regulator has replaced the nonzero-mode function space by a finite or Hilbert-space Gaussian problem. The zero mode is not determined by this inverse; it is the global Gauss-law constraint and must be supplied as part of the light-front regulator.

This is the three-dimensional analogue of the simplification used in the two-dimensional large-N 't Hooft model of Chapter 53. The kernel is ∂_-^{-1}, not $(-\partial_-^2)^{-1}$, because the Chern–Simons kinetic operator is first order rather than Yang–Mills second order. In both cases the light-front gauge field has no transverse propagating polarization and imposes a nonlocal current interaction after the constraint is solved.

Definition 95.1 (Planar Chern–Simons vector-model datum)**.** A planar Chern–Simons vector-model datum consists of:

(i) gauge group $U(N)$ or $SU(N)$, global form, spin structure, framing convention, and effective level k_{eff};

(ii) fundamental matter fields, such as scalars or Dirac fermions, with all relevant deformations tuned to the conformal point;

(iii) a parity-anomaly regulator fixing k_{eff} and all background Chern–Simons contact terms;

(iv) a large-N limit with $\lambda = N/k_{\text{eff}}$ fixed;

(v) a light-front or Euclidean regulator specifying zero modes, ∂_-^{-1}, and the renormalized gauge-invariant singlet operators whose correlators are computed.

For fundamental matter the color algebra uses the same trace-delta normalization as the QCD chapters,

$$(t^a)^i{}_j(t^a)^k{}_\ell = \delta^i{}_\ell\delta^k{}_j - \frac{1}{N}\delta^i{}_j\delta^k{}_\ell. \tag{95.7}$$

Consequently the interaction (95.6) preserves the single color trace at leading order, while the second term in (95.7) is suppressed by $1/N$ in connected singlet correlators. The trace-preserving leading contraction is the planar color algebra behind the bilocal closure of the two-dimensional 't Hooft model and the singlet light-front closure used here.

95.3.1 Finite-Regulator Singlet Closure

The preceding statement can be written as a finite-dimensional algebraic identity before taking the large-N limit. Let Ψ_i^A denote the regulated fundamental matter variables that remain after gauge fixing; the index $i = 1, \ldots, N$ is color, while A collects spin, flavor, position or momentum, and any regulator labels. Let $\overline{\Psi}{}^i_A$ be the conjugate variable in the antifundamental representation. Define the singlet bilocal matrix

$$B^A{}_B = \frac{1}{N}\overline{\Psi}{}^i_B\Psi_i^A. \tag{95.8}$$

A gauge current component that is bilinear in the matter variables has the form

$$J^{\mu a} = \overline{\Psi}{}^i_A\,(V^\mu)^A{}_B\,(t^a)^j{}_i\,\Psi_j^B,$$

where V^μ is the matrix, including all differential or regulator kernels, carried by the current component. The displayed algebraic identity below is written for ordinary complex variables. In fermionic vector models the current product is instead a graded product in the declared current order; the color contraction and its N-scaling are identical, while signs coming from moving Grassmann variables belong to the spinor and kernel convention of the particular Schwinger–Dyson equation rather than to the color algebra.

Planar color reduction of the light-cone kernel. Let K be any regulator kernel acting on the non-color labels; in the light-cone Chern–Simons problem $K = \partial_-^{-1}$ with the zero-mode prescription already declared. With the normalization (95.7),

$$\begin{aligned} J^{+a}KJ^{\perp a} &= N^2\,(V^+)^A{}_B\,B^B{}_C\,K^C{}_D\,(V^\perp)^D{}_E\,B^E{}_A \\ &\quad - N\,(V^+)^A{}_B\,B^B{}_A\,K^C{}_D\,(V^\perp)^D{}_E\,B^E{}_C. \end{aligned} \tag{95.9}$$

Consequently the light-cone interaction (95.6) has the form

$$S_{\text{eff}} = -2\pi N\lambda\,(V^+)^A{}_B B^B{}_C K^C{}_D (V^\perp)^D{}_E B^E{}_A + O(N^0)$$

at fixed $\lambda = N/k_{\text{eff}}$.

Insert the current expression and use (95.7):

$$(t^a)^j{}_i(t^a)^\ell{}_k = \delta^j{}_k\delta^\ell{}_i - \frac{1}{N}\delta^j{}_i\delta^\ell{}_k.$$

The first color contraction produces

$$\overline{\Psi}{}^i_A(V^+)^A{}_B\Psi_k^B\,K^C{}_D\,\overline{\Psi}{}^k_C(V^\perp)^D{}_E\Psi_i^E = N^2(V^+)^A{}_B B^B{}_C K^C{}_D (V^\perp)^D{}_E B^E{}_A.$$

The trace-subtraction term gives the second line of (95.9) with one fewer factor of N. Multiplication by the kernel coefficient $2\pi/k_{\text{eff}} = 2\pi\lambda/N$ then makes the first term order N and the trace-subtraction term order N^0.

Finite-regulator Schwinger–Dyson identity and large-N closure. At finite regulator, with $S_{\text{reg}}[\Psi, \overline{\Psi}]$ the matter action after the light-cone gauge field has been integrated out as in (95.6), the Schwinger–Dyson identity is the finite-dimensional integration-by-parts identity. For an ordinary complex bosonic fundamental vector,

$$0 = \int d\Psi \, d\overline{\Psi} \, \frac{\partial}{\partial \Psi_i^A} \left(\mathcal{O} \, \mathrm{e}^{-S_{\text{reg}}} \right),$$

with boundary terms removed by regulator damping or by compact contour deformation. Expanding the derivative gives

$$\left\langle \frac{\partial \mathcal{O}}{\partial \Psi_i^A} - \mathcal{O} \, \frac{\partial S_{\text{reg}}}{\partial \Psi_i^A} \right\rangle_{\text{reg}} = 0 \tag{95.10}$$

for every polynomial singlet observable $\mathcal{O}$. For a Grassmann fundamental vector the same equation holds with the ordinary derivative replaced by the left Berezin derivative and with the standard graded sign when the derivative crosses $\mathcal{O}$.

The part that matters for the planar vector model is not the integration by parts itself but the closure of the leading equations. After the gauge field has been integrated out, the bilocal color reduction above shows that the order-N interaction is a functional only of B. If the large-N singlet sector has factorization

$$\langle B_1 \cdots B_r \rangle_{\text{conn}} = O(N^{1-r}), \tag{95.11}$$

then the leading equations obtained from (95.10) close on the one-bilocal expectation $G^A{}_B := \langle B^A{}_B \rangle$. Indeed, taking $\mathcal{O}$ to be a product of singlet bilocals gives equations whose leading terms contain products of B's, and (95.11) replaces those products, at leading order, by products of G. Thus no nonsinglet correlator enters the leading large-N Schwinger–Dyson equation.

95.3.2 Planar Bilocal Saddle and Self-Energy Equation

The preceding closure becomes computational only after the leading planar equation is written as an equation for the exact two-point function. We give the finite-regulator derivation because it is the step often hidden behind the phrase "rainbow resummation" in the Chern–Simons vector-model literature.

Let $Q^A{}_B$ be the regulated inverse free propagator of the fundamental matter variable in the gauge-fixed theory, including all spinor, flavor, momentum, and regulator labels but no color index. Thus the quadratic part of the action is

$$S_0 = \overline{\Psi}^i_A Q^A{}_B \Psi^B_i.$$

Keeping the $O(N)$ term of the planar color reduction above and placing the $O(N^0)$ trace-subtraction in the $1/N$ expansion, the leading light-cone interaction is

$$S^{(0)}_{\text{int}} = N \, I_{\text{LC}}[B], \qquad I_{\text{LC}}[B] = -2\pi\lambda \, (V^+)^A{}_B B^B{}_C K^C{}_D (V^\perp)^D{}_E B^E{}_A, \tag{95.12}$$

where $K = \partial_-^{-1}$ with the same zero-mode prescription as in (95.6). Define the color-normalized two-point bilocal

$$G^A{}_B = \langle B^A{}_B \rangle = \frac{1}{N} \left\langle \overline{\Psi}{}^i_B \Psi^A_i \right\rangle .$$

Proposition 95.2 (Planar self-energy equation from the bilocal action)**.** *Assume the finite-regulator planar vector-model datum of Definition 95.1, the leading interaction* (95.12)*, and large-N singlet factorization in the form* (95.11)*. For ordinary complex bosonic fundamental variables, the leading large-N two-point function obeys*

$$G^A{}_C \left(Q^C{}_B + \Sigma^C{}_B[G] \right) = \delta^A{}_B, \tag{95.13}$$

where the self-energy is the functional derivative of the leading bilocal interaction:

$$\Sigma^C{}_M[G] = \frac{\partial I_{\mathrm{LC}}}{\partial G^M{}_C} = -2\pi\lambda \left[(V^+)^A{}_M K^C{}_D (V^\perp)^D{}_E G^E{}_A + (V^+)^C{}_B G^B{}_D K^D{}_E (V^\perp)^E{}_M \right]. \tag{95.14}$$

For fermionic vector models the same color and bilocal derivative formula is read in the declared graded order; the additional signs are those of the Berezin Schwinger–Dyson identity and of the spinor current matrices V^μ.

Proof. Apply the finite-regulator Schwinger–Dyson identity (95.10) with $\mathcal{O} = \Psi^A_i$. Summing over the color label and dividing by N gives

$$\delta^A{}_M = \frac{1}{N} \left\langle \Psi^A_i \frac{\partial S_{\mathrm{reg}}}{\partial \Psi^M_i} \right\rangle .$$

The quadratic action contributes

$$\frac{1}{N} \left\langle \Psi^A_i \overline{\Psi}{}^i_C \right\rangle Q^C{}_M = G^A{}_C Q^C{}_M .$$

Here the displayed equality uses that the variables in the proposition are ordinary bosonic complex variables, so the two fields in the finite regulator integral may be interchanged without a Koszul sign. For the leading planar interaction, the chain rule and the definition of B give

$$\frac{\partial}{\partial \Psi^M_i} \left(N I_{\mathrm{LC}}[B] \right) = \overline{\Psi}{}^i_C \frac{\partial I_{\mathrm{LC}}}{\partial B^M{}_C} .$$

Therefore the interaction contribution is

$$\frac{1}{N} \left\langle \Psi^A_i \overline{\Psi}{}^i_C \frac{\partial I_{\mathrm{LC}}}{\partial B^M{}_C} \right\rangle .$$

Large-N factorization replaces the product of the bilocal and the bilocal polynomial $\partial I_{\mathrm{LC}}/\partial B^M{}_C$ by the same expression evaluated at $B = G$, up to $O(N^{-1})$ corrections in this normalization. This gives

$$\delta^A{}_M = G^A{}_C \left(Q^C{}_M + \left. \frac{\partial I_{\mathrm{LC}}}{\partial B^M{}_C} \right|_{B=G} \right),$$

which is (95.13).

It remains only to compute the derivative of (95.12). Differentiating the first and second occurrence of B gives respectively

$$-2\pi\lambda(V^+)^A{}_M K^C{}_D(V^\perp)^D{}_E G^E{}_A, \qquad -2\pi\lambda(V^+)^C{}_B G^B{}_D K^D{}_E(V^\perp)^E{}_M.$$

Their sum is (95.14). The fermionic statement follows from the finite-regulator Berezin Schwinger–Dyson identity with the same color contraction; one must not infer the graded signs from the bosonic display without fixing the fermion-current ordering. □

In a translation-invariant continuum chart, the labels A, B include momenta and spin. Equation (95.13) then becomes the familiar planar Schwinger–Dyson integral equation for the exact fundamental propagator. Solving it requires the regulator-dependent mass and contact counterterms that tune the theory to the conformal point. The mathematical point of the derivation above is that the planar self-energy is the leading finite-regulator Schwinger–Dyson equation for gauge-invariant singlet bilocals after the light-cone Chern–Simons constraint has been integrated out.

95.3.3 Bilocal Hessian and the Planar Singlet Ladder

The planar two-point functions of singlet bilinears are controlled by the second variation of the same finite-regulator bilocal action. This is the precise object behind the solvability of Chern–Simons vector models: one does not sum diagrams by naming a "ladder", but derives an inverse Hessian on the finite bilocal space and only then passes, if possible, to a continuum integral equation.

For ordinary complex bosonic variables, after introducing the bilocal $B = N^{-1}\overline{\Psi}\Psi$ as a finite-regulator coordinate, the leading collective action has the form

$$\mathcal{S}_{\mathrm{col}}[B] = \mathrm{Tr}(QB) + I_{\mathrm{LC}}[B] - \mathrm{Tr}\log B. \tag{95.15}$$

Here the trace is over the finite non-color regulator labels, and the $-\,\mathrm{Tr}\log B$ term is the Jacobian of the change from N fundamental vectors to the positive bilocal matrix B, in the region where that finite-dimensional change of variables is nonsingular. The stationary equation of (95.15) is

$$Q + \Sigma[G] - G^{-1} = 0,$$

which is the matrix form of (95.13). For fermionic vector models the bilocal collective coordinate and the Jacobian are graded; the following Hessian formula must then be read after fixing the graded current order. The finite color and light-cone kernel algebra is unchanged.

Let X, Y be finite bilocal variations. The Hessian at a solution G is the bilinear form

$$\mathcal{H}_G(X,Y) = \mathrm{Tr}\big(G^{-1}XG^{-1}Y\big) + I''_{\mathrm{LC}}(X,Y), \tag{95.16}$$

where the interaction part is

$$I''_{\mathrm{LC}}(X,Y) = -2\pi\lambda\left[\mathrm{Tr}(V^+XKV^\perp Y) + \mathrm{Tr}(V^+YKV^\perp X)\right]. \tag{95.17}$$

This formula is a finite-dimensional identity: it is obtained by expanding $I_{\mathrm{LC}}[G+X+Y]-I_{\mathrm{LC}}[G+X]-I_{\mathrm{LC}}[G+Y]+I_{\mathrm{LC}}[G]$. The first term in (95.16) is not a free-field assumption; it is the second variation of the finite-regulator Jacobian $-\,\mathrm{Tr}\log B$.

Couple a normalized singlet source $s\ \mathrm{Tr}(WB)$ by replacing $\mathcal{S}_{\mathrm{col}}$ with $\mathcal{S}_{\mathrm{col}} - s\ \mathrm{Tr}(WB)$. Differentiating the saddle equation at $s = 0$ gives the finite-regulator Bethe–Salpeter equation

$$G^{-1}\delta_W G\, G^{-1} + \mathcal{K}_G(\delta_W G) = W, \qquad \mathrm{Tr}(X\,\mathcal{K}_G(Y)) = I''_{\mathrm{LC}}(X, Y), \tag{95.18}$$

for the source response $\delta_W G = (\partial G_s/\partial s)_{s=0}$. Equivalently, with the amputated vertex

$$\Gamma_W := G^{-1}\delta_W G\, G^{-1},$$

one has

$$\Gamma_W + \mathcal{K}_G(G\Gamma_W G) = W. \tag{95.19}$$

The connected planar two-point function of normalized bilocal sources is then

$$\langle \mathrm{Tr}(W_1 B)\, \mathrm{Tr}(W_2 B)\rangle_{\mathrm{conn}} = \frac{1}{N}\ \mathrm{Tr}(W_1\, \delta_{W_2} G) + O(N^{-2}). \tag{95.20}$$

In a translation-invariant continuum chart, (95.19) becomes the planar ladder integral equation for the exact singlet vertex. The equation is exact only inside the declared finite-regulator large-N construction; a continuum solution still requires the conformal mass/contact tuning and a proof that the zero-mode prescription and renormalized singlet operators have a limit.

Controlled Approximation 95.3 (Planar light-front closure)**.** For the regulated model of Definition 95.1, the large-N light-front perturbation series for single-trace singlet correlators is controlled by planar ladders and rainbow self-energies built from the kernel (95.6). The closure is a statement about the declared regulator and the singlet sector: it gives integral equations for exact planar two- and three-point functions once the mass and contact terms have been tuned to the conformal point. It is not a proof of nonperturbative existence of the continuum theory and does not identify colored fundamental fields as asymptotic states.

The useful lesson for the monograph is conceptual as well as computational. Chern–Simons vector models are conformal gauge theories in which conventional diagrams can be evaluated at the fixed point, and light-cone gauge converts the topological gauge field into an exactly solvable constraint kernel. This makes them a higher-dimensional counterpart of the 't Hooft model: not identical in dynamics, because the kernel, spin, parity anomaly, and operator spectrum are different, but similar in the precise sense that the gauge field can be integrated out before solving the planar singlet equations.

95.4 Planar Bosonization of Singlet Observables

The preceding light-cone construction is also the natural setting for the large-N Bose–Fermi duality tests in Chern–Simons vector models. In this monograph the word duality in this context will mean a statement about renormalized gauge-invariant observables after the Chern–Simons level convention, parity-anomaly counterterms, scalar mass/contact tuning, and operator normalization have all been declared. The matching concerns the gauge-invariant singlet sector; colored fundamental fields, line sectors, monopole sectors, and finite-N global refinements require additional objects beyond this planar observable chart.

Conjecture 95.4 (Planar Chern–Simons vector-model bosonization statement)**.** *Fix a positive Yang–Mills-regulator level coordinate $k_{\rm YM}$, a positive rank N, and the induced large-N Chern–Simons level coordinate*

$$K = N + k_{\rm YM}.$$

Let

$$\lambda_{\rm B} := \frac{N}{K}, \qquad N_{\rm F} := k_{\rm YM}, \qquad \lambda_{\rm F} := \frac{N_{\rm F}}{K} = 1 - \lambda_{\rm B}. \tag{95.21}$$

The planar bosonization statement asserts that the singlet sector of the regular bosonic $U(N)$ Chern–Simons vector model in its conformal chart is matched, after the scalar Legendre transform described below, to the singlet sector of the fermionic $U(N_{\rm F})$ Chern–Simons vector model with complementary coupling $\lambda_{\rm F}$. The orientation-reversed and negative-level versions are obtained by applying parity to all Chern–Simons contact terms. The statement is a conjectural equality of planar connected correlators of renormalized singlet currents and of their source functional, modulo the allowed local background Chern–Simons counterterms.

This formulation is deliberately weaker than a finite-N duality theorem. It records the observable statement that the planar calculations test. The rank–level map is the vector-model remnant of level–rank duality, but the matter coupling and the critical scalar/fermion Legendre transforms add singlet-source structures that ordinary pure-Chern–Simons level–rank duality does not contain.

Higher-spin parameters and the exact matching algebra. The planar singlet correlators of Chern–Simons vector models may be organized, following the slightly-broken higher-spin Ward identities, by an effective number of degrees of freedom and by a parity-odd deformation coordinate. In the convention of the light-cone calculation used here, the quasi-boson chart has

$$\widetilde{N}_{\rm B} = 2N\,\frac{\sin(\pi\lambda_{\rm B})}{\pi\lambda_{\rm B}}, \qquad \widetilde{\lambda}_{\rm qb} = \tan\left(\frac{\pi\lambda_{\rm B}}{2}\right), \tag{95.22}$$

while the complementary quasi-fermion chart is written with

$$\widetilde{N}_{\rm F} = 2N_{\rm F}\,\frac{\sin(\pi\lambda_{\rm F})}{\pi\lambda_{\rm F}}, \qquad \widetilde{\lambda}_{\rm qf} = \cot\left(\frac{\pi\lambda_{\rm F}}{2}\right). \tag{95.23}$$

Equations (95.21) give the exact identities

$$\widetilde{N}_{\rm B} = \widetilde{N}_{\rm F}, \qquad \widetilde{\lambda}_{\rm qb} = \widetilde{\lambda}_{\rm qf}. \tag{95.24}$$

Indeed $N/\lambda_{\rm B} = K = N_{\rm F}/\lambda_{\rm F}$, and $\sin(\pi\lambda_{\rm F}) = \sin(\pi(1-\lambda_{\rm B})) = \sin(\pi\lambda_{\rm B})$. The second identity is

$$\cot\left(\frac{\pi(1-\lambda_{\rm B})}{2}\right) = \tan\left(\frac{\pi\lambda_{\rm B}}{2}\right).$$

Thus the higher-spin parameter matching is not an additional dynamical assumption once the rank–level map and the operator normalizations (95.22)–(95.23) are chosen. The dynamical work is the derivation of the planar correlators and the proof that the continuum conformal charts realizing these parameters exist.

Legendre transform of the scalar singlet. The Bose–Fermi map relates regular and critical vector models. The large-N critical operation is most cleanly stated at the source-functional level. Let J_0 be the scalar singlet, for example $\phi^\dagger\phi$ in the bosonic chart or $\bar{\psi}\psi$ in the fermionic chart, and let $W_{\rm reg}[s]$ denote the planar connected generating functional with source s for J_0, after all contact terms have been fixed. On a source neighborhood where the Hessian

$$\mathcal{G}_{00}(x,y) = \frac{\delta^2 W_{\rm reg}}{\delta s(x)\delta s(y)}$$

has an inverse on the chosen test-function quotient, define the conjugate field

$$\sigma(x) = \frac{\delta W_{\rm reg}}{\delta s(x)}$$

and the Legendre-transformed planar functional

$$W_{\rm crit}[\sigma] = W_{\rm reg}[s] - \int s(x)\sigma(x)\,\mathrm{d}^3x, \qquad \sigma = \frac{\delta W_{\rm reg}}{\delta s}. \tag{95.25}$$

Differentiating (95.25) gives

$$\frac{\delta W_{\rm crit}}{\delta\sigma(x)} = -s(x), \qquad \frac{\delta^2 W_{\rm crit}}{\delta\sigma(x)\delta\sigma(y)} = -\mathcal{G}_{00}^{-1}(x,y).$$

Thus the critical scalar two-point kernel is the inverse, with the displayed sign convention, of the regular scalar-singlet two-point kernel. This operator inversion is the precise large-N content behind the informal phrase that the critical vector model is a Legendre transform of the regular one. Beyond the planar source neighborhood it is not a definition by slogan: one must specify the continuum limit, the test-function space, the contact-term quotient, and the domain on which the inverse exists.

Scope of the planar observable statement. The matching in Conjecture 95.4 concerns the renormalized singlet current algebra generated by the stress tensor, the higher-spin currents at leading large N, and the scalar singlet after the Legendre transform. Colored fundamental fields enter only as variables in a chosen gauge-redundant regulator; line operators, monopole operators, finite-N spectra, background Chern–Simons contact terms, and global forms are additional sectors whose matching must be specified by extra data. The light-front bilocal equations above give one finite-regulator route to the planar correlators; the higher-spin Ward classification gives an independent organization of the same planar correlators. Agreement of the two is strong evidence for the conjectural observable matching, but a proof of the full continuum duality would require a nonperturbative construction of both Chern–Simons–matter theories and an isomorphism of their gauge-invariant observable algebras or of their Kontsevich–Segal assignments.

The source lineage for this large-N matching includes the Maldacena–Zhiboedov higher-spin constraints and the explicit planar Chern–Simons vector-model correlator computations [70, 71, 72]. The citations identify the historical calculations; the status of the statement inside this monograph is the conjectural observable matching and the exact algebraic matching displayed above.

Hypothesis 95.5 (Planar bosonization matching hierarchy)**.** A proof of Conjecture 95.4 in the observable chart used in this chapter should supply a compatible hierarchy of matching data, not only the numerical identification (95.24). The hierarchy consists of:

1. the stress tensor and normalized single-trace higher-spin currents J_s, with the quasi-boson and quasi-fermion parameters identified as in (95.24);
2. the scalar-singlet source map, including the Legendre inversion $\mathcal{G}_{00} \mapsto -\mathcal{G}_{00}^{-1}$ on the contact-term quotient declared in (95.25);
3. the background flavor and topological $U(1)$ Chern–Simons contact coordinates, where integer counterterm shifts are scheme choices but the fractional parity-anomaly data and the relative contact terms must agree under the duality map;
4. the charged extended sector, including Wilson lines, monopole disorder operators, and the integral charge–flux pairing that any proposed line/monopole map has to preserve;
5. the finite-N and global-form completion, including the line lattice, allowed genuine line operators, baryon/monopole identifications, and level–rank contact phases;
6. the common continuum construction: a regulator, conformal tuning, source space, and isomorphism of gauge-invariant observable algebras, or equivalently of the relevant Kontsevich–Segal assignments.

Only the first two items, the integer-contact quotient, and the elementary charge–flux pairing invariant have been reduced in this section to explicit algebraic checks. The actual line lattice, monopole spectrum, finite-N operator map, and full continuum observable algebra are part of the remaining duality problem rather than consequences of the large-N rank–level identity.

This hierarchy is the working standard for future refinements of the Bose–Fermi duality discussion. A new computation of one planar three-point coefficient should be inserted here only if it is attached to the hierarchy: which source convention it fixes, which contact coordinate it shifts, which line/monopole datum it constrains, or which observable-algebra map it helps construct.

95.5 Supersymmetric Chern–Simons–Matter Action

The supersymmetric Chern–Simons action contains

$$S_{\mathrm{CS}}^{\mathcal{N}=2} = \frac{k}{4\pi} \int \mathrm{Tr} \left(A \wedge dA + \frac{2}{3} A^3 - \bar{\lambda}\lambda + 2D\sigma \right), \tag{95.26}$$

with the displayed signs defining the convention of this chapter. Let chiral multiplets Φ_i take values in a unitary representation R of G. Their scalar fields are ϕ_i. With real mass parameters m_i, FI coordinate ζ for an abelian factor, and superpotential W, the bosonic potential contains

$$\sum_i |(\sigma + m_i)\phi_i|^2 + \frac{1}{2} \left(\mu(\phi) + \frac{k}{2\pi}\sigma - \zeta \right)^2 + |dW(\phi)|^2, \tag{95.27}$$

in an abelian normalization where μ is the matter moment map. Supersymmetric classical vacua solve

$$dW = 0, \qquad (\sigma + m_i)\phi_i = 0, \qquad \mu(\phi) + \frac{k}{2\pi}\sigma - \zeta = 0, \tag{95.28}$$

modulo gauge transformations.

Remark 95.6 (Auxiliary-field origin of the shifted D-term). In the abelian convention of (95.26), the auxiliary equation shifts the moment-map equation by

$$\mu(\phi) - \zeta \longmapsto \mu(\phi) + \frac{k}{2\pi}\sigma - \zeta.$$

The part of the Lagrangian involving D has the form

$$\frac{1}{2}D^2 + D\big(\mu(\phi) - \zeta\big) + \frac{k}{2\pi}D\sigma,$$

where the last term is the $2D\sigma$ term in (95.26). Completing the square gives

$$\frac{1}{2}\left(D + \mu(\phi) + \frac{k}{2\pi}\sigma - \zeta\right)^2 - \frac{1}{2}\left(\mu(\phi) + \frac{k}{2\pi}\sigma - \zeta\right)^2.$$

Eliminating D leaves the displayed positive potential and hence the vacuum equation (95.28).

Non-abelian Chern–Simons auxiliary elimination. Let G be compact, let R be a unitary representation, and choose Hermitian representation matrices t^a_R with trace convention $\mathrm{Tr}(t^a t^b) = \delta^{ab}$ for the gauge action. Assume $k \neq 0$, no Yang–Mills kinetic term, no FI term, no real mass, and no superpotential. For one chiral multiplet in R, define the finite moment-map coordinates

$$\mu^a(\phi) = \overline{\phi}\, t^a_R \phi.$$

For several chiral multiplets, replace R by the direct sum of their representations, so that μ^a is the sum of the corresponding quadratic coordinates. Then the local non-derivative bosonic component terms involving the Chern–Simons auxiliary scalar $\sigma = \sigma^a t^a$ and the auxiliary field $D = D^a t^a$ may be written as

$$\mathcal{L}_{\text{bos,aux}} = \frac{k}{2\pi} D^a \sigma^a + D^a \mu^a(\phi) - \sigma^a \sigma^b\, \overline{\phi}\, t^a_R t^b_R \phi. \tag{95.29}$$

Eliminating D imposes

$$\sigma^a = -\frac{2\pi}{k}\mu^a(\phi), \tag{95.30}$$

and the remaining scalar interaction is

$$\mathcal{L}_{\text{bos,on-shell}} = -\frac{4\pi^2}{k^2}\big(\overline{\phi}\, t^a_R \phi\big)\big(\overline{\phi}\, t^b_R \phi\big)\big(\overline{\phi}\, t^a_R t^b_R \phi\big). \tag{95.31}$$

Equivalently, the scalar potential contains the positive contribution

$$V_{\text{sextic}} = \frac{4\pi^2}{k^2}\big(\overline{\phi}\, t^a_R \phi\big)\big(\overline{\phi}\, t^b_R \phi\big)\big(\overline{\phi}\, t^a_R t^b_R \phi\big)$$

in the same local coordinates.

The field D is a Lagrange multiplier in the pure Chern–Simons normalization. Varying (95.29) with respect to D^a gives

$$\frac{k}{2\pi}\sigma^a + \mu^a(\phi) = 0,$$

which is (95.30). Substitution into the last term of (95.29) gives

$$-\left(-\frac{2\pi}{k}\mu^a\right)\left(-\frac{2\pi}{k}\mu^b\right)\overline{\phi}\,t_R^a t_R^b\phi = -\frac{4\pi^2}{k^2}\mu^a\mu^b\overline{\phi}\,t_R^a t_R^b\phi.$$

No commutativity of the matrices t_R^a and t_R^b has been used; the ordered product in (95.31) is part of the component convention.

The same finite-dimensional elimination fixes the power of k and the relative sizes of the induced Yukawa terms. The signs of the terms carrying an explicit i depend on the three-dimensional spinor conjugation and on where the i is placed in the Chern–Simons gaugino mass. In the common component convention in which the non-derivative fermion terms are normalized as in the stringbook appendices, one obtains coefficient magnitudes

$$\frac{2\pi}{|k|}\quad\text{for}\quad(\overline{\phi}\,t_R^a\phi)(\overline{\psi}\,t_R^a\psi),\qquad\frac{4\pi}{|k|}\quad\text{for}\quad(\overline{\psi}\,t_R^a\phi)(\overline{\phi}\,t_R^a\psi).$$

Thus the convention-independent check is the $1:2$ relative coefficient and the single inverse power of k; the signed formula must be read only after the spinor bilinear convention has been fixed.

Remark 95.7 ($\mathcal{N}=3$ adjoint-chiral elimination)*.* Let Q and $\widetilde{Q}$ be conjugate chiral multiplets in representations R and $R^\vee$, and let $\varphi=\varphi^a t^a$ be an adjoint chiral multiplet. With the same trace convention as above, set

$$J^a=\widetilde{Q}\,t_R^a Q.$$

The $\mathcal{N}=3$ Chern–Simons-matter superpotential

$$W(\varphi,Q,\widetilde{Q})=-\frac{k}{8\pi}\varphi^a\varphi^a+J^a\varphi^a$$

has algebraic φ-equation

$$\varphi^a=\frac{4\pi}{k}J^a,$$

and the exact tree-level effective superpotential after eliminating φ is

$$W_{\text{eff}}(Q,\widetilde{Q})=\frac{2\pi}{k}J^aJ^a=\frac{2\pi}{k}\big(\widetilde{Q}\,t_R^aQ\big)\big(\widetilde{Q}\,t_R^aQ\big).$$

The F-term equation is

$$-\frac{k}{4\pi}\varphi^a+J^a=0.$$

Therefore $\varphi^a=4\pi J^a/k$. Substitution gives

$$-\frac{k}{8\pi}\left(\frac{4\pi}{k}\right)^2J^aJ^a+\frac{4\pi}{k}J^aJ^a=-\frac{2\pi}{k}J^aJ^a+\frac{4\pi}{k}J^aJ^a=\frac{2\pi}{k}J^aJ^a.$$

The calculation is an elimination in the polynomial chiral ring, so it does not use a string-theoretic realization of the same Lagrangian.

95.6 Parity Anomaly And Contact Terms

A massive three-dimensional Dirac fermion in representation R shifts the effective Chern–Simons level by

$$\Delta k = \frac{1}{2}\,\mathrm{sign}(m)\,T(R),$$

where

$$\mathrm{Tr}_R(t^a t^b) = T(R)\delta^{ab}$$

in the trace convention used for the gauge action. Therefore a massless fermion coupled to a gauge field requires a choice of regulator that fixes the integer or half-integer effective level. This choice is part of the definition of the continuum theory and of its time-reversal or parity properties.

Background flavor and R-symmetry Chern–Simons terms are equally important. Two proposed dual theories with the same local operator spectrum can still differ as QFTs if their background contact terms differ. In localization formulae these contact terms appear as phases or Gaussian factors in the Cartan variables; they are not disposable normalizations.

95.7 Monopole Operators

Let G have maximal torus T. A monopole operator at the origin is defined by imposing a magnetic singularity on a small sphere:

$$\frac{1}{2\pi}\int_{S^2} F = m \in \mathrm{Hom}(U(1), T)/W_G.$$

The local operator is gauge invariant only after its electric charges induced by Chern–Simons terms and fermion zero modes are accounted for. In an abelian theory with level k, Gauss law gives electric charge

$$q_{\mathrm{el}} = k\,m$$

for a bare monopole of magnetic charge m. Gauge-invariant monopole operators may require dressing by charged matter fields.

This point is not optional in supersymmetric Chern–Simons–matter theory. Monopole operators can carry flavor and R-charges, can complete multiplets that are invisible in a polynomial superfield presentation, and can change the apparent global symmetry of the interacting fixed point. Thus a Lagrangian written in $\mathcal{N} = 2$ variables may describe a theory with larger supersymmetry only after the monopole sectors are included.

95.8 ABJM Gauge-Theory Specification

Fix integers $N \geq 1$ and $k \neq 0$. The ABJM gauge group and levels are

$$G_{N,k} = U(N)_k \times U(N)_{-k},$$

with connections A and $\widehat{A}$. The subscript means that the Chern–Simons functional of A has level k, while that of $\widehat{A}$ has level $-k$, in the same trace normalization. The global form of the gauge group and the quotient by any diagonal one-form symmetry affect line operators and must be declared in a fully specified theory. In this chapter $U(N) \times U(N)$ denotes the standard ABJM choice used for the local-operator and S^3 partition-function formulae.

In $\mathcal{N} = 2$ language the matter consists of four chiral multiplets

$$A_a \in (\mathbf{N}, \overline{\mathbf{N}}), \qquad B_{\dot a} \in (\overline{\mathbf{N}}, \mathbf{N}), \qquad a, \dot a = 1, 2.$$

The classical superpotential is

$$W_{\text{ABJM}} = \frac{2\pi}{k} \varepsilon^{ab} \varepsilon^{\dot a \dot b} \operatorname{Tr}\big(A_a B_{\dot a} A_b B_{\dot b}\big). \tag{95.32}$$

The trace is over the first $U(N)$ gauge indices, so every product in (95.32) is an endomorphism of the first gauge factor. The displayed coefficient fixes the convention in which the Yukawa and sextic scalar terms assemble into the $\mathcal{N} = 6$ invariant potential after auxiliary fields are eliminated. The $\mathcal{N} = 2$ trial R-charge assignment

$$R(A_a) = R(B_{\dot a}) = \frac{1}{2}$$

gives $R(W_{\text{ABJM}}) = 2$, as required for a superpotential term.

The scalar variables can be grouped as

$$C_I = (A_1, A_2, B_1^\dagger, B_2^\dagger), \qquad I = 1, \dots, 4.$$

The interacting theory has an $SU(4)$ flavor symmetry rotating the C_I in the $\mathcal{N} = 6$ presentation. This is a field-theory statement about the completed gauge theory, including monopole sectors and the Chern–Simons Gauss law. The holomorphic $\mathcal{N} = 2$ superpotential variables display only one coordinate presentation of this symmetry.

Parity and the opposite levels. Let P be an orientation-reversing diffeomorphism of the three-manifold. The Chern–Simons functional changes sign under P. Therefore the pair of levels $(k, -k)$ is invariant under the combined operation

$$P: \quad A \longleftrightarrow \widehat{A}, \qquad A_a \longleftrightarrow B_{\dot a}^\dagger,$$

up to the corresponding exchange of flavor indices and spinor conventions: under an orientation reversal the integral of a three-form changes sign, so

$$S_{\text{CS,k}}[A] \mapsto -S_{\text{CS,k}}[P^* A].$$

Exchanging the two gauge factors sends the level k contribution to the sector that originally carried level $-k$, and conversely. Hence the sum

$$S_{\text{CS,k}}[A] + S_{\text{CS,}-\text{k}}[\widehat{A}]$$

is mapped to itself after relabeling the two connections. The matter exchange identifies the bifundamental and antibifundamental representations, so the gauge-covariant kinetic terms and the quartic superpotential are carried into their complex conjugate form with the same real action.

95.9 ABJM Conformal Locus And Marginal Couplings

The ABJM family is often described as if a Chern–Simons level were an ordinary coupling. For conformal-manifold questions this is the wrong category: the level is part of the topological gauge specification and is quantized by large gauge transformations. We first state the local notion being used.

Definition 95.8 (Local conformal manifold in a fixed QFT specification)**.** Fix a d-dimensional continuum QFT specification $\mathfrak{D}$, including the global form of the gauge group, line-operator lattice, spin and framing choices, background contact-term scheme, and the global or supersymmetry algebra required to be preserved. A local conformal manifold through a conformal theory $\mathcal{T}_0$ with specification $\mathfrak{D}$ is a germ of a smooth parameter space U and a family $\{\mathcal{T}_u\}_{u\in U}$ of conformal QFTs with the same specification such that tangent vectors at 0 are represented by integrated scalar primary deformations

$$\delta S = \sum_a \delta u^a \int d^d x\, \mathcal{O}_a(x), \qquad \Delta(\mathcal{O}_a) = d,$$

modulo redundant deformations coming from total derivatives, symmetry variations, and field-coordinate changes. A marginal Lagrangian parameter is exactly marginal only after it defines such a tangent vector with all beta functions and background contact terms remaining inside $\mathfrak{D}$.

Hypothesis 95.9 (Standard ABJM conformal class)**.** In this section "standard ABJM" means the local theory with gauge specification

$$U(N)_k \times U(N)_{-k}, \qquad k \in \mathbb{Z} \setminus \{0\},$$

the four bifundamental chiral multiplets of Section 95.8, zero real masses, zero FI coordinates, the standard line-operator lattice used in the S^3 localization formula, and the interacting $\mathcal{N} = 6$ superconformal symmetry with $SU(4)$ flavor symmetry. Within this class the quartic superpotential tensor is the one in (95.32); closure of the enhanced supersymmetry fixes its coefficient to

$$h_{\mathrm{ABJM}} = \frac{2\pi}{k}$$

in the component convention of this chapter. Deformations that keep only $\mathcal{N} = 2$, change the flavor symmetry, change the line lattice, or alter background contact terms are different QFT specifications and are not counted as tangent vectors to this standard ABJM conformal class.

Remark 95.10 (Standard ABJM has no continuous conformal coupling). Assume Hypothesis 95.9. At fixed (N, k) the local conformal-manifold germ of the standard ABJM specification is zero-dimensional. The standard ABJM theories form a discrete family indexed by the integer level k, with k and $-k$ related when one includes orientation reversal and exchange of the two gauge factors as described above.

The Chern–Simons levels are not coordinates on a real or complex manifold. Gauge invariance of $\exp(\mathrm{i}S_{\mathrm{CS}})$ under large gauge transformations restricts the levels to an integral lattice. Hence an infinitesimal deformation preserving the same gauge specification has

$$\delta k = 0.$$

The opposite-level condition in the standard ABJM specification also fixes the second level to be $-k$, so there is no independent continuous level direction.

The quartic superpotential is power-counting marginal in an $\mathcal{N}=2$ description: the chiral fields have protected $\Delta = R = \frac{1}{2}$ at the ABJM point, so the quartic monomial has dimension 2, and the $d^2\theta$ measure has dimension 1 in three dimensions. Power-counting marginality, however, is only a necessary test. Inside the standard $\mathcal{N}=6$ ABJM class the same coefficient is fixed by the supersymmetry-completion condition

$$h = h_{\mathrm{ABJM}} = \frac{2\pi}{k}.$$

Therefore

$$\delta h = -\frac{2\pi}{k^2}\,\delta k = 0$$

for a tangent vector that preserves the standard ABJM specification.

The remaining familiar Lagrangian coordinates do not give exactly marginal directions in this class. Real masses and FI coordinates have engineering dimension 1 in three dimensions and break part of the conformal or flavor specification. A Yang–Mills kinetic term, if introduced as a regulator or UV completion coordinate, carries a dimensionful coupling

$$[g_{\mathrm{YM}}^2] = 1$$

and is not a scalar-primary deformation of dimension 3 of the conformal theory. Background Chern–Simons contact terms are quantized scheme data, not continuous local-operator deformations. Thus every allowed tangent vector in the fixed standard ABJM specification is redundant or zero.

This remark is deliberately narrower than a classification theorem for all three-dimensional $\mathcal{N}=2$ Chern–Simons–matter theories with the same gauge group. If the $SU(4)$ symmetry, the $\mathcal{N}=6$ completion, or the standard line and contact-term data are relaxed, quartic marginal candidates must be analyzed by their own beta functions, current moment maps, and monopole-sector constraints. They are not continuous coordinates on the standard ABJM conformal locus just defined.

95.10 ABJM Classical Moduli Space

The vacuum equations of the $G_{N,k}$ theory are not those of an ordinary Yang–Mills quotient: the Chern–Simons auxiliary fields identify moment-map coordinates with vector-multiplet scalars. A useful branch is the commuting branch, where the four complex scalar matrices are simultaneously diagonal after gauge transformations:

$$C_I = \mathrm{diag}(c_{I,1}, \ldots, c_{I,N}), \qquad I = 1, \ldots, 4.$$

For generic diagonal eigenvalues the residual gauge group is the permutation group S_N, together with abelian gauge transformations acting on each eigenvalue.

Rank-one abelian quotient. For $N = 1$, set $C = (C_1, \ldots, C_4)$. The classical ABJM branch is

$$\mathcal{M}^{\mathrm{cl}}_{1,k} = \mathbb{C}^4/\mathbb{Z}_k,$$

where the generator of $\mathbb{Z}_k$ acts by simultaneous phase rotation

$$(C_1, \ldots, C_4) \longmapsto \big(\mathrm{e}^{2\pi\mathrm{i}/k} C_1, \ldots, \mathrm{e}^{2\pi\mathrm{i}/k} C_4\big).$$

For $U(1)_k \times U(1)_{-k}$, every matter scalar has charge $+1$ under the anti-diagonal gauge parameter and charge 0 under the diagonal gauge parameter. Put

$$A_D = A + \widehat{A}, \qquad B = A - \widehat{A}.$$

On a closed three-manifold the abelian Chern–Simons term is, up to the declared boundary convention,

$$\frac{k}{4\pi}\int (A \wedge dA - \widehat{A} \wedge d\widehat{A}) = \frac{k}{4\pi}\int B \wedge dA_D. \tag{95.33}$$

Thus the diagonal gauge field acts as a Lagrange multiplier for the anti-diagonal flux, while the anti-diagonal gauge field is the one seen by the matter. Dualizing the flat diagonal connection gives a periodic scalar. Large shifts of that scalar leave a residual anti-diagonal gauge transformation of order k. This residual transformation multiplies every C_I by $\mathrm{e}^{2\pi \mathrm{i}/k}$. Therefore the continuous anti-diagonal phase quotient is accompanied by the finite residual quotient $\mathbb{Z}_k$, giving $\mathbb{C}^4/\mathbb{Z}_k$.

Quoted Theorem 95.11 (ABJM commuting-branch moduli space)**.** *For the $U(N)_k \times U(N)_{-k}$ ABJM theory, the branch obtained from simultaneously diagonalizable scalar vacuum expectation values has classical moduli space*

$$\mathcal{M}^{\mathrm{cl}}_{N,k} = \big((\mathbb{C}^4/\mathbb{Z}_k)^N\big)/S_N.$$

For $k = 1, 2$ the interacting theory has enhanced $\mathcal{N} = 8$ supersymmetry; for $|k| > 2$ the standard interacting theory has $\mathcal{N} = 6$ supersymmetry. The enhancement statement uses monopole operators and requires data beyond the polynomial superpotential.

The theorem is quoted here because the full statement identifies quantum operator sectors, monopole operators, and the global form of the gauge group. The finite quotient in the rank-one calculation above is the intrinsic gauge-theory convention that fixes the k-dependent orbifold action used later in localization and protected-sector comparisons.

95.11 ABJM Localization Specialization

The S^3 localization formula of Chapter 101 specializes to ABJM when all four chiral multiplets have R-charge $1/2$ and zero real mass. For Cartan variables

$$\lambda_i, \mu_j \in \mathbb{R}, \qquad i, j = 1, \dots, N,$$

associated respectively to the two $U(N)$ factors, the matrix integral is

$$\begin{aligned} Z^{\mathrm{ABJM}}_{S^3}(N,k) = \frac{1}{(N!)^2}\int_{\mathbb{R}^{2N}} \prod_{i=1}^{N} d\lambda_i\, d\mu_i\, \exp\left[\mathrm{i}\pi k \sum_i (\lambda_i^2 - \mu_i^2)\right] \\ \times \frac{\prod_{i<j} 4\sinh^2\big(\pi(\lambda_i-\lambda_j)\big)\prod_{i<j} 4\sinh^2\big(\pi(\mu_i-\mu_j)\big)}{\prod_{i,j} 4\cosh^2\big(\pi(\lambda_i-\mu_j)\big)}. \end{aligned} \tag{95.34}$$

The numerator is the product of the two vector-multiplet determinants. The denominator is the product of two conjugate bifundamental chiral pairs: each pair contributes

$$\exp\left[\ell\Big(\frac{1}{2} + \mathrm{i}(\lambda_i - \mu_j)\Big) + \ell\Big(\frac{1}{2} - \mathrm{i}(\lambda_i - \mu_j)\Big)\right] = \frac{1}{2\cosh\big(\pi(\lambda_i - \mu_j)\big)}.$$

The two pairs give $1/(4\cosh^2(\pi(\lambda_i - \mu_j)))$ for each ordered pair (i, j). The oscillatory contour, framing phase, and contact terms are part of the localization input. Thus (95.34) is an exact protected observable once those inputs are supplied. A microscopic construction of the continuum theory requires independent definition of the underlying three-dimensional QFT.

95.12 Sphere Free Energy and F-Monotonicity

Three-dimensional conformal field theories have a distinguished finite observable on the round sphere. It is useful in supersymmetric Chern–Simons–matter theory because localization often reduces it to a finite-dimensional matrix integral, but its definition is not a localization artifact.

Definition 95.12 (Renormalized three-sphere free energy)**.** Let $\mathcal{T}$ be a reflection-positive three-dimensional Euclidean QFT with a specified regulator, background metric coupling, and local counterterm scheme. Its round-sphere partition function at radius r is denoted $Z_{S^3_r}(\mathcal{T})$. At a conformal fixed point, after removing power divergences by the declared local counterterms, the renormalized sphere free energy is

$$F(\mathcal{T}) = -\log |Z_{S^3}(\mathcal{T})| \,. \tag{95.35}$$

The absolute value removes pure phases produced by framing and background Chern–Simons contact terms. It does not remove the obligation to declare the real local counterterm scheme used to define the finite partition function.

In a supersymmetric theory the same definition is evaluated by the localized matrix integral only after the S^3 localization input of Chapter 101 has been supplied: Cartan contour, parity-anomaly regulator, framing phase, real-mass and FI coordinates, and background Chern–Simons counterterms. For ABJM,

$$F_{\mathrm{ABJM}}(N, k) = -\log \left| Z^{\mathrm{ABJM}}_{S^3}(N, k) \right| \tag{95.36}$$

with $Z^{\mathrm{ABJM}}_{S^3}$ defined by (95.34). This is a protected observable of the declared QFT specification; changing the line-operator lattice or contact terms can change phases and, in some schemes, finite background terms.

Remark 95.13 (Conditional status of entropic F-monotonicity)*.* The usual entropic route to three-dimensional F-monotonicity starts from vacuum entanglement for disks, not from a localized matrix integral. At the level of the continuum foundations used in this monograph, that argument is not yet a theorem: a sharp bounded-region local algebra is expected to be type III, has no faithful normal trace, and does not supply an unqualified reduced density matrix whose von Neumann entropy is $S(R)$. The object entering the standard physics argument must therefore be supplied by additional data such as a regulator with a proven continuum limit, a split inclusion with a controlled collar limit, or an AQFT replacement by relative entropy or mutual information.

Under such an additional datum, one may define the renormalized disk quantity

$$\mathcal{F}(R) = R\,\frac{dS(R)}{dR} - S(R), \tag{95.37}$$

where $S(R)$ denotes the regulated or split-limit disk entropy. If strong subadditivity survives the same limit, the Casini–Huerta deformation argument gives $\mathcal{F}(R)$ monotone nonincreasing as

R increases. If, in addition, the conformal map from the causal development of the disk to the hyperbolic thermal problem identifies the fixed-point value of $\mathcal{F}$ with the renormalized sphere free energy in Definition 95.12, then the endpoint consequence is

$$F_{\mathrm{UV}} \geq F_{\mathrm{IR}}. \tag{95.38}$$

Strict inequality, or a rigidity statement characterizing equality as a trivial flow modulo decoupled topological and contact-term data, is an additional input beyond monotone nonincrease.

Open Problem 95.14 (AQFT construction of three-dimensional F-monotonicity). Construct, from the local-algebraic data of a three-dimensional continuum QFT, a regulator-independent quantity that replaces the sharp-disk entropy in (95.37), proves the required monotonicity from relative entropy, split inclusions, or another intrinsic AQFT object, and identifies its conformal fixed-point value with $-\log|Z_{S^3}|$ in a declared background-counterterm scheme.

Remark 95.15 (F-monotonicity is not a matrix-model identity). The conditional monotonicity statement above is a continuum statement about unitary RG flows equipped with the needed entropic or algebraic construction. A localized S^3 matrix model computes F at a supersymmetric fixed point when its localization input is valid; it does not by itself prove monotonicity for flows between two fixed points. Conversely, the expected monotonicity phenomenon is not limited to supersymmetric flows for which a localization formula exists.

Remark 95.16 (Consequences for supersymmetric fixed-point comparisons). Suppose $\mathcal{T}_{\mathrm{UV}} \to \mathcal{T}_{\mathrm{IR}}$ is a three-dimensional unitary RG flow for which the conditional entropic construction in Remark 95.13 has been supplied, and suppose both endpoints have supersymmetric localization data whose finite parts compute the renormalized S^3 partition functions in the scheme of Definition 95.12. Then the localized endpoint comparison reads

$$-\log|Z^{\mathrm{loc}}_{S^3}(\mathcal{T}_{\mathrm{UV}})| \geq -\log|Z^{\mathrm{loc}}_{S^3}(\mathcal{T}_{\mathrm{IR}})|$$

with strictness only after an equality-rigidity input has also been supplied. For a product of decoupled fixed points $\mathcal{T}_1 \otimes \mathcal{T}_2$, the sphere free energy is additive:

$$F(\mathcal{T}_1 \otimes \mathcal{T}_2) = F(\mathcal{T}_1) + F(\mathcal{T}_2). \tag{95.39}$$

The first assertion is the conditional inequality (95.38) after identifying the endpoint F's with the localized finite parts. This identification is an endpoint statement: it uses the localization input at the fixed point and does not require a localized description of the entire RG trajectory.

For the product statement, the Hilbert space, operator algebra, and Euclidean functional integral factorize. With product regulators and counterterm schemes,

$$Z_{S^3}(\mathcal{T}_1 \otimes \mathcal{T}_2) = Z_{S^3}(\mathcal{T}_1) Z_{S^3}(\mathcal{T}_2).$$

Taking $-\log|\cdot|$ gives (95.39). Possible background contact terms also add because they are local functionals of the external fields.

Example 95.17 (Free chiral multiplet and a massive flow)**.** For a free $\mathcal{N}=2$ chiral multiplet of R-charge $1/2$ and no background mass, the round-sphere determinant convention of Chapter 101 gives

$$Z_{\mathrm{chiral}} = \exp \ell\left(\frac{1}{2}\right) = 2^{-1/2}, \qquad F_{\mathrm{chiral}} = \frac{1}{2}\log 2. \tag{95.40}$$

Giving this chiral multiplet a supersymmetric real mass and flowing to energies much smaller than $|m|$ leaves the empty gapped theory, whose non-topological sphere free energy is 0 in the same normalization. The flow therefore has

$$F_{\text{UV}} = \frac{1}{2}\log 2 > 0 = F_{\text{IR}},$$

consistent with the expected F-monotonicity pattern. This example is simple, but it fixes the normalization used when more complicated localized matrix integrals are compared.

Rank-one ABJM sphere integral. With the oscillatory contour and framing convention implicit in (95.34), the $N = 1$ ABJM matrix integral evaluates distributionally to

$$Z_{S^3}^{\text{ABJM}}(1, k) = \frac{1}{4|k|}, \qquad F_{\text{ABJM}}(1, k) = \log(4|k|). \tag{95.41}$$

For $N = 1$, the vector Vandermonde factors are absent and (95.34) becomes

$$Z = \int_{\mathbb{R}^2} d\lambda\, d\mu\, \frac{\exp\{\mathrm{i}\pi k(\lambda^2 - \mu^2)\}}{4\cosh^2(\pi(\lambda - \mu))}.$$

Set

$$x = \lambda - \mu, \qquad y = \lambda + \mu.$$

Then $d\lambda\, d\mu = \frac{1}{2}\, dx\, dy$ and $\lambda^2 - \mu^2 = xy$. In the oscillatory distribution sense,

$$\int_{-\infty}^{\infty} dy\, \mathrm{e}^{\mathrm{i}\pi kxy} = 2\,\delta(kx) = \frac{2}{|k|}\delta(x).$$

Therefore

$$Z = \frac{1}{2}\int dx\, \frac{1}{4\cosh^2(\pi x)}\frac{2}{|k|}\delta(x) = \frac{1}{4|k|}.$$

Taking the absolute value and then $-\log$ gives the displayed free energy.

95.13 ABJM Fermi-Gas Form and Airy Asymptotics

The ABJM sphere integral has a second exact representation which is useful because it separates three logically different statements:

1. the finite-dimensional localization integral (95.34);
2. an algebraic rewriting of that integral as the canonical partition function of N noninteracting fermions with a one-particle density operator;
3. the large-N Airy behavior that follows when the grand potential of that one-particle operator has a cubic perturbative part.

Only the second item is a change of variables in the localized integral. It does not define the continuum ABJM theory and it does not replace the localization hypotheses. Throughout this section take $k > 0$; negative level is obtained by orientation reversal and complex conjugation of the Chern–Simons phase.

Let $\hbar = 2\pi k$, let q denote multiplication on $L^2(\mathbb{R}, dq)$, and let $p = -\mathrm{i}\hbar\, d/dq$. Define

$$u(q) = \frac{1}{\sqrt{2\cosh(q/2)}}, \qquad v(p) = \frac{1}{2\cosh(p/2)}.$$

The ABJM one-particle density operator is

$$\rho_k = u(q)\, v(p)\, u(q). \tag{95.42}$$

Equivalently, in the q-representation,

$$\rho_k(q, q') = \frac{1}{2\pi k} \frac{1}{\sqrt{2\cosh(q/2)}} \frac{1}{2\cosh\left(\frac{q-q'}{2k}\right)} \frac{1}{\sqrt{2\cosh(q'/2)}}. \tag{95.43}$$

Lemma 95.18 (Trace-class normalization of the ABJM kernel)**.** *The operator ρ_k in* (95.42) *is positive and trace class. Its first trace is*

$$\mathrm{Tr}_{\mathcal{H}}\, \rho_k = \frac{1}{4k}. \tag{95.44}$$

Proof. Use the unitary Fourier transform normalized by

$$\langle q|p\rangle = (2\pi\hbar)^{-1/2} \mathrm{e}^{\mathrm{i}qp/\hbar}.$$

The multiplication operator $v(p)$ is positive. Hence

$$\rho_k = \left(v(p)^{1/2} u(q)\right)^* \left(v(p)^{1/2} u(q)\right)$$

is positive. The operator $v(p)^{1/2}u(q)$ is Hilbert–Schmidt, because its p, q kernel has squared norm

$$\int_{\mathbb{R}^2} \frac{dp\, dq}{2\pi\hbar} \frac{1}{2\cosh(p/2)} \frac{1}{2\cosh(q/2)} < \infty.$$

Therefore $\rho_k = T^*T$ with T Hilbert–Schmidt, so ρ_k is trace class. For a positive operator written in this form, $\mathrm{Tr}(T^*T) = \|T\|_{\mathrm{HS}}^2$; thus the trace can be computed either from the Hilbert–Schmidt integral or from the diagonal of the continuous kernel below. This avoids using any formal delta-function trace.

The Fourier integral

$$\int_{-\infty}^{\infty} \frac{dp}{2\pi\hbar} \frac{\mathrm{e}^{\mathrm{i}p(q-q')/\hbar}}{2\cosh(p/2)} = \frac{1}{2\pi k} \frac{1}{2\cosh\left(\frac{q-q'}{2k}\right)}$$

gives (95.43). On the diagonal,

$$\rho_k(q, q) = \frac{1}{2\pi k} \frac{1}{4\cosh(q/2)}.$$

Since

$$\int_{-\infty}^{\infty} \frac{dq}{\cosh(q/2)} = 2\pi,$$

the trace is $1/(4k)$. □

Quoted Theorem 95.19 (Finite ABJM Fermi-gas identity)**.** *For $k > 0$, the ABJM localization integral* (95.34) *is equal to*

$$Z_{S^3}^{\mathrm{ABJM}}(N,k) = \frac{1}{N!}\int_{\mathbb{R}^N}\prod_{i=1}^{N} dq_i \det_{1\le i,j\le N}\rho_k(q_i,q_j). \tag{95.45}$$

Remark 95.20 (What enters the finite identity)*.* The identity is finite-dimensional algebra inside the localized matrix model, not a holographic or continuum-existence statement. Its input is the change of variables

$$x_i = 2\pi\lambda_i, \qquad y_i = 2\pi\mu_i.$$

The main determinant step is the hyperbolic Cauchy identity

$$\frac{\prod_{i<j} 2\sinh\left(\frac{x_i-x_j}{2}\right)\prod_{i<j} 2\sinh\left(\frac{y_i-y_j}{2}\right)}{\prod_{i,j} 2\cosh\left(\frac{x_i-y_j}{2}\right)} = \det_{i,j}\frac{1}{2\cosh\left(\frac{x_i-y_j}{2}\right)}. \tag{95.46}$$

It follows from the ordinary Cauchy identity after setting $X_i = \mathrm{e}^{x_i}$, $Y_j = \mathrm{e}^{y_j}$ and extracting monomials from rows and columns. Squaring (95.46) rewrites the Vandermonde ratio in (95.34) as a product of determinants. The remaining step is a Gaussian Fourier transform plus a unitary phase conjugation. Phase conjugation leaves every Fredholm minor unchanged:

$$\det\left[\mathrm{e}^{\mathrm{i}kq_i^2/4\pi}\rho_k(q_i,q_j)\mathrm{e}^{-\mathrm{i}kq_j^2/4\pi}\right]_{i,j=1}^{N} = \det[\rho_k(q_i,q_j)]_{i,j=1}^{N}.$$

The Fourier normalization is the one displayed in the proof of Lemma 95.18. For $N=1$, (95.45) gives $\operatorname{Tr}\rho_k = 1/(4k)$, agreeing with the rank-one sphere-integral calculation above.

Definition 95.21 (ABJM grand partition function)**.** The grand partition function attached to the ABJM Fermi-gas operator is

$$\Xi_k(\kappa) = \det_{\mathcal{H}}(1+\kappa\rho_k), \qquad \kappa = \mathrm{e}^{\mu}, \tag{95.47}$$

and its logarithm

$$J_k(\mu) = \log \Xi_k(\mathrm{e}^{\mu}) \tag{95.48}$$

is the grand potential.

Lemma 95.22 (Fredholm expansion and canonical coefficients)**.** *Let ρ be a trace-class operator on $L^2(\mathbb{R}, dq)$ with integral kernel $\rho(q,q')$. Then*

$$\det(1+\kappa\rho) = \sum_{N=0}^{\infty}\kappa^N\frac{1}{N!}\int_{\mathbb{R}^N}\prod_{i=1}^{N} dq_i \det_{i,j}\rho(q_i,q_j). \tag{95.49}$$

Consequently the coefficient of κ^N in $\Xi_k(\kappa)$ is the ABJM sphere partition function $Z_{S^3}^{\mathrm{ABJM}}(N,k)$.

Proof. For finite-rank

$$\rho = \sum_{a=1}^{r} s_a\,|\psi_a\rangle\langle\phi_a|$$

with $\psi_a, \phi_a \in L^2(\mathbb{R})$, the nonzero spectrum of ρ is the spectrum of the $r \times r$ matrix

$$M_{ab} = s_a \langle \phi_a, \psi_b \rangle.$$

Thus $\det(1+\kappa\rho) = \det_r(1+\kappa M)$. On the other hand the N-th integral minor is the trace of the exterior power $\wedge^N \rho$:

$$\operatorname{Tr}(\wedge^N \rho) = \frac{1}{N!} \int_{\mathbb{R}^N} \prod_{i=1}^{N} dq_i \, \det_{i,j} \rho(q_i, q_j).$$

Indeed, expanding $\det_{i,j} \rho(q_i, q_j)$ in the displayed finite-rank decomposition and integrating the alternating products gives exactly the Cauchy–Binet formula for the $N \times N$ principal exterior minors of M. Summing these exterior traces gives the characteristic-polynomial identity

$$\det(1+\kappa\rho) = \sum_{N=0}^{r} \kappa^N \operatorname{Tr}(\wedge^N \rho).$$

For a general trace-class ρ, choose finite-rank $\rho_m \to \rho$ in trace norm. The Fredholm determinant is defined by the absolutely convergent series

$$\det(1+\kappa\rho) = \sum_{N=0}^{\infty} \kappa^N \operatorname{Tr}(\wedge^N \rho), \qquad |\operatorname{Tr}(\wedge^N \rho)| \le \frac{\|\rho\|_1^N}{N!},$$

and the same estimate, applied to multilinear differences of exterior powers, gives

$$\operatorname{Tr}(\wedge^N \rho_m) \to \operatorname{Tr}(\wedge^N \rho)$$

for each N, with a locally uniform majorant in κ. Hence the finite-rank identities pass to the trace-norm limit and yield (95.49). For the ABJM kernel, the unitary conjugation used above leaves all minors invariant and Quoted Theorem 95.19 identifies the N-th coefficient with the localized S^3 integral $Z_{S^3}^{\mathrm{ABJM}}(N,k)$. □

Leading phase-space coefficient. The leading Weyl coefficient of the ABJM Fermi gas is

$$C_k = \frac{2}{\pi^2 k}. \tag{95.50}$$

Equivalently, the leading large-μ contribution to the grand potential is

$$J_k(\mu) = \frac{C_k}{3}\mu^3 + \text{lower powers of } \mu \qquad (\mu \to +\infty) \tag{95.51}$$

at fixed k.

The classical symbol of ρ_k^{-1} is

$$\exp H(q,p), \qquad H(q,p) = \log\Bigl(2\cosh\frac{q}{2}\Bigr) + \log\Bigl(2\cosh\frac{p}{2}\Bigr).$$

For large E, the inequality $H(q,p) \le E$ has the same leading area as

$$\frac{|q|+|p|}{2} \le E,$$

namely a diamond of area $8E^2$. The phase-space cell is

$$2\pi\hbar = 4\pi^2 k.$$

Thus the integrated one-particle density has leading term

$$n(E) = \frac{8E^2}{4\pi^2 k} + O(E) = \frac{2}{\pi^2 k}E^2 + O(E).$$

At zero temperature the leading grand potential is

$$J_k(\mu) = \int_0^\mu n(E)\,dE + \text{lower powers of } \mu = \frac{2}{3\pi^2 k}\mu^3 + \text{lower powers},$$

which gives (95.50).

Quoted Theorem 95.23 (Perturbative ABJM grand potential)**.** *For the trace-class operator ρ_k above, the part of $J_k(\mu)$ which is polynomial in μ in the large-μ spectral expansion is*

$$J_k^{\text{pert}}(\mu) = \frac{C_k}{3}\mu^3 + B_k\mu + A_k, \qquad C_k = \frac{2}{\pi^2 k}, \qquad B_k = \frac{k}{24} + \frac{1}{3k}. \tag{95.52}$$

The remaining terms in $J_k(\mu)$ are exponentially small in μ within the spectral expansion and encode nonperturbative corrections to the canonical large-N answer. The coefficient A_k is independent of μ and depends on the normalization of the exact spectral determinant.

The inverse Laplace transform of a cubic perturbative grand potential is an Airy function. Assume

$$J^{\text{pert}}(\mu) = \frac{C}{3}\mu^3 + B\mu + A, \qquad C > 0. \tag{95.53}$$

Define the perturbative canonical coefficient by the inverse Laplace transform

$$Z_N^{\text{pert}} = \int_{\mathcal{C}} \frac{d\mu}{2\pi\mathrm{i}}\, \exp\bigl(J^{\text{pert}}(\mu) - N\mu\bigr)\,, \tag{95.54}$$

where $\mathcal{C}$ is the standard Airy contour. Then

$$Z_N^{\text{pert}} = \mathrm{e}^A\, C^{-1/3}\, \mathrm{Ai}\Bigl(C^{-1/3}(N-B)\Bigr)\,. \tag{95.55}$$

For ABJM this gives

$$Z_{S^3}^{\text{ABJM,pert}}(N,k) = \mathrm{e}^{A_k}\, C_k^{-1/3}\, \mathrm{Ai}\Bigl(C_k^{-1/3}(N-B_k)\Bigr)\,, \tag{95.56}$$

with C_k, B_k as in (95.52). Set $t = C^{1/3}\mu$. Then $d\mu = C^{-1/3}dt$ and

$$\frac{C}{3}\mu^3 - (N-B)\mu = \frac{t^3}{3} - C^{-1/3}(N-B)t.$$

Therefore

$$Z_N^{\text{pert}} = \mathrm{e}^A C^{-1/3} \int_{\mathcal{C}_A} \frac{dt}{2\pi\mathrm{i}}\, \exp\left(\frac{t^3}{3} - C^{-1/3}(N-B)t\right)\,.$$

The last integral is the contour definition of the Airy function $\mathrm{Ai}(z)$ with $z = C^{-1/3}(N-B)$. This proves (95.55). Substituting the ABJM coefficients gives (95.56).

Remark 95.24 (Status of the ABJM Airy formula). The Airy function in (95.56) is an exact consequence of the cubic perturbative grand potential; it is not by itself a statement about the full finite-N sphere partition function. The full Fredholm determinant $\Xi_k(\kappa)$ also contains exponentially small spectral corrections, and their inverse Laplace transforms correct the Airy expression. Thus the logically complete chain is: localization matrix integral, Fermi-gas Fredholm determinant, spectral large-μ expansion, Airy transform of the cubic part, and then nonperturbative corrections. Confusing the last two steps is a common source of overstated "exact large-N" claims.

95.14 Three-Dimensional Dynamics Summary

A complete $3D$ supersymmetric Chern–Simons–matter example must specify:

$$G, \quad k, \quad R, \quad W, \quad \zeta, \quad m_i,$$
$$\text{spin structure and framing data}, \quad \text{parity-anomaly regulator},$$
$$\text{monopole-operator dressing rules}, \quad \text{boundary and global-form choices}.$$

For ABJM this list becomes

$$U(N)_k \times U(N)_{-k}, \quad A_a, B_{\dot{a}}, \quad W_{\text{ABJM}},$$
$$\text{the discrete level } k \text{ and the zero-dimensional standard conformal locus},$$
$$\text{the chosen global form and line-operator lattice},$$
$$\text{the monopole sectors implementing the } SU(4) \text{ symmetry and possible } \mathcal{N}=8 \text{ enhancement},$$
$$\text{and the } S^3 \text{ contour, framing, and contact-term convention}.$$

Duality proposals compare these full data, including contact terms and background Chern–Simons counterterms.

Open Problem 95.25 (Three-dimensional supersymmetric continuum schemes). Construct $3D$ supersymmetric Chern–Simons–matter theories with a regulator preserving the stated gauge and supersymmetry Ward identities, including parity-anomaly choices, monopole-operator definitions, boundary conditions, contact-term coordinates, and global-form data.

Chapter 96

Six-Dimensional Superconformal Theories

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Six-dimensional superconformal theories are QFTs whose central examples do not admit ordinary Lagrangian descriptions at the fixed point. They therefore force a separation between the definition of a theory and the use of effective field variables on a branch of vacua. This chapter records the objects that can be defined without assuming a microscopic Lagrangian: superconformal representation data, tensor-branch effective actions, anomaly polynomials, BPS strings, compactification tests, and extended operators. The emphasis is the $(2,0)$ family and the adjacent $(1,0)$ tensor-branch formalism, because these are the supersymmetric field theories that most sharply test the distinction between protected data and construction of a local QFT.

Hypothesis 96.1 ($(2,0)$ theory of ADE type)**.** Let $\mathfrak{g}$ be a simply laced complex simple Lie algebra of type A, D, or E, with Cartan algebra $\mathfrak{t}_\mathfrak{g}$, Weyl group $W_\mathfrak{g}$, root lattice $Q_\mathfrak{g}$, weight lattice $P_\mathfrak{g}$, rank $r_\mathfrak{g}$, dimension $d_\mathfrak{g}$, and dual Coxeter number $h^\vee_\mathfrak{g}$. In this chapter the symbol

$$\mathfrak{T}_{6d}[\mathfrak{g}]$$

denotes an assumed six-dimensional local QFT with the following defining properties.

(i) It is a positive-energy $(2,0)$ superconformal QFT with superconformal algebra $OSp(8^*|4)$ and $USp(4)_R \simeq \mathrm{Spin}(5)_R$ R-symmetry.

(ii) Its generic tensor branch, after removing a decoupled center-of-mass free tensor multiplet when present, has scalar coordinate space

$$\big(\mathbb{R}^5 \otimes \mathfrak{t}_{\mathfrak{g}}\big)/W_{\mathfrak{g}},$$

and its low-energy abelian tensor fields are quantized with string-charge lattice $Q_{\mathfrak{g}}$ and Cartan pairing.

(iii) Its anomaly polynomial is

$$I_8[\mathfrak{g}] = r_{\mathfrak{g}} I_8(1) + \frac{d_{\mathfrak{g}} h_{\mathfrak{g}}^{\vee}}{24} p_2(N_R),$$

where N_R is the rank-five $SO(5)_R$ background bundle and $I_8(1)$ is as in (96.6). This is the protected anomaly datum used below; the quoted-theorem boundary is stated at Quoted Theorem 96.6.

(iv) Its BPS string charges on a generic tensor branch are the roots $\alpha \in Q_{\mathfrak{g}}$, with central-charge/tension normalization

$$T_\alpha = |\alpha(\phi)|$$

for $\phi \in \mathbb{R}^5 \otimes \mathfrak{t}_{\mathfrak{g}}$.

(v) Its finite surface-defect charge group is the discriminant group

$$A_{\mathfrak{g}} = P_{\mathfrak{g}}/Q_{\mathfrak{g}},$$

with finite Dirac pairing

$$b_{\mathfrak{g}}([\lambda],[\mu]) = (\lambda,\mu) \bmod \mathbb{Z}.$$

On a closed six-manifold Y, the corresponding finite flux sector is a self-dual finite Heisenberg system built from $H^3(Y; A_{\mathfrak{g}})$ and therefore requires a polarization before it gives a numerical partition function.

(vi) Circle compactification on a circle of radius R, on a branch where the compactified low-energy description is valid, gives five-dimensional maximally supersymmetric Yang–Mills with gauge algebra $\mathfrak{g}$ and trace-delta coupling

$$g_5^2 = 4\pi^2 R.$$

(vii) Riemann-surface compactification is used only with the partial twist and codimension-two defect data specified in Hypothesis 96.10, and only conditioned on existence of the resulting four-dimensional local QFT.

All subsequent uses of $\mathfrak{T}_{6d}[\mathfrak{g}]$ in this chapter are conditional on the existence of a local QFT with these properties. The chapter derives consistency checks and protected consequences from the listed data; it does not construct the interacting six-dimensional local QFT. Open Problem 96.14 records the missing construction problem.

96.1 Construction and Test Architecture

The data in Hypothesis 96.1 should be read as a layered construction target. The first layer is the local fixed-point object: a Hilbert space or equivalent local-QFT presentation with positive-energy $OSp(8^*|4)$ action, stress tensor, local operator algebra, $SO(5)_R$ background coupling, and anomaly functional. This is the layer that would deserve the name “the six-dimensional theory” without qualification. The chapter assumes it through $\mathfrak{T}_{6d}[\mathfrak{g}]$ and derives consequences from the specified properties.

The second layer is the tensor-branch effective description. It supplies abelian self-dual tensor fields, scalar coordinates, the root-lattice string charges, Green–Schwarz couplings, and BPS string central charges. This layer is a branch expansion around separated tensor-branch vacua. It gives local anomaly matching, string-tension formulae, and inflow tests for charged surfaces, and it also records precisely which data become singular at the tensionless origin.

The third layer is the finite global-form and defect datum. The discriminant group $A_{\mathfrak{g}} = P_{\mathfrak{g}}/Q_{\mathfrak{g}}$, the finite pairing $b_{\mathfrak{g}}$, the self-dual finite flux system $H^3(Y; A_{\mathfrak{g}})$, and a maximal-isotropic polarization determine which finite surface-defect charges are genuine in a chosen absolute theory. After circle compactification this same layer becomes the finite global-form choice for line and surface defects in the five-dimensional branch description.

The fourth layer consists of protected compactification tests. Circle compactification compares the BPS string/instanton particle normalization with five-dimensional maximally supersymmetric Yang–Mills and fixes $g_5^2 = 4\pi^2 R$ in trace-delta variables. Riemann-surface compactification, when the class-S datum is supplied, compares the six-dimensional anomaly polynomial with a four-dimensional anomaly pushforward and compares protected Coulomb coordinates with Hitchin-base sections. These compactification tests are stringent because they use the same $\mathfrak{g}$, anomaly, defect, and branch normalizations in lower dimensions.

The fifth layer is the construction frontier. A complete solution would produce the fixed-point local QFT together with the preceding branch, defect, and compactification structures as functorial outputs. The chapter therefore uses the following order: first state the assumed fixed-point datum, then derive branch and anomaly consequences, then expose the finite defect/global-form layer, and finally apply compactification tests. A calculation below should be evaluated by the layer it addresses: local representation theory, branch EFT, anomaly inflow, finite defect polarization, or compactification comparison.

96.2 Six-Dimensional Supersymmetry Data

Let the Lorentz group be $\mathrm{Spin}(1,5)$. A chiral spinor has four complex components before imposing a symplectic-Majorana-Weyl reality condition. Six-dimensional supersymmetry is labeled by $(\mathcal{N}_+, \mathcal{N}_-)$, the numbers of positive and negative chirality symplectic-Majorana-Weyl supercharges. The cases most central for interacting local fixed points are $(1,0)$ and $(2,0)$.

For $(1,0)$, the R-symmetry is $SU(2)_R$. For $(2,0)$, the R-symmetry is $USp(4)_R \simeq \mathrm{Spin}(5)_R$. Superconformal symmetry enlarges the bosonic group to

$$SO(2,6) \times G_R,$$

with $G_R = SU(2)_R$ or $USp(4)_R$. The full superconformal algebras are usually denoted $OSp(8^*|2)$ for $(1,0)$ and $OSp(8^*|4)$ for $(2,0)$. Local operators are organized as positive-energy representations of these algebras on the cylinder. This statement concerns Hilbert-space representation theory; it does not supply off-shell fields.

96.3 Why Six-Dimensional Gauge Fields Are Branch Variables

A six-dimensional vector multiplet can appear in an effective gauge-theory description on a tensor branch, but it is not a conformal gauge field at an interacting fixed point.

Remark 96.2 (Dimension of the Yang–Mills coupling). In a six-dimensional Yang–Mills presentation with canonical gauge field normalization, the gauge coupling has mass dimension -1, equivalently $[g_6^2] = -2$. Hence the local Yang–Mills interaction is not a marginal microscopic coupling of a six-dimensional conformal field theory.

In d spacetime dimensions a canonically normalized vector field has mass dimension

$$[A_\mu] = \frac{d-2}{2}.$$

The covariant derivative is $\partial_\mu + \mathrm{i} g_d A_\mu$, so

$$[g_d] + [A_\mu] = 1.$$

Therefore

$$[g_d] = 1 - \frac{d-2}{2} = \frac{4-d}{2}, \qquad [g_d^2] = 4 - d.$$

Putting $d = 6$ gives $[g_6] = -1$ and $[g_6^2] = -2$. A coupling with negative mass dimension is not a marginal conformal coordinate. Thus any six-dimensional vector-multiplet gauge theory used below is an effective description on a branch, not a Lagrangian definition of the fixed point.

96.4 Multiplets And Self-Dual Fields

A $(1,0)$ tensor multiplet contains a two-form gauge potential B with self-dual field strength, a real scalar ϕ, and fermions. A hypermultiplet contains scalars in quaternionic representations and fermions. A $(2,0)$ tensor multiplet contains

$$B, \qquad \phi^A \ (A = 1, \dots, 5), \qquad \text{fermions},$$

where the five real scalars transform as the vector representation of $\mathrm{Spin}(5)_R$. The field strength

$$H = dB + \cdots$$

obeys a self-duality constraint

$$H = *H.$$

For a nonabelian interacting $(2,0)$ theory there is no known ordinary local action written in terms of nonabelian two-form gauge potentials. Even for abelian self-dual fields, a covariant action requires auxiliary structure or a formulation in which self-duality is imposed after variation. Consequently this chapter states self-dual fields through their equations, charge lattice, anomaly inflow, and compactification tests.

Definition 96.3 (Abelian self-dual tensor system)**.** An abelian six-dimensional tensor sector on a spacetime M_6 consists, locally, of two-form gauge potentials B^I with gauge transformations

$$B^I \longmapsto B^I + d\Lambda^I,$$

three-form field strengths H^I, an integral string-charge lattice $\Lambda_{\rm str}$, and a nondegenerate symmetric Dirac pairing

$$\Omega : \Lambda_{\rm str} \times \Lambda_{\rm str} \longrightarrow \mathbb{Z}.$$

In a topologically trivial local chart without background Chern–Simons corrections, $H^I = dB^I$. Globally, B^I is better regarded as a differential-cohomology field, and the periods of $H^I/2\pi$ pair integrally with string charges. A self-dual tensor field is not specified by an unconstrained Gaussian integral over B^I. Its local equations include a chirality constraint of the form

$$H^I = *H^I$$

after choosing the self-dual subspace determined by the metric and the pairing convention. If both self-dual and anti-self-dual tensors are present, the chirality matrix is correspondingly part of the system.

On-shell degrees of freedom of a chiral two-form. For a free massless two-form gauge potential in six-dimensional Minkowski space, the gauge-invariant one-particle polarizations before imposing self-duality form $\Lambda^2\mathbb{R}^4$, where $\mathbb{R}^4$ is the transverse little-group vector representation. Thus an ordinary two-form has $\binom{4}{2} = 6$ physical bosonic polarizations. The Hodge star on $\Lambda^2\mathbb{R}^4$ squares to 1, so a chiral two-form has

$$\frac{1}{2}\binom{4}{2} = 3$$

physical bosonic polarizations. A free $(2,0)$ tensor multiplet therefore has $3 + 5 = 8$ bosonic on-shell polarizations.

For a massless particle choose momentum k and quotient by gauge transformations $B \mapsto B + k \wedge \lambda$. The physical polarizations are transverse two-forms modulo longitudinal ones; equivalently they are two-forms on the $D-2 = 4$ dimensional transverse representation of the massless little group $SO(4)$. This gives $\binom{4}{2} = 6$ polarizations for an unconstrained two-form. The six-dimensional self-duality condition on the field strength reduces, on the one-particle transverse space, to the four-dimensional decomposition

$$\Lambda^2\mathbb{R}^4 = \Lambda^2_+\mathbb{R}^4 \oplus \Lambda^2_-\mathbb{R}^4, \qquad \dim \Lambda^2_\pm\mathbb{R}^4 = 3.$$

Choosing one chirality leaves three polarizations. The five scalars of the $(2,0)$ tensor multiplet add five further bosonic polarizations, for a total of eight.

96.5 Tensor-Branch Datum

On a tensor branch, expectation values of tensor-multiplet scalars define coordinates

$$\phi^I \in \mathcal{C},$$

where $\mathcal{C}$ is a cone modulo any discrete identifications. The low-energy action contains

$$\frac{1}{2}G_{IJ}\, d\phi^I \wedge *d\phi^J + \frac{1}{2}G_{IJ}\, H^I \wedge *H^J + B^I \wedge X_I^{(4)} + \cdots , \tag{96.1}$$

with self-duality imposed as a constraint on the equations of motion or by a chosen self-dual-field formalism. The four-form $X_I^{(4)}$ encodes gauge, flavor, and gravitational couplings of the branch EFT.

Definition 96.4 (Tensor-branch effective theory)**.** A tensor-branch effective theory consists of

$$\Lambda_{\mathrm{str}}, \quad \Omega_{IJ}, \quad \phi \in \mathcal{C} \subset \Lambda_{\mathrm{str}}^{\vee} \otimes \mathbb{R}^n, \quad B^I,\ H^I,$$
$$X_I^{(4)} \in \Omega^4(M_6), \qquad \text{a self-dual tensor quantization prescription.}$$

The four-forms $X_I^{(4)}$ are built from background gauge, flavor, R-symmetry, and tangent-bundle curvatures. Here $n = 1$ for a $(1,0)$ tensor scalar and $n = 5$ for the $(2,0)$ $\mathrm{Spin}(5)_R$-vector scalar. The cone $\mathcal{C}$ is divided into chambers by loci where charged strings become tensionless. The low-energy formulae on one chamber are not a construction of the fixed point at the chamber wall.

For a $(2,0)$ theory of simply laced type $\mathfrak{g}$, with Cartan algebra $\mathfrak{t}_{\mathfrak{g}}$, rank $r_{\mathfrak{g}}$, and Weyl group $W_{\mathfrak{g}}$, the tensor-branch scalar space has the form

$$\mathcal{C}_{\mathfrak{g}}^{(2,0)} = \left(\mathbb{R}^5 \otimes \mathfrak{t}_{\mathfrak{g}}\right)/W_{\mathfrak{g}}$$

after removing any decoupled center-of-mass tensor multiplet. The five $\mathbb{R}^5$ directions are the $\mathrm{Spin}(5)_R$ vector directions. The Weyl quotient records the identification of Cartan coordinates that survive from the interacting nonabelian theory.

Example 96.5 (Type A_{N-1} tensor branch)**.** For type A_{N-1}, the rank is $N-1$, and the tensor branch may be written as

$$\left\{(x_1, \dots, x_N) \in (\mathbb{R}^5)^N : \sum_{i=1}^{N} x_i = 0\right\}/S_N.$$

The symmetric group permutes the N eigenvalue-like vectors. This formula is a tensor-branch coordinate chart; it is not a definition of the interacting fixed point at the origin $x_i = x_j$.

96.6 Anomaly Polynomial

For a six-dimensional theory coupled to background metric and R-symmetry fields, the anomaly is encoded by an eight-form

$$I_8 \in H^8(BG_{\mathrm{background}}, \mathbb{Q})$$

whose descent gives anomalous variations of the generating functional. For a $(1,0)$ theory one may write

$$I_8 = \alpha\, c_2(R)^2 + \beta\, c_2(R) p_1(T) + \gamma\, p_1(T)^2 + \delta\, p_2(T) + I_8^{\mathrm{flavor}},$$

where $c_2(R)$ is the second Chern class of the $SU(2)_R$ bundle and $p_i(T)$ are Pontryagin classes of the tangent bundle. The coefficients are part of the QFT structure.

On a tensor branch, anomaly matching takes the form

$$I_8^{\mathrm{UV}} - I_8^{\mathrm{branch}} = \frac{1}{2}\Omega^{IJ} X_I^{(4)} X_J^{(4)}, \tag{96.2}$$

where Ω^{IJ} is the inverse tensor pairing. The right-hand side is produced by Green–Schwarz couplings $B^I \wedge X_I^{(4)}$. This equation is a computable constraint on branch EFTs and on proposed fixed points.

Quadratic Green–Schwarz descent calculation. Fix the descent convention in which an anomaly polynomial P_8 contributes

$$\delta_\alpha \Gamma = 2\pi \mathrm{i} \int_{M_6} P_6^{(1)}(\alpha)$$

when $P_8 = dP_7^{(0)}$ and $\delta_\alpha P_7^{(0)} = dP_6^{(1)}(\alpha)$. Suppose

$$X_I^{(4)} = dX_I^{(3)}, \qquad \delta_\alpha X_I^{(3)} = dX_I^{(2)}(\alpha),$$

and Ω^{IJ} is symmetric. Then the quadratic polynomial

$$P_8^{\mathrm{GS}} = \frac{1}{2}\Omega^{IJ} X_I^{(4)} X_J^{(4)} \tag{96.3}$$

has descent representative

$$P_6^{(1),\mathrm{GS}}(\alpha) = \Omega^{IJ} X_I^{(2)}(\alpha) X_J^{(4)}. \tag{96.4}$$

With the sign of the local Green–Schwarz functional chosen so that its variation is $2\pi\mathrm{i}\int P_6^{(1),\mathrm{GS}}$, the tensor-branch matching condition is precisely

$$I_8^{\mathrm{UV}} = I_8^{\mathrm{branch}} + P_8^{\mathrm{GS}}.$$

The substantive six-dimensional input is the anomaly polynomial of the fixed point and of the branch theory; the following calculation fixes the local descent representative and sign convention. Because $X_I^{(4)}$ has even degree, the product $X_I^{(4)} X_J^{(4)}$ is symmetric under $I \leftrightarrow J$. A descent seven-form for (96.3) is

$$P_7^{(0),\mathrm{GS}} = \frac{1}{2}\Omega^{IJ}\big(X_I^{(3)} X_J^{(4)} + X_J^{(3)} X_I^{(4)}\big),$$

which equals $\Omega^{IJ} X_I^{(3)} X_J^{(4)}$ after using the symmetry of Ω^{IJ}. Varying gives

$$\delta_\alpha P_7^{(0),\mathrm{GS}} = \Omega^{IJ} dX_I^{(2)}(\alpha) X_J^{(4)} = d\Big(\Omega^{IJ} X_I^{(2)}(\alpha) X_J^{(4)}\Big),$$

since $dX_J^{(4)} = 0$. This proves (96.4). The final matching equation is the statement that the anomaly of the branch massless fields plus the anomaly generated by the Green–Schwarz term equals the anomaly of the fixed point.

Quoted Theorem 96.6 ($(2,0)$ anomaly polynomial)**.** *Let $\mathfrak{g}$ be simply laced, with rank $r_{\mathfrak{g}}$, dimension $d_{\mathfrak{g}}$, and dual Coxeter number $h_{\mathfrak{g}}^\vee$. The $(2,0)$ theory of type $\mathfrak{g}$ has anomaly polynomial*

$$I_8[\mathfrak{g}] = r_{\mathfrak{g}} I_8(1) + \frac{d_{\mathfrak{g}} h_{\mathfrak{g}}^\vee}{24}\, p_2(N_R), \tag{96.5}$$

where N_R is the rank-five $SO(5)_R$ background bundle and

$$I_8(1) = \frac{1}{48}\left[p_2(N_R) - p_2(T) + \frac{1}{4}\big(p_1(T) - p_1(N_R)\big)^2\right] \tag{96.6}$$

is the anomaly polynomial of one free $(2,0)$ tensor multiplet.

The theorem is quoted because the chapter does not reproduce a construction of the interacting six-dimensional local QFT together with a first-principles derivation of its anomaly functional. In this monograph it is used only as protected QFT data: it fixes anomaly coefficients that any proposed construction of the $(2,0)$ theory must reproduce. External geometric or string-theoretic derivations may motivate the formula, but they are not premises in the argument of this chapter.

What the quoted theorem asserts on the tensor branch. The generic tensor branch of the type $\mathfrak{g}$ theory contains $r_{\mathfrak{g}}$ free abelian $(2,0)$ tensor multiplets. Therefore

$$I_8[\mathfrak{g}] - r_{\mathfrak{g}} I_8(1) = \frac{d_{\mathfrak{g}} h_{\mathfrak{g}}^{\vee}}{24}\, p_2(N_R) \tag{96.7}$$

is the part of the anomaly which is invisible in the free branch spectrum. The background anomaly is an invertible field theory attached to the QFT, not to a particular coordinate chart on its vacuum branch. Thus the low-energy tensor-branch description must reproduce the same anomaly functional as the fixed point after its massless fields and its local Green–Schwarz terms are both included. This is the precise sense in which the formula is tested by branch data: the branch fields give $r_{\mathfrak{g}} I_8(1)$, while the remaining coefficient is a quadratic functional of the four-form couplings $X_I^{(4)}$ in (96.2). It is not an additional propagating degree-of-freedom count.

The following finite root-system facts are the arithmetic skeleton of this matching. Normalize the simply laced root system $\Delta_{\mathfrak{g}} \subset \mathfrak{t}_{\mathfrak{g}}^{\vee}$ so every root has squared length 2, and identify $\mathfrak{t}_{\mathfrak{g}} \simeq \mathfrak{t}_{\mathfrak{g}}^{\vee}$ by this inner product. Then

$$\sum_{\alpha \in \Delta_{\mathfrak{g}}^{+}} \alpha \otimes \alpha = h_{\mathfrak{g}}^{\vee}\, \mathrm{id}_{\mathfrak{t}_{\mathfrak{g}}}, \tag{96.8}$$

or, equivalently,

$$\sum_{\alpha \in \Delta_{\mathfrak{g}}^{+}} \langle \alpha, v\rangle^2 = h_{\mathfrak{g}}^{\vee}\, \langle v, v\rangle \qquad (v \in \mathfrak{t}_{\mathfrak{g}}).$$

Indeed the full sum over all roots defines a Weyl-invariant symmetric endomorphism of the irreducible reflection representation of $W_{\mathfrak{g}}$, hence is a scalar multiple of the identity. Taking the trace fixes the scalar: the trace of the full sum is $\sum_{\alpha \in \Delta_{\mathfrak{g}}} \langle \alpha, \alpha\rangle = 2|\Delta_{\mathfrak{g}}|$, while $|\Delta_{\mathfrak{g}}| = d_{\mathfrak{g}} - r_{\mathfrak{g}} = r_{\mathfrak{g}} h_{\mathfrak{g}}^{\vee}$. Dividing by $r_{\mathfrak{g}}$ gives $2h_{\mathfrak{g}}^{\vee}$ for the full root sum and $h_{\mathfrak{g}}^{\vee}$ for the positive-root sum. The identity

$$d_{\mathfrak{g}} = r_{\mathfrak{g}}(h_{\mathfrak{g}}^{\vee} + 1) \tag{96.9}$$

is the same root count with the Cartan directions restored.

These identities do not prove (96.5); they explain why, once the normalization of the self-dual tensor Green–Schwarz coupling is fixed on a long-root string, the Weyl-invariant quadratic contribution on a chamber has no remaining independent symmetric tensor structure. The nontrivial QFT datum in the quoted theorem is the overall coefficient with which the interacting fixed point realizes this rigid invariant.

Remark 96.7 (ADE arithmetic in the $(2,0)$ anomaly coefficient)*.* For simply laced $\mathfrak{g}$, the coefficient multiplying $p_2(N_R)$ in (96.5) is

$$\frac{d_{\mathfrak{g}} h_{\mathfrak{g}}^{\vee}}{24}.$$

For the ADE series this gives

$\mathfrak{g}$	$r_{\mathfrak{g}}$	$d_{\mathfrak{g}}$	$h_{\mathfrak{g}}^{\vee}$	$d_{\mathfrak{g}} h_{\mathfrak{g}}^{\vee}/24$
A_{N-1}	$N-1$	N^2-1	N	$N(N^2-1)/24$
D_N	N	$N(2N-1)$	$2N-2$	$N(N-1)(2N-1)/12$
E_6	6	78	12	39
E_7	7	133	18	399/4
E_8	8	248	30	310

The A_{N-1} coefficient grows as $N^3/24$ at large N, while the number of free tensor multiplets on the generic branch is only $N-1$.

The entries use the standard simply laced Lie-algebra dimensions and dual Coxeter numbers:

$$d_{A_{N-1}} = N^2 - 1, \quad h^\vee_{A_{N-1}} = N,$$

$$d_{D_N} = N(2N-1), \quad h^\vee_{D_N} = 2N-2,$$

and

$$(d_{E_6}, h^\vee_{E_6}) = (78, 12), \quad (d_{E_7}, h^\vee_{E_7}) = (133, 18), \quad (d_{E_8}, h^\vee_{E_8}) = (248, 30).$$

Substitution into $d_\mathfrak{g} h^\vee_\mathfrak{g}/24$ gives the table. For A_{N-1},

$$\frac{N(N^2-1)}{24} = \frac{N^3}{24} - \frac{N}{24},$$

which displays the cubic scaling.

96.7 BPS Strings

Tensor multiplets couple electrically to strings. A string with charge

$$q \in \Lambda_{\text{string}}$$

has tension

$$T_q = q_I \phi^I$$

on a $(1,0)$ tensor branch, in a chamber where this pairing is positive. For a $(2,0)$ theory of type $\mathfrak{g}$, the charge lattice is identified with the root lattice in the simply connected convention, and a root α gives a string with tension

$$T_\alpha = |\alpha(\phi)|, \qquad \phi \in \mathbb{R}^5 \otimes \mathfrak{t}_\mathfrak{g}.$$

At the superconformal point some tensions vanish, and the corresponding degrees of freedom are strings, not particles in six dimensions.

Proposition 96.8 (BPS string tension from the tensor-branch central charge)**.** *Normalize the tensor-branch scalar coordinates so that the string central charge vector of a charge $q \in \Lambda_{\text{str}}$ is*

$$Z_q^A = q_I \phi^{IA}, \qquad A = 1, \dots, n,$$

where $n = 1$ in $(1,0)$ notation and $n = 5$ in $(2,0)$ notation. The positive-energy representation of the six-dimensional supersymmetry algebra then gives the BPS inequality

$$T_q \geq |Z_q|,$$

and a BPS string saturates it:

$$T_q = |q_I \phi^I|.$$

For a $(2,0)$ theory of type $\mathfrak{g}$, taking Λ_{str} to be the root lattice gives

$$T_\alpha = |\alpha(\phi)|$$

for the string of root charge α.

Proof. Work per unit length for a straight static string oriented, say, along x^5. Translation invariance along the string replaces the energy by a tension T_q, and the tensor-multiplet central charge appears in the supercharge anticommutator as an R-symmetry vector Z_q^A. Choose the positive normalization of the supercharges so that the coefficient of the tension term in the anticommutator is T_q. After choosing a unit vector $\widehat{Z}^A = Z_q^A/|Z_q|$, the relevant Hermitian anticommutator on the subspace of spinor/R-symmetry parameters is

$$\{Q(\varepsilon), Q(\varepsilon)^\dagger\} = \varepsilon^\dagger\big(T_q\,\mathbf{1} - |Z_q|\,\Gamma_{\rm str}\Gamma_{\widehat{Z}}\big)\varepsilon.$$

Here $\Gamma_{\rm str}$ is the chirality matrix determined by the oriented two-plane swept out by the static string, and $\Gamma_{\widehat{Z}}$ is the unit R-symmetry Clifford multiplication in the direction $\widehat{Z}$. The two matrices square to 1 on the relevant spinor space and their product has eigenvalues ± 1. Positivity of the Hilbert-space norm of $Q(\varepsilon)^\dagger|\text{string}\rangle$ for every ε therefore requires

$$T_q - |Z_q| \geq 0.$$

If equality holds, the eigenspace with $\Gamma_{\rm str}\Gamma_{\widehat{Z}} = +1$ is annihilated by a nonzero linear combination of supercharges; conversely, preservation of a nonzero supercharge forces the smallest eigenvalue of the displayed matrix to vanish. This is the BPS saturation condition. With the scalar-coordinate normalization in the statement, $Z_q^A = q_I\phi^{IA}$, hence $T_q = |Z_q| = |q_I\phi^I|$ for a saturated string. In the $(2,0)$ ADE theory the charged strings on a generic tensor branch are labeled by roots, so $q = \alpha$ gives $T_\alpha = |\alpha(\phi)|$. □

Local anomaly inflow to a charged string. Let $\iota : \Sigma \hookrightarrow M_6$ be the oriented two-dimensional support of a string of charge q. Write N_Σ for its oriented rank-four normal bundle, and put

$$q^2 = q_I\Omega^{IJ}q_J.$$

The charge convention is the one used in the tensor-branch coupling: the surface insertion contains the differential-cohomology holonomy

$$\exp\left(2\pi\mathrm{i}\,q_I\int_\Sigma B^I\right).$$

In a local background chart with $X_I^{(4)} = dX_I^{(3)}$ and $\delta X_I^{(3)} = dX_I^{(2)}$, Green–Schwarz gauge invariance is expressed by the local transformation

$$\delta B^I = -X_I^{(2)}.$$

Consequently the charged insertion varies by

$$\delta \log W_q(\Sigma) = -2\pi\mathrm{i}\int_\Sigma q_I\,\iota^* X_I^{(2)}.$$

A two-dimensional quantum theory supported on Σ cancels this variation precisely when its anomaly polynomial contains the four-form

$$I^\Sigma_{4,\rm lin}(q) = q_I\,\iota^* X_I^{(4)}. \tag{96.10}$$

This is the linear inflow term. It is not optional: it is the descent of the same $B^I \wedge X_I^{(4)}$ coupling that appears in the six-dimensional Green–Schwarz matching equation.

There is also a local normal-bundle inflow from the self-pairing of the tensor field sourced by the string. It is best stated using the global angular-form regularization of a tubular neighborhood of Σ. Let $S(N_\Sigma) \to \Sigma$ be the unit-sphere bundle of the normal bundle. In the standard normalization the Chern–Simons form $e_3^{(0)}(N_\Sigma)$ on this sphere bundle satisfies

$$de_3^{(0)}(N_\Sigma) = \pi^* e(N_\Sigma), \qquad \delta e_3^{(0)}(N_\Sigma) = de_2^{(1)}(N_\Sigma),$$

where $e(N_\Sigma)$ is the Euler class of the rank-four normal bundle. Replacing the delta-current Poincare dual to Σ by the Thom form obtained from $e_3^{(0)}$ gives a smooth representative of the sourced Bianchi identity near the support. Substituting this representative into the quadratic tensor pairing produces, under a normal-bundle gauge transformation, the descent variation

$$2\pi \mathrm{i} \int_\Sigma \frac{1}{2}\, q^2\, e_2^{(1)}(N_\Sigma).$$

Equivalently the normal-bundle part of the inflow anomaly polynomial is

$$I^{\Sigma}_{4,\mathrm{norm}}(q) = \frac{1}{2}\, q_I \Omega^{IJ} q_J\, e(N_\Sigma). \tag{96.11}$$

The coefficient 1/2 is the same symmetric-pairing coefficient as in the quadratic Green–Schwarz descent: the self-interaction of one sourced tensor field is counted once, while distinct charge components are paired symmetrically by Ω^{IJ}.

Thus the bulk inflow polynomial for a string of charge q, in these normalizations, is

$$I^{\Sigma}_{4,\mathrm{inflow}}(q) = q_I\, \iota^* X_I^{(4)} + \frac{1}{2}\, q_I \Omega^{IJ} q_J\, e(N_\Sigma). \tag{96.12}$$

The actual anomaly polynomial of the two-dimensional degrees of freedom supported on the string must cancel this inflow after adding its intrinsic fermion, boson, and zero-mode contributions. Formula (96.12) is therefore a local consistency condition on a proposed BPS-string sector, not a construction of that sector.

Remark 96.9 (What the BPS-string statement does and does not construct)*.* The tension formula is a consequence of the supersymmetry algebra and tensor-branch normalization. It does not construct the interacting tensionless string theory at $T_q = 0$, nor does it define its operator algebra. The tensionless limit is part of the fixed-point data whose existence must be supplied by a construction or by a set of mutually consistent protected tests.

96.8 Defect Group, Finite Flux, And Polarization

The tensor-branch root lattice is not the whole discrete topological datum of a $(2,0)$ theory. Let $Q_{\mathfrak{g}}$ be the root lattice of the simply laced Lie algebra $\mathfrak{g}$, with its integral Cartan pairing $(\ ,\ )$. The dual lattice is

$$P_{\mathfrak{g}} = \{\lambda \in Q_{\mathfrak{g}} \otimes \mathbb{R} : (\lambda, \alpha) \in \mathbb{Z} \text{ for every } \alpha \in Q_{\mathfrak{g}}\},$$

the weight lattice. The finite discriminant group

$$A_{\mathfrak{g}} = P_{\mathfrak{g}}/Q_{\mathfrak{g}} \tag{96.13}$$

is the finite group detected by the center of the associated simply connected compact group. It carries the nondegenerate finite pairing

$$b_{\mathfrak{g}} : A_{\mathfrak{g}} \times A_{\mathfrak{g}} \longrightarrow \mathbb{Q}/\mathbb{Z}, \qquad b_{\mathfrak{g}}([\lambda],[\mu]) = (\lambda,\mu) \bmod \mathbb{Z}. \tag{96.14}$$

The expression is independent of the representatives because adding a root to either entry changes (λ,μ) by an integer. Nondegeneracy is the statement that $P_{\mathfrak{g}}$ is the full dual of $Q_{\mathfrak{g}}$.

The ADE values are

$$\begin{array}{c|c} \mathfrak{g} & A_{\mathfrak{g}} \\ \hline A_{N-1} & \mathbb{Z}_N \\ D_N,\ N \text{ odd} & \mathbb{Z}_4 \\ D_N,\ N \text{ even} & \mathbb{Z}_2 \oplus \mathbb{Z}_2 \\ E_6 & \mathbb{Z}_3 \\ E_7 & \mathbb{Z}_2 \\ E_8 & 0. \end{array} \tag{96.15}$$

The order of $A_{\mathfrak{g}}$ is $\det C_{\mathfrak{g}}$, where $C_{\mathfrak{g}}$ is the Cartan matrix. Thus the displayed table is the finite discriminant form of the root lattice, not a new continuous parameter of the tensor-branch metric.

The group $A_{\mathfrak{g}}$ is often described as the group of surface defect charges modulo dynamical root-lattice strings. This sentence should be read as a statement about a protected topological sector: the monograph is not using it as a construction of the interacting fixed point. The local tensor-branch EFT sees strings of charge $Q_{\mathfrak{g}}$. The quotient $P_{\mathfrak{g}}/Q_{\mathfrak{g}}$ records which finite surface charge can be measured at infinity after dynamical root-lattice strings have been allowed to end or screen.

On a closed oriented six-manifold Y, the finite flux sector associated with this discriminant group is

$$K_Y(\mathfrak{g}) = H^3(Y; A_{\mathfrak{g}}). \tag{96.16}$$

Cup product and (96.14) define a skew pairing

$$\langle x, y\rangle_Y = \int_Y b_{\mathfrak{g}}(x \smile y) \in \mathbb{Q}/\mathbb{Z}. \tag{96.17}$$

The skew sign comes from the odd degree:

$$x \smile y = -\, y \smile x \qquad \text{for } x, y \in H^3(Y; A_{\mathfrak{g}}),$$

while $b_{\mathfrak{g}}$ is symmetric. A self-dual finite flux theory is not an ordinary sum over all of $K_Y(\mathfrak{g})$. It determines a finite Heisenberg representation whose commutator is

$$U_x U_y U_x^{-1} U_y^{-1} = \exp\bigl(2\pi \mathrm{i} \langle x, y\rangle_Y\bigr).$$

To obtain an ordinary numerical partition function one must supply a polarization, namely a maximal isotropic subgroup

$$L \subset K_Y(\mathfrak{g}), \qquad \langle L, L\rangle_Y = 0, \qquad |L|^2 = |K_Y(\mathfrak{g})|$$

when the finite pairing on $K_Y(\mathfrak{g})$ is nondegenerate. Changing the polarization is a finite global datum, analogous in role to choosing a global form and discrete theta sector after compactification. It is not visible in the local superconformal algebra or in the real tensor-branch metric alone.

Cyclic finite-flux model. For type A_{N-1}, choose a generator of $A_{\mathfrak{g}} \simeq \mathbb{Z}_N$ and write the finite pairing as

$$b(a,b) = \frac{ab}{N} \bmod \mathbb{Z}$$

up to the harmless sign determined by the chosen generator. In the elementary finite model

$$K = \mathbb{Z}_N \oplus \mathbb{Z}_N$$

with coordinates (a,b), the skew pairing is

$$\langle (a,b),(a',b')\rangle = \frac{ab' - ba'}{N} \bmod \mathbb{Z}. \tag{96.18}$$

It is alternating and nondegenerate. The subgroup

$$L = \mathbb{Z}_N \oplus 0$$

is isotropic and has $|L| = N$, while $|K| = N^2$; hence L is a polarization. This finite model is the algebraic core of the polarization choice. In an actual six-dimensional compactification the cohomology group $H^3(Y; A_{\mathfrak{g}})$, torsion linking forms, boundary conditions, and defect insertions must all be specified before the phrase "the partition function" denotes a number rather than a vector in a finite Heisenberg representation.

Circle reduction of the finite Heisenberg system. The same finite datum explains why the global form of the five-dimensional gauge theory is not fixed by the local gauge algebra alone. Let

$$Y = S^1 \times X_5$$

with X_5 closed and oriented, and let $\eta \in H^1(S^1; \mathbb{Z})$ be normalized by $\int_{S^1} \eta = 1$. Pullback from X_5 and cup product with η give the canonical product splitting

$$H^3(S^1 \times X_5; A_{\mathfrak{g}}) \simeq H^3(X_5; A_{\mathfrak{g}}) \oplus \eta\, H^2(X_5; A_{\mathfrak{g}}). \tag{96.19}$$

Write an element as

$$x = u + \eta v, \qquad u \in H^3(X_5; A_{\mathfrak{g}}), \quad v \in H^2(X_5; A_{\mathfrak{g}}).$$

For $y = u' + \eta v'$, the six-dimensional pairing (96.17) becomes

$$\langle u + \eta v,\ u' + \eta v'\rangle_{S^1 \times X_5} = \int_{X_5} \big(b_{\mathfrak{g}}(v \smile u') - b_{\mathfrak{g}}(u \smile v')\big). \tag{96.20}$$

The sign is fixed by moving the degree-one class η past the degree-three class u:

$$u \smile \eta v' = -\,\eta\, u \smile v', \qquad \eta v \smile u' = \eta\, v \smile u'.$$

Thus $H^3(X_5; A_{\mathfrak{g}})$ and $\eta H^2(X_5; A_{\mathfrak{g}})$ are mutually dual isotropic subgroups whenever the resulting finite pairing is nondegenerate.

This is the finite topological core of the five-dimensional global-form choice. A six-dimensional surface defect wrapping S^1 becomes a five-dimensional line defect whose finite charge is measured by $H^2(X_5; A_{\mathfrak{g}})$; an unwrapped six-dimensional surface defect gives a five-dimensional surface defect measured by $H^3(X_5; A_{\mathfrak{g}})$. Equation (96.20) says that the two sets of charges are mutually nonlocal. A polarization chooses a maximal mutually local set before the compactified partition function is a number. In gauge theory language this is the global form and discrete finite-flux datum that supplements the Lie algebra $\mathfrak{g}$. It is not determined by the local $5d$ Yang–Mills Lagrangian density.

Genuine defects after choosing an absolute polarization. The preceding finite Heisenberg system is the point at which the phrase "the $(2,0)$ theory of type $\mathfrak{g}$" must be refined by a global finite datum. Let X_5 be a closed oriented five-manifold and set

$$K_{X_5} = H^3(X_5; A_{\mathfrak{g}}) \oplus H^2(X_5; A_{\mathfrak{g}}),$$

with the alternating pairing

$$\omega_{X_5}\big((u,v),(u',v')\big) = \int_{X_5} \big(b_{\mathfrak{g}}(v \smile u') - b_{\mathfrak{g}}(u \smile v')\big).$$

Poincare duality with finite coefficients, together with nondegeneracy of $b_{\mathfrak{g}}$, makes this pairing perfect modulo any explicitly removed radical. A finite absolute polarization on the circle compactified theory is a choice, functorial under the bordisms under consideration, of a maximal isotropic subgroup

$$L_{X_5} \subset K_{X_5}, \qquad L_{X_5} = L_{X_5}^{\perp_\omega}.$$

The equality with its orthogonal complement is the load-bearing condition, not the notation: it says that the chosen set of finite line/surface charges is maximal among mutually local charges.

Let U_κ, $\kappa \in K_{X_5}$, denote the finite topological operator that shifts the corresponding discrete flux sector. The finite Heisenberg relation is

$$U_\kappa U_\lambda = \exp(2\pi \mathrm{i}\, \omega_{X_5}(\kappa, \lambda))\, U_\lambda U_\kappa.$$

After the absolute polarization L_{X_5} is chosen, the charges $\kappa \in L_{X_5}$ define the mutually local genuine finite-charge defects of that absolute theory. If $\kappa \notin L_{X_5}$, then maximality gives some $\lambda \in L_{X_5}$ with $\omega_{X_5}(\kappa, \lambda) \neq 0$. The corresponding operators fail to commute by the displayed phase, so κ cannot be inserted as an ordinary genuine defect in the same absolute theory while preserving all defects in L_{X_5}. It is instead a relative operator: it changes the finite background sector, or it must end on a topological interface that changes the polarization.

In the elementary A_{N-1} cyclic model, $K = \mathbb{Z}_N \oplus \mathbb{Z}_N$ with pairing (96.18). The two coordinate subgroups

$$L_{\mathrm{surf}} = \mathbb{Z}_N \oplus 0, \qquad L_{\mathrm{line}} = 0 \oplus \mathbb{Z}_N$$

are maximal isotropic. The first choice makes the unwrapped finite surface charges genuine; the second makes the circle-wrapped finite line charges genuine. These are different absolute global forms of the same local compactified theory. A topological interface between two such choices is not an equality of absolute theories; at the finite level it is represented by a Lagrangian relation

$$R \subset K^{(1)}_{X_5} \oplus \overline{K}^{(2)}_{X_5},$$

where the bar reverses the sign of the pairing. The graph of a symplectic automorphism of K_{X_5} is the simplest example. This finite relation is the precise algebraic content behind the common statement that some defects are "non-genuine": the defect exists as a relative morphism between finite sectors, not as an autonomous operator in a fixed absolute polarization.

This paragraph is still only the finite topological layer. It does not construct the full six-dimensional category of codimension-two defects, nor the interacting two-dimensional theories supported on BPS strings. It fixes the consistency condition that any such construction must obey when reduced to the protected finite charge algebra.

96.9 Circle Compactification And Five-Dimensional Yang–Mills

Compactification on a smooth compact manifold Y gives lower-dimensional QFTs whose protected data test the six-dimensional proposal. For a circle of radius R, a $(2,0)$ theory of type $\mathfrak{g}$ yields five-dimensional maximally supersymmetric Yang–Mills on its Coulomb branch with gauge algebra $\mathfrak{g}$. This statement is a compactification test and an effective description, not a Lagrangian definition of the six-dimensional fixed point.

Five-dimensional instanton charge and coupling convention. Use the monograph trace-delta convention for the five-dimensional gauge action,

$$S_{5d} = \frac{1}{4g_5^2}\int \mathrm{tr}_\delta(F_{\mu\nu}F^{\mu\nu}) + \cdots .$$

The topological charge in the five-dimensional gauge theory is the Chern–Weil instanton number on a spatial four-cycle. Let $\mathrm{tr}_{\mathrm{top}}$ denote the characteristic-class trace normalized so that

$$\mathcal{Q}_4(F) := \frac{1}{8\pi^2}\,\mathrm{tr}_{\mathrm{top}}(F\wedge F)$$

has integral periods on principal G-bundles in the charge sector under discussion. On a five-dimensional spacetime X_5, the corresponding current is the one-form

$$j_{\mathrm{inst}} := *_5\,\mathcal{Q}_4(F), \qquad *_5 j_{\mathrm{inst}} = \mathcal{Q}_4(F).$$

The Bianchi identity gives

$$d(*_5 j_{\mathrm{inst}}) = d\mathcal{Q}_4(F) = 0.$$

Thus for a spatial Cauchy four-manifold Σ_4

$$Q_{\mathrm{inst}}(\Sigma_4) = \int_{\Sigma_4} *_5 j_{\mathrm{inst}} = \frac{1}{8\pi^2}\int_{\Sigma_4}\mathrm{tr}_{\mathrm{top}}(F\wedge F)\in\mathbb{Z}. \tag{96.21}$$

This is a conserved charge of the five-dimensional effective theory. In the circle compactification test it is identified with momentum around the compactification circle, not with a new microscopic six-dimensional Noether symmetry.

For a static configuration on spatial $\mathbb{R}^4$, with vanishing electric field and scalar gradients, the energy functional reduces to the Euclidean four-dimensional Yang–Mills action in the same trace-delta convention. The BPST normalization fixed in Lemma 51.23 and Equation (33.53) gives the bound

$$E_{\mathrm{inst}} \geq \frac{4\pi^2}{g_5^2}\,|Q_{\mathrm{inst}}|, \tag{96.22}$$

with equality for a self-dual or anti-self-dual spatial instanton in the appropriate charge sector. The square-completion input is the ordinary four-dimensional instanton bound; the five-dimensional statement is that this finite-action Euclidean saddle is read as a massive particle state when the four coordinates are spatial.

If one unit of five-dimensional instanton charge is identified with one unit of Kaluza–Klein momentum on a circle of radius R, then

$$g_5^2 = 4\pi^2 R.$$

In the common half-trace convention this same relation is

$$g_{5,\mathrm{ht}}^2 = 8\pi^2 R.$$

Kaluza–Klein momentum on a circle of radius R is quantized in units

$$E_{\mathrm{KK}} = \frac{1}{R}$$

for a BPS state with one unit of momentum and no additional transverse motion. Identifying a unit instanton-particle charge with one unit of KK momentum gives

$$\frac{4\pi^2}{g_5^2} = \frac{1}{R},$$

hence $g_5^2 = 4\pi^2 R$. The half-trace convention has $g_{5,\mathrm{ht}}^2 = 2g_5^2$, so $g_{5,\mathrm{ht}}^2 = 8\pi^2 R$.

This proportionality is often quoted without specifying the trace convention. Here it is tied to the instanton normalization used elsewhere in the monograph, so that compactification tests, instanton-soliton masses, and Yang–Mills kinetic terms use the same coupling coordinate. It still does not make five-dimensional Yang–Mills a microscopic six-dimensional definition: the statement uses the compactified branch description, the protected charge identification $Q_{\mathrm{inst}} = P_{S^1} R$, and the BPS mass formula.

Wrapped strings and Coulomb-branch masses. Compactify a $(2,0)$ theory of type $\mathfrak{g}$ on a circle of radius R, and work on a generic tensor branch. A BPS string of root charge α wrapped once on the circle gives a five-dimensional BPS particle with mass

$$M_\alpha^{\mathrm{wrap}} = 2\pi R\,|\alpha(\phi_{6d})|.$$

If this particle is identified with the W-boson of five-dimensional maximally supersymmetric Yang–Mills on its Coulomb branch, then the five-dimensional Coulomb scalar coordinate is normalized by

$$\phi_{5d} = 2\pi R\,\phi_{6d},$$

so that $M_\alpha^W = |\alpha(\phi_{5d})|$. This is a compactification matching condition; it is not an independent six-dimensional Lagrangian definition.

A static string of tension T_α wrapped on a circle of circumference $2\pi R$ has rest energy

$$M_\alpha^{\mathrm{wrap}} = (2\pi R)T_\alpha.$$

Using Proposition 96.8 gives the first displayed formula. In a five-dimensional gauge theory on the Coulomb branch, the mass of the vector multiplet associated with a root α is the absolute value of the corresponding root evaluated on the Coulomb scalar. Equating the wrapped-string particle with this W-boson therefore fixes the scalar normalization $\phi_{5d} = 2\pi R\,\phi_{6d}$.

96.10 Compactification On Riemann Surfaces

For compactification on a Riemann surface C, partial twists use an embedding of the spin connection of C into G_R. The resulting four-dimensional theory, when it exists, must have matching anomalies, flavor symmetries, defect descendants, and moduli-space dimensions. The following hypothesis records the compactification structure used below and gives the class-S notation a precise conditional meaning.

Hypothesis 96.10 (Class-S compactification structure)**.** Fix a six-dimensional theory $\mathfrak{T}_{6d}[\mathfrak{g}]$ satisfying Hypothesis 96.1. A class-S compactification structure is a tuple

$$\mathcal{C}_{\mathrm{S}} = \big(C, \{(p_a, \mathfrak{D}_a)\}_{a\in A}, \eta\big)$$

with the following constituents.

(i) C is a connected oriented smooth surface equipped with a complex structure. The marked points $p_a \in C$ are pairwise distinct. The unpunctured case is $A = \emptyset$.

(ii) The sign $\eta = \pm 1$ specifies the embedding of the $\mathrm{Spin}(2)_C$ connection into the $SO(2)_r \subset SO(5)_R$ R-symmetry factor used in the partial twist. In characteristic-class language this is the convention

$$e(N_2) = x + \eta\, t$$

of (96.23).

(iii) Each $\mathfrak{D}_a$ is a codimension-two defect or boundary condition for $\mathfrak{T}_{6d}[\mathfrak{g}]$ at p_a. At the level used in this chapter, such a defect input must specify at least its preserved four-dimensional supersymmetry, flavor symmetry F_a, contribution to the four-dimensional anomaly polynomial, its allowed defect descendants, and, when a Hitchin-system description is invoked, the corresponding allowed singularity or boundary condition for the Hitchin field at p_a.

(iv) The compactification is assumed to have a four-dimensional $\mathcal{N} = 2$ local QFT limit, denoted

$$\mathfrak{T}_{\mathfrak{g}}[C, \{(p_a, \mathfrak{D}_a)\}; \eta],$$

with local operator algebra, stress tensor multiplet, Hilbert-space radial-quantization data, and extended operators descending from the six-dimensional defect and finite-flux sectors after a choice of global polarization when such a choice is required.

(v) The protected data of this four-dimensional QFT agree with the six-dimensional reduction data: the preserved supercharges are those annihilated by the twist; the flavor symmetry includes the defect flavor groups and any continuous symmetry left by the compactification; the anomaly six-form is the fiber integral of $I_8[\mathfrak{g}]$ plus the local defect contributions; and the Coulomb-branch chiral coordinates are modeled by the appropriate Hitchin base associated with $(\mathfrak{g}, C, \{\mathfrak{D}_a\})$.

The notation $\mathfrak{T}_{\mathfrak{g}}[C, \{(p_a, \mathfrak{D}_a)\}; \eta]$ is therefore conditional: it denotes the four-dimensional QFT whose existence and properties are asserted in this hypothesis. The Hitchin base, anomaly push-forward, or Seiberg–Witten curve is not by itself a construction of the QFT. In the smooth unpunctured calculation below, $A = \emptyset$, C is closed, and the only compactification input beyond $\mathfrak{g}$ is the twist sign η.

The compactification is a test with substantial content. It must specify:

the twist homomorphism from $\mathrm{Spin}(2)_C$ to G_R,
the defect or puncture data,
the surviving supercharges and flavor symmetries,
the map of six-dimensional anomalies to four-dimensional anomalies,
and the descent of surface defects and BPS strings.

Supplying only a Seiberg–Witten curve in four dimensions does not by itself construct either the six-dimensional parent theory or the four-dimensional local QFT denoted by the class-S compactification symbol.

Twist convention and anomaly pushforward. Here is the local anomaly calculation behind the preceding list in the unpunctured case. Let

$$M_6 = M_4 \times C$$

with C a closed oriented Riemann surface. Denote by

$$t = e(TC) \in H^2(C; \mathbb{Z}), \qquad \int_C t = \chi(C) = 2 - 2g_C,$$

the Euler class of C. Decompose the rank-five $SO(5)_R$ background bundle as

$$N_R = N_2 \oplus N_3,$$

where N_2 is an oriented real rank-two bundle for the $SO(2)_r$ subgroup and N_3 is an oriented real rank-three bundle for the four-dimensional $SO(3)_R \simeq SU(2)_R/\mathbb{Z}_2$ background. Write

$$x = e(N_2|_{M_4}), \qquad u = p_1(N_3|_{M_4}).$$

The twist chooses a sign $\eta = \pm 1$, equivalently one of the two conjugate scalar supercharges after the compactification, and sets the rank-two Euler class on $M_4 \times C$ to

$$e(N_2) = x + \eta\, t. \tag{96.23}$$

The sign is a convention tied to the orientation of C and to the choice of $SO(2)_r$ weight. The formula below keeps η visible so that no physical conclusion is hidden in an unstated sign convention.

The characteristic classes entering (96.5) are then, modulo terms of degree too high on C,

$$p_1(N_R) = (x + \eta t)^2 + u = x^2 + u + 2\eta x\, t, \tag{96.24}$$
$$p_2(N_R) = (x + \eta t)^2 u = x^2 u + 2\eta x\, u\, t. \tag{96.25}$$

For the tangent bundle,

$$p_1(T(M_4 \times C)) = p_1(TM_4), \qquad p_2(T(M_4 \times C)) = p_2(TM_4),$$

after integration over C, because t^2 has no component on the two-dimensional surface. Let $p = p_1(TM_4)$. The coefficient of t in the free tensor polynomial (96.6) is

$$\big[I_8(1)\big]_t = \frac{\eta x\, t}{48}\,\big(x^2 - p + 3u\big), \tag{96.26}$$

and the coefficient of t in the interacting excess is

$$\left[\frac{d_{\mathfrak{g}} h^\vee_{\mathfrak{g}}}{24} p_2(N_R)\right]_t = \frac{\eta d_{\mathfrak{g}} h^\vee_{\mathfrak{g}}}{12}\, x\, u\, t. \tag{96.27}$$

Therefore the six-form anomaly polynomial of the smooth unpunctured four-dimensional compactification, in this normalization of x and u, is

$$I_6^{\text{class S}}(\mathfrak{g}, C; \eta) = \eta\, \chi(C)\, x \left[\frac{r_{\mathfrak{g}}}{48}(x^2 - p + 3u) + \frac{d_{\mathfrak{g}} h^\vee_{\mathfrak{g}}}{12} u\right]. \tag{96.28}$$

If one rewrites the $SO(3)_R$ class as $u = -4c_2(SU(2)_R)$, this becomes the conventional four-dimensional $\mathcal{N} = 2$ anomaly-polynomial coordinate system, up to the same η and orientation convention. Punctures, codimension-two defects, and irregular singularities add localized contributions; they are not contained in the smooth pushforward (96.28).

96.11 Hitchin-Base Coulomb Operators In Class-S Compactification

Assume Hypotheses 96.1 and 96.10 in the smooth unpunctured case: C is a closed Riemann surface of genus $g_C > 1$, $A = \emptyset$, and the partial twist preserves four-dimensional $\mathcal{N} = 2$ supersymmetry. Let

$$d_1, \ldots, d_{r_{\mathfrak{g}}}$$

be the degrees of the algebraically independent invariant polynomials on $\mathfrak{g}_{\mathbb{C}}$, ordered increasingly. Equivalently

$$d_i = m_i + 1$$

where m_i are the exponents of $\mathfrak{g}$. The protected Coulomb coordinates in the compactified theory are modeled by the Hitchin-base spaces

$$\varphi_{d_i} \in H^0(C, K_C^{d_i}), \qquad i = 1, \ldots, r_{\mathfrak{g}}, \tag{96.29}$$

where K_C is the holomorphic canonical bundle of C. In the four-dimensional theory the coordinate associated with φ_{d_i} has scaling dimension d_i.

Remark 96.11 (Status of the Hitchin-base statement)*.* Equation (96.29) is used here as a protected compactification test of the $(2,0)$ theory, not as a definition of the six-dimensional QFT. A construction of the compactified theory must also specify the Hilbert space, local operators, line and surface defects, metric dependence before twisting, and the treatment of degenerating complex structures. The Hitchin base records the Coulomb chiral ring coordinates visible in the twisted protected sector.

Riemann–Roch count for Hitchin differentials. Let C be a closed Riemann surface of genus $g_C > 1$. For every integer $d \geq 2$,

$$\dim_{\mathbb{C}} H^0(C, K_C^d) = (2d-1)(g_C - 1). \tag{96.30}$$

The degree of K_C is $2g_C - 2$, hence

$$\deg K_C^d = d(2g_C - 2).$$

Riemann–Roch gives

$$h^0(C, K_C^d) - h^0(C, K_C^{1-d}) = \deg K_C^d + 1 - g_C. \tag{96.31}$$

For $d \geq 2$, the line bundle K_C^{1-d} has negative degree, so it has no nonzero holomorphic sections. Therefore

$$h^0(C, K_C^d) = d(2g_C - 2) + 1 - g_C = (2d-1)(g_C - 1).$$

Degree sum for a simple Lie algebra. For a complex simple Lie algebra $\mathfrak{g}_{\mathbb{C}}$ with invariant polynomial degrees $d_i = m_i + 1$,

$$\sum_{i=1}^{r_{\mathfrak{g}}} (2d_i - 1) = d_{\mathfrak{g}}, \tag{96.32}$$

where $d_{\mathfrak{g}} = \dim_{\mathbb{C}} \mathfrak{g}_{\mathbb{C}}$.

The exponents m_i satisfy

$$\sum_{i=1}^{r_{\mathfrak{g}}} m_i = |\Delta_+|, \tag{96.33}$$

where Δ_+ is a set of positive roots. One way to see (96.33) is to compare the top degree in the standard Poincare polynomial identity for the flag variety:

$$\sum_{w \in W_{\mathfrak{g}}} t^{2\ell(w)} = \prod_{i=1}^{r_{\mathfrak{g}}} \frac{1 - t^{2(m_i+1)}}{1 - t^2}.$$

The maximal length of a Weyl-group element is $|\Delta_+|$, while the top degree on the product side is $2\sum_i m_i$. Hence $\sum_i m_i = |\Delta_+|$. Since

$$d_{\mathfrak{g}} = r_{\mathfrak{g}} + 2|\Delta_+|,$$

we obtain

$$\sum_i (2d_i - 1) = \sum_i (2m_i + 1) = 2|\Delta_+| + r_{\mathfrak{g}} = d_{\mathfrak{g}}.$$

Unpunctured class-S Coulomb-base dimension. For the conditional unpunctured genus-$g_C > 1$ compactification structure above, the protected Coulomb base modeled by the Hitchin base

$$\mathcal{B}_{\mathfrak{g},C} = \bigoplus_{i=1}^{r_{\mathfrak{g}}} H^0(C, K_C^{d_i}) \tag{96.34}$$

has complex dimension

$$\dim_{\mathbb{C}} \mathcal{B}_{\mathfrak{g},C} = (g_C - 1) d_{\mathfrak{g}}. \tag{96.35}$$

By the Riemann–Roch calculation above,

$$\dim_{\mathbb{C}} \mathcal{B}_{\mathfrak{g},C} = \sum_i (2d_i - 1)(g_C - 1). \tag{96.36}$$

Factoring out $g_C - 1$ and using the degree-sum calculation above gives

$$\dim_{\mathbb{C}} \mathcal{B}_{\mathfrak{g},C} = (g_C - 1)\sum_i (2d_i - 1) = (g_C - 1)d_{\mathfrak{g}}.$$

The assumptions $g_C > 1$ and $d_i \geq 2$ are used in the Riemann–Roch lemma to remove the $H^0(C, K_C^{1-d_i})$ term; punctures or genus zero/one surfaces require additional divisor data and obey different dimension formulae.

Regular-Puncture Pole-Order Chart. The preceding count has a controlled extension once the codimension-two defect data in Hypothesis 96.10 supply pole orders for the Hitchin differentials. For each invariant degree d_i, suppose the defect $\mathfrak{D}_a$ specifies an integer

$$p_{a,i} \geq 0$$

and let

$$D_i = \sum_{a \in A} p_{a,i} p_a$$

be the corresponding effective divisor on C. The protected Coulomb coordinate of degree d_i is then modeled by

$$\varphi_{d_i} \in H^0(C, K_C^{d_i}(D_i)). \tag{96.37}$$

This records only the pole orders supplied by the defect datum. It does not classify codimension-two defects and it does not assert that an arbitrary choice of the integers $p_{a,i}$ defines a four-dimensional QFT.

When the nonspeciality condition

$$H^0(C, K_C^{1-d_i}(-D_i)) = 0 \tag{96.38}$$

holds, Riemann–Roch gives the exact dimension

$$\dim_{\mathbb{C}} H^0(C, K_C^{d_i}(D_i)) = (2d_i - 1)(g_C - 1) + \deg D_i. \tag{96.39}$$

Indeed, Riemann–Roch applied to $L_i = K_C^{d_i}(D_i)$ gives

$$h^0(C, L_i) - h^0(C, K_C \otimes L_i^{-1}) = \deg L_i + 1 - g_C.$$

But $K_C \otimes L_i^{-1} = K_C^{1-d_i}(-D_i)$, so (96.38) removes the second term. Since $\deg L_i = d_i(2g_C - 2) + \deg D_i$, the displayed formula follows.

Thus a compactification datum with compatible pole orders has protected Coulomb-base dimension

$$\dim_{\mathbb{C}} \mathcal{B}_{\mathfrak{g},C,\{D_i\}} = (g_C - 1)d_{\mathfrak{g}} + \sum_{a \in A} \sum_{i=1}^{r_{\mathfrak{g}}} p_{a,i}, \tag{96.40}$$

under the same nonspeciality condition for all i. Residue constraints, mass parameters, nilpotent-orbit labels, irregular singularity slopes, and defect flavor anomalies are additional parts of the defect datum; they are not encoded by the pole-order sum alone.

Example 96.12 (Full regular punctures for type A_{N-1})**.** For the full regular puncture of type A_{N-1}, the standard pole-order list is

$$p_{a,d} = d - 1, \qquad d = 2, \dots, N.$$

If n_{full} such punctures are supplied and the nonspeciality condition holds, then

$$\begin{aligned} \dim_{\mathbb{C}} \mathcal{B}_{A_{N-1},C,n_{\text{full}}} &= (g_C - 1)(N^2 - 1) + n_{\text{full}} \sum_{d=2}^{N} (d - 1) \\ &= (g_C - 1)(N^2 - 1) + n_{\text{full}} \frac{N(N-1)}{2}. \end{aligned} \tag{96.41}$$

For the three-punctured sphere this gives

$$-(N^2 - 1) + \frac{3N(N-1)}{2} = \frac{(N-1)(N-2)}{2}.$$

This last number is the Coulomb-base dimension of the protected three-full-puncture fixture in this pole-order coordinate. It is still not a substitute for constructing the associated four-dimensional local QFT.

Example 96.13 (ADE degree checks)**.** For A_{N-1}, the degrees are $2, 3, \dots, N$. Therefore

$$\sum_{d=2}^{N} (2d - 1) = N^2 - 1 = d_{A_{N-1}},$$

and the unpunctured genus-g_C Coulomb-base dimension is

$$(g_C - 1)(N^2 - 1).$$

For D_N, the degrees are

$$2, 4, 6, \dots, 2N - 2, \; N,$$

with the usual repeated degree when N is even. Their contribution is

$$\sum_{j=1}^{N-1} (4j - 1) + (2N - 1) = N(2N - 1) = d_{D_N}.$$

For the exceptional algebras, the degree lists are

$\mathfrak{g}$	(d_i)
E_6	$2, 5, 6, 8, 9, 12$
E_7	$2, 6, 8, 10, 12, 14, 18$
E_8	$2, 8, 12, 14, 18, 20, 24, 30.$

In each case $\sum_i (2d_i - 1)$ gives 78, 133, and 248, respectively.

96.12 Status Summary

The assumed $(2,0)$ fixed-point object used in this chapter is $\mathfrak{T}_{6d}[\mathfrak{g}]$ of Hypothesis 96.1. The class-S four-dimensional object used in the compactification discussion is $\mathfrak{T}_{\mathfrak{g}}[C,\{(p_a,\mathfrak{D}_a)\};\eta]$ of Hypothesis 96.10. The substantive six-dimensional data are therefore not reconstructed retrospectively from this summary; they are the named assumptions and construction layers fixed in Section 96.1.

For orientation, the data invoked in this chapter are:

superconformal representation category,
tensor-branch EFT with self-dual fields and Green–Schwarz terms,
anomaly polynomial and flavor-background data,
BPS string charge lattice and inflow,
finite defect group, flux pairing, and polarization data,
compactification tests and extended-operator descendants.

For $(2,0)$ type $\mathfrak{g}$ this includes the simply laced Lie algebra, the tensor-branch quotient $(\mathbb{R}^5 \otimes \mathfrak{t}_{\mathfrak{g}})/W_{\mathfrak{g}}$, the anomaly polynomial (96.5), the root-lattice BPS strings, the discriminant group $A_{\mathfrak{g}} = P_{\mathfrak{g}}/Q_{\mathfrak{g}}$ with finite pairing (96.14), and the five-dimensional compactification normalization $g_5^2 = 4\pi^2 R$ in trace-delta variables. For class-S compactifications it includes the surface C, the marked points p_a, their defect data $\mathfrak{D}_a$, the twist sign η, the anomaly pushforward with defect contributions, the protected Hitchin-base description of Coulomb coordinates when available, and the descended extended operators and finite-flux polarization data.

No item in this list is dispensable. A proposed fixed point that supplies only a branch EFT is incomplete; a proposed non-Lagrangian object that supplies only protected indices is also incomplete. Likewise, a phrase such as “the class-S theory” has no standing in this chapter unless the tuple of compactification data in Hypothesis 96.10 has been supplied and the existence of the resulting four-dimensional local QFT is being assumed or constructed.

Open Problem 96.14 (Six-dimensional construction). Construct an interacting six-dimensional SCFT as a local QFT with Hilbert space, local operator algebra, stress tensor, anomaly polynomial, tensor branch, BPS strings, and compactification functoriality in one framework.

Chapter 97

Planar N=4 Supersymmetric Yang-Mills as a Spectral Problem

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The subject of this chapter is the spectrum of local gauge-invariant operators in four-dimensional $\mathcal{N} = 4$ supersymmetric Yang–Mills theory in the planar limit. The same theory appears earlier in this volume as a maximally supersymmetric conformal gauge theory. Here it is treated as a spectral problem: diagonalize the dilatation operator on the vector space of renormalized local operators.

97.1 SCFT Data Before Planarity

The planar spin chain is not the definition of the theory. It is a useful coordinate system on a particular large-N sector of a four-dimensional superconformal gauge theory. We first record the intrinsic quantum-field theory data that will remain meaningful at finite N, away from any spin-chain limit.

Definition 97.1 (N=4 Yang-Mills field content)**.** Fix a compact gauge group G, Lie algebra $\mathfrak{g}$, and invariant trace form tr as in Chapter 87. The four-dimensional $\mathcal{N} = 4$ Yang–Mills field variables consist of

$$A_\mu, \qquad \Phi^I \ (I = 1, \dots, 6), \qquad \lambda^A_\alpha,\ \bar{\lambda}_{A\dot{\alpha}} \ (A = 1, \dots, 4),$$

all valued in $\mathfrak{g}$. The six real scalars transform in the vector representation of $SO(6)_R$, equivalently in the antisymmetric representation of $SU(4)_R$; the four Weyl fermions transform in the fundamental of $SU(4)_R$. In the monograph coupling convention the bosonic part of the Lorentzian action is

$$\begin{aligned} S_{\rm bos} = \frac{1}{g_{\rm YM}^2}\int d^4x\ \mathrm{tr}\Big(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}D_\mu\Phi^I D^\mu\Phi^I + \frac{1}{4}[\Phi^I,\Phi^J][\Phi^I,\Phi^J]\Big) \\ + \frac{\theta}{32\pi^2}\int d^4x\ \mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}). \end{aligned}$$

together with the unique Yukawa couplings compatible with the same $\mathcal{N}=4$ supersymmetry and $SU(4)_R$ action. The complexified gauge coupling is

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi\mathrm{i}}{g_{\rm YM}^2}$$

in the present trace convention.

Equivalently, after choosing an $\mathcal{N}=1$ subalgebra, the same field field content is an $\mathcal{N}=1$ vector multiplet plus three adjoint chiral multiplets Φ_1,Φ_2,Φ_3, with cubic superpotential

$$W_{\mathcal{N}=4} = \sqrt{2}\ \mathrm{tr}\big(\Phi_1[\Phi_2,\Phi_3]\big),$$

after the superpotential coupling has been tied to $g_{\rm YM}$ by the enhanced supersymmetry. This $\mathcal{N}=1$ presentation is a bookkeeping device for an $SU(3)\times U(1)$ subgroup of the full $SU(4)_R$ theory.

Remark 97.2 (One-loop and holomorphic beta-function cancellation)*.* Let $C_2(\mathfrak{g})$ denote the adjoint quadratic Casimir in the chosen trace convention. The one-loop gauge beta-function coefficient of the $\mathcal{N}=4$ field content vanishes. In the $\mathcal{N}=1$ Wilsonian coordinate, the one-loop holomorphic coefficient vanishes as well.

In an ordinary four-dimensional gauge-theory convention, one vector, four adjoint Weyl fermions, and six adjoint real scalars contribute

$$b_0 = C_2(\mathfrak{g})\left(\frac{11}{3} - 4\frac{2}{3} - 6\frac{1}{6}\right) = 0.$$

In the $\mathcal{N}=1$ presentation, the holomorphic Wilsonian coefficient is

$$b_0^{\rm hol} = 3\,C_2(\mathfrak{g}) - \sum_{a=1}^{3} T(\mathrm{adj}) = 3\,C_2(\mathfrak{g}) - 3\,C_2(\mathfrak{g}) = 0.$$

The two calculations use different coupling coordinates, but both make the same elementary field-content cancellation explicit.

Hypothesis 97.3 (Finite $\mathcal{N}=4$ conformal-family input)**.** Fix the field content of Definition 97.1. Assume there is a local regulator and renormalization prescription with the following properties.

1. Gauge Ward identities, locality, Poincare covariance, all sixteen supercharges, and $SU(4)_R$ Ward identities are implemented without an anomaly.

2. The renormalized stress tensor and the $SU(4)_R$ currents belong to a conserved $\mathfrak{psu}(2,2|4)$ current multiplet, and the operator sourced by the complex coupling τ is the corresponding dimension-four top component.
3. After the gauge-field normalization and possible background contact terms are fixed, no additional $\mathcal{N}=4$-preserving marginal source or current-moment-map obstruction is generated in the τ direction.
4. In this source coordinate the beta function of τ vanishes.

This hypothesis is the precise status boundary. The one-loop calculation in Remark 97.2 is an elementary check on the required cancellation; the all-order assertion is a renormalization theorem with regulator-level input, independent of the spin-chain construction.

Definition 97.4 (Coupling chart and theta periodicity)**.** The local Yang–Mills coupling chart is the upper half-plane

$$\mathfrak{H}=\{\tau\in\mathbb{C}\mid \operatorname{Im}\tau>0\},\qquad \tau=\frac{\theta}{2\pi}+\frac{4\pi\mathrm{i}}{g_{\mathrm{YM}}^2}.$$

When the trace form and global gauge choice are chosen so that

$$k[A]=\frac{1}{32\pi^2}\int \operatorname{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu})\in\mathbb{Z}$$

on closed spin four-manifolds in the allowed bundle sectors, the path-integral phase $\exp(\mathrm{i}\theta k[A])$ is unchanged by

$$T:\tau\longmapsto\tau+1.$$

Thus $q=\exp(2\pi\mathrm{i}\tau)$ is the natural weak-coupling coordinate near the cusp $\operatorname{Im}\tau=\infty$.

Remark 97.5 (Local $\mathcal{N}=4$ conformal manifold)*.* Assume Hypothesis 97.3. Before imposing any duality identifications, the $\mathcal{N}=4$ Yang–Mills coupling chart $\mathfrak{H}$ is a one-complex-dimensional local conformal manifold. The tangent vector represented by $\partial/\partial\tau$ is the exactly marginal deformation obtained by integrating the protected dimension-four top component of the stress-tensor multiplet.

The source-coordinate system has one complex coordinate, namely τ. By Hypothesis 97.3, the beta map in this chart is identically zero after the declared normalization and contact-term choices are fixed, and no redundant source direction remains to be quotiented. The zero locus of the beta map is therefore all of $\mathfrak{H}$. Its complex dimension is

$$1-0=1.$$

The inserted operator $\partial S/\partial\tau$ is exactly marginal because it has protected dimension 4 and its source has zero beta function in the same chart. This proves the local statement. Global electric-magnetic identifications between different values of τ are additional data.

Hypothesis 97.6 (Duality-group input)**.** For a fixed global form of the gauge group and a fixed line-operator lattice, let $\Gamma\subset PSL(2,\mathbb{Z})$ be the subgroup of genuine four-dimensional $\mathcal{N}=4$ QFT equivalences acting on the same extended theory structure. Its elements act on the coupling by

$$\tau\longmapsto\frac{a\tau+b}{c\tau+d},\qquad \begin{pmatrix}a&b\\c&d\end{pmatrix}\in SL(2,\mathbb{Z}).$$

For the standard local-operator sector of $SU(N)$ Yang–Mills one often uses the full $PSL(2,\mathbb{Z})$ generated by

$$T:\tau\mapsto\tau+1,\qquad S:\tau\mapsto-\frac{1}{\tau},$$

but the precise global quotient depends on the line-operator and global-form structure. This hypothesis is an input about nonperturbative QFT equivalences, with no string-theory construction entering the argument.

Remark 97.7 (Global quotient, conditional on duality). Assume Hypotheses 97.3 and 97.6. The local conformal family $\mathfrak{H}$ descends, for the chosen extended theory structure, to the orbifold conformal parameter space

$$\mathfrak{H}/\Gamma.$$

The generator T is the theta-periodicity of Definition 97.4; the generator S, when it belongs to Γ, exchanges the weak-coupling cusp with a strong-coupling region.

The action of $SL(2,\mathbb{Z})$ on $\mathfrak{H}$ preserves $\operatorname{Im}\tau>0$ because

$$\operatorname{Im}\frac{a\tau+b}{c\tau+d}=\frac{\operatorname{Im}\tau}{|c\tau+d|^2}.$$

Matrices differing by an overall sign have the same fractional-linear action, so the effective group is a subgroup of $PSL(2,\mathbb{Z})$. If $\gamma\in\Gamma$ is a genuine QFT equivalence for the fixed extended structure, then τ and $\gamma\cdot\tau$ label the same point of the global conformal parameter space. Quotienting $\mathfrak{H}$ by these identifications gives $\mathfrak{H}/\Gamma$. The statement is conditional on the duality hypothesis; local exact marginality alone gives only $\mathfrak{H}$.

The spin-chain spectral problem below uses the local SCFT family and the planar single-trace dilatation operator, with the global duality quotient kept as separate input.

Conformal anomaly coefficients. For $G=SU(N)$,

$$a=c=\frac{\dim\mathfrak{su}(N)}{4}=\frac{N^2-1}{4}.$$

For $G=U(N)$, including the free abelian multiplet,

$$a=c=\frac{N^2}{4}.$$

The trace-anomaly coefficients of free conformal fields in four dimensions are

$$a_{\text{scalar}}=\frac{1}{360},\quad a_{\text{Weyl}}=\frac{11}{720},\quad a_{\text{vector}}=\frac{31}{180},$$

and

$$c_{\text{scalar}}=\frac{1}{120},\quad c_{\text{Weyl}}=\frac{1}{40},\quad c_{\text{vector}}=\frac{1}{10}.$$

One $\mathcal{N}=4$ vector multiplet per generator of $\mathfrak{g}$ contains six real scalars, four Weyl fermions, and one gauge vector. Hence

$$a_{\text{multiplet}}=6\frac{1}{360}+4\frac{11}{720}+\frac{31}{180}=\frac{1}{4},$$

and

$$c_{\text{multiplet}} = 6\frac{1}{120} + 4\frac{1}{40} + \frac{1}{10} = \frac{1}{4}.$$

Multiplying by $\dim\mathfrak{g}$ gives the displayed formulae. The use of the free-field coefficients is justified because a and c sit in the same protected stress-tensor multiplet data along the exactly marginal $\mathcal{N} = 4$ conformal manifold. Thus the calculation is a convention fixing and protected-data evaluation, conditional on the exact marginality and stress-tensor-multiplet protection stated above, rather than an independent derivation of the interacting theory from first principles.

97.2 Protected Operators and Elementary OPE Data

Let $C_{I_1\cdots I_k}$ be a symmetric traceless tensor of $SO(6)_R$:

$$C_{I_{\sigma(1)}\cdots I_{\sigma(k)}} = C_{I_1\cdots I_k}, \qquad \delta^{I_a I_b} C_{I_1\cdots I_k} = 0.$$

The corresponding single-trace half-BPS representative is

$$\mathcal{O}_C^{(k)} = C_{I_1\cdots I_k}\,\operatorname{tr}\!\big(\Phi^{I_1}\cdots\Phi^{I_k}\big),$$

with normal ordering and operator mixing understood in the renormalized local-operator space. It transforms in the $SU(4)_R$ representation $[0,k,0]$.

Quoted Theorem 97.8 (Half-BPS protection)**.** *In unitary four-dimensional $\mathcal{N} = 4$ superconformal Yang–Mills theory, a scalar superconformal primary in the half-BPS multiplet $[0,k,0]$ has exact scaling dimension*

$$\Delta = k.$$

The two- and three-point functions of half-BPS primaries, after fixing their normalization, are independent of the exactly marginal coupling.

This is a representation-theoretic and nonrenormalization statement. It does not say that every gauge-invariant word with classical dimension k is protected. The singlet

$$\mathcal{K} = \operatorname{tr}(\Phi^I\Phi^I)$$

is the Konishi primary; it belongs to a long multiplet and acquires an anomalous dimension. The traceless tensor

$$\mathcal{O}^{IJ} = \operatorname{tr}(\Phi^I\Phi^J) - \frac{1}{6}\delta^{IJ}\operatorname{tr}(\Phi^K\Phi^K)$$

is the bottom component of the stress-tensor multiplet and has $\Delta = 2$.

Remark 97.9 (Stress-tensor-multiplet normalization)*.* Choose an orthonormal basis T^a of $\mathfrak{g}$ with $\operatorname{tr}(T^aT^b) = \delta^{ab}$, and write

$$\langle \Phi^{I,a}(x)\Phi^{J,b}(0)\rangle_{\text{free}} = \frac{g_{\text{YM}}^2\,\delta^{IJ}\delta^{ab}}{4\pi^2 x^2}.$$

Define

$$B^{IJ} = g_{\text{YM}}^{-2} : \operatorname{tr}(\Phi^I\Phi^J) :, \qquad O^{IJ} = B^{IJ} - \frac{1}{6}\delta^{IJ}B^{KK},$$

and let

$$P^{IJ,KL} = \frac{1}{2}(\delta^{IK}\delta^{JL} + \delta^{IL}\delta^{JK}) - \frac{1}{6}\delta^{IJ}\delta^{KL}$$

be the projector onto symmetric traceless $SO(6)$ tensors. Then

$$\langle O^{IJ}(x)O^{KL}(0)\rangle_{\rm free} = \frac{\dim\mathfrak{g}}{8\pi^4 x^4}\, P^{IJ,KL}.$$

Consequently

$$\mathbb{O}^{IJ} = \frac{2\sqrt{2}\,\pi^2}{\sqrt{\dim\mathfrak{g}}}\, O^{IJ}$$

has the canonical two-point function

$$\langle\mathbb{O}^{IJ}(x)\mathbb{O}^{KL}(0)\rangle = \frac{P^{IJ,KL}}{x^4}.$$

Wick contraction gives

$$\langle B^{IJ}(x)B^{KL}(0)\rangle_{\rm free} = \frac{\dim\mathfrak{g}}{16\pi^4 x^4}\big(\delta^{IK}\delta^{JL}+\delta^{IL}\delta^{JK}\big).$$

Subtracting the trace in each pair projects both index pairs with P. The result is

$$\frac{\dim\mathfrak{g}}{16\pi^4 x^4}\left(\delta^{IK}\delta^{JL}+\delta^{IL}\delta^{JK}-\frac{1}{3}\delta^{IJ}\delta^{KL}\right) = \frac{\dim\mathfrak{g}}{8\pi^4 x^4}P^{IJ,KL}.$$

Multiplication by $(2\sqrt{2}\pi^2)^2/\dim\mathfrak{g}$ gives the normalized two-point function.

There is also a useful planar chiral-ring convention. For $G = SU(N)$, let

$$Z = \frac{1}{\sqrt{2}}(\Phi^5 + \mathrm{i}\Phi^6), \qquad \mathcal{Z}_J = \frac{(2\pi)^J}{g_{\rm YM}^J\sqrt{J\,N^J}}\ :\mathrm{tr}(Z^J):$$

in the leading large-N single-trace normalization.

Planar chiral-primary normalization. At leading planar order and for fixed J, J_1, J_2 as $N\to\infty$,

$$\langle\mathcal{Z}_J(x)\,\overline{\mathcal{Z}}_J(0)\rangle_{\rm pl} = \frac{1}{x^{2J}},$$

and, for $J = J_1 + J_2$,

$$\big\langle\mathcal{Z}_{J_1}(x_1)\mathcal{Z}_{J_2}(x_2)\overline{\mathcal{Z}}_J(x_3)\big\rangle_{\rm pl} = \frac{\sqrt{J_1J_2J}}{N}\,\frac{1}{x_{13}^{2J_1}x_{23}^{2J_2}}.$$

Equivalently, in the protected chiral OPE,

$$\mathcal{Z}_{J_1}(x)\mathcal{Z}_{J_2}(0) = \frac{\sqrt{J_1J_2(J_1+J_2)}}{N}\,\mathcal{Z}_{J_1+J_2}(0) + \text{descendants and multi-traces}$$

at leading planar order. The elementary propagator in the trace convention used above is

$$\langle Z^a(x)\overline{Z}^b(0)\rangle_{\rm free} = \frac{g_{\rm YM}^2}{4\pi^2x^2}\delta^{ab}.$$

For a single trace of length J, planar Wick contractions pair each Z with the corresponding $\overline{Z}$ in a cyclic order. There are J choices of the relative cyclic origin, and the double-line color graph has J leading color loops. Hence

$$\big\langle : \mathrm{tr}(Z^J) : (x) : \mathrm{tr}(\overline{Z}^J) : (0) \big\rangle_{\mathrm{pl}} = J N^J \left(\frac{g_{\mathrm{YM}}^2}{4\pi^2 x^2} \right)^J .$$

Multiplying by the two normalization factors in $\mathcal{Z}_J$ gives the unit two-point function.

For the extremal three-point function put $J = J_1 + J_2$. At leading planar order all fields in the first two holomorphic traces contract with the single anti-holomorphic trace. A planar embedding is fixed by choosing the cyclic origin of the anti-holomorphic trace, the cyclic origin of the J_1-trace, and the cyclic origin of the J_2-trace. This gives $J J_1 J_2$ planar embeddings. The connected double-line graph has $J-1$ leading color loops, because three trace boundaries have been joined into one planar pair-of-pants graph. Therefore

$$\begin{aligned} &\big\langle : \mathrm{tr}(Z^{J_1}) : (x_1) : \mathrm{tr}(Z^{J_2}) : (x_2) : \mathrm{tr}(\overline{Z}^J) : (x_3) \big\rangle_{\mathrm{pl}} \\ &\qquad = J J_1 J_2 \, N^{J-1} \left(\frac{g_{\mathrm{YM}}^2}{4\pi^2} \right)^J \frac{1}{x_{13}^{2J_1} x_{23}^{2J_2}}. \end{aligned}$$

The product of the three normalization constants contributes

$$\frac{(2\pi)^{2J}}{g_{\mathrm{YM}}^{2J} \sqrt{J_1 N^{J_1}} \sqrt{J_2 N^{J_2}} \sqrt{J N^J}},$$

which reduces the coefficient to $\sqrt{J_1 J_2 J}/N$. The displayed OPE coefficient follows by taking $x_1 \to x_2$ in this protected extremal three-point function with the two-point normalization already fixed.

Finite-N trace relations and multi-trace mixing are not suppressed in this formula; they are part of the exact protected-operator problem rather than the single-trace spin-chain problem.

97.3 Planar Limit and Operator Hilbert Space

Let the gauge group be $SU(N)$ or $U(N)$, with gauge coupling defined by the kinetic term

$$\frac{1}{4 g_{\mathrm{YM}}^2} \int \mathrm{d}^4 x \; \mathrm{tr}(F_{\mu\nu} F^{\mu\nu})$$

in the trace normalization used throughout the gauge-theory volumes. The 't Hooft coupling is

$$\lambda = g_{\mathrm{YM}}^2 N, \qquad g = \frac{\sqrt{\lambda}}{4\pi}.$$

The planar limit means

$$N \to \infty, \qquad \lambda \text{ fixed},$$

with local single-trace operators normalized so that their planar two-point functions remain finite.

Let $\mathfrak{V}$ denote the one-letter vector space spanned by the elementary adjoint fields of $\mathcal{N}=4$ SYM and their covariant derivatives, modulo the equations of motion and gauge-covariant descendant relations appropriate to local-operator renormalization. For fixed length L, a single-trace word is

$$\mathcal{O}_W(x) = \mathrm{tr}\big(W_1(x)W_2(x)\cdots W_L(x)\big), \qquad W_i \in \mathfrak{V}.$$

Cyclicity of the trace identifies

$$W_1 \otimes W_2 \otimes \cdots \otimes W_L \sim W_2 \otimes \cdots \otimes W_L \otimes W_1.$$

Thus the fixed-length single-trace space is the cyclic quotient

$$\mathcal{H}_L^{\mathrm{cyc}} = \mathfrak{V}^{\otimes L}/\langle \tau - 1\rangle,$$

where τ shifts all letters by one site. For planar perturbation theory this quotient is equipped with the large-N inner product obtained from the leading two-point function. If $W = W_1 \otimes \cdots \otimes W_L$ and $W' = W_1' \otimes \cdots \otimes W_L'$ are words of the same classical dimension, define

$$\langle W, W'\rangle_{\mathrm{pl}} = \frac{1}{L\,N^L}\sum_{r=0}^{L-1}\prod_{j=1}^{L}\langle W_j, W'_{j+r}\rangle_{\mathrm{letter}}, \qquad j+r \text{ read modulo } L.$$

The factor N^{-L} removes the leading color loop, while L^{-1} averages over the trace origin. Operators differing by equations of motion, gauge-covariant total derivatives, or Q-exact representatives in a chosen protected cohomological sector are first quotient out in the renormalized local-operator space; the spin-chain word is a representative of that quotient, not a new fundamental Hilbert space.

Definition 97.10 (Planar single-trace spectral problem)**.** Fix a renormalization scheme preserving the manifest $SU(4)_R$ and Poincare symmetries of $\mathcal{N}=4$ SYM and fix a closed sector $\mathcal{S} \subset \mathfrak{V}$ under planar mixing to a specified loop order. The planar single-trace spectral problem of length L is the cyclic quotient $\mathcal{S}^{\otimes L}/\langle \tau - 1\rangle$, its planar two-point inner product above, and the restriction of the anomalous-dimension matrix Γ to the quotient. A spin-chain Hamiltonian is a coordinate description of this problem after a basis of letters has been chosen.

The dilatation operator D is defined by the logarithmic scale dependence of renormalized local operators. In a basis $\mathcal{O}_A$,

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\mathcal{O}_A = -\Gamma_A{}^B\mathcal{O}_B, \qquad D_A{}^B = (D_0)_A{}^B + \Gamma_A{}^B.$$

At a conformal fixed point, diagonalizing D is equivalent to finding local operators of definite scaling dimension. Planarity implies that, at fixed loop order, D acts locally along the cyclic word: at ℓ loops the range is at most $\ell+1$ adjacent sites in the scalar subsectors described below.

97.4 The SU(2) Scalar Sector

Choose two complex scalar fields

$$Z = \frac{1}{\sqrt{2}}(\phi^5 + \mathrm{i}\phi^6), \qquad X = \frac{1}{\sqrt{2}}(\phi^1 + \mathrm{i}\phi^2).$$

The $SU(2)$ scalar sector is spanned by

$$\mathrm{tr}(W_1 \cdots W_L), \qquad W_i \in \{Z, X\}.$$

Identify Z with a down spin and X with an up spin:

$$Z \leftrightarrow |{\downarrow}\rangle, \qquad X \leftrightarrow |{\uparrow}\rangle.$$

Let $P_{j,j+1}$ be the operator that interchanges the two letters at sites j and $j+1$, with $L+1 \equiv 1$.

The local mixing calculation may be reduced to a two-site problem. In the scalar potential, the term relevant for holomorphic X, Z words is

$$V_{\mathrm{sc}} = -\frac{1}{2g_{\mathrm{YM}}^2}\,\mathrm{tr}\big([\Phi_I, \Phi_J][\Phi_I, \Phi_J]\big)$$

with the complex fields embedded in the six real scalars Φ_I. In double-line notation, a planar contraction of a quartic vertex with two adjacent letters has one color loop fewer than the free two-point function and hence contributes $g_{\mathrm{YM}}^2 N = \lambda$. Non-adjacent contractions are non-planar in a single trace and are suppressed by N^{-2}.

Two-site scalar mixing tensor. On adjacent holomorphic $SU(2)$ letters, the logarithmic one-loop mixing tensor has the form

$$\Gamma^{(1)}_{j,j+1} = \frac{\lambda}{8\pi^2}\,(1 - P_{j,j+1})$$

after adding the scalar self-energy and gauge-exchange contributions required by $\mathcal{N} = 4$ supersymmetry. The statement is perturbative in the chosen dimensional-regularization/minimal-subtraction convention.

The quartic vertex gives a local tensor generated by the two invariant maps on two $SU(2)$ letters: the identity I and the permutation P. The planar color contraction of $\mathrm{tr}(T^{a_1}T^{a_2}\cdots T^{a_L})$ with a vertex joining sites $j, j+1$ produces $N\,\mathrm{tr}(\cdots T^{b_j}T^{b_{j+1}}\cdots)$ at leading order, so the coefficient is proportional to λ and no operator with separated sites appears. The scalar bubble integral in $d = 4 - 2\varepsilon$ dimensions has logarithmic part

$$\frac{1}{(4\pi)^2}\frac{1}{\varepsilon} \quad \Longleftrightarrow \quad \frac{1}{8\pi^2}\log\mu$$

in the anomalous-dimension convention used for two-point functions. The exchange part is $-P$. The identity part receives the non-exchange scalar, gauge, and self-energy pieces. The protected operator $\mathrm{tr}(Z^L)$ is an eigenvector with zero anomalous dimension, so the total two-site tensor annihilates every symmetric ZZ pair. This fixes the identity coefficient to $+1$ in the same normalization, giving $(\lambda/8\pi^2)(I - P)$.

The argument just given is a convention-fixing derivation inside the one-loop planar regularization scheme. Its scope is the local Hamiltonian normalization used by the integrability construction below; a component-level diagrammatic derivation in a chosen gauge would require listing the scalar, gauge-exchange, and self-energy graphs in the same regularization convention.

97.5 The SO(6) Scalar Hamiltonian and the SU(2) Reduction

The $SU(2)$ calculation is a subsector of the full scalar one-loop Hamiltonian. Let the six real scalars be denoted Φ_I, $I = 1, \ldots, 6$, and write a two-site scalar state as $\Phi_I \otimes \Phi_J$. There are three invariant maps on $\mathbb{R}^6 \otimes \mathbb{R}^6$:

$$I(\Phi_I \otimes \Phi_J) = \Phi_I \otimes \Phi_J,$$
$$P(\Phi_I \otimes \Phi_J) = \Phi_J \otimes \Phi_I,$$
$$K(\Phi_I \otimes \Phi_J) = \delta_{IJ} \sum_{K=1}^{6} \Phi_K \otimes \Phi_K.$$

Here P permutes the two $SO(6)$ vector indices and K is the trace operator. In the normalization fixed by the $SU(2)$ Hamiltonian calculation below, the scalar one-loop nearest-neighbor density is

$$\Gamma^{(1)}_{j,j+1} = \frac{\lambda}{16\pi^2}\big(K_{j,j+1} + 2I_{j,j+1} - 2P_{j,j+1}\big). \tag{97.1}$$

The K-term is the place where the full $SO(6)$ chain differs from the holomorphic $SU(2)$ chain.

Remark 97.11 (Holomorphic reduction of the $SO(6)$ density). On the holomorphic sector generated by Z and X, the trace operator K in (97.1) vanishes, and the density reduces to

$$\frac{\lambda}{8\pi^2}(I - P).$$

The operator K contracts the $SO(6)$ indices of the two adjacent scalar letters. In the complex basis $Z = (\Phi_5 + \mathrm{i}\Phi_6)/\sqrt{2}$ and $X = (\Phi_1 + \mathrm{i}\Phi_2)/\sqrt{2}$, the holomorphic contractions are null:

$$Z \cdot Z = X \cdot X = Z \cdot X = 0.$$

Thus K annihilates $Z \otimes Z$, $Z \otimes X$, $X \otimes Z$, and $X \otimes X$. Substituting $K = 0$ in (97.1) gives

$$\frac{\lambda}{16\pi^2}(2I - 2P) = \frac{\lambda}{8\pi^2}(I - P).$$

This reduction is a useful convention check. If the overall coefficient or the sign of P is changed, the $SO(6)$ formula no longer restricts to the Konishi normalization computed below.

One-loop planar $SU(2)$ Hamiltonian. In the $SU(2)$ scalar sector, with the coupling convention above, the planar one-loop dilatation operator is

$$D = L + \frac{\lambda}{8\pi^2} H_{XXX} + O(\lambda^2), \qquad H_{XXX} = \sum_{j=1}^{L} (1 - P_{j,j+1})$$

on the cyclic spin-chain quotient. The only planar one-loop terms that move X and Z letters inside the holomorphic $SU(2)$ sector come from the scalar quartic interaction and the wavefunction and non-exchange diagrams required by supersymmetry. The exchange part is local on adjacent sites and is proportional to $-P_{j,j+1}$. The non-exchange part is proportional to the identity on the same pair. Hence the most general nearest-neighbor planar one-loop operator compatible with the two-letter $SU(2)$ symmetry and translation around the trace is

$$c\sum_{j=1}^{L}(1-P_{j,j+1})+c'\sum_{j=1}^{L}(1+P_{j,j+1}).$$

The half-BPS operators $\mathrm{tr}(Z^L)$ and their fully symmetrized scalar descendants have protected scaling dimension. On a symmetric pair, $P_{j,j+1}=1$, so protection forces $c'=0$. The coefficient c is fixed by the logarithmic part of the single adjacent exchange integral. In the present normalization the divergent scalar integral is

$$\frac{\lambda}{8\pi^2}\log(\mu|x|)+\text{finite},$$

and its color contraction is planar and nearest-neighbor. Substituting this coefficient gives the displayed H_{XXX}. The cyclic quotient implements the trace relation and imposes zero total spin-chain momentum on physical single-trace states.

This calculation uses supersymmetric protection only to fix the identity term after the exchange coefficient has been computed. This is a useful place where the field-theory origin of the spin chain matters: the spin-chain Hamiltonian is not introduced independently of the renormalized local-operator problem.

97.6 Coordinate Bethe Ansatz

We now diagonalize the dimensionless Hamiltonian

$$H_{XXX}=\sum_j(1-P_{j,j+1})$$

before reinstating the overall factor $\lambda/(8\pi^2)$. On an infinite chain let $|n\rangle$ denote the state with a single X at site $n\in\mathbb{Z}$ and Z's elsewhere. A one-magnon wavefunction is

$$\Psi_p(n)=\mathrm{e}^{\mathrm{i}pn}.$$

The Hamiltonian H_{XXX} gives

$$(H_{XXX}\Psi_p)(n)=2\Psi_p(n)-\Psi_p(n+1)-\Psi_p(n-1),$$

hence

$$E_1(p)=4\sin^2\frac{p}{2}.$$

The corresponding one-loop anomalous dimension contribution is

$$\gamma_1(p)=\frac{\lambda}{8\pi^2}E_1(p).$$

On a cyclic single trace a single magnon with nonzero momentum is not a physical operator, because trace cyclicity requires total momentum zero. The one-magnon calculation is nevertheless the local dispersion input for multi-magnon states.

Two-magnon contact matching for the one-loop chain. Let $z_a = \mathrm{e}^{\mathrm{i}p_a}$, $a = 1, 2$, and let

$$u_a = \frac{1}{2}\cot\frac{p_a}{2}, \qquad z_a = \frac{u_a + \mathrm{i}/2}{u_a - \mathrm{i}/2}.$$

In the two-magnon chamber $n_1 < n_2$, set

$$\Psi(n_1, n_2) = A_{12}\mathrm{e}^{\mathrm{i}(p_1 n_1 + p_2 n_2)} + A_{21}\mathrm{e}^{\mathrm{i}(p_2 n_1 + p_1 n_2)}.$$

The separated chamber equation makes Ψ an eigenfunction of H_{XXX} with energy $E_1(p_1)+E_1(p_2)$. The contact equation at $n_2 = n_1 + 1$ then fixes the exchange coefficient:

$$\frac{A_{21}}{A_{12}} = \mathsf{a}(p_1, p_2) = -\frac{1 - 2\mathrm{e}^{\mathrm{i}p_2} + \mathrm{e}^{\mathrm{i}(p_1+p_2)}}{1 - 2\mathrm{e}^{\mathrm{i}p_1} + \mathrm{e}^{\mathrm{i}(p_1+p_2)}} = \frac{u_1 - u_2 - \mathrm{i}}{u_1 - u_2 + \mathrm{i}}.$$

When the magnons are separated, $n_2 > n_1 + 1$, the Hamiltonian is the sum of two one-particle finite-difference operators:

$$\begin{aligned}(H_{XXX}\Psi)(n_1, n_2) = 4\Psi(n_1, n_2) &- \Psi(n_1 - 1, n_2) - \Psi(n_1 + 1, n_2)\\ &- \Psi(n_1, n_2 - 1) - \Psi(n_1, n_2 + 1).\end{aligned}$$

Hence the separated equation fixes the energy to $E_1(p_1) + E_1(p_2)$. The only remaining condition is the contact equation at $n_2 = n_1 + 1$. There the adjacent XX pair is symmetric, so the middle link contributes zero and only the two exterior links move a magnon:

$$(H_{XXX}\Psi)(n, n+1) = 2\Psi(n, n+1) - \Psi(n-1, n+1) - \Psi(n, n+2).$$

Substituting the plane-wave ansatz and the separated energy gives

$$0 = -A_{21}z_1z_2 + 2A_{21}z_1 - A_{21} - A_{12}z_1z_2 + 2A_{12}z_2 - A_{12}.$$

Solving this linear equation for A_{21}/A_{12} gives the rational expression in z_1, z_2. Rewriting $z_a = (u_a + \mathrm{i}/2)/(u_a - \mathrm{i}/2)$ gives the displayed rapidity form.

The coefficient $\mathsf{a}(p_1, p_2)$ is the exchange coefficient for the ordered chamber convention just chosen. The Bethe–Yang phase in the closed chain is its inverse:

$$S_{\mathrm{BY}}(u_j, u_k) = \mathsf{a}(p_j, p_k)^{-1} = \frac{u_j - u_k + \mathrm{i}}{u_j - u_k - \mathrm{i}}.$$

This inverse is a convention-sensitive point. For two magnons, trace periodicity says

$$\Psi(n_1, n_2) = \Psi(n_2, n_1 + L), \qquad n_1 < n_2.$$

Comparing the coefficient of $\mathrm{e}^{\mathrm{i}(p_1 n_1 + p_2 n_2)}$ gives

$$A_{12} = A_{21}\mathrm{e}^{\mathrm{i}p_1 L}, \qquad \mathrm{e}^{\mathrm{i}p_1 L} = S_{\mathrm{BY}}(u_1, u_2).$$

The second equation is obtained by moving the other magnon around the trace. For M magnons the one-loop $XXX_{1/2}$ chain has factorized scattering; equivalently, the transfer matrices constructed in Chapter 74 commute and the coordinate wavefunction is consistent in every ordering chamber. The finite periodic-chain equations are therefore

$$\left(\frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2}\right)^L = \prod_{\substack{k=1\\k\neq j}}^{M} \frac{u_j - u_k + \mathrm{i}}{u_j - u_k - \mathrm{i}}, \qquad \prod_{j=1}^{M} \frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2} = 1. \tag{97.2}$$

The second equation is the cyclicity condition. The energy is

$$E = \sum_{j=1}^{M} \frac{1}{u_j^2 + 1/4}.$$

At one loop these Bethe equations are exact for the finite $XXX_{1/2}$ Hamiltonian. The word "asymptotic" in later all-loop chapters has a different meaning: the long-range planar dilatation operator ceases to be finite-neighbor once the loop order grows, and finite trace length then requires wrapping corrections.

97.7 Free Magnons and BMN Double Scaling

The spin-chain excitation called a magnon is first a coordinate on the single-trace local-operator problem. Temporarily work in the unquotiented length-L spin-chain space

$$\mathcal{H}_L = (\mathbb{C}Z \oplus \mathbb{C}X)^{\otimes L}$$

and let τ be the translation by one site. The cyclic single-trace space is the quotient by $\tau - 1$, equivalently the image of the projection

$$\Pi_0 = \frac{1}{L} \sum_{r=0}^{L-1} \tau^r$$

when the planar two-point inner product is used. A one-magnon Fourier vector

$$|p\rangle = \sum_{n=1}^{L} \mathrm{e}^{\mathrm{i}pn} |n\rangle, \qquad |n\rangle = Z \otimes \cdots \otimes X_n \otimes \cdots \otimes Z,$$

has τ-eigenvalue $\mathrm{e}^{-\mathrm{i}p}$. On the finite periodic chain

$$H_{XXX}|p\rangle = 4\sin^2\frac{p}{2}\,|p\rangle, \qquad \mathrm{e}^{\mathrm{i}pL} = 1.$$

Its cyclic projection vanishes unless $p = 0$ modulo 2π. Thus a single nonzero-momentum magnon is not itself a physical single-trace local operator; it is the one-particle building block out of which total zero-momentum multi-magnon operators are assembled.

For M impurities with M fixed and $L \to \infty$, the integer

$$J = L - M$$

counts the number of Z's in the word. A two-impurity relative Fourier coordinate, for example with distinguishable scalar letters A, B whose planar two-point pairing with Z vanishes, is represented before diagonalization by the local-operator class

$$\mathcal{O}^{(L)}_{n;A,B} = \sum_{\ell=0}^{J} \exp\left(\frac{2\pi \mathrm{i} n\ell}{J+1}\right) \mathrm{tr}\Big(AZ^{\ell}BZ^{J-\ell}\Big).$$

This formula fixes the relative-momentum convention. Diagonalization of an interacting closed sector further requires the contact matching and the cyclic Bethe equations. In the $SU(2)$ chain those equations give an exact finite-L statement.

Example 97.12 (Two-magnon BMN quantization at one loop)**.** Consider the $SU(2)$ scalar sector at one loop with two magnons of momenta p and $-p$ on a cyclic trace of length $L \geq 3$. Let

$$u = \frac{1}{2}\cot\frac{p}{2}.$$

The cyclic Bethe equations imply

$$e^{ip(L-1)} = 1, \qquad p = \frac{2\pi n}{L-1}, \quad n \in \mathbb{Z}/(L-1)\mathbb{Z}.$$

For the non-protected modes $n \not\equiv 0$, the one-loop anomalous dimension is

$$\gamma^{(1)}_{n,L} = \frac{\lambda}{8\pi^2}\, 8\sin^2\frac{\pi n}{L-1} = \frac{\lambda}{\pi^2}\sin^2\frac{\pi n}{L-1}. \tag{97.3}$$

Consequently, in the double scaling limit

$$J = L-2 \to \infty, \qquad \lambda' = \frac{\lambda}{J^2} \text{ fixed}, \qquad n \text{ fixed},$$

one obtains

$$\gamma^{(1)}_{n,L} = \lambda' n^2 + O(J^{-1}).$$

For the two roots $u_1 = u$, $u_2 = -u$, the cyclicity equation in (97.2) is automatic:

$$\frac{u+i/2}{u-i/2}\,\frac{-u+i/2}{-u-i/2} = 1.$$

The first Bethe equation gives

$$\left(\frac{u+i/2}{u-i/2}\right)^L = \frac{2u+i}{2u-i}.$$

Since $u = \frac{1}{2}\cot(p/2)$,

$$\frac{u+i/2}{u-i/2} = \frac{2u+i}{2u-i} = e^{ip},$$

where the branch of p is understood modulo 2π. Hence $e^{ipL} = e^{ip}$, equivalently $e^{ip(L-1)} = 1$. The second Bethe equation is the inverse of the first.

The spin-chain energy is the sum of the two one-particle energies,

$$E = \frac{1}{u^2+1/4} + \frac{1}{u^2+1/4} = 8\sin^2\frac{p}{2}.$$

Multiplication by the overall field-theory coefficient $\lambda/(8\pi^2)$ gives (97.3). Finally set $L = J+2$ and use

$$\sin\frac{\pi n}{J+1} = \frac{\pi n}{J+1} + O(J^{-3})$$

for fixed n. Then

$$\frac{\lambda}{\pi^2}\sin^2\frac{\pi n}{J+1} = \frac{\lambda}{(J+1)^2}n^2 + O(\lambda J^{-4}) = \lambda' n^2 + O(J^{-1})$$

with $\lambda' = \lambda/J^2$ fixed.

The shift $J \mapsto J+1$ in the exact one-loop two-magnon formula is the finite-chain scattering phase written in relative-coordinate language. It disappears at leading order in the BMN double scaling limit but is essential for the finite-length normalization: the length-four Konishi state below is the case $L = 4$, $n = 1$, hence $p = 2\pi/3$, not $p = \pi/2$.

The all-loop dispersion relation in Chapter 98 is a much stronger statement. Its small-momentum limit reproduces the same one-loop normalization, but its derivation uses the centrally extended residual symmetry, crossing, and the dressing phase. The BMN calculation above is therefore a normalization anchor for the gauge-theory spin chain, not an external definition of the spectral problem.

97.8 Konishi at One Loop

The shortest non-BPS state in this sector is the length-four two-magnon descendant represented by $\mathrm{tr}([X, Z]^2)$. Set

$$L = 4, \qquad M = 2, \qquad u_1 = \frac{1}{2\sqrt{3}}, \qquad u_2 = -\frac{1}{2\sqrt{3}}.$$

Then

$$\frac{u_1 + \mathrm{i}/2}{u_1 - \mathrm{i}/2} = \mathrm{e}^{2\pi\mathrm{i}/3}, \qquad \frac{u_2 + \mathrm{i}/2}{u_2 - \mathrm{i}/2} = \mathrm{e}^{-2\pi\mathrm{i}/3},$$

so the cyclicity product is 1. The first Bethe equation reads

$$\mathrm{e}^{8\pi\mathrm{i}/3} = \frac{u_1 - u_2 + \mathrm{i}}{u_1 - u_2 - \mathrm{i}} = \mathrm{e}^{2\pi\mathrm{i}/3},$$

and the second is its inverse. The spin-chain energy is

$$E = \frac{1}{1/12 + 1/4} + \frac{1}{1/12 + 1/4} = 6.$$

Therefore the one-loop anomalous dimension is

$$\gamma_{\mathrm{K}}^{(1)} = \frac{\lambda}{8\pi^2}\,6 = \frac{3\lambda}{4\pi^2}.$$

97.9 Closed Sectors and Long-Range Data

The $SU(2)$ sector is the smallest laboratory, but the planar spectral problem uses a hierarchy of closed sectors. The $SL(2)$ sector is spanned by

$$\mathrm{tr}(D_+^{s_1} Z\, D_+^{s_2} Z \cdots D_+^{s_L} Z), \qquad s_j \in \mathbb{Z}_{\geq 0},$$

where D_+ is a fixed lightlike covariant derivative. Its one-loop chain has noncompact spin at each site. The $SU(2|3)$ sector contains three complex scalars and two fermions and is the smallest sector in which the dynamic length-changing character of the higher-loop planar chain is visible. The full one-letter module is the oscillator representation of $\mathfrak{psu}(2, 2|4)$, and a local operator is a cyclic tensor product of that module subject to gauge-invariant trace cyclicity.

At ℓ loops in the asymptotic regime, the planar dilatation operator has the range-$(\ell{+}1)$ asymptotic expansion

$$D = D_0 + \sum_{\ell\geq 1} g^{2\ell} \sum_{j=1}^{L} H^{(\ell)}_{j,\dots,j+\ell}, \qquad g = \frac{\sqrt{\lambda}}{4\pi}.$$

For example, the $SU(2)$ long-range Hamiltonian begins

$$H^{(1)} = 2\{\,\} - 2\{1\},$$

in the permutation-structure notation where $\{\,\}$ is the identity sum and $\{1\} = \sum_j P_{j,j+1}$. Higher orders contain $\{1,2\}$, $\{2,1\}$, and longer permutation words. Their coefficients are not fixed by $SU(2)$ symmetry alone; they are constrained by Feynman graphs, BMN-scaling tests, integrability, and supersymmetry. The all-loop Bethe ansatz in the next chapter should therefore be read as an asymptotic integrability reconstruction of this long-range operator, not as an independent definition of the four-dimensional theory.

97.10 Beyond One Loop

At higher loop order the planar dilatation operator becomes a long-range spin chain. In the $SU(2)$ sector its range grows with loop order, and for an operator of finite length L interactions can wrap around the trace once the perturbative range reaches L. Thus there are two different spectral problems:

asymptotic long chain and finite trace with wrapping.

The next chapter describes the asymptotic all-loop Bethe ansatz. The following chapter explains how mirror TBA and the Y-system incorporate finite-length effects.

Chapter 98

All-Loop Asymptotic Bethe Ansatz

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The one-loop spin chain of Chapter 97 is only the first term in the planar dilatation operator. The all-loop asymptotic Bethe ansatz is a proposal for the spectrum of long single-trace operators before wrapping effects become important. Its inputs are the residual centrally extended $\mathfrak{su}(2|2) \oplus \mathfrak{su}(2|2)$ symmetry of the BMN vacuum, factorized magnon scattering, crossing, and analyticity.

98.1 Conventions

The integrability literature uses several mutually inverse scalar-factor and Zhukovsky conventions. This chapter follows the convention used in the stringbook integrability chapters, with the notational replacement

$$h_{\text{stringbook}} = g_{\text{monograph}} = \frac{\sqrt{\lambda}}{4\pi}.$$

Thus

$$x^\pm + \frac{1}{x^\pm} = \frac{u \pm \mathrm{i}/2}{g}, \qquad \mathrm{e}^{\mathrm{i}p} = \frac{x^+}{x^-},$$

and the physical weak-coupling branch has

$$x^\pm = \frac{\mathrm{e}^{\pm \mathrm{i}p/2}}{2g\sin(p/2)} + O(g).$$

The one-magnon energy is written in the stringbook-compatible form

$$H = -1 - 2\mathrm{i}g(x^+ - x^-) = 1 + 2\mathrm{i}g\left(\frac{1}{x^+} - \frac{1}{x^-}\right).$$

The equality of the two expressions is not an additional assumption: it is obtained by multiplying the defining $x^\pm$ relation by $2\mathrm{i}g$.

The scalar factor convention is also fixed here. We write $\sigma = \mathrm{e}^{\mathrm{i}\theta}$ and use the crossing equation in the same orientation as the stringbook:

$$\sigma_{12}\sigma_{\bar{1}2} = \frac{x_2^+}{x_2^-}\frac{x_2^+ - x_1^+}{x_2^+ - x_1^-}\frac{x_2^- - 1/x_1^+}{x_2^- - 1/x_1^-}.$$

Authors who put the rational scalar factor into the opposite part of the S-matrix, or define the two-body operator as S_{21} rather than S_{12}, write the reciprocal of this expression. The companion check script enforces the $x^\pm$-energy and sheet conventions used here; crossing itself remains a monodromy input rather than a finite algebraic identity.

Warning 98.1 (Status of magnon crossing). The crossing equation used in this chapter is not the Wightman–LSZ crossing theorem for four-dimensional local scattering amplitudes. A spin-chain magnon is an excitation above the chosen BMN vacuum in a single-trace sector; it is not created by a gauge-invariant local field, its two-body scattering operator is asymptotic in the trace length, and its dispersion is encoded on the Zhukovsky rapidity surface rather than on a relativistic mass shell. Consequently ordinary four-dimensional QFT does not by itself give the replacement $x^\pm \mapsto 1/x^\pm$ or the scalar equation displayed above.

The conceptual origin of the condition is the two-dimensional relativistic integrable sigma-model description that appears in the strong-coupling AdS/CFT analysis. Before uniform light-cone gauge fixing, that auxiliary two-dimensional theory has a Euclidean continuation in which mirror-channel interchange of space and time and relativistic crossing are realized as analytic continuation in rapidity. Gauge fixing and decompactification turn the excitations into nonrelativistic magnons with the Zhukovsky dispersion, so the inherited crossing operation has no intrinsic explanation from the nonrelativistic spin chain alone. In the resulting integrable sigma-model/spin-chain framework the crossed magnon is the contragredient short representation of the centrally extended residual symmetry; equivalently, the antipode operation sends the Zhukovsky variables to the crossed sheet $x^\pm \mapsto 1/x^\pm$ along a specified path. Matrix consistency with this crossed representation fixes the rational multiplier below, while the dressing phase is required to have the corresponding scalar monodromy. Extending this strong-coupling crossing condition to all 't Hooft coupling, and using it as a statement about planar gauge-theory anomalous dimensions, is an integrability assumption constrained by symmetry, unitarity, Yang–Baxter factorization, perturbative gauge-theory checks, and finite-size TBA/QSC tests. It is not presently a theorem derived from a constructive four-dimensional definition of $\mathcal{N} = 4$ SYM.

98.2 BMN Vacuum and Central Extension

The vacuum spin-chain state is

$$\mathrm{tr}(Z^J).$$

It is protected and carries large R-charge J. A magnon is an insertion into this long chain. The residual symmetry preserving the vacuum contains two centrally extended $\mathfrak{su}(2|2)$ algebras. For one copy, write the bosonic states as $|\phi^a\rangle$, $a = 1, 2$, and the fermionic states as $|\psi^\alpha\rangle$, $\alpha = 1, 2$. The supercharges act in the form

$$\begin{aligned} Q^\alpha{}_a|\phi^b\rangle &= a\,\delta^b_a|\psi^\alpha\rangle, & Q^\alpha{}_a|\psi^\beta\rangle &= b\,\varepsilon^{\alpha\beta}\varepsilon_{ab}|\phi^b Z^+\rangle, \\ S^a{}_\alpha|\phi^b\rangle &= c\,\varepsilon^{ab}\varepsilon_{\alpha\beta}|\psi^\beta Z^-\rangle, & S^a{}_\alpha|\psi^\beta\rangle &= d\,\delta^\beta_\alpha|\phi^a\rangle. \end{aligned}$$

The symbols Z^+ and Z^- record insertion or removal of one vacuum letter. On a momentum eigenstate,

$$Z^+ \mapsto \mathrm{e}^{\mathrm{i}p} - 1, \qquad Z^- \mapsto \mathrm{e}^{-\mathrm{i}p} - 1$$

up to the chosen side convention. The representation constraint is

$$ad - bc = 1.$$

The central charges are

$$P = ab, \qquad K = cd, \qquad H = ad + bc.$$

The shortening condition of the centrally extended algebra is

$$H^2 - 4PK = 1.$$

The central extension is compatible with trace cyclicity. A single magnon is an excitation on an open segment of the BMN vacuum, so $Z^\pm$ records length data at the edge of that segment. On a closed trace the product of all such phases is $\mathrm{e}^{\mathrm{i}P_{\mathrm{tot}}} - 1$, where P_{tot} is the total spin-chain momentum. The physical single-trace projection sets $P_{\mathrm{tot}} = 0$, but individual magnons may still transform in centrally extended short representations before that projection is imposed. With

$$P = g(1 - \mathrm{e}^{\mathrm{i}p}), \qquad K = g(1 - \mathrm{e}^{-\mathrm{i}p}),$$

this gives the exact single-magnon dispersion relation

$$H(p) = \sqrt{1 + 16g^2 \sin^2 \frac{p}{2}}. \tag{98.1}$$

The anomalous contribution of a magnon is $H(p) - 1$.

Controlled Approximation 98.2 (BMN scaling check)**.** Let $J \to \infty$ with $\lambda' = \lambda/J^2$ fixed and take

$$p = \frac{2\pi n}{J} + O(J^{-2}), \qquad n \in \mathbb{Z}.$$

Then

$$H(p) = \sqrt{1 + \lambda' n^2} + O(J^{-1}).$$

This is the all-coupling continuation of the gauge-theory BMN normalization derived at one loop in Example 97.12. Here it is a convention check on the spectral problem, not an external derivation of the all-coupling Bethe ansatz.

Dispersion from shortening. Assuming the centrally extended residual symmetry acts on a one-magnon representation with central charges $P = g(1-\mathrm{e}^{\mathrm{i}p})$ and $K = g(1-\mathrm{e}^{-\mathrm{i}p})$, the shortening condition gives the dispersion relation by the finite algebraic check

$$PK = g^2(1-\mathrm{e}^{\mathrm{i}p})(1-\mathrm{e}^{-\mathrm{i}p}) = 4g^2 \sin^2 \frac{p}{2}.$$

Substitution into $H^2 - 4PK = 1$ gives

$$H^2 = 1 + 16g^2 \sin^2 \frac{p}{2}.$$

The positive square root is selected because $H = 1$ at $g = 0$ for an elementary magnon.

98.3 Zhukovsky Variables

Introduce the Zhukovsky map $x(u)$ by

$$x(u) + \frac{1}{x(u)} = \frac{u}{g}. \tag{98.2}$$

The physical branch is the solution satisfying $x(u) \sim u/g$ as $|u| \to \infty$. Equivalently,

$$x_{\mathrm{ph}}(u) = \frac{1}{2}\left(\frac{u}{g} + \sqrt{\frac{u}{g} - 2}\sqrt{\frac{u}{g} + 2}\right),$$

with the square-root cuts chosen so that the only short cut on the physical sheet is $[-2g, 2g]$. Crossing this cut replaces x by $1/x$. More precisely, the two boundary values on the cut satisfy

$$x_{\mathrm{above}}(u)x_{\mathrm{below}}(u) = 1, \qquad -2g < u < 2g.$$

The large-u expansion on the physical sheet is

$$x_{\mathrm{ph}}(u) = \frac{u}{g} - \frac{g}{u} - \frac{g^3}{u^3} - \frac{2g^5}{u^5} + O(u^{-7}).$$

For a one-magnon state set

$$x^{\pm} = x(u \pm \mathrm{i}/2).$$

Then

$$\mathrm{e}^{\mathrm{i}p} = \frac{x^+}{x^-}, \qquad x^+ + \frac{1}{x^+} - x^- - \frac{1}{x^-} = \frac{\mathrm{i}}{g}. \tag{98.3}$$

The physical region for real magnons is chosen so that $(x^+)^* = x^-$ and $|x^{\pm}| > 1$ at weak coupling. The magnon energy is the sheet-dependent but single-valued physical-branch expression

$$H = -1 - 2\mathrm{i}g(x^+ - x^-) = 1 + 2\mathrm{i}g\left(\frac{1}{x^+} - \frac{1}{x^-}\right), \tag{98.4}$$

with the same branch as (98.1). Conserved charges used in the dressing phase are

$$q_r(x^{\pm}) = \frac{\mathrm{i}}{r-1}\left[\frac{1}{(x^+)^{r-1}} - \frac{1}{(x^-)^{r-1}}\right], \qquad r \geq 2.$$

Zhukovsky form of the dispersion. Let $x^{\pm}$ obey (98.3) on the physical sheet. Then (98.4) is equal to

$$\sqrt{1+16g^2\sin^2\frac{p}{2}}.$$

For a physical real magnon write

$$x^{\pm}=\mathrm{e}^{\pm\mathrm{i}p/2}\frac{1+\sqrt{1+16g^2\sin^2(p/2)}}{4g\sin(p/2)}.$$

This parametrization gives $x^+/x^-=\mathrm{e}^{\mathrm{i}p}$. Substituting it into the difference $x^++1/x^+-x^--1/x^-$ gives i/g, so it is the physical solution of (98.3). The inverse difference is

$$\frac{1}{x^+}-\frac{1}{x^-}=\frac{\sqrt{1+16g^2\sin^2(p/2)}-1}{2\mathrm{i}g}.$$

Equation (98.4) therefore reproduces (98.1). The equality extends to the physical sheet by analytic continuation away from the branch points.

Warning 98.3 (Sheet data in crossing). The formal replacement $x^{\pm}\mapsto 1/x^{\pm}$ is not a harmless algebraic substitution. It records a path on the Zhukovsky surface. Poles crossed by that path alter the scalar factor through the crossing equation, and the answer depends on whether the continuation crosses the short cut of x^+, the short cut of x^-, or the composed path used in the mirror theory. In the u-plane, the cuts of $x^{\pm}=x(u\pm\mathrm{i}/2)$ have endpoints

$$u=\pm 2g-\mathrm{i}/2,\qquad u=\pm 2g+\mathrm{i}/2.$$

The physical crossing path for one magnon winds around the appropriate pair of these branch points so that both x^+ and x^- reach the crossed sheet. Applying the path twice generally lands on a different sheet of the scalar factor even though the projected $x^{\pm}$-values have returned.

Definition 98.4 (Crossing path for the Zhukovsky variables)**.** Let $x(u)$ be the physical Zhukovsky branch, $x(u)\sim u/g$ at infinity, and set $x^{\pm}(u)=x(u\pm\mathrm{i}/2)$. A crossing path for the first magnon is a path γ_{cr} on the u-surface, starting at a physical point and ending above the same projected u-value on a crossed sheet, with the following topological data:

$$w_{\gamma_{\mathrm{cr}}}\left(2g-\frac{\mathrm{i}}{2}\right)=1,\qquad w_{\gamma_{\mathrm{cr}}}\left(2g+\frac{\mathrm{i}}{2}\right)=1 \pmod 2,$$

and with even winding around the other two shifted endpoints in this right-endpoint convention. Equivalently, the lower shifted cut of x^+ and the upper shifted cut of x^- are each crossed once. When this path is used in the DHM contour formula below, it is also required to avoid the Cauchy and Gamma-function singularities except for the controlled residue crossings described in Proposition 98.5.

Zhukovsky monodromy along the crossing path. Analytic continuation of $x^{\pm}(u)$ along a path with the winding data of Definition 98.4 gives

$$\bar{x}^{\pm}(u)=\frac{1}{x^{\pm}(u)}.$$

Consequently,

$$\frac{\bar{x}^+}{\bar{x}^-} = \left(\frac{x^+}{x^-}\right)^{-1}, \qquad H(\bar{x}^\pm) = -H(x^\pm),$$

where $H(x^\pm) = 1 + 2\mathrm{i}g(1/x^+ - 1/x^-)$ is the physical-sheet energy continued as a meromorphic function of $x^\pm$.

Write

$$x(u) = \frac{1}{2}\left(\frac{u}{g} + \sqrt{\left(\frac{u}{g}\right)^2 - 4}\right)$$

on the physical branch. Encircling either endpoint of the short cut $[-2g, 2g]$ changes the sign of the square root and therefore sends

$$x(u) \longmapsto \frac{1}{2}\left(\frac{u}{g} - \sqrt{\left(\frac{u}{g}\right)^2 - 4}\right) = \frac{1}{x(u)}.$$

For $x^+(u) = x(u + \mathrm{i}/2)$, the relevant shifted cut has endpoint $2g - \mathrm{i}/2$; for $x^-(u) = x(u - \mathrm{i}/2)$, it has endpoint $2g + \mathrm{i}/2$. The mod-two winding data in Definition 98.4 therefore inverts both x^+ and x^-. The momentum ratio is then inverted immediately. Finally, using the stringbook energy identity

$$H(x^\pm) = -1 - 2\mathrm{i}g(x^+ - x^-) = 1 + 2\mathrm{i}g(1/x^+ - 1/x^-)$$

on the physical sheet, the crossed value is

$$H(\bar{x}^\pm) = 1 + 2\mathrm{i}g(x^+ - x^-) = -H(x^\pm).$$

98.4 Two-Body S-Matrix

The magnon two-body S-matrix is fixed by symmetry up to a scalar factor, after one has fixed the dynamic spin-chain frame. Let S_{12}^{mat} denote the matrix part determined by the two centrally extended $\mathfrak{su}(2|2)$ copies. The full two-magnon operator is

$$S_{12} = S_0(x_1^\pm, x_2^\pm)\, S_{12}^{\mathrm{mat}}.$$

For one $\mathfrak{su}(2|2)$ copy, write $S_{12}^{(1)}$ for the intertwiner before tensoring the left and right copies. It is defined by

$$S_{12}^{(1)}\Delta_{12}(J) = \Delta_{21}(J)S_{12}^{(1)}, \qquad J \in \mathfrak{su}(2|2)_{\mathrm{c}},$$

where the coproduct includes the braiding phase carried by the magnon momentum. The labels $1, 2$ on one-particle states record the momenta p_1, p_2, not extra internal indices. We use unnormalized symmetrization and antisymmetrization:

$$v^{\{a}w^{b\}} = v^a w^b + v^b w^a, \qquad v^{[a}w^{b]} = v^a w^b - v^b w^a.$$

With this convention the most general $\mathfrak{su}(2)\oplus\mathfrak{su}(2)$-covariant two-particle operator has ten amplitudes:

$$
\begin{aligned}
S^{(1)}_{12}|\phi^a_1\phi^b_2\rangle &= A_{12}|\phi^{\{a}_2\phi^{b\}}_1\rangle + B_{12}|\phi^{[a}_2\phi^{b]}_1\rangle + \frac{1}{2}C_{12}\varepsilon^{ab}\varepsilon_{\alpha\beta}|\psi^\alpha_2\psi^\beta_1\rangle,\\
S^{(1)}_{12}|\psi^\alpha_1\psi^\beta_2\rangle &= D_{12}|\psi^{\{\alpha}_2\psi^{\beta\}}_1\rangle + E_{12}|\psi^{[\alpha}_2\psi^{\beta]}_1\rangle + \frac{1}{2}F_{12}\varepsilon^{\alpha\beta}\varepsilon_{ab}|\phi^a_2\phi^b_1\rangle,\\
S^{(1)}_{12}|\phi^a_1\psi^\beta_2\rangle &= G_{12}|\psi^\beta_2\phi^a_1\rangle + H_{12}|\phi^a_2\psi^\beta_1\rangle,\\
S^{(1)}_{12}|\psi^\alpha_1\phi^b_2\rangle &= K_{12}|\psi^\alpha_2\phi^b_1\rangle + L_{12}|\phi^b_2\psi^\alpha_1\rangle.
\end{aligned}
$$

The length-changing markers are suppressed in this display but not forgotten: the C_{12} channel carries the Z^- marker and the F_{12} channel carries the Z^+ marker in the stringbook dynamic spin-chain frame. The markers are precisely the source of the braided coproduct.

For a magnon i write its representation coefficients as a_i, b_i, c_i, d_i. The stringbook frame relates them to $x^\pm_i$ by

$$
x^+_i = \mathrm{i}\frac{d_i}{b_i},\qquad x^-_i = -\mathrm{i}\frac{a_i}{c_i},\qquad a_id_i - b_ic_i = 1.
$$

The same $x^\pm_i$ obey the Zhukovsky relations already fixed in (98.3). At generic non-pole kinematics the tensor product of two fundamental short representations is long and the intertwiner is unique up to one scalar. Let this one-copy scalar be denoted $\mathcal{S}_{12}$. Solving the finite intertwining equations gives

$$
\begin{aligned}
A_{12} &= \mathcal{S}_{12}\frac{x^+_2 - x^-_1}{x^-_2 - x^+_1},\\
B_{12} &= A_{12}\left[1 - 2\frac{x^+_2 - x^+_1}{x^+_2 - x^-_1}\frac{x^-_2 - 1/x^+_1}{x^-_2 - 1/x^-_1}\right],\\
C_{12} &= \mathcal{S}_{12}\frac{2g^{-1}a_1a_2}{x^-_1x^-_2 - 1}\frac{x^+_2 - x^+_1}{x^-_2 - x^+_1},\\
D_{12} &= -\mathcal{S}_{12},\\
E_{12} &= -\mathcal{S}_{12}\left[1 - 2\frac{x^-_2 - x^-_1}{x^-_2 - x^+_1}\frac{x^+_2 - 1/x^-_1}{x^+_2 - 1/x^+_1}\right],\\
F_{12} &= -\mathcal{S}_{12}\frac{2g(x^+_1 - x^-_1)(x^+_2 - x^-_2)}{a_1a_2(x^+_1x^+_2 - 1)}\frac{x^-_2 - x^-_1}{x^-_2 - x^+_1},\\
G_{12} &= \mathcal{S}_{12}\frac{x^+_2 - x^+_1}{x^-_2 - x^+_1},\qquad H_{12} = \mathcal{S}_{12}\frac{a_1}{a_2}\frac{x^+_2 - x^-_2}{x^-_2 - x^+_1},\\
K_{12} &= \mathcal{S}_{12}\frac{a_2}{a_1}\frac{x^+_1 - x^-_1}{x^-_2 - x^+_1},\qquad L_{12} = \mathcal{S}_{12}\frac{x^-_2 - x^-_1}{x^-_2 - x^+_1}.
\end{aligned}
\tag{98.5}
$$

Generic row-rank computation for the one-copy intertwiner. Let

$$
d_{12} = x^-_2 - x^+_1,\qquad n_{12} = x^+_2 - x^-_1,
$$

and work on the generic open set where

$$
x^\pm_1x^\pm_2d_{12}n_{12}a_1a_2(x^-_1x^-_2 - 1)(x^+_1x^+_2 - 1) \neq 0.
$$

After the dynamic-frame Q- and S-intertwining equations are written on the ten $\mathfrak{su}(2)\oplus\mathfrak{su}(2)$-covariant amplitudes and the representation ratios

$$x_i^+ = \mathrm{i}d_i/b_i, \qquad x_i^- = -\mathrm{i}a_i/c_i, \qquad a_id_i - b_ic_i = 1$$

are used, the finite linear system has the following row-echelon chart:

$$\begin{gathered} B_{12} = \beta_{12}A_{12}, \qquad C_{12} = \gamma_{12}A_{12}, \qquad D_{12} = \delta_{12}A_{12}, \qquad E_{12} = \varepsilon_{12}A_{12}, \\ F_{12} = \varphi_{12}A_{12}, \qquad G_{12} = g_{12}A_{12}, \qquad H_{12} = h_{12}A_{12}, \qquad K_{12} = k_{12}A_{12}, \qquad L_{12} = \ell_{12}A_{12}, \end{gathered} \tag{98.6}$$

where

$$\begin{aligned} \beta_{12} &= 1 - 2\frac{x_2^+ - x_1^+}{n_{12}}\frac{x_2^- - 1/x_1^+}{x_2^- - 1/x_1^-}, & \gamma_{12} &= \frac{2a_1a_2}{g}\frac{x_2^+ - x_1^+}{(x_1^-x_2^- - 1)n_{12}}, \\ \delta_{12} &= -\frac{d_{12}}{n_{12}}, & \varepsilon_{12} &= -\frac{d_{12}}{n_{12}}\left[1 - 2\frac{x_2^- - x_1^-}{d_{12}}\frac{x_2^+ - 1/x_1^-}{x_2^+ - 1/x_1^+}\right], \\ \varphi_{12} &= -\frac{2g}{a_1a_2}\frac{(x_1^+ - x_1^-)(x_2^+ - x_2^-)(x_2^- - x_1^-)}{(x_1^+x_2^+ - 1)n_{12}}, & g_{12} &= \frac{x_2^+ - x_1^+}{n_{12}}, \\ h_{12} &= \frac{a_1}{a_2}\frac{x_2^+ - x_2^-}{n_{12}}, & k_{12} &= \frac{a_2}{a_1}\frac{x_1^+ - x_1^-}{n_{12}}, \\ \ell_{12} &= \frac{x_2^- - x_1^-}{n_{12}}. \end{aligned} \tag{98.7}$$

Consequently the solution space of the finite intertwining equations is one-dimensional on this open set. Choosing

$$A_{12} = \mathcal{S}_{12}\frac{n_{12}}{d_{12}}$$

recovers the ten amplitudes in (98.5).

This row chart is a finite row-reduction attached to the chosen dynamic frame; the theorem-level inputs are the representation and analytic construction of the planar magnon S-matrix. The unknown vector is

$$(A, B, C, D, E, F, G, H, K, L).$$

Each equation in (98.6) has the form

$$X - r_XA = 0, \qquad X \in \{B, C, D, E, F, G, H, K, L\},$$

with r_X one of the nine coefficients in (98.7). Ordered by $(B, C, D, E, F, G, H, K, L)$, these nine equations have pivot coefficient 1 in nine distinct columns. Their coefficient matrix therefore has rank nine, so its kernel in the ten-dimensional amplitude space is one dimensional and is parametrized by A.

It remains only to identify the parametrization with the convention used in (98.5). Substituting $A_{12} = \mathcal{S}_{12}n_{12}/d_{12}$ into (98.6) gives, term by term, the formulas for $B, C, D, E, F, G, H, K, L$ in (98.5). The genericity assumptions are exactly the denominators that appear in this chart and in the two channels where the $Z^{\pm}$ markers enter. At their zero loci the tensor product can become reducible or the chosen dynamic-frame chart can degenerate; those special loci are the bound-state and pole analyses, not the generic two-magnon scattering case.

Local intertwining and block-unitarity checks. Assume the denominators in (98.5) are non-zero. The amplitudes above satisfy the highest-weight Q-intertwining relation

$$A_{12} = \frac{a_1}{a_2} K_{12} + G_{12} = L_{12} + \frac{a_2}{a_1} H_{12}. \tag{98.8}$$

After factoring out $\mathcal{S}_{12}$, the same amplitudes also obey the local analytic unitarity identities

$$R_{12}^{\mathrm{BB}} R_{21}^{\mathrm{BB}} = 1, \qquad R_{12}^{\mathrm{BF}} R_{21}^{\mathrm{BF}} = 1,$$

on the boson-boson/fermion-fermion block R^{BB} and the mixed boson-fermion block R^{BF} written, for fixed distinct doublet indices, in the ordered bases

$$(\psi_1^1 \psi_2^2, \psi_1^2 \psi_2^1, \phi_1^1 \phi_2^2, \phi_1^2 \phi_2^1), \qquad (\phi_1^1 \psi_2^1, \psi_1^1 \phi_2^1).$$

The blocks are explicitly

$$R_{12}^{\mathrm{BB}} = \frac{1}{\mathcal{S}_{12}} \begin{pmatrix} (D-E)/2 & (D+E)/2 & -F/2 & F/2 \\ (D+E)/2 & (D-E)/2 & F/2 & -F/2 \\ -C/2 & C/2 & (A-B)/2 & (A+B)/2 \\ C/2 & -C/2 & (A+B)/2 & (A-B)/2 \end{pmatrix}_{12}, \qquad R_{12}^{\mathrm{BF}} = \frac{1}{\mathcal{S}_{12}} \begin{pmatrix} H & G \\ L & K \end{pmatrix}_{12}.$$

Equivalently, the matrix part has the correct inverse when the two magnons are exchanged; the scalar unitarity condition is the separate condition $\mathcal{S}_{12}\mathcal{S}_{21} = 1$, or, for the full two-copy scalar split, $S_{0,12} S_{0,21} = 1$.

The finite check is local in the chosen dynamic frame. The first identity is the result of applying ${Q^\alpha}_a$ to the symmetric boson state and comparing the independent $|\psi_2 \phi_1\rangle$ and $|\phi_2 \psi_1\rangle$ components. Substituting (98.5) gives

$$\begin{aligned} \frac{a_1}{a_2} K_{12} + G_{12} &= \mathcal{S}_{12} \frac{x_1^+ - x_1^- + x_2^+ - x_1^+}{x_2^- - x_1^+} = \mathcal{S}_{12} \frac{x_2^+ - x_1^-}{x_2^- - x_1^+} = A_{12}, \\ L_{12} + \frac{a_2}{a_1} H_{12} &= \mathcal{S}_{12} \frac{x_2^- - x_1^- + x_2^+ - x_2^-}{x_2^- - x_1^+} = A_{12}. \end{aligned}$$

The remaining Q- and S-intertwining equations are obtained in the same finite way from the other highest-weight components. Clearing denominators reduces them to polynomial identities in $x_i^\pm$ and the representation ratios a_1/a_2, together with the defining relations for b_i, c_i, d_i above. The companion calculation check constructs the two nontrivial finite blocks explicitly from (98.5) and verifies $R_{12}R_{21} = 1$ on nonsingular physical-branch samples. This check is not a substitute for crossing or Yang–Baxter factorization; it verifies the local frame and amplitude algebra used by the nested Bethe derivation below.

The convenient scalar split used in the stringbook is

$$\mathcal{S}_{12} = \left[\frac{x_2^- - x_1^+}{x_2^+ - x_1^-} \frac{1 - 1/(x_2^+ x_1^-)}{1 - 1/(x_2^- x_1^+)} \right]^{1/2} \sigma_{12}.$$

Consequently the compact $SU(2)$-sector two-copy amplitude $(A_{12})^2$ contains the rational factor

$$\frac{x_2^+ - x_1^-}{x_2^- - x_1^+} \frac{1 - 1/(x_2^+ x_1^-)}{1 - 1/(x_2^- x_1^+)}$$

times σ_{12}^2, in the orientation fixed by the conventions above. Equivalently, the full scalar factor may be written as

$$S_0(x_1^\pm, x_2^\pm) = S_{\rm rat}(x_1^\pm, x_2^\pm)\,\sigma(x_1^\pm, x_2^\pm)^2, \qquad \sigma = \mathrm{e}^{\mathrm{i}\theta}.$$

The rational part is fixed by the same representation data that determine the matrix part. The dressing phase has the antisymmetric charge expansion

$$\theta(x_1^\pm, x_2^\pm) = -\sum_{r=2}^{\infty} \sum_{\substack{s>r\\ r+s \text{ odd}}} c_{r,s}(g)\,\big(q_r(x_1^\pm)q_s(x_2^\pm) - q_s(x_1^\pm)q_r(x_2^\pm)\big).$$

Dressing unitarity from charge antisymmetry. Work in a domain of the physical sheet where the charge expansion for $\theta_{12} = \theta(x_1^\pm, x_2^\pm)$ is either finite after truncation or absolutely convergent, and use the same branch of the phase for the exchanged pair $\theta_{21} = \theta(x_2^\pm, x_1^\pm)$. Then

$$\theta_{21} = -\theta_{12}, \qquad \sigma_{12}\sigma_{21} = 1.$$

Consequently the squared dressing factor appearing in the $SU(2)$ asymptotic Bethe equation also satisfies

$$\sigma_{12}^2\sigma_{21}^2 = 1.$$

For each pair $r < s$ with $r+s$ odd, define

$$A_{r,s}(1,2) = q_r(x_1^\pm)q_s(x_2^\pm) - q_s(x_1^\pm)q_r(x_2^\pm).$$

Exchanging the two magnons gives

$$A_{r,s}(2,1) = q_r(x_2^\pm)q_s(x_1^\pm) - q_s(x_2^\pm)q_r(x_1^\pm) = -A_{r,s}(1,2).$$

The coefficients $c_{r,s}(g)$ are scalar functions of the coupling and the integer pair (r,s), not of the ordering of the two magnons. Therefore the whole series changes sign under $1 \leftrightarrow 2$, proving $\theta_{21} = -\theta_{12}$. Since $\sigma = \mathrm{e}^{\mathrm{i}\theta}$,

$$\sigma_{12}\sigma_{21} = \mathrm{e}^{\mathrm{i}(\theta_{12}+\theta_{21})} = 1.$$

Squaring gives the last statement.

This is only the scalar dressing part of physical two-body unitarity. The rational scalar factor and the matrix part must obey their own unitarity relations in the same frame. The computation also does not replace crossing: after analytic continuation around Zhukovsky cuts the charge expansion must be continued with its prescribed monodromy, and that monodromy is precisely the additional crossing datum discussed below.

Equivalently, the BES phase may be packaged as

$$\begin{aligned} \theta(x_1^\pm, x_2^\pm) &= \chi(x_1^+, x_2^+) - \chi(x_1^+, x_2^-) - \chi(x_1^-, x_2^+) + \chi(x_1^-, x_2^-), \\ \chi(x,y) &= -\mathrm{i}\oint \frac{\mathrm{d}z}{2\pi\mathrm{i}} \oint \frac{\mathrm{d}w}{2\pi\mathrm{i}} \frac{1}{x-z}\frac{1}{y-w} \log\frac{\Gamma\big(1+\mathrm{i}g(z+z^{-1}-w-w^{-1})\big)}{\Gamma\big(1-\mathrm{i}g(z+z^{-1}-w-w^{-1})\big)}. \end{aligned} \tag{98.9}$$

The contours encircle the unit circle singularities of z and w in the domain where $|x| > 1$, $|y| > 1$. Formula (98.9) is therefore not a global formula on all sheets by itself; outside that domain it is continued with the crossing prescription.

Proposition 98.5 (Local DHM residue continuation)**.** *Let*

$$\mathcal{L}_g(z,w) = \log \frac{\Gamma\big(1 + \mathrm{i}g(z + z^{-1} - w - w^{-1})\big)}{\Gamma(1 - \mathrm{i}g(z + z^{-1} - w - w^{-1}))}.$$

Fix x, y away from the unit circle and away from the Gamma-function poles. Denote by $\chi_{\mathrm{raw}}^{\epsilon_x,\epsilon_y}(x,y)$ the same unit-circle double contour integral as in (98.9), with the x-Cauchy kernel expanded radially outside the contour when $\epsilon_x = +$ and inside the contour when $\epsilon_x = -$, and similarly for y. Thus $\chi_{\mathrm{raw}}^{+,+}$ is the original DHM branch. With the stringbook orientation for crossing inward through $|x| = 1$ and $|y| = 1$, define the one-residue corrections by

$$\Psi_x^{\epsilon_y}(x,y) = \chi_{\mathrm{raw}}^{-,\epsilon_y}(x,y) - \chi_{\mathrm{raw}}^{+,\epsilon_y}(x,y), \qquad \Psi_y^{\epsilon_x}(x,y) = \chi_{\mathrm{raw}}^{\epsilon_x,+}(x,y) - \chi_{\mathrm{raw}}^{\epsilon_x,-}(x,y). \tag{98.10}$$

Equivalently, Ψ_x is the residue picked up by the pole $z = x$, with the sign fixed by the kernel $1/(x - z)$, while Ψ_y is written with the opposite sign because the stringbook formula adds the y-crossing correction. The analytic continuation of the original outside branch into the three neighboring radial chambers is

$$\chi_{\mathrm{cont}}^{-,+}(x,y) = \chi_{\mathrm{raw}}^{-,+}(x,y) - \Psi_x^{+}(x,y), \tag{98.11}$$

$$\chi_{\mathrm{cont}}^{+,-}(x,y) = \chi_{\mathrm{raw}}^{+,-}(x,y) + \Psi_y^{+}(x,y), \tag{98.12}$$

$$\chi_{\mathrm{cont}}^{-,-}(x,y) = \chi_{\mathrm{raw}}^{-,-}(x,y) - \Psi_x^{-}(x,y) + \Psi_y^{-}(x,y) - \mathcal{C}_g(x,y), \tag{98.13}$$

where the contact term is the double residue

$$\mathcal{C}_g(x,y) = \mathrm{i}\,\mathcal{L}_g(x,y). \tag{98.14}$$

These formulas are local sheet-transition rules for the DHM contour. The global Janik crossing equation additionally specifies a path on the u-surface, hence which of $x^{\pm}(u)$ winds around which shifted Zhukovsky branch point.

Proof. The only singularities that move when x crosses the unit circle are the Cauchy poles $z = x$; the Gamma-function poles are excluded by hypothesis. For fixed y-chamber, deforming the z-contour back to the unit circle therefore changes the raw integral exactly by the residue at $z = x$. This gives the first definition in (98.10) and hence the first continuation formula. The y-crossing computation is the same, except that the sign is recorded in Ψ_y so that the stringbook notebook convention reads as addition in the second formula.

If both variables cross, the two single residues count the intersection where $z = x$ and $w = y$ once too many. The product of the residue signs from $1/(x - z)$ and $1/(y - w)$, together with the overall factor $-\mathrm{i}$ in χ, gives the contact term $\mathrm{i}\mathcal{L}_g(x,y)$. Subtracting this double residue gives the final $-\mathrm{i}\mathcal{L}_g(x,y)$ contribution in (98.13). The planar $N = 4$ companion calculation check verifies these signs on finite Laurent models of the DHM kernel, including the failure that occurs if the contact term is omitted. ∎

DHM pole lattice and admissible local continuations. Let

$$\Delta(z,w) = z + z^{-1} - w - w^{-1}.$$

For fixed $g > 0$, the Gamma-kernel logarithm $\mathcal{L}_g(z, w)$ can develop singularities only on the divisors

$$\Delta(z,w) = \frac{\mathrm{i}n}{g} \quad \text{or} \quad \Delta(z,w) = -\frac{\mathrm{i}n}{g}, \qquad n \in \mathbb{Z}_{\geq 1}. \tag{98.15}$$

Consequently the one-residue correction $\Psi_x^{\epsilon_y}(x, y)$ is a single-valued holomorphic function on any connected region in which

$$x + x^{-1} - w - w^{-1} \notin \frac{\mathrm{i}}{g}\big(\mathbb{Z} \setminus \{0\}\big)$$

for every w on the chosen y-contour, and similarly for $\Psi_y^{\epsilon_x}$. The contact term $\mathcal{C}_g(x, y) = \mathrm{i}\mathcal{L}_g(x, y)$ is holomorphic precisely away from the endpoint divisors

$$x + x^{-1} - y - y^{-1} \in \frac{\mathrm{i}}{g}\big(\mathbb{Z} \setminus \{0\}\big).$$

If a crossing path for $x_1^\pm$ satisfies these avoidance conditions for each of the four pairs (x_1^α, x_2^β), $\alpha, \beta \in \{+, -\}$, except for the Cauchy-pole crossings already accounted for in Proposition 98.5, then the local BES-continuation formula below applies with no additional Gamma-pole residue terms.

The Gamma function has poles at $0, -1, -2, \ldots$. The numerator in $\mathcal{L}_g$ is singular only when

$$1 + \mathrm{i}g\Delta(z, w) = 1 - n, \qquad n \in \mathbb{Z}_{\geq 1},$$

which is equivalent to $\Delta(z, w) = \mathrm{i}n/g$. The denominator is singular only when

$$1 - \mathrm{i}g\Delta(z, w) = 1 - n, \qquad n \in \mathbb{Z}_{\geq 1},$$

which is equivalent to $\Delta(z, w) = -\mathrm{i}n/g$. This proves (98.15). The displayed conditions for Ψ_x, Ψ_y, and $\mathcal{C}_g$ are the same divisors evaluated after setting $z = x$, $w = y$, or one variable on the remaining contour. Away from those divisors the integrands are holomorphic in the external variables and the contour deformations are ordinary Cauchy deformations. Thus a path that crosses only the specified Cauchy poles has exactly the residue contributions listed above, with no further Gamma-kernel terms.

Local crossing of the BES phase. Assume $x_1^\pm$ and $x_2^\pm$ lie in the physical DHM chamber $|x_1^\pm| > 1$, $|x_2^\pm| > 1$, and continue the first magnon along the crossing path of Definition 98.4, so that $\bar{x}_1^\pm = 1/x_1^\pm$ lie in the inside radial chamber. Suppose the path avoids the Gamma-function poles of the DHM kernel. Define

$$\begin{aligned} \theta_{\mathrm{raw}}^{-,+}(\bar{x}_1^\pm, x_2^\pm) &= \chi_{\mathrm{raw}}^{-,+}(\bar{x}_1^+, x_2^+) - \chi_{\mathrm{raw}}^{-,+}(\bar{x}_1^+, x_2^-) - \chi_{\mathrm{raw}}^{-,+}(\bar{x}_1^-, x_2^+) + \chi_{\mathrm{raw}}^{-,+}(\bar{x}_1^-, x_2^-), \\ \Delta_{\bar{1}}\theta &= -\Psi_x^+(\bar{x}_1^+, x_2^+) + \Psi_x^+(\bar{x}_1^+, x_2^-) + \Psi_x^+(\bar{x}_1^-, x_2^+) - \Psi_x^+(\bar{x}_1^-, x_2^-). \end{aligned} \tag{98.16}$$

Then the local DHM continuation of the BES phase for the crossed first particle is

$$\theta_{\bar{1}2}^{\mathrm{loc}} = \theta_{\mathrm{raw}}^{-,+}(\bar{x}_1^\pm, x_2^\pm) + \Delta_{\bar{1}}\theta. \tag{98.17}$$

The BES phase is the alternating four-kernel combination

$$\chi(x_1^+, x_2^+) - \chi(x_1^+, x_2^-) - \chi(x_1^-, x_2^+) + \chi(x_1^-, x_2^-).$$

Along the stated crossing path, only the first argument of each kernel crosses from the outside chamber to the inside chamber; the second argument stays outside. The local DHM continuation formula for one kernel therefore adds the correction $-\Psi_x^+$ to each crossed kernel. Multiplying these four single-kernel corrections by the signs $+, -, -, +$ in the BES combination gives exactly the four terms defining $\Delta_{\bar{1}}\theta$. The raw continued kernels give $\theta_{\mathrm{raw}}^{-,+}$, so their sum is (98.17).

This local contour computation fixes how the DHM residue terms enter the four-kernel BES phase in the stringbook convention, but it does not by itself prove the scalar crossing equation (98.27). That equation also requires the global branch of the scalar factor, the homotopy class of the crossing path, and the absence or prescribed treatment of additional crossed singularities.

Inside this domain the weak-coupling coefficients can be extracted without any sheet continuation. Expand

$$\chi(x,y) = \sum_{r=2}^{\infty} \sum_{\substack{s>r \\ r+s \text{ odd}}} \frac{c_{r,s}(g)}{(r-1)(s-1)} \left(\frac{1}{x^{r-1}y^{s-1}} - \frac{1}{x^{s-1}y^{r-1}} \right). \tag{98.18}$$

The charge-expansion convention above uses the same $c_{r,s}$; the two forms agree because $q_r = \mathrm{i}A_r/(r-1)$, where $A_r = (x^+)^{1-r} - (x^-)^{1-r}$.

Proposition 98.6 (Weak coefficients from the DHM contour)**.** *For $N = 2m+1 \geq 3$, let $R_{r,s}^{(N)}$ be the Laurent coefficient*

$$R_{r,s}^{(N)} = [z^{-(r-1)}w^{-(s-1)}]\big(z + z^{-1} - w - w^{-1}\big)^N.$$

Then the contribution of order g^N to the coefficient $c_{r,s}(g)$ in (98.18) is

$$c_{r,s}(g)\big|_{g^N} = \frac{2(-1)^{m+1}}{N}(r-1)(s-1)R_{r,s}^{(N)}\,\zeta(N)g^N. \tag{98.19}$$

Equivalently, writing

$$c_{r,s}(g)\big|_{g^N} = C_{r,s}^{(N)}\zeta(N)g^N,$$

one has the closed coefficient

$$C_{r,s}^{(N)} = \frac{2(-1)^{(N-1)/2+s}(r-1)(s-1)N!(N-1)!}{\Gamma\big(\frac{N-r-s+4}{2}\big)\,\Gamma\big(\frac{N+r-s+2}{2}\big)\,\Gamma\big(\frac{N-r+s+2}{2}\big)\,\Gamma\big(\frac{N+r+s}{2}\big)}, \tag{98.20}$$

when the four Gamma-function arguments are positive integers; otherwise this order gives zero. In particular, the coefficient vanishes unless the Laurent powers can be supplied by the N factors in $(z+z^{-1}-w-w^{-1})^N$. In particular,

$$c_{2,3}(g) = 4\zeta(3)g^3 - 40\zeta(5)g^5 + O(g^7).$$

The first dressing contribution to a weak-coupling anomalous dimension is nevertheless order g^6, since $q_2 = O(g)$ and $q_3 = O(g^2)$ on the physical weak-coupling branch.

Proof. For $|x|, |y| > 1$,

$$\frac{1}{x-z}\frac{1}{y-w} = \sum_{a,b\geq 0} \frac{z^a w^b}{x^{a+1}y^{b+1}}.$$

The logarithm in (98.9) has the Taylor expansion at small g

$$\log \frac{\Gamma(1+\eta)}{\Gamma(1-\eta)} = -2\gamma\eta - 2\sum_{m=1}^{\infty} \frac{\zeta(2m+1)}{2m+1}\eta^{2m+1}, \qquad \eta = \mathrm{i}g(z+z^{-1}-w-w^{-1}).$$

The linear Euler-constant term is symmetric data that drops out of the antisymmetric coefficient family $s>r$. For $N=2m+1\geq 3$, multiplying by the outside factor $-\mathrm{i}$ gives

$$\frac{2(-1)^{m+1}\zeta(N)}{N} g^N \left(z+z^{-1}-w-w^{-1}\right)^N.$$

The two contour integrals pick the coefficient of $z^{-1}w^{-1}$. Thus the coefficient of $x^{1-r}y^{1-s}$ is the displayed constant times $R^{(N)}_{r,s}$. Comparing with (98.18) proves (98.19).

It remains to evaluate the Laurent coefficient. Put $a=r-1$ and $b=s-1$. Expanding first in the number ℓ of $(-w-w^{-1})$ factors gives the finite convolution

$$R^{(N)}_{r,s} = \sum_{\ell=0}^{N} (-1)^{\ell} \binom{N}{\ell} \binom{N-\ell}{(N-\ell+a)/2} \binom{\ell}{(\ell+b)/2}, \tag{98.21}$$

where a binomial coefficient is declared zero unless its lower entry is an integer between 0 and the upper entry. This convention records the parity and support restrictions.

For the closed evaluation, compare Fourier coefficients by writing $z=\mathrm{e}^{\mathrm{i}\theta}$, $w=\mathrm{e}^{\mathrm{i}\varphi}$, $A=(\theta+\varphi)/2$, and $B=(\theta-\varphi)/2$. Then

$$z+z^{-1}-w-w^{-1} = -4\sin A\,\sin B, \qquad z^{-a}w^{-b} = \mathrm{e}^{-\mathrm{i}(a+b)A}\mathrm{e}^{-\mathrm{i}(a-b)B}.$$

The elementary expansion

$$\sin^N t = (2\mathrm{i})^{-N}\sum_{j=0}^{N}(-1)^j\binom{N}{j}\mathrm{e}^{\mathrm{i}(N-2j)t}$$

therefore gives

$$R^{(N)}_{r,s} = (-1)^{N+a}\binom{N}{(N+a+b)/2}\binom{N}{(N+a-b)/2},$$

with the same convention that the binomial coefficient is zero unless its lower entry is an integer between 0 and N. Since N is odd and the nonzero cases have $a+b$ odd, the sign is $(-1)^{N+a} = (-1)^b = (-1)^{s-1}$. Rewriting the two binomial coefficients as Gamma functions gives

$$R^{(N)}_{r,s} = (-1)^{s-1}\frac{(N!)^2}{\Gamma\left(\frac{N-r-s+4}{2}\right)\Gamma\left(\frac{N+r-s+2}{2}\right)\Gamma\left(\frac{N-r+s+2}{2}\right)\Gamma\left(\frac{N+r+s}{2}\right)}.$$

The lower-entry bounds in the binomial formula are exactly the positivity conditions on the four Gamma-function arguments; outside those bounds the coefficient is zero. Substitution into (98.19) gives (98.20). For $r=2, s=3$,

$$R^{(3)}_{2,3} = 3, \qquad R^{(5)}_{2,3} = 50,$$

which gives $4\zeta(3)g^3 - 40\zeta(5)g^5$. □

A saddle expansion of the same contour representation gives the AFS and Hernandez–Lopez strong-coupling terms.

Crossing scalar convention algebra. Assume $x_1^\pm, x_2^\pm$ are non-zero and avoid the displayed denominators. In the stringbook scalar-factor orientation used in this chapter set

$$\Xi_{12} = \frac{x_2^+}{x_2^-}\frac{x_2^+ - x_1^+}{x_2^+ - x_1^-}\frac{x_2^- - 1/x_1^+}{x_2^- - 1/x_1^-}.$$

The common reciprocal orientation is

$$\Xi_{12}^{\mathrm{rec}} = \frac{x_2^-}{x_2^+}\frac{x_2^+ - x_1^-}{x_2^+ - x_1^+}\frac{x_2^- - 1/x_1^-}{x_2^- - 1/x_1^+},$$

and satisfies

$$\Xi_{12}^{\mathrm{rec}} = \Xi_{12}^{-1}.$$

This is the product of the three displayed inverse factors:

$$\Xi_{12}\Xi_{12}^{\mathrm{rec}} = \frac{x_2^+}{x_2^-}\frac{x_2^-}{x_2^+}\frac{x_2^+ - x_1^+}{x_2^+ - x_1^-}\frac{x_2^+ - x_1^-}{x_2^+ - x_1^+}\frac{x_2^- - 1/x_1^+}{x_2^- - 1/x_1^-}\frac{x_2^- - 1/x_1^-}{x_2^- - 1/x_1^+} = 1.$$

Moreover the algebraic substitution $x_1^\pm \mapsto 1/x_1^\pm$ does not leave this multiplier invariant as a rational function. At

$$x_1^+ = 2, \qquad x_1^- = 3, \qquad x_2^+ = 5, \qquad x_2^- = 7$$

one obtains $\Xi_{12} = 117/112$, whereas the same formula with $x_1^\pm$ replaced by $1/x_1^\pm$ gives 675/784. Hence equality can hold only on a proper algebraic sublocus, not as a convention-independent identity.

Matrix-channel origin of the scalar crossing multiplier. Let $\bar{x}_1^\pm = 1/x_1^\pm$ denote the crossed-sheet continuation of the first magnon and assume all displayed denominators are non-zero. Form the crossed amplitude $L_{\bar{1}2}$ from (98.5) by replacing (x_1^+, x_1^-) with $(\bar{x}_1^+, \bar{x}_1^-)$, and form G_{21} by exchanging the two magnons and using the analytic-unitarity convention $\mathcal{S}_{21} = \mathcal{S}_{12}^{-1}$. Then

$$\mathcal{R}_{\mathrm{cr}} := \frac{L_{\bar{1}2}}{G_{21}} = \mathcal{S}_{\bar{1}2}\mathcal{S}_{12}\frac{(x_2^- - 1/x_1^-)\,x_1^+(x_1^- - x_2^+)}{(x_2^- x_1^+ - 1)(x_1^+ - x_2^+)}. \tag{98.22}$$

Substitute the stringbook scalar split for $\mathcal{S}_{12}$ and $\mathcal{S}_{\bar{1}2}$. In the chamber

$$x_2^- > x_1^+ > 1, \qquad x_2^+ > x_1^- > 1,$$

where the square roots in the scalar split are taken continuously from positive real values, the finite channel condition $\mathcal{R}_{\mathrm{cr}} = 1$ is equivalent to

$$\sigma(\bar{x}_1^\pm, x_2^\pm)\sigma(x_1^\pm, x_2^\pm) = \Xi_{12}. \tag{98.23}$$

The formula for $L_{\bar{1}2}$ is

$$L_{\bar{1}2} = \mathcal{S}_{\bar{1}2}\frac{x_2^- - 1/x_1^-}{x_2^- - 1/x_1^+},$$

whereas the exchanged G-amplitude is

$$G_{21} = \mathcal{S}_{21}\frac{x_1^+ - x_2^+}{x_1^- - x_2^+}.$$

Using $\mathcal{S}_{21} = \mathcal{S}_{12}^{-1}$ and rewriting $(x_2^- - 1/x_1^+) = (x_2^- x_1^+ - 1)/x_1^+$ gives (98.22).

Now write

$$\mathcal{S}_{12} = R_{12}^{1/2}\sigma_{12}, \qquad \mathcal{S}_{\bar{1}2} = R_{\bar{1}2}^{1/2}\sigma_{\bar{1}2},$$

with R the rational factor displayed in the scalar split. Squaring the condition $\mathcal{R}_{\rm cr} = 1$ removes the square-root branch and gives a rational identity:

$$R_{12}R_{\bar{1}2}\Xi_{12}^2 \left[\frac{(x_2^- - 1/x_1^-)\, x_1^+ (x_1^- - x_2^+)}{(x_2^- x_1^+ - 1)(x_1^+ - x_2^+)}\right]^2 = 1.$$

After cancellation this identity is exactly the product of the factors in Ξ_{12}. The stated real chamber fixes the sign of the square root, so the unsquared channel equation is (98.23). This is the finite matrix-amplitude calculation behind the scalar multiplier; it does not construct the analytic crossed scalar branch.

Scalar crossing monodromy cocycle. Let C_1 denote a chosen crossing continuation of the first magnon along $\gamma_{\rm cr}$ from Definition 98.4. Fix a scalar branch $\sigma_{12}^{[\gamma]}$ on a sheet reached by a path γ from the physical chamber, and write

$$\sigma_{12}^{[\gamma\mathsf{C}_1]} := \sigma^{[\gamma\mathsf{C}_1]}(x_1^\pm, x_2^\pm)$$

for the same external coordinates evaluated after analytic continuation along the concatenated path. Assume that the crossed path avoids all poles and zeros in the denominators below and that Janik crossing holds on each crossed segment in the stringbook orientation,

$$\sigma_{12}^{[\gamma\mathsf{C}_1]}\,\sigma_{12}^{[\gamma]} = \Xi_{12}(x_1^\pm, x_2^\pm).$$

Then the second crossing is not an involution on the scalar branch. It acts by

$$\frac{\sigma_{12}^{[\gamma\mathsf{C}_1\mathsf{C}_1]}}{\sigma_{12}^{[\gamma]}} = \frac{\Xi_{12}(1/x_1^\pm, x_2^\pm)}{\Xi_{12}(x_1^\pm, x_2^\pm)} =: \mathcal{M}_{12}^{(2)}, \tag{98.24}$$

where explicitly

$$\mathcal{M}_{12}^{(2)} = \frac{x_2^+ - 1/x_1^+}{x_2^+ - 1/x_1^-}\,\frac{x_2^+ - x_1^-}{x_2^+ - x_1^+}\,\frac{x_2^- - x_1^+}{x_2^- - x_1^-}\,\frac{x_2^- - 1/x_1^-}{x_2^- - 1/x_1^+}. \tag{98.25}$$

In the reciprocal scalar convention $\sigma_{12}^{\rm rec} := \sigma_{12}^{-1}$, the one-crossing multiplier and the double-crossing multiplier are inverted. The first crossing gives

$$\sigma_{12}^{[\gamma\mathsf{C}_1]} = \frac{\Xi_{12}(x_1^\pm, x_2^\pm)}{\sigma_{12}^{[\gamma]}}.$$

After that continuation the projected Zhukovsky variables of the first magnon are $1/x_1^\pm$. Applying the same crossing equation once more therefore gives

$$\sigma_{12}^{[\gamma\mathsf{C}_1\mathsf{C}_1]} = \frac{\Xi_{12}(1/x_1^\pm, x_2^\pm)}{\sigma_{12}^{[\gamma\mathsf{C}_1]}} = \frac{\Xi_{12}(1/x_1^\pm, x_2^\pm)}{\Xi_{12}(x_1^\pm, x_2^\pm)}\sigma_{12}^{[\gamma]}.$$

Substituting the definition of Ξ_{12} cancels the common x_2^+/x_2^- factor and yields (98.25). If the reciprocal scalar factor is used, every displayed crossing equation is inverted because $\sigma_{12}^{\rm rec} = 1/\sigma_{12}$; the same two-line calculation then gives $(\mathcal{M}_{12}^{(2)})^{-1}$. The argument proves the rational branch displacement forced by the chosen crossing law. It does not classify all possible homotopy classes of scalar branches or prove that the BES branch is the unique minimal solution.

Proposition 98.7 (CDD ambiguity of the scalar crossing problem)**.** *Let $\mathcal{U}$ be a regular domain of scalar branches closed under particle exchange and under the chosen first-particle crossing segments C_1, with no zero or pole of the functions below on the crossed segments. Suppose σ_{12} and σ'_{12} are two non-zero meromorphic scalar branches on $\mathcal{U}$ satisfying the same Janik crossing equation*

$$\sigma_{12}^{[\gamma \mathsf{C}_1]}\,\sigma_{12}^{[\gamma]} = \sigma_{12}'^{[\gamma \mathsf{C}_1]}\,\sigma_{12}'^{[\gamma]} = \Xi_{12}$$

and the same scalar unitarity equation

$$\sigma_{21}^{[\gamma P]}\sigma_{12}^{[\gamma]} = \sigma_{21}'^{[\gamma P]}\sigma_{12}'^{[\gamma]} = 1,$$

where P exchanges the two particles. Set

$$\mathcal{F}_{12}^{[\gamma]} := \frac{\sigma_{12}'^{[\gamma]}}{\sigma_{12}^{[\gamma]}}.$$

Then $\mathcal{F}$ is a CDD factor in this convention:

$$\mathcal{F}_{12}^{[\gamma \mathsf{C}_1]}\mathcal{F}_{12}^{[\gamma]} = 1, \qquad \mathcal{F}_{21}^{[\gamma P]}\mathcal{F}_{12}^{[\gamma]} = 1, \qquad \mathcal{F}_{12}^{[\gamma \mathsf{C}_1 \mathsf{C}_1]} = \mathcal{F}_{12}^{[\gamma]}. \tag{98.26}$$

Conversely, multiplying any scalar solution by a non-zero meromorphic $\mathcal{F}$ satisfying the first two equations in (98.26) gives another scalar solution with the same crossing multiplier and scalar unitarity. If $\mathcal{F}$ has a zero of order m at a regular point of a branch, then it has a pole of order m at the corresponding crossed point and at the exchanged point. Thus a CDD-free BES branch is a minimality and analyticity choice, not a formal consequence of the finite matrix crossing multiplier.

Proof. Dividing the crossing equation for σ' by the crossing equation for σ gives

$$\mathcal{F}_{12}^{[\gamma \mathsf{C}_1]}\mathcal{F}_{12}^{[\gamma]} = 1.$$

Dividing scalar unitarity for σ' by scalar unitarity for σ gives

$$\mathcal{F}_{21}^{[\gamma P]}\mathcal{F}_{12}^{[\gamma]} = 1.$$

Applying the first identity on the already crossed branch gives

$$\mathcal{F}_{12}^{[\gamma \mathsf{C}_1 \mathsf{C}_1]}\mathcal{F}_{12}^{[\gamma \mathsf{C}_1]} = 1.$$

Together with $\mathcal{F}_{12}^{[\gamma \mathsf{C}_1]} = (\mathcal{F}_{12}^{[\gamma]})^{-1}$, this proves the double-crossing identity for the CDD factor. Conversely, if $\sigma'_{12} := \mathcal{F}_{12}\sigma_{12}$, then the first CDD equation cancels the extra factor in crossing and the second cancels the extra factor in scalar unitarity.

For the divisor statement, work in a coordinate chart where crossing and exchange are biholomorphic. The identity $\mathcal{F}^{[\gamma \mathsf{C}_1]} = 1/\mathcal{F}^{[\gamma]}$ changes an order-m zero into an order-m pole on the crossed branch, and the exchange identity does the same on the exchanged branch. In a simply connected pole-free subdomain one may write $\mathcal{F} = \exp \ell$, and the equations become

$$\ell_{12}^{[\gamma \mathsf{C}_1]} = -\ell_{12}^{[\gamma]}, \qquad \ell_{21}^{[\gamma P]} = -\ell_{12}^{[\gamma]}.$$

Hence setting $\ell = 0$, or equivalently $\mathcal{F} = 1$, is an additional minimal-branch normalization. $\square$

Crossing relates $\sigma(x_1^\pm, x_2^\pm)$ to the phase obtained by analytically continuing the first magnon through the crossed sheet $x_1^\pm \mapsto 1/x_1^\pm$. It supplies an additional analyticity condition on the scalar factor beyond factorized Yang–Baxter compatibility. The preceding propositions fix the rational convention carried by the chosen scalar split, its finite crossed-amplitude origin, and the double-crossing sheet displacement forced by that convention, while (98.26) identifies the remaining CDD freedom. They do not derive the global monodromy condition from finite algebra. Homotopic crossing paths inside a singularity-free chamber give the same rational cocycle, but paths separated by poles, zeros, or extra CDD singularities can land on different scalar branches. In the normalization where the rational matrix part is separated as above, Janik crossing has the convention-fixed form

$$\sigma(\bar{x}_1^\pm, x_2^\pm)\sigma(x_1^\pm, x_2^\pm) = \frac{x_2^+}{x_2^-}\frac{x_2^+ - x_1^+}{x_2^+ - x_1^-}\frac{x_2^- - 1/x_1^+}{x_2^- - 1/x_1^-}, \tag{98.27}$$

where $\bar{x}_1^\pm$ denotes the crossed-sheet continuation of the first magnon on the same rapidity surface. The equation is best read as a monodromy statement for the scalar factor on the Zhukovsky rapidity surface. The BES solution selects the minimal solution compatible with (98.27), unitarity $\sigma_{12}\sigma_{21} = 1$, and the prescribed weak/strong asymptotics.

Definition 98.8 (Dressing-phase data and status)**.** The scalar dressing phase used in the asymptotic equations is specified by the antisymmetric charge expansion above, the AFS leading strong-coupling term $c_{r,r+1}(g) = g + O(g^0)$, the Hernandez–Lopez semiclassical correction, Janik crossing on the Zhukovsky surface, and the BES integral representation selecting the minimal crossing-symmetric solution. Its weak series begins with

$$c_{2,3}(g) = 4\zeta(3)g^3 + O(g^5),$$

and, because the weak charges contribute $q_2q_3 - q_3q_2 = O(g^3)$, the first dressing contribution to Konishi enters at four loops. This is an exact datum of the planar-integrability framework and a highly tested claim about gauge theory, not a theorem here derived from a constructive definition of four-dimensional $\mathcal{N} = 4$ SYM.

Quoted Theorem 98.9 (Asymptotic planar magnon S-matrix)**.** *Assume planar integrability of the long $\mathcal{N} = 4$ SYM spin chain, the centrally extended residual symmetry, physical unitarity, Yang–Baxter factorization, crossing, and the minimal analyticity condition used in the BES construction. Then the asymptotic two-body magnon S-matrix is the $\mathfrak{su}(2|2)^2$-invariant matrix part multiplied by the scalar dressing factor whose phase has the charge expansion above with the BES coefficients $c_{r,s}(g)$.*

This is stated as a quoted theorem because the full claim is not a theorem of four-dimensional QFT derived from a nonperturbative definition of the gauge theory. It is an integrability theorem inside a framework whose physical identification with the planar gauge-theory dilatation operator has extensive checks and strong structural constraints.

98.5 Asymptotic Bethe Equations

Assumption 98.10 (Asymptotic Bethe–Yang regime)**.** The all-loop asymptotic Bethe ansatz is used as a long-chain Bethe–Yang quantization rule for planar single traces. More precisely, for a sequence of operators of length L, with either fixed magnon number or a controlled large-L distribution of Bethe roots, one assumes:

$$L \to \infty, \qquad \text{virtual propagation around the trace is suppressed,}$$

and the many-magnon quantization is obtained by factorizing scattering into the two-body S-matrix of the preceding section. This intended domain includes high-density configurations of roots in the $L \to \infty$ limit. The additional assumption is not thermodynamic Bethe ansatz; it is the asymptotic finite-volume quantization of the physical spin-chain problem. At finite L, or at a loop order where the long-range dilatation operator can wrap around the trace, the omitted mirror-winding effects are no longer parametrically absent.

The word "asymptotic" in this assumption refers to the spatial length of the spin chain, not to a sparse-gas approximation. The following elementary counting calculation is the precise large-L bridge used whenever an ABA solution is replaced by a macroscopic root distribution.

Finite-density ABA counting equation. Consider a diagonal component of a long-chain Bethe–Yang problem with rapidity variables u_j, $j = 1, \dots, M_L$, and logarithmic equations

$$Lp(u_j) + \sum_{\substack{k=1\\k\neq j}}^{M_L} \Theta(u_j, u_k) = 2\pi I_j, \qquad I_j \in \mathbb{Z}. \tag{98.28}$$

Assume a branch of the two-body phase Θ has been chosen on the rapidity interval under consideration; possible diagonal jumps have either been removed by this branch choice or are treated as explicit source terms. Assume also that M_L/L has a finite limit and that the empirical measures

$$\nu_L = \frac{1}{L}\sum_{j=1}^{M_L} \delta_{u_j}$$

converge weakly to $\rho(v)\mathrm{d}v$ on compact subintervals where p and Θ are C^1. Define

$$Z_L(u) = p(u) + \frac{1}{L}\sum_{j=1}^{M_L} \Theta(u, u_j), \qquad Z(u) = p(u) + \int \Theta(u, v)\rho(v)\,\mathrm{d}v.$$

Then $Z_L \to Z$ locally uniformly after differentiating under the stated regularity hypotheses, and the density of available Bethe levels is

$$\rho(u) + \rho_\mathrm{h}(u) = \frac{1}{2\pi}Z'(u) = \frac{p'(u)}{2\pi} + \int K(u, v)\rho(v)\,\mathrm{d}v, \qquad K(u, v) = \frac{1}{2\pi}\partial_u\Theta(u, v). \tag{98.29}$$

Here ρ_h is the macroscopic density of unoccupied integer levels in the chosen branch. In a completely filled interval $\rho_\mathrm{h} = 0$.

Weak convergence gives

$$\frac{1}{L}\sum_{j=1}^{M_L} \Theta(u, u_j) \longrightarrow \int \Theta(u, v)\rho(v)\,\mathrm{d}v$$

uniformly on compact subsets on which the phase is continuous and the root cloud stays away from unaccounted singularities. The same statement applied to $\partial_u\Theta$ gives the derivative formula for $Z'(u)$. Equation (98.28) says that the integer level attached to rapidity u is, to leading order, $LZ(u)/(2\pi)$. Hence the number of available integer levels in a small interval $[u, u + \mathrm{d}u]$ is

$$\frac{L}{2\pi} Z'(u)\,\mathrm{d}u + o(L\,\mathrm{d}u).$$

Dividing by $L\,\mathrm{d}u$ gives the density of occupied plus unoccupied available levels, namely $\rho + \rho_{\mathrm{h}}$. This proves (98.29).

This is a large-L continuum limit of the same physical ABA equations. It is not mirror TBA: no mirror energy, entropy functional, pseudoenergy, or thermal variational problem has been introduced. The additional dynamical assumption is exactly the one stated above, namely that the long-chain two-body magnon S-matrix continues to quantize the physical chain when $M_L/L = O(1)$, as long as mirror winding remains asymptotically suppressed.

For the rank-one equation in the stringbook $SL(2)$-vacuum orientation, the all-loop asymptotic Bethe equations can be written

$$\left(\frac{x_j^+}{x_j^-}\right)^L = \prod_{\substack{k=1\\k\neq j}}^{M} \frac{x_j^- - x_k^+}{x_j^+ - x_k^-}\,\frac{1 - 1/(x_j^+ x_k^-)}{1 - 1/(x_j^- x_k^+)}\,\sigma(x_j^\pm, x_k^\pm)^2, \tag{98.30}$$

with cyclicity

$$\prod_{j=1}^{M} \frac{x_j^+}{x_j^-} = 1.$$

The scaling dimension is

$$\Delta = L + \sum_{j=1}^{M} \left(\sqrt{1 + 16g^2 \sin^2 \frac{p_j}{2}} - 1\right),$$

up to wrapping corrections. The full $\mathfrak{psu}(2,2|4)$ asymptotic Bethe equations require nested roots for the residual symmetry algebra. Their algebraic role is the same as in Chapter 75; their physical meaning is different because these auxiliary roots organize polarizations of magnons on a gauge-theory trace.

The first nesting step can be derived without the full $\mathfrak{psu}(2,2|4)$ bookkeeping. Start with M level-I magnons $\phi_1(p_k)$. A single level-II fermionic excitation with spectral parameter y is a wave packet

$$|\Psi_{\mathrm{II}}(y)\rangle = \sum_{k=1}^{M} C_k(y;\{p_j\})\,|\phi_1(p_1)\cdots\psi(p_k)\cdots\phi_1(p_M)\rangle,$$

where the omitted flavour index of ψ is a spectator. Write $x_i^\pm = x^\pm(p_i)$ and let $\mathsf{a}_i = \mathsf{a}(p_i)$ be the nonzero representation coefficient which converts a bosonic vacuum magnon to the chosen fermionic excitation in the spin-chain basis. Normalize the two-body matrix part by the boson-boson amplitude A_{12}. In the stringbook orientation of the $\mathfrak{su}(2|2)_{\mathrm{c}}$ intertwiner, the four mixed amplitudes needed for one fermionic defect obey

$$\begin{aligned}\frac{G_{12}}{A_{12}} &= \frac{x_2^+ - x_1^+}{x_2^+ - x_1^-}, \quad \frac{K_{12}}{A_{12}} = \frac{\mathsf{a}_2}{\mathsf{a}_1}\frac{x_1^+ - x_1^-}{x_2^+ - x_1^-},\\ \frac{L_{12}}{A_{12}} &= \frac{x_2^- - x_1^-}{x_2^+ - x_1^-}, \quad \frac{H_{12}}{A_{12}} = \frac{\mathsf{a}_1}{\mathsf{a}_2}\frac{x_2^+ - x_2^-}{x_2^+ - x_1^-}.\end{aligned} \tag{98.31}$$

These are ratios of matrix amplitudes, so the scalar dressing factor and reciprocal scalar-factor conventions cancel.

Lemma 98.11 (Single level-II nesting step)**.** *Assume $x_i^\pm$, a_i, and the denominators in (98.31) are nonzero. The one-defect ansatz*

$$C_k(y;\{p_j\}) = f(y,p_k)\prod_{\ell<k} S^{\mathrm{II,I}}(y,p_\ell)$$

is compatible with every adjacent transposition of level-I magnons if

$$f(y,p) = \frac{\mathsf{a}(p)}{y - x^+(p)}, \qquad S^{\mathrm{II,I}}(y,p) = \frac{y - x^-(p)}{y - x^+(p)}. \tag{98.32}$$

The normalization of f may be multiplied by a function of y alone; this rescales the one-defect state and has no Bethe-equation content.

Proof. It is enough to check a two-site exchange, because the many-site statement is then obtained by multiplying the same local move along the chain. Let

$$f_i = f(y,p_i), \qquad S_i = S^{\mathrm{II,I}}(y,p_i).$$

Before exchanging magnons 1 and 2, the defect can sit on the first or second site with coefficients f_1 and f_2S_1. After applying the matrix S-matrix to this two-site block, equality with the ansatz in the exchanged order gives the two coefficient equations

$$\begin{aligned}f_1L_{12} + f_2S_1H_{12} &= S_2f_1A_{12},\\ f_1K_{12} + f_2S_1G_{12} &= f_2A_{12}.\end{aligned} \tag{98.33}$$

These equations are the algebraic content of moving a single auxiliary fermion through two neighbouring physical magnons.

Substitute (98.31) and (98.32). The second equation in (98.33) reduces, after clearing the common factor $A_{12}\mathsf{a}_2/((y - x_1^+)(x_2^+ - x_1^-))$, to

$$(x_1^+ - x_1^-)(y - x_2^+) + (x_2^+ - x_1^+)(y - x_1^-) = (x_2^+ - x_1^-)(y - x_1^+).$$

The first equation reduces in the same way to

$$(x_2^- - x_1^-)(y - x_2^+) + (x_2^+ - x_2^-)(y - x_1^-) = (x_2^+ - x_1^-)(y - x_2^-).$$

Both are polynomial identities in y. Therefore the local exchange is consistent, and repeated adjacent exchanges give the displayed coefficient formula for an arbitrary ordering of the level-I magnons. □

The square-root braiding factors of the string basis multiply (98.32) by convention-dependent one-body factors; they cancel from the local identities above after the corresponding basis change is applied to the mixed amplitudes.

With two level-II fermions there is a genuine contact term: when the two auxiliary fermions occupy the same level-I site, the local state can turn into the boson ϕ_2 accompanied by the length-changing marker Z^+. Let

$$v(y) = y + y^{-1}.$$

In the spin-chain basis, the local two-defect calculation fixes the $SU(2)$-invariant level-II matrix

$$(S^{\mathrm{II}})^{\delta\gamma}_{\alpha\beta}(y_1, y_2) = \delta^\delta_\alpha \delta^\gamma_\beta \, M(y_1, y_2) + \delta^\gamma_\alpha \delta^\delta_\beta \, N(y_1, y_2), \tag{98.34}$$

with

$$M + N = -1, \qquad M - N = \frac{v(y_1) - v(y_2) - \mathrm{i}/g}{v(y_1) - v(y_2) + \mathrm{i}/g}. \tag{98.35}$$

Equivalently,

$$M(y_1, y_2) = -\frac{\mathrm{i}/g}{v(y_1) - v(y_2) + \mathrm{i}/g}, \qquad N(y_1, y_2) = -\frac{v(y_1) - v(y_2)}{v(y_1) - v(y_2) + \mathrm{i}/g}.$$

The accompanying same-site coefficient is

$$\mathsf{f}^{(2)}(y_1, y_2; p) = \frac{g}{\mathsf{a}(p)^2} \left(\frac{x^-(p)}{x^+(p)} - 1 \right) \frac{y_1 y_2 - x^+(p) x^-(p)}{y_1 y_2} \frac{y_1 - y_2}{v(y_1) - v(y_2) + \mathrm{i}/g}. \tag{98.36}$$

The sign $M + N = -1$ is not a scalar scattering phase to be inserted into the level-III vacuum Bethe equation. It is the ungraded matrix eigenvalue on the state with two identical level-II fermions; after the fermionic exchange sign is accounted for, the level-III vacuum scattering is trivial. This is the local sign convention behind the sentence that the ψ_1-vacuum has $S^{\mathrm{II,II}} = 1$.

Single level-III nesting step. Assume $g \neq 0$, $v(y_1) - v(y_2) + \mathrm{i}/g \neq 0$, and the denominators below are nonzero. In the graded convention just described, a single level-III excitation with spectral parameter w has one-body coefficient

$$\gamma(w, y) = \frac{w + \mathrm{i}/(2g)}{w - v(y) + \mathrm{i}/(2g)}$$

and level-III/level-II scattering factor

$$S^{\mathrm{III,II}}(w, y) = \frac{w - v(y) - \mathrm{i}/(2g)}{w - v(y) + \mathrm{i}/(2g)}. \tag{98.37}$$

This is a two-site compatibility calculation after the level-II matrix, contact-term convention, and graded exchange sign have been fixed. Put $\eta = \mathrm{i}/(2g)$, $v_i = v(y_i)$, $D = v_1 - v_2$, and $a_i = w - v_i$. Then $a_2 = a_1 + D$, and (98.35) is equivalent to

$$M = -\frac{2\eta}{D + 2\eta}, \qquad N = -\frac{D}{D + 2\eta}.$$

For an adjacent exchange of two level-II sites, the two independent coefficients must obey

$$\begin{aligned}\gamma(w,y_1)M+\gamma(w,y_2)S^{\mathrm{III,II}}(w,y_1)N&=-\gamma(w,y_2),\\ \gamma(w,y_1)N+\gamma(w,y_2)S^{\mathrm{III,II}}(w,y_1)M&=-\gamma(w,y_1)S^{\mathrm{III,II}}(w,y_2).\end{aligned}\tag{98.38}$$

The minus signs are precisely the fermionic exchange signs which convert the ungraded $M+N=-1$ vacuum eigenvalue into trivial graded vacuum scattering.

Substituting $\gamma(w,y_i)=(w+\eta)/(a_i+\eta)$ and $S^{\mathrm{III,II}}(w,y_i)=(a_i-\eta)/(a_i+\eta)$, the first equation in (98.38) reduces to

$$-2\eta(a_2+\eta)-D(a_1-\eta)=-(D+2\eta)(a_1+\eta).$$

The second reduces to

$$-D(a_2+\eta)-2\eta(a_1-\eta)=-(D+2\eta)(a_2-\eta).$$

Both identities follow immediately from $a_2=a_1+D$. Therefore the one-level-III ansatz is compatible with every adjacent level-II exchange.

A pair of level-III roots scatters by

$$S^{\mathrm{III,III}}(w_1,w_2)=\frac{w_1-w_2+\mathrm{i}/g}{w_1-w_2-\mathrm{i}/g}.$$

Single-copy nested Bethe–Yang periodicity. In the spin-chain basis for one $\mathfrak{su}(2|2)_{\mathrm{c}}$ copy, let K^{I}, K^{II}, and K^{III} be the numbers of level-I momenta p_k, level-II roots y_a, and level-III roots w_b. Assume the asymptotic long-chain regime and the absence of singular coincidences among the displayed factors. The following equations are the closed-chain periodicity conditions for the three nested Zamolodchikov–Faddeev chains:

$$\begin{aligned}\mathrm{e}^{\mathrm{i}p_kL}&=\prod_{\substack{j=1\\ j\neq k}}^{K^{\mathrm{I}}}S^{\mathrm{I,I}}(p_j,p_k)\prod_{a=1}^{K^{\mathrm{II}}}S^{\mathrm{II,I}}(y_a,p_k),\qquad k=1,\ldots,K^{\mathrm{I}},\\ 1&=\prod_{j=1}^{K^{\mathrm{I}}}S^{\mathrm{II,I}}(y_a,p_j)^{-1}\prod_{b=1}^{K^{\mathrm{III}}}S^{\mathrm{III,II}}(w_b,y_a),\qquad a=1,\ldots,K^{\mathrm{II}},\\ 1&=\prod_{a=1}^{K^{\mathrm{II}}}S^{\mathrm{III,II}}(w_b,y_a)^{-1}\prod_{\substack{c=1\\ c\neq b}}^{K^{\mathrm{III}}}S^{\mathrm{III,III}}(w_c,w_b),\qquad b=1,\ldots,K^{\mathrm{III}}.\end{aligned}\tag{98.39}$$

For a state with N_1,N_2,N_3,N_4 excitations of types $\phi_1,\phi_2,\psi_1,\psi_2$, the nesting numbers are

$$K^{\mathrm{I}}=N_1+N_2+N_3+N_4,\qquad K^{\mathrm{II}}=2N_2+N_3+N_4,\qquad K^{\mathrm{III}}=N_2+N_4.$$

The first line records transport of the k-th level-I magnon once around the trace: it crosses every other physical magnon and every level-II defect. The second line records transport of the a-th level-II defect around the level-II chain; crossing a physical magnon in the reverse direction contributes $S^{\mathrm{II,I}}(y_a,p_j)^{-1}$, while crossing a level-III defect contributes $S^{\mathrm{III,II}}(w_b,y_a)$. The third line records transport of the b-th level-III defect around the level-III chain; each level-II defect is crossed in the reverse direction and each other level-III defect is crossed in the forward direction. Thus (98.39) is the Bethe–Yang periodicity bookkeeping for the already-defined nested scattering factors, not an additional theorem and not the thermodynamic Bethe ansatz. The relations among $K^{\mathrm{I}}, K^{\mathrm{II}}, K^{\mathrm{III}}$ and N_1, N_2, N_3, N_4 are the nesting content of the letters: ϕ_1 contributes only level I, ψ_1 contributes one level-II root, ψ_2 contributes one level-II and one level-III root, and ϕ_2 is the two-level-II contact state carrying one level-III root.

The dynamic spin-chain basis contains explicit $Z^{\pm}$ markers. In the string basis one absorbs half of this marker into each boson:

$$|\phi_A(p)\rangle_{\mathrm{str}} = |\phi_A(p)(Z^+)^{1/2}\rangle, \qquad |\psi_\alpha(p)\rangle_{\mathrm{str}} = |\psi_\alpha(p)\rangle.$$

The scattering factors which enter (98.39) are then replaced by

$$\begin{aligned} S^{\mathrm{I,I}}_{\mathrm{str}}(p_1,p_2) &= S^{\mathrm{I,I}}(p_1,p_2)\sqrt{\frac{x_1^+x_2^-}{x_1^-x_2^+}}, \\ S^{\mathrm{II,I}}_{\mathrm{str}}(y,p) &= \frac{y-x^-(p)}{y-x^+(p)}\sqrt{\frac{x^+(p)}{x^-(p)}}, \\ S^{\mathrm{III,II}}_{\mathrm{str}}(w,y) &= S^{\mathrm{III,II}}(w,y), \qquad S^{\mathrm{III,III}}_{\mathrm{str}}(w_1,w_2) = S^{\mathrm{III,III}}(w_1,w_2). \end{aligned} \tag{98.40}$$

For the full $\mathfrak{su}(2|2)_{\mathrm{c}} \otimes \mathfrak{su}(2|2)_{\mathrm{c}}$ magnon, the left and right copies have independent roots $y_a^{(\varsigma)}, w_b^{(\varsigma)}$, $\varsigma = L, R$. The physical level-I Bethe equation is obtained by multiplying the left and right auxiliary factors and by using the chosen physical vacuum scalar factor. In the $SU(2)$-vacuum nesting, the auxiliary insertion on the k-th physical magnon is

$$\prod_{\varsigma=L,R}\prod_{a=1}^{K^{\mathrm{II}}_{(\varsigma)}} \frac{y_a^{(\varsigma)}-x_k^-}{y_a^{(\varsigma)}-x_k^+}\sqrt{\frac{x_k^+}{x_k^-}}.$$

In the alternative $SL(2)$-vacuum nesting this factor is inverted:

$$\prod_{\varsigma=L,R}\prod_{a=1}^{K^{\mathrm{II}}_{(\varsigma)}} \frac{y_a^{(\varsigma)}-x_k^+}{y_a^{(\varsigma)}-x_k^-}\sqrt{\frac{x_k^-}{x_k^+}}.$$

This reciprocal relation is a nesting-frame statement, not a change of the BES dressing phase. Thus the nested Bethe equations are the diagonalization of a matrix-valued factorized scattering problem, not additional dynamical assumptions.

Controlled Approximation 98.12 (Asymptotic long-chain regime)**.** Equation (98.30) is an instance of Assumption 98.10. At weak coupling, wrapping effects begin at the loop order at which planar interactions can close around the trace. At finite coupling, the distinction is between the asymptotic Bethe–Yang problem and the finite-size problem solved by mirror TBA or the quantum spectral curve.

98.6 Weak-Coupling Reduction

The weak-coupling limit is a useful place to separate two conventions that are often conflated. The stringbook rank-one equation (98.30) uses the $SL(2)$-vacuum orientation. The compact $SU(2)$

coordinate Bethe equation of Chapter 97 uses the reciprocal Bethe–Yang phase, because there the scattering phase is the inverse of the ordered-chamber exchange coefficient. The two statements are compatible, but only after the orientation is named.

Weak limit of the rank-one ABA orientation. Let u_j, u_k stay away from the shifted branch points and from each other as $g \to 0$, and take the physical outside branch

$$x_j^{\pm} = x(u_j \pm \mathrm{i}/2), \qquad x(U) + x(U)^{-1} = \frac{U}{g}.$$

Then

$$x_j^{\pm} = \frac{u_j \pm \mathrm{i}/2}{g} - \frac{g}{u_j \pm \mathrm{i}/2} + O(g^3), \tag{98.41}$$

$$\frac{x_j^+}{x_j^-} = \frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2} + O(g^2), \tag{98.42}$$

$$\mathcal{R}_{jk} := \frac{x_j^- - x_k^+}{x_j^+ - x_k^-} \frac{1 - 1/(x_j^+ x_k^-)}{1 - 1/(x_j^- x_k^+)} = \frac{u_j - u_k - \mathrm{i}}{u_j - u_k + \mathrm{i}} + O(g^2), \tag{98.43}$$

$$H(u_j) - 1 = 2\mathrm{i}g \left(\frac{1}{x_j^+} - \frac{1}{x_j^-} \right) = \frac{2g^2}{u_j^2 + 1/4} + O(g^4). \tag{98.44}$$

The dressing phase satisfies $\sigma^2 = 1 + O(g^6)$ for fixed weak Bethe roots in the charge-expansion regime. Hence (98.30) reduces to

$$\left(\frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2} \right)^L = \prod_{\substack{k=1 \\ k \neq j}}^{M} \frac{u_j - u_k - \mathrm{i}}{u_j - u_k + \mathrm{i}}, \tag{98.45}$$

the $SL(2)$-orientation also used in (98.52) and in the weak-QSC Baxter degeneration of Chapter 100. The reciprocal factor $\mathcal{R}_{jk}^{-1}$ gives the compact $SU(2)$ one-loop phase of (97.2).

The convention check is an ordinary weak-branch expansion. The outside branch of $x + x^{-1} = U/g$ has the expansion

$$x(U) = \frac{U}{g} - \frac{g}{U} + O(g^3),$$

which gives (98.41). Dividing the two shifted expansions proves (98.42). In $\mathcal{R}_{jk}$, the first factor has numerator

$$\frac{u_j - \mathrm{i}/2}{g} - \frac{u_k + \mathrm{i}/2}{g} + O(g) = \frac{u_j - u_k - \mathrm{i}}{g} + O(g)$$

and denominator $(u_j - u_k + \mathrm{i})/g + O(g)$. The second factor equals $1 + O(g^2)$, because each $x^{\pm}$ is $O(g^{-1})$. This proves (98.43). The energy expansion is

$$2\mathrm{i}g^2 \left(\frac{1}{u_j + \mathrm{i}/2} - \frac{1}{u_j - \mathrm{i}/2} \right) + O(g^4) = \frac{2g^2}{u_j^2 + 1/4} + O(g^4).$$

Finally, the first weak BES coefficient is $c_{2,3}(g) = O(g^3)$, while the weak charges give $q_2 q_3 - q_3 q_2 = O(g^3)$; the dressing phase therefore cannot affect the one-loop equation. Substituting the limits into (98.30) gives (98.45).

Using $g^2 = \lambda/(16\pi^2)$, the leading energy term in (98.44) agrees with the one-loop normalization of Chapters 97 and 100.

98.7 Bound States and the Large-Spin Cusp Limit

A pole of the $SL(2)$ scalar factor at $x_1^- = x_2^+$ gives a Bethe string. More precisely, the displayed $SL(2)$-vacuum rational factor has a simple kinematic pole when the outgoing endpoint of the first constituent equals the incoming endpoint of the second constituent. A Q-magnon string with real center u is described by

$$u_j = u + \frac{\mathrm{i}}{2}(Q - 2j + 1), \qquad j = 1, \dots, Q,$$

with constituent variables $x_j^\pm = x(u_j \pm \mathrm{i}/2)$. Put

$$x_Q^+(u) = x\left(u + \frac{\mathrm{i}Q}{2}\right) = x_1^+, \qquad x_Q^-(u) = x\left(u - \frac{\mathrm{i}Q}{2}\right) = x_Q^-. \tag{98.46}$$

These are the only Zhukovsky endpoints that survive after fusion.

Construction 98.13 (Fusion of a Bethe string)**.** Assume a Q-string is formed by the adjacent-pole conditions

$$x_j^- = x_{j+1}^+, \qquad j = 1, \dots, Q-1,$$

and that the path to this pole does not cross a dressing-phase branch singularity. Then the total momentum and physical energy of the string are

$$\mathrm{e}^{\mathrm{i}p} = \frac{x_1^+}{x_Q^-}, \qquad H_Q(u) = Q + 2\mathrm{i}g\left(\frac{1}{x_Q^+(u)} - \frac{1}{x_Q^-(u)}\right), \tag{98.47}$$

and the shortening identity gives

$$H_Q(p) = \sqrt{Q^2 + 16g^2 \sin^2 \frac{p}{2}}. \tag{98.48}$$

If y is a level-II auxiliary root in the $SL(2)$-vacuum string basis, then its product of constituent scattering factors telescopes:

$$\prod_{j=1}^{Q} \frac{y - x_j^+}{y - x_j^-}\sqrt{\frac{x_j^-}{x_j^+}} = \frac{y - x_Q^+(u)}{y - x_Q^-(u)}\sqrt{\frac{x_Q^-(u)}{x_Q^+(u)}}. \tag{98.49}$$

For two bound states of charges Q_1, Q_2 and centers u, v, the fused $SL(2)$-vacuum scalar factor is

$$S_{SL(2)}^{Q_1,Q_2}(u,v) = \prod_{m=1}^{Q_1}\prod_{n=1}^{Q_2} S_{SL(2)}\left(u + \frac{\mathrm{i}}{2}(Q_1 - 2m + 1), v + \frac{\mathrm{i}}{2}(Q_2 - 2n + 1)\right), \tag{98.50}$$

followed by projection to the short Q_1- and Q_2-particle representations. The product formula is the scalar part of fusion; the matrix projection and the analytic continuation of the dressing factor are separate representation-theoretic and scalar-branch inputs.

Derivation. The condition $u_{j+1} = u_j - \mathrm{i}$ gives

$$x_j^- = x(u_j - \mathrm{i}/2) = x(u_{j+1} + \mathrm{i}/2) = x_{j+1}^+,$$

so all internal endpoints cancel in products. Hence

$$\prod_{j=1}^{Q} \frac{x_j^+}{x_j^-} = \frac{x_1^+}{x_Q^-} = \frac{x_Q^+(u)}{x_Q^-(u)}.$$

The same cancellation in the sum of the physical-sheet one-particle energies

$$H_j = 1 + 2\mathrm{i}g\left(\frac{1}{x_j^+} - \frac{1}{x_j^-}\right)$$

gives the energy formula in (98.47). The bound-state shortening relation is the centrally extended one with $H^2 - Q^2 = 4PK$, and the total central charges telescope to

$$P = g(1 - \mathrm{e}^{\mathrm{i}p}), \qquad K = g(1 - \mathrm{e}^{-\mathrm{i}p}).$$

Therefore

$$H_Q^2 = Q^2 + 4g^2(1 - \mathrm{e}^{\mathrm{i}p})(1 - \mathrm{e}^{-\mathrm{i}p}) = Q^2 + 16g^2 \sin^2 \frac{p}{2}.$$

Taking the positive physical branch gives (98.48).

The auxiliary factor (98.49) is the same endpoint cancellation:

$$\prod_{j=1}^{Q} \frac{y - x_j^+}{y - x_j^-} = \frac{y - x_1^+}{y - x_Q^-}, \qquad \prod_{j=1}^{Q} \frac{x_j^-}{x_j^+} = \frac{x_Q^-}{x_1^+}.$$

Finally, scalar fusion means that a distant probe scatters consecutively through every constituent of each string before the internal tensor product is projected to the short representation. This gives (98.50). Internal pole residues and the matrix projection determine the normalization of the bound-state intertwiner; they are not additional endpoint identities and are kept as fusion inputs here. The resulting asymptotic Bethe–Yang equations for K^{I} bound-state particles of charges Q_k are obtained from (98.39) by replacing the fundamental scalar factor with (98.50) and the fundamental auxiliary factors with (98.49). Thus the first equation in the $SL(2)$-vacuum frame reads

$$\mathrm{e}^{\mathrm{i}p_k L} = \prod_{\substack{j=1 \\ j\neq k}}^{K^{\mathrm{I}}} S_{SL(2)}^{Q_j,Q_k}(u_j, u_k) \prod_{\varsigma=L,R} \prod_{\ell=1}^{K_{(\varsigma)}^{\mathrm{II}}} \frac{y_\ell^{(\varsigma)} - x_{Q_k}^+(u_k)}{y_\ell^{(\varsigma)} - x_{Q_k}^-(u_k)} \sqrt{\frac{x_{Q_k}^-(u_k)}{x_{Q_k}^+(u_k)}}. \tag{98.51}$$

The level-II and level-III equations keep the same rational form as before, with each fundamental $x^\pm$ replaced by the appropriate bound-state endpoints $x_Q^\pm(u)$. This is the bound-state input used by the mirror TBA node $Q \mid \bullet$ in Chapter 99.

In the $SL(2)$ sector, twist-L spin-S operators are

$$\mathrm{tr}(D_+^{s_1} Z \cdots D_+^{s_L} Z), \qquad \sum_{j=1}^{L} s_j = S.$$

The one-loop Bethe equations in this noncompact sector are

$$\left(\frac{u_k+\mathrm{i}/2}{u_k-\mathrm{i}/2}\right)^L=\prod_{\substack{j=1\\ j\neq k}}^{S}\frac{u_k-u_j-\mathrm{i}}{u_k-u_j+\mathrm{i}},\qquad k=1,\dots,S, \tag{98.52}$$

with cyclicity selecting zero total spin-chain momentum. For the minimal large-spin trajectory the roots are real, symmetric under $u\mapsto -u$, and have mode numbers ±1 in the logarithmic equation. Write

$$u_k=S\bar{u}_k,\qquad \int_{-a}^{a}\bar{\rho}_0(\bar{u})\,\mathrm{d}\bar{u}=1,$$

where $\bar{\rho}_0$ is the continuum density of the scaled roots. In the limit $S\gg L$, the derivative of the continuum logarithmic equation is encoded by the resolvent

$$G_0(z)=\int_{-a}^{a}\frac{\bar{\rho}_0(v)}{z-v}\,\mathrm{d}v,\qquad G_0(z)=\frac{1}{z}+O(z^{-2}).$$

The discontinuity convention is

$$G_0(y+\mathrm{i}0)-G_0(y-\mathrm{i}0)=-2\pi\mathrm{i}\,\bar{\rho}_0(y).$$

After differentiating the continuum Bethe equation, the boundary values obey

$$G_0'(y+\mathrm{i}0)+G_0'(y-\mathrm{i}0)=4\pi\delta(y)\qquad(-a<y<a).$$

The endpoint is fixed by the same Riemann-Hilbert data. For a candidate symmetric cut $[-a,a]$, put

$$R_a(z)=\sqrt{z^2-a^2},\qquad R_a(z)\sim z\quad(z\to\infty),\qquad H_a(z)=-\frac{1}{zR_a(z)}.$$

On the cut,

$$R_{a,+}(y)=\mathrm{i}\sqrt{a^2-y^2},\qquad R_{a,-}(y)=-\mathrm{i}\sqrt{a^2-y^2}.$$

Thus $H_{a,+}+H_{a,-}=0$ away from the origin. At the origin the pole of $1/z$ must be read distributionally:

$$H_{a,+}(y)+H_{a,-}(y)=\frac{\mathrm{i}}{\sqrt{a^2-y^2}}\left(\frac{1}{y+\mathrm{i}0}-\frac{1}{y-\mathrm{i}0}\right)=\frac{2\pi}{a}\delta(y).$$

The source $4\pi\delta(y)$ therefore fixes $a=1/2$. The primitive with $G_0(\infty)=0$ is obtained by writing $Y=\sqrt{4z^2-1}=2R_{1/2}(z)$, because

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[-\mathrm{i}\log\frac{Y+\mathrm{i}}{Y-\mathrm{i}}\right]=-\frac{1}{zR_{1/2}(z)}.$$

The one-cut solution with the displayed normalization is consequently

$$G_0(z)=-\mathrm{i}\log\frac{\sqrt{4z^2-1}+\mathrm{i}}{\sqrt{4z^2-1}-\mathrm{i}},\qquad a=\frac{1}{2}, \tag{98.53}$$

where the square root is chosen so that $\sqrt{4z^2-1}\sim 2z$ at infinity. Its density is

$$\bar{\rho}_0(y)=\frac{1}{\pi}\log\frac{1+\sqrt{1-4y^2}}{1-\sqrt{1-4y^2}},\qquad |y|<\frac{1}{2}. \tag{98.54}$$

The normalization is not an additional assumption. With $y = (2\cosh t)^{-1}$, the evenness of the density gives

$$2\int_0^{1/2} \bar{\rho}_0(y)\,\mathrm{d}y = \frac{2}{\pi}\int_0^\infty t\ \mathrm{sech}\,t\ \tanh t\,\mathrm{d}t = \frac{2}{\pi}\int_0^\infty \mathrm{sech}\,t\,\mathrm{d}t = 1.$$

The one-loop anomalous dimension is

$$\gamma^{(1)} = 2g^2\sum_{k=1}^{S}\frac{1}{u_k^2+1/4}.$$

The singular region near $u=0$ is resolved by evaluating the resolvent at $z = \mathrm{i}/(2S)$:

$$\gamma^{(1)} = -4g^2\ \mathrm{Im}\,G_0\!\left(\frac{\mathrm{i}}{2S}\right) + O(S^0g^2).$$

Using the continuation

$$G_0\!\left(\frac{\mathrm{i}}{2S}\right) = -\mathrm{i}\log\frac{\sqrt{1+S^{-2}}+1}{\sqrt{1+S^{-2}}-1},$$

one obtains

$$\Delta - S - L = \gamma^{(1)} = 8g^2\log S + O(S^0g^2, g^4). \tag{98.55}$$

98.8 Finite-Density Bethe Roots and Spectral Curves

The phrase "finite gap" will be used here in a precise Bethe-ansatz sense. It is the statement that, in a large-charge limit, Bethe roots condense on finitely many arcs and that the Cauchy transform of the limiting measure extends to a meromorphic object on a finite-sheeted Riemann surface. This surface is the spectral curve of the state.

Definition 98.14 (Finite-density Bethe-root curve)**.** Let N be a scaling parameter and let $\{u_{j,N}\}_{j=1}^{M_N} \subset \mathbb{C}$ be Bethe roots with $M_N \to \infty$. Suppose that, after a scaling $u_{j,N} = N\bar{u}_{j,N}$, the empirical measures

$$\mu_N = \frac{1}{M_N}\sum_{j=1}^{M_N}\delta_{\bar{u}_{j,N}}$$

converge weakly to an absolutely continuous measure

$$\mathrm{d}\mu(u) = \rho(u)\,\mathrm{d}s$$

supported on a finite union of oriented analytic arcs $\mathcal{C} = \bigcup_{a=1}^r \mathcal{C}_a$. The Cauchy transform is

$$G(z) = \int_{\mathcal{C}}\frac{\rho(u)\,\mathrm{d}s(u)}{z-u}, \qquad z \in \mathbb{C}\setminus\mathcal{C}.$$

A finite-density Bethe-root curve is the data

$$\big(\Sigma, \pi : \Sigma \to \mathbb{P}^1_z, \lambda, \{A_a, B_a\}\big)$$

where Σ is a compact Riemann surface obtained by gluing sheets across the arcs $\mathcal{C}_a$, λ is a meromorphic differential whose physical-sheet primitive contains $G(z)\mathrm{d}z$, and the cycles A_a encircle the cuts. The filling fractions are the periods

$$\mathcal{S}_a = \frac{1}{2\pi\mathrm{i}}\oint_{A_a}\lambda.$$

The definition separates three pieces of information that are often blurred. The measure ρ is the continuum limit of roots. The Riemann surface Σ records analytic continuation across root cuts. The meromorphic differential λ records the logarithmic Bethe equation: mode numbers are period data, not decorations on a drawing of cuts.

Reconstruction ledger. A finite-density root curve is not yet a classical string solution and is not yet the quantum spectral curve. The reconstruction problem used in this chapter has the following rank-one spine. For each regulator N start with a logarithmic Bethe record

$$L_N p_N(u_{j,N}) = 2\pi n_{j,N} + \sum_{\substack{k=1\\k\neq j}}^{M_N} \theta_N(u_{j,N}, u_{k,N}) + r_{j,N}, \tag{98.56}$$

where p_N is the one-particle momentum, θ_N is the antisymmetric two-body phase, $n_{j,N} \in \mathbb{Z}$ fixes the logarithm branch, and $r_{j,N}$ denotes regulator terms whose limiting status must be declared. A rank-one finite-gap reconstruction record consists of:

1. a partition of the roots into macroscopic packets converging to the oriented cuts $\mathcal{C}_a$, with $n_{j,N} = n_a$ on each packet away from endpoint layers;
2. a limiting driving term $V'(z)$ and a limiting Cauchy transform $G(z)$ such that the scaled version of (98.56) gives a cutwise principal-value equation with source $V'(u) - 2\pi n_a$;
3. the quasimomentum $p = V'/2 - G$, whose cut gluing is $p_+ + p_- = 2\pi n_a$;
4. A-cycle fillings $\mathcal{S}_a = (2\pi\mathrm{i})^{-1} \oint_{A_a} p\,\mathrm{d}z$ and large-z moments $\mathcal{M}_k$, which are the finite-density charge coordinates before sector-specific constraints are imposed;
5. the global spin-chain closure condition obtained by summing the finite equations, namely the cyclicity/level-matching constraint on $\sum_a n_a \mathcal{S}_a$;
6. any extra reality involution, closed-period, pole-residue, Virasoro, or target-space reconstruction constraints required by the sigma-model sector under study.

The calculations below establish the first five items in the rank-one Cauchy-transform setting. They do not establish the final $\mathrm{AdS}_5 \times S^5$ finite-gap construction, where one must replace the single p by the full collection of quasimomenta, impose the inversion and reality symmetries of the coset curve, fix the pole residues at the distinguished spectral points, and reconstruct closed embedding coordinates. Nor do they prove the TBA-to-QSC equivalence. The QSC later in Chapter 100 is the quantum finite-difference/Riemann-Hilbert replacement for this classical curve record, with additional gluing and asymptotic data.

Jump data from the Cauchy transform. Let G be the Cauchy transform in Definition 98.14, and assume that ρ is Hölder continuous on the interior of a cut $\mathcal{C}_a$. For $u \in \mathcal{C}_a^\circ$, denote by $G_+(u)$ and $G_-(u)$ the boundary values from the two sides of the oriented arc. Then

$$G_+(u) - G_-(u) = -2\pi\mathrm{i}\,\rho(u), \qquad \frac{G_+(u) + G_-(u)}{2} = \mathrm{p.v.} \int_{\mathcal{C}} \frac{\rho(v)\,\mathrm{d}s(v)}{u - v}. \tag{98.57}$$

Consequently, whenever the continuum logarithmic Bethe equation has the form

$$2\ \mathrm{p.v.} \int_{\mathcal{C}} \frac{\rho(v)\,\mathrm{d}s(v)}{u - v} = V'(u) - 2\pi n_a, \qquad u \in \mathcal{C}_a^\circ, \tag{98.58}$$

with integer mode number n_a on $\mathcal{C}_a$, the quasimomentum

$$p(z) = \frac{1}{2}V'(z) - G(z) \tag{98.59}$$

satisfies the gluing condition

$$p_+(u) + p_-(u) = 2\pi n_a. \tag{98.60}$$

The two boundary-value identities are the Plemelj formulas for the Cauchy transform on a smooth oriented arc. Substituting the second identity into (98.58) gives

$$G_+(u) + G_-(u) = V'(u) - 2\pi n_a.$$

Using $p_\pm(u) = V'(u)/2 - G_\pm(u)$ gives

$$p_+(u) + p_-(u) = V'(u) - \big(G_+(u) + G_-(u)\big) = 2\pi n_a.$$

Multi-cut periods and sheet reflections. The same boundary values also recover the filling periods. In the rank-one finite-density normalization where $\lambda = p(z)\mathrm{d}z$ on the physical sheet, and where V' is holomorphic in a collar of the cut, one has

$$p_+(u) - p_-(u) = 2\pi\mathrm{i}\,\rho(u), \qquad u \in \mathcal{C}_a^\circ. \tag{98.61}$$

Thus a small A_a-cycle surrounding $\mathcal{C}_a$, with the first traverse along the $+$-side of the oriented cut and the return traverse on the $-$-side, gives

$$\mathcal{S}_a = \frac{1}{2\pi\mathrm{i}}\oint_{A_a} p(z)\mathrm{d}z = \int_{\mathcal{C}_a} \rho(u)\,\mathrm{d}s(u). \tag{98.62}$$

Indeed, the contribution of a shrinking collar is

$$\oint_{A_a} p(z)\mathrm{d}z = \int_{\mathcal{C}_a} \big(p_+(u) - p_-(u)\big)\mathrm{d}s(u),$$

and (98.61) gives (98.62). This is the first non-one-cut datum: each cut carries an independent filling fraction before global period and level-matching constraints are imposed.

Large-z moments as charge coordinates. The same Cauchy transform also records the global root moments. If the cuts lie in a compact set and $|z|$ is larger than the support, then

$$\frac{1}{z-u} = \sum_{k=0}^{K} \frac{u^k}{z^{k+1}} + \frac{u^{K+1}}{z^{K+1}(z-u)}.$$

Hence

$$G(z) = \sum_{k=0}^{K} \frac{\mathcal{M}_k}{z^{k+1}} + O(|z|^{-K-2}), \qquad \mathcal{M}_k := \int_{\mathcal{C}} u^k \rho(u)\,\mathrm{d}s(u). \tag{98.63}$$

The zeroth moment is the total filling,

$$\mathcal{M}_0 = \sum_a \mathcal{S}_a,$$

while higher moments are the large-z charge coordinates carried by the root distribution. In a particular sector one combines these coefficients with the large-z expansion of the driving term $V'(z)$ in (98.59) and with the chosen normalization of the spin-chain or sigma-model charges. The moment expansion is the finite-density Cauchy-coordinate statement; a complete classical string solution still imposes the additional Virasoro, reality, and closed-period constraints.

The finite regulator has the same bookkeeping before taking a limit. For scaled roots $\bar{u}_j$ with weights w_j,

$$G_N(z) = \sum_j \frac{w_j}{z - \bar{u}_j} = \sum_{k=0}^{K} \frac{\sum_j w_j \bar{u}_j^k}{z^{k+1}} + \sum_j \frac{w_j \bar{u}_j^{K+1}}{z^{K+1}(z - \bar{u}_j)}.$$

Thus the continuum charge coordinates are limits of finite weighted power sums, not independent decorations of the spectral curve.

Cyclicity as a global period constraint. The local cut-reflection algebra does not by itself impose the closed-chain condition. In a finite rank-one spin chain, write a logarithmic Bethe equation in the form

$$L\, p(u_j) = 2\pi n_j + \sum_{\substack{k=1 \\ k\neq j}}^{M} \theta(u_j, u_k), \qquad \theta(u,v) = -\theta(v,u), \tag{98.64}$$

where L is the chain length and $n_j \in \mathbb{Z}$ is the chosen logarithm branch. Summing (98.64) over j cancels the two-body phases pairwise:

$$L \sum_{j=1}^{M} p(u_j) = 2\pi \sum_{j=1}^{M} n_j.$$

Trace cyclicity requires $\exp(\mathrm{i} \sum_j p(u_j)) = 1$, hence

$$\sum_{j=1}^{M} n_j \in L\mathbb{Z}.$$

If the roots on $\mathcal{C}_a$ have a common mode number n_a and multiplicity M_a, this becomes the finite congruence

$$\sum_a n_a M_a \in L\mathbb{Z}. \tag{98.65}$$

In a simultaneous finite-density limit with $M_a/N \to \mathcal{S}_a$ and $L/N \to \ell$, (98.65) is the discrete origin of the level-matching condition

$$\sum_a n_a \mathcal{S}_a \in \ell\, \mathbb{Z}$$

whenever the integer multiple is stable along the regulator sequence. This is only the spin-chain global momentum constraint; closed classical strings also require the remaining period, reality, and Virasoro conditions of the sigma-model reconstruction.

The mode numbers are the affine gluing data. Equation (98.60) says that analytic continuation through $\mathcal{C}_a$ acts on the local quasimomentum coordinate by the reflection

$$R_a: \ p \longmapsto 2\pi n_a - p. \tag{98.66}$$

Hence $R_a^2 = 1$, while crossing first $\mathcal{C}_a$ and then $\mathcal{C}_b$ gives

$$R_b R_a(p) = p + 2\pi(n_b - n_a). \tag{98.67}$$

For integer mode numbers the exponentiated quasimomentum is therefore single-valued around such a two-cut continuation,

$$\exp\bigl(\mathrm{i} R_b R_a(p)\bigr) = \exp(\mathrm{i}p).$$

The unexponentiated p, however, has remembered the relative mode number. A full classical finite-gap solution also has Virasoro constraints, reconstruction of the embedding coordinates, and global period conditions; the calculation here is only the local multi-cut period and monodromy bookkeeping that those global constraints use.

For the large-spin one-loop cusp density above, the spectral curve is already visible. Put

$$y^2 = 4z^2 - 1, \qquad G_0(z) = -\mathrm{i}\log\frac{y+\mathrm{i}}{y-\mathrm{i}}, \qquad y \sim 2z \quad (z \to \infty).$$

The two sheets are exchanged by $y \mapsto -y$. Under this exchange,

$$\exp\bigl(\mathrm{i}G_0\bigr) = \frac{y+\mathrm{i}}{y-\mathrm{i}} \quad \longmapsto \quad \frac{y-\mathrm{i}}{y+\mathrm{i}} = \exp\bigl(-\mathrm{i}G_0\bigr),$$

so the cut $[-1/2, 1/2]$ is the branch locus of the two-sheeted curve. The density (98.54) is precisely the discontinuity of this primitive across the cut, and the endpoint square-root behavior is the local branch structure of Σ.

Classical S^2 Pohlmeyer cell. The finite-gap language also has a classical sigma-model origin. The smallest honest reduction, avoiding all worldsheet quantization, is the $\mathbb{R} \times S^2$ subsector. Let

$$n(\sigma^+, \sigma^-) \in S^2 \subset \mathbb{R}^3, \qquad n \cdot n = 1,$$

and impose the constant-stress conformal-gauge constraints

$$\partial_+ n \cdot \partial_+ n = \partial_- n \cdot \partial_- n = \mu^2.$$

The sphere equation of motion is

$$\partial_+\partial_- n + (\partial_+ n \cdot \partial_- n)\, n = 0. \tag{98.68}$$

On a patch where $\partial_+ n$ and $\partial_- n$ are not parallel, put

$$e_\pm = \mu^{-1}\partial_\pm n, \qquad e_\pm \cdot e_\pm = 1, \qquad e_+ \cdot e_- = \cos\alpha.$$

Then (98.68) gives

$$\partial_+ e_- = -\mu\cos\alpha\, n, \qquad \partial_- e_+ = -\mu\cos\alpha\, n.$$

The remaining frame derivatives are fixed by differentiating $e_\pm \cdot e_\pm = 1$, $e_\pm \cdot n = 0$, and $e_+ \cdot e_- = \cos\alpha$:

$$\partial_+ e_+ = -\mu n - \frac{\partial_+\alpha}{\sin\alpha}(e_- - \cos\alpha\, e_+),$$
$$\partial_- e_- = -\mu n - \frac{\partial_-\alpha}{\sin\alpha}(e_+ - \cos\alpha\, e_-).$$

Now compare the e_+-coefficient in $\partial_-(\partial_+ e_-) = \partial_+(\partial_- e_-)$. The left hand side has no e_+-component, while the right hand side has

$$-\mu^2 - \frac{\partial_+\partial_-\alpha}{\sin\alpha}.$$

Therefore the reduced field obeys

$$\partial_+\partial_-\alpha + \mu^2 \sin\alpha = 0. \tag{98.69}$$

This is the classical Pohlmeyer reduction in the S^2 subsector. It is a local equivalence of moving-frame data on nondegenerate patches; reconstructing n from α, gluing across $\sin\alpha = 0$, imposing closed-string periodicity, and quantizing the resulting finite-gap data are separate problems. Thus (98.69) explains why integrable PDE spectral curves appear in the strong-coupling classical story, but it is not a derivation of the planar gauge-theory quantum spectral curve.

The all-loop ABA refines this large-spin calculation rather than replacing it. The macroscopic roots at $|u| \sim S$ still produce the logarithmic enhancement, but the all-loop phase induces an $O(\log S)$ fluctuation of the density near $u = O(1)$. This near-origin fluctuation is the object described by the BES equation.

Assumption 98.15 (Large-spin BES scaling regime)**.** Use the $SL(2)$ all-loop asymptotic Bethe equation in the regime

$$S \to \infty, \qquad L \text{ fixed}, \qquad \log S \gg L,$$

with symmetric mode numbers $n_k = \pm 1$, zero total momentum, the physical Zhukovsky branch for the Bethe roots before crossing, and the BES dressing phase of Definition 98.8. Assume that the large-spin root density has a one-cut bulk part at $u = O(S)$ and a near-origin correction at $u = O(1)$ whose coefficient is proportional to the scaling function. The all-loop ABA is being used here in precisely the finite-density sense of Assumption 98.10; wrapping effects are outside this asymptotic calculation.

The derivative of the logarithmic all-loop $SL(2)$ Bethe equation has three types of terms. The rational XXX kernel gives the one-loop singular source already solved above. The rational all-loop Zhukovsky factors and the dressing phase give additional kernels. Finally, the $u = O(S)$ bulk roots feed a source into the $u = O(1)$ equation. After the bulk density has been used to solve the outer problem, the inner equation is closed for a single Fourier density. We use the standard BES normalization

$$\widehat{\sigma}(t) = \mathrm{e}^{-t/2} \int_{-\infty}^{\infty} \mathrm{d}u\, \mathrm{e}^{-\mathrm{i}tu} \sigma(u), \qquad t > 0,$$

where $\sigma(u)$ denotes the inner density correction in the present large-spin normalization.

Definition 98.16 (BES kernel in the stringbook convention)**.** Let J_n be the Bessel function of the first kind. Set

$$K_0(t, t') = \frac{J_1(t)J_0(t') - J_0(t)J_1(t')}{t - t'}, \tag{98.70}$$

with the diagonal value defined by analytic continuation. The dressing part is

$$K_{\mathrm{dr}}(t,t';g) = \frac{4}{tt'} \sum_{k,\ell\geq 1} (-1)^{k+\ell} c_{2k,2\ell+1}(g) J_{2\ell}(t) J_{2k-1}(t'), \tag{98.71}$$

where $c_{r,s}(g) = -c_{s,r}(g)$ for $r > s$. The BES kernel used below is

$$K_{\mathrm{BES}}(t,t';g) = K_0(t,t') + K_{\mathrm{dr}}(t,t';g), \qquad K_{\mathrm{BES}}(t,0;g) = \frac{J_1(t)}{t} + \frac{2}{t}\sum_{\ell\geq 1}(-1)^{\ell-1} c_{2,2\ell+1}(g) J_{2\ell}(t).$$

The Fourier transform which turns powers of $x^{\pm}(u)^{-1}$ into Bessel functions is the convention-sensitive step. We spell it out because replacing the signed ratio $J_m(2gt)/t$ by $J_m(2g|t|)/|t|$ loses a parity sign in the lower half-plane transform.

Lemma 98.17 (Zhukovsky Fourier transform)**.** *Let $g > 0$, let $m \in \mathbb{Z}_{>0}$, and let $x(z)$ be the physical outside branch of*

$$x(z) + x(z)^{-1} = \frac{z}{g}, \qquad x(z) \sim \frac{z}{g} \quad (z \to \infty),$$

with short cut $[-2g, 2g]$. Put

$$x^{\pm}(u) = x(u \pm \mathrm{i}/2), \qquad u \in \mathbb{R}.$$

For $t \neq 0$, with $\Theta(t) = 1$ for $t > 0$ and 0 for $t < 0$, the oscillatory Fourier transform is

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{2\pi}\, \mathrm{e}^{-\mathrm{i}tu} \frac{1}{x^{\pm}(u)^m} = (-\mathrm{i})^m m\, \mathrm{e}^{-|t|/2}\, \frac{\pm\Theta(\pm t) J_m(2gt)}{t}. \tag{98.72}$$

Here the upper sign is used for x^+, the lower sign for x^-, and J_m is the Bessel function of the first kind.

Proof. For the x^+ transform set $z = u + \mathrm{i}/2$. Then

$$\mathrm{e}^{-\mathrm{i}tu} = \mathrm{e}^{-t/2}\mathrm{e}^{-\mathrm{i}tz},$$

and the integration line is $\mathbb{R} + \mathrm{i}/2$. If $t < 0$, this line is closed in the upper half-plane and no cut is crossed, so the integral vanishes. If $t > 0$, close in the lower half-plane. The deformation crosses the Zhukovsky cut and gives the negatively oriented loop around it. Using $z = g(x + x^{-1})$ on this loop, with x running around $|x| = 1$, gives

$$I_m^+(t) = -\mathrm{e}^{-t/2} \oint_{|x|=1} \frac{\mathrm{d}x}{2\pi}\, g(1 - x^{-2}) x^{-m} \mathrm{e}^{-\mathrm{i}gt(x+x^{-1})}.$$

The generating function

$$\mathrm{e}^{-\mathrm{i}gt(x+x^{-1})} = \sum_{n\in\mathbb{Z}} (-\mathrm{i})^n J_n(2gt) x^n$$

therefore yields

$$I_m^+(t) = -\mathrm{i}g\mathrm{e}^{-t/2}\left[(-\mathrm{i})^{m-1} J_{m-1}(2gt) - (-\mathrm{i})^{m+1} J_{m+1}(2gt)\right].$$

Since $(-\mathrm{i})^{m+1} = -(-\mathrm{i})^{m-1}$, and

$$J_{m-1}(z) + J_{m+1}(z) = \frac{2m}{z} J_m(z),$$

this becomes

$$I_m^+(t) = (-\mathrm{i})^m m\, \mathrm{e}^{-t/2} \frac{J_m(2gt)}{t}, \qquad t > 0.$$

Together with the vanishing for $t < 0$, this proves the upper-sign formula.

For real u, the lower shifted branch satisfies

$$x^-(u) = \overline{x^+(u)}.$$

Hence

$$I_m^-(t) = \overline{I_m^+(-t)}.$$

Applying the upper-sign result to $-t > 0$ and using $J_m(-z) = (-1)^m J_m(z)$ gives the lower-sign formula in (98.72). The factor $J_m(2gt)/t$, with signed t, is therefore part of the convention rather than a cosmetic choice. □

The signs in Lemma 98.17 encode the same x^+-above, x^--below sheet convention as the mirror continuation in Chapter 99. Reversing them would invert the dressing-source contribution.

BES equation from the large-spin ABA. Under Assumption 98.15, and after applying the Fourier transform (98.72), the inner large-spin density obeys

$$\widehat{\sigma}(t) = \frac{t}{\mathrm{e}^t - 1} \left[K_{\mathrm{BES}}(2gt, 0; g) - 4g^2 \int_0^\infty \mathrm{d}t'\, K_{\mathrm{BES}}(2gt, 2gt'; g) \widehat{\sigma}(t') \right], \qquad t > 0. \tag{98.73}$$

The corresponding large-spin scaling function in the normalization of (98.55) is

$$\Delta - S - L = f(g) \log S + O(S^0), \qquad f(g) = 8g^2 - 64g^4 \int_0^\infty \mathrm{d}t\, \widehat{\sigma}(t) \frac{J_1(2gt)}{2gt}. \tag{98.74}$$

We spell out the logic because it is easy to hide a convention change in this calculation. Taking the logarithm of the $SL(2)$ all-loop Bethe equation and differentiating with respect to the target rapidity u gives the rational kernel

$$\frac{2}{(u-v)^2 + 1} - 2\pi \delta(u - v),$$

where the delta term records the branch of the arccotangent scattering phase. The all-loop rational Zhukovsky factor contributes

$$-2\mathrm{i}\, \partial_u \log \frac{1 - 1/(x^+(u) x^-(v))}{1 - 1/(x^-(u) x^+(v))},$$

and the scalar factor contributes

$$2\partial_u \theta(x^\pm(v), x^\pm(u)).$$

The $O(S)$ bulk roots are described by the one-cut density (98.54); expanding their contribution for fixed u produces the driving term

$$\partial_u \left[\frac{1}{2g} \left(\frac{1}{x^+(u)} + \frac{1}{x^-(u)} \right) + \frac{\mathrm{i}}{2g} \sum_{n \geq 1} \frac{c_{2,2n+1}(g)}{n} \left(\frac{1}{x^+(u)^{2n}} - \frac{1}{x^-(u)^{2n}} \right) \right]. \tag{98.75}$$

The explicit L-dependent terms in the logarithmic Bethe equation are smaller than this driving term in the scaling regime $\log S \gg L$, so they affect the $O(S^0)$ part rather than $f(g)$.

Applying $(2\pi)^{-1}\mathrm{e}^{-t/2} \int \mathrm{d}u\, \mathrm{e}^{-\mathrm{i}tu}$ to the inner equation gives (98.73). The Bessel transform (98.72) turns the all-loop rational factors into K_0, and turns the charge expansion of the dressing phase into K_{dr}. The factor $t/(\mathrm{e}^t - 1)$ is the inverse of the rational large-spin kernel in this Fourier normalization. The energy formula follows by expanding

$$\sum_k \left(\frac{1}{x_k^+} - \frac{1}{x_k^-} \right)$$

against the same bulk-plus-inner density and applying (98.72) with $m = 1$. The outer density gives the one-loop $8g^2 \log S$ term; the inner density gives the integral correction in (98.74).

Proposition 98.18 (Weak expansion of the BES scaling function)**.** *The BES equation* (98.73) *implies*

$$f(g) = 8g^2 - \frac{8\pi^2}{3} g^4 + \frac{88\pi^4}{45} g^6 - \left(\frac{584\pi^6}{315} + 64\zeta(3)^2 \right) g^8 + O(g^{10}). \tag{98.76}$$

The first dressing-phase contribution enters at order g^8 and supplies the $\zeta(3)^2$ term in this normalization.

Proof. Write

$$\widehat{\sigma}(t) = \sigma_0(t) + g^2\sigma_1(t) + g^4\sigma_2(t) + O(g^6).$$

Since

$$\frac{J_1(2gt)}{2gt} = \frac{1}{2} - \frac{g^2t^2}{4} + \frac{g^4t^4}{24} + O(g^6), \qquad K_0(2gt, 0) = \frac{1}{2} - \frac{g^2t^2}{4} + \frac{g^4t^4}{24} + O(g^6),$$

and

$$K_0(2gt, 2gt') = \frac{1}{2} - \frac{g^2}{4}(t^2 - tt' + t'^2) + O(g^4),$$

the first two density coefficients are

$$\sigma_0(t) = \frac{t}{2(\mathrm{e}^t - 1)}$$

and

$$\sigma_1(t) = \frac{t}{\mathrm{e}^t - 1} \left[-\frac{t^2}{4} - 4 \int_0^\infty \mathrm{d}t'\, \frac{1}{2} \sigma_0(t') \right] = -\frac{t}{\mathrm{e}^t - 1} \left(\frac{t^2}{4} + \frac{\pi^2}{6} \right),$$

because

$$\int_0^\infty \frac{t\,\mathrm{d}t}{\mathrm{e}^t - 1} = \frac{\pi^2}{6}.$$

For the g^4 density coefficient the dressing source is now required. Using

$$c_{2,3}(g) = 4\zeta(3)g^3 + O(g^5), \qquad J_2(2gt) = \frac{g^2t^2}{2} + O(g^4),$$

and the prefactor $2/(2gt)$ in $K_{\mathrm{BES}}(2gt, 0; g)$, one obtains

$$K_{\mathrm{dr}}(2gt, 0; g) = 2\zeta(3)g^4 t + O(g^6).$$

The g^4 part of the BES equation is therefore

$$\sigma_2(t) = \frac{t}{\mathrm{e}^t - 1}\left[\frac{t^4}{24} + 2\zeta(3)t - 4\int_0^\infty \mathrm{d}t' \left(\frac{1}{2}\sigma_1(t') - \frac{t^2 - tt' + t'^2}{4}\sigma_0(t')\right)\right].$$

The required moments are

$$\int_0^\infty \sigma_0(t')\,\mathrm{d}t' = \frac{\pi^2}{12}, \qquad \int_0^\infty t'\sigma_0(t')\,\mathrm{d}t' = \zeta(3), \qquad \int_0^\infty t'^2\sigma_0(t')\,\mathrm{d}t' = \frac{\pi^4}{30},$$
$$\int_0^\infty \sigma_1(t')\,\mathrm{d}t' = -\frac{2\pi^4}{45}.$$

Substitution gives the explicit coefficient

$$\sigma_2(t) = \frac{t}{\mathrm{e}^t - 1}\left(\frac{t^4}{24} + \frac{\pi^2 t^2}{12} + \zeta(3)t + \frac{11\pi^4}{90}\right).$$

Put

$$A(g) = \int_0^\infty \mathrm{d}t\,\widehat{\sigma}(t)\frac{J_1(2gt)}{2gt} = A_0 + g^2A_1 + g^4A_2 + O(g^6).$$

The two coefficients are

$$A_0 = \int_0^\infty \frac{t\,\mathrm{d}t}{4(\mathrm{e}^t - 1)} = \frac{\pi^2}{24},$$

and

$$A_1 = \int_0^\infty \frac{t\,\mathrm{d}t}{\mathrm{e}^t - 1}\left(-\frac{t^2}{4} - \frac{\pi^2}{12}\right) = -\frac{\pi^4}{60} - \frac{\pi^4}{72} = -\frac{11\pi^4}{360},$$

using

$$\int_0^\infty \frac{t^3\,\mathrm{d}t}{\mathrm{e}^t - 1} = \frac{\pi^4}{15}.$$

The next coefficient is

$$A_2 = \int_0^\infty \mathrm{d}t\,\left(\frac{1}{2}\sigma_2(t) - \frac{t^2}{4}\sigma_1(t) + \frac{t^4}{24}\sigma_0(t)\right).$$

Using

$$\int_0^\infty \frac{t^2\,\mathrm{d}t}{\mathrm{e}^t - 1} = 2\zeta(3), \qquad \int_0^\infty \frac{t^5\,\mathrm{d}t}{\mathrm{e}^t - 1} = \frac{8\pi^6}{63},$$

one finds

$$A_2 = \frac{73\pi^6}{2520} + \zeta(3)^2.$$

Substitution into (98.74) gives

$$f(g) = 8g^2 - 64g^4A_0 - 64g^6A_1 - 64g^8A_2 + O(g^{10}),$$

which is (98.76). $\square$

Quoted Theorem 98.19 (Strong solution of the BES equation)**.** *Let* $\Gamma_{\mathrm{BES}}(g)$ *denote the conventional BES cusp function whose weak expansion begins* $\Gamma_{\mathrm{BES}}(g) = 4g^2 + O(g^4)$. *Define the Dirichlet beta values*

$$\beta_{\mathrm{D}}(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s}, \qquad \mathrm{K} = \beta_{\mathrm{D}}(2),$$

where K *is Catalan's constant, and set*

$$c_1 = \frac{3\log 2}{4\pi}, \qquad c_2 = \frac{\mathrm{K}}{16\pi^2}, \qquad c_3 = \frac{27\zeta(3)}{2^{11}\pi^3}, \qquad c_4 = \frac{21\,\beta_{\mathrm{D}}(4)}{2^{10}\pi^4}.$$

The strong-coupling solution of the BES Riemann–Hilbert problem gives, as a Poincare asymptotic expansion on the positive real g*-axis,*

$$\Gamma_{\mathrm{BES}}(g) = 2g_{\mathrm{s}} \left[1 - \frac{c_2}{g_{\mathrm{s}}^2} - \frac{c_3}{g_{\mathrm{s}}^3} - \frac{c_4 + 2c_2^2}{g_{\mathrm{s}}^4} + O(g_{\mathrm{s}}^{-5})\right], \qquad g_{\mathrm{s}} = g - c_1. \tag{98.77}$$

The quoted theorem is a theorem about the analytic solution of the BES integral equation. Its use in the spectral problem is conditional on the large-spin BES bridge of Assumption 98.15.

Remark 98.20 (Strong BES expansion in the monograph normalization). With $f(g)$ defined by

$$\Delta - S - L = f(g)\log S + O(S^0)$$

in (98.74), the strong-coupling expansion corresponding to Quoted Theorem 98.19 is

$$f(g) = 4g_{\mathrm{s}} - \frac{4c_2}{g_{\mathrm{s}}} - \frac{4c_3}{g_{\mathrm{s}}^2} - \frac{4(c_4 + 2c_2^2)}{g_{\mathrm{s}}^3} + O(g_{\mathrm{s}}^{-4}), \qquad g_{\mathrm{s}} = g - \frac{3\log 2}{4\pi}. \tag{98.78}$$

In particular, expanded in the unshifted coupling,

$$f(g) = 4g - \frac{3\log 2}{\pi} - \frac{\mathrm{K}}{4\pi^2 g} + O(g^{-2}). \tag{98.79}$$

The weak expansion in Proposition 98.18 begins

$$f(g) = 8g^2 + O(g^4),$$

whereas $\Gamma_{\mathrm{BES}}(g) = 4g^2 + O(g^4)$. Therefore the present normalization is

$$f(g) = 2\Gamma_{\mathrm{BES}}(g).$$

Multiplying (98.77) by 2 gives (98.78). The first two unshifted terms are obtained from $4g_{\mathrm{s}} = 4g - 4c_1$, and

$$4c_1 = \frac{3\log 2}{\pi}, \qquad 4c_2 = \frac{\mathrm{K}}{4\pi^2}.$$

All further displayed terms are $O(g^{-2})$ or smaller in the unshifted coupling, giving (98.79).

This is the convention in which the large-spin anomalous dimension has coefficient $f(g)$. Some Wilson-line literature denotes by Γ_{cusp} half of this quantity; Chapter 100 states its Wilson-line convention at the point where cusped Wilson loops are introduced. The equality between the large-spin scaling function and the Wilson-line cusp observable is a structural matching within planar $\mathcal{N} = 4$ theory; a nonperturbative construction-level proof remains outside the present monograph.

The QCD counterpart of the same large-spin/Wilson-line mechanism was defined from a regulated cusped Wilson-line operator in Section 49.24: there the lightlike coefficient Γ^q_{cusp} is a perturbative anomalous dimension entering DIS endpoint evolution as $P_{\mathrm{ns},N} = -\Gamma^q_{\mathrm{cusp}} \log N + O(N^0)$. The comparison fixes the shared operator meaning of the word "cusp" while keeping the theories' logical status distinct: QCD uses factorization and perturbative renormalization of Wilson-line composites, whereas the present planar $\mathcal{N} = 4$ construction uses the integrable all-loop spectral problem.

Chapter 99

Mirror TBA and the Y-System

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The asymptotic Bethe ansatz determines long-chain states before wrapping effects. Finite single traces require a finite-size formalism. The planar $\mathcal{N} = 4$ construction uses a mirror model obtained by exchanging worldsheet space and Euclidean time in the integrable spin-chain scattering problem. This chapter states the logical structure without pretending that the mirror thermodynamic Bethe ansatz is a theorem of four-dimensional constructive QFT.

Warning 99.1 (Status of mirror magnons). A mirror magnon is not a new physical particle of four-dimensional $\mathcal{N} = 4$ SYM. It is the excitation obtained after analytically continuing the two-dimensional worldsheet/spin-chain kinematics so that Euclidean time and space are exchanged. In the relativistic string worldsheet before light-cone gauge fixing this exchange is the natural mirror-channel operation. After gauge fixing, the magnon dispersion is nonrelativistic and the same operation becomes a path on the Zhukovsky surface together with a specified dressing-phase branch. This is why the mirror relation is not explained by the nonrelativistic spin chain alone.

The reason to use mirror magnons in the gauge-theory spectral problem is therefore indirect. Once the long-trace magnon S-matrix is accepted as the asymptotic integrable description, finite trace length is computed by the standard mirror-channel finite-size construction: virtual mirror particles wrap around the trace, and their thermodynamics gives the TBA. This is the worldsheet and integrability explanation. There is no known purely four-dimensional derivation in which the mirror magnon appears directly as an ordinary local QFT asymptotic state.

99.1 Mirror Transformation

For a physical magnon with energy $E(p)$ and momentum p, the mirror variables are defined by the stringbook analytic continuation

$$E = \mathrm{i}\widetilde{p}, \qquad p = \mathrm{i}\widetilde{E}, \qquad \text{equivalently} \qquad \widetilde{E} = -\mathrm{i}p, \quad \widetilde{p} = -\mathrm{i}E.$$

Since

$$E(p) = \sqrt{1 + 16g^2 \sin^2 \frac{p}{2}},$$

the mirror dispersion is not the same function of $\widetilde{p}$ as the physical dispersion. This is the first major difference from relativistic two-dimensional QFT, where the mirror transformation is a rapidity shift.

Mirror dispersion from double Wick rotation. For a Q-magnon bound state with physical dispersion

$$E_Q(p) = \sqrt{Q^2 + 16g^2 \sin^2 \frac{p}{2}},$$

the mirror dispersion obtained from $p = \mathrm{i}\widetilde{E}$, $E_Q = \mathrm{i}\widetilde{p}$ is

$$\widetilde{E}_Q(\widetilde{p}) = 2 \,\operatorname{arsinh}\left(\frac{\sqrt{Q^2 + \widetilde{p}^2}}{4g} \right). \tag{99.1}$$

The positive branch is chosen for real mirror momentum. Set $p = \mathrm{i}\widetilde{E}$ and $E_Q = \mathrm{i}\widetilde{p}$ in the physical dispersion relation. Since $\sin(\mathrm{i}\widetilde{E}/2) = \mathrm{i}\sinh(\widetilde{E}/2)$, one obtains

$$-\widetilde{p}^2 = Q^2 - 16g^2 \sinh^2 \frac{\widetilde{E}}{2}.$$

Thus

$$\sinh^2 \frac{\widetilde{E}}{2} = \frac{Q^2 + \widetilde{p}^2}{16g^2}.$$

The mirror energy is positive on the real mirror line, which selects (99.1).

Equivalently one continues the Zhukovsky variables through a specified sequence of cuts. On the physical sheet a real magnon has $(x^+)^* = x^-$ and $|x^\pm| > 1$. A mirror particle of real mirror rapidity has one Zhukovsky variable inside and one outside the unit circle. The choice of path is part of the definition of the mirror S-matrix: crossing a different cut changes the scalar dressing factor by the crossing equation. For this reason the mirror TBA below always includes the contour and sheet data together with the functional formulae.

It is useful to keep the following sheet bookkeeping explicit. Let $x^{[\pm Q]}(u) = x(u \pm \mathrm{i}Q/2)$, with $x(u)$ the physical Zhukovsky branch of Chapter 98. Physical bound-state kinematics keeps both $x^{[+Q]}$ and $x^{[-Q]}$ on the outside sheet for real momentum. Mirror kinematics continues one of the two variables through the short cut, so the pair has opposite inside/outside status. The transfer matrix and the dressing factor must be continued along the same path. A formula copied from the physical sheet but evaluated with the wrong inside/outside assignment gives the wrong wrapping sign.

Bound states of Q physical magnons are described by pairs

$$x_Q^+, \qquad x_Q^-,$$

obeying the bound-state shortening relation

$$x_Q^+ + \frac{1}{x_Q^+} - x_Q^- - \frac{1}{x_Q^-} = \frac{\mathrm{i}Q}{g}.$$

In the mirror region the branch is chosen so that the mirror energy $\widetilde{E}_Q(u)$ is positive for real mirror rapidity u.

The preceding paragraph is often the place where sign mistakes enter. The following elementary parametrization fixes the sheet convention used in the rest of the chapter and in the stringbook mirror Bethe–Yang equations.

Mirror Zhukovsky sheet parametrization. Fix $g > 0$, $Q \in \mathbb{Z}_{>0}$, and real mirror momentum $\widetilde{p}$. Put

$$R_Q = \sqrt{Q^2 + \widetilde{p}^2}, \qquad \widetilde{E}_Q = 2 \operatorname{arsinh} \frac{R_Q}{4g}, \qquad r_Q = \mathrm{e}^{-\widetilde{E}_Q/2}, \qquad \xi_Q = \frac{\widetilde{p} - \mathrm{i}Q}{R_Q}.$$

Then

$$x_Q^+ = r_Q \xi_Q, \qquad x_Q^- = r_Q^{-1} \xi_Q \tag{99.2}$$

satisfy

$$x_Q^+ + \frac{1}{x_Q^+} - x_Q^- - \frac{1}{x_Q^-} = \frac{\mathrm{i}Q}{g}, \tag{99.3}$$

$$\log \frac{x_Q^-}{x_Q^+} = \widetilde{E}_Q, \tag{99.4}$$

$$\mathrm{i}\widetilde{p} = -Q - 2\mathrm{i}g(x_Q^+ - x_Q^-). \tag{99.5}$$

Moreover $|x_Q^+| < 1 < |x_Q^-|$. At weak coupling with $Q, \widetilde{p}$ fixed,

$$\mathrm{e}^{-L\widetilde{E}_Q} = \left(\frac{4g^2}{Q^2 + \widetilde{p}^2} \right)^L \bigl(1 + O(g^2)\bigr), \tag{99.6}$$

so one mirror wrapping around a length-L trace first appears at order g^{2L} before transfer-matrix and dressing factors are included.

The number ξ_Q has unit modulus and $0 < r_Q < 1$. Hence $|x_Q^+| < 1 < |x_Q^-|$, and $\log(x_Q^-/x_Q^+) = -2\log r_Q = \widetilde{E}_Q$ on the real mirror branch. Since

$$\sinh \frac{\widetilde{E}_Q}{2} = \frac{R_Q}{4g}, \qquad \xi_Q^{-1} - \xi_Q = \frac{2\mathrm{i}Q}{R_Q},$$

we get

$$x_Q^+ + \frac{1}{x_Q^+} - x_Q^- - \frac{1}{x_Q^-} = 2 \sinh \frac{\widetilde{E}_Q}{2} \, (\xi_Q^{-1} - \xi_Q) = \frac{\mathrm{i}Q}{g}.$$

Similarly,

$$-Q - 2\mathrm{i}g(x_Q^+ - x_Q^-) = -Q + 4\mathrm{i}g \sinh \frac{\widetilde{E}_Q}{2}\, \xi_Q = -Q + \mathrm{i}R_Q\xi_Q = \mathrm{i}\widetilde{p}.$$

Finally, $\mathrm{e}^{-\widetilde{E}_Q} = (\sqrt{1 + R_Q^2/(16g^2)} - R_Q/(4g))^2$, whose small-$g$ expansion gives (99.6).

The same mirror sheet parametrization fixes the large-rapidity fermionic-node ratio used later to identify the $Y_{1,1}Y_{2,2}$ energy exponent in the QSC bridge; the corresponding local expansion is written in the QSC chapter. This is a good example of the convention burden in this subject: the sign comes from the inside/outside assignment just checked here, but the ratio itself belongs to the later TBA-to-QSC asymptotic bridge rather than to the definition of the mirror kernel.

Definition 99.2 (Mirror-kernel datum)**.** A mirror kernel in this chapter means the triple

$$(S_{BA}^{\mathrm{mirror}}(v,u), \mathcal{C}_B, \mathcal{C}_A),$$

where S_{BA}^{mirror} is the analytically continued two-body scattering factor and $\mathcal{C}_B, \mathcal{C}_A$ are the source and target rapidity contours on the chosen sheets. The derivative kernel is then

$$K_{BA}(v,u) = \frac{1}{2\pi\mathrm{i}}\partial_v \log S_{BA}^{\mathrm{mirror}}(v,u).$$

This convention packages the branch choice with the formula and prevents the common mistake of treating mirror TBA kernels as functions on a single unbranched u-plane.

99.2 Mirror Bethe-String Species

The mirror TBA is nested because a physical magnon carries two $\mathfrak{su}(2|2)$ wings of polarization data. The canonical node families are as follows.

symbol	range	role
$\bullet_Q$	$Q = 1, 2, \ldots$	mirror bound states carrying energy
$y_\pm$	two sheets	fermionic auxiliary roots
$(v\vert M)$	$M = 1, 2, \ldots$	auxiliary strings in each wing
$(w\vert M)$	$M = 1, 2, \ldots$	isotopic auxiliary strings in each wing

Each wing has its own copy of the v- and w-string nodes. The $\bullet_Q$ nodes are the only nodes that enter the energy formula with the measure $\mathrm{d}\widetilde{p}_Q/\mathrm{d}u$. The auxiliary nodes enter through kernels and chemical potentials and enforce the nested Bethe constraints.

The string hypothesis used here is a thermodynamic statement on the mirror Bethe roots: in a large mirror volume, complex roots organize into regular arrays whose centers become real variables u. In a relativistic model this statement is already subtle; in planar $\mathcal{N} = 4$ it is further entangled with the Zhukovsky cuts. Consequently, the node table is part of the integrability construction, not a theorem about the original four-dimensional path integral.

The node table above abbreviates arrays selected by the Bethe-Yang equations. We now derive the auxiliary arrays from the large-volume pole-cancellation mechanism in one $\mathfrak{su}(2|2)_{\mathrm{c}}$ wing. Put

$$\eta = \frac{\mathrm{i}}{2g}, \qquad v(y) = y + \frac{1}{y}.$$

Let $x_j^\pm$, $j = 1, \dots, K^{\mathrm{I}}$, be mirror level-I rapidity data on the real mirror line, so that

$$x_j^+ = r_j \xi_j, \qquad x_j^- = r_j^{-1} \xi_j, \qquad 0 < r_j < 1, \quad |\xi_j| = 1.$$

The one-wing auxiliary Bethe-Yang subsystem in the stringbook mirror orientation has the form

$$\begin{aligned} -1 &= F(y_a) \prod_{b=1}^{K^{\mathrm{III}}} \frac{w_b - v(y_a) - \eta}{w_b - v(y_a) + \eta}, \qquad F(y) = \prod_{j=1}^{K^{\mathrm{I}}} \frac{y - x_j^+}{y - x_j^-} \sqrt{\frac{x_j^-}{x_j^+}}, \\ 1 &= \prod_{a=1}^{K^{\mathrm{II}}} \frac{w_b - v(y_a) + \eta}{w_b - v(y_a) - \eta} \prod_{\substack{c=1 \\ c \neq b}}^{K^{\mathrm{III}}} \frac{w_c - w_b + 2\eta}{w_c - w_b - 2\eta}. \end{aligned} \tag{99.7}$$

The sign in the first equation is the mirror fermion boundary-condition sign. It is not the graded sign in the physical nested ABA of Chapter 98.

Proposition 99.3 (Mirror auxiliary support and Bethe strings)**.** *Assume the mirror Bethe-Yang equations* (99.7)*, the mirror sheet data above, and a thermodynamic limit in which the logarithmic averages of* $\log|F(y)|$ *have the signs determined by the one-particle factor below. Then elementary* y*-roots have finite density only on* $|y| = 1$*, elementary* w*-roots have finite density only on the real* w*-axis, and the non-elementary auxiliary complexes have the following pole-cancellation arrays:*

$$\begin{aligned} v(y_{j,\mathrm{in}}) &= v_0 + (M + 2 - 2j)\eta, \quad |y_{j,\mathrm{in}}| < 1, \\ v(y_{j,\mathrm{out}}) &= v_0 - (M + 2 - 2j)\eta, \quad |y_{j,\mathrm{out}}| > 1, \\ w_j &= v_0 + (M + 1 - 2j)\eta, \qquad j = 1, \dots, M, \end{aligned}$$

with real thermodynamic center v_0*. These are the* $(v|M)$*, or* $M|yw$*, strings. The pure* $(w|M)$ *strings are*

$$w_j = w_0 + (M + 1 - 2j)\eta, \qquad j = 1, \dots, M,$$

with real center w_0*. The assertion is a statement about the mirror string hypothesis and the displayed large-volume equations; it is not a theorem of four-dimensional* $\mathcal{N} = 4$ *SYM.*

Proof. First isolate the source of exponential growth and decay. For a single mirror level-I root write $x^+ = r\xi$, $x^- = r^{-1}\xi$, with $0 < r < 1$ and $|\xi| = 1$. The magnitude of the one-particle factor

$$s(y; x^\pm) = \frac{y - x^+}{y - x^-} \sqrt{\frac{x^-}{x^+}}$$

obeys

$$|s(y; x^\pm)|^2 = \frac{|y - r\xi|^2}{|ry - \xi|^2}, \qquad |ry - \xi|^2 - |y - r\xi|^2 = (1 - r^2)(1 - |y|^2).$$

Thus $|s| < 1$ for $|y| < 1$, $|s| = 1$ for $|y| = 1$, and $|s| > 1$ for $|y| > 1$. Multiplying over K^{I} level-I roots gives the stated support of elementary y-roots in the thermodynamic limit, provided the level-I density has a nonzero logarithmic average.

Now take a y-root with $|y_1| < 1$. The factor $F(y_1)$ then tends to zero exponentially, so the first equation in (99.7) can remain finite only if one of the w-factors has a pole:

$$w_1 - v(y_1) + \eta = 0, \qquad \text{that is} \qquad w_1 = v(y_1) - \eta.$$

The corresponding w_1-equation has a zero from the same y_1. The local cancellation recursion remains finite by acquiring a pole from a second y-root,

$$w_1 - v(y_2) - \eta = 0, \qquad v(y_2) = v(y_1) - 2\eta.$$

If this second root lies outside the unit circle, the divergence of $F(y_2)$ is canceled by the zero of the same w_1-factor and the string closes. If it lies inside, $F(y_2)$ again decays exponentially and the second auxiliary equation must acquire another compensating pole, now from a neighboring w-root. Alternating these zero-pole cancellations creates the sequence

$$v(y_{j+1}) = v(y_j) - 2\eta, \qquad w_j = v(y_j) - \eta,$$

until the terminal y-root lies on the opposite support side. This iteration fixes the displayed imaginary spacings. The real center is the thermodynamic string-hypothesis input: the mirror densities are assigned to real centers, while the roots in the complex plane are the regular arrays above those centers.

For the w-only strings, set the background y-roots on $|y| = 1$, so their elementary factors have unit magnitude and do not supply exponential growth. A complex w-root in the second line of (99.7) then produces an uncanceled large-volume factor unless a neighboring w-root sits at distance $2\eta = \mathrm{i}/g$. Repeating the cancellation in both directions gives the finite array $w_j = w_0 + (M + 1 - 2j)\eta$ with real center w_0. □

99.3 Thermodynamic Bethe Ansatz

The mirror TBA is a thermodynamic limit of Bethe-Yang quantization, not an independent postulate. We first derive the one-species formula because it is the place where the signs, statistics, and kernel orientation are least forgiving. The derivation is also the convention bridge to the stringbook: the continuum density equation naturally differentiates the scattering phase in the target rapidity, while the multi-species formula below is written with the source-derivative kernels of Definition 99.2.

Proposition 99.4 (One-species mirror TBA from Bethe-Yang quantization)**.** *Consider a diagonal integrable mirror model with one particle species, real mirror rapidity u, mirror momentum $\widetilde{p}(u)$, mirror energy $\widetilde{\epsilon}(u)$, and two-body scattering factor $S(v, u)$. Assume:*

1. *a logarithm of $S(v, u)$ has been chosen with no self-scattering jump at $v = u$;*
2. *the thermodynamic mode number $n(u)$ is smooth and monotone on the rapidity interval under consideration;*
3. *the finite-volume Bethe levels have Fermi level statistics, so a level is either occupied or empty;*
4. *no additional Bethe-string species or bound-state poles cross the integration contour in the ground-state calculation.*

Let ρ_{p}, ρ_{L}, and $\rho_{\mathrm{h}} = \rho_{\mathrm{L}} - \rho_{\mathrm{p}}$ denote the particle, level, and hole densities with respect to u. Define the density kernel

$$K^{\mathrm{dens}}(v, u) = \frac{1}{2\pi\mathrm{i}} \partial_u \log S(v, u). \tag{99.8}$$

Then the thermodynamic limit of the Bethe-Yang equation gives

$$\rho_{\mathrm{L}}(u) = \frac{\widetilde{L}}{2\pi}\frac{\mathrm{d}\widetilde{p}(u)}{\mathrm{d}u} - \int \mathrm{d}v\, \rho_{\mathrm{p}}(v) K^{\mathrm{dens}}(v,u). \tag{99.9}$$

At inverse mirror temperature $\widetilde{\beta}$ and mirror chemical potential $\widetilde{\mu}$, extremizing

$$\mathfrak{s} - \widetilde{\beta}\int \mathrm{d}u\, \rho_{\mathrm{p}}(u)\widetilde{\epsilon}(u) - \widetilde{\mu}\int \mathrm{d}u\, \rho_{\mathrm{p}}(u)$$

subject to (99.9), where

$$\mathfrak{s} = \int \mathrm{d}u\, \left[\rho_{\mathrm{L}}\log\rho_{\mathrm{L}} - \rho_{\mathrm{p}}\log\rho_{\mathrm{p}} - \rho_{\mathrm{h}}\log\rho_{\mathrm{h}}\right],$$

gives the pseudoenergy equation

$$\zeta(u) = \widetilde{\beta}\,\widetilde{\epsilon}(u) + \widetilde{\mu} + \int \mathrm{d}v\, K^{\mathrm{dens}}(u,v)\log(1+\mathrm{e}^{-\zeta(v)}), \qquad \frac{\rho_{\mathrm{p}}(u)}{\rho_{\mathrm{L}}(u)} = \frac{1}{1+\mathrm{e}^{\zeta(u)}}. \tag{99.10}$$

At the stationary point,

$$\log Z = \widetilde{L}\int \frac{\mathrm{d}u}{2\pi}\frac{\mathrm{d}\widetilde{p}(u)}{\mathrm{d}u}\log(1+\mathrm{e}^{-\zeta(u)}). \tag{99.11}$$

After exchanging mirror space and Euclidean time, with $\widetilde{\beta} = L$ and no chemical potential, this gives

$$E_0(L) = -\int \frac{\mathrm{d}u}{2\pi}\frac{\mathrm{d}\widetilde{p}(u)}{\mathrm{d}u}\log(1+\mathrm{e}^{-\zeta(u)}). \tag{99.12}$$

Proof. For occupied rapidities u_k, choose the logarithmic Bethe-Yang equation in the stringbook orientation

$$\widetilde{p}(u_k)\widetilde{L} = 2\pi n_k - \mathrm{i}\sum_{j\neq k}\log S(u_j,u_k).$$

Replacing the sum by $\int \mathrm{d}v\, \rho_{\mathrm{p}}(v)$ and differentiating with respect to the target rapidity u gives

$$\widetilde{L}\,\widetilde{p}'(u) = 2\pi\rho_{\mathrm{L}}(u) + 2\pi\int \mathrm{d}v\, \rho_{\mathrm{p}}(v)K^{\mathrm{dens}}(v,u),$$

which is (99.9).

Now vary the entropy while preserving the density equation. Since $\rho_{\mathrm{h}} = \rho_{\mathrm{L}} - \rho_{\mathrm{p}}$,

$$\delta\mathfrak{s} = \int \mathrm{d}u\, \left[\log\frac{\rho_{\mathrm{L}}}{\rho_{\mathrm{h}}}\delta\rho_{\mathrm{L}} - \log\frac{\rho_{\mathrm{p}}}{\rho_{\mathrm{h}}}\delta\rho_{\mathrm{p}}\right].$$

The constraint implies

$$\delta\rho_{\mathrm{L}}(u) = -\int \mathrm{d}v\, K^{\mathrm{dens}}(v,u)\delta\rho_{\mathrm{p}}(v).$$

Collecting the coefficient of an arbitrary $\delta\rho_{\mathrm{p}}(u)$ gives

$$\log\frac{\rho_{\mathrm{h}}(u)}{\rho_{\mathrm{p}}(u)} = \widetilde{\beta}\,\widetilde{\epsilon}(u) + \widetilde{\mu} + \int \mathrm{d}v\, K^{\mathrm{dens}}(u,v)\log\frac{\rho_{\mathrm{L}}(v)}{\rho_{\mathrm{h}}(v)}.$$

With $\zeta = \log(\rho_{\rm h}/\rho_{\rm p})$, one has $\rho_{\rm p}/\rho_{\rm L} = 1/(1+\mathrm{e}^{\zeta})$ and $\rho_{\rm L}/\rho_{\rm h} = 1+\mathrm{e}^{-\zeta}$, proving (99.10).

It remains to evaluate the extremized functional. Substitute the stationary equation into the expression for $\mathfrak{s} - \widetilde{\beta}\widetilde{E} - \widetilde{\mu}\widetilde{N}$. The terms proportional to $\rho_{\rm p}$ cancel against the convolution term in the entropy variation, and the remaining density equation leaves

$$\log Z = \int \mathrm{d}u\, \rho_{\rm L}(u) \log(1+\mathrm{e}^{-\zeta(u)}) + \int \mathrm{d}u\mathrm{d}v\, \rho_{\rm p}(v) K^{\rm dens}(v,u) \log(1+\mathrm{e}^{-\zeta(u)}),$$

which reduces by (99.9) to (99.11). The ground-state formula (99.12) is the usual mirror space-time reinterpretation of $\log Z = -\widetilde{L}E_0(L)$. □

The kernel sign in (99.10) is the stringbook convention. If the scattering factor obeys the diagonal unitarity relation $S(v,u)S(u,v) = 1$, then

$$K^{\rm dens}(u,v) = \frac{1}{2\pi\mathrm{i}}\partial_v \log S(u,v) = -\frac{1}{2\pi\mathrm{i}}\partial_v \log S(v,u).$$

Thus the source-derivative kernel $K(v,u)$ of Definition 99.2 enters the same TBA equation with a minus sign. This is the convention used in (99.16); the displayed chemical potential μ_A there is the negative of $\widetilde{\mu}_A$ in the one-species statistical-mechanical derivation.

Excited states add one more honest hypothesis: the contour deformation must be specified. In the one-species model, inserting moving defects at rapidities u_j changes the pseudoenergy to

$$\zeta(u) = \widetilde{\beta}\,\widetilde{\epsilon}(u) + \widetilde{\mu} - \sum_j \log S(u,u_j) + \int \mathrm{d}v\, K^{\rm dens}(u,v) \log(1+\mathrm{e}^{-\zeta(v)}),$$

and the defect rapidities are fixed by

$$\zeta(u_j) = -2\pi\mathrm{i}\left(n_j + \frac{1}{2}\right), \qquad n_j \in \mathbb{Z}.$$

Equivalently $1+\mathrm{e}^{-\zeta}$ has a zero at each defect. In the planar $\mathcal{N}=4$ problem this condition becomes the exact Bethe-root regularity condition after analytic continuation to the physical sheet.

Lemma 99.5 (Excited-state contour deformation in one species)**.** *Let $S(u,v)$ be a meromorphic two-particle mirror scattering factor with no zero or pole on the undeformed real v-contour, and let*

$$K^{\rm dens}(u,v) = \frac{1}{2\pi\mathrm{i}}\partial_v \log S(u,v).$$

Let $Y(v) = \mathrm{e}^{-\zeta(v)}$, and suppose that

$$F(v) = 1 + Y(v)$$

has simple zeros $u_1, \ldots, u_K$ crossed by the excited-state contour deformation and no other singularities are crossed. Assume also that the boundary terms at infinity vanish in the integrations by parts below. With the contour orientation for which the crossed zeros are enclosed counterclockwise, the deformed TBA source is

$$\int_{\mathcal{C}_{\rm exc}} \mathrm{d}v\, K^{\rm dens}(u,v) \log F(v) = \int_{\mathbb{R}} \mathrm{d}v\, K^{\rm dens}(u,v) \log F(v) - \sum_{j=1}^{K} \log S(u,u_j). \qquad (99.13)$$

Moreover, the mirror energy integral obeys

$$-\int_{\mathcal{C}_{\mathrm{exc}}}\frac{\mathrm{d}v}{2\pi}\frac{\mathrm{d}\widetilde{p}(v)}{\mathrm{d}v}\log F(v)=\sum_{j=1}^{K}\mathrm{i}\,\widetilde{p}(u_j)-\int_{\mathbb{R}}\frac{\mathrm{d}v}{2\pi}\frac{\mathrm{d}\widetilde{p}(v)}{\mathrm{d}v}\log F(v). \tag{99.14}$$

The root condition is

$$F(u_j)=0\quad\Longleftrightarrow\quad\zeta(u_j)=-2\pi\mathrm{i}\left(n_j+\frac{1}{2}\right),\qquad n_j\in\mathbb{Z}. \tag{99.15}$$

Proof. For the source term, write the contour difference as the sum of small positively oriented loops around the crossed zeros and integrate by parts:

$$\oint\frac{\mathrm{d}v}{2\pi\mathrm{i}}\,\partial_v\log S(u,v)\log F(v)=-\oint\frac{\mathrm{d}v}{2\pi\mathrm{i}}\,\log S(u,v)\partial_v\log F(v).$$

Since each u_j is a simple zero of F, $\partial_v\log F(v)$ has residue 1 at $v=u_j$. The residue theorem therefore gives the contribution $-\sum_j\log S(u,u_j)$, proving (99.13). The energy term is the same calculation with $\log S(u,v)$ replaced by $\mathrm{i}\widetilde{p}(v)$:

$$-\oint\frac{\mathrm{d}v}{2\pi}\,\widetilde{p}'(v)\log F(v)=\oint\frac{\mathrm{d}v}{2\pi}\,\widetilde{p}(v)\partial_v\log F(v)=\sum_j\mathrm{i}\,\widetilde{p}(u_j).$$

Finally, $F(u_j)=0$ means $\mathrm{e}^{-\zeta(u_j)}=-1$. Choosing a logarithm branch gives $-\zeta(u_j)=\mathrm{i}\pi(2n_j+1)$, which is (99.15). □

This proposition fixes the signs in the one-species formula above. The minus sign in the driving term is not an optional convention: it is the integration-by-parts sign in the residue calculation. The plus sign in the discrete energy contribution is equally fixed. After the inverse mirror continuation of a crossed level-I root, the planar convention is $\mathrm{i}\widetilde{p}(u_j)=E(u_j)$. Thus the deformed mirror free-energy formula produces the physical-magnon energy term in (99.25), while the zero condition becomes the exact Bethe equation on the physical sheet.

Let A,B range over the mirror Bethe-string species: physical Q-particle bound states, auxiliary y-roots, and auxiliary w-string families for the two $\mathfrak{su}(2|2)$ wings. Denote the mirror pseudoenergy by $\epsilon_A(u)$ and set

$$Y_A(u)=\mathrm{e}^{-\epsilon_A(u)}.$$

The TBA equations have the form

$$\epsilon_A(u)=L\,\widetilde{E}_A(u)-\mu_A-\sum_B\int_{\mathcal{C}_B}\mathrm{d}v\,K_{BA}(v,u)\log(1+Y_B(v))\,, \tag{99.16}$$

with kernels

$$K_{BA}(v,u)=\frac{1}{2\pi\mathrm{i}}\frac{\partial}{\partial v}\log S_{BA}^{\mathrm{mirror}}(v,u).$$

The contours $\mathcal{C}_B$, signs for fermionic auxiliary roots, and chemical potentials μ_A are part of the mirror-model definition. They are not decorations: changing them changes the finite-size answer.

Mirror-TBA node and source inventory. Assume the mirror Bethe-string hypothesis of Proposition 99.3, diagonal scalar factors for the fused mirror strings satisfying

$$S^{\mathrm{sb}}_{AB}(u,v)S^{\mathrm{mirror}}_{BA}(v,u)=1$$

on the chosen contours, and no contour-crossing source in the ground-state calculation. Let

$$\mathcal{I}=\{\bullet_Q:Q\ge 1\}\cup\{y^{(\alpha)}_{\pm},(v|M)^{(\alpha)},(w|M)^{(\alpha)}:M\ge 1,\ \alpha=L,R\}.$$

The stringbook symbols are identified as

$$y^{(\alpha)}_{+}=\oplus^{(\alpha)},\qquad y^{(\alpha)}_{-}=\ominus^{(\alpha)},\qquad (v|M)^{(\alpha)}=M|yw^{(\alpha)},\qquad (w|M)^{(\alpha)}=M|w^{(\alpha)}.$$

Put $\nu_L=+1$, $\nu_R=-1$. The driving terms are

$$d_{\bullet_Q}=L\,\widetilde{E}_Q,\qquad d_{y^{(\alpha)}_{\pm}}=\nu_\alpha\pi\mathrm{i},\qquad d_{(v|M)^{(\alpha)}}=d_{(w|M)^{(\alpha)}}=0,$$

or equivalently $d_A=L\widetilde{E}_A-\mu_A$, with $\widetilde{E}_A=0$ for auxiliary nodes and $\mu_{y^{(\alpha)}_{\pm}}=-\nu_\alpha\pi\mathrm{i}$. If

$$K^{\mathrm{sb}}_{AB}(u,v)=\frac{1}{2\pi\mathrm{i}}\partial_v\log S^{\mathrm{sb}}_{AB}(u,v)$$

denotes the target-first stringbook kernel, then

$$K^{\mathrm{sb}}_{AB}(u,v)=-K_{BA}(v,u). \tag{99.17}$$

Consequently (99.16) is equivalent to the target-first system

$$\epsilon_A(u)=d_A(u)+\sum_{B\in\mathcal{I}}\int_{\mathcal{C}_B}\mathrm{d}v\,K^{\mathrm{sb}}_{AB}(u,v)\log(1+Y_B(v)). \tag{99.18}$$

Write $\mathcal{S}_A$ for the convolution part of (99.18). The nonzero source families are precisely

$$\begin{aligned}
\mathcal{S}_{\bullet_Q}&=\sum_{Q'\ge 1}K^{\bullet\bullet}_{QQ'}*\log(1+Y_{\bullet_{Q'}})+\sum_{\alpha=L,R}\sum_{M\ge 2}K^{\bullet v}_{QM}*\log(1+Y_{(v|M)^{(\alpha)}})\\
&\quad+\sum_{\alpha=L,R}\left[K^{\bullet y_+}_{Q}*\log(1+Y_{y^{(\alpha)}_{+}})+K^{\bullet y_-}_{Q}*\log(1+Y_{y^{(\alpha)}_{-}})\right],\\
\mathcal{S}_{y^{(\alpha)}_{\pm}}&=\sum_{Q\ge 1}K^{y_\pm\bullet}_{Q}*\log(1+Y_{\bullet_Q})+\sum_{M\ge 1}K_M*\log\frac{1+Y_{(v|M)^{(\alpha)}}}{1+Y_{(w|M)^{(\alpha)}}},\\
\mathcal{S}_{(v|M)^{(\alpha)}}&=\sum_{Q\ge 1}K^{v\bullet}_{MQ}*\log(1+Y_{\bullet_Q})+\sum_{N\ge 1}K_{MN}*\log(1+Y_{(v|N)^{(\alpha)}})\\
&\quad+K_M*\log\frac{1+Y_{y^{(\alpha)}_{+}}}{1+Y_{y^{(\alpha)}_{-}}},\\
\mathcal{S}_{(w|M)^{(\alpha)}}&=\sum_{N\ge 1}K_{MN}*\log(1+Y_{(w|N)^{(\alpha)}})+K_M*\log\frac{1+Y_{y^{(\alpha)}_{+}}}{1+Y_{y^{(\alpha)}_{-}}}.
\end{aligned}$$

The lower bound $M \geq 2$ in the $\bullet_Q$ equation is the stringbook indexing of the v-string source adjacent to the y-sheet boundary. The denominator entries are not optional signs: they encode the reversed ordering of the $\ominus$-sheet and w-string Bethe roots.

The one-species derivation above already gives the density-to-TBA transpose. For species A, the logarithmic Bethe-Yang equation is differentiated in the target rapidity. The entropy variation transposes the density kernel, so the target-first TBA contains $(2\pi\mathrm{i})^{-1}\partial_v \log S^{\mathrm{sb}}_{AB}(u,v)$. The diagonal unitarity relation displayed in the hypotheses gives

$$\frac{1}{2\pi\mathrm{i}}\partial_v \log S^{\mathrm{sb}}_{AB}(u,v) = -\frac{1}{2\pi\mathrm{i}}\partial_v \log S^{\mathrm{mirror}}_{BA}(v,u),$$

which is (99.17) and converts (99.18) to (99.16). Only the $\bullet_Q$ strings carry mirror energy, so the length term is $L\widetilde{E}_Q$ on those nodes and is zero on auxiliary nodes. The sign -1 in the level-II mirror Bethe-Yang equation gives the additive fermion chemical potential $\nu_\alpha\pi\mathrm{i}$ in each wing, equivalently $\mu_{y^{(\alpha)}_{\pm}} = -\nu_\alpha\pi\mathrm{i}$ in the source-kernel convention.

The remaining entries are the fused forms of (99.7). Fusing level-I roots gives the $\bullet$-kernels; fusing the auxiliary arrays gives the rational K_M and K_{MN} kernels used below. The roots on the $\ominus$-sheet and in the w-strings are ordered oppositely to the $\oplus$-sheet and v-string roots in the stringbook Bethe-Yang products. Their scattering factors therefore enter inverted, which is exactly the logarithmic ratio displayed in the y-, v-, and w-node equations.

Fused mirror kernels in the chosen orientation. Fix a mirror sheet and write

$$\mathcal{R}_m = \left\{-\frac{m-1}{2}, -\frac{m-1}{2}+1, \ldots, \frac{m-1}{2}\right\}.$$

For an integer a, $x^{[a]}(u)$ denotes the continuation of $x(u+\mathrm{i}a/2)$ on that sheet, with the side of the cut fixed by the contour datum. Away from zeros and poles, the fused rational $\bullet_m$-$\bullet_n$ phase is

$$\Phi^{\bullet\bullet}_{mn}(u,v) = \sum_{r\in\mathcal{R}_m}\sum_{s\in\mathcal{R}_n} \log\left[\frac{x^{[2s-1]}(v) - x^{[2r+1]}(u)}{x^{[2s+1]}(v) - x^{[2r-1]}(u)} \frac{1-\left(x^{[2s+1]}(v)x^{[2r-1]}(u)\right)^{-1}}{1-\left(x^{[2s-1]}(v)x^{[2r+1]}(u)\right)^{-1}}\right], \tag{99.19}$$

and the dressing contribution is the endpoint combination

$$\begin{aligned}\Theta^{\bullet\bullet}_{mn}(u,v) = 2\mathrm{i}\big(&\chi(x^{[m]}(u), x^{[n]}(v)) + \chi(x^{[-m]}(u), x^{[-n]}(v)) \\ &- \chi(x^{[-m]}(u), x^{[n]}(v)) - \chi(x^{[m]}(u), x^{[-n]}(v))\big).\end{aligned} \tag{99.20}$$

Thus

$$K^{\bullet\bullet}_{mn}(u,v) = \frac{1}{2\pi\mathrm{i}}\partial_v\left(\Phi^{\bullet\bullet}_{mn}(u,v) + \Theta^{\bullet\bullet}_{mn}(u,v)\right)$$

in the target-first convention. The y-sheet and v-string phases that occur in the source inventory are

$$\log S_n^{y+\bullet}(u,v) = \log\left[\frac{x(u)-x^{[n]}(v)}{x(u)-x^{[-n]}(v)}\sqrt{\frac{x^{[-n]}(v)}{x^{[n]}(v)}}\right], \tag{99.21}$$

$$\log S_n^{y-\bullet}(u,v) = \log\left[\frac{x(u)^{-1}-x^{[n]}(v)}{x(u)^{-1}-x^{[-n]}(v)}\sqrt{\frac{x^{[-n]}(v)}{x^{[n]}(v)}}\right], \tag{99.22}$$

$$\log S_{mn}^{(v|m)\bullet}(u,v) = \sum_{\sigma=\pm1}\sum_{r\in\mathcal{R}_m}\sum_{s\in\mathcal{R}_n}\log\left[\frac{x^{[2r+\sigma]}(u)-x^{[2s+1]}(v)}{x^{[2r+\sigma]}(u)-x^{[2s-1]}(v)}\sqrt{\frac{x^{[2s-1]}(v)}{x^{[2s+1]}(v)}}\right]. \tag{99.23}$$

Their opposite orientations are reciprocals:

$$\log S_n^{\bullet y_\pm}(v,u) = -\log S_n^{y\pm\bullet}(u,v), \qquad \log S_{nm}^{\bullet(v|m)}(v,u) = -\log S_{mn}^{(v|m)\bullet}(u,v).$$

Finally, the auxiliary rational string phase

$$\Phi_{mn}^{\mathrm{aux}}(u,v) = \sum_{r\in\mathcal{R}_m}\sum_{s\in\mathcal{R}_n}\log\frac{u-v+\mathrm{i}(r-s+1)}{u-v+\mathrm{i}(r-s-1)}$$

has kernel

$$\frac{1}{2\pi\mathrm{i}}\partial_v\Phi_{mn}^{\mathrm{aux}} = K_{m+n}+K_{|m-n|}+2\sum_{j=1}^{\min(m,n)-1}K_{|m-n|+2j}, \tag{99.24}$$

with $K_0=0$ and

$$K_N(u-v) = \frac{1}{2\pi\mathrm{i}}\partial_v\log\frac{u-v+\mathrm{i}N/2}{u-v-\mathrm{i}N/2}.$$

These formulae are obtained by fusing the elementary factors. The rational factors are finite products over the constituents of the two Bethe strings. The sets $\mathcal{R}_m$ and $\mathcal{R}_n$ are exactly the constituent shifts, so multiplying the elementary level-I rational factor over r and s gives (99.19). Reversing the order of the two strings in each elementary factor inverts the product, which is the reciprocal orientation stated for $y_\pm$ and $(v|m)$ sources. The distinction between y_+ and y_- is the replacement $x(u)\mapsto x(u)^{-1}$ on the two y-sheets.

For the purely auxiliary kernel put $z=u-v$. The derivative of $\log(z+\mathrm{i}a)$ with respect to v is $-(z+\mathrm{i}a)^{-1}$. Hence the double sum contributes the difference between the multiplicities of $r-s-1$ and $r-s+1$. As r and s run over two consecutive arithmetic strings, all interior multiplicities cancel in pairs except the two boundary terms and the doubled interior terms with the same parity as $|m-n|$. This gives precisely the right-hand side of (99.24); it is the fused A_∞ kernel used in the Y-system inversion below.

The dressing term is different in status. If $\chi(x,y)=-\chi(y,x)$ on the continued branches, the endpoint combination (99.20) is antisymmetric under $(m,u)\leftrightarrow(n,v)$, so it is compatible with scalar unitarity. The existence of the globally continued χ, its branch choice, and possible additional crossed singularities are part of the dressing-phase analytic-continuation input inherited from Chapter 98. The finite fusion algebra proved here supplies the rational products and the endpoint sign package.

For a state with physical Bethe roots $u_1, \ldots, u_K$, the excited-state TBA is obtained by contour deformation. The physical roots satisfy exact Bethe equations of the form

$$1 + Y_{\bullet_1}^{\text{phys}}(u_j) = 0, \qquad j = 1, \ldots, K,$$

and the energy is

$$\Delta - J = \sum_{j=1}^{K} E(u_j) - \sum_{Q=1}^{\infty} \int_{\mathbb{R}} \frac{\mathrm{d}u}{2\pi} \frac{\mathrm{d}\widetilde{p}_Q}{\mathrm{d}u} \log(1 + Y_{\bullet_Q}(u)). \tag{99.25}$$

The first term is the asymptotic magnon energy. The integral term is the finite-size wrapping correction from the thermal mirror bath.

99.4 Y-System

The passage from TBA integral equations to local Y-system relations is not a formal cancellation of symbols. It uses a special inverse for the fused Bethe-string kernels and an analyticity strip in which s^{-1} may be represented by shifts. We spell out the auxiliary-wing case because it is the cleanest part of the planar $\mathcal{N} = 4$ construction and fixes the normalization used later by the T-hook notation.

Lemma 99.6 (Auxiliary-string kernel inverse)**.** *Let*

$$(K * f)(u) = \int_{\mathbb{R}} \mathrm{d}v \, K(u - v) f(v)$$

and, for $n \geq 1$*,*

$$K_n(u - v) = \frac{1}{2\pi \mathrm{i}} \partial_v \log \frac{u - v + \mathrm{i}n/2}{u - v - \mathrm{i}n/2} = \frac{n/2}{\pi((u - v)^2 + n^2/4)}. \tag{99.26}$$

Set $K_0 = 0$*, and define the fused* A_∞*-string kernel*

$$K_{mn} = K_{m+n} + K_{|m-n|} + 2 \sum_{j=1}^{\min(m,n)-1} K_{|m-n|+2j}, \qquad m, n \geq 1,$$

with $A_{mn} = \delta_{mn}\delta + K_{mn}$*. Finally put*

$$\mathsf{s}(u - v) = \frac{1}{2\cosh \pi(u - v)}.$$

On the Fourier side, with $\widehat{f}(\omega) = \int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}\omega u} f(u) \, \mathrm{d}u$ *and* $q = \mathrm{e}^{-|\omega|/2}$*,*

$$\widehat{K}_n = q^n, \qquad \widehat{\mathsf{s}} = \frac{q}{1 + q^2}, \qquad \widehat{A}_{mn} = \frac{1 + q^2}{1 - q^2} \left(q^{|m-n|} - q^{m+n} \right). \tag{99.27}$$

Consequently

$$(A^{-1})_{mn} = \delta_{mn}\delta - (\delta_{m+1,n} + \delta_{m-1,n})\mathsf{s}, \qquad \delta_{0n} = 0. \tag{99.28}$$

Equivalently,

$$\sum_{\ell \geq 1} (A^{-1})_{m\ell} * K_{\ell n} = (\delta_{m+1,n} + \delta_{m-1,n})\mathsf{s}, \tag{99.29}$$

$$\sum_{\ell \geq 1} (A^{-1})_{m\ell} * K_\ell = \delta_{m1}\mathsf{s}. \tag{99.30}$$

Proof. The transform of (99.26) is the elementary Poisson-kernel identity $\widehat{K}_n(\omega) = \mathrm{e}^{-n|\omega|/2}$. The displayed expression for $\widehat{A}_{mn}$ follows by summing the finite geometric progression in the fused kernel. In Fourier variables the asserted inverse is the tridiagonal matrix

$$\delta_{mn} - \frac{q}{1+q^2}(\delta_{m+1,n} + \delta_{m-1,n}).$$

Multiplying it by the last expression in (99.27) gives

$$\widehat{A}_{mn} - \widehat{\mathsf{s}}\,(\widehat{A}_{m+1,n} + \widehat{A}_{m-1,n}) = \delta_{mn}, \qquad \widehat{A}_{0n} = 0.$$

For $m \neq n$ the two geometric tails cancel. For $m = n$ the remaining jump is one. Equations (99.29) and (99.30) follow as follows. For (99.29), replace A_{mn} by $(A-I)_{mn}$ in the Fourier identity just proved and transform back from ω-space. For the one-index kernel, multiplication by the tridiagonal inverse gives

$$q^m - \frac{q}{1+q^2}\big(q^{m+1} + q^{m-1}\big) = 0 \quad (m > 1),$$

while for $m = 1$ the missing $m = 0$ term leaves $q/(1+q^2) = \widehat{\mathsf{s}}$. This is (99.30). $\square$

One auxiliary wing of the Y-system. Fix one $\mathfrak{su}(2|2)$ wing. Let Y_n° denote the $n|w$ string Y-functions, and let $Y^\oplus, Y^\ominus$ denote the two y-sheet Y-functions in the stringbook orientation. Suppose the $n|w$ part of the mirror TBA can be written as

$$\sum_{m\geq 1} A_{nm} * \log Y_m^\circ = \sum_{m\geq 1} K_{nm} * \log(1+Y_m^\circ) + K_n * \log\frac{1+1/Y^\oplus}{1+1/Y^\ominus}, \tag{99.31}$$

with $Y_0^\circ = 0$. Assume all functions appearing after convolution by s are analytic in the strip $|\operatorname{Im} u| < 1/2$, decay enough at infinity, and have no pole crossed by the shifts $u \mapsto u \pm \mathrm{i}/2$. Then

$$\frac{(Y_n^\circ)^+(Y_n^\circ)^-}{(1+Y_{n+1}^\circ)(1+Y_{n-1}^\circ)} = \left(\frac{1+1/Y^\oplus}{1+1/Y^\ominus}\right)^{\delta_{n1}}, \qquad n \geq 1. \tag{99.32}$$

Apply A^{-1} to (99.31). Using (99.29) and (99.30) gives

$$\log Y_n^\circ = \mathsf{s} * \log(1+Y_{n+1}^\circ) + \mathsf{s} * \log(1+Y_{n-1}^\circ) + \delta_{n1}\mathsf{s} * \log\frac{1+1/Y^\oplus}{1+1/Y^\ominus}.$$

For functions analytic in the stated strip, $\mathsf{s}^{-1} * f = f^+ + f^-$. Applying this inverse and exponentiating gives (99.32). If a pole or branch cut crosses during the shift, the right-hand side acquires extra source terms; this is precisely why the analytic data below are part of the finite-volume spectral problem.

Data lost by the s-kernel inverse. Write

$$(\nabla_{\mathsf{s}} F)(u) = F(u+\mathrm{i}/2) + F(u-\mathrm{i}/2).$$

Suppose $F = \mathsf{s} * f$, where f is integrable, F is holomorphic in a neighbourhood of the closed strip $|\operatorname{Im} u| \leq 1/2$, and the two contour shifts meet no singularity of F. In logarithmic TBA applications this means that no zero or pole of the underlying Y-function, and no logarithmic cut, is crossed. Then

$$\nabla_{\mathsf{s}} F = f. \tag{99.33}$$

The converse datum is not contained in the local equation. For $\alpha \in \mathbb{C}$,

$$Z_\alpha(u) = \frac{1}{2\cosh \pi(u-\alpha)} \tag{99.34}$$

satisfies

$$Z_\alpha(u+\mathrm{i}/2) + Z_\alpha(u-\mathrm{i}/2) = 0$$

as a meromorphic identity away from $u = \alpha$, while its closest poles are at $u = \alpha \pm \mathrm{i}/2$. Thus applying s^{-1} determines only the regular part of a logarithmic TBA equation modulo such boundary data. If the logarithm instead contains the shifted zero-pole factor

$$B_\alpha(u) = \frac{u-\alpha+\mathrm{i}/2}{u-\alpha-\mathrm{i}/2},$$

then

$$B_\alpha^+(u) B_\alpha^-(u) = \frac{u-\alpha+\mathrm{i}}{u-\alpha-\mathrm{i}}. \tag{99.35}$$

Consequently the local Y-system knows neither the positions α, nor the integer powers of such factors, nor the side of the cut from which they come. These are extra analytic data supplied by the mirror-TBA contour and by exact-root regularity.

With the Fourier convention of Lemma 99.6, $\widehat{\mathsf{s}}(\omega) = 1/(2\cosh(\omega/2))$. Shifting the contour for $F(u+\mathrm{i}/2)$ and $F(u-\mathrm{i}/2)$ back to the real line multiplies the Fourier transform by $\mathrm{e}^{\omega/2}$ and $\mathrm{e}^{-\omega/2}$, respectively. Under the stated no-crossing hypothesis the sum therefore has transform

$$2\cosh(\omega/2)\,\widehat{\mathsf{s}}(\omega)\widehat{f}(\omega) = \widehat{f}(\omega),$$

which proves (99.33). If a singularity lies on a shifted contour, the same contour move has a residue or jump contribution; that contribution is not present in the pointwise local equation unless it is recorded as a source.

For the zero mode, use

$$\cosh \pi(u-\alpha+\mathrm{i}/2) = \mathrm{i}\sinh \pi(u-\alpha), \qquad \cosh \pi(u-\alpha-\mathrm{i}/2) = -\mathrm{i}\sinh \pi(u-\alpha).$$

The two shifted terms cancel away from their boundary pole. The last identity is the direct multiplication of $B_\alpha(u+\mathrm{i}/2)$ and $B_\alpha(u-\mathrm{i}/2)$.

The TBA equations imply functional relations among Y-functions after applying kernel identities and shifting the spectral parameter. The bulk of the planar $\mathcal{N} = 4$ T-hook is the subset

$$\mathfrak{T}_{\text{bulk}} = \{(a,s) : a \in \mathbb{Z}_{\geq 1},\ s \in \mathbb{Z},\ |s| \leq 2\} \cup \{(a,s) : a = 1,2,\ s \in \mathbb{Z}\}.$$

The row $a = 0$ is adjoined as a gauge boundary $T_{0,s}$; outside $\mathfrak{T}_{\text{bulk}}$ and this boundary row one sets $T_{a,s} = 0$. A Y-node is called interior when $a \geq 1$ and all four neighbouring T-functions appearing below belong to $\mathfrak{T}_{\text{bulk}}$ or to the boundary row, and are non-zero on the patch under consideration. On such nodes the local relation has the form

$$Y_{a,s}^+(u) Y_{a,s}^-(u) = \frac{(1+Y_{a,s+1}(u))(1+Y_{a,s-1}(u))}{(1+Y_{a+1,s}(u)^{-1})(1+Y_{a-1,s}(u)^{-1})}, \tag{99.36}$$

where

$$f^{\pm}(u) = f(u \pm \mathrm{i}/2).$$

Boundary nodes require modified relations because one of the four neighboring T-functions has left the hook.

Proposition 99.7 (Hirota origin of the interior T-hook Y-system)**.** *Let $T_{a,s}$ be non-zero on an interior patch of the T-hook and obey Hirota,*

$$T_{a,s}^{+}T_{a,s}^{-} = T_{a+1,s}T_{a-1,s} + T_{a,s+1}T_{a,s-1}.$$

Define

$$Y_{a,s} = \frac{T_{a,s+1}T_{a,s-1}}{T_{a+1,s}T_{a-1,s}}.$$

Then $Y_{a,s}$ obeys (99.36) at every interior node. Moreover $Y_{a,s}$ is invariant under the local Hirota gauge transformation

$$T_{a,s} \mapsto g_1^{[a+s]}g_2^{[a-s]}g_3^{[-a-s]}g_4^{[-a+s]}T_{a,s}, \tag{99.37}$$

where $f^{[n]}(u) = f(u + \mathrm{i}n/2)$.

Proof. Hirota at (a, s) gives the two useful identities

$$1 + Y_{a,s} = \frac{T_{a,s}^{+}T_{a,s}^{-}}{T_{a+1,s}T_{a-1,s}}, \qquad 1 + \frac{1}{Y_{a,s}} = \frac{T_{a,s}^{+}T_{a,s}^{-}}{T_{a,s+1}T_{a,s-1}}.$$

Now compute

$$Y_{a,s}^{+}Y_{a,s}^{-} = \frac{T_{a,s+1}^{+}T_{a,s+1}^{-}T_{a,s-1}^{+}T_{a,s-1}^{-}}{T_{a+1,s}^{+}T_{a+1,s}^{-}T_{a-1,s}^{+}T_{a-1,s}^{-}}.$$

Applying the two displayed identities to the four neighboring nodes turns the numerator into

$$(1 + Y_{a,s+1})(1 + Y_{a,s-1})T_{a+1,s+1}T_{a-1,s+1}T_{a+1,s-1}T_{a-1,s-1},$$

and the denominator into

$$(1 + Y_{a+1,s}^{-1})(1 + Y_{a-1,s}^{-1})T_{a+1,s+1}T_{a+1,s-1}T_{a-1,s+1}T_{a-1,s-1}.$$

The common T-product cancels, proving (99.36).

For the gauge statement, let

$$\mathcal{G}(a, s) = \{(1, a + s), (2, a - s), (3, -a - s), (4, -a + s)\}$$

record which g_i appears with which shift. The numerator of $Y_{a,s}$ has gauge multiset

$$\mathcal{G}(a, s + 1) \sqcup \mathcal{G}(a, s - 1),$$

and the denominator has

$$\mathcal{G}(a + 1, s) \sqcup \mathcal{G}(a - 1, s).$$

These two multisets are identical. Hence all gauge factors cancel. The calculation is local; analytic gauge choices in the QSC additionally fix cuts, asymptotics, and gluing data. □

Warning 99.8 (Y-system data are not only equations)*.* The functional relations (99.36) do not by themselves determine a spectrum. A solution also requires analyticity domains, large-u asymptotics, branch-cut discontinuity relations, exact Bethe-root conditions, and state labels. Treating the Y-system equations alone as a definition of the theory discards the information that distinguishes physical states.

Local source factor from a shifted zero-pole pair. Let $\alpha \in \mathbb{C}$, and define the elementary shifted source factor

$$B_\alpha(u) = \frac{u-\alpha+\mathrm{i}/2}{u-\alpha-\mathrm{i}/2}.$$

Thus B_α has a simple zero at $u=\alpha-\mathrm{i}/2$ and a simple pole at $u=\alpha+\mathrm{i}/2$. If a meromorphic Y-function is written locally as

$$Y_{a,s}(u) = B_\alpha(u)^\nu\, \widehat{Y}_{a,s}(u), \qquad \nu \in \mathbb{Z},$$

where $\widehat{Y}_{a,s}$ has no zero or pole crossed by the two shifts $u \mapsto u \pm \mathrm{i}/2$, then

$$Y_{a,s}^+(u)Y_{a,s}^-(u) = \left(\frac{u-\alpha+\mathrm{i}}{u-\alpha-\mathrm{i}}\right)^\nu \widehat{Y}_{a,s}^+(u)\widehat{Y}_{a,s}^-(u). \tag{99.38}$$

For a finite set of shifted zero-pole pairs, the source factor is the product of the corresponding powers of $(u-\alpha+\mathrm{i})/(u-\alpha-\mathrm{i})$. This is the algebra of the shift operation. Directly,

$$B_\alpha^+(u) = \frac{u-\alpha+\mathrm{i}}{u-\alpha}, \qquad B_\alpha^-(u) = \frac{u-\alpha}{u-\alpha-\mathrm{i}}.$$

Multiplying the two expressions gives

$$B_\alpha^+(u)B_\alpha^-(u) = \frac{u-\alpha+\mathrm{i}}{u-\alpha-\mathrm{i}}.$$

Raising to the integer power ν and multiplying by $\widehat{Y}_{a,s}^+\widehat{Y}_{a,s}^-$ proves (99.38). A finite product follows by commutativity.

This local calculation is the algebraic shadow of contour deformation. The homogeneous Y-system records what happens after s^{-1} is applied in a strip free of singularities. If a zero, pole, or logarithmic branch point crosses one of the two strip boundaries, the remaining regular part $\widehat{Y}_{a,s}$ still obeys the same shift algebra, but the singular part contributes the rational source factor in (99.38). The Y-system equations alone do not determine the positions α, the powers ν, or the sheet on which the factor lives; these data come from the excited-state contours and from the exact Bethe-root regularity conditions.

99.5 Analytic Y-System Data

The analytic Y-system supplements (99.36) by specifying where each shifted function is meromorphic and which discontinuities occur when a shift crosses a cut. The analytic data determine which local solution of the symbolic Y-system is the finite-volume spectrum.

We use the shifted notation

$$f^{[n]}(u) = f(u + \mathrm{i}n/2)$$

and write

$$Z_0^\vee = [-2g, 2g]$$

for the central mirror-sheet Zhukovsky cut. Let $\mathcal{A}_\ell$ denote the class of functions meromorphic in the open strip

$$-\frac{\ell}{2} < \operatorname{Im} u < \frac{\ell}{2}, \tag{99.39}$$

after removing state-dependent poles and zeros that are explicitly recorded by source factors. The basic stringbook-to-T-hook identification is

$$\begin{aligned} Y_n^{\bullet} = Y_{n,0}, \qquad Y^{\oplus,L/R} = Y_{1,\pm 1}^{-1}, \qquad Y^{\ominus,L/R} = Y_{2,\pm 2}, \\ Y_n^{\triangle,L/R} = Y_{n+1,\pm 1}^{-1}, \qquad Y_n^{\circ,L/R} = Y_{1,\pm(n+1)}^{-1}. \end{aligned} \tag{99.40}$$

Assumption 99.9 (Analytic Y-system strip and cut data)**.** Fix the mirror-TBA sheet and the stringbook node convention (99.40). For every $n \geq 1$,

$$Y_{n,0}, \qquad Y_{n+1,\pm 1}, \qquad Y_{1,\pm(n+1)} \in \mathcal{A}_n. \tag{99.41}$$

Their first Zhukovsky branch points occur at

$$u = \pm 2g + \frac{\mathrm{i}\ell}{2}, \qquad \ell = \pm n,\ \pm(n+2),\ \pm(n+4), \ldots. \tag{99.42}$$

The central fermionic nodes $Y_{1,\pm 1}$ and $Y_{2,\pm 2}$ have cuts $Z_0^{\vee} + \mathrm{i}m$, $m \in \mathbb{Z}$, and their two sides on the central cut are related by

$$Y_{2,\pm 2}(u + \mathrm{i}0) = \frac{1}{Y_{1,\pm 1}(u - \mathrm{i}0)}, \qquad u \in Z_0^{\vee}. \tag{99.43}$$

Finally, every zero or pole introduced by an excited-state contour deformation is part of the data and must be recorded by a finite product of local factors of the form (99.38).

The strip statement is an analytic input extracted from the mirror TBA and its asymptotic Bethe-ansatz limit; it is not a theorem of the local Hirota algebra. The same limitation is visible in the Arutyunov–Frolov derivation of the mirror TBA: applying the inverse s-kernel loses data, and the local Y-equations are controlled on the central mirror interval unless additional analytic-continuation assumptions are supplied. We do not use that literature statement as a proof. The assumption above is the independent monograph datum that must be checked against the chosen mirror sheet, source factors, and dressing branch. The branch-point lattice explains why the homogeneous Y-system can be shifted inside the strip $\mathcal{A}_n$ but acquires sources when an endpoint, pole, zero, or dressing cut is crossed. The central inversion (99.43) is the piece that later turns into the magic-sheet gluing condition for the one-row T-gauge and, eventually, into the $P\mu$-system. Chapter 100 makes this status boundary explicit in Proposition 100.8: the local Y-system supplies gauge-invariant ratios, while the QSC spectral problem also needs the restored source data, the central fermionic large-u coefficient, the P-power charge gap, and the physical gluing assignment of the characteristic roots.

With these conventions the exact Bethe equations are regularity conditions:

$$1 + Y_{\bullet_1}^{\mathrm{phys}}(u_j) = 0$$

removes would-be singularities at the physical roots after contour deformation. This formulation is deliberately more verbose than the symbolic Y-system. The shifts $u \mapsto u \pm \mathrm{i}/2$ can move a point from the physical sheet to the mirror sheet, and the operations "shift" and "continue through a cut" do not commute without recording the path.

99.6 Konishi as the First Wrapping Test

The Konishi multiplet contains an $SL(2)$ descendant represented by a length-two state with two derivative magnons. In the asymptotic Bethe ansatz, the two physical roots are opposite,

$$u_1 = -u_2.$$

At weak coupling the one-loop solution is

$$u_1 = \frac{1}{2\sqrt{3}}, \qquad u_2 = -\frac{1}{2\sqrt{3}},$$

in the normalization of the rational Bethe equations. The asymptotic dimension matches direct perturbative gauge theory through the orders before wrapping appears. The first wrapping correction is supplied by the mirror integral term in (99.25). Conceptually, this is the finite-length correction to the trace, not a correction to the two-magnon dispersion relation.

The exact finite-coupling Konishi dimension is obtained by solving the excited-state TBA or, more efficiently, the quantum spectral curve of the next chapter. In this monograph the TBA formula is used as a logically explicit construction of the finite-size correction: every input is an S-matrix kernel, a contour, a chemical potential, or a Bethe-root condition.

At weak coupling the leading finite-size effect is the Luescher or first wrapping correction. The $SL(2)$ Konishi descendant has $K = 2$ physical roots $u_1 = -u_2$ with

$$u_1 = \frac{1}{2\sqrt{3}}\big(1 + 8g^2 - 20g^4 + (112 + 48\zeta(3))g^6 + O(g^8)\big)$$

in the asymptotic equation. The asymptotic energy gives

$$\Delta_{\mathrm{K}}^{\mathrm{ABA}} = 4 + 12g^2 - 48g^4 + 336g^6 - (2820 + 288\zeta(3))g^8 + O(g^{10}).$$

Assumption 99.10 (Leading Konishi mirror input)**.** The leading weak-coupling wrapping computation uses the following mirror-TBA data. First, the physical roots in all factors contributing at order g^8 are the one-loop values

$$u_* = \frac{1}{2\sqrt{3}}, \qquad u_1 = u_*, \qquad u_2 = -u_*.$$

Second, the virtual particles are the mirror Q-particle bound states, $Q \in \mathbb{Z}_{>0}$, in the antisymmetric $SL(2)$-type mirror sector. Third, for real mirror rapidity u the weak mirror branch is

$$x^{[Q]}(u) = \frac{g}{u + \mathrm{i}Q/2} + O(g^3), \qquad x^{[\ell]}(u) = \frac{u + \mathrm{i}\ell/2}{g} + O(g), \qquad -Q \le \ell \le Q - 1.$$

Fourth, the leading contribution of the dressing phase to the asymptotic mirror Y_Q-function starts beyond the displayed g^8 coefficient. Finally, the excited-state contour deformation is the one for which (99.25) has no additional weak-coupling μ-term for Konishi.

Under this assumption the asymptotic mirror density has the stringbook form

$$Y_{\bullet_Q}(u) = g^8\, Y_Q^{(0)}(u) + O(g^{10}), \tag{99.44}$$

where

$$Y_Q^{(0)}(u) = \frac{4Q^2\left(u^2 - u_*^2 + (Q^2-1)/4\right)^2}{(u^2+Q^2/4)^4} \times \prod_{\varepsilon=\pm1} \frac{1}{((u-\varepsilon u_*)^2 + (Q+1)^2/4)\,((u-\varepsilon u_*)^2+(Q-1)^2/4)}. \tag{99.45}$$

All symbols in (99.45) are now fixed: Q is the positive integer mirror bound-state charge, $u \in \mathbb{R}$ is the mirror rapidity, and $g = \sqrt{\lambda}/(4\pi)$ is the coupling of Chapter 98.

The mirror momentum measure is also a branch-sensitive datum. In the same convention as the stringbook, the mirror Q-particle satisfies

$$\mathrm{i}\widetilde{p}_Q(u) = -Q - 2\mathrm{i}g\big(x^{[Q]}(u) - x^{[-Q]}(u)\big).$$

Substituting the weak mirror branch of Assumption 99.10 gives

$$\mathrm{i}\widetilde{p}_Q(u) = -Q - 2\mathrm{i}g\left(\frac{g}{u+\mathrm{i}Q/2} - \frac{u-\mathrm{i}Q/2}{g}\right) + O(g^2) = 2\mathrm{i}u + O(g^2),$$

and therefore

$$\frac{\mathrm{d}\widetilde{p}_Q}{\mathrm{d}u} = 2 + O(g^2). \tag{99.46}$$

This is the sign check that fixes the convention in the energy correction.

Konishi weak-density rationalization. Let $u_* = 1/(2\sqrt{3})$ and put $q = 2u$. For $Q \geq 1$, define

$$B_Q^-(q) = 9q^4 + 6(3(Q-2)Q+2)q^2 + (3(Q-2)Q+4)^2,$$
$$B_Q^+(q) = 9q^4 + 6(3Q(Q+2)+2)q^2 + (3Q(Q+2)+4)^2.$$

Then the weak mirror density (99.45) is equivalently

$$Y_Q^{(0)}(q/2) = \frac{147456Q^2(3q^2+3Q^2-4)^2}{(q^2+Q^2)^4 B_Q^-(q)B_Q^+(q)}. \tag{99.47}$$

The first denominator factor and the numerator become

$$u^2 + \frac{Q^2}{4} = \frac{q^2+Q^2}{4}, \qquad u^2 - u_*^2 + \frac{Q^2-1}{4} = \frac{3q^2+3Q^2-4}{12}.$$

It remains only to rationalize the two factors involving u_*. For any real a,

$$\prod_{\varepsilon=\pm1}\left(\left(\frac{q}{2} - \frac{\varepsilon}{2\sqrt{3}}\right)^2 + \frac{a^2}{4}\right) = \frac{1}{16}\left[q^4 + 2\left(a^2 - \frac{1}{3}\right)q^2 + \left(a^2+\frac{1}{3}\right)^2\right].$$

Multiplying the bracket by 9, and setting $a = Q-1$ and $a = Q+1$, gives respectively $B_Q^-(q)$ and $B_Q^+(q)$. Hence each such product is $B_Q^{\pm}(q)/144$. Substituting these three elementary identities in (99.45) gives

$$\frac{4Q^2\left((3q^2+3Q^2-4)/12\right)^2}{((q^2+Q^2)/4)^4\,B_Q^-(q)B_Q^+(q)/144^2},$$

which simplifies to (99.47).

Pole structure of the Konishi weak density. Let $Q \geq 1$, and let $Y_Q^{(0)}(q/2)$ be the rational function in (99.47). Then

$$B_Q^-(q) = 9 \prod_{\sigma=\pm 1} \left(\left(q - \frac{\sigma}{\sqrt{3}} \right)^2 + (Q-1)^2 \right),$$
$$B_Q^+(q) = 9 \prod_{\sigma=\pm 1} \left(\left(q - \frac{\sigma}{\sqrt{3}} \right)^2 + (Q+1)^2 \right).$$

After the removable $Q = 1$ cancellation described below, the function has no pole on the real q-axis and obeys $Y_Q^{(0)}(q/2) = O(q^{-12})$ as $|q| \to \infty$. Its upper-half-plane poles are the fourth-order pole $q = \mathrm{i}Q$, the simple poles

$$q = \pm \frac{1}{\sqrt{3}} + \mathrm{i}(Q+1),$$

and, for $Q > 1$, the simple poles

$$q = \pm \frac{1}{\sqrt{3}} + \mathrm{i}(Q-1).$$

For $Q = 1$ the latter two points are real removable points, not contour poles.

For any real a,

$$\prod_{\sigma=\pm 1} \left(\left(q - \frac{\sigma}{\sqrt{3}} \right)^2 + a^2 \right) = q^4 + 2\left(a^2 - \frac{1}{3} \right) q^2 + \left(a^2 + \frac{1}{3} \right)^2.$$

Multiplication by 9, followed by $a = Q - 1$ and $a = Q + 1$, gives the two displayed factorizations and therefore the zero locations. The factor $(q^2+Q^2)^4$ gives a fourth-order zero of the denominator at $q = \mathrm{i}Q$ and at $q = -\mathrm{i}Q$. For real q, the factors with $a > 0$ are strictly positive. The only exceptional case is $Q = 1$ in B_1^-, where

$$B_1^-(q) = (3q^2 - 1)^2.$$

The numerator of (99.47) contains the identical factor $(3q^2-1)^2$, so the singularities at $q = \pm 1/\sqrt{3}$ are removable. Finally, before this cancellation the numerator has degree 4 and the denominator has degree 16. For $Q = 1$, after cancellation the numerator has degree 0 and the denominator has degree 12. In both cases the decay is q^{-12}.

Lemma 99.11 (Residue reduction for the Konishi summand)**.** *For $Q \geq 1$, put*

$$\mathcal{Y}_Q(q) = \frac{147456 Q^2 (3q^2 + 3Q^2 - 4)^2}{(q^2 + Q^2)^4 B_Q^-(q) B_Q^+(q)}$$

and

$$I_Q = -\int_{\mathbb{R}} \frac{\mathrm{d}q}{2\pi} \, \mathcal{Y}_Q(q),$$

with the $Q = 1$ removable real points understood as in the pole-structure calculation above. Then

$$I_Q = R_Q + \frac{864}{Q^3} - \frac{1440}{Q^5}, \tag{99.48}$$

where

$$R_Q = -\frac{7776Q\,N(Q)}{(9Q^4 - 3Q^2 + 1)^4(27Q^6 - 27Q^4 + 36Q^2 + 16)}$$

and

$$N(Q) = 19683Q^{18} - 78732Q^{16} + 150903Q^{14} - 134865Q^{12} + 1458Q^{10} \\ + 48357Q^8 - 13311Q^6 - 1053Q^4 + 369Q^2 - 10.$$

Proof. Write

$$\mathcal{N}_Q(q) = 147456Q^2(3q^2 + 3Q^2 - 4)^2.$$

By the pole-structure calculation above, the upper contour encloses the fourth-order pole at $q = \mathrm{i}Q$, the two simple poles with $a = Q + 1$, and, when $Q > 1$, the two simple poles with $a = Q - 1$. Hence

$$I_Q = -\mathrm{i}\left(\rho_Q^{(4)} + \rho_Q^{(+)} + \mathbf{1}_{Q>1}\rho_Q^{(-)}\right),$$

where

$$\rho_Q^{(4)} = \frac{1}{3!}\,\frac{\mathrm{d}^3}{\mathrm{d}q^3}\frac{\mathcal{N}_Q(q)}{(q+\mathrm{i}Q)^4 B_Q^-(q)B_Q^+(q)}\bigg|_{q=\mathrm{i}Q}.$$

For the simple poles, set $a_\eta = Q + \eta$ for $\eta = \pm 1$, $q_{\sigma,\eta} = \sigma/\sqrt{3} + \mathrm{i}a_\eta$, and use the convention $B_Q^{+1} = B_Q^+$, $B_Q^{-1} = B_Q^-$. When $a_\eta > 0$,

$$\rho_Q^{(\eta)} = \sum_{\sigma=\pm 1} \frac{\mathcal{N}_Q(q_{\sigma,\eta})}{(q_{\sigma,\eta}^2 + Q^2)^4 B_Q^{-\eta}(q_{\sigma,\eta})\partial_q B_Q^{\eta}(q_{\sigma,\eta})}.$$

The derivative in the denominator is not an implicit convention. The factorization in the pole-structure calculation above gives

$$\partial_q B_Q^{\eta}(q_{\sigma,\eta}) = 24\mathrm{i}a_\eta(1 + \sigma\mathrm{i}\sqrt{3}\,a_\eta).$$

These displayed formulas reduce the contour calculation to arithmetic in the field $\mathbb{Q}(\sqrt{3}, \mathrm{i})$. Adding the $\sigma = +1$ and $\sigma = -1$ residues cancels the $\sqrt{3}$-odd part. Combining the two simple-pole pairs with the fourth-order Laurent coefficient over a common denominator then gives exactly (99.48). The companion calculation check performs this Laurent extraction literally, including the $Q = 1$ omission of the removable $a_- = 0$ pair. □

Leading Konishi wrapping integral. Assume the mirror input of Assumption 99.10. Then the order g^8 finite-size correction to the Konishi dimension is

$$\Delta_{\mathrm{K}}^{\mathrm{wrap}} = (324 + 864\zeta(3) - 1440\zeta(5))g^8 + O(g^{10}).$$

Since $Y_{\bullet_Q} = O(g^8)$,

$$\log(1 + Y_{\bullet_Q}) = Y_{\bullet_Q} + O(g^{16}).$$

The $O(g^{10})$ uncertainty in (99.44), rather than the logarithm square, controls the next uncomputed term. Combining (99.25), (99.44), and (99.46) gives

$$\Delta_{\mathrm{K}}^{\mathrm{wrap}} = -g^8 \sum_{Q=1}^{\infty} \int_{\mathbb{R}} \frac{\mathrm{d}u}{\pi}\, Y_Q^{(0)}(u) + O(g^{10}).$$

Set $q = 2u$ and use the weak-density rationalization above. The coefficient of g^8 becomes

$$\sum_{Q=1}^{\infty} I_Q,$$

with

$$I_Q = -\int_{-\infty}^{\infty} \frac{\mathrm{d}q}{2\pi} \frac{147456Q^2(3q^2 + 3Q^2 - 4)^2}{(q^2 + Q^2)^4 B_Q^-(q) B_Q^+(q)}, \tag{99.49}$$

where $B_Q^{\pm}$ are the polynomials in the weak-density rationalization above. By Lemma 99.11, this integral satisfies (99.48). The non-zeta part is not left as a numerical experiment: it telescopes. Define

$$F_Q = -\frac{648P(Q)}{(3Q^2+1)(3Q^2-6Q+4)(3Q^2-3Q+1)^4},$$

with

$$\begin{aligned} P(Q) = 486Q^{10} &- 2430Q^9 + 5184Q^8 - 6156Q^7 + 4752Q^6 - 2916Q^5 \\ &+ 1521Q^4 - 504Q^3 + 51Q^2 + 12Q - 2. \end{aligned}$$

A direct common-denominator calculation verifies

$$R_Q = F_Q - F_{Q+1}.$$

Moreover $F_1 = 324$ and $F_Q \to 0$ as $Q \to \infty$. Hence

$$\sum_{Q=1}^{\infty} R_Q = 324.$$

The remaining two sums in (99.48) are absolutely convergent and give $864\zeta(3) - 1440\zeta(5)$. This proves the displayed coefficient.

Combining the asymptotic and wrapping pieces gives the physical four-loop coefficient

$$-2496 + 576\zeta(3) - 1440\zeta(5).$$

The companion script `calculation-checks/planar_n4_integrability_checks.py` now checks the stringbook u-integrand against the $q = 2u$ rational form, numerically integrates the first four real-line rational integrals, and verifies the exact telescoping identity for the rational tail. The conceptual assumptions remain those isolated in Assumption 99.10; the finite residue and summation algebra is not conjectural.

99.7 Relation to Relativistic Integrable QFT

The nested TBA of Chapter 78 supplies the finite-density variational logic. The present system differs in three ways. First, the dispersion is not relativistic. Second, the mirror theory has an infinite family of bound-state and auxiliary nodes. Third, the spectral problem is a gauge-theory local-operator problem, so the circle length is the trace length L or a charge shifted from it by sector-dependent conventions. These differences are why the planar $\mathcal{N} = 4$ TBA is placed in the supersymmetric gauge-theory volume rather than in the relativistic integrable-QFT volume.

Chapter 100

Quantum Spectral Curve and Hexagon Form Factors

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The mirror TBA is exact but highly redundant. The quantum spectral curve compresses the same spectral information into a finite system of analytic functions. This chapter states the $P\mu$-system, explains its weak-coupling reduction to Baxter equations, and records the separate hexagon framework for three-point functions.

100.1 Analytic T-Hook Gauge

Before introducing the $P\mu$-system it is useful to isolate the analytic T-hook step that makes the later two-row Wronskians plausible. The input is not the full QSC; it is the analytic Y-system data of Chapter 99 together with a gauge choice for Hirota T-functions.

Assumption 100.1 (One-row analytic T-gauge)**.** Use the Hirota equation

$$T_{a,s}^{+}T_{a,s}^{-} = T_{a+1,s}T_{a-1,s} + T_{a,s+1}T_{a,s-1}, \qquad f^{\pm}(u) = f(u \pm \mathrm{i}/2).$$

In a horizontal wing of the planar $\mathcal{N}=4$ T-hook, choose a gauge with

$$\mathcal{T}_{0,m} = 1, \qquad \mathcal{T}_{1,m} \in \mathcal{A}_m, \qquad \mathcal{T}_{1,m}(u) \to m \quad (u \to \infty), \qquad m \geq 1,$$

where $\mathcal{A}_m = \{|\operatorname{Im} u| < m/2\}$ with the prescribed Zhukovsky cuts removed. Assume that the discontinuity data of the two fermionic nodes can be represented by a real density ρ through Cauchy transforms

$$G(u) = -\int \frac{\mathrm{d}v}{2\pi\mathrm{i}} \frac{\rho(v)}{u-v}, \qquad \operatorname{Im} u > 0, \qquad \overline{G}(u) = +\int \frac{\mathrm{d}v}{2\pi\mathrm{i}} \frac{\rho(v)}{u-v}, \qquad \operatorname{Im} u < 0.$$

The gauge is normalized so that

$$\mathcal{T}_{1,m} = m + G^{[m]} + \overline{G}^{[-m]}, \qquad m \geq 1. \tag{100.1}$$

This assumption is deliberately stated as an analytic reconstruction datum. It summarizes the information normally hidden in the phrase "choose the $\mathcal{T}$-gauge": asymptotics, analyticity strips, and the branch on which the Cauchy transforms are evaluated.

Under Assumption 100.1, the Hirota equation at $(1, m)$, $m \geq 2$, gives

$$\mathcal{T}_{2,m} = \left(1 + G^{[m+1]} - G^{[m-1]}\right)\left(1 + \overline{G}^{[-m-1]} - \overline{G}^{[-m+1]}\right). \tag{100.2}$$

Since $\mathcal{T}_{0,m} = 1$, Hirota gives

$$\mathcal{T}_{2,m} = \mathcal{T}_{1,m}^{+}\mathcal{T}_{1,m}^{-} - \mathcal{T}_{1,m+1}\mathcal{T}_{1,m-1}.$$

Put

$$a = G^{[m+1]}, \quad b = G^{[m-1]}, \quad c = \overline{G}^{[-m-1]}, \quad d = \overline{G}^{[-m+1]}.$$

Then the right-hand side is

$$(m + a + d)(m + b + c) - (m + 1 + a + c)(m - 1 + b + d).$$

Expanding and collecting terms gives

$$1 + a - b + c - d + (a - b)(c - d) = (1 + a - b)(1 + c - d),$$

which is (100.2).

Magic-sheet continuation algebra. Assume the one-row gauge of Assumption 100.1. Suppose that, across the mirror cut in the strip under consideration, the analytic Y-system gluing condition

$$Y_{2,\pm 2}(u + \mathrm{i}0) = \frac{1}{Y_{1,\pm 1}(u - \mathrm{i}0)}$$

imposes the Cauchy-transform boundary relation

$$G(u + \mathrm{i}0) = -\overline{G}(u - \mathrm{i}0).$$

Define the magic-sheet continuation by

$$\widehat{G} = \begin{cases} G, & \text{on the branch reached from the upper half-plane,} \\ -\overline{G}, & \text{on the branch reached from the lower half-plane.} \end{cases}$$

Then, on the continued sheet,

$$\widehat{\mathcal{T}}_{1,m} = m + \widehat{G}^{[m]} - \widehat{G}^{[-m]}, \qquad \widehat{\mathcal{T}}_{2,m} = \widehat{\mathcal{T}}_{1,1}^{[m]} \widehat{\mathcal{T}}_{1,1}^{[-m]}, \qquad m \geq 2. \tag{100.3}$$

The boundary relation says precisely that the upper branch G and the negative of the lower branch $-\overline{G}$ glue with the same boundary value across the mirror cut. Thus $\widehat{G}$ is a single continued Cauchy transform on the magic sheet; the remaining discontinuity data are carried by the Zhukovsky short cut rather than by two independent upper/lower transforms.

In the original strip,

$$\mathcal{T}_{1,m} = m + G^{[m]} + \overline{G}^{[-m]}.$$

After continuation through the mirror cut, the upper shifted term remains $\widehat{G}^{[m]}$, while the lower shifted term satisfies $\overline{G}^{[-m]} = -\widehat{G}^{[-m]}$. This gives the first formula in (100.3).

For the two-row function use the factorization (100.2). Its first factor becomes

$$1 + \widehat{G}^{[m+1]} - \widehat{G}^{[m-1]} = \widehat{\mathcal{T}}_{1,1}^{[m]},$$

and its second factor becomes

$$1 + \widehat{G}^{[-m+1]} - \widehat{G}^{[-m-1]} = \widehat{\mathcal{T}}_{1,1}^{[-m]}.$$

Multiplying the two factors gives the second formula. This calculation is local in the strip: if a pole or endpoint is crossed, the missing information is an extra source term rather than a failure of the algebra.

The hats emphasize that these are not new functions on the same sheet but analytic continuations through the Zhukovsky cut. In the strip $|\operatorname{Im} u| < m/2$, $\widehat{\mathcal{T}}_{1,m}$ agrees with the original $\mathcal{T}_{1,m}$. Outside that strip, the equality is a statement about a chosen continuation path. The $P\mu$-system below is a finite Riemann–Hilbert compression of this analytically continued T-hook data.

100.1.1 Gauge Chain and Fermionic Discontinuities

The stringbook derivation passes through two further analytic gauges before introducing P-functions. We spell out the finite algebra because the same formulae fix the signs in the $Y_{1,1}Y_{2,2}$ product and in the large-u energy exponent.

Assumption 100.2 (Fermionic $\mathbf{T}$-gauge and $\mathbb{T}$-gauge)**.** Choose a Hirota gauge $\mathbf{T}_{a,s}$ with the same Y-functions as Chapter 99. For $n \geq 1$, its central analyticity strips satisfy

$$\mathbf{T}_{n-1,0}, \quad \mathbf{T}_{n,\pm 1}, \quad \mathbf{T}_{n+1,\pm 2} \in \mathcal{A}_n.$$

The boundary and left-right identifications are

$$\mathbf{T}_{n,\pm 2} = \mathbf{T}_{2,\pm n} \quad (n \geq 3), \qquad \mathbf{T}_{0,0}^{[+1]} = \mathbf{T}_{0,0}^{[-1]}, \qquad \mathbf{T}_{0,s} = \mathbf{T}_{0,0}^{[s]}.$$

On a patch where the indicated square roots are fixed, define

$$\mathbb{T}_{a,s} = (-1)^{as} \mathbf{T}_{a,s} \left(\mathbf{T}_{0,0}^{[a+s]} \right)^{(a-2)/2}. \tag{100.4}$$

Then $\mathbb{T}_{0,s} = 1$, and the magic-sheet continuations obey

$$\widehat{\mathbb{T}}_{1,m} = \widehat{h}^{[m]}\widehat{h}^{[-m]}\widehat{\mathcal{T}}_{1,m}, \tag{100.5}$$
$$\widehat{\mathbb{T}}_{2,m} = \widehat{h}^{[m+1]}\widehat{h}^{[m-1]}\widehat{h}^{[-m+1]}\widehat{h}^{[-m-1]}\widehat{\mathcal{T}}_{2,m}, \qquad m \geq 2, \tag{100.6}$$

for a function $\widehat{h}$ with only the central short cut in this gauge.

The assumption records an analytic gauge choice, not a new dynamical equation. The gauge transformation leaves the local Y-functions unchanged; its role is to put the central row and the two horizontal wings in a form adapted to single-short-cut Wronskians.

Fermionic product and central-row telescope. In the $\mathbf{T}$-gauge of Assumption 100.2,

$$Y_{1,1}Y_{2,2} = \frac{\mathbf{T}_{1,0}}{\mathbf{T}_{0,0}^{[+1]}}. \tag{100.7}$$

Moreover, for every $n \geq 1$,

$$\prod_{a=1}^{n}\left(1 + Y_{a,0}^{[2n-a]}\right) = \frac{\mathbf{T}_{1,0}^{[2n]}\mathbf{T}_{n,0}^{[n-1]}}{\mathbf{T}_{0,0}^{[2n-1]}\mathbf{T}_{n+1,0}^{[n]}}. \tag{100.8}$$

Consequently the analytic-Y-system discontinuity input

$$\mathrm{Disc}\,(\log Y_{1,1}Y_{2,2})^{[2n]} = \sum_{a=1}^{n}\mathrm{Disc}\log\left(1 + Y_{a,0}^{[2n-a]}\right)$$

is equivalently written as

$$\mathrm{Disc}\,(\log Y_{1,1}Y_{2,2})^{[2n]} = \mathrm{Disc}\log\frac{\mathbf{T}_{1,0}^{[2n]}\mathbf{T}_{n,0}^{[n-1]}}{\mathbf{T}_{0,0}^{[2n-1]}\mathbf{T}_{n+1,0}^{[n]}}. \tag{100.9}$$

Use the local T-system definition

$$Y_{a,s} = \frac{\mathbf{T}_{a,s+1}\mathbf{T}_{a,s-1}}{\mathbf{T}_{a+1,s}\mathbf{T}_{a-1,s}}.$$

Then

$$Y_{1,1}Y_{2,2} = \frac{\mathbf{T}_{1,2}\mathbf{T}_{1,0}}{\mathbf{T}_{2,1}\mathbf{T}_{0,1}}\frac{\mathbf{T}_{2,3}\mathbf{T}_{2,1}}{\mathbf{T}_{3,2}\mathbf{T}_{1,2}} = \frac{\mathbf{T}_{1,0}\mathbf{T}_{2,3}}{\mathbf{T}_{0,1}\mathbf{T}_{3,2}}.$$

The $\mathbf{T}$-gauge identifications give $\mathbf{T}_{3,2} = \mathbf{T}_{2,3}$ and $\mathbf{T}_{0,1} = \mathbf{T}_{0,0}^{[+1]}$, proving (100.7).

For the telescope, Hirota gives

$$1 + Y_{a,0} = \frac{\mathbf{T}_{a,0}^{+}\mathbf{T}_{a,0}^{-}}{\mathbf{T}_{a+1,0}\mathbf{T}_{a-1,0}}.$$

After the shift by $[2n-a]$, the a-th factor is

$$\frac{\mathbf{T}_{a,0}^{[2n-a+1]}\mathbf{T}_{a,0}^{[2n-a-1]}}{\mathbf{T}_{a+1,0}^{[2n-a]}\mathbf{T}_{a-1,0}^{[2n-a]}}.$$

Multiplying from $a = 1$ to $a = n$, every interior $\mathbf{T}_{b,0}^{[2n-b+1]}$ and $\mathbf{T}_{b,0}^{[2n-b-1]}$ with $2 \le b \le n-1$ appears once in the numerator and once in the denominator. The surviving boundary terms are $\mathbf{T}_{1,0}^{[2n]}$, $\mathbf{T}_{n,0}^{[n-1]}$, $\mathbf{T}_{0,0}^{[2n-1]}$, and $\mathbf{T}_{n+1,0}^{[n]}$, which is (100.8). Taking logarithmic discontinuities along the central cut gives (100.9). The first displayed discontinuity relation is the analytic-Y-system input from the mirror TBA; the telescope above supplies its T-gauge form.

$\mathbb{T}$-gauge compatibility. Under Assumption 100.2 and the magic-sheet continuation algebra above,

$$\widehat{\mathbb{T}}_{2,m} = \widehat{\mathbb{T}}_{1,1}^{[m]}\widehat{\mathbb{T}}_{1,1}^{[-m]}, \qquad m \ge 2.$$

Thus the two-row Wronskian reconstruction used below may be normalized in the $\mathbb{T}$-gauge without changing the one-row magic-sheet algebra.

Substitute $\widehat{\mathcal{T}}_{2,m} = \widehat{\mathcal{T}}_{1,1}^{[m]}\widehat{\mathcal{T}}_{1,1}^{[-m]}$ from (100.3) into (100.6). The four $\widehat{h}$-factors become

$$\widehat{h}^{[m+1]}\widehat{h}^{[m-1]}\widehat{h}^{[-m+1]}\widehat{h}^{[-m-1]}.$$

Using (100.5) at arguments $u + \mathrm{i}m/2$ and $u - \mathrm{i}m/2$ gives

$$\widehat{\mathbb{T}}_{1,1}^{[m]} = \widehat{h}^{[m+1]}\widehat{h}^{[m-1]}\widehat{\mathcal{T}}_{1,1}^{[m]}, \qquad \widehat{\mathbb{T}}_{1,1}^{[-m]} = \widehat{h}^{[-m+1]}\widehat{h}^{[-m-1]}\widehat{\mathcal{T}}_{1,1}^{[-m]}.$$

Multiplication gives the claim. The statement is local in the chosen analytic gauge; zeros of $\widehat{h}$ or Wronskian degeneracies are handled by analytic continuation on the neighboring patch.

100.2 Pmu System

Let $u \in \mathbb{C}$ be the spectral parameter. The quantum spectral curve uses four functions $P_a(u)$, $a = 1, \ldots, 4$, and an antisymmetric matrix $\mu_{ab}(u) = -\mu_{ba}(u)$. The functions P_a have one short branch cut, conventionally on $[-2g, 2g]$, on their principal sheet. Analytic continuation through this cut is denoted by a tilde. Introduce

$$P^a = \chi^{ab} P_b,$$

where χ^{ab} is the fixed antisymmetric tensor with nonzero entries chosen so that $P_a P^a = 0$.

The $P\mu$-system is

$$\widetilde{P}_a(u) = \mu_{ab}(u) P^b(u), \tag{100.10}$$

$$\widetilde{\mu}_{ab}(u) - \mu_{ab}(u) = P_a(u)\widetilde{P}_b(u) - P_b(u)\widetilde{P}_a(u), \tag{100.11}$$

$$\mu_{ab}(u + \mathrm{i}) = \widetilde{\mu}_{ab}(u). \tag{100.12}$$

The normalization is completed by the Pfaffian condition

$$\operatorname{Pf}\mu = \mu_{12}\mu_{34} - \mu_{13}\mu_{24} + \mu_{14}\mu_{23} = 1, \tag{100.13}$$

up to the standard P-basis gauge transformations. The tilde operation is analytic continuation through the short cut of the corresponding function; it is not complex conjugation. On real states one imposes an additional reality or gluing condition relating the two sheets.

Definition 100.3 (QSC Riemann–Hilbert datum)**.** Within the planar integrability framework, a $P\mu$ Riemann–Hilbert datum consists of the spectral plane with parameter $u \in \mathbb{C}$, four single-short-cut functions P_a, an antisymmetric matrix μ_{ab}, the fixed antisymmetric tensor χ^{ab}, and the continuation operation $\widetilde{\ }$ through the short cut $[-2g, 2g]$. The datum is required to satisfy (100.10)–(100.12), (100.13), regularity on the defining sheet, and large-u asymptotics specifying the global charges. A physical solution adds state-dependent gluing conditions and cyclic single-trace constraints.

This is a Riemann–Hilbert problem, not a finite collection of ordinary functions on one copy of the plane. The data above tell the reader both what is evaluated on the defining sheet and how the functions are continued to the neighboring sheets.

Assumption 100.4 (Two-row T-hook reconstruction datum)**.** Use the shift notation

$$f^{[n]}(u) = f(u + n\mathrm{i}/2).$$

In the physical or magic T-hook gauge, assume that the two horizontal rows adjacent to the central node are reconstructed from two single-short-cut functions P_1, P_2 by

$$\mathbb{T}_{1,m} = P_1^{[m]} P_2^{[-m]} - P_2^{[m]} P_1^{[-m]}, \qquad m \geq 1, \tag{100.14}$$

and that Hirota continuation extends this expression to the central nodes. Let

$$\mathbb{T}_{0,1} = \mu_{12}^2, \qquad \mathbb{T}_{3,2} = \mathbb{T}_{2,3}\mu_{12}.$$

The functions $\mathbb{T}_{2,1}$ and $\mathbb{T}_{1,0}$ are assumed to be regular across the central short cut in the strip where the following calculation is made. Finally, the Wronskian

$$D = P_1^{[+2]} P_2^{[-2]} - P_2^{[+2]} P_1^{[-2]}$$

is not identically zero; any isolated zeros are handled by analytic continuation of the resulting identities.

Local T-hook algebra behind the $P\mu$ bridge. Under Assumption 100.4, regularity of the T-hook functions across the central short cut forces

$$\widetilde{\mu}_{12} - \mu_{12} = P_1 \widetilde{P}_2 - P_2 \widetilde{P}_1. \tag{100.15}$$

Moreover

$$\mathbb{T}_{1,0} = \mu_{12}\widetilde{\mu}_{12}, \qquad Y_{1,1}Y_{2,2} = \frac{\widetilde{\mu}_{12}}{\mu_{12}} = 1 + \frac{P_1\widetilde{P}_2 - P_2\widetilde{P}_1}{\mu_{12}}. \tag{100.16}$$

The preceding assumption is the nontrivial analytic input: it asserts that a two-row Wronskian reconstruction by single-short-cut P-functions exists in the strip, and that the relevant central T-hook nodes are regular there. The displayed formulae below are the local Pluecker calculation extracted from that input.

Write $P_a^+ = P_a^{[+2]}$, $P_a^- = P_a^{[-2]}$, and

$$D = P_1^+ P_2^- - P_2^+ P_1^-.$$

The Hirota equation at the $(2,2)$ square, expressed through the Wronskians in (100.14), gives in the upper half-plane sector

$$\mathbb{T}_{2,1} = (\widetilde{P}_1 P_2^- - \widetilde{P}_2 P_1^-)(P_1^+ P_2 - P_2^+ P_1) - \mu_{12} D.$$

Continuing across the central cut replaces P_1, P_2, μ_{12} in the unshifted slot by their tilded values, so the same node is

$$\widetilde{\mathbb{T}}_{2,1} = (P_1 P_2^- - P_2 P_1^-)(P_1^+ \widetilde{P}_2 - P_2^+ \widetilde{P}_1) - \widetilde{\mu}_{12} D.$$

Since $\mathbb{T}_{2,1}$ is regular in this strip, these two expressions are equal. Moving the μ-terms to one side gives

$$(\mu_{12} - \widetilde{\mu}_{12}) D = A - B,$$

where

$$A = (\widetilde{P}_1 P_2^- - \widetilde{P}_2 P_1^-)(P_1^+ P_2 - P_2^+ P_1),$$
$$B = (P_1 P_2^- - P_2 P_1^-)(P_1^+ \widetilde{P}_2 - P_2^+ \widetilde{P}_1).$$

Expanding both products and canceling pairwise gives the Pluecker identity

$$A - B = (\widetilde{P}_1 P_2 - \widetilde{P}_2 P_1) D.$$

Because D is not identically zero, division by D on the open set where it is nonzero gives

$$\mu_{12} - \widetilde{\mu}_{12} = \widetilde{P}_1 P_2 - \widetilde{P}_2 P_1,$$

and analyticity extends the identity through isolated zeros of D. This is (100.15).

The $(1,1)$ Hirota square gives the central-node relation

$$\frac{\mathbb{T}_{1,0}}{\mu_{12}} = \mu_{12} + \widetilde{P}_2 P_1 - \widetilde{P}_1 P_2.$$

Using the discontinuity identity just proved, the right side is $\widetilde{\mu}_{12}$, hence $\mathbb{T}_{1,0} = \mu_{12}\widetilde{\mu}_{12}$. In this gauge

$$Y_{1,1} Y_{2,2} = \frac{\mathbb{T}_{1,0}}{\mathbb{T}_{0,1}} = \frac{\mu_{12}\widetilde{\mu}_{12}}{\mu_{12}^2},$$

which proves the first equality in (100.16). The second equality is the same statement after substituting (100.15).

Fermionic-node asymptotics. Let $x_{\mathrm{f}}(u)$ be the fermionic-node Zhukovsky sheet used in the large-u expansion of $Y_{1,1}Y_{2,2}$, normalized by

$$x_{\mathrm{f}}(u) = \frac{g}{u} + O(u^{-3}), \qquad u \to \infty.$$

For a mirror Q-particle at rapidity v, write

$$a = x_Q^+(v), \qquad b = x_Q^-(v), \qquad |a| < 1 < |b|, \qquad \widetilde{E}_Q(v) = \log \frac{b}{a},$$

with the logarithm on the real mirror branch, and use the mirror momentum normalization of Chapter 99,

$$\mathrm{i}\widetilde{p}_Q(v) = -Q - 2\mathrm{i}g(a - b). \tag{100.17}$$

Define the fermionic-node scattering-ratio factor

$$R_Q(u,v) = \frac{x_{\mathrm{f}}(u) - a}{x_{\mathrm{f}}(u) - b}\,\frac{x_{\mathrm{f}}(u)^{-1} - b}{x_{\mathrm{f}}(u)^{-1} - a}.$$

On the branch of $\log R_Q$ which tends to $\log(a/b)$ at $u = \infty$,

$$\log R_Q(u,v) = -\widetilde{E}_Q(v) - \frac{\widetilde{p}_Q(v)}{u} + O(u^{-2}). \tag{100.18}$$

After the inverse mirror continuation of the second particle, $\widetilde{E} = -\mathrm{i}p$, $\widetilde{p} = -\mathrm{i}E$, the same local expansion becomes

$$\log R_1^{\mathrm{phys}}(u,u_j) = \mathrm{i}p(u_j) + \frac{\mathrm{i}E(u_j)}{u} + O(u^{-2}).$$

Indeed, set $x = x_{\mathrm{f}}(u)$. Since $x = g/u + O(u^{-3})$,

$$\frac{x-a}{x-b} = \frac{a}{b}\left(1 + x\left(\frac{1}{b} - \frac{1}{a}\right) + O(x^2)\right), \qquad \frac{x^{-1}-b}{x^{-1}-a} = 1 + x(a-b) + O(x^2).$$

Multiplying,

$$R_Q(u,v) = \frac{a}{b}\left[1 + \frac{g}{u}\left(a - b + \frac{1}{b} - \frac{1}{a}\right) + O(u^{-2})\right].$$

The constant term gives $\log(a/b) = -\widetilde{E}_Q$. For the coefficient of u^{-1}, use the bound-state shortening condition

$$a + \frac{1}{a} - b - \frac{1}{b} = \frac{\mathrm{i}Q}{g}.$$

It implies

$$g\left(a - b + \frac{1}{b} - \frac{1}{a}\right) = 2g(a-b) - \mathrm{i}Q = -\widetilde{p}_Q,$$

where the last equality is (100.17). Taking the logarithm gives (100.18). The physical expansion is the same identity written with the stringbook double-Wick convention $\widetilde{E} = -\mathrm{i}p$, $\widetilde{p} = -\mathrm{i}E$.

Assumption 100.5 (Fermionic-product energy normalization)**.** Work in the analytic T-hook gauge of Chapter 99, and consider an $SL(2)$ single-trace state of twist J, Lorentz spin S, and total physical momentum zero. Let $Y_{1,1}$ and $Y_{2,2}$ denote the two fermionic T-hook nodes adjacent to the central strip in this gauge. The local reconstruction algebra in the local T-hook algebra above supplies the $P\mu$ expression for $Y_{1,1}Y_{2,2}$. The remaining framework input is the energy normalization inherited from the excited-state mirror TBA:

$$\log\bigl(Y_{1,1}(u)Y_{2,2}(u)\bigr) = \frac{\mathrm{i}(\Delta - J)}{u} + O(u^{-2}), \qquad u \to \infty$$

in the upper-half-plane sector used for the $SL(2)$ large-u asymptotics. The fermionic-node asymptotics calculation above is the local sheet calculation behind this normalization: in the standard excited-state mirror-TBA formula, the fermionic product is assembled from the ratio R_Q, and its u^{-1} coefficient is the same mirror-momentum functional that appears in the energy formula (99.25). The assumption that remains is the global TBA contour formula for $Y_{1,1}Y_{2,2}$, not an independent sign choice in the QSC.

Dimension exponent of μ_{12}. Under Assumption 100.5 and the $P\mu$ equations,

$$Y_{1,1}(u)Y_{2,2}(u) = \frac{\mu_{12}(u+\mathrm{i})}{\mu_{12}(u)}. \tag{100.19}$$

If μ_{12} has power-law large-u behavior on this sheet, then

$$\mu_{12}(u) \sim u^{\Delta-J}.$$

The $(1,2)$ component of the $P\mu$ discontinuity equation gives

$$\widetilde{\mu}_{12}(u) - \mu_{12}(u) = P_1(u)\widetilde{P}_2(u) - P_2(u)\widetilde{P}_1(u).$$

Dividing by $\mu_{12}(u)$ and using the bridge formula (100.16) gives

$$Y_{1,1}(u)Y_{2,2}(u) = \frac{\widetilde{\mu}_{12}(u)}{\mu_{12}(u)}.$$

Pseudo-periodicity (100.12) then gives (100.19). Now suppose $\mu_{12}(u) = Cu^{\alpha}(1+O(u^{-1}))$ with $C \neq 0$. Then

$$\log\frac{\mu_{12}(u+\mathrm{i})}{\mu_{12}(u)} = \alpha\log\left(1+\frac{\mathrm{i}}{u}\right) + O(u^{-2}) = \frac{\mathrm{i}\alpha}{u} + O(u^{-2}).$$

Comparison with the large-u energy normalization in Assumption 100.5 gives $\alpha = \Delta - J$.

State quantum numbers also enter through large-u asymptotics of P_a and the remaining μ_{ab}. In the $SL(2)$ sector with twist J, spin S, and dimension Δ, these asymptotics encode the charges

$$(J, S, \Delta)$$

through powers and leading coefficients constrained by the algebraic relations of the QSC. With the $SL(2)$ conventions used in the previous chapter, the large-u powers of the P-functions are

$$(P_1, P_2, P_3, P_4) \sim (A_1u^{-J/2}, A_2u^{-J/2-1}, A_3u^{J/2}, A_4u^{J/2-1}).$$

Together with the dimension-exponent calculation above, the leading terms in

$$\widetilde{P}_a = \mu_{ab}P^b, \qquad (P^1, P^2, P^3, P^4) = (-P_4, P_3, -P_2, P_1)$$

then force the remaining μ-powers. For the characteristic calculation one should not insert the physical value of the exponent too early: keep a trial exponent α and write

$$(\mu_{12}, \mu_{13}, \mu_{14}, \mu_{23}, \mu_{24}, \mu_{34}) \sim (u^{\alpha-J}, u^{\alpha+1}, u^{\alpha}, u^{\alpha}, u^{\alpha-1}, u^{\alpha+J}), \tag{100.20}$$

provided the leading terms do not undergo accidental cancellations. The physical branch later has $\alpha = \Delta$.

Large-u power balance for the $P\mu$ sheet equation. Fix the $SL(2)$ P-powers displayed above and use $(P^1,P^2,P^3,P^4)=(-P_4,P_3,-P_2,P_1)$. Suppose that in the trial large-u sector $\widetilde{P}_a$ has the common sheet exponent shift

$$\widetilde{P}_a=O(u^{\alpha+p_a}),\qquad (p_1,p_2,p_3,p_4)=\left(-\frac{J}{2},-\frac{J}{2}-1,\frac{J}{2},\frac{J}{2}-1\right),$$

and that the μ_{12} exponent is $\alpha-J$. If no leading row of $\widetilde{P}_a=\mu_{ab}P^b$ is cancelled identically, then the six μ-exponents are exactly those in (100.20). With these exponents, every term kept in the leading monodromy system (100.22) has the same power as $\mu_{ab}(u+\mathrm{i})-\mu_{ab}(u)$ in its component.

Let $e_{ab}=e_{ba}$ be the large-u exponent of μ_{ab}, and let

$$(q_1,q_2,q_3,q_4)=\left(\frac{J}{2}-1,\frac{J}{2},-\frac{J}{2}-1,-\frac{J}{2}\right)$$

be the powers of (P^1,P^2,P^3,P^4). The row $a=1$ in $\widetilde{P}_a=\mu_{ab}P^b$ has target exponent $\alpha-J/2$. Since $e_{12}=\alpha-J$,

$$e_{12}+q_2=\alpha-\frac{J}{2},\qquad e_{13}+q_3=\alpha-\frac{J}{2},\qquad e_{14}+q_4=\alpha-\frac{J}{2},$$

and hence $e_{13}=\alpha+1$, $e_{14}=\alpha$. Applying the same balance to the second row, whose target exponent is $\alpha-J/2-1$, gives

$$e_{23}+q_3=\alpha-\frac{J}{2}-1,\qquad e_{24}+q_4=\alpha-\frac{J}{2}-1,$$

so $e_{23}=\alpha$ and $e_{24}=\alpha-1$. The third row has target $\alpha+J/2$, and therefore

$$e_{34}+q_4=\alpha+\frac{J}{2},$$

which gives $e_{34}=\alpha+J$. The fourth row gives the same value and is a consistency check. These are precisely the powers in (100.20).

For the monodromy recursion, the left side of the (a,b) component has power $e_{ab}-1$, since

$$\mu_{ab}(u+\mathrm{i})-\mu_{ab}(u)=\mathrm{i}e_{ab}m_{ab}u^{e_{ab}-1}+\cdots.$$

Each retained term on the right is of the form $P_a\mu_{bc}P^c$ or $P_b\mu_{ac}P^c$. The row balance just proved says

$$e_{bc}+q_c=\alpha+p_b,\qquad e_{ac}+q_c=\alpha+p_a.$$

Thus these terms have powers $p_a+\alpha+p_b$ and $p_b+\alpha+p_a$. A direct substitution of the exponents above gives

$$p_a+p_b+\alpha=e_{ab}-1$$

for the six pairs $(12),(13),(14),(23),(24),(34)$. Hence the leading linear system is homogeneous in powers of u, as required.

Large-u characteristic determinant. Let

$$X = A_1 A_4, \qquad Y = A_2 A_3.$$

Insert the $SL(2)$ P-powers and the trial μ-powers (100.20) into the one-step monodromy recursion (100.23). For a nonzero leading vector with $\alpha \neq 0$, the characteristic equation is

$$\Phi(\alpha) = (J+1)^2\alpha^2 - 4\mathrm{i}J(J+1)Y - \Big(\alpha^2 + J - \mathrm{i}(J+1)Y + \mathrm{i}(J-1)X\Big)^2 = 0. \tag{100.21}$$

Write

$$P_1 = A_1 u^{-J/2}, \quad P_2 = A_2 u^{-J/2-1}, \quad P_3 = A_3 u^{J/2}, \quad P_4 = A_4 u^{J/2-1}$$

at leading order, and use $(P^1, P^2, P^3, P^4) = (-P_4, P_3, -P_2, P_1)$. If $\mu_{ab} = m_{ab} u^{e_{ab}} + \cdots$, then $\mu_{ab}(u+\mathrm{i}) - \mu_{ab}(u) = \mathrm{i}e_{ab}m_{ab}u^{e_{ab}-1} + \cdots$. Keeping the leading power in each component gives the finite linear system

$$\begin{aligned}
\mathrm{i}(\alpha - J)m_{12} &= (X-Y)m_{12} + A_2^2 m_{13} - A_1A_2 m_{14} - A_1A_2 m_{23} + A_1^2 m_{24},\\
\mathrm{i}(\alpha+1)m_{13} &= -A_3^2 m_{12} + (X+Y)m_{13} - A_1A_3 m_{14} - A_1A_3 m_{23} + A_1^2 m_{34},\\
\mathrm{i}\alpha m_{14} &= -A_3A_4 m_{12} + A_2A_4 m_{13} - A_1A_3 m_{24} + A_1A_2 m_{34},\\
\mathrm{i}\alpha m_{23} &= -A_3A_4 m_{12} + A_2A_4 m_{13} - A_1A_3 m_{24} + A_1A_2 m_{34},\\
\mathrm{i}(\alpha-1)m_{24} &= -A_4^2 m_{12} + A_2A_4 m_{14} + A_2A_4 m_{23} - (X+Y)m_{24} + A_2^2 m_{34},\\
\mathrm{i}(\alpha+J)m_{34} &= -A_4^2 m_{13} + A_3A_4 m_{14} + A_3A_4 m_{23} - A_3^2 m_{24} + (-X+Y)m_{34}.
\end{aligned} \tag{100.22}$$

Let $\mathcal{M}_\alpha$ be the coefficient matrix obtained by moving the right-hand side to the left. Expanding the determinant and collecting only the invariant products $X = A_1A_4$ and $Y = A_2A_3$ gives

$$\det \mathcal{M}_\alpha = \alpha^2 \Phi(\alpha).$$

The factor α^2 records a degenerate zero-exponent subspace of this redundant six-coordinate leading system before imposing the Pfaffian and gauge reductions. The nonzero characteristic exponents used below therefore obey $\Phi(\alpha) = 0$.

Assumption 100.6 (Physical QSC characteristic roots)**.** For an $SL(2)$ single-trace state of twist J, Lorentz spin S, and dimension Δ, the physical large-u gluing/asymptotic data select, after quotienting the redundant α^2 factor above, the roots

$$\alpha = \Delta, \qquad \alpha = S - 1$$

of the characteristic polynomial (100.21); cases in which the spin-shadow root coincides with the zero sector are understood by continuity.

This root assignment is not a path-integral theorem. It is the large-u representation-theory and QSC-gluing input in this gauge. The local algebra above is included because it is a common source of sign errors: the products A_1A_4 and A_2A_3 carry the sign convention for χ, for $f^{[\pm 1]}(u) = f(u \pm \mathrm{i}/2)$, and for the short-cut sheet.

Large-u coefficient products. Under Assumption 100.6, the leading coefficient products are

$$A_1A_4 = \mathrm{i}\,\frac{((J+S-2)^2-\Delta^2)((J-S)^2-\Delta^2)}{16J(J-1)},$$
$$A_2A_3 = \mathrm{i}\,\frac{((J-S+2)^2-\Delta^2)((J+S)^2-\Delta^2)}{16J(J+1)}.$$

Put $\beta = S-1$. Since $\Phi(\Delta) = \Phi(\beta) = 0$, subtracting the two equations cancels the term proportional to Y and gives

$$\Delta^2+\beta^2+2J+2U = (J+1)^2, \qquad U = -\mathrm{i}(J+1)Y+\mathrm{i}(J-1)X,$$

as long as $\Delta^2 \neq \beta^2$; the degenerate cases are obtained by continuity. Thus

$$U = \frac{J^2+1-\Delta^2-\beta^2}{2}.$$

Substituting this value in $\Phi(\Delta)=0$ gives

$$4\mathrm{i}J(J+1)Y = (J+1)^2\Delta^2 - \frac{1}{4}\left(\Delta^2+(J+1)^2-\beta^2\right)^2.$$

Now

$$(J+1)^2-\beta^2 = (J-S+2)(J+S),$$

and the elementary factorization

$$4(J+1)^2\Delta^2 - \left(\Delta^2+(J-S+2)(J+S)\right)^2 = -\left(\Delta^2-(J-S+2)^2\right)\left(\Delta^2-(J+S)^2\right)$$

gives the displayed formula for $A_2A_3 = Y$. The formula for $A_1A_4 = X$ follows from

$$\mathrm{i}(J-1)X = U + \mathrm{i}(J+1)Y$$

and the same factorization with $(J-S+2, J+S)$ replaced by $(J-S, J+S-2)$.

Consistency of the $P\mu$ discontinuity. The discontinuity equation (100.11) preserves the Pfaffian normalization (100.13) if the P-functions satisfy

$$P_aP^a = 0.$$

Use the rank-two Pfaffian identity

$$\mathrm{Pf}(M+v\wedge w) = \mathrm{Pf}(M)\left(1+w^TM^{-1}v\right)$$

for an invertible antisymmetric 4×4 matrix M. Here $M=\mu$, $v=P$, and $w=\widetilde{P}=\mu P^\sharp$, where $(P^\sharp)^a = P^a = \chi^{ab}P_b$. Hence

$$w^T\mu^{-1}v = (\mu P^\sharp)^T\mu^{-1}P = -(P^\sharp)^TP = -P^aP_a = 0,$$

because χ^{ab} is antisymmetric. Therefore crossing the cut does not change $\mathrm{Pf}\,\mu$, so the normalization can be imposed on one sheet and transported consistently.

Monodromy recursion for μ. The $P\mu$ equations imply the one-step continuation formula

$$\mu_{ab}(u+\mathrm{i}) = \mu_{ab}(u) + P_a(u)\mu_{bc}(u)P^c(u) - P_b(u)\mu_{ac}(u)P^c(u). \tag{100.23}$$

Consequently, once the functions in one strip and the continued P-sheet are known, the shifted μ-sheets are generated recursively by algebraic substitution.

Use the pseudo-periodicity equation (100.12) and the discontinuity equation (100.11):

$$\mu_{ab}(u+\mathrm{i}) - \mu_{ab}(u) = P_a(u)\widetilde{P}_b(u) - P_b(u)\widetilde{P}_a(u).$$

Substituting $\widetilde{P}_b = \mu_{bc}P^c$ from (100.10) gives (100.23). Applying the same identity at successively shifted arguments gives the recursion through the tower of short cuts. The recursion is algebraic; it does not by itself prove that the regularity, gluing, and large-u conditions select a unique physical solution.

Quoted Theorem 100.7 (Quantum spectral curve spectral problem)**.** *Within the planar $\mathcal{N} = 4$ integrability framework, a physical single-trace state is specified by QSC analyticity, gluing, asymptotic, and regularity data. Solving the resulting $P\mu$-system determines the exact planar scaling dimension of the state as a function of λ.*

The theorem is quoted with its framework stated. The QSC is a powerful exact solution of the planar integrability system; it is not presently a constructive definition of four-dimensional $\mathcal{N} = 4$ SYM. In this chapter, the algebraic consequences of the $P\mu$ equations are derived when feasible: Pfaffian preservation, monodromy recursion, the weak-coupling Baxter degeneration, and the one-loop dimension extraction below. The existence of the single-short-cut P-functions, the normalization of the discontinuity coefficient in (100.11), the physical gluing conditions, and the equivalence to the exact planar gauge-theory spectrum remain QSC framework inputs.

100.2.1 TBA-to-QSC Compression Record

The mirror TBA, the analytic Y-system, the T-hook gauges, and the QSC are not four independent spectral claims. They are successive ways of packaging the same planar-integrability data after extra analytic information has been attached. The local Y-system by itself remembers Hirota-gauge invariant ratios, but it does not remember the source factors lost by the inverse s-kernel, the sheet on which the central fermionic discontinuity is read, or the large-u charge normalization. The $P\mu$-system is the finite Riemann–Hilbert package obtained only after those pieces have been restored.

Proposition 100.8 (TBA-to-QSC charge record)**.** *Fix an $SL(2)$ single-trace state in the analytic framework used above. Assume that the mirror-TBA analytic Y-system data of Assumption 99.9, the T-gauge chain of Assumption 100.2, and the local two-row $P\mu$ bridge of Assumption 100.4 have all been fixed. Suppose further that*

$$\log\bigl(Y_{1,1}(u)Y_{2,2}(u)\bigr) = \frac{\mathrm{i}r}{u} + O(u^{-2}), \qquad u \to \infty,$$

and that the P-function powers are

$$(p_1, p_2, p_3, p_4) = \left(-\frac{J}{2}, -\frac{J}{2} - 1, \frac{J}{2}, \frac{J}{2} - 1\right), \qquad P_a(u) = O(u^{p_a}).$$

If the physical gluing/asymptotic data select the nonzero characteristic roots $\alpha = \Delta$ and $\beta = S-1$ in Assumption 100.6, then the bridge recovers the charge triple by the finite record

$$J = p_3 - p_1 = p_4 - p_2, \qquad \Delta = J + r, \qquad S = \beta + 1. \tag{100.24}$$

Consequently the symbolic Y-system plus Hirota algebra is not, by itself, the QSC spectral problem: the fermionic large-u coefficient, the P-power gap, and the physical characteristic-root assignment are all separate inputs.

Proof. The local T-hook calculation above gives

$$Y_{1,1}Y_{2,2} = \frac{\widetilde{\mu}_{12}}{\mu_{12}} = \frac{\mu_{12}(u+\mathrm{i})}{\mu_{12}(u)}.$$

If $\mu_{12}(u) = Cu^{\gamma}(1 + O(u^{-1}))$, then

$$\log \frac{\mu_{12}(u+\mathrm{i})}{\mu_{12}(u)} = \frac{\mathrm{i}\gamma}{u} + O(u^{-2}).$$

Comparison with the assumed fermionic-product expansion gives $\gamma = r$. In the physical branch the dimension-exponent calculation above identifies $\gamma = \Delta - J$, hence $\Delta = J + r$.

The P-powers independently recover the twist:

$$p_3 - p_1 = \frac{J}{2} - \left(-\frac{J}{2}\right) = J, \qquad p_4 - p_2 = \frac{J}{2} - 1 - \left(-\frac{J}{2} - 1\right) = J.$$

Finally the physical characteristic-root input identifies the spin-shadow root $\beta = S - 1$, so $S = \beta + 1$. These three finite readings give (100.24). Omitting any of them leaves visible ambiguity: r is unchanged under the simultaneous shift $(J, \Delta) \mapsto (J + c, \Delta + c)$, the local Y-functions are invariant under T-gauge changes, and the unordered characteristic roots do not distinguish the dimension root from the spin-shadow root before the gluing/representation assignment is supplied. □

100.3 Q-Omega System and Gauge Choices

The $P\mu$-system is one gauge for the QSC. A dual description uses four functions $Q_i(u)$, $i = 1, \ldots, 4$, and an antisymmetric matrix $\omega_{ij}(u)$. In the standard convention the Q_i have long cuts, while the P_a have one short cut on the principal sheet. The equations mirror the $P\mu$ system. Introduce

$$Q^i = \eta^{ij} Q_j,$$

where η^{ij} is a fixed antisymmetric tensor, normalized in the same unimodular convention as the P-index tensor χ^{ab}. Then

$$\widetilde{Q}_i = \omega_{ij} Q^j, \qquad \widetilde{\omega}_{ij} - \omega_{ij} = Q_i \widetilde{Q}_j - Q_j \widetilde{Q}_i, \qquad \omega_{ij}(u + \mathrm{i}) = \widetilde{\omega}_{ij}(u).$$

The dual Pfaffian normalization is

$$\operatorname{Pf} \omega = \omega_{12}\omega_{34} - \omega_{13}\omega_{24} + \omega_{14}\omega_{23} = 1$$

after the same class of Q-basis gauge transformations used below. The two descriptions are related by the full $Q_{A|I}$ system of finite-difference relations. The useful point for this monograph is not the formal symmetry itself but the bookkeeping: choosing short cuts for P_a and long cuts for Q_i is an analytic gauge. Physical charges are read from large-u asymptotics only after this gauge and the gluing conditions are fixed.

For real left-right symmetric states, the gluing conditions may be written, after a basis choice, as relations between $\widetilde{P}_a(u)$ and complex conjugates of $P_a(u)$ on the real axis outside the cut. These conditions select the physical solution among many formal solutions of the same finite-difference equations. They play the same role as contour data in the mirror TBA.

Dual $Q\omega$ transport. Assume the local $Q\omega$ equations displayed above and the null contraction $Q_iQ^i = 0$. Then, as a local algebraic consequence of these equations, the ω-Pfaffian is transported consistently through a long cut, and the one-step shifted sheet obeys

$$\omega_{ij}(u + \mathrm{i}) = \omega_{ij}(u) + Q_i(u)\omega_{jk}(u)Q^k(u) - Q_j(u)\omega_{ik}(u)Q^k(u). \tag{100.25}$$

Write

$$\widetilde{Q}_i = \omega_{ij}Q^j.$$

The discontinuity equation says that crossing the cut changes ω by the rank-two antisymmetric update

$$\omega \longmapsto \omega + Q \wedge \widetilde{Q}.$$

For an invertible antisymmetric 4×4 matrix M,

$$\mathrm{Pf}(M + v \wedge w) = \mathrm{Pf}(M)\big(1 + w^T M^{-1} v\big).$$

With $M = \omega$, $v = Q$, and $w = \widetilde{Q} = \omega Q^\sharp$, where $(Q^\sharp)^i = Q^i = \eta^{ij}Q_j$, the correction factor is

$$w^T\omega^{-1}Q = (\omega Q^\sharp)^T\omega^{-1}Q = -(Q^\sharp)^T Q = -Q^iQ_i = 0.$$

Hence the Pfaffian normalization, once imposed on one sheet, is preserved by the cut transport. The pseudo-periodicity equation $\omega_{ij}(u + \mathrm{i}) = \widetilde{\omega}_{ij}(u)$ and the discontinuity equation give

$$\omega_{ij}(u + \mathrm{i}) - \omega_{ij}(u) = Q_i(u)\widetilde{Q}_j(u) - Q_j(u)\widetilde{Q}_i(u).$$

Substituting $\widetilde{Q}_j = \omega_{jk}Q^k$ gives (100.25).

The scope of the calculation is the local transport rule after the long-cut gauge has been chosen. A global $Q\omega$ Riemann–Hilbert construction requires the cut structure, gluing data, asymptotics, and gauge fixing. In particular, changing the antisymmetric tensor η, reversing the tilde continuation, or using the opposite $u \mapsto u - \mathrm{i}$ shift convention changes the displayed signs unless the same convention change is made everywhere in the $Q_{A|I}$ bridge.

100.3.1 Local P-Q Bridge

The two QSC gauges are not independent systems. Locally in a strip where the auxiliary Q-functions have no singularities, they are projections of a first-order finite-difference system. The following algebraic statement is the part of that system that is used repeatedly when changing between short-cut and long-cut descriptions.

Assumption 100.9 (Local P-Q bridge)**.** Let $Q_{a|i}(u)$, $a, i = 1, \dots, 4$, be a 4×4 matrix of auxiliary Q-functions with nonzero determinant in the strip under consideration. Let χ^{ab} and η^{ij} be the fixed antisymmetric tensors used to raise P- and Q-indices:

$$P^a = \chi^{ab} P_b, \qquad Q^i = \eta^{ij} Q_j.$$

In the local Q-system gauge assume

$$Q^+_{a|i} - Q^-_{a|i} = P_a Q_i, \tag{100.26}$$

$$Q_i = -P^a Q^-_{a|i}, \qquad P_a = -Q^i Q^-_{a|i}, \tag{100.27}$$

with the null contractions

$$P_a P^a = 0, \qquad Q_i Q^i = 0.$$

The null contractions are automatic once the upper indices are obtained from antisymmetric forms and the same lower vector is contracted with itself. They are nevertheless displayed because they are exactly what makes the choice of $+$ or $-$ shift harmless in the contraction formulae.

The assumption above is not a harmless notational convention. Its nontrivial content is the existence, in a chosen analytic strip and a chosen local Q-system gauge, of an invertible auxiliary matrix $Q_{a|i}$ whose finite-difference jump has rank one and whose projections reproduce the short-cut P_a and long-cut Q_i functions. In the global QSC this datum is part of the Riemann–Hilbert problem: one must specify the cut structure, gluing conditions, large-u asymptotics, possible zeros of $\det Q_{a|i}$, and the residual i-periodic gauge transformations. The local algebra below only records consequences of that datum after those analytic choices have been made.

Unimodular P-Q bridge algebra. Under Assumption 100.9, the contraction identities are independent of the shift:

$$Q_i = -P^a Q^+_{a|i}, \qquad P_a = -Q^i Q^+_{a|i}. \tag{100.28}$$

Moreover

$$\det(Q^+_{a|i}) = \det(Q^-_{a|i}). \tag{100.29}$$

Contract (100.26) with $-P^a$. Using (100.27) and $P_a P^a = 0$ gives

$$-P^a Q^+_{a|i} = -P^a Q^-_{a|i} - P^a P_a Q_i = Q_i.$$

Contracting the same rank-one shift with $-Q^i$ gives

$$-Q^i Q^+_{a|i} = -Q^i Q^-_{a|i} - P_a Q_i Q^i = P_a,$$

which proves (100.28).

For the determinant, write $M_{ai} = Q^-_{a|i}$, regard Q^i as a column vector $Q^\sharp$, and regard Q_i as a row vector $Q^\flat$. The second identity in (100.27) says

$$P = -MQ^\sharp.$$

Therefore

$$Q^+_{a|i} = M_{ai} + P_a Q_i = \big(M(1 - Q^\sharp Q^\flat)\big)_{ai}.$$

The matrix-determinant lemma gives

$$\det(1 - Q^\sharp Q^\flat) = 1 - Q^\flat Q^\sharp = 1 - Q_i Q^i = 1.$$

Thus $\det(Q^+_{a|i}) = \det M = \det(Q^-_{a|i})$.

This calculation is only local algebra. It does not prove the existence of global $Q_{a|i}$ functions with the required cut structure, nor does it fix the physical gluing conditions. It explains why the QSC can be written in a unimodular $Q_{a|i}$ gauge and why the formulae $Q_i = -P^a Q^\pm_{a|i}$, $P_a = -Q^i Q^\pm_{a|i}$ are conventionally quoted without specifying the shift: the rank-one difference has zero contraction in both null directions. Nonconstant i-periodic changes of $Q_{a|i}$ are still analytic gauge transformations; they must be fixed before large-u asymptotics or gluing relations are compared.

100.4 Weak-Coupling Baxter Limit

At weak coupling the short cut collapses to the origin in the rescaled spectral plane. In the $SL(2)$ sector, the QSC reduces to a finite-difference Baxter equation for a polynomial $Q(u)$ whose zeros are one-loop Bethe roots. The equation has the form

$$(u + \mathrm{i}/2)^J Q(u + \mathrm{i}) + (u - \mathrm{i}/2)^J Q(u - \mathrm{i}) = T(u)Q(u), \tag{100.30}$$

where $T(u)$ is a degree-J transfer polynomial fixed by charges and regularity. If

$$Q(u) = \prod_{k=1}^{S} (u - u_k),$$

then evaluating (100.30) at $u = u_j$ gives

$$\left(\frac{u_j + \mathrm{i}/2}{u_j - \mathrm{i}/2}\right)^J = \prod_{\substack{k=1\\k\neq j}}^{S} \frac{u_j - u_k - \mathrm{i}}{u_j - u_k + \mathrm{i}},$$

which is the one-loop $SL(2)$ Bethe equation in this convention. Thus the QSC contains the one-loop spin chain as its weak-coupling degeneration.

Assumption 100.10 (Weak-coupling regularity of the $SL(2)$ QSC)**.** Fix an $SL(2)$ state of twist $J \in \mathbb{Z}_{>0}$, Lorentz spin $S \in \mathbb{Z}_{\geq 0}$, and dimension $\Delta = J + S + O(g^2)$. In the weak-coupling degeneration of the $P\mu$-system assume the following analytic data.

$$P_2 P_3 = O(g^2), \qquad P_1(u) = A_1 u^{-J/2} + O(g^2),$$

and P_4/P_1 tends to a polynomial of degree $J - 1$. Moreover

$$\mu_{12}(u + \mathrm{i}/2) = Q(u) + g^2 R(u) + O(g^4),$$

where $Q(u)$ is a degree-S polynomial and the singular part of R is produced only by the short cuts $[-2g, 2g] + \mathrm{i}n$ collapsing to the points $\mathrm{i}n$. The defining sheet has no additional poles.

Weak $P\mu$ elimination before the Baxter equation. Work in the weak-coupling regularity class of Assumption 100.10. To leading order in g, neglect the P_2P_3 terms in the $(1,2)$ and $(2,4)$ components of the monodromy recursion (100.23). Then

$$\mu_{24} = \frac{\mu_{12}^{[+2]} - \mu_{12}}{P_1^2} - \mu_{12}\frac{P_4}{P_1}, \tag{100.31}$$

and eliminating μ_{24} gives

$$\begin{aligned} &\left(\frac{P_4^{[+1]}}{P_1^{[+1]}} - \frac{P_4^{[-1]}}{P_1^{[-1]}} + \frac{1}{(P_1^{[+1]})^2} + \frac{1}{(P_1^{[-1]})^2}\right)\mu_{12}^{[+1]} \\ &\qquad = \frac{\mu_{12}^{[+3]}}{(P_1^{[+1]})^2} + \frac{\mu_{12}^{[-1]}}{(P_1^{[-1]})^2}. \end{aligned} \tag{100.32}$$

Use $P^1 = -P_4$, $P^4 = P_1$, and discard terms proportional to P_2 or P_3. The $(1,2)$ component of (100.23) gives

$$\mu_{12}^{[+2]} - \mu_{12} = P_1(\mu_{21}P^1 + \mu_{24}P^4) = P_1(\mu_{12}P_4 + \mu_{24}P_1),$$

which is (100.31). The $(2,4)$ component gives

$$\mu_{24}^{[+2]} - \mu_{24} = -P_4(\mu_{21}P^1 + \mu_{24}P^4) = -\frac{P_4}{P_1}(\mu_{12}^{[+2]} - \mu_{12}), \tag{100.33}$$

where the last equality uses the preceding $(1,2)$ equation. Now write (100.31) once at u and once at $u + \mathrm{i}$, and substitute both expressions into (100.33). After shifting the resulting equation by $u \mapsto u - \mathrm{i}/2$, the terms proportional to $\mu_{12}^{[+1]}$ collect exactly into the coefficient displayed in (100.32). This gives the stated elimination formula.

The Baxter equation obtained in this degeneration is the following. Assume the hypotheses of the weak $P\mu$ elimination above. Normalize P_1 so that, at $g = 0$,

$$\frac{1}{(P_1^{[\pm 1]}(u))^2} = (u \pm \mathrm{i}/2)^J,$$

and assume the regular ratio P_4/P_1 is such that

$$T(u) = \frac{P_4^{[+1]}}{P_1^{[+1]}} - \frac{P_4^{[-1]}}{P_1^{[-1]}} + (u + \mathrm{i}/2)^J + (u - \mathrm{i}/2)^J$$

is a degree-J transfer polynomial. With

$$Q(u) = \mu_{12}^{[+1]}(u)\big|_{g=0},$$

the leading weak equation is the Baxter equation

$$(u + \mathrm{i}/2)^J Q(u + \mathrm{i}) + (u - \mathrm{i}/2)^J Q(u - \mathrm{i}) = T(u)Q(u).$$

If $Q(u) = \prod_{k=1}^{S}(u - u_k)$, regularity at the zeros of Q gives the one-loop $SL(2)$ Bethe equations. Equation (100.32), with the stated leading asymptotics of P_1, is exactly the displayed Baxter equation. The right side is regular at every zero u_j of Q, so the left side must vanish there:

$$(u_j + \mathrm{i}/2)^J Q(u_j + \mathrm{i}) + (u_j - \mathrm{i}/2)^J Q(u_j - \mathrm{i}) = 0.$$

Dividing by $(u_j - \mathrm{i}/2)^J Q(u_j - \mathrm{i})$ and using

$$\frac{Q(u_j - \mathrm{i})}{Q(u_j + \mathrm{i})} = -\prod_{\substack{k=1\\k\neq j}}^{S} \frac{u_j - u_k - \mathrm{i}}{u_j - u_k + \mathrm{i}}$$

gives the displayed Bethe equation.

Lemma 100.11 (Half-integer digamma primitive)**.** *Let*

$$\Phi(u) = \psi\left(\frac{1}{2} - \mathrm{i}u\right) + \psi\left(\frac{1}{2} + \mathrm{i}u\right), \qquad \psi = \Gamma'/\Gamma.$$

As a meromorphic function of u, Φ has only simple poles at

$$u = \pm\mathrm{i}\left(n + \frac{1}{2}\right), \qquad n \in \mathbb{Z}_{\geq 0}.$$

It satisfies the one-step shift identities

$$\Phi(u+\mathrm{i}) - \Phi(u) = \frac{2\mathrm{i}}{u + \mathrm{i}/2}, \qquad \Phi(u) - \Phi(u - \mathrm{i}) = \frac{2\mathrm{i}}{u - \mathrm{i}/2}, \tag{100.34}$$

and hence the central second-difference identity

$$\Phi(u+\mathrm{i}) - 2\Phi(u) + \Phi(u-\mathrm{i}) = \frac{2}{u^2 + 1/4}. \tag{100.35}$$

Moreover, if G is another meromorphic function with the same two one-step defects in (100.34)*, then $G - \Phi$ is* i*-periodic wherever both functions are holomorphic. If this difference extends to an entire function bounded on a vertical fundamental strip $0 \leq \operatorname{Im} u \leq 1$, then $G - \Phi$ is constant.*

Proof. The pole statement follows from the poles of ψ at $0, -1, -2, \ldots$. The recurrence relation

$$\psi(z+1) = \psi(z) + \frac{1}{z}$$

gives

$$\begin{aligned}\Phi(u+\mathrm{i}) &= \psi\left(\frac{3}{2} - \mathrm{i}u\right) + \psi\left(-\frac{1}{2} + \mathrm{i}u\right)\\ &= \psi\left(\frac{1}{2} - \mathrm{i}u\right) + \frac{1}{1/2 - \mathrm{i}u} + \psi\left(\frac{1}{2} + \mathrm{i}u\right) - \frac{1}{-1/2 + \mathrm{i}u}.\end{aligned}$$

Since $1/2 - \mathrm{i}u = -\mathrm{i}(u + \mathrm{i}/2)$, this is the first identity in (100.34). For the second identity, write

$$\begin{aligned}\Phi(u-\mathrm{i}) &= \psi\left(-\frac{1}{2} - \mathrm{i}u\right) + \psi\left(\frac{3}{2} + \mathrm{i}u\right)\\ &= \psi\left(\frac{1}{2} - \mathrm{i}u\right) - \frac{1}{-1/2 - \mathrm{i}u} + \psi\left(\frac{1}{2} + \mathrm{i}u\right) + \frac{1}{1/2 + \mathrm{i}u},\end{aligned}$$

which is the second displayed defect formula after rewriting the denominators as shifts of u. Subtracting the two one-step identities gives (100.35).

If G has the same one-step defects, then

$$(G - \Phi)(u + \mathrm{i}) = (G - \Phi)(u)$$

on the common holomorphy domain. If this difference is entire and bounded on a vertical fundamental strip, periodicity bounds it on all of $\mathbb{C}$; Liouville's theorem makes it constant. □

Collapsed-cut digamma package. Let $Q(u)$ be a polynomial with $Q(\mathrm{i}/2) \neq 0$, and set

$$r_0 = Q'(\mathrm{i}/2) - Q'(-\mathrm{i}/2).$$

Define

$$R_{\text{sing}}(u) = \mathrm{i}r_0\, \frac{Q(u)}{Q(\mathrm{i}/2)} \left(\psi\left(\frac{1}{2} - \mathrm{i}u\right) + \psi\left(\frac{1}{2} + \mathrm{i}u\right)\right), \qquad \psi = \Gamma'/\Gamma. \tag{100.36}$$

Then R_{sing} has only simple poles at

$$u = \pm\mathrm{i}\left(n + \frac{1}{2}\right), \qquad n \in \mathbb{Z}_{\geq 0},$$

and their residues are

$$\operatorname*{Res}_{u=\mathrm{i}(n+1/2)} R_{\text{sing}} = -r_0\, \frac{Q(\mathrm{i}(n+1/2))}{Q(\mathrm{i}/2)},$$
$$\operatorname*{Res}_{u=-\mathrm{i}(n+1/2)} R_{\text{sing}} = r_0\, \frac{Q(-\mathrm{i}(n+1/2))}{Q(\mathrm{i}/2)}.$$

In particular, if $Q(\mathrm{i}/2) = Q(-\mathrm{i}/2)$, the residues at $\mathrm{i}/2$ and $-\mathrm{i}/2$ are $-r_0$ and r_0. In the large-u sector used for the $SL(2)$ asymptotics,

$$\frac{R_{\text{sing}}(u)}{Q(u)} = 2\mathrm{i}\, \frac{r_0}{Q(\mathrm{i}/2)} \log u + O(u^{-1}).$$

Finally, for the meromorphic quotient $F_{\text{sing}}(u) = R_{\text{sing}}(u)/Q(u)$,

$$F_{\text{sing}}(u+\mathrm{i}) - F_{\text{sing}}(u) = -\frac{2r_0}{Q(\mathrm{i}/2)(u+\mathrm{i}/2)}, \qquad F_{\text{sing}}(u) - F_{\text{sing}}(u-\mathrm{i}) = -\frac{2r_0}{Q(\mathrm{i}/2)(u-\mathrm{i}/2)}. \tag{100.37}$$

The residue formulas follow from the pole structure of ψ. The digamma function has simple poles at $0, -1, -2, \ldots$, each with residue -1. The factor $\psi(1/2 + \mathrm{i}u)$ therefore has a pole at $u = \mathrm{i}(n+1/2)$; since

$$\frac{1}{2} + \mathrm{i}u + n = \mathrm{i}\big(u - \mathrm{i}(n+1/2)\big),$$

its residue as a function of u is $-1/\mathrm{i} = \mathrm{i}$. Multiplication by the prefactor $\mathrm{i}r_0 Q(u)/Q(\mathrm{i}/2)$ gives the first residue formula. The factor $\psi(1/2 - \mathrm{i}u)$ gives the second formula in the same way, because

$$\frac{1}{2} - \mathrm{i}u + n = -\mathrm{i}\big(u + \mathrm{i}(n+1/2)\big)$$

and hence the residue in u is $-1/(-\mathrm{i}) = -\mathrm{i}$.

Finally, Stirling's expansion gives

$$\psi(z) = \log z + O(z^{-1})$$

in sectors avoiding the negative real axis. Applying this to $z = 1/2 \pm \mathrm{i}u$ in the chosen large-u sector gives

$$\psi\left(\frac{1}{2} - \mathrm{i}u\right) + \psi\left(\frac{1}{2} + \mathrm{i}u\right) = 2\log u + O(u^{-1}),$$

with the branch inherited from the QSC large-u sheet. This proves the logarithmic coefficient. The quotient shift identities (100.37) follow by multiplying (100.34) by $\mathrm{i}r_0/Q(\mathrm{i}/2)$.

One-loop dimension from the weak QSC. Assume the regular weak-coupling degeneration of Assumption 100.10, and impose the cyclic single-trace condition

$$\frac{Q(\mathrm{i}/2)}{Q(-\mathrm{i}/2)} = 1.$$

Then the first anomalous-dimension coefficient extracted from the QSC is

$$\Delta = J + S + 2\mathrm{i}g^2\, \partial_u \log \frac{Q(u+\mathrm{i}/2)}{Q(u-\mathrm{i}/2)}\bigg|_{u=0} + O(g^4).$$

Equivalently, if $Q(u) = \prod_{k=1}^{S}(u-u_k)$, then

$$\Delta = J + S + 2g^2 \sum_{k=1}^{S} \frac{1}{u_k^2 + 1/4} + O(g^4).$$

The leading $P_2P_3 = O(g^2)$ scaling decouples the (μ_{12}, μ_{24}) equations of the $P\mu$-system. Eliminating μ_{24}, and using the regularity of P_1 and P_4/P_1 in Assumption 100.10, gives the Baxter equation derived above for $Q(u) = \mu_{12}(u + \mathrm{i}/2) + O(g^2)$. The same i-pseudoperiodicity of μ_{12}, evaluated at the collapsed branch point, gives the cyclicity condition $Q(\mathrm{i}/2) = Q(-\mathrm{i}/2)$.

It remains to extract the coefficient of g^2 in Δ. Since $\mu_{12}(u) \sim u^{\Delta-J}$ and $Q(u-\mathrm{i}/2) \sim u^S$, the ratio $\mu_{12}(u)/Q(u-\mathrm{i}/2)$ has the large-u expansion

$$1 + (\Delta - J - S)\log u + O(g^4) + O(u^{-1}).$$

The collapsed square-root cuts force the singular part of the g^2 correction to $\mu_{12}(u+\mathrm{i}/2)$ to be (100.36). The collapsed-cut calculation above shows that this expression has the prescribed simple poles at the collapsed cuts, with residues fixed by the branch-point relation and cyclic endpoint condition. More precisely, after division by the leading Baxter polynomial, the singular quotient has the one-step defects (100.37). These defects are the collapsed-cut remnant of the square-root decomposition of μ_{12}; by Lemma 100.11, they fix the non-regular meromorphic primitive up to an entire i-periodic ambiguity. In the weak-coupling regularity class of Assumption 100.10, that ambiguity belongs to the regular polynomial/asymptotic-power part of R, and it carries no $\log u$ term. Comparing the logarithmic coefficient in the collapsed-cut calculation with the large-u expansion of μ_{12}/Q gives

$$\Delta - J - S = 2\mathrm{i}g^2\, \frac{Q'(\mathrm{i}/2) - Q'(-\mathrm{i}/2)}{Q(\mathrm{i}/2)} + O(g^4).$$

Using $Q(\mathrm{i}/2) = Q(-\mathrm{i}/2)$ rewrites this as the derivative formula. Finally

$$\mathrm{i}\left(\frac{Q'(\mathrm{i}/2)}{Q(\mathrm{i}/2)} - \frac{Q'(-\mathrm{i}/2)}{Q(-\mathrm{i}/2)}\right) = \sum_{k=1}^{S} \frac{1}{u_k^2 + 1/4},$$

which proves the root-sum form.

Example 100.12 (Konishi Baxter polynomial)**.** For the twist-two spin-two representative of Konishi, take

$$J = 2, \qquad S = 2, \qquad Q(u) = u^2 - \frac{1}{12}.$$

Then

$$(u+\mathrm{i}/2)^2Q(u+\mathrm{i})+(u-\mathrm{i}/2)^2Q(u-\mathrm{i})=\left(2u^2-\frac{13}{2}\right)Q(u).$$

Thus the transfer polynomial is $T(u)=2u^2-13/2$, and the zeros

$$u=\pm\frac{1}{2\sqrt{3}}$$

are precisely the one-loop Konishi Bethe roots used in Chapter 97. This example is a finite calculation showing how the QSC weak-coupling limit remembers the spin-chain normalization.

Proposition 100.13 (Twist-two Baxter family)**.** *For twist $J=2$ and spin S, define the terminating polynomial*

$$Q_S(u)={}_3F_2\left(\begin{matrix}-S,\ S+1,\ \frac{1}{2}-\mathrm{i}u\\ 1,\ 1\end{matrix};1\right),$$

where $(a)_k$ is the rising Pochhammer symbol in the terminating hypergeometric series. It obeys

$$(u+\mathrm{i}/2)^2Q_S(u+\mathrm{i})+(u-\mathrm{i}/2)^2Q_S(u-\mathrm{i})=\left(2u^2-S(S+1)-\frac{1}{2}\right)Q_S(u).$$

Thus $T_S(u)=2u^2-S(S+1)-1/2$. The endpoint values and logarithmic derivative are the finite identities

$$Q_S(-\mathrm{i}/2)=1,\qquad Q_S(\mathrm{i}/2)=(-1)^S,\qquad \mathrm{i}\left(\frac{Q_S'(\mathrm{i}/2)}{Q_S(\mathrm{i}/2)}-\frac{Q_S'(-\mathrm{i}/2)}{Q_S(-\mathrm{i}/2)}\right)=4H_S,$$

with $H_S=\sum_{m=1}^{S}m^{-1}$. Hence the physical even-spin twist-two single-trace states have

$$\Delta_{J=2,S}=2+S+8g^2H_S+O(g^4).$$

For $S=2$, $Q_2(u)=1/4-3u^2$, which is the same root set as the monic Konishi polynomial $u^2-1/12$, and the formula gives $\Delta_{\mathrm{K}}=4+12g^2+O(g^4)$.

Proof. Put

$$z=\frac{1}{2}-\mathrm{i}u,\qquad Q_S(u)=\widehat{Q}_S(z),$$

so that $u+\mathrm{i}/2=\mathrm{i}z$, $u-\mathrm{i}/2=\mathrm{i}(z-1)$, and the shifts $u\mapsto u\pm\mathrm{i}$ become $z\mapsto z\pm 1$. The Baxter equation is equivalent to

$$z^2\widehat{Q}_S(z+1)+(z-1)^2\widehat{Q}_S(z-1)=\big(2z^2-2z+S(S+1)+1\big)\widehat{Q}_S(z).$$

Write the k-th summand of $\widehat{Q}_S$ as

$$\tau_k(z)=\frac{(-S)_k(S+1)_k(z)_k}{(k!)^3},\qquad 0\le k\le S,$$

and set $\tau_k=0$ outside this range. For $k\ge 1$,

$$(z+1)_k=(z)_k\frac{z+k}{z},\qquad (z-1)_k=(z)_k\frac{z-1}{z+k-1}.$$

A direct substitution gives the summand identity

$$\begin{aligned}z^2\tau_k(z+1)+(z-1)^2\tau_k(z-1)-\big(2z^2-2z+S(S+1)+1\big)\tau_k(z)\\=\frac{(k+1)^3}{z+k}\tau_{k+1}(z)-\frac{k^3}{z+k-1}\tau_k(z).\end{aligned}$$

The identity is first read away from the apparent poles $z=1-k$ and then extended across the removable singularities by polynomial continuation. It is a finite telescoping identity. The right side sums to zero because the lower boundary has the factor k^3 and the upper boundary has $\tau_{S+1}=0$. Hence the z-form of the Baxter equation holds, and the displayed Baxter equation follows after returning to u.

The endpoint $u=-\mathrm{i}/2$ corresponds to $z=0$, so $\widehat{Q}_S(0)=1$. The endpoint $u=\mathrm{i}/2$ corresponds to $z=1$, and Chu–Vandermonde gives the finite sum

$$\widehat{Q}_S(1)={}_2F_1(-S,S+1;1;1)=\frac{(-S)_S}{S!}=(-1)^S.$$

It remains to prove the logarithmic-derivative identity. Since

$$\frac{d}{du}Q_S(u)=-\mathrm{i}\,\widehat{Q}'_S(z),$$

we must compute the endpoint logarithmic derivatives of $\widehat{Q}_S$. First,

$$\widehat{Q}'_S(0)=\sum_{k=1}^{S}\frac{(-S)_k(S+1)_k}{(k!)^2\,k}. \tag{100.38}$$

Indeed $d(z)_k/dz|_{z=0}=(k-1)!$. The same sum may be written as

$$\widehat{Q}'_S(0)=\int_0^1\frac{{}_2F_1(-S,S+1;1;t)-1}{t}\,dt.$$

Using ${}_2F_1(-S,S+1;1;t)=\mathsf{P}_S(1-2t)$, where P_S is the Legendre polynomial, and setting $x=1-2t$, this is

$$I_S=\int_{-1}^1\frac{\mathsf{P}_S(x)-1}{1-x}\,dx.$$

The finite Christoffel–Darboux identity at one endpoint,

$$n\,\frac{\mathsf{P}_n(x)-\mathsf{P}_{n-1}(x)}{x-1}=\sum_{\ell=0}^{n-1}(2\ell+1)\mathsf{P}_\ell(x),$$

is the telescoping form of the Legendre three-term recurrence. It gives $I_n-I_{n-1}=-2/n$ after integration, because only the $\ell=0$ Legendre polynomial has nonzero integral over $[-1,1]$. Since $I_0=0$,

$$\widehat{Q}'_S(0)=-2H_S.$$

At $z=1$,

$$\widehat{Q}'_S(1)=\sum_{k=1}^{S}\frac{(-S)_k(S+1)_k}{(k!)^2}H_k.$$

This is obtained from $d(1)_k/dz = k!H_k$. Differentiate Chu–Vandermonde in the denominator parameter:

$$F(a) = {}_2F_1(-S, S+1; a; 1) = \frac{(a-S-1)_S}{(a)_S}.$$

At $a = 1$,

$$\frac{F'(1)}{F(1)} = \sum_{m=0}^{S-1} \frac{1}{1-S-1+m} - \sum_{m=0}^{S-1} \frac{1}{1+m} = -2H_S.$$

On the other hand differentiating the series in a gives

$$F'(1) = -\widehat{Q}'_S(1).$$

Since $F(1) = (-1)^S$, we obtain

$$\frac{\widehat{Q}'_S(1)}{\widehat{Q}_S(1)} = 2H_S.$$

Therefore

$$\mathrm{i}\left(\frac{Q'_S(\mathrm{i}/2)}{Q_S(\mathrm{i}/2)} - \frac{Q'_S(-\mathrm{i}/2)}{Q_S(-\mathrm{i}/2)}\right) = \frac{\widehat{Q}'_S(1)}{\widehat{Q}_S(1)} - \frac{\widehat{Q}'_S(0)}{\widehat{Q}_S(0)} = 4H_S.$$

The dimension formula is now the specialization of the weak-QSC one-loop dimension extraction above. □

100.5 Cusp Anomalous Dimension and Konishi

Two benchmark spectral quantities organize many tests of the framework. The first is the cusp anomalous dimension $f(\lambda)$, extracted from large spin twist operators:

$$\Delta - S = f(\lambda)\log S + O(S^0), \qquad S \to \infty.$$

The QSC reproduces the same function obtained from the BES equation. At weak coupling it has a perturbative series in g^2; at strong coupling the QSC asymptotics begin with

$$f(\lambda) = \frac{\sqrt{\lambda}}{\pi} + O(\lambda^0)$$

in the convention where $\Delta - S = f(\lambda)\log S + O(S^0)$. Comparisons with semiclassical string theory are historically important but are not used as inputs to the QFT-side spectral problem in this chapter.

The second is the Konishi dimension. The QSC treats it as an ordinary finite state with $J = 2$, $S = 2$, and zero total momentum. At weak coupling it matches the asymptotic Bethe ansatz until wrapping corrections enter; at strong coupling it gives the QSC strong-coupling expansion

$$\Delta_{\mathrm{K}}(\lambda) = 2\lambda^{1/4} + O(\lambda^0).$$

The numerical and analytic solution of the QSC is a calculation problem inside the finite functional system (100.10)–(100.12).

100.6 Small Spin Expansion

The QSC gives a controlled analytic expansion near $S = 0$, where the state approaches the protected vacuum. The finite coefficient calculation below begins after the following local analytic germ has been supplied.

Assumption 100.14 (Small-spin QSC germ)**.** Fix the twist $J \in \mathbb{Z}_{>0}$ and let S be a complex spin parameter near 0. Assume that the physical QSC solution admits a gauge in which the $S = 0$ limit is the protected vacuum and the first nonzero deformation has $\epsilon^2 = O(S)$. In this gauge the leading μ-matrix is

$$\mu_{12} = -1, \quad \mu_{13} = 0, \quad \mu_{14} = \mu_{23} = 1, \quad \mu_{24} = -\sinh(2\pi u), \quad \mu_{34} = 0.$$

Writing

$$x(u) = \frac{u}{2g} + \sqrt{\frac{u^2}{4g^2} - 1},$$

with the short-cut branch, the leading solution has

$$P_1 = \epsilon x^{-J/2}, \qquad P_3 = \epsilon(x^{-J/2} - x^{J/2}),$$

and the remaining two P-functions are obtained by solving the local $P\mu$ equations with the displayed μ-matrix and imposing single-short-cut regularity. Equivalently, the Laurent expansion of $\sinh(2\pi u)$ is split into the powers allowed in P_2 and P_4. Since

$$u = g(x + x^{-1}),$$

the modified-Bessel generating function gives

$$\sinh(2\pi u) = \sum_{n\in\mathbb{Z}} I_{2n+1}(4\pi g)x^{2n+1}. \tag{100.39}$$

Here I_m is the modified Bessel function of the first kind. The first large-x coefficients produced by the regularity split are

$$P_2 = -\epsilon I_{J+1}(4\pi g)x^{-J/2-1} + O(x^{-J/2-3}), \qquad P_4 = -\epsilon I_{J-1}(4\pi g)x^{J/2-1} + O(x^{J/2-3}).$$

In the large-u normalization of the coefficient-product calculation above,

$$A_1 = \epsilon g^{J/2}, \quad A_2 = -\epsilon g^{J/2+1} I_{J+1}(4\pi g), \quad A_3 = -\epsilon g^{-J/2}, \quad A_4 = -\epsilon g^{-J/2+1} I_{J-1}(4\pi g).$$

The corresponding large-u charge relations are

$$\Delta - J = -\mathrm{i}(A_1A_4 - A_2A_3) + O(S^2), \qquad S = -\mathrm{i}(A_1A_4 + A_2A_3) + O(S^2).$$

Small-spin QSC slope extraction. Under Assumption 100.14, the leading anomalous dimension satisfies

$$\Delta - J - S = \frac{4\pi g\, I_{J+1}(4\pi g)}{J\, I_J(4\pi g)}\, S + O(S^2).$$

The finite calculation is as follows. First prove the Laurent identity (100.39). The generating function

$$\exp\Big(\frac{z}{2}(t+t^{-1})\Big) = \sum_{m\in\mathbb{Z}} I_m(z)t^m$$

with $z = 4\pi g$ and $t = x$ gives $\exp(2\pi g(x+x^{-1})) = \sum_m I_m(4\pi g)x^m$. Replacing x by $-x$ gives the negative exponential. Taking half the difference selects exactly the odd powers and proves (100.39).

The displayed leading coefficients then give

$$A_1A_4 = -\epsilon^2 g\, I_{J-1}(4\pi g), \qquad A_2A_3 = \epsilon^2 g\, I_{J+1}(4\pi g).$$

Substituting these products into the large-u charge relations in Assumption 100.14 yields

$$\Delta - J = \mathrm{i}\epsilon^2 g\big(I_{J-1}(4\pi g) + I_{J+1}(4\pi g)\big) + O(S^2),$$

and

$$S = \mathrm{i}\epsilon^2 g\big(I_{J-1}(4\pi g) - I_{J+1}(4\pi g)\big) + O(S^2).$$

Use the modified-Bessel recurrence

$$I_{J-1}(z) - I_{J+1}(z) = \frac{2J}{z} I_J(z), \qquad z = 4\pi g.$$

It gives

$$\epsilon^2 = \frac{2\pi S}{\mathrm{i}J I_J(4\pi g)} + O(S^2).$$

Therefore

$$\Delta - J - S = 2\mathrm{i}\epsilon^2 g\, I_{J+1}(4\pi g) + O(S^2) = \frac{4\pi g\, I_{J+1}(4\pi g)}{J\, I_J(4\pi g)} S + O(S^2),$$

which is the stated slope.

Continuing the resulting strong-coupling expansion to $J = 2, S = 2$ gives the Konishi estimate

$$\Delta_{\mathrm{K}} = 2\lambda^{1/4} + 2\lambda^{-1/4} + O(\lambda^{-3/4}).$$

The analytic continuation from small spin to integer spin is a QSC input, not a theorem proved from the gauge-theory path integral.

100.7 Hexagon Form Factors

The spectral problem determines scaling dimensions. Three-point functions require additional data. In the planar limit, a single-trace three-point function can be represented by a pair of pants worldsheet. The hexagon program cuts this pair of pants into two hexagons and assigns a form factor to each half.

For three single-trace operators with spin-chain lengths L_1, L_2, L_3, the bridge lengths are

$$\ell_{12} = \frac{L_1 + L_2 - L_3}{2}, \qquad \ell_{23} = \frac{L_2 + L_3 - L_1}{2}, \qquad \ell_{31} = \frac{L_3 + L_1 - L_2}{2}.$$

Remark 100.15 (Planar bridge-length bookkeeping). Consider the planar tree-level pair-of-pants skeleton for three cyclic single-trace operators of lengths L_1, L_2, L_3. Let $\ell_{ij} = \ell_{ji}$ be the number of adjacent Wick-contraction slots shared by operators i and j. Then the bridge lengths are forced to be

$$\ell_{12} = \frac{L_1 + L_2 - L_3}{2}, \qquad \ell_{23} = \frac{L_2 + L_3 - L_1}{2}, \qquad \ell_{31} = \frac{L_3 + L_1 - L_2}{2}. \tag{100.40}$$

Conversely, a nonnegative integer planar skeleton of this type exists exactly when $L_1 + L_2 + L_3$ is even and the three triangle inequalities $L_i + L_j \geq L_k$ hold.

In the planar skeleton each trace is cut at two cyclic splitting points. The letters of operator 1 are divided into the bridge shared with operator 2 and the bridge shared with operator 3, and similarly for the other two operators. Hence

$$L_1 = \ell_{12} + \ell_{31}, \qquad L_2 = \ell_{12} + \ell_{23}, \qquad L_3 = \ell_{23} + \ell_{31}.$$

Adding the first two equations and subtracting the third gives $2\ell_{12} = L_1 + L_2 - L_3$. The other two formulae follow by cyclic permutation. Since ℓ_{ij} counts letters, the right-hand sides must be nonnegative integers; this is precisely even total length plus the triangle inequalities. Conversely, if these conditions hold, the three numbers in (100.40) define nonnegative integer arcs on the three cyclic words satisfying the displayed length-balance equations, which is the required pair-of-pants skeleton.

Asymptotically, the structure constant is a sum over partitions of magnons between the two hexagons:

$$C_{123}^{\text{asym}} = \sum_{\alpha \cup \bar{\alpha} = \{u\}} (-1)^{|\bar{\alpha}|} \mathrm{e}^{\mathrm{i}p(\bar{\alpha})\ell} H(\alpha) H(\bar{\alpha}) \mathcal{N}^{-1/2}.$$

Here the displayed formula is the specialization in which the magnons shown belong to one external operator: ℓ is the bridge crossed by the subset $\bar{\alpha}$, H is the hexagon form factor, and

$$p(\bar{\alpha}) = \sum_{u_j \in \bar{\alpha}} p(u_j).$$

The phase is the spin-chain translation phase. Moving a magnon of momentum p_j through ℓ vacuum letters multiplies the coordinate Bethe wave by $\mathrm{e}^{\mathrm{i}p_j\ell}$, and moving all magnons in $\bar{\alpha}$ gives

$$\prod_{u_j \in \bar{\alpha}} \mathrm{e}^{\mathrm{i}p_j\ell} = \mathrm{e}^{\mathrm{i}p(\bar{\alpha})\ell}. \tag{100.41}$$

The factor $\mathcal{N}$ is the Gaudin norm factor of the Bethe state. Mirror particles on the bridges add finite-size corrections, analogous in spirit to wrapping corrections in the spectral problem.

For scalar magnons the elementary two-particle scalar factor entering the hexagon can be written, in the same $x^{\pm}$ and σ conventions as Chapter 98, as

$$h(u,v) = \frac{x_u^- - x_v^-}{x_u^- - x_v^+} \frac{1 - 1/(x_u^+ x_v^-)}{1 - 1/(x_u^+ x_v^+)} \frac{1}{\sigma(u,v)}. \tag{100.42}$$

The matrix part is the same centrally extended symmetry intertwiner that appears in the magnon S-matrix, after the crossing moves needed to put excitations on a chosen hexagon edge. Thus the hexagon construction imports the same analytic-continuation obligations: specifying $h(u,v)$ without the crossing path of the moved excitation is incomplete.

Hexagon scalar Watson factor. Let $x_i^\pm$, $i=1,2$, be non-zero Zhukovsky variables in a sheet domain where the denominators below do not vanish, and write $\sigma_{ij}=\sigma(x_i^\pm,x_j^\pm)$. Assume scalar dressing unitarity $\sigma_{12}\sigma_{21}=1$ on this domain. If $h_{12}=h(u_1,u_2)$ is defined by (100.42), then

$$\frac{h_{12}}{h_{21}}=S^{\rm hex}_{\rm scal}(1,2),\qquad S^{\rm hex}_{\rm scal}(1,2)=\frac{x_1^+-x_2^-}{x_1^--x_2^+}\frac{1-1/(x_1^+x_2^-)}{1-1/(x_1^-x_2^+)}\frac{1}{\sigma_{12}^2}. \tag{100.43}$$

On the weak physical branch

$$x_i^\pm=\frac{u_i\pm {\rm i}/2}{g}+O(g),\qquad \sigma_{12}=1+O(g^6),$$

with $u_1-u_2\neq {\rm i}$, this becomes

$$S^{\rm hex}_{\rm scal}(1,2)=\frac{u_1-u_2+{\rm i}}{u_1-u_2-{\rm i}}+O(g^2). \tag{100.44}$$

From (100.42),

$$\frac{h_{12}}{h_{21}}=\frac{x_1^--x_2^-}{x_1^--x_2^+}\frac{x_2^--x_1^+}{x_2^--x_1^-}\frac{1-1/(x_1^+x_2^-)}{1-1/(x_1^+x_2^+)}\frac{1-1/(x_2^+x_1^+)}{1-1/(x_2^+x_1^-)}\frac{\sigma_{21}}{\sigma_{12}}.$$

The first two rational factors reduce to

$$\frac{x_1^--x_2^-}{x_2^--x_1^-}\frac{x_2^--x_1^+}{x_1^--x_2^+}=\frac{x_1^+-x_2^-}{x_1^--x_2^+}.$$

The middle denominator $1-1/(x_1^+x_2^+)$ cancels the identical factor $1-1/(x_2^+x_1^+)$, and the remaining denominator is $1-1/(x_1^-x_2^+)$. Finally scalar dressing unitarity gives $\sigma_{21}/\sigma_{12}=1/\sigma_{12}^2$. This proves (100.43).

For the weak limit, the first rational factor is

$$\frac{(u_1+{\rm i}/2)/g-(u_2-{\rm i}/2)/g+O(g)}{(u_1-{\rm i}/2)/g-(u_2+{\rm i}/2)/g+O(g)}=\frac{u_1-u_2+{\rm i}}{u_1-u_2-{\rm i}}+O(g^2).$$

The second rational factor is $1+O(g^2)$, because $(x_i^\pm x_j^\pm)^{-1}=O(g^2)$, and the dressing factor contributes $1+O(g^6)$. This proves (100.44).

The weak factor in (100.44) is the compact one-loop exchange phase. It should not be identified with the rank-one $SL(2)$ phase in (98.30) unless the vacuum, edge-crossing path, and scalar-factor split have all been transported to the same convention.

Open Problem 100.16 (Derivation of the hexagon expansion). Give a derivation of the hexagon expansion from a nonperturbatively defined planar gauge-theory path integral, including the emergence of the hexagon form factor, the measure for mirror corrections, and the treatment of operator mixing at finite length. Existing evidence is strong inside the integrability framework. A derivation at the level of a theorem in four-dimensional QFT remains open.

100.8 Maldacena–Wilson Line, Cusp, and Bremsstrahlung

Let C be an oriented smooth curve with parameter s, tangent $\dot{x}^\mu(s)$, and scalar-coupling unit vector $\theta^I(s) \in S^5$. The Maldacena–Wilson line in the fundamental representation is

$$W_{\mathrm{MW}}(C, \theta) = \mathrm{Tr}\ P \exp \int_C \big(\mathrm{i} A_\mu(x(s))\dot{x}^\mu(s) + \Phi_I(x(s))\theta^I(s)|\dot{x}(s)|\big)\,\mathrm{d}s.$$

It is a gauge-invariant line operator after taking the trace and imposing the usual endpoint prescription for closed curves or external heavy sources. A straight line with constant θ is half-BPS and has vanishing cusp anomalous dimension.

For two rays meeting with Euclidean deflection angle φ, define $\Gamma_{\mathrm{cusp}}(\varphi, \theta)$ by the logarithmic divergence of the renormalized cusped line:

$$\log\langle W_{\mathrm{cusp}}\rangle_{\mathrm{ren}} = -\Gamma_{\mathrm{cusp}}(\varphi) \log \frac{R}{\epsilon} + O(1),$$

where R is an infrared distance scale and ϵ is the short-distance cutoff at the cusp. At small geometric angle and fixed scalar coupling,

$$\Gamma_{\mathrm{cusp}}(\varphi) = -B(\lambda)\varphi^2 + O(\varphi^4).$$

The coefficient $B(\lambda)$ is the Bremsstrahlung function. On the straight-line defect, the transverse displacement operators $\mathbb{D}_i(t)$ are defined by the first shape variation of the line; $i = 1, 2, 3$ labels directions normal to the straight Euclidean line. Conformal invariance on the defect fixes their two-point function to

$$\langle \mathbb{D}_i(t)\mathbb{D}_j(0)\rangle = \frac{C_D \delta_{ij}}{(t - \mathrm{i}0)^4}.$$

Displacement normalization of the small cusp. Assume the straight half-BPS line admits the standard defect shape-variation Ward identity: for a small transverse deformation $X^i(t)$ of compact support, the universal nonlocal quadratic part of $\log\langle W_X\rangle$ is, with summation over the transverse index i,

$$\frac{1}{2}\int \frac{\mathrm{d}\omega}{2\pi}\, \frac{\pi C_D}{6} |\omega|^3 \widehat{X}^i(\omega)\widehat{X}^i(-\omega),$$

up to local counterterms polynomial in ω. Equivalently, this fixes the nonlocal part of the finite-part Fourier transform of the displacement two-point distribution,

$$\mathrm{Fp}\int \mathrm{d}t\, \frac{\mathrm{e}^{\mathrm{i}\omega t}}{(t - \mathrm{i}0)^4} = \frac{\pi}{6}|\omega|^3 + \text{a polynomial in } \omega.$$

Then the small-angle cusp coefficient satisfies

$$C_D = 12B(\lambda).$$

Represent a cusp of small angle φ by a deformation of the straight line for which

$$X^1(t) = \varphi t\, \Theta(t)$$

between the scales ϵ and R, with a smooth interpolation outside that window. Different smoothings change only power divergences and finite terms. In the logarithmic momentum window $R^{-1} \ll |\omega| \ll \epsilon^{-1}$,

$$\widehat{X}^1(\omega) = -\frac{\varphi}{(\omega + \mathrm{i}0)^2} + \text{endpoint and zero-frequency terms} + O(R^{-1}, \epsilon).$$

The endpoint terms depend on the infrared smoothing and give only power divergences or finite oscillatory contributions, while the polynomial/contact terms and the $\omega = 0$ terms do not affect the coefficient of $\log(R/\epsilon)$. Hence the universal logarithmic contribution is

$$\frac{1}{2}\int_{R^{-1}<|\omega|<\epsilon^{-1}} \frac{\mathrm{d}\omega}{2\pi}\,\frac{\pi C_D}{6}|\omega|^3\frac{\varphi^2}{\omega^4} = \frac{C_D}{12}\varphi^2 \log\frac{R}{\epsilon}.$$

By the definition of the cusp anomalous dimension,

$$\log\langle W_{\text{cusp}}\rangle_{\text{ren}}\big|_{\log} = -\Gamma_{\text{cusp}}(\varphi)\log\frac{R}{\epsilon}.$$

Using $\Gamma_{\text{cusp}}(\varphi) = -B(\lambda)\varphi^2 + O(\varphi^4)$ and comparing the quadratic logarithmic coefficient gives $B = C_D/12$.

Thus the radiated power of a slowly accelerating heavy probe is

$$\frac{\mathrm{d}E_{\text{rad}}}{\mathrm{d}t} = 2\pi B(\lambda)\,\dot{v}^2.$$

The protected circular-loop matrix model is derived in Theorem 101.22. In the convention where the circular-loop Gaussian coupling of Definition 101.21 is the planar coupling λ used in this chapter, the supersymmetric Ward identity relating the circular loop to the small-angle cusp gives

$$B(\lambda) = \frac{\sqrt{\lambda}}{4\pi^2}\frac{I_2(\sqrt{\lambda})}{I_1(\sqrt{\lambda})} \qquad (N = \infty).$$

The algebraic step from the circular-loop answer $2I_1(\sqrt{\lambda})/\sqrt{\lambda}$ to this expression is Remark 101.23. The genuinely field-theoretic input is the protected Ward identity (101.50); the QSC and mirror-TBA descriptions of the cusped line reproduce the same function with cusp-specific asymptotic data, providing a stringent normalization check on $g = \sqrt{\lambda}/(4\pi)$.

Weak and strong expansions of $B(\lambda)$. Let $z = \sqrt{\lambda}$ and use the planar convention

$$B(\lambda) = \frac{z}{4\pi^2}\frac{I_2(z)}{I_1(z)}.$$

At weak coupling,

$$B(\lambda) = \frac{\lambda}{16\pi^2} - \frac{\lambda^2}{384\pi^2} + \frac{\lambda^3}{6144\pi^2} - \frac{\lambda^4}{92160\pi^2} + O(\lambda^5). \tag{100.45}$$

Equivalently, in the spectral-problem coupling $g = \sqrt{\lambda}/(4\pi)$,

$$B(g) = g^2 - \frac{2\pi^2}{3}g^4 + \frac{2\pi^4}{3}g^6 - \frac{32\pi^6}{45}g^8 + O(g^{10}). \tag{100.46}$$

At strong coupling,

$$B(\lambda) = \frac{\sqrt{\lambda}}{4\pi^2} - \frac{3}{8\pi^2} + \frac{3}{32\pi^2\sqrt{\lambda}} + \frac{3}{32\pi^2\lambda} + \frac{63}{512\pi^2\lambda^{3/2}} + O(\lambda^{-2}). \tag{100.47}$$

The weak expansion follows from the defining power series

$$I_\nu(z) = \sum_{k=0}^{\infty} \frac{1}{k!\,\Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}.$$

For the required orders,

$$\begin{aligned} I_1(z) &= \frac{z}{2} + \frac{z^3}{16} + \frac{z^5}{384} + \frac{z^7}{18432} + O(z^9), \\ I_2(z) &= \frac{z^2}{8} + \frac{z^4}{96} + \frac{z^6}{3072} + \frac{z^8}{184320} + O(z^{10}). \end{aligned}$$

Dividing the two series gives

$$\frac{I_2(z)}{I_1(z)} = \frac{z}{4} - \frac{z^3}{96} + \frac{z^5}{1536} - \frac{z^7}{23040} + O(z^9).$$

Multiplication by $z/(4\pi^2)$ and $z^2 = \lambda$ gives (100.45); substituting $\lambda = 16\pi^2 g^2$ gives (100.46).

For the strong expansion use the large-positive-z asymptotic expansion

$$I_\nu(z) \sim \frac{e^z}{\sqrt{2\pi z}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(8z)^k} \prod_{j=1}^{k} \big(4\nu^2 - (2j-1)^2\big). \tag{100.48}$$

Let us recall the coefficient derivation, since the signs are a common source of normalization errors. The modified Bessel equation is

$$I_\nu''(z) + \frac{1}{z} I_\nu'(z) - \left(1 + \frac{\nu^2}{z^2}\right) I_\nu(z) = 0.$$

Insert

$$I_\nu(z) \sim \frac{e^z}{\sqrt{2\pi z}} S_\nu(z), \qquad S_\nu(z) = \sum_{k=0}^{\infty} a_k^{(\nu)} z^{-k}, \qquad a_0^{(\nu)} = 1.$$

After cancellation of the leading exponential part, S_ν must satisfy formally

$$2S_\nu'(z) + S_\nu''(z) + \left(\frac{1}{4} - \nu^2\right) z^{-2} S_\nu(z) = 0.$$

Equating the coefficient of z^{-k-1}, for $k \ge 1$, gives

$$a_k^{(\nu)} = -\frac{4\nu^2 - (2k-1)^2}{8k}\, a_{k-1}^{(\nu)},$$

which is precisely (100.48). The classical asymptotic theorem for this ordinary differential equation gives the corresponding remainder after truncation on every closed angular subsector containing the positive real axis and not crossing a Stokes ray; in this proposition only the positive real $z = \sqrt{\lambda}$ ray is used. The common factor $e^z/\sqrt{2\pi z}$ cancels in the ratio. For $\nu = 1, 2$,

$$I_1(z) \sim \frac{e^z}{\sqrt{2\pi z}}\left(1 - \frac{3}{8z} - \frac{15}{128z^2} - \frac{315}{3072z^3} - \frac{4725}{32768z^4} + O(z^{-5})\right),$$
$$I_2(z) \sim \frac{e^z}{\sqrt{2\pi z}}\left(1 - \frac{15}{8z} + \frac{105}{128z^2} + \frac{945}{3072z^3} + \frac{10395}{32768z^4} + O(z^{-5})\right).$$

Series division gives

$$\frac{I_2(z)}{I_1(z)} = 1 - \frac{3}{2z} + \frac{3}{8z^2} + \frac{3}{8z^3} + \frac{63}{128z^4} + O(z^{-5}).$$

Multiplying by $z/(4\pi^2)$ gives (100.47).

These expansions describe a near-BPS Wilson-line observable. The lightlike cusp coefficient of Section 49.24 and the planar large-spin scaling function $f(g)$ of Chapter 98 are cusp data in different analytic limits of the generalized cusp. Their common normalization check is the line-defect definition $C_D = 12B(\lambda)$ together with the protected circular-loop Ward identity.

100.9 Wilson-Line Flux Tube as a Conformal Spectral Datum

Pure Yang–Mills glueball spectroscopy, $\mathcal{N} = 1$ SYM chiral branches, softly broken $\mathcal{N} = 2$ monopole condensation, and planar $\mathcal{N} = 4$ Wilson-line integrability are related examples of gauge-theory spectral data, but the objects being measured differ. In planar $\mathcal{N} = 4$ SYM, conformal invariance supplies no intrinsic mass gap. The analog of a flux-tube spectrum is instead a defect spectrum: the Hamiltonian is the generator of translations along a line or of boosts around a null cusp, and the excitations are states of the defect Hilbert space associated to the Wilson line.

Definition 100.17 (Planar $\mathcal{N} = 4$ Wilson-line flux-tube datum)**.** Fix the planar limit of $\mathcal{N} = 4$ $SU(N)$ SYM with $\lambda = g_{\mathrm{YM}}^2 N$ and $g = \sqrt{\lambda}/(4\pi)$. A Wilson-line flux-tube datum consists of:

(i) a half-BPS straight Maldacena–Wilson line or its analytic continuation to a null cusp, with the scalar coupling normalized as in the preceding section;

(ii) the defect Hilbert space $\mathcal{H}_{\mathrm{def}}$ obtained by radial or boost quantization around the line or cusp, with Hamiltonian H_{def} and spatial momentum P_{def} along the defect;

(iii) local defect fields $\mathbb{O}_a(t)$, including the displacement operators $\mathbb{D}_i(t)$, scalar insertions, fermion insertions, and covariant derivatives along the line;

(iv) a spectral resolution of connected defect two-point functions,

$$\langle \mathbb{O}_a(t)\mathbb{O}_b(0)\rangle_{\mathrm{conn}} = \sum_\alpha \int d\mu_\alpha(u)\, F_{a,\alpha}(u) F_{b,\alpha}(u)^* e^{-E_\alpha(u)|t| + ip_\alpha(u)t}, \tag{100.49}$$

where u denotes rapidity-type spectral coordinates when the integrability description is available;

(v) transition data, such as pentagon form factors or mirror-TBA kernels, that reconstruct multi-particle defect correlators from the one-particle energies, momenta, measures, and scattering phases.

The cusp anomalous dimension is the defect vacuum energy density in the null-cusp frame, while the generalized scaling function and pentagon OPE data are higher spectral coordinates of the same defect system.

Remark 100.18 (Cusp anomalous dimension as defect ground-state energy). Assume the null-cusp Wilson line admits a defect Hamiltonian description on a strip of length $L = \log(R/\epsilon)$, where R and ϵ are the infrared and ultraviolet scales used in the cusp renormalization. If $E_0(\varphi, \theta)$ denotes the extensive ground-state energy in this strip normalization, then

$$E_0(\varphi, \theta) = \Gamma_{\mathrm{cusp}}(\varphi, \theta)\, L + O(1). \tag{100.50}$$

For the lightlike cusp, the coefficient of L is the lightlike cusp anomalous dimension $\Gamma_{\mathrm{cusp}}(\lambda)$. In the large-spin operator problem, the same coefficient controls the leading logarithmic growth

$$\Delta(S) - S = \Gamma_{\mathrm{cusp}}(\lambda) \log S + O(S^0) \qquad (S \to \infty) \tag{100.51}$$

in the normalization of Chapter 98.

The renormalized cusped-line expectation value has the defining logarithmic form

$$\log\langle W_{\mathrm{cusp}}\rangle_{\mathrm{ren}} = -\Gamma_{\mathrm{cusp}}(\varphi, \theta) \log \frac{R}{\epsilon} + O(1).$$

In the defect Hamiltonian representation, the same path integral on the logarithmic strip is

$$\langle W_{\mathrm{cusp}}\rangle_{\mathrm{ren}} = \langle B_{\mathrm{out}}| \mathrm{e}^{-LH_{\mathrm{def}}} |B_{\mathrm{in}}\rangle_{\mathrm{ren}}, \qquad L = \log \frac{R}{\epsilon},$$

with boundary states determined by the ultraviolet and infrared regularizations. If the lowest defect energy is isolated in the sector selected by the cusp boundary conditions, the large-L asymptotic is

$$\log\langle W_{\mathrm{cusp}}\rangle_{\mathrm{ren}} = -E_0(\varphi, \theta) + O(L^0) = -\varepsilon_0(\varphi, \theta)L + O(L^0),$$

where $E_0 = \varepsilon_0 L$ is extensive in the strip length. Comparing the coefficients of L gives $\varepsilon_0 = \Gamma_{\mathrm{cusp}}$, which is (100.50). The large-spin relation (100.51) follows from the same logarithmic renormalization of the lightlike Wilson line that appears in the factorized soft-collinear limit of twist-two operators; in the planar integrability description this coefficient is reproduced by the BES equation and by the large-spin limit of the asymptotic Bethe equations.

100.9.1 Pentagon Channel Expansion as a Defect Spectral Resolution

The pentagon OPE used for null polygonal Wilson loops is a spectral resolution in the defect Hilbert space of the null-cusp flux tube. The word "pentagon" does not replace the Hilbert-space statement: it names the transition amplitude between two adjacent channel decompositions of the same renormalized Wilson-line observable.

Fix a channel of a null polygon whose conformal cross-ratios are parametrized by (τ, σ, ϕ). Here τ is the Euclidean propagation length in the chosen flux-tube channel, σ is the translation coordinate along the defect, and ϕ is the angle conjugate to the transverse rotation charge. The channel evolution operator is

$$U(\tau, \sigma, \phi) = \exp\{-\tau H_{\mathrm{def}} + \mathrm{i}\sigma P_{\mathrm{def}} + \mathrm{i}\phi J_{\mathrm{def}}\}, \tag{100.52}$$

where H_{def}, P_{def}, and J_{def} are the energy, longitudinal momentum, and transverse angular-momentum generators of Definition 100.17. Let $\langle \mathcal{P}_{\mathrm{R}}|$ and $|\mathcal{P}_{\mathrm{L}}\rangle$ denote the right and left boundary states prepared by the adjacent Wilson-line segments in this channel, after the local cusp and edge renormalizations have been fixed. The channel contribution is the matrix element

$$\mathcal{W}_{\mathrm{ch}}(\tau, \sigma, \phi) = \langle \mathcal{P}_{\mathrm{R}}|U(\tau, \sigma, \phi)|\mathcal{P}_{\mathrm{L}}\rangle. \tag{100.53}$$

Pentagon OPE from the defect spectral resolution. Assume the planar Wilson-line defect datum of Definition 100.17 and assume that, in the channel under consideration, the identity on $\mathcal{H}_{\mathrm{def}}$ admits the generalized multiparticle resolution

$$\mathbf{1}_{\mathrm{def}} = |0\rangle\langle 0| + \sum_{k\geq 1} \sum_{a_1,\ldots,a_k} \frac{1}{|\operatorname{Aut}(a_1, \ldots, a_k)|} \int \prod_{r=1}^{k} \frac{\mathrm{d}u_r}{2\pi}\, \mu_{a_r}(u_r)\, |\mathbf{a}, \mathbf{u}\rangle\langle \mathbf{a}, \mathbf{u}|, \tag{100.54}$$

where a_r labels the one-particle species, u_r is its rapidity coordinate, $\mu_a(u)$ is the one-particle spectral measure in this normalization, and $|\operatorname{Aut}(a_1, \ldots, a_k)|$ is the product of factorials of the multiplicities of identical species. Suppose further that these states diagonalize H_{def}, P_{def}, and J_{def} additively:

$$E(\mathbf{a}, \mathbf{u}) = \sum_{r=1}^{k} E_{a_r}(u_r), \quad p(\mathbf{a}, \mathbf{u}) = \sum_{r=1}^{k} p_{a_r}(u_r), \quad m(\mathbf{a}) = \sum_{r=1}^{k} m_{a_r}.$$

Then (100.53) has the expansion

$$\begin{aligned}
\mathcal{W}_{\mathrm{ch}}(\tau, \sigma, \phi) &= \langle \mathcal{P}_{\mathrm{R}}|0\rangle\langle 0|\mathcal{P}_{\mathrm{L}}\rangle \\
&\quad + \sum_{k\geq 1} \sum_{a_1,\ldots,a_k} \frac{1}{|\operatorname{Aut}(a_1, \ldots, a_k)|} \int \prod_{r=1}^{k} \frac{\mathrm{d}u_r}{2\pi}\, \mu_{a_r}(u_r) \\
&\quad \times \exp\left[-\tau \sum_{r=1}^{k} E_{a_r}(u_r) + \mathrm{i}\sigma \sum_{r=1}^{k} p_{a_r}(u_r) + \mathrm{i}\phi \sum_{r=1}^{k} m_{a_r}\right] \\
&\quad \times P_{\mathrm{R}}(\mathbf{a}, \mathbf{u})\, P_{\mathrm{L}}(\mathbf{a}, \mathbf{u})^*,
\end{aligned} \tag{100.55}$$

where

$$P_{\mathrm{R}}(\mathbf{a}, \mathbf{u}) = \langle \mathcal{P}_{\mathrm{R}}|\mathbf{a}, \mathbf{u}\rangle, \qquad P_{\mathrm{L}}(\mathbf{a}, \mathbf{u}) = \langle \mathcal{P}_{\mathrm{L}}|\mathbf{a}, \mathbf{u}\rangle.$$

Insert (100.54) between $\langle \mathcal{P}_{\mathrm{R}}|$ and $U(\tau, \sigma, \phi)|\mathcal{P}_{\mathrm{L}}\rangle$ in (100.53). On the vacuum term, H_{def}, P_{def}, and J_{def} give the chosen vacuum normalization. On a multiparticle eigenstate, the assumed additivity gives

$$U(\tau, \sigma, \phi)|\mathbf{a}, \mathbf{u}\rangle = \exp\left[-\tau \sum_{r} E_{a_r}(u_r) + \mathrm{i}\sigma \sum_{r} p_{a_r}(u_r) + \mathrm{i}\phi \sum_{r} m_{a_r}\right] |\mathbf{a}, \mathbf{u}\rangle.$$

The overlap with the right and left boundary states gives the two transition amplitudes displayed in (100.55). The automorphism factor is the standard quotient by permutations of indistinguishable equal-species particles; without it, the same unordered spectral state would be integrated once for each such permutation. This proves the expansion.

In the integrability solution, the overlaps P_{R} and P_{L} are encoded by elementary pentagon transition functions. For a diagonal scalar component of two ordered one-particle excitations in a fixed rapidity normalization, Watson consistency is the identity

$$P_{a|b}(u|v) = S_{ab}(u, v)\, P_{b|a}(v|u), \tag{100.56}$$

with S_{ab} the flux-tube scattering phase or matrix element in the same rapidity normalization. Mirror moves relate a particle crossing an edge of the pentagon to an excitation in the adjacent channel. These relations are powerful constraints, but they are not substitutes for the spectral datum: the species, rapidity sheets, measures, transition normalizations, and renormalized Wilson-line observable must all be fixed before the formula has a mathematical meaning.

This defect datum is the $\mathcal{N} = 4$ member of the broader gauge-theory family discussed in Section 90.6: the protected local operators, line operators, and spectral measures must be defined before their integrable or semiclassical solution methods are invoked. The integrability problem in planar $\mathcal{N} = 4$ SYM gives an unusually complete solution of the defect spectral data; a derivation from a nonperturbatively defined four-dimensional planar QFT remains part of the proof boundary below.

100.10 Proof Boundary

Planar integrability gives a highly constrained exact spectral structure for four-dimensional interacting QFT. Symmetry fixes matrix scattering, crossing fixes the scalar factor, Bethe equations organize asymptotic spectra, mirror TBA incorporates finite size, and the QSC compresses the exact spectral problem into analytic functional equations. The remaining mathematical task is to derive this structure from a precise nonperturbative formulation of planar $\mathcal{N} = 4$ SYM, including the topology in which local operators, large-N limits, and analytic continuation in λ are controlled.

Chapter 101

Supersymmetric Localization On Compact Manifolds

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Supersymmetric localization is an exact-evaluation method only after the regulated functional integral has been specified with enough information to justify deformation of the integration cycle. This chapter separates the finite-dimensional theorem from the QFT formulae used in four-dimensional $\mathcal{N} = 2$ gauge theory on S^4 and three-dimensional $\mathcal{N} = 2$ Chern–Simons–matter theory on S^3. The formulae below are therefore not introduced as formal replacements for a definition of a QFT: they are consequences of a localization structure in the sense of Definition 86.7, together with the determinant and instanton-sector prescriptions explicitly stated here.

Definition 101.1 (Compact supersymmetric localization structure)**.** A compact supersymmetric localization structure is a tuple

$$\mathfrak{L}_M = \big(M, \mathfrak{B}, \{(\mathfrak{F}_\Lambda, C_\Lambda, \mu_\Lambda)\}_{\Lambda\in I}, Q_\Lambda, B_\Lambda, V_\Lambda, S_\Lambda, \mathcal{O}_\Lambda, \mathfrak{F}_\Lambda^{\mathrm{fix}}, \mathcal{N}_\Lambda^\bullet, \mathfrak{R}_\Lambda, \mathcal{C}_\Lambda\big)$$

with the following meaning.

(i) M is the compact oriented Riemannian manifold on which the QFT is placed, and $\mathfrak{B}$ denotes the chosen rigid-supergravity background: spin or spin-c structure, R-symmetry and flavor bundles, background connections, masses, and any metric or framing data used by the preserved supercharge.

(ii) I is a directed regulator set. For each $\Lambda \in I$, $\mathfrak{F}_\Lambda$ is the regulated super field-variable space, including gauge-fixing or BV variables when gauge symmetry is present; C_Λ is the declared integration cycle, including orientation, boundary, singular strata, and gauge quotient; and μ_Λ is the regulated Berezinian density or BV half-density. Fermionic variables are understood as coordinates on a locally super-ringed space, so μ_Λ is not a Borel measure.

(iii) Q_Λ is an odd derivation of the regulated algebra of observables, and

$$B_\Lambda := Q_\Lambda^2 = \mathcal{L}_{v_\Lambda} + \delta_{\mathrm{gauge}}(\Lambda_Q) + \delta_R(\Theta_R) + \delta_F(\Theta_F)$$

is the specified even transformation: an isometry of M, a gauge transformation, and R- or flavor rotations. In a gauge theory this statement is made in the BV complex; compatibility means that Q_Λ commutes with the gauge-fixed BV differential up to B_Λ-symmetry and that the regulated quantum master equation has the residual boundary terms specified below.

(iv) S_Λ is the regulated action and $\mathcal{O}_\Lambda$ is the regulated protected insertion. They obey

$$Q_\Lambda S_\Lambda = 0, \qquad Q_\Lambda \mathcal{O}_\Lambda = 0, \qquad \mathcal{L}_{Q_\Lambda}\mu_\Lambda = 0$$

in the regulated theory, or else the anomaly term is stated explicitly as a defect of these equations.

(v) V_Λ is an odd localizing functional such that $\mathrm{Re}(Q_\Lambda V_\Lambda) \geq 0$ on the bosonic body of C_Λ, with equality precisely on a stated fixed locus $\mathfrak{F}_\Lambda^{\mathrm{fix}}$. The Q_Λ-Stokes boundary term

$$\int_{\partial C_\Lambda} \mu_\Lambda \, \mathcal{O}_\Lambda V_\Lambda \exp\{-S_\Lambda - tQ_\Lambda V_\Lambda\}$$

is either proved to vanish, canceled by a declared relative-cycle or boundary counterterm, or recorded as an open obstruction.

(vi) $\mathcal{N}_\Lambda^\bullet$ is the normal Q_Λ-complex to the smooth part of $\mathfrak{F}_\Lambda^{\mathrm{fix}}$, with zero modes split off as coordinates or insertion modes. The one-loop determinant is the Berezin Gaussian/Euler class of the acyclic part of this complex, with orientation, phase, zeta or heat-kernel regularization, framing, and local counterterm conventions included in the structure.

(vii) $\mathfrak{R}_\Lambda$ is the residual finite-dimensional integration or summation prescription after normal integration: Cartan contour, residue chamber, Jeffrey–Kirwan vector, instanton compactification, polarization, or equivariant pushforward. Singular fixed strata such as zero-size instantons are part of $\mathfrak{R}_\Lambda$, not automatic consequences of the notation $\int \mathcal{D}\Phi$.

(viii) $\mathcal{C}_\Lambda$ is the continuum comparison structure: topology on protected observables, renormalization and contact-term scheme, convergence or matching statement for the limit $\Lambda \to \infty$, and the precise status of any equality between the regulated localization formula and the underlying QFT observable.

All localization formulae in this chapter are statements about consequences of a structure $\mathfrak{L}_M$. If one of the entries is supplied only by a standard prescription rather than by a regulator theorem, the corresponding formula is a controlled prescription or hypothesis at that entry, even when the finite-dimensional localization identity itself is a theorem.

101.1 Finite-Dimensional Deformation Identity

Let X be an oriented finite-dimensional supermanifold equipped with an integration cycle $C \subset X$ and Berezinian density μ. Let Q be an odd vector field on X, and let B be the even vector field Q^2.

For the following proposition all functions and densities are assumed smooth on a neighborhood of C, and C is assumed to be either compact without boundary or equipped with boundary conditions for which the displayed Stokes term vanishes.

Remark 101.2 (Localization deformation identity). Let $S, V, \mathcal{O}$ be functions on X with parities $|S| = |\mathcal{O}| = 0$, $|V| = 1$. Suppose

$$QS = 0, \qquad Q\mathcal{O} = 0, \qquad BS = B\mathcal{O} = BV = 0,$$

and suppose the Berezinian density is Q-invariant: $\mathcal{L}_Q\mu = 0$. Define

$$I(t) = \int_C \mu\, \mathcal{O}\, \exp\{-S - t\, QV\}.$$

If $Q(\mu\, \mathcal{O}\, V\, \exp\{-S - tQV\})$ has zero integral over C, then $I(t)$ is independent of t. Because Q is an odd derivation and $Q^2V = BV = 0$,

$$\frac{dI}{dt} = -\int_C \mu\, \mathcal{O}\, (QV)\, \exp\{-S - tQV\}.$$

The hypotheses $QS = 0$, $Q\mathcal{O} = 0$, and $Q(QV) = 0$ give

$$\begin{aligned} Q(\mathcal{O}\, V\, \exp\{-S - tQV\}) &= (Q\mathcal{O})V\mathrm{e}^{-S-tQV} + \mathcal{O}(QV)\mathrm{e}^{-S-tQV} \\ &\quad - \mathcal{O}V\, Q(S + tQV)\mathrm{e}^{-S-tQV} \\ &= \mathcal{O}(QV)\mathrm{e}^{-S-tQV}. \end{aligned}$$

Using $\mathcal{L}_Q\mu = 0$, the right hand side integrates to the assumed zero Stokes term. Hence $dI/dt = 0$.

Definition 101.3 (Clean fixed locus and normal complex)**.** Let $F \subset C_{\bar{0}}$ be the common zero locus of the bosonic part of QV on the body of the integration cycle. The fixed locus is called clean for the finite-dimensional localization quadruple (C, μ, Q, V) if F is a submanifold, if the Hessian of $\mathrm{Re}(QV)$ is nondegenerate on a chosen normal bundle $N_{F,\bar{0}}$, and if the linearization of Q in the directions normal to F gives a finite $\mathbb{Z}_2$-graded complex

$$N_F^\bullet : \qquad N_{F,\bar{0}} \xrightarrow{Q_{\mathrm{lin}}} N_{F,\bar{1}} \xrightarrow{Q_{\mathrm{lin}}} N_{F,\bar{0}}$$

whose cohomology has been split off as zero-mode coordinates on F. The one-loop determinant is the Berezin Gaussian of this acyclic normal complex, with orientation and square-root branches inherited from the integration cycle.

Assume, in addition to the deformation identity in Remark 101.2, that $\mathrm{Re}(QV) \geq 0$ on the bosonic body of C, that F is compact and clean, and that the large-t integral is dominated by a tubular neighborhood of F. In local coordinates y along F, bosonic normal coordinates x^i, and fermionic normal coordinates ϑ^A, the even quadratic normal model has the form

$$(QV)_{\mathrm{quad}} = \frac{1}{2}x^i A_{ij}(y)x^j - \frac{1}{2}\vartheta^A \mathsf{F}_{AB}(y)\vartheta^B. \tag{101.1}$$

Here A is symmetric and positive on the declared real normal cycle, and F is antisymmetric. The differential in the normal Q-complex is encoded in the fermion-fermion matrix F, not in an odd-even term: an ordinary even quadratic action cannot contain a term linear in one fermionic normal coordinate and one bosonic normal coordinate with an even numerical coefficient. If, after choosing splittings,

$$\mathsf{F} = \begin{pmatrix} 0 & D \\ -D^T & 0 \end{pmatrix}, \tag{101.2}$$

then the corresponding Pfaffian is $\pm \det D$, with the sign fixed by the orientation of the odd variables. With the conventional finite-dimensional Gaussian normalization absorbed into μ, the local normal factor is therefore

$$Z_{\text{normal}}(y) = \frac{\operatorname{Pf} \mathsf{F}(y)}{\sqrt{\det A(y)}}. \tag{101.3}$$

The finite normal Gaussian factor is fixed by the ordinary Gaussian integral and the defining coefficient of the Pfaffian. Let A be an invertible real symmetric matrix whose real part is positive on the chosen bosonic normal cycle, and let F be an invertible antisymmetric $2m \times 2m$ matrix. With the Berezin orientation $d\vartheta_{2m} \cdots d\vartheta_1$,

$$\int d^{2m}\vartheta \, \exp\!\left(\frac{1}{2}\vartheta^A \mathsf{F}_{AB} \vartheta^B\right) = \operatorname{Pf} \mathsf{F}. \tag{101.4}$$

After absorbing the ordinary factor $(2\pi)^{n/2}$ into the local normalization of the bosonic measure,

$$\int dx \, d\vartheta \, \exp\!\left(-\frac{1}{2}x^i A_{ij} x^j + \frac{1}{2}\vartheta^A \mathsf{F}_{AB} \vartheta^B\right) = \frac{\operatorname{Pf} \mathsf{F}}{\sqrt{\det A}}. \tag{101.5}$$

If A or F has a zero mode, that direction must be split off and integrated on the fixed locus or saturated by insertions; it is not part of the one-loop determinant. The Berezin integral extracts the coefficient of the ordered monomial $\vartheta^1 \cdots \vartheta^{2m}$. By definition of the Pfaffian, this coefficient in $\exp(\frac{1}{2}\vartheta^A \mathsf{F}_{AB} \vartheta^B)$ is $\operatorname{Pf} \mathsf{F}$, proving (101.4). The bosonic Gaussian on the declared cycle is

$$\int_{\mathbb{R}^n} d^n x \, \exp\!\left(-\frac{1}{2}x^i A_{ij} x^j\right) = \frac{(2\pi)^{n/2}}{\sqrt{\det A}}$$

after choosing the square-root branch determined by the cycle. Multiplying the bosonic and fermionic factors and absorbing $(2\pi)^{n/2}$ into the local measure gives (101.5).

If a bosonic zero mode is present, the Gaussian integral diverges or becomes a distribution along that direction; if a fermionic zero mode is present, the top Berezin coefficient is absent unless an insertion supplies the missing odd variable. Thus zero modes are coordinates on F, insertions to be saturated, or obstructions showing that the proposed localized observable vanishes. A determinant formula without this zero-mode statement is not a localization theorem.

Equation (101.3) is the finite model for the one-loop determinants in the compact-space QFT examples below.

101.2 QFT Localization Structure

For a supersymmetric QFT on a compact Riemannian manifold M, the symbol

$$Z_M = \int_{\mathfrak{F}} \mathcal{D}\Phi \, \mathrm{e}^{-S[\Phi]}$$

is shorthand for the regulated data in Definition 101.1. In the minimal notation one keeps only

$$(\mathfrak{F}_\Lambda, C_\Lambda, \mu_\Lambda, S_\Lambda, Q_\Lambda, V_\Lambda)$$

together with the limiting prescription $\Lambda \to \infty$, but the omitted entries of $\mathfrak{L}_M$ remain logically present. Thus the localized expression is justified only when the following conditions are verified or retained as hypotheses:

$$Q_\Lambda S_\Lambda = 0, \qquad \mathcal{L}_{Q_\Lambda}\mu_\Lambda = 0, \qquad Q_\Lambda \mathcal{O}_\Lambda = 0,$$
$$\int_{\partial C_\Lambda} \mu_\Lambda\, \mathcal{O}_\Lambda V_\Lambda \mathrm{e}^{-S_\Lambda - tQ_\Lambda V_\Lambda} = 0,$$
$$\mathrm{Re}(Q_\Lambda V_\Lambda) \geq 0 \quad \text{on the bosonic integration cycle.}$$

The fixed locus, zero modes, determinants, and singular strata must then be included in the limiting prescription. Noncompact Coulomb directions, Jeffrey–Kirwan residues, and point-instanton compactifications are not automatic outputs of the symbol $\int \mathcal{D}\Phi$. They are part of the regulated localization structure. In the formulae below this structure will be stated by naming the finite-dimensional variables, the contour, and the determinant or equivariant-integral prescription.

101.3 S^4 Localization In Four-Dimensional $\mathcal{N} = 2$ Gauge Theory

Let G be compact connected, with Lie algebra $\mathfrak{g}$, maximal torus T, Cartan algebra $\mathfrak{t}$, Weyl group W_G, and root system $\Delta \subset \mathfrak{t}^*$. The monograph convention for non-Abelian gauge theory is

$$\mathrm{tr}_\square(t^a t^b) = \delta^{ab}, \qquad S_{\mathrm{YM}} = \frac{1}{4g_{\mathrm{YM}}^2}\int \mathrm{tr}_\square(F_{\mu\nu}F^{\mu\nu}).$$

With $(a,a) := \mathrm{tr}_\square(a^2)$ for $a \in \mathfrak{t}$, the round S^4_r localization variable is the constant adjoint scalar $a \in \mathfrak{t}$ on the Coulomb locus. The classical Gaussian in this trace-delta convention is

$$Z_{\mathrm{cl}}(a) = \exp\left\{-\frac{4\pi^2 r^2}{g_{\mathrm{YM}}^2}(a,a)\right\}. \tag{101.6}$$

The coefficient equals the familiar half-trace coefficient because $t^a = \sqrt{2}\,T^a$ and $g_{\mathrm{ht}}^2 = 2g_{\mathrm{YM}}^2$, so

$$\frac{8\pi^2}{g_{\mathrm{ht}}^2}\,\mathrm{tr}_{\mathrm{ht}}(a_{\mathrm{ht}}^2) = \frac{4\pi^2}{g_{\mathrm{YM}}^2}(a,a).$$

The localization structure contains more than the Cartan variable. An off-shell $4D$ $\mathcal{N} = 2$ vector multiplet on the round sphere consists of a connection A_μ, complex adjoint scalars $\phi, \bar{\phi}$, gaugini $\lambda_A, \bar{\lambda}_A$ with $A = 1, 2$ an $SU(2)_R$ index, and auxiliary fields $D_{AB} = D_{BA}$. The chosen supercharge is specified by a pair of conformal Killing spinors $(\xi_A, \bar{\xi}_A)$. Its square has the form

$$Q^2 = \mathcal{L}_v + \delta_{\mathrm{gauge}}(\Lambda) + \delta_{SU(2)_R}(\Theta) + \delta_{\mathrm{flavor}}(m), \tag{101.7}$$

where $v^\mu = \xi^A\sigma^\mu\bar{\xi}_A$ is a Killing vector on S^4_r, Λ is an adjoint-valued gauge parameter, Θ is an $SU(2)_R$ rotation, and m denotes background flavor masses when present. For the standard round-sphere choice, v vanishes at the north and south poles and generates rotations in the two tangent planes near each pole.

The bosonic fixed-locus statement used in the matrix model is the following field-theoretic reduction. After choosing the real integration cycle for which the localizing term is nonnegative, all smooth bulk fields on the trivial bundle are gauge-equivalent to

$$A_\mu = 0, \qquad \phi, \bar{\phi} \text{ constant with real-cycle coordinate } a, \qquad D_{AB} = D_{AB}(a), \qquad a \in \mathfrak{t}, \quad (101.8)$$

with the auxiliary field determined algebraically by the Killing-spinor background. Nonzero instanton number is not represented by a smooth global bulk deformation of (101.8); in the large-localizing parameter limit it is concentrated in the two tangent-space models at the poles. The north-pole tangent space contributes an equivariant instanton integral on $\mathbb{C}^2$, and the south pole contributes the anti-instanton factor. This is why the compactification or contour defining Z_{inst} is part of Definition 101.5, rather than an optional interpretation added after the formula.

Define the even entire function

$$H(x) = \prod_{n=1}^{\infty} \left(1 + \frac{x^2}{n^2}\right)^n \exp\!\left(-\frac{x^2}{n}\right), \tag{101.9}$$

where the product is taken in the displayed counterterm convention.

Lemma 101.4 (Local uniform convergence and Barnes representation of H)**.** *The product in* (101.9) *converges locally uniformly on the complex x-plane, defines an even entire function, and satisfies*

$$\frac{d}{dx} \log H(x) = \sum_{n=1}^{\infty} \left(\frac{2nx}{n^2 + x^2} - \frac{2x}{n}\right) \tag{101.10}$$

away from its zeros. If G_B is the Barnes G-function normalized by $G_B(1) = 1$ and $G_B(z+1) = \Gamma(z)G_B(z)$, then

$$H(x) = \exp\!\big(-(1+\gamma_{\mathrm{E}})x^2\big)\, G_B(1 + \mathrm{i}x) G_B(1 - \mathrm{i}x). \tag{101.11}$$

Changing the exponential prefactor $\exp(cx^2)$ is a local quadratic counterterm for the background vector multiplet or the gauge coupling; it is therefore a convention that must be kept fixed when comparing Gaussian coefficients.

Proof. Let $K \subset \mathbb{C}$ be compact and choose N so that $|x|^2/n^2 \leq 1/2$ for $x \in K$ and $n \geq N$. For $n \geq N$, the logarithm of the n-th factor is

$$n \log\!\left(1 + \frac{x^2}{n^2}\right) - \frac{x^2}{n} = -\frac{x^4}{2n^3} + R_n(x), \qquad |R_n(x)| \leq C_K n^{-5},$$

uniformly on K. The series of logarithms therefore converges uniformly on K after the finitely many initial factors are separated. The resulting Weierstrass product converges locally uniformly and defines an entire function. Each factor depends only on x^2, so the function is even; its zeros are at $x = \pm \mathrm{i}n$ with the multiplicities encoded by the exponent n.

On a compact set avoiding these zeros, the differentiated logarithmic series is uniformly convergent because

$$\frac{d}{dx}\left[n \log\!\left(1 + \frac{x^2}{n^2}\right) - \frac{x^2}{n}\right] = \frac{2nx}{n^2 + x^2} - \frac{2x}{n} = O_K(n^{-3}).$$

Termwise differentiation is then justified by the Weierstrass theorem for locally uniformly convergent holomorphic series and gives (101.10).

For the Barnes representation, use the standard Weierstrass product

$$G_B(1+z) = (2\pi)^{z/2} \exp\left[-\frac{1}{2}\big(z + (1+\gamma_{\rm E})z^2\big)\right] \prod_{n=1}^{\infty} \left(1+\frac{z}{n}\right)^n \exp\left(\frac{z^2}{2n} - z\right),$$

which fixes the normalization $G_B(1) = 1$ and the functional equation $G_B(z+1) = \Gamma(z)G_B(z)$. Multiplying the products for $z = \mathrm{i}x$ and $z = -\mathrm{i}x$, the factors $(2\pi)^{\pm \mathrm{i}x/2}$ and $\exp(\mp \mathrm{i}x/2)$ cancel, while

$$\left(1+\frac{\mathrm{i}x}{n}\right)^n \left(1-\frac{\mathrm{i}x}{n}\right)^n \exp\left(-\frac{x^2}{n}\right) = \left(1+\frac{x^2}{n^2}\right)^n \exp\left(-\frac{x^2}{n}\right).$$

The remaining exponential is $\exp((1+\gamma_{\rm E})x^2)$, which gives (101.11). Multiplying H by $\exp(cx^2)$ changes only this local quadratic prefactor. □

The one-loop determinant formula below uses the following finite-part operation. Suppose that, after pairing all acyclic Q-doublets in a fixed root or weight sector, the unpaired finite-cutoff determinant has the form

$$D^{\rm raw}_{\kappa,N}(x) = \prod_{n=1}^{N} (n^2 + x^2)^{\kappa n}, \qquad \kappa \in \mathbb{Z}. \tag{101.12}$$

Here x is the dimensionless equivariant mass in that sector: $x = r\,\alpha(a)$ for a vector root, or $x = r\rho(a) + rm$ for a hypermultiplet weight. The exponent n is the residual cohomological multiplicity of the transversally elliptic complex after the paired boson–fermion modes have cancelled. The constant factor $\prod_{n=1}^{N} n^{2\kappa n}$ is independent of the Coulomb variable and is removed by the normalization of the partition function. The remaining Coulomb-dependent finite part is

$$D^{\rm fp}_{\kappa,N}(x) = \prod_{n=1}^{N} \left(1+\frac{x^2}{n^2}\right)^{\kappa n} \exp\left(-\kappa\frac{x^2}{n}\right) = H_N(x)^{\kappa}, \tag{101.13}$$

where H_N is the N-th partial product of (101.9). The exponential in (101.13) subtracts the logarithmically divergent local term $\kappa x^2 \sum_{n\le N} n^{-1}$. Since x is linear in the background scalar and mass parameters, this subtraction is a quadratic local counterterm in the rigid vector or flavor background. The finite determinant is therefore the finite part specified by the quadratic counterterm convention in Lemma 101.4.

With this convention the perturbative S^4 determinant is obtained from a short sector calculation. A vector root pair $\{\alpha, -\alpha\}$ contributes the Cartan gauge-fixing factor $\alpha(a)^2$ and the finite part with $\kappa = 2$, hence $H(r\alpha(a))^2$ for each positive root. A full hypermultiplet weight ρ contributes the inverse finite part with $\kappa = -1$, hence $H(r\rho(a) + rm)^{-1}$. The formula below is exactly this sector calculation multiplied over the root and weight multisets. A derivation from the infinite-dimensional field theory has to supply the regulated normal complex, the pairing of acyclic modes, the cohomological multiplicity n, and the absence of boundary contributions in the chosen field-variable cycle; those structures are part of Definition 101.1.

For a vector multiplet and hypermultiplets of masses m_ℓ in representations R_ℓ, the perturbative determinant on the round sphere is

$$Z^{S^4}_{1-\rm loop}(a) = \prod_{\alpha\in\Delta_+} \alpha(a)^2 H(r\,\alpha(a))^2 \prod_{\ell} \prod_{\rho\in{\rm wt}(R_\ell)} H(r\,\rho(a) + r\,m_\ell)^{-1}. \tag{101.14}$$

The factor $\prod_{\alpha>0}\alpha(a)^2$ is the Cartan gauge-fixing Jacobian. If the representation is real or pseudoreal, the weights are counted with the multiplicity appropriate to a full hypermultiplet in the chosen convention. A different convention for the hypermultiplet mass shifts the argument of H by the corresponding R-symmetry background; this is a coordinate change in the localization structure, not a change of the underlying determinant complex.

$\mathcal{N}=4$ determinant cancellation on S^4. Let G be compact connected with complexified Lie algebra $\mathfrak{g}_{\mathbb{C}}$, Cartan subalgebra $\mathfrak{t}_{\mathbb{C}}$, root system Δ, chosen positive roots Δ_+, and invariant inner product fixed by the Yang–Mills action $\frac{1}{4g_{\mathrm{YM}}^2}\int \mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$. Consider the $\mathcal{N}=2$ localization formula (101.14) for one vector multiplet and one massless adjoint hypermultiplet, with the hypermultiplet mass coordinate tuned to the point where the field content is an $\mathcal{N}=4$ vector multiplet. Then the non-polynomial H-factors cancel identically and the one-loop factor is only the Cartan gauge-fixing Jacobian:

$$Z_{1-\mathrm{loop}}^{\mathcal{N}=4}(a)=\prod_{\alpha\in\Delta_+}\alpha(a)^2. \tag{101.15}$$

The adjoint representation has weight multiset

$$\mathrm{wt}(\mathfrak{g}_{\mathbb{C}})=\Delta\ \sqcup\ \{0,\dots,0\},$$

where the zero weight occurs $\mathrm{rank}\,G$ times. By Lemma 101.4, H is an even entire function and $H(0)=1$. Therefore the zero weights of the adjoint hypermultiplet contribute the constant $H(0)^{-\mathrm{rank}\,G}=1$, while each pair $\{\alpha,-\alpha\}$ contributes

$$H(r\,\alpha(a))^{-1}H(-r\,\alpha(a))^{-1}=H(r\,\alpha(a))^{-2}.$$

The vector multiplet contributes

$$\prod_{\alpha\in\Delta_+}\alpha(a)^2H(r\,\alpha(a))^2.$$

Multiplying the vector and adjoint-hyper factors root pair by root pair leaves exactly (101.15). No regularization ambiguity is hidden in this cancellation: changing the normalization of H by an even quadratic exponential would multiply the vector and adjoint hypermultiplet factors by inverse local quadratic counterterms at the $\mathcal{N}=4$ point, so the cancellation holds in the determinant convention fixed in (101.11).

Definition 101.5 (Nekrasov factor in the S^4 formula)**.** For the four-dimensional localization formula, $Z_{\mathrm{inst}}(a,\tau;\epsilon_1,\epsilon_2)$ denotes the equivariant integral over the framed instanton moduli space on $\mathbb{C}^2$, with equivariant rotation weights (ϵ_1,ϵ_2), Coulomb parameter a, and

$$\tau=\frac{\theta}{2\pi}+\frac{4\pi\mathrm{i}}{g_{\mathrm{YM}}^2}.$$

For $U(N)$ this can be defined by integration over the smooth framed torsion-free-sheaf compactification. For other gauge groups and matter systems the formula requires the corresponding compactification or an equivalent contour prescription as part of the stated localization structure.

101.4 Singular Instantons And Compactification Choices

The factor Z_{inst} in the S^4 formula is a compactification choice, not a decorative notation. Smooth instantons on $\mathbb{C}^2$ can shrink to zero size. The limiting curvature is then no longer represented by a smooth connection; the lost action is recorded as a point with multiplicity. A localization formula that sums over point instantons must therefore say which compactified moduli space is being integrated over.

Definition 101.6 (Framed instanton compactifications on $\mathbb{C}^2$)**.** Let $\mathcal{M}^{\text{sm}}_{k,N}$ be the framed smooth $U(N)$ k-instanton moduli space on $\mathbb{C}^2$.

1. The Uhlenbeck compactification is stratified by
$$\overline{\mathcal{M}}^{U}_{k,N} = \bigsqcup_{\ell=0}^{k} \mathcal{M}^{\text{sm}}_{k-\ell,N} \times \text{Sym}^{\ell}(\mathbb{C}^2). \tag{101.16}$$
The integer ℓ is the amount of instanton number carried by ideal point instantons.
2. The Gieseker, or framed torsion-free-sheaf, compactification $\mathfrak{M}^{G}_{k,N}$ is the smooth stable ADHM quotient. It carries a proper morphism
$$\pi_{G\to U} : \mathfrak{M}^{G}_{k,N} \longrightarrow \overline{\mathcal{M}}^{U}_{k,N}$$
that resolves the small-instanton strata relevant to equivariant localization.
3. On a compact four-manifold X, the Donaldson–Uhlenbeck compactification of ASD connections is the analogous compactification by weak limits of connections away from finitely many points, together with the lost second Chern class recorded as an effective zero-cycle on X.

These spaces have different meanings. A field-theoretic path integral with a specified regulator must explain which compactification, or which equivalent contour, it produces.

Construction 101.7 (Gieseker-to-Uhlenbeck charge bookkeeping)**.** Work on the framed $U(N)$ instanton compactification on $\mathbb{C}^2$, equivalently on $\mathbb{P}^2$ with a fixed framing along the line at infinity, and take $c_1 = 0$. Let E be a framed torsion-free sheaf in the Gieseker compactification. Its double dual E^{**} is locally free, and there is an exact sequence
$$0 \longrightarrow E \longrightarrow E^{**} \longrightarrow Q \longrightarrow 0 \tag{101.17}$$
with Q a zero-dimensional sheaf of length ℓ. If the instanton number is $k(F) = -\int_{\mathbb{P}^2} \text{ch}_2(F)$ in the framed $c_1 = 0$ convention, then
$$k(E) = k(E^{**}) + \ell. \tag{101.18}$$
Thus the Uhlenbeck morphism sends E to the smooth framed instanton represented by E^{**}, together with the effective length-ℓ zero-cycle supporting Q.

The charge identity in (101.18) is a finite Chern-character calculation. On a smooth complex surface the double dual of a torsion-free sheaf is locally free, and the quotient in (101.17) is supported in complex codimension two. The Chern character is additive in exact sequences, so
$$\text{ch}(E^{**}) = \text{ch}(E) + \text{ch}(Q).$$

Since Q is zero-dimensional of length ℓ, its only nonzero Chern character component is $\mathrm{ch}_2(Q) = \ell[\mathrm{pt}]$. Therefore

$$\mathrm{ch}_2(E) = \mathrm{ch}_2(E^{**}) - \ell[\mathrm{pt}].$$

Integrating and multiplying by the instanton-number sign convention gives

$$k(E) = -\int \mathrm{ch}_2(E) = -\int \mathrm{ch}_2(E^{**}) + \ell = k(E^{**}) + \ell.$$

The support cycle of Q is an element of $\mathrm{Sym}^\ell(\mathbb{C}^2)$. Keeping only $(E^{**}, \operatorname{supp} Q)$ gives the corresponding Uhlenbeck stratum, while the Gieseker space retains the punctual quotient data that resolve that stratum.

Remark 101.8 (Dimension of Uhlenbeck strata). The complex dimension of the top stratum $\mathcal{M}^{\mathrm{sm}}_{k,N}$ is $2kN$. The stratum with ℓ ideal points in (101.16) has complex dimension

$$2(k-\ell)N + 2\ell.$$

Hence its complex codimension inside the top-dimensional instanton component is

$$2\ell(N-1). \tag{101.19}$$

For $N = 2$ and $\ell = 1$, this is real codimension four.

The ADHM dimension count in Chapter 91 gives complex dimension $2rN$ for the smooth framed charge-r moduli space. The symmetric product $\mathrm{Sym}^\ell(\mathbb{C}^2)$ has complex dimension 2ℓ. Adding the two dimensions gives the displayed stratum dimension with $r = k - \ell$. Subtracting from $2kN$ gives

$$2kN - \big(2(k-\ell)N + 2\ell\big) = 2\ell(N-1).$$

Definition 101.9 (ADHM stability and the small-instanton locus)**.** Let $K \simeq \mathbb{C}^k$, $W \simeq \mathbb{C}^N$, and let (B_1, B_2, I, J) obey the complex ADHM equation

$$[B_1, B_2] + IJ = 0.$$

The ADHM tuple is stable if no proper subspace $S \subsetneq K$ satisfies

$$B_1 S \subset S, \qquad B_2 S \subset S, \qquad I(W) \subset S.$$

The Gieseker compactification is the stable GIT quotient by $GL(K)$, or equivalently the positive real-moment-map hyperkähler quotient. The Uhlenbeck compactification is obtained from the affine quotient after discarding the stable framing data carried by punctual quotients. Failure of stability is the ADHM description of the small-instanton stratum.

The geometric content of this definition is as follows. If the cyclic span of $I(W)$ under B_1 and B_2 is all of K, the framing data generate the length-k torsion-free sheaf and the complexified gauge action has the expected stable orbit. If the cyclic span is a proper subspace S, then the quotient K/S carries ADHM length not seen by the framed smooth connection. In the affine quotient this quotient is recorded only by the eigenvalue data of B_1, B_2, hence by points of $\mathbb{C}^2$ with multiplicity. These are precisely the ideal point-instanton coordinates in the Uhlenbeck stratum. The positive real-moment-map quotient forbids this loss by forcing stability; the price is that the singular stratum is replaced by the Gieseker resolution.

Proposition 101.10 (Positive ADHM moment map excludes the small-instanton stratum)**.** *Let K and W be Hermitian vector spaces and let $(B_1, B_2, I, J) \in \mathrm{End}(K)^{\oplus 2} \oplus \mathrm{Hom}(W, K) \oplus \mathrm{Hom}(K, W)$ obey the ADHM equations*

$$[B_1, B_2] + IJ = 0, \tag{101.20}$$

$$[B_1, B_1^\dagger] + [B_2, B_2^\dagger] + II^\dagger - J^\dagger J = \zeta\, \mathbf{1}_K, \qquad \zeta > 0. \tag{101.21}$$

Then the ADHM configuration is stable in the sense of Definition 101.9. Equivalently, the positive real moment-map parameter forbids a quotient K/S that is invisible to the framing $I(W)$, hence forbids the finite-dimensional small-instanton degeneration represented by failure of stability.

Proof. Suppose, for contradiction, that there is a proper subspace $S \subsetneq K$ such that $B_i S \subset S$ for $i = 1, 2$ and $I(W) \subset S$. Let $T = S^\perp$, and write each B_i in block form with respect to $K = S \oplus T$:

$$B_i = \begin{pmatrix} A_i & C_i \\ 0 & D_i \end{pmatrix}.$$

The T-trace of the commutator block is

$$\mathrm{Tr}_T\big([B_i, B_i^\dagger]_{TT}\big) = \mathrm{Tr}(D_i D_i^\dagger) - \mathrm{Tr}(C_i^\dagger C_i + D_i^\dagger D_i) = -\,\mathrm{Tr}(C_i^\dagger C_i). \tag{101.22}$$

Since $I(W) \subset S$, the T-block of $II^\dagger$ is zero. The T-trace of $-J^\dagger J$ is $-\,\mathrm{Tr}_T(J^\dagger J) \le 0$. Taking the trace over T in (101.21) therefore gives

$$-\sum_{i=1}^{2} \mathrm{Tr}(C_i^\dagger C_i) - \mathrm{Tr}_T(J^\dagger J) = \zeta\, \dim T. \tag{101.23}$$

The left hand side is nonpositive and the right hand side is positive because $\zeta > 0$ and $T \neq 0$. This contradiction proves stability. □

Remark 101.11 (What this finite-dimensional statement does not prove)*.* Proposition 101.10 is a theorem inside the ADHM finite-dimensional reduction. It justifies the common statement that a positive ADHM/FI resolution, for example the finite-dimensional resolution induced by noncommutative deformation in the Nekrasov–Schwarz construction, removes the small-instanton stratum. It does not by itself prove that a four-dimensional supersymmetric gauge-theory path integral with a given regulator localizes to this resolved quotient. That identification is the separate regulator theorem, or hypothesis, stated in Hypothesis 101.17.

Definition 101.12 (Tangent Euler class at a Young-diagram fixed point)**.** Let a fixed point of $\mathfrak{M}^G_{k,N}$ be labeled by $\vec{Y} = (Y_1, \ldots, Y_N)$. For a box $s = (i, j) \in Y_\alpha$, define the arm and relative leg lengths by

$$A_{Y_\alpha}(s) = Y_{\alpha,i} - j, \qquad L_{Y_\beta}(s) = Y^T_{\beta,j} - i.$$

With Coulomb weights a_α and rotation weights ϵ_1, ϵ_2, use the orientation convention

$$\begin{aligned} e_T(T_{\vec{Y}}\mathfrak{M}^G_{k,N}) = \prod_{\alpha,\beta=1}^{N} \prod_{s\in Y_\alpha} &\Big(a_\alpha - a_\beta - \epsilon_1 L_{Y_\beta}(s) + \epsilon_2(A_{Y_\alpha}(s) + 1)\Big) \\ &\times \Big(a_\alpha - a_\beta + \epsilon_1(L_{Y_\beta}(s) + 1) - \epsilon_2 A_{Y_\alpha}(s)\Big). \end{aligned} \tag{101.24}$$

The pure vector fixed-point contribution is

$$Z^{\mathrm{vec}}_{\vec{Y}} = e_T(T_{\vec{Y}}\mathfrak{M}^G_{k,N})^{-1}$$

in this convention. Matter multiplets multiply this by the Euler classes of their corresponding equivariant index bundles.

Proposition 101.13 (Rank-one monomial-ideal tangent character)**.** *Let Y be a Young diagram and let $I_Y \subset R = \mathbb{C}[z_1, z_2]$ be the monomial ideal whose quotient has basis*

$$Q_Y = R/I_Y, \qquad \{z_1^{i-1}z_2^{j-1} : (i,j) \in Y\}.$$

Let z_1 and z_2 have torus weights ϵ_1 and ϵ_2, and write $t_a = e^{\epsilon_a}$. At the fixed point $I_Y \in \mathrm{Hilb}^{|Y|}(\mathbb{C}^2)$, the rank-one tangent character is

$$\mathrm{ch}_T\, T_{I_Y}\mathrm{Hilb}^{|Y|}(\mathbb{C}^2) = \sum_{s\in Y}\left(t_1^{-L_Y(s)}t_2^{A_Y(s)+1} + t_1^{L_Y(s)+1}t_2^{-A_Y(s)}\right). \tag{101.25}$$

Equivalently, the additive tangent weights are

$$-L_Y(s)\epsilon_1 + (A_Y(s)+1)\epsilon_2, \qquad (L_Y(s)+1)\epsilon_1 - A_Y(s)\epsilon_2, \qquad s \in Y.$$

Their product is the $N = 1$ specialization of (101.24).

Proof. Set

$$V_Y = \mathrm{ch}_T\, Q_Y = \sum_{(i,j)\in Y} t_1^{i-1}t_2^{j-1}, \qquad V_Y^\vee = \sum_{(i,j)\in Y} t_1^{1-i}t_2^{1-j}.$$

The monomial-ideal deformation complex, equivalently the rank-one ADHM deformation complex with framing character 1, gives the finite equivariant K-theory identity

$$\mathrm{ch}_T\, T_{I_Y}\mathrm{Hilb}^{|Y|}(\mathbb{C}^2) = V_Y^\vee + t_1t_2V_Y - (1-t_1)(1-t_2)V_YV_Y^\vee. \tag{101.26}$$

Here the last term subtracts the gauge and relation part of the two-arrow ADHM complex, while $V_Y^\vee$ and $t_1t_2V_Y$ are the two framing terms. Expanding $(1-t_1)(1-t_2)V_YV_Y^\vee$ is a finite second-difference calculation on the set of box displacements in $Y - Y$: every displacement whose four adjacent translates stay inside the diagram cancels, and the only uncancelled displacements are the two hooks that cross the boundary at each box s. The east hook contributes $t_1^{-L_Y(s)}t_2^{A_Y(s)+1}$, and the south hook contributes $t_1^{L_Y(s)+1}t_2^{-A_Y(s)}$. Summing over boxes gives (101.25). Passing from multiplicative characters to additive weights gives the displayed two weights per box, and their product is exactly (101.24) with $N = 1$ and $a_\alpha - a_\beta = 0$. □

For $k = 1$, the fixed points are the N choices in which one Young diagram Y_α consists of a single box. Formula (101.24) gives

$$Z_1^{U(N)} = \frac{1}{\epsilon_1\epsilon_2}\sum_{\alpha=1}^{N}\prod_{\beta\neq\alpha}\frac{1}{(a_\alpha - a_\beta)(a_\alpha - a_\beta + \epsilon_1 + \epsilon_2)},$$

which is the one-instanton formula used in Definition 91.7. Let $s = (1,1)$ be the single box of Y_α. For $\beta = \alpha$,

$$A_{Y_\alpha}(s) = 0, \qquad L_{Y_\alpha}(s) = 0,$$

so the two tangent weights are ϵ_2 and ϵ_1. For $\beta \neq \alpha$, the diagram Y_β is empty, hence

$$A_{Y_\alpha}(s) = 0, \qquad L_{Y_\beta}(s) = -1.$$

The two weights in (101.24) are then

$$a_\alpha - a_\beta + \epsilon_1 + \epsilon_2, \qquad a_\alpha - a_\beta.$$

Multiplying the weights and summing over the N possible positions of the one box gives the displayed expression.

Example 101.14 (The two-point Gieseker fixed-point laboratory)**.** The first place where the resolved punctual data are not just a projective orientation space is the rank-one length-two Gieseker sector

$$\mathfrak{M}^G_{2,1} \simeq \mathrm{Hilb}^2(\mathbb{C}^2).$$

This is a finite-dimensional model of the resolved ideal-point sector, not a claim that ordinary commutative $U(1)$ Yang–Mills has smooth finite-action instantons. The Uhlenbeck record over a double point at the origin sees only the support cycle $2[0] \in \mathrm{Sym}^2(\mathbb{C}^2)$. The Gieseker space retains the punctual quotient data; over the double point the exceptional fiber is $\mathbb{P}^1$. The torus $T^2_{\epsilon_1,\epsilon_2}$ has two fixed points, corresponding to the partitions (2) and $(1,1)$. Proposition 101.13 computes their tangent characters directly from the monomial ideals.

For the horizontal partition $Y = (2)$, the boxes have

$$(A, L) = ((1,0), (0,0)),$$

and (101.24) gives tangent weights

$$2\epsilon_2, \quad \epsilon_1 - \epsilon_2, \quad \epsilon_2, \quad \epsilon_1.$$

Hence

$$e_T(T_{(2)}\mathrm{Hilb}^2(\mathbb{C}^2)) = 2\epsilon_1\epsilon_2^2(\epsilon_1 - \epsilon_2).$$

For the vertical partition $Y = (1,1)$, the boxes have

$$(A, L) = ((0,1), (0,0)),$$

and the weights are

$$\epsilon_2 - \epsilon_1, \quad 2\epsilon_1, \quad \epsilon_2, \quad \epsilon_1,$$

so

$$e_T(T_{(1,1)}\mathrm{Hilb}^2(\mathbb{C}^2)) = 2\epsilon_1^2\epsilon_2(\epsilon_2 - \epsilon_1).$$

The equivariant integral of 1 over the resolved length-two sector is therefore

$$\begin{aligned}\int_{\mathrm{Hilb}^2(\mathbb{C}^2)} 1 &= \frac{1}{2\epsilon_1\epsilon_2^2(\epsilon_1 - \epsilon_2)} + \frac{1}{2\epsilon_1^2\epsilon_2(\epsilon_2 - \epsilon_1)} \\ &= \frac{1}{2\epsilon_1^2\epsilon_2^2}.\end{aligned} \tag{101.27}$$

This calculation displays the singular-instanton bookkeeping in its smallest nontrivial form. The Uhlenbeck compactification remembers the effective cycle $2[0]$; the Gieseker/Nekrasov resolution remembers which punctual monomial ideal lies over that cycle. A localization formula that uses the two Young-diagram residues is therefore using resolved punctual data, and a field-theoretic regulator must still explain why this resolved sector, rather than the collapsed Uhlenbeck point, is the integration cycle it selects.

Example 101.15 (The charge-one resolution)**.** For $k = 1$, the ADHM matrices B_1, B_2 are complex numbers, hence give the center point in $\mathbb{C}^2$. The remaining data

$$I \in W^*, \qquad J \in W, \qquad IJ = 0$$

modulo the $GL(1)$ action, with positive real moment map, give

$$\mathfrak{M}^G_{1,N} \simeq \mathbb{C}^2 \times T^*\mathbb{CP}^{N-1}.$$

The Uhlenbeck compactification collapses the resolved orientation/size data over the ideal point-instanton stratum. For $N = 2$, the collapsed stratum has real codimension four in the top moduli component, matching Remark 101.8.

Construction 101.16 (Charge-one Uhlenbeck map as a nilpotent-cone resolution)**.** The charge-one example admits a completely explicit map to the Uhlenbeck compactification. Let $W \simeq \mathbb{C}^N$. Since $K = \mathbb{C}$, the complex ADHM equation is the scalar condition

$$I(J) = 0, \qquad I \in W^*, \quad J \in W.$$

Stability is $I \neq 0$. Dividing by $GL(1) = \mathbb{C}^*$ gives

$$(I, J)/\mathbb{C}^* = \{([I], J) : [I] \in \mathbb{P}(W^*),\ I(J) = 0\} \simeq T^*\mathbb{P}(W^*) \simeq T^*\mathbb{CP}^{N-1}.$$

The Uhlenbeck map is the moment-map morphism

$$\mu_{\mathrm{Uh}} : T^*\mathbb{P}(W^*) \longrightarrow \mathfrak{sl}(W), \qquad ([I], J) \longmapsto \nu_{I,J} := J \otimes I, \tag{101.28}$$

where $\nu_{I,J}(w) = I(w)J$. The equation $I(J) = 0$ gives

$$\operatorname{Tr} \nu_{I,J} = I(J) = 0, \qquad \nu_{I,J}^2 = I(J)\nu_{I,J} = 0, \qquad \operatorname{rank} \nu_{I,J} \leq 1.$$

Thus the image is the closure of the minimal nilpotent orbit in $\mathfrak{sl}(W)$. The open subset $J \neq 0$ maps to the smooth rank-one nilpotent orbit; the zero section $J = 0$ maps to the apex $\nu = 0$, which is precisely the ideal point-instanton stratum in the charge-one Uhlenbeck compactification. Including the center $(B_1, B_2) \in \mathbb{C}^2$, one has

$$\mathfrak{M}^G_{1,N} \simeq \mathbb{C}^2 \times T^*\mathbb{CP}^{N-1} \longrightarrow \mathbb{C}^2 \times \overline{\mathcal{O}}_{\min},$$

and the small-instanton boundary is $\mathbb{C}^2 \times \{0\}$. This is the concrete geometric content hidden behind the phrase "point instanton" in the first nontrivial instanton sector.

Hypothesis 101.17 (Field-theoretic matching to the Gieseker integral)**.** The S^4 formula (101.29) uses the Gieseker/Nekrasov equivariant integral at each pole. A theorem-level derivation from the four-dimensional gauge-theory path integral requires a regulator and integration cycle whose localized small-instanton sector is identified with $\mathfrak{M}^G_{k,N}$, with the Uhlenbeck space $\overline{\mathcal{M}}^U_{k,N}$ serving as a boundary compactification rather than as the resolved integration domain. Noncommutative or positive ADHM-moment-map regularization supplies a finite-dimensional resolution of the small-instanton singularity, but the construction-level equivalence between this resolved equivariant integral and the continuum supersymmetric path integral is retained here as a hypothesis unless the regulator has been specified and the limiting argument has been proven.

Definition 101.18 (Regulator selection structure for the Gieseker sector)**.** Fix k, N, equivariant weights (ϵ_1, ϵ_2), Coulomb parameter a, and the trace normalization of this volume. A four-dimensional supersymmetric regulator is said to select the Gieseker instanton sector at charge k if it supplies the following structures.

1. A regulated BV localization problem

$$(\mathfrak{F}_\delta, C_\delta, \rho_\delta, Q_\delta, V_\delta), \qquad \delta > 0,$$

satisfying the finite-regulator quantum master equation and the Q_δ-Stokes condition on the integration cycle. Here δ denotes the regulator parameter, not an equivariant weight.

2. A Q_δ-fixed local model at the pole with coordinates

$$(B_1, B_2, I, J) \in \mathrm{End}(K)^{\oplus 2} \oplus \mathrm{Hom}(W, K) \oplus \mathrm{Hom}(K, W), \qquad \dim K = k, \quad \dim W = N,$$

whose equations reduce, after a controlled finite change of coordinates, to

$$[B_1, B_2] + IJ = 0, \qquad [B_1, B_1^\dagger] + [B_2, B_2^\dagger] + II^\dagger - J^\dagger J = \zeta_\delta \mathbf{1}_K + R_\delta,$$

with $\zeta_\delta > 0$ and $R_\delta \to 0$ in the norm in which the fixed-locus reduction is being taken. Equivalently, the regulator may supply another explicit resolution cycle together with an equivariant proper map to $\overline{\mathcal{M}}^U_{k,N}$; it must then prove that the resolved cycle is isomorphic to $\mathfrak{M}^G_{k,N}$ or give the exact contour replacing it.

3. An orientation and determinant-line comparison between the regulated normal Q_δ-complex and the equivariant tangent/index complex on $\mathfrak{M}^G_{k,N}$. In local fixed-point coordinates this means that the nonzero-mode Berezinian gives the Euler class (101.24), multiplied only by declared one-loop determinants and local counterterms already included in the localization structure.

4. A boundary-cancellation statement for the Uhlenbeck small-instanton collars. In the notation of Chapter 111, the regulator must prove either

$$\sum_\alpha B_{S_\alpha}(\rho_\delta) \longrightarrow 0$$

or an equality of this boundary functional with the difference between the smooth-stratum pushforward and the Gieseker-resolved pushforward. Thus the resolution is part of the BV pushforward structure, not an informal replacement made after the path integral has already been localized.

5. A limiting theorem for equivariantly regular test insertions $\mathcal{O}$:

$$\lim_{\delta \to 0} \int_{C_\delta} \rho_\delta \, \mathcal{O} \, \mathrm{e}^{-tQ_\delta V_\delta} = \int^{\mathrm{equiv}}_{\mathfrak{M}^G_{k,N}} \mathcal{O}_{\mathrm{red}} \, \frac{e_T(E_{\mathrm{matter}})}{e_T(T\mathfrak{M}^G_{k,N})},$$

with the order of the $t \to \infty$ localization limit and the $\delta \to 0$ regulator limit specified. If the result is stated only as a formal power series in the instanton counting parameter, the topology of that formal series must be part of the statement.

This hypothesis is the point at which the monograph refuses the common shortcut. Equivariant localization on $\mathfrak{M}^G_{k,N}$ is a mathematical theorem after the resolved moduli space is chosen. The assertion that the original QFT path integral localizes to that particular resolved integral is an additional field-theoretic statement. A BV pushforward formulation, as in Chapter 111, must include the boundary term at the singular stratum or a regulator that replaces the stratum by a cycle without boundary. The singular-stratum obstruction formula (111.12) is the finite-dimensional statement behind this requirement: a smooth-stratum pushforward satisfies the residual QME only after the small-instanton boundary functional has vanished or has been cancelled by a specified resolution, relative cycle, or residue prescription.

Definition 101.18 is deliberately more structured than the phrase "use the Nekrasov partition function". The finite-dimensional ADHM statement Proposition 101.10 proves the stability part of item 2 once the fixed-locus reduction has already been made. It proves neither the existence of the four-dimensional regulator in item 1, nor the determinant-line comparison in item 3, nor the small-instanton boundary cancellation in item 4, nor the limiting statement in item 5. Conversely, a regulator that selected the Uhlenbeck compactification without a compensating boundary functional would not give the Nekrasov factor appearing in (101.29); it would define a different localized problem.

Open Problem 101.19 (Continuum regulator theorem for Gieseker selection)*.* Construct, for four-dimensional $\mathcal{N}=2$ gauge theory on S^4, a BV-compatible supersymmetric regulator and integration cycle satisfying Definition 101.18 for every instanton charge k, or prove that the standard S^4 localization formula requires an additional nonperturbative definition whose instanton sectors are Gieseker/Nekrasov equivariant integrals by construction. The known finite-dimensional ADHM resolution and the noncommutative positive-moment-map mechanism are necessary local pieces of such a theorem, but they do not by themselves prove the continuum field-theoretic selection statement.

The localized partition function is

$$Z^{\mathcal{N}=2}_{S^4} = \frac{1}{|W_G|}\int_{\mathfrak{t}} da\, Z_{\mathrm{cl}}(a)\, Z^{S^4}_{1-\mathrm{loop}}(a)\, Z_{\mathrm{inst}}\left(a,\tau;\frac{1}{r},\frac{1}{r}\right) Z_{\mathrm{inst}}\left(a,\bar{\tau};\frac{1}{r},\frac{1}{r}\right). \tag{101.29}$$

The north pole contributes instantons and the south pole contributes anti-instantons. The two factors are complex conjugates only when the integration cycle and masses are chosen so that reflection positivity gives the corresponding conjugation property.

Protected-insertion comparison budget. Equation (101.29) is the scalar partition-function case of a functional statement on protected insertions. Fix a finite test space $\mathcal{P}_N$ of Q-closed regulated insertions containing the unit insertion and the protected Wilson-loop insertions under study. On $\mathcal{P}_N$ let

$$\mathcal{Z}_{\mathrm{reg}},\quad \mathcal{Z}_{\mathrm{def}},\quad \mathcal{Z}_{\mathrm{normal}},\quad \mathcal{Z}_{\mathrm{res}},\quad \mathcal{Z}_{\mathrm{Nek}},\quad \mathcal{Z}_{\mathrm{loc}} \in \mathcal{P}_N^\vee$$

denote, respectively, the regulated QFT functional, the functional after the Q-exact deformation, the functional after normal Gaussian integration, the residual Cartan-contour functional, the north/south ADHM–Nekrasov functional, and the final localized expression. Define

$$\begin{aligned} E_{\mathrm{Stokes}} &= \mathcal{Z}_{\mathrm{reg}} - \mathcal{Z}_{\mathrm{def}}, \\ E_{\mathrm{normal}} &= \mathcal{Z}_{\mathrm{def}} - \mathcal{Z}_{\mathrm{normal}}, \\ E_{\mathrm{res}} &= \mathcal{Z}_{\mathrm{normal}} - \mathcal{Z}_{\mathrm{res}}, \\ E_{\mathrm{inst}} &= \mathcal{Z}_{\mathrm{res}} - \mathcal{Z}_{\mathrm{Nek}}, \\ E_{\mathrm{cont}} &= \mathcal{Z}_{\mathrm{Nek}} - \mathcal{Z}_{\mathrm{loc}}. \end{aligned}$$

Then the comparison to the displayed localization formula is the exact telescope

$$\mathcal{Z}_{\mathrm{reg}} - \mathcal{Z}_{\mathrm{loc}} = E_{\mathrm{Stokes}} + E_{\mathrm{normal}} + E_{\mathrm{res}} + E_{\mathrm{inst}} + E_{\mathrm{cont}}. \tag{101.30}$$

Here E_{Stokes} records finite-regulator Q-Stokes, anomaly, and boundary defects; E_{normal} records normal-complex, zero-mode, and determinant-line defects; E_{res} records the residual Cartan contour, Weyl quotient, and fixed-locus restriction of the insertion; E_{inst} records the choice of Gieseker/Nekrasov instanton compactification at the poles; and E_{cont} records the continuum limit and contact-term comparison. For any norm on the finite-dimensional dual space,

$$\|\mathcal{Z}_{\mathrm{reg}} - \mathcal{Z}_{\mathrm{loc}}\| \leq \|E_{\mathrm{Stokes}}\| + \|E_{\mathrm{normal}}\| + \|E_{\mathrm{res}}\| + \|E_{\mathrm{inst}}\| + \|E_{\mathrm{cont}}\|.$$

Thus the usual formula is recovered for this test space only after these five residuals have been killed or bounded. For the half-BPS circular Wilson loop, the fixed-locus restriction is $W_{\square}|_{\mathrm{fix}} = N^{-1}\operatorname{Tr}_{\square} \mathrm{e}^{X}$; the Laguerre and Bessel computations below evaluate $\mathcal{Z}_{\mathrm{loc}}(W_{\square})/\mathcal{Z}_{\mathrm{loc}}(1)$. They do not by themselves prove the continuum field-theoretic vanishing of the five residuals above.

Example 101.20 ($\mathcal{N} = 4$ Yang–Mills degeneration)**.** For an $\mathcal{N} = 2$ vector multiplet plus an adjoint hypermultiplet with the mass coordinate tuned to the $\mathcal{N} = 4$ point, the determinant cancellation above reduces the perturbative one-loop determinant to the Cartan gauge-fixing Jacobian. The remaining matrix model is

$$Z_{S^4}^{\mathcal{N}=4} = \frac{1}{|W_G|} \int_{\mathfrak{t}} da \, \exp\left\{-\frac{4\pi^2 r^2}{g_{\mathrm{YM}}^2}(a,a)\right\} \prod_{\alpha \in \Delta_+} \alpha(a)^2.$$

For $G = U(1)$ there is no root factor and no finite-action instanton sector on $\mathbb{C}^2$. The partition function reduces to the ordinary Gaussian

$$\int_{-\infty}^{\infty} da \, \exp\left(-\frac{4\pi^2 r^2}{g_{\mathrm{YM}}^2} a^2\right) = \frac{g_{\mathrm{YM}}}{2\sqrt{\pi}\, r}. \tag{101.31}$$

Definition 101.21 (Circular Wilson-loop matrix-model coupling)**.** For $G = U(N)$, let X denote the Hermitian matrix whose eigenvalues x_i are the dimensionless arguments that appear in the localized fundamental half-BPS circular Wilson loop,

$$W_{\square}(X) = \frac{1}{N} \operatorname{Tr}_{\square} \mathrm{e}^{X} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{e}^{x_i}. \tag{101.32}$$

The circular-loop Gaussian coupling λ_{circ} is defined by writing the $\mathcal{N} = 4$ localized $U(N)$ Gaussian model in the form

$$\mathrm{d}\mu_{\lambda_{\mathrm{circ}},N}(X) = \frac{1}{Z_{\lambda_{\mathrm{circ}},N}} \exp\left(-\frac{2N}{\lambda_{\mathrm{circ}}} \operatorname{Tr}_{\square} X^2\right) \mathrm{d}X, \tag{101.33}$$

or, after diagonalizing X,

$$\mathrm{d}\mu_{\lambda_{\mathrm{circ}},N}(x) \propto \prod_{i=1}^{N} dx_i \, \exp\left(-\frac{2N}{\lambda_{\mathrm{circ}}}\sum_{i=1}^{N} x_i^2\right) \prod_{i<j}(x_i - x_j)^2. \tag{101.34}$$

This is a definition of the coupling that enters the exact circular-loop answer. Converting it to a Lagrangian symbol such as $g_{\mathrm{YM}}^2 N$ requires using the same fundamental-matrix normalization for the scalar in the Wilson-loop exponent and for the quadratic form in the localized Gaussian. Rescaling the Lie-algebra generators rescales both data; the matrix model (101.33) is the normalization-invariant statement.

Finite-N circular Wilson loop in the Gaussian model. For the $U(N)$ Gaussian matrix model (101.33), the normalized fundamental circular Wilson-loop insertion has the exact finite-N value

$$\left\langle \frac{1}{N}\operatorname{Tr}_{\square} \mathrm{e}^{X} \right\rangle_{\lambda_{\mathrm{circ}},N} = \frac{1}{N}\exp\left(\frac{\lambda_{\mathrm{circ}}}{8N}\right) L_{N-1}^{(1)}\left(-\frac{\lambda_{\mathrm{circ}}}{4N}\right), \tag{101.35}$$

where $L_n^{(1)}$ is the associated Laguerre polynomial.

To derive this formula, put

$$\alpha = \frac{2N}{\lambda_{\mathrm{circ}}}.$$

After diagonalizing X, the one-point density is expressed by the orthogonal-polynomial kernel for the weight $e^{-\alpha x^2}$:

$$\rho_N(x) = \frac{1}{N} e^{-\alpha x^2} \sum_{j=0}^{N-1} \frac{p_j(x)^2}{h_j}, \qquad h_j = \int_{\mathbb{R}} p_j(x)^2 e^{-\alpha x^2}\, dx, \tag{101.36}$$

where p_j is any degree-j polynomial family orthogonal for this weight. Therefore, for an auxiliary parameter s,

$$W_N(s) = \left\langle \frac{1}{N}\operatorname{Tr} \mathrm{e}^{sX} \right\rangle = \frac{1}{N}\sum_{j=0}^{N-1} \frac{1}{h_j}\int_{\mathbb{R}} e^{sx} p_j(x)^2 e^{-\alpha x^2}\, dx. \tag{101.37}$$

Choose $p_j(x) = \alpha^{-j/2} 2^{-j} H_j(\sqrt{\alpha}x)$, where H_j is the physicists' Hermite polynomial. The scale factors cancel in the ratio appearing in (101.37). With $y = \sqrt{\alpha}x$ and $t = s/\sqrt{\alpha}$, the required normalized Hermite integral is

$$\frac{\int_{\mathbb{R}} e^{-y^2+ty} H_j(y)^2\, dy}{\int_{\mathbb{R}} e^{-y^2} H_j(y)^2\, dy} = \exp\left(\frac{t^2}{4}\right) L_j\left(-\frac{t^2}{2}\right). \tag{101.38}$$

For completeness, recall the derivation. The Hermite generating function is

$$\exp(2uy - u^2) = \sum_{j\ge 0} H_j(y)\frac{u^j}{j!}.$$

Multiplying two generating functions and integrating against e^{-y^2+ty} gives

$$\sqrt{\pi}\, \exp\left(\frac{t^2}{4}\right) \exp(2uv + tu + tv)\,.$$

Taking the coefficient of $u^j v^j/(j!)^2$ and dividing by the same coefficient at $t = 0$, namely $2^j j!$, gives (101.38); the remaining finite coefficient sum is the standard Laguerre expansion

$$L_j(z) = \sum_{m=0}^{j} \binom{j}{m} \frac{(-z)^m}{m!}.$$

Substitution into (101.37) yields

$$W_N(s) = \frac{1}{N} \exp\left(\frac{s^2}{4\alpha}\right) \sum_{j=0}^{N-1} L_j\left(-\frac{s^2}{2\alpha}\right). \tag{101.39}$$

The elementary identity

$$\sum_{j=0}^{N-1} L_j(z) = L_{N-1}^{(1)}(z)$$

follows by summing the coefficient formula for L_j and using $\sum_{j=m}^{N-1} \binom{j}{m} = \binom{N}{m+1}$, which is precisely the coefficient formula for $L_{N-1}^{(1)}$. Setting $s = 1$ and $\alpha = 2N/\lambda_{\mathrm{circ}}$ gives (101.35).

Theorem 101.22 (Planar circular Wilson loop from the Gaussian model)**.** *In the large-N limit of* (101.34)*, with $\lambda_{\mathrm{circ}} > 0$ fixed, the normalized eigenvalue density is*

$$\rho_{\lambda_{\mathrm{circ}}}(x) = \frac{2}{\pi \lambda_{\mathrm{circ}}} \sqrt{\lambda_{\mathrm{circ}} - x^2}\, \mathbf{1}_{[-\sqrt{\lambda_{\mathrm{circ}}}, \sqrt{\lambda_{\mathrm{circ}}}]}(x). \tag{101.40}$$

Consequently the planar fundamental circular Wilson loop is

$$\lim_{N\to\infty} \left\langle \frac{1}{N} \operatorname{Tr}_{\Box} \mathrm{e}^X \right\rangle_{\lambda_{\mathrm{circ}},N} = \int dx\, \rho_{\lambda_{\mathrm{circ}}}(x) \mathrm{e}^x = \frac{2}{\sqrt{\lambda_{\mathrm{circ}}}} I_1\left(\sqrt{\lambda_{\mathrm{circ}}}\right), \tag{101.41}$$

where I_1 is the modified Bessel function.

Proof. For an empirical density ρ with $\int \rho(x)dx = 1$, the large-N functional obtained from (101.34) is, up to an additive constant,

$$\mathcal{F}[\rho] = \frac{2}{\lambda_{\mathrm{circ}}} \int x^2 \rho(x) dx - \iint \rho(x)\rho(y) \log|x - y|\, dx\, dy. \tag{101.42}$$

Varying ρ under the normalization constraint gives, on the support of the minimizing density,

$$\frac{2}{\lambda_{\mathrm{circ}}} x^2 - 2 \int \rho(y) \log|x - y|\, dy = \text{constant}. \tag{101.43}$$

Differentiating inside the support gives the singular-integral equation

$$\frac{4x}{\lambda_{\mathrm{circ}}} = 2 \text{ p. v.} \int \frac{\rho(y)}{x - y}\, dy. \tag{101.44}$$

The Cauchy transform of (101.40) is

$$G(z) = \int \frac{\rho_{\lambda_{\mathrm{circ}}}(y)}{z - y}\, dy = \frac{2}{\lambda_{\mathrm{circ}}} \left(z - \sqrt{z^2 - \lambda_{\mathrm{circ}}}\right), \tag{101.45}$$

where the branch is chosen so that $G(z) \sim z^{-1}$ at infinity. The average of the boundary values on the cut is

$$\frac{1}{2}\big(G(x+\mathrm{i}0)+G(x-\mathrm{i}0)\big)=\frac{2x}{\lambda_{\mathrm{circ}}},$$

which is precisely the principal value integral in (101.44). The normalization of (101.40) follows from $\int_{-a}^{a}\sqrt{a^2-x^2}\,dx=\pi a^2/2$, with $a^2=\lambda_{\mathrm{circ}}$. Because the quadratic potential is strictly convex, the logarithmic-energy minimizer is unique. The displayed density satisfies the Euler–Lagrange equation on its support. Outside the support, differentiating the difference between the left side of (101.43) and its constant value gives

$$\frac{4}{\lambda_{\mathrm{circ}}}\sqrt{x^2-\lambda_{\mathrm{circ}}}\quad(x>\sqrt{\lambda_{\mathrm{circ}}})$$

and the corresponding negative derivative on the left component $x<-\sqrt{\lambda_{\mathrm{circ}}}$. Hence the effective potential is larger off the support, so the candidate density is the equilibrium density.

It remains to evaluate the loop insertion. Put $z=\sqrt{\lambda_{\mathrm{circ}}}$ and $x=zt$. Then

$$\int dx\,\rho_{\lambda_{\mathrm{circ}}}(x)\mathrm{e}^{x}=\frac{2}{\pi}\int_{-1}^{1}\mathrm{e}^{zt}\sqrt{1-t^2}\,dt. \tag{101.46}$$

The integral representation

$$I_1(z)=\frac{z}{\pi}\int_{-1}^{1}\mathrm{e}^{zt}\sqrt{1-t^2}\,dt \tag{101.47}$$

gives (101.41). Equivalently, expanding e^{x} and using the even moments

$$\int dx\,\rho_{\lambda_{\mathrm{circ}}}(x)x^{2n}=\frac{1}{n+1}\binom{2n}{n}\left(\frac{\lambda_{\mathrm{circ}}}{4}\right)^{n} \tag{101.48}$$

gives the same series

$$\sum_{n=0}^{\infty}\frac{1}{n!(n+1)!}\left(\frac{\lambda_{\mathrm{circ}}}{4}\right)^{n}=\frac{2}{\sqrt{\lambda_{\mathrm{circ}}}}I_1\Big(\sqrt{\lambda_{\mathrm{circ}}}\Big). \tag{101.49}$$

□

Remark 101.23 (Bremsstrahlung derivative algebra). Assume the protected circular-loop Ward identity

$$B(\lambda_{\mathrm{circ}})=\frac{1}{2\pi^2}\lambda_{\mathrm{circ}}\frac{d}{d\lambda_{\mathrm{circ}}}\log\left\langle\frac{1}{N}\operatorname{Tr}_{\square}\mathrm{e}^{X}\right\rangle_{\mathrm{planar}}. \tag{101.50}$$

Then Theorem 101.22 gives

$$B(\lambda_{\mathrm{circ}})=\frac{\sqrt{\lambda_{\mathrm{circ}}}}{4\pi^2}\frac{I_2(\sqrt{\lambda_{\mathrm{circ}}})}{I_1(\sqrt{\lambda_{\mathrm{circ}}})}. \tag{101.51}$$

Let $z=\sqrt{\lambda_{\mathrm{circ}}}$ and

$$W_0(z)=\frac{2I_1(z)}{z}.$$

Since $\lambda_{\mathrm{circ}}\partial_{\lambda_{\mathrm{circ}}}=(z/2)\partial_z$,

$$\lambda_{\mathrm{circ}}\frac{d}{d\lambda_{\mathrm{circ}}}\log W_0=\frac{z}{2}\left(\frac{I_1'(z)}{I_1(z)}-\frac{1}{z}\right).$$

The modified-Bessel recurrence

$$I_1'(z) = I_2(z) + \frac{1}{z} I_1(z) \tag{101.52}$$

therefore gives

$$\lambda_{\rm circ} \frac{d}{d\lambda_{\rm circ}} \log W_0 = \frac{z}{2} \frac{I_2(z)}{I_1(z)}. \tag{101.53}$$

Substitution into (101.50) proves (101.51).

101.5 S^3 Localization In Three-Dimensional $\mathcal{N}=2$ Theories

Let G, T, $\mathfrak{t}$, W_G, and Δ be as above. On the round S^3_r, the Coulomb-branch localization variable is the constant real scalar $\sigma \in \mathfrak{t}$ in the vector multiplet. With Chern–Simons level k, FI parameter $\zeta \in \mathfrak{t}^*$, chiral multiplets Φ in representations R_Φ, real masses m_Φ, and R-charges Δ_Φ, the Kapustin–Willett–Yaakov matrix integral is

$$Z_{S^3} = \frac{1}{|W_G|} \int_{\mathfrak{t}} d\sigma \, e^{i\pi k r^2(\sigma,\sigma) + 2\pi i r(\zeta,\sigma)} \prod_{\alpha\in\Delta_+} 4\sinh^2\big(\pi r\, \alpha(\sigma)\big) \prod_\Phi Z_\Phi^{S^3}(\sigma), \tag{101.54}$$

where

$$Z_\Phi^{S^3}(\sigma) = \prod_{\rho\in \mathrm{wt}(R_\Phi)} \exp[\ell(1-\Delta_\Phi + i r\,\rho(\sigma) + i r\, m_\Phi)]\,.$$

The function ℓ is fixed, up to an additive constant that changes the overall normalization of Z_{S^3}, by

$$\ell'(z) = -\pi z \cot(\pi z), \qquad \ell\left(\frac{1}{2} + i x\right) + \ell\left(\frac{1}{2} - i x\right) = -\log\big(2\cosh(\pi x)\big). \tag{101.55}$$

Equivalently, the round-sphere determinant is the $b=1$ specialization of the double-sine convention

$$s_b(x) = \prod_{m,n\geq 0} \frac{mb + nb^{-1} + \frac{Q_b}{2} - i x}{mb + nb^{-1} + \frac{Q_b}{2} + i x}, \qquad Q_b = b + b^{-1}, \tag{101.56}$$

defined by zeta regularization or by the corresponding meromorphic continuation. In the convention of (101.54),

$$\exp[\ell(1-\Delta + i x)] = s_1(i(1-\Delta) - x)\,, \qquad x\in\mathbb{R}, \tag{101.57}$$

after fixing the same x-independent normalization on both sides. The reflection identity used for a conjugate R-charge 1/2 pair is

$$s_1\left(\frac{i}{2} - x\right) s_1\left(\frac{i}{2} + x\right) = \frac{1}{2\cosh(\pi x)}. \tag{101.58}$$

The integration contour is the real Cartan in the displayed unitary normalization. This statement includes a convergence hypothesis. If the large-$|\sigma|$ behavior is not damping in every Weyl chamber, then a real mass, FI chamber, analytic continuation, or residue prescription must be declared as part of the theory being evaluated. In abelianized descriptions the possible poles lie on complexified hyperplanes

$$\rho(\sigma) + m_\Phi + i\big(n + \Delta_\Phi\big)/r = 0, \qquad n\in\mathbb{Z}_{\geq 0},$$

up to the convention used for ℓ or s_b. A Jeffrey–Kirwan-type residue prescription is therefore not a decorative evaluation trick: it is a choice of chamber and integration cycle for the regulated localization structure.

Example 101.24 (Abelian Chern–Simons Gaussian)**.** For pure $U(1)_k$ Chern–Simons theory on S^3 with FI parameter ζ, the localized integral is the oscillatory Fresnel integral

$$Z_{U(1)_k}(\zeta) = \int_{\mathbb{R}} d\sigma\, \exp\bigl(\mathrm{i}\pi k\sigma^2 + 2\pi\mathrm{i}\zeta\sigma\bigr)\,.$$

With the contour obtained by rotating the positive quadratic direction by the usual i0 prescription, completing the square gives

$$Z_{U(1)_k}(\zeta) = \frac{\mathrm{e}^{\mathrm{i}\pi\,\mathrm{sgn}(k)/4}}{\sqrt{|k|}}\,\exp\biggl(-\frac{\mathrm{i}\pi\zeta^2}{k}\biggr)\,. \tag{101.59}$$

The phase depends on the framing convention of the Chern–Simons theory; the absolute value and FI dependence follow from the declared contour.

Example 101.25 (A conjugate pair of chirals)**.** For two chiral multiplets of charges $+1$ and -1, both with R-charge $1/2$ and zero real mass, the product of one-loop determinants on the $U(1)$ Coulomb parameter is

$$\exp\biggl[\ell\biggl(\frac{1}{2}+\mathrm{i}\sigma\biggr)+\ell\biggl(\frac{1}{2}-\mathrm{i}\sigma\biggr)\biggr] = \frac{1}{2\cosh(\pi\sigma)}.$$

The elementary integral

$$\int_{-\infty}^{\infty}\frac{d\sigma}{2\cosh(\pi\sigma)} = \frac{1}{2} \tag{101.60}$$

is a finite check on the determinant normalization in (101.54). It is not by itself a definition of a gauge theory; the gauge quotient, possible Chern–Simons contact terms, and background-field couplings remain part of the QFT definition.

101.6 Theorem Boundary And Singular Loci

Equations (101.29) and (101.54) have different mathematical status from Remark 101.2. That remark records the finite-dimensional deformation identity under stated hypotheses. The compact-space QFT formulae additionally require:

- S^4 : a regulator preserving the chosen supercharge, determinant regularization, and an instanton compactification or contour for singular point instantons;
- S^3 : a parity-anomaly regulator, framing convention, real contour or analytic continuation, and a treatment of noncompact Coulomb directions and singular hyperplanes.

When these data are supplied, localization turns a protected observable into an integral over fixed-locus variables. It does not erase the distinction between regularization and renormalization, nor does it make contact terms canonical. Background Chern–Simons terms in three dimensions and local curvature counterterms in four dimensions change phases or local source-dependent factors in the localized answer; they must be tracked as coordinates of the QFT, not discarded as normalization details.

Part VIII

Topological and Cohomological Quantum Field Theory

Chapter 102

Metric Independence and Cohomological Observables

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

A topological sector of a quantum field theory is not defined by the absence of formulas containing a metric. It is defined by correlation functionals whose variation under allowed changes of geometric representatives is controlled and, for the observables under consideration, vanishes in cohomology. The most common mechanism is cohomological: an odd symmetry Q acts on the regulated fields, sources, and insertions; metric variation of the action and of the insertions is Q-exact; and the regulated expectation functional obeys the corresponding Ward identity. This chapter develops that mechanism before any particular example is introduced.

102.1 Proof Status Map for Topological Claims

The topological volumes use several kinds of input, and the status labels are not interchangeable. A finite state-sum identity, a differential-topology classification theorem, and a physical RG comparison do not have the same proof burden. The reader should use the following map while moving through Volumes VIII and IX.

1. A *local finite proof* is a calculation inside a finite-dimensional or finite-groupoid model. Examples include the de Rham Q-Stokes model in this chapter, the ordinary Frobenius-algebra

classification in Chapter 103, the finite gauge groupoid gluing in Chapter 112, and the finite cochain higher-gauging identities in Chapter 123.

2. A *local mechanism under an explicit model hypothesis* proves the algebraic step after the model has been supplied. This is the status of the normal Gaussian/Euler-factor calculation in Chapter 106, the BV collar-boundary formula in Chapter 111, the Müger-center plaquette criterion in Chapter 110, and the finite gauging projector comparison in Chapter 122. These arguments explain what a model must do; they do not construct every continuum or higher-categorical model that could realize the algebra.

3. An *external mathematical theorem boundary* is used with stated hypotheses and a declared role, but is not reproved locally. This includes the fully extended cobordism hypothesis, the full Reshetikhin–Turaev/Crane–Yetter/Walker–Wang theorem packages, Gromov–Witten virtual-cycle technology in Chapter 107, and the Donaldson, Seiberg–Witten, Bauer–Furuta, and Furuta differential-topology inputs in Chapter 109.

4. A *controlled QFT hypothesis* asks for a regulator, BV or Hilbert space construction, anomaly-line convention, compactness statement, and determinant-line normalization. Chern–Simons theory in Chapter 105, continuum cohomological gauge theories, and symmetry-defect systems in Chapters 114 and 121 often sit here until the required construction data have been given.

5. An *open comparison problem* remains when the chapter has exposed the finite or local mechanisms but not built the single continuum framework that identifies them. The Donaldson–Seiberg–Witten comparison is the main example: the finite moduli-space theorems, u-plane wall-normal model, contact terms, and determinant phases test individual arrows, while the full Q-compatible Wilsonian flow from twisted non-Abelian gauge theory to the Abelian monopole theory remains a theorem target rather than a settled local derivation.

This map is a guard against a common source of false depth. Proving a finite cell identity can make a quoted theorem less mysterious, but it closes only the arrow whose hypotheses it actually checks. Conversely, leaving a theorem as external is acceptable only when the surrounding QFT statement has been separated from that theorem and the missing construction has been recorded as a hypothesis or open problem.

102.2 Metric Response as a Renormalized Insertion

Let M be a closed oriented smooth D-manifold and let $\mathcal{G}(M)$ denote the Fréchet manifold of smooth Riemannian metrics on M. A metric-dependent QFT presentation assigns, for $g \in \mathcal{G}(M)$, renormalized correlation functionals

$$\langle X \rangle_g$$

for insertions X in a specified algebra of local, extended, and source dependent observables. The notation includes the regulator, local counterterms, source-coordinate system, and contact-term convention. Different choices can present the same continuum theory, but a metric-variation formula is a statement inside one such presentation.

Definition 102.1 (Stress-tensor response)**.** Let X be a renormalized insertion whose support is disjoint from the support of a compactly supported metric variation $\delta g^{\mu\nu}$. The stress-tensor response of the presentation is the distributional identity

$$\delta_g\langle X\rangle_g = -\frac{1}{2}\int_M \mathrm{d}^D x\,\sqrt{g}\,\delta g^{\mu\nu}(x)\,\langle T_{\mu\nu}(x)X\rangle_g + \langle C_{\delta g}(X)\rangle_g. \tag{102.1}$$

Here $T_{\mu\nu}$ is the renormalized stress tensor and $C_{\delta g}(X)$ is the sum of contact insertions supported where the metric variation collides with the support or source-dependence of X.

If X is a product of metric-independent local operators at fixed points and δg is supported away from those points, then $C_{\delta g}(X) = 0$. If X includes integrated observables, line operators, boundary conditions, or metric-dependent representatives, the contact term is part of the definition of the observable. Ignoring it is equivalent to changing the problem.

For a one-parameter family g_s, (102.1) becomes

$$\frac{\mathrm{d}}{\mathrm{d}s}\langle X_s\rangle_{g_s} = -\frac{1}{2}\int_M \mathrm{d}^D x\sqrt{g_s}\,\dot{g}_s^{\mu\nu}\langle T_{\mu\nu}(x)X_s\rangle_{g_s} + \left\langle C_{\dot{g}_s}(X_s) + \dot{X}_s\right\rangle_{g_s}. \tag{102.2}$$

The term $\dot{X}_s$ records explicit variation of the chosen insertion representative. Metric independence is the assertion that the right-hand side vanishes for the chosen cohomology classes and for every allowed path g_s.

102.3 Cohomological Ward Systems

The letter Q will denote an odd operator of degree $+1$. If Y is a homogeneous insertion, write $|Y| \in \mathbb{Z}_2$ for its parity. The graded commutator with Q is

$$[Q, Y]_{\mathrm{gr}} = QY - (-1)^{|Y|}YQ.$$

For an odd Y, this is an anticommutator; for an even Y, it is a commutator. In a path-integral presentation it is often clearer to regard Q as an odd derivation δ_Q on the algebra of insertions.

Definition 102.2 (Cohomological Ward system)**.** A cohomological Ward system over a family of metrics g_s consists of:

(i) a graded algebra $\mathcal{A}_s$ of renormalized insertions;

(ii) an odd degree-$+1$ derivation $\delta_Q : \mathcal{A}_s \to \mathcal{A}_s$ satisfying $\delta_Q^2 = 0$;

(iii) expectation functionals $\langle\cdot\rangle_s : \mathcal{A}_s \to \mathbb{C}$;

(iv) a subspace $\mathcal{A}_s^{\mathrm{adm}} \subset \mathcal{A}_s$ of insertions for which metric derivatives and Q-variations are both defined;

(v) the Ward identity

$$\langle\delta_Q Y\rangle_s = 0, \qquad Y \in \mathcal{A}_s^{\mathrm{adm}}. \tag{102.3}$$

The system is anomaly-free on this family if (102.3) holds without additional local or boundary terms. If $\langle\delta_Q Y\rangle_s = \langle\mathcal{A}_Q(Y)\rangle_s$ for a nonzero local functional $\mathcal{A}_Q$, then $\mathcal{A}_Q$ is the cohomological anomaly in the chosen presentation.

The Ward identity is a theorem only after regularization and renormalization are specified. In finite-dimensional compact models it is Stokes' theorem. In gauge theory it is a statement about the BV or BRST measure, the quantum master equation, and boundary behavior at infinity or at singular strata.

Definition 102.3 (Cohomological topological sector)**.** A cohomological topological sector for the family g_s is a cohomological Ward system together with, for every s, a class of Q-closed insertions X_s and an odd insertion V_s such that

$$\frac{\mathrm{d}}{\mathrm{d}s}\langle X_s\rangle_s = \langle \delta_Q V_s\rangle_s\,. \tag{102.4}$$

If the metric response is expressed by a stress tensor, this condition is usually implemented by

$$-\frac{1}{2}\sqrt{g_s}\,\dot{g}_s^{\mu\nu}T_{\mu\nu}\,X_s + C_{\dot{g}_s}(X_s) + \dot{X}_s = \delta_Q V_s \tag{102.5}$$

as an integrated renormalized insertion.

This definition keeps two distinct statements separate. The local formula $T_{\mu\nu} = [Q, G_{\mu\nu}]_{\mathrm{gr}}$ is sufficient only when the contact terms and the explicit variation of X_s are also Q-exact. In examples, the latter terms are often where the nontrivial geometry of the observable enters.

Metric independence from a cohomological Ward system. Let g_s, $0 \le s \le 1$, be a smooth path of metrics on a closed manifold M. Suppose $X_s \in \mathcal{A}_s^{\mathrm{adm}}$ satisfies $\delta_Q X_s = 0$, the one-parameter metric response is defined for all s, and the response insertion is Q-exact as in (102.4). If the Ward identity (102.3) holds for the corresponding V_s, then

$$\langle X_1\rangle_{g_1} = \langle X_0\rangle_{g_0}.$$

The derivation is the integration of the Ward identity along the path:

$$\frac{\mathrm{d}}{\mathrm{d}s}\langle X_s\rangle_s = \langle \delta_Q V_s\rangle_s.$$

The right-hand side vanishes pointwise in s, so $s \mapsto \langle X_s\rangle_s$ is constant on the interval. The load-bearing hypotheses are the existence of the renormalized metric response, the construction of V_s, and the Ward identity for that insertion.

In local stress-tensor language, this criterion is often summarized by saying that $T_{\mu\nu}$ is Q-exact. The precise statement is (102.5): the whole renormalized metric response, including contact terms and moving insertion representatives, must be Q-exact.

102.4 Contact Terms, Anomalies, and Boundaries of Field Space

There are three common mechanisms by which the proof above can fail.

First, the Q-variation of the measure may be anomalous. In BV language, the anomaly is the obstruction to the quantum master equation for the renormalized action. If S is a regulated BV action and Δ is the regulated BV Laplacian, the formal Ward identity follows from

$$\frac{1}{2}(S,S) - \hbar\Delta S = 0.$$

Failure of this equation gives a local functional $\mathcal{A}_Q$ whose cohomology class controls whether a local counterterm can restore the Ward identity. This is why gauge-theoretic cohomological theories must be connected to the BV formalism developed in Chapter 48.

Second, the integration space may have boundaries or noncompact ends. A finite-dimensional model with an odd vector field Q obeys

$$\int_{\mathcal{F}} \delta_Q(\cdots) = \int_{\partial\mathcal{F}} \iota_Q(\cdots)$$

whenever Stokes' theorem applies. Thus the Ward identity requires either $\partial\mathcal{F} = \varnothing$, boundary conditions that kill the boundary term, or compactification data for which the boundary contribution is itself cancelled by other strata. In Donaldson–Witten theory this issue is reflected in compactification of instanton moduli spaces; in localization integrals it appears as contour, Jeffrey–Kirwan, or other cycle choices.

Third, contact terms can change the observable algebra. If a metric variation moves the representative of an integrated observable, its collision with other insertions can generate lower-dimensional descendant operators. A cohomological topological sector therefore contains not only the cohomology of isolated operators but also a contact-term prescription compatible with products and collisions.

Warning 102.4 (Metric independence is not automatic functoriality)*.* Metric-independent closed-manifold numbers do not by themselves define a topological field theory in the Atiyah–Segal sense. A functorial theory must also assign state spaces to codimension-one manifolds, linear maps to bordisms, and gluing isomorphisms. Cohomological Ward identities are a mechanism for producing such data under additional compactness, anomaly, boundary, and finite-dimensionality hypotheses; those hypotheses are part of the construction data.

102.5 Descent and Integrated Observables

Cohomological gauge theories often start from a local Q-closed scalar operator and produce observables supported on cycles. This is the descent construction. It is the source of Donaldson observables, topological sigma-model observables, and many protected line and surface insertions.

Definition 102.5 (Descent equations)**.** Let $O^{(p)}$ be a sequence of renormalized p-form-valued local operators, $0 \le p \le D$, with total parity including form degree. A descent system is

$$\delta_Q O^{(0)} = 0, \qquad \delta_Q O^{(p+1)} + \mathrm{d}O^{(p)} = 0, \quad 0 \le p \le D-1, \tag{102.6}$$

as an identity of renormalized local operator-valued forms, including contact terms when other insertions are present.

For a closed oriented p-cycle $\Sigma_p \subset M$, define

$$O(\Sigma_p) = \int_{\Sigma_p} O^{(p)}.$$

This integral is a renormalized insertion; if Σ_p intersects other operator supports, one must specify the collision prescription.

Homological invariance of descended observables. Assume (102.6) holds and that no contact terms are encountered during the deformations below. If Σ_p is closed, then $O(\Sigma_p)$ is Q-closed. If $\Sigma_p' - \Sigma_p = \partial B_{p+1}$, then

$$O(\Sigma_p') - O(\Sigma_p) = -\delta_Q \int_{B_{p+1}} O^{(p+1)}. \tag{102.7}$$

Thus the Q-cohomology class of $O(\Sigma_p)$ depends only on the homology class $[\Sigma_p]$.

For a closed cycle,

$$\delta_Q O(\Sigma_p) = \int_{\Sigma_p} \delta_Q O^{(p)} = -\int_{\Sigma_p} \mathrm{d}O^{(p-1)} = -\int_{\partial\Sigma_p} O^{(p-1)} = 0.$$

If $\Sigma_p' - \Sigma_p = \partial B_{p+1}$, then Stokes' theorem and the descent equation give

$$O(\Sigma_p') - O(\Sigma_p) = \int_{\partial B_{p+1}} O^{(p)} = \int_{B_{p+1}} \mathrm{d}O^{(p)} = -\delta_Q \int_{B_{p+1}} O^{(p+1)}.$$

The calculation is an identity in the renormalized insertion algebra under the stated no-contact hypothesis.

Example 102.6 (Donaldson-type descent form)**.** In a four-dimensional cohomological gauge theory obtained from a twisted $\mathcal{N} = 2$ vector multiplet, the scalar field ϕ is even and the connection A has an odd one-form partner ψ. With gauge-invariant quadratic polynomial

$$O^{(0)} = \frac{1}{8\pi^2} \operatorname{tr} \phi^2,$$

and with the convention

$$\delta_Q A = \psi, \qquad \delta_Q \psi = -D_A \phi, \qquad \delta_Q \phi = 0,$$

one may choose the first descendants to be

$$O^{(1)} = \frac{1}{4\pi^2} \operatorname{tr}(\phi\psi), \qquad O^{(2)} = \frac{1}{4\pi^2} \operatorname{tr}\left(\phi F + \frac{1}{2}\psi \wedge \psi\right),$$

with higher descendants determined by (102.6). Direct substitution gives

$$\delta_Q O^{(1)} = -\mathrm{d}O^{(0)}, \qquad \delta_Q O^{(2)} = -\mathrm{d}O^{(1)},$$

using the Bianchi identity $D_A F = 0$ and cyclicity of the trace. The normalizations depend on the trace convention and must match the instanton-number normalization of the gauge theory. The integral $\int_{\Sigma_2} O^{(2)}$ is the basic surface observable entering Donaldson–Witten theory; the homological-invariance calculation above explains why its cohomology class depends on the homology class of Σ_2, before compactification and contact-term issues are addressed.

102.6 A Finite-Dimensional Model of the Ward Identity

The following model is not a field theory, but it is the algebraic core of cohomological localization. Let x^i be even coordinates on a compact oriented finite-dimensional manifold X, and let ψ^i be odd coordinates corresponding to $\mathrm{d}x^i$. The de Rham differential is the odd derivation

$$Qx^i = \psi^i, \qquad Q\psi^i = 0.$$

An ordinary differential form α is represented as a function $\alpha(x, \psi)$, and $Q\alpha$ is $\mathrm{d}\alpha$. If X is closed, Stokes' theorem gives

$$\int_X Q\eta = \int_X \mathrm{d}\eta = 0.$$

If α is Q-closed and V_t is an odd family, then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_X \alpha \, \exp[-QV_t] = -\int_X \alpha \, Q(\dot{V}_t) \, \exp[-QV_t] = -\int_X Q\left(\alpha \, \dot{V}_t \, \exp[-QV_t]\right) = 0,$$

where $Q\alpha = 0$, $Q(QV_t) = 0$, and the graded Leibniz rule fix the sign. This is the finite-dimensional prototype of the cohomological metric-independence criterion above. Infinite-dimensional QFT applications require a regulator, a measure or BV half-density, and a proof that no boundary terms appear from noncompact field directions.

102.7 Functorial Target

The Atiyah–Segal form of a topological field theory assigns a vector space $Z(\Sigma)$ to a closed $(D-1)$-manifold Σ and a linear map

$$Z(M) : Z(\Sigma_{\mathrm{in}}) \to Z(\Sigma_{\mathrm{out}})$$

to a bordism $M : \Sigma_{\mathrm{in}} \to \Sigma_{\mathrm{out}}$, with compatibility under gluing:

$$Z(M_2 \circ M_1) = Z(M_2) Z(M_1).$$

Extended versions refine this assignment to manifolds with corners and higher-categorical targets. The cohomological construction above is one source of such functors after gauge fixing, anomaly cancellation, boundary conditions, compactness, and finite-dimensionality hypotheses have been supplied.

Open Problem 102.7 (From cohomological path integrals to extended functors). Given a cohomological gauge theory with a Q-exact metric response, determine the exact hypotheses under which its correlation functions define an extended topological functor: compactness or properness of the moduli problem, anomaly cancellation, boundary conditions, gluing analysis, orientation data, contact-term compatibility, and compatibility with local operator insertions. The objective is to construct the functorial QFT object from regulated cohomological field theory data while matching known invariants.

Chapter 103

Bordism Functoriality and Extended Topological Field Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The cohomological mechanism of Chapter 102 produces correlation functionals whose metric variation is controlled by a Ward identity. A functorial topological field theory is a stronger object. It assigns state spaces to closed spatial manifolds, linear maps to bordisms, and compatibility maps for cutting and gluing. The purpose of this chapter is to state the functorial data precisely enough that later claims about BF theory, Chern–Simons theory, finite gauge theory, cohomological gauge theory, and topological twists have a definite target.

Throughout this chapter $\Bbbk$ is a field of characteristic zero unless a finite-field example is explicitly declared. The finite-dimensional target is denoted $\mathbf{Vect}_{\Bbbk}^{\mathrm{fd}}$. Infinite-dimensional state spaces, bornological vector spaces, and Hilbert completions occur naturally in non-topological QFT and in some noncompact topological theories; whenever they are used later, the topology on the target is part of the datum and the gluing maps must be continuous for that topology.

103.1 Tangential Structures and Collared Bordisms

A tangential structure in dimension D is a map

$$\theta_\xi : B_\xi \longrightarrow BO(D).$$

Examples are $BSO(D) \to BO(D)$ for oriented manifolds, $BSpin(D) \to BO(D)$ for spin manifolds, and the point $* \to BO(D)$ classifying a framing of the tangent bundle. If N is a smooth d-manifold with $d \leq D$, a ξ-structure on N means a lift, up to homotopy through lifts, of the classifying map of the stabilized tangent bundle

$$TN \oplus \mathbb{R}^{D-d} \longrightarrow BO(D)$$

to B_ξ. This stabilized convention is the one compatible with extended bordisms: a hypersurface, a boundary corner, and a point are all regarded as lower-dimensional strata inside D-dimensional bordisms.

Definition 103.1 (Collared ξ-bordism)**.** Let Σ_0 and Σ_1 be closed compact smooth $(D-1)$-manifolds with ξ-structure. A collared ξ-bordism

$$M : \Sigma_0 \longrightarrow \Sigma_1$$

is a compact smooth D-manifold M with ξ-structure, together with boundary identifications

$$\partial M \simeq \bar{\Sigma}_0 \sqcup \Sigma_1$$

and collar embeddings near the incoming and outgoing components. Here $\bar{\Sigma}_0$ denotes the same underlying manifold with the tangential structure reversed in the sense appropriate to ξ. Two collared bordisms are equivalent if they are related by a ξ-diffeomorphism which respects the collars up to isotopy.

The collars are not decorative. They make composition a smooth operation: if $M_{01} : \Sigma_0 \to \Sigma_1$ and $M_{12} : \Sigma_1 \to \Sigma_2$, then

$$M_{12} \circ M_{01} = M_{01} \cup_{\Sigma_1} M_{12}$$

is obtained by identifying the outgoing collar of M_{01} with the incoming collar of M_{12}, followed by the canonical smoothing determined by the collar product coordinate. The identity of Σ is the cylinder $\Sigma \times [0,1]$ with product ξ-structure.

Definition 103.2 (Ordinary bordism category)**.** The category Bord_D^ξ has:

- objects: closed compact smooth $(D-1)$-manifolds with ξ-structure;
- morphisms: equivalence classes of collared ξ-bordisms;
- composition: collar gluing;
- symmetric monoidal product: disjoint union.

The monoidal unit is the empty manifold $\varnothing$.

Figure 103.1 records the collar-gluing operation in the bordism category: two bordisms sharing the collared boundary Σ_1 compose by identifying that boundary.

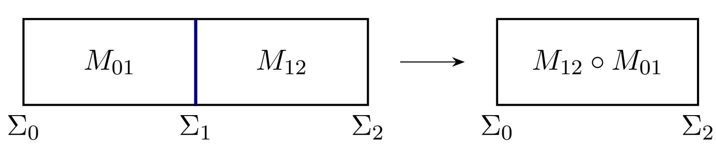

Figure 103.1: Composition in the ordinary bordism category is collar gluing. The intermediate boundary component is not retained as extra data after the composition; extended TQFT keeps track of such cuts through lower-codimension assignments.

Definition 103.2 is the 1-categorical version. A fully smooth functorial QFT normally uses a topologically enriched or higher bordism object in which smooth families of bordisms are also visible. The ordinary category is nevertheless the right first target for finite-dimensional Atiyah–Segal TQFT because it isolates the exact gluing identities on which every refinement must reduce.

103.2 Atiyah–Segal Functoriality

Definition 103.3 (Atiyah–Segal topological field theory)**.** An honest D-dimensional Atiyah–Segal TQFT with tangential structure ξ and finite-dimensional linear target is a symmetric monoidal functor

$$Z : \mathrm{Bord}_D^{\xi} \longrightarrow \mathbf{Vect}_{\Bbbk}^{\mathrm{fd}}.$$

Thus there are canonical isomorphisms

$$Z(\Sigma_1 \sqcup \Sigma_2) \simeq Z(\Sigma_1) \otimes Z(\Sigma_2), \qquad Z(\varnothing) \simeq \Bbbk,$$

and for composable bordisms

$$Z(M_{12} \circ M_{01}) = Z(M_{12})\, Z(M_{01}).$$

The cylinder maps to the identity on $Z(\Sigma)$.

For a closed D-manifold M, regarded as a bordism $\varnothing \to \varnothing$, the value $Z(M)$ is the scalar by which the map $\Bbbk \to \Bbbk$ multiplies. For a bordism $M : \Sigma_0 \to \Sigma_1$, $Z(M)$ is the transition amplitude from the state space $Z(\Sigma_0)$ to $Z(\Sigma_1)$. These words are justified by the gluing axiom, not by an analogy with time evolution: if M is cut along an intermediate closed hypersurface Σ, the amplitude factors through the state space assigned to Σ.

Definition 103.4 (Anomalous or relative functorial theory)**.** Let $\mathcal{L}$ be an invertible $(D+1)$-dimensional field theory, or equivalently in the finite-dimensional setting a compatible system of one-dimensional anomaly lines over boundary data. A D-dimensional theory relative to $\mathcal{L}$ assigns vectors or linear maps valued in those lines. It becomes an honest functor only after the anomaly line has been trivialized, or after the $(D+1)$-dimensional bulk $\mathcal{L}$ is kept as part of the system.

This distinction matters in gauge theory and in theories with gravitational or global anomalies. A path integral may satisfy topological Ward identities on closed manifolds while its boundary Hilbert spaces transform projectively. The functorial object is then relative, not an honest functor $\mathrm{Bord}_D^{\xi} \to \mathbf{Vect}_{\Bbbk}^{\mathrm{fd}}$.

The Atiyah–Segal datum is the metric-independent specialization of the Kontsevich–Segal viewpoint developed in Chapter 13. A Kontsevich–Segal theory keeps admissible complex metrics and analytic dependence as part of the bordism datum. An Atiyah–Segal TQFT is obtained when, after all anomaly and background-structure data have been included, the amplitude is locally constant on the geometric parameters.

103.3 Duality, Pairings, and Reflection

If $\bar{\Sigma}$ is the reversed ξ-manifold, the evaluation bordism

$$\mathrm{ev}_{\Sigma} : \bar{\Sigma} \sqcup \Sigma \longrightarrow \varnothing$$

and coevaluation bordism

$$\mathrm{coev}_\Sigma : \varnothing \longrightarrow \Sigma \sqcup \bar{\Sigma}$$

are the two caps obtained by viewing the cylinder on Σ sideways. Functoriality gives maps

$$Z(\mathrm{ev}_\Sigma) : Z(\bar{\Sigma}) \otimes Z(\Sigma) \to \Bbbk, \qquad Z(\mathrm{coev}_\Sigma) : \Bbbk \to Z(\Sigma) \otimes Z(\bar{\Sigma}).$$

Proposition 103.5 (State-space duality from caps and cylinders)**.** *Let Z be an honest finite-dimensional Atiyah–Segal TQFT. Then $Z(\bar{\Sigma})$ is canonically dual to $Z(\Sigma)$. Equivalently, the pairing $Z(\bar{\Sigma}) \otimes Z(\Sigma) \to \Bbbk$ from the evaluation bordism is nondegenerate, and the coevaluation vector is its inverse tensor.*

Proof. Put $V = Z(\Sigma)$ and $W = Z(\bar{\Sigma})$. The evaluation bordism gives a bilinear form $e : W \otimes V \to \Bbbk$, and the coevaluation bordism gives an element $c \in V \otimes W$. The point of the cap/cylinder argument is that (e, c) is not arbitrary bordism data; it obeys the categorical triangle identities.

The two zig-zag composites

$$\Sigma \xrightarrow{\mathrm{coev}_\Sigma \sqcup \mathrm{id}_\Sigma} \Sigma \sqcup \bar{\Sigma} \sqcup \Sigma \xrightarrow{\mathrm{id}_\Sigma \sqcup \mathrm{ev}_\Sigma} \Sigma$$

and

$$\bar{\Sigma} \xrightarrow{\mathrm{id}_{\bar{\Sigma}} \sqcup \mathrm{coev}_\Sigma} \bar{\Sigma} \sqcup \Sigma \sqcup \bar{\Sigma} \xrightarrow{\mathrm{ev}_\Sigma \sqcup \mathrm{id}_{\bar{\Sigma}}} \bar{\Sigma}$$

are diffeomorphic to the corresponding cylinders. Applying Z, and using the cylinder axiom, gives the triangular identities for a dual pair in $\mathbf{Vect}_\Bbbk^{\mathrm{fd}}$:

$$(\mathrm{id}_V \otimes e)(c \otimes \mathrm{id}_V) = \mathrm{id}_V, \qquad (e \otimes \mathrm{id}_W)(\mathrm{id}_W \otimes c) = \mathrm{id}_W.$$

Write $c = \sum_\alpha v_\alpha \otimes w_\alpha$. If $v \in V$ is annihilated by every functional $e(w, \cdot)$, then the first triangle identity gives

$$v = \sum_\alpha v_\alpha e(w_\alpha, v) = 0.$$

Thus the induced map $W \to V^\vee$ is injective. The second triangle identity similarly shows that $V \to W^\vee$ is injective. Since $\dim W \leq \dim V$ and $\dim V \leq \dim W$, both maps are isomorphisms. Under the resulting identification $W \simeq V^\vee$, the tensor c is the canonical inverse tensor to the pairing e, as expressed by the same two triangle identities. $\square$

A unitary topological theory requires more than the bilinear pairing above. Reflection of bordisms should be implemented by an antilinear involution on state spaces, and the pairing obtained by reflecting a state and gluing it to the original state should be positive. This is the TQFT analogue of reflection positivity in Euclidean QFT. Positivity is therefore not a consequence of Definition 103.3; it is extra structure on the functor.

103.4 The Complete Two-Dimensional Oriented Example

The first nontrivial test of the definition is the classification of ordinary two-dimensional oriented TQFTs by finite-dimensional commutative Frobenius algebras. This theorem gives the exact algebraic form of the gluing law for surfaces, with pair-of-pants multiplication as one piece of the structure.

Definition 103.6 (Commutative Frobenius algebra)**.** A commutative Frobenius algebra over $\Bbbk$ is a finite-dimensional unital associative commutative $\Bbbk$-algebra A, together with a linear functional

$$\epsilon : A \to \Bbbk$$

such that the bilinear form

$$\eta(a, b) = \epsilon(ab) \tag{103.1}$$

is nondegenerate. The unit is denoted 1_A, the multiplication is $m : A \otimes A \to A$, and the counit is ϵ.

Choose a basis $\{e_i\}_{i=1}^N$ of A. Let

$$\eta_{ij} = \epsilon(e_i e_j), \qquad (\eta^{ij}) = (\eta_{ij})^{-1},$$

and define the dual elements

$$e^i = \sum_j \eta^{ij} e_j.$$

Then $\epsilon(e_i e^j) = \delta_i^j$. The inverse of the Frobenius pairing is the tensor

$$C = \sum_i e_i \otimes e^i \in A \otimes A.$$

The comultiplication adjoint to multiplication is

$$\Delta(a) = \sum_i e_i \otimes e^i a = \sum_i a e_i \otimes e^i. \tag{103.2}$$

The equality of the two expressions follows from commutativity. The defining adjointness property is

$$(\epsilon \otimes \epsilon)\big(\Delta(a)(b \otimes c)\big) = \epsilon(abc), \qquad a, b, c \in A. \tag{103.3}$$

Cylinder and neck identities. For a commutative Frobenius algebra A, the inverse-pairing tensor $C = \sum_i e_i \otimes e^i$ and the comultiplication Δ give the two elementary algebraic relations used in the surface-cutting construction:

$$\sum_i e_i \, \epsilon(e^i a) = a, \tag{103.4}$$

$$(m \otimes \mathrm{id}_A)(\mathrm{id}_A \otimes \Delta) = \Delta m = (\mathrm{id}_A \otimes m)(\Delta \otimes \mathrm{id}_A). \tag{103.5}$$

These identities are not separate structure on A; they are the algebraic form of gluing a cap to a cup and sliding a pair-of-pants critical level past another one. For the cylinder identity, pair the left side with an arbitrary $b \in A$:

$$\epsilon\left(b \sum_i e_i \, \epsilon(e^i a)\right) = \sum_i \epsilon(b e_i)\epsilon(e^i a) = \epsilon(ba),$$

because $\{e_i\}$ and $\{e^i\}$ are dual bases for the pairing η. Nondegeneracy of η gives (103.4).

For the first equality in (103.5), the left side on $a \otimes b$ is

$$(m \otimes \mathrm{id})(a \otimes \Delta(b)) = \sum_i ae_i \otimes e^i b.$$

The middle term is

$$\Delta(ab) = \sum_i e_i \otimes e^i ab.$$

To prove equality, pair the first tensor factor with an arbitrary $x \in A$ and the second with an arbitrary $y \in A$. The two pairings give

$$\sum_i \epsilon(xae_i)\epsilon(ye^i b) = \epsilon(xyab) = \sum_i \epsilon(xe_i)\epsilon(ye^i ab),$$

again by the dual-basis identity. Nondegeneracy of $\eta \otimes \eta$ proves the equality. The second equality is identical, or follows by commutativity.

Theorem 103.7 (Oriented 2-dimensional TQFTs and Frobenius algebras)**.** *Finite-dimensional ordinary oriented* 2*-dimensional Atiyah–Segal TQFTs over* $\Bbbk$ *are equivalent to finite-dimensional commutative Frobenius algebras over* $\Bbbk$*. Under this equivalence,*

$$A = Z(S^1), \qquad m = Z(P_{2\to 1}), \qquad 1_A = Z(D_{\varnothing\to 1})(1), \qquad \epsilon = Z(D_{1\to\varnothing}),$$

where $P_{2\to 1}$ *is the oriented pair of pants with two incoming circles and one outgoing circle.*

Proof. First let Z be a 2-dimensional oriented TQFT and set $A = Z(S^1)$. The disk from $\varnothing$ to S^1 gives the unit 1_A, the reversed disk gives $\epsilon : A \to \Bbbk$, and the pair of pants gives $m : A \otimes A \to A$. Associativity follows because the two bordisms

$$P_{2\to 1} \circ (P_{2\to 1} \sqcup \mathrm{id}_{S^1}) \quad \text{and} \quad P_{2\to 1} \circ (\mathrm{id}_{S^1} \sqcup P_{2\to 1})$$

are diffeomorphic as oriented bordisms from three circles to one circle. Commutativity follows from the diffeomorphism of the pair of pants which interchanges the two incoming circles. The unit axiom follows by gluing a disk to one incoming circle of the pair of pants, producing a cylinder. Proposition 103.5, applied to the circle, shows that $\epsilon(ab)$ is nondegenerate. Thus A is a commutative Frobenius algebra.

Conversely, let (A, ϵ) be a commutative Frobenius algebra. Define the value on a disjoint union of n positively oriented circles to be $A^{\otimes n}$; reversed circles are assigned the dual vector space, and the Frobenius pairing identifies it with A when an explicit formula is written using the basis-independent tensor C. Assign to the elementary Morse bordisms the maps

$$D_{\varnothing\to 1} \mapsto 1_A, \qquad D_{1\to\varnothing} \mapsto \epsilon, \qquad P_{2\to 1} \mapsto m, \qquad P_{1\to 2} \mapsto \Delta.$$

For a general compact oriented surface with collared boundary, choose a Morse function constant near the boundary. The critical points decompose the surface into disks and pairs of pants, hence into a composite of the elementary maps above.

It remains to show that the resulting linear map is independent of the Morse decomposition. The differential-topological input is the standard Cerf description of decompositions of compact surfaces: two decompositions are related by isotopy, by creation or cancellation of adjacent critical points, and by interchange of neighboring critical levels. Isotopy gives no change. Cancellation of a birth-death pair gives precisely the cylinder identity (103.4) and the unit/counit identities. Interchange of two pair-of-pants critical levels gives associativity, commutativity, and the neck-exchange Frobenius identity (103.5). Therefore the map depends only on the oriented bordism class, and collar gluing is sent to composition because concatenating Morse functions concatenates the associated elementary maps. The construction is symmetric monoidal by tensor product on disjoint unions.

The two constructions are inverse up to the evident natural isomorphism: a TQFT is generated by disks and pairs of pants modulo the relations just checked, and a Frobenius algebra is recovered from the values of the constructed functor on those generating bordisms. □

Figure 103.2 displays the generating bordisms for the two-dimensional oriented TQFT/Frobenius-algebra equivalence: multiplication, unit, pairing, and the relations imposed by bordism invariance.

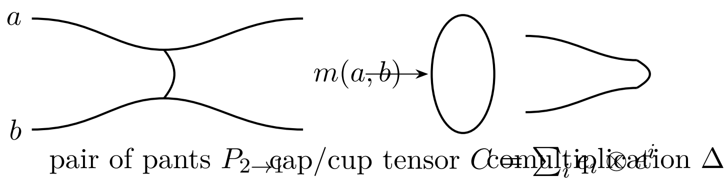

Figure 103.2: In two dimensions the gluing generators are disks, caps, cups, and pairs of pants. The inverse pairing tensor C is what makes gluing along an internal circle a contraction over the intermediate state space.

Example 103.8 (Semisimple idempotent basis)**.** Let

$$A = \Bbbk^r$$

with pointwise multiplication. Let p_α, $1 \le \alpha \le r$, be the primitive idempotents and choose nonzero weights $\lambda_\alpha \in \Bbbk^\times$. Define

$$\epsilon(p_\alpha) = \lambda_\alpha.$$

Then

$$\eta(p_\alpha, p_\beta) = \lambda_\alpha \delta_{\alpha\beta}, \qquad p^\alpha = \lambda_\alpha^{-1} p_\alpha,$$

and the Euler element

$$E = m\Delta(1_A) = \sum_{\alpha=1}^{r} \lambda_\alpha^{-1} p_\alpha$$

is the handle operator. For a closed connected oriented surface Σ_g of genus g,

$$Z(\Sigma_g) = \epsilon(E^g) = \sum_{\alpha=1}^{r} \lambda_\alpha^{1-g}. \tag{103.6}$$

The formula is obtained by cutting Σ_g into a sphere with g handles. Each handle inserts E; the final cap applies ϵ.

103.5 Extended Theories and What Extra Data They Contain

An extended TQFT assigns data not only to closed $(D-1)$-manifolds but also to lower-dimensional strata. Its source is an extended bordism (∞, D)-category

$$\mathrm{Bord}_D^{\xi,\mathrm{ext}},$$

whose objects are points with ξ-structure, whose 1-morphisms are collared intervals and 1-manifolds, whose 2-morphisms are surfaces with corners, and so on up to D-morphisms. The target is a symmetric monoidal higher category $\mathcal{C}$. A fully extended TQFT is a symmetric monoidal higher functor

$$Z : \mathrm{Bord}_D^{\xi,\mathrm{ext}} \to \mathcal{C}.$$

The phrase "locality at all codimensions" means that cutting along corners, defects, junctions, and points is represented by composition in $\mathcal{C}$.

Remark 103.9 (Framed cobordism-hypothesis boundary). The fully general framed cobordism hypothesis is an external higher-categorical classification theorem. In one standard formulation, for a symmetric monoidal (∞, D)-category $\mathcal{C}$ with the duals and adjoints required by framed bordism composition, framed fully extended D-dimensional TQFTs valued in $\mathcal{C}$ are classified by fully D-dualizable objects of $\mathcal{C}$. Here fully D-dualizable means that the object has a dual, its evaluation and coevaluation 1-morphisms have left and right adjoints, those adjunction data have the corresponding adjoints one level higher, and the same adjointability condition continues through level D. This theorem fixes the target data that must be named in a fully extended example: the target higher category, the point object, all duals and adjoints, the tangential-structure action, and any anomaly or boundary decoration. The local calculation below is the part of this structure used in the present chapter.

Definition 103.10 (Finite-dimensional Morita 2-category)**.** Let $\mathbf{Alg}_2^{\mathrm{fd}}$ denote the Morita 2-category over $\Bbbk$. Its objects are finite-dimensional associative unital $\Bbbk$-algebras. A 1-morphism $A \to B$ is a finite-dimensional (B, A)-bimodule. A 2-morphism is a bimodule homomorphism. Composition is relative tensor product over the intermediate algebra, and the symmetric monoidal product is $\otimes_{\Bbbk}$.

Theorem 103.11 (Separable algebras are two-dualizable Morita objects)**.** *Let A be a finite-dimensional separable $\Bbbk$-algebra. Thus there is an element*

$$e = \sum_r x_r \otimes y_r \in A \otimes A^{\mathrm{op}}$$

such that

$$\sum_r x_r y_r = 1_A, \qquad \sum_r a x_r \otimes y_r = \sum_r x_r \otimes y_r a \quad \textit{for every } a \in A. \tag{103.7}$$

In the Morita 2-category Definition 103.10, A has dual A^{op}, and the evaluation and coevaluation bimodules have both left and right adjoints. Hence A is fully 2-dualizable. If, in addition, $\epsilon : A \to \Bbbk$ is a symmetric Frobenius trace, meaning that

$$\epsilon(ab) = \epsilon(ba), \qquad (a, b) \mapsto \epsilon(ab)$$

is nondegenerate, then ϵ is the Calabi–Yau trace datum for the oriented two-dimensional refinement. The circle trace of the Morita object is

$$HH_0(A) = A/[A, A].$$

Proof. The dual object of A is A^{op}. The evaluation 1-morphism

$$\mathrm{ev} : A^{\mathrm{op}} \otimes A \longrightarrow \Bbbk$$

is the vector space A, regarded as a $(\Bbbk, A^{\mathrm{op}} \otimes A)$-bimodule by

$$z \cdot (a^{\mathrm{op}} \otimes b) = azb.$$

The coevaluation

$$\mathrm{coev} : \Bbbk \longrightarrow A \otimes A^{\mathrm{op}}$$

is the same vector space A, now regarded as an $(A \otimes A^{\mathrm{op}}, \Bbbk)$-bimodule by

$$(a \otimes b^{\mathrm{op}}) \cdot z = azb.$$

The two zig-zag composites reduce to the canonical associativity isomorphisms

$$A \otimes_A M \simeq M, \qquad M \otimes_A A \simeq M$$

for finite bimodules M. Thus A^{op} is a Morita dual of A.

It remains to justify adjointability of the evaluation and coevaluation bimodules. Put $A^e = A \otimes A^{\mathrm{op}}$. The multiplication map

$$\mu : A^e \to A, \qquad \mu(a \otimes b^{\mathrm{op}}) = ab$$

is a homomorphism of A-bimodules, equivalently of left A^e-modules. The separability element defines an A^e-linear splitting

$$s : A \to A^e, \qquad s(a) = \sum_r ax_r \otimes y_r = \sum_r x_r \otimes y_r a. \tag{103.8}$$

The equality of the two displayed expressions is precisely the centrality identity in (103.7), and

$$\mu s(a) = \sum_r ax_r y_r = a$$

uses the normalization identity in (103.7). Hence A is a direct summand of the free A^e-module A^e. In particular A is finite projective as a left A^e-module. Applying the same argument to A^{op}, equivalently using the right-module splitting determined by the second expression in (103.8), gives finite projectivity as a right A^e-module as well. These are exactly the enveloping-algebra projectivity statements needed by the coevaluation and evaluation bimodules.

For a finite projective bimodule M representing a 1-morphism $R \to S$, the usual dual modules

$$M^R = \mathrm{Hom}_S(M, S), \qquad M^L = \mathrm{Hom}_R(M, R)$$

carry the opposite bimodule structures and furnish right and left adjoints; the units and counits are the finite dual-basis coevaluation and evaluation maps. Applying this to the evaluation and coevaluation bimodules above gives their adjoints, so A is fully 2-dualizable.

A symmetric Frobenius trace identifies the left and right A-module duals by the nondegenerate pairing $(a, b) \mapsto \epsilon(ab)$. This is the finite-dimensional Calabi–Yau trace datum: it trivializes the Nakayama automorphism and supplies the oriented structure on the Morita object. The value assigned to the circle by the Morita trace is the shadow of the identity (A, A)-bimodule, namely

$$A \otimes_{A^e} A \simeq A/\langle ab - ba : \ a, b \in A\rangle = HH_0(A).$$

$\square$

Example 103.12 (The extended refinement of the semisimple 2-dimensional model)**.** For $A = \Bbbk^r$ with primitive idempotents p_α, the separability element is

$$e = \sum_{\alpha=1}^{r} p_\alpha \otimes p_\alpha.$$

Indeed,

$$\sum_\alpha p_\alpha p_\alpha = 1_A, \qquad a p_\alpha \otimes p_\alpha = p_\alpha \otimes p_\alpha a$$

for every $a \in A$. Choosing nonzero weights $\lambda_\alpha \in \Bbbk^\times$ and

$$\epsilon(p_\alpha) = \lambda_\alpha$$

gives the symmetric Frobenius trace of Example 103.8. Since A is commutative,

$$HH_0(A) = A/[A, A] = A,$$

and the ordinary theory obtained by forgetting point and interval data is the Frobenius-algebra TQFT of Example 103.8.

This example illustrates the conceptual difference between an ordinary and a fully extended theory. In the ordinary theory the circle state space is the primitive datum. In the fully extended Morita-theoretic description the point value is the algebra A, the interval value is the regular bimodule, and the circle state space is obtained by taking the trace of the identity A-bimodule.

103.6 From Local QFT to Functorial Data

A local QFT with a protected topological sector can produce a functorial TQFT only after several independent statements have been proved. The cohomological Ward identity must show metric independence of the selected observables. The state spaces obtained by quantizing on closed spatial manifolds must be identified and completed. Cutting a spacetime along a hypersurface must give an actual contraction over the intermediate state space. Gauge redundancies must be treated by a BV or lattice construction which supplies a gauge-independent integral and a boundary-condition category. Finally, possible anomaly lines must either be trivialized or kept as relative data.

Functorial extraction criterion. Let a regulated local QFT assign, for each collared ξ-bordism $M : \Sigma_0 \to \Sigma_1$, a multilinear amplitude

$$Z_{\mathrm{reg}}(M) : \mathcal{H}_{\mathrm{reg}}(\Sigma_0) \to \mathcal{H}_{\mathrm{reg}}(\Sigma_1)$$

depending on a regulator and auxiliary geometric choices. Suppose:

(i) the selected state spaces have regulator-independent finite-dimensional limits $\mathcal{H}(\Sigma)$;

(ii) for every smooth family of metrics or auxiliary choices, the renormalized variation of the selected amplitudes is zero or is valued in a specified anomaly line;

(iii) for every decomposition $M = M_{12} \circ M_{01}$, the regulated gluing identity converges to

$$Z(M) = Z(M_{12})Z(M_{01})$$

in the operator topology on the finite-dimensional state spaces, with the same anomaly-line convention on both sides;

(iv) the cylinder converges to the identity and disjoint union converges to tensor product.

Then the limiting amplitudes define an honest or relative Atiyah–Segal TQFT, according to whether the anomaly line has been trivialized. Indeed, the limiting assignments are $\Sigma \mapsto \mathcal{H}(\Sigma)$ on objects and $M \mapsto Z(M)$ on morphisms; the four hypotheses are precisely the well-definedness, composition, identity, and symmetric monoidal requirements in Definition 103.3, with the representative dependence recorded as the relative anomaly of Definition 103.4 when the anomaly line is kept.

Open Problem 103.13 (Functorial extraction from interacting QFT). Given a local interacting QFT with a proposed topological sector, construct the associated ordinary or extended bordism functor directly from its Hilbert spaces, local algebras, or Euclidean functional integrals. The construction should identify the precise completion of state spaces, the gluing topology, the anomaly line, the boundary category, and the status of gauge fixing or BV integration cycles.

103.7 Four-Dimensional Cohomological Gauge Theory As A Test Case

Witten–Donaldson theory will be a principal test case for the relation between cohomological path integrals, functorial TQFT data, and ordinary four-dimensional gauge theory. Its ultraviolet presentation is the topological twist of $\mathcal{N} = 2$ Yang–Mills theory. The Donaldson side is organized by the moduli space of anti-self-dual connections, its compactification and orientation data, and the observables obtained from the descent of gauge-invariant polynomials. To define the invariant one must state the gauge group and global form, the four-manifold and its orientation data, the metric dependence and chamber structure, the compactification of the instanton moduli space, and the treatment of reducible connections.

The Seiberg–Witten explanation compares this ultraviolet cohomological gauge theory to an Abelian monopole effective theory on the Coulomb branch. The physical mechanism is an RG statement: after flowing from the non-Abelian $\mathcal{N} = 2$ theory to its low-energy Wilsonian effective action, the singular loci where monopoles or dyons become light supply the Abelian fields and monopole equations that enter the low-energy topological path integral. For the monograph this comparison is not proved by naming the flow. A complete argument must construct the Wilsonian coordinate u, the effective coupling $\tau(u)$, the duality frame near each singular locus, the monopole or dyon hypermultiplet, the contact terms and gravitational couplings, the observable map from Donaldson classes to low-energy observables, and the wall-crossing or u-plane terms when they are present.

Open Problem 103.14 (Donaldson–SW RG comparison). Develop the Donaldson/Seiberg–Witten relation as a theorem-level QFT comparison. The target is to identify which parts follow from rigorous differential geometry of instanton and monopole moduli spaces, which parts follow from finite-dimensional localization or gluing, and which parts depend on a still-unproved RG theorem connecting the twisted non-Abelian $\mathcal{N} = 2$ Yang–Mills theory to the Abelian monopole effective theory. The desired output is an object-level map between correlation functions or cohomological observables, with stated metric, chamber, compactification, contact-term, and anomaly data.

Chapter 104

BF Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

BF theory is the first gauge-theoretic example in this volume. It already displays recurring ingredients of topological gauge theory: fields with form degree, ordinary gauge symmetry, higher-form gauge symmetry, reducibility, metric-free classical action, extended observables, and a BV complex. The abelian compact theory also has a finite cochain model, so some correlation functions reduce to exact finite sums.

Definition 104.1 (closed-manifold BF theory datum)**.** A closed-manifold BF theory datum in dimension D consists of the following objects.

(i) A closed oriented smooth D-manifold M.

(ii) A compact Lie group G, a principal G-bundle $P \to M$, and the adjoint bundle

$$\operatorname{ad} P = P \times_G \mathfrak{g}.$$

(iii) An invariant nondegenerate pairing between the coadjoint and adjoint bundles, written fiberwise as

$$\langle -, - \rangle : \mathfrak{g}^* \otimes \mathfrak{g} \longrightarrow \mathbb{R}.$$

(iv) The classical field space

$$\operatorname{Conn}(P) \times \Omega^{D-2}(M, \operatorname{ad} P^*),$$

with field coordinates (A, B). Here A is a connection on P, B is a $(D-2)$-form valued in the dual adjoint bundle, and $F_A \in \Omega^2(M, \operatorname{ad} P)$ denotes the curvature of A.

(v) The exponentiated classical action

$$\exp S_{\mathrm{BF}}[A,B], \qquad S_{\mathrm{BF}}[A,B] = 2\pi\mathrm{i}\int_M \langle B \wedge F_A\rangle.$$

The factor $2\pi\mathrm{i}$ fixes the convention matched to the compact abelian normalization below.

(vi) Gauge data consisting of the ordinary gauge group $\mathcal{G}(P)$ and the B-field shift parameters $\Omega^{D-3}(M, \mathrm{ad}\, P^*)$. In the nonabelian theory the shift reducibility chain is naturally a derived symmetry over the flat-connection locus $F_A = 0$; in the abelian specialization the same chain is an off-shell cochain complex because $\mathrm{d}^2 = 0$.

(vii) Extended observable data: Wilson observables associated with closed one-cycles and representations or charges, and surface observables associated with closed $(D-2)$-cycles and dual charges. The finite abelian model below makes these observables precise as cochain functionals and computes their linking phase exactly.

(viii) Quantization status. The preceding data define the classical and BV input for continuum BF theory on closed M. A functorial quantum theory additionally requires a BV or BV–BFV regularized semidensity, gauge fixing, anomaly control, and gluing data. The compact finite $\mathbb{Z}_N$ cochain model in Section 104.4 is a fully finite realization of the abelian sector.

104.1 Nonabelian Classical Fields

We first unpack Definition 104.1 in the smooth nonabelian continuum setting. Let M be a closed oriented smooth D-manifold. Let G be a compact Lie group, let $P \to M$ be a principal G-bundle, and let

$$\mathrm{ad}\, P = P \times_G \mathfrak{g}$$

be the adjoint bundle. Choose an invariant nondegenerate pairing

$$\langle -, - \rangle : \mathfrak{g}^* \otimes \mathfrak{g} \longrightarrow \mathbb{R}.$$

The classical fields of D-dimensional nonabelian BF theory are a connection A on P and a B-field

$$B \in \Omega^{D-2}(M, \mathrm{ad}\, P^*).$$

The curvature of A is

$$F_A = \mathrm{d}A + \frac{1}{2}[A, A] \in \Omega^2(M, \mathrm{ad}\, P),$$

where the bracket combines the Lie bracket on $\mathfrak{g}$ with exterior wedge product. The action is

$$S_{\mathrm{BF}}[A,B] = 2\pi\mathrm{i}\int_M \langle B \wedge F_A\rangle. \tag{104.1}$$

The coefficient is fixed so that the compact abelian specialization below has integer periods in the exponent.

Remark 104.2 (Euler–Lagrange equations). On a closed M, the stationary equations of (104.1) are

$$F_A = 0, \qquad \mathrm{d}_A B = 0.$$

Let $\alpha \in \Omega^1(M, \operatorname{ad} P)$ and $\beta \in \Omega^{D-2}(M, \operatorname{ad} P^*)$ be variations of A and B. The curvature variation is

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} F_{A+t\alpha} = \mathrm{d}_A\alpha.$$

Therefore

$$\delta S_{\mathrm{BF}} = 2\pi\mathrm{i}\int_M \left(\langle\beta \wedge F_A\rangle + \langle B \wedge \mathrm{d}_A\alpha\rangle\right).$$

The first term gives $F_A = 0$ because β is arbitrary. For the second term, use the covariant Leibniz rule

$$\mathrm{d}\langle B \wedge \alpha\rangle = \langle \mathrm{d}_A B \wedge \alpha\rangle + (-1)^{D-2}\langle B \wedge \mathrm{d}_A\alpha\rangle.$$

The integral of the left side vanishes on a closed manifold, so

$$\int_M \langle B \wedge \mathrm{d}_A\alpha\rangle = (-1)^{D-1}\int_M \langle \mathrm{d}_A B \wedge \alpha\rangle.$$

Since α is arbitrary, $\mathrm{d}_A B = 0$.

104.2 Gauge Symmetry And Reducibility

The ordinary gauge group $\mathcal{G}(P)$ acts by

$$A \longmapsto A^g = g^{-1}Ag + g^{-1}\mathrm{d}g, \qquad B \longmapsto B^g = \operatorname{Ad}^*_{g^{-1}} B,$$

where the second formula is the coadjoint action; after choosing an invariant inner product on $\mathfrak{g}$ it is written as $B^g = g^{-1}Bg$. Infinitesimally, for $c \in \Omega^0(M, \operatorname{ad} P)$,

$$\delta_c A = \mathrm{d}_A c, \qquad \delta_c B = [B, c].$$

There is also a B-field shift symmetry. For

$$\eta \in \Omega^{D-3}(M, \operatorname{ad} P^*)$$

define

$$\delta_\eta A = 0, \qquad \delta_\eta B = \mathrm{d}_A\eta.$$

Remark 104.3 (Invariance of the shift symmetry). On a closed manifold,

$$S_{\mathrm{BF}}[A, B + \mathrm{d}_A\eta] - S_{\mathrm{BF}}[A, B] = 2\pi\mathrm{i}\int_M \mathrm{d}\langle\eta \wedge F_A\rangle.$$

Hence the shift is a symmetry on closed M. On a manifold with boundary it produces the boundary term

$$2\pi\mathrm{i}\int_{\partial M} \langle\eta \wedge F_A\rangle.$$

The change in the action is

$$2\pi\mathrm{i}\int_M \langle \mathrm{d}_A\eta \wedge F_A\rangle.$$

The covariant Leibniz rule and Bianchi identity $\mathrm{d}_A F_A = 0$ give

$$\mathrm{d}\langle \eta \wedge F_A \rangle = \langle \mathrm{d}_A \eta \wedge F_A \rangle + (-1)^{D-3} \langle \eta \wedge \mathrm{d}_A F_A \rangle = \langle \mathrm{d}_A \eta \wedge F_A \rangle.$$

This proves the formula. On a closed manifold the integral of this exact top form is zero by Stokes' theorem. When $\partial M \neq \varnothing$, the same Stokes calculation gives the displayed boundary functional, which is the datum that must be cancelled or coupled to boundary degrees of freedom in a theory with boundary.

The shift symmetry is reducible. If $\eta = \mathrm{d}_A \eta_1$, then

$$\delta_\eta B = \mathrm{d}_A^2 \eta_1 = F_A \cdot \eta_1.$$

Thus the transformation vanishes on the flat-connection locus. Continuing,

$$\eta_1 \in \Omega^{D-4}(M, \operatorname{ad} P^*), \quad \eta_1 \sim \eta_1 + \mathrm{d}_A \eta_2, \quad \ldots,$$

ending in degree 0. In the abelian theory $\mathrm{d}_A = \mathrm{d}$ and the reducibility holds off shell because $\mathrm{d}^2 = 0$. The BV field space therefore contains the ordinary ghost c, the B-shift ghost η, the ghosts-for-ghosts $\eta_1, \eta_2, \ldots$, and antifields for all these variables. This is the finite-stage reducible gauge system whose BV integration is modeled by Chapter 111.

104.3 Compact Abelian Normalization

A precise continuum normalization of compact abelian BF theory uses differential characters. For a $U(1)$ one-form gauge field A, the holonomy around a closed one-cycle γ is

$$W_q(\gamma) = \exp\left(2\pi \mathrm{i} q \int_\gamma A\right), \qquad q \in \mathbb{Z}.$$

For a compact $(D-2)$-form gauge field B, the surface operator on a closed $(D-2)$-cycle Σ is

$$V_p(\Sigma) = \exp\left(2\pi \mathrm{i} p \int_\Sigma B\right), \qquad p \in \mathbb{Z}.$$

The local action is

$$S_{\mathrm{BF}}[A, B] = 2\pi \mathrm{i} N \int_M B \wedge \mathrm{d}A, \qquad N \in \mathbb{Z}_{>0}.$$

Large gauge transformations shift the periods of A and B by integers, and the factor N makes the exponent single-valued. The equation obtained by varying B is

$$N \, \mathrm{d}A = 0$$

as a differential character. Thus the theory describes a flat $\mathbb{Z}_N$ one-form gauge sector after the continuous modes and gauge redundancies are accounted for.

104.4 Finite $\mathbb{Z}_N$ Cochain Model

Let K be a finite cell decomposition of a closed connected oriented D-manifold M. Write $C^r = C^r(K;\mathbb{Z}_N)$, and let

$$\delta_r : C^r \longrightarrow C^{r+1}$$

be the cellular coboundary. The finite constraint form of $\mathbb{Z}_N$ BF theory has a one-cochain gauge field $a \in C^1$ and a Lagrange multiplier $b \in \mathrm{Hom}(C^2, \mathbb{Z}_N)$. Its normalized finite sum is

$$Z^{(N)}_{\mathrm{BF}}(K) = \frac{1}{|C^0|\,|C^2|} \sum_{a\in C^1} \sum_{b\in \mathrm{Hom}(C^2,\mathbb{Z}_N)} \exp\!\left(\frac{2\pi \mathrm{i}}{N}\, b(\delta_1 a)\right). \tag{104.2}$$

The factor $|C^0|^{-1}$ divides by ordinary 0-form gauge transformations $a \mapsto a + \delta_0\lambda$. The factor $|C^2|^{-1}$ normalizes the b-sum as a finite Fourier transform.

Finite BF partition function. For a connected K,

$$Z^{(N)}_{\mathrm{BF}}(K) = \frac{|Z^1(K;\mathbb{Z}_N)|}{|C^0(K;\mathbb{Z}_N)|} = \frac{|H^1(K;\mathbb{Z}_N)|}{|H^0(K;\mathbb{Z}_N)|} = \frac{|H^1(M;\mathbb{Z}_N)|}{N}.$$

The finite character identity

$$\frac{1}{|C^2|} \sum_{b\in \mathrm{Hom}(C^2,\mathbb{Z}_N)} \exp\!\left(\frac{2\pi \mathrm{i}}{N}\, b(x)\right) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0 \end{cases}$$

holds for every $x \in C^2$. Applying it to $x = \delta_1 a$ reduces (104.2) to

$$Z^{(N)}_{\mathrm{BF}}(K) = |C^0|^{-1}\,|\ker \delta_1|.$$

Since

$$H^1(K;\mathbb{Z}_N) = \ker \delta_1 / \operatorname{im} \delta_0,$$

we have $|\ker \delta_1| = |H^1|\,|\operatorname{im}\delta_0|$. Also

$$H^0(K;\mathbb{Z}_N) = \ker \delta_0$$

and $|C^0| = |\ker \delta_0|\,|\operatorname{im}\delta_0|$. Hence

$$|C^0|^{-1}|\ker \delta_1| = |H^1|/|H^0|.$$

For connected K, $H^0(K;\mathbb{Z}_N) \cong \mathbb{Z}_N$, giving the last formula.

The value $|H^1|/N$ is the groupoid cardinality of flat $\mathbb{Z}_N$-bundles on a connected manifold: each flat bundle has automorphism group $\mathbb{Z}_N$, and groupoid cardinality weights each isomorphism class by the reciprocal of the automorphism-group order.

104.5 Wilson Operators And Linking

Let $\gamma \in Z_1(K;\mathbb{Z})$ be a closed cellular one-cycle. Its Wilson operator of charge $q \in \mathbb{Z}_N$ is

$$W_q(\gamma)(a) = \exp\left(\frac{2\pi \mathrm{i} q}{N}\langle a, \gamma\rangle\right).$$

Gauge invariance follows from the cellular Stokes formula:

$$\langle a + \delta_0\lambda, \gamma\rangle = \langle a, \gamma\rangle + \langle \lambda, \partial\gamma\rangle = \langle a, \gamma\rangle.$$

Let Σ be a closed oriented $(D-2)$-cycle, represented on the dual cellulation by a two-cochain $\sigma_\Sigma \in C^2(K;\mathbb{Z}_N)$ through Poincare duality. The B-surface insertion of charge $p \in \mathbb{Z}_N$ shifts the finite Fourier constraint by

$$b(\delta a) \longmapsto b(\delta a + p\,\sigma_\Sigma).$$

Thus the b-sum imposes

$$\delta a = -p\,\sigma_\Sigma.$$

This equation is the finite cochain form of a prescribed surface singularity.

Linking phase in finite BF theory. Assume $M = S^D$ and let γ and Σ be disjoint closed cycles of dimensions 1 and $D-2$. Choose a two-chain C with $\partial C = \gamma$, and define

$$\mathrm{Lk}(\gamma, \Sigma) = \langle \sigma_\Sigma, C\rangle \in \mathbb{Z}.$$

With the sign convention above,

$$\langle W_q(\gamma)V_p(\Sigma)\rangle = \exp\left(-\frac{2\pi \mathrm{i} pq}{N}\,\mathrm{Lk}(\gamma, \Sigma)\right)\langle V_p(\Sigma)\rangle.$$

After inserting $V_p(\Sigma)$, the b-sum enforces

$$\delta a = -p\,\sigma_\Sigma.$$

The hypothesis $M = S^D$ removes the cohomological ambiguity in this constraint in the linking range $D \geq 3$: every closed one-cycle is a boundary, and changing the bounding two-chain changes $\langle \sigma_\Sigma, C\rangle$ by the intersection of Σ with a closed two-cycle, which vanishes because $H_2(S^D;\mathbb{Z}) = 0$. Thus the integer $\mathrm{Lk}(\gamma, \Sigma)$ is well defined.

For any allowed a, cellular Stokes gives

$$\langle a, \gamma\rangle = \langle a, \partial C\rangle = \langle \delta a, C\rangle = -p\,\langle \sigma_\Sigma, C\rangle.$$

Therefore the Wilson insertion contributes the phase

$$\exp\left(-\frac{2\pi \mathrm{i} pq}{N}\,\mathrm{Lk}(\gamma, \Sigma)\right)$$

for every allowed a. The remaining sum is the sum defining $\langle V_p(\Sigma)\rangle$, with the same normalization of the finite path integral. The result is therefore an exact ratio statement inside the same finite sum.

104.6 Metric Independence And Boundary Data

The classical action (104.1) uses the orientation of M, the exterior derivative, wedge product, bundle data, and invariant pairing. It uses no Riemannian metric. Quantum metric independence requires additional data:

BV field space, gauge-fixing Lagrangian, regularized semidensity, anomaly analysis.

If the metric variation of the gauge-fixed integrand is BV-exact and if the BV Stokes boundary term vanishes, then the corresponding BV-closed correlators are metric independent by Remark 111.3. On a manifold with boundary, Remark 104.3 shows that the shift symmetry produces boundary phase-space data. The boundary fields are the restrictions of A and B, and the boundary symplectic pairing is obtained from the boundary term in the variation:

$$\Omega_{\partial M} = 2\pi \mathrm{i} \int_{\partial M} \langle \delta B \wedge \delta A \rangle.$$

A functorial BF theory must choose a boundary polarization or boundary condition, then prove compatibility with gluing.

Open Problem 104.4 (Continuum nonabelian BF theory with boundaries)*.* Construct nonabelian continuum BF theory on manifolds with boundary as a functorial BV-BFV theory. The construction must specify the boundary phase space, residual fields, gauge fixing, anomaly line, gluing pairing, and the relationship between Wilson, surface, and boundary observables.

Chapter 105

Chern–Simons Theory

Conventions in this chapter.

- **Signature.** oriented smooth three-manifold M.
- **Metric sign.** no metric is part of the classical action.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** a vacuum vector appears only after choosing a Hamiltonian splitting or a boundary state space.
- **Generator convention.** gauge transformations act on connections by pullback and conjugation.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Let G be a compact Lie group with Lie algebra $\mathfrak{g}$. Fix an invariant symmetric bilinear form tr on $\mathfrak{g}$. For a connection A on a principal G-bundle over an oriented three-manifold M, the Chern–Simons functional in a local trivialization is

$$S_{\mathrm{CS}}(A) = \frac{k}{4\pi}\int_M \mathrm{tr}\Big(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\Big).$$

Large gauge transformations quantize the allowed values of the level k.

Definition 105.1 (Chern–Simons defining structure)**.** A Chern–Simons defining structure is a tuple

$$\mathfrak{C} = (G, \lambda, \mathfrak{f}, \mathfrak{L}, \mathfrak{b})$$

with the following constituents.

(i) G is a compact Lie group with Lie algebra $\mathfrak{g}$.

(ii) λ is an integral level: equivalently, a class in $H^4(BG;\mathbb{Z})$ together with a de Rham representative determined by an invariant symmetric bilinear form on $\mathfrak{g}$. In the simply connected compact simple normalization used below, this is the assertion that

$$\frac{1}{24\pi^2}\int_{S^3} \mathrm{tr}(g^{-1}dg)^3 \in \mathbb{Z}$$

for a generator $g : S^3 \to G$, and the level is an integer k multiplying this normalized bilinear form. For general compact G, the phrase "integral level" means the corresponding differential-cohomology lift $\check{\lambda} \in \widehat{H}^4(BG;\mathbb{Z})$ has been chosen, so that the exponentiated action is a globally defined $U(1)$-valued functional on connections on all principal G-bundles. The displayed local expression is the trivial-bundle representative of this global functional.

(iii) $\mathfrak{f}$ is the framing anomaly specification. Depending on the chosen normalization of the quantum theory, this is specified either by a three-dimensional framing or two-framing convention for closed manifolds, or equivalently by the gravitational Chern–Simons counterterm/central-charge anomaly needed to make partition functions and Wilson-line expectations unambiguous.

(iv) $\mathfrak{L}$ is the allowed line-operator sector: oriented framed link components are labelled by admissible representations or, after quantization, by the simple objects of the associated line category. The framing of each line is part of the operator insertion, because coincident point-splitting of the holonomy has a local self-linking ambiguity.

(v) $\mathfrak{b}$ is boundary and polarization structure. If manifolds with boundary are admitted, $\mathfrak{b}$ specifies the boundary condition or boundary theory, the polarization of the boundary phase space, and the allowed boundary line endpoints. On closed three-manifolds this entry is empty.

The structure $\mathfrak{C}$ is the object used below. The integral level controls large-gauge invariance; the framing specification controls the gravitational/framing anomaly; and the line and boundary structures specify which observables and state spaces are being assigned.

Hypothesis 105.2 (Quantum Chern–Simons theory associated to a defining structure)**.** Given a Chern–Simons defining structure $\mathfrak{C} = (G, \lambda, \mathfrak{f}, \mathfrak{L}, \mathfrak{b})$, the notation

$$Z_{\mathfrak{C}}$$

denotes a quantum Chern–Simons theory only when a construction or hypothesis supplies the following structures.

(i) To each closed oriented surface Σ, with the boundary data required by $\mathfrak{b}$, it assigns a finite-dimensional state space $\mathcal{H}_{\mathfrak{C}}(\Sigma)$.

(ii) To each compact oriented three-dimensional cobordism M, equipped with the required framing specification and with framed labelled links from $\mathfrak{L}$, it assigns a vector or linear map between the boundary state spaces.

(iii) The assignments satisfy functorial gluing: cutting along Σ corresponds to contraction with the pairing on $\mathcal{H}_{\mathfrak{C}}(\Sigma)$, including the anomaly phases fixed by $\mathfrak{f}$.

(iv) On a closed manifold, in a semiclassical regime where stationary phase is meaningful, the critical points are flat G-connections and the local phase is the exponentiated Chern–Simons action determined by $\check{\lambda}$.

(v) The line sector realizes the framed Wilson-line observables specified by $\mathfrak{L}$. In a modular case, the simple line labels, fusion, braiding, twists, and mapping-class-group matrices agree with the associated modular tensor category or modular functor.

(vi) If a holomorphic boundary polarization is used, the boundary variation is cancelled by the corresponding chiral Wess–Zumino–Witten boundary theory at the same integral level.

This hypothesis is intentionally conditional. Perturbative configuration space integrals, geometric quantization, WZW conformal blocks, and modular tensor categories provide constructions in important regimes. The comparison maps among these constructions are part of Open Problem 105.9.

105.1 Variation and Classical Equations

Let $F_A = dA + A \wedge A$. For a compactly supported variation δA on a closed manifold,

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M \mathrm{tr}(\delta A \wedge F_A).$$

Thus the Euler–Lagrange equation is

$$F_A = 0.$$

On a manifold with boundary, the integration by parts also gives

$$\frac{k}{4\pi} \int_{\partial M} \mathrm{tr}(A \wedge \delta A),$$

so a boundary condition, boundary action, or boundary phase space must be part of the definition.

105.2 Level Quantization

For simply connected compact simple G, choose the bilinear form so that

$$\frac{1}{24\pi^2} \int_{S^3} \mathrm{tr}\big(g^{-1}dg\big)^3 \in \mathbb{Z}$$

for maps $g : S^3 \to G$ representing the generator of $\pi_3(G)$. Under a gauge transformation $g : M \to G$, use the right action convention

$$A^g = gAg^{-1} - (dg)g^{-1}.$$

Put $\theta_g = dg\, g^{-1}$. The right Maurer–Cartan equation is

$$d\theta_g = \theta_g \wedge \theta_g.$$

Chern–Simons transgression under a finite gauge transformation. With

$$\mathrm{cs}(A) = \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right),$$

one has the identity of three-forms

$$\mathrm{cs}(A^g) - \mathrm{cs}(A) = d\, \mathrm{tr}(\theta_g \wedge A) + \frac{1}{3} \mathrm{tr}(\theta_g \wedge \theta_g \wedge \theta_g). \tag{105.1}$$

Write $B = gAg^{-1}$. Then $A^g = B - \theta_g$,

$$dB = g(dA)g^{-1} + \theta_g \wedge B - B \wedge \theta_g, \qquad d\theta_g = \theta_g \wedge \theta_g.$$

Substitute these two identities into $\mathrm{tr}((B-\theta_g)\wedge d(B-\theta_g)+\frac{2}{3}(B-\theta_g)^3)$. The terms containing only B give $\mathrm{cs}(A)$ by invariance of tr. The terms with two B's and one θ_g cancel after cyclic permutation inside the invariant trace. Conjugating the remaining one-θ_g mixed terms back to the original A-frame by invariance of tr, they combine as

$$\mathrm{tr}(d\theta_g\wedge A)-\mathrm{tr}(\theta_g\wedge dA)=d\ \mathrm{tr}(\theta_g\wedge A),$$

and the pure Maurer–Cartan terms give $\frac{1}{3}\,\mathrm{tr}(\theta_g^3)$. This is (105.1).

Therefore, on closed M,

$$S_{\mathrm{CS}}(A^g)-S_{\mathrm{CS}}(A)=2\pi k\,n(g)$$

where

$$n(g)=\frac{1}{24\pi^2}\int_M \mathrm{tr}(\theta_g\wedge\theta_g\wedge\theta_g)\in\mathbb{Z}$$

is the winding number with the chosen normalization. The two displayed three-forms define the same integral because $\theta_g=dg\,g^{-1}=g(g^{-1}dg)g^{-1}$ and the bilinear form is invariant. The sign in this formula is tied to the displayed right-action convention for A^g.

Remark 105.3 (Gauge invariance of the exponentiated action). With the integral normalization above, $\exp(\mathrm{i}S_{\mathrm{CS}})$ is invariant under all gauge transformations on closed M if $k\in\mathbb{Z}$.

The transformation formula gives $\exp(\mathrm{i}S_{\mathrm{CS}}(A^g))=\exp(\mathrm{i}S_{\mathrm{CS}}(A))\exp(2\pi\mathrm{i}kn(g))$. This equals $\exp(\mathrm{i}S_{\mathrm{CS}}(A))$ for all $n(g)\in\mathbb{Z}$ exactly when k is integral in the normalization used for the bilinear form. The closed-manifold hypothesis removes boundary terms from the gauge-variation formula. With a boundary, the gauge-variation formula contains the boundary term obtained by restricting the transgression form to ∂M; its cancellation requires boundary degrees of freedom, boundary conditions, or a specified anomalous boundary theory together with the integral level.

The same integrality controls the Wess–Zumino term that appears below. If $g:\Sigma\to G$ extends to two oriented three-manifolds B_1 and B_2 with $\partial B_i=\Sigma$, glue B_1 to $-B_2$ along the boundary. The resulting closed three-manifold $X=B_1\cup_\Sigma(-B_2)$, equipped with the glued map to G, gives

$$\frac{k}{12\pi}\left(\int_{B_1}\mathrm{tr}(g^{-1}dg)^3-\int_{B_2}\mathrm{tr}(g^{-1}dg)^3\right)=2\pi k\,\frac{1}{24\pi^2}\int_X \mathrm{tr}(g^{-1}dg)^3.$$

Thus the ambiguity of the Wess–Zumino functional is an integral multiple of $2\pi k$, and $\exp(\mathrm{i}S)$ is independent of the chosen extension for integer level. The boundary statement is the same integral class λ in Definition 105.1 viewed through a boundary extension.

105.3 Phase Space on a Surface

Let $M=\mathbb{R}\times\Sigma$. Write $A=A_0dt+A_\Sigma$. The term linear in time derivatives is

$$\frac{k}{4\pi}\int_{\mathbb{R}}dt\int_\Sigma \mathrm{tr}(A_\Sigma\wedge\partial_tA_\Sigma).$$

The field A_0 is a Lagrange multiplier imposing $F_{A_\Sigma}=0$. Therefore the classical phase space is the moduli space

$$\mathcal{M}_{\mathrm{flat}}(\Sigma,G)=\{A_\Sigma:F_{A_\Sigma}=0\}/\mathcal{G}_\Sigma$$

with symplectic form

$$\omega_{\rm CS} = \frac{k}{4\pi}\int_\Sigma \mathrm{tr}(\delta A_\Sigma \wedge \delta A_\Sigma).$$

In a quantum Chern–Simons theory satisfying Hypothesis 105.2, the quantization of this phase space is the state space assigned to Σ. In compact semisimple modular examples this state space is finite-dimensional after the integral level, framing convention, and boundary data are fixed.

105.4 Wilson Lines and Framing

For an oriented knot or link component $K \subset M$ and a representation R of G, the Wilson line is

$$W_R(K) = \mathrm{Tr}_R\,\mathrm{Pexp}\int_K A.$$

The quantum expectation value of Wilson links in Hypothesis 105.2 requires precisely the line-framing datum included in Definition 105.1: a nonvanishing normal vector field along each component, or equivalently a choice of point-splitting direction. The framing dependence is a local counterterm effect for the self-linking singularity of the line operator.

105.5 $SU(2)_k$ Modular Data and Link Invariants

The first fully explicit compact simple example of the conditional quantum datum in Hypothesis 105.2 is $G = SU(2)$ at positive integer level k. In the Hamiltonian/modular-functor normalization, the simple Wilson-line labels are

$$a = 0, 1, \ldots, k, \qquad \mathrm{spin}(a) = \frac{a}{2}.$$

These are the integrable highest weights of the affine algebra $\widehat{\mathfrak{su}}(2)_k$. Define

$$K = k+2, \qquad S_{ab} = \sqrt{\frac{2}{K}}\sin\left(\frac{(a+1)(b+1)\pi}{K}\right), \qquad 0 \le a, b \le k. \tag{105.2}$$

The matrix S is the modular S-transformation on the torus state space in the basis determined by core Wilson lines in a solid torus.

Remark 105.4 (Orthogonality of the $SU(2)_k$ modular S-matrix)*.* The matrix (105.2) is real, symmetric, and orthogonal:

$$\sum_{b=0}^{k} S_{ab}S_{cb} = \delta_{ac}.$$

Put $r = a+1$, $s = c+1$, and $K = k+2$. The required identity is

$$\frac{2}{K}\sum_{m=1}^{K-1}\sin\left(\frac{rm\pi}{K}\right)\sin\left(\frac{sm\pi}{K}\right) = \delta_{rs}, \qquad 1 \le r, s \le K-1.$$

Using $2\sin x \sin y = \cos(x-y) - \cos(x+y)$, the sum reduces to finite geometric sums of K-th roots of unity. For $r = s$, it gives $\sum_{m=1}^{K-1}\sin^2(rm\pi/K) = K/2$. For $r \neq s$, the two cosine sums cancel. This proves the orthogonality relation.

The quantum dimension of the label a is

$$d_a = \frac{S_{0a}}{S_{00}} = \frac{\sin((a+1)\pi/K)}{\sin(\pi/K)}. \tag{105.3}$$

This number is the normalized expectation value of a 0-framed unknot in S^3 colored by a, in the convention in which the empty normalized expectation value on S^3 is 1:

$$\langle W_a(\text{unknot})\rangle_{S^3} = d_a. \tag{105.4}$$

The corresponding 0-framed Hopf link with components colored by a and b is

$$\langle W_a(K_1)W_b(K_2)\rangle_{S^3,\text{Hopf}} = \frac{S_{ab}}{S_{00}}. \tag{105.5}$$

Equation (105.5) is the basic bridge between the path integral and the modular category: the two linked Wilson loops compute the matrix element of the modular S-transformation, normalized by the vacuum amplitude.

Example 105.5 ($SU(2)_1$)**.** At $k=1$, there are two labels $0,1$ and

$$S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The nontrivial line has quantum dimension $d_1 = 1$, and the Hopf link with both components colored by 1 has normalized value

$$\frac{S_{11}}{S_{00}} = -1.$$

Thus $SU(2)_1$ is an abelian topological order in this line basis: the nontrivial line is invertible and has nontrivial mutual braiding with itself as measured by the Hopf link.

Example 105.6 ($SU(2)_2$)**.** At $k=2$, the labels $0,1,2$ have quantum dimensions

$$d_0 = 1, \qquad d_1 = \sqrt{2}, \qquad d_2 = 1.$$

The label 1, the spin-$\frac{1}{2}$ Wilson line, is non-invertible. This is the Chern–Simons modular tensor category underlying the Ising fusion rules after the conventional identification of labels.

105.6 Verlinde Formula

The same modular data determine the fusion coefficients. For $SU(2)_k$, the Verlinde formula reads

$$N_{ab}{}^{c} = \sum_{x=0}^{k} \frac{S_{ax}S_{bx}S_{cx}}{S_{0x}}, \tag{105.6}$$

where all labels are self-dual. The result is the truncated $SU(2)$ Clebsch–Gordan rule:

$$N_{ab}{}^{c} = 1$$

exactly when

$$|a-b| \le c \le \min(a+b,\, 2k-a-b), \qquad a+b+c \in 2\mathbb{Z},$$

and $N_{ab}{}^{c} = 0$ otherwise. The truncation $c \le 2k-a-b$ is the finite-level effect: tensor products that would exceed the affine alcove are reflected back by the quantum Weyl symmetry.

Remark 105.7 (Verlinde dimensions from the modular-functor input). Let Σ_g be a closed oriented genus-g surface. The $SU(2)_k$ Chern–Simons state-space dimension assigned by the modular functor is

$$\dim \mathcal{H}_{SU(2)_k}(\Sigma_g) = \sum_{x=0}^{k} (S_{0x})^{2-2g}. \tag{105.7}$$

In particular,

$$\dim \mathcal{H}(S^2) = 1, \qquad \dim \mathcal{H}(T^2) = k+1.$$

The dimension formula is the zero-puncture specialization of the Verlinde formula for conformal blocks,

$$\dim \mathcal{H}(\Sigma_g; a_1, \dots, a_n) = \sum_{x=0}^{k} (S_{0x})^{2-2g-n} \prod_{j=1}^{n} S_{a_j x}.$$

For $g = 0, n = 0$, orthogonality of the first row of S gives $\sum_x S_{0x}^2 = 1$. For $g = 1, n = 0$, the exponent vanishes, so the sum has $k+1$ terms. Higher genus follows from pair-of-pants gluing: each handle contraction inserts a sum over an intermediate label weighted by the inverse square of S_{0x}, which is exactly the displayed exponent. This paragraph records the modular-functor consequence used below; it is not a derivation of the modular functor from the Chern–Simons path integral.

105.7 Boundary Theory

If M has boundary, the boundary term in the variation is the first datum. Let $\partial M = \Sigma$, choose a complex coordinate $z, \bar{z}$ on Σ, and write

$$A_\Sigma = A_z \, \mathrm{d}z + A_{\bar{z}} \, \mathrm{d}\bar{z}, \qquad \mathrm{d}^2 z := \mathrm{d}z \wedge \mathrm{d}\bar{z}.$$

On solutions of $F_A = 0$, the variation of the bulk Chern–Simons action is

$$\delta S_{\mathrm{CS}}\big|_{F=0} = \frac{k}{4\pi} \int_\Sigma \mathrm{tr}(A_\Sigma \wedge \delta A_\Sigma) = \frac{k}{4\pi} \int_\Sigma \mathrm{d}^2 z \ \mathrm{tr}\left(A_z \delta A_{\bar{z}} - A_{\bar{z}} \delta A_z\right).$$

The holomorphic polarization fixes $A_{\bar{z}}$ as the source and adds

$$S_{\mathrm{pol}} = \frac{k}{4\pi} \int_\Sigma \mathrm{d}^2 z \ \mathrm{tr}(A_z A_{\bar{z}}).$$

Then

$$\delta(S_{\mathrm{CS}} + S_{\mathrm{pol}})\big|_{F=0} = \frac{k}{2\pi} \int_\Sigma \mathrm{d}^2 z \ \mathrm{tr}(A_z \, \delta A_{\bar{z}}). \tag{105.8}$$

Thus $kA_z/(2\pi)$ is the current conjugate to $A_{\bar{z}}$ in this polarization.

The corresponding boundary field is a map $g : \Sigma \to G$. For an extension of g to a three-manifold B with $\partial B = \Sigma$, set

$$S_{\mathrm{WZW},k}(g) = \frac{k}{4\pi} \int_\Sigma \mathrm{d}^2 z \ \mathrm{tr}\left(g^{-1} \partial_z g \, g^{-1} \partial_{\bar{z}} g\right) + \frac{k}{12\pi} \int_B \mathrm{tr}(g^{-1} \mathrm{d}g)^3. \tag{105.9}$$

The Wess–Zumino term is defined modulo $2\pi\mathbb{Z}$ in $\exp(\mathrm{i}S)$ exactly when the same integral normalization that quantizes the Chern–Simons level is used. The extension-independence calculation in the level-quantization section is the global input behind this sentence; the Polyakov–Wiegmann identity below is the local product formula in the same normalization.

Lemma 105.8 (Polyakov–Wiegmann identity)**.** *With the convention* (105.9),

$$S_{\mathrm{WZW},k}(gh) = S_{\mathrm{WZW},k}(g) + S_{\mathrm{WZW},k}(h) + \frac{k}{2\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}\left(g^{-1}\partial_{\bar z} g\,\partial_z h\,h^{-1}\right).$$

Proof. The Maurer–Cartan form of the product is

$$(gh)^{-1}\mathrm{d}(gh) = h^{-1}(g^{-1}\mathrm{d}g)h + h^{-1}\mathrm{d}h.$$

Cyclicity and invariance of tr give the three-form identity

$$\mathrm{tr}\big((gh)^{-1}\mathrm{d}(gh)\big)^3 = \mathrm{tr}(g^{-1}\mathrm{d}g)^3 + \mathrm{tr}(h^{-1}\mathrm{d}h)^3 - 3\,\mathrm{d}\,\mathrm{tr}\left(g^{-1}\mathrm{d}g \wedge \mathrm{d}h\,h^{-1}\right).$$

Stokes' theorem converts the final term to a boundary integral on Σ. To see the coefficient and sign, put

$$X_z = g^{-1}\partial_z g, \qquad X_{\bar z} = g^{-1}\partial_{\bar z} g, \qquad Y_z = \partial_z h\,h^{-1}, \qquad Y_{\bar z} = \partial_{\bar z} h\,h^{-1}.$$

The kinetic term of gh has the cross contribution

$$\frac{k}{4\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}(X_z Y_{\bar z} + X_{\bar z} Y_z).$$

The boundary contribution from the Wess–Zumino three-form identity is

$$\frac{k}{4\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}(-X_z Y_{\bar z} + X_{\bar z} Y_z).$$

Adding the two cross contributions cancels the $X_z Y_{\bar z}$ term and doubles the $X_{\bar z} Y_z$ term:

$$\frac{k}{2\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}(g^{-1}\partial_{\bar z} g\,\partial_z h\,h^{-1}).$$

The non-cross terms are exactly $S_{\mathrm{WZW},k}(g)$ and $S_{\mathrm{WZW},k}(h)$. This proves the identity. ☐

The gauged chiral boundary action in the same polarization is

$$S_{\partial,k}(g; A_{\bar z}) = S_{\mathrm{WZW},k}(g) + \frac{k}{2\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}\left(\partial_z g\,g^{-1} A_{\bar z}\right). \tag{105.10}$$

If a flat bulk connection is written near the boundary as

$$A_z = -\partial_z g\,g^{-1},$$

then

$$\delta_{A_{\bar z}} S_{\partial,k} = -\frac{k}{2\pi}\int_\Sigma \mathrm{d}^2 z\ \mathrm{tr}(A_z\,\delta A_{\bar z}),$$

which cancels the polarized Chern–Simons variation (105.8). This is the precise sense in which the chiral Wess–Zumino–Witten boundary degree of freedom supplies the boundary term needed by the Chern–Simons variational problem.

The holomorphic current is

$$J(z) = -\frac{k}{2\pi}\,\partial_z g\,g^{-1}.$$

Writing $J = J^a t_a$, with $\mathrm{tr}(t_a t_b) = \kappa_{ab}$ and $[t_a, t_b] = \mathrm{i} f^c{}_{ab} t_c$, quantization gives the affine current algebra at level k,

$$J^a(z) J^b(w) \sim \frac{k\,\kappa^{ab}}{(z-w)^2} + \frac{\mathrm{i} f^{ab}{}_c J^c(w)}{z-w},$$

where the double pole is the local expression of the same integral level that entered the bulk action. The mode form is obtained by the contour-residue calculation, not by an additional convention. Define

$$J^a_m = \oint_0 \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, z^m J^a(z), \qquad J^a(z) = \sum_{m\in\mathbb{Z}} J^a_m z^{-m-1}.$$

With the z-contour around w and the w-contour around the origin, the double and simple poles give

$$\operatorname*{Res}_{z=w} \frac{z^m}{(z-w)^2} = m\, w^{m-1}, \qquad \operatorname*{Res}_{z=w} \frac{z^m}{z-w} = w^m.$$

Therefore

$$[J^a_m, J^b_n] = \mathrm{i} f^{ab}{}_c J^c_{m+n} + k\, m\, \kappa^{ab} \delta_{m+n,0}.$$

The central term is the loop-algebra cocycle

$$\Omega(X_m, Y_n) = k\, m\, \kappa(X, Y) \delta_{m+n,0}.$$

Its Jacobi compatibility follows from invariance of κ:

$$\Omega([X_m, Y_n], Z_\ell) + \Omega([Y_n, Z_\ell], X_m) + \Omega([Z_\ell, X_m], Y_n) = 0.$$

Indeed, when the Kronecker deltas are nonzero, $m+n+\ell = 0$, and the three invariant trace factors are cyclic permutations of $\kappa([X, Y], Z)$; their coefficient is $(m+n) + (n+\ell) + (\ell+m) = 0$. Thus the mode algebra is a genuine central extension of the boundary loop algebra with level k. The precise Hilbert space still depends on the boundary polarization and global boundary condition, but the local current algebra and its level are fixed by the same integer k that quantizes the bulk Chern–Simons action.

Open Problem 105.9 (Chern–Simons path integral as a QFT object)*.* Give a regulator-level construction of Chern–Simons theory that simultaneously controls gauge fixing, framing, Wilson-line renormalization, boundary conditions, and functorial gluing. Perturbative configuration-space integrals, geometric quantization, and modular tensor category constructions each capture part of this structure; a single local-QFT construction should state the comparison maps among them.

Chapter 106

Cohomological Field Theories

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

A cohomological field theory is a quantum field theory equipped with an odd symmetry Q whose cohomology controls a protected class of observables. Its data include the regulated fields, the algebra on which Q acts, the closure relation for Q^2, the integration cycle or state, and the anomaly-free Q-Ward identity. Once those entries are supplied, metric independence and localization are consequences of Stokes' theorem in the regulated field space or of the BV quantum master equation.

This chapter gives the abstract cohomological mechanism in a form usable by the later Donaldson–Witten, topological sigma-model, and twisted supersymmetric examples. The finite-dimensional statements are proved directly. Infinite-dimensional QFT examples are presented as constructions that must be implemented at regulator level; the later chapters identify which parts are theorem-level mathematics and which parts still depend on unproved continuum-QFT input.

106.1 Regulated Cohomological Datum

Let Λ denote a finite regulator: a lattice, a finite-dimensional mode cutoff, a finite-dimensional approximation to a BV complex, or any other scheme for which the integral is an actual finite-dimensional integral or a finite trace. The regulator is part of the notation because the Ward identity is false until it has been proved at some regulator and shown to survive the continuum limit.

Definition 106.1 (Regulated cohomological QFT data)**.** Regulated cohomological QFT data consist of:

(i) a $\mathbb{Z}_2$-graded algebra $\mathcal{A}_\Lambda$ of regulated insertions;

(ii) an odd derivation
$$Q_\Lambda : \mathcal{A}_\Lambda \to \mathcal{A}_\Lambda;$$

(iii) a distinguished even element S_Λ, or more generally a regulated state/weight, defining expectation functionals
$$\langle X \rangle_\Lambda = Z_\Lambda^{-1} \int_{\Gamma_\Lambda} X \, \exp(-S_\Lambda);$$

(iv) a declared integration cycle Γ_Λ, contour, BV Lagrangian, or Hilbert-space trace;

(v) an even derivation B_Λ such that
$$Q_\Lambda^2 = B_\Lambda \tag{106.1}$$
on $\mathcal{A}_\Lambda$;

(vi) a B_Λ-invariant subalgebra
$$\mathcal{A}_\Lambda^B = \ker B_\Lambda$$
on which $Q_\Lambda^2 = 0$;

(vii) the Ward identity
$$\langle Q_\Lambda Y \rangle_\Lambda = 0 \tag{106.2}$$
for every admissible insertion Y used in the argument.

The regulated cohomological observables are the classes
$$H_{Q_\Lambda}(\mathcal{A}_\Lambda^B) = \frac{\ker(Q_\Lambda : \mathcal{A}_\Lambda^B \to \mathcal{A}_\Lambda^B)}{\operatorname{im}(Q_\Lambda : \mathcal{A}_\Lambda^B \to \mathcal{A}_\Lambda^B)}.$$

The even derivation B_Λ may be zero. In important examples it is the infinitesimal action of a compact bosonic symmetry, a gauge transformation, a flavor transformation, or a combination of these. The restriction to $\mathcal{A}_\Lambda^B$ is the Cartan-model step: Q is a cohomological differential on invariant observables even when it is not nilpotent on all fields.

If the data are presented by a finite-dimensional supermanifold, the odd coordinates are coordinates in the structure sheaf of a locally super-ringed space; they are not points of an ordinary set. Berezin integration is the algebraic operation extracting the top odd coefficient, possibly after multiplication by a Berezinian density. It is not a Borel measure. This distinction is harmless in finite-dimensional algebraic manipulations but becomes important when one discusses continuum "fermionic field configurations."

Definition 106.2 (Anomaly and boundary term)**.** If instead of (106.2) one has
$$\langle Q_\Lambda Y \rangle_\Lambda = \langle \mathcal{A}_\Lambda(Y) \rangle_\Lambda + \langle \partial_\Lambda(Y) \rangle_\Lambda,$$
then $\mathcal{A}_\Lambda$ is the local cohomological anomaly and ∂_Λ is the boundary contribution from the integration cycle, compactification, contour endpoint, or singular stratum. A cohomological statement is anomaly-free for the chosen observables only when both terms vanish or cancel by a stated inflow or boundary prescription.

106.2 Ward Identities and Metric Independence

Let g be a background metric, and write δ_g for a tangent vector to the space of metrics. A metric variation formula is a statement in a fixed renormalization and contact-term scheme:

$$\delta_g\langle X\rangle_\Lambda = \langle \delta_g X - (\delta_g S_\Lambda)X\rangle_\Lambda - \langle X\rangle_\Lambda\langle -\delta_g S_\Lambda\rangle_\Lambda, \tag{106.3}$$

with additional contact insertions included in $\delta_g X$ when the representative of X depends on the metric.

Metric-independence criterion. Let $X_g \in \mathcal{A}_\Lambda^B$ be a smooth family of Q_Λ-closed insertions. Suppose the normalized metric variation can be written as

$$\delta_g\langle X_g\rangle_\Lambda = \langle Q_\Lambda Y_{\delta g}\rangle_\Lambda \tag{106.4}$$

for an admissible insertion $Y_{\delta g}$. If the Ward identity (106.2) holds for $Y_{\delta g}$, then

$$\delta_g\langle X_g\rangle_\Lambda = 0.$$

Indeed, equation (106.4) identifies the metric derivative with $\langle Q_\Lambda Y_{\delta g}\rangle_\Lambda$, and the regulated Ward identity evaluates this expectation value to zero. The substantive part of using the criterion is constructing $Y_{\delta g}$ with all contact terms and metric dependence of X_g included in the regulated insertion algebra.

The familiar condition that the stress tensor is Q-exact is a sufficient local mechanism, not the full statement. In the notation of Chapter 102, the complete condition is that the full renormalized metric response, including contact terms and explicit variation of the insertion representative, is a Q-variation. This is the condition expressed by (106.4).

106.3 Exact Deformations and Localization

Let $V \in \mathcal{A}_\Lambda$ be odd and B_Λ-invariant. If

$$S_{\Lambda,t} = S_\Lambda + t\, Q_\Lambda V \tag{106.5}$$

is an allowed family of regulated actions and X is Q_Λ-closed, then t-independence is the same Ward identity.

***Q*-exact deformation independence.** Assume $Q_\Lambda S_\Lambda = 0$, $Q_\Lambda X = 0$, $Q_\Lambda^2 V = 0$, and the Ward identity holds for

$$Y = V X \exp(-tQ_\Lambda V)$$

for every t in the interval under consideration. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle X\rangle_{\Lambda,t} = 0.$$

For the unnormalized integral

$$I_t(X) = \int_{\Gamma_\Lambda} X \exp(-S_\Lambda - tQ_\Lambda V),$$

one has

$$\frac{\mathrm{d}}{\mathrm{d}t}I_t(X) = -\int_{\Gamma_\Lambda} X(Q_\Lambda V)\exp(-S_\Lambda - tQ_\Lambda V).$$

Since $Q_\Lambda X = 0$, $Q_\Lambda S_\Lambda = 0$, and $Q_\Lambda^2 V = 0$, the integrand is

$$-Q_\Lambda(VX\exp(-S_\Lambda - tQ_\Lambda V))\,.$$

The Ward identity makes the derivative of $I_t(X)$ vanish. Applying the calculation above with $X = 1$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}I_t(1) = -\int_{\Gamma_\Lambda} Q_\Lambda(V\exp(-S_\Lambda - tQ_\Lambda V)) = 0,$$

so the normalization is also t-independent. The quotient $\langle X\rangle_{\Lambda,t} = I_t(X)/I_t(1)$ is therefore independent of t.

This calculation is the algebraic part of localization. It does not by itself identify the locus, the determinant, or the boundary contribution. Those require positivity or contour information for the bosonic part of $Q_\Lambda V$, a compactness statement, and a normal-mode analysis.

106.4 Finite-Dimensional Localization Theorem

The following theorem is the precise finite-dimensional model used by cohomological path integrals. It is intentionally stated with hypotheses that are checkable in finite dimension.

Theorem 106.3 (Compact nondegenerate localization)**.** *Let $\mathcal{X}$ be an oriented finite-dimensional supermanifold with compact even integration cycle Γ, Berezinian density μ, and an odd vector field Q. Let $S \in C^\infty(\mathcal{X})_{\bar 0}$ be an even action, $X \in C^\infty(\mathcal{X})$ an insertion, and $V \in C^\infty(\mathcal{X})_{\bar 1}$ an odd deformation functional. Suppose:*

- *(i) Q^2 generates a compact even symmetry whose invariant subalgebra is used for observables;*
- *(ii) $L_Q\mu = 0$ and Q has no boundary flux through $\partial\Gamma$;*
- *(iii) $QS = 0$, $QX = 0$, and S and X are invariant under Q^2;*
- *(iv) V is invariant under Q^2;*
- *(v) the even part $h = (QV)_{\mathrm{bos}}$ satisfies $h \geq 0$, and $F = h^{-1}(0) \subset \Gamma_{\mathrm{red}}$ is a compact smooth submanifold;*
- *(vi) in a tubular neighborhood of F, the normal expansion of QV has a nondegenerate quadratic normal super-Gaussian whose local inverse Euler factors patch to a smooth density-valued section $e_{Q^2}(N_F)^{-1}$ on F.*

More explicitly, the last condition means the following. For each $y \in F$, after choosing even normal coordinates $u \in N_{\bar 0,y}$ and odd normal coordinates $\eta \in N_{\bar 1,y}$, the quadratic normal part is

$$(QV)_2(y;u,\eta) = \frac{1}{2}u^i A_{ij}(y)u^j + \frac{1}{2}\eta^\alpha C_{\alpha\beta}(y)\eta^\beta, \tag{106.6}$$

where A_y is positive definite on the chosen real normal contour and C_y is nondegenerate on the odd normal variables. The leading Berezinian degree is assumed to be balanced: after the normal *rescaling $u = t^{-1/2}\xi$, $\eta = t^{-1/2}\zeta$, the product of the even Jacobian, the odd Berezinian, and the degree of the normal Berezin polynomial has a finite t^0 limit. If this balance fails, the right large-t statement is a different asymptotic expansion rather than the simple fixed-locus formula below. Under this balance condition, the normal Gaussian*

$$\mathcal{G}(y) = \int_{N_y} \exp[-(QV)_2(y; u, \eta)]\; Du\, D\eta = (2\pi)^{\dim N_{\bar{0}}/2} \frac{\mathrm{Pf}(C_y)}{\det(A_y)^{1/2}} \tag{106.7}$$

with the chosen orientation and square-root convention is the local value of $e_{Q^2}(N_F)^{-1}$. Changes of normal coordinates act on the numerator, denominator, and Berezinian in the compensating way, so the patched object is a density on F. Then

$$I_t(X) = \int_\Gamma X \exp(-S - tQV)\, \mu$$

is independent of $t \geq 0$, and its large-t value is

$$\int_F \frac{X|_F \, \exp(-S|_F)}{e_{Q^2}(N_F)} \tag{106.8}$$

with the orientation line and equivariant Euler class determined by the normal Gaussian datum (106.6).

Proof. The t-independence is the Q-exact deformation calculation above in finite dimension, with $\langle QY \rangle = 0$ following from Stokes' theorem for the divergence-free odd vector field Q. In detail,

$$\frac{\mathrm{d}I_t(X)}{\mathrm{d}t} = -\int_\Gamma X(QV)e^{-S-tQV}\,\mu = -\int_\Gamma Q\big(VXe^{-S-tQV}\big)\,\mu = 0,$$

because $QS = 0$, $QX = 0$, $Q^2V = 0$ on the invariant subalgebra, and the boundary flux of Q through $\partial\Gamma$ vanishes.

It remains to identify the common value by the limit $t \to +\infty$. Since Γ_{red} is compact and $h^{-1}(0) = F$, for every sufficiently small tubular neighborhood U of F there is a constant $c_U > 0$ with $h \geq c_U$ on $\Gamma_{\mathrm{red}} \setminus U$. The even body of the integrand outside U is therefore bounded by $\exp(-tc_U)$ times a fixed smooth density, and the odd part is a finite polynomial. Hence the complement of U does not contribute to the limit.

Work in one coordinate patch of the tubular neighborhood and write a point as (y, u, η), with $y \in F$, even normal coordinate u, and odd normal coordinate η. Taylor expansion gives

$$QV = (QV)_2 + R_3,$$

where $(QV)_2$ is the quadratic expression (106.6) and R_3 has total normal degree at least three. Put $u = t^{-1/2}\xi$ and $\eta = t^{-1/2}\zeta$. In Berezin integration the odd change of variables contributes the inverse determinant of the odd scaling, while the even change of variables contributes the ordinary Jacobian; these are exactly the factors included in the super-Gaussian density (106.7). The remainder satisfies

$$t\,R_3(y, t^{-1/2}\xi, t^{-1/2}\zeta) = O(t^{-1/2})$$

on every bounded normal set, and the positive quadratic form A_y gives a uniform Gaussian majorant after shrinking the tubular neighborhood and using a finite cover of F. Dominated convergence therefore replaces the normal integral by the fiberwise Gaussian $\mathcal{G}(y)$.

The local Gaussian factors transform by the Berezinian of changes of normal coordinates, so they glue to the inverse Euler density $e_{Q^2}(N_F)^{-1}$. Summing the finite partition of unity on F gives (106.8). The orientation line and the square root of $\det A_y$ are not decoration: without these choices the local Gaussian factors do not define a global integrand on F. □

106.5 Mathai–Quillen Form of a Cohomological Theory

Many cohomological field theories are infinite-dimensional analogues of the Mathai–Quillen representative of the Euler class of a vector bundle. The finite-dimensional model is as follows.

Let $\pi : E \to X$ be an oriented real vector bundle with metric $(\cdot,\cdot)_E$, compatible connection ∇^E, and smooth section

$$s : X \to E.$$

The zero locus is

$$Z(s) = \{x \in X : s(x) = 0\}.$$

Introduce local even coordinates x^i, odd tangent variables ψ^i representing dx^i, odd fiber variables χ_a, and even auxiliary fiber variables H_a. The cohomological differential is

$$Qx^i = \psi^i, \qquad Q\psi^i = 0, \qquad Q\chi_a = H_a, \qquad QH_a = 0 \tag{106.9}$$

in a local trivialization. The connection-covariant version adds the standard connection terms so that the expression below globalizes.

Choose the standard oscillatory auxiliary-field representative

$$V_{\mathrm{MQ}} = \frac{1}{2}(\chi, H)_E + \mathrm{i}(\chi, s)_E.$$

Then

$$QV_{\mathrm{MQ}} = \frac{1}{2}(H, H)_E + \mathrm{i}(H, s)_E - \mathrm{i}(\chi, \nabla^E_\psi s)_E + \text{curvature term.} \tag{106.10}$$

With the real Gaussian contour for H, completing the square gives

$$\int \mathrm{d}H \, \exp\left[-\frac{1}{2}(H, H)_E - \mathrm{i}(H, s)_E\right] = \text{const.} \exp\left[-\frac{1}{2}\|s\|^2\right].$$

The remaining Berezin polynomial, including the connection and curvature terms suppressed in (106.10), has top fiber-degree component equal to the Mathai–Quillen Thom form. Pulling this Thom form back by the section s gives a de Rham representative of $e(E)$. If the section is transverse, the large-deformation limit localizes the integral to $Z(s)$, with sign determined by the orientation of $\det(ds)$.

Proposition 106.4 (Mathai–Quillen localization for a transverse section)**.** *Let X be compact and s transverse to the zero section. For any closed differential form α on X of degree* $\dim X - \operatorname{rank} E$*,*

$$\int_X \alpha \wedge e(E) = \int_{Z(s)} \alpha|_{Z(s)}$$

where $Z(s)$ is oriented by the isomorphism

$$N_{Z(s)/X} \simeq s^*E$$

induced by ds and by the given orientations of X and E. When $\operatorname{rank} E = \dim X$ *it is the signed sum of zeros of s.*

Proof. Let $\tau_{\mathrm{MQ}} \in \Omega^{\operatorname{rank} E}_{\mathrm{cpt}}(E)$ denote the Mathai–Quillen Thom form in the convention described above. Its pullback $s^*\tau_{\mathrm{MQ}}$ represents $e(E)$. Replacing s by $t^{1/2}s$ does not change this cohomology class: differentiating the Gaussian representative with respect to t gives an exact form, equivalently a Q-exact variation in the finite-dimensional supersymmetric notation. Since X is compact and α is closed,

$$I(t) := \int_X \alpha \wedge (t^{1/2}s)^*\tau_{\mathrm{MQ}}$$

is independent of $t > 0$.

It remains to identify the common value by the limit $t \to +\infty$. Transversality makes $Z(s)$ a compact submanifold of codimension $\operatorname{rank} E$, and

$$ds : N_{Z(s)/X} \longrightarrow s^*E$$

is a vector-bundle isomorphism. Choose a tubular neighborhood of $Z(s)$ with base coordinate $y \in Z(s)$ and normal coordinate $n \in N_{Z(s)/X,y}$. In this neighborhood

$$s(y, n) = ds_y(n) + O(|n|^2).$$

On the complement of any smaller tubular neighborhood, $\|s\|$ is bounded below by a positive constant, so the Gaussian part of the pulled-back Mathai–Quillen form is exponentially small in t. Thus only $|n| = O(t^{-1/2})$ contributes. Put $n = t^{-1/2}\xi$. The leading normal part of the Mathai–Quillen density is

$$(2\pi)^{-r/2} \exp\left[-\frac{1}{2}\|ds_y(\xi)\|^2\right] \det(ds_y)\, d^r\xi, \qquad r = \operatorname{rank} E,$$

where the determinant is computed after oriented bases of $N_{Z(s)/X,y}$ and E_y are chosen. The ordinary Gaussian integral contributes $|\det(ds_y)|^{-1}$, while the Berezin fiber integration contributes $\det(ds_y)$. Their quotient is the sign of ds_y. The orientation on $Z(s)$ is exactly the orientation for which the decomposition

$$T_xX \simeq T_xZ(s) \oplus N_{Z(s)/X,x} \simeq T_xZ(s) \oplus E_x$$

has the prescribed orientations of X and E. With this convention the sign is absorbed into the oriented integration over $Z(s)$. The terms omitted in $s(y, t^{-1/2}\xi)$ and in the pulled-back form give negative powers of $t^{1/2}$ after the rescaling and vanish by the Gaussian bound. Therefore

$$\lim_{t\to+\infty} I(t) = \int_{Z(s)} \alpha|_{Z(s)}.$$

The equality follows because $I(t)$ is independent of t. When $\operatorname{rank} E = \dim X$, $Z(s)$ is a finite oriented zero-manifold, so the last integral is the signed sum over its points. □

The proposition is the finite-dimensional ancestor of statements such as "Donaldson–Witten theory localizes to anti-self-dual connections" and "the A-model localizes to holomorphic maps." In field theory, X is an infinite-dimensional field space, E is an infinite-dimensional bundle of equations, and $s = 0$ is the PDE defining the moduli problem. The theorem then requires regularization, compactness, orientation, and determinant-line control.

106.6 Descent and Integrated Observables

Let M be a smooth D-manifold. A local cohomological observable is not only a Q-closed local density. If it is to be integrated over cycles of different dimensions, it must come with a descent package.

Definition 106.5 (Descent package)**.** A descent package is a sequence of local p-form insertions

$$\mathcal{O}^{(p)} \in \Omega^p(M) \otimes \mathcal{A}, \qquad 0 \le p \le D,$$

such that

$$Q\mathcal{O}^{(0)} = 0, \qquad Q\mathcal{O}^{(p)} + \mathrm{d}\mathcal{O}^{(p-1)} = 0 \quad (1 \le p \le D). \tag{106.11}$$

If γ_p is a closed p-cycle, then

$$I_{\gamma_p}(\mathcal{O}) = \int_{\gamma_p} \mathcal{O}^{(p)} \tag{106.12}$$

is Q-closed: using (106.11) and $\partial\gamma_p = 0$,

$$QI_{\gamma_p}(\mathcal{O}) = \int_{\gamma_p} Q\mathcal{O}^{(p)} = -\int_{\gamma_p} \mathrm{d}\mathcal{O}^{(p-1)} = -\int_{\partial\gamma_p} \mathcal{O}^{(p-1)} = 0.$$

If γ_p is changed by a boundary, $\gamma_p \mapsto \gamma_p + \partial C_{p+1}$, then

$$I_{\gamma_p+\partial C_{p+1}}(\mathcal{O}) - I_{\gamma_p}(\mathcal{O}) = \int_{C_{p+1}} \mathrm{d}\mathcal{O}^{(p)} = -Q\int_{C_{p+1}} \mathcal{O}^{(p+1)}. \tag{106.13}$$

Thus integrated descendants depend only on the homology class of the cycle in Q-cohomology, subject to the specified contact-term qualification when cycles collide with other insertions.

In Donaldson–Witten theory, the descent package is generated from invariant polynomials in the universal equivariant curvature. With

$$QA = \psi, \qquad Q\psi = -d_A\phi, \qquad Q\phi = 0, \qquad Q^2 = \delta_{\text{gauge}}(-\phi), \tag{106.14}$$

gauge-invariant polynomials of ϕ define local Q-classes, and their descendants give observables integrated over cycles in the four-manifold. The detailed four-dimensional field content, instanton moduli space, and orientation data are developed in Chapter 109.

106.7 Cartan Model and Equivariant Closure

When Q^2 is an even symmetry, the finite-dimensional model is the Cartan model of equivariant cohomology. Let a compact Lie group K act on a manifold X with infinitesimal vector fields v_ξ, $\xi \in \mathfrak{k}$. On K-invariant polynomial maps

$$\alpha : \mathfrak{k} \to \Omega^\bullet(X)$$

define

$$d_K\alpha(\xi) = d\alpha(\xi) - \iota_{v_\xi}\alpha(\xi). \tag{106.15}$$

Then

$$d_K^2\alpha(\xi) = -\mathcal{L}_{v_\xi}\alpha(\xi), \tag{106.16}$$

so $d_K^2 = 0$ on invariant forms.

Example 106.6 (Hamiltonian S^1 model)**.** Let $X = \mathbb{R}^2$ with coordinates x, y, symplectic form $\omega = dx \wedge dy$, and rotation vector field

$$v = -y\partial_x + x\partial_y.$$

The moment map in the sign convention of (106.15) is

$$\mu = -\frac{1}{2}(x^2 + y^2), \qquad d\mu = \iota_v\omega.$$

Therefore

$$d_{S^1}(\omega + \mu) = 0.$$

This is the elementary algebra behind equivariant localization: the deformation can be made by an equivariantly exact term without changing the equivariant cohomology class.

106.8 Gauge Theories and BV Closure

For a gauge theory, replacing Q by a formal odd symmetry on fields is not enough. Gauge fixing, ghosts, antifields, and the integration cycle must be part of the construction. The correct compatibility condition is the BV master equation. In finite-dimensional notation, let $(\mathcal{F}, \omega_{\mathrm{BV}})$ be an odd symplectic supermanifold, let $(\,\cdot\,,\,\cdot\,)$ be the BV bracket, and let Δ be the regulated BV Laplacian. A quantum BV action satisfies

$$\frac{1}{2}(S, S) - \hbar\Delta S = 0. \tag{106.17}$$

The corresponding BV differential on insertions is

$$\widehat{Q}X = (S, X) - \hbar\Delta X. \tag{106.18}$$

If $X = \widehat{Q}Y$ and the BV Stokes theorem holds on the chosen Lagrangian integration cycle $\mathcal{L}$, then

$$\int_{\mathcal{L}} X\, \exp(-S/\hbar) = 0.$$

A cohomological gauge theory is therefore a BV theory together with an extra odd symmetry, or with a BV differential whose cohomology contains the desired topological observables. The equality $Q^2 = \delta_{\mathrm{gauge}}(\phi)$ must be interpreted inside the BV complex. A Q-exact deformation is legitimate only if it preserves the quantum master equation, or equivalently if it is generated by a BV-canonical deformation plus allowed gauge-fixing data.

106.9 Donaldson–Witten as the Guiding Gauge-Theory Model

The universal finite-dimensional template for Donaldson–Witten theory is the Mathai–Quillen model applied to the bundle of self-dual two-forms over the space of connections modulo gauge. Let X be a closed oriented Riemannian four-manifold and $P \to X$ a principal $SU(2)$ bundle. The configuration space before quotient is the affine space $\mathcal{A}(P)$. The equation bundle is

$$\Omega^{2,+}(X, \mathrm{ad}\, P) \longrightarrow \mathcal{A}(P),$$

and the section is

$$s(A) = F_A^+.$$

The zero locus $s^{-1}(0)$ is the space of anti-self-dual connections. After quotient by the gauge group, the virtual tangent-obstruction complex is

$$\Omega^0(X, \operatorname{ad} P) \xrightarrow{d_A} \Omega^1(X, \operatorname{ad} P) \xrightarrow{d_A^+} \Omega^{2,+}(X, \operatorname{ad} P).$$

The twisted fields

$$\psi \in \Omega^1(X, \operatorname{ad} P), \qquad \chi^+ \in \Omega^{2,+}(X, \operatorname{ad} P), \qquad H^+ \in \Omega^{2,+}(X, \operatorname{ad} P)$$

are the tangent odd variable, antighost, and auxiliary variable in this Mathai–Quillen construction. The scalar ϕ is the equivariant gauge parameter appearing in $Q^2 = \delta_{\mathrm{gauge}}(-\phi)$. Thus localization to $F_A^+ = 0$ is the infinite-dimensional, gauge-equivariant Mathai–Quillen localization mechanism, with all analytic and compactness issues still to be supplied.

This perspective also explains why contact terms are unavoidable. The Donaldson observables are descendants integrated over cycles. Products of integrated descendants require a prescription for collisions of cycles and for boundary strata of the instanton compactification. A cohomological Ward identity controls these contact terms only after they have been included in the regulated observable algebra.

106.10 What Must Be Proved in Infinite Dimension

The finite-dimensional statements above reduce the infinite-dimensional problem to a clear list of mathematical obligations:

(i) construct the regulated field/BV space and integration cycle;

(ii) prove the Q-Ward identity or quantum master equation at the regulator;

(iii) prove that the continuum limit exists for the selected observables;

(iv) identify the fixed locus and its virtual normal complex;

(v) prove compactness or describe boundary strata and their contributions;

(vi) orient the determinant line and compute the one-loop or Euler factor;

(vii) define contact terms for integrated descendants and show that the descent equations survive renormalization.

Only after these steps are complete does a formal localization computation become a theorem about a QFT.

Open Problem 106.7 (Cohomological QFT beyond formal localization)*.* Develop regulator-level cohomological QFT constructions in which Q-Stokes theorem, BV gauge fixing, compactness of fixed loci, orientation lines, one-loop determinants, contact terms, and boundary strata are all controlled. This is the prerequisite for using cohomological field theory as a theorem-level bridge between quantum field theory and topological invariants.

Chapter 107

Topological Sigma Models

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Let Σ be a compact oriented smooth surface equipped with a complex structure j_Σ, and let X be a compact smooth manifold. A two-dimensional sigma model has bosonic field $\phi : \Sigma \to X$. A topological sigma model is a cohomological sigma model in the sense of Chapter 106: it has an odd differential Q, Q-closed observables, and a stress-tensor or metric-response insertion which is Q-exact after the anomaly and boundary terms have been accounted for. The central examples are the A-model and the B-model obtained from two-dimensional $\mathcal{N} = (2, 2)$ sigma models by two distinct twists.

This chapter develops the finite-dimensional geometry which the path integral is expected to produce after localization. The identification of a Gromov–Witten integral with a continuum QFT is made at the level where the regulator, integration cycle, compactification, anomaly line, and gluing maps have been specified. Those data are part of the QFT problem surrounding the enumerative formula.

Hypothesis 107.1 (A-model cohomological sigma-model structure)**.** A closed A-model cohomological structure is a tuple

$$\mathfrak{A}_X = \big(\Sigma, j_\Sigma; X, \omega, J, g_X; [B + \mathrm{i}\omega]; \mathcal{F}_{\Sigma,X}, Q_A, \bar{\partial}_J, \mathfrak{V}_X, \Lambda_{\mathrm{Nov}}, \mathfrak{c}_A\big)$$

with the following entries.

(i) (Σ, j_Σ) is a compact marked Riemann surface, and (X, ω, J, g_X) is compact almost Kähler.

(ii) $\mathcal{F}_{\Sigma,X}$ is the locally super-ringed mapping field space whose bosonic body is $C^\infty(\Sigma, X)$ and whose odd directions are the twisted fermionic section spaces.

(iii) Q_A is an odd cohomological vector field whose scalar local observable complex is identified with $(\Omega^\bullet(X), d)$, and $\bar\partial_J \phi = 0$ is the localization equation for the bosonic zero modes.

(iv) The action splits into a Q_A-exact nonnegative term plus the complexified Kähler weight $Q^\beta = \exp(2\pi\mathrm{i} \int_\beta (B + \mathrm{i}\omega))$ on each curve class $\beta \in H_2(X; \mathbb{Z})$.

(v) $\mathfrak{V}_X$ is a virtual integration package for the stable-map compactifications $\overline{\mathcal{M}}_{g,n}(X, \beta)$, including evaluation maps, forgetful maps, deformation identifications, and gluing compatibility.

(vi) Λ_{Nov} is the Novikov completion in which the instanton sum is interpreted, and $\mathfrak{c}_A$ is the source, collision, and contact-term prescription used to define products of local insertions.

This is a hypothesis about the cohomological QFT layer. The finite-dimensional Gromov–Witten package $\mathfrak{V}_X$ is an output expected from localization once this structure is constructed. The continuum QFT construction itself is the separate problem recorded in Open Problem 107.11.

Hypothesis 107.2 (Closed B-model cohomological sigma-model structure)**.** A closed B-model cohomological structure is a tuple

$$\mathfrak{B}_X = \big(\Sigma, j_\Sigma; X, J_X, \Omega_X; \mathcal{F}_{\Sigma,X}, Q_B, \mathrm{PV}^{\bullet,\bullet}(X), \mathrm{Tr}_{\Omega_X}, \mathfrak{c}_B\big)$$

where X is a compact complex m-fold with trivial canonical bundle, Ω_X is a nowhere-vanishing holomorphic volume form, $\mathrm{PV}^{p,q}(X) = \Omega^{0,q}(X, \wedge^p T^{1,0}X)$, $Q_B = \bar\partial$ on scalar local observables, and

$$\mathrm{Tr}_{\Omega_X}(\mu) = \int_X \Omega_X \wedge \iota_\mu \Omega_X$$

is the trace pairing when the total polyvector and Dolbeault degrees make the integrand top degree. The structure also includes the anomaly-free axial twist, the regulated Q_B-invariant integration cycle, and the contact-term prescription $\mathfrak{c}_B$. Its first-order deformation complex is the Dolbeault polyvector differential graded Lie algebra

$$\big(\mathrm{PV}^{1,\bullet}(X), \bar\partial, [\cdot,\cdot]\big),$$

with Maurer–Cartan equation $\bar\partial\mu + \frac{1}{2}[\mu, \mu] = 0$.

107.1 The Mapping Field Space

Definition 107.3 (Mapping space and fermionic directions)**.** Let X be a smooth manifold. The smooth bosonic field space is the Fréchet manifold

$$\mathcal{F}^{\mathrm{bos}}_{\Sigma,X} = C^\infty(\Sigma, X).$$

At $\phi \in C^\infty(\Sigma, X)$, its tangent space is

$$T_\phi \mathcal{F}^{\mathrm{bos}}_{\Sigma,X} = \Gamma(\Sigma, \phi^* TX).$$

A fermionic field in a sigma model is a parity-reversed section of a vector bundle built functorially from $T\Sigma$, $T^*\Sigma$, and ϕ^*TX. Thus the full field space is a locally super-ringed Fréchet space over $C^\infty(\Sigma, X)$. Its structure sheaf is locally the completed exterior algebra on the continuous dual of the chosen fermionic section spaces.

The last sentence is part of the definition. Fermionic field variables are odd generators in a supercommutative algebra of functions on field space, rather than elements of the underlying bosonic mapping space. Fermionic field operators in Lorentzian QFT are linear operators on a Hilbert-space domain and are different objects.

For the A-model structure of Hypothesis 107.1, the target data are an almost Kähler manifold

$$(X, \omega, J, g_X),$$

where $J^2 = -1$,

$$\omega(u, v) = g_X(Ju, v), \qquad d\omega = 0,$$

and J is compatible with ω. For the B-model structure of Hypothesis 107.2, the target is a compact complex manifold X with trivial canonical bundle and a choice of nowhere-vanishing holomorphic volume form Ω_X; a Kähler metric is used to write a Lagrangian representative, but the cohomological observables depend on complex-structure data.

107.2 The A-Model Complex

The scalar A-model differential acts on local functions of ϕ as the de Rham differential on X. In local coordinates x^i on X, introduce an odd scalar field ψ^i transforming as dx^i. On zero-form observables,

$$Q_A\phi^i = \psi^i, \qquad Q_A\psi^i = 0.$$

For $\alpha \in \Omega^p(X)$, define

$$\mathcal{O}_\alpha(x) = \frac{1}{p!}\,\alpha_{i_1\cdots i_p}(\phi(x))\psi^{i_1}(x)\cdots\psi^{i_p}(x).$$

The Koszul sign rule gives

$$Q_A\mathcal{O}_\alpha = \mathcal{O}_{d\alpha}.$$

Therefore local scalar A-model observables are represented by

$$H^\bullet_{\mathrm{dR}}(X; \mathbb{C})$$

when contact terms are controlled by a specified source and collision prescription. The coordinate computation is part of the definition of this local complex: since $Q_A\phi^i = \psi^i$ and $Q_A\psi^i = 0$,

$$Q_A\mathcal{O}_\alpha = \frac{1}{p!}\,\partial_j\alpha_{i_1\cdots i_p}(\phi)\,\psi^j\psi^{i_1}\cdots\psi^{i_p}.$$

Antisymmetrizing the coefficient of the product of odd variables gives

$$\frac{1}{(p+1)!}(d\alpha)_{ji_1\cdots i_p}\psi^j\psi^{i_1}\cdots\psi^{i_p},$$

which is $\mathcal{O}_{d\alpha}$.

107.3 The Cauchy–Riemann Section and the Energy Identity

Let $J_\Sigma = j_\Sigma$. For a smooth map $\phi : \Sigma \to X$, define the $(0,1)$-part of its differential by

$$\bar{\partial}_J \phi = \frac{1}{2}\big(d\phi + J \circ d\phi \circ J_\Sigma\big) \in \Omega^{0,1}(\Sigma, \phi^* TX).$$

The A-model localization equation is

$$\bar{\partial}_J \phi = 0.$$

Its solutions are J-holomorphic maps.

Pointwise energy identity. Fix a conformal metric on Σ. Normalize the norm on the $(0,1)$-form bundle by

$$|\bar{\partial}_J \phi|^2 = \frac{1}{2}\big|d\phi(e) + J d\phi(J_\Sigma e)\big|^2$$

for any positively oriented unit vector e, one has the pointwise identity

$$\frac{1}{2}|d\phi|^2 \, d\mathrm{vol}_\Sigma = \phi^*\omega + |\bar{\partial}_J \phi|^2 \, d\mathrm{vol}_\Sigma.$$

Consequently

$$E(\phi) := \frac{1}{2}\int_\Sigma |d\phi|^2 \, d\mathrm{vol}_\Sigma = \int_\Sigma \phi^*\omega + \int_\Sigma |\bar{\partial}_J \phi|^2 \, d\mathrm{vol}_\Sigma.$$

Let $e_1 = e$ and $e_2 = J_\Sigma e$. Put $u = d\phi(e_1)$ and $v = d\phi(e_2)$. Compatibility of (ω, J, g_X) gives

$$\phi^*\omega(e_1, e_2) = \omega(u, v) = g_X(Ju, v).$$

Since J is g_X-orthogonal,

$$\frac{1}{2}|u + Jv|^2 = \frac{1}{2}|u|^2 + \frac{1}{2}|v|^2 + g_X(u, Jv) = \frac{1}{2}|u|^2 + \frac{1}{2}|v|^2 - \omega(u, v).$$

Multiplying by $d\mathrm{vol}_\Sigma$ gives the stated identity.

For a fixed homology class $\beta \in H_2(X; \mathbb{Z})$, the topological quantity $\int_\Sigma \phi^*\omega = \int_\beta \omega$ is constant on the component of maps with $\phi_*[\Sigma] = \beta$. Thus the positive part of the action is the squared norm of the section $\bar{\partial}_J$ plus a topological term. In a cohomological representative, the squared norm and its fermionic linearization arise as Q_A-variation of a gauge fermion, while the complexified Kähler weight is

$$Q^\beta := \exp\left(2\pi \mathrm{i} \int_\beta (B + \mathrm{i}\omega)\right) = \exp\left(2\pi \mathrm{i} \int_\beta B - 2\pi \int_\beta \omega\right),$$

for a closed real B-field. This fixes the convention used below for the Novikov weight Q^β.

107.4 The Linearized Equation and Virtual Dimension

Let $\phi : \Sigma \to X$ solve $\bar{\partial}_J \phi = 0$. If J is integrable and X is Kähler, the linearization is the Dolbeault operator

$$D_\phi = \bar{\partial}_{\phi^* T^{1,0} X} : \Gamma(\Sigma, \phi^* T^{1,0} X) \longrightarrow \Omega^{0,1}(\Sigma, \phi^* T^{1,0} X),$$

up to the standard connection expression which is complex-linear on the holomorphic locus. For an almost complex target, the linearization is a real Cauchy–Riemann type Fredholm operator with the same index.

Index of the fixed-domain deformation complex. Let $m = \dim_{\mathbb{C}} X$, g be the genus of Σ, and $\beta = \phi_*[\Sigma]$. The real Fredholm index calculation for the deformation complex

$$\Gamma(\Sigma, \phi^*TX) \xrightarrow{D_\phi} \Omega^{0,1}(\Sigma, \phi^*TX)$$

is

$$\operatorname{ind}_{\mathbb{R}}(D_\phi) = 2\left(m(1-g) + \int_\beta c_1(TX)\right).$$

For integrable J, the complex index of $\bar{\partial}_{\phi^*T^{1,0}X}$ is

$$\chi(\Sigma, \phi^*T^{1,0}X) = \operatorname{rank}_{\mathbb{C}}(T^{1,0}X)(1-g) + \deg(\phi^*T^{1,0}X)$$

by Riemann–Roch for holomorphic vector bundles on a compact Riemann surface. Here the rank is m, and

$$\deg(\phi^*T^{1,0}X) = \int_\Sigma \phi^* c_1(T^{1,0}X) = \int_\beta c_1(TX).$$

The real index is twice the complex index. For almost complex J, connect the operator through elliptic real Cauchy–Riemann type operators to a complex-linear representative; the Fredholm index is locally constant in such families.

When the complex structure of the domain and n marked points are also allowed to vary, the expected real dimension of the stable-map moduli space is

$$\operatorname{vdim}_{\mathbb{R}} \overline{\mathcal{M}}_{g,n}(X,\beta) = 2\left((1-g)(m-3) + n + \int_\beta c_1(TX)\right).$$

This is obtained by adding the real dimension $6g - 6 + 2n$ of the stable curve moduli space to the fixed-domain index above; the unstable cases are defined by quotienting by the automorphism group of the marked curve.

107.5 Stable Maps and the A-Model Correlator

Definition 107.4 (Stable maps)**.** Let X be compact almost Kähler. A genus-g, n-marked stable map to X in class β is a tuple

$$(C, p_1, \ldots, p_n, \phi)$$

where C is a connected nodal complex curve of arithmetic genus g, the p_a are distinct smooth marked points, $\phi : C \to X$ is continuous and J-holomorphic on every irreducible component after normalization, and $\phi_*[C] = \beta$. Stability means that every irreducible component on which ϕ is constant has finite automorphism group after retaining its marked and nodal special points.

The compact moduli space is denoted

$$\overline{\mathcal{M}}_{g,n}(X,\beta).$$

There are evaluation maps

$$\operatorname{ev}_a : \overline{\mathcal{M}}_{g,n}(X,\beta) \to X, \qquad \operatorname{ev}_a(C, p_1, \ldots, p_n, \phi) = \phi(p_a).$$

The rigorous finite-dimensional object used by the A-model is a virtual fundamental class

$$[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}} \in H_{\mathrm{vdim}_{\mathbb{R}}}\big(\overline{\mathcal{M}}_{g,n}(X,\beta);\mathbb{Q}\big),$$

or an equivalent virtual integration theory such as a Kuranishi atlas, polyfold Fredholm structure, or derived geometric perfect obstruction theory in the algebraic case. The present chapter records exactly which properties of this object enter the QFT formulas.

Virtual integration package used below. The virtual integration entry $\mathfrak{V}_X$ in Hypothesis 107.1 is the following package. The symbol $[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}$ denotes more than a homology class attached to a space. The formulas below require the following finite-dimensional package.

$$\mathfrak{V}_X = \big([\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}, \mathrm{ev}_a, \pi, \rho, \iota_\Gamma\big)_{g,n,\beta,\Gamma}.$$

Here ev_a are the evaluation maps, π denotes the forgetful maps that drop marked points and stabilize the curve, ρ denotes deformation identifications for smooth families of targets or almost complex structures, and ι_Γ denotes gluing maps associated to nodal boundary strata labelled by stable graphs Γ. The package is required to satisfy:

$$\dim_{\mathbb{R}}[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}} = 2\left((1-g)(m-3)+n+\int_\beta c_1(TX)\right),$$

compatibility with pullback of cohomology classes by the evaluation and forgetful maps, invariance under the deformation identifications ρ, and compatibility with gluing. In genus zero, the gluing compatibility used for associativity is the following boundary-stratum identity. For a four-point splitting S and a decomposition $\beta=\beta_1+\beta_2$, let

$$\mathcal{D}_S(\beta_1,\beta_2) = \overline{\mathcal{M}}_{0,3}(X,\beta_1)\times_X \overline{\mathcal{M}}_{0,3}(X,\beta_2),$$

where the fiber product uses the two evaluation maps at the node. The virtual class on this boundary stratum is required to be

$$[\mathcal{D}_S(\beta_1,\beta_2)]^{\mathrm{vir}} = \Delta_X^!\big([\overline{\mathcal{M}}_{0,3}(X,\beta_1)]^{\mathrm{vir}}\times[\overline{\mathcal{M}}_{0,3}(X,\beta_2)]^{\mathrm{vir}}\big)\,, \tag{107.1}$$

where $\Delta_X^!$ is the refined pullback along the diagonal $\Delta_X : X\to X\times X$. If $\iota_S : \mathcal{D}_S(\beta_1,\beta_2)\to \overline{\mathcal{M}}_{0,4}(X,\beta)$ is the gluing map, then

$$(\iota_S)_*\sum_{\beta_1+\beta_2=\beta}[\mathcal{D}_S(\beta_1,\beta_2)]^{\mathrm{vir}}$$

is the virtual class of the boundary fiber over the corresponding boundary point of $\overline{\mathcal{M}}_{0,4}\cong\mathbb{P}^1$. The two boundary points of $\overline{\mathcal{M}}_{0,4}$ used in the WDVV comparison are linearly equivalent divisors on $\mathbb{P}^1$. This is the nontrivial input from virtual geometry; it is not derived from the formal algebra of the QFT path integral in this chapter.

Definition 107.5 (Primary Gromov–Witten functional)**.** Given a virtual integration package $\mathfrak{V}_X$ with the properties specified above, the primary Gromov–Witten functional is the multilinear map defined for cohomology classes $\alpha_a\in H^\bullet(X;\mathbb{C})$ by

$$\langle\alpha_1,\ldots,\alpha_n\rangle^X_{g,n,\beta} = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}}\bigwedge_{a=1}^n \mathrm{ev}_a^*\alpha_a.$$

The A-model primary correlator at fixed genus is the Novikov series

$$\langle \alpha_1, \ldots, \alpha_n \rangle^A_{g,n} = \sum_{\beta \in H_2(X;\mathbb{Z})} Q^\beta \, \langle \alpha_1, \ldots, \alpha_n \rangle^X_{g,n,\beta}\,,$$

whenever the series is defined in the chosen Novikov completion.

Proposition 107.6 (Worldsheet-instanton sector as an amplitude datum)**.** *Fix the A-model datum of Hypothesis 107.1. For a curve class β, the finite-dimensional expression*

$$Q^\beta \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}} \bigwedge_{a=1}^{n} \mathrm{ev}_a^* \alpha_a \tag{107.2}$$

is the localized amplitude assigned to the β worldsheet-instanton sector only after the determinant, orientation, compactification, and contact-term data in $\mathfrak{V}_X$ and $\mathfrak{c}_A$ have been fixed. The bare space of J-holomorphic maps is not by itself the QFT amplitude.

Proof. The topological term in the action is constant on the component $\phi_*[\Sigma] = \beta$, giving the factor Q^β. The remaining Q_A-exact positive term localizes the bosonic zero locus to $\bar{\partial}_J \phi = 0$. Its linearization is the Fredholm operator D_ϕ. Even and odd nonzero modes contribute a determinant-line factor for the virtual index of D_ϕ, not an automatic scalar. Choosing the orientation, obstruction, gluing, and compactification data in $\mathfrak{V}_X$ identifies this determinant-line datum with the virtual integration cycle. Local observables $\mathcal{O}_{\alpha_a}$ restrict to the evaluation pullbacks $\mathrm{ev}_a^* \alpha_a$. Finally, coincident marks, nodal boundary strata, and source collisions are interpreted using the contact prescription $\mathfrak{c}_A$. These steps produce (107.2). ☐

Proposition 107.7 (Zero-mode saturation and the A-model selection rule)**.** *Let*

$$v_{g,n,\beta} = \mathrm{vdim}_{\mathbb{R}} \, \overline{\mathcal{M}}_{g,n}(X,\beta).$$

After the nonzero modes have been paired by the cohomological regulator and the determinant line has been oriented by $\mathfrak{V}_X$, the fixed β primary amplitude is a top-degree extraction on the remaining zero-mode space. Therefore

$$\langle \alpha_1, \ldots, \alpha_n \rangle^X_{g,n,\beta} = 0 \quad \textit{unless} \quad \sum_{a=1}^{n} \deg_{\mathbb{R}} \alpha_a = v_{g,n,\beta}.$$

For gravitational descendants the same zero-mode extraction gives

$$\langle \tau_{k_1}(\alpha_1) \cdots \tau_{k_n}(\alpha_n) \rangle^X_{g,n,\beta} = 0 \quad \textit{unless} \quad \sum_{a=1}^{n} \big(\deg_{\mathbb{R}} \alpha_a + 2k_a \big) = v_{g,n,\beta}. \tag{107.3}$$

Equivalently, in complex degree $|\alpha|_{\mathbb{C}} = \deg_{\mathbb{R}} \alpha / 2$,

$$\sum_{a=1}^{n} \big(|\alpha_a|_{\mathbb{C}} + k_a \big) = (1-g)(m-3) + n + \int_\beta c_1(TX). \tag{107.4}$$

Proof. On a regular component of the holomorphic-map locus, the odd scalar zero modes are identified with differentials on the finite-dimensional zero-mode manifold. The insertion $\mathcal{O}_{\alpha_a}$

restricts to the form $\mathrm{ev}_a^*\alpha_a$. Berezin integration over the odd zero modes is therefore ordinary top-form extraction: a form of total real degree smaller than $v_{g,n,\beta}$ leaves unsaturated odd zero modes, while a form of larger degree has no component of the required top degree. On a nonregular component the same statement is imposed on the virtual tangent-obstruction complex by the orientation and obstruction data in $\mathfrak{V}_X$. The classes $\psi_a^{k_a}$ have real degree $2k_a$, so descendants simply add the terms $2k_a$ in (107.3). Substituting the virtual dimension formula gives (107.4). $\square$

Thus the selection rule is not a slogan about holomorphic maps. It is the zero-mode saturation condition for the regulated instanton sector after the nonzero-mode determinant and virtual orientation have supplied a scalar measure. In the primary case, Proposition 107.7 specializes to the familiar dimension rule: the integral vanishes unless

$$\sum_{a=1}^{n} \deg_{\mathbb{R}} \alpha_a = \mathrm{vdim}_{\mathbb{R}} \overline{\mathcal{M}}_{g,n}(X,\beta).$$

Equivalently, in complex degree $|\alpha|_{\mathbb{C}} = \deg_{\mathbb{R}} \alpha/2$,

$$\sum_{a=1}^{n} |\alpha_a|_{\mathbb{C}} = (1-g)(m-3) + n + \int_\beta c_1(TX).$$

107.6 Quantum Cohomology from Genus-Zero Gluing

Let $H = H^\bullet(X;\mathbb{C})$, and choose a homogeneous basis $\{T_a\}$. The Poincaré pairing is

$$\eta_{ab} = \int_X T_a \wedge T_b, \qquad \eta^{ab}\eta_{bc} = \delta^a_c.$$

Definition 107.8 (Small quantum product)**.** The genus-zero small quantum product is the Novikov-linear product on $H \otimes \Lambda_{\mathrm{Nov}}$ determined by

$$(T_a \star T_b, T_c) = \sum_\beta Q^\beta \langle T_a, T_b, T_c \rangle^X_{0,3,\beta},$$

where $(\cdot,\cdot)$ is the Poincaré pairing. Equivalently,

$$T_a \star T_b = \sum_{\beta,c,d} Q^\beta \langle T_a, T_b, T_d \rangle^X_{0,3,\beta} \eta^{dc} T_c.$$

Associativity from the splitting axiom. With the genus-zero splitting identity (107.1) as part of the virtual integration package, the product $\star$ is associative. This is a conditional consequence of the virtual package, not an independent construction of that package from continuum QFT.

It is enough to prove

$$((T_a \star T_b) \star T_c, T_d) = (T_a \star (T_b \star T_c), T_d)$$

for basis elements. Expanding the left side by the definition of $\star$ gives

$$\sum_{\beta_1+\beta_2=\beta} \sum_{e,f} Q^\beta \langle T_a, T_b, T_e \rangle_{0,3,\beta_1} \eta^{ef} \langle T_f, T_c, T_d \rangle_{0,3,\beta_2}.$$

Expanding the right side gives

$$\sum_{\beta_1+\beta_2=\beta} \sum_{e,f} Q^\beta \langle T_a, T_f, T_d \rangle_{0,3,\beta_1} \eta^{fe} \langle T_b, T_c, T_e \rangle_{0,3,\beta_2}.$$

The two expressions are equal because they are the virtual integrals of $\mathrm{ev}_1^* T_a \,\mathrm{ev}_2^* T_b \,\mathrm{ev}_3^* T_c \,\mathrm{ev}_4^* T_d$ over two linearly equivalent boundary divisors in $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, pulled back to $\overline{\mathcal{M}}_{0,4}(X, \beta)$. The splitting axiom identifies the restriction of the virtual class to the boundary with the fiber product of two three-point virtual classes over X; the diagonal class of X is

$$[\Delta_X] = \sum_{e,f} \eta^{ef} \, T_e \otimes T_f.$$

Substituting this diagonal decomposition gives exactly the two displayed factorizations. Hence the pairings with all T_d agree, and nondegeneracy of η gives associativity.

The calculation shows precisely where the geometry enters: compactness of the stable-map space, construction of virtual classes, and compatibility of those classes with the two boundary factorizations of a four-marked rational curve. The algebraic conclusion is then formal.

107.7 Example: Projective Space

Let $X = \mathbb{P}^m$, let $H \in H^2(\mathbb{P}^m; \mathbb{Z})$ be the hyperplane class, and normalize

$$\int_{\mathbb{P}^m} H^m = 1.$$

The classical cohomology is $\mathbb{C}[H]/(H^{m+1})$. The small quantum cohomology is

$$QH^\bullet(\mathbb{P}^m) \cong \mathbb{C}[H, Q]/(H^{m+1} - Q),$$

where Q is the Novikov variable for the positive line class.

Proposition 107.9 (Degree-one projective instanton and the quantum relation)**.** *In the Gromov–Witten normalization used above,*

$$\left\langle H^a, H^b, H^c \right\rangle_{0,3,1}^{\mathbb{P}^m} = \begin{cases} 1, & a+b+c = 2m+1, \\ 0, & a+b+c \neq 2m+1 \end{cases}$$

for $0 \le a, b, c \le m$*. Consequently*

$$H^{m+1} = Q$$

in $QH^\bullet(\mathbb{P}^m)$*.*

Proof. The expected complex dimension of $\overline{\mathcal{M}}_{0,3}(\mathbb{P}^m, d)$ is

$$m + (m+1)d.$$

Thus

$$\left\langle H^a, H^b, H^c \right\rangle_{0,3,d}$$

can be nonzero only when $a+b+c = m+(m+1)d$. For $d = 1$, the moduli space can be described on its open degree-one stratum as a line $\ell \subset \mathbb{P}^m$ together with three marked points on ℓ:

$$\dim_{\mathbb{C}} \mathrm{Gr}(2, m+1) + 3 = (2m-2) + 3 = 2m+1.$$

Equivalently, use the automorphism group of the domain to put the three marked points at $0, \infty, 1 \in \mathbb{P}^1$. A degree-one map is represented by two independent vectors $A, B \in \mathbb{C}^{m+1}$, up to common scale, through

$$[s : t] \longmapsto [sA + tB].$$

Generic representatives of H^a, H^b, H^c are linear subspaces $L_a, L_b, L_c \subset \mathbb{P}^m$ of codimensions a, b, c. The conditions $A \in \widehat{L}_a$, $B \in \widehat{L}_b$, and $A+B \in \widehat{L}_c$ impose $a+b+c$ independent linear equations on $(A, B) \in \mathbb{C}^{m+1} \oplus \mathbb{C}^{m+1}$. If $a+b+c = 2m+1$, the solution space is one-dimensional and therefore gives one projective degree-one map. If the codimensions do not sum to $2m+1$, the dimension rule gives zero. This proves the displayed degree-one invariant.

For $d = 0$, the invariant is the classical intersection number and gives $H^a \star H^b = H^{a+b}$ when $a+b \leq m$. If $a+b = m+1$, pairing $H^a \star H^b$ with 1 receives exactly the degree-one contribution above, so $H^a \star H^b = Q$. Taking $a = m$ and $b = 1$ gives $H^{m+1} = Q$. Associativity then determines all products. ☐

For $m = 1$, this gives

$$QH^{\bullet}(\mathbb{P}^1) = \mathbb{C}[H, Q]/(H^2 - Q).$$

The quantum correction is detected by

$$(H \star H, H) = Q \, \langle H, H, H \rangle_{0,3,1} = Q,$$

so $H \star H = Q \cdot 1$. This example is the smallest exact check that the quantum product differs from the classical cup product.

107.8 Descendants and the Universal Curve

Let

$$\pi : \mathcal{C}_{g,n}(X, \beta) \to \overline{\mathcal{M}}_{g,n}(X, \beta)$$

be the universal curve and s_a the section corresponding to the a-th marked point. The cotangent line at the marked point is

$$L_a = s_a^* \omega_\pi,$$

where ω_π is the relative dualizing sheaf. Its first Chern class

$$\psi_a = c_1(L_a) \in H^2(\overline{\mathcal{M}}_{g,n}(X, \beta))$$

is the gravitational descendant class. The descendant invariant is

$$\langle \tau_{k_1}(\alpha_1) \cdots \tau_{k_n}(\alpha_n) \rangle_{g,n,\beta}^X = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}} \prod_{a=1}^n \psi_a^{k_a} \, \mathrm{ev}_a^* \alpha_a.$$

In the fixed-source-surface cohomological field theory, descendants arise from the descent equations of Chapter 106. After coupling to topological gravity, the ψ_a-classes are the finite-dimensional remnants of the marked-point cotangent directions. These are related constructions occupying different layers: the latter requires integration over the moduli of domains.

107.9 The B-Model Complex

Let X be a compact complex m-fold with $K_X \simeq \mathcal{O}_X$, and choose a holomorphic volume form Ω_X. Define the polyvector Dolbeault complex

$$\mathrm{PV}^{p,q}(X) = \Omega^{0,q}(X, \wedge^p T^{1,0}X), \qquad Q_B = \bar{\partial}.$$

If

$$\mu = \mu^{i_1\cdots i_p}{}_{\bar{j}_1\cdots\bar{j}_q} d\bar{z}^{\bar{j}_1}\cdots d\bar{z}^{\bar{j}_q} \otimes \partial_{i_1}\wedge\cdots\wedge\partial_{i_p},$$

then the corresponding local observable is represented by

$$\mathcal{O}_\mu = \mu^{i_1\cdots i_p}{}_{\bar{j}_1\cdots\bar{j}_q}(\phi)\eta^{\bar{j}_1}\cdots\eta^{\bar{j}_q}\theta_{i_1}\cdots\theta_{i_p},$$

where $\eta^{\bar{j}}$ is the odd variable corresponding to $d\bar{z}^{\bar{j}}$ and θ_i is the odd variable dual to ∂_i. The scalar Q_B-cohomology of local observables is

$$H^q(X, \wedge^p T^{1,0}X).$$

The holomorphic volume form converts polyvectors into forms by contraction,

$$\iota_\mu \Omega_X \in \Omega^{m-p,q}(X).$$

For $\mu_1, \ldots, \mu_k$ with total polyvector degree m and total Dolbeault degree m, the genus-zero classical B-model pairing is

$$\langle \mu_1, \ldots, \mu_k\rangle_B = \int_X \Omega_X \wedge \iota_{\mu_1\wedge\cdots\wedge\mu_k}\Omega_X.$$

If either total degree condition fails, the integral is zero by form degree.

Remark 107.10 (B-model dependence on complex structure). Infinitesimal deformations of the complex structure are represented by

$$\mu \in \Omega^{0,1}(X, T^{1,0}X).$$

The integrability equation for a finite deformation is the Maurer–Cartan equation

$$\bar{\partial}\mu + \frac{1}{2}[\mu, \mu] = 0,$$

where $[\cdot,\cdot]$ is the Schouten–Nijenhuis bracket extended to Dolbeault forms. Thus the B-model deformation problem is controlled by the differential graded Lie algebra

$$\big(\mathrm{PV}^{1,\bullet}(X), \bar{\partial}, [\cdot,\cdot]\big).$$

Write the deformed $(0,1)$-operator on functions as

$$\bar{\partial}_\mu = \bar{\partial} + \mu \cdot \partial.$$

A complex structure is integrable exactly when $\bar{\partial}_\mu^2 = 0$. Expanding the square on local functions gives

$$\bar{\partial}\mu + \frac{1}{2}[\mu, \mu],$$

with the bracket equal to the Schouten–Nijenhuis bracket in the vector index and the wedge product in the Dolbeault indices. Hence integrability is the displayed Maurer–Cartan equation.

The B-twist uses the axial $U(1)$ R-symmetry of the parent $\mathcal{N} = (2, 2)$ sigma model. The anomaly of this symmetry is measured by $c_1(TX)$, so the closed B-model requires $c_1(TX) = 0$, equivalently triviality of K_X after choosing the holomorphic volume form needed for the trace pairing. Boundary conditions and open B-models require additional Chan–Paton and anomaly data.

107.10 A-Model and B-Model Parameter Dependence

The two models organize target geometry in complementary ways.

- The A-model local scalar complex is $H^\bullet_{\mathrm{dR}}(X)$. Its instanton weights depend on the complexified Kähler class $[B + \mathrm{i}\omega]$ through Q^β. A change of the target complex structure changes the representative of the Cauchy–Riemann section, but under a Q_A-preserving regulator and compactness hypotheses the induced variation of primary correlators is Q_A-exact.
- The B-model local scalar complex is $H^q(X, \wedge^p T^{1,0}X)$. Its deformation theory is controlled by the complex-structure Maurer–Cartan equation above. Kähler variations occur in Q_B-exact metric-response terms, again provided the regulator and integration cycle preserve the Q_B-Ward identity.

These statements are Ward-identity statements. They require the hypotheses listed in Chapter 106: a Q-invariant regulated cycle, control of boundary terms at noncompact ends, and a treatment of contact terms for colliding insertions.

107.11 Continuum QFT Status

The A-model has a mathematically rich finite-dimensional output in Gromov–Witten theory. The B-model has a finite-dimensional classical closed sector expressed through polyvector Dolbeault cohomology and, at higher genus, through the Kodaira–Spencer/BCOV effective theory. These outputs are indispensable, and the continuum sigma-model QFT still demands a field-theoretic construction.

Open Problem 107.11 (Topological sigma model as continuum QFT)*.* Construct A- and B-models as continuum cohomological QFTs with explicit regulated field spaces, Q-invariant integration cycles, anomaly lines, compactified zero-mode spaces, compatible virtual integration, and gluing maps for cutting and sewing source surfaces. The construction should recover the Gromov–Witten and B-model finite-dimensional formulas as theorems rather than as definitions of the path integral.

This open problem is the precise interface where enumerative geometry, supersymmetric sigma-model regularization, and functorial QFT meet.

Chapter 108

Twists of Supersymmetric Theories

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

A topological twist is a change of the rotation background of a supersymmetric QFT by an R-symmetry background. Its first output is representation-theoretic: fields and supercharges are reinterpreted as sections of different geometric bundles. Its second output, when the twisted theory has a scalar odd symmetry Q satisfying the hypotheses of Chapter 106, is a cohomological QFT. The two steps are logically distinct. The twist supplies a candidate cohomological differential; the Ward identity, stress-tensor relation, contact prescription, and regulator decide whether it defines a metric-independent continuum theory.

108.1 Twisting as a Background Construction

Let the flat-space Euclidean theory have rotation group $\mathrm{Spin}(d)$ and compact internal R-symmetry group G_R. The odd supersymmetry generators form a finite-dimensional $\mathbb{Z}_2$-odd representation

$$\mathsf{Q} \quad \text{of} \quad \mathrm{Spin}(d) \times G_R.$$

The fields form representations of the same group together with gauge and flavor representations.

Definition 108.1 (Twisting homomorphism)**.** A twisting homomorphism is a continuous homomorphism

$$\rho : \mathrm{Spin}(d) \longrightarrow G_R.$$

The corresponding twisted rotation group is the diagonal subgroup

$$\mathrm{Spin}(d)_\rho = \{(s, \rho(s)) \mid s \in \mathrm{Spin}(d)\} \subset \mathrm{Spin}(d) \times G_R.$$

Every flat-space representation V of $\mathrm{Spin}(d) \times G_R$ becomes a representation

$$V_\rho := \mathrm{Res}^{\mathrm{Spin}(d)\times G_R}_{\mathrm{Spin}(d)_\rho} V$$

of the twisted rotation group.

On a curved oriented Riemannian d-manifold M, this restriction has the following geometric meaning. Let $P_{\mathrm{Spin}}(M)$ be a spin frame bundle when M is spin. Turning on the R-symmetry background determined by the twist means choosing the principal G_R-bundle

$$P_R(M) = P_{\mathrm{Spin}}(M) \times_\rho G_R.$$

If V is a representation of $\mathrm{Spin}(d) \times G_R$, the twisted associated bundle is

$$E^\rho_V(M) = P_{\mathrm{Spin}}(M) \times_{\mathrm{Spin}(d)} V_\rho.$$

Thus a field which was a spinor in the untwisted flat theory can become a differential form, a tensor, or a section of another natural bundle after restriction to $\mathrm{Spin}(d)_\rho$.

Definition 108.2 (Scalar twisting supercharge)**.** A supercharge $Q \in \mathsf{Q}$ is a scalar for the twist ρ if

$$Q \in \mathsf{Q}^{\mathrm{Spin}(d)_\rho}.$$

It is a cohomological candidate if, in the twisted algebra of fields or observables,

$$Q^2 = \delta_{\mathrm{gauge}}(\lambda_Q) + \delta_{\mathrm{flavor}}(\alpha_Q) + \mathcal{L}_{v_Q} \tag{108.1}$$

with specified field-dependent or background-dependent parameters. On a chosen observable sector, Q is a differential exactly when the right side acts trivially there.

For a topological twist on a closed manifold, the intended sector in the examples below is gauge-invariant, flavor-invariant, and invariant under the bosonic symmetry generated by v_Q. In the Donaldson-Witten twist below, $v_Q = 0$ and Q^2 is a gauge transformation by the adjoint scalar. In many localization problems on curved manifolds, v_Q is an isometry; the theory is then equivariant with respect to that isometry rather than purely topological.

108.2 Anomaly and Global-Form Conditions

The homomorphism ρ is a representation-theoretic input. A QFT twist requires additional conditions.

Definition 108.3 (Regulator-level twist datum)**.** A regulator-level twist datum on M consists of:

rotation geometry	$P_{\mathrm{Spin}}(M)$ or the corresponding spin^c/oriented replacement,
R-background	$P_R(M)$ induced by ρ or by a specified refinement,
field/BV space	$\mathcal{F}^\rho_\Lambda$,
odd symmetry	Q_Λ,
closure relation	Q^2_Λ as in (108.1),
regulated action or state	S^ρ_Λ or ω^ρ_Λ,
integration cycle	Γ^ρ_Λ,
observable algebra	$\mathcal{A}^\rho_\Lambda$ including contact representatives.

The datum is admissible when the R-symmetry used in the twist is nonanomalous for the chosen backgrounds and when Q_Λ preserves S^ρ_Λ, Γ^ρ_Λ, and the source/contact prescription.

The global condition may be weaker than the original spin condition. For example, after twisting, every field in a particular multiplet may transform as a tensor field on an oriented four-manifold. This tensorial appearance is a conclusion about the associated bundles after the twist; possible anomaly-line and global-form constraints still come from fermion determinants, gauge bundles, discrete theta angles, and background R-symmetry bundles. Each theory requires its own determinant-line and anomaly check.

108.3 Metric Independence from a Twisted Supercharge

Let M be compact and let

$$Z^\rho_g(\mathcal{O}_1, \dots, \mathcal{O}_n)$$

be a regulated correlator of Q_Λ-closed insertions in the twisted theory. The general cohomological criterion was proved in Chapter 106. In a supersymmetric twist, the required metric-response statement is supplied by the supercurrent multiplet together with the chosen curved-background coupling.

Remark 108.4 (Regulator-level metric-independence criterion). Assume the regulator-level twist datum above satisfies:

(i) the Q_Λ-Ward identity has no boundary anomaly on Γ^ρ_Λ;

(ii) $Q_\Lambda \mathcal{O}_i = 0$ for all insertions;

(iii) the integrated metric response has the form

$$\delta_g S^\rho_\Lambda = Q_\Lambda \mathcal{V}_{\Lambda,\delta g} + \mathcal{C}_{\Lambda,\delta g}, \tag{108.2}$$

where $\mathcal{C}_{\Lambda,\delta g}$ is the declared contact and anomaly representative;

(iv) the insertion prescription makes

$$\left\langle \mathcal{C}_{\Lambda,\delta g} \prod_i \mathcal{O}_i \right\rangle_\Lambda = 0.$$

Then $Z^\rho_g(\mathcal{O}_1, \dots, \mathcal{O}_n)$ is locally constant in the metric g at the regulator scale. If the continuum limit exists and the hypotheses are stable under that limit, the limiting correlator is locally metric independent. Differentiating the regulated path integral gives

$$\delta_g Z^\rho_g = -\left\langle \delta_g S^\rho_\Lambda \prod_i \mathcal{O}_i \right\rangle_\Lambda + \delta_g^{\text{contact}} \left\langle \prod_i \mathcal{O}_i \right\rangle_\Lambda .$$

Insert (108.2). The term with $Q_\Lambda \mathcal{V}_{\Lambda,\delta g}$ is, up to the vanishing $Q_\Lambda \mathcal{O}_i$ terms, the Q_Λ-variation of $\mathcal{V}_{\Lambda,\delta g} \prod_i \mathcal{O}_i$. The Ward identity sets its expectation value to zero. The remaining contact representative is zero by hypothesis. Hence the first variation vanishes.

The criterion is deliberately stated at regulator level. A formal identity $\delta_g S = Q(\cdots)$ in a continuum Lagrangian is one ingredient; it becomes a theorem about a QFT only after the regulated Ward identity, contact terms, and boundary behavior are controlled.

108.4 Four-Dimensional $\mathcal{N} = 2$ Supercharges

In four Euclidean dimensions,

$$\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-.$$

The pure $\mathcal{N} = 2$ vector multiplet has R-symmetry

$$G_R = SU(2)_R \times U(1)_r$$

at the classical level; the subgroup used in the Donaldson twist is $SU(2)_R$. The supercharges transform as

$$Q_\alpha{}^A \in (2, 1; 2)_{+1}, \qquad \bar{Q}_{\dot{\alpha}A} \in (1, 2; 2)_{-1}, \tag{108.3}$$

where α is an $SU(2)_+$ spinor index, $\dot{\alpha}$ is an $SU(2)_-$ spinor index, A is an $SU(2)_R$ index, and the subscript denotes $U(1)_r$ charge in the normalization used in this chapter. The Donaldson twisting homomorphism uses the $SU(2)_R$ factor.

The Donaldson twisting homomorphism identifies $SU(2)_+$ with $SU(2)_R$:

$$\rho_{\mathrm{DW}} : SU(2)_+ \times SU(2)_- \longrightarrow SU(2)_R, \qquad \rho_{\mathrm{DW}}(s_+, s_-) = s_+.$$

The twisted rotation group is

$$\mathrm{Spin}(4)' = SU(2)'_+ \times SU(2)_-, \qquad SU(2)'_+ = \mathrm{diag}(SU(2)_+ \times SU(2)_R).$$

Under $SU(2)'_+ \times SU(2)_-$,

$$(2, 1; 2) \longmapsto (1, 1) \oplus (3, 1), \qquad (1, 2; 2) \longmapsto (2, 2).$$

Thus the twisted supercharges consist of a scalar Q, a self-dual two-form supercharge $Q^+_{\mu\nu}$, and a one-form supercharge Q_μ. The first representation has an $SU(2)_+$ doublet index and an $SU(2)_R$ doublet index. Restriction to the diagonal $SU(2)'_+$ gives

$$2 \otimes 2 = 1 \oplus 3.$$

The singlet is represented by contraction with the invariant tensor $\epsilon^{\alpha A}$, hence

$$Q = \epsilon^{\alpha A} Q_{\alpha A}$$

is a scalar. The triplet is the symmetric traceless part and is identified with self-dual two-forms because the adjoint representation of $SU(2)_+$ is $\Lambda^{2,+}$. The second representation has no $SU(2)_+$ spinor index; its $SU(2)_R$ doublet becomes an $SU(2)'_+$ doublet, while the $SU(2)_-$ doublet remains. Therefore it becomes $(2, 2)$, the vector representation T^*M.

108.5 Donaldson-Witten Field Decomposition

The untwisted vector multiplet contains a connection A_μ, complex adjoint scalars $\phi, \bar{\phi}$, gauginos $\lambda_\alpha{}^A$, $\bar{\lambda}_{\dot{\alpha}A}$, and auxiliary fields. After the twist, the gauginos decompose by the same representation calculation:

$$\lambda_\alpha{}^A \longmapsto \eta \oplus \chi^+_{\mu\nu}, \qquad \bar{\lambda}_{\dot{\alpha}A} \longmapsto \psi_\mu.$$

On a four-manifold M and a principal gauge bundle $P \to M$, the twisted fields are therefore

$$A \in \mathcal{A}(P), \qquad \phi, \bar{\phi}, \eta \in \Omega^0(M, \mathrm{ad}\, P),$$
$$\psi \in \Omega^1(M, \mathrm{ad}\, P), \qquad \chi^+, H^+ \in \Omega^{2,+}(M, \mathrm{ad}\, P).$$

The fields η, ψ, χ^+ are odd; $A, \phi, \bar{\phi}, H^+$ are even. The auxiliary field H^+ is included so that the scalar symmetry closes off shell.

Use the infinitesimal gauge convention

$$\delta_\epsilon A = d_A \epsilon, \qquad \delta_\epsilon X = -[\epsilon, X] \quad \text{for } X \in \Omega^\bullet(M, \mathrm{ad}\, P). \tag{108.4}$$

The Donaldson-Witten scalar supercharge acts by

$$\begin{aligned} &QA = \psi, && Q\psi = -d_A\phi, && Q\phi = 0, \\ &Q\bar{\phi} = \eta, && Q\eta = [\phi, \bar{\phi}], && \\ &Q\chi^+ = H^+, && QH^+ = [\phi, \chi^+]. && \end{aligned} \tag{108.5}$$

Off-shell closure. With the convention (108.4), the transformations (108.5) satisfy

$$Q^2 = \delta_{-\phi}$$

on all twisted vector-multiplet fields. Hence $Q^2 = 0$ on gauge-invariant observables.

For the connection,

$$Q^2 A = Q\psi = -d_A\phi = \delta_{-\phi} A.$$

For an adjoint field X, $\delta_{-\phi} X = [\phi, X]$. The transformations give

$$Q^2\bar{\phi} = Q\eta = [\phi, \bar{\phi}], \qquad Q^2\chi^+ = QH^+ = [\phi, \chi^+], \qquad Q^2\phi = 0 = [\phi, \phi].$$

It remains to check ψ and H^+. Since $QA = \psi$ and $Q\phi = 0$,

$$Q(d_A\phi) = [\psi, \phi],$$

where the bracket includes the wedge product of forms and the Lie bracket in $\mathrm{ad}\, P$. Hence

$$Q^2\psi = -[\psi, \phi] = [\phi, \psi] = \delta_{-\phi}\psi.$$

Finally,

$$Q^2 H^+ = Q[\phi, \chi^+] = [\phi, H^+] = \delta_{-\phi} H^+.$$

Thus the closure relation holds on each generator and therefore on the graded algebra they generate.

108.6 The Donaldson-Witten Gauge Fermion

Let the monograph gauge trace convention be $\mathrm{tr}(t^a t^b) = \delta^{ab}$. The Yang-Mills action is written with

$$\frac{1}{4g_{\mathrm{YM}}^2} \int_M d\mathrm{vol}_g \; \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}) = \frac{1}{2g_{\mathrm{YM}}^2} \int_M \mathrm{tr}(F \wedge *F).$$

The topological term uses

$$k(P) = -\frac{1}{8\pi^2}\int_M \operatorname{tr}(F_A \wedge F_A).$$

A convenient cohomological representative of the Donaldson-Witten action has the form

$$S_{\mathrm{DW}} = \frac{1}{g_{\mathrm{YM}}^2} Q\Psi - \frac{i\theta}{8\pi^2}\int_M \operatorname{tr}(F_A \wedge F_A), \tag{108.6}$$

where the gauge fermion contains the Mathai-Quillen term

$$\Psi_{\mathrm{MQ}} = \int_M \operatorname{tr}\left[\chi^+ \wedge *\left(F_A^+ + \frac{1}{2}H^+\right)\right] + \text{terms fixing the scalar/gauge directions.} \tag{108.7}$$

Applying Q to the displayed term gives

$$Q\Psi_{\mathrm{MQ}} = \int_M \operatorname{tr}\left[H^+ \wedge *\left(F_A^+ + \frac{1}{2}H^+\right) - \chi^+ \wedge * d_A^+\psi\right],$$

up to the sign convention for moving the odd field χ^+ through Q. The algebraic equation of motion for H^+ is

$$H^+ = -F_A^+.$$

Substitution gives the bosonic square

$$\frac{1}{2}\int_M \operatorname{tr}(F_A^+ \wedge *F_A^+)$$

with the convention of (108.7). Rescaling Ψ_{MQ} rescales this square; the localization locus is the zero-set of the section

$$F_A^+ = 0.$$

The cohomological fields (ψ, χ^+, H^+) are precisely the gauge-equivariant Mathai-Quillen variables for the section

$$F^+ : \mathcal{A}(P) \longrightarrow \Omega^{2,+}(M, \operatorname{ad} P).$$

After quotienting by the gauge group, the virtual tangent-obstruction complex at an anti-self-dual connection is

$$\Omega^0(M, \operatorname{ad} P) \xrightarrow{d_A} \Omega^1(M, \operatorname{ad} P) \xrightarrow{d_A^+} \Omega^{2,+}(M, \operatorname{ad} P),$$

which is the complex used in the Witten-Donaldson chapter to compute the expected dimension.

108.7 Donaldson Observables from Descent

Let p be an invariant polynomial on $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{su}(2)$ the basic example in the monograph trace convention is

$$p(\phi) = \frac{1}{2}\operatorname{tr}(\phi^2).$$

Using the universal curvature combination

$$\mathcal{F}_Q = F_A + \psi + \phi$$

graded by form degree plus Q-degree, the Bianchi identity and (108.5) imply the equivariant closure

$$(Q + d_A)\mathcal{F}_Q = 0 \quad \text{modulo the gauge action.}$$

Consequently

$$(Q + d)\,\mathrm{tr}(\mathcal{F}_Q^2) = 0$$

on gauge-invariant expressions. Writing

$$\frac{1}{2}\,\mathrm{tr}(\mathcal{F}_Q^2) = \mathcal{O}^{(0)} + \mathcal{O}^{(1)} + \mathcal{O}^{(2)} + \mathcal{O}^{(3)} + \mathcal{O}^{(4)}$$

by form degree gives the descent equations

$$Q\mathcal{O}^{(0)} = 0, \qquad Q\mathcal{O}^{(r)} + d\mathcal{O}^{(r-1)} = 0 \quad (1 \le r \le 4).$$

For a closed r-cycle $\gamma \subset M$,

$$\mathcal{O}_\gamma = \int_\gamma \mathcal{O}^{(r)}$$

is Q-closed. If $\gamma - \gamma' = \partial C$, then

$$\mathcal{O}_\gamma - \mathcal{O}_{\gamma'} = -Q \int_C \mathcal{O}^{(r+1)}.$$

Thus Donaldson observables depend on homology classes in Q-cohomology, subject to the contact-term prescription for products of integrated descendants.

108.8 Two-Dimensional $\mathcal{N} = (2,2)$ A and B Twists

Let the Euclidean rotation group in two dimensions be $U(1)_{\mathrm{rot}}$. Use supercharges

$$Q_+, \quad Q_-, \quad \bar{Q}_+, \quad \bar{Q}_-,$$

with spins

$$s(Q_+) = s(\bar{Q}_+) = +\frac{1}{2}, \qquad s(Q_-) = s(\bar{Q}_-) = -\frac{1}{2}.$$

Assume both vector and axial R-symmetries exist as nonanomalous symmetries. Fix charges

	Q_+	Q_-	$\bar{Q}_+$	$\bar{Q}_-$
q_V	$+1$	$+1$	-1	-1
q_A	$+1$	-1	-1	$+1.$

With the convention

$$s_\rho = s + \frac{1}{2} q_R,$$

the A-twist uses $q_R = q_V$ and has scalar supercharges

$$\bar{Q}_+, \qquad Q_-.$$

The B-twist uses $q_R = q_A$ and has scalar supercharges

$$\bar{Q}_+, \qquad \bar{Q}_-.$$

The standard BRST choices are

$$Q_A = \bar{Q}_+ + Q_-, \qquad Q_B = \bar{Q}_+ + \bar{Q}_-.$$

The A-model and B-model complexes developed in Chapter 107 arise from these two scalar supercharges after twisting the $\mathcal{N} = (2, 2)$ sigma-model fields. The vector or axial R-symmetry used in the twist must be nonanomalous for the chosen target and background. In sigma models this condition becomes the familiar restriction: the A-twist is controlled by symplectic data, while the closed B-twist requires the Calabi-Yau condition for the axial R-symmetry and trace pairing.

108.9 What the Twist Does and What Remains to Construct

The twist provides a representation-theoretic and geometric conversion:

$$\text{spinors with } R\text{-indices} \longrightarrow \text{forms or tensors on } M.$$

It also identifies candidate scalar supercharges and candidate descent complexes. A theorem-level cohomological QFT requires the additional regulator-level closure statement (108.1) and Remark 108.4: Q-invariant regularization, anomaly cancellation, off-shell closure or BV closure, contact representatives, compactification of fixed loci, and gluing.

Open Problem 108.5 (Twisted continuum construction). Construct four-dimensional twisted supersymmetric gauge theories as continuum QFTs with Q-preserving regulator or BV regularization, off-shell or BV closure, determinant-line and global-form control, contact term prescriptions for descendant observables, compactified fixed loci, and a derivation of the Q-exact metric-response relation. For Donaldson-Witten theory this construction should recover the anti-self-dual-moduli-space description and the Donaldson observables from the regulated cohomological gauge theory.

Chapter 109

Witten-Donaldson Theory And The Seiberg-Witten Comparison

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops the four-dimensional cohomological gauge theory whose correlation functions are Donaldson invariants and explains the precise mathematical content of the comparison with Seiberg-Witten monopole theory. There are three logically distinct layers. The first layer is finite dimensional differential geometry: anti-self-dual connections, monopole equations, deformation complexes, orientations, compactifications, and intersection pairings. The second layer is cohomological field theory: twisted supersymmetry, descent, Q-exact metric variation, and contact terms. The third layer is the four-dimensional RG statement: the twisted non-Abelian ultraviolet theory has a Coulomb-branch Wilsonian description whose singular fibers are Abelian monopole theories. The chapter keeps these layers separate because they have different hypotheses and different mathematical status.

How the comparison argument is organized. The local mechanisms in this chapter should be read as a dependency ladder, not as interchangeable derivations of the final comparison formula. The Donaldson moduli-space layer defines the ASD invariants, the μ-classes, simple type, compactification faces, and wall links. The cohomological field-theory layer explains why these finite-dimensional objects are the localization targets of a twisted gauge theory, and why metric changes produce BV boundary terms rather than arbitrary chamber jumps. The Coulomb-branch layer supplies a different description of the same putative four-dimensional QFT: a regular u-plane integral, contact terms, electromagnetic-duality determinant lines, and singular-fiber monopole theories. The comparison statement at the end is the assertion that these three layers can be glued by one Q-compatible Wilsonian construction. Consequently a wall-link identity, a monopole dimension count, a u-plane boundary model, or a blow-up coefficient table is only a test of one arrow in the ladder; none of them alone is a proof of the Donaldson–Seiberg-Witten comparison.

109.1 Four-Manifold And Gauge Conventions

Let X be a closed connected oriented smooth four-manifold equipped with a Riemannian metric g. The Hodge star operator satisfies

$$*_g^2 = 1 \quad \text{on } \Omega^2(X),$$

and the bundle of two-forms decomposes as

$$\Lambda^2 T^*X = \Lambda^{2,+}T^*X \oplus \Lambda^{2,-}T^*X.$$

For a two-form ω, write

$$\omega^\pm = \frac{1}{2}(\omega \pm *_g\omega).$$

The integer $b_2^+(X)$ is the dimension of a maximal positive subspace of the intersection form on $H^2(X;\mathbb{R})$, equivalently the dimension of the space of harmonic self-dual two-forms for any metric g. The Euler characteristic and signature are

$$\chi(X) = 2 - 2b_1(X) + b_2^+(X) + b_2^-(X), \qquad \sigma(X) = b_2^+(X) - b_2^-(X).$$

Let $P \to X$ be a principal $SU(2)$ bundle and let $\operatorname{ad} P = P \times_{\mathrm{Ad}} \mathfrak{su}(2)$. The gauge group is

$$\mathcal{G}_P = \Gamma(X, P \times_{\mathrm{Ad}} SU(2)).$$

The monograph uses the gauge trace convention fixed in the gauge-theory volumes:

$$\operatorname{tr}(t^a t^b) = \delta^{ab}$$

for the fundamental $SU(N)$ generators. With this convention the Euclidean Yang–Mills action is written

$$S_{\mathrm{YM}}(A) = \frac{1}{4g_{\mathrm{YM}}^2}\int_X d\,\mathrm{vol}_g \ \operatorname{tr}(F_{A,\mu\nu}F_A^{\mu\nu}) - \frac{i\theta}{8\pi^2}\int_X \operatorname{tr}(F_A \wedge F_A).$$

Equivalently,

$$S_{\mathrm{YM}}(A) = \frac{1}{2g_{\mathrm{YM}}^2}\int_X \operatorname{tr}(F_A \wedge *_g F_A) - \frac{i\theta}{8\pi^2}\int_X \operatorname{tr}(F_A \wedge F_A).$$

The instanton number is

$$k(P) = -\frac{1}{8\pi^2}\int_X \operatorname{tr}(F_A \wedge F_A).$$

It is independent of the connection A. For $SU(2)$, this integer is the second Chern number $c_2(P)$ in the present trace convention.

The curvature decomposes as $F_A = F_A^+ + F_A^-$. Define

$$\|F_A^\pm\|^2 = \int_X \mathrm{tr}(F_A^\pm \wedge *_g F_A^\pm).$$

Then

$$\int_X \mathrm{tr}(F_A \wedge *_g F_A) = \|F_A^+\|^2 + \|F_A^-\|^2, \qquad \int_X \mathrm{tr}(F_A \wedge F_A) = \|F_A^+\|^2 - \|F_A^-\|^2.$$

Hence

$$\frac{1}{2g_{\mathrm{YM}}^2}\int_X \mathrm{tr}(F_A \wedge *_g F_A) = \frac{1}{g_{\mathrm{YM}}^2}\|F_A^+\|^2 + \frac{4\pi^2}{g_{\mathrm{YM}}^2}k(P).$$

The anti-self-dual equation

$$F_A^+ = 0$$

therefore saturates the topological bound in the sector $k(P) \geq 0$. The normalization $4\pi^2 k/g_{\mathrm{YM}}^2$ is the trace-delta version of the common half-trace expression $8\pi^2 k/g_{\mathrm{ht}}^2$, with $g_{\mathrm{ht}}^2 = 2g_{\mathrm{YM}}^2$.

109.2 Donaldson-Witten Cohomological Fields

The Donaldson twist of the pure four-dimensional $\mathcal{N} = 2$ vector multiplet was defined in Chapter 108. After twisting, the fields are organized as differential forms on X:

$$A \in \mathcal{A}(P), \qquad \phi, \bar\phi \in \Omega^0(X, \mathrm{ad}\, P), \qquad \psi \in \Omega^1(X, \mathrm{ad}\, P),$$
$$\chi^+ \in \Omega^{2,+}(X, \mathrm{ad}\, P), \qquad \eta \in \Omega^0(X, \mathrm{ad}\, P),$$

together with an auxiliary self-dual field

$$H^+ \in \Omega^{2,+}(X, \mathrm{ad}\, P).$$

The scalar odd symmetry Q acts by

$$\begin{aligned} &QA = \psi, && Q\psi = -d_A\phi, && Q\phi = 0, && \\ &Q\bar\phi = \eta, && Q\eta = [\phi, \bar\phi], && Q\chi^+ = H^+, && QH^+ = [\phi, \chi^+]. \end{aligned}$$

Thus Q^2 is the infinitesimal gauge transformation with parameter $-\phi$ on A and with the corresponding adjoint action on all adjoint fields. On gauge-invariant observables, $Q^2 = 0$.

The bosonic localization term may be chosen so that its H^+-dependent part has the form

$$\int_X \mathrm{tr}\left(\frac{1}{2}H^+ \wedge *_g H^+ + H^+ \wedge *_g F_A^+\right).$$

The algebraic equation of motion for H^+ gives $H^+ = -F_A^+$, and the corresponding bosonic square is proportional to

$$\|F_A^+\|^2.$$

Hence the weak-coupling localization locus in a fixed instanton sector is the anti-self-dual equation $F_A^+ = 0$. Metric independence of Q-closed correlators follows from the Ward identity of Remark 108.4 once a regulator, contact prescription, and compactification boundary prescription preserving the required Q-relations have been fixed.

109.3 The Anti-Self-Dual Moduli Space

The anti-self-dual moduli space in the sector P is

$$\mathcal{M}(P,g) = \{A \in \mathcal{A}(P) \mid F_A^+ = 0\}/\mathcal{G}_P.$$

A connection A is irreducible when its stabilizer in $\mathcal{G}_P$ is the center of $SU(2)$. At an irreducible anti-self-dual connection, the linearized equation and infinitesimal gauge transformations form the elliptic complex

$$0 \longrightarrow \Omega^0(X, \operatorname{ad} P) \xrightarrow{d_A} \Omega^1(X, \operatorname{ad} P) \xrightarrow{d_A^+} \Omega^{2,+}(X, \operatorname{ad} P) \longrightarrow 0.$$

Its cohomology groups are denoted

$$H_A^0, \qquad H_A^1, \qquad H_A^2.$$

Irreducibility gives $H_A^0 = 0$. The Zariski tangent space is H_A^1, and H_A^2 is the obstruction space. When $H_A^2 = 0$, the moduli space is a smooth manifold near $[A]$ of dimension $\dim H_A^1$.

The gauge-fixed elliptic operator is

$$D_A = d_A^* \oplus d_A^+ : \Omega^1(X, \operatorname{ad} P) \longrightarrow \Omega^0(X, \operatorname{ad} P) \oplus \Omega^{2,+}(X, \operatorname{ad} P).$$

Its index equals the virtual dimension

$$\dim_{\mathrm{vir}} \mathcal{M}(P,g) = \dim H_A^1 - \dim H_A^0 - \dim H_A^2.$$

Index of the anti-self-dual complex. For $P \to X$ a principal $SU(2)$ bundle with instanton number k, the virtual dimension of the anti-self-dual moduli space is

$$\dim_{\mathrm{vir}} \mathcal{M}(P,g) = 8k - 3(1 - b_1(X) + b_2^+(X)).$$

The Atiyah–Singer index theorem applied to

$$D_A = d_A^* \oplus d_A^+$$

gives

$$\operatorname{ind} D_A = -2p_1(\operatorname{ad} P) - \frac{\dim SU(2)}{2}\big(\chi(X) + \sigma(X)\big).$$

For a principal $SU(2)$ bundle,

$$p_1(\operatorname{ad} P) = -4c_2(P) = -4k.$$

Also

$$\frac{\chi(X) + \sigma(X)}{2} = 1 - b_1(X) + b_2^+(X).$$

Since $\dim SU(2) = 3$, substitution gives

$$\operatorname{ind} D_A = 8k - 3(1 - b_1(X) + b_2^+(X)).$$

This index is $\dim H^1_A - \dim H^0_A - \dim H^2_A$, the virtual dimension of the elliptic complex.

The determinant line of the complex is

$$\mathcal{L}_A = \det H^1_A \otimes (\det H^0_A)^{-1} \otimes (\det H^2_A)^{-1}.$$

An orientation of the instanton moduli problem is an orientation of this line over the irreducible moduli space. For simply connected X, such an orientation is fixed by an orientation of $H^2_+(X;\mathbb{R})$. For general X, the homology orientation is an orientation of

$$H^1(X;\mathbb{R}) \oplus H^2_+(X;\mathbb{R}).$$

The Uhlenbeck compactification $\overline{\mathcal{M}}(P,g)$ adds ideal instantons:

$$([A'], x_1, \dots, x_\ell), \qquad k(P') + \ell = k(P).$$

Geometrically, curvature concentration at finitely many points carries the missing second Chern number. Donaldson invariants are defined by extending the relevant cohomology classes to this compactification or to an equivalent smooth compactified model, with reducible strata and wall crossing treated separately when they occur.

109.4 Donaldson Series, Simple Type, And Uhlenbeck Boundary Terms

For the comparison formula it is useful to use the standard $SO(3)$ or $U(2)$ notation. Fix an integral lift $w \in H^2(X;\mathbb{Z})$ of the second Stiefel–Whitney class of the adjoint bundle; the earlier $SU(2)$ case is the special case $w = 0$. Let $x \in H_0(X;\mathbb{Z})$ denote the positive generator represented by a point and let $h \in H_2(X;\mathbb{R})$. The Donaldson generating function is the degree-selected formal series

$$D^w_X(p,h) = \sum_{r,n\geq 0} \frac{p^r}{r!\,n!} D^w_X(x^r h^n), \tag{109.1}$$

where terms whose total Donaldson degree does not equal the relevant virtual dimension are zero. The point class has Donaldson degree four and a surface class has Donaldson degree two.

Definition 109.1 (Donaldson simple type and Donaldson series)**.** Assume $b_1(X) = 0$ and $b_2^+(X) > 1$, so that the Donaldson invariants are independent of the metric chamber after the orientation convention has been fixed. The pair (X,w) has Donaldson simple type if

$$D^w_X\big((x^2 - 4)z\big) = 0 \tag{109.2}$$

for every homogeneous element z of the Donaldson homology algebra. In that case the reduced Donaldson series is

$$\mathbb{D}^w_X(h) = D^w_X\left(\left(1 + \frac{x}{2}\right)\exp(h)\right). \tag{109.3}$$

The normalization $x^2 = 4$ in (109.2) is part of the Donaldson-invariant convention used in this chapter; changing the normalization of the point observable rescales the numerical constant but not the structure of the comparison.

Quoted Theorem 109.2 (Donaldson simple-type structure theorem). *Let X be a closed connected oriented smooth four-manifold with $b_1(X) = 0$ and $b_2^+(X) > 1$. If (X, w) has Donaldson simple type, then there is a finite set*

$$\mathcal{B}_D(X, w) \subset H^2(X; \mathbb{Z})$$

and rational coefficients a_K^w such that

$$\mathbb{D}_X^w(h) = \exp\left(\frac{1}{2}Q_X(h, h)\right) \sum_{K \in \mathcal{B}_D(X,w)} a_K^w \, \exp\bigl(\langle K, h\rangle\bigr). \tag{109.4}$$

The classes K are characteristic:

$$K \equiv w_2(TX) \pmod 2. \tag{109.5}$$

They are called Donaldson basic classes. The theorem is a theorem of the finite-dimensional Donaldson-invariant construction, not a theorem that the four-dimensional twisted QFT has been constructed non-perturbatively.

What the structure theorem supplies. The theorem contains two different kinds of information, and they should not be confused. The first is geometric. Donaldson invariants define a linear functional on the graded algebra generated by the point class x and homology classes of X. The simple-type relation (109.2) says that the point operator has minimal polynomial $T^2 - 4$ on this functional. Passing to the reduced series (109.3) selects the $+2$-eigensummand in the normalization used here. The structure theorem then proves, using the finite-dimensional Donaldson moduli-space construction, gluing, blow-up, and compactification control, that the remaining surface-source dependence has only finitely many characteristic exponential weights after a universal quadratic factor is removed. The finiteness of the set $\mathcal{B}_D(X, w)$, the characteristic parity (109.5), and the fact that no polynomial prefactors multiply the exponentials are part of this deep theorem.

The second kind of information is algebraic and follows once the finite set is known. Put

$$F_X^w(h) = \exp\left(-\frac{1}{2}Q_X(h, h)\right) \mathbb{D}_X^w(h). \tag{109.6}$$

Then (109.4) is equivalent to saying that F_X^w is a finite exponential sum on the vector space $H_2(X; \mathbb{R})$. For every $v \in H_2(X; \mathbb{R})$, it obeys the constant-coefficient ordinary differential equation

$$\prod_{K \in \mathcal{B}_D(X,w)} \left(\frac{d}{dt} - \langle K, v\rangle\right) F_X^w(h + tv) = 0. \tag{109.7}$$

Conversely, a finite-dimensional space of translation derivatives with simultaneously diagonal characters K gives a sum of exponentials $\exp\langle K, h\rangle$. This converse is linear algebra over the commuting derivative operators; the hard Donaldson input is the geometric statement that the derivative representation is finite and semisimple with characteristic weights.

The exponential form in (109.4) is not a slogan. It says that all dependence on the ordinary intersection form is the universal Gaussian factor $\exp(Q_X/2)$, while all differentiable information in the simple-type sector is carried by finitely many characteristic linear functionals K. A coefficient a_K^w is defined only after the orientation, lift w, point-observable normalization, and compactification extension of the μ-classes have been fixed.

Definition 109.3 (Donaldson simple-type normalization convention)**.** The symbol a_K^w in (109.4) is used only after the following data have been declared.

homology orientation	an orientation of $H^1(X;\mathbb{R}) \oplus H^{2,+}(X;\mathbb{R})$,
lift	$w \in H^2(X;\mathbb{Z}), \quad w \equiv w_2(\mathrm{ad}\, P) \pmod 2$,
point class	$x \in H_0(X;\mathbb{Z})$ normalized by $D_X^w((x^2-4)z) = 0$,
surface source	$h \in H_2(X;\mathbb{R})$ paired with $K \in H^2(X;\mathbb{Z})$,
compactification	an extension of the relevant μ-classes to the boundary.

Changing any one of these data changes the numerical coefficients by a controlled convention: reversing the homology orientation reverses all oriented counts, changing w changes the chamber and phase convention of the $SO(3)$ or $U(2)$ invariant, and rescaling the point observable changes the constant 4 in (109.2).

Remark 109.4 (Coefficient extraction from the reduced series)*.* Assume the hypotheses and conventions of Theorem 109.2. Define

$$F_X^w(h) = \exp\left(-\frac{1}{2}Q_X(h,h)\right)\mathbb{D}_X^w(h).$$

Then

$$F_X^w(h) = \sum_{K \in \mathcal{B}_D(X,w)} a_K^w e^{\langle K,h\rangle}. \tag{109.8}$$

For any $v \in H_2(X;\mathbb{R})$ and $m \geq 0$,

$$\left.\frac{d^m}{dt^m}\right|_{t=0} F_X^w(tv) = \sum_{K \in \mathcal{B}_D(X,w)} a_K^w \,\langle K, v\rangle^m. \tag{109.9}$$

Consequently, after the finite set $\mathcal{B}_D(X,w)$ is known, the coefficients a_K^w are determined by finitely many Donaldson invariants: choose vectors $v_1,\ldots,v_N$ such that the numbers $\langle K, v_i\rangle$ separate the basic classes, and invert the resulting finite Vandermonde system.

Multiplying (109.4) by $\exp(-Q_X/2)$ gives (109.8). Differentiating the finite sum in the direction v gives (109.9). If the linear functionals K are separated by the chosen directions, the matrix of moments is a product of ordinary Vandermonde matrices. Its determinant is nonzero precisely because the exponential weights have distinct evaluations. Thus the a_K^w are not extra formal symbols: they are the finite Fourier coefficients of the reduced Donaldson series in the basis of characteristic exponentials.

Uhlenbeck boundary as BV boundary data. Let $\mathcal{M}_k^{\mathrm{sm}}$ be a smooth irreducible ASD stratum in instanton number k, and let

$$\overline{\mathcal{M}}_k^U = \mathcal{M}_k^{\mathrm{sm}} \sqcup \bigsqcup_{\ell \geq 1} \mathcal{M}_{k-\ell}^{\mathrm{sm}} \times \mathrm{Sym}^\ell(X) \sqcup \{\text{reducible and obstructed strata}\} \tag{109.10}$$

denote the relevant part of the Uhlenbeck compactification. If the Donaldson-Witten path integral is represented as a BV pushforward over $\mathcal{M}_k^{\mathrm{sm}}$, then the compactification strata contribute boundary functionals:

$$\Delta_{\mathrm{res}}(\pi_*^{\mathcal{M}_k^{\mathrm{sm}}}\rho) = -\sum_{\ell\geq 1} \mathcal{B}_{\mathcal{M}_{k-\ell}^{\mathrm{sm}} \times \mathrm{Sym}^\ell(X)}(\rho) - \mathcal{B}_{\mathrm{red/obs}}(\rho). \tag{109.11}$$

Consequently a Donaldson invariant obtained from the compactified moduli space is a QME-preserving cohomological correlator only after the boundary terms in (109.11) have been cancelled by codimension, by an extension of the insertion classes, by a chamber wall-crossing term, or by an explicitly supplied compactification datum.

This is the singular-stratum obstruction formula (111.12) applied to the finite-dimensional localization locus of Donaldson-Witten theory. The strata $\mathcal{M}^{\mathrm{sm}}_{k-\ell} \times \mathrm{Sym}^{\ell}(X)$ are the small-instanton boundary strata: ℓ units of second Chern number have left the smooth connection and are recorded as unordered points of X. Reducible and obstructed strata are additional singular strata of the same BV-boundary type. Removing collar neighborhoods of all these strata gives a manifold with boundary; BV Stokes produces exactly the right hand side of (109.11). If the sum vanishes or is cancelled, the residual semidensity satisfies the QME; otherwise the formal localization expression is not yet a cohomological invariant.

Remark 109.5 (Donaldson–Uhlenbeck versus Gieseker policy). The small-instanton charge bookkeeping in this paragraph is the compact four-manifold analogue of the local map from the Gieseker compactification to the Uhlenbeck one in Construction 101.7. In a holomorphic $\mathbb{C}^2$ chart, a torsion-free sheaf E has a double dual E^{**} and a zero-dimensional quotient Q of length ℓ, and the instanton number splits as

$$k(E) = k(E^{**}) + \ell.$$

The Donaldson–Uhlenbeck compactification keeps the lower-charge smooth connection together with the unordered support of the lost charge. The Gieseker/Nekrasov compactification instead keeps the punctual quotient data that resolve the same boundary stratum and then integrates its equivariant Euler class.

Thus the two compactifications are not interchangeable names for the same localized integral. A Donaldson invariant uses Uhlenbeck compactification, extensions of the μ-classes, and possible wall or reducible boundary terms. A Nekrasov factor uses the resolved Gieseker fixed-point cycle and therefore needs the regulator-selection datum of Definition 101.18 before it can be claimed as the output of a continuum S^4 gauge-theory path integral. In BV language the distinction is exactly the choice between cancelling the boundary functional $\mathcal{B}_{\mathcal{M}^{\mathrm{sm}}_{k-\ell} \times \mathrm{Sym}^{\ell}(X)}$ on the Uhlenbeck side and replacing it by a specified resolved pushforward on the Gieseker side.

109.5 Donaldson Observables And The μ-Map

The basic degree-four gauge-invariant polynomial in the twisted scalar is

$$\mathcal{O}^{(0)} = \frac{1}{8\pi^2} \operatorname{tr} \phi^2.$$

With the Q-rules above, the first descendants may be chosen as

$$\mathcal{O}^{(1)} = \frac{1}{4\pi^2} \operatorname{tr}(\phi\psi),$$

$$\mathcal{O}^{(2)} = \frac{1}{4\pi^2} \operatorname{tr}\left(\phi F_A + \frac{1}{2}\psi \wedge \psi\right),$$

and the sequence continues to four-forms. They obey the descent equations

$$d\mathcal{O}^{(j)} = \{Q, \mathcal{O}^{(j+1)}\}_{\mathrm{gr}}, \qquad j = 0, 1, 2, 3.$$

For a cycle $\gamma_j \in H_j(X;\mathbb{Z})$, define

$$I(\gamma_j) = \int_{\gamma_j} \mathcal{O}^{(j)}.$$

The Q-cohomological degree of $I(\gamma_j)$ is $4-j$. In the finite dimensional instanton description this insertion represents the Donaldson μ-class

$$\mu(\gamma_j) \in H^{4-j}(\mathcal{M}(P,g);\mathbb{Q}).$$

If $\mathbb{P} \to X \times \mathcal{M}(P,g)$ denotes a universal bundle over a smooth irreducible locus, then the class can be written as

$$\mu(\gamma_j) = -\frac{1}{4} p_1(\operatorname{ad}\mathbb{P})/\gamma_j,$$

where the slash denotes slant product with γ_j. This formula is a differential-geometric version of the descent construction. In particular, for a point $x \in X$, $\mu(x)$ has degree four, and for a surface $\Sigma \subset X$, $\mu(\Sigma)$ has degree two.

Given homology classes $\gamma_{j_1}, \ldots, \gamma_{j_r}$, the Donaldson correlator in the smooth compact case is the pairing

$$\langle I(\gamma_{j_1}) \cdots I(\gamma_{j_r}) \rangle_{P,g} = \int_{\overline{\mathcal{M}}(P,g)} \mu(\gamma_{j_1}) \smile \cdots \smile \mu(\gamma_{j_r})$$

when

$$\sum_{\alpha=1}^{r} (4 - j_\alpha) = \dim_{\mathrm{vir}} \mathcal{M}(P,g).$$

If the total degree differs, the correlator vanishes after the usual degree-selection rule, apart from contact-term and reducible-wall phenomena specified by the compactified theory.

109.6 Spin^c Structures And Monopole Equations

A Spin^c structure $\mathfrak{s}$ on X consists of rank-two Hermitian bundles $S^\pm_{\mathfrak{s}}$, a determinant line

$$L_{\mathfrak{s}} = \det S^+_{\mathfrak{s}} = \det S^-_{\mathfrak{s}},$$

and Clifford multiplication

$$\rho : T^*X \longrightarrow \operatorname{Hom}(S^+_{\mathfrak{s}}, S^-_{\mathfrak{s}})$$

compatible with the metric. The characteristic class

$$c_1(\mathfrak{s}) = c_1(L_{\mathfrak{s}}) \in H^2(X;\mathbb{Z})$$

is characteristic:

$$c_1(\mathfrak{s}) \equiv w_2(TX) \pmod 2.$$

Proposition 109.6 (Spin^c structures as characteristic lifts)**.** *On an oriented smooth four-manifold X, the set* Spin$^c(X)$ *of isomorphism classes of* Spin^c *structures is a torsor for $H^2(X;\mathbb{Z})$. Tensoring $\mathfrak{s}$ by a Hermitian line bundle $M \to X$ gives a new* Spin^c *structure $\mathfrak{s} \otimes M$ with*

$$c_1(\mathfrak{s} \otimes M) = c_1(\mathfrak{s}) + 2c_1(M). \tag{109.12}$$

The image of the map

$$c_1 : \mathrm{Spin}^c(X) \longrightarrow H^2(X;\mathbb{Z})$$

is the affine set of integral characteristic classes

$$\{c \in H^2(X;\mathbb{Z}) \mid c \equiv w_2(TX) \pmod 2\}. \tag{109.13}$$

If $H^2(X;\mathbb{Z})$ has two-torsion, distinct Spin^c *structures can have the same first Chern class; the structure, not only c_1, is then part of the data.*

Proof. Existence of Spin^c structures on oriented smooth four-manifolds is the topological fact $W_3(TX) = \beta(w_2(TX)) = 0$; this is equivalent to the existence of an integral characteristic lift of $w_2(TX)$. The group $\mathrm{Spin}^c(4)$ is

$$\mathrm{Spin}^c(4) = (\mathrm{Spin}(4) \times U(1))/\{\pm(1,1)\}.$$

A Spin^c structure lifting the oriented frame bundle can therefore be tensored by any principal $U(1)$-bundle, equivalently by a Hermitian line bundle M. This action is free and transitive because the difference between two lifts of the same oriented frame bundle is exactly a $U(1)$-bundle. The determinant representation squares the additional $U(1)$ factor, so tensoring by M changes the determinant line by

$$L_{\mathfrak{s}\otimes M} = L_{\mathfrak{s}} \otimes M^{\otimes 2},$$

which proves (109.12). The reduction of $c_1(L_{\mathfrak{s}})$ modulo two is the obstruction $w_2(TX)$ to a spin lift, so every $c_1(\mathfrak{s})$ is characteristic. Conversely, if c is another integral lift of $w_2(TX)$, then $c - c_1(\mathfrak{s})$ lies in the kernel of reduction modulo two. The exact sequence associated to

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}_2 \longrightarrow 0$$

identifies this kernel with $2H^2(X;\mathbb{Z})$, so $c = c_1(\mathfrak{s}) + 2c_1(M)$ for some line bundle M. If $2c_1(M) = 0$, tensoring by M leaves the first Chern class unchanged, which is the stated two-torsion caveat. $\square$

For a unitary connection B on $L_{\mathfrak{s}}$, the Dirac operator is

$$\not{D}_B : \Gamma(S^+_{\mathfrak{s}}) \longrightarrow \Gamma(S^-_{\mathfrak{s}}).$$

Let $q \in \Gamma(S^+_{\mathfrak{s}})$. The quadratic map

$$\sigma : S^+_{\mathfrak{s}} \longrightarrow \Lambda^{2,+}T^*X \otimes i\mathbb{R}$$

is defined by

$$\rho(\sigma(q)) = (qq^\dagger)_0,$$

where $(qq^\dagger)_0$ is the traceless Hermitian endomorphism of $S^+_{\mathfrak{s}}$ obtained from the rank-one endomorphism $v \mapsto q\,\langle q, v\rangle$. This equation fixes the normalization of σ used below.

The unperturbed Seiberg-Witten equations are

$$\not{D}_B q = 0, \qquad F^+_B = \sigma(q).$$

A perturbation by a self-dual imaginary two-form η^+ gives

$$\not{D}_B q = 0, \qquad F^+_B = \sigma(q) + \eta^+.$$

The gauge group is

$$\mathcal{G}_L = C^\infty(X, U(1)),$$

acting by

$$u \cdot (B, q) = (B - u^{-1}du,\, uq).$$

The moduli space is

$$\mathcal{M}_{\mathrm{SW}}(\mathfrak{s}, g, \eta^+) = \{(B, q) \mid \not{D}_B q = 0,\ F_B^+ = \sigma(q) + \eta^+\}/\mathcal{G}_L.$$

The linearized, gauge-fixed elliptic operator at a solution is

$$\mathcal{D}_{(B,q)} : \Omega^1(X; i\mathbb{R}) \oplus \Gamma(S_{\mathfrak{s}}^+) \longrightarrow \Omega^0(X; i\mathbb{R}) \oplus \Omega^{2,+}(X; i\mathbb{R}) \oplus \Gamma(S_{\mathfrak{s}}^-).$$

Its index is the expected dimension of the monopole moduli space.

Index of the monopole deformation problem. For a Spin^c structure $\mathfrak{s}$, the expected dimension of the Seiberg-Witten moduli space is

$$d_{\mathrm{SW}}(\mathfrak{s}) = \frac{c_1(\mathfrak{s})^2 - \big(2\chi(X) + 3\sigma(X)\big)}{4}.$$

The real index of the Dirac part is twice the complex index:

$$2 \operatorname{ind}_{\mathbb{C}} \not{D}_B = 2\int_X \widehat{A}(TX) e^{c_1(\mathfrak{s})/2} = \frac{c_1(\mathfrak{s})^2 - \sigma(X)}{4}.$$

The Abelian gauge-fixing and self-dual curvature operator

$$d^* \oplus d^+ : \Omega^1(X; i\mathbb{R}) \longrightarrow \Omega^0(X; i\mathbb{R}) \oplus \Omega^{2,+}(X; i\mathbb{R})$$

has index

$$-(1 - b_1(X) + b_2^+(X)).$$

Adding the two real indices gives

$$\frac{c_1(\mathfrak{s})^2 - \sigma(X)}{4} - (1 - b_1(X) + b_2^+(X)).$$

The identity

$$2\chi(X) + 3\sigma(X) = 4(1 - b_1(X) + b_2^+(X)) + \sigma(X)$$

converts this expression to

$$\frac{c_1(\mathfrak{s})^2 - \big(2\chi(X) + 3\sigma(X)\big)}{4}.$$

109.7 Compactness Input From The Weitzenbock Identity

The monopole equations have compactness properties stronger than the instanton equations because the spinor obeys a nonlinear elliptic equation with a quartic term. The Dirac Weitzenbock identity is

$$\not{D}_B^* \not{D}_B q = \nabla_B^* \nabla_B q + \frac{s_g}{4} q + \frac{1}{2}\rho(F_B^+)q,$$

where s_g is the scalar curvature. Pairing with q, integrating over X, and using $\not{D}_B q = 0$ gives

$$0 = \int_X \left(|\nabla_B q|^2 + \frac{s_g}{4}|q|^2 + \frac{1}{2}\langle \rho(F_B^+)q, q\rangle \right) d\,\mathrm{vol}_g\,.$$

With the normalization $\rho(\sigma(q)) = (qq^\dagger)_0$, the unperturbed curvature equation gives

$$\frac{1}{2}\langle \rho(F_B^+)q, q\rangle = \frac{1}{4}|q|^4.$$

Thus solutions obey

$$\int_X \left(|\nabla_B q|^2 + \frac{s_g}{4}|q|^2 + \frac{1}{4}|q|^4 \right) d\,\mathrm{vol}_g = 0$$

in the unperturbed case. With perturbation η^+, the right side is controlled by the L^2-norm of η^+ and the negative part of scalar curvature. Elliptic bootstrapping then bounds q and F_B^+, while the topological identity fixing $c_1(L_{\mathfrak{s}})$ controls the anti-self-dual component modulo gauge. This is the analytic source of compactness for the monopole moduli space after excluding reducible solutions by a generic perturbation when the usual chamber hypotheses allow it.

109.8 Seiberg-Witten Invariants And Basic Classes

Assume $b_2^+(X) > 1$, choose a homology orientation, and choose generic metric/perturbation data so that the monopole moduli spaces are cut out transversely and contain no reducible solutions in the relevant classes. If $\mathfrak{s}$ has expected dimension $d_{\mathrm{SW}}(\mathfrak{s}) = 0$, its Seiberg-Witten invariant is the signed count

$$\mathrm{SW}_X(\mathfrak{s}) = \#_{\mathrm{or}}\, \mathcal{M}_{\mathrm{SW}}(\mathfrak{s}, g, \eta^+). \tag{109.14}$$

For positive expected dimension, the invariant pairs the fundamental cycle of the compact oriented monopole moduli space with the standard cohomology classes obtained from the universal $U(1)$ bundle and, when $b_1(X) > 0$, from $H_1(X)$. In this chapter the zero-dimensional case is the main one because it is the simple-type sector appearing in Witten's comparison.

Definition 109.7 (Seiberg-Witten basic class and simple type)**.** An integral class $K \in H^2(X;\mathbb{Z})$ is a Seiberg-Witten basic class if $K = c_1(\mathfrak{s})$ for a Spin^c structure $\mathfrak{s}$ whose Seiberg-Witten invariant is nonzero. The four-manifold has Seiberg-Witten simple type if every basic class satisfies

$$d_{\mathrm{SW}}(\mathfrak{s}) = 0, \qquad \text{equivalently} \qquad K^2 = 2\chi(X) + 3\sigma(X). \tag{109.15}$$

The adjective "basic" is not a synonym for "allowed." Allowed Spin^c structures are the characteristic lifts of Proposition 109.6. Basic classes are the much smaller subset detected by a nonzero monopole count or by the higher-dimensional monopole pairings. The Weitzenbock estimate gives the compactness input for each fixed characteristic class; the finiteness of nonzero basic classes is a global theorem of the Seiberg-Witten invariant construction.

Example 109.8 (K3, elliptic surfaces, and blow-up bookkeeping)**.** For a K3 surface,

$$\chi = 24, \qquad \sigma = -16, \qquad 2\chi + 3\sigma = 0.$$

The only Seiberg-Witten basic class is $K = 0$, with invariant ± 1 after a homology-orientation choice. Hence the simple-type dimension condition is saturated exactly:

$$K^2 = 0 = 2\chi + 3\sigma.$$

For the relatively minimal simply connected elliptic surface $E(n)$ with no multiple fibers and $n \geq 2$,

$$\chi(E(n)) = 12n, \qquad \sigma(E(n)) = -8n, \qquad 2\chi + 3\sigma = 0.$$

Let F be the fiber class. Since $F^2 = 0$, the classes

$$K_j = (n - 2 - 2j)F, \qquad j = 0, \ldots, n-2, \tag{109.16}$$

all have square zero. With an orientation convention fixed, their Seiberg-Witten coefficients are encoded by

$$\sum_K \mathrm{SW}_{E(n)}(K)\, \mathrm{e}^K = \left(\mathrm{e}^F - \mathrm{e}^{-F}\right)^{n-2}. \tag{109.17}$$

The case $n = 2$ is K3. The formula records both the location of the basic classes and their binomial coefficients; changing the homology orientation multiplies all coefficients by a common sign.

If $\widetilde{X} = X \# \overline{\mathbb{CP}}^2$ and E is the exceptional class with $E^2 = -1$, then

$$\chi(\widetilde{X}) = \chi(X) + 1, \qquad \sigma(\widetilde{X}) = \sigma(X) - 1.$$

Thus $2\chi + 3\sigma$ decreases by one. If $K^2 = 2\chi(X) + 3\sigma(X)$, then

$$(K \pm E)^2 = K^2 - 1 = 2\chi(\widetilde{X}) + 3\sigma(\widetilde{X}). \tag{109.18}$$

The blow-up formula for Seiberg-Witten invariants gives

$$\mathrm{SW}_{\widetilde{X}}(K + E) = \mathrm{SW}_{\widetilde{X}}(K - E) = \mathrm{SW}_X(K) \tag{109.19}$$

up to the same global orientation convention. This example is a useful normalization check: the dimension formula, basic-class positions, and coefficient doubling are independent pieces of data.

109.9 Stable Cohomotopy Refinement And Furuta's Constraint

The ordinary Seiberg-Witten invariant is an integer obtained after choosing perturbations and counting or pairing with a finite-dimensional moduli space. There is a sharper object before this integer is extracted. After Sobolev completion and Coulomb-slice gauge fixing, the monopole equations define a Fredholm map with compact nonlinear part, called the monopole map. Its finite-dimensional approximations define a stable cohomotopy class. The ordinary Seiberg-Witten invariant is obtained from this stable cohomotopy class by applying the appropriate Hurewicz-type map and evaluating on the orientation class.

Monopole map before taking its degree. Fix a background connection B_0 on $L_{\mathfrak{s}}$, write $B = B_0 + b$, and impose the Coulomb condition $d^*b = 0$. After Sobolev completion, the Seiberg-Witten equations are encoded by a map

$$\mathcal{F} : L^2_k\big(i\Omega^1(X) \oplus \Gamma(S^+_{\mathfrak{s}})\big) \longrightarrow L^2_{k-1}\big(i\Omega^0(X) \oplus i\Omega^{2,+}(X) \oplus \Gamma(S^-_{\mathfrak{s}})\big) \tag{109.20}$$

of the form

$$\mathcal{F}(b,q) = \Big(d^*b,\ d^+b - \sigma(q) + F^+_{B_0} - \eta^+,\ \ \not{D}_{B_0} q + \rho(b)q\Big). \tag{109.21}$$

Thus

$$\mathcal{F} = L + C, \qquad L = d^* \oplus d^+ \oplus\ \not{D}_{B_0}, \tag{109.22}$$

where L is Fredholm and C is compact on bounded sets after Sobolev embedding and Rellich compactness. The Weitzenbock estimate in the previous section supplies the a priori bound needed to restrict to a large ball whose boundary contains no zeroes of $\mathcal{F}$, after the perturbation has been chosen away from reducible walls. Choose finite-dimensional subspaces

$$W_\Lambda \subset L^2_{k-1}\big(i\Omega^0 \oplus i\Omega^{2,+} \oplus \Gamma(S^-_{\mathfrak{s}})\big)$$

spanned by low eigenmodes of LL^*, and set

$$V_\Lambda = L^{-1}(W_\Lambda).$$

For sufficiently large Λ, the projected map

$$\mathcal{F}_\Lambda : V_\Lambda \longrightarrow W_\Lambda \tag{109.23}$$

has the same zero set as $\mathcal{F}$ in the chosen ball and gives a based map between one-point compactifications

$$S^{V_\Lambda} \longrightarrow S^{W_\Lambda}. \tag{109.24}$$

Increasing Λ suspends this map by finite-dimensional pieces of the Fredholm operator. The stable homotopy class of the resulting map is the object refined by Bauer and Furuta. The ordinary integer invariant is obtained only after applying a cohomological orientation map to this stable class; it is therefore a shadow, not the primary object.

Quoted Theorem 109.9 (Bauer-Furuta)**.** *For a closed oriented smooth four-manifold with a fixed* Spin^c *structure, the finite-dimensional approximation of the Seiberg-Witten monopole map defines a stable cohomotopy invariant*

$$\mathrm{BF}_X(\mathfrak{s}). \tag{109.25}$$

The ordinary Seiberg-Witten invariant is the image of $\mathrm{BF}_X(\mathfrak{s})$ *under the corresponding stable cohomotopy to cohomology map, with the usual orientation conventions.*

The theorem packages the independence of all auxiliary choices in the construction above: Sobolev index, background connection, radius of the isolating ball, spectral cutoff, and generic perturbation inside a chamber. The proof is not a formal consequence of the integer Seiberg-Witten invariant. It uses properness on the chosen bounded region, homotopy invariance of finite-dimensional approximations, and compatibility of the suspension coordinates with the Fredholm index. The monograph uses this theorem as a pure four-manifold input adjacent to QFT, while the displayed monopole-map construction records exactly what additional structure an eventual non-perturbative twisted-QFT construction would have to recover.

Quoted Theorem 109.10 (Furuta $10/8 + 2$ inequality)**.** *Let X be a closed smooth spin four-manifold with $b_2^+(X) > 0$. Then*

$$b_2(X) \geq \frac{10}{8}\,|\sigma(X)| + 2. \tag{109.26}$$

Why stable cohomotopy can imply a smooth constraint. When X is spin, the spinor bundles carry a quaternionic structure and the monopole map is equivariant for the group $\mathrm{Pin}(2) \subset \mathbb{H}^\times$ generated by $U(1)$ and an element j with $jzj^{-1} = \bar{z}$. The finite-dimensional approximation therefore produces a Pin(2)-equivariant stable map between representation spheres. The virtual difference of these representation spheres is fixed by the Dirac index and by $b_2^+(X)$; for a spin four-manifold the Dirac index is $-\sigma(X)/8$ as a quaternionic index. Furuta's theorem applies equivariant stable homotopy, in particular the Pin(2) representation-theoretic obstruction to such a stable map, to force (109.26). Thus the inequality is not derived from a numerical monopole count. It uses the extra equivariant information retained by the Bauer-Furuta refinement.

The relevance for the present chapter is conceptual and structural. The Donaldson-Witten path integral suggests that four-manifold invariants should arise from a cohomological field theory, but ordinary integer invariants can forget stable homotopy information in the monopole map. The Bauer-Furuta class records part of that information. Furuta's inequality is a theorem extracted from this stronger structure in the spin case; for K3,

$$b_2 = 22, \qquad |\sigma| = 16,$$

and (109.26) is saturated:

$$22 = \frac{10}{8} \cdot 16 + 2.$$

The monograph treats these as rigorous four-manifold theorems adjacent to the QFT comparison. A future theorem-level QFT construction should explain which refinement of the twisted path integral sees the stable cohomotopy class rather than only its integer shadow.

109.10 The Coulomb-Branch Description

The Donaldson-Witten theory is the twist of pure $\mathcal{N} = 2$ Yang–Mills theory. Its local supersymmetric Coulomb-branch data for gauge algebra $\mathfrak{su}(2)$ were developed in Chapter 91. The gauge-invariant coordinate is

$$u = \langle \operatorname{tr} \phi^2 \rangle,$$

and the special coordinates $(a_D(u), a(u))$ determine the Abelian coupling

$$\tau(u) = \frac{da_D/du}{da/du}.$$

The Coulomb branch has singular fibers where a BPS monopole or dyon becomes massless. Near such a fiber, the Wilsonian effective theory contains an Abelian vector multiplet together with the light charged hypermultiplet. The topological twist of this local Abelian theory has localization equations of Seiberg-Witten monopole type.

The cohomological path integral on a four-manifold receives three kinds of terms:

instanton description	intersection pairing on ASD moduli spaces,
Coulomb branch	integral over u with Abelian zero modes and contact terms,
singular fibers	monopole or dyon Seiberg-Witten invariants.

For $b_2^+(X) > 1$, the Coulomb-branch integral has no chamber-wall contribution in the generic situation used for the simple-type comparison. For $b_2^+(X) = 1$, the u-plane contribution and wall crossing are part of the invariant. The comparison requires the contact terms of the Abelian effective action because products of descent observables collide in the low-energy description.

109.11 Chambers, Reducibles, And Wall Crossing

When $b_2^+(X) = 1$, a metric g determines a ray

$$\mathbb{R}_{>0}\,\omega_g \subset \{v \in H^2(X;\mathbb{R}) \mid Q_X(v,v) > 0\} \tag{109.27}$$

by the self-dual harmonic line. A chamber is a connected component of the positive cone after removing the walls on which reducible connections can occur. To describe those walls without hiding the bundle data, take a rank-two Hermitian bundle $E \to X$ with fixed determinant and write

$$w = c_1(E), \qquad p = c_1(E)^2 - 4c_2(E) = p_1(\mathfrak{su}(E)).$$

A reducible splitting has the form

$$E = L \oplus (\det E \otimes L^{-1}),$$

and the associated characteristic wall class is

$$\lambda = 2c_1(L) - c_1(E). \tag{109.28}$$

It satisfies

$$\lambda \equiv w \pmod 2, \qquad \lambda^2 = p. \tag{109.29}$$

The reducible connection is anti-self-dual precisely when the self-dual harmonic projection of λ vanishes, equivalently

$$\langle \lambda, \omega_g \rangle = 0 \tag{109.30}$$

in the $b_2^+ = 1$ chamber picture.

Quoted Theorem 109.11 (Donaldson wall crossing in link form)**.** *Let a generic one-parameter path of metrics cross exactly one wall W_λ transversely and let z be a homogeneous Donaldson insertion whose degree matches the virtual dimension. The jump of Donaldson invariants is the oriented link pairing at the reducible stratum:*

$$D^w_{X,+}(z) - D^w_{X,-}(z) = \varepsilon(\lambda) \int_{\mathbb{L}_\lambda} \mu_\lambda(z). \tag{109.31}$$

Here $\mathbb{L}_\lambda$ is the link of the reducible in the parameterized compactified moduli space, $\mu_\lambda(z)$ is the restriction of the universal μ-class expression to this link, and $\varepsilon(\lambda)$ is the orientation sign determined by the direction in which the period point crosses $\langle \lambda, \omega \rangle = 0$. Equivalently, the right hand side is the Kotschick-Morgan wall-crossing polynomial

$$\mathrm{WC}^X_\lambda(z) = \varepsilon(\lambda) \int_{\mathbb{L}_\lambda} \mu_\lambda(z), \tag{109.32}$$

whose coefficients are universal functions of the intersection form, the class λ, the lift w, and the degree of z.

Mechanism of the link formula. Let

$$\mathcal{W} = \{(t, A) \mid F^+_{A,g_t} = 0\}/\mathcal{G}$$

be the parameterized moduli problem for the metric path g_t. Away from reducibles and compactification strata, $\mathcal{W}$ is the zero set of a Fredholm section whose index is one larger than the index of the fixed-metric ASD problem. Truncate small neighborhoods of the reducible stratum and of the other Uhlenbeck boundary strata. The resulting oriented space has boundary of the form

$$\partial\mathcal{W}^{\mathrm{tr}} = \mathcal{M}_+ \sqcup (-\mathcal{M}_-) \sqcup \mathbb{L}_\lambda \sqcup \{\text{other compactification faces}\}, \tag{109.33}$$

with the last faces absent or cancelled under the hypotheses of the theorem. The universal Donaldson class representing z extends over $\mathcal{W}^{\mathrm{tr}}$ and is closed there. Stokes' theorem therefore gives

$$0 = \int_{\partial\mathcal{W}^{\mathrm{tr}}} \mu(z) = D^w_{X,+}(z) - D^w_{X,-}(z) + \varepsilon_{\mathrm{link}} \int_{\mathbb{L}_\lambda} \mu_\lambda(z),$$

which is (109.31) after fixing the convention $\varepsilon(\lambda) = -\varepsilon_{\mathrm{link}}$. The sign is the orientation comparison between the outward normal of the deleted tubular neighborhood and the direction in which the period point crosses $\langle\lambda, \omega\rangle = 0$.

The local model of the deleted neighborhood is obtained by decomposing the bundle near a reducible splitting

$$E = L \oplus (\det E \otimes L^{-1}).$$

The diagonal part gives the Abelian stabilizer and the wall equation; the off-diagonal part gives the normal ASD complex. Its finite-dimensional kernel, cokernel, and stabilizer quotient determine the link $\mathbb{L}_\lambda$. Restricting the universal μ-classes to this normal model leaves only the intersection form, the wall class λ, the lift w, and the topological numbers entering the index. This is the geometric reason that the right hand side can be written as the universal Kotschick-Morgan polynomial rather than as a new invariant attached to each metric path.

This is the form of the wall-crossing theorem that is most useful for QFT. It identifies the jump with an actual boundary integral. In BV language, the two chamber invariants are the two sides of a one-parameter family of pushforwards, and the link $\mathbb{L}_\lambda$ is the boundary component where the pushforward fails to be closed. Thus wall crossing is not an exception to metric independence; it is the boundary term required by the Uhlenbeck BV-boundary calculation above.

Remark 109.12 (Kotschick-Morgan polynomial normal form). In the situation of Theorem 109.11, restrict for simplicity to insertions generated by a point class x and a surface source $h \in H_2(X;\mathbb{R})$. For each pair (r, n) with the correct Donaldson degree, the wall-crossing functional has the universal form

$$\mathrm{WC}^X_\lambda(x^r h^n) = \varepsilon(\lambda) \sum_{\substack{j,\ell\geq 0\\ j+2\ell\leq n}} c_{r,n;j,\ell}(p, w^2, \lambda^2, \chi, \sigma)\, \langle\lambda, h\rangle^{\,n-j-2\ell} Q_X(h,h)^\ell \langle w, h\rangle^j. \tag{109.34}$$

The coefficients are universal rational numbers after the Donaldson normalization and orientation convention have been fixed. The allowed monomials are exactly the scalar polynomials in h built from the link data (Q_X, λ, w) with total Donaldson surface degree $2n$.

The link of a reducible is constructed from the normal complex to the reducible stratum and the universal μ-classes restricted to it. After the determinant line and orientation of the crossing are fixed, the only homological tensors available on $H_2(X;\mathbb{R})$ are the intersection form Q_X, the wall class λ, and the fixed lift w. Therefore the restriction of a homogeneous insertion $x^r h^n$ to the link is an ordinary polynomial in the three scalar contractions

$$Q_X(h,h), \qquad \langle\lambda,h\rangle, \qquad \langle w,h\rangle.$$

The exponent of $Q_X(h,h)$ consumes two surface insertions, while each linear contraction consumes one. This gives the degree constraint $j+2\ell \le n$. Universality follows because the local model of the reducible link depends on the four-manifold only through the topological numbers $p, w^2, \lambda^2, \chi, \sigma$ and the intersection pairing. The theorem of Kotschick and Morgan computes these universal coefficients by evaluating the link integral; (109.34) is the coordinate form of that computation needed for comparison with the u-plane boundary.

Chamber jump as a BV boundary term. Let ρ_t be a regulated Donaldson-Witten BV semidensity along a generic metric path g_t, $t \in [-1,1]$, crossing a single reducible wall λ at $t=0$. Assume the parameterized moduli problem is smooth away from the reducible link and all insertion classes have been extended to the truncated compactification. Then

$$D^w_{X,+}(z) - D^w_{X,-}(z) = \mathcal{B}_{\mathbb{L}_\lambda}(\rho z), \tag{109.35}$$

with the sign convention of (109.31). If the path crosses several walls, the jump is the signed sum of the corresponding link functionals.

Under these hypotheses, Stokes' theorem applies to the parameterized compactified moduli space with small collar neighborhoods of reducibles removed. The ordinary boundary at $t=1$ gives $D^w_{X,+}(z)$, the ordinary boundary at $t=-1$ gives $-D^w_{X,-}(z)$, and the new boundary is the link $\mathbb{L}_\lambda$. In the finite-dimensional BV model, include the metric parameter among the residual variables and apply (111.12) to the residual cycle with the reducible collar removed. Its boundary functional is precisely the localized integral over $\mathbb{L}_\lambda$. The link orientation is exactly the sign $\varepsilon(\lambda)$ in the wall-crossing theorem.

109.12 Observable Map And Contact Terms

The Donaldson generating function is organized by a point variable p and a two-cycle source $S \in H_2(X;\mathbb{R})$:

$$Z_D(p,S) = \left\langle \exp\bigl(p\, I(x) + I(S)\bigr) \right\rangle.$$

Here $I(x)$ has degree four and $I(S)$ has degree two. In the Abelian Coulomb-branch description, the point observable maps to a function of the Coulomb coordinate,

$$I(x) \longmapsto u,$$

whereas the surface observable has a linear term in the Abelian field strength and a quadratic contact term:

$$I(S) \longmapsto \int_S \Omega_u + \frac{1}{2}(S,S)\, T(u).$$

The two-form Ω_u is the appropriate descent two-form in the Abelian effective theory, and $T(u)$ is fixed by compatibility of the low-energy Ward identities, duality transformations, and short-distance collision of two surface descendants. The symbol (S,S) denotes the intersection pairing on $H_2(X;\mathbb{R})$.

At a monopole singularity, the Abelian surface insertion evaluates on the monopole line bundle and gives an exponential weight

$$\exp\big((S,\lambda)\big)$$

for a characteristic class $\lambda = c_1(\mathfrak{s})$ associated with the corresponding Spin^c structure, multiplied by the contact factor coming from $T(u)$. The final comparison formula in simple-type chambers has the general form

$$Z_D(p,S) = e^{Q(S)/2} \sum_{\mathfrak{s}} C_{\mathfrak{s}}\ \mathrm{SW}(\mathfrak{s})\ \exp\big(p\,u_{\mathfrak{s}} + (S,\lambda_{\mathfrak{s}})\big),$$

where $Q(S) = (S,S)$. The constants $C_{\mathfrak{s}}$, the phases, and the choice of chamber depend on the global form of the gauge group, the orientation convention, and the contact-term normalization. In this monograph the displayed expression is treated as a target comparison statement: the rigorous QFT problem is to derive every term from a constructed twisted continuum theory and its Wilsonian flow, rather than to declare the formula from moduli-space geometry alone.

Hypothesis 109.13 (Witten simple-type comparison statement)**.** Let X be a closed connected oriented smooth four-manifold with $b_1(X) = 0$, $b_2^+(X) > 1$, and Seiberg-Witten simple type. Fix the lift w, homology orientation, point-observable convention (109.2), and the trace-delta gauge normalization used in this chapter. The Donaldson-Seiberg-Witten comparison predicts that the reduced Donaldson series has the form

$$\mathbb{D}_X^w(h) = \exp\left(\frac{1}{2}Q_X(h,h)\right) \sum_{K\in\mathcal{B}_{\mathrm{SW}}(X)} A_X^w(K)\,\mathrm{SW}_X(K)\ \exp\big(\langle K,h\rangle\big), \tag{109.36}$$

where $\mathcal{B}_{\mathrm{SW}}(X)$ is the set of Seiberg-Witten basic classes. The phase has the characteristic form

$$A_X^w(K) = C_X\,(-1)^{\frac{1}{2}(w^2+\langle w,K\rangle)} \tag{109.37}$$

when w is chosen so that the exponent is integral. The universal constant C_X depends only on the Euler characteristic, signature, and normalization of Donaldson observables; in the common Moore-Witten normalization it is written as

$$C_X = 2^{\,2+(7\chi(X)+11\sigma(X))/4}. \tag{109.38}$$

Equation (109.36) is a theorem in the mathematical Donaldson/SW comparison under the hypotheses of the relevant comparison theorems, but in this monograph it is also a QFT target: a non-perturbative construction of the twisted four-dimensional gauge theory should derive the same coefficient, phase, contact, and boundary data from Wilsonian flow.

The distinction between (109.4) and (109.36) is important. The first is the Donaldson simple-type structure: there are some Donaldson basic classes with some coefficients. The second identifies those classes and coefficients with Seiberg-Witten monopole invariants. The QFT explanation of that identification is the central RG statement.

Factor ledger for the simple-type comparison. Equation (109.36) contains several pieces with different origins. Keeping them separate prevents the comparison from being read as a single formal substitution:

1. The Donaldson simple-type theorem supplies only the finite exponential shape

$$\exp(Q_X/2)\sum_K a_K^w \exp\langle K,h\rangle$$

after the point-operator normalization is fixed. It does not identify the labels K with monopole basic classes or compute the coefficients a_K^w.

2. The Seiberg-Witten finite-dimensional theory supplies the candidate labels $K = c_1(\mathfrak{s})$, the oriented counts $\mathrm{SW}_X(K)$, and the simple-type vanishing of nonzero-dimensional monopole moduli spaces. It does not by itself produce the Donaldson contact Gaussian, the w-dependent phase, or the universal power of 2.

3. The Abelian low-energy observable map supplies the Gaussian $\exp(Q_X(h,h)/2)$ through the surface-contact term $T(u)$. This is a collision-of-descendants datum in the twisted Abelian theory, not a consequence of the finite monopole count.

4. Electromagnetic-duality determinant lines and the lift w supply the phase $(-1)^{(w^2+\langle w,K\rangle)/2}$. This phase must be matched with the orientation conventions for Donaldson and monopole determinant lines; changing the lift changes the exceptional-class parity as in Example 109.14.

5. The numerical constant C_X comes from the regularized Abelian measure factors, one-loop gravitational terms, and the normalization of Donaldson observables. Its dependence on $\chi(X)$ and $\sigma(X)$ is therefore a determinant-line and contact-normalization claim, not a topological fact following from simple type.

6. Chamber dependence is controlled separately. For $b_2^+ > 1$ the simple-type comparison uses the absence of a residual regular u-plane bulk contribution under the stated hypotheses. For $b_2^+ = 1$, the regularized u-plane boundary and the Donaldson link wall-crossing formula must be matched before a chamber-specific comparison is meaningful.

The open problem below is precisely to construct a single QFT framework in which all six entries arise from the same regulator, anomaly line, Wilsonian flow, and compactification boundary convention.

Proof obligations hidden by the comparison formula. A derivation of Hypothesis 109.13 from QFT would have to prove the following chain of constructions and identifications. First, one needs a non-perturbative construction of the twisted pure $\mathcal{N} = 2$ gauge theory on X, including a regulator or BV/lattice replacement for the gauge-fixed measure, the scalar charge Q, the descent observables $I(x)$ and $I(S)$, and compactification boundary data for the ASD localization locus. Second, the Wilsonian flow must be defined on this cohomological theory in a way compatible with Q and with the anomaly line, and must produce the Seiberg-Witten Coulomb-branch family away from singular fibers with the Abelian contact term $T(u)$. Third, the regular u-plane integral has to be shown to supply only the stated chamber-dependent boundary contribution, with no hidden bulk term for the $b_2^+ > 1$ simple-type situation. Fourth, a neighborhood of each singular fiber must be identified with the twisted Abelian monopole theory, including orientation lines, determinant phases, and the map from surface observables to the monopole determinant line. This is the step that turns the singular-fiber boundary contribution into $\mathrm{SW}_X(K)\exp\langle K,h\rangle$. Finally, the normalization of the point observable, the Gaussian surface contact term, the phase $(-1)^{(w^2+\langle w,K\rangle)/2}$, and the power of 2 in (109.38) must be derived from the same regularized determinant and contact-term conventions, rather than fitted after the fact.

Thus (109.36) is not used below as an unproved physical theorem. It is a precise target statement: each factor in the formula names a separate analytic, topological, or Wilsonian ingredient that a future theorem-level construction must supply.

Finite comparison residual budget. The proof obligations above are not only a list of words. On any finite set of Donaldson insertions they define a comparison budget: a partial construction is meaningful only to the extent that each arrow in the comparison ladder has a controlled residual.

Fix a finite-dimensional test space $\mathcal{T}_N$ generated by finitely many point and surface Donaldson insertions, with norm $\|\cdot\|_{\mathcal{T}}$. Let

$$\mathcal{D}_{\mathrm{UV}}, \quad \mathcal{D}_{\mathrm{ASD}}, \quad \mathcal{D}_{\mathrm{Coul}}, \quad \mathcal{D}_{\mathrm{split}}, \quad \mathcal{D}_{\mathrm{mon}}, \quad \mathcal{D}_{\mathrm{SW}} \in \mathcal{T}_N^{\vee}$$

be the finite linear functionals obtained, respectively, from a regulated twisted UV theory, its ASD localization formula, the regular Coulomb-branch description before singular neighborhoods are separated, the regular u-plane plus raw singular-fiber boundary split, the monopole replacement with raw determinant-line normalization, and the normalized Donaldson–Seiberg-Witten comparison formula. Define

$$R_Q = \mathcal{D}_{\mathrm{UV}} - \mathcal{D}_{\mathrm{ASD}}, \qquad R_{\mathrm{RG}} = \mathcal{D}_{\mathrm{ASD}} - \mathcal{D}_{\mathrm{Coul}}, \tag{109.39}$$

$$R_u = \mathcal{D}_{\mathrm{Coul}} - \mathcal{D}_{\mathrm{split}}, \qquad R_{\mathrm{sing}} = \mathcal{D}_{\mathrm{split}} - \mathcal{D}_{\mathrm{mon}}, \tag{109.40}$$

$$R_{\mathrm{norm}} = \mathcal{D}_{\mathrm{mon}} - \mathcal{D}_{\mathrm{SW}}. \tag{109.41}$$

Then for every $T \in \mathcal{T}_N$,

$$\mathcal{D}_{\mathrm{UV}}(T) - \mathcal{D}_{\mathrm{SW}}(T) = R_Q(T) + R_{\mathrm{RG}}(T) + R_u(T) + R_{\mathrm{sing}}(T) + R_{\mathrm{norm}}(T). \tag{109.42}$$

Consequently, if $|R_\bullet(T)| \le \varepsilon_\bullet \|T\|_{\mathcal{T}}$ for the five residuals above, then

$$|\mathcal{D}_{\mathrm{UV}}(T) - \mathcal{D}_{\mathrm{SW}}(T)| \le (\varepsilon_Q + \varepsilon_{\mathrm{RG}} + \varepsilon_u + \varepsilon_{\mathrm{sing}} + \varepsilon_{\mathrm{norm}})\|T\|_{\mathcal{T}}. \tag{109.43}$$

Indeed, subtract and add the intermediate functionals:

$$\begin{aligned}\mathcal{D}_{\mathrm{UV}} - \mathcal{D}_{\mathrm{SW}} = {} & (\mathcal{D}_{\mathrm{UV}} - \mathcal{D}_{\mathrm{ASD}}) + (\mathcal{D}_{\mathrm{ASD}} - \mathcal{D}_{\mathrm{Coul}}) + (\mathcal{D}_{\mathrm{Coul}} - \mathcal{D}_{\mathrm{split}}) \\ & + (\mathcal{D}_{\mathrm{split}} - \mathcal{D}_{\mathrm{mon}}) + (\mathcal{D}_{\mathrm{mon}} - \mathcal{D}_{\mathrm{SW}}).\end{aligned}$$

Evaluating on T gives (109.42). The bound follows from the triangle inequality and the five declared residual estimates.

The five residuals have separate meanings:

- R_Q measures nonperturbative construction and Q-localization error.
- R_{RG} measures Q-compatible Wilsonian-flow error.
- R_u measures regular u-plane compactification and chamber-boundary error.
- R_{sing} measures singular-fiber replacement by monopole neighborhoods.
- R_{norm} measures contact-term, determinant-phase, and universal-constant normalization error.

A theorem proving (109.36) is the special case in which these five residuals vanish on every finite test space and the finite estimates are compatible as N increases. A controlled physical approximation must state the five residuals rather than quoting the final comparison formula as a single black-box identity.

Example 109.14 (Donaldson-side tables from the comparison statement)**.** The following table is not an independent derivation of the Donaldson–Seiberg-Witten comparison. It is the explicit Donaldson-side output of Hypothesis 109.13 in examples where the Seiberg-Witten basic classes were fixed in Example 109.8. We take $w = 0$ when this is compatible with the displayed parity and choose the common Moore–Witten constant (109.38).

X	$\sum_K \mathrm{SW}_X(K)e^K$	$\mathbb{D}^0_X(h)$
$K3$	1	$\exp(Q_X(h,h)/2)$
$E(n)$	$(e^F - e^{-F})^{n-2}$	$2^{2-n}\exp(Q_X(h,h)/2)(e^{\langle F,h\rangle} - e^{-\langle F,h\rangle})^{n-2}$

For $E(2) = K3$, the second row reduces to the first. For $E(3)$, the Donaldson basic classes predicted by the comparison are $\pm F$ with opposite coefficients in this $w = 0$ convention:

$$\mathbb{D}^0_{E(3)}(h) = \frac{1}{2}\, e^{Q(h,h)/2}\big(e^{\langle F,h\rangle} - e^{-\langle F,h\rangle}\big). \tag{109.44}$$

If $\widetilde{X} = X\#\overline{\mathbb{CP}}^2$, let E be the exceptional cohomology class and choose $e \in H_2(\widetilde{X};\mathbb{R})$ with $Q_{\widetilde{X}}(e,e) = -1$ and $\langle E, e\rangle = 1$. With the lift chosen so that the two exceptional basic classes have the same phase, the comparison gives

$$\mathbb{D}^{\widetilde{w}}_{\widetilde{X}}(h + te) = e^{-t^2/2}\cosh(t)\,\mathbb{D}^w_X(h). \tag{109.45}$$

With the adjacent lift convention the two exceptional phases differ and the factor $\cosh(t)$ is replaced by $\sinh(t)$, up to the common orientation sign. In the exponential generating series this gives an explicit parity table:

$$[t^{2m}]\, e^{-t^2/2}\cosh t = \sum_{r=0}^{m} \frac{(-1)^r}{2^r r!(2m-2r)!}, \qquad [t^{2m+1}]\, e^{-t^2/2}\cosh t = 0, \tag{109.46}$$

$$[t^{2m+1}]\, e^{-t^2/2}\sinh t = \sum_{r=0}^{m} \frac{(-1)^r}{2^r r!(2m+1-2r)!}, \qquad [t^{2m}]\, e^{-t^2/2}\sinh t = 0. \tag{109.47}$$

Thus the even lift has coefficients $1, 0, -1/12, \ldots$ in degrees $0, 2, 4, \ldots$, while the adjacent lift has coefficients $1, -1/3, 1/20, \ldots$ in degrees $1, 3, 5, \ldots$. These numbers are not new invariants; they are the exact exceptional-insertion Taylor coefficients obtained by multiplying the contact Gaussian by the two exceptional basic-class phases. The point of writing the table is that the constants, phases, Gaussian factor, basic-class locations, and blow-up square shift are separate pieces of information; none is determined by the word "simple type" alone.

109.13 The u-Plane Integral And Boundary Contributions

For $b_2^+(X) = 1$, the Coulomb-branch contribution cannot be suppressed by the simple-type argument. The low-energy path integral over the regular part of the u-plane is written, after fixing the Donaldson observable normalization used in (109.36), as

$$Z^w_u(p, S; \omega) = \int_{\mathcal{B}^{\mathrm{reg}}_u} da \wedge d\bar{a}\, A(u)^{\chi(X)} B(u)^{\sigma(X)} \Psi_{\Gamma,w,\omega}(\tau,\bar{\tau}; S)\exp\big(2pu + S^2 T(u)\big)\,. \tag{109.48}$$

Here $\mathcal{B}_u^{\mathrm{reg}}$ is the Coulomb branch with the singular fibers removed, a is a local special coordinate,

$$\tau(u) = \frac{da_D/du}{da/du},$$

$\Gamma = H^2(X;\mathbb{Z})/\mathrm{torsion}$ with intersection pairing Q_X, and ω is the chamber period point in (109.27). The functions

$$A(u) = \alpha\left(\frac{du}{da}\right)^{1/2}, \qquad B(u) = \beta\,\Delta(u)^{1/8} \tag{109.49}$$

encode the gravitational measure factors of the Abelian twisted effective theory; $\Delta(u)$ is the discriminant of the Seiberg-Witten family and α, β are convention-dependent constants. The factor $\Psi_{\Gamma,w,\omega}$ is the Siegel-Narain theta kernel obtained by summing over Abelian fluxes shifted by $w/2$, with the positive and negative projections defined by ω. Its dependence on ω is the analytic source of wall crossing. The contact term $T(u)$ is the same surface-observable collision term that appears in the observable map above.

Definition 109.15 (Regularized u-plane integral)**.** Let $\Omega_u^w(p,S;\omega)$ denote the two-form integrand in (109.48). Choose small disjoint disks

$$D_\epsilon(u_\alpha)$$

around the singular fibers u_α of the Seiberg-Witten family and, in a coordinate $q = e^{2\pi i\tau}$ near the weak-coupling cusp, remove the region $|q| < \epsilon_\infty$. The regulated domain is

$$\mathcal{B}_{u,\epsilon}^{\mathrm{reg}} = \mathcal{B}_u \setminus \left(\bigcup_\alpha D_\epsilon(u_\alpha) \cup \{|q| < \epsilon_\infty\}\right).$$

The regulated Coulomb contribution is

$$Z_{u,\epsilon}^w(p,S;\omega) = \int_{\mathcal{B}_{u,\epsilon}^{\mathrm{reg}}} \Omega_u^w(p,S;\omega). \tag{109.50}$$

A renormalized u-plane contribution means a limit

$$Z_u^w = \lim_{\epsilon\to 0}\left[Z_{u,\epsilon}^w + \sum_\alpha C_{\alpha,\epsilon}^{\mathrm{sing}} + C_{\infty,\epsilon}^{\mathrm{cusp}}\right], \tag{109.51}$$

when the boundary counterterms $C_{\alpha,\epsilon}^{\mathrm{sing}}$ and $C_{\infty,\epsilon}^{\mathrm{cusp}}$ have been derived from the corresponding monopole/dyon and weak-coupling effective theories. They are not arbitrary subtractions; they are part of the declared low-energy regularization datum.

Wall-normal factor in the theta kernel. Only one elementary analytic mechanism is needed to understand why the u-plane wall crossing is localized on fluxes satisfying the wall equation. Fix a lattice flux

$$\lambda \in \Gamma + \frac{w}{2}$$

and write

$$x_\lambda(\omega) = \langle \lambda, \omega\rangle$$

for its component along the positive period ray. The complete Siegel-Narain kernel contains smooth factors in u, S, and a wall-normal factor whose local model is

$$E_t(x) = \mathrm{erf}(\sqrt{\pi t}\, x), \qquad t > 0. \tag{109.52}$$

The parameter t is proportional to the positive imaginary part of the Abelian coupling in the normal direction. This model records the only feature relevant for the wall jump: as $t \to \infty$,

$$E_t(x) \to \mathrm{sgn}(x) \qquad (x \neq 0),$$

while

$$\frac{dE_t}{dx} = 2\sqrt{t}\, \exp(-\pi t x^2) \tag{109.53}$$

is a delta sequence of total mass 2:

$$\int_{-\infty}^{\infty} 2\sqrt{t}\, \exp(-\pi t x^2)\, \mathrm{d}x = 2.$$

Thus changing the chamber by moving ω through $x_\lambda(\omega) = 0$ changes the corresponding completed sign factor by exactly 2. Flux sectors with $x_\lambda(\omega) \neq 0$ remain smooth in ω; the only singular chamber dependence comes from the codimension one locus $x_\lambda = 0$. This is the local analytic content behind the phrase that wall crossing is supported on reducible fluxes. The global statement still requires the remaining factors in the u-plane integrand, the singular-fiber counterterms, and the determinant-line orientation matching to be controlled.

Controlled Approximation 109.16 (Wall crossing from the u-plane boundary)**.** Assume that the Abelian low-energy description is valid on $\mathcal{B}_u^{\mathrm{reg}}$, that the singular-fiber contributions are computed by the corresponding monopole or dyon Seiberg-Witten theories, and that the theta kernel has the standard Gaussian decay away from walls. When ω crosses the wall $\langle \lambda, \omega \rangle = 0$, the discontinuity of Z_u^w is a boundary term supported by the flux sector λ, and equals the link wall-crossing functional WC_λ^X after the contact normalization and determinant-line phase are matched.

Only the positive/negative decomposition of the lattice vector in $\Psi_{\Gamma,w,\omega}$ changes discontinuously at a wall. The local model (109.52)–(109.53) shows that the chamber jump of a completed sign factor is a delta-supported normal contribution with total jump 2. Away from $\lambda_+ = 0$, the Gaussian factor in the theta kernel gives a smooth ω-dependent integral. Crossing the wall changes the prescription for the sector with $\lambda_+ = 0$; differentiating with respect to the period point produces a total $\bar{\partial}$-term on the u-plane. By Stokes' theorem the change is therefore a sum of boundary contributions from the cusps and singular fibers. The singular-fiber boundaries are precisely the monopole/dyon effective theories, while the cusp and reducible boundary combine into the Donaldson wall link. Matching determinant-line orientations and the contact term $T(u)$ identifies the result with (109.32). This proves the conditional statement: the analytic equality rests on the stated validity of the Abelian Wilsonian description and on the matching of its measure factors.

109.14 Comparison As A QFT Statement

The comparison can be formulated as a precise datum. Fix:

UV theory	twisted pure $\mathcal{N} = 2$ $SU(2)$ gauge theory on X,
observable algebra	Q-cohomology generated by Donaldson descent classes,
metric chamber	a chamber for $b_2^+(X) = 1$ or the $b_2^+ > 1$ regime,
Wilsonian coordinate	u and the Abelian special geometry near each singular fiber,
contact prescription	$T(u)$ and all collision counterterms for descent insertions,
IR localization	Spin^c monopole moduli spaces with orientations.

The QFT comparison asserts equality between the ultraviolet cohomological correlators and the sum of the Coulomb-branch integral plus monopole contributions computed from this datum. A theorem-level derivation would therefore contain the following ingredients:

construction	a four-dimensional twisted continuum gauge theory,
Ward identities	Q-invariance, descent, and metric variation including contacts,
RG control	flow from the UV description to the Abelian Coulomb chart,
singular fiber analysis	derivation of the monopole effective theory and its measure,
compactness	finite-dimensional compactified moduli spaces and boundary terms,
orientation and phases	matching of determinant lines and electromagnetic duality phases.

The finite-dimensional Donaldson and Seiberg-Witten invariants are rigorous objects once their compactifications, orientations, perturbations, and wall crossing conventions are fixed. The physical comparison supplies a deep explanation by RG, and the present formulation isolates the additional analytic QFT construction needed to turn that explanation into a theorem inside four-dimensional quantum field theory.

Open Problem 109.17 (Donaldson-SW QFT comparison). Construct the Donaldson-Witten theory and its Coulomb-branch Wilsonian flow with enough analytic control to derive the Seiberg-Witten comparison formula, including the ultraviolet twisted gauge theory, the Abelian monopole effective theories at singular fibers, the u-plane contribution, descent contact terms, determinant-line phases, chamber dependence, wall crossing, and compactification boundary contributions.

Chapter 110

Boundaries, Defects, And Categories In Topological QFT

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops boundary and defect data in topological QFT through finite-dimensional models where the algebra can be written explicitly. The central mechanism is that cutting a topological theory along strata produces state spaces, module categories, defect bimodules, junction spaces, and gluing pairings, and these structures must satisfy concrete associativity and nondegeneracy identities.

110.1 Boundary Conditions As Modules

Let Z be an extended n-dimensional topological theory with a chosen target (∞, n)-category. A boundary condition B for Z assigns compatible data to manifolds with boundary labelled by B. In a fully local formulation, if the value of Z on a point is an algebraic object $\mathcal{A}$, then a boundary condition is represented by a module object over $\mathcal{A}$, in the target appropriate to the codimension.

Definition 110.1 (Boundary category in dimension one)**.** Let Z be a one-dimensional topological theory with a set of boundary labels $\mathcal{B}$. For $B_0, B_1 \in \mathcal{B}$, let

$$\mathcal{H}(B_0, B_1).$$

be the state space assigned to an interval whose incoming endpoint is labelled by B_0 and outgoing endpoint by B_1. Gluing intervals gives linear maps

$$\mathcal{H}(B_0, B_1) \otimes \mathcal{H}(B_1, B_2) \longrightarrow \mathcal{H}(B_0, B_2).$$

If these maps are associative and if the interval of zero length gives identity elements in $\mathcal{H}(B, B)$, then the labels B are the objects of a category and $\mathcal{H}(B_0, B_1)$ are its morphism spaces.

Remark 110.2 (Associativity from one-dimensional gluing). The interval amplitudes of a one-dimensional topological theory with labelled endpoints satisfy the associativity condition in Definition 110.1.

Let $I = I_1 \cup I_2 \cup I_3$ be an interval decomposed into three subintervals with intermediate labels B_1, B_2. Topological invariance says that the linear map assigned to I depends only on the labelled bordism, not on the choice of cut positions. The two parenthesizations

$$(I_1 \cup I_2) \cup I_3, \qquad I_1 \cup (I_2 \cup I_3)$$

are the same labelled bordism. Functoriality therefore identifies the two composite multiplication maps from $\mathcal{H}(B_0, B_1) \otimes \mathcal{H}(B_1, B_2) \otimes \mathcal{H}(B_2, B_3)$ to $\mathcal{H}(B_0, B_3)$. The zero-length interval is the identity bordism, hence gives identity morphisms.

110.2 BF_2 and Finite Gauge Boundary States

The most concrete two-dimensional example is untwisted finite gauge theory, the finite counterpart of BF_2. Let G be a finite group. The closed state space on the circle is the center of the group algebra

$$\mathcal{H}(S^1) = Z(\mathbb{C}[G]).$$

The pair-of-pants product is multiplication in $Z(\mathbb{C}[G])$. For each irreducible complex representation ρ of G, with character χ_ρ and dimension d_ρ, define

$$e_\rho = \frac{d_\rho}{|G|} \sum_{g \in G} \chi_\rho(g^{-1})\, g \in Z(\mathbb{C}[G]). \tag{110.1}$$

The elements e_ρ, as ρ ranges over irreducible representations of G, are the primitive central idempotents of $\mathbb{C}[G]$:

$$e_\rho e_\sigma = \delta_{\rho\sigma} e_\rho, \qquad \sum_{\rho \in \mathrm{Irr}(G)} e_\rho = 1.$$

The elementary closed-channel topological boundary states of this semisimple model are therefore labelled by $\mathrm{Irr}(G)$. Equivalently, the elementary modules over the closed Frobenius algebra $Z(\mathbb{C}[G])$ are

$$M_\rho = e_\rho Z(\mathbb{C}[G]), \qquad \mathrm{Hom}_{Z(\mathbb{C}[G])}(M_\rho, M_\sigma) \cong \begin{cases} \mathbb{C}, & \rho = \sigma, \\ 0, & \rho \neq \sigma. \end{cases}$$

The regular representation decomposes as

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \mathrm{Irr}(G)} V_\rho \otimes V_\rho^*.$$

The center acts on the summand $V_\rho \otimes V_\rho^*$ by a scalar. Schur orthogonality gives the central projector onto that summand as (110.1). Projectors onto distinct summands are orthogonal and their sum is the identity. Since $Z(\mathbb{C}[G])$ is a finite direct product of copies of $\mathbb{C}$, its simple modules are exactly the one-dimensional summands $e_\rho Z(\mathbb{C}[G])$, with the displayed Hom spaces.

This calculation is deliberately a closed-sector statement. A fully extended finite-gauge theory has a richer boundary classification by module categories over $\mathrm{Rep}(G)$, and twisted theories replace $\mathrm{Rep}(G)$ by the corresponding cocycle-twisted tensor category. The idempotents above are the part of that story visible to the ordinary two-dimensional closed TQFT.

110.3 Defect Fusion and Junction Spaces

A codimension-one defect D_{12} from a theory Z_1 to a theory Z_2 is placed on a separating hypersurface. Fusion is defined by bringing two parallel defects together:

$$D_{12} \circ D_{23} = D_{13}.$$

For a topological defect, correlation functions are unchanged under deformations that keep the ordering of defects and avoid operator insertions. The fusion product is associative when the three-defect collision limit is independent of the relative distances used to take the limit. In a regulated QFT this independence is a theorem to be proved; in a bordism presentation it is part of the functorial data.

Example 110.3 (Finite gauge defects as bisets)**.** Let G and H be finite groups. A topological interface from two-dimensional finite G-gauge theory to finite H-gauge theory can be modelled by a finite (G, H)-biset X, namely a finite set with commuting left G- and right H-actions. If Y is an (H, K)-biset, the fused interface is the balanced product

$$X \otimes_H Y = (X \times Y)/\{(xh, y) \sim (x, hy) : h \in H\},$$

which is a (G, K)-biset. Junction fields between two (G, H)-bisets X and X' are the equivariant functions

$$\mathrm{Hom}_{G,H}(X, X') = \{f : X \to X' : f(gxh) = gf(x)h\}.$$

Associativity of finite defect fusion. For finite biset defects the balanced-product convention gives

$$(X \otimes_H Y) \otimes_K Z \cong X \otimes_H (Y \otimes_K Z)$$

canonically as (G, L)-bisets, whenever X, Y, Z are composable (G, H)-, (H, K)-, and (K, L)-bisets. Both sides are quotients of $X \times Y \times Z$ by the equivalence relation generated by

$$(xh, y, z) \sim (x, hy, z), \qquad (x, yk, z) \sim (x, y, kz), \qquad h \in H,\ k \in K.$$

The identity map on $X \times Y \times Z$ therefore descends to a bijection between the two iterated quotients. The left G- and right L-actions commute with the two generating equivalences, so the bijection is a (G, L)-biset isomorphism. This is quotient bookkeeping, not a theorem about the existence of defect fusion in an arbitrary continuum QFT; the latter requires the topological-defect sector and its fusion operation to have been constructed independently.

110.4 Line Defects and Modular Tensor Data

In a three-dimensional oriented TQFT, line defects form a monoidal category $\mathcal{C}$. The tensor product is parallel fusion:

$$X \otimes Y = \lim_{\epsilon \to 0} X(\epsilon) Y(0).$$

Braiding is obtained by exchanging two topological lines in $\mathbb{R}^3$:

$$c_{X,Y} : X \otimes Y \longrightarrow Y \otimes X.$$

The hexagon identities are equalities of isotopies of three braided lines. If every line has a dual line and the vacuum line has endomorphism ring $\mathbb{C}$, then $\mathcal{C}$ is rigid and has a distinguished unit object. Additional finiteness, semisimplicity, and nondegenerate braiding hypotheses give a modular tensor category; these hypotheses are part of the claimed three-dimensional theory, not consequences of topology alone.

Definition 110.4 (Modular tensor category data)**.** For the purposes of this volume, a semisimple modular tensor category $\mathcal{C}$ consists of a finite set I of simple objects, a tensor unit $0 \in I$, duals $a \mapsto a^*$, fusion multiplicities ${N_{ab}}^c$, associativity isomorphisms F, braiding isomorphisms R, twists θ_a, and a nondegenerate matrix

$$S_{ab} = \frac{1}{\mathcal{D}} \operatorname{tr}_{a \otimes b}(c_{b,a} c_{a,b}) \,, \qquad \mathcal{D}^2 = \sum_{a \in I} d_a^2,$$

subject to the pentagon, hexagon, ribbon, and nondegeneracy identities. The numbers d_a are categorical dimensions. A three-dimensional TQFT whose line category is $\mathcal{C}$ must produce these data from line fusion, line exchange, and framed line rotation.

Example 110.5 ($SU(2)_k$ line data)**.** For $SU(2)$ Chern–Simons theory at positive integer level k, the line labels are $a = 0, \ldots, k$, corresponding to spin $a/2$. The modular matrix is

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\!\left(\frac{(a+1)(b+1)\pi}{k+2}\right),$$

and the fusion rule is

$${N_{ab}}^c = 1 \quad \Longleftrightarrow \quad |a-b| \le c \le \min(a+b, 2k-a-b), \quad a+b+c \in 2\mathbb{Z},$$

with ${N_{ab}}^c = 0$ otherwise. The derivation and the Hopf-link normalization are given in Chapter 105; here the same data are read as the input controlling boundaries, condensable algebras, and line-defect junctions.

110.5 Boundary Categories for Chern–Simons Theory

For compact G Chern–Simons theory at level k, a Wilson line labelled by a representation R is

$$W_R(\gamma) = \operatorname{tr}_R \mathcal{P} \exp\!\left(\int_\gamma A\right).$$

Quantum gauge invariance restricts the admissible line labels. For simply connected compact simple G, the expected finite label set is the set of integrable highest weights at level k. Fusion of line defects is then encoded by the corresponding representation category of the affine algebra or quantum group at a root of unity, depending on the construction chosen.

The boundary chiral algebra is not an extra decoration. It is the algebraic object that describes states on a surface with boundary and line endpoints. The equality between bulk Chern–Simons amplitudes and boundary conformal blocks is a gluing statement relating two constructions of the same surface-state spaces.

Definition 110.6 (Lagrangian subgroup of an abelian Chern–Simons theory)**.** Let K be a non-degenerate integral symmetric $r \times r$ matrix defining an abelian Chern–Simons theory. The finite group of line labels is

$$A_K = \mathbb{Z}^r / K\mathbb{Z}^r.$$

For charge representatives $\ell, \ell' \in \mathbb{Z}^r$, define

$$b(\ell, \ell') = \ell^T K^{-1} \ell' \mod \mathbb{Z}, \qquad q(\ell) = \frac{1}{2} \ell^T K^{-1} \ell \mod \mathbb{Z}.$$

A subgroup $L \subset A_K$ is Lagrangian if $q|_L = 0$ and

$$L = L^\perp, \qquad L^\perp = \{a \in A_K : b(a, \ell) = 0 \text{ for every } \ell \in L\}.$$

Abelian Chern–Simons boundary algebra. In the pointed modular category of an abelian Chern–Simons theory, a Lagrangian subgroup $L \subset A_K$ determines a commutative separable algebra object

$$A_L = \bigoplus_{\ell \in L} \ell.$$

The condition $q|_L = 0$ is exactly the condition that the summands have trivial topological spin, and $L = L^\perp$ is the condition that no nontrivial bulk line has trivial monodromy with every condensed line. This follows from the finite semisimple algebra of the pointed category: the simple objects form an abelian group under fusion, here A_K. The braiding monodromy of charges ℓ, ℓ' is $\exp(2\pi i b(\ell, \ell'))$, and the twist of ℓ is $\exp(2\pi i q(\ell))$. The direct sum over a subgroup carries multiplication induced by the group law $L \times L \to L$. Commutativity of this multiplication requires trivial mutual braiding among condensed summands; this follows from $q|_L = 0$ by the quadratic identity

$$b(\ell, \ell') = q(\ell + \ell') - q(\ell) - q(\ell').$$

Separability is the finite semisimple averaging idempotent for the finite group L. A line a can pass through the boundary condensate without monodromy precisely when $a \in L^\perp$. If $L = L^\perp$, the only such transparent bulk lines are the condensed ones, so the boundary is a boundary to the trivial bulk rather than to another residual topological order.

Example 110.7 (Toric-code boundary subgroups)**.** For

$$K = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix},$$

the line group is $A_K \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2$. Writing the generators as $e = (1, 0)$ and $m = (0, 1)$, one has

$$q(e) = q(m) = 0, \qquad b(e, m) = \frac{1}{2}.$$

Thus

$$L_e = \{0, e\}, \qquad L_m = \{0, m\}$$

are Lagrangian subgroups, while $\{0, e+m\}$ is not because $q(e+m) = 1/2$. The two Lagrangian subgroups are the two elementary gapped boundaries: one condenses the electric line and the other condenses the magnetic line.

110.6 BV-BFV Boundary Mechanism

For a gauge theory on a manifold M with boundary, the BV variation has the form

$$\delta S_{\mathrm{BV}} = \iota_{Q_{\mathrm{BV}}}\omega_{\mathrm{BV}} + \pi^*_{\partial M}\alpha_\partial.$$

The boundary one-form α_∂ has differential

$$\omega_\partial = \delta\alpha_\partial.$$

The pair $(\mathcal{F}_\partial, \omega_\partial)$ is the BFV phase space, and the bulk BV vector field induces a boundary cohomological vector field when the master equation is compatible with the boundary term. A boundary condition is a Lagrangian subspace or derived Lagrangian object

$$\mathcal{L}_\partial \subset \mathcal{F}_\partial$$

on which the boundary variation vanishes. This is the local mechanism behind boundary categories in gauge-theoretic TQFTs.

110.7 Crane–Yetter and Walker–Wang Boundaries

Four-dimensional state-sum and lattice-Hamiltonian constructions make the boundary role of braided tensor categories especially visible. The fully general Crane–Yetter state sum and Walker–Wang Hamiltonian construction belong to a broad categorical theorem package. The local mechanism needed for the present monograph can be proved directly in the pointed case, which is already rich enough to distinguish bulk deconfinement from boundary topological order.

Definition 110.8 (Pointed braided input and its radical)**.** Let A be a finite abelian group. A pointed braided input is a quadratic function

$$q : A \to \mathbb{Q}/\mathbb{Z}$$

whose polarization

$$b(a, x) = q(a+x) - q(a) - q(x) \tag{110.2}$$

is a symmetric biadditive pairing $A \times A \to \mathbb{Q}/\mathbb{Z}$. The topological spin of the simple line a is $\theta_a = \exp(2\pi \mathrm{i} q(a))$, and the double braiding of a around x is

$$M(a, x) = \exp(2\pi \mathrm{i} b(a, x)).$$

The radical, or transparent subgroup, is

$$\mathrm{Rad}(A, b) = \{a \in A : \ b(a, x) = 0 \text{ in } \mathbb{Q}/\mathbb{Z} \text{ for every } x \in A\}. \tag{110.3}$$

The pointed input is nondegenerate when $\mathrm{Rad}(A, b) = 0$.

Construction 110.9 (Pointed Walker–Wang boundary mechanism)**.** Fix the pointed braided finite input of Definition 110.8. The finite Walker–Wang mechanism used here is the following algebraic lattice model. Oriented edge labels take values in A. Vertex terms enforce the group-law fusion constraint, and a plaquette move inserts a closed loop of label $x \in A$ and recouples it through the existing labels. When a candidate line of label a crosses that loop, the local move multiplies the state by the double-braiding phase

$$M(a,x) = \exp(2\pi \mathrm{i} b(a,x)). \tag{110.4}$$

In this finite mechanism a bulk line label a is called deconfined when its line insertion commutes with every plaquette recoupling away from its endpoints; equivalently, translating a segment of the line through the bulk leaves no string of plaquette violations. The construction therefore gives the finite criterion

$$a \text{ is deconfined} \quad \Longleftrightarrow \quad M(a,x) = 1 \quad \text{for every } x \in A \quad \Longleftrightarrow \quad a \in \mathrm{Rad}(A,b). \tag{110.5}$$

If b is nondegenerate, this finite algebraic model has no nontrivial deconfined bulk line. On a boundary where strings are allowed to end, the endpoint labels still carry the pointed braided category (A,q): fusion is addition in A, spins are $\exp(2\pi \mathrm{i} q(a))$, and double braiding is $\exp(2\pi \mathrm{i} b(a,x))$.

Verification of the finite criterion. The vertex constraint only uses the group law of A, so a closed plaquette loop can be slid through a candidate line one elementary crossing at a time. Each crossing contributes the phase (110.4). If $M(a,x) \neq 1$ for some x, the plaquette term with loop label x detects the line segment; moving the segment changes the local plaquette eigenvalue along its path, which is the finite-lattice meaning of confinement in this mechanism. If $M(a,x) = 1$ for all x, all plaquette recouplings away from the endpoints commute with the line insertion, so the energy cost is localized at the endpoints and the line is deconfined. The equality $M(a,x) = 1$ for every x is precisely $b(a,x) = 0$ in $\mathbb{Q}/\mathbb{Z}$ for every x, hence (110.5). A boundary removes the plaquette relation that would continue the detection surface outside the material, so a string may terminate there; the same local crossing calculation records the boundary spin and double-braiding data. Thus the finite mechanism separates a radical bulk spectrum from a boundary pointed braided category.

Example 110.10 (Toric-code input as a modular boundary order)**.** For $A = \mathbb{Z}_2 e \oplus \mathbb{Z}_2 m$, set

$$q(e) = q(m) = 0, \qquad q(e+m) = \frac{1}{2}.$$

Then $b(e,m) = 1/2$, while $b(e,e) = b(m,m) = 0$. The radical is zero:

$$\mathrm{Rad}(A,b) = \{0\}.$$

The pointed Walker–Wang mechanism therefore has no nontrivial deconfined bulk line, but the boundary still supports the toric-code anyons $\{1, e, m, e+m\}$ with the displayed mutual semionic braiding.

Definition 110.11 (Müger center)**.** Let $\mathcal{C}$ be a finite semisimple braided tensor category over $\mathbb{C}$, with braiding

$$c_{a,x} : a \otimes x \longrightarrow x \otimes a.$$

For simple objects a, x, the monodromy is the natural automorphism

$$M_{a,x} = c_{x,a}c_{a,x} : a \otimes x \longrightarrow a \otimes x.$$

The Müger center $\mathcal{Z}_2(\mathcal{C})$ is the full fusion subcategory generated by the simple objects a such that

$$M_{a,x} = \mathrm{id}_{a\otimes x} \qquad \text{for every simple } x \in \mathcal{C}. \tag{110.6}$$

The braided category is nondegenerate when $\mathcal{Z}_2(\mathcal{C})$ is generated only by the tensor unit.

Operational Müger-center criterion. The categorical input supplied by the Walker–Wang plaquette test is the local crossing datum between plaquette-loop recouplings and transverse line insertions. Assume a local plaquette algebra whose simple string labels are the simple objects of $\mathcal{C}$, and assume that dragging a plaquette loop labelled by x through a transverse simple line a acts on $a \otimes x$ by the monodromy

$$M_{a,x} = c_{x,a}c_{a,x}.$$

Let P_x be the elementary plaquette recoupling in which a small loop labelled by x is created, slid across the candidate line a, and absorbed. The part of P_x detected by the crossing with a is exactly $M_{a,x}$. Therefore a commutes with every local plaquette-loop recoupling precisely when

$$M_{a,x} = \mathrm{id}_{a\otimes x} \qquad \text{for every simple } x,$$

which is the defining condition (110.6) for

$$a \in \mathcal{Z}_2(\mathcal{C}).$$

Equivalently, the categorical candidates for bulk deconfined lines are the simple objects in the Müger center. If $\mathcal{C}$ is nondegenerate, this local plaquette-commutation test leaves only the tensor unit as a bulk deconfined line. Thus the criterion isolates the local algebraic datum that any concrete non-pointed Hamiltonian realization must implement before the global excitation-sector statement can be made.

Example 110.12 (Non-pointed check: the Ising braided category)**.** The Ising modular category has simple labels

$$1, \qquad \psi, \qquad \sigma,$$

with quantum dimensions $1, 1, \sqrt{2}$ and total dimension $\mathcal{D} = 2$. In the ordered basis $(1, \psi, \sigma)$, its modular matrix is

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}.$$

For a transparent simple object a in a modular category, the Hopf-link amplitude factorizes as

$$S_{ax} = \frac{d_a d_x}{\mathcal{D}} \qquad \text{for every simple } x. \tag{110.7}$$

The unit row satisfies this identity. The ψ row fails at $x = \sigma$, because

$$S_{\psi\sigma} = -\frac{\sqrt{2}}{2} \neq \frac{d_\psi d_\sigma}{\mathcal{D}} = \frac{\sqrt{2}}{2}.$$

The σ row fails already at $x = \psi$. Hence the Müger center is trivial. The Ising input is non-pointed and nondegenerate, so the local Walker–Wang plaquette criterion leaves no nontrivial bulk deconfined line, while the non-pointed Ising anyons may still appear as boundary line data when the boundary construction is supplied.

Remark 110.13 (General categorical status). For a general spherical, ribbon, or modular fusion category $\mathcal{C}$, the Crane–Yetter and Walker–Wang constructions require the full categorical state-sum or lattice-Hamiltonian theorem: pentagon and hexagon identities, Pachner-move invariance, spherical trace normalization, and the analysis of bulk excitation sectors. The monograph uses the general theorem as external mathematics until those categorical proofs are developed in full. The pointed theorem above proves the mechanism that is used conceptually here: bulk deconfined lines are controlled by the Müger center, and a nondegenerate braided input leaves its nontrivial anyons as boundary, not bulk, topological order.

110.8 Extended Assignment

An extended n-dimensional TQFT assigns:

$$Z(M^n) \in \mathbb{C}, \qquad Z(\Sigma^{n-1}) \in \mathrm{Vect},$$

$$Z(Y^{n-2}) \in \mathrm{Cat},$$

and higher categorical objects to lower-dimensional strata, in a target (∞, n)-category chosen before the assignment is made. Dualizability conditions ensure that gluing along all strata is represented by composition and evaluation maps. In a QFT realization, the target objects must be constructed from regulated state spaces, boundary operator algebras, and defect junction spaces.

Open Problem 110.14 (Gauge-theoretic extended construction). Construct the extended boundary and defect categories of four-dimensional cohomological gauge theories directly from regulated BV-BFV data, including instanton and monopole boundary strata, line and surface defects, gluing maps, orientations, and anomaly lines.

Chapter 111

BV Integration And Finite-Dimensional Localization

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Topological and cohomological field theories use two related operations. The first is BV integration: a semidensity on an odd symplectic space is restricted to a Lagrangian cycle and integrated. The second is cohomological localization: an integral of a closed density is deformed by an exact term and evaluated from the zero locus of the deformation. This chapter proves the finite-dimensional identities with all cycle, boundary, and determinant data visible. Infinite-dimensional gauge-theoretic applications in later chapters must supply a regulator, a replacement for compactness, determinant-line orientations, and control of singular strata.

111.1 Odd Symplectic BV Data

Let $\mathcal{F}$ be a finite-dimensional smooth graded manifold. Its fermionic coordinates are coordinates of the structure sheaf; they are not points of an underlying ordinary set. An odd symplectic form of cohomological degree -1 is a closed nondegenerate two-form

$$\omega \in \Omega^2(\mathcal{F}), \qquad |\omega| = -1.$$

For a homogeneous function $F \in C^\infty(\mathcal{F})$, the Hamiltonian vector field X_F is defined by

$$\iota_{X_F}\omega = \delta F.$$

The BV bracket is the odd Poisson bracket

$$\{F,G\} = X_F(G).$$

It has cohomological degree 1, satisfies the graded Jacobi identity, and obeys the graded Leibniz rule in each argument. If S is an even function of degree zero, the vector field

$$Q_S = \{S,-\}$$

has degree 1. The classical master equation is

$$\{S,S\} = 0. \tag{111.1}$$

The Jacobi identity gives

$$Q_S^2 F = \frac{1}{2}\{\{S,S\},F\},$$

so (111.1) is equivalent to $Q_S^2 = 0$ on functions.

In local Darboux coordinates (x^i,ξ_i), where

$$|x^i| + |\xi_i| = 1, \qquad \omega = \sum_i \delta x^i\,\delta\xi_i,$$

the bracket is the coordinate expression determined by the preceding Hamiltonian convention. A choice of Berezinian density μ compatible with these coordinates gives a BV Laplacian on functions,

$$\Delta_\mu = \sum_i (-1)^{|x^i|}\frac{\partial^2}{\partial x^i\,\partial\xi_i}. \tag{111.2}$$

Intrinsic BV integration is most cleanly stated for semidensities. In a coordinate chart a semidensity is written $f\,\mu^{1/2}$; the canonical odd-symplectic operator on semidensities is denoted Δ. After the choice of μ, it is represented by Δ_μ on the coefficient function f. The price of working with functions is that a change of μ changes the displayed operator.

111.2 Quantum Master Equation And Observables

Let ρ be a semidensity on $\mathcal{F}$. The finite-dimensional quantum master equation is

$$\Delta\rho = 0. \tag{111.3}$$

If $\rho = \exp(S/\hbar)\mu^{1/2}$ in a chosen compatible Berezinian trivialization, then (111.3) becomes

$$\frac{1}{2}\{S,S\} + \hbar\Delta_\mu S = 0. \tag{111.4}$$

Indeed, applying the second-order operator Δ_μ to $\exp(S/\hbar)$ gives

$$\Delta_\mu\left(\mathrm{e}^{S/\hbar}\right) = \hbar^{-1}\left(\Delta_\mu S + \frac{1}{2\hbar}\{S,S\}\right)\mathrm{e}^{S/\hbar},$$

which is equivalent to (111.4).

An observable is a function $\mathcal{O}$ for which

$$\Delta(\mathcal{O}\rho) = 0.$$

In the same trivialization, if S obeys (111.4), this condition reads

$$\Delta_\mu \mathcal{O} + \hbar^{-1}\{S, \mathcal{O}\} = 0. \tag{111.5}$$

The classical limit of (111.5) is $\{S, \mathcal{O}\} = 0$, the Q_S-closedness condition.

Remark 111.1 (BV product identity). For a homogeneous function F and an action S satisfying (111.4),

$$\Delta_\mu(F e^{S/\hbar}) = \left(\Delta_\mu F + \hbar^{-1}\{S, F\}\right) e^{S/\hbar}.$$

The defining second-order identity for Δ_μ is

$$\Delta_\mu(FG) = (\Delta_\mu F)G + (-1)^{|F|} F \Delta_\mu G + (-1)^{|F|}\{F, G\}.$$

Take $G = e^{S/\hbar}$. Since S is even,

$$\{F, e^{S/\hbar}\} = \hbar^{-1}\{F, S\} e^{S/\hbar}.$$

The graded antisymmetry of the odd bracket converts the last term to $\hbar^{-1}\{S, F\}e^{S/\hbar}$ with the displayed convention. The term containing $\Delta_\mu G$ vanishes by the quantum master equation for $e^{S/\hbar}\mu^{1/2}$. This proves the identity.

111.3 Lagrangian Cycles And BV Stokes

A Lagrangian submanifold $\mathcal{L} \subset \mathcal{F}$ is a graded submanifold for which the pullback of ω vanishes and whose dimension is half the dimension of $\mathcal{F}$ in the graded sense. The restriction of a semidensity ρ to $\mathcal{L}$ is an ordinary Berezinian density on $\mathcal{L}$, denoted $\rho|_{\mathcal{L}}$. The BV expectation of an observable is

$$\langle \mathcal{O} \rangle_{\mathcal{L}} = \int_{\mathcal{L}} (\mathcal{O}\rho)|_{\mathcal{L}}.$$

The Lagrangian $\mathcal{L}$ is the gauge-fixing cycle.

Theorem 111.2 (Finite-dimensional BV Stokes theorem)**.** *Let σ be a compactly supported semidensity on an odd symplectic finite-dimensional graded manifold $\mathcal{F}$, and let $\mathcal{L} \subset \mathcal{F}$ be an oriented Lagrangian submanifold without boundary. Then*

$$\int_{\mathcal{L}} (\Delta\sigma)|_{\mathcal{L}} = 0.$$

If $\mathcal{L}$ has boundary, the same calculation gives the induced boundary integral on $\partial\mathcal{L}$.

Proof. Cover the compact support of $\sigma|_{\mathcal{L}}$ by Darboux–Weinstein charts adapted to $\mathcal{L}$. In such a chart the coordinates are (x^i, ξ_i), the odd symplectic form is $\sum_i dx^i\, d\xi_i$, and $\mathcal{L}$ is given by $\xi_i = 0$. Write the semidensity as

$$\sigma = f(x, \xi)\, \mu^{1/2}.$$

In this chart the canonical BV operator on semidensities is

$$\Delta\sigma = \left(\sum_i (-1)^{|x^i|} \frac{\partial^2 f}{\partial x^i\, \partial \xi_i}\right) \mu^{1/2}.$$

After restriction to $\mathcal{L}$, this is the ordinary divergence of the vector density

$$V = \sum_i (-1)^{|x^i|} \left.\frac{\partial f}{\partial \xi_i}\right|_{\xi=0} \iota_{\partial/\partial x^i}\, \mathrm{d}x,$$

because

$$\mathrm{d}_{\mathcal{L}} V = \sum_i (-1)^{|x^i|} \left.\frac{\partial^2 f}{\partial x^i\, \partial \xi_i}\right|_{\xi=0} \mathrm{d}x.$$

Choose a partition of unity on $\mathcal{L}$ subordinate to the adapted charts. Applying the preceding calculation to each partitioned semidensity gives an ordinary divergence on each chart. The local boundary contributions on chart overlaps cancel because the canonical BV operator on semidensities and the restricted density are invariant under adapted canonical coordinate changes. Hence the only remaining boundary term is the geometric boundary of $\mathcal{L}$. If $\partial\mathcal{L} = \varnothing$ the integral vanishes. If $\partial\mathcal{L}$ is present, the same calculation gives the induced ordinary Stokes integral $\int_{\partial\mathcal{L}} V$. □

Remark 111.3 (Gauge-fixing independence). Let ρ satisfy $\Delta\rho = 0$, and let $\mathcal{O}$ satisfy $\Delta(\mathcal{O}\rho) = 0$. Let $\mathcal{L}_s$, $s \in [0,1]$, be a smooth family of compact Lagrangian cycles such that the swept chain has no boundary except $\mathcal{L}_1 - \mathcal{L}_0$. Then

$$\int_{\mathcal{L}_1} (\mathcal{O}\rho)|_{\mathcal{L}_1} = \int_{\mathcal{L}_0} (\mathcal{O}\rho)|_{\mathcal{L}_0}.$$

The family $\mathcal{L}_s$ sweeps a Lagrangian cobordism, so this is the direct finite-dimensional consequence of Theorem 111.2 applied to the closed semidensity $\mathcal{O}\rho$. The hypotheses $\Delta\rho = 0$ and $\Delta(\mathcal{O}\rho) = 0$ are the finite-dimensional BV analogues of the quantum master equation and the observable Ward identity; they are exactly what makes the integrand closed for the Stokes argument.

111.4 BV Pushforward

Let the odd symplectic space split as

$$\mathcal{F} = \mathcal{F}_{\mathrm{res}} \times \mathcal{F}_{\mathrm{fluc}}, \qquad \omega = \omega_{\mathrm{res}} + \omega_{\mathrm{fluc}}.$$

The first factor contains residual fields and the second factor contains fluctuation fields to be integrated out. Let $\mathcal{L}_{\mathrm{fluc}} \subset \mathcal{F}_{\mathrm{fluc}}$ be a Lagrangian cycle. For a semidensity ρ on $\mathcal{F}$, define the BV pushforward

$$(\pi_*\rho)(\phi_{\mathrm{res}}) = \int_{\mathcal{L}_{\mathrm{fluc}}} \rho(\phi_{\mathrm{res}}, \phi_{\mathrm{fluc}})\big|_{\mathcal{L}_{\mathrm{fluc}}}.$$

The result is a semidensity on $\mathcal{F}_{\mathrm{res}}$.

Theorem 111.4 (QME preservation by BV pushforward)**.** *Assume that ρ has compact support in the fluctuation directions, or that the chosen fluctuation cycle has decay strong enough to eliminate the boundary term in Theorem 111.2. If*

$$\Delta_{\mathcal{F}}\rho = 0,$$

then

$$\Delta_{\mathrm{res}}(\pi_*\rho) = 0.$$

The same statement holds for observables:

$$\Delta_{\mathcal{F}}(\mathcal{O}\rho) = 0 \quad \Longrightarrow \quad \Delta_{\mathrm{res}}\pi_*(\mathcal{O}\rho) = 0.$$

Proof. Work in product Darboux coordinates

$$(x^i, \xi_i; y^a, \eta_a)$$

adapted to $\mathcal{F}_{\mathrm{res}} \times \mathcal{F}_{\mathrm{fluc}}$, with (x^i, ξ_i) residual and (y^a, η_a) fluctuation coordinates. The canonical BV semidensity Laplacian is local in these coordinates, and the product Berezinian contains no mixed residual-fluctuation term. Hence

$$\Delta_{\mathcal{F}} = \Delta_{\mathrm{res}} + \Delta_{\mathrm{fluc}}.$$

The residual Laplacian differentiates only (x, ξ), so, under the compact support or decay hypothesis, it may be moved through the fluctuation integral. In a local trivialization of the semidensity line this is ordinary differentiation under the integral sign; coordinate independence follows from the transformation law of the BV Laplacian on semidensities. Therefore

$$\Delta_{\mathrm{res}}\pi_*\rho = \int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{res}}\rho)|_{\mathcal{L}_{\mathrm{fluc}}} = -\int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{fluc}}\rho)|_{\mathcal{L}_{\mathrm{fluc}}}.$$

The last integral is zero by Theorem 111.2 applied to the fluctuation BV manifold and the cycle $\mathcal{L}_{\mathrm{fluc}}$. The hypotheses in the statement are exactly the conditions needed for this Stokes step: either there is no boundary contribution because the fluctuation support is compact, or the chosen noncompact cycle has decay strong enough for the boundary functional to vanish.

If $\Delta_{\mathcal{F}}(\mathcal{O}\rho) = 0$, the same argument is applied to the semidensity $\mathcal{O}\rho$. No new algebraic input is used: $\mathcal{O}\rho$ is simply the semidensity being pushed forward, and the product decomposition gives

$$\Delta_{\mathrm{res}}\pi_*(\mathcal{O}\rho) = -\int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{fluc}}(\mathcal{O}\rho))|_{\mathcal{L}_{\mathrm{fluc}}},$$

and the fluctuation BV Stokes theorem again makes the right hand side vanish. □

Definition 111.5 (Boundary functional of a fluctuation cycle)**.** Let $\mathcal{L}_{\mathrm{fluc}}$ be a fluctuation Lagrangian cycle with boundary. For a semidensity $\sigma = f\,\mu^{1/2}$ on $\mathcal{F}_{\mathrm{fluc}}$, define $\mathcal{B}_{\partial\mathcal{L}_{\mathrm{fluc}}}(\sigma)$ by

$$\int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{fluc}}\sigma)|_{\mathcal{L}_{\mathrm{fluc}}} = \mathcal{B}_{\partial\mathcal{L}_{\mathrm{fluc}}}(\sigma). \tag{111.6}$$

In a Darboux chart in which $\mathcal{L}_{\mathrm{fluc}} = \{\xi_i = 0\}$, this boundary functional is the ordinary Stokes boundary integral of the vector density with components

$$(-1)^{|x^i|} \left.\frac{\partial f}{\partial \xi_i}\right|_{\xi=0}. \tag{111.7}$$

The coordinate-independence is precisely the boundary version of Theorem 111.2.

Proposition 111.6 (BV pushforward with fluctuation boundary)**.** *Let $\mathcal{F} = \mathcal{F}_{\mathrm{res}} \times \mathcal{F}_{\mathrm{fluc}}$ and let $\mathcal{L}_{\mathrm{fluc}} \subset \mathcal{F}_{\mathrm{fluc}}$ be a fluctuation Lagrangian cycle with boundary. Assume that all integrations converge and that ρ obeys the full finite-dimensional QME $\Delta_{\mathcal{F}}\rho = 0$. Then the pushed semidensity satisfies*

$$\Delta_{\mathrm{res}}(\pi_*\rho) = -\mathcal{B}_{\partial\mathcal{L}_{\mathrm{fluc}}}(\rho). \tag{111.8}$$

Thus BV pushforward preserves the QME if and only if the displayed boundary functional vanishes or is cancelled by an additional boundary, residue, or regulator contribution. If $\mathcal{O}$ is a quantum observable in the finite-dimensional sense that $\Delta_{\mathcal{F}}(\mathcal{O}\rho) = 0$, the same formula holds with ρ replaced by $\mathcal{O}\rho$.

Proof. As in Theorem 111.4, the product Darboux/Berezinian convention gives

$$\Delta_{\mathcal{F}} = \Delta_{\mathrm{res}} + \Delta_{\mathrm{fluc}}.$$

The residual BV Laplacian differentiates only the residual variables, so the convergence hypothesis permits it to pass through the fluctuation integral:

$$\Delta_{\mathrm{res}}\pi_*\rho = \int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{res}}\rho)|_{\mathcal{L}_{\mathrm{fluc}}} = -\int_{\mathcal{L}_{\mathrm{fluc}}} (\Delta_{\mathrm{fluc}}\rho)|_{\mathcal{L}_{\mathrm{fluc}}}. \tag{111.9}$$

The second equality uses the full QME to replace $\Delta_{\mathrm{res}}\rho$ by $-\Delta_{\mathrm{fluc}}\rho$. Definition 111.5 identifies the last integral with the fluctuation-boundary functional, including its Darboux-coordinate Stokes expression and its coordinate-independent semidensity meaning. This proves (111.8). The observable statement is identical after replacing ρ by $\mathcal{O}\rho$ and using $\Delta_{\mathcal{F}}(\mathcal{O}\rho) = 0$. □

In field theory this theorem is the finite-dimensional model for Wilsonian integration of BV fields. A regulator must provide a finite-dimensional replacement or a trace-class analogue of the split above. If $\Delta\rho = \mathcal{A}\rho$, the semidensity $\mathcal{A}\rho$ is the finite-dimensional anomaly insertion; the pushforward preserves the anomaly class rather than producing a solution of the quantum master equation.

111.5 Singular Strata As BV Boundary Data

Gauge-theoretic localization often integrates over a smooth moduli stratum whose natural compactification has additional strata. Let $\mathcal{M}^{\mathrm{sm}}$ denote the smooth locus and let

$$\overline{\mathcal{M}} = \mathcal{M}^{\mathrm{sm}} \sqcup \bigsqcup_{\alpha} S_\alpha \tag{111.10}$$

be a stratified compactification. In applications S_α may be a reducible-connection stratum, a singular-monopole stratum, or the small-instanton stratum of an Uhlenbeck compactification. A BV localization argument over $\mathcal{M}^{\mathrm{sm}}$ is not complete until it states the boundary functional produced by the strata S_α.

Definition 111.7 (Boundary-cancellation data)**.** For a regulated BV localization problem with smooth fixed locus $\mathcal{M}^{\mathrm{sm}}$ and compactification (111.10), boundary-cancellation data consists of one of the following:

1. vanishing of every stratum functional, $\mathcal{B}_{S_\alpha}(\rho) = 0$;
2. a relative integration cycle whose boundary cancels $\sum_\alpha \mathcal{B}_{S_\alpha}(\rho)$;
3. a resolved regulated fixed locus $\widetilde{\mathcal{M}} \to \overline{\mathcal{M}}$ with an induced semidensity $\widetilde{\rho}$;
4. a finite-dimensional contour or residue rule whose Stokes boundary equals $-\sum_\alpha \mathcal{B}_{S_\alpha}(\rho)$.

In the resolution case the necessary equality is the identity of pushed residual semidensities

$$\pi_*^{\widetilde{\mathcal{M}}}\widetilde{\rho} = \pi_*^{\mathcal{M}^{\mathrm{sm}}}\rho + \sum_\alpha C_\alpha, \qquad \Delta_{\mathrm{res}} C_\alpha = \mathcal{B}_{S_\alpha}(\rho), \tag{111.11}$$

where C_α is the contribution supported over the exceptional or relative piece replacing S_α. Equation (111.11) is data to be proved from the regulator; it is not implied by writing a compactification symbol.

Singular-stratum obstruction to the localized QME. Assume that a regulated gauge-theoretic BV integral has been localized to a smooth finite-dimensional fixed locus $\mathcal{M}^{\mathrm{sm}}$ and that the full regulated semidensity obeys the QME before the limiting compactification is taken. If the compactification of $\mathcal{M}^{\mathrm{sm}}$ contains strata S_α, then the residual localized semidensity obeys

$$\Delta_{\mathrm{res}}(\pi_*^{\mathcal{M}^{\mathrm{sm}}}\rho) = -\sum_\alpha \mathcal{B}_{S_\alpha}(\rho). \tag{111.12}$$

Therefore the localized expression is a BV pushforward satisfying the QME only after boundary-cancellation data in the sense of Definition 111.7 have been supplied and verified.

Choose a collar neighborhood of each singular stratum in the compactified moduli problem and remove a tubular neighborhood of radius ϵ. The resulting truncated fixed locus $\mathcal{M}_\epsilon^{\mathrm{sm}}$ is a manifold with boundary. Applying Proposition 111.6 gives

$$\Delta_{\mathrm{res}}(\pi_*^{\mathcal{M}_\epsilon^{\mathrm{sm}}}\rho) = -\sum_\alpha \mathcal{B}_{\partial_\alpha \mathcal{M}_\epsilon^{\mathrm{sm}}}(\rho). \tag{111.13}$$

If the limit $\epsilon \to 0$ exists as a distributional or semidensity limit, the boundary functionals converge to the stratum functionals $\mathcal{B}_{S_\alpha}(\rho)$, giving (111.12). If the limit does not exist, the localization formula itself is not yet defined. A vanishing, relative-cycle, resolution, or residue input is exactly the additional input that makes the right hand side vanish or cancel in the residual QME.

Collar form of the stratum functional. The preceding obstruction is useful only when the stratum functional is a specified object. A collar description supplies such an object. Assume that, near a stratum S_α, the compactified fixed locus is equipped with a defining function $r \geq 0$, a link fibration

$$L_\alpha \longrightarrow S_\alpha,$$

and boundary faces

$$L_{\alpha,\epsilon} = \{r = \epsilon\} \subset \mathcal{M}^{\mathrm{sm}}$$

with the boundary orientation inherited from the truncated space $\mathcal{M}_\epsilon^{\mathrm{sm}} = \{r \geq \epsilon\}$. Let V_ρ^{bdry} be the ordinary boundary density obtained from Definition 111.5. The stratum functional is then the residual semidensity limit

$$\mathcal{B}_{S_\alpha}(\rho) = \lim_{\epsilon \downarrow 0} \int_{L_{\alpha,\epsilon}} V_\rho^{\mathrm{bdry}}, \tag{111.14}$$

whenever this limit exists in the topology used for residual semidensities. Thus the words "compactification", "resolution", and "residue prescription" acquire content only after the collar, orientation, boundary density, and limiting topology have been stated.

The collar also gives a practical vanishing/residue test. Suppose, in a norm on residual semidensities, that the boundary integral has an asymptotic expansion

$$\int_{L_{\alpha,\epsilon}} V_\rho^{\mathrm{bdry}} = \epsilon^{\eta_\alpha} B_{\alpha,0} + O(\epsilon^{\eta_\alpha+\delta}), \qquad \delta > 0. \tag{111.15}$$

If $\eta_\alpha > 0$, the stratum contributes zero to the residual QME. If $\eta_\alpha = 0$, the coefficient $B_{\alpha,0}$ is the finite boundary residue that must vanish, cancel against other strata, or be represented by a relative/resolved contribution C_α. If $\eta_\alpha < 0$, the localized expression has a divergent boundary term; the proposed localization formula is then undefined until the regulator or integration cycle is changed. Logarithmic terms are treated by the same rule: every coefficient of $\epsilon^0(\log\epsilon)^k$ is boundary data, and powers with negative η_α signal divergent boundary data.

For the S^4 instanton factors of Chapter 101, the Uhlenbeck small-instanton strata are precisely such S_α, with the instanton scale playing the role of the collar coordinate r. Replacing the Uhlenbeck compactification by the Gieseker/Nekrasov resolution supplies a candidate C_α in (111.11). The statement that the original four-dimensional path integral selects those resolved data is therefore a regulator theorem or a hypothesis, as stated in Hypothesis 101.17, rather than a consequence of formal Q-exact deformation alone.

111.6 Cohomological Localization

Let M be an oriented finite-dimensional manifold and let $\mathcal{A}$ be a finite-dimensional graded-commutative algebra. An integration functional

$$\int_M : \Omega^\bullet(M) \otimes \mathcal{A} \longrightarrow \mathbb{C}$$

is obtained by taking the top ordinary form degree and the chosen Berezin coefficient in $\mathcal{A}$. Let Q be an odd derivation of $\Omega^\bullet(M) \otimes \mathcal{A}$ such that $Q^2 = 0$. Assume that the cycle and boundary conditions give

$$\int_M Q\beta = 0 \tag{111.16}$$

for every integrand β in the class being integrated.

Let S and $\mathcal{O}$ be even Q-closed elements and let V be odd. Define

$$I(t) = \int_M \mathcal{O}\, \mathrm{e}^{-S-tQV}.$$

Remark 111.8 (Q-exact deformation identity). Under the hypotheses above,

$$\frac{dI}{dt} = 0.$$

Differentiate under the integral:

$$\frac{dI}{dt} = -\int_M \mathcal{O}\,(QV)\,\mathrm{e}^{-S-tQV}.$$

Since $Q\mathcal{O} = QS = Q^2V = 0$, the graded Leibniz rule gives

$$Q\big(V\mathcal{O}\,\mathrm{e}^{-S-tQV}\big) = (QV)\mathcal{O}\,\mathrm{e}^{-S-tQV}.$$

Thus dI/dt is minus the integral of a Q-exact term, and (111.16) gives the result.

The identity has content only after (111.16) has been verified. For compact M and $Q = \mathrm{d}$, it is Stokes' theorem. For noncompact cycles it is a decay estimate. For contour prescriptions it is a statement about the chosen relative homology class.

111.7 One-Loop Normal Form

Assume the real part of QV is nonnegative and its zero locus is a compact submanifold $F \subset M$. Let N_F be the normal bundle of F. In a tubular neighborhood write y for even normal coordinates and χ for their odd partners. Suppose the quadratic normal part is

$$QV = \frac{1}{2}\,y^T Ay + \frac{1}{2}\,\chi^T B\chi + O(3), \tag{111.17}$$

where A is a positive definite symmetric operator on the even normal bundle and B is a nondegenerate antisymmetric operator on the odd normal bundle. The orientation of the normal integration cycle fixes the branch of $\sqrt{\det A}$ and the sign of $\mathrm{Pf}(B)$.

Normal Gaussian factor. If the limit $t \to \infty$ may be exchanged with the integral after the standard rescaling $y = t^{-1/2}u$, then the normal variables contribute

$$\frac{\mathrm{Pf}(B)}{\sqrt{\det A}}.$$

Consequently the localized integral has the finite-dimensional form

$$I = \int_F \frac{\iota_F^*(\mathcal{O}\,\mathrm{e}^{-S})}{e_Q(N_F)},$$

where $e_Q(N_F)$ is the equivariant Euler class represented by the normal complex.

The even normal integral is

$$\int_{\mathbb{R}^r} \exp\!\Big(-\frac{t}{2}y^T Ay\Big)\,\mathrm{d}^r y = (2\pi)^{r/2} t^{-r/2} (\det A)^{-1/2}.$$

The odd normal integral is

$$\int \mathrm{d}\chi_1 \cdots \mathrm{d}\chi_{2s}\, \exp\!\Big(-\frac{t}{2}\chi^T B\chi\Big) = t^s\,\mathrm{Pf}(B)$$

with the Berezin orientation fixed by the ordering of the χ's. In a cohomological normal complex the odd rank equals the even rank after pairing by Q, so the powers of t cancel once the ordinary normalization of the Thom density is included. The remaining determinant is $\mathrm{Pf}(B)/\sqrt{\det A}$. The higher-order terms in (111.17) vanish in the rescaled limit under the stated dominated-convergence hypothesis. Gluing these local normal factors over F gives the inverse equivariant Euler class of N_F.

111.8 Mathai–Quillen Rank-One Model

The following model fixes the normalization used in localization. Let

$$s_a : \mathbb{R} \longrightarrow \mathbb{R}, \qquad s_a(x) = ax, \qquad a \neq 0,$$

be a section of the oriented trivial real line bundle over $\mathbb{R}$. For $t > 0$, define the one-form

$$\Theta_{t,a} = \left(\frac{t}{2\pi}\right)^{1/2} a \exp\left(-\frac{ta^2x^2}{2}\right) \mathrm{d}x.$$

It is the rank-one Mathai–Quillen Thom density pulled back by s_a. Its integral is

$$\int_{\mathbb{R}} \Theta_{t,a} = \operatorname{sign}(a). \tag{111.18}$$

Indeed, setting $u = \sqrt{t}\,|a|x$ gives

$$\left(\frac{t}{2\pi}\right)^{1/2} a\,\mathrm{d}x = \operatorname{sign}(a)\frac{\mathrm{d}u}{\sqrt{2\pi}},$$

and

$$\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-u^2/2} \mathrm{d}u = 1.$$

This proves (111.18). The sign is the local degree of the section at its zero $x = 0$, namely the orientation sign of the derivative $\mathrm{d}s_a = a\,\mathrm{d}x$.

For a rank-r oriented Euclidean vector bundle $E \to M$ with connection ∇, section s, and curvature R_∇, the Mathai–Quillen representative is obtained by replacing the single odd variable above by odd fiber variables χ^a and using the exponent

$$-\frac{t}{2}\|s\|^2 + \mathrm{i}\sqrt{t}\,\chi_a(\nabla s)^a + \frac{1}{2}\,\chi_a(R_\nabla)^{ab}\chi_b.$$

The Berezin integral over the χ's gives a closed differential form whose cohomology class is independent of t. When the zero locus of s is transverse, the $t \to \infty$ limit is the signed integration current on $s^{-1}(0)$. The rank-one calculation above is the local normal calculation at a transverse zero.

111.9 Atiyah–Bott Fixed-Point Model on the Two-Sphere

The preceding normal determinant has a visible fixed-point form in the two-sphere model. Write S^2 with coordinates

$$x \in [-1, 1], \qquad \varphi \in \mathbb{R}/2\pi\mathbb{Z}$$

away from the two poles, and orient it by

$$\omega = \mathrm{d}\varphi \wedge \mathrm{d}x.$$

Then $\int_{S^2} \omega = 4\pi$. Let the circle act by $\varphi \mapsto \varphi + \alpha$, and let $V = \partial_\varphi$ be its generating vector field. With equivariant parameter λ, define

$$\mathrm{d}_\lambda := \mathrm{d} - \lambda\, \iota_V, \qquad H = x.$$

Since $\iota_V \omega = \mathrm{d}x$, the equivariant two-form

$$\Omega_\lambda := \omega + \lambda H$$

is closed:

$$\mathrm{d}_\lambda \Omega_\lambda = \lambda\, \mathrm{d}x - \lambda\, \iota_V \omega = 0.$$

Fixed-point formula in the S^2 model. For $\lambda \neq 0$,

$$\int_{S^2} \mathrm{e}^{\Omega_\lambda} = 2\pi\, \frac{\mathrm{e}^{\lambda} - \mathrm{e}^{-\lambda}}{\lambda}. \tag{111.19}$$

The same value is obtained from the two fixed points of the circle action:

$$2\pi \left(\frac{\mathrm{e}^{\lambda H(N)}}{\lambda w_N} + \frac{\mathrm{e}^{\lambda H(S)}}{\lambda w_S} \right), \qquad H(N) = 1, \quad H(S) = -1, \quad w_N = 1, \quad w_S = -1.$$

Only the top ordinary form degree contributes to the integral:

$$\mathrm{e}^{\Omega_\lambda} = \mathrm{e}^{\lambda x}(1 + \omega), \qquad [\mathrm{e}^{\Omega_\lambda}]_{\mathrm{top}} = \mathrm{e}^{\lambda x} \omega.$$

Therefore

$$\int_{S^2} \mathrm{e}^{\Omega_\lambda} = \int_0^{2\pi} \mathrm{d}\varphi \int_{-1}^{1} \mathrm{e}^{\lambda x} \mathrm{d}x = 2\pi\, \frac{\mathrm{e}^{\lambda} - \mathrm{e}^{-\lambda}}{\lambda}.$$

Near the north pole, use a complex coordinate z in which the circle acts by $z \mapsto \mathrm{e}^{\mathrm{i}\alpha} z$. The tangent weight is $w_N = 1$, and $H(N) = 1$. Near the south pole the coordinate z^{-1} has tangent weight $w_S = -1$, and $H(S) = -1$. The one-dimensional normal Gaussian factor is the inverse equivariant Euler factor $(\lambda w_p)^{-1}$ at each fixed point p, multiplied by the angular normalization 2π. Summing the two contributions gives exactly (111.19).

This model is finite-dimensional and compact, so no boundary term is hidden. In gauge-theory localization the fixed set is replaced by a moduli space of solutions to field equations, the weights become determinant lines of elliptic complexes, and compactness or a relative-cycle prescription must be proved before the fixed-point formula has the status of a field-theoretic identity.

111.10 Boundary And Noncompact Terms

If the integration cycle has boundary or noncompact ends, the derivative in Remark 111.8 is the corresponding boundary integral:

$$\frac{dI}{dt} = -\int_{\partial M_{\mathrm{eff}}} \text{the boundary value of } V\mathcal{O}\, \mathrm{e}^{-S - tQV}.$$

A localization statement must therefore state one of the following pieces of data:

a compact integration cycle,	a decay estimate at every noncompact end,
a boundary condition cancelling the term,	a specified relative cycle or residue prescription.

In gauge-theoretic localization, Coulomb-branch ends, reducible connections, small instantons, singular monopole strata, and boundaries of moduli spaces are exactly the loci where this analysis enters. A residue prescription such as a Jeffrey–Kirwan rule is part of the integration-cycle data; it is a finite-dimensional replacement for a boundary analysis, and its validity in a field-theoretic construction must be proved from the chosen regulator and cycle.

Open Problem 111.9 (Infinite-dimensional BV localization)*.* Give theorem-level localization statements for cohomological gauge theories from a regulated BV-BFV functional integral, including integration cycles, compactness or boundary estimates, zero modes, determinant lines, singular strata, and gluing compatibility.

Chapter 112

Finite Gauge Theory and State-Sum TQFT

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Finite gauge theory is the cleanest nonperturbative laboratory for topological path integrals. The field spaces are finite groupoids, the "measure" is groupoid cardinality, the state spaces are finite-dimensional vector spaces of functions or sections over boundary groupoids, and gluing is Fubini's theorem for finite homotopy fibers. This chapter develops the untwisted theory and its Dijkgraaf–Witten cocycle twists with enough detail that later references to finite topological gauge theory have a precise meaning.

Throughout this chapter G is a finite group. A principal G-bundle means a right principal bundle. All manifolds are compact, smooth, oriented when orientation is needed, and equipped with collars at boundary components. If X is a finite groupoid, $\pi_0 X$ denotes the set of isomorphism classes of objects and $\mathrm{Aut}_X(x)$ denotes the automorphism group of an object x.

Definition 112.1 (Finite gauge state-sum datum)**.** Fix a dimension D. A finite G-gauge state-sum datum is a tuple

$$\mathfrak{Z}^{(D)}_{G,\omega} = \big(G, [\omega], M \mapsto \mathrm{Bun}_G(M), \mathcal{L}_\omega, \mathcal{H}_{G,\omega}, Z_{G,\omega}, K \mapsto \mathrm{Flat}_G(K)\big)$$

with the following entries.

(i) G is a finite group, and $[\omega] \in H^D(BG, U(1))$ is the Dijkgraaf–Witten twist. The untwisted theory is the case $[\omega] = 0$.

(ii) For every compact collared D-manifold M and closed $(D-1)$-manifold Σ, $\mathrm{Bun}_G(M)$ and $\mathrm{Bun}_G(\Sigma)$ are the finite groupoids of right principal G-bundles and bundle isomorphisms.

(iii) $\mathcal{L}_\omega(\Sigma)$ is the transgressed boundary line over $\mathrm{Bun}_G(\Sigma)$. In the untwisted theory this line is canonically trivial.

(iv) The boundary state space is

$$\mathcal{H}_{G,\omega}(\Sigma) = \Gamma\big(\mathrm{Bun}_G(\Sigma), \mathcal{L}_\omega(\Sigma)\big),$$

with the groupoid-cardinality pairing. In the untwisted case this is the vector space of functions on $\pi_0 \mathrm{Bun}_G(\Sigma)$.

(v) A bordism $M : \Sigma_0 \to \Sigma_1$ defines the restriction span

$$\mathrm{Bun}_G(\Sigma_0) \xleftarrow{r_0} \mathrm{Bun}_G(M) \xrightarrow{r_1} \mathrm{Bun}_G(\Sigma_1),$$

and $Z_{G,\omega}(M)$ is the corresponding push-pull operator, weighted by the cocycle phase and by automorphism denominators.

(vi) For a branched triangulation K of M, $\mathrm{Flat}_G(K)$ is the finite set of flat edge labelings whose action groupoid $\mathrm{Flat}_G(K)//G^{K_0}$ presents $\mathrm{Bun}_G(M)$. The simplicial state sum is a representative of the intrinsic groupoid integral.

The gluing theorem below proves that these entries define a symmetric monoidal TQFT on the corresponding collared bordism category.

112.1 Finite Groupoids and the Gauge Volume

Definition 112.2 (Finite groupoid cardinality)**.** A finite groupoid is a groupoid with finitely many isomorphism classes and finite automorphism groups. Its cardinality is

$$|X| = \sum_{[x]\in\pi_0 X} \frac{1}{|\operatorname{Aut}_X(x)|}. \tag{112.1}$$

If $F : X \to \mathbb{C}$ is a function constant on isomorphism classes, its integral over X is

$$\int_X F = \sum_{[x]\in\pi_0 X} \frac{F(x)}{|\operatorname{Aut}_X(x)|}. \tag{112.2}$$

The factor $1/|\operatorname{Aut}(x)|$ is the finite gauge-volume factor. It is forced by gluing. Omitting it gives the wrong cylinder and the wrong answer for a quotient with stabilizers.

Action groupoid cardinality. Let a finite group H act on a finite set X. The action groupoid $X//H$ has objects X and morphisms $x \to hx$ labeled by $h \in H$. Its groupoid cardinality is

$$|X//H| = \frac{|X|}{|H|}. \tag{112.3}$$

The isomorphism classes are the orbits $\mathcal{O} \subset X$. For $x \in \mathcal{O}$, the automorphism group is the stabilizer H_x. Thus

$$|X//H| = \sum_{\mathcal{O}} \frac{1}{|H_x|} = \sum_{\mathcal{O}} \frac{|\mathcal{O}|}{|H|} = \frac{|X|}{|H|},$$

where the orbit-stabilizer theorem was used in the second equality. This calculation is the finite model for dividing by gauge volume: summing one over each object and then weighting by automorphisms gives the same result as averaging over the acting group before passing to orbits.

112.2 Fields as Principal-Bundle Groupoids

For the datum $\mathfrak{Z}^{(D)}_{G,\omega}$, and in particular for the untwisted theory used first, define for a compact manifold M

$$\mathrm{Bun}_G(M)$$

to be the groupoid of principal G-bundles $P \to M$ and bundle isomorphisms. Equivalently, $\mathrm{Bun}_G(M)$ is the mapping groupoid from M to the classifying stack BG. In this chapter this phrase has the concrete meaning supplied by principal bundles and their isomorphisms; no analytic mapping-space structure is being assumed.

Definition 112.3 (Untwisted finite gauge partition function)**.** For a closed D-manifold M, the untwisted finite G-gauge theory partition function is

$$Z_G(M) = |\operatorname{Bun}_G(M)| = \sum_{[P] \in \pi_0 \operatorname{Bun}_G(M)} \frac{1}{|\operatorname{Aut}(P)|}. \tag{112.4}$$

If M is connected and $m \in M$ is a basepoint, a principal G-bundle is classified by a monodromy homomorphism

$$\rho : \pi_1(M, m) \to G$$

up to conjugation by G. The automorphism group of the corresponding bundle is

$$C_G(\rho) = \{h \in G : h\rho(\gamma)h^{-1} = \rho(\gamma) \text{ for all } \gamma \in \pi_1(M, m)\}.$$

For connected closed M,

$$Z_G(M) = \frac{|\operatorname{Hom}(\pi_1(M), G)|}{|G|}. \tag{112.5}$$

The groupoid $\mathrm{Bun}_G(M)$ is equivalent to the action groupoid

$$\operatorname{Hom}(\pi_1(M), G)//G,$$

where G acts by conjugation. The stabilizer of ρ is $C_G(\rho)$, which is the automorphism group of the corresponding bundle. The action-groupoid cardinality calculation gives (112.5). This argument uses connectedness to choose one basepoint and identify every flat finite G-bundle with a homomorphism $\pi_1(M, m) \to G$ modulo change of trivialization at the basepoint, namely conjugation by G.

For a disconnected $M = \bigsqcup_{\alpha=1}^{c} M_\alpha$, groupoid cardinality is multiplicative under disjoint union:

$$Z_G(M) = \prod_{\alpha=1}^{c} Z_G(M_\alpha).$$

The empty manifold has value 1, as required for a symmetric monoidal field theory.

112.3 State Spaces and Push-Pull Along Spans

Let Σ be a closed $(D-1)$-manifold. The boundary field groupoid in Definition 112.1 is

$$\mathcal{B}_\Sigma = \mathrm{Bun}_G(\Sigma).$$

In the untwisted theory the state space is

$$\mathcal{H}_G(\Sigma) = \mathrm{Fun}(\pi_0\mathcal{B}_\Sigma, \mathbb{C}), \tag{112.6}$$

the vector space of complex-valued functions on isomorphism classes of boundary bundles. Its canonical bilinear pairing is the groupoid integral

$$\langle f_1, f_2\rangle_\Sigma = \int_{\mathcal{B}_\Sigma} f_1(Q)f_2(Q) = \sum_{[Q]} \frac{f_1(Q)f_2(Q)}{|\operatorname{Aut}(Q)|}. \tag{112.7}$$

The pairing identifies $\mathcal{H}_G(\bar{\Sigma})$ with the dual of $\mathcal{H}_G(\Sigma)$, in agreement with Proposition 103.5.

If

$$M : \Sigma_0 \to \Sigma_1$$

is a collared bordism, restriction to the boundary gives a span of finite groupoids

$$\mathcal{B}_{\Sigma_0} \xleftarrow{r_0} \mathrm{Bun}_G(M) \xrightarrow{r_1} \mathcal{B}_{\Sigma_1}. \tag{112.8}$$

The bordism operator is the push-pull map

$$Z_G(M) = (r_1)_!\, r_0^* : \mathcal{H}_G(\Sigma_0) \to \mathcal{H}_G(\Sigma_1). \tag{112.9}$$

Here r_0^* is pullback of functions, and $(r_1)_!$ is integration over the finite homotopy fiber of r_1.

Concretely, if $Q_1 \in \mathcal{B}_{\Sigma_1}$, then

$$\big(Z_G(M)f\big)(Q_1) = \int_{P\in \mathrm{Bun}_G(M)_{Q_1}} f(r_0P), \tag{112.10}$$

where $\mathrm{Bun}_G(M)_{Q_1}$ is the groupoid of pairs (P,α) with $P \in \mathrm{Bun}_G(M)$ and an isomorphism $\alpha : r_1P \simeq Q_1$.

Lemma 112.4 (Finite groupoid Fubini with automorphism weights)**.** *Let*

$$p : E \to B$$

be a functor between finite groupoids, and let $F : E \to \mathbb{C}$ be constant on isomorphism classes. For $b \in B$, let E_b be the homotopy fiber groupoid whose objects are pairs (e, ϕ), with $e \in E$ and $\phi : p(e) \simeq b$ in B. Then

$$\int_E F = \int_{b\in B}\int_{E_b} F. \tag{112.11}$$

Consequently, if $X \xrightarrow{f} Y \xrightarrow{g} Z$ are finite groupoids, then

$$(g\circ f)_!F = g_!f_!F. \tag{112.12}$$

Proof. Choose representatives b for the isomorphism classes of B. The right hand side of (112.11) is

$$\sum_{[b]\in\pi_0 B}\frac{1}{|\operatorname{Aut}_B(b)|}\sum_{[(e,\phi)]\in\pi_0 E_b}\frac{F(e)}{|\operatorname{Aut}_{E_b}(e,\phi)|}.$$

We compare the contribution of one isomorphism class $[e]\in\pi_0 E$. After replacing b by an isomorphic representative, assume

$$b=p(e).$$

Let

$$I_e=\operatorname{im}\bigl(\operatorname{Aut}_E(e)\to\operatorname{Aut}_B(p(e))\bigr),\qquad K_e=\ker\bigl(\operatorname{Aut}_E(e)\to\operatorname{Aut}_B(p(e))\bigr).$$

The possible identifications $\phi : p(e)\simeq b$ form a torsor for $\operatorname{Aut}_B(b)$. Passing to the homotopy fiber divides this torsor by the image I_e. Hence the class $[e]$ gives

$$\frac{|\operatorname{Aut}_B(b)|}{|I_e|}$$

isomorphism classes in E_b, and the automorphism group of each such fiber object is K_e. Its total contribution to the iterated integral is

$$\frac{1}{|\operatorname{Aut}_B(b)|}\frac{|\operatorname{Aut}_B(b)|}{|I_e|}\frac{F(e)}{|K_e|}=\frac{F(e)}{|I_e|\,|K_e|}=\frac{F(e)}{|\operatorname{Aut}_E(e)|}.$$

This is exactly the contribution of $[e]$ to $\int_E F$. Summing over $\pi_0 E$ proves (112.11).

For pushforwards, the homotopy fiber of $g\circ f$ over $z\in Z$ is equivalent to the homotopy fiber product

$$X\times_Y Y_z.$$

Applying (112.11) to the projection

$$X\times_Y Y_z\to Y_z$$

gives the equality between integration over this single homotopy fiber and the iterated integration first over X_y and then over Y_z. This is (112.12). ☐

Theorem 112.5 (Finite groupoid gluing)**.** *Let $M_{01}:\Sigma_0\to\Sigma_1$ and $M_{12}:\Sigma_1\to\Sigma_2$ be collared bordisms. The natural restriction functor gives an equivalence of finite groupoids*

$$\operatorname{Bun}_G(M_{12}\circ M_{01})\simeq\operatorname{Bun}_G(M_{01})\times_{\operatorname{Bun}_G(\Sigma_1)}\operatorname{Bun}_G(M_{12}),\tag{112.13}$$

where the fiber product is the homotopy fiber product of groupoids. With the push-pull definition (112.9),

$$Z_G(M_{12}\circ M_{01})=Z_G(M_{12})\,Z_G(M_{01}).\tag{112.14}$$

Proof. The equivalence (112.13) is descent for principal bundles over the collar cover

$$M_{12}\circ M_{01}=M_{01}\cup_{\Sigma_1}M_{12}.$$

An object on the right is a bundle $P_{01}\to M_{01}$, a bundle $P_{12}\to M_{12}$, and an isomorphism

$$P_{01}|_{\Sigma_1}\simeq P_{12}|_{\Sigma_1}.$$

Gluing the total spaces along this isomorphism gives a principal bundle on the glued manifold. Conversely, restricting a bundle on the glued manifold gives such data. A bundle isomorphism on the glued manifold restricts to a pair of isomorphisms on M_{01} and M_{12} compatible with the chosen boundary identification; conversely, any compatible pair glues uniquely to an isomorphism of the glued bundles. Thus the equivalence includes the morphism groupoids, not only the object sets.

The bordism operator is the linear map associated to the restriction span. Composition of such span operators is composition of finite-groupoid pullbacks and pushforwards. By Lemma 112.4, the pushforward along a composite finite-groupoid map is the iterated pushforward, with the automorphism denominators matching by the image-kernel calculation in the lemma. Combining this Fubini identity with the descent equivalence (112.13) gives

$$Z_G(M_{12} \circ M_{01}) = Z_G(M_{12})\, Z_G(M_{01}),$$

as an equality of linear maps $\mathcal{H}_G(\Sigma_0) \to \mathcal{H}_G(\Sigma_2)$. □

Figure 112.1 records the span-level gluing proof: restriction from bulk G-bundles to boundary G-bundles composes by homotopy pullback, and groupoid cardinality supplies the Fubini identity.

$$\begin{tikzcd}
 & & \mathrm{Bun}_G(M_{01}) \times_{\mathrm{Bun}_G(\Sigma_1)} \mathrm{Bun}_G(M_{12}) \arrow[dl, dashed] \arrow[dr, dashed] & & \\
 & \mathrm{Bun}_G(M_{01}) \arrow[dl] \arrow[dr] & & \mathrm{Bun}_G(M_{12}) \arrow[dl] \arrow[dr] & \\
\mathrm{Bun}_G(\Sigma_0) & & \mathrm{Bun}_G(\Sigma_1) & & \mathrm{Bun}_G(\Sigma_2)
\end{tikzcd}$$

Figure 112.1: Gluing finite gauge theory is composition of spans of finite groupoids. The middle groupoid is retained as a homotopy fiber product, so automorphisms of the intermediate boundary bundle are divided out exactly once.

112.4 Triangulated State Sums

Let K be a finite triangulation of a closed oriented D-manifold M, equipped with a branching, meaning an ordering of vertices on every simplex compatible on faces. A lattice G-connection assigns an element

$$g_{ij} \in G$$

to every oriented edge $i \to j$, with $g_{ji} = g_{ij}^{-1}$. It is flat when

$$g_{ij} g_{jk} = g_{ik} \tag{112.15}$$

on every oriented triangle. Gauge transformations are vertex labels

$$h_i \in G,$$

acting by

$$g_{ij} \longmapsto h_i g_{ij} h_j^{-1}. \tag{112.16}$$

Let $\mathrm{Flat}_G(K)$ be the finite set of flat edge labelings. The action groupoid

$$\mathrm{Flat}_G(K)//G^{K_0}$$

is equivalent to $\mathrm{Bun}_G(M)$. Therefore

$$Z_G(M) = \frac{|\,\mathrm{Flat}_G(K)|}{|G|^{|K_0|}}. \tag{112.17}$$

For connected K, choose a spanning tree T in the one-skeleton. Every flat connection is gauge equivalent to a unique flat connection with

$$g_{ij} = 1 \qquad \text{for every edge } (ij) \in T,$$

up to the remaining constant conjugation by G. This gauge fixing gives the equality between (112.17) and (112.5).

112.5 Dijkgraaf–Witten Twists

The Dijkgraaf–Witten entry $[\omega]$ in Definition 112.1 is represented by

$$\omega \in Z^D(BG, U(1))$$

normalized as a group cocycle. In an inhomogeneous cochain convention this means a function

$$\omega : G^D \to U(1)$$

which is 1 when any argument is the identity and satisfies

$$(\delta\omega)(g_1, \ldots, g_{D+1}) = 1. \tag{112.18}$$

For example, for $D = 2$,

$$\omega(g_2, g_3)\,\omega(g_1 g_2, g_3)^{-1}\,\omega(g_1, g_2 g_3)\,\omega(g_1, g_2)^{-1} = 1. \tag{112.19}$$

For a branched oriented D-simplex $\sigma = [i_0 \cdots i_D]$, define

$$\omega_\sigma(g) = \omega(g_{i_0 i_1}, g_{i_1 i_2}, \ldots, g_{i_{D-1} i_D})^{\varepsilon(\sigma)}, \tag{112.20}$$

where $\varepsilon(\sigma) = +1$ if the branching order agrees with the orientation of M on σ, and -1 otherwise. The twisted state-sum is

$$Z_{G,\omega}(M; K) = \frac{1}{|G|^{|K_0|}} \sum_{g \in \mathrm{Flat}_G(K)} \prod_{\sigma^D \in K_D} \omega_\sigma(g). \tag{112.21}$$

Proposition 112.6 (Triangulation independence of the cocycle weight)**.** *For closed M, the quantity (112.21) is invariant under branching-preserving Pachner moves and changes by the expected boundary coboundary factor if ω is replaced by $\omega\,\delta\lambda$. Hence on closed manifolds it depends only on the cohomology class*

$$[\omega] \in H^D(BG, U(1)).$$

Proof. A flat edge labeling of a branched triangulation is equivalently a simplicial map from the triangulation to the nerve model of BG: the ordered edge labels on a simplex are composable group elements, and flatness says that the label on a longer edge is their product. The factor $\omega_\sigma(g)$ is therefore the pullback of the normalized D-cochain ω evaluated on the oriented D-simplex σ.

A branching-preserving Pachner move replaces a collection of oriented D-faces in the boundary of one branched $(D+1)$-simplex by the complementary collection of D-faces. For a fixed flat labeling, all boundary labels are restrictions of one $(D+1)$-simplex label $(g_1, \dots, g_{D+1})$. The ratio between the two products of ω-weights is the alternating product over the D-faces of this $(D+1)$-simplex, namely

$$(\delta\omega)(g_1, \dots, g_{D+1}),$$

where the exponent sign records whether the face orientation agrees with the boundary orientation. The cocycle condition (112.18) makes this ratio equal to 1. Since a Pachner move is local and gives a bijection between flat labelings that agree outside the moved ball, the whole state-sum is unchanged.

Now replace ω by $\omega\,\delta\lambda$, with λ a $(D-1)$-cochain on G. On each D-simplex the extra factor is the alternating product of the λ-weights on its $(D-1)$-faces. When one multiplies over all D-simplices, every interior $(D-1)$-face appears twice with opposite induced orientations and cancels. The surviving product is supported on ∂M. Thus a closed M has no residual factor, while a manifold with boundary changes by exactly the boundary coboundary term. The closed state-sum therefore depends only on $[\omega] \in H^D(BG, U(1))$. $\square$

The intrinsic form of the same definition is the following. A bundle $P \to M$ has a classifying map

$$f_P : M \to BG$$

well-defined up to homotopy. The cocycle supplies a phase

$$\exp\left(2\pi \mathrm{i} \int_M f_P^* \omega\right).$$

Then

$$Z_{G,\omega}(M) = \sum_{[P] \in \pi_0 \operatorname{Bun}_G(M)} \frac{1}{|\operatorname{Aut}(P)|} \exp\left(2\pi \mathrm{i} \int_M f_P^* \omega\right). \tag{112.22}$$

Equation (112.21) is the simplicial representative of (112.22).

On a manifold with boundary, the same transgression produces a line over $\operatorname{Bun}_G(\partial M)$. The state space is therefore not ordinary functions but sections of that line:

$$\mathcal{H}_{G,\omega}(\Sigma) = \Gamma\big(\operatorname{Bun}_G(\Sigma), \mathcal{L}_\omega(\Sigma)\big). \tag{112.23}$$

The gluing theorem continues to hold because the two transgressed boundary lines on a cut hypersurface are dual and contract under gluing.

112.6 Explicit Closed-Surface Partition Functions

Let Σ_g be a closed connected oriented surface of genus g. Its fundamental group has presentation

$$\pi_1(\Sigma_g) = \left\langle a_1, b_1, \dots, a_g, b_g \;\middle|\; \prod_{r=1}^{g} [a_r, b_r] = 1 \right\rangle.$$

Thus

$$Z_G(\Sigma_g) = \frac{1}{|G|}\#\left\{(A_r, B_r)_{r=1}^g \in G^{2g} : \prod_{r=1}^{g}[A_r, B_r] = 1\right\}. \tag{112.24}$$

Proposition 112.7 (Character formula for the $2D$ untwisted theory)**.** *Let* $\mathrm{Irr}(G)$ *be the set of irreducible complex representations of* G*, and let* $d_R = \dim R$*. Then*

$$Z_G(\Sigma_g) = |G|^{2g-2} \sum_{R\in\mathrm{Irr}(G)} d_R^{2-2g} = \sum_{R\in\mathrm{Irr}(G)} \left(\frac{d_R}{|G|}\right)^{2-2g}. \tag{112.25}$$

Proof. Let

$$\delta_e(x) = \begin{cases} 1, & x = e, \\ 0, & x \neq e \end{cases}$$

be the delta function at the identity. The character expansion of the central delta function is

$$\delta_e(x) = \frac{1}{|G|} \sum_{R\in\mathrm{Irr}(G)} d_R\, \chi_R(x), \tag{112.26}$$

which follows from the orthogonality of irreducible characters. Define the commutator-sum operator on class functions by

$$(Tf)(x) = \sum_{a,b\in G} f(x[a,b]).$$

Since T commutes with conjugation and convolution by central functions, Schur orthogonality gives

$$T\chi_R = \frac{|G|^2}{d_R^2}\,\chi_R. \tag{112.27}$$

To see this, write the representation matrix also as $R(\cdot)$. Schur averaging says that for every endomorphism A of R,

$$\sum_{b\in G} R(b)AR(b)^{-1} = \frac{|G|\ \mathrm{tr}_R\, A}{d_R}\,\mathbf{1}_R. \tag{112.28}$$

With $A = R(a)^{-1}$,

$$\sum_{b\in G} R([a,b]) = R(a)\sum_{b\in G} R(b)R(a)^{-1}R(b)^{-1} = \frac{|G|\chi_R(a^{-1})}{d_R}R(a).$$

The central-idempotent form of character orthogonality gives

$$\sum_{a\in G} \chi_R(a^{-1})R(a) = \frac{|G|}{d_R}\,\mathbf{1}_R.$$

Therefore

$$\sum_{a,b\in G} R([a,b]) = \frac{|G|^2}{d_R^2}\,\mathbf{1}_R,$$

and multiplying by $R(x)$ and taking the trace proves (112.27).

The number of solutions of $\prod_{r=1}^{g}[A_r, B_r] = 1$ is $T^g\delta_e(e)$. Applying (112.26) and (112.27) gives

$$\#\,\mathrm{Hom}(\pi_1(\Sigma_g), G) = |G|^{2g-1}\sum_R d_R^{2-2g}.$$

Dividing by the residual global gauge volume $|G|$ gives (112.25). □

For $g = 0$, the formula gives

$$Z_G(S^2) = \frac{1}{|G|}$$

because $\sum_R d_R^2 = |G|$. For $g = 1$,

$$Z_G(T^2) = |\,\mathrm{Irr}(G)|,$$

which is also the number of conjugacy classes of G. If $G = A$ is finite abelian, all irreducible representations are one-dimensional, so

$$Z_A(\Sigma_g) = |A|^{2g-1}. \tag{112.29}$$

112.7 Circle State Space and the Two-Dimensional Algebra

For the circle,

$$\mathrm{Bun}_G(S^1) \simeq G//G,$$

where G acts on itself by conjugation. Therefore

$$\mathcal{H}_G(S^1) = \{\text{class functions } G \to \mathbb{C}\}. \tag{112.30}$$

The pairing (112.7) becomes

$$\langle f_1, f_2\rangle_{S^1} = \frac{1}{|G|}\sum_{g\in G} f_1(g)f_2(g). \tag{112.31}$$

The pair of pants defines a commutative Frobenius algebra on class functions. In the orientation convention where the outgoing holonomy is the product of the incoming holonomies, the product is convolution

$$(f_1 * f_2)(k) = \sum_{\substack{a,b\in G\\ ab=k}} f_1(a)f_2(b). \tag{112.32}$$

Associativity is associativity of group multiplication plus finite Fubini. Commutativity on class functions follows by the change of variables

$$(a, b) \mapsto (aba^{-1}, a),$$

which preserves the product ab up to conjugacy; class functions cannot distinguish conjugate outgoing holonomies. The counit is the disk amplitude, and the nondegenerate pairing is (112.31). This identifies the ordinary $2D$ finite gauge theory with the Frobenius algebra construction of Theorem 103.7.

112.8 Three-Dimensional Finite Gauge Theory

For a closed connected 3-manifold M, the untwisted partition function is again

$$Z_G(M) = \frac{|\operatorname{Hom}(\pi_1(M), G)|}{|G|}.$$

In particular,

$$Z_G(T^3) = \frac{1}{|G|}\#\{(a,b,c) \in G^3 : ab = ba,\ ac = ca,\ bc = cb\}. \tag{112.33}$$

The state space on T^2 is the vector space of functions on commuting pairs (a, b) modulo simultaneous conjugation. The extended three-dimensional theory assigns to the circle a category. In the untwisted case this category is equivalent to representations of the Drinfeld double

$$D(G),$$

and in the ω-twisted case to representations of the twisted quantum double. Equivalently, the line operators form the Drinfeld center

$$Z(\mathrm{Vect}_G^\omega).$$

The simple objects are labeled by a conjugacy class $\mathcal{C} \subset G$ and an irreducible projective representation of the centralizer $C_G(c)$, with the projective 2-cocycle obtained by transgressing ω to the flux $c \in \mathcal{C}$. Fusion and braiding are finite tensor-category operations. This is a theorem of finite tensor categories, but the path-integral origin is already visible: a line insertion fixes magnetic monodromy around a small linking circle, while electric labels are representations of the residual stabilizer of that monodromy.

112.9 Finite Abelian Boundaries and Line Condensation

The preceding construction gives a finite bulk theory. To test boundary and defect statements without hiding behind categorical terminology, it is useful to write one family of boundaries entirely in finite algebra. Let A be a finite abelian gauge group in the untwisted three-dimensional theory. Its simple bulk line labels form the finite abelian group

$$\mathcal{C}_A = A \oplus \widehat{A}, \qquad \widehat{A} = \operatorname{Hom}(A, U(1)).$$

An element

$$c = (a, \chi) \in \mathcal{C}_A$$

has magnetic flux $a \in A$ and electric character $\chi \in \widehat{A}$. The fusion law is addition in $\mathcal{C}_A$. The nondegenerate braiding pairing is

$$B\big((a,\chi),(b,\psi)\big) = \chi(b)\psi(a), \tag{112.34}$$

and the topological spin is

$$\theta(a,\chi) = \chi(a). \tag{112.35}$$

These formulas are the Abelian specialization of the Drinfeld-double line description above: a charge line $(0,\chi)$ evaluates the flux linked by the line, and a flux line $(a,1)$ is detected by linked electric characters.

For a subgroup $L \subset \mathcal{C}_A$, define its braiding orthogonal by

$$L^{\perp} = \{c \in \mathcal{C}_A \mid B(c,\ell) = 1 \text{ for every } \ell \in L\}. \tag{112.36}$$

The subgroup L is Lagrangian if

$$L = L^{\perp} \qquad \text{and} \qquad \theta(\ell) = 1 \quad \text{for every } \ell \in L.$$

Equivalently, the spin is trivial on L, the pairing B is trivial on $L \times L$, and

$$|L| = |A|.$$

Indeed, nondegeneracy of B identifies $\mathcal{C}_A/L^{\perp}$ with $\widehat{L}$, hence

$$|L|\,|L^{\perp}| = |\mathcal{C}_A| = |A|^2.$$

Thus an isotropic subgroup with $|L| = |A|$ has $|L^{\perp}| = |L|$ and $L \subset L^{\perp}$, so equality follows. The spin condition is the locality condition for endpoint operators: trivial-spin lines supply bosonic local boundary endpoints, while a line with nontrivial topological spin carries intrinsic boundary statistics even when its mutual braiding with the other condensed lines is trivial.

Definition 112.8 (Finite Abelian line-condensation boundary datum)**.** For the untwisted three-dimensional finite A-gauge theory, a finite Abelian line-condensation boundary datum is a Lagrangian subgroup

$$L \subset \mathcal{C}_A = A \oplus \widehat{A}.$$

The boundary B_L is the finite topological boundary model in which a bulk line $c \in \mathcal{C}_A$ has a topological endpoint on the boundary exactly when $c \in L$. Lines in L are called condensed at B_L. Lines not in L terminate only after adjoining an additional boundary excitation external to this datum.

The Lagrangian condition is the finite algebraic form of mutual locality for condensed endpoints. If $\ell_1, \ell_2 \in L$, moving one endpoint around the other at the boundary produces the bulk braiding phase $B(\ell_1, \ell_2)$. A boundary vacuum containing both endpoints as local topological operators requires this phase to be 1. Maximality, $L = L^{\perp}$, says that every line mutually local with the condensed endpoint algebra has already been declared condensable; otherwise the boundary datum was not complete.

Theorem 112.9 (Cylinder sectors for finite Abelian boundaries)**.** *Let B_{L_0} and B_{L_1} be two finite Abelian line-condensation boundaries of the untwisted three-dimensional finite A-gauge theory. The topological line sectors on an interval with left boundary L_0 and right boundary L_1 are canonically labeled by*

$$\mathcal{C}_A/(L_0 + L_1). \tag{112.37}$$

Consequently the cylinder ground-sector vector space in this finite topological model is

$$\mathcal{H}_{\mathrm{cyl}}(L_0, L_1) = \mathbb{C}[\mathcal{C}_A/(L_0 + L_1)], \qquad \dim \mathcal{H}_{\mathrm{cyl}}(L_0, L_1) = |\mathcal{C}_A/(L_0 + L_1)| = |L_0 \cap L_1|. \tag{112.38}$$

Proof. Place a representative bulk line $c \in \mathcal{C}_A$ along the interval. If $\ell_0 \in L_0$, then ℓ_0 may end on the left boundary. Fusing ℓ_0 to the interval line therefore changes the representative

$$c \longmapsto c + \ell_0$$

without changing the topological sector. Similarly, every $\ell_1 \in L_1$ can be absorbed at the right boundary, so

$$c \longmapsto c + \ell_1$$

is the same sector. Hence the sector label depends only on the coset of c in $\mathcal{C}_A/(L_0 + L_1)$.

Conversely, if two interval lines c and c' define the same sector in the finite line-condensation model, their difference can be removed by endpoint operations at the two boundaries. The available endpoint operations are precisely fusion with elements of L_0 and L_1, by Definition 112.8. Therefore

$$c' - c \in L_0 + L_1,$$

and the quotient above is neither too coarse nor too fine.

The vector space statement is the finite state-space assignment for this topological sector set: one basis vector for each simple sector. For the dimension, use

$$|\mathcal{C}_A/(L_0 + L_1)| = \frac{|\mathcal{C}_A|}{|L_0 + L_1|} = \frac{|A|^2}{|L_0||L_1|/|L_0 \cap L_1|} = |L_0 \cap L_1|,$$

because L_0 and L_1 are Lagrangian and therefore both have cardinality $|A|$. □

For $A = \mathbb{Z}_N$, write line labels as

$$(m, e) \in \mathbb{Z}_N^{\mathrm{m}} \oplus \mathbb{Z}_N^{\mathrm{e}}, \qquad B((m, e), (m', e')) = \exp\left(\frac{2\pi \mathrm{i}}{N}(em' + e'm)\right).$$

The electric and magnetic boundaries are

$$L_{\mathrm{e}} = \{(0, e) : e \in \mathbb{Z}_N\}, \qquad L_{\mathrm{m}} = \{(m, 0) : m \in \mathbb{Z}_N\}.$$

They are Lagrangian. Therefore

$$\dim \mathcal{H}_{\mathrm{cyl}}(L_{\mathrm{e}}, L_{\mathrm{e}}) = \dim \mathcal{H}_{\mathrm{cyl}}(L_{\mathrm{m}}, L_{\mathrm{m}}) = N, \qquad \dim \mathcal{H}_{\mathrm{cyl}}(L_{\mathrm{e}}, L_{\mathrm{m}}) = 1.$$

The first two cases have one nontrivial topological line label running along the cylinder, while in the mixed case every line can be absorbed by one of the two boundaries. This is the finite algebraic version of the rough/smooth boundary distinction in the $\mathbb{Z}_N$ gauge-code laboratory. It supplies an exact finite calibration for claims about line endpoints and boundary topological sectors; continuum boundary theorems require their own regulator, locality, and limiting hypotheses.

112.10 Subgroup Boundaries and Relative Dijkgraaf–Witten Cocycles

For a nonabelian finite group the most direct boundary condition is a reduction of the gauge group at the boundary. Let $H \subset G$ be a subgroup. If $P \to M$ is a principal G-bundle and $N \subset \partial M$ is a boundary component, an H-reduction of P along N is a principal H-bundle $Q \to N$ together with an isomorphism of principal G-bundles

$$Q \times_H G \cong P|_N.$$

The subgroup boundary B_H is the boundary theory whose boundary fields are such reductions. In a triangulated representative, after choosing a boundary trivialization, this says that boundary edge labels lie in H, while bulk edge labels lie in G. Boundary gauge transformations are H-valued and bulk gauge transformations are G-valued.

This definition is phrased as a condition on bundles. The boundary datum is the restriction functor

$$\mathrm{Bun}_H(N) \longrightarrow \mathrm{Bun}_G(N), \qquad Q \longmapsto Q \times_H G,$$

and all stabilizers are the automorphism groups of these finite groupoids.

Interval sector groupoid for subgroup boundaries. Consider the untwisted finite G-gauge theory on an interval with left boundary subgroup $H_0 \subset G$ and right boundary subgroup $H_1 \subset G$. Choose a trivialization of the G-bundle at one point of the interval. Flatness identifies the bundle with a single parallel-transport element $g \in G$. Changing the left boundary H_0-trivialization by $h_0 \in H_0$ multiplies g on the left, while changing the right boundary H_1-trivialization by $h_1 \in H_1$ multiplies g on the right by h_1^{-1}, because parallel transport is read from the left endpoint to the right endpoint. Thus the boundary field groupoid is the action groupoid

$$G//(H_0 \times H_1), \qquad (h_0, h_1) : g \longmapsto h_0 g h_1^{-1}.$$

Its isomorphism classes, hence the simple topological sectors, are the double cosets

$$H_0 \backslash G / H_1, \tag{112.39}$$

and the finite action-groupoid cardinality formula (112.3) gives

$$\big|G//(H_0 \times H_1)\big| = \frac{|G|}{|H_0|\,|H_1|}. \tag{112.40}$$

For a representative $g \in G$, the automorphism group of the corresponding sector is the stabilizer of g:

$$\mathrm{Aut}(g) = \{(h_0, h_1) \in H_0 \times H_1 : h_0 g = g h_1\} \cong H_0 \cap g H_1 g^{-1}. \tag{112.41}$$

Projection to the first factor identifies this stabilizer with

$$H_0 \cap g H_1 g^{-1},$$

and projection to the second factor identifies it with

$$g^{-1} H_0 g \cap H_1.$$

The vector space assigned to an interval in this boundary sector is therefore

$$\mathrm{Fun}(H_0 \backslash G / H_1, \mathbb{C}),$$

with the groupoid pairing weighted by the stabilizers in (112.41). For example, $H_0 = H_1 = G$ has one sector, while $H_0 = H_1 = \{1\}$ has sectors indexed by all elements of G. If $G = S_3$ and $H = \langle(12)\rangle$, then

$$H\backslash S_3/H$$

has two sectors, represented by the identity and by a three-cycle; the stabilizer weights distinguish the sector containing H from the sector with trivial stabilizer.

Proposition 112.10 (Subgroup-boundary junction convolution)**.** *Let $H_0, H_1, H_2 \subset G$ be subgroup boundary data in the untwisted finite G-gauge theory. Put*

$$\mathcal{V}(H_i, H_j) = \mathrm{Fun}(G, \mathbb{C})^{H_i \times H_j}, \qquad f(h_i g h_j^{-1}) = f(g).$$

The finite pair-of-intervals junction

$$(B_{H_0}, B_{H_1}) \times (B_{H_1}, B_{H_2}) \longrightarrow (B_{H_0}, B_{H_2})$$

defines the bilinear product

$$(f \star_{H_1} k)(g) = \frac{1}{|H_1|} \sum_{\substack{x,y\in G\\ xy=g}} f(x)k(y), \qquad f \in \mathcal{V}(H_0, H_1), \quad k \in \mathcal{V}(H_1, H_2). \tag{112.42}$$

This product lands in $\mathcal{V}(H_0, H_2)$, is associative under successive junction composition, and has unit

$$\mathbf{1}_H(g) = \begin{cases} 1, & g \in H, \\ 0, & g \notin H, \end{cases} \qquad \mathbf{1}_H \in \mathcal{V}(H, H).$$

If $A = H_0 a H_1$, $B = H_1 b H_2$, and $C = H_0 c H_2$, then the double-coset basis structure constants are

$$\mathbf{1}_A \star_{H_1} \mathbf{1}_B = \sum_C N_{AB}^C \mathbf{1}_C, \qquad N_{AB}^C = \frac{1}{|H_1|} \#\{(x,y) \in A \times B : xy = c\}, \tag{112.43}$$

where the integer N_{AB}^C is independent of the representative $c \in C$.

Proof. The junction field groupoid is the finite action groupoid

$$G^2//(H_0 \times H_1 \times H_2),$$

where

$$(h_0, h_1, h_2) : (x, y) \longmapsto (h_0 x h_1^{-1}, h_1 y h_2^{-1}).$$

The incoming restriction remembers x and y, while the outgoing restriction remembers the composite parallel transport

$$g = xy.$$

Thus the junction is the span

$$G//(H_0 \times H_1) \times G//(H_1 \times H_2) \leftarrow G^2//(H_0 \times H_1 \times H_2) \xrightarrow{xy} G//(H_0 \times H_2).$$

Push-pull along this finite span gives (112.42). The factor $|H_1|^{-1}$ is the groupoid-volume factor for the intermediate boundary gauge transformation h_1; it is the same automorphism denominator that appears in the finite gluing theorem.

The result is H_0-left and H_2-right invariant. For instance,

$$(f \star_{H_1} k)(h_0 g h_2^{-1}) = \frac{1}{|H_1|} \sum_{xy=g} f(h_0 x) k(y h_2^{-1}) = (f \star_{H_1} k)(g),$$

using the bi-invariance of f and k. Associativity follows from finite Fubini and associativity in G:

$$((f \star_{H_1} k) \star_{H_2} \ell)(t) = \frac{1}{|H_1|\,|H_2|} \sum_{\substack{x,y,z \in G \\ xyz=t}} f(x)k(y)\ell(z) = (f \star_{H_1} (k \star_{H_2} \ell))(t).$$

The unit statement follows because, for $f \in \mathcal{V}(H, K)$,

$$(\mathbf{1}_H \star_H f)(g) = \frac{1}{|H|} \sum_{h \in H} f(h^{-1} g) = f(g),$$

and similarly $f \star_K \mathbf{1}_K = f$.

For the structure constants, first note that $\mathbf{1}_A \star_{H_1} \mathbf{1}_B$ is (H_0, H_2)-bi-invariant, so it is constant on every output double coset C. If $c' \in C$, say $c' = h_0 c h_2^{-1}$, multiplication by (h_0, h_2) gives a bijection between the factorization sets with product c and c'. Finally, H_1 acts freely on

$$\{(x, y) \in A \times B : xy = c\}$$

by

$$h : (x, y) \longmapsto (xh^{-1}, hy).$$

The quotient by this free action is exactly the intermediate boundary gauge identification in the span, so the cardinality is divisible by $|H_1|$ and N_{AB}^{C} is a nonnegative integer. $\square$

For $G = S_3$ and $H = \langle (12) \rangle$, the H-H double cosets are

$$A_0 = H, \qquad A_1 = H(123)H.$$

The characteristic function of A_0 is the unit, and the nontrivial sector $X = \mathbf{1}_{A_1}$ obeys

$$X \star_H X = 2\,\mathbf{1}_{A_0} + X.$$

This small example shows why the stabilizer and intermediate gauge-volume factors cannot be dropped: the finite path integral produces the Hecke-type integer algebra with the normalization fixed by gluing, not by a later choice of basis.

Now include a three-dimensional Dijkgraaf–Witten twist. Let

$$\omega \in Z^3(G, U(1))$$

be the normalized inhomogeneous cocycle used above. A subgroup boundary supports a local finite boundary counterterm exactly when the pulled-back bulk cocycle is a coboundary on H.

Definition 112.11 (Relative Dijkgraaf–Witten boundary datum)**.** Let $i : H \hookrightarrow G$ be a subgroup inclusion and let $\omega \in Z^3(G, U(1))$ be normalized. A relative Dijkgraaf–Witten boundary datum for B_H is a normalized 2-cochain

$$\beta : H \times H \to U(1)$$

such that

$$(\delta\beta)(h_1,h_2,h_3)=\beta(h_2,h_3)\,\beta(h_1h_2,h_3)^{-1}\,\beta(h_1,h_2h_3)\,\beta(h_1,h_2)^{-1}=\omega(h_1,h_2,h_3) \qquad (112.44)$$

for every $h_1,h_2,h_3\in H$. Equivalently,

$$(\omega,\beta)\in C^3(G,U(1))\oplus C^2(H,U(1))$$

is a relative cocycle for the pair (BG,BH) with relative differential

$$d_{\mathrm{rel}}(\omega,\beta)=\big(\delta\omega,\ i^*\omega\,(\delta\beta)^{-1}\big).$$

In the triangulated state sum, write the bulk 3-simplex weight as before and multiply by the inverse boundary 2-cochain weight

$$W_{\omega,\beta}(K,g)=\prod_{\sigma^3\subset K}\omega(g_\sigma)^{\varepsilon(\sigma)}\prod_{\tau^2\subset\partial K}\beta(h_\tau)^{-\varepsilon_\partial(\tau)}. \qquad (112.45)$$

Here $h_\tau=(h_{i_0i_1},h_{i_1i_2})$ for an oriented boundary triangle $\tau=[i_0i_1i_2]$, and $\varepsilon_\partial(\tau)$ is computed using the induced boundary orientation. The inverse sign is part of the convention; with this convention the condition (112.44) is the cancellation equation.

Boundary cocycle cancellation. Let G be finite, $H\subset G$, and let (ω,β) be a relative Dijkgraaf–Witten boundary datum in the sense of Definition 112.11. The weight (112.45) is invariant under elementary triangulation changes that preserve the boundary H-reduction. Interior Pachner moves use $\delta\omega=1$. Boundary Pachner moves use exactly the identity $\delta\beta=i^*\omega$.

The verification is local in the moved simplex. Interior Pachner moves are the closed-manifold argument in Proposition 112.6; the ratio of the two bulk products is $(\delta\omega)(g_1,g_2,g_3,g_4)$, hence 1.

It remains to check the boundary elementary move. A boundary 2-2 or 1-3 move is represented, after adding the collar direction, by the boundary of a single oriented 3-simplex whose edge labels lie in H. For a branched simplex with ordered edge labels (h_1,h_2,h_3), the ratio of the boundary β^{-1}-weights on the two sides of the move is

$$(\delta\beta)(h_1,h_2,h_3)^{-\varepsilon},$$

where ε is the sign of the corresponding oriented 3-simplex. The missing bulk collar contribution would be

$$\omega(h_1,h_2,h_3)^{\varepsilon}.$$

Multiplying the two ratios gives

$$\omega(h_1,h_2,h_3)^{\varepsilon}(\delta\beta)(h_1,h_2,h_3)^{-\varepsilon}=1$$

by (112.44). The argument is local and the set of flat labelings outside the moved ball is unchanged, so the whole finite state-sum weight is unchanged.

The obstruction to such a boundary is therefore the restriction class

$$i^*[\omega]\in H^3(BH,U(1)).$$

If this class vanishes, choices of β form a torsor for

$$H^2(BH, U(1)).$$

In the untwisted case $\omega = 1$, the condition says that β is a normalized 2-cocycle on H; it supplies a projective boundary counterterm, and different cohomology classes give different finite boundary theories. Thus the subgroup boundary data in the twisted finite theory are the finite path-integral counterparts of the module-category labels $(H, [\beta])$, but the present formulation is only the concrete state-sum construction needed here: boundary fields are H-reductions and the equality $\delta\beta = i^*\omega$ is the local gauge-invariance and triangulation-invariance equation.

Relative gluing and the boundary line. The preceding local check proves invariance of one relative state-sum weight. The gluing statement uses a second, independent sign fact: a boundary component that is outgoing for one bordism is incoming for the other, so the same β-trivialization appears with reciprocal exponent. Let

$$M_L : \Sigma_{\text{in}} \to \Sigma \qquad \text{and} \qquad M_R : \Sigma \to \Sigma_{\text{out}}$$

be collared triangulated three-dimensional finite G-gauge bordisms, with an H-reduction on the common boundary Σ. Choose compatible branched triangulations whose restrictions to Σ agree. For a fixed flat boundary H-labeling h on Σ, let $\varepsilon_\Sigma(\tau)$ be the orientation sign of a boundary triangle τ as seen from M_L. The same triangle has sign $-\varepsilon_\Sigma(\tau)$ as seen from M_R. Hence the interface part of the product of the two relative weights is

$$\prod_{\tau \subset \Sigma} \beta(h_\tau)^{-\varepsilon_\Sigma(\tau)} \prod_{\tau \subset \Sigma} \beta(h_\tau)^{+\varepsilon_\Sigma(\tau)} = 1. \tag{112.46}$$

All remaining factors are exactly the bulk ω-weights and the external-boundary β-weights of the glued triangulation. Thus, before any summation over fields,

$$W_{\omega,\beta}(K_L, g_L)\, W_{\omega,\beta}(K_R, g_R) = W_{\omega,\beta}(K_L \cup_\Sigma K_R, g_L \cup_h g_R), \tag{112.47}$$

for every pair of bulk labelings whose H-boundary restrictions are the chosen h. The subsequent sum over h, with the automorphism denominator of the boundary H-bundle, is precisely finite groupoid Fubini:

$$Z_{G,\omega,\beta}(M_R \circ M_L) = \int_{Q \in \mathrm{Bun}_H(\Sigma)} Z_{G,\omega,\beta}(M_R; Q)\, Z_{G,\omega,\beta}(M_L; Q). \tag{112.48}$$

Equation (112.48) is the concrete finite version of pairing a section of the transgressed boundary line with a section of the dual line over the oppositely oriented boundary. If β is changed by a coboundary $\delta\lambda$, each open boundary state is multiplied by the corresponding transgressed 1-cochain factor. The two factors on a glued interface are reciprocal by the same orientation argument as (112.46). Thus β is a choice of boundary-line trivialization; the obstruction to making such a choice remains the relative class $i^*[\omega]$.

112.11 Defects, Boundaries, and Anomaly Inflow

Finite gauge theories also provide exact models of defects. A codimension-one defect between G- and H-gauge theories is described, in the finite path-integral language, by a finite correspondence of boundary field groupoids

$$\mathrm{Bun}_G(\Sigma) \leftarrow \mathcal{D}(\Sigma) \rightarrow \mathrm{Bun}_H(\Sigma), \tag{112.49}$$

possibly decorated by a cocycle line. A concrete subgroup defect is obtained from

$$K \subset G \times H,$$

by requiring the two gauge fields to reduce on the wall to a common principal K-bundle. Composition of such defects is again a homotopy fiber product of finite groupoids.

In a D-dimensional theory with finite global symmetry G, an anomaly is represented by a class

$$[\alpha] \in H^{D+1}(BG, U(1)).$$

For a background G-bundle A on a closed D-manifold M, the would-be anomalous partition function is not a number canonically attached to A. It is a vector in the transgressed line determined by α. Keeping a $(D+1)$-dimensional bulk with action α makes the combined bulk-boundary amplitude gauge invariant. Gauging the boundary symmetry alone requires a trivialization of this anomaly line. This is the finite model of the relative field theory language in Definition 103.4.

Open Problem 112.12 (Extended finite models). Construct extended finite-gauge TQFTs with boundaries, defects, junctions, twists, and anomaly inflow as functors on a chosen bordism (∞, D)-category, including a direct comparison between the higher-categorical functor and the finite groupoid state-sum formulas in this chapter.

Part IX

Global Structure, Phases, and Extended Operators

Chapter 113

Global Forms and Higher-Form Symmetry

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The local Lie algebra of gauge transformations does not by itself define a quantum gauge theory. A continuum gauge theory also contains global bundle data, a spectrum of genuine line and surface operators, and topological defects that act on those extended operators. This chapter gives the finite-dimensional algebra that controls these data for compact connected semisimple gauge groups. The analytic definition of a renormalized 't Hooft line is given in Chapter 115; here the purpose is to define the electric and magnetic charge lattices on which that construction depends.

113.1 Character and Cocharacter Lattices

Let $\mathfrak{g}$ be a compact semisimple Lie algebra and let G_{sc} be the connected simply connected compact Lie group with Lie algebra $\mathfrak{g}$. Fix a maximal torus $T_{\mathrm{sc}} \subset G_{\mathrm{sc}}$, with Lie algebra $\mathfrak{t}$. The character and cocharacter groups of a compact torus T are

$$X^*(T) = \mathrm{Hom}_{\mathrm{Lie\,grp}}(T, U(1)), \qquad X_*(T) = \mathrm{Hom}_{\mathrm{Lie\,grp}}(U(1), T).$$

They are free abelian groups of rank $\dim T$. A character $\lambda \in X^*(T)$ and a cocharacter $m \in X_*(T)$ have an integer pairing

$$\lambda \circ m : U(1) \longrightarrow U(1), \qquad z \longmapsto z^{\langle \lambda, m\rangle}, \qquad \langle \lambda, m\rangle \in \mathbb{Z}.$$

This pairing is part of the definition of $X^*(T)$ and $X_*(T)$; no metric on $\mathfrak{t}$ is needed.

For T_{sc}, write

$$\Lambda_{\mathrm{wt}} = X^*(T_{\mathrm{sc}}), \qquad \Lambda_{\mathrm{crt}} = X_*(T_{\mathrm{sc}}).$$

These are the weight lattice and the coroot lattice. The root lattice $\Lambda_{\mathrm{rt}} \subset \Lambda_{\mathrm{wt}}$ is generated by the roots of $(G_{\mathrm{sc}}, T_{\mathrm{sc}})$. The coweight lattice is

$$\Lambda_{\mathrm{cwt}} = X_*(T_{\mathrm{ad}}), \qquad T_{\mathrm{ad}} = T_{\mathrm{sc}}/Z(G_{\mathrm{sc}}),$$

and contains Λ_{crt}. The finite quotients

$$\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}}, \qquad \Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}}$$

are naturally dual finite abelian groups. The first quotient records central characters of representations of G_{sc}; the second records cocharacters modulo ordinary coroot magnetic charges.

Definition 113.1 (Global form)**.** A global form of the compact gauge group with Lie algebra $\mathfrak{g}$ is a connected compact group

$$G = G_{\mathrm{sc}}/\Gamma, \qquad \Gamma \subset Z(G_{\mathrm{sc}}).$$

The maximal torus of G is $T_G = T_{\mathrm{sc}}/\Gamma$. Its character and cocharacter lattices are

$$X^*(T_G) = \{\lambda \in \Lambda_{\mathrm{wt}} \mid \lambda(\gamma) = 1 \text{ for every } \gamma \in \Gamma\}, \tag{113.1}$$

and

$$X_*(T_G) = \{m \in \Lambda_{\mathrm{cwt}} \mid \exp_{G_{\mathrm{sc}}}(2\pi \mathrm{i} m) \in \Gamma\}. \tag{113.2}$$

The exponential in the second formula uses the identification of a cocharacter with an element of $\mathfrak{t}$ whose exponential closes in the adjoint torus. It closes in $G = G_{\mathrm{sc}}/\Gamma$ precisely when its endpoint in G_{sc} lies in Γ.

The local curvature term in the Yang–Mills Lagrangian depends only on $\mathfrak{g}$ and on the normalization of the invariant quadratic form. The global form changes the allowed bundles and the admissible line operators. These changes are invisible in ordinary perturbation theory around a trivial bundle unless extended operators or background fields for higher-form symmetries are included.

Let V_λ be an irreducible representation of G_{sc} with highest weight $\lambda \in \Lambda_{\mathrm{wt}}$. It defines a representation of $G = G_{\mathrm{sc}}/\Gamma$ if and only if $\lambda \in X^*(T_G)$, equivalently if and only if every element of Γ acts trivially on V_λ. The representation V_λ factors through the quotient map $G_{\mathrm{sc}} \to G_{\mathrm{sc}}/\Gamma$ exactly when Γ is contained in the kernel of the representation. Since Γ is central, Schur's lemma says that every $\gamma \in \Gamma$ acts on the irreducible module V_λ by a scalar. That scalar is the central character determined by λ, namely the restriction of λ to $Z(G_{\mathrm{sc}}) \subset T_{\mathrm{sc}}$. Hence the action of every $\gamma \in \Gamma$ is trivial exactly when $\lambda(\gamma) = 1$ for all γ, which is (113.1).

Example 113.2 ($SU(N)/\mathbb{Z}_k$)**.** Let k divide N, let $z = \exp(2\pi \mathrm{i}/N)\mathbf{1}_N$, and let $\mathbb{Z}_k \subset Z(SU(N))$ be the subgroup generated by $z^{N/k}$. An irreducible $SU(N)$-representation has an N-ality

$$e \in \mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z},$$

defined as the number of boxes of its Young diagram modulo N, or equivalently as its central character $z \mapsto \exp(2\pi \mathrm{i} e/N)$. The representation descends to $SU(N)/\mathbb{Z}_k$ if and only if

$$\exp(2\pi \mathrm{i} e/k) = 1, \qquad \text{that is} \qquad e \in k\mathbb{Z}_N.$$

Thus the quotient removes Wilson lines whose N-ality is not divisible by k. Dually, the cocharacter lattice is enlarged: modulo the coroot lattice, the allowed magnetic classes are

$$m \in (N/k)\mathbb{Z}_N.$$

Indeed, after choosing the first fundamental coweight as the generator of $\Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}} \cong \mathbb{Z}_N$, the cocharacter represented by m times this generator closes in $SU(N)/\mathbb{Z}_k$ precisely when its endpoint z^m lies in $\mathbb{Z}_k$, i.e. precisely when m is a multiple of N/k modulo N.

113.2 Genuine Lines and the Dirac Pairing

Definition 113.3 (Genuine line operator)**.** A line operator $L(C)$ supported on an embedded oriented curve C is genuine if its definition does not require a choice of an auxiliary surface ending on C. If a surface S with $\partial S = C$ is part of the definition, then the insertion is a boundary of a higher-dimensional defect rather than an autonomous line operator. In gauge theory, being genuine is a property of the chosen global form, discrete theta data, and spectrum of allowed charged defects; it is not a property of the local gauge algebra alone.

Before dividing by screening, a dyonic line in a gauge theory with local algebra $\mathfrak{g}$ has an electric weight and a magnetic cocharacter. For the finite center-sensitive part of the discussion one may record only the classes

$$\overline{\lambda} \in \Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}}, \qquad \overline{m} \in \Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}}.$$

The root and coroot quotients remove charges that can be screened by adjoint gluonic fields or shifted by ordinary smooth monopole cores. The remaining finite data are precisely the charges detected by center and magnetic one-form symmetries.

For two dyonic labels

$$\gamma = (\lambda, m), \qquad \gamma' = (\lambda', m'),$$

define the Dirac pairing

$$B(\gamma, \gamma') = \lambda(m') - \lambda'(m) \quad \in \mathbb{Q}. \tag{113.3}$$

The value may be fractional when λ is a weight and m' is a coweight of the universal center-sensitive lattice. Its class in $\mathbb{Q}/\mathbb{Z}$ is independent of representatives modulo $\Lambda_{\mathrm{rt}} \oplus \Lambda_{\mathrm{crt}}$. Two lines can be simultaneously genuine in the same theory only if

$$B(\gamma, \gamma') \in \mathbb{Z}. \tag{113.4}$$

Equivalently, the finite Aharonov–Bohm phase $\exp(2\pi \mathrm{i} B(\gamma, \gamma'))$ must be trivial.

Definition 113.4 (Finite line-spectrum datum)**.** For a fixed local gauge algebra, the finite line-spectrum datum is a subgroup

$$L \subset \big(\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}}\big) \oplus \big(\Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}}\big)$$

on which the Dirac pairing (113.3) vanishes modulo $\mathbb{Z}$. It is maximal if it equals its orthogonal complement with respect to that finite pairing. Maximality is the finite-charge statement that no further center-sensitive Wilson–'t Hooft charge can be added while preserving mutual locality.

The definition is deliberately finite and algebraic. A complete continuum definition of the corresponding line operator also requires singular boundary conditions, renormalization of the defect, and a regulator or constructive framework; these analytic data are separated in Definition 115.2.

Proposition 113.5 ($\mathfrak{su}(N)$ finite line lattices)**.** *For local algebra $\mathfrak{su}(N)$, identify*

$$\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}} \cong \Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}} \cong \mathbb{Z}_N.$$

Write a finite dyonic charge as $(e, m) \in \mathbb{Z}_N \oplus \mathbb{Z}_N$. The finite Dirac pairing is

$$\overline{B}\big((e, m), (e', m')\big) = \frac{em' - e'm}{N} \quad \in \mathbb{Q}/\mathbb{Z}. \tag{113.5}$$

Let $k \mid N$ and let $p \in \mathbb{Z}_k$. The subgroup

$$L_{N,k,p} = \big\langle (k, 0), (p, N/k) \big\rangle \subset \mathbb{Z}_N \oplus \mathbb{Z}_N \tag{113.6}$$

is maximal isotropic. In the Yang–Mills convention used in Chapter 117, it is the finite Wilson–'t Hooft line lattice of the $SU(N)/\mathbb{Z}_k$ theory with discrete theta parameter p, before adding matter that could screen center charges.

Proof. The quotient $\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}} \cong \mathbb{Z}_N$ is generated by the fundamental Wilson N-ality. The dual quotient $\Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}} \cong \mathbb{Z}_N$ is generated by the fundamental coweight magnetic class. The generator pair has pairing $1/N$, which gives (113.5) by bilinearity and antisymmetry.

The two generators in (113.6) pair as

$$\overline{B}((k, 0), (p, N/k)) = 1 = 0 \quad \text{in } \mathbb{Q}/\mathbb{Z},$$

and each generator pairs trivially with itself by antisymmetry. Hence $L_{N,k,p}$ is isotropic. Its first generator gives N/k distinct electric classes. Modulo this subgroup, the second generator has magnetic component N/k, which has order k. Thus $|L_{N,k,p}| = N$.

It remains to see that no larger isotropic subgroup exists. If $(x, y) \in (\mathbb{Z}_N)^2$ pairs trivially with $(k, 0)$, then $-ky/N \in \mathbb{Z}$, so $y = (N/k)b$ modulo N. Trivial pairing with $(p, N/k)$ then gives

$$\frac{x(N/k) - py}{N} = \frac{x - pb}{k} \in \mathbb{Z},$$

so $x = pb + ka$ modulo N. Therefore

$$(x, y) = a(k, 0) + b(p, N/k),$$

and the orthogonal complement of $L_{N,k,p}$ is $L_{N,k,p}$ itself. This proves maximality. The identification with the global form follows from Example 113.2: Wilson charges are multiples of k, magnetic charges are multiples of N/k, and the discrete theta coordinate shifts the minimal magnetic line by electric charge p. □

Figure 113.1 displays the finite line-charge lattice in the $SU(5)$-type example and marks the maximal mutually local sublattice selected by the global form and discrete theta coordinate.

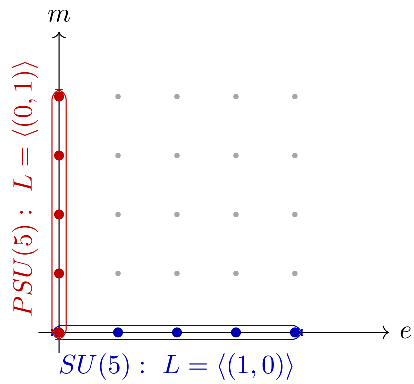

Figure 113.1: The finite $\mathfrak{su}(5)$ charge group $\mathbb{Z}_5^{\mathrm{e}} \oplus \mathbb{Z}_5^{\mathrm{m}}$. A line spectrum is not the whole grid but a maximal isotropic subgroup for (113.5). The $SU(5)$ and $PSU(5)$ choices shown here are the two coordinate axes; discrete theta choices tilt the magnetic axis as in (113.6).

113.3 Abelian Group-Like Higher-Form Symmetry

Definition 113.6 (Abelian group-like p-form global symmetry)**.** Let D be the spacetime dimension and let $0 \leq p \leq D-2$. A group-like abelian p-form global symmetry with finite or compact abelian group A assigns to every closed oriented codimension-$(p+1)$ submanifold M^{D-p-1} and every $a \in A$ an invertible topological defect

$$U_a(M).$$

The defects obey

$$U_a(M)U_b(M) = U_{a+b}(M), \qquad U_0(M) = 1,$$

inside correlation functions, and their correlation functions are invariant under smooth deformations of M that avoid charged insertions, boundaries, and singularities. A charged p-dimensional operator $W_\chi(\Sigma^p)$ has charge $\chi \in A^\vee = \mathrm{Hom}_{\mathrm{cont}}(A, U(1))$ if

$$U_a(M)W_\chi(\Sigma) = \chi(a)^{\mathrm{Link}(M,\Sigma)}W_\chi(\Sigma) \tag{113.7}$$

inside correlation functions whenever M and Σ are disjoint and their linking number is defined.

Here is the chain-level content of the linking phase. Work in an oriented region where the closed codimension-$(p+1)$ submanifold M bounds an oriented $(D-p)$-chain B, and assume that B meets the closed p-dimensional support Σ transversely. The linking number is the oriented intersection number

$$\mathrm{Link}(M,\Sigma) = B \cdot \Sigma = \sum_{x \in B \cap \Sigma} \mathrm{sgn}_x(B,\Sigma). \tag{113.8}$$

Let M' be obtained from M by a smooth deformation, and let C be the oriented $(D-p)$-dimensional sweep, so

$$\partial C = M' - M.$$

Choose bounding chains B and B' with $\partial B = M$ and $\partial B' = M'$. In a local ball, or more generally after fixing the homology class of the bounding chain in the complement of Σ, the change in linking is the intersection of the sweep with the charged support,

$$\mathrm{Link}(M', \Sigma) - \mathrm{Link}(M, \Sigma) = C \cdot \Sigma.$$

Thus a topological defect can be moved without changing its action as long as the sweep does not cross the charged insertion. If the sweep crosses Σ once with sign $\varepsilon = \pm 1$, the linking number changes by ε, and the correlator is multiplied by $\chi(a)^\varepsilon$. Iterating this local crossing rule gives (113.7). This is the topological part of the statement only: the existence of the defect $U_a(M)$ as a renormalized QFT operator and the allowed set of charged $W_\chi(\Sigma)$ remain part of the theory data.

For $p = 0$, this definition covers the abelian part of ordinary internal symmetry. Nonabelian zero-form symmetry requires topological defects labelled by group elements with noncommutative fusion; the one-form symmetries used below are abelian in precisely the sense of Definition 113.6.

For a continuous $U(1)$ p-form symmetry, a charge is an integer $q \in \mathbb{Z}$, the character is $\chi_q(e^{\mathrm{i}\alpha}) = e^{\mathrm{i}q\alpha}$, and (113.7) becomes

$$U_\alpha(M) W_q(\Sigma) = \exp\bigl(\mathrm{i}\alpha q \ \mathrm{Link}(M, \Sigma)\bigr) W_q(\Sigma).$$

The linking formula is not an analogy: it is the operator definition of the symmetry action on extended charged objects.

113.4 Electric-Magnetic One-Form Symmetries

Let G be a compact connected gauge group and assume for the moment that there is no dynamical matter carrying nontrivial center charge and no dynamical magnetic object that allows 't Hooft lines to end. The group of electric one-form symmetry defects is

$$Z(G).$$

Its characters act on Wilson lines. A Wilson line in representation R has electric one-form charge given by the central character

$$Z(G) \longrightarrow U(1), \qquad z \longmapsto R(z).$$

If $G = SU(N)$, this is the usual N-ality. If matter of nonzero N-ality is present, the corresponding Wilson line can end on a dynamical field and the electric one-form symmetry is explicitly reduced or broken.

For a non-simply connected G, there is also a magnetic one-form symmetry whose background fields measure the obstruction class of G-bundles. The topological magnetic surface defects are labelled by

$$\pi_1(G)^\vee = \mathrm{Hom}(\pi_1(G), U(1)).$$

Equivalently, a background two-form field for this symmetry is valued in $\pi_1(G)$. The distinction between $\pi_1(G)$ and its Pontryagin dual is only invisible after choosing a noncanonical perfect pairing; the definition keeps them separate. For $G = SU(N)/\mathbb{Z}_k$,

$$Z(G) \cong \mathbb{Z}_{N/k}, \qquad \pi_1(G) \cong \mathbb{Z}_k.$$

The line lattice $L_{N,k,p}$ in Proposition 113.5 is the finite algebraic record of which Wilson–'t Hooft probes are genuine for the chosen global form and discrete theta parameter.

Screening and confinement are therefore statements about the behavior of renormalized extended operators with charges in a specified line spectrum. They are developed in Chapter 116. Gauging a finite one-form symmetry and tracking its anomaly require cochain-level background fields and are developed in Chapters 117 and 123.

Open Problem 113.7 (Global structure from continuum local data). Determine, for a continuum QFT specified by local fields and local correlators, what additional data recover the full lattice of genuine line, surface, and defect operators. A Wightman field presentation determines local correlators and charged local sectors under stated hypotheses; global form, higher-form symmetries, and extended-operator completeness require further information about bundles, boundary conditions around defects, screening, and topological defect fusion.

Chapter 114

Extended Operators and Topological Defects

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Extended operators are QFT data supported on submanifolds. Their definition fixes allowed singularities, endpoint conditions, defect degrees of freedom, counterterms along the support, and the correlation functionals obtained after the regulator is removed. Local operator algebras control the behavior near points. A full theory also contains a completion by line, surface, wall, and higher-codimension objects, together with rules for their collision, crossing, and fusion.

114.1 Submanifold Support And Tubular Neighborhood Data

Let M be a smooth D-dimensional spacetime manifold, Euclidean or Lorentzian according to the context. Let

$$i : \Sigma^p \hookrightarrow M$$

be a properly embedded oriented p-manifold. Its normal bundle is

$$N_{\Sigma/M} = i^*TM/T\Sigma, \qquad \mathrm{rank}\, N_{\Sigma/M} = D - p.$$

When a framing, spin structure, orientation of the normal bundle, or background gauge field is required, that structure is part of the support datum. For sufficiently small $\varepsilon > 0$, a tubular neighborhood is a disk bundle

$$T_\varepsilon(\Sigma) \subset N_{\Sigma/M}$$

embedded in M. The punctured neighborhood

$$T_\varepsilon(\Sigma) \setminus \Sigma$$

records how bulk fields approach the support.

Definition 114.1 (Extended operator datum)**.** An extended operator $D(\Sigma)$ supported on $\Sigma \subset M$ is a renormalized correlation-functional datum consisting of:

support data	$(\Sigma, N_{\Sigma/M})$ with the declared auxiliary structures,
bulk condition	allowed fields on $M \setminus T_\varepsilon(\Sigma)$,
normal model	singular or boundary behavior on $\partial T_\varepsilon(\Sigma)$,
defect fields	degrees of freedom living on Σ when present,
coupling	local terms coupling bulk and defect variables,
counterterms	local functionals on Σ used in the limit $\varepsilon \to 0$,
correlators	distributions with insertions away from Σ.

Two representatives define the same extended operator when all renormalized correlation distributions with external insertions away from the support agree.

For local fields $\mathcal{O}_j(x_j)$ inserted at points

$$x_j \in M \setminus \Sigma,$$

the defining distributions are

$$\langle D(\Sigma)\, \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle\,.$$

They are defined on the configuration space where the x_j are pairwise distinct and avoid Σ, and they are extended to collision loci by a chosen bulk and defect renormalization prescription. The bulk-to-defect operator product expansion is the local expansion, in a normal coordinate

$$x = \exp_y(r\hat{n}), \qquad y \in \Sigma, \quad \hat{n} \in S(N_{\Sigma/M,y}),$$

of a bulk operator near the defect:

$$\mathcal{O}(x)\, D(\Sigma) \sim \sum_\alpha C_{\mathcal{O}\alpha}(r, \hat{n}; \partial_y)\, \widehat{\mathcal{O}}_\alpha(y)\, D(\Sigma).$$

The operators $\widehat{\mathcal{O}}_\alpha$ are local operators of the defect theory on Σ. The coefficients are distributions in the normal variables and differential operators along Σ. This expansion is part of the defect's short-distance data, just as the ordinary OPE is part of the local operator algebra.

Figure 114.1 records the basic operations on extended defects used in this chapter: tubular neighborhoods, defect-local operators, fusion, and deformation through topological regions.

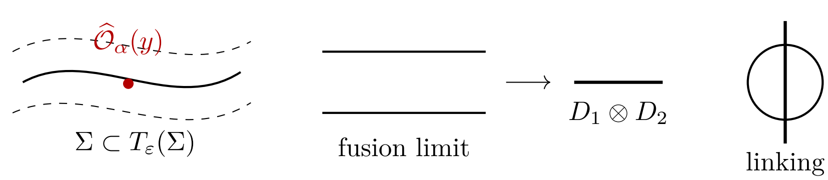

Figure 114.1: Tubular-neighborhood data for a defect Σ, including a defect-local insertion $\widehat{\mathcal{O}}_\alpha(y)$, fusion of parallel defects, and a topological linking configuration.

114.2 Genuine Defects, Boundaries, And Junctions

Let $D(\Sigma)$ have codimension $c = D - p$. A defect is genuine when its definition uses only data on an arbitrarily small tubular neighborhood of Σ and the complement $M \setminus \Sigma$. If an additional $(p+1)$-dimensional submanifold B with

$$\partial B = \Sigma$$

is required, then the insertion is a boundary of a higher defect rather than an autonomous p-dimensional defect. This distinction is part of the operator spectrum of the theory.

If two p-dimensional defects D_a and D_b meet along a $(p-1)$-dimensional submanifold J, the junction is a lower-dimensional defect operator

$$\Psi_{ab}(J) \in \mathrm{Hom}(D_a, D_b)$$

in the defect category appropriate to the theory. For a line operator, a point endpoint is a local operator in such a Hom space. In gauge theory, a Wilson line in representation R ending at a point requires an endpoint operator transforming in $R^\vee$ or an equivalent charged state. The existence of such an endpoint is the operator statement of screening.

114.3 The Displacement Operator

Let $D(\Sigma)$ be a renormalized defect in a QFT with stress tensor $T_{\mu\nu}$. Near a point of Σ, choose coordinates

$$(y^a, n^i), \qquad a = 1, \ldots, p, \quad i = 1, \ldots, D - p,$$

where y^a are tangent to Σ and n^i are normal coordinates. Translation perpendicular to the defect is broken by the insertion. The corresponding Ward identity has the form

$$\nabla_\mu T^{\mu i}(y, n) = \delta_\Sigma(n)\, \mathcal{D}^i(y)$$

inside correlation functions with the defect inserted, up to contact terms with other insertions. The defect local operator

$$\mathcal{D}^i(y)$$

is the displacement operator. Its index is a section of

$$N_{\Sigma/M}.$$

The identity is a definition of the normalization of $\mathcal{D}^i$.

Proposition 114.2 (Shape variation and displacement). *Let Σ_t be a smooth one-parameter family of embedded supports with $\Sigma_0 = \Sigma$. Let*

$$v^i(y) = \left.\frac{d\Sigma_t}{dt}\right|_{t=0}$$

be the normal deformation vector. Assume the external insertions are kept away from the swept tubular neighborhood and that no boundary term is produced at infinity or at $\partial\Sigma$. Then

$$\left.\frac{d}{dt}\right|_{t=0} \langle D(\Sigma_t)\, \mathcal{X} \rangle = - \int_\Sigma d^p y\, \sqrt{h}\, v_i(y) \left\langle \mathcal{D}^i(y)\, D(\Sigma)\, \mathcal{X} \right\rangle,$$

where h is the induced metric on Σ and $\mathcal{X}$ is the product of external insertions.

Proof. Represent the deformation by a diffeomorphism generated by a vector field v^μ supported in the tubular neighborhood and normal to Σ on the support. The variation of a correlation functional under this diffeomorphism is the Ward variation generated by the stress tensor:

$$\delta_v\langle D(\Sigma)\mathcal{X}\rangle = -\int_M d^D x\,\sqrt{g}\,\nabla_\mu v_\nu \langle T^{\mu\nu}(x) D(\Sigma)\mathcal{X}\rangle.$$

Integrating by parts gives

$$\delta_v\langle D(\Sigma)\mathcal{X}\rangle = \int_M d^D x\,\sqrt{g}\,v_\nu \langle \nabla_\mu T^{\mu\nu}(x) D(\Sigma)\mathcal{X}\rangle,$$

with no boundary contribution by hypothesis. Tangential components generate reparametrizations of the support, so only the normal components contribute to the shape. Inserting the defect Ward identity

$$\nabla_\mu T^{\mu i} = \delta_\Sigma \mathcal{D}^i$$

and using

$$\int_M d^D x\,\sqrt{g}\,\delta_\Sigma(n) f(x) = \int_\Sigma d^p y\,\sqrt{h}\,f(y,0)$$

gives the stated formula, with the sign fixed by the convention that moving the support rather than the coordinates gives the opposite diffeomorphism. □

Definition 114.3 (Topological defect)**.** A defect $D(\Sigma)$ is topological relative to a declared class of background structures when its renormalized correlation functionals are invariant under every isotopy of Σ that preserves those background structures and avoids the other insertions.

Displacement criterion. Under the hypotheses of Proposition 114.2, if the displacement operator vanishes inside the physical correlation functionals, then $D(\Sigma)$ is topological. In a cohomological theory with odd differential Q, the same conclusion holds for Q-closed external insertions when

$$\mathcal{D}^i = \{Q, \mathcal{V}^i\}_{\mathrm{gr}}$$

and the Q-Ward identity has no boundary contribution.

The shape derivative is the integral of the displacement insertion. If the displacement insertion vanishes in all relevant correlators, the derivative along every isotopy path is zero. If $\mathcal{D}^i = \{Q, \mathcal{V}^i\}_{\mathrm{gr}}$, the integrand is a Q-variation after moving Q through the Q-closed external insertions. The Ward identity sets the expectation value of this Q-variation to zero. Hence the correlation functional is constant along the isotopy.

114.4 Fusion

Let $D_1(\Sigma)$ and $D_2(\Sigma)$ be topological defects of the same dimension and compatible normal structures. Choose two nearby parallel copies $\Sigma_0, \Sigma_\varepsilon$ inside a tubular neighborhood. Their fusion is defined by the correlation-function limit

$$(D_1 \otimes D_2)(\Sigma) = \lim_{\varepsilon \downarrow 0} D_1(\Sigma_\varepsilon) D_2(\Sigma_0),$$

with external insertions kept outside the tubular neighborhood. The limit is taken after subtracting local defect counterterms supported on Σ. The result can be a single defect, a finite direct sum, an integral over continuous labels, or a distributional kernel. The precise target depends on the analytic category in which the QFT and its defects are defined.

Fusion is associative when the triple-collision limits are compatible:

$$(D_1 \otimes D_2) \otimes D_3 \cong D_1 \otimes (D_2 \otimes D_3).$$

The isomorphism is part of the defect data. In a fully extended topological theory these associators, unit defects, adjoints, and higher junctions form a higher category. In a non-topological QFT, the same language applies only to the topological subsector of defects or to protected cohomological defects for which the required collision limits exist.

Example 114.4 (Group-like topological defects)**.** Let A be a finite abelian group. A set of invertible topological defects

$$\{U_a\}_{a \in A}$$

is group-like when

$$U_a \otimes U_b = U_{a+b}, \qquad U_0 = \mathbf{1}, \qquad U_a^{-1} = U_{-a}.$$

For $A = \mathbb{Z}_N$, this says

$$U_r \otimes U_s = U_{r+s \bmod N}.$$

The associator may be trivial or may carry a cocycle. A nontrivial cocycle is part of the anomaly data of the symmetry-defect system.

114.5 Higher-Form Ward Defects

Let A be an abelian p-form symmetry group in a D-dimensional QFT. For

$$a \in A$$

there is a topological symmetry defect

$$U_a(N)$$

supported on an oriented closed $(D-p-1)$-manifold

$$N^{D-p-1} \subset M.$$

A charged p-dimensional operator

$$W_q(\Sigma^p)$$

has charge

$$q \in \widehat{A} = \mathrm{Hom}(A, U(1)).$$

When N and Σ are disjoint, their linking number is the integer

$$\mathrm{Link}(N, \Sigma) \in \mathbb{Z}$$

defined as follows. If $N = \partial B$ for an oriented $(D-p)$-chain B transverse to Σ, then

$$\mathrm{Link}(N, \Sigma) = B \cdot \Sigma,$$

the signed intersection number. In general the same integer is the Alexander-duality pairing between the homology class of N in

$$H_{D-p-1}(M \setminus \Sigma; \mathbb{Z})$$

and the class of Σ.

Higher-form Ward action. For an abelian p-form symmetry defect $U_a(N)$ and a charged p-operator $W_q(\Sigma)$, the Ward action inside correlation functions whose other insertions are outside the isotopy region is

$$U_a(N)\, W_q(\Sigma) = q(a)^{\mathrm{Link}(N,\Sigma)} W_q(\Sigma)$$

inside correlation functions whose other insertions are outside the isotopy region. For $A = \mathbb{Z}_N$, with $a = r$ and $q = s$ represented by integers modulo N,

$$q(a) = \exp\left(\frac{2\pi i\, rs}{N}\right).$$

This is the topological definition of the action written in linking-number form. Because U_a is topological, move N through the complement of Σ without changing the correlator. The only change occurs when the swept chain crosses Σ. A single positive crossing implements the symmetry action on W_q, which is multiplication by the character

$$q(a) \in U(1).$$

A negative crossing contributes $q(a)^{-1}$. Summing the signed crossings gives the intersection number $B \cdot \Sigma$ for a bounding chain $\partial B = N$. This is the linking number. Homological invariance of the intersection pairing extends the formula to the Alexander-duality definition of linking. For $\mathbb{Z}_N$, the character group is canonically $\mathbb{Z}_N$ after choosing the generator

$$s(r) = \exp(2\pi i\, rs/N),$$

which gives the displayed phase.

The dimensions satisfy

$$\dim N + \dim \Sigma = (D - p - 1) + p = D - 1,$$

which is the dimension condition for linking in D dimensions. If N crosses an endpoint or junction of W_q, the Ward identity includes the action on that endpoint or junction operator as well. Charge conservation at a junction is the statement that the product of all character factors associated with a small linking symmetry defect around the junction is one.

114.6 Locality And Braiding

In Lorentzian signature, two extended operators with spacelike separated supports admit an exchange operation. For ordinary bosonic local operators this exchange is commutation. For line, surface, and defect operators the exchange can carry a braiding determined by topology, spin, and gauge charge. The braiding is measured by transporting one support around another in the configuration space of disjoint submanifolds.

For abelian Wilson–'t Hooft line charges

$$\gamma = (\lambda, m), \qquad \gamma' = (\lambda', m'),$$

the center-sensitive part of the braiding phase is

$$\exp\bigl(2\pi i\, B(\gamma,\gamma')\bigr), \qquad B(\gamma,\gamma') = \lambda(m') - \lambda'(m),$$

with the pairing defined in Chapter 113. A line spectrum in a four-dimensional gauge theory is mutually local when this phase is one for every pair of genuine lines in the declared spectrum. The analytic construction of the corresponding magnetic and dyonic lines is developed in Chapter 115.

114.7 Defect Completeness Data

The local observable net or Wightman field content determines many constraints on defects: covariance, locality, stress-tensor Ward identities, superselection sectors, and possible endpoint charges. A QFT also contains additional completion data specifying which extended operators are genuine, which higher defects they may bound, which junctions exist, and which fusion limits are part of the theory. Distinct global forms of a gauge theory have the same local Lie algebra and perturbative local vertices but different line spectra and higher-form symmetry defects. This is the basic reason that extended operators must be included as primary QFT data in any complete nonperturbative definition.

Open Problem 114.5 (Extended operators from local reconstruction)*.* Develop reconstruction theorems that recover the extended-operator completion of a QFT from local observables, phase-space or split-property data, superselection sectors, and specified global-form input. The target is a theorem that identifies the genuine line, surface, wall, endpoint, junction, and fusion data determined by locality and duality, and separates that determined part from additional global completion choices.

Chapter 115

Line, Surface, and Domain-Wall Operators

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Extended operators are QFT data supported on submanifolds. They are not determined by local operators alone. This chapter defines the basic objects needed for later discussions of confinement, screening, duality, and anomaly inflow.

115.1 Defect Data

Let X be a spacetime manifold and let $Y \subset X$ be an embedded submanifold. A defect supported on Y is specified by:

(i) a choice of fields or boundary conditions on the complement $X \setminus Y$;

(ii) singular behavior or extra degrees of freedom along Y;

(iii) a coupling between bulk and defect variables;

(iv) a renormalization prescription for correlation functions with the defect inserted;

(v) an equivalence relation identifying defect representatives with the same correlation functionals.

The codimension of Y determines whether the object is a local operator, line, surface, domain wall, boundary condition, or higher defect.

115.2 Wilson Lines

Let G be a compact gauge group and R a finite-dimensional representation. For an oriented curve γ, the Wilson line is the parallel-transport functional

$$W_R(\gamma) = \mathrm{Tr}_R \ \mathrm{Pexp} \int_\gamma A.$$

In a regulated quantum gauge theory this expression is a coordinate for a renormalized line operator. Its definition requires the global form of G, the representation R, framing or endpoint data when needed, and a renormalization scheme for the short-distance singularity along the line. If γ is open, the endpoints must carry charged states or additional operators so that the full insertion is gauge invariant.

For a closed curve, choose a base point $p \in \gamma$ and write $\mathrm{Hol}_\gamma(A) \in G$ for parallel transport from p back to p. Under a gauge transformation $u : X \to G$,

$$\mathrm{Hol}_\gamma(A^u) = u(p)^{-1} \, \mathrm{Hol}_\gamma(A) u(p),$$

so $\mathrm{Tr}_R \, \mathrm{Hol}_\gamma(A)$ is gauge invariant. For an interval $\gamma = [p_-, p_+]$, parallel transport is a map

$$U_\gamma(A) : R_{p_-} \longrightarrow R_{p_+}, \qquad U_\gamma(A^u) = R(u(p_+))^{-1} U_\gamma(A) R(u(p_-)).$$

Gauge invariance is then obtained after contracting the endpoint indices with charged endpoint vectors or endpoint local operators in the dual representations. This endpoint bookkeeping is part of the line definition because it determines whether the line is genuine, a boundary of a surface operator, or an operator that changes the charged sector of the Hilbert space.

115.3 Magnetic and Dyonic Lines

A magnetic line in four-dimensional gauge theory is defined by a prescribed singularity of the gauge field near a curve. Around a small sphere linking the line, the magnetic flux is labelled by a cocharacter

$$m : U(1) \longrightarrow G$$

modulo conjugation, or equivalently by an element of the magnetic cocharacter lattice modulo the Weyl group. A dyonic line carries both the magnetic label m and an electric representation of the stabilizer of m. The allowed electric and magnetic labels obey the Dirac pairing condition: the phase obtained by moving one line around another must be trivial for mutually local operators in the chosen theory.

The global form of the gauge group changes the allowed labels. For example, the electric lattice of $SU(N)$ differs from that of $PSU(N)$, and the dual magnetic lattice changes accordingly. Therefore the line-operator spectrum is part of the definition of a gauge theory: it is fixed by the global gauge group, the allowed bundles, and the chosen mutually local electric-magnetic charge lattice.

115.3.1 The Local Monopole Model

Let $m : U(1) \to T \subset G$ be a cocharacter and let

$$m_* : \mathrm{i}\mathbb{R} \longrightarrow \mathfrak{t}$$

be its differential. We identify m_* with the element of $\mathfrak{t}$ obtained by evaluating the differential on i. On a small oriented two-sphere linking the candidate line, use spherical coordinates (ϑ, φ). Cover the sphere by the northern and southern patches $U_N = \{\vartheta \neq \pi\}$ and $U_S = \{\vartheta \neq 0\}$. The embedded Dirac monopole connection is

$$A_N = \frac{m_*}{2}(1 - \cos\vartheta)\,\mathrm{d}\varphi, \qquad A_S = -\frac{m_*}{2}(1 + \cos\vartheta)\,\mathrm{d}\varphi. \tag{115.1}$$

On the overlap,

$$A_N - A_S = m_*\,\mathrm{d}\varphi.$$

The transition function

$$g_{NS}(\varphi) = m(e^{\mathrm{i}\varphi})$$

satisfies

$$A_N = g_{NS}^{-1} A_S g_{NS} + g_{NS}^{-1} \mathrm{d} g_{NS}$$

in the abelianized T-valued model, and in a nonabelian theory after the choice of local reduction to T. Its curvature is

$$F_m = \frac{m_*}{2} \sin\vartheta\,\mathrm{d}\vartheta \wedge \mathrm{d}\varphi, \qquad \int_{S^2} \frac{F_m}{2\pi} = m_*. \tag{115.2}$$

The equality in the second formula means equality in the cocharacter lattice after pairing with characters of T.

Lemma 115.1 (Cocharacter criterion for the monopole singularity)**.** *The local model* (115.1) *defines a G-connection on the punctured ball modulo the chosen singularity exactly when $m \in X_*(T) = \mathrm{Hom}(U(1), T)$. For every character $\lambda \in X^*(T)$, a charge-λ Wilson probe transported across the Dirac ray changes by the phase*

$$\exp\bigl(2\pi\mathrm{i}\langle\lambda, m\rangle\bigr),$$

which is one by the definition of the character-cocharacter pairing.

Proof. The northern and southern gauge potentials define a single connection on the punctured sphere precisely when the transition map on the equatorial overlap is a single-valued smooth map $S^1 \to T$. Such maps are exactly cocharacters $m : U(1) \to T$. The relation $g_{NS}^{-1}\mathrm{d}g_{NS} = m_*\,\mathrm{d}\varphi$ gives the patching equation above.

Let a Wilson probe have T-weight λ. In the one-dimensional weight space, $g_{NS}(\varphi)$ acts by

$$\lambda(g_{NS}(\varphi)) = \lambda(m(e^{\mathrm{i}\varphi})) = e^{\mathrm{i}\langle\lambda, m\rangle\varphi}.$$

Taking φ once around the equator gives $\exp(2\pi\mathrm{i}\langle\lambda, m\rangle)$. Since $\lambda \circ m : U(1) \to U(1)$ is a group homomorphism, it has integer degree $\langle\lambda, m\rangle$. Hence the phase is trivial. Conversely, a T-valued transition map with this single-valuedness property for all characters is an element of $X_*(T)$, because characters separate points of the compact torus. $\square$

Two cocharacters related by the Weyl group determine gauge-equivalent local singularities: a gauge transformation whose value on the linking sphere implements the Weyl representative conjugates one Cartan reduction to the other. The underlying topological class of the G-bundle on the linking two-sphere records only the image of m in $\pi_1(G)$. The full cocharacter class is finer data: it fixes an asymptotic reduction of the connection and thereby determines the stabilizer group and electric labels of a dyonic refinement.

Definition 115.2 (Renormalized 't Hooft line)**.** Let X be an oriented four-manifold, let $\gamma \subset X$ be an oriented embedded curve, and let G be a compact connected gauge group with maximal torus T and Weyl group W. A bare magnetic line of magnetic charge

$$m \in \mathrm{Hom}(U(1), T)/W$$

is defined by removing a tubular neighborhood $N_\epsilon(\gamma)$ and integrating over G-connections on $X \setminus N_\epsilon(\gamma)$ whose restriction to each small linking sphere S^2_p lies in the topological class determined by m. In a local trivialization over $\gamma \times (S^2_p \setminus \{\text{Dirac ray}\})$, this means that the connection is gauge-equivalent near the boundary to the embedded Dirac monopole

$$F \sim \frac{m_*}{2} \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\varphi, \qquad \int_{S^2_p} \frac{F}{2\pi} = m_*,$$

where $m_* \in \mathfrak{t}$ is the differential of the cocharacter in the chosen normalization. Gauge transformations are required to preserve the cocharacter class on the boundary. Let $\mathcal{A}_m$ denote the resulting space of regulated connections and let $\mathcal{G}_m$ denote the corresponding boundary-preserving gauge group. The renormalized 't Hooft line $H_m(\gamma)$ is the limit

$$H_m(\gamma) = \lim_{\epsilon \to 0} Z_{\text{line}}(\epsilon) \int_{\mathcal{A}_m(X \setminus N_\epsilon(\gamma))/\mathcal{G}_m} \mathcal{D}A \, \exp(-S[A])$$

as a correlation functional with the same additional insertions, after multiplication by local line counterterms $Z_{\text{line}}(\epsilon)$. The definition exists only in a specified regulator or constructive framework in which the limit and the allowed counterterms are part of the line-operator data.

115.3.2 Dyonic Refinement and the Stabilizer

The magnetic singularity m breaks the gauge transformations along the line to the centralizer

$$G_m = Z_G(m(U(1))).$$

Equivalently, G_m is the compact subgroup whose Lie algebra is the sum of $\mathfrak{t}$ and the root spaces with roots α satisfying $\langle \alpha, m \rangle = 0$. The tangential component of the connection along γ, restricted to the boundary condition above, is a G_m-connection up to terms fixed by the singular normal model. A dyonic Wilson–'t Hooft line is therefore specified by

$$(m, \rho), \qquad m \in X_*(T)/W, \quad \rho \in \mathrm{Rep}(G_m),$$

with the convention that ρ is replaced by its Weyl-conjugate representation when m is Weyl-conjugated. The insertion is the magnetic boundary condition together with the Wilson parallel transport in ρ for the residual G_m-connection along γ, followed by the same line-counterterm renormalization appearing in Definition 115.2. If γ has endpoints, the endpoint states must be modules for the corresponding residual stabilizer groups.

For center-sensitive questions it is often enough to abelianize the label to

$$(\lambda, m) \in \big(\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}}\big) \oplus \big(\Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}}\big),$$

as in Chapter 113. This quotient records the charge detected by electric and magnetic one-form symmetries. The full operator also contains the stabilizer representation, defect counterterms, and the choice of regularization.

Linking phase and mutual locality. Let $L_{\lambda,m}$ and $L_{\lambda',m'}$ be two abelianized dyonic labels in the universal center-sensitive lattice. In a configuration where the first line links once with the second, the finite topological phase obtained by moving the first line around the second is

$$\exp\big[2\pi\mathrm{i}\big(\lambda(m') - \lambda'(m)\big)\big]\,.$$

Consequently two line labels can belong to the same mutually local genuine line spectrum only if

$$\lambda(m') - \lambda'(m) \in \mathbb{Z}. \tag{115.3}$$

Choose disjoint tubular neighborhoods of the two curves. Near $L_{\lambda',m'}$, the first line's Wilson factor of weight λ sees the transition function $m'(e^{\mathrm{i}\varphi})$ when its path crosses the Dirac ray of the second magnetic singularity. By the computation in Lemma 115.1, the phase is $\exp(2\pi\mathrm{i}\lambda(m'))$. Interchanging the roles of the two lines gives $\exp(2\pi\mathrm{i}\lambda'(m))$. The orientation convention for linking assigns the second contribution with the opposite sign, because dragging the first line around the second is inverse to dragging the second around the first. Multiplying the two contributions gives the displayed phase. The line operators are mutually local precisely when the phase is one for all allowed linking moves, which is (115.3).

Screening equivalence of center-sensitive charges. Adjoint dynamical particles shift the electric weight of a line by an element of the root lattice Λ_{rt}. Smooth monopole cores shift the magnetic cocharacter by an element of the coroot lattice Λ_{crt}. Therefore the finite charge of a genuine Wilson–'t Hooft line is a class in

$$\big(\Lambda_{\mathrm{wt}}/\Lambda_{\mathrm{rt}}\big) \oplus \big(\Lambda_{\mathrm{cwt}}/\Lambda_{\mathrm{crt}}\big),$$

subject to the mutual-locality condition imposed by the chosen theory.

An adjoint field transforms with weights in the root lattice together with zero weights. Attaching a massive adjoint particle worldline to a Wilson line changes the line's electric weight by one of these roots. The endpoint of the adjoint worldline is a local charged insertion in the screened sector, so the two Wilson lines differ by a locally creatable charge and have the same finite one-form charge.

For the magnetic statement, a smooth finite-energy monopole embedded in the $\mathfrak{su}(2)$ subalgebra of a root α has magnetic charge the coroot $\alpha^\vee$. Bringing such a monopole worldline to the 't Hooft line changes the singular cocharacter by $\alpha^\vee$ while leaving the asymptotic bundle class modulo smooth monopole creation unchanged. The coroots generate Λ_{crt}, so magnetic charges are screened modulo this lattice.

115.3.3 Theta Angle and the Witten Effect

In an abelian sector with curvature normalized by $\int_\Sigma F/(2\pi) \in \mathbb{Z}$, the theta term changes the electric charge measured at infinity in a magnetic background. With the convention

$$S_\theta = \frac{\mathrm{i}\theta}{8\pi^2} \int_X F \wedge F$$

in Euclidean signature, varying the action with respect to the boundary value of A shows that the electric displacement receives the magnetic contribution $(\theta/2\pi)m$. Thus a line whose integral electric label is λ and magnetic label is m has theta-dependent electric charge

$$q_\theta = \lambda + \frac{\theta}{2\pi} m$$

after pairing with any probe character. Under $\theta \mapsto \theta + 2\pi$, the integral label changes by the lattice automorphism

$$T : (\lambda, m) \longmapsto (\lambda + m, m). \tag{115.4}$$

For a nonabelian gauge theory the theta term contains a chosen invariant integral quadratic form. Its polarization gives a symmetric homomorphism

$$I : X_*(T) \longrightarrow X^*(T), \qquad I(m)(m') = I(m, m').$$

The Witten-effect automorphism is then

$$T_I : (\lambda, m) \longmapsto (\lambda + I(m), m).$$

The abelian formula (115.4) is the special case in which I is the identity under the chosen electric-magnetic identification. The automorphism T_I preserves the Dirac pairing:

$$(\lambda + I(m))(m') - (\lambda' + I(m'))(m) = \lambda(m') - \lambda'(m),$$

because $I(m, m') = I(m', m)$. Hence theta periodicity maps mutually local line spectra to mutually local line spectra.

For $G = U(1)$, the cocharacter lattice is $\mathbb{Z}$. The minimal 't Hooft line $H(\gamma)$ has magnetic charge 1, so the boundary condition is

$$\int_{S^2_p} \frac{F}{2\pi} = 1$$

for every small linking sphere. If the line also carries electric charge q, one multiplies the magnetic boundary condition by a Wilson factor for the unbroken gauge group along γ; this gives a dyonic line. A dynamical monopole operator is an endpoint for such a magnetic line. Hence a theory containing dynamical monopoles does not have the magnetic one-form symmetry under which closed 't Hooft lines are charged.

115.4 Surface Operators

A surface operator supported on a two-dimensional submanifold Σ may be defined by singular behavior of the gauge field, by coupling a two- dimensional QFT on Σ to the bulk, or by inserting a higher-form charged object. In four-dimensional gauge theory an abelianized surface singularity has local model

$$A \sim \alpha\, d\varphi$$

near the normal plane to Σ, where α is valued in a Cartan subalgebra modulo the cocharacter lattice and Weyl group. Additional parameters can couple to the magnetic flux through Σ. A complete definition must state the singularity, defect counterterms, gauge equivalence, and how local operators approaching the surface are renormalized.

Here is the corresponding local definition. Let N_Σ be the oriented real rank-two normal bundle of Σ, and on each punctured normal disk write polar coordinates (r, φ). A Cartan-type surface singularity is a reduction of the gauge field near Σ to a maximal torus T with

$$A = \alpha\, \mathrm{d}\varphi + A_{\mathrm{reg}}, \qquad \alpha \in \mathfrak{t}, \tag{115.5}$$

where A_{reg} has a finite limit as $r \to 0$ in the chosen regulator. The holonomy around a small positively oriented normal circle is

$$\exp(2\pi\alpha) \in T$$

after the convention that $\mathfrak{t}$ is the compact real Lie algebra. If $u \in X_*(T)$, the gauge transformation

$$g_u(\varphi) = u(e^{\mathrm{i}\varphi})$$

shifts α by u_*. Thus the singular holonomy parameter is

$$\alpha \in \mathfrak{t}/X_*(T)$$

modulo the Weyl group. In distributional notation the singular part of the curvature is

$$F_{\mathrm{sing}} = 2\pi\alpha\, \delta_\Sigma,$$

where δ_Σ is the two-form current Poincare dual to Σ; the formula is an instruction for the regulated boundary condition and for the boundary values of correlation functions with the surface insertion.

An abelianized surface operator may also contain a two-dimensional theta parameter

$$\eta \in X^*(T) \otimes_{\mathbb{Z}} \mathbb{R}/X^*(T),$$

paired with magnetic flux through Σ. If F_T denotes the regularized T-curvature on Σ, the phase is

$$\exp\left(2\pi\mathrm{i}\,\eta\left(\int_\Sigma \frac{F_T}{2\pi}\right)\right),$$

with the flux interpreted in the cocharacter lattice. The pair (α, η), modulo cocharacter shifts and Weyl transformations, is the center-sensitive datum of the abelianized surface singularity. Coupling a genuine two-dimensional QFT on Σ refines this datum by adding a defect Hilbert space, a defect stress tensor, and defect counterterms.

Gauge equivalence of Cartan surface singularities. Two Cartan surface singularities with parameters $\alpha, \alpha' \in \mathfrak{t}$ have the same normal-circle holonomy datum exactly when

$$\alpha' = w(\alpha) + u_* \qquad \text{for some } w \in W,\ u \in X_*(T).$$

A boundary-preserving gauge transformation may conjugate the chosen Cartan reduction by an element of the normalizer $N_G(T)$; modulo T, this is the Weyl action $W = N_G(T)/T$. A gauge transformation on the punctured normal disk whose restriction to the normal circle is a map $S^1 \to T$ is a cocharacter u, and its logarithmic derivative is $u_* \, \mathrm{d}\varphi$. It shifts the coefficient of $\mathrm{d}\varphi$ by u_*. These two operations generate the displayed equivalence.

Conversely, equal holonomies in the compact torus mean $\exp(2\pi\alpha') = \exp(2\pi w(\alpha))$ for some Weyl representative after matching Cartan reductions. The kernel of the exponential map $\mathfrak{t} \to T$ is the cocharacter lattice, so $\alpha' - w(\alpha) = u_*$ for some $u \in X_*(T)$.

115.5 Domain Walls and Interfaces

A domain wall between theories T_- and T_+ is a codimension-one defect whose correlation functional takes bulk fields of T_- on one side and bulk fields of T_+ on the other. A topological wall is one whose correlators are invariant under deformations of the wall that avoid other insertions. Such a wall acts on local and extended operators by bringing an operator to the wall and moving it through, provided the required limits exist.

Definition 115.3 (Fusion of topological defects)**.** Let D_1 and D_2 be parallel topological defects of the same codimension. Their fusion $D_1 \otimes D_2$, when it exists, is the defect defined by the limit in which the separation between D_1 and D_2 goes to zero, after subtracting or renormalizing local defect counterterms.

Fusion is an operation only after the limiting procedure and equivalence relation have been supplied. In favorable topological sectors the resulting objects form a category or higher category; in a general QFT the short-distance limit can produce new defect degrees of freedom or fail to exist.

Open Problem 115.4 (Extended-operator reconstruction). Given the local operator algebra and symmetry data of a QFT, determine what additional input is required to reconstruct its extended operators. The answer must include global forms, higher-form symmetries, singular boundary conditions, defect renormalization, and the topology in which defect fusion limits are taken.

Chapter 116

Confinement, Screening, and Oblique Confinement

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Confinement is a property of gauge-invariant extended-observable data in a specified infinite-volume or continuum limit. Its basic inputs are external probes, endpoint maps for dynamical charged objects, renormalized large-distance asymptotics of line insertions, and the Hilbert-space sectors generated by those probes. This chapter develops the finite charge-lattice and line-operator part of that structure. Chapter 119 then embeds it in the broader definition of a phase of a gauge theory.

116.1 Line-Charge Data

Let G be the global form of the compact gauge group and let $\mathfrak{g}$ be its Lie algebra. Fix a maximal torus $T \subset G$. Electric Wilson charges are integral weights of T that exponentiate to representations of G. Magnetic 't Hooft charges are cocharacters

$$U(1) \longrightarrow T$$

allowed by the same global form, modulo the Weyl group. We write the electric and magnetic charge groups as

$$\Gamma_{\mathrm{e}} \subset X^*(T), \qquad \Gamma_{\mathrm{m}} \subset X_*(T),$$

and a dyonic line label as

$$\gamma = (\lambda_{\mathrm{e}}, \lambda_{\mathrm{m}}) \in \Gamma_{\mathrm{e}} \oplus \Gamma_{\mathrm{m}}.$$

The unscreened center-sensitive part is often finite; for an $\mathfrak{su}(N)$ theory it is

$$\mathcal{C} = \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}.$$

The full charge lattice and the finite center-sensitive quotient should not be confused. The former remembers weights, cocharacters, and masses of dynamical states; the latter remembers precisely the data detected by center-linked topological surfaces and by the global form.

The Dirac pairing of two line labels is

$$\langle \gamma, \gamma' \rangle_D = \lambda_{\mathrm{e}}(\lambda'_{\mathrm{m}}) - \lambda'_{\mathrm{e}}(\lambda_{\mathrm{m}}) \in \mathbb{Z} \tag{116.1}$$

for simultaneously genuine lines. On the finite $\mathfrak{su}(N)$ charge group this becomes the alternating pairing

$$B((e,m),(e',m')) = \frac{em' - e'm}{N} \in \mathbb{Q}/\mathbb{Z}. \tag{116.2}$$

The phase acquired by moving one line once around the other is $\exp(2\pi \mathrm{i} B)$. Thus a set of line operators can be simultaneously genuine only when the relevant pairings vanish modulo integers.

Definition 116.1 (External charge system)**.** For the purposes of confinement diagnostics, an external charge system is a triple

$$(\mathcal{C}, B, S),$$

where $\mathcal{C}$ is a finite abelian group of center-sensitive line charges, $B : \mathcal{C} \times \mathcal{C} \to \mathbb{Q}/\mathbb{Z}$ is a nondegenerate alternating pairing, and $S \subset \mathcal{C}$ is the subgroup of charges screened by finite-energy dynamical objects. The quotient

$$\mathcal{C}_{\mathrm{ext}} = \mathcal{C}/S$$

is the group of static external charges modulo screening. If S is isotropic, the residual topological charge group is

$$\mathcal{C}_{\mathrm{top}} = S^{\perp}/S, \qquad S^{\perp} = \{\gamma \in \mathcal{C} \mid B(\gamma, s) = 0 \text{ for every } s \in S\}.$$

The distinction between $\mathcal{C}_{\mathrm{ext}}$ and $\mathcal{C}_{\mathrm{top}}$ is important. $\mathcal{C}_{\mathrm{ext}}$ classifies external sources after allowing dynamical endpoints. $\mathcal{C}_{\mathrm{top}}$ classifies charges with well-defined finite braiding after quotienting by screening.

Pairing on the screened topological quotient. If $S \subset \mathcal{C}$ is isotropic, then B descends to a nondegenerate alternating pairing on $S^{\perp}/S$. The verification is the linear algebra of orthogonal quotients. Let $\gamma, \eta \in S^{\perp}$ and change representatives by

$$\gamma \mapsto \gamma + s, \qquad \eta \mapsto \eta + t, \qquad s, t \in S.$$

Then

$$B(\gamma + s, \eta + t) - B(\gamma, \eta) = B(s, \eta) + B(\gamma, t) + B(s, t).$$

The first two terms vanish because $\gamma, \eta \in S^{\perp}$, and the last vanishes because S is isotropic. Hence the value of B is independent of representatives. The descended pairing is alternating because B is. For nondegeneracy, suppose $\bar{\gamma} \in S^{\perp}/S$ pairs trivially with all of $S^{\perp}/S$. Then γ pairs trivially with all of $S^{\perp}$, so $\gamma \in (S^{\perp})^{\perp}$. Nondegeneracy of B on $\mathcal{C}$ gives $(S^{\perp})^{\perp} = S$. Therefore $\bar{\gamma} = 0$.

116.2 Renormalized Line Laws

Let $L_\gamma(C)$ be a Euclidean line operator with charge $\gamma \in \mathcal{C}$ supported on a closed contour C. A regulator is part of the definition: on a lattice, C is a lattice path; in a continuum scheme, C is a smooth embedded loop together with a short-distance definition of the singular line insertion. A line has self-energy and possible curvature/contact counterterms. We therefore write

$$L_\gamma^{\mathrm{ren}}(C) = Z_\gamma(C,a)\, L_\gamma^{(a)}(C),$$

where a is the cutoff and $Z_\gamma(C,a)$ is local along the line in the chosen scheme. Statements about area or perimeter laws are statements about the renormalized object and about an order of limits.

Definition 116.2 (Area, perimeter, and zero-tension laws)**.** Let $\{C_R\}_{R\geq 1}$ be a family of planar loops whose linear size is R, with minimal spanning area A_R and length ℓ_R. Assume

$$A_R \to \infty, \qquad \ell_R \to \infty, \qquad A_R/\ell_R \to \infty.$$

For a renormalized line L_γ^{ren}:

1. an area law with string tension $\sigma_\gamma > 0$ means
$$-\log |\langle L_\gamma^{\mathrm{ren}}(C_R)\rangle| = \sigma_\gamma A_R + o(A_R);$$
2. a perimeter law means
$$-\log |\langle L_\gamma^{\mathrm{ren}}(C_R)\rangle| = \mu_\gamma \ell_R + o(\ell_R)$$
for some scheme-dependent μ_γ;
3. a zero-tension law means
$$\lim_{R\to\infty} \frac{-\log |\langle L_\gamma^{\mathrm{ren}}(C_R)\rangle|}{A_R} = 0.$$

The perimeter coefficient is not universal because it changes under local line counterterms. The statement $\sigma_\gamma > 0$, when defined by the area coefficient after perimeter renormalization, is the robust asymptotic claim. In a gapless Coulomb regime neither the area/perimeter dichotomy nor the word "string tension" is by itself a complete asymptotic description.

Definition 116.3 (Static potential from rectangular lines)**.** For a rectangular Euclidean loop $C_{T,L}$ of time length T and spatial separation L, define

$$V_\gamma(L) = -\lim_{T\to\infty} \frac{1}{T} \log\langle L_\gamma^{\mathrm{ren}}(C_{T,L})\rangle$$

when the limit exists after line renormalization. Linear confinement of the external charge γ means

$$V_\gamma(L) = \sigma_\gamma L + o(L), \qquad \sigma_\gamma > 0.$$

Screening means that $V_\gamma(L)$ remains bounded as $L \to \infty$.

Transfer-matrix extraction of the static potential. Assume a reflection-positive regulator with a transfer matrix, and assume the rectangular line creates a nonzero vector $\Psi_{\gamma,L}$ in the Hilbert space with static external sources separated by L. If

$$\langle L_\gamma^{\mathrm{ren}}(C_{T,L})\rangle = \langle \Psi_{\gamma,L}, e^{-T(H_{\gamma,L}-E_0)}\Psi_{\gamma,L}\rangle$$

and the spectral measure of $\Psi_{\gamma,L}$ has bottom $E_\gamma(L)$, then $V_\gamma(L) = E_\gamma(L)$.

Let $\rho_{\gamma,L}$ be the positive finite spectral measure of $H_{\gamma,L} - E_0$ in the vector $\Psi_{\gamma,L}$. The loop expectation is

$$\int e^{-TE}\, d\rho_{\gamma,L}(E).$$

If $E_\gamma(L)$ is the bottom of the support of this measure, then every interval $[E_\gamma(L), E_\gamma(L)+\varepsilon]$ has positive measure. The elementary Laplace principle for a finite positive measure gives

$$-\lim_{T\to\infty}\frac{1}{T}\log\int e^{-TE}\, d\rho_{\gamma,L}(E) = E_\gamma(L).$$

Indeed, the upper and lower bounds follow respectively from $E \geq E_\gamma(L)$ and from restricting the integral to $[E_\gamma(L), E_\gamma(L)+\varepsilon]$, then letting $\varepsilon \downarrow 0$.

116.3 Screening and String Breaking

Screening is a statement about endpoint maps. If a dynamical finite-energy operator or state carries line charge $s \in S$, then a line of charge $\gamma + s$ and a line of charge γ differ by an endpoint that can be supplied from the Hilbert space. The static probe cannot distinguish the two charges at arbitrarily long distance.

Endpoint screening bound on the static energy. Assume a charge γ can be screened by a pair of dynamical particles of finite mass M_{scr}, in the sense that the external-source sector at separation L contains a trial state equal outside bounded neighborhoods of the two sources to the vacuum, with energy at most $2M_{\mathrm{scr}} + o(1)$ above the vacuum. Then

$$\limsup_{L\to\infty} V_\gamma(L) \leq 2M_{\mathrm{scr}}.$$

In particular this external charge cannot have a positive linear static potential. This is a variational consequence of the trial-state hypothesis. In the transfer-matrix construction, $V_\gamma(L)$ is the infimum of the static-source Hamiltonian spectrum in the sector with external charge γ at separation L. For any normalized trial state Ψ_L in that sector,

$$V_\gamma(L) \leq \langle \Psi_L, H_{\gamma,L}\Psi_L\rangle - E_{\mathrm{vac}}.$$

The assumed screened trial state has the right charge at the endpoints and has energy $2M_{\mathrm{scr}}+o(1)$ above the vacuum. Taking the limsup gives the displayed bound. A function bounded along large L cannot satisfy $V_\gamma(L) = \sigma L + o(L)$ with $\sigma > 0$. The nontrivial physical input is not the inequality but the construction of the screened endpoint state in the regulated theory and its control under the large-volume and large-L limits.

This bound gives the precise endpoint-screening content needed to interpret Wilson loops in theories with dynamical fundamental matter. The closed loop may display a large intermediate string regime, while the asymptotic static energy is bounded once the string can break by pair creation. Fredenhagen–Marcu type ratios, defined in Chapter 119, measure a different overlap after dividing out the universal flux cost.

116.4 Strong-Coupling Lattice Area Law

Consider a hypercubic Euclidean lattice gauge theory with compact group G. Let each plaquette Boltzmann weight be a central function with character expansion

$$w(U_p) = \sum_R c_R \chi_R(U_p), \qquad c_{\mathbf{1}} > 0.$$

Write

$$q_R = \frac{c_R}{c_{\mathbf{1}}}.$$

At strong coupling the nontrivial q_R are small. The Wilson loop in an irreducible representation R_0 is

$$W_{R_0}(C) = \operatorname{tr}_{R_0} \prod_{\ell \in C} U_\ell.$$

The key algebraic fact is Haar projection: an integral over a link vanishes unless the tensor product of all representation matrices incident on that link contains the trivial representation.

Surface selection by Haar projection. In the character expansion of a Wilson-loop expectation, every nonzero term contains a collection of plaquettes whose oriented boundary carries the representation flux inserted by the Wilson loop. In particular a Wilson loop with nonscreened charge cannot be saturated by plaquettes supported away from a spanning surface.

Expand each plaquette weight in characters and integrate link by link. A link on the Wilson contour carries one matrix factor in R_0 or R_0^*. Haar integration over that link is the projection onto the invariant subspace of the tensor product of all representation factors incident on the link. Unless adjacent plaquette characters supply flux whose oriented boundary cancels the Wilson factor, the tensor product has no invariant vector in the center-sensitive sector and the integral is zero. Repeating this for every link on the contour forces a two-chain of plaquettes whose boundary is C, up to screened charges. If the charge is nonscreened, the boundary cannot be removed by local endpoints.

Controlled Approximation 116.4 (Area mechanism)**.**

Let R_0 have nonscreened center-sensitive charge. For a family of planar loops C_R, let $A_{\min}(C_R)$ be the minimal lattice area of a plaquette surface spanning C_R. A strong-coupling proof of an area mechanism consists of the following uniform polymer-expansion estimates. First, the Haar-projection selection rule permits no center-sensitive contribution of area $< A_{\min}(C_R)$. Second, the sum of all minimal-area contributions is nonzero and has magnitude at least

$$m_0 \exp[-\sigma_+ A_{\min}(C_R)]$$

for some $m_0 > 0$ and finite σ_+. Third, the total contribution of all larger decorated surfaces is bounded by

$$M_0 \exp[-(\sigma_+ + \delta)A_{\min}(C_R)]$$

for some $M_0 > 0$ and $\delta > 0$, after possibly restricting to a sufficiently small strong-coupling neighborhood. Once these estimates have actually been proved for the chosen lattice action, there are constants $a, b > 0$ such that

$$e^{-bA_{\min}(C_R)} \leq |\langle W_{R_0}(C_R)\rangle| \leq e^{-aA_{\min}(C_R)}$$

for all sufficiently large loops in the family, up to subexponential factors. Consequently every subsequential exponential rate $-A_{\min}(C_R)^{-1} \log |\langle W_{R_0}(C_R)\rangle|$ is strictly positive and finite.

The first hypothesis is the kinematic selection part: link integration forces the center-sensitive flux to be carried by a plaquette two-chain with boundary C_R, so no term of smaller area can occur. The third hypothesis gives the upper bound after choosing $a < \sigma_+$ and absorbing the subexponential number of local decorations into the prefactor. For the lower bound, write the expectation as $M_R + E_R$, where M_R is the minimal-area sum and $|E_R| \leq M_0 e^{-(\sigma_+ + \delta)A_{\min}}$. The lower estimate on $|M_R|$ gives

$$|M_R + E_R| \geq m_0 e^{-\sigma_+ A_{\min}} \left(1 - \frac{M_0}{m_0} e^{-\delta A_{\min}}\right),$$

which is at least $\frac{1}{2} m_0 e^{-\sigma_+ A_{\min}}$ for sufficiently large loops. This gives the lower exponential bound with any $b > \sigma_+$, after absorbing constants into the allowed subexponential factors. The logarithmic rate statement follows immediately. The exact string tension is not obtained from these estimates alone; it depends on the complete convergent polymer sum and on the lattice action.

This becomes a theorem about a controlled lattice regime only when the displayed area estimates are proved for the chosen action. Its continuum analogue requires a continuum limit, a mass gap or other infrared control, and renormalized line operators whose asymptotics survive the limit.

116.5 Three-Dimensional Monopole Gas Area Mechanism

There is one important continuum semiclassical setting where magnetic topological events do produce a Wilson-loop area law by a direct calculation: compact $U(1)$ gauge theory in three Euclidean dimensions, or the long-distance $U(1)$ sector of the three-dimensional Georgi–Glashow model after $SU(2)$ is Higgsed to $U(1)$. The magnetic events are monopole-instantons, not four-dimensional BPST instantons. Their physical effect is not a theta-vacuum cumulant; it is a mass for the dual photon and a domain-wall cost for an electric Wilson loop.

Controlled Approximation 116.5 (3D monopole gas area law)**.** Assume a three-dimensional compact Abelian infrared window in which the heavy charged fields are separated from the monopole scale, primitive monopoles and anti-monopoles have a finite fugacity ζ_{M}, and the dilute Coulomb-gas expansion is controlled on distances large compared with the monopole core. Let $\varphi \sim \varphi + 2\pi$ be the dual photon and choose the local normalization

$$S_{\mathrm{dual}}[\varphi] = \int_{\mathbb{R}^3} \left[\frac{\kappa_{\mathrm{d}}}{2}(\partial\varphi)^2 + 2\zeta_{\mathrm{M}}(1 - \cos\varphi)\right] \mathrm{d}^3x. \tag{116.3}$$

This is the sine–Gordon representative of the monopole and anti-monopole grand canonical sum in this normalization. Expanding at a vacuum $\varphi = 2\pi n$ gives the dual-photon mass

$$m_\gamma^2 = \frac{2\zeta_{\mathrm{M}}}{\kappa_{\mathrm{d}}}. \tag{116.4}$$

For a primitive electric Wilson loop $W(C)$ which is not screened by dynamical endpoints in the declared low-energy theory, the dual description imposes a 2π discontinuity of φ across an oriented spanning surface Σ with $\partial\Sigma = C$. For a large nearly planar loop, the leading saddle is a one-dimensional sine–Gordon wall normal to Σ. It satisfies

$$\kappa_{\mathrm{d}}\varphi'' = 2\zeta_{\mathrm{M}} \sin\varphi, \qquad \varphi(-\infty) = 0, \quad \varphi(+\infty) = 2\pi,$$

or, equivalently, the first-order equation

$$\varphi' = 2m_\gamma \sin\frac{\varphi}{2}. \tag{116.5}$$

The wall tension is therefore

$$\sigma_{\mathrm{P}} = \int_0^{2\pi} \sqrt{2\kappa_{\mathrm{d}}\, 2\zeta_{\mathrm{M}}(1-\cos\varphi)}\,\mathrm{d}\varphi = 8\sqrt{2\kappa_{\mathrm{d}}\zeta_{\mathrm{M}}} = 8\kappa_{\mathrm{d}} m_\gamma. \tag{116.6}$$

After local line counterterms have removed the perimeter self-energy, the Wilson loop has the semiclassical large-area form

$$-\log|\langle W(C)\rangle| = \sigma_{\mathrm{P}}\,\mathrm{Area}(\Sigma_{\min}) + o(\mathrm{Area}) \tag{116.7}$$

for loop families whose curvature radii and linear sizes are large compared with m_γ^{-1}, provided no lower-energy screened endpoint channel is available.

The derivation is the following finite chain of physical inputs. The monopole fugacity inserts $\mathrm{e}^{\mathrm{i}\varphi}$, the anti-monopole fugacity inserts $\mathrm{e}^{-\mathrm{i}\varphi}$, and the dilute grand canonical sum gives the cosine potential in (116.3). The cosine pins the dual photon and opens the mass gap (116.4). The Wilson loop is not evaluated by a monopole moduli-space volume; it changes the dual-photon boundary condition across Σ. The minimum action configuration with that discontinuity is the sine–Gordon wall, and its action per unit area is (116.6).

The scope of this mechanism is the three-dimensional compact-Abelian semiclassical window. Here the semiclassical objects live in three Euclidean dimensions, the dual photon is a local scalar in the infrared effective theory, and the Wilson loop is converted into a calculable wall problem. The dilute BPST instanton gas of Chapter 51 supplies theta-dependent amplitude data in four dimensions rather than this dual-photon wall mechanism.

116.6 Condensation and Oblique Confinement

The finite part of a gapped line-operator response can often be organized by an isotropic subgroup

$$K \subset \mathcal{C}$$

of condensed charges. The isotropy condition

$$B(k,k') = 0, \qquad k, k' \in K,$$

is mutual locality of the condensed objects. It is a consistency condition, not a dynamical proof that such objects actually condense.

Definition 116.6 (Finite confinement diagnostic from a condensate)**.** Given a finite charge system $(\mathcal{C}, B)$ and an isotropic subgroup $K \subset \mathcal{C}$, define

$$K^{\perp} = \{\gamma \in \mathcal{C} \mid B(\gamma, k) = 0 \text{ for every } k \in K\}.$$

The finite condensate diagnostic declares γ unconfined only if $\gamma \in K^{\perp}$. Charges outside $K^{\perp}$ have nontrivial finite braiding with the condensate and are confined in this diagnostic.

Assume a gapped regime in which perimeter-law lines must braid trivially with all condensed charges in an isotropic subgroup $K \subset \mathcal{C}$. Then only charges in $K^{\perp}$ can have perimeter behavior. If K is maximal isotropic, then $K^{\perp} = K$. Moving a test line of charge γ once around a condensed object of charge k multiplies the state by

$$\exp(2\pi \mathrm{i} B(\gamma, k)).$$

A perimeter-law line compatible with the condensate background must be single-valued under this operation for every $k \in K$. This is exactly the condition $\gamma \in K^{\perp}$.

If $K \subsetneq K^{\perp}$, choose $\eta \in K^{\perp} \setminus K$. Since the pairing is alternating, $B(\eta, \eta) = 0$, and by definition $B(\eta, K) = 0$. Therefore the subgroup generated by K and η is isotropic and strictly larger than K, contradicting maximal isotropy. Hence a maximal isotropic K satisfies $K = K^{\perp}$.

Example 116.7 ($\mathfrak{su}(N)$ finite charges)**.** For

$$\mathcal{C} = \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}, \qquad B((e, m), (e', m')) = \frac{em' - e'm}{N},$$

magnetic condensation is $K = \langle (0, 1) \rangle$. A charge (e, m) lies in $K^{\perp}$ exactly when

$$B((e, m), (0, 1)) = \frac{e}{N} = 0 \quad \text{in } \mathbb{Q}/\mathbb{Z},$$

so only $e = 0$ survives the finite diagnostic. Pure electric probes with nonzero N-ality are confined. Electric condensation $K = \langle (1, 0) \rangle$ gives the dual statement: purely magnetic probes with nonzero magnetic center charge are confined.

For a dyonic condensate

$$K_p = \langle (p, 1) \rangle,$$

the unconfined finite charges obey

$$B((e, m), (p, 1)) = \frac{e - pm}{N} = 0, \qquad \text{so} \qquad e \equiv pm \pmod{N}.$$

Thus the perimeter-law direction is the dyonic line generated by the condensate. Pure electric probes with $e \neq 0$ are confined, and pure magnetic probes are confined unless $pm \equiv 0 \pmod{N}$. This is oblique confinement: the unconfined direction is neither the electric nor the magnetic axis.

Figure 116.1 displays the finite charge lattice for this oblique-confinement example: the dyonic condensate selects the perimeter-law direction $e \equiv pm \pmod{N}$.

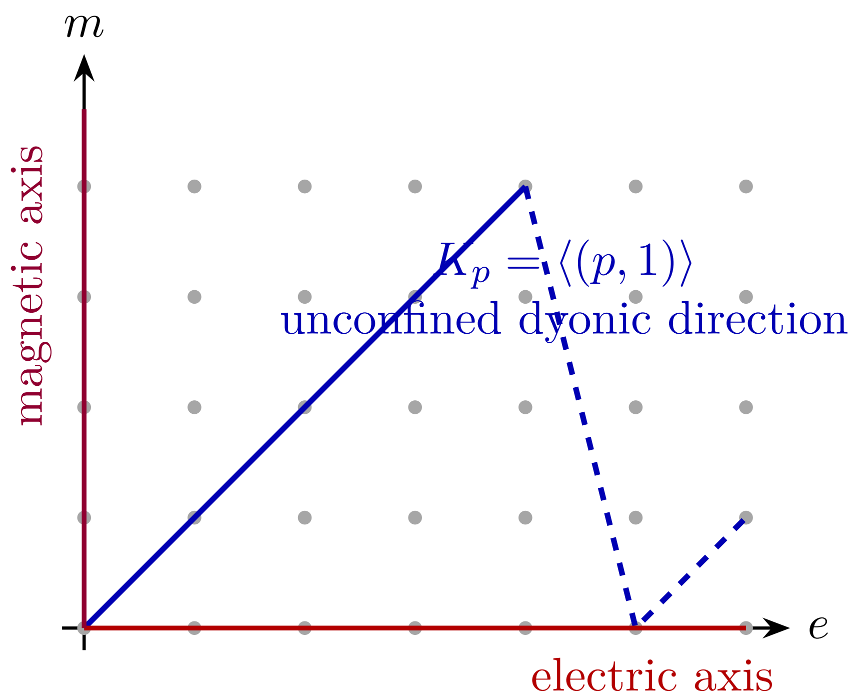

Figure 116.1: Finite charge-lattice picture of oblique confinement. The unconfined finite direction is the dyonic condensate line; charges with nontrivial finite Dirac pairing against it are confined in the condensate diagnostic.

116.7 Logical Status of the Diagnostic

The finite charge-lattice criterion is kinematic plus an explicit dynamical input. The kinematic part is the nondegenerate Dirac pairing and the screening quotient. The dynamical input is the assertion that a particular subgroup K is condensed and that nontrivial braiding with K forces an area law. In a lattice strong-coupling domain the dynamical input can be verified by a convergent expansion, as in the controlled strong-coupling estimate 116.4. In a continuum four-dimensional Yang–Mills theory it is a major theorem-level target rather than a definition.

The following implication is therefore the correct logical form:

$$\begin{array}{c}\text{regulator and continuum limit}\\ \text{renormalized line operators}\\ \text{mass gap or controlled infrared behavior}\\ \text{screening and condensate hypotheses}\end{array} \Longrightarrow \begin{array}{c}\text{area/perimeter laws}\\ \text{electric, magnetic, or oblique}\\ \text{confinement diagnostics.}\end{array}$$

The converse is not automatic. Observing an area law for one family of Wilson loops does not by itself reconstruct the infrared topological sector, the complete screening subgroup, or the full spectrum of stable flux tubes.

Open Problem 116.8 (Continuum confinement criterion). Formulate and prove a continuum QFT confinement criterion that uses renormalized line operators, screening quotients, cluster properties, a mass gap or another controlled infrared spectral hypothesis, and a precise endpoint/condensate structure. The criterion should reproduce the lattice strong-coupling area mechanism where a lattice regulator is available, and it should distinguish electric, magnetic, and oblique confinement through the full line-charge lattice rather than through a single Wilson-loop diagnostic.

Chapter 117

Discrete Theta Terms

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Discrete theta terms are topological phases depending on global bundle data and finite higher-form gauge fields. They are part of the definition of a gauge theory with a specified global form: the same local gauge algebra and the same curvature Lagrangian can have distinct sums over bundles, distinct topological weights, and distinct spectra of genuine line operators. This chapter gives the cochain-level definitions needed for the four-dimensional gauge-theory applications in Volume IX.

Definition 117.1 (Discrete theta structure)**.** Fix a spacetime category $\mathcal{C}_D$, for example closed oriented triangulated D-manifolds or closed spin D-manifolds with compatible triangulations. A finite-cochain discrete theta structure is a tuple

$$\mathfrak{d}_{\mathrm{fin}} = (\mathcal{C}_D, A, q, [\omega], \omega_{\mathrm{loc}})$$

where A is a finite abelian coefficient group for a q-form background, $[\omega]$ is a class in the appropriate group of local topological actions on $B^{q+1}A$, and $\omega_{\mathrm{loc}}(B) \in C^D(M, \mathbb{R}/\mathbb{Z})$ is a natural cocycle representative for each flat background

$$B \in Z^{q+1}(M, A), \qquad B \sim B + \delta\Lambda.$$

The exponentiated phase is

$$S_{\mathfrak{d}_{\mathrm{fin}}}[M, B] = \exp\left(2\pi\mathrm{i}\int_M \omega_{\mathrm{loc}}(B)\right),$$

and changing ω_{loc} by a natural coboundary gives the same closed-manifold phase.

For a four-dimensional Yang–Mills theory with a fixed local Lie algebra, a global-form discrete theta structure additionally specifies

$$\mathfrak{d}_{\mathrm{YM}} = \big(\mathfrak{g}, G, \mathcal{B}_G(M), w_2^\Gamma, \theta, \tau_{\mathrm{disc}}, L_{\mathrm{line}}\big).$$

Here $G = \widetilde{G}/\Gamma$ is the chosen global form of the compact gauge group, $\mathcal{B}_G(M)$ is the groupoid of principal G-bundles, w_2^Γ is the obstruction to lifting a G-bundle to a $\widetilde{G}$-bundle, θ is the continuous theta coordinate, τ_{disc} is the finite topological weight assigned to the obstruction and any higher-form backgrounds, and L_{line} is the maximal mutually local lattice of genuine Wilson–'t Hooft line charges. For $G = SU(N)/\mathbb{Z}_k$ this last entry is denoted

$$L_{\mathrm{line}} = L_{N,k,p} = \langle (k,0), (p, N/k)\rangle \subset \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}, \qquad p \in \mathbb{Z}_k.$$

Thus a discrete theta coordinate is part of the global definition of the theory: it is not visible in perturbation theory around the trivial bundle, but it changes the bundle weights and the genuine line spectrum.

117.1 Finite Cochains And Topological Actions

Let M be a closed oriented triangulated D-manifold. For an abelian group A, write

$$C^r(M,A), \qquad Z^r(M,A), \qquad B^r(M,A), \qquad H^r(M,A)$$

for r-cochains, cocycles, coboundaries, and cohomology classes. The coboundary is denoted δ. A finite A-valued zero-form gauge field is a one-cocycle

$$a \in Z^1(M,A),$$

with gauge equivalence

$$a \sim a + \delta\lambda, \qquad \lambda \in C^0(M,A).$$

For a finite p-form symmetry or gauge field, the background is instead a $(p+1)$-cocycle

$$B \in Z^{p+1}(M,A), \qquad B \sim B + \delta\Lambda, \quad \Lambda \in C^p(M,A).$$

The finite-cochain part of Definition 117.1 is implemented by an exponentiated finite topological action, namely a function

$$S_{\mathrm{top}} : Z^{p+1}(M,A) \longrightarrow U(1)$$

that is invariant under subdivision and gauge transformation, and is local in the sense that it is obtained by evaluating a D-cocycle built from cochains on the fundamental class of M. For a zero-form finite gauge field this structure is equivalently represented by a class

$$[\omega] \in H^D(BA, U(1)).$$

For a p-form field it is represented by a class on $B^{p+1}A$. Explicitly, if $\omega(a) \in C^D(M, \mathbb{R}/\mathbb{Z})$ is a cocycle depending naturally on a, then

$$S_{\mathrm{top}}[a] = \exp\left(2\pi \mathrm{i} \int_M \omega(a)\right).$$

Here

$$\int_M : C^D(M, \mathbb{R}/\mathbb{Z}) \longrightarrow \mathbb{R}/\mathbb{Z}$$

is evaluation on the fundamental cycle.

Remark 117.2 (Coboundary invariance on closed manifolds). If two finite topological Lagrangians differ by a coboundary,

$$\omega'(a) - \omega(a) = \delta\alpha(a),$$

then their exponentiated actions agree on every closed D-manifold for every flat finite background field a.

Let c_M be a fundamental D-cycle of M. Since M is closed,

$$\partial c_M = 0.$$

The evaluation pairing satisfies

$$\langle \delta\alpha(a), c_M \rangle = \langle \alpha(a), \partial c_M \rangle = 0 \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Therefore

$$\exp\left(2\pi \mathrm{i} \int_M \omega'(a)\right) = \exp\left(2\pi \mathrm{i} \int_M \omega(a)\right).$$

For a manifold with boundary, $\partial c_M = c_{\partial M}$, so the coboundary contribution is

$$\langle \delta\alpha(a), c_M \rangle = \langle \alpha(a), c_{\partial M} \rangle = \int_{\partial M} \alpha(a).$$

This boundary evaluation is the boundary term used in anomaly inflow.

117.2 Two-Form $\mathbb{Z}_N$ Fields

The four-dimensional examples use a two-form finite field

$$B \in Z^2(M, \mathbb{Z}_N).$$

Choose an integral lift

$$\widetilde{B} \in C^2(M, \mathbb{Z})$$

whose reduction modulo N is B. Since B is closed modulo N, there is an integral three-cochain C such that

$$\delta\widetilde{B} = NC.$$

The Pontryagin square is the cohomology operation

$$\mathcal{P} : H^2(M, \mathbb{Z}_N) \longrightarrow H^4(M, \mathbb{Z}_{2N})$$

characterized by

$$\mathcal{P}(B) \equiv B \smile B \pmod{N}$$

and by the quadratic-refinement identity

$$\mathcal{P}(B + B') = \mathcal{P}(B) + \mathcal{P}(B') + 2\, B \smile B' \quad \text{in } H^4(M, \mathbb{Z}_{2N}).$$

A cochain representative is

$$\mathcal{P}(B) = \widetilde{B} \smile \widetilde{B} - N\,\widetilde{B} \smile_1 C \mod 2N,$$

where $\smile_1$ is Steenrod's first cup product. The second term is what makes the expression independent of the chosen lift and invariant under $B \mapsto B + \delta\Lambda$. When N is odd, 2 is invertible modulo N, and the same information can be represented by the ordinary square

$$B \smile B \in H^4(M, \mathbb{Z}_N).$$

When N is even, the lift to $\mathbb{Z}_{2N}$ is essential.

The corresponding four-dimensional local counterterms for a background $\mathbb{Z}_N$ one-form symmetry have the form

$$S_p[B] = \exp\left(\frac{2\pi \mathrm{i} p}{2N} \int_M \mathcal{P}(B)\right).$$

On oriented manifolds the coefficient is valued in

$$p \in \begin{cases} \mathbb{Z}_N, & N \text{ odd}, \\ \mathbb{Z}_{2N}, & N \text{ even}, \end{cases}$$

using the identifications of $H^4(B^2\mathbb{Z}_N, U(1))$. On spin manifolds some terms that differ on a general oriented manifold can coincide, because the Wu-class relations constrain the integral of the Pontryagin square. For odd N, the displayed formula is read as

$$\frac{p}{2N}\mathcal{P}(B) = \frac{p}{N}\,\frac{1}{2} B \smile B \quad \text{in } \mathbb{R}/\mathbb{Z},$$

where $1/2$ denotes the inverse of 2 in $\mathbb{Z}_N$. Thus the spacetime category is part of the definition of the available counterterm group.

Remark 117.3 (Quadratic refinement phase). Let $B, B' \in H^2(M, \mathbb{Z}_N)$. The ratio of counterterms obeys

$$\frac{S_p[B + B']}{S_p[B] S_p[B']} = \exp\left(\frac{2\pi \mathrm{i} p}{N} \int_M B \smile B'\right).$$

Use the defining identity of the Pontryagin square:

$$\mathcal{P}(B + B') - \mathcal{P}(B) - \mathcal{P}(B') = 2B \smile B' \mod 2N.$$

Substituting this into the exponent gives

$$\frac{2\pi \mathrm{i} p}{2N} \int_M 2B \smile B' = \frac{2\pi \mathrm{i} p}{N} \int_M B \smile B'.$$

Exponentiation gives the stated ratio. Thus S_p is a quadratic refinement of the bilinear pairing $(B, B') \mapsto \exp[(2\pi \mathrm{i} p/N) \int_M B \smile B']$. The formula is independent of the chosen cochain representatives because the Pontryagin square identity is a cohomological identity modulo $2N$.

117.3 Global Form And Obstruction Classes

Let

$$G = SU(N)/\mathbb{Z}_k, \qquad k \mid N,$$

where $\mathbb{Z}_k \subset Z(SU(N))$ is generated by

$$\exp(2\pi\mathrm{i}/k)\mathbf{1}_N = \exp(2\pi\mathrm{i}\, N/kN)\mathbf{1}_N.$$

A principal G-bundle $P_G \to M$ has an obstruction class

$$w_2^{(k)}(P_G) \in H^2(M, \mathbb{Z}_k)$$

to lifting P_G to an $SU(N)$-bundle. This class is obtained from the central extension

$$1 \longrightarrow \mathbb{Z}_k \longrightarrow SU(N) \longrightarrow SU(N)/\mathbb{Z}_k \longrightarrow 1$$

by the connecting homomorphism in nonabelian cohomology. Concretely, choose local lifts of the G-transition functions to $SU(N)$. On triple overlaps the lifted transition functions multiply to a $\mathbb{Z}_k$-valued two-cocycle. Its cohomology class is $w_2^{(k)}(P_G)$. Changing the local lifts changes this two-cocycle by a coboundary.

For $k = N$, $G = PSU(N)$ and the obstruction class is usually written

$$w_2(P_{PSU(N)}) \in H^2(M, \mathbb{Z}_N).$$

Proposition 117.4 (Fractional $PSU(N)$ instanton number)**.** *With the instanton-number convention compatible with*

$$k = c_2(E)$$

for an $SU(N)$-bundle E, the fractional part of the instanton number of a $PSU(N)$-bundle is fixed by its obstruction class:

$$\nu(P_{PSU(N)}) = -\frac{N-1}{2N}\int_M \mathcal{P}(w_2) \quad \mathit{mod}\ 1.$$

Proof. A projective $PSU(N)$-bundle can be described locally by $SU(N)$ transition functions that close on triple overlaps up to a central $\mathbb{Z}_N$-valued two-cocycle representing w_2. Equivalently, for the purpose of computing the fractional characteristic class, choose local $U(N)$ lifts and write x for the integral Cech two-cocycle that reduces to w_2 modulo N. If the projective bundle is represented by an ordinary $U(N)$-bundle E, then $x = c_1(E)$; the local Cech computation gives the same expression in general.

The projective second Chern coordinate is

$$\nu(E) = c_2(E) - \frac{N-1}{2N}c_1(E)^2 \quad \in \mathbb{Q}.$$

This combination is invariant under tensoring E by a line bundle. Indeed, if $\ell = c_1(L)$, then

$$c_1(E \otimes L) = c_1(E) + N\ell$$

and

$$c_2(E \otimes L) = c_2(E) + (N-1)c_1(E)\ell + \frac{N(N-1)}{2}\ell^2.$$

Substitution gives

$$\begin{aligned} c_2(E \otimes L) - \frac{N-1}{2N}c_1(E \otimes L)^2 \\ = c_2(E) - \frac{N-1}{2N}c_1(E)^2. \end{aligned}$$

Thus ν is a characteristic class of the projective bundle. Since $c_2(E)$ is integral, its fractional part is

$$-\frac{N-1}{2N}c_1(E)^2 \mod 1.$$

The Pontryagin square is the lift of w_2^2 modulo $2N$; hence

$$c_1(E)^2 \equiv \mathcal{P}(w_2) \pmod{2N}.$$

The displayed formula follows. □

This characteristic-class formula is the global reason that shifting the ordinary continuous theta angle by 2π changes the discrete theta coordinate in a theory whose bundles fail to lift to $SU(N)$.

117.4 Discrete Theta Coordinate For $SU(N)/\mathbb{Z}_k$

The pure $SU(N)/\mathbb{Z}_k$ Yang–Mills theories have a discrete theta coordinate

$$p \in \mathbb{Z}_k$$

in addition to the continuous theta angle. This coordinate can be detected without writing a continuum path integral: it is the electric tilt of the minimal magnetic line. With the finite charge notation of Chapter 113, write a dyonic line label as

$$(e, m) \in \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}.$$

The genuine line lattice is

$$L_{N,k,p} = \langle (k, 0), (p, N/k) \rangle.$$

The generator $(k, 0)$ records the Wilson lines that descend to $SU(N)/\mathbb{Z}_k$. The generator $(p, N/k)$ records the minimal magnetic line, whose magnetic charge is N/k and whose electric charge is shifted by the discrete theta coordinate p. The identification

$$p \sim p + k$$

holds because

$$(p + k, N/k) = (p, N/k) + (k, 0).$$

Thus p is naturally a class in $\mathbb{Z}_k$.

Remark 117.5 (Discrete theta tilt and maximal locality). For each $p \in \mathbb{Z}_k$, the line lattice

$$L_{N,k,p} = \langle (k,0), (p, N/k) \rangle \subset \mathbb{Z}_N \oplus \mathbb{Z}_N$$

is maximal isotropic for the finite Dirac pairing

$$\overline{B}((e,m),(e',m')) = \frac{em' - e'm}{N} \quad \in \mathbb{Q}/\mathbb{Z}.$$

Changing p changes the electric charge of the minimal magnetic line by a Witten-effect shift.

The isotropy and maximality are proved in Proposition 113.5. The new point is the interpretation of p. The minimal magnetic generator has magnetic component N/k, the generator of the magnetic subgroup allowed by the global form. Its electric component is p. Replacing p by $p+k$ adds the Wilson generator $(k,0)$, so the subgroup is unchanged. Therefore the inequivalent tilts are indexed by $p \in \mathbb{Z}_k$. A shift of the continuous theta angle by 2π changes the electric charge of a magnetic line by its magnetic charge in the finite pairing convention; in these coordinates this sends

$$p \longmapsto p+1 \quad \text{in } \mathbb{Z}_k.$$

For $N = 2$ this gives the familiar three finite line spectra:

$$SU(2): \ \langle (1,0) \rangle, \qquad SO(3)_+: \ \langle (0,1) \rangle, \qquad SO(3)_-: \ \langle (1,1) \rangle.$$

The last two have the same local Lie algebra and the same quotient gauge group $SO(3) = SU(2)/\mathbb{Z}_2$, but different discrete theta coordinate and different dyonic line spectrum.

117.5 Gauging One-Form Symmetry And Counterterms

Start with the $SU(N)$ theory and its electric $\mathbb{Z}_N$ one-form symmetry. Coupling this symmetry to a background two-form field

$$B \in Z^2(M, \mathbb{Z}_N)$$

modifies the allowed bundles by allowing transition functions to close only up to the center. Gauging a subgroup $\mathbb{Z}_k \subset \mathbb{Z}_N$ means summing over the corresponding two-form gauge field, with a choice of Pontryagin-square counterterm. The result is an $SU(N)/\mathbb{Z}_k$ theory, and the counterterm choice is the discrete theta coordinate p seen by the line lattice above.

The counterterm also controls anomalies. If the generating functional in background B transforms under a one-form gauge transformation by a phase

$$Z[B + \delta\Lambda] = \exp\left(2\pi \mathrm{i} \int_M \alpha(B,\Lambda)\right) Z[B],$$

then the phase is removed by a local counterterm exactly when α is the gauge variation of a D-dimensional local cochain. If the phase is the boundary variation of a $(D+1)$-dimensional invertible theory, it is an anomaly. Discrete theta terms and anomaly inflow are therefore the same cohomological mechanism viewed in neighboring dimensions: in D dimensions the cochain changes the definition of the theory, and in $D+1$ dimensions its variation supplies an inflow phase for a boundary.

117.6 Continuum Gauge-Theory Status

A continuum Yang–Mills theory with nontrivial global form is specified by the global-form discrete theta datum of Definition 117.1. For the $SU(N)/\mathbb{Z}_k$ family the entries are:

local data	$\mathfrak{su}(N)$ fields and curvature action,
global form	$SU(N)/\mathbb{Z}_k$,
bundle sum	principal $SU(N)/\mathbb{Z}_k$ bundles on M,
topological weights	θ and $p \in \mathbb{Z}_k$,
line spectrum	$L_{N,k,p} \subset \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}$,
background fields	two-form backgrounds for remaining one-form symmetries.

A lattice regulator implements the same data by plaquette variables and finite cochains for higher-form backgrounds. Matching a lattice construction to the continuum description requires matching the obstruction class $w_2^{(k)}$, the Pontryagin-square phase, the continuous theta coordinate, and the genuine line spectrum. Perturbation theory around the trivial bundle sees only the local Lie algebra and curvature vertices; the discrete theta data are detected by global bundle sectors and extended operators.

Open Problem 117.6 (Discrete theta terms in interacting continuum QFT)*.* Give a nonperturbative construction of four-dimensional gauge theories with all global forms and discrete theta terms in which bundles, one-form backgrounds, line operators, anomalies, and continuum limits are treated in a single framework, and prove the equivalence between continuum characteristic classes, lattice finite-cochain definitions, and the operator line spectra.

Chapter 118

Anomaly Inflow and Invertible Field Theories

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

An anomaly is a precise failure of background-field covariance. The partition function of an anomalous D-dimensional theory is not an ordinary function on the moduli groupoid of background fields; it is a vector in a one-dimensional complex vector space depending on the background. An invertible $(D+1)$-dimensional theory packages these lines and their gluing laws. Inflow is the statement that the anomalous D-dimensional theory is a boundary condition, or relative theory, for this invertible bulk.

Definition 118.1 (Anomaly-inflow structure)**.** Fix a spacetime dimension D. An anomaly-inflow structure is a tuple

$$\mathfrak{A} = \left(\mathcal{B}, \mathrm{Bord}^{\mathcal{B}}_{D+1}, \mathcal{Z}_{\mathrm{bulk}}, \mathcal{L}, Z_\partial\right)$$

with the following entries.

1. $\mathcal{B}$ assigns to each closed D-manifold M, with the declared spacetime structure, a groupoid $\mathcal{B}(M)$ of background fields and background isomorphisms.
2. $\mathrm{Bord}^{\mathcal{B}}_{D+1}$ is the corresponding background bordism category. Its objects are pairs (M, A), $A \in \mathcal{B}(M)$; its morphisms are $(D+1)$-dimensional bordisms $X : M_0 \to M_1$ equipped with background fields restricting to the prescribed boundary backgrounds.
3. $\mathcal{Z}_{\mathrm{bulk}}$ is an invertible $(D+1)$-dimensional field theory,

$$\mathcal{Z}_{\mathrm{bulk}} : \mathrm{Bord}^{\mathcal{B}}_{D+1} \longrightarrow \mathrm{Lines}_{\mathbb{C}},$$

namely a symmetric monoidal functor to one-dimensional complex vector spaces and nonzero linear maps. On a closed $(D+1)$-manifold X it gives a nonzero complex number.

4. $\mathcal{L}$ is the boundary line functor induced by $\mathcal{Z}_{\mathrm{bulk}}$. For each closed M it is a functor

$$\mathcal{L}_M : \mathcal{B}(M) \longrightarrow \mathrm{Lines}_{\mathbb{C}},$$

with the orientation convention that a boundary partition vector $Z_{\partial,M}(A)$ lies in $\mathcal{L}_M(A)$. When fillings exist, $\mathcal{L}_M(A)$ may be constructed by the filling quotient described in Section 118.2; otherwise the functor $\mathcal{L}$ is taken as the boundary value of the invertible bulk.

5. Z_∂ is a relative D-dimensional QFT partition assignment: for each closed (M, A),

$$Z_{\partial,M}(A) \in \mathcal{L}_M(A),$$

with covariance under background isomorphisms and with the usual gluing law after pairing with compatible bulk states.

A trivialization of the anomaly is a monoidal natural isomorphism $\mathcal{L} \simeq \mathbf{1}$, possibly after tensoring with a local counterterm invertible theory or with an inverse anomalous sector. A coordinate anomaly formula, such as a descent form or a finite cochain cocycle, is a representative of this structure after choices of local frames and local representatives have been made.

This definition is the object controlled throughout the chapter. The connected Chern–Weil descent, the finite higher-form cochain models, the Weyl-fermion index representative, and the noninvertible wall examples below are different coordinate presentations of the same kind of background-line and invertible-bulk structure.

Coordinate comparison. The anomaly line has three coordinate shadows used in different volumes. In a finite-regulator chart such as the one used in Chapter 51, a background transformation $g : A \to A^g$ is represented by the additive cochain

$$C_\Lambda(g; A) = W_\Lambda[A^g] - W_\Lambda[A] \quad \in \mathbb{R}/\mathbb{Z}, \qquad C_\Lambda(gh; A) = C_\Lambda(g; A) + C_\Lambda(h; A^g). \tag{118.1}$$

After a local frame s_A for the anomaly line is chosen, the same information is the multiplicative line cocycle

$$\alpha_\Lambda(A, g) = \exp\bigl(2\pi\mathrm{i} C_\Lambda(g; A)\bigr), \qquad \mathcal{L}(g) s_A = \alpha_\Lambda(A, g) s_{A^g}. \tag{118.2}$$

Adding a local finite-regulator counterterm $B_\Lambda(A)$ shifts the effective-action representative by

$$C_\Lambda(g; A) \longmapsto C_\Lambda(g; A) + B_\Lambda(A^g) - B_\Lambda(A). \tag{118.3}$$

Equivalently, moving that counterterm from the scalar coordinate into the line frame uses the inverse frame change $s_A \mapsto \exp(-2\pi\mathrm{i} B_\Lambda(A)) s_A$, giving the same transformed α_Λ.

The inflow bulk carries the inverse coordinate. If a cylinder or mapping cylinder realizes the transformation g, then the bulk transition phase on that bordism is

$$\mathcal{Z}_{\mathrm{bulk}}([0, 1] \times_g M) = \alpha_\Lambda(A, g)^{-1} = \exp\bigl(-2\pi\mathrm{i} C_\Lambda(g; A)\bigr), \tag{118.4}$$

so that the boundary vector and the bulk state pair to a background-invariant number. In the free-fermion global-anomaly chapter the same line cocycle is computed analytically by determinant-line transport,

$$\alpha_{\det}(A, g) = \mathrm{Hol}_g(\mathrm{Det}\, D) = \exp\bigl(-2\pi\mathrm{i}\, \xi(B_{Y_g})\bigr)\,,$$

or by the corresponding Pfaffian sign in the real case. Thus the finite cochain, the inflow boundary line, and the determinant/Pfaffian holonomy are not separate anomaly notions; they are regulator, invertible-bulk, and global-analysis coordinates on the same obstruction to descent. The analytic theorems quoted in Chapter 153 compute this coordinate for Dirac families, while the algebra above is the QFT-facing descent and cancellation bookkeeping.

118.1 Background Fields and Relative Partition Functions

Let M be a closed D-manifold with the spacetime structure required by the theory: orientation, spin structure, metric, conformal class, or another declared structure. Let

$$\mathcal{B}(M)$$

be the background-field groupoid in the anomaly-inflow structure $\mathfrak{A}$ of Definition 118.1. An object $A \in \mathcal{B}(M)$ may contain ordinary gauge fields, higher-form gauge fields, metric data, spin connections, and background defect data. A morphism

$$u : A \longrightarrow A'$$

is a background-field isomorphism, for example a gauge transformation, higher-form gauge transformation, or diffeomorphism preserving the declared spacetime structure.

For a nonanomalous theory the partition function is a function on isomorphism classes:

$$Z_M(A') = Z_M(A) \quad \text{whenever } A \simeq A'.$$

For an anomalous theory the correct object is a complex line over each background. Write

$$\mathcal{L}_M : \mathcal{B}(M) \longrightarrow \mathrm{Lines}_{\mathbb{C}}$$

for a functor from the background groupoid to the groupoid of one-dimensional complex vector spaces and nonzero linear maps. The anomalous partition function is a section

$$Z_M(A) \in \mathcal{L}_M(A).$$

For a morphism $u : A \to A'$, covariance means

$$Z_M(A') = \mathcal{L}_M(u) Z_M(A),$$

not equality of complex numbers. The functor $\mathcal{L}_M$ is the anomaly line.

Choosing a nonzero vector $s_A \in \mathcal{L}_M(A)$ for every object A trivializes the line as a collection of vector spaces, but generally not as a functor. In such a trivialization,

$$Z_M(A) = z_M(A) s_A, \qquad z_M(A) \in \mathbb{C},$$

and every morphism $u : A \to A'$ acts by a nonzero phase or scalar

$$\mathcal{L}_M(u) s_A = \alpha_M(u; A) s_{A'}.$$

Thus

$$z_M(A') = \alpha_M(u; A) z_M(A).$$

The scalar α_M is not itself the anomaly; it is the anomaly written in a chosen local trivialization.

Remark 118.2 (Functorial cocycle condition). Let $u : A \to A'$ and $v : A' \to A''$ be composable morphisms in $\mathcal{B}(M)$. In any choice of nonzero local frame s_A for the anomaly line,

$$\alpha_M(vu; A) = \alpha_M(v; A')\alpha_M(u; A).$$

If the frame is changed by

$$s'_A = c(A)s_A, \qquad c(A) \in \mathbb{C}^\times,$$

then

$$\alpha'_M(u; A) = c(A')^{-1}\alpha_M(u; A)c(A).$$

Functoriality gives

$$\mathcal{L}_M(vu) = \mathcal{L}_M(v)\mathcal{L}_M(u).$$

Applying both sides to s_A gives

$$\alpha_M(vu; A)s_{A''} = \alpha_M(u; A)\alpha_M(v; A')s_{A''},$$

which is the stated cocycle identity. If $s'_A = c(A)s_A$, then

$$\mathcal{L}_M(u)s'_A = c(A)\alpha_M(u; A)s_{A'} = c(A)\alpha_M(u; A)c(A')^{-1}s'_{A'}.$$

This is exactly the displayed transformation law.

A local counterterm is a choice of local functional $K_M(A)$ whose exponential changes the frame:

$$s_A \longmapsto \exp(2\pi \mathrm{i} K_M(A))s_A.$$

It changes the representative cocycle by a coboundary. Therefore an anomaly class is not a particular variation formula; it is the isomorphism class of the line functor together with its locality and gluing structure.

118.2 Invertible Bulk Theory and Boundary Lines

For the same structure $\mathfrak{A}$, the category $\mathrm{Bord}^{\mathcal{B}}_{D+1}$ has closed D-manifolds with background fields as objects and $(D+1)$-dimensional bordisms with compatible background fields as morphisms. Its invertible bulk entry is a symmetric monoidal functor

$$\mathcal{Z}_{\mathrm{bulk}} : \mathrm{Bord}^{\mathcal{B}}_{D+1} \longrightarrow \mathrm{Lines}_{\mathbb{C}}$$

such that every closed $(D+1)$-manifold X is assigned a nonzero complex number and every D-manifold M is assigned a line. If

$$X = X_1 \cup_M X_2,$$

then the value on X is obtained by pairing the line assigned to M from X_1 with the dual line assigned from X_2. This gluing axiom is the mathematical content behind the word “inflow”.

It is useful to name the two line variances separately. For a boundary background $A \in \mathcal{B}(M)$, write

$$\mathcal{B}^{\mathrm{bulk}}_M(A) := \mathcal{Z}_{\mathrm{bulk}}(M, A)$$

for the bulk state line assigned to the closed D-manifold M. With the convention of Definition 118.1, the boundary anomaly line is the dual line

$$\mathcal{L}_M(A) = \big(\mathcal{B}_M^{\mathrm{bulk}}(A)\big)^{\vee}. \tag{118.5}$$

Thus a relative boundary partition vector is a covector on the bulk state line, while a filling of M supplies a vector in the bulk state line. For a background isomorphism $u : A \to A'$, the bulk line map

$$\mathcal{B}_M^{\mathrm{bulk}}(u) : \mathcal{B}_M^{\mathrm{bulk}}(A) \longrightarrow \mathcal{B}_M^{\mathrm{bulk}}(A')$$

induces the anomaly-line map by inverse duality,

$$\mathcal{L}_M(u) = \big(\mathcal{B}_M^{\mathrm{bulk}}(u)^{-1}\big)^{\vee} : \mathcal{L}_M(A) \longrightarrow \mathcal{L}_M(A').$$

Orientation reversal dualizes these lines:

$$\mathcal{B}_{-M}^{\mathrm{bulk}}(A) \simeq \big(\mathcal{B}_M^{\mathrm{bulk}}(A)\big)^{\vee}, \qquad \mathcal{L}_{-M}(A) \simeq \mathcal{L}_M(A)^{\vee}.$$

This is the same variance used in the Dai–Freed construction of Chapter 153: the boundary fermion determinant is a vector in the anomaly line, and the filling supplies the inverse-line vector with which it is evaluated.

When the bulk is represented locally by a $(D+1)$-cocycle

$$I_{D+1}(\widetilde{A}) \in C^{D+1}(X, \mathbb{R}/\mathbb{Z})$$

or by a differential-cocycle representative, its closed-manifold value is

$$\mathcal{Z}_{\mathrm{bulk}}(X, \widetilde{A}) = \exp\left(2\pi \mathrm{i} \int_X I_{D+1}(\widetilde{A})\right).$$

The periods are integral in the normalization used, so changing a local representative by a globally defined coboundary leaves the closed value unchanged.

There is a concrete construction of the boundary anomaly line from the bulk. Fix $A \in \mathcal{B}(M)$. A filling

$$(X, \widetilde{A}), \qquad \partial X = M, \quad \widetilde{A}|_M = A,$$

is a bordism $\emptyset \to (M, A)$, so the bulk functor supplies a state

$$b_X(\widetilde{A}) := \mathcal{Z}_{\mathrm{bulk}}(X, \widetilde{A})(1) \in \mathcal{B}_M^{\mathrm{bulk}}(A),$$

not a number. Since the bulk is invertible this state is nonzero, hence it determines a dual frame

$$b_X(\widetilde{A})^{-1} \in \big(\mathcal{B}_M^{\mathrm{bulk}}(A)\big)^{\vee} = \mathcal{L}_M(A), \qquad b_X(\widetilde{A})^{-1}\big(b_X(\widetilde{A})\big) = 1.$$

A vector in the anomaly line may therefore be represented as

$$z\, b_X(\widetilde{A})^{-1}, \qquad z \in \mathbb{C}.$$

Two representatives $(X, \widetilde{A}, z)$ and $(X', \widetilde{A}', z')$ are identified if

$$z' = z\, \mathcal{Z}_{\mathrm{bulk}}\big((-X) \cup_M X',\, (-\widetilde{A}) \cup_A \widetilde{A}'\big).$$

Equivalently, the closed phase relates the two bulk states by

$$b_{X'}(\widetilde{A}') = \mathcal{Z}_{\mathrm{bulk}}\big((-X) \cup_M X',\, (-\widetilde{A}) \cup_A \widetilde{A}'\big)\, b_X(\widetilde{A}),$$

so the dual frames in $\mathcal{L}_M(A)$ transform by the inverse phase and the coordinate z transforms by the displayed phase. The quotient is therefore the coordinate law for dual frames, not a declaration that the filling itself is a scalar. This quotient is one-dimensional whenever the background admits the relevant extensions and the usual bordism-locality assumptions hold. A relative boundary theory assigns a vector in this line. The scalar attached to a chosen compatible filling is the evaluation pairing

$$Z_{\mathrm{glued}}(X; A) = \mathrm{ev}_{\mathcal{B}^{\mathrm{bulk}}_M(A)}\big(Z_{\partial,M}(A) \otimes b_X(\widetilde{A})\big) \in \mathbb{C}. \tag{118.6}$$

Thus the filling supplies the bulk state that is evaluated by the boundary covector; a filling with nonempty boundary is not itself a numerical bulk partition function.

Bulk gluing and the anomaly line. The construction above defines a line functor over the background groupoid of M. If the bulk invertible theory is trivializable, the line functor is trivializable; if a boundary theory has partition vectors in this line, then the combined bulk-boundary partition function on a closed glued manifold is an ordinary complex number invariant under background isomorphisms. The mathematical content rests on the functorial gluing law of the bulk invertible theory, the existence of the allowed fillings or the corresponding colimit over fillings, and the lifting of background isomorphisms to the filling data. Under these inputs, transitivity of the equivalence relation follows from the closed-manifold gluing axiom:

$$\mathcal{Z}((-X) \cup_M X'') = \mathcal{Z}((-X) \cup_M X')\, \mathcal{Z}((-X') \cup_M X'')$$

with the common boundary X' paired with its orientation reverse. A background isomorphism $u : A \to A'$ maps a filling of A to the same filling with boundary background identified by u, hence gives a linear map between the corresponding quotient lines. Functoriality of these maps is again the gluing axiom. A trivialization of the bulk functor supplies canonical nonzero vectors in all bulk state lines, hence dual frames in all boundary anomaly lines, and therefore trivializes the line functor. In the absence of such a trivialization, the boundary partition function is a vector in the anomaly line, and evaluation on the compatible bulk filling state produces a number. Gauge covariance of the number follows because the bulk state and boundary covector transform by dual line maps under this pairing. Scalar phase checks after gluing can verify the resulting number, but by themselves they cannot detect a mistaken global dualization of $\mathcal{B}^{\mathrm{bulk}}_M(A)$ and $\mathcal{L}_M(A)$; the line variance in (118.5) is part of the datum.

Figure 118.1 records the gluing statement for anomaly inflow: a boundary partition vector evaluates on a compatible bulk filling state to give a gauge-covariant number.

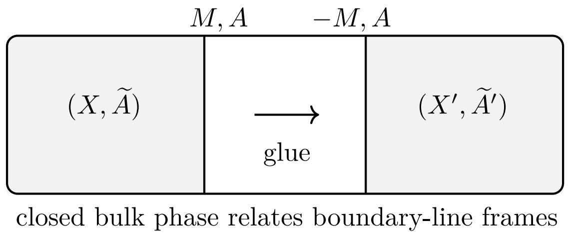

Figure 118.1: A boundary anomaly line is obtained by comparing the dual frames defined by fillings. Gluing two fillings along their common boundary produces the closed invertible-bulk phase that changes the frame.

118.3 Descent for Connected Symmetries

Let A be a connection for a connected Lie symmetry and let

$$F = dA + A \wedge A$$

be its curvature. Let $P_{D+2}(F)$ be a closed invariant polynomial of degree $D+2$, normalized so that its integral on every closed $(D+2)$-manifold with the corresponding bundle is integral. Locally one can choose a Chern–Simons form $I^{(0)}_{D+1}(A)$ satisfying

$$dI^{(0)}_{D+1}(A) = P_{D+2}(F).$$

For the common case $D = 2n$ and

$$P_{2n+2}(F) = \langle F^{n+1} \rangle,$$

where $\langle \cdots \rangle$ is an invariant symmetric $(n+1)$-linear form on the Lie algebra, this local primitive is constructed by a path of connections. Put

$$A_t = tA, \qquad F_t = dA_t + A_t \wedge A_t = t\, dA + t^2 A \wedge A.$$

Define

$$I^{(0)}_{2n+1}(A) = (n+1) \int_0^1 dt\, \langle A\, F_t^n \rangle. \tag{118.7}$$

Chern–Weil transgression. The form (118.7) satisfies

$$dI^{(0)}_{2n+1}(A) = \langle F^{n+1} \rangle.$$

Let $D_t = d + [A_t, \cdot]$ be the covariant exterior derivative for A_t. Since $\dot{A}_t = A$, the curvature variation is

$$\dot{F}_t = D_t \dot{A}_t = D_t A.$$

Using invariance of $\langle \cdots \rangle$ and the Bianchi identity $D_t F_t = 0$,

$$\frac{d}{dt} \langle F_t^{n+1} \rangle = (n+1) \langle D_t A\, F_t^n \rangle = (n+1) d\langle A\, F_t^n \rangle.$$

Integrating this identity from $t = 0$ to $t = 1$ gives

$$\langle F^{n+1} \rangle - \langle F_0^{n+1} \rangle = d\left((n+1) \int_0^1 dt \, \langle A \, F_t^n \rangle \right).$$

Because $F_0 = 0$, the left side is $\langle F^{n+1} \rangle$, proving the claim.

In the Abelian case $F = dA$ and (118.7) reduces to

$$I_{2n+1}^{(0)}(A) = A \wedge F^n.$$

For a gauge variation $\delta_\lambda A = d\lambda$,

$$\delta_\lambda I_{2n+1}^{(0)} = d\lambda \wedge F^n = d(\lambda F^n), \qquad dF = 0,$$

so $I_{2n}^{(1)}(\lambda, A) = \lambda F^n$ is an explicit descent representative in this normalization. For an infinitesimal gauge parameter α,

$$\delta_\alpha A = D_A \alpha = d\alpha + [A, \alpha].$$

Since $P_{D+2}(F)$ is gauge invariant,

$$d \, \delta_\alpha I_{D+1}^{(0)}(A) = 0.$$

Locally, and globally after passing to the descent complex, this gives a D-form $I_D^{(1)}(\alpha, A)$ such that

$$\delta_\alpha I_{D+1}^{(0)}(A) = dI_D^{(1)}(\alpha, A).$$

Therefore, for X^{D+1} with boundary M,

$$\delta_\alpha 2\pi\mathrm{i} \int_X I_{D+1}^{(0)} = 2\pi\mathrm{i} \int_M I_D^{(1)}(\alpha, A).$$

The boundary generating functional must transform by the negative of this phase for the combined bulk-boundary object to be gauge invariant.

Proposition 118.3 (Wess–Zumino consistency from descent)**.** *Let M be a closed D-manifold and let*

$$\delta_\alpha W_M[A] = 2\pi\mathrm{i} \int_M I_D^{(1)}(\alpha, A)$$

be obtained from a gauge-invariant polynomial by descent. Then

$$\delta_\alpha \delta_\beta W_M - \delta_\beta \delta_\alpha W_M = \delta_{[\alpha,\beta]} W_M.$$

Proof. Introduce a ghost c valued in the Lie algebra and the BRST differential $\mathsf{s}A = D_A c$, $\mathsf{s}c = -\frac{1}{2}[c, c]$. Gauge invariance of $P(F)$ and the Chern–Weil transgression give a local descent tower

$$\mathsf{s}I_{D+1}^{(0)} + dI_D^{(1)} = 0, \qquad \mathsf{s}I_D^{(1)} + dI_{D-1}^{(2)} = 0,$$

with the sign convention chosen so that the component of the first equation linear in a ghost value α is

$$\delta_\alpha I_{D+1}^{(0)}(A) = dI_D^{(1)}(\alpha, A).$$

We now extract the ghost-number-two coefficient explicitly. Introduce two odd scalar tags ε, η, put

$$c = \varepsilon\alpha + \eta\beta, \qquad \varepsilon\eta = -\eta\varepsilon, \qquad \varepsilon^2 = \eta^2 = 0.$$

The BRST rule gives

$$[c, c] = 2\varepsilon\eta[\alpha, \beta], \qquad \mathsf{s}c = -\varepsilon\eta[\alpha, \beta],$$

while

$$\mathsf{s}A = \varepsilon D_A\alpha + \eta D_A\beta.$$

Since $I_D^{(1)}(c, A)$ is linear in c, the coefficient of $\varepsilon\eta$ in $\mathsf{s}I_D^{(1)}(c, A)$ is

$$\delta_\alpha I_D^{(1)}(\beta, A) - \delta_\beta I_D^{(1)}(\alpha, A) - I_D^{(1)}([\alpha, \beta], A). \tag{118.8}$$

The first two terms come from the variation of A; the relative minus sign is the Koszul sign obtained when the odd tag of the varied parameter crosses the odd BRST differential. The last term comes from $\mathsf{s}c = -\varepsilon\eta[\alpha, \beta]$. Thus the second descent equation gives

$$\delta_\alpha I_D^{(1)}(\beta, A) - \delta_\beta I_D^{(1)}(\alpha, A) - I_D^{(1)}([\alpha, \beta], A) = -d\, I_{D-1}^{(2)}(\alpha, \beta, A), \tag{118.9}$$

where $I_{D-1}^{(2)}(\alpha, \beta, A)$ denotes the $\varepsilon\eta$-coefficient of $I_{D-1}^{(2)}(c, A)$.

Integrating (118.9) over closed M annihilates the right side. Multiplication by $2\pi\mathrm{i}$ gives

$$\delta_\alpha\delta_\beta W_M - \delta_\beta\delta_\alpha W_M - \delta_{[\alpha,\beta]}W_M = 0.$$

This is the Chevalley–Eilenberg cocycle condition for the local anomaly one-cochain

$$\mathcal{A}_M(\alpha; A) = 2\pi\mathrm{i}\int_M I_D^{(1)}(\alpha, A).$$

A change of descent representative by a local counterterm adds a Chevalley–Eilenberg coboundary, so the identity survives the representative change. If M has a boundary, the right side of (118.9) is the boundary descent term that must be included in the corresponding relative anomaly-line structure. □

Remark 118.4 (Cocycle ambiguity and local counterterms). Changing $I_{D+1}^{(0)}$ by

$$I_{D+1}^{(0)} \longmapsto I_{D+1}^{(0)} + dK_D(A)$$

changes the boundary anomaly by the variation of the local D-dimensional functional

$$2\pi\mathrm{i}\int_M K_D(A).$$

Thus the anomaly invariant is the cohomology class of the bulk invertible theory, while a representative anomaly current depends on a boundary counterterm coordinate.

On a manifold with boundary,

$$\int_X dK_D = \int_M K_D.$$

Under a background gauge transformation, the additional boundary variation is exactly the variation of this D-dimensional local functional. Hence two bulk cocycles differing by a coboundary define anomaly representatives that are related by a local counterterm. The same statement in the anomaly-line language is the frame change in Remark 118.2.

118.4 Finite Higher-Form Inflow

The connected descent formalism is not the whole theory of anomalies. Finite symmetries and higher-form symmetries are naturally described by cochains. Let A be a finite abelian group and let a p-form background on a $(D+1)$-manifold X be

$$B \in Z^{p+1}(X, A), \qquad B \sim B + \delta\Lambda.$$

An invertible finite response is represented by a cocycle

$$\omega(B) \in Z^{D+1}(X, \mathbb{R}/\mathbb{Z})$$

natural under subdivision and pullback. Its value is

$$\exp\left(2\pi \mathrm{i} \int_X \omega(B)\right).$$

If X is closed and ω is a cocycle on the classifying space $B^{p+1}A$, gauge invariance follows by the same evaluation-on-cycles argument as Remark 117.2. If X has boundary, gauge variation leaves a boundary cochain. The boundary QFT is then a section of the corresponding finite anomaly line.

For one-form symmetries in four dimensions the bulk is five-dimensional. A particularly useful local model is the finite Heisenberg response with

$$b, c \in C^2(X, \mathbb{Z}_N)$$

and

$$S_H[X; b, c] = \exp\left(\frac{2\pi \mathrm{i}}{N} \int_X b \smile \delta c\right).$$

Here b and c are the two finite two-form fields paired by the $\mathbb{Z}_N$ Pontryagin duality. On a closed X the action is invariant under

$$b \mapsto b + \delta\lambda, \qquad c \mapsto c + \delta\kappa, \qquad \lambda, \kappa \in C^1(X, \mathbb{Z}_N).$$

On a manifold with boundary M, the first transformation changes the exponent by

$$\frac{2\pi \mathrm{i}}{N} \int_M \lambda \smile \delta c.$$

This is an anomaly for a boundary condition that keeps $c|_M$ as a background. Different boundary conditions correspond to different choices of which finite field is fixed, which is summed, and which boundary counterterm has been attached. This is the cochain model behind one-form gauging, duality interfaces, and the symmetry-TQFT discussion in Chapter 123.

Remark 118.5 (Finite bulk variation). For the five-dimensional action

$$\frac{2\pi \mathrm{i}}{N} \int_X b \smile \delta c,$$

the variation under $b \mapsto b + \delta\lambda$ is trivial on closed X and equals

$$\frac{2\pi \mathrm{i}}{N} \int_{\partial X} \lambda \smile \delta c$$

when X has boundary. The variation under $c \mapsto c + \delta\kappa$ is zero.

Since $\delta^2 = 0$, changing c by $\delta\kappa$ does not change δc. Under $b \mapsto b + \delta\lambda$, the exponent changes by

$$\frac{2\pi\mathrm{i}}{N}\int_X \delta\lambda \smile \delta c.$$

The cup-product differential satisfies

$$\delta(\lambda \smile \delta c) = \delta\lambda \smile \delta c - \lambda \smile \delta^2 c = \delta\lambda \smile \delta c,$$

because $\deg\lambda = 1$. Evaluation of a coboundary on a closed fundamental cycle is zero. On a manifold with boundary, the cochain Stokes formula gives the displayed boundary term.

118.5 Weyl-Fermion Inflow

The local descent formalism of Chapter 51 becomes an invertible five-dimensional theory after the anomaly polynomial is placed in its index normalization. Let X be an oriented five-manifold with boundary

$$M = \partial X.$$

Let A_X be a background connection on X, and let R_X be the curvature of a Riemannian connection on TX. For one left-handed four-dimensional Weyl fermion in a representation R, the index-normalized six-form is

$$\mathcal{I}_6(R) = \left[\widehat{A}(TX)\,\mathrm{ch}_R(F_X)\right]_6, \tag{118.10}$$

with the notation of Section 51.9. Choose locally a five-form $\mathcal{I}_5^{(0)}$ satisfying

$$d\mathcal{I}_5^{(0)} = \mathcal{I}_6, \qquad \delta_\lambda \mathcal{I}_5^{(0)} = d\mathcal{I}_4^{(1)}(\lambda).$$

The five-dimensional response

$$W_X^{\mathrm{inv}}[A_X, g_X] = -2\pi\mathrm{i}\int_X \mathcal{I}_5^{(0)} \tag{118.11}$$

has boundary variation

$$\delta_\lambda W_X^{\mathrm{inv}} = -2\pi\mathrm{i}\int_M \mathcal{I}_4^{(1)}(\lambda).$$

It cancels a boundary left-handed determinant with

$$\delta_\lambda W_M = +2\pi\mathrm{i}\int_M \mathcal{I}_4^{(1)}(\lambda).$$

The sign in (118.11) is a boundary-condition choice: reversing the orientation of X, or using a right-handed boundary fermion, reverses the phase.

Coefficient of abelian inflow. Let the boundary field be one left-handed Weyl fermion of integer $U(1)$ charge q. Let $F = dA$, and normalize the first Pontryagin form by

$$p_1(TX) = -\frac{1}{8\pi^2}\,\mathrm{tr}_{TX}(\mathsf{R}^2).$$

Then a local representative of the inflow action is

$$W_X^{\mathrm{inv}} = -2\pi\mathrm{i}\int_X \left[\frac{q^3}{6(2\pi)^3}AF^2 - \frac{q}{24(2\pi)}A\,p_1(TX)\right], \tag{118.12}$$

and under $A \mapsto A + d\lambda$,

$$\delta_\lambda W_X^{\mathrm{inv}} = -2\pi\mathrm{i}\int_M \lambda\left[\frac{q^3}{6(2\pi)^3}F^2 - \frac{q}{24(2\pi)}p_1(TM)\right]. \tag{118.13}$$

For $U(1)$, the index-normalized polynomial is

$$\mathcal{I}_6(q) = \frac{q^3}{6(2\pi)^3}F^3 - \frac{q}{24(2\pi)}p_1(TX)F.$$

Because $dF = 0$ and $dp_1 = 0$ for the Chern–Weil representative, a primitive is the bracketed five-form in (118.12). Varying A gives

$$\delta_\lambda\left[\frac{q^3}{6(2\pi)^3}AF^2 - \frac{q}{24(2\pi)}A\,p_1\right] = d\lambda\left[\frac{q^3}{6(2\pi)^3}F^2 - \frac{q}{24(2\pi)}p_1\right].$$

The expression in square brackets is closed, hence this variation is the exterior derivative of

$$\lambda\left[\frac{q^3}{6(2\pi)^3}F^2 - \frac{q}{24(2\pi)}p_1\right].$$

Stokes' theorem gives (118.13) with the boundary orientation $M = \partial X$.

For several left-handed Weyl fermions, the local $U(1)^3$ and mixed gravitational-$U(1)$ anomaly coefficients are

$$\sum_i q_i^3, \qquad \sum_i q_i.$$

If the $U(1)$ symmetry is dynamical, the anomaly line for the dynamical gauge group must be canonically trivial. In the absence of a four-dimensional local counterterm that shifts these representatives, the local conditions are

$$\sum_i q_i^3 = 0, \qquad \sum_i q_i = 0,$$

together with the mixed nonabelian-$U(1)$ conditions when other gauge factors are present. If the $U(1)$ symmetry is a background global symmetry, the same polynomial is retained as the response that must be matched by any infrared description. Local cancellation does not by itself settle possible global anomalies; those are statements about the full invertible theory on the relevant bordism category.

118.6 From Inflow to Noninvertible Chiral Defects

Inflow also defines certain noninvertible codimension-one operators. Start with a transformation g of a D-dimensional QFT. Suppose that, in the presence of background fields A, implementing g across a hypersurface

$$\Sigma \subset M$$

leaves an anomaly line described by an invertible $(D+1)$-dimensional response $\mathcal{Z}_{\mathrm{anom}}(g; A)$. The bare wall for g is then not an ordinary topological symmetry wall. It is a relative theory for the restriction of the anomaly to a small tubular neighborhood of Σ.

Let $\mathcal{D}_g$ be the transformation wall. A wall theory $\mathcal{T}_\Sigma$ produces a genuine defect precisely when there is a trivialization

$$\mathcal{L}_{\mathcal{D}_g} \otimes \mathcal{L}_{\mathcal{T}_\Sigma} \simeq \mathbf{1}$$

of anomaly lines over the wall-background groupoid. The honest defect is

$$D_g = \mathcal{D}_g \otimes \mathcal{T}_\Sigma.$$

If $\mathcal{T}_\Sigma$ is obtained by gauging a finite higher-form symmetry on the wall, fusion of D_g with its orientation reverse contains a sum over the gauged sectors. That fusion is not a single copy of the identity defect, and D_g is noninvertible.

Remark 118.6 (Wall trivialization criterion). Assume the transformation wall $\mathcal{D}_g$ and the auxiliary wall theory $\mathcal{T}_\Sigma$ are topological away from their anomaly lines. The tensor product $D_g = \mathcal{D}_g \otimes \mathcal{T}_\Sigma$ is a genuine topological defect exactly when the tensor product of their anomaly-line functors is trivializable. If the trivializing wall theory contains a nontrivial finite gauge sum, then $D_g \otimes D_g^\vee$ contains the corresponding condensation projector and need not be the identity defect.

Topological invariance of a defect means that its correlation functional is a number invariant under isotopy, with no uncancelled dependence on a choice of background-field frame. The product wall has anomaly line $\mathcal{L}_{\mathcal{D}_g} \otimes \mathcal{L}_{\mathcal{T}_\Sigma}$. A trivialization of this line converts its partition vector into an ordinary number and removes the frame dependence, so the wall is genuine. Conversely, if the product line is nontrivial, changing the background by a morphism in the wall groupoid changes the wall vector by a nontrivial line map; no ordinary topological defect has this transformation law. When $\mathcal{T}_\Sigma$ is a finite gauge theory, fusing with the orientation reverse performs the finite gauge sum on the wall twice with the dual orientation. The result is the projector onto invariant wall sectors, not necessarily the invisible wall. This is the categorical source of noninvertibility.

Example 118.7 (Axial defects in compact QED)**.** Consider compact four-dimensional QED with a unit-charge massless Dirac fermion and with no dynamical monopoles. The axial rotation by angle α changes the effective action by the abelian anomaly

$$\delta_\alpha W = \mathrm{i}\alpha \, \frac{1}{4\pi^2} \int_M F \wedge F$$

in the convention of (51.6). For $\alpha = 2\pi/N$, the phase is sensitive to the magnetic one-form background when the theory is placed in sectors with fractional $F/2\pi$ induced by that background. The codimension-one axial wall therefore carries a mixed anomaly with the magnetic one-form symmetry. Attaching a three-dimensional finite gauge theory whose Pontryagin-square variation cancels this mixed phase gives the defects $D_{N,p}$ constructed in Chapter 123. The role of the three-dimensional theory is fixed by the inflow equation above: it is the boundary trivialization of the anomaly line of the axial wall, not an additional optional decoration.

118.7 One-Form Symmetry Example

Let a four-dimensional theory have a finite abelian one-form symmetry A. A background is a two-cocycle

$$B \in Z^2(M, A).$$

An anomaly involving this symmetry is represented by a five-dimensional cohomology class

$$\omega \in H^5(B^2A, U(1))$$

or, with spacetime structure included, by the corresponding generalized cohomology or bordism invariant. For $A = \mathbb{Z}_N$, expressions built from cup products and the Pontryagin square give concrete representatives after the manifold orientation and spin structure have been specified. The boundary theory then has line operators whose fusion and braiding with surface defects are acted on by the anomaly line; changing a four-dimensional discrete theta term shifts the representative but not the inflow class unless the bulk cohomology class changes.

The finite one-form anomaly is detected operationally as follows. Couple the theory to B. If a one-form gauge transformation

$$B \mapsto B + \delta\Lambda$$

changes the generating functional by

$$Z_M[B + \delta\Lambda] = \exp\left(2\pi \mathrm{i} \int_M \eta(B, \Lambda)\right) Z_M[B],$$

then η is a representative of the anomaly-line functor. If there is a local four-dimensional cochain $K(B)$ such that

$$\eta(B, \Lambda) = K(B + \delta\Lambda) - K(B) \mod \mathbb{Z},$$

the anomaly is removed by the counterterm $\exp(-2\pi\mathrm{i} \int_M K(B))$. If no such local cochain exists in the chosen background-field theory, gauging the one-form symmetry is obstructed unless another sector supplies the inverse anomaly.

118.8 Gauging Criterion

To gauge a global symmetry, one must sum or integrate over its backgrounds. For a finite symmetry this means a groupoid cardinality weighted sum; for a continuous symmetry it means a regularized path integral over connections and gauge transformations. In either case the integrand must be an ordinary number on the background groupoid. If the partition function is a section of a nontrivial anomaly line, there is no canonical summand to add.

In the invertible-language formulation the criterion is

$$\mathscr{Z}_{\text{bulk}} \otimes \mathscr{Z}_{\text{counter}} \simeq 1$$

as an invertible theory over the background-field bordism category. A local counterterm supplies such a trivialization exactly when the anomaly class is a coboundary in the chosen background-field cohomology theory. A second QFT sector can also supply the inverse line; this is anomaly cancellation by tensor product.

Remark 118.8 (Gauging requires a trivialized anomaly line). Let a finite global symmetry have background groupoid $\mathcal{B}(M)$. The gauged partition function

$$Z_{\mathrm{gauged}}(M) = \sum_{[A]\in\pi_0\mathcal{B}(M)} \frac{1}{|\operatorname{Aut}(A)|} Z_M(A)$$

is canonically defined only after the anomaly line $\mathcal{L}_M$ has been trivialized compatibly with morphisms in $\mathcal{B}(M)$. If two trivializations differ by a local counterterm, the two gauged theories differ by that counterterm before summation.

The summation formula adds complex numbers. If $Z_M(A)$ lies in $\mathcal{L}_M(A)$, then terms belonging to different isomorphism classes live in unrelated one-dimensional vector spaces. A trivialization identifies all these vectors with complex numbers. Compatibility with morphisms is required because the factor $1/|\operatorname{Aut}(A)|$ divides by automorphisms; if an automorphism acts nontrivially on the line, the would-be summand is frame dependent. A second trivialization differs from the first by a nowhere-zero function on the background groupoid. If this function is the exponential of a local cochain, it is exactly a local counterterm, and it weights each background sector before the groupoid sum is performed.

Open Problem 118.9 (Operator-level inflow)*.* Develop a reconstruction theorem that starts from an anomalous local QFT, constructs its anomaly line over background fields, and derives the action of the inflow theory on extended operators, defect junctions, and Hilbert spaces on manifolds with boundary.

Chapter 119

Phases Of Gauge Theories

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

A phase of a gauge theory is a statement about infinite-volume gauge-invariant QFT data. Gauge-variant fields, gauge-fixed expectation values, and coordinates on a weakly coupled Lagrangian chart are not phase invariants by themselves. The invariant data are states on observable algebras, spectra, local correlators, line and surface asymptotics, background-field responses, and the realization of ordinary and higher-form symmetries.

119.1 Observable Systems And Thermodynamic Limits

Fix a regulator class. On a finite spatial region Λ, let

$$\mathcal{A}_\Lambda$$

be the algebra generated by the declared gauge-invariant observables in that region. The word "declared" matters: one may study only local operators, or one may enlarge the observable system by including line operators, surface operators, boundary-condition sectors, and background-field responses. If $\Lambda \subset \Lambda'$, there is an inclusion

$$\iota_{\Lambda\Lambda'} : \mathcal{A}_\Lambda \hookrightarrow \mathcal{A}_{\Lambda'}.$$

The quasi-local observable algebra is the norm completion, or the analogous Euclidean distributional completion, of the directed union

$$\mathcal{A}_{\mathrm{obs}} = \overline{\bigcup_{\Lambda} \mathcal{A}_{\Lambda}}.$$

A thermodynamic-limit state is a positive normalized linear functional

$$\omega : \mathcal{A}_{\mathrm{obs}} \longrightarrow \mathbb{C}$$

obtained as a limit of finite-volume states along a specified boundary condition and regulator trajectory. In a Euclidean formulation the same data are encoded by compatible Schwinger distributions for all operators in the declared observable system, together with OS positivity when a Hilbert space reconstruction is intended.

Definition 119.1 (Declared phase problem)**.** A phase problem consists of:

regulator class	lattice, continuum cutoff, constructive scheme, or another regulator,
parameter domain	$\mathcal{P}$ of couplings, masses, theta angles, and counterterm data,
observable system	$\mathcal{A}_{\mathrm{obs}}$ including the chosen extended probes,
limit prescription	thermodynamic and continuum limits with boundary conditions,
comparison topology	the topology in which states, spectra, and operator data are compared.

A phase is always a phase of such a declared problem, not an absolute label attached to a Lagrangian density.

Definition 119.2 (Phase)**.** Let $\mathcal{U} \subset \mathcal{P}$ be a connected region where the declared limits exist. The points of $\mathcal{U}$ belong to one phase if the following data extend continuously over $\mathcal{U}$ in the declared comparison topology:

states	ω_p on $\mathcal{A}_{\mathrm{obs}}$,
local spectrum	mass gap, isolated particles, or specified gapless spectral type,
local correlators	renormalized distributions of local gauge-invariant operators,
extended operators	line/surface/wall fusion, endpoint maps, and asymptotic laws,
symmetry realization	ordinary, higher-form, higher-group, and anomalous responses,
topological sector	Hilbert spaces on spatial manifolds and topological operator maps.

A phase boundary is a locus where at least one declared datum fails to extend continuously or where the declared limit ceases to exist.

This definition is intentionally conditional. It does not assert that all gauge theories have a known complete phase invariant. It specifies what must be held fixed before statements such as "the Higgs and confinement regions are analytically connected" or "two regions are distinct" can be evaluated.

119.2 Finite-Regulator Locality Machinery For Phase Stability

The phase definition above uses continuity of local and extended data along a declared parameter region. In a regulated Hamiltonian system this continuity has mathematical content only after locality and spectral stability are named. This section records the finite machinery used later when a gapped lattice regularization is treated as a proof laboratory for gauge-theory phase data.

Let Λ be a finite metric lattice and let

$$\mathcal{A}_\Lambda = \bigotimes_{x\in\Lambda} \mathcal{B}(\mathcal{H}_x)$$

with $\dim \mathcal{H}_x < \infty$. For $X \subset \Lambda$, write $\mathcal{A}_X \subset \mathcal{A}_\Lambda$ for the subalgebra supported in X. A finite interaction is a map

$$\Phi : \{Z \subset \Lambda\} \longrightarrow \mathcal{A}_Z, \qquad \Phi(Z) = \Phi(Z)^*,$$

with Hamiltonian

$$H_\Lambda = \sum_{Z\subset\Lambda} \Phi(Z).$$

Let

$$\mathcal{I}_\Lambda = \{Z \subset \Lambda \mid \Phi(Z) \neq 0\}.$$

The finite overlap constants used below are

$$J_* = \sup_{Z\in\mathcal{I}_\Lambda} \|\Phi(Z)\|, \qquad N_* = \sup_{Z\in\mathcal{I}_\Lambda} \#\{Z' \in \mathcal{I}_\Lambda \mid Z' \cap Z \neq \varnothing\}.$$

For regions $X, Y \subset \Lambda$, define $d_\Phi(X, Y)$ to be the smallest integer $n \geq 1$ for which there are $Z_1, \ldots, Z_n \in \mathcal{I}_\Lambda$ with

$$Z_1 \cap X \neq \varnothing, \qquad Z_i \cap Z_{i+1} \neq \varnothing, \qquad Z_n \cap Y \neq \varnothing.$$

If no such chain exists, set $d_\Phi(X, Y) = \infty$. Also write

$$\partial_\Phi X = \{Z \in \mathcal{I}_\Lambda \mid Z \cap X \neq \varnothing\}.$$

Theorem 119.3 (Finite path-count Lieb–Robinson bound)**.** *Let*

$$\tau_t(A) = e^{\mathrm{i}tH_\Lambda} A e^{-\mathrm{i}tH_\Lambda}.$$

For $A_X \in \mathcal{A}_X$, $B_Y \in \mathcal{A}_Y$, and $X \cap Y = \varnothing$,

$$\|[\tau_t(A_X), B_Y]\| \leq 2\|A_X\| \, \|B_Y\| \, |\partial_\Phi X| \sum_{n\geq d_\Phi(X,Y)} \frac{(2J_*N_*|t|)^n}{n!}. \tag{119.1}$$

Consequently, for every $\mu > 0$,

$$\|[\tau_t(A_X), B_Y]\| \leq 2\|A_X\| \, \|B_Y\| \, |\partial_\Phi X| \, \exp\{-\mu d_\Phi(X, Y) + 2J_*N_*e^\mu|t|\}. \tag{119.2}$$

If the finite systems form an exhausting bounded-degree finite-range family, with J_*, N_*, and the conversion from d_Φ to the metric distance bounded uniformly, this gives a uniform light cone for local observables in the thermodynamic sequence.

Proof. For a finite region $S \subset \Lambda$, set

$$C_B(t, S) = \sup_{\substack{A\in\mathcal{A}_S \\ A\neq 0}} \frac{\|[\tau_t(A), B_Y]\|}{\|A\|}.$$

If $S \cap Y \neq \varnothing$, then

$$C_B(t,S) \le 2\|B_Y\|$$

by $\|[A, B_Y]\| \le 2\|A\|\,\|B_Y\|$. Suppose now that $S \cap Y = \varnothing$. Decompose the Hamiltonian as

$$H_\Lambda = H_{S^c}^{\mathrm{far}} + \sum_{Z \cap S \neq \varnothing} \Phi(Z),$$

where $H_{S^c}^{\mathrm{far}}$ is the sum of terms whose supports are disjoint from S. The terms in $H_{S^c}^{\mathrm{far}}$ commute with the initial observable $A \in \mathcal{A}_S$. Applying Duhamel's formula to the perturbation by the remaining terms and using the unitary invariance of the operator norm gives

$$C_B(t,S) \le 2 \sum_{Z \cap S \neq \varnothing} \|\Phi(Z)\| \int_0^{|t|} C_B(s, S \cup Z)\,\mathrm{d}s.$$

The factor 2 is the elementary commutator estimate $\|[\Phi(Z), A]\| \le 2\|\Phi(Z)\|\,\|A\|$. Iterating the inequality generates chains $Z_1, \ldots, Z_n$ of overlapping interaction supports. A chain can reach Y only when $n \ge d_\Phi(X,Y)$. The first step has at most $|\partial_\Phi X|$ choices, and each later step has at most N_* choices. The interaction norms contribute at most J_*^n, the commutator estimates contribute 2^n, and the ordered time simplex contributes $|t|^n/n!$. The final contact with Y contributes $2\|B_Y\|$, yielding (119.1), with N_*^n used as a harmless uniform upper bound for the chain count. Finally,

$$\sum_{n \ge d} \frac{a^n}{n!} \le e^{-\mu d} \sum_{n \ge 0} \frac{(ae^\mu)^n}{n!} = \exp(-\mu d + ae^\mu),$$

with $a = 2J_* N_* |t|$. This proves (119.2). □

The estimate is finite and quantitative. Its thermodynamic use requires the constants in the last sentence of the theorem. A volume-dependent interaction norm or an interaction graph with growing overlap degree can erase the uniform light cone.

Spectral transport before locality. Let $s \mapsto H_\Lambda(s)$ be a C^1 finite-volume Hamiltonian path and let $P_\Lambda(s)$ be the spectral projection onto an isolated band. The projection identity

$$P_\Lambda(s)^2 = P_\Lambda(s)$$

implies, after differentiating and multiplying by P_Λ on both sides, that

$$P_\Lambda(\partial_s P_\Lambda) P_\Lambda = 0.$$

Define the anti-self-adjoint finite spectral generator

$$K_\Lambda(s) = [\partial_s P_\Lambda(s), P_\Lambda(s)].$$

Then

$$\partial_s P_\Lambda(s) = [K_\Lambda(s), P_\Lambda(s)]. \tag{119.3}$$

Indeed,

$$[K_\Lambda, P_\Lambda] = (\partial_s P_\Lambda) P_\Lambda + P_\Lambda(\partial_s P_\Lambda) = \partial_s P_\Lambda.$$

Thus the finite ODE

$$\partial_s U_\Lambda(s) = K_\Lambda(s) U_\Lambda(s), \qquad U_\Lambda(0) = 1,$$

transports the band:

$$P_\Lambda(s) = U_\Lambda(s) P_\Lambda(0) U_\Lambda(s)^*.$$

This is exact spectral algebra, but the operator $K_\Lambda(s)$ is built from the global projection $P_\Lambda(s)$. It carries no locality estimate by itself.

Quasi-adiabatic continuation is the locality replacement for this projection generator. Suppose the chosen band is separated from the rest of the finite spectrum by a gap $\gamma > 0$. Choose a real odd filter

$$F_\gamma : \mathbb{R} \longrightarrow \mathbb{R}$$

whose Fourier transform, with convention

$$\widehat{F}_\gamma(\omega) = \int_{\mathbb{R}} e^{\mathrm{i}\omega t} F_\gamma(t)\,\mathrm{d}t,$$

satisfies

$$\widehat{F}_\gamma(\omega) = \frac{\mathrm{i}}{\omega}, \qquad |\omega| \geq \gamma,$$

and whose decay is fast enough for the integrals below. Let

$$V_\Lambda(s) = \partial_s H_\Lambda(s)$$

and define the self-adjoint finite quasi-adiabatic generator

$$D_\Lambda(s) = \int_{\mathbb{R}} F_\gamma(t)\, e^{\mathrm{i}tH_\Lambda(s)} V_\Lambda(s) e^{-\mathrm{i}tH_\Lambda(s)}\,\mathrm{d}t. \tag{119.4}$$

In an eigenbasis of $H_\Lambda(s)$, with a in the band and b outside the band,

$$\langle b|\partial_s P_\Lambda|a\rangle = -\frac{\langle b|V_\Lambda|a\rangle}{E_b - E_a}.$$

On the other hand,

$$\langle b|D_\Lambda|a\rangle = \widehat{F}_\gamma(E_b - E_a)\langle b|V_\Lambda|a\rangle = \frac{\mathrm{i}\,\langle b|V_\Lambda|a\rangle}{E_b - E_a}.$$

Therefore

$$\langle b|\mathrm{i}[D_\Lambda, P_\Lambda]|a\rangle = -\frac{\langle b|V_\Lambda|a\rangle}{E_b - E_a},$$

and the same calculation in the opposite off-diagonal block gives

$$\partial_s P_\Lambda(s) = \mathrm{i}[D_\Lambda(s), P_\Lambda(s)]. \tag{119.5}$$

The diagonal blocks are zero on both sides. Thus the quasi-adiabatic generator transports the isolated band exactly at finite volume.

Now assume the Hamiltonian path is local:

$$V_\Lambda(s) = \sum_{X \subset \Lambda} V_X(s), \qquad V_X(s) \in \mathcal{A}_X,$$

with a uniform interaction norm. Let X_R be the R-neighborhood of X, and let $\mathbb{E}_{X_R}$ be the conditional expectation from $\mathcal{A}_\Lambda$ onto $\mathcal{A}_{X_R}$ obtained by normalized partial trace over the complement. In finite dimension this conditional expectation can be written as Haar averaging over unitaries U supported in $\Lambda \setminus X_R$:

$$\mathbb{E}_{X_R}(A) = \int UAU^* \, \mathrm{d}U.$$

Therefore

$$A - \mathbb{E}_{X_R}(A) = \int (A - UAU^*) \, \mathrm{d}U, \qquad \|A - \mathbb{E}_{X_R}(A)\| \leq \sup_U \|[A, U]\|.$$

Apply this with $A = e^{\mathrm{i}tH_\Lambda} V_X e^{-\mathrm{i}tH_\Lambda}$. The unitaries U in the last supremum are supported at distance at least R from X, and the Lieb–Robinson estimate gives

$$\left\| e^{\mathrm{i}tH_\Lambda} V_X e^{-\mathrm{i}tH_\Lambda} - \mathbb{E}_{X_R}\left(e^{\mathrm{i}tH_\Lambda} V_X e^{-\mathrm{i}tH_\Lambda}\right) \right\| \leq C_X \|V_X\| \, e^{-\mu(R - v|t|)}$$

for $R \geq v|t|$, with constants inherited from (119.2). Inserting this estimate into (119.4) gives a quasi-local tail bound:

$$\|D_{\Lambda,X}(s) - D^{(R)}_{\Lambda,X}(s)\| \leq C_X \|V_X(s)\| \left(e^{-\mu R/2} \int_{|t| \leq R/(2v)} |F_\gamma(t)| \, \mathrm{d}t + 2 \int_{|t| > R/(2v)} |F_\gamma(t)| \, \mathrm{d}t \right),$$

where $D^{(R)}_{\Lambda,X}(s) \in \mathcal{A}_{X_R}$. The first term is the Lieb–Robinson leakage inside the time window, and the second term is the filter tail outside that window. This is the mechanism behind quasi-local spectral flow: the gap fixes the filter in frequency space, the filter tail controls long times, and the Lieb–Robinson estimate controls the spatial spread during the retained times.

Consequently, a uniformly gapped local path supplies a quasi-local automorphism of the quasi-local observable algebra in the thermodynamic limit, provided the interaction norms, gap, boundary convention, filter tail, and volume exhaustion are all uniform. This automorphism is the regulated mathematical object that transports local phase data. It is distinct from both physical time evolution and from an arbitrary finite-dimensional unitary interpolation of ground spaces.

119.3 Finite Flux-Torus Hall Response

Hall response in an interacting system is not a single-particle Chern number unless the theory has first been reduced to a free one-particle problem. The finite many-body object is a family of Hamiltonians over a flux torus. This formulation is the appropriate finite-regulator laboratory for quantized topological response, and it is also the place where the order of statements must be kept straight: curvature over the finite flux torus is exact; thermodynamic Hall conductance is obtained only after a separate locality, gap, and limit argument.

Let Λ be a finite two-dimensional torus with a conserved $U(1)$ charge. Choose two oriented cuts, one dual to each primitive cycle, and introduce dimensionless fluxes

$$\phi = (\phi_x, \phi_y) \in \mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2.$$

The finite flux Hamiltonian

$$H_\Lambda(\phi)$$

is obtained by coupling charged terms crossing the a-cut to the flat background holonomy $e^{\mathrm{i}\mathfrak{q}\phi_a}$, with $\mathfrak{q}$ the $U(1)$ charge of the term crossing the cut. Equivalently, $H_\Lambda(\phi)$ is a C^2 family of self-adjoint matrices which is periodic in ϕ, or periodic up to a large background-gauge transformation. In the latter case the formulas below are read as formulas for the spectral vector bundle over $\mathbb{T}^2$ with the corresponding transition functions. The flux-current operators in this normalization are

$$I_a(\phi) = \partial_{\phi_a} H_\Lambda(\phi), \qquad a = x, y.$$

If a physical electric field is produced by a time-dependent flux, its sign relative to I_a is fixed by the convention used for $\phi_a(t)$; the finite curvature formula below is independent of that conversion convention.

Definition 119.4 (Finite flux-torus response structure)**.** A finite flux-torus response structure consists of:

finite Hilbert space	$\mathcal{H}_\Lambda$,
flux Hamiltonian	$H_\Lambda : \mathbb{T}^2 \to \mathrm{End}(\mathcal{H}_\Lambda)_{\mathrm{sa}}$,
isolated band	$P_\Lambda(\phi)$ of constant rank r_Λ,
uniform finite gap	$\mathrm{dist}\big(\mathrm{Spec}\, H_\Lambda\vert_{P_\Lambda}, \mathrm{Spec}\, H_\Lambda\vert_{1-P_\Lambda}\big) \geq \gamma_\Lambda > 0.$

The band curvature is the two-form

$$\mathcal{F}_{xy,\Lambda}(\phi)\, \mathrm{d}\phi_x \wedge \mathrm{d}\phi_y = \mathrm{i}\, \mathrm{Tr}_{\mathcal{H}_\Lambda}\big(P_\Lambda[\partial_{\phi_x} P_\Lambda, \partial_{\phi_y} P_\Lambda]\big) \mathrm{d}\phi_x \wedge \mathrm{d}\phi_y. \tag{119.6}$$

The flux-averaged dimensionless Hall coefficient of this finite structure is

$$\overline{\kappa}_{xy,\Lambda} = \frac{1}{r_\Lambda (2\pi)^2} \int_{\mathbb{T}^2} \mathcal{F}_{xy,\Lambda}(\phi)\, \mathrm{d}\phi_x\, \mathrm{d}\phi_y. \tag{119.7}$$

In units $e = \hbar = 1$, the corresponding physical Hall conductivity uses the same number when the flux/electric-field convention is the one stated above; restoring units multiplies by $e^2/\hbar$ and the conventional 2π normalization is already contained in (119.7).

Theorem 119.5 (Finite Hall curvature theorem)**.** *Let $H_\Lambda(\phi)$ and $P_\Lambda(\phi)$ be a finite flux-torus response structure. Then*

$$C_\Lambda = \frac{1}{2\pi} \int_{\mathbb{T}^2} \mathcal{F}_{xy,\Lambda}(\phi)\, \mathrm{d}\phi_x\, \mathrm{d}\phi_y \tag{119.8}$$

is an integer depending only on the spectral vector bundle $\mathrm{Ran}\, P_\Lambda \to \mathbb{T}^2$. It is unchanged under every C^1 deformation of the finite structure for which the rank of P_Λ and the spectral gap remain fixed.

If, in addition, the isolated band is an exact r_Λ-fold degenerate eigenvalue,

$$H_\Lambda(\phi) P_\Lambda(\phi) = E_0(\phi) P_\Lambda(\phi),$$

and

$$Q_\Lambda = 1 - P_\Lambda, \qquad R_\Lambda(\phi) = Q_\Lambda(\phi)\big(H_\Lambda(\phi) - E_0(\phi)\big)^{-1} Q_\Lambda(\phi),$$

then the same curvature has the finite Kubo form

$$\mathcal{F}_{xy,\Lambda} = \mathrm{i}\, \mathrm{Tr}\big(P_\Lambda I_x R_\Lambda^2 I_y P_\Lambda - P_\Lambda I_y R_\Lambda^2 I_x P_\Lambda\big)\,. \tag{119.9}$$

Thus

$$\overline{\kappa}_{xy,\Lambda} = \frac{C_\Lambda}{2\pi r_\Lambda}.$$

Proof. The integrality statement is the Chern-Weil formula for the determinant line of the finite-rank bundle $\operatorname{Ran} P_\Lambda$, and in this finite setting the proof can be written explicitly. On a coordinate patch $U_\alpha \subset \mathbb{T}^2$, choose a smooth orthonormal frame

$$\Psi_\alpha = (\psi_{\alpha,1}, \ldots, \psi_{\alpha,r_\Lambda})$$

for $\operatorname{Ran} P_\Lambda$. The trace Berry connection in this frame is

$$A_\alpha = \mathrm{i}\,\mathrm{Tr}\big(\Psi_\alpha^* \mathrm{d}\Psi_\alpha\big).$$

Using $P_\Lambda = \Psi_\alpha \Psi_\alpha^*$ and $\Psi_\alpha^* \Psi_\alpha = 1_q$, one obtains

$$\mathrm{d}A_\alpha = \mathrm{i}\,\mathrm{Tr}\big(P_\Lambda\, \mathrm{d}P_\Lambda \wedge \mathrm{d}P_\Lambda\big),$$

whose xy-component is (119.6). On an overlap $U_\alpha \cap U_\beta$, the two frames obey

$$\Psi_\beta = \Psi_\alpha g_{\alpha\beta}, \qquad g_{\alpha\beta} : U_\alpha \cap U_\beta \to U(r_\Lambda).$$

Therefore

$$A_\beta - A_\alpha = \mathrm{i}\,\mathrm{Tr}(g_{\alpha\beta}^{-1} \mathrm{d}g_{\alpha\beta}) = \mathrm{i}\,\mathrm{d}\log\det g_{\alpha\beta}.$$

Integrating $\mathrm{d}A_\alpha$ over a fundamental polygon for the torus and using Stokes' theorem reduces the integral to the winding of the transition function $\det g_{\alpha\beta} : S^1 \to U(1)$. Hence

$$\frac{1}{2\pi} \int_{\mathbb{T}^2} \mathrm{i}\,\mathrm{Tr}(P_\Lambda\, \mathrm{d}P_\Lambda \wedge \mathrm{d}P_\Lambda) \in \mathbb{Z}.$$

The same argument applies when the flux family is only covariantly periodic: the large-gauge transition unitaries are precisely the transition functions of the bundle. Constancy under a gapped deformation follows because the integer $C_\Lambda(s)$ depends continuously on s while the projection bundle is defined throughout the deformation.

It remains to prove the Kubo expression. Differentiate

$$H_\Lambda P_\Lambda = E_0 P_\Lambda$$

and multiply by Q_Λ on the left and P_Λ on the right. Since $Q_\Lambda P_\Lambda = 0$, this gives

$$Q_\Lambda (\partial_a P_\Lambda) P_\Lambda = -R_\Lambda Q_\Lambda I_a P_\Lambda, \qquad a = x, y.$$

Taking adjoints gives

$$P_\Lambda (\partial_a P_\Lambda) Q_\Lambda = -P_\Lambda I_a Q_\Lambda R_\Lambda.$$

The diagonal identities $P_\Lambda (\partial_a P_\Lambda) P_\Lambda = 0$, obtained from $P_\Lambda^2 = P_\Lambda$, imply

$$\mathrm{Tr}\big(P_\Lambda [\partial_x P_\Lambda, \partial_y P_\Lambda]\big) = \mathrm{Tr}\big(P_\Lambda (\partial_x P_\Lambda) Q_\Lambda (\partial_y P_\Lambda) P_\Lambda - P_\Lambda (\partial_y P_\Lambda) Q_\Lambda (\partial_x P_\Lambda) P_\Lambda\big).$$

Substituting the two off-diagonal formulas above gives (119.9). The final displayed equality follows by inserting (119.8) into (119.7). □

For a finite interacting system the coefficient at a fixed flux value need not be quantized. The quantized object in Theorem 119.5 is the flux average of the band curvature. A thermodynamic Hall theorem requires additional hypotheses: the flux family must be local in the sense that $I_a = \partial_{\phi_a} H$ is supported near the corresponding cut up to uniformly local tails; the gap above the relevant band must remain uniform over ϕ and over the volume sequence; the quasi-adiabatic flow from Section 119.2 must control the change of local current response under moving the cuts; and the thermodynamic limit of the response must exist. Under such hypotheses the fixed-flux response becomes independent of ϕ up to errors vanishing in the thermodynamic limit, and the finite integer C_Λ becomes the phase-stable Hall datum. Without these hypotheses, writing a continuum integer Hall conductance directly from a finite Kubo formula hides the substance of the theorem.

119.4 Gauge Invariance And Elitzur Averaging

Gauge-variant local fields cannot be order parameters for a gauge theory unless a gauge has been fixed and the residual meaning of the statement has been specified. The finite-regulator reason is elementary.

Finite-volume gauge averaging. Let a finite lattice gauge theory have compact gauge group G, gauge invariant action, and gauge invariant measure. Let O be an observable which transforms nontrivially under the gauge transformation at one site x. Then

$$\langle O\rangle_\Lambda = 0$$

in finite volume with gauge-invariant boundary conditions.

Average over gauge transformations supported at x. Gauge invariance of the measure and action gives

$$\langle O\rangle_\Lambda = \int_G dh\, \langle h\cdot O\rangle_\Lambda .$$

If O transforms in a representation ρ with no invariant vector in the component being averaged, Haar orthogonality gives

$$\int_G dh\, \rho(h) = 0$$

on that component. Therefore the group average of O is zero, and so is its expectation value. For a finite collection of gauge-variant insertions, perform the Haar average independently at each site touched by the product. The integral at a given site projects the tensor product of the local representation factors onto its invariant subspace. The expectation can survive only if this projection is nonzero at every site. The load-bearing input here is the finite-regulator Haar projection with gauge-invariant boundary conditions; the calculation is the local averaging step used in Elitzur-type statements. Continuum phase statements require the separate limiting data specified in Definition 119.1.

Consequently a Higgs regime must be described by gauge-invariant data: screening of external charges, massive vector states in gauge-invariant channels, boundary-condition response, or nonlocal diagnostics. A gauge-fixed scalar expectation value may be a useful coordinate in a weakly coupled chart, but it is not the invariant phase datum.

119.5 Static Potentials From Transfer Matrices

Let $W_R(C)$ be a renormalized Wilson loop in representation R. For a rectangle $C_{T,L}$ with temporal length T and spatial separation L, define the static potential by

$$V_R(L) = -\lim_{T\to\infty} \frac{1}{T} \log \langle W_R(C_{T,L})\rangle_{\text{ren}}\,, \tag{119.10}$$

when the limit exists. The subscript indicates that the line self-energy has been subtracted or divided out in a specified line-renormalization scheme.

Reflection positivity gives a Hilbert-space interpretation. In a finite spatial box, a pair of static external sources separated by L defines a sector Hilbert space $\mathcal{H}_{R,\bar{R};L}$ or, equivalently, a state $\Psi_{R,L}$ in the transfer-matrix construction with a Wilson transporter between the two sources. If $H_{R,L}$ is the Hamiltonian in this sector and E_0 is the vacuum energy, then

$$\langle W_R(C_{T,L})\rangle_{\text{ren}} = \langle \Psi_{R,L}, e^{-T(H_{R,L}-E_0)}\Psi_{R,L}\rangle.$$

By the spectral theorem this has the form

$$\int_0^\infty e^{-TE}\, d\rho_{R,L}(E)$$

for a positive finite spectral measure $\rho_{R,L}$.

Large-time spectral extraction. The static-potential limit uses the elementary Laplace principle for a positive spectral measure. Let ρ be a nonzero finite positive measure on $[E_*,\infty)$, and assume every interval $[E_*, E_*+\varepsilon]$ has positive ρ-measure. Then

$$-\lim_{T\to\infty}\frac{1}{T}\log\int e^{-TE}\, d\rho(E) = E_*.$$

Since $E \geq E_*$,

$$\int e^{-TE}\, d\rho(E) \leq \rho([E_*,\infty))e^{-TE_*}.$$

Thus the liminf of the left side is at least E_*. For every $\varepsilon > 0$,

$$\int e^{-TE}\, d\rho(E) \geq \rho([E_*, E_*+\varepsilon])e^{-T(E_*+\varepsilon)}.$$

The measure factor is positive and independent of T, so the limsup is at most $E_*+\varepsilon$. Letting $\varepsilon \to 0$ proves the claim. This calculation supplies only the last step after a reflection-positive transfer-matrix sector and a renormalized Wilson-loop spectral measure have already been constructed.

Thus $V_R(L)$ is the bottom of the static-source spectrum in the chosen sector. Its large-L behavior gives several distinct responses:

confining	$V_R(L) = \sigma_R L + o(L), \quad \sigma_R > 0,$
screened	$V_R(L)$ is bounded as $L \to \infty$,
Coulombic in four spacetime dimensions	$V_R(L) = V_\infty - \kappa_R/L + o(L^{-1}).$

These are responses of the chosen external probe. A theory with dynamical fundamental matter can screen the fundamental Wilson charge even when other probes or boundary conditions still diagnose nontrivial long-distance structure.

119.6 Loop Laws And Higher-Form Symmetry

For a closed contour C, let $L_\gamma(C)$ be a renormalized line operator with finite charge γ. For a sequence of large planar loops, an area law means

$$-\log\langle L_\gamma(C)\rangle = \sigma_\gamma\ \mathrm{Area}_{\min}(C) + o(\mathrm{Area}_{\min}(C)), \qquad \sigma_\gamma > 0,$$

after perimeter renormalization. A perimeter law means

$$-\log\langle L_\gamma(C)\rangle = \mu_\gamma\ \mathrm{Length}(C) + o(\mathrm{Length}(C)).$$

In a gapless Coulomb regime, logarithmic and power corrections can replace both pure laws; the phase datum is then the full asymptotic class, not a binary area/perimeter label.

Let A be a finite one-form symmetry and let

$$U_a(\Sigma), \qquad a \in A,$$

be its topological symmetry surface on a closed codimension-two surface Σ. If a line $L_\chi(C)$ carries character

$$\chi \in \widehat{A} = \mathrm{Hom}(A, U(1)),$$

then linking gives

$$U_a(\Sigma)L_\chi(C) = \chi(a)^{\mathrm{Link}(\Sigma,C)}L_\chi(C).$$

In a gapped phase with a homogeneous vacuum and no boundary contribution, the unbroken realization of a finite electric one-form symmetry is diagnosed by area laws for charged Wilson lines. A perimeter law for a charged line is the characteristic response of a broken one-form symmetry or a deconfined topological sector. This statement requires the line to be genuine and charged under an actual one-form symmetry; if dynamical matter lets the line end, the one-form symmetry is explicitly absent or reduced.

119.7 Screening, Condensation, And Oblique Confinement

Let $\mathcal{C}$ be the finite abelian group of center-sensitive dyonic line charges, with nondegenerate alternating pairing

$$B : \mathcal{C} \times \mathcal{C} \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

For $\mathfrak{su}(N)$ this is

$$\mathcal{C} = \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}}, \qquad B((e,m),(e',m')) = \frac{em' - e'm}{N}.$$

Let $S \subset \mathcal{C}$ be the subgroup screened by dynamical finite-energy excitations: a line whose charge lies in S can end on a dynamical operator or state. The asymptotic external-probe labels are then identified modulo screened charges,

$$\mathcal{C}_{\mathrm{ext}} = \mathcal{C}/S$$

for questions about static sources. The finite braiding form does not, in general, descend to all of this quotient. If the screened subgroup is isotropic, then the genuine residual finite topological charges are

$$\mathcal{C}_{\mathrm{top}} = S^\perp/S, \qquad S^\perp = \{\gamma \in \mathcal{C} \mid B(\gamma, s) = 0 \text{ for every } s \in S\},$$

and the pairing descends to $\mathcal{C}_{\mathrm{top}}$ because changing a representative by $s \in S$ shifts $B(\gamma, \gamma')$ by $B(s, \gamma') = 0$ whenever $\gamma' \in S^{\perp}$.

A condensate of line or particle charges is encoded, at the finite center-sensitive level, by an isotropic subgroup

$$K \subset \mathcal{C}, \qquad B|_{K \times K} = 0.$$

The isotropy condition is mutual locality of condensed charges. The charges with trivial finite braiding against the condensate form

$$K^{\perp} = \{\gamma \in \mathcal{C} \mid B(\gamma, k) = 0 \text{ for every } k \in K\}.$$

Assume a gapped regime in which the long-distance condensate is described by an isotropic finite subgroup $K \subset \mathcal{C}$, and assume a line of charge γ has perimeter behavior only if it has trivial braiding with every condensed charge. Then the unconfined finite charges lie in

$$K^{\perp},$$

and every $\gamma \notin K^{\perp}$ is confined in this finite-charge diagnostic. If K is maximal isotropic, then $K^{\perp} = K$. Moving a test line of charge γ around a condensed object of charge k multiplies the state by

$$\exp(2\pi \mathrm{i} B(\gamma, k)).$$

A perimeter-law line in a condensate background must have a nonzero overlap with the condensate sector without introducing a branch-dependent phase. By the hypothesis this requires the displayed phase to be one for every $k \in K$, which is exactly $\gamma \in K^{\perp}$. If $K \subsetneq K^{\perp}$, an element $\eta \in K^{\perp} \setminus K$ would generate with K a strictly larger isotropic subgroup, because $B(\eta, K) = 0$ and $B(\eta, \eta) = 0$. Hence a maximal isotropic subgroup obeys $K = K^{\perp}$.

Electric confinement corresponds to magnetic condensation: purely electric charges pair nontrivially with the magnetic condensate and acquire area law. A Higgs response from electric condensation screens electric probes and confines appropriate magnetic probes. Oblique confinement is the same criterion with a dyonic condensate

$$K = \langle (e_c, m_c) \rangle$$

whose generator has both electric and magnetic components. The word "oblique" is therefore not a new mechanism; it is the finite Dirac pairing applied to a dyonic condensate.

119.8 Gauge-Higgs Analytic Connectivity

Consider a lattice gauge theory with compact gauge group and matter in a representation that screens the fundamental Wilson charge. At finite lattice volume the partition function is an analytic function of the couplings because it is an integral of an analytic bounded integrand over a compact domain. A cluster expansion gives open regions in the strong-gauge and large-Higgs coupling domains where connected correlations of local gauge-invariant operators decay exponentially and are analytic in the couplings.

Gauge-Higgs analytic corridor. Assume the cluster expansions in the strong-gauge and large-Higgs domains converge in a common connected corridor of lattice couplings, uniformly in volume for local gauge-invariant observables. Then those local observables have a single analytic infinite-volume continuation throughout the corridor.

Uniform convergence of the cluster expansion gives an absolutely convergent series for each connected local correlator with coefficients analytic in the couplings. The thermodynamic limit may therefore be taken term by term in the corridor. Analytic continuation along any path inside the connected corridor gives the same value because the local correlator is represented by one convergent analytic series on overlapping neighborhoods. Hence the two coordinate descriptions belong to one local-observable phase in the declared observable set.

This analytic-corridor argument concerns the declared local observable system. It does not say that every possible extended-probe datum is identical. Boundary conditions, heavy external charges, remnant higher-form backgrounds, and topological sectors can refine the phase problem. This is why the observable system in Definition 119.1 must be specified before claiming analytic connectivity or separation.

119.9 Fredenhagen–Marcu Type Ratios

When dynamical matter screens a Wilson charge, the Wilson loop itself need not distinguish a confinement-like region from a Higgs-like region. A more refined gauge-invariant probe uses an open transporter with charged matter fields at its endpoints. Let ϕ be a matter field in representation R, let γ_{xy} be a path from x to y, and define the gauge-invariant open-line correlator

$$G_R(\gamma_{xy}) = \left\langle \phi^\dagger(x)\, U_R(\gamma_{xy})\, \phi(y) \right\rangle .$$

Let C_γ be a closed contour built from γ_{xy} and a reference return path of comparable size. A Fredenhagen–Marcu type ratio is

$$\rho_R(\gamma) = \frac{G_R(\gamma)}{\sqrt{\langle W_R(C_\gamma)\rangle_{\rm ren}}},$$

with the regulator, endpoint renormalization, and order of limits specified. The denominator removes the bulk string contribution associated with the closed flux sheet. The resulting ratio asks whether the open charged configuration has nonzero overlap with the vacuum or with a charged superselection sector after the universal flux cost has been divided out.

Exponential scaling of the ratio. Suppose a family of paths has size R, and suppose

$$-\log|G_R(\gamma_R)| = aR + o(R), \qquad -\log\langle W_R(C_{\gamma_R})\rangle_{\rm ren} = bR + o(R)$$

with $b \geq 0$. Then

$$-\log|\rho_R(\gamma_R)| = \left(a - \frac{b}{2}\right) R + o(R).$$

In particular the ratio has a nonzero exponential limit only when the endpoint-open-line decay matches one half of the closed-loop decay at leading order. This is the logarithm of the definition:

$$-\log|\rho_R| = -\log|G_R| + \frac{1}{2}\log\langle W_R\rangle_{\rm ren}.$$

Substituting the two assumed asymptotic formulas gives

$$aR + o(R) - \frac{b}{2}R + o(R),$$

which is the displayed expression.

The ratio is therefore not a slogan for "Higgs" or "confinement" without the accompanying limit definition. Its interpretation depends on the matter representation, the boundary conditions, the renormalization of the endpoints, and the sector whose overlap is being measured. Its virtue is that all of these choices are gauge-invariant and can be stated at finite regulator.

119.10 Topological Order And Deconfined Gauge Theory

A gapped gauge theory can have a finite-dimensional Hilbert space on spatial manifolds with nontrivial topology, nontrivial line-operator braiding, and surface operators whose correlation functions depend on homology classes. These data form the long-distance topological sector. In continuum language they are encoded by an infrared TQFT together with a map

$$\Phi_{\mathrm{IR}} : \begin{matrix}\{\text{microscopic topological} \\ \text{extended operators}\}\end{matrix} \longrightarrow \begin{matrix}\{\text{topological operators} \\ \text{of the IR TQFT}\}\end{matrix} .$$

The map is part of the phase definition. Two microscopic theories may have the same emergent TQFT but differ in massive local spectra, symmetry fractionalization, anomaly inflow, or the embedding of microscopic line operators into the topological category.

119.11 Finite One-Form Laboratory: The $\mathbb{Z}_2$ Gauge Code

The preceding paragraph is abstract because a continuum gapped phase rarely comes with an exactly known Hilbert space on every spatial manifold. A finite commuting-projector gauge code supplies a useful laboratory, provided its logical status is kept clear. It is not a definition of continuum QFT and it does not replace the infinite-volume phase problem. It gives an exact operator algebra in which one-form charge, topological line operators, local indistinguishability, and energy-barrier limitations can be checked without asymptotic or semiclassical language.

Let Λ be an $L_x \times L_y$ square cellulation of the spatial torus, with $L_x, L_y \geq 3$. Let V, E, P be its vertices, edges, and plaquettes, and put a two-dimensional Hilbert space

$$\mathcal{H}_\Lambda = \bigotimes_{e\in E} \mathbb{C}^2_e$$

on the edges. Write X_e, Z_e for Pauli operators on the edge e. The star and plaquette stabilizers are

$$A_v = \prod_{e\ni v} X_e, \qquad B_p = \prod_{e\subset\partial p} Z_e. \tag{119.11}$$

The orientation of ∂p is irrelevant because all coefficients are in $\mathbb{Z}_2$. Each A_v and B_p squares to one, and all stabilizers commute: a star and a plaquette share either no edge or two edges, so the two $XZ = -ZX$ signs cancel. The Hamiltonian

$$H_\Lambda = -J \sum_{v \in V} A_v - J \sum_{p \in P} B_p, \qquad J > 0, \tag{119.12}$$

has ground projector

$$P_0 = \prod_{v \in V} \frac{1 + A_v}{2} \prod_{p \in P} \frac{1 + B_p}{2}. \tag{119.13}$$

The two products of all stars and all plaquettes are identities,

$$\prod_{v \in V} A_v = 1, \qquad \prod_{p \in P} B_p = 1,$$

and these are the only stabilizer relations on the connected torus. Hence the number of independent stabilizer constraints is

$$(|V| - 1) + (|P| - 1).$$

Since $|E| - |V| - |P| = 0$ for the square torus, the ground space has dimension

$$2^{|E| - (|V|-1) - (|P|-1)} = 4.$$

Equivalently, the two encoded qubits are the two $\mathbb{Z}_2$-homology cycles of the torus.

It is useful to express the logical algebra in chain-complex notation. For a $\mathbb{Z}_2$ edge chain $c \in C_1(\Lambda; \mathbb{Z}_2)$ and an edge cochain $\eta \in C^1(\Lambda; \mathbb{Z}_2)$, define

$$Z(c) = \prod_{e : c_e = 1} Z_e, \qquad X(\eta) = \prod_{e : \eta(e) = 1} X_e. \tag{119.14}$$

Their commutation relation is the finite intersection pairing

$$X(\eta) Z(c) = (-1)^{\langle \eta, c \rangle} Z(c) X(\eta), \qquad \langle \eta, c \rangle = \sum_{e \in E} \eta(e) c_e \pmod 2. \tag{119.15}$$

The operator $Z(c)$ commutes with every star iff $\partial c = 0$. If $c = \partial s$ is the boundary of a plaquette two-chain s, then

$$Z(c) = \prod_{p : s_p = 1} B_p$$

and $Z(c)$ acts as the identity on $P_0 \mathcal{H}_\Lambda$. Thus nontrivial Z-type logical lines are represented by

$$[c] \in H_1(T^2; \mathbb{Z}_2).$$

Dually, $X(\eta)$ commutes with every plaquette iff $\delta \eta = 0$; if $\eta = \delta f$ is exact, then $X(\eta)$ is a product of star stabilizers and is trivial on the ground space. The nontrivial X-type logical lines are represented by

$$[\eta] \in H^1(T^2; \mathbb{Z}_2).$$

Equation (119.15) is then the exact finite version of the one-form linking phase: a primal Z-loop and a dual X-loop anticommute precisely when their mod-two intersection number is one.

Local indistinguishability. Let $R \subset E$ be an edge set contained in a contractible disk of the cellulation, and let O_R be any operator supported on $\bigotimes_{e\in R} \mathbb{C}_e^2$. Expanding O_R in the Pauli basis, it is enough to consider a monomial

$$W = i^\ell X(\eta)Z(c), \qquad \mathrm{supp}(\eta, c) \subset R.$$

If $\partial c \neq 0$, then $Z(c)$ anticommutes with at least one star stabilizer; if $\delta\eta \neq 0$, then $X(\eta)$ anticommutes with at least one plaquette stabilizer. In either case

$$P_0 W P_0 = 0,$$

because the anticommuting stabilizer S obeys

$$P_0 W P_0 = P_0 S W P_0 = -P_0 W S P_0 = -P_0 W P_0.$$

If $\partial c = 0$ and $\delta\eta = 0$, contractibility of R implies that c is a plaquette boundary and η is a star coboundary inside the same disk. Therefore W is a phase times a product of stabilizers, and

$$P_0 W P_0 = \lambda(W) P_0, \qquad \lambda(W) \in \{0, \pm 1, \pm \mathrm{i}\}.$$

By linearity,

$$P_0 O_R P_0 = c(O_R) P_0 \tag{119.16}$$

for every operator supported in a contractible region R. This is the finite local-indistinguishability mechanism. It is stronger than saying that particular expectation values agree: the full local operator algebra acts projectively as scalars after projection to the ground sector. It is also weaker than a continuum phase theorem: stability under perturbations and passage to an infinite-volume gapped phase require separate locality and gap input, such as Lieb–Robinson and quasi-adiabatic-continuation estimates.

Open strings and the finite energy barrier. For an open chain c, the boundary ∂c is the set of star violations created by $Z(c)$. More precisely, for a ground vector ψ,

$$\langle Z(c)\psi, H_\Lambda Z(c)\psi\rangle - \langle \psi, H_\Lambda \psi\rangle = 2J\,|\partial c|. \tag{119.17}$$

A simple partial string has two endpoints, so the energy cost is $4J$, independent of its length. When the string closes into a noncontractible cycle, the endpoints disappear and the operator becomes a logical line inside the ground sector. The same statement holds for dual X-strings with plaquette violations. Thus this exact laboratory simultaneously exhibits topological logical lines and fixes the two-dimensional $\mathbb{Z}_2$ gauge-code barrier scale at order J, independent of system size. Any finite-temperature memory statement must additionally specify the bath, rates, decoder or absence of decoder, and the limiting notion of lifetime.

The lesson for gauge theory is the logical separation. The finite stabilizer model makes the microscopic line algebra and the one-form intersection phase completely explicit. A continuum deconfined gauge theory may have the same long-distance topological sector when a regulator family, phase classification, and infrared limit connect the microscopic model to that sector.

Open Problem 119.6 (Complete phase invariant for gauge theories). Define a tractable invariant of gauge-theory phases that includes local correlators, extended-operator categories, higher-form symmetry realization, background responses, boundary-condition sectors, and massive spectra, and prove its stability under finite local deformations inside a gapped phase.

Chapter 120

Boundaries And Defects

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Boundaries and defects are QFT data supported on submanifolds. They modify operator algebras, symmetry realization, anomaly inflow, and phase structure. This chapter formulates them as local data rather than as afterthoughts added to a bulk theory. The examples below are elementary enough that their fusion and anomaly data can be checked directly.

120.1 Boundary QFT Datum

Definition 120.1 (Boundary QFT datum)**.** Let M be a spacetime manifold with boundary ∂M. A boundary condition for a local QFT assigns the following data:

$$\begin{array}{ll} \text{bulk observables} & \mathcal{A}(U), \quad U \subset M^\circ, \\ \text{boundary observables} & \mathcal{B}(V), \quad V \subset \partial M, \\ \text{bulk-to-boundary maps} & \mathcal{A}(U) \to \mathcal{B}(V) \text{ for } U \to V, \\ \text{state or functional} & \omega_\partial \text{ on the combined algebra.} \end{array}$$

Locality requires that observables with spacelike separated supports commute or graded-commute after their supports are interpreted in the manifold with boundary.

The bulk-to-boundary maps in Definition 120.1 are not optional. They specify what happens when a bulk operator is brought to the boundary and therefore determine boundary operator expansions, boundary Ward identities, and the possible endpoints of bulk extended operators.

120.2 Boundary Operator Expansion

Definition 120.2 (Boundary operator expansion)**.** For a scalar bulk local operator $\mathcal{O}(x_\perp, y)$, with $x_\perp \geq 0$ the distance to the boundary and $y \in \partial M$, the boundary expansion has the form

$$\mathcal{O}(x_\perp, y) \sim \sum_\alpha c_{\mathcal{O}}^\alpha(x_\perp)\, \widehat{\mathcal{O}}_\alpha(y),$$

where $\widehat{\mathcal{O}}_\alpha$ are boundary local operators. The symbol means convergence inside the declared correlation-function topology after smearing away from coincident boundary insertions. Boundary critical exponents and boundary anomalies are properties of this expansion and of the boundary stress tensor.

Example 120.3 (Free scalar half-space boundary expansion)**.** Let ϕ be a free massless scalar field on the Euclidean half-space $\mathbb{R}_{\geq 0} \times \mathbb{R}^{D-1}$, with coordinates $(x_\perp, y)$. For Neumann boundary condition $\partial_\perp \phi|_{x_\perp=0} = 0$, the method of images gives

$$\langle \phi(x_\perp, y)\phi(x'_\perp, y')\rangle_N = G_D(x_\perp - x'_\perp, y - y') + G_D(x_\perp + x'_\perp, y - y').$$

For Dirichlet boundary condition $\phi|_{x_\perp=0} = 0$, the sign of the image term is reversed. Consequently the leading boundary operator in the Neumann case is $\widehat{\phi}(y) = \phi(0, y)$, whereas in the Dirichlet case the leading nonzero boundary operator is $\widehat{\partial_\perp \phi}(y) = \partial_\perp \phi(0, y)$. This example shows that the boundary operator expansion is a statement about a boundary condition together with the corresponding boundary field content.

120.3 Interfaces And Domain Walls

Definition 120.4 (Interface and domain wall)**.** An interface between two QFTs $\mathcal{T}_-$ and $\mathcal{T}_+$ is a codimension-one defect whose local neighborhoods have $\mathcal{T}_-$ on one side and $\mathcal{T}_+$ on the other. A domain wall is an interface between two phases or vacua of the same microscopic theory. Topological interfaces can be moved without changing separated correlation functions, up to contact terms specified on the wall. Non-topological interfaces carry ordinary localized excitations and have their own stress-tensor Ward identities.

Example 120.5 (Chern–Simons boundary and chiral WZW currents)**.** For Chern–Simons theory on a three-manifold with boundary, the variation of the bulk action contains the boundary one-form

$$\frac{k}{4\pi} \int_{\partial M} \mathrm{tr}(A \wedge \delta A).$$

After choosing the holomorphic polarization on the boundary surface Σ, the boundary field $g : \Sigma \to G$ has the chiral Wess–Zumino–Witten action displayed in Chapter 105. The boundary current

$$J(z) = -\frac{k}{2\pi} \partial_z g\, g^{-1}$$

obeys the affine current algebra at level k. Thus the boundary condition does not simply remove a boundary term; it creates a boundary operator algebra whose representation theory is the state-space input for Wilson-line endpoints and boundary conformal blocks.

120.4 Fusion

Definition 120.6 (Renormalized defect fusion)**.** Let D_1 and D_2 be parallel defects separated by distance ℓ. Fusion asks for the limit

$$D_2 \circ_\ell D_1 \quad \text{as} \quad \ell \to 0.$$

The limit is a defect $D_2 \star D_1$ only if correlation functions with separated external insertions converge after renormalizing defect-local operators. In a topological sector, fusion gives a monoidal category of defects. In a general QFT, fusion is a renormalization problem for codimension-one local data.

Example 120.7 (Ising topological defect fusion)**.** At the critical two-dimensional Ising point, the basic topological defects are

$$\mathbf{1}, \qquad \eta, \qquad \mathcal{N},$$

where η is the invertible spin-flip defect and $\mathcal{N}$ is the Kramers–Wannier duality defect. Their fusion rules are

$$\eta \otimes \eta \simeq \mathbf{1}, \qquad \eta \otimes \mathcal{N} \simeq \mathcal{N} \otimes \eta \simeq \mathcal{N}, \qquad \mathcal{N} \otimes \mathcal{N} \simeq \mathbf{1} \oplus \eta.$$

The last equation is the first test that the word "duality" is being used as a defect statement rather than as a slogan: $\mathcal{N}$ has no inverse, because its square is a direct sum. Chapter 121 derives the corresponding action on primary sectors from the Ising modular S-matrix.

Remark 120.8 (Fusion detects noninvertibility). In the fusion algebra of Example 120.7, the defect $\mathcal{N}$ is not invertible and has Frobenius–Perron dimension $\sqrt{2}$.

If $\mathcal{N}$ had an inverse X, then $\mathcal{N} \otimes X \simeq \mathbf{1}$ would be simple. But fusing $\mathcal{N} \otimes \mathcal{N} \simeq \mathbf{1} \oplus \eta$ with X gives

$$\mathcal{N} \simeq X \oplus (\eta \otimes X),$$

which expresses the simple defect $\mathcal{N}$ as a direct sum of two nonzero summands, a contradiction in the semisimple defect category. The positive dimension homomorphism satisfies

$$d_{\mathcal{N}}^2 = d_{\mathbf{1}} + d_\eta = 2,$$

because $d_{\mathbf{1}} = d_\eta = 1$. Hence $d_{\mathcal{N}} = \sqrt{2}$.

Example 120.9 (Electric-magnetic duality surface in three-dimensional BF)**.** The $\mathbb{Z}_N$ topological gauge theory can be written in continuum notation as the $U(1) \times U(1)$ BF action

$$S_{\mathrm{BF}} = \frac{N}{2\pi} \int B \wedge dA,$$

with compact one-form gauge fields A and B. Its simple line labels are pairs $(e, m) \in \mathbb{Z}_N \oplus \mathbb{Z}_N$, represented by

$$W_{e,m}(\gamma) = \exp\!\left(\mathrm{i}e \int_\gamma A + \mathrm{i}m \int_\gamma B \right).$$

The codimension-one surface defect S that exchanges A and B acts by

$$S:\ (e, m) \longmapsto (m, e).$$

It is topological because it preserves the BF action and the braiding pairing

$$\langle (e,m),(e',m')\rangle = \exp\left(\frac{2\pi \mathrm{i}}{N}(em'+me')\right).$$

Fusion satisfies $S \otimes S \simeq \mathbf{1}$. This is an invertible surface defect; unlike the Ising $\mathcal{N}$ defect, it is an ordinary duality wall because the induced map on the finite line-label group is an automorphism.

120.5 Anomalies And Inflow

Definition 120.10 (Anomaly line of a boundary or defect)**.** A boundary or defect can carry an anomaly line for background fields restricted to its support. If A_D denotes those restricted background fields, then a defect partition function with anomaly is a section

$$Z_D(A_D) \in \Gamma(\mathcal{L}_D),$$

where $\mathcal{L}_D$ is a complex line over the background-field space.

An invertible bulk theory with boundary D supplies a section of $\mathcal{L}_D^{-1}$. The product is then a complex number. If a boundary condition preserves a global symmetry, its boundary anomaly must match the restriction of the bulk inflow. If it breaks the symmetry, the broken background fields are not part of the boundary datum.

Remark 120.11 (Inflow cancellation as line trivialization). The combined bulk-defect amplitude is an ordinary complex-valued functional of background fields if and only if the tensor product of the defect anomaly line and the bulk inflow line is canonically trivialized.

If $Z_D(A_D)$ is a section of $\mathcal{L}_D$ and the bulk inflow amplitude $Z_{\mathrm{bulk}}(A)$ is a section of $\mathcal{L}_{\mathrm{bulk}}$, then their product is a section of $\mathcal{L}_D \otimes \mathcal{L}_{\mathrm{bulk}}$. It is a number precisely after choosing a natural isomorphism

$$\mathcal{L}_D \otimes \mathcal{L}_{\mathrm{bulk}} \cong \underline{\mathbb{C}}.$$

Conversely, if the product is a complex-valued functional for all background fields and is compatible with gluing of backgrounds, it defines exactly such a trivialization of the tensor product line.

120.6 Gauge-Theory Boundaries

For gauge theories, a boundary condition must specify which gauge transformations are redundancies at the boundary and which become global symmetries acting on boundary operators. Electric boundary conditions fix the tangential connection or electric flux according to the chosen Hamiltonian polarization. Magnetic boundary conditions fix the dual data. Line operators may end on the boundary precisely when boundary degrees of freedom carry the corresponding charge and cancel the endpoint gauge variation.

Example 120.12 (Endpoint gauge variation of a Wilson line)**.** Let γ be an oriented path from p to q, and let

$$U_\gamma(A) = \mathrm{Pexp}\int_\gamma A$$

be parallel transport in a representation R. Under a gauge transformation g,

$$U_\gamma(A^g) = R(g(p))\, U_\gamma(A)\, R(g(q))^{-1}.$$

Thus an open Wilson line is not gauge invariant by itself. It can end on a boundary only if boundary operators at p and q transform in the dual representations needed to contract these endpoint factors, or if the boundary condition converts the relevant gauge transformations into fixed global symmetry transformations and the line is interpreted as a charged operator under that boundary symmetry.

120.7 Defects And Phase Data

Boundary and defect categories refine the phase invariant of Chapter 119. A gapped phase includes the fusion category of topological line defects, higher-codimension defect operators, boundary conditions, and their anomaly responses. Two theories with the same local massive spectrum can be distinguished by these extended data.

Open Problem 120.13 (Defect-complete phase classification)*.* Construct a phase invariant that includes boundary conditions, defect fusion, endpoint maps for line and surface operators, anomaly inflow, and local operator data, and prove stability of this invariant under finite gapped deformations.

Chapter 121

Categorical Symmetry And Defect Fusion

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter formulates categorical symmetry through topological defects and their fusion. The object replacing a group is a category or higher category of topological operators, and the action on local or extended observables is defined by surrounding, ending, or fusing defects.

121.1 Topological Defect Category

Let $\mathcal{Q}$ be a D-dimensional local QFT. A codimension-one topological defect D is an operator supported on a hypersurface such that correlation functions are invariant under compactly supported deformations of the hypersurface avoiding insertions. Defects form a category $\mathcal{D}_{\mathcal{Q}}$ whose objects are boundary labels on small transverse intervals and whose morphisms are junction operators. Fusion is a monoidal product

$$\otimes : \mathcal{D}_{\mathcal{Q}} \times \mathcal{D}_{\mathcal{Q}} \longrightarrow \mathcal{D}_{\mathcal{Q}}.$$

If $D_g \otimes D_h \simeq D_{gh}$ and every D_g is invertible, the data reduce to an ordinary group symmetry. Categorical symmetry keeps the same fusion operation but allows noninvertible simple objects and nontrivial junction spaces.

Definition 121.1 (Topological defect datum in correlation functions)**.** Fix the class of Euclidean backgrounds and insertions on which $\mathcal{Q}$ is defined. A codimension-one defect label a assigns to every oriented framed hypersurface Σ, together with a tubular neighborhood disjoint from the other insertions, multilinear distributions

$$Z_{\Sigma,a}(\mathfrak{A}_1, \dots, \mathfrak{A}_n) = \big\langle D_a(\Sigma)\, \mathfrak{A}_1 \cdots \mathfrak{A}_n \big\rangle \tag{121.1}$$

on separated smeared local and extended insertions. The assignment is called topological if, for every smooth isotopy Σ_t fixed near the boundary of the chosen coordinate patch and avoiding the supports of $\mathfrak{A}_i$,

$$\frac{d}{dt} Z_{\Sigma_t,a}(\mathfrak{A}_1, \dots, \mathfrak{A}_n) = 0 \tag{121.2}$$

as a distribution in the insertion variables. The datum is local if it is compatible with restriction to sub-balls and gluing along collars. It is renormalized if the limit defining (121.1) is independent of auxiliary cutoff choices up to the declared finite defect counterterms.

When the response to a normal displacement is represented by a local defect operator $\mathfrak{D}_{\perp,a}$, called the displacement operator, the infinitesimal Ward identity has the form

$$\frac{d}{dt} \big\langle D_a(\Sigma_t)\, \mathfrak{A}_1 \cdots \mathfrak{A}_n \big\rangle = \left\langle D_a(\Sigma_t) \int_{\Sigma_t} \nu_t\, \mathfrak{D}_{\perp,a}\, \mathfrak{A}_1 \cdots \mathfrak{A}_n \right\rangle, \tag{121.3}$$

where ν_t is the scalar normal velocity of the deformation.

Remark 121.2 (Displacement criterion for topological defects). Assume the displacement Ward identity (121.3) holds for a renormalized defect label a. The defect is topological in the sense of Definition 121.1 if $\mathfrak{D}_{\perp,a} = 0$ as a defect local operator. Conversely, if (121.2) holds for all compactly supported normal variations and separated test insertions, then $\mathfrak{D}_{\perp,a}$ is zero in the defect-operator quotient by correlation functions. If $\mathfrak{D}_{\perp,a}$ is zero in all defect correlation functions, the right hand side of (121.3) vanishes for every normal velocity ν_t, hence every correlation functional is constant along the isotopy. This is exactly (121.2).

Conversely, suppose (121.2) holds. Choose ν_t supported in an arbitrarily small coordinate patch of Σ_t. The left hand side of (121.3) is then zero for every such ν_t and every separated test configuration. Since test functions ν_t separate distributions on the defect worldvolume, the distribution defined by inserting $\mathfrak{D}_{\perp,a}$ has zero pairing with every test function and every exterior insertion. Therefore $\mathfrak{D}_{\perp,a}$ lies in the null subspace of defect local operators, which is the asserted quotient statement.

Definition 121.3 (Physical defect category)**.** For a finite ordered list $X = (D_{a_1}, \dots, D_{a_m})$ of defect sheets entering a small ball and a finite ordered list Y exiting it, let $\mathcal{J}(X,Y)$ be the vector space spanned by renormalized junction insertions in the ball with those boundary defect labels. Let $\mathcal{N}(X,Y) \subset \mathcal{J}(X,Y)$ be the subspace of junctions whose ball correlation functions vanish after gluing to every compatible exterior defect network and every separated bulk insertion. The physical junction space is

$$\mathrm{Hom}_{\mathcal{D}_{\mathcal{Q}}}(X,Y) = \mathcal{J}(X,Y)/\mathcal{N}(X,Y). \tag{121.4}$$

Vertical composition is gluing of balls; tensor product is juxtaposition of defect sheets in a collar; fusion of parallel topological sheets is the correlation-function limit in which the collar thickness tends to zero. Whenever these operations exist and are associative up to specified junction isomorphisms, the quotient spaces form the physical defect category $\mathcal{D}_{\mathcal{Q}}$.

This definition is deliberately not a declaration that every local QFT has a finite semisimple defect category. Semisimplicity, rigidity, and finite-dimensionality of junction spaces are additional hypotheses. They hold in many rational examples and in finite topological models, but in a general continuum QFT the null quotient, the fusion limit, and the topology of the junction spaces are part of the mathematical problem. In the reflection-positive finite-dimensional case the quotient is made explicit by the Gram-radical construction in Proposition 121.9; that construction is the algebraic bridge between the correlation-function definition above and the dagger formulas for junctions below.

121.2 Action On Local Operators

The phrase "action on local operators" hides a limiting operation. It is not enough to draw a small sphere around an insertion; one must specify the topology in which the limit of the defect-surrounded insertion is taken.

Definition 121.4 (Correlation topology on local operators)**.** Let $\mathcal{V}_{\mathrm{loc}}(0)$ be the vector space of renormalized local operators at the origin, modulo the relation that an operator is zero if all its Euclidean correlation functions with separated test insertions vanish. For a finite list of smeared local or extended insertions $\mathfrak{A}_1, \ldots, \mathfrak{A}_n$ supported in the complement of a ball around the origin, define the seminorm

$$p_{\mathfrak{A}_1,\ldots,\mathfrak{A}_n}(\mathcal{O}) = \left|\langle \mathfrak{A}_1 \cdots \mathfrak{A}_n \, \mathcal{O}(0)\rangle\right| . \tag{121.5}$$

The correlation topology is the locally convex topology generated by these seminorms as the separated insertions vary. In a unitary conformal theory with radial quantization, the state-operator map $\mathcal{O} \mapsto \mathcal{O}(0)\Omega$ gives a stronger Hilbert topology on the image in $\mathcal{H}(S^{D-1})$; convergence in that Hilbert topology implies convergence of all seminorms (121.5) for bounded separated insertions.

Definition 121.5 (Defect action on local operators)**.** Let D be a topological codimension-one defect in Euclidean space. For $r > 0$, let S_r^{D-1} be the sphere of radius r centered at the origin, carrying D with the outward normal convention. The defect action on $\mathcal{O} \in \mathcal{V}_{\mathrm{loc}}(0)$ is defined if the limit

$$\rho_D(\mathcal{O}) = \lim_{r \downarrow 0} \left[D(S_r^{D-1}) \, \mathcal{O}(0) \right] \tag{121.6}$$

exists in the correlation topology, or in the radial Hilbert topology when that stronger structure is available. The local-operator space acted on by the defect category is the common domain on which these limits exist for the defects under discussion.

Remark 121.6 (Fusion acts by composition)*.* Assume D_1, D_2 and $D_1 \otimes D_2$ have shrinking limits on a common local-operator domain in the sense of Definition 121.5. Then

$$\rho_{D_1 \otimes D_2} = \rho_{D_1} \rho_{D_2} \tag{121.7}$$

on that domain. Choose radii $0 < r_1 < r_2$, put D_2 on $S_{r_1}^{D-1}$ and D_1 on $S_{r_2}^{D-1}$. Because the defects are topological, any deformation of the two spheres through configurations avoiding separated insertions leaves all seminorms (121.5) unchanged. First take $r_1 \downarrow 0$; by definition this gives $\rho_{D_2}(\mathcal{O})$. Then take $r_2 \downarrow 0$, giving $\rho_{D_1}\rho_{D_2}(\mathcal{O})$. Alternatively, keep the two spheres at fixed small separation and collapse the annular slab between them to the fused defect $D_1 \otimes D_2$, then take the common shrinking limit. The two procedures give the same correlation seminorms with every separated test configuration, hence the same element of $\mathcal{V}_{\mathrm{loc}}(0)$.

Proposition 121.7 (Reflection adjoint of a topological-defect action)**.** *Assume radial quantization is available and let $\mathcal{H}_{\mathrm{loc}}$ be the Hilbert completion of local states obtained from the reflection-positive BPZ inner product. Suppose D and its reflection-reversed defect $D^\dagger$ have shrinking actions on a common dense local-operator domain $\mathcal{D}_{\mathrm{loc}} \subset \mathcal{H}_{\mathrm{loc}}$, and suppose these actions extend to bounded operators on $\mathcal{H}_{\mathrm{loc}}$. Then*

$$\langle \rho_D v, w\rangle_{\mathrm{BPZ}} = \langle v, \rho_{D^\dagger} w\rangle_{\mathrm{BPZ}}, \qquad v, w \in \mathcal{H}_{\mathrm{loc}}. \tag{121.8}$$

Thus $\rho_{D^\dagger} = \rho_D^*$. In particular, a reflection-invariant defect acts by a self-adjoint bounded operator, but it need not act by a unitary operator unless the defect is invertible.

Proof. It is enough first to prove the identity on $\mathcal{D}_{\mathrm{loc}}$. Choose local-operator representatives $\mathcal{O}_v, \mathcal{O}_w$ of v, w, with all insertions placed in the unit ball and with the BPZ inner product written as the reflection-positive pairing

$$\langle v, w\rangle_{\mathrm{BPZ}} = \lim_{R\to\infty} \langle \Theta_R \mathcal{O}_w(R)\, \mathcal{O}_v(0)\rangle\,,$$

where Θ_R denotes radial reflection together with the anti-linear adjunction on operator labels. For $\epsilon > 0$ small, let $\rho_{D,\epsilon}$ be the operation of inserting D on the sphere S_ϵ^{D-1} around the origin. By the definition of the shrinking action,

$$\rho_{D,\epsilon} v \longrightarrow \rho_D v \quad \text{and} \quad \rho_{D^\dagger,\epsilon} w \longrightarrow \rho_{D^\dagger} w$$

in the Hilbert topology on the common domain. Hence

$$\langle \rho_D v, w\rangle_{\mathrm{BPZ}} = \lim_{\epsilon\downarrow 0} \langle \rho_{D,\epsilon} v, w\rangle_{\mathrm{BPZ}}.$$

The finite-ϵ inner product is the ball correlator with D on S_ϵ^{D-1}, $\mathcal{O}_v$ inside it, and the reflected insertion representing w outside it. Radial reflection reverses the coorientation of the defect sheet, so the reflected defect label is $D^\dagger$. The reflected sphere surrounds the reflected w-insertion and no other operator. Because the defect is topological, this sphere may be isotoped, inside the complement of all operator insertions, to the standard small sphere used in Definition 121.5 for the action of $D^\dagger$ on w. Therefore, at fixed ϵ,

$$\langle \rho_{D,\epsilon} v, w\rangle_{\mathrm{BPZ}} = \langle v, \rho_{D^\dagger,\epsilon} w\rangle_{\mathrm{BPZ}},$$

with the same orientation convention as in the definition of $D^\dagger$. Taking $\epsilon \downarrow 0$ gives

$$\langle \rho_D v, w\rangle_{\mathrm{BPZ}} = \langle v, \rho_{D^\dagger} w\rangle_{\mathrm{BPZ}}$$

for $v, w \in \mathcal{D}_{\mathrm{loc}}$. Since $\mathcal{D}_{\mathrm{loc}}$ is dense and both ρ_D and $\rho_{D^\dagger}$ are bounded by hypothesis, the identity extends uniquely to all of $\mathcal{H}_{\mathrm{loc}}$, which is precisely $\rho_{D^\dagger} = \rho_D^*$.

If $D \simeq D^\dagger$ through the chosen reflection structure, the operator is self-adjoint. Unitarity is stronger: it would require

$$\rho_D^* \rho_D = \rho_{D^\dagger} \rho_D = \rho_{D^\dagger \otimes D} = \mathbf{1}$$

on the Hilbert space, where the middle equality is the fusion-action identity (121.7). Thus unitarity follows from an invertible defect with $D^\dagger \otimes D \simeq \mathbf{1}$. Reflection symmetry by itself gives self-adjointness; invertibility supplies the inverse required for unitarity. $\square$

When D is not invertible, ρ_D may be a projection, a sum of endomorphisms, or a map with a nontrivial kernel. Selection rules follow from eigenvectors and images of these maps, not from characters of a group alone.

121.3 Fusion Coefficients And Junctions

Assume the defect category is semisimple with simple objects D_a. Fusion has coefficients

$$D_a \otimes D_b \simeq \bigoplus_c N_{ab}{}^c D_c, \qquad N_{ab}{}^c \in \mathbb{Z}_{\geq 0}.$$

A choice of basis in the junction spaces

$$\mathrm{Hom}(D_a \otimes D_b, D_c)$$

gives associator matrices F. The equality of the two ways of fusing four defects is the pentagon equation. This equation is obtained by comparing two isotopic networks of topological junctions in a ball.

Definition 121.8 (Adjoint junction and reflection pairing)**.** Assume the QFT is reflection positive. Let Θ denote reflection through an equatorial hyperplane in a small Euclidean ball, together with the anti-linear operation on operator labels induced by Hilbert-space adjunction after radial quantization. If

$$\psi \in \mathrm{Hom}(D_a \otimes D_b, D_c)$$

is a junction operator inserted at the center of the ball, its reflected adjoint is

$$\psi^\dagger \in \mathrm{Hom}(D_c, D_a \otimes D_b), \tag{121.9}$$

with orientations reversed as dictated by the reflected defect sheets. For two junctions ψ, χ with the same boundary defect labels, define

$$\langle \psi, \chi \rangle_{\mathrm{jun}} = Z\big(B^D; \Theta(\psi) \text{ in the reflected half-ball}, \chi \text{ in the original half-ball}\big), \tag{121.10}$$

where the boundary defect labels are connected by straight topological sheets. The notation means the ball correlation function obtained after gluing the two half-balls along the reflection plane.

Proposition 121.9 (Gram radical model for the physical null quotient)**.** *Fix boundary defect labels X, Y, and suppose the renormalized junction space*

$$J(X,Y) = \mathcal{J}(X,Y)$$

is finite dimensional. Assume reflection positivity gives the half-ball pairing $\langle -, - \rangle_{X,Y}$ of (121.10). Let

$$R(X,Y) = \{\psi \in J(X,Y) : \langle \psi, \psi \rangle_{X,Y} = 0\}. \tag{121.11}$$

Then $R(X,Y)$ is the radical

$$R(X,Y) = \{\psi \in J(X,Y) : \langle \psi, \chi \rangle_{X,Y} = 0 \textit{ for every } \chi \in J(X,Y)\},$$

and the quotient $J(X,Y)/R(X,Y)$ has a positive definite Hermitian pairing. If vertical composition and tensor product of junctions have reflected adjoints on these finite spaces, then both operations preserve the radicals and therefore descend to the quotients. When reflected half-ball states generated by compatible exterior defect networks separate correlation functionals, this Gram-radical quotient agrees with the correlation-null quotient of Definition 121.3.

Proof. The half-ball construction assigns to each junction ψ a vector $|\psi\rangle$ in the pre-Hilbert space obtained by radial quantization of the ball, with

$$\langle \psi, \chi \rangle_{X,Y} = \langle |\psi\rangle, |\chi\rangle \rangle.$$

Reflection positivity says this form is positive semidefinite. If $\langle \psi, \psi \rangle_{X,Y} = 0$, then $|\psi\rangle$ is zero in the Hilbert quotient, so $\langle \psi, \chi \rangle_{X,Y} = 0$ for every χ. Thus the zero-norm subspace is exactly the radical. Modding out by it removes precisely the vectors with zero pairing against all states, and the induced Hermitian form is positive definite.

Let $C_\phi : J(X,Y) \to J(X,Z)$ be postcomposition with a fixed junction $\phi : Y \to Z$. Its reflected adjoint is precomposition with $\phi^\dagger$ in the reflected ball, so

$$\langle C_\phi \psi, \eta \rangle_{X,Z} = \langle \psi, C_\phi^\dagger \eta \rangle_{X,Y}.$$

If $\psi \in R(X,Y)$, the right hand side vanishes for every η, hence $C_\phi \psi \in R(X,Z)$. The same argument applies to precomposition and to tensoring with a fixed junction, using the reflected adjoint of the corresponding gluing map. Therefore changing a representative by a radical element does not change the composite or tensor product in the quotient.

Finally, a junction lies in the correlation-null subspace of Definition 121.3 exactly when all gluings to compatible exterior networks vanish. Under the stated separation hypothesis, those exterior networks span the same test states as the reflected half-ball vectors. Vanishing against all exterior networks is then the same condition as orthogonality to every $\chi \in J(X,Y)$, which is the radical condition above. ☐

Proposition 121.10 (Reflection positivity gives a dagger category)**.** *After replacing raw junction spaces by the Gram-radical quotients of Proposition 121.9, suppose the reflected gluing maps are defined on the junction spaces under consideration. Then reflection defines an anti-linear contravariant involution on junctions,*

$$(\chi \circ \psi)^\dagger = \psi^\dagger \circ \chi^\dagger, \qquad (\psi^\dagger)^\dagger = \psi, \tag{121.12}$$

and the pairings are positive:

$$\langle \psi, \psi \rangle_{\mathrm{jun}} \geq 0, \qquad \langle \psi, \psi \rangle_{\mathrm{jun}} = 0 \implies \psi = 0$$

on the quotient. In orthonormal bases for these pairings, isotopy maps between two bases of the same junction network are unitary matrices.

Proof. Place the junction network in a ball cut by the reflection hyperplane into incoming and outgoing half-balls. Vertical composition of junctions is gluing of such half-balls in the radial time direction. Reflection exchanges the two halves and reverses that gluing order; hence the reflected adjoint of $\chi \circ \psi$ is $\psi^\dagger \circ \chi^\dagger$. Applying the same reflection a second time restores the original oriented ball and the original defect-sheet orientations, which gives $(\psi^\dagger)^\dagger = \psi$.

For positivity, let $|\psi\rangle$ denote the state on the equatorial boundary Hilbert space produced by the lower half-ball with junction ψ. The pairing (121.10) is exactly $\langle \psi | \psi \rangle$ in the reflection-positive Hilbert space obtained from the ball. Reflection positivity gives $\langle \psi, \psi \rangle_{\mathrm{jun}} \geq 0$. By Proposition 121.9, the zero-norm junctions form the radical of this pairing, and quotienting by them gives a positive definite Hermitian pairing.

Finally, an isotopy of a topological junction network inside the ball changes the chosen basis vector but not the ball correlation function with reflected adjoint boundary conditions. Therefore the linear isotopy map preserves the Hermitian form (121.10). In bases orthonormal for that form, preservation of the Hermitian form is exactly the matrix identity $U^\dagger U = 1$. □

121.4 Gauging A Categorical Symmetry

For a finite group symmetry, gauging sums over flat bundles. For a finite categorical symmetry, the analogous operation is a condensation of a commutative separable algebra object

$$A \in \mathcal{D}_{\mathcal{Q}}.$$

The algebra maps are

$$m : A \otimes A \to A, \qquad \eta : \mathbf{1} \to A,$$

and the Frobenius and separability identities encode the local moves of the defect network. The gauged theory has topological defects given by A-modules or A-bimodules, depending on the codimension and the placement. This construction is a QFT operation only when the defect-network sum or condensation is defined as a regulated insertion in correlation functions.

121.5 Finite Regular Algebra And Self-Dual Gauging

The finite-group case is the minimal example in which the abstract word "condensation" becomes an explicit algebra object. Let H be a finite group acting by distinct invertible topological defects

$$\{D_h : h \in H\}, \qquad D_g \otimes D_h \simeq D_{gh}.$$

Assume for the moment that the associator phases in the pointed defect subcategory have been trivialized. A nontrivial 3-cocycle associator is an anomaly or discrete-torsion datum; it modifies the junction maps below and must be carried explicitly.

Definition 121.11 (Regular algebra of a finite symmetry)**.** The regular algebra object of the untwisted pointed defect subcategory generated by H is

$$A_H = \bigoplus_{h \in H} D_h.$$

Its multiplication is the direct sum of the group-law junctions

$$m_{g,h} : D_g \otimes D_h \longrightarrow D_{gh},$$

and its unit is the inclusion $\eta_A : \mathbf{1} = D_e \hookrightarrow A_H$. Choose the normalized comultiplication

$$\Delta(D_k) = \frac{1}{|H|} \sum_{\substack{g,h \in H \\ gh = k}} D_g \otimes D_h, \tag{121.13}$$

where the displayed formula means the corresponding sum of dual junction maps in the chosen bases.

Regular algebra identities. In the untwisted pointed defect category generated by H, the regular algebra object satisfies

$$A_H \otimes A_H \simeq |H|\, A_H, \qquad m \circ \Delta = \mathrm{id}_{A_H}, \tag{121.14}$$

and the Frobenius identity

$$\Delta \circ m = (m \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \Delta) = (\mathrm{id} \otimes m) \circ (\Delta \otimes \mathrm{id}) \tag{121.15}$$

holds on every simple summand $D_g \otimes D_h$. These are the finite group-algebra identities encoded by the multiplication and normalized comultiplication, not a separate structural theorem. The object identity follows from

$$A_H \otimes A_H \simeq \bigoplus_{g,h\in H} D_{gh}.$$

For fixed $k \in H$, the equation $gh = k$ has exactly $|H|$ solutions: choose g freely and set $h = g^{-1}k$. Hence each D_k occurs $|H|$ times.

For separability, apply $m \circ \Delta$ to D_k. The summands in (121.13) all multiply back to D_k, and there are $|H|$ of them, so the normalized coefficient gives one copy of D_k.

For the Frobenius identity, evaluate on $D_g \otimes D_h$. The left side is

$$\Delta(D_{gh}) = \frac{1}{|H|} \sum_{\substack{c,d\in H \\ cd=gh}} D_c \otimes D_d.$$

The middle expression gives

$$\frac{1}{|H|} \sum_{\substack{a,b\in H \\ ab=h}} D_{ga} \otimes D_b.$$

The map $(a, b) \mapsto (c, d) = (ga, b)$ is a bijection from the set $\{(a, b) : ab = h\}$ to the set $\{(c, d) : cd = gh\}$. Thus the two sums agree. The right expression is identical by the bijection $(a, b) \mapsto (c, d) = (a, bh)$ from $\{(a, b) : ab = g\}$ to $\{(c, d) : cd = gh\}$.

Self-dual finite gauging fusion algebra. Suppose anomaly-free H-gauging of $\mathcal{Q}$ admits an equivalence $\Phi : \mathcal{Q}/H \to \mathcal{Q}$. Let $I_H : \mathcal{Q} \to \mathcal{Q}/H$ be the gauging interface, and let $\mathcal{N}_H = \Phi \circ I_H$ be the resulting gauging defect. The nontrivial inputs are the existence of this gauging interface, the equivalence Φ, and a choice of equivalence for which $\mathcal{N}_H^\dagger \simeq \mathcal{N}_H$ and the H-defects are absorbed on both sides of the gauging interface:

$$D_h \otimes \mathcal{N}_H \simeq \mathcal{N}_H \otimes D_h \simeq \mathcal{N}_H.$$

After these assumptions have been supplied, the object-level fusion algebra generated by D_h and $\mathcal{N}_H$ is

$$D_g D_h = D_{gh}, \qquad D_h \mathcal{N}_H = \mathcal{N}_H D_h = \mathcal{N}_H, \qquad \mathcal{N}_H^2 = \bigoplus_{h\in H} D_h. \tag{121.16}$$

The first rule is the ordinary symmetry-defect fusion. The second is the absorption property at an H-gauging interface. The third is the self-adjoint version of the slab fusion

$$\mathcal{N}_H^\dagger \otimes \mathcal{N}_H \simeq A_H$$

from the finite groupoid sum over the slab between the interface and its reverse. Thus the formula is not a construction of the self-duality; it is the object-level algebra forced by that self-duality once the categorical data have been chosen.

Associativity is then an object-level check. If only D_h's occur, it is associativity of the group H. If exactly one $\mathcal{N}_H$ occurs, both bracketings reduce to $\mathcal{N}_H$. If two $\mathcal{N}_H$'s occur, for example

$$(\mathcal{N}_H^2)D_g = \bigoplus_{h\in H} D_{hg} = \bigoplus_{k\in H} D_k = \mathcal{N}_H(\mathcal{N}_H D_g),$$

where right multiplication by g permutes H; the other placements are the same. If three $\mathcal{N}_H$'s occur, then

$$(\mathcal{N}_H^2)\mathcal{N}_H = \bigoplus_{h\in H} D_h\mathcal{N}_H = |H|\,\mathcal{N}_H = \mathcal{N}_H\big(\mathcal{N}_H^2\big).$$

Frobenius–Perron dimension is additive on direct sums and multiplicative on fusion products. Thus $d(D_h) = 1$, and the last fusion rule gives

$$d(\mathcal{N}_H)^2 = \sum_{h\in H} d(D_h) = |H|.$$

The positive solution is $\sqrt{|H|}$. If $\mathcal{N}_H$ were invertible, its square would be a single simple invertible object rather than a direct sum of $|H| > 1$ distinct simple defects. Therefore

$$d(D_h) = 1, \qquad d(\mathcal{N}_H) = \sqrt{|H|}.$$

For $|H| > 1$, $\mathcal{N}_H$ is not invertible.

Example 121.12 (Cyclic self-dual gauging fusion rings)**.** For $H = \mathbb{Z}_N = \langle a : a^N = e\rangle$, the finite self-dual gauging fusion ring has simple invertible objects D_{a^j} and one noninvertible object $\mathcal{N}_N$ obeying

$$D_{a^i}D_{a^j} = D_{a^{i+j}}, \qquad D_{a^j}\mathcal{N}_N = \mathcal{N}_N D_{a^j} = \mathcal{N}_N, \qquad \mathcal{N}_N^2 = \bigoplus_{j=0}^{N-1} D_{a^j}.$$

The $N = 2$ case is the Ising Kramers–Wannier fusion rule. For $N > 2$ the same equations are only the fusion-ring part of a possible defect category: associators, junction pairings, and the existence of a local QFT realizing the required self-duality are additional data, not consequences of the fusion ring alone.

121.6 The Ising/Kramers–Wannier Defect

The simplest noninvertible example is the defect category of the critical two-dimensional Ising model. We describe the categorical data abstractly, because these data are what survive as the symmetry object. Let the simple codimension-one topological defects be

$$\mathbf{1}, \qquad \eta, \qquad \mathcal{N}.$$

Here $\mathbf{1}$ is the invisible defect, η is the invertible $\mathbb{Z}_2$ spin-flip defect, and $\mathcal{N}$ is the Kramers–Wannier duality defect. Their fusion rules are

$$\begin{gathered} \eta\otimes\eta \simeq \mathbf{1}, \qquad \eta\otimes\mathcal{N} \simeq \mathcal{N}\otimes\eta \simeq \mathcal{N}, \\ \mathcal{N}\otimes\mathcal{N} \simeq \mathbf{1}\oplus\eta. \end{gathered} \tag{121.17}$$

The last line is the decisive one: $\mathcal{N}$ has no inverse, because fusing it with itself gives two simple summands rather than the identity.

Remark 121.13 (Quantum dimensions of the Ising defects)*.* The Frobenius–Perron dimensions of $\mathbf{1}, \eta, \mathcal{N}$ are

$$d_{\mathbf{1}} = 1, \qquad d_\eta = 1, \qquad d_{\mathcal{N}} = \sqrt{2}.$$

Frobenius–Perron dimension is the positive homomorphism from the fusion semiring to $\mathbb{R}_{>0}$. Thus $d_{\mathbf{1}} = 1$, and $\eta^2 = \mathbf{1}$ with positivity gives $d_\eta = 1$. The last fusion rule in (121.17) gives

$$d_{\mathcal{N}}^2 = d_{\mathbf{1}} + d_\eta = 2,$$

hence $d_{\mathcal{N}} = \sqrt{2}$. The negative square root is excluded by the defining positivity of the Frobenius–Perron dimension. These dimensions also solve the eigenvector equation for fusion by $\mathcal{N}$, whose matrix has Perron eigenvalue $\sqrt{2}$.

The defect is therefore noninvertible in a quantitative sense: surrounding an insertion by $\mathcal{N}$ is not an automorphism of the local-operator space. At the critical point the diagonal rational CFT has three primary families, which we denote

$$\mathbf{1}, \qquad \varepsilon, \qquad \sigma.$$

The topological line action on these primary sectors is diagonalized by the Ising modular S-matrix

$$S_{\text{Ising}} = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}_{\mathbf{1},\varepsilon,\sigma}. \tag{121.18}$$

Here the row label ε is the same invertible topological line as the spin-flip defect η, and the row label σ is the same topological line as the Kramers–Wannier defect $\mathcal{N}$. We keep the notations distinct because ε, σ are also conventional names for primary sectors, whereas $\eta, \mathcal{N}$ emphasize the defect interpretation. The line D_a acts on the primary sector i by the scalar

$$\lambda_a(i) = \frac{S_{ai}}{S_{\mathbf{1}i}}. \tag{121.19}$$

For $a = \mathcal{N}$, this gives

$$\lambda_{\mathcal{N}}(\mathbf{1}) = \sqrt{2}, \qquad \lambda_{\mathcal{N}}(\varepsilon) = -\sqrt{2}, \qquad \lambda_{\mathcal{N}}(\sigma) = 0.$$

The zero on the spin sector is not a paradox: a noninvertible defect need not act as a bijection on local fields.

Remark 121.14 (Fusion action on Ising local sectors)*.* The eigenvalues (121.19) furnish a representation of the fusion algebra:

$$\lambda_a(i)\lambda_b(i) = \sum_c N_{ab}{}^c \lambda_c(i)$$

for every defect label a, b and every primary sector i.

For η the eigenvalues are

$$\lambda_\eta(\mathbf{1}) = 1, \qquad \lambda_\eta(\varepsilon) = 1, \qquad \lambda_\eta(\sigma) = -1,$$

so $\lambda_\eta^2 = 1$, matching $\eta \otimes \eta \simeq \mathbf{1}$. Multiplication by η leaves the $\mathcal{N}$-eigenvalues unchanged, because $\mathcal{N}$ has zero eigenvalue on σ, where η differs from $\mathbf{1}$. Finally,

$$\lambda_{\mathcal{N}}(i)^2 = \begin{cases} 2, & i = \mathbf{1}, \varepsilon, \\ 0, & i = \sigma, \end{cases} = 1 + \lambda_\eta(i),$$

which is exactly the action of $\mathbf{1} \oplus \eta$. These identities are the three fusion rules (121.17) evaluated on local sectors.

Remark 121.15 (Duality defect from gauging interface). Let $\mathcal{N}_{\mathbb{Z}_2}$ be the interface that gauges the spin-flip $\mathbb{Z}_2$ symmetry. At the self-dual critical Ising point the gauged theory is equivalent to the original theory after choosing the duality identification. Composing the gauging interface with this identification gives the Kramers–Wannier defect $\mathcal{N}$. The general interface identity of Chapter 122 then gives

$$\mathcal{N}^\dagger \otimes \mathcal{N} \simeq \mathbf{1} \oplus \eta,$$

which is the noninvertible fusion rule above. Thus the noninvertibility is the algebraic memory of summing over $\mathbb{Z}_2$-bundles in the gauging interface.

121.7 SymTFT Interpretation In The Rational Example

For a rational two-dimensional CFT, the categorical symmetry can be packaged by a three-dimensional topological theory whose bulk line category contains the defect category and whose boundary condition is the CFT. In the Ising case the relevant bulk topological line category has simple lines $\mathbf{1}, \eta, \mathcal{N}$ with the same fusion rules (121.17). A topological defect in the CFT is obtained by ending the corresponding bulk line on the boundary. Fusion of defects is then inherited from fusion of bulk lines before they end.

This statement is deliberately precise but limited. It does not assert that an arbitrary continuum QFT with a categorical symmetry has already been constructed from a fully extended topological bulk and boundary condition. Rather, in rational examples it supplies a concrete comparison map:

$$\text{bulk topological lines} \longrightarrow \text{boundary topological defects} \longrightarrow \mathrm{End}(\mathcal{V}_{\mathrm{loc}}).$$

The first arrow is the boundary endpoint operation, and the second is the surrounding action on local operators. The Ising computation above verifies the last map explicitly on the three primary sectors.

Semisimple modular-category mechanism. Here is the precise finite mechanism behind the rational examples. Let $\mathcal{C}$ be a semisimple modular tensor category with finite set $\mathcal{I}$ of simple labels, vacuum label $\mathbf{1}$, fusion coefficients $N_{ab}{}^c$, and modular matrix S_{ai}. When a Cardy diagonal full CFT with chiral category $\mathcal{C}$ has been constructed, the topological defects preserving the chosen chiral algebra are labelled by the same simple objects $a \in \mathcal{I}$. The defect D_a acts diagonally on the bulk sector labelled by $i \in \mathcal{I}$ by

$$\lambda_a(i) = \frac{S_{ai}}{S_{\mathbf{1}i}}. \tag{121.20}$$

The content of this formula is not that every categorical symmetry must be modular. It is the finite semisimple mechanism by which the rational SymTFT comparison map becomes a computable operator on local sectors.

The compatibility with fusion is the Verlinde diagonalization. For each simple label a, let N_a be the matrix

$$(N_a)_b{}^c = N_{ab}{}^c.$$

The Verlinde formula says

$$N_{ab}{}^c = \sum_{x\in\mathcal{I}} \frac{S_{ax}S_{bx}S_{c^\vee x}}{S_{\mathbf{1}x}}, \tag{121.21}$$

where $c^\vee$ is the dual label and the displayed convention assumes the standard unitary normalization of S, so $S_{c^\vee x} = S^*_{cx}$. Multiplying (121.21) by S_{ci} and summing over c, unitarity of S gives

$$\sum_c N_{ab}{}^c S_{ci} = \frac{S_{ai}}{S_{\mathbf{1}i}}\, S_{bi}. \tag{121.22}$$

Thus the i-th column of S is an eigenvector of N_a with eigenvalue $\lambda_a(i)$. Equivalently,

$$\lambda_a(i)\lambda_b(i) = \sum_c N_{ab}{}^c \lambda_c(i). \tag{121.23}$$

This is the general version of the Ising calculation (121.19): the surrounding action is a representation of the fusion algebra because the modular S-matrix diagonalizes fusion.

The boundary endpoint step is separate. A three-dimensional Reshetikhin– Turaev theory built from $\mathcal{C}$ supplies bulk topological lines labelled by $\mathcal{I}$. A choice of rational boundary condition supplies endpoint data turning those bulk lines into CFT defects. The formula (121.20) describes the Cardy diagonal case after this endpoint functor has been chosen. A non-diagonal modular invariant, a logarithmic theory, or an interacting continuum QFT without a known semisimple modular category requires different boundary and defect data; its defect category must be constructed from those data directly.

121.8 Anomaly As Obstruction To Condensation

An anomaly of a categorical symmetry is an obstruction to defining the defect networks independently of choices of triangulation, ordering, framing, or extension to one higher dimension. Algebraically it appears as failure of the pentagon, braiding, or trace identities in the proposed physical category. Geometrically it is encoded by an invertible theory in one higher dimension whose boundary is the original QFT with its defect category.

If the anomaly is cancelled by coupling to an invertible bulk, a defect network on the boundary is evaluated together with a bulk topological term. If the anomaly is trivialized by a local counterterm, the defect category can be modified by an equivalent associator or braiding datum.

Pointed finite mechanism. For ordinary finite invertible defects this obstruction is completely visible. Let G be a finite group and let D_g be invertible defects with

$$D_g \otimes D_h \simeq D_{gh}.$$

Choose junction isomorphisms

$$\mu_{g,h} : D_g \otimes D_h \to D_{gh}.$$

The two ways of identifying $((D_g \otimes D_h) \otimes D_k) \otimes D_l$ with D_{ghkl} can differ by a phase. Equivalently, the associator is a function

$$\alpha(g,h,k) \in U(1)$$

defined by

$$\mu_{gh,k} \circ (\mu_{g,h} \otimes \mathrm{id}_{D_k}) = \alpha(g,h,k)\, \mu_{g,hk} \circ (\mathrm{id}_{D_g} \otimes \mu_{h,k}). \tag{121.24}$$

Coherent evaluation of a four-fold junction requires the pentagon identity. Substituting (121.24) into the pentagon gives

$$\alpha(h,k,l)\, \alpha(g,hk,l)\, \alpha(g,h,k) = \alpha(gh,k,l)\, \alpha(g,h,kl). \tag{121.25}$$

This is precisely the inhomogeneous 3-cocycle condition $\delta\alpha = 1$. Thus the anomaly class of the pointed defect symmetry is

$$[\alpha] \in H^3(G, U(1))$$

for a two-dimensional QFT, or more generally the corresponding degree $D+1$ class for a D-dimensional finite symmetry defect network.

The dependence on the choice of junction basis is also explicit. Replacing

$$\mu_{g,h} \longmapsto \beta(g,h)\, \mu_{g,h}, \qquad \beta(g,h) \in U(1),$$

changes the associator by

$$\alpha(g,h,k) \longmapsto \alpha(g,h,k)\, \frac{\beta(g,h)\beta(gh,k)}{\beta(h,k)\beta(g,hk)}. \tag{121.26}$$

Hence only the cohomology class is invariant. A local counterterm or redefinition of junction normalization can remove the anomaly exactly when $[\alpha] = 0$. If the class is nonzero, the same local formulas define a relative defect theory: the missing phase is supplied by an invertible bulk whose boundary variation is the cocycle α. This finite pointed calculation is the model for the higher categorical obstruction, but it does not prove that a general interacting continuum QFT has finite junction spaces or a finite cohomological anomaly representative.

121.9 Beyond The Rational Example

The Ising example is a controlled rational model. The general interacting problem requires additional construction. Chapter 122 develops two further finite or algebraic examples: noninvertible defects obtained from gauging a finite normal subgroup at a self-dual point, and duality walls acting on electric-magnetic line-operator lattices. These examples still leave the genuinely hard continuum problem open: construct the defect category, junction operators, fusion rules, and anomaly or condensation data directly in interacting QFTs rather than importing them from a topological or rational model.

The next chapter continues this development for finite one-form symmetries. It writes the five-dimensional $\mathbb{Z}_N$ symmetry TQFT as an explicit cochain theory, and evaluates the finite higher-gauging condensation defect by an actual cochain sum. The point of separating the chapters is logical: this chapter defines the categorical action and its simplest rational example, while Chapter 123 constructs the higher-gauging mechanism that produces four-dimensional noninvertible condensation defects when the necessary continuum surface operators are available.

Open Problem 121.16 (Interacting categorical symmetries). Construct categorical-symmetry defect categories in interacting continuum QFTs with local algebras, reflection-positive state spaces, fusion of extended operators, and anomaly inflow, beyond finite topological models and rational two-dimensional examples.

Chapter 122

Duality Defects, Gauging, And Orbifold Data

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Gauging a finite symmetry is a finite path integral over background fields. A duality defect is an interface obtained from such a gauging operation, possibly followed by an equivalence between the gauged theory and another theory. The definitions in this chapter are formulated in terms of background-field groupoids, topological symmetry defects, and line-charge lattices. This keeps separate three pieces of data which are often compressed in the literature: the finite sum over bundles, the anomaly or counterterm choice, and the identification of the resulting theory with a theory already under study.

Hypothesis 122.1 (Finite gauging and duality-defect structure)**.** A finite gauging and duality-defect structure in dimension D is a tuple

$$\mathfrak{D}_{\mathcal{Q},G} = \big(\mathcal{Q}, G, \mathrm{Bun}_G, \alpha, \tau, \{D_g\}_{g\in G}, \mathcal{N}_G, \mathcal{N}_G^\dagger, \Phi, \Gamma_{\mathrm{line}}\big)$$

with the following entries.

(i) $\mathcal{Q}$ is a D-dimensional local QFT admitting a finite internal symmetry G, and $M \mapsto \mathrm{Bun}_G(M)$ is the finite groupoid of background G-bundles on collared spacetimes.

(ii) For every background $P \to M$, the background partition function $Z_{\mathcal{Q}}[M; P]$ and the corresponding correlators are defined as vectors in the anomaly line determined by the cohomology class

$$\alpha \in H^{D+1}(BG, U(1)).$$

(iii) τ is a chosen anomaly trivialization and degree-D Dijkgraaf–Witten counterterm. After τ is fixed, the finite groupoid sum over $\mathrm{Bun}_G(M)$ is an ordinary complex amplitude with a specified normalization.

(iv) D_g is the topological symmetry defect associated to $g \in G$, with junctions implementing $D_g \otimes D_h \simeq D_{gh}$ and with the orientation convention used below.

(v) $\mathcal{N}_G : \mathcal{Q} \to \mathcal{Q}/G$ and $\mathcal{N}_G^\dagger : \mathcal{Q}/G \to \mathcal{Q}$ are the gauging interface and its reverse, defined with the same groupoid normalization and anomaly trivialization τ.

(vi) $\Phi : \mathcal{Q}/G \to \mathcal{Q}'$ is optional equivalence data. When $\mathcal{Q}' = \mathcal{Q}$, the composite $\Phi \circ \mathcal{N}_G$ is the self-dual-gauging defect studied below.

(vii) In gauge-theory applications, Γ_{line} is the lattice or finite discriminant group of genuine line charges, equipped with its Dirac pairing and with the action induced by any proposed duality wall.

The finite algebraic conclusions in this chapter are consequences of this structure. The continuum construction of the interfaces and junctions is the problem isolated in Open Problem 122.10.

122.1 Finite Symmetry Backgrounds

Let G be the finite internal symmetry in Hypothesis 122.1. A background G-field on a spacetime M is a principal G-bundle

$$P \to M.$$

On a good cover $\{U_i\}$ it may be represented by locally constant transition functions

$$g_{ij} : U_i \cap U_j \longrightarrow G$$

obeying

$$g_{ij} g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

Two representatives related by functions $u_i : U_i \to G$,

$$g_{ij} \mapsto u_i g_{ij} u_j^{-1},$$

define isomorphic bundles. Equivalently, after choosing a triangulation, a background is a flat G-valued one-cocycle on edges, modulo vertex gauge transformations.

The QFT with background P has partition function

$$Z_{\mathcal{Q}}[M; P],$$

and correlation functions with insertions carry the same background field. The collection of bundles and their isomorphisms is the finite groupoid

$$\mathrm{Bun}_G(M).$$

For a representative P, the automorphism group

$$\mathrm{Aut}(P)$$

is the group of gauge transformations preserving P. On connected M and for the trivial bundle it is G, while for a general flat bundle it is the centralizer of the holonomy representation.

Definition 122.2 (Finite gauging groupoid sum)**.** Assume the anomaly-trivialization and counterterm entry τ in Hypothesis 122.1 has been fixed. The gauged partition function is the groupoid cardinality sum

$$Z_{\mathcal{Q}/G}(M) = \sum_{[P]\in\pi_0 \operatorname{Bun}_G(M)} \frac{Z_{\mathcal{Q}}[M;P]}{|\operatorname{Aut}(P)|}. \tag{122.1}$$

Different choices of an invertible finite gauge theory multiplying (122.1) are different normalization conventions. Once a convention is fixed, the same convention must be used for closed manifolds, manifolds with boundary, interfaces, and operator insertions.

The groupoid factor is part of the finite path integral. It is the finite analogue of dividing by the gauge group volume in a continuous gauge theory.

Remark 122.3 (Disjoint-union multiplicativity). With the groupoid convention (122.1), finite gauging is multiplicative under disjoint union:

$$Z_{\mathcal{Q}/G}(M_1 \sqcup M_2) = Z_{\mathcal{Q}/G}(M_1)Z_{\mathcal{Q}/G}(M_2),$$

provided the original background partition functions are multiplicative.

A principal G-bundle on $M_1 \sqcup M_2$ is a pair (P_1, P_2) with $P_i \to M_i$. Its automorphism group is the product

$$\operatorname{Aut}(P_1, P_2) = \operatorname{Aut}(P_1) \times \operatorname{Aut}(P_2).$$

Thus the groupoid sum factorizes:

$$\begin{aligned} Z_{\mathcal{Q}/G}(M_1 \sqcup M_2) &= \sum_{[P_1],[P_2]} \frac{Z_{\mathcal{Q}}[M_1;P_1]Z_{\mathcal{Q}}[M_2;P_2]}{|\operatorname{Aut}(P_1)|\,|\operatorname{Aut}(P_2)|} \\ &= Z_{\mathcal{Q}/G}(M_1)Z_{\mathcal{Q}/G}(M_2). \end{aligned}$$

122.2 Dijkgraaf–Witten Twists And Anomaly Data

A class

$$\omega \in H^D(BG, U(1))$$

defines an invertible topological action for a G-bundle $P \to M$. If Ω is a cocycle representative, the exponentiated action is

$$\exp(\mathrm{i}S_\omega(P)) = \exp\left(2\pi\mathrm{i}\int_M f_P^*\Omega\right),$$

where $f_P : M \to BG$ classifies P. On a triangulated manifold this is the evaluation of the finite group cocycle on the flat edge labels of P. The twisted gauging is

$$Z_{\mathcal{Q}/G,\omega}(M) = \sum_{[P]} \frac{Z_{\mathcal{Q}}[M;P]\exp(\mathrm{i}S_\omega(P))}{|\operatorname{Aut}(P)|}. \tag{122.2}$$

If $\mathcal{Q}$ has a G-anomaly

$$\alpha \in H^{D+1}(BG, U(1)),$$

then $Z_{\mathcal{Q}}[M;P]$ is naturally a vector in a one-dimensional anomaly line rather than an ordinary complex number. Gauging requires a trivialization of this anomaly line over the groupoid of G-bundles, or a $(D+1)$-dimensional invertible bulk whose boundary is $\mathcal{Q}$. A Dijkgraaf–Witten twist in degree D changes the gauged theory; it cancels an anomaly only through its coboundary when a cochain-level trivialization of the degree $D+1$ anomaly has been chosen.

122.3 The Gauging Interface

The interface entry of Hypothesis 122.1 is a codimension-one interface

$$\mathcal{N}_G : \mathcal{Q} \longrightarrow \mathcal{Q}/G$$

obtained by gauging G on one side of the interface. More concretely, if a spacetime is cut along a hypersurface Y, then on the gauged side one sums over G-bundles whose restriction to Y is allowed to vary, while on the ungauged side the boundary value is interpreted as a background for $\mathcal{Q}$. The reverse interface is denoted

$$\mathcal{N}_G^\dagger : \mathcal{Q}/G \longrightarrow \mathcal{Q}.$$

For every $g \in G$ let D_g be the invertible symmetry defect of $\mathcal{Q}$. It is supported on Y and imposes G-holonomy g across a small transverse interval. The orientation convention is

$$D_g \otimes D_h \simeq D_{gh}.$$

Slab fusion calculation. Assume the anomaly of the G-symmetry has been trivialized, so that the gauging interface is an honest defect rather than a section of an anomaly line, and use the groupoid normalization of Definition 122.2 for both interfaces. Then the fused defect on the ungauged side is

$$\mathcal{N}_G^\dagger \circ \mathcal{N}_G \simeq \bigoplus_{g\in G} D_g \tag{122.3}$$

as defects of $\mathcal{Q}$.

Fuse $\mathcal{N}_G$ with $\mathcal{N}_G^\dagger$ and separate the two interfaces by a thin slab $Y \times [0,\epsilon]$. Inside the slab the G-field is flat. Choose trivializations on the two boundary components of the slab. A flat bundle on an interval with these endpoint trivializations is classified by a single holonomy $g \in G$. In the sector with holonomy g, transporting an operator across the slab inserts the symmetry defect D_g on Y. Summing over the finite gauge field in the slab therefore produces the direct sum over all holonomies.

The only normalization issue is the volume of gauge transformations in the slab that are identity on the two boundary components. With the interface normalization induced from Definition 122.2, these factors cancel between the two sides of the cut, leaving one copy of each holonomy sector. Hence the fused interface is $\oplus_{g\in G} D_g$. Its scope is the finite groupoid-sum calculation after the gauging construction and normalization have been chosen; existence of the interface is part of the gauging construction itself.

The opposite fusion

$$\mathcal{N}_G \circ \mathcal{N}_G^\dagger$$

is a defect of the gauged theory. It is the algebra object generated by the quantum symmetry defects acting on twisted sectors.

Definition 122.4 (Regular algebra defect of a finite symmetry)**.** The regular algebra object associated to a finite anomaly-free symmetry G is

$$A_G = \bigoplus_{g \in G} D_g$$

with multiplication induced by the defect junctions

$$D_g \otimes D_h \longrightarrow D_{gh}$$

and unit $\mathbf{1} = D_e \hookrightarrow A_G$. As an object in the fusion category generated by the symmetry defects,

$$A_G \otimes A_G \simeq |G|\, A_G. \tag{122.4}$$

The displayed square follows directly from $D_g \otimes D_h \simeq D_{gh}$:

$$A_G \otimes A_G \simeq \bigoplus_{g,h \in G} D_{gh}.$$

For each fixed $k \in G$, the equation $gh = k$ has exactly $|G|$ solutions, one for each choice of g, with $h = g^{-1}k$. Therefore every simple summand D_k appears with multiplicity $|G|$, which gives $|G| A_G$.

122.4 Mechanism Map For Finite Gauging

The same finite gauging operation appears in four coordinate systems in this chapter. In the background-field coordinate it is the groupoid-cardinality sum (122.1). In the interface coordinate it is the wall $\mathcal{N}_G$ and its reverse, whose slab fusion gives the regular algebra A_G. In the local-operator coordinate it is the passage from punctured-disk holonomy g to conjugacy classes and centralizer invariants. In the symmetry-TFT coordinate it is a boundary-changing operation: the ungauged boundary remembers G-backgrounds, while the gauged boundary condenses the finite symmetry algebra and exposes the quantum symmetry of the gauged theory. These are not four independent principles; they are four readings of the same finite sum, after the anomaly trivialization and normalization have been chosen.

For a finite Abelian subgroup H, the comparison can be written without any continuum input. Let $V_H = \mathbb{C}[H]$ have basis e_h, interpreted as a finite slab holonomy state, and let the original symmetry defect D_a act by translation,

$$D_a e_h = e_{a+h}.$$

Then the regular algebra

$$A_H = \sum_{a \in H} D_a$$

satisfies

$$A_H e_h = \sum_{k \in H} e_k, \qquad A_H^2 = |H| A_H, \qquad \Pi_H := |H|^{-1} A_H, \qquad \Pi_H^2 = \Pi_H. \tag{122.5}$$

Thus the normalized slab sum is the orbit-sum projector onto the H-invariant part. This is the finite linear-algebra shadow of $\mathcal{N}_H^\dagger \mathcal{N}_H = A_H$: fusing the two interfaces on the ungauged side projects by summing over the original symmetry line.

The gauged side has the Fourier-dual description. Its ordinary quantum symmetry is the character group $\widehat{H}$. On a holonomy sector e_h, the character defect $\widehat{D}_\chi$ acts by multiplication by $\chi(h)$. The dual regular algebra therefore acts by

$$\widehat{A}_H e_h = \sum_{\chi \in \widehat{H}} \chi(h) e_h = \begin{cases} |H| e_0, & h = 0, \\ 0, & h \neq 0. \end{cases} \tag{122.6}$$

Equations (122.5) and (122.6) are the two boundary coordinates of the same interface pair. The first condenses the original H-symmetry defects; the second resolves the holonomy sectors by the dual character symmetry. A self-duality equivalence $\Phi : \mathcal{Q}/H \to \mathcal{Q}$, when it exists, identifies these two boundary descriptions as defects of one QFT. The noninvertibility conclusion then follows from the regular algebra. The existence of Φ, topological invariance of the interface, and reflection-positive state spaces are separate QFT inputs, not consequences of the finite Fourier calculation.

122.5 Orbifold Local Operators In Two Dimensions

For a two-dimensional QFT with finite internal symmetry G, a local operator of the gauged theory is described by a punctured disk carrying a flat G-bundle. Choose a small positively oriented circle around the insertion. Its holonomy is an element

$$g \in G.$$

Changing the trivialization at the base point conjugates g, so the topological sector is a conjugacy class $[g]$. Let

$$\mathcal{V}_g$$

be the local operator space in the g-twisted sector. Gauge transformations preserving the chosen holonomy form the centralizer

$$C_G(g) = \{h \in G \mid hg = gh\}.$$

Thus the local operator space of the gauged theory is

$$\mathcal{V}_{\mathcal{Q}/G} = \bigoplus_{[g] \subset G} \mathcal{V}_g^{C_G(g)}. \tag{122.7}$$

In a two-dimensional orbifold by a finite group G, an OPE coefficient between twisted sectors can receive a contribution from a flat bundle on a pair of pants only if, after choosing compatible base paths,

$$g_1 g_2 g_3^{-1} = e. \tag{122.8}$$

Equivalently, the outgoing holonomy is the product $g_3 = g_1 g_2$, up to simultaneous conjugation. The fundamental group of a pair of pants has generators represented by the three boundary circles with the relation

$$\gamma_1 \gamma_2 \gamma_3^{-1} = 1$$

for the convention in which γ_3 is the outgoing boundary. A flat G-bundle is a homomorphism from this fundamental group to G, modulo conjugation. Therefore the boundary holonomies

$$g_i = \mathrm{Hol}(\gamma_i)$$

must satisfy $g_1 g_2 g_3^{-1} = e$. Conversely any triple satisfying this relation defines a flat bundle on the pair of pants after choosing a trivialization at the base point. Changing the base trivialization conjugates all three holonomies simultaneously.

Formula (122.7) and the pair-of-pants monodromy condition (122.8) give the finite topological part of the orbifold OPE. The analytic OPE coefficients inside each allowed twisted sector are still QFT data; they are not determined by the group G alone.

122.6 Duality Defects From Self-Dual Gauging

A duality defect is a topological interface whose fusion with its reverse is the regular algebra defect A_G rather than a single copy of the invisible defect. The construction is especially transparent when a gauged theory is equivalent to the original theory.

Self-dual gauging and the resulting defect. The substantial input in this construction is the existence of a fully specified self-duality of the gauged theory. Let H be finite, assume that H acts internally on a QFT $\mathcal{Q}$ and can be gauged without an anomaly. Suppose there is an equivalence

$$\Phi : \mathcal{Q}/H \longrightarrow \mathcal{Q}$$

after all background choices, discrete torsion choices, and local counterterms have been fixed. Then the composite

$$D_H := \Phi \circ \mathcal{N}_H$$

is a topological defect of $\mathcal{Q}$ satisfying

$$D_H^\dagger \otimes D_H \simeq A_H = \bigoplus_{h \in H} D_h. \tag{122.9}$$

If H is nontrivial and the D_h are distinct simple invertible defects, then D_H is noninvertible and has quantum dimension $\sqrt{|H|}$.

This last conclusion is finite defect algebra once the self-duality equivalence has been supplied. The equivalence Φ is invertible as an interface, so

$$\Phi^\dagger \circ \Phi \simeq \mathbf{1}_{\mathcal{Q}/H}.$$

Therefore

$$D_H^\dagger \otimes D_H = \mathcal{N}_H^\dagger \circ \Phi^\dagger \circ \Phi \circ \mathcal{N}_H \simeq \mathcal{N}_H^\dagger \circ \mathcal{N}_H.$$

The slab fusion calculation (122.3) identifies the last defect with $\oplus_{h\in H} D_h = A_H$, proving (122.9).

If D_H were invertible, $D_H^\dagger \otimes D_H$ would be a single copy of the invisible defect. For nontrivial H with distinct simple D_h, the right hand side of (122.9) has $|H|$ simple summands, so D_H is not invertible.

Quantum dimension is multiplicative under fusion and additive under direct sum. Since each D_h is invertible, $d(D_h) = 1$. Hence

$$d(D_H)^2 = d(D_H^\dagger \otimes D_H) = \sum_{h \in H} d(D_h) = |H|.$$

Positivity gives $d(D_H) = \sqrt{|H|}$.

Example 122.5 (The Ising Kramers–Wannier defect)**.** At the critical Ising point the anomaly-free spin-flip symmetry is $H = \mathbb{Z}_2 = \{e, \eta\}$. The gauged theory is equivalent to the original critical Ising CFT after the Kramers–Wannier self-duality identification is chosen. The construction above gives

$$\mathcal{N}^\dagger \otimes \mathcal{N} \simeq \mathbf{1} \oplus \eta, \qquad d(\mathcal{N}) = \sqrt{2},$$

which is the categorical fusion rule used in Chapter 121. The statement is conditional on the self-duality equivalence, while the fusion calculation itself is the finite slab calculation above.

122.7 Normal-Subgroup Gauging

Let G be a finite internal symmetry of $\mathcal{Q}$, and let

$$H \triangleleft G$$

be a normal subgroup. Assume the H-symmetry is anomaly-free after the restriction of all anomaly and counterterm data has been fixed. Gauging H produces $\mathcal{Q}/H$.

If the G-action on $\mathcal{Q}$ is anomaly-compatible with the chosen H-gauging, then $\mathcal{Q}/H$ has a residual ordinary symmetry

$$G/H.$$

For $g \in G$, conjugation maps an H-bundle P to another H-bundle

$$P^g := P \times_{H,c_g} H, \qquad c_g(h) = ghg^{-1}.$$

This is well-defined because H is normal. The original G-symmetry identifies the H-background partition functions for P and P^g, up to the specified anomaly-compatible counterterm. Therefore g acts on the groupoid sum defining $\mathcal{Q}/H$. If $g \in H$, this action is a gauge transformation of the summed H-field and hence acts trivially on gauge-invariant observables of the gauged theory. The action therefore factors through G/H. Compatibility of the anomaly and counterterm choices ensures that the factorized action is associative rather than projective.

Example 122.6 (Normal-subgroup interface and a finite test case)**.** Let $G = S_3$ and let

$$H = A_3 = \{e, r, r^2\} \triangleleft S_3.$$

After anomaly-free H-gauging, the residual ordinary symmetry is

$$S_3/A_3 \simeq \mathbb{Z}_2.$$

The gauging interface obeys

$$D_h \otimes \mathcal{N}_H \simeq \mathcal{N}_H, \qquad h \in H,$$

because an H-defect ending on the gauged side can be absorbed into the summed H-bundle. At a self-dual point with an equivalence $\Phi : \mathcal{Q}/H \to \mathcal{Q}$, the defect

$$D_H = \Phi \circ \mathcal{N}_H$$

satisfies

$$D_H^\dagger \otimes D_H \simeq \mathbf{1} \oplus D_r \oplus D_{r^2}, \qquad d(D_H) = \sqrt{3}.$$

This example separates the finite algebra from the dynamical question of whether a particular QFT possesses the required self-duality equivalence.

122.8 Gauge-Theory Duality Walls

In gauge theories, duality walls can act on electric-magnetic line-charge lattices. Let

$$\Gamma$$

be a free abelian group of line charges equipped with an integral antisymmetric Dirac pairing

$$\langle\ ,\ \rangle : \Gamma \times \Gamma \to \mathbb{Z}.$$

A line-lattice duality wall is a topological wall whose induced map

$$M : \Gamma \longrightarrow \Gamma$$

is an automorphism preserving this pairing:

$$\langle Mx, My \rangle = \langle x, y \rangle \qquad x, y \in \Gamma. \tag{122.10}$$

If the global form of the gauge group or a discrete theta angle selects a proper set of genuine lines, the wall is a wall between the corresponding theories only when M maps the genuine line data on one side to the genuine line data on the other side. Boundary degrees of freedom may be required to cancel an anomaly mismatch.

Example 122.7 (Four-dimensional abelian line-lattice duality walls)**.** For four-dimensional $U(1)$ gauge theory with electric and magnetic line charges

$$\Gamma = \mathbb{Z}^2, \qquad x = (q_e, q_m),$$

take the Dirac pairing

$$\langle x, y \rangle = q_e q_m' - q_m q_e'.$$

The two line-lattice generators used here act on column vectors by

$$S(q_e, q_m) = (q_m, -q_e), \qquad T(q_e, q_m) = (q_e + q_m, q_m). \tag{122.11}$$

The S-wall exchanges Wilson and 't Hooft charges with the orientation shown in (122.11). The T-wall implements the Witten-effect shift of electric charge by magnetic charge.

Remark 122.8 (Pairing preservation for the S and T walls). The transformations in Example 122.7 preserve the Dirac pairing on Γ.

For $Sx = (q_m, -q_e)$ and $Sy = (q'_m, -q'_e)$,

$$\langle Sx, Sy\rangle = q_m(-q'_e) - (-q_e)q'_m = q_e q'_m - q_m q'_e = \langle x, y\rangle.$$

For $Tx = (q_e + q_m, q_m)$ and $Ty = (q'_e + q'_m, q'_m)$,

$$\langle Tx, Ty\rangle = (q_e + q_m)q'_m - q_m(q'_e + q'_m) = q_e q'_m - q_m q'_e = \langle x, y\rangle.$$

Thus both maps are automorphisms of the charge lattice with its antisymmetric Dirac pairing.

Example 122.9 (Finite center-sensitive line lattice)**.** For a theory whose center-sensitive electric and magnetic charges reduce to

$$\mathcal{C}_N = \mathbb{Z}_N^{\mathrm{e}} \oplus \mathbb{Z}_N^{\mathrm{m}},$$

the finite pairing is

$$B((e,m),(e',m')) = \frac{em' - me'}{N} \in \mathbb{Q}/\mathbb{Z}.$$

The reductions of S and T modulo N,

$$S(e,m) = (m, -e), \qquad T(e,m) = (e+m, m),$$

preserve B. More generally

$$T^p(e,m) = (e + pm, m)$$

preserves B and maps a magnetic condensate direction to the dyonic direction used in oblique confinement:

$$(0,1) \mapsto (p,1).$$

Thus the finite line-lattice action gives a precise algebraic bridge between duality walls and the oblique-confinement diagnostic of Chapter 116.

122.9 Logical Scope

The finite statements above are theorem-level statements about groupoids of finite bundles, topological symmetry defects, and line-charge lattices. To apply them to an interacting continuum QFT one must supply the QFT input: existence of the symmetry defects as genuine topological extended operators, reflection-positive state spaces with the required defect sectors, anomaly trivializations, and any claimed equivalence $\Phi : \mathcal{Q}/H \to \mathcal{Q}$. Once those data are supplied, the slab fusion $\mathcal{N}_H^\dagger \mathcal{N}_H = A_H$, the orbifold twisted-sector decomposition, and the line-lattice pairing preservation are finite consequences.

Open Problem 122.10 (Continuum gauging interfaces). Construct gauging interfaces and duality defects in interacting continuum QFTs from background-field sums, local operator sectors, defect junctions, anomaly trivializations, and reflection-positive state spaces. The construction should prove topological invariance of the interface, the existence and composition law of junction operators, and compatibility with the Hilbert spaces obtained by cutting spacetime along the defect.

Chapter 123

Higher-Group Symmetry And Symmetry TQFT

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Higher-form symmetries and ordinary symmetries may combine into a single background-field structure. Its gauge transformations and anomalies may be linked by Postnikov classes, so the combined object carries more structure than a direct product of groups. This chapter defines higher-group backgrounds and then explains the symmetry-TQFT organization of boundary conditions and gauging operations.

Hypothesis 123.1 (Higher-group symmetry presentation)**.** A D-dimensional QFT with finite two-group symmetry and symmetry-TQFT presentation is specified, for the purposes of this chapter, by the tuple

$$\mathfrak{S}_{\mathcal{Q},\mathbb{G}} = \big(\mathcal{Q}, \mathbb{G}, \mathcal{B}_{\mathbb{G}}, \mathcal{L}_{\mathrm{anom}}, \mathfrak{T}_{\mathrm{sym}}, \mathfrak{B}_{\mathcal{Q}}, \mathcal{D}_{\mathrm{top}}, \tau\big).$$

The entries are as follows.

(i) $\mathcal{Q}$ is a D-dimensional local QFT in the sense used in the earlier foundational volumes.

(ii) $\mathbb{G} = (G_0, A, \rho, \beta)$ is a finite two-group: G_0 is a finite zero-form symmetry group, A is a finite abelian one-form symmetry group, $\rho : G_0 \to \mathrm{Aut}(A)$ is the action on one-form charges, and $\beta \in H^3(BG_0, A_\rho)$ is the Postnikov class.

(iii) $\mathcal{B}_{\mathbb{G}}(M)$ is the finite higher groupoid of backgrounds on a collared D-manifold M. In a triangulated chart its objects are pairs (g, B) satisfying the twisted cocycle equation $\delta_g B = \beta(g)$, and its higher morphisms are the zero-form, one-form, and gauge-for-gauge transformations preserving this equation.

(iv) $\mathcal{L}_{\mathrm{anom}}$ is the anomaly-line functor over $\mathcal{B}_{\mathbb{G}}$, equivalently the boundary value of an invertible $(D+1)$-dimensional theory for the same background fields.

(v) $\mathfrak{T}_{\mathrm{sym}}$ is the $(D+1)$-dimensional symmetry TQFT whose bulk fields are the $\mathbb{G}$-backgrounds and whose boundary conditions include background-fixing and summing/gauging boundaries.

(vi) $\mathfrak{B}_{\mathcal{Q}}$ is the boundary condition of $\mathfrak{T}_{\mathrm{sym}}$ representing the QFT $\mathcal{Q}$. Coupling $\mathcal{Q}$ to a background means placing $\mathfrak{B}_{\mathcal{Q}}$ opposite a background boundary of $\mathfrak{T}_{\mathrm{sym}}$.

(vii) $\mathcal{D}_{\mathrm{top}}$ is the corresponding topological defect system in $\mathcal{Q}$: symmetry surfaces, walls, junction lines, endpoint operators, and their isotopy and Pachner-move identifications.

(viii) τ is any chosen anomaly trivialization or discrete counterterm needed to turn a proposed gauging or condensation operation into an ordinary complex-valued path integral.

All finite cochain calculations below are consequences of such a presentation after a triangulated model for $\mathcal{B}_{\mathbb{G}}$ has been chosen. The existence of $\mathfrak{T}_{\mathrm{sym}}$, $\mathfrak{B}_{\mathcal{Q}}$, and $\mathcal{D}_{\mathrm{top}}$ for an interacting continuum theory is a separate QFT construction problem; see Open Problem 123.7.

123.1 Two-Group Backgrounds

Let $\mathfrak{S}_{\mathcal{Q},\mathbb{G}}$ be a presentation as in Hypothesis 123.1. The two-group entry $\mathbb{G} = (G_0, A, \rho, \beta)$ consists of an ordinary finite internal symmetry G_0, an abelian one-form symmetry A, an action of G_0 on A, and a Postnikov class

$$\beta \in H^3(BG_0, A_\rho).$$

On a triangulated spacetime M, a background consists of a G_0-valued one-cocycle g and an A-valued two-cochain B obeying

$$\delta_g B = \beta(g).$$

Here δ_g is the coboundary twisted by the G_0-action on A. Thus B is not closed when the ordinary background g has nontrivial Postnikov obstruction.

123.2 Gauge Transformations

A G_0 gauge transformation changes g by a coboundary. The two-form background B must transform at the same time so that $\delta_g B = \beta(g)$ remains valid. A one-form gauge transformation with A-valued one-cochain λ acts by

$$B \mapsto B + \delta_g \lambda.$$

If $\beta \neq 0$, the ordinary and one-form transformations do not commute as independent operations on the background fields. The physical statement is that charged line operators and local operators transform in a single higher-group representation problem.

123.3 Anomaly Functionals

An anomaly of a two-group symmetry in D dimensions is an invertible $(D+1)$-dimensional action for (g, B). A basic term is

$$S_{\mathrm{anom}}[g, B] = 2\pi i \int_X \langle B \smile \gamma(g)\rangle,$$

where $\gamma(g)$ is a G_0-dependent cochain with values in the Pontryagin dual of A. Gauge invariance on closed X imposes a cocycle condition using $\delta_g B = \beta(g)$. If X has boundary, the boundary QFT partition function is a section of the anomaly line over background fields.

123.4 Symmetry TQFT

For the presentation in Hypothesis 123.1, the object $\mathfrak{T}_{\mathrm{sym}}$ packages background fields, gauging, and duality defects in a $(D+1)$-dimensional topological theory. The physical QFT $\mathcal{Q}$ is represented by the boundary condition

$$\mathfrak{B}_{\mathcal{Q}}$$

for the symmetry TQFT. Coupling to a background corresponds to ending the topological theory on a background boundary. Gauging corresponds to replacing that boundary condition by a summing boundary condition.

For an ordinary finite group G, the symmetry TQFT is finite G-gauge theory in one higher dimension. For higher groups, the bulk field space is the groupoid of pairs (g, B) satisfying the Postnikov constraint. The fusion of topological defects in the QFT is inherited from the fusion of topological boundary-changing operators in the bulk.

123.5 $\mathbb{Z}_N$ One-Form Symmetry TQFT

We now write the finite cochain model that will be used as the local model for one-form gauging and condensation defects. Let X be an oriented triangulated five-manifold. Let

$$b, c \in C^2(X, \mathbb{Z}_N)$$

be two $\mathbb{Z}_N$-valued two-cochains. Their gauge transformations are

$$b \mapsto b + \delta\lambda, \qquad c \mapsto c + \delta\kappa, \qquad \lambda, \kappa \in C^1(X, \mathbb{Z}_N).$$

The finite Heisenberg, or BF, symmetry TQFT has exponentiated action

$$Z_H(X; b, c) = \exp\left(\frac{2\pi i}{N}\int_X b \smile \delta c\right). \tag{123.1}$$

The integral means evaluation of the resulting $\mathbb{Z}_N$-valued five-cochain on the fundamental class and then embedding $\mathbb{Z}_N \hookrightarrow \mathbb{R}/\mathbb{Z}$. Equivalently one chooses integer lifts and keeps the result modulo N; changing the lift changes the exponent by an integral multiple of $2\pi i$.

Remark 123.2 (Closed-manifold gauge invariance). If X is closed, (123.1) is invariant under the gauge transformations above.

The transformation $c \mapsto c + \delta\kappa$ does not change δc, because $\delta^2 = 0$. Under $b \mapsto b + \delta\lambda$, the exponent changes by

$$\frac{2\pi i}{N} \int_X \delta\lambda \smile \delta c.$$

Since $\deg \lambda = 1$,

$$\delta(\lambda \smile \delta c) = \delta\lambda \smile \delta c - \lambda \smile \delta^2 c = \delta\lambda \smile \delta c.$$

Thus the change is the integral of a coboundary. Evaluation of a coboundary on the fundamental cycle of a closed triangulated manifold is zero modulo N. The exponent is therefore unchanged.

If X has boundary $M = \partial X$, the last computation leaves the boundary term

$$\frac{2\pi i}{N} \int_M \lambda \smile \delta c.$$

A boundary condition must therefore specify which combination of b and c is fixed as a background and which is summed as a dynamical higher gauge field. A four-dimensional QFT $\mathcal{Q}$ with a $\mathbb{Z}_N$ one-form symmetry is a boundary condition for this five-dimensional theory after the choice of background boundary has been made. Gauging the one-form symmetry is the operation of replacing the boundary condition that fixes b by the boundary condition that sums over b and fixes the dual background c.

Example 123.3 (Four-dimensional $\mathbb{Z}_N$ line data)**.** In a four-dimensional theory whose electric and magnetic one-form backgrounds are both finite $\mathbb{Z}_N$ backgrounds, a line charge may be labelled abstractly by

$$(e, m) \in \mathbb{Z}_N \oplus \mathbb{Z}_N.$$

The pairing controlling mutual locality is the finite Dirac pairing

$$\langle (e, m), (e', m') \rangle = em' - me' \quad \in \mathbb{Z}_N.$$

A set of simultaneously genuine line operators is a subgroup on which this pairing vanishes. The five-dimensional action (123.1) packages the two one-form backgrounds and their finite symplectic pairing. The wall that exchanges b and c acts on line charges by

$$(e, m) \mapsto (m, -e),$$

the finite analogue of the S-wall discussed in Chapter 122. This statement is about the line-charge lattice and its mutual-locality pairing; it is not an assertion that arbitrary closed loops in four Euclidean dimensions have a topological pairwise braiding invariant.

123.6 Condensation And Gauging

Let $\mathcal{C}$ be the category or higher category of topological defects describing a finite symmetry. Gauging a sub-symmetry is condensation of a commutative algebra object

$$A_{\text{cond}} \in \mathcal{C}.$$

The gauged theory has boundary condition obtained by allowing endpoints of the condensed defects. If the algebra object is obstructed by an anomaly, the condensation exists only after coupling to an appropriate bulk or after choosing a trivialization of the anomaly.

Definition 123.4 (Triangulated higher-gauging defect)**.** Let $\mathcal{Q}$ be a four-dimensional QFT with an anomaly-free finite $\mathbb{Z}_N$ one-form symmetry. Assume that for every closed oriented surface Σ there are topological symmetry surfaces

$$U_n(\Sigma), \qquad n \in \mathbb{Z}_N,$$

with junctions implementing $U_nU_m \simeq U_{n+m}$, and assume these surface networks are invariant under triangulation moves. For a triangulated oriented three-manifold Y supporting a codimension-one defect, define the groupoid-normalized condensation insertion

$$\mathfrak{C}(Y) = \frac{|C^0(Y,\mathbb{Z}_N)|}{|C^1(Y,\mathbb{Z}_N)|} \sum_{b\in Z^2(Y,\mathbb{Z}_N)} \mathsf{U}_b(Y). \tag{123.2}$$

Here b is a closed two-cochain on Y. The symbol $\mathsf{U}_b(Y)$ denotes the topological wall network obtained from the Poincare-dual one-cycle of b: locally one thickens each dual edge in a small normal collar of Y and inserts the corresponding one-form symmetry surface segment. The condition $\delta b = 0$ is the conservation law at the vertices of the dual one-cycle, equivalently the absence of an uncancelled endpoint for the wall network. The factor $|C^0|/|C^1|$ is the finite groupoid volume of the two-form gauge parameter $C^1(Y,\mathbb{Z}_N)$ together with its degree-zero gauge-for-gauge redundancy. Omitting the $|C^0|$ factor gives a cellularly dependent representative; changing to a reduced convention divides by the constant factor $|H^0(Y,\mathbb{Z}_N)|$ on each connected component and changes the defect by an invertible normalization convention rather than by a dynamical QFT operation.

Finite two-form gauge groupoid behind the normalization. The coefficient in (123.2) is the homotopy cardinality of the finite two-form gauge groupoid, written before quotienting the field variables. Let $\mathcal{Z}^2_N(Y)$ be the finite 2-groupoid whose objects are closed two-cochains

$$b \in Z^2(Y,\mathbb{Z}_N),$$

whose 1-morphisms are one-cochains $\lambda \in C^1(Y,\mathbb{Z}_N)$ acting by

$$b \longmapsto b + \delta\lambda,$$

and whose 2-morphisms are zero-cochains $\xi \in C^0(Y,\mathbb{Z}_N)$ acting on gauge parameters by

$$\lambda \longmapsto \lambda + \delta\xi.$$

For a closed connected component this is the finite model of a $\mathbb{Z}_N$ two-form gauge field on Y. The homotopy cardinality of a finite 2-groupoid is

$$\sum_{[b]\in\pi_0} \frac{|\pi_2(\mathcal{Z}^2_N(Y),b)|}{|\pi_1(\mathcal{Z}^2_N(Y),b)|}.$$

Here

$$\pi_0 \simeq H^2(Y,\mathbb{Z}_N), \qquad \pi_1 \simeq H^1(Y,\mathbb{Z}_N), \qquad \pi_2 \simeq H^0(Y,\mathbb{Z}_N),$$

so

$$|\mathcal{Z}^2_N(Y)| = \frac{|H^0(Y,\mathbb{Z}_N)|\,|H^2(Y,\mathbb{Z}_N)|}{|H^1(Y,\mathbb{Z}_N)|}. \tag{123.3}$$

Equivalently, if one sums over the unquotiented set of closed cochains, the same cardinality is

$$|\mathcal{Z}^2_N(Y)| = \frac{|C^0(Y,\mathbb{Z}_N)|\,|Z^2(Y,\mathbb{Z}_N)|}{|C^1(Y,\mathbb{Z}_N)|}. \tag{123.4}$$

This is exactly the normalization used in (123.2): the summand $\mathsf{U}_b(Y)$ is written for every closed cochain b, while the factor $|C^0|/|C^1|$ divides by one-form gauge redundancy and restores the zero-form gauge-for-gauge volume. Thus the normalization is not an optional cellular convention; it is the finite path-integral measure for the two-form gauge field.

Theorem 123.5 (Finite higher gauging and condensation fusion)**.** *Under the hypotheses of Definition 123.4, the insertion $\mathfrak{C}(Y)$ is a topological codimension-one defect implementing finite $\mathbb{Z}_N$ one-form gauging on the hypersurface Y. The fusion of two parallel copies is*

$$\mathfrak{C}(Y)\circ\mathfrak{C}(Y) = Z^{(2)}_{\mathbb{Z}_N}(Y)\,\mathfrak{C}(Y), \qquad Z^{(2)}_{\mathbb{Z}_N}(Y) = \frac{|C^0(Y,\mathbb{Z}_N)|\,|Z^2(Y,\mathbb{Z}_N)|}{|C^1(Y,\mathbb{Z}_N)|}. \tag{123.5}$$

Equivalently,

$$Z^{(2)}_{\mathbb{Z}_N}(Y) = \frac{|H^0(Y,\mathbb{Z}_N)|\,|H^2(Y,\mathbb{Z}_N)|}{|H^1(Y,\mathbb{Z}_N)|}, \tag{123.6}$$

so the coefficient is the closed-three-manifold partition value of the finite three-dimensional $\mathbb{Z}_N$ two-form gauge theory in this normalization. With corners or boundaries the same state sum assigns the corresponding Hilbert spaces and operators; hence the fusion coefficient of condensation defects is, in general, a three-dimensional topological theory rather than an integer.

If the half-space gauged theory is identified with $\mathcal{Q}$ after all discrete theta terms and background boundary conditions are fixed, the half-space gauging interface is a duality defect of $\mathcal{Q}$. For $N>1$ it is noninvertible whenever the associated two-form gauge theory has non-one-dimensional state spaces, for example on a connected closed spatial surface where the flux label in $H^2(\Sigma,\mathbb{Z}_N)$ survives.

Proof. Topological invariance of $\mathfrak{C}(Y)$ has two logically distinct inputs. The QFT input is that the surface networks $U_n(\Sigma)$, their junctions, and their Pachner-move identifications have already been constructed as anomaly-free topological operators. The finite gauge-theory input is the groupoid measure (123.4). The cochain condition $\delta b = 0$ is exactly the junction conservation law for the Poincare-dual network, so a closed b produces a network without residual endpoints. A change $b \mapsto b+\delta\lambda$ inserts the boundary of a one-cochain of symmetry surfaces; by the assumed topological junction relations this is a gauge-equivalent representative of the same condensed configuration. A change $\lambda \mapsto \lambda + \delta\xi$ is the gauge-for-gauge redundancy of that boundary filling. Therefore the sum in (123.2) is precisely the unquotiented expression for the finite path integral over $\mathcal{Z}^2_N(Y)$. In a collar $(-\epsilon,\epsilon)\times Y$, replacing the fixed one-form background by this groupoid sum is the finite gauging operation on the hypersurface. This proves the gauging interpretation after the QFT surface-operator input has been supplied.

Place two copies of the defect on nearby parallel copies of Y. Their product is

$$\left(\frac{|C^0(Y,\mathbb{Z}_N)|}{|C^1(Y,\mathbb{Z}_N)|}\right)^2 \sum_{b,b'\in Z^2(Y,\mathbb{Z}_N)} \mathsf{U}_b(Y)\,\mathsf{U}_{b'}(Y).$$

Using the junction isomorphism $U_n U_m \simeq U_{n+m}$, change variables to

$$s = b + b', \qquad t = b.$$

For each closed s, the variable t runs freely over $Z^2(Y, \mathbb{Z}_N)$. The product therefore becomes

$$\frac{|C^0|\,|Z^2|}{|C^1|}\left(\frac{|C^0|}{|C^1|}\sum_{s\in Z^2(Y,\mathbb{Z}_N)} \mathsf{U}_s(Y)\right),$$

where all cochain groups are on Y. The second parenthesis is $\mathfrak{C}(Y)$, and the first factor is (123.5). It remains to see that this factor is topological. From the short exact sequences

$$0 \to Z^1 \to C^1 \to B^2 \to 0, \qquad 0 \to B^1 \to Z^1 \to H^1 \to 0, \qquad 0 \to H^0 \to C^0 \to B^1 \to 0,$$

and

$$0 \to B^2 \to Z^2 \to H^2 \to 0,$$

finite cardinalities multiply, giving

$$\frac{|C^0|\,|Z^2|}{|C^1|} = \frac{|H^0|\,|H^2|}{|H^1|}.$$

This proves (123.6). The extension to manifolds with boundary is the same finite gauge-theory state sum with boundary cochains left unsummed or summed according to the chosen boundary condition; it assigns vector spaces and linear maps, with closed three-manifold numbers as the special case. If half-space gauging lands back on $\mathcal{Q}$, the interface has the same theory on both sides and is therefore an internal duality defect. Its product with the opposite interface contains the nontrivial two-form gauge-theory summation just computed, so it is invertible only when that finite topological theory is the trivial invertible one. $\square$

Relative state space of the finite factor. The last sentence of the theorem has concrete content. Let Σ be a closed oriented triangulated surface obtained by cutting the three-dimensional defect worldvolume. A boundary value of the finite two-form gauge field is a closed two-cochain

$$z \in Z^2(\Sigma, \mathbb{Z}_N).$$

Since $\dim \Sigma = 2$, this is the same set as $C^2(\Sigma, \mathbb{Z}_N)$. The boundary one-form gauge transformations identify

$$z \sim z + \delta\lambda, \qquad \lambda \in C^1(\Sigma, \mathbb{Z}_N).$$

Thus the flux sectors on the cut surface are

$$[z] \in H^2(\Sigma, \mathbb{Z}_N), \qquad \mathcal{H}_N^{(2)}(\Sigma) = \mathbb{C}\big[H^2(\Sigma, \mathbb{Z}_N)\big], \tag{123.7}$$

with the automorphism groups H^0 and H^1 entering the gluing pairing through the same homotopy cardinality weights as in (123.3). More explicitly, a cobordism $W : \Sigma_0 \to \Sigma_1$ gives a matrix whose $([z_1], [z_0])$-entry is the homotopy cardinality of the finite groupoid of closed two-cochains $b \in Z^2(W, \mathbb{Z}_N)$ restricting to the prescribed boundary classes, with one-cochain gauge transformations and zero-cochain gauge-for-gauge transformations fixed on the boundary. For the cylinder $\Sigma \times$

$[0,1]$, restriction to the two ends forces $[z_0]=[z_1]$, so the operator is diagonal on the basis (123.7) up to the conventional automorphism scalar fixed by the gluing normalization.

This relative description is what the closed-manifold number $Z^{(2)}_{\mathbb{Z}_N}(Y)$ becomes when a cut is present. On a connected closed oriented surface,

$$H^2(\Sigma,\mathbb{Z}_N)\simeq\mathbb{Z}_N,$$

so the finite factor has an N-dimensional flux-sector space for $N>1$. Consequently fusion with the reverse gauging interface contains actual finite topological degrees of freedom. It is an invertible scalar line only when the relevant finite state spaces are all one-dimensional.

For Yang–Mills theories this construction becomes a QFT statement only after the $\mathbb{Z}_N$ one-form symmetry surfaces, their junctions, their correlation topology, and their anomaly trivialization have been constructed inside the theory or by a regulator such as the lattice. Once that input is available, the cochain calculation above is the local algebraic mechanism behind higher gauging and the resulting noninvertible condensation defects.

123.7 Chiral Defects In Abelian Gauge Theory

We now give the explicit abelian construction. Work on an oriented spin four-manifold and take compact unit-charge massless QED with gauge field A, curvature $F=\mathrm{d}A$, and flux normalization

$$\int_\Sigma \frac{F}{2\pi}\in\mathbb{Z}$$

for every closed two-cycle Σ. Let j_5 be the axial current in the normalization of Chapter 51: the axial rotation by angle α acts on the Dirac field by

$$\psi\mapsto\exp(\mathrm{i}\alpha\gamma_5/2)\psi,$$

and, in the massless theory,

$$\mathrm{d}\star j_5=\frac{1}{4\pi^2}F\wedge F. \tag{123.8}$$

This is the differential-form version of (51.6) with $e=q=1$. The magnetic one-form symmetry has current

$$j_m=\frac{1}{2\pi}\star F,\qquad \mathrm{d}\star j_m=\frac{1}{2\pi}\mathrm{d}F=0.$$

It is part of the QFT definition only when no dynamical monopole operator has been included; an inserted or dynamical monopole is an endpoint for the charged ’t Hooft line and explicitly breaks this symmetry.

For a closed oriented three-manifold M supporting a defect, the naive axial wall

$$U_\alpha(M)=\exp\left(\frac{\mathrm{i}\alpha}{2}\int_M\star j_5\right)$$

is not topological. If M is moved across a four-chain W, (123.8) changes it by

$$\exp\left(\frac{\mathrm{i}\alpha}{8\pi^2}\int_W F\wedge F\right).$$

It is also not a well-defined compact-gauge-field operator after replacing $\star j_5$ by the formally conserved Chern–Simons-improved current, because the required fractional Chern–Simons level is not invariant under large gauge transformations on a general compact M.

123.7.1 The Fractional Hall Dressing

Let $N > 0$, let $p \in \mathbb{Z}$ with $\gcd(p, N) = 1$, and put

$$\alpha = \frac{2\pi p}{N}.$$

Let $\mathcal{A}_{N,p}$ denote the minimal abelian three-dimensional TQFT with a $\mathbb{Z}_N$ one-form symmetry whose generating line ℓ has spins

$$h(\ell^s) = \frac{ps^2}{2N} \quad \mathrm{mod}\ 1.$$

The spin refinement of $\mathcal{A}_{N,p}$ is included when the pair (N, p) requires it; the spin structure on M is inherited from the ambient spin four-manifold. Equivalently, coupling this one-form symmetry to a $\mathbb{Z}_N$-valued two-cocycle B has four-dimensional inflow

$$\exp\left(-\frac{2\pi \mathrm{i} p}{2N}\int_Y P(B)\right),$$

on a four-manifold Y extending the background; P is the Pontryagin-square cohomology operation. When the integral curvature class of $A|_M$ is reduced modulo N, it defines

$$B_A = \left[\frac{F}{2\pi}\right] \mathrm{mod}\ N \in H^2(M, \mathbb{Z}_N).$$

The dressed chiral defect is

$$D_{N,p}(M) = \exp\left(\frac{\pi \mathrm{i} p}{N}\int_M \star j_5\right) Z_{\mathcal{A}_{N,p}}(M; B_A). \tag{123.9}$$

For $p = 1$, the minimal theory is the $U(1)_N$ Chern–Simons theory, and (123.9) becomes the more concrete path integral

$$D_{N,1}(M) = \int \mathcal{D}a \, \exp\left[\frac{\pi \mathrm{i}}{N}\int_M \star j_5 + \mathrm{i}\int_M \left(\frac{N}{4\pi} a \wedge \mathrm{d}a + \frac{1}{2\pi} a \wedge \mathrm{d}A\right)\right], \tag{123.10}$$

where a is a compact $U(1)$ gauge field living only on the defect worldvolume. The field a is not a new bulk particle; it is part of the operator insertion.

Topological cancellation for $D_{N,1}$. In the massless compact QED path integral without dynamical monopoles, the operator (123.10) is invariant under smooth deformations of M that avoid charged insertions.

Let M_0 and M_1 be two homologous placements of the defect and let W be the oriented four-chain swept out by the deformation. The axial factor in (123.10) changes by

$$\exp\left(\frac{\pi \mathrm{i}}{N}\int_W \mathrm{d} \star j_5\right) = \exp\left(\frac{\mathrm{i}}{4\pi N}\int_W F \wedge F\right).$$

The $U(1)_N$ defect theory is gauge invariant as a compact Chern–Simons theory. To compute its local dependence on the ambient field A, one may integrate out a at the level of the response action: the equation of motion is $N\mathrm{d}a + \mathrm{d}A = 0$, so locally

$$\frac{N}{4\pi} a \wedge \mathrm{d}a + \frac{1}{2\pi} a \wedge \mathrm{d}A \quad \rightsquigarrow \quad -\frac{1}{4\pi N} A \wedge \mathrm{d}A.$$

Therefore the change of the Hall response under the same deformation is

$$\exp\left(-\frac{\mathrm{i}}{4\pi N}\int_W F\wedge F\right),$$

which cancels the axial-anomaly phase. The argument uses only the local response to a deformation; the compact a-path integral in (123.10) is what makes the operator globally well-defined under large gauge transformations.

For general p/N, the same cancellation is expressed by the one-form anomaly of $\mathcal{A}_{N,p}$. The Pontryagin-square response of $\mathcal{A}_{N,p}(B_A)$ is the compact, gauge-invariant replacement for the fractional Chern–Simons counterterm whose differential-form shadow is

$$-\frac{\mathrm{i}p}{4\pi N}\int A\wedge \mathrm{d}A.$$

Its deformation phase cancels the axial phase from (123.8). Thus (123.9) is the compact topological defect associated with the rational axial angle $2\pi p/N$.

123.7.2 Half-Space Gauging Proof

There is a second construction that explains why the defect is tied to the magnetic one-form symmetry. Let $X_+ = \{x \geq 0\}$ and let $M = \partial X_+$. To gauge the $\mathbb{Z}_N$ subgroup of the magnetic $U(1)^{(1)}$ symmetry in X_+, introduce a compact two-form gauge field b and a compact one-form gauge field c on X_+. For $k\in\mathbb{Z}$ satisfying

$$pk \equiv 1 \pmod N,$$

add

$$S_{\mathrm{mag},N,k} = \mathrm{i}\int_{X_+}\left(\frac{1}{2\pi}b\wedge F + \frac{N}{2\pi}b\wedge \mathrm{d}c + \frac{Nk}{4\pi}b\wedge b\right). \tag{123.11}$$

The c-equation imposes the flatness condition that makes b a $\mathbb{Z}_N$ two-form gauge field after the finite gauge quotient. The last term is the discrete-torsion choice. In the smooth sector, completing the square gives the theta-angle shift

$$\Delta S_\theta = -\frac{\mathrm{i}p}{4\pi N}\int_{X_+} F\wedge F,$$

with the integral quantization fixed by the compact b, c path integral. The massless axial change of variables with

$$\alpha(x) = \frac{2\pi p}{N}\Theta(x)$$

contributes the inverse phase and a boundary insertion

$$\exp\left(\frac{\pi \mathrm{i}p}{N}\int_M \star j_5\right).$$

After imposing the topological boundary condition $b|_M = 0$, the remaining boundary theory produced by the b, c path integral is precisely $\mathcal{A}_{N,p}(B_A)$. The interface between the ungauged side and the half-space-gauged side is therefore (123.9). Since the operation is gauging on a half-space followed by a change of variables, the interface is topological as long as the magnetic one-form symmetry is unbroken.

123.7.3 Fusion And Action On Operators

The defect acts on local fermion operators by the axial rotation

$$\psi \mapsto \exp(\pi \mathrm{i} p \gamma_5 / N)\psi.$$

Wilson lines are not charged under the magnetic one-form symmetry, so the half-space gauging part leaves them unchanged. A minimal 't Hooft line $H(\gamma)$, in the sense of Definition 115.2, is charged. If $\gamma = \partial\Sigma$ while the defect is swept past the line, gauge invariance attaches the surface factor

$$H(\gamma) \longmapsto H(\gamma)\, \exp\left(\frac{\mathrm{i}p}{N}\int_\Sigma F\right), \tag{123.12}$$

with Σ stretching between the line and the displaced defect. This is the operational reason the defect is not an ordinary axial symmetry operator: its action on extended operators remembers the magnetic one-form symmetry.

The orientation-reverse fusion already shows noninvertibility. For $p = 1$,

$$D_{N,1}(M)D^\dagger_{N,1}(M) = \int \mathcal{D}a\, \mathcal{D}\bar{a}\, \exp\left[\mathrm{i}\int_M \left(\frac{N}{4\pi} a \wedge \mathrm{d}a - \frac{N}{4\pi}\bar{a} \wedge \mathrm{d}\bar{a} + \frac{1}{2\pi}(a - \bar{a}) \wedge \mathrm{d}A\right)\right]. \tag{123.13}$$

The axial factors cancel, but the Hall theories do not cancel to the trivial three-dimensional theory. Equation (123.13) is the magnetic $\mathbb{Z}_N$ condensation operator. In the finite cochain language used above, its general p analogue is a sum over $\mathbb{Z}_N$-valued two-cocycles on M, weighted by the Pontryagin-square phase and by the corresponding magnetic one-form symmetry surface network. Thus the fusion coefficient is a three-dimensional TQFT or finite state sum, not a number.

Controlled Approximation 123.6 (QED noninvertible chiral defects)**.** In compact unit-charge massless QED on oriented spin four-manifolds, with no dynamical monopoles and with the magnetic one-form symmetry retained in the QFT definition, every rational axial angle $2\pi p/N$ with $\gcd(p, N) = 1$ defines the topological defect $D_{N,p}$ in (123.9). The defect acts on local fermions by the corresponding axial rotation, acts on 't Hooft lines by (123.12), and is noninvertible because $D_{N,p}D^\dagger_{N,p}$ is a magnetic condensation defect rather than the invisible defect.

This statement is the compact path-integral construction just given. A fully axiomatic theorem would still require constructing compact QED, its charged line sectors, and the defect Hilbert spaces in a nonperturbative reconstruction framework. If dynamical monopoles are added, the magnetic one-form symmetry has endpoints and the half-space gauging step is no longer topological; the defect then fails to define an exact symmetry.

Let a more general four-dimensional abelian gauge theory contain massless fermions with a classical chiral transformation R_α. Suppose the ABJ anomaly changes the action by

$$\Delta S_\alpha = \frac{i\alpha q}{8\pi^2}\int_M F \wedge F,$$

where F is the $U(1)$ field strength and $q \in \mathbb{Z}$ is the anomaly coefficient in the chosen charge normalization. The codimension-one defect implementing R_α is topological only if this phase is cancelled or trivialized as the defect crosses insertions. In the absence of dynamical monopoles, the theory also has a magnetic one-form symmetry, and the anomaly phase can be represented by coupling the defect worldvolume to a three-dimensional topological theory whose anyons attach to magnetic background flux sectors. The resulting dressed defect is generally noninvertible: fusing it with its orientation reverse leaves a sum or a three-dimensional topological factor rather than the invisible defect alone.

123.8 Higher Fusion Categories And Truncations

The categorical level depends on which defects are retained. In a four-dimensional QFT, the full wall-junction system includes walls, junction surfaces, junction lines, and point junctions. It is therefore at least a three-categorical defect object. A fusion two-category appears only after a controlled truncation, for example by focusing on topological surfaces and their line and point junctions, or by studying the three-dimensional theory living on a wall. A finite semisimple fusion two-category has objects, one-morphisms, and two-morphisms, a monoidal product, duals, and finite semisimple Hom categories. Condensation is then formulated by separable algebra objects and their module two-categories.

This distinction matters in four-dimensional generalized symmetry. A one-form symmetry generated by topological surfaces is not fully described by a one-category of simple surfaces if its line junctions carry nontrivial operator spaces. The symmetry TQFT keeps this extra information because bulk boundary-changing defects, their junctions, and their endpoint operators live one categorical level higher than the ordinary group law. When a chapter later uses only a fusion ring of surfaces, the omission is a truncation and not a definition of the full symmetry object.

123.9 Example: Mixed Zero- And One-Form Data

Consider a four-dimensional gauge theory with gauge group $SU(N)$ and a charge conjugation symmetry acting on the center one-form symmetry

$$A = \mathbb{Z}_N$$

by inversion. A background for the ordinary symmetry changes the local system in which the A-valued two-form background lives. If a theta-angle or discrete theta term changes under charge conjugation, the anomaly is a five-dimensional functional of the combined background. The line-operator lattice then transforms as a module over the two-group rather than as a separate representation of each symmetry.

Comparison Summary

The symmetry-TQFT description is not a replacement for constructing the QFT. It records:

$$\begin{array}{rl} \text{allowed background fields,} & \text{anomaly lines,} \\ \text{gauging and condensation operations,} & \text{topological defect fusion.} \end{array}$$

Local correlators, Hilbert space, stress tensor, and dynamical phase structure must still be supplied by the QFT itself.

Open Problem 123.7 (Higher-group examples). Construct interacting continuum QFT examples whose higher-group backgrounds, symmetry TQFT, defect fusion, anomaly inflow, and local-operator action are all derived from a single regulator or reconstruction framework.

Part X

Thermal Quantum Field Theory, Hydrodynamics, and Nonequilibrium Dynamics

Chapter 124

KMS States and Thermal Correlators

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Thermal equilibrium is a property of a state together with a chosen time evolution. The KMS condition expresses this property as an analytic boundary condition on correlation functions. It agrees with the Gibbs trace in a finite system, but it does not require the trace formula to exist. This is why it is the correct starting point for infinite-volume thermal QFT, where the partition function per se is not an observable.

Statistical mechanics as a QFT absorption route. The statistical-mechanics material in Volumes X and XI is used as input for QFT only after it has been translated into observables, regulators, limits, and framework data. The finite Gibbs trace below is therefore the first rung of a route, not the template for every later thermal field theory. The route has five gates.

First, equilibrium data enter through a C^*-dynamical system and a KMS state. A finite Hamiltonian calculation may verify the strip boundary condition, detailed balance, and fluctuation–dissipation signs, but the QFT statement is the existence of a state on an infinite-volume observable algebra with a declared time evolution. This is the layer developed in the present chapter and used by the Kubo analysis of Chapter 127.

Second, transport and hydrodynamics use equilibrium correlation functions only after the order of limits and the retained slow variables are declared. Kubo formulae, the Ward-identity hydrodynamic family of Chapter 128, the Schwinger–Keldysh hydrodynamic effective theory of Chapter 129, and the long-time-tail analysis of Chapter 134 are linked by this datum: conserved densities and sources define the observable map, while the low-frequency and long-wavelength limits remain extra hypotheses or theorems in a specified model.

Third, nonequilibrium statistical mechanics contributes finite probability identities and representation changes before it contributes continuum QFT. The proof architecture of Chapter 133 separates path-measure fluctuation relations, MSRJD and Doi–Peliti rewritings, tilted generators, empirical-flow costs, GKSL limits, and Schwinger–Keldysh influence kernels from the later scaling claim that produces a stochastic or open-system field theory. Dynamic-critical labels such as Models A–H live at that scaling layer until a model-specific renormalization and observable limit have been supplied.

Fourth, finite-locality and topological laboratories test mechanisms without settling continuum existence. Lieb–Robinson bounds, quasi-adiabatic continuation, flux-torus Hall responses, toric-code sectors, and finite gauge or TQFT boundary models give controlled finite-regulator identities for locality, stability, and generalized-symmetry claims. Their role in the QFT spine is to expose which hypotheses a continuum theorem must retain, as in the lattice and local-QFT routes of Chapters 138, 140, and 143.

Fifth, the constructive and numerical material of Volume XI asks whether the regulated families actually produce QFT data. Cluster expansions, rigorous RG, stochastic quantization, continuum scaling windows, Monte Carlo evidence, transfer matrices, Hamiltonian truncation, tensor networks, and neural variational states are admissible coordinates only when accompanied by regulator families, renormalized observables, error budgets, and an OS/Wightman/AQFT reconstruction or comparison target, as organized in Chapters 136, 139, 142, 144, and 145.

This route prevents a common compression. A finite lattice identity, a stochastic action, a transfer-matrix spectrum, or a tensor-network fit is not yet a continuum QFT theorem. To be used at that level it must name the observable map, the thermodynamic or infinite-volume limit, the scaling window and topology, the positivity and locality conditions, and the framework in which the limiting object is reconstructed or compared.

124.1 Dynamical Algebras and Analytic Elements

Definition 124.1 (C^*-dynamical system)**.** A C^*-dynamical system is a pair $(\mathcal{A}, \alpha)$, where $\mathcal{A}$ is a C^*-algebra and

$$\alpha : \mathbb{R} \longrightarrow \mathrm{Aut}(\mathcal{A}), \qquad t \longmapsto \alpha_t,$$

is a strongly continuous one-parameter group of $*$-automorphisms. Thus $\alpha_{t+s} = \alpha_t \alpha_s$, $\alpha_0 = \mathrm{id}$, and for each $A \in \mathcal{A}$ the function $t \mapsto \alpha_t(A)$ is norm-continuous.

The algebra $\mathcal{A}$ is the algebra of bounded observables. Unbounded local fields, stress tensors, and currents are used later through smeared operators, affiliated operators, or distributional limits. The KMS condition is first stated on bounded observables because products and analytic continuation are then unambiguous.

Definition 124.2 (Analytic element)**.** An element $A \in \mathcal{A}$ is entire analytic for α if the map $t \mapsto \alpha_t(A)$ extends to an entire $\mathcal{A}$-valued function

$$z \longmapsto \alpha_z(A)$$

whose restriction to the real axis is the original orbit. If $w \in \mathbb{R}$, then $\alpha_w(A)$ is again entire analytic and its continuation is

$$z \longmapsto \alpha_{z+w}(A).$$

A subalgebra $\mathcal{A}_{\mathrm{an}} \subset \mathcal{A}$ is an analytic core if it is norm dense, α_t-invariant, and consists of entire analytic elements.

Lemma 124.3 (Existence of analytic elements by Gaussian smearing)**.** *Let $(\mathcal{A}, \alpha)$ be a C^*-dynamical system. For $A \in \mathcal{A}$ and $n > 0$, define*

$$A_n = \sqrt{\frac{n}{\pi}} \int_{\mathbb{R}} \mathrm{d}t \, \mathrm{e}^{-nt^2} \alpha_t(A), \tag{124.1}$$

where the integral is the Bochner integral in the norm of $\mathcal{A}$. Then A_n is entire analytic and $A_n \to A$ in norm as $n \to \infty$. Hence entire analytic elements form a norm-dense $*$-subalgebra.

Proof. Strong continuity implies that the integral in (124.1) exists and that

$$\|A_n - A\| \le \sqrt{\frac{n}{\pi}} \int_{\mathbb{R}} \mathrm{d}t \, \mathrm{e}^{-nt^2} \|\alpha_t(A) - A\|.$$

After the change of variables $u = \sqrt{n}\, t$, the integrand is bounded by $\pi^{-1/2}\mathrm{e}^{-u^2} 2\|A\|$ and converges to zero for each fixed u; dominated convergence gives $A_n \to A$. For complex z, define

$$\alpha_z(A_n) = \sqrt{\frac{n}{\pi}} \int_{\mathbb{R}} \mathrm{d}t \, \mathrm{e}^{-n(t-z)^2} \alpha_t(A).$$

On every compact subset of the z-plane, the norm of the integrand is bounded by $\|A\| \exp[-nt^2 + C(1 + |t|)]$, which is integrable in t. Differentiation under the integral is therefore justified by dominated convergence, so $z \mapsto \alpha_z(A_n)$ is entire. If A and B are entire analytic, the functions

$$z \longmapsto \alpha_z(A)\alpha_z(B), \qquad z \longmapsto \alpha_{\bar{z}}(A)^*$$

are respectively holomorphic and holomorphic as Banach-space-valued maps; on the real axis they equal $\alpha_t(AB)$ and $\alpha_t(A^*)$. The identity theorem gives the analytic continuations of AB and A^*. For the adjoint statement, if ℓ is any continuous linear functional on $\mathcal{A}$ and $\ell^{\#}(X) = \overline{\ell(X^*)}$, then $\ell(\alpha_{\bar{z}}(A)^*) = \overline{\ell^{\#}(\alpha_{\bar{z}}(A))}$; this is holomorphic in z because $w \mapsto \ell^{\#}(\alpha_w(A))$ is entire. Finite linear combinations then give a dense $*$-subalgebra. ∎

124.2 The KMS Boundary Condition

Definition 124.4 (KMS state)**.** Let $\beta > 0$. A state $\omega : \mathcal{A} \to \mathbb{C}$ is a β-KMS state for α if, for every $A, B \in \mathcal{A}_{\mathrm{an}}$, the function

$$t \longmapsto \omega(A\alpha_t(B))$$

is the lower boundary value of a function $F_{A,B}(z)$ that is holomorphic on the open strip

$$S_\beta = \{z \in \mathbb{C} : 0 < \operatorname{Im} z < \beta\},$$

continuous and bounded on the closed strip, and satisfies

$$F_{A,B}(t) = \omega(A\alpha_t(B))\,, \qquad F_{A,B}(t+\mathrm{i}\beta) = \omega(\alpha_t(B)A) \tag{124.2}$$

for all $t \in \mathbb{R}$. The analytic core may be replaced by any norm-dense analytic core; the resulting condition extends by continuity to the usual KMS condition on $\mathcal{A}$.

The inverse temperature β has dimensions of time when $\hbar = 1$. The time evolution α_t is part of the datum. Changing the time evolution, for example by adding a chemical-potential generator, changes the KMS condition.

Figure 124.1 displays the analytic strip in the KMS condition: the two boundary values differ by the time evolution through imaginary time $\mathrm{i}\beta$.

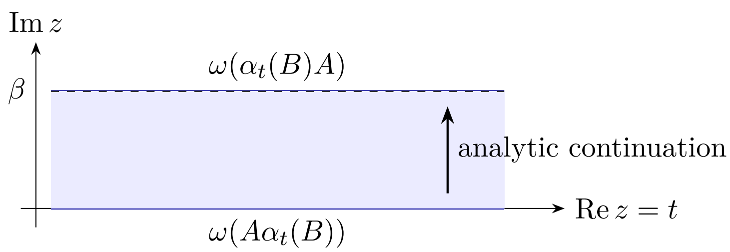

Figure 124.1: The KMS condition is a strip-analytic boundary condition. The lower edge is the ordered correlator $A\alpha_t(B)$; the upper edge is obtained by moving B past the thermal state.

Finite Gibbs traces as KMS states. Let $\mathcal{H}$ be finite dimensional, let $H = H^*$, and let $\alpha_t(A) = \mathrm{e}^{\mathrm{i}Ht}A\mathrm{e}^{-\mathrm{i}Ht}$ on $\mathcal{A} = \mathrm{End}(\mathcal{H})$. For

$$\omega_\beta(A) = \frac{\mathrm{Tr}(\mathrm{e}^{-\beta H}A)}{\mathrm{Tr}(\mathrm{e}^{-\beta H})}, \tag{124.3}$$

ω_β is a β-KMS state. To see the strip function explicitly, choose an orthonormal eigenbasis $H|m\rangle = E_m|m\rangle$, and write $A_{mn} = \langle m|A|n\rangle$, $B_{mn} = \langle m|B|n\rangle$. Then

$$F_{A,B}(z) = \frac{1}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_m} A_{mn}B_{nm}\,\mathrm{e}^{\mathrm{i}(E_n-E_m)z}, \qquad Z_\beta = \sum_m \mathrm{e}^{-\beta E_m}, \tag{124.4}$$

is entire because the sum is finite. At $z = t$ it equals $\omega_\beta(A\alpha_t(B))$. At $z = t + \mathrm{i}\beta$,

$$F_{A,B}(t+\mathrm{i}\beta) = \frac{1}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_n} A_{mn}B_{nm}\,\mathrm{e}^{\mathrm{i}(E_n-E_m)t}.$$

Relabeling $m \leftrightarrow n$ gives

$$F_{A,B}(t+\mathrm{i}\beta) = \frac{1}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_m} B_{mn}A_{nm}\,\mathrm{e}^{\mathrm{i}(E_m-E_n)t} = \omega_\beta(\alpha_t(B)A),$$

which is the boundary condition (124.2). This finite-dimensional calculation is the model for the general KMS definition. Infinite-volume thermal states require the strip analyticity and boundary condition to be verified in the chosen operator-algebraic or distributional framework.

Remark 124.5 (Chemical potentials). If Q_A are commuting conserved charges and μ_A are chemical potentials, the finite trace

$$Z^{-1}\,\mathrm{Tr}\exp[-\beta(H-\mu_A Q_A)]\,(\cdot)$$

is KMS for the automorphism generated by $H-\mu_A Q_A$. Equivalently, it is KMS for the original Hamiltonian time evolution together with a simultaneous imaginary-time internal-symmetry twist. The two descriptions must not be mixed without declaring which automorphism defines equilibrium.

124.3 Spectral Form of Detailed Balance

In this section the Fourier transform of a tempered distribution $G(t)$ is

$$\widetilde{G}(\omega) = \int_{\mathbb{R}} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t} G(t). \tag{124.5}$$

With this convention, positive ω is the energy absorbed by the operator insertion in the finite-volume spectral formulas below.

Let A, B be observables whose time-dependent correlation functions have tempered boundary values. Define

$$G^{>}_{AB}(t) = \omega(A(t)B), \qquad G^{<}_{AB}(t) = \omega(BA(t)), \qquad A(t) = \alpha_t(A). \tag{124.6}$$

The KMS condition implies, in the distributional boundary-value sense,

$$G^{>}_{AB}(t-\mathrm{i}\beta) = G^{<}_{AB}(t), \tag{124.7}$$

or equivalently the frequency-space detailed-balance relation

$$\widetilde{G}^{<}_{AB}(\omega) = \mathrm{e}^{-\beta\omega}\widetilde{G}^{>}_{AB}(\omega). \tag{124.8}$$

Finite-volume spectral detailed balance. In the same finite-dimensional setting, the distributions $\widetilde{G}^{>}_{AB}$ and $\widetilde{G}^{<}_{AB}$ are finite sums of delta functions and obey (124.8). Using the eigenbasis above,

$$G^{>}_{AB}(t) = \frac{1}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_m} A_{mn} B_{nm} \mathrm{e}^{\mathrm{i}(E_m-E_n)t}.$$

Therefore

$$\widetilde{G}^{>}_{AB}(\omega) = \frac{2\pi}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_m} A_{mn} B_{nm} \delta(\omega - E_n + E_m). \tag{124.9}$$

Similarly,

$$\widetilde{G}^{<}_{AB}(\omega) = \frac{2\pi}{Z_\beta}\sum_{m,n} \mathrm{e}^{-\beta E_n} A_{mn} B_{nm} \delta(\omega - E_n + E_m), \tag{124.10}$$

where the second formula follows by relabeling the indices in $\omega(BA(t))$. On the support of the displayed delta function, $\mathrm{e}^{-\beta E_n} = \mathrm{e}^{-\beta\omega}\mathrm{e}^{-\beta E_m}$, proving (124.8).

Define the thermal spectral distribution

$$\rho_{AB}(\omega) = \widetilde{G}^{>}_{AB}(\omega) - \widetilde{G}^{<}_{AB}(\omega). \tag{124.11}$$

When $1-\mathrm{e}^{-\beta\omega}$ is invertible as a multiplier on the relevant test-function class, detailed balance gives

$$\widetilde{G}^{>}_{AB}(\omega)=\frac{\rho_{AB}(\omega)}{1-\mathrm{e}^{-\beta\omega}},\qquad \widetilde{G}^{<}_{AB}(\omega)=\frac{\mathrm{e}^{-\beta\omega}\rho_{AB}(\omega)}{1-\mathrm{e}^{-\beta\omega}}. \tag{124.12}$$

For conserved densities, the $\omega=0$ sector may contain delta functions or derivatives of delta functions. Those pieces have to be separated before dividing by $1-\mathrm{e}^{-\beta\omega}$. This is the same mathematical issue that later appears as susceptibilities, hydrodynamic poles, and contact terms.

Remark 124.6 (Bosonic fluctuation–dissipation identity). Assume $A=A^*$, $B=B^*$, and no singular zero-frequency term in the frequency window under consideration. Define the symmetrized spectrum

$$\widetilde{G}^{\mathrm{sym}}_{AB}(\omega)=\frac{1}{2}\left(\widetilde{G}^{>}_{AB}(\omega)+\widetilde{G}^{<}_{AB}(\omega)\right).$$

Then

$$\widetilde{G}^{\mathrm{sym}}_{AB}(\omega)=\frac{1}{2}\coth\left(\frac{\beta\omega}{2}\right)\rho_{AB}(\omega). \tag{124.13}$$

This identity follows directly from detailed balance. Insert (124.12) into the definition of $\widetilde{G}^{\mathrm{sym}}_{AB}$:

$$\widetilde{G}^{\mathrm{sym}}_{AB}=\frac{1}{2}\,\frac{1+\mathrm{e}^{-\beta\omega}}{1-\mathrm{e}^{-\beta\omega}}\rho_{AB}=\frac{1}{2}\coth\left(\frac{\beta\omega}{2}\right)\rho_{AB}.$$

The stated zero-frequency hypothesis is precisely the condition that this division is an ordinary distributional operation in the chosen window.

For odd operators in a $\mathbb{Z}_2$-graded theory, the KMS boundary condition includes the parity sign obtained by moving an odd operator around the thermal circle. The same proof gives

$$\widetilde{G}^{<}_{AB}(\omega)=-\mathrm{e}^{-\beta\omega}\widetilde{G}^{>}_{AB}(\omega)$$

for two odd operators in the convention where $G^{<}$ is not itself graded by an additional sign. Equivalently, one may absorb the sign into the definition of the fermionic lesser function. The convention must be declared before writing Fermi factors.

124.4 Linear Response and Retarded Correlators

Let $B(t)$ be coupled to a compactly supported real source $f(t)$ by

$$H_f(t)=H-f(t)B(t) \tag{124.14}$$

in a finite system, or by the corresponding time-dependent derivation on an analytic algebra. In the interaction picture,

$$U_f(t,-\infty)=1+\mathrm{i}\int_{-\infty}^{t}\mathrm{d}s\,f(s)B(s)+O(f^2). \tag{124.15}$$

Thus

$$U_f(t,-\infty)^{-1}A(t)U_f(t,-\infty)=A(t)+\mathrm{i}\int_{-\infty}^{t}\mathrm{d}s\,f(s)[A(t),B(s)]+O(f^2).$$

Stationarity of ω gives

$$\delta\omega(A(t)) = \int_{-\infty}^{t} \mathrm{d}s\, f(s)\, G^R_{AB}(t-s), \qquad G^R_{AB}(t) = -\mathrm{i}\,\theta(t)\,\omega([A(t),B(0)]). \tag{124.16}$$

The retarded distribution is causal: its support in the time variable lies in $t \geq 0$. Consequently, under the usual polynomial-growth hypotheses for tempered distributions, its Fourier transform is the boundary value from the upper half ω-plane.

Figure 124.2 records the causal support property of the retarded response function, which is the input for its upper-half-plane analytic Fourier transform.

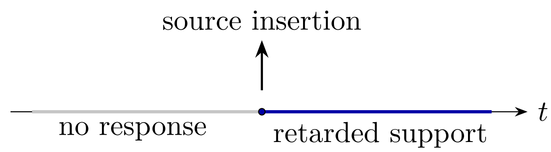

Figure 124.2: The retarded kernel is supported only after the source insertion. This causal support, not Euclidean periodicity alone, is what gives the upper-half-plane analyticity of $G^R(\omega)$.

For translation-invariant thermal QFT, the retarded two-point function of local densities is

$$G^R_{AB}(\omega,\mathbf{k}) = -\mathrm{i}\int_0^\infty \mathrm{d}t \int \mathrm{d}^{D-1}x\, \mathrm{e}^{\mathrm{i}\omega t - \mathrm{i}\mathbf{k}\cdot\mathbf{x}}\omega([A(t,\mathbf{x}),B(0,\mathbf{0})]). \tag{124.17}$$

With the Fourier convention (124.5), the spectral function associated to G^R is

$$\rho_{AB}(\omega,\mathbf{k}) = -2\,\operatorname{Im} G^R_{AB}(\omega+\mathrm{i}0,\mathbf{k}). \tag{124.18}$$

For Hermitian $A=B$, positivity of the finite-volume spectral weights gives $\rho_{AA}(\omega,\mathbf{k}) \geq 0$ for $\omega > 0$ before the thermodynamic limit. The infinite-volume statement is obtained by distributional limits, provided the local algebra and state have been constructed.

124.5 Conserved Densities and Hydrodynamic Variables

Hydrodynamics begins with conserved local currents in the microscopic theory. Let $T^{\mu\nu}$ be the stress tensor and let J^μ_A be conserved currents for ordinary internal symmetries:

$$\partial_\mu T^{\mu\nu} = 0, \qquad \partial_\mu J^\mu_A = 0. \tag{124.19}$$

In a homogeneous isotropic thermal state with unit future-timelike vector u^μ, temperature $T = \beta^{-1}$, and chemical potentials μ_A, the one-point functions define thermodynamic densities by

$$\omega(T^{\mu\nu}) = \varepsilon u^\mu u^\nu + p\Delta^{\mu\nu}, \qquad \omega(J^\mu_A) = n_A u^\mu, \qquad \Delta^{\mu\nu} = \eta^{\mu\nu} + u^\mu u^\nu. \tag{124.20}$$

Here ε is the energy density, p the pressure, and n_A the charge densities. These are numbers attached to a homogeneous equilibrium state. They become hydrodynamic fields only after one assumes a family of near-equilibrium states whose long-wavelength connected correlators are controlled by slowly varying local values of

$$T(x), \qquad \mu_A(x), \qquad u^\mu(x).$$

The derivative expansion is a statement about constitutive relations:

$$T^{\mu\nu} = T^{\mu\nu}_{(0)} + T^{\mu\nu}_{(1)} + T^{\mu\nu}_{(2)} + \cdots, \qquad J^\mu_A = J^\mu_{A,(0)} + J^\mu_{A,(1)} + J^\mu_{A,(2)} + \cdots, \tag{124.21}$$

where the subscript records the number of derivatives acting on T, μ_A, u^μ and on background sources. Existence of such an expansion requires hypotheses beyond the KMS boundary condition: it is a long-time, long-distance property of the QFT state.

First hydrodynamic poles from conservation. Let a homogeneous parity-even thermal state in $D = d_s + 1$ spacetime dimensions have enthalpy $w = \varepsilon + p > 0$, speed of sound

$$c_s^2 = \left(\frac{\partial p}{\partial \varepsilon}\right)_{n_A/s},$$

shear viscosity $\eta_{\rm sh}$, bulk viscosity ζ, and a decoupled charge-diffusion channel with conductivity σ and susceptibility $\chi > 0$. Linearizing the first-order constitutive relations around the homogeneous state gives the following small-$|\mathbf{k}|$ hydrodynamic denominators, and therefore the corresponding retarded poles in every conserved-current correlator whose residue on the mode is nonzero:

$$\omega_{\rm shear}(\mathbf{k}) = -\,\mathrm{i}\,\frac{\eta_{\rm sh}}{w}\,\mathbf{k}^2 + O(|\mathbf{k}|^4), \tag{124.22}$$

$$\omega_{\rm charge}(\mathbf{k}) = -\,\mathrm{i}\,\frac{\sigma}{\chi}\,\mathbf{k}^2 + O(|\mathbf{k}|^4), \tag{124.23}$$

and

$$\omega^{\pm}_{\rm sound}(\mathbf{k}) = \pm c_s|\mathbf{k}| - \frac{\mathrm{i}}{2w}\left(\zeta + \frac{2(d_s-1)}{d_s}\eta_{\rm sh}\right)\mathbf{k}^2 + O(|\mathbf{k}|^3). \tag{124.24}$$

Take $\mathbf{k}$ in the x-direction and use the Fourier convention $\mathrm{e}^{-\mathrm{i}\omega t + \mathrm{i}kx}$. In the shear channel a transverse velocity perturbation v_y obeys

$$-\mathrm{i}\omega\, w\, v_y + \eta_{\rm sh} k^2 v_y = 0,$$

which gives (124.22). In a decoupled charge channel, Fick's law $J^x_A = -\sigma\,\partial_x(\delta\mu_A)$ and $\delta n_A = \chi\,\delta\mu_A$ give

$$-\mathrm{i}\omega\,\delta n_A + \frac{\sigma}{\chi}k^2\delta n_A = 0,$$

which gives (124.23). For longitudinal sound, the linearized conservation equations are

$$-\mathrm{i}\omega\,\delta\varepsilon + \mathrm{i}k\,w\,v_x = 0,$$

$$-\mathrm{i}\omega\,w\,v_x + \mathrm{i}k\,c_s^2\delta\varepsilon + \left(\zeta + \frac{2(d_s-1)}{d_s}\eta_{\rm sh}\right)k^2 v_x = 0.$$

Eliminating v_x gives

$$\omega^2 - c_s^2k^2 + \frac{\mathrm{i}}{w}\left(\zeta + \frac{2(d_s-1)}{d_s}\eta_{\rm sh}\right)\omega k^2 = 0.$$

Solving this quadratic at small k gives (124.24). This calculation identifies the hydrodynamic poles of the retarded functions once the linearized constitutive relations, the long-distance derivative expansion, and the existence of the corresponding retarded correlators have been established.

Figure 124.3 depicts this analytic content of the hydrodynamic limit. The hydrodynamic variables are not new microscopic operators; they are the coordinates whose linearized equations describe the poles of retarded conserved-current correlators that approach the origin as $|\mathbf{k}| \to 0$.

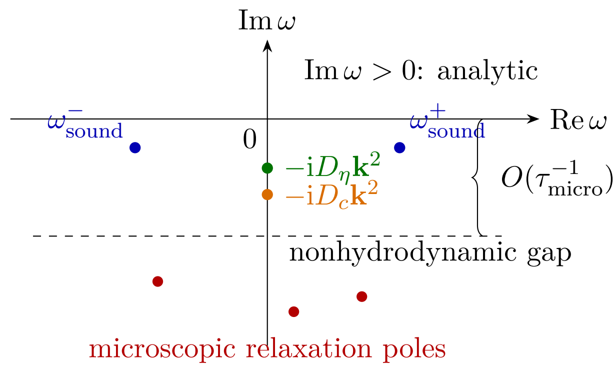

Figure 124.3: At fixed small $|\mathbf{k}|$, hydrodynamics appears in retarded thermal correlators as poles that approach the origin: two sound poles, a shear-diffusion pole with $D_\eta = \eta_{\text{sh}}/w$, and charge-diffusion poles with $D_c = \sigma/\chi$ in a decoupled channel. Nonhydrodynamic relaxation poles remain separated by a microscopic gap in a regime where the derivative expansion is controlled.

At first order, parity-even relativistic hydrodynamics in flat spacetime contains shear viscosity η_{sh}, bulk viscosity ζ, and charge-conductivity data. In the shear channel, couple a metric perturbation $h_{xy}(t)$ to T^{xy}. The linear-response formula gives

$$\delta\omega(T^{xy}(\omega)) = -\frac{1}{2} G^R_{T^{xy}T^{xy}}(\omega, \mathbf{0}) h_{xy}(\omega)$$

with the factor $1/2$ inherited from the metric coupling $\frac{1}{2}\int T^{ij} h_{ij}$. Comparing with the constitutive relation fixes

$$\eta_{\text{sh}} = \lim_{\omega \downarrow 0} \frac{\rho_{T^{xy}T^{xy}}(\omega, \mathbf{0})}{2\omega} = -\lim_{\omega \downarrow 0} \frac{\operatorname{Im} G^R_{T^{xy}T^{xy}}(\omega, \mathbf{0})}{\omega}. \tag{124.25}$$

This is a definition of the transport coefficient in terms of QFT correlators once the zero-frequency limit exists. Later chapters derive the constitutive relations, entropy-current constraints, Schwinger–Keldysh effective action, anomaly-induced transport, and kinetic limits from these correlator-level objects.

Open Problem 124.7 (Thermal real-time reconstruction). Give constructive hypotheses under which Euclidean finite-temperature Schwinger functions on $S^1_\beta \times \mathbb{R}^{D-1}$ reconstruct a real-time KMS QFT with local observables, spectral functions, and transport coefficients. The required analytic continuation is more delicate than the vacuum OS reconstruction because thermal correlators have periodic Euclidean time and no vacuum spectrum condition.

Chapter 125

Finite-Temperature Path Integrals

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter derives the Euclidean thermal path integral from the operator definition of equilibrium correlation functions. The derivation is first a finite-dimensional statement about a trace-class Gibbs operator. A field theory path integral on $S^1_\beta \times \mathbb{R}^{D-1}$ is obtained only after a spatial regulator, an operator-domain convention, and a limiting procedure have been specified. The circle, spin structure, Matsubara frequencies, and chemical-potential twists all follow from the trace formula.

125.1 Regulated Meaning of the Thermal Path Integral

The notation

$$\int [D\Phi]\, \mathrm{e}^{-S_E[\Phi]}$$

has a precise meaning only after one has specified the class of variables Φ, the algebra in which they take values, and the limiting operation. In this chapter it is used in the following restricted sense. One begins with a regulated quantum system with Hilbert space $\mathcal{H}_{\mathrm{reg}}$, a self-adjoint Hamiltonian H_{reg}, and a trace-class Gibbs operator $\mathrm{e}^{-\beta H_{\mathrm{reg}}}$. The thermal path integral is the finite time-sliced expression obtained by inserting resolutions of the identity in

$$\mathrm{Tr}_{\mathcal{H}_{\mathrm{reg}}}\, \mathrm{e}^{-\beta H_{\mathrm{reg}}}, \tag{125.1}$$

or in trace insertions with operators. Removing the regulator is a separate problem. It may be a lattice thermodynamic limit, a constructive continuum limit, a Hamiltonian limit, or a perturbative renormalized limit; the symbol $[D\Phi]$ alone does not specify which one is intended.

There are two algebraically different kinds of variables. Bosonic variables are ordinary commuting coordinates or fields at the regulator scale. Fermionic variables in a coherent-state path integral are generators of a finite Grassmann algebra at the regulator scale. Their "integration" is a Berezin linear functional. It is not a Borel measure on a set of fermionic field configurations. The continuum notation for fermionic fields therefore denotes the limit of a sequence of finite Grassmann-algebra calculations, not an ordinary measure space of anticommuting points.

125.2 Gibbs Traces and Euclidean Correlators

Let $\mathcal{H}$ be a Hilbert space and let H be a self-adjoint Hamiltonian bounded below. Fix $\beta > 0$ and assume, in this finite-volume discussion, that

$$Z(\beta) = \mathrm{Tr}_{\mathcal{H}}\, \mathrm{e}^{-\beta H} < \infty.$$

The Gibbs state is the positive normalized functional

$$\omega_\beta(A) = Z(\beta)^{-1}\, \mathrm{Tr}_{\mathcal{H}}(\mathrm{e}^{-\beta H} A)$$

on bounded operators A, and on suitable unbounded operators after a common trace-class domain has been declared. For Heisenberg operators

$$A(t) = \mathrm{e}^{\mathrm{i}Ht} A \mathrm{e}^{-\mathrm{i}Ht},$$

their Euclidean-time translates are

$$A_E(\tau) = \mathrm{e}^{\tau H} A \mathrm{e}^{-\tau H}, \qquad 0 \le \tau \le \beta,$$

whenever the displayed expression is defined as a quadratic form or as a trace insertion. If

$$0 < \tau_1 < \tau_2 < \cdots < \tau_n < \beta,$$

the Euclidean thermal correlator is

$$G_E(\tau_1, \ldots, \tau_n) = Z(\beta)^{-1}\, \mathrm{Tr}\Big(\mathrm{e}^{-\beta H} A_{n,E}(\tau_n) \cdots A_{1,E}(\tau_1)\Big)\,. \tag{125.2}$$

Moving the factor $\mathrm{e}^{-\beta H}$ cyclically through the trace shifts one operator by β in imaginary time. This is the finite-volume operator origin of the KMS boundary condition in Chapter 124.

Remark 125.1 (Trace cyclicity gives the thermal circle). Let A and B be bounded operators such that the traces below exist. For $0 < \tau < \beta$,

$$\omega_\beta(A_E(\tau)B) = \omega_\beta(B A_E(\tau - \beta)).$$

Equivalently, after analytic continuation $t = -\mathrm{i}\tau$, the two boundary values of the strip correlator differ by a cyclic permutation of the operators.

Using $A_E(\tau) = \mathrm{e}^{\tau H} A \mathrm{e}^{-\tau H}$,

$$\begin{aligned}
\mathrm{Tr}(\mathrm{e}^{-\beta H} A_E(\tau) B) &= \mathrm{Tr}(\mathrm{e}^{-(\beta-\tau)H} A \mathrm{e}^{-\tau H} B) \\
&= \mathrm{Tr}(B \mathrm{e}^{-(\beta-\tau)H} A \mathrm{e}^{-\tau H}) \\
&= \mathrm{Tr}(\mathrm{e}^{-\beta H} B\, \mathrm{e}^{(\tau-\beta)H} A \mathrm{e}^{-(\tau-\beta)H}).
\end{aligned}$$

Dividing by $Z(\beta)$ gives the stated identity.

125.3 Time Slicing and Bosonic Boundary Conditions

Let $q = (q^1, \dots, q^r)$ be a configuration coordinate for a regulated bosonic quantum system. Let $\{|q\rangle\}$ be a generalized coordinate resolution of the identity. For

$$\beta = N\epsilon,$$

the partition function is

$$Z(\beta) = \int \prod_{j=1}^{N} dq_j \prod_{j=1}^{N} K_\epsilon(q_{j+1}, q_j), \qquad q_{N+1} = q_1, \tag{125.3}$$

where

$$K_\epsilon(q', q) = \langle q'|\mathrm{e}^{-\epsilon H}|q\rangle.$$

The equality is the trace written as

$$\int dq_1 \, \langle q_1|(\mathrm{e}^{-\epsilon H})^N|q_1\rangle$$

with $N-1$ intermediate identity insertions. The condition $q_{N+1} = q_1$ is therefore exact at finite N.

For a local bosonic field with a spatial regulator, the same formula has one coordinate q_j for every spatial lattice degree of freedom at Euclidean time slice j. A formal continuum notation for the result is

$$Z(\beta) = \int_{\phi(\tau+\beta,\mathbf{x})=\phi(\tau,\mathbf{x})} [D\phi]_{\mathrm{reg}} \, \mathrm{e}^{-S_E[\phi]}.$$

The subscript reg is part of the formula: it denotes the particular lattice, constructive, Hamiltonian, or perturbative regulator from which the expression is obtained. The Euclidean time direction is a circle because the trace identifies the final and initial bosonic coordinates.

125.4 Fermionic Coherent States and Spin Structure

Let $c_i, c_i^\dagger$, $i = 1, \dots, r$, be fermionic annihilation and creation operators with

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0.$$

Let $\eta_i, \bar\eta_i$ be independent Grassmann variables. The coherent state

$$|\eta\rangle = \mathrm{e}^{-\sum_i \eta_i c_i^\dagger}|0\rangle$$

satisfies

$$c_i|\eta\rangle = \eta_i|\eta\rangle,$$

where the sign comes from moving the odd operator c_i through the odd coefficient η_i. The resolution of identity has the form

$$\mathbf{1} = \int \prod_i d\bar\eta_i \, d\eta_i \, \mathrm{e}^{-\bar\eta\eta} \, |\eta\rangle\langle\eta|, \qquad \bar\eta\eta = \sum_i \bar\eta_i \eta_i.$$

Definition 125.2 (Berezin integral at finite regulator)**.** Let Λ be the Grassmann algebra generated by $\eta_1, \dots, \eta_r, \bar\eta_1, \dots, \bar\eta_r$. The ordered Berezin integral

$$\int \prod_{i=1}^{r} d\bar\eta_i \, d\eta_i : \Lambda \longrightarrow \mathbb{C}$$

is the linear functional fixed by

$$\int \prod_{i=1}^{r} d\bar\eta_i \, d\eta_i \prod_{i=1}^{r} \bar\eta_i \eta_i = (-1)^r \tag{125.4}$$

and by the requirement that every monomial missing at least one generator integrates to zero. Equivalently,

$$\int \prod_{i=1}^{r} d\bar\eta_i \, d\eta_i \, \exp\left(- \sum_i \bar\eta_i \eta_i \right) = 1. \tag{125.5}$$

The variables $\eta_i, \bar\eta_i$ are algebra generators, not complex numbers. The integral is a linear functional on Λ, not a positive measure.

Fermionic trace identity. For every operator O on the finite fermionic Fock space,

$$\operatorname{Tr} O = \int d\bar\eta \, d\eta \, \mathrm{e}^{-\bar\eta\eta} \, \langle -\eta | O | \eta \rangle,$$

whereas

$$\operatorname{Tr}((-1)^F O) = \int d\bar\eta \, d\eta \, \mathrm{e}^{-\bar\eta\eta} \, \langle \eta | O | \eta \rangle.$$

It is enough to check one fermionic mode, since the r-mode formula is the ordered tensor product of one-mode identities with the ordering in Definition 125.2. In the basis $|0\rangle, |1\rangle = c^\dagger |0\rangle$, expand O as a 2×2 matrix and use

$$|\eta\rangle = |0\rangle - \eta |1\rangle, \qquad \langle \eta | = \langle 0 | - \langle 1 | \bar\eta.$$

Write $O_{ab} = \langle a | O | b \rangle$. Since $\langle -\eta | = \langle 0 | + \langle 1 | \bar\eta$,

$$\mathrm{e}^{-\bar\eta\eta} \langle -\eta | O | \eta \rangle = O_{00} - O_{01}\eta + \bar\eta O_{10} - (O_{00} + O_{11}) \bar\eta\eta.$$

With the convention $\int d\bar\eta \, d\eta \, \bar\eta\eta = -1$, the integral is $O_{00} + O_{11} = \operatorname{Tr} O$. Similarly,

$$\mathrm{e}^{-\bar\eta\eta} \langle \eta | O | \eta \rangle = O_{00} - O_{01}\eta - \bar\eta O_{10} - (O_{00} - O_{11}) \bar\eta\eta,$$

whose integral is $O_{00} - O_{11} = \operatorname{Tr}((-1)^F O)$.

Applying this finite coherent-state trace identity to $(\mathrm{e}^{-\epsilon H})^N$ gives the thermal Grassmann time-sliced integral. The first and last coherent-state labels obey

$$\eta_{N+1} = -\eta_1, \qquad \bar\eta_{N+1} = -\bar\eta_1$$

for the ordinary thermal partition function. Thus fermionic variables in the thermal trace are antiperiodic on S^1_β:

$$\psi(\tau + \beta, \mathbf{x}) = -\psi(\tau, \mathbf{x}), \qquad \bar\psi(\tau + \beta, \mathbf{x}) = -\bar\psi(\tau, \mathbf{x}).$$

The graded trace $\mathrm{Tr}((-1)^F e^{-\beta H}\cdots)$ uses the periodic spin structure. This is the operator origin of the distinction between a thermal partition function and a supersymmetric index.

Figure 125.1 records the trace-gluing convention: bosons are periodic on S^1_β, fermions are antiperiodic for the thermal trace, and the graded trace uses the periodic spin structure.

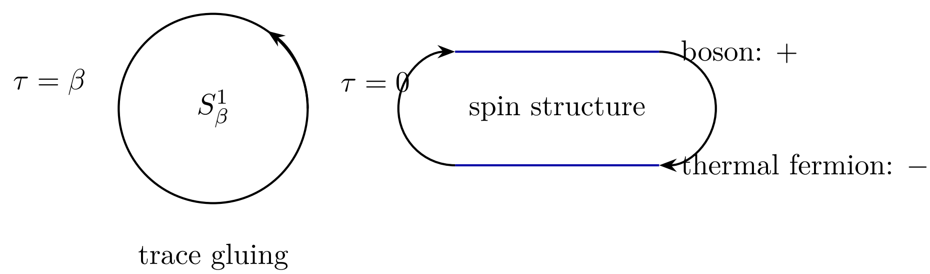

Figure 125.1: The thermal trace glues the end of the Euclidean time interval to its beginning. Coordinate variables are periodic, while ordinary thermal fermionic coherent-state variables are antiperiodic because the trace identity contains $\langle -\eta|$.

125.5 Matsubara Modes as Circle Eigenfunctions

Let $f(\tau)$ be a field component on the Euclidean thermal circle obeying

$$f(\tau+\beta) = s f(\tau), \qquad s = \pm 1.$$

An eigenfunction of $-i\partial_\tau$ has the form $e^{i\omega\tau}$. The boundary condition requires

$$e^{i\omega\beta} = s.$$

Hence

$$\omega_n^{\mathrm{B}} = \frac{2\pi n}{\beta}, \qquad n \in \mathbb{Z},$$

for periodic bosonic variables, and

$$\omega_n^{\mathrm{F}} = \frac{(2n+1)\pi}{\beta}, \qquad n \in \mathbb{Z},$$

for antiperiodic fermionic variables.

Let

$$K = -\partial_\tau^2 - \Delta_{\mathbf{x}} + m^2$$

on periodic functions on $S^1_\beta \times \mathbb{R}^{D-1}$. The translation invariant inverse of K, as a distribution, is

$$G_E(\tau,\mathbf{x}) = \frac{1}{\beta}\sum_{n\in\mathbb{Z}} \int \frac{d^{D-1}k}{(2\pi)^{D-1}} \frac{e^{i\omega_n^{\mathrm{B}}\tau + i\mathbf{k}\cdot\mathbf{x}}}{(\omega_n^{\mathrm{B}})^2 + \mathbf{k}^2 + m^2}.$$

The functions

$$\beta^{-1/2} e^{i\omega_n^{\mathrm{B}}\tau}$$

form the Fourier basis of $L^2(S^1_\beta)$, and plane waves form the Fourier basis in the spatial directions. Acting with K on one mode multiplies it by

$$(\omega_n^{\mathrm{B}})^2 + \mathbf{k}^2 + m^2.$$

The displayed kernel is therefore the spectral inverse of K on each Fourier mode. Applying K to it gives the periodic delta distribution on S^1_β times $\delta^{D-1}(\mathbf{x})$.

For fermions, the same diagonalization applies to the chosen Euclidean Dirac operator, but the frequency set is shifted to ω_n^{F}. The shift is a spin-structure effect, not a perturbative convention.

125.6 Thermal Spectral Representation and Analytic Continuation

The Euclidean circle values determine real-time response only after the analytic continuation problem has been specified. The finite-volume spectral formula makes the required data explicit. Let A and B be bosonic operators, and let $|n\rangle$ be an eigenbasis of H with

$$H|n\rangle = E_n|n\rangle.$$

For $0 < \tau < \beta$,

$$G_E^{AB}(\tau) = Z(\beta)^{-1} \sum_{m,n} \mathrm{e}^{-\beta E_n} \mathrm{e}^{-\tau(E_m - E_n)} A_{nm} B_{mn}.$$

The terms with $E_m = E_n$ are independent of τ. Denote this zero-frequency contribution by

$$G_0^{AB} = Z(\beta)^{-1} \sum_{m,n:E_m=E_n} \mathrm{e}^{-\beta E_n} A_{nm} B_{mn}.$$

For the remaining transitions define the thermal spectral distribution

$$\rho_{AB}(\omega) = Z(\beta)^{-1} \sum_{m,n:E_m \neq E_n} \left(\mathrm{e}^{-\beta E_n} - \mathrm{e}^{-\beta E_m}\right) A_{nm} B_{mn}\, \delta(\omega - (E_m - E_n)). \tag{125.6}$$

Then

$$G_E^{AB}(\tau) = G_0^{AB} + \int_{\mathbb{R}} \frac{\mathrm{e}^{-\omega\tau}}{1 - \mathrm{e}^{-\beta\omega}}\, \rho_{AB}(\omega)\, d\omega, \qquad 0 < \tau < \beta. \tag{125.7}$$

Indeed, on the support $\omega = E_m - E_n$,

$$\frac{\mathrm{e}^{-\beta E_n} - \mathrm{e}^{-\beta E_m}}{1 - \mathrm{e}^{-\beta\omega}} = \mathrm{e}^{-\beta E_n}.$$

The explicit G_0^{AB} term is necessary: it is invisible in the commutator spectral density because the numerator in (125.6) vanishes at $\omega = 0$.

The bosonic Matsubara transform is

$$G_E^{AB}(\mathrm{i}\omega_\ell^{\mathrm{B}}) = \int_0^\beta d\tau\, \mathrm{e}^{\mathrm{i}\omega_\ell^{\mathrm{B}}\tau} G_E^{AB}(\tau).$$

Using $\mathrm{e}^{\mathrm{i}\omega_\ell^{\mathrm{B}}\beta} = 1$ in (125.7) gives

$$G_E^{AB}(\mathrm{i}\omega_\ell^{\mathrm{B}}) = \beta G_0^{AB}\delta_{\ell 0} + \int_{\mathbb{R}} \frac{\rho_{AB}(\omega)}{\omega - \mathrm{i}\omega_\ell^{\mathrm{B}}}\, d\omega. \tag{125.8}$$

For fermionic operators the graded spectral density and the denominator $1 + \mathrm{e}^{-\beta\omega}$ replace the bosonic ones; the antiperiodic identity $\mathrm{e}^{\mathrm{i}\omega_\ell^{\mathrm{F}}\beta} = -1$ gives the same Cauchy-transform form with ω_ℓ^{F}.

The retarded correlator is the boundary value of a function obtained from the same spectral density,

$$G_R^{AB}(z) = \int_{\mathbb{R}} \frac{\rho_{AB}(\omega)}{\omega - z}\, d\omega, \qquad \operatorname{Im} z > 0,$$

under the growth and distributional hypotheses required for the integral. Thus the rule

$$\mathrm{i}\omega_n \mapsto \omega + \mathrm{i}0$$

is a shorthand for a spectral reconstruction problem. It is justified when the Euclidean data determine a spectral distribution with the stated analytic control; the discrete Matsubara values alone do not constitute such a proof.

125.7 Chemical Potentials, Twists, and Holonomies

Let Q be the self-adjoint generator of a global $U(1)$ symmetry and assume $[H, Q] = 0$. The grand-canonical Gibbs operator is

$$\mathrm{e}^{-\beta(H-\mu Q)}.$$

If an operator Φ_q has charge q,

$$[Q, \Phi_q] = q\Phi_q,$$

then moving $\mathrm{e}^{\beta\mu Q}$ through Φ_q gives

$$\mathrm{e}^{\beta\mu Q}\Phi_q \mathrm{e}^{-\beta\mu Q} = \mathrm{e}^{\beta\mu q}\Phi_q.$$

Consequently the Euclidean coordinate representation may be written in two equivalent ways:

$$D_\tau \longmapsto D_\tau - \mu q$$

with the original thermal boundary condition, or as a twisted boundary condition

$$\Phi_q(\tau + \beta, \mathbf{x}) = s\, \mathrm{e}^{\beta\mu q}\Phi_q(\tau, \mathbf{x}), \qquad s = +1 \text{ for bosons}, \quad s = -1 \text{ for fermions}.$$

For imaginary chemical potential $\mu = \mathrm{i}\nu$, the twist is a unitary background holonomy

$$\exp(\mathrm{i}\beta\nu q)$$

around S^1_β. If the $U(1)$ is compact and the charges are integers in the chosen normalization, large background gauge transformations identify

$$\nu \sim \nu + \frac{2\pi}{\beta}.$$

For real μ, the Euclidean action of a relativistic theory is generally complex in common field coordinates; this is the operator origin of the finite-density sign problem discussed later in this volume.

125.8 Gauge Fields on the Thermal Circle

For a gauge theory, the Euclidean thermal path integral integrates over connections on $S^1_\beta \times \Sigma$ modulo gauge transformations whose periodicity is compatible with the chosen bundle and matter boundary conditions. The component A_τ has a gauge-invariant zero-mode datum: the thermal holonomy

$$\mathcal{P}(\mathbf{x}) = \mathrm{Pexp}\left(\int_0^\beta A_\tau(\tau, \mathbf{x})\, d\tau\right),$$

defined up to conjugation. In a compact lattice regulator this holonomy is the Polyakov loop variable developed in Chapter 46. In a continuum gauge-fixed coordinate chart it should not be discarded as a pure-gauge variable: a gauge transformation that removes a constant A_τ may fail to be periodic around the thermal circle.

The gauge-theory thermal trace is over the physical Hilbert space, or equivalently over gauge-invariant states after imposing Gauss constraints. In the Euclidean path integral this appears as integration over gauge holonomies rather than fixing a chemical potential for a local gauge charge. External background holonomies may be fixed for global symmetries. Dynamical gauge holonomies are integrated, and their center transformations encode the thermal one-form symmetry data discussed in the gauge-theory volumes.

Open Problem 125.3 (Euclidean thermal reconstruction). Give constructive hypotheses under which Schwinger functions on $S^1_\beta \times \mathbb{R}^{D-1}$, with the appropriate spin structure and background holonomies, reconstruct a real-time KMS state with local observables, spectral functions, and transport correlators. The hypotheses must replace the vacuum spectrum condition by precise thermal analyticity and growth bounds and must specify the operator topology in which the infinite-volume limit is taken.

Chapter 126

Real-Time Schwinger–Keldysh Formalism

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The Schwinger–Keldysh formalism is the real-time generating functional for correlation and response functions in a specified initial state. It is an operator identity before it is a path integral. The doubled fields and sources are a bookkeeping device for the two sides of a trace, and the fundamental constraints come from unitarity, positivity of the density operator, and, in thermal equilibrium, the KMS condition.

126.1 Finite-System Definition

Let $\mathcal{H}$ be a Hilbert space, let H be a self-adjoint Hamiltonian, and let ρ_0 be a positive trace-class density operator with

$$\operatorname{Tr} \rho_0 = 1.$$

Let O_α, $\alpha \in \mathcal{I}$, be a finite or regulated family of Hermitian operators. For real compactly supported sources $J^\alpha(t)$, define the source-dependent Hamiltonian

$$H_J(t) = H - \sum_{\alpha \in \mathcal{I}} J^\alpha(t) O_\alpha.$$

The evolution operator $U_J(t,t_i)$ is the solution of

$$\mathrm{i}\frac{d}{dt}U_J(t,t_i) = H_J(t)U_J(t,t_i), \qquad U_J(t_i,t_i) = \mathbf{1},$$

equivalently

$$U_J(t_f,t_i) = \mathrm{T}\exp\left[-\mathrm{i}\int_{t_i}^{t_f} dt\,\left(H - \sum_\alpha J^\alpha(t)O_\alpha\right)\right].$$

The closed-time-path generating functional is

$$Z[J_+,J_-] = \mathrm{Tr}\Big(U_{J_+}(t_f,t_i)\rho_0 U_{J_-}(t_f,t_i)^\dagger\Big)\,. \tag{126.1}$$

The $+$ and $-$ labels record whether a source acts on the forward or backward side of the trace. The physical source is obtained on the diagonal $J_+ = J_-$.

126.2 Closed Contour Notation as a Trace Identity

It is often convenient to replace (126.1) by a single contour-ordered expression. This notation is shorthand for an operator identity, not an additional axiom. Let $\mathcal{C}$ be the oriented contour that runs from t_i to t_f on the $+$ branch and then from t_f back to t_i on the $-$ branch. Define

$$\int_{\mathcal{C}} dt_{\mathcal{C}}\, J^\alpha_{\mathcal{C}} O_\alpha = \int_{t_i}^{t_f} dt\, J^\alpha_+(t)O_\alpha(t) - \int_{t_i}^{t_f} dt\, J^\alpha_-(t)O_\alpha(t). \tag{126.2}$$

The minus sign is the orientation of the backward branch. The contour ordering operator $\mathrm{T}_{\mathcal{C}}$ orders later contour points to the left. With the interaction-picture convention determined by the unperturbed Hamiltonian,

$$Z[J_+,J_-] = \mathrm{Tr}\left(\rho_0\ \mathrm{T}_{\mathcal{C}}\exp\left[\mathrm{i}\int_{\mathcal{C}} dt_{\mathcal{C}}\, J^\alpha_{\mathcal{C}}(t_{\mathcal{C}})O_\alpha(t_{\mathcal{C}})\right]\right). \tag{126.3}$$

Expanding the contour exponential reproduces the four derivative rules below because insertions on the backward branch are anti-time ordered and carry the orientation sign in (126.2).

Figure 126.1 fixes the closed-time-path orientation used in the Schwinger–Keldysh generating functional: the forward branch is time-ordered, the backward branch is anti-time-ordered, and the source integral carries the contour orientation.

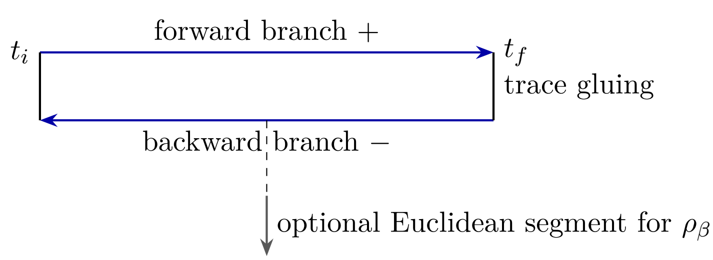

Figure 126.1: The closed time path is a compact notation for $\mathrm{Tr}(U_{J_+}\rho_0 U^\dagger_{J_-})$. The backward branch has the opposite orientation. For a thermal initial state, the density matrix may itself be represented by a Euclidean segment of length β, but that segment is the representation of ρ_β, not part of the real-time evolution.

Definition 126.1 (Closed-time-path correlators)**.** The Schwinger–Keldysh correlators of the regulated system are the functional derivatives of $Z[J_+, J_-]$ at $J_+ = J_- = 0$. Differentiation with respect to J_+ inserts operators on the forward branch with time ordering; differentiation with respect to J_- inserts operators on the backward branch with anti-time ordering and an opposite sign from the adjoint evolution.

For example, in the unperturbed Heisenberg convention

$$O_\alpha(t) = \mathrm{e}^{\mathrm{i}H(t-t_i)} O_\alpha \mathrm{e}^{-\mathrm{i}H(t-t_i)},$$

one has

$$\left.\frac{\delta Z}{\delta J_+^\alpha(t)}\right|_0 = \mathrm{i}\ \mathrm{Tr}(\rho_0 O_\alpha(t)), \qquad \left.\frac{\delta Z}{\delta J_-^\alpha(t)}\right|_0 = -\mathrm{i}\ \mathrm{Tr}(\rho_0 O_\alpha(t)).$$

Higher derivatives give the four contour orderings. With two insertions,

$$\frac{1}{\mathrm{i}^2} \left.\frac{\delta^2 Z}{\delta J_+^\alpha(t) \delta J_+^\beta(t')}\right|_0 = \mathrm{Tr}\big(\rho_0\ \mathrm{T}\, O_\alpha(t) O_\beta(t')\big)\,, \tag{126.4}$$

$$\frac{1}{(-\mathrm{i})^2} \left.\frac{\delta^2 Z}{\delta J_-^\alpha(t) \delta J_-^\beta(t')}\right|_0 = \mathrm{Tr}\Big(\rho_0\ \widetilde{\mathrm{T}} O_\alpha(t) O_\beta(t')\Big)\,, \tag{126.5}$$

$$\frac{1}{\mathrm{i}(-\mathrm{i})} \left.\frac{\delta^2 Z}{\delta J_+^\alpha(t) \delta J_-^\beta(t')}\right|_0 = \mathrm{Tr}\big(\rho_0\, O_\beta(t') O_\alpha(t)\big)\,, \tag{126.6}$$

$$\frac{1}{(-\mathrm{i})\mathrm{i}} \left.\frac{\delta^2 Z}{\delta J_-^\alpha(t) \delta J_+^\beta(t')}\right|_0 = \mathrm{Tr}\big(\rho_0\, O_\alpha(t) O_\beta(t')\big)\,. \tag{126.7}$$

These formulae are finite trace identities. In a continuum QFT the same symbols require a regulator, an operator topology, and a limiting procedure.

126.3 Unitarity, Reality, and Positivity Constraints

Setting the two sources equal gives

$$Z[J, J] = \mathrm{Tr}(U_J \rho_0 U_J^\dagger) = \mathrm{Tr}\,\rho_0 = 1. \tag{126.8}$$

This is the closed-time-path normalization constraint.

Basic Schwinger–Keldysh constraints. For real sources in the finite system above, unitarity and positivity give

$$Z[J, J] = 1, \tag{126.9}$$

$$Z[J_+, J_-]^* = Z[J_-, J_+], \tag{126.10}$$

$$|Z[J_+, J_-]| \leq 1. \tag{126.11}$$

If $Z = \exp(\mathrm{i}W)$ is defined by the branch with $W[0,0] = 0$, then near the origin

$$W[J, J] = 0, \qquad W[J_+, J_-]^* = -W[J_-, J_+], \qquad \mathrm{Im}\, W[J_+, J_-] \geq 0.$$

The first identity is (126.8). For real sources U_J is unitary, and complex conjugation of the trace gives

$$Z[J_+, J_-]^* = \mathrm{Tr}(U_{J_-}\rho_0 U_{J_+}^\dagger) = Z[J_-, J_+].$$

For the bound, write

$$Z[J_+, J_-] = \mathrm{Tr}\Big(\rho_0\, U_{J_-}^\dagger\, U_{J_+}\Big)\,.$$

The operator $V = U_{J_-}^\dagger\, U_{J_+}$ is unitary. Since ρ_0 is a positive normalized density matrix, $Z = \omega_{\rho_0}(V)$ is an expectation value of a unitary in a state. Cauchy–Schwarz for the positive inner product

$$(A, B)_{\rho_0} = \mathrm{Tr}(\rho_0 A^\dagger B)$$

gives

$$|Z|^2 \le \mathrm{Tr}(\rho_0 \mathbf{1})\,\mathrm{Tr}(\rho_0 V^\dagger V) = 1.$$

The statements about W follow from $Z = \exp(\mathrm{i}W)$ on the branch connected to $Z = 1$.

Equation (126.8) implies that all derivatives of Z or W with respect only to a common physical source vanish on the diagonal. This is the origin of the rule that every local term in a Schwinger–Keldysh effective action contains at least one difference field.

Largest-time cancellation. Let $T \in (t_i, t_f)$. If $J_+(t) = J_-(t) = J(t)$ for $T \le t \le t_f$, then $Z[J_+, J_-]$ is independent of the common future source J on that interval. Consequently every contour correlator in which the latest-time insertion is generated by variation of the common physical source cancels after summing over the two branches.

Factor the two evolution operators at T:

$$U_{J_\pm}(t_f, t_i) = U_J(t_f, T) U_{J_\pm}(T, t_i),$$

because the two sources agree on $[T, t_f]$. Substitution in (126.1) gives

$$Z[J_+, J_-] = \mathrm{Tr}\Big(U_J(t_f, T) U_{J_+}(T, t_i)\rho_0 U_{J_-}(T, t_i)^\dagger U_J(t_f, T)^\dagger\Big)\,.$$

Cyclicity of the trace and $U_J(t_f, T)^\dagger U_J(t_f, T) = \mathbf{1}$ give

$$Z[J_+, J_-] = \mathrm{Tr}\Big(U_{J_+}(T, t_i)\rho_0 U_{J_-}(T, t_i)^\dagger\Big)\,,$$

which contains no source on $[T, t_f]$. Functional differentiation with respect to such a common future source therefore gives zero. Taking T just below the latest insertion gives the stated largest-time identity. This is the finite trace form of the largest-time cancellation rule; the continuum use of the rule inherits the regulator and limiting assumptions of the closed-time-path functional.

126.4 The r/a Basis and Retarded Response

Introduce average and difference sources

$$J_{\mathrm{r}}^\alpha = \frac{1}{2}(J_+^\alpha + J_-^\alpha), \qquad J_{\mathrm{a}}^\alpha = J_+^\alpha - J_-^\alpha.$$

Equivalently,

$$J_+^\alpha = J_\mathrm{r}^\alpha + \frac{1}{2}J_\mathrm{a}^\alpha, \qquad J_-^\alpha = J_\mathrm{r}^\alpha - \frac{1}{2}J_\mathrm{a}^\alpha.$$

The connected generating functional is

$$W[J_\mathrm{r}, J_\mathrm{a}] = -\mathrm{i}\log Z[J_\mathrm{r}, J_\mathrm{a}], \qquad W[0,0] = 0.$$

The expectation value in the physical source J_r is

$$\langle O_\alpha(t)\rangle_{J_\mathrm{r}} = \left.\frac{\delta W}{\delta J_\mathrm{a}^\alpha(t)}\right|_{J_\mathrm{a}=0}. \tag{126.12}$$

At zero source this follows from the one-point identities above:

$$\left.\frac{\delta Z}{\delta J_\mathrm{a}^\alpha(t)}\right|_0 = \frac{1}{2}\left(\mathrm{i}\langle O_\alpha(t)\rangle + \mathrm{i}\langle O_\alpha(t)\rangle\right) = \mathrm{i}\langle O_\alpha(t)\rangle.$$

Source-response kernel from r/a differentiation. With the source convention $H_J = H - J^\alpha O_\alpha$,

$$\frac{\delta}{\delta J_\mathrm{r}^\beta(t')}\left.\frac{\delta W}{\delta J_\mathrm{a}^\alpha(t)}\right|_{J_\mathrm{r}=J_\mathrm{a}=0} = K_{\alpha\beta}^R(t,t'), \tag{126.13}$$

where

$$K_{\alpha\beta}^R(t,t') = \mathrm{i}\,\theta(t-t')\ \mathrm{Tr}\big(\rho_0[O_\alpha(t), O_\beta(t')]\big). \tag{126.14}$$

If the commutator-retarded convention is

$$G_{\alpha\beta}^{R,\mathrm{comm}}(t,t') = -\mathrm{i}\,\theta(t-t')\ \mathrm{Tr}\big(\rho_0[O_\alpha(t), O_\beta(t')]\big),$$

then $K_{\alpha\beta}^R = -G_{\alpha\beta}^{R,\mathrm{comm}}$. The sign is fixed by the source coupling $H_J = H - J^\gamma O_\gamma$, with summation over $\gamma \in \mathcal{I}$.

Let J_r be the same physical source on both branches. To first order in the source, the interaction-picture evolution gives

$$U_J(t,t_i) = U_0(t,t_i)\left[1 + \mathrm{i}\int_{t_i}^t ds\, J_\mathrm{r}^\beta(s) O_\beta(s) + O(J_\mathrm{r}^2)\right].$$

Inserting this in the Heisenberg expectation of $O_\alpha(t)$ gives

$$\begin{aligned}\delta\langle O_\alpha(t)\rangle &= \mathrm{i}\int_{t_i}^t ds\, J_\mathrm{r}^\beta(s)\ \mathrm{Tr}(\rho_0[O_\alpha(t), O_\beta(s)]) \\ &= \int ds\, J_\mathrm{r}^\beta(s)\ \big[\mathrm{i}\theta(t-s)\,\mathrm{Tr}(\rho_0[O_\alpha(t), O_\beta(s)])\big].\end{aligned}$$

Combining this linear-response identity with (126.12) gives the displayed derivative of W. The derivation is a finite-system sign check for the source convention. Its continuum counterpart is the regulated linear-response identity for the renormalized source and operator pair.

The same statement can be read directly from contour ordering. The derivative with respect to J_a inserts the physical operator whose expectation value is measured; the derivative with respect to J_r inserts the perturbation. The cancellation between the two contour branches removes all terms in which the perturbation is later than the measured operator, leaving the causal source-response commutator $\mathrm{i}\theta(t-t')\,\mathrm{Tr}(\rho_0[O_\alpha(t), O_\beta(t')])$.

126.5 Two-Point Matrix

For a bosonic Hermitian operator O, define

$$G^{>}(t,t') = \mathrm{Tr}(\rho_0 O(t) O(t')), \qquad G^{<}(t,t') = \mathrm{Tr}(\rho_0 O(t') O(t)).$$

The contour-ordered functions are

$$G_{++}(t,t') = \theta(t-t')G^{>}(t,t') + \theta(t'-t)G^{<}(t,t'), \tag{126.15}$$
$$G_{--}(t,t') = \theta(t'-t)G^{>}(t,t') + \theta(t-t')G^{<}(t,t'), \tag{126.16}$$
$$G_{-+}(t,t') = G^{>}(t,t'), \tag{126.17}$$
$$G_{+-}(t,t') = G^{<}(t,t'). \tag{126.18}$$

In the r/a basis the independent linear-response and fluctuation data are

$$G^{R,\mathrm{comm}}(t,t') = -\mathrm{i}\theta(t-t')\big(G^{>}(t,t') - G^{<}(t,t')\big),$$

$$G^{A,\mathrm{comm}}(t,t') = \mathrm{i}\theta(t'-t)\big(G^{>}(t,t') - G^{<}(t,t')\big),$$

$$G^{K}(t,t') = G^{>}(t,t') + G^{<}(t,t').$$

The component with two difference-type operator insertions vanishes:

$$G^{\mathrm{aa}}(t,t') = 0.$$

This is another form of $Z[J_\mathrm{r}, J_\mathrm{a} = 0] = 1$. The rr component is the symmetrized fluctuation, while ra and ar contain the causal response kernels. With the source convention $H - J^\alpha O_\alpha$, the physical source-response kernel is the negative of $G^{R,\mathrm{comm}}$.

126.6 Thermal KMS Constraint on the Contour

Let

$$\rho_\beta = Z(\beta)^{-1}\mathrm{e}^{-\beta H}.$$

For operators whose thermal strip analytic continuation is controlled, cyclicity of the trace gives

$$\mathrm{Tr}(\rho_\beta A(t) B(0)) = \mathrm{Tr}(\rho_\beta B(0) A(t+\mathrm{i}\beta)).$$

In frequency space this becomes detailed balance:

$$G^{<}(\omega) = \mathrm{e}^{-\beta\omega} G^{>}(\omega)$$

for bosonic operators in the convention of Chapter 124. Therefore

$$G^{K}(\omega) = \coth\!\left(\frac{\beta\omega}{2}\right)\big(G^{>}(\omega) - G^{<}(\omega)\big).$$

Since

$$G^{R,\mathrm{comm}}(\omega) - G^{A,\mathrm{comm}}(\omega) = -\mathrm{i}\big(G^{>}(\omega) - G^{<}(\omega)\big)$$

with the commutator-retarded convention above, the fluctuation-dissipation relation is

$$G^K(\omega) = \mathrm{i}\coth\left(\frac{\beta\omega}{2}\right)\left(G^{R,\mathrm{comm}}(\omega) - G^{A,\mathrm{comm}}(\omega)\right).$$

For fermionic operators the graded KMS condition replaces the hyperbolic cotangent by the hyperbolic tangent in the corresponding anticommutator fluctuation function.

Thermal Schwinger–Keldysh path integrals often implement this condition by adding an imaginary Euclidean segment of length β to the contour or, in effective theory, by imposing a dynamical KMS symmetry. Both formulations express the same operator statement: the initial density matrix is the Gibbs operator and the required strip analyticity exists.

126.7 Coordinate Path Integrals

A Schwinger–Keldysh path integral is a coordinate representation of (126.1). For a regulated bosonic configuration variable q, inserting coordinate resolutions of identity on both sides of the trace gives two histories

$$q_+(t), \qquad q_-(t),$$

with a final-time gluing condition implementing the trace:

$$q_+(t_f) = q_-(t_f).$$

The initial density matrix contributes its coordinate kernel

$$\rho_0(q_+(t_i), q_-(t_i)).$$

In formal continuum notation,

$$Z[J_+, J_-] = \int_{\mathrm{SK}} Dq_+ Dq_-\ \rho_0(q_+(t_i), q_-(t_i))\ \exp\left\{\mathrm{i}S[q_+] - \mathrm{i}S[q_-] + \mathrm{i}\int J_+ O[q_+] - \mathrm{i}\int J_- O[q_-]\right\}.$$

This formula has the same logical status as every path integral in the monograph: its meaning is supplied by the finite regulator or constructive definition from which it is obtained. For thermal initial data, the kernel ρ_β may be represented by the Euclidean thermal path integral of Chapter 125, producing the familiar contour with an imaginary thermal segment.

126.8 Gauge Theories and BV Data on the Closed Time Path

For a gauge theory, branch doubling must be applied to the full gauge-fixed or BV field space. A perturbative BV representation has two copies

$$\Phi_+, \Phi_+^*, \qquad \Phi_-, \Phi_-^*,$$

and a regulated exponent whose local part has the form

$$\mathrm{i}S_{\mathrm{BV}}[\Phi_+, \Phi_+^*] - \mathrm{i}S_{\mathrm{BV}}[\Phi_-, \Phi_-^*] + \mathrm{i}J_+ \cdot O[\Phi_+] - \mathrm{i}J_- \cdot O[\Phi_-]$$

plus the initial density matrix and the final gluing of physical variables. Gauge consistency is the statement that the doubled measure, action, sources, and boundary conditions satisfy the appropriate quantum BV master equation and preserve the diagonal identity $Z[J, J] = 1$. In a lattice gauge regulator the same physical constraints are expressed without gauge fixing by two copies of link variables on the closed time path, projection to gauge-invariant states or Gauss-law sectors, and final gluing inside the trace.

The real-time continuum construction is therefore stronger than writing a formal doubled action. It must define the initial state, operator domains, source space, contour gluing, gauge constraints, and limiting topology in which unitarity, positivity, KMS relations, and retarded support survive.

Open Problem 126.2 (Real-time continuum construction)*.* Construct Schwinger–Keldysh generating functionals for interacting continuum QFTs in a topology strong enough to derive retarded correlators, KMS relations, and transport limits. The construction must state the initial state, operator domains, regulator, contour matching, gauge or BV data when present, and the sense in which unitarity and positivity survive the continuum limit.

Chapter 127

Spectral Functions, Kubo Formulae, and Transport

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Transport coefficients are low-frequency limits of real-time response functions. The definitions use a thermal state, a conserved current or stress tensor, and a source coupled to the corresponding operator. The derivation is linear response from the Schwinger–Keldysh generating functional of Chapter 126. This chapter keeps separate three pieces that are often conflated: the commutator spectral density, possible local contact terms in the source response, and the order of limits needed to define a transport coefficient.

127.1 Retarded Functions and Spectral Densities

Let ρ_β be a KMS state at inverse temperature β. For bosonic Hermitian operators $A(t, \mathbf{x})$ and $B(t, \mathbf{x})$, define the retarded function

$$G^R_{AB}(t, \mathbf{x}) = -\mathrm{i}\,\Theta(t)\,\langle [A(t, \mathbf{x}), B(0, \mathbf{0})]\rangle_\beta \tag{127.1}$$

and the commutator spectral density

$$\rho_{AB}(t, \mathbf{x}) = \langle [A(t, \mathbf{x}), B(0, \mathbf{0})]\rangle_\beta. \tag{127.2}$$

Fourier transforms are defined distributionally by

$$G^R_{AB}(\omega,\mathbf{k}) = \int dt\, d^{D-1}x\, e^{\mathrm{i}\omega t - \mathrm{i}\mathbf{k}\cdot\mathbf{x}} G^R_{AB}(t,\mathbf{x}). \tag{127.3}$$

With this convention, a retarded distribution that obeys the required tempered-growth hypothesis is the boundary value of a function holomorphic for $\operatorname{Im}\omega > 0$.

Finite-volume Lehmann representation. Work in a finite spatial volume. Let $H|n\rangle = E_n|n\rangle$, let $Z = \sum_n e^{-\beta E_n} < \infty$, and let $A_{\mathbf{k}}$ obey $A^\dagger_{\mathbf{k}} = A_{-\mathbf{k}}$. Then

$$G^>_{AA}(\omega,\mathbf{k}) = \frac{2\pi}{Z}\sum_{m,n} e^{-\beta E_m} |\langle m|A_{\mathbf{k}}|n\rangle|^2 \delta(\omega - E_n + E_m), \tag{127.4}$$

and

$$\rho_{AA}(\omega,\mathbf{k}) = \frac{2\pi}{Z}\sum_{m,n} e^{-\beta E_m}\left(1 - e^{-\beta\omega}\right)|\langle m|A_{\mathbf{k}}|n\rangle|^2 \delta(\omega - E_n + E_m). \tag{127.5}$$

Consequently $\omega\,\rho_{AA}(\omega,\mathbf{k}) \geq 0$ as a distributional positivity statement. Indeed, inserting a complete set of energy eigenstates into

$$G^>_{AA}(t,\mathbf{k}) = Z^{-1}\operatorname{Tr}\left(e^{-\beta H} A_{\mathbf{k}}(t) A_{-\mathbf{k}}(0)\right).$$

Since

$$A_{\mathbf{k}}(t) = e^{\mathrm{i}Ht} A_{\mathbf{k}} e^{-\mathrm{i}Ht},$$

the matrix element carries the phase

$$e^{\mathrm{i}(E_m - E_n)t}\langle m|A_{\mathbf{k}}|n\rangle\langle n|A_{-\mathbf{k}}|m\rangle.$$

The adjoint relation makes the product the squared modulus displayed in (127.4), and the Fourier integral gives the delta function $\delta(\omega - E_n + E_m)$.

The lesser function has the same delta support with the Boltzmann weight $e^{-\beta E_n}$ after relabelling $m \leftrightarrow n$. On the support of the delta function,

$$E_n - E_m = \omega, \qquad e^{-\beta E_n} = e^{-\beta\omega} e^{-\beta E_m}.$$

Subtracting $G^<$ from $G^>$ gives (127.5). For $\omega > 0$ every coefficient in (127.5) is nonnegative; for $\omega < 0$ it is nonpositive. This is exactly $\omega\rho_{AA} \geq 0$.

Retarded boundary value and sign convention. Assume the finite-volume spectral measure above has a thermodynamic limit for which the displayed integrals remain meaningful as tempered boundary values. With the retarded convention (127.1), the nonlocal part of the retarded function is

$$G^R_{AA}(z,\mathbf{k}) = \int_{\mathbb{R}} \frac{d\omega'}{2\pi}\,\frac{\rho_{AA}(\omega',\mathbf{k})}{z - \omega'}, \qquad \operatorname{Im} z > 0, \tag{127.6}$$

up to real polynomial contact terms. The corresponding boundary-value convention is

$$\rho_{AA}(\omega,\mathbf{k}) = -2\,\operatorname{Im} G^R_{AA}(\omega + \mathrm{i}0,\mathbf{k}) \tag{127.7}$$

for the nonlocal part of the response. This sign is fixed by writing the inverse Fourier representation of ρ in (127.1): for $\operatorname{Im} z > 0$,

$$-\mathrm{i}\int_0^\infty dt\, e^{\mathrm{i}zt} e^{-\mathrm{i}\omega' t} = \frac{1}{z-\omega'}.$$

This gives (127.6). Taking $z = \omega + \mathrm{i}0$ and using

$$\frac{1}{\omega + \mathrm{i}0 - \omega'} = \mathrm{PV}\,\frac{1}{\omega-\omega'} - \mathrm{i}\pi\delta(\omega-\omega')$$

gives (127.7). Local source counterterms add real polynomials in ω and $\mathbf{k}$ to the real-time kernel in the absence of dissipation; they are not part of the commutator spectral measure.

127.2 KMS and Fluctuation–Dissipation

Define the Wightman functions

$$G^>_{AB}(t) = \langle A(t)B(0)\rangle_\beta, \qquad G^<_{AB}(t) = \langle B(0)A(t)\rangle_\beta.$$

The KMS condition gives

$$G^>_{AB}(t-\mathrm{i}\beta) = G^<_{AB}(t)$$

for operators in the analytic domain of the thermal state. In frequency space this implies detailed balance. For $A = B^\dagger$, the finite-volume formula above gives

$$G^<_{AA}(\omega,\mathbf{k}) = e^{-\beta\omega} G^>_{AA}(\omega,\mathbf{k}), \qquad \rho_{AA} = G^>_{AA} - G^<_{AA}. \tag{127.8}$$

Therefore the symmetrized correlator

$$G^{\mathrm{sym}}_{AA}(\omega,\mathbf{k}) = \frac{1}{2}\big(G^>_{AA}(\omega,\mathbf{k}) + G^<_{AA}(\omega,\mathbf{k})\big)$$

satisfies

$$G^{\mathrm{sym}}_{AA}(\omega,\mathbf{k}) = \frac{1}{2}\coth\!\left(\frac{\beta\omega}{2}\right)\rho_{AA}(\omega,\mathbf{k}). \tag{127.9}$$

The apparent singularity at $\omega = 0$ is interpreted distributionally: the product is meaningful when the small-ω behavior of ρ_{AA} is specified. In a diffusive or viscous channel, $\rho_{AA}(\omega) = O(\omega)$, so the symmetrized noise has a finite zero-frequency limit. If a ballistic channel produces a Drude delta function, this limit is a different observable and the dc transport coefficient is infinite rather than a finite number.

127.3 Linear Response from the Closed Time Path

Perturb the Hamiltonian by a source $h_B(t,\mathbf{x})$ coupled to B:

$$H(t) = H_0 - \int d^{D-1}x\, h_B(t,\mathbf{x})B(t,\mathbf{x}). \tag{127.10}$$

To first order in h_B, the interaction-picture evolution operator is

$$U_h(t,-\infty) = 1 + \mathrm{i}\int_{-\infty}^{t} dt' \int d^{D-1}x'\, h_B(t',\mathbf{x}')B(t',\mathbf{x}') + O(h_B^2).$$

Consequently

$$U_h(t,-\infty)^{-1}A(t,\mathbf{x})U_h(t,-\infty) = A(t,\mathbf{x}) + \mathrm{i}\int_{-\infty}^{t} dt'\, d^{D-1}x'\, h_B(t',\mathbf{x}')[A(t,\mathbf{x}),B(t',\mathbf{x}')] + O(h_B^2).$$

Taking the thermal expectation value gives

$$\delta\langle A(t,\mathbf{x})\rangle = -\int dt' d^{D-1}x'\, G^R_{AB}(t-t',\mathbf{x}-\mathbf{x}')h_B(t',\mathbf{x}'). \tag{127.11}$$

Equivalently, the source-response kernel for the convention $H = H_0 - h_B B$ is

$$K^R_{AB} = \mathrm{i}\Theta(t)\langle[A(t),B(0)]\rangle_\beta = -G^R_{AB}.$$

This is the branch-difference identity obtained from the Schwinger–Keldysh functional in Chapter 126: differentiating with respect to the a-type source and then setting $J_a = 0$ leaves the causal source-response commutator. Transport formulae below are stated in terms of the commutator spectral density ρ_{AB}, which is independent of this source-sign convention.

127.4 Contact Terms and Response Kernels

The commutator retarded function is not always the full response kernel to a background source. If the operator itself depends on the source, or if the regulated generating functional contains local source counterterms, then

$$\delta\langle A\rangle = (K^R_{AB} + C_{AB})\, h_B$$

in Fourier space, where $C_{AB}(\omega,\mathbf{k})$ is local in spacetime and therefore polynomial, or a distributional limit of polynomials, in $\omega,\mathbf{k}$. For gauge and metric sources these terms are not optional: they are fixed by Ward identities, equilibrium thermodynamics, and the chosen operator scheme.

The dissipative transport coefficient is extracted from the spectral slope of the nonlocal commutator part. Real local contact terms can change static susceptibilities and Euclidean zero-mode correlators, but they do not change the positive spectral measure ρ_{AA} at nonzero frequency. This is why Kubo formulae below are stated using ρ, with a separate warning about Drude weights and conserved-density projections.

127.5 Transport Limits and Zero-Frequency Singular Terms

Transport is a statement about an infinite system. In finite spatial volume the spectrum of the Hamiltonian is discrete, so real-time correlators are almost periodic sums of phases rather than decaying functions. The thermodynamic limit is therefore part of the definition of a finite dissipative coefficient.

Definition 127.1 (Transport limit datum)**.** A transport limit datum for a pair of local operators A, B consists of

$$(\Sigma_L, \rho_{\beta,L}, A_L, B_L, \mathcal{R}, \mathcal{O}),$$

where Σ_L is a sequence of spatial regions or lattices of volume V_L, $\rho_{\beta,L}$ is a KMS or finite Gibbs state on the regulated system, A_L, B_L are regulated extensive or density operators with their normalization declared, $\mathcal{R}$ is the prescription for subtracting local contact terms and conserved-density projections, and $\mathcal{O}$ is the order of limits. For ordinary dc transport the order is

$$L \to \infty, \qquad \mathbf{k} \to \mathbf{0} \text{ in the channel specified by symmetry}, \qquad \omega \downarrow 0. \tag{127.12}$$

Changing $\mathcal{O}$ can change the answer; for example static susceptibilities and dc conductivities use different zero-frequency limits.

The most elementary obstruction to a finite dc coefficient is an exactly conserved operator that has nonzero overlap with the current. Let I_i be the total current in direction i in finite volume V, and let $\bar{I}_i = I_i - \langle I_i \rangle_\beta$. Suppose $\{Q_a\}_{a=1}^N$ are Hermitian conserved operators with zero thermal expectation and positive covariance matrix

$$C_{ab} = \frac{1}{2V} \langle Q_a Q_b + Q_b Q_a \rangle_\beta. \tag{127.13}$$

Define the current–charge overlap vector

$$v_{ia} = \frac{1}{2V} \langle \bar{I}_i Q_a + Q_a \bar{I}_i \rangle_\beta. \tag{127.14}$$

The projection of $\bar{I}_i$ onto the span of conserved charges is

$$P\bar{I}_i = Q_a (C^{-1})^{ab} v_{ib}. \tag{127.15}$$

Lemma 127.2 (Finite-volume Mazur projection)**.** *In the finite-dimensional Gibbs setting above, for every real vector u_i, the infinite-time averaged connected symmetrized current autocorrelation satisfies*

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T dt \, \frac{1}{2V} \left\langle \{u_i \bar{I}_i(t), u_j \bar{I}_j(0)\} \right\rangle_\beta \geq u_i v_{ia} (C^{-1})^{ab} v_{jb} u_j. \tag{127.16}$$

If the right hand side has a nonzero thermodynamic limit, the conductivity has a zero-frequency singular contribution and the ordinary dc limit is not a finite dissipative number.

Proof. Use the real Hilbert-space inner product on Hermitian operators

$$(X, Y)_{\beta,\mathrm{sym}} = \frac{1}{2V} \langle XY + YX \rangle_\beta.$$

Time evolution is orthogonal for this inner product because ρ_β commutes with H. For $X = u_i \bar{I}_i$, let

$$X_\parallel = Q_a (C^{-1})^{ab} (Q_b, X)_{\beta,\mathrm{sym}}.$$

Then $X_\parallel$ is conserved and is the orthogonal projection of X onto the conserved subspace spanned by the Q_a. The time average of the symmetrized correlation is the squared norm of the component of X in the zero-frequency subspace of the Liouville operator $L = [H, \cdot]$. This zero-frequency subspace contains $X_\parallel$, hence the time average is at least

$$(X_\parallel, X_\parallel)_{\beta,\mathrm{sym}} = u_i v_{ia} (C^{-1})^{ab} v_{jb} u_j.$$

If this quantity survives per unit volume as $V \to \infty$, the symmetrized current spectrum contains a term

$$\frac{2\pi}{\beta} D_{ij} \delta(\omega), \qquad D_{ij} = \beta \lim_{t\to\infty} \lim_{V\to\infty} \frac{1}{2V} \langle \{\bar{I}_i(t), \bar{I}_j(0)\} \rangle_\beta, \tag{127.17}$$

in the convention of (127.9). Equivalently $\operatorname{Re}\sigma_{ij}$ contains $\pi D_{ij}\delta(\omega)$. This is a singular transport sector, not the regular slope appearing in the finite dc Kubo formula. □

After the thermodynamic limit has been taken, the low-frequency real conductivity in a fixed tensor channel must be regarded as a distribution. The separation used below is

$$\operatorname{Re}\sigma_{ij}(\omega) = \pi D_{ij}\delta(\omega) + \sigma^{\mathrm{reg}}_{ij}(\omega), \qquad \sigma^{\mathrm{reg}}_{ij}(\omega) = \frac{1}{2\omega} \rho^{\mathrm{reg}}_{J_i J_j}(\omega, \mathbf{0}) \quad (\omega \neq 0), \tag{127.18}$$

where D_{ij} is the Drude matrix produced by conserved projections and $\sigma^{\mathrm{reg}}_{ij}$ is locally integrable near $\omega = 0$. The ordinary dissipative dc coefficient is

$$\sigma^{\mathrm{dc}}_{ij} = \lim_{\omega\downarrow 0} \sigma^{\mathrm{reg}}_{ij}(\omega) = \lim_{\omega\downarrow 0} \frac{1}{2\omega} \rho^{\mathrm{reg}}_{J_i J_j}(\omega, \mathbf{0}), \tag{127.19}$$

when this limit exists. The singular term is a separate distributional summand, while σ^{reg} is the locally integrable part used in the finite dc limit. Figure 127.1 records this order-of-limits prescription.

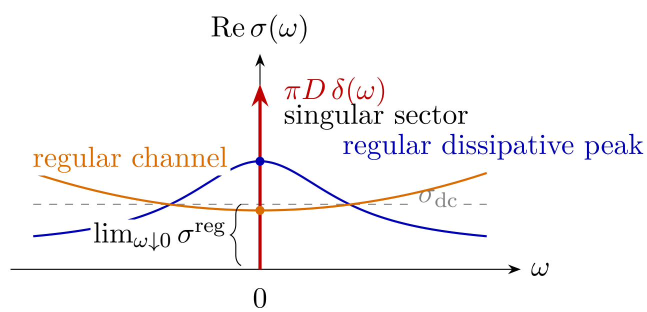

Figure 127.1: Distributional low-frequency content of a conductivity after the thermodynamic limit. The finite dc coefficient is the zero-frequency limit of the regular part σ^{reg}, equivalently the slope of $\rho^{\mathrm{reg}}_{JJ}(\omega, \mathbf{0})$ divided by 2ω. A Drude or superfluid weight is represented by the separate $\delta(\omega)$ summand.

127.6 Projection Identities and Memory Kernels

Kubo formulae identify transport coefficients as low-frequency response data. A second finite-regulator identity explains where hydrodynamic closures enter. Work in finite volume and let

$\mathcal{A}_L$ be the finite-dimensional operator space generated by the regulated degrees of freedom in the channel under study. Time evolution acts on this space by

$$X(t) = e^{t\mathcal{L}} X, \qquad \mathcal{L}X = \mathrm{i}[H_L, X],$$

so that $dX(t)/dt = \mathcal{L}X(t)$. For linearly independent Hermitian operators A_a with zero thermal expectation define the Kubo–Mori inner product

$$(X, Y)_\beta = \frac{1}{\beta} \int_0^\beta d\lambda \left\langle X^\dagger(-\mathrm{i}\lambda) Y \right\rangle_\beta, \qquad X(-\mathrm{i}\lambda) = e^{\lambda H_L} X e^{-\lambda H_L}. \tag{127.20}$$

Let

$$\chi_{ab} = (A_a, A_b)_\beta$$

and assume that χ is positive definite on the chosen slow subspace. The orthogonal projection onto this subspace is

$$PX = A_a (\chi^{-1})^{ab} (A_b, X)_\beta, \qquad Q = 1 - P. \tag{127.21}$$

The projection depends on the regulated state and on the declared slow operators. In hydrodynamic applications A_a are Fourier modes of conserved densities; in a transport calculation they may also include exactly conserved charges whose overlap creates the Drude sector isolated in Lemma 127.2.

Exact finite-regulator block identity. Let $X(t) = e^{t\mathcal{L}} X_0$. On the finite operator space above,

$$\begin{aligned} \frac{d}{dt} PX(t) = P\mathcal{L}P\, PX(t) + P\mathcal{L}e^{tQ\mathcal{L}Q} QX_0 \\ + \int_0^t ds\, P\mathcal{L}e^{(t-s)Q\mathcal{L}Q} Q\mathcal{L}P\, PX(s). \end{aligned} \tag{127.22}$$

Consequently, if X_0 lies in the range of P, its slow coordinates

$$\alpha^a(t) = (\chi^{-1})^{ab} (A_b, PX(t))_\beta$$

obey

$$\dot{\alpha}^b(t) = \Omega_a{}^b\, \alpha^a(t) - \int_0^t ds\, M_a{}^b(t-s) \alpha^a(s), \tag{127.23}$$

where

$$\Omega_a{}^b = (\chi^{-1})^{bc} (A_c, \mathcal{L}A_a)_\beta, \tag{127.24}$$

$$M_a{}^b(t) = -(\chi^{-1})^{bc} \left(A_c, \mathcal{L}e^{tQ\mathcal{L}Q} Q\mathcal{L}A_a\right)_\beta. \tag{127.25}$$

For a general X_0, the additional term

$$f_X^b(t) = (\chi^{-1})^{bc} \left(A_c, \mathcal{L}e^{tQ\mathcal{L}Q} QX_0\right)_\beta$$

must be added to the right hand side of (127.23).

To obtain the identity, write $x(t) = PX(t)$ and $y(t) = QX(t)$. Applying P and Q to $\dot{X} = \mathcal{L}X$ gives the exact block system

$$\dot{x} = P\mathcal{L}P\, x + P\mathcal{L}Q\, y, \qquad \dot{y} = Q\mathcal{L}P\, x + Q\mathcal{L}Q\, y.$$

The second equation is a finite-dimensional inhomogeneous linear equation. Duhamel's formula gives

$$y(t) = e^{tQ\mathcal{L}Q}QX_0 + \int_0^t ds\, e^{(t-s)Q\mathcal{L}Q}Q\mathcal{L}P\, x(s).$$

Substitution into the first block equation gives (127.22). If $X_0 = A_a\alpha^a(0)$, then $QX_0 = 0$ and

$$P\mathcal{L}P\, PX(t) = A_b\Omega_a{}^b\alpha^a(t).$$

The integral term is written in the same A_b basis by applying $(\chi^{-1})^{bc}(A_c,\cdot)_\beta$, which gives (127.23) with the definition (127.25). For a general initial operator the Duhamel term involving QX_0 is projected in the same way and produces $f_X^b(t)$.

The identity is algebraic at finite regulator. It has no decay statement in it: the unresolved evolution $e^{tQ\mathcal{L}Q}$ can be almost periodic in finite volume, and the memory kernel can fail to have a long-time integral before the thermodynamic limit is taken. A hydrodynamic closure is therefore an additional scaling statement about the kernel $M(t)$. In a conserved density channel the Ward identity gives

$$\partial_t n_A(t,\mathbf{k}) = -\mathrm{i}k_i J_A^i(t,\mathbf{k}).$$

If the slow variables are the densities $n_A(\mathbf{k})$, then $Q\mathcal{L}n_A(\mathbf{k})$ is proportional to k_i times the part of the current orthogonal to the density subspace. Hence the memory term begins at order k^2:

$$\partial_t n_A(t,\mathbf{k}) = -k^2\int_0^t ds\, \mathsf{D}_{AB}(t-s,\mathbf{k})n_B(s,\mathbf{k}) + \text{reactive terms and higher-gradient corrections.} \tag{127.26}$$

Replacing the kernel by a local diffusion matrix,

$$\int_0^t ds\, \mathsf{D}_{AB}(t-s,\mathbf{k})n_B(s,\mathbf{k}) \rightsquigarrow D_{AB}n_B(t,\mathbf{k}), \qquad D_{AB} = \int_0^\infty dt\, \mathsf{D}_{AB}(t,\mathbf{0}),$$

requires a thermodynamic limit, decay of the projected current kernel, control of the $k \to 0$ limit, and prior separation of Drude sectors. When these hypotheses hold, the resulting matrix D is the diffusion matrix that appears in the hydrodynamic pole $\omega = -\mathrm{i}\lambda k^2 + \cdots$ and in the Einstein relation $D = \Sigma\chi^{-1}$ derived in Chapter 128. If the projected kernel has a persistent component or a nonintegrable long-time tail, the local first-order constitutive relation is replaced by a singular or scale-dependent low-frequency response. Chapter 134 develops the hydrodynamic version of this infrared phenomenon.

127.6.1 The Closure Datum Behind a Diffusion Pole

The finite-regulator identity (127.22) becomes a transport statement only after a limiting datum is supplied. Let Λ_L be a sequence of spatial volumes tending to $\mathbb{R}^{d-1}$, let $n_A^L(\mathbf{k})$ be the regulated Fourier modes of conserved densities at momenta compatible with Λ_L, and let $\mathsf{D}_{AB}^L(t,\mathbf{k})$ denote the projected kernel obtained from (127.25) after the Drude sector and exact zero modes have been removed. A local diffusive closure consists of the following data.

Hypothesis 127.3 (Projected-current hydrodynamic closure datum)**.** For $\mathbf{k}$ in a neighborhood of 0 there is a matrix-valued locally integrable function $\mathsf{D}_{AB}(t,\mathbf{k})$ such that:

(i) for every compact time interval $[0,T]$ and every test matrix $u^{AB}(t,\mathbf{k})$ smooth in t and $\mathbf{k}$,

$$\lim_{L\to\infty}\int_0^T dt\,u^{AB}(t,\mathbf{k})\mathsf{D}^L_{AB}(t,\mathbf{k})=\int_0^T dt\,u^{AB}(t,\mathbf{k})\mathsf{D}_{AB}(t,\mathbf{k});$$

(ii) the Laplace transform

$$\widehat{\mathsf{D}}_{AB}(z,\mathbf{k})=\int_0^\infty e^{-zt}\mathsf{D}_{AB}(t,\mathbf{k})\,dt,\qquad \operatorname{Re}z>0,\tag{127.27}$$

has a finite limit at $(z,\mathbf{k})=(0,\mathbf{0})$ after the prescribed subtraction of singular conserved pieces, and admits a holomorphic continuation to a neighborhood of this point in the z-plane with

$$\widehat{\mathsf{D}}(z,\mathbf{k})=\widehat{\mathsf{D}}(0,\mathbf{0})+O(|z|+|\mathbf{k}|)$$

in the finite-dimensional charge-coordinate norm;

(iii) the finite-volume initial-force term $P\mathcal{L}e^{tQ\mathcal{L}Q}QX_0$ has no contribution to the hydrodynamic pole under the same thermodynamic and small-$\mathbf{k}$ limit, either because the initial perturbation lies in the slow subspace or because the projected force term decays in the limiting topology;

(iv) the susceptibility matrix $\chi_{AB}(\mathbf{k})=(n_A(\mathbf{k}),n_B(-\mathbf{k}))_\beta$ has a nonsingular limit $\chi_{AB}=\chi_{AB}(\mathbf{0})$ on the declared hydrodynamic charge sector.

These conditions are the theorem boundary for transport: in a concrete QFT one must prove them, or else keep them as hypotheses. Under Hypothesis 127.3, the memory equation has a precise small-momentum meaning. For simplicity suppress reactive terms and write the density equation in the form

$$\partial_t n_A(t,\mathbf{k})=-k^2\int_0^t ds\,\mathsf{D}_{AB}(t-s,\mathbf{k})n_B(s,\mathbf{k})+O(k^3)\tag{127.28}$$

in the thermodynamic limit.

In the following displays A,B are charge-coordinate labels. The position of an index marks row or column placement in a matrix product and is not a metric raising operation. The Laplace transform of the density memory equation is

$$\Big[z\delta_A{}^B+k^2\widehat{\mathsf{D}}_A{}^B(z,\mathbf{k})\Big]\,\widetilde{n}_B(z,\mathbf{k})=n_A(0,\mathbf{k})+O(k^3),\qquad \operatorname{Re}z>0.\tag{127.29}$$

If $D_A{}^B=\widehat{\mathsf{D}}_A{}^B(0,\mathbf{0})$ is diagonalizable on the charge sector and λ_r is one of its eigenvalues, analytic perturbation of the finite matrix

$$z\mathbf{1}+k^2\widehat{\mathsf{D}}(z,\mathbf{k})$$

gives a pole of the density response at

$$z_r(\mathbf{k})=-k^2\lambda_r+o(k^2).\tag{127.30}$$

With the real-time Fourier convention of (127.3), $z=-\mathrm{i}\omega$, this is the hydrodynamic pole

$$\omega_r(\mathbf{k})=-\mathrm{i}\lambda_r k^2+o(k^2).\tag{127.31}$$

The construction of the diffusion pole therefore has three distinct steps: construct the thermodynamic projected-current kernel, prove the zero-frequency regularity of its Laplace transform, and then expand the resolvent of the closed slow-coordinate equation.

The conductivity matrix entering the Einstein relation is the same datum expressed in current variables. Let Σ_{AB} denote the regular zero-frequency limit of the current response after subtracting the Drude sector and contact terms. In a purely diffusive charge channel,

$$\Sigma_{AB} = D_A{}^C \chi_{CB}. \tag{127.32}$$

This relation follows from the continuity equation: an external static source changes the equilibrium charge density by $\delta n_A = \chi_{AB}\delta\mu^B$, while the hydrodynamic constitutive response to the slowly varying source is

$$J^i_A = -D_A{}^B \partial_i \delta n_B = -\Sigma_{AB}\partial_i\delta\mu^B.$$

Equation (127.32) is therefore a statement about the same limiting kernel seen in two coordinate systems, not an independent postulate.

127.7 Conductivity

For a conserved current J^μ coupled to a background gauge field $A^{\rm ext}_\mu$, the electric field in the Fourier convention (127.3) is

$$E_i(\omega,\mathbf{k}) = \mathrm{i}\omega A^{\rm ext}_i(\omega,\mathbf{k}) - \mathrm{i}k_i A^{\rm ext}_0(\omega,\mathbf{k}).$$

At zero spatial momentum and in a gauge with $A^{\rm ext}_0 = 0$, define the full transverse response kernel K_{ij} by the conventional background-gauge-field sign

$$\langle J_i(\omega)\rangle = -K_{ij}(\omega,\mathbf{0})A^{\rm ext}_j(\omega) = \sigma_{ij}(\omega)E_j(\omega).$$

Thus

$$\sigma_{ij}(\omega) = -\frac{1}{\mathrm{i}\omega}K_{ij}(\omega,\mathbf{0}), \qquad K_{ij} = G^R_{J_iJ_j} + C_{ij}.$$

For $\omega \neq 0$, the dissipative real part is determined by the spectral density:

$$\operatorname{Re}\sigma_{ij}(\omega) = \frac{1}{2\omega}\,\rho_{J_iJ_j}(\omega,\mathbf{0}) \quad \text{in a transverse channel after contact terms are separated.} \tag{127.33}$$

The dc conductivity, when finite, is therefore

$$\sigma^{\rm dc}_{ij} = \lim_{\omega\downarrow 0}\frac{1}{2\omega}\rho_{J_iJ_j}(\omega,\mathbf{0}),$$

with the thermodynamic limit taken before the $\omega\to 0$ limit. If momentum conservation or another exactly conserved quantity overlaps with the current, a $\delta(\omega)$ Drude contribution is present and the dc conductivity is infinite unless the conserved contribution is relaxed or projected out.

127.8 Shear and Bulk Viscosity

Couple the theory to a spatial metric perturbation

$$g_{ij}(t, \mathbf{x}) = \delta_{ij} + h_{ij}(t, \mathbf{x}).$$

The stress tensor is defined by the first variation of the regulated generating functional with respect to the metric source. In an isotropic thermal state, take a shear perturbation $h_{xy}(t)$ at zero spatial momentum. The dissipative part of the stress response is characterized by the shear viscosity η. With the retarded sign convention (127.1) and the spectral convention (127.7),

$$\eta = \lim_{\omega \downarrow 0} \frac{1}{2\omega}\, \rho_{T_{xy}T_{xy}}(\omega, \mathbf{0}) = -\lim_{\omega \downarrow 0} \frac{1}{\omega}\, \operatorname{Im} G^R_{T_{xy}T_{xy}}(\omega, \mathbf{0}). \tag{127.34}$$

The equality between the two expressions is exactly (127.7). Positivity of $\omega \rho_{T_{xy}T_{xy}}$ gives $\eta \geq 0$ when the limit exists and no singular zero-frequency contribution is present.

The bulk channel requires a projection. For a neutral relativistic fluid in $d = D - 1$ spatial dimensions, define

$$\mathcal{P} = \frac{1}{d} T^i{}_i - c_s^2 T^{00}, \qquad c_s^2 = \left(\frac{\partial p}{\partial \varepsilon}\right)_{\mathrm{eq}}.$$

The subtraction removes the overlap of the pressure fluctuation with the conserved energy density. In charged fluids one must project out all conserved densities using the susceptibility matrix. After this projection, the bulk viscosity is

$$\zeta = \lim_{\omega \downarrow 0} \frac{1}{2\omega} \rho_{\mathcal{P}\mathcal{P}}(\omega, \mathbf{0}),$$

again with the thermodynamic and zero-momentum limits taken before the zero-frequency limit.

127.9 Dispersion Relations and Sum-Rule Boundaries

Retardedness gives analyticity in the upper half ω-plane. If the response kernel is sufficiently bounded after subtracting its local polynomial terms, then Cauchy's theorem gives a subtracted dispersion relation. A typical once-subtracted form is

$$G^R(\omega) - G^R(0) = \omega \int_{\mathbb{R}} \frac{d\omega'}{2\pi}\, \frac{\rho(\omega')}{\omega'(\omega - \omega' + \mathrm{i}0)}. \tag{127.35}$$

This formula is not a definition. It follows only after the large semicircle is controlled and after all contact terms have been declared. Sum rules for conductivity or viscosity compare the small-ω transport slope with Euclidean or equal-time data only under these additional analytic and ultraviolet assumptions.

127.10 Hydrodynamic Pole Boundary

Kubo formulae define transport coefficients when the indicated limits exist and commute with the thermodynamic and zero-momentum limits in the stated order. Hydrodynamics begins after these coefficients are inserted into the long-wavelength Ward identities for conserved densities. The microscopic Kubo formula by itself does not prove local equilibration, the derivative expansion, or the dominance of hydrodynamic poles; those are additional dynamical statements developed in the following chapters.

Open Problem 127.4 (Rigorous Kubo-to-hydrodynamics bridge). Give sufficient QFT hypotheses under which retarded stress-tensor and current correlators have the analyticity, spectral positivity, clustering, and low-frequency limits required by the Kubo formulae, and prove that the same coefficients control the long-time hydrodynamic effective theory.

Chapter 128

Hydrodynamics from Ward Identities

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Hydrodynamics is the long-wavelength effective theory of conserved densities. Its variables are local thermodynamic parameters attached to a family of states whose correlators are controlled, at long distances and long times, by Ward identities and by an expansion in gradients. This chapter develops the deterministic first-order theory in flat spacetime and fixes the sign and normalization conventions that connect it to the Kubo formulae of Chapter 127.

Let $D = d + 1$, where d is the number of spatial dimensions. The Minkowski metric is mostly plus, $\eta_{\mu\nu} = (-, +, \dots, +)$. Repeated ordinary charge labels A, B are summed. All formulas in this chapter are written for parity-even fluids without anomalies; anomalous transport is developed separately in Chapter 132.

128.1 Background Sources and Ward Identities

The conserved operators that enter hydrodynamics are defined by their response to background sources. Let $g_{\mu\nu}$ be a Lorentzian metric and let A_μ^A be background gauge fields for ordinary Abelian internal symmetries. For the regulated Schwinger–Keldysh generating functional

$$W[g_+, A_+; g_-, A_-],$$

the physical one-point functions are obtained by differentiating with respect to the difference sources and then setting the two branches equal, as in Chapter 126. On the diagonal source

$g_+ = g_- = g$, $A_+ = A_- = A$, we use the convention

$$\delta W_{\mathrm{phys}} = \frac{1}{2}\int d^D x\,\sqrt{-g}\,\langle T^{\mu\nu}\rangle\,\delta g_{\mu\nu} + \int d^D x\,\sqrt{-g}\,\langle J^\mu_A\rangle\,\delta A^A_\mu. \tag{128.1}$$

Equation (128.1) is a definition of the stress-tensor and current normalization in this chapter. Local contact terms may change the local polynomial part of the response kernels, but they do not change the operator Ward identities derived from gauge and diffeomorphism invariance.

Gauge invariance under $\delta_\lambda A^A_\mu = \partial_\mu \lambda^A$ gives

$$\nabla_\mu J^\mu_A = 0 \tag{128.2}$$

when the corresponding symmetry is not anomalous. Diffeomorphism invariance under the infinitesimal vector field ξ^μ gives

$$\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu, \qquad \delta_\xi A^A_\mu = \xi^\lambda F^A_{\lambda\mu} + \partial_\mu(\xi^\lambda A^A_\lambda),$$

where $F^A_{\mu\nu} = \partial_\mu A^A_\nu - \partial_\nu A^A_\mu$. Combining this variation with (128.2) and integrating by parts gives

$$\nabla_\mu T^\mu{}_\nu = F^A_{\nu\lambda} J^\lambda_A. \tag{128.3}$$

Thus the flat, source-free conservation laws are the specialization of the QFT Ward identities to $g_{\mu\nu} = \eta_{\mu\nu}$ and $A^A_\mu = 0$.

The same source formulation fixes the thermodynamic forces used below. For a fluid velocity u^μ, define the electric covector measured in the local rest frame by

$$\mathcal{E}^A_\mu = F^A_{\mu\nu} u^\nu, \qquad u^\mu \mathcal{E}^A_\mu = 0.$$

The source-free formulas in most of this chapter are recovered by setting $\mathcal{E}^A_\mu = 0$.

128.2 Hydrodynamic Scaling Families

The hydrodynamic expansion is an asymptotic statement about a family of states and sources whose variation scale is taken large compared with all microscopic relaxation scales except the modes protected by exact conservation laws. The following definition records the data that are being approximated.

Definition 128.1 (Hydrodynamic scaling family)**.** A hydrodynamic scaling family near a homogeneous KMS state consists of regulated real-time states ρ_λ, background sources $(g_{\lambda\mu\nu}, A^A_{\lambda\mu})$, and local fields $(T_\lambda, \mu_{\lambda A}, u^\mu_\lambda)$, indexed by $\lambda > 0$, such that in local inertial coordinates

$$\partial_{\alpha_1}\cdots\partial_{\alpha_m} T_\lambda, \quad \partial_{\alpha_1}\cdots\partial_{\alpha_m}\mu_{\lambda A}, \quad \partial_{\alpha_1}\cdots\partial_{\alpha_m} u^\mu_\lambda, \quad \partial_{\alpha_1}\cdots\partial_{\alpha_{m-1}} F^A_{\lambda\alpha_m\alpha_{m+1}} = O(\lambda^m) \qquad (m \geq 1),$$

while the zeroth-order fields remain in a fixed compact subset of the equilibrium state manifold with $T > 0$. A constitutive relation through order N is an asymptotic expansion of the one-point functions $\langle T^{\mu\nu}\rangle_{\rho_\lambda}$ and $\langle J^\mu_A\rangle_{\rho_\lambda}$ whose error is $O(\lambda^{N+1})$ after the thermodynamic limit and after subtraction of declared local contact terms.

This definition separates the kinematic part of hydrodynamics from the microscopic QFT assertion. The kinematic part is the classification of all local tensor structures compatible with the Ward identities and the chosen state variables. The microscopic assertion is that nonconserved excitations produce analytic terms in the scaling region, so that the nonanalytic long-distance response is exhausted by the conservation laws.

Definition 128.2 (First-order hydrodynamic constitutive structure)**.** A parity-even first-order hydrodynamic constitutive structure for ordinary Abelian charges is a tuple

$$\mathfrak{D}_{\rm hydro} = \big(W[g_+, A_+; g_-, A_-], T^{\mu\nu}, J^\mu_A, \mathcal{M}_{\rm eq}, T, \mu_A, u^\mu, \mathsf{f}, \mathsf{C}^{(0)}, \mathsf{C}^{(1)}, \mathsf{E}\big)$$

with the following entries.

(i) $W[g_+, A_+; g_-, A_-]$ is the regulated Schwinger–Keldysh generating functional of the microscopic QFT with background metric $g_{\mu\nu}$ and Abelian background gauge fields A^A_μ. The operators $T^{\mu\nu}$ and J^μ_A use the source normalization of (128.1).

(ii) The exact QFT input is the Ward-identity pair

$$\nabla_\mu J^\mu_A = 0, \qquad \nabla_\mu T^\mu{}_\nu = F^A_{\nu\lambda} J^\lambda_A,$$

with $F^A_{\mu\nu} = \partial_\mu A^A_\nu - \partial_\nu A^A_\mu$.

(iii) $\mathcal{M}_{\rm eq}$ is the thermodynamic manifold of homogeneous KMS states in the thermodynamic limit. Its local coordinates are temperature $T > 0$, chemical potentials μ_A, and a future-timelike unit vector u^μ satisfying $u^\mu u_\mu = -1$.

(iv) f is the hydrodynamic frame, namely the prescription for extending (T, μ_A, u^μ) away from exact equilibrium. This chapter uses Landau matching:

$$u_\mu T^{\mu\nu} = -\varepsilon u^\nu, \qquad n_A = -u_\mu J^\mu_A.$$

(v) $\mathsf{C}^{(0)}$ is the ideal constitutive map

$$T^{\mu\nu}_{(0)} = \varepsilon u^\mu u^\nu + p\Delta^{\mu\nu}, \qquad J^\mu_{A(0)} = n_A u^\mu, \qquad \Delta^{\mu\nu} = \eta^{\mu\nu} + u^\mu u^\nu.$$

The thermodynamic functions obey $dp = s\,dT + n_A\,d\mu_A$ and $\varepsilon + p = Ts + \mu_A n_A$.

(vi) $\mathsf{C}^{(1)}$ is the first-gradient constitutive map in the declared frame. In the source-free parity-even sector treated here its independent transport data are shear viscosity η, bulk viscosity ζ, and conductivity matrix Σ_{AB}, appearing as

$$\pi^{\mu\nu}_{(1)} = -\eta\sigma^{\mu\nu} - \zeta\Delta^{\mu\nu}\partial_\lambda u^\lambda, \qquad \nu^\mu_{A(1)} = -T\Sigma_{AB}\Delta^{\mu\nu}\partial_\nu(\mu_B/T).$$

(vii) E is the entropy-current and positivity condition. It consists of $S^\mu = su^\mu + O(\partial)$ and the requirement that the quadratic entropy-production form be nonnegative for all first gradients. In this sector the condition is

$$\eta \geq 0, \qquad \zeta \geq 0, \qquad \Sigma_{(AB)} v^A v^B \geq 0 \quad \text{for all } v^A \in \mathbb{R}.$$

Here $\Sigma_{(AB)} = (\Sigma_{AB} + \Sigma_{BA})/2$.

The structure $\mathfrak{D}_{\rm hydro}$ is evaluated on a hydrodynamic scaling family in the sense of Definition 128.1. The sections below derive the ideal equations, frame transformations, first-order entropy production, and hydrodynamic poles from this structure.

128.3 Conserved Densities and Local State Variables

Let $T^{\mu\nu}$ be the stress tensor and let J_A^μ be conserved currents for ordinary internal symmetries. In flat space without external sources,

$$\partial_\mu T^{\mu\nu} = 0, \qquad \partial_\mu J_A^\mu = 0. \tag{128.4}$$

A local equilibrium state is parametrized by a future-timelike velocity field $u^\mu(x)$, temperature $T(x) > 0$, and chemical potentials $\mu_A(x)$, with

$$u^\mu u_\mu = -1.$$

The spatial projector in the local rest frame is

$$\Delta^{\mu\nu} = \eta^{\mu\nu} + u^\mu u^\nu, \qquad \Delta^{\mu\nu} u_\nu = 0.$$

The ideal constitutive relations are

$$T_{(0)}^{\mu\nu} = \varepsilon u^\mu u^\nu + p\,\Delta^{\mu\nu}, \qquad J_{A(0)}^\mu = n_A u^\mu. \tag{128.5}$$

Here $\varepsilon(T,\mu)$, $p(T,\mu)$, and $n_A(T,\mu)$ are local thermodynamic functions of the underlying QFT. The frame used below is the Landau frame:

$$u_\mu T^{\mu\nu} = -\varepsilon u^\nu, \tag{128.6}$$

so the nonideal stress tensor carries no energy flux in the local rest frame. Charge diffusion is allowed to have components orthogonal to u^μ.

128.4 Hydrodynamic Frames

Hydrodynamic variables are local coordinates on the manifold of equilibrium states and its derivative expansion. They are not uniquely defined away from strict equilibrium. For any chosen u^μ, decompose the one-point functions as

$$T^{\mu\nu} = \varepsilon u^\mu u^\nu + p\Delta^{\mu\nu} + u^\mu q^\nu + u^\nu q^\mu + \pi^{\mu\nu}, \tag{128.7}$$
$$J_A^\mu = n_A u^\mu + \nu_A^\mu, \tag{128.8}$$

with

$$u_\mu q^\mu = u_\mu \nu_A^\mu = 0, \qquad u_\mu \pi^{\mu\nu} = 0.$$

The scalar quantities ε, p, n_A, the velocity u^μ, and the transverse tensors $q^\mu, \nu_A^\mu, \pi^{\mu\nu}$ depend on the chosen frame. Under a first-derivative redefinition

$$u^\mu \mapsto u^\mu + \delta u^\mu, \qquad T \mapsto T + \delta T, \qquad \mu_A \mapsto \mu_A + \delta\mu_A, \qquad u_\mu \delta u^\mu = 0,$$

while the physical one-point functions are held fixed, the transverse vectors change at first derivative order as

$$q^\mu \mapsto q^\mu - (\varepsilon + p)\delta u^\mu, \qquad \nu_A^\mu \mapsto \nu_A^\mu - n_A \delta u^\mu. \tag{128.9}$$

The scalar redefinitions change the separation between ideal scalar data and first-order scalar corrections. Landau matching is the choice

$$\varepsilon = u_\mu u_\nu T^{\mu\nu}, \qquad n_A = -u_\mu J_A^\mu,$$

so that no first-order correction is hidden inside the definitions of ε and n_A. To derive (128.9), keep $T^{\mu\nu}$ and J_A^μ fixed and vary only the ideal pieces. Since $\delta\Delta^{\mu\nu} = u^\mu\delta u^\nu + \delta u^\mu u^\nu$, the ideal stress changes by

$$\delta T_{(0)}^{\mu\nu} = (\varepsilon + p)(u^\mu \delta u^\nu + \delta u^\mu u^\nu) + \text{scalar variations}.$$

Its transverse energy-flux part is therefore $(\varepsilon+p)\delta u^\mu$. The ideal current changes by $\delta J_{A(0)}^\mu = n_A \delta u^\mu$ in the transverse sector. The nonideal vectors must shift by the negatives of these transverse changes. The Landau frame condition (128.6) is the choice $q^\mu = 0$. If $\varepsilon + p \neq 0$, it is reached by $\delta u^\mu = q^\mu/(\varepsilon + p)$ to the order considered here. This coordinate choice changes the displayed form of charge diffusion but cannot change pole locations, Kubo coefficients, or source-response functions.

128.5 Thermodynamic Closure

The equation of state supplies a function $p = p(T, \mu_A)$. In a finite spatial volume, before the thermodynamic limit, it is obtained from the grand-canonical partition function

$$Z_{\beta,\mu} = \operatorname{Tr} \exp[-\beta(H - \mu_A Q_A)], \qquad p_{\beta,\mu,V} = \frac{T}{V} \log Z_{\beta,\mu}.$$

The pressure $p(T, \mu)$ used in hydrodynamics is the thermodynamic limit of $p_{\beta,\mu,V}$, assuming this limit exists and is differentiable in the state under consideration. Differentiating the finite-volume expression gives

$$\left(\frac{\partial p}{\partial \mu_A}\right)_T = \frac{\langle Q_A \rangle}{V} \equiv n_A,$$

and

$$\left(\frac{\partial p}{\partial T}\right)_\mu = \frac{p + \varepsilon - \mu_A n_A}{T} \equiv s,$$

where $\varepsilon = \langle H \rangle / V$. Thus

$$p = p(T, \mu_A), \qquad s = \left(\frac{\partial p}{\partial T}\right)_\mu, \qquad n_A = \left(\frac{\partial p}{\partial \mu_A}\right)_T$$

are not independent definitions: they are the differentiability data of the equilibrium KMS state. Consequently,

$$s = \left(\frac{\partial p}{\partial T}\right)_\mu, \qquad n_A = \left(\frac{\partial p}{\partial \mu_A}\right)_{T,\mu_{B\neq A}},$$

and

$$\varepsilon + p = Ts + \mu_A n_A. \tag{128.10}$$

Then

$$dp = s\,dT + n_A\,d\mu_A, \qquad d\varepsilon = T\,ds + \mu_A\,dn_A. \tag{128.11}$$

In the rest frame of a homogeneous equilibrium state, spatial rotations give

$$\langle T^{00}\rangle = \varepsilon, \qquad \langle T^{0i}\rangle = 0, \qquad \langle T^{ij}\rangle = p\,\delta^{ij}, \qquad \langle J_A^0\rangle = n_A, \qquad \langle J_A^i\rangle = 0.$$

Lorentz covariance then gives the ideal tensors (128.5) in a state whose local rest frame velocity is u^μ. This derivation is local in the sense of the gradient expansion: it assumes that, on each hydrodynamic cell, the state is approximated by a homogeneous KMS state with the displayed values of T, μ_A, u^μ. These identities are the local thermodynamic closure of the ideal Ward system. They are imposed pointwise on the slowly varying local state variables; the microscopic QFT input is the equilibrium equation of state and the assumption that the hydrodynamic scaling limit probes only this local state manifold.

128.6 Ideal Equations and Entropy Conservation

Remark 128.3 (Ideal hydrodynamic equations). Substituting (128.5) into the Ward identities gives

$$u^\mu\partial_\mu\varepsilon + (\varepsilon+p)\partial_\mu u^\mu = 0, \tag{128.12}$$
$$(\varepsilon+p)u^\mu\partial_\mu u^\alpha + \Delta^{\alpha\mu}\partial_\mu p = 0, \tag{128.13}$$
$$u^\mu\partial_\mu n_A + n_A\partial_\mu u^\mu = 0. \tag{128.14}$$

Contract $\partial_\mu T^{\mu\nu}_{(0)} = 0$ with u_ν. Since

$$u_\nu\partial_\mu(\varepsilon u^\mu u^\nu) = -\partial_\mu(\varepsilon u^\mu), \qquad u_\nu\partial_\mu(p\Delta^{\mu\nu}) = -p\,\partial_\mu u^\mu,$$

using $u_\nu\partial_\mu u^\nu = \frac{1}{2}\partial_\mu(u_\nu u^\nu) = 0$, one obtains (128.12). Projecting the same Ward identity with $\Delta^\alpha{}_\nu$ gives

$$\Delta^\alpha{}_\nu\partial_\mu(\varepsilon u^\mu u^\nu) = \varepsilon u^\mu\partial_\mu u^\alpha, \qquad \Delta^\alpha{}_\nu\partial_\mu(p\Delta^{\mu\nu}) = p\,u^\mu\partial_\mu u^\alpha + \Delta^{\alpha\mu}\partial_\mu p,$$

which is (128.13). Current conservation gives (128.14) directly.

Remark 128.4 (Ideal entropy current). The ideal entropy current

$$S^\mu_{(0)} = su^\mu$$

is conserved on ideal hydrodynamic solutions:

$$\partial_\mu S^\mu_{(0)} = 0.$$

Using the first law,

$$Tu^\mu\partial_\mu s = u^\mu\partial_\mu\varepsilon - \mu_A u^\mu\partial_\mu n_A.$$

Substitute the ideal energy equation (128.12) and the ideal charge equations (128.14). This gives

$$Tu^\mu\partial_\mu s = -(\varepsilon+p)\partial_\mu u^\mu + \mu_A n_A\partial_\mu u^\mu = -Ts\,\partial_\mu u^\mu,$$

where $\varepsilon + p = Ts + \mu_A n_A$ was used. Therefore

$$\partial_\mu(su^\mu) = u^\mu\partial_\mu s + s\partial_\mu u^\mu = 0.$$

128.7 First-Order Constitutive Relations

The first-order derivative data must be defined before the transport coefficients are introduced. Define the expansion, acceleration, and shear tensor by

$$\vartheta = \partial_\mu u^\mu, \qquad a^\mu = u^\nu \partial_\nu u^\mu,$$

and

$$\sigma^{\mu\nu} = 2\Delta^{\mu\alpha}\Delta^{\nu\beta}\left(\partial_{(\alpha}u_{\beta)} - \frac{1}{d}\Delta_{\alpha\beta}\vartheta\right). \tag{128.15}$$

Then $\sigma^{\mu\nu}u_\nu = 0$, $\sigma^\mu{}_\mu = 0$, and $\sigma^{\mu\nu} = \sigma^{\nu\mu}$. The antisymmetric transverse derivative is the vorticity

$$\omega^{\mu\nu} = \Delta^{\mu\alpha}\Delta^{\nu\beta}\partial_{[\alpha}u_{\beta]}, \qquad \omega^{\mu\nu} = -\omega^{\nu\mu}.$$

It does not contribute to the parity-even dissipative stress tensor at first order because the stress tensor is symmetric and the entropy production is quadratic in the symmetric thermodynamic forces.

With sources, the ideal Euler equation is obtained by projecting (128.3) orthogonally to u^μ:

$$(\varepsilon + p)a^\mu + \Delta^{\mu\nu}\partial_\nu p = n_A \mathcal{E}^{A\mu}. \tag{128.16}$$

Using $dp = s\,dT + n_A\,d\mu_A$ and $\varepsilon + p = Ts + \mu_A n_A$, this equation is equivalently

$$a^\mu + \Delta^{\mu\nu}\partial_\nu \log T = \frac{Tn_A}{\varepsilon + p}V_A^\mu, \qquad V_A^\mu = \Delta^{\mu\nu}\left[\frac{\mathcal{E}_\nu^A}{T} - \partial_\nu\left(\frac{\mu_A}{T}\right)\right]. \tag{128.17}$$

Thus acceleration and the transverse temperature gradient are not independent first-order data once the ideal equations are used. The independent parity-even dissipative structures in Landau frame are therefore the symmetric traceless tensor $\sigma^{\mu\nu}$, the scalar ϑ, and the transverse charge forces V_A^μ. This is the classification statement behind the constitutive relations below; it is not an additional dynamical assumption.

In Landau frame, the parity-even first-order constitutive relations are

$$T^{\mu\nu} = \varepsilon u^\mu u^\nu + p\Delta^{\mu\nu} - \eta\,\sigma^{\mu\nu} - \zeta\,\Delta^{\mu\nu}\vartheta + O(\partial^2), \tag{128.18}$$
$$J_A^\mu = n_A u^\mu + \nu_A^\mu + O(\partial^2). \tag{128.19}$$

In the absence of external electric fields,

$$\nu_A^\mu = -T\,\Sigma_{AB}\Delta^{\mu\nu}\partial_\nu\left(\frac{\mu_B}{T}\right). \tag{128.20}$$

With background gauge fields, the same convention is

$$\nu_A^\mu = T\,\Sigma_{AB}\Delta^{\mu\nu}\left[\frac{\mathcal{E}_\nu^B}{T} - \partial_\nu\left(\frac{\mu_B}{T}\right)\right]. \tag{128.21}$$

The matrix Σ_{AB} is the conductivity matrix in the normalization where, at constant T and zero background electric field, Fick diffusion has diffusion matrix

$$D_{AB} = \Sigma_{AC}(\chi^{-1})_{CB}, \qquad \chi_{AB} = \left(\frac{\partial n_A}{\partial \mu_B}\right)_T.$$

The entropy analysis below constrains the symmetric part of Σ_{AB}. In a parity-even state with no time-reversal-breaking background, the KMS condition together with microscopic time reversal gives the Onsager reciprocity $\Sigma_{AB} = \Sigma_{BA}$. If time reversal or parity is broken, an antisymmetric part may appear; it is nondissipative because $\Sigma_{[AB]}X_A X_B = 0$ for any real thermodynamic force X_A, and it belongs with Hall-type transport rather than with the dissipative parity-even sector treated in this chapter.

128.8 Entropy Production at First Order

The first-order entropy current in Landau frame is

$$S^\mu = su^\mu - \frac{\mu_A}{T}\nu_A^\mu + O(\partial^2), \qquad \nu_A^\mu = -T\Sigma_{AB}\Delta^{\mu\nu}\partial_\nu\Big(\frac{\mu_B}{T}\Big) \quad (A_\mu^A = 0). \tag{128.22}$$

Entropy production fixes positivity. Using (128.18) and (128.19), the entropy production through second order in gradients is

$$\partial_\mu S^\mu = \frac{\eta}{2T}\sigma^{\mu\nu}\sigma_{\mu\nu} + \frac{\zeta}{T}\vartheta^2 + T\,\Sigma_{AB}\Delta^{\mu\nu}\partial_\mu\Big(\frac{\mu_A}{T}\Big)\,\partial_\nu\Big(\frac{\mu_B}{T}\Big) + O(\partial^3). \tag{128.23}$$

Hence local entropy production is nonnegative for arbitrary first gradients if

$$\eta \geq 0, \qquad \zeta \geq 0, \qquad \Sigma_{AB} \text{ is positive semidefinite.}$$

Write the dissipative stress tensor as

$$\tau^{\mu\nu} = -\eta\sigma^{\mu\nu} - \zeta\Delta^{\mu\nu}\vartheta, \qquad T^{\mu\nu} = T^{\mu\nu}_{(0)} + \tau^{\mu\nu}.$$

Using the Ward identities, the thermodynamic identities, and $u_\mu\tau^{\mu\nu} = u_\mu\nu_A^\mu = 0$, the divergence of (128.22) through second order reduces to

$$\partial_\mu S^\mu = -\frac{1}{T}\tau^{\mu\nu}\partial_\mu u_\nu - \nu_A^\mu\partial_\mu\Big(\frac{\mu_A}{T}\Big) + O(\partial^3).$$

With background gauge fields, the second term is replaced by

$$\nu_A^\mu\left[\frac{\mathcal{E}_\mu^A}{T} - \partial_\mu\Big(\frac{\mu_A}{T}\Big)\right],$$

which follows from the force term (128.3) in the energy equation. The source-free formula used in (128.23) is the special case $\mathcal{E}_\mu^A = 0$. The decomposition of $\partial_\mu u_\nu$ into shear, trace, vorticity, and acceleration gives

$$\sigma^{\mu\nu}\partial_\mu u_\nu = \frac{1}{2}\sigma^{\mu\nu}\sigma_{\mu\nu}, \qquad \Delta^{\mu\nu}\partial_\mu u_\nu = \vartheta.$$

Substituting $\tau^{\mu\nu}$ and ν_A^μ gives (128.23).

128.9 Charge Diffusion and Susceptibility Geometry

Consider a state with $u^\mu = (1, \mathbf{0})$, constant T, and vanishing background flow. Linearize first a single conserved charge around equilibrium. The constitutive relation gives

$$J^0 = n_0 + \delta n, \qquad J^i = -\Sigma\, \partial_i \delta\mu, \qquad \delta n = \chi\, \delta\mu.$$

Charge conservation gives

$$\partial_t \delta n - D\nabla^2 \delta n = 0, \qquad D = \frac{\Sigma}{\chi}. \tag{128.24}$$

For a plane wave $\delta n \propto e^{-\mathrm{i}\omega t + \mathrm{i}\mathbf{k}\cdot\mathbf{x}}$, the pole is

$$\omega = -\mathrm{i}Dk^2 + O(k^4). \tag{128.25}$$

The Einstein relation $D = \Sigma/\chi$ is the hydrodynamic form of the same coefficient matching that appears in the Kubo conductivity formula.

For several charges, the susceptibility matrix is the Hessian of the equilibrium pressure with respect to chemical potentials,

$$\chi_{AB} = \left(\frac{\partial n_A}{\partial \mu_B}\right)_T = \frac{\partial^2 p}{\partial \mu_A \partial \mu_B}.$$

Thermodynamic stability of the homogeneous KMS state means that χ is positive definite on the subspace of charges under consideration. In a parity-even time-reversal-invariant fluid, Σ is symmetric positive semidefinite. The linearized equations at constant temperature are

$$\partial_t \delta n_A - D_{AB}\nabla^2 \delta n_B = 0, \qquad D = \Sigma\chi^{-1}. \tag{128.26}$$

Although D need not be symmetric in the ordinary Euclidean inner product, its spectrum is nonnegative. Indeed D is similar to

$$\chi^{-1/2}\Sigma\chi^{-1/2},$$

because

$$D = \chi^{1/2}\left(\chi^{-1/2}\Sigma\chi^{-1/2}\right)\chi^{-1/2}.$$

The middle matrix is symmetric positive semidefinite. Hence the diffusive normal modes have

$$\omega_i(\mathbf{k}) = -\mathrm{i}\lambda_i k^2 + O(k^4), \qquad \lambda_i \geq 0,$$

where λ_i are the eigenvalues of D. Strictly positive λ_i require that no linear combination of the charges has zero conductivity.

The density source-response matrix also fixes the order of matrix multiplication in diffusion formulae. Couple a source h_A to the density with the Hamiltonian convention $H = H_0 - \int h_A n_A$, so that the electrochemical force is proportional to $\nabla(\delta\mu_A - h_A)$. In Fourier space the linearized equation is

$$\left(Dk^2 - \mathrm{i}\omega\,\mathbf{1}\right)_{AB}\delta n_B = k^2\Sigma_{AB}h_B.$$

Therefore the nonlocal hydrodynamic part of the retarded density source-response kernel is

$$K^R_{n_A n_B}(\omega, \mathbf{k}) = k^2\left(Dk^2 - \mathrm{i}\omega\,\mathbf{1}\right)^{-1}_{AC}\Sigma_{CB} + \text{analytic contact terms}. \tag{128.27}$$

At $\omega = 0$ and then $k \to 0$, the nonlocal term gives

$$D^{-1}\Sigma = \chi,$$

assuming Σ is invertible on the diffusive subspace. This is the matrix version of the static susceptibility matching in (128.35).

128.10 Neutral Linearized Modes

Linearize a neutral fluid around a homogeneous rest state:

$$u^\mu = (1, \mathbf{v}) + O(\mathbf{v}^2), \qquad \varepsilon = \varepsilon_0 + \delta\varepsilon, \qquad p = p_0 + \delta p, \qquad \delta p = c_s^2 \delta\varepsilon,$$

where

$$c_s^2 = \left(\frac{\partial p}{\partial \varepsilon}\right)_{\rm eq}, \qquad w = \varepsilon_0 + p_0.$$

The stress tensor components to first order in fluctuations and first order in gradients are

$$T^{00} = \varepsilon_0 + \delta\varepsilon, \tag{128.28}$$

$$T^{0i} = w v^i, \tag{128.29}$$

$$T^{ij} = p_0\delta^{ij} + c_s^2 \delta\varepsilon\, \delta^{ij} - \eta\left(\partial_i v_j + \partial_j v_i - \frac{2}{d}\delta_{ij}\partial_\ell v^\ell\right) - \zeta \delta_{ij}\partial_\ell v^\ell. \tag{128.30}$$

The Ward identities become

$$\partial_t \delta\varepsilon + w\partial_i v^i = 0, \tag{128.31}$$

$$w\partial_t v_i + \partial_i \delta p - \eta\nabla^2 v_i - \left(\zeta + \eta\left(1 - \frac{2}{d}\right)\right)\partial_i\partial_j v^j = 0. \tag{128.32}$$

Shear and sound poles. For fluctuations proportional to $e^{-\mathrm{i}\omega t + \mathrm{i}\mathbf{k}\cdot\mathbf{x}}$, the transverse velocity mode has

$$\omega_{\rm shear} = -\mathrm{i}\,\frac{\eta}{w}k^2 + O(k^4). \tag{128.33}$$

The longitudinal energy-momentum modes obey

$$\omega_{{\rm sound},\pm} = \pm c_s k - \frac{\mathrm{i}}{2}\Gamma_s k^2 + O(k^3), \qquad \Gamma_s = \frac{\zeta + 2\eta\frac{d-1}{d}}{w}. \tag{128.34}$$

For a transverse mode, $k_i v^i = 0$ and (128.32) becomes

$$-\mathrm{i}\omega w v_i + \eta k^2 v_i = 0,$$

which gives (128.33).

For a longitudinal mode take $\mathbf{k} = k\hat{x}$ and $v_i = v_L \hat{x}_i$. Equations (128.31) and (128.32) give

$$-\mathrm{i}\omega\delta\varepsilon + \mathrm{i}wkv_L = 0,$$

and

$$-\mathrm{i}\omega w v_L + \mathrm{i}kc_s^2\delta\varepsilon + \left(\zeta + 2\eta\frac{d-1}{d}\right)k^2 v_L = 0.$$

Eliminating v_L gives

$$\omega^2 - c_s^2 k^2 + \mathrm{i}\frac{\zeta + 2\eta(d-1)/d}{w}\omega k^2 = 0.$$

Solving perturbatively at small k gives (128.34).

128.11 Retarded Source-Response Pole Matching

Hydrodynamic poles are singularities of retarded source-response kernels after the thermodynamic limit and the hydrodynamic scaling limit have been taken in the specified order. For the source convention $H = H_0 - \int h\, n$, the density source-response kernel is the negative of the commutator-retarded function of Chapter 127. For a single diffusive charge with susceptibility χ, it has the universal long-wavelength form

$$K^R_{nn}(\omega, \mathbf{k}) = \chi \, \frac{Dk^2}{Dk^2 - \mathrm{i}\omega} + \text{terms analytic at } \omega, k = 0. \tag{128.35}$$

so that $K^R_{nn}(0, \mathbf{k} \to 0) = \chi$. The pole is $\omega = -\mathrm{i}Dk^2$. Taking $\mathbf{k} \to 0$ before $\omega \to 0$ removes the diffusive denominator; taking the hydrodynamic scaling limit $\omega \sim k^2$ keeps it. This is the same order-of-limits issue that appears in dc conductivity and static susceptibility.

In the shear channel, the momentum-density correlator has a pole at (128.33). In the sound channel, the stress-tensor and energy-density correlators have the pair of poles (128.34). The residues are fixed by thermodynamic susceptibilities and Ward identities, while the imaginary parts of the pole locations are fixed by the Kubo coefficients η, ζ, and Σ_{AB}.

128.12 Hydrodynamic Limit as a QFT Claim

The deterministic equations above are consequences of a proposed hydrodynamic scaling limit of the microscopic QFT. The limit has the following structure. First take the thermodynamic limit of the KMS state. Then consider sources and correlators with

$$\omega\tau_{\text{micro}} \ll 1, \qquad |\mathbf{k}|\ell_{\text{micro}} \ll 1,$$

while keeping the ratios appropriate to the mode under study: $\omega \sim k^2$ for diffusion and shear, and $\omega \sim k$ for sound. The QFT assertion is that all nonconserved excitations produce analytic contributions in this scaling region, and the nonanalyticities are the conservation-law poles derived above. Proving this assertion requires clustering, real-time analyticity, local equilibration, and control of long-time tails beyond the deterministic approximation.

Open Problem 128.5 (Hydrodynamic emergence theorem). Find physically useful QFT hypotheses under which local thermalization, clustering, analyticity of retarded functions, positivity of entropy production, and the hydrodynamic pole expansion can be proved from the microscopic theory.

Chapter 129

Schwinger–Keldysh Hydrodynamic Effective Actions

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Hydrodynamic Schwinger–Keldysh effective actions are real-time doubled generating functionals for conserved densities in a long-wavelength state. Their defining constraints are inherited from the closed time path: normalization at coincident sources, a reality relation, positivity of the imaginary part, gauge or diffeomorphism Ward identities, and the KMS condition when the initial state is thermal. This chapter develops the quadratic multi-charge diffusion action explicitly, because it is the simplest place where all of these constraints can be seen without hiding the source-response signs or the matrix structure of susceptibilities and conductivities.

Let spacetime be $\mathbb{R}_t \times \mathbb{R}^d$. Spatial indices i, j are raised with δ^{ij}, and $\nabla^2 = \partial_i\partial_i$. Fourier modes are written as $\exp(-\mathrm{i}\omega t + \mathrm{i}\mathbf{k}\cdot\mathbf{x})$ in the derivations below. The microscopic retarded commutator convention is the one fixed in Chapter 127. In this chapter K^R denotes the source-response kernel obtained by differentiating the Schwinger–Keldysh functional with respect to the a-type source. The conversion between K^R and the commutator retarded function includes the sign convention of the microscopic source coupling and possible local contact terms.

129.1 Closed-Time-Path Constraints

Let $A_{1\mu}$ and $A_{2\mu}$ be background gauge fields on the two branches of the Schwinger–Keldysh contour. For an initial density operator ρ_0, define

$$Z[A_1, A_2] = \mathrm{Tr}\Big(U[A_1]\rho_0 U[A_2]^\dagger\Big)\,. \tag{129.1}$$

Introduce average and difference variables

$$A_{r\mu} = \frac{1}{2}(A_{1\mu} + A_{2\mu}), \qquad A_{a\mu} = A_{1\mu} - A_{2\mu}. \tag{129.2}$$

If $Z = \exp(\mathrm{i}I_{\mathrm{SK}})$, unitarity and Hermiticity imply

$$I_{\mathrm{SK}}[A_r, A_a = 0] = 0, \tag{129.3}$$

$$I_{\mathrm{SK}}[A_r, A_a]^* = -I_{\mathrm{SK}}[A_r, -A_a], \tag{129.4}$$

$$\mathrm{Im}\, I_{\mathrm{SK}}[A_r, A_a] \geq 0 \tag{129.5}$$

for source configurations in the domain of the effective description. Equation (129.3) says that every term in the effective action contains at least one a-type field. The positivity condition is the damping condition in the path-integral weight $\exp(\mathrm{i}I_{\mathrm{SK}})$.

Finite trace constraints. Suppose first that the Hilbert space is finite-dimensional, that $\rho_0 \geq 0$ is a density matrix with $\mathrm{Tr}\,\rho_0 = 1$, and that the branch evolutions $U[A_s]$ are unitary. Then

$$Z[A, A] = 1, \qquad Z[A_1, A_2]^* = Z[A_2, A_1], \qquad |Z[A_1, A_2]| \leq 1. \tag{129.6}$$

Consequently, on the branch of $I_{\mathrm{SK}} = -\mathrm{i}\log Z$ continuous from $I_{\mathrm{SK}}[A_r, 0] = 0$, equations (129.3)–(129.5) follow.

The diagonal identity is

$$Z[A, A] = \mathrm{Tr}(U[A]\rho_0 U[A]^\dagger) = \mathrm{Tr}\,\rho_0 = 1.$$

Complex conjugation gives

$$Z[A_1, A_2]^* = \mathrm{Tr}\Big(U[A_2]\rho_0 U[A_1]^\dagger\Big) = Z[A_2, A_1], \tag{129.7}$$

where cyclicity of the finite trace and $\rho_0^\dagger = \rho_0$ are used. For the bound, introduce the Hilbert–Schmidt inner product $(X, Y)_{\mathrm{HS}} = \mathrm{Tr}(X^\dagger Y)$ and set

$$X = U[A_2]\rho_0^{1/2}, \qquad Y = U[A_1]\rho_0^{1/2}.$$

Then $Z[A_1, A_2] = (X, Y)_{\mathrm{HS}}$. Cauchy–Schwarz gives

$$|Z[A_1, A_2]|^2 \leq \mathrm{Tr}(\rho_0)\,\mathrm{Tr}(\rho_0) = 1. \tag{129.8}$$

Since $Z = \exp(\mathrm{i}I_{\mathrm{SK}})$, the last inequality is $\exp(-\,\mathrm{Im}\, I_{\mathrm{SK}}) \leq 1$, hence $\mathrm{Im}\, I_{\mathrm{SK}} \geq 0$. The reality relation for I_{SK} follows from (129.7) after rewriting branch exchange as $A_a \mapsto -A_a$ at fixed A_r. This is the same finite trace algebra as the microscopic Schwinger–Keldysh construction of Chapter 126. In this hydrodynamic chapter it serves as the finite-system origin of the local effective-action constraints (129.3)–(129.5).

For continuum QFT the same algebra is first applied with an ultraviolet regulator and finite volume, and only then passed to the thermodynamic and continuum limits. Without this order of limits the inequality $|Z| \le 1$ is a statement about the regulated closed-time-path functional, not a theorem about an unconstructed real-time path integral.

129.2 Hydrodynamic Phase Field and Gauge Invariance

For N commuting conserved $U(1)$ charges, labelled by $A, B = 1, \dots, N$, the hydrodynamic modes are represented by phase fields

$$\varphi_1^A, \qquad \varphi_2^A.$$

The branchwise gauge transformations are

$$A_{s\mu}^A \mapsto A_{s\mu}^A + \partial_\mu \lambda_s^A, \qquad \varphi_s^A \mapsto \varphi_s^A - \lambda_s^A, \qquad s = 1, 2.$$

Thus the gauge-invariant combinations are

$$B_{s\mu}^A = A_{s\mu}^A + \partial_\mu \varphi_s^A. \tag{129.9}$$

In r/a variables,

$$B_{r\mu}^A = A_{r\mu}^A + \partial_\mu \varphi_r^A, \qquad B_{a\mu}^A = A_{a\mu}^A + \partial_\mu \varphi_a^A. \tag{129.10}$$

The field φ_a^A is the Lagrange-multiplier partner of the conservation law for the A-th charge. Varying the action with respect to φ_a^A is equivalent to the corresponding Ward identity because $\delta B_{a\mu}^A = \partial_\mu \delta \varphi_a^A$.

Remark 129.1 (Gauge invariance and the hydrodynamic Ward identity). Let I_{eff} be a local functional of the gauge-invariant variables $B_{r\mu}^A, B_{a\mu}^A$. Define the expectation-value current on the physical slice $A_a = \varphi_a = 0$ by

$$J_A^\mu(x) = \left. \frac{\delta I_{\text{eff}}}{\delta A_{a\mu}^A(x)} \right|_{A_a = \varphi_a = 0}. \tag{129.11}$$

Then the Euler–Lagrange equation for φ_a^A is the distributional Ward identity

$$\partial_\mu J_A^\mu = 0 \tag{129.12}$$

in the absence of external anomaly terms. Equivalently, the a-phase is not an additional physical scalar; it is the doubled-field implementation of current conservation.

On the physical slice, $B_{a\mu}^A = A_{a\mu}^A + \partial_\mu \varphi_a^A$. Thus the variation of the action under a compactly supported $\delta\varphi_a^A$ is

$$\delta I_{\text{eff}} = \int dt\, d^d x\, J_A^\mu\, \partial_\mu \delta\varphi_a^A = -\int dt\, d^d x\, (\partial_\mu J_A^\mu) \delta\varphi_a^A, \tag{129.13}$$

with no boundary term. Since $\delta\varphi_a^A$ is arbitrary, the Euler–Lagrange equation is $\partial_\mu J_A^\mu = 0$. If the microscopic current is anomalous, the branch gauge transformation of the regulated generating functional contains the corresponding local anomaly functional; that case belongs to the anomalous-transport chapter, not to the neutral diffusion action treated here.

129.3 Quadratic Diffusion Action

Consider a homogeneous thermal state at temperature $T > 0$. Let χ_{AB} be the static susceptibility matrix

$$\chi_{AB} = \left.\frac{\partial n_A}{\partial \mu_B}\right|_T .$$

At a thermodynamically stable point χ is real, symmetric, and positive definite on the space of non-redundant conserved charges. Let Σ_{AB} be the dissipative conductivity matrix in the convention of Chapter 128; for time-reversal-even charges and in the absence of parity-odd nondissipative transport, Σ is real, symmetric, and positive semidefinite. The diffusion endomorphism acting on densities is

$$D_A{}^B = \Sigma_{AC}(\chi^{-1})^{CB}. \tag{129.14}$$

Stability of the multi-charge diffusion matrix. Assume χ is real symmetric positive definite and Σ is real symmetric positive semidefinite. Then

$$D = \Sigma\chi^{-1} \tag{129.15}$$

is diagonalizable over $\mathbb{R}$ and all its eigenvalues are nonnegative. It is self-adjoint for the density inner product

$$\langle u, v\rangle_{\chi^{-1}} = u_A(\chi^{-1})^{AB}v_B. \tag{129.16}$$

If Σ is positive definite, all eigenvalues of D are positive. Indeed, let $\chi^{1/2}$ be the positive square root of χ. The matrix

$$S = \chi^{-1/2}\Sigma\chi^{-1/2} \tag{129.17}$$

is real symmetric positive semidefinite. Moreover

$$D = \Sigma\chi^{-1} = \chi^{1/2}S\chi^{-1/2}, \tag{129.18}$$

so D is similar to S. Therefore D has the same spectrum as S, is diagonalizable over $\mathbb{R}$, and has nonnegative eigenvalues. For self-adjointness,

$$\langle u, Dv\rangle_{\chi^{-1}} = u^T\chi^{-1}\Sigma\chi^{-1}v = (Du)^T\chi^{-1}v = \langle Du, v\rangle_{\chi^{-1}}, \tag{129.19}$$

because χ^{-1} and Σ are symmetric. If Σ is positive definite then S is positive definite, so its eigenvalues, and hence the eigenvalues of D, are strictly positive.

The leading quadratic local action for the diffusive modes is

$$I_{\text{diff}} = \int dt\, d^dx\, \left[\chi_{AB}B^A_{a0}B^B_{r0} - \Sigma_{AB}B^A_{ai}\partial_t B^B_{ri} + \mathrm{i}T\Sigma_{AB}B^A_{ai}B^B_{ai}\right], \tag{129.20}$$

up to terms with more derivatives or more powers of a-fields. The one-charge formula is recovered by setting $\chi_{11} = \chi$, $\Sigma_{11} = \sigma$, and $D = \sigma/\chi$. The current obtained by varying with respect to the a-type source at $A^A_a = \varphi^A_a = 0$ is

$$n_A = \frac{\delta I_{\text{diff}}}{\delta A^A_{a0}} = \chi_{AB}B^B_{r0}, \qquad j^i_A = \frac{\delta I_{\text{diff}}}{\delta A^A_{ai}} = -\Sigma_{AB}\partial_t B^B_{ri}. \tag{129.21}$$

Using $B^A_{r0} = A^A_{r0} + \partial_t \varphi^A_r$ and $B^A_{ri} = A^A_{ri} + \partial_i \varphi^A_r$, this current may be written as

$$j^i_A = -D_A{}^B \partial_i n_B + \Sigma_{AB} E^B_i, \qquad E^A_i = \partial_i A^A_{r0} - \partial_t A^A_{ri}. \tag{129.22}$$

The sign of E^A_i is fixed by the source convention in which a static scalar-source perturbation changes the density by $\delta n_A = \chi_{AB} \delta A^B_{r0}$. Equation (129.22) is the sourceful first-order constitutive relation

$$j^i_A = \Sigma_{AB} \partial_i A^B_{r0} - \Sigma_{AB} \partial_i \mu_B, \qquad \mu_A = B^A_{r0} = (\chi^{-1})^{AB} n_B,$$

written with the hydrodynamic phase made explicit.

Remark 129.2 (Diffusion equation from the a-phase). At $A^A_a = \varphi^A_a = 0$, varying (129.20) with respect to φ^A_a gives

$$\partial_t n_A + \partial_i j^i_A = 0, \tag{129.23}$$

with n_A and j^i_A as in (129.21). With external sources set to zero this becomes

$$\partial_t n_A - D_A{}^B \nabla^2 n_B = 0. \tag{129.24}$$

Under $\varphi^A_a \mapsto \varphi^A_a + \delta\varphi^A_a$,

$$\delta B^A_{a0} = \partial_t \delta\varphi^A_a, \qquad \delta B^A_{ai} = \partial_i \delta\varphi^A_a.$$

The terms linear in a-fields therefore vary as

$$\delta I_{\text{diff}} = \int dt\, d^d x\, \left[\chi_{AB} (\partial_t \delta\varphi^A_a) B^B_{r0} - \Sigma_{AB} (\partial_i \delta\varphi^A_a) \partial_t B^B_{ri}\right],$$

because the quadratic $B_a B_a$ term does not contribute at $A^A_a = \varphi^A_a = 0$. After integrating by parts and discarding boundary terms compatible with the closed-time-path variational problem,

$$\delta I_{\text{diff}} = -\int dt\, d^d x\, \delta\varphi^A_a \left[\partial_t (\chi_{AB} B^B_{r0}) - \partial_i (\Sigma_{AB} \partial_t B^B_{ri})\right].$$

Using (129.21), the Euler-Lagrange equation is exactly (129.23). If $A^A_{r\mu} = 0$, then $n_A = \chi_{AB} \partial_t \varphi^B_r$ and

$$j^i_A = -\Sigma_{AB} \partial_i \partial_t \varphi^B_r = -\Sigma_{AB} (\chi^{-1})^{BC} \partial_i n_C = -D_A{}^C \partial_i n_C,$$

which gives (129.24).

129.4 Density Response Kernel

The density response to a scalar source follows from the same saddle equation. Set $A^A_{ri} = 0$ and use a Fourier mode $\exp(-\mathrm{i}\omega t + \mathrm{i}\mathbf{k}\cdot\mathbf{x})$. Write $\mathbf{A}_0$, $\boldsymbol{\varphi}$, and $\mathbf{n}$ for the column vectors with components A^A_{r0}, φ^A_r, and n_A. The continuity equation gives

$$\mathbf{n} + k^2 \Sigma \boldsymbol{\varphi} = 0, \qquad \mathbf{n} = \chi(\mathbf{A}_0 - \mathrm{i}\omega\boldsymbol{\varphi}).$$

Equivalently,

$$(k^2 \Sigma - \mathrm{i}\omega\chi)\boldsymbol{\varphi} = -\chi \mathbf{A}_0.$$

Eliminating $\boldsymbol{\varphi}$ and using $D = \Sigma\chi^{-1}$ gives the matrix source-response kernel

$$K^R_{n_A n_B}(\omega, \mathbf{k}) = \frac{\delta n_A}{\delta A^B_{r0}} = \left[(Dk^2 - \mathrm{i}\omega\mathbf{1})^{-1} k^2 \Sigma\right]_{AB}. \tag{129.25}$$

It has the diffusive pole
$$\det(Dk^2 - \mathrm{i}\omega\mathbf{1}) = 0. \tag{129.26}$$
At fixed nonzero k, the static limit is
$$K^R_{n_A n_B}(0, \mathbf{k}) = \chi_{AB},$$
provided D is invertible on the charge sector being probed. If Σ has an exact null vector, the corresponding charge does not diffuse at this derivative order and must be treated by retaining the next allowed term. At $k = 0$ and nonzero ω, the kernel vanishes, expressing conservation of the total charges. These two limits are distinct; the hydrodynamic poles interpolate between them.

For a transverse vector source, $k_i A^A_{ri} = 0$ and $A^A_{r0} = 0$, the phase decouples at quadratic order and
$$j^i_A = -\Sigma_{AB}\partial_t A^B_{ri}.$$
With the electric-field convention (129.22), this is Ohm response $j^i_A = \Sigma_{AB}E^B_i$. The same matrix Σ therefore appears in the diffusion endomorphism, the vector-source response, and the noise covariance below.

129.5 Noise Positivity and Langevin Form

The imaginary part of (129.20) is
$$\operatorname{Im} I_{\text{diff}} = T \int dt\, d^d x\, \Sigma_{AB} B^A_{ai} B^B_{ai}. \tag{129.27}$$
It is nonnegative precisely when Σ is positive semidefinite. The same term has a stochastic representation on the image of Σ. If Σ is positive definite, then for each spatial component
$$\exp\left[-T\int \Sigma_{AB} B^A_{ai} B^B_{ai}\right] = \int D\xi\, \exp\left[-\int \frac{1}{4T}\xi_{Ai}(\Sigma^{-1})^{AB}\xi_{Bi} + \mathrm{i}\int B^A_{ai}\xi_{Ai}\right], \tag{129.28}$$
with normalization independent of B_a. For positive semidefinite Σ, the same formula is understood after restricting to the support of Σ, or as a limit from positive definite matrices. Thus the noise satisfies
$$\langle \xi_{Ai}(t, \mathbf{x})\xi_{Bj}(t', \mathbf{x}')\rangle = 2T\Sigma_{AB}\,\delta_{ij}\delta(t - t')\delta^{(d)}(\mathbf{x} - \mathbf{x}'). \tag{129.29}$$
After the Hubbard–Stratonovich transformation, varying φ^A_a gives the Langevin equation
$$\partial_t n_A + \partial_i\big(-D_A{}^B\partial_i n_B + \Sigma_{AB}E^B_i + \xi_{Ai}\big) = 0. \tag{129.30}$$
This derivation fixes the factor $2T\Sigma_{AB}$ in the noise correlator from the same matrix that governs the dissipative current.

Theorem 129.3 (Density fluctuation–dissipation theorem from the diffusion action)**.** *Assume χ is positive definite and Σ is positive semidefinite. For real ω, $\mathbf{k}$, let*
$$M(\omega, \mathbf{k}) = (Dk^2 - \mathrm{i}\omega\mathbf{1})^{-1}, \qquad k^2 = \mathbf{k}\cdot\mathbf{k}, \tag{129.31}$$

on the subspace where the inverse exists. The Langevin equation (129.30) at zero external source gives the symmetrized density correlator

$$G^S_{nn}(\omega,\mathbf{k}) = 2T\,k^2\,M(\omega,\mathbf{k})\,\Sigma\,M(-\omega,\mathbf{k})^T. \tag{129.32}$$

This matrix is positive semidefinite as a Hermitian kernel. If

$$K^A_{nn}(\omega,\mathbf{k}) = K^R_{nn}(-\omega,\mathbf{k})^T = k^2\Sigma\,M(-\omega,\mathbf{k})^T, \tag{129.33}$$

then, for $\omega \neq 0$,

$$G^S_{nn}(\omega,\mathbf{k}) = \frac{2T}{\omega}\,\frac{K^R_{nn}(\omega,\mathbf{k}) - K^A_{nn}(\omega,\mathbf{k})}{2\mathrm{i}}. \tag{129.34}$$

The equality is the classical low-frequency form of the KMS fluctuation–dissipation relation in the density sector.

Proof. Set the external source to zero. With the Fourier convention fixed at the beginning of the chapter, (129.30) becomes

$$(Dk^2 - \mathrm{i}\omega\mathbf{1})\,\mathbf{n} = -\mathrm{i}k_i\,\boldsymbol{\xi}_i. \tag{129.35}$$

Thus $\mathbf{n} = -\mathrm{i}k_i M\boldsymbol{\xi}_i$. Using (129.29) and summing over the spatial noise index gives exactly (129.32). For real ω and real D, Σ, $M(-\omega)^T = M(\omega)^\dagger$ after lowering and raising only ordinary finite charge indices, so

$$u^\dagger G^S_{nn} u = 2Tk^2\,(M^\dagger u)^\dagger \Sigma (M^\dagger u) \geq 0 \tag{129.36}$$

for every complex vector u.

It remains to prove the matrix fluctuation–dissipation identity without assuming that D and Σ commute. The response kernel from (129.25) is

$$K^R_{nn} = Mk^2\Sigma. \tag{129.37}$$

Since $D = \Sigma\chi^{-1}$ and χ, Σ are symmetric,

$$(Dk^2 - \mathrm{i}\omega)\chi = k^2\Sigma - \mathrm{i}\omega\chi, \qquad \chi(D^Tk^2 + \mathrm{i}\omega) = k^2\Sigma + \mathrm{i}\omega\chi. \tag{129.38}$$

Therefore

$$K^R_{nn} = M\big[(Dk^2 - \mathrm{i}\omega)\chi + \mathrm{i}\omega\chi\big] = \chi + \mathrm{i}\omega M\chi, \tag{129.39}$$

and similarly

$$K^A_{nn} = \big[\chi(D^Tk^2 + \mathrm{i}\omega) - \mathrm{i}\omega\chi\big]M(-\omega)^T = \chi - \mathrm{i}\omega\chi M(-\omega)^T. \tag{129.40}$$

Subtracting gives

$$K^R_{nn} - K^A_{nn} = \mathrm{i}\omega\big(M\chi + \chi M(-\omega)^T\big). \tag{129.41}$$

Multiplying (129.38) on the left and right by the two resolvents gives

$$M\chi + \chi M(-\omega)^T = 2k^2 M\Sigma M(-\omega)^T. \tag{129.42}$$

Combining the last two displays yields

$$\frac{K^R_{nn} - K^A_{nn}}{2\mathrm{i}} = \omega k^2 M\Sigma M(-\omega)^T, \tag{129.43}$$

which is equivalent to (129.34) after using (129.32). □

129.6 Dynamical KMS at Quadratic Order

Assume the initial state is thermal with inverse temperature $\beta = 1/T$ and that the long-wavelength effective action has the classical dynamical KMS symmetry appropriate to the derivative order under consideration. In the diffusion sector, and for time-reversal-even charge densities, the transformation needed at this order is

$$B^A_{ai}(t, \mathbf{x}) \mapsto B^A_{ai}(-t, \mathbf{x}) + \mathrm{i}\beta\, \partial_t B^A_{ri}(-t, \mathbf{x}), \tag{129.44}$$

with the time-reversal parity of the spatial vector field included in $\partial_t B^A_{ri}(-t, \mathbf{x})$. To see the coefficient constraint, keep one spatial component and write the dissipative-noise part as

$$L = -a^A \Sigma_{AB} x^B + \mathrm{i}a^A C_{AB} a^B, \qquad a^A = B^A_{ai}, \qquad x^A = \partial_t B^A_{ri},$$

where C_{AB} is the unknown symmetric noise matrix. Time reversal sends $x \mapsto -x$, while the KMS shift sends $a \mapsto a + \mathrm{i}\beta x$. The transformed expression is

$$L' = a^A(\Sigma_{AB} - 2\beta C_{AB})x^B + \mathrm{i}a^A C_{AB} a^B + \mathrm{i}\beta x^A(\Sigma_{AB} - \beta C_{AB})x^B.$$

Equality with L, up to total derivatives and terms beyond the retained order, requires the coefficient of $a^A x^B$ to match:

$$\Sigma_{AB} - 2\beta C_{AB} = -\Sigma_{AB}.$$

The $x^A x^B$ terms then cancel by the same condition. Therefore

$$C_{AB} = \frac{\Sigma_{AB}}{\beta} = T\Sigma_{AB}, \tag{129.45}$$

which is the coefficient in (129.20). This is the local effective-action form of the fluctuation–dissipation relation of Chapter 127.

129.7 Relation to Stress-Tensor Hydrodynamics

For stress-tensor hydrodynamics, the charge phase fields are replaced by doubled maps from a fluid worldvolume into spacetime, together with doubled metrics and, when charges are present, the phases above. The structural logic is the same:

$$\begin{aligned}
\text{gauge/diffeomorphism invariance} &\Longrightarrow \text{Ward identities},\\
I[A, A] = 0 &\Longrightarrow \text{closed-time-path normalization},\\
\operatorname{Im} I \geq 0 &\Longrightarrow \text{positive noise kernel},\\
\text{dynamical KMS} &\Longrightarrow \text{dissipation–fluctuation relations}.
\end{aligned}$$

At first derivative order this reproduces the viscous stress tensor and noise kernel used in Chapter 134. At higher derivative order the classification becomes a local invariant problem for doubled fields subject to the constraints above. A term belongs to the hydrodynamic effective action only after its frame dependence, contact-term ambiguity, KMS transformation, and positivity properties have been specified.

129.8 The Microscopic Theorem Boundary

The construction in this chapter is an effective-field-theory statement. A microscopic derivation from QFT would have to construct the real-time generating functional in a thermal state, take the thermodynamic limit, identify the hydrodynamic scaling limit, prove that the only nonanalytic long-wavelength modes are the conserved densities, and show that the local functional above is the asymptotic expansion of connected Schwinger–Keldysh correlators in a controlled topology. The local positivity and KMS constraints are necessary consequences of the microscopic closed time path; they are not by themselves a proof of thermalization.

Open Problem 129.4 (Microscopic derivation of SK hydrodynamic action). Starting from a continuum QFT with a KMS state, derive the hydrodynamic Schwinger–Keldysh effective action as a controlled long-wavelength limit of connected real-time correlators, including the topology in which the derivative expansion is asymptotic, the treatment of contact terms, and the relation between source-response kernels and commutator spectral functions.

Chapter 130

Thermal Gauge Theory And Screening

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter studies gauge theories at nonzero temperature through static correlators and their long-distance screening behavior. Let d denote the number of spatial dimensions, so the thermal Euclidean spacetime is $S^1_\beta \times \mathbb{R}^d$ before spatial infrared regularization. The coupling and trace conventions are those of the gauge chapters: the Yang–Mills action is written with

$$\frac{1}{4g^2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}),$$

and group invariants are computed using the same invariant bilinear form.

130.1 Static Correlators

Let the Euclidean time circle have length $\beta = 1/T$. A bosonic field has Matsubara frequencies $\omega_n = 2\pi nT$. Static screening is controlled by the $n = 0$ sector of gauge-invariant correlators. For an operator $\mathcal{O}(\tau, \mathbf{x})$, define its static projection by

$$\mathcal{O}_0(\mathbf{x}) = T\int_0^\beta d\tau\,\mathcal{O}(\tau, \mathbf{x}).$$

The screening masses are inverse correlation lengths extracted from

$$\langle \mathcal{O}_0(\mathbf{x})\mathcal{O}_0(0)\rangle_c \sim \sum_\alpha C_\alpha \frac{e^{-M_\alpha|\mathbf{x}|}}{|\mathbf{x}|^{(d-1)/2}}$$

when the static transfer matrix has isolated lowest eigenvalues in the chosen channel.

The power of $|\mathbf{x}|$ in this formula is fixed by the continuum spatial Fourier transform, not by a convention for the operator. If the static momentum-space correlator has an isolated scalar pole

$$\widetilde{C}(\mathbf{k}) = \frac{Z}{\mathbf{k}^2 + M^2} + \text{terms less singular at } \mathbf{k}^2 = -M^2,$$

then

$$C(\mathbf{x}) = Z\,\frac{M^{d/2-1}}{(2\pi)^{d/2}|\mathbf{x}|^{d/2-1}}\,K_{d/2-1}(M|\mathbf{x}|) + \cdots, \tag{130.1}$$

and the large-distance asymptotic of the modified Bessel function gives

$$C(\mathbf{x}) \sim Z\,\frac{M^{(d-3)/2}}{2(2\pi)^{(d-1)/2}}\,\frac{e^{-M|\mathbf{x}|}}{|\mathbf{x}|^{(d-1)/2}}.$$

If instead one projects to zero momentum in the transverse directions and keeps a single separation coordinate z, the same pole gives

$$C_{\text{proj}}(z) = \int d^{d-1}x_\perp\, C(z,\mathbf{x}_\perp) = \frac{Z}{2M}e^{-M|z|} + \cdots.$$

This projected form is the one most directly read from a spatial transfer matrix. With spatial volume first regulated in a box transverse to the z-direction, reflection positivity in z gives a Hilbert space on the slice $S^1_\beta \times \mathbb{T}^{d-1}$, a positive transfer Hamiltonian H_z, and

$$C_{\text{proj}}(z) = \sum_{\alpha\neq 0} |\langle \Omega_z, \widehat{\mathcal{O}}_0\alpha\rangle|^2 e^{-M_\alpha|z|}, \qquad M_\alpha = E^{(z)}_\alpha - E^{(z)}_0.$$

The sum is over states in the static, gauge-invariant channel created by $\mathcal{O}_0$. In infinite transverse volume the isolated eigenvalues may become thresholds, but the leading exponential rate is still a property of gauge-invariant static correlators, not of gauge-fixed propagator poles.

130.2 Electric Screening

In a weakly coupled high-temperature regime, the temporal gauge field A_0 behaves in the static sector as an adjoint scalar. The electric screening mass is defined perturbatively by the pole of the static longitudinal propagator, or nonperturbatively by the lowest gauge-invariant state in the electric channel after the relevant operator has been specified.

These two definitions should not be conflated. The perturbative Debye mass is a matching coefficient of the static effective theory in a specified gauge, regulator, and operator convention. A physical screening mass is extracted from a gauge-invariant static correlator, for example from operators built from A_0 inside the dimensionally reduced theory or from renormalized Polyakov loops in the original thermal theory. At weak coupling the Debye coefficient controls the electric scale gT and organizes the gauge-invariant spectrum, but the spectrum itself is defined by the transfer matrix construction above.

Choose a basis T_R^a for each representation R such that

$$\mathrm{tr}_R(T_R^a T_R^b) = T_R \delta^{ab}, \qquad f^{acd} f^{bcd} = C_A \delta^{ab}.$$

With the same coupling g in the covariant derivative and in the kinetic term convention above, the one-loop static polarization gives

$$m_D^2 = g^2 T^2 \left(\frac{C_A}{3} + \frac{1}{3} \sum_{\text{Dirac } f} T_{R_f} + \frac{1}{3} \sum_{\text{complex } s} T_{R_s} + \frac{1}{6} \sum_{\text{real } r} T_{R_r} \right). \tag{130.2}$$

For $SU(N)$ with the fundamental trace convention $\mathrm{tr}_{\mathbf{N}}(T^a T^b) = \delta^{ab}$, this means $C_A = 2N$ and $T_{\mathbf{N}} = 1$.

Thermal susceptibility integrals. For $n_B(x) = (e^x - 1)^{-1}$ and $n_F(x) = (e^x + 1)^{-1}$,

$$\frac{1}{T} \int \frac{\mathrm{d}^3 p}{(2\pi)^3} n_B(|p|/T)(1 + n_B(|p|/T)) = \frac{T^2}{6}, \tag{130.3}$$

$$\frac{1}{T} \int \frac{\mathrm{d}^3 p}{(2\pi)^3} n_F(|p|/T)(1 - n_F(|p|/T)) = \frac{T^2}{12}. \tag{130.4}$$

Set $x = |p|/T$. Then

$$\frac{1}{T} \int \frac{\mathrm{d}^3 p}{(2\pi)^3} n(1 \pm n) = \frac{T^2}{2\pi^2} \int_0^\infty x^2 n(x)(1 \pm n(x))\, \mathrm{d}x.$$

Since

$$n_B(1 + n_B) = -\frac{\mathrm{d}n_B}{\mathrm{d}x}, \qquad n_F(1 - n_F) = -\frac{\mathrm{d}n_F}{\mathrm{d}x},$$

integration by parts gives

$$\int_0^\infty x^2 n_B(1 + n_B)\, \mathrm{d}x = 2 \int_0^\infty \frac{x}{e^x - 1}\, \mathrm{d}x = \frac{\pi^2}{3},$$

and

$$\int_0^\infty x^2 n_F(1 - n_F)\, \mathrm{d}x = 2 \int_0^\infty \frac{x}{e^x + 1}\, \mathrm{d}x = \frac{\pi^2}{6}.$$

Substitution gives (130.3) and (130.4).

Lemma 130.1 (One-loop origin and coefficient of the Debye mass term)**.** *At one loop, the static quadratic term for A_0 in the dimensionally reduced action is the zero-frequency, zero-momentum limit of the Euclidean current-current correlator coupled to A_0, and its value is* (130.2)*.*

Proof. Introduce a background temporal field A_0^a that is independent of Euclidean time and space. Expanding the finite-temperature generating functional to second order gives

$$\Delta W^{(2)} = \frac{1}{2} A_0^a A_0^b \int_0^\beta d\tau \int d^{d-1}x \, \left\langle J_0^a(\tau, \mathbf{x}) J_0^b(0, 0) \right\rangle_c . \tag{130.5}$$

Translation invariance identifies the integral with the static polarization $\Pi^{ab}_{00}(0,\mathbf{0})$. Gauge symmetry in the unbroken thermal state sets this tensor proportional to δ^{ab}. The coefficient is the mass matrix of the adjoint scalar A_0 in the static effective action, which is $m_D^2\delta^{ab}$.

It remains to compute the coefficient. A constant $A_0 = A_0^a t^a$ shifts the Matsubara frequencies exactly as an imaginary color chemical potential. For a weight w of a representation R, the quadratic operator depends on

$$\omega_n \longmapsto \omega_n + w(A_0),$$

for bosons, with the analogous anti-periodic frequencies for fermions. This statement can be checked directly from the covariant derivative $D_0 = \partial_\tau + \mathrm{ad}_{A_0}$, and then extended from a Cartan-valued A_0 to arbitrary A_0 by G-invariance of the quadratic form.

For one complex bosonic oscillator of energy $E = |p|$ and charge q, the thermal part of the free-energy density is

$$T\log\left(1-e^{-\beta(E-\mu q)}\right) + T\log\left(1-e^{-\beta(E+\mu q)}\right).$$

Putting $\mu = ia$ and differentiating twice at $a = 0$ gives

$$\left.\frac{\partial^2 f_B(ia)}{\partial a^2}\right|_{a=0} = \frac{2q^2}{T}\, n_B(E/T)(1+n_B(E/T)).$$

Thus a real bosonic Gaussian determinant, which is one half of the complex determinant in a real representation, contributes

$$\frac{q^2}{T}\, n_B(E/T)(1+n_B(E/T)).$$

For one Dirac oscillator, the two spin polarizations and the particle-antiparticle pair give

$$\left.\frac{\partial^2 f_F(ia)}{\partial a^2}\right|_{a=0} = \frac{4q^2}{T}\, n_F(E/T)(1-n_F(E/T)).$$

After summing over weights, q^2 is replaced by $w^a w^b A_0^a A_0^b$, and $\sum_w w^a w^b = \mathrm{tr}_R(T_R^a T_R^b) = T_R\delta^{ab}$.

A single real bosonic degree of freedom in a real representation therefore contributes

$$\chi^{ab}_{\text{real boson}} = T_R\delta^{ab}\frac{1}{T}\int\frac{\mathrm{d}^3p}{(2\pi)^3} n_B(1+n_B) = \frac{T^2}{6}T_R\delta^{ab}. \tag{130.6}$$

A complex scalar has two real thermal degrees of freedom in R, and hence contributes $T^2 T_R\delta^{ab}/3$. A Dirac fermion has two spin polarizations and particle/antiparticle excitations. The anti-periodic Matsubara sum gives the fermionic integral in the susceptibility calculation above, so

$$\chi^{ab}_{\text{Dirac}} = 4T_R\delta^{ab}\frac{1}{T}\int\frac{\mathrm{d}^3p}{(2\pi)^3} n_F(1-n_F) = \frac{T^2}{3}T_R\delta^{ab}. \tag{130.7}$$

For gauge fields, use background-field gauge around the flat background A_0. Since the background field strength vanishes, the quadratic vector operator is $-D^2\delta_{\mu\nu}$ and the ghost operator is $-D^2$, both in the adjoint representation. The one-loop contribution to the thermal free energy is therefore

$$\frac{1}{2}\mathrm{Tr}_{\text{adj, vec}}\log(-D^2) - \mathrm{Tr}_{\text{adj, gh}}\log(-D^2) = \left(\frac{4}{2}-1\right)\mathrm{Tr}_{\text{adj}}\log(-D^2)$$

in four spacetime dimensions. The right side is the determinant of two real periodic bosonic thermal degrees of freedom in the adjoint representation: the temporal and longitudinal gauge-dependent pieces have cancelled against the periodic complex ghost pair. Using $T_{\rm adj} = C_A$, this gives

$$\chi^{ab}_{\rm gauge} = 2\frac{T^2}{6}C_A\delta^{ab} = \frac{T^2}{3}C_A\delta^{ab}. \tag{130.8}$$

The four-dimensional one-loop free-energy density contains $\frac{1}{2}\chi^{ab}A_0^aA_0^b$. Multiplying by the thermal circle length $\beta = 1/T$ gives the three-dimensional effective action. Since the static action is written as

$$\int \mathrm{d}^3x\,\frac{m_D^2}{2g_3^2}\,\mathrm{tr}(A_0^2), \qquad g_3^2 = g^2T,$$

and $\mathrm{tr}(A_0^2) = \delta^{ab}A_0^aA_0^b$ in the default trace convention, matching gives $m_D^2 = g^2\chi$. Adding the gauge, Dirac, complex-scalar, and real-scalar susceptibilities gives (130.2). □

Controlled Approximation 130.2 (HTL bridge from Debye matching to response)**.** The Debye coefficient above is the static face of a real-time soft response. The hard-thermal-loop approximation is the controlled leading response when the external frequency and momentum satisfy

$$\omega,\ |\mathbf{k}| = O(gT)$$

while the particles carrying the loop momentum have

$$p = O(T).$$

The calculation keeps the collisionless angular motion of the hard particles and expands only in the ratio of soft external scale to hard momentum. It is therefore not a local Lorentz-invariant mass term for the gauge field, and it does not determine the nonperturbative magnetic scale.

Lemma 130.3 (Static Debye term as the zero-frequency HTL limit)**.** *In the rest frame of the equilibrium state, let*

$$v^\mu = (1,\mathbf{v}), \qquad |\mathbf{v}| = 1, \qquad \int_{\mathbf{v}}(\cdots) = \int_{S^2}\frac{\mathrm{d}\Omega_{\mathbf{v}}}{4\pi}(\cdots).$$

With mostly-plus Lorentzian convention and

$$K_\mu = (-\omega,\mathbf{k}),$$

the retarded HTL induced-current kernel

$$\mathcal{K}_R^{\mu\nu,ab}(\omega,\mathbf{k}) = \delta^{ab}m_D^2\left[\delta_0^\mu\delta_0^\nu - \omega\int_{\mathbf{v}}\frac{v^\mu v^\nu}{\omega - \mathbf{v}\cdot\mathbf{k} + i0}\right] \tag{130.9}$$

is transverse,

$$K_\mu\mathcal{K}_R^{\mu\nu,ab} = 0,$$

and its static response is

$$\mathcal{K}_R^{00,ab}(0,\mathbf{k}) = \delta^{ab}m_D^2, \qquad \mathcal{K}_R^{0i,ab}(0,\mathbf{k}) = \mathcal{K}_R^{ij,ab}(0,\mathbf{k}) = 0.$$

Thus the one-loop coefficient in (130.2) is the zero-frequency electric susceptibility of a conserved retarded response, not a gauge-field mass assigned independently of the response kernel.

Proof. The mechanism is most transparent in the gauge-covariant kinetic form. Let $W(x, \mathbf{v})$ be an adjoint auxiliary field and set

$$(v \cdot D)W(x, \mathbf{v}) = v^i F_{i0}(x), \qquad j^\mu(x) = m_D^2 \int_{\mathbf{v}} v^\mu W(x, \mathbf{v}).$$

The current is covariantly conserved before choosing a gauge:

$$D_\mu j^\mu = m_D^2 \int_{\mathbf{v}} (v \cdot D)W = m_D^2 \int_{\mathbf{v}} v^i F_{i0} = 0,$$

because F_{i0} is independent of $\mathbf{v}$ and $\int_{\mathbf{v}} v^i = 0$.

Linearize around $A_\mu = 0$ and take fields proportional to $\exp(-i\omega t + i\mathbf{k} \cdot \mathbf{x})$. The retarded solution of $(\partial_t + \mathbf{v} \cdot \nabla)W = v^i F_{i0}$ is

$$W(K, \mathbf{v}) = -\frac{\mathbf{v} \cdot \mathbf{k}\, A_0(K) + \omega\, v^i A_i(K)}{\omega - \mathbf{v} \cdot \mathbf{k} + i0},$$

where $F_{i0} = \partial_i A_0 - \partial_0 A_i$. Substitution in $j^\mu = m_D^2 \int_{\mathbf{v}} v^\mu W$ gives

$$j^\mu = m_D^2 \left[\delta_0^\mu \delta_0^\nu - \omega \int_{\mathbf{v}} \frac{v^\mu v^\nu}{\omega - \mathbf{v} \cdot \mathbf{k} + i0} \right] A_\nu.$$

The A_0 coefficient uses the identity

$$-\frac{\mathbf{v} \cdot \mathbf{k}}{\omega - \mathbf{v} \cdot \mathbf{k} + i0} = 1 - \frac{\omega}{\omega - \mathbf{v} \cdot \mathbf{k} + i0}$$

inside the angular integral and $\int_{\mathbf{v}} v^\mu = \delta_0^\mu$. Restoring the color factor gives (130.9).

Transversality follows directly. Since

$$K_\mu v^\mu = -(\omega - \mathbf{v} \cdot \mathbf{k}),$$

contracting K_μ into (130.9) cancels the denominator and leaves

$$K_\mu \mathcal{K}_R^{\mu\nu,ab} = \delta^{ab} m_D^2 \left[-\omega \delta_0^\nu + \omega \int_{\mathbf{v}} v^\nu \right].$$

The angular integral equals 1 for $\nu = 0$ and 0 for each spatial index, so the contraction vanishes.

Finally set $\omega = 0$ in the retarded static response at fixed $\mathbf{k}$. The angular term in (130.9) carries an explicit factor of ω, and the remaining tensor is $\delta^{ab} m_D^2 \delta_0^\mu \delta_0^\nu$. This is exactly the electric mass coefficient matched into the static effective theory. □

130.3 Magnetic Sector

The spatial gauge fields A_i have no gauge-invariant local mass term in the dimensionally reduced action. The high-temperature static magnetic sector is therefore a three-dimensional Yang–Mills theory with coupling

$$g_3^2 = g^2 T$$

in four spacetime dimensions. Its infrared scale is of order g_3^2, and the corresponding magnetic screening lengths are nonperturbative objects of the three-dimensional gauge theory. Perturbative expansions of static magnetic quantities encounter this scale because massless spatial gauge fields remain in the reduced action.

130.4 Polyakov Loop And Center Symmetry

The thermal Wilson line at spatial point $\mathbf{x}$ is

$$P(\mathbf{x}) = \mathcal{P}\exp\left(\mathrm{i}\int_0^\beta d\tau\, A_0(\tau,\mathbf{x})\right).$$

In a pure gauge theory, its trace in a representation charged under the center transforms under the center one-form symmetry wrapped on the thermal circle. The free energy of an external static charge is defined by

$$e^{-\beta F_Q} = \langle \mathrm{tr}_R\, P(\mathbf{x})\rangle$$

after renormalizing the line. Correlators of two Polyakov loops define static color-singlet free energies; their large-distance decay belongs to the screening spectrum rather than to a zero-temperature particle spectrum.

Definition 130.4 (Renormalized static-source free-energy datum)**.** At finite cutoff Λ, let $P_R^\Lambda(\mathbf{x})$ be the regularized thermal line in representation R. A static-source renormalization scheme consists of additive line energies $\delta m_R(\Lambda)$, equivalently multiplicative factors

$$Z_R(\Lambda,\beta) = \exp\{\beta\,\delta m_R(\Lambda)\}, \qquad P_R^{\mathrm{ren}}(\mathbf{x}) = Z_R(\Lambda,\beta)P_R^\Lambda(\mathbf{x}).$$

For a conjugate pair at separation $r = |\mathbf{x}-\mathbf{y}|$, define

$$\begin{aligned} e^{-\beta F_R^{\mathrm{ren}}} &:= \left\langle d_R^{-1}\,\mathrm{tr}_R\, P_R^{\mathrm{ren}}(\mathbf{x})\right\rangle, \\ e^{-\beta F_{R\bar{R}}^{\mathrm{ren}}(r)} &:= \left\langle d_R^{-2}\,\mathrm{tr}_R\, P_R^{\mathrm{ren}}(\mathbf{x})\ \mathrm{tr}_{\bar{R}}\, P_{\bar{R}}^{\mathrm{ren}}(\mathbf{y})\right\rangle. \end{aligned}$$

Changing the finite part of δm_R shifts F_R^{ren} by an additive constant. The datum therefore includes the line scheme whenever an absolute one-source free energy is quoted.

Lemma 130.5 (Line self-energy cancellation in static-source ratios)**.** *In the datum of Definition 130.4, assume the expectation values are taken in a fixed physical state and center-sector prescription, or in a source-selected infinite-volume limit, for which the one-source expectations satisfy $C_R^{\mathrm{ren}}C_{\bar{R}}^{\mathrm{ren}} \neq 0$. Then the pair excess free energy*

$$\Delta F_{R\bar{R}}(r) = F_{R\bar{R}}^{\mathrm{ren}}(r) - F_R^{\mathrm{ren}} - F_{\bar{R}}^{\mathrm{ren}}$$

is independent of the multiplicative line factors $Z_R, Z_{\bar{R}}$. So is the static force

$$\mathcal{F}_{R\bar{R}}(r,T) = -\partial_r F_{R\bar{R}}^{\mathrm{ren}}(r,T),$$

which requires no division by the one-source expectations, provided the pair correlator defines $F_{R\bar{R}}^{\mathrm{ren}}$ and the line counterterms are local energies of the individual wrapped sources and hence independent of r.

Proof. Write the cutoff expectations as

$$C_R^\Lambda = \langle P_R^\Lambda\rangle, \qquad C_{R\bar{R}}^\Lambda(r) = \langle P_R^\Lambda(\mathbf{x})P_{\bar{R}}^\Lambda(\mathbf{y})\rangle$$

with the displayed normalized traces suppressed. Renormalization gives

$$C_R^{\mathrm{ren}} = Z_R C_R^\Lambda, \qquad C_{R\bar{R}}^{\mathrm{ren}}(r) = Z_R Z_{\bar{R}} C_{R\bar{R}}^\Lambda(r).$$

When $C_R^\Lambda C_{\bar R}^\Lambda \neq 0$, equivalently after finite renormalization $C_R^{\rm ren} C_{\bar R}^{\rm ren} \neq 0$, therefore

$$\exp\{-\beta \Delta F_{R\bar R}(r)\} = \frac{C_{R\bar R}^{\rm ren}(r)}{C_R^{\rm ren} C_{\bar R}^{\rm ren}} = \frac{C_{R\bar R}^{\Lambda}(r)}{C_R^{\Lambda} C_{\bar R}^{\Lambda}},$$

so all one-line factors cancel before the continuum limit is discussed. This is a conditional ratio statement, not a definition in a sector where either one-point function vanishes. For the force, no one-point divisor is used:

$$F_{R\bar R}^{\rm ren}(r,T) = F_{R\bar R}^{\Lambda}(r,T) - \delta m_R(\Lambda) - \delta m_{\bar R}(\Lambda),$$

and differentiating with respect to r removes the same local constants. □

Thus the absolute number assigned to a single static-source free energy is a line-scheme coordinate, while the conditional connected Polyakov-loop ratio and the source-pair force are the quantities that can be compared across regulators after the same physical state and center-sector limit have been specified. In a center-symmetric finite-volume pure gauge theory the nonzero-N-ality one-point function still vanishes before any finite multiplicative renormalization; the source-selected infinite-volume limit is the separate order-parameter limit used in phase discussions. In the center-symmetric sector itself one instead uses the neutral pair correlator

$$G_{R\bar R}^{\rm ren}(r) = \left\langle d_R^{-2}\, \mathrm{tr}_R\, P_R^{\rm ren}(\mathbf{x})\ \mathrm{tr}_{\bar R}\, P_{\bar R}^{\rm ren}(\mathbf{y})\right\rangle$$

or, where this correlator is nonzero and differentiable in r, the direct pair force

$$-\partial_r F_{R\bar R}^{\rm ren}(r,T) = \beta^{-1}\partial_r \log G_{R\bar R}^{\rm ren}(r),$$

without assigning finite one-source free energies.

130.5 Dimensional Reduction

At distances much larger than β, nonzero Matsubara modes can be integrated out into a three-dimensional effective theory. In four spacetime dimensions the bosonic part begins as

$$S_{\rm EQCD} = \int d^3x\, \left[\frac{1}{4g_3^2}\,\mathrm{tr}(F_{ij}F_{ij}) + \frac{1}{2g_3^2}\,\mathrm{tr}(D_iA_0D_iA_0) + \frac{m_E^2}{2g_3^2}\,\mathrm{tr}(A_0^2) + \lambda_E\ \mathrm{tr}(A_0^4) + \cdots\right].$$

The ellipsis denotes higher local operators ordered by the static derivative and field expansion. A quantitative use of this action requires matching the coefficients $g_3^2, m_E^2, \lambda_E, \ldots$ in the same regulator and operator scheme as the original thermal theory.

Open Problem 130.6 (Gauge-invariant screening spectrum). Give a continuum construction of the full static screening spectrum of thermal non-Abelian gauge theory from gauge-invariant operators, including the relation between perturbative electric matching, nonperturbative magnetic dynamics, and Polyakov-loop sectors.

Chapter 131

Kinetic Theory As A Controlled Limit

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Kinetic theory is the long-wavelength, weakly off-equilibrium description in which selected real-time QFT correlators are represented by on-shell phase-space densities. The construction has three inputs: a quasiparticle spectral regime, an on-shell collision kernel extracted from microscopic scattering or finite-temperature amplitudes, and a hierarchy separating the microscopic correlation scale from the spacetime scale of variation. This chapter develops the relativistic Boltzmann framework with the assumptions kept explicit, derives detailed balance and the entropy inequality, and records the precise point at which kinetic theory becomes hydrodynamics.

Let spacetime dimension be $D = d + 1$, with spatial momentum $\mathbf{p} \in \mathbb{R}^d$. For species a,

$$p^\mu = (E_a(\mathbf{p}), \mathbf{p}), \qquad E_a(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m_a^2}.$$

The Lorentz-invariant on-shell measure used below is

$$d\Pi_a(p) = \frac{d^d\mathbf{p}}{(2\pi)^d \, 2E_a(\mathbf{p})}. \tag{131.1}$$

The factor $2p^\mu d\Pi_a$ has the standard equal-time interpretation:

$$2E_a \, d\Pi_a = \frac{d^d\mathbf{p}}{(2\pi)^d}.$$

Internal degeneracies may be included by replacing the species label a with a refined label, or by inserting an explicit degeneracy g_a.

131.1 Quasiparticle Distribution From Spectral Data

Let $G_a^<(x,y)$ be a Wightman two-point function for a microscopic field or for a gauge-invariant quasiparticle interpolating operator in a regulated theory. Introduce center and relative coordinates

$$X = \frac{x+y}{2}, \qquad s = x - y,$$

and define the Wigner transform

$$G_a^<(X,p) = \int d^D s\, e^{\mathrm{i}p\cdot s} G_a^<(X+s/2, X-s/2).$$

The quasiparticle distribution $f_a(X,\mathbf{p})$ is the coefficient of the positive-energy pole contribution when the spectral density has the form

$$\rho_a(X,p) = 2\pi Z_a(X,\mathbf{p})\,\mathrm{sgn}(p^0)\delta\big((p^0)^2 - E_a(\mathbf{p})^2\big) + \rho_a^{\mathrm{off}}(X,p), \tag{131.2}$$

with the off-shell remainder small in the topology used for the gradient expansion. Equivalently, near $p^0 = E_a(\mathbf{p})$,

$$G_a^<(X,p) = 2\pi Z_a(X,\mathbf{p}) f_a(X,\mathbf{p})\delta(p^0 - E_a(\mathbf{p})) + \text{controlled off-shell remainder}.$$

The pole approximation requires

$$\Gamma_a(\mathbf{p}) \ll E_a(\mathbf{p}), \qquad \Gamma_a(\mathbf{p})\,\ell_{\mathrm{variation}} \ll 1, \tag{131.3}$$

where Γ_a is the damping width and $\ell_{\mathrm{variation}}$ is the spacetime scale over which f_a changes. These inequalities are hypotheses on the state and on the coupling regime; they are not consequences of conservation laws alone.

131.2 Quasiparticle Projection of the Two-Point Equation

The Boltzmann drift term is the on-shell projection of the leading Schwinger–Keldysh two-point equation. We record the projection explicitly because it is the point where an operator statement about correlators becomes a phase-space equation.

Controlled Approximation 131.1 (Force-free quasiparticle projection)**.** Consider one scalar species in flat spacetime, with constant mass and a single narrow positive-energy peak. Let

$$F(p) = (p^0)^2 - E(\mathbf{p})^2, \qquad E(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m^2}, \tag{131.4}$$

and assume that the leading-gradient Kadanoff–Baym equation has the form

$$\frac{1}{2}\{F, G^<\}_{\mathrm{kin}} = I^<[G], \tag{131.5}$$

where $I^<$ is the collision functional and

$$\{A,B\}_{\mathrm{kin}} = \partial_{p^0} A\,\partial_t B - \partial_t A\,\partial_{p^0} B - \partial_{\mathbf{p}} A \cdot \nabla_{\mathbf{x}} B + \nabla_{\mathbf{x}} A \cdot \nabla_{\mathbf{p}} B. \tag{131.6}$$

Use the positive-energy ansatz

$$G^{<}(X,p) = 2\pi Z(\mathbf{p}) f(X,\mathbf{p}) \delta(p^0 - E(\mathbf{p})) \tag{131.7}$$

with Z independent of X at this order. Terms from mean-field forces, Berry curvature, spatially varying masses, and off-shell memory kernels are omitted in this controlled approximation; each of them modifies the left-hand side by additional explicitly computable phase-space terms.

Remark 131.2 (Projection of the drift term). Under Controlled Approximation 131.1, the positive-energy shell projection of (131.5) is

$$p^\mu \partial_\mu f(X,\mathbf{p}) = \mathcal{C}[f](X,\mathbf{p}), \qquad p^\mu = (E(\mathbf{p}),\mathbf{p}), \tag{131.8}$$

with

$$\mathcal{C}[f](X,\mathbf{p}) = \frac{1}{Z(\mathbf{p})} \int_0^\infty \frac{dp^0}{2\pi}\, I^{<}[G](X,p). \tag{131.9}$$

Since F is independent of X,

$$\{F, G^{<}\}_{\mathrm{kin}} = 2p^0 \partial_t G^{<} + 2\mathbf{p}\cdot\nabla_{\mathbf{x}} G^{<}. \tag{131.10}$$

Using (131.7) and the X-independence of Z and E,

$$\frac{1}{2}\{F, G^{<}\}_{\mathrm{kin}} = 2\pi Z(\mathbf{p})\big(p^0\partial_t + \mathbf{p}\cdot\nabla_{\mathbf{x}}\big) f(X,\mathbf{p})\, \delta(p^0 - E(\mathbf{p})). \tag{131.11}$$

Integrating over $p^0 > 0$ and dividing by $Z(\mathbf{p})$ gives

$$E(\mathbf{p})\partial_t f + \mathbf{p}\cdot\nabla_{\mathbf{x}} f = \frac{1}{Z(\mathbf{p})}\int_0^\infty \frac{dp^0}{2\pi} I^{<}[G](X,p), \tag{131.12}$$

which is (131.8). Dividing by E gives the coordinate-time form $(\partial_t + \mathbf{v}\cdot\nabla_{\mathbf{x}})f = C^{(t)}[f]$ with $\mathbf{v} = \mathbf{p}/E$.

131.3 Boltzmann Equation and Collision Kernel

The covariant Boltzmann equation is

$$p^\mu \partial_\mu f_a(X,\mathbf{p}) = \mathcal{C}_a[f](X,\mathbf{p}). \tag{131.13}$$

In coordinate time this is

$$(\partial_t + \mathbf{v}_a\cdot\nabla_{\mathbf{x}})\, f_a = C_a^{(t)}[f], \qquad \mathbf{v}_a = \nabla_{\mathbf{p}} E_a, \qquad C_a^{(t)} = \frac{\mathcal{C}_a}{E_a}.$$

For $a(p_1)b(p_2) \to c(p_3)d(p_4)$, define

$$\eta_a = \begin{cases} +1, & a \text{ bosonic}, \\ -1, & a \text{ fermionic}. \end{cases}$$

The $2 \to 2$ contribution to $\mathcal{C}_a$ is

$$\mathcal{C}_a(p_1) = \frac{1}{2}\sum_{bcd}\int d\Pi_b(p_2)d\Pi_c(p_3)d\Pi_d(p_4)\,(2\pi)^D\delta^{(D)}(p_1+p_2-p_3-p_4)|\mathcal{M}_{ab\to cd}|^2$$
$$\times\Big[f_c(p_3)f_d(p_4)(1+\eta_a f_a(p_1))(1+\eta_b f_b(p_2))$$
$$- f_a(p_1)f_b(p_2)(1+\eta_c f_c(p_3))(1+\eta_d f_d(p_4))\Big]. \tag{131.14}$$

The factor $1/2$ converts the coordinate-time normalization to the covariant equation (131.13); equivalently, $C_a^{(t)}$ carries the familiar prefactor $1/(2E_1)$. Symmetry factors for identical particles are included in the definition of $|\mathcal{M}|^2$ and in the channel sum.

The matrix element in (131.14) is the on-shell scattering amplitude in the normalization of the scattering chapters, after the regulator and medium corrections relevant at the kinetic scale have been specified. In gauge theories this last clause is essential because soft exchange and collinear splitting can contribute at the same parametric order as $2 \to 2$ hard scattering.

131.4 Detailed Balance

The local equilibrium distribution is

$$f_a^{(0)}(X,\mathbf{p}) = \frac{1}{\exp[\beta(X)(-u_\mu(X)p^\mu - \mu_A(X)q_a^A)] - \eta_a}. \tag{131.15}$$

At a fixed point X, the identity

$$1+\eta_a f_a^{(0)} = e^{\beta(-u_\mu p^\mu - \mu_A q_a^A)} f_a^{(0)} \tag{131.16}$$

implies detailed balance. Indeed, for an allowed reaction satisfying

$$p_1+p_2 = p_3+p_4, \qquad q_a^A+q_b^A = q_c^A+q_d^A,$$

the loss product

$$X_{12\to34} = f_a^{(0)}(p_1)f_b^{(0)}(p_2)(1+\eta_c f_c^{(0)}(p_3))(1+\eta_d f_d^{(0)}(p_4))$$

equals the gain product

$$Y_{34\to12} = f_c^{(0)}(p_3)f_d^{(0)}(p_4)(1+\eta_a f_a^{(0)}(p_1))(1+\eta_b f_b^{(0)}(p_2)).$$

This is an algebraic consequence of (131.16): the exponential multiplying $f_1f_2f_3f_4$ is the same on the two sides by energy-momentum and charge conservation. Therefore the collision term vanishes on global equilibrium, and on local equilibrium it vanishes up to gradients of βu^μ and $\beta\mu_A$.

131.5 Kinetic Entropy and the H Theorem

Define the entropy current

$$S^\mu_{\text{kin}} = -\sum_a \int 2d\Pi_a\, p^\mu \left[f_a \log f_a - \eta_a(1+\eta_a f_a)\log(1+\eta_a f_a)\right]. \tag{131.17}$$

Using the Boltzmann equation,

$$\partial_\mu S^\mu_{\text{kin}} = -\sum_a \int 2d\Pi_a\, \mathcal{C}_a[f]\, \log\left(\frac{f_a}{1+\eta_a f_a}\right). \tag{131.18}$$

For the $2 \to 2$ kernel, symmetrizing over the four external legs gives

$$\begin{aligned}\partial_\mu S^\mu_{\text{kin}} = \frac{1}{4}\sum_{abcd}\int & d\Pi_1 d\Pi_2 d\Pi_3 d\Pi_4\, (2\pi)^D \delta^{(D)}(p_1+p_2-p_3-p_4)|\mathcal{M}_{ab\to cd}|^2 \\ & \times (X_{12\to 34} - Y_{34\to 12})\log\left(\frac{X_{12\to 34}}{Y_{34\to 12}}\right) \geq 0. \end{aligned}\tag{131.19}$$

The inequality follows from

$$(x-y)\log(x/y) \geq 0 \qquad (x>0,\ y>0).$$

Equality holds, within the set of reactions included in the kernel, exactly when each allowed gain product equals the corresponding loss product. When the reactions connect the species space sufficiently, this condition forces $\log[f_a/(1+\eta_a f_a)]$ to be a linear combination of the conserved collision invariants 1, p^μ, and q_a^A, giving (131.15).

131.6 Hydrodynamic Moments

The kinetic stress tensor and conserved currents are

$$T^{\mu\nu}(X) = \sum_a \int 2d\Pi_a\, p^\mu p^\nu f_a(X,\mathbf{p}), \tag{131.20}$$

$$J^\mu_A(X) = \sum_a q_{A,a}\int 2d\Pi_a\, p^\mu f_a(X,\mathbf{p}). \tag{131.21}$$

Multiplying (131.13) by $2p^\nu$ and integrating gives

$$\partial_\mu T^{\mu\nu} = \sum_a \int 2d\Pi_a\, p^\nu \mathcal{C}_a[f].$$

The right side vanishes because each collision conserves total four-momentum. Similarly,

$$\partial_\mu J^\mu_A = \sum_a q_{A,a}\int 2d\Pi_a\, \mathcal{C}_a[f] = 0$$

when the collision channels conserve charge q_A. Thus the hydrodynamic Ward identities are the collision-invariant moments of the kinetic equation.

At local equilibrium (131.15), the moments have the ideal-fluid form

$$T^{\mu\nu}_{(0)} = \varepsilon u^\mu u^\nu + p\Delta^{\mu\nu}, \qquad J^\mu_{A(0)} = n_A u^\mu,$$

with p, ε, and n_A obtained from the phase-space integrals. First-order transport arises from the first gradient correction to f_a.

131.7 Linearized Collision Operator

Write

$$f_a = f_a^{(0)} + f_a^{(0)}(1 + \eta_a f_a^{(0)})\chi_a. \tag{131.22}$$

To first order,

$$\mathcal{C}_a[f] = -\sum_b \mathcal{L}_{ab}\chi_b + O(\chi^2).$$

The natural inner product on perturbations is

$$\langle \phi, \psi \rangle = \sum_a \int 2d\Pi_a\, f_a^{(0)}(1 + \eta_a f_a^{(0)})\, \phi_a(p)\psi_a(p). \tag{131.23}$$

For the $2 \to 2$ collision kernel,

$$\begin{aligned} \langle \chi, \mathcal{L}\chi \rangle = \frac{1}{4} \sum_{abcd} \int & d\Pi_1 d\Pi_2 d\Pi_3 d\Pi_4\, (2\pi)^D \delta^{(D)}(p_1 + p_2 - p_3 - p_4) |\mathcal{M}_{ab\to cd}|^2 \\ & \times f_1^{(0)} f_2^{(0)} (1 + \eta_c f_3^{(0)})(1 + \eta_d f_4^{(0)}) \big(\chi_1 + \chi_2 - \chi_3 - \chi_4\big)^2 \geq 0. \end{aligned} \tag{131.24}$$

The null space consists of perturbations for which $\chi_1 + \chi_2 - \chi_3 - \chi_4 = 0$ for every included reaction. Under the same connectivity assumption used in the entropy argument, this null space is spanned by conserved collision invariants. Transport coefficients are quadratic forms involving the inverse of $\mathcal{L}$ on the orthogonal complement of this null space.

131.8 Finite Collision Algebra Behind the Integral Formula

The continuum formulas above are compact notations for a positive measure on reaction kinematics. The sign, null-space, and detailed-balance statements are already present at the finite algebraic level, so we record that level explicitly. This also fixes the precise content of the phrase "collision invariant" before functional analysis is added.

Definition 131.3 (Finite reversible collision datum)**.** A finite reversible collision datum consists of:

1. a finite set I of one-particle cells, each with a statistics sign $\eta_i \in \{+1, -1\}$;
2. an open state domain

$$\mathcal{D} = \{f \in \mathbb{R}^I : \ f_i > 0,\ 1 + \eta_i f_i > 0 \text{ for all } i\};$$

3. a finite set $\mathcal{R}$ of reversible reactions $r = (i, j; k, l)$, interpreted as $i + j \rightleftarrows k + l$, with weights $w_r > 0$;
4. additive labels P_i taking values in a real vector space $\mathcal{P}$, such that

$$P_i + P_j = P_k + P_l \tag{131.25}$$

for every reaction $r = (i, j; k, l)$.

For $f \in \mathcal{D}$, define

$$L_r(f) = f_i f_j(1+\eta_k f_k)(1+\eta_l f_l), \qquad G_r(f) = f_k f_l(1+\eta_i f_i)(1+\eta_j f_j), \tag{131.26}$$

and the stoichiometric vector

$$\nu_r = -e_i - e_j + e_k + e_l \in \mathbb{R}^I. \tag{131.27}$$

The collision vector field is

$$\dot{f} = \sum_{r\in\mathcal{R}} w_r\big(L_r(f) - G_r(f)\big)\nu_r. \tag{131.28}$$

Proposition 131.4 (Finite collision conservation and entropy production)**.** *For every finite reversible collision datum, the vector field (131.28) preserves every additive conserved quantity*

$$Q_A(f) = \sum_{i\in I} A(P_i) f_i, \qquad A \in \mathcal{P}^*. \tag{131.29}$$

Moreover the entropy

$$S(f) = -\sum_{i\in I}\left[f_i \log f_i - \eta_i(1+\eta_i f_i)\log(1+\eta_i f_i)\right] \tag{131.30}$$

satisfies

$$\dot{S}(f) = \sum_{r\in\mathcal{R}} w_r\big(L_r(f) - G_r(f)\big)\log\left(\frac{L_r(f)}{G_r(f)}\right) \geq 0. \tag{131.31}$$

Equality holds if and only if $L_r(f) = G_r(f)$ for every reaction with $w_r > 0$.

Proof. The finite state space lets us work entirely inside the open domain $\mathcal{D}$, where every factor in L_r and G_r is positive. Thus the logarithms below are honest finite-dimensional functions, and the boundary of $\mathcal{D}$ is not used in the argument.

For the conserved quantity,

$$\frac{d}{dt}Q_A(f) = \sum_r w_r(L_r - G_r)\big[-A(P_i) - A(P_j) + A(P_k) + A(P_l)\big],$$

and the bracket vanishes by (131.25).

For the entropy, differentiate the one-cell entropy. The same formula covers the classical, Bose, and Fermi signs because

$$\frac{\mathrm{d}}{\mathrm{d}f}\left[-f\log f + \eta(1+\eta f)\log(1+\eta f)\right] = -\log\left(\frac{f}{1+\eta f}\right),$$

with the $\eta = 0$ case understood by taking the limit. Hence

$$\frac{\partial S}{\partial f_i} = -\log\left(\frac{f_i}{1+\eta_i f_i}\right). \tag{131.32}$$

For $r = (i,j;k,l)$,

$$\nu_r \cdot \nabla S = \log\left(\frac{f_i f_j(1+\eta_k f_k)(1+\eta_l f_l)}{f_k f_l(1+\eta_i f_i)(1+\eta_j f_j)}\right) = \log\left(\frac{L_r}{G_r}\right).$$

Substitution into $\dot{S} = \nabla S \cdot \dot{f}$ gives (131.31). The sign is fixed reaction by reaction: for $x, y > 0$,

$$(x - y)\log(x/y) \geq 0, \qquad x > 0, \quad y > 0.$$

Indeed $x > y$ implies $\log x - \log y > 0$, while $x < y$ makes both factors negative. Equality in the sum holds exactly when every positive-weight reaction has $L_r = G_r$, because all summands are nonnegative. □

Definition 131.5 (Finite local equilibrium and collision invariants)**.** A state $f^{(0)} \in \mathcal{D}$ is a finite equilibrium for the datum when

$$\frac{f_i^{(0)}}{1 + \eta_i f_i^{(0)}} = \exp[-\alpha - A(P_i)] \tag{131.33}$$

for some $\alpha \in \mathbb{R}$ and $A \in \mathcal{P}^*$. A function $\chi : I \to \mathbb{R}$ is a collision invariant when

$$\chi_i + \chi_j - \chi_k - \chi_l = 0 \tag{131.34}$$

for every reaction $i + j \rightleftarrows k + l$.

Finite linearized collision operator. Let $f^{(0)}$ be a finite equilibrium. Write

$$f_i = f_i^{(0)} + f_i^{(0)}(1 + \eta_i f_i^{(0)})\chi_i. \tag{131.35}$$

To first order in χ,

$$L_r(f) - G_r(f) = M_r\big(\chi_i + \chi_j - \chi_k - \chi_l\big) + O(\chi^2), \tag{131.36}$$

where

$$M_r = L_r(f^{(0)}) = G_r(f^{(0)}) > 0. \tag{131.37}$$

Consequently the linearized entropy production is

$$\dot{S}^{(2)} = \sum_{r \in \mathcal{R}} w_r M_r \big(\chi_i + \chi_j - \chi_k - \chi_l\big)^2 \geq 0. \tag{131.38}$$

Its null space is precisely the space of collision invariants. If the reaction graph is connected enough that every collision invariant is of the form $\chi_i = \alpha + A(P_i)$, then the only zero modes are the conserved hydrodynamic variables.

The linearization is finite-dimensional algebra. At equilibrium, detailed balance follows from (131.33) and additive conservation:

$$\frac{L_r(f^{(0)})}{G_r(f^{(0)})} = \exp[-A(P_i) - A(P_j) + A(P_k) + A(P_l)] = 1.$$

For a variation $\delta f_i = f_i^{(0)}(1 + \eta_i f_i^{(0)})\chi_i$,

$$\delta \log\left(\frac{f_i}{1 + \eta_i f_i}\right) = \chi_i.$$

Therefore

$$\delta \log(L_r/G_r) = \chi_i + \chi_j - \chi_k - \chi_l.$$

Since $L_r = G_r = M_r$ at equilibrium, this is equivalent to (131.36): if $R_r = L_r/G_r$, then

$$\delta(L_r - G_r) = M_r\,\delta\log R_r.$$

The second variation of the entropy-production summand is obtained from

$$(L_r - G_r)\log(L_r/G_r) = M_r(\delta\log R_r)^2 + O(\chi^3),$$

because both $L_r - G_r$ and $\log(L_r/G_r)$ vanish to first order at equilibrium with the same linear coefficient $M_r^{-1}\delta(L_r - G_r)$. Summing the reactions gives (131.38). Its null space is the intersection of the kernels of the linear forms $\chi_i + \chi_j - \chi_k - \chi_l$, which is precisely the space of collision invariants by Definition 131.5. Under the stated connectivity hypothesis this space is the span of the constant invariant and the additive conserved quantities $A(P_i)$.

The continuum $2 \to 2$ formulas are obtained from Definition 131.3 by replacing the finite reaction sum by the invariant positive measure

$$d\Pi_1 d\Pi_2 d\Pi_3 d\Pi_4\,(2\pi)^D\delta^{(D)}(p_1 + p_2 - p_3 - p_4)|\mathcal{M}_{ab\to cd}|^2.$$

The finite theorem is not a derivation of this measure from QFT; it is the exact algebraic part of the kinetic construction once the collision measure has been derived or declared.

131.9 Worked Example: Relaxation-Time Shear Viscosity

Controlled Approximation 131.6 (Constant relaxation time)**.** Consider a single massless classical species in $d = 3$, with $f^{(0)} = e^{-p/T}$, no conserved charge, and a collision operator replaced on the shear channel by

$$C^{(t)}_{\mathrm{RTA}}[\delta f] = -\frac{\delta f}{\tau_R},$$

where τ_R is a constant. This replacement preserves the shear tensor structure and gives a closed worked example. It is a model for the positive inverse collision operator on the shear subspace, not a derivation of a microscopic collision kernel.

Let

$$\delta f = -\tau_R f^{(0)}\frac{p^i p^j}{2TE}\sigma_{ij}, \qquad \sigma_{ij} = \partial_i v_j + \partial_j v_i - \frac{2}{3}\delta_{ij}\partial_k v_k.$$

Substitution in the kinetic stress tensor gives

$$\delta T^{ij} = -\eta\,\sigma^{ij},$$

where angular averaging in three dimensions uses

$$\int d\Omega\,\hat{p}_i\hat{p}_j\hat{p}_k\hat{p}_l = \frac{4\pi}{15}(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Since $\sigma^i{}_i = 0$, the coefficient is

$$\eta_{\mathrm{RTA}} = \frac{\tau_R}{15T}\int\frac{d^3\mathbf{p}}{(2\pi)^3}p^2 e^{-p/T} = \frac{4}{5}p_{\mathrm{therm}}\tau_R, \tag{131.39}$$

with

$$p_{\text{therm}} = \frac{1}{3}\int \frac{d^3\mathbf{p}}{(2\pi)^3} p\, e^{-p/T} = \frac{T^4}{\pi^2}.$$

This example displays the general structure of kinetic transport: a source term from gradients, inversion of the collision operator on a nonconserved subspace, and a positive quadratic form for the coefficient.

131.10 Gauge Theories and Soft Scales

In weakly coupled gauge theories, hard particles with momentum of order T interact through soft gauge exchange of order gT. Electric fields are screened by the Debye scale of Chapter 130; magnetic fields at scale g^2T remain nonperturbative in the static sector. Kinetic theory for gauge plasmas therefore contains several coupled effective descriptions:

$$\text{hard quasiparticles} \quad \leftrightarrow \quad \text{soft collective gauge fields} \quad \leftrightarrow \quad \text{ultrasoft hydrodynamic modes.}$$

Small-angle scattering, Landau damping, and collinear splitting processes must be matched in the same source and regulator convention before a transport coefficient is a well-defined weak-coupling prediction.

131.11 Controlled-Limit Requirements

A kinetic derivation from QFT requires:

spectral input	narrow quasiparticle peaks with a controlled off-shell remainder,
scale separation	$\ell_{\text{micro}} \ll \ell_{\text{variation}}$ and $\Gamma\ell_{\text{variation}} \ll 1$,
collision input	on-shell amplitudes with regulator and medium corrections,
conservation laws	collision invariants matching QFT Ward identities,
error estimate	bounds on neglected gradients, memory terms, and off-shell terms.

Open Problem 131.7 (QFT derivation of kinetic theory with errors)*.* Derive relativistic kinetic theory from Schwinger–Keldysh QFT with explicit error bounds, including quasiparticle projection, collision kernels, screening, collinear processes in gauge theory, and the passage from kinetic theory to hydrodynamic transport.

Chapter 132

Anomalous and Topological Transport

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- **$\hbar$.** 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter studies transport terms fixed by anomalies and topology. The logic is real-time response first: a transport coefficient is defined by a correlator or by the response of a generating functional to background sources, and anomaly equations constrain the allowed terms.

132.1 Hydrodynamic Source Datum

Let $W[g, A]$ be the equilibrium generating functional for a QFT with stress tensor $T^{\mu\nu}$ and current J^μ. Variations define

$$\delta W = \frac{1}{2}\int \mathrm{d}^D x\,\sqrt{-g}\,\langle T^{\mu\nu}\rangle\,\delta g_{\mu\nu} + \int \mathrm{d}^D x\,\sqrt{-g}\,\langle J^\mu\rangle\,\delta A_\mu.$$

Hydrodynamic variables are a temperature T, velocity u^μ, and chemical potential μ, with

$$u^\mu u_\mu = -1.$$

The ideal constitutive relations are

$$T^{\mu\nu}_{(0)} = \epsilon u^\mu u^\nu + p\Delta^{\mu\nu}, \qquad J^\mu_{(0)} = n u^\mu,$$

where

$$\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu.$$

132.2 Anomaly Equation

In four dimensions, a $U(1)^3$ anomaly has the local equation

$$\nabla_\mu J^\mu = \frac{C}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}.$$

The coefficient C is part of the current normalization. The magnetic field and vorticity in the local rest frame are

$$B^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}u_\nu F_{\rho\sigma}, \qquad \omega^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}u_\nu \nabla_\rho u_\sigma.$$

The first-derivative parity-odd current terms have the form

$$J^\mu_{\text{odd}} = \xi_B B^\mu + \xi_\omega \omega^\mu.$$

The coefficients ξ_B, ξ_ω are transport coefficients. They must be defined either by Kubo formulae or by differentiating an equilibrium generating functional.

For the explicit four-dimensional formulas below, let J_R and J_L be the source-normalized right- and left-handed number currents of one massless Dirac fermion, and set

$$J_V = J_R + J_L, \qquad J_A = J_R - J_L, \qquad \mu_R = \mu_V + \mu_A, \qquad \mu_L = \mu_V - \mu_A.$$

The vector source is A_V. In the convention of Chapter 51, preserving the vector Ward identity gives

$$\nabla_\mu J^\mu_A = \frac{1}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}(F_V)_{\mu\nu}(F_V)_{\rho\sigma} \qquad \text{for a unit source charge.}$$

If a physical electromagnetic source A_{em} couples with charge $Q = eq$, then $A_V = QA_{\text{em}}$. The current conjugate to A_{em} is QJ_V, and the magnetic response acquires the factor $Q^2 = e^2q^2$.

132.3 Anomalous Source Protocols and Current Representatives

Anomaly-induced transport is meaningful only after the source protocol and the current representative have been specified. This point is not a matter of taste: anomalous Ward identities, Chern–Simons contact terms, and zero-frequency limits mix under local counterterms.

Definition 132.1 (Hydrostatic anomalous-source datum)**.** A hydrostatic anomalous-source datum consists of the following objects:

1. a stationary Lorentzian background (g, A_I) with a timelike Killing field K;
2. chemical potentials μ_I defined as the gauge-invariant thermal holonomies of A_I around the Euclidean thermal circle, after choosing the time-independent gauge used in the hydrostatic functional;
3. a choice of local counterterms, equivalently a choice of consistent current representatives;
4. a declaration of which gauge symmetries are exact constraints on the background functional and which sources are allowed to have anomalous Ward identities.

For an anomalous charge, μ_I is a background source parameter in this datum. It is a genuine grand-canonical chemical potential for a conserved operator only in source sectors for which the corresponding anomaly density vanishes, or after passing to a conserved combination of charges.

Definition 132.2 (Consistent and covariant currents)**.** Let $W[A]$ be a renormalized connected generating functional. The consistent current is the variational current

$$J^\mu_{\text{cons}}(x) = \frac{1}{\sqrt{-g(x)}} \frac{\delta W}{\delta A_\mu(x)}.$$

With the sign convention

$$\delta_\alpha W = -\int d^4x \sqrt{-g}\, \alpha\, \mathcal{A}_{\text{cons}}, \qquad \delta_\alpha A_\mu = \nabla_\mu \alpha, \tag{132.1}$$

integration by parts gives

$$\nabla_\mu J^\mu_{\text{cons}} = \mathcal{A}_{\text{cons}}.$$

A covariant current is a shifted local current

$$J^\mu_{\text{cov}} = J^\mu_{\text{cons}} + J^\mu_{\text{BZ}}, \tag{132.2}$$

where J^μ_{BZ} is a local Bardeen–Zumino polynomial in the background fields chosen so that J_{cov} transforms covariantly under the background gauge symmetry being represented. The shift (132.2) changes local contact terms in current correlators; it is not a change of the Hilbert-space operator algebra.

Remark 132.3 (Vector-preserving convention in the Dirac example). The Dirac formulas in this chapter use the Bardeen convention in which the vector source A_V is an exact background gauge field and the axial source may have an anomalous Ward identity. Thus the physical electromagnetic response is obtained by varying the vector-preserving functional with respect to A_{em}. The axial chemical potential in (132.6) is a hydrostatic source parameter. A conserved axial charge in a sector with nonzero $\epsilon F_V F_V$ would require additional anomaly-free data.

132.4 Kubo Definitions

In flat space at temperature T, define retarded correlators

$$G^R_{AB}(t, \mathbf{x}) = -\mathrm{i}\theta(t)\langle[A(t, \mathbf{x}), B(0)]\rangle_\beta.$$

For a current response to a weak magnetic field, the chiral magnetic coefficient is

$$\xi_B = \lim_{k\to 0} \frac{\mathrm{i}}{2k} \epsilon_{ijn}\, G^R_{J^i J^j}(\omega = 0, \mathbf{k} = k\hat{e}_n).$$

This formula is a definition of the transport coefficient in the declared static, homogeneous limit. If the limits $\omega \to 0$ and $k \to 0$ fail to commute, the chosen order is part of the observable.

132.5 Equilibrium Derivation

Place the system on a stationary background

$$\mathrm{d}s^2 = -\mathrm{e}^{2\sigma}(dt + a_i dx^i)^2 + h_{ij}dx^i dx^j, \qquad A = A_0(dt + a_i dx^i) + A_i dx^i.$$

Gauge and Kaluza–Klein invariant three-dimensional fields are

$$\mathcal{A}_i = A_i - A_0 a_i, \qquad A_0.$$

An anomalous equilibrium functional contains Chern–Simons-type terms

$$W_{\text{odd}} = \int_{\Sigma_3} \left[c_1 A_0\, \mathcal{A} \wedge \mathrm{d}\mathcal{A} + c_2 A_0^2\, \mathcal{A} \wedge \mathrm{d}a + c_3 A_0^3\, a \wedge \mathrm{d}a\right],$$

with coefficients constrained by the anomaly and by large gauge transformations. Differentiating this functional with respect to $\mathcal{A}_i$ and a_i gives the parity-odd equilibrium pieces of J^i and T^{0i}. The derivation is algebraic once the equilibrium functional and its gauge-variation convention are fixed.

Remark 132.4 (Hydrostatic Chern–Simons variation). Let Σ_3 be closed, and let A and a be ordinary spatial one-forms on Σ_3. For constant coefficients c_{AA}, c_{Aa}, and c_{aa}, set

$$W_{\mathrm{CS}}[A, a] = \int_{\Sigma_3} \left[c_{AA} A \wedge \mathrm{d}A + c_{Aa} A \wedge \mathrm{d}a + c_{aa} a \wedge \mathrm{d}a\right]. \tag{132.3}$$

In local coordinates with orientation d^3x and $\epsilon^{123} = 1$, define

$$B_A^i = \epsilon^{ijk}\partial_j A_k, \qquad B_a^i = \epsilon^{ijk}\partial_j a_k. \tag{132.4}$$

Then

$$\frac{\delta W_{\mathrm{CS}}}{\delta A_i} = 2c_{AA}B_A^i + c_{Aa}B_a^i, \qquad \frac{\delta W_{\mathrm{CS}}}{\delta a_i} = c_{Aa}B_A^i + 2c_{aa}B_a^i. \tag{132.5}$$

If the coefficients are functions of A_0 or of thermodynamic scalars, (132.5) is the part of the variation with respect to the spatial one-forms at fixed scalar sources; variations of the scalar sources produce the corresponding charge and energy densities.

On a closed Σ_3,

$$\delta \int A \wedge \mathrm{d}A = \int \delta A \wedge \mathrm{d}A + \int A \wedge \mathrm{d}(\delta A) = 2\int \delta A \wedge \mathrm{d}A,$$

because $0 = \int \mathrm{d}(A \wedge \delta A) = \int \mathrm{d}A \wedge \delta A - \int A \wedge \mathrm{d}(\delta A)$ and $\mathrm{d}A \wedge \delta A = \delta A \wedge \mathrm{d}A$. Similarly,

$$\delta \int A \wedge \mathrm{d}a = \int \delta A \wedge \mathrm{d}a + \int \mathrm{d}A \wedge \delta a, \qquad \delta \int a \wedge \mathrm{d}a = 2\int \delta a \wedge \mathrm{d}a.$$

Writing $\delta A \wedge \mathrm{d}A = \delta A_i \epsilon^{ijk}\partial_j A_k\, \mathrm{d}^3x$ and the analogous formulas for a gives (132.5).

132.6 Chiral Magnetic Coefficient

The anomaly fixes a Chern–Simons term in the three-dimensional equilibrium functional. Hold the axial source constant in Euclidean time,

$$(A_A)_0 = \mu_A, \qquad \partial_i \mu_A = 0,$$

and set the spatial axial field to zero. The inflow functional for the vector-vector-axial anomaly contains the five-dimensional local term

$$W_5[A_A, A_V] = -\frac{1}{4\pi^2} \int_{X_5} A_A \wedge F_V \wedge F_V,$$

whose axial gauge variation is

$$\delta_\beta W_5 = -\frac{1}{4\pi^2} \int_{\partial X_5} \beta\, F_V \wedge F_V.$$

Since

$$F_V \wedge F_V = \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} (F_V)_{\mu\nu} (F_V)_{\rho\sigma}\, \mathrm{d}^4 x,$$

this is the inflow form of the axial anomaly displayed above, with the sign fixed by the convention that the combined bulk-boundary system is invariant. Reducing the same local term on the thermal circle gives the parity-odd equilibrium functional

$$W_{\mathrm{CME}}[A_V; \mu_A] = \frac{\mu_A}{4\pi^2} \int_{\Sigma_3} A_V \wedge \mathrm{d}A_V.$$

In components,

$$\int_{\Sigma_3} A_V \wedge \mathrm{d}A_V = \int \mathrm{d}^3 x\, \epsilon^{ijk} (A_V)_i \partial_j (A_V)_k.$$

Varying with respect to the spatial vector source gives

$$\frac{\delta W_{\mathrm{CME}}}{\delta (A_V)_i} = \frac{\mu_A}{2\pi^2} \epsilon^{ijk} \partial_j (A_V)_k = \frac{\mu_A}{2\pi^2} B_V^i.$$

Thus the covariant equilibrium chiral magnetic response of the source-normalized vector current is

$$J_V^i = \frac{\mu_A}{2\pi^2} B_V^i. \tag{132.6}$$

For the physical electromagnetic current of a Dirac fermion of charge $Q = eq$, $B_V = QB_{\mathrm{em}}$ and $J_{\mathrm{em}} = QJ_V$, hence

$$J_{\mathrm{em}}^i = \frac{e^2 q^2 \mu_A}{2\pi^2} B_{\mathrm{em}}^i. \tag{132.7}$$

The same coefficient is obtained without invoking hydrodynamics by counting lowest Landau levels. For $B_V = B\hat{z}$, the degeneracy per unit transverse area is $|B|/(2\pi)$. The right-handed lowest Landau level has group velocity along B, while the left-handed lowest Landau level has group velocity opposite to B. Filling momenta from 0 to μ_R and 0 to μ_L gives

$$J_R^z = \frac{\mu_R}{4\pi^2} B, \qquad J_L^z = -\frac{\mu_L}{4\pi^2} B.$$

Therefore

$$J_V^z = J_R^z + J_L^z = \frac{\mu_R - \mu_L}{4\pi^2} B = \frac{\mu_A}{2\pi^2} B,$$

which agrees with (132.6). This derivation also shows why the order of limits in the Kubo formula is part of the observable: the current is an equilibrium response to a static magnetic source and a specified axial chemical potential.

132.7 Chiral Vortical Coefficients

For one massless Dirac fermion, the covariant first-derivative vortical currents in the same source-normalized convention are

$$J_V^\mu\big|_\omega = \frac{\mu_V \mu_A}{\pi^2}\,\omega^\mu, \tag{132.8}$$

$$J_A^\mu\big|_\omega = \left[\frac{\mu_V^2 + \mu_A^2}{2\pi^2} + \frac{T^2}{6}\right]\omega^\mu. \tag{132.9}$$

These formulas follow by adding the two Weyl contributions

$$J_R^\mu\big|_\omega = \left(\frac{\mu_R^2}{4\pi^2} + \frac{T^2}{12}\right)\omega^\mu, \qquad J_L^\mu\big|_\omega = -\left(\frac{\mu_L^2}{4\pi^2} + \frac{T^2}{12}\right)\omega^\mu.$$

The vector current is the sum $J_R + J_L$, while the axial current is the difference $J_R - J_L$. Hence the $T^2/12$ terms cancel in the vector current and add in the axial current. There is therefore no universal source-normalized vector-current term proportional to $\mu_A T$ in this Dirac example; the universal temperature-squared contribution belongs to the axial vortical current and is tied to the mixed axial-gravitational anomaly.

132.8 Entropy Constraint

Let

$$\nabla_\mu T^{\mu\nu} = F^{\nu\lambda} J_\lambda, \qquad \nabla_\mu J^\mu = \mathcal{A}$$

with anomaly density $\mathcal{A}$. The canonical entropy current is

$$S^\mu_{\text{can}} = p\beta^\mu - \beta_\nu T^{\mu\nu} - \alpha J^\mu, \qquad \beta^\mu = \frac{u^\mu}{T}, \quad \alpha = \frac{\mu}{T}.$$

The following derivation fixes the signs in this formula. It is independent of any derivative expansion.

Canonical entropy divergence with anomaly. Let

$$T^{\mu\nu} = \epsilon u^\mu u^\nu + p\Delta^{\mu\nu} + \tau^{\mu\nu}, \qquad J^\mu = n u^\mu + \nu^\mu, \tag{132.10}$$

with $u_\mu \tau^{\mu\nu} = 0$ and $u_\mu \nu^\mu = 0$. Assume the thermodynamic relations

$$\mathrm{d}p = s\,\mathrm{d}T + n\,\mathrm{d}\mu, \qquad \epsilon + p = Ts + \mu n. \tag{132.11}$$

Then the canonical entropy current satisfies

$$\nabla_\mu S^\mu_{\text{can}} = -\tau^{\mu\nu}\nabla_\mu\beta_\nu - \nu^\mu \left(\nabla_\mu\alpha - \beta^\nu F_{\mu\nu}\right) - \alpha\,\mathcal{A}. \tag{132.12}$$

The calculation is as follows. Differentiate

$$S^\mu_{\text{can}} = p\beta^\mu - \beta_\nu T^{\mu\nu} - \alpha J^\mu.$$

Using the anomalous Ward identities gives

$$\nabla_\mu S^\mu_{\text{can}} = \beta^\mu\nabla_\mu p + p\nabla_\mu\beta^\mu - T^{\mu\nu}\nabla_\mu\beta_\nu - J^\mu\nabla_\mu\alpha - \beta_\nu F^{\nu\lambda}J_\lambda - \alpha\mathcal{A}. \tag{132.13}$$

The last source term can be rewritten as

$$-J^\mu\nabla_\mu\alpha - \beta_\nu F^{\nu\lambda}J_\lambda = -J^\mu\left(\nabla_\mu\alpha - \beta^\nu F_{\mu\nu}\right), \tag{132.14}$$

using antisymmetry of $F_{\mu\nu}$. It remains to show that the ideal part cancels. Substituting

$$T^{\mu\nu}_{\text{id}} = \epsilon u^\mu u^\nu + p(g^{\mu\nu} + u^\mu u^\nu), \qquad J^\mu_{\text{id}} = nu^\mu,$$

and $\beta^\mu = u^\mu/T$, $\alpha = \mu/T$, the ideal contribution to (132.13) is

$$\frac{u^\mu\nabla_\mu p}{T} + p\nabla_\mu\left(\frac{u^\mu}{T}\right) - T^{\mu\nu}_{\text{id}}\nabla_\mu\left(\frac{u_\nu}{T}\right) - nu^\mu\nabla_\mu\left(\frac{\mu}{T}\right). \tag{132.15}$$

Since $u^\nu u_\nu = -1$, $u^\nu\nabla_\mu u_\nu = 0$. A direct expansion of (132.15) reduces it to

$$\frac{u^\mu}{T}\big(\nabla_\mu p - s\nabla_\mu T - n\nabla_\mu\mu\big),$$

which vanishes by $\mathrm{d}p = s\,\mathrm{d}T + n\,\mathrm{d}\mu$. The remaining terms are exactly those involving $\tau^{\mu\nu}$, ν^μ, and $\mathcal{A}$, giving (132.12).

Equation (132.12) explains how anomaly-induced transport can be nondissipative. The anomaly contributes the fixed term $-\alpha\mathcal{A}$; parity-odd additions to the entropy current, built from B^μ, ω^μ, and thermodynamic scalars, can move this term into a total divergence and thereby relate ξ_B, ξ_ω to anomaly coefficients. Coefficients invisible to this entropy analysis must instead be fixed by the hydrostatic functional or by Kubo formulae with the current representative and contact terms specified.

132.9 Topological Terms and Contact Terms

Equilibrium Chern–Simons terms may change local current correlators by contact terms. A transport coefficient must therefore state whether the current is covariant or consistent, which background functional is used, and which contact terms are included in the retarded correlator. The physically measured response is attached to the source protocol and to the contact-term convention, as well as to the local operator symbol.

Open Problem 132.5 (Nonperturbative anomalous transport). Derive anomaly-induced transport coefficients in interacting QFT directly from nonperturbative KMS states, local current algebras, Ward identities, contact terms, and hydrodynamic scaling limits, with the order of limits and source protocol stated as part of the theorem.

Chapter 133

Nonequilibrium Steady States And Open-System Limits

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Nonequilibrium steady states and open systems are real-time QFT objects. A steady state is a positive invariant state on an algebra of observables. An open-system limit is a scaling limit of a larger closed system in which a subalgebra evolves by a completely positive trace-preserving semigroup. This chapter fixes these definitions, derives the reservoir entropy-production formula, records finite path-measure fluctuation identities before taking a continuum limit, derives the finite-step Langevin/MSRJD representation, derives the finite reaction-network Doi–Peliti representation, records the tilted-generator construction of dynamical large deviations, records the empirical-flow level-2.5 cost behind finite dynamical large deviations, records the weak-coupling route to the GKSL generator, and connects the Markovian limit to a Schwinger–Keldysh influence functional and to hydrodynamic relaxation with noise.

133.1 Proof Architecture For Nonequilibrium Claims

The chapter has many finite-regulator identities because nonequilibrium QFT has several inequivalent entry points. The entry points serve different roles in one dependency ladder.

At the first layer are exact finite probability or operator-algebra statements. These include finite reservoir jump processes, labelled path measures and their Radon–Nikodym ratios, finite Gaussian transition kernels, finite reaction-network master equations, tilted finite Markov generators, empirical-flow costs, finite GKSL maps, and finite Gaussian Schwinger–Keldysh kernels. Each statement at this layer has a specified state space, measure, kernel, or matrix generator. The checks in this chapter live at this layer: they protect signs, normalizations, positivity, and finite algebra.

The second layer is the representation layer. Here a finite statement is rewritten in a field-theoretic coordinate system: a Fourier-dual MSRJD response variable for Gaussian increments, creation and annihilation variables for Doi–Peliti occupation dynamics, Keldysh r/a fields and noise kernels for influence functionals, or source variables for tilted generators. A representation at this layer is useful only because it is connected to the first layer by an explicit finite identity. It is not yet a continuum stochastic field theory.

The third layer is the scaling-hypothesis layer. One must declare the spatial regulator, local observables, field and rate scalings, time scaling, source normalization, topology of convergence, and which Ward identities or conservation laws survive the limit. Dynamic-critical names such as Model A, Model B, Model C, and Model H belong here: they are hypotheses about retained slow variables, reversible and dissipative structures, and continuum renormalization, not finite theorems.

The fourth layer is the continuum QFT output layer. A construction at this layer supplies real-time correlation functions, response functions, entropy production, fluctuation or large-deviation functionals, hydrodynamic coefficients, and open-system semigroups as limits of regulated objects. A Schwinger–Keldysh action, an MSRJD action, or a Doi–Peliti action counts as a QFT object only after this limiting statement and its observable map have been specified.

This ladder is the organizing rule for the rest of the chapter. For example, the finite fluctuation theorem proves a path-measure identity; it does not construct a continuum entropy-production operator. The finite MSRJD and Doi–Peliti rewritings prove coordinate changes from regulated stochastic processes; they do not prove a renormalized nonequilibrium fixed point. The finite SK Gaussian bridge identifies a noise coordinate in a quadratic influence functional; it does not by itself prove Markovianity, locality, or complete positivity of a reduced continuum dynamics.

133.2 Steady-State Datum

Let $\mathcal{A}$ be a $*$-algebra of observables for a real-time QFT, or a local C^*-algebra when a norm-completed formulation is available. A state is a positive normalized linear functional

$$\omega : \mathcal{A} \to \mathbb{C}, \qquad \omega(A^*A) \geq 0, \qquad \omega(\mathbf{1}) = 1.$$

Let τ_t be the time-evolution automorphism group. A steady state is a state ω_{ss} such that

$$\omega_{\mathrm{ss}}(\tau_t(A)) = \omega_{\mathrm{ss}}(A) \tag{133.1}$$

for all A in the common domain of the state and the time evolution. Local stationarity means that the distributional correlation functions of local observables are invariant under simultaneous time translation. A nonequilibrium steady state is a steady state whose current expectation values or reservoir boundary data are not those of a single KMS state.

Equation (133.1) is only the kinematic definition. Existence, uniqueness, clustering, stability under changes of the preparation protocol, and local normality relative to reference states are separate properties. They must be proved or assumed in any concrete model.

133.3 Reservoir Construction

Consider a QFT on a line prepared at $t = 0$ by joining two half-systems. The left and right initial states are KMS states with parameters

$$(\beta_L, \mu_L), \qquad (\beta_R, \mu_R).$$

For a local observable A supported near the junction, a central steady state is a weak limit

$$\omega_{\rm ss}(A) = \lim_{t\to\infty} \omega_0(\tau_t(A)), \tag{133.2}$$

if the limit exists. The limit is local: it is not a statement that the entire infinite-volume density matrix converges in trace norm. When the large-time state has a ray dependence, one instead studies limits of $\omega_0(\tau_t(A_{x=vt}))$, with $v = x/t$. The central steady state is the ray $v = 0$.

The physical currents in the central steady state are defined by the expectation values of local current operators:

$$J_E = \omega_{\rm ss}(j_E), \qquad J_Q = \omega_{\rm ss}(j_Q),$$

with the convention that positive current flows from left to right.

133.4 Reservoir Entropy Production

Let the right reservoir gain energy and charge at rates

$$\frac{dE_R}{dt} = J_E, \qquad \frac{dQ_R}{dt} = J_Q,$$

while the left reservoir loses the same amounts. The reservoir entropy change is computed using the equilibrium reservoir first laws

$$dS_\alpha = \beta_\alpha(dE_\alpha - \mu_\alpha dQ_\alpha), \qquad \alpha = L, R.$$

Therefore

$$\begin{aligned}\dot{S}_{\rm res} &= \beta_R(J_E - \mu_R J_Q) + \beta_L(-J_E + \mu_L J_Q)\\ &= (\beta_R - \beta_L)J_E - (\beta_R\mu_R - \beta_L\mu_L)J_Q.\end{aligned} \tag{133.3}$$

This formula is an identity once the currents and reservoir parameters are defined. Positivity is a dynamical statement about the state and the reservoir construction. In linear response, if

$$X = \begin{pmatrix} \beta_R - \beta_L \\ -(\beta_R\mu_R - \beta_L\mu_L) \end{pmatrix}, \qquad \begin{pmatrix} J_E \\ J_Q \end{pmatrix} = LX,$$

then $\dot{S}_{\rm res} = X^T L X$. The second law in this linear regime is equivalent to the positive semidefiniteness of the symmetric part of the Onsager matrix L.

133.5 Finite Local Detailed Balance

The reservoir formula above is the steady-state remnant of a stronger finite statement. We record it because it fixes the sign convention for entropy production before any continuum or hydrodynamic limit is taken.

Definition 133.1 (Finite reservoir jump datum)**.** A finite reservoir jump datum consists of:

1. a finite set I of system configurations;
2. real energy and charge functions E_i, Q_i on I;
3. reservoirs α with parameters $(\beta_\alpha, \mu_\alpha)$;
4. transition rates $W_{ij}^{(\alpha)} \geq 0$, $i \neq j$, for jumps $i \to j$ induced by reservoir α.

The rates obey local detailed balance when

$$W_{ij}^{(\alpha)} > 0 \quad \Longleftrightarrow \quad W_{ji}^{(\alpha)} > 0, \qquad \log \frac{W_{ij}^{(\alpha)}}{W_{ji}^{(\alpha)}} = -\beta_\alpha \left[E_j - E_i - \mu_\alpha(Q_j - Q_i)\right]. \tag{133.4}$$

The master equation is

$$\dot{p}_i = \sum_\alpha \sum_{j \neq i} \left[p_j W_{ji}^{(\alpha)} - p_i W_{ij}^{(\alpha)}\right], \qquad p_i \geq 0, \qquad \sum_i p_i = 1. \tag{133.5}$$

Finite entropy production. For a finite reservoir jump datum satisfying (133.4), define the Shannon entropy

$$S_{\mathrm{sys}}(p) = -\sum_i p_i \log p_i. \tag{133.6}$$

The reservoir entropy-production rate is

$$\dot{S}_{\mathrm{res}} = \sum_\alpha \sum_{i \neq j} p_i W_{ij}^{(\alpha)} \log \frac{W_{ij}^{(\alpha)}}{W_{ji}^{(\alpha)}}, \tag{133.7}$$

and the total entropy-production rate is

$$\dot{S}_{\mathrm{tot}} = \frac{1}{2} \sum_\alpha \sum_{i \neq j} \left(p_i W_{ij}^{(\alpha)} - p_j W_{ji}^{(\alpha)}\right) \log \frac{p_i W_{ij}^{(\alpha)}}{p_j W_{ji}^{(\alpha)}} \geq 0. \tag{133.8}$$

Moreover

$$\dot{S}_{\mathrm{tot}} = \frac{d}{dt} S_{\mathrm{sys}}(p) + \dot{S}_{\mathrm{res}}. \tag{133.9}$$

At a stationary distribution p^{ss}, the system entropy derivative vanishes, hence $\dot{S}_{\mathrm{res}} = \dot{S}_{\mathrm{tot}} \geq 0$. The local detailed-balance relation says that a jump $i \to j$ changes the reservoir entropy by

$$-\beta_\alpha \left[E_j - E_i - \mu_\alpha(Q_j - Q_i)\right] = \log \frac{W_{ij}^{(\alpha)}}{W_{ji}^{(\alpha)}}, \tag{133.10}$$

which gives (133.7) after averaging over jumps. Differentiating S_{sys} and using $\sum_i \dot{p}_i = 0$ gives

$$\frac{dS_{\text{sys}}}{dt} = -\sum_i \dot{p}_i \log p_i. \tag{133.11}$$

Substituting the master equation and symmetrizing over i, j,

$$\frac{dS_{\text{sys}}}{dt} = \frac{1}{2}\sum_\alpha \sum_{i\neq j} \left(p_i W_{ij}^{(\alpha)} - p_j W_{ji}^{(\alpha)}\right) \log \frac{p_i}{p_j}. \tag{133.12}$$

Similarly, (133.7) can be written as

$$\dot{S}_{\text{res}} = \frac{1}{2}\sum_\alpha \sum_{i\neq j} \left(p_i W_{ij}^{(\alpha)} - p_j W_{ji}^{(\alpha)}\right) \log \frac{W_{ij}^{(\alpha)}}{W_{ji}^{(\alpha)}}. \tag{133.13}$$

Adding the last two equations gives (133.8). Each summand is nonnegative because

$$(x-y)\log(x/y) \geq 0, \qquad x > 0, \quad y > 0. \tag{133.14}$$

Terms with $x = y = 0$ are omitted by continuity. This proves both (133.9) and the inequality.

The two-reservoir current formula (133.3) is obtained from this finite statement when the stationary currents have well-defined macroscopic energy and charge rates. The finite theorem does not prove existence of the long-time QFT state ω_{ss}; it proves the algebraic positivity that any Markovian reservoir reduction with local detailed balance must satisfy.

133.6 Finite Path Measures and Fluctuation Relations

Fluctuation relations are identities between probability measures on histories. They become mathematical statements after the path space and the reversed path measure have been constructed. We state the finite continuous-time version because it is the regulator-level statement that survives, if the corresponding continuum limit exists, in nonequilibrium QFT.

Let I be a finite configuration set. For $0 \leq t \leq T$, let $W_{ij}^{(\alpha)}(t) \geq 0$, $i \neq j$, be time-dependent jump rates labelled by reservoirs α, and let

$$r_i(t) = \sum_\alpha \sum_{j\neq i} W_{ij}^{(\alpha)}(t)$$

be the escape rate from i. A labelled trajectory is

$$\gamma = (i_0; t_1, \alpha_1, i_1; \ldots; t_n, \alpha_n, i_n), \qquad 0 < t_1 < \cdots < t_n < T,$$

where the state is i_m on $t_m < t < t_{m+1}$, with $t_0 = 0$ and $t_{n+1} = T$. Given an initial distribution p_0, the forward path density with respect to counting measure over labels and Lebesgue measure over ordered jump times is

$$\begin{aligned}\mathbb{P}_F(\gamma) = p_0(i_0) \exp&\left[-\int_0^{t_1} r_{i_0}(s)\,ds\right] \\ &\times \prod_{m=1}^{n} \left[W_{i_{m-1}i_m}^{(\alpha_m)}(t_m) \exp\left[-\int_{t_m}^{t_{m+1}} r_{i_m}(s)\,ds\right]\right]. \end{aligned}\tag{133.15}$$

The reversed protocol has rates

$$\widetilde{W}_{ij}^{(\alpha)}(t) = W_{ij}^{(\alpha)}(T - t)$$

and initial distribution q_0. The time-reversed trajectory is

$$\Theta\gamma = (i_n; T - t_n, \alpha_n, i_{n-1}; \dots; T - t_1, \alpha_1, i_0).$$

Assume absolute continuity: whenever a forward path has positive probability, the reversed path has positive probability under the reversed protocol. Define the trajectory entropy production by the Radon–Nikodym logarithm

$$\Sigma(\gamma) = \log \frac{\mathbb{P}_F(\gamma)}{\mathbb{P}_R(\Theta\gamma)}. \tag{133.16}$$

Proposition 133.2 (Finite path-measure fluctuation identity)**.** *For the finite jump process above,*

$$\Sigma(\gamma) = \log \frac{p_0(i_0)}{q_0(i_n)} + \sum_{m=1}^{n} \log \frac{W_{i_{m-1}i_m}^{(\alpha_m)}(t_m)}{W_{i_m i_{m-1}}^{(\alpha_m)}(t_m)}, \tag{133.17}$$

$$\left\langle e^{-\Sigma} \right\rangle_F = 1. \tag{133.18}$$

If q_0 is chosen to be the forward final distribution p_T, then $\langle \Sigma \rangle_F \geq 0$. More generally, $\langle \Sigma \rangle_F$ is the relative entropy

$$D(\mathbb{P}_F \| \Theta^{-1}\mathbb{P}_R)$$

between the forward history measure and the pullback of the reversed history measure.

Proof. Insert the path densities. The waiting-time factors in the denominator are the same factors as in the numerator after the change of variables $s \mapsto T - s$ on each reversed interval. They therefore cancel. What remains is exactly the initial-distribution ratio and the product of jump rate ratios in (133.17).

The integral identity is the normalization of the reversed path measure:

$$\left\langle e^{-\Sigma} \right\rangle_F = \int e^{-\Sigma(\gamma)} \, d\mathbb{P}_F(\gamma) = \int d\mathbb{P}_R(\Theta\gamma) = 1.$$

The map Θ is an involution on the labelled path space, so the last integral is the total mass of $\mathbb{P}_R$. Finally,

$$\langle \Sigma \rangle_F = \int \log \frac{d\mathbb{P}_F}{d(\Theta^{-1}\mathbb{P}_R)} \, d\mathbb{P}_F = D(\mathbb{P}_F \| \Theta^{-1}\mathbb{P}_R) \geq 0,$$

with equality precisely when the two history measures agree almost surely. When $q_0 = p_T$, the boundary term in (133.17) is the stochastic system entropy change $-\log p_T(i_n) + \log p_0(i_0)$, and the jump-rate term is the reservoir entropy change under local detailed balance. Hence the average is the mean total entropy production. $\square$

The detailed fluctuation relation is a direct disintegration of the same identity. If P_F is the forward law of Σ and P_R is the reversed law of the reversed entropy production, then

$$P_F(ds) = e^s \, P_R(-ds)$$

on every Borel set on which both measures are defined. This form is often the most convenient one experimentally, but the Radon–Nikodym statement (133.16) is the mathematical content.

Jarzynski identity as a special case. Take one thermal reservoir with inverse temperature β, no chemical potential, and a time-dependent finite energy function $E_i(t)$. Assume local detailed balance at each jump time:

$$\log \frac{W_{ij}(t)}{W_{ji}(t)} = -\beta\big(E_j(t) - E_i(t)\big).$$

Let $p_0(i) = Z(0)^{-1}e^{-\beta E_i(0)}$ and $q_0(i) = Z(T)^{-1}e^{-\beta E_i(T)}$. For a trajectory γ, define the protocol work

$$W_{\text{ext}}(\gamma) = \int_0^T \partial_t E_{i(t)}(t)\, dt, \tag{133.19}$$

where $i(t)$ is the piecewise-constant state path. The total system energy change decomposes as

$$E_{i_n}(T) - E_{i_0}(0) = W_{\text{ext}}(\gamma) + \sum_{m=1}^{n} \big(E_{i_m}(t_m) - E_{i_{m-1}}(t_m)\big).$$

Combining this identity with local detailed balance gives

$$\Sigma(\gamma) = \beta\big(W_{\text{ext}}(\gamma) - \Delta F\big), \qquad \Delta F = F(T) - F(0), \qquad F(t) = -\beta^{-1}\log Z(t).$$

Therefore (133.18) becomes

$$\left\langle e^{-\beta W_{\text{ext}}} \right\rangle_F = e^{-\beta \Delta F}. \tag{133.20}$$

For a continuum QFT quench, the corresponding use of (133.20) requires a specified finite regulator, a measured work protocol or two-time energy measurement replacement, a time-reversal operation, and a limit in which the history measures converge.

133.7 Finite-Step Langevin Measures and MSRJD Variables

The Martin–Siggia–Rose–Janssen–de Dominicis (MSRJD) response-field path integral used in classical stochastic field theory is an exact rewriting of a finite-dimensional path measure before it is a continuum notation. Let $x_n = (x_n^1, \ldots, x_n^N) \in \mathbb{R}^N$ be the variables of a finite regulator at $t_n = n\Delta t$, $0 \le n \le M$. The index $a = 1, \ldots, N$ may include internal labels and lattice sites. Fix a smooth drift vector field $F^a(x,t)$ and a real symmetric positive definite matrix D^{ab}. The Ito finite-step process is

$$x_{n+1}^a = x_n^a + \Delta t\, F^a(x_n, t_n) + \eta_n^a, \qquad \mathbb{E}(\eta_n^a \eta_m^b) = 2\Delta t\, D^{ab}\delta_{nm}. \tag{133.21}$$

With x_0 fixed and

$$\Delta x_n^a = x_{n+1}^a - x_n^a, \qquad F_n^a = F^a(x_n, t_n),$$

the conditional path density is

$$\mathbb{P}_\Delta[x_1, \ldots, x_M \mid x_0] = \prod_{n=0}^{M-1} \frac{\exp\big[-\frac{1}{4\Delta t}(\Delta x_n - \Delta t F_n)_a (D^{-1})^{ab} (\Delta x_n - \Delta t F_n)_b\big]}{\sqrt{(4\pi\Delta t)^N \det D}}. \tag{133.22}$$

This formula is the regulated stochastic theory. A continuum stochastic field theory is a limit of such finite measures, together with convergence of the observables under study.

The Gaussian kernel in (133.22) has the Fourier representation

$$\frac{\exp\left[-\frac{1}{4\Delta t}y_a(D^{-1})^{ab}y_b\right]}{\sqrt{(4\pi\Delta t)^N \det D}} = \int_{\mathbb{R}^N} \frac{d^N\hat{x}}{(2\pi)^N} \exp\left[-\mathrm{i}\hat{x}_a y^a - \Delta t\, \hat{x}_a D^{ab}\hat{x}_b\right]. \tag{133.23}$$

Indeed, the right side is the elementary Gaussian integral

$$\frac{1}{(2\pi)^N}\int_{\mathbb{R}^N} d^N\hat{x}\, \exp\left[-\hat{x}_a(\Delta t D^{ab})\hat{x}_b - \mathrm{i}\hat{x}_a y^a\right],$$

and completing the square gives the determinant and exponent on the left. The finite-step MSRJD identity follows by applying (133.23) to $y^a = \Delta x_n^a - \Delta t F_n^a$ at each time step:

$$\begin{aligned}\mathbb{E}_\Delta[\mathcal{O}] = \int_{\mathbb{R}^{NM}} \prod_{n=1}^{M} d^N x_n\, \mathcal{O}(x_0,\dots,x_M) \prod_{n=0}^{M-1} \int_{\mathbb{R}^N} \frac{d^N\hat{x}_n}{(2\pi)^N} \\ \times \exp\left\{\sum_{n=0}^{M-1}\left[-\mathrm{i}\hat{x}_{n,a}\big(\Delta x_n^a - \Delta t F_n^a\big) - \Delta t\, \hat{x}_{n,a}D^{ab}\hat{x}_{n,b}\right]\right\}.\end{aligned} \tag{133.24}$$

Here $\mathcal{O}$ is any bounded measurable functional of the finite path. If the initial condition has density $\rho_0(x_0)$, one also integrates over x_0 with weight ρ_0. The auxiliary variable $\hat{x}_{n,a}$ is the Fourier-dual variable to the finite increment constraint. The frequently used variable $\widetilde{x}_{n,a} = \mathrm{i}\hat{x}_{n,a}$ is a contour notation for the same finite integral; its convergence and allowed deformations are inherited from (133.23).

The same finite construction fixes the generator and the Fokker–Planck equation from finite moments. For $f \in C_b^4(\mathbb{R}^N)$, Taylor expansion of $f(x + \Delta t F(x,t) + \eta)$ and the Gaussian moments $\mathbb{E}\eta^a = 0$, $\mathbb{E}(\eta^a\eta^b) = 2\Delta t D^{ab}$ give

$$\mathbb{E}[f(x_{n+1}) \mid x_n = x] = f(x) + \Delta t\left(F^a(x,t_n)\partial_a f(x) + D^{ab}\partial_a\partial_b f(x)\right) + O((\Delta t)^2) \tag{133.25}$$

uniformly on compact sets where the drift and the required derivatives of f are bounded. The limiting Markov generator on test functions is therefore

$$L_t f = F^a(x,t)\partial_a f + D^{ab}\partial_a\partial_b f. \tag{133.26}$$

The adjoint equation for a smooth density $p(x,t)$ is

$$\partial_t p = -\partial_a(F^a p) + D^{ab}\partial_a\partial_b p. \tag{133.27}$$

For stochastic PDEs and QFT applications, $\mathbb{R}^N$ is replaced first by a finite spatial regulator and only then by a continuum limit. The data to be controlled are the regulated drift, the regulated covariance, the choice of stochastic discretization, ultraviolet renormalizations of drift and noise, and convergence of correlation functions or path measures.

133.8 Finite Data for Model A and Model B

The dynamic-critical labels usually called Model A and Model B designate finite-regulator stochastic field data: a slow variable, a free-energy functional, a mobility operator, a noise covariance tied to that mobility, and a specified scaling limit. This finite formulation is the part that can be stated before any continuum renormalization group analysis.

Let $x = (x^1, \dots, x^N) \in \mathbb{R}^N$ be a finite spatial regulator for a real order parameter and let $\mathcal{F} : \mathbb{R}^N \to \mathbb{R}$ be a smooth finite-regulator free-energy functional. Fix a symmetric positive-semidefinite matrix M^{ab}. The mobility-gradient Langevin datum is

$$dx^a(t) = -\mathsf{M}^{ab}\partial_b\mathcal{F}(x(t))\,dt + \sqrt{2T}\,\sigma^a{}_r\,dB_t^r, \qquad \sigma\sigma^T = \mathsf{M}, \tag{133.28}$$

where $T > 0$ is the temperature parameter and B_t^r are independent standard Brownian motions. If M has a kernel, the stochastic equation is understood on affine subspaces parallel to $\operatorname{ran}\mathsf{M}$; directions in $\ker\mathsf{M}$ are exact conserved linear coordinates. The generator on smooth test functions is

$$Lf = -\mathsf{M}^{ab}\partial_b\mathcal{F}\,\partial_a f + T\mathsf{M}^{ab}\partial_a\partial_b f. \tag{133.29}$$

The adjoint Fokker–Planck equation can be written as a continuity equation

$$\partial_t p = -\partial_a J^a, \qquad J^a = -\mathsf{M}^{ab}\left(\partial_b\mathcal{F}\,p + T\partial_b p\right). \tag{133.30}$$

Consequently the finite Gibbs density

$$p_{\mathrm{eq}}(x) = Z^{-1}\exp[-\mathcal{F}(x)/T] \tag{133.31}$$

has zero probability current on each affine sector on which it is normalizable, because

$$\partial_b p_{\mathrm{eq}} = -T^{-1}(\partial_b\mathcal{F})p_{\mathrm{eq}}.$$

This is the finite fluctuation–dissipation relation: the same mobility M determines the dissipative drift and the noise covariance

$$\mathbb{E}(dx^a dx^b) = 2T\mathsf{M}^{ab}\,dt.$$

Model A is the case in which the order parameter is not constrained by a conserved charge. On a finite lattice one typically takes

$$\mathsf{M}^{ab} = \Gamma\,\delta^{ab}, \qquad \Gamma > 0,$$

up to local positive mobility factors. The drift is then the ordinary gradient flow of $\mathcal{F}$ with independent local thermal noise:

$$dx^a = -\Gamma\,\partial_a\mathcal{F}\,dt + \sqrt{2T\Gamma}\,dB_t^a. \tag{133.32}$$

Model B is the case in which the total order parameter is conserved. Let $G = (V, E)$ be a finite graph approximating space, orient each edge ℓ, and write $s(\ell)$ and $t(\ell)$ for its source and target vertices. Let

$$B_{\ell v} = \delta_{v,t(\ell)} - \delta_{v,s(\ell)}$$

be the incidence matrix, so $(Bx)_\ell = x_{t(\ell)} - x_{s(\ell)}$. With positive edge mobilities $c_\ell > 0$, set

$$\mathsf{M} = B^T C B, \qquad C_{\ell\ell'} = c_\ell\delta_{\ell\ell'}. \tag{133.33}$$

Then $\mathsf{M} \geq 0$ and $\mathsf{M}\mathbf{1} = 0$, hence

$$\frac{d}{dt}\sum_{i\in V} x_i = -\mathbf{1}^T\mathsf{M}\nabla\mathcal{F} = 0$$

and the noise has no constant component. Equivalently, write

$$dx_i = -\sum_{\ell} B_{\ell i} c_\ell \sum_j B_{\ell j} \partial_j \mathcal{F}\, dt + \sqrt{2T} \sum_{\ell} B_{\ell i} \sqrt{c_\ell}\, dW_t^\ell.$$

The noise is the discrete divergence of an edge noise, so the conserved zero mode is absent before any continuum notation is introduced. In a regular lattice limit this is the finite source of the formal expression

$$\partial_t \phi = \nabla \cdot \left(\lambda \nabla \frac{\delta \mathcal{F}}{\delta \phi} \right) + \nabla \cdot \xi, \qquad \langle \xi_i(t,x) \xi_j(t',y) \rangle = 2T\lambda\, \delta_{ij} \delta(t-t') \delta(x-y),$$

with the continuum formula understood only after the spatial regulator and renormalization have been specified.

The finite path weight is $\exp[-S^{\mathrm{fin}}_{\mathrm{MSRJD}}]$, where the MSRJD action for the finite mobility-gradient datum follows from Section 133.7 with $F^a = -\mathsf{M}^{ab} \partial_b \mathcal{F}$ and $D^{ab} = T\mathsf{M}^{ab}$. In response-field notation it is

$$S^{\mathrm{fin}}_{\mathrm{MSRJD}} = \sum_n \Delta t \left[\mathrm{i} \hat{x}_{n,a} \left(\frac{x^a_{n+1} - x^a_n}{\Delta t} + \mathsf{M}^{ab} \partial_b \mathcal{F}(x_n) \right) + T \hat{x}_{n,a} \mathsf{M}^{ab} \hat{x}_{n,b} \right], \tag{133.34}$$

with the finite Fourier-contour convention of (133.24). This expression is a finite action for a specified regulator. A dynamic critical field theory additionally requires: the scaling of $\mathcal{F}$, M, fields, and time; the list of slow variables retained; the renormalized drift and noise terms allowed by the exact symmetries and conservation laws; and a theorem or controlled calculation showing convergence of correlation and response functions. The dynamic exponent z, aging exponents, and scaling functions are outputs of that construction rather than entries in the finite datum itself.

133.9 Coupled Slow Variables Beyond Model A/B

The Hohenberg–Halperin labels for dynamic critical phenomena are not definitions of continuum QFTs. They are shorthand for a finite-regulator choice of slow variables, reversible brackets, dissipative mobilities, noise covariances, and scaling hypotheses. The finite datum below is the part of that language that can be stated without assuming a continuum fixed point.

Let $y = (y^1, \ldots, y^N)$ collect all retained slow variables at a finite spatial regulator. Some components may be nonconserved order-parameter coordinates, while others may be graph densities for conserved charges. Let $\mathcal{F}(y)$ be the finite free energy and consider the Ito process

$$dy^a(t) = \Big[R^a(y(t)) - \mathsf{M}^{ab} \partial_b \mathcal{F}(y(t)) \Big]\, dt + \sqrt{2T}\, \sigma^a{}_r\, dB^r_t, \qquad \sigma\sigma^T = \mathsf{M}, \tag{133.35}$$

where M is a constant symmetric positive-semidefinite mobility matrix and R is the reversible drift. The noise covariance is again fixed by the dissipative mobility. The condition that the reversible part preserve the Gibbs density is the finite divergence identity

$$\partial_a \left(R^a e^{-\mathcal{F}/T} \right) = 0. \tag{133.36}$$

Under this condition the Fokker–Planck current at $p_{\rm eq} = Z^{-1}e^{-\mathcal{F}/T}$ is

$$J^a_{\rm eq} = R^a p_{\rm eq} - \mathsf{M}^{ab}\left(\partial_b\mathcal{F}\, p_{\rm eq} + T\partial_b p_{\rm eq}\right) = R^a p_{\rm eq}, \tag{133.37}$$

and $\partial_a J^a_{\rm eq} = 0$. Thus the reversible current may circulate on a finite energy shell, while the dissipative part has zero Gibbs current. The common Hamiltonian example is

$$R^a = \mathsf{J}^{ab}\partial_b\mathcal{F}, \qquad \mathsf{J}^{ab} = -\mathsf{J}^{ba}, \qquad \partial_a\mathsf{J}^{ab} = 0, \tag{133.38}$$

for which $R^a\partial_a\mathcal{F} = 0$ and $\partial_a R^a = 0$ when J is constant. A state-dependent Poisson tensor requires the same divergence identity (133.36); writing a formal Poisson bracket is not by itself a definition of the regulated stochastic theory.

Exact finite conserved linear coordinates are also part of the datum. If $c_I = (c_{I,a})$ are covectors satisfying

$$c_{I,a}\mathsf{M}^{ab} = 0, \qquad c_{I,a}R^a(y) = 0, \tag{133.39}$$

then $Q_I = c_{I,a}y^a$ is conserved pathwise, including the noise term. For graph densities this is the finite statement that the mobility and noise are discrete divergences, as in (133.33).

This finite formulation separates three commonly conflated inputs. First, a Model C type system retains a nonconserved order parameter ϕ and at least one conserved scalar density n, with static coupling in $\mathcal{F}(\phi, n)$ and a block mobility such as

$$\mathsf{M}_{\rm C} = \begin{pmatrix} \Gamma\mathbf{1} & 0 \\ 0 & B^T C B \end{pmatrix}.$$

The conservation law is a statement about the lower block, not about the static coupling in $\mathcal{F}$. Second, a Model H type system retains a conserved order parameter and a momentum density; the advective nonlinearities belong to the reversible drift R and must obey (133.36) at the regulator where the path measure is defined. Third, any dynamic universality claim requires a renormalized continuum limit of correlation and response functions for this entire coupled datum. The label of the model is therefore a hypothesis about which slow variables and couplings survive the scaling limit, not a theorem about the microscopic QFT.

The finite MSRJD action associated to (133.35) is obtained from the Gaussian identity of Section 133.7:

$$S^{\rm coupled}_{\rm MSRJD} = \sum_n \Delta t\left[\mathrm{i}\hat{y}_{n,a}\left(\frac{y^a_{n+1} - y^a_n}{\Delta t} - R^a(y_n) + \mathsf{M}^{ab}\partial_b\mathcal{F}(y_n)\right) + T\hat{y}_{n,a}\mathsf{M}^{ab}\hat{y}_{n,b}\right]. \tag{133.40}$$

If M has a kernel, the response variables are understood on the quotient by the deterministic conserved directions, exactly as in the finite Model B construction. Continuum response-field actions for dynamic critical models are shorthand for a limit of (133.40), together with counterterms and operator renormalizations fixed by the regulator.

133.10 Finite Reaction Networks and Doi–Peliti Variables

Doi–Peliti variables are the occupation-number analogue of the response variables above, but the construction is more than a notation change. It is the finite-regulator bridge from a classical stochastic many-body semigroup to a local non-Hermitian QFT representation. A master equation for probabilities is rewritten as a linear evolution equation

$$\partial_t|P_t\rangle = \mathcal{L}_{\mathrm{DP}}|P_t\rangle$$

on a bosonic occupation Fock space, and coherent states then turn $\exp(T\mathcal{L}_{\mathrm{DP}})$ into a time-sliced path integral. The operator $\mathcal{L}_{\mathrm{DP}}$ is not a Hamiltonian generating unitary time evolution. Its structural conservation law is instead the left-null-vector identity

$$\langle \mathbf{1}|\mathcal{L}_{\mathrm{DP}} = 0,$$

or, in coherent-state symbols, $H_{\mathrm{DP}}(1, z) = 0$. This is the Markov analogue of the Ward identity that prevents probability from leaking out of the state space.

The reason this belongs in a QFT monograph is that local reaction networks, after spatial regularization, give local polynomial non-Hermitian actions. When a continuum limit exists, renormalization can be applied to these actions just as it is applied to unitary or Euclidean field theories, but the meaning of "same critical theory" must be stated operationally: two regulated stochastic systems are in the same nonequilibrium scaling class only after a common scaling limit, a matching of relevant perturbations, and agreement of scaling correlation and response functions have been specified. Examples usually described in this language include annihilation $A + A \to 0$, directed-percolation or Reggeon-field-theory reactions, parity-preserving branching-annihilating walks, and pair-contact processes with diffusion. The finite construction below is the algebraic input; the continuum fixed-point construction is part of the theorem boundary at the end of the chapter.

Let $s = 1, \dots, S$ label a finite set of species, and let $n = (n_1, \dots, n_S) \in \mathbb{N}^S$ be the occupation vector of a finite spatial regulator. For a multi-index $\alpha \in \mathbb{N}^S$, set

$$z^\alpha = \prod_s z_s^{\alpha_s}, \qquad (n)_\alpha = \prod_s n_s(n_s - 1)\cdots(n_s - \alpha_s + 1),$$

with $(n)_\alpha = 0$ if some $n_s < \alpha_s$. A reaction r has input ν_r, output ν_r', stoichiometric vector $v_r = \nu_r' - \nu_r$, and rate constant $\kappa_r \geq 0$. The mass-action jump rate is

$$w_r(n) = \kappa_r (n)_{\nu_r}, \qquad n \longmapsto n + v_r.$$

The master equation for the probability $P_t(n)$ is

$$\partial_t P_t(m) = \sum_r \left[\kappa_r(m - v_r)_{\nu_r} P_t(m - v_r) - \kappa_r(m)_{\nu_r} P_t(m)\right], \tag{133.41}$$

where terms with $m - v_r \notin \mathbb{N}^S$ vanish.

Let $\mathcal{F}_{\mathrm{alg}}$ be the algebraic bosonic Fock space with basis $\{|n\rangle : n \in \mathbb{N}^S\}$. Define

$$a_s|n\rangle = n_s|n - e_s\rangle, \qquad a_s^\dagger|n\rangle = |n + e_s\rangle, \qquad [a_s, a_t^\dagger] = \delta_{st}.$$

Here $a_s|n\rangle = 0$ when $n_s = 0$. For $\alpha \in \mathbb{N}^S$, write

$$a^\alpha = \prod_s a_s^{\alpha_s}, \qquad (a^\dagger)^\alpha = \prod_s (a_s^\dagger)^{\alpha_s}.$$

Encode a probability distribution as

$$|P_t\rangle = \sum_{n\in\mathbb{N}^S} P_t(n)|n\rangle.$$

The Doi–Peliti generator is the normally ordered operator

$$\mathcal{L}_{\mathrm{DP}} = \sum_r \kappa_r \left[(a^\dagger)^{\nu_r'} - (a^\dagger)^{\nu_r}\right] a^{\nu_r}. \tag{133.42}$$

The passage from (133.41) to (133.42) is an algebraic dictionary, not a separate continuum theorem. Its coefficient check is the following. For every finitely supported probability vector P, the coefficient of $|m\rangle$ in $\mathcal{L}_{\mathrm{DP}}|P\rangle$ is the right side of (133.41). Indeed, for every multi-index α,

$$a^\alpha|n\rangle = (n)_\alpha|n-\alpha\rangle. \tag{133.43}$$

Hence a single reaction term acts by

$$\kappa_r\left[(a^\dagger)^{\nu_r'} - (a^\dagger)^{\nu_r}\right] a^{\nu_r}|n\rangle = \kappa_r(n)_{\nu_r}\left(|n+v_r\rangle - |n\rangle\right). \tag{133.44}$$

After multiplying by $P(n)$ and summing over n, the coefficient of $|m\rangle$ is the gain from $n = m - v_r$ minus the loss from $n = m$, which is exactly (133.41). Therefore $\partial_t|P_t\rangle = \mathcal{L}_{\mathrm{DP}}|P_t\rangle$ is the reaction-network master equation on the algebraic occupation domain.

The left vector encoding total probability is

$$\langle\mathbf{1}|n\rangle = 1 \qquad (n\in\mathbb{N}^S).$$

For every multi-index α,

$$\langle\mathbf{1}|(a^\dagger)^\alpha = \langle\mathbf{1}|,$$

because both sides assign value 1 to every basis vector. Each bracket in (133.42) is consequently annihilated by $\langle\mathbf{1}|$, and

$$\langle\mathbf{1}|\mathcal{L}_{\mathrm{DP}} = 0. \tag{133.45}$$

The nontrivial analytic questions begin only after this dictionary: whether $\mathcal{L}_{\mathrm{DP}}$ has a closed Markov generator on a chosen weighted sequence space, whether a spatial continuum scaling exists, and whether the coherent-state time-slicing below converges to a stochastic field theory. Those are model-dependent domain and limit statements, not consequences of the finite coefficient computation.

The coherent-state form is a controlled time-slicing of the same operator. Let

$$|z\rangle = \exp\left(\sum_s z_s a_s^\dagger\right)|0\rangle, \qquad \langle\bar z| = \langle 0|\exp\left(\sum_s \bar z_s a_s\right),$$

where $z, \bar z \in \mathbb{C}^S$. Then

$$a_s|z\rangle = z_s|z\rangle, \qquad \langle\bar z|a_s^\dagger = \bar z_s\langle\bar z|, \qquad \langle\bar z|z\rangle = e^{\bar z\cdot z}.$$

The normal symbol of the generator is

$$H_{\mathrm{DP}}(\bar z, z) = e^{-\bar z\cdot z}\langle\bar z|\mathcal{L}_{\mathrm{DP}}|z\rangle = \sum_r \kappa_r\left(\bar z^{\nu_r'} - \bar z^{\nu_r}\right) z^{\nu_r}. \tag{133.46}$$

In any finite-dimensional truncation whose projected generator and boundary rules have been specified, and in particular in any finite total-number sector preserved by the reactions, the coherent-state resolution of the identity gives an ordinary finite-dimensional time-sliced integral for matrix elements of $\exp(T\mathcal{L}_{\mathrm{DP}})$. Replacing each short-time factor by

$$\langle \bar{z}_{j+1}|e^{\Delta t \mathcal{L}_{\mathrm{DP}}}|z_j\rangle = \exp(\bar{z}_{j+1}\cdot z_j)\left[1+\Delta t\, H_{\mathrm{DP}}(\bar{z}_{j+1}, z_j) + O((\Delta t)^2)\right]$$

gives the continuum Doi–Peliti action whenever the time-slicing limit is controlled. The shift $\bar{z} = 1+\widetilde{z}$ is useful because

$$H_{\mathrm{DP}}(1, z) = 0,$$

which is the symbol form of probability conservation. This identity is the non-Hermitian Markov replacement for unitary norm conservation: expectation values are computed with the projection state $\langle \mathbf{1}|$, not with a Hilbert-space adjoint bra, and the distinguished left vector is fixed by normalization of probabilities rather than by an inner product. The shifted variable $\bar{z} = 1+\widetilde{z}$ therefore expands around the correct conservation point of the stochastic theory. Differentiating at $\bar{z} = 1$ gives the deterministic rate equation for the mean-field density variables:

$$\dot{z}_s = \left.\frac{\partial H_{\mathrm{DP}}}{\partial \bar{z}_s}\right|_{\bar{z}=1} = \sum_r \kappa_r(\nu'_{r,s} - \nu_{r,s}) z^{\nu_r}. \tag{133.47}$$

The same finite generator also exposes the dynamical large-deviation Hamiltonian. For an exponential test function

$$f_p(n) = e^{p\cdot n},$$

the backward generator L associated with (133.41) satisfies

$$\frac{(Lf_p)(n)}{f_p(n)} = \sum_r \kappa_r(n)_{\nu_r}\left(e^{p\cdot v_r} - 1\right). \tag{133.48}$$

This follows by applying the backward generator to f_p: the r-th jump changes $f_p(n)$ to $f_p(n+v_r) = e^{p\cdot v_r} f_p(n)$, while the loss term subtracts $f_p(n)$. If a density scaling $n = \Omega x$ is chosen together with rate constants scaled so that $\kappa_r(n)_{\nu_r} = \Omega\, \lambda_r(x) + o(\Omega)$, then

$$\mathcal{H}_{\mathrm{LD}}(x, p) = \sum_r \lambda_r(x)\left(e^{p\cdot v_r} - 1\right)$$

is the finite-regulator Hamiltonian for the path rate function. Proving the large-deviation principle itself requires tightness, exponential estimates, and a specified scaling limit; (133.48) is the exact algebraic input.

The Doi–Peliti and MSRJD constructions are complementary finite-regulator routes into stochastic field theory. MSRJD starts from continuous variables and Gaussian increments; Doi–Peliti starts from integer occupations and jump rates, so it retains the discreteness that controls absorbing states and rare-event tails. The coherent-state action obtained from H_{DP}, together with the conservation identity $H_{\mathrm{DP}}(1, z) = 0$, is the object on which perturbative RG, saddle-point large-deviation analysis, and fixed-point comparisons act. Calling a continuum stochastic model a directed-percolation, annihilation, or parity-preserving branching-annihilating scaling theory is therefore not a definition by name; it is a claim that the regulated Doi–Peliti actions converge, after the declared tunings and field rescalings, to the same fixed-point data and the same scaling observables.

133.11 Tilted Generators and Dynamical Large Deviations

Finite path measures also give the precise source of dynamical large-deviation generators. Let I be a finite state space and let $W_{ij} \geq 0$, $i \neq j$, be time-independent jump rates, with

$$r_i = \sum_{j \neq i} W_{ij}.$$

For a trajectory $i(t)$, define an additive observable

$$A_T(\gamma) = \int_0^T h_{i(t)}\, dt + \sum_{m=1}^{N_T} g_{i_{m-1} i_m}, \tag{133.49}$$

where $h_i \in \mathbb{R}$ is a state weight, $g_{ij} \in \mathbb{R}$ is a jump weight, and N_T is the number of jumps up to time T. For $\lambda \in \mathbb{C}$, the tilted backward generator on functions $f : I \to \mathbb{C}$ is

$$(\mathsf{L}_\lambda f)_i = \lambda h_i f_i + \sum_{j \neq i} W_{ij} \left(e^{\lambda g_{ij}} f_j - f_i \right). \tag{133.50}$$

Proposition 133.3 (Finite Feynman–Kac identity). *For $i \in I$, set*

$$u_i(t) = \mathbb{E}_i[e^{\lambda A_t} f(i(t))],$$

where the chain starts from i at time 0. *Then*

$$\partial_t u(t) = \mathsf{L}_\lambda u(t), \qquad u(0) = f.$$

Consequently, for an initial distribution p_0,

$$\mathbb{E}_{p_0}\left[e^{\lambda A_T}\right] = \sum_{i \in I} p_0(i) \left(e^{T \mathsf{L}_\lambda} \mathbf{1}\right)_i. \tag{133.51}$$

Proof. Condition on the evolution during a short interval $[0, \Delta t]$. With probability $1 - r_i \Delta t + O((\Delta t)^2)$, the chain stays at i, and the additive functional contributes $h_i \Delta t + O((\Delta t)^2)$. With probability $W_{ij} \Delta t + O((\Delta t)^2)$, the chain jumps once from i to j, contributing the jump weight g_{ij} and the state weight over a time interval of length Δt. In the jump branch the latter changes the contribution only by $1 + O(\Delta t)$, and hence only by $O((\Delta t)^2)$ after multiplication by the jump probability. Therefore

$$u_i(t + \Delta t) = (1 + \lambda h_i \Delta t - r_i \Delta t)\, u_i(t) + \sum_{j \neq i} W_{ij} \Delta t\, e^{\lambda g_{ij}} u_j(t) + O((\Delta t)^2),$$

uniformly on the finite state space. Dividing by Δt and sending $\Delta t \to 0$ gives $\partial_t u = \mathsf{L}_\lambda u$. The final identity follows by taking $f = \mathbf{1}$ and averaging over p_0. □

If the original Markov chain is irreducible, then $\mathsf{L}_\lambda + c\mathbf{1}$ is an irreducible nonnegative matrix for real λ and all sufficiently large c. Perron–Frobenius therefore gives a simple real eigenvalue with strictly positive left and right eigenvectors and with maximal real part. It identifies the long-time scaled cumulant generating function

$$\psi(\lambda) = \lim_{T \to \infty} \frac{1}{T} \log \mathbb{E}_{p_0}\left[e^{\lambda A_T}\right]$$

with the eigenvalue of L_λ whose real part is maximal, independently of any nonzero nonnegative initial distribution. The positivity of the Perron eigenvectors is what prevents the leading spectral coefficient in (133.51) from vanishing. A path large-deviation principle and a Legendre-dual rate function require the separate analytic and tightness hypotheses of large-deviation theory. The finite tilted generator is the exact microscopic input to those hypotheses.

There is a useful exact symmetry when A_T is entropy production in a stationary finite jump process. Assume $W_{ij} > 0$ if and only if $W_{ji} > 0$, and let $\pi_i > 0$ be the stationary distribution. Define the entropy increment of one jump by

$$\sigma_{ij} = \log \frac{\pi_i W_{ij}}{\pi_j W_{ji}}. \tag{133.52}$$

Let L_q^Σ be the tilted generator for $\mathbb{E}[e^{-q\Sigma_T}]$:

$$(\mathsf{L}_q^\Sigma f)_i = \sum_{j\neq i} W_{ij} e^{-q\sigma_{ij}} f_j - r_i f_i. \tag{133.53}$$

With $\Pi = \mathrm{diag}(\pi_i)$, one has the finite matrix identity

$$\mathsf{L}_q^\Sigma = \Pi^{-1} \big(\mathsf{L}_{1-q}^\Sigma\big)^T \Pi. \tag{133.54}$$

Indeed, the off-diagonal i, j entry of the right side is

$$\frac{\pi_j}{\pi_i} W_{ji} e^{-(1-q)\sigma_{ji}} = W_{ij} \left(\frac{\pi_j W_{ji}}{\pi_i W_{ij}} \right)^q = W_{ij} e^{-q\sigma_{ij}},$$

and the diagonal entries are both $-r_i$. Hence L_q^Σ and L_{1-q}^Σ have the same spectrum, and the entropy-production scaled cumulant generating function obeys

$$\psi_\Sigma(q) = \psi_\Sigma(1-q).$$

This is the finite-generator form of the Gallavotti–Cohen symmetry. The finite-time fluctuation relation in Section 133.6 is the corresponding path-measure identity; the spectral symmetry is its time-homogeneous stationary long-time shadow.

133.12 Empirical Measures, Flows, and the Level-2.5 Cost

The tilted generator computes exponential moments of additive observables. The path-space object behind it is the joint empirical occupation and empirical flow. For a trajectory of the same finite jump process, define

$$\rho_i^T(\gamma) = \frac{1}{T} \int_0^T \mathbf{1}_{\{i(t)=i\}}\, dt, \qquad q_{ij}^T(\gamma) = \frac{1}{T} N_{ij}(T), \tag{133.55}$$

where $N_{ij}(T)$ is the number of jumps $i \to j$ before time T. Thus ρ^T is a probability vector and $q_{ij}^T \geq 0$ for $i \neq j$. A limiting pair (ρ, q) can occur only when

$$\rho_i \geq 0, \qquad \sum_i \rho_i = 1, \qquad q_{ij} \geq 0, \qquad \sum_{j\neq i} q_{ij} = \sum_{j\neq i} q_{ji}. \tag{133.56}$$

The last equation is current conservation at the vertex i. At finite time it is violated only by the boundary term

$$\sum_{j\neq i} N_{ij}(T) - \sum_{j\neq i} N_{ji}(T) = \mathbf{1}_{\{i(0)=i\}} - \mathbf{1}_{\{i(T)=i\}},$$

which disappears after division by T.

For a pair satisfying (133.56), define

$$I_{2.5}(\rho, q) = \sum_i \sum_{j\neq i} \left[q_{ij} \log \frac{q_{ij}}{\rho_i W_{ij}} - q_{ij} + \rho_i W_{ij} \right], \tag{133.57}$$

with the conventions $0\log(0/a) = 0$, $q\log(q/0) = +\infty$ for $q > 0$, and $q_{ij} = 0$ whenever $W_{ij} = 0$. If the constraints fail, set $I_{2.5}(\rho, q) = +\infty$. This is the relative entropy rate between the observed jump intensities and the original Markov intensities.

Proposition 133.4 (Finite empirical-flow cost)**.** *Let the finite jump process be irreducible on its support. The exponential cost of observing a limiting empirical pair (ρ, q) is (133.57). More precisely, if $\rho_i > 0$ on the communicating class, $q_{ij} > 0$ whenever $W_{ij} > 0$, and (ρ, q) satisfies (133.56), then the auxiliary rates*

$$R_{ij} = \frac{q_{ij}}{\rho_i} \tag{133.58}$$

make ρ stationary and make $q_{ij} = \rho_i R_{ij}$ the typical empirical flow. The Radon–Nikodym density of the original path measure $\mathbb{P}_W$ relative to the auxiliary path measure $\mathbb{P}_R$ obeys

$$\frac{1}{T} \log \frac{d\mathbb{P}_W}{d\mathbb{P}_R}(\gamma) = \sum_{i,j\neq i} q^T_{ij}(\gamma) \log \frac{W_{ij}}{R_{ij}} - \sum_{i,j\neq i} \rho^T_i(\gamma)(W_{ij} - R_{ij}). \tag{133.59}$$

On path sets where $(\rho^T, q^T) \to (\rho, q)$, this expression converges to $-I_{2.5}(\rho, q)$. Consequently, for sufficiently small neighborhoods U of an admissible continuity point of $I_{2.5}$,

$$\lim_{U\downarrow(\rho,q)} \lim_{T\to\infty} \frac{1}{T} \log \mathbb{P}_W\big((\rho^T, q^T) \in U\big) = -I_{2.5}(\rho, q).$$

Proof. The stationarity of ρ for R is exactly the conservation equation:

$$\rho_i \sum_{j\neq i} R_{ij} = \sum_{j\neq i} q_{ij} = \sum_{j\neq i} q_{ji} = \sum_{j\neq i} \rho_j R_{ji}.$$

For a path with jump counts $N_{ij}(T)$ and occupation times $T_i = T\rho^T_i$, the ratio of the two continuous-time jump-process densities is

$$\log \frac{d\mathbb{P}_W}{d\mathbb{P}_R} = \sum_i \sum_{j\neq i} N_{ij}(T) \log \frac{W_{ij}}{R_{ij}} - \sum_i T_i \sum_{j\neq i} (W_{ij} - R_{ij}).$$

Dividing by T, substituting $q^T_{ij} = N_{ij}(T)/T$, and then letting the empirical pair tend to the candidate pair gives, since $R_{ij} = q_{ij}/\rho_i$,

$$\sum_{i,j\neq i} q_{ij} \log \frac{\rho_i W_{ij}}{q_{ij}} - \sum_{i,j\neq i} \rho_i W_{ij} + \sum_{i,j\neq i} q_{ij} = -I_{2.5}(\rho, q),$$

which is the limiting value of (133.59). Under the auxiliary irreducible finite chain, the ergodic theorem gives $(\rho^T, q^T) \to (\rho, q)$ almost surely. Hence the same identity gives the lower exponential bound for every sufficiently small neighborhood.

For the matching upper bound, cover a compact closed set of admissible pairs by finitely many small neighborhoods and use the same Radon–Nikodym identity in each neighborhood with the corresponding auxiliary rates. The boundary cases with some $\rho_i = 0$ or $q_{ij} = 0$ are obtained by replacing the pair by strictly positive admissible pairs on the same support and then taking the lower-semicontinuous limit of (133.57). The finite state space is what makes this covering argument elementary; in continuum QFT a corresponding statement would require compactness and tightness estimates that are not contained in the finite calculation. □

Contracting over the empirical flow gives the level-2 occupation cost

$$I_2(\rho) = \inf_{\substack{q \geq 0 \\ \nabla \cdot q = 0}} I_{2.5}(\rho, q), \tag{133.60}$$

where $\nabla \cdot q = 0$ denotes the conservation condition in (133.56). For a two-state chain with rates $W_{01} = a$, $W_{10} = b$, conservation forces $q_{01} = q_{10} = s$. If $\rho_0 = p$ and $\rho_1 = 1 - p$, minimizing over s gives

$$I_2(p) = \left(\sqrt{pa} - \sqrt{(1-p)b}\right)^2. \tag{133.61}$$

The minimum vanishes exactly at the stationary occupation $p = b/(a + b)$.

The tilted-generator formula is the Legendre side of the same finite construction. The additive observable (133.49) can be written as

$$\frac{A_T}{T} = \sum_i h_i \rho_i^T + \sum_{i, j \neq i} g_{ij} q_{ij}^T.$$

Therefore the scaled cumulant generating function is equivalently

$$\psi(\lambda) = \sup_{\rho, q} \left[\lambda \left(\sum_i h_i \rho_i + \sum_{i, j \neq i} g_{ij} q_{ij} \right) - I_{2.5}(\rho, q) \right], \tag{133.62}$$

with the supremum over admissible empirical pairs. In finite dimension this variational formula is equivalent to the Perron eigenvalue statement for L_λ. The equality is a useful diagnostic: if a continuum hydrodynamic or field-theoretic large-deviation claim does not identify the finite empirical pair whose contraction is being taken, then the limiting claim has not yet been mathematically formulated.

133.13 Weak-Coupling Open-System Limit

Let a system Hilbert space couple to an environment by

$$H = H_S + H_E + \lambda \sum_a S_a \otimes E_a. \tag{133.63}$$

We take $S_a = S_a^\dagger$ and $E_a = E_a^\dagger$; a non-Hermitian coupling is rewritten in this form by separating Hermitian and anti-Hermitian parts. This convention fixes the adjoints in the spectral matrices below. Assume the environment state ρ_E is stationary, $[H_E, \rho_E] = 0$, and that connected environment correlators decay fast enough for their Fourier transforms to exist as positive matrices:

$$\gamma_{ab}(\omega) = \int_{-\infty}^{\infty} dt\, e^{\mathrm{i}\omega t}\, \mathrm{Tr}_E(\rho_E E_a(t) E_b(0))\,. \tag{133.64}$$

Positivity follows because, for any complex vector v_a,

$$C_v(t) = v_a^* v_b\ \mathrm{Tr}_E(\rho_E E_a(t) E_b(0)) = \mathrm{Tr}_E\Big(\rho_E F(t)^\dagger F(0)\Big)\,, \qquad F = v_b E_b,$$

is a function of positive type. Indeed, for any times t_i and complex numbers c_i,

$$\sum_{i,j} c_i^* c_j\, C_v(t_i - t_j) = \mathrm{Tr}_E\left(\rho_E \left(\sum_i c_i F(t_i)\right)^\dagger \left(\sum_j c_j F(t_j)\right)\right) \ge 0.$$

The distributional Bochner theorem then says that its Fourier transform is a positive measure. Hence $v_a^* \gamma_{ab}(\omega) v_b \ge 0$ at frequencies where the density $\gamma(\omega)$ exists, and more generally as a positive matrix-valued measure.

Let P_ϵ be the spectral projections of H_S, and define Bohr-frequency components

$$S_a(\omega) = \sum_{\epsilon' - \epsilon = \omega} P_\epsilon S_a P_{\epsilon'}. \tag{133.65}$$

Under the weak-coupling scaling

$$\lambda \to 0, \qquad t = \lambda^{-2}\tau,$$

the interaction-picture density matrix satisfies the integral equation

$$\rho_I(t) = \rho_I(0) - \mathrm{i}\lambda \int_0^t ds\, [V_I(s), \rho_I(s)], \qquad V_I(s) = \sum_a S_a(s) \otimes E_a(s).$$

Expanding once more, tracing over the environment, assuming $\mathrm{Tr}_E(\rho_E E_a) = 0$ after absorbing one-point terms into H_S, and using fast decay of connected environment correlators gives a second-order memory integral. The van Hove limit keeps $\lambda^2 t = \tau$ fixed. Oscillatory factors $e^{\mathrm{i}(\omega - \omega')s}$ then project onto equal Bohr frequencies, while the time integral of the bath correlator supplies $\gamma_{ab}(\omega)$ and its principal-value Hilbert transform. For a finite-dimensional system with discrete Bohr spectrum, bounded system couplings, a stationary environment state, and absolutely integrable environment connected correlators with the mixing estimates needed to discard higher cumulants in the van Hove limit, this limit gives

$$\frac{d\rho_S}{d\tau} = \mathcal{L}(\rho_S), \tag{133.66}$$

where

$$\begin{aligned}\mathcal{L}(\rho) = {}&-\mathrm{i}[H_S + H_{\mathrm{LS}}, \rho] \\ &+ \sum_\omega \sum_{a,b} \gamma_{ab}(\omega) \left[S_b(\omega)\rho S_a(\omega)^\dagger - \frac{1}{2}\{S_a(\omega)^\dagger S_b(\omega), \rho\}\right].\end{aligned} \tag{133.67}$$

H_{LS} is the Lamb-shift Hamiltonian from the principal-value part of the bath correlator. The dissipative part has the GKSL form because each matrix $\gamma_{ab}(\omega)$ is positive semidefinite and hence can be diagonalized into jump operators.

Proposition 133.5 (Trace preservation and complete positivity)**.** *For any finite-dimensional system, the generator* (133.67) *preserves* $\operatorname{Tr}\rho$*, preserves Hermiticity, and generates a completely positive trace-preserving semigroup.*

Proof. Trace preservation follows term by term:

$$\operatorname{Tr}(L\rho L^\dagger) - \frac{1}{2}\operatorname{Tr}(L^\dagger L\rho) - \frac{1}{2}\operatorname{Tr}(\rho L^\dagger L) = 0$$

by cyclicity of the trace. The commutator has zero trace and preserves Hermiticity. Since $\gamma(\omega)$ is positive, write $\gamma_{ab}(\omega) = \sum_\alpha v_{\alpha a}(\omega)^* v_{\alpha b}(\omega)$ and define

$$L_{\alpha\omega} = \sum_b v_{\alpha b}(\omega) S_b(\omega).$$

Then the dissipator is a sum of standard Lindblad terms

$$L_{\alpha\omega}\rho L_{\alpha\omega}^\dagger - \frac{1}{2}\{L_{\alpha\omega}^\dagger L_{\alpha\omega}, \rho\}.$$

Set $H = H_S + H_{\mathrm{LS}}$,

$$\Gamma = \sum_{\alpha,\omega} L_{\alpha\omega}^\dagger L_{\alpha\omega}, \qquad H_{\mathrm{eff}} = H - \frac{\mathrm{i}}{2}\Gamma,$$

and define the completely positive maps

$$\mathcal{K}_t(\rho) = e^{-\mathrm{i}H_{\mathrm{eff}}t}\rho e^{\mathrm{i}H_{\mathrm{eff}}^\dagger t}, \qquad \mathcal{J}(\rho) = \sum_{\alpha,\omega} L_{\alpha\omega}\rho L_{\alpha\omega}^\dagger.$$

In finite dimension the Dyson–Phillips series

$$T_t = \mathcal{K}_t + \sum_{n\geq 1}\int_{0<s_1<\cdots<s_n<t} \mathcal{K}_{t-s_n}\mathcal{J}\mathcal{K}_{s_n-s_{n-1}}\cdots\mathcal{J}\mathcal{K}_{s_1}\,ds_1\cdots ds_n \tag{133.68}$$

converges in operator norm on the trace-class matrices. Each summand is completely positive, so T_t is completely positive. Differentiating the series gives $dT_t/dt = \mathcal{L}T_t = T_t\mathcal{L}$ and $T_0 = \mathrm{id}$. Since the trace of $\mathcal{L}(\rho)$ vanishes for every ρ,

$$\frac{d}{dt}\operatorname{Tr} T_t(\rho) = \operatorname{Tr}\mathcal{L}(T_t(\rho)) = 0,$$

and therefore T_t is trace preserving. This proves complete positivity and trace preservation for the semigroup generated by (133.67). □

For comparison with the infinitesimal Kraus form, a short time step $d\tau$ has

$$K_0 = \mathbf{1} - d\tau\left(\mathrm{i}H + \frac{1}{2}\Gamma\right), \qquad K_{\alpha\omega} = \sqrt{d\tau}\,L_{\alpha\omega}.$$

This is a first-order local representation of the same generator; the identity

$$K_0^\dagger K_0 + \sum_{\alpha,\omega} K_{\alpha\omega}^\dagger K_{\alpha\omega} = \mathbf{1} + O(d\tau^2)$$

is the infinitesimal trace-preservation statement, while the norm-convergent series (133.68) is the finite-time object.

133.14 Thermal Bath and Detailed Balance

If the environment is a KMS state at inverse temperature β, its spectral matrix obeys

$$\gamma_{ab}(-\omega) = e^{-\beta\omega}\gamma_{ba}(\omega) \tag{133.69}$$

for Hermitian environment operators with the Fourier convention in (133.64). To see the sign and the order of indices, insert a spectral resolution of H_E. If $H_E|m\rangle = E_m|m\rangle$, then

$$\gamma_{ab}(\omega) = \frac{2\pi}{Z}\sum_{m,n} e^{-\beta E_m}\delta(\omega - (E_n - E_m))\,(E_a)_{mn}(E_b)_{nm}.$$

The same expression for $\gamma_{ba}(-\omega)$, with m and n exchanged, gives

$$\gamma_{ba}(-\omega) = e^{-\beta\omega}\gamma_{ab}(\omega),$$

which is equivalent to (133.69). For a two-level system with energy gap $\Omega > 0$, let

$$\Gamma_\downarrow = \gamma(\Omega), \qquad \Gamma_\uparrow = \gamma(-\Omega) = e^{-\beta\Omega}\Gamma_\downarrow.$$

The population master equation is

$$\dot{p}_e = -\Gamma_\downarrow p_e + \Gamma_\uparrow p_g, \qquad p_g = 1 - p_e.$$

The stationary ratio is

$$\frac{p_e}{p_g} = \frac{\Gamma_\uparrow}{\Gamma_\downarrow} = e^{-\beta\Omega},$$

which is the Gibbs ratio. This derivation is the open-system counterpart of the detailed-balance identity in kinetic theory.

133.15 Schwinger–Keldysh Influence Functional

For a field-theoretic system variable ϕ linearly coupled to an environment operator $\mathcal{O}_E$, the environment can be integrated out on the closed time path:

$$e^{\mathrm{i}S_{\mathrm{IF}}[\phi_+,\phi_-]} = \mathrm{Tr}_E\Big(U_E[\phi_+]\rho_E U_E[\phi_-]^\dagger\Big).$$

Closed-time-path normalization gives

$$S_{\mathrm{IF}}[\phi,\phi] = 0.$$

In r/a variables,

$$\phi_r = \frac{1}{2}(\phi_+ + \phi_-), \qquad \phi_a = \phi_+ - \phi_-,$$

the quadratic influence action has the form

$$S_{\mathrm{IF}}^{(2)} = \int \phi_a D_R \phi_r + \frac{\mathrm{i}}{2}\int \phi_a N \phi_a + O(\phi_a^3, \phi_a^2\phi_r). \tag{133.70}$$

with

$$D_R(x-y) = -\mathrm{i}\theta(x^0-y^0)\omega_E([\mathcal{O}_E(x),\mathcal{O}_E(y)]), \qquad N(x-y) = \frac{1}{2}\omega_E(\{\mathcal{O}_E(x),\mathcal{O}_E(y)\})$$

for a stationary environment state ω_E. The kernel D_R is retarded because variations of the forward and backward branches cancel for earlier times, as in the Schwinger–Keldysh derivation of retarded response. The kernel N is positive in the sense

$$\int fNf \geq 0$$

for real test functions f, so that $\exp(\mathrm{i}S_{\mathrm{IF}})$ is damped rather than amplified. If the environment is KMS, N and the anti-Hermitian part of D_R obey the fluctuation–dissipation relation of Chapter 127.

Finite Gaussian bridge to MSRJD. The relation between the Schwinger–Keldysh influence functional and a classical stochastic equation is a finite-regulator Gaussian identity, not a formal analogy. Let $A, B = 1, \ldots, m$ label a finite collection of regulated time, space, and internal indices, and let

$$E_A(\phi_r) = 0$$

be the deterministic retarded equation produced by the real part of the closed-time-path effective action. At quadratic order in the difference field, the finite effective weight has the form

$$\exp\left[\mathrm{i}\phi_a^A E_A(\phi_r) - \frac{1}{2}\phi_a^A N_{AB}\phi_a^B\right], \qquad N = N^T \geq 0. \tag{133.71}$$

If $N > 0$, the second factor is the characteristic function of a real Gaussian random vector ξ_A with covariance N_{AB}:

$$\exp\left[-\frac{1}{2}\phi_a^A N_{AB}\phi_a^B\right] = \frac{1}{\sqrt{(2\pi)^m \det N}} \int_{\mathbb{R}^m} d^m\xi \, \exp\left[-\frac{1}{2}\xi_A (N^{-1})^{AB}\xi_B + \mathrm{i}\phi_a^A\xi_A\right]. \tag{133.72}$$

Substituting (133.72) into (133.71) and integrating over ϕ_a gives the finite constraint

$$E_A(\phi_r) + \xi_A = 0. \tag{133.73}$$

Equivalently one may replace ξ by $-\xi$; the covariance is unchanged. Thus the SK a-field is the same finite response variable as the Fourier dual $\hat{x}$ in (133.24), and the imaginary quadratic a-field kernel is the covariance of the induced noise. If N has null directions, the same statement is made on the quotient by $\ker N$, while the null components of ϕ_a impose exact deterministic constraints. This is the finite form appropriate for conserved densities: the noise covariance can be degenerate because exact conservation forbids noise in the spatially constant charge mode.

For a linear equation

$$E_A(\phi_r) = K_{AB}\phi_r^B - J_A$$

with invertible retarded matrix K, the induced Gaussian law for ϕ_r has mean $K^{-1}J$ and covariance

$$\left\langle(\phi_r^A - \langle\phi_r^A\rangle)(\phi_r^B - \langle\phi_r^B\rangle)\right\rangle = (K^{-1})^{AC}N_{CD}(K^{-T})^{DB}. \tag{133.74}$$

This formula is the finite-regulator source of the Langevin covariance in hydrodynamic effective theory. It also fixes the order of logic: a Markovian white-noise Langevin equation is obtained only after the environment correlator has been approximated by a local kernel in the chosen scaling limit. Before that limit, N is a nonlocal finite covariance kernel and (133.73) is a colored-noise equation.

133.16 Hydrodynamic Open Systems

Consider a single approximately conserved density $n(t, \mathbf{x})$ whose conservation is weakly broken by a KMS environment. The Markovian hydrodynamic equation is

$$\partial_t n + \nabla \cdot j = -\gamma(n - n_{\mathrm{eq}}) + \xi. \tag{133.75}$$

For one homogeneous mode with susceptibility χ, the quadratic equilibrium free energy is

$$F(n) = \frac{(n - n_{\mathrm{eq}})^2}{2\chi}.$$

The Fokker–Planck equation associated with

$$\dot{n} = -\gamma(n - n_{\mathrm{eq}}) + \xi, \qquad \langle \xi(t)\xi(t') \rangle = 2D_n \delta(t - t')$$

has stationary distribution proportional to $e^{-\beta F(n)}$ precisely when

$$D_n = \gamma \chi T. \tag{133.76}$$

Indeed, the Fokker–Planck equation is

$$\partial_t P(n,t) = \partial_n[\gamma(n - n_{\mathrm{eq}})P(n,t)] + D_n \partial_n^2 P(n,t) = -\partial_n J_P(n,t),$$

where

$$J_P(n,t) = -\gamma(n - n_{\mathrm{eq}})P(n,t) - D_n \partial_n P(n,t).$$

For the equilibrium candidate

$$P_{\mathrm{eq}}(n) = Z^{-1} \exp\left[-\frac{(n - n_{\mathrm{eq}})^2}{2\chi T} \right],$$

the stationary probability current vanishes for every n exactly when $D_n = \gamma \chi T$. Thus

$$\langle \xi(t, \mathbf{x}) \xi(t', \mathbf{x}') \rangle = 2\gamma \chi T \, \delta(t - t') \delta^{(d)}(\mathbf{x} - \mathbf{x}') \tag{133.77}$$

in the local Markovian limit. The relation is the Einstein relation for a nonconserved relaxation mode; it is distinct from diffusive current noise, where the noise enters through a divergence.

133.17 Theorem Boundary

The proof architecture in §133.1 is part of the theorem boundary: a continuum result must say which finite layer is being used, which field-theoretic representation is only a coordinate rewrite, and which scaling hypotheses turn that coordinate rewrite into a QFT limit. For continuum QFT, a complete nonequilibrium construction must state the local algebra, the reservoir states, the topology of the long-time local limit, the domains of the current operators, and the order in which weak coupling, thermodynamic, Markovian, hydrodynamic, and continuum limits are taken. For open systems, complete positivity is a statement about the reduced dynamics of a subalgebra after tracing over an environment and taking a specified scaling limit. A nonlocal truncation of a closed QFT becomes an open-system dynamics only after such an environment trace and limit have been constructed. For hydrodynamic open systems, the noise kernel and relaxation matrix must be derived from the same influence functional that produces the dissipative terms. For Doi–Peliti reaction networks, the corresponding continuum problem must construct the spatial regulator, the scaling of rates and fields, the renormalized non-Hermitian action, the projection-state Ward identity $H_{\mathrm{DP}}(1, z) = 0$, and the scaling local observables whose limits define the proposed nonequilibrium fixed point.

Open Problem 133.6 (Continuum nonequilibrium states)*.* Construct nonequilibrium steady states and open-system limits in interacting continuum QFT with local algebras, KMS reservoirs, real-time correlators, entropy production, hydrodynamic scaling limits, and complete-positivity constraints derived from the microscopic dynamics. Include continuum limits of spatially regulated Doi–Peliti reaction networks when the goal is a non-Hermitian stochastic field theory for absorbing-state or reaction-diffusion critical behavior.

Chapter 134

Hydrodynamic Fluctuations And Long-Time Tails

Conventions in this chapter.

- **Signature.** real time; no spacetime metric unless a field-theory example is explicitly introduced.
- **Metric sign.** field-theory examples use $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ when a vacuum vector is present.
- **Generator convention.** time evolution is $\exp(-\mathrm{i}Ht)$, and spacetime translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$ when introduced.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Hydrodynamics describes the long-wavelength evolution of conserved densities. In a thermal state those densities also fluctuate. The fluctuations are fixed at the linear level by static susceptibilities and the KMS fluctuation–dissipation relation; their nonlinear couplings then produce nonanalytic low-frequency terms in real-time correlators. These terms are the hydrodynamic long-time tails. They are universal within the hydrodynamic effective theory because their nonanalytic part is determined by the conserved-mode propagators and by the local nonlinear vertices.

Throughout this chapter spacetime is $\mathbb{R}_t \times \mathbb{R}^d$, with d spatial dimensions. Fourier modes are $\exp(-\mathrm{i}\omega t + \mathrm{i}\mathbf{k}\cdot\mathbf{x})$. The diffusion normalizations are those of Chapter 129: the charge susceptibility is $\chi = \partial n/\partial\mu$, the conductivity is σ, and $D = \sigma/\chi$. Equal-time thermal fluctuations of a coarse-grained density carry the covariance $T\chi$.

134.1 Static Fluctuation Data

Let $q_A(t, \mathbf{x})$ be conserved densities in a homogeneous KMS state at temperature T and chemical potentials μ_A. Write

$$\delta q_A = q_A - \omega_\beta(q_A),$$

where ω_β denotes the thermal state. The static susceptibility matrix in the chemical-potential convention is

$$\chi_{AB} = \left(\frac{\partial q_A}{\partial \mu_B}\right)_T .$$

For hydrodynamic variables averaged over a cell whose size is large compared with microscopic correlation lengths, the equal-time connected covariance is

$$\int d^d x\, \omega_\beta(\delta q_A(0,\mathbf{x})\delta q_B(0,\mathbf{0}))_{\mathrm{conn}} = T\chi_{AB}. \tag{134.1}$$

Equivalently, if one uses the dimensionless thermodynamic force $\alpha_A = \mu_A/T$, the covariance equals $\partial q_A/\partial \alpha_B$. This convention removes a common source of factor-of-T errors in stochastic hydrodynamics.

For one diffusive charge density n, the Gaussian equilibrium probability functional for long-wavelength configurations is

$$P_{\mathrm{eq}}[n] \propto \exp\left[-\frac{1}{2T\chi}\int d^d x\,(\delta n)^2\right], \qquad \delta n = n - \omega_\beta(n). \tag{134.2}$$

This formula is a local central-limit statement inside the hydrodynamic window. The continuum delta function in (134.1) is an effective-theory notation for a cell-averaged covariance; the cell scale is kept between the microscopic correlation length and the external wavelength.

134.2 Linear Stochastic Diffusion

The first-order stochastic constitutive relation for one conserved charge is

$$\partial_t n + \partial_i j^i = 0, \qquad j^i = -D\partial_i n + \xi^i, \tag{134.3}$$

with Gaussian white current noise

$$\langle \xi^i(t,\mathbf{x})\xi^j(t',\mathbf{x}')\rangle = 2T\sigma\,\delta^{ij}\delta(t-t')\delta^{(d)}(\mathbf{x}-\mathbf{x}'). \tag{134.4}$$

The coefficient $2T\sigma$ is the same coefficient derived from the dynamical KMS symmetry of the diffusion action in Chapter 129. The stochastic equation gives

$$n(\omega,\mathbf{k}) = \frac{\mathrm{i}k_i\xi^i(\omega,\mathbf{k})}{-\mathrm{i}\omega + Dk^2}, \qquad k^2 = \mathbf{k}\cdot\mathbf{k}.$$

Hence the symmetrized density correlator is

$$G^{\mathrm{sym}}_{nn}(\omega,\mathbf{k}) = \frac{2T\sigma k^2}{\omega^2 + (Dk^2)^2}. \tag{134.5}$$

Integrating over frequency gives

$$\int \frac{d\omega}{2\pi} G^{\mathrm{sym}}_{nn}(\omega,\mathbf{k}) = \frac{T\sigma}{D} = T\chi,$$

which is exactly the static covariance (134.1). Thus the Einstein relation $D = \sigma/\chi$ is forced by compatibility of the deterministic diffusion constant, the conductivity, and the equilibrium fluctuation measure.

The retarded density source-response kernel from the Schwinger–Keldysh action is

$$K^R_{nn}(\omega, \mathbf{k}) = \chi \, \frac{Dk^2}{Dk^2 - \mathrm{i}\omega}. \tag{134.6}$$

Its imaginary part obeys the low-frequency fluctuation–dissipation relation

$$G^{\mathrm{sym}}_{nn}(\omega, \mathbf{k}) = \frac{2T}{\omega} \, \mathrm{Im}\, K^R_{nn}(\omega, \mathbf{k}) + O(\omega/T)$$

inside the classical hydrodynamic regime. The exact quantum formula is the KMS relation of Chapter 127; the displayed expression is its $|\omega| \ll T$ limit.

134.3 Local Schwinger–Keldysh Origin

The stochastic equation (134.3) is the Hubbard–Stratonovich form of the local diffusion action

$$I_{\mathrm{diff}} = \int dt \, d^d x \, \left[\chi B_{a0} B_{r0} - \sigma B_{ai} \partial_t B_{ri} + \mathrm{i} T \sigma B_{ai} B_{ai}\right]. \tag{134.7}$$

Here $B_{r\mu} = A_{r\mu} + \partial_\mu \varphi_r$ and $B_{a\mu} = A_{a\mu} + \partial_\mu \varphi_a$ are the gauge-invariant hydrodynamic source combinations. The a-phase variation imposes the continuity equation, and the imaginary term has the Gaussian representation

$$\exp\left[-T\sigma \int B_{ai} B_{ai}\right] = \int D\xi \, \exp\left[-\int \frac{\xi_i \xi_i}{4T\sigma} + \mathrm{i} \int B_{ai} \xi_i\right].$$

The same coefficient therefore controls current response, current noise, and diffusive relaxation. This is the local effective-action statement of KMS: one chooses the dissipative term and the noise term together.

134.4 Finite-Cell Current Fluctuations and the Macroscopic Cost

The white current noise in (134.4) can be obtained from a finite empirical-flow calculation before any continuum stochastic field notation is introduced. This is the finite origin of the Gaussian current cost used in macroscopic fluctuation theory. The continuum functional is a hydrodynamic scaling statement; the exact statement at this stage is the contraction of a finite jump-flow relative entropy.

Consider one oriented mesoscopic bond. During a time block, let q_+ and q_- be the empirical rates of jumps across the bond in the positive and negative directions. At fixed local occupation data the corresponding reference jump intensities are positive numbers r_+ and r_-. The local contribution to the finite level-2.5 cost of Section 133.12 is

$$\Phi(q_+, q_-; r_+, r_-) = q_+ \log \frac{q_+}{r_+} - q_+ + r_+ + q_- \log \frac{q_-}{r_-} - q_- + r_-, \tag{134.8}$$

with $q_\pm \geq 0$. The net empirical current through the bond is

$$j = q_+ - q_-.$$

For fixed j, strict convexity gives a unique minimizer. Introducing a Lagrange multiplier ℓ for $q_+ - q_- = j$ gives

$$\log \frac{q_+}{r_+} + \ell = 0, \qquad \log \frac{q_-}{r_-} - \ell = 0,$$

hence

$$q_+ q_- = r_+ r_-, \qquad q_+ - q_- = j.$$

Writing

$$\Delta(j) = \sqrt{j^2 + 4r_+ r_-},$$

the minimizer is

$$q_+^*(j) = \frac{\Delta(j) + j}{2}, \qquad q_-^*(j) = \frac{\Delta(j) - j}{2}. \tag{134.9}$$

Substitution into (134.8) gives the exact finite current cost

$$\Phi_{\mathrm{cur}}(j; r_+, r_-) = j \log \frac{j + \sqrt{j^2 + 4r_+ r_-}}{2r_+} - \sqrt{j^2 + 4r_+ r_-} + r_+ + r_-. \tag{134.10}$$

It vanishes at the typical current

$$j_0 = r_+ - r_-, \qquad \Delta(j_0) = r_+ + r_-.$$

The derivative of the contracted cost is the multiplier conjugate to the current,

$$\frac{d\Phi_{\mathrm{cur}}}{dj} = \log \frac{q_+^*(j)}{r_+},$$

and therefore

$$\left.\frac{d^2\Phi_{\mathrm{cur}}}{dj^2}\right|_{j=j_0} = \frac{1}{r_+ + r_-}.$$

The current cost near its minimum is consequently

$$\Phi_{\mathrm{cur}}(j; r_+, r_-) = \frac{(j - j_0)^2}{2(r_+ + r_-)} + O\big((j - j_0)^3\big)\,. \tag{134.11}$$

The same exact expression also records the entropy-production asymmetry:

$$\Phi_{\mathrm{cur}}(j; r_+, r_-) - \Phi_{\mathrm{cur}}(-j; r_+, r_-) = -j \log \frac{r_+}{r_-}. \tag{134.12}$$

Thus the symmetric traffic $r_+ + r_-$ fixes the Gaussian current variance, while the ratio r_+/r_- fixes the antisymmetric entropy-production part. No continuum limit has entered this calculation.

To pass from the finite current cost to hydrodynamics, one must add scaling input. Suppose a sequence of finite cell regulators has a local-equilibrium diffusive scaling limit in which the conserved density is $n(t, \mathbf{x})$ and the empirical current is $j^i(t, \mathbf{x})$. Suppose moreover that the finite typical current and finite traffic converge in the local hydrodynamic window to the deterministic current and covariance density. In a local basis where the positive symmetric conductivity matrix is diagonal, this componentwise statement is

$$j^i_{\rm det}(n) = -D^{ij}(n)\partial_j n + O(\partial^2), \qquad r^i_+ + r^i_- \longrightarrow 2T\,\sigma^{ij}(n)$$

in the normalization of (134.4). Then the quadratic Taylor term (134.11) becomes the local Gaussian current cost

$$\mathcal{I}_{\rm cur}[n,j] = \frac{1}{4T}\int dt\,d^dx\,\big(j^i + D^{ik}\partial_k n\big)(\sigma^{-1})_{ij}\big(j^j + D^{j\ell}\partial_\ell n\big), \tag{134.13}$$

subject to the continuity equation

$$\partial_t n + \partial_i j^i = 0. \tag{134.14}$$

Equation (134.13) is the Onsager–Machlup current form of the stochastic diffusion theory (134.3). Integrating over j with the constraint (134.14), or equivalently introducing a Lagrange multiplier for the constraint, gives the same quadratic Schwinger–Keldysh action (134.7) after the Hubbard–Stratonovich step above. The non-Gaussian terms in the exact finite expression (134.10) are irrelevant in the central-limit hydrodynamic fluctuation regime but are essential for far-from-typical current large deviations when the underlying finite regulator is retained.

The logical status is therefore sharp. The finite contraction (134.10) is a direct finite consequence of the regulated jump-flow rate function. The macroscopic functional (134.13) is obtained from it only after one assumes, or proves in a specified model, the local-equilibrium hydrodynamic scaling limit, the convergence of traffic to $2T\sigma$, and the Einstein relation between D, σ, and χ.

134.5 Hydrodynamic Nonlinearities

Hydrodynamic variables are local coordinates on the manifold of equilibrium states. Expanding the currents in fluctuations around a homogeneous state gives

$$J^i_A = J^i_{A,\rm lin} + \frac{1}{2}\,\lambda^i{}_{A,BC}\,\phi_B\phi_C + O(\phi^3, \phi\,\partial\phi, \partial^2\phi), \tag{134.15}$$

where $\phi_A = \delta q_A$ after diagonalizing the linear hydrodynamic modes, and $\lambda^i{}_{A,BC}$ is symmetric in B, C. The coefficient λ is determined by the equation of state and by the chosen hydrodynamic frame. For example, the convective part of the stress tensor contains $\pi^i\pi^j/(\varepsilon+p)$, where $\pi^i = T^{0i}$ is momentum density and $\varepsilon + p$ is the enthalpy density.

The quadratic term in (134.15) produces a loop of hydrodynamic modes in current-current correlators. The calculation is an effective-theory calculation with an ultraviolet cutoff $\Lambda_{\rm hyd}$ below the inverse microscopic length. Local analytic terms depend on this cutoff and renormalize local transport coefficients. The low-frequency nonanalytic part is independent of the detailed cutoff profile.

134.6 Time-Domain Tail From A Diffusive Pair

Consider one scalar diffusive mode ϕ with

$$\langle\phi(t,\mathbf{k})\phi(0,-\mathbf{k})\rangle = T\chi\,e^{-Dk^2 t}, \qquad t > 0.$$

Let the long-wavelength current contain the normal-ordered quadratic piece

$$J(t,\mathbf{x}) = \lambda :\phi(t,\mathbf{x})^2: .$$

At Gaussian order, Wick contraction gives

$$\begin{aligned}
C_{JJ}(t) &= \int d^dx \, \langle J(t,\mathbf{x})J(0,\mathbf{0})\rangle_{\text{conn}} \\
&= 2\lambda^2(T\chi)^2 \int \frac{d^dk}{(2\pi)^d}\, e^{-2Dk^2t} \\
&= 2\lambda^2(T\chi)^2\,(8\pi Dt)^{-d/2}, \qquad t>0.
\end{aligned} \tag{134.16}$$

This is the basic long-time tail. More general hydrodynamic systems give sums of terms of the same form, with diffusion constants replaced by sums of the damping rates of the two modes in the loop and with tensor projectors from momentum or sound modes. The exponent is fixed by the Gaussian integration over small loop momentum.

134.7 Frequency-Domain Nonanalyticity

The same tail appears in the retarded correlator as a nonanalytic low-frequency term. The scalar integral that controls a pair of diffusive modes is

$$I_d(z;D) = \int \frac{d^dk}{(2\pi)^d}\frac{1}{z+2Dk^2}, \qquad z = -\mathrm{i}\omega + 0^+. \tag{134.17}$$

The analytic terms are cutoff dependent. The nonanalytic part is obtained by subtracting local Taylor terms, or equivalently by dimensional regularization of the infrared integral:

$$I_d^{\text{nonan}}(z;D) = \frac{\Gamma(1-d/2)}{(4\pi)^{d/2}(2D)^{d/2}}\, z^{d/2-1}, \qquad d \notin 2\mathbb{Z}_{\geq 1}. \tag{134.18}$$

Infrared loop integral by Schwinger parameter. For $\operatorname{Re} z > 0$ and $D > 0$, the dimensionally regularized infrared integral

$$\int \frac{d^dk}{(2\pi)^d}\frac{1}{z+2Dk^2} \tag{134.19}$$

is

$$\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}(2D)^{d/2}} z^{d/2-1}. \tag{134.20}$$

The retarded correlator is obtained by analytic continuation to

$$z = -\mathrm{i}\omega + 0^+, \tag{134.21}$$

with the principal logarithm, so that the result is analytic for $\operatorname{Im}\omega > 0$.

For $\operatorname{Re} z > 0$,

$$\frac{1}{z+2Dk^2} = \int_0^\infty ds\, e^{-s(z+2Dk^2)}. \tag{134.22}$$

The Gaussian integral in d dimensions is

$$\int \frac{d^d k}{(2\pi)^d} e^{-2Dsk^2} = (4\pi\, 2Ds)^{-d/2}. \tag{134.23}$$

Therefore

$$I_d(z; D) = \frac{1}{(4\pi)^{d/2}(2D)^{d/2}} \int_0^\infty ds\, s^{-d/2} e^{-zs}. \tag{134.24}$$

The remaining integral is the defining analytic continuation of the gamma function:

$$\int_0^\infty ds\, s^{\alpha-1} e^{-zs} = \Gamma(\alpha) z^{-\alpha}, \qquad \alpha = 1 - \frac{d}{2}. \tag{134.25}$$

Substituting $\alpha = 1 - d/2$ gives (134.20). Dimensional regularization drops power-divergent local terms and retains the nonanalytic infrared dependence. The retarded prescription (134.21) follows from the original denominator $z + 2Dk^2 = -\mathrm{i}\omega + 0^+ + 2Dk^2$, which is nonzero for $\mathrm{Im}\,\omega > 0$.

In even spatial dimension the pole of $\Gamma(1-d/2)$ is replaced, after local subtraction, by $z^{d/2-1}\log z$. The first cases are

d	$I_d^{\mathrm{nonan}}(z; D)$
1	$\dfrac{1}{2(2D)^{1/2}}\, z^{-1/2}$
2	$-\dfrac{1}{8\pi D}\, \log z$
3	$-\dfrac{1}{4\pi(2D)^{3/2}}\, z^{1/2}$

up to scheme-dependent additive constants in $d = 2$. The branch is fixed by $z = -\mathrm{i}\omega + 0^+$, so the retarded function is analytic in the upper half ω-plane.

For $d = 3$, an explicit cutoff computation displays the separation between local and nonlocal terms:

$$I_3^\Lambda(z; D) = \frac{1}{2\pi^2(2D)} \left[\Lambda - \left(\frac{z}{2D}\right)^{1/2} \arctan\!\left(\frac{\Lambda}{(z/2D)^{1/2}}\right)\right].$$

As $z/\Lambda^2 \to 0$,

$$I_3^\Lambda(z; D) = \frac{\Lambda}{4\pi^2 D} - \frac{z^{1/2}}{4\pi(2D)^{3/2}} + O(z/\Lambda).$$

The first term renormalizes a local transport coefficient. The square-root term is the long-time tail written in frequency space.

134.8 Transport Coefficients As Scale-Dependent Local Data

Let κ be a transport coefficient multiplying a local term in the hydrodynamic constitutive relation or Schwinger–Keldysh action. Hydrodynamic loops with momenta below Λ_{hyd} shift the coefficient:

$$\kappa_{\mathrm{phys}}(\omega) = \kappa(\Lambda_{\mathrm{hyd}}) + \Delta\kappa_{\mathrm{loc}}(\Lambda_{\mathrm{hyd}}) + \Delta\kappa_{\mathrm{nonan}}(\omega).$$

The sum of the first two terms is the local coefficient appropriate to the chosen hydrodynamic cutoff. The last term is fixed by hydrodynamic mode propagation and is a genuinely nonlocal-in-time low-frequency contribution, distinct from a finite local derivative coefficient. This is the finite-temperature analogue of ordinary effective-field theory renormalization: local coefficients are scheme dependent, while nonanalytic infrared dependence is physical within the regime of validity.

The order of limits is part of the definition. One first takes the thermodynamic limit of the KMS state, then probes frequencies and momenta inside the hydrodynamic scaling regime, and only then studies the small $\omega, \mathbf{k}$ asymptotics. Interchanging this order can erase the hydrodynamic pole, mix static and dynamic susceptibilities, or replace retarded functions by finite-volume quasiperiodic sums.

134.9 Stress-Tensor Noise

For a relativistic fluid in the local rest frame, write the stochastic spatial stress as

$$T^{ij} = p\,\delta^{ij} - \eta\left(\partial^i v^j + \partial^j v^i - \frac{2}{d}\delta^{ij}\partial_k v^k\right) - \zeta\,\delta^{ij}\partial_k v^k + \Xi^{ij} + O(v^2, \partial^2).$$

The white-noise covariance required by local KMS is

$$\langle \Xi^{ij}(x)\Xi^{kl}(x')\rangle = 2T\Big[\eta\left(\delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk} - \frac{2}{d}\delta^{ij}\delta^{kl}\right) + \zeta\,\delta^{ij}\delta^{kl}\Big] \times \delta(t-t')\delta^{(d)}(\mathbf{x}-\mathbf{x}'). \tag{134.26}$$

For any real symmetric test tensor s_{ij}, decompose

$$s_{ij} = s^{\mathrm{T}}_{ij} + \frac{1}{d}\delta_{ij}s^k{}_k, \qquad \delta^{ij}s^{\mathrm{T}}_{ij} = 0.$$

Contracting the tensor in (134.26) with $s_{ij}s_{kl}$ gives

$$4T\eta\, s^{\mathrm{T}}_{ij}s^{\mathrm{T}}_{ij} + 2T\zeta\,(s^k{}_k)^2.$$

Thus positivity of the stochastic stress covariance is equivalent, at this order, to $\eta \geq 0$ and $\zeta \geq 0$. The same inequalities are obtained from nonnegative entropy production in deterministic hydrodynamics.

134.10 Momentum Tails

Momentum density $\pi^i = T^{0i}$ has static covariance

$$\langle \pi^i(\mathbf{k})\pi^j(-\mathbf{k})\rangle_{\mathrm{eq}} = T(\varepsilon + p)\delta^{ij}$$

in the local rest frame. Transverse momentum diffuses with kinematic viscosity

$$\nu = \frac{\eta}{\varepsilon + p}.$$

Because the stress tensor contains the convective term

$$T^{ij}_{\mathrm{conv}} = \frac{\pi^i\pi^j}{\varepsilon + p},$$

the shear-stress correlator contains a two-momentum hydrodynamic loop. Its late-time behavior is again of the form

$$\langle T^{ij}(t)T^{kl}(0)\rangle_{\text{conn}} \sim T^2\,\mathcal{P}^{ij,kl}\,(8\pi\nu t)^{-d/2},$$

with $\mathcal{P}^{ij,kl}$ determined by the transverse projectors and by angular averaging. Sound modes add oscillatory pieces with damping controlled by the sound attenuation constant. The existence of the tail therefore follows from the convective nonlinearity and the conserved momentum pole; the exact tensor coefficient requires the full hydrodynamic mode basis and frame convention.

134.11 Microscopic Theorem Boundary

A microscopic derivation from QFT must specify the KMS state, conserved currents, thermodynamic susceptibilities, the thermodynamic limit, clustering estimates, and the topology in which hydrodynamic correlators emerge from real-time QFT correlators. It must also identify the slow subspace of conserved densities, prove local equilibration in the scaling regime being used, and derive the Schwinger–Keldysh positivity and KMS constraints on the effective action. Hydrodynamic fluctuation theory then computes the consequences of these inputs.

The long-time-tail calculation is rigorous as an internal statement of the stochastic hydrodynamic effective theory with a stated cutoff and Gaussian leading measure. Promoting it to a theorem about a particular interacting continuum QFT requires a proof that the QFT has the hydrodynamic scaling limit assumed above.

Open Problem 134.1 (Hydrodynamic limit). Derive fluctuating hydrodynamics and long-time tails directly from an interacting QFT with a KMS state, including the hydrodynamic scaling limit, noise kernel, nonlinear couplings, transport-coefficient renormalization, and control of the order of thermodynamic, real-time, and low-frequency limits.

Chapter 135

QCD Phase Structure, Plasma, and Dense Matter

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$; Euclidean $\delta_{\mu\nu}$.
- **$\hbar$.** 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter records the finite-temperature and finite-density phase questions of QCD in the same convention as the Yang–Mills chapters:

$$S_{\mathrm{YM}} = \frac{1}{4g^2}\int \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}), \qquad \mathrm{tr}_{\square}(T^aT^b) = \delta^{ab}$$

for $SU(N_c)$ unless another invariant bilinear form is explicitly stated. The chapter is deliberately not a catalogue of phase-diagram lore. Each order parameter is tied to a symmetry, a limiting procedure, or a specified source. Each weak-coupling statement is separated from the nonperturbative questions that remain after the weak-coupling scale hierarchy has been identified.

135.1 Thermal QCD Data

Let N_f Dirac quarks transform in the fundamental representation of $SU(N_c)$. Put the Euclidean theory on

$$M_\beta = S^1_\beta \times \mathbb{R}^3, \qquad \beta = T^{-1},$$

with periodic gauge fields and antiperiodic quark fields around S^1_β. A baryon chemical potential μ_B couples to the conserved baryon number B, normalized so that a quark has baryon charge

$1/N_c$. Equivalently, every quark flavor has quark chemical potential

$$\mu_q = \frac{\mu_B}{N_c}.$$

At finite spatial volume V, a regulator a, and quark masses $m = (m_1, \ldots, m_{N_f})$, define

$$Z_{a,V}(T, \mu_B, m) = \mathrm{Tr}_{\mathcal{H}_{a,V}} \exp\{-\beta(H_{a,V} - \mu_B B_{a,V})\}, \tag{135.1}$$

when the Hamiltonian regulator is used. In a Euclidean lattice regulator the same object is represented by the corresponding gauge integral with the temporal links weighted by $e^{\pm a\mu_q}$ for forward and backward quark propagation. The pressure is the thermodynamic limit

$$p(T, \mu_B, m) = \lim_{a\to 0} \lim_{V\to\infty} \frac{T}{V} \log Z_{a,V}(T, \mu_B, m), \tag{135.2}$$

provided the limit exists along the chosen continuum trajectory.

Definition 135.1 (QCD phase specification)**.** A QCD phase specification at fixed N_c, N_f consists of:

1. the thermodynamic pressure $p(T, \mu_B, m)$ as a function on the domain where the continuum thermal state is defined;
2. a list of symmetries that are exact at the chosen masses and chemical potentials;
3. order parameters or source derivatives associated to those exact symmetries;
4. the limiting prescription used to detect spontaneous symmetry breaking;
5. a status label for each nonanalytic locus: theorem, lattice continuum-extrapolation claim, controlled weak-coupling claim, or conjectural continuation.

The word "phase" is used in this chapter only through such a specification.

Operational spine of the chapter. The rest of the chapter uses Definition 135.1 as a repeated test, not as introductory bookkeeping. First, finite-regulator analyticity and source curvatures identify what can and cannot become nonanalytic in the thermodynamic continuum limit. Second, center and chiral sections attach candidate order parameters to exact symmetries and to specified source limits. Third, the plasma and transport sections extract observable response data from pressure derivatives, spectral limits, and controlled hard-thermal-loop windows, while keeping those data separate from claims about real-time hydrodynamic poles. Fourth, the baryon-density sections explain the sign problem, imaginary-source diagnostics, and high-density effective theory before any dense phase is named. Finally, the CFL analysis proceeds from attractive Fermi-surface channels to gauge-invariant order diagnostics, screening response, collective fields, and anomaly matching. Thus a named region in the QCD phase diagram is accepted only after the corresponding pressure, symmetry, source, response, and limiting data have been displayed or explicitly marked as uncontrolled.

135.2 Finite-Regulator Analyticity and Thermodynamic Phases

A phase transition is not a property of a single finite lattice integral. At finite cutoff and finite volume the partition function is an analytic function of its sources away from its zeros. Nonanalyticity can only emerge from a limiting procedure: thermodynamic volume, continuum cutoff, chiral mass, or a source-selection limit. This elementary point is important because it prevents a phase diagram from being treated as a drawing whose curves exist before the limiting observables have been defined.

Definition 135.2 (Finite lattice thermal gauge setup)**.** Fix a finite Euclidean lattice

$$\Lambda = (N_\tau, N_s^3, a)$$

with compact gauge group G, Haar measure dU on each link, and antiperiodic temporal boundary condition for quarks. Let $\kappa = (\kappa_\alpha)$ denote finitely many gauge couplings multiplying bounded continuous class functions $S_\alpha(U)$, let $\zeta = e^{a\mu_q}$ be the quark fugacity, and let

$$M_\Lambda(U; m, \zeta)$$

be the finite fermion matrix in the chosen lattice discretization, with entries polynomial in the masses m_i and Laurent-polynomial in ζ. For an integer number of unrooted Dirac flavors, the finite-regulator partition function is

$$Z_\Lambda(\kappa, m, \zeta) = \int_{G^{E(\Lambda)}} \prod_\ell dU_\ell \, \exp\left[-\sum_\alpha \kappa_\alpha S_\alpha(U)\right] \prod_{i=1}^{N_f} \det M_{\Lambda,i}(U; m_i, \zeta), \tag{135.3}$$

where $E(\Lambda)$ is the finite link set. Rooted or otherwise non-polynomial fermion prescriptions are not included in this setup unless a separate branch and locality construction has been supplied.

Finite-regulator analyticity. For the setup of Definition 135.2, $Z_\Lambda(\kappa, m, \zeta)$ is entire in the gauge couplings and masses and Laurent-polynomial in the fugacity ζ. Consequently the finite pressure

$$p_\Lambda(\kappa, m, \zeta) = \frac{T}{V_\Lambda} \log Z_\Lambda(\kappa, m, \zeta)$$

is holomorphic on every simply connected parameter domain that does not meet the zero locus of Z_Λ, after a branch of the logarithm has been chosen. This finite-dimensional analytic fact becomes a statement about a thermodynamic or continuum pressure only after a separate compact-convergence input. The lattice has finitely many links, and $G^{E(\Lambda)}$ is compact. For each fixed gauge field U, the determinant factors are polynomials in the masses and Laurent polynomials in ζ. The exponential of the gauge action has an absolutely and uniformly convergent power series in κ on compact subsets of complex coupling space, because every $S_\alpha(U)$ is bounded on the compact integration domain. Termwise integration is justified by uniform convergence on compact parameter sets. Thus the integral defines an entire function in κ and m, with only finitely many positive and negative fugacity powers. Wherever this function is nonzero on a simply connected domain, a holomorphic logarithm exists, and the displayed finite pressure is holomorphic there.

Definition 135.3 (Thermodynamic phase transition and Lee–Yang obstruction)**.** Let x denote a real collection of thermal parameters, masses, and sources, and let $p_\Lambda(x)$ be the finite-regulator pressure along a specified sequence of lattices and continuum trajectories. A thermodynamic phase transition at x_0 means that the limiting pressure

$$p(x) = \lim_{\Lambda\to\infty} p_\Lambda(x)$$

exists in a real neighborhood of x_0 but is not real analytic at x_0, or that the limiting local Gibbs/KMS state is not unique after the source-selection prescription has been removed. The finite-regulator Lee–Yang locus is

$$\mathcal{Z}_\Lambda = \{z \in \mathcal{U}_\mathbb{C} \mid Z_\Lambda(z) = 0\}.$$

Remark 135.4 (Zero-free neighborhoods exclude pressure singularities). Let $\mathcal{U}_\mathbb{C}$ be a complex neighborhood of a real point x_0. Suppose that, for all sufficiently large Λ, Z_Λ has no zeros in $\mathcal{U}_\mathbb{C}$, that the logarithm branches are chosen compatibly, and that p_Λ converges uniformly on compact subsets of $\mathcal{U}_\mathbb{C}$. Then the limiting pressure p is holomorphic on $\mathcal{U}_\mathbb{C}$, hence real analytic near x_0. Therefore a pressure nonanalyticity at x_0 requires either Lee–Yang zeros approaching x_0 in every complex neighborhood, or failure of the stated uniform analytic convergence. By the finite-regulator analyticity discussion above, each p_Λ is holomorphic on $\mathcal{U}_\mathbb{C}$ after the stated choice of logarithm. The uniform compact convergence hypothesis then makes the limit holomorphic by the Weierstrass theorem for holomorphic functions. Restricting a holomorphic function to the real slice gives a real-analytic function. The contrapositive gives the final statement: if the real pressure has a nonanalyticity at x_0, the hypotheses just listed cannot all hold on any complex neighborhood of x_0. Thus either zeros approach x_0, or the finite-regulator analytic functions fail to converge in the compact-open holomorphic topology there.

Source curvature as an integrated connected correlator. Let h be a real source coupled at finite regulator to an integrated gauge-invariant local observable

$$X_\Lambda = \sum_{x\in\Lambda_s} a^3\, \mathcal{O}_x$$

through the Euclidean weight $\exp(hX_\Lambda)$. With

$$p_\Lambda(h) = \frac{T}{V_\Lambda}\log Z_\Lambda(h), \qquad \overline{\mathcal{O}}_\Lambda = V_\Lambda^{-1} X_\Lambda,$$

one has

$$\frac{\partial p_\Lambda}{\partial h} = T\langle \overline{\mathcal{O}}_\Lambda\rangle_h, \qquad \frac{\partial^2 p_\Lambda}{\partial h^2} = T\,V_\Lambda\left(\langle \overline{\mathcal{O}}_\Lambda^2\rangle_h - \langle \overline{\mathcal{O}}_\Lambda\rangle_h^2\right). \tag{135.4}$$

Thus a divergent susceptibility is a statement about the scaling of an integrated connected correlator in the thermodynamic limit.

At finite regulator X_Λ is an ordinary measurable function on the finite configuration space or finite lattice field space, and $Z_\Lambda(h)$ is a finite-dimensional integral with positive radius of analyticity in h around any real value for which the weight is integrable. Differentiating under the integral therefore gives

$$\partial_h \log Z_\Lambda(h) = \langle X_\Lambda\rangle_h.$$

For the second derivative, write $\langle Y\rangle_h = Z_\Lambda(h)^{-1}\int Y\, e^{hX_\Lambda} d\mu_\Lambda$. Then

$$\partial_h\langle Y\rangle_h = \langle YX_\Lambda\rangle_h - \langle Y\rangle_h\langle X_\Lambda\rangle_h.$$

Taking $Y = X_\Lambda$ yields the connected two-point formula

$$\partial_h^2 \log Z_\Lambda(h) = \langle X_\Lambda^2\rangle_h - \langle X_\Lambda\rangle_h^2.$$

Multiplying by T/V_Λ and writing $X_\Lambda = V_\Lambda \overline{\mathcal{O}}_\Lambda$ gives (135.4). Thus the possible thermodynamic singularity is not a pointwise singularity of $\mathcal{O}_x$; it is the volume growth of the integrated connected correlator after the infrared limit is taken.

For physical QCD with fundamental quarks of nonzero mass, the most familiar finite-temperature change near $T \sim \Lambda_{\mathrm{QCD}}$ is not attached to an exact fundamental center symmetry and not attached to an exact chiral symmetry. It is therefore a crossover unless a nonanalyticity is established in the thermodynamic continuum theory. Pure Yang–Mills theory and chiral limits have sharper order parameters, and those are treated first.

135.3 Center Deconfinement in Pure Yang–Mills

Let $G = SU(N_c)$ and first remove all dynamical fundamental matter. The thermal Wilson line at spatial point $\mathbf{x}$ is

$$\mathcal{P}(\mathbf{x}) = \mathcal{P}_\tau \exp\left(\mathrm{i}\int_0^\beta A_0(\tau,\mathbf{x})\,d\tau\right). \tag{135.5}$$

For an irreducible representation R define the normalized Polyakov loop

$$P_R(\mathbf{x}) = \frac{1}{d_R}\,\mathrm{tr}_R\,\mathcal{P}(\mathbf{x}). \tag{135.6}$$

If R has N-ality k_R, then the thermal center transformation by $z \in Z(SU(N_c))$ sends

$$P_R(\mathbf{x}) \longmapsto z^{k_R} P_R(\mathbf{x}).$$

The lattice construction of the transformation and the static-source interpretation were given in Chapter 46; here we use them as the phase-ordering language of the thermal theory.

Definition 135.5 (Center-order deconfinement)**.** In pure $SU(N_c)$ Yang–Mills, choose a small source h that selects a center sector and set

$$\overline{P}_\square = V^{-1}\int_{\mathbb{R}^3} P_\square(\mathbf{x})\,d^3x$$

in a finite box before the infinite-volume limit. The center order parameter is

$$\ell(T) = \lim_{h\downarrow 0}\lim_{V\to\infty}\left\langle \overline{P}_\square \right\rangle_{T,h,V}. \tag{135.7}$$

The thermal state is center symmetric when $\ell(T) = 0$ and center broken when $\ell(T) \neq 0$. A center deconfinement transition is a nonanalytic change of the infinite-volume free energy across which this symmetry realization changes.

Remark 135.6 (Center symmetry forces the finite-volume one-point loop). At finite spatial volume and zero center-breaking source in pure $SU(N_c)$ Yang–Mills,

$$\langle P_R(\mathbf{x})\rangle = 0$$

for every representation R whose N-ality is nonzero.

The finite-volume regulated measure and action are invariant under the thermal center transformation. Averaging the expectation value over the finite group $Z(SU(N_c))$ gives

$$\langle P_R \rangle = \frac{1}{N_c} \sum_{j=0}^{N_c-1} e^{2\pi i j k_R / N_c} \langle P_R \rangle.$$

If $k_R \not\equiv 0 \pmod{N_c}$, the finite geometric sum vanishes. Hence $\langle P_R \rangle = 0$. A nonzero Polyakov-loop order parameter is therefore necessarily a spontaneous-symmetry-breaking limit, not a finite-volume one-point function.

After multiplicative line renormalization, the one-point function has the static-source interpretation

$$\langle P_R \rangle_{\text{ren}} = \exp\{-\beta(F_R - F_0)\}. \tag{135.8}$$

Thus $\ell = 0$ is equivalent to an infinite free-energy cost for an isolated fundamental external charge, in the center-symmetric limiting state. This statement is logically distinct from the zero-temperature Wilson-loop area law. The latter diagnoses the large-distance static potential of a source-antisource pair; the former diagnoses the realization of thermal center symmetry and the free energy of one wrapped external line.

Remark 135.7 (Dynamical fundamental quarks). Fundamental quarks explicitly break the fundamental thermal center symmetry. Consequently $P_{\square}$ remains a useful renormalized static-source observable, but not a strict order parameter. In physical QCD the rapid change of the Polyakov loop is evidence for liberation of color degrees of freedom in a crossover regime. Existence of an exact phase transition requires a separate thermodynamic-limit statement.

135.3.1 Polyakov-Loop Effective Measures at Finite Regulator

A Polyakov-loop effective theory is not, by definition, a new continuum QCD. At finite regulator it is a marginal measure obtained from the thermal lattice gauge theory by retaining only temporal Wilson lines. Its usefulness depends on whether the resulting nonlocal measure can be approximated by a controlled local or rapidly decaying interaction.

Definition 135.8 (Exact finite-regulator Polyakov-loop effective measure)**.** Fix a finite thermal lattice $\Lambda = N_\tau \times \Lambda_s$ with spacing a, compact gauge group $SU(N_c)$, and thermal circle length $\beta = aN_\tau$. For a gauge configuration U, define the temporal Wilson line at a spatial site $\mathbf{x}$ by

$$\mathcal{P}_{\mathbf{x}}(U) = \prod_{\tau=0}^{N_\tau-1} U_0(\tau, \mathbf{x}),$$

with the product ordered along the Euclidean time circle. For every bounded class function F of the collection $\{\mathcal{P}_{\mathbf{x}}\}$, define

$$\int F(P)\, d\nu_\Lambda(P) = \frac{1}{Z_\Lambda} \int d\mu_\Lambda(U)\, F(\{\mathcal{P}_{\mathbf{x}}(U)\})\, e^{-S_g[U]} \prod_f \det D_f[U].$$

When the regulated weight is positive, this is a probability measure on $SU(N_c)^{\Lambda_s}$, restricted to class functions at each site. At real chemical potential, or in any other situation where the determinant is complex, the same formula defines a normalized complex linear functional rather than a probability measure. In either case it is the exact finite-regulator Polyakov-loop effective object. If it is absolutely continuous with respect to the product Haar measure $dP = \prod_{\mathbf{x}} dP_{\mathbf{x}}$, one may write

$$d\nu_\Lambda(P) = Z_{\text{eff}}^{-1} e^{-S_{\text{eff},\Lambda}(P)} dP,$$

where $S_{\text{eff},\Lambda}$ is defined up to an additive constant and up to changes on Haar-null sets.

Character coordinates and center charge. Assume $d\nu_\Lambda$ has an L^2 density with respect to product Haar measure. Then the density admits an L^2 character expansion

$$\frac{d\nu_\Lambda}{dP} = \sum_{\{\lambda_{\mathbf{x}}\}} c_{\{\lambda_{\mathbf{x}}\}} \prod_{\mathbf{x}\in\Lambda_s} \chi_{\lambda_{\mathbf{x}}}(P_{\mathbf{x}}),$$

where only finitely many sites carry nontrivial representations in each basis vector. In pure Yang–Mills, the coefficient $c_{\{\lambda_{\mathbf{x}}\}}$ vanishes unless the total N-ality

$$\sum_{\mathbf{x}} k_{\lambda_{\mathbf{x}}}$$

is 0 modulo N_c. With dynamical fundamental quarks this selection rule is explicitly broken by terms of nonzero total N-ality.

The Peter–Weyl theorem gives an orthonormal basis of class functions on $SU(N_c)$ consisting of irreducible characters. Tensoring these bases over the finite set Λ_s gives the displayed expansion in $L^2(SU(N_c)^{\Lambda_s})$.

For pure Yang–Mills the finite-regulator measure is invariant under the thermal center transformation

$$P_{\mathbf{x}} \mapsto zP_{\mathbf{x}}, \qquad z \in Z(SU(N_c)),$$

applied simultaneously at all spatial sites. Since $\chi_\lambda(zP) = z^{k_\lambda}\chi_\lambda(P)$, invariance of the density forces

$$c_{\{\lambda_{\mathbf{x}}\}} = z^{\sum_{\mathbf{x}} k_{\lambda_{\mathbf{x}}}} c_{\{\lambda_{\mathbf{x}}\}} \quad \text{for all } z \in Z(SU(N_c)).$$

This is possible only when the total N-ality vanishes modulo N_c, or when the coefficient is zero. Fundamental quark determinants contain closed fermion paths winding once around the thermal circle; such paths carry fundamental N-ality and therefore break the selection rule.

Controlled Approximation 135.9 (Local Polyakov-loop action)**.** A local Polyakov-loop action is a truncation of the exact effective measure to finitely many character monomials, usually organized by spatial range, representation size, winding number, and powers of a strong-coupling or hopping parameter. The truncation is a controlled approximation only in a domain where the omitted character coefficients are bounded by a convergent cluster or hopping expansion, or by a separately stated numerical error estimate.

Leading strong-coupling tube in pure Yang–Mills. Let the pure-gauge lattice action have a plaquette character expansion whose fundamental coefficient is $u_\Box$, normalized so that $u_\Box = 0$ at zero plaquette coupling. In a finite lattice strong-coupling expansion, the leading center-invariant nearest-neighbor interaction between Polyakov loops has the form

$$S^{(1)}_{\text{eff},\Lambda}(P) = -J_1 \sum_{\langle\mathbf{xy}\rangle} \left[\chi_\Box(P_{\mathbf{x}})\chi_{\overline{\Box}}(P_{\mathbf{y}}) + \chi_{\overline{\Box}}(P_{\mathbf{x}})\chi_\Box(P_{\mathbf{y}})\right],$$

with

$$J_1 = u_{\Box}^{N_\tau} + O(u_{\Box}^{N_\tau+2})$$

in the same expansion. The displayed term is center neutral and therefore compatible with the character-coordinate center-selection rule above.

Expand each plaquette Boltzmann factor in irreducible characters. A nonzero contribution to the Polyakov-loop marginal measure must leave no unpaired spatial link after Haar integration. For two neighboring spatial sites $\mathbf{x}, \mathbf{y}$, the minimal surface that connects their temporal Wilson lines is a tube of N_τ temporal-spatial plaquettes wrapping once around the thermal circle. Placing the fundamental character on each plaquette of this tube and integrating the shared spatial links contracts the representation matrices into

$$\chi_{\Box}(P_{\mathbf{x}})\chi_{\overline{\Box}}(P_{\mathbf{y}})$$

or its complex conjugate. The weight of the minimal tube is $u_{\Box}^{N_\tau}$. Corrections either decorate the tube or use larger representations and therefore carry higher powers of the plaquette character coefficients; in the fundamental channel the first plaquette decorations occur at the indicated higher order. The product has total N-ality $1-1=0$, so it is invariant under the simultaneous center transformation.

Leading heavy-quark center-breaking term. For Wilson fermions with hopping parameter κ and quark chemical potential μ_q, the leading temporal-winding term in the hopping expansion of N_f degenerate fundamental Dirac quarks is

$$S^{\mathrm{hq}}_{\mathrm{eff},\Lambda}(P) = -h\sum_{\mathbf{x}}\left[e^{\beta\mu_q}\chi_{\Box}(P_{\mathbf{x}}) + e^{-\beta\mu_q}\chi_{\overline{\Box}}(P_{\mathbf{x}})\right] + \cdots ,$$

where, in the standard Wilson hopping normalization,

$$h = 2N_f(2\kappa)^{N_\tau}.$$

The ellipsis denotes longer closed fermion paths, multiple windings, spatial hops, and higher powers of κ. The displayed term has nonzero N-ality and is the finite-regulator origin of explicit center breaking by dynamical fundamental quarks.

Write the Wilson Dirac operator as $D = 1 - H$, where H is the hopping matrix. At finite lattice volume,

$$\log\det D = \mathrm{Tr}\log(1-H) = -\sum_{n\geq 1}\frac{1}{n}\,\mathrm{Tr}\,H^n$$

as a formal hopping expansion. Each trace is a sum over closed lattice paths. A path that winds once forward around the Euclidean time circle at fixed $\mathbf{x}$ has N_τ forward temporal hops and gauge factor $\mathrm{tr}_{\Box} P_{\mathbf{x}}$. The spin projectors along a straight temporal Wilson loop leave two spin states, giving the factor 2; the N_f degenerate flavors multiply the result by N_f. The antiperiodic fermion boundary condition supplies an additional minus sign for a path with odd thermal winding. This sign cancels the minus sign in the logarithmic expansion, so the leading forward-winding contribution to $\log\det D$ is

$$2N_f(2\kappa)^{N_\tau}e^{\beta\mu_q}\chi_{\Box}(P_{\mathbf{x}}).$$

Here $\beta = aN_\tau$, so the N_τ factors of $e^{a\mu_q}$ combine to $e^{\beta\mu_q}$. The backward winding gives the conjugate term with $e^{-\beta\mu_q}$ and $\chi_{\overline{\square}}$. Since $e^{-S_{\rm eff}}$ contains $\det D = \exp(\log\det D)$, the same positive term in $\log\det D$ appears with a minus sign in $S_{\rm eff}$, as displayed.

Under $P_{\mathbf{x}} \mapsto zP_{\mathbf{x}}$, the first term is multiplied by z and the second by z^{-1}. Hence the heavy-quark determinant is a center-breaking source unless the coefficient h is set to zero.

135.4 Controlled High-Temperature Holonomy Potential

The Polyakov loop becomes a calculable order-coordinate in a regime where a constant temporal holonomy is a valid background-field variable. Choose a spatially constant commuting background on $S^1_\beta \times \mathbb{R}^3$ and write its fundamental holonomy as

$$P = \mathrm{diag}\left(e^{\mathrm{i}\theta_1}, \ldots, e^{\mathrm{i}\theta_{N_c}}\right), \qquad \sum_{j=1}^{N_c} \theta_j \equiv 0 \pmod{2\pi}. \tag{135.9}$$

The variables θ_j are defined modulo permutations and common shifts compatible with $\det P = 1$. A center transformation sends $P \mapsto zP$, $z^{N_c} = 1$.

Controlled Approximation 135.10 (One-loop holonomy regime)**.** The one-loop holonomy potential is the leading weak-coupling thermal effective potential for the zero Matsubara mode of A_0, computed in a background whose spatial gradients and noncommuting components are set to zero. Its expansion parameter is the running gauge coupling at scale T, and its infrared completion still requires the electric and magnetic static sectors discussed in Section 135.7 below.

One-loop Polyakov-holonomy potential. For pure $SU(N_c)$ Yang–Mills, the one-loop gluonic free-energy density as a function of the holonomy P is

$$f_g(P) = -\frac{2}{\pi^2\beta^4}\sum_{n=1}^{\infty}\frac{1}{n^4}\,\mathrm{tr}_{\rm adj}(P^n) = -\frac{2}{\pi^2\beta^4}\sum_{n=1}^{\infty}\frac{|\,\mathrm{tr}_\square P^n|^2 - 1}{n^4}. \tag{135.10}$$

For N_f massless Dirac fundamental quarks at $\mu_q = 0$, the one-loop fermion contribution is

$$f_q(P) = \frac{2N_f}{\pi^2\beta^4}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^4}\left(\mathrm{tr}_\square P^n + \mathrm{tr}_\square P^{-n}\right). \tag{135.11}$$

Consider first one bosonic oscillator with thermal phase φ, meaning that its Matsubara frequencies are shifted by the holonomy eigenvalue $e^{\mathrm{i}\varphi}$. Its thermal free-energy density is

$$T\int\frac{d^3p}{(2\pi)^3}\log\left(1 - e^{-\beta|\mathbf{p}|+\mathrm{i}\varphi}\right).$$

Expanding the logarithm and integrating term by term gives

$$-T\sum_{n\geq 1}\frac{e^{\mathrm{i}n\varphi}}{n}\int\frac{d^3p}{(2\pi)^3}e^{-n\beta|\mathbf{p}|} = -\frac{1}{\pi^2\beta^4}\sum_{n\geq 1}\frac{e^{\mathrm{i}n\varphi}}{n^4}, \tag{135.12}$$

because

$$\int \frac{d^3p}{(2\pi)^3} e^{-n\beta|\mathbf{p}|} = \frac{1}{2\pi^2}\int_0^\infty p^2 e^{-n\beta p}\,dp = \frac{1}{\pi^2 n^3\beta^3}. \tag{135.13}$$

In background gauge, ghosts and unphysical vector polarizations cancel so that two physical bosonic polarizations remain for each adjoint weight. Summing (135.12) over adjoint weights gives the first expression in (135.10). The identity

$$\operatorname{tr}_{\mathrm{adj}}(P^n) = \operatorname{tr}_{\square}(P^n)\operatorname{tr}_{\overline{\square}}(P^n) - 1 = |\operatorname{tr}_{\square} P^n|^2 - 1$$

uses $\square \otimes \overline{\square} = \mathbf{1} \oplus \mathrm{adj}$.

For a Dirac fundamental quark at zero chemical potential the two spin polarizations and the quark-antiquark pair give

$$-2T\int \frac{d^3p}{(2\pi)^3}\left[\log\left(1+e^{-\beta|\mathbf{p}|+\mathrm{i}\theta_j}\right) + \log\left(1+e^{-\beta|\mathbf{p}|-\mathrm{i}\theta_j}\right)\right]$$

for each color eigenphase θ_j. The plus sign inside the logarithm is the antiperiodic thermal boundary condition. Expanding $\log(1+x) = \sum_{n\geq 1}(-1)^{n+1}x^n/n$, using (135.13), and summing over colors and flavors gives (135.11).

Perturbative center-broken minima in pure Yang–Mills. For pure $SU(N_c)$ Yang–Mills, the one-loop potential (135.10) is minimized exactly at center holonomies

$$P = z\,\mathbf{1}_{N_c}, \qquad z^{N_c} = 1. \tag{135.14}$$

At these minima

$$f_g(z\mathbf{1}) = -\frac{\pi^2 T^4}{45}(N_c^2 - 1), \tag{135.15}$$

which is the free gluon free-energy density. For every unitary $N_c \times N_c$ matrix,

$$|\operatorname{tr}_{\square} P^n| \leq N_c.$$

Hence $\operatorname{tr}_{\mathrm{adj}}(P^n) \leq N_c^2 - 1$ for every n. All coefficients in (135.10) are negative, so the minimum is obtained by maximizing each $\operatorname{tr}_{\mathrm{adj}}(P^n)$. The $n = 1$ term already forces $|\operatorname{tr}_{\square} P| = N_c$, which for a unitary matrix means all eigenvalues are equal. The determinant-one condition then gives $P = z\mathbf{1}_{N_c}$ with $z^{N_c} = 1$. Substitution in (135.10) gives

$$f_g(z\mathbf{1}) = -\frac{2}{\pi^2\beta^4}(N_c^2-1)\sum_{n\geq 1}\frac{1}{n^4} = -\frac{2}{\pi^2\beta^4}(N_c^2-1)\frac{\pi^4}{90},$$

which is (135.15).

Example 135.11 (Uniform center-symmetric holonomy)**.** Let P_{cs} have eigenvalues uniformly spaced around the unit circle, with an overall phase chosen so that $P_{\mathrm{cs}} \in SU(N_c)$. Then

$$\operatorname{tr}_{\square} P_{\mathrm{cs}}^n = 0 \quad (N_c \nmid n), \qquad |\operatorname{tr}_{\square} P_{\mathrm{cs}}^n|^2 = N_c^2 \quad (N_c \mid n).$$

Therefore

$$f_g(P_{\mathrm{cs}}) = \frac{\pi^2 T^4}{45}\left(1 - \frac{1}{N_c^2}\right), \qquad f_g(P_{\mathrm{cs}}) - f_g(\mathbf{1}) = \frac{\pi^2 T^4}{45}\left(N_c^2 - \frac{1}{N_c^2}\right). \tag{135.16}$$

The perturbative high-temperature gluon determinant therefore energetically selects a center sector rather than the uniformly distributed holonomy.

Remark 135.12 (Fundamental quarks as a center source). The gluon potential depends on P through adjoint traces and is invariant under $P \mapsto zP$. The quark potential (135.11) contains fundamental traces:

$$\mathrm{tr}_{\square}(zP)^n = z^n \, \mathrm{tr}_{\square} \, P^n.$$

Thus dynamical fundamental quarks supply an explicit center-breaking source. At $P = \mathbf{1}$, (135.11) gives

$$f_q(\mathbf{1}) = -\frac{7\pi^2 T^4}{180} N_c N_f, \tag{135.17}$$

the negative of the free massless Dirac-quark pressure in the free-plasma calculation below.

135.5 Chiral Symmetry and the Dirac Spectrum

Now set $\mu_B = 0$ and take N_f Dirac quarks with a common mass m. In the chiral limit $m = 0$, and ignoring the anomalous axial $U(1)$, the classical flavor symmetry is

$$SU(N_f)_L \times SU(N_f)_R \times U(1)_B.$$

Let V_3 be the spatial volume and let

$$\mathcal{V}_4 = \beta V_3$$

be the Euclidean thermal volume. For degenerate quark mass, the pressure derivative is the flavor-summed scalar source response. We use the positive per-flavor condensate convention

$$\Sigma_m(T) = \frac{1}{N_f} \frac{\partial p(T, m)}{\partial m}. \tag{135.18}$$

Equivalently, after integrating the fermions in a fixed gauge background, $\Sigma_m(T)$ is the thermal average of $\mathcal{V}_4^{-1} \, \mathrm{Tr}(D_E + m)^{-1}$ for the one-flavor Euclidean Dirac operator D_E. With the Grassmann-ordering convention in which the local scalar density is $\bar{q}q = \sum_{i=1}^{N_f} \bar{q}_i q_i$, its flavor-summed expectation is

$$\langle \bar{q}q \rangle_{\mathrm{sum},T,m} = -N_f \Sigma_m(T).$$

Thus $\Sigma_m \geq 0$ in a phase with the usual sign of the scalar condensate. The chiral order parameter is the limit $\Sigma(T) = \lim_{m \downarrow 0} \lim_{V_3 \to \infty} \Sigma_m(T)$, when this limit exists. A nonzero condensate in the chiral limit transforms as $(\overline{\mathbf{N}_f}, \mathbf{N}_f) \oplus (\mathbf{N}_f, \overline{\mathbf{N}_f})$ and breaks

$$SU(N_f)_L \times SU(N_f)_R \longrightarrow SU(N_f)_V.$$

Definition 135.13 (Chiral order parameter)**.** The chiral condensate used as an order parameter is

$$\Sigma(T) = \lim_{m \downarrow 0} \lim_{V_3 \to \infty} \frac{1}{\mathcal{V}_4} \left\langle \mathrm{Tr}(D_E + m)^{-1} \right\rangle_{V_3,T,m}, \tag{135.19}$$

where D_E is the one-flavor massless Euclidean Dirac operator in a gauge background at $\mu_B = 0$. The infinite-volume limit precedes the chiral limit.

Banks–Casher relation under a spectral-density hypothesis. The nontrivial input is the existence of a thermodynamic Dirac spectral density continuous at the origin. Assume that the positive Dirac eigenvalue density

$$\rho_{\mathcal{V}_4}(\lambda) = \frac{1}{\mathcal{V}_4} \left\langle \sum_{\lambda_n > 0} \delta(\lambda - \lambda_n) \right\rangle$$

has a thermodynamic limit $\rho(\lambda)$ that is continuous at $\lambda = 0$, and that zero-mode contributions have zero density in the same limit. Then

$$\Sigma(T) = \pi\rho(0) \tag{135.20}$$

with the normalization in (135.19). If $\rho(0) = 0$, this particular condensate vanishes.

At $\mu_B = 0$, D_E is anti-Hermitian in the usual positive Euclidean inner product and anticommutes with γ_5 in the massless theory. Nonzero eigenvalues occur in pairs $\pm i\lambda_n$, $\lambda_n > 0$. For fixed positive mass,

$$\frac{1}{\mathcal{V}_4} \operatorname{Tr}(D_E + m)^{-1} = \frac{N_0}{\mathcal{V}_4 m} + \frac{1}{\mathcal{V}_4} \sum_{\lambda_n > 0} \left(\frac{1}{m + i\lambda_n} + \frac{1}{m - i\lambda_n} \right),$$

where N_0 is the number of zero modes. The nonzero pair contribution is

$$\frac{2m}{\mathcal{V}_4} \sum_{\lambda_n > 0} \frac{1}{\lambda_n^2 + m^2}.$$

By the zero-density hypothesis the first term drops out after $V_3 \to \infty$. The condensate definition therefore gives

$$\Sigma(T) = \lim_{m \downarrow 0} \int_0^\infty \frac{2m\,\rho(\lambda)}{\lambda^2 + m^2}\, d\lambda.$$

The kernels

$$K_m(\lambda) = \frac{2m}{\lambda^2 + m^2}, \qquad \lambda \geq 0,$$

form an approximate identity on the half-line with total mass π:

$$\int_0^\infty K_m(\lambda)\, d\lambda = \pi.$$

Continuity of ρ at the origin gives the limit $\pi\rho(0)$, and hence (135.20). If instead one defines the spectral density on all of $\mathbb{R}$, counting both positive and negative eigenvalues, the same formula holds with that two-sided density evaluated at zero.

Remark 135.14 (Order of limits)*.* The relation is a statement about the thermodynamic spectral density near zero. Setting $m = 0$ at finite volume first gives no condensate from paired nonzero eigenvalues. The density of eigenvalues at spacing $O(\mathcal{V}_4^{-1})$ near the origin is precisely the infinite-volume order parameter that can survive the subsequent chiral limit.

135.5.1 Axial Ward Identity and Thermal GMOR Logic

The condensate is tied to integrated pseudoscalar correlations by an axial Ward identity. This is the cleanest finite-temperature form of the Gell-Mann–Oakes–Renner logic, because at $T > 0$

Lorentz invariance is absent and one must specify which pion pole, screening pole, or spectral residue is being used.

Let τ^a, $a = 1, \ldots, N_f^2 - 1$, be Hermitian generators of $\mathfrak{su}(N_f)$ normalized by

$$\mathrm{tr}_f(\tau^a \tau^b) = \delta^{ab}. \tag{135.21}$$

Define the nonsinglet Euclidean axial current and pseudoscalar density by

$$A_\mu^a = \bar{q}\gamma_\mu\gamma_5\tau^a q, \qquad P^a = \bar{q}\,\mathrm{i}\gamma_5\tau^a q. \tag{135.22}$$

The factor of i in P^a is part of the Euclidean source convention: a real pseudoscalar source p^a appears in the mass matrix as $m\mathbf{1} + \mathrm{i}\gamma_5 p^a\tau^a$. With this convention the Euclidean pseudoscalar susceptibility is

$$\chi_\pi^{ab}(T, m) = -\int_0^\beta d\tau \int_{\mathbb{R}^3} d^3x\, \langle P^a(\tau, \mathbf{x}) P^b(0)\rangle_{c,T,m}, \tag{135.23}$$

with the finite-volume definition obtained by replacing $\mathbb{R}^3$ by the spatial torus and taking the thermodynamic limit after the integral.

Lemma 135.15 (Integrated nonsinglet axial Ward identity)**.** *Assume a regulator and renormalization prescription preserving the nonsinglet vector flavor symmetry and the nonsinglet axial Ward identity up to the explicit mass term; there is no nonsinglet axial anomaly. Then*

$$2m\,\chi_\pi^{ab}(T, m) = -\left\langle \bar{q}\{\tau^a, \tau^b\}q \right\rangle_{T,m}. \tag{135.24}$$

In a flavor-symmetric thermal state this becomes

$$m\,\chi_\pi^{ab}(T, m) = \delta^{ab}\Sigma_m(T). \tag{135.25}$$

Proof. The proof is a finite-regulator Ward-identity argument, followed by the thermodynamic limit in the order specified in the definition of the susceptibility. Perform the local nonsinglet axial change of variables

$$\delta q = \mathrm{i}\alpha^a(x)\tau^a\gamma_5 q, \qquad \delta\bar{q} = \mathrm{i}\alpha^a(x)\bar{q}\gamma_5\tau^a.$$

The regulator hypothesis says that the fermion measure has no nonsinglet Jacobian anomaly. The variation of the mass term is fixed by the Euclidean source convention in (135.22):

$$\delta_\alpha\big(m\bar{q}q\big) = 2m\,\alpha^a\,\bar{q}\mathrm{i}\gamma_5\tau^a q = 2m\,\alpha^a P^a.$$

The kinetic term gives the divergence of the current A_μ^a. With the active-variation convention used here, differentiating the change-of-variables identity with respect to $\alpha^a(x)$ and applying it to the insertion $P^b(0)$ gives the distributional identity

$$\partial_\mu\langle A_\mu^a(x)P^b(0)\rangle_{T,m} + 2m\langle P^a(x)P^b(0)\rangle_{T,m} + \delta_\beta^{(4)}(x)\langle\delta^a P^b(0)\rangle_{T,m} = 0. \tag{135.26}$$

Here $\delta_\beta^{(4)}$ is the periodic delta distribution on the thermal spacetime torus before the thermodynamic limit, and $\delta^a P^b$ denotes the variation of the inserted operator, not a second source variation. The contact term is

$$\delta^a P^b(0) = \mathrm{i}\,\bar{q}\gamma_5\tau^a\,\mathrm{i}\gamma_5\tau^b q + \bar{q}\,\mathrm{i}\gamma_5\tau^b\,\mathrm{i}\tau^a\gamma_5 q = -\bar{q}\{\tau^a, \tau^b\}q(0).$$

Integrating (135.26) over the thermal spacetime torus removes the divergence term by periodicity of the bosonic current. Vector flavor invariance gives $\langle P^a \rangle_{T,m} = 0$ for nonsinglet a, so the connected and ordinary two-point functions coincide in this channel. Using the susceptibility definition (135.23), one obtains

$$2m\, \chi_\pi^{ab}(T,m) = \langle \delta^a P^b \rangle_{T,m} = -\left\langle \bar{q}\{\tau^a, \tau^b\} q \right\rangle_{T,m},$$

which is (135.24). No statement about a pion pole has entered; the pole-saturation input is a separate infrared hypothesis.

In a flavor-symmetric state

$$\left\langle \bar{q}\{\tau^a, \tau^b\} q \right\rangle_{T,m} = \frac{2}{N_f} \delta^{ab} \langle \bar{q} q \rangle_{\mathrm{sum},T,m} = -2\delta^{ab} \Sigma_m(T),$$

where (135.21) fixes the singlet part of the anticommutator. Substitution gives (135.25). □

GMOR as a pole-saturation consequence. The integrated Ward identity becomes the finite-temperature GMOR relation only after one supplies a spectral hypothesis about the small-mass broken phase. Suppose, in addition to the hypotheses of Lemma 135.15, that for small $m > 0$ the zero-spatial-momentum pseudoscalar susceptibility in a broken phase is saturated at leading order by an isolated pion pole of mass $M_\pi(T,m)$. Let $G_\pi(T,m)$ and $f_{t,\delta}(T,m)$ be defined in the same flavor normalization (135.21) by the pole matrix elements

$$\langle \Omega_T | P^a(0) | \pi^b \rangle = \delta^{ab} G_\pi, \qquad \langle \Omega_T | A_0^a(0) | \pi^b \rangle = \delta^{ab} f_{t,\delta} M_\pi,$$

or by the corresponding KMS spectral residues when a thermal Hilbert-space representative is used. If the pole obeys the axial Ward relation

$$2m\, G_\pi = f_{t,\delta} M_\pi^2 \tag{135.27}$$

and the non-pole part of χ_π^{aa} is $O(1)$ as $m \downarrow 0$, then pole saturation gives

$$\chi_\pi^{ab}(T,m) = \delta^{ab} \frac{G_\pi(T,m)^2}{M_\pi(T,m)^2} + O(1).$$

The Ward identity (135.25) gives the same susceptibility as

$$\chi_\pi^{ab}(T,m) = \delta^{ab} \frac{\Sigma(T)}{m} + o(m^{-1})$$

when $\Sigma_m(T) \to \Sigma(T)$. Combining these two leading terms with (135.27),

$$\frac{G_\pi^2}{M_\pi^2} = \frac{f_{t,\delta}^2 M_\pi^2}{4m^2},$$

and equating the coefficients of $1/m$ gives

$$f_{t,\delta}(T,m)^2 M_\pi(T,m)^2 = 4m\Sigma(T) + o(m). \tag{135.28}$$

If instead one uses generators t^a normalized by $\mathrm{tr}_f(t^a t^b) = \frac{1}{2}\delta^{ab}$, the decay constant is smaller by $\sqrt{2}$ and the same statement is

$$f_{t,1/2}(T,m)^2 M_\pi(T,m)^2 = 2m\Sigma(T) + o(m). \tag{135.29}$$

Here rescaling the generators from $\mathrm{tr}_f(\tau^a \tau^b) = \delta^{ab}$ to $\mathrm{tr}_f(t^a t^b) = \frac{1}{2}\delta^{ab}$ rescales both the current and the pseudoscalar density by $1/\sqrt{2}$, hence $f_{t,1/2} = f_{t,\delta}/\sqrt{2}$, giving (135.29).

135.5.2 Low-Temperature Chiral Effective Theory

The Ward identity identifies the source derivative of the pressure with the integrated pion channel. In the low-temperature phase this identity can be evaluated in a controlled effective theory. The control assumptions are part of the statement: the spatial thermodynamic limit has been taken, the temperature is small compared with the first non-Goldstone mass scale, and the derivative expansion is organized at momenta $p \ll 4\pi F$.

Definition 135.16 (Leading chiral effective theory)**.** For N_f degenerate light quarks, the leading Euclidean chiral effective theory consists of a field $U : \mathbb{R}^3 \times S^1_\beta \to SU(N_f)$, a constant $F > 0$, and a constant $B > 0$, with leading action

$$S^{(2)}_\chi = \int_0^\beta d\tau \int d^3x \left[\frac{F^2}{4} \operatorname{tr}_f(\partial_\mu U^\dagger \partial_\mu U) - \frac{F^2 B}{2} \operatorname{tr}_f\big(MU^\dagger + UM^\dagger\big) \right]. \tag{135.30}$$

Here $M = m\mathbf{1}_{N_f}$ for the degenerate-mass discussion, and tr_f is the ordinary fundamental flavor trace. We parameterize the pion fields by

$$U = \exp\left(\frac{\mathrm{i}\sqrt{2}}{F} \pi^a \tau^a \right), \qquad \operatorname{tr}_f(\tau^a \tau^b) = \delta^{ab}. \tag{135.31}$$

With this convention F is the decay constant in the half-trace flavor normalization; the trace-delta decay constant used in the pole-saturation discussion above is $f_{t,\delta} = \sqrt{2}F + O(m, T^2)$.

Remark 135.17 (Tree-level source normalization)*.* The action (135.30) gives

$$\Sigma = F^2 B, \qquad M_\pi^2 = 2Bm \tag{135.32}$$

at tree level, where Σ is the positive per-flavor condensate in (135.18). Equivalently,

$$F^2 M_\pi^2 = 2m\Sigma, \qquad f_{t,\delta}^2 M_\pi^2 = 4m\Sigma$$

at this order. For a constant field $U = \mathbf{1}$, the Euclidean free-energy density from (135.30) is

$$f_0(m) = -N_f F^2 B m, \qquad p_0(m) = N_f F^2 B m.$$

Therefore $N_f^{-1} \partial p_0 / \partial m = F^2 B$, proving $\Sigma = F^2 B$. Expanding (135.31),

$$U + U^\dagger = 2\mathbf{1} - \frac{2}{F^2} \pi^a \pi^b \tau^a \tau^b + O(\pi^4),$$

and substituting $M = m\mathbf{1}$ gives the quadratic Euclidean Lagrangian

$$\frac{1}{2} \partial_\mu \pi^a \partial_\mu \pi^a + Bm\, \pi^a \pi^a.$$

Comparing with $\frac{1}{2}(\partial\pi)^2 + \frac{1}{2} M_\pi^2 \pi^2$ gives $M_\pi^2 = 2Bm$. The two GMOR forms follow from $f_{t,\delta} = \sqrt{2}F$ at leading order.

Lemma 135.18 (One-loop pion-gas correction to the condensate)**.** *Under the leading chiral effective assumptions above, and in the chiral limit after the spatial thermodynamic limit, the low-temperature condensate obeys*

$$\frac{\Sigma(T)}{\Sigma(0)} = 1 - \frac{N_f^2 - 1}{12 N_f} \frac{T^2}{F^2} + O\left(\frac{T^4}{F^4} \right) \tag{135.33}$$

within the chiral derivative expansion. For $N_f = 2$, this is

$$\Sigma(T)/\Sigma(0) = 1 - \frac{T^2}{8F^2} + O(T^4/F^4).$$

Proof. To one loop in the leading effective theory the only m-dependent thermal term at order T^2 is the free pressure of $d_\pi = N_f^2 - 1$ pion fields with $\omega_k = (k^2 + M_\pi^2)^{1/2}$:

$$p_\pi(T, M_\pi) = -d_\pi T \int \frac{d^3k}{(2\pi)^3} \log\bigl(1 - e^{-\beta\omega_k}\bigr). \tag{135.34}$$

The vacuum part is absorbed into the zero-temperature low-energy constants; the displayed thermal part is finite. Differentiating at fixed T,

$$\frac{\partial p_\pi}{\partial M_\pi^2} = -\frac{d_\pi}{2} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\omega_k} \frac{1}{e^{\beta\omega_k} - 1}. \tag{135.35}$$

Since $M_\pi^2 = 2Bm$, the pion contribution to the per-flavor condensate is

$$\frac{1}{N_f}\frac{\partial p_\pi}{\partial m} = -\frac{d_\pi B}{N_f} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\omega_k} \frac{1}{e^{\beta\omega_k} - 1}.$$

In the chiral limit the integral is

$$\int \frac{d^3k}{(2\pi)^3} \frac{1}{|k|} \frac{1}{e^{\beta|k|} - 1} = \frac{T^2}{12}. \tag{135.36}$$

Thus

$$\Sigma(T) = F^2 B - \frac{N_f^2 - 1}{N_f} B \frac{T^2}{12} + O(T^4/F^2).$$

Dividing by $\Sigma(0) = F^2 B$ gives (135.33). The next terms require the $O(p^4)$ chiral Lagrangian and two-loop pion interactions; they are not fixed by symmetry and F, B alone. □

Remark 135.19 (What the low-temperature formula establishes). Equation (135.33) is a controlled asymptotic statement inside the broken phase. It does not by itself prove a chiral restoration temperature or determine the order of a transition. Such questions require the nonperturbative thermal QCD limit, or an effective description whose domain includes the critical region.

135.6 Free Plasma Limit and Thermodynamic Observables

At asymptotically high temperature and fixed μ_B/T, asymptotic freedom makes the leading term in the pressure the free massless quark-gluon pressure. It is the leading term in a controlled high-temperature expansion whose subleading terms require screening and infrared resummation.

Massless free QCD pressure. For $SU(N_c)$ with N_f massless Dirac fundamental quarks at $\mu_B = 0$, the free pressure is

$$p_{\mathrm{SB}}(T) = \frac{\pi^2 T^4}{45}(N_c^2 - 1) + \frac{7\pi^2 T^4}{180} N_c N_f. \tag{135.37}$$

At nonzero baryon chemical potential, the quark part becomes

$$p_{q,\mathrm{free}}(T, \mu_B) = N_c N_f \left[\frac{7\pi^2 T^4}{180} + \frac{\mu_B^2 T^2}{6N_c^2} + \frac{\mu_B^4}{12\pi^2 N_c^4}\right]. \tag{135.38}$$

A massless bosonic degree of freedom contributes

$$-T \int \frac{d^3p}{(2\pi)^3} \log(1 - e^{-|p|/T}) = \frac{\pi^2 T^4}{90},$$

using $\int_0^\infty x^3(e^x - 1)^{-1} dx = \pi^4/15$. A gluon has two physical polarizations and $N_c^2 - 1$ color states, giving the first term.

A massless fermionic degree of freedom contributes 7/8 times the bosonic answer at zero chemical potential. A Dirac quark has four particle-antiparticle and spin degrees for each color and flavor, giving $(7/8)(4N_c N_f)\pi^2 T^4/90$, which is the second term of (135.37). The standard Fermi integral with chemical potential μ_q gives, for one Dirac quark of one color,

$$\frac{7\pi^2 T^4}{180} + \frac{\mu_q^2 T^2}{6} + \frac{\mu_q^4}{12\pi^2}.$$

Substituting $\mu_q = \mu_B/N_c$ and multiplying by $N_c N_f$ gives (135.38).

The energy density, entropy density, and trace anomaly are then defined by

$$s = \frac{\partial p}{\partial T}, \qquad n_B = \frac{\partial p}{\partial \mu_B}, \qquad \epsilon = -p + Ts + \mu_B n_B,$$

and

$$\Delta(T, \mu_B) = \epsilon - 3p.$$

For a conformal thermal state in four dimensions, $\Delta = 0$. In QCD, Δ measures the breaking of scale invariance by quark masses, by the running coupling, and by the thermal state itself.

Remark 135.20 (Thermodynamic identities). Let $p(T, \mu_B)$ be the thermodynamic pressure of a QCD phase on an open domain in which it is twice continuously differentiable. Define

$$s = \partial_T p, \qquad n_B = \partial_{\mu_B} p, \qquad \epsilon = -p + Ts + \mu_B n_B.$$

Then

$$dp = s\,dT + n_B\,d\mu_B,$$

and

$$d\epsilon = T\,ds + \mu_B\,dn_B,$$

and the interaction measure is

$$\Delta(T, \mu_B) = \epsilon - 3p = T\partial_T p + \mu_B \partial_{\mu_B} p - 4p.$$

Equivalently, at fixed $x = \mu_B/T$,

$$\frac{\Delta(T,\mu_B)}{T^4} = T\,\partial_T \left(\frac{p(T,xT)}{T^4}\right)_x .$$

At $\mu_B = 0$, if charge conjugation gives $n_B(T,0) = 0$ and $\partial_T \epsilon(T,0) > 0$, the adiabatic sound speed determined by the pressure is

$$c_s^2(T) = \frac{dp(T,0)/dT}{d\epsilon(T,0)/dT} = \frac{\partial_T p(T,0)}{T\,\partial_T^2 p(T,0)}.$$

These formulae are definitions and thermodynamic consequences of the pressure; they are not model assumptions about quasiparticles.

The first identity is the definition of s and n_B as derivatives of the limiting pressure. Differentiating $\epsilon = -p + Ts + \mu_B n_B$ gives

$$d\epsilon = -dp + T\,ds + s\,dT + \mu_B\,dn_B + n_B\,d\mu_B.$$

Substituting $dp = s\,dT + n_B\,d\mu_B$ leaves $d\epsilon = T\,ds + \mu_B\,dn_B$. The displayed formula for Δ follows by substituting $s = \partial_T p$ and $n_B = \partial_{\mu_B} p$ into $\epsilon - 3p$.

For the fixed-x identity, write $\mu_B = xT$. The chain rule gives

$$T\,\partial_T \left(\frac{p(T,xT)}{T^4}\right)_x = \frac{T\partial_T p + \mu_B \partial_{\mu_B} p - 4p}{T^4}.$$

At $\mu_B = 0$, charge conjugation removes the baryon-density term. Along that one-parameter equilibrium curve,

$$\frac{d\epsilon}{dT} = T\,\frac{ds}{dT} = T\,\partial_T^2 p(T,0), \qquad \frac{dp}{dT} = \partial_T p(T,0),$$

which gives the sound-speed formula whenever the denominator is nonzero.

Example 135.21 (Conformal Stefan–Boltzmann benchmark)**.** If

$$p(T,\mu_B) = aT^4 + b\mu_B^2 T^2 + c\mu_B^4$$

with a, b, c independent of T and μ_B, then p is homogeneous of degree four, so Euler's identity gives

$$T\partial_T p + \mu_B \partial_{\mu_B} p = 4p, \qquad \Delta = 0.$$

At $\mu_B = 0$, one obtains $\epsilon = 3aT^4$ and

$$c_s^2 = \frac{4aT^3}{12aT^3} = \frac{1}{3}.$$

The free plasma is therefore a normalization benchmark for lattice and resummed calculations of Δ/T^4 and c_s^2; departures from it measure interactions, masses, running couplings, and nonperturbative thermal physics.

135.7 Hard Thermal Loops and the Linde Obstruction

Chapter 130 derived the one-loop Debye mass in the present trace convention:

$$m_D^2 = g^2T^2\left(\frac{C_A}{3} + \frac{1}{3}\sum_{\text{Dirac } f} T_{R_f}\right)$$

for QCD without scalars. For $SU(N_c)$ with $\mathrm{tr}_\square(T^aT^b) = \delta^{ab}$, this gives

$$C_A = 2N_c, \qquad T_\square = 1.$$

The scale gT is electric. It must be resummed in static longitudinal propagators before a weak-coupling expansion of many thermal quantities is well-defined.

Controlled Approximation 135.22 (Hard-thermal-loop regime)**.** The HTL approximation applies to external frequencies and momenta

$$\omega,\ |\mathbf{k}| = O(gT)$$

in a plasma whose internal loop momenta are hard, $p = O(T)$. The leading soft gauge-field effective action is the gauge-invariant functional whose second variation reproduces the transverse and longitudinal one-loop polarization tensors in this kinematic regime. A calculation outside this scale separation is not an HTL calculation.

Static HTL polarization and Debye mass. Let $a_0 = a_0^aT^a$ be a slowly varying canonically normalized temporal gauge potential, and normalize the generators by

$$\mathrm{tr}_R(T_R^aT_R^b) = T_R\,\delta^{ab}, \qquad f^{acd}f^{bcd} = C_A\,\delta^{ab}.$$

For a weakly coupled thermal plasma at $\mu_B = 0$, the static HTL color susceptibility is

$$\Pi_{00}^{ab}(0,\mathbf{k}\to 0) = \delta^{ab}m_D^2, \qquad m_D^2 = g^2T^2\left(\frac{C_A}{3} + \frac{1}{3}\sum_{\text{Dirac } f} T_{R_f}\right).$$

In the present $SU(N_c)$ convention $\mathrm{tr}_\square(T^aT^b) = \delta^{ab}$, one has

$$C_A = 2N_c, \qquad T_\square = 1,$$

and hence

$$m_D^2 = g^2T^2\left(\frac{2N_c}{3} + \frac{N_f}{3}\right).$$

To leading static order the longitudinal electric propagator is screened as

$$D_{00}^{ab}(0,\mathbf{k}) = \frac{\delta^{ab}}{\mathbf{k}^2 + m_D^2},$$

with the understanding that this formula is the static electric HTL limit, not a gauge-invariant definition of a particle pole.

The statement is a color-susceptibility calculation for hard particles. Let a hard excitation of momentum $p = |\mathbf{p}|$ and color weight w couple to the soft potential by the energy shift $-ga_0^a w^a$. Linearizing its distribution gives

$$\delta n(p) = ga_0^b w^b \, [-n'(p)].$$

Multiplying by the color charge gw^a, summing over weights, and integrating over hard momenta gives the induced color density. Therefore a species in representation R contributes the positive susceptibility

$$g^2 \, d_{\text{spin/particle}} \, \mathrm{tr}_R(T_R^a T_R^b) \int \frac{d^3p}{(2\pi)^3} \, [-n'(p)].$$

For gluons the representation is the adjoint and there are two transverse hard polarizations. For a massless Dirac fermion there are two spin states and independent particle and antiparticle excitations; the antiparticle charge is opposite, but the susceptibility is quadratic in the charge, so the total degeneracy factor is 4.

The required thermal integrals are elementary but convention-sensitive:

$$I_B = \int \frac{d^3p}{(2\pi)^3} \, [-n_B'(p)] = \frac{1}{\pi^2} \int_0^\infty dp \, p \, n_B(p) = \frac{T^2}{6},$$

$$I_F = \int \frac{d^3p}{(2\pi)^3} \, [-n_F'(p)] = \frac{1}{\pi^2} \int_0^\infty dp \, p \, n_F(p) = \frac{T^2}{12}.$$

The integration by parts has no boundary term: at $p = \infty$ the distributions decay exponentially, while at $p = 0$ one has $p^2 n_B(p) \to 0$ and $p^2 n_F(p) \to 0$; moreover $p^2[-n_B'(p)]$ and $p^2[-n_F'(p)]$ are locally integrable. The last equalities use

$$\int_0^\infty \frac{x \, dx}{e^x - 1} = \frac{\pi^2}{6}, \qquad \int_0^\infty \frac{x \, dx}{e^x + 1} = \frac{\pi^2}{12}.$$

Thus

$$m_D^2 = g^2 \big(2C_A I_B + 4 \sum_f T_{R_f} I_F \big) = g^2 T^2 \left(\frac{C_A}{3} + \frac{1}{3} \sum_f T_{R_f} \right).$$

The resummed static longitudinal propagator is the geometric series obtained by inserting this static self-energy into the soft electric propagator. The derivation has used only the HTL separation between hard loop momentum and soft external field; it does not determine the magnetic sector.

Definition 135.23 (Retarded HTL induced-current kernel)**.** Work in the rest frame of the equilibrium state, with mostly-plus Lorentzian metric and $K = (\omega, \mathbf{k})$. Let

$$v^\mu = (1, \mathbf{v}), \qquad |\mathbf{v}| = 1, \qquad \int_{\mathbf{v}} (\cdots) = \int_{S^2} \frac{\mathrm{d}\Omega_{\mathbf{v}}}{4\pi} (\cdots).$$

The retarded HTL induced-current kernel is the distribution $\mathcal{K}_R^{\mu\nu,ab}(\omega, \mathbf{k})$ defined by

$$j_a^\mu(\omega, \mathbf{k}) = \mathcal{K}_R^{\mu\nu,ab}(\omega, \mathbf{k}) A_\nu^b(\omega, \mathbf{k}),$$

where

$$\mathcal{K}_R^{\mu\nu,ab}(\omega, \mathbf{k}) = \delta^{ab} m_D^2 \left[\delta_0^\mu \delta_0^\nu - \omega \int_{\mathbf{v}} \frac{v^\mu v^\nu}{\omega - \mathbf{v} \cdot \mathbf{k} + i0} \right].$$

The $+i0$ prescription is part of the definition: it is the retarded boundary condition for the collisionless response of hard particles. This kernel is normalized so that

$$\mathcal{K}_R^{00,ab}(0,\mathbf{k}) = \delta^{ab} m_D^2,$$

matching the static HTL calculation above.

Gauge-covariant HTL equations and transversality. Let $W_a(x,\mathbf{v})$ be an adjoint auxiliary field on spacetime times the unit velocity sphere. The gauge-covariant linear-response equations

$$(v\cdot D)W(x,\mathbf{v}) = v^i F_{i0}(x), \qquad j^\mu(x) = m_D^2 \int_{\mathbf{v}} v^\mu W(x,\mathbf{v}),$$

where D is the adjoint covariant derivative, are covariantly conserved:

$$D_\mu j^\mu = 0.$$

Linearizing around $A_\mu = 0$ and imposing the retarded prescription gives the kernel of Definition 135.23. Consequently, in momentum space,

$$K_\mu \mathcal{K}_R^{\mu\nu,ab}(K) = 0, \qquad K_\mu = (-\omega, \mathbf{k}),$$

so the HTL response is transverse in the mostly-plus convention.

The verification is explicit. The auxiliary field W transforms in the adjoint representation. Applying the covariant divergence to the current gives

$$D_\mu j^\mu = m_D^2 \int_{\mathbf{v}} (v\cdot D)W = m_D^2 \int_{\mathbf{v}} v^i F_{i0}.$$

Since F_{i0} is independent of $\mathbf{v}$ and $\int_{\mathbf{v}} v^i = 0$, the right side vanishes. This proves covariant current conservation before choosing a gauge.

For the linearized kernel, take fields proportional to $\exp(-i\omega t + i\mathbf{k}\cdot\mathbf{x})$. The retarded solution of $(\partial_t + \mathbf{v}\cdot\nabla)W = v^i F_{i0}$ is

$$W(K,\mathbf{v}) = -\frac{\mathbf{v}\cdot\mathbf{k}\, A_0(K) + \omega\, v^i A_i(K)}{\omega - \mathbf{v}\cdot\mathbf{k} + i0},$$

using $F_{i0} = \partial_i A_0 - \partial_0 A_i$. Inserting this solution into $j^\mu = m_D^2 \int_{\mathbf{v}} v^\mu W$ and using $\int_{\mathbf{v}} v^i = 0$ gives

$$j^\mu = m_D^2 \left[\delta_0^\mu \delta_0^\nu - \omega \int_{\mathbf{v}} \frac{v^\mu v^\nu}{\omega - \mathbf{v}\cdot\mathbf{k} + i0} \right] A_\nu,$$

which is precisely the stated kernel, with the color factor δ^{ab} restored.

Transversality is an algebraic consequence of the same formula. With $K_\mu = (-\omega, \mathbf{k})$,

$$K_\mu v^\mu = -(\omega - \mathbf{v}\cdot\mathbf{k}).$$

Therefore

$$K_\mu \mathcal{K}_R^{\mu\nu} = m_D^2 \left[-\omega\, \delta_0^\nu + \omega \int_{\mathbf{v}} v^\nu \right].$$

The integral of v^0 is 1, and the integral of v^i is 0. Thus the expression vanishes for $\nu = 0$ and for each spatial $\nu = i$.

Longitudinal HTL response and Landau cut. For $k = |\mathbf{k}| > 0$,

$$\mathcal{K}_R^{00,ab}(\omega, \mathbf{k}) = \delta^{ab} m_D^2 \left[1 - \frac{\omega}{2k} \log \frac{\omega + k + i0}{\omega - k + i0}\right].$$

Equivalently,

$$\int_{\mathbf{v}} \frac{1}{\omega - \mathbf{v} \cdot \mathbf{k} + i0} = \frac{1}{2k} \log \frac{\omega + k + i0}{\omega - k + i0}.$$

For $|\omega| < k$, the logarithm has an imaginary part. This is the Landau cut of the retarded thermal response, caused by hard particles whose velocity satisfies $\omega = \mathbf{v} \cdot \mathbf{k}$.

Choose the z-axis along $\mathbf{k}$ and write $\mu = \cos\theta$. Then

$$\int_{\mathbf{v}} \frac{1}{\omega - \mathbf{v} \cdot \mathbf{k} + i0} = \frac{1}{2} \int_{-1}^{1} \frac{\mathrm{d}\mu}{\omega - k\mu + i0}.$$

The antiderivative is $-k^{-1} \log(\omega - k\mu + i0)$. Evaluating it at $\mu = 1$ and $\mu = -1$ gives the displayed logarithm. Substitution in Definition 135.23 gives $\mathcal{K}_R^{00}$.

The distribution identity

$$\frac{1}{\omega - k\mu + i0} = \mathrm{PV}\, \frac{1}{\omega - k\mu} - i\pi\delta(\omega - k\mu)$$

shows that an imaginary part is present exactly when the equation $\mu = \omega/k$ has a solution in $[-1, 1]$, namely when $|\omega| < k$. This proves the Landau-cut statement and identifies its kinematic origin.

Remark 135.24 (Magnetic sector at g^6T^4). In four-dimensional high-temperature Yang–Mills theory, the static magnetic sector has three-dimensional coupling

$$g_3^2 = g^2 T.$$

If the magnetic sector develops a mass gap of order g_3^2, its contribution to the four-dimensional pressure begins at order

$$T(g_3^2)^3 = g^6 T^4.$$

The coefficient of this contribution is not determined by ordinary perturbation theory around the massless magnetic theory.

After the nonzero Matsubara modes and the electric scale have been matched into a three-dimensional static effective theory, the remaining magnetic Yang–Mills action has the form

$$\frac{1}{4g_3^2} \int d^3x\ \mathrm{tr}(F_{ij} F_{ij}).$$

In three dimensions g_3^2 has mass dimension one. A vacuum free-energy density of the magnetic theory has mass dimension three. If no scale other than g_3^2 remains in the confining magnetic sector, dimensional analysis forces the nonperturbative contribution to be $c(g_3^2)^3$, where c is a dimensionless number defined by the three-dimensional Yang–Mills theory. The four-dimensional pressure is T times the static three-dimensional free-energy density, so the order is g^6T^4.

Perturbation theory cannot determine c, because it expands around a massless three-dimensional gauge theory whose infrared behavior is precisely the magnetic theory being asked to generate the scale. This dimensional argument does not prove the magnetic mass gap; it identifies the order at which the nonperturbative magnetic input first enters the high-temperature expansion.

135.8 Transport and Quark–Gluon Plasma Observables

The quark–gluon plasma is not defined by the presence of free quarks and gluons. In this monograph it is the thermal regime of QCD in which the dominant excitations and response functions are better organized by color degrees of freedom, screening, and hydrodynamic transport than by a dilute gas of hadrons. The precise observables are correlation functions.

The shear viscosity, bulk viscosity, and electrical conductivity are defined by the Kubo limits of Chapter 127. For example, with T_{xy} a spatial off-diagonal stress tensor component,

$$\eta = \lim_{\omega \downarrow 0} \frac{1}{2\omega}\, \rho_{T_{xy}T_{xy}}(\omega, \mathbf{0})$$

when the limit exists after the thermodynamic limit. The definition is nonperturbative; a kinetic or holographic computation is an approximation to this spectral limit, not the definition of the coefficient.

Controlled Approximation 135.25 (QCD hydrodynamic response window)**.** Fix a thermal QCD phase at (T, μ_B) for which the continuum real-time retarded kernels of the gauge-invariant stress tensor and global currents exist. A claim that QCD is described by first-order hydrodynamics in a specified channel consists of more than the Kubo limits. It requires a compact low-frequency and long-wavelength window W_ϵ, a subtraction of declared local contact terms, and a comparison

$$K^R_{\mathrm{QCD}}(\omega, \mathbf{k}) = K^R_{\mathrm{hydro}}(\omega, \mathbf{k}; \mathcal{T}_{\mathrm{QCD}}) + P_{\mathrm{loc}}(\omega, \mathbf{k}) + R_\epsilon(\omega, \mathbf{k}), \tag{135.39}$$

where $\mathcal{T}_{\mathrm{QCD}}$ is the QCD transport datum

$$\mathcal{T}_{\mathrm{QCD}} = \big(p(T, \mu_B),\ \chi_{ab}(T, \mu_B),\ \eta(T, \mu_B),\ \zeta(T, \mu_B),\ \Sigma_{ab}(T, \mu_B), \ldots\big)$$

extracted from thermodynamic derivatives and Kubo spectral slopes. Here χ_{ab} and Σ_{ab} are the susceptibility and conductivity matrices for the conserved charge sector after the stress-tensor variables have been projected into the chosen hydrodynamic frame. The remainder statement must be in the same scaling topology as the hydrodynamic mode: $\omega \sim k^2$ for shear and diffusion, and $\omega \sim k$ for sound. Thus the shear channel uses

$$D_\eta = \frac{\eta}{w}, \qquad \omega_{\mathrm{shear}} = -\mathrm{i} D_\eta k^2 + O(k^4), \qquad w = \epsilon + p,$$

a decoupled baryon diffusion channel uses

$$D_B = \frac{\sigma_B}{\chi_B}, \qquad \omega_{\mathrm{diff}} = -\mathrm{i} D_B k^2 + O(k^4),$$

while at nonzero density or with several conserved charges D_B is replaced by the relevant eigenvalue of $\Sigma\,\chi^{-1}$ after the sound variables have been separated, and the sound channel uses

$$\omega_{\mathrm{sound},\pm} = \pm c_s k - \frac{\mathrm{i}}{2}\Gamma_s k^2 + O(k^3), \qquad \Gamma_s = \frac{\zeta + 2\eta(d-1)/d}{w}.$$

Here d is the number of spatial dimensions and c_s^2 is computed from the QCD pressure with the stated thermodynamic constraint. A bound such as

$$\sup_{(\omega, \mathbf{k}) \in W_\epsilon} |R_\epsilon(\omega, \mathbf{k})| \le C\, \epsilon^{N+1} \tag{135.40}$$

is the additional microscopic input. It asserts that nonconserved QCD excitations and long-time tails do not produce extra singular terms in the window being approximated. Without a statement of this form, Kubo formulae define transport coefficients, but they do not by themselves derive hydrodynamics from QCD.

Proposition 135.26 (Momentum-projected baryon diffusion current). *In a translationally invariant QCD phase at nonzero baryon density, the raw baryon current is not itself the diffusive current. Let*

$$n_B = \frac{\partial p}{\partial \mu_B}, \qquad w = \epsilon + p, \qquad P^i = T^{0i},$$

and let J_B^i be the spatial baryon-number current. In a homogeneous fluid state boosted by a small velocity δu^i,

$$\delta\langle P^i\rangle = w\,\delta u^i, \qquad \delta\langle J_B^i\rangle = n_B\,\delta u^i.$$

Thus the static susceptibility Gram matrix in the vector sector contains

$$\chi_{P^iP^j} = w\,\delta^{ij}, \qquad \chi_{J_B^iP^j} = n_B\,\delta^{ij}.$$

The momentum-orthogonal baryon current is therefore

$$J^i_{\rm inc} = J^i_B - \frac{n_B}{w}T^{0i}. \tag{135.41}$$

Writing $\chi_{J_B^iJ_B^j} = \chi_{JJ}\delta^{ij}$, it obeys

$$\chi_{J^i_{\rm inc}P^j} = 0, \qquad \chi_{J^i_{\rm inc}J^j_{\rm inc}} = \left(\chi_{JJ} - \frac{n_B^2}{w}\right)\delta^{ij}. \tag{135.42}$$

Consequently the baryon conductivity entering the diffusive part of $\mathcal{T}_{\rm QCD}$ is the regular Kubo slope of $J_{\rm inc}$,

$$\Sigma_B^{\rm inc} = \lim_{\omega\downarrow 0}\frac{1}{2\omega}\rho^{\rm reg}_{J^i_{\rm inc}J^i_{\rm inc}}(\omega,\mathbf{0}), \tag{135.43}$$

with the Drude sector separated as in (127.18). At $n_B = 0$, $J_{\rm inc} = J_B$. At $n_B \neq 0$, the raw J_BJ_B response contains a convective contribution from the exactly conserved momentum. If translations are weakly broken this contribution becomes a narrow momentum-relaxation peak; that peak is a property of the relaxation mechanism, not the intrinsic QCD baryon-diffusion coefficient.

Proof. The ideal constitutive relations in the local rest frame are

$$T^{0i} = w\,u^i + O(\partial), \qquad J_B^i = n_B\,u^i + O(\partial).$$

Differentiating the expectation values with respect to a uniform velocity source gives the displayed static overlaps. Equation (135.41) is then ordinary Gram–Schmidt projection in the static susceptibility inner product, so

$$\chi_{J_{\rm inc}P} = \chi_{J_BP} - \frac{n_B}{w}\chi_{PP} = n_B - \frac{n_B}{w}w = 0.$$

The second identity in (135.42) is the same projection applied to the norm of the current. Positivity is the Schur complement positivity of the static susceptibility matrix. The last statement is then the Drude separation of (127.18): any current with a nonzero static overlap with a conserved momentum has a singular zero-frequency sector, while the projected current is the object whose regular spectral slope can be matched to the diffusive hydrodynamic coefficient. □

Remark 135.27 (The η/s lower-bound claim). The value $1/(4\pi)$ is an important result in a class of strongly coupled large-N theories with classical gravity duals. It is not a theorem of QCD. When this number is used as a comparison scale in QCD phenomenology, the statement is a quantitative comparison of transport data or models, not a rigorous lower bound derived from the QCD axioms.

135.9 Finite Baryon Density and the Sign Problem

At real μ_B, the Euclidean Dirac operator is no longer anti-Hermitian up to the mass term. In continuum notation the temporal derivative is shifted by $-\mu_q$, and on the lattice forward and backward temporal hops are weighted asymmetrically. The determinant need not be real and nonnegative.

At $\mu_B = 0$, a vectorlike pair of equal-mass quark flavors gives a nonnegative Euclidean determinant. At real $\mu_B \neq 0$, the same positivity argument fails because the Dirac operator loses the γ_5-Hermiticity relation that pairs eigenvalues into complex conjugates at fixed gauge field. At $\mu_B = 0$, the lattice or continuum Euclidean Dirac operator obeys

$$\gamma_5(D_E + m)\gamma_5 = (D_E + m)^\dagger$$

for real mass m. Therefore $\det(D_E + m)$ is real, and for two degenerate flavors the product is $|\det(D_E + m)|^2 \geq 0$. This is the measure-positivity input behind importance sampling.

A real quark chemical potential changes the temporal part of the operator by a term that is Hermitian rather than anti-Hermitian in Euclidean signature. Equivalently, the lattice temporal hopping factors $e^{a\mu_q}$ and $e^{-a\mu_q}$ are not related by the same unitary conjugation that holds at $\mu_q = 0$. The displayed γ_5-Hermiticity relation is replaced by a relation connecting μ_q to $-\mu_q$. At fixed real μ_q the determinant is generally complex, so it cannot be used directly as a positive probability weight.

135.9.1 Imaginary Chemical Potential and Roberge–Weiss Periodicity

Imaginary chemical potential is a different, exactly defined thermal boundary condition from real finite density. It is useful because it restores positivity in vectorlike pairs and has an exact center periodicity at finite regulator.

Definition 135.28 (Imaginary quark and baryon chemical-potential angles)**.** Let

$$\mu_q = iT\theta_q, \qquad \mu_B = iT\theta_B, \qquad \theta_B = N_c\theta_q.$$

Equivalently, the fundamental quark field obeys the Euclidean thermal boundary condition

$$q(\beta, \mathbf{x}) = -e^{i\theta_q} q(0, \mathbf{x}).$$

For a finite lattice regulator, write the corresponding partition function as $Z_{a,V}(\theta_q)$. The regulator is assumed to use compact $SU(N_c)$ links, a gauge action invariant under center-twisted temporal gauge transformations, and a gauge-covariant Dirac operator whose imaginary chemical potential is this boundary twist.

Lemma 135.29 (Finite-regulator Roberge–Weiss periodicity)**.** *At finite regulator and finite spatial volume,*

$$Z_{a,V}\left(\theta_q + \frac{2\pi k}{N_c}\right) = Z_{a,V}(\theta_q), \qquad k \in \mathbb{Z}.$$

Equivalently, as a function of the baryon angle,

$$Z_{a,V}(\theta_B + 2\pi k) = Z_{a,V}(\theta_B).$$

For a vectorlike pair of equal-mass flavors, the Euclidean determinant at real θ_q is nonnegative. Therefore imaginary-μ_B simulations probe a positive finite-regulator measure, while analytic continuation to real μ_B remains a separate limiting and analytic-control problem.

Proof. Let $z_k = e^{2\pi ik/N_c} \in Z(SU(N_c))$. A center-twisted temporal gauge transformation is a gauge transformation $g(\tau, \mathbf{x})$ satisfying

$$g(\beta, \mathbf{x}) = z_k\, g(0, \mathbf{x}).$$

Gauge fields in the adjoint representation are periodic under this transformation, because the center acts trivially on adjoint variables. The compact-link Haar measure and gauge action are invariant.

For a fundamental quark field with $q(\beta, \mathbf{x}) = -e^{i\theta_q} q(0, \mathbf{x})$, the transformed field has boundary condition

$$q'(\beta, \mathbf{x}) = g(\beta, \mathbf{x})q(\beta, \mathbf{x}) = -e^{i\theta_q} z_k\, g(0, \mathbf{x})q(0, \mathbf{x}) = -e^{i(\theta_q+2\pi k/N_c)} q'(0, \mathbf{x}).$$

Thus the change of integration variables by the center-twisted gauge transformation maps the finite-regulator path integral at θ_q to the one at $\theta_q + 2\pi k/N_c$. Since the transformation is invertible and leaves the measure and gauge action invariant, the two partition functions are equal. Multiplying the quark-angle period by N_c gives the baryon-angle period 2π.

For $\mu_q = iT\theta_q$, the Euclidean chemical-potential term is anti-Hermitian in the same sense as the ordinary Euclidean Dirac operator. The massive Dirac operator obeys the γ_5-Hermiticity relation at fixed gauge field,

$$\gamma_5 D_{\theta_q,m} \gamma_5 = D^\dagger_{\theta_q,m}.$$

Hence its determinant is real, and a pair of equal-mass Dirac flavors gives $|\det D_{\theta_q,m}|^2 \geq 0$. This proves the positivity statement for vectorlike pairs. None of these equalities implies analyticity at real μ_B; analytic continuation is controlled only inside zero-free domains of the complex fugacity or chemical-potential plane. □

Remark 135.30 (Roberge–Weiss lines)*.* The finite-volume function $Z_{a,V}(\theta_q)$ is analytic in the imaginary fugacity. A Roberge–Weiss transition is therefore a thermodynamic-limit phenomenon: in high-temperature $SU(N_c)$ QCD, different center-related holonomy sectors may exchange dominance at

$$\theta_q = \frac{(2r+1)\pi}{N_c}, \qquad r \in \mathbb{Z}.$$

At finite volume this appears as Lee–Yang zero organization, not as a literal singularity of $Z_{a,V}$. The statement is about the imaginary chemical-potential problem and does not by itself locate a real-μ_B critical endpoint.

Consequently, the conjectural critical endpoint and the first-order line in the (T, μ_B) plane are not presently consequences of direct first-principles continuum control at large real μ_B. Taylor expansions around $\mu_B = 0$, imaginary-μ_B continuation, reweighting, and effective models are valuable but must carry their own domains of validity.

135.10 Baryon Susceptibilities and Critical-Point Diagnostics

At $\mu_B = 0$ the Euclidean measure of vectorlike QCD is positive in the standard lattice discretizations, so derivatives with respect to μ_B give first-principles thermal observables even though direct importance sampling at real $\mu_B \neq 0$ is obstructed. The natural dimensionless source is

$$x = \frac{\mu_B}{T} = \beta\mu_B.$$

At finite regulator and finite volume,

$$Z_{a,V}(x) = \mathrm{Tr}_{\mathcal{H}_{a,V}} \exp\{-\beta H_{a,V} + xB_{a,V}\}. \tag{135.44}$$

Here $B_{a,V}$ is the finite-regulator baryon charge, normalized so that a baryon has charge 1, and it is assumed to be an exactly conserved self-adjoint charge of the regulated theory:

$$[H_{a,V}, B_{a,V}] = 0.$$

Definition 135.31 (Baryon susceptibilities)**.** The finite-regulator baryon cumulants and susceptibilities at $x = 0$ are

$$\kappa^B_{n,a,V}(T) = \frac{\partial^n}{\partial x^n} \log Z_{a,V}(x)\bigg|_{x=0}, \qquad \chi^B_{n,a,V}(T) = \frac{1}{VT^3}\kappa^B_{n,a,V}(T). \tag{135.45}$$

The continuum thermodynamic susceptibility $\chi^B_n(T)$ is the limit of $\chi^B_{n,a,V}(T)$ along the chosen continuum trajectory, when the limit exists.

Baryon susceptibilities as baryon-number cumulants. At finite regulator and finite volume,

$$\chi^B_{1,a,V} = \frac{\langle B\rangle}{VT^3}, \qquad \chi^B_{2,a,V} = \frac{\langle B^2\rangle - \langle B\rangle^2}{VT^3}, \tag{135.46}$$

and

$$\begin{aligned}\chi^B_{4,a,V} = \frac{1}{VT^3}\big(\langle B^4\rangle - 4\langle B^3\rangle\langle B\rangle - 3\langle B^2\rangle^2 \\ + 12\langle B^2\rangle\langle B\rangle^2 - 6\langle B\rangle^4\big).\end{aligned} \tag{135.47}$$

If charge conjugation is an exact symmetry at $x = 0$ and sends $B \mapsto -B$, then all odd finite-volume cumulants vanish.

Because $B_{a,V}$ is conserved, $H_{a,V}$ and $B_{a,V}$ can be simultaneously decomposed by the spectral theorem, and differentiating (135.44) with respect to x is the ordinary differentiation of the finite-regulator moment generating function of the baryon charge in the thermal state. For any observable Y commuting with B, in particular for $Y = B^k$, one has

$$\frac{\partial}{\partial x}\langle Y\rangle_x = \langle YB\rangle_x - \langle Y\rangle_x\langle B\rangle_x. \tag{135.48}$$

Applying this identity to $Y = 1, B, B^2, B^3$, or equivalently differentiating $\log Z(x)$, gives the displayed cumulants. If charge conjugation is an exact symmetry at $x = 0$, the spectral measure of B in the thermal state is invariant under $B \mapsto -B$. The moment generating function then satisfies

$$Z(x) = Z(-x),$$

so $\log Z(x)$ is even near the origin and its odd derivatives vanish.

Definition 135.32 (Taylor expansion and critical diagnostic)**.** When the thermodynamic continuum pressure is analytic at $x = 0$, write

$$\frac{p(T,x)}{T^4} = \sum_{n=0}^{\infty} \frac{\chi_n^B(T)}{n!} x^n \tag{135.49}$$

inside its disk of convergence. The Cauchy–Hadamard radius of this germ is

$$R(T) = \left[\limsup_{n\to\infty} \left|\frac{\chi_n^B(T)}{n!}\right|^{1/n}\right]^{-1}. \tag{135.50}$$

Remark 135.33 (What a susceptibility radius establishes)*.* Assume that $p(T,x)/T^4$ is the germ of a holomorphic function of complex x at the origin and has Taylor coefficients $\chi_n^B(T)/n!$. Then $R(T)$ in (135.50) is the radius of convergence of the Taylor series. The pressure is analytic for $|x| < R(T)$. If independent input shows that the nearest limiting singularity is located at a positive real point x_c, then $x_c = R(T)$ and this singularity is the critical-point candidate seen by the Taylor expansion. Without that location input, the nearest singularity may lie at complex x and the radius alone does not identify a real critical endpoint.

The first two assertions are the Cauchy–Hadamard theorem applied to the power series (135.49). A real critical point at positive x_c must be a singularity of the same analytic germ after the thermodynamic and continuum limits have been taken. If it is independently known to be the nearest singularity to the origin, its distance from the origin is the convergence radius. The final statement is the complementary case: a power series records distance to the nearest complex obstruction, whereas the physical critical endpoint is a statement about the real (T, μ_B) plane and the limiting Gibbs/KMS states.

Remark 135.34 (Ratio estimators)*.* For a charge-conjugation invariant theory the odd susceptibilities vanish at $x = 0$, and one often forms even-order estimators

$$R_{2n}(T) = \sqrt{\frac{(2n+2)(2n+1)\chi_{2n}^B(T)}{\chi_{2n+2}^B(T)}}. \tag{135.51}$$

If the even Taylor coefficients are eventually positive and have a single dominant singular scale, these estimators can converge to the radius. The positivity and single-scale hypotheses are part of the diagnostic; they are not consequences of the cumulant definitions alone.

135.11 High-Density Effective Theory and Fermi-Surface Instability

Real baryon density is difficult nonperturbatively, but asymptotically large quark chemical potential has a separate weak-coupling control parameter. The running coupling is evaluated at a scale of order μ_q, while the low-energy excitations live in thin shells around the Fermi surface. The resulting effective theory is not an expansion around empty space; it is an expansion around a filled Fermi sea with soft residual momenta.

Controlled Approximation 135.35 (High-density effective-theory regime)**.** Let

$$\mu_q = \frac{\mu_B}{N_c}.$$

The high-density effective theory applies to modes whose residual energy and momentum are bounded by a scale $\Lambda_{\rm H}$ satisfying

$$\Delta,\ T,\ |\ell^\mu|,\ \Lambda_{\rm QCD} \ll \Lambda_{\rm H} \ll \mu_q,$$

where Δ denotes any superconducting gap scale that is being resolved. Gauge fields retained in the effective theory have momenta much smaller than μ_q. Modes far from the Fermi surface and antiparticle branches are integrated out into local operators and into hard-dense-loop response terms.

Definition 135.36 (Patch decomposition near the quark Fermi surface)**.** For each unit vector $\mathbf{v} \in S^2$, write a quark momentum near the Fermi surface as

$$\mathbf{p} = \mu_q \mathbf{v} + \boldsymbol{\ell}, \qquad |\boldsymbol{\ell}| \le \Lambda_{\rm H}.$$

Decompose the residual momentum into longitudinal and transverse parts,

$$\ell_\parallel = \mathbf{v}\cdot\boldsymbol{\ell}, \qquad \boldsymbol{\ell}_\perp = \boldsymbol{\ell} - \mathbf{v}\,\ell_\parallel.$$

The patch field $\psi_{\mathbf{v}}$ is the positive-energy quark field whose momenta lie in this shell and whose group velocity is near $\mathbf{v}$. More invariantly, it is obtained by applying the positive spectral projector of the massless one-particle Dirac Hamiltonian in the direction $\mathbf{v}$, multiplying by the phase $e^{-i\mu_q\mathbf{v}\cdot\mathbf{x}}$, and then restricting residual momenta to the shell. A finite regulator is obtained by choosing an angular partition of S^2 whose cells have diameter $O(\Lambda_{\rm H}/\mu_q)$; every occurrence of $\sum_{\mathbf{v}}$ below denotes this regulated patch sum, with the continuum angular integral recovered by refining the partition inside the scale separation $\Lambda_{\rm H} \ll \mu_q$.

Fermi-surface measure and density of states. For functions supported in the shell $|\boldsymbol{\ell}| \le \Lambda_{\rm H} \ll \mu_q$,

$$\int \frac{d^3p}{(2\pi)^3} f(\mathbf{p}) = \frac{\mu_q^2}{(2\pi)^3}\int_{S^2} d\Omega_{\mathbf{v}} \int d\ell_\parallel\, f(\mu_q\mathbf{v} + \ell_\parallel \mathbf{v}) + O\!\left(\frac{\Lambda_{\rm H}}{\mu_q}\right)$$

after angular patches are refined so that transverse residual momenta are counted once. The density of states at the Fermi surface, per spin state and per internal state, is therefore

$$N_{\text{one spin}}(0) = \frac{\mu_q^2}{2\pi^2}.$$

Including the two spin states of a massless Dirac quark gives

$$N_{\rm Dirac}(0) = \frac{\mu_q^2}{\pi^2}.$$

Write $\mathbf{p} = (\mu_q + \ell_\parallel)\mathbf{v} + \boldsymbol{\ell}_\perp$ inside a patch. To leading order in $\Lambda_{\mathrm{H}}/\mu_q$, the radial Jacobian is

$$p^2 dp\, d\Omega_{\mathbf{v}} = (\mu_q + \ell_\parallel)^2 d\ell_\parallel\, d\Omega_{\mathbf{v}} = \mu_q^2 \left(1 + O\left(\frac{\Lambda_{\mathrm{H}}}{\mu_q}\right)\right) d\ell_\parallel\, d\Omega_{\mathbf{v}}.$$

The shell decomposition is a partition of momentum space near the Fermi surface after a choice of angular patches; the patch refinement only avoids double counting transverse residual momenta. Integrating the leading measure over S^2 gives

$$\frac{\mu_q^2}{(2\pi)^3}\, 4\pi = \frac{\mu_q^2}{2\pi^2}$$

per spin and internal state. A massless Dirac quark has two spin states on the positive-energy branch, giving the final formula.

Tree-level high-density quark action. For a massless quark near the Fermi surface, the positive-energy dispersion relative to the chemical potential is

$$|\mu_q \mathbf{v} + \boldsymbol{\ell}| - \mu_q = \ell_\parallel + \frac{\boldsymbol{\ell}_\perp^2}{2\mu_q} + O\left(\frac{|\boldsymbol{\ell}|^3}{\mu_q^2}\right).$$

Consequently the tree-level inverse propagator in a patch is

$$G_{\mathbf{v}}^{-1}(\ell) = \ell_0 - \ell_\parallel - \frac{\boldsymbol{\ell}_\perp^2}{2\mu_q} + O\left(\frac{|\boldsymbol{\ell}|^3}{\mu_q^2}\right),$$

with soft gauge fields introduced by replacing residual derivatives by covariant derivatives. The leading local patch Lagrangian has the form

$$\mathcal{L}^{(0)}_{\mathrm{HDET}} = \sum_{\mathbf{v}} \psi_{\mathbf{v}}^\dagger \left(iD_t + i\mathbf{v}\cdot\mathbf{D} + \frac{D_\perp^2}{2\mu_q} + \cdots\right)\psi_{\mathbf{v}},$$

where $D_\perp^2 = D_i D_i - (\mathbf{v}\cdot\mathbf{D})^2$, and the ellipsis contains higher powers of residual momentum divided by μ_q, quark masses, and local interactions generated by integrating out hard modes.

The expansion follows from

$$|\mu_q \mathbf{v} + \boldsymbol{\ell}| = \mu_q \sqrt{1 + \frac{2\ell_\parallel}{\mu_q} + \frac{\ell_\parallel^2 + \boldsymbol{\ell}_\perp^2}{\mu_q^2}}.$$

Taylor expansion of the square root gives

$$\mu_q + \ell_\parallel + \frac{\boldsymbol{\ell}_\perp^2}{2\mu_q} + O(|\boldsymbol{\ell}|^3/\mu_q^2).$$

Subtracting the chemical potential gives the displayed energy relative to the filled Fermi sea. The inverse propagator is ℓ_0 minus this relative energy. In coordinate space, residual momentum acts as $-iD$; hence the terms $-\ell_\parallel$ and $-\boldsymbol{\ell}_\perp^2/(2\mu_q)$ become $i\mathbf{v}\cdot\mathbf{D}$ and $D_\perp^2/(2\mu_q)$, with the sign fixed by the displayed momentum-space inverse propagator.

Hard-dense-loop Debye mass at zero temperature. At $T = 0$, for N_f massless Dirac quarks with common quark chemical potential μ_q, the static dense-quark contribution to the electric screening mass is

$$m_{D,\mu}^2 = g^2 \sum_{\text{Dirac } f} T_{R_f} \frac{\mu_q^2}{\pi^2}.$$

For $SU(N_c)$ fundamentals in the monograph trace convention $\mathrm{tr}_\square(T^a T^b) = \delta^{ab}$, this becomes

$$m_{D,\mu}^2 = g^2 N_f \frac{\mu_q^2}{\pi^2}.$$

This is the one-line susceptibility calculation entering the HDL approximation. The static electric response is the color susceptibility of occupied quark states. At $T = 0$, differentiating the occupation number $\Theta(\mu_q - p)$ with respect to a color chemical potential localizes the response on the Fermi surface:

$$-\frac{\partial}{\partial p}\Theta(\mu_q - p) = \delta(p - \mu_q).$$

For one spin state in representation R,

$$\int \frac{d^3p}{(2\pi)^3} \delta(p - \mu_q) = \frac{\mu_q^2}{2\pi^2}.$$

The two positive-energy spin states of a Dirac quark give μ_q^2/π^2. Multiplying by the color trace $\mathrm{tr}_R(T_R^a T_R^b) = T_R \delta^{ab}$, by g^2, and by the number of flavors gives the displayed result. Antiquark states are far from the positive-μ_q Fermi surface and do not contribute to this leading $T = 0$ dense response.

135.12 Dense Non-Fermi-Liquid Scaling

The high-density effective theory also controls the quark two-point function near an unpaired Fermi surface. A local four-fermion interaction would give a regular Fermi-liquid expansion in the external energy. Dense QCD is more singular because static magnetic gluons are not Debye screened; their Landau-damped propagator produces a logarithmic derivative of the quark self-energy. This is a perturbative high-density statement, not a nonperturbative theorem about all finite-density QCD states.

Controlled Approximation 135.37 (HDL non-Fermi-liquid self-energy regime)**.** Work at $T = 0$, in an unpaired high-density patch of Definition 135.36, with

$$|p_0|,\ |\ell_\parallel| \ll \Lambda_{\mathrm{H}} \ll \mu_q, \qquad g(\mu_q) \ll 1.$$

Use the trace-delta color convention $\mathrm{tr}_\square(T^a T^b) = \delta^{ab}$, so that

$$C_F = T^a T^a = \frac{N_c^2 - 1}{N_c} \tag{135.52}$$

on a fundamental quark. Retain the HDL transverse propagator in the Landau-damped region

$$|\omega| \ll k_\perp \ll \mu_q, \qquad D_T(\omega, k_\perp) = \frac{1}{k_\perp^2 + \frac{\pi}{4} m_{D,\mu}^2 \frac{|\omega|}{k_\perp}} \left(1 + o(1)\right), \tag{135.53}$$

with $m_{D,\mu}$ given by the zero-temperature HDL susceptibility calculation above. Electric exchange, non-collinear transverse exchange, and hard matching contributions are kept only to the extent that they define the finite matching scale $\Lambda_{\rm NFL}$; they do not affect the coefficient of the logarithm derived below.

Definition 135.38 (Leading non-Fermi-liquid self-energy)**.** Let $\Sigma(ip_0, \ell_\parallel)$ be the one-particle-irreducible quark self-energy of a patch field $\psi_{\mathbf{v}}$, projected onto the positive-energy branch and evaluated in Euclidean frequency. The leading non-Fermi-liquid part is the nonanalytic term in

$$G_{\mathbf{v}}^{-1}(ip_0, \ell_\parallel) = ip_0 - \ell_\parallel - \Sigma(ip_0, \ell_\parallel)$$

which is proportional to $ip_0 \log |p_0|$ at $\ell_\parallel = 0$. Local analytic terms proportional to ip_0, $\ell_\parallel$, or higher powers of residual momentum are scheme dependent HDET counterterms and are not included in $\Sigma_{\rm NFL}$.

Leading logarithmic cold dense quark self-energy. In Controlled Approximation 135.37, the on-Fermi-surface Euclidean self-energy has the leading nonanalytic form

$$\Sigma_{\rm NFL}(ip_0, 0) = ip_0\, \lambda_{\rm NFL} \log \frac{\Lambda_{\rm NFL}}{|p_0|} + O(ip_0), \qquad \lambda_{\rm NFL} = \frac{g^2 C_F}{12\pi^2}. \tag{135.54}$$

For $SU(3)$ in the monograph trace convention,

$$C_F = \frac{8}{3}, \qquad \lambda_{\rm NFL} = \frac{2g^2}{9\pi^2}. \tag{135.55}$$

The one-loop patch self-energy from transverse exchange is obtained by contracting the patch quark current with the transverse HDL propagator. The color projection gives C_F. Choose the patch normal $\mathbf{v} = \mathbf{e}_3$. The leading transverse vertex is $gT^a v^i$, and the transverse projector gives

$$v^i \left(\delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2} \right) v^j = 1 + O\!\left(\frac{k_\parallel^2}{k_\perp^2} \right)$$

in the Landau-damped region after the internal quark pole restricts $|k_\parallel| \lesssim |\omega| \ll k_\perp$. Thus the singular part is independent of the detailed spinor wave functions, while analytic terms are absorbed into local HDET counterterms.

With the external residual momentum set to $\ell_\parallel = 0$, subtract $\Sigma(0,0)$ and integrate first over the normal residual momentum of the internal quark. For $p_0 > 0$, the residue theorem gives the reduced singular expression

$$\Sigma_{\rm sing}(ip_0, 0) - \Sigma_{\rm sing}(0,0) = i\, \frac{g^2 C_F}{4\pi^2} \int_0^{p_0} d\nu\, \mathcal{L}(\nu) + O(ip_0), \tag{135.56}$$

where ν is the Euclidean energy carried by the nearly on-shell internal quark and $\mathcal{L}(\nu)$ is the remaining transverse Landau-damped logarithm. The prefactor in (135.56) is the product of the color factor, the patch phase-space measure, the transverse-current projection, and the $2\pi i$ residue of the normal propagator. For $p_0 < 0$ the same formula holds with the integral oriented from 0 to p_0; this is why the final answer is proportional to ip_0.

The only nonanalytic part comes from the Landau-damped transverse region. For fixed ν, the HDL denominator in (135.53) cuts off the transverse integral at

$$k_{\perp,\min}^3 \asymp m_{D,\mu}^2|\nu|, \qquad k_{\perp,\max} \asymp \mu_q,$$

and therefore

$$\int_{k_{\perp,\min}}^{k_{\perp,\max}} \frac{dk_\perp}{k_\perp} = \frac{1}{3}\log\frac{\Omega_{\mathrm{NFL}}}{|\nu|} + O(1), \qquad \Omega_{\mathrm{NFL}} \asymp \frac{\mu_q^3}{m_{D,\mu}^2}. \tag{135.57}$$

The $O(1)$ part and the precise constant in Ω_{NFL} are absorbed into the matching scale Λ_{NFL}. Equivalently,

$$\mathcal{L}(\nu) = \frac{1}{3}\log\frac{\Lambda_{\mathrm{NFL}}}{|\nu|} + O(1). \tag{135.58}$$

Substitution into (135.56) gives

$$\Sigma_{\mathrm{NFL}}(ip_0, 0) = i\,\frac{g^2C_F}{4\pi^2}\int_0^{p_0} d\nu\, \left(\frac{1}{3}\log\frac{\Lambda_{\mathrm{NFL}}}{|\nu|}\right) + O(ip_0). \tag{135.59}$$

The endpoint singularity is integrable and

$$\int_0^{p_0} d\nu\, \log\frac{\Lambda_{\mathrm{NFL}}}{\nu} = p_0\log\frac{\Lambda_{\mathrm{NFL}}}{p_0} + p_0 \qquad (p_0 > 0).$$

The second term is analytic and is included in $O(ip_0)$. Odd continuation in p_0, written with $|p_0|$, gives

$$ip_0\,\frac{g^2C_F}{12\pi^2}\log\frac{\Lambda_{\mathrm{NFL}}}{|p_0|} + O(ip_0),$$

which is (135.54). The $SU(3)$ specialization follows from (135.52).

Remark 135.39 (Logarithmic vanishing of the patch residue). Under the assumptions of the leading logarithmic self-energy calculation above, the coefficient of ip_0 in the inverse patch propagator behaves as

$$Z^{-1}(p_0) = 1 + \lambda_{\mathrm{NFL}}\log\frac{\Lambda_{\mathrm{NFL}}}{|p_0|} + O(1), \tag{135.60}$$

and hence the perturbative patch residue satisfies

$$Z(p_0) \sim \left[\lambda_{\mathrm{NFL}}\log\frac{\Lambda_{\mathrm{NFL}}}{|p_0|}\right]^{-1} \longrightarrow 0 \qquad (p_0 \to 0) \tag{135.61}$$

within the unpaired high-density regime.

Insert (135.54) into the inverse propagator of Definition 135.38. Analytic wave-function renormalizations change the $O(1)$ part of (135.60) but cannot remove the logarithm. Inverting the coefficient of ip_0 gives (135.61). The statement is perturbative and patch-local: it identifies the logarithmic loss of a quasiparticle pole inside the unpaired high-density effective description, with pairing gaps, confinement-scale physics, and regulator matching held outside the displayed asymptotic regime.

Remark 135.40 (Relation to the superconducting gap). The logarithm in (135.54) is generated by the same Landau-damped magnetic gluons that produce the leading magnetic pairing kernel. It does not change the leading $1/g$ exponent in Proposition 135.51; it changes subleading matching data and prefactors. Thus the phrase "non-Fermi liquid" here has a precise meaning: the derivative of the inverse quark propagator has a logarithmic singularity in the unpaired high-density effective theory. It is not a claim that gauge-invariant observables in every dense phase have been nonperturbatively constructed.

BCS logarithm from the Fermi surface. Consider a specified antisymmetric two-quark channel of the high-density effective theory and approximate its attractive interaction by a local four-fermion coupling $G > 0$ inside a shell $|\xi| \le \Lambda_{\mathrm{H}}$, where $\xi = |\mathbf{p}| - \mu_q$. Let $N_{\mathrm{ch}}(0)$ be the density of states in that channel at the Fermi surface, including the relevant spin, color, and flavor multiplicity. The zero-temperature mean-field gap equation is

$$1 = G N_{\mathrm{ch}}(0) \int_0^{\Lambda_{\mathrm{H}}} \frac{d\xi}{\sqrt{\xi^2 + \Delta^2}}.$$

For $\Delta \ll \Lambda_{\mathrm{H}}$,

$$\int_0^{\Lambda_{\mathrm{H}}} \frac{d\xi}{\sqrt{\xi^2 + \Delta^2}} = \log \frac{2\Lambda_{\mathrm{H}}}{\Delta} + O\left(\frac{\Delta^2}{\Lambda_{\mathrm{H}}^2}\right),$$

and hence

$$\Delta = 2\Lambda_{\mathrm{H}} \exp\left[-\frac{1}{G N_{\mathrm{ch}}(0)}\right] \left(1 + O\left(\frac{\Delta^2}{\Lambda_{\mathrm{H}}^2}\right)\right).$$

In the paired mean-field Hamiltonian, the quasiparticle energy in the shell is

$$E(\xi) = \sqrt{\xi^2 + \Delta^2}.$$

The anomalous expectation value is proportional to $\Delta/(2E(\xi))$. Dividing the self-consistency equation by $\Delta \neq 0$ gives the displayed integral equation after the angular and internal sums are collected into $N_{\mathrm{ch}}(0)$. The integral is exactly

$$\int_0^{\Lambda_{\mathrm{H}}} \frac{d\xi}{\sqrt{\xi^2 + \Delta^2}} = \operatorname{arsinh} \frac{\Lambda_{\mathrm{H}}}{\Delta} = \log\left(\frac{\Lambda_{\mathrm{H}}}{\Delta} + \sqrt{1 + \frac{\Lambda_{\mathrm{H}}^2}{\Delta^2}}\right).$$

Expanding for $\Delta/\Lambda_{\mathrm{H}} \to 0$ gives $\log(2\Lambda_{\mathrm{H}}/\Delta) + O(\Delta^2/\Lambda_{\mathrm{H}}^2)$, and solving the gap equation gives the stated exponential scale.

Remark 135.41 (What is and is not contained in the BCS logarithm). The preceding calculation proves the kinematic Fermi-surface instability for a local attractive channel. In QCD the microscopic attraction, the color-flavor structure, and the weak-coupling gap prefactor also involve electric screening and dynamically screened magnetic gluons. Those ingredients refine the value of Δ; they do not remove the basic logarithmic instability once an attractive channel is present.

135.13 Neutrality Constraints and Gauge Sources

Dense QCD calculations often introduce several quantities called chemical potentials. The logically primary distinction is between sources for gauge-invariant global charges and variables that enforce gauge constraints. The two have different status already at finite regulator.

Definition 135.42 (Finite-regulator global-charge ensemble)**.** Let $\mathcal{H}_\Lambda$ be the regulated Hilbert space of a finite spatial lattice or another finite gauge regulator, let H_Λ be the regulated Hamiltonian, and let $\mathcal{G}_\Lambda$ be the compact group of time-independent gauge transformations. Write

$$\Pi_{\mathrm{phys}} = \int_{\mathcal{G}_\Lambda} d\eta\, U(\eta)$$

for the normalized Haar projector onto gauge-invariant states. If Q_A are mutually commuting conserved global charges which commute with $\mathcal{G}_\Lambda$, the finite-regulator grand partition function is

$$Z_\Lambda(\boldsymbol{\mu}) = \mathrm{Tr}_{\mathcal{H}_\Lambda}\left[\Pi_{\mathrm{phys}} \exp\left(-\beta H_\Lambda + \beta \sum_A \mu_A Q_A\right)\right]. \tag{135.62}$$

The intensive charge density conjugate to μ_A is

$$n_A = \frac{1}{\beta V}\frac{\partial}{\partial \mu_A} \log Z_\Lambda(\boldsymbol{\mu}), \tag{135.63}$$

with the thermodynamic and continuum limits taken only after the finite quantity has been defined.

Gauss-law annihilation of gauge charge. Let $G_\Lambda(f)$ be the infinitesimal generator of a one-parameter family of gauge transformations $U(e^{tf}) \subset \mathcal{G}_\Lambda$. Then

$$G_\Lambda(f)\Pi_{\mathrm{phys}} = 0, \qquad \Pi_{\mathrm{phys}} G_\Lambda(f) = 0. \tag{135.64}$$

Consequently every finite-regulator state defined by (135.62) has zero gauge charge:

$$\langle G_\Lambda(f)\rangle_{\boldsymbol{\mu}} = 0. \tag{135.65}$$

Thus a color charge cannot be assigned an independent physical chemical potential in the same sense as baryon number or another gauge-invariant global charge.

The Haar projector satisfies $U(\eta)\Pi_{\mathrm{phys}} = \Pi_{\mathrm{phys}}$ and $\Pi_{\mathrm{phys}}U(\eta) = \Pi_{\mathrm{phys}}$ for every $\eta \in \mathcal{G}_\Lambda$, by left and right invariance of Haar measure. Apply this identity to the one-parameter subgroup e^{tf} and differentiate at $t = 0$. This gives (135.64). Inserting $G_\Lambda(f)$ inside the trace defining the expectation value, and using either equality in (135.64) together with cyclicity of the finite trace, gives (135.65).

Remark 135.43 (Meaning of color-neutrality variables). In a gauge-fixed high-density calculation one may parametrize a homogeneous temporal gauge background by Cartan components A_0^a, or equivalently by symbols often denoted μ_3, μ_8. These symbols are not external thermodynamic chemical potentials for physical color charge. They are coordinates on the gauge-field zero mode left after gauge fixing. If $\Omega_{\mathrm{gf}}(A_0)$ is the gauge-fixed finite-volume effective potential obtained after the nonzero modes have been integrated out, a homogeneous saddle representing a physical bulk state must obey

$$\frac{\partial \Omega_{\mathrm{gf}}}{\partial A_0^a} = 0 \tag{135.66}$$

for every retained Cartan component. Equation (135.66) is the saddle form of the Gauss-law constraint. Its content is color neutrality; it is not the definition of a new grand-canonical ensemble.

Definition 135.44 (Neutrality problem for long-range global or gauged charges)**.** Let Q_{ext} be a conserved charge which is gauge invariant with respect to color, for example the electromagnetic charge after coupling QCD matter to a nondynamical or dynamical $U(1)$ gauge field. If the bulk state is required to be neutral with respect to this charge, introduce a source μ_{ext} in the finite-regulator ensemble and solve

$$n_{\mathrm{ext}}(T, \mu_B, \mu_{\mathrm{ext}}, \ldots) = \frac{\partial p}{\partial \mu_{\mathrm{ext}}} = 0 \tag{135.67}$$

after all charged species included in the physical problem, such as electrons in electrically neutral matter, have been included in the pressure. If Q_{ext} is itself gauged and long-ranged in the phase under consideration, the same condition is the thermodynamic form of the zero-mode Gauss law: a nonzero uniform charge density would require an electric field whose energy density is not compatible with an infinite homogeneous neutral bulk limit.

Remark 135.45 (Ideal CFL is electrically neutral at zero electric source). Assume $N_c = N_f = 3$, massless u, d, s quarks, a common quark chemical potential μ_q, and the ideal CFL symmetry realization described below: the gauge-invariant long-distance order preserves the diagonal vector $SU(3)$ flavor symmetry. Let

$$Q_{\mathrm{em}} = \mathrm{diag}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right) \in \mathfrak{su}(3)_V \tag{135.68}$$

act on the three flavor labels. Then the electric charge density conjugate to a source $\mu_Q Q_{\mathrm{em}}$ satisfies

$$n_Q(T, \mu_B, \mu_Q = 0) = 0 \tag{135.69}$$

inside the ideal CFL effective theory. Moreover the CFL orientation $U = 1$ in Controlled Approximation 135.57 is invariant under the simultaneous color and flavor transformation generated by Q_{em}, with the same matrix embedded in a Cartan subalgebra of color. The corresponding photon-gluon mixture is the unbroken rotated $U(1)$ gauge field of ideal CFL, with the relative sign fixed by the displayed transformation law $U \mapsto hUg^{-1}$.

The source μ_Q couples to the gauge-invariant vector flavor charge whose one-particle matrix is Q_{em}. At $\mu_Q = 0$, the assumed ideal CFL state is invariant under the diagonal vector $SU(3)$ flavor symmetry. Therefore the expectation value of every traceless generator of $\mathfrak{su}(3)_V$ vanishes. Since $\mathrm{tr}\, Q_{\mathrm{em}} = 2/3 - 1/3 - 1/3 = 0$, the density conjugate to $\mu_Q Q_{\mathrm{em}}$ is zero. This proves (135.69).

For the second assertion, recall the transformation law of the CFL orientation coordinate,

$$U \longmapsto h\, U\, g^{-1},$$

where h is a color gauge transformation and g is a locked vector flavor transformation. With $U = 1$, choosing

$$h = e^{i\alpha Q_{\mathrm{em}}}, \qquad g = e^{i\alpha Q_{\mathrm{em}}}$$

gives $hUg^{-1} = 1$. Hence this simultaneous transformation is unbroken. Equivalently, the infinitesimal action on U is $\delta U = iQ_{\mathrm{color}}U - iUQ_{\mathrm{flavor}}$, so $U = 1$ is fixed by the diagonal pair $Q_{\mathrm{color}} = Q_{\mathrm{flavor}} = Q_{\mathrm{em}}$. Because the color factor is gauged and the electromagnetic flavor factor is gauged when QED is included, the massless gauge boson is the corresponding linear combination of the photon and the two Cartan gluons. The statement is independent of the gauge choice used to display $U = 1$; the gauge-invariant content is the existence of the unbroken charge acting trivially on the CFL order.

Remark 135.46 (When neutrality becomes dynamical). Quark masses, weak equilibrium constraints, finite temperature, boundary conditions, and possible crystalline or meson-condensed order can break the ideal assumptions of Remark 135.45. Then neutrality is not a slogan attached to a named phase. It is the coupled problem consisting of the global neutrality equations (135.67), the gauge stationarity equations (135.66), and the variational equations for the order parameters retained in the controlled effective theory.

135.14 Fermi-Surface Stress from Masses and Neutrality

The ideal CFL limit has equal massless quarks and a common quark chemical potential. Real dense matter introduces flavor-dependent stresses through quark masses, electric neutrality, and weak-equilibrium constraints. The first controlled step is kinematic: determine how these deformations shift the location of the Fermi surface before asking which paired state minimizes the effective potential.

Definition 135.47 (Effective Fermi-surface mismatch)**.** Consider two quark species i, j with masses m_i, m_j and chemical potentials μ_i, μ_j. Assume

$$m_i^2,\ m_j^2,\ |\mu_i - \mu_j|\,\bar{\mu} \ll \bar{\mu}^2, \qquad \bar{\mu} = \frac{\mu_i + \mu_j}{2}.$$

For momenta in a common patch near $p = \bar{\mu}$, define the residual single-particle energies

$$\xi_a(p) = \sqrt{p^2 + m_a^2} - \mu_a, \qquad a = i, j. \tag{135.70}$$

The average residual energy and half-mismatch are

$$\bar{\xi}_{ij}(p) = \frac{\xi_i(p) + \xi_j(p)}{2}, \qquad \delta_{ij}^{\text{eff}} = \frac{\xi_i(p) - \xi_j(p)}{2} \quad \text{evaluated to leading order at } p = \bar{\mu}. \tag{135.71}$$

Mismatch expansion. The mass and chemical-potential contributions to the effective Fermi-surface mismatch are fixed by the following elementary expansion. In the regime of Definition 135.47,

$$p_{F,a} = \sqrt{\mu_a^2 - m_a^2} = \mu_a - \frac{m_a^2}{2\mu_a} + O\left(\frac{m_a^4}{\mu_a^3}\right), \tag{135.72}$$

and

$$\delta_{ij}^{\text{eff}} = -\frac{\mu_i - \mu_j}{2} + \frac{m_i^2 - m_j^2}{4\bar{\mu}} + O\left(\frac{m^4}{\bar{\mu}^3}, \frac{|\mu_i - \mu_j|\,m^2}{\bar{\mu}^2}\right). \tag{135.73}$$

Equivalently, the leading Fermi-momentum splitting is

$$p_{F,i} - p_{F,j} = (\mu_i - \mu_j) - \frac{m_i^2 - m_j^2}{2\bar{\mu}} + O\left(\frac{m^4}{\bar{\mu}^3}, \frac{|\mu_i - \mu_j|\,m^2}{\bar{\mu}^2}\right). \tag{135.74}$$

For a strange quark paired with a massless light quark at common chemical potential μ_q, the Fermi-momentum splitting scale is

$$|p_{F,s} - p_{F,\ell}| = \frac{m_s^2}{2\mu_q} + O(m_s^4/\mu_q^3), \tag{135.75}$$

while the pairwise half-mismatch is

$$|\delta_{s\ell}^{\text{eff}}| = \frac{m_s^2}{4\mu_q} + O(m_s^4/\mu_q^3). \tag{135.76}$$

The Fermi momentum solves $\sqrt{p_{F,a}^2 + m_a^2} = \mu_a$, giving the first identity in (135.72). Expanding the square root gives

$$\sqrt{\mu_a^2 - m_a^2} = \mu_a \sqrt{1 - \frac{m_a^2}{\mu_a^2}} = \mu_a - \frac{m_a^2}{2\mu_a} + O(m_a^4/\mu_a^3).$$

For the mismatch, expand (135.70) at $p = \bar{\mu}$:

$$\xi_a(\bar{\mu}) = \bar{\mu} - \mu_a + \frac{m_a^2}{2\bar{\mu}} + O(m_a^4/\bar{\mu}^3).$$

Taking half the difference between $a = i$ and $a = j$ gives (135.73); keeping the dependence of the denominator on μ_a rather than $\bar{\mu}$ produces the displayed mixed remainder. Subtracting (135.72) for i and j gives (135.74). The strange-light specializations set $m_\ell = 0$ and $\mu_s = \mu_\ell = \mu_q$.

Controlled Approximation 135.48 (Two-species stressed BCS comparison)**.** To isolate the local pairing stress, consider two species with a common density of states $N(0)$, a momentum-independent attractive interaction in a shell, zero temperature, and half-mismatch δ. Let Δ_0 be the BCS gap at $\delta = 0$, computed in the same cutoff scheme. Compare only the homogeneous paired state with equal paired Fermi momenta to the unpaired normal state with split Fermi surfaces. This is a local diagnostic for the onset of stress; it is not a complete variational analysis of CFL, gapless CFL, meson-condensed phases, or crystalline phases.

Clogston scale in the controlled two-species model. In Controlled Approximation 135.48, the zero-temperature grand-potential difference between the homogeneous paired state and the split normal state has the leading form

$$\Omega_{\text{pair}} - \Omega_{\text{normal}} = -\frac{1}{2}N(0)\Delta_0^2 + N(0)\delta^2 + O(\Delta_0^4/\Lambda_{\text{H}}^2). \tag{135.77}$$

Thus the homogeneous paired state ceases to win against the split normal state in this model when

$$|\delta| > \delta_{\text{C}}, \qquad \delta_{\text{C}} = \frac{\Delta_0}{\sqrt{2}}. \tag{135.78}$$

At $\delta = 0$, the BCS condensation energy in the constant-density shell is

$$\Omega_{\text{pair}}(\delta = 0) - \Omega_{\text{normal}}(\delta = 0) = -\frac{1}{2}N(0)\Delta_0^2 + O(\Delta_0^4/\Lambda_{\text{H}}^2),$$

which follows by substituting the gap equation into the mean-field energy functional: the logarithmic part of the interaction energy cancels the logarithmic part of the quasiparticle energy, leaving the displayed finite term. A normal state with half-mismatch δ fills one species up to energy $+\delta$ relative to the average Fermi surface and empties the other down to $-\delta$. With constant density of states, the energy lowering of the split normal state relative to the unsplit normal state is

$$-2N(0)\int_0^{|\delta|} \xi\, d\xi = -N(0)\delta^2,$$

where the factor 2 counts the filled particle interval for one species and the emptied hole interval for the other. Therefore, when the paired state remains unsplit, its grand potential relative to the split normal state is shifted upward by $N(0)\delta^2$, giving (135.77). Setting the leading difference to zero gives (135.78).

Remark 135.49 (Use of the stress scale in dense QCD). Equations (135.75) and (135.78) should be used with their hypotheses attached. The scale $m_s^2/(2\mu_q)$ is the leading strange-light Fermi-momentum splitting, while $m_s^2/(4\mu_q)$ is the pairwise half-mismatch in the two-species reduction. Full dense QCD also requires the neutrality equations of Section 135.13, the coupled gap equations in all pairing channels, and the possibility that the minimizing state is not spatially homogeneous.

135.15 Color Superconductivity and CFL Order

At asymptotically large μ_B, quarks near the Fermi surface interact through channels that can be attractive. The BCS mechanism is then a controlled weak-coupling instability of the Fermi surface, though the precise gap prefactor is sensitive to magnetic gluon exchange and screening.

Color factor of one-gluon exchange between two quarks. Let the $SU(N_c)$ fundamental generators be normalized as throughout the monograph,

$$\mathrm{tr}_\square(T^aT^b) = \delta^{ab}.$$

In the tensor product

$$\square \otimes \square = \mathrm{Sym}^2 \square \oplus \wedge^2\square,$$

the color operator $T_1^aT_2^a$ has eigenvalues

$$(T_1 \cdot T_2)_{\mathrm{Sym}^2 \square} = \frac{N_c - 1}{N_c}, \qquad (T_1 \cdot T_2)_{\wedge^2\square} = -\frac{N_c + 1}{N_c}.$$

With the Born-kernel sign convention in which the static exchange energy is proportional to

$$\mathcal{V}_R(\mathbf{q}) = \frac{g^2}{\mathbf{q}^2}\,(T_1 \cdot T_2)_R\,,$$

the two-index antisymmetric channel is attractive and the symmetric channel is repulsive. For $SU(3)$, the attractive channel is $\wedge^2\square = \overline{\square}$, with color factor $-4/3$, while the symmetric sextet has color factor $2/3$.

Let $C_2(R)$ denote the quadratic Casimir in the present trace convention. On an irreducible summand $R \subset \square \otimes \square$,

$$(T_1^a + T_2^a)(T_1^a + T_2^a) = C_2(R)\mathbf{1}.$$

Since each tensor factor is fundamental,

$$2T_1^aT_2^a = C_2(R) - 2C_2(\square).$$

The trace-delta convention doubles the more common half-trace Casimirs, so

$$C_2(\square) = \frac{N_c^2 - 1}{N_c},$$

$$C_2(\mathrm{Sym}^2 \square) = \frac{2(N_c + 2)(N_c - 1)}{N_c}, \qquad C_2(\wedge^2\square) = \frac{2(N_c - 2)(N_c + 1)}{N_c}.$$

Substitution gives

$$\frac{1}{2}\left[\frac{2(N_c + 2)(N_c - 1)}{N_c} - \frac{2(N_c^2 - 1)}{N_c}\right] = \frac{N_c - 1}{N_c},$$

and

$$\frac{1}{2}\left[\frac{2(N_c - 2)(N_c + 1)}{N_c} - \frac{2(N_c^2 - 1)}{N_c}\right] = -\frac{N_c + 1}{N_c}.$$

The sign statement follows from the displayed exchange-energy convention.

Controlled Approximation 135.50 (Leading-log magnetic gap problem)**.** Work at $T = 0$, $\mu_q \gg \Lambda_{\rm QCD}$, and $g(\mu_q) \ll 1$ in the high-density effective theory. Retain the attractive two-index antisymmetric quark-quark channel and the HDL-resummed soft gluon propagator. The leading logarithm in the gap equation is the part generated by nearly collinear transverse magnetic exchange with frequency transfer

$$\omega = p_0 - q_0$$

and transverse momentum $k_\perp$ satisfying

$$|\omega| \ll k_\perp \ll \mu_q.$$

In this region the Euclidean transverse propagator has the form

$$D_T(\omega, k_\perp) = \frac{1}{k_\perp^2 + \frac{\pi}{4} m_{D,\mu}^2 \frac{|\omega|}{k_\perp}} \left(1 + o(1)\right), \tag{135.79}$$

where $m_{D,\mu}$ is the dense HDL Debye mass computed above. Electric exchange and non-collinear magnetic exchange are not discarded; in the leading-log problem they contribute to the matching scale and to the prefactor, not to the coefficient of the $1/g$ exponent.

HDL magnetic leading-log gap equation. In the regime specified by Controlled Approximation 135.50, the gap in the attractive antisymmetric quark-quark channel obeys, to leading logarithmic accuracy,

$$\Delta(p_0) = \lambda_{\rm mag} \int_0^{\Lambda_{\rm M}} dq_0 \, \frac{\Delta(q_0)}{\sqrt{q_0^2 + \Delta(q_0)^2}} \log \frac{\Lambda_{\rm M}}{|p_0 - q_0|} + \mathcal{R}(p_0), \tag{135.80}$$

where $\mathcal{R}$ has no additional large collinear logarithm. The leading-log coefficient is

$$\lambda_{\rm mag} = \frac{g^2 C_{\rm qq}}{12\pi^2}, \qquad C_{\rm qq} = -\left(T_1 \cdot T_2\right)_{\wedge^2\square} = \frac{N_c + 1}{N_c}. \tag{135.81}$$

For $SU(3)$ in the monograph trace convention,

$$C_{\rm qq} = \frac{4}{3}, \qquad \lambda_{\rm mag} = \frac{g^2}{9\pi^2}. \tag{135.82}$$

The scale $\Lambda_{\rm M}$ is a matching scale of order μ_q times powers of g and numerical constants fixed only by next-to-leading matching. It is not determined by the leading-log equation alone.

The gap equation is the off-diagonal Schwinger–Dyson equation in Nambu–Gorkov variables, restricted to the shell of Definition 135.36. The radial residual momentum integral across the Fermi surface gives the BCS denominator

$$\int d\ell_\parallel \, \frac{1}{q_0^2 + \ell_\parallel^2 + \Delta(q_0)^2} = \frac{\pi}{\sqrt{q_0^2 + \Delta(q_0)^2}},$$

up to terms without a large logarithm. The angular integral is dominated by small scattering angle, equivalently by transverse momentum transfer

$$k_\perp \simeq \mu_q \theta.$$

For transverse gluons the HDL denominator (135.79) cuts off the collinear singularity in the infrared at

$$k_{\perp,\min}^3 \asymp m_{D,\mu}^2 |\omega|,$$

while the upper end is fixed by the edge of the patch or by hard matching, $k_{\perp,\max} \asymp \mu_q$. Thus the logarithmic part of the transverse integral is

$$\int_{k_{\perp,\min}}^{k_{\perp,\max}} \frac{dk_\perp}{k_\perp} = \frac{1}{3} \log \frac{\Omega_{\mathrm{M}}}{|\omega|} + O(1), \qquad \Omega_{\mathrm{M}} \asymp \frac{\mu_q^3}{m_{D,\mu}^2},$$

where the $O(1)$ and the precise constant multiplying Ω_{M} are not leading-log data. The factor $1/3$ is kept explicitly in the coefficient below. After this factor has been absorbed into λ_{mag}, the remaining energy matching scale is denoted Λ_{M} in (135.80).

The remaining coefficient is fixed by the Fermi-surface measure, the spin-current projection for transverse exchange, and the color projection. The shell measure and the radial propagator integral give the common prefactor $g^2 C_{\mathrm{qq}}/(4\pi^2)$ multiplying the transverse collinear integral. The Landau-damped magnetic integral contributes only one third of the logarithm of the energy transfer,

$$\int \frac{dk_\perp}{k_\perp} = \frac{1}{3} \log \frac{1}{|\omega|} + \text{terms independent of } |\omega|$$

at leading logarithmic accuracy. Thus the coefficient of the logarithmic energy integral is

$$\frac{g^2 C_{\mathrm{qq}}}{4\pi^2} \cdot \frac{1}{3} = \frac{g^2 C_{\mathrm{qq}}}{12\pi^2}.$$

The color factor is not an independent convention: by the two-quark color calculation above,

$$C_{\mathrm{qq}} = -\left(T_1 \cdot T_2\right)_{\wedge^2 \Box} = \frac{N_c + 1}{N_c}.$$

For $N_c = 3$ this gives $C_{\mathrm{qq}} = 4/3$, hence $\lambda_{\mathrm{mag}} = g^2/(9\pi^2)$. Terms from screened electric exchange, from non-collinear transverse exchange, and from quark self-energy do not change this leading collinear logarithm; they are included in $\mathcal{R}$ and in the matching scale.

Proposition 135.51 (Leading exponent of the magnetic color-superconducting gap)**.** *Consider the leading-log integral equation*

$$\Delta(p) = \lambda \int_{\Delta_0}^{\Lambda} \frac{dq}{q}\, \Delta(q) \log \frac{\Lambda}{\max(p,q)}, \qquad \Delta_0 \ll p, q \ll \Lambda, \tag{135.83}$$

with $\lambda > 0$. If the solution is nontrivial and varies only on the logarithmic scale, then

$$\log \frac{\Lambda}{\Delta_0} = \frac{\pi}{2\sqrt{\lambda}} + O(1). \tag{135.84}$$

Applying this to the $SU(3)$ HDL magnetic kernel in the monograph trace-delta convention gives the leading exponent

$$\Delta_0 = \Lambda_{\mathrm{M}} \exp\left[-\frac{3\pi^2}{2g} + O(1)\right]. \tag{135.85}$$

Equivalently, if $g_{\mathrm{ht}} = \sqrt{2}\, g$ denotes the coupling in the common half-trace convention, the exponent is

$$\exp\left[-\frac{3\pi^2}{\sqrt{2}\, g_{\mathrm{ht}}}\right].$$

Proof. Set

$$x = \log \frac{\Lambda}{p}, \qquad y = \log \frac{\Lambda}{q}, \qquad X = \log \frac{\Lambda}{\Delta_0}.$$

The leading-log kernel becomes

$$\log \frac{\Lambda}{\max(p,q)} = \min(x,y).$$

Equation (135.83) becomes

$$\Delta(x) = \lambda \int_0^X dy\, \min(x,y)\,\Delta(y) = \lambda \left[\int_0^x dy\, y\, \Delta(y) + x \int_x^X dy\, \Delta(y) \right].$$

Differentiating once gives

$$\Delta'(x) = \lambda \int_x^X dy\, \Delta(y),$$

and differentiating again gives the ordinary differential equation

$$\Delta''(x) + \lambda \Delta(x) = 0.$$

The integral equation imposes

$$\Delta(0) = 0, \qquad \Delta'(X) = 0.$$

The lowest nontrivial solution is therefore

$$\Delta(x) = A \sin(\sqrt{\lambda}\, x), \qquad \cos(\sqrt{\lambda}\, X) = 0,$$

so

$$X = \frac{\pi}{2\sqrt{\lambda}}$$

at leading logarithmic accuracy. Higher nodes give smaller gaps and do not describe the leading instability. Substituting $\lambda = \lambda_{\text{mag}} = g^2/(9\pi^2)$ from (135.82) gives

$$\frac{\pi}{2\sqrt{\lambda_{\text{mag}}}} = \frac{3\pi^2}{2g}.$$

The $O(1)$ in (135.85) records terms outside leading-log accuracy. The power of g multiplying μ_q in the prefactor and the numerical constant multiplying it are next-to-leading data, not consequences of this proposition. □

Exchange symmetry and the CFL tensor. For a local s-wave spin-zero pairing of two left-handed Weyl quarks, the coefficient multiplying

$$q_L^{ai\alpha} q_L^{bj\beta} \epsilon_{\alpha\beta}$$

is symmetric under the simultaneous interchange $(a,i) \leftrightarrow (b,j)$. Therefore a pair in the attractive color-antisymmetric channel must also be antisymmetric in flavor. For $N_c = N_f = 3$, the nondegenerate antisymmetric tensor

$$\epsilon^{abA} \epsilon_{ijA}$$

is invariant precisely under the diagonal subgroup obtained by identifying color and flavor $SU(3)$ rotations.

The fermion fields anticommute, while $\epsilon_{\alpha\beta}$ is antisymmetric. Hence

$$q_L^{bj\alpha} q_L^{ai\beta} \epsilon_{\alpha\beta} = -q_L^{ai\beta} q_L^{bj\alpha} \epsilon_{\alpha\beta} = q_L^{ai\alpha} q_L^{bj\beta} \epsilon_{\alpha\beta},$$

where the last equality relabels $\alpha \leftrightarrow \beta$ and uses $\epsilon_{\beta\alpha} = -\epsilon_{\alpha\beta}$. Thus the color-flavor coefficient must be symmetric under simultaneous exchange. Since the two-quark color calculation above identifies the attractive color channel as antisymmetric, the flavor tensor must be antisymmetric as well.

Let $U \in SU(3)_{\text{color}}$ and $V \in SU(3)_{\text{flavor}}$. The tensor $\epsilon^{abA}\epsilon_{ijA}$ may be viewed as the identity map from the color $\overline{\Box}$ carried by $\wedge^2\Box_{\text{color}}$ to the flavor $\overline{\Box}$ carried by $\wedge^2\Box_{\text{flavor}}$. It is preserved exactly when these two $\overline{\Box}$ actions agree, which is the diagonal $SU(3)$ condition after the chosen color-flavor identification. This is the color-flavor locking statement.

Definition 135.52 (CFL condensate data)**.** For $N_c = N_f = 3$, color-flavor-locked Higgs data are a source-selected or gauge-fixed representative of the gauge-covariant pairing tensor

$$\Phi^{ab}_{L,ij} = \langle q^a_{Li} C q^b_{Lj} \rangle_J = \Delta_L\, \epsilon^{abA}\epsilon_{ijA}, \qquad \Phi^{ab}_{R,ij} = \langle q^a_{Ri} C q^b_{Rj} \rangle_J = \Delta_R\, \epsilon^{abA}\epsilon_{ijA},$$

with color indices a, b, flavor indices i, j, and spinor charge conjugation matrix C. The subscript J means that an infinitesimal pairing source and the infinite-volume limit are used before the source is removed, exactly as in any spontaneous-order definition. The tensor is not a gauge-invariant local operator. Its gauge-invariant content is the Higgs pattern of the color gauge field, the residual global diagonal symmetry, and the long-distance order of color-singlet composites built from the pairing tensor. The displayed tensor locks color and flavor: simultaneous color and flavor rotations preserving the same diagonal $SU(3)$ leave the representative invariant.

Definition 135.53 (Gauge-invariant CFL diagnostics)**.** Let

$$\varphi_L \in \overline{\Box}_{\text{color}} \otimes \overline{\Box}_L, \qquad \varphi_R \in \overline{\Box}_{\text{color}} \otimes \overline{\Box}_R$$

denote the antisymmetric spin-zero diquark composite obtained from two left-handed or two right-handed quark fields:

$$(\varphi_L)^A{}_I(x) = \frac{1}{2}\epsilon^A{}_{ab}\epsilon_I{}^{ij}\, q^a_{Li}(x) C q^b_{Lj}(x), \qquad (\varphi_R)^A{}_{\bar{I}}(x) = \frac{1}{2}\epsilon^A{}_{ab}\epsilon_{\bar{I}}{}^{ij}\, q^a_{Ri}(x) C q^b_{Rj}(x). \tag{135.86}$$

Here A is an anti-fundamental color index, I is an anti-fundamental $SU(3)_L$ flavor index, and $\bar{I}$ is an anti-fundamental $SU(3)_R$ flavor index. Each quark has baryon charge $1/3$, hence φ_L and φ_R have baryon charge $2/3$.

The following operators are gauge invariant and carry the physical symmetry information of the CFL phase:

$$\mathcal{B}_L = \frac{1}{3!}\epsilon_{ABC}\epsilon^{IJK}(\varphi_L)^A{}_I(\varphi_L)^B{}_J(\varphi_L)^C{}_K, \qquad \mathcal{B}_R = \frac{1}{3!}\epsilon_{ABC}\epsilon^{\bar{I}\bar{J}\bar{K}}(\varphi_R)^A{}_{\bar{I}}(\varphi_R)^B{}_{\bar{J}}(\varphi_R)^C{}_{\bar{K}}, \tag{135.87}$$

and

$$\Sigma^I{}_{\bar{J}} = (\varphi_L^\dagger)^I{}_A(\varphi_R)^A{}_{\bar{J}}. \tag{135.88}$$

The operators $\mathcal{B}_L, \mathcal{B}_R$ are color and flavor singlets with baryon charge 2. The operator Σ is color invariant, baryon neutral, and transforms as

$$\Sigma \longmapsto g_L \, \Sigma \, g_R^{-1} \qquad (g_L, g_R) \in SU(3)_L \times SU(3)_R.$$

Finally, for a path γ from y to x, the Wilson-line completion

$$G^I_{L\,J}(x, y; \gamma) = \left\langle (\varphi_L^\dagger)^I{}_A(x) \ \left[W_{\overline{\square}}(\gamma)\right]^A{}_B \, (\varphi_L)^B{}_J(y) \right\rangle \tag{135.89}$$

is gauge invariant but path dependent. It is not a new local order parameter; it is a gauge-invariant way to test the long-distance order encoded by a gauge-covariant diquark field.

Proposition 135.54 (Gauge averaging and physical CFL order)**.** *In a finite-volume regularized gauge theory with a gauge-invariant action, measure, boundary condition, and source, the expectation value of any local operator transforming in a nontrivial irreducible representation of the color gauge group is zero. Consequently the gauge-covariant diquark one-point function $\langle \varphi_L \rangle$ is not itself a physical local order parameter. The CFL symmetry realization is instead detected by the gauge-invariant composites of Definition 135.53: long-distance order in $\mathcal{B}_{L,R}$ diagnoses baryon superfluid order, while long-distance order in Σ diagnoses*

$$SU(3)_L \times SU(3)_R \longrightarrow SU(3)_{\text{diagonal}}.$$

Proof. Let O be a local operator whose color indices span a finite-dimensional representation ρ of $SU(3)_{\text{color}}$. Gauge invariance of the regularized functional integral gives, for every constant color transformation $h \in SU(3)$,

$$\langle O \rangle = \rho(h) \langle O \rangle.$$

Averaging over h with normalized Haar measure gives

$$\langle O \rangle = \int_{SU(3)} dh \, \rho(h) \langle O \rangle.$$

The Haar average is the projection onto the invariant subspace of ρ. If ρ has no singlet component, this projection is zero. The diquark field φ_L carries the $\overline{\square}_{\text{color}}$ representation, so its gauge-invariant finite-volume expectation vanishes unless one first fixes a gauge or uses a gauge-covariant source-selected representative as in Definition 135.52.

The composites in Definition 135.53 are different. For $\mathcal{B}_L$, the color indices are contracted with ϵ_{ABC}, and $\det h = 1$ for $h \in SU(3)$, so the color factor is invariant. The flavor contraction is likewise invariant under $SU(3)_L$, and the baryon charge is

$$3 \cdot \frac{2}{3} = 2.$$

For $\mathcal{B}_R$, the right-handed color indices are contracted by the same ϵ_{ABC} tensor and the flavor indices by the right-handed ϵ^{ijk}, so the operator is again an $SU(3)_{\text{color}}$ singlet of baryon charge 2. Therefore a nonzero source-selected long-distance limit of the two-point function of $\mathcal{B}_L$ or $\mathcal{B}_R$ is a gauge-invariant statement that $U(1)_B$ is spontaneously broken.

For Σ, the color anti-fundamental index of φ_R is contracted with the conjugate color index of $\varphi_L^\dagger$, so the operator is color invariant. Its baryon charge is

$$-\frac{2}{3} + \frac{2}{3} = 0.$$

Since $\varphi_L^\dagger$ transforms in the fundamental of $SU(3)_L$ and φ_R in the anti-fundamental of $SU(3)_R$, Σ transforms as $g_L \Sigma g_R^{-1}$. A source-selected expectation proportional to the identity leaves only $g_L = g_R$, which is the diagonal chiral subgroup. The Wilson-line correlator (135.89) is gauge invariant because the endpoint transformations of the Wilson line cancel the color transformations of the two diquark insertions. Its path dependence is the price of using a nonlocal completion of a gauge-covariant field, not an additional physical symmetry. □

Remark 135.55 (Physical CFL Goldstone count). Assume $N_c = N_f = 3$, ignore the anomalous axial $U(1)$, and suppose the infinite-volume dense state has the gauge-invariant CFL long-distance order described in Definition 135.53 and Proposition 135.54. The physical global symmetry breaking pattern is

$$SU(3)_L \times SU(3)_R \times U(1)_B \longrightarrow SU(3)_{\mathrm{diagonal}},$$

and the number of physical Nambu–Goldstone modes before quark-mass effects are included is

$$(8 + 8 + 1) - 8 = 9.$$

The color $SU(3)$ is gauged. Directions that rotate the condensate by a pure color transformation are Higgs directions of the gauge field, not generators of physical global particles. The remaining exact global generators, in the massless and anomaly-idealized limit stated in the proposition, span

$$\mathfrak{su}(3)_L \oplus \mathfrak{su}(3)_R \oplus \mathfrak{u}(1)_B.$$

The CFL tensor identifies a diagonal $\mathfrak{su}(3)$ subgroup with the unbroken simultaneous flavor rotation, as shown in the exchange-symmetry calculation above. The dimension of the broken global coset is therefore $8 + 8 + 1 - 8 = 9$. The conclusion is only a group-theoretic consequence of the source-selected CFL order. Establishing which region of QCD realizes this order requires control of dense QCD, where direct lattice importance sampling is obstructed by the finite-density sign-problem mechanism described above.

Definition 135.56 (Gauge-invariant static screening diagnostics in dense QCD)**.** Let $x = (0, \mathbf{x})$, $y = (0, \mathbf{y})$, and let γ be a spatial path from y to x. For the canonically normalized color gauge potential $a_\mu = a_\mu^a T^a$, with field strength $\mathfrak{f}_{\mu\nu} = \mathfrak{f}_{\mu\nu}^a T^a$, define the Wilson-line completed static electric and magnetic correlators

$$C_E(r;\gamma) = \sum_{i=1}^{3} \big\langle \mathrm{tr}\big[\mathfrak{f}_{0i}(x)\, W_\square(\gamma)\, \mathfrak{f}_{0i}(y)\, W_\square(\gamma)^{-1}\big]\big\rangle_{\mathrm{conn}}\,,$$

$$C_M(r;\gamma) = \frac{1}{2} \sum_{i,j=1}^{3} \big\langle \mathrm{tr}\big[\mathfrak{f}_{ij}(x)\, W_\square(\gamma)\, \mathfrak{f}_{ij}(y)\, W_\square(\gamma)^{-1}\big]\big\rangle_{\mathrm{conn}}\,, \qquad r = |\mathbf{x} - \mathbf{y}|. \tag{135.90}$$

The static electric and magnetic screening exponents are the leading large-r exponential decay rates, when they exist after the thermodynamic and continuum limits,

$$m_E = -\lim_{r\to\infty} \frac{1}{r} \log|C_E(r;\gamma_r)|, \qquad m_M = -\lim_{r\to\infty} \frac{1}{r} \log|C_M(r;\gamma_r)|, \tag{135.91}$$

for a specified family of paths γ_r. In weakly coupled phases these exponents are computed by a gauge-fixed long-distance effective action, but equations (135.90)–(135.91) are the gauge-invariant diagnostic.

Controlled Approximation 135.57 (CFL Higgs screening effective action)**.** In a dense state with the CFL long-distance order of Definition 135.53, introduce a local gauge-covariant coordinate $U(x) \in SU(3)$ for the color-flavor locked orientation. Under a color gauge transformation $h(x)$ and a global locked flavor transformation g, it transforms as

$$U(x) \longmapsto h(x)U(x)g^{-1}.$$

The leading static screening part of the long-distance effective action is

$$\mathcal{L}_{\mathrm{H}} = \frac{F_H^2}{2}\,\mathrm{tr}\Big[(D_0U)^\dagger D_0U\Big] - \frac{F_H^2 v_H^2}{2}\,\mathrm{tr}\Big[(D_iU)^\dagger D_iU\Big]\,, \tag{135.92}$$

where

$$D_\mu U = \partial_\mu U - ig\,a_\mu^a T^a U, \qquad \mathrm{tr}_\square(T^aT^b) = \delta^{ab}.$$

The constants $F_H > 0$ and $0 < v_H \le 1$ are low-energy coefficients of the dense medium. Symmetry fixes the tensor structure of (135.92), not the microscopic numerical value of F_H or v_H.

CFL Higgs screening mass matrix. In the CFL Higgs-screening effective action, the color-flavor-locked background gives equal screening masses to all eight color gauge fields. More precisely, within Controlled Approximation 135.57, one has

$$(m_E^2)^{ab} = g^2F_H^2\,\delta^{ab}, \qquad (m_M^2)^{ab} = g^2F_H^2v_H^2\,\delta^{ab}. \tag{135.93}$$

These masses are coefficients in the gauge-fixed quadratic effective action whose gauge-invariant content is the exponential screening described by Definition 135.56.

Use the gauge freedom to set $U = 1$ locally; this is unitary gauge for the Higgs-coordinate chart, not a definition of a gauge-invariant operator. Then

$$D_\mu U = -ig\,a_\mu^a T^a.$$

Substitution into (135.92) gives

$$\mathcal{L}_{\mathrm{H}}^{(2)} = \frac{g^2F_H^2}{2}a_0^a a_0^b\,\mathrm{tr}(T^aT^b) - \frac{g^2F_H^2v_H^2}{2}a_i^a a_i^b\,\mathrm{tr}(T^aT^b).$$

The trace convention of the chapter makes $\mathrm{tr}(T^aT^b) = \delta^{ab}$. Therefore the quadratic coefficients are exactly the matrices displayed in (135.93). The equality of the eight eigenvalues follows from the unbroken diagonal $SU(3)$: the gluons form the adjoint representation of this diagonal group, and the quadratic mass operator is an invariant bilinear form on the adjoint. Since the adjoint of $SU(3)$ is simple, this invariant form is proportional to δ^{ab}, and the direct computation fixes the proportionality constant.

Proposition 135.58 (Rotated electromagnetic response in ideal CFL)**.** *Couple the ideal CFL effective coordinate to a dynamical electromagnetic field A_μ with coupling e and flavor generator*

$$Q_{\mathrm{em}} = \mathrm{diag}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right).$$

In the same trace-delta convention as Controlled Approximation 135.57, embed Q_{em} in the color Cartan subalgebra and write

$$q_Q^2 = \mathrm{tr}(Q_{\mathrm{em}}^2) = \frac{2}{3}, \qquad \widehat{Q} = \frac{Q_{\mathrm{em}}}{q_Q}, \qquad a_\mu^Q = \mathrm{tr}(a_\mu\widehat{Q}).$$

The gauge-covariant derivative is

$$D_\mu U = \partial_\mu U - ig\, a_\mu U + ie\, A_\mu U Q_{\rm em}. \tag{135.94}$$

In unitary gauge $U = 1$, the electric quadratic form in the (a_0^Q, A_0) sector is

$$\frac{F_H^2}{2} \begin{pmatrix} a_0^Q & A_0 \end{pmatrix} \begin{pmatrix} g^2 & -geq_Q \\ -geq_Q & e^2 q_Q^2 \end{pmatrix} \begin{pmatrix} a_0^Q \\ A_0 \end{pmatrix}. \tag{135.95}$$

The determinant of this matrix is zero. The massless long-range gauge field is the rotated photon

$$\widetilde{A}_\mu = \frac{gA_\mu + eq_Q a_\mu^Q}{\sqrt{g^2 + e^2 q_Q^2}}, \tag{135.96}$$

while the orthogonal combination

$$\widetilde{G}_\mu = \frac{ga_\mu^Q - eq_Q A_\mu}{\sqrt{g^2 + e^2 q_Q^2}} \tag{135.97}$$

has electric screening mass squared

$$m^2_{\widetilde{G},E} = F_H^2 (g^2 + e^2 q_Q^2).$$

The spatial response has the same null and massive eigenvectors, with the massive eigenvalue multiplied by v_H^2. The seven color directions orthogonal to $\widehat{Q}$ retain the CFL Higgs screening masses in (135.93).

Thus ideal CFL is not a superconductor for the unbroken rotated $U(1)$ gauge field. It is a medium in which all eight color directions are screened, one of them after mixing with the ordinary photon, while the orthogonal photon-gluon combination remains long-ranged.

Proof. The right action in (135.94) is the gauged version of the locked flavor transformation $U \mapsto hUg^{-1}$. In the local chart $U = 1$,

$$D_\mu U = -iga_\mu + ieA_\mu Q_{\rm em}.$$

Decompose the color gauge field as

$$a_\mu = a_\mu^\perp + a_\mu^Q \widehat{Q}, \qquad \mathrm{tr}(a_\mu^\perp \widehat{Q}) = 0.$$

Then

$$\mathrm{tr}\big[(ga_\mu - eA_\mu Q_{\rm em})^2\big] = g^2\, \mathrm{tr}\Big[(a_\mu^\perp)^2\Big] + (ga_\mu^Q - eq_Q A_\mu)^2.$$

This gives (135.95) for the electric part of the effective action and the corresponding factor of v_H^2 for the spatial part. The two-by-two matrix has

$$\det \begin{pmatrix} g^2 & -geq_Q \\ -geq_Q & e^2 q_Q^2 \end{pmatrix} = g^2 e^2 q_Q^2 - g^2 e^2 q_Q^2 = 0,$$

and trace $g^2 + e^2 q_Q^2$. Its null vector is proportional to (eq_Q, g), giving $\widetilde{A}_\mu$, and its nonzero eigenvector is proportional to $(g, -eq_Q)$, giving $\widetilde{G}_\mu$. The remaining color components are orthogonal to $Q_{\rm em}$, so they do not mix with the photon and keep the color-Higgs screening coefficients of (135.93). $\square$

Definition 135.59 (CFL collective fields)**.** Ignoring quark-mass effects and the anomalous axial $U(1)$, the gauge-invariant low-energy collective fields of the CFL phase are:

$$\Sigma(x) \in SU(3), \qquad \Sigma \longmapsto g_L \Sigma g_R^{-1},$$

for the chiral octet, and

$$e^{i\phi_B(x)} \in U(1)_B, \qquad \phi_B \longmapsto \phi_B + \alpha_B,$$

for the baryon superfluid phase. The leading real-time effective Lagrangian in the rest frame of the dense medium is

$$\mathcal{L}_{\mathrm{CFL,NGB}} = \frac{F_\pi^2}{4}\,\mathrm{tr}\Big[\partial_0\Sigma\,\partial_0\Sigma^\dagger - v_\pi^2\partial_i\Sigma\,\partial_i\Sigma^\dagger\Big] + \frac{F_B^2}{2}\left[(\partial_0\phi_B)^2 - v_B^2(\partial_i\phi_B)^2\right] + \cdots. \qquad (135.98)$$

The constants F_π, F_B, v_π, v_B are dense-medium low-energy coefficients. The ellipsis contains higher derivatives, mass-source terms, and anomaly-suppressed effects.

Physical collective-mode content of the ideal CFL coordinate chart. Under the assumptions encoded in Definition 135.59, the ideal CFL effective coordinate chart contains eight chiral Nambu–Goldstone modes and one baryon superfluid phonon. To quadratic order their dispersion relations are

$$\omega^2 = v_\pi^2\mathbf{k}^2 \quad \text{for the chiral octet,} \qquad \omega^2 = v_B^2\mathbf{k}^2 \quad \text{for the baryon phonon.} \qquad (135.99)$$

The eight color-orientation modes are not additional physical Nambu–Goldstone particles; in the screening effective action they are the fields removed by the Higgs mechanism and are represented physically by the eight screened color gauge sectors of the CFL Higgs-screening calculation above.

The logical input is the effective coordinate chart; the independent dense-QCD problem is the construction and control of that chart from a regulated QCD definition. The finite calculation is as follows. Write

$$\Sigma(x) = \exp\left(\frac{2i\pi^a(x)\tau^a}{F_\pi}\right), \qquad \mathrm{tr}(\tau^a\tau^b) = \delta^{ab},$$

where $a = 1, \ldots, 8$. Expanding (135.98) to quadratic order gives

$$\mathcal{L}_\pi^{(2)} = \frac{1}{2}\sum_{a=1}^{8}\left[(\partial_0\pi^a)^2 - v_\pi^2(\partial_i\pi^a)^2\right],$$

after the displayed normalization of the exponential is used. The Euler equation is

$$\partial_0^2\pi^a - v_\pi^2\nabla^2\pi^a = 0,$$

so a plane wave has $\omega^2 = v_\pi^2\mathbf{k}^2$. The field $\Sigma \in SU(3)$ has dimension 8, hence the chiral sector contains eight modes.

The baryon phase part of (135.98) gives

$$\partial_0^2\phi_B - v_B^2\nabla^2\phi_B = 0,$$

and therefore the second dispersion relation in (135.99). This is one real compact phase field. Finally, the color orientation U in Controlled Approximation 135.57 has eight local coordinates, but it transforms under local color. The unitary-gauge computation in the CFL Higgs-screening calculation above shows that those coordinates are not gauge-invariant massless particles; they supply the longitudinal screening components of the eight color gauge sectors. Thus the physical gapless count is $8 + 1 = 9$, matching Remark 135.55.

135.16 Anomaly Matching in the CFL Phase

The preceding CFL discussion identified gauge-invariant order diagnostics and the physical collective fields. A further consistency test is the response to background fields for exact global symmetries. This test is independent of weak-coupling quasiparticle language: if the same microscopic QCD theory is followed to long distances in a CFL phase, the background anomaly class of the exact symmetry must be represented by the CFL infrared fields.

Definition 135.60 (Backgrounds and normalization for CFL anomaly matching)**.** Work with massless $N_f = N_c = 3$ QCD and with the connected continuous global symmetry generated by

$$SU(3)_L \times SU(3)_R \times U(1)_B,$$

excluding the anomalous singlet axial rotation. Let A_L, A_R be background $SU(3)_L$ and $SU(3)_R$ connections, and let B be the $U(1)_B$ background connection. The quark baryon charge is normalized as

$$q_B(q) = \frac{1}{3}.$$

In this anomaly-matching subsection, flavor traces are taken in the half-trace convention

$$\mathrm{tr_f}(t^a t^b) = \frac{1}{2}\delta^{ab},$$

the same convention used in the WZW level-matching argument leading to (55.71). This is only a convention for anomaly traces; the trace-delta convention used elsewhere in this chapter is related by $\tau^a = \sqrt{2}\, t^a$.

Lemma 135.61 (UV continuous flavor anomaly polynomial)**.** *With the conventions of Definition 135.60, the local six-form anomaly polynomial of the massless quarks, restricted to $SU(3)_L \times SU(3)_R \times U(1)_B$ backgrounds and omitting purely gravitational terms, is*

$$I_6^{\mathrm{UV}} = \frac{N_c}{6(2\pi)^3}\left[\mathrm{tr_f}(F_L^3) - \mathrm{tr_f}(F_R^3)\right] + \frac{N_c q_B}{2(2\pi)^3} F_B \wedge \left[\mathrm{tr_f}(F_L^2) - \mathrm{tr_f}(F_R^2)\right], \tag{135.100}$$

where F_L, F_R, F_B are the corresponding curvatures. For $N_c = 3$ and $q_B = 1/3$, the mixed coefficient is

$$N_c q_B/2 = \frac{1}{2}. \tag{135.101}$$

Proof. The left-handed quark field consists of N_c color copies of a left-handed Weyl fermion in the fundamental representation of $SU(3)_L$ with baryon charge q_B. For a single left-handed Weyl fermion in a flavor bundle with connection A, tensored with a $U(1)$ line of charge q_B, the relevant degree-six Chern-character terms are

$$\frac{1}{6(2\pi)^3}\,\mathrm{tr_f}(F^3) + \frac{q_B}{2(2\pi)^3} F_B \wedge \mathrm{tr_f}(F^2). \tag{135.102}$$

The terms proportional to $F_B^2\,\mathrm{tr_f}\,F$ vanish because the flavor connection is $SU(3)$-valued, and the pure F_B^3 term cancels between the left- and right-handed components of a vectorlike Dirac quark. Multiplication by the N_c color copies gives the left contribution. A right-handed Weyl fermion with the same representation and charge contributes with the opposite sign to the four-dimensional chiral anomaly polynomial, giving the minus signs in (135.100). Substituting $q_B = 1/3$ gives (135.101). ∎

Proposition 135.62 (CFL realization of the continuous anomaly)**.** *Assume the ideal CFL infrared field content of Definition 135.59: a chiral field $\Sigma \in SU(3)$ with $\Sigma \mapsto g_L \Sigma g_R^{-1}$, and a compact baryon phase ϕ_B which transforms under a $U(1)_B$ background gauge transformation by*

$$B \mapsto B + d\lambda_B, \qquad \phi_B \mapsto \phi_B + \lambda_B. \tag{135.103}$$

Then the CFL infrared effective action can represent the full local anomaly polynomial (135.100) as follows.

First, the gauged Wess–Zumino–Witten functional of Σ has level

$$n = N_c = 3 \tag{135.104}$$

and matches the pure nonabelian chiral part

$$\frac{N_c}{6(2\pi)^3}\left[\mathrm{tr}_{\mathrm{f}}(F_L^3) - \mathrm{tr}_{\mathrm{f}}(F_R^3)\right]. \tag{135.105}$$

Second, define the invariant one-form

$$D_B\phi_B = d\phi_B - B. \tag{135.106}$$

On a five-dimensional extension X_5 of spacetime, the local inflow representative

$$\omega_{5,B}^{\mathrm{CFL}} = -\frac{D_B\phi_B}{2\pi} \wedge \frac{\mathrm{tr}_{\mathrm{f}}(F_L^2) - \mathrm{tr}_{\mathrm{f}}(F_R^2)}{2(2\pi)^2} \tag{135.107}$$

satisfies

$$d\omega_{5,B}^{\mathrm{CFL}} = \frac{F_B}{2\pi} \wedge \frac{\mathrm{tr}_{\mathrm{f}}(F_L^2) - \mathrm{tr}_{\mathrm{f}}(F_R^2)}{2(2\pi)^2}. \tag{135.108}$$

This is exactly the mixed $U(1)_B$-flavor term in (135.100) for $N_c q_B = 1$.

Proof. The field Σ has the same transformation law as the ordinary chiral field used in the WZW level-matching construction of Chapter 55. The anomaly-matching argument uses only the number of microscopic color copies and the $SU(N_f)_L$ background anomaly coefficient; it does not use the zero-density chiral condensate as the microscopic order parameter. Therefore the same level condition applies in the CFL phase:

$$n = N_c.$$

For $N_c = 3$ this gives (135.104). The gauged WZW functional at this level has precisely the pure nonabelian chiral descent class (135.105).

It remains to check the terms involving the baryon background. Under (135.103), the combination $D_B\phi_B = d\phi_B - B$ is invariant. Its exterior derivative is

$$d(D_B\phi_B) = -F_B. \tag{135.109}$$

The Bianchi identities for the background flavor curvatures give

$$d\ \mathrm{tr}_{\mathrm{f}}(F_L^2) = 0, \qquad d\ \mathrm{tr}_{\mathrm{f}}(F_R^2) = 0.$$

Taking the exterior derivative of (135.107) and using (135.109) gives (135.108). Comparing with (135.100), and using $N_c q_B = 1$, shows equality of the mixed anomaly polynomial. The vector diagonal subgroup is anomaly-free in both descriptions: setting $F_L = F_R$ makes both (135.105) and (135.108) vanish. $\square$

Remark 135.63 (Goldstone normalization and local order parameters). The scalar ϕ_B in (135.103) is normalized by the background $U(1)_B$ gauge parameter. This normalization should not be confused with the charge of a particular microscopic composite used to diagnose the phase. For example, with $q_B(q) = 1/3$, a local gauge-invariant determinant built from three diquark factors has baryon charge 2. Such an operator is locally represented by an appropriate integer-charge function of the compact Goldstone coordinate. The anomaly matching statement uses the coupling of the compact Goldstone circle to the background $U(1)_B$ bundle; it is not a statement that a chosen local order operator has unit baryon charge.

135.17 Comparison of Confinement Criteria

Several observables are often described by the same word "confinement." In this monograph they are kept distinct.

1. The zero-temperature Wilson-loop area law is a statement about the large-loop asymptotics of a gauge-invariant loop operator and the static source potential.
2. The thermal Polyakov-loop order parameter is a statement about center symmetry and the free energy of one static external charge.
3. The 't Hooft loop probes magnetic or center-twist sectors and is dual, in favorable regimes, to Wilson-loop ordering.
4. The center-vortex picture is a proposed infrared mechanism in which center-valued magnetic disorder controls the area law.
5. The dual-superconductor picture is a proposed infrared mechanism in which condensation of magnetically charged objects squeezes electric flux.

The first two items have clean operator definitions in lattice gauge theory. The last three can be developed into sharp statements only after the relevant line operators, disorder sectors, and continuum limits have been specified. Their value in this monograph is not as slogans, but as candidate structures to be tested in examples and connected to the Wilsonian and BV definitions of gauge theory developed earlier.

This comparison is the endpoint of the chapter's phase-specification discipline. A Polyakov loop, a Wilson loop, a diquark diagnostic, a transport coefficient, and a susceptibility are not interchangeable signals: they live in different limiting procedures and respond to different sources. When two such signals are correlated in a regime of QCD, that correlation is a dynamical conclusion to be proved, simulated with controlled errors, or recorded as a conjectural continuation. It is not part of the definition of the phase.

Open Problem 135.64 (Continuum QCD phase diagram with controlled errors). Construct the continuum QCD phase diagram in the (T, μ_B, m) variables with mathematically controlled thermodynamic and continuum limits, including the order of the pure-Yang–Mills deconfinement transition, the chiral transition in massless limits, the physical crossover region, the existence or nonexistence of a critical endpoint, and the color-superconducting regions at high baryon density.

Part XI

Constructive, Lattice, and Numerical Quantum Field Theory

Chapter 136

Constructive Status and Routes to Continuum QFT

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Constructive QFT asks for existence theorems for the objects used elsewhere in the monograph: Schwinger functions, Wightman functions, Hilbert spaces, local fields or local observable algebras, symmetry actions, positivity, and continuum limits. A regulator is part of a construction only after a limit is specified and the limiting object is shown to satisfy the required axioms or replacement framework.

136.1 Three Constructive Routes

Definition 136.1 (Constructive realization)**.** A constructive realization of a QFT in a specified framework consists of: correlation functions or an observable net; positivity or reflection positivity; covariance or the declared background-functorial replacement; locality; a reconstruction theorem or direct Hilbert-space construction; and control of the limiting procedure in a topology strong enough to verify those properties.

The main routes developed in this volume are cluster and phase-cell expansions, which construct measures and Schwinger functions by estimates; lattice constructions, which define finite systems exactly and study scaling limits; and stochastic quantization or singular SPDE, which construct invariant Euclidean measures through dynamics. These routes overlap in low-dimensional scalar examples. Their comparison is part of the content: agreement of independent constructions requires a separate identification theorem in the framework where the limiting objects are compared.

136.2 Status Catalog

Theory or regime	Constructive status	Framework boundary
Free scalar, spinor, and generalized-free fields	Constructed Wightman and Euclidean examples with explicit correlation functions and positive Hilbert-space reconstruction.	They calibrate the frameworks; interacting local dynamics require additional construction.
$P(\phi)_2$, including ϕ_2^4	Constructed non-Gaussian Euclidean and Wightman models for bounded-below polynomial interactions after Wick ordering and local renormalization.	Cluster expansion and Euclidean measure methods verify OS/Wightman properties in two spacetime dimensions.
ϕ_3^4, massive regime	Constructed non-Gaussian scalar models with mass, coupling, and vacuum-energy renormalization.	This is the fundamental three-dimensional scalar constructive example; the continuum measure is singular with respect to the Gaussian reference.
Sine–Gordon in two dimensions	Constructed in parameter ranges where the vertex interaction is renormalizable and the infrared setup is controlled.	The bosonization relation to massive Thirring-type descriptions requires its own operator and Hilbert-space identifications.
Yukawa$_2$ and related scalar–fermion models	Constructive results exist for specific two-dimensional models after fixing fermionic determinants or Pfaffians, Wick ordering, and ultraviolet renormalization.	Each theorem has model-dependent hypotheses; the Lagrangian label alone does not define a completed construction.
Gross–Neveu$_2$, massive asymptotically free regime	Constructive renormalization-group constructions exist for massive two-dimensional fermionic models.	They are important tests of rigorous asymptotic freedom in a non-gauge setting.
Two-dimensional Yang–Mills	Constructed by heat-kernel, holonomy, and projective-limit methods for compact gauge group.	This is a gauge-theory construction with no local propagating gluons; it does not settle four-dimensional Yang–Mills.
Stochastic quantization of ϕ_2^4 and ϕ_3^4	Constructed through Da Prato–Debussche, regularity structures, paracontrolled distributions, and related singular-SPDE methods, with renormalized invariant measures.	The SPDE route gives a dynamic construction of Euclidean measures and a precise algebraic meaning to the counterterms in the stochastic equation.
Three-dimensional Ising scaling limit as a CFT	Open as a constructive conformal field theory with full operator algebra and OS/Wightman reconstruction.	Numerical and bootstrap data are extremely constraining, but a continuum CFT construction remains a separate theorem.
ϕ_D^4, $D \geq 5$, standard positive lattice route	Triviality theorems force Gaussian scaling limits in the standard critical Ising/$\lambda\phi^4$ family.	This rules out that constructive route to an interacting scalar continuum limit above four dimensions.
ϕ_4^4, standard critical Ising/$\lambda\phi^4$ route	Marginal triviality theorems give Gaussian scaling limits for the standard reflection-positive critical problem.	The logically different possibility of another four-dimensional UV-complete QFT agreeing with formal ϕ_4^4 perturbation theory to all orders remains a separate open question.
Four-dimensional pure Yang–Mills	Finite Wilson lattices are rigorous compact-group integrals; continuum existence with a mass gap is the Clay Millennium problem.	The desired limit must construct gauge-invariant Schwinger functions or an equivalent local observable theory satisfying positivity, locality, and spectral properties.
QCD and the Standard Model	Available nonperturbative definitions are hybrid: lattice gauge theory for QCD-like sectors, perturbative and effective descriptions for electroweak and heavy-scale observables, and experimentally fixed parameters.	A single complete constructive continuum theorem for the full Standard Model is presently an open construction.

Table 136.1: Constructive status of representative QFT regimes. The entries separate completed constructions, triviality theorems, finite-regulator definitions, and open continuum-limit problems.

Reference anchors for Table 136.1. The scalar constructive line runs through Nelson, Glimm–Jaffe, Glimm–Jaffe–Spencer, Simon, Feldman–Osterwalder, Magnen–Seneor, Brydges, Fröhlich, and Sokal. The two-dimensional fermionic and asymptotically free examples include work of Feldman, Magnen, Rivasseau, and Seneor. Rigorous two-dimensional Yang–Mills may be approached through Driver, Sengupta, Levy, and related heat-kernel/holonomy constructions. The stochastic route uses Da Prato–Debussche, Hairer, Gubinelli–Imkeller–Perkowski, Kupiainen, Mourrat–Weber, and later singular-SPDE developments. Scalar triviality above and at four dimensions is represented by Aizenman, Fröhlich, and Aizenman–Duminil-Copin.

136.3 Stochastic Quantization as a Constructive Route

For Euclidean scalar ϕ^4, stochastic quantization starts from the formal Langevin equation

$$\partial_t \Phi = \Delta\Phi - m^2\Phi - \lambda\Phi^3 + \xi,$$

where t is an auxiliary stochastic time and ξ is spacetime-time white noise in the stochastic variables. In $D = 2, 3$, Φ is a distribution, so Φ^3 is defined only after renormalization. The renormalized equation has the form

$$\partial_t \Phi = \Delta\Phi - m^2\Phi - \lambda\Phi^3 + c_1(\varepsilon)\Phi + c_0(\varepsilon) + \xi_\varepsilon$$

before the cutoff $\varepsilon \downarrow 0$ is removed, with the precise counterterms determined by the chosen model and symmetry. Regularity structures, paracontrolled distributions, and rigorous RG methods provide the local expansion and reconstruction technology that make this limit a theorem in the constructed cases.

Open Problem 136.2 (Four-dimensional constructive gauge theory). Construct four-dimensional pure Yang–Mills, or QCD in a stated parameter regime, as a continuum local QFT with gauge-invariant observables, reflection positivity or Hilbert-space positivity, Euclidean and Lorentzian covariance, locality, and the spectral properties needed for particle and flux-tube questions. Lattice finite-volume integrals, perturbative asymptotic freedom, and numerical evidence are indispensable inputs, but the continuum existence and mass-gap theorem remains open.

Chapter 137

Constructive Scalar Models and OS Data

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
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- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Constructive scalar models are the basic interacting examples in which the objects used throughout this monograph are produced by estimates rather than postulated: Schwinger functions, reflection positivity, Euclidean covariance, growth bounds, and the Hilbert-space fields reconstructed from them. The word "constructive" will always mean a theorem about a limiting object in a specified topology. A finite cutoff integral, by itself, is only an input.

This chapter focuses on the scalar line beginning with $P(\phi)_2$ and Φ^4_3. It does not treat these theories as formal path integrals with unspecified counterterms. It records the regulated variables, the Wick and local counterterm coordinates appearing in the single regulated action, the estimates that control the infinite-volume and ultraviolet limits, and the OS data produced at the end.

137.1 Gaussian Reference Field and Wick Powers

Let $m > 0$, let

$$C = (-\Delta + m^2)^{-1}$$

be the massive Euclidean covariance on $\mathbb{R}^d$, and let μ_C be the centered Gaussian generalized field with covariance

$$\mathbb{E}_{\mu_C}[\phi(f)\phi(g)] = \int_{\mathbb{R}^d\times\mathbb{R}^d} f(x)C(x-y)g(y)\,\mathrm{d}^d x\,\mathrm{d}^d y. \tag{137.1}$$

The field ϕ is not a pointwise function. The expression $\phi(x)^k$ is therefore not defined until a regularization and a limiting procedure have been supplied.

Definition 137.1 (Wick powers by Gaussian regularization)**.** Let ρ_ε be a smooth approximate identity and set $\phi_\varepsilon = \rho_\varepsilon * \phi$. Write

$$C_\varepsilon(0) = \mathbb{E}_{\mu_C}[\phi_\varepsilon(x)^2],$$

which is translation independent. The Wick powers used below are the Hermite polynomials with variance $C_\varepsilon(0)$:

$$:\phi_\varepsilon^2: = \phi_\varepsilon^2 - C_\varepsilon(0), \tag{137.2}$$

$$:\phi_\varepsilon^3: = \phi_\varepsilon^3 - 3C_\varepsilon(0)\phi_\varepsilon, \tag{137.3}$$

$$:\phi_\varepsilon^4: = \phi_\varepsilon^4 - 6C_\varepsilon(0)\phi_\varepsilon^2 + 3C_\varepsilon(0)^2. \tag{137.4}$$

After smearing against a test function, the Wick power $:\phi^k:(f)$ is the $L^p(\mu_C)$-limit of $\int f(x) :\phi_\varepsilon(x)^k:\,\mathrm{d}^d x$ when that limit exists.

Origin of the Wick coefficients. For a centered Gaussian variable X with variance c, the Wick polynomial is

$$:X^n:= c^{n/2}H_n(X/\sqrt{c}),$$

where H_n is the probabilists' Hermite polynomial. In particular,

$$:X^2:= X^2 - c, \qquad :X^3:= X^3 - 3cX, \qquad :X^4:= X^4 - 6cX^2 + 3c^2.$$

The generating function

$$\exp(tX - \tfrac{1}{2}ct^2) = \sum_{n\geq 0} \frac{t^n}{n!} :X^n:$$

is characterized by the property that its expectation against $\exp(sX)$ equals

$$\mathbb{E}\big[\exp(tX - \tfrac{1}{2}ct^2)\exp(sX)\big] = \exp(cst + \tfrac{1}{2}cs^2).$$

Thus the coefficient of t^n is orthogonal to all polynomials of degree less than n and has leading term X^n, which is the defining orthogonality property of the Hermite polynomial. Expanding the generating function through order four gives the displayed coefficients.

The calculation-check companion `calculation-checks/constructive_scalar_spde_checks.py` verifies these coefficients and the cutoff-asymptotic coefficients used below.

137.2 Finite-Volume $P(\phi)_2$

Let $\Lambda \subset \mathbb{R}^2$ be a square or torus, and let

$$P(q) = \sum_{j=0}^{2r} p_j q^j, \qquad p_{2r} > 0,$$

be bounded below. Wick ordering is always relative to the Gaussian covariance used in the finite regulator. The regulated interaction is

$$V_{\varepsilon,\Lambda}(\phi) = \int_\Lambda :P(\phi_\varepsilon(x)): \, \mathrm{d}^2 x. \tag{137.5}$$

The finite-volume Schwinger functional is

$$S_{\varepsilon,\Lambda,n}(f_1, \dots, f_n) = \frac{1}{Z_{\varepsilon,\Lambda}} \int \prod_{i=1}^n \phi_\varepsilon(f_i) \, \exp[-V_{\varepsilon,\Lambda}(\phi)] \, \mathrm{d}\mu_C(\phi). \tag{137.6}$$

This formula is meaningful because the cutoff field is a finite-variance Gaussian process after smearing. The theorem is the existence and identification of the limits in which $\varepsilon \downarrow 0$ and $\Lambda \uparrow \mathbb{R}^2$.

Definition 137.2 (Negative Sobolev control norm)**.** Fix a bounded open set Λ_1 containing Λ and the supports reached by $\rho_\varepsilon * f$ for all sufficiently small ε. For $s > 0$, define

$$\|\phi\|^2_{H^{-s}(\Lambda_1)} := \sum_\nu (1 + \lambda_\nu)^{-s} \, |\phi(e_\nu)|^2, \tag{137.7}$$

where $\{e_\nu\}$ is any smooth orthonormal eigenbasis of a positive self-adjoint realization of $-\Delta$ on Λ_1 with eigenvalues λ_ν. Different standard boundary conditions give equivalent norms on smaller compact subsets, so the estimate below is independent of that auxiliary choice after the constants are changed.

Quoted Theorem 137.3 (Nelson–Glimm–Jaffe–Spencer stability in $P(\phi)_2$)**.** *Let $\Lambda \subset \mathbb{R}^2$ be bounded, let $P(q)$ be a real polynomial of even degree with positive leading coefficient and bounded below, and let*

$$V_{\varepsilon,\Lambda} = \int_\Lambda :P(\phi_\varepsilon(x)): \, \mathrm{d}^2 x$$

be Wick ordered with respect to the same massive covariance as the Gaussian measure. There are $s > 0$, $0 < \alpha < 2$, constants $A_\Lambda, B_\Lambda < \infty$, and $\varepsilon_0 > 0$, independent of $\varepsilon \in (0, \varepsilon_0]$, such that for μ_C-almost every field

$$V_{\varepsilon,\Lambda}(\phi) \geq -A_\Lambda - B_\Lambda \|\phi\|^\alpha_{H^{-s}(\Lambda_1)}. \tag{137.8}$$

Consequently, for every finite $p \geq 1$,

$$\sup_{0<\varepsilon\leq\varepsilon_0} \mathbb{E}_{\mu_C}[\exp(-pV_{\varepsilon,\Lambda})] < \infty. \tag{137.9}$$

Together with the Gaussian-chaos convergence of the Wick monomials, this implies that $\exp[-V_{\varepsilon,\Lambda}]$ *converges in $L^p(\mu_C)$, for every finite p, to the finite-volume $P(\phi)_2$ density.*

The quoted theorem is a compact formulation of the stability and exponential integrability estimates used in the Nelson, Glimm–Jaffe–Spencer, and Guerra–Rosen–Simon constructions of $P(\phi)_2$. Its proof is not the pointwise observation that P is bounded below; pointwise lower bounds for $:P(X):$ contain covariance-dependent negative constants that diverge as $\varepsilon \downarrow 0$. The constructive proof decomposes the covariance into scales, estimates the Wick-contraction combinatorics in each cell, absorbs lower-degree Wick terms into the positive leading part by large-field regulators, and uses the two-dimensional logarithmic covariance to obtain the subquadratic Sobolev exponent $\alpha < 2$. The theorem is therefore imported here as a quoted constructive estimate, not hidden inside the formal proof below.

Theorem 137.4 (Finite-volume ultraviolet limit in $P(\phi)_2$)**.** *For bounded Λ, bounded-below polynomial P, and Schwartz test functions supported in the interior of Λ, the limits*

$$S_{\Lambda,n}(f_1,\ldots,f_n) = \lim_{\varepsilon\downarrow 0} S_{\varepsilon,\Lambda,n}(f_1,\ldots,f_n)$$

exist after Wick ordering. The limiting finite-volume measure is absolutely continuous with respect to the massive Gaussian free field restricted to Λ, with density represented by $\exp[-\int_\Lambda :P(\phi(x)):\mathrm{d}^2x]$ as an L^p-limit for every finite p.

Proof. The L^p-convergence of the smeared Wick monomials follows from the covariance estimate

$$\mathbb{E}[:\phi_\varepsilon^k:(f):\phi_{\varepsilon'}^k:(f)] = k!\int f(x)C_{\varepsilon,\varepsilon'}(x-y)^k f(y)\,\mathrm{d}^2x\,\mathrm{d}^2y,$$

where the right-hand side converges because $C(x)$ has only logarithmic short-distance singularity in two dimensions and $C(x)^k$ is locally integrable for each fixed k. Hypercontractivity of Gaussian chaoses upgrades L^2-convergence to L^p-convergence for finite p.

The nontrivial point is uniform integrability of the exponential density. This is exactly the content of Quoted Theorem 137.3. To see how it enters, choose the Sobolev index $s > 0$ in (137.7). The massive Gaussian free field restricted to Λ_1 belongs to $H^{-s}(\Lambda_1)$ almost surely, because

$$\mathbb{E}_{\mu_C}\|\phi\|^2_{H^{-s}(\Lambda_1)} = \sum_\nu (1+\lambda_\nu)^{-s}\langle e_\nu, Ce_\nu\rangle < \infty$$

for $s > 0$ in two dimensions; the last inequality uses the elliptic bound $\langle e_\nu, Ce_\nu\rangle \le C_0(1+\lambda_\nu)^{-1}$ and Weyl's law for $-\Delta$ on a bounded planar region. Fernique's theorem gives some $\eta > 0$ with

$$\mathbb{E}_{\mu_C}\exp\{\eta\|\phi\|^2_{H^{-s}(\Lambda_1)}\} < \infty.$$

Since $\alpha < 2$, Young's inequality gives, for every $p < \infty$,

$$pB_\Lambda\|\phi\|^\alpha_{H^{-s}} \le \eta\|\phi\|^2_{H^{-s}} + C_{p,\eta,\alpha,B_\Lambda}.$$

Combining this with (137.8) proves (137.9). The L^q-convergence of the Wick polynomial integrals gives convergence of $V_{\varepsilon,\Lambda}$ in probability. Applying (137.9) with an exponent strictly larger than p gives uniform integrability of $\exp[-pV_{\varepsilon,\Lambda}]$, hence L^p-convergence of the densities. Finally, the smeared insertions $\phi_\varepsilon(f_i)$ converge in every Gaussian L^q space, and Holder's inequality combines this convergence with the uniform exponential moments. Therefore the normalized Schwinger functions have the stated finite-volume ultraviolet limits. The argument is local in Λ, so test functions supported away from the boundary are unaffected by the boundary condition used to define the finite covariance. $\square$

137.3 Cluster Expansion and Infinite Volume

The ultraviolet limit at fixed Λ does not yet construct the infinite-volume QFT. For massive $P(\phi)_2$ the large-volume limit is controlled by a cluster expansion. The following formulation isolates the estimate that turns local finite-volume data into global Schwinger functions.

Definition 137.5 (Polymer activity and cluster norm)**.** Partition $\mathbb{R}^2$ into unit cells Δ. A polymer X is a finite connected union of cells. Fix a finite-range symmetric incompatibility relation $X \nsim Y$; in the cell decompositions used below this means that the R_0-cell neighbourhoods of X and Y intersect, for a fixed integer R_0. Write $X \sim Y$ for compatibility. A polymer activity assigns to X a functional

$$K(X, \phi)$$

depending only on the field near X. For constants $a, b > 0$, define the numerical majorant

$$\kappa(X) := \sup_{\phi} \left(|K(X, \phi)| e^{-b\|\phi\|_X^{2r}} \right) \tag{137.10}$$

and the weighted cluster norm

$$\|K\|_{a,b} = \sup_{\Delta_0} \sum_{X \ni \Delta_0} e^{a|X|} \kappa(X), \tag{137.11}$$

where $\|\phi\|_X$ is the chosen local large-field norm. The exact constructive proof fixes the Sobolev regularity and large-field regulator; (137.11) records the structural requirement.

Lemma 137.6 (Penrose tree-graph bound)**.** *Let $X_1, \ldots, X_n$ be polymers and set*

$$u_{ij} := \begin{cases} 1, & X_i \nsim X_j, \\ 0, & X_i \sim X_j. \end{cases}$$

For $n \geq 2$, define the hard-core Ursell coefficient

$$\varphi_{\mathrm{c}}(X_1, \ldots, X_n) := \sum_{G \in \mathcal{C}_n} \prod_{\{i,j\} \in E(G)} (-u_{ij}), \tag{137.12}$$

where $\mathcal{C}_n$ is the set of connected simple graphs on $\{1, \ldots, n\}$. For $n = 1$, set $\varphi_{\mathrm{c}}(X_1) = 1$. Then

$$|\varphi_{\mathrm{c}}(X_1, \ldots, X_n)| \leq \sum_{T \in \mathcal{T}_n} \prod_{\{i,j\} \in E(T)} u_{ij}, \tag{137.13}$$

where $\mathcal{T}_n$ is the set of spanning trees on $\{1, \ldots, n\}$.

Proof. Fix an ordering of the edges of the complete graph on $\{1, \ldots, n\}$. For a connected graph G, let $T(G)$ be its lexicographically first spanning tree. Penrose's partition scheme assigns to every tree T a graph $R(T)$, containing T, such that the connected graphs are the disjoint union of intervals

$$\{G : \ T \subseteq G \subseteq R(T)\}.$$

Explicitly, an edge $e \notin T$ is put in $R(T)$ precisely when adding e to T does not change the lexicographically first spanning tree. Therefore

$$\sum_{\substack{G \in \mathcal{C}_n \\ T(G) = T}} \prod_{e \in E(G)} (-u_e) = (-1)^{|T|} \prod_{e \in E(T)} u_e \prod_{e \in E(R(T)) \setminus E(T)} (1 - u_e).$$

Since each u_e is 0 or 1, the absolute value of the right-hand side is at most $\prod_{e\in E(T)} u_e$. Summing over T proves (137.13). $\square$

Theorem 137.7 (Kotecky–Preiss/Brydges–Kennedy convergence criterion)**.** *Let $\mathcal{P}_\Lambda$ be the finite set of polymers contained in a finite cell region Λ. Let*

$$Z_\Lambda(K) = \sum_{\mathcal{X}\subset\mathcal{P}_\Lambda \text{ pairwise compatible}} \prod_{X\in\mathcal{X}} K(X)$$

be the abstract hard-core polymer partition function. Suppose there are numbers $a > a' > 0$ and $\epsilon > 0$ such that

$$\sup_{\Delta_0} \sum_{X\ni\Delta_0} \kappa(X) e^{a|X|} \le \epsilon \tag{137.14}$$

and $B_{R_0}\epsilon \le a'$, where B_{R_0} is the cell-neighbourhood constant defined by

$$|\{\Delta : \mathrm{dist}_{\mathrm{cell}}(\Delta, X) \le R_0\}| \le B_{R_0}|X|.$$

Then

$$\log Z_\Lambda(K) = \sum_{n\ge1} \frac{1}{n!} \sum_{X_1,\dots,X_n\in\mathcal{P}_\Lambda} \varphi_{\mathrm{c}}(X_1,\dots,X_n) \prod_{i=1}^n K(X_i) \tag{137.15}$$

is absolutely convergent uniformly in Λ. More precisely, for each root polymer X_0,

$$\sum_{n\ge1} \frac{1}{(n-1)!} \sum_{X_2,\dots,X_n\in\mathcal{P}_\Lambda} |\varphi_{\mathrm{c}}(X_0, X_2,\dots,X_n)| \prod_{i=2}^n \kappa(X_i) \le e^{a'|X_0|}. \tag{137.16}$$

Consequently connected correlations of observables supported in separated regions A and B, whose source activities obey the same local majorant, satisfy

$$|\langle F_A G_B\rangle - \langle F_A\rangle\langle G_B\rangle| \le C_{F,G} e^{-m_{\mathrm{cl}}\,\mathrm{dist}(A,B)} \tag{137.17}$$

for a positive cluster mass m_{cl} fixed by the exponential weight in the norm and by the covariance decay.

Proof. For finite Λ, the Mayer identity is a finite algebraic identity. It expands the logarithm of the hard-core gas in connected incompatibility graphs and gives (137.15), with the coefficient φ_{c} defined in (137.12). Equivalently, it is the formal Taylor expansion of the analytic branch of $\log Z_\Lambda$ satisfying $\log Z_\Lambda(0) = 0$; the absolute convergence proved below is what makes this formal expansion a uniformly convergent function in the stated polydisc of activities.

Apply Lemma 137.6. Root each tree at the vertex carrying X_0 and orient its edges away from the root. Let $F_N(X)$ be the total majorant of rooted incompatibility trees of depth at most N, with root polymer X, excluding the activity of the root itself. The tree bound gives the recursive domination

$$F_{N+1}(X) \le \exp\left(\sum_{Y\nsim X} \kappa(Y) F_N(Y)\right), \qquad F_0(X) = 1. \tag{137.18}$$

Assume inductively that $F_N(Y) \le e^{a'|Y|}$. Since $a > a'$,

$$\sum_{Y\nsim X} \kappa(Y) e^{a'|Y|} \le \sum_{\Delta\in N_{R_0}(X)} \sum_{Y\ni\Delta} \kappa(Y) e^{a|Y|} \le B_{R_0}\epsilon|X| \le a'|X|. \tag{137.19}$$

Thus $F_{N+1}(X) \le e^{a'|X|}$. Letting $N \to \infty$ proves (137.16). Multiplying by the root majorant $\kappa(X_0)$ and summing over X_0 with (137.14) proves absolute convergence of (137.15).

For two separated observables, the connected part can only be produced by a connected cluster intersecting both supports. Every such cluster contains a path of incompatible polymers joining the source near A to the source near B. Along such a path,

$$\sum_{X \text{ on the path}} |X| \ge c_{R_0} \operatorname{dist}(A, B) \tag{137.20}$$

for a positive cell-metric constant c_{R_0}. In the rooted-tree estimate one may use the a'-weighted recursion while retaining the unused factor $e^{-(a-a')|X|}$ from the a-weighted smallness norm. Equation (137.20) then gives

$$m_{\mathrm{cl}} \ge c_{R_0}(a - a')$$

up to the additional exponential-decay rate supplied by the massive Gaussian covariance. The source activity norms for F_A and G_B supply $C_{F,G}$. This proves (137.17). □

In the Glimm–Jaffe–Spencer $P(\phi)_2$ construction, the activities $K(X, \phi)$ arise by expanding around cell-local interaction pieces while retaining large-field regulators. The theorem above is the abstract polymer estimate of Kotecky–Preiss and Brydges–Kennedy,[1] written here in the hard-core finite-range form needed for the constructive scalar models. It is not an optional heuristic: it is the analytic step that justifies taking $\Lambda \uparrow \mathbb{R}^2$, proves exponential clustering in the massive phase, and gives uniform bounds on the Schwinger functions.

Source derivatives and connected clusters. The polymer estimate becomes a QFT statement only after one explains how local insertions are read from the polymer gas. Let $\mathfrak{S}$ be a finite set of source labels. For each $\alpha \in \mathfrak{S}$, let A_α be the cell neighbourhood of the support of the corresponding test insertion and let $b_\alpha(X)$ be a source weight that vanishes unless X meets A_α. At finite volume set

$$K_\lambda(X) = K(X) \exp\left(\sum_{\alpha \in \mathfrak{S}} \lambda_\alpha b_\alpha(X) \right)$$

and define $Z_\Lambda(\lambda)$ by replacing K with K_λ in the hard-core partition function. The connected source coefficient is the finite identity

$$C^{(\Lambda)}_{\alpha_1 \cdots \alpha_m} = \partial_{\lambda_{\alpha_1}} \cdots \partial_{\lambda_{\alpha_m}} \log Z_\Lambda(\lambda) \Big|_{\lambda=0}.$$

Substituting K_λ into the absolutely convergent cluster expansion gives

$$C^{(\Lambda)}_{\alpha_1 \cdots \alpha_m} = \sum_{n \ge 1} \frac{1}{n!} \sum_{X_1, \ldots, X_n \in \mathcal{P}_\Lambda} \varphi_{\mathrm{c}}(X_1, \ldots, X_n) \prod_{i=1}^{n} K(X_i)$$
$$\times \sum_{\tau : \{1, \ldots, m\} \to \{1, \ldots, n\}} \prod_{r=1}^{m} b_{\alpha_r}(X_{\tau(r)}).$$

[1]See R. Kotecky and D. Preiss, "Cluster expansion for abstract polymer models," *Commun. Math. Phys.* 103 (1986), and D. C. Brydges and T. Kennedy, "Mayer expansions and the Hamilton-Jacobi equation," *J. Stat. Phys.* 48 (1987).

Thus a nonzero term in a connected m-point source coefficient contains a connected incompatibility graph whose vertices collectively meet all source neighbourhoods. If two groups of sources lie in cell regions that cannot be joined by an incompatibility path, the finite partition function factorizes and the mixed cumulant vanishes. If they can be joined only through a path of total polymer size at least proportional to their cell distance, the unused exponential weight in (137.14) gives precisely the decay used in (137.17). For local fields in $P(\phi)_2$, the source weights b_α are obtained by expanding the regulated source functional together with the local interaction pieces; the cluster estimate controls the resulting connected Schwinger distributions after the finite-volume ultraviolet limit has already been taken.

Quoted Theorem 137.8 (Massive $P(\phi)_2$ OS output from the cluster expansion)**.** *Let P be an even-degree real polynomial bounded below with positive leading coefficient, let $m > 0$, and choose a reflection-positive finite-volume ultraviolet regulator. Assume the finite-volume ultraviolet limit of Theorem 137.4 has been taken, and that the resulting polymer activities satisfy the uniform cluster norm bound* (137.14) *in the massive phase. Then the thermodynamic limits*

$$S_n(f_1, \ldots, f_n) = \lim_{\Lambda \uparrow \mathbb{R}^2} S_{\Lambda,n}(f_1, \ldots, f_n)$$

exist for Schwartz test functions. The hierarchy $S = \{S_n\}_{n \geq 0}$ satisfies Euclidean covariance, permutation symmetry, reflection positivity, distributional regularity, the OS growth hypotheses required for reconstruction, and exponential clustering of connected Schwinger functions.

The quoted theorem is the constructive $P(\phi)_2$ output statement. The source lineage is the Glimm–Jaffe–Spencer constructive cluster-expansion program for $P(\phi)_2$, as presented for example in Glimm–Jaffe's functional-integral account. The inputs include the polynomial interaction, Wick ordering, stability, ultraviolet convergence, the polymer norm estimate, and preservation of reflection positivity along the regulator sequence.

Spectral consequence of Euclidean clustering. The constructive input is the exponential clustering statement in the quoted theorem; the passage from such a bound to a Hamiltonian gap is a spectral support calculation. Assume OS reconstruction has been performed from a scalar Schwinger hierarchy satisfying Quoted Theorem 137.8. Let A be a local Euclidean-time-zero observable with $\langle \Omega, A\Omega \rangle = 0$, and suppose its two-point function obeys, for some $m_{\mathrm{E}} > 0$,

$$0 \leq \langle \Omega, A^* e^{-\tau H} A\Omega \rangle \leq C_A e^{-m_{\mathrm{E}}\tau}, \qquad \tau \geq 0.$$

Then the spectral measure of H in the vector $A\Omega$ is supported in $[m_{\mathrm{E}}, \infty)$. If the set of such local vectors is dense in the orthogonal complement of the vacuum, the reconstructed Hamiltonian has mass gap at least m_{E}.

By the spectral theorem,

$$\langle \Omega, A^* e^{-\tau H} A\Omega \rangle = \int_{[0,\infty)} e^{-\tau E} \, \mathrm{d}\mu_A(E), \qquad \mathrm{d}\mu_A(E) = \mathrm{d}\langle A\Omega, P_H(E) A\Omega \rangle.$$

If $\mu_A([0, m_{\mathrm{E}} - \delta]) > 0$ for some $\delta > 0$, then

$$\int e^{-\tau E} \, \mathrm{d}\mu_A(E) \geq e^{-(m_{\mathrm{E}} - \delta)\tau} \mu_A([0, m_{\mathrm{E}} - \delta]).$$

This contradicts the assumed upper bound $C_A e^{-m_{\mathrm{E}}\tau}$ as $\tau \to \infty$. Therefore no spectral weight lies below m_{E} in the vector $A\Omega$. If such local vectors are dense in $\Omega^{\perp}$, the spectral projection $P_H([0, m_{\mathrm{E}}))$ annihilates a dense subspace of $\Omega^{\perp}$ and hence vanishes there. This is the claimed gap.

137.4 Hamiltonian Viewpoint and Number-Operator Bounds

The original construction of ϕ_2^4 can also be formulated in the Hamiltonian representation. Let $\mathcal{F}$ be the massive one-particle Fock space on a spatial line or circle, H_0 the free Hamiltonian, and N the number operator. With spatial cutoff $g \geq 0$, the interaction is

$$V(g) = \lambda \int g(x) :\phi(x)^4: \, \mathrm{d}x.$$

The decisive estimates are number-operator bounds of the form

$$\|(N+1)^{-2} V(g) (N+1)^{-2}\| \leq C_g |\lambda|, \qquad \|V(g)\psi\| \leq \alpha \|H_0 \psi\| + \beta \|\psi\| \tag{137.21}$$

with $\alpha < 1$ after localization and choice of constants. These estimates imply self-adjointness and lower boundedness of

$$H(g) = H_0 + V(g) + E(g),$$

where $E(g)$ is the vacuum-energy coordinate in the regulated Hamiltonian. The removal of the spatial cutoff is not a formal limit of operators on a fixed dense domain; it is controlled by spatial cluster estimates and by convergence of local vacuum expectations.

Remark 137.9 (Relation between Hamiltonian and Euclidean constructions). The Hamiltonian and Euclidean $P(\phi)_2$ constructions produce the same local QFT only after a comparison theorem identifies the Schwinger functions generated by $e^{-tH(g)}$ with the Euclidean measure construction and then passes to the infinite-volume limit. The common notation ϕ_2^4 is not itself such an identification.

137.5 Φ_3^4, Phase Cells, and Local Coordinates

In three Euclidean dimensions the quartic scalar theory is superrenormalizable, but Wick ordering alone does not remove every local divergence. A regulator may be given by a momentum cutoff, lattice spacing, or mollifier. A typical finite-volume action is

$$S_{\varepsilon,\Lambda}(\phi) = \frac{1}{2}\langle \phi, (-\Delta + m^2)\phi \rangle + \int_{\Lambda} \left[\lambda \phi_\varepsilon^4 + a_\varepsilon \phi_\varepsilon^2 + b_\varepsilon\right] \mathrm{d}^3 x. \tag{137.22}$$

The coefficients a_ε and b_ε are local coordinates of the regulated action. They are not a second Lagrangian and are not optional decorations. Their values are fixed by the renormalization conditions or by the constructive normalization scheme.

For connected scalar ϕ^4 graphs in d Euclidean dimensions with E external legs, V quartic vertices, and L loops, the superficial degree of divergence is

$$\omega_d = dL - 2I = d - \frac{d-2}{2} E + (d-4)V. \tag{137.23}$$

In $d = 3$, only finitely many local divergences occur: vacuum-energy and two-point local counterterms. The four-point and higher connected kernels are superficially convergent after subdivergences are removed. The graph identities are

$$4V = 2I + E, \qquad L = I - V + 1.$$

Substituting $I = 2V - E/2$ into $dL - 2I$ gives

$$d(I - V + 1) - 2I = (d-2)I - dV + d = d - \frac{d-2}{2}E + (d-4)V.$$

For $d = 3$, this is $3 - E/2 - V$. At fixed E, only finitely many V can satisfy $\omega_3 \geq 0$, and $E \geq 4$ is convergent except for subdivergences. Thus all primitive local divergences are proportional to the identity or to ϕ^2.

With sharp momentum cutoff $|p| \leq \Lambda$,

$$I_d(\Lambda, m) = \int_{|p|\leq\Lambda} \frac{\mathrm{d}^d p}{(2\pi)^d} \frac{1}{p^2 + m^2}$$

satisfies

$$I_2(\Lambda, m) = \frac{1}{2\pi}\log\Lambda + O(1), \qquad I_3(\Lambda, m) = \frac{\Lambda}{2\pi^2} + O(1). \tag{137.24}$$

In polar coordinates,

$$I_d(\Lambda, m) = \frac{|S^{d-1}|}{(2\pi)^d} \int_0^\Lambda \frac{r^{d-1}}{r^2 + m^2}\,\mathrm{d}r.$$

For $d = 2$, $|S^1| = 2\pi$ and the radial integral is $\frac{1}{2}\log(\Lambda^2 + m^2) - \frac{1}{2}\log m^2$, giving the coefficient $(2\pi)^{-1}$. For $d = 3$, $|S^2| = 4\pi$ and the radial integral is $\Lambda - m\arctan(\Lambda/m)$, giving the coefficient $(2\pi^2)^{-1}$.

In the normalization $\lambda\phi^4$, Wick ordering the quartic interaction produces

$$\lambda :\phi_\varepsilon^4: = \lambda\phi_\varepsilon^4 - 6\lambda C_\varepsilon(0)\phi_\varepsilon^2 + 3\lambda C_\varepsilon(0)^2. \tag{137.25}$$

Thus the leading mass-coordinate divergence is fixed by the six single contractions in ϕ^4. In three dimensions a logarithmic two-loop local two-point divergence also appears. In a BPHZ or phase-cell scheme it is encoded in the finite-regulator coordinate a_ε, while the vacuum diagrams are encoded in b_ε.

It is useful to separate the Wick-coordinate bookkeeping from the remaining logarithmic local coordinate. Write the same cutoff action temporarily in normal-ordered coordinates as

$$S_{\varepsilon,\Lambda} = S_{0,\varepsilon} + \int_\Lambda \left[\lambda :\phi_\varepsilon^4:_\varepsilon + \alpha_\varepsilon :\phi_\varepsilon^2:_\varepsilon + \beta_\varepsilon\right] \mathrm{d}^3x. \tag{137.26}$$

Here the subscript on $:\cdot:_\varepsilon$ records that Wick ordering is relative to the same cutoff covariance used in the Gaussian reference measure. Relative to the un-Wick-ordered coordinates in (137.22),

$$a_\varepsilon = \alpha_\varepsilon - 6\lambda C_\varepsilon(0), \qquad b_\varepsilon = \beta_\varepsilon - \alpha_\varepsilon C_\varepsilon(0) + 3\lambda C_\varepsilon(0)^2. \tag{137.27}$$

The linear divergence in a_ε is therefore the one-loop Wick conversion; the logarithmic divergence is the leading divergent part of α_ε.

This is also the coordinate in which the comparison with stochastic quantization must be made. To avoid overloading the symbol λ, rename the Euclidean quartic coefficient in (137.22) temporarily as g_4. If the same cutoff action is used to generate a Langevin equation in the convention of Chapter 144, and if that equation is written with dynamic cubic coefficient λ_{dyn} and local linear drift coefficient M_ε, then

$$\lambda_{\mathrm{dyn}} = 4g_4, \qquad M_\varepsilon = -2a_\varepsilon.$$

Therefore the normal-ordered mass coordinate is

$$\alpha_\varepsilon = -\frac{1}{2}M_\varepsilon + \frac{3}{2}\lambda_{\mathrm{dyn}}C_\varepsilon(0).$$

Proposition 144.90 expands this identity in the dynamic BPHZ coordinates. The comparison is made in α_ε, not in the raw drift coefficient, because the one-loop Wick conversion is a coordinate change between the two presentations.

Proposition 137.10 (Vacuum coordinate and normalized Schwinger data)**.** *Fix a cutoff ε, a bounded volume Λ, and a source J supported in Λ. Let $Z^{(\beta)}_{\varepsilon,\Lambda}[J]$ be the finite-volume source partition function formed from (137.26). If β_ε is replaced by $\beta_\varepsilon + c_\varepsilon$, with λ, α_ε, and the Gaussian reference measure fixed, then*

$$Z^{(\beta+c)}_{\varepsilon,\Lambda}[J] = e^{-c_\varepsilon|\Lambda|}Z^{(\beta)}_{\varepsilon,\Lambda}[J]. \tag{137.28}$$

Hence every normalized finite-volume Schwinger function is unchanged:

$$S^{(\beta+c)}_{n,\varepsilon,\Lambda}(f_1,\dots,f_n) = S^{(\beta)}_{n,\varepsilon,\Lambda}(f_1,\dots,f_n). \tag{137.29}$$

The connected source functional

$$W^{(\beta)}_{\varepsilon,\Lambda}[J] := \log Z^{(\beta)}_{\varepsilon,\Lambda}[J] - \log Z^{(\beta)}_{\varepsilon,\Lambda}[0]$$

is also unchanged, while the unnormalized thermodynamic coordinates shift by

$$p^{(\beta+c)}_{\varepsilon,\Lambda} = p^{(\beta)}_{\varepsilon,\Lambda} - c_\varepsilon, \qquad f^{(\beta+c)}_{\varepsilon,\Lambda} = f^{(\beta)}_{\varepsilon,\Lambda} + c_\varepsilon, \tag{137.30}$$

where $p_{\varepsilon,\Lambda} = |\Lambda|^{-1}\log Z_{\varepsilon,\Lambda}[0]$ and $f_{\varepsilon,\Lambda} = -|\Lambda|^{-1}\log Z_{\varepsilon,\Lambda}[0]$. At fixed α_ε and λ, the same shift changes the un-Wick-ordered coordinate in (137.27) by

$$b_\varepsilon \longmapsto b_\varepsilon + c_\varepsilon, \qquad a_\varepsilon \longmapsto a_\varepsilon.$$

Thus the stochastic-quantization drift chart determines the mass coordinate but not the vacuum coordinate; the latter is fixed only after specifying the partition-function, pressure, or free-energy normalization.

Proof. The replacement $\beta_\varepsilon \mapsto \beta_\varepsilon + c_\varepsilon$ adds the constant $c_\varepsilon|\Lambda|$ to the finite-volume action. Pulling this factor out of the integral gives (137.28). Differentiating the source partition function, or equivalently inserting finitely many smeared cutoff fields before normalization, multiplies numerator and denominator by the same factor $e^{-c_\varepsilon|\Lambda|}$, proving (137.29). Taking the difference $\log Z[J] - \log Z[0]$ cancels the same additive logarithmic term, while dividing $\log Z[0] - c_\varepsilon|\Lambda|$ by $|\Lambda|$ gives the pressure and free-energy shifts in (137.30). Finally, the coordinate conversion (137.27) is affine in β_ε and independent of β_ε in a_ε. The functional derivative of an additive constant in the action is zero, so the Langevin drift sees no c_ε term. $\square$

Lemma 137.11 (Universal logarithm in the three-dimensional sunset kernel)**.** *Let*

$$C_m(x) = \frac{e^{-m|x|}}{4\pi|x|}$$

be the massive covariance in $\mathbb{R}^3$*, and let* C_ε *be obtained from* C_m *by a smooth ultraviolet mollifier at length* ε*. For every fixed* $R > 0$*,*

$$J_\varepsilon(R) := \int_{|x|\le R} C_\varepsilon(x)^3 \,\mathrm{d}^3x = \frac{1}{16\pi^2}\log\frac{1}{\varepsilon} + O_R(1). \tag{137.31}$$

The coefficient of the logarithm is independent of the mollifier.

Proof. Near the origin,

$$C_m(x) = \frac{1}{4\pi|x|} + O(1).$$

On the annulus $A_{\varepsilon,R} = \{c\varepsilon \le |x| \le R\}$, with c fixed large enough for the chosen mollifier, standard estimates on smooth approximate identities give

$$C_\varepsilon(x) = \frac{1}{4\pi|x|} + O(1) + O(\varepsilon|x|^{-2}).$$

The last error contributes $O(1)$ to the integral of C_ε^3 over $A_{\varepsilon,R}$. The ball $|x| \le c\varepsilon$ contributes only a mollifier-dependent bounded term after the logarithmic part has been extracted, because its volume is $O(\varepsilon^3)$ and the mollified covariance is $O(\varepsilon^{-1})$. Hence the divergent part is

$$\int_{c\varepsilon}^{R} 4\pi r^2 \left(\frac{1}{4\pi r}\right)^3 \mathrm{d}r = \frac{1}{16\pi^2}\log\frac{R}{c\varepsilon},$$

which proves (137.31). □

Lemma 137.12 (Two-loop local mass coordinate in the chapter's normalization)**.** *Use the normal-ordered coordinates* (137.26)*. Through order* λ^2 *and first order in* α_ε*, the logarithmically divergent local part of the connected two-point Schwinger function is*

$$\big(96\lambda^2 J_\varepsilon(R) - 2\alpha_\varepsilon\big)\int f(u)C_m(u-z)C_m(z-v)g(v)\,\mathrm{d}^3u\,\mathrm{d}^3z\,\mathrm{d}^3v, \tag{137.32}$$

up to finite terms and derivative local terms. Therefore the logarithmic mass coordinate is

$$\alpha_\varepsilon = 48\lambda^2 J_\varepsilon(R) + \alpha_{\mathrm{fin}} + O(\lambda^3) = \frac{3\lambda^2}{\pi^2}\log\frac{1}{\varepsilon} + \alpha_{\mathrm{fin}} + O(1) + O(\lambda^3), \tag{137.33}$$

with α_{fin} *fixed by the chosen finite mass normalization.*

Proof. The normal-ordered product identity for two quartic vertices is

$$:\phi_\varepsilon(x)^4::\phi_\varepsilon(y)^4:= \sum_{k=0}^{4}\binom{4}{k}^2 k!\,C_\varepsilon(x-y)^k :\phi_\varepsilon(x)^{4-k}\phi_\varepsilon(y)^{4-k}: . \tag{137.34}$$

The logarithmic two-point local term is the $k = 3$ term:

$$\binom{4}{3}^2 3!\, C_\varepsilon(x-y)^3 :\phi_\varepsilon(x)\phi_\varepsilon(y): = 96\, C_\varepsilon(x-y)^3 :\phi_\varepsilon(x)\phi_\varepsilon(y): .$$

The second order in the expansion of $\exp[-\lambda \int :\phi^4:]$ has the factor $\lambda^2/2$. Contracting $:\phi(x)\phi(y):$ with the two external fields gives

$$C_m(u-x)C_m(v-y) + C_m(u-y)C_m(v-x).$$

In the singular region $x - y$ small, both terms have the same local part. Thus the quartic vertices contribute

$$\frac{\lambda^2}{2} \cdot 96 \cdot 2\, J_\varepsilon(R) \int f(u)C_m(u-z)C_m(z-v)g(v)\, \mathrm{d}^3u\, \mathrm{d}^3z\, \mathrm{d}^3v,$$

which is the $96\lambda^2 J_\varepsilon(R)$ term in (137.32). The insertion $\alpha_\varepsilon \int :\phi^2:$ contributes at first order

$$-\alpha_\varepsilon \int \langle \phi(u)\phi(v) :\phi(z)^2: \rangle_0\, \mathrm{d}^3z = -2\alpha_\varepsilon \int C_m(u-z)C_m(z-v)\, \mathrm{d}^3z.$$

Cancellation of the logarithmic coefficient gives $\alpha_\varepsilon = 48\lambda^2 J_\varepsilon(R) + \alpha_{\mathrm{fin}}$. The coefficient in (137.33) follows from Lemma 137.11. The dependence on the radius R, on the mollifier, and on the subtraction convention is finite and is absorbed into α_{fin}. □

Finite-cutoff stability of the local normal-ordered polynomial. Fix a cutoff covariance with $c_\varepsilon = C_\varepsilon(0) < \infty$. Let $\lambda > 0$, and define the one-site polynomial

$$p_\varepsilon(q) = \lambda :q^4:_\varepsilon + \alpha_\varepsilon :q^2:_\varepsilon + \beta_\varepsilon.$$

Equivalently,

$$p_\varepsilon(q) = \lambda y^2 + A_\varepsilon y + B_\varepsilon, \qquad y = q^2, \quad A_\varepsilon = \alpha_\varepsilon - 6\lambda c_\varepsilon, \quad B_\varepsilon = 3\lambda c_\varepsilon^2 - \alpha_\varepsilon c_\varepsilon + \beta_\varepsilon.$$

For all $q \in \mathbb{R}$,

$$p_\varepsilon(q) \geq B_\varepsilon - \frac{(A_\varepsilon^-)^2}{4\lambda}, \qquad A_\varepsilon^- := \max\{-A_\varepsilon, 0\}. \tag{137.35}$$

Consequently the finite-cutoff interaction

$$\int_\Lambda p_\varepsilon(\phi_\varepsilon(x))\, \mathrm{d}^3x$$

is bounded below by

$$|\Lambda| \left(B_\varepsilon - \frac{(A_\varepsilon^-)^2}{4\lambda} \right).$$

The Wick identities in Definition 137.1 give

$$:q^4:_\varepsilon = q^4 - 6c_\varepsilon q^2 + 3c_\varepsilon^2, \qquad :q^2:_\varepsilon = q^2 - c_\varepsilon.$$

With $y = q^2 \geq 0$, the polynomial becomes $\lambda y^2 + A_\varepsilon y + B_\varepsilon$. If $A_\varepsilon \geq 0$, its minimum on $y \geq 0$ is attained at $y = 0$ and equals B_ε. If $A_\varepsilon < 0$, the unconstrained minimum lies at $y = -A_\varepsilon/(2\lambda) > 0$

and equals $B_\varepsilon - A_\varepsilon^2/(4\lambda)$. These two cases give (137.35). Integrating the pointwise bound over Λ proves the final assertion.

This finite-cutoff lower bound does not by itself prove the continuum Φ^4_3 theorem, because the lower bound contains divergent local constants that must be absorbed into β_ε and because uniform polymer smallness requires scale-by-scale estimates on fluctuations rather than only a one-site lower bound. It is nevertheless the first stability inequality in the constructive chain: before phase cells are introduced, the regulated density is an ordinary integrable function after the local vacuum-energy coordinate is chosen.

Reflection positivity under split local interactions. Let $(\mathfrak{A}, \theta, \langle\cdot\rangle_0)$ be a reflection-positive Euclidean probability space, and let $\mathfrak{A}_+ \subset \mathfrak{A}$ be the algebra generated by fields supported at positive Euclidean time. Thus

$$\langle \theta F\, F\rangle_0 \geq 0, \qquad F \in \mathfrak{A}_+.$$

Let V_+ be a real measurable functional affiliated with $\mathfrak{A}_+$, and suppose $e^{-V_+}F \in L^2$ for the positive-time test polynomials under consideration. Define the interacting expectation by the unnormalized density

$$\mathrm{d}\nu = e^{-V_+ - \theta V_+}\,\mathrm{d}\mu_0.$$

Then the unnormalized interacting form is reflection positive:

$$\langle \theta F\, F\rangle_\nu = \int \theta F\, F\, e^{-V_+ - \theta V_+}\,\mathrm{d}\mu_0 \geq 0, \qquad F \in \mathfrak{A}_+. \tag{137.36}$$

Because θ is an anti-linear algebra involution and V_+ is real,

$$\theta(e^{-V_+}F) = e^{-\theta V_+}\theta F.$$

Therefore

$$\int \theta F\, F\, e^{-V_+ - \theta V_+}\,\mathrm{d}\mu_0 = \int \theta(e^{-V_+}F)\,(e^{-V_+}F)\,\mathrm{d}\mu_0.$$

The right hand side is nonnegative by reflection positivity of the reference measure, applied to $e^{-V_+}F \in \mathfrak{A}_+$.

For a lattice or finite-mode regulator, the split-interaction calculation above applies directly when the interaction is written as a positive-time part plus its reflection. Terms sitting on the reflection plane are either assigned half weight before a limiting argument or are treated by a separate reflection-positive boundary kernel. Thus reflection positivity in the constructive theorem is not a formal property of the expression $\exp[-S]$; it is a property of the chosen regulator and of the way the local interaction is split relative to the reflection.

Quoted Theorem 137.13 (Constructive Φ^4_3 output theorem)**.** *Let the ultraviolet regulator, finite-volume boundary condition, and mass coordinate be chosen in one of the massive small-coupling regimes covered by the constructive Φ^4_3 phase-cell theorems. Then there are choices of local coordinates*

$$a_\varepsilon = a_{\mathrm{div}}(\varepsilon) + a_{\mathrm{fin}}, \qquad b_\varepsilon = b_{\mathrm{div}}(\varepsilon) + b_{\mathrm{fin}},$$

such that the regulated Schwinger functions associated with (137.22) converge, as $\varepsilon \downarrow 0$ and $\Lambda \uparrow \mathbb{R}^3$, to Euclidean Schwinger distributions satisfying symmetry, Euclidean covariance, reflection positivity, regularity, clustering in the massive phase, and the growth hypotheses needed for OS reconstruction.

This is a quoted constructive theorem, not a theorem proved in this paragraph. Foundational sources include the Glimm–Jaffe–Spencer constructive program, Magnen–Seneor's infinite-volume construction of Φ^4_3, the Brydges–Federbush phase-cell expansion, and the presentation in Glimm–Jaffe's *Quantum Physics: A Functional Integral Point of View*.[2] The proof is not the formal perturbative cancellation of the diagrams in (137.25); the constructive proof has four analytic components.

In the final standard of this monograph, the quoted theorem is a proof obligation rather than an authority claim. The cited constructive literature identifies the estimates to be checked, but a theorem about the existence of an interacting QFT should ultimately be accompanied here by a self-contained derivation of the regulator choice, stability bounds, phase-cell estimates, uniform infinite-volume convergence, reflection positivity, and OS growth conditions.

First, the covariance is decomposed into phase cells or scales. Field configurations are separated into small-field regions, where perturbative Taylor expansion is controlled, and large-field regions, where the positive quartic term and local regulators give stability.

Second, all local divergent pieces allowed by the power-counting formula (137.23) are extracted into the two coordinates a_ε and b_ε. The remaining polymer activities satisfy bounds of the type

$$\|K_j\|_{a,b} \le C|\lambda|^{1+\delta} L^{-j\alpha} \tag{137.37}$$

on scale j, for positive δ, α in the constructed regime.

Third, the cluster or phase-cell expansion is summed uniformly in the volume and cutoff. The convergence criterion is the scale-dependent version of Theorem 137.7. It gives limits of connected Schwinger functions and exponential clustering.

Fourth, reflection positivity and Euclidean covariance are maintained by the choice of regulator or recovered in the limiting Schwinger functions. The OS growth conditions are checked from uniform moment bounds. OS reconstruction then gives the Lorentzian Wightman theory. Each step is a theorem with hypotheses on the regulator, mass coordinate, coupling size, and volume sequence; the Lagrangian label Φ^4_3 abbreviates this entire construction.

Definition 137.14 (Abstract multiscale phase-cell system)**.** Fix a scale factor $L > 1$. A multiscale phase-cell system consists of:

(i) cell complexes $\mathcal{D}_j$ of side L^{-j}, $j \ge 0$;

(ii) polymers $X \subset \mathcal{D}_j$, with size $|X|_j$ equal to their number of j-cells;

(iii) a finite-range incompatibility relation $(j, X) \not\sim (k, Y)$, whose cell-neighbourhood constant is B_R;

(iv) polymer activities $K_j(X)$ obtained after extracting the local coordinates $a_\varepsilon, b_\varepsilon$;

(v) numerical majorants $\kappa_j(X) \ge |K_j(X)|$ after the declared large-field weights have been applied.

The system has a scale-decay bound with constants (C_0, δ, α, a), $\delta, \alpha, a > 0$, if

$$\sup_{\Delta \in \mathcal{D}_j} \sum_{X \ni \Delta} e^{a|X|_j} \kappa_j(X) \le C_0 |\lambda|^{1+\delta} L^{-\alpha j} \quad \text{for every } j \ge 0. \tag{137.38}$$

[2] For the constructive lineage relevant here, see J. Glimm and A. Jaffe's constructive-field-theory papers and book, J. Magnen and R. Seneor on the infinite-volume Φ^4_3 model, and D. Brydges and P. Federbush on phase-cell expansions. The statement above is intentionally quoted at theorem-boundary level because reproducing those multi-scale estimates would require a separate chapter-length proof.

Theorem 137.15 (Summability from the phase-cell scale-decay bound)**.** *Assume a multiscale phase-cell system satisfies* (137.38)*. If $a > a' > 0$ and*

$$B_R \frac{C_0|\lambda|^{1+\delta}}{1 - L^{-\alpha}} \le a', \tag{137.39}$$

then the combined polymer expansion over all scales is absolutely convergent. For a root polymer (j_0, X_0),

$$\sum_{n\ge 1} \frac{1}{(n-1)!} \sum_{(j_2,X_2),\dots,(j_n,X_n)} |\varphi_c((j_0,X_0),\dots,(j_n,X_n))| \prod_{i=2}^{n} \kappa_{j_i}(X_i) \le e^{a'|X_0|_{j_0}}. \tag{137.40}$$

Moreover the contribution of scales $j \ge J$ to the one-root majorant is bounded by

$$\frac{C_0|\lambda|^{1+\delta} L^{-\alpha J}}{1 - L^{-\alpha}}. \tag{137.41}$$

Therefore the ultraviolet cutoff sequence is Cauchy in every Schwinger seminorm whose source activities obey the same bound.

Proof. Regard a pair $\widehat{X} = (j, X)$ as a single polymer in an enlarged polymer set. Its size is $|\widehat{X}| := |X|_j$ and its majorant is $\widehat{\kappa}(\widehat{X}) := \kappa_j(X)$. For a fixed cell Δ at any scale, the scale-decay hypothesis gives

$$\sum_{j\ge 0} \sum_{X \ni \Delta} e^{a|X|_j} \kappa_j(X) \le \sum_{j\ge 0} C_0|\lambda|^{1+\delta} L^{-\alpha j} = \frac{C_0|\lambda|^{1+\delta}}{1 - L^{-\alpha}}.$$

The finite-range constant B_R gives the neighbourhood estimate (137.19), with ϵ replaced by this scale sum. Condition (137.39) is the hypothesis needed for the rooted-tree induction in the proof of Theorem 137.7. Applying that proof to the enlarged polymer set gives (137.40).

For the ultraviolet tail, the majorant entering the rooted-tree induction is the restricted scale sum

$$\sum_{j\ge J} C_0|\lambda|^{1+\delta} L^{-\alpha j}.$$

Evaluating this geometric series gives

$$\sum_{j\ge J} C_0|\lambda|^{1+\delta} L^{-\alpha j} = \frac{C_0|\lambda|^{1+\delta} L^{-\alpha J}}{1 - L^{-\alpha}}.$$

The rooted-tree recursion is monotone in the majorants. Thus removing all scales below J bounds the remaining ultraviolet tail by (137.41). A cutoff change from J to J' changes the Schwinger function only by clusters touching at least one scale $j \ge J$, so the same tail estimate proves Cauchy convergence in the declared source seminorms. $\square$

Construction 137.16 (Source-decorated phase-cell seminorms)**.** The abstract scale-decay bound becomes a statement about Schwinger distributions only after local insertions are included in the same norm. Fix source labels $1, \dots, n$, test functions f_r, and seminorms $p_r(f_r)$ controlling the corresponding finite-regulator insertion weights. A source extension of the phase-cell system consists of functions $b_{r,j}(X)$, one for each source label and scale-j polymer, such that for every subset $I \subset \{1, \ldots, n\}$

$$\sup_{\Delta \in \mathcal{D}_j} \sum_{X \ni \Delta} e^{a|X|_j} \kappa_j(X) \prod_{r \in I} |b_{r,j}(X)| \leq C_I |\lambda|^{1+\delta} L^{-\alpha j} \prod_{r \in I} p_r(f_r). \tag{137.42}$$

Here $I = \varnothing$ recovers (137.38). At finite volume define

$$K_{j,\lambda}(X) = K_j(X) \exp\left(\sum_{r=1}^{n} \lambda_r b_{r,j}(X) \right).$$

Because the cluster expansion is absolutely convergent in a polydisc around $\lambda = 0$, finite source derivatives may be taken term by term. The coefficient

$$C_{1\cdots n}^{(J,\Lambda)} = \partial_{\lambda_1} \cdots \partial_{\lambda_n} \log Z_{J,\Lambda}(\lambda)|_{\lambda=0}$$

is a sum over connected clusters together with an assignment of each source label to one cluster vertex. The product over labels assigned to a vertex is precisely one of the factors controlled in (137.42). Applying the tree-graph proof of Theorem 137.15 to these decorated vertices gives constants $A_n < \infty$, depending on the neighbourhood constant and on the finite collection $\{C_I\}_{I \subset \{1,\ldots,n\}}$, such that

$$|C_{1\cdots n}^{(J,\Lambda)}| \leq A_n \prod_{r=1}^{n} p_r(f_r) \tag{137.43}$$

uniformly in the volume and cutoff. Moreover, clusters containing at least one scale $j \geq J_0$ obey the ultraviolet tail estimate

$$|C_{1\cdots n}^{(J,\Lambda)} - C_{1\cdots n}^{(J_0,\Lambda)}| \leq A_n \frac{|\lambda|^{1+\delta} L^{-\alpha J_0}}{1 - L^{-\alpha}} \prod_{r=1}^{n} p_r(f_r), \qquad J \geq J_0. \tag{137.44}$$

Thus (137.42) is the missing source-level hypothesis behind the phrase "Cauchy in Schwinger seminorms." If the constants A_n can be chosen with an OS-admissible growth envelope, for instance $A_n \leq AB^n (n!)^\gamma$ in a nuclear test-space seminorm family, then the limiting connected Schwinger distributions have the corresponding OS growth bound. Proving (137.42) and such an n-growth envelope for the actual Φ_3^4 phase-cell activities is part of the still-open model-specific construction.

Connected-to-full hierarchy bookkeeping. The OS input is the full Schwinger hierarchy, whereas the cluster expansion first produces connected source coefficients. This passage is exact but it is not free at the level of growth constants. Let Π_n be the set of partitions of $\{1, \ldots, n\}$. If the finite-regulator connected coefficients $C_B^{(J,\Lambda)}$, $B \subset \{1, \ldots, n\}$, are defined by differentiating $\log Z_{J,\Lambda}$, then the corresponding full source coefficient is

$$S_{1\cdots n}^{(J,\Lambda)} = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} C_B^{(J,\Lambda)}. \tag{137.45}$$

This is the finite moment–cumulant identity applied to $Z_{J,\Lambda}(\lambda) = \exp \log Z_{J,\Lambda}(\lambda)$. Taking the ultraviolet and volume limits term by term is justified only after the connected coefficients have uniform bounds and Cauchy estimates such as (137.43) and (137.44).

The cost in the OS growth envelope is explicit. Suppose, for one declared test-seminorm family, that every connected block obeys

$$|C_B^{(J,\Lambda)}| \le AB_0^{|B|}(|B|!)^\gamma \prod_{r\in B} p_r(f_r) \tag{137.46}$$

uniformly in J, Λ, with A, B_0, γ independent of the block. Then (137.45) gives

$$|S_{1\cdots n}^{(J,\Lambda)}| \le B_0^n \left(\prod_{r=1}^n p_r(f_r)\right) \sum_{\pi\in\Pi_n} A^{|\pi|} \prod_{B\in\pi} (|B|!)^\gamma. \tag{137.47}$$

For every partition π, $\prod_{B\in\pi} |B|! \le n!$. Also $|\Pi_n| \le n^n \le e^n n!$. Hence

$$|S_{1\cdots n}^{(J,\Lambda)}| \le \big(eB_0 \max\{1, A\}\big)^n (n!)^{\gamma+1} \prod_{r=1}^n p_r(f_r). \tag{137.48}$$

Thus a connected envelope of the OS-II form produces a full hierarchy with the same linear seminorm-order control but with one additional factorial power from the partition sum. This is still admissible for the OS-II condition (12.4) after replacing γ by $\gamma+1$, provided the test-seminorm family is stable under the translation reduction used in Chapter 12. The statement does not prove the model-specific connected envelope; it records the exact growth loss that a constructive proof must budget before invoking OS reconstruction.

Definition 137.17 (Model-specific phase-cell derivation budget)**.** A model-specific phase-cell derivation budget for the massive Φ^4_3 construction is the data needed to turn the regulated scalar measure into the abstract system of Definition 137.14. It consists of:

(i) a reflection-compatible finite-volume regulator, a finite-range covariance decomposition into scale pieces C_j, and cell algebras on the scale-j decomposition $\mathcal{D}_j$;

(ii) small-field norms, large-field weights, and a local extraction map $\mathcal{L}_j$ onto the vacuum and mass coordinates $\mathcal{O}_{0,j}$ and $\mathcal{O}_{2,j}$;

(iii) counterterm increments $\delta a_j, \delta b_j$ and a remainder

$$R_j(X) = (1-\mathcal{L}_j)K_j(X)$$

for each scale-j activity;

(iv) source weights $b_{r,j}(X)$ and test seminorms $p_r(f_r)$ for local insertions;

(v) regulator-comparison, reflection-positivity, and moment-growth defects which measure the distance from the finite phase-cell regulator to the OS hierarchy asserted in Quoted Theorem 137.13.

The budget is admissible with exponents $\alpha_{\mathrm{act}}, \alpha_{\mathrm{loc}}, \alpha_{\mathrm{OS}} > 0$ if, after the chosen local coordinates have been extracted, the actual estimates imply

$$\sup_{\Delta\in\mathcal{D}_j} \sum_{X\ni\Delta} e^{a|X|_j} \big(\kappa_j^{\mathrm{sf}}(X) + \kappa_j^{\mathrm{lf}}(X) + \kappa_j^{\mathrm{irr}}(X)\big) \le C_{\mathrm{act}} |\lambda|^{1+\delta} L^{-\alpha_{\mathrm{act}} j}, \tag{137.49}$$

for the small-field, large-field, and irrelevant-remainder parts of the activity, together with the source-decorated estimates

$$\sup_{\Delta\in\mathcal{D}_j} \sum_{X\ni\Delta} e^{a|X|_j} \kappa_j(X) \prod_{r\in I} |b_{r,j}(X)| \le C_I |\lambda|^{1+\delta} L^{-\alpha_{\mathrm{act}} j} \prod_{r\in I} p_r(f_r) \tag{137.50}$$

for every finite source set I. The local coordinates and OS-regulator defects must have summable tails:

$$\sum_{m\geq j}\big(|\delta a_m-\delta a_m^{\mathrm{target}}|+|\delta b_m-\delta b_m^{\mathrm{target}}|\big)\leq C_{\mathrm{loc}}L^{-\alpha_{\mathrm{loc}}j},\qquad \mathcal{R}_j^{\mathrm{OS}}\leq C_{\mathrm{OS}}L^{-\alpha_{\mathrm{OS}}j}. \tag{137.51}$$

Here $\delta a_m^{\mathrm{target}}$ and $\delta b_m^{\mathrm{target}}$ denote the declared mass and vacuum-energy coordinate increments, and $\mathcal{R}_j^{\mathrm{OS}}$ includes the regulator-comparison, reflection positivity, and moment-growth errors which survive after truncating at scale j. The definition is a proof budget: it records the estimates whose model-specific proof produces (137.49)– (137.51) from the actual finite-regulator scalar measure.

Proposition 137.18 (From phase-cell budget to constructive output inputs)**.** *Assume an admissible model-specific phase-cell budget. Then* (137.49) *implies the abstract scale-decay hypothesis* (137.38)*, with $C_0=C_{\mathrm{act}}$ and $\alpha=\alpha_{\mathrm{act}}$. Equation* (137.50) *implies the source-decorated Schwinger-seminorm bound* (137.42)*. For every cutoff scale J, the remaining constructive defect is bounded by*

$$\mathcal{E}_J^{\mathrm{constr}}\leq\frac{C_{\mathrm{act}}|\lambda|^{1+\delta}L^{-\alpha_{\mathrm{act}}J}}{1-L^{-\alpha_{\mathrm{act}}}}+C_{\mathrm{loc}}L^{-\alpha_{\mathrm{loc}}J}+C_{\mathrm{OS}}L^{-\alpha_{\mathrm{OS}}J}. \tag{137.52}$$

Consequently, if the Kotecky–Preiss smallness condition (137.39) *also holds, the local summability, source-seminorm Cauchy property, and finite-window OS limits needed by Quoted Theorem 137.13 reduce to proving the model-specific estimates in Definition 137.17.*

Proof. The first assertion is just the identification

$$\kappa_j(X)=\kappa_j^{\mathrm{sf}}(X)+\kappa_j^{\mathrm{lf}}(X)+\kappa_j^{\mathrm{irr}}(X)$$

inside (137.49). The second assertion is the same estimate with the source weights retained, hence it is exactly (137.42). Summing the activity bound over $m\geq J$ gives the geometric tail

$$\sum_{m\geq J}C_{\mathrm{act}}|\lambda|^{1+\delta}L^{-\alpha_{\mathrm{act}}m}=\frac{C_{\mathrm{act}}|\lambda|^{1+\delta}L^{-\alpha_{\mathrm{act}}J}}{1-L^{-\alpha_{\mathrm{act}}}}.$$

The local-coordinate and OS-regulator tails are the two terms stated in (137.51). Adding these three independent errors gives (137.52). Theorem 137.15, Construction 137.16, and the connected-to-full hierarchy bookkeeping above then supply the abstract summation and OS-growth consequences once the displayed model-specific inputs are proved. □

The difficult model-specific part of the Φ_3^4 construction is now located precisely: one must construct Definition 137.17 for the actual phase-cell activities obtained from the regulated scalar field after the local coordinates have been extracted. Proposition 137.18 proves that these bounds, once established with reflection-positive regulators and uniform moment estimates, are sufficient for ultraviolet convergence of the Schwinger functions. Construction 137.16 records the additional source-decorated estimate needed to obtain distributional seminorm bounds and OS growth rather than only convergence of the vacuum polymer partition function.

Open Problem 137.19 (Self-contained constructive proof of Φ_3^4). Give a complete monograph-internal proof of Quoted Theorem 137.13. The proof must derive the actual phase-cell activities from the regulated Φ^4_3 measure, state the small-field and large-field norms, prove the model-specific stability estimates beyond the finite-cutoff one-site bound, identify the local coordinate b_ε to the same precision as the mass coordinate in Lemma 137.12, and prove (137.38) and (137.42) for those activities, with OS-admissible growth in the number of source insertions. The abstract summation, reflection-positivity preservation, and ultraviolet Cauchy steps are supplied above; the remaining proof obligation is the model-specific multiscale estimate producing their hypotheses.

137.6 OS Data Produced by a Construction

The output of a constructive scalar theorem is a hierarchy

$$S_n(f_1, \dots, f_n) = \lim_{\Lambda \uparrow \mathbb{R}^d,\ \varepsilon \downarrow 0} S_{\varepsilon,\Lambda,n}(f_1, \dots, f_n) \tag{137.53}$$

on Schwartz test functions or on a declared nuclear test space. The data must include the following.

1. **Distributional regularity.** Each S_n is a continuous multilinear functional in the chosen topology.
2. **Euclidean covariance and symmetry.** The Schwinger functions are invariant under the Euclidean group and symmetric under permutation of scalar insertions.
3. **Reflection positivity.** For the time reflection $\theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x})$ and test polynomials supported at positive Euclidean time,
$$\sum_{\alpha,\beta} \overline{c_\alpha} c_\beta S_{\alpha+\theta\beta} \geq 0.$$
4. **Growth.** The OS-II regularity and linear-growth hypotheses needed for analytic continuation and Hilbert-space reconstruction are verified, not assumed from formal manipulations.
5. **Clustering or phase decomposition.** A unique massive vacuum requires exponential clustering; a phase coexistence construction must specify the phase sector.

Once these statements are proved, OS reconstruction gives a Hilbert space, a cyclic vacuum, Wightman fields, and local commutativity. The constructive status table in Chapter 136 records representative examples. The lattice, stochastic, and numerical chapters below must match the same data format when they claim to approximate or construct a continuum QFT.

137.7 Comparison of Constructive Routes

The same theory label can arise from cluster expansions, lattice scaling limits, rigorous RG, and singular SPDE. Equality of the resulting QFTs is a statement about Schwinger functions or local observable algebras. A comparison theorem must provide a parameter map

$$(m, \lambda, a_{\mathrm{fin}}, b_{\mathrm{fin}})_{\mathrm{cluster}} \longleftrightarrow (m, \lambda, c_1, c_2)_{\mathrm{SPDE}} \longleftrightarrow (m_0^2, \lambda_0, a)_{\mathrm{lattice}}$$

and prove convergence to the same S_n's in a common topology.

Open Problem 137.20 (Explicit comparison of constructive routes). For a model such as Φ^4_3, compare the Schwinger functions produced by cluster/phase-cell expansions, stochastic quantization, rigorous RG, and lattice scaling limits in one common topology. The comparison should identify the OS Hilbert space, the local fields, and the finite coordinate map between the constructions.

Chapter 138

Lattice Reflection Positivity

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Reflection positivity is the lattice analogue of the positivity axiom used in Euclidean reconstruction. It is a property of a regulated measure and a chosen reflection, not an automatic property of every discretization. This chapter states the condition and verifies it for the basic nearest-neighbor scalar lattice action.

138.1 Reflection and Positive Algebra

Let $\Lambda \subset a\mathbb{Z}^D$ be a finite reflection-invariant lattice with reflection ϑ across the hyperplane $x_0 = 0$. Let ϕ_x be a real scalar variable at $x \in \Lambda$. Let $\mathcal{A}_+$ be the algebra of polynomial functions of the variables with $x_0 > 0$. The reflection acts antilinearly on functions by

$$(\Theta F)(\phi) = \overline{F(\phi_{\vartheta x})}.$$

For a probability measure $d\mu(\phi)$, define

$$\langle F \rangle = \int F(\phi)\, d\mu(\phi).$$

Definition 138.1 (Lattice reflection positivity)**.** The measure $d\mu$ is reflection positive with respect to ϑ if

$$\langle \Theta F\, F \rangle \geq 0$$

for every $F \in \mathcal{A}_+$.

138.2 Nearest-Neighbor Scalar Measure

Consider the finite-volume lattice measure

$$d\mu(\phi) = Z^{-1} e^{-S(\phi)} \prod_{x\in\Lambda} d\phi_x$$

with

$$S(\phi) = \frac{1}{2}\sum_{\langle xy\rangle} c_{xy}(\phi_x - \phi_y)^2 + \sum_x V(\phi_x), \qquad c_{xy} > 0,$$

where V is a real potential for which the finite-volume integral exists. Split the action as

$$S = S_+ + \Theta S_+ + S_0$$

where S_+ contains terms strictly in the positive half, ΘS_+ its reflected copy, and S_0 the terms on or crossing the reflection plane. The crossing terms have the form

$$\frac{1}{2}c(\phi_x - \phi_{\vartheta x})^2 = \frac{1}{2}c\phi_x^2 + \frac{1}{2}c\phi_{\vartheta x}^2 - c\phi_x\phi_{\vartheta x}.$$

The first two terms can be assigned to S_+ and ΘS_+. The coupling across the plane contributes the positive kernel

$$\exp(c\phi_x\phi_{\vartheta x}) = \sum_{n=0}^{\infty} \frac{c^n}{n!}\phi_x^n\phi_{\vartheta x}^n.$$

Proposition 138.2 (Scalar lattice reflection positivity)**.** *Assume the finite lattice, the coefficients c_{xy}, and the one-site potential V are invariant under the reflection ϑ, and assume the finite-volume integral exists. Then the nearest-neighbor scalar lattice measure above is reflection positive on the polynomial algebra $\mathcal{A}_+$ generated by fields in the positive half-lattice:*

$$\int \overline{\Theta F(\phi)}\, F(\phi)\, d\mu(\phi) \ge 0, \qquad F \in \mathcal{A}_+.$$

Proof. After assigning all non-crossing terms and the square pieces in $\frac{1}{2}c(\phi_x - \phi_{\vartheta x})^2$ to S_+ and ΘS_+, the only factors coupling reflected variables have the form

$$K_i(\phi_i, \phi_{\vartheta i}) = \exp(c_i\phi_i\phi_{\vartheta i}), \qquad c_i > 0,$$

one for each bond crossing the reflection plane. Their product admits the absolutely convergent expansion

$$\prod_i K_i = \sum_{\mathbf{n}} \left(\prod_i \frac{c_i^{n_i}}{n_i!}\right) \prod_i \phi_i^{n_i}\phi_{\vartheta i}^{n_i}.$$

Let

$$d\nu_+(\phi_+) = \exp[-S_+(\phi_+)] \prod_{x\in\Lambda_+} d\phi_x$$

including the positive-half one-site factors and the assigned half of the crossing square terms. Reflection invariance gives the reflected measure $\Theta d\nu_+$ on the negative half. For a polynomial $F \in \mathcal{A}_+$, Fubini's theorem and the finite-volume integrability assumption allow termwise integration of the nonnegative-coefficient series, or equivalently first truncation of the series and then monotone convergence for the positive kernels. Each multi-index contribution is

$$\left(\prod_i \frac{c_i^{n_i}}{n_i!}\right)\left|\int F(\phi_+)\prod_i \phi_i^{n_i}\,d\nu_+(\phi_+)\right|^2$$

up to the positive normalization constant Z^{-1}. The coefficient is nonnegative and the remaining factor is an absolute square. Summing over $\mathbf{n}$ gives $\int \overline{\Theta F} F\,d\mu \geq 0$. □

138.3 Free Lattice Fermions

Reflection positivity for Grassmann variables is a positivity property of the Berezin functional together with an antilinear anti-automorphism. Let $\psi_A(x)$ and $\bar\psi^A(x)$ be independent Grassmann-odd variables on $\Lambda \subset \mathbb{Z}^D$, and let C_μ be Euclidean Clifford matrices,

$$\{C_\mu, C_\nu\}_A{}^B = 2\delta_{\mu\nu}\delta_A{}^B.$$

The nearest-neighbor free lattice action is

$$-S = \sum_x \left[m\,\bar\psi^A(x)\psi_A(x) - \frac{1}{2}\sum_{\mu=1}^{D}\bar\psi^A(x)(C_\mu)_A{}^B\big(\psi_B(x-e_\mu)-\psi_B(x+e_\mu)\big)\right].$$

The corresponding two-point function is

$$\langle\bar\psi^B(y)\psi_A(x)\rangle = \int_{[-\pi,\pi]^D}\frac{d^Dp}{(2\pi)^D}\,e^{\mathrm{i}p\cdot(x-y)}\,\frac{m\delta_A{}^B - \mathrm{i}\sum_\mu (C_\mu)_A{}^B\sin p_\mu}{m^2+\sum_\nu \sin^2 p_\nu}.$$

Adding a Wilson term changes the numerator by a chirality-matrix term and the denominator by the corresponding Wilson square; it removes doublers at the price of explicitly coupling the reflection-positivity question to the chosen Wilson operator and mass range.

For the naive action above, the reflection compatible with the simple positivity proof is the reflection through the plane halfway between two time slices. Take $v = e_1$, $c = 1/2$, and

$$R(x) = x - 2(x\cdot e_1 - c)e_1.$$

Let $H = \{x : x_1 \geq 1\}$. Define the antilinear anti-automorphism

$$\Theta(AB) = \Theta(B)\Theta(A),$$

$$\Theta(\psi_A(x)) = (C_1)_B{}^A\,\bar\psi^B(Rx), \qquad \Theta(\bar\psi^A(x)) = (C_1)_A{}^B\,\psi_B(Rx).$$

Reflection positivity means

$$\langle\Theta(\mathcal{O})\mathcal{O}\rangle \geq 0$$

for every polynomial $\mathcal{O}$ in the Grassmann variables supported in H, where positivity is the coefficient positivity obtained after the Berezin integral is reduced to a sum of squares.

The placement of the reflection plane is essential. A two-point test with $\mathcal{O} = \zeta^A \psi_A(x)$ gives, for the mid-link plane,

$$\int_{[-\pi,\pi]^D} \frac{d^D p}{(2\pi)^D} \frac{|\zeta|^2 \sin p_1 \, \sin\bigl(p_1(2x_1 - 1)\bigr)}{m^2 + \sum_\nu \sin^2 p_\nu}, \qquad x_1 \geq 1,$$

which has the positive one-dimensional transfer-kernel expansion in the normal direction. For a site reflection $c = 0$, the same test contains the term

$$\int_{[-\pi,\pi]^D} \frac{d^D p}{(2\pi)^D} \frac{m\,(\zeta C_1 \zeta^*) \cos(2p_1 x_1)}{m^2 + \sum_\nu \sin^2 p_\nu},$$

whose sign is not fixed for arbitrary spinor ζ. A diagonal reflection such as $v = e_1 + e_2$ similarly fails the same elementary two-point positivity test. Thus the fermion reflection structure is not interchangeable among lattice hyperplanes.

Proposition 138.3 (Mid-link free-fermion reflection positivity)**.** *For the nearest-neighbor free lattice fermion action above, reflection positivity holds for the mid-link reflection $v = e_1, c = 1/2$ with the Grassmann reflection Θ displayed above.*

Proof. Write $P_+ = \{x : x_1 = 1\}$ and $P_- = \{x : x_1 = 0\}$. Split the action into positive-half, reflected-half, and cross-plane terms. The cross-plane factor is a finite Grassmann polynomial over links crossing the plane:

$$\prod_{x \in P_+} \exp\Bigl[-\frac{1}{2}\bigl(\bar\psi^A(x)(C_1)_A{}^B \psi_B(Rx) - \bar\psi^A(Rx)(C_1)_A{}^B \psi_B(x)\bigr)\Bigr].$$

Using the displayed definition of Θ, each cross factor can be written as a product of terms of the form

$$1 + \frac{1}{2}\left[\Theta(\psi_A(x))\,\bar\psi^A(x) + \Theta(\bar\psi^A(x))\,\psi_A(x)\right].$$

Integrating the variables strictly inside H against $\mathcal{O}e^{-S_H}$ produces a boundary Grassmann polynomial Ψ in the variables on P_+. The reflected half gives $\Theta(\Psi)$. The remaining Berezin integral over $P_+ \cup P_-$ is a finite sum of products $\Theta(c_I)c_I$ of the coefficients of Ψ, with nonnegative numerical weights supplied by the cross factors. This is the Grassmann analogue of the scalar sum-of-squares argument. $\square$

138.4 Staggered Fermions and a Thirring Interaction

In two dimensions with $C_1 = \sigma_3$ and $C_2 = \sigma_1$, a closed staggered subsector of the Dirac lattice fermion can be written in terms of one-component Grassmann variables $\eta(x), \bar\eta(x)$. In the convention of the staggered reduction,

$$-S = \sum_{\langle xy \rangle} M(x,y)\bar\eta(x)\eta(y) + m\sum_x \bar\eta(x)\eta(x),$$

with nearest-neighbor coefficients

$$M(x, x \pm e_1) = \pm\frac{(-1)^{x_2}}{2}, \qquad M(x, x \pm e_2) = \pm 1.$$

The mid-link reflection acts by

$$\Theta(\eta(x)) = (-1)^{x_2}\bar{\eta}(Rx), \qquad \Theta(\bar{\eta}(x)) = (-1)^{x_2}\eta(Rx).$$

The same boundary-polynomial proof gives reflection positivity for the free staggered action.

Add the nearest-neighbor Thirring interaction

$$-\Delta S = -U\sum_x \sum_{\mu=1,2} \bar{\eta}(x)\eta(x+e_\mu)\bar{\eta}(x+e_\mu)\eta(x).$$

It is invariant under the displayed reflection. The cross-plane factor has the form

$$1 + U\,\Theta(\eta(x)\bar{\eta}(x))\,\eta(x)\bar{\eta}(x)$$

on each crossing link. Therefore the same coefficient-square argument extends to the interacting lattice Thirring model for

$$U \geq 0.$$

This is a finite-regulator reflection-positivity statement. Continuum existence, scaling limits, and identification of the resulting QFT require the additional compactness and renormalization estimates discussed in Chapter 139.

138.5 Transfer Matrix Interpretation

When the lattice is periodic in Euclidean time, reflection positivity permits the construction of a pre-Hilbert space from $\mathcal{A}_+$ by quotienting the null space of the sesquilinear form

$$(F, G) = \langle \Theta F\, G \rangle.$$

Time translation by one lattice spacing acts as a contraction on this pre-Hilbert space. Under the additional hypotheses required for a transfer matrix, this contraction is e^{-aH} for a positive self-adjoint lattice Hamiltonian H. This is the finite-regulator version of the positivity step in OS reconstruction.

138.6 Gauge Fields

For lattice gauge theory the configuration variables are group elements $U_\ell \in G$ on oriented links ℓ, with $U_{-\ell} = U_\ell^{-1}$. Here G is a compact Lie group and the finite-regulator measure is the product Haar measure multiplied by local class functions of plaquette holonomies. If

$$p = (\ell_1, \ell_2, \ell_3, \ell_4), \qquad U_p = U_{\ell_1}U_{\ell_2}U_{\ell_3}U_{\ell_4},$$

then a one-plaquette gauge action is specified by a continuous positive class function $w : G \to \mathbb{R}_{>0}$:

$$d\mu(U) = Z^{-1}\prod_\ell dU_\ell \prod_p w(U_p).$$

The reflection ϑ maps links to reflected links with reflected orientation. The positive algebra $\mathcal{A}_+$ is generated by matrix elements of link variables supported in the positive half-lattice, and Θ is the antilinear anti-automorphism obtained by reflecting links and complex conjugating coefficients.

The useful finite-volume theorem is not special to the defining representation. Let $\widehat{G}$ be the set of equivalence classes of finite-dimensional irreducible unitary representations of G, let χ_λ be the character of λ, and write D^λ_{ij} for matrix elements in an orthonormal basis.

Theorem 138.4 (Osterwalder-Seiler character criterion)**.** *Assume that the plaquette weight has an absolutely convergent character expansion*

$$w(g) = \sum_{\lambda\in\widehat{G}} c_\lambda \chi_\lambda(g), \qquad c_\lambda \geq 0.$$

Then the finite-volume gauge measure defined above is reflection positive with respect to the link-reflection plane.

Proof. Plaquettes lying strictly in the positive half-lattice and in the negative half-lattice are assigned to W_+ and ΘW_+, respectively. A plaquette crossing the reflection plane is cut into a positive half-path c and its reflected half. With a choice of orientation its holonomy has the form

$$U_p = V_c\,\Theta(V_c)^{-1}, \qquad V_c = \prod_{\ell\in c} U_\ell.$$

Changing the orientation replaces g by g^{-1}, which only conjugates the character expansion and does not affect the positivity argument.

For one crossing plaquette, Peter-Weyl gives

$$w\big(V_c\Theta(V_c)^{-1}\big) = \sum_\lambda c_\lambda \sum_{i,j} D^\lambda_{ij}(V_c)\,\Theta\Big(D^\lambda_{ij}(V_c)\Big).$$

The product over crossing plaquettes is therefore a finite or absolutely convergent sum

$$\prod_{p\ \text{crossing}} w(U_p) = \sum_\alpha C_\alpha\,\Theta(B_\alpha)B_\alpha, \qquad C_\alpha \geq 0,$$

where each B_α is a polynomial in positive-half link matrix elements. For $F \in \mathcal{A}_+$,

$$\langle \Theta F\,F\rangle = Z^{-1}\sum_\alpha C_\alpha \int \Theta(FW_+B_\alpha)\,(FW_+B_\alpha)\prod_\ell dU_\ell.$$

Product Haar measure is invariant under reflection and factorizes between the two halves once the crossing-plaquette expansion has been made. Thus each summand is the modulus square of the same half-lattice integral:

$$\int \Theta(FW_+B_\alpha)\,(FW_+B_\alpha)\prod_\ell dU_\ell = \left|\int_{\Lambda_+} FW_+B_\alpha \prod_{\ell\subset\Lambda_+} dU_\ell\right|^2 \geq 0.$$

Since $C_\alpha \geq 0$, the full sum is nonnegative. $\square$

Wilson plaquette action. For $G = SU(N)$ or $U(N)$, the Wilson plaquette weight

$$w_W(g) = \exp\biggl(\frac{\beta}{N}\,\mathrm{Re}\,\mathrm{Tr}_F\, g\biggr), \qquad \beta \geq 0,$$

is reflection positive. This is the character-expansion check required by Theorem 138.4. Let F be the defining representation and write $\chi_{\overline{F}}(g) = \overline{\chi_F(g)} = \chi_F(g^{-1})$. Then

$$w_W(g) = \exp\biggl(\frac{\beta}{2N}\,\bigl(\chi_F(g) + \chi_{\overline{F}}(g)\bigr)\biggr) = \sum_{m,n\geq 0} \frac{(\beta/2N)^{m+n}}{m!\,n!}\,\chi_F(g)^m \chi_{\overline{F}}(g)^n.$$

The product $\chi_F^m \chi_{\overline{F}}^n$ is the character of $F^{\otimes m} \otimes \overline{F}^{\otimes n}$. Decomposing this finite representation into irreducibles gives

$$\chi_F^m \chi_{\overline{F}}^n = \sum_{\lambda \in \widehat{G}} N_{m,n}^{\lambda} \chi_\lambda, \qquad N_{m,n}^{\lambda} \in \mathbb{Z}_{\geq 0}.$$

Hence the Wilson weight has nonnegative character coefficients. Absolute convergence follows from the uniform convergence of the exponential series on compact G, so Theorem 138.4 applies.

For $G = U(1)$, the same statement is the elementary Fourier expansion

$$e^{\beta\cos\theta} = \sum_{n\in\mathbb{Z}} I_n(\beta) e^{\mathrm{i}n\theta}, \qquad I_n(\beta) = \sum_{r\geq 0} \frac{(\beta/2)^{2r+|n|}}{r!\,(r+|n|)!} \geq 0.$$

This formula is the abelian form of the same positive-type condition.

$SU(2)$ **Wilson character coefficients.** Let $G = SU(2)$, and write every conjugacy class as

$$g(\theta) \sim \begin{pmatrix} e^{\mathrm{i}\theta} & 0 \\ 0 & e^{-\mathrm{i}\theta} \end{pmatrix}, \qquad 0 \leq \theta \leq \pi.$$

For the Wilson weight

$$w_W(\theta) = \exp(\beta\cos\theta),$$

the character expansion

$$w_W(\theta) = \sum_{\ell=0}^{\infty} a_\ell(\beta)\,\chi_\ell(\theta), \qquad \chi_\ell(\theta) = \frac{\sin((\ell+1)\theta)}{\sin\theta},$$

has coefficients

$$a_\ell(\beta) = I_\ell(\beta) - I_{\ell+2}(\beta) = \frac{2(\ell+1)}{\beta} I_{\ell+1}(\beta) > 0 \qquad (\beta > 0), \tag{138.1}$$

with the limiting value at $\beta = 0$ understood as $a_0(0) = 1$, $a_{\ell>0}(0) = 0$. The irreducible $SU(2)$ characters are indexed here by $\ell = 2j \in \mathbb{Z}_{\geq 0}$, where j is the spin. The class-function form of normalized Haar measure is

$$\mathrm{d}\mu_{\text{class}}(\theta) = \frac{2}{\pi}\sin^2\theta\,\mathrm{d}\theta.$$

Orthogonality gives

$$a_\ell(\beta) = \frac{2}{\pi}\int_0^{\pi} e^{\beta\cos\theta}\sin\theta\,\sin((\ell+1)\theta)\,\mathrm{d}\theta. \tag{138.2}$$

Using

$$\sin\theta\,\sin((\ell+1)\theta) = \frac{1}{2}\left(\cos(\ell\theta) - \cos((\ell+2)\theta)\right)$$

and

$$I_n(\beta) = \frac{1}{\pi}\int_0^\pi e^{\beta\cos\theta}\cos(n\theta)\,\mathrm{d}\theta \qquad (n \geq 0),$$

we obtain $a_\ell = I_\ell - I_{\ell+2}$. The Bessel recurrence

$$I_\ell(\beta) - I_{\ell+2}(\beta) = \frac{2(\ell+1)}{\beta} I_{\ell+1}(\beta)$$

follows by differentiating the Fourier series for $e^{\beta\cos\theta}$ or, equivalently, from integration by parts in the defining integral. Since

$$I_{\ell+1}(\beta) = \sum_{r=0}^{\infty} \frac{(\beta/2)^{2r+\ell+1}}{r!(r+\ell+1)!}$$

has strictly positive terms when $\beta > 0$, the coefficient (138.1) is strictly positive.

`calculation-checks/lattice_reflection_positivity_checks.py` verifies the $U(1)$ Bessel coefficients, the $SU(2)$ coefficient identity (138.1), and finite $SU(2)$ tensor-product character multiplicities used in the proof of the Wilson plaquette positivity statement above.

The proof also explains why reflection positivity is not an automatic consequence of gauge invariance. Gauge fixing may destroy the product-Haar factorization used above. Improved gauge actions must be tested loop by loop. A sufficient condition is that every crossing-loop weight admit a nonnegative character expansion after the loop is split into a positive half-path and its reflected half. The standard tree-level Symanzik or Luscher-Weisz action and the Iwasaki action contain rectangle terms with negative coefficient c_1 in

$$S_{\mathrm{imp}} = -\frac{\beta}{N}\left(c_0 \sum_{\text{plaquettes}} \operatorname{Re}\operatorname{Tr} U_p + c_1 \sum_{\text{rectangles}} \operatorname{Re}\operatorname{Tr} U_R + \cdots\right),$$

with $c_1 = -1/12$ at tree-level Symanzik improvement and $c_1 \simeq -0.331$ for Iwasaki. For a single crossing rectangle with coefficient $\alpha = \beta c_1/N < 0$,

$$\exp(\alpha \operatorname{Re}\operatorname{Tr}_F g) = 1 + \frac{\alpha}{2}\big(\chi_F(g) + \chi_{\overline{F}}(g)\big) + O(\alpha^2),$$

so the fundamental character coefficient is negative to first order. Such a factor is not of positive type and cannot enter the sum-of-squares proof. Consequently the Wilson-action proof does not extend to these improved one-step transfer matrices; any use of such actions in an argument requiring reflection positivity must supply a separate positivity theorem, often after blocking the time direction or changing the regulator.

Open Problem 138.5 (Reflection-positive continuum limits). Give verifiable criteria ensuring that a sequence of reflection-positive lattice measures converges to OS-positive continuum Schwinger functions with nontrivial local fields. The criteria must include tightness or correlation bounds, control of counterterms, and compatibility of reflection with the continuum scaling limit.

Chapter 139

Continuum Limits and Scaling Windows

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

A regulated model becomes a continuum QFT only after its correlation functions converge as distributions and the limiting Schwinger functions satisfy the hypotheses needed for reconstruction. The continuum limit is therefore a statement about a family of measures, a scaling trajectory in the regulated action, and a coordinate system for local observables. The object produced is the limiting system of Schwinger distributions together with its reconstruction data.

Throughout this chapter D is the Euclidean spacetime dimension, $\Lambda_a = a\mathbb{Z}^D$, and $a > 0$ is the lattice spacing. The infinite-volume limit is taken before $a \downarrow 0$, except where a finite-volume scaling limit is explicitly stated.

139.1 Regulated Families and Schwinger Distributions

For each a, let $\mathcal{E}_a$ be a space of lattice fields and let

$$d\mu_a(\phi) = Z_a^{-1} \exp[-S_a(\phi)] D\phi$$

be the regulated expectation functional. The notation $D\phi$ denotes the finite-dimensional Lebesgue or Berezin product before infinite volume; after an infinite-volume limit it denotes the resulting cylindrical expectation, understood through its values on local observables.

A scaling trajectory is a map

$$a \longmapsto c(a)$$

from lattice spacing to the local coefficients appearing in S_a. These coefficients include all regulator-dependent local terms. In this monograph the word counterterm means such a local term in the regulated action. It is part of the action defining the path integral, not a second action added afterward.

Let $O_a^I(n)$ be local lattice observables. A renormalized smeared observable is

$$O_a^I(f) = a^D \sum_{n\in\mathbb{Z}^D} f(an)\, Z^I{}_J(a)\, O_a^J(n), \qquad f \in C_c^\infty(\mathbb{R}^D), \tag{139.1}$$

where $Z^I{}_J(a)$ may mix all observables with the same quantum numbers and dimension bounded by the chosen truncation. A continuum Schwinger distribution is the multilinear distribution

$$S^{I_1\ldots I_k}(f_1,\ldots,f_k) = \lim_{a\downarrow 0} \left\langle O_a^{I_1}(f_1)\cdots O_a^{I_k}(f_k)\right\rangle_a, \tag{139.2}$$

when the limit exists for all test functions in the stated topology.

The topology matters. A useful constructive statement gives uniform bounds on the regulated distributions in a negative Sobolev or Schwartz seminorm, then proves convergence on a dense test-function space. Pointwise convergence of unsmeared lattice correlators at separated lattice sites is a weaker statement; it does not determine contact terms and does not by itself give an operator-valued distribution after reconstruction.

139.2 Free Lattice Scalar as a Model Calculation

The massive free scalar illustrates the exact normalization of the continuum limit. Let

$$S_a(\phi) = \frac{1}{2} a^D \sum_{x\in a\mathbb{Z}^D} \left[\sum_{\mu=1}^{D} \frac{(\phi_{x+a\hat{\mu}} - \phi_x)^2}{a^2} + m_a^2\phi_x^2\right], \tag{139.3}$$

with $m_a \to m > 0$. The infinite-volume covariance is

$$C_a(x-y) = \int_{B_a} \frac{d^D p}{(2\pi)^D} \frac{e^{\mathrm{i}p\cdot(x-y)}}{\hat{p}_a^2 + m_a^2}, \qquad \hat{p}_a^2 = \frac{4}{a^2}\sum_{\mu=1}^{D} \sin^2\!\left(\frac{ap_\mu}{2}\right), \tag{139.4}$$

where $B_a = (-\pi/a, \pi/a]^D$ is the Brillouin zone. Define the smeared field

$$\phi_a(f) = a^D \sum_{x\in a\mathbb{Z}^D} f(x)\phi_x.$$

Proposition 139.1 (Massive free lattice scalar continuum limit)**.** *If $m_a \to m > 0$, then for every $f, g \in C_c^\infty(\mathbb{R}^D)$,*

$$\lim_{a\downarrow 0}\langle \phi_a(f)\phi_a(g)\rangle_a = \int_{\mathbb{R}^D} \frac{d^D p}{(2\pi)^D} \frac{\widehat{f}(-p)\widehat{g}(p)}{p^2+m^2}. \tag{139.5}$$

All higher Schwinger functions converge to the Gaussian Wick contractions of this covariance.

Proof. For the covariance,

$$\langle \phi_a(f)\phi_a(g)\rangle_a = \int_{B_a} \frac{d^D p}{(2\pi)^D} \frac{\widehat{f}_a(-p)\widehat{g}_a(p)}{\hat{p}_a^2 + m_a^2},$$

where

$$\widehat{f}_a(p) = a^D \sum_{x\in a\mathbb{Z}^D} f(x)e^{-\mathrm{i}p\cdot x}.$$

Poisson summation gives

$$\widehat{f}_a(p) = \sum_{\ell\in\mathbb{Z}^D} \widehat{f}(p + 2\pi\ell/a).$$

Since f is smooth and compactly supported, $\widehat{f}$ is rapidly decreasing; hence $\widehat{f}_a(p) \to \widehat{f}(p)$ uniformly on every fixed compact momentum set and has uniform rapid-decay bounds after periodization. Also

$$\hat{p}_a^2 = p^2 + O(a^2|p|^4)$$

on compact sets, so the integrand converges pointwise to the continuum integrand. Split the Brillouin zone into $|p| \le R$ and its complement. On $|p| \le R$, dominated convergence applies. On the complement, rapid decay of the periodized test functions and positivity of $\hat{p}_a^2 + m_a^2$ give a bound whose $R \to \infty$ limit is uniform for small a. This proves the two-point convergence. The finite-dimensional Gaussian measures are determined by their covariance, so Wick's theorem gives convergence of every smeared k-point function. □

This proof is the prototype of a constructive continuum statement: regulated objects are smeared first, the limiting object is a distribution, and the proof supplies a uniform estimate that lets the limit pass through the integral.

139.3 Tree-Level Symanzik Expansion and Cutoff Effects

The phrase lattice artifact is precise only after one has chosen continuum coordinates for both sources and operators. At fixed lattice spacing the theory has hypercubic symmetry, not full $O(D)$ invariance. The continuum limit may recover $O(D)$ invariance because hypercubic but non-$O(D)$ local operators are irrelevant along the scaling trajectory, or because they have been canceled by improvement terms. This is the content, in its tree-level form, of the Symanzik expansion.

Definition 139.2 (Symanzik expansion data)**.** Fix a regulated lattice family, a scaling trajectory $c(a)$, and renormalized observable coordinates $O_a^I(f)$. A Symanzik expansion to order a^r for a class of smeared correlators is a finite list of continuum local distributions $S_{(n)}$, built from local operators allowed by the lattice symmetries, such that

$$\left\langle O_a^{I_1}(f_1)\cdots O_a^{I_k}(f_k)\right\rangle_a = \sum_{0\le n<r} a^n S_{(n)}^{I_1\ldots I_k}(f_1,\ldots,f_k) + R_{r,a}(f_1,\ldots,f_k), \tag{139.6}$$

with a stated seminorm q on the test functions and constants $C, a_0 > 0$ such that

$$|R_{r,a}(f_1,\ldots,f_k)| \le Ca^r\, q(f_1)\cdots q(f_k), \qquad 0 < a < a_0,$$

or with another explicitly stated remainder estimate. In an interacting theory this is a theorem only after the uniform distributional bounds, operator-mixing coordinates, and irrelevant operator estimates have been proved. The formal list of allowed local operators alone is not a continuum-limit proof.

Free scalar tree-level cutoff expansion. For the free lattice scalar of (139.3), write

$$\hat{p}_{\mu,a} = \frac{2}{a}\sin\Big(\frac{ap_\mu}{2}\Big), \qquad \hat{p}_a^2 = \sum_{\mu=1}^{D} \hat{p}_{\mu,a}^2.$$

Assume

$$m_a^2 = m^2 + a^2\delta m_2 + O(a^4) \tag{139.7}$$

with $m > 0$. Then for every $f, g \in C_c^\infty(\mathbb{R}^D)$,

$$\langle \phi_a(f)\phi_a(g)\rangle_a = \int_{\mathbb{R}^D} \frac{\mathrm{d}^D p}{(2\pi)^D}\,\frac{\widehat{f}(-p)\widehat{g}(p)}{p^2+m^2} + a^2\int_{\mathbb{R}^D} \frac{\mathrm{d}^D p}{(2\pi)^D}\,\widehat{f}(-p)\widehat{g}(p)\frac{\frac{1}{12}\sum_\mu p_\mu^4 - \delta m_2}{(p^2+m^2)^2} + O(a^4), \tag{139.8}$$

where the $O(a^4)$ remainder is bounded by a finite Schwartz seminorm of f and g. The term $\sum_\mu p_\mu^4$ is hypercubic invariant but not $O(D)$-invariant for $D > 1$; it is the leading rotation-breaking cutoff artifact of the nearest-neighbor action. The Taylor expansion of the lattice momentum is

$$\hat{p}_{\mu,a}^2 = p_\mu^2 - \frac{a^2}{12}p_\mu^4 + \frac{a^4}{360}p_\mu^6 + O(a^6 p_\mu^8) \tag{139.9}$$

on every compact momentum set. Hence the inverse kernel has the form

$$\hat{p}_a^2 + m_a^2 = p^2 + m^2 + a^2\left(\delta m_2 - \frac{1}{12}\sum_\mu p_\mu^4\right) + O(a^4(1+|p|^6)).$$

For $A = p^2 + m^2$ and

$$B = \delta m_2 - \frac{1}{12}\sum_\mu p_\mu^4,$$

the elementary identity

$$\frac{1}{A + a^2 B + O(a^4)} = \frac{1}{A} - a^2\frac{B}{A^2} + O(a^4)$$

gives the displayed coefficient. The replacement of the Brillouin-zone integral by the integral over $\mathbb{R}^D$, and the uniform control of the high-momentum tail, are the same Poisson-summation and rapid-decay estimates used in the proof of Proposition 139.1. The additional polynomial factors in p are harmless because $\widehat{f}$ and $\widehat{g}$ are Schwartz functions. Thus the expansion holds as a distributional expansion on smeared fields, with the remainder bounded by a Schwartz seminorm.

Remark 139.3 (Tree-level scalar improvement)*.* Define an improved free scalar quadratic kernel by

$$K_a^{\mathrm{imp}}(p) = m_a^2 + \hat{p}_a^2 + \frac{a^2}{12}\sum_{\mu=1}^{D}\hat{p}_{\mu,a}^4. \tag{139.10}$$

If $m_a^2 = m^2 + O(a^4)$, then

$$K_a^{\mathrm{imp}}(p) = m^2 + p^2 + O(a^4(1+|p|^6)) \tag{139.11}$$

on compact momentum sets. Consequently the $O(a^2)$ correction in (139.8) is canceled for the two-point function. By (139.9),

$$\hat{p}_a^2 = p^2 - \frac{a^2}{12}\sum_\mu p_\mu^4 + O(a^4|p|^6), \qquad \sum_\mu \hat{p}_{\mu,a}^4 = \sum_\mu p_\mu^4 + O(a^2|p|^6).$$

The $a^2 \sum_\mu p_\mu^4/12$ improvement term cancels the nearest-neighbor a^2 artifact, and the mass assumption removes the δm_2 part. The conclusion for smeared two-point functions follows from the previous proposition.

Remark 139.4 (Status in interacting lattice theories). For an interacting lattice theory, a Symanzik expansion is not obtained by writing all local operators compatible with hypercubic symmetry. One must also prove that the regulated correlators admit the expansion after renormalized operator mixing, and that the remainder is controlled in the distribution topology used for reconstruction. Gauge theories add the further requirement that improvement terms and operator coordinates be gauge-invariant or be formulated inside a gauge-fixed BV/lattice framework.

139.4 Correlation Length and Scaling Window

The lattice correlation length is a property of a regulated state. In a massive translation-invariant theory, a connected two-point function at large separation behaves like

$$\langle O(n)O(0)\rangle_{\text{conn}} \sim e^{-|n|/\xi_{\text{lat}}},$$

where ξ_{lat} is measured in lattice units. A massive continuum limit with physical mass $M > 0$ requires

$$\xi_{\text{lat}}(a) \to \infty, \qquad a\,\xi_{\text{lat}}(a) \to M^{-1}. \tag{139.12}$$

For a scale-invariant continuum limit, $a\xi_{\text{lat}}(a) \to \infty$. The phrase scaling window means that the lattice spacing, observation length, and correlation length are separated in the order

$$a \ll r \lesssim a\xi_{\text{lat}}(a),$$

with the last inequality replaced by $r \ll a\xi_{\text{lat}}(a)$ when one is studying critical short-distance scaling inside a massive deformation.

For the free lattice scalar, the zero-spatial-momentum pole gives an exact correlation length. The pole in the Euclidean time direction is

$$\frac{4}{a^2}\sinh^2\Big(\frac{a\kappa_a}{2}\Big) = m_a^2, \qquad \kappa_a = \frac{2}{a}\operatorname{arcsinh}\Big(\frac{am_a}{2}\Big).$$

Thus

$$\xi_{\text{lat}}(a) = \frac{1}{a\kappa_a} = \frac{1}{2\operatorname{arcsinh}(am_a/2)}, \qquad a\xi_{\text{lat}}(a) \to m^{-1}. \tag{139.13}$$

The expansion

$$\kappa_a = m_a - \frac{a^2 m_a^3}{24} + O(a^4 m_a^5)$$

shows explicitly how the lattice mass approaches the physical pole mass.

139.5 Finite-Volume Scaling and Step-Scaling Coordinates

Finite volume is an infrared regulator and can be used as a renormalized coordinate on the continuum theory through finite-volume observables. To separate this coordinate from the ultraviolet cutoff, keep three lengths separate:

$$a, \qquad L = Na, \qquad \xi_{\text{phys}} = a\,\xi_{\text{lat}}.$$

Here N is the number of lattice spacings across the box, L is the physical size, and ξ_{phys} is the physical correlation length. A continuum finite-volume observable is obtained by taking

$$a \to 0, \qquad L \text{ fixed}, \qquad \text{renormalized relevant coordinates fixed.}$$

The infinite-volume limit $L \to \infty$ is a subsequent infrared limit. Interchanging these limits is an additional theorem, not part of the definition of the finite-volume continuum object.

Definition 139.5 (Finite-size scaling data)**.** Fix a block scale $b > 1$, lattice tori $(a\mathbb{Z}/Na\mathbb{Z})^D$ with $N = b^n$, and an RG chart near a fixed point with one relevant coordinate t, irrelevant coordinates $w = (w_\alpha)$, and local operators O^I of RG dimensions Δ_I. Finite-size scaling data on a compact set $\mathcal{K} \subset \mathbb{R}$ of scaling variables consists of:

(i) the integer $n = \log_b N$;

(ii) an endpoint coordinate

$$u_N = t\, N^{y_t} + E_N(t, w), \qquad |E_N(t, w)| \le CN^{-\omega}$$

whenever $tN^{y_t} \in \mathcal{K}$;

(iii) an irrelevant-coordinate estimate

$$\|w_N\| \le CN^{-\omega}$$

for some $\omega > 0$;

(iv) endpoint kernels $F^{I_1 \dots I_k}(u; z_1, \dots, z_k)$, locally Lipschitz in u, and renormalized separated-point kernels $G^{I_1 \dots I_k}_{N,t,w}(z_1, \dots, z_k)$, with $z_i \in (\mathbb{R}/\mathbb{Z})^D$ and $z_i \ne z_j$, such that after n RG steps

$$\left| N^{\Delta_{I_1} + \cdots + \Delta_{I_k}} G^{I_1 \dots I_k}_{N,t,w}(z_1, \dots, z_k) - F^{I_1 \dots I_k}(u_N; z_1, \dots, z_k) \right| \le CN^{-\omega}$$

uniformly on compact subsets of the configuration space with pairwise separation bounded below.

The constants C may depend on the compact set of noncoincident configuration points and on the bounded microscopic set of irrelevant coordinates, but not on N.

Finite-size scaling from an RG chart. Assume the finite-size scaling data of Definition 139.5. Then

$$N^{\Delta_{I_1} + \cdots + \Delta_{I_k}}\, G^{I_1 \dots I_k}_{N,t,w}(z_1, \dots, z_k) = F^{I_1 \dots I_k}(tN^{y_t}; z_1, \dots, z_k) + O(N^{-\omega}) \tag{139.14}$$

uniformly for $tN^{y_t} \in \mathcal{K}$. If the continuum massive coordinate is parameterized by a physical mass M through

$$t(a) = (aM)^{y_t} + O(a^{y_t + \epsilon})$$

for some $\epsilon > 0$, then the scaling variable has the continuum finite-volume limit

$$t(a)N^{y_t} \to (ML)^{y_t}. \tag{139.15}$$

After n blocking steps, lengths have been divided by $b^n = N$, and the torus has order-one diameter in the blocked lattice units. The renormalized operator coordinate O^I contributes the factor $b^{-n\Delta_I} = N^{-\Delta_I}$. The relevant coordinate at the endpoint is $u_N = tN^{y_t} + O(N^{-\omega})$, and the irrelevant coordinates have norm $O(N^{-\omega})$. The endpoint estimate in the data gives the correlator with $F(u_N; z_1, \dots, z_k)$ up to $O(N^{-\omega})$. Since the endpoint kernel is locally Lipschitz in u on $\mathcal{K}$, replacing u_N by tN^{y_t} changes the right-hand side by another $O(N^{-\omega})$. This proves (139.14). Finally, if $N = L/a$ and $t(a) = (aM)^{y_t} + O(a^{y_t+\epsilon})$, then

$$t(a)N^{y_t} = (aM)^{y_t}(L/a)^{y_t} + O(a^\epsilon (L/a)^{y_t} a^{y_t}) = (ML)^{y_t} + O(a^\epsilon L^{y_t}),$$

which proves (139.15).

This scaling consequence states exactly what must be proved before finite-size scaling is a theorem: an RG chart, endpoint control, irrelevant decay, and operator-coordinate stability. The common formula $G_N = N^{-\sum \Delta_I} F(tN^{y_t})$ is the conclusion of these estimates, not their replacement.

Definition 139.6 (Step-scaling coordinate)**.** Let $\mathcal{O}_{a,L}$ be a finite-volume renormalized observable whose regulator-level definition is gauge-invariant, or is a specified coordinate in a gauge-fixed BV formulation, and whose continuum limit exists at fixed physical size L. Suppose a real coordinate g_L is defined by an injective normalization map

$$\mathcal{O}_{\mathrm{cont}}(L) = \Phi(g_L)$$

on an interval of couplings. For $s > 1$, the continuum step-scaling function associated to this coordinate is

$$\sigma_s(u) = \lim_{a/L \to 0} g_{sL}(a) \quad \text{with } g_L(a) = u, \tag{139.16}$$

provided the tuning $g_L(a) = u$ and the continuum limit exist. If the cutoff effects are described by a Symanzik expansion, the numerical extrapolation has the form

$$g_{sL}(a) = \sigma_s(u) + \sum_{j=1}^{r-1} c_j(u,s)(a/L)^{\omega_j} + R_{r,a}(u,s), \tag{139.17}$$

with explicitly stated exponents and a stated bound on $R_{r,a}$.

This definition is the precise finite-volume version of a running coupling. Different choices of $\mathcal{O}$ define different schemes. Scheme dependence is not a defect: it is the coordinate dependence of the renormalized theory. What is scheme independent is the existence of the continuum finite-volume observable, the consistency of the coordinate changes where they overlap, and universal data such as fixed-point critical exponents when the relevant theorem has been proved.

For lattice sizes not exactly of the form b^n, the residual size $N/b^{\lfloor \log_b N \rfloor} \in [1, b)$ is an additional finite endpoint coordinate. It must be included in the scaling data or eliminated by an explicit interpolation estimate; ignoring it is harmless only after that bounded-coordinate dependence has been controlled.

Definition 139.7 (Numerical scaling-window evidence package)**.** Fix a finite-volume continuum coordinate such as Definition 139.6, and write $q_i = a_i/L$ for the cutoff coordinate on a chosen set of regulator values. A numerical scaling-window evidence package for a continuum value c_0 consists of finite data Y_i, a covariance matrix Σ, remainder envelopes $\epsilon_i \geq 0$, accepted fit windows W, and declared exponents

$$0 = \omega_0 < \omega_1 < \cdots < \omega_p,$$

such that, on each accepted window,

$$Y_i = \sum_{\ell=0}^{p} c_\ell q_i^{\omega_\ell} + R_i + \eta_i, \qquad |R_i| \le \epsilon_i, \qquad \mathbb{E}\eta_i = 0, \quad \mathrm{Cov}(\eta) = \Sigma. \tag{139.18}$$

For a window W, let X_W be the design matrix $(X_W)_{i\ell} = q_i^{\omega_\ell}$. The package must record that X_W has full column rank, that Σ_W is positive definite, and that the weighted least-squares propagator

$$P_W = (X_W^T \Sigma_W^{-1} X_W)^{-1} X_W^T \Sigma_W^{-1} \tag{139.19}$$

is the map used to report $\widehat{c}_W = P_W Y_W$.

Lemma 139.8 (Finite numerical evidence propagation)**.** *For every accepted window in Definition 139.7,*

$$\widehat{c}_W - c = P_W R_W + P_W \eta_W, \qquad \mathrm{Cov}(\widehat{c}_W) = P_W \Sigma_W P_W^T. \tag{139.20}$$

In particular the continuum-coordinate component satisfies

$$|(\widehat{c}_W)_0 - c_0| \le \sum_{i \in W} |(P_W)_{0i}|\, \epsilon_i + |(P_W \eta_W)_0|. \tag{139.21}$$

If the accepted windows are collected in $\mathcal{W}$, a finite evidence budget may be reported as

$$B_{\mathrm{num}} = \left(\max_{W \in \mathcal{W}} (\widehat{c}_W)_0 - \min_{W \in \mathcal{W}} (\widehat{c}_W)_0 \right) + \max_{W \in \mathcal{W}} \delta_{\mathrm{stat}}(W) + \max_{W \in \mathcal{W}} \delta_{\mathrm{sys}}(W), \tag{139.22}$$

where

$$\delta_{\mathrm{sys}}(W) = \sum_{i \in W} |(P_W)_{0i}|\, \epsilon_i$$

and $\delta_{\mathrm{stat}}(W)$ is a stated statistical convention, for example $\sqrt{(P_W \Sigma_W P_W^T)_{00}}$ or a block-resampling confidence envelope.

Proof. Equation (139.18) on the selected window is $Y_W = X_W c + R_W + \eta_W$. Since

$$P_W X_W = I$$

by the full-rank assumption and the definition of P_W, applying P_W gives the first identity in (139.20). The covariance identity follows from linear propagation of the centered fluctuation η_W. Taking the zeroth component and applying the triangle inequality to the deterministic remainder term gives (139.21); the statistical term remains probabilistic until a confidence or resampling convention is declared. □

The evidence package also records the observable coordinate and tuning condition, the finite-volume geometry, the regulator schedule, autocorrelation or blocking diagnostics for Σ, the chosen exponent list and remainder hypothesis, rank and condition-number diagnostics for each window, residuals and χ^2 coordinates, and independent action or observable cross-checks when a universality claim is being made. These data make a finite numerical result reproducible and checkable. They still do not replace the continuum theorem: the Symanzik or RG expansion, finite-volume control, operator-coordinate estimates, and reconstruction target remain the mathematical inputs that turn the finite calculation into a continuum statement. The companion script

`qft_scripts/finite_regulator_extrapolation.py`

implements the finite linear algebra in Lemma 139.8 and reports the budget (139.22).

139.6 Operator Coordinates and Contact Terms

Renormalized operator coordinates are part of the continuum-limit input. For composite observables, the matrix $Z^I{}_J(a)$ in (139.1) may mix an operator with derivatives of lower-dimensional operators and with the identity. The identity and derivative mixings are the lattice form of contact terms. They are visible only after smearing test functions and studying coincident-point limits.

For example, the lattice square ϕ_x^2 in the free scalar model has

$$\langle \phi_x^2 \rangle_a = C_a(0),$$

which diverges for $D \geq 2$. The Wick coordinate

$$:\phi_x^2:_a = \phi_x^2 - C_a(0)$$

has finite separated correlators and a well-defined smeared distributional limit in the superrenormalizable dimensions where the corresponding continuum Wick square exists. If one changes the subtraction by a finite local constant, the smeared operator changes by a multiple of $\int f(x)\mathrm{d}^D x$, namely by an identity contact coordinate. Thus contact terms are part of the coordinate chart on local distributions.

139.7 Osterwalder–Schrader Reconstruction Target

Definition 139.9 (Constructive continuum limit)**.** A constructive continuum limit of a regulated family is a collection of limiting Schwinger distributions satisfying Euclidean covariance, permutation symmetry, reflection positivity, clustering, regularity, and the growth assumptions required by the reconstruction theorem, together with the OS-reconstructed Hilbert space, vacuum, fields, local domains, and translation generators.

Reflection positivity can pass to the limit by a closed-cone argument: if

$$\sum_{i,j} \overline{c_i} c_j S_a(\Theta F_i, F_j) \geq 0$$

for every finite set of positive-time test functions and the distributions converge on these tests, then the limiting quadratic form is nonnegative. Euclidean invariance is a separate limit theorem. Hypercubic invariance at fixed a becomes full Euclidean invariance only after one proves that rotation-breaking local operators are irrelevant or have been tuned away.

The reconstruction target also includes growth bounds. In Euclidean QFT the Schwinger functions must be sufficiently controlled as tempered distributions, and the analytic continuation to Wightman functions requires the hypotheses discussed in Chapter 12. A continuum limit that produces reflection-positive distributions but lacks the needed growth control has not yet produced the full Lorentzian QFT object used elsewhere in this monograph.

139.8 Wilsonian Coordinates and Scaling Manifolds

Let $\mathcal{B}$ be a Banach space of local interactions near a fixed point S_*, and let $\mathcal{R}_b : \mathcal{U} \subset \mathcal{B} \to \mathcal{B}$ be a block-spin map with scale factor $b > 1$. If $\mathcal{R}_b(S_*) = S_*$ and

$$D\mathcal{R}_b|_{S_*}\, v_i = b^{y_i} v_i,$$

then $y_i > 0$, $y_i = 0$, and $y_i < 0$ define relevant, marginal, and irrelevant directions in this specified Banach norm and blocking scheme. The classification is therefore tied to the chosen RG map and norm.

Suppose for simplicity that there is one relevant coordinate t with eigenvalue b^{y_t}, $y_t > 0$, and that all irrelevant coordinates are controlled by a stable-manifold estimate. Under N blocking steps,

$$t_N \approx b^{N y_t} t_0.$$

The correlation length in lattice units is the number of lattice spacings needed until the running coordinate becomes order one, so

$$b^{N y_t} |t_0| \sim 1, \qquad \xi_{\text{lat}} \sim b^N \sim |t_0|^{-1/y_t}.$$

Thus the exponent relation

$$\nu = \frac{1}{y_t} \tag{139.23}$$

is a consequence of a controlled one-relevant-coordinate RG chart. Without the chart, the nonlinear estimates, and the identification of the correlation length, the same formula is a scaling hypothesis rather than a proof.

139.9 Status of Scalar Examples

Superrenormalizable scalar theories such as $P(\phi)_2$ and ϕ^4_3 have constructive continuum realizations with OS or Wightman reconstruction under their stated hypotheses. The existence theorem for ϕ^4_3 as an interacting massive QFT is distinct from the assertion that the three-dimensional Ising critical point is a conformal field theory with the full expected operator spectrum and OPE. The latter requires a critical scaling limit with conformal covariance and operator-product control, not only a massive constructive realization.

In four dimensions, perturbative renormalizability of ϕ^4_4 does not settle nonperturbative ultraviolet existence. The current rigorous status is that no construction is known of a nonperturbative four-dimensional UV-complete QFT whose perturbation series agrees with ϕ^4_4 to all orders around the Gaussian fixed point. Perturbative formal power series and constructive continuum measures are different mathematical objects.

139.10 Theorem Boundary

The minimal data for a continuum-limit theorem are:

$$\begin{gathered}\text{a regulated expectation functional for each } a,\\ \text{a scaling trajectory } c(a),\\ \text{renormalized observable coordinates } Z^I{}_J(a),\\ \text{uniform distributional bounds and convergence,}\\ \text{reflection positivity, covariance, clustering, and growth control,}\\ \text{and an OS reconstruction of the limiting data.}\end{gathered}$$

For Wilsonian fixed-point arguments, the theorem must also specify the function space, the blocking map, the fixed point, the relevant and irrelevant projections, and estimates that control nonlinear remainders. This is the line between a precise construction and a physically motivated scaling picture.

Open Problem 139.10 (Nonperturbative fixed points from Wilsonian maps). Develop a Wilsonian Banach-space framework for physically relevant four-dimensional and three-dimensional QFTs in which fixed points, renormalized operator coordinates, contact terms, scaling windows, and OS reconstruction can all be controlled in the same theorem. The framework should make precise when a fixed-point argument is a proof and when it is a well-supported construction principle.

Chapter 140

Wilson Lattice Gauge Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Wilson lattice gauge theory gives a gauge-invariant finite-regulator definition of compact gauge fields. Its continuum use requires a scaling trajectory, reflection positivity, locality, and a reconstruction or continuum-limit argument. The lattice theory is a nonperturbative regulated path integral with compact gauge variables.

140.1 Link Variables and Gauge Transformations

Let $\Lambda \subset a\mathbb{Z}^D$ be a finite hypercubic lattice and G a compact Lie group. To each oriented link $\ell = (x, x + a\hat{\mu})$, assign

$$U_\mu(x) \in G,$$

with $U_{-\mu}(x + a\hat{\mu}) = U_\mu(x)^{-1}$. A lattice gauge transformation is a function $g : \Lambda^0 \to G$ acting by

$$U_\mu(x) \mapsto g(x)U_\mu(x)g(x + a\hat{\mu})^{-1}.$$

The plaquette holonomy is

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x + a\hat{\mu})U_\mu(x + a\hat{\nu})^{-1}U_\nu(x)^{-1}.$$

140.2 Wilson Action

Fix a faithful unitary representation R and normalize the trace in the continuum convention of the gauge-theory volume. A Wilson plaquette action has the form

$$S_W[U] = \beta \sum_p \left(1 - \frac{1}{\dim R} \operatorname{Re} \operatorname{Tr}_R U_p\right).$$

The finite-volume partition function is the compact group integral

$$Z = \int \prod_\ell dU_\ell \, e^{-S_W[U]},$$

where dU is Haar measure.

140.3 A Finite $\mathbb{Z}_2$ Gauge Benchmark

Before taking any continuum limit, a lattice gauge theory is an ordinary finite-dimensional integral. The smallest nontrivial pure gauge benchmark is the $\mathbb{Z}_2$ gauge model on a finite periodic cubic lattice. Let E be the set of oriented positive links, let P be the set of elementary plaquettes, and assign

$$U_e \in \{\pm 1\}, \qquad e \in E.$$

For a plaquette p, define

$$U_p = \prod_{e \in \partial p} U_e,$$

where orientation is irrelevant because $U_e^{-1} = U_e$. The finite partition function at parameter $\beta \in \mathbb{R}$ is

$$Z_\Lambda(\beta) = \sum_{U : E \to \{\pm 1\}} \exp\left(\beta \sum_{p \in P} U_p\right).$$

A gauge transformation is a function $g : V \to \{\pm 1\}$ acting on a link $e = (x, y)$ by

$$U_e \longmapsto g_x U_e g_y.$$

The plaquette variables and Wilson loop variables

$$W(C) = \prod_{e \in C} U_e$$

for closed lattice contours C are gauge invariant.

The finite $\mathbb{Z}_2$ character expansion is an exact algebraic rewriting of this finite gauge integral. Let $t = \tanh \beta$. For a closed contour C, regard C as a $\mathbb{Z}_2$-one-chain, and regard a subset $A \subset P$ as a $\mathbb{Z}_2$-two-chain with boundary ∂A. For each plaquette variable $U_p = \pm 1$,

$$e^{\beta U_p} = \cosh \beta \, (1 + t U_p) = \cosh \beta \sum_{n_p = 0}^{1} t^{n_p} U_p^{n_p}.$$

Multiplying over plaquettes gives

$$\exp\left(\beta \sum_p U_p\right) = (\cosh\beta)^{|P|} \sum_{A\subset P} t^{|A|} \prod_{p\in A} \prod_{e\in\partial p} U_e.$$

Summing over a single link variable gives $\sum_{U_e=\pm 1} U_e^m = 0$ if m is odd and 2 if m is even. Therefore the partition function keeps precisely those plaquette subsets for which every link appears an even number of times, namely $\partial A = 0$, and contributes the overall factor $2^{|E|}$. Thus

$$Z_\Lambda(\beta) = 2^{|E|}(\cosh\beta)^{|P|} \sum_{\substack{A\subset P\\ \partial A=0}} t^{|A|}. \tag{140.1}$$

For the Wilson-loop numerator, the additional factor $\prod_{e\in C} U_e$ changes the parity condition on each link from $(\partial A)_e = 0$ to $(\partial A)_e = C_e$. Dividing the resulting finite numerator by the finite partition function gives the displayed expectation value.

$$\langle W(C)\rangle_{\Lambda,\beta} = \frac{\displaystyle\sum_{\substack{A\subset P\\ \partial A=C}} t^{|A|}}{\displaystyle\sum_{\substack{A\subset P\\ \partial A=0}} t^{|A|}}. \tag{140.2}$$

The sums are finite sums over plaquette subsets. Thus the strong-coupling area mechanism in this model is a finite-cutoff identity: the Wilson-loop numerator is exactly a weighted sum over lattice surfaces whose boundary is the loop.

The companion script

```
qft_scripts/z2_gauge_3d_metropolis.py
```

samples this finite model by single-link Metropolis updates and measures the plaquette average and rectangular Wilson loops. Its output is finite-cutoff data. An infinite-volume transition, a continuum limit, or a continuum confinement statement requires additional limiting hypotheses and estimates.

140.4 Surface Counting and the Strong-Coupling Area Bound

The exact finite character expansion (140.2) becomes an area-law estimate only after one controls how many surfaces of a given area have the same boundary. This counting step is often hidden in the phrase "strong-coupling expansion." We spell out the finite statement.

Fix a finite lattice and a closed contour C which is a boundary in the $\mathbb{Z}_2$ plaquette chain complex. Define the lattice area

$$A_{\min}(C) = \min\{|A| : A\subset P,\ \partial A = C\}. \tag{140.3}$$

For $n \geq 0$, let

$$N_C(n) = \#\{A\subset P : \partial A = C,\ |A| = A_{\min}(C) + n\}. \tag{140.4}$$

Similarly let $N_0(n)$ count closed plaquette subsets of area n.

Finite surface expansion and area estimate. Let $\beta \geq 0$ and $t = \tanh\beta$. If C is a boundary, then

$$\langle W(C)\rangle_{\Lambda,\beta} = t^{A_{\min}(C)} \frac{\sum_{n\geq 0} N_C(n)t^n}{\sum_{n\geq 0} N_0(n)t^n}. \tag{140.5}$$

In particular, since the denominator is at least 1,

$$0 \leq \langle W(C)\rangle_{\Lambda,\beta} \leq t^{A_{\min}(C)} \sum_{n\geq 0} N_C(n)t^n. \tag{140.6}$$

If, for a family of contours, the surface entropy obeys

$$N_C(n) \leq K(C)\rho^n \qquad (n \geq 0) \tag{140.7}$$

with $0 \leq \rho t < 1$, then

$$\langle W(C)\rangle_{\Lambda,\beta} \leq \frac{K(C)}{1-\rho t} \exp\{-[-\log t]A_{\min}(C)\}. \tag{140.8}$$

This estimate is a reorganization of the finite character expansion followed by a geometric-series bound. The numerator in (140.2) is a finite sum over plaquette subsets satisfying $\partial A = C$. Grouping these subsets by the excess area $n = |A| - A_{\min}(C)$ gives the numerator

$$t^{A_{\min}(C)} \sum_{n\geq 0} N_C(n)t^n.$$

The same grouping with $C = 0$ gives the denominator. For $\beta \geq 0$, all terms in the denominator are nonnegative and the empty closed surface gives the term 1; hence the denominator is at least 1, proving (140.6). The final estimate is the geometric-series bound

$$\sum_{n\geq 0} N_C(n)t^n \leq K(C) \sum_{n\geq 0} (\rho t)^n = \frac{K(C)}{1-\rho t}.$$

For a rectangular loop $C_{R,T}$ in a coordinate plane, $A_{\min}(C_{R,T}) = RT/a^2$ when the rectangle is smaller than the chosen finite box and the flat spanning rectangle is the unique minimal surface in that homology class. Equation (140.8) then gives a finite-regulator area bound at sufficiently small t. If the constants in the surface-counting estimate have at most perimeter growth, $K(C_{R,T}) \leq \exp\{c(R+T)/a\}$, the area coefficient in lattice units is bounded below by

$$\sigma_{\text{lat}}(\beta) \geq -\log(\tanh\beta)$$

with the displayed perimeter and finite entropy prefactors kept separately. This is a strong-coupling theorem at fixed cutoff. It is not a proof of continuum confinement, because a continuum Yang–Mills limit in four dimensions would require control along a scaling trajectory where the Wilson parameter does not remain in this small-β domain.

Example 140.1 (One-cube surface polynomial)**.** On the boundary of one elementary cube, let C be the boundary of one face. There are exactly two plaquette subsets with boundary C: the face itself, with area 1, and the complementary five faces, with area 5. The closed subsets are the empty surface and the six-face cube boundary. Therefore

$$\langle W(C)\rangle = \frac{t+t^5}{1+t^6}. \tag{140.9}$$

This example shows explicitly why the leading strong-coupling term is the minimal spanning surface and why closed surfaces correct both numerator and denominator.

140.5 Compact-Group Character Expansion

The $\mathbb{Z}_2$ expansion has a nonabelian analogue. The analogue is not a sum over unoriented plaquette subsets; it is a finite tensor-network sum over irreducible representations and Haar projectors. This distinction is important: in a nonabelian gauge theory the local degrees of freedom carried by a surface include representation labels and invariant tensors at links.

Let G be a compact Lie group and let $\widehat{G}$ be the set of equivalence classes of finite-dimensional irreducible unitary representations. For $\lambda \in \widehat{G}$, write

$$D_\lambda : G \to U(V_\lambda), \qquad \chi_\lambda(U) = \mathrm{Tr}_{V_\lambda} D_\lambda(U).$$

Let $K : G \to \mathbb{C}$ be a continuous class function. Its character coefficients are

$$c_\lambda[K] = \int_G dU\, K(U)\overline{\chi_\lambda(U)}. \tag{140.10}$$

Peter–Weyl orthogonality gives the $L^2(G)$ expansion

$$K(U) = \sum_{\lambda \in \widehat{G}} c_\lambda[K]\chi_\lambda(U), \tag{140.11}$$

with rapid coefficient decay when K is smooth. For the Wilson plaquette weight with the irrelevant constant removed one may take

$$K_\beta(U) = \exp\left(\frac{\beta}{\dim R}\,\mathrm{Re}\,\mathrm{Tr}_R\, U\right). \tag{140.12}$$

Compact-group spin-foam expansion at finite cutoff. On a finite lattice, replace each plaquette Boltzmann factor by a smooth class function $K(U_p)$. Then the finite-volume partition function is represented by

$$Z_\Lambda[K] = \sum_{\lambda : P \to \widehat{G}} \left(\prod_{p \in P} c_{\lambda_p}[K]\right) \mathcal{A}_\Lambda(\{\lambda_p\}), \tag{140.13}$$

where $\mathcal{A}_\Lambda(\{\lambda_p\})$ is the tensor contraction obtained by assigning V_{λ_p} or $V_{\lambda_p}^*$ to the four oriented sides of each plaquette and, for every link ℓ, inserting the Haar projector

$$P_\ell(\{\lambda_p\}_{p \supset \ell}) = \int_G dU_\ell \bigotimes_{p \supset \ell} D_{\lambda_p}(U_\ell)^{\epsilon(p,\ell)}, \tag{140.14}$$

with $\epsilon(p, \ell) = +1$ if the plaquette orientation uses U_ℓ and $\epsilon(p, \ell) = -1$ if it uses U_ℓ^{-1}. A Wilson loop in representation S is represented by inserting the additional boundary tensor $D_S(U_\ell)^{\pm 1}$ on each link of the loop before taking the same Haar projectors. The derivation is the Peter–Weyl expansion performed link by link. Insert the character expansion (140.11) independently for every plaquette. After choosing bases of the representation spaces, every plaquette character is a cyclic contraction of four matrix factors, one for each link in its boundary. Each link variable appears only in the factors associated to plaquettes incident on that link, and in the Wilson-loop insertion if the link belongs to the loop. Integrating over this link gives precisely the Haar average

(140.14). Performing the finitely many link integrals leaves a finite tensor contraction for each plaquette-label assignment. Truncating the representation labels gives finite tensor network sums. For smooth K, in particular for (140.12), the character coefficients decay rapidly enough that the truncated sums converge to the original finite-dimensional integral. The L^2 Peter–Weyl statement gives the same formula for less regular class functions after the corresponding convergence mode has been specified.

For $G = SU(N)$, $N \geq 3$, with R the fundamental representation and with the conventional Wilson weight

$$K_\beta(U) = \exp\left(\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}U\right), \tag{140.15}$$

the small-β coefficients of the fundamental and antifundamental characters are

$$c_F[K_\beta] = c_{\overline{F}}[K_\beta] = \frac{\beta}{2N} + O(\beta^2). \tag{140.16}$$

This follows by expanding $K_\beta = 1 + \frac{\beta}{2N}(\chi_F + \chi_{\bar{F}}) + O(\beta^2)$ and using character orthogonality. The coefficient changes for $SU(2)$, because the fundamental representation is self-conjugate and $\operatorname{Re}\chi_F = \chi_F$.

Example 140.2 (Fundamental plaquette normalization in $SU(N)$)**.** For $N \geq 3$, let

$$w_p(U) = \frac{1}{N}\operatorname{Re}\operatorname{Tr}U_p.$$

At $\beta = 0$, the plaquette holonomy U_p is Haar distributed, so $\langle w_p\rangle_{\beta=0} = 0$. Differentiating the normalized expectation at $\beta = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\langle w_p\rangle_\beta\bigg|_{\beta=0} = \int_G dU\,\frac{\chi_F(U)+\chi_{\bar{F}}(U)}{2N}\,\frac{\chi_F(U)+\chi_{\bar{F}}(U)}{2N} = \frac{1}{2N^2}. \tag{140.17}$$

Thus, for $SU(3)$,

$$\langle w_p\rangle_\beta = \frac{\beta}{18} + O(\beta^2). \tag{140.18}$$

The calculation fixes the normalization used in the nonabelian strong-coupling expansion. Higher-area Wilson loops are obtained by the same Haar-projector contractions, but a theorem-level area estimate also requires bounds on representation sums, invariant-tensor contractions, and surface entropy.

140.6 Continuum Expansion

For a smooth continuum connection $A_\mu(x)$, set

$$U_\mu(x) = \exp(aA_\mu(x)).$$

The plaquette has expansion

$$U_{\mu\nu}(x) = \exp\big(a^2F_{\mu\nu}(x) + O(a^3)\big)\,.$$

If the representation trace obeys

$$\mathrm{Tr}_R(XY) = \kappa_R \, \mathrm{tr}(XY),$$

then

$$1 - \frac{1}{\dim R} \operatorname{Re} \mathrm{Tr}_R \, U_{\mu\nu} = -\frac{a^4 \kappa_R}{2 \dim R} \mathrm{tr}(F_{\mu\nu} F_{\mu\nu}) + O(a^6)$$

for anti-Hermitian A_μ. Matching to

$$\frac{1}{4g^2} \int d^D x \; \mathrm{tr}(F_{\mu\nu} F_{\mu\nu})$$

fixes the relation between β, g, and the trace normalization.

140.7 Improved Gauge Actions at Tree Level

The lattice action is part of the regulator definition. Two local gauge-invariant actions may define the same continuum theory only after a scaling trajectory and a comparison of renormalized coordinates have been specified. Improvement is the finite-regulator problem of choosing the local action so that selected irrelevant coordinates are absent, or smaller, in a chosen coordinate system.

For $G = SU(N)$ in the fundamental trace convention used above, let $U_{\mu\nu}^{1\times 1}(x)$ be the elementary plaquette and let $U_{\mu\nu}^{1\times 2}(x)$ denote the oriented rectangle with side lengths a and $2a$ in the μ, ν plane. The plaquette-plus-rectangle family is

$$S_{c_0,c_1}[U] = \frac{\beta}{N} \sum_x \sum_{\mu<\nu} \left[c_0 \big(N - \operatorname{Re} \mathrm{Tr} \, U_{\mu\nu}^{1\times 1}(x)\big) + c_1 \sum_{\rho \in \{(\mu,\nu),(\nu,\mu)\}} \big(N - \operatorname{Re} \mathrm{Tr} \, U_{\rho}^{1\times 2}(x)\big) \right]. \tag{140.19}$$

The two rectangles in the last sum have their long side in the first or second direction of the ordered pair. Since a 1×2 rectangle has twice the area of a plaquette, its leading $F_{\mu\nu}^2$ contribution is four times the plaquette contribution. Thus the continuum normalization is

$$c_0 + 8c_1 = 1. \tag{140.20}$$

The Wilson action is $c_0 = 1, c_1 = 0$.

Lemma 140.3 (Rectangle coefficients)**.** *For the family* (140.19)*, the tree-level cancellation of the abelian quadratic $O(a^2)$ derivative artifact is equivalent to*

$$c_0 + 20c_1 = 0. \tag{140.21}$$

Together with (140.20)*, this gives*

$$c_1 = -\frac{1}{12}, \qquad c_0 = \frac{5}{3}. \tag{140.22}$$

Proof. It suffices for the tree-level quadratic calculation to work in an abelian Cartan direction. Let $F = F_{\mu\nu}$ be a smooth scalar field strength and consider a rectangle centered at x with side lengths ra in the μ-direction and sa in the ν-direction. Its flux is

$$\Phi_{r,s}(x) = \int_{-ra/2}^{ra/2} \mathrm{d}u \int_{-sa/2}^{sa/2} \mathrm{d}v \, F(x + u\hat{\mu} + v\hat{\nu}).$$

Taylor expansion around the center gives

$$\Phi_{r,s} = rsa^2 F + \frac{rsa^4}{24} \left(r^2 \partial_\mu^2 + s^2 \partial_\nu^2 \right) F + O(a^6). \tag{140.23}$$

The small-field loop weight is proportional to $\frac{1}{2}\Phi_{r,s}^2$. After summing over lattice sites, the $F\partial^2 F$ term may be integrated by parts up to boundary terms or finite-volume periodic identities:

$$\frac{1}{2}\Phi_{r,s}^2 = \frac{1}{2} r^2 s^2 a^4 F^2 - \frac{r^2 s^2 a^6}{24} \left[r^2 (\partial_\mu F)^2 + s^2 (\partial_\nu F)^2 \right] + O(a^8). \tag{140.24}$$

For the plaquette $(r,s) = (1,1)$, the derivative coefficient in one $\mu\nu$-plane is proportional to 1. For the two 1×2 rectangles, $(r,s) = (1,2)$ and $(2,1)$, the corresponding coefficients add to

$$4(1^2 + 2^2) = 20$$

in the symmetric combination $(\partial_\mu F_{\mu\nu})^2 + (\partial_\nu F_{\mu\nu})^2$. Therefore the quadratic $O(a^2)$ derivative artifact is proportional to $c_0 + 20c_1$. Setting this coefficient to zero and imposing the leading continuum normalization $c_0 + 8c_1 = 1$ gives (140.22). □

The preceding lemma is deliberately a tree-level statement. In a nonabelian interacting theory, the dimension-six gauge-invariant operator basis contains covariant-derivative and cubic-curvature terms, modulo Bianchi identities and field redefinitions. Loop-level improvement is therefore a renormalized-coordinate problem whose conditions extend the two tree-level algebraic equations above. The full Luescher–Weisz class also allows chair and parallelogram six-link loops; with coefficients (c_0, c_1, c_2, c_3) in the standard orientation-counting convention, the leading normalization is

$$c_0 + 8c_1 + 16c_2 + 8c_3 = 1, \tag{140.25}$$

where the numerical factors count orientations and leading squared areas in one coordinate plane. The commonly used Iwasaki one-parameter action belongs to the plaquette-plus-rectangle subfamily with $c_1 \simeq -0.331$ and $c_0 = 1 - 8c_1$; this coefficient is fixed by a renormalization-group criterion, not by the tree-level cancellation (140.21).

Reflection positivity is an additional constraint. The compact Euclidean integral (140.19) is well-defined whenever the Boltzmann weight is integrable, but a transfer-matrix reconstruction requires a positivity argument for the chosen action. Negative rectangle coefficients, useful for improvement, need not preserve the elementary reflection-positivity proof available for the Wilson plaquette action. Thus improvement, continuum normalization, and Hilbert-space reconstruction are separate requirements of the regulator datum.

140.8 Gauge-Covariant Link Smearing

Smearing is a regulator-level map from link fields to link fields. It is used to build extended observables, improve overlap with low-energy states, and, in some actions, define the local lattice Lagrangian itself. Its mathematical datum is a map

$$\mathcal{S} : \mathcal{U}_\Lambda = G^E \to G^E$$

with a locality radius and a gauge-covariance law. For a link $\ell = (x, y)$, gauge covariance means

$$\mathcal{S}_\ell(g \cdot U) = g(x)\mathcal{S}_\ell(U)g(y)^{-1}. \tag{140.26}$$

When a Wilson loop is built from $\mathcal{S}(U)$, it is a different regularized line operator from the loop built from U. When an action is built from $\mathcal{S}(U)$, it is a different regulator action. The continuum comparison is again a scaling-trajectory and operator-scheme question.

Let $G = SU(N)$. The oriented staple sum $C_\ell(U)$ associated with a link $\ell = (x, y)$ is the sum over the $2(D-1)$ shortest three-link paths from x to y that, together with ℓ^{-1}, form an elementary plaquette. It transforms as a link:

$$C_\ell(g \cdot U) = g(x)C_\ell(U)g(y)^{-1}. \tag{140.27}$$

The APE preprojection matrix is

$$X_\ell(U) = (1-\alpha)U_\ell + \frac{\alpha}{2(D-1)}C_\ell(U), \qquad \alpha \in \mathbb{R}. \tag{140.28}$$

It has the same covariance law as a link, but it is generally a complex matrix rather than an element of $SU(N)$. A projection back to the group is therefore part of the regulator definition.

Equivariance of polar link projection. Let $X \in GL(N, \mathbb{C})$, and write its polar decomposition as

$$X = U_X H_X, \qquad U_X \in U(N), \qquad H_X = (X^\dagger X)^{1/2} > 0.$$

For $g, h \in SU(N)$,

$$U_{gXh^{-1}} = gU_X h^{-1}. \tag{140.29}$$

Consequently the locally defined special-unitary projection

$$\Pi_{SU(N)}(X) = (\det U_X)^{-1/N} U_X \tag{140.30}$$

is gauge covariant on any open set where a branch of the N-th root is chosen continuously. Since

$$(gXh^{-1})^\dagger(gXh^{-1}) = h(X^\dagger X)h^{-1},$$

functional calculus gives

$$H_{gXh^{-1}} = hH_X h^{-1}.$$

Therefore

$$gXh^{-1} = (gU_X h^{-1})(hH_X h^{-1})$$

is the polar decomposition of gXh^{-1}, because the first factor is unitary and the second is positive. This proves (140.29). The determinant of $gU_X h^{-1}$ equals $\det U_X$, since $g, h \in SU(N)$, so the determinant-normalized projection has the same covariance law whenever the root branch is fixed on the chosen connected domain.

The projection map is smooth on $GL(N, \mathbb{C})$ after choosing the determinant branch, and it becomes singular when X loses rank or crosses a branch cut. This is why differentiable algorithms often use stout smearing, which stays on the group without a polar projection.

Let
$$\Pi_{\mathfrak{su}(N)}(M) = \frac{1}{2}(M - M^\dagger) - \frac{1}{2N}\operatorname{Tr}(M - M^\dagger)\mathbf{1}. \tag{140.31}$$
This is the orthogonal projection, with respect to the real Hilbert–Schmidt inner product, onto anti-Hermitian traceless matrices. Define
$$\Omega_\ell(U) = C_\ell(U)U_\ell^{-1}, \qquad Q_\ell(U) = \rho\,\Pi_{\mathfrak{su}(N)}(\Omega_\ell(U)), \qquad \mathcal{S}_\ell^{\text{stout}}(U) = e^{Q_\ell(U)}U_\ell. \tag{140.32}$$

Stout smearing is a smooth gauge-covariant link map. For every real ρ, the map (140.32) is a smooth map $\mathcal{U}_\Lambda \to SU(N)^E$ and satisfies the gauge-covariance law (140.26). The staple covariance (140.27) gives
$$\Omega_\ell(g\cdot U) = g(x)\Omega_\ell(U)g(x)^{-1}.$$
The projection $\Pi_{\mathfrak{su}(N)}$ commutes with unitary conjugation, so
$$Q_\ell(g\cdot U) = g(x)Q_\ell(U)g(x)^{-1}.$$
Hence
$$e^{Q_\ell(g\cdot U)} = g(x)e^{Q_\ell(U)}g(x)^{-1}.$$
Multiplying by the transformed link $(g\cdot U)_\ell = g(x)U_\ell g(y)^{-1}$ gives
$$\mathcal{S}_\ell^{\text{stout}}(g\cdot U) = g(x)\mathcal{S}_\ell^{\text{stout}}(U)g(y)^{-1}.$$
Because Q_ℓ is anti-Hermitian and traceless, $e^{Q_\ell} \in SU(N)$, and therefore $e^{Q_\ell}U_\ell \in SU(N)$. Smoothness follows from the polynomial staple map, the linear projection (140.31), and the matrix exponential.

Hypercubic smearing is an iterated local construction in which staples are restricted to lower-dimensional faces before the next level is formed. Its definition is a finite composition of maps of the preceding type; hence its gauge covariance follows by induction once every intermediate projection or exponential map is specified. Locality is regulator locality: after k smearing steps the smeared link depends on links in a bounded neighborhood whose radius grows at most linearly in k. A continuum theorem using smeared operators must keep this radius fixed in lattice units, or specify how it scales with a, before the operator scheme has mathematical meaning.

The companion script

`qft_scripts/su3_ape_smearing_hdf5.py`

implements the finite $SU(3)$ APE map (140.28) for raw, flowed, or already-smeared HDF5 checkpoints. In its spatial mode the active directions are $1, 2, 3$, and temporal links are left unchanged; this is the finite-data convention appropriate for constructing smeared spatial transporters in static-source correlators. In its all-direction mode all four Euclidean directions are active. The paired calculation check verifies the cold fixed point, preservation of temporal links in spatial mode, gauge covariance of the smearing map, and the HDF5 checkpoint convention `checkpoint/smeared_links`. As above, this is a regulator-level operator construction, not an assertion about continuum improvement by itself.

140.9 Lattice Perturbative Coordinates and Tadpole Normalization

Perturbation theory on a compact gauge lattice begins after choosing a chart near the trivial connection. For $G = SU(N)$, use Hermitian generators t^a normalized by

$$\operatorname{tr}(t^a t^b) = \delta^{ab}, \tag{140.33}$$

and write a link near the identity as

$$U_\mu(x) = \exp\{-\mathrm{i}g_0 a A_\mu(x + \tfrac{a}{2}\widehat{\mu})\}, \qquad A_\mu = A_\mu^a t^a. \tag{140.34}$$

With the Wilson action in the convention

$$S_W[U] = \frac{\beta}{N} \sum_{x,\mu<\nu} \operatorname{Re}\operatorname{tr}\big(\mathbf{1} - U_{\mu\nu}(x)\big), \qquad \beta = \frac{N}{g_0^2}, \tag{140.35}$$

the parameter g_0 is the lattice coupling coordinate matching the continuum trace-delta convention $\frac{1}{4g_0^2}\int \operatorname{tr}(F_{\mu\nu}F_{\mu\nu})$ at tree level.

Let

$$\nabla_\mu^+ f(x) = \frac{f(x + a\widehat{\mu}) - f(x)}{a}$$

be the forward difference. The linearized lattice curvature is

$$f_{\mu\nu}(x) = \nabla_\mu^+ A_\nu(x + \tfrac{a}{2}\widehat{\nu}) - \nabla_\nu^+ A_\mu(x + \tfrac{a}{2}\widehat{\mu}). \tag{140.36}$$

Expanding the plaquette gives

$$U_{\mu\nu}(x) = \exp\{-\mathrm{i}g_0 a^2 f_{\mu\nu}(x) + O(g_0 a^3 A, g_0^2 a^3 A^2)\}. \tag{140.37}$$

Therefore the quadratic Wilson action is

$$S_W^{(2)}[A] = \frac{1}{2}\sum_x a^D \sum_{\mu<\nu} \operatorname{tr}\big(f_{\mu\nu}(x) f_{\mu\nu}(x)\big). \tag{140.38}$$

All higher vertices come from the same compact-link exponential and from the Baker–Campbell–Hausdorff expansion of the plaquette. The finite-regulator definition remains the Haar integral; the perturbative chart is a local coordinate system for asymptotic expansion around $U_\ell = \mathbf{1}$.

On a periodic lattice, use Fourier modes with lattice momentum

$$\widehat{p}_\mu = \frac{2}{a}\sin\frac{ap_\mu}{2}, \qquad \widehat{p}^2 = \sum_\rho \widehat{p}_\rho^2. \tag{140.39}$$

After absorbing the harmless mid-link phases into the Fourier components, (140.38) becomes

$$S_W^{(2)} = \frac{1}{2}\sum_p A_\mu^a(-p)\left[\delta_{\mu\nu}\widehat{p}^2 - \widehat{p}_\mu\widehat{p}_\nu\right] A_\nu^a(p). \tag{140.40}$$

The zero modes and gauge orbits are treated by a finite-volume gauge-fixing or by restricting to gauge-invariant observables. In a covariant perturbative calculation one adds the lattice gauge-fixing quadratic form

$$S^{(2)}_{\mathrm{gf}} = \frac{1}{2\xi}\sum_p A^a_\mu(-p)\,\widehat{p}_\mu\widehat{p}_\nu\,A^a_\nu(p), \qquad \xi > 0. \tag{140.41}$$

The resulting kernel is

$$K^{(\xi)}_{\mu\nu}(p) = \delta_{\mu\nu}\widehat{p}^2 - \left(1 - \frac{1}{\xi}\right)\widehat{p}_\mu\widehat{p}_\nu, \tag{140.42}$$

with inverse on nonzero momenta

$$D^{ab}_{\mu\nu}(p) = \delta^{ab}\frac{1}{\widehat{p}^2}\left[\delta_{\mu\nu} - (1-\xi)\frac{\widehat{p}_\mu\widehat{p}_\nu}{\widehat{p}^2}\right]. \tag{140.43}$$

As $a \to 0$ at fixed physical p,

$$\widehat{p}^2 = p^2 - \frac{a^2}{12}\sum_\mu p_\mu^4 + O(a^4p^6). \tag{140.44}$$

This displayed artifact is one elementary source of the tree-level Symanzik corrections discussed above.

Remark 140.4 (Tree-level lattice propagator). For every nonzero lattice momentum, the tensor (140.43) is the inverse of (140.42):

$$K^{(\xi)}_{\mu\rho}(p)D^{ab}_{\rho\nu}(p) = \delta^{ab}\delta_{\mu\nu}. \tag{140.45}$$

Let P_L be the rank-one projector

$$(P_L)_{\mu\nu} = \frac{\widehat{p}_\mu\widehat{p}_\nu}{\widehat{p}^2}, \qquad P_T = \mathbf{1} - P_L.$$

Then

$$K^{(\xi)} = \widehat{p}^2(P_T + \xi^{-1}P_L), \qquad D = \widehat{p}^{-2}(P_T + \xi P_L).$$

Since $P_TP_L = 0$ and $P_T^2 = P_T$, $P_L^2 = P_L$, their product is the identity tensor. The color factor is diagonal by (140.33).

Mean-link or tadpole normalization is best understood as a finite coordinate change for perturbative expansions of link observables and action coefficients. Define the plaquette mean link, when the expectation is positive, by

$$u_0(\beta) = \left\langle \frac{1}{N}\,\mathrm{Re}\,\mathrm{tr}\,U_p \right\rangle_\beta^{1/4}. \tag{140.46}$$

If a Wilson-line observable contains L elementary links, its plaquette-tadpole-normalized coordinate is

$$O_{\mathrm{TI}} = u_0^{-L}O. \tag{140.47}$$

For the Wilson plaquette action, the same bookkeeping rewrites the plaquette coefficient as

$$\frac{\beta}{N}\,\mathrm{Re}\,\mathrm{tr}\,U_p = \frac{\beta u_0^4}{N}\,\mathrm{Re}\,\mathrm{tr}\big(u_0^{-4}U_p\big),$$

so the boosted coupling coordinate is

$$g_{\mathrm{TI}}^2 = \frac{g_0^2}{u_0^4}. \tag{140.48}$$

Equations (140.47) and (140.48) define a perturbative coordinate convention. They acquire predictive content only after specifying which renormalized observable is being matched and which finite-order truncation is being compared.

Example 140.5 (First terms of the boosted coupling)**.** Suppose the mean plaquette has a weak-coupling expansion

$$\left\langle \frac{1}{N} \operatorname{Re}\operatorname{tr} U_p \right\rangle_\beta = 1 - c_P g_0^2 + O(g_0^4). \tag{140.49}$$

Then $u_0^4 = 1 - c_P g_0^2 + O(g_0^4)$ and

$$g_{\mathrm{TI}}^2 = g_0^2 \left(1 + c_P g_0^2 + O(g_0^4)\right). \tag{140.50}$$

The coefficient c_P depends on the action, gauge group, and regulator volume used to define the expectation value. The formula itself is the finite algebra of the coordinate change.

140.10 Wilson Loops

For a closed oriented lattice contour C,

$$W_R(C) = \mathrm{Tr}_R \prod_{\ell \in C} U_\ell$$

is gauge invariant. Wilson loops are the basic gauge-invariant probes of line charges. Their strong-coupling character expansion gives the area-law mechanism described in Chapter 116. Their continuum limit requires line renormalization and an operator scheme.

140.11 Static Potential and Creutz Ratios

A rectangular Wilson loop has a Hamiltonian meaning when the transfer matrix exists. Let $C_{n,m}$ be a rectangle with spatial side $R = na$ and Euclidean time side $T = ma$. Use the normalized loop

$$W_S(n,m) = \frac{1}{\dim S} \left\langle \mathrm{Tr}_S \prod_{\ell \in C_{n,m}} U_\ell \right\rangle. \tag{140.51}$$

In the transfer-matrix Hilbert space, the spatial transporter between two static sources defines a vector $\Psi_{S,n}$ in the sector containing a source and an antisource separated by n lattice spacings. Reflection positivity gives

$$W_S(n,m) = \langle \Psi_{S,n}, T^m \Psi_{S,n} \rangle. \tag{140.52}$$

If the spectral measure of $\Psi_{S,n}$ for $T = e^{-aH}$ has an isolated largest transfer-matrix eigenvalue $e^{-aE_0(n,S)}$ with nonzero weight, then

$$aV_{a,S}(n) = -\lim_{m\to\infty} \log \frac{W_S(n,m+1)}{W_S(n,m)} = aE_0(n,S). \tag{140.53}$$

This is the finite-cutoff static potential in the chosen static-source sector. In a continuum limit it contains an additive line self-energy: the force, potential differences, or a specified subtraction scheme must be used before comparing continuum quantities.

For the finite spectral model, assume

$$W_m = \sum_{\alpha=0}^{r} A_\alpha e^{-amE_\alpha}, \qquad A_0 > 0, \quad A_\alpha \geq 0, \quad E_0 < E_1 \leq \cdots \leq E_r. \tag{140.54}$$

Then

$$-\log \frac{W_{m+1}}{W_m} = aE_0 + O(e^{-am(E_1-E_0)}). \tag{140.55}$$

Write

$$W_m = A_0 e^{-amE_0} \left(1 + \sum_{\alpha>0} \frac{A_\alpha}{A_0} e^{-am(E_\alpha - E_0)} \right).$$

The ratio W_{m+1}/W_m is e^{-aE_0} times a quotient of two factors which both tend to 1, and the difference from 1 is bounded by a constant times $e^{-am(E_1-E_0)}$. Taking the logarithm gives (140.55).

The same static-source sector admits a finite variational analysis. Fix the separation n, representation S, and a finite list of gauge-covariant spatial transporters $P_i^{(n,S)}$, for example different smearing levels or path shapes joining the static source to the antisource. They define vectors $\Psi_{i,n,S}$ in the same static sector and a Hermitian matrix

$$C_{ij}^{(n,S)}(m) = \langle \Psi_{i,n,S}, T^m \Psi_{j,n,S} \rangle. \tag{140.56}$$

If the chosen source space is exactly r-dimensional and has spectral form

$$C^{(n,S)}(m) = Z_{n,S} \operatorname{diag}(\lambda_\alpha(n,S)^m) Z_{n,S}^\dagger, \qquad \lambda_\alpha(n,S) = e^{-aE_\alpha(n,S)}, \tag{140.57}$$

with $Z_{n,S}$ invertible, then the generalized eigenvalue problem

$$C^{(n,S)}(m) v_\alpha = \Lambda_\alpha^{(n,S)}(m, m_0) C^{(n,S)}(m_0) v_\alpha \tag{140.58}$$

has eigenvalues

$$\Lambda_\alpha^{(n,S)}(m, m_0) = \lambda_\alpha(n,S)^{m-m_0} = e^{-a(m-m_0)E_\alpha(n,S)}. \tag{140.59}$$

Indeed

$$C^{(n,S)}(m_0)^{-1} C^{(n,S)}(m) = (Z_{n,S}^\dagger)^{-1} \operatorname{diag}(\lambda_\alpha(n,S)^{m-m_0}) Z_{n,S}^\dagger,$$

so the finite GEVP is only a similarity transform of the diagonal transfer-matrix ratios. Under an invertible change of static-source basis the matrices transform by congruence and these generalized eigenvalues are unchanged. With higher-state contamination, the residual norm after whitening by $C^{(n,S)}(m_0)$, the separation n, the representation S, the smearing or path recipe, and the chosen line self-energy subtraction are all part of the finite observable datum.

The Creutz ratio is a finite-difference observable designed to cancel the leading perimeter and constant terms in a rectangular Wilson-loop ansatz. For positive Wilson loops define

$$\chi(n,m) = -\log \frac{W(n,m)W(n-1,m-1)}{W(n,m-1)W(n-1,m)}. \tag{140.60}$$

If, over a range of lattice sizes,

$$W(n,m) = \exp\{-\sigma_{\text{lat}} nm - \mu_{\text{lat}}(n+m) - c_{\text{lat}} + \varepsilon(n,m)\}, \tag{140.61}$$

then

$$\chi(n,m) = \sigma_{\text{lat}} - \varepsilon(n,m) - \varepsilon(n-1,m-1) + \varepsilon(n,m-1) + \varepsilon(n-1,m). \tag{140.62}$$

Thus an observed Creutz-ratio plateau is a statement about a controlled two-dimensional finite difference of the correction term ε. A continuum string-tension theorem requires control of the scaling trajectory, the infinite-volume limit, and line-operator renormalization.

The companion analysis script

`qft_scripts/static_potential_from_wilson_loops.py`

implements exactly the finite formulas in (140.53) and (140.60) for positive rectangular Wilson-loop data. It reads a table of $W(R,T)$ values, returns effective-mass ratios and Creutz ratios, and includes a synthetic area-plus-perimeter smoke test for (140.61). It also has a sample-level mode for tables with columns `sample`, R, T, W: this mode forms block means, deletes or resamples entire Monte Carlo blocks, and recomputes the logarithmic ratios themselves. The same sample-level analysis can be fed directly from the HDF5 sampler dataset

`measurements/wilson_loops[sample,R-1,T-1]`,

which is the finite-data coordinate used by `su3_gauge_4d_metropolis_hdf5.py`. This is the finite-regulator implementation of the correlated jackknife construction in Chapter 141. The optional elementary error propagation in aggregate mode is only the independent-error algebra of the displayed logarithms. The script also has a static-source matrix mode for rows `R`, `T`, i, j, C or `R`, `T`, i, j, `real`, `imag`. For each fixed separation it solves (140.58) by positive whitening of $C(m_0)$ and reports `static_gevp_energy` levels; this is finite variational source-space analysis of smeared transporters, not continuum string-tension evidence.

140.12 Finite-Volume Correlator Matrices and the GEVP

The generalized eigenvalue problem is a finite-regulator spectral-analysis method. It is not tied to any particular dynamics: once a reflection-positive Euclidean-time regulator supplies a transfer matrix and a finite family of physical time-slice sources in a fixed symmetry channel, gauge-invariant when there is gauge redundancy, the same linear-algebra problem applies to pure Yang–Mills glueballs, to $\mathcal{N}=1$ supersymmetric Yang–Mills channel matrices, and to any other lattice channel with a positive transfer matrix. The present section uses pure Yang–Mills glueball sources as the running example because their operator construction is especially concrete.

Pure Yang–Mills spectroscopy on a lattice begins with gauge-invariant time-slice operators, not with constituent gluons. Fix a finite spatial lattice, a reflection-positive Euclidean-time regulator, and a symmetry channel $\mathfrak{c}$ for the finite lattice symmetry group. On a cubic lattice $\mathfrak{c}$ may include an irreducible representation of the octahedral group, parity, and charge conjugation. A glueball source in this channel is a gauge-invariant function $O_i[U(\tau)]$ of the spatial links on a time slice, usually built from traced closed Wilson loops after smearing and projection to $\mathfrak{c}$. Its vacuum-subtracted version is

$$\widetilde{O}_i(\tau) = O_i(\tau) - \langle O_i \rangle_\Lambda. \tag{140.63}$$

The connected correlator matrix is

$$C_{ij}(n) = \langle \widetilde{O}_i(na)\, \widetilde{O}_j(0)^\dagger \rangle_\Lambda, \qquad n \in \mathbb{Z}_{\geq 0}. \tag{140.64}$$

If the transfer matrix $T = e^{-aH_\Lambda}$ exists in the sector selected by the channel, reflection positivity identifies vectors $\psi_i = \widetilde{O}_i \Omega_\Lambda$ and gives

$$C_{ij}(n) = \langle \psi_i, T^n \psi_j \rangle. \tag{140.65}$$

Thus $C(n)$ is a finite-regulator spectral object. The continuum glueball statement is obtained only after the thermodynamic, continuum, and channel identification limits have been controlled.

The spectral resolution in finite volume gives

$$C_{ij}(n) = \sum_\alpha Z_i^{(\alpha)} \overline{Z_j^{(\alpha)}}\, \lambda_\alpha^n, \qquad Z_i^{(\alpha)} = \langle \psi_i, \alpha \rangle, \qquad \lambda_\alpha = e^{-aE_\alpha}. \tag{140.66}$$

The finite cubic channel is part of the regulator. For example, a continuum $J = 0$ channel restricts to an A_1 channel, while a continuum $J = 2$ channel restricts to the $E \oplus T_2$ channels. Degeneracy across the split lattice channels is therefore a continuum-restoration test, not an input to the finite calculation.

In the exact finite-state idealization, the GEVP reduces to an ordinary similarity transformation. Let $\mathcal{S}$ be an r-dimensional source space in a fixed finite lattice channel. Suppose that, on this source space, the correlator matrix has the exact finite spectral form

$$C(n) = Z\, \mathrm{diag}(\lambda_1^n, \ldots, \lambda_r^n)\, Z^\dagger, \qquad 1 > \lambda_1 > \cdots > \lambda_r > 0, \tag{140.67}$$

where Z is an invertible $r \times r$ matrix. Then for every $n > n_0 \geq 0$ the generalized eigenvalue problem

$$C(n) v_\alpha(n, n_0) = \Lambda_\alpha(n, n_0)\, C(n_0) v_\alpha(n, n_0) \tag{140.68}$$

has generalized eigenvalues

$$\Lambda_\alpha(n, n_0) = \lambda_\alpha^{n-n_0} = e^{-a(n-n_0)E_\alpha}. \tag{140.69}$$

Since Z is invertible and all λ_α are positive, $C(n_0)$ is positive definite. Multiplying (140.68) by $C(n_0)^{-1}$ reduces the generalized problem to the ordinary eigenvalue problem for

$$C(n_0)^{-1} C(n) = (Z^\dagger)^{-1}\, \mathrm{diag}(\lambda_1^{n-n_0}, \ldots, \lambda_r^{n-n_0}) Z^\dagger. \tag{140.70}$$

The displayed matrix is similar to the diagonal matrix of $\lambda_\alpha^{n-n_0}$. Its eigenvalues are therefore precisely the numbers in (140.69), with multiplicity.

The generalized eigenvalues are invariant under changes of source coordinates. If $R : \mathcal{S} \to \mathcal{S}$ is invertible and $O_i' = R_i{}^j O_j$, then

$$C(n) \longmapsto C'(n) = R\, C(n)\, R^\dagger. \tag{140.71}$$

The transformed GEVP,

$$RC(n)R^{\dagger}v' = \Lambda' RC(n_0)R^{\dagger}v',$$

is equivalent, after multiplication by R^{-1} and the substitution $v = R^{\dagger}v'$, to

$$C(n)v = \Lambda' C(n_0)v.$$

Hence the multiset of generalized eigenvalues is source-basis invariant. Equivalently,

$$C'(n_0)^{-1}C'(n) = (R^{\dagger})^{-1}C(n_0)^{-1}C(n)R^{\dagger}$$

is similar to $C(n_0)^{-1}C(n)$. This invariance concerns the extracted finite-regulator energy estimates. Statements such as "mostly glueball", "mostly two-hadron", or "mostly hybrid" also depend on a declared decomposition of the source space and on the inner product used to compare overlap covectors; those labels are not invariant consequences of the GEVP alone.

Controlled Approximation 140.6 (Whitened residual criterion for a GEVP window)**.** Fix $n > n_0$ and suppose $C(n_0)$ is positive definite. Define the Hermitian whitened matrix

$$B(n, n_0) = C(n_0)^{-1/2}C(n)C(n_0)^{-1/2}. \tag{140.72}$$

Assume that in the selected time window

$$B(n, n_0) = B_{\text{low}}(n, n_0) + R(n, n_0), \qquad \|R(n, n_0)\|_{\text{op}} \leq \epsilon, \tag{140.73}$$

where B_{low} has eigenvalues $\lambda_\alpha^{n-n_0} = e^{-a(n-n_0)E_\alpha}$ for the states intended to be resolved. Let $\widehat{\Lambda}_\alpha$ be the corresponding eigenvalues of B, ordered decreasingly. Weyl's finite-dimensional eigenvalue inequality gives

$$|\widehat{\Lambda}_\alpha - \lambda_\alpha^{n-n_0}| \leq \epsilon. \tag{140.74}$$

If $\epsilon < \lambda_\alpha^{n-n_0}$, then the effective energy

$$\widehat{E}_\alpha(n, n_0) = -\frac{1}{a(n-n_0)} \log \widehat{\Lambda}_\alpha \tag{140.75}$$

obeys

$$|\widehat{E}_\alpha(n, n_0) - E_\alpha| \leq \frac{1}{a(n-n_0)} \max\left\{ \log \frac{\lambda_\alpha^{n-n_0}}{\lambda_\alpha^{n-n_0} - \epsilon}, \log \frac{\lambda_\alpha^{n-n_0} + \epsilon}{\lambda_\alpha^{n-n_0}} \right\}. \tag{140.76}$$

A quoted GEVP plateau is therefore a controlled finite-matrix statement only after such a residual bound, or a stronger replacement, has been supplied for the selected operator basis and time window.

Derivation of the finite-matrix residual bound. The matrix $B(n, n_0)$ is Hermitian because $C(n)$ and $C(n_0)$ are Hermitian and $C(n_0)^{-1/2}$ is Hermitian. Weyl's inequality for Hermitian matrices gives

$$|\widehat{\Lambda}_\alpha - \Lambda_\alpha^{\text{low}}| \leq \|R(n, n_0)\|_{\text{op}} \leq \epsilon,$$

with $\Lambda_\alpha^{\text{low}} = \lambda_\alpha^{n-n_0}$, after matching the ordered eigenvalues. The map $x \mapsto -(a(n-n_0))^{-1} \log x$ is smooth and monotone on the positive interval $(\lambda_\alpha^{n-n_0} - \epsilon, \lambda_\alpha^{n-n_0} + \epsilon)$. Evaluating the largest possible logarithmic displacement at the interval endpoints gives (140.76).

The companion script

```
qft_scripts/glueball_gevp_from_correlators.py
```

implements exactly the finite matrix problem (140.68). It reads Hermitian correlator matrices from a CSV table with columns `t,i,j,C` or `t,i,j,real,imag`, whitens by $C(n_0)$, solves the ordinary Hermitian eigenvalue problem for (140.72), and reports (140.75). Its smoke test uses an exact two-state correlator matrix of the form (140.67); it is a check of the finite algebra, not a glueball-mass computation.

140.13 Wilson Flow, Smoothing, and Topological Charge

Gradient flow is a finite-regulator operation before it is a continuum renormalization tool. Let

$$\mathcal{U}_\Lambda = G^E$$

be the compact manifold of link variables on the finite lattice, and equip each copy of G with the bi-invariant Riemannian metric obtained from an invariant positive inner product $(X,Y)_{\mathfrak{g}}$ on $\mathfrak{g} = \mathrm{Lie}(G)$. For a smooth gauge-invariant lattice action $S : \mathcal{U}_\Lambda \to \mathbb{R}$, define the left-link gradient $\nabla_\ell S(U) \in \mathfrak{g}$ by

$$(X, \nabla_\ell S(U))_{\mathfrak{g}} = \frac{\mathrm{d}}{\mathrm{d}s} S(U^{(s)}_{\ell'})\bigg|_{s=0}, \qquad U^{(s)}_\ell = e^{sX} U_\ell, \quad U^{(s)}_{\ell'} = U_{\ell'} \text{ for } \ell' \neq \ell. \tag{140.77}$$

The Wilson-flow trajectory $V_t \in \mathcal{U}_\Lambda$ starting at $U \in \mathcal{U}_\Lambda$ is the solution of

$$\frac{\mathrm{d}V_t(\ell)}{\mathrm{d}t} V_t(\ell)^{-1} = -\nabla_\ell S(V_t), \qquad V_0(\ell) = U_\ell. \tag{140.78}$$

The parameter t has engineering dimension a^2 if one keeps physical units. The normalized flow time $\tau = t/a^2$ is often used in finite-lattice computations, but the continuum smoothing interpretation concerns fixed physical $t > 0$ as $a \to 0$.

Finite-lattice Wilson flow. For every finite lattice and every smooth action S, the flow (140.78) exists for all $t \geq 0$. It is gauge covariant, and the action decreases according to

$$\frac{\mathrm{d}}{\mathrm{d}t} S(V_t) = -\sum_{\ell \in E} \|\nabla_\ell S(V_t)\|^2_{\mathfrak{g}} \leq 0. \tag{140.79}$$

The right-hand side of (140.78) is a smooth vector field on the compact manifold G^E. Ordinary differential-equation existence gives a local solution, and compactness prevents finite-time escape, so the solution is global. Gauge covariance follows because S and the metric are invariant under the lattice gauge group: differentiating $S(g \cdot U) = S(U)$ along a left variation of one link shows that the gradient is transported by the same endpoint conjugations as the link tangent vector. Finally, by the chain rule and (140.77),

$$\frac{\mathrm{d}}{\mathrm{d}t} S(V_t) = \sum_{\ell \in E} \left(\frac{\mathrm{d}V_t(\ell)}{\mathrm{d}t} V_t(\ell)^{-1}, \nabla_\ell S(V_t) \right)_{\mathfrak{g}} = -\sum_{\ell \in E} \|\nabla_\ell S(V_t)\|^2_{\mathfrak{g}}.$$

For the pure $SU(3)$ Wilson action it is useful to display the finite force rather than hide it behind the abstract gradient notation. Use the score coordinate

$$Q(U) = \sum_x \sum_{0\le\mu<\nu\le 3} \frac{1}{3}\operatorname{Re}\operatorname{Tr}_3 U_{\mu\nu}(x), \qquad S_{\text{flow}}(U) = 6|\Lambda^0| - Q(U), \tag{140.80}$$

so that flowing downward for S_{flow} is the same as flowing upward for the normalized plaquette score Q. For a link $U_\mu(x)$, define the staple matrix

$$\begin{aligned} C_\mu(x;U) = \sum_{\nu\ne\mu}\Big[&U_\nu(x+\widehat{\mu})U_\mu(x+\widehat{\nu})^{-1}U_\nu(x)^{-1} \\ &+ U_\nu(x-\widehat{\nu}+\widehat{\mu})^{-1}U_\mu(x-\widehat{\nu})^{-1}U_\nu(x-\widehat{\nu})\Big]. \end{aligned} \tag{140.81}$$

Then the part of Q depending on $U_\mu(x)$, with the other links held fixed, is

$$Q_{x,\mu}(U) = \frac{1}{3}\operatorname{Re}\operatorname{Tr}_3\big(U_\mu(x)C_\mu(x;U)\big).$$

For $A\in M_3(\mathbb{C})$, write

$$[A]_{\mathfrak{su}(3)} = \frac{1}{2}(A-A^\dagger) - \frac{1}{6}\operatorname{Tr}(A-A^\dagger)\mathbf{1}_3. \tag{140.82}$$

This is the anti-Hermitian traceless projection of A. With the inner product

$$(X,Y)_{\mathfrak{su}(3)} = -\operatorname{Re}\operatorname{Tr}_3(XY),$$

the left-gradient vector for the score Q is

$$Z_\mu(x;U) = -\frac{1}{3}\big[U_\mu(x)C_\mu(x;U)\big]_{\mathfrak{su}(3)}. \tag{140.83}$$

Thus the normalized finite $SU(3)$ Wilson-flow equation used below is

$$\frac{\mathrm{d}V_\mu(t,x)}{\mathrm{d}t} = Z_\mu(x;V_t)V_\mu(t,x), \qquad V_\mu(0,x) = U_\mu(x). \tag{140.84}$$

Including the gauge coupling in the action only rescales t in this finite pure-gauge flow.

Explicit $SU(3)$ Wilson-score gradient. For $X\in\mathfrak{su}(3)$ and $U_\mu^{(s)}(x) = e^{sX}U_\mu(x)$,

$$\frac{\mathrm{d}}{\mathrm{d}s}Q(U^{(s)})\bigg|_{s=0} = (X, Z_\mu(x;U))_{\mathfrak{su}(3)}.$$

Consequently the flow (140.84) satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}Q(V_t) = \sum_{x,\mu}\|Z_\mu(x;V_t)\|^2_{\mathfrak{su}(3)} \ge 0.$$

The two summands in (140.81) are precisely the forward and backward plaquettes containing the oriented link $U_\mu(x)$, rewritten so that their total contribution has the form

$$\frac{1}{3}\operatorname{Re}\operatorname{Tr}\big(U_\mu(x)C_\mu(x;U)\big).$$

Therefore the left variation gives

$$\frac{\mathrm{d}}{\mathrm{d}s}Q(U^{(s)})\bigg|_{s=0} = \frac{1}{3}\operatorname{Re}\operatorname{Tr}\big(X\,U_\mu(x)C_\mu(x;U)\big).$$

If X is anti-Hermitian and traceless, the Hermitian part and the scalar trace part of $U_\mu C_\mu$ have zero real trace against X. Hence the last expression equals

$$\frac{1}{3}\operatorname{Re}\operatorname{Tr}\big(X[U_\mu C_\mu]_{\mathfrak{su}(3)}\big) = -\operatorname{Re}\operatorname{Tr}\big(XZ_\mu\big) = (X, Z_\mu)_{\mathfrak{su}(3)}.$$

Substituting $X = Z_\mu(x;V_t)$ for every link gives the displayed monotonicity identity for Q.

The continuum reason that positive flow time is useful is already visible in the linearized equation. Let $B_\mu(t,x)$ be a smooth continuum connection and write its curvature as $G_{\mu\nu}(t)$. The Yang–Mills gradient flow is

$$\partial_t B_\mu = D_\nu G_{\nu\mu}, \qquad B_\mu(0,x) = A_\mu(x), \tag{140.85}$$

in the anti-Hermitian connection convention used in the lattice expansion. Linearizing about $A_\mu = 0$ gives

$$\partial_t B_\mu = \partial^2 B_\mu - \partial_\mu\partial_\nu B_\nu.$$

In the transverse gauge $\partial_\mu B_\mu = 0$, a Fourier mode of momentum k is multiplied by e^{-tk^2}. Thus positive flow time damps momenta $k^2 \gg t^{-1}$, and the heat-kernel radius is of order $\sqrt{8t}$ in four Euclidean dimensions.

Definition 140.7 (Flowed energy density and scale coordinates)**.** Let $G_{\mu\nu}(t,x)$ be a chosen gauge-covariant lattice curvature constructed from the flowed links V_t, for example the clover curvature. The flowed energy density is represented by

$$E(t,x) = -\frac{1}{2}\operatorname{tr}\big(G_{\mu\nu}(t,x)G_{\mu\nu}(t,x)\big),$$

for anti-Hermitian curvature. Its volume average is

$$\overline{E}(t) = \frac{a^4}{V_\Lambda}\sum_x E(t,x), \qquad V_\Lambda = a^4\#\Lambda^0.$$

Given a positive reference number c, the finite-cutoff scale coordinates t_0 and w_0, if they exist uniquely, are defined by

$$t_0^2\,\langle\overline{E}(t_0)\rangle = c, \qquad t\frac{\mathrm{d}}{\mathrm{d}t}\big(t^2\langle\overline{E}(t)\rangle\big)\bigg|_{t=w_0^2} = c.$$

Existence, uniqueness, and a continuum limit of these coordinates are properties of a scaling trajectory; they are not consequences of the finite-lattice ODE alone.

The same distinction is essential for topological charge. A generic finite lattice gauge field is a collection of group elements, not a smooth bundle connection with a canonical integer Q. A topological sector on the lattice requires an additional construction, such as an admissibility condition plus a geometric interpolation, or a chiral lattice Dirac operator whose index is well-defined.

Definition 140.8 (Admissible geometric topological charge)**.** Let $G = SU(N)$. On a periodic lattice in four dimensions, an admissibility condition is the bound

$$\|1 - U_p\| < \epsilon \qquad \text{for every plaquette } p$$

with ϵ small enough that plaquette holonomies stay in a coordinate neighborhood of the identity. Together with a specified interpolation of the link data to transition functions on the continuum torus, this determines a principal $SU(N)$-bundle $P_U \to T^4$. The geometric topological charge is

$$Q_{\mathrm{geom}}(U) = \frac{1}{8\pi^2} \int_{T^4} \mathrm{tr}(F_{\widetilde{A}} \wedge F_{\widetilde{A}}) \in \mathbb{Z},$$

where $\widetilde{A}$ is any smooth connection on P_U. Independence of $\widetilde{A}$ is the Chern–Weil theorem; independence of the interpolation requires the stated admissibility construction and is part of the regulator datum.

Definition 140.9 (Clover topological-charge diagnostic)**.** For a finite periodic $SU(3)$ lattice, write

$$U_{+\mu}(x) = U_\mu(x), \qquad U_{-\mu}(x) = U_\mu(x - \widehat{\mu})^{-1}.$$

For signed directions α, β, let

$$P_{\alpha\beta}(x) = U_\alpha(x)\, U_\beta(x + \widehat{\alpha})\, U_{-\alpha}(x + \widehat{\alpha} + \widehat{\beta})\, U_{-\beta}(x + \widehat{\beta})$$

be the oriented plaquette based at x. For $0 \le \mu < \nu \le 3$, define the clover matrix and clover curvature by

$$C^{\mathrm{cl}}_{\mu\nu}(x) = P_{\mu\nu}(x) + P_{\nu,-\mu}(x) + P_{-\mu,-\nu}(x) + P_{-\nu,\mu}(x), \tag{140.86}$$

$$F^{\mathrm{cl}}_{\mu\nu}(x) = \frac{1}{4}\big[C^{\mathrm{cl}}_{\mu\nu}(x)\big]_{\mathfrak{su}(3)}. \tag{140.87}$$

Set $F^{\mathrm{cl}}_{\nu\mu} = -F^{\mathrm{cl}}_{\mu\nu}$. The clover diagnostic for the Chern–Weil density is

$$Q_{\mathrm{clover}}(U) = \frac{1}{32\pi^2} \sum_x \sum_{\mu,\nu,\rho,\sigma=0}^{3} \epsilon_{\mu\nu\rho\sigma} \,\mathrm{Re}\,\mathrm{Tr}_3\big(F^{\mathrm{cl}}_{\mu\nu}(x) F^{\mathrm{cl}}_{\rho\sigma}(x)\big). \tag{140.88}$$

The associated clover action-density diagnostic is

$$e_{\mathrm{clover}}(U) = \frac{1}{|\Lambda^0|} \sum_x \sum_{\mu<\nu} -\mathrm{Re}\,\mathrm{Tr}_3\big(F^{\mathrm{cl}}_{\mu\nu}(x) F^{\mathrm{cl}}_{\mu\nu}(x)\big).$$

These are gauge-invariant finite-regulator observables. They are not, by themselves, an integer topological charge or a definition of a theta sector.

Remark 140.10 (Gauge invariance of the clover diagnostics). Under a lattice gauge transformation $U_\mu(x) \mapsto g(x) U_\mu(x) g(x + \widehat{\mu})^{-1}$,

$$F^{\mathrm{cl}}_{\mu\nu}(x) \mapsto g(x) F^{\mathrm{cl}}_{\mu\nu}(x) g(x)^{-1}.$$

Consequently $Q_{\mathrm{clover}}(U)$, $e_{\mathrm{clover}}(U)$, and $\max_p \|1 - U_p\|$ are gauge invariant. The directed link $U_\alpha(y)$ transforms by conjugation with the gauge matrices at the initial and terminal vertices of the directed edge. In the oriented plaquette $P_{\alpha\beta}(x)$, all gauge matrices at intermediate vertices cancel, leaving

$$P_{\alpha\beta}(x) \mapsto g(x)P_{\alpha\beta}(x)g(x)^{-1}.$$

The clover sum therefore transforms by the same conjugation. The projection $[\cdot]_{\mathfrak{su}(3)}$ commutes with unitary conjugation, hence $F^{\mathrm{cl}}_{\mu\nu}(x)$ transforms by conjugation at x. Traces of products at the same base point are invariant. The plaquette-deviation norm is invariant because $1 - U_p$ is conjugated by a unitary matrix at the base point.

Continuum flow preserves Chern–Weil charge. Let $B(t)$ be a smooth one-parameter family of connections on a fixed principal $SU(N)$-bundle over a closed oriented four-manifold M, with curvature $G(t)$. Then

$$Q(t) = \frac{1}{8\pi^2}\int_M \mathrm{tr}(G(t) \wedge G(t))$$

is independent of t. In particular, smooth Yang–Mills gradient flow cannot change the bundle topological charge unless the flow leaves the smooth fixed-bundle setting.

Write $\dot{B} = \partial_t B$. The curvature variation is $\dot{G} = D_B\dot{B}$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{tr}(G \wedge G) = 2\,\mathrm{tr}(D_B\dot{B} \wedge G).$$

Using invariance of the trace and the Bianchi identity $D_BG = 0$,

$$\mathrm{tr}(D_B\dot{B} \wedge G) = \mathrm{d}\ \mathrm{tr}(\dot{B} \wedge G).$$

Thus

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{1}{4\pi^2}\int_M \mathrm{d}\ \mathrm{tr}(\dot{B} \wedge G) = 0$$

because M is closed.

Controlled Approximation 140.11 (Using flow in continuum extrapolations)**.** A flowed observable at positive physical flow time is a composite observable with its own regulator definition. Statements such as "the flow removes ultraviolet noise" mean the following precise finite-cutoff procedure: first solve (140.78), then construct a gauge-invariant observable from V_t, and only then study its infinite-volume and continuum limits. Integer topological charge plateaux observed after flow are numerical evidence for stability within an admissible sector; without an admissibility or index-theoretic definition they are not, by themselves, a definition of a continuum theta sector.

The companion script

`qft_scripts/`

`su3_wilson_flow_hdf5.py`

implements (140.84) by a finite-dimensional Lie–Euler step

$$V_\mu((n+1)\delta t, x) = \exp\bigl(\delta t\, Z_\mu(x; V_{n\delta t})\bigr)V_\mu(n\delta t, x), \tag{140.89}$$

with all link forces evaluated before the update. It can read the `checkpoint/links` dataset written by the $SU(3)$ subgroup-Metropolis generator, write an HDF5 trajectory containing τ, plaquette score, action density, and final flowed links, and expose a smoke test that checks score monotonicity and group preservation at finite cutoff. The corresponding calculation check verifies (140.83) by directional derivatives, tests one-step gauge covariance, and runs the sampler-to-flow HDF5 pipeline. The Lie–Euler step is a numerical integrator for the finite ODE; its step-size error is separate from the continuum-limit questions in Definition 140.7 and Definition 140.8.

The companion diagnostic script

`qft_scripts/`
`su3_topological_charge_diagnostics_hdf5.py`

implements Definition 140.9. It reads either the `checkpoint/flowed_links` dataset from the Wilson-flow script or the raw `checkpoint/links` dataset from the sampler, reports Q_{clover}, e_{clover}, and $\max_p \|1 - U_p\|$, and can write local density arrays to HDF5. The paired calculation check verifies the oriented plaquette convention against the plaquette used in the Wilson action, proves numerically that the clover field is anti-Hermitian, traceless, and antisymmetric in μ, ν, tests gauge invariance of the scalar diagnostics, and runs the full sampler-to-flow-to-topology HDF5 pipeline. The output name Q_{clover} is deliberate: it is a finite diagnostic to be compared along a scaling trajectory, not a replacement for Definition 140.8.

140.14 Reflection Positivity and Transfer Matrix

For the Wilson plaquette action, reflection positivity follows by assigning plaquettes to the two halves of the lattice and expanding cross-plane plaquette Boltzmann weights in positive characters. The reflection-positive inner product on positive-time functions gives a transfer matrix

$$T = e^{-aH}$$

with $H \geq 0$ after quotienting null vectors. Gauge constraints are implemented by the gauge-invariant subspace or by a gauge-covariant transfer matrix followed by projection.

140.15 Fermions and Chiral Difficulties

Adding lattice fermions requires a fermion action whose reflection positivity, locality, spectrum, and continuum chiral properties are checked separately. The mid-link fermion reflection structure of Chapter 138 is one such finite-regulator positivity mechanism. Chiral gauge theories require more: gauge invariance, locality, anomaly cancellation, and a definition of the fermion determinant phase compatible with the target continuum theory.

Open Problem 140.12 (Lattice continuum QCD construction). Construct four-dimensional continuum Yang–Mills or QCD from Wilson-type lattice gauge theory with a mass gap, OS reconstruction, local gauge-invariant operator limits, and controlled line-operator and theta-angle sectors.

Chapter 141

Monte Carlo Methods and Sign Problems

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Monte Carlo methods evaluate finite-regulator Euclidean expectation values by sampling a probability distribution. Their mathematical content is a finite-dimensional integration problem plus a convergence theorem for a Markov chain. Their limitation in many QFTs is the absence of a positive sampling weight compatible with the target observable.

141.1 Production Evidence Architecture

A lattice Monte Carlo output becomes evidence through a layered finite record. The first layer is the target finite measure: the regulator space, boundary conditions, action, source insertions, observables, and normalization convention. This layer fixes the mathematical object

$$(\Omega_\Lambda, d\nu_\Lambda; O_1, \ldots, O_k)$$

before any algorithm is discussed. A quoted number without this datum is not yet an estimate of a QFT observable; it is at most an entry in an unidentified finite computation.

The second layer is the invariant Markov kernel. For a Metropolis, heat-bath, overrelaxation-assisted, HMC, or RHMC computation, the chapter separates the proof that a finite kernel preserves $d\nu_\Lambda$ from the empirical question of how well a particular run explores that measure. The theorem layer records the exact proposal law, the acceptance rule or conditional distribution, the reversibility or detailed-balance identity, and the irreducibility or Harris-recurrence hypotheses invoked by the estimator.

The third layer is the implementation-defect record. A production lattice run records the random-number state, checkpoint format, code revision, floating-point precision, linear-solver tolerance, integrator step data, Hamiltonian-change distribution, reversibility defect, rational approximation interval, and any reweighting convention. On a cluster this layer also contains the job manifest: array coordinate, input deck, environment module set, executable hash when available, wall-clock limits, restart policy, and the HDF5 or table schema for outputs. These data make the finite Markov chain reproducible enough for an independent rerun and make approximation defects visible as finite numbers.

The fourth layer is the estimator and error-analysis record. It contains thermalization cuts, stationarity diagnostics, autocorrelation estimates, blocking choices, jackknife or bootstrap covariance estimates, and the rule by which nonlinear observables are recomputed on deleted or resampled blocks. Wilson-loop effective masses, Creutz ratios, static-source GEVP energies, and glueball GEVP energies belong to this layer as functions of correlated finite samples, not as independent scalar measurements.

The final layer is the scaling-window transfer to continuum language. A finite production ensemble enters a continuum claim only as input to a scaling-window evidence package in the sense of Definition 139.7 of Chapter 139: selected regulator points, covariance data, finite-volume and cutoff coordinates, fit ansatz, stability tests, and an evidence-budget decomposition. The output of this chapter is therefore an admissible finite evidence record; the continuum interpretation is a further structured inference with its own hypotheses.

This architecture governs the examples below. The small companion scripts illustrate individual layers, while the HDF5 samplers and cluster templates show how the layers can be serialized. A full numerical claim must name the layer at which each assertion lives: invariant measure, implementation defect, estimator error, observable analysis, or scaling-window inference.

Proposition 141.1 (Independent-chain aggregation record)**.** *Suppose a cluster job array produces K chain-level summaries for the same finite-regulator observable:*

$$(\widehat{O}_a, s_a, N_a^{\mathrm{eff}}), \qquad a = 1, \dots, K,$$

where $s_a > 0$ is the declared standard error of the a-th chain after its thermalization cut and autocorrelation analysis, and N_a^{eff} is an optional effective-sample-size coordinate. Conditional on the finite target measure, implementation record, and the hypothesis that the chain-level errors are independent with variances s_a^2, define

$$w_a = s_a^{-2}, \qquad \widehat{O}_{\mathrm{ens}} = \frac{\sum_a w_a \widehat{O}_a}{\sum_a w_a}.$$

Then the internal variance of the weighted estimator is

$$\operatorname{Var}(\widehat{O}_{\mathrm{ens}}) = \frac{1}{\sum_a w_a}. \tag{141.1}$$

The finite between-chain disagreement coordinate is

$$\chi^2_{\mathrm{chain}} = \sum_{a=1}^{K} w_a \big(\widehat{O}_a - \widehat{O}_{\mathrm{ens}}\big)^2, \qquad \nu_{\mathrm{chain}} = K - 1. \tag{141.2}$$

It is an estimator-record diagnostic: large $\chi^2_{\mathrm{chain}}/\nu_{\mathrm{chain}}$ indicates either underestimated within-chain errors, insufficient equilibration, hidden correlations between tasks, or a mismatch in the finite target measure. Inflating the internal error by

$$\max\{1, \sqrt{\chi^2_{\mathrm{chain}}/\nu_{\mathrm{chain}}}\}$$

is therefore a conservative finite reporting convention, not a proof that the underlying chains have mixed.

Proof. The estimator is a linear combination

$$\widehat{O}_{\mathrm{ens}} = \sum_a \alpha_a \widehat{O}_a, \qquad \alpha_a = \frac{w_a}{\sum_b w_b}.$$

Under the stated independence and variance hypotheses,

$$\mathrm{Var}(\widehat{O}_{\mathrm{ens}}) = \sum_a \alpha_a^2 s_a^2 = \sum_a \frac{w_a^2}{(\sum_b w_b)^2}\frac{1}{w_a} = \frac{\sum_a w_a}{(\sum_b w_b)^2} = \frac{1}{\sum_b w_b},$$

which proves (141.1). Equation (141.2) is the corresponding weighted residual norm after fitting one common mean parameter, hence the finite residual coordinate has $K-1$ formal degrees of freedom when the usual independent Gaussian error model is adopted. Outside that model it is still a well-defined finite discrepancy number, but its interpretation is diagnostic rather than inferential. □

The companion utility `qft_scripts/cluster/chain_ensemble_summary.py` implements exactly this finite estimator record for CSV rows

```
chain,estimate,standard_error
```

with an optional `effective_sample_size` column. It belongs to the estimator layer of the production-evidence architecture. The script does not replace the target-measure declaration, the Markov-kernel theorem, the autocorrelation analysis inside each chain, or the scaling-window evidence package.

141.2 Finite-Regulator Expectation Values

Let Ω_Λ be a finite-dimensional regulator space and let

$$d\nu_\Lambda(\Phi) = \frac{1}{Z_\Lambda} e^{-S_\Lambda(\Phi)} d\Phi$$

be a probability measure on it. For an observable $O : \Omega_\Lambda \to \mathbb{C}$,

$$\langle O \rangle_\Lambda = \int_{\Omega_\Lambda} O(\Phi)\, d\nu_\Lambda(\Phi).$$

On a compact gauge lattice, $d\Phi$ is a product of Haar measures and finite-dimensional fermionic integrals have already been performed into a determinant or Pfaffian when a purely bosonic sampling problem is used.

141.3 Markov Chains and Detailed Balance

A Markov transition kernel $P(\Phi, d\Phi')$ preserves $d\nu_\Lambda$ if

$$\int_{\Omega_\Lambda} d\nu_\Lambda(\Phi)\, P(\Phi, A) = \nu_\Lambda(A)$$

for every measurable set A. Detailed balance is the stronger condition

$$d\nu_\Lambda(\Phi)P(\Phi, d\Phi') = d\nu_\Lambda(\Phi')P(\Phi', d\Phi).$$

If the chain is irreducible and aperiodic in the finite-state case, or satisfies the corresponding Harris recurrence hypotheses in a continuous state space, the estimator

$$\overline{O}_N = \frac{1}{N}\sum_{n=1}^{N} O(\Phi_n)$$

converges to $\langle O\rangle_\Lambda$ in the sense made precise below.

The finite-state statement is elementary enough that it should not be used as a black box. Let Ω be a finite set, let $P(x, y)$ be a stochastic matrix, and let $\pi(x) > 0$ be a probability vector with $\pi P = \pi$. A state y is reachable from x if $P^n(x, y) > 0$ for some n. The chain is irreducible if every state is reachable from every other state, and aperiodic if

$$\gcd\{n \geq 1 : P^n(x, x) > 0\} = 1$$

for one, hence every, state x.

Proposition 141.2 (Finite-state ergodic theorem)**.** *Let P be an irreducible and aperiodic stochastic matrix on a finite set Ω. Then P has a unique stationary probability distribution π with strictly positive entries, and*

$$\lim_{n\to\infty} P^n(x, y) = \pi(y) \qquad (x, y \in \Omega). \tag{141.3}$$

For every function $f : \Omega \to \mathbb{C}$ and every initial state X_0,

$$\frac{1}{N}\sum_{n=0}^{N-1} f(X_n) \longrightarrow \sum_{x\in\Omega} \pi(x)f(x) \tag{141.4}$$

in probability. If the chain is run in stationarity, the same convergence holds in L^2.

Proof. For an irreducible aperiodic finite chain there is an integer m for which

$$P^m(x, y) > 0 \qquad \text{for all } x, y \in \Omega. \tag{141.5}$$

Indeed, irreducibility gives closed walks based at any state, and aperiodicity makes the semigroup of return times have greatest common divisor one; adding fixed paths from x to a reference state and from that reference state to y gives positivity for all sufficiently large common times. Let

$$\epsilon = \min_{x,y} P^m(x, y) > 0.$$

For probability vectors μ, ν, write $\alpha = \mu - \nu$, so $\sum_x \alpha(x) = 0$. Decompose

$$P^m(x,y) = \epsilon + R(x,y), \qquad R(x,y) \geq 0, \qquad \sum_y R(x,y) = 1 - |\Omega|\epsilon.$$

The constant part drops out after multiplication by α, and therefore

$$\|\alpha P^m\|_1 = \|\alpha R\|_1 \leq (1 - |\Omega|\epsilon)\|\alpha\|_1. \tag{141.6}$$

Thus P^m is a strict contraction on the affine space of probability vectors modulo total mass. It has a unique fixed probability vector π, and every μP^{km} converges to π. Since πP is also fixed by P^m, uniqueness gives $\pi P = \pi$. The finitely many intermediate subsequences μP^{km+r}, $0 \leq r < m$, converge to $\pi P^r = \pi$, proving (141.3). The positivity of π follows from

$$\pi(y) = \sum_x \pi(x) P^m(x,y) \geq \epsilon.$$

For convergence of time averages in stationarity, it is enough to treat real-valued f, since the real and imaginary parts can be handled separately. Set $g = f - \sum_x \pi(x) f(x)$. Since $P^n(x,y) \to \pi(y)$, the stationary covariance

$$C_g(n) = \sum_{x,y} \pi(x) g(x) P^n(x,y) g(y)$$

tends to zero. Hence

$$\mathbb{E}_\pi \left| \frac{1}{N} \sum_{n=0}^{N-1} g(X_n) \right|^2 = \frac{1}{N^2} \sum_{r,s=0}^{N-1} C_g(|r-s|) \longrightarrow 0, \tag{141.7}$$

because a Cesaro average of a sequence tending to zero tends to zero. This gives L^2 convergence in stationarity. For a nonstationary initial state, discard a fixed burn-in time M; by (141.3) the law of X_M can be made arbitrarily close to π, while the first M terms have vanishing weight as $N \to \infty$. The convergence in probability follows. $\square$

141.4 Metropolis Updates as a Finite-Regulator Theorem

The simplest exact check of the preceding definitions is a finite spin system. Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^2$ be the periodic square lattice and let a configuration be a map

$$s : \Lambda_L \to \{-1, +1\}.$$

For real J, h, define

$$H_L(s) = -J \sum_{\langle xy \rangle} s_x s_y - h \sum_x s_x, \qquad \pi_\beta(s) = Z_{\beta,L}^{-1} e^{-\beta H_L(s)}.$$

Here $\langle xy \rangle$ denotes an unoriented nearest-neighbor bond and $Z_{\beta,L}$ is the finite sum over all 2^{L^2} configurations.

The single-spin Metropolis chain chooses a site x uniformly, proposes

$$s \longmapsto s^x, \qquad s^x_y = \begin{cases} -s_x, & y = x, \\ s_y, & y \neq x, \end{cases}$$

and accepts the proposal with probability

$$a(s,x) = \min\{1, e^{-\beta(H_L(s^x) - H_L(s))}\}.$$

The energy difference is local:

$$H_L(s^x) - H_L(s) = 2s_x \left(J \sum_{y:\langle xy\rangle} s_y + h \right).$$

Detailed balance for the finite Ising Metropolis chain. The transition kernel P defined by the preceding update satisfies

$$\pi_\beta(s)P(s,s') = \pi_\beta(s')P(s',s)$$

for all configurations s, s'. If $0 < \beta < \infty$, the chain is irreducible. If additionally $(J,h) \neq (0,0)$, it is aperiodic on the finite set $\{-1,+1\}^{\Lambda_L}$. For the degenerate infinite-temperature Hamiltonian $J = h = 0$, adding a nonzero holding probability gives the usual lazy aperiodic variant without changing the stationary measure.

If s' is not equal to s or to a one-spin flip of s, both sides of the detailed-balance identity vanish. For $s' = s^x$, write $\Delta H = H_L(s^x) - H_L(s)$. The off-diagonal transition probabilities are

$$P(s,s^x) = \frac{1}{L^2} \min\{1, e^{-\beta\Delta H}\}, \qquad P(s^x,s) = \frac{1}{L^2} \min\{1, e^{\beta\Delta H}\}.$$

If $\Delta H \geq 0$, then

$$e^{-\beta H_L(s)} P(s,s^x) = \frac{1}{L^2} e^{-\beta H_L(s)} e^{-\beta\Delta H} = \frac{1}{L^2} e^{-\beta H_L(s^x)} = e^{-\beta H_L(s^x)} P(s^x,s).$$

The case $\Delta H < 0$ is the same identity with s and s^x interchanged. Dividing by the finite partition function gives detailed balance.

Irreducibility follows because any configuration can be reached from any other by flipping the finite set of sites on which they differ, and every single spin flip has positive proposal probability and positive acceptance probability. If $(J,h) \neq (0,0)$, there is a configuration s and a site x with $H_L(s^x) > H_L(s)$, so the chain has a positive holding probability at s. In a finite irreducible Markov chain, one state with a self-loop forces every state to have period one. When $J = h = 0$, every proposal is accepted and the one-spin-flip walk on the hypercube is bipartite; the standard lazy modification inserts a holding probability and restores aperiodicity.

The companion script

`qft_scripts/ising2d_metropolis.py`

implements exactly this finite chain and reports the acceptance rate, energy per site, absolute magnetization, and a windowed estimate of τ_{int}. Its smoke mode is a finite-regulator check of the algorithmic identities above. It does not estimate critical exponents, prove the Onsager scaling limit, or construct the continuum Ising CFT. The deterministic calculation check

`calculation-checks/ising_metropolis_finite_checks.py`

enumerates the 2×2 periodic chain and verifies both the local energy difference and the detailed-balance identity used by the script.

141.5 Scalar ϕ^4 Metropolis Warmup

The Ising example has a finite state space. A scalar lattice field already has the other feature that appears in continuum constructions: the finite-regulator measure is a genuine finite-dimensional integral rather than a finite sum. Let

$$\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^2, \qquad \phi = (\phi_x)_{x \in \Lambda_L} \in \mathbb{R}^{\Lambda_L}.$$

For $m^2 \in \mathbb{R}$ and $\lambda > 0$, define the finite Euclidean action

$$S_L(\phi) = \sum_{x \in \Lambda_L} \left[\frac{1}{2} \sum_{\mu=1}^{2} (\phi_{x+\hat{\mu}} - \phi_x)^2 + \frac{1}{2} m^2 \phi_x^2 + \frac{\lambda}{24} \phi_x^4 \right], \tag{141.8}$$

where the directions $\hat{\mu}$ are the two positive coordinate directions on the periodic lattice. The target probability measure is

$$\pi_L(d\phi) = Z_L^{-1} e^{-S_L(\phi)} \prod_{x \in \Lambda_L} d\phi_x.$$

This is a finite-dimensional probability measure. Indeed, the kinetic term is nonnegative and the one-site potential

$$V(u) = \frac{1}{2} m^2 u^2 + \frac{\lambda}{24} u^4$$

is bounded below. More explicitly, if $m^2 < 0$, then

$$V(u) \geq -\frac{3(m^2)^2}{2\lambda},$$

with equality at $u^2 = -6m^2/\lambda$; if $m^2 \geq 0$, then $V(u) \geq 0$. The quartic term gives integrability at infinity in every coordinate.

The single-site Metropolis proposal chooses $x \in \Lambda_L$ uniformly and sets

$$\phi'_x = \phi_x + \eta, \qquad \phi'_y = \phi_y \quad (y \neq x),$$

where η is drawn from a symmetric compactly supported density $r(\eta)d\eta$, for instance the uniform density on $[-\delta, \delta]$. The proposal is accepted with probability

$$a(\phi, \eta, x) = \min\{1, e^{-[S_L(\phi') - S_L(\phi)]}\}.$$

The action difference is local. If $u = \phi_x$ and $u' = \phi'_x$, then

$$S_L(\phi') - S_L(\phi) = V(u') - V(u) + \frac{1}{2} \sum_{b:\, b \text{ touches } x} \left[(\phi'_{b_+} - \phi'_{b_-})^2 - (\phi_{b_+} - \phi_{b_-})^2 \right]. \tag{141.9}$$

Here the sum is over the four oriented positive-direction kinetic bonds in (141.8) whose endpoint is x. On lattices so small that periodicity identifies some neighbors, the multiplicities are inherited from the action itself.

Detailed balance for the finite scalar Metropolis chain. The preceding single-site kernel satisfies

$$\pi_L(d\phi)\, P(\phi, d\psi) = \pi_L(d\psi)\, P(\psi, d\phi)$$

as measures on $\mathbb{R}^{\Lambda_L} \times \mathbb{R}^{\Lambda_L}$. It is enough to check the off-diagonal pair $\psi = \phi'$ related by one proposed single-site displacement. The proposal density from ϕ to ψ is $|\Lambda_L|^{-1} r(\psi_x - \phi_x)$, while the reverse proposal density is $|\Lambda_L|^{-1} r(\phi_x - \psi_x)$. These are equal by symmetry of r. Writing $\Delta S = S_L(\psi) - S_L(\phi)$, the Metropolis identity

$$e^{-S_L(\phi)} \min\{1, e^{-\Delta S}\} = e^{-S_L(\psi)} \min\{1, e^{\Delta S}\}$$

proves pairwise balance. The holding part is then fixed by total probability. This statement is entirely finite-regulator: it neither proves a continuum ϕ^4_2 construction nor determines the critical surface.

The companion script

`qft_scripts/phi4_2d_metropolis.py`

implements the finite action (141.8) with a symmetric uniform single-site proposal. It reports acceptance, action density, $\langle \phi^2 \rangle$, $\langle \phi^4 \rangle$, a susceptibility-like zero-momentum second moment, and a windowed estimate of τ_{int} for ϕ^2. The paired calculation check

`calculation-checks/phi4_lattice_metropolis_checks.py`

verifies the local action difference against the total action, checks the pairwise detailed-balance identity, and checks the pointwise quartic stability bound used above.

141.6 Single-Link Metropolis for Finite Gauge Fields

The same finite-regulator theorem should be stated for gauge variables, because lattice gauge Monte Carlo samples compact group data rather than coordinate fields. For the finite $\mathbb{Z}_2$ gauge model whose exact surface expansion is (140.2), write

$$Q(U) = \sum_p U_p, \qquad \pi_\beta(U) = Z_\Lambda(\beta)^{-1} e^{\beta Q(U)}.$$

For a link e, let U^e be the configuration obtained by flipping U_e. The only plaquettes whose values change are the plaquettes containing e. If

$$B_e(U) = \sum_{p \supset e} U_p,$$

then

$$Q(U^e) - Q(U) = -2B_e(U).$$

The single-link Metropolis chain chooses a link uniformly and accepts $U \mapsto U^e$ with probability

$$a(U, e) = \min\{1, \exp(\beta[Q(U^e) - Q(U)])\}.$$

Detailed balance for the finite $\mathbb{Z}_2$ gauge chain. The preceding single-link update satisfies

$$\pi_\beta(U)P(U,U') = \pi_\beta(U')P(U',U)$$

for all finite lattice gauge configurations U, U'. On a connected finite lattice, the chain is irreducible on the full link-configuration space. Gauge-invariant observables may be averaged either on this full space or after quotienting by gauge transformations; the full-space chain preserves the gauge-invariant expectation values because π_β is gauge invariant.

If U' differs from U by more than one link, the off-diagonal transition probabilities in both directions vanish. If $U' = U^e$, set $\Delta Q = Q(U^e) - Q(U)$. The off-diagonal transition probabilities are

$$P(U,U^e) = \frac{1}{|E|}\min\{1, e^{\beta\Delta Q}\}, \qquad P(U^e,U) = \frac{1}{|E|}\min\{1, e^{-\beta\Delta Q}\}.$$

For $\Delta Q \geq 0$,

$$e^{\beta Q(U)}P(U,U^e) = \frac{1}{|E|}e^{\beta Q(U)} = \frac{1}{|E|}e^{\beta Q(U^e)}e^{-\beta\Delta Q} = e^{\beta Q(U^e)}P(U^e,U).$$

The case $\Delta Q < 0$ is the same identity with U and U^e interchanged. Division by $Z_\Lambda(\beta)$ gives detailed balance.

Irreducibility is immediate on the finite link hypercube: any configuration can be reached from any other by flipping the finite set of links on which they differ, and each proposed flip has positive acceptance probability for finite β. Gauge invariance follows because each U_p and each closed Wilson loop is unchanged under $U_{xy} \mapsto g_x U_{xy} g_y$. Therefore averaging over the full link space includes gauge orbits with equal weights and gives the same gauge-invariant expectations as any exact gauge-orbit average with the corresponding orbit weights.

The companion script

```
qft_scripts/z2_gauge_3d_metropolis.py
```

implements this update for a periodic three-dimensional cubic lattice. The associated deterministic check

```
calculation-checks/z2_gauge_metropolis_checks.py
```

verifies the local score change, pairwise detailed balance, gauge invariance of the plaquette action and Wilson loops, and the equality between the 1×1 Wilson-loop average and the plaquette average.

141.7 Compact $SU(2)$ Link Variables

The finite $\mathbb{Z}_2$ model is useful because it is exactly enumerable on small lattices, but nonabelian gauge Monte Carlo must also confront the continuous compact group structure. On a finite periodic 4-dimensional hypercubic lattice, assign to each oriented link $e = (x,\mu)$ a group element

$$U_{x,\mu} \in SU(2), \qquad U_{x+\hat{\mu},-\mu} = U_{x,\mu}^{-1}.$$

The product Haar measure is finite and normalized. For an oriented plaquette

$$U_{x;\mu\nu} = U_{x,\mu}U_{x+\hat{\mu},\nu}U_{x+\hat{\nu},\mu}^{-1}U_{x,\nu}^{-1},$$

define the finite Wilson score

$$Q(U) = \sum_x \sum_{\mu<\nu} \frac{1}{2} \operatorname{Re}\operatorname{tr} U_{x;\mu\nu}. \tag{141.10}$$

The probability measure used by the companion script is

$$d\nu_{\beta,\Lambda}(U) = Z_{\beta,\Lambda}^{-1} \exp[\beta Q(U)] \prod_{(x,\mu)} dU_{x,\mu}, \tag{141.11}$$

where dU is normalized Haar measure on $SU(2)$. This is the finite probability problem corresponding to the Wilson action convention; continuum Yang–Mills normalization and the $a \to 0$ scaling trajectory are separate questions.

A local compact-group Metropolis proposal chooses a link $e = (x, \mu)$ and an element $R \in SU(2)$, and replaces

$$U_{x,\mu} \longmapsto R\, U_{x,\mu}.$$

Let $\kappa(dR)$ be the proposal law. The crucial algebraic hypothesis is inversion symmetry,

$$\kappa(dR) = \kappa(dR^{-1}). \tag{141.12}$$

For example, representing $SU(2)$ by unit quaternions

$$R = \cos\alpha\, 1 + \mathrm{i} \sin\alpha\, \hat{n}\cdot\sigma, \qquad \hat{n} \in S^2,$$

with an even distribution of α and a rotation-invariant distribution of $\hat{n}$, satisfies (141.12).

Lemma 141.3 (Pairwise detailed balance for compact $SU(2)$ link Metropolis)**.** *For the finite measure* (141.11)*, suppose the link proposal chooses e uniformly and R with a law $\kappa(dR) = \kappa(dR^{-1})$. Accept the proposed configuration U' with probability*

$$a(U \to U') = \min\{1, \exp[\beta(Q(U') - Q(U))]\}. \tag{141.13}$$

Then the resulting transition kernel satisfies detailed balance with respect to $d\nu_{\beta,\Lambda}$. If the support of κ generates $SU(2)$ as a closed subgroup, the chain is irreducible with respect to the product Haar measure on the compact link space; rejection probabilities give the usual holding component.

Proof. The left multiplication $U_e \mapsto RU_e$ preserves Haar measure on the updated link. Its inverse move is $U_e' \mapsto R^{-1}U_e' = U_e$. Because the link is chosen with the same probability in both directions and $\kappa(dR) = \kappa(dR^{-1})$, the proposal density for the forward move and the proposal density for the reverse move agree. It remains only to check the Metropolis factor. Set $\Delta Q = Q(U') - Q(U)$. If $\Delta Q \geq 0$, then

$$e^{\beta Q(U)} a(U \to U') = e^{\beta Q(U)} = e^{\beta Q(U')} e^{-\beta \Delta Q} = e^{\beta Q(U')} a(U' \to U).$$

The case $\Delta Q < 0$ is the same identity with U and U' exchanged. Multiplying by the common proposal density and normalizing by $Z_{\beta,\Lambda}$ gives detailed balance. If the support of κ generates $SU(2)$, finite products of allowed local left multiplications can approximate any assignment of link variables by compactness and the group generation hypothesis. This gives irreducibility relative to product Haar measure. The holding component is present whenever a proposal is rejected; one may also insert an explicit lazy step without changing the invariant measure. $\square$

Gauge transformations are maps $g_x \in SU(2)$ at sites, acting by

$$U_{x,\mu} \longmapsto g_x U_{x,\mu} g_{x+\hat{\mu}}^{-1}.$$

They leave (141.10) invariant because plaquettes transform by conjugation:

$$U_{x;\mu\nu} \longmapsto g_x U_{x;\mu\nu} g_x^{-1}.$$

The rectangular Wilson-loop observable in the (μ,ν)-plane is

$$W_{R,T}(x;\mu,\nu) = \frac{1}{2} \operatorname{Re}\operatorname{tr}\left[\prod_{\ell \in \partial\mathcal{R}_{R,T}(x;\mu,\nu)} U_\ell\right], \tag{141.14}$$

with inverse links used when the path orientation is opposite to the positive lattice orientation. The companion script averages (141.14) over translations and coordinate planes.

The script

```
qft_scripts/su2_gauge_4d_metropolis.py
```

implements the preceding finite Markov chain using the unit-quaternion model of $SU(2)$. The deterministic check

```
calculation-checks/su2_gauge_metropolis_checks.py
```

verifies quaternion group operations, the local plaquette-score change, pairwise detailed balance, gauge invariance of $Q(U)$, gauge invariance of Wilson loops, and the equality between the 1×1 Wilson-loop average and the average plaquette. This script is a compact nonabelian finite-regulator Metropolis demonstration. Heat-bath, overrelaxation, and continuum extrapolation require additional regulator data and are not proved by this example.

141.8 Heat-Bath and Overrelaxation Updates

Heat-bath and overrelaxation updates are most cleanly defined as statements about conditional probability measures at fixed regulator. Let

$$\Omega = X \times Y, \qquad d\nu(x,y) = d\nu(x \mid y)\, d\nu_Y(y) \tag{141.15}$$

be a probability space with regular conditional probability $d\nu(x \mid y)$. A heat-bath update of the X-coordinate leaves y fixed and samples x' from $d\nu(x' \mid y)$:

$$K_{\mathrm{HB}}\big((x,y), dx'dy'\big) = d\nu(x' \mid y)\, \delta_y(dy'). \tag{141.16}$$

The kernel (141.16) satisfies detailed balance with respect to $d\nu$. Therefore it preserves $d\nu$. A random scan of finitely many such coordinate heat-bath kernels, with scan probabilities independent of the present configuration, also preserves $d\nu$. The measure on pairs of configurations generated by one heat-bath move is

$$d\nu_Y(y)\, d\nu(x \mid y)\, d\nu(x' \mid y)\, \delta_y(dy'). \tag{141.17}$$

This expression is symmetric under $x \leftrightarrow x'$. That is exactly the detailed-balance identity. A convex linear combination of invariant kernels is invariant, so a random scan preserves $d\nu$.

For the finite $SU(2)$ Wilson measure (141.11), fix every link except one link U_e. The part of the score depending on U_e can be written

$$Q(U) = \frac{1}{2}\operatorname{Re}\operatorname{tr}(U_e V_e) + Q_{\widehat{e}}(U_{\widehat{e}}), \tag{141.18}$$

where $U_{\widehat{e}}$ denotes all links other than U_e, and V_e is the sum of oriented staples multiplying U_e in the plaquettes that contain e. In the fundamental representation of $SU(2)$, the set of matrices

$$a_0\,1 + \mathrm{i}\sum_{j=1}^{3} a_j\sigma_j, \qquad a_j \in \mathbb{R},$$

is closed under addition and multiplication. Hence, if $V_e \neq 0$, there are unique $c_e > 0$ and $W_e \in SU(2)$ such that

$$V_e = c_e W_e. \tag{141.19}$$

If $V_e = 0$, the conditional distribution of U_e is Haar measure.

For $V_e \neq 0$, left or right Haar invariance reduces the conditional density to the one-parameter form

$$d\nu(U_e \mid U_{\widehat{e}}) = Z_e^{-1}\exp\left[\beta c_e \frac{1}{2}\operatorname{Re}\operatorname{tr}(U_e W_e)\right] dU_e. \tag{141.20}$$

Writing

$$X = U_e W_e = x_0\,1 + \mathrm{i}\,\vec{x}\cdot\vec{\sigma}, \qquad x_0^2 + \|\vec{x}\|^2 = 1,$$

the Haar measure on $SU(2) \simeq S^3$ gives the marginal density

$$p_{\beta c_e}(x_0)\,dx_0 = \mathcal{N}_{\beta c_e}^{-1}\sqrt{1-x_0^2}\,e^{\beta c_e x_0}\,dx_0, \qquad -1 \le x_0 \le 1, \tag{141.21}$$

and, conditional on x_0, the direction $\vec{x}/\sqrt{1-x_0^2}$ is uniformly distributed on S^2. An exact $SU(2)$ heat-bath update is precisely an exact sampler of (141.21) and this uniform direction, followed by

$$U_e' = X W_e^{-1}. \tag{141.22}$$

Different practical heat-bath algorithms are different exact or approximate ways of sampling (141.21); the invariant-measure statement is the conditional-sampling theorem above.

Overrelaxation is deterministic rather than heat-bath sampling. At fixed staple W_e, define

$$\mathcal{O}_{W_e}(U_e) = W_e^{-1}U_e^{-1}W_e^{-1}. \tag{141.23}$$

Then

$$\mathcal{O}_{W_e}(\mathcal{O}_{W_e}(U_e)) = U_e, \qquad \frac{1}{2}\operatorname{Re}\operatorname{tr}\big(\mathcal{O}_{W_e}(U_e)W_e\big) = \frac{1}{2}\operatorname{Re}\operatorname{tr}(U_e W_e). \tag{141.24}$$

The first identity says that the map is an involution. The second follows because $\mathcal{O}_{W_e}(U_e)W_e = (U_e W_e)^{-1}$, and an $SU(2)$ matrix has the same real trace as its inverse. The map preserves Haar measure because it is a composition of inversion and left/right multiplication by fixed group elements.

Finite $SU(2)$ heat-bath and overrelaxation invariance. At fixed finite lattice and finite β, an exact single-link heat-bath update with conditional density (141.20) satisfies detailed balance for (141.11). The single-link overrelaxation map (141.23) preserves (141.11). If used as a deterministic involutive kernel, it is reversible with respect to the conditional measure at fixed staple. Any finite composition or random mixture of these updates with Metropolis updates preserves the finite Wilson measure. Indeed, the heat-bath statement is the detailed-balance argument for (141.16) applied with $X = SU(2)$ equal to the selected link and Y equal to all other links. The local conditional density is (141.20) because every plaquette containing e is linear in U_e, and all other plaquettes are absorbed into $Q_{\widehat{e}}$.

For overrelaxation, fix $U_{\widehat{e}}$. Equation (141.24) says that the local Boltzmann factor is unchanged, and Haar preservation says that the conditional Haar measure is unchanged. Thus the conditional probability measure is preserved. Because the map is an involution, the measure on pairs $(U_e, \mathcal{O}_{W_e}(U_e))$ is symmetric, which is detailed balance for the deterministic conditional kernel. Integrating over $U_{\widehat{e}}$ gives preservation of the full finite Wilson measure. Compositions and random mixtures of invariant kernels remain invariant.

Overrelaxation alone is not an ergodic sampling algorithm: it preserves the local score and is deterministic at fixed staple. Its role in practice is to move along approximately constant-action directions between stochastic updates. The mathematical invariant-measure statement above is therefore only one ingredient in a numerical method; it does not by itself establish mixing, error bars, or continuum scaling.

The companion script

```
qft_scripts/su2_gauge_4d_heatbath_overrelaxation.py
```

implements the finite $SU(2)$ Wilson model with exact single-link heat-bath conditionals for (141.21), interleaved with a user-chosen number of deterministic overrelaxation sweeps. It reports finite-cutoff plaquette and Wilson-loop estimates in JSON format. Its smoke mode checks only the finite algorithmic range of the measured observables; statistical autocorrelation analysis, thermodynamic limits, and continuum extrapolation require additional runs and estimates.

The deterministic check

```
calculation-checks/heatbath_overrelaxation_checks.py
```

verifies the finite conditional heat-bath balance identity, the quaternionic $SU(2)$ staple reduction $V = cW$, the overrelaxation involution, local score preservation, orthogonality of the overrelaxation map on S^3, and the local-staple identities used by the companion script.

141.9 $SU(3)$ Link Updates and Subgroup Kernels

Four-dimensional QCD uses $SU(3)$ rather than $SU(2)$. The finite Wilson measure is still an ordinary compact group integral, but the special quaternionic simplification used in (141.19) no longer holds. Let $U_e \in SU(3)$ be a selected link in a finite pure-gauge Wilson lattice system and write the local part of the plaquette score as

$$Q_e(U_e; U_{\widehat{e}}) = \frac{1}{3}\,\mathrm{Re}\,\mathrm{Tr}_3(U_e V_e) + Q_{\widehat{e}}(U_{\widehat{e}}). \tag{141.25}$$

Here V_e is the sum of oriented staples. It is a complex 3×3 matrix determined by the other links and is not, in general, a positive scalar times an element of $SU(3)$. Consequently an exact $SU(3)$ one-link heat-bath does not reduce to the elementary scalar-density (141.21).

The standard finite-regulator substitute is to use $SU(2)$ subgroups of $SU(3)$. For $1 \leq i < j \leq 3$, let $H_{ij} \subset SU(3)$ be the subgroup which acts as $SU(2)$ on the span of the i-th and j-th color basis vectors and acts trivially on the orthogonal color line. Its Lie algebra is spanned by

$$X_{ij} = E_{ij} - E_{ji}, \qquad Y_{ij} = \mathrm{i}(E_{ij} + E_{ji}), \qquad H_{ij} = \mathrm{i}(E_{ii} - E_{jj}),$$

where E_{ij} is the matrix unit. These are anti-Hermitian traceless matrices.

The connected subgroup of $SU(3)$ generated by

$$H_{12}, \qquad H_{13}, \qquad H_{23}$$

is all of $SU(3)$. The six off-diagonal matrices X_{ij}, Y_{ij} for $(ij) = (12), (13), (23)$ are linearly independent. The two diagonal matrices H_{12} and H_{23} are linearly independent and span the diagonal anti-Hermitian traceless subalgebra. Together these eight matrices form a basis of $\mathfrak{su}(3)$. Therefore the Lie algebra generated by the three embedded $\mathfrak{su}(2)$ subalgebras is $\mathfrak{su}(3)$. The subgroup generated by connected subgroups is connected, and the connected Lie subgroup with Lie algebra $\mathfrak{su}(3)$ is $SU(3)$ itself.

Fix a link e and one of the subgroups H_α, where $\alpha \in \{12, 13, 23\}$. A left subgroup update replaces

$$U_e \longmapsto hU_e, \qquad h \in H_\alpha,$$

and leaves all other links fixed. At fixed $U_{\widehat{e}}$, the induced conditional density on H_α is

$$d\nu_{\alpha,e}(h \mid U) = Z_{\alpha,e}(U)^{-1} \exp\left[\frac{\beta}{3} \operatorname{Re} \operatorname{Tr}_3(hU_eV_e)\right] dh, \tag{141.26}$$

where dh is normalized Haar measure on the embedded $SU(2)$.

Finite $SU(3)$ subgroup-update invariance. At fixed finite lattice and finite β, an exact heat-bath draw from (141.26) satisfies detailed balance for the finite $SU(3)$ Wilson measure restricted to the selected subgroup orbit. The same is true for any Metropolis kernel on H_α whose proposal measure is invariant under inversion $h \mapsto h^{-1}$ and whose acceptance probability is computed from the local score (141.25). A random scan over finitely many links and over the three subgroups preserves the finite Wilson measure.

The heat-bath statement is exactly the detailed-balance argument for (141.16) applied to the coordinate $h \in H_\alpha$ on the subgroup orbit hU_e. The conditional density is (141.26) because all plaquettes not containing e are absorbed into $Q_{\widehat{e}}$, while the plaquettes containing e contribute the linear staple term $\frac{1}{3} \operatorname{Re} \operatorname{Tr}_3(hU_eV_e)$.

For the Metropolis statement, let $r(dh)$ be the proposal law on H_α with $r(A) = r(A^{-1})$. From a present subgroup coordinate h_0, propose $h'h_0$ with $h' \sim r$, and accept with

$$a(h_0, h'h_0) = \min\{1, \exp[\beta(Q_e(h'h_0U_e) - Q_e(h_0U_e))]\}.$$

The off-diagonal pairwise detailed-balance identity is the same two-case identity as in the finite Ising Metropolis calculation above, with the symmetric proposal measure supplied by inversion invariance of r and Haar measure. Rejected proposals provide the diagonal holding term. Finally, finite compositions and random convex combinations of invariant kernels preserve the measure.

Controlled Approximation 141.4 (Status of subgroup algorithms)**.** The subgroup update calculation above proves exact finite-cutoff invariance of the Wilson measure for the specified heat-bath or Metropolis kernels. The algorithmic performance questions require additional data: proposal support, scan schedule, boundary conditions, autocorrelation estimates, possible topological-sector barriers, and a continuum-extrapolation analysis. The subgroup Lie-algebra calculation supplies the local group-theoretic reach; the ergodicity and mixing assertions for a concrete Markov chain are separate finite-state or compact-state Markov-chain problems.

The calculation check

`calculation-checks/su3_lattice_update_checks.py`

verifies the embedded $SU(2)$ determinants and unitarity, the $\mathfrak{su}(3)$ span, the subgroup commutator normalizations, gauge covariance of the local staple score, and the finite Metropolis detailed-balance identity used in the preceding proposition.

The companion data-generator

`qft_scripts/`
`su3_gauge_4d_metropolis_hdf5.py`

implements the finite subgroup-update Metropolis kernel described above for the periodic four-dimensional pure $SU(3)$ Wilson lattice model. Its normalized plaquette score is

$$Q(U) = \sum_p \frac{1}{3} \operatorname{Re} \operatorname{Tr}_3 U_p, \qquad d\nu_\beta(U) = Z_{\beta,\Lambda}^{-1} e^{\beta Q(U)} \prod_\ell dU_\ell. \tag{141.27}$$

The script writes, when the optional package `h5py` is available, an HDF5 file containing the run parameters, per-sweep plaquette and rectangular Wilson-loop measurements, the final link field, and the state of the pseudorandom generator. This is a checkpoint at the level of the finite Markov chain: resuming from it continues a regulator computation, not a continuum construction. The script can also export the Wilson-loop measurements as a table with columns

`sample`, $\quad R, \quad T, \quad W,$

which is the sample-level input expected by

`qft_scripts/`
`static_potential_from_wilson_loops.py`.

The deterministic check

`calculation-checks/`
`su3_hdf5_sampler_checks.py`

verifies the embedded-subgroup proposals, the local score change against a direct total-score difference, gauge invariance of the score and Wilson loop, the 1×1 Wilson-loop/plaquette identity, and the HDF5 measurement/checkpoint layout.

The cluster wrapper

`qft_scripts/cluster/slurm/`
`su3_small_pipeline.sbatch`

runs a small finite-regulator vertical slice: the $SU(3)$ sampler above, the Wilson-flow post-processing script of Chapter 140, and the static-potential analysis script of Chapter 140. The surrounding `qft_scripts/cluster` shell templates record the SSH-control, source-sync, result-pull, and SLURM-submission protocol. These files are execution wrappers only. They do not change the finite Markov chain, do not prove mixing, and do not turn the small $L = 2$ smoke job into a continuum calculation.

The array wrapper

`qft_scripts/cluster/slurm/`

`su3_parameter_sweep_array.sbatch`

adds only deterministic parameter bookkeeping. Its helper

`qft_scripts/cluster/su3_sweep_grid.py`

maps a SLURM array index to a finite Cartesian-product coordinate (β, `seed`), writes that coordinate into the task manifest, and then runs the same finite-regulator sampler, Wilson-flow, and static-potential pipeline in a task-local directory. The calculation check

`calculation-checks/cluster_sweep_grid_checks.py`

verifies the Cartesian ordering, out-of-range rejection, and shell-assignment output. The array index is not a physical or statistical index: it is a reproducibility coordinate for a family of finite Markov chains. Mixing, thermalization, and continuum scaling remain separate mathematical and statistical claims.

141.10 Hybrid Monte Carlo and Pseudofermions

Hybrid Monte Carlo is a finite-regulator construction before it is a continuum algorithm. The construction begins with a finite-dimensional configuration manifold Q_Λ, a positive reference measure dq, and a target probability density

$$d\nu(q) = Z^{-1} e^{-S(q)}\, dq, \qquad Z = \int_{Q_\Lambda} e^{-S(q)}\, dq < \infty. \tag{141.28}$$

For a scalar finite lattice, Q_Λ is typically $\mathbb{R}^n$ or a torus. For a compact gauge lattice, Q_Λ is a finite product of compact Lie groups and dq is product Haar measure. The following theorem is stated on an abstract finite-dimensional phase space because this is the part of HMC that is independent of the chosen regulator.

Let $(\Gamma_\Lambda, d\lambda)$ be a finite-dimensional phase space with a positive reference measure, let $H : \Gamma_\Lambda \to \mathbb{R}$ be a measurable Hamiltonian for which

$$d\mu_H(z) = Z_H^{-1} e^{-H(z)}\, d\lambda(z) \tag{141.29}$$

is a probability measure, and let $R : \Gamma_\Lambda \to \Gamma_\Lambda$ be an involution preserving $d\lambda$. In the usual cotangent-bundle example

$$\Gamma_\Lambda = T^* Q_\Lambda, \qquad z = (q, p), \qquad R(q, p) = (q, -p).$$

For $Q_\Lambda = \mathbb{R}^n$ with a positive mass matrix M,

$$H(q,p) = S(q) + \frac{1}{2}p^T M^{-1} p, \qquad d\lambda = dq\,dp. \tag{141.30}$$

The q-marginal of $d\mu_H$ is then (141.28) after normalizing the Gaussian momentum integral.

Definition 141.5 (Reversible volume-preserving trajectory)**.** A numerical trajectory map $\Phi : \Gamma_\Lambda \to \Gamma_\Lambda$ is called reversible and volume preserving, with respect to R, if

$$\Phi_*(d\lambda) = d\lambda, \qquad R\Phi R = \Phi^{-1}. \tag{141.31}$$

The corresponding Metropolis proposal is

$$G = R\Phi. \tag{141.32}$$

The explicit momentum flip in G is not a convention with physical content; it is the algebraic device that makes the proposal an involution. Indeed (141.31) gives

$$G^2 = R\Phi R\Phi = \Phi^{-1}\Phi = 1.$$

HMC Metropolis correction at finite regulator. Assume that Φ is reversible and volume preserving in the sense of Definition 141.5. Let

$$a(z) = \min\{1, \exp[-H(Gz) + H(z)]\} \tag{141.33}$$

and define the Markov kernel

$$K(z,B) = a(z)\mathbf{1}_B(Gz) + (1 - a(z))\mathbf{1}_B(z) \tag{141.34}$$

for measurable $B \subset \Gamma_\Lambda$. Then K satisfies detailed balance with respect to $d\mu_H$. Consequently $d\mu_H$ is invariant.

Since G is an involution and preserves $d\lambda$, every nontrivial proposed move is paired with its reverse. It is enough to prove the pairwise identity

$$e^{-H(z)}a(z) = e^{-H(Gz)}a(Gz). \tag{141.35}$$

Write $\Delta H = H(Gz) - H(z)$. If $\Delta H \geq 0$, then

$$e^{-H(z)}a(z) = e^{-H(z)}e^{-\Delta H} = e^{-H(Gz)} = e^{-H(Gz)}a(Gz),$$

because the reverse move has energy change $-\Delta H \leq 0$ and is accepted with probability 1. The case $\Delta H < 0$ is the same calculation with z and Gz interchanged. Multiplying (141.35) by the common volume element on the paired points gives detailed balance for the moving part of the kernel. The holding part is diagonal and is therefore symmetric with respect to $d\mu_H$. This proves detailed balance for K.

The usual HMC transition first refreshes p from its Gaussian conditional distribution and then applies (141.34). Momentum refreshment preserves $d\mu_H$ because it leaves q fixed and replaces the momentum by an independent sample from the p-conditional density. The composition therefore preserves $d\mu_H$, and its q-marginal preserves $d\nu$. Ergodicity, mixing time, acceptance rate, and scaling toward a continuum limit are separate claims; they do not follow from detailed balance alone.

Leapfrog reversibility and volume preservation. Let $Q_\Lambda = \mathbb{R}^n$, let $S \in C^2(\mathbb{R}^n)$, and let M be a positive definite symmetric matrix. Define the kick and drift maps

$$K_a(q,p) = (q, p - a\nabla S(q)), \qquad D_\epsilon(q,p) = (q + \epsilon M^{-1}p, p), \tag{141.36}$$

and the one-step leapfrog map

$$L_\epsilon = K_{\epsilon/2} D_\epsilon K_{\epsilon/2}. \tag{141.37}$$

Then L_ϵ preserves $dq\,dp$ and satisfies

$$RL_\epsilon R = L_\epsilon^{-1}, \qquad R(q,p) = (q, -p). \tag{141.38}$$

Every finite power L_ϵ^N, $N \in \mathbb{Z}_{\geq 0}$, has the same two properties. These properties are the finite-dimensional algebra needed by the HMC Metropolis step. The Jacobian matrix of K_a is block triangular,

$$\begin{pmatrix} 1 & 0 \\ -a \operatorname{Hess} S(q) & 1 \end{pmatrix},$$

and therefore has determinant 1. The Jacobian matrix of D_ϵ is

$$\begin{pmatrix} 1 & \epsilon M^{-1} \\ 0 & 1 \end{pmatrix},$$

also of determinant 1. Hence L_ϵ preserves $dq\,dp$. Moreover

$$RK_aR = K_{-a} = K_a^{-1}, \qquad RD_\epsilon R = D_{-\epsilon} = D_\epsilon^{-1}.$$

Using the symmetric ordering in (141.37),

$$RL_\epsilon R = K_{-\epsilon/2} D_{-\epsilon} K_{-\epsilon/2} = L_\epsilon^{-1}.$$

A finite product of equal leapfrog steps is again reversible in the sense needed for HMC, because

$$RL_\epsilon^N R = (RL_\epsilon R)^N = L_\epsilon^{-N}.$$

For a compact gauge field, the same proof is expressed on the cotangent bundle of a compact Lie group product. The drift is group multiplication by the exponential of a Lie-algebra momentum, the kick changes the momentum by the derivative of the gauge action, and the reference measure is the Liouville measure built from Haar measure and Lebesgue measure on the Lie algebra. The finite-regulator statement remains exactly the same: volume preservation, reversibility, and the Metropolis correction imply invariance of the finite target measure. The construction of globally smooth forces, the treatment of constraints, and the proof of ergodicity depend on the chosen lattice action and are additional data.

Pseudofermions enter when the fermion determinant is real and positive after flavor pairing. Let A be a positive Hermitian $N \times N$ matrix and write

$$d^{2N}\phi = \prod_{j=1}^{N} d(\operatorname{Re}\phi_j)\, d(\operatorname{Im}\phi_j)$$

for Lebesgue measure on $\mathbb{C}^N$. The elementary identity is the following finite-dimensional Gaussian integral.

If $A > 0$ is Hermitian, then

$$\int_{\mathbb{C}^N} \exp[-\phi^\dagger A^{-1}\phi]\, d^{2N}\phi = \pi^N \det A. \tag{141.39}$$

Consequently a positive determinant factor $\det A$ in a finite Euclidean integral may be represented, up to the A-independent constant π^N, by the pseudofermion action $\phi^\dagger A^{-1}\phi$. Choose a unitary matrix U such that

$$A = U\ \mathrm{diag}(\lambda_1, \dots, \lambda_N)\, U^\dagger, \qquad \lambda_j > 0.$$

The real Jacobian of the unitary change of variables $\psi = U^\dagger\phi$ has absolute value 1. Hence the integral factorizes:

$$\prod_{j=1}^{N} \int_{\mathbb{C}} \exp[-|\psi_j|^2/\lambda_j]\, d(\mathrm{Re}\,\psi_j) d(\mathrm{Im}\,\psi_j) = \prod_{j=1}^{N} \pi\lambda_j = \pi^N \det A.$$

For a lattice Dirac operator $D(U)$, two degenerate flavors commonly give the positive factor

$$\det(D^\dagger D) = |\det D|^2, \qquad A(U) = D(U)^\dagger D(U) > 0$$

after zero modes have been excluded or regulated. Equation (141.39) then turns the determinant into a bosonic Gaussian integral with a nonlocal action involving $A(U)^{-1}$. The nonlocality is not a conceptual defect; it is a finite linear-algebra statement. It becomes an algorithmic problem because each force evaluation requires solving a linear system.

For N_f degenerate flavors one formally needs

$$\det A^\alpha, \qquad \alpha = N_f/2,$$

and the exact pseudofermion action would contain the matrix function $A^{-\alpha}$. Rational hybrid Monte Carlo replaces $x^{-\alpha}$ on a spectral interval $[\lambda_-, \lambda_+]$ containing the spectrum of $A(U)$ by a positive rational function $r_m(x)$. At finite regulator this replacement defines the exactly sampled determinant

$$\pi^{-N} \int_{\mathbb{C}^N} \exp[-\phi^\dagger r_m(A)\phi]\, d^{2N}\phi = \frac{1}{\det r_m(A)}. \tag{141.40}$$

It equals $\det A^\alpha$ only if $r_m(A) = A^{-\alpha}$ on the spectrum, or if an additional exact correction or reweighting factor is included. Thus RHMC is not a slogan for "fractional fermions"; it is an exact Markov-chain algorithm for a rationalized finite measure, together with a separate approximation or correction statement comparing that measure to the desired determinant.

The elementary spectral error estimate used in such a comparison is pointwise. If

$$\|r_m(A) - A^{-\alpha}\|_{\mathrm{op}} \le \delta, \tag{141.41}$$

then for every pseudofermion vector ϕ,

$$\left|\phi^\dagger\big(r_m(A) - A^{-\alpha}\big)\phi\right| \le \delta\, \|\phi\|^2. \tag{141.42}$$

This bound controls the error in the pseudofermion action for a fixed configuration and a fixed pseudofermion vector. Turning it into a uniform observable bound requires further hypotheses, for example a uniform spectral gap, a chosen rational approximation, and control of the pseudofermion tail or an exact reweighting step. These hypotheses are finite-regulator algorithmic data, not consequences of QFT axioms.

Solver, rationalization, and reversibility checks. A production HMC or RHMC run must record more than the target action and the measured observables. This is the implementation-defect layer of Section 141.1. At finite regulator the algorithmic check record contains the integrator step size ϵ, the number of leapfrog or Omelyan steps, the momentum-refresh law, the linear solver, the rational approximation interval $[\lambda_-, \lambda_+]$, the solver tolerance, and the finite diagnostics

$$\Delta H(z) = H(Gz) - H(z), \qquad \Delta_{\rm rev}(z) = d_\Gamma(z, R\Phi R\Phi z), \qquad \rho_{\rm lin} = \|b - Ax\|.$$

The exact detailed-balance argument above applies to the exact finite map actually used by the Markov chain. Therefore finite-precision code must make its reversibility and solver errors visible as part of the data record before the output is interpreted as evidence for the intended lattice measure.

The basic linear-solver bounds are finite spectral estimates. Let $A = A^\dagger > 0$ with $A \geq \lambda_- I$, let $y = A^{-1}\phi$, and let x be an approximate solve with residual

$$r = \phi - Ax.$$

Then

$$y - x = A^{-1}r, \qquad \|y - x\| \leq \lambda_-^{-1}\|r\|.$$

Consequently the pseudofermion-action coordinate $\phi^\dagger A^{-1}\phi$ obeys

$$\left|\phi^\dagger A^{-1}\phi - \phi^\dagger x\right| \leq \lambda_-^{-1}\|\phi\|\,\|r\|. \tag{141.43}$$

If $\dot{A}$ is the directional derivative of A along a configuration variation and $\|\dot{A}\|_{\rm op} \leq L$, the exact force term $-y^\dagger \dot{A}\, y$ and the force computed from x differ by

$$\left|y^\dagger \dot{A}\, y - x^\dagger \dot{A}\, x\right| \leq L\left(2\|x\|\lambda_-^{-1}\|r\| + \lambda_-^{-2}\|r\|^2\right). \tag{141.44}$$

This is the finite estimate that connects Krylov residual logs to Hamiltonian-force accuracy. A solver tolerance quoted without λ_-, force-norm, and reversibility diagnostics is only an implementation setting.

For RHMC the determinant comparison is also finite. Suppose

$$r_m(\lambda) = \lambda^{-\alpha} e^{\eta(\lambda)} \quad \text{for} \quad \lambda \in [\lambda_-, \lambda_+], \qquad |\eta(\lambda)| \leq \eta_*.$$

For a finite matrix A of size N with spectrum in this interval, the rationalized determinant sampled by (141.40) differs from the desired $\det A^\alpha$ by the reweighting factor

$$W_{\rm rat}(A) = \det(A)^\alpha \det r_m(A) = \exp\left(\sum_{j=1}^{N} \eta(\lambda_j)\right), \qquad |\log W_{\rm rat}(A)| \leq N\eta_*. \tag{141.45}$$

Thus an exact rationalized Markov chain, an exact reweighting chain, and an unreweighted approximation are three different finite probability measures. The simulation record must specify which one is being used.

A small public-facing implementation of these coordinates is

`qft_scripts/hmc_rhmc_finite_demo.py`.

It samples a finite periodic scalar lattice action by reversible leapfrog HMC and evaluates a positive rational pseudofermion action by shifted conjugate gradient solves. Its JSON output reports the acceptance coordinate, $\max|\Delta H|$, a reversibility defect, the positive-matrix spectral edge, the rational action, and the maximum solver residual. These are finite-regulator diagnostics; continuum and infinite-volume claims require the additional limit data described in Chapter 145.

The deterministic companion check

```
calculation-checks/hmc_pseudofermion_checks.py
```

verifies the leapfrog determinant and reversibility identities for a one-dimensional quadratic action, the pairwise Metropolis balance identity, the pseudofermion determinant identity after diagonalization, and the spectral action-error bound (141.42). It also checks (141.43)– (141.45) and imports the finite HMC/RHMC smoke module as a companion-script regression.

141.11 Autocorrelation and Effective Sample Size

For a stationary chain and a real observable O, define the connected autocorrelation function

$$C_O(t) = \langle O(\Phi_t)O(\Phi_0)\rangle - \langle O\rangle^2.$$

The integrated autocorrelation time is

$$\tau_{\text{int}}(O) = \frac{1}{2} + \sum_{t=1}^{\infty} \frac{C_O(t)}{C_O(0)}$$

when the sum converges. The variance of the sample mean is then

$$\mathrm{Var}(\overline{O}_N) = \frac{2\tau_{\text{int}}(O)}{N}\,\mathrm{Var}(O) + o(N^{-1}).$$

Critical slowing down is the growth of τ_{int} along a scaling trajectory toward a continuum limit.

Exact finite-N variance identity. Assume the chain is stationary and O is real with $C_O(0) > 0$. Then

$$\mathrm{Var}(\overline{O}_N) = \frac{1}{N^2}\left[NC_O(0) + 2\sum_{t=1}^{N-1}(N-t)C_O(t)\right]. \tag{141.46}$$

If

$$\sum_{t=1}^{\infty} |C_O(t)| < \infty, \tag{141.47}$$

then

$$\mathrm{Var}(\overline{O}_N) = \frac{2\tau_{\text{int}}(O)}{N}\,C_O(0) + o(N^{-1}). \tag{141.48}$$

Write $O_n = O(\Phi_n)$ and $\mu_O = \langle O\rangle$. Stationarity gives

$$\mathrm{Cov}(O_r, O_s) = C_O(|r-s|).$$

Therefore

$$\mathrm{Var}(\overline{O}_N) = \frac{1}{N^2} \sum_{r,s=0}^{N-1} \mathrm{Cov}(O_r, O_s).$$

There are N pairs with $r = s$. For each lag $t \geq 1$, there are $N - t$ ordered pairs with $s = r + t$ and $N - t$ ordered pairs with $r = s + t$. This proves (141.46). Dividing by $N^{-1}C_O(0)$ and using absolute summability to replace

$$\sum_{t=1}^{N-1} \left(1 - \frac{t}{N}\right) \frac{C_O(t)}{C_O(0)}$$

by its infinite sum gives (141.48).

141.12 Blocking, Jackknife, and Bootstrap Error Coordinates

Autocorrelation estimates enter numerical QFT through finite data-analysis coordinates. Let $O_0, \ldots, O_{N-1}$ be a real time series sampled after thermalization. Choose a block length b and write

$$B = \left\lfloor \frac{N}{b} \right\rfloor.$$

Discarding the final incomplete block for definiteness, define block means

$$Y_j = \frac{1}{b} \sum_{r=0}^{b-1} O_{jb+r}, \qquad j = 0, \ldots, B - 1, \tag{141.49}$$

and

$$\overline{Y} = \frac{1}{B} \sum_{j=0}^{B-1} Y_j.$$

If b is large compared with the integrated autocorrelation scale, the block means are closer to independent samples than the original time series. This sentence is a diagnostic hypothesis about the Markov chain and the chosen observable; it is checked by varying b, by monitoring slow modes, and by comparing independent chains.

Remark 141.6 (Delete-one-block jackknife for the mean). Assume $B \geq 2$. Let

$$s_Y^2 = \frac{1}{B-1} \sum_{j=0}^{B-1} (Y_j - \overline{Y})^2. \tag{141.50}$$

The blocked standard-error coordinate for $\overline{Y}$ is

$$\mathrm{se}_{\mathrm{block}}(\overline{Y}) = \sqrt{\frac{s_Y^2}{B}}. \tag{141.51}$$

Let $\overline{Y}_{(j)}$ be the mean of the $B - 1$ block means with the j-th block removed:

$$\overline{Y}_{(j)} = \frac{1}{B-1} \sum_{k \neq j} Y_k. \tag{141.52}$$

Then the delete-one-block jackknife variance for the mean,

$$\mathrm{Var}_{\mathrm{jack}}(\overline{Y}) = \frac{B-1}{B} \sum_{j=0}^{B-1} \left(\overline{Y}_{(j)} - \frac{1}{B} \sum_{k=0}^{B-1} \overline{Y}_{(k)} \right)^2, \tag{141.53}$$

equals s_Y^2/B. Since

$$\overline{Y}_{(j)} = \frac{B\overline{Y} - Y_j}{B-1},$$

the average of the delete-one estimates is

$$\frac{1}{B} \sum_{j=0}^{B-1} \overline{Y}_{(j)} = \overline{Y}.$$

Therefore

$$\overline{Y}_{(j)} - \overline{Y} = -\frac{Y_j - \overline{Y}}{B-1}.$$

Substitution in (141.53) gives

$$\mathrm{Var}_{\mathrm{jack}}(\overline{Y}) = \frac{B-1}{B} \frac{1}{(B-1)^2} \sum_{j=0}^{B-1} (Y_j - \overline{Y})^2 = \frac{s_Y^2}{B}.$$

The preceding remark is the scalar case of the error-coordinate needed for Wilson-loop ratios, flowed-energy couplings, and hadron effective masses. Let $Y_j \in \mathbb{R}^m$ be block means for m elementary observables, let

$$\overline{Y} = \frac{1}{B} \sum_{j=0}^{B-1} Y_j, \qquad d_j = Y_j - \overline{Y}, \qquad S_Y = \frac{1}{B-1} \sum_{j=0}^{B-1} d_j d_j^{\mathsf{T}}, \tag{141.54}$$

and let

$$\overline{Y}_{(j)} = \frac{B\overline{Y} - Y_j}{B-1}. \tag{141.55}$$

For a smooth scalar analysis function $F : U \to \mathbb{R}$, where $U \subset \mathbb{R}^m$ is open and contains the convex hull of the $\overline{Y}_{(j)}$, define

$$F_{(j)} = F(\overline{Y}_{(j)}), \qquad \widehat{F}_\bullet = \frac{1}{B} \sum_{j=0}^{B-1} F_{(j)},$$

and

$$\widehat{V}_{\mathrm{jack}}(F) = \frac{B-1}{B} \sum_{j=0}^{B-1} (F_{(j)} - \widehat{F}_\bullet)^2. \tag{141.56}$$

Correlated delete-one-block jackknife for smooth ratios. Assume $B \geq 2$. Let $g = DF_{\overline{Y}}$ be the differential of F at $\overline{Y}$, regarded as a row vector. Suppose that the operator norm of the Hessian satisfies

$$\|D^2 F_x\|_{\mathrm{op}} \leq M$$

on the convex hull of the delete-one means. Put

$$D_2 = \left(\sum_{j=0}^{B-1} \|d_j\|^2 \right)^{1/2}, \qquad D_4 = \left(\sum_{j=0}^{B-1} \|d_j\|^4 \right)^{1/2},$$

and

$$A = \frac{\|g\| D_2}{B-1}, \qquad E = \frac{M D_4}{(B-1)^2}.$$

Then

$$\left| \widehat{V}_{\text{jack}}(F) - \frac{1}{B}\, g S_Y g^{\mathsf{T}} \right| \le \frac{B-1}{B}(2AE + E^2). \tag{141.57}$$

Taylor's theorem gives

$$F(\overline{Y}_{(j)}) = F(\overline{Y}) + g(\overline{Y}_{(j)} - \overline{Y}) + R_j, \qquad |R_j| \le \frac{M}{2} \|\overline{Y}_{(j)} - \overline{Y}\|^2.$$

Since

$$\overline{Y}_{(j)} - \overline{Y} = -\frac{d_j}{B-1}$$

and $\sum_j d_j = 0$, the average of the linear terms is zero. Therefore the linear part of $\widehat{V}_{\text{jack}}(F)$ is

$$\frac{B-1}{B} \sum_{j=0}^{B-1} \left(\frac{g d_j}{B-1} \right)^2 = \frac{1}{B}\, g S_Y g^{\mathsf{T}}.$$

Let

$$a_j = -\frac{g d_j}{B-1}, \qquad e_j = R_j - \frac{1}{B} \sum_{k=0}^{B-1} R_k.$$

Then the difference between the full jackknife variance and its linear part is

$$\frac{B-1}{B} \left(2 \sum_{j=0}^{B-1} a_j e_j + \sum_{j=0}^{B-1} e_j^2 \right).$$

Cauchy's inequality bounds

$$\left(\sum_j a_j^2 \right)^{1/2} \le A.$$

Also

$$\left(\sum_j e_j^2 \right)^{1/2} \le 2 \left(\sum_j R_j^2 \right)^{1/2} \le E.$$

Substituting these two bounds proves (141.57).

For the static potential, the elementary block vector contains the positive Wilson-loop coordinates $W(n,m)$ needed for a chosen finite list of rectangles. The analysis functions are

$$F^{\text{eff}}_{n,m}(W) = -a^{-1} \log \frac{W(n,m+1)}{W(n,m)} \tag{141.58}$$

and

$$F^{\chi}_{n,m}(W) = -\log \frac{W(n,m)W(n-1,m-1)}{W(n,m-1)W(n-1,m)}. \tag{141.59}$$

They are smooth on the positive orthant. Applying the jackknife to $F^{\mathrm{eff}}_{n,m}$ or $F^{\chi}_{n,m}$ after deleting an entire Monte Carlo block is therefore a correlated calculation: the covariance among the rectangles is the matrix S_Y in (141.54). Propagating the four Wilson-loop errors in a Creutz ratio independently is a different approximation, justified only after a separate argument that the relevant off-diagonal covariance entries are negligible in the chosen data-analysis scheme.

Block bootstrap resampling similarly resamples the block means or the original contiguous blocks and recomputes the full nonlinear analysis function on every resampled data set. Its consistency requires an asymptotic regime in which $b \to \infty$, $B \to \infty$, and the block length grows slowly enough compared with the total chain length; the finite script described below reports the coordinate values and leaves the asymptotic claim to the stated Markov-chain hypotheses.

The companion script

`qft_scripts/autocorrelation_resampling.py`

computes the biased autocorrelation estimate, a windowed τ_{int}, blocked standard errors, delete-one-block jackknife errors for the mean, and block-bootstrap errors from a one-column CSV time series. It is intended as the upstream error-analysis layer for Wilson-loop and flowed-observable measurements before ratios such as (140.53) are formed.

141.13 Reweighting and the Sign Problem

Suppose the target finite-regulator integral has complex weight

$$e^{-S(\Phi)} = e^{-S_R(\Phi)} e^{\mathrm{i}\Theta(\Phi)}$$

with positive reference measure

$$d\nu_R = \frac{1}{Z_R} e^{-S_R(\Phi)} d\Phi.$$

Then

$$\langle O \rangle = \frac{\langle O e^{\mathrm{i}\Theta} \rangle_R}{\langle e^{\mathrm{i}\Theta} \rangle_R}.$$

The denominator is the average phase. If the free-energy densities of the target and phase-quenched systems differ by Δf in volume V, then

$$\left| \langle e^{\mathrm{i}\Theta} \rangle_R \right| = \exp[-V\Delta f].$$

Here

$$Z = \int e^{-S_R(\Phi)} e^{\mathrm{i}\Theta(\Phi)}\, d\Phi, \qquad Z_R = \int e^{-S_R(\Phi)}\, d\Phi, \qquad \langle e^{\mathrm{i}\Theta} \rangle_R = \frac{Z}{Z_R},$$

and Δf is the real number defined by

$$\Delta f = -\frac{1}{V} \log \left| \frac{Z}{Z_R} \right|. \tag{141.60}$$

Estimating a ratio with a denominator of this size requires a number of independent samples growing as $\exp[2V\Delta f]$ for fixed relative error, unless the numerator and denominator are computed by a method that avoids phase reweighting.

Finite-volume phase-reweighting variance bound. Let

$$m = \langle e^{\mathrm{i}\Theta} \rangle_R \neq 0.$$

For N independent samples from $d\nu_R$, the unbiased estimator

$$\widehat{m}_N = \frac{1}{N} \sum_{j=1}^{N} e^{\mathrm{i}\Theta(\Phi_j)}$$

satisfies

$$\frac{\mathbb{E}_R |\widehat{m}_N - m|^2}{|m|^2} = \frac{1 - |m|^2}{N|m|^2}. \tag{141.61}$$

Consequently, if $|m| = \exp[-V\Delta f]$, any direct phase-reweighting estimation of the denominator with relative mean-square error at most δ^2 requires

$$N \geq \delta^{-2} \left(e^{2V\Delta f} - 1 \right). \tag{141.62}$$

Since $|e^{\mathrm{i}\Theta}| = 1$,

$$\mathbb{E}_R |e^{\mathrm{i}\Theta} - m|^2 = \mathbb{E}_R |e^{\mathrm{i}\Theta}|^2 - |m|^2 = 1 - |m|^2.$$

Independence divides this variance by N, proving (141.61). Substitution of $|m| = e^{-V\Delta f}$ gives (141.62). The estimate concerns the denominator alone. A ratio estimator for $\langle O \rangle$ can have additional covariance between numerator and denominator, but it cannot remove the need to determine a denominator of exponentially small magnitude unless the method changes the representation of the finite-regulator integral.

141.14 Fermion Determinants

For lattice fermions with quadratic action

$$\bar{\psi} D[U] \psi,$$

Berezin integration gives

$$\int d\bar{\psi}\, d\psi\, e^{-\bar{\psi} D[U] \psi} = \det D[U].$$

If D obeys a positivity relation, such as

$$\gamma_5 D \gamma_5 = D^\dagger$$

together with flavor pairing, the determinant can be nonnegative. The single-flavor statement and the paired-flavor statement are different.

Remark 141.7 (Determinant reality from γ_5-Hermiticity). Let D be a finite lattice Dirac matrix and let Γ be an involution with $\Gamma^\dagger = \Gamma$ and $\Gamma^2 = 1$. If

$$\Gamma D \Gamma = D^\dagger, \tag{141.63}$$

then $\det D \in \mathbb{R}$. For two degenerate flavors with fermion integral $(\det D)^2$, the finite weight is nonnegative because the real determinant is squared. For a conjugate pair of flavors with determinant

$$\det D \ \det D^\dagger = |\det D|^2, \tag{141.64}$$

the finite-regulator weight is nonnegative before possible zero-mode subtleties. Thus a positivity claim must state the actual flavor and operator pairing together with the formal γ_5-Hermiticity relation. Taking determinants in (141.63) gives

$$\det D^\dagger = \det(\Gamma D \Gamma) = \det(\Gamma)^2 \det D = \det D.$$

Since $\det D^\dagger = \overline{\det D}$, the determinant is real. A real number can have either sign, so γ_5-Hermiticity alone does not give a positive single-flavor measure. For an even number of exactly degenerate copies, $(\det D)^{2k} \geq 0$. The conjugate-pair identity (141.64) is the elementary determinant identity $\det D^\dagger = \overline{\det D}$. If $\det D = 0$, the nonnegative weight vanishes and sampling remains mathematically defined, but algorithmic ergodicity and condition-number issues become separate finite regulator questions.

At finite chemical potential or in chiral gauge theories this positivity may fail. The finite regulator then has no declared positive probability measure whose expectations are the target correlators without reweighting or contour deformation.

The deterministic check

```
calculation-checks/monte_carlo_sign_problem_checks.py
```

verifies the finite-N autocorrelation identity, the phase-reweighting variance bound, and the determinant-reality-versus-positivity distinction in small finite examples.

141.15 Continuum Interpretation

Monte Carlo data become QFT data only after the regulator, source normalization, operator renormalization, scaling trajectory, and error limits are specified. A numerical estimate at finite lattice spacing is a statement about $d\nu_\Lambda$. A continuum claim requires convergence of a family of finite-regulator expectations in the topology needed for OS reconstruction, Wightman reconstruction, or the intended observable sector.

Open Problem 141.8 (Sign-problem-preserving continuum construction). Classify QFTs with complex Euclidean weights for which a nonperturbative continuum limit can be constructed by a positive enlarged-variable measure, an exact contour deformation, or another finite-regulator method with controlled variance and locality.

Chapter 142

Rigorous Renormalization Group

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter formulates renormalization group transformations as maps on normed spaces of interactions or probability measures. The aim is to replace informal terminology about flows by explicit maps, estimates, fixed points, and coordinate decompositions.

142.1 The Wilsonian Reconstruction Problem

The word "Wilsonian" will be used in a strict sense. It refers first to an exact finite-regulator operation that integrates or blocks degrees of freedom, and only second to a coordinate description of the resulting object. A coordinate flow is useful only when it is tied to the finite operation and to reconstruction estimates for observables.

Definition 142.1 (Wilsonian RG reconstruction problem)**.** A Wilsonian RG reconstruction problem consists of the following data.

1. A directed regulator family indexed by k, with finite-regulator configuration spaces, field spaces, measures, densities, or BV integration data, denoted collectively by $\mathcal{X}_k$. The ordering convention is part of the datum: increasing k means removing or refining the ultraviolet regulator.
2. Exact finite-regulator maps $T_{k,k-1}$ obtained by Gaussian fluctuation integration, block-spin pushforward, tensor blocking, gauge-equivariant lattice blocking, or BV pushforward. These

maps act on the finite-regulator objects themselves before any projection to finitely many couplings is made.

3. Localization maps

$$\mathrm{Loc}_k : T_{k,k-1}(\mathcal{X}_k) \longrightarrow \mathcal{B}_{k-1}$$

into declared Banach, Fréchet, polymer, tensor, or BV charts. The image separates retained local coordinates from the irrelevant or nonlocal tail measured in the chosen norm.

4. Source extensions for finite test windows E. These include finite-regulator source functionals $Z_{k,E}$ or $W_{k,E} = \log Z_{k,E}$, local source and contact coordinates, and the source-decorated irrelevant tail. The source radius and the topology in which source derivatives are taken are part of the data.

5. Reconstruction maps from the regulated or chart-level objects to the intended observable topology: Schwinger distributions, Wightman distributions, local algebras, gauge-invariant line or loop observables, tensor-network observables, or BV cohomology classes.

6. Estimates, in the stated norms and seminorms, proving the domain stability of the exact maps, localization remainders, contraction or summability of irrelevant tails, tuning of unstable coordinates, convergence of source windows, reflection positivity or its declared replacement, distributional growth bounds, and a common directed regulator schedule on which the estimates needed for each finite reconstruction demand hold together.

A fixed point, stable trajectory, or continuum limit is a theorem about this package, not about the zero of a beta function by itself. Perturbative beta functions, functional-RG truncations, and tensor-network numerics are useful coordinate shadows when comparison maps and residual estimates relate them to Definition 142.1. The later sections unpack the items in the definition: exact Gaussian pushforward, Banach-space linearization, projection residuals, polymer estimates, source windows, OS/Wightman reconstruction, auxiliary-to-short-range transfer, gauge-compatible blocking, and tensor or functional-RG comparison.

142.2 Gaussian Block Integration

Let C be a positive covariance on a finite regulator space of fields and suppose it decomposes as

$$C = C_< + \Gamma.$$

Let $d\mu_C$, $d\mu_{C_<}$, and $d\mu_\Gamma$ be the corresponding Gaussian measures. For an interaction V, define the single-step effective interaction

$$e^{-V_<(\phi)} = \int d\mu_\Gamma(\zeta)\, e^{-V(\phi+\zeta)}.$$

This identity is exact at finite regulator. A scale transformation S_L then gives an RG map

$$\mathcal{R}_L(V) = S_L V_<.$$

The definition contains three inputs: covariance decomposition, integration domain, and rescaling convention.

142.3 Several RG Maps And Their Mathematical Objects

The name "renormalization group" is used for several closely related, but mathematically distinct, maps. In this chapter the object being transformed is always part of the statement. Let C_Λ be a regulated covariance and let $C_{\Lambda_1} - C_{\Lambda_0}$ be positive for $\Lambda_1 > \Lambda_0$ in the chosen cutoff ordering. The exact Wilsonian pushforward from Λ_1 to Λ_0 is the identity

$$e^{-V_{\Lambda_0}(\phi)} = \int d\mu_{C_{\Lambda_1}-C_{\Lambda_0}}(\zeta)\, e^{-V_{\Lambda_1}(\phi+\zeta)}, \tag{142.1}$$

with additive normalization fixed separately. Its semigroup property is the Gaussian convolution identity:

$$\int d\mu_{\Gamma_1}(\zeta_1) \int d\mu_{\Gamma_2}(\zeta_2)\, F(\phi+\zeta_1+\zeta_2) = \int d\mu_{\Gamma_1+\Gamma_2}(\zeta)\, F(\phi+\zeta), \tag{142.2}$$

valid whenever Γ_1, Γ_2, and $\Gamma_1+\Gamma_2$ are the corresponding positive covariances. Thus the exact Wilsonian map transforms interactions or measures.

The Wilson–Polchinski equation is the infinitesimal form of the same Gaussian pushforward in a smooth-cutoff coordinate chart. Let t be a cutoff parameter such that lowering t by $\delta t > 0$ integrates the positive shell covariance $\delta t\, \dot{C}_t + O(\delta t^2)$, where $\dot{C}_t = \partial_t C_t$. Thus

$$e^{-V_{t-\delta t}[\phi]} = \left(1 + \frac{\delta t}{2}\Delta_{\dot{C}_t}\right) e^{-V_t[\phi]} + O(\delta t^2). \tag{142.3}$$

Writing

$$\Delta_{\dot{C}_t} = \int_{x,y} \dot{C}_t(x,y) \frac{\delta}{\delta\phi(x)} \frac{\delta}{\delta\phi(y)},$$

one obtains, equivalently, at finite regulator

$$\partial_t e^{-V_t[\phi]} = -\frac{1}{2}\, \Delta_{\dot{C}_t} e^{-V_t[\phi]}, \tag{142.4}$$

or, equivalently,

$$\partial_t V_t = \frac{1}{2}\langle \delta V_t, \dot{C}_t\, \delta V_t\rangle - \frac{1}{2}\,\mathrm{Tr}(\dot{C}_t\, \delta^2 V_t), \tag{142.5}$$

with this convention for t. Reparametrizing the scale variable multiplies both sides by the corresponding chain-rule factor. The equation is a differential equation for the Wilsonian interaction. Its theorem-level use requires a function space, bounds on the functional derivatives, and a construction of the trajectory.

The effective-average-action or Wetterich flow transforms a different functional. At finite regulator, introduce an infrared quadratic term

$$\Delta S_k[\phi] = \frac{1}{2}\langle \phi, R_k \phi\rangle$$

and define

$$W_k[J] = \log \int d\mu(\phi)\, \exp(\langle J, \phi\rangle - \Delta S_k[\phi] - V[\phi])\,.$$

For $\varphi = \delta W_k/\delta J$, the effective average action is the modified Legendre transform

$$\Gamma_k[\varphi] = \langle J, \varphi\rangle - W_k[J] - \Delta S_k[\varphi]. \tag{142.6}$$

Where the finite-dimensional Hessians are invertible, direct differentiation gives

$$\partial_t \Gamma_k[\varphi] = \frac{1}{2}\,\mathrm{Tr}\Big[\big(\Gamma_k^{(2)}[\varphi] + R_k\big)^{-1}\partial_t R_k\Big]\,. \tag{142.7}$$

This is an exact identity for a regulated Legendre transform. A fixed point of a finite truncation of Γ_k is a computational approximation to this functional flow; a theorem-level statement requires an embedding of the truncation into a complete function space and a bound on the omitted directions.

Constructive polymer RG transforms a pair consisting of local coordinates and polymer activities. Its single step has the form

$$(g, K) \longmapsto (g', K')$$

where g' is obtained by a localization map extracting finitely many local Taylor coefficients, while K' is the remaining nonlocal activity measured in a polymer norm. The decisive estimate is a contraction or summability bound for K', together with Lipschitz dependence on g. This is the form in which many rigorous scalar and fermionic constructions are proved.

Tensor RG starts from a tensor-network representation of a partition function or state sum. Its map acts on local tensors T rather than on a continuum interaction:

$$T \longmapsto \mathcal{T}(T), \tag{142.8}$$

with blocking, truncation or disentangling, and a declared Hilbert, Hilbert–Schmidt, or Banach topology on the tensor space. A tensor-RG fixed point theorem therefore proves contraction or stable-manifold estimates for $\mathcal{T}$ in that tensor topology, plus a reconstruction map if the claim concerns continuum QFT observables.

Definition 142.2 (Comparison of two RG descriptions)**.** Let $(\mathcal{B}_1, \mathcal{R}_1)$ and $(\mathcal{B}_2, \mathcal{R}_2)$ be two RG descriptions, each with its own domain, norm, and reconstruction maps. A comparison from the first to the second consists of maps $\Phi : \mathcal{U}_1 \to \mathcal{B}_2$ and $\Psi : \mathcal{O}_1 \to \mathcal{O}_2$, where $\mathcal{O}_i$ are locally convex spaces of observables with declared seminorm families $\mathfrak{P}_i$, together with estimates

$$\|\Phi(\mathcal{R}_1 V) - \mathcal{R}_2(\Phi V)\|_{\mathcal{B}_2} \le E_{\mathrm{RG}}(V), \tag{142.9}$$

and

$$p\Big(\Psi\ \mathrm{Rec}_k^{(1)}(V), \mathrm{Rec}_k^{(2)}(\Phi V)\Big) \le E_{\mathrm{obs},p}(k, V), \tag{142.10}$$

for every $p \in \mathfrak{P}_2$, where E_{RG} and $E_{\mathrm{obs},p}$ are explicitly bounded errors in the stated domains. An exact equivalence is the special case in which both errors vanish and Φ identifies the relevant stable manifold neighborhoods. A controlled approximation is the case in which the errors are shown to vanish in the continuum or long-distance limit relevant to the claim.

This definition is the standard used below when relating a perturbative Polchinski trajectory, a BPHZ coordinate system, a functional-RG truncation, or a tensor-network calculation to a Wilsonian fixed point. The comparison map and its error estimates are part of the mathematical claim.

142.4 Banach Space Of Interactions

Let $\mathcal{B}$ be a Banach space of local functionals with norm $\|\cdot\|_{\mathcal{B}}$. A typical constructive choice separates relevant local coordinates from a polymer remainder:

$$V(\phi) = \sum_i g_i \mathcal{O}_i(\phi) + K(\phi),$$

where $g_i \in \mathbb{R}$ or $\mathbb{C}$ and K is measured by a norm that includes large-field weights and spatial decay. The RG map is controlled on an open set $U \subset \mathcal{B}$ if

$$\mathcal{R}_L : U \longrightarrow \mathcal{B}$$

is differentiable and its derivative satisfies explicit bounds in the chosen norm.

142.5 Linearization And Coordinates

Let V_* be a fixed point:

$$\mathcal{R}_L(V_*) = V_*.$$

The linearized map is

$$D\mathcal{R}_L|_{V_*} : \mathcal{B} \to \mathcal{B}.$$

An eigenvector e_i with eigenvalue L^{y_i} is relevant, marginal, or irrelevant according as $y_i > 0$, $y_i = 0$, or $y_i < 0$. These words refer to the declared linearized map and norm. A coordinate g_i is a scaling coordinate only after nonlinear terms have been put into a normal form or controlled by a stable-manifold argument.

Theorem 142.3 (Local stable graph for an RG map)**.** *Let $\mathcal{R}$ be a C^1 map on a Banach space $\mathcal{B}$ with fixed point V_*. Translate V_* to the origin and suppose*

$$\mathcal{B} = E_u \oplus E_s$$

where E_u is finite dimensional and E_s is closed. In product norm write

$$\mathcal{R}(u, s) = (Au + F_u(u, s),\, Bs + F_s(u, s)), \tag{142.11}$$

where $A : E_u \to E_u$ is invertible, $B : E_s \to E_s$, $F(0) = 0$, and $DF(0) = 0$. Assume that, on a closed ball of radius r,

$$\|A^{-1}\| \le a < 1, \qquad \|B\| \le b < 1, \qquad \|F(x) - F(y)\| \le \kappa \|x - y\|. \tag{142.12}$$

Choose γ with $\max(a, b) < \gamma < 1$ and suppose

$$q_u := \frac{\kappa a}{1 - a\gamma} < 1, \qquad q_s := \frac{b + \kappa}{\gamma} < 1, \qquad q := \max(q_u, q_s) < 1. \tag{142.13}$$

Then, after possibly decreasing r, for every sufficiently small $s_0 \in E_s$ there is a unique forward orbit $(u_n, s_n)_{n \ge 0}$ satisfying s_0 as its stable coordinate,

$$(u_{n+1}, s_{n+1}) = \mathcal{R}(u_n, s_n), \qquad \sup_{n \ge 0} \gamma^{-n} \|(u_n, s_n)\| < \infty.$$

The initial unstable coordinate u_0 is a Lipschitz function $h(s_0)$. The graph $u = h(s)$ is forward invariant, consists exactly of the points whose forward orbit stays in the chosen γ-decay neighborhood, and is tangent to E_s at the fixed point in the sense $\|h(s)\|/\|s\| \to 0$ as $s \to 0$.

Proof. Fix $\rho \le r$ and $\|s_0\| \le \rho$. Let

$$\mathcal{X}_{\gamma,\rho}(s_0) = \left\{ x = (u_n, s_n)_{n\ge 0} : s_0 \text{ is fixed}, \ \|x\|_\gamma := \sup_{n\ge 0} \gamma^{-n} \|(u_n, s_n)\| \le \rho \right\}.$$

With the metric induced by $\|\cdot\|_\gamma$, this is a closed subset of the Banach space of γ-weighted sequences, hence is complete. Define the Lyapunov–Perron map $\mathcal{T}_{s_0}$ by

$$(\mathcal{T}_{s_0} x)_0^s = s_0, \tag{142.14}$$

$$(\mathcal{T}_{s_0} x)_{n+1}^s = B s_n + F_s(u_n, s_n), \tag{142.15}$$

$$(\mathcal{T}_{s_0} x)_n^u = -\sum_{j=0}^{\infty} A^{-(j+1)} F_u(u_{n+j}, s_{n+j}). \tag{142.16}$$

The series in (142.16) converges absolutely because

$$\gamma^{-n} \sum_{j\ge 0} \|A^{-(j+1)}\| \, \|F_u(u_{n+j}, s_{n+j})\| \le \frac{\kappa a}{1 - a\gamma} \|x\|_\gamma. \tag{142.17}$$

Here $F(0) = 0$ was used. Similarly,

$$\gamma^{-(n+1)} \|B s_n + F_s(u_n, s_n)\| \le \frac{b + \kappa}{\gamma} \|x\|_\gamma. \tag{142.18}$$

Equations (142.13), (142.17), and (142.18) show that $\mathcal{T}_{s_0}$ maps $\mathcal{X}_{\gamma,\rho}(s_0)$ into itself for every $\rho \le r$ with $\|s_0\| \le \rho$.

For two sequences $x, y \in \mathcal{X}_{\gamma,\rho}(s_0)$, the same estimates applied to differences give

$$\|\mathcal{T}_{s_0} x - \mathcal{T}_{s_0} y\|_\gamma \le q \|x - y\|_\gamma, \tag{142.19}$$

with $q < 1$. Banach's fixed-point theorem gives a unique fixed sequence $x(s_0)$. The s-part of the fixed-point equation is exactly the stable coordinate recurrence. The u-part is equivalent to the unstable coordinate recurrence because multiplying (142.16) by A and shifting the summation gives

$$A u_n + F_u(u_n, s_n) = u_{n+1}.$$

Thus the fixed sequence is precisely a forward orbit of $\mathcal{R}$ with γ-decay. Conversely, any forward orbit with this decay must satisfy (142.16); otherwise the growing homogeneous solution of $u_{n+1} = A u_n + F_u(u_n, s_n)$ would violate $\sup_n \gamma^{-n} \|u_n\| < \infty$.

Define $h(s_0) = u_0(s_0)$. If s_0, t_0 are two stable initial coordinates, the fixed-point equations give

$$\|x(s_0) - x(t_0)\|_\gamma \le \|s_0 - t_0\| + q \|x(s_0) - x(t_0)\|_\gamma,$$

and hence

$$\|x(s_0) - x(t_0)\|_\gamma \le \frac{1}{1 - q} \|s_0 - t_0\|. \tag{142.20}$$

Therefore h is Lipschitz. Since $DF(0) = 0$, the Lipschitz constant κ for F on a ball of radius ρ tends to zero as $\rho \downarrow 0$. The bound (142.17) then implies $\|h(s)\|/\|s\| \to 0$, so the graph is tangent to E_s at the fixed point. Forward invariance is built into the recurrence, and the γ-weighted bound gives convergence to the fixed point. $\square$

Theorem 142.4 (C^1 stable graph from the Lyapunov–Perron equation)**.** *Assume the hypotheses and notation of Theorem 142.3. Suppose in addition that F is C^1 on the ball used there and that DF is uniformly continuous on that ball. Let*

$$\kappa_\rho = \sup_{\|(u,s)\|\le\rho} \|DF(u,s)\|_{\mathcal{B}\to\mathcal{B}}.$$

After restricting to a smaller ball if necessary, assume

$$q_\rho = \max\left\{\frac{a\kappa_\rho}{1-a\gamma}, \frac{b+\kappa_\rho}{\gamma}\right\} < 1. \tag{142.21}$$

Then the orbit map

$$s_0 \longmapsto x(s_0) = (u_n(s_0), s_n(s_0))_{n\ge0} \in \mathcal{X}_{\gamma,\rho}(s_0)$$

constructed in Theorem 142.3 is C^1 as a map from the stable-coordinate ball into the γ-weighted sequence space. Its derivative in the direction $v \in E_s$ is the unique γ-weighted sequence $Y(v) = (Y_n^u(v), Y_n^s(v))_{n\ge0}$ satisfying

$$Y_0^s(v) = v, \tag{142.22}$$

$$Y_{n+1}^s(v) = B\,Y_n^s(v) + DF_s(x_n(s_0))\,Y_n(v), \tag{142.23}$$

$$Y_n^u(v) = -\sum_{j=0}^{\infty} A^{-(j+1)} DF_u(x_{n+j}(s_0))\,Y_{n+j}(v). \tag{142.24}$$

Consequently the stable graph $h(s_0) = u_0(s_0)$ is C^1, with

$$Dh(s_0)v = Y_0^u(v), \qquad Dh(0) = 0.$$

Proof. Let X_γ be the Banach space of sequences $(u_n, s_n)_{n\ge0}$ with finite norm

$$\|x\|_\gamma = \sup_n \gamma^{-n}\|(u_n, s_n)\|.$$

The Lyapunov–Perron map used in the proof of Theorem 142.3 may be written

$$x = \mathcal{T}(s_0, x).$$

For fixed s_0, its derivative with respect to x is obtained by replacing each occurrence of F in (142.15)–(142.16) by $DF(x_n)$. The same estimates as (142.17) and (142.18), with κ replaced by κ_ρ, give

$$\|D_x\mathcal{T}(s_0, x)\|_{X_\gamma\to X_\gamma} \le q_\rho < 1. \tag{142.25}$$

The derivative of $\mathcal{T}$ with respect to the parameter s_0 is the sequence whose stable component at $n = 0$ is the input vector and whose other components are zero before the contraction equation is solved.

For $v \in E_s$, define $Y(v)$ to be the fixed point of the affine linear equation

$$Y = D_{s_0}\mathcal{T}(s_0, x(s_0))v + D_x\mathcal{T}(s_0, x(s_0))Y. \tag{142.26}$$

Banach's fixed-point theorem applies because of (142.25). Written in components, (142.26) is exactly (142.22)–(142.24). It also gives the bound

$$\|Y(v)\|_\gamma \le \frac{\|v\|}{1-q_\rho}.$$

It remains to check that $Y(v)$ is the derivative of $x(s_0)$. Put

$$\Delta_t = \frac{x(s_0 + tv) - x(s_0)}{t}.$$

Subtract (142.26) from the difference quotient of

$$x(s_0 + tv) = \mathcal{T}(s_0 + tv, x(s_0 + tv)), \qquad x(s_0) = \mathcal{T}(s_0, x(s_0)).$$

The result is

$$\Delta_t - Y(v) = D_x\mathcal{T}(s_0, x(s_0))(\Delta_t - Y(v)) + r_t,$$

where $r_t \to 0$ in X_γ. The convergence of r_t is the Fréchet differentiability of F on each finite component together with the uniform continuity of DF, the uniform γ-weighted bound on the two orbits, and the dominated geometric sums in (142.17). Taking norms and using (142.25) gives

$$\|\Delta_t - Y(v)\|_\gamma \le q_\rho \|\Delta_t - Y(v)\|_\gamma + \|r_t\|_\gamma,$$

so $\Delta_t \to Y(v)$. Thus $x(\cdot)$ is differentiable.

Continuity of $Dx(s_0)$ follows from the same contraction equation: if s_0 and s_0' are close, the corresponding orbits are close in X_γ, and uniform continuity of DF makes

$$D_x\mathcal{T}(s_0, x(s_0)) - D_x\mathcal{T}(s_0', x(s_0'))$$

small as an operator on X_γ. The inverse $(1 - D_x\mathcal{T})^{-1}$ is uniformly bounded by $(1 - q_\rho)^{-1}$, so the fixed solution of (142.26) depends continuously on s_0.

Finally, at $s_0 = 0$ the orbit is identically zero and $DF(0) = 0$. The derivative equations reduce to

$$Y_n^s = B^n v, \qquad Y_n^u = 0.$$

Therefore $Dh(0)v = Y_0^u(v) = 0$ for every $v \in E_s$. $\square$

142.6 Critical Surfaces And Continuum Tuning

The local stable graph is the mathematical object behind continuum tuning. Let the hypotheses and notation of Theorem 142.3 hold. Write

$$\Pi_u : \mathcal{B} = E_u \oplus E_s \longrightarrow E_u, \qquad \Pi_s : \mathcal{B} = E_u \oplus E_s \longrightarrow E_s$$

for the continuous projections. In a sufficiently small neighborhood of V_*, define the local critical surface

$$\mathcal{W}^s_{\text{loc}}(V_*) = \{V_* + h(s) + s : \ s \in E_s, \ \|s\| \text{ small}\}. \tag{142.27}$$

The word "critical" here means attraction to the fixed point under the declared RG map. It is a property of the pair $(\mathcal{R}, \mathcal{B})$, together with the reconstruction maps if a continuum QFT limit is claimed.

Definition 142.5 (Tuning map)**.** In the setting above, the local tuning map is

$$\tau(V) = \Pi_u(V - V_*) - h(\Pi_s(V - V_*)) \,, \tag{142.28}$$

defined for V in the local product neighborhood on which the graph h exists. A point V is tuned to the local critical surface precisely when

$$\tau(V) = 0. \tag{142.29}$$

If $\dim E_u = r$ and a basis of E_u has been chosen, this is an r-component equation. The components of τ, rather than a chosen list of formal Lagrangian coefficients, are the intrinsic relevant tuning coordinates in this RG chart.

Where the analytic burden lies. Theorem 142.3 proves the existence of a Lipschitz stable graph under the displayed Lyapunov–Perron contraction hypotheses. Theorem 142.4 gives one self-contained route to C^1 regularity: prove uniform derivative bounds for the chosen Banach RG map so that the differentiated Lyapunov–Perron equation remains a contraction. A model-specific RG theorem must establish these derivative bounds in its polymer, kernel, tensor, lattice, or BV norm. The geometric consequence after this analytic input is a standard submersion calculation, not an RG theorem in its own right.

Submanifold form after differentiability. Assume the stable graph $h : E_s \to E_u$ in Theorem 142.3 is C^1 on a neighborhood of the origin in E_s, and let $r = \dim E_u < \infty$. Then $\mathcal{W}^s_{\mathrm{loc}}(V_*)$ is a C^1 submanifold of $\mathcal{B}$ of codimension r. Equivalently, the equation $\tau(V) = 0$ cuts out the local critical surface by r independent conditions.

Let P be a finite-dimensional C^1 parameter manifold of microscopic finite-regulator theories, and let

$$\iota : P \longrightarrow \mathcal{B}$$

be a C^1 chart map into the RG Banach space. If $p_0 \in P$ satisfies $\tau(\iota(p_0)) = 0$ and

$$D(\tau \circ \iota)_{p_0} : T_{p_0}P \longrightarrow E_u \tag{142.30}$$

has rank r, then the tuned microscopic parameters near p_0,

$$P_{\mathrm{tuned}} = \{p \in P : \tau(\iota(p)) = 0\}, \tag{142.31}$$

form a C^1 submanifold of P of codimension r.

For $V = V_* + u + s$, with $u \in E_u$ and $s \in E_s$, the derivative of τ in the E_u-direction is the identity map on E_u:

$$D_u\tau(V_* + u + s) = 1_{E_u}. \tag{142.32}$$

Thus τ is a C^1 submersion onto the finite-dimensional space E_u. Its zero set is exactly

$$\{V_* + u + s : \; u = h(s)\},$$

which is the graph (142.27). The submersion theorem for Banach spaces with finite-dimensional target gives a C^1 submanifold of codimension $\dim E_u = r$. In this product chart the submanifold statement is also direct: the map

$$s \longmapsto V_* + h(s) + s$$

parametrizes the zero set, and the transverse coordinates are precisely $\tau \in E_u$.

For the microscopic parameter space, apply the finite-dimensional implicit function theorem to $\tau \circ \iota : P \to E_u$. The rank condition (142.30) makes $0 \in E_u$ a regular value near p_0. Hence the inverse image $(\tau \circ \iota)^{-1}(0)$ is a C^1 submanifold of P of codimension r, which is (142.31).

Quantitative finite-dimensional tuning estimate. The rank condition is not the estimate used in a finite-regulator construction. After a choice of r microscopic tuning coordinates, one needs a ball on which the nonlinear tuning equation is solved with controlled constants. Let X and Y be r-dimensional normed spaces, let $F : \overline{B}_\rho(0) \subset X \to Y$ be C^1 on a neighborhood of the closed ball, and let

$$e = F(0), \qquad A = DF(0).$$

Assume $A : X \to Y$ is invertible and, for some $0 \le \kappa < 1/2$,

$$\|A^{-1}e\| \le \frac{\rho}{2}, \qquad \sup_{\|x\| \le \rho} \|A^{-1}(DF(x) - A)\| \le \kappa. \tag{142.33}$$

Then $F(x) = 0$ has a unique solution in the closed ball $\overline{B}_\rho(0)$. Indeed define

$$T(x) = x - A^{-1}F(x).$$

For $\|x\|, \|y\| \le \rho$, the fundamental theorem of calculus gives

$$T(x) - T(y) = -\int_0^1 A^{-1}\big(DF(y + t(x-y)) - A\big)(x-y)\,\mathrm{d}t, \tag{142.34}$$

so $\|T(x) - T(y)\| \le \kappa\|x - y\|$. Also

$$\|T(x)\| \le \|T(0)\| + \kappa\|x\| = \|A^{-1}e\| + \kappa\|x\| \le \rho.$$

Thus T is a contraction of the complete metric space $\overline{B}_\rho(0)$ into itself, and Banach's fixed-point theorem gives a unique fixed point in that closed ball. The fixed-point equation $T(x) = x$ is exactly $F(x) = 0$.

In the RG application F is the local coordinate expression for $\tau \circ \iota$ in the chosen r microscopic tuning variables, after any spectator parameters have been fixed or placed in a small auxiliary ball. The residual $\|A^{-1}e\|$ measures how far the proposed finite microscopic point is from the local critical surface in intrinsic relevant coordinates; the second bound in (142.33) is the quantitative transversality estimate. This estimate still does not construct a fixed point or a continuum QFT. It only says that, once the stable graph and the microscopic parameter chart have already been established with these constants, the finite-dimensional relevant-coordinate equation can actually be solved in the declared neighborhood.

Linear finite-scale tuning exponents. Let E_u be a finite-dimensional normed vector space with basis $\{e_a\}_{a=1}^r$. Suppose the linearized unstable map is diagonal in this basis,

$$Ae_a = \lambda_a e_a, \qquad |\lambda_a| > 1.$$

For the linear RG map

$$\mathcal{R}_{\text{lin}}(u, s) = (Au, Bs),$$

the unstable component of an initial condition

$$u_0 = \sum_{a=1}^{r} u_a(0)e_a$$

stays in the coordinate box

$$|u_a(n)| \le \rho \qquad (a = 1, \dots, r,\ 0 \le n \le N) \tag{142.35}$$

if and only if

$$|u_a(0)| \le \rho\,|\lambda_a|^{-N} \qquad (a = 1, \dots, r). \tag{142.36}$$

Writing $\lambda_a = L^{y_a}$, this finite-depth accuracy is ρL^{-Ny_a} in the a-th relevant coordinate.

The linear recurrence gives

$$u_n = A^n u_0 = \sum_{a=1}^{r} \lambda_a^n u_a(0)e_a.$$

Therefore

$$|u_a(n)| = |\lambda_a|^n |u_a(0)|.$$

Since $|\lambda_a| > 1$, the maximum over $0 \le n \le N$ occurs at $n = N$. The bound (142.35) is therefore equivalent to

$$|\lambda_a|^N |u_a(0)| \le \rho$$

for every a, which is (142.36). The notation $\lambda_a = L^{y_a}$ gives $|\lambda_a|^{-N} = L^{-Ny_a}$ when $\lambda_a > 0$; for complex or negative eigenvalues the same estimate uses $|\lambda_a| = L^{y_a}$.

For a nonlinear RG map, the finite-scale tuning calculation above is the linear model for the precision required before N RG steps. The stable graph replaces the coordinate plane $u = 0$ by the intrinsic equation $\tau(V) = 0$. A theorem assigning the same exponents beyond the linear approximation must provide normal-form or graph-transform estimates controlling the nonlinear remainder.

The diagonal estimate is not a convention-independent statement about finite-depth tuning. For a general finite-dimensional unstable block $A : E_u \to E_u$, the intrinsic quantity is

$$M_A(N) = \max_{0 \le n \le N} \|A^n\|_{E_u \to E_u}. \tag{142.37}$$

The sufficient uniform tuning condition for the linear unstable component to remain in the ball $\|u_n\| \le \rho$ for $0 \le n \le N$ is

$$\|u_0\| \le \frac{\rho}{M_A(N)}. \tag{142.38}$$

For a Jordan block $A = \lambda 1 + N_J$ of size m, with $N_J^m = 0$,

$$A^n = \lambda^n \sum_{p=0}^{m-1} \binom{n}{p} \lambda^{-p} N_J^p. \tag{142.39}$$

Thus non-diagonal unstable coordinates can carry a polynomial amplification factor of degree $m-1$ in addition to the exponential factor $|\lambda|^n$. A proof that quotes only the eigenvalue exponent y has therefore not specified the finite-depth tuning precision unless it also controls the conditioning of the unstable coordinate chart and any Jordan or near-resonant polynomial losses.

The same point appears for finite comparison or truncation errors. If the unstable coordinate obeys

$$u_{n+1} = Au_n + e_n,$$

then

$$u_N = A^N u_0 + \sum_{j=0}^{N-1} A^{N-1-j} e_j. \tag{142.40}$$

The error sequence must be summable after multiplication by the same unstable powers. Otherwise a trajectory can stay close to the stable graph in its stable coordinates while drifting out of the relevant-coordinate window at the finite depth needed for the reconstructed observable.

142.7 Polymer Representation

On a lattice, a polymer is a finite connected union of blocks in a fixed block decomposition. Let $\mathbb{B}_k$ be the set of scale-k blocks, let $\mathcal{P}_k$ be the family of connected finite unions of elements of $\mathbb{B}_k$, and let $X \sim Y$ mean that the polymers $X, Y \in \mathcal{P}_k$ are disjoint and separated by at least one block. A polymer activity is a map

$$K_k : \mathcal{P}_k \times \Phi_k \to \mathbb{C}, \qquad (X, \phi) \mapsto K_k(X, \phi),$$

where Φ_k is the finite-dimensional scale-k field space of the regulated model. The polymer form of a finite-volume partition function is

$$Z_k = \int d\mu_{C_k}(\phi)\, \exp\{-V_k(\phi)\} \sum_{\mathcal{X} \text{ compatible}} \prod_{X \in \mathcal{X}} K_k(X, \phi). \tag{142.41}$$

Here V_k is the local part, C_k is the fluctuation covariance assigned to the current scale, and the sum is over finite compatible families $\mathcal{X} \subset \mathcal{P}_k$. Compatibility is the locality input: it is what permits a disconnected family of small activities to exponentiate through the cluster expansion rather than behaving as one large nonlocal object.

Definition 142.6 (Finite-range fluctuation covariance decomposition)**.** Let Λ_k be the finite scale-k regulator lattice with graph distance d_k. A fluctuation covariance Γ_k has range R_k if

$$\Gamma_k(x, y) = 0 \qquad \text{whenever} \qquad d_k(x, y) > R_k. \tag{142.42}$$

A finite-range decomposition for a scalar block-spin RG is a decomposition

$$C_{\le k} = \Gamma_0 + \Gamma_1 + \cdots + \Gamma_k + C_{\mathrm{rem,k}}$$

on each finite regulator, with every Γ_j positive semidefinite, Γ_j satisfying a range bound of order L^j in microscopic distance, and derivative or difference bounds in the field norm used by the polymer activity. Positivity is needed to realize Γ_j as a Gaussian fluctuation covariance; the range bound is the input that turns separated polymer activities into independent fluctuation integrations; the derivative bounds enter the localization and Taylor-remainder estimates.

Lemma 142.7 (Finite-range Gaussian factorization)**.** *Let Φ_Λ be a finite real field space and let μ_Γ be a centered Gaussian measure with positive semidefinite covariance Γ. If $A, B \subset \Lambda$ satisfy*

$$\Gamma(x,y) = 0 \qquad (x \in A,\ y \in B),$$

then the Gaussian fields $\zeta_A = (\zeta_x)_{x\in A}$ and $\zeta_B = (\zeta_y)_{y\in B}$ are independent. Consequently, for bounded measurable F_A and F_B depending only on the fields in A and B,

$$\int F_A(\zeta_A)F_B(\zeta_B)\,\mathrm{d}\mu_\Gamma(\zeta) = \left(\int F_A(\zeta_A)\,\mathrm{d}\mu_\Gamma(\zeta)\right)\left(\int F_B(\zeta_B)\,\mathrm{d}\mu_\Gamma(\zeta)\right). \tag{142.43}$$

Proof. The joint characteristic function of (ζ_A, ζ_B) is

$$\exp\left\{-\frac{1}{2}t_A\Gamma_{AA}t_A - t_A\Gamma_{AB}t_B - \frac{1}{2}t_B\Gamma_{BB}t_B\right\}.$$

The hypothesis gives $\Gamma_{AB} = 0$, so the characteristic function is the product of the characteristic functions of ζ_A and ζ_B. For finite-dimensional probability measures, characteristic functions determine the law, and this factorization is exactly independence of the two random vectors. The displayed integral identity is then the defining factorization property of independent random variables. If Γ is only positive semidefinite, the same argument applies to the induced Gaussian law on the quotient by the null directions, or equivalently after adding $\epsilon 1$ and letting $\epsilon \downarrow 0$ in finite dimension. $\square$

Applied with A and B equal to R_k-neighborhoods of two separated polymers, Lemma 142.7 is the exact finite-regulator reason why fluctuation integration preserves polymer locality. If a covariance decomposition has only exponential decay rather than finite range, this factorization is replaced by a cluster estimate with decay constants in the polymer norm; the conclusion is weaker and must be proved separately. Thus a short-range scalar Wilsonian RG theorem cannot begin from the phrase "integrate a momentum shell" alone. It must specify the covariance shell, its locality or decay estimate, and the norm in which the residual inter-polymer correlations are summable.

Lemma 142.8 (Summable covariance-tail bridge estimate)**.** *Let Λ be a finite lattice with graph distance d. Assume that there are constants $D_{\mathrm{vol}} \geq 1$ and $C_{\mathrm{sh}} < \infty$ such that for every $x \in \Lambda$ and $n \geq 0$*

$$\#\{y \in \Lambda :\ d(x,y) = n\} \leq C_{\mathrm{sh}}(1+n)^{D_{\mathrm{vol}}-1}. \tag{142.44}$$

Let Γ be a positive semidefinite covariance satisfying

$$|\Gamma(x,y)| \leq A_\Gamma(1+d(x,y))^{-s}, \qquad s > D_{\mathrm{vol}}. \tag{142.45}$$

If $A, B \subset \Lambda$ are finite sets with $d(A,B) \geq R$, then the cross block $\Gamma_{AB} : \ell^2(B) \to \ell^2(A)$ obeys

$$\|\Gamma_{AB}\|_{\ell^2(B)\to\ell^2(A)} \leq A_\Gamma C_{\mathrm{sh}}\left(1 + \frac{1}{s - D_{\mathrm{vol}}}\right)(1+R)^{D_{\mathrm{vol}}-s}. \tag{142.46}$$

Consequently, if F_A and F_B are C^1 bounded functions depending only on the fields in A and B, with

$$\|\nabla_A F_A\|_{2,\infty} = \sup_\phi\left(\sum_{x\in A}|\partial_x F_A(\phi)|^2\right)^{1/2}$$

and similarly for F_B, then

$$\begin{aligned}&|\mathbb{E}_\Gamma[F_A(\zeta_A)F_B(\zeta_B)] - \mathbb{E}_\Gamma[F_A(\zeta_A)]\,\mathbb{E}_\Gamma[F_B(\zeta_B)]|\\ &\qquad \le A_\Gamma C_{\mathrm{sh}}\left(1+\frac{1}{s-D_{\mathrm{vol}}}\right)(1+R)^{D_{\mathrm{vol}}-s}\|\nabla_A F_A\|_{2,\infty}\|\nabla_B F_B\|_{2,\infty}.\end{aligned}\tag{142.47}$$

Proof. For $x \in A$, the row sum over B is bounded by the full tail outside distance R:

$$\sum_{y\in B}|\Gamma(x,y)| \le A_\Gamma C_{\mathrm{sh}}\sum_{n\ge R}(1+n)^{D_{\mathrm{vol}}-1-s}.$$

The same bound holds for column sums by symmetry of Γ. Since $s > D_{\mathrm{vol}}$,

$$\sum_{n\ge R}(1+n)^{D_{\mathrm{vol}}-1-s} \le (1+R)^{D_{\mathrm{vol}}-1-s} + \int_R^\infty (1+t)^{D_{\mathrm{vol}}-1-s}\,\mathrm{d}t \le \left(1+\frac{1}{s-D_{\mathrm{vol}}}\right)(1+R)^{D_{\mathrm{vol}}-s}.$$

Schur's test gives (142.46).

It remains to connect the operator bound to the Gaussian expectation. For $0 \le t \le 1$, let μ_t be the centered Gaussian law on $\mathbb{R}^A \oplus \mathbb{R}^B$ with covariance

$$\begin{pmatrix}\Gamma_{AA} & t\Gamma_{AB}\\ t\Gamma_{BA} & \Gamma_{BB}\end{pmatrix}.$$

The covariance is positive because it is obtained from the original Gaussian law by multiplying the B-component by an independent sign and averaging: $t = 2p-1$ for $p = (1+t)/2$. Differentiating the finite-dimensional Gaussian expectation with respect to t gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_{\mu_t}[F_A F_B] = \sum_{x\in A, y\in B}\Gamma(x,y)\mathbb{E}_{\mu_t}[(\partial_x F_A)(\partial_y F_B)].$$

At $t=0$ the A and B variables are independent; at $t=1$ the law is the original marginal of μ_Γ on $A \cup B$. Integrating in t, applying Cauchy–Schwarz in the finite index sets, and using (142.46) gives (142.47). □

Lemma 142.8 is the replacement for exact factorization when the covariance shell is not finite range. The decay exponent, shell-count constant, derivative norm, and polymer separation enter the RG theorem as actual constants. A smooth momentum-shell covariance can supply such constants after integration by parts in momentum space; a sharp cutoff or a long-range covariance may fail to give the summability needed by the chosen polymer norm. This is why a rigorous Wilsonian proof states the covariance decomposition and cluster norm before using locality language.

Lemma 142.9 (Finite localization remainder and scaling gain)**.** *Let E be a finite-dimensional normed real vector space, let $F : E \to \mathbb{R}$ be C^{p+1} on the closed ball $\{\xi : \|\xi\| \le r\}$, and let*

$$(\mathcal{L}_{\le p}F)(\xi) = \sum_{n=0}^{p}\frac{1}{n!}D^nF(0)[\xi,\dots,\xi]$$

be the Taylor-localization map at the origin. If

$$M_{p+1}(r) = \sup_{\|\eta\|\le r}\|D^{p+1}F(\eta)\|_{\mathrm{op}},$$

then, for every $\|\xi\| \le r$,

$$|F(\xi) - (\mathcal{L}_{\le p}F)(\xi)| \le \frac{M_{p+1}(r)}{(p+1)!}\,\|\xi\|^{p+1}. \tag{142.48}$$

In a scalar block-spin chart with scale factor L*, dimension* D*, field scaling dimension* Δ_ϕ*, and one lattice difference carrying one inverse power of* L*, a local density monomial with* ν *fields and* m *lattice differences has canonical scaling exponent*

$$y_{\nu,m} = D - \nu\Delta_\phi - m. \tag{142.49}$$

Thus a Taylor-localization choice gives a linear irrelevant gain $L^{-\alpha}$ *only after the first omitted local monomials satisfy*

$$\alpha = \inf_{\text{omitted } (\nu,m)} \big(\nu\Delta_\phi + m - D\big) > 0. \tag{142.50}$$

Proof. The integral Taylor formula in finite dimension gives

$$F(\xi) - (\mathcal{L}_{\le p}F)(\xi) = \frac{1}{p!}\int_0^1 (1-s)^p D^{p+1}F(s\xi)[\xi,\dots,\xi]\,\mathrm{d}s.$$

Taking the operator norm and using $\int_0^1 (1-s)^p \mathrm{d}s = (p+1)^{-1}$ gives (142.48). For the scaling formula, write a fine field inside one coarse block as $\phi = L^{-\Delta_\phi}\phi'$. The sum or integral over a coarse block contains L^D fine cells, while each discrete difference contributes an additional factor L^{-1}. Hence the coefficient of a monomial with ν fields and m differences is multiplied, at the canonical linearized level, by

$$L^D L^{-\nu\Delta_\phi} L^{-m} = L^{D-\nu\Delta_\phi - m}.$$

The first omitted local monomial is contractive by a factor at most $L^{-\alpha}$ exactly when the exponents of all omitted monomials are $\le -\alpha$, which is (142.50). □

This lemma is finite-regulator bookkeeping, but it removes an important ambiguity. The localization map $\mathcal{L}_k$ in a polymer RG proof is a Taylor jet with a declared center, derivative norm, and retained local coordinate set. It must be paired with the derivative norm in (142.48), the field rescaling used in the chart, and a proof that the omitted Taylor coordinates are contractive in that chart. In canonical scalar bookkeeping at $D = 4$, $\Delta_\phi = 1$ makes ϕ^4 marginal and ϕ^6 irrelevant by two powers. In canonical bookkeeping at $D = 3$, $\Delta_\phi = \frac{1}{2}$ makes the ϕ^6 density marginal by (142.49); it cannot be discarded into the irrelevant remainder on engineering grounds alone. An interacting fixed-point proof may replace the canonical monomial coordinate list by scaling-field coordinates, but that replacement is itself part of the model-specific RG theorem.

Definition 142.10 (Local-coordinate extraction budget)**.** At scale k, let $E_{\mathrm{loc},k}$ be the finite-dimensional local coordinate space retained by the RG chart. A local-coordinate extraction datum consists of local basis elements

$$M_{a,k} \in E_{\mathrm{loc},k}, \qquad a \in A_k,$$

and continuous coordinate functionals

$$\ell_{a,k} : E_{\mathrm{loc},k} + \mathcal{K}_k \longrightarrow \mathbb{R}$$

such that

$$\ell_{a,k}(M_{b,k}) = \delta_{ab}. \tag{142.51}$$

The localization operator is

$$\mathcal{L}_k F = \sum_{a \in A_k} \ell_{a,k}(F)\, M_{a,k}.$$

The condition number of this coordinate extraction in the polymer norm is the finite quantity

$$\chi_k = \sum_{a \in A_k} \|\ell_{a,k}\|_{\mathcal{K}_k'}\, \|M_{a,k}\|_{\mathcal{K}_k}. \tag{142.52}$$

A uniform RG theorem must bound χ_k on the scale range on which the coordinate chart is used.

Let $\mathcal{A}_k$ denote one exact finite-regulator fluctuation step, including rescaling and reblocking, before the localization projection is applied. After the purely local part of the input has been propagated by the declared finite-dimensional local map $\mathcal{M}_k$, write the remaining one-step contribution as $F_k(g_k, K_k)$. The local-coordinate extraction budget is the pair of estimates

$$|\ell_{a,k+1}(F_k(g_k, K_k))| \le C^{(1)}_{a,k}\|K_k\|_{\mathcal{K}_k} + C^{(2)}_{a,k}\|K_k\|^2_{\mathcal{K}_k} + \varepsilon^g_{a,k}, \tag{142.53}$$

and

$$\|(1 - \mathcal{L}_{k+1})F_k(g_k, K_k)\|_{\mathcal{K}_{k+1}} \le A_{\mathrm{irr},k} L^{-\alpha_k}\|K_k\|_{\mathcal{K}_k} + B_{\mathrm{rem},k}\|K_k\|^2_{\mathcal{K}_k} + \varepsilon^K_k. \tag{142.54}$$

The budget separates two tasks that are often conflated. Equation (142.53) says how fluctuation integration and the polymer tail change the retained relevant or marginal coordinates. Equation (142.54) says how the omitted part becomes a controlled irrelevant activity. The first estimate feeds the finite-dimensional tuning equations; the second feeds the polymer contraction recursion. The constants are not formal labels: in a model proof they must be obtained from the finite-range or summable covariance bounds, the Taylor-localization derivative norms, the large-field regulator stability, and the finite polymer-overlap estimates.

The condition number χ_k is part of the same theorem. A basis of formal monomials and a list of coordinate functionals do not define a stable RG chart if the functionals become large in the polymer norm. In that case small errors in the exact finite-regulator step may produce uncontrolled local-coordinate errors after projection. Changing the retained coordinate basis is harmless only when the corresponding change-of-basis maps and their inverses are uniformly bounded in the norms used for (142.53) and (142.54).

Lemma 142.11 (Finite Gaussian stability of a quadratic large-field regulator)**.** *Let E be a finite-dimensional real Hilbert space, let $\Gamma : E \to E$ be positive semidefinite and self-adjoint, and let μ_Γ be the centered Gaussian law with covariance Γ. For $\kappa > 0$ assume*

$$2\kappa\|\Gamma\|_{\mathrm{op}} < 1.$$

Then, for every $\phi \in E$,

$$\int_E \exp\{\kappa\|\phi + \zeta\|^2\}\, \mathrm{d}\mu_\Gamma(\zeta) = \det(1 - 2\kappa\Gamma)^{-1/2} \exp\{\kappa\langle\phi, (1 - 2\kappa\Gamma)^{-1}\phi\rangle\}. \tag{142.55}$$

Consequently

$$\int_E \exp\{\kappa\|\phi + \zeta\|^2\}\, \mathrm{d}\mu_\Gamma(\zeta) \le C_\Gamma(\kappa)\, \exp\left\{\frac{\kappa}{1 - 2\kappa\|\Gamma\|_{\mathrm{op}}}\|\phi\|^2\right\}, \tag{142.56}$$

where $C_\Gamma(\kappa) = \det(1 - 2\kappa\Gamma)^{-1/2}$ on $(\ker\Gamma)^\perp$.

Proof. It is enough to work on $(\ker\Gamma)^\perp$. Choose an orthonormal basis diagonalizing Γ, with eigenvalues $\lambda_i > 0$. The integral is a product of one-dimensional integrals

$$\frac{1}{\sqrt{2\pi\lambda_i}}\int_{\mathbb{R}} \exp\left\{\kappa(\phi_i+\zeta_i)^2 - \frac{\zeta_i^2}{2\lambda_i}\right\} \mathrm{d}\zeta_i.$$

Since $2\kappa\lambda_i < 1$, completing the square gives

$$\kappa(\phi_i+\zeta_i)^2 - \frac{\zeta_i^2}{2\lambda_i} = -\left(\frac{1}{2\lambda_i} - \kappa\right)\left(\zeta_i - \frac{\kappa\phi_i}{\frac{1}{2\lambda_i}-\kappa}\right)^2 + \frac{\kappa}{1-2\kappa\lambda_i}\phi_i^2.$$

The Gaussian integral in the shifted variable contributes $(1-2\kappa\lambda_i)^{-1/2}$. Multiplying over i gives (142.55). Since $(1-2\kappa\lambda_i)^{-1} \le (1-2\kappa\|\Gamma\|_{\mathrm{op}})^{-1}$, the quadratic form in the exponent is bounded by the exponent in (142.56). □

Thus a large-field regulator such as $\exp\{\kappa\|\phi\|_X^2\}$ is stable under fluctuation integration only with a declared covariance spectral bound and with the next-scale regulator large enough to absorb the enlarged coefficient $\kappa(1-2\kappa\|\Gamma\|_{\mathrm{op}})^{-1}$. In a polymer RG theorem this finite estimate is combined with finite range or cluster decay and with Taylor localization. The constants $C_\Gamma(\kappa)$, the covariance spectral bound, and the regulator strengthening are part of the proof data entering A_{irr}, B_{pol}, $B_{\mathrm{tail},k}$, and the extraction defect.

Definition 142.12 (One-step polymer RG contraction estimates)**.** A one-step polymer RG contraction structure at scale k consists of the following finite-regulator data.

1. A scale factor $L > 1$, a reblocking map $X \mapsto \overline{X}$ from scale-k polymers to the smallest scale-$k+1$ polymer containing X, and a finite overlap constant

$$\mathfrak{C}_a = \sup_{\Delta\in\mathbb{B}_{k+1}} \sum_{\substack{X\in\mathcal{P}_k\\ \overline{X}\supset\Delta}} \exp\{-a|X|_k\} < \infty. \tag{142.57}$$

2. Large-field regulators $G_k(X,\phi) \ge 1$, analytic radii h_k, and a weighted polymer norm

$$\|K_k\|_{k,a,h} = \sup_{\Delta\in\mathbb{B}_k} \sum_{\substack{X\in\mathcal{P}_k\\ X\supset\Delta}} \exp\{a|X|_k\} \sup_{\phi\in\Phi_k} \frac{|K_k(X,\phi)|_{h_k}}{G_k(X,\phi)}. \tag{142.58}$$

   The seminorm $|\cdot|_{h_k}$ is the finite Taylor or analytic norm in the field variables used by the model. Its precise definition is part of the contraction structure, because the contraction constant depends on the allowed derivatives and on the large-field weights.

3. A localization map $\mathcal{L}_k$ from polymer functionals supported in one reblocked polymer to the chosen finite-dimensional local-coordinate space. The complementary projection $1-\mathcal{L}_k$ is the irrelevant remainder. After fluctuation integration and rescaling, denoted here by $\mathcal{A}_k$, the contraction structure includes constants $A_{\mathrm{irr}} < \infty$ and $\alpha > 0$ such that

$$\|(1-\mathcal{L}_k)\mathcal{A}_k K_k\|_{k+1,a,h} \le A_{\mathrm{irr}} L^{-\alpha} \|K_k\|_{k,a,h}. \tag{142.59}$$

   This is the analytic place where Taylor remainder estimates, covariance finite-range bounds, field rescaling, and large-field stability enter.

4. A quadratic product estimate for incompatible small polymers. For two activities define the scale-$k+1$ circle product by

$$(K_1 \circ_k K_2)(Y,\phi) = \sum_{\substack{X_1,X_2\in\mathcal{P}_k \\ \overline{X_1\cup X_2}=Y \\ X_1 \not\sim X_2}} K_1(X_1,\phi)\, K_2(X_2,\phi), \tag{142.60}$$

with the product understood after pulling both activities to the same reblocked field variables. The contraction structure requires a constant $B_{\mathrm{pol}} < \infty$ such that

$$\|K_1 \circ_k K_2\|_{k+1,a,h} \le B_{\mathrm{pol}} \|K_1\|_{k,a,h} \|K_2\|_{k,a,h}. \tag{142.61}$$

The constant B_{pol} is a consequence of the finite overlap constant $\mathfrak{C}_a$, the compatibility convention, and the multiplicative behavior of G_k. It is not a universal number.

The finite-overlap content can be made explicit at the level of scalar majorants. Put

$$M_i(X) = \sup_{\phi\in\Phi_k} \frac{|K_i(X,\phi)|_{h_k}}{G_k(X,\phi)}.$$

Suppose that the analytic norm and large-field regulator satisfy the submultiplicative estimate

$$\frac{|K_1(X_1,\phi)K_2(X_2,\phi)|_{h_{k+1}}}{G_{k+1}(\overline{X_1\cup X_2},\phi)} \le C_G\, M_1(X_1) M_2(X_2) \tag{142.62}$$

for every incompatible pair entering (142.60). Define the pair-overlap majorant

$$\mathfrak{C}_a^\circ = \sup_{\Delta\in\mathbb{B}_{k+1}} \sum_{\substack{X_1,X_2\in\mathcal{P}_k \\ X_1\not\sim X_2,\, \overline{X_1\cup X_2}\supset\Delta}} \exp\bigl\{a|\overline{X_1\cup X_2}|_{k+1} - a|X_1|_k - a|X_2|_k\bigr\}. \tag{142.63}$$

If $\mathfrak{C}_a^\circ < \infty$, then $B_{\mathrm{pol}} = C_G \mathfrak{C}_a^\circ$ is admissible in (142.61). Indeed, from (142.58),

$$M_i(X_i) \le e^{-a|X_i|_k} \|K_i\|_{k,a,h}$$

because the norm sum contains the term X_i for each block contained in X_i. Therefore, for each coarse block Δ,

$$\begin{aligned}
&\sum_{Y\supset\Delta} e^{a|Y|_{k+1}} \sup_\phi \frac{|(K_1\circ_k K_2)(Y,\phi)|_{h_{k+1}}}{G_{k+1}(Y,\phi)} \\
&\quad\le C_G \|K_1\|_{k,a,h}\|K_2\|_{k,a,h} \sum_{\substack{X_1,X_2 \\ X_1\not\sim X_2,\, \overline{X_1\cup X_2}\supset\Delta}} e^{a|\overline{X_1\cup X_2}|_{k+1} - a|X_1|_k - a|X_2|_k} \\
&\quad\le C_G \mathfrak{C}_a^\circ \|K_1\|_{k,a,h}\|K_2\|_{k,a,h}.
\end{aligned}$$

The model-specific proof must bound $\mathfrak{C}_a^\circ$ uniformly in the volume and scale. On a finite-range block lattice this is a geometric counting estimate; with long-range covariances or nonlocal blockings it may fail unless the polymer norm is strengthened.

5. If the fluctuation covariance is not finite range, separated compatible polymers also produce a connected bridge term. Let

$$\omega_k(R) = A_{\Gamma,k} C_{\mathrm{sh},k} \left(1 + \frac{1}{s_k - D_{\mathrm{vol},k}}\right) (1+R)^{D_{\mathrm{vol},k} - s_k}$$

be the covariance-tail bound from Lemma 142.8. Suppose the analytic polymer norm controls the derivative majorants

$$M_i^{\nabla}(X) = \sup_{\phi} \frac{\|\nabla_X K_i(X,\phi)\|_{2,h_k}}{G_k(X,\phi)} \le D_{\nabla,k} e^{-a|X|_k} \|K_i\|_{k,a,h}.$$

Define the separated-tail overlap constant

$$\mathfrak{C}^{\mathrm{tail}}_{a,\omega,k} = \sup_{\Delta \in \mathbb{B}_{k+1}} \sum_{\substack{X_1, X_2 \in \mathcal{P}_k \\ X_1 \sim X_2,\, \overline{X_1 \cup X_2} \supset \Delta}} e^{a|\overline{X_1 \cup X_2}|_{k+1} - a|X_1|_k - a|X_2|_k} \omega_k(d_k(X_1, X_2)) \,. \tag{142.64}$$

The connected Gaussian-interpolation bridge between separated polymers then obeys

$$\|\,\mathrm{Bridge}_{\Gamma_k}(K_1, K_2)\|_{k+1,a,h} \le C_G D^2_{\nabla,k} \mathfrak{C}^{\mathrm{tail}}_{a,\omega,k} \|K_1\|_{k,a,h} \|K_2\|_{k,a,h}. \tag{142.65}$$

This term is zero in the exact finite-range case once the compatibility separation exceeds the covariance range. For a smooth shell it is not zero: it is a second quadratic constant,

$$B_{\mathrm{tail},k} = C_G D^2_{\nabla,k} \mathfrak{C}^{\mathrm{tail}}_{a,\omega,k}, \qquad B_k^{(2)} = B_{\mathrm{pol}} + B_{\mathrm{tail},k}, \tag{142.66}$$

that must enter the one-step contraction budget. Treating a summably decaying covariance shell as if it were finite range is exactly the error of dropping $B_{\mathrm{tail},k}$.

6. A local-coordinate extraction error $\varepsilon_k \ge 0$, measured in the same polymer norm, accounting for the difference between the exact finite-regulator RG step and the chosen local chart. In a fully exact chart $\varepsilon_k = 0$; in a truncated chart it is a theorem-level remainder and must be estimated.

The preceding datum turns contraction of the irrelevant activity into a finite inequality. If

$$x_k = \|K_k\|_{k,a,h}, \qquad q = A_{\mathrm{irr}} L^{-\alpha}, \qquad B_k^{(2)} = B_{\mathrm{pol}} + B_{\mathrm{tail},k},$$

then the linear estimate (142.59), the quadratic product estimate (142.61), the tail bridge (142.65), and the extraction error give the one-step bound

$$x_{k+1} \le q x_k + B_k^{(2)} x_k^2 + \varepsilon_k. \tag{142.67}$$

Consequently a closed radius-r polymer ball is invariant with contraction margin $\theta < 1$ whenever

$$q + B_k^{(2)} r + \frac{\varepsilon_k}{r} \le \theta < 1 \qquad \text{uniformly in } k. \tag{142.68}$$

Indeed, $x_k \le r$ gives

$$x_{k+1} \le \left(q + B_k^{(2)} r + \frac{\varepsilon_k}{r}\right) r \le \theta r.$$

This is elementary arithmetic, but it is the arithmetic that fixes the meaning of the polymer estimate: every proof must exhibit the norm, the large-field regulator, the localization operator, the finite overlap bound, the covariance-tail bridge bound when exact finite range is absent, and the uniform smallness inequality. Without these data the displayed contraction is a slogan, not a Wilsonian RG theorem.

The same data must be controlled as a sequence of scales, not only at one scale. Suppose that, for $0 \le j < n$, the trajectory stays in the radius-r polymer ball and

$$q + B_j^{(2)} r \le \theta < 1 \qquad (0 \le j < n).$$

Then (142.67) implies the forced linear recursion

$$x_{j+1} \le \theta x_j + \varepsilon_j \qquad (0 \le j < n),$$

and iteration gives

$$x_n \le \theta^n x_0 + \sum_{j=0}^{n-1} \theta^{\,n-1-j} \varepsilon_j. \tag{142.69}$$

Thus a scale-dependent extraction error is harmless only after its weighted sum is controlled. If $\varepsilon_j \le E\sigma^j$ with $0 < \sigma < 1$, then

$$\sum_{j=0}^{n-1} \theta^{\,n-1-j} \varepsilon_j \le E\frac{\theta^n - \sigma^n}{\theta - \sigma} \quad (\theta \ne \sigma), \qquad \le En\theta^{n-1} \quad (\theta = \sigma). \tag{142.70}$$

For a nondecaying uniform defect $\varepsilon_j \le E$, the corresponding bound is

$$x_n \le \theta^n x_0 + \frac{E(1-\theta^n)}{1-\theta}.$$

Consequently the ball of radius r is invariant for all scales under the stronger smallness condition

$$x_0 + \frac{E}{1-\theta} \le r \tag{142.71}$$

in the uniform-forcing case, or under the analogous supremum of the weighted geometric sum in the decaying case. These estimates are still only bookkeeping; the model-specific constructive theorem must prove the bounds on q, $B_j^{(2)}$, r, and ε_j from the covariance, localization, and large-field regulators.

Lemma 142.13 (Finite source-window extraction and propagated cumulant error)**.** *Let B_j be Banach spaces and let $R_j(J) \in B_j$ be holomorphic on a neighborhood of the closed polydisc*

$$P_{\rho_j} = \{J \in \mathbb{C}^m : \ |J_a| \le \rho_{j,a}, \ 1 \le a \le m\}.$$

Set

$$\eta_j = \sup_{J \in P_{\rho_j}} \|R_j(J)\|_{B_j}.$$

For an integer $s_j \ge 0$, let $\mathcal{T}^{\mathrm{src}}_{\le s_j} R_j$ be the Taylor polynomial in J containing all monomials of total source degree at most s_j. If $\beta = (\beta_1, \dots, \beta_m)$ is a multi-index and

$$\rho_j^\beta = \prod_{a=1}^m \rho_{j,a}^{\beta_a}, \qquad \beta! = \prod_{a=1}^m \beta_a!,$$

then

$$\partial_J^\beta\big(R_j - \mathcal{T}^{\mathrm{src}}_{\le s_j} R_j\big)(0) = 0 \qquad (|\beta| \le s_j), \tag{142.72}$$

and, for arbitrary β,

$$\Big\|\partial_J^\beta\big(R_j - \mathcal{T}^{\mathrm{src}}_{\le s_j} R_j\big)(0)\Big\|_{B_j} \le \frac{\beta!}{\rho_j^\beta}\,\eta_j. \tag{142.73}$$

Suppose moreover that the error in a reconstructed connected source window at scale N is obtained by propagating the source-extraction tails through bounded linear maps

$$U_{j\to N,\beta} : B_j \longrightarrow \mathcal{O}_\beta, \qquad p_\beta(U_{j\to N,\beta} v) \le P_{j,N,\beta}\|v\|_{B_j},$$

where p_β is the seminorm of the declared β-source observable window. Then the accumulated β-window error satisfies

$$p_\beta(\Delta O_{N,\beta}) \le \sum_{\substack{0\le j<N \\ s_j<|\beta|}} P_{j,N,\beta}\,\frac{\beta!}{\rho_j^\beta}\,\eta_j. \tag{142.74}$$

Proof. The Taylor polynomial contains exactly the source monomials of total degree at most s_j, hence the derivative in (142.72) vanishes for $|\beta| \le s_j$. For $|\beta| > s_j$, the β-derivative of the tail at the origin is the β-derivative of R_j. Use the Banach-valued Cauchy integral formula on the distinguished boundary $|Z_a| = \rho_{j,a}$:

$$\partial_J^\beta R_j(0) = \frac{\beta!}{(2\pi i)^m}\int_{|Z_1|=\rho_{j,1}}\cdots\int_{|Z_m|=\rho_{j,m}} \frac{R_j(Z)}{Z^{\beta+1}}\,\mathrm{d}Z_1\cdots\mathrm{d}Z_m.$$

Taking the B_j-norm and using the boundary length factors gives (142.73). The propagated estimate follows by applying $U_{j\to N,\beta}$ to each scale-j source-tail derivative, using its operator bound, and summing over scales. Terms with $s_j \ge |\beta|$ vanish by (142.72). □

Thus a source-extended polymer RG chart must declare which source monomials are retained and which source derivatives of the connected generating functional are being reconstructed. To control all n-point Schwinger distribution windows in a fixed test-function family, the source-coordinate system must retain source degree at least n, or it must prove the Cauchy-tail sum (142.74). A scalar fixed-point trajectory without this source-extension estimate is a theorem about the partition function or un-sourced activity only; it is not yet a theorem about the reconstructed local QFT correlators.

For ordinary short-range critical scalar models, the unresolved work is to construct this datum with constants uniform along the tuned critical trajectory and with source extensions strong enough for reconstructed correlation functions. For gauge theories, the same structure must be built on gauge-invariant lattice activities or in a BV-compatible finite-regulator chart; a gauge-fixed formal activity without the quantum master equation or a gauge-invariant blocking map is not a Wilsonian gauge-theory construction in the sense used here.

142.8 Fixed Points And Continuum Limits

A continuum limit is obtained by tuning the unstable coordinates so that the RG trajectory remains near a fixed point for arbitrarily many steps and then converges, after reconstruction, to regulator-independent correlation functions. The existence of a non-Gaussian fixed point as a Banach-space fixed point is a theorem only after the RG map, norm, domain, and estimates are supplied. Perturbative beta functions and numerical fixed-point searches are evidence for a coordinate chart; they do not replace the fixed-point construction.

Definition 142.14 (Theorem-level nonperturbative RG fixed point)**.** A nonperturbative Wilsonian fixed-point theorem consists of the following declared data.

1. A regulator class and a concrete RG map $\mathcal{R} : \mathcal{U} \to \mathcal{B}$ on an open subset $\mathcal{U}$ of a Banach space $(\mathcal{B}, \|\cdot\|)$ of interactions, measures, tensors, or polymer activities.
2. A proof that the map is defined on the claimed domain and is differentiable to the order used in the argument.
3. A point $V_* \in \mathcal{U}$ satisfying $\mathcal{R}(V_*) = V_*$, or a periodic orbit when the intended RG structure is not a fixed point.
4. A spectral or graph-transform analysis of $D\mathcal{R}|_{V_*}$, including the relevant, marginal, irrelevant, and redundant subspaces in the chosen norm.
5. A reconstruction map from the tuned sequence of finite-cutoff theories to correlation functions, local algebras, or Euclidean Schwinger functions, with convergence estimates sufficient for the claimed QFT framework.

The word "fixed point" in a theorem of this kind refers to the specified map $\mathcal{R}$ and norm. A zero of a truncated beta function, a solution of a finite-dimensional functional-RG truncation, or a numerical tensor-RG fixed point is evidence for such a theorem only after an embedding into this list has been supplied.

Theorem 142.15 (Validated fixed point by a Newton–Kantorovich estimate)**.** *Let $\mathcal{B}$ be a Banach space, let $F : \mathcal{U} \to \mathcal{B}$ be a C^1 map on an open ball $B_R(V_0) \subset \mathcal{U}$, and let*

$$F(V) = \mathcal{R}(V) - V.$$

Assume that $A : \mathcal{B} \to \mathcal{B}$ is bounded and injective, and that there are constants $\eta, \rho, L \geq 0$ with

$$\|AF(V_0)\| \leq \eta, \tag{142.75}$$
$$\|1 - A\,DF(V_0)\|_{\mathcal{B}\to\mathcal{B}} \leq \rho, \tag{142.76}$$
$$\|A(DF(V) - DF(W))\|_{\mathcal{B}\to\mathcal{B}} \leq L\|V - W\| \qquad (V, W \in B_R(V_0)). \tag{142.77}$$

If there is $r \leq R$ such that

$$\eta + \rho r + \frac{L}{2} r^2 \leq r, \qquad \rho + Lr < 1, \tag{142.78}$$

then $F(V) = 0$ has a unique solution V_* in the closed ball $\overline{B}_r(V_0)$. Equivalently, $\mathcal{R}(V_*) = V_*$.

Proof. Define the Newton-like map

$$\mathcal{N}(V) = V - AF(V).$$

For $V \in \overline{B}_r(V_0)$, the fundamental theorem of calculus gives

$$\mathcal{N}(V) - V_0 = -AF(V_0) + \int_0^1 \big(1 - A\,DF(V_0 + s(V - V_0))\big)(V - V_0)\,\mathrm{d}s.$$

Using (142.75) and (142.77), the norm of the right-hand side is bounded by

$$\eta + \rho r + \frac{L}{2} r^2,$$

which is at most r by (142.78). Thus $\mathcal{N}$ maps $\overline{B}_r(V_0)$ into itself. For $V, W \in \overline{B}_r(V_0)$,

$$\mathcal{N}(V) - \mathcal{N}(W) = \int_0^1 \big(1 - A\, DF(W + s(V - W))\big)(V - W)\, \mathrm{d}s,$$

and the same estimates give

$$\|\mathcal{N}(V) - \mathcal{N}(W)\| \le (\rho + Lr)\|V - W\|.$$

The coefficient is strictly less than one, so Banach's fixed-point theorem gives a unique fixed point of $\mathcal{N}$ in the ball. If $\mathcal{N}(V_*) = V_*$, then $AF(V_*) = 0$. Since A is injective, $F(V_*) = 0$. Conversely any zero of F is a fixed point of $\mathcal{N}$, so uniqueness follows. □

This theorem isolates the final functional-analytic step after RG estimates have been supplied. The difficult part in constructive RG is to choose $\mathcal{B}$, V_0, and A, and then prove (142.75)–(142.77) with constants small enough that (142.78) holds. In polymer RG this means summing tree or cluster expansions with large-field regulators; in tensor RG it means proving that the disentangled coarse-graining map is a contraction on a neighborhood that is stable under the infinite-dimensional map.

142.9 Projection Truncations And Residual Control

Finite-dimensional RG calculations enter the monograph only through a declared projection. Let $P_N : \mathcal{B} \to \mathcal{B}_N$ be a bounded projection onto a finite-dimensional subspace and put $Q_N = 1 - P_N$. For

$$F(V) = \mathcal{R}(V) - V,$$

the projected fixed-point equation is

$$P_N F(V_N) = 0, \qquad V_N \in \mathcal{B}_N. \tag{142.79}$$

The complement residual is

$$\epsilon_N(V_N; A) = \|AQ_N F(V_N)\|, \tag{142.80}$$

where A is the approximate inverse used in the Newton–Kantorovich estimate.

Example 142.16 (Spurious projected zeros)**.** Projection residuals must be controlled in the omitted directions. The finite-dimensional model $\mathcal{B} = \mathbb{R}^2$, $P(x, y) = (x, 0)$, and

$$F(x, y) = (0, 1).$$

has $PF(x, 0) = 0$ for every $x \in \mathbb{R}$, while the second component of $F(x, y)$ is identically one. Thus the entire projected line solves (142.79), and no point of the full space solves $F(V) = 0$. The residual $\epsilon_N(V_N; A) = \|AQ_N F(V_N)\|$ is the datum that detects this failure in the Newton–Kantorovich lifting problem.

Lifting a projected RG zero. Let $V_N \in \mathcal{B}_N$ solve (142.79). Suppose the hypotheses of Theorem 142.15 hold with $V_0 = V_N$. Then a true fixed point V_* of $\mathcal{R}$ exists in $\overline{B}_r(V_N)$ whenever

$$\epsilon_N(V_N; A) + \rho r + \frac{L}{2} r^2 \leq r, \qquad \rho + Lr < 1. \tag{142.81}$$

Since $P_N F(V_N) = 0$, the full residual is

$$F(V_N) = Q_N F(V_N).$$

Therefore

$$\|AF(V_N)\| = \|AQ_N F(V_N)\| = \epsilon_N(V_N; A).$$

The conditions in (142.81) are exactly the Newton–Kantorovich conditions (142.78) with $\eta = \epsilon_N(V_N; A)$. Theorem 142.15 then gives a unique zero of F in $\overline{B}_r(V_N)$, equivalently a fixed point of $\mathcal{R}$.

Controlled Approximation 142.17 (Projected fixed points as observable approximants)**.** The projected point V_N becomes an approximation to QFT data only after the lift estimate is paired with reconstruction estimates. Let $\mathcal{O}$ be a locally convex observable space with seminorms $\{p_\alpha\}$, and let

$$\mathrm{Rec} : \overline{B}_r(V_N) \longrightarrow \mathcal{O}$$

be the reconstruction map for the exact RG chart on the same ball as in (142.81). Suppose that for each finite observable window α there is a Lipschitz constant $L_\alpha(r)$ such that

$$p_\alpha(\mathrm{Rec}(V) - \mathrm{Rec}(W)) \leq L_\alpha(r)\|V - W\|_{\mathcal{B}}, \qquad V, W \in \overline{B}_r(V_N). \tag{142.82}$$

If the finite-dimensional calculation also reports a projected observable $\mathrm{Rec}_N(V_N)$ with chart error

$$p_\alpha(\mathrm{Rec}_N(V_N) - \mathrm{Rec}(V_N)) \leq \delta_{N,\alpha}, \tag{142.83}$$

then the lifted fixed point V_* supplied by (142.81) obeys

$$p_\alpha(\mathrm{Rec}_N(V_N) - \mathrm{Rec}(V_*)) \leq \delta_{N,\alpha} + L_\alpha(r) r. \tag{142.84}$$

The estimate is only a controlled approximation on the displayed finite window. It does not prove a continuum limit, an OS-positive hierarchy, or a universality statement unless the residual bound, the Newton–Kantorovich constants, the Lipschitz constants, and the observable chart errors are controlled on the directed family of windows used for reconstruction.

Indeed, the lifted fixed point satisfies $\|V_* - V_N\|_{\mathcal{B}} \leq r$. The triangle inequality and (142.82) give

$$p_\alpha(\mathrm{Rec}_N(V_N) - \mathrm{Rec}(V_*)) \leq p_\alpha(\mathrm{Rec}_N(V_N) - \mathrm{Rec}(V_N)) + L_\alpha(r)\|V_N - V_*\|_{\mathcal{B}},$$

which is (142.84) by (142.83). The load-bearing data are the residual and reconstruction constants that a finite-dimensional fixed point computation must report: (142.81) and (142.84). With those constants controlled on the directed observable family, projected observables become finite-window approximants to the lifted fixed-point reconstruction; without them, the projected calculation remains a finite-dimensional coordinate calculation.

The companion calculation

`calculation-checks/rg_projection_checks.py` checks these finite-dimensional algebraic examples with exact rational arithmetic, including the residual-lift and finite-window observable-error budget. In an actual Wilsonian RG proof, the hard work is to bound $\epsilon_N(V_N; A)$, ρ, L, the reconstruction constants $L_\alpha(r)$, and the chart errors $\delta_{N,\alpha}$ in the infinite-dimensional norm and in the directed observable topology, not to solve the projected equation.

142.10 Constructive Fixed-Point Output Data

The most useful rigorous fixed-point examples produce a fixed interaction together with correlation and response functions whose remainders are controlled in a declared norm. The following output record is the minimum structure recorded in this monograph when using the recent long-range fermionic RG constructions as a benchmark.

Definition 142.18 (Constructive fermionic fixed-point output)**.** A constructive fermionic fixed-point output consists of:

1. a finite ultraviolet cutoff and a Grassmann Gaussian covariance C_Λ, including its scaling law;
2. a Banach space $\mathcal{B}_{\mathrm{ker}}$ of antisymmetric interaction kernels, with a norm controlling decay, differentiability, and scale weights;
3. an exact RG map $\mathcal{R}$ on an open subset of $\mathcal{B}_{\mathrm{ker}}$, defined by Grassmann Gaussian integration, localization of relevant coordinates, and rescaling;
4. a fixed interaction $V_* \in \mathcal{B}_{\mathrm{ker}}$, satisfying $\mathcal{R}(V_*) = V_*$;
5. source coordinates for the field and for the density, with scaling dimensions $[\psi]$ and $[J]$;
6. response functions constructed from the fixed-point generating functional, together with scale-covariance estimates and explicit remainders.

This is fixed-cutoff constructive RG output. A continuum-QFT theorem requires an additional removal or interpretation of the cutoff in the observable topology being claimed.

Quoted Theorem 142.19 (Long-range fermionic ψ^4_d fixed point)**.** *For the long-range fermionic ψ^4_d model in dimensions $d = 1, 2, 3$, with fractional kinetic term chosen so that the quartic interaction is weakly relevant, there is a small-parameter regime in which the constructive RG map has a nontrivial fixed interaction V_* in the Banach space of kernels. With a fixed ultraviolet cutoff, the field and density response functions at V_* are constructed by convergent tree expansions. The field exponent is the naive exponent, while the density exponent is anomalous and analytic in the small parameter. The associated scaling operators have two-point functions that are scale covariant up to a stretched-exponential large-distance remainder controlled by the ultraviolet cutoff.*

Status of the quoted theorem. This is the theorem-level output of the GMR and GMRS constructive RG works just quoted. Its proof constructs V_* in an infinite-dimensional kernel space and controls the irrelevant tail by a convergent multiscale tree expansion. The theorem is quoted here as a state-of-the-art benchmark for what a nonperturbative Wilsonian fixed-point statement must contain. The monograph uses the theorem to calibrate the form of a complete RG claim, while separate scalar and gauge theory claims in this book require their own estimates.

Kernel-space mechanism in the fermionic benchmark. The fermionic example supplies a constructive fixed point by building an invariant infinite-dimensional kernel space and proving that the RG map closes on it. Write a source-extended monomial by a label

$$\alpha = (n, m, \ell, p), \qquad p = (p_1, \dots, p_\ell), \quad p_i \in \{0, 1\},$$

where n counts field sources φ, m counts density sources J, ℓ counts fermion fields ψ, and $|p|_1 = \sum_i p_i$ counts derivatives on the fermion fields. The associated coordinate is a translation-invariant kernel

$$H_\alpha(x_1, \dots, x_n; y_1, \dots, y_m; z_1, \dots, z_\ell)$$

with the internal antisymmetries imposed by the Grassmann variables and with finite pinned L^1 norm, obtained by integrating the absolute value over all but one spacetime variable and then taking the supremum over internal indices. The fixed point is therefore a sequence $H = \{H_\alpha\}_\alpha$, not a finite polynomial in local fields.

Let the fermion scaling dimension be

$$[\psi] = \frac{d}{4} - \frac{\epsilon}{2},$$

and let the source fields φ and J scale with dimensions $d - \Delta_1$ and $d - \Delta_2$. Equivalently, Δ_1 and Δ_2 are the dimensions of the scaling operators whose response functions are generated by differentiating with respect to φ and J. If the scale factor is γ, the dilatation of a pinned L^1 kernel has exponent

$$D_{\mathrm{sc}}(\alpha) = d - n(d - \Delta_1) - m(d - \Delta_2) - \ell[\psi] - |p|_1. \tag{142.85}$$

Thus $\|DH_\alpha\|_1 = \gamma^{D_{\mathrm{sc}}(\alpha)} \|H_\alpha\|_1$ in the kernel norm before the additional Gevrey or stretched-exponential weights are applied. With $\Delta_1 = [\psi] + O(\epsilon)$ and $\Delta_2 = 2[\psi] + O(\epsilon)$, the exceptional local block separated by the trimming chart consists of the mass-type bilinear $(0, 0, 2, 0)$, the quartic $(0, 0, 4, 0)$, the source-field coordinate $(1, 0, 1, 0)$, the density-source coordinate $(0, 1, 2, 0)$, and the first-derivative bilinears $(0, 0, 2, p)$ with $|p|_1 = 1$. This list is not being introduced as a finite truncation of the theory: $(0, 1, 2, 0)$ may be weakly irrelevant depending on the sign of $\Delta_2 - 2[\psi]$, and the first-derivative bilinears are irrelevant in $d = 1$ for small ϵ. They are kept in the local block because their scaling exponents are close enough to zero, or because a uniform trimmed coordinate system for $d = 1, 2, 3$ makes the localization/interpolation estimates stable. For example,

$$D_{\mathrm{sc}}(0, 0, 4, 0) = d - 4[\psi] = 2\epsilon, \qquad D_{\mathrm{sc}}(0, 0, 2, 0) = d - 2[\psi] = \frac{d}{2} + \epsilon. \tag{142.86}$$

The quartic coordinate is weakly relevant for $\epsilon > 0$, which is why a small non-Gaussian fixed point can be searched for near the Gaussian one.

The invariant space is obtained by a trimming operation. The kernels in the finite exceptional local list are represented by local delta-functions with constant coefficients. The nonlocal part of those same kernels is subtracted and rewritten in derivative coordinates, so that it enters the irrelevant Banach component. All remaining coordinates live in a kernel space whose norm controls pinned L^1 size, internal index sums, scale weights, and short-range decay. Schematically, one writes

$$H = (g, K), \qquad g \in \mathbb{C}^r, \quad K \in \mathcal{K},$$

where g is the finite vector of local trimmed coordinates and $\mathcal{K}$ is the Banach space of irrelevant nonlocal kernels. One RG step has the form

$$(g, K) \longmapsto \big(A_{\mathrm{loc}} g + B_{\mathrm{loc}}(g, K),\ A_{\mathrm{irr}} K + B_{\mathrm{irr}}(g, K)\big), \tag{142.87}$$

after fluctuation integration, trimming, and rescaling. The theorem's analytic burden is to prove this map on a ball in $\mathbb{C}^r \oplus \mathcal{K}$, with estimates of the form

$$\|A_{\mathrm{irr}}\| \le q < 1, \qquad \|B_{\mathrm{irr}}(g, K) - B_{\mathrm{irr}}(g', K')\| \le c\,(|g| + |g'| + \|K\| + \|K'\|)\,(|g - g'| + \|K - K'\|), \tag{142.88}$$

and corresponding finite-dimensional equations for g. The fixed point is the simultaneous solution of both the local equation and

$$K = A_{\mathrm{irr}} K + B_{\mathrm{irr}}(g, K). \tag{142.89}$$

Dropping (142.89) would reduce the argument to a projected beta-function calculation and would miss exactly the part of the theorem that makes the construction nonperturbative.

The determinant expansion used in the fermionic construction is also part of the mechanism. The Grassmann Gaussian integration reorganizes the multiscale series into determinants of covariance matrices, and Gram-type determinant bounds give absolute convergence in the kernel norm. This is why the fermionic example presently reaches a level of rigor not available for the ordinary short-range scalar critical point: the invariant kernel space and the convergent determinant/tree expansion are proved together. The monograph uses this as a proof template, not as evidence that the scalar or gauge targets are settled.

Remark 142.20 (Irrelevant tails are part of the theorem). Let $\mathcal{B} = \mathbb{R}^m \oplus \mathcal{K}$, where $\mathcal{K}$ is a Banach space of irrelevant kernels, and let $P : \mathcal{B} \to \mathbb{R}^m$ be the coordinate projection. A solution of the projected equation

$$P(\mathcal{R}(g, 0) - (g, 0)) = 0$$

does not define a fixed point of $\mathcal{R}$ unless the irrelevant residual

$$Q(\mathcal{R}(g, 0) - (g, 0)) \in \mathcal{K}$$

is either zero or is absorbed by a controlled graph $K = h(g)$ satisfying the full fixed-point equation. Write $Q = 1 - P$. If $V = (g, 0)$ solves only the projected equation, then

$$\mathcal{R}(V) - V = (0, Q(\mathcal{R}(V) - V)).$$

Thus V is a fixed point if and only if the displayed irrelevant residual vanishes. More generally, a fixed point with the same local coordinates must have the form $V_h = (g, h(g))$ and satisfy

$$P(\mathcal{R}(V_h) - V_h) = 0, \qquad Q(\mathcal{R}(V_h) - V_h) = 0.$$

The second equation is the irrelevant-tail equation. In a constructive RG proof it is solved by contraction, graph transform, or Newton validation in the norm of $\mathcal{K}$; without such a bound, the projected equation is only a coordinate calculation.

142.11 Hierarchical Scalar Fixed-Point Benchmark

Scalar non-Gaussian fixed points provide the most familiar physical target for Wilsonian RG, but the mathematical status depends strongly on the map being studied. The hierarchical scalar models form a class in which the block-spin map is an explicitly defined nonlinear integral operator. They are not the short-range nearest-neighbor lattice ϕ^4 models, and a theorem for the hierarchical map is not, by itself, a theorem for the ordinary three-dimensional Ising or ϕ^4 fixed point. They are nevertheless an essential benchmark because the fixed point, unstable directions, and irrelevant remainder can be treated in a genuine infinite-dimensional RG problem rather than in a finite beta-function truncation.

Definition 142.21 (Hierarchical scalar RG system)**.** A one-component hierarchical scalar RG system consists of:

1. a scale factor $L > 1$, a dimension parameter $D > 0$, and $b = L^D$;
2. a field scaling exponent $\Delta > 0$, a number $a = L^{-\Delta}$, and a fluctuation variance $\gamma > 0$;
3. a Banach space $\mathcal{B}_{\text{hier}}$ of even real-analytic functions $V : \mathbb{R} \to \mathbb{R}$, modulo additive constants, with a norm controlling a complex strip or disk and large-field growth;
4. the nonlinear map $\mathcal{R}_{\text{hier}}$ defined, whenever the integral is finite and analytic in the chosen domain, by
$$e^{-(\mathcal{R}_{\text{hier}}V)(\Phi)} = Z_V^{-1} \int_{\mathbb{R}} \exp[-b\,V(a\Phi + \zeta)]\, \frac{e^{-\zeta^2/(2\gamma)}}{\sqrt{2\pi\gamma}}\, \mathrm{d}\zeta, \tag{142.90}$$
where Z_V is fixed by the normalization $(\mathcal{R}_{\text{hier}}V)(0) = 0$.

This is a mathematical RG system. The hierarchical covariance and the normalization convention are part of the definition; they are not inferred from a continuum path integral after the fact.

The linearization at the Gaussian point already shows the local-coordinate arithmetic that underlies the usual relevance labels. The following calculation is exact for the hierarchical Gaussian kernel and does not use a formal momentum shell.

Gaussian linearization in Wick coordinates. Let $0 < a < 1$, $\gamma > 0$, and
$$c_* = \frac{\gamma}{1 - a^2}.$$
For $n \geq 0$, define the Wick polynomial $: x^n :_{c_*}$ by
$$\exp\left(tx - \frac{c_* t^2}{2}\right) = \sum_{n=0}^{\infty} \frac{t^n}{n!}\ : x^n :_{c_*}\ . \tag{142.91}$$
The derivative of $\mathcal{R}_{\text{hier}}$ at $V = 0$, before quotienting the additive constant, acts on the nonconstant Wick coordinates by
$$D\mathcal{R}_{\text{hier}}|_0\big(: \Phi^n :_{c_*}\big) = b\, a^n : \Phi^n :_{c_*}, \qquad n \geq 1. \tag{142.92}$$
In the canonical scalar bookkeeping
$$\Delta = \frac{D-2}{2}, \qquad b = L^D,$$
the even local coordinate $: \Phi^{2r} :$ has engineering eigenvalue
$$\lambda_{2r} = L^{y_{2r}}, \qquad y_{2r} = D - r(D-2). \tag{142.93}$$

Thus $y_2 = 2$, $y_4 = 4 - D$, and $y_6 = 6 - 2D$. For the first variation of (142.90), additive normalization affects only the constant coordinate. Hence for a nonconstant test direction f,

$$D\mathcal{R}_{\mathrm{hier}}|_0 f = b \int_{\mathbb{R}} f(a\Phi + \zeta)\, \mathrm{d}\mu_\gamma(\zeta) \quad \text{mod constants,} \tag{142.94}$$

where $\mathrm{d}\mu_\gamma$ is the centered Gaussian measure of variance γ. Apply this to the generating function (142.91). Since $c_* - \gamma = a^2 c_*$,

$$\begin{aligned}\int \exp\left(t(a\Phi + \zeta) - \frac{c_* t^2}{2}\right) \mathrm{d}\mu_\gamma(\zeta) &= \exp\left(ta\Phi - \frac{(c_* - \gamma)t^2}{2}\right)\\ &= \exp\left(ta\Phi - \frac{a^2 c_* t^2}{2}\right)\\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!}\, a^n :\Phi^n:_{c_*}\ .\end{aligned}$$

Comparing coefficients gives the factor a^n; the prefactor b in (142.94) gives (142.92). Substituting $a = L^{-(D-2)/2}$ and $n = 2r$ gives

$$b\, a^{2r} = L^D L^{-r(D-2)} = L^{D - r(D-2)},$$

which is (142.93).

Quoted Theorem 142.22 (Hierarchical scalar non-Gaussian fixed point)**.** *For the Dyson–Wilson hierarchical scalar RG maps in the Bleher–Sinai and Collet–Eckmann class, and for the closely related $\phi^4_{3,\epsilon}$ RG construction of Brydges–Mitter–Scoppola,*[1] *there are parameter regimes in which a non-Gaussian fixed point exists as an element of a Banach space of analytic or polymer interactions. In these regimes the theorem supplies an RG map, a domain and norm, a fixed interaction V_*, a finite-dimensional unstable subspace, contraction or graph estimates for the stable directions, and scaling estimates for the corresponding hierarchical or $\phi^4_{3,\epsilon}$ observable data.*

Status of the quoted theorem. The theorem is quoted as a benchmark, not as an imported proof for the ordinary short-range three-dimensional scalar fixed point. Its mathematical content is that the non-Gaussian fixed point is an element of the specified infinite-dimensional RG domain and that the irrelevant coordinates are controlled by estimates in that domain. The hierarchical proofs exploit the recursive structure of the covariance; the $\phi^4_{3,\epsilon}$ construction uses a multiscale decomposition and polymer estimates adapted to the fractional covariance. In both cases the result is stronger than locating a zero of a finite beta function and weaker than a construction of the short-range ϕ^4_3 Wilson–Fisher CFT. That last comparison would require a theorem relating the short-range block-spin map and reconstruction data to one of these fixed-point systems.

The companion calculation

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calculation-checks/rg_hierarchical_scalar_checks.py
```

checks (142.92) and (142.93) with exact rational arithmetic for the even Wick coordinates used above.

[1] See P. M. Bleher and Ya. G. Sinai, "Investigation of the critical point in models of the type of Dyson's hierarchical models," *Commun. Math. Phys.* **33** (1973); P. Collet and J.-P. Eckmann, *A Renormalization Group Analysis of the Hierarchical Model in Statistical Mechanics*, Lecture Notes in Physics **74**, Springer (1978); and D. C. Brydges, P. K. Mitter, and B. Scoppola, "Critical $(\Phi^4)_{3,\epsilon}$," arXiv:hep-th/0206040.

142.12 Current Rigorous Fixed-Point Benchmarks

The preceding definition deliberately separates what has been proved from what is inferred from formal or numerical RG. Recent rigorous work supplies benchmarks for both the constructive-field-theory and tensor-network versions of the problem.

The hierarchical scalar results just described are the cleanest scalar examples of theorem-level non-Gaussian Wilsonian fixed points. Their role in this chapter is deliberately calibrated: they prove that scalar non-Gaussian fixed points can be constructed in rigorous RG systems with irrelevant tails, but they do not settle the ordinary short-range three-dimensional scalar fixed point.

In the long-range fermionic ψ_d^4 models analyzed by Giuliani, Mastropietro, Rychkov, and collaborators,[2] the Banach space contains all short-ranged irrelevant kernels, not only a finite list of couplings. The fixed interaction is constructed by a convergent RG scheme with a small parameter controlling the relevance of the quartic interaction. The important lesson for this monograph is structural: the fixed point is an element of an infinite-dimensional space, the irrelevant tail is part of the theorem rather than a truncation error, and scaling operators are constructed as response functions with stated remainder estimates.

In the rigorous tensor-RG work of Kennedy and Rychkov,[3] the map acts on infinite-dimensional tensors. The theorem is not a statement about the naive contraction of four tensors, which fails to be contractive near the high-temperature fixed point; it is a statement about a modified RG map with a disentangling step and a Hilbert–Schmidt neighborhood on which contraction estimates are proved. The low-temperature analysis gives stable ordered fixed points and a discontinuity fixed point for first-order transition behavior. These examples illustrate why the RG map itself is part of the theorem: changing the coarse-graining prescription changes the mathematical assertion being made.

Functional RG equations such as the Wetterich equation are exact identities at finite regulator, as derived in Chapter 36. A finite-dimensional truncation of such an equation can be an excellent computational scheme, but the theorem-level fixed-point claim requires an error estimate comparing the truncated subspace with the full Banach space. The same distinction applies to perturbative epsilon expansions: the formal fixed point organizes local coordinates and critical exponents order by order, while a nonperturbative fixed-point theorem additionally proves existence, stability, and reconstruction.

Controlled Approximation 142.23 (Functional-RG truncations as projected flows)**.** Let $\mathcal{B}_\Gamma$ be a Banach space of regulated effective-average actions, and write the exact finite-regulator Wetterich equation as

$$\dot{\Gamma}_t = \mathcal{F}_t(\Gamma_t), \qquad \mathcal{F}_t(\Gamma) = \frac{1}{2}\,\mathrm{Tr}\Big[\big(\Gamma^{(2)} + R_t\big)^{-1}\dot{R}_t\Big]$$

on a domain where the inverse Hessian and trace define an element of $\mathcal{B}_\Gamma$. Let $P_N : \mathcal{B}_\Gamma \to \mathcal{B}_N$ be

[2] See A. Giuliani, V. Mastropietro, and S. Rychkov, "Gentle introduction to rigorous Renormalization Group: a worked fermionic example," JHEP **01** (2021) 026, arXiv:2008.04361; and A. Giuliani, V. Mastropietro, S. Rychkov, and G. Scola, "Non-trivial fixed point of a ψ_d^4 fermionic theory, II. Anomalous exponent and scaling operators," arXiv:2404.14904.

[3] See T. Kennedy and S. Rychkov, "Tensor RG approach to high-temperature fixed point," J. Stat. Phys. **187** (2022) 33, arXiv:2107.11464; and "Tensor Renormalization Group at Low Temperatures: Discontinuity Fixed Point," arXiv:2210.06669.

a bounded finite-rank projection, $Q_N = 1 - P_N$, and suppose the truncated trajectory $\gamma_N(t) \in \mathcal{B}_N$ solves

$$\dot{\gamma}_N(t) = P_N \mathcal{F}_t(\gamma_N(t)). \tag{142.95}$$

The omitted vector-field component along this projected trajectory is

$$r_N(t) = Q_N \mathcal{F}_t(\gamma_N(t)). \tag{142.96}$$

Assume that the exact trajectory $\Gamma(t)$ and the projected trajectory remain in a common tube on $0 \le t \le T$, that $\Gamma(0) = \gamma_N(0)$, and that

$$\|\mathcal{F}_t(X) - \mathcal{F}_t(Y)\|_{\mathcal{B}_\Gamma} \le M(t)\|X - Y\|_{\mathcal{B}_\Gamma}$$

there. Then

$$\|\Gamma(T) - \gamma_N(T)\|_{\mathcal{B}_\Gamma} \le \int_0^T \exp\left(\int_s^T M(u)\,\mathrm{d}u\right) \|r_N(s)\|_{\mathcal{B}_\Gamma}\,\mathrm{d}s. \tag{142.97}$$

If the reconstruction map into a finite observable window satisfies

$$p_\alpha(\mathrm{Rec}_T(X) - \mathrm{Rec}_T(Y)) \le L_{\alpha,T}\|X - Y\|_{\mathcal{B}_\Gamma}$$

on the same tube, then the right-hand side of (142.97), multiplied by $L_{\alpha,T}$, is the corresponding observable-window error.

To derive (142.97), put $e(t) = \Gamma(t) - \gamma_N(t)$. Using (142.95) and (142.96),

$$\dot{e}(t) = \mathcal{F}_t(\Gamma(t)) - \mathcal{F}_t(\gamma_N(t)) + r_N(t).$$

The upper Dini derivative of $\|e(t)\|_{\mathcal{B}_\Gamma}$ is therefore bounded by $M(t)\|e(t)\| + \|r_N(t)\|$, and (142.97) is Gronwall's inequality with zero initial error. This is the finite-regulator statement behind the functional-RG truncation boundary: the projected vector field must be accompanied by an omitted-direction residual in the same function space as the exact effective action, and by observable reconstruction estimates in the seminorms actually used.

Controlled Approximation 142.24 (Functional-RG projected fixed points)**.** In dimensionless autonomous variables, write the exact regulated functional-RG vector field as

$$\dot{\Gamma} = \beta(\Gamma), \qquad \beta : \mathcal{U}_\Gamma \subset \mathcal{B}_\Gamma \to \mathcal{B}_\Gamma.$$

Let $P_N : \mathcal{B}_\Gamma \to \mathcal{B}_N$ be a finite ansatz projection, $Q_N = 1 - P_N$, and let $\Gamma_N^* \in \mathcal{B}_N$ solve only the projected fixed-point equation

$$P_N \beta(\Gamma_N^*) = 0. \tag{142.98}$$

The omitted beta-function component is

$$r_N^* = Q_N \beta(\Gamma_N^*). \tag{142.99}$$

Suppose an approximate inverse $A : \mathcal{B}_\Gamma \to \mathcal{B}_\Gamma$ for $D\beta(\Gamma_N^*)$ and constants ρ, L satisfy the Newton–Kantorovich estimates of Theorem 142.15, with

$$\eta_N^{\mathrm{FRG}} := \|A r_N^*\|_{\mathcal{B}_\Gamma}.$$

If, for some r_N,

$$\eta_N^{\mathrm{FRG}} + \rho r_N + \frac{L}{2} r_N^2 \le r_N, \qquad \rho + L r_N < 1, \tag{142.100}$$

then there is an exact fixed point Γ_* of the full vector field in the ball $\|\Gamma_* - \Gamma_N^*\| \le r_N$. If a finite observable window p_α is reconstructed by a map Rec_α that is $L_{\alpha,N}$-Lipschitz on the same ball, and if the ansatz observable itself has chart error $\delta_{\alpha,N}$, then

$$p_\alpha(\mathrm{Rec}_{\alpha,N}(\Gamma_N^*) - \mathrm{Rec}_\alpha(\Gamma_*)) \le \delta_{\alpha,N} + L_{\alpha,N} r_N. \tag{142.101}$$

Thus a zero of the beta functions retained in a derivative expansion is not yet a Wilsonian fixed-point theorem. It is a controlled finite-window approximation only to the extent that (142.99), (142.100), and (142.101) are supplied in the full regulated effective-action norm and in the declared observable seminorms.

The proof is the projected-zero lifting argument above, applied to $F = \beta$. Since (142.98) kills only the retained coordinates, $\beta(\Gamma_N^*) = r_N^*$. The Newton–Kantorovich theorem gives the nearby exact zero once the omitted component is small after applying A. The observable estimate is the same triangle inequality as (142.84). The important point is negative: a critical exponent matrix computed inside the ansatz subspace does not control the discarded beta-function directions, the existence of a full fixed point, or the reconstruction of Schwinger or Wightman data.

142.13 Fixed-Point Claim Content

The preceding benchmarks should be compared through the objects they actually construct. A fixed-point statement in this chapter is compared through four questions:

1. what space the RG map acts on, such as local interactions, polymer activities, Grassmann kernels, tensors, lattice measures, or BV actions;
2. what estimates prove that the map is defined on its domain, is differentiable or analytic there, has the claimed fixed point, and has the stated stable/unstable decomposition;
3. which source extensions, contact-term subspaces, and reconstruction estimates turn the fixed point into distributions or observable algebras;
4. which comparison map identifies the constructed RG system with the physical short-range, gauge, or continuum QFT target.

The strength of a result is the first question in this list for which the necessary objects and estimates are missing. This convention prevents an auxiliary fixed point from being silently promoted to a theorem about a different QFT.

Hierarchical scalar systems. For the hierarchical scalar maps of Definition 142.21, the RG object and map theorem are part of the rigorous theorem: the nonlinear integral map is defined on a Banach space of analytic or polymer interactions, the non-Gaussian fixed point is constructed in that space, and stable/unstable directions are controlled by the corresponding estimates. Observable statements are hierarchical observable statements unless accompanied by a short-range reconstruction estimates and a transfer map of the form Definitions 142.31 and 142.40. The target identification with the ordinary nearest-neighbor or finite-range three-dimensional scalar critical point is therefore an additional problem, requiring its own comparison map and reconstruction estimates beyond shared coordinate names such as scalar, relevant, or quartic.

Long-range fermionic models. For the long-range fermionic ψ_d^4 systems, the rigorous result constructs a fixed interaction in a Banach space of Grassmann kernels and constructs selected response functions with anomalous scaling and controlled remainders. The source layer is part of the theorem for the field and density response functions that are actually constructed. The target is the long-range fermionic model with the specified fractional kinetic term and fixed ultraviolet cutoff interpretation. Using this result as evidence for ordinary short-range scalar critical behavior requires a separately stated auxiliary-to-target comparison; it is not an identification of universality classes by analogy.

Tensor RG fixed points. For tensor RG, the RG object is an infinite-dimensional tensor in a declared Hilbert–Schmidt or Banach topology. The theorem is a contraction statement for a modified tensor map, including its disentangling or truncation rule, on a specified neighborhood of the high-temperature, low-temperature, or discontinuity fixed point. The reconstructed object is initially a lattice statistical model or tensor-network partition function. A continuum-QFT claim requires a further scaling-window reconstruction and comparison to Schwinger functions, local algebras, or another QFT observable topology. Thus tensor RG supplies theorem-level fixed points of tensor maps and a valuable laboratory for RG proof technology; it becomes a continuum QFT fixed-point theorem only after the reconstruction layer is supplied.

Controlled Approximation 142.25 (Tensor-RG truncation windows)**.** Let $(\mathcal{B}_T, \|\cdot\|_T)$ be the declared tensor Banach or Hilbert–Schmidt space, let $\mathcal{T}_j : \mathcal{U}_j \to \mathcal{B}_T$ be the exact tensor blocking map at step j, and let $\widehat{\mathcal{T}}_j$ be the algorithmic map actually used, including its finite-rank truncation or disentangling prescription. Starting from the same initial tensor $T_0 = \widehat{T}_0$, define

$$T_{j+1} = \mathcal{T}_j(T_j), \qquad \widehat{T}_{j+1} = \widehat{\mathcal{T}}_j(\widehat{T}_j).$$

Suppose both orbits remain in a common tube on which

$$\|\mathcal{T}_j(X) - \mathcal{T}_j(Y)\|_T \leq L_j \|X - Y\|_T,$$

and suppose the one-step algorithmic residual along the computed orbit obeys

$$\delta_j = \|\mathcal{T}_j(\widehat{T}_j) - \widehat{\mathcal{T}}_j(\widehat{T}_j)\|_T. \tag{142.102}$$

Then

$$\|T_n - \widehat{T}_n\|_T \leq \sum_{j=0}^{n-1} \left(\prod_{\ell=j+1}^{n-1} L_\ell \right) \delta_j. \tag{142.103}$$

If a normalized finite observable window W_α is evaluated by a map $\mathcal{O}_{\alpha,n}$ satisfying

$$p_\alpha(\mathcal{O}_{\alpha,n}(X) - \mathcal{O}_{\alpha,n}(Y)) \leq C_{\alpha,n} \|X - Y\|_T$$

on the same tube, and if the normalization or boundary-window error is $\eta_{\alpha,n}$, then

$$p_\alpha\Big(\mathcal{O}_{\alpha,n}(T_n) - \mathcal{O}_{\alpha,n}(\widehat{T}_n)\Big) \leq C_{\alpha,n} \sum_{j=0}^{n-1} \left(\prod_{\ell=j+1}^{n-1} L_\ell \right) \delta_j + \eta_{\alpha,n}. \tag{142.104}$$

This is the tensor-network analogue of the auxiliary-transfer estimate: finite bond dimension, singular-value truncation, or a disentangling step is part of the RG map. Its effect on a claimed observable window is controlled by the residual in (142.102), the exact-map amplification factors, and the window map $\mathcal{O}_{\alpha,n}$.

The derivation is the same finite recursion as (142.185). Put $e_j = T_j - \widehat{T}_j$. Then

$$e_{j+1} = \mathcal{T}_j(T_j) - \mathcal{T}_j(\widehat{T}_j) + \mathcal{T}_j(\widehat{T}_j) - \widehat{\mathcal{T}}_j(\widehat{T}_j),$$

so

$$\|e_{j+1}\|_T \le L_j \|e_j\|_T + \delta_j.$$

Iteration from $e_0 = 0$ gives (142.103); the finite-window estimate (142.104) then follows from the Lipschitz bound for $\mathcal{O}_{\alpha,n}$ and the added normalization error. A computed finite-bond-dimension flow is therefore a controlled tensor-network result precisely on the windows for which these residuals and Lipschitz constants are supplied.

Ordinary short-range scalar targets. For a short-range three-dimensional scalar Wilson–Fisher target, the RG object must be the block-spin/polymer structure of Definition 142.31, or a proven equivalent structure connected to it by Definition 142.2. The theorem would have to construct the stable critical trajectory or fixed point in that structure, including the irrelevant polymer tail, source extensions, contact terms, and distributional reconstruction estimates. The massive constructive ϕ^4_3 theorems and the critical hierarchical or long-range fixed-point theorems supply important pieces of the landscape, but the ordinary short-range critical reconstruction theorem is a separate target-identification problem. Perturbative epsilon expansions and finite-dimensional functional-RG truncations are then used as coordinate and numerical evidence for this target, not as substitutes for the map theorem and reconstruction estimates.

Gauge targets. For gauge theories, the RG object is not a list of gauge-fixed coefficients. The object is either a gauge-invariant lattice measure and observable system, or a regularized BV action and observable complex satisfying the quantum master equation and its RG pushforward. The source layer must use gauge-invariant local observables, Wilson-line or Wilson-loop observables with their geometric data, or BV cohomology classes. A gauge-theory fixed-point or continuum-limit theorem therefore includes preservation of the gauge structure under blocking, reconstruction of physical observables, and comparison of line or extended sectors when those sectors are part of the claim. Beta functions in a gauge-fixed coordinate chart are useful local coordinates only after this gauge-compatible RG structure has been declared.

Model-by-model status synthesis. The examples above should be compared by the stage at which a theorem has actually been proved. The stages are: construction of the RG map in its declared norm; construction of a fixed point or stable critical trajectory; source extension and operator-coordinate control; reconstruction of Schwinger, Wightman, or observable-algebra data; and identification of the constructed object with a specified physical target. These stages are logically ordered. A proof at an earlier stage supplies important mathematics, but it does not automatically supply the later stages.

For massive scalar models with superrenormalizable interactions, constructive methods can supply Euclidean Schwinger functions and, after the OS reconstruction hypotheses are checked, Wightman theories. The theorem-level output is a local QFT in a massive regime. It is not a construction of a scale-invariant non-Gaussian fixed point of the ordinary short-range critical block-spin map, because the critical stable trajectory and its source-coordinate spectrum are not part of that massive theorem.

For hierarchical scalar models and related auxiliary scalar RG systems, the theorem-level output is a fixed point or critical trajectory for the auxiliary map in its own Banach space. If that theorem also constructs hierarchical observables, those observables belong first to the hierarchical model. To turn the result into a statement about the ordinary short-range scalar target one must additionally supply Definition 142.40: an intertwining map, relevant-coordinate comparison, observable-window comparison, and a summable target-orbit defect. Shared exponents or shared local coordinate names are not such a transfer map.

For long-range fermionic fixed-point constructions, the theorem-level output is the fixed interaction in a Grassmann-kernel Banach space together with the response functions and remainders actually constructed. This is a genuine nonperturbative fixed-point theorem in its declared fermionic model. It is not a scalar Wilson–Fisher theorem, nor a gauge-theory theorem, because the microscopic covariance, field variables, source space, and reconstruction maps are different.

For tensor RG, the theorem-level output is a fixed point or contraction of a specified tensor map in a Hilbert–Schmidt or Banach tensor topology. When a finite-bond or disentangled computation is used, the controlled statement is only the residual-bound window of (142.104). A continuum QFT claim requires a separate scaling-window reconstruction from tensor data to Schwinger functions, local algebras, or another declared observable topology.

For functional-RG calculations, the exact finite-regulator statement is the Wetterich identity for a regulated effective average action. A finite ansatz or derivative expansion is a projected flow, and the meaningful comparison with the exact flow is the omitted-vector-field estimate (142.97). A fixed point of the projected ordinary differential equation is therefore a controlled approximation only on those finite windows for which the residual and reconstruction Lipschitz constants have been supplied.

For the ordinary short-range critical scalar target, the theorem-level output would be a tuned trajectory

$$(g_k, K_k, W_{k,E}) \longrightarrow (g_*, K_*, W_{*,E})$$

in the short-range block-spin and source-extended polymer structure. The precise reconstruction target is the assembly package of Definition 142.39: the source-window budget (142.178), uniform distribution bounds, cofinal window compatibility, directed OS positivity, positive-time semigroup estimates, corrected OS-II growth, and a common scale schedule on which these estimates hold together. This is the point at which the OS reconstruction theorem can be invoked. Existing benchmark systems identify pieces of this target structure, but the ordinary short-range critical scalar construction remains open until these estimates are proved in that model.

For gauge targets, the corresponding output is either a continuum limit of gauge-invariant lattice measures and Wilson-line or Wilson-loop observable windows, or a regularized BV flow preserving the quantum master equation and the observable cochain maps. The reconstruction data are gauge-invariant observables or BV cohomology classes. A beta function in a gauge-fixed local coordinate chart is one coordinate shadow of this structure; it is not the structure itself.

This comparison also fixes how this monograph treats the phrase "state of the art." The current rigorous fixed-point frontier contains theorem-level auxiliary and model-specific systems with impressive control of irrelevant coordinates. It does not yet contain a general theorem turning every physically expected Wilsonian fixed point into a local QFT, nor a theorem constructing the ordinary three-dimensional Ising CFT from a short-range block-spin map with all source and reconstruction estimates. The monograph uses the proved systems as proof templates and laboratories, while keeping the physical target-identification problems explicit.

142.14 Gauge-Compatible RG And Reconstruction Estimates

A Wilsonian RG theorem for a scalar probability measure can be formulated as a pushforward of measures, followed by a localization step and a rescaling. Gauge theories require richer structure, because the variables used in a calculation contain redundancy. The theorem-level object is therefore an RG map on gauge-invariant lattice measures and observables, or an RG map in the BV category whose effective actions and observable complexes satisfy the quantum master equation at every scale. The BV framework used here is developed in Chapter 48; the lattice Yang–Mills and reflection positivity inputs are developed in Chapters 46, 138, and 140.

Definition 142.26 (Gauge-compatible Wilsonian RG structure)**.** Fix a spacetime dimension D, a scale factor $L > 1$, and a regulator sequence indexed by $k \in \mathbb{Z}_{\geq 0}$. A gauge-compatible Wilsonian RG structure is one of the following two forms.

1. In the lattice form, one has finite cell complexes Λ_k, a compact Lie group G_k, configuration spaces $\mathcal{A}_k = G_k^{E(\Lambda_k)}$ of link variables, gauge groups $\mathcal{G}_k = G_k^{V(\Lambda_k)}$ acting on $\mathcal{A}_k$, and finite measures $\mathrm{d}\mu_k$ on $\mathcal{A}_k$. The observables are functions or distributions on $\mathcal{A}_k$ invariant under $\mathcal{G}_k$, with a declared norm controlling Wilson loops, local plaquette fields, and any matter insertions with their Wilson-line connectors. The RG step is a gauge-equivariant blocking kernel
$$B_{k,k-1}(A_{k-1}, A_k)\, \mathrm{d}\mu_k(A_k)$$
whose pushforward gives a measure $\mathrm{d}\mu_{k-1}$ on $\mathcal{A}_{k-1}$, followed by a localization map into a Banach chart of gauge-invariant polymer activities. Reflection positivity, when claimed, is part of the structure and is expressed as positivity of the blocked quadratic form on the positive-time observable algebra.

2. In the BV form, one has finite-regulator graded vector spaces $\mathcal{F}_k$ of fields, ghosts, antifields, and possible sources; odd symplectic forms of degree -1; BV Laplacians Δ_k associated with the chosen finite-dimensional Berezin-Lebesgue density; and actions S_k satisfying the quantum master equation
$$\frac{1}{2}(S_k, S_k)_k - \hbar\, \Delta_k S_k = 0, \tag{142.105}$$
or equivalently $\Delta_k e^{-S_k/\hbar} = 0$. The RG step is BV pushforward along a fluctuation subspace with gauge-fixing Lagrangian $\mathcal{L}_k$:
$$e^{-S_{k-1}(\phi_{k-1})/\hbar} = \int_{\mathcal{L}_k} e^{-S_k(\phi_{k-1}+\xi_k)/\hbar}\, \mathrm{d}\xi_k, \tag{142.106}$$
together with a localization map into a Banach space of local BV functionals. The induced observable map $P_{k,k-1}$ sends Q_{S_k}-closed observables to $Q_{S_{k-1}}$-closed observables, where $Q_{S_k} = (S_k, \cdot)_k - \hbar\Delta_k$.

In either form, a fixed point or continuum limit also includes reconstruction maps for gauge-invariant local and extended observables. A flow of finitely many gauge-fixed coefficients is a coordinate shadow of this structure; the theorem-level statement is the lattice or BV structure together with its norm estimates.

The definition deliberately includes Wilson-line and loop observables. A colored field variable in a gauge-fixed calculation is useful for local coordinate work, but a nonperturbative observable statement is formulated in terms of gauge-invariant insertions or BV cohomology classes. Thus a charged-particle or partonic reconstruction theorem must specify the line, asymptotic dressing, or cohomology class used to define the observable before one speaks of an RG limit.

Construction 142.27 (Gauge-equivariant path blocking at finite regulator)**.** The lattice half of Definition 142.26 has a minimal exact example. Let Γ_f and Γ_c be finite oriented graphs, interpreted as fine and coarse lattices. Let $i : V(\Gamma_c) \hookrightarrow V(\Gamma_f)$ embed coarse vertices into fine vertices. For every oriented coarse edge $e : v \to w$, choose an oriented fine path

$$P_e = (\ell_1, \dots, \ell_m), \qquad s(\ell_1) = i(v), \quad t(\ell_m) = i(w),$$

and require $P_{\bar{e}} = \overline{P_e}$, the same path with reversed orientation. A fine gauge field assigns $U_\ell \in G$ to oriented fine edges with $U_{\bar{\ell}} = U_\ell^{-1}$. Define the blocked coarse link by

$$B(A)_e = U_{\ell_1} U_{\ell_2} \cdots U_{\ell_m}. \tag{142.107}$$

For a fine gauge transformation $g \in G^{V(\Gamma_f)}$,

$$U_\ell^g = g_{s(\ell)} U_\ell g_{t(\ell)}^{-1}.$$

The internal endpoint factors along P_e cancel, so

$$B(A^g)_e = g_{i(v)}\, B(A)_e\, g_{i(w)}^{-1}. \tag{142.108}$$

Thus B is equivariant from the fine gauge action to the coarse gauge action obtained by restricting g to embedded coarse vertices.

Let μ_f be any finite measure on $G^{E(\Gamma_f)}$ invariant under the fine gauge group, and let $\mu_c = B_* \mu_f$. For a coarse gauge transformation $h \in G^{V(\Gamma_c)}$, choose any extension $\tilde{h} \in G^{V(\Gamma_f)}$ with $\tilde{h}_{i(v)} = h_v$. Equation (142.108) gives

$$B(A^{\tilde{h}}) = B(A)^h.$$

Since μ_f is invariant under $A \mapsto A^{\tilde{h}}$, the pushforward μ_c is invariant under the coarse gauge group. If $C = e_1 \cdots e_r$ is a closed coarse loop and R is a finite-dimensional unitary representation of G, then

$$W_R(C; B(A)) = \frac{1}{\dim R} \operatorname{tr}_R\bigl(B(A)_{e_1} \cdots B(A)_{e_r}\bigr) \tag{142.109}$$

is the Wilson loop of the concatenated fine path and is gauge invariant.

This construction proves only the exact finite-regulator covariance and observable pushforward. It does not prove that the logarithm of the blocked measure is a local Wilson action, nor that a continuum local QFT follows. Those are precisely the localization, polymer-decay, reflection-positivity, and reconstruction estimates required in the rest of Definition 142.26.

The companion script `calculation-checks/lattice_gauge_blocking_checks.py` verifies the finite group algebra behind (142.108) and (142.109) for an S_3 plaquette example, including invariance of the blocked pushforward weights under coarse gauge transformations.

142.15 Gauge-Blocking Locality, Positivity, And Reconstruction Estimates

The path-blocking construction above is an exact gauge-covariance statement. It is not yet a Wilsonian RG theorem for a continuum gauge theory. A continuum claim needs three further pieces of analysis: locality of the blocked measure in a declared norm, reflection positivity or a stated replacement for it, and reconstruction estimates for gauge-invariant observables. This section records the estimates required for a theorem-level four-dimensional Yang–Mills statement.

Let Γ_k be a sequence of Euclidean lattice complexes with lattice spacing $a_k \to 0$. For a connected set X of coarse plaquettes, links, or cells, let $\operatorname{diam}_k(X)$ be its diameter in physical units and let $\operatorname{dist}_k(X, Y)$ be the physical distance between two cell sets. Let $\operatorname{supp} F$ be the smallest connected cell set on which a cylinder observable F depends. The norm $\|\cdot\|_\infty$ is the supremum norm on the corresponding compact link-configuration space.

Definition 142.28 (Gauge-blocking continuum-control estimates)**.** A continuum-control estimate package for a gauge-compatible blocking sequence consists of the finite path-blocking data of Construction 142.27, together with the following estimates.

1. A gauge-invariant polymer representation of the blocked measure at scale k:

$$\frac{\mathrm{d}\mu_k^{\mathrm{blk}}}{\mathrm{d}\nu_k}(A) = Z_k^{-1} \exp\left(- \sum_{X \subset \Gamma_k} K_{k,X}(A_X) \right), \tag{142.110}$$

   where ν_k is a reflection-positive local reference measure, $K_{k,X}$ is gauge invariant and depends only on links meeting X, and the sum is over connected polymers X.

2. A quasi-locality estimate: for some $\zeta > 0$,

$$\|K_k\|_\zeta = \sup_{e \in E(\Gamma_k)} \sum_{X \ni e} e^{\zeta \operatorname{diam}_k(X)} \|K_{k,X}\|_\infty \le \varepsilon_k, \qquad \varepsilon_k \longrightarrow 0 \tag{142.111}$$

   after subtracting the finitely many local coordinates retained in the Wilsonian chart. In addition, for every bounded gauge-invariant cylinder observable F, the cluster expansion or polymer estimate supplies truncated expectations $\langle F \rangle_{k,R}$, obtained by keeping only connected polymer clusters contained in the R-neighborhood of $\operatorname{supp} F$, and constants C_F such that

$$|\langle F \rangle_k - \langle F \rangle_{k,R}| \le C_F e^{-\zeta R} \|K_k\|_\zeta + \chi_{F,k}(R), \qquad \lim_{R\to\infty} \sup_k \chi_{F,k}(R) = 0. \tag{142.112}$$

   This cluster-tail estimate is part of the continuum-control package, not a consequence of gauge covariance alone.

3. Reflection-positivity control. Let $\mathcal{A}_{k,+}^{\mathrm{inv}}$ be the algebra of gauge-invariant cylinder observables supported in the positive-time half-lattice, and let Θ be time reflection with orientation reversal on links crossing reflected edges. For every finite list $F_1, \ldots, F_m \in \mathcal{A}_{k,+}^{\mathrm{inv}}$ and coefficients $c_i \in \mathbb{C}$,

$$\sum_{i,j=1}^{m} \bar{c}_i c_j \int \Theta F_i(A)\, F_j(A)\, \mathrm{d}\mu_k^{\mathrm{blk}}(A) \ge -\rho_k(F, c), \tag{142.113}$$

   with $\rho_k(F, c) \to 0$ for each fixed finite family. Exact reflection positivity is the special case $\rho_k = 0$.

4. Gauge-invariant reconstruction maps for a declared class of observables $\mathcal{W}$, such as finite Wilson-loop products, smeared small-loop field-strength representatives, or matter bilinears with their Wilson-line connectors. For each $O \in \mathcal{W}$, the reconstructed finite-scale observable $\mathrm{Rec}_{O,k}$ takes values in a locally convex distribution or cylinder-observable topology and satisfies a bound of the form

$$p(\mathrm{Rec}_{O,k} - \mathrm{Rec}_{O,*}) \leq C_{O,p}\varepsilon_k + \eta_{O,p,k}, \qquad \eta_{O,p,k} \longrightarrow 0, \tag{142.114}$$

for every seminorm p used in the continuum statement.

The first item is not a harmless rewriting. The logarithm of a blocked gauge-invariant measure can contain arbitrarily large Wilson loops. The locality estimate (142.111) is the assertion that those large connected contributions are summable in a norm strong enough to define a Wilsonian chart. In gauge theory this estimate is one of the main constructive burdens.

Consequences of the estimates. Once Definition 142.28 is proved in a model, every $O \in \mathcal{W}$ has the declared continuum limit by (142.114). Gauge invariance passes to the limit because it holds at finite regulator by path-blocking equivariance and because the reconstruction topology tests gauge-invariant cylinder data. The estimate (142.112) is the quasi-locality statement: distant connected polymer clusters have a uniformly vanishing effect on each bounded local gauge-invariant observable after the retained local Wilsonian coordinates have been removed.

Reflection positivity is also a closed-cone statement once the finite quadratic forms have been identified. For $F_1, \ldots, F_m \in \mathcal{A}^{\mathrm{inv}}_{k,+}$ and $c_i \in \mathbb{C}$, write

$$Q_k(F, c) = \sum_{i,j} \bar{c}_i c_j \int \Theta F_i \, F_j \, \mathrm{d}\mu_k^{\mathrm{blk}}.$$

The reconstruction estimate gives convergence of the finitely many correlators entering $Q_k(F, c)$, while (142.113) gives $Q_k(F, c) \geq -\rho_k(F, c)$. Taking $k \to \infty$ gives the limiting positive-time quadratic form $Q_*(F, c) \geq 0$. This is why the estimate package must include the reflection-positive observable algebra and the defect ρ_k; it is not enough to know that the blocked links transform covariantly.

Gauge-invariant source windows. The source-window layer in a gauge RG theorem is part of the datum, because the source variables select which continuum distributions are being reconstructed. In the lattice form of Definition 142.26, let

$$E = \mathrm{span}_{\mathbb{C}}\{O_1, \ldots, O_N\}$$

be a finite-dimensional space of gauge-invariant cylinder observables on $\mathcal{A}_k$. Typical basis elements are closed Wilson loops, products of small-loop field-strength representatives, or matter bilinears whose color indices are contracted by a declared Wilson-line connector. The path, representation, endpoint contraction, and smearing data are part of the observable label. For $z = (z_1, \ldots, z_N)$ in a fixed polydisc P_ρ, define

$$W^{\mathrm{inv}}_{k,E}(z) = \log \int_{\mathcal{A}_k} \exp\left(\sum_{a=1}^{N} z_a O_a(A) \right) \mathrm{d}\mu_k(A), \tag{142.115}$$

whenever the integral is nonzero and the logarithm branch has been fixed at $z = 0$. Since each O_a is invariant under $\mathcal{G}_k$, the source-deformed density in (142.115) is invariant pointwise and therefore descends to the orbit space or, equivalently, to the invariant observable algebra. An admissible source for a colored transport operator includes the endpoint color contractions, Wilson-line geometry, or asymptotic dressing that makes the observable gauge invariant; the uncontracted open link belongs to a gauge-fixed coordinate chart.

The endpoint contraction can be checked at the same finite-regulator level. Let $R : G \to U(V)$ be a finite-dimensional unitary representation and let $P : x \to y$ be an oriented lattice path with holonomy $U_P(A)$. If $\psi_y \in V$ and $\bar{\psi}_x \in V^*$ transform as

$$\psi_y \mapsto R(g_y)\psi_y, \qquad \bar{\psi}_x \mapsto \bar{\psi}_x R(g_x)^{-1},$$

then the transported bilinear

$$O_{P,R}(A, \bar{\psi}_x, \psi_y) = \bar{\psi}_x \, R(U_P(A)) \, \psi_y \tag{142.116}$$

is invariant, because

$$U_P(A^g) = g_x U_P(A) g_y^{-1}.$$

The same two endpoint vectors without the transporter transform by $\bar{\psi}_x R(g_x)^{-1} R(g_y)\psi_y$, so they do not define a source on the gauge-orbit space when $x \neq y$. The path P is also part of the observable label: replacing P by another path with the same endpoints inserts the Wilson loop around the closed contour $P'\bar{P}$. A source-window RG estimate must therefore specify the transporter geometry and its renormalized path class, or else prove a reconstruction estimate that controls the difference between the chosen paths.

Path-deformation ledger for transported sources. The preceding path dependence has an exact finite-regulator algebra. Let $P, P' : x \to y$ be two oriented paths and let $\bar{P} : y \to x$ be the reverse of P. Put

$$C_{P',P} := P'\bar{P}, \qquad U_{C_{P',P}}(A) = U_{P'}(A) U_P(A)^{-1}.$$

Then

$$U_{P'}(A) = U_{C_{P',P}}(A) \, U_P(A), \tag{142.117}$$

and the two transported sources differ by the endpoint-local loop insertion

$$O_{P',R} - O_{P,R} = \bar{\psi}_x \left(R(U_{C_{P',P}}) - \mathbf{1} \right) R(U_P)\psi_y. \tag{142.118}$$

The closed-loop holonomy in (142.118) transforms by conjugation at x, and the endpoint vector $\bar{\psi}_x$ transforms by the dual representation, so the whole difference is still gauge invariant. If $C_{P',P}$ is a small plaquette loop in a continuum scaling window, then $R(U_{C_{P',P}}) - \mathbf{1}$ is the finite-regulator representative of a local curvature insertion plus higher local Wilson-loop coordinates. Thus identifying P and P' in a continuum reconstruction is a theorem only after the source-window estimate supplies, in the declared seminorms, a bound of the form

$$p\Big(\mathrm{Rec}_{O_{P',R},k} - \mathrm{Rec}_{O_{P,R},k}\Big) \leq C_{P,P',p} \, \varepsilon_{C_{P',P},k} + \eta_{P,P',p,k}, \tag{142.119}$$

where $\varepsilon_{C_{P',P},k}$ is the local small-loop or curvature coordinate controlled by the gauge-invariant Wilsonian chart and $\eta_{P,P',p,k} \to 0$. Without such a bound, different connector paths are different source labels, not alternate notations for the same observable.

In the BV form the same statement is slightly sharper. A finite source window is a holomorphic family $S_k(z)$ with $S_k(0) = S_k$ satisfying the scale-k quantum master equation and equipped with its composite-operator source coordinates. The first source derivatives $\partial_{z_a} S_k(0)$ are quantum observables, meaning $Q_{S_k}\partial_{z_a} S_k(0) = 0$. The higher source derivatives of $S_k(z)$ are the contact and composite-operator coordinates needed for the source family itself to remain in the BV chart. Thus the distinction between regularizing the path-integral measure and regularizing composite operators is built into the RG source datum rather than added after correlators have been computed.

The gauge source window is useful only if it obeys the same analytic estimates as the scalar source windows above, now in the invariant or BV cohomological chart. A theorem-facing estimate has the form

$$W^{\mathrm{inv}}_{k,E}(z) = C^{\mathrm{inv}}_{k,E}(z) + \mathfrak{W}^{\mathrm{inv}}_E\big(g^{\mathrm{inv}}_{k,E}(z), K^{\mathrm{inv}}_{k,E}(z)\big) + r^{\mathrm{inv}}_{k,E}(z), \tag{142.120}$$

on a fixed polydisc, with all three terms defined in the gauge-invariant lattice chart or in a BV chart satisfying the quantum master equation. If the right side lies in a common Lipschitz neighborhood of the limiting source chart, then

$$\sup_{z\in P_\rho}\big|W^{\mathrm{inv}}_{k,E}(z) - W^{\mathrm{inv}}_{*,E}(z)\big| \le \delta^{\mathrm{inv}}_{C,E} + L^{\mathrm{inv}}_E\big(\delta^{\mathrm{inv}}_{g,E} + \delta^{\mathrm{inv}}_{K,E}\big) + \delta^{\mathrm{inv}}_{r,E}. \tag{142.121}$$

The Cauchy estimates then extract connected correlators of the gauge observables O_a. Equation (142.121) is the place where a finite-regulator gauge computation becomes a reconstruction statement. A finite list of Wilson-loop expectation values, or a beta function in a gauge-fixed coordinate chart, becomes reconstruction input only after the source radius, local coordinates, polymer tail, contact/source coordinates, and thermodynamic boundary tails have been controlled.

The companion script `calculation-checks/lattice_gauge_blocking_checks.py` includes the finite S_3 check behind this paragraph: class functions of the blocked closed loop give source polynomials constant on coarse gauge orbits, while a source attached to a single open link fails the same invariance test. It also checks (142.116) in the permutation representation: the Wilson-line transported endpoint bilinear is invariant under all independent endpoint gauge transformations, the untransported endpoint pairing is not, and changing the connecting path changes the observable for a configuration with nontrivial holonomy. It also verifies the exact loop-insertion identity (142.117)–(142.118) and the finite reconstruction error budget (142.119). The same check file also tests the finite locality-tail arithmetic, reflection positive Gram compression, and the reconstruction error budget in (142.114); these finite checks are not substitutes for the continuum estimates, but they keep the algebraic interfaces of the estimate package explicit.

Common-scale schedule for gauge reconstruction. The estimates above are not independent badges. A gauge-compatible continuum claim needs them on one directed schedule. Let $\mathfrak{A}_{\mathrm{gauge}}$ be a directed family of finite gauge observation windows. A window $\alpha \in \mathfrak{A}_{\mathrm{gauge}}$ consists of a finite set $\mathcal{W}_\alpha$ of gauge-invariant Wilson-loop, small-loop, transported-matter, or BV-cohomology source labels; a finite positive-time test family $\mathcal{F}_\alpha$; a seminorm p_α on the reconstructed observable window; a source polydisc radius $\rho_\alpha > 0$; and, when connector paths are to be identified, a finite set $\mathcal{P}_\alpha$ of path-deformation pairs (P, P'). The window remembers the representation, path, endpoint contraction, smearing, orientation, and contact/source-coordinate convention for every label.

For a scale k, a polymer truncation radius R, and a finite volume Ω, the gauge reconstruction error for α is recorded by the budget

$$\begin{aligned}\mathcal{E}^{\mathrm{gauge}}_{\alpha,k,R,\Omega} &= C_\alpha e^{-\zeta_\alpha R}\|K_k\|_\zeta + \chi_{\alpha,k}(R) + \delta^{\mathrm{vol}}_{\alpha,k,\Omega} + \delta^{\mathrm{src}}_{\alpha,k} \\ &\quad + \eta^{\mathrm{rec}}_{\alpha,k} + \sum_{(P,P')\in\mathcal{P}_\alpha} \big(C_{P,P',\alpha}\varepsilon_{C_{P',P},k} + \eta_{P,P',\alpha,k}\big).\end{aligned} \tag{142.122}$$

The first two terms are the quasi-local polymer tail from (142.112); the third is the finite volume or boundary-tail defect; the fourth is the holomorphic source-window error in (142.121); the fifth is the observable reconstruction defect in (142.114); and the last line is the connector-path identification budget from (142.119). Reflection positivity has its own finite Gram-window defect:

$$\mathcal{R}^{\mathrm{gauge}}_{\alpha,k} = \rho_k(\mathcal{F}_\alpha) + |\mathcal{F}_\alpha|\,\epsilon^{\mathrm{Gram}}_{\alpha,k}, \tag{142.123}$$

where $\epsilon^{\mathrm{Gram}}_{\alpha,k}$ denotes the entrywise error between the regulated and limiting positive-time Gram matrices in that window, or is replaced by the corresponding operator-norm error if that stronger estimate is available.

A theorem-level gauge continuum limit must provide a cofinal schedule for these two budgets. For a nested cofinal sequence $\alpha_1 \preceq \alpha_2 \preceq \cdots$, this means choosing

$$N \longmapsto (k_N, R_N, \Omega_N)$$

so that, for every fixed j,

$$\mathcal{E}^{\mathrm{gauge}}_{\alpha_j,k_N,R_N,\Omega_N} \longrightarrow 0, \qquad \mathcal{R}^{\mathrm{gauge}}_{\alpha_j,k_N} \longrightarrow 0 \qquad (N\to\infty,\ N\ge j). \tag{142.124}$$

Equivalently, after increasing the schedule if necessary, one may demand a diagonal bound such as

$$\mathcal{E}^{\mathrm{gauge}}_{\alpha_j,k_N,R_N,\Omega_N} + \mathcal{R}^{\mathrm{gauge}}_{\alpha_j,k_N} \le 2^{-N}, \qquad j \le N,$$

with constants and radii already fixed by the windows. This is the gauge-theory version of the common-scale requirement in Definition 142.39. Separate statements that the locality tail vanishes for one list of Wilson loops, the source Cauchy estimate holds on another source polydisc, and reflection positivity holds on a third positive-time family do not yet define a projective gauge observable germ. The theorem is the existence of one schedule controlling all windows in the directed family, followed by OS, Wightman, or BV/cohomological reconstruction in the topology declared by those windows.

Warning 142.29 (Finite gauge blocking is not a continuum Yang–Mills proof)**.** Construction 142.27 and the finite S_3 companion calculation prove exact gauge covariance and exact pushforward invariance at finite regulator. They do not by themselves prove the quasi-local polymer bound (142.111), reflection positivity (142.113), or reconstruction (142.114). A theorem that a gauge-blocked Wilsonian trajectory defines a continuum local QFT must supply those estimates in a specific regulator class and observable topology.

Definition 142.30 (RG reconstruction estimate)**.** Let $\mathcal{S}_n = \mathcal{S}((\mathbb{R}^D)^n)$ be the Schwartz space and let $\mathcal{S}'_n$ be its continuous dual. For a bounded set $B \subset \mathcal{S}_n$, write

$$p_B(T) = \sup_{f\in B} |\langle T, f\rangle|, \qquad T \in \mathcal{S}'_n.$$

Let $\mathcal{R} : \mathcal{U} \to \mathcal{B}$ be an RG map with fixed point V_*. An n-point reconstruction estimate consists of distributions $\mathcal{S}_{n,*} \in \mathcal{S}'_n$, maps

$$\mathrm{Rec}_{n,k} : \mathcal{U} \longrightarrow \mathcal{S}'_n, \qquad k \in \mathbb{Z}_{\geq 0}, \tag{142.125}$$

and constants $C_B < \infty$, $0 < \theta < 1$, and $\varepsilon_{B,k} \to 0$, such that every tuned initial point V in the declared stable manifold satisfies

$$p_B(\mathrm{Rec}_{n,k}(V) - \mathcal{S}_{n,*}) \leq C_B\, \theta^k \, \|V - V_*\|_{\mathcal{B}} + \varepsilon_{B,k} \tag{142.126}$$

for all bounded $B \subset \mathcal{S}_n$. The maps $\mathrm{Rec}_{n,k}$ include the lattice spacing, coordinate rescaling, field-strength normalization, source normalization, and possible Wilson-line or BV-observable representative.

Immediate consequences of a reconstruction estimate. Assume Definition 142.30 for every $n \geq 1$ and every bounded set $B \subset \mathcal{S}_n$. This assumption already contains strong-dual convergence of the renormalized n-point functions. Indeed, strong convergence in $\mathcal{S}'_n$ means convergence with respect to the seminorms p_B as B ranges over bounded subsets of $\mathcal{S}_n$, and (142.126) gives

$$p_B(\mathrm{Rec}_{n,k}(V) - \mathcal{S}_{n,*}) \longrightarrow 0$$

because $\theta^k \to 0$ and $\varepsilon_{B,k} \to 0$. Thus the distributional convergence is not an additional theorem beyond the estimate; it is the meaning of the estimate in the chosen topology.

The same distinction matters for properties carried to the limit. For symmetry and Euclidean covariance, let U denote a permutation or Euclidean transformation acting continuously on $\mathcal{S}_n$. If the finite-regulator distributions satisfy $\langle \mathrm{Rec}_{n,k}(V), f\rangle = \langle \mathrm{Rec}_{n,k}(V), Uf\rangle$, then passing to the limit in the two scalar sequences gives the same identity for $\mathcal{S}_{n,*}$.

For reflection positivity, take a finite family of positive-time test functions f_i and coefficients c_i. At finite regulator the corresponding quadratic expression has the form

$$Q_k(c, f) = \sum_{i,j} \bar{c}_i c_j \left\langle \mathrm{Rec}_{m_i+m_j,k}(V), \Theta f_i \otimes f_j \right\rangle \geq 0, \tag{142.127}$$

where Θ is Euclidean time reflection. Each term converges by the distributional convergence encoded in (142.126), hence $Q_k(c, f) \to Q_*(c, f)$. The cone $[0, \infty) \subset \mathbb{R}$ is closed, so $Q_*(c, f) \geq 0$. Thus the limiting Schwinger functions inherit reflection positivity in the declared test-function domain whenever the finite-regulator positivity statement is formulated as a finite positive-semidefinite quadratic form continuous in the seminorms p_B. This closed-cone passage does not provide the missing model-specific estimates; it only records what follows after those estimates have already been proved.

142.16 Ordinary Short-Range Scalar Reconstruction

The scalar benchmark in the previous sections is hierarchical. An ordinary short-range scalar lattice model has a different microscopic object: fields on $a_k\mathbb{Z}^D$, finite-range interactions, and block-spin maps whose range is geometric in the lattice. The purpose of this section is to record the reconstruction structure for that ordinary setting. It is the structure a proof of a short-range Wilson–Fisher fixed point would have to supply; it is not replaced by a hierarchical or tensor-network theorem unless a comparison map with reconstruction estimates is also proved.

Definition 142.31 (Short-range scalar block-spin structure)**.** Fix an integer scale factor $L \geq 2$, spacetime dimension D, and lattice spacings $a_k = L^{-k}a_0$. A short-range scalar block-spin structure consists of the following objects.

1. Periodic lattice regions $\Lambda_k \subset a_k\mathbb{Z}^D$, scalar configuration spaces $\mathbb{R}^{\Lambda_k}$, and finite Gibbs measures

$$\mathrm{d}\mu_k(\phi) = Z_k^{-1} \exp\left(- \sum_{\substack{X\subset\Lambda_k \\ \mathrm{diam}(X)\leq Ra_k}} U_{k,X}(\phi_X) \right) \prod_{x\in\Lambda_k} \mathrm{d}\phi_x. \tag{142.128}$$

   Here $R < \infty$ is fixed before the continuum limit, and the potentials are assumed measurable with enough coercivity to make all polynomial moments used below finite.

2. A block kernel $q : \{0, \ldots, L-1\}^D \to \mathbb{R}$ satisfying

$$\sum_{r\in\{0,\ldots,L-1\}^D} q(r) = 1, \tag{142.129}$$

   and a field scaling exponent Δ_ϕ. The blocked field on Λ_{k-1} is

$$(B_k\phi)(y) = L^{\Delta_\phi} \sum_{r\in\{0,\ldots,L-1\}^D} q(r)\, \phi(y + a_kr). \tag{142.130}$$

   Periodic representatives are used when $y + a_kr$ crosses the boundary. The normalization makes a constant microscopic field transform by the declared field-scaling factor L^{Δ_ϕ}.

3. A localization map which writes the pushforward $(B_k)_\#\mu_k$ in a Banach RG chart

$$(g_k, K_k) \in E_{\mathrm{loc}} \oplus \mathcal{K}.$$

   The finite-dimensional coordinates g_k record relevant and marginal local terms. The polymer activity K_k records the infinite-dimensional irrelevant tail in a norm controlling both spatial decay and large-field behavior.

The structure is short-range only before the block step. After integrating out fluctuations, the effective interaction is generally not finite range; the polymer norm and its contraction estimates are the mathematical replacement for the microscopic finite-range property.

Construction 142.32 (Block-spin cumulants)**.** Let $f \in C_c^\infty(\mathbb{R}^D)$. The normalized lattice field at spacing a_k is the random variable

$$\Phi_k(f) = a_k^D\, a_k^{-\Delta_\phi} \sum_{x\in\Lambda_k} f(x)\phi_x. \tag{142.131}$$

For labels $f_1, \ldots, f_n$, define the connected reconstructed distribution by the joint cumulant

$$\langle S_{n,k}, f_1 \otimes \cdots \otimes f_n\rangle = \kappa_{\mu_k}\big(\Phi_k(f_1), \ldots, \Phi_k(f_n)\big). \tag{142.132}$$

The cumulant is computed from

$$\log \mathbb{E}_{\mu_k} \exp\left(\sum_{i=1}^n t_i\Phi_k(f_i)\right)$$

by taking $\partial_{t_1} \cdots \partial_{t_n}$ at the origin. The normalization in (142.131) is fixed by the source pairing: if $x = Ly$, then a source coupled to a field of dimension Δ_ϕ scales with exponent $D - \Delta_\phi$, as in (142.196).

The finite-regulator identity behind this normalization is the adjoint source-blocking relation. For a coarse test function f, define the fine-lattice pullback $B_k^\dagger f$ on one block by

$$(B_k^\dagger f)(y + a_k r) = L^D q(r) f(y), \qquad r \in \{0, \dots, L-1\}^D. \tag{142.133}$$

Then, with $a_{k-1} = La_k$,

$$a_{k-1}^{D-\Delta_\phi} \sum_{y \in \Lambda_{k-1}} f(y)(B_k\phi)(y) = a_k^{D-\Delta_\phi} \sum_{y \in \Lambda_{k-1}} \sum_{r \in \{0,\dots,L-1\}^D} (B_k^\dagger f)(y + a_k r)\phi_{y+a_k r}. \tag{142.134}$$

Thus the source normalization, field-strength normalization, and block-spin normalization are one datum, not three independent conventions. A source-window RG estimate for Φ_k must use the same $B_k^\dagger$ that is adjoint to the blocking map appearing in the measure pushforward. If one changes the field normalization while keeping the same symbol for the source window, the reconstructed Schwinger distributions have been rescaled; the comparison then belongs to the normalization group in Definition 142.45, not to equality of un-normalized observable germs.

The ordinary short-range scalar reconstruction problem is to prove, for each n, convergence of $S_{n,k}$ in $\mathcal{S}'((\mathbb{R}^D)^n)$ or in a declared local distribution topology. A Wilsonian RG proof supplies this convergence through estimates in the RG chart:

$$p_B(S_{n,k} - S_{n,*}) \le C_{n,B}\big(\|g_k - g_*\|_{E_{\rm loc}} + \|K_k - K_*\|_{\mathcal{K}}\big) + \epsilon_{n,B,k}, \tag{142.135}$$

where $B \subset \mathcal{S}((\mathbb{R}^D)^n)$ is bounded and $\epsilon_{n,B,k} \to 0$ is the discretization and finite-volume error. If the RG trajectory is tuned so that

$$\|g_k - g_*\|_{E_{\rm loc}} + \|K_k - K_*\|_{\mathcal{K}} \le C_0\theta^k, \qquad 0 < \theta < 1, \tag{142.136}$$

then (142.135) is precisely a reconstruction estimate of the form (142.126). The point is that (142.136) is the hard RG estimate, not the vocabulary of fixed points or universality.

Finite source windows to cumulant distributions. The cumulant notation in (142.132) is useful only after the source-extension estimates have been connected to actual test-function windows. Fix a finite-dimensional test space

$$E = \operatorname{span}\{f_1, \dots, f_m\} \subset C_c^\infty(\mathbb{R}^D)$$

and write $J_z = \sum_{a=1}^m z_a f_a$. At scale k, define the finite-window connected source functional

$$W_{k,E}(z) = \log \mathbb{E}_{\mu_k} \exp\{\Phi_k(J_z)\}, \qquad z \in \mathbb{C}^m, \tag{142.137}$$

whenever the expectation is finite and holomorphic in the chosen polydisc. If

$$\beta = (\beta_1, \dots, \beta_m), \qquad |\beta| = \beta_1 + \cdots + \beta_m,$$

then

$$\partial_z^\beta W_{k,E}(0) = \kappa_{\mu_k}\big(\underbrace{\Phi_k(f_1), \dots, \Phi_k(f_1)}_{\beta_1 \text{ times}}, \dots, \underbrace{\Phi_k(f_m), \dots, \Phi_k(f_m)}_{\beta_m \text{ times}}\big). \tag{142.138}$$

This identity is just the definition of joint cumulants through the logarithm of the moment-generating function, but it fixes the exact derivative whose convergence is needed for the Schwinger distribution window.

Suppose now that $W_{*,E}$ is holomorphic on the same closed polydisc $P_\rho = \{|z_a| \le \rho_a\}$ and that

$$\sup_{z\in P_\rho} |W_{k,E}(z) - W_{*,E}(z)| \le \epsilon_{E,k}, \qquad \epsilon_{E,k} \longrightarrow 0. \tag{142.139}$$

The multidimensional Cauchy formula gives, for every multi-index β,

$$\left|\partial_z^\beta W_{k,E}(0) - \partial_z^\beta W_{*,E}(0)\right| \le \frac{\beta!}{\rho^\beta}\,\epsilon_{E,k}, \qquad \rho^\beta = \prod_a \rho_a^{\beta_a}. \tag{142.140}$$

Thus finite-source-window convergence in a complex neighborhood gives quantitative convergence of every cumulant whose test functions lie in E. If $E \subset F$, compatibility of the source windows means

$$W_{k,F}(z_E, 0) = W_{k,E}(z_E), \qquad W_{*,F}(z_E, 0) = W_{*,E}(z_E),$$

so the cumulants extracted from larger windows restrict to the smaller ones.

The passage from these finite windows to a distribution is a further continuity statement, not a formal consequence of (142.140). For each n, the RG construction must also provide an integer r_n and a constant C_n, independent of the finite test space E and of k after the stated scale threshold. Applied to any ordered n-tuple $(h_1, \ldots, h_n) \subset E$, with source coordinates $z_1, \ldots, z_n$ chosen in those directions, the required bound is

$$|\partial_{z_1} \cdots \partial_{z_n} W_{k,E}(0)| \le C_n\, q_{r_n}(h_1 \otimes \cdots \otimes h_n) \tag{142.141}$$

for the declared Schwartz seminorm q_{r_n} on $\mathcal{S}((\mathbb{R}^D)^n)$, and the same bound for the limiting window. Compatibility plus (142.141) is precisely the input of Lemma 142.48. It produces the limiting tempered distribution $S_{n,*}$; the Cauchy estimate (142.140) supplies the finite-window convergence rate. Local source counterterms are handled as additional polynomial source coordinates whose derivatives are distributions supported on collision diagonals. They are part of the source window, not an after-the-fact modification of separated cumulants.

Source majorants for the OS-II growth input. The preceding paragraph controls fixed cumulant windows. The corrected OS-II reconstruction theorem needs a uniform many-insertion growth estimate for the Schwinger hierarchy itself, not only convergence at each fixed n. The corresponding Wilsonian estimate is most naturally stated for the moment source functional

$$Z_{k,E}(z) = \mathbb{E}_{\mu_k} \exp\{\Phi_k(J_z)\}, \qquad W_{k,E}(z) = \log Z_{k,E}(z), \tag{142.142}$$

on the same finite test space E. Let

$$N_n = N_0 + C_{\mathrm{OS}} n$$

be the seminorm order allowed in the OS-II linear-growth condition, and let q_{1,N_n} be the corresponding one-variable Schwartz seminorm. A source-window estimate strong enough for OS-II has the following form. For every ordered n-tuple $h_1, \ldots, h_n$ in the test space, the functional $Z_{k,E}$, $E = \mathrm{span}\{h_1, \ldots, h_n\}$, is holomorphic on the polydisc

$$|z_i| \le \frac{\rho}{q_{1,N_n}(h_i)}, \qquad i = 1, \ldots, n,$$

with constants $\rho, A, B, \gamma > 0$, independent of k, n, and of the chosen tuple, such that

$$\sup_{P_\rho(h_1,\dots,h_n)} |Z_{k,E}(z)| \le A\, B^n\, (n!)^\gamma. \tag{142.143}$$

Cauchy's formula in the n independent source variables then gives

$$|S_{n,k}(h_1 \otimes \cdots \otimes h_n)| \le A \left(\frac{B}{\rho}\right)^n (n!)^\gamma \prod_{i=1}^n q_{1,N_n}(h_i). \tag{142.144}$$

For a general zero-diagonal test function F, let $\pi_{n,N_n}(F)$ be the projective tensor seminorm obtained by decomposing F into finite sums of simple tensors and taking the infimum of the corresponding products of one-variable q_{1,N_n}-seminorms. Linearity gives

$$|S_{n,k}(F)| \le A \left(\frac{B}{\rho}\right)^n (n!)^\gamma \pi_{n,N_n}(F). \tag{142.145}$$

Thus the OS-II bound (12.4) follows from the source majorant whenever the chosen Schwartz seminorm family satisfies a comparison estimate

$$\pi_{n,N_n}(F) \le C_\pi^n q_{n,N_0'+C_{\mathrm{OS}}'n}(F)$$

on the zero-diagonal test class. The constants in the OS-II growth condition are then $B_{\mathrm{OS}} = C_\pi B/\rho$, with the same factorial exponent γ. This is a stricter demand than fixed-n source convergence: if the available source radius shrinks with n, if the supremum in (142.143) has uncontrolled combinatorial growth, or if only connected cumulants are bounded without a summable partition estimate, the OS-II analytic input has not been obtained.

From connected cumulants to moments. The source functional most directly produced by a polymer or cluster RG construction is often $W_{k,E} = \log Z_{k,E}$, hence connected cumulants. The corrected OS-II growth hypothesis is a bound on Schwinger functions themselves. The passage from one to the other is the moment–cumulant partition sum, and its constants must be controlled uniformly in the number of insertions.

Lemma 142.33 (Cumulant partition bound for OS-II moments)**.** *Fix $n \ge 1$, test functions $h_1, \dots, h_n$, and the seminorm q_{1,N_n} used in the preceding paragraph. For each nonempty $B \subset \{1, \dots, n\}$, let $\kappa_B(h_B)$ be the connected cumulant with insertion set B, where $h_B = (h_i)_{i\in B}$. Assume that constants $A_{\mathrm{c}}, C_{\mathrm{c}}, \gamma_{\mathrm{c}}$, independent of n and of the chosen test functions, satisfy*

$$|\kappa_B(h_B)| \le A_{\mathrm{c}}\, C_{\mathrm{c}}^{|B|}\, (|B|!)^{\gamma_{\mathrm{c}}} \prod_{i\in B} q_{1,N_n}(h_i) \tag{142.146}$$

for every nonempty B. Let

$$M_n(h_1, \dots, h_n) = \sum_{\pi\in\Pi_n} \prod_{B\in\pi} \kappa_B(h_B)$$

be the corresponding Schwinger moment, with Π_n the set of set partitions of $\{1, \dots, n\}$. Then

$$|M_n(h_1, \dots, h_n)| \le \big(2C_{\mathrm{c}} \max(1, A_{\mathrm{c}})\big)^n (n!)^{\gamma_{\mathrm{c}}+1} \prod_{i=1}^n q_{1,N_n}(h_i). \tag{142.147}$$

Proof. The moment–cumulant identity is the finite identity obtained by expanding $\exp W$. Taking absolute values and applying (142.146) gives

$$|M_n(h_1,\dots,h_n)| \le C_{\mathrm{c}}^n \prod_{i=1}^n q_{1,N_n}(h_i) \sum_{\pi\in\Pi_n} A_{\mathrm{c}}^{|\pi|} \prod_{B\in\pi} (|B|!)^{\gamma_{\mathrm{c}}}.$$

For every partition π, one has $\prod_{B\in\pi} |B|! \le n!$, since the left hand side counts permutations internal to the blocks and is bounded by all permutations of the n labels. Also $A_{\mathrm{c}}^{|\pi|} \le \max(1, A_{\mathrm{c}})^n$. Finally, $|\Pi_n| \le 2^{n-1}n!$: every partition can be obtained by choosing a permutation of the n labels and then choosing cuts between consecutive labels, so the stated number is a surjective overcount. Combining these three bounds gives (142.147). $\square$

Consequently a connected-cumulant RG theorem can feed the corrected OS-II hypothesis only after the constants in (142.146) are obtained uniformly in the insertion number and with the declared seminorm order $N_n = N_0 + C_{\mathrm{OS}}n$. The estimate loses one factorial power when no sharper partition control is available. This loss is part of the theorem statement, not a matter of notation: if the connected bound carries constants depending on the number of clusters, on the source-window dimension, or on an n-dependent radius, then (142.147) does not supply the OS-II growth input.

Polymer derivative norm behind the connected input. In a polymer RG proof the connected estimate (142.146) should be produced before the moment–cumulant partition sum is used. Let $h_1,\dots,h_m$ be local test directions in a finite source window, and let K_i be the block support of the discretized source h_i at scale k. For $\emptyset \ne B \subset \{1,\dots,m\}$, write

$$D_B = \prod_{i\in B} \frac{\partial}{\partial z_i}\bigg|_{z=0}, \qquad K_B = \bigcup_{i\in B} K_i.$$

If the connected source functional has a cluster expansion

$$W_{k,E}(z) = \sum_{\mathcal{C}} w_{\mathcal{C}}(z),$$

then the model-specific source theorem must control the derivatives $D_B w_{\mathcal{C}}(0)$ as part of the source extension of the polymer activity bounds. A useful form of the required estimate is the anchored derivative norm

$$\mathcal{N}_{k,B}(h_B) = \sum_{\mathcal{C}\in\mathfrak{C}_k(B)} e^{\zeta\, \mathrm{diam}_k(\mathrm{supp}\,\mathcal{C}\cup K_B)} \left| D_B w_{\mathcal{C}}(0)\right|, \tag{142.148}$$

where $\mathfrak{C}_k(B)$ is the family of connected clusters whose source dependence can attach to every K_i, $i\in B$. For a strictly local source coupling this means that the support of the cluster meets each K_i; for a block-smeared source it means the corresponding enlarged block support. Source-local contact coordinates are included separately in the local source chart, so that (142.148) is the estimate for the declared nonlocal connected tail.

The polymer input strong enough for the connected OS-II bound is

$$\mathcal{N}_{k,B}(h_B) \le A_{\mathrm{p}}\, C_{\mathrm{p}}^{|B|}\, (|B|!)^{\gamma_{\mathrm{p}}} \prod_{i\in B} q_{1,N_{|B|}}(h_i), \tag{142.149}$$

with $A_{\mathrm{p}}, C_{\mathrm{p}}, \gamma_{\mathrm{p}}, \zeta$ independent of k, of the finite window E, and of $|B|$ except through the displayed factors. The connected cumulant is

$$\kappa_B(h_B) = \sum_{\mathcal{C}\in\mathfrak{C}_k(B)} D_B w_{\mathcal{C}}(0),$$

after the contact-source coordinates assigned to B have been separated. Since the exponential weight in (142.148) is at least one, (142.149) gives (142.146) with

$$A_{\mathrm{c}} = A_{\mathrm{p}}, \qquad C_{\mathrm{c}} = C_{\mathrm{p}}, \qquad \gamma_{\mathrm{c}} = \gamma_{\mathrm{p}}.$$

Keeping the exponential weight also records separated-cluster decay: if

$$\tau_k(K_B) = \inf\{\mathrm{diam}_k(Y) : Y \text{ is a connected block set meeting every } K_i,\ i \in B\},$$

then

$$|\kappa_B(h_B)| \le e^{-\zeta\tau_k(K_B)} A_{\mathrm{p}} C_{\mathrm{p}}^{|B|} (|B|!)^{\gamma_{\mathrm{p}}} \prod_{i\in B} q_{1,N_{|B|}}(h_i).$$

Thus the load-bearing model estimate is the uniform derivative cluster norm (142.149). A cluster expansion whose constants depend on the dimension of the source window, on an uncontrolled choice of source radius, or on the number of possible source attachments has not supplied the connected input needed by the corrected OS-II reconstruction theorem.

RG-chart control of a holomorphic source window. The hypothesis (142.139) is not supplied by knowing that the unsourced local coordinates converge. A Wilsonian proof must control the source-extended chart itself. For a fixed finite test space E and a fixed closed polydisc P_ρ, the source extension of the RG chart should give a decomposition of the connected source functional of the form

$$W_{k,E}(z) = C_{k,E}(z) + \mathfrak{W}_E\big(g_{k,E}(z), K_{k,E}(z)\big) + r_{k,E}(z), \qquad z \in P_\rho. \tag{142.150}$$

Here $C_{k,E}$ is the local normalizing coordinate, including source dependent vacuum and contact coordinates, $g_{k,E}$ are the retained local coordinates, $K_{k,E}$ is the source-decorated polymer tail, and $r_{k,E}$ is the declared sampling, boundary, and finite-step remainder. The map $\mathfrak{W}_E$ is the finite-window evaluation map of the RG chart. It is not assumed to be a universal function independent of the chosen chart; it is part of the reconstruction datum for this finite source window.

Suppose that all chart points occurring in (142.150) lie in a common tube $\mathcal{T}_E$ on which $\mathfrak{W}_E$ obeys

$$\big|\mathfrak{W}_E(g,K) - \mathfrak{W}_E(g',K')\big| \le L_E\big(\|g-g'\|_{E_{\mathrm{loc}}} + \|K-K'\|_{\mathcal{K}_E}\big). \tag{142.151}$$

If limiting source-chart data $(C_{*,E}, g_{*,E}, K_{*,E}, r_{*,E})$ are holomorphic on the same polydisc and

$$\delta^C_{E,k} := \sup_{P_\rho} |C_{k,E} - C_{*,E}|, \tag{142.152}$$

$$\delta^g_{E,k} := \sup_{P_\rho} \|g_{k,E} - g_{*,E}\|_{E_{\mathrm{loc}}}, \tag{142.153}$$

$$\delta^K_{E,k} := \sup_{P_\rho} \|K_{k,E} - K_{*,E}\|_{\mathcal{K}_E}, \tag{142.154}$$

$$\delta^r_{E,k} := \sup_{P_\rho} |r_{k,E} - r_{*,E}| \tag{142.155}$$

all tend to zero, then

$$\sup_{z\in P_\rho} |W_{k,E}(z) - W_{*,E}(z)| \le \delta^C_{E,k} + L_E(\delta^g_{E,k} + \delta^K_{E,k}) + \delta^r_{E,k}. \tag{142.156}$$

Inserting this bound into (142.140) gives the cumulant convergence rate for the finite window. The fixed polydisc is load-bearing: a bound on P_{ρ_k} with $\rho_k \downarrow 0$ does not give a uniform Cauchy estimate for derivatives. Thus source-radius control, local normalizing coordinates, source-decorated irrelevant tails, and finite-step remainders are all part of the Wilsonian reconstruction theorem; they cannot be recovered from an unsourced fixed-point statement.

Source-stable trajectory estimate. The source-extended chart also has its own stable and unstable directions. This is where local contact terms enter the RG theorem rather than being added after the continuum limit. For a finite test space E, write

$$g_{k,E}(z) - g_{*,E}(z) = u_{k,E}(z) + s_{k,E}(z)$$

according to the unstable/retained and stable/irrelevant splitting of the source-local coordinate space. The unstable part includes ordinary relevant couplings and any source-local or contact coordinates whose engineering or anomalous exponent makes them non-decaying in the source window. For a fixed polydisc P_ρ, a finite-depth RG estimate has the form

$$\sup_{P_\rho} \|s_{k+1,E}\| \le \theta_E \sup_{P_\rho} \|s_{k,E}\| + \varepsilon^s_{k,E}, \qquad 0 < \theta_E < 1, \tag{142.157}$$

$$\sup_{P_\rho} \|u_{k+1,E}\| \le \Lambda_E \sup_{P_\rho} \|u_{k,E}\| + \varepsilon^u_{k,E}, \qquad \Lambda_E > 1. \tag{142.158}$$

The first recursion gives

$$\sup_{P_\rho} \|s_{N,E}\| \le \theta_E^N \sup_{P_\rho} \|s_{0,E}\| + \sum_{j=0}^{N-1} \theta_E^{N-1-j} \varepsilon^s_{j,E}. \tag{142.159}$$

The second recursion gives the finite-depth amplification bound

$$\sup_{P_\rho} \|u_{N,E}\| \le \Lambda_E^N \sup_{P_\rho} \|u_{0,E}\| + \sum_{j=0}^{N-1} \Lambda_E^{N-1-j} \varepsilon^u_{j,E}. \tag{142.160}$$

Thus a source-local coordinate whose unstable component is not tuned to order Λ_E^{-N}, up to the accumulated source-local defects, can spoil the finite-window limit even if the unsourced scalar trajectory is on its critical surface.

Combining (142.159)–(142.160) with (142.156) gives the actual holomorphic source-window error bound

$$\epsilon^{\mathrm{src}}_{E,N} \le \delta^C_{E,N} + L_E\left(\sup_{P_\rho} \|u_{N,E}\| + \sup_{P_\rho} \|s_{N,E}\| + \delta^K_{E,N} \right) + \delta^r_{E,N}. \tag{142.161}$$

This is the source-window version of stable-manifold tuning. The source-local counterterms and collision-diagonal coordinates must either be retained and tuned in $u_{k,E}$, shown to contract in $s_{k,E}$, or bounded in the explicit remainder. If they are simply omitted from the chart, the Wilsonian RG estimate has not yet produced the Schwinger distribution window used in OS reconstruction.

Finite volume to infinite volume for a source window. The thermodynamic step at fixed lattice spacing is another estimate, not a notation change. At fixed lattice spacing a_k, let $\Omega \subset a_k\mathbb{Z}^D$ be a finite connected region and write $W_{k,\Omega,E}(z)$ for the connected source functional (142.137) computed with the finite-volume Gibbs measure on Ω and the declared local boundary convention. Let

$$K_{E,k} = \Omega \cap \bigcup_{a=1}^{m} \operatorname{supp} f_a$$

be the lattice source support, after the continuum test functions are restricted to $a_k\mathbb{Z}^D$, and let

$$R_\Omega = \operatorname{dist}_k(K_{E,k}, a_k\mathbb{Z}^D \setminus \Omega)$$

be the lattice distance from the source support to the boundary of Ω, in physical units or in lattice units according to the declared cluster norm. Suppose the source-window cluster expansion gives, on a fixed closed polydisc P_ρ, the boundary-tail estimate

$$\sup_{z\in P_\rho} |W_{k,\Omega',E}(z) - W_{k,\Omega,E}(z)| \le A_E\, \Xi(R_\Omega) \tag{142.162}$$

whenever $\Omega \subset \Omega'$ and both regions contain $K_{E,k}$, where $\Xi(R) \to 0$ as $R \to \infty$. In a finite-range or exponentially clustering polymer estimate one expects $\Xi(R) \le C\exp(-\zeta R)$; in a critical construction the available Ξ is whatever the RG polymer norm actually proves. Equation (142.162) is the statement that clusters touching the source window and clusters that feel the boundary have a summable connected bridge.

Lemma 142.34 (Polymer bridge criterion for source-window boundary tails)**.** *Work at a fixed scale k and a fixed finite source window E. Let $\Lambda_k = a_k\mathbb{Z}^D$ be the ambient lattice, let $K = K_{E,k}$ be the lattice source support, and let d_k be the block distance. Suppose the connected source functional in a finite region $\Omega \supset K$ has an absolutely convergent connected-cluster expansion on P_ρ,*

$$W_{k,\Omega,E}(z) = \sum_{\substack{\mathcal{C}\subset\Omega\\ \operatorname{supp}\mathcal{C}\cap K\neq\emptyset}} w_{\mathcal{C}}(z), \qquad z \in P_\rho, \tag{142.163}$$

where $\operatorname{supp}\mathcal{C}$ *is a finite connected block set. Assume that the cluster weights obey the bridge majorant*

$$\sum_{\substack{\mathcal{C}:\,\operatorname{supp}\mathcal{C}\cap K\neq\emptyset\\ b\in\operatorname{supp}\mathcal{C}}} \sup_{z\in P_\rho} |w_{\mathcal{C}}(z)| \le B_E\, e^{-\zeta d_k(b,K)} \tag{142.164}$$

for every block $b \in \Lambda_k$, with $B_E < \infty$ and $\zeta > 0$, uniformly in the finite volume. Then, whenever $\Omega \subset \Omega'$ and both regions contain K,

$$\sup_{z\in P_\rho} |W_{k,\Omega',E}(z) - W_{k,\Omega,E}(z)| \le B_E \sum_{b\in\Lambda_k\setminus\Omega} e^{-\zeta d_k(b,K)}. \tag{142.165}$$

In particular, if

$$R_\Omega = d_k(K, \Lambda_k \setminus \Omega)$$

and the source-centered shells obey

$$\#\{b \in \Lambda_k :\ d_k(b,K) = n\} \le C_E(1+n)^{D_E-1}, \qquad n \ge 0,$$

then (142.162) *holds with*

$$A_E\,\Xi(R) = B_E C_E \sum_{n\geq R} (1+n)^{D_E-1} e^{-\zeta n}. \tag{142.166}$$

The right side tends to zero as $R\to\infty$.

Proof. Subtract the two cluster expansions. The terms whose support lies in Ω cancel. Every remaining cluster has $\operatorname{supp}\mathcal{C}\cap K\neq\emptyset$ and contains at least one block $b\in\Lambda_k\setminus\Omega$. Absolute convergence permits the union bound

$$\begin{aligned}\sup_{z\in P_\rho}|W_{k,\Omega',E}(z)-W_{k,\Omega,E}(z)| &\leq \sum_{\substack{\mathcal{C}\subset\Omega'\\ \operatorname{supp}\mathcal{C}\cap K\neq\emptyset\\ \operatorname{supp}\mathcal{C}\not\subset\Omega}} \sup_{z\in P_\rho}|w_{\mathcal{C}}(z)| \\ &\leq \sum_{b\in\Lambda_k\setminus\Omega}\ \sum_{\substack{\mathcal{C}:\,\operatorname{supp}\mathcal{C}\cap K\neq\emptyset\\ b\in\operatorname{supp}\mathcal{C}}} \sup_{z\in P_\rho}|w_{\mathcal{C}}(z)|.\end{aligned}$$

The bridge majorant (142.164) gives (142.165). If $b\in\Lambda_k\setminus\Omega$, then $d_k(b,K)\geq R_\Omega$. Grouping the boundary sum by the distance $n=d_k(b,K)$ and using the shell-count bound gives (142.166). The final series is the tail of an exponentially weighted polynomial series, hence it converges and its tail tends to zero. □

The bridge majorant (142.164) is the model-specific cluster estimate that must be proved from the polymer norm and the finite range or summable covariance bounds. Once it is available, Lemma 142.34 turns it into the thermodynamic source-window bound used in the rest of this subsection.

For every multi-index β, Cauchy's formula on the same polydisc gives

$$\left|\partial_z^\beta W_{k,\Omega',E}(0)-\partial_z^\beta W_{k,\Omega,E}(0)\right| \leq \frac{\beta!}{\rho^\beta}\,A_E\,\Xi(R_\Omega). \tag{142.167}$$

Thus the finite-volume cumulant windows are Cauchy as the boundary recedes from the source support. Their limit is denoted

$$\partial_z^\beta W_{k,\infty,E}(0) := \lim_{\Omega\uparrow a_k\mathbb{Z}^D} \partial_z^\beta W_{k,\Omega,E}(0),$$

provided the directed family of regions is chosen so that $R_\Omega\to\infty$. The resulting window is independent of the chosen cofinal exhaustion whenever the same tail estimate controls any two exhaustions through a common larger region. This is the finite-volume part of item 1 in Definition 142.36. Without a bound of the form (142.162), a sequence of finite volumes may have plausible local numerical stability while failing to define the thermodynamic source window required by OS reconstruction.

The independence of the exhaustion is also part of the same estimate. Let $\{\Omega_i\}$ and $\{\widetilde{\Omega}_j\}$ be two directed exhaustions whose boundary distances from $K_{E,k}$ tend to infinity. For fixed i,j, choose a connected region $\Omega^\sharp_{ij}$ containing both Ω_i and $\widetilde{\Omega}_j$. Applying (142.162) twice gives

$$\sup_{z\in P_\rho}|W_{k,\Omega_i,E}(z)-W_{k,\widetilde{\Omega}_j,E}(z)| \leq A_E\Xi(R_{\Omega_i}) + A_E\Xi(R_{\widetilde{\Omega}_j}). \tag{142.168}$$

The right side tends to zero as $i, j \to \infty$, so the holomorphic limit on P_ρ is independent of the cofinal volume sequence. Cauchy's formula then yields the derivative comparison

$$\left|\partial_z^\beta W_{k,\Omega_i,E}(0) - \partial_z^\beta W_{k,\widetilde{\Omega}_j,E}(0)\right| \le \frac{\beta!}{\rho^\beta}\big(A_E \Xi(R_{\Omega_i}) + A_E \Xi(R_{\widetilde{\Omega}_j})\big). \tag{142.169}$$

The same bookkeeping fixes the joint thermodynamic and continuum limit used later in the RG-to-OS assembly package. Suppose that, for the same finite source window E, the RG-chart comparison to a limiting source functional $W_{*,E}$ gives an error $\epsilon_{E,k}^{\mathrm{RG}}$ on P_ρ, while the chosen finite volume Ω_k obeys the boundary distance $R_{\Omega_k} \to \infty$. Combining (142.156) with (142.162) gives

$$\sup_{z \in P_\rho} |W_{k,\Omega_k,E}(z) - W_{*,E}(z)| \le \epsilon_{E,k}^{\mathrm{RG}} + A_E \Xi(R_{\Omega_k}). \tag{142.170}$$

Consequently every extracted cumulant in the E-window satisfies

$$\left|\partial_z^\beta W_{k,\Omega_k,E}(0) - \partial_z^\beta W_{*,E}(0)\right| \le \frac{\beta!}{\rho^\beta}\left(\epsilon_{E,k}^{\mathrm{RG}} + A_E \Xi(R_{\Omega_k})\right). \tag{142.171}$$

Thus the finite-volume simulation or finite-volume constructive estimate may be used as thermodynamic Schwinger data only through a boundary-tail function and a scale schedule that makes the combined right side tend to zero for each fixed source window.

Adjoint source blocking and smooth-test error. The source normalization in (142.131) is not a convention that can be chosen independently at each scale. Let F be a real-valued test function on the coarse lattice Λ_{k-1}. The source paired with the blocked field is

$$a_{k-1}^{D-\Delta_\phi} \sum_{y \in \Lambda_{k-1}} F(y)(B_k\phi)(y).$$

Using $a_{k-1} = La_k$ and the definition (142.130), this expression is equal to

$$a_k^{D-\Delta_\phi} \sum_{y \in \Lambda_{k-1}} \sum_{r \in \{0,\dots,L-1\}^D} L^D q(r) F(y)\, \phi_{y+a_k r}. \tag{142.172}$$

Thus the adjoint source map associated with the normalized pairings is

$$(B_k^\dagger F)(y + a_k r) = L^D q(r) F(y), \qquad r \in \{0, \dots, L-1\}^D, \tag{142.173}$$

with periodic representatives at the boundary. Equivalently,

$$a_{k-1}^{D-\Delta_\phi} \sum_y F(y)(B_k\phi)(y) = a_k^{D-\Delta_\phi} \sum_x (B_k^\dagger F)(x)\phi_x. \tag{142.174}$$

The exponent D, rather than $D - \Delta_\phi$, in (142.173) is forced by the factor L^{Δ_ϕ} already present in the blocked field. This is the finite-regulator identity behind source-window extraction in an ordinary short-range scalar RG chart.

For comparison with continuum test functions, take the uniform kernel $q(r) = L^{-D}$ and let $f \in C_c^1(\mathbb{R}^D)$. Put $F(y) = f(y)$ on the coarse lattice and let Y_k be a finite set of coarse sites containing all blocks that meet the support of f. Then

$$a_k^D \sum_{y \in Y_k} \sum_{r \in \{0,\dots,L-1\}^D} \left|(B_k^\dagger F)(y + a_k r) - f(y + a_k r)\right| \le C_{D,L}\, \|\nabla f\|_\infty\, a_{k-1}\, \mathrm{Vol}_{k-1}(Y_k), \tag{142.175}$$

where

$$\mathrm{Vol}_{k-1}(Y_k) = a_{k-1}^D |Y_k|, \qquad C_{D,L} = L^{-(D+1)} \sum_{r\in\{0,\ldots,L-1\}^D} |r|.$$

Indeed, for the uniform kernel $B_k^\dagger F$ is constant on each block and equal to $f(y)$, while $|f(y) - f(y+a_k r)| \le a_k |r| \, \|\nabla f\|_\infty$. Multiplication by a_k^D and summation over y and r gives (142.175). Higher-order block kernels may improve this sampling error when their moments are matched to the chosen cell coordinates; without those moment conditions the exact identity (142.174), not pointwise approximation of smooth tests, is the invariant source statement.

This construction also fixes the role of the block kernel. A different kernel q, a different field-strength normalization, or a different localization map may give an equivalent continuum theory only after a comparison map with estimates in the sense of Definition 142.2 is supplied. Finite-range microscopic interactions alone do not identify the continuum normalization of the field.

Lemma 142.35 (Reflection-positive block-spin pullback)**.** *Let Λ_k be split by a time reflection θ into positive, negative, and fixed-time sites, and let $\mathcal{A}_{k,+}$ be the algebra of polynomials in the positive-time fine fields. Suppose the finite measure μ_k is reflection positive:*

$$\int_{\mathbb{R}^{\Lambda_k}} \overline{\theta P(\phi)}\, P(\phi)\, \mathrm{d}\mu_k(\phi) \ge 0, \qquad P \in \mathcal{A}_{k,+}.$$

Assume that the block-spin map B_k is reflection compatible in the following finite sense: every coarse positive-time field $\Phi_y = (B_k\phi)(y)$, $y^0 > 0$, is a real linear combination of fine fields in the positive-time half-lattice, and $(B_k\phi)(\theta y) = (B_k(\theta\phi))(y)$. Let $\nu_{k-1} := (B_k)_\#\mu_k$ be the blocked measure. Then ν_{k-1} is reflection positive for the coarse positive-time polynomial algebra.

Proof. Let F be a polynomial in coarse positive-time fields, and define its pullback

$$B_k^* F(\phi) = F(B_k\phi).$$

By the support assumption on the positive-time block stencils, $B_k^*F \in \mathcal{A}_{k,+}$. Reflection compatibility gives

$$\theta(B_k^*F) = B_k^*(\theta F),$$

because reflecting the fine configuration and then blocking is the same, on the reflected coarse lattice, as blocking and then reflecting. Therefore

$$\begin{aligned} \int \overline{\theta F(\Phi)}\, F(\Phi)\, \mathrm{d}\nu_{k-1}(\Phi) &= \int \overline{B_k^*(\theta F)(\phi)}\, B_k^*F(\phi)\, \mathrm{d}\mu_k(\phi) \\ &= \int \overline{\theta(B_k^*F)(\phi)}\, B_k^*F(\phi)\, \mathrm{d}\mu_k(\phi) \ge 0. \end{aligned}$$

Thus every finite coarse reflection-positivity Gram matrix is the compression of a fine reflection-positivity Gram form under the pullback B_k^*. The lemma is finite-regulator only. A continuum OS statement still requires the directed-family convergence and growth estimates described in Definition 142.47. □

Requirements for OS reconstruction. The preceding construction and lemma are finite pieces. A critical short-range scalar limit reaches a local QFT through a directed family of estimates in exactly the form required by Definition 12.2.

Definition 142.36 (Short-range scalar OS reconstruction requirements)**.** Fix a short-range scalar block-spin structure as in Definition 142.31 and a tuned trajectory $V_k = (g_k, K_k)$ in its RG chart. A short-range scalar OS reconstruction argument consists of the following additional objects and estimates.

1. An infinite-volume prescription for the cumulants $S_{n,k}$. Either the finite-volume limit is taken at fixed lattice spacing before $k \to \infty$, or a joint volume–continuum limit is specified, with estimates uniform enough to give the same limiting distributions on every compactly supported test window.

2. Distributional reconstruction estimates: for every n and every bounded set $B \subset \mathcal{S}((\mathbb{R}^D)^n)$,

$$p_B(S_{n,k} - S_{n,*}) \longrightarrow 0,$$

   together with a uniform temperedness bound of the form

$$|S_{n,k}(f)| \le C_n\, q_{n,r_n}(f), \qquad f \in \mathcal{S}((\mathbb{R}^D)^n),$$

   for a declared Schwartz seminorm q_{n,r_n}. This bound is the continuity estimate that turns a projective family of finite windows into a tempered distribution, as in Lemma 142.48.

3. Euclidean covariance, permutation symmetry, normalization $S_{0,k} = 1$, and Hermiticity at finite regulator, or regulator defects for these identities that vanish in the same distributional topology.

4. Reflection positivity in a directed positive-time test family. More precisely, for every finite positive-time polynomial test family $\mathcal{F} = \{F_1, \ldots, F_m\}$, the regulated Gram matrices

$$G_{k,\mathcal{F};ij} = S_k((\Theta F_i) F_j)$$

   are positive semidefinite, or obey a defect inequality whose defect tends to zero, and their entries converge to the limiting Gram matrix. If one uses quantitative entrywise error estimates for increasing families, the estimates must be compatible with the family size in the sense explained below (142.213).

5. Positive-time semigroup regularity on the OS quotient. This means that the positive-time translations defined from the limiting Schwinger hierarchy obey the semigroup law and strong continuity conditions (12.5)–(12.8). In an RG proof this is an estimate on translated positive-time test windows, additional to the finite Gram-form positivity estimates.

6. Corrected OS-II analytic regularity. A sufficient form is the uniform linear-growth estimate

$$|S^{\mathrm{red}}_{n,k}(F)| \le A\, B^n\, (n!)^\gamma\, q_{n,N_0 + C_{\mathrm{OS}} n}(F)$$

   for all sufficiently large k, with constants $A, B, \gamma, N_0, C_{\mathrm{OS}}$ independent of n and k. The same bound then passes to the limiting reduced hierarchy by distributional convergence. Equivalently, one may provide a different model-specific analytic theorem that yields the Lorentzian boundary-value package of Definition 12.8. Ordinary temperedness at each fixed n supplies fixed-n continuity; the OS-II item controls the many-variable analytic continuation uniformly in the insertion number. In a Wilsonian source construction, a usable route to this item is a moment-source majorant of the form (142.143), together with the projective-seminorm comparison following (142.145).

7. The ordered-insertion adjoint identity required in Theorem 12.16; for a single real scalar field this is the compatibility of Hermiticity with the positive-time multiplication convention, while for a labelled field family it is an identity of the declared label reflection.

Clustering estimates are added when the target claim includes uniqueness of the vacuum.

Positive-time translation continuity from finite windows. Item 5 of Definition 142.36 is a continuity estimate in the limiting OS Hilbert norm. That norm is not the Schwartz norm of a Euclidean test function; it is the limiting positive-time Gram norm. Thus strong continuity of the positive-time translation semigroup is proved by estimates on translated Gram windows.

Let F be a polynomial positive-time test vector whose Euclidean-time support has distance $d_F > 0$ from the reflection plane. For $0 \le t < d_F/2$, let $\tau_t F$ denote translation of every Euclidean-time argument of F forward by t. Suppose the regulated Schwinger hierarchy gives, uniformly for all sufficiently large k,

$$S_k(\Theta(\tau_t F - F)(\tau_t F - F)) \le \omega_F(t) + \varepsilon_{F,k}, \qquad \lim_{t\downarrow 0} \omega_F(t) = 0, \qquad \varepsilon_{F,k} \to 0. \tag{142.176}$$

Here the left side is a finite positive-time Gram-window entry, and the support restriction ensures that $\tau_t F - F$ still belongs to the positive-time test algebra. If the corresponding Gram entries converge to the limiting OS form, then

$$\|[\tau_t F] - [F]\|^2_{\mathcal{H}_{\mathrm{OS}}} = S_*(\Theta(\tau_t F - F)(\tau_t F - F)) \le \omega_F(t),$$

after taking $k \to \infty$. Therefore $t \mapsto [\tau_t F]$ is continuous at $t = 0$ on every such vector. Translation covariance gives $\tau_t \tau_s F = \tau_{t+s} F$ while the support remains positive-time; after passing to the OS quotient this supplies the semigroup law on the dense positive-time domain. Extending the semigroup from this domain to the OS Hilbert-space closure requires the same bound on a directed family of positive-time vectors whose classes are dense. A Wilsonian RG proof must therefore supply (142.176), or an equivalent model-specific equicontinuity estimate, in addition to finite reflection positivity.

Lemma 142.37 (Positive-time quotient semigroup criterion)**.** *Let P_∞ be the algebraic positive-time test space and let*

$$Q(F, G) = S_*((\Theta F)G)$$

be the limiting positive semidefinite OS form on P_∞. Put

$$\mathcal{N}_Q = \{F \in P_\infty \mid Q(F, F) = 0\}, \qquad \|[F]\|^2_{\mathrm{OS}} = Q(F, F).$$

Assume that for $0 \le t \le t_0$ the positive-time translations $\tau_t : P_\infty \to P_\infty$ satisfy $\tau_0 = 1$ and $\tau_t \tau_s = \tau_{t+s}$ whenever $s, t, s + t \in [0, t_0]$. Assume also the quotient-stability estimate

$$Q(\tau_t F, \tau_t F) \le C_t\, Q(F, F), \qquad C_t < \infty, \tag{142.177}$$

with $\sup_{0\le t\le t_0} C_t < \infty$, and the limiting Gram-continuity estimate

$$Q(\tau_t F - F, \tau_t F - F) \longrightarrow 0 \qquad (t \downarrow 0)$$

for every $F \in P_\infty$. Then

$$T_t[F] = [\tau_t F]$$

is a well-defined strongly continuous local semigroup on the OS Hilbert-space completion of $P_\infty/\mathcal{N}_Q$ for $0 \le t \le t_0$, with $\|T_t\| \le C_t^{1/2}$ on its initial domain of definition. If the same estimates hold on every compact positive-time interval, these local semigroups assemble to a strongly continuous semigroup for $t \ge 0$. If $C_t \le 1$, the corresponding semigroup is contractive.

Proof. The first load-bearing point is descent to the quotient. If $F \in \mathcal{N}_Q$, then (142.177) gives

$$Q(\tau_t F, \tau_t F) \le C_t Q(F, F) = 0,$$

so $\tau_t F \in \mathcal{N}_Q$. Hence the class $[\tau_t F]$ depends only on the class $[F]$. The same estimate gives

$$\|T_t[F]\|_{\mathrm{OS}}^2 = Q(\tau_t F, \tau_t F) \le C_t Q(F, F) = C_t \|[F]\|_{\mathrm{OS}}^2,$$

so T_t is bounded on the quotient norm and extends uniquely to the Hilbert completion.

The semigroup law is inherited from the algebraic identity $\tau_t \tau_s F = \tau_{t+s} F$ on the common positive-time domain and then extends by continuity. It remains to prove strong continuity on the completion. On the dense quotient $P_\infty/\mathcal{N}_Q$, the assumed Gram-continuity estimate is exactly

$$\|T_t[F] - [F]\|_{\mathrm{OS}}^2 = Q(\tau_t F - F, \tau_t F - F) \longrightarrow 0.$$

For a general Hilbert vector v, choose $[F]$ with $\|v - [F]\|_{\mathrm{OS}} < \alpha$. If $M = \sup_{0 \le t \le t_0} C_t^{1/2}$, then

$$\|T_t v - v\|_{\mathrm{OS}} \le (M + 1)\alpha + \|T_t[F] - [F]\|_{\mathrm{OS}}.$$

First choose α small and then t small. This proves strong continuity at $t = 0$, and the semigroup law gives strong continuity at every positive time in the interval. □

This lemma isolates an estimate that is logically prior to writing $[\tau_t F]$ as an OS Hilbert-space vector. The finite-window bound (142.176) supplies the continuity term after the regulator is removed. The quotient-stability bound (142.177) supplies invariance of the OS null space. In a model proof both estimates should be derived from the same directed Gram-window control.

Given Definition 142.36, the passage to a Wightman theory is the reconstruction stage following the RG estimates. The distributional estimates give tempered limiting Schwinger functions. The covariance, symmetry, Hermiticity, and normalization identities pass to the limit because they are continuous linear identities on the test-function spaces. The reflection-positive Gram forms pass to the limit by the closed cone argument following Definition 142.30, provided the directed finite-family estimates have actually been supplied. The positive-time semigroup regularity and the corrected OS-II growth package are then exactly the remaining hypotheses of Theorem 12.16. Applying that theorem constructs the Hilbert space, vacuum vector, translation representation, and operator-valued tempered distribution whose Euclidean boundary values reproduce the non-coincident limiting Schwinger functions.

These reconstruction requirements contain more information than the statement that (g_k, K_k) approaches a fixed point. Finite-regulator reflection positivity plus convergence of a few correlators supplies only the finite windows under control. OS reconstruction asks for the whole directed positive-time Gram family, the positive-time translation semigroup, and the OS-II analytic growth estimates. These are the model-specific burdens that remain for the ordinary short-range scalar Wilson–Fisher target.

Remark 142.38 (Current theorem status for the ordinary scalar target). For massive superrenormalizable scalar theories, constructive methods can produce Euclidean measures and hence Wightman theories after OS reconstruction. For an ordinary short-range three-dimensional critical scalar model, a theorem at the Wilsonian fixed-point level would have to construct Definition 142.31, prove a Banach-space fixed point or stable trajectory for the corresponding short-range block-spin RG, establish (142.135) for local fields and their source extensions, prove reflection-compatible blocking as in Lemma 142.35 or an equivalent positivity-preserving construction, and then supply the full OS reconstruction requirements of Definition 142.36. This is stronger than identifying a candidate fixed point in an epsilon expansion, in a hierarchical model, or in a finite projection truncation.

Proof dependency ledger for the ordinary short-range target. The assembly package below is reconstruction-facing; it is not the whole ordinary short-range scalar fixed-point theorem. The missing theorem has to close the following dependency chain in this order.

1. Construct the exact finite short-range block-spin map, including the microscopic Gibbs measures, reflection-compatible block kernel, source adjoint, finite-volume prescription, and localization map into the declared polymer coordinates.
2. Prove that the localized RG map is defined on a Banach polymer chart with uniformly bounded coordinate extraction, large-field regulators, finite-range or summable covariance tails, and irrelevant-activity contraction.
3. Build the critical trajectory or stable graph for the ordinary short-range map itself, including relevant-coordinate tuning and the nonsemisimple or marginal normal forms when those occur.
4. Extend the same trajectory to source windows: retained local source coordinates, connected polymer derivative norms, thermodynamic boundary tails, and Cauchy estimates must all use common source radii and compatible finite test windows.
5. Feed those estimates into the RG-to-OS package: cofinal distributional limits, directed reflection positivity, quotient-stable positive-time translations, OS-II growth, and one common scale schedule for every finite reconstruction demand.
6. Identify the target. Hierarchical, long-range, fermionic, functional-RG, tensor, or numerical fixed points count as ordinary short-range scalar evidence only after auxiliary-to-short-range transfer estimates compare the RG maps, observables, normalizations, and reconstruction topology.

A failure at any earlier layer is not repaired by a later finite identity. For example, a finite OS Gram-window estimate cannot supply the stable graph; a source Cauchy bound cannot supply reflection positivity; and a list of critical exponents from an auxiliary model cannot identify the ordinary short-range Wilson–Fisher QFT without the transfer estimates of Definition 142.40.

The ordinary short-range scalar RG-to-OS assembly package. The preceding estimates can be organized into one theorem-facing package. This is not an additional theorem about the three-dimensional Ising or Wilson–Fisher target; it is the precise list of estimates a model-specific proof must supply before the OS reconstruction theorem can be invoked.

Definition 142.39 (Short-range scalar RG-to-OS assembly package)**.** Fix a short-range scalar block-spin structure and source normalization as in Definitions 142.31 and 142.36. An RG-to-OS assembly package for a critical short-range scalar target consists of the following scale-indexed estimates.

1. A tuned RG trajectory (g_k, K_k) in a declared Banach polymer chart and a limiting source-extended chart $(C_{*,E}, g_{*,E}, K_{*,E}, r_{*,E})$ for every finite test space E, all defined on a fixed source polydisc P_{ρ_E}.

2. For every E, a source-window error budget

$$\begin{aligned}\epsilon^{\mathrm{src}}_{E,k} = \delta^{C}_{E,k} + L_E(\delta^{g}_{E,k} + \delta^{K}_{E,k}) + \delta^{r}_{E,k} \\ + A_E \Xi_E(R_{E,k}) + \sigma_{E,k},\end{aligned} \tag{142.178}$$

   with $\epsilon^{\mathrm{src}}_{E,k} \to 0$. The first line is the RG-chart error of (142.156); $A_E \Xi_E(R_{E,k})$ is the thermodynamic boundary tail of (142.162); and $\sigma_{E,k}$ is the remaining cofinal-window, sampling, or finite-step error at scale k.

3. Uniform distribution bounds: for each n there are r_n and C_n, independent of k and of the finite window E, such that the n-source derivatives extracted from $W_{k,E}$ satisfy (142.141). Consequently, for every multi-index β in the E-window,

$$\left|\partial_z^{\beta} W_{k,E}(0) - \partial_z^{\beta} W_{*,E}(0)\right| \le \frac{\beta!}{\rho_E^{\beta}}\, \epsilon^{\mathrm{src}}_{E,k}. \tag{142.179}$$

4. Cofinal compatibility: the controlled finite test spaces are cofinal in the Schwartz test space, their restriction maps commute with the source-window functionals, and the fixed-window Cauchy estimates have the form required by Lemma 142.49.

5. Directed OS positivity and time-regularity: the positive-time Gram windows satisfy the directed positivity assembly hypotheses of Lemma 142.51, the quotient stability and Gram-continuity estimates of Lemma 142.37, and the corrected OS-II growth package in Definition 142.36.

6. A common directed scale schedule. For every finite reconstruction demand

$$\mathcal{D} = (E, \beta, \mathcal{F}, F, t, H, n),$$

   consisting of a source window and multi-index, a finite positive-time Gram family, a translated positive-time vector, and an n-point test H used in the OS-II bound, there is $k_{\mathcal{D}}$ such that all estimates in the preceding items hold for $k \ge k_{\mathcal{D}}$ with the same constants assigned to $\mathcal{D}$. The assignment is directed: if $\mathcal{D}_1 \le \mathcal{D}_2$ means that the second demand contains the first by enlarging the test windows or increasing the insertion order, then the constants and thresholds restrict compatibly from $\mathcal{D}_2$ to $\mathcal{D}_1$.

When these estimates are supplied by a specific short-range scalar model, the logic is fixed. Equation (142.179) gives convergence of every finite source window. Cofinal compatibility and the uniform Schwartz bounds produce limiting tempered Schwinger distributions. Directed positivity, quotient-stable time translations, and OS-II growth provide the hypotheses of Theorem 12.16. The conclusion is then a Wightman QFT whose Euclidean Schwinger functions are the reconstructed limits. The unproved part for the ordinary critical scalar target is the model-specific construction of the estimates in Definition 142.39; the implication from those estimates to reconstruction is the functional-analytic assembly just described. The common-scale item prevents a frequent logical gap: source convergence, finite Gram positivity, translation equicontinuity, and OS-II growth must be available on the same cofinal regulator tail for the same reconstruction demand. Separate estimates along incompatible scale subsequences give separate finite statements; the OS reconstruction theorem uses their joint directed limit.

142.17 Transfer Estimates From Auxiliary Fixed-Point Models

The theorem-level fixed points known in hierarchical scalar models, fractional or long-range models, fermionic small-parameter models, and tensor-network RG are not automatically fixed points of the ordinary short-range scalar block-spin problem of Definition 142.31. Their use for the ordinary target is controlled by an auxiliary-to-target comparison, not by a shared list of exponents. This section records the estimate that such a comparison must contain.

Definition 142.40 (Auxiliary-to-short-range RG transfer estimates)**.** Let

$$(\mathcal{B}_{\mathrm{aux}}, \mathcal{R}_{\mathrm{aux}}, V_*^{\mathrm{aux}}) \qquad \text{and} \qquad (\mathcal{B}_{\mathrm{sr}}, \mathcal{R}_{\mathrm{sr}}, V_*^{\mathrm{sr}})$$

be an auxiliary RG system and an ordinary short-range target RG system, with Banach spaces, domains, fixed points, stable neighborhoods, source extensions, and reconstruction maps already specified. Transfer estimates from the auxiliary system to the short-range target consist of:

1. a map $\Phi : \mathcal{U}_{\mathrm{aux}} \to \mathcal{B}_{\mathrm{sr}}$, defined on an auxiliary neighborhood that contains the orbit under consideration;
2. a map $\Psi : \mathcal{O}_{\mathrm{aux}} \to \mathcal{O}_{\mathrm{sr}}$ between the corresponding locally convex observable-data spaces;
3. one-step RG defects

$$d_j(y_0) = \mathcal{R}_{\mathrm{sr}}(\Phi(y_j)) - \Phi(y_{j+1}), \qquad y_j = \mathcal{R}_{\mathrm{aux}}^j y_0, \tag{142.180}$$

   with explicit norm bounds $\|d_j(y_0)\|_{\mathcal{B}_{\mathrm{sr}}} \le \delta_j(y_0)$;
4. Lipschitz constants M_j for $\mathcal{R}_{\mathrm{sr}}$ on a tube in $\mathcal{B}_{\mathrm{sr}}$ containing both the true short-range orbit $\mathcal{R}_{\mathrm{sr}}^j \Phi(y_0)$ and the transferred auxiliary orbit $\Phi(y_j)$;
5. comparison of tuning maps for the unstable coordinates,

$$\tau_{\mathrm{sr}}(\Phi y) = A\, \tau_{\mathrm{aux}}(y) + r(y), \tag{142.181}$$

   where A is a specified linear map between unstable coordinate spaces and $r(y)$ is bounded in the target unstable norm;
6. observable comparison estimates

$$p\big(\Psi\ \mathrm{Rec}_{n,k}^{\mathrm{aux}}(y_0) - \mathrm{Rec}_{n,k}^{\mathrm{sr}}(\Phi y_0)\big) \le \varepsilon_{n,p,k}(y_0) \tag{142.182}$$

   for every declared observable seminorm p relevant to the claimed continuum statement.

The transfer estimates refine Definition 142.2 in the direction needed for ordinary short-range critical models. The finite RG-orbit estimate below is elementary as a recurrence, but it is not a decorative calculation: it is the point at which an auxiliary theorem either does or does not transfer to the ordinary short-range target. The model-specific work is to construct the comparison map, the common tube, the constants M_j, and the defects δ_j. Let

$$x_j = \mathcal{R}_{\mathrm{sr}}^j \Phi(y_0), \qquad z_j = x_j - \Phi(y_j).$$

Then $z_0 = 0$ and

$$z_{j+1} = \mathcal{R}_{\mathrm{sr}}(x_j) - \mathcal{R}_{\mathrm{sr}}(\Phi(y_j)) + d_j(y_0). \tag{142.183}$$

The Lipschitz estimate on the tube gives

$$\|z_{j+1}\|_{\mathcal{B}_{\mathrm{sr}}} \le M_j \|z_j\|_{\mathcal{B}_{\mathrm{sr}}} + \delta_j(y_0). \tag{142.184}$$

Iterating this scalar inequality yields

$$\|z_n\|_{\mathcal{B}_{\mathrm{sr}}} \le \sum_{j=0}^{n-1} \left(\prod_{\ell=j+1}^{n-1} M_\ell \right) \delta_j(y_0). \tag{142.185}$$

Thus an auxiliary fixed-point theorem becomes a controlled statement about the short-range target only when the one-step defects are summable after the target RG amplification factors are included.

On a stable subspace, suppose $M_j \le \theta < 1$ and $\delta_j(y_0) \le Cq^j$ with $0 < q < 1$. Then

$$\|z_n\| \le C \sum_{j=0}^{n-1} \theta^{n-1-j} q^j = \begin{cases} C \dfrac{\theta^n - q^n}{\theta - q}, & \theta \ne q, \\ Cn\theta^{n-1}, & \theta = q. \end{cases} \tag{142.186}$$

Both terms tend to zero when $\theta, q < 1$. This is the stable-direction part of the comparison problem.

Relevant directions have the opposite behavior. If a target unstable coordinate has linear eigenvalue $\lambda_a = L^{y_a}$, $y_a > 0$, and one-step comparison errors $e_{a,j}$, then the finite-depth bound is

$$|u_{a,N}| \le |\lambda_a|^N |u_{a,0}| + \sum_{j=0}^{N-1} |\lambda_a|^{N-1-j} |e_{a,j}|. \tag{142.187}$$

Consequently, if y lies on the auxiliary critical surface $\tau_{\mathrm{aux}}(y) = 0$, the target initial point Φy lies on the target critical surface only when the remainder $r(y)$ in (142.181) is itself tuned away, or is small to the finite-depth accuracy demanded by (142.187). This is why a comparison between a hierarchical or long-range theorem and an ordinary short-range scalar theorem must include the unstable tuning map, not only the stable contraction estimate.

Finally, the observable estimate (142.182) must be combined with (142.185). If $\mathrm{Rec}_{n,k}^{\mathrm{sr}}$ is locally Lipschitz in the RG chart with respect to a seminorm p, then the observable error is bounded by the explicit auxiliary-observable defect plus the Lipschitz image of the orbit error z_k. Only after both terms vanish in the chosen observable topology does an auxiliary theorem transfer to the ordinary short-range QFT claim. Without this transfer structure, hierarchical, long-range, tensor, and functional-RG fixed points should be read as theorem-level benchmarks or computational evidence in their own frameworks, not as a proof of the ordinary short-range Wilson–Fisher QFT.

Transfer to a projective observable germ. The comparison must be made window by window, because a continuum local-QFT claim is a projective statement about all reconstruction windows. Fix a finite observation window α in the directed family of the short-range target. Suppose the auxiliary theorem supplies a corresponding auxiliary window and an auxiliary limiting observable $O^{\mathrm{aux}}_{*,\alpha}$ such that, after its normalization,

$$p^{\mathrm{sr}}_{\alpha}\big(\Psi\ \mathrm{Rec}^{\mathrm{aux}}_{\alpha,k}(y_0) - \Psi O^{\mathrm{aux}}_{*,\alpha}\big) \le a_{\alpha,k}. \tag{142.188}$$

Assume further that the comparison of observable reconstructions has the finite-window defect

$$p^{\mathrm{sr}}_{\alpha}\big(\mathrm{Rec}^{\mathrm{sr}}_{\alpha,k}(\Phi y_0) - \Psi\ \mathrm{Rec}^{\mathrm{aux}}_{\alpha,k}(y_0)\big) \le \varepsilon_{\alpha,k}, \tag{142.189}$$

and that the short-range reconstruction map is Lipschitz on the target tube:

$$p^{\mathrm{sr}}_{\alpha}\big(\mathrm{Rec}^{\mathrm{sr}}_{\alpha,k}(x_0) - \mathrm{Rec}^{\mathrm{sr}}_{\alpha,k}(\Phi y_0)\big) \le L_{\alpha,k}\|x_k - \Phi(y_k)\|_{\mathcal{B}_{\mathrm{sr}}}. \tag{142.190}$$

Here x_0 is the actual short-range initial point whose window is being claimed; in the simplest transferred construction $x_0 = \Phi y_0$, but the separate notation is useful when the target microscopic regulator is only close to the transferred auxiliary initial condition. If the allowed normalization of the short-range window differs from the pushed-forward auxiliary normalization by at most $n_{\alpha,k}$, the triangle inequality and (142.185) give the finite projective-window transfer estimate

$$\begin{aligned} p^{\mathrm{sr}}_{\alpha}\big(g_{\mathrm{sr}}\,\mathrm{Rec}^{\mathrm{sr}}_{\alpha,k}(x_0) - \Psi O^{\mathrm{aux}}_{*,\alpha}\big) &\le n_{\alpha,k} + a_{\alpha,k} + \varepsilon_{\alpha,k} \\ &\quad + L_{\alpha,k}\sum_{j=0}^{k-1}\left(\prod_{\ell=j+1}^{k-1} M_\ell\right)\delta_j(y_0). \end{aligned} \tag{142.191}$$

Thus an auxiliary fixed point transfers to the ordinary short-range observable germ only after the right-hand side of (142.191) tends to zero for every window α in the directed family. The estimate separates four different analytic tasks: convergence of the auxiliary observable window, comparison of auxiliary and target reconstructions, control of the target orbit defect, and compatibility of the normalization group. A shared critical exponent controls at most one contribution to one family of windows; it does not provide the projective observable-germ transfer.

142.18 Source-Extended RG, Scaling Operators, And Contact Terms

A fixed interaction is not yet a QFT spectrum. Scaling operators are obtained by extending the RG map to sources and by reconstructing the resulting functional derivatives as distributions. This section records the minimal theorem-level structure, because otherwise the phrase "operator dimension" may mean only an eigenvalue of a finite list of couplings.

Let A be a set of source labels. For each $\alpha \in A$, let $\mathcal{J}_\alpha$ be a locally convex space of test sources on $\mathbb{R}^D$, for instance $C^\infty_c(\mathbb{R}^D)$ or $\mathcal{S}(\mathbb{R}^D)$. A regulated source-dependent generating functional has the form

$$Z_k(V;J) = \int \exp\left(-S_k(\phi;V) + \sum_{\alpha\in A}\langle J_\alpha, \mathcal{O}_{\alpha,k}(\phi)\rangle\right) \mathrm{d}\mu_k(\phi), \tag{142.192}$$

where $\langle J_\alpha, \mathcal{O}_{\alpha,k} \rangle$ denotes the pairing between a test source and a regulated insertion. In a gauge theory, the source direction must either couple to a gauge-invariant lattice observable, to a Wilson-line or Wilson-loop observable with its path data included, or to a class in the BV observable cohomology. A source coupled to a gauge-fixed colored field variable is not, by itself, a nonperturbative observable.

Definition 142.41 (Source-extended RG datum)**.** A source-extended RG datum near a fixed point V_* consists of:

1. a Banach RG chart $\mathcal{B}$, an RG map $\mathcal{R} : \mathcal{U} \to \mathcal{B}$, and a fixed point $\mathcal{R}(V_*) = V_*$;
2. a locally convex source space $\mathcal{J} = \bigoplus_{\alpha \in A} \mathcal{J}_\alpha$;
3. a map
$$\widehat{\mathcal{R}} : \mathcal{U} \times \mathcal{V}_J \longrightarrow \mathcal{B} \times \mathcal{J}, \qquad \widehat{\mathcal{R}}(V, J) = \big(\mathcal{R}(V),\, \mathcal{M}_V J + \mathcal{N}_V(J)\big), \tag{142.193}$$
where $\mathcal{V}_J$ is a neighborhood of $0 \in \mathcal{J}$, $\mathcal{M}_V$ is linear in J, and $\mathcal{N}_V(J) = O(J^2)$ in the declared source topology;
4. an exact finite-regulator identity
$$Z_k(V; J) = e^{A_k(V;J)} Z_{k-1}\Big(\widehat{\mathcal{R}}(V, J)\Big), \tag{142.194}$$
with $A_k(V; J)$ a local normalization functional in the sources;
5. reconstruction maps taking n source derivatives of $\log Z_k(V; J)$ at $J = 0$ to connected distributions on $(\mathbb{R}^D)^n$, or to distributions on configuration space away from collision diagonals.

The local functional $A_k(V; J)$ is part of the datum. Its source derivatives are precisely where contact terms generated by the RG step are recorded.

At a fixed point, the linear source map
$$\mathcal{M}_* = \mathcal{M}_{V_*} = D_J \widehat{\mathcal{R}}(V_*, 0) \tag{142.195}$$
contains the scaling data. If a source J_α couples to a local field whose scaling dimension is Δ_α, then the constant-source coordinate has RG exponent
$$y_\alpha = D - \Delta_\alpha. \tag{142.196}$$
This relation is fixed by the invariant pairing $\int J_\alpha(x) \mathcal{O}_\alpha(x)\, \mathrm{d}^D x$: under $x = Lx'$, a field with $\mathcal{O}_\alpha(Lx') = L^{-\Delta_\alpha} \mathcal{O}_\alpha(x')$ requires
$$J'_\alpha(x') = L^{D-\Delta_\alpha} J_\alpha(Lx'). \tag{142.197}$$
Thus the dimension is not assigned to a formal monomial alone; it is read from the source direction after redundant directions, null fields, and contact redefinitions have been quotiented as appropriate.

Definition 142.42 (Scaling source and scaling field)**.** Let $\mathcal{M}_*$ be the linearized source RG map. A scaling source direction is a nonzero equivalence class $[j_\alpha]$ in the source space modulo null source directions and source redefinitions whose functional derivatives vanish at separated points, such that

$$\mathcal{M}_*[j_\alpha] = L^{D-\Delta_\alpha}[j_\alpha]. \tag{142.198}$$

The corresponding scaling field is the distribution-valued functional derivative of the connected generating functional in the direction $[j_\alpha]$, reconstructed in the continuum limit. In a BV formulation, the equivalence class is taken in Q_{S_*}-cohomology of observables; in a lattice gauge formulation, it is taken among gauge-invariant source directions.

The quotient in Definition 142.42 is not a matter of taste. If two source directions differ by a local source counterterm, their separated-point connected correlators agree while their extensions across coincident points may differ. The following definition fixes the vocabulary used in the rest of the monograph.

Definition 142.43 (Contact term)**.** Let $n \geq 2$, and let

$$\mathfrak{D}_n = \bigcup_{1 \leq i < j \leq n} \{(x_1, \dots, x_n) \in (\mathbb{R}^D)^n : \; x_i = x_j\} \tag{142.199}$$

be the union of pairwise collision diagonals. A contact term in an n-point distribution is a distribution

$$C \in \mathcal{D}'((\mathbb{R}^D)^n)$$

whose support is contained in $\mathfrak{D}_n$. Equivalently, $\langle C, f \rangle = 0$ for every $f \in C_c^\infty((\mathbb{R}^D)^n)$ whose support is disjoint from $\mathfrak{D}_n$. Contact terms are not ordinary separated-point functions. In a source formalism they are generated by local functionals of the sources and their derivatives.

Let

$$\mathrm{Conf}_n(\mathbb{R}^D) = (\mathbb{R}^D)^n \setminus \mathfrak{D}_n \tag{142.200}$$

be the configuration space of ordered distinct points. For $f \in C_c^\infty(\mathrm{Conf}_n(\mathbb{R}^D))$, define the dilation of the test function by

$$(S_L f)(x_1, \dots, x_n) = L^{-nD} f(L^{-1}x_1, \dots, L^{-1}x_n). \tag{142.201}$$

This convention is chosen so that

$$\langle G, S_L f \rangle = \int G(Ly_1, \dots, Ly_n) f(y_1, \dots, y_n) \, \mathrm{d}^D y_1 \cdots \mathrm{d}^D y_n \tag{142.202}$$

whenever G is represented by an integrable function on the support of the test function.

Separated-point scaling from source-extended RG. The nontrivial step in a source-extended RG construction is proving the finite-scale identity below for actual source derivatives of the regulated theory. Once that identity is available with a vanishing error, its separated-point consequence is a distributional limit statement. Fix labels $\alpha_1, \dots, \alpha_n$ with source eigenvalues $L^{D-\Delta_{\alpha_i}}$. Suppose that the source-extended RG datum gives connected finite-regulator distributions

$$G_{k;\alpha_1 \dots \alpha_n} \in \mathcal{D}'(\mathrm{Conf}_n(\mathbb{R}^D))$$

which converge to

$$G_{*;\alpha_1 \dots \alpha_n} \in \mathcal{D}'(\mathrm{Conf}_n(\mathbb{R}^D))$$

in the usual distribution topology on every compact subset of $\mathrm{Conf}_n(\mathbb{R}^D)$. Assume moreover that for every $f \in C_c^\infty(\mathrm{Conf}_n(\mathbb{R}^D))$ the finite-scale RG identity implies

$$\left| \langle G_{k;\alpha_1\ldots\alpha_n}, S_L f\rangle - L^{-\sum_i \Delta_{\alpha_i}} \langle G_{k-1;\alpha_1\ldots\alpha_n}, f\rangle \right| \le \varepsilon_k(f), \tag{142.203}$$

with $\varepsilon_k(f) \to 0$. Then the limiting separated-point distribution is homogeneous under scale L:

$$\langle G_{*;\alpha_1\ldots\alpha_n}, S_L f\rangle = L^{-\sum_i \Delta_{\alpha_i}} \langle G_{*;\alpha_1\ldots\alpha_n}, f\rangle \tag{142.204}$$

for every $f \in C_c^\infty(\mathrm{Conf}_n(\mathbb{R}^D))$.

If $T_{*;\alpha_1\ldots\alpha_n} \in \mathcal{D}'((\mathbb{R}^D)^n)$ is any extension of $G_{*;\alpha_1\ldots\alpha_n}$ to all of $(\mathbb{R}^D)^n$, then the distribution

$$H_L(f) = \langle T_{*;\alpha_1\ldots\alpha_n}, S_L f\rangle - L^{-\sum_i \Delta_{\alpha_i}} \langle T_{*;\alpha_1\ldots\alpha_n}, f\rangle \tag{142.205}$$

is a contact term.

The verification is a support calculation for distributions, not a separate RG theorem. The dilation S_L maps $C_c^\infty(\mathrm{Conf}_n(\mathbb{R}^D))$ continuously to itself. By distributional convergence on configuration space,

$$\langle G_{k;\alpha_1\ldots\alpha_n}, S_L f\rangle \longrightarrow \langle G_{*;\alpha_1\ldots\alpha_n}, S_L f\rangle$$

and

$$\langle G_{k-1;\alpha_1\ldots\alpha_n}, f\rangle \longrightarrow \langle G_{*;\alpha_1\ldots\alpha_n}, f\rangle.$$

Taking $k \to \infty$ in (142.203) gives (142.204), because $\varepsilon_k(f) \to 0$.

For the last assertion, let $f \in C_c^\infty((\mathbb{R}^D)^n)$ have support disjoint from $\mathfrak{D}_n$. Then f and $S_L f$ are test functions on $\mathrm{Conf}_n(\mathbb{R}^D)$, and the extension $T_{*;\alpha_1\ldots\alpha_n}$ agrees with $G_{*;\alpha_1\ldots\alpha_n}$ on both of them. Therefore (142.204) gives $H_L(f) = 0$. A distribution that vanishes on every test function supported in the open set $(\mathbb{R}^D)^n \setminus \mathfrak{D}_n$ has support contained in $\mathfrak{D}_n$, by the definition of support of a distribution. Hence H_L is a contact term in the sense of Definition 142.43.

This is the logical order. Source eigenvalues determine candidate scaling dimensions. Reconstruction estimates make the source derivatives into distributions. The finite-scale RG identity then proves homogeneity at separated points. Extension across collision diagonals is an additional renormalization problem; different extensions differ by contact terms, and local source counterterms change precisely those contact terms. This is why anomalous dimensions, OPE coefficients at separated points, and contact terms must be kept as distinct pieces of data.

Remark 142.44 (Status of short-range scalar and gauge targets)*.* Constructive massive ϕ^4_3 and related superrenormalizable models give Wightman QFTs after their Euclidean constructions and OS reconstruction are completed. That achievement is logically distinct from constructing the critical Wilson–Fisher CFT as a non-Gaussian fixed point of a short-range block-spin map with stable manifold, operator spectrum, and reconstruction estimates of the form (142.126). Hierarchical scalar models, fractional long-range models, fermionic small-parameter models, and rigorous tensor RG supply theorem-level benchmarks for parts of this structure.

For gauge theories the corresponding theorem must be formulated through Definition 142.26. A lattice proof would construct the RG map on gauge-invariant measures and Wilson-line observables with reflection-positivity control. A continuum perturbative proof would construct the same flow in a regularized BV complex, prove the quantum master equation after each RG step, and identify the physical observables as cohomology classes before reconstruction. These are stronger requirements than the computation of beta functions for a finite list of local coefficients.

In four-dimensional scalar ϕ^4, perturbation theory supplies a formal renormalized coordinate system. Existing triviality results and expectations belong to particular regulator classes and limiting procedures. The existence of a UV-complete nonperturbative QFT whose short-distance expansion agrees with four-dimensional ϕ^4 perturbation theory to all orders remains outside the theorem-level results recorded here.

142.19 Relation To Universality Classes

In this monograph, the phrase *universality class* is not used as a synonym for "same symmetries and same dimension." It denotes a convergence statement in a declared topology. The minimal statement is the following.

Definition 142.45 (Wilsonian universality statement)**.** A Wilsonian universality statement consists of:

1. an index set I of microscopic regulators or bare finite-cutoff models;
2. for each $i \in I$, a tuned initial point V_i in a common Banach RG chart $(\mathcal{B}, \|\cdot\|)$, or in charts identified by specified coordinate maps;
3. an RG map $\mathcal{R} : \mathcal{U} \to \mathcal{B}$, a neighborhood $\mathcal{N} \subset \mathcal{U}$, and a fixed point $V_* \in \mathcal{N}$;
4. a topological vector space $\mathcal{O}$ of normalized long-distance observable data, for example finite lists of smeared Schwinger functions in a distribution topology;
5. reconstruction maps
$$\mathrm{Rec}_{i,n} : \mathcal{U} \to \mathcal{O}$$
giving the n-step long-distance observable data of the microscopic model i;
6. a normalization group G acting continuously on $\mathcal{O}$, encoding allowed metric factors, field normalizations, and coordinate rescalings.

This is a theorem-level universality statement if, for every $i \in I$, there are $g_i \in G$ and $O_* \in \mathcal{O}$ such that
$$\mathcal{R}^n(V_i) \in \mathcal{N} \quad (n \gg 1), \qquad \mathcal{R}^n(V_i) \longrightarrow V_* \quad \text{in } \mathcal{B},$$
and
$$g_i \, \mathrm{Rec}_{i,n}(V_i) \longrightarrow O_* \qquad \text{in } \mathcal{O}.$$
The common limit O_* is part of the assertion; without the reconstruction convergence, convergence of local couplings alone is not a QFT universality theorem.

Observable-germ formulation. For an actual QFT the reconstructed observable is rarely a single Banach-space coordinate. It is normally a compatible family of finite tests: finitely many smeared n-point functions, finite lists of Wilson loops, finite momentum windows, finite source derivatives, or finite spectral windows. Thus the topology in Definition 142.45 is most naturally given by a projective system.

Definition 142.46 (Observable-germ universality class)**.** An observable-germ universality structure is a Wilsonian universality statement together with the following refinement of $\mathcal{O}$.

1. A directed set $\mathfrak{A}$ of finite observation windows.
2. For each $\alpha \in \mathfrak{A}$, a Banach space $\mathcal{O}_\alpha$ with norm p_α, and for $\alpha \preceq \beta$ a continuous projection $\pi_{\alpha\beta} : \mathcal{O}_\beta \to \mathcal{O}_\alpha$ satisfying
$$\pi_{\alpha\gamma} = \pi_{\alpha\beta}\pi_{\beta\gamma}, \qquad \alpha \preceq \beta \preceq \gamma.$$
3. The projective-limit observable space
$$\mathcal{O}_{\mathrm{germ}} = \varprojlim_{\alpha\in\mathfrak{A}} \mathcal{O}_\alpha = \{(O_\alpha)_\alpha : \pi_{\alpha\beta}O_\beta = O_\alpha \text{ for } \alpha \preceq \beta\},$$
equipped with the seminorms $p_\alpha(O) = p_\alpha(O_\alpha)$.
4. Reconstruction maps
$$\mathrm{Rec}^\alpha_{i,n} : \mathcal{U} \longrightarrow \mathcal{O}_\alpha$$
compatible with the projections:
$$\pi_{\alpha\beta}\,\mathrm{Rec}^\beta_{i,n} = \mathrm{Rec}^\alpha_{i,n}\,.$$
5. A normalization group G acting by compatible continuous maps on all $\mathcal{O}_\alpha$.

The microscopic regulator i belongs to the observable-germ universality class represented by the normalized germ $O_* \in \mathcal{O}_{\mathrm{germ}}$ if there exists $g_i \in G$ such that, for every finite observation window α,
$$p_\alpha\big(g_i\ \mathrm{Rec}^\alpha_{i,n}(V_i) - (O_*)_\alpha\big) \longrightarrow 0. \tag{142.206}$$
The word *class* therefore refers to an equivalence class of normalized observable germs in this projective topology. Agreement of finitely many critical exponents, amplitude ratios, or correlation functions is a finite-window statement until the remaining windows are controlled.

The preceding definition deliberately uses observable windows rather than the phrase "the same theory." Equality of a few numerical quantities, or even of a large but finite list of reconstructed correlators, is not equality of local QFT data. The bridge to a local quantum field theory requires that the projective family contain enough test-function windows, positivity conditions, covariance identities, and growth bounds to feed one of the reconstruction frameworks developed in Chapters 12 and 2.

Definition 142.47 (Local-QFT-strength observable germ)**.** An observable-germ structure is *local-QFT-strength* for an Euclidean scalar or gauge-invariant observable sector if its directed family $\mathfrak{A}$ contains, as compatible subwindows, the following data.

1. For each finite list of observable labels $a_1, \dots, a_n$ and each finite-dimensional subspace $E \subset \mathcal{S}((\mathbb{R}^D)^n)$ of test functions, a window that records the values of a distribution $S_{a_1\dots a_n} \in \mathcal{S}'((\mathbb{R}^D)^n)$ on E. The projective limit over E recovers the full tempered distribution only after the compatibility and seminorm-bound conditions of Lemma 142.48 are supplied; finite pointwise or moment data alone are a smaller collection of tests.

2. Windows recording the algebraic identities required of the sector: permutation or graded permutation covariance, Euclidean covariance, reflection compatibility for the chosen time direction, and the local source-counterterm subspace whose derivatives are distributions supported on collision diagonals.
3. Windows recording the positivity forms used in reconstruction. In the OS route these are the finite reflection-positivity Gram matrices for test functions supported at positive Euclidean time. In a directly Lorentzian route they are the Wightman positivity forms on the polynomial test-field algebra.
4. The growth or regularity hypotheses needed by the chosen reconstruction theorem. For the OS-to-Wightman route this includes the corrected OS-II growth hypothesis used in Chapter 12; for the Wightman route it includes temperedness, the spectrum condition, locality, and the common invariant domain structure.
5. If vacuum uniqueness or particle data are part of the claim, windows for cluster properties or spectral projections in the declared observable topology.

The normalization group is restricted to transformations of the observable labels, coordinates, metric factors, and source normalizations that preserve these identities and positivity forms.

Projective distribution-window extension. The first item in Definition 142.47 contains a functional analytic obligation that is easy to hide if one speaks only of many finite windows. Fix n and write $X = (\mathbb{R}^D)^n$. For an integer $r \geq 0$, set

$$q_r(f) = \max_{|\alpha| \leq r} \sup_{x \in X} (1 + |x|)^r \, |\partial^\alpha f(x)|, \qquad f \in \mathcal{S}(X), \tag{142.207}$$

where α is a multi-index on X. These seminorms generate the Schwartz topology up to the usual harmless replacement by an equivalent countable family.

Lemma 142.48 (Projective finite windows define a distribution)**.** *Let $\mathcal{E}$ be a directed family of finite-dimensional subspaces of $\mathcal{S}(X)$, ordered by inclusion, and suppose that $E_\infty = \bigcup_{E \in \mathcal{E}} E$ is dense in $\mathcal{S}(X)$. For each $E \in \mathcal{E}$, let $S_E : E \to \mathbb{C}$ be linear. Assume:*

1. *whenever $E \subset F$, $S_F|_E = S_E$;*
2. *there exist $r \in \mathbb{Z}_{\geq 0}$ and $C < \infty$ such that*

$$|S_E(f)| \leq C \, q_r(f) \qquad (E \in \mathcal{E}, \; f \in E). \tag{142.208}$$

Then there is a unique tempered distribution $S \in \mathcal{S}'(X)$ satisfying $S|_E = S_E$ for every $E \in \mathcal{E}$. Moreover S obeys the same bound

$$|S(f)| \leq C \, q_r(f), \qquad f \in \mathcal{S}(X).$$

Proof. Compatibility first produces a single algebraic functional on E_∞. If $f \in E \cap F$, choose $H \in \mathcal{E}$ with $E \subset H$ and $F \subset H$. Then

$$S_E(f) = S_H(f) = S_F(f),$$

so $S_\infty(f) := S_E(f)$ for $f \in E$ is well defined on E_∞. Linearity follows after passing to a common upper window containing the finitely many vectors being combined. The bound (142.208) gives

$$|S_\infty(f)| \le Cq_r(f), \qquad f \in E_\infty.$$

Let $f \in \mathcal{S}(X)$. Choose $f_j \in E_\infty$ with $f_j \to f$ in the Schwartz topology. Then $q_r(f_j - f_k) \to 0$, hence

$$|S_\infty(f_j) - S_\infty(f_k)| \le Cq_r(f_j - f_k),$$

so $S_\infty(f_j)$ is Cauchy in $\mathbb{C}$. If $g_j \in E_\infty$ is another approximating sequence, then $q_r(f_j - g_j) \to 0$, so the two limits agree. Define

$$S(f) = \lim_{j\to\infty} S_\infty(f_j).$$

The definition is linear by applying the same construction to approximating sequences for finite linear combinations, and the estimate $|S(f)| \le Cq_r(f)$ follows by taking the limit of $|S_\infty(f_j)| \le Cq_r(f_j)$. Thus S is continuous on $\mathcal{S}(X)$, hence $S \in \mathcal{S}'(X)$, and it extends each S_E. If $T \in \mathcal{S}'(X)$ is another continuous extension, then $T - S$ vanishes on the dense subspace E_∞; continuity gives $T = S$. □

This lemma isolates the exact theorem-facing content of the phrase "distribution window." Compatibility supplies a single algebraic functional; the uniform Schwartz-seminorm bound supplies tempered continuity. In a genuine Wilsonian RG theorem the estimates must be produced for the Schwinger hierarchy and for the source derivatives that generate the desired local observables. After this distributional object exists, the finite reflection-positivity windows below test whether it is eligible for OS reconstruction.

Cofinal assembly from regulated finite windows. The previous lemma starts from limiting window functionals. An RG proof usually produces regulated window functionals $S_{k,N}$ at scale k, where N records how large a finite test window has been controlled. The following criterion is the diagonal step needed before the limiting functionals in Lemma 142.48 exist.

Lemma 142.49 (Cofinal finite-window assembly)**.** *Let*

$$E_1 \subset E_2 \subset \cdots \subset \mathcal{S}(X)$$

be finite-dimensional subspaces whose union is dense in $\mathcal{S}(X)$. Let $N_k \to \infty$, and suppose that for every k and every $N \le N_k$ there is a linear functional

$$S_{k,N} : E_N \to \mathbb{C}.$$

Assume:

1. *finite-scale compatibility: $S_{k,M}|_{E_N} = S_{k,N}$ whenever $N \le M \le N_k$;*
2. *a uniform seminorm bound: there exist r and C, independent of k and N, such that*

$$|S_{k,N}(f)| \le Cq_r(f), \qquad f \in E_N, \quad N \le N_k;$$

3. cofinal Cauchy control: for each fixed N,

$$\eta_{N,k} = \sup_{\ell \geq k} \sup_{\substack{f \in E_N \\ q_r(f) \leq 1}} |S_{k,N}(f) - S_{\ell,N}(f)| \longrightarrow 0 \tag{142.209}$$

as $k \to \infty$, with the supremum restricted to k, ℓ for which $N \leq N_k, N_\ell$.

Then there are compatible limiting functionals

$$S_N : E_N \to \mathbb{C}, \qquad S_M|_{E_N} = S_N \quad (N \leq M),$$

with $|S_N(f)| \leq C q_r(f)$. Hence Lemma 142.48 produces a unique tempered distribution $S \in \mathcal{S}'(X)$ extending all S_N. Moreover one may choose a strictly increasing scale subsequence k_j with $N_{k_j} \geq j$ and

$$\eta_{N,k_j} \leq 2^{-j}, \qquad 1 \leq N \leq j. \tag{142.210}$$

Along this subsequence every fixed finite window E_N converges to its limit at a quantified rate after $j \geq N$.

Proof. Fix N. Since $N_k \to \infty$, the functionals $S_{k,N}$ are defined for all sufficiently large k. The Cauchy control (142.209) says that $S_{k,N}$ is Cauchy in the operator seminorm

$$\|T\|_{E_N,r} = \sup_{\substack{f \in E_N \\ q_r(f) \leq 1}} |T(f)|.$$

The target space is finite dimensional, so this normed space of linear functionals is complete. Define S_N to be the limit of $S_{k,N}$ in that norm. The bound $|S_N(f)| \leq C q_r(f)$ follows by taking the limit of the uniform bound for $S_{k,N}$.

If $N \leq M$, choose k so large that $M \leq N_k$. Finite-scale compatibility gives

$$S_{k,M}|_{E_N} = S_{k,N}.$$

Taking $k \to \infty$ in the operator seminorms on E_M and E_N gives

$$S_M|_{E_N} = S_N.$$

Thus the limiting windows satisfy the hypotheses of Lemma 142.48, and the tempered distribution follows.

It remains to spell out the diagonal schedule. For a fixed j, the set $\{1, \ldots, j\}$ is finite. Since each $\eta_{N,k} \to 0$, and since $N_k \to \infty$, there exists k_j larger than k_{j-1} such that

$$N_{k_j} \geq j, \qquad \eta_{N,k_j} \leq 2^{-j} \quad (1 \leq N \leq j).$$

Choosing these k_j recursively gives (142.210). For fixed N and $j \geq N$, the definition of η_{N,k_j} bounds the distance from $S_{k_j,N}$ to the limiting functional S_N by 2^{-j} in the same operator seminorm. □

This lemma identifies a common hidden gap in RG reconstruction arguments. Controlling one finite window at each scale is insufficient: the number of controlled windows must be cofinal, the already-controlled windows must be Cauchy as the regulator is removed, and the seminorm bound must be uniform in the window. The lemma supplies the diagonal functional-analytic step. It does not supply the model-specific polymer or cluster estimates that prove the hypotheses.

Lemma 142.50 (Moving finite windows approximate fixed test functions)**.** *Assume the hypotheses of Lemma 142.49. Let $\Pi_N : \mathcal{S}(X) \to E_N$ be linear approximation maps such that, for each $f \in \mathcal{S}(X)$,*

$$q_r(\Pi_N f - f) \longrightarrow 0.$$

Let $S \in \mathcal{S}'(X)$ be the limiting distribution constructed in Lemma 142.49. Then, for every fixed $f \in \mathcal{S}(X)$,

$$S_{k,N_k}(\Pi_{N_k} f) \longrightarrow S(f) \qquad (k \to \infty). \tag{142.211}$$

Proof. The point is to avoid confusing a moving finite window with convergence on a fixed test function. Fix N, and take k so large that $N \le N_k$. By finite-scale compatibility,

$$S_{k,N_k}(\Pi_N f) = S_{k,N}(\Pi_N f).$$

Therefore

$$\begin{aligned}\big|S_{k,N_k}(\Pi_{N_k} f) - S(f)\big| \le{}& \big|S_{k,N_k}(\Pi_{N_k} f - \Pi_N f)\big| \\ &+ \big|S_{k,N}(\Pi_N f) - S_N(\Pi_N f)\big| + \big|S(\Pi_N f - f)\big|.\end{aligned}$$

The uniform seminorm bound gives

$$\big|S_{k,N_k}(\Pi_{N_k} f - \Pi_N f)\big| \le C q_r(\Pi_{N_k} f - \Pi_N f) \le C q_r(\Pi_{N_k} f - f) + C q_r(f - \Pi_N f).$$

The limiting distribution satisfies the same bound $|S(g)| \le C q_r(g)$, because it is the continuous extension of the compatible finite-window limits. Hence

$$|S(\Pi_N f - f)| \le C q_r(\Pi_N f - f).$$

For fixed N, the middle term tends to zero as $k \to \infty$, by the fixed-window convergence in Lemma 142.49. Taking the upper limit in k gives

$$\limsup_{k\to\infty} \big|S_{k,N_k}(\Pi_{N_k} f) - S(f)\big| \le 2C q_r(\Pi_N f - f).$$

Finally let $N \to \infty$. The assumed approximation property of Π_N gives (142.211). ☐

This lemma is the distributional diagonal step used when the RG construction controls only a scale-dependent finite test space. It still assumes the uniform seminorm bound and fixed-window Cauchy control from the preceding lemma. If the projection error $q_r(\Pi_N f - f)$ is not controlled in the same seminorm that bounds the finite functionals, or if N_k does not tend to infinity, the moving-window values do not define the continuum action of a tempered distribution on the fixed test function f.

Finite OS-positivity Gram-matrix bounds. The positivity windows in Definition 142.47 are not decorative. They are finite Gram matrices whose limits must be controlled before Osterwalder–Schrader reconstruction can be invoked. Fix a finite family $\mathcal{F} = \{F_1, \dots, F_m\}$ of polynomial Euclidean observable test vectors supported at positive Euclidean time, and let Θ be the Euclidean time reflection with the conjugation appropriate to the observable sector. A regulated Schwinger hierarchy gives the finite matrix

$$G_{k,\mathcal{F};ij} = S_k\big((\Theta F_i) F_j\big), \qquad 1 \le i, j \le m. \tag{142.212}$$

Reflection positivity at scale k is the assertion that

$$c^* G_{k,\mathcal{F}} c \geq 0 \qquad \text{for every } c \in \mathbb{C}^m.$$

If the window is part of a reconstruction-strength RG estimate, the entries of $G_{k,\mathcal{F}}$ converge to entries

$$G_{*,\mathcal{F};ij} = S_*\big((\Theta F_i) F_j\big).$$

For each fixed c, continuity of the finite quadratic form gives

$$c^* G_{*,\mathcal{F}} c = \lim_{k\to\infty} c^* G_{k,\mathcal{F}} c \geq 0.$$

Thus positivity passes to the limit for this finite family. The quantitative finite-window form is also useful. If $G_{k,\mathcal{F}}$ is Hermitian, $G_{k,\mathcal{F}} \geq \ell\, 1_m$, and the limiting-window error satisfies

$$|G_{*,\mathcal{F};ij} - G_{k,\mathcal{F};ij}| \leq \epsilon \qquad (1 \leq i, j \leq m),$$

then

$$c^* G_{*,\mathcal{F}} c \geq \big(\ell - m\epsilon\big) \|c\|_2^2. \tag{142.213}$$

Indeed,

$$|c^*(G_{*,\mathcal{F}} - G_{k,\mathcal{F}})c| \leq \epsilon \Big(\sum_{i=1}^m |c_i|\Big)^2 \leq m\epsilon\, \|c\|_2^2.$$

Equation (142.213) is a finite Gram-matrix lower bound, not an OS reconstruction theorem. For this entrywise form to give a positive lower bound on a family of size m, one needs $\epsilon < \ell/m$. Hence a directed-family OS theorem cannot be obtained by fixing an error tolerance independently of the number of test vectors: if $\mathcal{F}_N$ is an increasing family with $m_N = |\mathcal{F}_N|$, the estimate must supply $\epsilon_{N,k}$ and $\ell_{N,k}$ with $\ell_{N,k} - m_N \epsilon_{N,k} \geq 0$ in the limit being taken, or else use a stronger operator-norm estimate $\|G_{*,\mathcal{F}_N} - G_{k,\mathcal{F}_N}\|_{\ell^2\to\ell^2} \leq \epsilon^{\mathrm{op}}_{N,k}$ with $\ell_{N,k} - \epsilon^{\mathrm{op}}_{N,k} \geq 0$. Producing such family-size-compatible error control from RG estimates is a load-bearing part of the theorem. OS positivity for the continuum hierarchy requires this bound, or the corresponding limiting argument, for every finite positive-time test family in the directed observable germ. Agreement of separated two-point entries, critical exponents, or a finite list of amplitudes does not supply the missing Gram windows.

There is a closely related semidefinite version, which is often the form available in an RG proof with regulator defects. Suppose that the regulated window is not strictly bounded below but satisfies

$$c^* G_{k,\mathcal{F}} c \geq -\delta_{k,\mathcal{F}} \|c\|_2^2 \qquad (c \in \mathbb{C}^m)$$

and that the entrywise error from this regulated window to the candidate limit is at most $\epsilon_{k,\mathcal{F}}$. Then the same estimate gives

$$c^* G_{*,\mathcal{F}} c \geq -\big(\delta_{k,\mathcal{F}} + m\epsilon_{k,\mathcal{F}}\big) \|c\|_2^2. \tag{142.214}$$

For a fixed family $\mathcal{F}$, exact finite reflection positivity and entrywise convergence imply positivity of the limiting window by closedness of the finite positive cone. For a directed sequence of growing windows, a proof using the quantitative defect route must instead produce a common scale schedule such that

$$\delta_{k_N,\mathcal{F}_N} + |\mathcal{F}_N| \epsilon_{k_N,\mathcal{F}_N} \longrightarrow 0.$$

An operator-norm estimate replaces the factor $|\mathcal{F}_N|\epsilon$ by the corresponding operator-norm error. The distinction matters: exact reflection positivity is a finite-regulator structural property, while approximate positivity with defects is an estimate whose constants must scale with the observation window.

Lemma 142.51 (Directed OS-positive form assembly)**.** *Let*

$$P_1 \subset P_2 \subset \cdots$$

be finite-dimensional subspaces of the positive-time polynomial test algebra, with dense algebraic union P_∞ in the positive-time test topology used for OS reconstruction. Let $N_k \to \infty$. For every $N \le N_k$, suppose the regulated Schwinger hierarchy supplies a Hermitian sesquilinear form

$$Q_{k,N} : P_N \times P_N \longrightarrow \mathbb{C}$$

such that:

1. *$Q_{k,M}|_{P_N \times P_N} = Q_{k,N}$ whenever $N \le M \le N_k$;*
2. *$Q_{k,N}(F, F) \ge 0$ for every $F \in P_N$;*
3. *after choosing any norm $\|\cdot\|_N$ on P_N,*

$$\eta_{N,k} = \sup_{\ell \ge k} \sup_{\substack{\|F\|_N \le 1 \\ \|G\|_N \le 1}} |Q_{k,N}(F, G) - Q_{\ell,N}(F, G)| \longrightarrow 0 \tag{142.215}$$

   for each fixed N, with the supremum restricted to scales for which $N \le N_k, N_\ell$.

Then there is a unique compatible positive Hermitian form

$$Q_* : P_\infty \times P_\infty \longrightarrow \mathbb{C}$$

whose restriction to P_N is the limit of $Q_{k,N}$ for every fixed N. If, in addition, the limiting forms obey the continuity estimate required by the OS chapter on a dense positive-time family, then Q_* extends to the positive-time OS pre-Hilbert form. The lemma by itself does not supply strong continuity of the time-translation semigroup or the OS-II growth hypothesis.

Proof. For fixed N, the space of sesquilinear forms on P_N is finite dimensional and complete in the operator norm induced by $\|\cdot\|_N$. Condition (142.215) therefore gives a limit form $Q_{*,N}$ on P_N. Positivity passes to the limit because

$$Q_{*,N}(F, F) = \lim_{k \to \infty} Q_{k,N}(F, F) \ge 0$$

for each fixed $F \in P_N$. Hermiticity passes in the same way.

If $N \le M$, finite-scale compatibility gives

$$Q_{k,M}|_{P_N \times P_N} = Q_{k,N}$$

for all sufficiently large k. Passing to the finite-dimensional limits on P_M and P_N gives

$$Q_{*,M}|_{P_N \times P_N} = Q_{*,N}.$$

Thus one defines $Q_*(F,G) = Q_{*,N}(F,G)$ whenever $F, G \in P_N$; the preceding compatibility makes this independent of N. Positivity and Hermiticity on P_∞ follow by choosing a single P_N that contains the finitely many vectors involved. The final extension statement is exactly the continuity step in OS reconstruction: without a bound in the positive-time topology, a positive algebraic form on the dense union need not define a continuous form on the desired test completion. □

The quantitative entrywise estimate (142.213) is one way to prove the Cauchy and positivity inputs of Lemma 142.51. For a nested family with m_N basis vectors it must be used with a scale-dependent error schedule satisfying

$$\liminf_{k\to\infty}\bigl(\ell_{N,k} - m_N\epsilon_{N,k}\bigr) \geq 0 \qquad \text{for each fixed } N.$$

If the goal is a uniform lower bound on an increasing family, the right condition is correspondingly uniform in N; a fixed entrywise tolerance is not enough. This is the precise mathematical content behind the warning that finite OS Gram windows must be controlled as a directed family.

When universality reconstructs the same local QFT. Suppose a Wilsonian universality statement has a local-QFT-strength observable germ and that the normalized reconstructions of all microscopic regulators converge to a common germ O_*. The common germ then determines a common Schwinger or Wightman hierarchy in the sense of the chosen reconstruction framework. Applying the reconstruction theorem proved earlier in the monograph gives the corresponding Hilbert space, vacuum vector, fields or observable algebra, and spacetime symmetry representation. Thus the phrase "the regulators define the same local QFT" means: after the declared normalizations, their full local-QFT-strength observable germs converge to one reconstruction input, and the reconstructed presentations are identified by the uniqueness clause of that reconstruction theorem. Contact-term choices are part of the source-local extension data; they may change extensions across collision diagonals without changing the separated-point germ.

This paragraph is not an additional fixed-point theorem. It records the dependency chain that a model-specific universality theorem must establish: RG attraction in a Banach chart, convergence of the full projective observable germ, verification of OS or Wightman positivity and growth hypotheses, and then reconstruction. A calculation of critical exponents, a finite bootstrap table, a finite tensor-window comparison, or a lattice finite-size scaling plot supplies only the finite windows it actually controls.

The finite-window form is still useful because it is the object produced by simulations, rigorous finite-volume estimates, and perturbative calculations. If $\mathfrak{A}_0 \subset \mathfrak{A}$ is finite and, for each $\alpha \in \mathfrak{A}_0$,

$$p_\alpha\bigl(g_i\ \mathrm{Rec}^\alpha_{i,n}(V_i) - (O_*)_\alpha\bigr) \leq e_{\alpha,i,n}, \qquad p_\alpha\bigl(g_j\ \mathrm{Rec}^\alpha_{j,n}(V_j) - (O_*)_\alpha\bigr) \leq e_{\alpha,j,n}, \tag{142.216}$$

then the triangle inequality gives the finite-window estimate

$$p_\alpha\bigl(g_i\ \mathrm{Rec}^\alpha_{i,n}(V_i) - g_j\ \mathrm{Rec}^\alpha_{j,n}(V_j)\bigr) \leq e_{\alpha,i,n} + e_{\alpha,j,n}, \qquad \alpha \in \mathfrak{A}_0. \tag{142.217}$$

A finite-window estimate becomes substantially more informative when its error is expressed in RG coordinates. Fix a finite window $\alpha \in \mathfrak{A}$ and a scale n_0 after which two normalized microscopic data lie in a common stable chart at V_*. Write $u \in E_u$ for the relevant coordinate and $s \in E_s$ for the stable coordinate in that chart. Suppose that, after the allowed normalizations, the relevant mismatch is bounded by

$$\|u_i(n_0) - u_j(n_0)\| \leq \delta_u,$$

and that the stable mismatch d_ℓ after ℓ further RG steps obeys the finite recurrence

$$d_{\ell+1} \le \theta d_\ell + \eta_\ell, \qquad d_0 \le \delta_s, \qquad 0 \le \theta < 1. \tag{142.218}$$

Here η_ℓ is the one-step comparison defect: it may come from different microscopic blocking kernels, a finite auxiliary-to-target intertwining error, or a controlled truncation of the stable coordinate. Assume also that the reconstructed α-window is Lipschitz in the chart, with constants $C_{u,\alpha}$ and $C_{s,\alpha}$, and that the source-tail and normalization errors in this window at depth m are bounded by $\Xi_{\alpha,m}$. Then the finite observable-window bound is

$$\begin{aligned} p_\alpha\big(g_i \operatorname{Rec}^\alpha_{i,n_0+m}(V_i) - g_j \operatorname{Rec}^\alpha_{j,n_0+m}(V_j)\big) &\le C_{u,\alpha}\delta_u + C_{s,\alpha}\theta^m\delta_s \\ &\quad + C_{s,\alpha}\sum_{\ell=0}^{m-1}\theta^{m-1-\ell}\eta_\ell + \Xi_{\alpha,m}. \end{aligned} \tag{142.219}$$

This is the finite-scale form of a Wilsonian universality comparison. Its derivation is the iteration of (142.218), followed by the Lipschitz bound for the reconstruction map in the chosen window. Exact tuning means $\delta_u = 0$. A theorem-level universality result must prove constants for every window in a directed family and show that the right-hand side of (142.219) tends to zero along the claimed continuum or long-distance limit. Without that directed-window limit, (142.219) remains a precise finite observation bound.

The theorem-level burden is the passage from these finite bounds to (142.206) for the directed family of observation windows.

Quantifier order in the directed-window limit. The finite estimate (142.219) becomes a universality theorem only with the projective-limit quantifiers in the right order. For every window $\alpha \in \mathfrak{A}$ and every $\varepsilon > 0$, after the window-dependent constants $C_{u,\alpha}$, $C_{s,\alpha}$, the normalization convention, and the source-tail bound have been fixed, one must exhibit a depth $m_{\alpha,\varepsilon}$ such that for all $m \ge m_{\alpha,\varepsilon}$

$$C_{u,\alpha}\delta_u + C_{s,\alpha}\theta^m\delta_s + C_{s,\alpha}\sum_{\ell=0}^{m-1}\theta^{m-1-\ell}\eta_\ell + \Xi_{\alpha,m} < \varepsilon. \tag{142.220}$$

If exact relevant tuning has not been proved, the first term in (142.220) is a nondecaying obstruction for the window α. If the one-step defects do not have a summable tail, the convolution term has a positive error floor even when $\theta < 1$. If the source-tail and normalization errors are controlled only for a fixed finite collection of windows, the conclusion is exactly that finite collection, not the projective germ.

For a cofinal nested family α_N, the theorem must provide a schedule

$$N \longmapsto m_N$$

such that the estimates for α_N hold on compatible regulator tails and the right-hand side of (142.220) tends to zero for each fixed α_N. The constants may grow with N; the proof must show that the chosen depth and regulator tail outpace that growth. This is the same directed-family issue as in the OS Gram bounds above: a finite numerical window, or a family of windows chosen after the calculation, is not a substitute for a cofinal projective-limit estimate.

Remark 142.52 (Universality as an equivalence relation after reconstruction). Assume the data of Definition 142.45, and suppose the displayed convergence to a common O_* holds. Define $i \sim j$ if there exist $g_i, g_j \in G$ such that

$$g_i \operatorname{Rec}_{i,n}(V_i) - g_j \operatorname{Rec}_{j,n}(V_j) \longrightarrow 0 \qquad \text{in } \mathcal{O}.$$

Then $\sim$ is an equivalence relation on I. Reflexivity holds by taking the same normalization on both sides. Symmetry follows because $\mathcal{O}$ is a topological vector space: if a sequence of differences tends to zero, the sequence with the opposite sign also tends to zero.

For transitivity, suppose $i \sim j$ and $j \sim k$. Then there are normalizations $g_i, g_j, h_j, h_k \in G$ for which

$$g_i \operatorname{Rec}_{i,n}(V_i) - g_j \operatorname{Rec}_{j,n}(V_j) \to 0, \qquad h_j \operatorname{Rec}_{j,n}(V_j) - h_k \operatorname{Rec}_{k,n}(V_k) \to 0.$$

In the theorem-level situation above, each normalized sequence also converges to the same orbit representative O_* after some allowed normalization. Choose the normalizations $a_i, a_j, a_k \in G$ that appear in Definition 142.45. Then

$$a_i \operatorname{Rec}_{i,n}(V_i) \to O_*, \qquad a_k \operatorname{Rec}_{k,n}(V_k) \to O_*,$$

so

$$a_i \operatorname{Rec}_{i,n}(V_i) - a_k \operatorname{Rec}_{k,n}(V_k) \to 0.$$

Thus $i \sim k$. The proof uses the existence of a common reconstructed limit and therefore controls more than pairwise similarity of formal coupling lists.

The common-limit hypothesis is essential to the way this remark is used in the monograph. Pairwise finite-window comparisons with unrelated choices of normalization do not by themselves define a theorem-level universality relation: the normalization of j in the i-j comparison need not be compatible with the normalization of j in the j-k comparison, and agreement in one finite window does not control the remaining directed windows. The object that carries the equivalence relation is the reconstructed observable germ modulo the declared normalization group, not a list of matched exponents or a chain of pairwise plots.

Thus a universality class may be summarized by a triple

$$(\mathcal{R}, \mathcal{N}, V_*)$$

together with the observable topology, reconstruction maps, and normalization group of Definition 142.45. Symmetry and dimension may guide the choice of I and the list of relevant coordinates, but they do not by themselves define the class. The theorem is the attraction and reconstruction statement.

Correction-to-scaling data. The limiting reconstructed datum is only the first layer of a Wilsonian universality theorem. A sharper theorem also identifies the controlled approach to the limit. This matters because the amplitudes of irrelevant directions depend on the microscopic regulator, while the decay exponents and the reconstructed correction distributions are fixed by the RG chart and the fixed point.

Definition 142.53 (Correction-to-scaling datum)**.** Assume the data of Definition 142.45. A correction-to-scaling datum of order A consists of:

1. a locally convex topology on $\mathcal{O}$, specified by seminorms $\{p_\beta\}$;
2. a stable coordinate chart near V_* in which the tuned trajectories have stable coordinates $w_i(n) \in E_s$;
3. vectors $e_1, \dots, e_A \in E_s$ and positive numbers $\omega_1 \leq \cdots \leq \omega_A$ such that the linearized RG map sends
$$e_a \longmapsto L^{-\omega_a} e_a;$$
4. a number $\omega_{A+1} > \omega_A$, or another declared remainder rate, bounding the omitted stable directions;
5. constants $c_{i,a} \in \mathbb{C}$, depending on the microscopic regulator i, and remainders $\rho_i(n) \in E_s$ satisfying
$$w_i(n) = \sum_{a=1}^{A} c_{i,a} L^{-n\omega_a} e_a + \rho_i(n), \qquad \|\rho_i(n)\|_{E_s} \leq C_i\big(L^{-n\omega_{A+1}} + L^{-2n\omega_1}\big); \tag{142.221}$$
6. differentiability of the normalized reconstruction map at V_*, with first derivative
$$D\,\mathrm{Rec}_* : E_s \longrightarrow \mathcal{O}$$
and a quadratic remainder estimate in every seminorm p_β.

The associated correction distributions are

$$\mathcal{C}_a = D\,\mathrm{Rec}_*(e_a) \in \mathcal{O}.$$

Under these hypotheses the normalized reconstructed observables have the expansion

$$g_i\,\mathrm{Rec}_{i,n}(V_i) = O_* + \sum_{a=1}^{A} c_{i,a} L^{-n\omega_a} \mathcal{C}_a + \mathcal{E}_i(n), \tag{142.222}$$

where, for every seminorm p_β, there is a constant $C_{\beta,i}$ such that

$$p_\beta(\mathcal{E}_i(n)) \leq C_{\beta,i}\big(L^{-n\omega_{A+1}} + L^{-2n\omega_1}\big). \tag{142.223}$$

Indeed, write the normalized reconstruction map in the stable chart as

$$\mathrm{Rec}_*(V_* + w) = O_* + D\,\mathrm{Rec}_*(w) + Q(w), \qquad p_\beta(Q(w)) \leq C_\beta \|w\|_{E_s}^2$$

for w in a sufficiently small stable neighborhood. Substitution of (142.221) gives the linear correction terms and puts the stable-coordinate remainder together with $Q(w_i(n))$ into (142.223).

This is the precise content behind the common phrase "same universality class but different corrections to scaling." It is not a theorem about a model until the stable-coordinate expansion, the reconstruction derivative, and the seminorm estimates have all been proved in the chosen RG framework. The coefficients $c_{i,a}$ are microscopic coordinates. The exponents ω_a and correction distributions $\mathcal{C}_a$ are fixed-point data in the declared Wilsonian chart and observable topology.

Second-order reconstruction corrections. The linear correction ledger above is sufficient only when the quadratic reconstruction term is truly below the retained accuracy. Suppose, in the same stable chart, that the normalized reconstruction map has a second Frechet derivative and a cubic seminorm remainder,

$$\mathrm{Rec}_*(V_*+w)=O_*+D\,\mathrm{Rec}_*(w)+\frac{1}{2}D^2\,\mathrm{Rec}_*(w,w)+R_3(w),\qquad p_\beta(R_3(w))\le C_\beta\|w\|_{E_s}^3.$$

Substituting the stable-coordinate expansion gives the contact-free second-order correction terms

$$\frac{1}{2}\sum_{a,b=1}^{A}c_{i,a}c_{i,b}L^{-n(\omega_a+\omega_b)}\mathcal{C}_{ab}^{(2)},\qquad \mathcal{C}_{ab}^{(2)}:=D^2\,\mathrm{Rec}_*(e_a,e_b).\tag{142.224}$$

These terms are not optional bookkeeping if their exponents fall within the declared error window. If $2\omega_1<\omega_2$, the first subleading observable correction after the e_1 term is the nonlinear $\mathcal{C}_{11}^{(2)}$ contribution, not the next linear stable eigendirection. If $2\omega_1=\omega_2$, the coefficient at the common exponent is

$$c_{i,2}\mathcal{C}_2+\frac{1}{2}c_{i,1}^2\mathcal{C}_{11}^{(2)}.$$

Thus a fit that identifies the $L^{-2n\omega_1}$ amplitude with only the linear coordinate $c_{i,2}$ has silently assumed either vanishing quadratic reconstruction or a coordinate choice that removes it with a controlled normal-form argument. A theorem-level correction-to-scaling statement must declare which quadratic and higher nonlinear reconstruction terms are retained, and must put the rest into the same seminorm remainder estimate as the omitted stable coordinates.

Nonsemisimple stable blocks. The definition above is the semisimple case. If the stable linearization is not diagonalizable, the correction-to-scaling datum must record the Jordan chains; otherwise the theorem silently drops polynomial factors. Suppose a stable block has $\lambda_b=L^{-\omega_b}$, $\omega_b>0$, and a chain $e_{b,0},\dots,e_{b,m_b-1}$ with

$$D\mathcal{R}|_{V_*}e_{b,r}=\lambda_b e_{b,r}+\lambda_b e_{b,r-1},\qquad e_{b,-1}:=0.\tag{142.225}$$

Equivalently, on the block $D\mathcal{R}|_{V_*}=\lambda_b(1+N_b)$ with $N_b^{m_b}=0$. The exact iterate is

$$(D\mathcal{R}|_{V_*})^n e_{b,r}=L^{-n\omega_b}\sum_{q=0}^{r}\binom{n}{q}e_{b,r-q}.\tag{142.226}$$

Thus a stable-coordinate expansion with amplitudes $c_{i,b,r}$ gives

$$w_i(n)=\sum_b L^{-n\omega_b}\sum_{r=0}^{m_b-1}c_{i,b,r}\sum_{q=0}^{r}\binom{n}{q}e_{b,r-q}+\rho_i(n),\tag{142.227}$$

with a remainder bounded in the declared stable norm. Applying the linearized reconstruction map gives

$$g_i\,\mathrm{Rec}_{i,n}(V_i)=O_*+\sum_b L^{-n\omega_b}\sum_{r=0}^{m_b-1}c_{i,b,r}\sum_{q=0}^{r}\binom{n}{q}\mathcal{C}_{b,r-q}+\mathcal{E}_i(n),\qquad \mathcal{C}_{b,r}:=D\,\mathrm{Rec}_*(e_{b,r}).\tag{142.228}$$

The factor $n^{m_b-1}L^{-n\omega_b}$ is still decaying, but it is not a pure power law. In finite-volume or numerical RG comparisons, fitting corrections only by $L^{-n\omega_b}$ therefore tests a semisimplicity hypothesis about the stable block. A theorem-level correction-to-scaling statement must either prove semisimplicity in the chosen chart or include the binomial factors in (142.228), together with seminorm bounds for the nilpotent-chain remainder.

Marginally irrelevant normal forms. A zero linear exponent is even more delicate. Calling a coordinate "marginal" only records the linearized RG map. Logarithmic corrections come from a nonlinear normal form and from a remainder estimate in the same Banach chart. The basic one-dimensional mechanism is the following.

Lemma 142.54 (Marginally irrelevant reciprocal estimate)**.** *Let $m_n \in (0,\rho]$ be a scalar RG coordinate satisfying*

$$m_{n+1} = m_n - \beta m_n^2 + R_n, \qquad |R_n| \le c m_n^3, \tag{142.229}$$

with $\beta > 0$ and $(\beta + c\rho)\rho \le \frac{1}{2}$. Then $m_{n+1} > 0$ and

$$\left| \frac{1}{m_{n+1}} - \frac{1}{m_n} - \beta \right| \le 2(\beta^2 + c + \beta c\rho) m_n. \tag{142.230}$$

If also $c\rho < \beta$, then

$$\frac{1}{m_n} = \frac{1}{m_0} + \beta n + O\left(\sum_{j<n} m_j \right), \qquad m_j \le \frac{1}{m_0^{-1} + (\beta - c\rho) j}.$$

Hence the nonlinear normal form gives

$$m_n = \frac{1}{\beta n + O(\log n)}.$$

Proof. Write $R_n = m_n^3 \eta_n$, $|\eta_n| \le c$. Then

$$m_{n+1} = m_n(1 - \beta m_n + \eta_n m_n^2).$$

The denominator is at least $1 - (\beta + c\rho)\rho \ge \frac{1}{2}$, so $m_{n+1} > 0$. Moreover

$$\frac{1}{m_{n+1}} - \frac{1}{m_n} = \frac{\beta - \eta_n m_n}{1 - \beta m_n + \eta_n m_n^2},$$
$$\frac{1}{m_{n+1}} - \frac{1}{m_n} - \beta = \frac{\beta^2 m_n - \eta_n m_n - \beta \eta_n m_n^2}{1 - \beta m_n + \eta_n m_n^2}.$$

Taking absolute values gives (142.230). The last assertion follows by summing (142.230). For the displayed bound on m_j, use

$$m_{j+1} \le m_j - (\beta - c\rho) m_j^2$$

and invert. The resulting harmonic majorant has logarithmic partial sums. ☐

Thus a logarithmic correction in a Wilsonian theorem is not supplied by the word marginal. The theorem must identify the marginal coordinate, put its flow into a normal form such as (142.229), prove the sign of β, bound the cubic and irrelevant-tail remainders, and transport the resulting reciprocal estimate through the same reconstruction seminorms used for power-law corrections. An exactly marginal coordinate is the different case $\beta = 0$ together with a flat normal form or a symmetry/geometric argument that produces a genuine fixed-point manifold.

Open Problem 142.55 (Nonperturbative Wilsonian fixed points). Construct non-Gaussian Wilsonian fixed points in three and four dimensions as fixed points of explicit RG maps on Banach spaces of interactions, with stable manifolds, operator spectra, and reconstruction of the associated local QFT data. The open directions include: short-range scalar critical fixed points with distributional reconstruction; gauge-compatible fixed points in the lattice or BV sense of Definition 142.26; and estimates that identify the limiting local, extended, and cohomological observables rather than only the local coordinate flow.

Chapter 143

Relation Between Lattice And Continuum Local QFT

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter explains how finite-lattice data become continuum local QFT data. The passage requires scaling maps for observables, convergence of correlation functions as distributions, reflection positivity, and reconstruction or local-net comparison.

143.1 Scaling Maps

Let $a > 0$ be the lattice spacing and let $\Lambda_a = a\mathbb{Z}^d \cap V$ inside a finite region V. A lattice field $\phi_a(x)$ becomes a continuum distribution only after a scaling map is specified:

$$\Phi_a(f) = Z_a^{1/2} a^d \sum_{x \in \Lambda_a} f(x)\phi_a(x),$$

where $f \in C_c^\infty(V)$ and Z_a is the field normalization. Composite operators require a mixing matrix

$$\mathcal{O}_{A,a}^{\mathrm{ren}} = \sum_B Z_{AB}(a)\mathcal{O}_{B,a}.$$

The operator basis and normalization are part of the continuum-limit datum.

A useful way to remove a common normalization ambiguity is to use cell averages. Let $Q_a(x)$ be the half-open cube of side a centered at $x \in a\mathbb{Z}^d$, and define

$$(I_a f)(x) = a^{-d} \int_{Q_a(x)} f(u)\, d^d u. \tag{143.1}$$

The smeared lattice field is then

$$\Phi_a(f) = Z_a^{1/2} a^d \sum_{x \in a\mathbb{Z}^d} (I_a f)(x) \phi_a(x). \tag{143.2}$$

This is exactly the pairing of the lattice field with the piecewise-constant test function

$$f_a(u) = \sum_{x \in a\mathbb{Z}^d} (I_a f)(x) \mathbf{1}_{Q_a(x)}(u). \tag{143.3}$$

Cell averages approximate test functions. Let $K \subset \mathbb{R}^d$ be compact and let $f, h \in C_c^1(\mathbb{R}^d)$ have support contained in K. Then

$$\left| a^d \sum_{x \in a\mathbb{Z}^d} (I_a f)(x)(I_a h)(x) - \int_{\mathbb{R}^d} f(u) h(u)\, d^d u \right| \le C_K a \big(\|f\|_{C^1} \|h\|_{C^0} + \|f\|_{C^0} \|h\|_{C^1} \big), \tag{143.4}$$

where C_K depends only on a fixed compact enlargement of K. In particular $f_a \to f$ in L^2_{loc}. On each cell $Q_a(x)$,

$$|(I_a f)(x) - f(u)| \le c_d a \|f\|_{C^1}, \qquad |(I_a h)(x) - h(u)| \le c_d a \|h\|_{C^1}$$

for all $u \in Q_a(x)$, by the mean-value theorem and $\operatorname{diam} Q_a(x) = \sqrt{d}\, a$. Therefore

$$|(I_a f)(x)(I_a h)(x) - f(u)h(u)| \le c_d' a \big(\|f\|_{C^1} \|h\|_{C^0} + \|f\|_{C^0} \|h\|_{C^1} \big) + c_d'' a^2 \|f\|_{C^1} \|h\|_{C^1}.$$

Integrating this estimate over the cells meeting the compact enlargement of K gives (143.4), after increasing C_K.

Definition 143.1 (Scaling-limit datum)**.** A scaling-limit datum for a lattice regulator family consists of:

regulator spaces	Ω_a and probability states $\langle - \rangle_a$,
scaling trajectory	couplings and boundary conditions as functions of a,
test spaces	$\mathcal{D}_i$ for each intended continuum field or operator,
operator maps	$\mathcal{O}_{i,a}(f)$ for $f \in \mathcal{D}_i$,
normalizations	$Z_{ij}(a)$ and operator-mixing matrices,
topology	the seminorms in which all correlators converge.

Without these data the phrase "the continuum limit of a lattice operator" does not define a mathematical object.

143.2 Distributional Convergence

The n-point Schwinger distributions are

$$S_{n,a}(f_1, \ldots, f_n) = \langle \Phi_a(f_1) \cdots \Phi_a(f_n) \rangle_a .$$

A continuum Schwinger function S_n is obtained when

$$\lim_{a \to 0} S_{n,a}(f_1, \ldots, f_n) = S_n(f_1, \ldots, f_n)$$

for all test functions in the chosen class, with compatible infinite-volume and continuum limits. Pointwise lattice correlators are auxiliary data; the continuum field is distributional.

The next criterion is the practical form in which distributional convergence is often verified. Let $\mathcal{D}$ be a test-function space equipped with an increasing family of seminorms $p_1, p_2, \ldots$. For functions supported in a compact set K, a typical seminorm is

$$p_{K,r}(f) = \max_{|\alpha| \le r} \sup_{x \in K} |\partial^\alpha f(x)|.$$

Dense-test convergence plus uniform seminorm control. Let T_a be a family of n-linear functionals on $\mathcal{D}$, and let $D_0 \subset \mathcal{D}$ be dense in the seminorm used below. Assume there are a constant C and a continuous seminorm p such that

$$|T_a(f_1, \ldots, f_n)| \le C\, p(f_1) \cdots p(f_n) \tag{143.5}$$

for all sufficiently small a, and assume $T_a(g_1, \ldots, g_n)$ converges for all $g_i \in D_0$. Then there is a unique continuous n-linear functional T satisfying

$$\lim_{a \to 0} T_a(f_1, \ldots, f_n) = T(f_1, \ldots, f_n) \qquad (f_i \in \mathcal{D}). \tag{143.6}$$

For $g_i \in D_0$, define $T(g_1, \ldots, g_n)$ as the limit. The bound (143.5) passes to the limit on D_0, so T is uniformly continuous with respect to p on D_0^n. It therefore has a unique continuous extension to $\mathcal{D}^n$. For general $f_i \in \mathcal{D}$, choose $g_i \in D_0$ with

$$p(f_i - g_i) < \epsilon.$$

Multilinearity and (143.5) give, uniformly in a,

$$|T_a(f_1, \ldots, f_n) - T_a(g_1, \ldots, g_n)| \le C' \epsilon \prod_i (1 + p(f_i))$$

after replacing one entry at a time. The same estimate holds for T. Taking first $a \to 0$ on the dense tuple $g_1, \ldots, g_n$, and then $\epsilon \to 0$, proves (143.6).

This argument isolates the countability step after the uniform seminorm bound has been proved: convergence may be checked on a countable dense family of test functions. The hard constructive work is the bound (143.5), not the extension argument.

143.3 Random-Walk Representation of a Massive Lattice Covariance

The most concrete bridge from lattice locality to continuum estimates is the path expansion of a massive lattice Green function. Let $G = (V, E)$ be a finite undirected r-regular graph without self-loops, let A be its adjacency matrix, and let $m^2 > 0$. Define

$$K_m = (m^2 + r)I - A. \tag{143.7}$$

For vertices x, y, let $N_n(x, y)$ be the number of length-n nearest neighbor paths from x to y, equivalently $N_n(x, y) = (A^n)_{xy}$.

Finite-graph random-walk resolvent. The matrix K_m is positive definite and

$$(K_m^{-1})_{xy} = \sum_{n=0}^{\infty} \frac{N_n(x, y)}{(m^2 + r)^{n+1}}. \tag{143.8}$$

Equivalently, if $P = A/r$ is the simple-random-walk transition matrix and

$$\rho = \frac{r}{m^2 + r} < 1,$$

then

$$(K_m^{-1})_{xy} = \frac{1}{m^2 + r} \sum_{n=0}^{\infty} \rho^n P^n(x, y). \tag{143.9}$$

For any vector u,

$$u^* K_m u = m^2 \sum_x |u_x|^2 + \frac{1}{2} \sum_{x \sim y} |u_x - u_y|^2, \tag{143.10}$$

where each unoriented edge is counted once. Thus K_m is positive definite. Since the operator norm of A on $\ell^2(V)$ is at most r,

$$\left\| \frac{A}{m^2 + r} \right\| < 1.$$

The Neumann series converges absolutely in operator norm:

$$K_m^{-1} = \frac{1}{m^2 + r} \left(I - \frac{A}{m^2 + r} \right)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{(m^2 + r)^{n+1}}.$$

Taking the (x, y) matrix entry gives (143.8). Replacing A^n by $r^n P^n$ gives (143.9).

For a cubic lattice with spacing a, the free covariance is obtained from the same formula after the replacement

$$K_{m,a} = m^2 I - a^{-2} \sum_{\mu=1}^{d} (T_\mu + T_\mu^{-1} - 2I) = (m^2 + 2da^{-2})I - a^{-2} A. \tag{143.11}$$

The path expansion then weights an n-step path by

$$a^{-2n} \big(m^2 + 2da^{-2} \big)^{-(n+1)}$$

in the covariance matrix $K_{m,a}^{-1}$. In an interacting scalar construction the free path expansion is replaced by a random-walk or polymer expansion with local weights. The proof obligations are the same in structure as in the finite-graph resolvent calculation above: identify a positive reference operator, expand the perturbation in local objects, and prove a norm or polymer-sum bound that survives the intended limit.

143.4 Reflection Positivity And OS Reconstruction

Assume the lattice measures are reflection positive with respect to a hyperplane and that the scaling limit preserves reflection positivity:

$$\sum_{i,j} \bar{c}_i c_j \, S(F_i^{\Theta} F_j) \geq 0$$

for all finite linear combinations of positive-time test functionals. If the limit Schwinger functions also satisfy Euclidean covariance, symmetry, regularity, and the OS growth hypotheses, OS reconstruction produces a Hilbert space, vacuum, Hamiltonian, and Wightman distributions. Thus the lattice-to-continuum bridge is a theorem only after these hypotheses are verified for the limiting distributions.

Closedness of reflection positivity. Let S_a be reflection-positive Schwinger functionals on a common positive-time test algebra $\mathfrak{A}_+$. Suppose that, for every finite list $F_1, \ldots, F_N \in \mathfrak{A}_+$, the limits

$$\lim_{a \to 0} S_a(F_i^{\Theta} F_j) = S(F_i^{\Theta} F_j) \tag{143.12}$$

exist. Then the limiting functional S is reflection positive:

$$\sum_{i,j=1}^{N} \bar{c}_i c_j \, S(F_i^{\Theta} F_j) \geq 0 \tag{143.13}$$

for all $c_i \in \mathbb{C}$.

For each a, the matrix

$$M_a(F)_{ij} = S_a(F_i^{\Theta} F_j)$$

is positive semidefinite by reflection positivity. Hence

$$\sum_{i,j} \bar{c}_i c_j M_a(F)_{ij} \geq 0.$$

Taking the entrywise limit (143.12) gives (143.13). No compactness or reconstruction input is hidden here; the only point is that positivity of finite Gram matrices is a closed condition under entrywise convergence.

143.5 Local Algebras From Lattice Algebras

For a bounded continuum region O, let $O_a \subset \Lambda_a$ be the set of lattice sites or links whose cells lie in O. The lattice observable algebra $\mathcal{A}_a(O_a)$ is finite dimensional in spin systems and finite-regulator gauge systems after gauge constraints are imposed. A continuum local algebra is approached by embeddings or correlation-function limits

$$\mathcal{A}_a(O_a) \longrightarrow \mathcal{A}(O)$$

that preserve inclusion, locality, and the state in the scaling limit. The construction is delicate for gauge theories because Gauss-law constraints and center degrees of freedom affect the algebra assigned to a region with boundary.

For spin systems the algebraic limit can be stated directly. Let $\mathcal{H}_x$ be a finite-dimensional Hilbert space at each site and set

$$\mathcal{A}_a(X) = \bigotimes_{x\in X} \mathrm{End}(\mathcal{H}_x) \tag{143.14}$$

for a finite set $X \subset \Lambda_a$. If $X \subset Y$, the inclusion is

$$A \longmapsto A \otimes \mathbf{1}_{Y\setminus X}.$$

For disjoint X, Y, the images of $\mathcal{A}_a(X)$ and $\mathcal{A}_a(Y)$ commute. The quasi-local algebra at fixed a is the norm closure of the union over finite X.

Lemma 143.2 (Locality passes through state convergence for spin nets)**.** *Assume that for every bounded continuum region O and every sufficiently small a there is a finite set $O_a \subset \Lambda_a$, with isotony $O_1 \subset O_2 \Rightarrow O_{1,a} \subset O_{2,a}$. Suppose a collection of lattice observables $A_{i,a}(O_i) \in \mathcal{A}_a(O_{i,a})$ has limiting correlation functions in a state ω on the abstract algebra generated by the symbols $A_i(O_i)$. If O_1 and O_2 have positive separation, then for all sufficiently small a,*

$$[A_{1,a}(O_1), A_{2,a}(O_2)] = 0,$$

and the limiting GNS representation obeys

$$[\pi_\omega(A_1(O_1)), \pi_\omega(A_2(O_2))] = 0$$

on the dense domain generated by finite products of local symbols.

Proof. Positive separation of O_1 and O_2 implies $O_{1,a} \cap O_{2,a} = \varnothing$ for all sufficiently small a, after the cell approximation is chosen inside a fixed small thickening. The finite tensor-product construction then gives exact commutation at lattice spacing a. For any finite product B_a of additional local lattice observables whose correlations converge,

$$\omega\big(B[A_1(O_1), A_2(O_2)]\big) = \lim_{a\to 0} \omega_a\big(B_a[A_{1,a}(O_1), A_{2,a}(O_2)]\big) = 0.$$

Moreover, for every such B, set

$$C = [A_1(O_1), A_2(O_2)], \qquad C_a = [A_{1,a}(O_1), A_{2,a}(O_2)].$$

Then

$$\|\pi_\omega(C)\pi_\omega(B)\Omega_\omega\|^2 = \omega(B^*C^*CB) = \lim_{a\to 0} \omega_a(B_a^* C_a^* C_a B_a) = 0.$$

Thus the commutator annihilates the GNS dense domain generated by local symbols. Extension to bounded local von Neumann algebras, when they exist, is a further closure statement and requires the chosen topology of the scaling limit. □

Gauge theories are not obtained by the same tensor-factor argument after imposing Gauss constraints. The gauge-invariant algebra of a region need not factor over a boundary; electric center, magnetic center, and extended Hilbert-space prescriptions assign different boundary algebras. A continuum local-net claim for a lattice gauge scaling limit must therefore state which regional algebra is being scaled and how charged or line-operator sectors are represented.

143.6 Gauge-Invariant Operators

In lattice gauge theory, link variables are regulator coordinates. Continuum local gauge-invariant operators arise from small Wilson loops, covariant lattice differences, and matter bilinears after subtracting and mixing all operators with the same symmetries. For example, the continuum action density is represented by a plaquette combination

$$\mathcal{P}_a(x) = \frac{1}{a^4}\left(1 - \frac{1}{\dim R}\,\mathrm{Re}\,\mathrm{Tr}_R\, U_{\Box_x}\right) + \text{local mixings.}$$

The mixings include identity and lower-dimensional terms allowed by the regulator symmetries.

143.7 Universality As A Convergence Statement

Two lattice regulator families define the same continuum QFT when there are scaling maps for their observables such that all reconstructed local QFT data agree: Schwinger distributions, Wightman distributions or local nets, symmetry actions, stress tensor normalization, line or defect sectors when included, and background responses. Equality of critical exponents is a useful diagnostic but is not the full equivalence relation.

143.8 Continuum Limit Checklist

A lattice construction intended to define a continuum QFT must record:

regulator family	lattice actions, measures, boundary conditions,
scaling trajectory	couplings as functions of a,
operator maps	normalization and mixing,
limit topology	distributional, OS, Wightman, or local-net,
positivity	reflection positivity or replacement structure,
symmetries	continuum symmetry recovery and anomalies.

Open Problem 143.3 (Local nets from interacting lattice gauge limits). Construct continuum local algebras for interacting lattice gauge theory scaling limits, including Gauss-law center choices, charged sectors, extended operators, reflection positivity, and comparison with Wightman or OS reconstruction.

Chapter 144

Stochastic Quantization and Singular SPDE

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Stochastic quantization constructs Euclidean QFT data as the invariant law of a Markov process on fields. The auxiliary time of the process is not Lorentzian time and is not an extra physical dimension. Its purpose is to replace a formal functional integral by a dynamical construction whose stationary distribution can sometimes be proved to exist.

In singular dimensions the stochastic equation itself must be renormalized. The counterterms are local coordinates in the regulated stochastic drift and must be fixed before the cutoff is removed. The resulting invariant measure still has to be checked against the OS hypotheses before it becomes a Lorentzian QFT by reconstruction.

144.1 Finite-Dimensional Langevin Identity

Finite-dimensional invariant density. Let $S : \mathbb{R}^n \to \mathbb{R}$ be smooth and confining enough that

$$Z = \int_{\mathbb{R}^n} \mathrm{e}^{-S(x)}\,\mathrm{d}x < \infty.$$

The overdamped Langevin equation

$$\mathrm{d}X_t = -\nabla S(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t \tag{144.1}$$

has generator

$$Lf = -\nabla S \cdot \nabla f + \Delta f.$$

For smooth compactly supported f, g,

$$\int (Lf) g \,\mathrm{e}^{-S} \mathrm{d}x = -\int \nabla f \cdot \nabla g \,\mathrm{e}^{-S} \mathrm{d}x. \tag{144.2}$$

Therefore L is symmetric in $L^2(\mathrm{e}^{-S}\mathrm{d}x)$, and the probability measure $Z^{-1}\mathrm{e}^{-S}\mathrm{d}x$ is invariant. The calculation is ordinary finite-dimensional integration by parts:

$$\int (-\partial_i S)(\partial_i f) g \,\mathrm{e}^{-S} \mathrm{d}x = \int \partial_i(\mathrm{e}^{-S})(\partial_i f) g \,\mathrm{d}x.$$

Adding the Laplacian contribution and integrating by parts once more,

$$\int \left[\partial_i(\mathrm{e}^{-S})(\partial_i f) g + (\partial_i^2 f) g \mathrm{e}^{-S}\right] \mathrm{d}x = -\int (\partial_i f)(\partial_i g) \mathrm{e}^{-S} \mathrm{d}x.$$

Summing over i gives (144.2). With $g = 1$, the right-hand side vanishes, so the adjoint generator annihilates $\mathrm{e}^{-S}\mathrm{d}x$. This is the infinitesimal invariant-density identity; actual stationarity of the stochastic process is the semigroup statement in Lemma 144.1.

Lemma 144.1 (Finite-cutoff reversibility from the gradient Langevin datum)**.** *Let*

$$\mathrm{d}\pi(x) = Z^{-1} \mathrm{e}^{-S(x)} \mathrm{d}x.$$

Assume that the symmetric operator

$$L_0 f = -\nabla S \cdot \nabla f + \Delta f, \qquad f \in C_c^\infty(\mathbb{R}^n),$$

is essentially self-adjoint as an operator in $L^2(\pi)$, and that its self-adjoint closure $\overline{L}$ generates a conservative Markov semigroup

$$P_t = \mathrm{e}^{t\overline{L}}, \qquad t \geq 0,$$

on $L^2(\pi)$. Then P_t is reversible with respect to π:

$$\int (P_t f) g \,\mathrm{d}\pi = \int f (P_t g) \,\mathrm{d}\pi, \qquad f, g \in L^2(\pi), \tag{144.3}$$

and π is invariant:

$$\int P_t f \,\mathrm{d}\pi = \int f \,\mathrm{d}\pi, \qquad f \in L^2(\pi) \cap L^1(\pi). \tag{144.4}$$

For $f, g \in C_c^\infty(\mathbb{R}^n)$, the Dirichlet form of the closure is the closure of

$$\mathcal{E}(f, g) = \int \nabla f \cdot \nabla g \,\mathrm{d}\pi.$$

Proof. Equation (144.2) says precisely that

$$\langle L_0 f, g \rangle_{L^2(\pi)} = -\mathcal{E}(f, g) = \langle f, L_0 g \rangle_{L^2(\pi)}$$

on the core $C_c^\infty(\mathbb{R}^n)$. Hence the self-adjoint closure $\overline{L}$ is nonpositive. By the spectral theorem, $e^{t\overline{L}}$ is a self-adjoint contraction on $L^2(\pi)$, which gives (144.3). Conservativeness is the statement $P_t 1 = 1$. Taking $g = 1$ in (144.3) gives

$$\int P_t f \, d\pi = \langle P_t f, 1\rangle = \langle f, P_t 1\rangle = \int f \, d\pi,$$

first for bounded L^2 functions and then for $L^1(\pi) \cap L^2(\pi)$ by truncation. The Dirichlet form statement is the definition of the closed quadratic form associated with the nonpositive self-adjoint operator $\overline{L}$, restricted to the displayed core where the integration-by-parts calculation identifies the form explicitly. □

The lemma records the finite-regulator theorem actually used below. The integration-by-parts identity proves the symmetric Dirichlet form on a core; the passage to a stationary Markov process also requires the finite-dimensional diffusion to be conservative and the generator closure to produce the semigroup. These hypotheses are elementary for the smooth compact cutoff systems used as finite regulators, but they are logically distinct from the singular infinite-dimensional limit.

144.2 Regularized Field Equation

Let M be a compact Euclidean spatial manifold, let C_Λ be a smooth ultraviolet cutoff covariance on M, and let ξ_Λ be white noise in stochastic time with spatial covariance chosen so that the linear equation has invariant Gaussian law μ_{C_Λ}. A regularized scalar Φ^4 Langevin equation has the form

$$\partial_\tau \phi_\Lambda = -C_\Lambda^{-1}\phi_\Lambda - \lambda_\Lambda \phi_\Lambda^3 + c_\Lambda \phi_\Lambda + \xi_\Lambda. \tag{144.5}$$

The drift is the negative L^2-gradient of the regulated action

$$S_\Lambda(\phi) = \frac{1}{2}(\phi, C_\Lambda^{-1}\phi) + \int_M \left[\frac{\lambda_\Lambda}{4}\phi^4 - \frac{c_\Lambda}{2}\phi^2\right] d^d x. \tag{144.6}$$

At finite Λ, the argument of the previous section applies on the finite-dimensional cutoff field space. Thus

$$Z_\Lambda^{-1} e^{-S_\Lambda(\phi)} D\phi$$

is invariant. The constructive question is whether the laws of ϕ_Λ, with tuned local coordinates, converge as $\Lambda \to \infty$.

144.3 Renormalized Products

For the Gaussian free field in spatial dimension $d \geq 2$, $\phi(x)$ is a distribution. Products in the drift are therefore defined through renormalized local functionals. If $\phi_\epsilon = \rho_\epsilon * \phi$ and

$$C_\epsilon = \mathbb{E}[\phi_\epsilon(x)^2],$$

then

$$:\phi_\epsilon^3: = \phi_\epsilon^3 - 3C_\epsilon \phi_\epsilon. \tag{144.7}$$

The coefficient 3 is not conventional: it counts the possible contractions of two of the three fields. The same convention gives

$$:\phi_\epsilon^4:= \phi_\epsilon^4 - 6C_\epsilon\phi_\epsilon^2 + 3C_\epsilon^2.$$

144.4 Stationary Stochastic Convolution and Wick Powers

The first analytic object in the Da Prato–Debussche construction is the stationary solution of the linear stochastic heat equation. We record the Fourier construction because it fixes the normalization of the noise and proves, in a Sobolev form, the existence of the Wick powers used later.

Let $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, and let $\{e_k\}_{k\in\mathbb{Z}^2}$ be the orthonormal Fourier basis of $L^2(\mathbb{T}^2)$, normalized so that $-\Delta e_k = |k|^2 e_k$. The spacetime white noise in the stochastic quantization equation is normalized by

$$\mathbb{E}[\xi(\tau,x)\xi(\tau',y)] = 2\,\delta(\tau-\tau')\delta(x-y).$$

Equivalently, in Fourier coordinates,

$$\mathrm{d}X_k(\tau) = -(|k|^2+m^2)X_k(\tau)\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_k(\tau), \qquad X_{-k} = \overline{X_k}, \tag{144.8}$$

where the complex Brownian motions are paired so that the reconstructed field is real.

Lemma 144.2 (Stationary covariance of the linear SPDE)**.** *The stationary solution of* (144.8) *satisfies*

$$\mathbb{E}[X_k\overline{X_\ell}] = \frac{\delta_{k\ell}}{|k|^2+m^2}. \tag{144.9}$$

Consequently $X(\tau,\cdot)$ has the massive Gaussian free-field law with covariance $(-\Delta+m^2)^{-1}$, and

$$\mathbb{E}\|X(\tau)\|^2_{H^{-s}(\mathbb{T}^2)} = \sum_{k\in\mathbb{Z}^2} \frac{(1+|k|^2)^{-s}}{|k|^2+m^2} < \infty \qquad (s>0). \tag{144.10}$$

Proof. For the real one-dimensional Ornstein–Uhlenbeck equation $\mathrm{d}Z_t = -aZ_t\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $a>0$, the stationary solution is obtained by solving from time $-\infty$:

$$Z_t = \sqrt{2}\int_{-\infty}^t e^{-a(t-r)}\,\mathrm{d}B_r,$$

where the integral is well-defined in $L^2(\Omega)$ because $\int_0^\infty e^{-2au}\mathrm{d}u < \infty$. Ito isometry gives

$$\mathbb{E}|Z_t|^2 = 2\int_0^\infty e^{-2au}\,\mathrm{d}u = a^{-1}.$$

For the complex Fourier mode X_k one applies the same calculation to the paired real and imaginary coordinates, with the covariance convention in (144.8) chosen so that $\mathbb{E}[X_k\overline{X_\ell}]=0$ for $k\neq\ell$ and $\mathbb{E}|X_k|^2 = (|k|^2+m^2)^{-1}$. Taking $a=|k|^2+m^2$ gives (144.9).

For a finite Fourier cutoff N, Parseval gives

$$\mathbb{E}\|P_NX(\tau)\|^2_{H^{-s}} = \sum_{|k|\le N}\frac{(1+|k|^2)^{-s}}{|k|^2+m^2}.$$

The summands are nonnegative, so monotone convergence passes to the infinite sum. In two dimensions the large-$|k|$ summand is comparable to $|k|^{-2-2s}$, and the lattice count $\#\{k \in \mathbb{Z}^2 : R \le |k| < R+1\} = O(R)$ reduces convergence to $\sum_R R^{-1-2s} < \infty$, namely $s > 0$. Thus the stationary field is an H^{-s}-valued Gaussian random distribution for every $s > 0$, with covariance $(-\Delta + m^2)^{-1}$. ☐

Let P_N be the Fourier projection to $|k| \le N$, set $X_N = P_N X$, and write

$$C_N(x-y) = \mathbb{E}[X_N(x)X_N(y)].$$

The Wick powers are

$$:X_N^2 := X_N^2 - C_N(0), \qquad :X_N^3 := X_N^3 - 3C_N(0)X_N.$$

Lemma 144.3 (Two-dimensional massive-convolution bound)**.** *Let*

$$g(k) = \frac{1}{1+|k|^2}, \qquad L(k) = 1 + \log(2+|k|), \qquad k \in \mathbb{Z}^2.$$

For every integer $n \ge 1$ there is $C_n < \infty$ such that the n-fold lattice convolution satisfies

$$g^{*n}(q) \le C_n \frac{L(q)^{n-1}}{1+|q|^2}, \qquad q \in \mathbb{Z}^2. \tag{144.11}$$

Proof. The case $n = 1$ is the definition. The induction step is the elementary estimate

$$\sum_{p\in\mathbb{Z}^2} \frac{L(p)^a}{(1+|p|^2)(1+|q-p|^2)} \le C_a \frac{L(q)^{a+1}}{1+|q|^2}, \qquad a \ge 0. \tag{144.12}$$

For $|q| \le 4$ the left side is finite and the estimate is absorbed into the constant. Assume $|q| > 4$. Split the sum into three regions:

$$A = \{|p| \le |q|/2\}, \qquad B = \{|q-p| \le |q|/2\}, \qquad C = (A \cup B)^c.$$

On A, $1 + |q-p|^2 \ge c(1+|q|^2)$, and

$$\sum_{|p|\le|q|/2} \frac{L(p)^a}{1+|p|^2} \le C_a L(q)^{a+1},$$

because the two-dimensional shell of radius r contributes $\lesssim r^{-1}(1+\log r)^a$. This proves the desired bound on A. The region B is identical after replacing $r = q - p$: then $|p| \ge |q|/2$, $L(p) \le CL(q)$, and the remaining $\sum_{|r|\le|q|/2}(1+|r|^2)^{-1}$ contributes one additional logarithm. On C, both $|p|$ and $|q-p|$ are $\ge |q|/2$. Decompose C into dyadic annuli $|p| \sim 2^j|q|$ and $|q-p| \sim 2^k|q|$. The number of lattice points in a two-dimensional dyadic annulus of radius R is $O(R^2)$, while the summand is bounded by $CL(p)^a/(|p|^2|q-p|^2)$. Summing first over the smaller of the two annuli and then over the larger gives

$$\sum_{p\in C} \frac{L(p)^a}{(1+|p|^2)(1+|q-p|^2)} \le C_a \frac{L(q)^a}{1+|q|^2}.$$

Combining the three regions proves (144.12). Applying it with $a = n-1$ to $g * g^{*n}$ proves (144.11) by induction. ☐

Proposition 144.4 (Smeared Wick powers of the two-dimensional stochastic convolution)**.** *For every $n \geq 1$ and every $f \in C^\infty(\mathbb{T}^2)$, the random variables*

$$:X_N^n:(f) = \int_{\mathbb{T}^2} f(x) :X_N(x)^n: \, \mathrm{d}^2x$$

converge in $L^2(\Omega)$ as $N \to \infty$. Moreover, for every $s > 0$, the fields $:X_N^n:$ converge in $L^2(\Omega; H^{-s}(\mathbb{T}^2))$.

Proof. The Wick product covariance is the finite-dimensional Gaussian identity

$$\mathbb{E}[:X_N(x)^n::X_M(y)^n:] = n!\, C_{N,M}(x-y)^n, \qquad C_{N,M}(x-y) = \mathbb{E}[X_N(x)X_M(y)]. \tag{144.13}$$

This follows from the generating function $\exp(tX_N - \frac{1}{2}t^2 C_N(0))$ and differentiating n times in each variable. Hence

$$\mathbb{E}\left[:X_N^n:(f)\ :X_M^n:(f)\right] = n! \int_{\mathbb{T}^2\times\mathbb{T}^2} f(x)f(y)\, C_{N,M}(x-y)^n \, \mathrm{d}^2x \, \mathrm{d}^2y.$$

The massive two-dimensional covariance has the local form

$$C(x) = -\frac{1}{2\pi} \log|x| + O(1) \qquad (x \to 0),$$

and powers of $\log|x|$ are locally integrable in two dimensions:

$$\int_0^r \rho\, |\log\rho|^n \, \mathrm{d}\rho < \infty.$$

The kernels $C_{N,M}$ converge to C away from the diagonal, and the standard Fourier cutoff bounds are dominated by an integrable logarithmic majorant. Dominated convergence therefore proves the L^2-Cauchy property for every smeared Wick power.

For the Sobolev statement, use the dual norm

$$\|u\|_{H^{-s}}^2 = \sum_{q\in\mathbb{Z}^2} (1+|q|^2)^{-s} |\widehat{u}(q)|^2.$$

Applying (144.13) to Fourier test functions shows that

$$\mathbb{E}|\widehat{:X_N^n:}(q)|^2 = n! \sum_{\substack{k_1+\cdots+k_n=q\\ |k_i|\leq N}} \prod_{i=1}^n \frac{1}{|k_i|^2+m^2}. \tag{144.14}$$

The right side is bounded by $C_{n,m}L(q)^{n-1}(1+|q|^2)^{-1}$ by Lemma 144.3. Therefore

$$\sum_{q\in\mathbb{Z}^2} (1+|q|^2)^{-s} \mathbb{E}|\widehat{:X_N^n:}(q)|^2 \leq C_{n,m} \sum_{q\in\mathbb{Z}^2} \frac{L(q)^{n-1}}{(1+|q|^2)^{1+s}} < \infty$$

for every $s > 0$. For fixed q, the cutoff variances and mixed covariances converge by dominated convergence of the convolution sums. The same convolution bound dominates $\mathbb{E}|\widehat{:X_N^n:}(q) - \widehat{:X_M^n:}(q)|^2$, up to a harmless constant, so dominated convergence over q proves convergence in $L^2(\Omega; H^{-s})$. In particular the stochastic input needed by the Sobolev DPD fixed point below is available for any prescribed $0 < \kappa < 1/4$ after taking $s = \kappa$. $\square$

Proposition 144.5 (Smooth Fourier multiplier covariance convergence)**.** *Let $m^2 > 0$, and let*

$$C(z) = \sum_{k\in\mathbb{Z}^2} \frac{e^{ik\cdot z}}{|k|^2 + m^2}$$

be the massive covariance on $\mathbb{T}^2$, understood through the dyadic Fourier decomposition used below. For $\sigma \in \{A, B\}$, let $\mu^\sigma \in C_c^\infty(\mathbb{R}^2)$ be real, even, and satisfy $\mu^\sigma(0) = 1$. Define the cutoff Gaussian field on the same Gaussian probability space by

$$X_N^\sigma = \mu^\sigma(D/N)X,$$

so that the mixed and cutoff covariances are translation invariant kernels

$$C_N^{\sigma\infty}(z) = \sum_{k\in\mathbb{Z}^2} \frac{\mu^\sigma(k/N)}{|k|^2 + m^2} e^{ik\cdot z}, \qquad C_N^{\sigma\sigma}(z) = \sum_{k\in\mathbb{Z}^2} \frac{\mu^\sigma(k/N)^2}{|k|^2 + m^2} e^{ik\cdot z}.$$

Then, for every $1 \le p < \infty$,

$$C_N^{\sigma\infty} \longrightarrow C, \qquad C_N^{\sigma\sigma} \longrightarrow C \quad \text{in } L^p(\mathbb{T}^2). \tag{144.15}$$

Consequently the hypotheses of Lemma 144.6 hold for every finite Wick order n.

Proof. Choose a smooth radial dyadic partition of unity on frequency space,

$$\chi_{-1}(\xi) + \sum_{j\ge 0} \chi_j(\xi) = 1, \qquad \chi_j(\xi) = \chi(2^{-j}\xi),$$

where χ_{-1} is supported near the origin and χ is supported in an annulus. Let

$$a(k) = |k|^2 + m^2,$$

and let $\nu_N(k)$ denote any one of the three multipliers

$$1, \qquad \mu^\sigma(k/N), \qquad \mu^\sigma(k/N)^2.$$

For $j \ge 0$, set

$$K_{j,N}^\nu(z) = \sum_{k\in\mathbb{Z}^2} \chi_j(k) \frac{\nu_N(k)}{a(k)} e^{ik\cdot z}.$$

We first prove the uniform dyadic kernel bound

$$|K_{j,N}^\nu(z)| \le C_M \big(1 + 2^j d_{\mathbb{T}^2}(z, 0)\big)^{-M}, \qquad M \ge 0, \tag{144.16}$$

with a constant independent of j, N, ν. Indeed, put

$$q_{j,N}^\nu(\xi) = \chi_j(\xi) \frac{\nu_N(\xi)}{|\xi|^2 + m^2}.$$

On the support of χ_j, $|\xi| \simeq 2^j$. If $\nu_N(\xi) = \mu^\sigma(\xi/N)$ or $\mu^\sigma(\xi/N)^2$, the support of $q_{j,N}^\nu$ is empty unless $2^j \le CN$; on the nonempty support, derivatives of ν_N satisfy

$$|\partial^\alpha \nu_N(\xi)| \le C_\alpha N^{-|\alpha|} \le C_\alpha 2^{-j|\alpha|}.$$

Together with the corresponding derivative bounds for χ_j and $(|\xi|^2 + m^2)^{-1}$, this gives

$$|\partial^\alpha q^\nu_{j,N}(\xi)| \le C_\alpha \, 2^{-2j} 2^{-j|\alpha|}. \tag{144.17}$$

Let $\check{q}^\nu_{j,N}$ be the inverse Fourier transform on $\mathbb{R}^2$. Integrating by parts in the oscillatory integral and using (144.17) gives

$$|\check{q}^\nu_{j,N}(x)| \le C_M (1 + 2^j |x|)^{-M}.$$

Poisson summation identifies $K^\nu_{j,N}$ with the periodization of $\check{q}^\nu_{j,N}$. Summing over lattice translates gives (144.16) when $M > 2$; decreasing M if necessary gives the stated form for all $M \ge 0$. The low-frequency block $j = -1$ is bounded uniformly because it contains only finitely many lattice modes.

Summing (144.16) over j with $M > 2$ gives the logarithmic majorant

$$\left| \sum_{j \ge -1} K^\nu_{j,N}(z) \right| \le C\big(1 + |\log d_{\mathbb{T}^2}(z, 0)|\big), \tag{144.18}$$

where the right side is interpreted as a bounded function away from $z = 0$. The logarithmic singularity belongs to $L^p(\mathbb{T}^2)$ for every finite p, because in polar coordinates

$$\int_0^{1/2} r \, |\log r|^p \, \mathrm{d}r < \infty.$$

It remains to prove pointwise convergence away from the diagonal. Fix $z \ne 0$. For any fixed dyadic level j, the block $K^\nu_{j,N}$ is a finite sum, and the multiplier $\nu_N(k)$ tends to 1 for every fixed k. Hence

$$K^\nu_{j,N}(z) \longrightarrow K_{j,\infty}(z)$$

for each fixed j. The tail is uniform in N:

$$\sum_{j > J} |K^\nu_{j,N}(z)| \le C_M \sum_{j > J} (1 + 2^j d_{\mathbb{T}^2}(z, 0))^{-M} \le C_{M,z} \, 2^{-JM}.$$

Therefore the full dyadic sums converge pointwise for every $z \ne 0$. Together with the logarithmic majorant (144.18), dominated convergence gives (144.15). Taking $\nu_N = \mu^\sigma(k/N)$ gives the mixed covariance, while taking $\nu_N = \mu^\sigma(k/N)^2$ gives the cutoff covariance. Since this holds for $p = n$, and since a translation-invariant kernel $K(x-y)$ has $L^n(\mathbb{T}^2 \times \mathbb{T}^2)$-norm equal to the volume factor times its $L^n(\mathbb{T}^2)$-norm, the hypotheses of Lemma 144.6 hold for every finite Wick order n. ☐

Lemma 144.6 (Mixed-covariance criterion for common Wick limits)**.** *Let X be a centered Gaussian generalized field on $\mathbb{T}^2$, and let X^σ_N, $\sigma \in \{A, B\}$, be centered Gaussian cutoff fields on the same probability space. Write*

$$C(x, y) = \mathbb{E}[X(x)X(y)], \qquad C^{\sigma\sigma}_N(x, y) = \mathbb{E}[X^\sigma_N(x) X^\sigma_N(y)],$$

and

$$C^{\sigma\infty}_N(x, y) = \mathbb{E}[X^\sigma_N(x) X(y)].$$

Fix $n \ge 1$, and assume that these covariance kernels are represented by functions in $L^n(\mathbb{T}^2 \times \mathbb{T}^2)$ and that, for $\sigma = A, B$,

$$\|C^{\sigma\sigma}_N - C\|_{L^n(\mathbb{T}^2 \times \mathbb{T}^2)} + \|C^{\sigma\infty}_N - C\|_{L^n(\mathbb{T}^2 \times \mathbb{T}^2)} \longrightarrow 0. \tag{144.19}$$

Then, for every real $f \in C^\infty(\mathbb{T}^2)$,

$$:(X_N^\sigma)^n:(f) \longrightarrow :X^n:(f) \quad \textit{in } L^2(\Omega), \qquad \sigma = A, B, \tag{144.20}$$

where each Wick product is normal ordered with respect to the covariance of the Gaussian field appearing in it.

Proof. Let

$$Y_N^\sigma =:(X_N^\sigma)^n:(f), \qquad Y =:X^n:(f).$$

For two jointly Gaussian centered random variables U, V, Wick ordering with respect to the individual variances gives

$$\mathbb{E}\big[:U^n::V^n:\big] = n!\,\mathbb{E}[UV]^n.$$

This follows by differentiating the identity

$$\mathbb{E}\Big[e^{tU-\frac{1}{2}t^2\mathbb{E}[U^2]}e^{sV-\frac{1}{2}s^2\mathbb{E}[V^2]}\Big] = e^{ts\mathbb{E}[UV]}$$

n times in t and n times in s at the origin. Therefore

$$\mathbb{E}|Y_N^\sigma - Y|^2 = n! \int_{\mathbb{T}^2\times\mathbb{T}^2} f(x)f(y)\Big[\big(C_N^{\sigma\sigma}(x,y)\big)^n - 2\big(C_N^{\sigma\infty}(x,y)\big)^n + C(x,y)^n\Big]\,\mathrm{d}^2x\,\mathrm{d}^2y.$$

It remains to show that the two n-th powers converge in L^1 to C^n. If $a, b \in L^n(\mathbb{T}^2\times\mathbb{T}^2)$, then

$$|a^n - b^n| \le n|a-b|\big(|a|^{n-1} + |b|^{n-1}\big).$$

Holder's inequality gives

$$\|a^n - b^n\|_{L^1} \le n\|a-b\|_{L^n}\left(\|a\|_{L^n}^{n-1} + \|b\|_{L^n}^{n-1}\right).$$

The convergence assumption (144.19) implies that the L^n-norms of $C_N^{\sigma\sigma}$ and $C_N^{\sigma\infty}$ are bounded and that both kernels converge to C in L^n. Hence

$$\big(C_N^{\sigma\sigma}\big)^n \to C^n, \qquad \big(C_N^{\sigma\infty}\big)^n \to C^n \quad \text{in } L^1.$$

Since f is bounded, the displayed formula for $\mathbb{E}|Y_N^\sigma - Y|^2$ tends to zero. This proves (144.20). $\square$

Remark 144.7 (Common limits of finite Wick-coordinate vectors). Fix finitely many pairs

$$(n_j, f_j), \qquad 1 \le j \le r,$$

where $n_j \ge 1$ and $f_j \in C^\infty(\mathbb{T}^2)$. For one cutoff sequence X_N, set

$$\mathcal{X}_N = \big(:X_N^{n_1}:(f_1), \dots, :X_N^{n_r}:(f_r)\big) \in \mathbb{R}^r.$$

Then $\mathcal{X}_N$ converges in $L^2(\Omega;\mathbb{R}^r)$, hence in probability, to

$$\mathcal{X} = \big(:X^{n_1}:(f_1), \dots, :X^{n_r}:(f_r)\big).$$

More generally, suppose X_N^A and X_N^B are two cutoff approximations of the same massive Gaussian field on a common probability space, and suppose that, for every j and $\sigma \in \{A, B\}$,

$$:(X_N^\sigma)^{n_j}:(f_j) \longrightarrow :X^{n_j}:(f_j) \quad \text{in } L^2(\Omega). \tag{144.21}$$

Then the coordinate vectors

$$\mathcal{X}_N^\sigma = \big(:(X_N^\sigma)^{n_1}:(f_1), \dots, :(X_N^\sigma)^{n_r}:(f_r)\big)$$

satisfy

$$\mathcal{X}_N^\sigma \longrightarrow \mathcal{X} \quad \text{in } L^2(\Omega; \mathbb{R}^r) \quad \text{and in probability,} \qquad \sigma = A, B. \tag{144.22}$$

For every bounded continuous $F : \mathbb{R}^r \to \mathbb{C}$,

$$F(\mathcal{X}_N^\sigma) \longrightarrow F(\mathcal{X}) \quad \text{in probability.} \tag{144.23}$$

For the single Fourier cutoff sequence, each coordinate converges in $L^2(\Omega)$ by Proposition 144.4. Therefore

$$\mathbb{E}|\mathcal{X}_N - \mathcal{X}|^2_{\mathbb{R}^r} = \sum_{j=1}^r \mathbb{E}\left|:X_N^{n_j}:(f_j) - :X^{n_j}:(f_j)\right|^2 \longrightarrow 0.$$

The two-regulator statement is identical after replacing the preceding coordinate convergence by (144.21); a concrete covariance criterion for this hypothesis is Lemma 144.6. L^2-convergence implies convergence in probability by Chebyshev's inequality. Finally, if $\mathcal{X}_N^\sigma \to \mathcal{X}$ in probability and F is continuous, then $F(\mathcal{X}_N^\sigma) \to F(\mathcal{X})$ in probability because the inverse image of the complement of any small neighborhood of $F(\mathcal{X})$ is controlled by continuity of F at the realized value of $\mathcal{X}$; equivalently, every almost surely convergent subsequence has a further subsequence on which the conclusion follows pointwise.

Remark 144.8 (Convergence of the Wick quartic potential)*.* Let $\lambda \in \mathbb{R}$, and define the cutoff Wick quartic functional on the Gaussian probability space of Proposition 144.4 by

$$V_N = \frac{\lambda}{4} \int_{\mathbb{T}^2} :X_N(x)^4: \, \mathrm{d}^2 x = \frac{\lambda}{4} \, :X_N^4:(1).$$

Then there is an $L^2(\Omega)$ random variable

$$V = \frac{\lambda}{4} \, :X^4:(1)$$

such that $V_N \to V$ in $L^2(\Omega)$, and hence in probability.

More generally, suppose X_N^A and X_N^B are two cutoff approximations of the same massive Gaussian field for which the conclusion of Proposition 144.4 holds for $:(X_N^\sigma)^4:(1)$, $\sigma = A, B$, with the same limiting Wick functional $:X^4:(1)$. Set

$$V_N^\sigma = \frac{\lambda}{4} \, :(X_N^\sigma)^4:(1), \qquad V = \frac{\lambda}{4} \, :X^4:(1).$$

Then $V_N^A \to V$ and $V_N^B \to V$ in $L^2(\Omega)$. Consequently, if the exponential weights $e^{-V_N^\sigma}$ obey a uniform L^p bound for some $p > 1$ and if the cylinder coordinates are finite Wick-coordinate vectors satisfying Remark 144.7, then the bounded-cylinder comparison conclusion of Corollary 144.33 applies. The first statement is Proposition 144.4 with $n = 4$ and with the smooth test function $f \equiv 1$. Multiplication by the scalar $\lambda/4$ preserves L^2-convergence. The two cutoff approximations are handled separately by the same Wick-power convergence proposition; because both sequences have the same L^2 limit,

$$\|V_N^\sigma - V\|_{L^2(\Omega)} \longrightarrow 0, \qquad \sigma = A, B.$$

The last assertion follows from Corollary 144.33, since L^2-convergence implies convergence in probability of the potentials.

Proposition 144.9 (Path-space convergence of the two-dimensional enhanced noise)**.** *Let*

$$W_N^{(n)}(\tau) =: X_N(\tau)^n :, \qquad n = 1, 2, 3.$$

For every $T < \infty$, every $s > 0$, every $0 < \vartheta < \min\{s, 1\}$, and every integer $p \geq 1$ with $p\vartheta > 1$, there is a constant $C = C(T, s, \vartheta, p, n, m) < \infty$, independent of N, such that

$$\mathbb{E}\, \|W_N^{(n)}(\tau) - W_N^{(n)}(\sigma)\|_{H^{-s}}^{2p} \leq C|\tau - \sigma|^{p\vartheta} \qquad (0 \leq \sigma, \tau \leq T). \tag{144.24}$$

Consequently the triple

$$\mathbb{X}_N = (X_N, :X_N^2:, :X_N^3:)$$

converges in probability in

$$C([0, T]; H^{-s}(\mathbb{T}^2))^3$$

to a continuous enhanced noise $\mathbb{X} = (X, :X^2:, :X^3:)$.

Proof. Write $a_k = |k|^2 + m^2$. The stationary Ornstein–Uhlenbeck formula gives, for $\tau, \sigma \in \mathbb{R}$,

$$\mathbb{E}\Big[X_k(\tau)\overline{X_\ell(\sigma)}\Big] = \delta_{k\ell}\frac{e^{-a_k|\tau-\sigma|}}{a_k}. \tag{144.25}$$

Indeed, with $\tau \geq \sigma$,

$$X_k(\tau) = e^{-a_k(\tau-\sigma)}X_k(\sigma) + \sqrt{2}\int_\sigma^\tau e^{-a_k(\tau-r)}\,\mathrm{d}B_k(r),$$

and the stochastic integral is independent of $X_k(\sigma)$.

Let $\widehat{W}_{N,q}^{(n)}(\tau)$ denote the q-th Fourier coefficient of $W_N^{(n)}(\tau)$. Wick's theorem and (144.25) give

$$\mathbb{E}\Big[\widehat{W}_{N,q}^{(n)}(\tau)\overline{\widehat{W}_{N,q}^{(n)}(\sigma)}\Big] = n! \sum_{\substack{k_1+\cdots+k_n=q \\ |k_i|\leq N}} \prod_{i=1}^n \frac{e^{-a_{k_i}|\tau-\sigma|}}{a_{k_i}}. \tag{144.26}$$

Therefore

$$\begin{aligned} &\mathbb{E}\left|\widehat{W}_{N,q}^{(n)}(\tau) - \widehat{W}_{N,q}^{(n)}(\sigma)\right|^2 \\ &\qquad = 2n! \sum_{\substack{k_1+\cdots+k_n=q \\ |k_i|\leq N}} \prod_{i=1}^n \frac{1}{a_{k_i}}\left(1 - e^{-|\tau-\sigma|\sum_i a_{k_i}}\right). \end{aligned} \tag{144.27}$$

For $0 < \vartheta < 1$, $1 - e^{-r} \leq r^\vartheta$, and concavity gives $(\sum_i a_{k_i})^\vartheta \leq \sum_i a_{k_i}^\vartheta$. Hence the right side of (144.27) is bounded by

$$C_{n,\vartheta}|\tau - \sigma|^\vartheta \sum_{j=1}^n \sum_{k_1+\cdots+k_n=q} a_{k_j}^{-1+\vartheta} \prod_{i\neq j} a_{k_i}^{-1}.$$

Let $g_\vartheta(k) = (1 + |k|^2)^{-1+\vartheta}$. Repeating the three-region proof of Lemma 144.3, with the factor $(1 + |p|^2)^{-1}$ replaced once by $(1 + |p|^2)^{-1+\vartheta}$, gives

$$(g_\vartheta * g^{*(n-1)})(q) \leq C_{n,\vartheta}\,\frac{L(q)^n}{(1 + |q|^2)^{1-\vartheta}}. \tag{144.28}$$

The only changed part of the induction is the region in which the weakened factor is the small momentum: there $\sum_{|p|\le |q|/2}(1+|p|^2)^{-1+\vartheta}\le C_\vartheta(1+|q|^2)^{\vartheta}$, which changes $(1+|q|^2)^{-1}$ into $(1+|q|^2)^{-1+\vartheta}$. All other regions are bounded by the same or a smaller power, up to logarithms. Thus

$$\mathbb{E}\left|\widehat{W}^{(n)}_{N,q}(\tau)-\widehat{W}^{(n)}_{N,q}(\sigma)\right|^2\le C|\tau-\sigma|^{\vartheta}\frac{L(q)^n}{(1+|q|^2)^{1-\vartheta}}. \tag{144.29}$$

After multiplying by the H^{-s} weight and summing over q, the large-shell exponent is

$$1-2s-2(1-\vartheta)=-1-2s+2\vartheta.$$

This is strictly smaller than -1 because $\vartheta<s$. Hence

$$\mathbb{E}\,\|W^{(n)}_N(\tau)-W^{(n)}_N(\sigma)\|^2_{H^{-s}}\le C|\tau-\sigma|^{\vartheta}.$$

Each scalar Fourier coefficient of the increment lies in the n-th homogeneous Wiener chaos. Applying Lemma 144.51 to scalar Fourier coordinates and using Minkowski's inequality in $L^p(\Omega)$ for the series defining the squared H^{-s}-norm gives

$$\mathbb{E}\,\|W^{(n)}_N(\tau)-W^{(n)}_N(\sigma)\|^{2p}_{H^{-s}}\le C_{p,n}\left(\sum_q(1+|q|^2)^{-s}\mathbb{E}\left|\widehat{W}^{(n)}_{N,q}(\tau)-\widehat{W}^{(n)}_{N,q}(\sigma)\right|^2\right)^p,$$

which proves (144.24).

It remains to pass from equal-time convergence to path convergence. Fix $\epsilon,\eta>0$. By Proposition 144.4, for every finite time grid $\mathcal{G}\subset[0,T]$,

$$\max_{\tau\in\mathcal{G}}\|W^{(n)}_N(\tau)-W^{(n)}_M(\tau)\|_{H^{-s}}\longrightarrow 0\quad\text{in probability as } N,M\to\infty.$$

The uniform moment bound (144.24) and the Kolmogorov dyadic argument imply that, after choosing the mesh of $\mathcal{G}$ sufficiently small,

$$\sup_N\mathbb{P}\left(\sup_{|\tau-\sigma|\le\mathrm{mesh}(\mathcal{G})}\|W^{(n)}_N(\tau)-W^{(n)}_N(\sigma)\|_{H^{-s}}>\epsilon\right)<\eta.$$

The same estimate holds with M in place of N. The triangle inequality between a time and the nearest grid point then shows that $\{W^{(n)}_N\}_N$ is Cauchy in probability in $C([0,T];H^{-s})$. Its limit has continuous paths because this Banach space is complete. The finite-dimensional time marginals agree with the equal-time Wick-power limits, so the limiting process is precisely $:X^n:$. Applying the argument to $n=1,2,3$ proves the convergence of $\mathbb{X}_N$. □

Heat-kernel smoothing estimate. Let $A=\Delta-m^2$ on $\mathbb{T}^d$. The parabolic gain used below is the following Fourier estimate: for $r\in\mathbb{R}$, $\theta\ge 0$, and $t>0$,

$$\|e^{tA}u\|_{H^{r+2\theta}}\le C_\theta t^{-\theta}e^{-m^2t/2}\|u\|_{H^r}. \tag{144.30}$$

Consequently, if $F\in L^\infty([0,T];H^r)$, then

$$\left\|\int_0^t e^{(t-s)A}F(s)\,\mathrm{d}s\right\|_{H^{r+2\theta}}\le C_{\theta,T}\|F\|_{L^\infty([0,T];H^r)}\quad(0\le\theta<1). \tag{144.31}$$

Indeed, in Fourier coordinates,

$$\|e^{tA}u\|_{H^{r+2\theta}}^2 = \sum_k (1+|k|^2)^{r+2\theta} e^{-2t(|k|^2+m^2)} |\widehat{u}(k)|^2.$$

The elementary bound

$$(1+|k|^2)^{2\theta} e^{-t(|k|^2+m^2)} \le C_\theta t^{-2\theta} e^{-m^2 t/2}$$

gives (144.30). The Duhamel estimate follows by integrating $(t-s)^{-\theta}$ over $0<s<t$, which is finite exactly for $\theta<1$. The restriction $\theta<1$ is therefore not a stochastic effect; it is the endpoint singularity of the deterministic heat kernel in this L_t^∞ estimate. Sharper parabolic Besov–Holder versions distribute time regularity differently, but the fixed-point argument below only needs the displayed Sobolev form.

144.5 Da Prato–Debussche in Two Dimensions

On a two-dimensional torus, consider the stochastic quantization equation for the Wick-ordered measure

$$\partial_\tau \Phi = \Delta\Phi - m^2\Phi - \lambda :\Phi^3: +\xi. \tag{144.32}$$

Let X solve the stationary linear equation

$$\partial_\tau X = (\Delta - m^2)X + \xi. \tag{144.33}$$

The field $X(\tau,\cdot)$ has the massive Gaussian free-field law on the spatial torus and is almost surely a spatial distribution of regularity $-\kappa$ for every $\kappa>0$. Write

$$\Phi = X + Y.$$

Then the nonlinear equation for Y is

$$\partial_\tau Y = (\Delta - m^2)Y - \lambda\big(Y^3 + 3Y^2X + 3Y :X^2: + :X^3:\big). \tag{144.34}$$

In two dimensions the products in (144.34) are well-defined once $:X^2:$ and $:X^3:$ are constructed as Wick powers, because the unknown Y gains positive parabolic regularity from the heat operator.

144.5.1 Local Product Estimates and Mild Fixed Points

Lemma 144.10 (Sobolev products for the DPD fixed point)**.** *Let $0<\kappa<1/4$ and set $\beta = 1+2\kappa$. On $\mathbb{T}^2$, there is $C_\kappa<\infty$ such that*

$$\|uv\|_{H^\beta} \le C_\kappa \|u\|_{H^\beta}\|v\|_{H^\beta}, \tag{144.35}$$

$$\|uv\|_{H^{-\kappa}} \le C_\kappa \|u\|_{H^\beta}\|v\|_{H^{-\kappa}}, \tag{144.36}$$

for u,v in the indicated spaces. Consequently, if $\mathbb{X} = (X, X^{(2)}, X^{(3)}) \in C([0,T];H^{-\kappa})^3$ and

$$\mathcal{N}(Y,\mathbb{X}) = Y^3 + 3Y^2X + 3YX^{(2)} + X^{(3)}, \tag{144.37}$$

then $\mathcal{N}(Y,\mathbb{X}) \in C([0,T];H^{-\kappa})$ for every $Y \in C([0,T];H^\beta)$. We write

$$\|\mathbb{X}\|_{C_TH^{-\kappa}} = \max_{1\le j\le 3} \|X^{(j)}\|_{C([0,T];H^{-\kappa})}, \qquad X^{(1)} = X,$$

and for $\|Y\|_{C_TH^\beta}, \|Z\|_{C_TH^\beta} \le R$,

$$\|\mathcal{N}(Y,\mathbb{X}) - \mathcal{N}(Z,\mathbb{X})\|_{C_TH^{-\kappa}} \le C_{\kappa,R}\big(1+\|\mathbb{X}\|_{C_TH^{-\kappa}}\big)\|Y-Z\|_{C_TH^\beta}. \tag{144.38}$$

Proof. We recall the dyadic proof because this is the exact multiplication input in the fixed point. Let Δ_j be a Littlewood–Paley partition on the torus and $S_j = \sum_{i<j} \Delta_i$. Bony's decomposition writes

$$uv = u \prec v + u \circ v + u \succ v, \qquad u \prec v = \sum_j S_{j-1}u\, \Delta_j v,$$

with $u \circ v = \sum_{|i-j|\le 1} \Delta_i u\, \Delta_j v$. Bernstein's inequality in two dimensions gives

$$\|\Delta_j u\|_{L^\infty} \le C 2^{j(1-\beta)} \|u\|_{H^\beta} \qquad (\beta > 1), \tag{144.39}$$

and summing in j gives $H^\beta \hookrightarrow L^\infty$. The paraproduct estimates following from the almost-disjoint frequency supports are

$$\|u \prec v\|_{H^s} + \|u \succ v\|_{H^s} \le C\|u\|_{L^\infty}\|v\|_{H^s} + C\|v\|_{L^\infty}\|u\|_{H^s}, \tag{144.40}$$

and, for the resonant term,

$$\|u \circ v\|_{H^{s+t-1}} \le C\|u\|_{H^s}\|v\|_{H^t} \quad \text{when } s+t>1. \tag{144.41}$$

Indeed, $\Delta_\ell(u \prec v)$ only receives terms with $|j-\ell| \le O(1)$, so

$$2^{\ell s}\|\Delta_\ell(u \prec v)\|_{L^2} \le C\|u\|_{L^\infty} \sum_{|j-\ell|\le O(1)} 2^{js}\|\Delta_j v\|_{L^2},$$

and square summation in ℓ gives the first paraproduct bound; the bound for $u \succ v$ is identical. For $u \circ v$, the product $\Delta_i u\, \Delta_j v$ with $|i-j| \le 1$ has frequencies $\lesssim 2^j$. Bernstein and Cauchy–Schwarz give $\|\Delta_i u\, \Delta_j v\|_{L^2} \le C2^{j(1-s-t)}a_i b_j$, where $a_j = 2^{js}\|\Delta_j u\|_{L^2}$ and $b_j = 2^{jt}\|\Delta_j v\|_{L^2}$. Multiplication by the H^{s+t-1} weight cancels the factor $2^{j(1-s-t)}$, and the condition $s+t>1$ makes the finite-overlap high-frequency convolution summable. Taking $s=t=\beta$ gives the algebra estimate (144.35), since $2\beta - 1 > \beta$. Taking $s=\beta$ and $t=-\kappa$ gives $\beta-\kappa>1$, so the resonant term lands in $H^{\beta-\kappa-1}$. Because $\beta-\kappa-1 = \kappa > -\kappa$, this term is in $H^{-\kappa}$, and the two paraproduct terms are controlled by the L^∞ embedding of H^β. This proves (144.36).

The bounds on $\mathcal{N}$ follow by applying (144.35) and (144.36) term by term:

$$\|Y^3\|_{H^\beta} \le C\|Y\|_{H^\beta}^3, \quad \|Y^2X\|_{H^{-\kappa}} \le C\|Y\|_{H^\beta}^2\|X\|_{H^{-\kappa}},$$

and similarly for $YX^{(2)}$. For the Lipschitz estimate, use

$$Y^3 - Z^3 = (Y-Z)(Y^2+YZ+Z^2), \qquad Y^2X - Z^2X = (Y-Z)(Y+Z)X,$$

and the same two product estimates. This gives (144.38). $\square$

Proposition 144.11 (Besov–Holder product continuity for the DPD nonlinearity)**.** *Let* $0<\kappa<\alpha$*, and let*

$$\mathcal{C}^\gamma(\mathbb{T}^2) = B^\gamma_{\infty,\infty}(\mathbb{T}^2), \qquad \|u\|_{\mathcal{C}^\gamma} = \sup_{j\ge -1} 2^{j\gamma}\|\Delta_j u\|_{L^\infty}.$$

Here $(\Delta_j)_{j\ge-1}$ *is a fixed smooth dyadic partition of unity on the torus, with* Δ_{-1} *the low-frequency block. Then Bony's formula defines canonical continuous bilinear maps extending ordinary multiplication of smooth functions,*

$$\mathcal{C}^\alpha \times \mathcal{C}^{-\kappa} \longrightarrow \mathcal{C}^{-\kappa}, \qquad \mathcal{C}^\alpha \times \mathcal{C}^\alpha \longrightarrow \mathcal{C}^\alpha.$$

Consequently, for

$$\mathbb{X} = (X_1, X_2, X_3) \in C([0,T]; \mathcal{C}^{-\kappa})^3 \quad \textit{and} \quad Y \in C([0,T]; \mathcal{C}^{\alpha}),$$

the expression

$$\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X}) = Y^3 + 3Y^2 X_1 + 3Y X_2 + X_3$$

is a well-defined element of $C([0,T]; \mathcal{C}^{-\kappa})$. *Here*

$$\|\mathbb{X}\|_{C_T \mathcal{C}^{-\kappa}} = \max_{1 \le i \le 3} \|X_i\|_{C([0,T]; \mathcal{C}^{-\kappa})}.$$

The same convention is used for any other triple $\mathbb{Z} = (Z_1, Z_2, Z_3)$. *More precisely, for every* $R > 0$ *there is* $C_{\alpha,\kappa,R}$ *such that, whenever* $\mathbb{Z} = (Z_1, Z_2, Z_3)$ *and*

$$\|Y\|_{C_T \mathcal{C}^{\alpha}} + \|Z\|_{C_T \mathcal{C}^{\alpha}} + \|\mathbb{X}\|_{C_T \mathcal{C}^{-\kappa}} + \|\mathbb{Z}\|_{C_T \mathcal{C}^{-\kappa}} \le R,$$

one has

$$\begin{aligned} &\|\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X}) - \mathcal{N}_{\mathcal{C}}(Z, \mathbb{Z})\|_{C_T \mathcal{C}^{-\kappa}} \\ &\qquad \le C_{\alpha,\kappa,R} \left(\|Y - Z\|_{C_T \mathcal{C}^{\alpha}} + \max_{1 \le i \le 3} \|X_i - Z_i\|_{C_T \mathcal{C}^{-\kappa}} \right). \end{aligned} \tag{144.42}$$

Proof. It is enough to prove the two multiplication statements, since the continuity of $\mathcal{N}_{\mathcal{C}}$ then follows by expanding differences:

$$Y^3 - Z^3 = (Y - Z)(Y^2 + YZ + Z^2), \qquad Y^2 X_1 - Z^2 Z_1 = (Y^2 - Z^2) X_1 + Z^2 (X_1 - Z_1),$$

and similarly for $Y X_2 - Z Z_2$.

Use Bony's decomposition

$$fg = f \prec g + f \circ g + f \succ g, \qquad f \prec g = \sum_j S_{j-1} f \, \Delta_j g,$$

where $S_j = \sum_{i<j} \Delta_i$ and

$$f \circ g = \sum_{|i-j| \le 1} \Delta_i f \, \Delta_j g.$$

The frequency supports imply that $\Delta_\ell(f \prec g)$ receives only $|j - \ell| \le C_0$. Hence, for any $\beta \in \mathbb{R}$,

$$2^{\ell\beta} \|\Delta_\ell (f \prec g)\|_{L^\infty} \le C \|f\|_{L^\infty} \sup_{|j-\ell| \le C_0} 2^{j\beta} \|\Delta_j g\|_{L^\infty},$$

and therefore

$$\|f \prec g\|_{\mathcal{C}^\beta} \le C \|f\|_{L^\infty} \|g\|_{\mathcal{C}^\beta}. \tag{144.43}$$

If $\beta < 0$, then the low-frequency sum satisfies

$$\|S_{j-1} g\|_{L^\infty} \le C 2^{-j\beta} \|g\|_{\mathcal{C}^\beta},$$

because $\sum_{i<j} 2^{-i\beta} \le C 2^{-j\beta}$. Thus, for $\alpha > 0$ and $\beta < 0$,

$$\|g \prec f\|_{\mathcal{C}^{\alpha+\beta}} \le C \|g\|_{\mathcal{C}^\beta} \|f\|_{\mathcal{C}^\alpha}. \tag{144.44}$$

For the resonant term, the block $\Delta_\ell(f \circ g)$ receives only indices $i, j \geq \ell - C_1$ with $|i-j| \leq 1$. Therefore

$$2^{\ell(\alpha+\beta)} \|\Delta_\ell(f \circ g)\|_{L^\infty} \leq C \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta} \sum_{i \geq \ell - C_1} 2^{(\ell-i)(\alpha+\beta)}.$$

The final geometric sum is bounded exactly when $\alpha + \beta > 0$, and then

$$\|f \circ g\|_{\mathcal{C}^{\alpha+\beta}} \leq C \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}. \tag{144.45}$$

Now take $\beta = -\kappa$. Since $\alpha - \kappa > 0$, (144.43), (144.44), and (144.45) give

$$\|fg\|_{\mathcal{C}^{-\kappa}} \leq C_{\alpha,\kappa} \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^{-\kappa}},$$

using the continuous embedding $\mathcal{C}^{\alpha-\kappa} \hookrightarrow \mathcal{C}^{-\kappa}$. Taking instead $f, g \in \mathcal{C}^\alpha$, the paraproduct terms are controlled by $\mathcal{C}^\alpha \hookrightarrow L^\infty$, and the resonant term belongs to $\mathcal{C}^{2\alpha}$, hence to $\mathcal{C}^\alpha$. This proves that $\mathcal{C}^\alpha$ is an algebra. The bilinear maps are therefore well-defined by the displayed paraproduct and resonant sums. On smooth inputs the dyadic partition is a partition of unity, so Bony's formula equals the ordinary pointwise product; on rough inputs the formula is the canonical product used in this Besov scale. If a sequence of smooth inputs converges in a slightly stronger Besov norm, the bilinear estimates show that its ordinary products converge to the same paraproduct value.

Applying these two bilinear estimates at each time t, taking suprema in time, and expanding the displayed differences gives (144.42). □

Proposition 144.12 (Besov heat smoothing on the torus)**.** *Let $m^2 > 0$, $0 < \theta < 1$, and $A = \Delta - m^2$ on $\mathbb{T}^2$. For every $\rho \in \mathbb{R}$ there is $C_{\rho,\theta,m} < \infty$ such that*

$$\|e^{tA} f\|_{\mathcal{C}^{\rho+2\theta}} \leq C_{\rho,\theta,m}\, t^{-\theta}\, \|f\|_{\mathcal{C}^\rho}, \qquad 0 < t \leq 1. \tag{144.46}$$

Consequently, if $F \in C([0,T]; \mathcal{C}^\rho)$ and $T \leq 1$, then

$$\left\| \int_0^t e^{(t-s)A} F(s)\, \mathrm{d}s \right\|_{C([0,T];\mathcal{C}^{\rho+2\theta})} \leq C_{\rho,\theta,m}\, T^{1-\theta} \|F\|_{C([0,T];\mathcal{C}^\rho)}.$$

Proof. Let $(\Delta_j)_{j \geq -1}$ be the dyadic partition used in Proposition 144.11. For $j \geq 0$, let $K_{j,t}$ be the convolution kernel on $\mathbb{T}^2$ of the multiplier

$$k \longmapsto \varphi(2^{-j} k) \exp(-t(|k|^2 + m^2)),$$

where φ is the annular cutoff defining Δ_j. The rescaled multiplier

$$\xi \longmapsto \varphi(\xi) \exp(-t(2^{2j}|\xi|^2 + m^2))$$

has derivatives of every fixed order bounded by

$$C_N (1 + t2^{2j})^N \exp(-c\, t2^{2j}),$$

hence by $C_N \exp(-c_N t 2^{2j})$. The standard integration-by-parts estimate for Fourier kernels with compactly supported smooth multipliers, applied after this rescaling and then periodized to $\mathbb{T}^2$, gives

$$\|K_{j,t}\|_{L^1(\mathbb{T}^2)} \leq C_N (1 + t2^{2j})^{-N} \leq C_\theta t^{-\theta} 2^{-2\theta j}, \qquad j \geq 0, \quad 0 < t \leq 1,$$

after choosing $N > \theta$. The low block Δ_{-1} has a smooth compactly supported multiplier, so its kernel has uniformly bounded L^1-norm and is absorbed by increasing the constant. Since Δ_j commutes with e^{tA}, Young's inequality gives

$$2^{j(\rho+2\theta)}\|\Delta_j e^{tA} f\|_{L^\infty} \le C_{\theta,m} t^{-\theta} 2^{j\rho}\|\Delta_j f\|_{L^\infty}.$$

Taking the supremum over $j \ge -1$ proves (144.46).

For the Duhamel term, apply (144.46) to $e^{(t-s)A}F(s)$ and integrate:

$$\left\|\int_0^t e^{(t-s)A}F(s)\,\mathrm{d}s\right\|_{\mathcal{C}^{\rho+2\theta}} \le C_{\rho,\theta,m}\|F\|_{C_T\mathcal{C}^\rho}\int_0^t (t-s)^{-\theta}\,\mathrm{d}s.$$

Since $\theta < 1$, the last integral is $t^{1-\theta}/(1-\theta)$, uniformly bounded by a constant times $T^{1-\theta}$. $\square$

Theorem 144.13 (Deterministic DPD local fixed point in Besov–Holder coordinates)**.** *Let $m^2 > 0$, $\lambda > 0$, and choose exponents*

$$0 < \kappa < \alpha < 2 - \kappa.$$

Let

$$\mathbb{X} = (X_1, X_2, X_3) \in C([0,T_0];\mathcal{C}^{-\kappa}(\mathbb{T}^2))^3, \qquad Y_0 \in \mathcal{C}^\alpha(\mathbb{T}^2).$$

Set

$$M_{\mathbb{X}} = \max_{1\le i\le 3}\|X_i\|_{C([0,T_0];\mathcal{C}^{-\kappa})}.$$

For another enhanced noise $\mathbb{Z} = (Z_1, Z_2, Z_3)$, the notation $\|\mathbb{Z}\|_{C_T\mathcal{C}^{-\kappa}}$ below always means the same maximum over the three components restricted to $[0,T]$. Then there is $T \in (0, T_0]$, depending only on

$$\alpha, \kappa, m, \lambda, \|Y_0\|_{\mathcal{C}^\alpha}, \quad \textit{and} \quad M_{\mathbb{X}},$$

such that the mild equation

$$Y(t) = e^{t(\Delta-m^2)}Y_0 - \lambda\int_0^t e^{(t-s)(\Delta-m^2)}\mathcal{N}_{\mathcal{C}}(Y(s), \mathbb{X}(s))\,\mathrm{d}s \tag{144.47}$$

has a unique solution $Y \in C([0,T];\mathcal{C}^\alpha)$. On every ball where

$$\|Y_0\|_{\mathcal{C}^\alpha} + \|Z_0\|_{\mathcal{C}^\alpha} + \|\mathbb{X}\|_{C_T\mathcal{C}^{-\kappa}} + \|\mathbb{Z}\|_{C_T\mathcal{C}^{-\kappa}} \le R,$$

and for T chosen depending only on $\alpha, \kappa, m, \lambda, R$, the solution map

$$(Y_0, \mathbb{X}) \longmapsto Y$$

is Lipschitz from

$$\mathcal{C}^\alpha \times \left(C_T\mathcal{C}^{-\kappa}\right)^3$$

to $C_T\mathcal{C}^\alpha$.

Proof. Put

$$\theta = \frac{\alpha+\kappa}{2}.$$

The hypothesis $\alpha < 2 - \kappa$ is exactly $\theta < 1$, which is the integrability condition for the Duhamel kernel. Let $A = \Delta - m^2$ and define, for $Y \in C_T\mathcal{C}^\alpha$,

$$(\Psi Y)(t) = e^{tA}Y_0 - \lambda \int_0^t e^{(t-s)A}\mathcal{N}_{\mathcal{C}}(Y(s), \mathbb{X}(s))\, \mathrm{d}s.$$

By Proposition 144.11, $\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X}) \in C_T\mathcal{C}^{-\kappa}$. Applying Proposition 144.12 with $\rho = -\kappa$ and $2\theta = \alpha + \kappa$ gives

$$\left\| \int_0^t e^{(t-s)A}\mathcal{N}_{\mathcal{C}}(Y(s), \mathbb{X}(s))\, \mathrm{d}s \right\|_{C_T\mathcal{C}^\alpha} \leq CT^{1-\theta}\|\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X})\|_{C_T\mathcal{C}^{-\kappa}}. \tag{144.48}$$

The heat semigroup is bounded on $\mathcal{C}^\alpha$. Choose

$$R_0 = 2C\|Y_0\|_{\mathcal{C}^\alpha} + 1.$$

For $\|Y\|_{C_T\mathcal{C}^\alpha} \leq R_0$, the product-continuity proposition gives

$$\|\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X})\|_{C_T\mathcal{C}^{-\kappa}} \leq C_{\alpha,\kappa,R_0+M_{\mathbb{X}}}.$$

Taking $T \leq T_0$ so that the right side of (144.48), multiplied by λ, is at most $R_0/2$, the map Ψ sends the closed ball

$$B_{R_0} = \{Y \in C_T\mathcal{C}^\alpha : \|Y\|_{C_T\mathcal{C}^\alpha} \leq R_0\}$$

to itself.

For $Y, Z \in B_{R_0}$, equations (144.42) and (144.48) give

$$\|\Psi Y - \Psi Z\|_{C_T\mathcal{C}^\alpha} \leq \lambda C_{\alpha,\kappa,m,R_0+M_{\mathbb{X}}}T^{1-\theta}\|Y - Z\|_{C_T\mathcal{C}^\alpha}.$$

Decreasing T if necessary makes the coefficient strictly smaller than one. Since $C_T\mathcal{C}^\alpha$ is complete, Banach's fixed-point theorem gives a unique fixed point in B_{R_0}. If another solution exists on the same interval with finite $C_T\mathcal{C}^\alpha$ norm, the same contraction argument on a smaller ball containing both solutions, possibly after partitioning the time interval into finitely many subintervals, forces equality.

For Lipschitz dependence, let Y and Z solve (144.47) with data $(Y_0, \mathbb{X})$ and $(Z_0, \mathbb{Z})$, all bounded by R. The semigroup bound, the Duhamel estimate, and (144.42) give

$$\begin{aligned}\|Y - Z\|_{C_T\mathcal{C}^\alpha} &\leq C\|Y_0 - Z_0\|_{\mathcal{C}^\alpha} \\ &\quad + \lambda C_{\alpha,\kappa,m,R}T^{1-\theta}\left(\|Y - Z\|_{C_T\mathcal{C}^\alpha} + \max_i \|X_i - Z_i\|_{C_T\mathcal{C}^{-\kappa}}\right).\end{aligned}$$

Choose T so the coefficient of $\|Y - Z\|_{C_T\mathcal{C}^\alpha}$ on the right is at most $1/2$, and absorb it to the left. This proves the stated Lipschitz bound. □

Theorem 144.14 (Deterministic DPD local fixed point in Sobolev coordinates). *Let* $0 < \kappa < 1/4$, $\beta = 1 + 2\kappa$, $m^2 > 0$, $\lambda > 0$, *and*

$$\mathbb{X} = (X, X^{(2)}, X^{(3)}) \in C([0, T_0]; H^{-\kappa}(\mathbb{T}^2))^3.$$

Set

$$M_{\mathbb{X}} = \max_{1 \leq j \leq 3} \|X^{(j)}\|_{C([0,T_0];H^{-\kappa})}, \qquad X^{(1)} = X.$$

For every $Y_0 \in H^\beta(\mathbb{T}^2)$ there is $T \in (0, T_0]$, depending only on $\|Y_0\|_{H^\beta}$, λ, m, κ, and $M_{\mathbb{X}}$, such that the mild equation

$$Y(t) = e^{t(\Delta - m^2)} Y_0 - \lambda \int_0^t e^{(t-s)(\Delta - m^2)} \mathcal{N}(Y(s), \mathbb{X}(s))\, \mathrm{d}s \tag{144.49}$$

has a unique solution $Y \in C([0,T]; H^\beta)$. The solution depends locally Lipschitz continuously on Y_0 and on $\mathbb{X}$ in these norms.

Proof. Let $C_T H^r = C([0,T]; H^r)$ and put $A = \Delta - m^2$. Define

$$(\Psi Y)(t) = e^{tA} Y_0 - \lambda \int_0^t e^{(t-s)A} \mathcal{N}(Y(s), \mathbb{X}(s))\, \mathrm{d}s.$$

The forcing belongs to $C_T H^{-\kappa}$ by Lemma 144.10. To estimate the Duhamel term in H^β, use the heat-kernel estimate (144.30) with

$$r = -\kappa, \qquad r + 2\theta = \beta, \qquad \theta = \frac{\beta + \kappa}{2} = \frac{1 + 3\kappa}{2}.$$

Since $\kappa < 1/4$, $\theta < 1$. Hence

$$\left\| \int_0^t e^{(t-s)A} F(s)\, \mathrm{d}s \right\|_{C_T H^\beta} \le C_{\kappa,m} T^{1-\theta} \|F\|_{C_T H^{-\kappa}}. \tag{144.50}$$

Choose $R = 2C\|Y_0\|_{H^\beta} + 1$, where C bounds the heat semigroup on H^β. Lemma 144.10 gives, for $\|Y\|_{C_T H^\beta} \le R$,

$$\|\mathcal{N}(Y, \mathbb{X})\|_{C_T H^{-\kappa}} \le C_{\kappa,R}\big(1 + \|\mathbb{X}\|_{C_T H^{-\kappa}}\big).$$

By (144.50), T can be chosen so small that Ψ maps the closed ball $B_R \subset C_T H^\beta$ into itself. For $Y, Z \in B_R$, equations (144.38) and (144.50) give

$$\|\Psi Y - \Psi Z\|_{C_T H^\beta} \le |\lambda| C_{\kappa,R,m} T^{1-\theta} \big(1 + \|\mathbb{X}\|_{C_T H^{-\kappa}}\big) \|Y - Z\|_{C_T H^\beta}.$$

Reducing T once more makes the coefficient < 1. Starting from any $Y_0^\sharp \in B_R$ and setting $Y_{n+1}^\sharp = \Psi Y_n^\sharp$, the differences obey

$$\|Y_{n+1}^\sharp - Y_n^\sharp\|_{C_T H^\beta} \le q^n \|Y_1^\sharp - Y_0^\sharp\|_{C_T H^\beta}, \qquad q < 1.$$

The resulting sequence is Cauchy in the complete space $C_T H^\beta$; its limit is a fixed point, and the same inequality applied to two fixed points forces equality. Applying the contraction estimate to two sets of data gives local Lipschitz dependence, after taking the smaller of the two local existence times. □

This theorem is deliberately weaker than the sharp Da Prato–Debussche theorem in Holder–Besov spaces, but it proves the exact deterministic logic: enhanced noise first makes every term in the nonlinearity meaningful, then heat smoothing supplies a positive time power, and the local solution is obtained by contraction. The remaining global and invariant-measure steps require estimates beyond this local theorem.

144.5.2 Energy Inequalities and Compactness of Smooth Approximations

Lemma 144.15 (Smooth enhanced-noise energy estimate)**.** *Let X_1, X_2, X_3 be smooth real functions on $[0,T] \times \mathbb{T}^2$, let $m^2 > 0$, $\lambda > 0$, and let Y be a smooth solution of*

$$\partial_t Y = (\Delta - m^2)Y - \lambda\big(Y^3 + 3Y^2 X_1 + 3Y X_2 + X_3\big). \tag{144.51}$$

Then, for a numerical constant C,

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\|Y(t)\|_{L^2}^2 + \|\nabla Y(t)\|_{L^2}^2 + m^2\|Y(t)\|_{L^2}^2 + \frac{\lambda}{4}\|Y(t)\|_{L^4}^4 \\ \leq C\lambda\left(\|X_1(t)\|_{L^4}^4 + \|X_2(t)\|_{L^2}^2 + \|X_3(t)\|_{L^{4/3}}^{4/3}\right).\end{aligned} \tag{144.52}$$

Consequently no finite-time blowup can occur in a smooth Galerkin approximation whose enhanced noise norms on the right-hand side stay integrable on compact stochastic-time intervals.

Proof. Multiply (144.51) by Y and integrate over the torus. Periodicity gives

$$\int_{\mathbb{T}^2} Y \Delta Y \, \mathrm{d}^2 x = -\|\nabla Y\|_{L^2}^2.$$

Thus

$$\begin{aligned}&\frac{1}{2}\frac{d}{dt}\|Y\|_{L^2}^2 + \|\nabla Y\|_{L^2}^2 + m^2\|Y\|_{L^2}^2 + \lambda\|Y\|_{L^4}^4 \\ &\qquad = -3\lambda\int Y^3 X_1 - 3\lambda\int Y^2 X_2 - \lambda\int Y X_3.\end{aligned}$$

Young's inequality with conjugate exponents $(4/3, 4)$, $(2, 2)$, and $(4, 4/3)$ gives pointwise bounds

$$\begin{aligned}3|Y|^3|X_1| &\leq \frac{1}{4}|Y|^4 + C|X_1|^4, \\ 3|Y|^2|X_2| &\leq \frac{1}{4}|Y|^4 + C|X_2|^2, \\ |Y||X_3| &\leq \frac{1}{4}|Y|^4 + C|X_3|^{4/3}.\end{aligned}$$

Substituting these three estimates into the energy identity and retaining the remaining positive $\lambda\|Y\|_{L^4}^4/4$ proves (144.52). Integrating the inequality in time bounds $\|Y(t)\|_{L^2}^2$, the accumulated H^1 norm, and the accumulated L^4 norm by the initial L^2 norm and the three enhanced noise norms. A smooth Galerkin solution can be continued as long as its finite-dimensional norm is finite; the displayed bound prevents blowup on intervals where the right-hand side is integrable. □

Proposition 144.16 (Closedness of the DPD energy inequality under approximation)**.** *Let $m^2 > 0$, $\lambda > 0$, and $T > 0$. For $n \geq 1$, let $(X_{1,n}, X_{2,n}, X_{3,n})$ be smooth real functions on $[0,T] \times \mathbb{T}^2$, and let Y_n be a smooth solution of (144.51) with X_i replaced by $X_{i,n}$. Assume*

$$\begin{aligned}X_{1,n} &\to X_1 \quad \text{in } L^4([0,T] \times \mathbb{T}^2), \\ X_{2,n} &\to X_2 \quad \text{in } L^2([0,T] \times \mathbb{T}^2), \\ X_{3,n} &\to X_3 \quad \text{in } L^{4/3}([0,T] \times \mathbb{T}^2),\end{aligned}$$

and suppose that $Y_n(0) \to Y_0$ strongly in $L^2(\mathbb{T}^2)$. Let Y be a function such that, after passing to a subsequence,

$$\begin{aligned} Y_n &\rightharpoonup Y && \text{weakly in } L^2([0,T];H^1(\mathbb{T}^2)),\\ Y_n &\rightharpoonup Y && \text{weakly in } L^4([0,T]\times\mathbb{T}^2),\\ Y_n(T) &\rightharpoonup Y(T) && \text{weakly in } L^2(\mathbb{T}^2). \end{aligned}$$

Then Y satisfies the integrated energy inequality

$$\begin{aligned} \frac{1}{2}\|Y(T)\|_{L^2}^2 &+ \int_0^T \left[\|\nabla Y(t)\|_{L^2}^2 + m^2\|Y(t)\|_{L^2}^2 + \frac{\lambda}{4}\|Y(t)\|_{L^4}^4\right] \mathrm{d}t \\ &\le \frac{1}{2}\|Y_0\|_{L^2}^2 + C\lambda \int_0^T \left[\|X_1(t)\|_{L^4}^4 + \|X_2(t)\|_{L^2}^2 + \|X_3(t)\|_{L^{4/3}}^{4/3}\right] \mathrm{d}t. \qquad (144.53) \end{aligned}$$

Proof. Integrating (144.52) for the n-th smooth solution gives

$$\begin{aligned} \frac{1}{2}\|Y_n(T)\|_{L^2}^2 &+ \int_0^T \left[\|\nabla Y_n(t)\|_{L^2}^2 + m^2\|Y_n(t)\|_{L^2}^2 + \frac{\lambda}{4}\|Y_n(t)\|_{L^4}^4\right] \mathrm{d}t \\ &\le \frac{1}{2}\|Y_n(0)\|_{L^2}^2 + C\lambda \int_0^T \left[\|X_{1,n}(t)\|_{L^4}^4 + \|X_{2,n}(t)\|_{L^2}^2 + \|X_{3,n}(t)\|_{L^{4/3}}^{4/3}\right] \mathrm{d}t. \qquad (144.54) \end{aligned}$$

The right hand side converges to the right hand side of (144.53): the initial data converge strongly in L^2, and convergence in L^p implies convergence of $\|X_{1,n}\|_{L^4}^4$, $\|X_{2,n}\|_{L^2}^2$, and $\|X_{3,n}\|_{L^{4/3}}^{4/3}$ by the elementary inequality

$$\big||a|^p - |b|^p\big| \le p\,|a-b|(|a|+|b|)^{p-1}, \qquad p \ge 1,$$

followed by Hölder's inequality.

The left hand side is lower semicontinuous under the assumed weak convergences. For the quadratic terms this is the lower semicontinuity of the norm in Hilbert space, applied in

$$L^2([0,T];H^1(\mathbb{T}^2)), \qquad L^2([0,T];L^2(\mathbb{T}^2)),$$

and also at the endpoint T. For the quartic term, the map

$$u \longmapsto \|u\|_{L^4([0,T]\times\mathbb{T}^2)}^4$$

is convex and norm-continuous on L^4, hence weakly lower semicontinuous. Taking the liminf of the left hand side of (144.54) and the limit of its right hand side proves (144.53). ☐

Proposition 144.17 (Energy compactness for smooth DPD remainders)**.** *Let $m^2 > 0$, $\lambda > 0$, $T > 0$, and set $Q_T = [0,T]\times\mathbb{T}^2$. For $n \ge 1$, let $(X_{1,n}, X_{2,n}, X_{3,n})$ be smooth real functions on Q_T, and let Y_n be a smooth solution of (144.51) with X_i replaced by $X_{i,n}$. Assume that*

$$M_T := \sup_n \left[\|Y_n(0)\|_{L^2(\mathbb{T}^2)} + \|X_{1,n}\|_{L^4(Q_T)} + \|X_{2,n}\|_{L^2(Q_T)} + \|X_{3,n}\|_{L^{4/3}(Q_T)}\right] < \infty.$$

Then there is a constant C_T, depending on T, m^2, λ, and M_T, but not on n, such that

$$\begin{aligned} \sup_n \Big[\|Y_n\|_{L^\infty([0,T];L^2)} &+ \|Y_n\|_{L^2([0,T];H^1)} + \|Y_n\|_{L^4(Q_T)} \\ &+ \|\partial_t Y_n\|_{L^{4/3}([0,T];H^{-1})}\Big] \le C_T. \qquad (144.55) \end{aligned}$$

Consequently, after passing to a subsequence, there is a function Y such that

$$\begin{aligned} Y_n &\rightharpoonup^* Y \quad \text{weakly-* in } L^\infty([0,T];L^2(\mathbb{T}^2)),\\ Y_n &\rightharpoonup Y \quad \text{weakly in } L^2([0,T];H^1(\mathbb{T}^2)),\\ Y_n &\rightharpoonup Y \quad \text{weakly in } L^4(Q_T), \end{aligned}$$

and, after a further subsequence if necessary,

$$Y_n(T) \rightharpoonup Y(T) \qquad \text{weakly in } L^2(\mathbb{T}^2),$$

where Y is represented by its weakly L^2-continuous time representative.

Proof. Apply (144.52) on every interval $[0,t]$, $0 \le t \le T$. Since the right hand side is bounded uniformly by the definition of M_T, the $L^\infty_t L^2_x$, $L^2_t H^1_x$, and $L^4_{t,x}$ parts of (144.55) follow.

It remains to prove the time-derivative bound. Let $\|v\|_{H^1(\mathbb{T}^2)} \le 1$. In two dimensions the Sobolev embedding $H^1(\mathbb{T}^2) \hookrightarrow L^4(\mathbb{T}^2)$ gives

$$\|f\|_{H^{-1}} \le C\|f\|_{L^{4/3}} \qquad \text{for } f \in L^{4/3}(\mathbb{T}^2).$$

Testing the smooth equation against v therefore gives, in the dual space H^{-1},

$$\begin{aligned} \|\partial_t Y_n\|_{H^{-1}} \le C\Big(&\|Y_n\|_{H^1} + \|Y_n\|_{L^2} + \|Y_n\|_{L^4}^3 + \|Y_n\|_{L^4}^2\|X_{1,n}\|_{L^4} \\ &+ \|Y_n\|_{L^4}\|X_{2,n}\|_{L^2} + \|X_{3,n}\|_{L^{4/3}}\Big). \end{aligned} \tag{144.56}$$

Here and below the spatial norms are taken on $\mathbb{T}^2$ at the fixed time under consideration. Taking the $L^{4/3}$-norm in time and using Hölder's inequality on Q_T gives

$$\begin{aligned} \|Y_n^3\|_{L^{4/3}(Q_T)} &= \|Y_n\|_{L^4(Q_T)}^3,\\ \|Y_n^2 X_{1,n}\|_{L^{4/3}(Q_T)} &\le \|Y_n\|_{L^4(Q_T)}^2\|X_{1,n}\|_{L^4(Q_T)},\\ \|Y_n X_{2,n}\|_{L^{4/3}(Q_T)} &\le \|Y_n\|_{L^4(Q_T)}\|X_{2,n}\|_{L^2(Q_T)}. \end{aligned}$$

The already established energy bounds and the definition of M_T now prove the $L^{4/3}_t H^{-1}_x$ estimate for $\partial_t Y_n$.

The compactness assertions are consequences of standard weak compactness, but we spell out the endpoint identification because it is the point needed in Proposition 144.16. The spaces $L^2([0,T];H^1)$ and $L^4(Q_T)$ are reflexive, while $L^\infty([0,T];L^2)$ is the dual of $L^1([0,T];L^2)$ because $L^2(\mathbb{T}^2)$ is separable. Banach–Alaoglu and a diagonal subsequence therefore give the displayed weak and weak-* convergences. The derivative bound passes to the same limit in $\mathcal{D}'((0,T);H^{-1})$, so the limit has an H^{-1}-valued absolutely continuous representative.

Let $D \subset H^1(\mathbb{T}^2)$ be countable and dense in L^2. For each $v \in D$, the scalar functions

$$a_{n,v}(t) := \langle Y_n(t), v\rangle_{L^2}$$

are bounded in $W^{1,4/3}([0,T])$: the $L^\infty_t L^2_x$ bound controls $a_{n,v}$, and the $L^{4/3}_t H^{-1}_x$ bound controls its derivative. The compact embedding $W^{1,4/3}([0,T]) \hookrightarrow C([0,T])$, together with the distributional convergence already obtained, implies

$$\langle Y_n(T), v\rangle_{L^2} \longrightarrow \langle Y(T), v\rangle \qquad (v \in D)$$

after a diagonal extraction. The sequence $Y_n(T)$ is bounded in L^2, so a further subsequence has a weak L^2 limit; the preceding identity on the dense set D identifies that limit with $Y(T)$. Hence $Y_n(T) \rightharpoonup Y(T)$ weakly in $L^2(\mathbb{T}^2)$. $\square$

Lemma 144.18 (Fourier Aubin–Lions compactness on the torus)**.** *Let (u_n) be a sequence such that*

$$\sup_n \Big[\|u_n\|_{L^2([0,T];H^1(\mathbb{T}^2))} + \|\partial_t u_n\|_{L^{4/3}([0,T];H^{-1}(\mathbb{T}^2))} \Big] < \infty.$$

Then a subsequence converges strongly in $L^2([0,T];L^2(\mathbb{T}^2))$.

Proof. Let P_M denote the Fourier projection onto modes $|k| \le M$. The high-frequency tail is uniformly small:

$$\|(1-P_M)u_n\|_{L^2_t L^2_x} \le (1+M^2)^{-1/2} \|u_n\|_{L^2_t H^1_x}.$$

It remains to prove compactness of $P_M u_n$ for fixed M. The range $E_M = P_M L^2(\mathbb{T}^2)$ is finite dimensional. On E_M all Hilbert norms are equivalent, and the hypotheses imply

$$\sup_n \Big[\|P_M u_n\|_{L^2([0,T];E_M)} + \|\partial_t P_M u_n\|_{L^{4/3}([0,T];E_M)} \Big] < \infty.$$

Since $[0,T]$ has finite measure, the first term also bounds $\|P_M u_n\|_{L^{4/3}([0,T];E_M)}$. Thus $P_M u_n$ is bounded in $W^{1,4/3}([0,T];E_M)$. For $s,t \in [0,T]$,

$$\|P_M u_n(t) - P_M u_n(s)\|_{E_M} \le |t-s|^{1/4} \, \|\partial_t P_M u_n\|_{L^{4/3}([0,T];E_M)},$$

so the family is equicontinuous. The $L^{4/3}$-bound also gives a uniform pointwise bound: for each n, choose t_n with

$$\|P_M u_n(t_n)\|_{E_M} \le T^{-3/4} \|P_M u_n\|_{L^{4/3}([0,T];E_M)},$$

and use the preceding Hölder estimate to bound $P_M u_n(t)$ for every t. Arzela–Ascoli in the finite-dimensional space E_M now gives a subsequence converging in $C([0,T];E_M)$, hence strongly in $L^2([0,T];E_M)$. Taking $M = 1,2,\dots$ and using a diagonal subsequence, the finite-mode parts converge for every M. The uniform tail estimate then shows that the same subsequence is Cauchy in $L^2([0,T];L^2(\mathbb{T}^2))$, proving the lemma. $\square$

Proposition 144.19 (Identification of the DPD distributional limit)**.** *Let the hypotheses of Proposition 144.17 hold, and in addition assume that, for some functions X_1, X_2, X_3,*

$$\begin{aligned} X_{1,n} &\to X_1 \quad \textit{strongly in } L^4(Q_T), \\ X_{2,n} &\to X_2 \quad \textit{strongly in } L^2(Q_T), \\ X_{3,n} &\to X_3 \quad \textit{strongly in } L^{4/3}(Q_T), \end{aligned}$$

and that $Y_n(0) \to Y_0$ strongly in $L^2(\mathbb{T}^2)$. Let Y be the subsequential limit supplied by Proposition 144.17. After passing to a subsequence, $Y_n \to Y$ strongly in $L^r(Q_T)$ for every $1 \le r < 4$. Moreover Y satisfies

$$\partial_t Y = \Delta Y - m^2 Y - \lambda\big(Y^3 + 3Y^2 X_1 + 3Y X_2 + X_3\big) \tag{144.57}$$

as an identity in $\mathcal{D}'((0,T)\times\mathbb{T}^2)$. Equivalently, for every $\varphi\in C^\infty([0,T]\times\mathbb{T}^2)$,

$$\begin{aligned}\langle Y(T),\varphi(T)\rangle - \langle Y_0,\varphi(0)\rangle - \int_0^T \langle Y(t),\partial_t\varphi(t)\rangle\,\mathrm{d}t \\ = \int_0^T \langle Y(t),\Delta\varphi(t) - m^2\varphi(t)\rangle\,\mathrm{d}t \\ - \lambda\int_{Q_T}\big(Y^3 + 3Y^2X_1 + 3YX_2 + X_3\big)\varphi\,\mathrm{d}t\,\mathrm{d}x. \end{aligned}\tag{144.58}$$

Proof. The compactness lemma applied to $u_n = Y_n$ gives, after passing to a subsequence,

$$Y_n \to Y \qquad \text{strongly in } L^2(Q_T).$$

The same sequence is bounded in $L^4(Q_T)$. Interpolation between L^2 convergence and the uniform L^4 bound gives strong convergence in $L^r(Q_T)$ for every $2\le r<4$, and finite measure gives strong convergence for $1\le r<2$ as well.

We next pass the nonlinearities. The cubic term satisfies

$$\|Y_n^3 - Y^3\|_{L^1} \le C\|Y_n - Y\|_{L^2}\big(\|Y_n\|_{L^4}^2 + \|Y\|_{L^4}^2\big) \longrightarrow 0.$$

For the Y^2X_1 term, choose the strong convergence $Y_n\to Y$ in $L^{8/3}(Q_T)$. Then

$$\|Y_n^2 - Y^2\|_{L^{4/3}} \le C\|Y_n - Y\|_{L^{8/3}}\big(\|Y_n\|_{L^{8/3}} + \|Y\|_{L^{8/3}}\big) \longrightarrow 0.$$

Hence, using Hölder's inequality and the strong L^4 convergence of $X_{1,n}$,

$$\begin{aligned}\|Y_n^2X_{1,n} - Y^2X_1\|_{L^1} &\le \|Y_n^2 - Y^2\|_{L^{4/3}}\|X_{1,n}\|_{L^4} \\ &\quad + \|Y^2\|_{L^{4/3}}\|X_{1,n} - X_1\|_{L^4} \longrightarrow 0.\end{aligned}$$

The mixed term with X_2 is simpler:

$$\|Y_nX_{2,n} - YX_2\|_{L^1} \le \|Y_n - Y\|_{L^2}\|X_{2,n}\|_{L^2} + \|Y\|_{L^2}\|X_{2,n} - X_2\|_{L^2} \longrightarrow 0.$$

Finally $X_{3,n}\to X_3$ in $L^{4/3}(Q_T)$, hence in $L^1(Q_T)$ because Q_T has finite measure.

For every smooth test function φ, the smooth equation for Y_n gives

$$\begin{aligned}\langle Y_n(T),\varphi(T)\rangle - \langle Y_n(0),\varphi(0)\rangle - \int_0^T\langle Y_n,\partial_t\varphi\rangle\,\mathrm{d}t \\ = \int_0^T\langle Y_n,\Delta\varphi - m^2\varphi\rangle\,\mathrm{d}t - \lambda\int_{Q_T}\big(Y_n^3 + 3Y_n^2X_{1,n} + 3Y_nX_{2,n} + X_{3,n}\big)\varphi\,\mathrm{d}t\,\mathrm{d}x.\end{aligned}$$

The initial term converges by the strong L^2 convergence of $Y_n(0)$. The terminal term converges by the weak L^2 convergence $Y_n(T)\rightharpoonup Y(T)$ from Proposition 144.17. The linear spacetime terms converge by the strong $L^2(Q_T)$ convergence of Y_n, and the nonlinear terms converge by the L^1 convergence just proved. Passing to the limit gives (144.58), which is precisely (144.57) in distributional form. □

Proposition 144.20 (Compatibility of the Besov local solution with smooth energy approximation)**.** *Let $m^2 > 0$, $\lambda > 0$, and*

$$0 < \kappa < \alpha < 2 - \kappa.$$

Let $T_0 > 0$, let

$$Y_{0,n} \in C^\infty(\mathbb{T}^2), \qquad Y_0 \in \mathcal{C}^\alpha(\mathbb{T}^2),$$

with $Y_{0,n} \to Y_0$ in $\mathcal{C}^\alpha$, and let

$$\mathbb{X}_n = (X_{1,n}, X_{2,n}, X_{3,n}) \in C^\infty([0, T_0] \times \mathbb{T}^2)^3$$

converge to $\mathbb{X} = (X_1, X_2, X_3)$ in

$$C([0, T_0]; \mathcal{C}^{-\kappa}(\mathbb{T}^2))^3.$$

Let Y_n be the smooth solution of

$$\partial_t Y_n = (\Delta - m^2)Y_n - \lambda\big(Y_n^3 + 3Y_n^2 X_{1,n} + 3Y_n X_{2,n} + X_{3,n}\big), \qquad Y_n(0) = Y_{0,n}.$$

Let Y be the Besov–Holder mild solution obtained from Theorem 144.13 for the data $(Y_0, \mathbb{X})$. Then there is a time $T \in (0, T_0]$, depending only on a bounded neighborhood of the convergent data, such that

$$Y_n \longrightarrow Y \qquad \text{in } C([0, T]; \mathcal{C}^\alpha). \tag{144.59}$$

Assume in addition that, on $Q_T = [0, T] \times \mathbb{T}^2$,

$$\begin{aligned} X_{1,n} &\to X_1 \quad \text{in } L^4(Q_T), \\ X_{2,n} &\to X_2 \quad \text{in } L^2(Q_T), \\ X_{3,n} &\to X_3 \quad \text{in } L^{4/3}(Q_T), \end{aligned}$$

and that

$$\sup_n \Big[\|Y_{0,n}\|_{L^2} + \|X_{1,n}\|_{L^4(Q_T)} + \|X_{2,n}\|_{L^2(Q_T)} + \|X_{3,n}\|_{L^{4/3}(Q_T)}\Big] < \infty.$$

Then Y is the energy-compactness limit of the same smooth approximations. More precisely, $Y \in L^2([0, T]; H^1) \cap L^4(Q_T)$ and for every $0 \le \tau \le T$,

$$\begin{aligned} &\frac{1}{2}\|Y(\tau)\|_{L^2}^2 + \int_0^\tau \left[\|\nabla Y(t)\|_{L^2}^2 + m^2\|Y(t)\|_{L^2}^2 + \frac{\lambda}{4}\|Y(t)\|_{L^4}^4\right] \mathrm{d}t \\ &\qquad \le \frac{1}{2}\|Y_0\|_{L^2}^2 + C\lambda \int_0^\tau \left[\|X_1(t)\|_{L^4}^4 + \|X_2(t)\|_{L^2}^2 + \|X_3(t)\|_{L^{4/3}}^{4/3}\right] \mathrm{d}t. \end{aligned} \tag{144.60}$$

Proof. Choose R so that, for all sufficiently large n,

$$\|Y_{0,n}\|_{\mathcal{C}^\alpha} + \|Y_0\|_{\mathcal{C}^\alpha} + \|\mathbb{X}_n\|_{C([0,T_0];\mathcal{C}^{-\kappa})} + \|\mathbb{X}\|_{C([0,T_0];\mathcal{C}^{-\kappa})} \le R.$$

The local Lipschitz part of Theorem 144.13 gives a common time $T \in (0, T_0]$ and a Lipschitz constant L_R on this ball. The smooth solution Y_n satisfies the Duhamel formula with ordinary products. Since Bony's product agrees with ordinary multiplication on smooth inputs, this is the same fixed-point equation as (144.47) for the data $(Y_{0,n}, \mathbb{X}_n)$. The uniqueness statement in Theorem 144.13 therefore identifies the smooth solution with the Besov fixed point on $[0, T]$. The Lipschitz estimate gives

$$\|Y_n - Y\|_{C([0,T];\mathcal{C}^\alpha)} \le L_R \left(\|Y_{0,n} - Y_0\|_{\mathcal{C}^\alpha} + \max_i \|X_{i,n} - X_i\|_{C([0,T];\mathcal{C}^{-\kappa})}\right),$$

which proves (144.59).

For the energy statement, fix $\tau \in [0,T]$. The additional hypotheses on $[0,\tau]$ are exactly the hypotheses of Proposition 144.17, so every subsequence of (Y_n) has a further subsequence converging weakly in

$$L^2([0,\tau];H^1), \qquad L^4([0,\tau]\times\mathbb{T}^2), \qquad \text{and at time } \tau \text{ in } L^2.$$

But the already proved convergence in $C([0,T];\mathcal{C}^\alpha)$ implies convergence in $C([0,T];L^2)$, because $\mathbb{T}^2$ has finite volume and $\mathcal{C}^\alpha \hookrightarrow L^\infty \hookrightarrow L^2$. Thus every such weak limit is the same function Y. In particular

$$Y \in L^2([0,\tau];H^1) \cap L^4([0,\tau]\times\mathbb{T}^2).$$

Applying Proposition 144.16 on $[0,\tau]$ gives (144.60). Since τ was arbitrary, the inequality holds for every $\tau \le T$. □

The logical role of this proposition is to identify the two approximation mechanisms on their common domain of validity. The Besov topology defines the canonical rough products and gives a unique local mild solution; the Lebesgue hypotheses are exactly those needed by the elementary energy inequality. For the Wick-enhanced stochastic convolution in two dimensions, the corresponding global theorem requires a rough energy-continuity estimate in which the Lebesgue pairings above are replaced by the appropriate paracontrolled or Besov energy pairings.

The rough Da Prato–Debussche global argument is therefore reduced to two precise analytic tasks. First, construct smooth approximations whose enhanced noises converge in the integrability spaces appearing in Proposition 144.16, or in Besov/paracontrolled spaces that imply those bounds after pairing with the energy solution. Proposition 144.19 then identifies the compactness limit as a distributional solution whenever the products are represented in these strong L^p spaces. In the full rough stochastic application, where the enhanced noise is controlled in Besov or paracontrolled coordinates rather than in ordinary Lebesgue spaces, the same role is played, at the elementary Besov level, by Proposition 144.11. The quartic term supplies the coercive bound, compactness extracts a weak energy limit, product continuity identifies the equation in the topology where the enhanced noise is built, and the closedness proposition passes the energy inequality to that rough limit.

144.5.3 Global Continuity, Invariant Laws, and Tightness

Lemma 144.21 (Global compact-continuity criterion for DPD solution maps)**.** *Fix $T > 0$, $0 < \kappa < 1$, and an exponent $\alpha \in (0, 2-\kappa)$ for which the local Besov fixed-point theorem applies. Let*

$$\mathfrak{D}_T := (\mathcal{C}^\alpha(\mathbb{T}^2) \cap L^2(\mathbb{T}^2)) \times C([0,T];\mathcal{C}^{-\kappa}(\mathbb{T}^2))^3$$

with its product topology, and write a datum as

$$D = (Y_0, \mathbb{X}_1, \mathbb{X}_2, \mathbb{X}_3).$$

For each D let Y_D be the maximal local DPD solution produced by Theorem 144.13, and denote its maximal existence time in $[0,T]$ by $\tau(D)$.

Let $K \subset \mathfrak{D}_T$ be compact. Assume the following two statements hold for every $D \in K$.

1. Blow-up alternative. *If $\tau(D) < T$, then*

$$\limsup_{t\uparrow\tau(D)} \|Y_D\|_{C([0,t];\mathcal{C}^\alpha)} = \infty.$$

2. Rough energy-to-Besov bound. *There is a finite constant $B_{K,T}$, depending on K and T but not on D, such that*

$$\sup_{0<s<\tau(D)} \|Y_D\|_{C([0,s];\mathcal{C}^\alpha)} \le B_{K,T}.$$

Then every $D \in K$ has a solution on the full interval $[0,T]$, and the solution map

$$\mathcal{S}_T : K \longrightarrow C([0,T];\mathcal{C}^\alpha), \qquad D \longmapsto Y_D,$$

is uniformly continuous. Consequently, if $D_N \to D$ in $\mathfrak{D}_T$, and if $\{D\} \cup \{D_N : N \ge N_0\}$ is contained in a compact set on which the two hypotheses above hold with a common constant, then

$$Y_{D_N} \longrightarrow Y_D \quad \textit{in } C([0,T];\mathcal{C}^\alpha).$$

Proof. The full-time existence statement is the contradiction between the two assumptions. If $\tau(D) < T$, the rough energy-to-Besov estimate gives a finite bound for $\|Y_D\|_{C([0,s];\mathcal{C}^\alpha)}$ as $s \uparrow \tau(D)$, whereas the blow-up alternative requires this quantity to be unbounded. Hence $\tau(D) = T$.

It remains to prove continuity with constants that are uniform on K. Compactness of K gives a finite noise bound

$$M_{K,T} := \sup_{D\in K} \sum_{j=1}^{3} \|\mathbb{X}_j\|_{C([0,T];\mathcal{C}^{-\kappa})} < \infty.$$

Inspecting the contraction estimate in Theorem 144.13, one obtains a time $\delta > 0$ and a constant $L < \infty$, depending only on $B_{K,T}$, $M_{K,T}$, T, and the fixed exponents, with the following property. If two solutions are known on a time interval beginning at t_0 and both remainders have $\mathcal{C}^\alpha$-norm at most $B_{K,T}$, then on $[t_0, t_0+\delta] \cap [0,T]$

$$\|Y_D - Y_{\widetilde{D}}\|_{C([t_0,t_0+\delta]\cap[0,T];\mathcal{C}^\alpha)} \le L\Big(\|Y_D(t_0) - Y_{\widetilde{D}}(t_0)\|_{\mathcal{C}^\alpha} + d_{\mathfrak{D}_T}(D,\widetilde{D})\Big),$$

where $d_{\mathfrak{D}_T}$ is any metric inducing the product topology on bounded subsets of $\mathfrak{D}_T$. The same δ works at every t_0 because the heat-kernel estimates depend only on the length of the interval, while the nonlinear estimates depend only on the preceding uniform bounds.

Choose an integer m with $m\delta \ge T$. Iterating the last estimate over the grid $0, \delta, 2\delta, \dots, m\delta$ gives, for D and $\widetilde{D}$ sufficiently close in K,

$$\|Y_D - Y_{\widetilde{D}}\|_{C([0,T];\mathcal{C}^\alpha)} \le C_{K,T}\, d_{\mathfrak{D}_T}(D,\widetilde{D})$$

with finite $C_{K,T}$. Thus $\mathcal{S}_T$ is continuous at each point of K, and compactness of K upgrades pointwise continuity to uniform continuity. The convergence statement is the preceding uniform-continuity claim applied to the compact set containing the tail of the sequence and its limit. □

This criterion separates a soft continuation argument from the genuinely hard stochastic estimate. Once the local Besov theory is known, global cutoff convergence on compact enhanced-noise sets is reduced to proving a rough energy-to-Besov a priori bound. The latter is precisely where the Wick-renormalized coercivity of the two-dimensional equation must enter.

Proposition 144.22 (Local rough-forcing bootstrap for the energy-to-Besov step)**.** *Keep the notation and exponents of Theorem 144.13, and put*

$$\theta = \frac{\alpha + \kappa}{2} \in (0,1), \qquad A = \Delta - m^2.$$

Let C_0 and C_1 be constants such that, for all $0 \le t \le 1$ and all intervals I of length at most 1,

$$\|e^{tA} f\|_{\mathcal{C}^\alpha} \le C_0 \|f\|_{\mathcal{C}^\alpha}, \qquad \left\| \int_{t_0}^t e^{(t-s)A} F(s)\, \mathrm{d}s \right\|_{C(I;\mathcal{C}^\alpha)} \le C_1 |I|^{1-\theta} \|F\|_{C(I;\mathcal{C}^{-\kappa})}. \tag{144.61}$$

Let $Y \in C([0,T];\mathcal{C}^\alpha)$ solve the Besov mild equation (144.47) on $[0,T]$. Choose a partition

$$0 = t_0 < t_1 < \cdots < t_M = T, \qquad I_\ell = [t_\ell, t_{\ell+1}], \qquad |I_\ell| \le \delta \le 1.$$

Assume that on every I_ℓ the renormalized DPD nonlinearity satisfies a local rough-forcing estimate

$$\|\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X})\|_{C(I_\ell;\mathcal{C}^{-\kappa})} \le A_\ell + \varepsilon \|Y\|_{C(I_\ell;\mathcal{C}^\alpha)} \tag{144.62}$$

with $A_\ell \ge 0$ and

$$|\lambda| C_1 \delta^{1-\theta} \varepsilon \le \frac{1}{2}. \tag{144.63}$$

Then, writing

$$D_0 = 2C_0 \|Y(0)\|_{\mathcal{C}^\alpha} + 2|\lambda| C_1 \delta^{1-\theta} A_0,$$

and recursively

$$D_\ell = 2C_0 D_{\ell-1} + 2|\lambda| C_1 \delta^{1-\theta} A_\ell, \qquad 1 \le \ell \le M-1,$$

one has

$$\|Y\|_{C(I_\ell;\mathcal{C}^\alpha)} \le D_\ell, \qquad 0 \le \ell \le M-1. \tag{144.64}$$

Equivalently,

$$\|Y\|_{C([0,T];\mathcal{C}^\alpha)} \le (2C_0)^M \|Y(0)\|_{\mathcal{C}^\alpha} + 2|\lambda| C_1 \delta^{1-\theta} \sum_{j=0}^{M-1} (2C_0)^{M-1-j} A_j. \tag{144.65}$$

In particular, if the numbers A_ℓ are bounded uniformly on a compact set of enhanced noises and initial conditions, and if the same δ, ε satisfy (144.63) throughout that compact set, then the rough energy-to-Besov hypothesis in Lemma 144.21 follows.

Proof. For $t \in I_\ell$, the mild equation restarted at time t_ℓ is

$$Y(t) = e^{(t-t_\ell)A} Y(t_\ell) - \lambda \int_{t_\ell}^t e^{(t-s)A} \mathcal{N}_{\mathcal{C}}(Y(s), \mathbb{X}(s))\, \mathrm{d}s. \tag{144.66}$$

Taking the $C(I_\ell;\mathcal{C}^\alpha)$ norm and using (144.61) gives

$$R_\ell \le C_0 \|Y(t_\ell)\|_{\mathcal{C}^\alpha} + |\lambda| C_1 \delta^{1-\theta} \|\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X})\|_{C(I_\ell;\mathcal{C}^{-\kappa})}, \qquad R_\ell := \|Y\|_{C(I_\ell;\mathcal{C}^\alpha)}.$$

Insert the rough-forcing estimate (144.62). The absorption condition (144.63) implies

$$R_\ell \le 2C_0 \|Y(t_\ell)\|_{\mathcal{C}^\alpha} + 2|\lambda| C_1 \delta^{1-\theta} A_\ell. \tag{144.67}$$

For $\ell = 0$ this is exactly $R_0 \le D_0$. If $\ell \ge 1$, then $\|Y(t_\ell)\|_{\mathcal{C}^\alpha} \le R_{\ell-1}$. Induction using (144.67) proves (144.64). Iterating the recursion for D_ℓ gives the explicit bound (144.65). The compact-set consequence follows by taking the supremum of the right hand side over the compact family. $\square$

Proposition 144.23 (Energy low-mode forcing estimate)**.** *Let $0 < \kappa < \alpha$ and put*

$$\theta = \frac{\alpha + \kappa}{2} < \frac{1}{4}.$$

Let $P_{\le R}$ be a smooth Fourier multiplier on $\mathbb{T}^2$ whose range is contained in the span of the modes $|k| \le R$, with $R \ge 1$. For an interval $I = [a, b]$ of length at most one and smooth fields Y, X_1, X_2, X_3 on $I \times \mathbb{T}^2$, define

$$\mathcal{N}(Y, \mathbb{X}) = Y^3 + 3Y^2 X_1 + 3Y X_2 + X_3$$

and

$$\begin{aligned} L_I(Y, \mathbb{X}) := \|Y\|^3_{L^4(I\times\mathbb{T}^2)} &+ 3\|Y\|^2_{L^4(I\times\mathbb{T}^2)}\|X_1\|_{L^4(I\times\mathbb{T}^2)} \\ &+ 3\|Y\|_{L^4(I\times\mathbb{T}^2)}\|X_2\|_{L^2(I\times\mathbb{T}^2)} + \|X_3\|_{L^{4/3}(I\times\mathbb{T}^2)}. \end{aligned}$$

Then there are constants $C_{\kappa,R}$ and $C_{\alpha,\kappa,R,m}$ such that

$$\|P_{\le R}\mathcal{N}(Y, \mathbb{X})\|_{L^{4/3}(I;\mathcal{C}^{-\kappa})} \le C_{\kappa,R} L_I(Y, \mathbb{X}) \tag{144.68}$$

and, with $A = \Delta - m^2$,

$$\left\| \int_a^t e^{(t-s)A} P_{\le R}\mathcal{N}(Y(s), \mathbb{X}(s))\, \mathrm{d}s \right\|_{C(I;\mathcal{C}^\alpha)} \le C_{\alpha,\kappa,R,m}|I|^{1/4-\theta} L_I(Y, \mathbb{X}). \tag{144.69}$$

Proof. The only finite-dimensional input is the low-mode embedding

$$\|P_{\le R} f\|_{\mathcal{C}^{-\kappa}} \le C_{\kappa,R}\|f\|_{L^{4/3}(\mathbb{T}^2)}. \tag{144.70}$$

Indeed, only dyadic blocks with $2^j \lesssim R$ can contribute. Bernstein's inequality on $\mathbb{T}^2$, with $p = 4/3$, gives

$$2^{-j\kappa}\|\Delta_j P_{\le R} f\|_{L^\infty} \le C 2^{j(3/2-\kappa)}\|f\|_{L^{4/3}},$$

and the supremum over the finitely many contributing j's is absorbed into $C_{\kappa,R}$.

Holder's inequality in spacetime gives the exact energy exponents

$$\begin{aligned} \|Y^3\|_{L^{4/3}(I\times\mathbb{T}^2)} &= \|Y\|^3_{L^4(I\times\mathbb{T}^2)}, \\ \|Y^2 X_1\|_{L^{4/3}(I\times\mathbb{T}^2)} &\le \|Y\|^2_{L^4(I\times\mathbb{T}^2)}\|X_1\|_{L^4(I\times\mathbb{T}^2)}, \\ \|Y X_2\|_{L^{4/3}(I\times\mathbb{T}^2)} &\le \|Y\|_{L^4(I\times\mathbb{T}^2)}\|X_2\|_{L^2(I\times\mathbb{T}^2)}. \end{aligned}$$

Together with (144.70), these estimates prove (144.68). Proposition 144.12, with $\rho = -\kappa$ and $\rho + 2\theta = \alpha$, gives

$$\|e^{(t-s)A}F(s)\|_{\mathcal{C}^\alpha} \le C_{\alpha,\kappa,m}(t-s)^{-\theta}\|F(s)\|_{\mathcal{C}^{-\kappa}}.$$

Applying this to $F = P_{\le R}\mathcal{N}(Y, \mathbb{X})$ and using Holder's inequality in time with conjugate exponents 4 and 4/3 gives

$$\int_a^t (t-s)^{-\theta}\|F(s)\|_{\mathcal{C}^{-\kappa}}\, \mathrm{d}s \le \left(\int_0^{|I|} r^{-4\theta}\, \mathrm{d}r\right)^{1/4} \|F\|_{L^{4/3}(I;\mathcal{C}^{-\kappa})}.$$

The time integral is finite exactly because $4\theta < 1$, and it contributes a constant times $|I|^{1/4-\theta}$. Inserting (144.68) proves (144.69). $\square$

Corollary 144.24 (Low–high form of the rough-forcing input)**.** *Keep the notation of Proposition 144.22 and assume also $\theta < 1/4$. Let $P_{\le R}$ be as in Proposition 144.23. Assume that the low-frequency Duhamel term is represented by an energy-compatible smooth approximation, so that (144.69) holds on the intervals below with the displayed $L^4, L^2, L^{4/3}$ norms. On every interval $I_\ell = [t_\ell, t_{\ell+1}]$, suppose the high-mode remainder satisfies*

$$\|(1 - P_{\le R})\mathcal{N}_{\mathcal{C}}(Y, \mathbb{X})\|_{C(I_\ell;\mathcal{C}^{-\kappa})} \le B_{\ell,R} + \varepsilon \|Y\|_{C(I_\ell;\mathcal{C}^\alpha)} \tag{144.71}$$

and that the absorption condition (144.63) holds. Then, writing

$$S_\ell = \|Y\|_{C(I_\ell;\mathcal{C}^\alpha)}$$

and using the quantity $L_{I_\ell}(Y, \mathbb{X})$ from Proposition 144.23, one has

$$\begin{aligned} S_\ell \le\ & 2C_0 \|Y(t_\ell)\|_{\mathcal{C}^\alpha} + 2|\lambda| C_{\alpha,\kappa,R,m} |I_\ell|^{1/4-\theta} L_{I_\ell}(Y, \mathbb{X}) \\ & + 2|\lambda| C_1 |I_\ell|^{1-\theta} B_{\ell,R}. \end{aligned} \tag{144.72}$$

Consequently, a uniform energy bound on the displayed $L^4, L^2, L^{4/3}$ norms and a high-mode residual of the form (144.71) give the same global compact-continuity input as the local estimate (144.62).

Proof. Restart the mild equation at t_ℓ, split the nonlinearity into $P_{\le R}\mathcal{N}_{\mathcal{C}}$ and $(1 - P_{\le R})\mathcal{N}_{\mathcal{C}}$, and estimate the low-frequency Duhamel term by Proposition 144.23, using the stated energy-compatible approximation. The high-frequency term is estimated by (144.61) and (144.71). Thus

$$\begin{aligned} S_\ell \le\ & C_0 \|Y(t_\ell)\|_{\mathcal{C}^\alpha} + |\lambda| C_{\alpha,\kappa,R,m} |I_\ell|^{1/4-\theta} L_{I_\ell}(Y, \mathbb{X}) \\ & + |\lambda| C_1 |I_\ell|^{1-\theta} \big(B_{\ell,R} + \varepsilon S_\ell\big). \end{aligned}$$

The absorption condition moves the last term to the left and gives (144.72). Iterating this single-step estimate over the partition is identical to the recursion in Proposition 144.22, with the low-mode energy contribution and $B_{\ell,R}$ replacing the single abstract A_ℓ. □

Proposition 144.25 (High-frequency tail criterion for global DPD continuity)**.** *Keep the notation of Lemma 144.21 and Corollary 144.24, and assume $\theta = (\alpha+\kappa)/2 < 1/4$. Let $K \subset \mathfrak{D}_T$ be compact, and suppose the Besov local blow-up alternative holds on K. Assume that there is a finite energy constant $E_{K,T}$ with the following property: for every $D \in K$, for the smooth energy-compatible approximants used to construct the DPD solution, and for every interval $I \subset [0, T]$ of length at most one, the low-frequency forcing quantity of Proposition 144.23* obeys

$$L_I(Y_D, \mathbb{X}_D) \le E_{K,T}. \tag{144.73}$$

Assume further that the high-frequency paracontrolled residual has a compact tail: there are nonnegative functions $b_K(R)$ and $\varepsilon_K(R)$, with

$$b_K(R) \longrightarrow 0, \qquad \varepsilon_K(R) \longrightarrow 0 \qquad (R \to \infty),$$

such that, for every $D \in K$, every partition interval $I_\ell \subset [0, T]$, and every $R \ge 1$,

$$\|(1 - P_{\le R})\mathcal{N}_{\mathcal{C}}(Y_D, \mathbb{X}_D)\|_{C(I_\ell;\mathcal{C}^{-\kappa})} \le b_K(R) + \varepsilon_K(R) \|Y_D\|_{C(I_\ell;\mathcal{C}^\alpha)}. \tag{144.74}$$

Then the rough energy-to-Besov hypothesis in Lemma 144.21 holds on K. Consequently every $D \in K$ has a DPD solution on $[0, T]$, and the solution map is uniformly continuous on K.

Proof. Choose R so large that

$$|\lambda|C_1\varepsilon_K(R) \le 1.$$

Then choose a partition mesh $0 < \delta \le 1$ with

$$|\lambda|C_1\delta^{1-\theta}\varepsilon_K(R) \le \frac{1}{2}. \tag{144.75}$$

For a partition with $|I_\ell| \le \delta$, Corollary 144.24, (144.73), and (144.74) give the single-step bound

$$\begin{aligned}\|Y_D\|_{C(I_\ell;\mathcal{C}^\alpha)} \le{}& 2C_0\|Y_D(t_\ell)\|_{\mathcal{C}^\alpha}\\ &+ 2|\lambda|C_{\alpha,\kappa,R,m}\delta^{1/4-\theta}E_{K,T} + 2|\lambda|C_1\delta^{1-\theta}b_K(R).\end{aligned}$$

This is exactly Proposition 144.22 with

$$A_\ell = \frac{C_{\alpha,\kappa,R,m}}{C_1}\delta^{-3/4}E_{K,T} + b_K(R),$$

or, more directly, with the displayed low-mode and high-mode contributions kept separate. Iterating over

$$M = \lceil T/\delta \rceil$$

intervals gives

$$\begin{aligned}\|Y_D\|_{C([0,T];\mathcal{C}^\alpha)} \le{}& (2C_0)^M \sup_{\widetilde{D}\in K}\|Y_{\widetilde{D}}(0)\|_{\mathcal{C}^\alpha}\\ &+ \sum_{j=0}^{M-1}(2C_0)^{M-1-j}\Big[2|\lambda|C_{\alpha,\kappa,R,m}\delta^{1/4-\theta}E_{K,T} + 2|\lambda|C_1\delta^{1-\theta}b_K(R)\Big].\end{aligned}$$

The right hand side is finite and independent of $D \in K$. This proves the rough energy-to-Besov bound on K. The global existence and uniform continuity conclusions now follow from Lemma 144.21. □

The preceding propositions sharpen the remaining analytic obligation. The energy estimate and finite-mode smoothing already control the low-frequency part of the DPD forcing from the same $L^4, L^2, L^{4/3}$ quantities used in Proposition 144.17. What is still missing is not another finite exponent identity, but a genuinely rough high-frequency estimate of the compact-tail form (144.74). This is the paracontrolled commutator/noise-tail step in the rough energy-to-Besov bound. Once that estimate is proved on compact enhanced-noise sets, Proposition 144.25 converts it into global Besov compact-continuity.

Lemma 144.26 (Passing invariance through a cutoff limit)**.** *Let E be a Polish space, let $P_t^{(n)}$ and P_t be Markov semigroups on bounded continuous functions on E, and let μ_n be an invariant probability measure for $P_t^{(n)}$. Suppose $\mu_n \Rightarrow \mu$ weakly and that, for every bounded continuous function f and every fixed $t \ge 0$,*

$$P_t^{(n)}f \longrightarrow P_t f \quad \text{uniformly on compact subsets of } E.$$

Assume also that $\{\mu_n\}$ is tight and that the convergence above is uniform on a compact set whose μ_n-mass is arbitrarily close to one, uniformly in n. Then μ is invariant for P_t:

$$\int_E P_t f\,\mathrm{d}\mu = \int_E f\,\mathrm{d}\mu.$$

Proof. Fix $f \in C_b(E)$, $t \geq 0$, and $\eta > 0$. Choose a compact set $K \subset E$ such that $\mu_n(K) \geq 1 - \eta$ for all sufficiently large n and such that the convergence $P_t^{(n)} f \to P_t f$ is uniform on this same K. The latter requirement is not implied by tightness alone; it is the high-probability compact-set convergence hypothesis in the statement. Since Markov semigroups are contractions on bounded functions,

$$\|P_t^{(n)} f\|_\infty \leq \|f\|_\infty, \qquad \|P_t f\|_\infty \leq \|f\|_\infty.$$

Therefore

$$\left| \int P_t^{(n)} f \, \mathrm{d}\mu_n - \int P_t f \, \mathrm{d}\mu_n \right| \leq \sup_{x \in K} |P_t^{(n)} f(x) - P_t f(x)| + 2\|f\|_\infty \eta,$$

which tends to at most $2\|f\|_\infty \eta$. Letting $\eta \downarrow 0$ gives

$$\int P_t^{(n)} f \, \mathrm{d}\mu_n - \int P_t f \, \mathrm{d}\mu_n \longrightarrow 0.$$

The limiting semigroup is assumed to map $C_b(E)$ to $C_b(E)$, so weak convergence of μ_n gives

$$\int P_t f \, \mathrm{d}\mu_n \longrightarrow \int P_t f \, \mathrm{d}\mu.$$

The cutoff invariance gives

$$\int P_t^{(n)} f \, \mathrm{d}\mu_n = \int f \, \mathrm{d}\mu_n,$$

and weak convergence also gives

$$\int f \, \mathrm{d}\mu_n \longrightarrow \int f \, \mathrm{d}\mu.$$

Combining the three limits yields

$$\int P_t f \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Since f and t were arbitrary, μ is invariant for the limiting Markov semigroup. $\square$

For singular SPDE, the hypotheses of Lemma 144.26 are not automatic. They require tightness of the stationary cutoff laws in the chosen field topology, convergence of the renormalized solution maps on high-probability compact sets of enhanced noises and initial data, and continuity of the limiting Markov semigroup on the same topology. This is the exact invariant-law step that sits between cutoff Langevin stationarity and a Euclidean measure for QFT.

Lemma 144.27 (Sobolev compactness criterion for cutoff-field tightness)**.** *Let $0 < \eta < \kappa$, $p > 0$, and let $\{\nu_N\}$ be Borel probability measures on $H^{-\kappa}(\mathbb{T}^2)$. Suppose that each ν_N is carried by $H^{-\eta}(\mathbb{T}^2)$ and that*

$$\sup_N \int \|\phi\|_{H^{-\eta}}^p \, \mathrm{d}\nu_N(\phi) \leq C < \infty. \tag{144.76}$$

Then $\{\nu_N\}$ is tight as a family of probability measures on $H^{-\kappa}(\mathbb{T}^2)$. More explicitly, for every $R > 0$ the set

$$K_R = \{\phi \in H^{-\eta}(\mathbb{T}^2) : \|\phi\|_{H^{-\eta}} \leq R\}$$

is compact in $H^{-\kappa}(\mathbb{T}^2)$, and

$$\nu_N(K_R^c) \leq CR^{-p}.$$

Proof. Write the Fourier coefficients of ϕ as $\widehat{\phi}(k)$, $k \in \mathbb{Z}^2$, and set $\langle k\rangle = (1+|k|^2)^{1/2}$. The Sobolev norms are

$$\|\phi\|_{H^{-\rho}}^2 = \sum_{k\in\mathbb{Z}^2} \langle k\rangle^{-2\rho}|\widehat{\phi}(k)|^2, \qquad \rho \geq 0. \tag{144.77}$$

Let P_M be the projection to Fourier modes $|k| \leq M$, and write $\langle M\rangle = (1+M^2)^{1/2}$. If $\phi \in K_R$, then

$$\begin{aligned}\|(1-P_M)\phi\|_{H^{-\kappa}}^2 &= \sum_{|k|>M} \langle k\rangle^{-2(\kappa-\eta)}\langle k\rangle^{-2\eta}|\widehat{\phi}(k)|^2 \\ &\leq \langle M\rangle^{-2(\kappa-\eta)}\|\phi\|_{H^{-\eta}}^2 \leq \langle M\rangle^{-2(\kappa-\eta)}R^2.\end{aligned}$$

Thus the high-frequency tail of K_R is uniformly small in $H^{-\kappa}$. The set $P_M K_R$ is bounded in a finite-dimensional space and is therefore totally bounded. Given $\varepsilon > 0$, choose M so that $\langle M\rangle^{-(\kappa-\eta)}R < \varepsilon/3$, cover $P_M K_R$ by finitely many $H^{-\kappa}$-balls of radius $\varepsilon/3$, and add the tail estimate. This proves that K_R is totally bounded in $H^{-\kappa}$.

It remains only to see that K_R is closed in $H^{-\kappa}$. If $\phi_n \in K_R$ and $\phi_n \to \phi$ in $H^{-\kappa}$, then each Fourier coefficient converges. Fatou's lemma gives

$$\|\phi\|_{H^{-\eta}}^2 = \sum_k \langle k\rangle^{-2\eta}|\widehat{\phi}(k)|^2 \leq \liminf_{n\to\infty} \sum_k \langle k\rangle^{-2\eta}|\widehat{\phi}_n(k)|^2 \leq R^2.$$

Therefore $\phi \in K_R$, so K_R is compact in $H^{-\kappa}$. Finally, Markov's inequality applied to (144.76) gives

$$\nu_N(K_R^c) = \nu_N\{\|\phi\|_{H^{-\eta}} > R\} \leq R^{-p}\int \|\phi\|_{H^{-\eta}}^p \,\mathrm{d}\nu_N \leq CR^{-p}.$$

Choosing R large proves tightness. □

Proposition 144.28 (Massive Gaussian cutoff tightness in negative Sobolev topology)**.** *Let $A = -\Delta$ on $\mathbb{T}^2$, and let $\{e_\alpha\}_{\alpha\in\mathcal{I}}$ be a real orthonormal eigenbasis of $L^2(\mathbb{T}^2)$, with $Ae_\alpha = \lambda_\alpha e_\alpha$, $\lambda_\alpha \geq 0$. Fix $m^2 > 0$, and for $N \geq 1$ set*

$$\mathcal{I}_N = \{\alpha \in \mathcal{I} : \lambda_\alpha \leq N^2\}.$$

Let γ_N be the centered Gaussian law of

$$\phi_N = \sum_{\alpha\in\mathcal{I}_N} g_\alpha(\lambda_\alpha + m^2)^{-1/2}e_\alpha,$$

where $\{g_\alpha\}$ are independent real standard Gaussian random variables. Then for every $\eta > 0$,

$$\sup_N \int \|\phi\|_{H^{-\eta}(\mathbb{T}^2)}^2 \,\mathrm{d}\gamma_N(\phi) < \infty.$$

Consequently, if $0 < \eta < \kappa$, the family $\{\gamma_N\}$ is tight as a family of probability measures on $H^{-\kappa}(\mathbb{T}^2)$.

Proof. By definition of the Hilbert Sobolev norm associated with the Laplacian,

$$\|\phi_N\|_{H^{-\eta}}^2 = \sum_{\alpha\in\mathcal{I}_N} (1+\lambda_\alpha)^{-\eta}|(\phi_N, e_\alpha)_{L^2}|^2.$$

Taking expectation and using $\mathbb{E}g_\alpha^2 = 1$ gives

$$\mathbb{E}\|\phi_N\|_{H^{-\eta}}^2 = \sum_{\alpha\in\mathcal{I}_N} (1+\lambda_\alpha)^{-\eta}(\lambda_\alpha + m^2)^{-1}. \tag{144.78}$$

There is a constant $c_m < \infty$ such that $(\lambda + m^2)^{-1} \le c_m(1+\lambda)^{-1}$ for all $\lambda \ge 0$. Thus the right hand side of (144.78) is bounded by

$$c_m \sum_{\alpha\in\mathcal{I}} (1+\lambda_\alpha)^{-1-\eta}.$$

For the flat two-torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$, the complex Fourier basis has eigenvalues $|k|^2$, $k \in \mathbb{Z}^2$. Passing from the complex basis to a real sine-cosine eigenbasis changes the counting by a bounded factor. It therefore suffices to prove convergence of

$$\sum_{k\in\mathbb{Z}^2} (1+|k|^2)^{-1-\eta}.$$

Write $\langle k\rangle = (1+|k|^2)^{1/2}$. Let $A_0 = \{k : \langle k\rangle < 2\}$ and, for $j \ge 1$,

$$A_j = \{k \in \mathbb{Z}^2 : 2^j \le \langle k\rangle < 2^{j+1}\}.$$

There is a universal constant C such that the cardinality of A_j is at most $C2^{2j}$, while each summand on A_j is at most $2^{-2(1+\eta)j}$. Hence

$$\sum_{k\in A_j} (1+|k|^2)^{-1-\eta} \le C2^{-2\eta j}.$$

The geometric series over j converges because $\eta > 0$. This proves the uniform second-moment bound. Since each γ_N is carried by the finite-dimensional space spanned by $\{e_\alpha : \alpha \in \mathcal{I}_N\}$, it is carried by $H^{-\eta}$. Lemma 144.27 with $p = 2$ now gives tightness in $H^{-\kappa}$. □

Lemma 144.29 (L^q-density criterion for interacting negative-Sobolev moments)**.** *Let $0 < \eta < \kappa$. For each N, let γ_N be a centered Gaussian probability measure carried by a finite-dimensional subspace of $H^{-\eta}(\mathbb{T}^2)$, and assume*

$$\sigma^2 := \sup_N \int \|\phi\|_{H^{-\eta}}^2 \, \mathrm{d}\gamma_N(\phi) < \infty.$$

Let $\mu_N = \rho_N\gamma_N$, where $\rho_N \ge 0$, $\int \rho_N \, \mathrm{d}\gamma_N = 1$, and, for some $q > 1$,

$$A_q := \sup_N \|\rho_N\|_{L^q(\gamma_N)} < \infty.$$

Then, for every $r < \infty$,

$$\sup_N \int \|\phi\|_{H^{-\eta}}^r \, \mathrm{d}\mu_N(\phi) < \infty.$$

Consequently $\{\mu_N\}$ is tight in $H^{-\kappa}(\mathbb{T}^2)$.

Proof. Let $a = q/(q-1)$. Holder's inequality gives

$$\int \|\phi\|_{H^{-\eta}}^r \, \mathrm{d}\mu_N(\phi) = \int \|\phi\|_{H^{-\eta}}^r \rho_N(\phi) \, \mathrm{d}\gamma_N(\phi) \le A_q \left(\int \|\phi\|_{H^{-\eta}}^{ar} \, \mathrm{d}\gamma_N(\phi) \right)^{1/a}. \tag{144.79}$$

It remains to bound the Gaussian moment uniformly. Since γ_N is a finite-dimensional centered Gaussian law on the Hilbert space $H^{-\eta}$, its covariance operator is a nonnegative symmetric operator of finite rank. Diagonalize this covariance in an $H^{-\eta}$-orthonormal basis:

$$\phi_N = \sum_i \lambda_{N,i}^{1/2} g_i e_{N,i}, \qquad \|\phi_N\|_{H^{-\eta}}^2 = \sum_i \lambda_{N,i} g_i^2,$$

where the g_i are independent standard real Gaussians and $\sum_i \lambda_{N,i} \le \sigma^2$. For an integer $m \ge 1$,

$$\begin{aligned} \mathbb{E}_{\gamma_N} \|\phi\|_{H^{-\eta}}^{2m} &= \sum_{i_1,\dots,i_m} \lambda_{N,i_1} \cdots \lambda_{N,i_m} \mathbb{E}(g_{i_1}^2 \cdots g_{i_m}^2) \\ &\le (2m-1)!! \left(\sum_i \lambda_{N,i} \right)^m \le (2m-1)!! \, \sigma^{2m}. \end{aligned}$$

Choose m with $2m \ge ar$. The elementary inequality $x^{ar} \le 1 + x^{2m}$ for $x \ge 0$ gives a uniform bound for the Gaussian moment in (144.79). Thus the $H^{-\eta}$ r-th moments under μ_N are uniformly bounded. Applying Lemma 144.27 with any positive r proves tightness in $H^{-\kappa}$. □

Remark 144.30 (Density route to Φ_2^4 stationary-law tightness). Let $\mu_{0,N}$ be the massive Gaussian cutoff law in Proposition 144.28, and let

$$\mathrm{d}\mu_N(\phi) = Z_N^{-1} \exp\left[-\frac{\lambda}{4} \int_{\mathbb{T}^2} :(P_N\phi)^4: \, \mathrm{d}^2 x \right] \mathrm{d}\mu_{0,N}(\phi)$$

be a finite-dimensional Wick-ordered Φ_2^4 cutoff measure. If there exists $q > 1$ such that the normalized densities

$$\rho_N(\phi) = Z_N^{-1} \exp\left[-\frac{\lambda}{4} \int_{\mathbb{T}^2} :(P_N\phi)^4: \, \mathrm{d}^2 x \right]$$

satisfy

$$\sup_N \|\rho_N\|_{L^q(\mu_{0,N})} < \infty,$$

then $\{\mu_N\}$ is tight in $H^{-\kappa}(\mathbb{T}^2)$ for every $\kappa > 0$. Choose any $0 < \eta < \kappa$. The Gaussian reference law has a uniform $H^{-\eta}$ second moment by Proposition 144.28. The assumed L^q density bound then gives a uniform interacting $H^{-\eta}$ moment by Lemma 144.29, and Lemma 144.27 turns that moment bound into tightness in $H^{-\kappa}$.

Lemma 144.31 (Nelson stability implies normalized L^q density bounds)**.** *Let $\mu_{0,N}$ be a finite-dimensional massive Gaussian cutoff law on $\mathbb{T}^2$, and set*

$$V_N(\phi) = \frac{\lambda}{4} \int_{\mathbb{T}^2} :(P_N\phi)^4: \, \mathrm{d}^2 x, \qquad \lambda > 0,$$

where Wick ordering is taken with respect to $\mu_{0,N}$. Suppose that the cutoff family satisfies the Nelson stability exponential estimate

$$\sup_N \int \exp[-pV_N(\phi)] \, \mathrm{d}\mu_{0,N}(\phi) < \infty \qquad \text{for every finite } p \ge 1. \tag{144.80}$$

Let

$$Z_N = \int e^{-V_N(\phi)}\, \mathrm{d}\mu_{0,N}(\phi), \qquad \rho_N = Z_N^{-1} e^{-V_N}.$$

Then, for every finite $q \geq 1$,

$$\sup_N \|\rho_N\|_{L^q(\mu_{0,N})} < \infty.$$

Consequently, for every $0 < \eta < \kappa$, the interacting cutoff measures $\mu_N = \rho_N \mu_{0,N}$ are tight in $H^{-\kappa}(\mathbb{T}^2)$, provided the Gaussian reference laws have the uniform $H^{-\eta}$ second-moment bound of Proposition 144.28.

Proof. The normalization cannot be allowed to hide a cutoff-dependent loss. Wick ordering gives

$$\int :(P_N\phi(x))^4:\, \mathrm{d}\mu_{0,N}(\phi) = 0 \qquad \text{for every } x,$$

and hence, by Fubini in the finite-dimensional cutoff space,

$$\int V_N(\phi)\, \mathrm{d}\mu_{0,N}(\phi) = 0.$$

Jensen's inequality for the convex function $u \mapsto e^{-u}$ gives

$$Z_N = \int e^{-V_N}\, \mathrm{d}\mu_{0,N} \geq \exp\left[-\int V_N\, \mathrm{d}\mu_{0,N}\right] = 1. \tag{144.81}$$

Therefore, for $q \geq 1$,

$$\|\rho_N\|^q_{L^q(\mu_{0,N})} = Z_N^{-q} \int e^{-qV_N}\, \mathrm{d}\mu_{0,N} \leq \int e^{-qV_N}\, \mathrm{d}\mu_{0,N}.$$

The right-hand side is bounded uniformly in N by (144.80) with $p = q$. This proves the normalized L^q density bound. The tightness conclusion is then exactly Remark 144.30, or, equivalently, Lemma 144.29 followed by Lemma 144.27. □

The preceding proposition proves the probabilistic consequence of Nelson stability without losing constants in the normalization. For cutoff families covered by Quoted Theorem 137.3, the hypothesis (144.80) is exactly the $P(\phi)_2$ stability and exponential-integrability estimate. For a sharp Fourier-Galerkin stochastic regulator, the same conclusion requires either a direct Nelson stability proof for that regulator or a comparison with a smooth ultraviolet regulator for which the stability theorem has already been established. This is a regulator-specific constructive estimate, not a formal property of Wick ordering alone.

Lemma 144.32 (Bounded-cylinder comparison from a common density limit)**.** *Let $(\Omega, \mathcal{F}, \gamma)$ be a probability space. For $\sigma \in \{A, B\}$ and $N \geq 1$, let*

$$X_N^\sigma : \Omega \to \mathbb{R}^r, \qquad W_N^\sigma : \Omega \to [0, \infty)$$

be measurable, and let $X : \Omega \to \mathbb{R}^r$, $W : \Omega \to [0, \infty)$ be measurable limits. Assume

$$X_N^\sigma \longrightarrow X \quad \textit{in } \gamma\textit{-probability}, \qquad W_N^\sigma \longrightarrow W \quad \textit{in } L^1(\gamma) \tag{144.82}$$

for $\sigma = A, B$. Assume also that there is $p > 1$ such that

$$\sup_{N,\sigma} \|W_N^\sigma\|_{L^p(\gamma)} < \infty. \tag{144.83}$$

Set

$$Z_N^\sigma = \int W_N^\sigma \,\mathrm{d}\gamma, \qquad Z = \int W \,\mathrm{d}\gamma,$$

and suppose $Z > 0$. Then, for every bounded continuous $F : \mathbb{R}^r \to \mathbb{C}$,

$$\lim_{N\to\infty} \frac{\int F(X_N^A) W_N^A \,\mathrm{d}\gamma}{Z_N^A} = \frac{\int F(X) W \,\mathrm{d}\gamma}{Z} = \lim_{N\to\infty} \frac{\int F(X_N^B) W_N^B \,\mathrm{d}\gamma}{Z_N^B}. \tag{144.84}$$

Proof. The L^1-convergence of W_N^σ gives

$$Z_N^\sigma \longrightarrow Z$$

for $\sigma = A, B$. Thus it is enough to prove convergence of the unnormalized numerators. Fix σ and write

$$\begin{aligned} &\int F(X_N^\sigma) W_N^\sigma \,\mathrm{d}\gamma - \int F(X) W \,\mathrm{d}\gamma \\ &= \int \big(F(X_N^\sigma) - F(X)\big) W_N^\sigma \,\mathrm{d}\gamma + \int F(X) \big(W_N^\sigma - W\big) \,\mathrm{d}\gamma. \end{aligned}$$

The second term tends to zero because F is bounded and $W_N^\sigma \to W$ in L^1. For the first term, the convergence $X_N^\sigma \to X$ in probability and continuity of F imply

$$F(X_N^\sigma) - F(X) \longrightarrow 0 \quad \text{in probability.}$$

The difference is bounded by $2\|F\|_\infty$, while (144.83) implies uniform integrability of $\{W_N^\sigma\}$ by Holder's inequality:

$$\int_E W_N^\sigma \,\mathrm{d}\gamma \le \|W_N^\sigma\|_{L^p(\gamma)} \gamma(E)^{1-1/p}.$$

Given $\varepsilon > 0$, choose $\delta > 0$ so small that the right-hand side is at most ε for every N, σ whenever $\gamma(E) < \delta$. On the event $\{|F(X_N^\sigma) - F(X)| \le \varepsilon\}$, the first term is bounded by $\varepsilon \int W_N^\sigma \mathrm{d}\gamma$, and the integrals of W_N^σ are uniformly bounded by the L^p hypothesis. On the complement, the uniform-integrability estimate applies because the complement has γ-probability tending to zero. Hence the first term tends to zero for each fixed regulator type σ. Applying this conclusion for $\sigma = A$ and for $\sigma = B$ gives the two unnormalized limits, and the normalized limits follow from $Z_N^\sigma \to Z > 0$. □

Corollary 144.33 (Bounded-cylinder comparison from a common potential)**.** *In the setting of Lemma 144.32, replace the L^1-convergence assumption on the densities by the following potential-level assumptions. Let*

$$V_N^\sigma : \Omega \to \mathbb{R}, \qquad V : \Omega \to \mathbb{R}, \qquad \sigma \in \{A, B\},$$

be measurable real-valued interaction functionals, put

$$W_N^\sigma = e^{-V_N^\sigma}, \qquad W = e^{-V},$$

and assume

$$X_N^\sigma \longrightarrow X, \qquad V_N^\sigma \longrightarrow V \quad \textit{in } \gamma\textit{-probability} \tag{144.85}$$

for $\sigma = A, B$. Assume further that, for some $p > 1$,

$$\sup_{N,\sigma} \|e^{-V_N^\sigma}\|_{L^p(\gamma)} < \infty, \qquad Z := \int e^{-V}\, \mathrm{d}\gamma > 0. \tag{144.86}$$

Then the normalized bounded cylinder expectations of the two regulators have the common limit

$$\lim_{N\to\infty} \frac{\int F(X_N^\sigma) e^{-V_N^\sigma}\, \mathrm{d}\gamma}{\int e^{-V_N^\sigma}\, \mathrm{d}\gamma} = \frac{\int F(X) e^{-V}\, \mathrm{d}\gamma}{\int e^{-V}\, \mathrm{d}\gamma}, \qquad \sigma = A, B, \tag{144.87}$$

for every bounded continuous $F : \mathbb{R}^r \to \mathbb{C}$.

Proof. It is enough to verify the density hypotheses of Lemma 144.32. Since the exponential map is continuous,

$$e^{-V_N^\sigma} \longrightarrow e^{-V} \quad \text{in } \gamma\text{-probability.}$$

We first record explicitly that the limiting weight belongs to L^p. From any subsequence of $e^{-V_N^\sigma}$ one may choose a further subsequence converging to e^{-V} almost surely. Fatou's lemma gives

$$\|e^{-V}\|_{L^p(\gamma)}^p \le \liminf_{k\to\infty} \|e^{-V_{N_k}^\sigma}\|_{L^p(\gamma)}^p \le \sup_{N,\sigma} \|e^{-V_N^\sigma}\|_{L^p(\gamma)}^p.$$

Thus the family

$$\{e^{-V}\} \cup \{e^{-V_N^\sigma} : N \ge 1,\ \sigma = A, B\}$$

is uniformly integrable. Indeed, for every measurable $E \subset \Omega$, Holder's inequality gives

$$\int_E e^{-V_N^\sigma}\, \mathrm{d}\gamma \le \|e^{-V_N^\sigma}\|_{L^p(\gamma)} \gamma(E)^{1-1/p}, \qquad \int_E e^{-V}\, \mathrm{d}\gamma \le \|e^{-V}\|_{L^p(\gamma)} \gamma(E)^{1-1/p}.$$

We now prove the L^1-convergence rather than citing Vitali's theorem. Fix $\varepsilon > 0$. By the preceding uniform-integrability estimate, choose $\delta > 0$ so that the integral of any member of the displayed family over a set of γ-measure less than δ is at most ε. Let

$$E_N^\sigma = \{|e^{-V_N^\sigma} - e^{-V}| > \varepsilon\}.$$

Convergence in probability gives $\gamma(E_N^\sigma) < \delta$ for all sufficiently large N. Then

$$\begin{aligned}\int |e^{-V_N^\sigma} - e^{-V}|\, \mathrm{d}\gamma &\le \varepsilon + \int_{E_N^\sigma} e^{-V_N^\sigma}\, \mathrm{d}\gamma + \int_{E_N^\sigma} e^{-V}\, \mathrm{d}\gamma \\ &\le 3\varepsilon\end{aligned}$$

for all sufficiently large N. Hence

$$e^{-V_N^\sigma} \longrightarrow e^{-V} \quad \text{in } L^1(\gamma), \qquad \sigma = A, B.$$

The hypotheses of Lemma 144.32 are now satisfied, and its conclusion is exactly (144.87). $\square$

Bounded Wick-cylinder comparison for stable quartic cutoffs. Let X_N^A and X_N^B be two cutoff approximations of the same massive two-dimensional Gaussian field on a common probability space $(\Omega, \mathcal{F}, \gamma)$. Fix $\lambda \in \mathbb{R}$ and define

$$V_N^\sigma = \frac{\lambda}{4} :(X_N^\sigma)^4: (1), \qquad \sigma \in \{A, B\}.$$

Assume that the finite Wick coordinates

$$:(X_N^\sigma)^{n_j}: (f_j), \qquad 1 \le j \le r,$$

converge in $L^2(\Omega)$, for both $\sigma = A, B$, to common limits $:X^{n_j}: (f_j)$, and that

$$V_N^\sigma \longrightarrow V := \frac{\lambda}{4} :X^4: (1) \quad \text{in } L^2(\Omega), \qquad \sigma = A, B. \tag{144.88}$$

Assume also that there is $p > 1$ such that

$$\sup_{N,\sigma} \left\| e^{-V_N^\sigma} \right\|_{L^p(\gamma)} < \infty. \tag{144.89}$$

Let

$$\mathcal{X}_N^\sigma = \big(:(X_N^\sigma)^{n_1}: (f_1), \dots, :(X_N^\sigma)^{n_r}: (f_r)\big), \qquad \mathcal{X} = \big(:X^{n_1}: (f_1), \dots, :X^{n_r}: (f_r)\big).$$

Then, for every bounded continuous $G : \mathbb{R}^r \to \mathbb{C}$,

$$\lim_{N\to\infty} \frac{\int G(\mathcal{X}_N^A) e^{-V_N^A} \, \mathrm{d}\gamma}{\int e^{-V_N^A} \, \mathrm{d}\gamma} = \frac{\int G(\mathcal{X}) e^{-V} \, \mathrm{d}\gamma}{\int e^{-V} \, \mathrm{d}\gamma} = \lim_{N\to\infty} \frac{\int G(\mathcal{X}_N^B) e^{-V_N^B} \, \mathrm{d}\gamma}{\int e^{-V_N^B} \, \mathrm{d}\gamma}. \tag{144.90}$$

Remark 144.7 gives

$$\mathcal{X}_N^\sigma \longrightarrow \mathcal{X} \quad \text{in probability}, \qquad \sigma = A, B.$$

The assumed L^2-convergence (144.88) gives

$$V_N^\sigma \longrightarrow V \quad \text{in probability}.$$

From every subsequence one may choose a further subsequence along which $V_N^\sigma \to V$ almost surely. Fatou's lemma and (144.89) imply

$$\int e^{-pV} \, \mathrm{d}\gamma \le \liminf_{k\to\infty} \int e^{-pV_{N_k}^\sigma} \, \mathrm{d}\gamma < \infty.$$

Thus $e^{-V} \in L^p(\gamma)$, while $e^{-V} > 0$ almost surely because V is real-valued and finite almost surely. Hence

$$0 < \int e^{-V} \, \mathrm{d}\gamma < \infty.$$

All hypotheses of Corollary 144.33 are satisfied, and its conclusion is precisely (144.90).

144.5.4 Regulators, Reflection Positivity, and the Φ^4_2 Assembly

Proposition 144.34 (Smooth regulator consequences of Nelson stability)**.** *Let γ be the massive Gaussian free-field law on $\mathbb{T}^2$. For $\sigma \in \{A, B\}$, let*

$$X^\sigma_N = \mu^\sigma(D/N)X$$

be smooth Fourier multiplier cutoffs as in Proposition 144.5, and let

$$V^\sigma_N = \frac{\lambda}{4}\int_{\mathbb{T}^2} :(X^\sigma_N(x))^4: \, \mathrm{d}^2x, \qquad \lambda > 0,$$

with Wick ordering taken with respect to the cutoff covariance of X^σ_N. Assume the Nelson stability input

$$\sup_{N,\sigma} \int \exp[-pV^\sigma_N] \, \mathrm{d}\gamma < \infty \qquad \textit{for every finite } p \geq 1. \tag{144.91}$$

Set

$$Z^\sigma_N = \int e^{-V^\sigma_N} \, \mathrm{d}\gamma, \qquad \rho^\sigma_N = (Z^\sigma_N)^{-1} e^{-V^\sigma_N}, \qquad \mathrm{d}\nu^\sigma_N = \rho^\sigma_N \, \mathrm{d}\gamma.$$

Then the following statements hold.

1. *For every finite $q \geq 1$,*
$$\sup_{N,\sigma} \|\rho^\sigma_N\|_{L^q(\gamma)} < \infty. \tag{144.92}$$
2. *For every $\kappa > 0$, the two families $\{\nu^A_N\}$ and $\{\nu^B_N\}$ are tight in $H^{-\kappa}(\mathbb{T}^2)$.*
3. *For every finite list of Wick coordinates $:(X^\sigma_N)^{n_j}:(f_j)$, $1 \leq j \leq r$, and every bounded continuous $G : \mathbb{R}^r \to \mathbb{C}$, the normalized expectations of G under the two cutoff families have the common continuum limit stated in (144.90).*

Proof. Wick centering gives

$$\int V^\sigma_N \, \mathrm{d}\gamma = 0.$$

Jensen's inequality therefore gives

$$Z^\sigma_N = \int e^{-V^\sigma_N} \, \mathrm{d}\gamma \geq \exp\left[-\int V^\sigma_N \, \mathrm{d}\gamma\right] = 1.$$

Consequently

$$\|\rho^\sigma_N\|^q_{L^q(\gamma)} = (Z^\sigma_N)^{-q} \int e^{-qV^\sigma_N} \, \mathrm{d}\gamma \leq \int e^{-qV^\sigma_N} \, \mathrm{d}\gamma,$$

and (144.92) follows from (144.91).

For tightness, choose $0 < \eta < \kappa$. The massive Gaussian field has finite $H^{-\eta}$-moments of every order: diagonalizing the covariance in Fourier modes gives

$$\mathbb{E}_\gamma \|X\|^2_{H^{-\eta}} = \sum_{k \in \mathbb{Z}^2} \frac{(1 + |k|^2)^{-\eta}}{|k|^2 + m^2} < \infty,$$

and the higher moments are bounded by the same Gaussian Hilbert-norm argument used in Lemma 144.29. Holder's inequality with the uniform L^q bound for ρ_N^σ gives

$$\sup_{N,\sigma} \int \|\phi\|_{H^{-\eta}}^r \, \mathrm{d}\nu_N^\sigma(\phi) < \infty \qquad (r < \infty).$$

The compact embedding $H^{-\eta} \hookrightarrow H^{-\kappa}$, quantified in Lemma 144.27, proves tightness in $H^{-\kappa}$.

It remains to prove the comparison statement. By Proposition 144.5 and Lemma 144.6, the finite Wick-coordinate vectors of the two smooth regulators have common L^2 limits. Applying the fourth Wick-power convergence statement with $f \equiv 1$ gives

$$V_N^\sigma \longrightarrow \frac{\lambda}{4} :X^4:(1) \quad \text{in } L^2(\gamma), \qquad \sigma = A, B.$$

The stability input (144.91) supplies the uniform L^p bound on $e^{-V_N^\sigma}$. Hence the bounded Wick-cylinder comparison (144.90) applies and gives the stated comparison. □

In the regulator-comparison problem, X_N^σ is the finite list of smeared field or Wick-polynomial coordinates entering a bounded positive-time cylinder observable, and W_N^σ is the finite-cutoff interaction density. Lemma 144.32 therefore proves the bounded-observable comparison once both regulators are constructed on a common Gaussian probability space and their Wick densities are shown to have the same L^1 limit. Corollary 144.33 gives the common route used in cutoff constructions: it is enough to prove that the renormalized interaction functionals converge in probability to the same limiting potential and that their exponential weights obey a uniform L^p stability estimate. In the Wick quartic case, the comparison (144.90) is obtained once the common Wick-coordinate limits and exponential-weight stability have been proved. For smooth Fourier multiplier regulators satisfying the Nelson stability input, Proposition 144.34 derives the normalized L^q density bounds, tightness, and bounded Wick-cylinder comparison in one statement. Unbounded polynomial observables still require the separate uniform-integrability argument in Remark 144.41.

Proposition 144.35 (Brascamp–Lieb covariance domination for convex cutoffs)**.** *Let H_N be a finite-dimensional real Hilbert space, let A_N be a positive self-adjoint operator on H_N, and let $S_N \in C^2(H_N)$ satisfy*

$$\nabla^2 S_N(\phi) \geq A_N \tag{144.93}$$

as quadratic forms on H_N, for every $\phi \in H_N$. Define

$$\mathrm{d}\nu_N(\phi) = Z_N^{-1} e^{-S_N(\phi)} \, \mathrm{d}\phi$$

where $\mathrm{d}\phi$ is Lebesgue measure in an orthonormal coordinate system and $Z_N = \int e^{-S_N(\phi)} \mathrm{d}\phi$. Then $Z_N < \infty$, the integration-by-parts identities used below hold for smooth functions of polynomial growth, and every linear functional $\ell_v(\phi) = (v, \phi)_{H_N}$ satisfies

$$\mathrm{Var}_{\nu_N}(\ell_v) \leq (v, A_N^{-1} v)_{H_N}. \tag{144.94}$$

Suppose in addition that L_N is a nonnegative self-adjoint operator with orthonormal eigenvectors $e_{N,\alpha}$ and eigenvalues $\lambda_{N,\alpha}$, that $A_N \geq L_N + m^2 \mathbf{1}$ for some $m^2 > 0$, and that ν_N is invariant under $\phi \mapsto -\phi$. Define

$$\|\phi\|_{H_N^{-\eta}}^2 = \sum_\alpha (1 + \lambda_{N,\alpha})^{-\eta} |(\phi, e_{N,\alpha})_{H_N}|^2. \tag{144.95}$$

Then, for every $\eta > 0$,

$$\int \|\phi\|^2_{H_N^{-\eta}} \,\mathrm{d}\nu_N(\phi) \le \sum_\alpha (1+\lambda_{N,\alpha})^{-\eta}(\lambda_{N,\alpha}+m^2)^{-1}. \tag{144.96}$$

Consequently, any cutoff family for which the right hand side of (144.96) is uniformly bounded and whose $H_N^{-\eta}$ spaces embed into a common $H^{-\eta}$ approximation scheme satisfies the moment hypothesis in Lemma 144.27.

Proof. Let $a_N > 0$ be the smallest eigenvalue of A_N. Taylor's formula gives

$$S_N(\phi) \ge S_N(0) + (\nabla S_N(0), \phi)_{H_N} + \frac{a_N}{2}\|\phi\|^2_{H_N} \ge \frac{a_N}{4}\|\phi\|^2_{H_N} - C_N$$

for a finite constant C_N. Hence e^{-S_N} has Gaussian decay. This proves $Z_N < \infty$ and justifies the integration by parts below after inserting radial cutoffs and sending their radii to infinity.

We first prove the covariance estimate. Write

$$\mathcal{L} = \Delta - \nabla S_N \cdot \nabla.$$

This operator is symmetric in $L^2(\nu_N)$ on smooth compactly supported functions, and

$$-\int f\,\mathcal{L}g\,\mathrm{d}\nu_N = \int (\nabla f, \nabla g)_{H_N}\,\mathrm{d}\nu_N. \tag{144.97}$$

Let f be a smooth centered test function. On the centered subspace of $L^2(\nu_N)$, let $u_\varepsilon = (\varepsilon - \mathcal{L})^{-1} f$, $\varepsilon > 0$, where the inverse is the Friedrichs resolvent of the closed Dirichlet form in (144.97). The resolvent satisfies

$$\varepsilon u_\varepsilon - \mathcal{L}u_\varepsilon = f. \tag{144.98}$$

Pairing (144.98) with f and using (144.97) gives

$$\int f^2\,\mathrm{d}\nu_N = \varepsilon \int f u_\varepsilon \,\mathrm{d}\nu_N + \int (\nabla f, \nabla u_\varepsilon)_{H_N}\,\mathrm{d}\nu_N. \tag{144.99}$$

The spectral calculus of the nonnegative operator $-\mathcal{L}$ on centered $L^2(\nu_N)$ gives $\varepsilon u_\varepsilon \to 0$ and $-\mathcal{L}u_\varepsilon \to f$ in $L^2(\nu_N)$. Hence the first term on the right of (144.99) vanishes as $\varepsilon \downarrow 0$.

For smooth compactly supported u, differentiating $\mathcal{L}u$ and integrating by parts gives the finite-dimensional Bochner identity

$$\int (\mathcal{L}u)^2\,\mathrm{d}\nu_N = \int \|\nabla^2 u\|^2_{\mathrm{HS}}\,\mathrm{d}\nu_N + \int (\nabla u, \nabla^2 S_N\,\nabla u)_{H_N}\,\mathrm{d}\nu_N. \tag{144.100}$$

Indeed, expanding $\int (\Delta u - \nabla S_N \cdot \nabla u)^2 e^{-S_N}\,\mathrm{d}\phi$, moving one derivative from each second-derivative term, and collecting the mixed terms leaves precisely the Hessian term displayed above. Applying (144.100) to resolvent approximants and using (144.93) yields

$$\int (\nabla u_\varepsilon, A_N \nabla u_\varepsilon)_{H_N}\,\mathrm{d}\nu_N \le \int (\mathcal{L}u_\varepsilon)^2\,\mathrm{d}\nu_N. \tag{144.101}$$

Taking $\varepsilon \downarrow 0$ gives

$$\limsup_{\varepsilon\downarrow 0} \int (\nabla u_\varepsilon, A_N \nabla u_\varepsilon)_{H_N}\,\mathrm{d}\nu_N \le \int f^2\,\mathrm{d}\nu_N. \tag{144.102}$$

Therefore

$$\begin{aligned}\int f^2\,\mathrm{d}\nu_N &= \lim_{\varepsilon\downarrow 0}\int(\nabla f,\nabla u_\varepsilon)_{H_N}\,\mathrm{d}\nu_N\\ &\le \left[\int(\nabla f,A_N^{-1}\nabla f)_{H_N}\,\mathrm{d}\nu_N\right]^{1/2}\left[\limsup_{\varepsilon\downarrow 0}\int(\nabla u_\varepsilon,A_N\nabla u_\varepsilon)_{H_N}\,\mathrm{d}\nu_N\right]^{1/2}\\ &\le \left[\int(\nabla f,A_N^{-1}\nabla f)_{H_N}\,\mathrm{d}\nu_N\right]^{1/2}\left[\int f^2\,\mathrm{d}\nu_N\right]^{1/2}.\end{aligned}$$

Dividing by $(\int f^2\,\mathrm{d}\nu_N)^{1/2}$, and then applying this estimate to $f-\int f\,\mathrm{d}\nu_N$, gives

$$\mathrm{Var}_{\nu_N}(f)\le\int(\nabla f,A_N^{-1}\nabla f)_{H_N}\,\mathrm{d}\nu_N. \tag{144.103}$$

Linear functions are obtained by multiplying ℓ_v by smooth radial cutoffs and sending the cut-off radius to infinity; the quadratic lower bound following from (144.93) supplies the needed dominated-convergence bounds. Since $\nabla\ell_v=v$, (144.94) follows.

For the negative-Sobolev moment, the symmetry $\phi\mapsto-\phi$ gives $\int(\phi,e_{N,\alpha})_{H_N}\mathrm{d}\nu_N=0$. Applying (144.94) with $v=e_{N,\alpha}$ and using $A_N\ge L_N+m^2\mathbf{1}$ gives

$$\int|(\phi,e_{N,\alpha})_{H_N}|^2\,\mathrm{d}\nu_N(\phi)\le(e_{N,\alpha},A_N^{-1}e_{N,\alpha})_{H_N}\le(\lambda_{N,\alpha}+m^2)^{-1}.$$

Multiplying by $(1+\lambda_{N,\alpha})^{-\eta}$ and summing over α proves (144.96). $\square$

Remark 144.36 (Closedness of OS reflection positivity under weak convergence). Let E be a Hausdorff locally convex real vector space of Euclidean fields, let $\Theta:E\to E$ be a continuous involution, and let $\mu_N\Rightarrow\mu$ be weak convergence of Borel probability measures on E. Let $\mathcal{A}_+$ be an algebra of bounded continuous cylinder functions on E such that

$$\Theta F(\phi)=F(\Theta\phi)$$

is again a bounded continuous cylinder function whenever $F\in\mathcal{A}_+$. If

$$\int_E\overline{F(\Theta\phi)}\,F(\phi)\,\mathrm{d}\mu_N(\phi)\ge 0\qquad(F\in\mathcal{A}_+,\ N\ge 1), \tag{144.104}$$

then

$$\int_E\overline{F(\Theta\phi)}\,F(\phi)\,\mathrm{d}\mu(\phi)\ge 0\qquad(F\in\mathcal{A}_+). \tag{144.105}$$

If F is an unbounded polynomial cylinder observable, the same conclusion holds provided the random variables

$$\overline{F(\Theta\phi)}\,F(\phi)$$

are uniformly integrable under $\{\mu_N\}$ and integrable under μ, and F is approximated by bounded continuous cylinder cutoffs in $\mathcal{A}_+$. For $F\in\mathcal{A}_+$, define

$$Q_F(\phi)=\overline{F(\Theta\phi)}\,F(\phi).$$

Continuity of Θ and bounded continuity of F imply that Q_F is a bounded continuous function on E. Weak convergence therefore gives

$$\int_E Q_F \, \mathrm{d}\mu = \lim_{N\to\infty} \int_E Q_F \, \mathrm{d}\mu_N.$$

Each term in the sequence on the right is nonnegative by (144.104); the limit is therefore nonnegative, giving (144.105).

For an unbounded polynomial cylinder observable, choose bounded continuous functions χ_R on the finite-dimensional cylinder coordinate space such that $\chi_R = 1$ on the radius-R ball, $\chi_R = 0$ outside the radius-$2R$ ball, and $0 \le \chi_R \le 1$. Set $F_R = \chi_R F$. The bounded part of this observation gives

$$\int_E \overline{F_R(\Theta\phi)}\, F_R(\phi) \, \mathrm{d}\mu \ge 0.$$

Uniform integrability of $\overline{F(\Theta\phi)}F(\phi)$ under $\{\mu_N\}$, together with integrability under μ, permits $R \to \infty$ in the cutoff quadratic form. Hence the unbounded polynomial cylinder inequality follows whenever the stated moment hypothesis is available. This final step is substantive: in constructive field theory it is exactly where uniform moment or exponential-integrability estimates for the cutoff measures enter.

Remark 144.37 (Fourier cutoffs and reflection-positive regulators). The Fourier-Galerkin cutoff used above is an analytically convenient way to construct the stochastic convolution and the local Da Prato–Debussche solution map. OS reflection positivity, however, is not a formal consequence of being a reflection-symmetric ultraviolet cutoff. It is a positivity property of the Euclidean regulator with respect to a chosen time reflection. Thus the reflection-positivity hypothesis in Theorem 144.44 must be read as a condition on the actual cutoff family used to define the Euclidean measures, or else one must prove a regulator-independence comparison between a reflection-positive cutoff family and the Fourier-Galerkin SPDE approximation. The theorem below deliberately keeps this point visible: Fourier cutoffs are used for the path-space stochastic estimates, while OS positivity remains a separate regulator property until such a comparison is proved.

Proposition 144.38 (Lattice scalar cutoffs as OS-positive Euclidean regulators)**.** *Let $\Lambda_{a,L}$ be a finite two-dimensional periodic lattice with spacing $a > 0$, side length L, and an involutive time reflection ϑ whose reflection plane is either a set of sites or a set of mid-links. Let*

$$\mathrm{d}\mu_{a,L}(\phi) = Z_{a,L}^{-1} e^{-S_{a,L}(\phi)} \prod_{x\in\Lambda_{a,L}} \mathrm{d}\phi_x$$

where

$$S_{a,L}(\phi) = \frac{1}{2}\sum_{\langle xy\rangle} c_{xy}\,(\phi_x - \phi_y)^2 + \sum_{x\in\Lambda_{a,L}} V_x(\phi_x).$$

Assume $c_{xy} = c_{\vartheta x,\vartheta y} > 0$, the nearest-neighbor graph is invariant under ϑ, $V_{\vartheta x} = V_x$, and the finite integral defining $Z_{a,L}$ exists. Let the reflected field be $(\vartheta\phi)_x = \phi_{\vartheta x}$. Then $\mu_{a,L}$ is OS reflection positive on bounded observables depending only on the chosen nonnegative-time half-lattice, including the reflection plane when present:

$$\int \overline{F(\vartheta\phi)}\, F(\phi) \, \mathrm{d}\mu_{a,L}(\phi) \ge 0.$$

In particular, a lattice Φ^4_2 cutoff with reflection-invariant local coordinates,

$$V_x(u) = a^2\left[\frac{m_{a,L}^2}{2}u^2 + \frac{\lambda}{4}u^4 + E_{a,L}\right], \qquad \lambda > 0,$$

and positive nearest-neighbor kinetic coefficients is reflection positive at each finite cutoff whenever the displayed finite-dimensional integral is finite.

Proof. This is the scalar lattice reflection-positivity mechanism of Proposition 138.2, specialized to the regulators used in the constructive scalar theory. We spell out the assignment because it is the point at which the sign of the regulator is used. Split the lattice into the positive half, the reflected negative half, and the reflection plane if the plane contains sites. Terms contained strictly in one half are assigned to S_+ and ϑS_+. Reflection-plane on-site terms, and one half of every square term adjacent to the plane, are split symmetrically. A link crossing a mid-plane contributes

$$\frac{1}{2}c_{x,\vartheta x}(\phi_x - \phi_{\vartheta x})^2 = \frac{1}{2}c_{x,\vartheta x}\phi_x^2 + \frac{1}{2}c_{x,\vartheta x}\phi_{\vartheta x}^2 - c_{x,\vartheta x}\phi_x\phi_{\vartheta x}.$$

The first two summands have already been assigned to the two halves. In the Boltzmann weight the remaining cross-plane factor is therefore

$$\exp(c_{x,\vartheta x}\phi_x\phi_{\vartheta x}) = \sum_{n=0}^{\infty}\frac{c_{x,\vartheta x}^n}{n!}\,\phi_x^n\phi_{\vartheta x}^n,$$

with nonnegative coefficients. Multiplying these expansions for all crossing links expresses the reflection quadratic form $\int \overline{F(\vartheta\phi)}F(\phi)\mathrm{d}\mu_{a,L}$ as a sum of nonnegative coefficients times absolute squares of half-lattice integrals. Hence the quadratic form is nonnegative. The local potential and local counterterm coordinates do not enter this sign argument except through reflection invariance, reality, and integrability, because they are assigned to the two halves before the cross-plane kernel is expanded. $\square$

Proposition 144.38 proves the finite-cutoff OS-positivity input for a concrete lattice Euclidean regulator. Identifying that lattice family with the Fourier-Galerkin family used in the stochastic convolution estimates above is a separate comparison theorem. After matching the relevant local coordinates, one must prove convergence of the lattice and Fourier-Galerkin Schwinger functions, or construct a single regulator whose Langevin estimates and OS positivity are both available. Such a theorem must supply analytic estimates beyond either weak convergence or stochastic stationarity.

Remark 144.39 (Regulator comparison criterion for OS-positive stochastic limits)*.* Let E be a Hausdorff locally convex real vector space of Euclidean fields, let $\Theta : E \to E$ be a continuous involution, and let $\{\mu_N^{\mathrm{F}}\}$ and $\{\mu_N^{\mathrm{L}}\}$ be two cutoff families of Borel probability measures on E. Think of μ_N^{F} as the Fourier-Galerkin stochastic cutoff family and μ_N^{L} as a lattice cutoff family, but assume only the following properties.

First, suppose that

$$\mu_N^{\mathrm{F}} \Rightarrow \mu \qquad \text{weakly on } E.$$

Second, let $\mathcal{A}_+$ be an algebra of bounded continuous cylinder observables supported in nonnegative Euclidean time, and assume finite-cutoff OS positivity for the lattice family:

$$\int_E \overline{F(\Theta\phi)}\,F(\phi)\,\mathrm{d}\mu_N^{\mathrm{L}}(\phi) \geq 0, \qquad F \in \mathcal{A}_+. \tag{144.106}$$

Third, assume comparison of bounded cylinder observables in the following literal sense: for every bounded continuous cylinder function G in the algebra generated by $\mathcal{A}_+$ and its reflected observables,

$$\int_E G(\phi)\,\mathrm{d}\mu_N^{\mathrm{F}}(\phi) - \int_E G(\phi)\,\mathrm{d}\mu_N^{\mathrm{L}}(\phi) \longrightarrow 0. \tag{144.107}$$

Then the Fourier-Galerkin limiting measure μ is OS reflection positive on $\mathcal{A}_+$:

$$\int_E \overline{F(\Theta\phi)}\,F(\phi)\,\mathrm{d}\mu(\phi) \geq 0, \qquad F \in \mathcal{A}_+. \tag{144.108}$$

Let now P be a polynomial cylinder observable in finitely many continuous linear coordinates. Assume that bounded continuous cutoffs P_R of P can be chosen in the comparison algebra and that

$$|P_R(\phi) - P(\phi)| \leq 2|P(\phi)|\,\mathbf{1}_{\{|P(\phi)|>R\}}, \qquad P_R(\phi) \to P(\phi)$$

pointwise. If the random variables $|P|^2$ are uniformly integrable with respect to both cutoff families $\{\mu_N^{\mathrm{F}}\}$, $\{\mu_N^{\mathrm{L}}\}$, and are integrable with respect to μ, then both cutoff limits exist and

$$\int_E P\,\mathrm{d}\mu = \lim_{N\to\infty} \int_E P\,\mathrm{d}\mu_N^{\mathrm{F}} = \lim_{N\to\infty} \int_E P\,\mathrm{d}\mu_N^{\mathrm{L}}$$

holds. If, in addition, F is a positive-time polynomial observable and $|\overline{F\circ\Theta}F|$ has the same uniform-integrability property, then the OS inequality also holds for F. For bounded $F \in \mathcal{A}_+$, set

$$Q_F(\phi) = \overline{F(\Theta\phi)}\,F(\phi).$$

The function Q_F is a bounded continuous cylinder observable in the comparison algebra. Weak convergence of μ_N^{F} gives

$$\int_E Q_F\,\mathrm{d}\mu = \lim_{N\to\infty} \int_E Q_F\,\mathrm{d}\mu_N^{\mathrm{F}}.$$

The comparison hypothesis (144.107) gives

$$\lim_{N\to\infty} \int_E Q_F\,\mathrm{d}\mu_N^{\mathrm{F}} = \lim_{N\to\infty} \int_E Q_F\,\mathrm{d}\mu_N^{\mathrm{L}}$$

because the Fourier limit exists and the difference tends to zero. Each lattice integral is nonnegative by (144.106); hence its limit inferior is nonnegative. This proves (144.108).

For a polynomial cylinder observable P, fix R. Since P_R is bounded and belongs to the comparison algebra,

$$\lim_{N\to\infty} \left[\int_E P_R\,\mathrm{d}\mu_N^{\mathrm{F}} - \int_E P_R\,\mathrm{d}\mu_N^{\mathrm{L}}\right] = 0,$$

and weak convergence of μ_N^{F} gives

$$\int_E P_R\,\mathrm{d}\mu = \lim_{N\to\infty} \int_E P_R\,\mathrm{d}\mu_N^{\mathrm{F}}.$$

Uniform integrability allows $R \to \infty$ uniformly in N:

$$\left|\int (P - P_R)\,\mathrm{d}\mu_N^{\mathrm{F}}\right| \leq 2\int |P|\,\mathbf{1}_{\{|P|>R\}}\mathrm{d}\mu_N^{\mathrm{F}},$$

$$\left|\int (P - P_R)\,\mathrm{d}\mu_N^{\mathrm{L}}\right| \leq 2\int |P|\,\mathbf{1}_{\{|P|>R\}}\mathrm{d}\mu_N^{\mathrm{L}}.$$

The two L^1-tails tend to zero uniformly in N, because uniform integrability of $|P|^2$ implies uniform integrability of $|P|$ by Cauchy's inequality. Dominated convergence gives $\int P_R \, d\mu \to \int P \, d\mu$. Taking first $N \to \infty$ at fixed R, then $R \to \infty$, proves equality of the polynomial Schwinger-function limits. Applying the same truncation argument to $P = Q_F = \overline{F \circ \Theta F}$, after using the bounded part for each truncated F_R, gives the polynomial OS inequality.

The analytic comparison (144.107) is a separate constructive estimate; the criterion records what such a comparison must deliver. In Φ^4_2, the comparison is a constructive statement about two renormalized cutoff families in the same finite-volume local-coordinate chart. It must identify the mass and vacuum coordinates, prove convergence of bounded smeared-field observables, and provide the uniform-integrability estimates needed to remove the bounded truncations. Once these estimates are proved, lattice reflection positivity from Proposition 144.38 transfers to the stochastic Fourier-Galerkin limit by the criterion above.

Proposition 144.40 (Finite-volume Φ^4_2 SPDE/constructive route identification)**.** *Work on a fixed finite two-dimensional torus, with massive Gaussian reference law γ. Let $\sigma \in \{\mathrm{spde}, \mathrm{cons}\}$ label two ultraviolet cutoff families in the same field normalization and Wick-ordering chart. Let*

$$\mathcal{X}^\sigma_N$$

be the finite vector of smeared fields and Wick-polynomial coordinates entering a bounded cylinder observable, and let

$$V^\sigma_N = \frac{\lambda}{4} \int_{\mathbb{T}^2} :(R^\sigma_N \phi)^4: \, d^2x, \qquad W^\sigma_N = e^{-V^\sigma_N}, \qquad \lambda > 0,$$

where R^σ_N is the cutoff operator in the two regulator families. Assume that, for every finite cylinder-coordinate list under consideration,

$$\mathcal{X}^\sigma_N \to \mathcal{X} \quad \text{in } \gamma\text{-probability}, \qquad W^\sigma_N \to W \quad \text{in } L^1(\gamma), \qquad \sigma \in \{\mathrm{spde}, \mathrm{cons}\},$$

and that $\{W^\sigma_N\}$ is bounded in $L^p(\gamma)$ for some $p > 1$, uniformly in N and σ. Suppose also that the constructive family is one of the finite-volume $P(\phi)_2$ ultraviolet regulators covered by Theorem 137.4, specialized to $P(q) = \lambda q^4/4$, and that the chosen finite-volume constructive regulator is reflection positive in the sense of Proposition 144.38.

Then the bounded Wick-cylinder Schwinger functions of the stochastic cutoff family and of the constructive $P(\phi)_2$ cutoff family have the same finite-volume ultraviolet limit:

$$\lim_{N\to\infty} \frac{\int G(\mathcal{X}^{\mathrm{spde}}_N) W^{\mathrm{spde}}_N \, d\gamma}{\int W^{\mathrm{spde}}_N \, d\gamma} = \lim_{N\to\infty} \frac{\int G(\mathcal{X}^{\mathrm{cons}}_N) W^{\mathrm{cons}}_N \, d\gamma}{\int W^{\mathrm{cons}}_N \, d\gamma}$$

for every bounded continuous G of the displayed finite coordinate vector. If the polynomial cylinder observables used in the OS quadratic forms satisfy the uniform-integrability hypothesis of Remark 144.39, then the same common limit holds for those polynomial observables and the stochastic finite-volume limit is OS reflection positive on the positive-time polynomial cylinder algebra.

Consequently, in the massive finite-volume theory, the stochastic quantization invariant-law route and the constructive $P(\phi)_2$ route feed the same local Schwinger hierarchy into the later OS reconstruction problem whenever the displayed common-density and polynomial-tail estimates have been proved. Passing from this finite-volume identification to the infinite-volume scalar QFT is exactly the separate cluster-expansion input in Quoted Theorem 137.8; no thermodynamic-limit estimate is obtained merely from stochastic stationarity.

Proof. For a bounded continuous cylinder function G, apply Lemma 144.32 with $X_N^\sigma = \mathcal{X}_N^\sigma$ and $W_N^\sigma = e^{-V_N^\sigma}$. The lemma gives equality of the two normalized limiting expectations. The constructive side is the finite-volume $P(\phi)_2$ ultraviolet limit of Theorem 137.4, because the hypothesis places the constructive cutoff in the same Wick chart and supplies the L^1 density limit and uniform L^p integrability used there.

For polynomial cylinder observables, use bounded truncations. The uniform-integrability hypothesis permits the truncation radius to be sent to infinity uniformly in the two cutoff families, exactly as in Remark 144.39. Since the constructive finite regulator is reflection positive by Proposition 144.38, the bounded OS quadratic forms are nonnegative before the limit; the polynomial truncation passage transfers the inequality to the common ultraviolet limit. Finally, the infinite-volume statement is only a restatement of Quoted Theorem 137.8: once the finite-volume Schwinger hierarchies have been identified in every local window, the large-volume OS output still requires the cluster norm estimate and its consequences. □

Remark 144.41 (Polynomial OS observables from exponential moments). Let E be a Hausdorff locally convex real vector space, and let $\mu_N \Rightarrow \mu$ be Borel probability measures on E. Let $\ell_1, \dots, \ell_r$ be continuous real linear functionals on E, and fix $q > 0$, $\alpha > 0$. Put

$$V(\phi) = 1 + \sum_{j=1}^{r} |\ell_j(\phi)|^q.$$

Assume

$$\sup_N \int_E e^{\alpha V(\phi)}\, \mathrm{d}\mu_N(\phi) < \infty, \qquad \int_E e^{\alpha V(\phi)}\, \mathrm{d}\mu(\phi) < \infty. \tag{144.109}$$

Then for every polynomial $P \in \mathbb{C}[u_1, \dots, u_r]$, the random variables

$$|P(\ell_1(\phi), \dots, \ell_r(\phi))|^2$$

are uniformly integrable with respect to $\{\mu_N\}$ and integrable with respect to μ. Hence the unbounded polynomial-cylinder step in Remark 144.36 follows for such P whenever the bounded-cylinder reflection-positivity inequality is available.

There is a constant $C_P < \infty$ such that, for all $u \in \mathbb{R}^r$,

$$|P(u)|^2 \le C_P \exp\left(\frac{\alpha}{2}\left[1 + \sum_{j=1}^{r} |u_j|^q\right]\right). \tag{144.110}$$

This is the elementary domination of polynomial growth by exponential growth: each monomial is bounded by a constant times $\exp[(\alpha/4r)|u_j|^q]$ in each variable appearing in it, and the finite sum of monomials is absorbed into C_P. Let

$$A(\phi) = \exp\Big(\frac{\alpha}{2} V(\phi)\Big).$$

If $|P(\ell(\phi))|^2 > R$, then $C_P A(\phi) > R$. Therefore

$$|P(\ell(\phi))|^2 \mathbf{1}_{\{|P(\ell(\phi))|^2 > R\}} \le C_P A(\phi)\, \mathbf{1}_{\{A(\phi) > R/C_P\}} \le \frac{C_P^2}{R} A(\phi)^2.$$

Integrating and using $A^2 = e^{\alpha V}$ gives

$$\sup_N \int |P(\ell(\phi))|^2 \mathbf{1}_{\{|P(\ell(\phi))|^2 > R\}} \, d\mu_N(\phi) \le \frac{C_P^2}{R} \sup_N \int e^{\alpha V} \, d\mu_N \longrightarrow 0.$$

This is uniform integrability. The same estimate with μ instead of μ_N gives integrability under the limiting measure. Applying Remark 144.36 to bounded truncations of $P(\ell_1, \ldots, \ell_r)$ proves the final claim.

Lemma 144.42 (Finite lattice quartic tails)**.** *Let Λ be a finite set and let μ be a finite-dimensional scalar lattice measure*

$$d\mu(\phi) = Z^{-1} e^{-S(\phi)} \prod_{x \in \Lambda} d\phi_x, \qquad S(\phi) = K(\phi) + \sum_{x \in \Lambda} V_x(\phi_x),$$

where $K(\phi) \ge 0$. Suppose that, for constants $a_x > 0$, $b_x \ge 0$, and $d_x \in \mathbb{R}$,

$$V_x(u) \ge a_x u^4 - b_x u^2 - d_x. \tag{144.111}$$

Then $Z < \infty$, and for every $0 < \varepsilon < 1$,

$$\int \exp\left(\varepsilon \sum_{x \in \Lambda} a_x \phi_x^4 \right) d\mu(\phi) < \infty. \tag{144.112}$$

Consequently every polynomial cylinder observable has finite moments at this finite cutoff. A uniform version of (144.112) along a continuum cutoff family is a separate estimate; if it is available for the smeared coordinates entering an OS polynomial cylinder observable, then Remark 144.41 supplies the uniform integrability needed to pass that observable to the limit.

Proof. Since $K \ge 0$, the density is bounded above by

$$\exp\left[-\sum_x a_x \phi_x^4 + \sum_x b_x \phi_x^2 + \sum_x d_x \right].$$

For $A_x = (1 - \varepsilon) a_x$, the elementary inequality

$$b_x u^2 \le \frac{A_x}{2} u^4 + \frac{b_x^2}{2A_x}$$

gives

$$-A_x u^4 + b_x u^2 \le -\frac{A_x}{2} u^4 + \frac{b_x^2}{2A_x}.$$

Therefore

$$e^{\varepsilon \sum_x a_x \phi_x^4} e^{-S(\phi)} \le \exp\left[\sum_x d_x + \sum_x \frac{b_x^2}{2(1-\varepsilon) a_x} \right] \prod_x \exp\left[-\frac{(1-\varepsilon) a_x}{2} \phi_x^4 \right].$$

The product on the right is integrable over $\mathbb{R}^\Lambda$. Taking $\varepsilon = 0$ in the same estimate proves $Z < \infty$, while $Z > 0$ because the density is positive on a set of positive Lebesgue measure. Dividing by Z proves (144.112). Polynomial moments follow because every power of $|u|$ is bounded by a constant times $e^{\delta u^4}$ for any fixed $\delta > 0$. The final sentence is exactly Remark 144.41 applied to the finite list of smeared cylinder coordinates. $\square$

Hypothesis 144.43 (Φ^4_2 stochastic-quantization assembly inputs on the torus)**.** Fix $m^2 > 0$, $\lambda > 0$, $0 < \kappa < 1/4$, and $\beta = 1 + 2\kappa$. Let P_N be a reflection-symmetric Fourier cutoff on $\mathbb{T}^2$. Let μ_N be the finite-dimensional Wick-ordered Gibbs measure

$$\mathrm{d}\mu_N(\phi) = Z_N^{-1} \exp\left[-\frac{\lambda}{4}\int_{\mathbb{T}^2} :(P_N\phi)^4:\ \mathrm{d}^2x\right] \mathrm{d}\mu_{0,N}(\phi),$$

where $\mu_{0,N}$ is the massive Gaussian law with covariance $P_N(-\Delta + m^2)^{-1}P_N$. Let $P_t^{(N)}$ be the Markov semigroup of the corresponding finite-dimensional stochastic quantization equation, and embed all laws in $H^{-\kappa}(\mathbb{T}^2)$. By Proposition 144.9, the enhanced Ornstein–Uhlenbeck fields

$$\mathbb{X}_N = (X_N, :X_N^2:, :X_N^3:)$$

converge in probability in $C([0,T]; H^{-\kappa})^3$, for every $T < \infty$, to a limit $\mathbb{X} = (X, :X^2:, :X^3:)$.

Assume the following three further analytic facts.

(a) For every $Y_0 \in H^\beta$, the local solution map of Theorem 144.14 extends globally in stochastic time, and the cutoff solution maps converge to it uniformly on compact subsets of the enhanced-noise/initial-data space. Lemma 144.21 reduces this assertion, in the Besov formulation, to a blow-up alternative plus a rough energy-to-Besov a priori bound on compact enhanced-noise sets.

(b) The stationary laws $\{\mu_N\}$ are tight in $H^{-\kappa}$, and every weak limit μ is carried by fields admitting the decomposition $\Phi = X + Y$ with Y given by the global DPD solution map. By Lemma 144.27, this tightness follows from a cutoff-uniform moment bound in $H^{-\eta}$ for some $0 < \eta < \kappa$; the interacting stationary cutoff family therefore requires an independent constructive estimate establishing that moment bound before the weak-convergence passage is invoked. The corresponding free massive Gaussian estimate is Proposition 144.28. Lemma 144.29 and Remark 144.30 gives one precise route: a cutoff-uniform L^q bound for the normalized Wick-ordered density with respect to the massive Gaussian reference law implies the required interacting negative-Sobolev moments. If the cutoff Wick quartic satisfies the Nelson stability estimate (144.80), then Lemma 144.31 proves this L^q density bound.

(c) Reflection positivity of the cutoff measures μ_N holds for bounded cylinder observables depending on positive Euclidean time, and the polynomial Wick observables used to identify Schwinger functions obey the uniform-integrability bounds needed for the truncation step in Remark 144.36. This is a property of the chosen Euclidean regulator and must be proved for the regulator used in the limiting construction. For the lattice scalar regulator of Proposition 144.38, the bounded-observable positivity statement is proved at finite cutoff, and Lemma 144.42 proves finite-cutoff polynomial integrability. Passing polynomial observables uniformly through the continuum limit requires the exponential moment criterion of Remark 144.41. Using lattice positivity in the present Fourier-Galerkin stochastic assembly requires the bounded-cylinder and uniform-integrability comparison hypotheses stated in Remark 144.39. A sufficient condition for the bounded-cylinder comparison part is the common L^1 density-limit criterion of Lemma 144.32; a still more primitive sufficient condition, stated in terms of convergence of the renormalized interaction potentials and a uniform L^p stability estimate for their exponential weights, is Corollary 144.33. For the two-dimensional Wick quartic interaction, the potential convergence part of this condition follows from Remark 144.8; the remaining regulator-specific input is the uniform exponential-weight stability. The common finite cylinder-coordinate convergence for smeared Wick polynomials is supplied by Remark 144.7. Equivalently, under these common Wick-limit hypotheses and the uniform L^p stability bound, the bounded Wick-cylinder comparison is precisely (144.90). The common Wick-limit hypothesis itself can be checked from mixed covariance convergence by Lemma 144.6; for smooth compactly supported Fourier multiplier cutoffs, that covariance convergence is proved by Proposition 144.5. If, in addition, those smooth regulators satisfy the Nelson stability input, the density bounds, tightness, and bounded Wick-cylinder comparison are collected in Proposition 144.34.

Theorem 144.44 (Φ^4_2 stochastic-quantization assembly on the torus)**.** *Under Hypothesis 144.43, every weak limit μ of $\{\mu_N\}$ is an invariant law of the limiting DPD Markov semigroup. Its Schwinger functions are the limits of the finite-cutoff Wick-ordered Schwinger functions, and μ satisfies OS reflection positivity for cylinder observables. If, in addition, the resulting Schwinger hierarchy satisfies the moment-continuity, positive-time semigroup, and corrected OS-II linear-growth hypotheses in Remark 144.101, then the hierarchy determines a two-dimensional Wightman theory by Theorem 12.16.*

Proof. At finite cutoff the Langevin generator is the gradient generator of the finite-dimensional action defining μ_N. The integration-by-parts identity (144.2), together with the finite-dimensional semigroup hypotheses recorded in Lemma 144.1, therefore gives

$$\int P_t^{(N)} F \,\mathrm{d}\mu_N = \int F \,\mathrm{d}\mu_N$$

for every bounded smooth cylinder function F. Thus the word "stationary" at finite cutoff means reversibility and invariance of the Markov semigroup in the cutoff Hilbert space, not only a formal adjoint calculation for the infinitesimal generator.

The stochastic input is Proposition 144.9, which strengthens the equal-time Wick-power construction of Proposition 144.4 to path-space convergence and fixes the Wick normalization. Given this enhanced noise, the products in the DPD nonlinearity are defined by Lemma 144.10, and Theorem 144.14 gives the local solution map. Hypothesis 144.43(a) is precisely the global rough-energy and compact-continuity upgrade of the smooth estimate Lemma 144.15; in the Besov topology this upgrade is organized by Lemma 144.21. Thus the limiting Markov semigroup P_t is well-defined on the support of every weak stationary limit in Hypothesis 144.43(b), and $P_t^{(N)} F \to P_t F$ uniformly on the high-probability compact sets supplied by Hypothesis 144.43(a),(b).

Lemma 144.26 now applies with $E = H^{-\kappa}$. Therefore

$$\int P_t F \,\mathrm{d}\mu = \int F \,\mathrm{d}\mu$$

for every bounded continuous cylinder observable F, and hence μ is invariant. The convergence of Schwinger functions follows by applying the same weak convergence to products of smeared fields and Wick powers after approximating the polynomial observables by bounded cylinder observables; the uniform moment bounds needed for this approximation are part of the tightness and rough-energy input in Hypothesis 144.43(b).

For OS positivity, let Θ be Euclidean time reflection and let F be a cylinder observable supported in positive Euclidean time. The finite-cutoff reflection-positivity assumption gives

$$\int \overline{\Theta F}\, F \,\mathrm{d}\mu_N \geq 0.$$

Remark 144.36 passes this inequality to the weak limit, so

$$\int \overline{\Theta F}\, F \,\mathrm{d}\mu \geq 0.$$

This is OS reflection positivity on the cylinder algebra. Once the remaining OS growth and regularity estimates are verified, the OS reconstruction theorem of Chapter 12 applies. □

The theorem separates the logical assembly from the analytic estimates. In the chapter we have proved the finite-dimensional Langevin identity, the Sobolev Wick-power construction, the path-space enhanced-noise convergence, the Sobolev DPD local solution map, the smooth enhanced-noise energy inequality, and the abstract invariant-law passage. The sharp global two-dimensional theorem still requires four further inputs: a rough energy-to-Besov bound strong enough, by Lemma 144.21, to give global compact-continuity of the DPD solution map; a cutoff-uniform negative-Sobolev moment strong enough to imply tightness by Lemma 144.27; for Wick Φ^4_2 this moment follows from the density criterion Remark 144.30 once the uniform Nelson-type L^q density bound is proved, and Lemma 144.31 proves that bound from the Nelson exponential stability estimate; uniform integrability; and either finite-cutoff OS positivity for the stochastic regulator itself or a regulator-comparison theorem relating it to the OS-positive lattice regulator of Proposition 144.38. The abstract form of the latter implication is Remark 144.39; the bounded cylinder comparison follows from Lemma 144.32 if the two regulators have a common L^1 interaction-density limit and identical limiting cylinder coordinates; by Corollary 144.33, the common L^1 density limit follows from convergence in probability of the renormalized interaction potentials plus a uniform L^p stability bound on the exponential weights. For the Wick quartic potential, the convergence in probability is supplied by Remark 144.8, so the remaining constructive estimate is the uniform L^p stability of the weights in the chosen regulator family. The matching convergence of finite Wick-coordinate cylinder vectors is Remark 144.7; combined with the stability estimate, these inputs give the bounded comparison statement (144.90). The common Wick-limit hypothesis is reduced to a covariance estimate by Lemma 144.6, and for smooth Fourier multiplier cutoffs the needed covariance estimate follows from Proposition 144.5. The uniform-integrability input can be supplied by the exponential cylinder-moment criterion in Remark 144.41; finite quartic tails at each lattice cutoff are proved by Lemma 144.42, while uniformity as the cutoff is removed remains part of the constructive estimate. These inputs are listed explicitly in Hypothesis 144.43.

Sharp two-dimensional DPD closure status. The classical two-dimensional Da Prato–Debussche construction packages a local pathwise solution, global extension from the dissipative quartic drift, and identification of the invariant law with the Wick-ordered Euclidean Φ^4_2 measure on a finite torus. In this chapter that package is used through the explicit ingredients above, rather than as an imported black box. The mild form of (144.34) is

$$Y(\tau) = e^{\tau(\Delta-m^2)}Y(0) - \lambda \int_0^\tau e^{(\tau-s)(\Delta-m^2)}\mathcal{N}(Y(s), \mathbb{X}(s))\,\mathrm{d}s,$$

where $\mathcal{N}$ is the cubic expression in (144.34). Parabolic Schauder estimates increase spatial regularity by two derivatives relative to the forcing; the Sobolev version is the heat-kernel estimate (144.30), while the sharp two-dimensional fixed point uses the corresponding parabolic Besov–Holder estimates. Proposition 144.4 constructs the Wick powers in a Sobolev topology; the sharper fixed-point input places $:X^2:$ and $:X^3:$ in negative Holder/Besov spaces of arbitrarily small negative regularity. Lemma 144.10 and Theorem 144.14 give a self-contained Sobolev version of the local fixed-point argument. The sharp proof upgrades these Sobolev exponents to the scale-invariant parabolic Besov–Holder spaces. Multiplying the equation by Y, integrating over space, and using the sign of λY^4 gives an energy inequality with random lower-order forcing; the local rough-forcing estimate (144.62) is the precise remaining estimate that turns this energy mechanism into the global compact-continuity input of Hypothesis 144.43. The invariant law is identified through finite-cutoff Langevin reversibility, tightness, and regulator comparison; at fixed volume this comparison is assembled explicitly in Proposition 144.40. Those steps have been reduced above to Nelson stability for the chosen Wick quartic cutoff family, common Wick-coordinate limits, uniform integrability of polynomial cylinders, and the OS-II growth conditions in Remark 144.101. Thus the remaining two-dimensional work is a finite list of constructive estimates, not a separate theorem imported under the name Da Prato–Debussche.

144.5.5 Why the Two-Dimensional Decomposition Fails in Three Dimensions

Remark 144.45 (Sobolev DPD closure and the three-dimensional obstruction)*.* The Sobolev fixed-point mechanism used above for Φ^4_2 cannot close the dynamic Φ^4_3 equation by replacing 2 with 3. More precisely, let X be the three-dimensional stationary stochastic convolution. Its spatial regularity is

$$X \in H^{-1/2-\kappa-\varepsilon}(\mathbb{T}^3) \quad \text{for every } \varepsilon > 0,$$

and the cubic Wick power has the formal Sobolev regularity

$$:X^3: \in H^{-3/2-3\kappa-\varepsilon}(\mathbb{T}^3) \quad \text{for every } \varepsilon > 0.$$

If the remainder Y is sought in H^β, then even the minimal Sobolev multiplier requirement for the product YX is

$$\beta > \frac{1}{2} + \kappa. \tag{144.113}$$

The Duhamel map from the cubic forcing $:X^3:$ to H^β would require

$$\theta = \frac{\beta + \frac{3}{2} + 3\kappa}{2} > 1. \tag{144.114}$$

But the heat-kernel Duhamel estimate (144.31) is locally integrable at the upper time endpoint only for $\theta < 1$. Therefore the two-dimensional Sobolev DPD argument has no three-dimensional analogue.

The covariance of the massive three-dimensional Gaussian field has short-distance singularity $\|x-y\|^{-1}$. Equivalently, the field has Sobolev regularity below $-1/2$, and the k-th Wick power has the additive formal regularity below $-k/2$. With the small loss parameter κ, this gives the displayed regularities for X and $:X^3:$.

For multiplication in three spatial dimensions, a distribution of regularity $-1/2-\kappa$ can be multiplied by a Sobolev function of regularity β only when the sum of regularities is positive; this gives (144.113). If the forcing is in $H^{-3/2-3\kappa-\varepsilon}$, reaching H^β by the heat equation requires, after letting $\varepsilon \downarrow 0$, gain

$$2\theta > \beta + \frac{3}{2} + 3\kappa.$$

Combining this with (144.113) gives

$$\theta > \frac{\frac{1}{2} + \kappa + \frac{3}{2} + 3\kappa}{2} = 1 + 2\kappa.$$

This contradicts the condition $\theta < 1$ needed for the time singularity $(t-s)^{-\theta}$ to be integrable. If one instead requires H^β to be an algebra, then $\beta > 3/2$, and the obstruction is even stronger: $\theta > 3/2 + \frac{3}{2}\kappa$. Thus the failure is structural, not an artifact of a nonoptimal exponent choice.

144.6 Regularity Structures and Φ^4_3

In three spatial dimensions the decomposition $\Phi = X + Y$ is not sufficient: X has regularity $-\frac{1}{2} - \kappa$, and the products generated by the cubic drift fall below the classical multiplication threshold. Regularity structures supply the missing local expansion calculus.

144.6.1 Models, Modelled Distributions, and Reconstruction

Definition 144.46 (Regularity structure)**.** A regularity structure is a triple (A, T, G), where $A \subset \mathbb{R}$ is a locally finite set of homogeneities bounded below, $T = \bigoplus_{\alpha \in A} T_\alpha$ is a graded model space, and G is a structure group acting on T such that for $\tau \in T_\alpha$,

$$\Gamma\tau - \tau \in \bigoplus_{\beta<\alpha} T_\beta \qquad (\Gamma \in G).$$

A model on spacetime assigns to every point z a linear map $\Pi_z : T \to \mathcal{D}'(\mathbb{R}^{1+d})$ and to every pair z, z' an element $\Gamma_{zz'} \in G$, satisfying

$$\Pi_z = \Pi_{z'}\Gamma_{z'z}, \qquad \Gamma_{zz'}\Gamma_{z'z''} = \Gamma_{zz''},$$

together with analytic bounds compatible with the homogeneities.

Definition 144.47 (Parabolic model and modelled-distribution norms)**.** Let $\mathfrak{s} = (2, 1, \dots, 1)$ be the parabolic scaling on $\mathbb{R}^{1+d}$, let $|\mathfrak{s}| = d + 2$, and set

$$\|(\tau, x)\|_{\mathfrak{s}} = |\tau|^{1/2} + |x_1| + \cdots + |x_d|.$$

For $0 < \delta \le 1$, $z \in \mathbb{R}^{1+d}$, and $\varphi \in C^r_c(\mathbb{R}^{1+d})$, define

$$\varphi^\delta_z(w) = \delta^{-|\mathfrak{s}|}\varphi\big(\delta^{-2}(w_0 - z_0), \delta^{-1}(w_1 - z_1), \dots, \delta^{-1}(w_d - z_d)\big).$$

Let $\mathcal{B}_r$ be the set of such test functions supported in the unit $\mathfrak{s}$-ball and satisfying $\|\varphi\|_{C^r} \le 1$. Fix a compact set K and $\gamma > 0$. The model seminorm below γ is finite when

$$\|\Pi\|_{K;\gamma} = \sup_{\substack{z\in K,\ \alpha<\gamma \\ \tau\in T_\alpha,\ \|\tau\|_\alpha=1 \\ \varphi\in\mathcal{B}_r,\ 0<\delta\le 1}} \delta^{-\alpha}\left|\langle \Pi_z\tau, \varphi^\delta_z\rangle\right| < \infty$$

and

$$\|\Gamma\|_{K;\gamma} = \sup_{\substack{z,z'\in K,\ 0<\|z-z'\|_{\mathfrak{s}}\le 1 \\ \beta<\alpha<\gamma,\ \tau\in T_\alpha,\ \|\tau\|_\alpha=1}} \frac{\|(\Gamma_{zz'}\tau)_\beta\|_\beta}{\|z-z'\|_{\mathfrak{s}}^{\alpha-\beta}} < \infty.$$

Here $(\cdot)_\beta$ denotes projection to T_β. A modelled distribution of regularity γ on K is a function

$$F : K \longrightarrow T_{<\gamma} := \bigoplus_{\alpha<\gamma} T_\alpha$$

with finite norm

$$\|F\|_{\gamma;K} = \sup_{z\in K,\ \alpha<\gamma} \|F_\alpha(z)\|_\alpha + \sup_{\substack{z,z'\in K,\ 0<\|z-z'\|_{\mathfrak{s}}\le 1\\ \alpha<\gamma}} \frac{\|F_\alpha(z) - (\Gamma_{zz'}F(z'))_\alpha\|_\alpha}{\|z-z'\|_{\mathfrak{s}}^{\gamma-\alpha}}.$$

Lemma 144.48 (Coherence of a modelled local germ)**.** *Let F be a modelled distribution in the sense of Definition 144.47, and set $\zeta_z = \Pi_z F(z)$. If $z, z' \in K$, $\|z-z'\|_{\mathfrak{s}} \le \delta \le 1$, and $\varphi \in \mathcal{B}_r$, then*

$$\left|\left\langle \zeta_z - \zeta_{z'}, \varphi_z^\delta \right\rangle\right| \le C_K \|\Pi\|_{K;\gamma} \|F\|_{\gamma;K}\, \delta^\gamma.$$

Proof. The model identity gives

$$\Pi_{z'}F(z') = \Pi_z \Gamma_{zz'} F(z'),$$

hence

$$\zeta_z - \zeta_{z'} = \Pi_z\big(F(z) - \Gamma_{zz'}F(z')\big).$$

Decompose the vector in parentheses into homogeneous components. For each $\alpha < \gamma$, the modelled-distribution norm gives

$$\|(F(z) - \Gamma_{zz'}F(z'))_\alpha\|_\alpha \le \|F\|_{\gamma;K} \|z-z'\|_{\mathfrak{s}}^{\gamma-\alpha}.$$

The model bound gives

$$\left|\left\langle \Pi_z(F(z) - \Gamma_{zz'}F(z'))_\alpha, \varphi_z^\delta \right\rangle\right| \le \|\Pi\|_{K;\gamma} \|F\|_{\gamma;K} \delta^\alpha \|z-z'\|_{\mathfrak{s}}^{\gamma-\alpha}.$$

Since $\|z-z'\|_{\mathfrak{s}} \le \delta$ and the set $\{\alpha \in A : \alpha < \gamma\}$ is finite on compact homogeneity windows, the sum over α is bounded by the displayed constant times δ^γ. □

Theorem 144.49 (Compact finite-sector reconstruction theorem)**.** *Assume that the set $A_{<\gamma} = \{\alpha \in A : \alpha < \gamma\}$ is finite, assume $0 < \gamma < |\mathfrak{s}|$, let*

$$r > \max\{-\min A_{<\gamma},\, \gamma\},$$

and let the model seminorms of Definition 144.47 be finite on a compact parabolic neighborhood of K. For every modelled distribution F of regularity $\gamma > 0$ on that neighborhood, there is a unique distribution $\mathcal{R}F$ on the interior of K such that, for every $\varphi \in \mathcal{B}_r$, $z \in K$, and $0 < \delta \le 1$ with $\operatorname{supp} \varphi_z^\delta \subset K$,

$$\left|\left\langle \mathcal{R}F - \Pi_z F(z), \varphi_z^\delta \right\rangle\right| \le C_K (1 + \|\Pi\|_{K;\gamma})(1 + \|\Gamma\|_{K;\gamma}) \|F\|_{\gamma;K}\, \delta^\gamma. \tag{144.115}$$

Proof. Choose a compactly supported parabolic multiresolution system of class C^r: scaling functions φ_y^0 at scale 1, wavelets ψ_y^n at scale 2^{-n}, $y \in \Lambda_n$, with vanishing moments against all parabolic polynomials of degree $< r$, finite overlap at each scale, and the expansion formula

$$\eta = \sum_{y\in\Lambda_0} \langle \eta, \varphi_y^0 \rangle \varphi_y^0 + \sum_{n\ge 0} \sum_{y\in\Lambda_n} \langle \eta, \psi_y^n \rangle \psi_y^n$$

for every test function η supported in the interior of K. The wavelets are L^2-normalized, so $\psi_y^n = 2^{n|\mathfrak{s}|/2} \psi_y^{2^{-n}}$ in the notation of Definition 144.47.

Define a distribution by prescribing its coefficients

$$\langle \mathcal{R}F, \varphi_y^0\rangle = \langle \Pi_y F(y), \varphi_y^0\rangle, \qquad \langle \mathcal{R}F, \psi_y^n\rangle = \langle \Pi_y F(y), \psi_y^n\rangle.$$

The coefficient estimate for ψ_y^n is

$$|\langle \Pi_y F(y), \psi_y^n\rangle| \le C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K} 2^{-n(\alpha_0+|\mathfrak{s}|/2)}, \qquad \alpha_0 = \min A_{<\gamma}.$$

For a C^r test function η, the wavelet moment cancellation gives

$$|\langle \eta, \psi_y^n\rangle| \le C_\eta 2^{-n(r+|\mathfrak{s}|/2)}.$$

Only $O(2^{n|\mathfrak{s}|})$ wavelets at scale 2^{-n} meet $\operatorname{supp}\eta$, hence the absolute value of the scale-n contribution is bounded by

$$C_\eta\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K} 2^{n|\mathfrak{s}|} 2^{-n(\alpha_0+|\mathfrak{s}|/2)} 2^{-n(r+|\mathfrak{s}|/2)} = C_\eta\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K} 2^{-n(\alpha_0+r)}.$$

The hypothesis $r > -\alpha_0$ makes the series converge. Thus $\mathcal{R}F$ is a well-defined distribution.

It remains to prove the local estimate. Fix z, δ, and φ_z^δ. Put $\ell_n = 2^{-n}$. The coefficient of $\mathcal{R}F - \Pi_z F(z)$ against a wavelet ψ_y^n equals

$$\langle \Pi_y F(y) - \Pi_z F(z), \psi_y^n\rangle.$$

If ψ_y^n meets the support of φ_z^δ, then $\|y-z\|_{\mathfrak{s}} \le C(\ell_n+\delta)$. Split the wavelet expansion into coarse scales $\ell_n \ge \delta$ and fine scales $\ell_n < \delta$.

At a coarse scale, $\|y-z\|_{\mathfrak{s}} \le C\ell_n$, so Lemma 144.48, tested at scale ℓ_n, gives

$$|\langle \Pi_y F(y) - \Pi_z F(z), \psi_y^n\rangle| \le C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\ell_n^{\gamma+|\mathfrak{s}|/2}.$$

The r vanishing moments of ψ_y^n and the C^r bound of φ_z^δ give

$$|\langle \varphi_z^\delta, \psi_y^n\rangle| \le C\,\delta^r \ell_n^{-r-|\mathfrak{s}|/2}.$$

Only a bounded number of coarse-scale wavelets meet the support of φ_z^δ. Therefore the coarse contribution is bounded by

$$C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\delta^\gamma,$$

because

$$\sum_{\ell_n \ge \delta} \delta^r \ell_n^{\gamma-r} \le C\delta^\gamma$$

when $r > \gamma$. The scale-one scaling functions are handled in the same coarse block except that no moment cancellation is needed: their coefficients are bounded and their pairing with φ_z^δ is $O(\delta^{|\mathfrak{s}|})$, which is $O(\delta^\gamma)$ under the hypothesis $\gamma < |\mathfrak{s}|$.

At a fine scale, write the modelled-distribution difference by homogeneity:

$$\langle \mathcal{R}F - \Pi_z F(z), \psi_y^n\rangle = \sum_{\alpha<\gamma} \left\langle \Pi_z\big(\Gamma_{zy}F(y) - F(z)\big)_\alpha, \psi_y^n\right\rangle.$$

For wavelets meeting the support of φ_z^δ, $\|y-z\|_{\mathfrak{s}} \le C\delta$. The α-component of the modelled-distribution difference is therefore bounded by

$$C\|F\|_{\gamma;K}\delta^{\gamma-\alpha}.$$

Testing the corresponding model component at wavelet scale ℓ_n gives

$$\big|\big\langle \Pi_z\big(\Gamma_{zy}F(y) - F(z)\big)_\alpha, \psi_y^n\big\rangle\big| \le C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\delta^{\gamma-\alpha}\ell_n^{\alpha+|\mathfrak{s}|/2}.$$

The wavelet moment estimate in the opposite scale regime gives

$$|\langle \varphi_z^\delta, \psi_y^n\rangle| \le C\,\delta^{-|\mathfrak{s}|-r}\ell_n^{r+|\mathfrak{s}|/2}.$$

There are $O((\delta/\ell_n)^{|\mathfrak{s}|})$ fine-scale wavelets meeting the support of φ_z^δ. The scale-n contribution of a fixed homogeneity α is thus bounded by

$$C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\delta^{\gamma-\alpha-r}\ell_n^{\alpha+r}.$$

Since $r > -\alpha$, the sum over $\ell_n < \delta$ is bounded by

$$C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\delta^{\gamma-\alpha-r}\delta^{\alpha+r} = C\|\Pi\|_{K;\gamma}\|F\|_{\gamma;K}\delta^{\gamma}.$$

There are only finitely many $\alpha < \gamma$, so summing the homogeneities proves (144.115).

For uniqueness, let U be the difference of two distributions satisfying (144.115). Choose a non-negative $\chi \in C_c^r$ with integral one. For any test function η with support in the interior of K,

$$\langle U, \eta\rangle = \lim_{\delta\downarrow 0}\int \eta(z)\langle U, \chi_z^\delta\rangle\,\mathrm{d}z.$$

The bound $|\langle U, \chi_z^\delta\rangle| \le C\delta^\gamma$ and $\gamma > 0$ force the limit to vanish. Hence $U = 0$. $\square$

Theorem 144.49 supplies the reconstruction component of the finite-sector theory used by the local Φ^4_3 regularity structure. The still-open part for dynamic Φ^4_3 is not the reconstruction step itself; it is the construction of the BPHZ-renormalized random model with uniform analytic bounds, the convergence of those models as the mollifier is removed, and the nonlinear fixed-point argument in the corresponding modelled-distribution spaces.

144.6.2 Random Model Convergence and Finite-Chaos Coordinates

Lemma 144.50 (Dyadic Cauchy criterion for random models)**.** *Fix a compact parabolic set K, a regularity threshold $\gamma > 0$, and a finite homogeneity set $A_{<\gamma}$. Choose the test regularity*

$$r > -\min A_{<\gamma}.$$

Let $Z_n = (\Pi^{(n)}, \Gamma^{(n)})$, $n \ge 1$, be random models on this finite sector. Define

$$N_n = 1 + \|\Pi^{(n)}\|_{K;\gamma} + \|\Gamma^{(n)}\|_{K;\gamma},$$

where the seminorms are those of Definition 144.47. Let $d_{K;\gamma}(Z, Z')$ denote the corresponding model distance, obtained by replacing Π and Γ in those seminorms by $\Pi - \Pi'$ and $\Gamma - \Gamma'$. Suppose that, for some $p \ge 1$, $\rho > 0$, and constants $C_N, C_D < \infty$,

$$\sup_{n\ge 1}\mathbb{E}N_n^p \le C_N, \qquad \mathbb{E}\,d_{K;\gamma}(Z_{n+1}, Z_n)^p \le C_D 2^{-p\rho n}. \tag{144.116}$$

Then there exists a random model

$$Z_\infty = (\Pi^{(\infty)}, \Gamma^{(\infty)})$$

on the same finite sector such that

$$d_{K;\gamma}(Z_n, Z_\infty) \longrightarrow 0 \quad \textit{in } L^p(\Omega) \textit{ and almost surely.}$$

More precisely,

$$(\mathbb{E}\, d_{K;\gamma}(Z_n, Z_\infty)^p)^{1/p} \le \frac{C_D^{1/p}}{1-2^{-\rho}}\, 2^{-\rho n}. \tag{144.117}$$

For every $0 < \rho' < \rho$, there is an almost surely finite random constant $C_{\rho'}(\omega)$ such that

$$d_{K;\gamma}(Z_n(\omega), Z_\infty(\omega)) \le C_{\rho'}(\omega) 2^{-\rho' n} \tag{144.118}$$

for all n. The limiting model satisfies

$$\mathbb{E}\big(1 + \|\Pi^{(\infty)}\|_{K;\gamma} + \|\Gamma^{(\infty)}\|_{K;\gamma}\big)^p \le C_N.$$

Proof. Put

$$D_n = d_{K;\gamma}(Z_{n+1}, Z_n).$$

Minkowski's inequality and (144.116) give, for $m > n$,

$$(\mathbb{E}\, d_{K;\gamma}(Z_m, Z_n)^p)^{1/p} \le \sum_{j=n}^{m-1} (\mathbb{E} D_j^p)^{1/p} \le C_D^{1/p} \sum_{j=n}^{m-1} 2^{-\rho j}.$$

The geometric series is Cauchy, so Z_n is Cauchy in L^p for the model distance. Letting $m \to \infty$ after constructing the limit below will give (144.117).

We now construct the pointwise limiting model on the event where $\sum_{n\ge 1} D_n < \infty$. This event has probability one, because

$$\mathbb{E} \sum_{n\ge 1} D_n \le C_D^{1/p} \sum_{n \ge 1} 2^{-\rho n} < \infty.$$

Fix ω in this event. For each $z \in K$, $\alpha < \gamma$, and $\tau \in T_\alpha$, the numbers

$$\Big\langle \Pi_z^{(n)} \tau, \varphi_z^\delta \Big\rangle$$

are Cauchy uniformly over $\varphi \in \mathcal{B}_r$ and $\delta \in (0,1]$ after multiplication by $\delta^{-\alpha}$. Since $r > -\min A_{<\gamma}$, the dyadic wavelet expansion used in the proof of Theorem 144.49 implies that this Cauchy property determines a unique distribution, denoted $\Pi_z^{(\infty)}\tau$, on the interior of K. The same distance bound gives Cauchy convergence of each finite-dimensional component $(\Gamma_{zz'}^{(n)}\tau)_\beta$; define its limit to be $(\Gamma_{zz'}^{(\infty)}\tau)_\beta$.

The algebraic model identities pass to the limit. For instance, at every finite cutoff,

$$\Pi_{z'}^{(n)}\tau = \Pi_z^{(n)} \Gamma_{zz'}^{(n)} \tau.$$

Pair this identity with an arbitrary C^r test function supported in the interior of K, expand the right-hand side into the finitely many homogeneous components of $\Gamma_{zz'}^{(n)}\tau$, and take $n \to \infty$. The distributional convergence of the Π-components and the finite-dimensional convergence of the Γ-components give

$$\Pi_{z'}^{(\infty)}\tau = \Pi_z^{(\infty)}\Gamma_{zz'}^{(\infty)}\tau.$$

The identities $\Gamma_{zz}^{(n)} = \mathbf{1}$ and $\Gamma_{zz'}^{(n)}\Gamma_{z'z''}^{(n)} = \Gamma_{zz''}^{(n)}$ are polynomial identities among finitely many matrix entries on $\bigoplus_{\alpha<\gamma} T_\alpha$, so their limits satisfy the same identities.

The analytic bounds also pass to the limit. For every fixed normalized test configuration and every fixed reexpansion component, the limiting value is the limit of the finite-cutoff values. Taking the supremum gives

$$1 + \|\Pi^{(\infty)}\|_{K;\gamma} + \|\Gamma^{(\infty)}\|_{K;\gamma} \le \liminf_{n\to\infty} N_n.$$

Fatou's lemma therefore gives the displayed L^p bound on the limiting model. The same telescoping estimate gives, on the probability-one event,

$$d_{K;\gamma}(Z_n, Z_\infty) \le \sum_{j=n}^{\infty} D_j.$$

Taking the L^p norm of this inequality proves (144.117).

It remains to prove the almost-sure rate. Let $0 < \rho' < \rho$. Markov's inequality gives

$$\mathbb{P}(D_n > 2^{-\rho' n}) \le C_D\, 2^{-p(\rho-\rho')n}.$$

The right-hand side is summable in n. Borel–Cantelli implies that $D_n \le 2^{-\rho' n}$ for all sufficiently large n, almost surely. The finitely many earlier terms are absorbed into $C_{\rho'}(\omega)$, and the tail bound

$$\sum_{j=n}^{\infty} 2^{-\rho' j} = \frac{2^{-\rho' n}}{1 - 2^{-\rho'}}$$

proves (144.118). □

Lemma 144.50 reduces the convergence of a renormalized random model to two explicit analytic estimates: a uniform moment bound for the model seminorm and a dyadic Cauchy moment estimate for successive cutoff scales. In a BPHZ construction these estimates are proved tree by tree, after the forest character has removed every divergent local subtree and after the remaining kernels have been bounded in the parabolic model norm. The lemma above proves what those estimates imply; it does not prove the estimates for the concrete dynamic Φ^4_3 trees.

Lemma 144.51 (Finite Wiener-chaos moment bound)**.** *Let H be a real Hilbert space and let $\xi : H \to L^2(\Omega)$ be an isonormal Gaussian process:*

$$\mathbb{E}\xi(h) = 0, \qquad \mathbb{E}\xi(h)\xi(k) = \langle h, k\rangle_H.$$

For $q \ge 1$ and $f \in H^{\odot q}$, let $I_q(f)$ denote the q-fold Wick multiple integral, normalized by

$$I_q(h^{\otimes q}) = \|h\|_H^q H_q\left(\frac{\xi(h)}{\|h\|_H}\right) \quad (h \neq 0),$$

where H_q is the probabilists' Hermite polynomial and extension to $H^{\odot q}$ is by polarization and completion. Set $I_0(c) = c$. Then

$$\mathbb{E}\, I_q(f) I_{q'}(g) = \delta_{qq'} q!\, \langle f, g\rangle_{H^{\otimes q}}. \tag{144.119}$$

For every integer $m \geq 1$ there is a finite constant $C_{q,m}$, depending only on q and m, such that

$$\mathbb{E}|I_q(f)|^{2m} \leq C_{q,m}\|f\|_{H^{\otimes q}}^{2m}. \tag{144.120}$$

Proof. The exponential definition of Wick ordering is

$$:\exp \xi(h) := \exp\left(\xi(h) - \frac{1}{2}\|h\|_H^2\right).$$

For $s, t \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}\big(:\exp t\xi(h)::\exp s\xi(k):\big) &= \exp\left(\frac{1}{2}\|th+sk\|_H^2 - \frac{1}{2}t^2\|h\|_H^2 - \frac{1}{2}s^2\|k\|_H^2\right) \\ &= \exp(ts\langle h,k\rangle_H).\end{aligned}$$

Comparing the coefficient of $t^q s^{q'}$ gives

$$\mathbb{E}\, I_q(h^{\otimes q}) I_{q'}(k^{\otimes q'}) = \delta_{qq'} q!\, \langle h,k\rangle_H^q.$$

Polarization gives (144.119) for finite sums of pure symmetric tensors, and completion gives the general case.

For the even moment, first take f to be a finite-rank symmetric tensor. The needed pairing formula is derived from the same generating function. For vectors $h_1, \ldots, h_N \in H$,

$$\mathbb{E}\prod_{\ell=1}^{N} :\exp t_\ell \xi(h_\ell) := \exp\left(\sum_{1\leq i<j\leq N} t_i t_j \langle h_i, h_j\rangle_H\right).$$

Expanding the right-hand side and comparing coefficients gives pairings only between distinct Wick blocks. By multilinearity, the product of $2m$ finite-rank Wick monomials therefore obeys

$$\mathbb{E}\, I_q(f)^{2m} = \sum_{\pi\in\mathcal{P}_{q,2m}} C_\pi(f, \ldots, f).$$

Here $\mathcal{P}_{q,2m}$ is the finite set of pairings of the $\{1, \ldots, 2m\} \times \{1, \ldots, q\}$ labels that pair no two labels in the same block $\{\ell\} \times \{1, \ldots, q\}$, and C_π is the complete tensor contraction prescribed by π. Every C_π is a composition of tensor permutations and Hilbert contractions. Tensor permutations are unitary and each Hilbert contraction has operator norm ≤ 1, hence

$$|C_\pi(f, \ldots, f)| \leq \|f\|_{H^{\otimes q}}^{2m}.$$

Therefore

$$\mathbb{E}|I_q(f)|^{2m} = \mathbb{E}I_q(f)^{2m} \leq |\mathcal{P}_{q,2m}|\, \|f\|_{H^{\otimes q}}^{2m}.$$

Take $C_{q,m} = |\mathcal{P}_{q,2m}|$ for finite-rank tensors. Approximate a general $f \in H^{\odot q}$ by finite-rank tensors in $H^{\otimes q}$. The isometry (144.119) gives convergence in $L^2(\Omega)$, hence a subsequence converges almost surely. Fatou's lemma applied to that subsequence gives the same $2m$-moment bound for the limit. This proves (144.120). $\square$

Kernel bounds imply stochastic coordinate moments. Let $\mathcal{A}$ be an index set and let $\sigma : \mathcal{A} \to (0, \infty)$ be a deterministic scale weight. Suppose that random variables $Y_n(a)$, $a \in \mathcal{A}$, have finite chaos expansions

$$Y_n(a) = \sum_{q=0}^{Q} I_q(f_{n,a,q}), \qquad f_{n,a,q} \in H^{\odot q},$$

and that, for constants $B_q, \widetilde{B}_q < \infty$,

$$\|f_{n,a,q}\|_{H^{\otimes q}} \le B_q \sigma(a), \qquad \|f_{n+1,a,q} - f_{n,a,q}\|_{H^{\otimes q}} \le \widetilde{B}_q 2^{-\rho n} \sigma(a).$$

Then, for every integer $m \ge 1$, there are finite constants $M_m, \widetilde{M}_m$, depending only on m, Q, and the displayed kernel constants, such that

$$\mathbb{E}|Y_n(a)|^{2m} \le M_m \sigma(a)^{2m}, \qquad \mathbb{E}|Y_{n+1}(a) - Y_n(a)|^{2m} \le \widetilde{M}_m 2^{-2m\rho n} \sigma(a)^{2m}.$$

Use the elementary convexity inequality

$$\left| \sum_{q=0}^{Q} x_q \right|^{2m} \le (Q+1)^{2m-1} \sum_{q=0}^{Q} |x_q|^{2m}.$$

Apply Lemma 144.51 to each chaos component. The deterministic $q = 0$ component is included with $C_{0,m} = 1$. This gives

$$M_m = (Q+1)^{2m-1} \sum_{q=0}^{Q} C_{q,m} B_q^{2m}$$

and the same expression with B_q replaced by $\widetilde{B}_q$ for $\widetilde{M}_m$. The factor $2^{-2m\rho n}$ comes from raising the dyadic kernel-difference estimate to the $2m$-th power.

The kernel-to-moment calculation above is the stochastic part of a BPHZ model estimate. A concrete tree calculation must still prove the deterministic kernel bounds for every tested model coordinate and must upgrade the resulting pointwise coordinate estimates to the full model seminorm by spatial and scale regularity estimates. Once those deterministic steps are done, Lemma 144.50 turns the coordinate estimates into convergence of the random model.

Proposition 144.52 (Dual-norm finite-chaos estimate)**.** *Let E be a separable Banach space with dual E'. For a Hilbert space H, write $H^{\odot q} \widehat{\otimes}_\pi E'$ for the projective tensor completion of the algebraic tensor product, with norm*

$$\|F\|_\pi = \inf \left\{ \sum_{j=1}^{N} \|f_j\|_{H^{\otimes q}} \|\ell_j\|_{E'} : F = \sum_{j=1}^{N} f_j \otimes \ell_j \right\},$$

where the infimum is over all finite representations; the completed norm is obtained by taking limits. If $F \in H^{\odot q} \widehat{\otimes}_\pi E'$, define the E'-valued chaos integral $I_q(F)$ by completion from

$$I_q\left(\sum_{j=1}^{N} f_j \otimes \ell_j \right) = \sum_{j=1}^{N} I_q(f_j) \ell_j.$$

Then, for every integer $m \geq 1$,

$$\| \|I_q(F)\|_{E'} \|_{L^{2m}(\Omega)} \leq C_{q,m} \|F\|_\pi, \tag{144.121}$$

where $C_{q,m}$ is any constant satisfying (144.120) and $C_{q,m} \geq 1$.

Consequently, suppose that E'-valued random variables $Y_n(a)$, $a \in \mathcal{A}$, have finite chaos expansions

$$Y_n(a) = \sum_{q=0}^{Q_0} I_q(F_{n,a,q}), \qquad F_{n,a,q} \in H^{\odot q} \widehat{\otimes}_\pi E',$$

and that deterministic weights $\sigma(a)$ and $\omega(a,b)$ obey

$$\|F_{n,a,q}\|_\pi \leq B_q \sigma(a), \qquad \|F_{n,b,q} - F_{n,a,q}\|_\pi \leq B_q^{\mathrm{ed}} \omega(a,b).$$

Then

$$\mathbb{E}\|Y_n(a)\|_{E'}^{2m} \leq M_m \sigma(a)^{2m}, \qquad \mathbb{E}\|Y_n(b) - Y_n(a)\|_{E'}^{2m} \leq M_m^{\mathrm{ed}} \omega(a,b)^{2m},$$

with

$$M_m = \left(\sum_{q=0}^{Q_0} C_{q,m} B_q \right)^{2m}, \qquad M_m^{\mathrm{ed}} = \left(\sum_{q=0}^{Q_0} C_{q,m} B_q^{\mathrm{ed}} \right)^{2m}.$$

The same conclusion holds with an additional factor $2^{-\rho n}$ in the projective-norm bounds for cutoff differences; the $2m$-th moments then carry the factor $2^{-2m\rho n}$.

Proof. First take $F = \sum_{j=1}^N f_j \otimes \ell_j$. The triangle inequality in E' and Minkowski's inequality in $L^{2m}(\Omega)$ give

$$\begin{aligned} \| \|I_q(F)\|_{E'} \|_{L^{2m}} &\leq \sum_{j=1}^N \| |I_q(f_j)| \, \|\ell_j\|_{E'} \|_{L^{2m}} \\ &\leq C_{q,m} \sum_{j=1}^N \|f_j\|_{H^{\otimes q}} \|\ell_j\|_{E'}. \end{aligned}$$

Taking the infimum over finite representations proves (144.121) on the algebraic tensor product. If $F^{(N)} \to F$ in the projective norm, the same inequality applied to $F^{(N)} - F^{(M)}$ shows that $I_q(F^{(N)})$ is Cauchy in $L^{2m}(\Omega; E')$; its limit is the stated completion, and the bound passes to the limit.

For a finite chaos expansion, apply (144.121) to each q and use Minkowski:

$$\| \|Y_n(a)\|_{E'} \|_{L^{2m}} \leq \sum_{q=0}^{Q_0} C_{q,m} \|F_{n,a,q}\|_\pi \leq \left(\sum_{q=0}^{Q_0} C_{q,m} B_q \right) \sigma(a).$$

Raising to the $2m$-th power gives the first estimate. Replacing $F_{n,a,q}$ by $F_{n,b,q} - F_{n,a,q}$ gives the edge estimate. Replacing that difference by a cutoff difference with projective-norm bound $2^{-\rho n} B_q \sigma(a)$, or by an edge cutoff difference with $2^{-\rho n} B_q^{\mathrm{ed}} \omega(a,b)$, gives the stated cutoff factor. $\square$

Apply Proposition 144.52 with

$$E = E_r := \{\varphi \in C^r(\mathbb{R}^{1+3}) : \operatorname{supp}\varphi \subset B_{\mathfrak{s}}(0,1)\}, \qquad \|\varphi\|_{E_r} = \|\varphi\|_{C^r}.$$

Then E_r' is the space in which a tested model coordinate acts before the supremum over $\varphi \in \mathcal{B}_r$ is taken. For a fixed physical parameter a, an E_r'-valued chaos expansion of the functional

$$\varphi \longmapsto \delta^{-|\tau|}\langle \Pi_z^{(n)}\tau, \varphi_z^\delta\rangle$$

has dual norm exactly

$$\sup_{\varphi\in\mathcal{B}_r} \delta^{-|\tau|}|\langle \Pi_z^{(n)}\tau, \varphi_z^\delta\rangle|.$$

Thus the correct route to the $Q_{n,\tau}^{\#}$ estimates is not to net the test-function ball. It is to prove projective tensor bounds for the E_r'-valued BPHZ kernels and then apply the dual-norm chaos estimate above.

Lemma 144.53 (Projective kernel estimate for dual-norm bounds)**.** *Let E be a separable Banach space, let H be a Hilbert space, and let (S,μ) be a σ-finite measure space. Fix $q \geq 0$. Suppose that $f : S \to H^{\odot q}$ is strongly measurable and that $\ell : S \to E'$ is strongly measurable, with*

$$\int_S \|f(s)\|_{H^{\otimes q}}\|\ell(s)\|_{E'}\,\mathrm{d}\mu(s) < \infty.$$

Then the Bochner integral

$$F = \int_S f(s)\otimes \ell(s)\,\mathrm{d}\mu(s)$$

defines an element of $H^{\odot q}\widehat{\otimes}_\pi E'$, and

$$\|F\|_\pi \leq \int_S \|f(s)\|_{H^{\otimes q}}\|\ell(s)\|_{E'}\,\mathrm{d}\mu(s). \tag{144.122}$$

Let X be a parameter set. Suppose $f_x : S \to H^{\odot q}$ and $\ell_x : S \to E'$ satisfy the preceding hypotheses for every $x \in X$. If deterministic functions $\Sigma : X \to [0,\infty)$ and $\Omega : X\times X \to [0,\infty)$ obey

$$\int_S \|f_x(s)\|\,\|\ell_x(s)\|\,\mathrm{d}\mu(s) \leq B\Sigma(x) \tag{144.123}$$

and

$$\begin{aligned}\int_S &\|f_x(s)-f_y(s)\|\,\|\ell_x(s)\|\,\mathrm{d}\mu(s)\\ &+\int_S \|f_y(s)\|\,\|\ell_x(s)-\ell_y(s)\|\,\mathrm{d}\mu(s) \leq B_{\mathrm{ed}}\Omega(x,y),\end{aligned} \tag{144.124}$$

then

$$\|F_x\|_\pi \leq B\Sigma(x), \qquad \|F_x - F_y\|_\pi \leq B_{\mathrm{ed}}\Omega(x,y),$$

where

$$F_x = \int_S f_x(s)\otimes \ell_x(s)\,\mathrm{d}\mu(s).$$

Proof. The algebraic tensor norm is a crossnorm:

$$\|u \otimes \lambda\|_\pi = \|u\|_{H^{\otimes q}} \|\lambda\|_{E'}.$$

Hence

$$\|f(s) \otimes \ell(s)\|_\pi = \|f(s)\|_{H^{\otimes q}} \|\ell(s)\|_{E'}.$$

The assumed integrability therefore makes $s \mapsto f(s) \otimes \ell(s)$ Bochner integrable as a map into $H^{\odot q} \widehat{\otimes}_\pi E'$. Approximate this map by simple functions G_N in $L^1(S; H^{\odot q} \widehat{\otimes}_\pi E')$. For a simple function $G_N = \sum_a \mathbf{1}_{S_a} u_a \otimes \lambda_a$,

$$\left\| \int_S G_N \, \mathrm{d}\mu \right\|_\pi \le \sum_a \mu(S_a) \|u_a\| \, \|\lambda_a\| = \int_S \|G_N(s)\|_\pi \, \mathrm{d}\mu(s).$$

Passing to the L^1 limit gives (144.122).

The base estimate is (144.122) applied to (f_x, ℓ_x). For the edge estimate write

$$F_x - F_y = \int_S (f_x(s) - f_y(s)) \otimes \ell_x(s) \, \mathrm{d}\mu(s) + \int_S f_y(s) \otimes (\ell_x(s) - \ell_y(s)) \, \mathrm{d}\mu(s).$$

Applying (144.122) to the two terms and using (144.124) proves the displayed bound. □

The criterion has a concrete meaning for tested model kernels. With

$$E_r = \{\varphi \in C^r(\mathbb{R}^{1+3}) : \operatorname{supp} \varphi \subset B_{\mathfrak{s}}(0,1)\},$$

the point-evaluation functional

$$\delta_u(\varphi) = \varphi(u)$$

has $\|\delta_u\|_{E_r'} \le 1$. If $0 < \theta \le \min\{r, 1\}$, then

$$\|\delta_u - \delta_v\|_{E_r'} = \sup_{\|\varphi\|_{C^r} \le 1} |\varphi(u) - \varphi(v)| \le C_r \|u - v\|_{\mathfrak{s}}^\theta. \tag{144.125}$$

The same bound holds for a finite sum of derivative evaluations $\partial^a \delta_u$ whenever $|a|_{\mathfrak{s}} + \theta \le r$, after replacing C_r by a constant depending on r and a. Thus a BPHZ-tested chaos kernel represented as

$$F_{x,q} = \int_S f_{x,q}(s) \otimes \delta_{u_x(s)} \, \mathrm{d}\mu(s)$$

is controlled by

$$\|F_{x,q}\|_\pi \le \int_S \|f_{x,q}(s)\|_{H^{\otimes q}} \, \mathrm{d}\mu(s),$$

and its parameter increments are controlled by the two visible mechanisms: variation of the Hilbert kernel $f_{x,q}$, and variation of the evaluation point $u_x(s)$ measured by (144.125). These are the deterministic estimates that the tree-by-tree BPHZ analysis must prove for the six Π-coordinates in the strict negative sector.

Lemma 144.54 (Dyadic parabolic convolution bound)**.** *Let* $(\mathbb{R}^D, \mathfrak{s})$ *be a parabolically scaled space with homogeneous dimension*

$$Q = |\mathfrak{s}|.$$

Assume the parabolic balls satisfy

$$|B_{\mathfrak{s}}(0,r)| \le V r^Q, \qquad 0<r\le 1.$$

Let $a,b>0$. For $i,j\ge 0$, let K_i, L_j be measurable kernels such that, for constants A, C_K, C_L,

$$\operatorname{supp} K_i, \operatorname{supp} L_i \subset B_{\mathfrak{s}}(0, A2^{-i}),$$

and

$$|K_i(x)| \le C_K 2^{(Q-a)i}, \qquad |L_i(x)| \le C_L 2^{(Q-b)i}.$$

Define the k-th output scale of the convolution by

$$M_k = \sum_{\min(i,j)=k} K_i * L_j.$$

Then the series defining M_k converges absolutely in L^∞,

$$\operatorname{supp} M_k \subset B_{\mathfrak{s}}(0, 2A2^{-k}),$$

and

$$\|M_k\|_{L^\infty} \le C\, C_K C_L\, 2^{(Q-a-b)k}, \tag{144.126}$$

where

$$C = VA^Q \left(\frac{1}{1-2^{-b}} + \frac{2^{-a}}{1-2^{-a}} \right).$$

Consequently

$$\|M_k\|_{L^1} \le C'\, C_K C_L\, 2^{-(a+b)k} \tag{144.127}$$

for a constant C' depending only on A, V, a, b, Q.

Proof. For fixed i,j, the convolution is supported in

$$B_{\mathfrak{s}}(0, A2^{-i}) + B_{\mathfrak{s}}(0, A2^{-j}) \subset B_{\mathfrak{s}}(0, 2A2^{-\min(i,j)}).$$

The pointwise bound follows by estimating the volume of the smaller support:

$$\begin{aligned} |(K_i * L_j)(x)| &\le C_K C_L 2^{(Q-a)i+(Q-b)j} \left| B_{\mathfrak{s}}\big(0, A2^{-\max(i,j)}\big) \right| \\ &\le VA^Q C_K C_L 2^{(Q-a)i+(Q-b)j-Q\max(i,j)}. \end{aligned}$$

Fix k. First take $i=k$ and $j=k+r$, $r\ge 0$. The preceding estimate gives

$$|(K_k * L_{k+r})(x)| \le VA^Q C_K C_L 2^{(Q-a-b)k} 2^{-br}.$$

Summing $r\ge 0$ gives the first geometric factor in C. Next take $j=k$ and $i=k+r$, $r\ge 1$; the term $r=0$ has already been counted. Then

$$|(K_{k+r} * L_k)(x)| \le VA^Q C_K C_L 2^{(Q-a-b)k} 2^{-ar}.$$

Summing $r\ge 1$ gives the second geometric factor. This proves absolute convergence in L^∞ and (144.126).

The support statement for M_k follows from the support statement for each summand with $\min(i,j)=k$. Multiplying the L^∞ estimate by

$$|B_{\mathfrak{s}}(0, 2A2^{-k})| \le V(2A)^Q 2^{-Qk}$$

gives (144.127). ∎

The exponents in Lemma 144.54 are the deterministic scale arithmetic behind many model-coordinate estimates. A kernel of order a has scale-i size $2^{(Q-a)i}$ and L^1-size 2^{-ai}. After convolution with an order-b kernel, the output scale $k=\min(i,j)$ has size $2^{(Q-a-b)k}$, which is exactly the size of an order-$(a+b)$ kernel. In a BPHZ proof, Taylor subtraction at divergent subgraphs supplies additional powers of the small scale; the proposition above is the unsubtracted convolution estimate to which those extra powers are attached.

Proposition 144.55 (Gaussian Wick-coordinate scale estimates)**.** *Let $Q=5$ and let $\mathfrak{s}=(2,1,1,1)$. Fix $0<\theta\le 1$, $r>\theta$, and $\kappa>0$. For*

$$x=(z,\delta),\qquad 0<\delta\le 1,$$

let

$$\mathsf{D}_\delta(t,x_1,x_2,x_3)=(\delta^2 t,\delta x_1,\delta x_2,\delta x_3),$$

write

$$\psi_x^\varphi(u)=\delta^{-Q}\varphi(\mathsf{D}_\delta^{-1}(u-z)),\qquad \varphi\in E_r,\quad \|\varphi\|_{C^r}\le 1. \tag{144.128}$$

First let ξ be white noise on $\mathbb{R}^{1+3}$, normalized by

$$\mathbb{E}\,\xi(f)\xi(g)=\int f(u)g(u)\,\mathrm{d}u.$$

Then

$$\left\|\delta^{Q/2+\kappa}\xi(\psi_x^\varphi)\right\|_{L^{2m}(\Omega)}\le C_{m,r}\delta^\kappa \tag{144.129}$$

uniformly in x and φ. If $x=(z,\delta)$ and $y=(z',\delta')$ have $\delta,\delta'\in[2^{-M-1},2^{-M}]$ and

$$\eta=2^M\|z-z'\|_{\mathfrak{s}}+2^M|\delta-\delta'|\le 1,$$

then

$$\left\|\delta^{Q/2+\kappa}\xi(\psi_x^\varphi)-\delta'^{Q/2+\kappa}\xi(\psi_y^\varphi)\right\|_{L^{2m}(\Omega)}\le C_{m,r,\kappa}\,2^{-\kappa M}\eta^\theta. \tag{144.130}$$

Next let X be a centered Gaussian field with covariance $C(u,v)$ such that, on the one-neighborhood of the compact region under consideration,

$$|C(u,v)|\le A\|u-v\|_{\mathfrak{s}}^{-1}. \tag{144.131}$$

For $k=1,2,3$, let $:X^k:$ be the Wick power defined by this covariance. Then

$$\left\|\delta^{k/2+k\kappa}\langle :X^k:,\psi_x^\varphi\rangle\right\|_{L^{2m}(\Omega)}\le C_{k,m,r,A}\delta^{k\kappa}. \tag{144.132}$$

If x,y lie in the same dyadic scale annulus as above, then

$$\left\|\delta^{k/2+k\kappa}\langle :X^k:,\psi_x^\varphi\rangle-\delta'^{k/2+k\kappa}\langle :X^k:,\psi_y^\varphi\rangle\right\|_{L^{2m}(\Omega)}\le C_{k,m,r,A,\kappa}\,2^{-k\kappa M}\eta^\theta. \tag{144.133}$$

Consequently, for the Gaussian coordinates

$$\Xi,\quad X,\quad X^2,\quad X^3$$

in the strict negative sector, the physical scale slack appearing in the negative-sector coordinate-control consequence is

$$\sigma_\Xi=\kappa,\qquad \sigma_{X^k}=k\kappa\quad(k=1,2,3),$$

before the remaining finite-net entropy is paid by taking a sufficiently high moment.

Proof. For white noise,

$$\|\psi_x^\varphi\|_{L^2}^2 = \delta^{-2Q} \int \left|\varphi(\mathsf{D}_\delta^{-1}(u-z))\right|^2 \,\mathrm{d}u = \delta^{-Q}\|\varphi\|_{L^2}^2 \le C_r \delta^{-Q}.$$

The Gaussian moment formula gives

$$\|\xi(\psi_x^\varphi)\|_{L^{2m}} \le C_m \|\psi_x^\varphi\|_{L^2},$$

and multiplication by $\delta^{Q/2+\kappa}$ proves (144.129).

For the edge estimate, compare the two rescaled test functions on the common parabolic ball of radius $C2^{-M}$ containing their supports. The C^r-bound and $r > \theta$ give

$$\left\|\delta^{Q/2+\kappa}\psi_x^\varphi - \delta'^{Q/2+\kappa}\psi_y^\varphi\right\|_{L^2} \le C_{r,\kappa} 2^{-\kappa M}\eta^\theta. \tag{144.134}$$

Indeed, in the scale 2^{-M} variables the normalized amplitudes are bounded, the displacement of the arguments is $\le C\eta$, and the prefactor $\delta^{Q/2+\kappa}\delta^{-Q}$ has L^2-size comparable to δ^κ. Applying the Gaussian moment formula to the difference proves (144.130).

For the Wick powers, Wick covariance gives, for every test function ψ,

$$\mathbb{E}\left|\langle : X^k :, \psi\rangle\right|^2 = k! \iint \psi(u)\psi(v) C(u,v)^k \,\mathrm{d}u\,\mathrm{d}v. \tag{144.135}$$

Because $k \le 3 < Q = 5$, the kernel $\|u-v\|_{\mathfrak{s}}^{-k}$ is locally integrable in parabolic dimension Q. Using (144.131) and changing variables

$$u = z + \mathsf{D}_\delta a, \qquad v = z + \mathsf{D}_\delta b,$$

one obtains

$$\begin{aligned}\iint |\psi_x^\varphi(u)|\,|\psi_x^\varphi(v)||C(u,v)|^k \,\mathrm{d}u\,\mathrm{d}v &\le A^k\delta^{-k} \iint_{B_{\mathfrak{s}}(0,1)^2} \frac{|\varphi(a)|\,|\varphi(b)|}{\|a-b\|_{\mathfrak{s}}^k}\,\mathrm{d}a\,\mathrm{d}b \\ &\le C_{k,r,A}\delta^{-k}.\end{aligned} \tag{144.136}$$

Thus the $L^2(\Omega)$-norm of $\delta^{k/2+k\kappa}\langle : X^k :, \psi_x^\varphi\rangle$ is bounded by $C\delta^{k\kappa}$. Lemma 144.51 upgrades this k-th chaos L^2-bound to L^{2m}, proving (144.132).

For the edge estimate, apply the same covariance identity to

$$\Psi_{x,y}^\varphi = \delta^{k/2+k\kappa}\psi_x^\varphi - \delta'^{k/2+k\kappa}\psi_y^\varphi.$$

In scale 2^{-M} variables, the two supports lie in a fixed parabolic ball, and the C^r-bound gives

$$|\Psi_{x,y}^\varphi(u)| \le C_{r,k,\kappa}\, 2^{M(Q-k/2-k\kappa)}\eta^\theta. \tag{144.137}$$

Changing variables as above and using local integrability of $\|a-b\|_{\mathfrak{s}}^{-k}$ gives

$$\mathbb{E}\left|\delta^{k/2+k\kappa}\langle : X^k :, \psi_x^\varphi\rangle - \delta'^{k/2+k\kappa}\langle : X^k :, \psi_y^\varphi\rangle\right|^2 \le C\, 2^{-2k\kappa M}\eta^{2\theta}.$$

The finite-chaos moment bound again upgrades this to L^{2m}, proving (144.133). The displayed values of σ_Ξ and σ_{X^k} are exactly the powers of $\delta \sim 2^{-M}$ in (144.129) and (144.132). $\square$

Proposition 144.56 (Dual-norm Gaussian Wick-coordinate estimates)**.** *Let the hypotheses and notation of Proposition 144.55 hold. Assume in addition that*

$$0 < \theta < r - \frac{Q}{2}.$$

Choose s with

$$\frac{Q}{2} + \theta < s < r.$$

Let E_r be the test-function space

$$E_r = \{\varphi \in C^r(\mathbb{R}^{1+3}) : \operatorname{supp}\varphi \subset B_{\mathfrak{s}}(0,1)\}, \qquad \|\varphi\|_{E_r} = \|\varphi\|_{C^r}.$$

Define the E_r'-valued random functionals

$$\mathcal{W}_x(\varphi) = \delta^{Q/2+\kappa}\xi(\psi_x^\varphi)$$

and, for $k = 1, 2, 3$,

$$\mathcal{X}_{k,x}(\varphi) = \delta^{k/2+k\kappa}\langle : X^k :, \psi_x^\varphi\rangle.$$

Then, for every integer $m \geq 1$,

$$\big\|\|\mathcal{W}_x\|_{E_r'}\big\|_{L^{2m}(\Omega)} \leq C\delta^\kappa, \qquad \big\|\|\mathcal{X}_{k,x}\|_{E_r'}\big\|_{L^{2m}(\Omega)} \leq C\delta^{k\kappa}. \tag{144.138}$$

If $x = (z, \delta)$, $y = (z', \delta')$, $\delta, \delta' \in [2^{-M-1}, 2^{-M}]$, and

$$\eta = 2^M\|z - z'\|_{\mathfrak{s}} + 2^M|\delta - \delta'| \leq 1,$$

then

$$\big\|\|\mathcal{W}_x - \mathcal{W}_y\|_{E_r'}\big\|_{L^{2m}(\Omega)} \leq C2^{-\kappa M}\eta^\theta, \tag{144.139}$$

and

$$\big\|\|\mathcal{X}_{k,x} - \mathcal{X}_{k,y}\|_{E_r'}\big\|_{L^{2m}(\Omega)} \leq C2^{-k\kappa M}\eta^\theta. \tag{144.140}$$

The constants depend on the compact region, the chosen wavelet system, r, s, θ, κ, m, and on the covariance constant A for the Wick powers, but not on x, y, δ, δ', or the particular test function.

Proof. Use a compactly supported parabolic wavelet system with regularity larger than r, as in the reconstruction proof. Write e_λ, $\lambda \in \Lambda_j$, for the L^2-normalized wavelets at unit-domain scale 2^{-j}, and include the finitely many scaling functions in a separate set Λ_{-1}. The system has finite overlap,

$$|\Lambda_j| \leq C2^{Qj} \qquad (j \geq 0),$$

and the coefficient estimate

$$\sum_{\lambda\in\Lambda_{-1}} |\langle\varphi, e_\lambda\rangle|^2 + \sum_{j\geq 0}\sum_{\lambda\in\Lambda_j} 2^{2sj}|\langle\varphi, e_\lambda\rangle|^2 \leq C\|\varphi\|_{C^r}^2 \tag{144.141}$$

for $0 < s < r$. For a distribution T supported in a fixed enlargement of the unit ball set

$$\|T\|_{-s,*}^2 = \sum_{\lambda\in\Lambda_{-1}} |\langle T, e_\lambda\rangle|^2 + \sum_{j\geq 0}\sum_{\lambda\in\Lambda_j} 2^{-2sj}|\langle T, e_\lambda\rangle|^2.$$

By Cauchy–Schwarz and (144.141),

$$\|T\|_{E'_r} = \sup_{\|\varphi\|_{C^r} \le 1} |T(\varphi)| \le C\|T\|_{-s,*}. \tag{144.142}$$

It is therefore enough to estimate the weighted wavelet coefficient norm.

For white noise, put $T = \mathcal{W}_x$. For every L^2-normalized unit-domain wavelet e_λ,

$$\mathbb{E}|\langle T, e_\lambda\rangle|^2 = \delta^{2\kappa}.$$

Consequently

$$\mathbb{E}\|T\|^2_{-s,*} \le C\delta^{2\kappa}\left(1 + \sum_{j\ge 0} 2^{Qj}2^{-2sj}\right) \le C\delta^{2\kappa}, \tag{144.143}$$

because $s > Q/2$.

For the edge estimate, the coefficient tested against a scale-j wavelet is the white-noise pairing with the difference

$$\delta^{Q/2+\kappa}\psi_x^{e_\lambda} - \delta'^{Q/2+\kappa}\psi_y^{e_\lambda}.$$

Rescale the common physical ball of radius $C2^{-M}$ to unit size. The unit-domain wavelet has C^θ-size $O(2^{j(Q/2+\theta)})$, while its L^2-size is one; translation and dilation of the outer parameter by η therefore give

$$\left\|\delta^{Q/2+\kappa}\psi_x^{e_\lambda} - \delta'^{Q/2+\kappa}\psi_y^{e_\lambda}\right\|_{L^2} \le C2^{-\kappa M}\min\{1, 2^{j\theta}\eta^\theta\}. \tag{144.144}$$

The finite set Λ_{-1} obeys the same bound with $j = 0$ after changing the constant. Hence

$$\mathbb{E}\|\mathcal{W}_x - \mathcal{W}_y\|^2_{-s,*} \le C2^{-2\kappa M}\sum_{j\ge 0} 2^{(Q-2s)j}\min\{1, 2^{j\theta}\eta^\theta\}^2.$$

Let J be chosen so that $2^{-J-1} < \eta \le 2^{-J}$. The part $j \le J$ is bounded by

$$C\eta^{2\theta}\sum_{j\le J} 2^{(Q-2s+2\theta)j} \le C\eta^{2\theta},$$

because $Q - 2s + 2\theta < 0$. The part $j > J$ is bounded by

$$C\sum_{j>J} 2^{(Q-2s)j} \le C2^{-(2s-Q)J} \le C\eta^{2\theta},$$

again because $2s - Q > 2\theta$. This proves the $L^2(\Omega)$ version of (144.139).

For the Wick powers, let $T = \mathcal{X}_{k,x}$. Wick covariance gives

$$\mathbb{E}|\langle T, e_\lambda\rangle|^2 \le C\delta^{2k\kappa}2^{(k-Q)j}, \qquad \lambda \in \Lambda_j, \quad j \ge 0. \tag{144.145}$$

Indeed, after the outer scale $u = z + \mathsf{D}_\delta a$ is removed, the coefficient is tested by the unit-domain wavelet e_λ, and

$$\iint |e_\lambda(a)|\,|e_\lambda(b)|\|a-b\|_{\mathfrak{s}}^{-k}\,\mathrm{d}a\,\mathrm{d}b \le C2^{(k-Q)j}.$$

The right-hand side follows from the support diameter $O(2^{-j})$, the L^∞-size $O(2^{Qj/2})$, and the local integrability of $\|a-b\|_{\mathfrak{s}}^{-k}$. Summing the weighted coefficients gives

$$\mathbb{E}\|T\|_{-s,*}^2 \le C\delta^{2k\kappa}\left(1+\sum_{j\ge 0} 2^{Qj}2^{-2sj}2^{(k-Q)j}\right) \le C\delta^{2k\kappa}, \tag{144.146}$$

because $s > Q/2 \ge k/2$ for $k \le 3$.

For x, y in the same dyadic annulus, apply the covariance identity to the difference of the two normalized test functions. The same rescaling used in (144.144) gives

$$\mathbb{E}\,|\langle \mathcal{X}_{k,x} - \mathcal{X}_{k,y}, e_\lambda\rangle|^2 \le C2^{-2k\kappa M}2^{(k-Q)j}\min\{1, 2^{j\theta}\eta^\theta\}^2. \tag{144.147}$$

Therefore

$$\mathbb{E}\|\mathcal{X}_{k,x} - \mathcal{X}_{k,y}\|_{-s,*}^2 \le C2^{-2k\kappa M}\sum_{j\ge 0} 2^{(k-2s)j}\min\{1, 2^{j\theta}\eta^\theta\}^2.$$

Since $k \le 3 < Q$ and $2s - Q > 2\theta$, the same split at $2^{-J} \simeq \eta$ bounds this sum by $C\eta^{2\theta}$. This proves the $L^2(\Omega)$ versions of (144.138), (144.139), and (144.140).

The random variables $\mathcal{W}_x$ and $\mathcal{W}_x - \mathcal{W}_y$ lie in the first Wiener chaos with values in the Hilbert space defined by $\|\cdot\|_{-s,*}$. The variables $\mathcal{X}_{k,x}$ and $\mathcal{X}_{k,x} - \mathcal{X}_{k,y}$ lie in the k-th Wiener chaos with values in the same Hilbert space. We use the Hilbert-valued form of the finite-chaos moment bound. For a finite wavelet truncation F in the q-th homogeneous chaos, let P_t be the Ornstein–Uhlenbeck semigroup. Then $P_tF = e^{-qt}F$, and Jensen's inequality gives

$$e^{-qt}\|F\|_{-s,*} = \|P_tF\|_{-s,*} \le P_t(\|F\|_{-s,*}).$$

The scalar hypercontractive estimate $\|P_tg\|_{L^{2m}} \le \|g\|_{L^2}$ when $e^{2t} \ge 2m-1$ therefore gives

$$\|\|F\|_{-s,*}\|_{L^{2m}} \le (2m-1)^{q/2}\|\|F\|_{-s,*}\|_{L^2}.$$

Passing from finite wavelet truncations to the limit by monotone convergence for the squared coefficient norm gives

$$\|\|F\|_{-s,*}\|_{L^{2m}} \le C_{k,m}\,\|\|F\|_{-s,*}\|_{L^2}$$

for every one of these Hilbert-valued chaos variables. Combining this with (144.142) proves the claimed L^{2m} dual-norm estimates. □

144.6.3 Multiscale Sector Estimates and Local Subtractions

Parabolic Taylor-subtraction gain. The local scale gain used below is an elementary L^1 estimate, so it is recorded as a calculation. Let K_i be a dyadic kernel on $(\mathbb{R}^D, \mathfrak{s})$ such that

$$\operatorname{supp} K_i \subset B_{\mathfrak{s}}(0, A2^{-i}), \qquad \|K_i\|_{L^1} \le C_K 2^{-ai},$$

with $a > 0$. Let $r > 0$, let F be a function on $B_{\mathfrak{s}}(0, A2^{-i})$, and let P be a polynomial such that

$$|F(h) - P(h)| \le H\|h\|_{\mathfrak{s}}^r \tag{144.148}$$

for all h in that ball. Then

$$\left|\int K_i(h)\big(F(h)-P(h)\big)\,\mathrm{d}h\right| \le C_K H A^r 2^{-(a+r)i}. \tag{144.149}$$

On the support of K_i, the remainder hypothesis gives

$$|F(h)-P(h)| \le HA^r 2^{-ri}.$$

Therefore

$$\begin{aligned}\left|\int K_i(h)\big(F(h)-P(h)\big)\,\mathrm{d}h\right| &\le HA^r 2^{-ri}\int |K_i(h)|\,\mathrm{d}h\\ &\le C_K H A^r 2^{-(a+r)i}.\end{aligned}$$

The extra factor 2^{-ri} is the gain produced by subtracting the local Taylor polynomial P; without the subtraction the L^1-size of the kernel would only give the scale 2^{-ai}.

The estimate (144.149) deliberately separates this elementary local gain from the harder BPHZ bookkeeping: one must still specify which polynomial is subtracted for each divergent subgraph, prove that the forest formula does not double count overlapping local parts, and combine the gain with the dyadic convolution bound in Lemma 144.54.

One-loop relative-scale estimate. The preceding local gain becomes a usable dyadic estimate only after the two relative-scale sectors are separated. Let $(\mathbb{R}^D,\mathfrak{s})$ be a parabolically scaled space with homogeneous dimension $Q=|\mathfrak{s}|$, and let $a,b,\theta>0$. Suppose that dyadic kernels K_i, L_j, $i,j\ge 0$, obey

$$\operatorname{supp} K_i \subset B_{\mathfrak{s}}(0,A2^{-i}), \qquad |K_i(h)| \le C_K 2^{(Q-a)i}, \qquad \|K_i\|_{L^1} \le C_K' 2^{-ai},$$

and

$$\operatorname{supp} L_j \subset B_{\mathfrak{s}}(0,A2^{-j}), \qquad |L_j(h)| \le C_L 2^{(Q-b)j}.$$

Assume also that, whenever $j\le i$ and $\|h\|_{\mathfrak{s}} \le A2^{-i}$,

$$|L_j(h)-L_j(0)| \le C_L' 2^{(Q-b)j}2^{-\theta(i-j)}. \tag{144.150}$$

Define

$$T_{i,j}^{\mathrm{loc}} = \int K_i(h)\big(L_j(h)-L_j(0)\big)\,\mathrm{d}h \quad (j\le i),$$

and

$$T_{i,j}^{\mathrm{off}} = \int K_i(h)L_j(h)\,\mathrm{d}h \quad (j>i).$$

Then the following two sector estimates hold:

$$|T_{i,j}^{\mathrm{loc}}| \le C_K' C_L'\, 2^{(Q-a-b)j}2^{-(a+\theta)(i-j)} \quad (j\le i), \tag{144.151}$$

and

$$|T_{i,j}^{\mathrm{off}}| \le C_K C_L V_A\, 2^{(Q-a-b)i}2^{-b(j-i)} \quad (j>i), \tag{144.152}$$

where

$$V_A = \sup_{0<\rho\le A} |B_{\mathfrak{s}}(0,\rho)|\,\rho^{-Q}.$$

These bounds are the explicit scale bookkeeping. In the local sector $j \le i$, the Holder-scale bound (144.150) applies on the support of K_i, and therefore

$$\begin{aligned}|T^{\mathrm{loc}}_{i,j}| &\le \|K_i\|_{L^1} C'_L 2^{(Q-b)j} 2^{-\theta(i-j)} \\ &\le C'_K C'_L 2^{-ai} 2^{(Q-b)j} 2^{-\theta(i-j)} \\ &= C'_K C'_L 2^{(Q-a-b)j} 2^{-(a+\theta)(i-j)}.\end{aligned}$$

This is (144.151). In the off-diagonal sector $j > i$, the integrand is supported where L_j is supported, so

$$\begin{aligned}|T^{\mathrm{off}}_{i,j}| &\le C_K C_L 2^{(Q-a)i+(Q-b)j} |B_{\mathfrak{s}}(0, A2^{-j})| \\ &\le C_K C_L V_A 2^{(Q-a)i+(Q-b)j-Qj} \\ &= C_K C_L V_A 2^{(Q-a-b)i} 2^{-b(j-i)}.\end{aligned}$$

This is (144.152). The estimate isolates the relative-scale summability gain; the nontrivial BPHZ content is the subsequent forest organization of all local subtractions, not this one-loop norm calculation.

For the one-loop dynamic Φ^4_3 contraction, $Q = 5$ and the heat and covariance kernels both have order 2, so $a = b = 2$. The base factor $2^{(Q-a-b)k} = 2^k$, where $k = \min(i, j)$, is the local mass-coordinate divergence. The estimates above show that the two relative-scale sectors have strictly positive gap exponents: $2 + \theta$ in the subtracted sector $j \le i$ and 2 in the off-diagonal sector $j > i$. Thus the relative scales are summable once the local coordinate has been separated.

Proposition 144.57 (Two-loop sunset gaps and nested forest cancellation)**.** *Let $(\mathbb{R}^D, \mathfrak{s})$ be a parabolically scaled space with homogeneous dimension $Q = |\mathfrak{s}|$, and assume*

$$|B_{\mathfrak{s}}(0, r)| \le V r^Q, \qquad 0 < r \le A.$$

Let $a, b, c \in (0, Q)$ satisfy

$$a + b + c = 2Q.$$

For $i, j, \ell \ge 0$, let K_i, L_j, M_ℓ be measurable kernels such that

$$\operatorname{supp} K_i \subset B_{\mathfrak{s}}(0, A2^{-i}), \quad \operatorname{supp} L_j \subset B_{\mathfrak{s}}(0, A2^{-j}), \quad \operatorname{supp} M_\ell \subset B_{\mathfrak{s}}(0, A2^{-\ell}),$$

and

$$|K_i(h)| \le C_K 2^{(Q-a)i}, \quad |L_j(h)| \le C_L 2^{(Q-b)j}, \quad |M_\ell(h)| \le C_M 2^{(Q-c)\ell}.$$

Define

$$S_{i,j,\ell} = \int K_i(h) L_j(h) M_\ell(h)\, \mathrm{d}h, \qquad n_* = \max(i, j, \ell),$$

and the non-negative relative gaps

$$r_K = n_* - i, \qquad r_L = n_* - j, \qquad r_M = n_* - \ell.$$

Then

$$|S_{i,j,\ell}| \le V A^Q C_K C_L C_M 2^{-(Q-a)r_K - (Q-b)r_L - (Q-c)r_M}. \tag{144.153}$$

Consequently, if

$$S_N = \sum_{\max(i,j,\ell) \le N} S_{i,j,\ell},$$

then

$$|S_{N+1} - S_N| \le VA^Q C_K C_L C_M \sum_{\nu \in \{K,L,M\}} \prod_{\mu \ne \nu} \frac{1}{1 - 2^{-\eta_\mu}}, \tag{144.154}$$

where

$$\eta_K = Q - a, \qquad \eta_L = Q - b, \qquad \eta_M = Q - c.$$

For dynamic Φ^4_3, $Q = 5$. The heat kernel in the retarded edge has order $a = 2$. Each stationary covariance edge has order 4: it is the time convolution of two order-2 heat edges, so Lemma 144.54 gives the order $2 + 2 = 4$ dyadic bound. Thus $b = c = 4$ and

$$\eta_K = 3, \qquad \eta_L = \eta_M = 1,$$

and the non-nested two-loop local sunset sectors obey

$$|S_{i,j,\ell}| \le C\, 2^{-3r_K - r_L - r_M}, \qquad |S_{N+1} - S_N| \le \frac{60}{7} C, \tag{144.155}$$

with $C = VA^5 C_K C_L C_M$.

In the same finite cutoff chart, write

$$G = \sum_{\ell \ge 0} G_\ell, \qquad C_{1,\ell} = G_\ell(0).$$

At fixed scales i, j, ℓ, the nested local coefficient in

$$X(z)^2 \int K_i(z - w) X(w)^3 \,\mathrm{d}w$$

is

$$6C_{1,\ell} \int K_i(h) G_j(h) \,\mathrm{d}h.$$

After the recursive one-loop forest subtraction

$$X(w)^3 \longmapsto X(w)^3 - 3C_{1,\ell} X(w),$$

the corresponding local coefficient from the subtracted term is

$$-6C_{1,\ell} \int K_i(h) G_j(h) \,\mathrm{d}h.$$

Therefore the renormalized nested two-loop local coefficient is zero for each triple (i, j, ℓ). The only new two-loop local coordinate left by the forest operation is the non-nested sunset coefficient bounded above.

Proof. The product $K_i L_j M_\ell$ is supported in the smallest of the three supports. Hence

$$\mathrm{supp}(K_i L_j M_\ell) \subset B_{\mathfrak{s}}(0, A 2^{-n_*}).$$

Therefore

$$\begin{aligned} |S_{i,j,\ell}| &\le C_K C_L C_M 2^{(Q-a)i + (Q-b)j + (Q-c)\ell} |B_{\mathfrak{s}}(0, A2^{-n_*})| \\ &\le VA^Q C_K C_L C_M 2^{(Q-a)i + (Q-b)j + (Q-c)\ell - Qn_*}. \end{aligned}$$

Substitute

$$i = n_* - r_K, \qquad j = n_* - r_L, \qquad \ell = n_* - r_M.$$

The coefficient of n_* is

$$(Q-a) + (Q-b) + (Q-c) - Q = 2Q - (a+b+c) = 0,$$

so the remaining factor is exactly the right-hand side of (144.153).

The increment $S_{N+1} - S_N$ is the sum over triples with $\max(i,j,\ell) = N+1$. Equivalently, at least one of the gaps r_K, r_L, r_M is zero. This shell is contained in the union of the three sets $r_\nu = 0$. Summing the geometric majorant from (144.153) over that union gives (144.154). For $(Q,a,b,c) = (5,2,4,4)$, this shell constant is

$$\frac{1}{(1-2^{-1})^2} + 2\,\frac{1}{(1-2^{-3})(1-2^{-1})} = \frac{60}{7}.$$

It remains to verify the nested cancellation. In

$$X(z)^2 X(w)^3$$

leave one of the three fields at w uncontracted, contract the two fields at z with the two remaining fields at w, and then Taylor-expand the uncontracted field at w to its local value at z. This is the non-nested sunset and gives $3 \cdot 2 = 6$ bridge pairings. For the nested coefficient, instead leave one of the two fields at z uncontracted, contract the other field at z with one of the three fields at w, and contract the two remaining fields at w with each other. This gives

$$2 \cdot 3 = 6$$

pairings and the coefficient

$$6C_{1,\ell} \int K_i(h) G_j(h)\,\mathrm{d}h.$$

The inner one-loop forest subtraction contributes

$$-3C_{1,\ell} X(w).$$

Multiplying by $X(z)^2$, leaving one of the two fields at z uncontracted, and contracting the other one with $X(w)$ gives

$$-3 \cdot 2\, C_{1,\ell} \int K_i(h) G_j(h)\,\mathrm{d}h.$$

This is the negative of the nested coefficient above. The cancellation is scale-wise because $C_{1,\ell} = G_\ell(0)$ was kept at fixed tadpole scale ℓ throughout the calculation. □

Theorem 144.58 (Multiscale sector summability for coordinate kernels)**.** *Let $R \geq 1$ and let $\eta_1, \ldots, \eta_R$ be positive real numbers. Here $\mathbb{N} = \{0,1,2,\ldots\}$. Set*

$$\eta_* = \min_{1 \leq h \leq R} \eta_h, \qquad G = \prod_{h=1}^{R} \frac{1}{1 - 2^{-\eta_h}},$$

and

$$\widetilde{G} = \sum_{h=1}^{R} \prod_{\substack{1 \leq k \leq R \\ k \neq h}} \frac{1}{1 - 2^{-\eta_k}}.$$

Let $\mathcal{H}$ be a real Hilbert space. Let $\mathcal{A}$ be an index set with a positive scale weight

$$\sigma : \mathcal{A} \to (0, \infty).$$

For $q = 0, \dots, Q$, $a \in \mathcal{A}$, and $\mathbf{r} = (r_1, \dots, r_R) \in \mathbb{N}^R$, let

$$f_{\mathbf{r},a,q} \in \mathcal{H}^{\odot q}$$

with $\mathcal{H}^{\odot 0} = \mathbb{R}$. Suppose that, for constants $B_q < \infty$,

$$\|f_{\mathbf{r},a,q}\|_{\mathcal{H}^{\otimes q}} \le B_q \sigma(a) \prod_{h=1}^{R} 2^{-\eta_h r_h} \tag{144.156}$$

for all displayed indices. Define the cutoff kernel

$$f_{n,a,q} = \sum_{\substack{\mathbf{r} \in \mathbb{N}^R \\ \max_h r_h \le n}} f_{\mathbf{r},a,q}.$$

Then the sum defining $f_{n,a,q}$ is absolutely convergent in $\mathcal{H}^{\otimes q}$, and

$$\|f_{n,a,q}\|_{\mathcal{H}^{\otimes q}} \le B_q G \sigma(a). \tag{144.157}$$

Moreover,

$$\|f_{n+1,a,q} - f_{n,a,q}\|_{\mathcal{H}^{\otimes q}} \le B_q \widetilde{G}\, 2^{-\eta_*(n+1)} \sigma(a) \le B_q \widetilde{G}\, 2^{-\eta_* n} \sigma(a). \tag{144.158}$$

Consequently, for any isonormal Gaussian process $\xi : \mathcal{H} \to L^2(\Omega)$, the random variables

$$Y_n(a) = \sum_{q=0}^{Q} I_q(f_{n,a,q})$$

satisfy the hypotheses of the kernel-to-moment calculation above with the replacement

$$B_q \mapsto B_q G, \qquad \widetilde{B}_q \mapsto B_q \widetilde{G}, \qquad \rho = \eta_*.$$

Proof. The triangle inequality and (144.156) give

$$\begin{aligned} \|f_{n,a,q}\|_{\mathcal{H}^{\otimes q}} &\le B_q \sigma(a) \sum_{\substack{\mathbf{r} \in \mathbb{N}^R \\ \max_h r_h \le n}} \prod_{h=1}^{R} 2^{-\eta_h r_h} \\ &\le B_q \sigma(a) \prod_{h=1}^{R} \sum_{r_h=0}^{\infty} 2^{-\eta_h r_h} = B_q G \sigma(a). \end{aligned}$$

This proves absolute convergence and (144.157).

The increment $f_{n+1,a,q} - f_{n,a,q}$ is the sum of those scale-gap assignments for which at least one r_h is equal to $n+1$ and no r_h is larger than $n+1$. This set is contained in the union, over

$h = 1, \dots, R$, of the sets with $r_h = n+1$. Therefore

$$\begin{aligned}\|f_{n+1,a,q} - f_{n,a,q}\|_{\mathcal{H}^{\otimes q}} &\le B_q \sigma(a) \sum_{h=1}^{R} 2^{-\eta_h(n+1)} \prod_{\substack{1\le k\le R\\ k\ne h}} \sum_{r_k=0}^{\infty} 2^{-\eta_k r_k} \\ &\le B_q \sigma(a)\, 2^{-\eta_*(n+1)} \sum_{h=1}^{R} \prod_{\substack{1\le k\le R\\ k\ne h}} \frac{1}{1-2^{-\eta_k}}.\end{aligned}$$

This is (144.158). The final assertion is obtained by comparing (144.157) and (144.158) with the deterministic kernel hypotheses of the kernel-to-moment calculation above. ☐

Theorem 144.58 is the multiscale summation step behind a BPHZ coordinate estimate. In an actual tree calculation, a sector decomposition rewrites relative scale assignments as non-negative gap variables r_h. Taylor subtraction and the forest formula must then prove positive deficits η_h for every gap variable. Once that has been done, the theorem above converts those deficits into uniform kernel bounds and dyadic cutoff-increment bounds for the finite chaos coordinates. The difficult part left for dynamic Φ^4_3 is therefore the derivation of (144.156) for each negative tree and each coordinate entering the model seminorm.

Theorem 144.59 (Dyadic-net upgrade from coordinates to a supremum)**.** *Let $(\mathcal{X}, d_{\mathcal{X}})$ be a compact metric space. Assume that there are finite sets $\mathcal{X}_\ell \subset \mathcal{X}$, $\ell \ge 0$, and maps*

$$\pi_\ell : \mathcal{X} \to \mathcal{X}_\ell$$

such that $\pi_\ell x \to x$ for every $x \in \mathcal{X}$, and such that the edge sets

$$\mathcal{E}_\ell = \{(\pi_\ell x, \pi_{\ell+1} x) : x \in \mathcal{X}\}$$

satisfy

$$|\mathcal{X}_0| \le C_0, \qquad |\mathcal{E}_\ell| \le C_1 2^{d\ell} \quad (\ell \ge 0)$$

for constants $C_0, C_1, d > 0$. Let $Y : \mathcal{X} \to L^p(\Omega)$ be a random field with almost surely continuous sample paths, where $p \ge 1$. Assume that there are constants $B_0, B, \varepsilon > 0$ such that

$$\mathbb{E}|Y(u)|^p \le B_0^p \quad (u \in \mathcal{X}_0),$$

and

$$\mathbb{E}|Y(v) - Y(u)|^p \le B^p 2^{-(d+\varepsilon)\ell} \quad ((u,v) \in \mathcal{E}_\ell). \tag{144.159}$$

Then

$$\Big\| \sup_{x\in\mathcal{X}} |Y(x)| \Big\|_{L^p(\Omega)} \le C_0^{1/p} B_0 + \frac{C_1^{1/p} B}{1 - 2^{-\varepsilon/p}}. \tag{144.160}$$

If $Y = Y_n$ depends on a cutoff index n and both B_0 and B in the two hypotheses are replaced by $2^{-\rho n} B_0$ and $2^{-\rho n} B$, then the right-hand side of (144.160) is multiplied by $2^{-\rho n}$.

Proof. For every $x \in \mathcal{X}$, continuity of the sample path and $\pi_\ell x \to x$ give

$$Y(x) = Y(\pi_0 x) + \sum_{\ell \ge 0} \big(Y(\pi_{\ell+1} x) - Y(\pi_\ell x) \big)$$

on the event of continuity. Taking the supremum over x gives the pointwise bound

$$\sup_{x\in\mathcal{X}} |Y(x)| \le \max_{u\in\mathcal{X}_0} |Y(u)| + \sum_{\ell\ge 0} \max_{(u,v)\in\mathcal{E}_\ell} |Y(v) - Y(u)|.$$

The L^p triangle inequality gives

$$\left\|\sup_{x\in\mathcal{X}} |Y(x)|\right\|_{L^p} \le \left\|\max_{u\in\mathcal{X}_0} |Y(u)|\right\|_{L^p} + \sum_{\ell\ge 0} \left\|\max_{(u,v)\in\mathcal{E}_\ell} |Y(v) - Y(u)|\right\|_{L^p}.$$

For the base net,

$$\left\|\max_{u\in\mathcal{X}_0} |Y(u)|\right\|_{L^p}^p \le \sum_{u\in\mathcal{X}_0} \mathbb{E}|Y(u)|^p \le C_0 B_0^p.$$

For each edge level, (144.159) gives

$$\begin{aligned}\left\|\max_{(u,v)\in\mathcal{E}_\ell} |Y(v) - Y(u)|\right\|_{L^p}^p &\le \sum_{(u,v)\in\mathcal{E}_\ell} \mathbb{E}|Y(v) - Y(u)|^p \\ &\le C_1 B^p 2^{-\varepsilon\ell}.\end{aligned}$$

Taking p-th roots and summing the geometric series gives (144.160). The cutoff-dependent statement uses the base and edge estimates with the additional factor $2^{-\rho n p}$. This factor is independent of the dyadic level ℓ, so it passes through the union bound, the p-th root, and the geometric series as the factor $2^{-\rho n}$ in the final supremum estimate. $\square$

Theorem 144.59 is the analytic bridge from pointwise model-coordinate estimates to compact model-seminorm estimates. In an application, $\mathcal{X}$ is a finite product of compact base-point, scale, test-function, and normalized-symbol parameter spaces after a finite regularity sector has been fixed. The integer d records the metric entropy of these parameters. The missing BPHZ estimates must therefore include not only a moment bound for each fixed coordinate, but also the increment bound (144.159) in the parameters that enter the model seminorm.

Corollary 144.60 (Scale-summed dyadic-net coordinate upgrade)**.** *Let $p \ge 1$. For every $m \in \mathbb{N} = \{0, 1, 2, \ldots\}$, let $(\mathcal{X}_m, d_m)$ be a compact metric space with finite nets $\mathcal{X}_{m,\ell}$, $\ell \ge 0$, projections*

$$\pi_{m,\ell} : \mathcal{X}_m \to \mathcal{X}_{m,\ell},$$

and edge sets

$$\mathcal{E}_{m,\ell} = \{(\pi_{m,\ell}x, \pi_{m,\ell+1}x) : x \in \mathcal{X}_m\}.$$

Assume

$$|\mathcal{X}_{m,0}| \le C_0 2^{Dm}, \qquad |\mathcal{E}_{m,\ell}| \le C_1 2^{Dm} 2^{d\ell} \quad (m, \ell \ge 0), \tag{144.161}$$

for constants $C_0, C_1, D, d \ge 0$. Let $Y_m : \mathcal{X}_m \to L^p(\Omega)$ have almost surely continuous sample paths. Suppose there are constants $B_0, B, \sigma, \varepsilon > 0$ such that

$$\mathbb{E}|Y_m(u)|^p \le B_0^p 2^{-p\sigma m} \quad (u \in \mathcal{X}_{m,0})$$

and

$$\mathbb{E}|Y_m(v) - Y_m(u)|^p \le B^p 2^{-p\sigma m} 2^{-(d+\varepsilon)\ell} \quad ((u, v) \in \mathcal{E}_{m,\ell}).$$

If

$$\sigma > \frac{D}{p}, \tag{144.162}$$

then

$$\left\| \sup_{m\geq 0} \sup_{x\in\mathcal{X}_m} |Y_m(x)| \right\|_{L^p(\Omega)} \leq \frac{C_0^{1/p} B_0 + C_1^{1/p} B(1-2^{-\varepsilon/p})^{-1}}{1-2^{-(\sigma-D/p)}}. \tag{144.163}$$

If $Y_m = Y_{n,m}$ depends on a cutoff index n and the two moment hypotheses are multiplied by $2^{-p\rho n}$, with $\rho > 0$, then the right-hand side of (144.163) is multiplied by $2^{-\rho n}$.

Proof. Fix m. Apply Theorem 144.59 to the compact space $\mathcal{X}_m$. The base entropy and edge entropy in (144.161) replace C_0 and C_1 by $C_0 2^{Dm}$ and $C_1 2^{Dm}$, while the coordinate moment constants are multiplied by $2^{-\sigma m}$. Hence

$$\left\| \sup_{x\in\mathcal{X}_m} |Y_m(x)| \right\|_{L^p} \leq \left(C_0^{1/p} B_0 + \frac{C_1^{1/p} B}{1-2^{-\varepsilon/p}} \right) 2^{-(\sigma-D/p)m}.$$

The L^p triangle inequality gives

$$\left\| \sup_{m\geq 0} \sup_{x\in\mathcal{X}_m} |Y_m(x)| \right\|_{L^p} \leq \sum_{m\geq 0} \left\| \sup_{x\in\mathcal{X}_m} |Y_m(x)| \right\|_{L^p}.$$

The geometric series in m converges exactly under (144.162) and gives (144.163). The cutoff-dependent statement follows by carrying the factor $2^{-\rho n}$ through the same inequalities. □

Corollary 144.61 (Shell-separated cutoff bridge)**.** *Use the entropy data of Corollary 144.60. Thus, for every m, the compact space $\mathcal{X}_m$ is equipped with finite nets $\mathcal{X}_{m,\ell}$, edge sets $\mathcal{E}_{m,\ell}$, and entropy constants C_0, C_1, D, d satisfying (144.161). Let*

$$Y_{n,m} : \mathcal{X}_m \to L^p(\Omega), \qquad n, m \geq 0,$$

have almost surely continuous sample paths. Suppose that, for constants $B_0, B, \sigma, \varepsilon, \rho > 0$,

$$\mathbb{E}|Y_{n,m}(u)|^p \leq B_0^p 2^{-p\sigma m} 2^{-p\rho(n-m)_+} \quad (u \in \mathcal{X}_{m,0}) \tag{144.164}$$

and

$$\mathbb{E}|Y_{n,m}(v) - Y_{n,m}(u)|^p \leq B^p 2^{-p\sigma m} 2^{-p\rho(n-m)_+} 2^{-(d+\varepsilon)\ell} \quad ((u,v) \in \mathcal{E}_{m,\ell}). \tag{144.165}$$

Assume

$$a := \sigma - \frac{D}{p} > 0$$

and choose

$$0 < \rho_* < \min\{\rho, a\}.$$

Define

$$H_{\rho_*,\rho,a} := \sup_{n\geq 0} 2^{\rho_* n} \sum_{m\geq 0} 2^{-am} 2^{-\rho(n-m)_+}. \tag{144.166}$$

Then $H_{\rho_*,\rho,a} < \infty$, and

$$\left\| \sup_{m\geq 0} \sup_{x\in\mathcal{X}_m} |Y_{n,m}(x)| \right\|_{L^p(\Omega)} \leq H_{\rho_*,\rho,a} \left(C_0^{1/p} B_0 + C_1^{1/p} B(1-2^{-\varepsilon/p})^{-1} \right) 2^{-\rho_* n}. \tag{144.167}$$

Proof. For fixed n, m, apply Theorem 144.59 to $\mathcal{X}_m$ with the base and edge constants multiplied by

$$2^{-\sigma m} 2^{-\rho(n-m)_+}.$$

The entropy factor 2^{Dm} is taken to the p-th root, so

$$\left\| \sup_{x \in \mathcal{X}_m} |Y_{n,m}(x)| \right\|_{L^p} \le \left(C_0^{1/p} B_0 + C_1^{1/p} B (1 - 2^{-\varepsilon/p})^{-1} \right) 2^{-am} 2^{-\rho(n-m)_+}.$$

The L^p triangle inequality over m gives the same constant times

$$S_n := \sum_{m \ge 0} 2^{-am} 2^{-\rho(n-m)_+}.$$

It remains to prove that $S_n \le H_{\rho_*,\rho,a} 2^{-\rho_* n}$ with a finite $H_{\rho_*,\rho,a}$.

Split the sum at $m = n$. For $0 \le m \le n$,

$$2^{-am} 2^{-\rho(n-m)} = 2^{-\rho_* n} 2^{-(a-\rho_*)m} 2^{-(\rho-\rho_*)(n-m)}.$$

Both $a - \rho_*$ and $\rho - \rho_*$ are positive. Therefore

$$\sum_{m=0}^{n} 2^{-am} 2^{-\rho(n-m)} \le 2^{-\rho_* n} \frac{1}{1 - 2^{-(a-\rho_*)}} \frac{1}{1 - 2^{-(\rho-\rho_*)}}.$$

For $m > n$,

$$\sum_{m > n} 2^{-am} = 2^{-a(n+1)} (1 - 2^{-a})^{-1} \le 2^{-\rho_* n} \frac{2^{-a}}{1 - 2^{-a}},$$

because $a > \rho_*$. These two estimates prove

$$H_{\rho_*,\rho,a} \le \frac{1}{(1 - 2^{-(a-\rho_*)})(1 - 2^{-(\rho-\rho_*)})} + \frac{2^{-a}}{1 - 2^{-a}} < \infty,$$

and hence (144.167). $\square$

Corollary 144.62 (Projective shell-separated coordinate criterion)**.** *Use the entropy data of Corollary 144.60. Let $p = 2\mathfrak{m}$ be an even integer. Let E be a separable Banach space, let H be a Hilbert space, and fix a finite chaos order Q_0. For $n, m \ge 0$, let*

$$Y_{n,m}(x) = \sum_{q=0}^{Q_0} I_q(F_{n,m,x,q}), \qquad F_{n,m,x,q} \in H^{\odot q} \widehat{\otimes}_\pi E', \qquad x \in \mathcal{X}_m,$$

be E'-valued random fields whose sample paths are almost surely continuous in the E'-norm. Suppose that there are constants

$$B_q,\ B_q^{\mathrm{ed}},\ \sigma,\ \varepsilon,\ \rho > 0$$

such that, for every base-net point $u \in \mathcal{X}_{m,0}$,

$$\|F_{n,m,u,q}\|_\pi \le B_q\, 2^{-\sigma m} 2^{-\rho(n-m)_+}, \tag{144.168}$$

and, for every edge $(u, v) \in \mathcal{E}_{m,\ell}$,

$$\|F_{n,m,v,q} - F_{n,m,u,q}\|_\pi \le B_q^{\mathrm{ed}}\, 2^{-\sigma m} 2^{-\rho(n-m)_+} 2^{-(d+\varepsilon)\ell/p}. \tag{144.169}$$

Set

$$B_\pi = \sum_{q=0}^{Q_0} C_{q,\mathfrak{m}} B_q, \qquad B_\pi^{\mathrm{ed}} = \sum_{q=0}^{Q_0} C_{q,\mathfrak{m}} B_q^{\mathrm{ed}},$$

where $C_{q,\mathfrak{m}}$ is the finite-chaos constant in (144.120). If

$$a := \sigma - \frac{D}{p} > 0, \qquad 0 < \rho_* < \min\{\rho, a\},$$

then

$$\begin{aligned} &\left\| \sup_{m\geq 0} \sup_{x\in\mathcal{X}_m} \|Y_{n,m}(x)\|_{E'} \right\|_{L^p(\Omega)} \\ &\qquad \leq H_{\rho_*,\rho,a} \left(C_0^{1/p} B_\pi + C_1^{1/p} B_\pi^{\mathrm{ed}} (1 - 2^{-\varepsilon/p})^{-1} \right) 2^{-\rho_* n}. \end{aligned} \tag{144.170}$$

In particular, an E_r'-valued BPHZ kernel estimate satisfying (144.168) and (144.169) gives the scale-summed cutoff-Cauchy estimate needed for the corresponding dualized model coordinate. If the kernels are written as Bochner integrals

$$F_{n,m,x,q} = \int_S f_{n,m,x,q}(s) \otimes \ell_{m,x}(s)\,\mathrm{d}\mu(s),$$

the displayed projective hypotheses are exactly the integral estimates obtained from Lemma 144.53.

Proof. Apply Proposition 144.52 to the base-net variables. The triangle inequality in $L^p(\Omega; E')$ gives

$$\big\|\|Y_{n,m}(u)\|_{E'}\big\|_{L^p} \leq B_\pi\, 2^{-\sigma m} 2^{-\rho(n-m)_+}.$$

For an edge $(u,v) \in \mathcal{E}_{m,\ell}$, apply the same proposition to

$$Y_{n,m}(v) - Y_{n,m}(u) = \sum_{q=0}^{Q_0} I_q(F_{n,m,v,q} - F_{n,m,u,q}).$$

Using (144.169) gives

$$\big\|\|Y_{n,m}(v) - Y_{n,m}(u)\|_{E'}\big\|_{L^p} \leq B_\pi^{\mathrm{ed}} 2^{-\sigma m} 2^{-\rho(n-m)_+} 2^{-(d+\varepsilon)\ell/p}.$$

Since

$$\big| \|Y_{n,m}(v)\|_{E'} - \|Y_{n,m}(u)\|_{E'} \big| \leq \|Y_{n,m}(v) - Y_{n,m}(u)\|_{E'},$$

the scalar random field

$$x \longmapsto \|Y_{n,m}(x)\|_{E'}$$

satisfies the base and edge moment hypotheses of Corollary 144.61 with the constants $B_0 = B_\pi$ and $B = B_\pi^{\mathrm{ed}}$. That corollary gives (144.170). The final sentence is the projective-kernel bound (144.122) applied to the displayed Bochner integral, and similarly to its edge difference. $\square$

The parameter m in Corollary 144.60 is the physical testing scale, $\delta \simeq 2^{-m}$, in a model seminorm. The exponent D is the entropy cost of resolving base points and test functions at that scale. The exponent σ is the regularity slack between the assigned homogeneity and the moment regularity proved for the stochastic coordinate. Thus the condition $\sigma > D/p$ is the precise high-moment requirement: by choosing p large enough, a positive regularity slack beats the scale entropy. Corollary 144.61 is the corresponding cutoff-Cauchy bridge for estimates whose ultraviolet gain is visible only after comparing the cutoff shell with the physical testing scale. It is the deterministic step used by scalar shell bounds such as (144.239): the factor $2^{-\rho(n-m)_+}$ becomes a summable $2^{-\rho_* n}$ cutoff rate whenever the physical-scale slack $a = \sigma - D/p$ and the shell gain ρ are both positive.

Hypothesis 144.63 (Finite-sector coordinate data)**.** Fix K, γ, $A_{<\gamma}$, and r as in Lemma 144.50. Let

$$Z_n = (\Pi^{(n)}, \Gamma^{(n)}), \qquad n \geq 1,$$

be random models on the finite sector, and define

$$N_n = 1 + \|\Pi^{(n)}\|_{K;\gamma} + \|\Gamma^{(n)}\|_{K;\gamma}.$$

Let $d_{K;\gamma}$ be the corresponding model distance. Let

$$I = \{1, \ldots, M\}$$

be a finite coordinate index set. For every $i \in I$, let $(\mathcal{X}_i, d_{\mathcal{X}_i})$ be a compact metric space with finite nets $\mathcal{X}_{i,\ell}$, projections

$$\pi_{i,\ell} : \mathcal{X}_i \to \mathcal{X}_{i,\ell},$$

with $\pi_{i,\ell} x \to x$ for every $x \in \mathcal{X}_i$, and edge sets

$$\mathcal{E}_{i,\ell} = \{(\pi_{i,\ell} x, \pi_{i,\ell+1} x) : x \in \mathcal{X}_i\}$$

such that

$$|\mathcal{X}_{i,0}| \leq C_{i,0}, \qquad |\mathcal{E}_{i,\ell}| \leq C_{i,1} 2^{D_i \ell}$$

for constants $C_{i,0}, C_{i,1}, D_i > 0$. Fix $p \geq 1$, $\rho > 0$, and $\varepsilon_i > 0$ for $i \in I$. Suppose there are real random fields

$$U_{n,i}, V_{n,i} : \mathcal{X}_i \to L^p(\Omega), \qquad n \geq 1, \quad i \in I,$$

with almost surely continuous sample paths, and a deterministic constant $A_{\mathrm{ctrl}} < \infty$, such that, almost surely,

$$N_n \leq A_{\mathrm{ctrl}} \left(1 + \sum_{i \in I} \sup_{x \in \mathcal{X}_i} |U_{n,i}(x)| \right), \tag{144.171}$$

and

$$d_{K;\gamma}(Z_{n+1}, Z_n) \leq A_{\mathrm{ctrl}} \sum_{i \in I} \sup_{x \in \mathcal{X}_i} |V_{n,i}(x)|. \tag{144.172}$$

For the uniform bound, assume

$$\mathbb{E}|U_{n,i}(u)|^p \leq B_{i,0}^p \quad (n \geq 1,\ u \in \mathcal{X}_{i,0}),$$

and

$$\mathbb{E}|U_{n,i}(v) - U_{n,i}(u)|^p \leq B_i^p 2^{-(D_i + \varepsilon_i)\ell} \quad (n \geq 1,\ (u,v) \in \mathcal{E}_{i,\ell}).$$

For the cutoff increment, assume

$$\mathbb{E}|V_{n,i}(u)|^p \leq \widetilde{B}_{i,0}^p 2^{-p\rho n} \quad (u \in \mathcal{X}_{i,0}),$$

and

$$\mathbb{E}|V_{n,i}(v) - V_{n,i}(u)|^p \leq \widetilde{B}_i^p 2^{-p\rho n} 2^{-(D_i + \varepsilon_i)\ell} \quad ((u,v) \in \mathcal{E}_{i,\ell}).$$

Define

$$S_i = C_{i,0}^{1/p} B_{i,0} + \frac{C_{i,1}^{1/p} B_i}{1 - 2^{-\varepsilon_i/p}}, \qquad \widetilde{S}_i = C_{i,0}^{1/p} \widetilde{B}_{i,0} + \frac{C_{i,1}^{1/p} \widetilde{B}_i}{1 - 2^{-\varepsilon_i/p}}.$$

144.6.4 Finite-Sector Model Convergence from Coordinate Estimates

Under Hypothesis 144.63,

$$\sup_{n\geq 1} \|N_n\|_{L^p(\Omega)} \leq A_{\mathrm{ctrl}} \left(1 + \sum_{i\in I} S_i\right),$$

and

$$\|d_{K;\gamma}(Z_{n+1}, Z_n)\|_{L^p(\Omega)} \leq A_{\mathrm{ctrl}} \left(\sum_{i\in I} \widetilde{S}_i\right) 2^{-\rho n}. \tag{144.173}$$

Consequently, Z_n satisfies (144.116) with

$$C_N = A_{\mathrm{ctrl}}^p \left(1 + \sum_{i\in I} S_i\right)^p, \qquad C_D = A_{\mathrm{ctrl}}^p \left(\sum_{i\in I} \widetilde{S}_i\right)^p.$$

Thus the limiting random model and the convergence estimates of Lemma 144.50 follow.

Apply Theorem 144.59 to each $U_{n,i}$. The hypotheses for $U_{n,i}$ give

$$\left\|\sup_{x\in\mathcal{X}_i} |U_{n,i}(x)|\right\|_{L^p} \leq S_i \quad (n \geq 1).$$

The L^p triangle inequality and (144.171) therefore give

$$\|N_n\|_{L^p} \leq A_{\mathrm{ctrl}} \left(1 + \sum_{i\in I} S_i\right).$$

Taking the supremum over n gives the uniform model-seminorm moment bound.

Apply the cutoff-dependent part of Theorem 144.59 to $V_{n,i}$. The base and edge bounds for $V_{n,i}$ give

$$\left\|\sup_{x\in\mathcal{X}_i} |V_{n,i}(x)|\right\|_{L^p} \leq \widetilde{S}_i 2^{-\rho n}.$$

Together with (144.172), this gives (144.173). Raising the two displayed L^p bounds to the p-th power gives the constants C_N and C_D in (144.116). The final assertion is exactly Lemma 144.50.

The finite-sector coordinate-control consequence fixes the logical boundary of the remaining BPHZ task. A finite-dimensional reexpansion component or a finite basis component of T_α is included by enlarging the finite index set I. The model construction for dynamic Φ^4_3 must therefore provide, for each negative tree and each coordinate chart entering the model seminorm, the two estimates appearing in the theorem: the uniform coordinate moment estimate and the dyadic cutoff increment estimate with enough excess exponent ε_i to dominate the metric entropy of the coordinate parameters.

For dynamic Φ^4_3, the symbols are generated by the noise Ξ, the abstract integration map $\mathcal{I}$, multiplication, and polynomials in the spacetime coordinates. Negative-homogeneity trees encode precisely the subdivergent stochastic diagrams. Renormalization acts on these trees before the reconstruction theorem is applied.

Lemma 144.64 (Finite negative-sector coordinates for dynamic Φ^4_3)**.** *Let $0 < \kappa < 1/14$, and set*

$$X = \mathcal{I}(\Xi), \qquad Y = \mathcal{I}(X^3), \qquad |X| = -\frac{1}{2} - \kappa, \qquad |Y| = \frac{1}{2} - 3\kappa.$$

Put

$$\gamma_- = \frac{1}{2} - 7\kappa > 0.$$

Among the monomials $X^a Y^b$ with $a + b \le 3$, the non-polynomial symbols of homogeneity $< \gamma_-$ are

$$X, \quad X^2, \quad X^3, \quad XY, \quad X^2 Y. \tag{144.174}$$

Together with the noise symbol Ξ and the constant symbol $\mathbf{1}$, these form the finite sector

$$\mathcal{T}_- = \{\mathbf{1}, \Xi, X, X^2, X^3, XY, X^2Y\}.$$

Their homogeneities are

τ	$\lvert\tau\rvert$
$\mathbf{1}$	0
Ξ	$-\frac{5}{2} - \kappa$
X	$-\frac{1}{2} - \kappa$
X^2	$-1 - 2\kappa$
X^3	$-\frac{3}{2} - 3\kappa$
XY	-4κ
X^2Y	$-\frac{1}{2} - 5\kappa.$

The excluded cubic monomial XY^2 has homogeneity

$$|XY^2| = \gamma_-,$$

and is therefore not part of the strict sector $T_{<\gamma_-}$.

Let $Z_n = (\Pi^{(n)}, \Gamma^{(n)})$ be cutoff models whose restriction to $\mathcal{T}_-$ is multiplicative and whose reexpansion of Y, modulo terms of homogeneity $\ge \gamma_-$, has the form

$$\Gamma^{(n)}_{zz'} Y = Y + c_n(z, z')\mathbf{1}. \tag{144.175}$$

Then, on $\mathcal{T}_-$, the only strict lower stochastic reexpansion coefficients are

$$\Gamma^{(n)}_{zz'}(XY) = XY + c_n(z, z')X, \qquad \Gamma^{(n)}_{zz'}(X^2Y) = X^2Y + c_n(z, z')X^2. \tag{144.176}$$

In particular, for a compact K, the model seminorm below γ_- is controlled by the following explicit coordinate fields:

$$P_{n,\tau}(z, \delta, \varphi) = \delta^{-|\tau|} \left| \langle \Pi^{(n)}_z \tau, \varphi^\delta_z \rangle \right|, \qquad \tau \in \{\Xi, X, X^2, X^3, XY, X^2Y\},$$

and

$$G_n(z, z') = \frac{|c_n(z, z')|}{\|z - z'\|_{\mathfrak{s}}^{|Y|}} \qquad (0 < \|z - z'\|_{\mathfrak{s}} \le 1).$$

More precisely, with the basis vectors above assigned norm one,

$$1 + \|\Pi^{(n)}\|_{K;\gamma_-} + \|\Gamma^{(n)}\|_{K;\gamma_-} \le A_- \left(1 + \sum_{\tau \in \mathcal{T}_- \setminus \{\mathbf{1}\}} \sup_{z,\delta,\varphi} P_{n,\tau}(z, \delta, \varphi) + \sup_{z,z'} G_n(z, z') \right) \tag{144.177}$$

for a numerical constant $A_- < \infty$ depending only on this finite basis. The same inequality holds for the model distance $d_{K;\gamma_-}(Z_{n+1}, Z_n)$ after replacing $P_{n,\tau}$ and G_n by the corresponding cutoff differences.

Proof. The homogeneity of $X^a Y^b$ is

$$a\left(-\frac{1}{2} - \kappa\right) + b\left(\frac{1}{2} - 3\kappa\right).$$

For $a + b \leq 3$, direct evaluation gives

$$\begin{array}{c|c}
X & -\frac{1}{2} - \kappa \\
X^2 & -1 - 2\kappa \\
X^3 & -\frac{3}{2} - 3\kappa \\
Y & \frac{1}{2} - 3\kappa \\
XY & -4\kappa \\
X^2Y & -\frac{1}{2} - 5\kappa \\
Y^2 & 1 - 6\kappa \\
XY^2 & \frac{1}{2} - 7\kappa \\
Y^3 & \frac{3}{2} - 9\kappa.
\end{array}$$

Since $0 < \kappa < 1/14$, the entries in (144.174) are strictly below γ_-, while Y, Y^2, and Y^3 are above γ_-, and XY^2 is exactly at γ_-. The strict truncation $T_{<\gamma_-}$ therefore contains the displayed non-polynomial monomials and not XY^2.

The symbol Y has homogeneity $|Y| = \frac{1}{2} - 3\kappa$, which lies in $(0, 1)$. Below γ_-, the only polynomial symbol with lower homogeneity than Y that can appear in its reexpansion is the constant symbol. This is the coefficient $c_n(z, z')$ in (144.175). The negative symbols Ξ, X, X^2, X^3 have no lower polynomial component in the sector $\mathcal{T}_-$. Multiplicativity of the model then gives

$$\Gamma^{(n)}_{zz'}(XY) = \Gamma^{(n)}_{zz'} X \, \Gamma^{(n)}_{zz'} Y = X(Y + c_n(z, z')\mathbf{1}),$$

and similarly

$$\Gamma^{(n)}_{zz'}(X^2 Y) = X^2(Y + c_n(z, z')\mathbf{1}).$$

This proves (144.176). The homogeneity gaps are

$$|XY| - |X| = |Y|, \qquad |X^2 Y| - |X^2| = |Y|.$$

Thus the normalized Γ-seminorms of these two strict lower components are both bounded by $\sup G_n$; all other strict lower components in this sector vanish.

For Π, the seminorm in Definition 144.47 is the supremum over unit vectors in a finite-dimensional graded space. In the chosen basis, the unit-vector supremum is bounded by a finite multiple of the sum of the basis-vector suprema. The constant symbol contributes the deterministic bound

$$|\langle 1, \varphi^\delta_z \rangle| \leq \|\varphi\|_{L^1} \leq C_r.$$

Combining this deterministic constant, the six non-polynomial Π coordinates, and the single c_n-coordinate proves (144.177). Applying the same finite-dimensional norm comparison to $\Pi^{(n+1)} - \Pi^{(n)}$ and to the coefficient $c_{n+1} - c_n$ gives the stated model-distance version. $\square$

Hypothesis 144.65 (Negative-sector scale-summed coordinate scheme)**.** Fix $0 < \kappa < 1/14$, set

$$\gamma_- = \frac{1}{2} - 7\kappa,$$

and let $K \subset \mathbb{R}^{1+3}$ be compact. Choose

$$r > \frac{5}{2} + \kappa.$$

Let $Z_n = (\Pi^{(n)}, \Gamma^{(n)})$, $n \geq 1$, be random cutoff models on the finite sector $\mathcal{T}_-$ of Lemma 144.64, with the same basis norms and with reexpansion coefficient $c_n(z, z')$ as in (144.175). Put

$$N_n^- = 1 + \|\Pi^{(n)}\|_{K;\gamma_-} + \|\Gamma^{(n)}\|_{K;\gamma_-}, \qquad D_n^- = d_{K;\gamma_-}(Z_{n+1}, Z_n).$$

For

$$\mathfrak{C}_\Pi = \{\Xi, X, X^2, X^3, XY, X^2Y\}, \qquad \mathfrak{C} = \mathfrak{C}_\Pi \sqcup \{\mathfrak{g}\},$$

define the physical coordinate functions

$$Q_{n,\tau}(z, \delta, \varphi) = \delta^{-|\tau|} \left| \langle \Pi_z^{(n)} \tau, \varphi_z^\delta \rangle \right|, \qquad \tau \in \mathfrak{C}_\Pi,$$

$$\Delta Q_{n,\tau}(z, \delta, \varphi) = \delta^{-|\tau|} \left| \langle (\Pi_z^{(n+1)} - \Pi_z^{(n)}) \tau, \varphi_z^\delta \rangle \right|,$$

for $z \in K$, $0 < \delta \leq 1$, $\varphi \in \mathcal{B}_r$, and

$$Q_{n,\mathfrak{g}}(z, z') = \frac{|c_n(z, z')|}{\|z - z'\|_{\mathfrak{s}}^{|Y|}}, \qquad \Delta Q_{n,\mathfrak{g}}(z, z') = \frac{|c_{n+1}(z, z') - c_n(z, z')|}{\|z - z'\|_{\mathfrak{s}}^{|Y|}},$$

for $z, z' \in K$ with $0 < \|z - z'\|_{\mathfrak{s}} \leq 1$. For every $i \in \mathfrak{C}$ and $m \geq 0$, let $(\mathcal{X}_{i,m}, d_{i,m})$ be a compact metric space with finite nets $\mathcal{X}_{i,m,\ell}$, projections

$$\pi_{i,m,\ell} : \mathcal{X}_{i,m} \to \mathcal{X}_{i,m,\ell},$$

and edge sets

$$\mathcal{E}_{i,m,\ell} = \{(\pi_{i,m,\ell} x, \pi_{i,m,\ell+1} x) : x \in \mathcal{X}_{i,m}\}.$$

Assume that there are constants $C_{i,0}, C_{i,1}, D_i, d_i > 0$ such that

$$|\mathcal{X}_{i,m,0}| \leq C_{i,0} 2^{D_i m}, \qquad |\mathcal{E}_{i,m,\ell}| \leq C_{i,1} 2^{D_i m} 2^{d_i \ell} \tag{144.178}$$

for all $m, \ell \geq 0$. Suppose further that there are real random fields

$$U_{n,i,m}, V_{n,i,m} : \mathcal{X}_{i,m} \to L^p(\Omega), \qquad n \geq 1, \quad i \in \mathfrak{C}, \quad m \geq 0,$$

with almost surely continuous sample paths, such that the coordinate domination inequalities

$$\sup Q_{n,i} \leq \sup_{m \geq 0} \sup_{x \in \mathcal{X}_{i,m}} |U_{n,i,m}(x)|, \qquad \sup \Delta Q_{n,i} \leq \sup_{m \geq 0} \sup_{x \in \mathcal{X}_{i,m}} |V_{n,i,m}(x)| \tag{144.179}$$

hold almost surely. In (144.179), the first supremum is over the displayed physical variables of $Q_{n,i}$, and the second supremum is over the displayed physical variables of $\Delta Q_{n,i}$.

Fix $p \geq 1$ and $\rho > 0$. For each $i \in \mathfrak{C}$, assume there are positive constants $B_{i,0}, B_i, \widetilde{B}_{i,0}, \widetilde{B}_i, \sigma_i$, and ε_i, with

$$\sigma_i > \frac{D_i}{p}, \tag{144.180}$$

such that, for every n, m, ℓ,

$$\mathbb{E}|U_{n,i,m}(u)|^p \leq B_{i,0}^p 2^{-p\sigma_i m} \quad (u \in \mathcal{X}_{i,m,0}),$$

$$\mathbb{E}|U_{n,i,m}(v) - U_{n,i,m}(u)|^p \leq B_i^p 2^{-p\sigma_i m} 2^{-(d_i+\varepsilon_i)\ell} \quad ((u,v) \in \mathcal{E}_{i,m,\ell}),$$

and

$$\mathbb{E}|V_{n,i,m}(u)|^p \leq \widetilde{B}_{i,0}^p 2^{-p\rho n} 2^{-p\sigma_i m} \quad (u \in \mathcal{X}_{i,m,0}),$$

$$\mathbb{E}|V_{n,i,m}(v) - V_{n,i,m}(u)|^p \leq \widetilde{B}_i^p 2^{-p\rho n} 2^{-p\sigma_i m} 2^{-(d_i+\varepsilon_i)\ell} \quad ((u,v) \in \mathcal{E}_{i,m,\ell}).$$

Define

$$S_i^{\mathrm{sc}} = \frac{C_{i,0}^{1/p} B_{i,0} + C_{i,1}^{1/p} B_i (1 - 2^{-\varepsilon_i/p})^{-1}}{1 - 2^{-(\sigma_i - D_i/p)}},$$

and

$$\widetilde{S}_i^{\mathrm{sc}} = \frac{C_{i,0}^{1/p} \widetilde{B}_{i,0} + C_{i,1}^{1/p} \widetilde{B}_i (1 - 2^{-\varepsilon_i/p})^{-1}}{1 - 2^{-(\sigma_i - D_i/p)}}.$$

Negative-sector model convergence from scale-summed coordinates. Under Hypothesis 144.65,

$$\sup_{n \geq 1} \|N_n^-\|_{L^p(\Omega)} \leq A_- \left(1 + \sum_{i \in \mathfrak{C}} S_i^{\mathrm{sc}}\right),$$

and

$$\|D_n^-\|_{L^p(\Omega)} \leq A_- \left(\sum_{i \in \mathfrak{C}} \widetilde{S}_i^{\mathrm{sc}}\right) 2^{-\rho n}. \tag{144.181}$$

Consequently $Z_n|_{\mathcal{T}_-}$ converges in L^p and almost surely, in the model distance $d_{K;\gamma_-}$, to a limiting random model on $\mathcal{T}_-$. More precisely,

$$\left(\mathbb{E}\, d_{K;\gamma_-}(Z_n, Z_\infty)^p\right)^{1/p} \leq \frac{A_- \sum_{i \in \mathfrak{C}} \widetilde{S}_i^{\mathrm{sc}}}{1 - 2^{-\rho}} 2^{-\rho n}.$$

For a fixed coordinate label i, the entropy hypothesis (144.178) and the four moment hypotheses are exactly the hypotheses of Corollary 144.60, first for $U_{n,i,m}$ and then for $V_{n,i,m}$. Hence

$$\left\| \sup_{m \geq 0} \sup_{x \in \mathcal{X}_{i,m}} |U_{n,i,m}(x)| \right\|_{L^p} \leq S_i^{\mathrm{sc}},$$

uniformly in n, and

$$\left\| \sup_{m \geq 0} \sup_{x \in \mathcal{X}_{i,m}} |V_{n,i,m}(x)| \right\|_{L^p} \leq \widetilde{S}_i^{\mathrm{sc}} 2^{-\rho n}.$$

Lemma 144.64 gives the finite-dimensional coordinate domination

$$N_n^- \le A_- \left(1 + \sum_{i\in\mathfrak{C}} \sup Q_{n,i}\right)$$

and the corresponding distance domination

$$D_n^- \le A_- \sum_{i\in\mathfrak{C}} \sup \Delta Q_{n,i}.$$

Insert (144.179) into these two inequalities and apply the L^p triangle inequality. This proves the two displayed bounds for N_n^- and D_n^-.

The hypotheses of Lemma 144.50 are therefore satisfied with

$$C_N = A_-^p \left(1 + \sum_{i\in\mathfrak{C}} S_i^{\mathrm{sc}}\right)^p, \qquad C_D = A_-^p \left(\sum_{i\in\mathfrak{C}} \widetilde{S}_i^{\mathrm{sc}}\right)^p.$$

Applying that theorem on the finite sector $\mathcal{T}_-$ gives the limiting model and the stated L^p tail estimate.

Proposition 144.66 (Physical parameter entropy for the strict negative sector)**.** *Let $K \subset \mathbb{R}^{1+3}$ be compact, fix $r > 0$, let*

$$\mathfrak{C}_\Pi = \{\Xi, X, X^2, X^3, XY, X^2Y\},$$

let $|Y| > 0$ be the exponent used to normalize the c_n-coordinate, let $\mathfrak{s} = (2,1,1,1)$, and let

$$Q = |\mathfrak{s}| = 5.$$

For $\lambda > 0$, write

$$\mathsf{D}_\lambda(t, x_1, x_2, x_3) = (\lambda^2 t, \lambda x_1, \lambda x_2, \lambda x_3).$$

For $\tau \in \mathfrak{C}_\Pi$, set

$$Q^{\#}_{n,\tau}(z,\delta) = \sup_{\varphi\in\mathcal{B}_r} \delta^{-|\tau|} \left|\langle \Pi_z^{(n)}\tau, \varphi_z^\delta\rangle\right|.$$

Then the supremum over $z \in K$ and $0 < \delta \le 1$ is the supremum over the dyadic-scale spaces

$$\mathcal{X}_{\Pi,m} = K \times [1/2, 1], \qquad (z,a) \longmapsto Q^{\#}_{n,\tau}(z, 2^{-m}a), \qquad m \ge 0.$$

Equip $\mathcal{X}_{\Pi,m}$ with

$$d_{\Pi,m}((z,a),(z',a')) = 2^m \|z - z'\|_{\mathfrak{s}} + |a - a'|.$$

There are finite nets and projections as in Corollary 144.60 for which

$$|\mathcal{X}_{\Pi,m,0}| \le C_{\Pi,0} 2^{Qm}, \qquad |\mathcal{E}_{\Pi,m,\ell}| \le C_{\Pi,1} 2^{Qm} 2^{(Q+1)\ell}. \tag{144.182}$$

For the Γ-coordinate, let

$$A_{\mathfrak{s}} = \{h \in \mathbb{R}^{1+3} : 1/2 \le \|h\|_{\mathfrak{s}} \le 1\},$$

and, for the reexpansion coefficient $c_n(z,z')$, set

$$Q_{n,\mathfrak{g}}(z,z') = \frac{|c_n(z,z')|}{\|z - z'\|_{\mathfrak{s}}^{|Y|}}.$$

For $m \geq 0$, define

$$\mathcal{X}_{\Gamma,m} = \{(z,h) \in K \times A_{\mathfrak{s}} : z + \mathsf{D}_{2^{-m}} h \in K\}.$$

The map

$$(z,h) \longmapsto Q_{n,\mathfrak{g}}(z, z + \mathsf{D}_{2^{-m}} h)$$

parametrizes the part of the Γ-seminorm with

$$2^{-(m+1)} \leq \|z - z'\|_{\mathfrak{s}} \leq 2^{-m}.$$

Equip $\mathcal{X}_{\Gamma,m}$ with

$$d_{\Gamma,m}((z,h),(z',h')) = 2^m \|z - z'\|_{\mathfrak{s}} + \|h - h'\|_{\mathfrak{s}}.$$

There are finite nets and projections for which

$$|\mathcal{X}_{\Gamma,m,0}| \leq C_{\Gamma,0} 2^{Qm}, \qquad |\mathcal{E}_{\Gamma,m,\ell}| \leq C_{\Gamma,1} 2^{Qm} 2^{2Q\ell}. \tag{144.183}$$

The constants in (144.182) and (144.183) depend on K and on the chosen parabolic norm, but not on m, ℓ, n, or the coordinate label τ.

Proof. First record the covering estimate used twice. Since K is compact, it is contained in a rectangular box

$$[-R,R] \times [-R,R]^3.$$

A grid with temporal mesh η^2 and spatial mesh η covers this box by at most $C_R \eta^{-5}$ parabolic balls of radius $C\eta$, for $0 < \eta \leq 1$. After changing the constant, every compact subset of $\mathbb{R}^{1+3}$ therefore has parabolic covering number at scale η bounded by $C_K \eta^{-Q}$. The interval $[1/2,1]$ has covering number $\leq C\eta^{-1}$ in the ordinary metric, and the annulus $A_{\mathfrak{s}}$ has covering number $\leq C\eta^{-Q}$ in $\|\cdot\|_{\mathfrak{s}}$.

For $\mathcal{X}_{\Pi,m}$, a ball of radius $2^{-\ell}$ in $d_{\Pi,m}$ requires resolution $2^{-m-\ell}$ in K and resolution $2^{-\ell}$ in $[1/2,1]$. Hence

$$N(\mathcal{X}_{\Pi,m}, d_{\Pi,m}; 2^{-\ell}) \leq C\, 2^{Qm} 2^{(Q+1)\ell}.$$

Choose a $2^{-\ell}$-net $\mathcal{X}_{\Pi,m,\ell}$ with this cardinality, increasing in ℓ after replacing it by the union of the previous nets if needed. Let $\pi_{\Pi,m,\ell}$ be any nearest-net projection. The base bound in (144.182) is the case $\ell = 0$, and the edge set

$$\mathcal{E}_{\Pi,m,\ell} = \{(\pi_{\Pi,m,\ell} x, \pi_{\Pi,m,\ell+1} x) : x \in \mathcal{X}_{\Pi,m}\}$$

has cardinality no larger than $|\mathcal{X}_{\Pi,m,\ell+1}|$. Absorbing the extra factor 2^{Q+1} into $C_{\Pi,1}$ gives the displayed edge estimate.

For $\mathcal{X}_{\Gamma,m}$, cover K by $O(2^{Q(m+\ell)})$ balls of radius $2^{-m-\ell}$ and $A_{\mathfrak{s}}$ by $O(2^{Q\ell})$ balls of radius $2^{-\ell}$. The product cover has $d_{\Gamma,m}$-radius $2^{-\ell}$, so

$$N(\mathcal{X}_{\Gamma,m}, d_{\Gamma,m}; 2^{-\ell}) \leq C\, 2^{Qm} 2^{2Q\ell}.$$

The subset condition $z + \mathsf{D}_{2^{-m}} h \in K$ only decreases the covering number. The same projection and edge-set construction proves (144.183).

It remains to check the parametrization. If $0 < \delta \leq 1$, choose $m \geq 0$ and $a \in [1/2,1]$ with $\delta = 2^{-m} a$. This gives the Π-coordinate decomposition. If $0 < \|z - z'\|_{\mathfrak{s}} \leq 1$, choose $m \geq 0$ so that

$$2^{-(m+1)} \leq \|z - z'\|_{\mathfrak{s}} \leq 2^{-m}$$

and set

$$h = \mathsf{D}_{2^m}(z' - z).$$

The parabolic norm is homogeneous under D_λ, hence $h \in A_{\mathfrak{s}}$ and $z' = z + \mathsf{D}_{2^{-m}}h$. This proves the Γ-coordinate decomposition. $\square$

Proposition 144.67 (Gaussian Π-coordinate input for scale-summed model convergence)**.** *Let $K \subset \mathbb{R}^{1+3}$ be compact, let*

$$Q = 5, \qquad 0 < \kappa < \frac{1}{14},$$

and choose r, θ with

$$0 < \theta < r - \frac{Q}{2}.$$

Let ξ be a centered Gaussian space-time white noise and

$$X = G * \xi$$

be the massive heat-convolution field used in Proposition 144.55. For

$$\mathfrak{C}_{\mathrm{G}} = \{\Xi, X, X^2, X^3\}$$

interpret X^k as the Wick power $: X^k :$, $k = 1, 2, 3$, and assign

$$|\Xi| = -\frac{Q}{2} - \kappa, \qquad |X^k| = -\frac{k}{2} - k\kappa.$$

For $m \geq 0$ and $a \in [1/2, 1]$ set

$$x = (z, 2^{-m}a)$$

and define the random fields on $\mathcal{X}_{\Pi,m} = K \times [1/2, 1]$ by

$$U_{\Xi,m}(z,a) = \|\mathcal{W}_x\|_{E_r'}, \qquad U_{X^k,m}(z,a) = \|\mathcal{X}_{k,x}\|_{E_r'}, \quad k = 1,2,3, \tag{144.184}$$

where $\mathcal{W}_x$ and $\mathcal{X}_{k,x}$ are the E_r'-valued functionals of Proposition 144.56. Equivalently,

$$U_{\Xi,m}(z,a) = \sup_{\varphi \in \mathcal{B}_r} (2^{-m}a)^{Q/2+\kappa} |\xi(\varphi_z^{2^{-m}a})|,$$

and

$$U_{X^k,m}(z,a) = \sup_{\varphi \in \mathcal{B}_r} (2^{-m}a)^{k/2+k\kappa} |\langle : X^k :, \varphi_z^{2^{-m}a} \rangle|.$$

Let $p = 2m_0$ be an even integer satisfying

$$p\kappa > Q, \qquad p\theta > Q + 1. \tag{144.185}$$

Then, after choosing the net system of Proposition 144.66, the fields $U_{\tau,m}$, $\tau \in \mathfrak{C}_{\mathrm{G}}$, have continuous modifications on each $\mathcal{X}_{\Pi,m}$ and satisfy the Π-coordinate hypotheses of the negative-sector coordinate-control consequence with

$$D_\tau = Q, \qquad d_\tau = Q + 1,$$

$$\sigma_\Xi = \kappa, \qquad \sigma_{X^k} = k\kappa, \qquad \varepsilon_\tau = p\theta - (Q+1) > 0. \tag{144.186}$$

More explicitly, there are constants $B_{\tau,0}, B_\tau < \infty$ such that, for every $m, \ell \geq 0$,

$$\mathbb{E}|U_{\tau,m}(u)|^p \leq B_{\tau,0}^p 2^{-p\sigma_\tau m} \qquad (u \in \mathcal{X}_{\Pi,m,0}) \tag{144.187}$$

and

$$\mathbb{E}|U_{\tau,m}(v) - U_{\tau,m}(u)|^p \leq B_\tau^p 2^{-p\sigma_\tau m} 2^{-(Q+1+\varepsilon_\tau)\ell} \qquad ((u,v) \in \mathcal{E}_{\Pi,m,\ell}). \tag{144.188}$$

Consequently the Gaussian coordinate labels Ξ, X, X^2, X^3 already feed into the scale-summed negative-sector model theorem without an additional stochastic supremum argument.

If a cutoff or mollifier sequence $(\xi_n, X_n, : X_n^2 :, : X_n^3 :)$ satisfies the same dual-norm bounds for the increments of the corresponding fields $U_{\tau,m}^{(n+1)} - U_{\tau,m}^{(n)}$, with an additional factor $2^{-\rho n}$ in L^p for some $\rho > 0$, then the associated cutoff-difference variables satisfy the V-hypotheses of the negative-sector coordinate-control consequence with the same $D_\tau, d_\tau, \sigma_\tau, \varepsilon_\tau$ and with that convergence rate ρ.

Proof. We first identify the scale-normalized physical coordinates with the dual-norm fields. For $\tau = \Xi$,

$$\delta^{-|\Xi|}|\langle \Pi_z \Xi, \varphi_z^\delta \rangle| = \delta^{Q/2+\kappa}|\xi(\varphi_z^\delta)|,$$

and this is $|\mathcal{W}_{(z,\delta)}(\varphi)|$. For $\tau = X^k$,

$$\delta^{-|X^k|}|\langle \Pi_z X^k, \varphi_z^\delta \rangle| = \delta^{k/2+k\kappa}|\langle : X^k :, \varphi_z^\delta \rangle|,$$

which is $|\mathcal{X}_{k,(z,\delta)}(\varphi)|$. Taking the supremum over $\mathcal{B}_r$, which is the unit ball of the same test space E_r with the support convention used in the dual-norm proposition, gives (144.184). This is the promised map from physical Π-coordinates to the abstract coordinate fields of the negative-sector coordinate-control consequence.

Set $\delta = 2^{-m}a$ with $a \in [1/2, 1]$. The factor a is bounded above and below by numerical constants, hence

$$\delta^{\sigma_\tau} \asymp 2^{-\sigma_\tau m}$$

for $\sigma_\Xi = \kappa$ and $\sigma_{X^k} = k\kappa$. Applying (144.138) and raising the L^p-bound to a p-th moment proves (144.187).

Let $(u,v) \in \mathcal{E}_{\Pi,m,\ell}$, say

$$u = (z,a), \qquad v = (z',a').$$

The nearest-net construction in Proposition 144.66 gives

$$2^m\|z - z'\|_{\mathfrak{s}} + |a - a'| \leq C_\Pi 2^{-\ell}. \tag{144.189}$$

With $x = (z, 2^{-m}a)$ and $y = (z', 2^{-m}a')$, the parameter η of Proposition 144.56 satisfies

$$\eta \leq C2^{-\ell}.$$

The edge estimates (144.139) and (144.140), together with

$$\big|\|A\|_{E_r'} - \|B\|_{E_r'}\big| \leq \|A - B\|_{E_r'},$$

therefore give

$$\|U_{\tau,m}(v) - U_{\tau,m}(u)\|_{L^p} \leq C_\tau 2^{-\sigma_\tau m} 2^{-\theta\ell}.$$

Raising to the p-th power yields

$$\mathbb{E}|U_{\tau,m}(v) - U_{\tau,m}(u)|^p \le C_\tau^p 2^{-p\sigma_\tau m} 2^{-p\theta\ell}.$$

Since $\varepsilon_\tau = p\theta - (Q+1)$, this is exactly (144.188).

The entropy dimensions are

$$D_\tau = Q, \qquad d_\tau = Q+1$$

by Proposition 144.66. The moment condition (144.185) implies

$$\sigma_\Xi = \kappa > \frac{Q}{p}, \qquad \sigma_{X^k} = k\kappa \ge \kappa > \frac{Q}{p},$$

and it also implies $\varepsilon_\tau > 0$. These are precisely the scale-slack and edge-slack inequalities required in the negative-sector coordinate-control consequence.

It remains only to justify the continuity hypothesis. The same edge estimate, before restriction to the dyadic edge set, holds for arbitrary $(z,a),(z',a') \in \mathcal{X}_{\Pi,m}$ with

$$d_{\Pi,m}((z,a),(z',a')) \le 1,$$

with $2^{-\ell}$ replaced by that metric distance. Since $\mathcal{X}_{\Pi,m}$ has metric covering dimension $Q+1$ uniformly in m, the Kolmogorov–Chentsov dyadic-net proof gives, for every

$$0 < \alpha < \theta - \frac{Q+1}{p},$$

a modification that is α-Holder continuous on $\mathcal{X}_{\Pi,m}$. This proves the required sample-path continuity.

For a cutoff sequence, apply the preceding Kolmogorov-net construction to

$$U_{\tau,m}^{(n+1)} - U_{\tau,m}^{(n)}$$

in place of $U_{\tau,m}$. The assumed extra factor $2^{-\rho n}$ is carried unchanged through the base estimate, the edge estimate, and the dyadic-net continuity argument, and hence gives the V-hypotheses of the scale-summed theorem. □

For the nonlinear Π-coordinates XY and X^2Y, and for the Γ-coordinate $c_n(z,z')$, the same assembly theorem applies but the analytic inputs are different. The XY and X^2Y coordinates require the concrete BPHZ kernel estimates for products involving $Y = \mathcal{I}(X^3)$. The Γ-coordinate has no test-function dual norm; it is a scalar finite-chaos estimate on $\mathcal{X}_{\Gamma,m}$, with entropy exponent $d_{\mathfrak{g}} = 10$ by Proposition 144.66. Thus the stochastic and entropy mechanism is now completely explicit, while the remaining Φ^4_3 work is the model-specific BPHZ kernel analysis for XY, X^2Y, and the reexpansion coefficient of Y.

Proposition 144.68 (Heat-integration reexpansion estimate)**.** *Let* $\mathfrak{s} = (2,1,1,1)$ *and* $Q = 5$. *Let*

$$-2 < \alpha < 0, \qquad 0 < \sigma, \qquad \beta = \alpha + 2, \qquad \gamma = \beta + \sigma,$$

and assume

$$0 < \beta < 1, \qquad 0 < \gamma < 1.$$

Fix $r > -\alpha - \sigma$. Let $K = \sum_{j\geq 0} K_j$ be a dyadic decomposition of a parabolic heat kernel, with the following properties. There are constants A, C_K and a number $\vartheta \in (\gamma, 1]$ such that, after enlarging A if necessary, $A \geq 1$ and

$$\operatorname{supp} K_j \subset B_{\mathfrak{s}}(0, A2^{-j}), \qquad |K_j(u)| \leq C_K 2^{(Q-2)j}, \qquad j \geq 0, \tag{144.190}$$

and, for every a, b with $\|a-b\|_{\mathfrak{s}} \leq A2^{-j}$, the translated kernel difference admits the test-function representation

$$K_j(a-\cdot) - K_j(b-\cdot) = C_K 2^{-2j}(2^j\|a-b\|_{\mathfrak{s}})^{\vartheta}\psi_{a,b,j}^{2^{-j}}, \tag{144.191}$$

where $\psi_{a,b,j} \in C_c^r$ is supported in a fixed $\mathfrak{s}$-ball and $\|\psi_{a,b,j}\|_{C^r} \leq 1$. The translated kernel itself has the analogous representation

$$K_j(a-\cdot) = C_K 2^{-2j}\psi_{a,j}^{2^{-j}}, \qquad \|\psi_{a,j}\|_{C^r} \leq 1. \tag{144.192}$$

Let T be a distribution on a fixed compact neighborhood U such that

$$\|T\|_{\alpha+\sigma;U,r} := \sup_{\substack{x\in U,\ 0<\lambda\leq 1\\ \varphi\in\mathcal{B}_r}} \lambda^{-(\alpha+\sigma)}|\langle T, \varphi_x^\lambda\rangle| < \infty. \tag{144.193}$$

For $a \in U$ whose unit parabolic neighborhood lies in U, define

$$(\mathcal{K}T)(a) = \sum_{j\geq 0}\langle T, K_j(a-\cdot)\rangle. \tag{144.194}$$

Then the increment series is absolutely convergent and is given by

$$\mathfrak{c}_T(a,b) = (\mathcal{K}T)(b) - (\mathcal{K}T)(a) = \sum_{j\geq 0}\langle T, K_j(b-\cdot) - K_j(a-\cdot)\rangle. \tag{144.195}$$

Moreover, whenever $0 < \|a-b\|_{\mathfrak{s}} \leq 1$,

$$|\mathfrak{c}_T(a,b)| \leq C\|T\|_{\alpha+\sigma;U,r}\|a-b\|_{\mathfrak{s}}^{\beta+\sigma}. \tag{144.196}$$

Consequently the coefficient normalized with the model homogeneity β has scale slack σ:

$$\frac{|\mathfrak{c}_T(a,b)|}{\|a-b\|_{\mathfrak{s}}^{\beta}} \leq C\|T\|_{\alpha+\sigma;U,r}\|a-b\|_{\mathfrak{s}}^{\sigma}. \tag{144.197}$$

If (a,b) and (a',b') satisfy

$$h = \|a-b\|_{\mathfrak{s}}, \qquad h' = \|a'-b'\|_{\mathfrak{s}}, \qquad h, h' \in [2^{-m-1}, 2^{-m}],$$

and

$$\eta = 2^m\big(\|a-a'\|_{\mathfrak{s}} + \|b-b'\|_{\mathfrak{s}}\big) \leq 1,$$

then, for every $0 < \theta \leq \gamma$,

$$\left|\frac{\mathfrak{c}_T(a,b)}{\|a-b\|_{\mathfrak{s}}^{\beta}} - \frac{\mathfrak{c}_T(a',b')}{\|a'-b'\|_{\mathfrak{s}}^{\beta}}\right| \leq C_\theta\|T\|_{\alpha+\sigma;U,r}2^{-\sigma m}\eta^{\theta}. \tag{144.198}$$

Proof. First prove the Schauder increment estimate for $\mathcal{K}T$. Set

$$M = \|T\|_{\alpha+\sigma;U,r}.$$

The representation (144.192) and the definition of M give

$$|\langle T, K_j(a-\cdot)\rangle| \le C_K 2^{-2j} M (2^{-j})^{\alpha+\sigma} = C_K M 2^{-\gamma j}. \tag{144.199}$$

This bound is summable for $j \to \infty$, since $\gamma > 0$. The possible large-scale additive ambiguity is the usual harmless constant in the parabolic Schauder estimate; it cancels in increments. Hence $\mathfrak{c}_T(a,b)$ is given by the absolutely convergent difference series in (144.195).

Let

$$h = \|a-b\|_{\mathfrak{s}}$$

and choose $J \ge 0$ so that

$$2^{-J-1} < h \le 2^{-J}.$$

For the fine scales $j > J$, use (144.199) separately at a and b:

$$\sum_{j>J} |\langle T, K_j(b-\cdot) - K_j(a-\cdot)\rangle| \le CM \sum_{j>J} 2^{-\gamma j} \le CMh^{\gamma}. \tag{144.200}$$

For the coarse scales $j \le J$, the kernel-difference representation (144.191) gives

$$\begin{aligned} |\langle T, K_j(b-\cdot) - K_j(a-\cdot)\rangle| &\le C_K 2^{-2j} (2^j h)^{\vartheta} M (2^{-j})^{\alpha+\sigma} \\ &= CMh^{\vartheta} 2^{(\vartheta-\gamma)j}. \end{aligned}$$

Since $\vartheta > \gamma$,

$$\sum_{j \le J} h^{\vartheta} 2^{(\vartheta-\gamma)j} \le Ch^{\vartheta} 2^{(\vartheta-\gamma)J} \le Ch^{\gamma}. \tag{144.201}$$

Combining (144.200) and (144.201) proves (144.196); dividing by h^{β} proves (144.197).

For the normalized edge estimate, write

$$F(a) = (\mathcal{K}T)(a).$$

The estimate just proved says

$$|F(x) - F(y)| \le CM\|x-y\|_{\mathfrak{s}}^{\gamma} \tag{144.202}$$

on the compact region under consideration. The assumptions on (a,b) and (a',b') imply $h' \simeq h \simeq 2^{-m}$ and

$$\|a-a'\|_{\mathfrak{s}} + \|b-b'\|_{\mathfrak{s}} \le C\eta h. \tag{144.203}$$

Therefore

$$\begin{aligned} |(F(b)-F(a)) - (F(b')-F(a'))| &\le CM\big(\|b-b'\|_{\mathfrak{s}}^{\gamma} + \|a-a'\|_{\mathfrak{s}}^{\gamma}\big) \\ &\le CMh^{\gamma}\eta^{\gamma}. \end{aligned}$$

Dividing by h^{β} gives $CMh^{\sigma}\eta^{\gamma}$. The remaining effect is the change of denominator:

$$|h^{-\beta} - h'^{-\beta}| \le C_{\beta} h^{-\beta} \frac{|h-h'|}{h} \le C_{\beta} h^{-\beta} \eta,$$

while $|F(b') - F(a')| \le CMh^{\gamma}$. This contribution is bounded by $CMh^{\sigma}\eta$. Since $0 < \eta \le 1$, $h \simeq 2^{-m}$, and $\theta \le \gamma < 1$, the sum of the two contributions is bounded by the right-hand side of (144.198). □

Apply Proposition 144.68 to the symbol $Y = \mathcal{I}(X^3)$ in dynamic Φ^4_3. Here

$$\alpha = |X^3| = -\frac{3}{2} - 3\kappa, \qquad \beta = |Y| = \alpha + 2 = \frac{1}{2} - 3\kappa.$$

The primitive Gaussian estimate for X^3 proved above gives the additional regularity slack

$$\sigma = 3\kappa, \qquad \alpha + \sigma = -\frac{3}{2}, \qquad \beta + \sigma = \frac{1}{2}.$$

Thus the constant reexpansion coefficient

$$c_n(z,z') = \big(\mathcal{K}\Pi_z^{(n)} X^3\big)(z') - \big(\mathcal{K}\Pi_z^{(n)} X^3\big)(z) \tag{144.204}$$

has the deterministic normalization

$$\frac{|c_n(z,z')|}{\|z-z'\|_{\mathfrak{s}}^{|Y|}} \le C\, \|\Pi_z^{(n)} X^3\|_{-3/2;U,r} \|z-z'\|_{\mathfrak{s}}^{3\kappa}. \tag{144.205}$$

Proposition 144.69 (Γ-coordinate input from heat integration of X^3)**.** *Let $K \subset \mathbb{R}^{1+3}$ be compact and let U be a compact neighborhood containing the unit parabolic neighborhoods of the points of K. Fix*

$$0 < \kappa < \frac{1}{14}, \qquad \beta = \frac{1}{2} - 3\kappa, \qquad \sigma_{\mathfrak{g}} = 3\kappa.$$

Let T_n, $n \ge 1$, be random distributions on U. Assume that the heat-kernel decomposition satisfies the hypotheses of Proposition 144.68 with

$$\alpha = -\frac{3}{2} - 3\kappa, \qquad \sigma = 3\kappa, \qquad \gamma = \beta + \sigma = \frac{1}{2}.$$

For $z, z' \in K$, define

$$c_n(z,z') = (\mathcal{K}T_n)(z') - (\mathcal{K}T_n)(z).$$

Choose $0 < \theta \le 1/2$ and an even integer p such that

$$3p\kappa > 5, \qquad p\theta > 10. \tag{144.206}$$

Suppose that, for some constants $B_{\mathfrak{g}}$, $\widetilde{B}_{\mathfrak{g}}$, and $\rho > 0$,

$$\sup_{n\ge1} \big\| \|T_n\|_{-3/2;U,r} \big\|_{L^p(\Omega)} \le B_{\mathfrak{g}}, \tag{144.207}$$

and

$$\big\| \|T_{n+1} - T_n\|_{-3/2;U,r} \big\|_{L^p(\Omega)} \le \widetilde{B}_{\mathfrak{g}} 2^{-\rho n}. \tag{144.208}$$

Here $\|\cdot\|_{-3/2;U,r}$ is the norm (144.193) with exponent $-3/2$.

For $m \ge 0$, let $\mathcal{X}_{\Gamma,m}$ be the physical Γ-coordinate parameter space of Proposition 144.66. For $(z,h) \in \mathcal{X}_{\Gamma,m}$, set

$$z_h = z + \mathsf{D}_{2^{-m}} h, \qquad U_{n,\mathfrak{g},m}(z,h) = \frac{|c_n(z,z_h)|}{\|z-z_h\|_{\mathfrak{s}}^{\beta}},$$

and

$$V_{n,\mathfrak{g},m}(z,h) = \frac{|(c_{n+1}-c_n)(z,z_h)|}{\|z-z_h\|_{\mathfrak{s}}^{\beta}}. \tag{144.209}$$

Then the $\mathfrak{g}$-coordinate hypotheses of the negative-sector coordinate-control consequence hold with

$$D_{\mathfrak{g}} = 5, \qquad d_{\mathfrak{g}} = 10, \qquad \sigma_{\mathfrak{g}} = 3\kappa, \qquad \varepsilon_{\mathfrak{g}} = p\theta - 10 > 0. \tag{144.210}$$

More explicitly, after choosing the nets of Proposition 144.66, there are constants C_0, C_1, independent of n, m, ℓ, such that

$$\mathbb{E}|U_{n,\mathfrak{g},m}(u)|^p \le C_0^p B_{\mathfrak{g}}^p 2^{-3p\kappa m} \qquad (u \in \mathcal{X}_{\Gamma,m,0}), \tag{144.211}$$

$$\mathbb{E}|U_{n,\mathfrak{g},m}(v) - U_{n,\mathfrak{g},m}(u)|^p \le C_1^p B_{\mathfrak{g}}^p 2^{-3p\kappa m} 2^{-(10+\varepsilon_{\mathfrak{g}})\ell} \qquad ((u,v) \in \mathcal{E}_{\Gamma,m,\ell}), \tag{144.212}$$

and the same two estimates hold for $V_{n,\mathfrak{g},m}$ with $B_{\mathfrak{g}}^p$ replaced by $\widetilde{B}_{\mathfrak{g}}^p 2^{-p\rho n}$.

For the primitive dynamic Φ^4_3 reexpansion coefficient, one takes

$$T_n = \Pi_z^{(n)} X^3,$$

which is independent of the base point z for this primitive Wick-cubic symbol. When T_n is the cutoff Wick cubic $: X_n^3 :$, the uniform bound (144.207) is the scale-summed X^3 consequence of Proposition 144.67 applied on the enlarged compact region U. The cutoff-Cauchy input (144.208) is the corresponding cutoff-increment estimate for the Wick cubic.

Proof. Let $(z,h) \in \mathcal{X}_{\Gamma,m}$ and put

$$z_h = z + \mathsf{D}_{2^{-m}} h, \qquad h_m = \|z - z_h\|_{\mathfrak{s}}.$$

Since $h \in A_{\mathfrak{s}}$, one has

$$2^{-m-1} \le h_m \le 2^{-m}. \tag{144.213}$$

Apply (144.197) with $T = T_n$, $\beta = 1/2 - 3\kappa$, and $\sigma = 3\kappa$. This gives

$$U_{n,\mathfrak{g},m}(z,h) \le C\|T_n\|_{-3/2;U,r} h_m^{3\kappa} \le C\|T_n\|_{-3/2;U,r} 2^{-3\kappa m}. \tag{144.214}$$

Taking L^p-norms and using (144.207) proves (144.211).

Let $u = (z,h)$ and $v = (z',h')$ be an edge in $\mathcal{E}_{\Gamma,m,\ell}$. The net construction gives

$$2^m \|z - z'\|_{\mathfrak{s}} + \|h - h'\|_{\mathfrak{s}} \le C 2^{-\ell}. \tag{144.215}$$

Set

$$a = z, \quad b = z + \mathsf{D}_{2^{-m}} h, \qquad a' = z', \quad b' = z' + \mathsf{D}_{2^{-m}} h'.$$

Then $a, b, a', b' \in U$, their separations are in $[2^{-m-1}, 2^{-m}]$, and the edge parameter

$$\eta = 2^m (\|a - a'\|_{\mathfrak{s}} + \|b - b'\|_{\mathfrak{s}})$$

is bounded by $C2^{-\ell}$. The normalized edge estimate (144.198) therefore gives

$$\left| \frac{c_n(a,b)}{\|a-b\|_{\mathfrak{s}}^{\beta}} - \frac{c_n(a',b')}{\|a'-b'\|_{\mathfrak{s}}^{\beta}} \right| \le C_\theta \|T_n\|_{-3/2;U,r} 2^{-3\kappa m} 2^{-\theta\ell}.$$

Since $\big||A| - |B|\big| \le |A - B|$, this implies

$$|U_{n,\mathfrak{g},m}(v) - U_{n,\mathfrak{g},m}(u)| \le C_\theta \|T_n\|_{-3/2;U,r} 2^{-3\kappa m} 2^{-\theta\ell}. \tag{144.216}$$

After taking the p-th moment, this is exactly (144.212), because

$$10 + \varepsilon_{\mathfrak{g}} = p\theta.$$

The same deterministic inequalities apply to $T_{n+1} - T_n$, and (144.208) supplies the additional factor $2^{-\rho n}$ in L^p. The entropy exponents

$$D_{\mathfrak{g}} = 5, \qquad d_{\mathfrak{g}} = 10$$

are those of Proposition 144.66. The conditions (144.206) say precisely that

$$\sigma_{\mathfrak{g}} = 3\kappa > \frac{D_{\mathfrak{g}}}{p}, \qquad \varepsilon_{\mathfrak{g}} = p\theta - d_{\mathfrak{g}} > 0.$$

Thus the scale-slack and edge-slack hypotheses in the negative-sector coordinate-control consequence hold. The deterministic Holder estimate (144.216) also gives continuous sample paths whenever the norm $\|T_n\|_{-3/2;U,r}$ is finite, which proves the required continuity on each compact coordinate space. $\square$

Hence the c_n-coordinate does not require an independent singular stochastic contraction: its model-convergence input is a heat-integration consequence of the X^3 coordinate estimates. What remains genuinely new in the strict negative sector is the BPHZ analysis of the nonlinear products XY and X^2Y, where multiplication with the integrated field introduces additional contraction sectors.

144.6.5 Nonlinear Negative Coordinates and Local Shell Bounds

Finite-cutoff Wick decomposition for nonlinear negative coordinates. Let X_ϵ be a smooth centered Gaussian field on spacetime, with covariance

$$G_\epsilon(a,b) = \mathbb{E}[X_\epsilon(a)X_\epsilon(b)],$$

and let $K_\epsilon(a,b)$ be a smooth retarded heat kernel with the same ultraviolet cutoff. Define

$$Y_\epsilon(a) = \int K_\epsilon(a,b) :X_\epsilon(b)^3: \,\mathrm{d}b.$$

The Wick product is always taken with respect to the joint Gaussian family $\{X_\epsilon(c)\}_c$. Then

$$\begin{aligned} X_\epsilon(a)Y_\epsilon(a) = &\int K_\epsilon(a,b) :X_\epsilon(a)X_\epsilon(b)^3: \,\mathrm{d}b \\ &+ 3\int K_\epsilon(a,b)G_\epsilon(a,b) :X_\epsilon(b)^2: \,\mathrm{d}b. \end{aligned} \tag{144.217}$$

Furthermore, if

$$C_{2,\epsilon}(a) = 2\int K_\epsilon(a,b)G_\epsilon(a,b)^2 \,\mathrm{d}b, \tag{144.218}$$

then the two-loop locally subtracted coordinate is

$$
\begin{aligned}
&{:}X_\epsilon(a)^2{:}\,Y_\epsilon(a) - 3C_{2,\epsilon}(a)X_\epsilon(a)\\
&= \int K_\epsilon(a,b)\,:X_\epsilon(a)^2X_\epsilon(b)^3:\,\mathrm{d}b\\
&\quad + 6\int K_\epsilon(a,b)G_\epsilon(a,b)\,:X_\epsilon(a)X_\epsilon(b)^2:\,\mathrm{d}b\\
&\quad + 6\int K_\epsilon(a,b)G_\epsilon(a,b)^2\big(X_\epsilon(b)-X_\epsilon(a)\big)\,\mathrm{d}b.
\end{aligned}
\tag{144.219}
$$

Thus XY has only fourth- and second-chaos pieces, while the BPHZ coordinate X^2Y has fifth-, third-, and locally subtracted first-chaos pieces. The last line of (144.219) is the exact place where the local Taylor-subtraction gain of (144.149) enters the multiscale estimate.

The verification is the two-variable Wick product identity. For two centered Gaussian variables A, B, with $C = \mathbb{E}[AB]$, the Wick product identity is

$$
{:}A^m{:}\,{:}B^n{:} = \sum_{r=0}^{\min(m,n)} \binom{m}{r}\binom{n}{r} r!\,C^r\ {:}A^{m-r}B^{n-r}{:}\,. \tag{144.220}
$$

This identity follows by comparing coefficients of $s^mt^n/(m!n!)$ in

$$
\exp\Big(sA - \frac{s^2}{2}\mathbb{E}[A^2]\Big)\exp\Big(tB - \frac{t^2}{2}\mathbb{E}[B^2]\Big) = e^{stC}\exp\Big(sA + tB - \frac{1}{2}\mathbb{E}[(sA+tB)^2]\Big).
$$

Apply (144.220) first with $(m,n) = (1,3)$, $A = X_\epsilon(a)$, $B = X_\epsilon(b)$. The terms $r = 0,1$ have coefficients 1 and 3, and integration against $K_\epsilon(a,b)$ gives (144.217).

Next apply the same identity with $(m,n) = (2,3)$. The coefficients are

$$
\binom{2}{0}\binom{3}{0}0! = 1, \qquad \binom{2}{1}\binom{3}{1}1! = 6, \qquad \binom{2}{2}\binom{3}{2}2! = 6.
$$

Therefore

$$
\begin{aligned}
{:}X_\epsilon(a)^2{:}\,Y_\epsilon(a) &= \int K_\epsilon(a,b)\,:X_\epsilon(a)^2X_\epsilon(b)^3:\,\mathrm{d}b\\
&\quad + 6\int K_\epsilon(a,b)G_\epsilon(a,b)\,:X_\epsilon(a)X_\epsilon(b)^2:\,\mathrm{d}b\\
&\quad + 6\int K_\epsilon(a,b)G_\epsilon(a,b)^2X_\epsilon(b)\,\mathrm{d}b.
\end{aligned}
$$

By (144.218),

$$
3C_{2,\epsilon}(a)X_\epsilon(a) = 6\int K_\epsilon(a,b)G_\epsilon(a,b)^2X_\epsilon(a)\,\mathrm{d}b.
$$

Subtracting this local part from the preceding identity proves (144.219).

The formal homogeneity bookkeeping is now tied to concrete finite-cutoff coordinates. The analytic estimates still to be proved for (144.217) and (144.219) are not generic Holder-product estimates: the sums of the classical regularities are borderline or negative. They are finite-chaos kernel estimates after Wick expansion and BPHZ local subtraction. The target regularity slacks are

$$
0 - |XY| = 4\kappa, \qquad \Big(-\frac{1}{2}\Big) - |X^2Y| = 5\kappa.
$$

Once the fifth-, fourth-, third-, second-, and locally subtracted first-chaos kernels in the two displayed decompositions are bounded with these slacks, the dual-norm finite-chaos estimate, physical-parameter entropy theorem, and negative-sector model convergence theorem above apply without further stochastic input.

Finite-chaos kernels for tested nonlinear coordinates. Let $\mathcal{H}$ be a real Hilbert space and let $\xi : \mathcal{H} \to L^2(\Omega)$ be an isonormal Gaussian process. Let

$$p_\epsilon(a) \in \mathcal{H}$$

be a smooth cutoff noise-to-field kernel and set

$$X_\epsilon(a) = I_1(p_\epsilon(a)), \qquad G_\epsilon(a,b) = \langle p_\epsilon(a), p_\epsilon(b)\rangle_{\mathcal{H}}.$$

Let $K_\epsilon(a,b)$ be the cutoff retarded heat kernel and let χ be a smooth compactly supported test density in the a-variable. Define

$$Y_\epsilon(a) = \int K_\epsilon(a,b) : X_\epsilon(b)^3 : \,\mathrm{d}b.$$

Write Sym_q for orthogonal symmetrization in $\mathcal{H}^{\otimes q}$. Assume that the Hilbert-valued integrands displayed below are Bochner integrable; for the smooth compactly supported cutoffs used here this follows from the L^1-integrability of $|K_\epsilon(a,b)|$, $|K_\epsilon(a,b)G_\epsilon(a,b)|$, and $|K_\epsilon(a,b)G_\epsilon(a,b)^2|$ against the relevant products of $\|p_\epsilon(a)\|_{\mathcal{H}}$, $\|p_\epsilon(b)\|_{\mathcal{H}}$, and $\|p_\epsilon(b) - p_\epsilon(a)\|_{\mathcal{H}}$. Define

$$\begin{aligned} F^\chi_{XY,4} &= \mathrm{Sym}_4 \int \chi(a) K_\epsilon(a,b)\, p_\epsilon(a) \otimes p_\epsilon(b)^{\otimes 3}\,\mathrm{d}a\,\mathrm{d}b, \\ F^\chi_{XY,2} &= 3\ \mathrm{Sym}_2 \int \chi(a) K_\epsilon(a,b) G_\epsilon(a,b)\, p_\epsilon(b)^{\otimes 2}\,\mathrm{d}a\,\mathrm{d}b. \end{aligned} \tag{144.221}$$

Then

$$\int \chi(a) X_\epsilon(a) Y_\epsilon(a)\,\mathrm{d}a = I_4(F^\chi_{XY,4}) + I_2(F^\chi_{XY,2}). \tag{144.222}$$

For the locally subtracted X^2Y coordinate, set

$$\begin{aligned} F^\chi_{X^2Y,5} &= \mathrm{Sym}_5 \int \chi(a) K_\epsilon(a,b)\, p_\epsilon(a)^{\otimes 2} \otimes p_\epsilon(b)^{\otimes 3}\,\mathrm{d}a\,\mathrm{d}b, \\ F^\chi_{X^2Y,3} &= 6\ \mathrm{Sym}_3 \int \chi(a) K_\epsilon(a,b) G_\epsilon(a,b)\, p_\epsilon(a) \otimes p_\epsilon(b)^{\otimes 2}\,\mathrm{d}a\,\mathrm{d}b, \\ F^\chi_{X^2Y,1} &= 6 \int \chi(a) K_\epsilon(a,b) G_\epsilon(a,b)^2 \big(p_\epsilon(b) - p_\epsilon(a)\big)\,\mathrm{d}a\,\mathrm{d}b. \end{aligned} \tag{144.223}$$

Then

$$\begin{aligned} &\int \chi(a) \big(: X_\epsilon(a)^2 : Y_\epsilon(a) - 3C_{2,\epsilon}(a) X_\epsilon(a)\big)\,\mathrm{d}a \\ &\qquad = I_5(F^\chi_{X^2Y,5}) + I_3(F^\chi_{X^2Y,3}) + I_1(F^\chi_{X^2Y,1}). \end{aligned} \tag{144.224}$$

The identities are finite-chaos bookkeeping. The identity

$$: X_\epsilon(a) X_\epsilon(b)^3 := I_4\big(\mathrm{Sym}_4\big(p_\epsilon(a) \otimes p_\epsilon(b)^{\otimes 3}\big)\big)$$

is the definition of the fourth Wick chaos associated to the vector family $p_\epsilon(\cdot)$. Likewise

$$:X_\epsilon(b)^2 := I_2(p_\epsilon(b)^{\otimes 2}).$$

Multiplying the pointwise identity (144.217) by $\chi(a)$, integrating in a, and using the linearity and continuity of finite multiple Wiener integrals gives (144.222) with the kernels in (144.221). The stated Bochner integrability assumption permits approximation of the Hilbert-valued integrals by simple functions. Passing to the Hilbert-space limit and using $\mathbb{E}[I_q(f)I_q(g)] = q!\langle f,g\rangle_{\mathcal{H}^{\otimes q}}$ justifies the interchange with I_q.

For X^2Y, the three Wick monomials in (144.219) are

$$:X_\epsilon(a)^2X_\epsilon(b)^3 := I_5\big(\mathrm{Sym}_5(p_\epsilon(a)^{\otimes 2}\otimes p_\epsilon(b)^{\otimes 3})\big)\,,$$
$$:X_\epsilon(a)X_\epsilon(b)^2 := I_3\big(\mathrm{Sym}_3(p_\epsilon(a)\otimes p_\epsilon(b)^{\otimes 2})\big)\,,$$

and

$$X_\epsilon(b) - X_\epsilon(a) = I_1(p_\epsilon(b) - p_\epsilon(a)).$$

Multiplying by $\chi(a)$, integrating, and again using linearity of I_q gives (144.224). The coefficient of the first-chaos kernel is precisely the local subtraction from (144.218); no additional contraction is hidden in $F^\chi_{X^2Y,1}$.

Proposition 144.70 (Covariance integral formulas for nonlinear chaos kernels)**.** *Use the notation and hypotheses of the preceding finite-chaos kernel calculation. Write*

$$G_{uv} = G_\epsilon(u,v), \qquad K_{ab} = K_\epsilon(a,b),$$

and let all integrals below run over two copies of the spacetime variables (a,b) and (a',b'). Then the Hilbert norms of the tested kernels are

$$\begin{aligned}\|F^\chi_{XY,4}\|^2_{\mathcal{H}^{\otimes 4}} &= \int \chi(a)\chi(a')K_{ab}K_{a'b'}\left(\frac{1}{4}G_{aa'}G^3_{bb'} + \frac{3}{4}G_{ab'}G_{ba'}G^2_{bb'}\right)\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}a'\,\mathrm{d}b',\\ \|F^\chi_{XY,2}\|^2_{\mathcal{H}^{\otimes 2}} &= 9\int \chi(a)\chi(a')K_{ab}K_{a'b'}G_{ab}G_{a'b'}G^2_{bb'}\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}a'\,\mathrm{d}b'.\end{aligned} \tag{144.225}$$

For the locally subtracted X^2Y coordinate,

$$\begin{aligned}\|F^\chi_{X^2Y,5}\|^2_{\mathcal{H}^{\otimes 5}} &= \int \chi(a)\chi(a')K_{ab}K_{a'b'}\Big(\frac{1}{10}G^2_{aa'}G^3_{bb'} + \frac{3}{5}G_{aa'}G_{ab'}G_{ba'}G^2_{bb'}\\ &\qquad\qquad + \frac{3}{10}G^2_{ab'}G^2_{ba'}G_{bb'}\Big)\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}a'\,\mathrm{d}b',\\ \|F^\chi_{X^2Y,3}\|^2_{\mathcal{H}^{\otimes 3}} &= 36\int \chi(a)\chi(a')K_{ab}K_{a'b'}G_{ab}G_{a'b'}\left(\frac{1}{3}G_{aa'}G^2_{bb'} + \frac{2}{3}G_{ab'}G_{ba'}G_{bb'}\right)\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}a'\,\mathrm{d}b',\\ \|F^\chi_{X^2Y,1}\|^2_{\mathcal{H}} &= 36\int \chi(a)\chi(a')K_{ab}K_{a'b'}G^2_{ab}G^2_{a'b'}\\ &\qquad\qquad \times\big(G_{bb'} - G_{ba'} - G_{ab'} + G_{aa'}\big)\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}a'\,\mathrm{d}b'.\end{aligned} \tag{144.226}$$

Consequently, if

$$\begin{aligned}Z^\chi_{XY} &= \int \chi(a)X_\epsilon(a)Y_\epsilon(a)\,\mathrm{d}a,\\ Z^\chi_{X^2Y} &= \int \chi(a)\big(:X_\epsilon(a)^2:Y_\epsilon(a) - 3C_{2,\epsilon}(a)X_\epsilon(a)\big)\,\mathrm{d}a,\end{aligned}$$

then

$$\mathbb{E}\big[|Z^{\chi}_{XY}|^2\big] = 4!\,\|F^{\chi}_{XY,4}\|^2_{\mathcal{H}^{\otimes 4}} + 2!\,\|F^{\chi}_{XY,2}\|^2_{\mathcal{H}^{\otimes 2}},$$
$$\mathbb{E}\big[|Z^{\chi}_{X^2Y}|^2\big] = 5!\,\|F^{\chi}_{X^2Y,5}\|^2_{\mathcal{H}^{\otimes 5}} + 3!\,\|F^{\chi}_{X^2Y,3}\|^2_{\mathcal{H}^{\otimes 3}} + \|F^{\chi}_{X^2Y,1}\|^2_{\mathcal{H}}. \tag{144.227}$$

Proof. For $h_1,\dots,h_q,k_1,\dots,k_q \in \mathcal{H}$,

$$\big\langle \mathrm{Sym}_q(h_1\otimes\cdots\otimes h_q), \mathrm{Sym}_q(k_1\otimes\cdots\otimes k_q)\big\rangle = \frac{1}{q!}\sum_{\sigma\in S_q}\prod_{i=1}^{q}\langle h_i, k_{\sigma(i)}\rangle_{\mathcal{H}}. \tag{144.228}$$

This is the orthogonal projection formula for symmetrization: $\mathrm{Sym}_q^* = \mathrm{Sym}_q$ and $\mathrm{Sym}_q^2 = \mathrm{Sym}_q$.

Apply (144.228) first to $p(a)\otimes p(b)^{\otimes 3}$ and $p(a')\otimes p(b')^{\otimes 3}$. In a random permutation of the four right-hand factors, the single $p(a')$ occupies the single $p(a)$-slot with probability $1/4$, and one of the three $p(b)$-slots with probability $3/4$. This gives the two coefficients in the first line of (144.225). The second line follows because $p(b)^{\otimes 2}$ is already symmetric.

For $p(a)^{\otimes 2}\otimes p(b)^{\otimes 3}$, let k be the number of right-hand $p(a')$-factors landing in the two $p(a)$-slots. The hypergeometric weights are

$$\frac{\binom{2}{2}\binom{3}{0}}{\binom{5}{2}} = \frac{1}{10}, \qquad \frac{\binom{2}{1}\binom{3}{1}}{\binom{5}{2}} = \frac{3}{5}, \qquad \frac{\binom{2}{0}\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10},$$

for $k = 2,1,0$, respectively. These are the three coefficients in $\|F^{\chi}_{X^2Y,5}\|^2$. The same count for $p(a)\otimes p(b)^{\otimes 2}$ gives weights $1/3$ and $2/3$, producing $\|F^{\chi}_{X^2Y,3}\|^2$.

Finally,

$$\langle p(b)-p(a), p(b')-p(a')\rangle_{\mathcal{H}} = G_{bb'} - G_{ba'} - G_{ab'} + G_{aa'},$$

which proves the first-chaos formula. The second-moment identities (144.227) follow from orthogonality of distinct homogeneous Wiener chaoses and the isometry $\mathbb{E}[I_q(f)I_q(g)] = q!\langle f,g\rangle_{\mathcal{H}^{\otimes q}}$. ☐

Proposition 144.71 (Logarithmic power-counting bound for tested XY covariance graphs)**.** *Let $Q=5$. Assign to a heat edge the singular exponent $3 = Q-2$ and to one covariance edge the singular exponent $1 = Q-4$. Consider the three four-vertex multigraphs on*

$$V = \{a,b,a',b'\}$$

which occur in the two XY covariance expressions (144.225):

$$\begin{aligned}
\mathcal{G}_{4,A}:&\quad (a,b)_3,\ (a',b')_3,\ (a,a')_1,\ 3(b,b')_1,\\
\mathcal{G}_{4,B}:&\quad (a,b)_3,\ (a',b')_3,\ (a,b')_1,\ (b,a')_1,\ 2(b,b')_1,\\
\mathcal{G}_{2}:&\quad (a,b)_3,\ (a,b)_1,\ (a',b')_3,\ (a',b')_1,\ 2(b,b')_1.
\end{aligned}$$

The subscript is the singular exponent of the corresponding dyadic kernel; parallel covariance edges are listed with multiplicity. Let

$$s_{\mathcal{G}}(W) = \sum_{\substack{e=(u,v)\in E(\mathcal{G})\\ u,v\in W}} s_e$$

for $W\subset V$, where s_e is the edge singular exponent. Then, for every graph $\mathcal{G}\in\{\mathcal{G}_{4,A},\mathcal{G}_{4,B},\mathcal{G}_2\}$,

$$\sum_{e\in E(\mathcal{G})} s_e = Q(|V|-2) = 10, \tag{144.229}$$

and every proper subset $W \subsetneq V$ with $|W| \geq 2$ obeys the strict subgraph inequality

$$s_{\mathcal{G}}(W) < Q(|W| - 1). \tag{144.230}$$

More precisely, the smallest deficit

$$\Delta_{\mathcal{G}}(W) = Q(|W| - 1) - s_{\mathcal{G}}(W)$$

over proper W is 2 for $\mathcal{G}_{4,A}$, 2 for $\mathcal{G}_{4,B}$, and 1 for $\mathcal{G}_2$.

Consequently the only possible scale loss in the scalar second-moment power counting for the tested XY kernels is an overall logarithm. After dyadic decomposition, all relative-scale sums are geometrically summable; if $2^{-M-1} < \delta \leq 2^{-M}$, the remaining overall-scale sum is bounded by $C(1 + M)$. Hence any scalar covariance integral associated with the three graphs above and tested by two L^1-normalized functions supported in parabolic balls of radius $O(\delta)$ satisfies a power bound

$$|I_\delta| \leq C_\eta \delta^{-\eta} \qquad (\eta > 0), \tag{144.231}$$

provided the dyadic heat and covariance blocks satisfy the support and size bounds used in Lemma 144.54.

Proof. The total singular exponent is $3 + 3 + 1 + 3 = 10$ for $\mathcal{G}_{4,A}$, $3 + 3 + 1 + 1 + 2 = 10$ for $\mathcal{G}_{4,B}$, and $(3 + 1) + (3 + 1) + 2 = 10$ for $\mathcal{G}_2$. This proves (144.229).

It remains to check proper subgraphs. For two-vertex subsets, the largest internal singular exponents are

$$\max_{|W|=2} s_{\mathcal{G}_{4,A}}(W) = 3, \qquad \max_{|W|=2} s_{\mathcal{G}_{4,B}}(W) = 3, \qquad \max_{|W|=2} s_{\mathcal{G}_2}(W) = 4.$$

The two fourth-chaos graphs therefore have two-vertex deficit at least 2, while the second-chaos graph has the smaller deficit 1, coming from the parallel heat-covariance pairs $\{a, b\}$ and $\{a', b'\}$. For three vertices, inspection of the three displayed edge lists gives the following maximal internal singular exponents:

$$\max_{|W|=3} s_{\mathcal{G}_{4,A}}(W) = 6, \qquad \max_{|W|=3} s_{\mathcal{G}_{4,B}}(W) = 6, \qquad \max_{|W|=3} s_{\mathcal{G}_2}(W) = 6.$$

Because $Q(|W| - 1) = 10$ when $|W| = 3$, these deficits are at least 4. Thus (144.230) holds, with the stated minimum deficits $2, 2, 1$.

Now decompose every heat and covariance factor into dyadic blocks. A relative-scale sector is obtained by choosing one overall scale n for a spanning cluster of all four vertices and non-negative gap variables recording how much smaller each proper connected subcluster is. The support and size assumptions give, in that sector, a factor

$$2^{-\sum_W \Delta_{\mathcal{G}}(W) r_W}$$

up to a constant depending only on the dyadic partition and on the finite graph. The strict inequalities just proved imply $\Delta_{\mathcal{G}}(W) \geq 1$ for every proper cluster which can occur in the sector decomposition, so Theorem 144.58 applies to the relative gap variables.

After those relative sums have been performed, the exponent of the remaining overall scale is zero, precisely because of (144.229) and because the two external test functions are L^1-normalized. Thus the contribution of all overall scales $0 \le n \le M + O(1)$ is $O(1 + M)$; larger scales are killed by the test-function support and the same proper-cluster estimates. Finally, for every $\eta > 0$,

$$1 + M \le C_\eta 2^{\eta M} \le C_\eta \delta^{-\eta}.$$

This proves (144.231). □

Proposition 144.72 (Scalar tested XY coordinate bound)**.** *Let the hypotheses of Proposition 144.71 hold uniformly for the cutoff heat and covariance blocks on a compact spacetime set. Let $\varphi \in C_c^r(\mathbb{R}^{1+3})$ be supported in the unit parabolic ball, let $\|\varphi\|_{C^r} \le 1$, and define the L^1-normalized test density*

$$\varphi_z^\delta(a) = \delta^{-Q} \varphi(\mathsf{D}_\delta^{-1}(a - z)), \qquad Q = 5.$$

For the finite-cutoff field, set

$$Z_{\epsilon,\delta,z}^{XY}(\varphi) = \int \varphi_z^\delta(a) X_\epsilon(a) Y_\epsilon(a) \, \mathrm{d}a. \tag{144.232}$$

Then, for every $\eta > 0$, there is a constant C_η, uniform for ϵ and for z in the chosen compact set, such that

$$\mathbb{E}\big|Z_{\epsilon,\delta,z}^{XY}(\varphi)\big|^2 \le C_\eta \delta^{-\eta}. \tag{144.233}$$

Consequently, for every even $p \ge 2$ and every

$$0 < \zeta_{XY} < 4\kappa,$$

there is a constant $C_{p,\zeta_{XY}}$ such that

$$\big\|\delta^{4\kappa} Z_{\epsilon,\delta,z}^{XY}(\varphi)\big\|_{L^p(\Omega)} \le C_{p,\zeta_{XY}} \delta^{4\kappa - \zeta_{XY}}. \tag{144.234}$$

This is the scalar tested base estimate for the XY coordinate. The full input required by Proposition 144.79 additionally requires the same multiscale estimate after dualizing the test-function variable and after taking differences in the parameters (z, δ).

Proof. Insert $\chi = \varphi_z^\delta$ in (144.225). The density φ_z^δ is L^1-normalized, and its support lies in a parabolic ball of radius $O(\delta)$. Each of the three scalar covariance graphs in (144.225) is therefore bounded by Proposition 144.71. The numerical coefficients $1/4$, $3/4$, and 9, and the chaos factors $4!$ and $2!$, are finite fixed constants. Thus (144.227) gives (144.233).

The random variable $Z_{\epsilon,\delta,z}^{XY}(\varphi)$ is the sum of a fourth-chaos and a second-chaos variable by (144.222). Applying Lemma 144.51 to these two homogeneous chaos components gives

$$\big\|Z_{\epsilon,\delta,z}^{XY}(\varphi)\big\|_{L^p} \le C_p \big\|Z_{\epsilon,\delta,z}^{XY}(\varphi)\big\|_{L^2}.$$

Choose $\eta = 2\zeta_{XY}$ in (144.233). Multiplication by the model normalization

$$\delta^{-|XY|} = \delta^{4\kappa}$$

then gives (144.234). □

Proposition 144.73 (Scalar tested XY parameter-edge bound)**.** *Keep the hypotheses of Proposition 144.72. Let $r \geq 1$, $0 < \theta \leq 1$, and let*

$$\mathcal{X}_{\Pi,m} = K \times [1/2, 1], \qquad d_{\Pi,m}((z,a),(z',a')) = 2^m \|z - z'\|_{\mathfrak{s}} + |a - a'|.$$

For $u = (z,a) \in \mathcal{X}_{\Pi,m}$, put

$$\delta_u = 2^{-m} a, \qquad \varphi_u(a_0) = \delta_u^{-Q} \varphi(\mathsf{D}_{\delta_u}^{-1}(a_0 - z)), \qquad \|\varphi\|_{C^r} \leq 1.$$

Define

$$Z^{XY}_{\epsilon,m,u}(\varphi) = \int \varphi_u(a_0) X_\epsilon(a_0) Y_\epsilon(a_0) \, \mathrm{d}a_0.$$

Then, whenever $d_{\Pi,m}(u,v) \leq 1$, for every even $p \geq 2$ and every $0 < \zeta_{XY} < 4\kappa$,

$$\left\|2^{-4\kappa m}\big(Z^{XY}_{\epsilon,m,v}(\varphi) - Z^{XY}_{\epsilon,m,u}(\varphi)\big)\right\|_{L^p(\Omega)} \leq C_{p,\zeta_{XY},\theta} 2^{-(4\kappa - \zeta_{XY})m} d_{\Pi,m}(u,v)^\theta. \tag{144.235}$$

The constant is uniform in the cutoff and in u, v with centers in the fixed compact set.

Proof. First record the deterministic variation of the transported test density. For $r \geq 1$, the map

$$(z,a) \longmapsto (2^{-m}a)^{-Q} \varphi(\mathsf{D}_{(2^{-m}a)^{-1}}(\,\cdot - z))$$

is Lipschitz from $\mathcal{X}_{\Pi,m}$ to L^1, uniformly in m and in $\|\varphi\|_{C^1} \leq 1$. Indeed, translation by z differentiates the test by $\nabla_{\mathfrak{s}}\varphi$ and produces the dimensionless factor $2^m\|z - z'\|_{\mathfrak{s}}$; differentiating in the scale parameter $a \in [1/2, 1]$ differentiates the map

$$a^{-Q} \varphi(\mathsf{D}_{a^{-1}} w)$$

on a compact interval. Therefore

$$\|\varphi_v - \varphi_u\|_{L^1} \leq C\, d_{\Pi,m}(u,v) \leq C\, d_{\Pi,m}(u,v)^\theta, \tag{144.236}$$

because $d_{\Pi,m}(u,v) \leq 1$. The supports of φ_u and φ_v lie in a common parabolic ball of radius $C2^{-m}$.

Set $\Delta\varphi = \varphi_v - \varphi_u$. By linearity,

$$Z^{XY}_{\epsilon,m,v}(\varphi) - Z^{XY}_{\epsilon,m,u}(\varphi) = \int \Delta\varphi(a_0) X_\epsilon(a_0) Y_\epsilon(a_0) \, \mathrm{d}a_0.$$

The covariance-graph estimate used in Proposition 144.72 is homogeneous in the two external test densities. Applying it to $\Delta\varphi$ in both slots, using (144.236), and taking $\delta = 2^{-m}$ gives, for every $\eta > 0$,

$$\mathbb{E}\left|Z^{XY}_{\epsilon,m,v}(\varphi) - Z^{XY}_{\epsilon,m,u}(\varphi)\right|^2 \leq C_\eta d_{\Pi,m}(u,v)^{2\theta} 2^{\eta m}. \tag{144.237}$$

The difference is again a sum of fourth- and second-chaos variables, so the finite-chaos moment bound gives its L^p-norm from the square root of (144.237). Choose $\eta = 2\zeta_{XY}$ and multiply by the model normalization $2^{-4\kappa m}$. This proves (144.235). $\square$

Proposition 144.74 (Scalar tested XY cutoff-shell gain)**.** *Keep the notation and graph hypotheses of Propositions 144.71 and 144.72. Let (X_n, Y_n) be a dyadic ultraviolet cutoff sequence, and write*

$$Z^{XY}_{n,m,u}(\varphi) = \int \varphi_u(a_0) X_n(a_0) Y_n(a_0)\, \mathrm{d}a_0$$

with φ_u as in Proposition 144.73. Assume the cutoff difference $Z^{XY}_{n+1,m,u} - Z^{XY}_{n,m,u}$ has the following admissible shell representation: after expanding the second moment by the finite-chaos formulas (144.222) and (144.225), every term is a finite linear combination of the three XY graph integrals in Proposition 144.71, with at least one heat or covariance edge replaced by a cutoff-shell block whose dyadic scale is $\geq n - c_0$. The shell block satisfies the same support and size bounds as the corresponding ordinary dyadic block, with constants independent of n.

Then, for every $\eta > 0$, every $0 < \rho < 1/2$, and every $u \in \mathcal{X}_{\Pi,m}$,

$$\mathbb{E}\left|Z^{XY}_{n+1,m,u}(\varphi) - Z^{XY}_{n,m,u}(\varphi)\right|^2 \leq C_{\eta,\rho}\, 2^{\eta m}\, 2^{-2\rho(n-m)_+}. \tag{144.238}$$

Consequently, for every even $p \geq 2$ and every

$$0 < \zeta_{XY} < 4\kappa,$$

$$\left\|2^{-4\kappa m}\big(Z^{XY}_{n+1,m,u}(\varphi) - Z^{XY}_{n,m,u}(\varphi)\big)\right\|_{L^p(\Omega)} \leq C_{p,\zeta_{XY},\rho} 2^{-(4\kappa-\zeta_{XY})m} 2^{-\rho(n-m)_+}. \tag{144.239}$$

The estimate is uniform for u with center in a fixed compact set.

Proof. Let $M = m + O(1)$ be the largest overall graph scale that can interact with two L^1-normalized test densities supported in parabolic balls of radius $O(2^{-m})$. The unconstrained graph estimate in Proposition 144.71 leaves only the overall-scale sum $0 \leq N \leq M + O(1)$, producing the logarithmic loss recorded there.

Now impose the shell condition from the cutoff increment. In the dyadic sector decomposition used in the proof of Proposition 144.71, the edge at scale $\geq n - c_0$ either lies at the overall scale or creates a proper subcluster whose relative-scale gap is at least $(n - M - C)_+$. The first alternative can occur only when $n \leq M + C$. In the second alternative, the sector carries the additional factor

$$2^{-\Delta r}$$

with $r \geq n - M - C$ and with $\Delta \geq 1$, because the smallest proper-subgraph deficit among the three XY graphs is 1. Therefore every sector with $n > M + C$ gains at least $2^{-(n-M-C)}$. Enlarging the constant absorbs the fixed C and gives, for every graph contribution,

$$|I^{\Delta XY}_{n,m}| \leq C_\eta 2^{\eta m} 2^{-(n-m)_+}.$$

Since $0 < \rho < 1/2$, the weaker bound

$$2^{-(n-m)_+} \leq C_\rho 2^{-2\rho(n-m)_+}$$

holds uniformly in m, n. Summing the finite list of graph terms in the admissible shell representation gives (144.238).

The cutoff difference is still a finite sum of homogeneous Wiener-chaos variables of orders 4 and 2. Lemma 144.51 therefore bounds its L^p-norm by a constant times the square root of (144.238). Choose $\eta = 2\zeta_{XY}$ and multiply by the model normalization $2^{-4\kappa m}$. This proves (144.239). $\square$

Proposition 144.75 (Scalar tested high-chaos X^2Y bound)**.** *Let $Q = 5$. Use the same dyadic heat and covariance block hypotheses as in Proposition 144.71. Consider the five four-vertex multigraphs on $V = \{a, b, a', b'\}$ which occur in the fifth- and third-chaos parts of (144.226):*

$$\begin{aligned}
\mathcal{H}_{5,A}: &\quad (a,b)_3,\ (a',b')_3,\ 2(a,a')_1,\ 3(b,b')_1,\\
\mathcal{H}_{5,B}: &\quad (a,b)_3,\ (a',b')_3,\ (a,a')_1,\ (a,b')_1,\ (b,a')_1,\ 2(b,b')_1,\\
\mathcal{H}_{5,C}: &\quad (a,b)_3,\ (a',b')_3,\ 2(a,b')_1,\ 2(b,a')_1,\ (b,b')_1,\\
\mathcal{H}_{3,A}: &\quad (a,b)_3,\ (a,b)_1,\ (a',b')_3,\ (a',b')_1,\ (a,a')_1,\ 2(b,b')_1,\\
\mathcal{H}_{3,B}: &\quad (a,b)_3,\ (a,b)_1,\ (a',b')_3,\ (a',b')_1,\ (a,b')_1,\ (b,a')_1,\ (b,b')_1.
\end{aligned}$$

The subscript again denotes the singular exponent of the edge. For every one of these graphs,

$$\sum_{e\in E(\mathcal{H})} s_e = Q(|V|-2)+1 = 11. \tag{144.240}$$

Every proper subset $W \subsetneq V$ with $|W| \geq 2$ obeys

$$s_{\mathcal{H}}(W) < Q(|W|-1).$$

The minimum proper-subgraph deficit is 2 for each fifth-chaos graph and 1 for each third-chaos graph.

Consequently, if χ_z^δ is an L^1-normalized test density supported in a parabolic ball of radius $O(\delta)$, then the scalar covariance integrals belonging to the fifth- and third-chaos X^2Y kernels obey

$$|I_\delta^{X^2Y,\geq 3}| \leq C\delta^{-1}, \qquad 0 < \delta \leq 1. \tag{144.241}$$

Equivalently, if

$$Z_{\epsilon,\delta,z}^{X^2Y,\geq 3}(\varphi) = I_5(F^{\chi_z^\delta}_{X^2Y,5}) + I_3(F^{\chi_z^\delta}_{X^2Y,3}),$$

then, uniformly in the cutoff and for z in a fixed compact set,

$$\mathbb{E}\left|Z_{\epsilon,\delta,z}^{X^2Y,\geq 3}(\varphi)\right|^2 \leq C\delta^{-1}. \tag{144.242}$$

For every even $p \geq 2$,

$$\left\|\delta^{1/2+5\kappa} Z_{\epsilon,\delta,z}^{X^2Y,\geq 3}(\varphi)\right\|_{L^p(\Omega)} \leq C_p\delta^{5\kappa}. \tag{144.243}$$

Proof. The edge lists give total degree $3+3+5 = 11$ for the three fifth-chaos graphs and $3+1+3+1+3 = 11$ for the two third-chaos graphs, which proves (144.240). For two-vertex subsets, the largest internal singular exponent is 3 in the fifth-chaos graphs and 4 in the third-chaos graphs. These give deficits 2 and 1, respectively, relative to $Q(|W|-1) = 5$. For three-vertex subsets, inspection of the displayed edge lists gives maximal internal singular exponent 6 in all five graphs, hence deficit 4 relative to $Q(|W|-1) = 10$. This proves the strict proper-subgraph inequalities and the stated minimum deficits.

The dyadic sector argument uses the same multiscale sector decomposition as the XY graph bound above. The strict proper-subgraph deficits make every relative-scale gap geometrically summable by the multiscale sector summability theorem. After the relative sums have been performed, the remaining overall scale has exponent one, because the total graph degree is $Q(|V|-2)+1$ rather than $Q(|V|-2)$. If $2^{-M-1} < \delta \le 2^{-M}$, the relevant overall scales therefore contribute

$$\sum_{0\le n\le M+O(1)} 2^n \le C2^M \le C\delta^{-1},$$

and scales larger than $M+O(1)$ are excluded by the test supports together with the same relative-cluster estimates. This proves (144.241).

Insert $\chi = \chi_z^\delta$ in the fifth- and third-chaos lines of (144.226). The fixed numerical coefficients and the chaos factors 5! and 3! are absorbed into C, and (144.227) gives (144.242). The variable $Z^{X^2Y,\ge 3}_{\epsilon,\delta,z}$ is a finite sum of homogeneous Wiener chaoses of orders 5 and 3; the finite-chaos moment estimate converts the L^2-bound to L^p. Multiplication by the model normalization $\delta^{1/2+5\kappa}$, corresponding to true target regularity $-1/2$ and assigned regularity $-1/2-5\kappa$, gives (144.243). $\square$

Proposition 144.76 (Scalar tested high-chaos X^2Y parameter-edge bound)**.** *Keep the hypotheses of Proposition 144.75. Let*

$$\mathcal{X}_{\Pi,m} = K\times[1/2,1], \qquad d_{\Pi,m}((z,a),(z',a')) = 2^m\|z-z'\|_{\mathfrak{s}} + |a-a'|,$$

and let

$$\varphi_u(a_0) = \delta_u^{-Q}\varphi(\mathsf{D}_{\delta_u}^{-1}(a_0-z)), \qquad \delta_u = 2^{-m}a, \qquad \|\varphi\|_{C^r}\le 1,$$

with $r\ge 1$. Define

$$Z^{X^2Y,\ge 3}_{\epsilon,m,u}(\varphi) = I_5(F^{\varphi_u}_{X^2Y,5}) + I_3(F^{\varphi_u}_{X^2Y,3}).$$

Then, whenever $d_{\Pi,m}(u,v)\le 1$,

$$\mathbb{E}\left|Z^{X^2Y,\ge 3}_{\epsilon,m,v}(\varphi) - Z^{X^2Y,\ge 3}_{\epsilon,m,u}(\varphi)\right|^2 \le C_\theta\, 2^m\, d_{\Pi,m}(u,v)^{2\theta}, \qquad 0<\theta\le 1. \tag{144.244}$$

Consequently, for every even $p\ge 2$,

$$\left\|2^{-(1/2+5\kappa)m}\big(Z^{X^2Y,\ge 3}_{\epsilon,m,v}(\varphi) - Z^{X^2Y,\ge 3}_{\epsilon,m,u}(\varphi)\big)\right\|_{L^p(\Omega)} \le C_{p,\theta}\, 2^{-5\kappa m} d_{\Pi,m}(u,v)^\theta. \tag{144.245}$$

Proof. The transported test-density estimate (144.236) is deterministic and applies here unchanged:

$$\|\varphi_v-\varphi_u\|_{L^1} \le Cd_{\Pi,m}(u,v)^\theta,$$

and the two supports lie in one parabolic ball of radius $C2^{-m}$. The graph estimate proved inside Proposition 144.75 is homogeneous in the two external test densities: replacing the L^1-normalized test in that proof by a signed test ψ supported in the same parabolic ball adds the factor $\|\psi\|^2_{L^1}$ to the covariance integral and changes no scale exponent. Applying this homogeneous form to $\psi=\varphi_v-\varphi_u$, with $\delta\simeq 2^{-m}$, gives (144.244). The difference is still a finite sum of homogeneous Wiener chaoses of orders 5 and 3. The finite-chaos moment estimate converts the square root of (144.244) to an L^p-bound. Multiplication by the model normalization $2^{-(1/2+5\kappa)m}$ leaves the scale factor $2^{-5\kappa m}$, proving (144.245). $\square$

Proposition 144.77 (Scalar tested high-chaos X^2Y cutoff-shell gain)**.** *Keep the notation and graph hypotheses of Proposition 144.75. Let*

$$Z^{X^2Y,\ge 3}_{n,m,u}(\varphi) = I_5(F^{\varphi_u}_{n,X^2Y,5}) + I_3(F^{\varphi_u}_{n,X^2Y,3})$$

be the high-chaos scalar tested coordinate at dyadic ultraviolet cutoff n. Assume the cutoff difference $Z^{X^2Y,\geq 3}_{n+1,m,u} - Z^{X^2Y,\geq 3}_{n,m,u}$ has the following admissible shell representation: after expanding the second moment by (144.224) and (144.226), every term is a finite linear combination of the five high-chaos graph integrals in Proposition 144.75, with at least one heat or covariance edge replaced by a cutoff-shell block whose dyadic scale is $\geq n - c_0$. The shell block satisfies the same support and size bounds as the corresponding ordinary dyadic block, with constants independent of n.

Then, for every $0 < \rho < 1/2$,

$$\mathbb{E}\left|Z^{X^2Y,\geq 3}_{n+1,m,u}(\varphi) - Z^{X^2Y,\geq 3}_{n,m,u}(\varphi)\right|^2 \leq C_\rho\, 2^m\, 2^{-2\rho(n-m)_+}. \tag{144.246}$$

Consequently, for every even $p \geq 2$,

$$\left\|2^{-(1/2+5\kappa)m}\big(Z^{X^2Y,\geq 3}_{n+1,m,u}(\varphi) - Z^{X^2Y,\geq 3}_{n,m,u}(\varphi)\big)\right\|_{L^p(\Omega)} \leq C_{p,\rho}\, 2^{-5\kappa m} 2^{-\rho(n-m)_+}. \tag{144.247}$$

Proof. Let $M = m + O(1)$ be the largest overall scale interacting with the two scale-2^{-m} test densities. Without the cutoff-shell constraint, Proposition 144.75 leaves the overall sum

$$\sum_{0\leq N\leq M+O(1)} 2^N = O(2^m).$$

Now impose the shell condition. If the shell edge is not at the overall scale, it lies inside a proper subcluster whose relative-scale gap is at least $(n-M-C)_+$. The smallest proper-subgraph deficit among the five high-chaos X^2Y graphs is 1, so that sector gains

$$2^{-(n-M-C)_+}.$$

If the shell edge is at the overall scale, then $n \leq M + C$, and the same bound is absorbed into the constant. Thus every graph contribution is bounded by

$$C2^m 2^{-(n-m)_+}.$$

Since $0 < \rho < 1/2$, this implies the weaker bound (144.246). The cutoff difference is a finite sum of homogeneous chaoses of orders 5 and 3, so the finite-chaos moment estimate and the normalization $2^{-(1/2+5\kappa)m}$ give (144.247). $\square$

Remark 144.78 (Scalar shell gain and projective cutoff input). Estimate (144.239) is a scale-separated scalar Cauchy estimate. It identifies where the ultraviolet shell gain enters the XY graph analysis. Proposition 144.75 supplies the parallel scalar base estimate for the fifth- and third-chaos pieces of X^2Y, while Propositions 144.76 and 144.77 supply the corresponding scalar edge and shell layers. The remaining X^2Y first-chaos component is different: its local subtraction produces covariance increments, treated below in Propositions 144.81 and 144.82. Corollary 144.61 then converts this scalar factor into a scale-summed cutoff rate whenever the normalization slack exceeds the parameter entropy. The projective cutoff-increment hypothesis (144.254) asks for a uniform factor $2^{-\rho n}$ in the full E'_r-dualized kernel norm. The scalar estimate gives the finite-coordinate shell mechanism; the dualized cutoff-increment estimate and its parameter-edge version supply the corresponding projective-kernel input needed by the Φ^4_3 model-convergence theorem.

Proposition 144.79 (Nonlinear Π-coordinate input from projective kernel bounds)**.** *Let $K \subset \mathbb{R}^{1+3}$ be compact and let*

$$\mathcal{X}_{\Pi,m} = K \times [1/2, 1], \qquad d_{\Pi,m}((z,a),(z',a')) = 2^m \|z - z'\|_{\mathfrak{s}} + |a - a'|$$

be the physical Π*-coordinate spaces of Proposition 144.66. Let*

$$E_r = \{\varphi \in C^r(\mathbb{R}^{1+3}) : \operatorname{supp}\varphi \subset B_{\mathfrak{s}}(0,1)\}, \qquad \|\varphi\|_{E_r} = \|\varphi\|_{C^r}.$$

Fix

$$0 < \kappa < \frac{1}{14}, \qquad 0 \le \zeta_{XY} < 4\kappa, \qquad 0 < \zeta_{X^2Y} < 5\kappa,$$

and set

$$\begin{gathered} |XY| = -4\kappa, \qquad |X^2Y| = -\frac{1}{2} - 5\kappa, \\ \sigma_{XY} = 4\kappa - \zeta_{XY}, \qquad \sigma_{X^2Y} = 5\kappa - \zeta_{X^2Y}. \end{gathered} \tag{144.248}$$

The loss ζ_{XY} is optional: it may be set to zero only after the concrete XY projective kernel estimate proves that the marginal overall-scale logarithm in Proposition 144.71 is absent or harmless in the chosen norm. The loss ζ_{X^2Y} is reserved for logarithmic factors in the locally subtracted first-chaos component of X^2Y.

For

$$\mathfrak{C}_{\mathrm{nl}} = \{XY, X^2Y\}, \qquad \mathcal{Q}_{XY} = \{4, 2\}, \qquad \mathcal{Q}_{X^2Y} = \{5, 3, 1\},$$

let $F_{n,\tau,q,m}(u) \in \mathcal{H}^{\odot q} \widehat{\otimes}_\pi E_r'$, where

$$n \ge 1, \qquad \tau \in \mathfrak{C}_{\mathrm{nl}}, \qquad q \in \mathcal{Q}_\tau, \qquad m \ge 0, \qquad u = (z,a) \in \mathcal{X}_{\Pi,m}.$$

Assume that the E_r'-valued chaos expansion

$$Y_{n,\tau,m}(u) = \sum_{q \in \mathcal{Q}_\tau} I_q(F_{n,\tau,q,m}(u)) \tag{144.249}$$

is the scale-normalized tested Π*-coordinate:*

$$Y_{n,\tau,m}(z,a)(\varphi) = (2^{-m}a)^{-|\tau|} \left\langle \Pi_z^{(n)}\tau, \varphi_z^{2^{-m}a} \right\rangle. \tag{144.250}$$

For X^2Y, the kernels are those of the locally subtracted coordinate (144.219); equivalently, after pairing with a test density, they reduce to (144.223). For XY, they reduce to (144.221).

Choose $0 < \theta \le 1$, an even integer $p = 2m_0$, and $\rho > 0$ such that

$$p\sigma_\tau > 5 \quad (\tau \in \mathfrak{C}_{\mathrm{nl}}), \qquad p\theta > 6. \tag{144.251}$$

Suppose that there are constants

$$B_{\tau,q,0},\ B_{\tau,q,1},\ \widetilde{B}_{\tau,q,0},\ \widetilde{B}_{\tau,q,1} < \infty$$

such that, for every n, m, τ, q and $u, v \in \mathcal{X}_{\Pi,m}$ with $d_{\Pi,m}(u,v) \le 1$,

$$\|F_{n,\tau,q,m}(u)\|_\pi \le B_{\tau,q,0} 2^{-\sigma_\tau m}, \tag{144.252}$$

$$\|F_{n,\tau,q,m}(v) - F_{n,\tau,q,m}(u)\|_\pi \le B_{\tau,q,1} 2^{-\sigma_\tau m} d_{\Pi,m}(u,v)^\theta, \tag{144.253}$$

and the cutoff increments obey

$$\|F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u)\|_\pi \le \widetilde{B}_{\tau,q,0} 2^{-\rho n} 2^{-\sigma_\tau m}, \tag{144.254}$$

$$\begin{aligned} &\|(F_{n+1,\tau,q,m}(v) - F_{n,\tau,q,m}(v)) - (F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u))\|_\pi \\ &\quad \le \widetilde{B}_{\tau,q,1} 2^{-\rho n} 2^{-\sigma_\tau m} d_{\Pi,m}(u,v)^\theta. \end{aligned} \tag{144.255}$$

Then the nonlinear coordinates XY and X^2Y satisfy the Π-coordinate hypotheses of the negative-sector coordinate-control consequence with

$$D_\tau = 5, \qquad d_\tau = 6, \qquad \varepsilon_\tau = p\theta - 6. \tag{144.256}$$

In particular, if

$$U_{n,\tau,m}(u) = \|Y_{n,\tau,m}(u)\|_{E'_r}$$

and $V_{n,\tau,m}$ is defined by replacing $Y_{n,\tau,m}$ with $Y_{n+1,\tau,m} - Y_{n,\tau,m}$, then, on the nets of Proposition 144.66,

$$\mathbb{E}|U_{n,\tau,m}(u)|^p \le A_{\tau,0}^p 2^{-p\sigma_\tau m}, \qquad \mathbb{E}|V_{n,\tau,m}(u)|^p \le \widetilde{A}_{\tau,0}^p 2^{-p\rho n} 2^{-p\sigma_\tau m},$$

and

$$\mathbb{E}|U_{n,\tau,m}(v) - U_{n,\tau,m}(u)|^p \le A_{\tau,1}^p 2^{-p\sigma_\tau m} 2^{-(6+\varepsilon_\tau)\ell},$$

$$\mathbb{E}|V_{n,\tau,m}(v) - V_{n,\tau,m}(u)|^p \le \widetilde{A}_{\tau,1}^p 2^{-p\rho n} 2^{-p\sigma_\tau m} 2^{-(6+\varepsilon_\tau)\ell}$$

for all $(u,v) \in \mathcal{E}_{\Pi,m,\ell}$. The constants are

$$A_{\tau,i} = \sum_{q\in\mathcal{Q}_\tau} C_{q,m_0} B_{\tau,q,i}, \qquad \widetilde{A}_{\tau,i} = \sum_{q\in\mathcal{Q}_\tau} C_{q,m_0} \widetilde{B}_{\tau,q,i}, \qquad i = 0,1, \tag{144.257}$$

where C_{q,m_0} is the finite-chaos constant in Proposition 144.52.

Proof. By (144.250),

$$\|Y_{n,\tau,m}(z,a)\|_{E'_r} = \sup_{\varphi\in\mathcal{B}_r} (2^{-m}a)^{-|\tau|} \left| \left\langle \Pi_z^{(n)}\tau, \varphi_z^{2^{-m}a} \right\rangle \right|.$$

Thus $U_{n,\tau,m}$ is exactly the physical Π-coordinate $Q^{\#}_{n,\tau}$ restricted to dyadic scale m, and the analogous statement holds for cutoff differences.

Apply Proposition 144.52 to (144.249). The projective base bound (144.252) gives

$$\|U_{n,\tau,m}(u)\|_{L^p(\Omega)} \le \sum_{q\in\mathcal{Q}_\tau} C_{q,m_0} B_{\tau,q,0} 2^{-\sigma_\tau m} = A_{\tau,0} 2^{-\sigma_\tau m}. \tag{144.258}$$

The cutoff base estimate follows in the same way from (144.254). For increments, use

$$\big| \|A\|_{E'_r} - \|B\|_{E'_r} \big| \le \|A - B\|_{E'_r}$$

and then apply the finite-chaos estimate to

$$Y_{n,\tau,m}(v) - Y_{n,\tau,m}(u) = \sum_{q\in\mathcal{Q}_\tau} I_q(F_{n,\tau,q,m}(v) - F_{n,\tau,q,m}(u)).$$

This proves

$$\|U_{n,\tau,m}(v) - U_{n,\tau,m}(u)\|_{L^p} \le A_{\tau,1} 2^{-\sigma_\tau m} d_{\Pi,m}(u,v)^\theta, \tag{144.259}$$

and the cutoff edge estimate follows from (144.255).

For an edge $(u,v) \in \mathcal{E}_{\Pi,m,\ell}$, the construction in Proposition 144.66 gives

$$d_{\Pi,m}(u,v) \le C2^{-\ell}.$$

Absorbing the fixed constant C^θ into $A_{\tau,1}$ and $\widetilde{A}_{\tau,1}$, and then raising the L^p-bounds to the p-th power, gives the displayed edge moments because

$$p\theta = 6 + \varepsilon_\tau.$$

The same entropy proposition gives $D_\tau = 5$ and $d_\tau = 6$. The conditions (144.251) are precisely

$$\sigma_\tau > \frac{D_\tau}{p}, \qquad \varepsilon_\tau > 0.$$

The required sample continuity follows from the same finite-net estimate and does not require an additional stochastic input. For fixed n, τ, m, choose nested $2^{-\ell}$-nets in $\mathcal{X}_{\Pi,m}$ as in Proposition 144.66. The edge family at level ℓ has cardinality at most

$$C2^{5m}2^{6\ell}.$$

By (144.259),

$$\sum_{\ell\ge 0} \mathbb{E} \sum_{(u,v)\in\mathcal{E}_{\Pi,m,\ell}} |U_{n,\tau,m}(v) - U_{n,\tau,m}(u)|^p \le C_{n,\tau,m} \sum_{\ell\ge 0} 2^{-(p\theta-6)\ell} < \infty.$$

Fubini therefore gives almost sure summability of the p-th powers of the dyadic edge increments. Since a summable nonnegative sequence tends to zero along the levels, the usual nearest-net telescoping sequence is almost surely Cauchy uniformly along every chain of nested approximants. This defines a continuous modification on the compact metric space $\mathcal{X}_{\Pi,m}$. Applying the identical dyadic-edge summability estimate to $V_{n,\tau,m}$, with the V-moment bounds in place of the U-moment bounds, gives the required continuous modification for the cutoff-difference field. Hence all hypotheses needed for the negative-sector coordinate-control consequence are satisfied for the nonlinear Π-coordinates. □

Corollary 144.80 (Shell-separated nonlinear Π-coordinate cutoff input)**.** *Keep the notation and hypotheses of Proposition 144.79 for the uniform base and parameter-edge bounds* (144.252) *and* (144.253)*. Replace the cutoff assumptions* (144.254) *and* (144.255) *by the shell-separated estimates*

$$\|F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u)\|_\pi \le \widehat{B}_{\tau,q,0} 2^{-\sigma_\tau m} 2^{-\rho_\tau (n-m)_+}, \tag{144.260}$$

and

$$\begin{aligned}\|(F_{n+1,\tau,q,m}(v) - F_{n,\tau,q,m}(v)) - (F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u))\|_\pi \\ \le \widehat{B}_{\tau,q,1} 2^{-\sigma_\tau m} 2^{-\rho_\tau (n-m)_+} d_{\Pi,m}(u,v)^\theta,\end{aligned} \tag{144.261}$$

for constants $\rho_\tau > 0$. Let $p = 2m_0$ be the chosen even moment order, and assume

$$p\sigma_\tau > 5, \qquad p\theta > 6.$$

For any

$$0 < \rho_\tau^* < \min\left\{\rho_\tau,\, \sigma_\tau - \frac{5}{p}\right\}, \tag{144.262}$$

there is a finite constant $\widehat{S}_\tau^{\mathrm{sh}}$ *such that the cutoff-difference field*

$$V_{n,\tau,m}(u) = \left\| \sum_{q\in\mathcal{Q}_\tau} I_q(F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u)) \right\|_{E'_\tau}$$

satisfies the scale-summed estimate

$$\left\| \sup_{m\geq 0} \sup_{u\in\mathcal{X}_{\Pi,m}} V_{n,\tau,m}(u) \right\|_{L^p(\Omega)} \leq \widehat{S}_\tau^{\mathrm{sh}}\, 2^{-\rho_\tau^* n}. \tag{144.263}$$

One may take

$$\begin{aligned} \widehat{S}_\tau^{\mathrm{sh}} &= H_{\rho_\tau^*,\rho_\tau,\sigma_\tau-5/p}\left(C_{\Pi,0}^{1/p}\widehat{A}_{\tau,0} + C_{\Pi,1}^{1/p}\widehat{A}_{\tau,1}(1-2^{-(p\theta-6)/p})^{-1}\right), \\ \widehat{A}_{\tau,i} &= \sum_{q\in\mathcal{Q}_\tau} C_{q,m_0}\widehat{B}_{\tau,q,i}, \qquad i=0,1. \end{aligned} \tag{144.264}$$

Proof. For the cutoff-difference chaos kernels, apply Proposition 144.52 exactly as in the proof of Proposition 144.79. The base estimate gives

$$\|V_{n,\tau,m}(u)\|_{L^p} \leq \widehat{A}_{\tau,0} 2^{-\sigma_\tau m} 2^{-\rho_\tau (n-m)_+}.$$

For an edge $(u,v)\in\mathcal{E}_{\Pi,m,\ell}$, use the norm inequality

$$|V_{n,\tau,m}(v) - V_{n,\tau,m}(u)| \leq \|\Delta_n Y_{\tau,m}(v) - \Delta_n Y_{\tau,m}(u)\|_{E'_\tau}$$

where

$$\Delta_n Y_{\tau,m}(u) = \sum_{q\in\mathcal{Q}_\tau} I_q(F_{n+1,\tau,q,m}(u) - F_{n,\tau,q,m}(u)).$$

The shell-separated edge hypothesis (144.261) and the finite-chaos estimate give

$$\|V_{n,\tau,m}(v) - V_{n,\tau,m}(u)\|_{L^p} \leq \widehat{A}_{\tau,1} 2^{-\sigma_\tau m} 2^{-\rho_\tau (n-m)_+} d_{\Pi,m}(u,v)^\theta.$$

On the dyadic edge set,

$$d_{\Pi,m}(u,v) \leq C 2^{-\ell}.$$

Absorb C^θ into $\widehat{A}_{\tau,1}$. Since the edge entropy is six, the edge moment has excess exponent

$$\varepsilon_\tau = p\theta - 6 > 0.$$

The entropy in the physical scale m is $D=5$, and the effective scale slack is

$$a_\tau = \sigma_\tau - \frac{5}{p} > 0.$$

Corollary 144.61, or equivalently the projective version Corollary 144.62 with $D=5$, $d=6$, $B_\pi = \widehat{A}_{\tau,0}$, and $B_\pi^{\mathrm{ed}} = \widehat{A}_{\tau,1}$, gives (144.263) with the constant (144.264). ☐

Proposition 144.81 (Scale bound for the locally subtracted first-chaos kernel)**.** *Let* $Q = 5$, $\mathfrak{s} = (2, 1, 1, 1)$, *and*

$$\mathsf{D}_\delta(t, x) = (\delta^2 t, \delta x).$$

Fix $0 < \theta \le 1$. *Let*

$$\chi_z^\delta(a) = \delta^{-Q}\chi(\mathsf{D}_\delta^{-1}(a - z)), \qquad 0 < \delta \le 1,$$

where χ *is supported in a fixed unit parabolic ball and* $\|\chi\|_{L^\infty} \le 1$*. For* $n \ge 0$*, let* $A_n(a, h)$ *be the shell packet of the local two-loop factor* $K(a, a+h)G(a, a+h)^2$*. Assume there are constants* R, A_0 *such that*

$$A_n(a, h) = 0 \quad \textit{unless } \|h\|_{\mathfrak{s}} \le R2^{-n}, \qquad \sup_a \int |A_n(a, h)|\,\mathrm{d}h \le A_0. \tag{144.265}$$

Let C_s*,* $s \ge 0$*, be dyadic covariance blocks of order* 4*. Assume that, whenever* h *and* h' *lie in the supports of* A_n *and* $A_{n'}$*, the double increment*

$$\begin{aligned}\Delta_{h,h'}C_s(a, a') = C_s(a+h, a'+h') &- C_s(a+h, a') \\ &- C_s(a, a'+h') + C_s(a, a')\end{aligned}$$

satisfies

$$\begin{aligned}|\Delta_{h,h'}C_s(a, a')| \le A_0\, 2^s\, 2^{-\theta(n-s)_+} &2^{-\theta(n'-s)_+} \\ &\times \sum_{\rho\in\{0,h\}}\sum_{\rho'\in\{0,h'\}} \mathbf{1}_{\{\|a+\rho-a'-\rho'\|_{\mathfrak{s}}\le R2^{-s}\}}.\end{aligned} \tag{144.266}$$

For a finite ultraviolet cutoff N*, define the scalar sector sum*

$$\begin{aligned}V_{\delta,N}^{(1)} = \sum_{0\le n,n',s\le N} \int \chi_z^\delta(a)\chi_z^\delta(a')A_n(a, h)A_{n'}(a', h') \\ \times \Delta_{h,h'}C_s(a, a')\,\mathrm{d}h\,\mathrm{d}h'\,\mathrm{d}a\,\mathrm{d}a'.\end{aligned} \tag{144.267}$$

Then there is a constant C*, independent of* z*,* δ*, and* N*, such that*

$$|V_{\delta,N}^{(1)}| \le C\,\delta^{-1}\big(1 + |\log_2 \delta|\big)^2. \tag{144.268}$$

Consequently, if $Z_{\delta,N}^{(1)}$ *is the first-chaos part of the tested locally subtracted* X^2Y *coordinate whose second moment is bounded by a finite sum of terms of the form* (144.267)*, then, for every* $\eta > 0$*,*

$$\Big\|\delta^{1/2+5\kappa} Z_{\delta,N}^{(1)}\Big\|_{L^2(\Omega)} \le C_\eta\, \delta^{5\kappa-\eta}. \tag{144.269}$$

Proof. For a fixed shift r, the test-function overlap obeys

$$\int |\chi_z^\delta(a)\chi_z^\delta(a')|\mathbf{1}_{\{\|a-a'-r\|_{\mathfrak{s}}\le R2^{-s}\}}\,\mathrm{d}a\,\mathrm{d}a' \le C_\chi \min\{1, \delta^{-Q}2^{-Qs}\}. \tag{144.270}$$

Indeed, if $2^{-s} \ge c\delta$, the left side is at most $\|\chi_z^\delta\|_{L^1}^2$. If $2^{-s} < c\delta$, then for each a the allowed a'-set has volume $O(2^{-Qs})$, while $\|\chi_z^\delta\|_{L^\infty} \le C\delta^{-Q}$ and $\|\chi_z^\delta\|_{L^1} \le C$.

Insert (144.266) into (144.267), use (144.265) for the h and h' integrals, and then use (144.270). This gives

$$|V_{\delta,N}^{(1)}| \le C\sum_{s=0}^N B_s(\delta)\left(\sum_{n=0}^N 2^{-\theta(n-s)_+}\right)^2,$$

where

$$B_s(\delta) = 2^s \min\{1, \delta^{-Q} 2^{-Qs}\}.$$

The local two-loop packets are shell-summable but not summable over all shells; this is exactly why the square of the shell count remains. Since

$$\sum_{n=0}^{N} 2^{-\theta(n-s)_+} \le (s+1) + \frac{1}{1-2^{-\theta}} \le C_\theta(1+s),$$

we have

$$|V^{(1)}_{\delta,N}| \le C \sum_{s \ge 0} B_s(\delta)(1+s)^2.$$

Choose $m \ge 0$ such that $2^{-m-1} < \delta \le 2^{-m}$. For $s \le m$, $B_s(\delta) \le 2^s$, hence

$$\sum_{s \le m} B_s(\delta)(1+s)^2 \le C 2^m (1+m)^2 \le C\delta^{-1}(1+|\log_2 \delta|)^2.$$

For $s > m$,

$$B_s(\delta) \le C 2^{Qm} 2^{-(Q-1)s}.$$

Because $Q - 1 = 4 > 0$,

$$\sum_{s > m} B_s(\delta)(1+s)^2 \le C 2^{Qm} 2^{-(Q-1)m}(1+m)^2 \le C\delta^{-1}(1+|\log_2 \delta|)^2.$$

This proves (144.268).

The first-chaos sector is a centered Gaussian variable, so its L^2-norm is the square root of its variance. Multiplying by $\delta^{1/2+5\kappa}$ gives

$$\left\| \delta^{1/2+5\kappa} Z^{(1)}_{\delta,N} \right\|_{L^2} \le C\delta^{5\kappa}(1+|\log_2 \delta|).$$

For every $\eta > 0$,

$$\sup_{0<\delta\le 1} \delta^\eta (1+|\log_2 \delta|) < \infty,$$

which gives (144.269). □

Proposition 144.82 (Cutoff-shell gain for the locally subtracted first-chaos kernel)**.** *Keep the hypotheses and notation of Proposition 144.81, and choose $m_\delta \ge 0$ by*

$$2^{-m_\delta - 1} < \delta \le 2^{-m_\delta}.$$

For $N \ge 0$, set

$$\Delta_N V^{(1)}_\delta = V^{(1)}_{\delta,N+1} - V^{(1)}_{\delta,N}.$$

Then, for every $0 < \rho < \theta/2$, there is a constant C_ρ, independent of z, δ, N, such that

$$|\Delta_N V^{(1)}_\delta| \le C_\rho\, \delta^{-1}(1+m_\delta)^2 2^{-2\rho(N-m_\delta)_+}. \tag{144.271}$$

Consequently, if the cutoff increment $Z^{(1)}_{\delta,N+1} - Z^{(1)}_{\delta,N}$ of the first-chaos part of the tested locally subtracted X^2Y coordinate has its second moment bounded by a fixed finite sum of terms of the form $\Delta_N V^{(1)}_\delta$, then for every $\eta > 0$ and every $0 < \rho < \theta/2$,

$$\left\| \delta^{1/2+5\kappa}\big(Z^{(1)}_{\delta,N+1} - Z^{(1)}_{\delta,N}\big) \right\|_{L^2(\Omega)} \le C_{\eta,\rho}\, \delta^{5\kappa-\eta} 2^{-\rho(N-m_\delta)_+}. \tag{144.272}$$

Proof. The proof keeps the shell location visible. Write

$$\alpha_{j,s} = 2^{-\theta(j-s)_+}, \qquad B_s(\delta) = 2^s \min\{1, \delta^{-Q} 2^{-Qs}\}.$$

The estimate in the proof of Proposition 144.81 gives the majorant

$$|V^{(1)}_{\delta,N}| \leq C \sum_{0\leq s\leq N} B_s(\delta) \left(\sum_{0\leq n\leq N} \alpha_{n,s} \right)^2 .$$

Subtracting the N-cutoff sum from the $N+1$-cutoff sum leaves exactly the sectors in which at least one of the three indices n, n', s is $N+1$. Hence

$$\begin{aligned} |\Delta_N V^{(1)}_\delta| \leq C\, B_{N+1}(\delta) \left(\sum_{0\leq j\leq N+1} \alpha_{j,N+1} \right)^2 \\ + C \sum_{0\leq s\leq N+1} B_s(\delta)\alpha_{N+1,s} \left(\sum_{0\leq j\leq N+1} \alpha_{j,s} \right) . \end{aligned} \tag{144.273}$$

The second line accounts for the two symmetric cases $n = N+1$ and $n' = N+1$, with the factor 2 absorbed into C.

Let $L = (N - m_\delta)_+$. If $N \leq m_\delta$, then the right side of (144.271) has no shell gain, and the total bound (144.268) already gives the claim. Assume from now on that $N > m_\delta$. The covariance-shell term in (144.273) has $s = N+1 > m_\delta$, so

$$B_{N+1}(\delta) \leq C 2^{m_\delta} 2^{-4(N-m_\delta)}.$$

Moreover $\sum_{j\leq N+1} \alpha_{j,N+1} \leq C(1+N)$. Since $2\rho < 4$, the polynomial factor is absorbed by weakening the exponential:

$$B_{N+1}(\delta) \left(\sum_{j\leq N+1} \alpha_{j,N+1} \right)^2 \leq C_\rho 2^{m_\delta}(1+m_\delta)^2 2^{-2\rho(N-m_\delta)}.$$

It remains to bound the local-shell line in (144.273). Use $\sum_{j\leq N+1} \alpha_{j,s} \leq C(1+s)$. For $s \leq m_\delta$,

$$\begin{aligned} \sum_{s\leq m_\delta} B_s(\delta)\alpha_{N+1,s}(1+s) &\leq C(1+m_\delta)2^{-\theta(N-m_\delta)} \sum_{s\leq m_\delta} 2^s \\ &\leq C 2^{m_\delta}(1+m_\delta)2^{-\theta(N-m_\delta)}. \end{aligned}$$

For $s > m_\delta$, write $s = m_\delta + t$. Then

$$B_s(\delta) \leq C 2^{m_\delta} 2^{-4t}.$$

The part $1 \leq t \leq N - m_\delta$ is bounded by

$$C 2^{m_\delta}(1+m_\delta) \sum_{1\leq t\leq N-m_\delta} (1+t)2^{-4t}2^{-\theta(N-m_\delta-t)} \leq C 2^{m_\delta}(1+m_\delta)2^{-\theta(N-m_\delta)},$$

because $4-\theta>0$. The tail $t>N-m_\delta$ has $\alpha_{N+1,s}=1$ and is bounded by

$$C2^{m_\delta}(1+m_\delta)\sum_{t>N-m_\delta}(1+t)2^{-4t}\le C2^{m_\delta}(1+m_\delta)2^{-4(N-m_\delta)}.$$

Since $2\rho<\theta\le 1<4$, the two local-shell estimates are bounded by

$$C_\rho 2^{m_\delta}(1+m_\delta)^2 2^{-2\rho(N-m_\delta)}.$$

Combining the covariance-shell and local-shell estimates, and using $2^{m_\delta}\simeq\delta^{-1}$, proves (144.271).

The first-chaos increment is Gaussian, so the L^2-norm is the square root of its variance. Multiplication by $\delta^{1/2+5\kappa}$ gives

$$C_\rho\delta^{5\kappa}(1+m_\delta)2^{-\rho(N-m_\delta)_+}.$$

The logarithm is absorbed as before: for every $\eta>0$,

$$(1+m_\delta)\le C_\eta\delta^{-\eta}.$$

This proves (144.272). $\square$

Proposition 144.83 (Parameter-edge bound for the locally subtracted first-chaos kernel)**.** *Keep the hypotheses and notation of Propositions 144.81 and 144.82. Let*

$$\mathcal{X}_{\Pi,m}=K\times[1/2,1],\qquad \mathrm{d}_{\Pi,m}((z,a),(z',a'))=2^m\|z-z'\|_{\mathfrak{s}}+|a-a'|,$$

where $K\subset\mathbb{R}^{1+3}$ is compact. For $\varphi\in C_c^1(B_{\mathfrak{s}}(0,1))$, $\|\varphi\|_{C^1}\le 1$, set

$$\varphi_u(x)=\delta_u^{-Q}\varphi(\mathsf{D}_{\delta_u}^{-1}(x-z)),\qquad u=(z,a),\qquad \delta_u=2^{-m}a.$$

Let $Z^{(1)}_{N,m,u}(\varphi)$ be the first-chaos scalar coordinate obtained from the locally subtracted X^2Y kernel by replacing the test density χ_z^δ in (144.267) by φ_u. More precisely, assume its second moment is bounded by a fixed finite sum of the sector integrals used in Proposition 144.81, with constants independent of m,N,u,φ, and assume the cutoff difference has the shell representation used in Proposition 144.82.

Then, whenever $\mathrm{d}_{\Pi,m}(u,v)\le 1$, for every $0<\vartheta\le 1$ there is a constant C_ϑ such that

$$\mathbb{E}\left|Z^{(1)}_{N,m,v}(\varphi)-Z^{(1)}_{N,m,u}(\varphi)\right|^2\le C_\vartheta\,2^m(1+m)^2\,\mathrm{d}_{\Pi,m}(u,v)^{2\vartheta}.\tag{144.274}$$

For every even $p\ge 2$ and every $\eta>0$,

$$\left\|2^{-(1/2+5\kappa)m}\big(Z^{(1)}_{N,m,v}(\varphi)-Z^{(1)}_{N,m,u}(\varphi)\big)\right\|_{L^p(\Omega)}\le C_{p,\eta,\vartheta}\,2^{-(5\kappa-\eta)m}\mathrm{d}_{\Pi,m}(u,v)^{\vartheta}.\tag{144.275}$$

Moreover, for every $0<\rho<\theta/2$, where θ is the covariance double-increment exponent in (144.266),

$$\begin{aligned}\Big\|2^{-(1/2+5\kappa)m}\big[(Z^{(1)}_{N+1,m,v}-Z^{(1)}_{N,m,v})-(Z^{(1)}_{N+1,m,u}-Z^{(1)}_{N,m,u})\big]\Big\|_{L^p(\Omega)}\\ \le C_{p,\eta,\vartheta,\rho}\,2^{-(5\kappa-\eta)m}2^{-\rho(N-m)_+}\mathrm{d}_{\Pi,m}(u,v)^{\vartheta}.\end{aligned}\tag{144.276}$$

Proof. The proof is the first-chaos analogue of the high-chaos edge estimate, but the local subtraction makes the overlap estimate worth recording explicitly. Put

$$d = d_{\Pi,m}(u,v), \qquad \psi = \varphi_v - \varphi_u.$$

The transported test map is Lipschitz from $\mathcal{X}_{\Pi,m}$ to both L^1 and the scale-normalized L^∞ norm. Differentiating in the base point produces the dimensionless factor $2^m\|z-z'\|_{\mathfrak{s}}$, and differentiating in $a \in [1/2,1]$ differentiates the compact family

$$a^{-Q}\varphi(\mathsf{D}_{a^{-1}}w).$$

Hence, because $d \le 1$ and $0 < \vartheta \le 1$,

$$\|\psi\|_{L^1} \le Cd^\vartheta, \qquad \|\psi\|_{L^\infty} \le C2^{Qm}d^\vartheta, \tag{144.277}$$

and the supports of φ_u, φ_v, and ψ lie in one parabolic ball of radius $C2^{-m}$.

Let r be any of the finitely many shifts $0, \pm h, \pm h'$, or $h-h'$, appearing in the four support indicators of the covariance double increment. For every $s \ge 0$,

$$\int |\psi(a)\psi(a')|\mathbf{1}_{\{\|a-a'-r\|_{\mathfrak{s}} \le R2^{-s}\}}\,\mathrm{d}a\,\mathrm{d}a' \le Cd^{2\vartheta}\min\{1, 2^{Qm}2^{-Qs}\}. \tag{144.278}$$

Indeed, the first bound is $\|\psi\|_{L^1}^2$. The second is obtained by integrating in a' over a parabolic ball of volume $O(2^{-Qs})$, using $\|\psi\|_{L^\infty}$ for the a'-factor and $\|\psi\|_{L^1}$ for the a-factor. Equation (144.277) then gives the displayed factor.

Now repeat the proof of Proposition 144.81 with χ_z^δ replaced by the signed density ψ and with $2^{-m}a$ in place of δ. The shell L^1 bounds on A_n and $A_{n'}$, the covariance double-increment estimate (144.266), and the signed overlap estimate (144.278) give

$$\mathbb{E}|Z^{(1)}_{N,m,v} - Z^{(1)}_{N,m,u}|^2 \le Cd^{2\vartheta}\sum_{s\ge 0} B_s(m)(1+s)^2,$$

where

$$B_s(m) = 2^s\min\{1, 2^{Qm}2^{-Qs}\}.$$

The same split at $s = m$ as before gives

$$\sum_{s\ge 0} B_s(m)(1+s)^2 \le C2^m(1+m)^2,$$

which proves (144.274).

For the cutoff edge, repeat the proof of Proposition 144.82 with the same signed density ψ. The decomposition into the covariance-shell term and the two local-shell terms is unchanged; every occurrence of the test overlap now carries the extra factor $d^{2\vartheta}$ from (144.278). Therefore

$$\mathbb{E}\left|(Z^{(1)}_{N+1,m,v} - Z^{(1)}_{N,m,v}) - (Z^{(1)}_{N+1,m,u} - Z^{(1)}_{N,m,u})\right|^2 \le C_{\rho,\vartheta}2^m(1+m)^2 2^{-2\rho(N-m)_+}d^{2\vartheta}.$$

The first-chaos variables are Gaussian, so their L^p-norms are bounded by a p-dependent constant times their L^2-norms. Multiplying by $2^{-(1/2+5\kappa)m}$ leaves

$$2^{-5\kappa m}(1+m)d^\vartheta$$

in the same-scale estimate and the additional $2^{-\rho(N-m)_+}$ in the cutoff-edge estimate. The logarithm is absorbed by $1+m \le C_\eta 2^{\eta m}$, giving (144.275) and (144.276). $\square$

Together with Proposition 144.75, this identifies the scalar base estimate for the whole locally subtracted X^2Y coordinate. The fifth- and third-chaos pieces have the sharp δ^{-1} variance dictated by target regularity $-1/2$; only the first-chaos covariance-increment sector consumes the logarithmic slack that is recorded in the projective X^2Y input. The cutoff increment of this first-chaos sector also has the scale-separated gain of Proposition 144.82; its logarithmic factor is harmless only after one chooses a positive slack ζ_{X^2Y} in the nonlinear Π-coordinate input. The same-scale and cutoff parameter-edge estimates in Proposition 144.83 show that the first-chaos scalar sector has the same physical-parameter continuity structure as the high-chaos sector, again with the logarithmic factor paid from ζ_{X^2Y}. What remains for the full nonlinear X^2Y Π-coordinate theorem is the E_r'-projective uniformization of these scalar estimates over the whole unit ball of tests.

Corollary 144.84 (Local KG^2 shell bound in dynamic Φ^4_3)**.** *Let $Q = 5$. Let K_i, $i \geq 0$, be dyadic heat-kernel pieces of order 2, and let G_j, $j \geq 0$, be dyadic covariance pieces of order 4. Thus, for a fixed parabolic ball constant R, with*

$$\operatorname{supp} K_i \subset B_{\mathfrak{s}}(0, R2^{-i}), \qquad \operatorname{supp} G_j \subset B_{\mathfrak{s}}(0, R2^{-j}),$$

and

$$|K_i(h)| \leq C_K 2^{(Q-2)i}, \qquad |G_j(h)| \leq C_G 2^{(Q-4)j}.$$

For $n \geq 0$, define the local two-loop shell packet

$$A_n(h) = \sum_{\max(i,j,\ell)=n} K_i(h) G_j(h) G_\ell(h).$$

Then

$$\int |A_n(h)| \,\mathrm{d}h \leq C \tag{144.279}$$

uniformly in n. The same estimate holds uniformly in a base point after translating the kernels, and hence supplies (144.265) for the first-chaos proposition.

Proof. For a single triple (i, j, ℓ), set

$$n_* = \max(i, j, \ell), \qquad r_K = n_* - i, \qquad r_G = n_* - j, \qquad r_{G'} = n_* - \ell.$$

The product is supported in a ball of radius $O(2^{-n_*})$. Therefore

$$\begin{aligned}\int |K_i(h) G_j(h) G_\ell(h)| \,\mathrm{d}h &\leq C\, 2^{(Q-2)i+(Q-4)j+(Q-4)\ell} 2^{-Qn_*} \\ &= C\, 2^{-3r_K - r_G - r_{G'}}.\end{aligned}$$

Here the coefficient of n_* vanishes because

$$(Q-2) + (Q-4) + (Q-4) - Q = 3 + 1 + 1 - 5 = 0.$$

Summing over the shell $\max(i, j, \ell) = n$ gives

$$\int |A_n(h)| \,\mathrm{d}h \leq C \sum_{\substack{r_K, r_G, r_{G'} \geq 0 \\ \min(r_K, r_G, r_{G'})=0}} 2^{-3r_K - r_G - r_{G'}} \leq C \left(1 + \frac{1}{1-2^{-3}}\right) \left(\frac{1}{1-2^{-1}}\right)^2.$$

This proves the uniform shell L^1 estimate. Translation of all kernels does not change either the support radius or the integral bound. $\square$

Corollary 144.85 (Covariance double increment for compact order-four blocks)**.** *Let $Q = 5$, $0 < \theta \le 1$, and let $\mathsf{D}_\lambda(t,x) = (\lambda^2 t, \lambda x)$. Suppose the dyadic covariance blocks have the rescaled form*

$$C_s(a,a') = 2^s c_s\big(\mathsf{D}_{2^s}(a-a')\big), \qquad s \ge 0,$$

where the functions c_s are supported in a fixed parabolic unit-scale ball and are bounded uniformly in C^2. Let h, h' satisfy

$$\|h\|_{\mathfrak{s}} \le R_1 2^{-n}, \qquad \|h'\|_{\mathfrak{s}} \le R_1 2^{-n'}.$$

Then there are constants A, R, depending only on R_1, θ, and the uniform block bounds, such that

$$\begin{aligned} |\Delta_{h,h'} C_s(a,a')| &\le A\, 2^s\, 2^{-\theta(n-s)_+} 2^{-\theta(n'-s)_+} \\ &\qquad \times \sum_{\rho \in \{0,h\}} \sum_{\rho' \in \{0,h'\}} \mathbf{1}_{\{\|a+\rho-a'-\rho'\|_{\mathfrak{s}} \le R2^{-s}\}}. \end{aligned} \tag{144.280}$$

In particular the covariance-block hypothesis (144.266) follows for compact dyadic order-four decompositions of the cutoff covariance.

Proof. Write

$$r = \mathsf{D}_{2^s}(a-a'), \qquad u = \mathsf{D}_{2^s} h, \qquad u' = \mathsf{D}_{2^s} h'.$$

Then

$$\begin{aligned} \Delta_{h,h'} C_s(a,a') = 2^s \{ & c_s(r+u-u') - c_s(r+u) \\ & - c_s(r-u') + c_s(r) \}. \end{aligned} \tag{144.281}$$

The uniform C^2 bound gives the scale-one finite-difference estimate

$$|c_s(r+u-u') - c_s(r+u) - c_s(r-u') + c_s(r)| \le C_\theta \min\{1, \|u\|_{\mathfrak{s}}\}^\theta \min\{1, \|u'\|_{\mathfrak{s}}\}^\theta. \tag{144.282}$$

Indeed, if $\|u\|_{\mathfrak{s}} \le 1$ and $\|u'\|_{\mathfrak{s}} \le 1$, the second-order integral form of the increment bounds the left side by $C\|u\|_{\mathfrak{s}}\|u'\|_{\mathfrak{s}}$, which is stronger than (144.282) because $\theta \le 1$. If only one increment is small, apply the first-order mean-value bound to the small increment and the triangle inequality in the other variable. If both increments are large, the uniform C^0 bound gives the estimate.

Since

$$\|u\|_{\mathfrak{s}} = 2^s \|h\|_{\mathfrak{s}} \le R_1 2^{s-n}, \qquad \|u'\|_{\mathfrak{s}} = 2^s \|h'\|_{\mathfrak{s}} \le R_1 2^{s-n'},$$

we have

$$\min\{1, \|u\|_{\mathfrak{s}}\}^\theta \le C_{R_1} 2^{-\theta(n-s)_+}, \qquad \min\{1, \|u'\|_{\mathfrak{s}}\}^\theta \le C_{R_1} 2^{-\theta(n'-s)_+}.$$

Combining these estimates with (144.281) gives the numerical factor in (144.280).

It remains only to record support. Each of the four terms in (144.281) vanishes unless the corresponding separation $a+\rho-a'-\rho'$, with $\rho \in \{0,h\}$ and $\rho' \in \{0,h'\}$, lies in a fixed parabolic ball of radius $O(2^{-s})$. If all four indicators in (144.280) vanish, then all four terms vanish. Otherwise the indicator sum is at least 1, and the previous numerical bound applies. Enlarging the support constant gives the displayed estimate. $\square$

144.7 BPHZ Renormalization of the Dynamic Equation

For the normalization

$$\partial_\tau \Phi = \Delta\Phi - m^2\Phi - \lambda\Phi^3 + \xi,$$

the parabolic dimension is $|\mathfrak{s}| = 5$. A mollified white noise ξ_ϵ is assigned homogeneity

$$|\Xi| = -\frac{5}{2} - \kappa,$$

where $\kappa > 0$ is chosen small. The heat integration map raises homogeneity by 2. Thus the stochastic convolution symbol $X = \mathcal{I}(\Xi)$ has

$$|X| = -\frac{1}{2} - \kappa, \qquad |X^2| = -1 - 2\kappa, \qquad |X^3| = -\frac{3}{2} - 3\kappa.$$

The one-loop subtraction is already forced by X^3. The first nonlinear response $\mathcal{I}(X^3)$ has homogeneity

$$|\mathcal{I}(X^3)| = \frac{1}{2} - 3\kappa,$$

and the product $X^2\mathcal{I}(X^3)$ has

$$|X^2\mathcal{I}(X^3)| = -\frac{1}{2} - 5\kappa.$$

The next remark records exactly which cubic drift monomials have negative homogeneity in the first nonlinear expansion. This finite drift-level bookkeeping determines the one-loop and two-loop local mass coordinates in the equation displayed below; the full regularity-structure construction then adds the model, reexpansion, and analytic-coordinate estimates for these symbols.

Remark 144.86 (Drift-level negative homogeneity bookkeeping). Let $0 < \kappa < 1/14$. Put

$$\alpha = -\frac{1}{2} - \kappa, \qquad \beta = \frac{1}{2} - 3\kappa.$$

Here α is the homogeneity of $X = \mathcal{I}(\Xi)$ and β is the homogeneity of the first nonlinear response $Y = \mathcal{I}(X^3)$. Among the four cubic monomials

$$X^3, \qquad X^2Y, \qquad XY^2, \qquad Y^3,$$

the monomials of negative homogeneity are exactly X^3 and X^2Y. Their homogeneities are

$$|X^3| = -\frac{3}{2} - 3\kappa, \qquad |X^2Y| = -\frac{1}{2} - 5\kappa.$$

The remaining two have positive homogeneity:

$$|XY^2| = \frac{1}{2} - 7\kappa > 0, \qquad |Y^3| = \frac{3}{2} - 9\kappa > 0.$$

Homogeneity is additive under multiplication and is raised by 2 under $\mathcal{I}$. Therefore

$$|Y| = |\mathcal{I}(X^3)| = 3\alpha + 2 = \frac{1}{2} - 3\kappa.$$

The four cubic monomials in the first nonlinear expansion have homogeneities

monomial	homogeneity
X^3	$3\alpha = -\frac{3}{2} - 3\kappa$
X^2Y	$2\alpha + \beta = -\frac{1}{2} - 5\kappa$
XY^2	$\alpha + 2\beta = \frac{1}{2} - 7\kappa$
Y^3	$3\beta = \frac{3}{2} - 9\kappa.$

The first two numbers are negative for every $\kappa > 0$. The inequality $\kappa < 1/14$ gives $1/2 - 7\kappa > 0$, and the weaker inequality $\kappa < 1/6$ gives $3/2 - 9\kappa > 0$. Hence the displayed list contains all and only the negative cubic drift monomials at this stage.

The coordinate attached to X^3 is the one-loop Wick contraction $C_{1,\epsilon}$. The coordinate attached to $X^2\mathcal{I}(X^3)$ is the two-loop local contraction $C_{2,\epsilon}$ after the nested one-loop subgraph has already been subtracted.

With these conventions, the renormalized cutoff equation in the BPHZ regularity-structure convention has the form

$$\partial_\tau \Phi_\epsilon = \Delta\Phi_\epsilon - m^2\Phi_\epsilon - \lambda\Phi_\epsilon^3 + \big(3\lambda C_{1,\epsilon} - 9\lambda^2 C_{2,\epsilon} + c_{\text{fin}}\big)\Phi_\epsilon + \xi_\epsilon. \tag{144.283}$$

Here $C_{1,\epsilon}$ is the one-loop contraction already visible in $:\Phi^3:$, $C_{2,\epsilon}$ is the two-loop local contraction arising from the negative tree with two cubic vertices, and c_{fin} is the finite mass coordinate. Different smooth mollifiers or subtraction schemes change $C_{1,\epsilon}$, $C_{2,\epsilon}$, and c_{fin} separately, but not the statement that only the local linear drift and vacuum-energy coordinates are needed in this superrenormalizable model.

Example 144.87 (Sharp spatial cutoff one-loop coordinate)**.** Work on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with Fourier basis

$$e_n(x) = (2\pi)^{-3/2} e^{in\cdot x}, \qquad n \in \mathbb{Z}^3.$$

Let $m > 0$, and let X_M be the stationary solution of the linear equation

$$\partial_\tau X_M = (\Delta - m^2)X_M + \xi_M$$

obtained by projecting the noise to modes $|n|_\infty \le M$, with normalization

$$\mathbb{E}\,\xi(\tau,x)\xi(\tau',x') = 2\delta(\tau-\tau')\delta(x-x').$$

Then the one-loop local coordinate is

$$C_1(M) = \mathbb{E}X_M(\tau,x)^2 = (2\pi)^{-3} \sum_{\substack{n\in\mathbb{Z}^3\\ |n|_\infty \le M}} \frac{1}{|n|^2 + m^2}, \tag{144.284}$$

independent of (τ, x). There are constants $0 < c_m < C_m < \infty$ such that, for all $M \ge 1$,

$$C_1(M) \le C_m(1+M),$$

and, for all $M \ge \max(1, \lceil m \rceil)$,

$$C_1(M) \ge c_m M.$$

Consequently, for dyadic cutoffs $M_N = 2^N$,

$$0 \le C_1(M_{N+1}) - C_1(M_N) \le C_m' 2^N$$

with $C_m' < \infty$.

For a fixed Fourier mode n, the linear equation is the Ornstein–Uhlenbeck equation

$$dX_n = -(|n|^2 + m^2)X_n\, d\tau + \sqrt{2}\, dB_n(\tau),$$

with the real-field constraint $X_{-n} = \overline{X_n}$. The stationary variance is the ratio between the noise variance 2 and twice the damping rate:

$$\mathbb{E}|X_n|^2 = \frac{1}{|n|^2 + m^2}.$$

Since $|e_n(x)|^2 = (2\pi)^{-3}$, summing the retained modes gives (144.284).

Let

$$S_r = \{n \in \mathbb{Z}^3 : |n|_\infty = r\}.$$

For $r \geq 1$,

$$|S_r| = (2r+1)^3 - (2r-1)^3 = 24r^2 + 2.$$

If $n \in S_r$, then $r^2 \leq |n|^2 \leq 3r^2$. Hence

$$\sum_{n\in S_r} \frac{1}{|n|^2 + m^2} \leq \frac{24r^2 + 2}{r^2} \leq 26.$$

Adding the zero mode gives the upper bound $C_1(M) \leq C_m(1 + M)$.

For $r \geq \max(1, \lceil m \rceil)$, the same shell satisfies

$$|n|^2 + m^2 \leq 3r^2 + r^2 = 4r^2.$$

Therefore

$$\sum_{n\in S_r} \frac{1}{|n|^2 + m^2} \geq \frac{24r^2}{4r^2} = 6.$$

Summing these shell lower bounds from $\max(1, \lceil m \rceil)$ to M gives $C_1(M) \geq c_m M$ after adjusting c_m to cover the finite initial range. The dyadic increment bound follows from the upper shell estimate applied to $M_N < |n|_\infty \leq M_{N+1}$, whose shell radii are $M_N + 1, \ldots, M_{N+1}$.

Lemma 144.88 (Spatial Fourier cutoff two-loop coordinate)**.** *Use the notation of Example 144.87, and put*

$$a_n = |n|^2 + m^2, \qquad n \in \mathbb{Z}^3.$$

Let

$$K_M(t, x) = \mathbf{1}_{t>0}(2\pi)^{-3} \sum_{\substack{k\in\mathbb{Z}^3\\ |k|_\infty \leq M}} e^{-a_k t} e^{ik\cdot x}$$

be the retarded heat kernel with the same spatial cutoff, and let

$$G_M(t, x) = (2\pi)^{-3} \sum_{\substack{p\in\mathbb{Z}^3\\ |p|_\infty \leq M}} \frac{e^{-a_p|t|} e^{ip\cdot x}}{a_p}$$

be the stationary covariance. Define

$$C_2(M) = 2 \int_0^\infty \int_{\mathbb{T}^3} K_M(t, x) G_M(t, x)^2 \, \mathrm{d}x\, \mathrm{d}t. \tag{144.285}$$

Then

$$C_2(M) = 2(2\pi)^{-6} \sum_{\substack{p,q\in\mathbb{Z}^3 \\ |p|_\infty,|q|_\infty,|p+q|_\infty\le M}} \frac{1}{a_p a_q(a_p + a_q + a_{p+q})}. \tag{144.286}$$

There are constants $0 < c_m < C_m < \infty$, depending on m but not on M, such that

$$C_2(M) \le C_m\big(1 + \log(1 + M)\big) \qquad (M \ge 1),$$

and

$$C_2(M) \ge c_m \log M \qquad (M \ge 16).$$

Proof. Substituting the Fourier series for K_M and G_M into (144.285) gives

$$C_2(M) = 2(2\pi)^{-9} \sum_{\substack{k,p,q\in\mathbb{Z}^3 \\ |k|_\infty,|p|_\infty,|q|_\infty\le M}} \frac{1}{a_p a_q} \int_0^\infty e^{-(a_k+a_p+a_q)t}\,\mathrm{d}t \int_{\mathbb{T}^3} e^{i(k+p+q)\cdot x}\,\mathrm{d}x.$$

The spatial integral is $(2\pi)^3$ when $k + p + q = 0$ and is zero otherwise. Eliminating k gives (144.286).

For the upper bound, discard the restriction $|p + q|_\infty \le M$. Let

$$B_0 = \{n \in \mathbb{Z}^3 : |n|_\infty < 2\}, \qquad B_\ell = \{n \in \mathbb{Z}^3 : 2^\ell \le |n|_\infty < 2^{\ell+1}\} \quad (\ell \ge 1).$$

There is a constant $A < \infty$ such that

$$|B_\ell| \le A2^{3\ell} \qquad (\ell \ge 0).$$

For $\ell \ge 1$ this follows from

$$|B_\ell| \le |\{n : |n|_\infty < 2^{\ell+1}\}| = (2^{\ell+2} - 1)^3,$$

and B_0 is finite. There is also $b_m > 0$, for instance

$$b_m = \min(m^2, 1),$$

such that, for $n \in B_\ell$,

$$a_n \ge b_m 2^{2\ell} \qquad (\ell \ge 0).$$

If $p \in B_\ell$ and $q \in B_s$, then

$$a_p + a_q + a_{p+q} \ge a_p + a_q \ge b_m 2^{2\max(\ell,s)}.$$

Thus the contribution of the block $B_\ell \times B_s$ is bounded by

$$C_m \frac{2^{3\ell}2^{3s}}{2^{2\ell}2^{2s}2^{2\max(\ell,s)}} = C_m 2^{-|\ell-s|}.$$

If $|p|_\infty, |q|_\infty \le M$, then only $\ell, s \le L_M = \lceil \log_2(1 + M)\rceil$ occur. Hence

$$C_2(M) \le C_m \sum_{\ell=0}^{L_M} \sum_{s=0}^{L_M} 2^{-|\ell-s|} \le 3C_m(L_M + 1),$$

which is the stated upper bound after increasing C_m.

For the lower bound, write $N = \lfloor \log_2 M \rfloor$. For $2 \le \ell \le N-2$, let

$$P_\ell = \{n \in \mathbb{Z}^3 : 2^\ell \le n_i < 2^\ell + 2^{\ell-2},\ i = 1,2,3\}.$$

Then $|P_\ell| \ge 2^{3\ell-6}$. If $p, q \in P_\ell$, then

$$|p|_\infty,\ |q|_\infty,\ |p+q|_\infty \le 2^{\ell+2} \le M,$$

and

$$a_p,\ a_q,\ a_{p+q} \le A_m 2^{2\ell}$$

with $A_m < \infty$ independent of ℓ. The summands in (144.286) are therefore bounded below by $A_m^{-3} 2^{-6\ell}$ on $P_\ell \times P_\ell$. The contribution of this block is at least

$$2(2\pi)^{-6} |P_\ell|^2 A_m^{-3} 2^{-6\ell} \ge c_m.$$

Summing the disjoint lower-bound blocks for $\ell = 2, \dots, N-2$ gives $C_2(M) \ge c_m(N-3)$. Since there is an absolute constant $c_0 > 0$ with

$$N - 3 \ge c_0 \log M \qquad (M \ge 16),$$

the lower bound follows after decreasing c_m. $\square$

Proposition 144.89 (Dyadic shell bound for the two-loop coordinate)**.** *Let $C_2(M)$ be the sharp spatial Fourier-cutoff two-loop coordinate in Lemma 144.88, and set*

$$M_N = 2^N, \qquad N \in \mathbb{N}.$$

There is a constant $C_m^{\mathrm{shell}} < \infty$, depending on m but not on N, such that, for all $N \ge 2$,

$$0 \le C_2(M_{N+1}) - C_2(M_N) \le C_m^{\mathrm{shell}}.$$

Proof. Write

$$D_N = \{(p,q) \in \mathbb{Z}^3 \times \mathbb{Z}^3 : |p|_\infty, |q|_\infty, |p+q|_\infty \le 2^N\}.$$

The summand in (144.286) is non-negative, and $D_N \subset D_{N+1}$, so the increment is non-negative.

Use the dyadic annuli B_ℓ from the proof of Lemma 144.88. The same block estimate gives a constant $A_m < \infty$ such that, for every $\ell, s \ge 0$, the contribution of

$$(B_\ell \times B_s) \cap D_{N+1}$$

to $C_2(M_{N+1})$ is bounded by

$$A_m 2^{-|\ell - s|}.$$

If $(p,q) \in D_{N+1} \setminus D_N$ and

$$p \in B_\ell, \qquad q \in B_s,$$

then $\ell, s \le N+1$. Moreover, $\max(\ell, s) \ge N-1$. Indeed, if $\ell, s \le N-2$, then

$$|p|_\infty < 2^{N-1}, \qquad |q|_\infty < 2^{N-1},$$

hence $|p+q|_\infty < 2^N$, so $(p,q) \in D_N$, a contradiction. Therefore

$$\begin{aligned} C_2(M_{N+1}) - C_2(M_N) &\le A_m \sum_{\substack{0\le \ell,s\le N+1\\ \max(\ell,s)\ge N-1}} 2^{-|\ell-s|} \\ &\le A_m \sum_{t=N-1}^{N+1} \left(1 + 2\sum_{r=0}^{t-1} 2^{-(t-r)}\right) \\ &\le 9A_m. \end{aligned}$$

Thus $C_m^{\text{shell}} = 9A_m$ is admissible. □

Finite-cutoff local counterterm calculation. Let X_ϵ be the smooth stationary Gaussian solution of the linearized cutoff equation and write

$$G_\epsilon(z,w) = \mathbb{E}\big[X_\epsilon(z)X_\epsilon(w)\big].$$

Let K_ϵ denote the retarded heat kernel with the same ultraviolet mollification and define the local constants

$$C_{1,\epsilon} = G_\epsilon(z,z), \qquad C_{2,\epsilon} = 2\int K_\epsilon(z,w)G_\epsilon(z,w)^2\,\mathrm{d}w, \tag{144.287}$$

where translation invariance on the spacetime torus makes the right-hand sides independent of z. Consider the perturbative fixed-point expansion

$$\Phi_\epsilon = X_\epsilon - \lambda K_\epsilon(X_\epsilon^3) + O(\lambda^2),$$

and let $\mathcal{L}$ denote the projection that retains only the local linear part produced by Wick contractions and the zeroth Taylor coefficient of the remaining field at $w = z$. Then

$$\mathcal{L}\big[-\lambda X_\epsilon^3 + 3\lambda^2 X_\epsilon^2 K_\epsilon(X_\epsilon^3)\big] = \big(-3\lambda C_{1,\epsilon} + 9\lambda^2 C_{2,\epsilon}\big)X_\epsilon.$$

Consequently the counterterm inserted in the cutoff drift is the negative of this local singular part, namely

$$\big(3\lambda C_{1,\epsilon} - 9\lambda^2 C_{2,\epsilon}\big)\Phi_\epsilon,$$

up to the independent finite mass coordinate $c_{\text{fin}}\Phi_\epsilon$.

The one-loop term is the Wick contraction of two of the three fields in X_ϵ^3:

$$X_\epsilon(z)^3 = :X_\epsilon(z)^3: + 3G_\epsilon(z,z)X_\epsilon(z).$$

Thus

$$\mathcal{L}[-\lambda X_\epsilon^3] = -3\lambda C_{1,\epsilon}X_\epsilon.$$

For the two-loop term, expand the nonlinearity to first order in the nonlinear response:

$$-\lambda\Phi_\epsilon^3 = -\lambda X_\epsilon^3 + 3\lambda^2 X_\epsilon^2 K_\epsilon(X_\epsilon^3) + O(\lambda^3).$$

The coefficient 3 is the number of choices for the factor of Φ_ϵ in Φ_ϵ^3 that is replaced by $-\lambda K_\epsilon(X_\epsilon^3)$; the two minus signs give the positive sign in the displayed term. At a spacetime point z,

$$X_\epsilon(z)^2 K_\epsilon(X_\epsilon^3)(z) = \int K_\epsilon(z,w)X_\epsilon(z)^2 X_\epsilon(w)^3\,\mathrm{d}w.$$

Its local linear part is obtained by contracting four of the five Gaussian fields and replacing the remaining field at w by its zeroth Taylor term at z. There are two possible forms of such contractions.

First leave one of the three $X_\epsilon(w)$ factors uncontracted and contract the two $X_\epsilon(z)$ factors with the two remaining $X_\epsilon(w)$ factors. This gives $3 \cdot 2 = 6$ pairings and contributes

$$6 \int K_\epsilon(z,w) G_\epsilon(z,w)^2 \, \mathrm{d}w \; X_\epsilon(z) = 3C_{2,\epsilon} X_\epsilon(z).$$

Second leave one of the two $X_\epsilon(z)$ factors uncontracted, contract the other $X_\epsilon(z)$ with one of the three $X_\epsilon(w)$ factors, and contract the remaining two $X_\epsilon(w)$ factors with each other. This produces

$$6 \int K_\epsilon(z,w) G_\epsilon(z,w) G_\epsilon(w,w) \, \mathrm{d}w \; X_\epsilon(z).$$

This is the one-loop tadpole inserted into the inner cubic vertex. The finite-cutoff forest subtraction used here is the following recursive local operation: before extracting the new local part of a graph, replace every proper divergent subgraph by its already computed local counterterm. For the present tree this means replacing the inner $X_\epsilon(w)^3$ by its one-loop Wick-ordered cubic, so the displayed nested tadpole is removed before the two-loop coordinate is defined. Therefore the new two-loop local coordinate is the first contribution only, with the normalization in (144.287). Multiplying by the outer factor $3\lambda^2$ gives

$$\mathcal{L}[3\lambda^2 X_\epsilon^2 K_\epsilon(X_\epsilon^3)] = 9\lambda^2 C_{2,\epsilon} X_\epsilon.$$

Combining the one-loop and two-loop local parts proves the displayed formula. The finite coordinate c_{fin} is not fixed by singular subtraction; it labels the chosen renormalized mass coordinate.

Proposition 144.90 (Static-dynamic mass-coordinate translation)**.** *Fix a finite cutoff and write the Euclidean local density of Chapter 137 as*

$$g_4 \phi_\epsilon^4 + a_\epsilon \phi_\epsilon^2 + b_\epsilon.$$

Let the cutoff Langevin equation use the convention of (144.1)*, so that the drift is the negative L^2-gradient of the regulated action. If the same dynamics is written as*

$$\partial_\tau \Phi_\epsilon = \Delta \Phi_\epsilon - m^2 \Phi_\epsilon - \lambda_{\mathrm{dyn}} \Phi_\epsilon^3 + M_\epsilon \Phi_\epsilon + \xi_\epsilon,$$

then

$$\lambda_{\mathrm{dyn}} = 4g_4, \qquad M_\epsilon = -2a_\epsilon. \tag{144.288}$$

Consequently, relative to the Wick covariance $C_\epsilon(0)$ used in (137.27)*, the normal-ordered mass coordinate is*

$$\alpha_\epsilon = a_\epsilon + 6g_4 C_\epsilon(0) = -\frac{1}{2} M_\epsilon + \frac{3}{2} \lambda_{\mathrm{dyn}} C_\epsilon(0). \tag{144.289}$$

In the dynamic BPHZ chart

$$M_\epsilon = 3\lambda_{\mathrm{dyn}} C_{1,\epsilon} - 9\lambda_{\mathrm{dyn}}^2 C_{2,\epsilon} + c_{\mathrm{fin}}.$$

If the Wick covariance in the constructive chart is the same local covariance, $C_\epsilon(0) = C_{1,\epsilon}$, then the one-loop coordinate cancels in (144.289) *and*

$$\alpha_\epsilon^{\mathrm{dyn}} = \frac{9}{2} \lambda_{\mathrm{dyn}}^2 C_{2,\epsilon} - \frac{1}{2} c_{\mathrm{fin}}. \tag{144.290}$$

Thus the common-chart comparison with (137.33) is a comparison of normal-ordered mass coordinates, together with a chosen finite mass normalization, not a comparison of the raw a_ϵ and M_ϵ coefficients. The vacuum coordinate b_ϵ, or equivalently β_ϵ, remains a separate free-energy normalization because an additive constant in the action is invisible to the Langevin drift.

Proof. For the finite-cutoff local part

$$\int \left[g_4\phi_\epsilon^4 + a_\epsilon\phi_\epsilon^2 + b_\epsilon\right] \mathrm{d}^3x,$$

the variational derivative contributes

$$4g_4\Phi_\epsilon^3 + 2a_\epsilon\Phi_\epsilon.$$

The gradient Langevin drift is its negative, added to the Gaussian drift $\Delta\Phi_\epsilon - m^2\Phi_\epsilon$. Comparing the cubic and linear terms with the displayed dynamic equation gives (144.288). The first equality in (144.289) is precisely the coordinate conversion (137.27), with the Euclidean quartic coefficient renamed g_4. Substituting $g_4 = \lambda_{\mathrm{dyn}}/4$ and $a_\epsilon = -M_\epsilon/2$ gives the second equality. Finally substituting the BPHZ expression for M_ϵ and identifying $C_\epsilon(0)$ with $C_{1,\epsilon}$ cancels the one-loop term and leaves (144.290). The constant term b_ϵ drops out of the variational derivative, so it cannot be recovered from the drift comparison. □

Theorem 144.91 (Abstract local fixed point in modelled distributions)**.** *Let $0 < T \leq 1$. Let E_T be a Banach space of modelled distributions on the parabolic time slab $[0,T] \times \mathbb{T}^3$, let F_T be a Banach space of modelled distributional forcing terms on the same slab, and assume the following data and estimates.*

(i) A zero-initial-time abstract heat integration operator $\mathcal{K}_T : F_T \to E_T$ satisfies, for constants $A > 0$ and $\theta > 0$,

$$\|\mathcal{K}_T H\|_{E_T} \leq AT^\theta \|H\|_{F_T}. \tag{144.291}$$

The power T^θ encodes the positive time gain obtained after the singular local model has been fixed.

(ii) A reconstructed stochastic convolution $X_T \in E_T$ and an initial-data lift $P_T u_0 \in E_T$ are given. Put

$$R_0 = \|P_T u_0 + X_T\|_{E_T}.$$

(iii) The cubic product is a locally Lipschitz map $E_T \to F_T$: for a constant $B > 0$,

$$\|U^3\|_{F_T} \leq B\|U\|_{E_T}^3,$$

and

$$\|U^3 - V^3\|_{F_T} \leq B(\|U\|_{E_T}^2 + \|V\|_{E_T}^2)\|U - V\|_{E_T}. \tag{144.292}$$

(iv) The local linear term is bounded as a map $E_T \to F_T$:

$$\|U\|_{F_T} \leq D\|U\|_{E_T}$$

for a constant $D > 0$.

Fix real parameters λ and M. Define

$$\Psi_T(U) = P_T u_0 + X_T + \mathcal{K}_T\big(-\lambda U^3 + MU\big). \tag{144.293}$$

Let $R = 2R_0$. If

$$AT^\theta\big(B|\lambda|R^3 + D|M|R\big) \le \frac{R}{2}, \tag{144.294}$$

$$AT^\theta\big(2B|\lambda|R^2 + D|M|\big) < 1, \tag{144.295}$$

then Ψ_T has a unique fixed point in the closed ball

$$\mathbb{B}_R = \{U \in E_T : \|U\|_{E_T} \le R\}.$$

The fixed point depends Lipschitz-continuously on $P_T u_0 + X_T$ as long as the same constants and the same ball are used.

Proof. For $U \in \mathbb{B}_R$, the Schauder estimate (144.291), the cubic bound, and the linear bound give

$$\begin{aligned}\|\Psi_T(U)\|_{E_T} &\le R_0 + AT^\theta\big(|\lambda|\|U^3\|_{F_T} + |M|\|U\|_{F_T}\big)\\ &\le \frac{R}{2} + AT^\theta\big(B|\lambda|R^3 + D|M|R\big) \le R.\end{aligned}$$

Thus Ψ_T maps $\mathbb{B}_R$ into itself. For $U, V \in \mathbb{B}_R$, the Lipschitz estimate (144.292) gives

$$\begin{aligned}\|\Psi_T(U) - \Psi_T(V)\|_{E_T} &\le AT^\theta\left(|\lambda|\|U^3 - V^3\|_{F_T} + |M|\|U - V\|_{F_T}\right)\\ &\le AT^\theta\big(2B|\lambda|R^2 + D|M|\big)\|U - V\|_{E_T}.\end{aligned}$$

Condition (144.295) makes this Lipschitz constant strictly smaller than one; call it q. Choose $U_0 \in \mathbb{B}_R$ and set $U_{n+1} = \Psi_T(U_n)$. Since $\Psi_T(\mathbb{B}_R) \subset \mathbb{B}_R$,

$$\|U_{n+1} - U_n\|_{E_T} \le q^n\|U_1 - U_0\|_{E_T}.$$

For $m > n$,

$$\|U_m - U_n\|_{E_T} \le \sum_{j=n}^{m-1} q^j\|U_1 - U_0\|_{E_T} \le \frac{q^n}{1-q}\|U_1 - U_0\|_{E_T}.$$

The space E_T is complete, so U_n converges to some $U_* \in \mathbb{B}_R$. Continuity of Ψ_T gives $\Psi_T(U_*) = U_*$. If V_* is another fixed point in $\mathbb{B}_R$, then

$$\|U_* - V_*\|_{E_T} \le q\|U_* - V_*\|_{E_T},$$

and $q < 1$ forces $U_* = V_*$.

For dependence on the inhomogeneous term, let $Y = P_T u_0 + X_T$ and $\widetilde{Y} = \widetilde{P}_T\widetilde{u}_0 + \widetilde{X}_T$, and suppose the two corresponding fixed points $U, \widetilde{U}$ lie in the same ball $\mathbb{B}_R$. The same estimate gives

$$\|U - \widetilde{U}\|_{E_T} \le \|Y - \widetilde{Y}\|_{E_T} + q\|U - \widetilde{U}\|_{E_T}, \qquad q = AT^\theta(2B|\lambda|R^2 + D|M|) < 1.$$

Therefore

$$\|U - \widetilde{U}\|_{E_T} \le \frac{1}{1-q}\|Y - \widetilde{Y}\|_{E_T}.$$

□

In the dynamic Φ^4_3 application, E_T is a weighted modelled-distribution space whose lowest components encode the stochastic convolution and whose positive components encode the unknown remainder. The space F_T is the corresponding forcing space after subtracting the singular local parts of the cubic. The operator $\mathcal{K}_T$ is the abstract heat integration map, and

$$M = 3\lambda C_{1,\epsilon} - 9\lambda^2 C_{2,\epsilon} + c_{\mathrm{fin}}$$

in the finite-cutoff equation before passing to the renormalized model. The theorem proves the fixed-point step once the model bounds, product bounds, and Schauder bound have been established. It does not prove convergence of the BPHZ-renormalized models; that remains a separate stochastic estimate.

Proposition 144.92 (Fixed-point-sector comparison criterion). *Let E_T, F_T, R, $0 < q < 1$, and $\mathbb{B}_R \subset E_T$ be fixed. For $n \in \mathbb{N} \cup \{\infty\}$, suppose*

$$\Psi_n : \mathbb{B}_R \to \mathbb{B}_R$$

is a contraction with Lipschitz constant at most q. Let U_n be its unique fixed point in $\mathbb{B}_R$. If

$$\epsilon_n := \sup_{U \in \mathbb{B}_R} \|\Psi_n(U) - \Psi_\infty(U)\|_{E_T} \longrightarrow 0, \tag{144.296}$$

then

$$\|U_n - U_\infty\|_{E_T} \le \frac{\epsilon_n}{1-q}. \tag{144.297}$$

In the modelled Φ^4_3 chart, a sufficient way to produce (144.296) is the following common-space comparison. Write

$$\Psi_n(U) = Y_n + \mathcal{K}_n\big(-\lambda\, \mathcal{P}_{3,n}(U) + M_n \mathcal{L}_n U\big), \qquad Y_n = P_T^{(n)} u_0^{(n)} + X_T^{(n)},$$

and similarly for Ψ_∞. Suppose on $\mathbb{B}_R$

$$\|\mathcal{P}_{3,\infty}(U)\|_{F_T} \le BR^3, \qquad \|\mathcal{L}_\infty U\|_{F_T} \le DR,$$

and suppose the comparison estimates

$$\begin{aligned} \|Y_n - Y_\infty\|_{E_T} &\le \delta_{Y,n}, \\ \sup_{\|H\|_{F_T} \le H_R} \|(\mathcal{K}_n - \mathcal{K}_\infty)H\|_{E_T} &\le \delta_{K,n} H_R, \\ \sup_{\|U\|_{E_T} \le R} \|\mathcal{P}_{3,n}(U) - \mathcal{P}_{3,\infty}(U)\|_{F_T} &\le \delta_{P,n} R^3, \\ \sup_{\|U\|_{E_T} \le R} \|\mathcal{L}_n U - \mathcal{L}_\infty U\|_{F_T} &\le \delta_{L,n} R \end{aligned}$$

hold with

$$H_R = |\lambda| BR^3 + |M_\infty| DR.$$

If $\|\mathcal{K}_n H\|_{E_T} \le AT^\theta \|H\|_{F_T}$ and $|M_n - M_\infty| \le \delta_{M,n}$, then

$$\epsilon_n \le \delta_{Y,n} + \delta_{K,n} H_R + AT^\theta\big(|\lambda|\delta_{P,n} R^3 + |M_\infty|\delta_{L,n} R + D\delta_{M,n} R + \delta_{M,n}\delta_{L,n} R\big). \tag{144.298}$$

Proof. The fixed-point comparison is an a posteriori contraction estimate:

$$\begin{aligned}\|U_n - U_\infty\|_{E_T} &= \|\Psi_n(U_n) - \Psi_\infty(U_\infty)\|_{E_T} \\ &\le \|\Psi_n(U_n) - \Psi_\infty(U_n)\|_{E_T} + \|\Psi_\infty(U_n) - \Psi_\infty(U_\infty)\|_{E_T} \\ &\le \epsilon_n + q\|U_n - U_\infty\|_{E_T}.\end{aligned}$$

Moving the second term to the left gives (144.297).

For the displayed modelled-SPDE budget, add and subtract the terms obtained by successively replacing Y_n, $\mathcal{K}_n$, $\mathcal{P}_{3,n}$, $\mathcal{L}_n$, and M_n by their limiting counterparts. The $\mathcal{K}_n - \mathcal{K}_\infty$ term is tested on the limiting forcing $-\lambda\mathcal{P}_{3,\infty}(U) + M_\infty\mathcal{L}_\infty U$, whose norm is bounded by H_R. The remaining terms are controlled by the common Schauder bound for $\mathcal{K}_n$ and the product, linear, and mass-coordinate comparison estimates. The mass-linear part must be expanded as a product:

$$\begin{aligned}M_n\mathcal{L}_n U - M_\infty\mathcal{L}_\infty U = M_\infty(\mathcal{L}_n - \mathcal{L}_\infty)U &+ (M_n - M_\infty)\mathcal{L}_\infty U \\ &+ (M_n - M_\infty)(\mathcal{L}_n - \mathcal{L}_\infty)U.\end{aligned}$$

Hence its F_T-norm is bounded on $\mathbb{B}_R$ by

$$|M_\infty|\delta_{L,n}R + D\delta_{M,n}R + \delta_{M,n}\delta_{L,n}R.$$

Taking the supremum over $U \in \mathbb{B}_R$ gives (144.298). □

For dynamic Φ^4_3, this proposition is the fixed-point-sector layer following random-model convergence. The negative-sector model estimates control Y_n and the singular model coordinates. Schauder, multiplication, and reexpansion estimates control the comparison errors $\delta_{K,n}$, $\delta_{P,n}$, and $\delta_{L,n}$. The renormalized mass coordinate supplies $\delta_{M,n}$; the product $M_n\mathcal{L}_n$ also pays the mixed error $\delta_{M,n}\delta_{L,n}R$ when both coordinates vary at once. Once these quantities tend to zero while the contraction constant q stays below one on a common ball, the renormalized modelled solutions converge in E_T. This still leaves the invariant-law and OS-comparison layers as separate analytic steps.

Lemma 144.93 (Coupled stationary-law comparison after SPDE convergence)**.** *Let (E, d) be a Polish metric space, and let P_t be a Markov semigroup on E. For a fixed $t \ge 0$, suppose $U_n(0), U_n(t), U_*(0), U_*(t)$ are E-valued random variables on one probability space such that:*

(i) $U_n(0)$ and $U_n(t)$ have the same law μ_n;

(ii) $U_*(0)$ has law μ_*, and

$$\mathbb{E}[f(U_*(t))] = \int_E P_t f \,\mathrm{d}\mu_*$$

for every bounded Lipschitz $f : E \to \mathbb{C}$;

(iii) for some $\delta_n, \varepsilon_{n,t} \ge 0$ and $a_t < \infty$,

$$\mathbb{E}\, d(U_n(0), U_*(0)) \le \delta_n, \qquad \mathbb{E}\, d(U_n(t), U_*(t)) \le a_t\delta_n + \varepsilon_{n,t}. \tag{144.299}$$

Then every bounded Lipschitz f obeys

$$\left|\int_E f \,\mathrm{d}\mu_n - \int_E f \,\mathrm{d}\mu_*\right| \le \mathrm{Lip}(f)\,\delta_n, \tag{144.300}$$

and

$$\left|\int_E P_t f \,\mathrm{d}\mu_* - \int_E f \,\mathrm{d}\mu_*\right| \le \mathrm{Lip}(f)\big((a_t+1)\delta_n + \varepsilon_{n,t}\big). \tag{144.301}$$

If, for each fixed t, the quantity $(a_t+1)\delta_n+\varepsilon_{n,t}$ tends to zero along the comparison sequence, then μ_* is invariant for P_t in the sense of equality of probability measures.

Proof. For (144.300), write the two integrals as expectations of $f(U_n(0))$ and $f(U_*(0))$, and apply the Lipschitz bound:

$$|\mathbb{E}f(U_n(0)) - \mathbb{E}f(U_*(0))| \le \mathrm{Lip}(f)\,\mathbb{E}d(U_n(0), U_*(0)).$$

For the invariance defect, use the representation of P_t in (ii), insert the stationary cutoff variables, and use (i):

$$\begin{aligned}\left|\int_E P_t f \,\mathrm{d}\mu_* - \int_E f \,\mathrm{d}\mu_*\right| &= |\mathbb{E}f(U_*(t)) - \mathbb{E}f(U_*(0))| \\ &\le |\mathbb{E}f(U_*(t)) - \mathbb{E}f(U_n(t))| \\ &\quad + |\mathbb{E}f(U_n(t)) - \mathbb{E}f(U_n(0))| \\ &\quad + |\mathbb{E}f(U_n(0)) - \mathbb{E}f(U_*(0))| \\ &\le \mathrm{Lip}(f)\big(\mathbb{E}d(U_*(t), U_n(t)) + \mathbb{E}d(U_n(0), U_*(0))\big),\end{aligned}$$

because the middle term vanishes by stationarity of U_n. Substituting (144.299) gives (144.301). If the right side tends to zero for every bounded Lipschitz f, then $\int P_t f \,\mathrm{d}\mu_* = \int f \,\mathrm{d}\mu_*$ for every bounded Lipschitz f. Bounded Lipschitz functions separate probability measures on a Polish metric space, so $P_t^*\mu_* = \mu_*$. ☐

Lemma 144.94 (Polynomial stationarity by truncation)**.** *Keep the hypotheses and notation of Lemma 144.93. Let $\mathcal{O} : E \to \mathbb{C}$ be a measurable cylinder observable. Suppose there are bounded Lipschitz functions $\mathcal{O}_R : E \to \mathbb{C}$, $R \ge 1$, such that*

$$T_R(t) := \mathbb{E}|\mathcal{O}(U_*(t)) - \mathcal{O}_R(U_*(t))|, \qquad T_R(0) := \mathbb{E}|\mathcal{O}(U_*(0)) - \mathcal{O}_R(U_*(0))|$$

tend to zero as $R \to \infty$. Then

$$\begin{aligned}|\mathbb{E}\mathcal{O}(U_*(t)) - \mathbb{E}\mathcal{O}(U_*(0))| &\le T_R(t) + T_R(0) \\ &\quad + \mathrm{Lip}(\mathcal{O}_R)\big((a_t+1)\delta_n + \varepsilon_{n,t}\big).\end{aligned} \tag{144.302}$$

Consequently, if for each fixed R the comparison defect $(a_t+1)\delta_n+\varepsilon_{n,t}$ tends to zero and the tails $T_R(t)+T_R(0)$ tend to zero, then

$$\mathbb{E}\mathcal{O}(U_*(t)) = \mathbb{E}\mathcal{O}(U_*(0)).$$

For polynomial cylinder observables this reduces stationarity to tail estimates: the truncations $\mathcal{O}_R$ must converge in L^1 at times 0 and t.

Proof. Insert $\mathcal{O}_R$ and use the triangle inequality:

$$\begin{aligned}|\mathbb{E}\mathcal{O}(U_*(t)) - \mathbb{E}\mathcal{O}(U_*(0))| &\le T_R(t) + T_R(0) \\ &\quad + |\mathbb{E}\mathcal{O}_R(U_*(t)) - \mathbb{E}\mathcal{O}_R(U_*(0))|\,.\end{aligned}$$

The last term is controlled by Lemma 144.93, applied to the bounded Lipschitz test $\mathcal{O}_R$. This proves (144.302). Taking $n \to \infty$ at fixed R, then $R \to \infty$, proves the equality. ☐

In the renormalized Φ^4_3 problem, Lemma 144.93 is the invariant-law layer following the fixed-point-sector comparison. A stationary cutoff solution started from μ_n supplies $U_n(0)$ and $U_n(t)$. The random-model and fixed-point estimates must then be strengthened to a coupling of the time-zero and time-t fields in the topology E used for the Markov process. This proves invariance of a limiting SPDE law only for bounded Lipschitz tests on E. Lemma 144.94 records the additional truncation step needed before an unbounded polynomial cylinder observable can be treated as stationary. The polynomial Schwinger moments, reflection positivity, and corrected OS-II growth needed for QFT reconstruction still come from the hierarchy comparison in Construction 144.102.

Proposition 144.95 (Finite-window OS assembly defect budget)**.** *Let*

$$\mathcal{W} = \{F_1, \dots, F_M\}$$

be a finite family of positive-time polynomial cylinder observables. Let Θ denote physical Euclidean time reflection. For a cutoff stationary SPDE law μ_N, define the finite OS Gram matrix

$$G^{(N)}_{ij} = \int \overline{F_i(\Theta\Phi)}\, F_j(\Phi)\, \mathrm{d}\mu_N(\Phi), \qquad 1 \le i, j \le M,$$

and let G_{ij} be the corresponding entries of a candidate limiting Schwinger hierarchy. Suppose that, for this window, the cutoff Gram form has a lower OS defect

$$\sum_{i,j=1}^{M} \overline{c_i} c_j G^{(N)}_{ij} \ge -\delta_{\mathcal{W},N} \sum_{i=1}^{M} |c_i|^2 \tag{144.303}$$

for every $c \in \mathbb{C}^M$, and that the limiting entries are controlled entrywise by

$$\max_{1 \le i,j \le M} |G_{ij} - G^{(N)}_{ij}| \le \varepsilon_{\mathcal{W},N}. \tag{144.304}$$

Then the limiting window obeys

$$\sum_{i,j=1}^{M} \overline{c_i} c_j G_{ij} \ge -\big(\delta_{\mathcal{W},N} + M\varepsilon_{\mathcal{W},N}\big) \sum_{i=1}^{M} |c_i|^2. \tag{144.305}$$

Consequently, a directed family of finite positive-time windows gives polynomial OS positivity for the limiting hierarchy if one can choose a cutoff $N(\mathcal{W})$ for each window such that

$$\delta_{\mathcal{W},N(\mathcal{W})} + |\mathcal{W}|\varepsilon_{\mathcal{W},N(\mathcal{W})} \longrightarrow 0$$

along the directed window net.

Now let H_{ij} be the OS Gram entries of a constructive Euclidean hierarchy in the same field-normalization and local-counterterm chart. If

$$\max_{i,j} |G_{ij} - H_{ij}| \le \eta_{\mathcal{W}},$$

then

$$\left| \sum_{i,j=1}^{M} \overline{c_i} c_j (G_{ij} - H_{ij}) \right| \le M\eta_{\mathcal{W}} \sum_{i=1}^{M} |c_i|^2. \tag{144.306}$$

Thus equality of all finite-window entries identifies the positive-time pre-Hilbert inner products and their null spaces. If the comparison is first obtained with finite-window defects, the same directed-window schedule must send $M\eta_{\mathcal{W}}$ to zero before one may identify the OS quotients.

Finally, suppose the constructive hierarchy satisfies a corrected OS-II growth estimate

$$|S_n^{\mathrm{cons}}(F)| \le AB^n (n!)^\gamma q_{n,N_0+C_{\mathrm{OS}}n}(F),$$

and the SPDE/constructive difference satisfies, for the same or a dominated seminorm family,

$$|S_n^{\mathrm{spde}}(F) - S_n^{\mathrm{cons}}(F)| \le A_\Delta B_\Delta^n (n!)^{\gamma_\Delta} q_{n,N_\Delta+C_\Delta n}(F).$$

Then S_n^{spde} satisfies a corrected OS-II estimate with constants obtained by replacing A by $A+A_\Delta$, B by $\max\{B, B_\Delta\}$, γ by $\max\{\gamma,\gamma_\Delta\}$, and the linear seminorm order by a common dominating order. In particular, if the hierarchy comparison error tends to zero in the stated seminorms, the OS-II growth package transfers from the constructive hierarchy to the SPDE hierarchy.

Proof. Subtract the cutoff Gram form from the limiting Gram form:

$$\sum_{i,j} \overline{c_i} c_j G_{ij} = \sum_{i,j} \overline{c_i} c_j G_{ij}^{(N)} + \sum_{i,j} \overline{c_i} c_j (G_{ij} - G_{ij}^{(N)}).$$

The first term is bounded from below by (144.303). The second satisfies

$$\left| \sum_{i,j} \overline{c_i} c_j (G_{ij} - G_{ij}^{(N)}) \right| \le \varepsilon_{\mathcal{W},N} \left(\sum_i |c_i| \right)^2 \le M\varepsilon_{\mathcal{W},N} \sum_i |c_i|^2,$$

by Cauchy's inequality. This proves (144.305). If the directed-window schedule drives the displayed defect to zero, every finite positive-time polynomial OS quadratic form is nonnegative in the limit, which is exactly polynomial OS positivity on the directed cylinder algebra.

The constructive comparison estimate (144.306) is the same calculation with $G-H$ in place of $G-G^{(N)}$. When every finite-window entry is equal, the Gram matrices are equal, so the positive-time inner products and null spaces are equal before completion. With defects, the factor M is the price of entrywise control; a growing OS window therefore requires a schedule strong enough to make $M\eta_{\mathcal{W}}$ vanish, or else an operator-norm comparison replacing the entrywise bound.

For the growth statement, use the triangle inequality and dominate the two Schwartz seminorms by one member of the increasing seminorm family. Since

$$B^n (n!)^\gamma, \quad B_\Delta^n (n!)^{\gamma_\Delta} \le \max\{B, B_\Delta\}^n (n!)^{\max\{\gamma,\gamma_\Delta\}},$$

the displayed constants give the asserted OS-II bound for S_n^{spde}. If the difference term tends to zero in the relevant seminorms, the constructive constants alone control the limiting SPDE hierarchy. ◻

Proposition 144.96 (Finite-rate assembly schedule for dynamic Φ_3^4)**.** *Let*

$$\mathcal{W}_k = \{F_{k,1}, \dots, F_{k,m_k}\}, \qquad k = 1, 2, \dots,$$

be a directed sequence of finite positive-time polynomial OS windows. Suppose that the model estimates, fixed-point-sector comparison, stationary-law coupling, and polynomial truncation have been assembled into the entrywise window bound

$$\max_{i,j\le m_k} |G^{\mathrm{spde}}_{k,ij} - G^{(n)}_{k,ij}| \le C_k e_n + T_{k,R}. \tag{144.307}$$

Here $G^{(n)}$ is the cutoff OS Gram matrix, G^{spde} is the limiting stationary-SPDE Gram matrix, e_n is the projective model/fixed-point defect, C_k is the finite-window amplification constant, and $T_{k,R}$ is the polynomial-tail truncation error. Assume also that the cutoff window has lower OS defect $\delta_{k,n}$, and that the constructive hierarchy is entrywise comparable to the limiting SPDE hierarchy:

$$\max_{i,j\le m_k} |G^{\mathrm{spde}}_{k,ij} - H^{\mathrm{cons}}_{k,ij}| \le \eta_k. \tag{144.308}$$

Then for every cutoff and truncation schedule $n = n(k)$, $R = R(k)$,

$$\begin{aligned}\sum_{i,j\le m_k} \overline{c_i} c_j H^{\mathrm{cons}}_{k,ij} \ge -\Delta_k \sum_{i\le m_k} |c_i|^2,\\ \Delta_k := \delta_{k,n(k)} + m_k\big(C_k e_{n(k)} + T_{k,R(k)} + \eta_k\big).\end{aligned} \tag{144.309}$$

If $\Delta_k \to 0$, the directed union has the same positive-time OS pre-Hilbert form as the constructive hierarchy. The OS-II growth constants transfer whenever the truncation and comparison errors are dominated by a common OS-II seminorm envelope.

For example, if $m_k \le 2^{\alpha k}$ and $C_k \le 2^{\beta k}$, choose

$$e_{n(k)} \le 2^{-(\alpha+\beta+3)k}, \qquad T_{k,R(k)} \le 2^{-(\alpha+3)k}, \qquad \eta_k \le 2^{-(\alpha+3)k}, \qquad \delta_{k,n(k)} \le 2^{-k}.$$

Then

$$\Delta_k \le 2^{-k} + 2^{-3k} + 2^{-3k} + 2^{-3k} \le 4\,2^{-k}.$$

Proof. Apply Proposition 144.95 to the k-th window with $\varepsilon_{\mathcal{W}_{k,n,R}} = C_k e_n + T_{k,R}$. This gives a limiting SPDE lower defect $\delta_{k,n} + m_k(C_k e_n + T_{k,R})$. Comparing the SPDE Gram matrix with the constructive Gram matrix by (144.308) adds at most $m_k \eta_k \sum_i |c_i|^2$ to the lower defect, proving (144.309). The sample schedule is the displayed arithmetic: the model/fixed-point error must beat both the window cardinality m_k and the window amplification C_k, while truncation and hierarchy-comparison errors only pay the window-cardinality loss. The directed-union and OS-II statements are the corresponding conclusions of Proposition 144.95 applied along the schedule. □

Proposition 144.97 (Cross-route phase-cell/SPDE assembly budget)**.** *Work in one finite-volume chart: the same field normalization, Wick-ordering convention, renormalized mass coordinate, and local counterterm normalization are used for the phase-cell construction of Chapter 137 and for the dynamic Φ^4_3 SPDE construction. For a positive-time polynomial OS window*

$$\mathcal{W}_k = \{F_{k,1}, \dots, F_{k,m_k}\}$$

let $G^{\mathrm{spde}}_{k,ij}$ be the limiting stationary-SPDE Gram entries, $G^{(n)}_{k,ij}$ a cutoff SPDE Gram matrix, $H^{(J)}_{k,ij}$ the corresponding phase-cell regulator at constructive scale J, and $H^{\mathrm{cons}}_{k,ij}$ the limiting constructive

Gram matrix. Suppose that

$$\max_{i,j\le m_k} |G^{\mathrm{spde}}_{k,ij} - G^{(n)}_{k,ij}| \le C^{\mathrm{spde}}_k e_n + T_{k,R}, \tag{144.310}$$

$$\max_{i,j\le m_k} |G^{(n)}_{k,ij} - H^{(J)}_{k,ij}| \le \chi_{k,n,J}, \tag{144.311}$$

$$\max_{i,j\le m_k} |H^{(J)}_{k,ij} - H^{\mathrm{cons}}_{k,ij}| \le D_k\, \mathcal{E}^{\mathrm{constr}}_J. \tag{144.312}$$

Here e_n is the projective model/fixed-point SPDE defect, $T_{k,R}$ is the polynomial truncation tail, $\chi_{k,n,J}$ is the finite-chart comparison error between the stochastic and phase-cell regulators, and $\mathcal{E}^{\mathrm{constr}}_J$ is the constructive phase-cell defect bounded in (137.52). If the phase-cell window has lower OS defect

$$\sum_{i,j\le m_k} \overline{c_i} c_j H^{(J)}_{k,ij} \ge -\delta_{k,n,J} \sum_{i\le m_k} |c_i|^2, \tag{144.313}$$

then the limiting SPDE Gram matrix satisfies

$$\sum_{i,j\le m_k} \overline{c_i} c_j G^{\mathrm{spde}}_{k,ij} \ge -\Xi_{k,n,J,R} \sum_{i\le m_k} |c_i|^2, \tag{144.314}$$

where

$$\Xi_{k,n,J,R} = \delta_{k,n,J} + m_k\big(C^{\mathrm{spde}}_k e_n + T_{k,R} + \chi_{k,n,J}\big). \tag{144.315}$$

Moreover

$$\max_{i,j\le m_k} |G^{\mathrm{spde}}_{k,ij} - H^{\mathrm{cons}}_{k,ij}| \le C^{\mathrm{spde}}_k e_n + T_{k,R} + \chi_{k,n,J} + D_k \mathcal{E}^{\mathrm{constr}}_J. \tag{144.316}$$

Consequently, if there is a directed schedule $(n(k), J(k), R(k))$ for which

$$\Xi_{k,n(k),J(k),R(k)} \to 0, \qquad m_k\big(C^{\mathrm{spde}}_k e_{n(k)} + T_{k,R(k)} + \chi_{k,n(k),J(k)} + D_k \mathcal{E}^{\mathrm{constr}}_{J(k)}\big) \to 0,$$

then the directed SPDE hierarchy and the constructive hierarchy have the same positive-time OS pre-Hilbert form. The corrected OS-II growth package transfers from the constructive hierarchy whenever the entrywise comparison errors are dominated by a common OS-II seminorm envelope.

For instance, if

$$m_k \le 2^{\alpha k}, \qquad C^{\mathrm{spde}}_k \le 2^{\beta k}, \qquad D_k \le 2^{\gamma k},$$

then the sample choices

$$e_{n(k)} \le 2^{-(\alpha+\beta+4)k}, \qquad T_{k,R(k)} \le 2^{-(\alpha+4)k},$$
$$\chi_{k,n(k),J(k)} \le 2^{-(\alpha+4)k}, \qquad \mathcal{E}^{\mathrm{constr}}_{J(k)} \le 2^{-(\alpha+\gamma+4)k},$$
$$\delta_{k,n(k),J(k)} \le 2^{-k}$$

make the combined finite-window defect at most $2\,2^{-k}$.

Proof. By the triangle inequality,

$$|G^{\mathrm{spde}}_{k,ij} - H^{(J)}_{k,ij}| \le C^{\mathrm{spde}}_k e_n + T_{k,R} + \chi_{k,n,J}.$$

Substituting this estimate into the quadratic form and using $(\sum_i |c_i|)^2 \le m_k \sum_i |c_i|^2$, exactly as in Proposition 144.95, gives (144.314) and (144.315). Adding (144.312) to the same triangle estimate proves (144.316).

If the directed schedule sends the two displayed defects to zero, every finite positive-time polynomial OS quadratic form in the SPDE hierarchy is the limit of the corresponding constructive form, and the finite-window comparison identifies the null spaces before completion. The OS-II growth statement follows from the growth-transfer part of Proposition 144.95.

For the sample rate schedule, multiply each entrywise error by the worst window factor m_k. The model/fixed-point contribution is bounded by

$$2^{\alpha k}2^{\beta k}2^{-(\alpha+\beta+4)k} = 2^{-4k},$$

the truncation and finite-chart comparison terms by 2^{-4k}, and the phase-cell tail by

$$2^{\alpha k}2^{\gamma k}2^{-(\alpha+\gamma+4)k} = 2^{-4k}.$$

Together with the lower OS defect 2^{-k}, this is $2^{-k} + 4\,2^{-4k} \le 2\,2^{-k}$ for $k \ge 1$. □

Proposition 144.98 (Directed OS pre-Hilbert comparison)**.** *Let*

$$V_1 \subset V_2 \subset \cdots, \qquad V_k = \operatorname{span}\{F_{k,1}, \ldots, F_{k,m_k}\},$$

be an increasing exhaustion of the positive-time polynomial cylinder algebra $\mathfrak{A}_+^{\mathrm{poly}} = \bigcup_k V_k$. Let G^{spde} and H^{cons} be the sesquilinear OS Gram forms determined by the limiting stationary-SPDE hierarchy and by the constructive Euclidean hierarchy, respectively. Suppose that the schedules of Proposition 144.97 provide numbers $\Xi_k \to 0$ and $E_k \to 0$ such that, on V_k,

$$G^{\mathrm{spde}}(v,v) \ge -\Xi_k \|c(v)\|_{\ell^2}^2, \tag{144.317}$$

$$|G^{\mathrm{spde}}(v,w) - H^{\mathrm{cons}}(v,w)| \le E_k \|c(v)\|_{\ell^1} \|c(w)\|_{\ell^1}. \tag{144.318}$$

Here $c(v)$ denotes the coefficient vector of v in the listed window; when $v \in V_r \subset V_k$, the vector is padded by zeroes. Assume also that H^{cons} is OS positive on $\mathfrak{A}_+^{\mathrm{poly}}$. Then G^{spde} is OS positive on $\mathfrak{A}_+^{\mathrm{poly}}$, the two forms agree on the algebraic cylinder algebra, and the identity map descends to an isometric isomorphism

$$\mathfrak{A}_+^{\mathrm{poly}}/\mathcal{N}_{\mathrm{spde}} \cong \mathfrak{A}_+^{\mathrm{poly}}/\mathcal{N}_{\mathrm{cons}}, \qquad \mathcal{N}_\bullet := \{v : \langle v,v\rangle_\bullet = 0\}.$$

After completion, the OS Hilbert spaces generated by these positive-time cylinder vectors are canonically unitarily identified. This conclusion uses only the finite-window route-comparison estimates. It does not prove the model estimates, the finite-chart comparison, OS-II growth, Euclidean covariance, locality, or domain statements needed for the final Wightman field presentation; those remain the analytic inputs named in Theorem 144.99.

Proof. Fix $v \in \mathfrak{A}_+^{\mathrm{poly}}$. Choose r with $v \in V_r$. For every $k \ge r$, (144.317) gives

$$G^{\mathrm{spde}}(v,v) \ge -\Xi_k \|c(v)\|_{\ell^2}^2.$$

Letting $k \to \infty$ proves $G^{\mathrm{spde}}(v,v) \ge 0$. Thus the stationary-SPDE hierarchy has a positive algebraic OS form once the directed defects vanish.

For $v, w \in \mathfrak{A}_+^{\mathrm{poly}}$, choose r with $v, w \in V_r$. The same padding argument and (144.318) give, for all $k \geq r$,

$$|G^{\mathrm{spde}}(v, w) - H^{\mathrm{cons}}(v, w)| \leq E_k \|c(v)\|_{\ell^1} \|c(w)\|_{\ell^1}.$$

Letting $k \to \infty$ gives equality of the two sesquilinear forms on the directed union. Hence v has zero SPDE norm if and only if it has zero constructive norm; the null spaces are equal. The identity map on $\mathfrak{A}_+^{\mathrm{poly}}$ therefore descends to an isometry of the quotient pre-Hilbert spaces, and an isometry between dense pre-Hilbert spaces extends uniquely to a unitary between their Hilbert completions. □

Quoted Theorem 144.99 (Renormalized dynamic Φ^4_3 theorem package)**.** *Let the spatial manifold be a finite three-dimensional torus, let $\xi_\epsilon = \rho_\epsilon * \xi$ be a smooth space-time mollification of white noise, and fix the normalization of* (144.283)*. In the BPHZ regularity-structure chart there are local constants $C_{1,\epsilon}$, $C_{2,\epsilon}$, and a finite mass coordinate c_{fin}, such that the following statements hold.*

(i) For each fixed $\epsilon > 0$ and smooth initial field, the cutoff equation (144.283) *has a unique smooth local solution. The solution map is Markovian and depends continuously on the initial field and on the smooth noise.*

(ii) After enhancing the mollified noise by its BPHZ-renormalized negative trees, the cutoff solutions converge in probability, on compact stochastic-time intervals before explosion and in negative spatial Holder or Besov topologies, to a limit independent of the mollifier up to finite local coordinate changes.

(iii) The positive quartic drift, together with the renormalized mass coordinate, gives global solutions and invariant probability measures for the limiting Markov process in the massive phase. Starting the cutoff equation in its finite-cutoff Gibbs law gives stationary solutions, and every limiting stationary law is invariant for the limiting Markov semigroup.

(iv) In a common finite-volume regulator and renormalization chart, this invariant law has the same Schwinger moments as the constructive Φ^4_3 Euclidean measure with the same renormalized mass coordinate. Equivalently, after the field and mass normalizations are matched, the SPDE stationary construction and the constructive Euclidean construction determine the same OS input hierarchy.

This renormalized Φ^4_3 SPDE theorem is a quoted theorem-boundary for the regularity-structures construction, not a proof completed here. The quoted source lineage is Hairer's regularity-structures construction and the Hairer–Mourrat–Weber treatments of dynamic Φ^4_3 renormalization. The four clauses are logically distinct. Local cutoff well-posedness is an ordinary smooth parabolic theorem. Renormalized convergence is the singular SPDE theorem: one first builds the canonical model from the mollified noise; its negative tree components diverge as $\epsilon \downarrow 0$; the BPHZ character subtracts the expectations of all divergent negative subtrees, producing a renormalized model satisfying uniform analytic bounds; compactness and Kolmogorov estimates then give convergence of the model.

After the renormalized model has been constructed, the local fixed-point step has the form proved in Theorem 144.91. The missing analytic input is the verification, for the actual BPHZ model, of the Schauder, product, and small-time estimates used in that theorem. Reconstructing the fixed point then gives the distribution-valued solution by Theorem 144.49. The explicit local term in (144.283) is the image, in the concrete equation, of the BPHZ action on the negative trees. Invariant measures are identified by finite-cutoff stationarity, tightness, and convergence of stationary solutions. Identification with the constructive Euclidean Φ^4_3 measure is still another step: both constructions must be placed in the same finite-volume cutoff chart, the mass and field coordinates must be matched, and equality of the limiting Schwinger moments must be proved. Each sentence in this paragraph names a theorem-level estimate; none is to be treated as proved merely by being quoted.

The proof boundary has narrowed. For the Φ^4_2 equation, the construction of the stochastic convolution and its Wick powers, the path-space enhanced-noise convergence, the Sobolev and Besov local DPD maps, the smooth energy estimate, the compactness and closedness steps for smooth approximants, the regulator comparison criteria, and the abstract SPDE-to-OS reconstruction criterion have been proved above. The remaining two-dimensional work is not a generic appeal to the Da Prato–Debussche theorem. It consists of specific constructive estimates: the local rough-forcing estimate in (144.62), Nelson stability for the chosen Wick quartic cutoff family, and the moment-continuity, positive-time semigroup regularity, and corrected OS-II growth estimates needed by Remark 144.101.

For dynamic Φ^4_3, the self-contained proof boundary is still larger. The finite negative sector, its homogeneity bookkeeping, its reexpansion coordinates, and the scale-summed criterion that would imply convergence of that sector have been proved in Lemma 144.64 and the negative-sector coordinate-control consequence. What remains is to prove, for the actual BPHZ-renormalized model, the coordinate estimates required by that scale-summed criterion for

$$\Xi, \quad X, \quad X^2, \quad X^3, \quad XY, \quad X^2Y, \quad \text{and} \quad c_n(z,z'),$$

including both uniform model-seminorm moments and dyadic cutoff-increment moments with enough scale and parameter slack. One must then extend the negative-sector convergence to the full modelled-distribution space used by the fixed point, prove the Schauder and multiplication estimates for the concrete model, identify the BPHZ counterterms with the local constants in (144.283), and compare the stationary SPDE law with the Euclidean constructive measure. [1] The cited works identify deeper analytic machinery that remains to be reconstructed in this monograph; citation supplies orientation, while the proof obligations remain the estimates just listed.

Open Problem 144.100 (Remaining self-contained singular-SPDE proof stack). Complete the proof stack left open by the preceding analysis. The two-dimensional part requires: a proof of the rough-forcing estimate (144.62) for the Wick-enhanced Φ^4_2 stochastic convolution; a direct Nelson stability theorem, or a fully controlled comparison with a regulator for which Nelson stability has been proved, for the cutoff family used in the stochastic assembly; and the OS reconstruction inputs listed in Remark 144.101.

The three-dimensional part requires: construction of the BPHZ-renormalized dynamic Φ^4_3 model; proof of the coordinate moment and cutoff-Cauchy estimates in the negative-sector coordinate-control consequence; extension of those estimates from the strict negative sector to the full sector used in the fixed-point problem; verification that the concrete BPHZ model satisfies the Schauder and product estimates required by Theorem 144.91; derivation of the local counterterms $C_{1,\epsilon}$,

[1] See G. Da Prato and A. Debussche, "Strong solutions to the stochastic quantization equations," *Ann. Probab.* 31 (2003); M. Hairer, "A theory of regularity structures," *Invent. Math.* 198 (2014); and the Hairer–Mourrat–Weber regularity-structures treatment of Φ^4_3 renormalization. For recent singular-SPDE work in gauge contexts, see A. Chandra, I. Chevyrev, M. Hairer, and H. Shen, "Stochastic quantisation of Yang–Mills," Invent. Math. **230** (2022) 1025–1096.

$C_{2,\epsilon}$, and c_{fin} from the BPHZ character in the convention of (144.283); invariant-law construction and comparison with the constructive Euclidean Φ^4_3 measure; and the SPDE-to-OS reconstruction inputs for the resulting Schwinger hierarchy. Until these arguments are supplied, the remaining quoted theorem boundary is a documented proof boundary rather than an authority claim.

144.8 Paracontrolled and RG Routes

Regularity structures are not the only analytic language for singular SPDE. Paracontrolled distributions decompose products using Bony's paraproduct

$$fg = f \prec g + f \circ g + f \succ g,$$

and control the resonant product $f \circ g$ by enhancing the noise with the few ill-defined resonant objects needed by the equation. For Φ^4_3, this route uses the same local counterterm content as (144.283), expressed in Fourier/Besov language rather than in a tree Hopf algebra.

Kupiainen-style RG treats the stochastic convolution and nonlinear response by scale integration. It is closer in form to the constructive RG of Chapter 142. The common structure is that no finite truncation of diagrams is being trusted by itself: the theorem estimates the infinite irrelevant tail in a norm and shows convergence of the renormalized limit.

144.9 Invariant Measure and OS Data

If the renormalized Markov process exists and is ergodic with invariant measure μ, the Euclidean Schwinger functions are

$$S_n(f_1, \dots, f_n) = \int \Phi(f_1) \cdots \Phi(f_n) \, \mathrm{d}\mu(\Phi). \tag{144.319}$$

A complete QFT construction must verify:

$$\begin{array}{rl} \text{Euclidean covariance,} & \text{reflection positivity,} \\ \text{regularity and OS growth bounds,} & \text{clustering or phase decomposition.} \end{array}$$

Stochastic quantization is especially powerful because it gives a dynamic construction of the measure and a precise algebraic account of stochastic counterterms. Reflection positivity, however, is a property of the Euclidean field measure under physical time reflection; it is not automatic from Markov positivity in the auxiliary stochastic time.

Remark 144.101 (SPDE invariant law as an OS reconstruction input). Let $E = \mathcal{S}'(\mathbb{R}^d)$ be the real tempered-distribution space, equipped with its Borel σ-algebra, and let μ be a probability measure on E. For $f \in \mathcal{S}(\mathbb{R}^d)$ write $\Phi(f)(\varphi) = \varphi(f)$. Suppose that the following conditions hold.

(i) For every $n \geq 0$, the multilinear functional initially defined on finite tensor sums by

$$S_n\left(\sum_a c_a f_{a,1} \otimes \cdots \otimes f_{a,n}\right) = \sum_a c_a \int_E \Phi(f_{a,1}) \cdots \Phi(f_{a,n}) \, \mathrm{d}\mu(\Phi)$$

is well-defined and obeys a Schwartz-seminorm estimate

$$|S_n(F)| \le C_n\, q_{n,N_n}(F), \qquad F \in \mathcal{S}((\mathbb{R}^d)^n), \tag{144.320}$$

after extending from finite tensor sums. Here $q_{n,N}$ is any fixed increasing defining family of Schwartz seminorms on $\mathcal{S}((\mathbb{R}^d)^n)$. Equivalently, μ has enough polynomial cylinder moments, with enough seminorm continuity, to make each S_n a tempered Euclidean distribution.

(ii) The hierarchy $\{S_n\}$ is normalized, Euclidean covariant, permutation symmetric, and Hermitian in the sense of Definition 12.2. In a scalar real theory these conditions follow from $S_0 = 1$, invariance of μ under the Euclidean group action on E, and reality of the field coordinates; for labelled or graded fields the label reflection and Koszul signs must be part of the declared field presentation.

(iii) Let Θ denote physical Euclidean time reflection. For every polynomial cylinder observable

$$F(\Phi) = \sum_a c_a\, \Phi(f_{a,1}) \cdots \Phi(f_{a,r_a})$$

whose test functions are supported in the positive-time half-space,

$$\int_E \overline{F(\Theta\Phi)}\, F(\Phi)\, \mathrm{d}\mu(\Phi) \ge 0. \tag{144.321}$$

If this inequality is first proved only for bounded cylinder functions, then the passage to polynomial cylinders must be justified by the uniform integrability or truncation estimates appropriate to the cutoff limit.

(iv) The positive-time translation operators on the OS quotient satisfy the semigroup law, symmetry, and strong-continuity hypotheses (12.5)–(12.8).

(v) The translation-reduced hierarchy satisfies the corrected OS-II linear-growth estimate (12.4); explicitly, for some $A, B, \gamma > 0$ and integers N_0, C_{OS},

$$|S_n^{\mathrm{red}}(F)| \le A\, B^n\, (n!)^\gamma\, q_{n,N_0 + C_{\mathrm{OS}} n}(F)$$

on the zero-diagonal reduced Schwartz test space of Chapter 12.

Then the Schwinger hierarchy obtained from μ satisfies the hypotheses of Theorem 12.16. Consequently it reconstructs Lorentzian Wightman field data whose Euclidean time-ordered boundary values recover the Schwinger functions at noncoincident Euclidean configurations. If the hierarchy is clustering in the sense of Corollary 12.19, the reconstructed vacuum is unique; without clustering one obtains the corresponding phase-decomposed Wightman presentation. Condition (i) is the distributional regularity step. Finite tensor sums are dense in the Schwartz tensor product, and the estimate (144.320) makes the cylinder moment functional continuous for the Schwartz topology. Thus each S_n extends uniquely to a tempered distribution on $(\mathbb{R}^d)^n$.

Condition (ii) gives normalization, Euclidean covariance, permutation symmetry, and Hermiticity. Condition (iii) is exactly OS reflection positivity on the positive-time polynomial cylinder algebra; the warning in the second sentence records the point used repeatedly above, namely that polynomial OS positivity cannot be read off from bounded-cylinder positivity without a tail estimate. Conditions (iv) and (v) are the corrected analytic inputs emphasized in Chapter 12:

reflection positivity constructs the Hilbert space and positive Hamiltonian, while the OS-II semigroup and linear-growth package supplies the many-variable Fourier–Laplace continuation with tempered Lorentzian boundary values.

All hypotheses of Theorem 12.16 are therefore present. Applying that theorem gives

$$(\mathcal{H}_{\mathrm{OS}}, \Omega_{\mathrm{OS}}, U, \mathcal{D}_{\mathrm{OS}}, \{\widehat{\Phi}_\alpha\})$$

and the Wightman distributions obtained as boundary values of the analytic continuations of the S_n. The clustering statement is precisely Corollary 12.19.

In stochastic quantization the preceding remark is a checklist, not an automatic corollary of stationarity. The Markov invariant law supplies a candidate μ. The constructive proof must still establish moment continuity, physical-time reflection positivity, the OS-II positive-time semigroup regularity, and the corrected linear-growth estimate. For Φ^4_2 and Φ^4_3, these conditions should ultimately be verified by comparing the SPDE invariant law with the corresponding constructive Euclidean measure in a common renormalization chart, because the constructive measure is where reflection positivity and OS growth are naturally controlled. Proposition 144.95 is the finite-window bookkeeping for this comparison: it records the exact loss incurred when cutoff OS positivity, polynomial truncation, and hierarchy comparison are known only with window-dependent defects.

Construction 144.102 (Common Schwinger-hierarchy comparison)**.** Let

$$E = \mathcal{S}'(\mathbb{R}^d)$$

with its cylinder σ-algebra. Let μ_{spde} be a candidate stationary law produced by a renormalized stochastic quantization construction, and let μ_{cons} be a constructive Euclidean measure in the same finite-volume, field-normalization, mass-coordinate, and local-counterterm chart. For $f \in \mathcal{S}(\mathbb{R}^d)$, write

$$\Phi(f)(\varphi) = \varphi(f).$$

The comparison datum needed for OS reconstruction is the equality of Schwinger hierarchies, not equality of auxiliary stochastic processes. A precise sufficient datum is the following.

(i) For each n, both moment functionals

$$S_n^{\mathrm{spde}}(f_1, \dots, f_n) = \int_E \prod_{j=1}^n \Phi(f_j)\, \mathrm{d}\mu_{\mathrm{spde}}(\Phi), \qquad S_n^{\mathrm{cons}}(f_1, \dots, f_n) = \int_E \prod_{j=1}^n \Phi(f_j)\, \mathrm{d}\mu_{\mathrm{cons}}(\Phi)$$

extend continuously to $\mathcal{S}((\mathbb{R}^d)^n)$, with declared Schwartz seminorm bounds

$$|S_n^\bullet(F)| \le C_n^\bullet q_{n,N_n^\bullet}(F), \qquad \bullet \in \{\mathrm{spde}, \mathrm{cons}\}.$$

(ii) There is a dense linear subspace $\mathcal{D} \subset \mathcal{S}(\mathbb{R}^d)$ such that

$$S_n^{\mathrm{spde}}(f_1, \dots, f_n) = S_n^{\mathrm{cons}}(f_1, \dots, f_n), \qquad f_j \in \mathcal{D},$$

for every n. In applications $\mathcal{D}$ is often generated by the test functions used by the two regulator families, after matching the field and mass coordinates.

(iii) The constructive hierarchy $S^{\mathrm{cons}}_\bullet$ satisfies the OS normalization, Euclidean covariance, symmetry, Hermiticity, polynomial reflection positivity, positive-time semigroup regularity, corrected OS-II linear growth, and any declared clustering or phase-decomposition hypothesis.

Then the SPDE hierarchy and the constructive hierarchy coincide as tempered Schwinger distributions. Indeed, finite sums of tensors $f_1 \otimes \cdots \otimes f_n$ with $f_j \in \mathcal{D}$ are dense in $\mathcal{S}((\mathbb{R}^d)^n)$. The difference

$$S^{\mathrm{spde}}_n - S^{\mathrm{cons}}_n$$

is continuous by (i) and vanishes on that dense tensor subspace by (ii), hence vanishes on all of $\mathcal{S}((\mathbb{R}^d)^n)$. Every OS quadratic form is a finite linear combination of these distributions evaluated on reflected positive-time tensor products. Reflection positivity, covariance, symmetry, Hermiticity, the semigroup regularity inputs, the corrected OS-II growth bound, and clustering therefore transfer from $S^{\mathrm{cons}}_\bullet$ to $S^{\mathrm{spde}}_\bullet$ with the same constants after replacing the seminorm family by a common dominating family.

Consequently the OS reconstruction of the SPDE hierarchy is canonically identified with the OS reconstruction of the constructive hierarchy: the positive-time polynomial pre-Hilbert inner products are identical, the null spaces are identical, and the completed Hilbert spaces and reconstructed Wightman distributions agree under the induced unitary identification. If the comparison is first proved only for bounded cylinder functions, the bounded-to-polynomial passage must use the uniform-integrability and truncation criteria proved above; bounded-cylinder equality by itself supplies only the bounded part of the comparison datum. Equality of probability laws is a stronger route to the same conclusion when a moment-determinacy theorem is available, but OS reconstruction uses the Schwinger hierarchy itself as its input.

Remark 144.103 (Gauge theories and BV compatibility). For gauge theories, stochastic quantization must be reconciled with gauge symmetry, gauge fixing, and the BV master equation. A stochastic evolution for gauge potentials before quotienting is not by itself a construction of a gauge-invariant local QFT. The constructive target is a measure or observable theory for gauge-invariant fields, or a gauge-fixed/BV construction whose quantum master equation and positivity replacement are controlled. Two- and three-dimensional Yang–Mills SPDE programs provide important laboratories; four-dimensional Yang–Mills remains a separate open problem.

Open Problem 144.104 (SPDE-to-Wightman construction). Starting from a renormalized singular SPDE construction of an interacting Euclidean field, prove the inputs of Remark 144.101, identify the resulting Wightman distributions, and compare the stochastic counterterms with Wilsonian local coordinates in a common renormalization chart.

Chapter 145

Hamiltonian Truncation, DLCQ, And Benchmark Protocols

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Hamiltonian numerical methods are regulator frameworks for QFT. This chapter formulates Hamiltonian truncation, light-front discretization, and benchmark scripts as controlled approximations with explicitly declared Hilbert spaces, cutoffs, counterterms, and convergence tests.

Definition 145.1 (Hamiltonian truncation regulator specification)**.** A Hamiltonian truncation regulator specification is a tuple

$$\mathfrak{H}_{\mathrm{trunc}} = \big(\mathcal{H}, \mathcal{Q}_H, \{\mathcal{H}_\Lambda, P_\Lambda\}_{\Lambda \in I}, \mathcal{B}_\Lambda, \{\mathcal{O}_a, c_a(\Lambda)\}_{a \in \mathcal{A}_{\mathrm{ct}}}, \mathcal{O}_{\mathrm{obs}}, \mathcal{E}_{\mathrm{sys}}\big)$$

with the following entries.

(i) $\mathcal{H}$ is the target Hilbert space and $\mathcal{Q}_H$ is a lower-bounded closed quadratic form on a dense form domain $\mathrm{Dom}\,\mathcal{Q}_H$. When the limiting Hamiltonian is known as a self-adjoint operator H, $\mathcal{Q}_H$ is its associated form.

(ii) I is a directed cutoff set. For each $\Lambda \in I$, $\mathcal{H}_\Lambda \subset \mathrm{Dom}\,\mathcal{Q}_H$ is a finite-dimensional trial space and P_Λ is the Hilbert-space projection onto $\mathcal{H}_\Lambda$. The specification states whether the intended density statement is strong, form-norm, graph-norm, or another declared topology.

(iii) $\mathcal{B}_\Lambda$ is the ordered computational basis used to build finite matrices. The basis choice includes inner-product conventions, quantum-number sectors, symmetry projections, and normalization of finite volume states.

(iv) $\mathcal{O}_a$ are local counterterm or improvement operators admitted by the regulator, and $c_a(\Lambda)$ are their cutoff-dependent coordinates in a declared scheme. The finite Hamiltonian matrix is

$$H_\Lambda^{\text{ren}} = P_\Lambda H P_\Lambda + \sum_{a\in\mathcal{A}_{\text{ct}}} c_a(\Lambda) P_\Lambda \mathcal{O}_a P_\Lambda$$

when the formal operator expression is meaningful; otherwise the same formula denotes the finite quadratic form specified in the basis $\mathcal{B}_\Lambda$.

(v) $\mathcal{O}_{\text{obs}}$ is the list of finite-regulator observables to be extracted from H_Λ^{ren}, together with their normalization and symmetry-sector labels.

(vi) $\mathcal{E}_{\text{sys}}$ is the systematic-error model: cutoff extrapolation ansatz, finite-volume correction model, counterterm-fitting prescription, residual-bound tolerance, and benchmark scripts used to check finite-matrix algebra.

This records a regulator family. A continuum or infinite-volume claim requires an additional convergence theorem or controlled approximation that states the topology, the limiting quadratic form or observable, and an error bound.

145.1 Projected Hamiltonian Regulator

The elementary version of Definition 145.1 starts with a known self-adjoint Hamiltonian H on $\mathcal{H}$, bounded below. Choose finite-dimensional subspaces

$$\mathcal{H}_\Lambda \subset \mathcal{H}, \qquad P_\Lambda : \mathcal{H} \to \mathcal{H}_\Lambda,$$

with $P_\Lambda \to I$ strongly. The projected Hamiltonian is

$$H_\Lambda = P_\Lambda H P_\Lambda$$

on $\mathcal{H}_\Lambda$. If H is defined by a QFT regulator, H_Λ must be supplemented by local counterterms:

$$H_\Lambda^{\text{ren}} = P_\Lambda H P_\Lambda + \sum_a c_a(\Lambda) P_\Lambda \mathcal{O}_a P_\Lambda.$$

The pair $(\mathcal{H}_\Lambda, H_\Lambda^{\text{ren}})$ is the numerical regulator.

145.2 Rayleigh–Ritz Bound

Lemma 145.2 (Rayleigh–Ritz bound at fixed Hamiltonian)**.** *Let H be a self-adjoint operator on $\mathcal{H}$, bounded below, and suppose that*

$$E_0 \le E_1 \le \cdots \le E_n$$

are isolated eigenvalues below the bottom of the essential spectrum, counted with multiplicity. Let $\mathcal{Q}_H$ be the closed quadratic form of H. For a finite-dimensional subspace $\mathcal{H}_\Lambda \subset \operatorname{Dom}\mathcal{Q}_H$, define

$$E_n(\Lambda) = \min_{\substack{W\subset\mathcal{H}_\Lambda \\ \dim W = n+1}} \max_{\substack{\psi\in W \\ \|\psi\|=1}} \mathcal{Q}_H(\psi,\psi).$$

Then $E_n(\Lambda) \geq E_n$. If $\{\mathcal{H}_\Lambda\}$ is an increasing net and, for some orthonormal eigenbasis $\varphi_0, \ldots, \varphi_n$ of the first $n+1$ eigenspaces, there are $\eta_j^\Lambda \in \mathcal{H}_\Lambda$ with

$$\eta_j^\Lambda \to \varphi_j \quad \textit{in the form norm} \quad \|\psi\|_{\mathcal{Q}}^2 = \|\psi\|^2 + \mathcal{Q}_H(\psi,\psi) - c\|\psi\|^2$$

for a constant $c < \inf \operatorname{spec} H$, then $E_n(\Lambda) \downarrow E_n$.

Proof. The min–max characterization of the n-th discrete eigenvalue gives

$$E_n = \inf_{\substack{W \subset \operatorname{Dom} \mathcal{Q}_H \\ \dim W = n+1}} \sup_{\substack{\psi \in W \\ \|\psi\|=1}} \mathcal{Q}_H(\psi,\psi).$$

The variational problem defining $E_n(\Lambda)$ restricts the same infimum to the smaller family of subspaces contained in $\mathcal{H}_\Lambda$, hence $E_n(\Lambda) \geq E_n$. By the approximation hypothesis, Gram–Schmidt in the ordinary Hilbert norm produces an $(n+1)$-dimensional subspace $W_\Lambda \subset \mathcal{H}_\Lambda$ whose unit sphere converges in form norm to the unit sphere of $\operatorname{span}\{\varphi_0, \ldots, \varphi_n\}$. The Gram–Schmidt coefficients are rational functions of inner products whose limiting denominators are nonzero, so the form-norm convergence is preserved. The supremum of $\mathcal{Q}_H$ on that unit sphere therefore converges to E_n. The upper bound $E_n(\Lambda) \leq \sup_{\psi \in W_\Lambda, \|\psi\|=1} \mathcal{Q}_H(\psi,\psi)$ gives $\limsup_\Lambda E_n(\Lambda) \leq E_n$. Together with the lower bound this proves convergence. Monotonic decrease follows because increasing $\mathcal{H}_\Lambda$ enlarges the family of trial subspaces. ☐

This proposition concerns a fixed self-adjoint operator H. It does not assert monotonicity for a family H_Λ^{ren} whose counterterms change with Λ, nor for a sequence of finite matrices obtained from different regulators. Those comparisons require their own cutoff-error estimates.

Proposition 145.3 (Gap-stable Ritz counting for a fixed Hamiltonian)**.** *Let H and $\mathcal{Q}_H$ be as in Lemma 145.2. Let $\{\mathcal{H}_\Lambda\}$ be an increasing net of finite-dimensional subspaces of* $\operatorname{Dom} \mathcal{Q}_H$ *which is dense in the form norm. Let*

$$E_0 \leq E_1 \leq \cdots$$

be the discrete eigenvalues of H below the bottom of the essential spectrum, counted with multiplicity, and let

$$E_0(\Lambda) \leq E_1(\Lambda) \leq \cdots \leq E_{\dim \mathcal{H}_\Lambda - 1}(\Lambda)$$

be the Ritz eigenvalues obtained by restricting $\mathcal{Q}_H$ to $\mathcal{H}_\Lambda$. Suppose $a < b < \inf \operatorname{spec}_{\mathrm{ess}} H$ and that $a, b \notin \operatorname{spec} H$. Define

$$N_H(c) = \#\{j : E_j < c\}, \qquad c \in \{a, b\},$$

and $m = N_H(b) - N_H(a)$. Then, for all sufficiently large Λ, exactly m Ritz eigenvalues lie in the interval (a,b).

Proof. Since b lies below the essential spectrum and is not an eigenvalue, there are only finitely many eigenvalues below b. Put

$$M = N_H(b), \qquad r = N_H(a).$$

For each $j < M$, the form-norm density of $\bigcup_\Lambda \mathcal{H}_\Lambda$ implies the approximation hypothesis in Lemma 145.2 for the first $j+1$ eigenspaces; hence

$$E_j(\Lambda) \downarrow E_j. \tag{145.1}$$

If $j < r$, then $E_j < a$, and (145.1) gives

$$E_j(\Lambda) < a$$

for all sufficiently large Λ. If $r \le j < M$, then

$$a < E_j < b.$$

The lower inequality $E_j(\Lambda) \ge E_j > a$ follows from Lemma 145.2; convergence gives

$$E_j(\Lambda) < b$$

for all sufficiently large Λ. Thus, eventually, at least $M - r = m$ Ritz values lie in (a, b), namely those with labels

$$r, \ldots, M - 1.$$

It remains to exclude extra Ritz values below b. If $E_M(\Lambda) < b$ for some Λ, then the min–max definition of $E_M(\Lambda)$ supplies an $(M+1)$-dimensional subspace of $\operatorname{Dom} \mathcal{Q}_H$ on whose unit sphere the quadratic form is strictly less than b. Applying the min–max characterization in the full form domain would give $E_M < b$, contradicting $M = N_H(b)$. Therefore no Ritz value with label $j \ge M$ can lie below b. Combining the two statements gives exactly m Ritz values in (a, b) for all sufficiently large Λ. □

The proposition is a theorem about Galerkin approximation to one fixed self-adjoint Hamiltonian and one fixed quadratic form. Its hypotheses are visible: a lower-bounded self-adjoint operator, an isolated finite spectral window, and form-norm density of the trial spaces. It does not justify a counterterm-dependent truncation unless one has first proved that the renormalized finite matrices converge to a fixed limiting quadratic form in a sense strong enough to replace the form-norm density assumption.

145.3 Truncated Conformal Space

For a two-dimensional QFT built as a relevant deformation of a CFT, choose the CFT Hilbert space on a circle and keep states satisfying

$$L_0 + \bar{L}_0 - \frac{c}{12} \le E_{\mathrm{cut}} \frac{L}{2\pi}.$$

The Hamiltonian matrix elements are

$$\langle i|H|j\rangle = \frac{2\pi}{L}\left(\Delta_i - \frac{c}{12}\right)\delta_{ij} + \sum_a \lambda_a \left(\frac{L}{2\pi}\right)^{1-\Delta_a} \langle i|\mathcal{O}_a(1)|j\rangle.$$

High-energy states generate cutoff dependence. The local Hamiltonian counterterms are determined by matching low-energy observables as E_{cut} is varied.

145.4 Exactly Solvable TCSA Benchmark: Ising Energy Deformation

A useful benchmark must distinguish the finite Hamiltonian matrix from the continuum theory it approximates. The thermal deformation of the two dimensional Ising CFT is special because the deformed theory is a free massive Majorana fermion. On a circle of circumference L, in the Neveu–Schwarz sector, positive modes are labelled by

$$r \in \mathbb{Z}_{\geq 0} + \frac{1}{2}, \qquad p_r = \frac{2\pi r}{L}.$$

For each r, the finite-mode Hamiltonian reduces to the real symmetric Bogoliubov block

$$H_r = \begin{pmatrix} -p_r & m \\ m & p_r \end{pmatrix},$$

where m is the Majorana mass induced by the energy-operator coupling in a fixed normalization. The corresponding finite truncation with N modes is the direct sum

$$H_N = \bigoplus_{r=1/2}^{N-1/2} H_r.$$

Remark 145.4 (Finite-mode Majorana benchmark). The eigenvalues of H_N are

$$\left\{ \pm\sqrt{p_r^2 + m^2} \;:\; r = \frac{1}{2}, \frac{3}{2}, \ldots, N - \frac{1}{2} \right\}.$$

For a fixed mode r,

$$\det(H_r - \lambda I) = (-p_r - \lambda)(p_r - \lambda) - m^2 = \lambda^2 - (p_r^2 + m^2).$$

Thus the two eigenvalues of H_r are $\lambda = \pm\sqrt{p_r^2 + m^2}$. The matrix H_N is a direct sum of the blocks H_r, so its characteristic polynomial is the product of the block characteristic polynomials. The benchmark is therefore exact at every finite cutoff N; any numerical deviation from these values measures an implementation or diagonalization error rather than a truncation effect.

The script

`qft_scripts/tcsa_ising_energy_benchmark.py`

diagonalizes H_N and checks Remark 145.4. Its role is narrow and rigorous: it is a public finite-matrix benchmark for Hamiltonian-truncation conventions before nonintegrable perturbations, counterterm fitting, and cutoff extrapolation are introduced. The magnetic-field Ising TCSA problem and the E_8 mass-spectrum computation require additional sectors and perturbing operators.

145.5 Sine-Gordon Zero-Mode Truncation As A Vertex Benchmark

The sine-Gordon model is a useful Hamiltonian-truncation test because the perturbing operator is a pair of vertex operators. A full finite-volume sine-Gordon TCSA computation requires the compact-boson Hilbert space with oscillator descendants, winding sectors, vertex-operator normal ordering, topological-charge choices, cutoff counterterms, and an extrapolation scheme. Before those ingredients are assembled, there is a small finite benchmark that checks the zero-mode vertex selection rule without pretending to compute the complete QFT spectrum.

Let θ be a 2π-periodic coordinate and let

$$\mathcal{H}_0 = L^2(S^1, \mathrm{d}\theta/2\pi), \qquad |n\rangle(\theta) = e^{\mathrm{i}n\theta}, \qquad n \in \mathbb{Z}.$$

The zero-mode coordinate is related to the sine-Gordon field by $\theta = \beta q$, where q is the spatial zero mode of ϕ. All normalization factors depending on L, the compactification radius, and β are absorbed below into two positive finite-regulator coordinates: $\kappa > 0$, the coefficient of the zero-mode kinetic energy, and $g \in \mathbb{R}$, the coefficient of the vertex perturbation. For $N \in \mathbb{Z}_{\geq 0}$, let

$$P_N : \mathcal{H}_0 \to \operatorname{span}\{|n\rangle : -N \leq n \leq N\}$$

be the Fourier cutoff. The finite matrix is

$$H_N^{\mathrm{SG},0} = P_N \left(-\kappa \frac{\mathrm{d}^2}{\mathrm{d}\theta^2} + g\cos\theta \right) P_N. \tag{145.2}$$

In the ordered basis $|-N\rangle, \ldots, |N\rangle$,

$$\left(H_N^{\mathrm{SG},0}\right)_{nm} = \kappa n^2 \delta_{nm} + \frac{g}{2}\left(\delta_{n,m+1} + \delta_{n,m-1}\right), \qquad -N \leq n, m \leq N. \tag{145.3}$$

Remark 145.5 (Benchmark scope). The matrix (145.3) checks three finite facts that are sensitive to conventions. First, $\cos\theta = (e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta})/2$ shifts the compact momentum by ± 1 and has no diagonal zero-mode matrix element. Second, the projected finite matrix is Hermitian. Third, if $N \geq 1$, ordinary finite-dimensional perturbation theory around the nondegenerate state $|0\rangle$ gives the second-order ground-state shift

$$\Delta E_0^{(2)} = \sum_{n \neq 0} \frac{|\langle n|\, g\cos\theta\, |0\rangle|^2}{0 - \kappa n^2} = -\frac{g^2}{2\kappa}. \tag{145.4}$$

Only $n = \pm 1$ contributes. None of these finite identities asserts the existence or accuracy of a full sine-Gordon TCSA spectrum. The full two-dimensional calculation must add oscillator states and all regulator data listed at the beginning of this section.

The companion script

`qft_scripts/sine_gordon_zero_mode_truncation.py`

builds (145.3), diagonalizes it at a declared finite N, checks Hermiticity and reflection symmetry $n \mapsto -n$, and reports the perturbative coefficient (145.4). The calculation check

`calculation-checks/hamiltonian_truncation_dlcq_checks.py`

verifies the $\Delta n = \pm 1$ selection rule, the zero-coupling spectrum, finite Hermiticity, and the second-order shift directly from the finite matrix elements.

145.6 Sine-Gordon TCSA Vertex Assembly

The zero-mode benchmark checks only the compact momentum shift. A genuine TCSA matrix for the sine-Gordon perturbation also needs the oscillator descendants of the compact boson. This section records the finite algebraic object that must be assembled before any discussion of spectra, counterterms, or extrapolation can begin.

Fix a compact free-boson radius coordinate $R > 0$. The finite regulator uses integer momentum and winding labels

$$n \in \{-N_{\text{mom}}, \dots, N_{\text{mom}}\}, \qquad w \in \{-N_{\text{wind}}, \dots, N_{\text{wind}}\},$$

left and right oscillator occupation numbers

$$\mathbf{m} = (m_1, \dots, m_M), \qquad \bar{\mathbf{m}} = (\bar{m}_1, \dots, \bar{m}_M), \qquad m_r, \bar{m}_r \in \mathbb{Z}_{\geq 0},$$

and the total oscillator-level cutoff

$$|\mathbf{m}|_{\text{osc}} + |\bar{\mathbf{m}}|_{\text{osc}} = \sum_{r=1}^{M} r(m_r + \bar{m}_r) \leq L_{\text{osc}}.$$

The finite basis vector is denoted $|n, w; \mathbf{m}, \bar{\mathbf{m}}\rangle$. In the convention used by the companion script, its dimensionless free cylinder energy and spin are

$$E_0(n, w; \mathbf{m}, \bar{\mathbf{m}}) = \frac{1}{2}\left(\frac{n}{R}\right)^2 + \frac{1}{2}(wR)^2 + |\mathbf{m}|_{\text{osc}} + |\bar{\mathbf{m}}|_{\text{osc}}, \tag{145.5}$$

$$S(n, w; \mathbf{m}, \bar{\mathbf{m}}) = nw + |\mathbf{m}|_{\text{osc}} - |\bar{\mathbf{m}}|_{\text{osc}}. \tag{145.6}$$

The additive Casimir energy is irrelevant for the finite vertex-normalization check and has been omitted. If the calculation is performed in the spatial momentum-zero sector, one keeps $S = 0$; equivalently, the spatial integral of a spin-zero perturbing operator kills matrix elements whose bra and ket spins differ.

Let $b_r, \bar{b}_r$ be normalized oscillator annihilation operators, $[b_r, b_s^\dagger] = [\bar{b}_r, \bar{b}_s^\dagger] = \delta_{rs}$. For a real finite vertex charge coordinate α, define the normal-ordered oscillator factor

$$\mathcal{E}_r(\alpha) = \exp\!\left(\frac{\alpha}{\sqrt{r}} b_r^\dagger\right) \exp\!\left(-\frac{\alpha}{\sqrt{r}} b_r\right), \qquad \bar{\mathcal{E}}_r(\alpha) = \exp\!\left(\frac{\alpha}{\sqrt{r}} \bar{b}_r^\dagger\right) \exp\!\left(-\frac{\alpha}{\sqrt{r}} \bar{b}_r\right). \tag{145.7}$$

The finite matrix element of $\mathcal{E}_r(\alpha)$ between normalized number states is the finite sum

$$\langle a|\mathcal{E}_r(\alpha)|b\rangle = \sum_{\ell=0}^{b} \mathbf{1}_{a-b+\ell\geq 0} \, \frac{(-s)^\ell}{\ell!} \sqrt{\frac{b!}{(b-\ell)!}} \, \frac{s^{a-b+\ell}}{(a-b+\ell)!} \sqrt{\frac{a!}{(b-\ell)!}}, \qquad s = \frac{\alpha}{\sqrt{r}}. \tag{145.8}$$

The formula is an exact finite matrix element of the normal-ordered exponential. The sum is finite because the annihilation exponential can remove at most b quanta, and the subsequent creation exponential must create exactly $a - b + \ell$ quanta.

The finite local vertex shifting the compact momentum by $\varepsilon = \pm 1$ has the matrix element

$$\begin{aligned} &\langle n', w'; \mathbf{a}, \bar{\mathbf{a}}|\, V_\varepsilon(0)\, |n, w; \mathbf{b}, \bar{\mathbf{b}}\rangle \\ &\quad = \delta_{n', n+\varepsilon}\, \delta_{w', w} \prod_{r=1}^{M} \langle a_r|\mathcal{E}_r(\varepsilon\alpha)|b_r\rangle \prod_{r=1}^{M} \langle \bar{a}_r|\bar{\mathcal{E}}_r(\varepsilon\alpha)|\bar{b}_r\rangle. \end{aligned} \tag{145.9}$$

The spatially integrated sine-Gordon perturbation in this finite convention is

$$V_{\mathrm{SG}}^{\mathrm{fin}} = \frac{1}{2}\,\Pi_{\Delta S=0}\left(V_+(0) + V_-(0)\right), \tag{145.10}$$

where $\Pi_{\Delta S=0}$ means that the matrix element is set to zero unless the spin coordinate (145.6) is the same on the bra and ket. The finite TCSA matrix assembled by the script is

$$H_{\mathrm{SG}}^{\mathrm{fin}} = E_0 + \lambda V_{\mathrm{SG}}^{\mathrm{fin}}. \tag{145.11}$$

Remark 145.6 (What this finite construction verifies). Equations (145.5)–(145.11) verify the finite matrix-assembly problem for the compact-boson vertex operator. For example, between oscillator vacua one recovers $\langle n+1, w; 0, 0|V_{\mathrm{SG}}^{\mathrm{fin}}|n, w; 0, 0\rangle = 1/2$. Between a vacuum ket and a bra with one left and one right oscillator in the first mode, the integrated matrix element is $\alpha^2/2$, while a bra with only one left oscillator has zero integrated matrix element because the spin changes. The winding label is preserved. These are finite algebraic checks. A continuum sine-Gordon spectrum also requires the physical normalization connecting α to the usual coupling β, the choice of radius and topological sectors, relevant counterterms generated by the cutoff, and a controlled $L_{\mathrm{osc}}, N_{\mathrm{mom}}, M \to \infty$ extrapolation.

The companion script

```
qft_scripts/sine_gordon_tcsa_vertex.py
```

constructs the basis above, assembles (145.10), checks Hermiticity, verifies spin and winding selection, and diagonalizes the resulting finite matrix for a declared cutoff. The calculation-check harness verifies the displayed oscillator-vacuum, one-oscillator, and two-oscillator matrix elements directly from (145.8).

145.7 Nonintegrable Scalar ϕ^4 Truncation

The previous benchmark is exactly solvable. It is useful for checking mode normalizations, but it does not test the matrix assembly problem that appears in a genuinely interacting truncation. The simplest finite QFT benchmark is the massive scalar field on a spatial circle, with the quartic interaction normal ordered relative to the massive free basis. This is still only a finite regulator; the continuum two-dimensional ϕ^4 theory requires the counterterm and cutoff analysis developed in the constructive chapters.

Construction 145.7 (Finite normal-ordered scalar truncation)**.** Fix $L > 0$, $m > 0$, a Fourier cutoff $N \in \mathbb{Z}_{\geq 0}$, a particle cutoff $P_{\max}$, a free-energy cutoff E_{cut}, and an integer total momentum label P_{tot}. For

$$r \in \{-N, -N+1, \dots, N\}, \qquad k_r = \frac{2\pi r}{L}, \qquad \omega_r = \sqrt{m^2 + k_r^2},$$

let $a_r, a_r^\dagger$ be finite-mode creation and annihilation operators with $[a_r, a_s^\dagger] = \delta_{rs}$. The truncated basis is the finite set of occupation vectors $\mathbf{n} = (n_{-N}, \dots, n_N)$ satisfying

$$\sum_{|r|\leq N} n_r \leq P_{\max}, \qquad \sum_{|r|\leq N} r n_r = P_{\mathrm{tot}}, \qquad \sum_{|r|\leq N} \omega_r n_r \leq E_{\mathrm{cut}}. \tag{145.12}$$

The corresponding normalized Fock vector is

$$|\mathbf{n}\rangle = \prod_{|\mathbf{r}|\leq \mathbf{N}} \frac{(\mathbf{a}_\mathbf{r}^\dagger)^{\mathbf{n}_\mathbf{r}}}{\sqrt{\mathbf{n}_\mathbf{r}!}}|\mathbf{0}\rangle.$$

The free Hamiltonian matrix is diagonal:

$$(H_0)_{\mathbf{nm}} = \delta_{\mathbf{nm}} \sum_{|r|\leq N} \omega_r n_r.$$

Define the finite free field at time zero by

$$\phi_N(x) = \sum_{|r|\leq N} \frac{1}{\sqrt{2L\omega_r}} \left(a_r e^{\mathrm{i}k_r x} + a_r^\dagger e^{-\mathrm{i}k_r x} \right),$$

and normal order with respect to the free vacuum. The finite interaction matrix is

$$V_N = P_\Lambda \int_0^L :\phi_N(x)^4: \,\mathrm{d}x\, P_\Lambda, \tag{145.13}$$

where P_Λ is the orthogonal projection onto the span of (145.12). Equivalently,

$$\int_0^L :\phi_N(x)^4: \,\mathrm{d}x = \sum_{\substack{s_i=\pm 1,\ |r_i|\leq N \\ \sum_i s_i r_i = 0}} L \prod_{i=1}^4 \frac{1}{\sqrt{2L\omega_{r_i}}} :B_{s_1,r_1} B_{s_2,r_2} B_{s_3,r_3} B_{s_4,r_4}:, \tag{145.14}$$

with $B_{+1,r} = a_r$ and $B_{-1,r} = a_r^\dagger$. The Kronecker condition in (145.14) is precisely the integral over the circle. The finite benchmark Hamiltonian is

$$H_\Lambda^{(\phi^4)} = H_0 + \frac{\lambda}{4!} V_N. \tag{145.15}$$

No continuum statement is attached to (145.15) unless a counterterm scheme and a cutoff-removal theorem or controlled approximation have also been stated.

The matrix $H_\Lambda^{(\phi^4)}$ is Hermitian at each finite cutoff. The field $\phi_N(x)$ is self-adjoint on the finite Fock space generated by the retained modes, normal ordering preserves the adjoint of the polynomial, and P_Λ is an orthogonal projection. In the explicit sum (145.14), taking the adjoint changes each s_i to $-s_i$ and reverses the operator order before normal ordering; the coefficient is real and the constraint $\sum_i s_i r_i = 0$ is invariant. Thus the restricted matrix is equal to its adjoint. This finite Hermiticity is a minimal consistency check; it does not estimate the dependence on N, $P_{\max}$, or E_{cut}.

A useful normalization check comes from retaining only the zero mode. If $N = 0$, then $\omega_0 = m$ and

$$\int_0^L :\phi_0(x)^4: \,\mathrm{d}x = \frac{1}{4Lm^2} :(a_0 + a_0^\dagger)^4: .$$

The vacuum diagonal entry is zero. In the two-particle zero-mode state $|2\rangle = (a_0^\dagger)^2|0\rangle/\sqrt{2}$, the only diagonal monomial is $6(a_0^\dagger)^2 a_0^2$, hence

$$\langle 2| \int_0^L :\phi_0(x)^4: \,\mathrm{d}x |2\rangle = \frac{6}{4Lm^2} \langle 2|(a_0^\dagger)^2 a_0^2|2\rangle = \frac{3}{Lm^2}. \tag{145.16}$$

This is an elementary finite-regulator identity, but it is convention sensitive: changing the field normalization, the 4! convention, or the normal-ordering prescription changes the displayed number.

The script

```
qft_scripts/phi4_hamiltonian_truncation.py
```

implements Construction 145.7. Its smoke mode builds a small total-momentum-zero matrix, checks Hermiticity, and prints the lowest finite eigenvalues. The calculation check

```
calculation-checks/hamiltonian_truncation_dlcq_checks.py
```

verifies the momentum-sector condition, the free-energy cutoff condition, the zero-coupling reduction to H_0, and the zero-mode identity (145.16). These checks are not a substitute for the constructive ϕ_2^4 continuum theorem; they verify the finite matrix that a reader can diagonalize before studying cutoff renormalization.

145.8 Ising TCSA And TFFSA With Spin Deformation

The two-dimensional Ising field theory with both thermal and magnetic deformations is the basic test case in which Hamiltonian truncation is useful and conceptually delicate. Start from the critical Ising CFT on a spatial circle of circumference L. Let $H_{\mathrm{CFT}}(L)$ be the self-adjoint Hamiltonian generated by cylinder time translations,

$$H_{\mathrm{CFT}}(L) = \frac{2\pi}{L}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right), \qquad c = \frac{1}{2}.$$

Let ε be the energy operator of scaling dimension $\Delta_\varepsilon = 1$, and let σ be the spin operator of scaling dimension $\Delta_\sigma = 1/8$. A TCSA regulator for the two-coupling Ising family is the finite matrix

$$H_{\mathrm{TCSA}}(L; E_{\mathrm{cut}}) = P_{E_{\mathrm{cut}}}\left[H_{\mathrm{CFT}}(L) + \tau\int_0^L \varepsilon_{\mathrm{cyl}}(0,x)\,\mathrm{d}x + h\int_0^L \sigma_{\mathrm{cyl}}(0,x)\,\mathrm{d}x\right]P_{E_{\mathrm{cut}}} + H_{\mathrm{ct}}(E_{\mathrm{cut}}). \tag{145.17}$$

Here $P_{E_{\mathrm{cut}}}$ projects onto CFT states satisfying $(2\pi/L)(L_0 + \bar{L}_0 - c/12) \le E_{\mathrm{cut}}$. The symbol $\mathcal{O}_{\mathrm{cyl}}$ means the cylinder field obtained from the plane field by the conformal map. After fixing a plane normalization, the finite-volume matrix elements contain the factors $(2\pi/L)^{\Delta_a}$ dictated by the state-operator correspondence. The couplings have engineering dimensions

$$[\tau] = 1, \qquad [h] = \frac{15}{8}.$$

The counterterm matrix $H_{\mathrm{ct}}(E_{\mathrm{cut}})$ is part of the regulator. It is determined by a declared matching prescription; omitting it is a choice of truncation scheme, not a theorem that no high-energy effects exist.

The same physical family admits a second truncation. Treat the thermal deformation exactly. For $\tau \neq 0$, the unperturbed theory is the free massive Majorana theory with mass m proportional to τ in the chosen normalization. The magnetic perturbation is then

$$H_{\mathrm{IFT}}(L) = H_{\mathrm{Maj}}(m, L) + h\int_0^L \sigma(0,x)\,\mathrm{d}x. \tag{145.18}$$

Truncating (145.18) in the finite-volume fermion Fock space is the truncated free-fermion-space approach. Its advantage is not a new definition of the continuum theory; rather, it moves the relevant thermal deformation from the perturbation into the basis. This can reduce cutoff sensitivity when the low-energy states are better approximated by massive fermion states than by a small number of conformal states.

We now spell out the finite matrix. Let $\mathfrak{s}$ denote a fermion spin-structure sector on the circle, with allowed momenta

$$p_n = \frac{2\pi n}{L}, \qquad n \in Q_{\mathfrak{s}},$$

where $Q_{\mathfrak{s}}$ is either $\mathbb{Z} + \frac{1}{2}$ or $\mathbb{Z}$, depending on the sector. A finite-volume rapidity θ_n is defined by

$$p_n = m \sinh \theta_n, \qquad \omega_n = m \cosh \theta_n.$$

For a finite set $I = \{n_1, \ldots, n_k\} \subset Q_{\mathfrak{s}}$, write

$$|I\rangle_{\mathfrak{s}} = a^\dagger_{n_1} \cdots a^\dagger_{n_k} |0\rangle_{\mathfrak{s}}, \qquad E^{(0)}_I(L) = E^{\mathrm{vac}}_{\mathfrak{s}}(L) + \sum_{n\in I} \omega_n, \qquad P_I = \sum_{n \in I} p_n.$$

The TFFSA cutoff is an energy cutoff in this massive basis:

$$\mathcal{H}_{\mathrm{FF}}(L, E_{\mathrm{cut}}) = \mathrm{span}\{|I\rangle_{\mathfrak{s}} : E^{(0)}_I(L) \le E_{\mathrm{cut}}\}.$$

The diagonal part of $H_{\mathrm{Maj}}(m, L)$ is exact in this basis. The nontrivial matrix is the spin perturbation.

Definition 145.8 (TFFSA spin matrix element)**.** For two finite-volume fermion states I, J, the connected infinite-volume asymptotic spin matrix element is

$$V^{\mathrm{conn}}_{IJ} = hL\, \delta_{P_I,P_J}\, \frac{F^\sigma_{\mathrm{conn}}(\{\theta_i : i \in I\} \mid \{\theta_j : j \in J\})}{\sqrt{\rho_I \rho_J}}, \qquad \rho_I = \prod_{i\in I} mL \cosh\theta_i. \tag{145.19}$$

Here F^σ_{conn} is the crossed spin-field form factor derived in Chapter 76, and ρ_I is the free Bethe-state density of states. The full finite-volume matrix element also contains disconnected contraction terms when I and J share rapidities, and exponentially small finite-size corrections of order e^{-mL} in a massive theory. A numerical TFFSA scheme must specify whether it uses exact finite-volume Ising form factors, the asymptotic formula (145.19), or a hybrid prescription.

This definition makes the status of the computation explicit. At fixed (L, E_{cut}), after all matrix elements have been declared, TFFSA is ordinary finite-dimensional spectral theory. The physical assertions concern the removal or extrapolation of the cutoff and, if (145.19) is used, the large-mL control of the finite-volume form-factor approximation.

Definition 145.9 (Zero-momentum connected TFFSA block)**.** Fix $m > 0$, $L > 0$, an integer $N_{\mathrm{pair}} \ge 0$, a spin-field normalization $\bar{\sigma}$, and a magnetic coupling h. For

$$r_n = n + \frac{1}{2}, \qquad p_n = \frac{2\pi r_n}{L}, \qquad p_n = m \sinh\theta_n, \qquad \omega_n = m\cosh\theta_n, \quad n = 0, \ldots, N_{\mathrm{pair}} - 1,$$

let

$$|0\rangle, \qquad |n\rangle := |\theta_n, -\theta_n\rangle, \quad n = 0, \ldots, N_{\mathrm{pair}} - 1,$$

be the finite zero-total-momentum basis consisting of the vacuum and two-particle pairs. Set

$$\rho_n = (mL\cosh\theta_n)^2.$$

The connected-block TFFSA matrix is the Hermitian $(N_{\mathrm{pair}}+1)\times(N_{\mathrm{pair}}+1)$ matrix

$$H_{ij} = E_i^{(0)}\delta_{ij} + hL\,V_{ij},$$

where $E_0^{(0)} = 0$, $E_n^{(0)} = 2\omega_n$ for $n \geq 1$, and

$$V_{ii} = \bar{\sigma}.$$

For $i < j$, V_{ij} is defined by the crossed connected form factor

$$V_{ij} = \frac{F^{\sigma}_{\mathrm{conn}}(\{\theta_a + \mathrm{i}\pi : a \in I_i\} \mid \{\theta_b : b \in I_j\})}{\sqrt{\rho_i\rho_j}}, \qquad V_{ji} = \overline{V_{ij}}, \tag{145.20}$$

with $\rho_0 = 1$ and $I_0 = \varnothing$. The diagonal convention $V_{ii} = \bar{\sigma}$ keeps only the disconnected vacuum-expectation contribution. The omitted diagonal connected finite parts are singular crossed limits and require a separate diagonal finite-volume prescription.

Remark 145.10 (Connected TFFSA block)*.* The matrix in Definition 145.9 is Hermitian. If $h = 0$, its eigenvalues are

$$0, \quad 2\omega_0, \quad 2\omega_1, \quad \ldots, \quad 2\omega_{N_{\mathrm{pair}}-1}.$$

The diagonal entries $E_i^{(0)} + hL\bar{\sigma}$ are real. For $i < j$ the lower-triangular entry is defined to be $\overline{V_{ij}}$, hence

$$H_{ji} = hL\,\overline{V_{ij}} = \overline{hL\,V_{ij}} = \overline{H_{ij}}$$

for real h and L. Thus $H^\dagger = H$. At $h = 0$, all off-diagonal entries vanish and the matrix is diagonal with the displayed free finite-volume energies.

The script

`qft_scripts/tffsa_ising_spin_connected.py`

implements Definition 145.9. It checks Hermiticity in smoke mode and prints the finite eigenvalues together with the corresponding free energies. The purpose is to make the form-factor normalization, density factors, and diagonal convention executable. It is not a claim that the small connected block is the full magnetic-Ising spectrum.

Lemma 145.11 (Finite TFFSA spectral-flow derivative)**.** *Let*

$$H(h) = H_0 + hW, \qquad H_0 = H_0^\dagger, \quad W = W^\dagger, \tag{145.21}$$

be a finite Hermitian matrix family, with $h \in \mathbb{R}$. Suppose $\lambda_*(h_*)$ is a simple eigenvalue of $H(h_*)$, and let u_* be a normalized eigenvector. If

$$g = \operatorname{dist}\bigl(\lambda_*(h_*), \operatorname{spec} H(h_*) \setminus \{\lambda_*(h_*)\}\bigr) > 0, \tag{145.22}$$

then for $|h - h_*|\|W\| < g/2$ there is a unique eigenvalue branch $\lambda_*(h)$ near $\lambda_*(h_*)$, counted with its normalized eigenvector $u(h)$ after fixing a differentiable phase. Its derivative at $h = h_*$ is

$$\frac{\mathrm{d}\lambda_*}{\mathrm{d}h}(h_*) = u_*^\dagger W u_*. \tag{145.23}$$

If all eigenvalues of $H(h_*)$ are simple and $\lambda_a'(h_*) = u_a^\dagger W u_a$, then

$$\sum_a \lambda_a'(h_*) = \operatorname{tr} W. \tag{145.24}$$

Proof. Choose a positively oriented circle C centered at $\lambda_*(h_*)$ with radius smaller than $g/2$. For $z \in C$,

$$\|(h-h_*)W(H(h_*)-z)^{-1}\| \le |h-h_*|\|W\|\,\|(H(h_*)-z)^{-1}\| < 1,$$

when $|h-h_*|\|W\| < g/2$. Thus the resolvent identity and the Neumann series give a well-defined resolvent $(H(h)-z)^{-1}$ on C, analytic in h. The Riesz projection

$$P(h) = \frac{1}{2\pi \mathrm{i}} \oint_C (z-H(h))^{-1}\,\mathrm{d}z$$

therefore has rank one for h in this neighborhood and depends differentiably on h. Taking $u(h) = P(h)u_*/\|P(h)u_*\|$, with the phase chosen so that $u_*^\dagger u(h)$ is positive real near h_*, gives a differentiable normalized eigenvector for the unique eigenvalue branch inside C.

Differentiate

$$H(h)u(h) = \lambda_*(h)u(h)$$

at $h = h_*$. Since $H'(h) = W$,

$$Wu_* + H(h_*)u'(h_*) = \lambda_*'(h_*)u_* + \lambda_*(h_*)u'(h_*).$$

Taking the inner product with u_*, and using $u_*^\dagger H(h_*) = \lambda_*(h_*)u_*^\dagger$, cancels the terms containing $u'(h_*)$ and gives (145.23). If all eigenvalues are simple, their normalized eigenvectors form an orthonormal basis, so

$$\sum_a \lambda_a'(h_*) = \sum_a u_a^\dagger W u_a = \operatorname{tr} W.$$

$\square$

For the connected Ising TFFSA block, (145.21) is exactly the finite matrix

$$H(h) = H_0 + hL\,V.$$

The companion script

`qft_scripts/tffsa_ising_spectral_flow.py`

diagonalizes this finite block on a small grid of h-values and checks (145.23) against centered finite differences at a point where the finite spectrum is separated. This is a useful nonintegrable benchmark because it checks the actual finite spectral flow generated by spin-field form factors. It remains a finite-matrix check: comparison with the continuum magnetic-Ising spectrum requires the cutoff, finite-volume, diagonal-form-factor, and counterterm estimates described around (145.19).

145.8.1 Exact E_8 Target Data For Magnetic-Ising Benchmarks

The magnetic Ising ray $\tau = 0$, $h \neq 0$, is the standard case in which Hamiltonian truncation is compared with exact integrable-QFT target data. The target data are not obtained by diagonalizing a finite truncation. They come from the continuum factorized-scattering description of the integrable massive theory. In the numerical chapter we use only the part of that description needed for a reproducible benchmark: the dimensionless mass ratios

$$\rho_a = \frac{m_a}{m_1}, \qquad a = 1, \ldots, 8,$$

where m_1 sets the overall mass scale determined by the normalization of the magnetic coupling h. The eight ratios are

$$\begin{aligned}
&\rho_1 = 1, && \rho_2 = 2\cos\frac{\pi}{5}, && \rho_3 = 2\cos\frac{\pi}{30}, \\
&\rho_4 = 2\rho_2\cos\frac{7\pi}{30}, && \rho_5 = 2\rho_2\cos\frac{2\pi}{15}, && \rho_6 = 2\rho_2\cos\frac{\pi}{30}, \\
&\rho_7 = 4\rho_2\cos\frac{\pi}{5}\cos\frac{7\pi}{30}, && \rho_8 = 4\rho_2\cos\frac{\pi}{5}\cos\frac{2\pi}{15}.
\end{aligned} \tag{145.25}$$

Numerically this is approximately

$$\begin{gathered}
1, \quad 1.6180339887, \quad 1.9890437907, \quad 2.4048671724, \\
2.9562952015, \quad 3.2183404585, \quad 3.8911568233, \quad 4.7833861168.
\end{gathered}$$

There is a finite algebraic check for the convention used in (145.25). Let A_{E_8} be the adjacency matrix of the E_8 Dynkin graph with edges

$$(0,1), (1,2), (2,3), (3,4), (4,5), (5,6), (2,7).$$

The largest eigenvalue is

$$\lambda_{\max}(A_{E_8}) = 2\cos\frac{\pi}{30}.$$

If v is the positive Perron-Frobenius eigenvector and is normalized by $\min_i v_i = 1$, then the unordered component ratios of v are precisely the eight numbers in (145.25). For the node labeling above the normalized vector is

$$(v_0, \ldots, v_7) = (\rho_2, \rho_6, \rho_8, \rho_7, \rho_5, \rho_3, \rho_1, \rho_4).$$

The eigenvector equations $A_{E_8} v = \lambda_{\max} v$ reduce the statement to elementary trigonometric identities such as

$$v_5 + v_7 = \lambda_{\max} v_6, \qquad v_1 + v_3 + v_7 = \lambda_{\max} v_2,$$

after substituting $\lambda_{\max} = 2\cos(\pi/30)$ and the ratios above. This check fixes the table and its ordering; it does not replace the continuum integrable-QFT derivation that the magnetic-Ising scattering theory has this E_8 particle content.

The companion script

`qft_scripts/e8_ising_mass_ratios.py`

prints (145.25) and verifies the Perron–Frobenius check. In a TCSA or TFFSA comparison the observable being compared should be a set of finite-volume energy gaps $\Delta E_a(L, E_{\mathrm{cut}})$, divided by the lightest extracted gap, and then extrapolated in L and E_{cut} with a declared systematic model. Agreement with (145.25) is a target-data comparison only after those regulator coordinates have been specified.

Lemma 145.12 (Feshbach–Schur reduction with resolvent control)**.** *In a finite regulated Hilbert space, let $H = H^\dagger$ be a Hermitian matrix and let $\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}$, where P is an orthogonal projection and $Q = 1 - P$. Suppose E is not an eigenvalue of the finite matrix $QHQ : Q\mathcal{H} \to Q\mathcal{H}$. Then E is an eigenvalue of H with $P\psi \neq 0$ only if*

$$\left[PHP - PHQ(QHQ - E)^{-1}QHP\right] P\psi = E\,P\psi. \tag{145.26}$$

Conversely, if $u \in P\mathcal{H}$ *satisfies* (145.26)*, then*

$$\psi = u - (QHQ - E)^{-1}QHPu$$

is an eigenvector of H *with eigenvalue* E*. Moreover, if*

$$d(E) = \operatorname{dist}(E, \operatorname{spec}(QHQ)) > 0,$$

then the eliminated component obeys the finite-cutoff estimate

$$\|Q\psi\| \le d(E)^{-1} \, \|QHP\,P\psi\|. \tag{145.27}$$

Proof. Write the eigenvalue equation $H\psi = E\psi$ as two block equations:

$$(PHP - E)P\psi + PHQ\,Q\psi = 0, \qquad QHP\,P\psi + (QHQ - E)Q\psi = 0.$$

Since E is in the resolvent set of QHQ, the second equation gives

$$Q\psi = -(QHQ - E)^{-1}QHP\,P\psi.$$

Substituting this into the first equation gives (145.26). This proves that any eigenvector with nonzero P-projection gives a vector in the kernel of the energy-dependent effective operator on $P\mathcal{H}$. Conversely, suppose $u \in P\mathcal{H}$ obeys the Schur-complement equation and define

$$\psi = u - (QHQ - E)^{-1}QHPu.$$

Then $P\psi = u$ and the second block equation is satisfied by construction, because $(QHQ - E)(QHQ - E)^{-1} = Q$ on $Q\mathcal{H}$. Substituting the Schur-complement equation into the first block equation gives zero there as well. Hence $H\psi = E\psi$.

Since QHQ is a finite Hermitian matrix, the spectral theorem gives

$$\|(QHQ - E)^{-1}\| = d(E)^{-1}.$$

Applying this to the explicit formula for $Q\psi$ gives (145.27). This estimate is the part of the finite-dimensional identity that survives as useful information in truncation theory: a high-energy block can be eliminated only together with control of the resolvent distance and the coupling QHP. □

Equation (145.26) is the precise operator identity behind cutoff counterterms in TCSA and TFFSA. In practice one expands the Q-space resolvent at high energy and replaces its low-energy action by local or quasilocal operators allowed by the symmetries. For the Ising spin perturbation this expansion is governed by the short- distance operator product of σ with itself; for the free-fermion basis it is equivalently encoded in the high-rapidity behavior of the spin form factors. A claimed numerical spectrum is therefore meaningful only after the truncation scheme states the cutoff, the finite-volume form-factor prescription, the symmetry sector, the counterterm coordinates, and the extrapolation model.

145.9 Finite-Matrix Spectral Bounds

Every Hamiltonian-truncation or DLCQ computation ultimately diagonalizes a finite Hermitian matrix. This elementary fact is useful only if the finite linear-algebra output is verified separately from the continuum claim. We therefore record the minimal residual bound used by the companion scripts.

Definition 145.13 (Residual bound for a finite truncation)**.** Let $A = A^\dagger$ be a finite Hermitian matrix on a finite-dimensional Hilbert space V. A residual bound for a proposed eigenpair is a triple

$$(\lambda, u, \rho), \qquad \|u\| = 1, \qquad \rho = \|(A - \lambda)u\|. \tag{145.28}$$

It is a bound for the finite matrix A only. The number λ becomes a QFT energy or invariant mass only after the regulator label, counterterm prescription, symmetry sector, finite-volume convention, and continuum-extrapolation hypothesis have also been supplied.

Spectral meaning of a finite residual. Let $A = A^\dagger$ be a finite Hermitian matrix, let $\|u\| = 1$, and set $\rho = \|(A - \lambda)u\|$. Then

$$\operatorname{dist}(\lambda, \operatorname{spec} A) \le \rho. \tag{145.29}$$

More generally, let $S \subset \operatorname{spec} A$ and let P_S be the orthogonal spectral projection onto the span of eigenvectors whose eigenvalues lie in S. If

$$g = \operatorname{dist}\big(\lambda, \operatorname{spec} A \setminus S\big) > 0, \tag{145.30}$$

then

$$\|(1 - P_S)u\| \le \frac{\rho}{g}. \tag{145.31}$$

Choose an orthonormal eigenbasis e_a of A,

$$Ae_a = E_a e_a, \qquad u = \sum_a c_a e_a, \qquad \sum_a |c_a|^2 = 1.$$

Then

$$\rho^2 = \|(A - \lambda)u\|^2 = \sum_a |c_a|^2 |E_a - \lambda|^2. \tag{145.32}$$

If every eigenvalue obeyed $|E_a - \lambda| > \rho$, the right hand side of (145.32) would be strictly larger than ρ^2, a contradiction. This proves (145.29). For the projection bound, split the sum into $E_a \in S$ and $E_a \notin S$. On the complement, $|E_a - \lambda| \ge g$, hence

$$\rho^2 \ge g^2 \sum_{E_a \notin S} |c_a|^2 = g^2 \|(1 - P_S)u\|^2.$$

Taking square roots gives (145.31).

The first inequality says that a small residual forces the existence of some finite-matrix eigenvalue near λ. The second says that, when a target spectral cluster is separated from the rest of the finite spectrum, the approximate eigenvector has quantitatively small leakage out of that cluster. These statements do not estimate the distance between different cutoffs. Cutoff comparison belongs to the Feshbach identity above, to counterterm matching, and to the extrapolation bounds below.

For a Hermitian block decomposition $V = PV \oplus QV$, the same finite linear algebra gives a practical diagnostic for a truncation counterterm. At a chosen low-energy value E, the omitted-space correction is exactly

$$\Sigma_P(E) = PHQ(QHQ - E)^{-1}QHP. \tag{145.33}$$

Approximating $\Sigma_P(E)$ by a low-dimensional local counterterm is a separate approximation; the identity itself is exact only before this replacement is made. A controlled truncation must therefore state a norm, an energy window, and a bound on

$$\Sigma_P(E) - \Sigma_P^{\mathrm{ct}}(E) \tag{145.34}$$

in that window.

The companion check

`calculation-checks/hamiltonian_truncation_dlcq_checks.py`

verifies the finite residual bound (145.29), the projector leakage estimate (145.31), the Feshbach determinant identity behind (145.26), the Ising-energy Bogoliubov benchmark, the connected Ising TFFSA block normalization, the finite Ising TFFSA spectral-flow derivative identity, the E_8 magnetic-Ising target-ratio Perron–Frobenius check, the finite large-N QCD DLCQ quadratic-form identity, and the exact-intercept cases of the finite large-K extrapolation diagnostic.

145.10 Krylov Data as Finite Spectral Evidence

Large truncation matrices are rarely diagonalized by forming a complete eigenbasis. The usual numerical output is a Krylov subspace calculation: Lanczos vectors, a tridiagonal compression, residuals, and sometimes a spectral density with respect to a chosen seed vector. These objects have exact finite-matrix meanings. They become evidence for a QFT spectrum only after the finite regulator, symmetry sector, seed overlap, counterterms, and continuum extrapolation are separately controlled.

Definition 145.14 (Krylov spectral data)**.** Let $A = A^\dagger$ be a finite Hermitian matrix on a finite-dimensional Hilbert space V, and let $v \in V$ be a normalized seed vector. For $m \geq 1$, the m-th Krylov subspace is

$$\mathcal{K}_m(A, v) = \operatorname{span}\{v, Av, \dots, A^{m-1}v\}.$$

A Krylov spectral datum consists of an orthonormal basis

$$V_m = (q_0, \dots, q_{m-1}) \quad \text{of} \quad \mathcal{K}_m(A, v), \qquad q_0 = v,$$

the compression

$$T_m = V_m^\dagger A V_m,$$

and the residual matrix

$$R_m = AV_m - V_m T_m.$$

When the basis is the exact Lanczos basis before breakdown, T_m is tridiagonal and

$$AV_m = V_m T_m + \beta_m q_m e_m^\dagger, \tag{145.35}$$

where $q_m \perp \mathcal{K}_m(A, v)$, $\|q_m\| = 1$, e_m is the last coordinate vector in $\mathbb{C}^m$, and $\beta_m \geq 0$. Breakdown means $\beta_m = 0$, in which case $\mathcal{K}_m(A, v)$ is already an invariant subspace for A.

Suppose $T_m y = \theta y$, $\|y\| = 1$, and set $u = V_m y$. Multiplying (145.35) by y gives the exact finite residual identity

$$(A - \theta)u = \beta_m\, y_m\, q_m, \qquad \|(A - \theta)u\| = \beta_m |y_m|. \tag{145.36}$$

Thus a reported Lanczos Ritz value is accompanied by a finite-matrix residual bound of the kind defined in Definition 145.13. Its content is finite and sectorwise: the regulated matrix in the declared sector has spectrum within $\beta_m|y_m|$ of θ, and the approximate eigenvector leakage into a separated finite spectral cluster is bounded by (145.31). A continuum QFT spectral claim then requires the regulator and extrapolation data specified in Definition 145.1.

The same Krylov data carry finite spectral-measure information. Let

$$\mu_v = \sum_a |\langle e_a, v\rangle|^2 \delta_{E_a}$$

be the finite spectral measure of A in the seed state, where $\{e_a\}$ is an orthonormal eigenbasis of A, $Ae_a = E_a e_a$. If θ_r are the eigenvalues of T_m and $s^{(r)}$ are normalized eigenvectors of T_m, define

$$w_r = |\langle e_1, s^{(r)}\rangle|^2, \qquad \mu_{v,m}^{\mathrm{Krylov}} = \sum_{r=1}^{m} w_r \delta_{\theta_r}.$$

For every polynomial p of degree at most $2m - 1$,

$$\int p(\lambda)\, \mathrm{d}\mu_v(\lambda) = \sum_{r=1}^{m} w_r p(\theta_r). \tag{145.37}$$

To see the mechanism, first note that if $\deg b \le m - 1$, then

$$b(A)v = V_m b(T_m) e_1.$$

Let $\pi_m(\lambda) = \det(\lambda I - T_m)$. The Lanczos recurrence implies that $\pi_m(A)v$ is orthogonal to $\mathcal{K}_m(A, v)$, while $\pi_m(T_m) = 0$. Any polynomial p with $\deg p \le 2m - 1$ decomposes as $p = a\pi_m + b$ with $\deg a, \deg b \le m - 1$. Since $a(A)v \in \mathcal{K}_m(A, v)$,

$$\langle v, p(A)v\rangle = \langle v, b(A)v\rangle = e_1^\dagger b(T_m) e_1 = e_1^\dagger p(T_m) e_1,$$

and diagonalizing T_m gives (145.37). The finite Krylov measure is therefore an exact quadrature rule for the seed spectral measure, not a complete list of all finite eigenvalues unless the chosen seeds have overlap with the relevant spectral subspaces.

Symmetry sectors must be part of the same finite spectral datum. If a finite group or compact group acts by unitary matrices commuting with A, and P_α is the exact projector onto a declared irreducible sector, then the Krylov calculation for that sector is the calculation for the finite matrix

$$A_\alpha = P_\alpha A P_\alpha \quad \text{on} \quad P_\alpha V.$$

A seed v with $P_\alpha v = 0$ contains no spectral information about that sector, and a single nonzero seed can miss degenerate eigenvectors orthogonal to its cyclic subspace. A numerical claim about absence of states in a finite window must therefore use either full diagonalization in the sector, a block-Krylov seed family with a rank check for the target subspace, or an independent variational lower bound. Without such sector coverage, Lanczos data are reliable spectral evidence for the cyclic subspace generated by the stated seed, not for the whole Hilbert space.

145.11 Variational Ansatz Bounds

Tensor-network, DMRG, MERA, and neural-state computations enter QFT as finite-regulator variational methods. The ansatz may be sophisticated, but the first mathematical object is still a finite Hilbert space, a finite Hermitian operator, and a family of trial vectors. The following finite bounds are the common core behind those methods.

Definition 145.15 (Finite variational ansatz)**.** A finite variational ansatz for a regulated Hamiltonian consists of

$$(V, A, \Theta, \Psi, \mathcal{S}, \mathcal{E}_{\text{samp}}).$$

Here V is a finite-dimensional Hilbert space, $A = A^\dagger$ is the finite Hamiltonian or invariant-mass matrix in a declared sector, Θ is the parameter space, and

$$\Psi : \Theta \to V \setminus \{0\}$$

is the trial-vector map. The entry $\mathcal{S}$ records exact constraints imposed on the ansatz, such as gauge invariance, global charge, momentum, parity, tensor-network bond dimension, locality of tensor contractions, or a neural architecture. The entry $\mathcal{E}_{\text{samp}}$ records the sampling law and estimator if the Rayleigh quotient is estimated stochastically rather than contracted exactly.

For $\psi \neq 0$, write

$$E_\psi = \frac{\langle \psi, A\psi \rangle}{\langle \psi, \psi \rangle}, \qquad r_\psi = (A - E_\psi)\psi, \qquad \Delta_\psi^2 = \frac{\|r_\psi\|^2}{\|\psi\|^2}.$$

The number Δ_ψ^2 is the energy variance in the trial state:

$$\Delta_\psi^2 = \frac{\langle \psi, A^2\psi \rangle}{\langle \psi, \psi \rangle} - E_\psi^2.$$

Consequently

$$\operatorname{dist}(E_\psi, \operatorname{spec} A) \leq \Delta_\psi. \tag{145.38}$$

This is exactly the residual bound (145.29) applied to $\psi/\|\psi\|$. If $S \subset \operatorname{spec} A$ is a separated finite spectral cluster and

$$g = \operatorname{dist}(E_\psi, \operatorname{spec} A \setminus S) > 0,$$

then

$$\|(1 - P_S)\psi\| \leq \frac{\Delta_\psi}{g}\,\|\psi\|. \tag{145.39}$$

Thus variance is a finite-matrix eigenstate diagnostic. It measures spectral concentration of the vector already produced by the ansatz; it does not by itself measure the expressive power of the ansatz family.

The Rayleigh quotient supplies a different bound when an exact lower spectral edge is available. Let

$$E_0 < E_1$$

be the lowest two distinct eigenvalues of A, and let P_0 be the ground-space projection. Expanding a normalized vector in an eigenbasis gives

$$E_\psi - E_0 = \sum_a |\langle e_a, \psi \rangle|^2 (E_a - E_0) \geq (E_1 - E_0)\|(1 - P_0)\psi\|^2.$$

Therefore

$$\|(1 - P_0)\psi\|^2 \le \frac{E_\psi - E_0}{E_1 - E_0}. \tag{145.40}$$

This bound is useful only after E_0 and a finite gap are known by another method, or after a rigorous lower bound replaces the exact ground energy. The variational upper bound $E_\psi \ge E_0$ alone is insufficient to bound distance to the ground-state subspace.

For a smooth normalized ansatz $\psi(\theta)$, the differential of the Rayleigh quotient exposes the precise meaning of a variational optimum. If $\|\psi(\theta)\| = 1$ and $\xi \in T_\theta\Theta$, put $\dot\psi_\xi = d\psi_\theta(\xi)$. Differentiating $\langle\psi, A\psi\rangle$ and using $\mathrm{Re}\langle\dot\psi_\xi, \psi\rangle = 0$ gives

$$dE_\theta(\xi) = 2\ \mathrm{Re}\langle\dot\psi_\xi, (A - E_\theta)\psi(\theta)\rangle. \tag{145.41}$$

A stationary variational parameter therefore makes the residual orthogonal to the tangent space of the ansatz manifold. An eigenvector error bound is stronger: it requires the residual norm $\|(A-E_\theta)\psi(\theta)\|$ to be small, or a separate argument that the tangent directions cover the missing finite directions.

For neural-state and Monte Carlo variational calculations one also needs the finite estimator identity. Choose an orthonormal computational basis $\{|x\rangle\}_{x\in X}$ and assume $\psi_x = \langle x|\psi\rangle \ne 0$ on the sampled support. Define

$$p_\psi(x) = \frac{|\psi_x|^2}{\sum_y |\psi_y|^2}, \qquad E_{\mathrm{loc}}(x) = \frac{(A\psi)_x}{\psi_x}.$$

Then

$$E_\psi = \sum_x p_\psi(x) E_{\mathrm{loc}}(x), \tag{145.42}$$

and

$$\Delta_\psi^2 = \sum_x p_\psi(x) |E_{\mathrm{loc}}(x) - E_\psi|^2. \tag{145.43}$$

The second identity uses Hermiticity of A:

$$\sum_x p_\psi(x)|E_{\mathrm{loc}}(x) - E_\psi|^2 = \frac{\|(A - E_\psi)\psi\|^2}{\|\psi\|^2}.$$

Sampling estimates of (145.42) and (145.43) require the Markov-chain or independent-sample error analysis of Section 141.12; optimization bias, autocorrelation, and architecture bias are additional entries of $\mathcal{E}_{\mathrm{samp}}$, not properties of the finite identities.

For a sampled variational calculation, the local-energy time series has its own finite error coordinate. Let $X_0, \dots, X_{N-1}$ be a stationary finite Markov chain with invariant distribution p_ψ, and let

$$F_t = \mathrm{Re}\, E_{\mathrm{loc}}(X_t), \qquad \mu = \mathbb{E} F_t.$$

Define

$$\gamma_t = \mathrm{Cov}(F_0, F_t), \qquad \gamma_0 > 0, \qquad \rho_t = \gamma_t/\gamma_0.$$

For

$$\overline{F}_N = \frac{1}{N}\sum_{t=0}^{N-1} F_t,$$

stationarity gives the exact finite identity

$$\mathrm{Var}(\overline{F}_N) = \frac{\gamma_0}{N}\left[1 + 2\sum_{t=1}^{N-1}\left(1 - \frac{t}{N}\right)\rho_t\right]. \tag{145.44}$$

Indeed,

$$\mathrm{Var}(\overline{F}_N) = \frac{1}{N^2}\sum_{r,s=0}^{N-1}\mathrm{Cov}(F_r, F_s) = \frac{1}{N^2}\left(N\gamma_0 + 2\sum_{t=1}^{N-1}(N-t)\gamma_t\right),$$

because there are $N - t$ ordered pairs at separation t in each direction. Thus the finite variance-inflation coordinate is

$$I_N = 1 + 2\sum_{t=1}^{N-1}\left(1 - \frac{t}{N}\right)\rho_t, \qquad N_{\mathrm{eff}} = N/I_N.$$

For neural-state and VMC energy estimates, (145.44) is the object reported by an error bar after a window or blocking prescription has been declared. Thermalization, slow-mode monitoring, truncation of the autocorrelation window, independent-chain comparison, and the same analysis for score–local-energy force coordinates remain part of the sampling-error datum, not consequences of the local-energy identity.

The force used in a smooth neural-state or variational-Monte-Carlo optimization is an equally finite identity. Let $\theta = (\theta^a)$ be real local coordinates on a patch of Θ, and assume $\psi_x(\theta) \neq 0$ on the sampled support. Define the score coordinate

$$\mathsf{s}_a(x;\theta) = \partial_a \log\psi_x(\theta).$$

Put $p_\theta = p_{\psi(\theta)}$ and $E_{\mathrm{loc}}(x;\theta) = ((A\psi(\theta))_x)/\psi_x(\theta)$. Without normalizing the vector $\psi(\theta)$, differentiation of the Rayleigh quotient gives

$$\partial_a E_\theta = 2\ \mathrm{Re}\,\frac{\langle\partial_a\psi, (A - E_\theta)\psi\rangle}{\langle\psi,\psi\rangle}.$$

In the computational basis this becomes the score–local-energy covariance identity

$$\partial_a E_\theta = 2\ \mathrm{Re}\sum_x p_\theta(x)\Big(\overline{\mathsf{s}_a(x;\theta)} - \sum_y p_\theta(y)\overline{\mathsf{s}_a(y;\theta)}\Big)\big(E_{\mathrm{loc}}(x;\theta) - E_\theta\big). \tag{145.45}$$

The centering of the score does not change the value because $\sum_x p_\theta(x)(E_{\mathrm{loc}}(x;\theta) - E_\theta) = 0$. Equation (145.45) is what a stochastic force estimator estimates after a sampling law has been declared. It is not an optimizer theorem: a small empirical force may indicate a tangent-space stationary point while the residual norm, the energy variance, the integrated autocorrelation time of the Markov chain, or the ansatz bias remains large. For QFT applications the finite force is therefore reported together with (145.43), sector constraints, and the regulator extrapolation data of Section 145.16.

Tensor-network terminology fits this definition by specifying Ψ. A finite matrix-product state on a spin chain of length L, physical local dimension d, and bond dimension χ is a vector whose amplitudes are finite contractions of matrices

$$\psi_{i_1\ldots i_L} = \mathrm{Tr}\Big(A_1^{i_1}A_2^{i_2}\cdots A_L^{i_L}\Big), \qquad A_\ell^i \in \mathrm{Mat}_{\chi_{\ell-1}\times\chi_\ell}(\mathbb{C}).$$

For an infinite translation-invariant MPS, a transfer-operator spectrum can estimate a correlation length, but a QFT continuum claim still requires a sequence of finite regulators, a scaling map for operators, and convergence of the corresponding Schwinger or Wightman data. Bond dimension, discarded weight, transfer-operator gap, local-energy variance, and Krylov residuals are finite evidence coordinates. The continuum claim is the additional statement that those coordinates control the regulator limit in a declared topology.

Finite transfer-operator correlation length. The most common tensor-network continuum diagnostic is the transfer-operator gap. Here is the finite statement in the form needed for QFT use. For a translation-invariant infinite MPS with tensors A^i, define the finite transfer operator

$$\mathbb{E}(X) = \sum_i A^i X (A^i)^\dagger$$

on $\mathrm{Mat}_{\chi\times\chi}(\mathbb{C})$. Assume, for this finite ansatz, that $\mathbb{E}$ has a simple eigenvalue $\lambda_0 = 1$, positive left and right eigenvectors ℓ, r normalized by $\ell(r) = 1$, and remaining eigenvalues λ_α with $|\lambda_\alpha| < 1$. A local lattice operator O is represented by the insertion map

$$\mathbb{E}_O(X) = \sum_{i,j} O_{ij} A^i X (A^j)^\dagger.$$

The connected two-point function at lattice separation $n \geq 1$ is

$$C_O(n) = \ell(\mathbb{E}_O \mathbb{E}^n \mathbb{E}_O r) - \ell(\mathbb{E}_O r)^2. \tag{145.46}$$

If $\mathbb{E}$ is diagonalizable on the subspace complementary to the fixed vector, then

$$C_O(n) = \sum_\alpha A_\alpha(O) \lambda_\alpha^n, \tag{145.47}$$

where the coefficients are products of the left and right spectral projections of $\mathbb{E}_O$. Thus a single isolated contributing eigenvalue with $\lambda_\alpha > 0$ gives a finite lattice correlation length

$$\xi_\alpha^{\mathrm{lat}} = -\frac{1}{\log \lambda_\alpha}.$$

If the lattice spacing is a, the corresponding finite effective mass is

$$m_\alpha(a) = -\frac{1}{a} \log \lambda_\alpha. \tag{145.48}$$

This is a finite transfer-operator identity, not yet a continuum QFT statement. To interpret $m_\alpha(a)$ as a physical mass one needs a sequence of regulator data $(a, \chi(a), A^i(a), O_a)$, a normalization of the operator O_a, and estimates of the form

$$\lambda_\alpha(a) = \exp[-am + O(a^{1+\delta})], \qquad Z_O(a)^2 A_\alpha(O_a) \longrightarrow A_{\mathrm{phys}}, \tag{145.49}$$

together with convergence of the corresponding multipoint Schwinger or Wightman distributions. Excited transfer modes are not nuisances in this language; they are visible finite terms in (145.47). If

$$C_O(n) = A_1 e^{-am_1 n} + A_2 e^{-am_2 n}, \qquad m_2 > m_1, \quad A_1, A_2 > 0,$$

then the finite effective mass

$$m_{\text{eff}}(a,n) = -\frac{1}{a}\log\frac{C_O(n+1)}{C_O(n)}$$

lies between m_1 and m_2 and approaches m_1 only after the second transfer mode is suppressed. A finite transfer gap therefore tells one which exponential appears in the finite ansatz; it does not, by itself, prove the continuum operator map, reflection positivity, locality, or cutoff-removal estimate needed for QFT.

Calculation check. The companion check

`calculation-checks/hamiltonian_truncation_dlcq_checks.py`

also verifies the finite variational identities (145.38)– (145.45): energy variance as residual norm, spectral distance bounded by variance, ground-projection leakage from a known finite gap, the tangent-gradient formula, the local-energy Monte Carlo identities, the Markov-chain variance-inflation formula (145.44), and the score-force identity in a finite basis. It also checks the finite transfer-operator spectral expansion (145.47), the single-mode effective-mass identity (145.48), the two-mode contamination bound for $m_{\text{eff}}(a,n)$, and the elementary logarithmic remainder bound implicit in (145.49).

145.12 Light-Front Kinematics And Constraints

Light-front Hamiltonian methods use a null hyperplane as the time slice. In the mostly-plus convention, choose one spatial direction x^1 and set

$$x^{\pm} = \frac{x^0 \pm x^1}{\sqrt{2}}, \qquad p^{\pm} = \frac{p^0 \pm p^1}{\sqrt{2}}. \tag{145.50}$$

For D-dimensional Minkowski space,

$$\mathrm{d}s^2 = -2\,\mathrm{d}x^+\mathrm{d}x^- + \mathrm{d}x_\perp^2, \qquad p\cdot x = -p^-x^+ - p^+x^- + p_\perp\cdot x_\perp, \tag{145.51}$$

and x^+ is the evolution parameter. The generator conjugate to x^+ is P^-; the longitudinal momentum P^+ and transverse momenta $P_\perp$ label superselection sectors for the finite light-front Hamiltonians used below. These identities are kinematic; they do not define an interacting QFT until the constraint and zero-mode sectors have also been specified.

Definition 145.16 (Light-front Hamiltonian regulator specification)**.** A light-front Hamiltonian regulator specification is a Hamiltonian truncation regulator specification $\mathfrak{H}_{\text{trunc}}$ together with the following additional structure.

(i) A null-coordinate chart (145.50), the sign convention (145.51), and a declared evolution parameter x^+.

(ii) A positive-longitudinal nonzero-mode sector, with one-particle labels carrying $p^+ > 0$ and whatever transverse, spin, flavor, color, and statistics labels are kept by the regulator.

(iii) A zero-mode and constraint sector

$$\mathcal{Z}_\Lambda, \qquad \mathcal{C}_{\Lambda,\alpha} = 0, \qquad \mathcal{G}_{\Lambda,\mathrm{res}},$$

recording the $p^+ = 0$ modes, finite global Gauss-law constraints, residual gauge transformations, and any additional compact or topological degrees of freedom. Eliminating, quantizing, or imposing these variables are different regulator choices and must be part of the regulator specification.

(iv) A finite physical Hilbert space

$$\mathcal{H}_\Lambda^{\mathrm{phys}} \subset \mathcal{H}_\Lambda^{p^+>0} \otimes \mathcal{H}_{\mathcal{Z},\Lambda}$$

obtained after the stated constraint prescription, together with commuting finite operators P_Λ^+ and $P_{\perp,\Lambda}$ on the physical sector.

(v) A finite self-adjoint light-front Hamiltonian P_Λ^- on $\mathcal{H}_\Lambda^{\mathrm{phys}}$, commuting with the retained translation generators on the finite physical sector. In a simultaneous $P^+ = p^+ > 0$, $P_\perp = p_\perp$ sector, the finite invariant-mass operator is

$$M_\Lambda^2 = 2p^+ P_\Lambda^- - p_\perp^2. \tag{145.52}$$

Outside a fixed momentum sector the same formula is read as a joint functional calculus expression on the common spectral representation of the commuting momenta and P_Λ^-.

(vi) A continuum diagnostic specifying how p^+-cutoffs, harmonic resolution, transverse cutoffs, counterterms, principal-value prescriptions, and zero-mode constraints are taken to their limiting values.

One-particle measure. The free massive representation fixes the basic measure behind the positive-p^+ Fock construction. For a test function F on momentum space and $m > 0$,

$$\int_{\mathbb{R}^D} \mathrm{d}^D p\, \theta(p^0)\delta(p^2 + m^2) F(p) = \int_{p^+>0} \frac{\mathrm{d}p^+\, \mathrm{d}^{D-2}p_\perp}{2p^+}\, F\left(p^+, \frac{m^2 + p_\perp^2}{2p^+}, p_\perp\right). \tag{145.53}$$

Indeed,

$$p^2 + m^2 = -2p^+p^- + p_\perp^2 + m^2,$$

so the p^--integration localizes at

$$p^- = \frac{m^2 + p_\perp^2}{2p^+} \tag{145.54}$$

with Jacobian $|\partial(p^2 + m^2)/\partial p^-| = 2p^+$. The positive-energy condition selects $p^+ > 0$ on the massive shell. The massless boundary $p^+ = 0$, and the constrained zero modes of gauge theories, are not included in this one-particle measure; they are entries of $\mathcal{Z}_\Lambda$ and $\mathcal{C}_{\Lambda,\alpha}$ in Definition 145.16.

Spectral interpretation. At fixed p^+ and $p_\perp$, M_Λ^2 in (145.52) and P_Λ^- have the same finite eigenvectors, and their eigenvalues are related by

$$M^2 = 2p^+ P^- - p_\perp^2.$$

This is the finite-sector reason light-front calculations often diagonalize P^- while reporting invariant masses. A gauge-theory light-front calculation is therefore specified only after the nonzero-mode Hamiltonian, zero-mode/constraint prescription, and residual gauge data have all been declared in the same regulator scheme.

145.13 DLCQ Regulator

DLCQ is the compact positive-longitudinal-momentum specialization of Definition 145.16. It compactifies

$$x^- \sim x^- + 2\pi R$$

so that

$$p^+ = \frac{n}{R}, \qquad n \in \mathbb{Z}_{\geq 0}.$$

The total longitudinal momentum is

$$P^+ = \frac{K}{R},$$

where K is the harmonic resolution. The finite DLCQ Hilbert space at fixed K consists of partitions of K among partons, with transverse and internal cutoffs added as needed. Zero modes with $n = 0$ are constrained variables or additional degrees of freedom depending on the theory; their treatment is part of the regulator.

Definition 145.17 (DLCQ regulator specification)**.** A DLCQ regulator specification is a Hamiltonian truncation regulator specification together with the following light-front entries.

(i) The spatial light-front coordinate is compactified as $x^- \sim x^- + 2\pi R$, and the total longitudinal momentum sector is fixed by $P^+ = K/R$ with harmonic resolution $K \in \mathbb{Z}_{>0}$.

(ii) The finite basis consists of all declared parton partitions $K = n_1 + \cdots + n_N$, with $n_i > 0$, together with the transverse, internal, gauge, and statistics labels kept by the regulator.

(iii) The zero-mode sector is specified as part of the regulator: constrained variables, eliminated modes, residual global constraints, and additional zero-mode degrees of freedom are all recorded before diagonalization.

(iv) The finite operator is the light-front Hamiltonian P_K^-, or equivalently the invariant mass operator $M_K^2 = 2P^+P_K^- - P_\perp^2$, with counterterms and principal-value prescriptions stated in the same regulator scheme.

(v) The continuum diagnostic specifies the ordered finite labels followed as $K \to \infty$, the extrapolation variable, and the remainder model used to interpret finite-K data.

145.14 Scalar ϕ^4 DLCQ Warmup

Before the large-N gauge-theory example, it is useful to record a nonintegrable scalar light-front matrix whose finite definition is completely explicit. This is a warmup for DLCQ, not a definition of the continuum ϕ_2^4 theory. The $p^+ = 0$ mode is omitted in the construction below; in light-front scalar theory that omission is a regulator choice and is precisely where vacuum and symmetry-breaking information can hide. A continuum claim therefore needs a zero-mode prescription and a counterterm/extrapolation theorem in addition to the finite matrix.

Construction 145.18 (Finite scalar DLCQ matrix)**.** Fix a harmonic resolution $K \in \mathbb{Z}_{>0}$, a mass $m > 0$, and a finite coupling coordinate g. The finite bosonic basis consists of occupation vectors

$$\mathbf{N} = (N_1, \ldots, N_K) \in \mathbb{Z}_{\geq 0}^K, \qquad \sum_{n=1}^{K} nN_n = K. \tag{145.55}$$

The parton with label n carries longitudinal momentum $p_n^+ = n/R$, and the total sector has $P^+ = K/R$. The free invariant mass is

$$M_{0,K}^2(\mathbf{N}) = Km^2 \sum_{n=1}^{K} \frac{N_n}{n}. \tag{145.56}$$

This is the finite-sector form of

$$M^2 = 2P^+P^-, \qquad P_0^- = \sum_{n=1}^{K} \frac{m^2}{2p_n^+} N_n,$$

with zero transverse momentum.

Let $a_n, a_n^\dagger$ be bosonic creation and annihilation operators with $[a_n, a_m^\dagger] = \delta_{nm}$. The dimensionless normal-ordered quartic operator Q_K is

$$Q_K = \sum_{\substack{\epsilon_i = \pm 1,\ 1 \le n_i \le K \\ \sum_i \epsilon_i n_i = 0}} \frac{:B_{\epsilon_1,n_1}B_{\epsilon_2,n_2}B_{\epsilon_3,n_3}B_{\epsilon_4,n_4}:}{\sqrt{n_1 n_2 n_3 n_4}}, \qquad B_{+1,n} = a_n, \quad B_{-1,n} = a_n^\dagger. \tag{145.57}$$

The Kronecker condition is longitudinal momentum conservation. If one uses the field normalization

$$\phi(x^-) = \sum_{n>0} \frac{a_n e^{-inx^-/R} + a_n^\dagger e^{inx^-/R}}{\sqrt{4\pi n}}, \qquad x^- \sim x^- + 2\pi R,$$

then $g = \lambda/(96\pi)$ for the interaction $\lambda \int :\phi^4:/4!$ when written in invariant-mass units. The chapter treats g as the finite regulator coordinate, because a cutoff-dependent coupling and possible zero-mode counterterms are part of a continuum scheme.

The finite DLCQ invariant-mass matrix is

$$M_K^2 = M_{0,K}^2 + gK\, P_K Q_K P_K, \tag{145.58}$$

where P_K is the projection onto the finite partition basis (145.55). Optional particle-number cutoffs may be added as further finite-regulator coordinates, but they must then be removed or extrapolated separately from the harmonic resolution.

The matrix (145.58) is Hermitian. The coefficient in (145.57) is real, and taking the adjoint sends every ordered monomial with signs ϵ_i to the corresponding monomial with signs $-\epsilon_i$. The constraint $\sum_i \epsilon_i n_i = 0$ is invariant under this sign change, and normal ordering restores the displayed ordered form. Since P_K is an orthogonal projection onto a fixed total-resolution sector, the projected finite matrix is Hermitian.

A small matrix element fixes the convention. At $K = 3$, the states

$$|1,1,1\rangle = \frac{(a_1^\dagger)^3}{\sqrt{3!}}|0\rangle, \qquad |3\rangle = a_3^\dagger|0\rangle$$

both lie in the same harmonic-resolution sector. The only term in Q_K that connects them is the ordered family with one $a_3^\dagger$ and three a_1's, together with its four possible positions in the quartic product. Thus

$$\langle 3|Q_3|1,1,1\rangle = \frac{4}{\sqrt{3}}\sqrt{3!} = 4\sqrt{2}. \tag{145.59}$$

For the three-particle state itself, the two-creation two-annihilation normal-ordered term with all momenta equal to one gives

$$\langle 1,1,1|Q_3|1,1,1\rangle = 6\,\langle 1,1,1|(a_1^\dagger)^2 a_1^2|1,1,1\rangle = 36. \tag{145.60}$$

These finite identities are not physical mass predictions; they are normalization checks for the regulator matrix.

The script

```
qft_scripts/phi4_dlcq.py
```

implements Construction 145.18. Its smoke mode builds a small positive-p^+ matrix, checks finite Hermiticity, and prints the lowest finite invariant masses. The companion calculation check

```
calculation-checks/hamiltonian_truncation_dlcq_checks.py
```

verifies total harmonic resolution, zero-coupling reduction to (145.56), Hermiticity, and the two convention-sensitive matrix elements (145.59) and (145.60).

145.15 Large-N Two-Dimensional QCD: A Finite DLCQ Matrix

The standard large-N two-dimensional QCD meson equation involves the principal-value operator

$$(\mathcal{K}\phi)(x) = \mathrm{PV}\int_0^1 \frac{\phi(x)-\phi(y)}{(x-y)^2}\,\mathrm{d}y, \qquad 0<x<1,$$

together with endpoint mass terms

$$\frac{m_1^2}{x}\phi(x) + \frac{m_2^2}{1-x}\phi(x).$$

This section records a finite regulator, not a continuum theorem. Fix a harmonic resolution $K \geq 3$ and grid points $x_n = n/K$, $n = 1,\dots,K-1$. For $\gamma > 0$, define the real symmetric matrix

$$(M_K^2)_{nm} = \delta_{nm}\left[\frac{m_1^2}{x_n} + \frac{m_2^2}{1-x_n} + \gamma K \sum_{\substack{j=1\\ j\neq n}}^{K-1} \frac{1}{(n-j)^2}\right] - (1-\delta_{nm})\frac{\gamma K}{(n-m)^2}.$$

The normalization of γ depends on the convention for g^2N/π. The finite object above is sufficient to illustrate harmonic resolution, positivity, and extrapolation obligations.

Finite principal-value quadratic form. For $m_1^2, m_2^2, \gamma > 0$ and any $v = (v_1,\dots,v_{K-1}) \in \mathbb{R}^{K-1}$,

$$v^T M_K^2 v = \sum_{n=1}^{K-1}\left(\frac{m_1^2}{x_n} + \frac{m_2^2}{1-x_n}\right)v_n^2 + \gamma K \sum_{1\leq n<m\leq K-1} \frac{(v_n-v_m)^2}{(n-m)^2}. \tag{145.61}$$

In particular M_K^2 is positive definite.

Substitute the definition of M_K^2 into $v^T M_K^2 v$. The mass terms give the first displayed sum. The off-diagonal kernel gives

$$\gamma K \sum_{n=1}^{K-1} \sum_{\substack{m=1 \\ m \neq n}}^{K-1} \frac{v_n^2 - v_n v_m}{(n-m)^2}.$$

Pair the term (n, m) with (m, n). For each unordered pair $\{n, m\}$,

$$\frac{v_n^2 - v_n v_m}{(n-m)^2} + \frac{v_m^2 - v_m v_n}{(m-n)^2} = \frac{(v_n - v_m)^2}{(n-m)^2}.$$

This gives the stated quadratic form. Since $0 < x_n < 1$, the mass coefficients are positive; hence $v^T M_K^2 v > 0$ for $v \neq 0$. The calculation is a finite-regulator algebra check: it proves positivity of the displayed matrix. Positivity and self-adjointness of the continuum principal-value operator require a separate continuum-domain construction.

The script

```
qft_scripts/thooft_dlcq.py
```

constructs this matrix and diagonalizes it at finite K. A continuum claim would require an analysis of the $K \to \infty$ limit, endpoint behavior, zero modes, normalization of γ, and comparison with a Lorentz-covariant reconstruction. The script therefore prints finite eigenvalues and a quadratic-form positivity check rather than presenting the numbers as physical meson masses.

145.16 Finite-Regulator Extrapolation

The output of a lattice, Hamiltonian-truncation, or DLCQ script is a number attached to a regulator. A continuum claim is a separate mathematical statement about a limiting procedure. This section records the minimal finite-dimensional facts used in the companion scripts and explains what an extrapolation hypothesis must contain.

Definition 145.19 (Finite-regulator observable record)**.** A finite-regulator observable record consists of:

$$\big(I, \preceq;\ X_i,\ A_i,\ \sigma_i,\ \mathcal{E}_i\big).$$

Here $(I, \preceq)$ is a directed set of regulator labels, X_i is the finite object being computed at label i, $A_i \in \mathbb{R}$ is the finite observable extracted from X_i, $\sigma_i \geq 0$ is the statistical standard error if A_i is estimated by a random algorithm, and $\mathcal{E}_i$ is the declared systematic error model. For deterministic TCSA or DLCQ diagonalization one has $\sigma_i = 0$, but $\mathcal{E}_i$ is not zero: it contains cutoff, finite-volume, counterterm, endpoint, or zero-mode errors. For a Markov-chain computation, A_i itself is an estimator and σ_i must include autocorrelation.

Remark 145.20 (Interpolation ambiguity of finite cutoff data)*.* Let $K_1, \ldots, K_N$ be distinct positive regulator values and let $A_1, \ldots, A_N \in \mathbb{R}$ be finite outputs. For every proposed continuum value $A_\infty \in \mathbb{R}$, there is a polynomial $p(t)$ of degree at most N such that

$$p(0) = A_\infty, \qquad p(K_j^{-1}) = A_j, \quad j = 1, \ldots, N.$$

Consequently finitely many cutoff values, by themselves, cannot prove a continuum limit or its value. The $N+1$ points

$$0, \quad K_1^{-1}, \ldots, K_N^{-1}$$

are distinct. Lagrange interpolation gives a unique polynomial of degree at most N taking the prescribed values $A_\infty, A_1, \ldots, A_N$ at those points. Since A_∞ was arbitrary, the same finite data are compatible with arbitrarily many candidate limits unless additional information is supplied. The obstruction is the missing analytic ansatz, monotonicity estimate, remainder bound, or convergence theorem that would restrict the allowed interpolating functions near $K^{-1} = 0$.

Definition 145.21 (Power-law cutoff expansion with a remainder bound)**.** Let K be a positive regulator parameter tending to infinity. A deterministic sequence A_K has an m-term power-law cutoff expansion with base exponent $\omega > 0$ and controlled remainder if there are real numbers $A_\infty, c_1, \ldots, c_m$, a constant C, and K_0 such that

$$\left| A_K - A_\infty - \sum_{a=1}^{m} c_a K^{-a\omega} \right| \le C K^{-(m+1)\omega}, \qquad K \ge K_0. \tag{145.62}$$

The same definition applies to a subsequence, for example $K, Ks_1, \ldots, Ks_m$, provided these are admissible regulator labels and the displayed bound holds on that subsequence. In a TCSA computation the exponent ω and the counterterm coordinates should be derived from the high-energy tail or OPE; in a DLCQ computation they should be derived from endpoint and zero-mode analysis. A fit without (145.62) is a diagnostic, not a theorem.

For a two-cutoff Richardson extrapolant, the leading correction cancels only under a stated asymptotic expansion with a controlled remainder. Assume

$$A_K = A_\infty + cK^{-\omega} + r_K, \qquad |r_K| \le CK^{-\eta}, \qquad \eta > \omega > 0,$$

for $K \ge K_0$. Define

$$R_K = \frac{2^\omega A_{2K} - A_K}{2^\omega - 1}. \tag{145.63}$$

Then

$$|R_K - A_\infty| \le \frac{2^\omega 2^{-\eta} + 1}{2^\omega - 1} CK^{-\eta}. \tag{145.64}$$

Substitute the assumed expansion into (145.63):

$$2^\omega A_{2K} - A_K = (2^\omega - 1)A_\infty + 2^\omega c(2K)^{-\omega} - cK^{-\omega} + 2^\omega r_{2K} - r_K.$$

The two terms proportional to c cancel because $2^\omega (2K)^{-\omega} = K^{-\omega}$. Dividing by $2^\omega - 1$ gives

$$R_K - A_\infty = \frac{2^\omega r_{2K} - r_K}{2^\omega - 1}.$$

Using $|r_{2K}| \le C(2K)^{-\eta}$ gives (145.64).

Integer-power extrapolation with explicit weights. Let $s_0, \ldots, s_m$ be distinct positive real numbers and suppose A_K satisfies (145.62) on the subsequence $s_j K$, $j = 0, \ldots, m$. Let $w_0, \ldots, w_m$ be the unique solution of

$$\sum_{j=0}^{m} w_j = 1, \qquad \sum_{j=0}^{m} w_j s_j^{-a\omega} = 0, \quad a = 1, \ldots, m. \tag{145.65}$$

Then the extrapolated value

$$R_K^{(m)} = \sum_{j=0}^{m} w_j A_{s_j K}$$

satisfies

$$|R_K^{(m)} - A_\infty| \le C K^{-(m+1)\omega} \sum_{j=0}^{m} |w_j| \, s_j^{-(m+1)\omega}. \tag{145.66}$$

Set $q_j = s_j^{-\omega}$. The numbers $q_0, \dots, q_m$ are distinct, so the Vandermonde matrix $(q_j^a)_{0 \le a,j \le m}$ is invertible. Hence (145.65) has a unique solution. Insert (145.62) into $R_K^{(m)}$. The coefficient of A_∞ is $\sum_j w_j = 1$. For $a = 1, \dots, m$, the coefficient of $c_a K^{-a\omega}$ is $\sum_j w_j s_j^{-a\omega} = 0$. The remaining error is bounded by the triangle inequality:

$$\left| \sum_{j=0}^{m} w_j r_{s_j K} \right| \le \sum_{j=0}^{m} |w_j| \, C (s_j K)^{-(m+1)\omega}.$$

This is (145.66). The finite algebra is exact, but its use in a continuum claim is only as good as the expansion and remainder estimate in Definition 145.21. The weights cancel declared integer powers; the QFT problem separately supplies the evidence that these powers are the correct cutoff expansion.

Large-K diagnostic for the finite 't Hooft matrices. For the finite matrices M_K^2 whose quadratic form is displayed in (145.61), write their eigenvalues in nondecreasing order as

$$\lambda_0(K) \le \lambda_1(K) \le \cdots \le \lambda_{K-2}(K).$$

For a fixed label ℓ, the numerical sequence

$$A_K = \lambda_\ell(K)$$

is a finite-regulator sequence in the sense of Definition 145.19. Keeping the label ℓ fixed is only a diagnostic convention until one proves that the corresponding continuum state is isolated and that nearby finite-K levels do not interchange in a way that changes the limiting object. Choose distinct resolutions $K_1 < \cdots < K_N$, a base exponent $\omega > 0$, and an integer $p < N$. Set

$$q_r = K_r^{-\omega}, \qquad X_{ra} = q_r^a, \qquad r = 1, \dots, N, \quad a = 0, \dots, p.$$

The least-squares polynomial fit is

$$\widehat{c} = (X^T X)^{-1} X^T A, \qquad A = (A_{K_1}, \dots, A_{K_N})^T, \tag{145.67}$$

provided X has full column rank. The reported continuum diagnostic is $\widehat{c}_0$, the fitted intercept at $q = 0$.

Finite large-K fit with an explicit remainder amplifier. Assume that the finite data used in (145.67) have the form

$$A_{K_r} = A_\infty + \sum_{a=1}^{p} c_a K_r^{-a\omega} + \rho_r, \qquad r = 1, \dots, N, \tag{145.68}$$

with X of full column rank. Let

$$B = (X^T X)^{-1} X^T.$$

Then the fitted intercept satisfies

$$|\widehat{c}_0 - A_\infty| \le \left(\sum_{r=1}^{N} |B_{0r}| \right) \max_{1 \le r \le N} |\rho_r|. \tag{145.69}$$

In particular, if all ρ_r vanish, the fit recovers A_∞ exactly, independently of the overdetermination $N > p + 1$.

Let

$$c = (A_\infty, c_1, \dots, c_p)^T, \qquad \rho = (\rho_1, \dots, \rho_N)^T.$$

Equation (145.68) is the vector identity

$$A = Xc + \rho.$$

Since X has full column rank, $BX = I_{p+1}$. Hence

$$\widehat{c} = BA = B(Xc + \rho) = c + B\rho.$$

Taking the zeroth component gives

$$\widehat{c}_0 - A_\infty = \sum_{r=1}^{N} B_{0r} \rho_r.$$

The triangle inequality gives (145.69). If $\rho = 0$, then $\widehat{c} = c$ and the intercept is exact.

The companion script

`qft_scripts/thooft_dlcq_extrapolation.py`

implements this finite procedure for the matrices M_K^2: it prints the finite eigenvalues, the fitted intercepts, the residuals, the condition number of X, and the factor $\sum_r |B_{0r}|$ appearing in (145.69). These quantities measure how a chosen finite data set responds to a chosen ansatz. They do not prove that the large-K limit of the large-N QCD meson equation exists or that the selected exponent ω is correct. Those conclusions require separate endpoint and zero-mode estimates producing a bound of the form (145.68).

The companion calculation

`calculation-checks/numerical_extrapolation_checks.py`

verifies the algebra in (145.63)–(145.66) with exact rational arithmetic. It does not verify that a particular QFT observable obeys (145.62); that remains a problem-specific analytic estimate.

The companion calculation

`calculation-checks/hamiltonian_truncation_dlcq_checks.py`

also verifies the exact-intercept and constant-sequence cases of the finite large-K fit calculation above.

Correlated fits and window stability. Production lattice, tensor-network, and Hamiltonian-truncation data are rarely independent scalar data. A continuum extrapolation specification at a chosen fit window $W = \{i_1, \ldots, i_N\}$ therefore contains

$$A = (A_{i_1}, \ldots, A_{i_N})^T, \qquad \Sigma_W, \qquad f_0, \ldots, f_p, \qquad R_1, \ldots, R_N.$$

Here A_i are the finite-regulator outputs, Σ_W is the covariance matrix of their statistical errors on the window, $f_a(i)$ are the declared fit functions with $f_0(i) = 1$, and R_r is a deterministic bound or declared systematic envelope for the part of A_{i_r} not represented by the chosen fit functions. Put

$$X_{ra} = f_a(i_r), \qquad X \in \mathrm{Mat}_{N\times(p+1)}(\mathbb{R}).$$

Assume in this paragraph that Σ_W is positive definite and that X has full column rank. If an estimated covariance matrix is singular, the shrinkage, projection, or diagonal approximation used to replace it is a new part of the fit specification, not an invisible numerical detail.

The correlated least-squares coordinate is

$$\widehat{c} = (X^T \Sigma_W^{-1} X)^{-1} X^T \Sigma_W^{-1} A. \tag{145.70}$$

Writing

$$G = X^T \Sigma_W^{-1} X, \qquad B = G^{-1} X^T \Sigma_W^{-1},$$

one has $BX = I_{p+1}$. If the data on the window decompose as

$$A = Xc + \rho + \epsilon, \qquad \mathbb{E}\epsilon = 0, \qquad \mathbb{E}(\epsilon\epsilon^T) = \Sigma_W, \tag{145.71}$$

then

$$\widehat{c} - c = B\rho + B\epsilon. \tag{145.72}$$

Thus the intercept $\widehat{A}_\infty = \widehat{c}_0$ has the exact finite error decomposition

$$\widehat{A}_\infty - A_\infty = \sum_{r=1}^{N} B_{0r}\rho_r + \sum_{r=1}^{N} B_{0r}\epsilon_r.$$

If $|\rho_r| \le R_r$, the deterministic fit-window systematic coordinate is

$$S_W = \sum_{r=1}^{N} |B_{0r}| R_r. \tag{145.73}$$

The stochastic variance of the intercept is

$$\sigma^2_{\infty,W} = e_0^T B \Sigma_W B^T e_0 = e_0^T G^{-1} e_0, \tag{145.74}$$

where $e_0 = (1, 0, \ldots, 0)^T$. The second equality follows from

$$B\Sigma_W B^T = G^{-1} X^T \Sigma_W^{-1} \Sigma_W \Sigma_W^{-1} X G^{-1} = G^{-1}.$$

The correlated residual

$$\chi^2_W = (A - X\widehat{c})^T \Sigma_W^{-1} (A - X\widehat{c}) \tag{145.75}$$

is a finite diagnostic for the chosen window and model. Its probabilistic interpretation requires the sampling law of the estimator, the covariance estimator, and the treatment of autocorrelation described in Section 141.12.

A window-stability analysis is the finite collection

$$\mathcal{W} = \{(W_\alpha, X_\alpha, \Sigma_\alpha, R_\alpha)\}_{\alpha \in A}$$

together with the resulting coordinates

$$\widehat{A}_{\infty,\alpha}, \qquad \sigma_{\infty,\alpha}, \qquad S_\alpha, \qquad \chi^2_\alpha.$$

Agreement of these coordinates across windows is evidence about the chosen finite-regulator family only after the windows use the same physical observable, the same renormalization convention, and compatible operator normalizations. The analytic estimate producing the envelopes R_α remains the source of continuum control. For tensor-network and neural-state calculations the window label must also record bond dimension or architecture, optimizer termination criterion, local-energy variance or discarded weight, and any symmetry projection used to select the finite sector. Otherwise a stable sequence of numbers may be comparing different finite problems rather than a single regulated QFT observable.

The reader-facing implementation

```
qft_scripts/finite_regulator_extrapolation.py
```

takes this paragraph literally. Its input data consist of regulator labels, finite observable values, an optional covariance matrix, nonnegative remainder envelopes, and half-open fit windows. For each window it constructs

$$X_{ra} = K_r^{-a\omega}, \qquad B = (X^T \Sigma_W^{-1} X)^{-1} X^T \Sigma_W^{-1},$$

then reports $\widehat{A}_{\infty,W}$, the propagated statistical error $(e_0^T G^{-1} e_0)^{1/2}$, the deterministic coordinate $\sum_r |B_{0r}| R_r$, the correlated residual, the condition number, and the spread of fitted intercepts across the declared windows. Supplying the identity matrix as a covariance, or using diagonal errors, is part of the finite fit specification and is printed as such. The script therefore checks the bookkeeping of a proposed extrapolation. The cutoff exponent, counterterm scheme, and remainder estimate are mathematical inputs whose justification comes from the regulated theory, not from the fit output alone.

145.17 Light-Front Hamiltonian

The light-front Hamiltonian convention used in this chapter is the regulator specification of Definition 145.16. The finite numerical sections above use its DLCQ specialization; the analytic large-N QCD construction in Chapter 53 and the Chern–Simons–matter light-cone construction in Chapter 95 use the same coordinate convention, but with model-specific constraint equations instead of a universal finite-matrix recipe.

145.18 Cross-Method Benchmark Comparisons

Lattice Monte Carlo, Hamiltonian truncation, TCSA/TFFSA, and DLCQ produce different finite mathematical objects. A numerical agreement between their outputs is meaningful only after a common QFT observable and a common normalization have been specified. The comparison itself is therefore part of the regulated calculation.

Definition 145.22 (Cross-method benchmark specification)**.** A cross-method benchmark specification consists of the following information.

1. A target QFT observable Q. Examples are a dimensionless mass ratio, a finite-volume energy gap in units of the physical mass, a matrix element normalized by a declared two-point function, or a Euclidean distribution paired with a specified test function.
2. A finite set of methods $m \in \mathcal{M}$. For each method one specifies the finite regulator object X_m: for Euclidean lattice Monte Carlo this includes lattice spacing, finite volume, boundary conditions, action, update kernel, and measured estimator; for Hamiltonian truncation it includes the finite Hilbert space, Hamiltonian matrix, counterterm coordinates, symmetry sector, and extraction map; for DLCQ it includes harmonic resolution, zero-mode prescription, constraint sector, and finite invariant-mass matrix.
3. A normalization map
$$N_m : X_m \longrightarrow \mathbb{R}^r$$
which converts the method-specific finite output to the same coordinate system for Q. Field-strength conventions, finite-volume units, operator normalizations, and sector labels are part of N_m.
4. Statistical error coordinates s_m, deterministic finite-regulator envelopes R_m, and a scheme-matching coordinate b_m measuring any residual mismatch between the method's finite observable and the declared target observable.
5. A limiting or extrapolation hypothesis, when a continuum statement is claimed, stating that
$$N_m(X_m) = Q + \epsilon_m, \qquad |\epsilon_m| \leq s_m + R_m + b_m$$
in the declared norm or componentwise in $\mathbb{R}^r$.

The comparison check between two methods is then finite arithmetic. If two methods m, n are claimed to compute the same coordinate of Q, their finite outputs must satisfy

$$|N_m(X_m) - N_n(X_n)| \leq s_m + s_n + R_m + R_n + b_m + b_n \tag{145.76}$$

for the claimed error coordinates. Failure of (145.76) means that at least one of the finite estimates, error models, normalization maps, or target identifications is wrong. Satisfaction of (145.76) supplies a finite compatibility check conditional on the stated extrapolation hypotheses. The continuum statement is the separate proof that the hypotheses and envelopes follow from the regulated QFT.

The main finite objects in the present numerical layer are as follows.

Euclidean lattice Monte Carlo.
The finite object is a positive measure, a Markov kernel, and sampled correlators. The usual spectral coordinate is a transfer-matrix gap extracted from a correlator matrix or effective mass. The continuum burden is reflection positivity, volume control, autocorrelation analysis, continuum scaling, and operator matching.

Hamiltonian truncation/TCSA/TFFSA.
The finite object is a sector of a Hilbert space together with a projected Hamiltonian matrix. The usual spectral coordinates are Ritz eigenvalues, residual bounds, and finite matrix elements. The continuum burden is counterterm determination, cutoff convergence, finite-volume control, and operator normalization.

DLCQ/light-front truncation.
The finite object is a fixed harmonic-resolution physical sector together with a finite M^2 matrix. The usual spectral coordinate is an ordered invariant-mass eigenvalue at fixed K. The continuum burden is zero-mode constraints, endpoint behavior, large-K estimates, and Lorentz-covariant reconstruction.

Definition 145.23 (Benchmark manifest)**.** A benchmark manifest is a finite, machine-readable representative of Definition 145.22. It consists of:

1. the target observable label Q, the dimension r of its reported coordinate vector, and the norm in which the comparison is to be made;
2. for each method m, a regulator description, a normalization identifier, the coordinate vector $x_m \in \mathbb{R}^r$, the componentwise statistical error σ_m, regulator envelope ρ_m, and matching error β_m;
3. optional covariance data for the statistical part, with the convention that covariance data never replace the declared deterministic envelopes ρ_m, β_m;
4. provenance data: script name or analysis notebook, input dataset, random seed where relevant, regulator parameters, and any hash or version identifier needed to reproduce the finite number x_m.

The declared componentwise error vector is

$$e_m = \sigma_m + \rho_m + \beta_m. \tag{145.77}$$

The manifest is consistent at the finite-regulator level when, for every pair of methods m, n,

$$|x_m - x_n|_{\text{comp}} \leq e_m + e_n, \tag{145.78}$$

where the absolute value and inequality are componentwise. The script `qft_scripts/benchmark_manifest_consistency.py` implements exactly (145.78) for JSON manifests. A passing manifest verifies only that the finite reported coordinates are mutually compatible with the declared error envelopes and normalization identifiers. It does not verify that the envelopes are correct, that the regulator limits exist, that the methods compute the same continuum QFT observable, or that the target observable is nonperturbatively constructed.

This manifest discipline is intentionally stricter than a table of numerical values. If a lattice correlator fit and a Hamiltonian-truncation diagonalization report the same mass ratio but use different finite-volume normalizations, the mismatch belongs in β_m or in a failed normalization identifier before any comparison is made. If two finite coordinates disagree beyond (145.78), the failure has a definite meaning: at least one coordinate, uncertainty model, normalization map, or target identification is not the one recorded in the manifest. If the manifest passes, the continuum conclusion still begins only after the separate estimates on $\sigma_m, \rho_m, \beta_m$ are proved or justified in the relevant regulator framework.

The companion scripts instantiate this comparison at finite regulator. The Euclidean side contains finite Ising, scalar ϕ^4, $\mathbb{Z}_2$, $SU(2)$, and $SU(3)$ samplers, plus autocorrelation, static-potential, Wilson-flow, smearing, topological-charge, HMC/RHMC, and glueball-GEVP diagnostics. The Hamiltonian side contains exactly solvable Ising thermal truncation, compact-boson sine-Gordon vertex assembly, nonintegrable scalar ϕ^4 truncation, connected Ising TFFSA blocks, spectral-flow diagnostics, and the exact E_8 target table. The light-front side contains scalar ϕ^4 DLCQ, the large-N two-dimensional QCD matrix, and a large-K extrapolation diagnostic. The generic finite-regulator extrapolation script records the common window/covariance/remainder-envelope data used by all three classes of methods, while the benchmark-manifest script records pairwise compatibility of normalized outputs across methods.

Public Finite-Regulator Check Index

The companion-script layer is part of the mathematical presentation only when each script is read through a finite verification check. A smoke run by itself is an implementation regression; it is not a QFT result. The finite check attached to a script is the tuple

$$\mathcal{C}_{\text{num}} = \big(\mathcal{D}_{\text{fin}}, F_{\text{fin}}, \mathcal{I}_{\text{check}}, \mathcal{S}_{\text{scope}}\big),$$

where $\mathcal{D}_{\text{fin}}$ is the declared finite regulator specification, F_{fin} is the finite output map computed by the script, $\mathcal{I}_{\text{check}}$ is the list of algebraic, spectral, measure, or statistical identities checked against the chapter, and $\mathcal{S}_{\text{scope}}$ is the boundary between the finite assertion and any continuum or infinite-volume assertion. The smoke harness checks that the implementation still produces its declared finite output. The corresponding calculation-check scripts, when present, check the identities in $\mathcal{I}_{\text{check}}$. Neither layer replaces the estimates needed to pass from $\mathcal{D}_{\text{fin}}$ to a continuum QFT observable.

The public scripts currently have the following finite-check meanings.

Spin and scalar Metropolis.
The Ising and scalar ϕ^4 Metropolis scripts compute finite periodic lattice expectations for explicitly declared Boltzmann weights. Their check is invariance of the finite target measure under the displayed single-site Metropolis kernels, together with local energy or action-difference identities. They do not prove Onsager scaling, constructive ϕ^4_2, a critical surface, or an Ising CFT limit.

Compact gauge samplers.
The finite $\mathbb{Z}_2$, $SU(2)$ Metropolis, $SU(2)$ heat-bath and overrelaxation, and $SU(3)$ HDF5 sampler scripts compute finite compact link-integrals and Wilson-loop estimators. Their checks are gauge invariance of the measured finite observables, Haar measure preservation, detailed

balance or conditional heat-bath invariance, and the declared data coordinate convention. They do not prove continuum Yang–Mills, mass gaps, confinement, topological sector sampling, or a production continuum extrapolation.

Flow, smearing, topology, and static-potential analysis.

The Wilson flow, APE smearing, clover topology diagnostic, static potential, static-source GEVP, and glueball GEVP scripts compute finite analysis functionals of saved lattice data. Their checks are finite group projection, covariance of smearing, the declared clover diagnostic, nonlinear jackknife and bootstrap propagation for Wilson-loop ratios, exact static-source matrix spectral identities, and finite generalized eigenvalue residual control. They do not define a continuum topological charge, a continuum static potential, or a physical glueball spectrum without the continuum estimates of Chapters 140 and 143.

HMC, RHMC, and Markov-chain error analysis.

The HMC and RHMC smoke module and the autocorrelation and resampling script compute finite algorithmic diagnostics: HMC reversibility defects, Hamiltonian changes, positive-matrix spectral edges, rational pseudofermion actions, solver residuals, autocorrelations, block errors, and jackknife and bootstrap coordinates. Their checks are the finite symplectic and reversibility identities, determinant or rational-determinant bookkeeping, and finite time-series formulae. They do not prove ergodic sampling of a large production ensemble or a continuum-determinant limit.

Hamiltonian truncation and TCSA/TFFSA matrices.

The Ising energy benchmark, sine-Gordon zero-mode, sine-Gordon oscillator vertex, scalar ϕ^4 truncation, connected Ising TFFSA, TFFSA spectral-flow, and E_8 target-data scripts compute finite matrices or exact finite target data. Their checks are finite Hermiticity, selection rules, normal-ordering conventions, residual or spectral-flow identities, and the Perron–Frobenius convention check for the E_8 target table. They do not prove cutoff-extrapolated spectra. For the sine-Gordon, nonintegrable ϕ^4, and magnetic-Ising TCSA/TFFSA E_8 comparisons, the counterterm, finite-volume, and cutoff-error data specified above remain separate hypotheses.

DLCQ and light-front finite matrices.

The scalar ϕ^4 DLCQ, large-N two-dimensional QCD DLCQ, and large-K extrapolation scripts compute finite matrices at harmonic resolution K, together with diagnostics at fixed spectral label. Their checks are positive longitudinal partition bookkeeping, finite principal value or normal ordered matrix identities, positivity of the finite quadratic form in the declared model, and finite least squares amplification for a declared large-K ansatz. They do not prove the $K \to \infty$ continuum spectrum without zero-mode, endpoint, constraint, and remainder estimates.

Extrapolation, manifests, and cluster wrappers.

The finite-regulator extrapolation, benchmark-manifest, and cluster-grid scripts compute finite bookkeeping checks: correlated-window intercepts with declared covariance and remainder envelopes, componentwise benchmark-manifest compatibility, and deterministic grid-to-task assignment for cluster sweeps. They do not create a cutoff ansatz, validate a systematic error model, or promote a small cluster run to a continuum QFT calculation.

This index is deliberately redundant with the comments printed by the scripts and with `qft_scripts/README.md`, where exact filenames and smoke commands are listed. Its role in the monograph is to make the finite mathematical object visible before the reader interprets any number produced by a public script.

For the magnetic Ising theory, the present public scripts deliberately stop at finite checks: the TFFSA connected-block matrix fixes the spin-field normalization convention, the spectral-flow script checks finite Hellmann–Feynman slopes, and the E_8 script fixes the continuum target mass-ratio table. A full comparison between a finite TCSA/TFFSA spectrum, Euclidean lattice correlators, and the exact E_8 masses still requires a declared finite-volume convention, cutoff counterterms, extrapolation envelopes, and comparison estimates in the sense of Definition 145.22.

145.19 Benchmark Protocol

A benchmark for any truncation method should specify:

$$\text{regulated Hilbert space}, \qquad \text{operator basis}, \qquad \text{counterterm coordinates},$$
$$\text{observables used for matching}, \qquad \text{extrapolation model}, \qquad \text{independent checks}.$$

Independent checks include Ward identities, known finite-volume limits, reflection positivity where applicable, spectral sum rules, and comparison with Euclidean lattice or exact integrable data in overlapping regimes.

The public companion-script policy is recorded in `planning/14_code_policy.md`. The script smoke harness

```
tools/run_qft_scripts_smoke.sh
```

runs the finite lattice, Hamiltonian-truncation, TCSA/TFFSA, DLCQ, finite-extrapolation, and benchmark-manifest smoke examples. Passing that harness verifies only the finite computations and internal consistency checks just described.

Open Problem 145.24 (Controlled QFT truncation)*.* Develop Hamiltonian truncation and DLCQ schemes with computable error bounds for interacting QFT observables, including counterterm determination, constraint preservation, zero-mode treatment, continuum extrapolation, and public benchmark scripts.

Chapter 146

Lattice Fermions And Chiral Symmetry

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Fermions on a lattice require two separate pieces of structure. The variables integrated in the Euclidean functional integral are generators of a finite Grassmann algebra, so the finite-cutoff integral is a Berezin functional rather than a Borel measure. The fields reconstructed after Osterwalder–Schrader positivity are operator-valued distributions satisfying graded locality. The purpose of this chapter is to state the finite algebraic regulator precisely, to identify where chiral symmetry is lost or modified at nonzero lattice spacing, and to record the reflection-positive fermion constructions that can serve as honest nonperturbative cutoffs.

146.1 Finite Lattice Grassmann Integral

Let $\Lambda \subset a\mathbb{Z}^D$ be a finite lattice with a chosen ordering of the pairs (x, α), where $x \in \Lambda$ and $\alpha = 1, \ldots, N_s$ is a spinor, flavor, and internal index. The fermionic field variables are odd generators

$$\psi_\alpha(x), \qquad \bar{\psi}_\alpha(x)$$

of the complex Grassmann algebra $\bigwedge(\mathbb{C}^{2N_s|\Lambda|})$. They are not points of a configuration space. A lattice fermion action quadratic in the fields is specified by a finite matrix D with entries

$$D_{xy}^{\alpha\beta} : \mathbb{C}^{N_s|\Lambda|} \longrightarrow \mathbb{C}^{N_s|\Lambda|},$$

and is written

$$S_D(\bar\psi, \psi) = \sum_{x,y\in\Lambda} \bar\psi_\alpha(x) D_{xy}^{\alpha\beta} \psi_\beta(y).$$

The Berezin functional is normalized by

$$\int \prod_{(x,\alpha)} d\bar\psi_\alpha(x) d\psi_\alpha(x) \prod_{(x,\alpha)} \bar\psi_\alpha(x)\psi_\alpha(x) = 1.$$

Changing this ordering changes both numerator and denominator by the same sign in normalized correlation functions, but a fixed ordering must be chosen before signs in reflection positivity or anomalies are meaningful.

Finite Berezin Gaussian identities. Let D be an invertible $N \times N$ complex matrix and write

$$S_D = \sum_{i,j=1}^{N} \bar\psi_i D_{ij} \psi_j.$$

With the Berezin ordering fixed above,

$$Z_D = \int \prod_i d\bar\psi_i \, d\psi_i \, e^{-S_D} = \det D.$$

For every i, j,

$$\frac{1}{Z_D} \int \prod_k d\bar\psi_k \, d\psi_k \, \bar\psi_j \psi_i \, e^{-S_D} = (D^{-1})_{ij}.$$

More generally, for equal numbers of ψ's and $\bar\psi$'s, the normalized expectation is the determinant of the matrix of two-point functions, with the sign fixed by the displayed order of the insertions.

Introduce independent odd sources $\eta_i, \bar\eta_i$. Completing the square is an identity in the finite Grassmann algebra:

$$-\bar\psi D\psi + \bar\eta\psi + \bar\psi\eta = -(\bar\psi - \bar\eta D^{-1}) D (\psi - D^{-1}\eta) + \bar\eta D^{-1}\eta.$$

Berezin translation invariance gives

$$\int d\bar\psi \, d\psi \, e^{-\bar\psi D\psi + \bar\eta\psi + \bar\psi\eta} = \det D \, e^{\bar\eta D^{-1}\eta}.$$

Differentiating with respect to η_i and $\bar\eta_j$, using left Grassmann derivatives in the fixed order, gives the two-point identity. Higher identities follow by differentiating the same generating functional; the exponential is quadratic in sources, so the coefficient of a monomial is the corresponding determinant.

146.2 Naive Dirac Operator And Doublers

Let γ_μ, $\mu = 1, \dots, D$, be Hermitian Euclidean gamma matrices,

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}\mathbf{1}.$$

On a periodic lattice, the naive massive Dirac operator is

$$D_{\text{naive}}(p) = m\mathbf{1} + \frac{\mathrm{i}}{a}\sum_{\mu=1}^{D} \gamma_\mu \sin(ap_\mu), \qquad p_\mu \in (-\pi/a, \pi/a].$$

For $m \neq 0$ it is invertible and

$$D_{\text{naive}}(p)^{-1} = \frac{m\mathbf{1} - \frac{\mathrm{i}}{a}\sum_\mu \gamma_\mu \sin(ap_\mu)}{m^2 + a^{-2}\sum_\mu \sin^2(ap_\mu)}. \tag{146.1}$$

The massless symbol vanishes at every corner of the Brillouin zone:

$$p_\mu = \frac{\pi}{a}\epsilon_\mu, \qquad \epsilon_\mu \in \{0, 1\}.$$

The existence of these 2^D zeros is the elementary doubling phenomenon. The next proposition is a precise finite-dimensional version of the topological obstruction in the translation-invariant nearest-continuum setting.

Proposition 146.1 (Brillouin-torus index cancellation)**.** *Let $f = (f_1, \dots, f_D) : \mathbb{T}^D \to \mathbb{R}^D$ be a smooth periodic map whose zeros are isolated and nondegenerate. Consider the massless translation-invariant Dirac symbol*

$$D_f(p) = \mathrm{i}\sum_{\mu=1}^{D} \gamma_\mu f_\mu(p).$$

The chirality sign of a zero p_* is

$$\chi(p_*) = \operatorname{sign}\det\left(\frac{\partial f_\mu}{\partial p_\nu}(p_*)\right).$$

Then

$$\sum_{f(p_*)=0} \chi(p_*) = 0.$$

For $f_\mu(p) = a^{-1}\sin(ap_\mu)$, the zero at the corner $\epsilon \in \{0, 1\}^D$ has

$$\chi_\epsilon = (-1)^{\epsilon_1 + \cdots + \epsilon_D},$$

so the 2^D naive species split equally between the two chirality signs.

Proof. For a regular value $y \in \mathbb{R}^D$ of f, define

$$I(y) = \sum_{f(p)=y} \operatorname{sign}\det\left(\frac{\partial f_\mu}{\partial p_\nu}(p)\right).$$

The set of regular values is open and dense by Sard's theorem. On a connected component of regular values, $I(y)$ is locally constant: if $y(s)$ is a small path of regular values and $f(p_0) = y(0)$, the inverse function theorem gives a unique smooth branch $p(s)$ with $f(p(s)) = y(s)$, and the sign of the nonzero Jacobian determinant cannot change along the branch. No signed zero can appear or disappear while the value remains regular.

Since $f(\mathbb{T}^D)$ is compact, choose y_∞ outside the image. Then $I(y_\infty) = 0$. Connect 0 to y_∞ by a piecewise smooth path and perturb the path, keeping its endpoints fixed, so that it meets the critical-value set only at isolated crossing parameters. At such a crossing, the standard local normal form for a birth–death critical point creates or annihilates a pair of simple preimages with opposite Jacobian signs; hence the signed count does not change. Therefore $I(0) = I(y_\infty) = 0$. This is the degree-zero statement for a map from the torus to $\mathbb{R}^D$.

For the naive symbol,

$$\frac{\partial f_\mu}{\partial p_\nu} = \delta_{\mu\nu} \cos(ap_\mu),$$

and at $p_\mu = \pi\epsilon_\mu/a$ the determinant is $(-1)^{\epsilon_1+\cdots+\epsilon_D}$. □

Remark 146.2. The proposition proves the topological core of doubling for symbols of the form $D_f = \mathrm{i}\gamma \cdot f$. The full Nielsen–Ninomiya theorem allows a broader class of local, translation-invariant, chirally symmetric lattice Dirac operators. Its proof is the same degree-theoretic obstruction after one constructs the appropriate normalized vector field on the Brillouin torus. The hypotheses matter: dropping exact lattice chiral symmetry, ultralocality, or the finite-component Dirac symbol changes the conclusion.

146.3 Wilson Fermions

Wilson's construction changes the massless symbol by adding a spin-scalar lattice Laplacian:

$$D_W(p) = m\mathbf{1} + \frac{\mathrm{i}}{a}\sum_{\mu=1}^{D} \gamma_\mu \sin(ap_\mu) + \frac{r}{a}\sum_{\mu=1}^{D}(1 - \cos(ap_\mu))\mathbf{1}, \qquad r > 0. \tag{146.2}$$

In position space, with gauge links $U_\mu(x)$ in a unitary representation of the gauge group,

$$\begin{aligned}(D_W\psi)(x) &= (m + rDa^{-1})\psi(x) \\ &\quad - \frac{1}{2a}\sum_{\mu}\Big[(r - \gamma_\mu)U_\mu(x)\psi(x + a\hat{\mu}) + (r + \gamma_\mu)U_\mu(x - a\hat{\mu})^{-1}\psi(x - a\hat{\mu})\Big].\end{aligned} \tag{146.3}$$

The Wilson term is gauge covariant and local at fixed a. It is not chirally symmetric: when D is even and $\gamma_5 = \mathrm{i}^{D/2}\gamma_1 \cdots \gamma_D$, the scalar Wilson term does not anticommute with γ_5.

Wilson lifting of the naive species. Let $p = p_\epsilon + k$, where

$$(p_\epsilon)_\mu = \frac{\pi}{a}\epsilon_\mu, \qquad \epsilon_\mu \in \{0, 1\},$$

and ak_μ is small. Put

$$n_\epsilon = \epsilon_1 + \cdots + \epsilon_D.$$

Then the Wilson symbol has the expansion

$$D_W(p_\epsilon + k) = \left(m + \frac{2rn_\epsilon}{a}\right)\mathbf{1} + \mathrm{i}\sum_\mu (-1)^{\epsilon_\mu}\gamma_\mu k_\mu + O(a|k|^2).$$

Thus the physical corner $n_\epsilon = 0$ has mass m, while every doubler corner has mass $m + 2rn_\epsilon/a$, which diverges in lattice units as $a \to 0$ when m is held in the scaling window.

For $p_\mu = \pi\epsilon_\mu/a + k_\mu$,

$$\sin(ap_\mu) = (-1)^{\epsilon_\mu} ak_\mu + O(a^3|k_\mu|^3),$$

and

$$1 - \cos(ap_\mu) = 1 - (-1)^{\epsilon_\mu}\cos(ak_\mu) = 2\epsilon_\mu + O(a^2k_\mu^2).$$

Substitution in (146.2) gives the displayed expansion. The term $2rn_\epsilon/a$ is present exactly at corners with $n_\epsilon > 0$. Thus the Wilson term distinguishes the physical corner from the doubler corners by a regulator-scale mass shift while preserving the linear Dirac behavior of the physical corner.

146.4 Mid-link Reflection Positivity

Reflection positivity for fermions is a statement about an antilinear anti-automorphism of the Grassmann algebra and about the sign of the factor coupling the two halves of the lattice. It is not automatic from positivity of a determinant. We record the finite algebraic mechanism because this is where many informal discussions lose the distinction between a Berezin functional and a measure.

Let the reflection plane be the mid-link plane

$$x_0 = \frac{a}{2}.$$

Let

$$\Lambda_+ = \{x \in \Lambda : x_0 \geq a\}, \qquad \Lambda_- = \{x \in \Lambda : x_0 \leq 0\},$$

and let $\vartheta(t, \mathbf{x}) = (a - t, \mathbf{x})$. Choose the Euclidean time gamma matrix so that $\gamma_0^\dagger = \gamma_0$ and $\gamma_0^2 = \mathbf{1}$. The reflection Θ is defined on positive-time generators by

$$\Theta\big(\psi_\alpha(x)\big) = (\gamma_0)_\beta{}^\alpha\bar{\psi}_\beta(\vartheta x), \qquad \Theta\big(\bar{\psi}_\alpha(x)\big) = (\gamma_0)_\alpha{}^\beta\psi_\beta(\vartheta x),$$

extended antilinearly and with reversed order,

$$\Theta(AB) = \Theta(B)\Theta(A).$$

Figure 146.1 records the midlink reflection used in the lattice-fermion reflection-positivity argument and the induced exchange of fermion and antifermion variables under Θ.

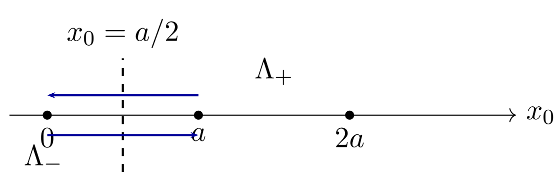

Figure 146.1: The mid-link reflection plane lies between neighboring time slices. The crossing hopping terms are the only terms that do not belong entirely to one half of the lattice.

Definition 146.3 (Fermionic reflection-positive crossing factor)**.** Let $\mathcal{A}_+$ be the Grassmann algebra generated by positive-time variables. A crossing factor is reflection-positive if it can be written, with a fixed ordering of the odd generators $F_1, \dots, F_M \in \mathcal{A}_+$, as

$$B_\times = \prod_{j=1}^{M} \bigl(1 + c_j \Theta(F_j) F_j\bigr), \qquad c_j \geq 0.$$

Proposition 146.4 (Finite Grassmann reflection-positivity criterion)**.** *Suppose a finite fermion functional has the form*

$$\langle A \rangle = \frac{1}{Z} \int d\mu_-\, d\mu_+\, \Theta(e^{-S_+})\, e^{-S_+}\, B_\times\, A,$$

where $S_+ \in \mathcal{A}_+$, $B_\times$ is a reflection-positive crossing factor in the sense of Definition 146.3, and $Z = \langle 1 \rangle > 0$. Then

$$\langle \Theta(O) O \rangle \geq 0$$

for every $O \in \mathcal{A}_+$.

Proof. Set

$$\Psi = e^{-S_+} O.$$

Because the F_j have a fixed order, expanding the crossing factor gives

$$B_\times = \sum_{I \subset \{1,\dots,M\}} c_I\, \Theta(F_I) F_I, \qquad c_I = \prod_{j \in I} c_j \geq 0,$$

where the ordered monomial F_I includes the sign convention needed to put $\Theta(F_I)F_I$ in reflected-positive order. The factorization of the Berezin volume into negative and positive halves and the anti-linear reflection convention give

$$\int d\mu_-\, d\mu_+\, \Theta(\Psi)\, B_\times\, \Psi = \sum_I c_I \int d\mu_-\, d\mu_+\, \Theta(F_I \Psi)\, F_I \Psi.$$

To see the sign and positivity explicitly, choose the ordered monomial basis $\{e_A\}$ of $\mathcal{A}_+$ normalized so that

$$\int d\mu_-\, d\mu_+\, \Theta(e_A) e_B = \delta_{AB}.$$

This normalization is exactly the finite Grassmann analogue of choosing an orthonormal basis in the positive-time algebra; it is fixed once the order of the generators and of their reflected partners has been fixed. Write

$$F_I \Psi = \sum_A z_{I,A} e_A.$$

Anti-linearity of Θ gives

$$\int d\mu_-\, d\mu_+\, \Theta(F_I \Psi)\, F_I \Psi = \sum_A |z_{I,A}|^2.$$

All possible signs from moving negative-time reflected generators past positive-time generators have already been absorbed into the definition of the ordered monomial F_I; no further indefinite form appears. Since each $c_I \geq 0$, the finite sum is nonnegative. Division by the positive normalization Z preserves the inequality. □

Proposition 146.5 (Free nearest-neighbor fermion: only the mid-link cut)**.** *For the free nearest-neighbor Dirac action with Hermitian Euclidean gamma matrices and reflection $x_0 \mapsto a - x_0$, the hopping terms crossing the plane $x_0 = a/2$ can be written as a reflection-positive crossing factor. The site reflection $x_0 \mapsto -x_0$ and the diagonal reflection generated by $e_0 + e_1$ fail already on one-fermion two-point tests for generic spinors.*

Proof. For the mid-link reflection, the only cross-plane nearest-neighbor terms are the links between $x_0 = 0$ and $x_0 = a$. In a basis diagonalizing the spin reflection pairing, the cross-plane part of the Boltzmann factor is a finite product of factors

$$1 + c\,\Theta(\chi_j)\chi_j, \qquad c > 0,$$

where each χ_j is one of the positive-time boundary Grassmann generators or its reflected spinor linear combination. Nilpotence makes the exponential of each bilinear equal to $1 + c\Theta(\chi_j)\chi_j$. Thus Definition 146.3 applies, and Proposition 146.4 proves reflection positivity.

The failure of the other cuts can be seen without a general theorem. For a one-particle positive-side test $O = \zeta^\alpha \psi_\alpha(x)$, the reflected two-point function is obtained from (146.1). For a site reflection in the time direction, the odd part of the $\sin p_0$ numerator integrates to zero and the remaining term contains

$$m\,\zeta^\dagger \gamma_0 \zeta \int_{-\pi/a}^{\pi/a} \frac{\cos(2p_0 x_0)\,\mathrm{d}p_0}{m^2 + a^{-2}\sin^2(ap_0) + B(\mathbf{p})},$$

where $B(\mathbf{p}) \geq 0$ is independent of p_0. The Hermitian form $\zeta^\dagger\gamma_0\zeta$ is indefinite, so the expression cannot be nonnegative for all spinors. For the diagonal reflection, the analogous test produces the indefinite Hermitian form $\zeta^\dagger(\gamma_0 + \gamma_1)\zeta$ after the symmetric momentum integral. Hence neither reflection gives OS positivity on the one-fermion subspace. $\square$

Wilson mid-link positivity at $r = 1$. For the gauge-covariant Wilson operator (146.3) with $r = 1$, unitary gauge links, and the mid-link time reflection, the cross-plane Wilson hopping term is a reflection-positive crossing factor after diagonalizing the positive semidefinite projectors

$$P_\pm = \frac{1}{2}(1 \pm \gamma_0).$$

The time-direction part of the Wilson action contains the cross-plane terms

$$-\frac{1}{a}\bar\psi(0,\mathbf{x})P_+ U_0(0,\mathbf{x})\psi(a,\mathbf{x}) - \frac{1}{a}\bar\psi(a,\mathbf{x})P_- U_0(0,\mathbf{x})^{-1}\psi(0,\mathbf{x}),$$

with all spin, color, and spatial indices suppressed. Reflection sends the second term to the first, with the unitary link reflected to its inverse. Choose orthonormal bases of the ranges of P_+ and P_-. In these bases the two crossing bilinears are a finite sum of terms

$$c_j\,\Theta(F_j)F_j, \qquad c_j = a^{-1} > 0,$$

where the link matrix is absorbed into the positive-side linear combination F_j. Nilpotence again turns the exponential of each bilinear into $1 + c_j\Theta(F_j)F_j$. Hence Definition 146.3 and Proposition 146.4 apply.

146.5 Staggered Fermions And A Reflection-positive Thirring Coupling

In two dimensions choose $\gamma_0 = \sigma_3$, $\gamma_1 = \sigma_1$. The staggered one-component field $\eta(x), \bar{\eta}(x)$ has kinetic matrix

$$M(x, x + a\hat{\mu}) = \frac{1}{2a}\eta_\mu(x), \qquad M(x, x - a\hat{\mu}) = -\frac{1}{2a}\eta_\mu(x - a\hat{\mu}),$$

with

$$\eta_0(x) = 1, \qquad \eta_1(x) = (-1)^{x_0/a}.$$

The action is

$$S_{\text{stag}} = \sum_{x,y} \bar{\eta}(x)M(x,y)\eta(y) + m\sum_x \bar{\eta}(x)\eta(x).$$

The mid-link reflection acts by

$$\Theta(\eta(x)) = (-1)^{x_1/a}\bar{\eta}(\vartheta x), \qquad \Theta(\bar{\eta}(x)) = (-1)^{x_1/a}\eta(\vartheta x),$$

with order reversal.

Proposition 146.6 (Staggered Thirring reflection positivity)**.** *Add the nearest-neighbor four-fermion interaction*

$$\Delta S = U\sum_{x,\mu} \bar{\eta}(x)\eta(x + a\hat{\mu})\bar{\eta}(x + a\hat{\mu})\eta(x).$$

For $U \geq 0$, the two-dimensional staggered fermion functional with action $S_{\text{stag}} + \Delta S$ is reflection positive for the mid-link time reflection.

Proof. The free staggered kinetic and mass terms have the mid-link crossing form already described for the nearest-neighbor fermion functional: all terms strictly on the positive half-lattice are placed in S_+, their reflected partners are placed in $\Theta(S_+)$, and the hopping terms crossing the reflection plane have a positive crossing-factor expansion.

It remains to check the quartic interaction at the reflection plane. Links entirely inside one half again belong to S_+ or to $\Theta(S_+)$. The only interaction terms crossing the plane are, for x on the positive boundary,

$$U\,\bar{\eta}(\vartheta x)\eta(x)\bar{\eta}(x)\eta(\vartheta x).$$

Using the displayed reflection rule and the order reversal, this monomial is the reflected-positive bilinear in the even positive-side generator $F_x = \eta(x)\bar{\eta}(x)$. More explicitly, with the fixed ordering of the positive generators,

$$e^{-\Delta S_\times} = \prod_{x \in P_+} \left(1 + U\,\Theta(\eta(x)\bar{\eta}(x))\,\eta(x)\bar{\eta}(x)\right),$$

because the square of each Grassmann quartic monomial vanishes. The different boundary links involve distinct Grassmann generators, so the finite product expands into a sum $\sum_I U^{|I|}\Theta(F_I)F_I$ with nonnegative coefficients precisely when $U \geq 0$. The full Boltzmann factor is therefore in the form required by Definition 146.3, and the finite reflection-positivity criterion, Proposition 146.4, applies. $\square$

146.6 Ginsparg–Wilson Relation

The Wilson term removes doublers by breaking the naive chiral symmetry. Ginsparg–Wilson fermions replace the naive anticommutation relation by a finite-cutoff identity. A lattice Dirac operator D satisfies the Ginsparg–Wilson relation with lattice spacing a if

$$\gamma_5 D + D\gamma_5 = aD\gamma_5 D. \tag{146.4}$$

Assume also γ_5-Hermiticity,

$$D^\dagger = \gamma_5 D\gamma_5.$$

Lemma 146.7 (Exact lattice chiral symmetry and the finite Berezinian)**.** *Let D satisfy* (146.4) *and γ_5-Hermiticity. The finite action*

$$S = \bar\psi D\psi$$

is invariant under the infinitesimal transformation

$$\delta\psi = \varepsilon\,\gamma_5(1-aD)\psi, \qquad \delta\bar\psi = \varepsilon\,\bar\psi\gamma_5.$$

If the finite spinor space has $\operatorname{Tr}\gamma_5 = 0$, the Berezin measure transforms by

$$d\bar\psi'\,d\psi' = \exp\Big(-2\varepsilon\ \operatorname{Tr}\gamma_5\left(1-\frac{a}{2}D\right) + O(\varepsilon^2)\Big)\,d\bar\psi\,d\psi.$$

The integer

$$\operatorname{index}(D) = \operatorname{Tr}\gamma_5\left(1-\frac{a}{2}D\right) \tag{146.5}$$

is the finite-cutoff index associated with the Ginsparg–Wilson operator.

Proof. The variation of the action is

$$\begin{aligned}\delta S &= \varepsilon\,\bar\psi\gamma_5 D\psi + \varepsilon\,\bar\psi D\gamma_5(1-aD)\psi\\ &= \varepsilon\,\bar\psi\big(\gamma_5 D + D\gamma_5 - aD\gamma_5 D\big)\psi = 0.\end{aligned}$$

For odd variables, the Berezin measure transforms with the inverse determinant. Therefore

$$d\psi' = \det\big(1+\varepsilon\gamma_5(1-aD)\big)^{-1}d\psi, \qquad d\bar\psi' = \det(1+\varepsilon\gamma_5)^{-1}d\bar\psi.$$

Taking logarithms gives

$$\log J = -\varepsilon\operatorname{Tr}\gamma_5(1-aD) - \varepsilon\operatorname{Tr}\gamma_5 + O(\varepsilon^2).$$

When $\operatorname{Tr}\gamma_5 = 0$, this becomes

$$\log J = \varepsilon a\ \operatorname{Tr}\gamma_5 D = -2\varepsilon\operatorname{Tr}\gamma_5\left(1-\frac{a}{2}D\right),$$

which is the displayed formula.

The integrality of the trace follows from a finite spectral argument. Define

$$\widehat{\gamma}_5 = \gamma_5(1-aD).$$

γ_5-Hermiticity gives

$$\widehat{\gamma}_5^\dagger = (1 - aD^\dagger)\gamma_5 = (1 - a\gamma_5 D\gamma_5)\gamma_5 = \gamma_5(1 - aD) = \widehat{\gamma}_5.$$

Multiplying the Ginsparg–Wilson relation by γ_5 on the left gives

$$D + \gamma_5 D\gamma_5 = a\gamma_5 D\gamma_5 D,$$

and therefore

$$\widehat{\gamma}_5^2 = (1 - a\gamma_5 D\gamma_5)(1 - aD) = 1.$$

Thus $\widehat{\gamma}_5$ is a Hermitian involution on the finite spinor space. Its eigenvalues are ± 1, so $\operatorname{Tr}\widehat{\gamma}_5 = \dim E_+ - \dim E_-$. Since $\operatorname{Tr}\gamma_5 = 0$, the finite spinor space is even-dimensional, and $\dim E_+ - \dim E_-$ is even. Finally,

$$\operatorname{Tr}\gamma_5\left(1 - \frac{a}{2}D\right) = \frac{1}{2}\operatorname{Tr}(\gamma_5 + \widehat{\gamma}_5) = \frac{1}{2}\operatorname{Tr}\widehat{\gamma}_5 \in \mathbb{Z}.$$

□

146.7 Overlap Operator

Let D_W be the Wilson operator with a negative mass parameter in the usual overlap window, and define the Hermitian kernel

$$H_W = \gamma_5(D_W - m_0/a).$$

Assume H_W has no zero eigenvalues. Let

$$\epsilon(H_W) = \operatorname{sign}(H_W) = H_W(H_W^2)^{-1/2}.$$

The overlap operator is

$$D_{\text{ov}} = \frac{1}{a}\left(\mathbf{1} + \gamma_5\epsilon(H_W)\right). \tag{146.6}$$

Overlap operator satisfies Ginsparg–Wilson. If $H_W = H_W^\dagger$ has no zero modes, then D_{ov} satisfies

$$\gamma_5 D_{\text{ov}} + D_{\text{ov}}\gamma_5 = aD_{\text{ov}}\gamma_5 D_{\text{ov}}.$$

Put

$$V = \gamma_5\epsilon(H_W).$$

Since $\epsilon(H_W)^\dagger = \epsilon(H_W)$ and $\epsilon(H_W)^2 = \mathbf{1}$, one has

$$V^\dagger = \epsilon(H_W)\gamma_5, \qquad V^\dagger V = \mathbf{1}.$$

Moreover $V^\dagger = \gamma_5 V\gamma_5$. Thus

$$D_{\text{ov}} = \frac{1}{a}(1 + V), \qquad D_{\text{ov}}^\dagger = \gamma_5 D_{\text{ov}}\gamma_5.$$

Unitarity of V gives

$$D_{\text{ov}} + D_{\text{ov}}^\dagger = \frac{1}{a}(2 + V + V^\dagger) = aD_{\text{ov}}^\dagger D_{\text{ov}}.$$

Multiplying this identity on the left by γ_5 and using $D_{\text{ov}}^\dagger = \gamma_5 D_{\text{ov}}\gamma_5$ gives the Ginsparg–Wilson relation.

Overlap index as spectral asymmetry. For the overlap operator,

$$\mathrm{index}(D_{\mathrm{ov}}) = -\frac{1}{2}\,\mathrm{Tr}\,\epsilon(H_W).$$

Using (146.6),

$$\gamma_5\left(1-\frac{a}{2}D_{\mathrm{ov}}\right) = \gamma_5\left(1-\frac{1}{2}(1+\gamma_5\epsilon(H_W))\right) = \frac{1}{2}\gamma_5 - \frac{1}{2}\epsilon(H_W).$$

The finite spinor-lattice trace of γ_5 vanishes, hence

$$\mathrm{Tr}\,\gamma_5\left(1-\frac{a}{2}D_{\mathrm{ov}}\right) = -\frac{1}{2}\,\mathrm{Tr}\,\epsilon(H_W).$$

The left-hand side is the Ginsparg–Wilson lattice index written as a finite trace, so the equality identifies the index with the spectral asymmetry of the Hermitian Wilson kernel H_W.

146.8 Domain-wall Fermions And The Overlap Limit

Domain-wall fermions realize the same chiral mechanism by adding a finite fifth lattice direction $s = 1, \dots, L_s$ with a mass profile whose sign changes across the wall. The low-energy modes are localized near opposite walls and have opposite chirality. At finite L_s their overlap gives a residual mass. In the limit $L_s \to \infty$, after integrating out the bulk fifth-dimensional modes, the effective four-dimensional operator is the overlap operator (146.6). Thus domain-wall fermions and overlap fermions should not be viewed as two unrelated solutions: one is a local higher-dimensional regulator whose infinite-wall limit produces the four-dimensional Ginsparg–Wilson operator.

The price paid by overlap fermions is that D_{ov} is not ultralocal. Locality means exponential decay of the kernel,

$$\|D_{\mathrm{ov}}(x,y)\| \le Ce^{-c|x-y|/a}.$$

This is a theorem only under hypotheses excluding near-zero modes of H_W, for instance an admissibility condition on the gauge plaquettes or a spectral-gap estimate in the gauge ensemble. Without such an estimate, the formula (146.6) still defines a finite matrix when H_W is invertible, but it does not by itself prove locality of the regulator.

146.9 Continuum Requirements

A lattice fermion construction used as a nonperturbative QFT regulator must specify the following data.

finite Grassmann algebra and Berezin ordering,

Dirac operator, gauge covariance, spin structure, and boundary conditions,

reflection positivity or a replacement Hilbert-space construction,

chiral symmetry, anomaly, and index convention,

locality estimates uniform enough for a scaling limit,

operator regularization and renormalization separate from the cutoff action.

For chiral gauge theories, the fermion determinant or Pfaffian is a section of a determinant line over the finite lattice gauge-field space. A nonperturbative definition requires a gauge-invariant, local, and reflection-compatible choice of phase. Anomaly cancellation is the local cohomological condition that makes such a choice possible perturbatively; it is not by itself a complete nonperturbative construction of the interacting continuum theory.

Open Problem 146.8 (Chiral gauge lattice)*.* Construct four-dimensional interacting chiral gauge theories on the lattice with locality, reflection positivity or an equivalent reconstruction route, exact gauge invariance, anomaly cancellation, and a continuum limit with the desired local operator algebra.

Part XII

Quantum Field Theory in Curved Spacetime and Background Fields

Chapter 147

Locally Covariant QFT and Hadamard States

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Curved-spacetime QFT is a background-field problem. The Lorentzian metric, orientation, time orientation, spin structure, background gauge bundle, and external sources are held fixed, while the quantum observables are assigned locally and covariantly to each background. The absence of a preferred translation-invariant vacuum is replaced by a condition on states: their short-distance singularity must have the same positive-frequency geometric form as the Minkowski vacuum when examined in a sufficiently small convex normal neighborhood.

This chapter fixes the functorial language and then constructs the free Klein–Gordon example in detail. The example is important because the time-slice axiom, causal propagation, commutation relations, and Hadamard subtraction are visible in one model without using a perturbative expansion.

147.1 Control Levels In Curved-Spacetime QFT

The chapters in this volume use four different levels of control. They are kept separate because the hypotheses that prove a fixed-background free-field statement are not the hypotheses that prove an interacting backreaction statement. The control-level matrix is:

Fixed-background free theory. Input data are a globally hyperbolic background, a linear hyperbolic operator, a CCR or CAR algebra, and a Hadamard state class. The controlled outputs are propagation, commutators, the two-point singularity, Wick powers, and the point-split stress tensor after finite local choices; see Chapters 147–155.

Perturbative interacting theory. Input data are the free Hadamard algebra, a local interaction, time-ordered products, causal factorization, and a renormalization prescription. The controlled outputs are relative S-matrices, interacting fields order by order, local counterterm freedom, and finite-order stress-tensor source terms; see Chapter 156.

Conditional semiclassical theory. Input data are an interacting or free stress tensor in a fixed scheme, an admissible state, gravitational EFT coordinates, and stability and response assumptions. The controlled outputs are the semiclassical Einstein equation, retarded metric response, noise kernel, Einstein–Langevin covariance, and validity windows; see Chapter 157.

Nonperturbative interacting curved QFT. Input data would be a genuine locally covariant interacting theory with controlled states and composite fields on the relevant background class. The controlled outputs would be state-independent structural claims beyond perturbation theory and exact horizon, cosmological, or interacting thermal statements. In this volume those claims appear only as open problems or as explicitly marked conditional statements.

Moving from one row to the next is a mathematical and physical strengthening: perturbative locality does not by itself give a nonperturbative state space, and a finite stress-tensor calculation does not by itself solve the semiclassical Einstein equation. The later chapters therefore state their state class, renormalization scheme, and response assumptions at the point where those assumptions first become load bearing.

147.2 Spacetimes And Causal Embeddings

Let M be a smooth D-manifold with Lorentzian metric g of mostly-plus signature. A tangent vector $v \in T_xM$ is causal when $g_x(v,v) \le 0$, timelike when $g_x(v,v) < 0$, and null when $g_x(v,v) = 0$ and $v \neq 0$. A time orientation is a continuous choice of future cone in the causal cone at every point. For a subset $S \subset M$, write $J^{\pm}_M(S)$ for the causal future and causal past of S.

Definition 147.1 (Globally hyperbolic spacetime)**.** An oriented and time-oriented Lorentzian manifold (M, g) is globally hyperbolic when it has no closed causal curve and when

$$J^+_M(p) \cap J^-_M(q)$$

is compact for every $p, q \in M$. A Cauchy surface is a subset $\Sigma \subset M$ intersected exactly once by every inextendible timelike curve. For globally hyperbolic M, smooth spacelike Cauchy surfaces exist, and M is diffeomorphic to $\mathbb{R} \times \Sigma$.

The compactness condition is the analytic input behind finite propagation statements. If $K \subset M$ is compact, then

$$J^+_M(K) \cap J^-_M(K')$$

is compact for every compact $K' \subset M$. This property is used below when a solution is cut off between two Cauchy surfaces.

Definition 147.2 (Causally convex subset)**.** An open subset $O \subset M$ is causally convex when every causal curve in M whose endpoints lie in O is entirely contained in O.

Causal convexity is the condition that lets a field equation be restricted to a subspacetime without importing causal propagation from the complement.

Definition 147.3 (Background category)**.** Let **Loc** be the category defined as follows.

An object is an oriented, time-oriented, globally hyperbolic Lorentzian manifold (M, g), possibly equipped with fixed background bundles and smooth background sources. A morphism

$$\chi : (M, g) \longrightarrow (N, h)$$

is a smooth embedding with open causally convex image such that

$$\chi^* h = g$$

and such that χ preserves the declared orientation, time orientation, and background bundle/source data. A morphism $\chi : M \to N$ is a Cauchy morphism when $\chi(M)$ contains a Cauchy surface of N.

Composition of morphisms is ordinary composition of embeddings. The image of a composite is causally convex: if a causal curve in the final spacetime has endpoints in the composite image, causal convexity of the second image first keeps the curve in the intermediate image, and causal convexity of the first image then keeps its pullback in the original image. Thus the above data form a category.

147.3 Algebra-Valued Functors

Let $\mathbf{Alg}_*$ denote a category of unital complex $*$-algebras. Its morphisms are unit-preserving injective $*$-homomorphisms. In analytic versions one uses locally convex topological $*$-algebras and continuous morphisms; the algebraic definition is enough for the functorial axioms in this chapter.

Definition 147.4 (Locally covariant QFT)**.** A locally covariant QFT is a covariant functor

$$\mathfrak{A} : \mathbf{Loc} \longrightarrow \mathbf{Alg}_*$$

with the following additional properties.

(i) If $\chi : M \to N$ is a Cauchy morphism, then

$$\mathfrak{A}(\chi) : \mathfrak{A}(M) \longrightarrow \mathfrak{A}(N)$$

is an isomorphism. This is the time-slice axiom.

(ii) If $\chi_i : M_i \to N$, $i = 1, 2$, have causally disjoint images, then

$$[\, \mathfrak{A}(\chi_1)(A_1), \mathfrak{A}(\chi_2)(A_2) \,] = 0$$

for all $A_i \in \mathfrak{A}(M_i)$. This is Einstein causality.

The injectivity of morphisms in $\mathbf{Alg}_*$ records isotony: an admissible spacetime inclusion identifies the algebra of the smaller background with a subalgebra of the larger one.

Definition 147.5 (Kinematic local algebra)**.** Let $O \subset M$ be open, causally convex, and globally hyperbolic with the induced metric and orientations. Let $\iota_{M;O} : O \hookrightarrow M$ be the inclusion morphism in **Loc**. The kinematic algebra of O inside M is

$$\mathfrak{A}^{\mathrm{kin}}(M;O) = \mathfrak{A}(\iota_{M;O})(\mathfrak{A}(O)) \subset \mathfrak{A}(M).$$

Remark 147.6 (Local net associated to a locally covariant theory). For each fixed M, the assignment

$$O \longmapsto \mathfrak{A}^{\mathrm{kin}}(M;O)$$

on causally convex globally hyperbolic open subsets is isotone and satisfies Einstein causality.

If $O_1 \subset O_2$, the inclusion factors as

$$O_1 \xrightarrow{\iota_{O_2;O_1}} O_2 \xrightarrow{\iota_{M;O_2}} M.$$

Functoriality gives

$$\mathfrak{A}(\iota_{M;O_1}) = \mathfrak{A}(\iota_{M;O_2}) \circ \mathfrak{A}(\iota_{O_2;O_1}),$$

so the image for O_1 is contained in the image for O_2. If O_1 and O_2 are causally disjoint in M, the two inclusion morphisms into M satisfy the hypothesis of Einstein causality in Definition 147.4, and the commutator of the two image algebras vanishes.

Locally covariant fields are defined by naturality. Let $\mathbf{Test}_E$ be the functor assigning to M the vector space $\Gamma_c(M, E_M)$ of compactly supported smooth sections of a natural bundle E_M, and to a morphism $\chi : M \to N$ the pushforward $\chi_* : \Gamma_c(M, E_M) \to \Gamma_c(N, E_N)$.

Definition 147.7 (Locally covariant field)**.** A locally covariant field of type E is a natural transformation

$$\Phi : \mathbf{Test}_E \Longrightarrow \mathfrak{A}.$$

Equivalently, for every M and every $f \in \Gamma_c(M, E_M)$ there is an element $\Phi_M(f) \in \mathfrak{A}(M)$, and every morphism $\chi : M \to N$ satisfies

$$\mathfrak{A}(\chi)\Phi_M(f) = \Phi_N(\chi_* f). \tag{147.1}$$

The condition is a commutative square:

$$\begin{tikzcd}
\Gamma_c(M, E_M) \arrow[r, "\Phi_M"] \arrow[d, "\chi_*"'] & \mathfrak{A}(M) \arrow[d, "\mathfrak{A}(\chi)"] \\
\Gamma_c(N, E_N) \arrow[r, "\Phi_N"'] & \mathfrak{A}(N).
\end{tikzcd}$$

147.4 States As Contravariant Data

A state on a unital $*$-algebra $\mathcal{A}$ is a linear functional $\omega : \mathcal{A} \to \mathbb{C}$ satisfying

$$\omega(\mathbf{1}) = 1, \qquad \omega(A^*A) \geq 0 \quad \text{for all } A \in \mathcal{A}.$$

Every morphism $\alpha : \mathcal{A} \to \mathcal{B}$ pulls states on $\mathcal{B}$ back to states on $\mathcal{A}$ by composition:

$$\alpha^*\omega = \omega \circ \alpha.$$

Thus a state assignment for a locally covariant theory is contravariant.

Definition 147.8 (Locally covariant state space)**.** A locally covariant state space for $\mathfrak{A}$ is a choice of subsets

$$\mathcal{S}(M) \subset \{\text{states on } \mathfrak{A}(M)\}$$

such that for every morphism $\chi : M \to N$,

$$\omega \in \mathcal{S}(N) \quad \Longrightarrow \quad \omega \circ \mathfrak{A}(\chi) \in \mathcal{S}(M).$$

Hadamard states provide the state spaces used for free scalar fields on curved backgrounds. Their definition is local in spacetime and stable under restriction to causally convex subregions.

147.5 The Free Klein–Gordon Functor

Fix real constants $m^2 \geq 0$ and $\xi \in \mathbb{R}$. For every $(M, g) \in \mathbf{Loc}$, define the real differential operator

$$P_M = -\nabla^\mu \nabla_\mu + m^2 + \xi R[g]$$

on $C^\infty(M, \mathbb{R})$. It is formally self-adjoint with respect to the pairing

$$(f, h)_M = \int_M f h \, \mathrm{d}\,\mathrm{vol}_g \, .$$

Global hyperbolicity gives unique retarded and advanced Green operators

$$E_M^{\mathrm{ret}}, E_M^{\mathrm{adv}} : C_c^\infty(M) \longrightarrow C^\infty(M)$$

with

$$\begin{aligned} P_M E_M^{\mathrm{ret}} f &= f, \qquad & P_M E_M^{\mathrm{adv}} f &= f, \\ E_M^{\mathrm{ret}} P_M f &= f, \qquad & E_M^{\mathrm{adv}} P_M f &= f, \end{aligned}$$

and support restrictions

$$\operatorname{supp} E_M^{\mathrm{ret}} f \subset J_M^+(\operatorname{supp} f), \qquad \operatorname{supp} E_M^{\mathrm{adv}} f \subset J_M^-(\operatorname{supp} f).$$

The causal propagator is

$$E_M = E_M^{\mathrm{ret}} - E_M^{\mathrm{adv}}.$$

Lemma 147.9 (Kernel of the causal propagator)**.** *For $f \in C_c^\infty(M)$,*

$$E_M f = 0 \quad \Longleftrightarrow \quad f = P_M h \textit{ for some } h \in C_c^\infty(M).$$

Proof. If $f = P_M h$ with compactly supported h, then

$$E_M f = E_M P_M h = h - h = 0,$$

using the defining identities for retarded and advanced Green operators.

Conversely assume $E_M f = 0$. Then

$$u := E_M^{\mathrm{ret}} f = E_M^{\mathrm{adv}} f.$$

The equality is an equality of smooth solutions, because both Green operators send compactly supported test functions to smooth functions and their difference is $E_M f$. The retarded representative satisfies

$$P_M u = f,$$

and the same identity holds for the advanced representative. The retarded expression has support in $J_M^+(\operatorname{supp} f)$, and the advanced expression has support in $J_M^-(\operatorname{supp} f)$. Thus

$$\operatorname{supp} u \subset J_M^+(\operatorname{supp} f) \cap J_M^-(\operatorname{supp} f),$$

which is compact by global hyperbolicity: the causal diamond of a compact set with itself is compact in a globally hyperbolic spacetime. Hence $u \in C_c^\infty(M)$, and $f = P_M u$ lies in $P_M C_c^\infty(M)$. $\square$

Define the real vector space of smeared solutions by

$$\mathcal{V}(M) = C_c^\infty(M, \mathbb{R}) / P_M C_c^\infty(M, \mathbb{R}), \qquad [f]_M \text{ for the class of } f.$$

Define

$$\sigma_M([f]_M, [h]_M) = \int_M f \, E_M h \, \mathrm{d}\,\mathrm{vol}_g \,. \tag{147.2}$$

This is well-defined because $E_M P_M = 0$ on compactly supported test functions and because P_M is formally self-adjoint. It is antisymmetric:

$$\int_M f \, E_M h \, \mathrm{d}\,\mathrm{vol}_g = -\int_M h \, E_M f \, \mathrm{d}\,\mathrm{vol}_g,$$

which follows by applying formal self-adjointness to retarded and advanced Green operators and using

$$(E_M^{\mathrm{ret}})^* = E_M^{\mathrm{adv}}.$$

Lemma 147.9 implies that σ_M is nondegenerate on $\mathcal{V}(M)$.

Definition 147.10 (CCR algebra of the scalar field)**.** The algebra $\mathfrak{A}_{\mathrm{KG}}(M)$ is the unital $*$-algebra generated by symbols $\Phi_M([f]_M)$, linear in $[f]_M$, with relations

$$\Phi_M([f]_M)^* = \Phi_M([f]_M), \qquad [\Phi_M([f]_M), \Phi_M([h]_M)] = \mathrm{i}\, \sigma_M([f]_M, [h]_M)\mathbf{1}.$$

The quotient by $P_M C_c^\infty(M)$ is the equation of motion:

$$\Phi_M(P_M h) = 0.$$

The commutator is determined by the causal propagator and therefore by the hyperbolic equation, not by a choice of state.

Proposition 147.11 (Functoriality of the free scalar field)**.** *For every morphism $\chi : M \to N$, the pushforward of test functions induces a symplectic linear map*

$$\mathcal{V}(\chi) : \mathcal{V}(M) \longrightarrow \mathcal{V}(N), \qquad [f]_M \longmapsto [\chi_* f]_N.$$

Consequently there is an injective ∗-homomorphism

$$\mathfrak{A}_{\mathrm{KG}}(\chi) : \mathfrak{A}_{\mathrm{KG}}(M) \to \mathfrak{A}_{\mathrm{KG}}(N)$$

defined by

$$\mathfrak{A}_{\mathrm{KG}}(\chi)\Phi_M([f]_M) = \Phi_N([\chi_* f]_N).$$

Proof. The embedding is isometric and preserves background scalar data, so

$$P_N \chi_* f = \chi_* P_M f.$$

Therefore χ_* maps $P_M C_c^\infty(M)$ into $P_N C_c^\infty(N)$, giving a map on quotients. Causal convexity gives compatibility of causal propagators:

$$E_N \chi_* f = \chi_* E_M f. \tag{147.3}$$

Indeed, both sides solve $P_N u = \chi_* f$ in $\chi(M)$, both vanish outside the same retarded-minus-advanced causal support determined by $\chi(\operatorname{supp} f)$, and uniqueness of retarded and advanced solutions gives equality. The volume form is preserved by χ, hence

$$\sigma_N([\chi_* f]_N, [\chi_* h]_N) = \int_N \chi_* f \, E_N \chi_* h \, \mathrm{d}\,\mathrm{vol}_h = \int_M f \, E_M h \, \mathrm{d}\,\mathrm{vol}_g = \sigma_M([f]_M, [h]_M).$$

The universal property of the CCR algebra then gives a ∗-homomorphism preserving the defining commutator. Injectivity follows from injectivity of the induced symplectic map; if $[\chi_* f]_N = 0$, then $E_N \chi_* f = 0$, so $\chi_* E_M f = 0$, hence $E_M f = 0$ and $[f]_M = 0$ by Lemma 147.9. □

Proposition 147.12 (Time-slice property for the free scalar)**.** *If $\chi : M \to N$ is a Cauchy morphism, then*

$$\mathfrak{A}_{\mathrm{KG}}(\chi) : \mathfrak{A}_{\mathrm{KG}}(M) \to \mathfrak{A}_{\mathrm{KG}}(N)$$

is an isomorphism.

Proof. It suffices to prove that $\mathcal{V}(\chi) : \mathcal{V}(M) \to \mathcal{V}(N)$ is surjective, since Proposition 147.11 already gives injectivity and the CCR algebra is generated by the smeared fields.

Let $f \in C_c^\infty(N)$ and set $u = E_N f$. Choose two smooth spacelike Cauchy surfaces Σ_- and Σ_+ contained in $\chi(M)$ such that Σ_+ lies to the future of Σ_- and the compact set

$$J_N(\operatorname{supp} f) \cap J_N^+(\Sigma_-) \cap J_N^-(\Sigma_+)$$

contains the part of u between the two surfaces. Choose a smooth function $\eta : N \to [0,1]$ that is 0 to the past of Σ_-, is 1 to the future of Σ_+, and whose derivative is supported in $\chi(M)$. Define

$$g = P_N(\eta u).$$

Since $P_N u = 0$, the support of g lies where derivatives of η are nonzero, hence $g \in C_c^\infty(\chi(M))$. Set

$$w = \eta u.$$

The section w vanishes to the past of Σ_- and satisfies $P_N w = g$, so uniqueness of the retarded Green solution gives

$$E_N^{\mathrm{ret}} g = w.$$

The section $w - u = (\eta - 1)u$ vanishes to the future of Σ_+ and also satisfies $P_N(w-u) = g$, so uniqueness of the advanced Green solution gives

$$E_N^{\mathrm{adv}} g = w - u.$$

Therefore

$$E_N g = E_N^{\mathrm{ret}} g - E_N^{\mathrm{adv}} g = w - (w-u) = u = E_N f.$$

Thus $E_N(g-f) = 0$. By Lemma 147.9, $g - f = P_N k$ for some $k \in C_c^\infty(N)$. Thus $[f]_N = [g]_N$. Since $\operatorname{supp} g \subset \chi(M)$, there is $\widetilde{g} \in C_c^\infty(M)$ with $g = \chi_* \widetilde{g}$, and $[f]_N = \mathcal{V}(\chi)[\widetilde{g}]_M$. □

147.6 Quasifree States

A state ω on $\mathfrak{A}_{\mathrm{KG}}(M)$ has n-point distributions

$$\omega_n(f_1, \ldots, f_n) = \omega(\Phi_M([f_1]) \cdots \Phi_M([f_n])) .$$

It is quasifree with vanishing one-point function when all odd n-point distributions vanish and all even n-point distributions are sums over pairings:

$$\omega_{2n}(f_1, \ldots, f_{2n}) = \sum_{\text{pairings } P} \prod_{\{i,j\} \in P} \omega_2(f_i, f_j).$$

The two-point distribution $\omega_2 \in \mathcal{D}'(M \times M)$ must satisfy

$$P_{M,x}\omega_2 = 0, \qquad P_{M,y}\omega_2 = 0,$$

$$\omega_2(f,h) - \omega_2(h,f) = \mathrm{i}\,\sigma_M([f],[h]),$$

and positivity:

$$\omega_2(\bar{f}, f) \geq 0 \quad \text{for all } f \in C_c^\infty(M, \mathbb{C}).$$

Conversely, a bidistribution with these properties defines a quasifree state on the CCR algebra.

147.7 Hadamard Form

Let x and y lie in a convex normal neighborhood $U \subset M$. Let

$$\sigma(x,y)$$

be Synge's world function: one half of the signed squared geodesic distance from x to y. Let T be any smooth time function on U, and set

$$\sigma_\epsilon(x,y) = \sigma(x,y) + \mathrm{i}\epsilon\big(T(x) - T(y)\big) + \epsilon^2, \qquad \epsilon > 0.$$

The limit $\epsilon \downarrow 0$ is taken in the distributional sense.

In four spacetime dimensions, the local Hadamard singular distribution for P_M has the form

$$H_{\epsilon,\mu}(x,y) = \frac{1}{8\pi^2}\left[\frac{U(x,y)}{\sigma_\epsilon(x,y)} + V(x,y)\log\bigl(\mu^2\sigma_\epsilon(x,y)\bigr)\right], \tag{147.4}$$

where $\mu > 0$ is a length inverse. The coefficient U is

$$U(x,y) = \Delta(x,y)^{1/2},$$

with Δ the van Vleck determinant. It is characterized by the transport equation

$$2\sigma^{;\mu}\nabla_\mu U + (\Box_g\sigma - D)U = 0, \qquad U(x,x) = 1.$$

The function V is locally determined by g, m, ξ through the requirement that $P_{M,x}H_{\epsilon,\mu}(x,y)$ have no singular part away from the diagonal. Equivalently, expanding

$$V(x,y) \sim \sum_{n\geq 0} V_n(x,y)\sigma(x,y)^n$$

gives transport equations along the geodesic from y to x, with initial data fixed recursively by P_M and the previous coefficients. The smooth part of a state is not included in $H_{\epsilon,\mu}$.

Definition 147.13 (Hadamard quasifree state)**.** A quasifree state ω of the Klein–Gordon field is Hadamard when, in every convex normal neighborhood,

$$\omega_2(x,y) - H_{\epsilon,\mu}(x,y)$$

has a smooth distributional limit as $\epsilon \downarrow 0$. The resulting smooth function is denoted $W_\omega(x,y)$.

Changing the time function T in σ_ϵ changes only the boundary-value prescription used to define the same positive-frequency singularity. Changing μ changes the subtraction by

$$\frac{1}{8\pi^2}V(x,y)\log(\mu'^2/\mu^2),$$

which is smooth near the diagonal because V is smooth.

Remark 147.14 (State-independent singular part)*.* Let ω and ω' be Hadamard quasifree states for the same operator P_M on the same globally hyperbolic spacetime. Then

$$\omega_2 - \omega_2' \in C^\infty(M \times M).$$

On each convex normal neighborhood, Definition 147.13 gives

$$\omega_2 = H_{\epsilon,\mu} + W_\omega, \qquad \omega_2' = H_{\epsilon,\mu} + W_{\omega'}$$

after taking the distributional boundary value. The same $H_{\epsilon,\mu}$ appears because U and V are fixed by (M,g), m, and ξ. Therefore

$$\omega_2 - \omega_2' = W_\omega - W_{\omega'}$$

is smooth on that neighborhood. Smoothness is local on $M \times M$, so the difference is smooth globally.

147.8 Microlocal Form And Wick Products

The local formula above is equivalent to the microlocal Hadamard condition developed in Chapter 155:

$$\mathrm{WF}(\omega_2) = \{(x, k_x; y, -k_y) : (x, k_x) \sim (y, k_y),\ k_x \in \overline{V}_+^*\}.$$

Here $(x, k_x) \sim (y, k_y)$ means that x and y are connected by a null geodesic and that k_y is the parallel transport of k_x. This condition is the curved-spacetime replacement for positive energy support.

For a Hadamard state, define the Wick square by point splitting:

$$:\phi^2:_H (f) = \int_M f(x) \lim_{y\to x} (\phi(x)\phi(y) - H_{\epsilon,\mu}(x, y)\mathbf{1})\, \mathrm{d}\,\mathrm{vol}_g(x).$$

Remark 147.14 implies that the subtraction removes the state-independent singularity. The remaining freedom is finite and local. In four dimensions,

$$:\phi^2:' = :\phi^2: + c_1 m^2 + c_2 R$$

for constants c_1, c_2, once covariance, scaling dimension, and the field equation are imposed. Stress-tensor point splitting in the next chapter is the same construction with the bidifferential operator dictated by the classical stress tensor.

147.9 Renormalization Freedom Of The Stress Tensor

The stress tensor is obtained by applying a bidifferential operator $D_{\mu\nu}(x, y)$ to the subtracted two-point expression and then taking the diagonal limit:

$$\langle T_{\mu\nu}(x)\rangle_\omega = \lim_{y\to x} D_{\mu\nu}(x, y)\big(\omega_2(x, y) - H_{\epsilon,\mu}(x, y)\big) + C_{\mu\nu}[g](x).$$

The tensor $C_{\mu\nu}[g]$ is local and covariant in the background. In four dimensions, conservation and engineering dimension restrict it to a linear combination of metric variations of

$$\int \sqrt{|g|}, \qquad \int \sqrt{|g|}R, \qquad \int \sqrt{|g|}R^2, \qquad \int \sqrt{|g|}R_{\mu\nu}R^{\mu\nu}, \qquad \int \sqrt{|g|}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma},$$

with constants specifying the finite local curvature coordinates of the renormalized theory. This statement is a covariance and conservation classification of finite local terms; the explicit point-splitting construction and the trace-anomaly consequences are developed in Chapters 148 and 149.

Open Problem 147.15 (Interacting curved-spacetime QFT). Construct interacting locally covariant QFTs with nonperturbative control of state spaces, composite operators, stress tensors, and background dependence. The construction should compare perturbative algebraic QFT, constructive methods, lattice methods on discretized backgrounds, and Hamiltonian approaches in examples where more than one construction exists.

Chapter 148

Point Splitting and Stress Tensor Renormalization

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The previous chapter introduced Hadamard states as the admissible short-distance class for free fields on curved backgrounds. Point splitting uses the state-independent singular part of a Hadamard two-point function to define composite local fields.

148.1 Local Hadamard Subtraction Datum

Throughout this chapter the scalar operator is the one fixed in Chapter 147,

$$P_M = -\nabla^\mu \nabla_\mu + m^2 + \xi R[g], \qquad m^2 \geq 0, \qquad \xi \in \mathbb{R}. \tag{148.1}$$

Let $\mathcal{U} \subset M$ be a geodesically convex normal neighborhood. For $x, y \in \mathcal{U}$, let $\sigma(x, y)$ be Synge's world function: one half of the signed squared geodesic distance from x to y. Choose a smooth time function T on $\mathcal{U}$ and put

$$\sigma_\epsilon(x, y) = \sigma(x, y) + \mathrm{i}\epsilon\big(T(x) - T(y)\big) + \epsilon^2, \qquad \epsilon > 0. \tag{148.2}$$

The limit $\epsilon \downarrow 0$ is always distributional.

Definition 148.1 (Hadamard subtraction)**.** In four dimensions, a Hadamard subtraction for P_M on $\mathcal{U}$ is the distributional boundary value

$$H_{\epsilon,\mu}(x,y) = \frac{1}{8\pi^2}\left[\frac{U_H(x,y)}{\sigma_\epsilon(x,y)} + V_H(x,y)\log\bigl(\mu^2\sigma_\epsilon(x,y)\bigr)\right], \qquad \mu > 0, \tag{148.3}$$

where $U_H = \Delta^{1/2}$, Δ is the van Vleck determinant, and V_H is a smooth representative of the Hadamard coefficient whose jet at the diagonal is determined recursively by P_M. Different smooth representatives with the same jet, and additional smooth local biscalars added to the subtraction, change only the finite local subtraction datum; they are not part of the state.

The word "determined" in Definition 148.1 means a local transport construction of the jet of V_H at the diagonal. It is not a global Green-function prescription. To spell out the local object, write

$$V_H(x,y) \sim \sum_{j\geq 0} v_j(x,y)\sigma(x,y)^j \tag{148.4}$$

as an asymptotic expansion of smooth biscalars in a normal neighborhood. For each integer $N \geq 0$, the partial sum

$$V_H^{[N]}(x,y) = \sum_{j=0}^{N} v_j(x,y)\sigma(x,y)^j \tag{148.5}$$

is chosen so that

$$P_{M,x}\left[\frac{U_H(x,y)}{\sigma(x,y)} + V_H^{[N]}(x,y)\log\bigl(\mu^2\sigma(x,y)\bigr)\right] \tag{148.6}$$

has no negative powers of σ and no logarithmic terms through the order fixed by N. Expanding (148.6) in powers of σ gives ordinary differential equations along the unique geodesic from y to x. Since the coefficient multiplying the unknown diagonal value at the j-th step is nonzero, the transport equation fixes v_j from lower coefficients. A smooth V_H with this prescribed infinite jet exists by Borel's lemma. Replacing this representative by another one with the same jet changes $H_{\epsilon,\mu}$ by a smooth biscalar, because the difference is flat at the diagonal before multiplication by $\log\sigma$; adding any allowed smooth local biscalar is precisely a finite renormalization choice.

Lemma 148.2 (Leading Hadamard transport equation)**.** *The coefficient U_H in (148.3) is fixed by*

$$2\sigma^{;\mu}\nabla_\mu U_H + (\Box_g\sigma - 4)U_H = 0, \qquad U_H(x,x) = 1. \tag{148.7}$$

Proof. The calculation is local and may be performed for $x \neq y$, before taking the distributional boundary value. The defining identities of Synge's function in a normal neighborhood are

$$\sigma^{;\mu}\sigma_{;\mu} = 2\sigma, \qquad \Box_g\sigma = 4 + O(\sigma). \tag{148.8}$$

For a smooth biscalar U,

$$\nabla_\mu(\sigma^{-1}) = -\sigma_{;\mu}\sigma^{-2}, \qquad \Box_g(\sigma^{-1}) = -(\Box_g\sigma)\sigma^{-2} + 2\sigma^{;\mu}\sigma_{;\mu}\sigma^{-3}.$$

Using $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$, this becomes

$$\Box_g(\sigma^{-1}) = (4 - \Box_g\sigma)\sigma^{-2}.$$

Hence the most singular term in $-\Box_{g,x}(U/\sigma)$ is

$$\frac{2\sigma^{;\mu}\nabla_\mu U + (\Box_g\sigma - 4)U}{\sigma^2}.$$

The zeroth-order part $m^2+\xi R$ of P_M contributes only σ^{-1} singularities, so the σ^{-2} coefficient must vanish. This is exactly (148.7). The initial value $U_H(x,x) = 1$ gives the flat-space normalization and uniquely solves the transport equation along the geodesic from y to x; the solution is $\Delta^{1/2}$. $\square$

Lemma 148.3 (First logarithmic Hadamard coefficient). *Let $V_H = v_0 + O(\sigma)$. With the operator convention* (148.1)*, the first coefficient is fixed by*

$$2\sigma^{;\mu}\nabla_\mu v_0 + (\Box_g\sigma - 2)v_0 = P_{M,x}U_H. \tag{148.9}$$

In particular,

$$v_0(x,x) = \frac{1}{2}\left(m^2 + \left(\xi - \frac{1}{6}\right)R[g](x)\right). \tag{148.10}$$

Proof. After Lemma 148.2, the σ^{-2} coefficient in $P_{M,x}(U_H/\sigma)$ vanishes. The next singular term is therefore

$$\frac{P_{M,x}U_H}{\sigma}. \tag{148.11}$$

For a smooth biscalar v,

$$\nabla_\mu \log\sigma = \frac{\sigma_{;\mu}}{\sigma}, \qquad \Box_g \log\sigma = \frac{\Box_g\sigma - 2}{\sigma}, \tag{148.12}$$

where the second identity uses $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$. Hence

$$P_{M,x}\big(v\log\sigma\big) = (P_{M,x}v)\log\sigma - \frac{2\sigma^{;\mu}\nabla_\mu v + (\Box_g\sigma - 2)v}{\sigma}. \tag{148.13}$$

The zeroth-order part of P_M multiplies $v\log\sigma$ and contributes only the displayed logarithmic term, not a negative power of σ. Canceling the coefficient of σ^{-1} between (148.11) and (148.13) gives (148.9). Taking the coincidence limit, $\sigma^{;\mu} = 0$ and $\Box_g\sigma = 4$. Also

$$U_H(x,x) = 1, \qquad \Box_{g,x}U_H(x,y)|_{y=x} = \frac{1}{6}R[g](x), \tag{148.14}$$

which follows from the normal-coordinate expansion of $\Delta^{1/2}$. Therefore

$$P_{M,x}U_H(x,y)|_{y=x} = m^2 + \xi R - \frac{1}{6}R,$$

and (148.10) follows. $\square$

Proposition 148.4 (Hadamard logarithmic recursion). *Let*

$$V_H(x,y) \sim \sum_{j\geq 0} v_j(x,y)\sigma(x,y)^j$$

be the local Hadamard logarithmic coefficient in four dimensions. After U_H has been fixed by (148.7)*, the coefficients v_j are determined recursively along the geodesic from y to x by*

$$\big(2\sigma^{;\mu}\nabla_\mu + \Box_g\sigma - 2\big)v_0 = P_{M,x}U_H, \tag{148.15}$$

and, for $j \ge 0$,

$$\big(2\sigma^{;\mu}\nabla_\mu + \Box_g\sigma + 2j\big)v_{j+1} = \frac{1}{j+1}\,P_{M,x}v_j. \tag{148.16}$$

At coincidence this gives

$$v_{j+1}(x,x) = \frac{1}{2(j+1)(j+2)}\,P_{M,x}v_j(x,y)\big|_{y=x}. \tag{148.17}$$

The recursion fixes the complete diagonal jet of V_H. Choosing a smooth representative with that jet is a Borel-extension choice and changes the parametrix only by a smooth biscalar flat at the diagonal, hence only by a finite local subtraction in point-split composite fields.

Proof. Write $s = \sigma(x,y)$ and compute away from the diagonal. For a smooth biscalar v,

$$P_{M,x}(v\,s^k\log s) = \big(P_{M,x}(v\,s^k)\big)\log s - s^{k-1}\big(2\sigma^{;\mu}\nabla_\mu v + (\Box_g\sigma + 4k-2)v\big). \tag{148.18}$$

This follows by applying the product rule and $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$, exactly as in the proof of Lemma 148.3. The zeroth-order part $m^2 + \xi R$ contributes to the logarithmic term $(P_{M,x}(v\,s^k))\log s$, but not to the displayed nonlogarithmic coefficient.

The σ^{-1} nonlogarithmic singularity in

$$P_{M,x}\left(\frac{U_H}{s} + v_0\log s\right)$$

is $P_{M,x}U_H$ minus the $k=0$ coefficient in (148.18); setting it to zero gives (148.15). For $j \ge 0$, inspect the coefficient of $s^j\log s$ in

$$P_{M,x}\left(\sum_{\ell\ge 0} v_\ell s^\ell\right).$$

The term $P_{M,x}(v_j s^j)$ contributes $s^j P_{M,x}v_j$, while the lower-derivative part of $P_{M,x}(v_{j+1}s^{j+1})$ contributes

$$-(j+1)s^j\big(2\sigma^{;\mu}\nabla_\mu v_{j+1} + (\Box_g\sigma + 2j)v_{j+1}\big).$$

Vanishing of the coefficient of $s^j\log s$ gives (148.16). At coincidence $\sigma^{;\mu} = 0$ and $\Box_g\sigma = 4$, so (148.16) gives (148.17). The transport operator has nonzero coincidence coefficient at each step, so the value at $x = y$ and then all transverse derivatives are fixed recursively by differentiating the transport equation. Borel's lemma supplies a smooth biscalar with the prescribed infinite jet, and two such choices differ by a smooth biscalar flat at the diagonal. □

Changing the time function in (148.2) changes only the boundary-value prescription for the same local positive frequency singularity. Changing μ changes the subtraction by the smooth local biscalar

$$\frac{1}{8\pi^2}V_H(x,y)\log(\mu'^2/\mu^2). \tag{148.19}$$

Thus the singular part is state-independent, while the scale and smooth local terms are renormalization coordinates.

148.2 Wick Square

Definition 148.5 (Wick square by point splitting)**.** Let ω be a Hadamard state of the real Klein–Gordon field. Let $H(x,y)$ denote the locally covariant Hadamard parametrix, containing the singular U/σ_ϵ and $V\log\sigma_\epsilon$ terms. The point-split Wick square in state ω is the distribution defined by

$$\omega\big(:\Phi^2:(f)\big) = \int_M f(x) \lim_{y\to x}\big(\omega_2(x,y) - H(x,y)\big)\, d\mathrm{vol}_g(x).$$

Remark 148.6 (State dependence and subtraction dependence)*.* Fix a normal convex neighborhood in which the Hadamard parametrix H is defined. If ω and ω' are Hadamard states of the same Klein–Gordon field, then

$$W_{\omega,\omega'}(x,y) = \omega_2(x,y) - \omega_2'(x,y)$$

is smooth near the diagonal, and the corresponding Wick-square expectation values obey

$$\omega\big(:\Phi^2:_H(f)\big) - \omega'\big(:\Phi^2:_H(f)\big) = \int_M f(x)\, W_{\omega,\omega'}(x,x)\, d\mathrm{vol}_g(x).$$

If $H' = H + S$, where $S(x,y)$ is a smooth symmetric locally constructed biscalar, then

$$:\Phi^2:_{H'}(f) =:\Phi^2:_H(f) - \int_M f(x)S(x,x)\, d\mathrm{vol}_g(x)\,\mathbf{1}$$

as a relation between expectation values in every Hadamard state.

The defining property of a Hadamard two-point function is that, in each normal convex neighborhood, its singular part is the universal local parametrix H, while the remainder is smooth. Thus

$$\omega_2 = H + R_\omega, \qquad \omega_2' = H + R_{\omega'}$$

with R_ω and $R_{\omega'}$ smooth near the diagonal. Their difference $W_{\omega,\omega'} = R_\omega - R_{\omega'}$ is smooth, so the diagonal value $W_{\omega,\omega'}(x,x)$ is an ordinary smooth function. Subtracting the two point-splitting definitions gives the first displayed formula. For the second formula,

$$\omega_2 - H' = \omega_2 - H - S.$$

Taking the diagonal limit and smearing with f subtracts $\int f(x)S(x,x)d\mathrm{vol}_g(x)$, independent of the state. A state-independent scalar multiple of every expectation value is represented by the same scalar multiple of the identity field.

For higher Wick powers, one subtracts the parametrix in every contraction. The result is an operator-valued distribution in perturbative algebraic QFT and a quadratic form in representations where the required domains are available.

The subtraction has finite local freedom. For the Wick square in four dimensions,

$$:\Phi^2: \mapsto\, :\Phi^2: + c_1 R + c_2 m^2$$

with constants c_1, c_2, subject to the chosen covariance and scaling conditions.

148.3 Stress-Tensor Prescription and Local Freedom

The stress tensor has a sharper renormalization problem than the Wick square because a choice of finite local terms must also respect conservation and background covariance.

Definition 148.7 (Wald-type stress-tensor renormalization conditions)**.** For the scalar theory with operator P_M in (148.1), an expectation-value stress-tensor renormalization is an assignment

$$(M, g; m^2, \xi, \omega) \longmapsto \omega(T^{\text{ren}}_{\mu\nu}) \in \mathcal{D}'(M, S^2T^*M)$$

for every globally hyperbolic spacetime (M, g) and every Hadamard state ω. In this chapter the renormalization conditions are the following.

1. *Local covariance.* If $\chi : (M, g) \to (N, h)$ is an orientation- and time-orientation-preserving isometric embedding with causally convex image, and if ω_N is a Hadamard state on N, then the expectation value in the pulled-back state $\chi^*\omega_N$ equals the pullback of the expectation value on N.
2. *State dependence only through the smooth remainder.* For any two Hadamard states ω, ω', the difference $\omega(T^{\text{ren}}_{\mu\nu}) - \omega'(T^{\text{ren}}_{\mu\nu})$ is obtained by applying a fixed local bidifferential operator to the smooth biscalar $\omega_2 - \omega'_2$ and then restricting to the diagonal.
3. *Conservation.* The distributional divergence vanishes,
$$\nabla^\mu \omega(T^{\text{ren}}_{\mu\nu}) = 0, \tag{148.20}$$
for all Hadamard states in the declared state class.
4. *Metric dimension and almost homogeneous scaling.* In local formulae we assign engineering dimensions
$$[\nabla] = 1, \qquad [m] = 1, \qquad [R_{\mu\nu\rho\sigma}] = 2, \qquad [g_{\mu\nu}] = 0. \tag{148.21}$$
The state-independent local terms in $T^{\text{ren}}_{\mu\nu}$ have total dimension four. A change of the Hadamard length scale is allowed to produce only local logarithmic terms of the same dimension; this is the curved-space form of the trace-anomaly ambiguity.
5. *Flat normalization.* On Minkowski spacetime the massless vacuum expectation value is zero after the finite cosmological term is set to zero.

These are renormalization conditions on expectation values. A stronger operator-valued statement additionally requires a common domain or an algebraic definition of the composite field; that extra domain problem is not hidden in the word "stress tensor".

Definition 148.8 (Point-split stress-tensor prescription)**.** A point-split realization of Definition 148.7 for the scalar operator (148.1) assigns to every globally hyperbolic background (M, g), every Hadamard state ω, and every test tensor $f^{\mu\nu} \in C^\infty_c(M, S^2TM)$ a number $\omega(T(f))$. The prescription is required to satisfy the following conditions.

1. It is obtained locally from
$$\lim_{y \to x} \mathcal{D}_{\mu\nu}(x, y)\big(\omega_2(x, y) - H_{\epsilon,\mu}(x, y)\big) + C_{\mu\nu}[g](x), \tag{148.22}$$
where $\mathcal{D}_{\mu\nu}$ is the bidifferential operator obtained by polarizing the classical Hilbert stress tensor and $C_{\mu\nu}[g]$ is a state-independent local curvature tensor.

2. It is locally covariant: if $\chi : (M, g) \to (N, h)$ is an admissible isometric embedding in the locally covariant category and ω_N is a Hadamard state on N, then the pullback state on M has stress-tensor expectation equal to the pullback of the N-expectation.
3. Its dependence on the state is exactly through the smooth biscalar $\omega_2 - H_{\epsilon,\mu}$. Equivalently, for two Hadamard states ω, ω', the difference $\omega(T_{\mu\nu}) - \omega'(T_{\mu\nu})$ is obtained by applying $\mathcal{D}_{\mu\nu}$ to $\omega_2 - \omega_2'$ and then taking the diagonal limit.
4. It is conserved after the finite local tensor has been fixed:
$$\nabla^\mu \omega(T_{\mu\nu}) = 0 \tag{148.23}$$
for all Hadamard states in the declared state space.
5. It has engineering dimension four under constant rescaling of the metric and fields in the sense of (148.21), and it is normalized so that the Minkowski vacuum stress tensor vanishes when $m = 0$ and the finite cosmological term is set to zero.

Proposition 148.9 (Finite local freedom of the scalar stress tensor)**.** *Let $T_{\mu\nu}$ and $T'_{\mu\nu}$ be two prescriptions satisfying Definition 148.8. In four dimensions their difference is state-independent and has the form*
$$T'_{\mu\nu} - T_{\mu\nu} = a_0 m^4 g_{\mu\nu} + a_1 m^2 G_{\mu\nu} + a_2 I_{\mu\nu} + a_3 J_{\mu\nu}, \tag{148.24}$$
with real constants a_i. Here
$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}, \tag{148.25}$$
and
$$I_{\mu\nu} = \frac{1}{\sqrt{|g|}} \frac{\delta}{\delta g^{\mu\nu}} \int \sqrt{|g|}\, R^2 \,\mathrm{d}^4x, \qquad J_{\mu\nu} = \frac{1}{\sqrt{|g|}} \frac{\delta}{\delta g^{\mu\nu}} \int \sqrt{|g|}\, R_{\alpha\beta} R^{\alpha\beta} \,\mathrm{d}^4x, \tag{148.26}$$
with compactly supported metric variations and boundary terms discarded.

Proof. Set $D_{\mu\nu} := T'_{\mu\nu} - T_{\mu\nu}$. The state-dependence clause in Definition 148.8 implies that for any two Hadamard states ω, ω',
$$\omega(D_{\mu\nu}) - \omega'(D_{\mu\nu}) = 0.$$
Thus $D_{\mu\nu}$ is a state-independent local covariant tensor constructed from $g_{\mu\nu}$, the constant mass parameter m, and curvature. The conservation clause gives $\nabla^\mu D_{\mu\nu} = 0$. The engineering dimension clause says that every monomial in $D_{\mu\nu}$ has total dimension four, counting m as dimension one, $g_{\mu\nu}$ as dimension zero, and each curvature tensor as dimension two.

The dimension-four local covariant candidates split by powers of m. The only m^4 tensor with two covariant indices and no derivatives is $m^4 g_{\mu\nu}$, and it is conserved because m is constant and $\nabla g = 0$. At order m^2 the possible curvature tensors are $m^2 R_{\mu\nu}$ and $m^2 R g_{\mu\nu}$. Their divergence is
$$\nabla^\mu (\alpha R_{\mu\nu} + \beta R g_{\mu\nu}) = \left(\frac{\alpha}{2} + \beta\right) \nabla_\nu R,$$
using the contracted Bianchi identity. Conservation therefore forces $\beta = -\alpha/2$, giving the Einstein tensor $G_{\mu\nu}$.

It remains to classify the curvature-squared part. In four dimensions the scalar local curvature densities of dimension four are spanned, modulo total derivatives, by

$$R^2, \qquad R_{\alpha\beta}R^{\alpha\beta}, \qquad R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}.$$

Their metric Euler derivatives are symmetric conserved tensors because the corresponding integrated counterterms are diffeomorphism invariant. The dimensionless parameter ξ in the scalar stress tensor can change the constants multiplying these local tensors but cannot create a new dimension-four covariant tensor. The four-dimensional Gauss–Bonnet combination

$$E_4 = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} - 4R_{\alpha\beta}R^{\alpha\beta} + R^2$$

has vanishing metric variation on compactly supported variations in the interior, since its integral is topological. Hence the Euler derivative of $R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ is a linear combination of the Euler derivatives of R^2 and $R_{\alpha\beta}R^{\alpha\beta}$. These two remaining Euler derivatives are precisely $I_{\mu\nu}$ and $J_{\mu\nu}$ in (148.26).

No further independent local covariant conserved tensor of this dimension is available inside the stated finite local freedom: terms with explicit derivatives, such as $\nabla_\mu\nabla_\nu R$, $g_{\mu\nu}\Box R$, and $\Box R_{\mu\nu}$, occur exactly in the Euler derivatives just described; total derivative scalar counterterms give boundary contributions under compactly supported metric variations; and tensors with more curvatures or more derivatives have dimension larger than four. Therefore $D_{\mu\nu}$ has the displayed four-parameter form. $\Box$

Proposition 148.10 (Curvature-squared Euler tensors)**.** *With the convention of* (148.26)*, the two independent curvature-squared Euler tensors are*

$$\begin{aligned} I_{\mu\nu} &= 2RR_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R^2 + 2g_{\mu\nu}\Box_g R - 2\nabla_\mu\nabla_\nu R, \\ J_{\mu\nu} &= 2R_{\mu\alpha\nu\beta}R^{\alpha\beta} - \frac{1}{2}g_{\mu\nu}R_{\alpha\beta}R^{\alpha\beta} + \nabla_\mu\nabla_\nu R - \Box_g R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\Box_g R. \end{aligned} \tag{148.27}$$

They are conserved,

$$\nabla^\mu I_{\mu\nu} = 0, \qquad \nabla^\mu J_{\mu\nu} = 0, \tag{148.28}$$

and in four dimensions their traces are

$$g^{\mu\nu}I_{\mu\nu} = 6\Box_g R, \qquad g^{\mu\nu}J_{\mu\nu} = -2\Box_g R. \tag{148.29}$$

Proof. For the R^2 functional, use

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}\delta g^{\mu\nu}, \qquad \delta R = R_{\mu\nu}\delta g^{\mu\nu} + g_{\mu\nu}\Box_g\delta g^{\mu\nu} - \nabla_\mu\nabla_\nu\delta g^{\mu\nu}. \tag{148.30}$$

After two integrations by parts with compactly supported metric variation,

$$\delta\int\sqrt{|g|}R^2 = \int\sqrt{|g|}\left(2RR_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R^2 + 2g_{\mu\nu}\Box_g R - 2\nabla_\mu\nabla_\nu R\right)\delta g^{\mu\nu},$$

which is the first line of (148.27). For $\int\sqrt{|g|}R_{\alpha\beta}R^{\alpha\beta}$, the variation starts from

$$\delta\Big(\sqrt{|g|}R_{\alpha\beta}R^{\alpha\beta}\Big) = \sqrt{|g|}\left(-\frac{1}{2}g_{\mu\nu}R_{\alpha\beta}R^{\alpha\beta} + 2R_{\mu\alpha}R_\nu{}^\alpha\right)\delta g^{\mu\nu} + 2\sqrt{|g|}R^{\mu\nu}\delta R_{\mu\nu},$$

using

$$\delta R_{\mu\nu} = \frac{1}{2} g_{\mu\alpha} g_{\nu\beta} \Box_g \delta g^{\alpha\beta} + \frac{1}{2} g_{\alpha\beta} \nabla_\mu \nabla_\nu \delta g^{\alpha\beta} - \frac{1}{2} g_{\mu\alpha} \nabla_\beta \nabla_\nu \delta g^{\alpha\beta} - \frac{1}{2} g_{\nu\alpha} \nabla_\beta \nabla_\mu \delta g^{\alpha\beta}, \tag{148.31}$$

commuting covariant derivatives where needed, gives the second line. These are ordinary variational identities for compactly supported variations; no field equation is used.

Conservation follows from diffeomorphism invariance of the two functionals. For a compactly supported infinitesimal diffeomorphism generated by ζ^μ,

$$\delta_\zeta g^{\mu\nu} = -\nabla^\mu \zeta^\nu - \nabla^\nu \zeta^\mu.$$

If $E_{\mu\nu}$ denotes either variational derivative, then

$$0 = \delta_\zeta \mathcal{F} = -2 \int \sqrt{|g|}\, E_{\mu\nu} \nabla^\mu \zeta^\nu = 2 \int \sqrt{|g|}\, (\nabla^\mu E_{\mu\nu}) \zeta^\nu,$$

so $\nabla^\mu E_{\mu\nu} = 0$ distributionally, hence smoothly. Taking the trace of the two displayed formulas and using $g^{\mu\nu} \Box_g R_{\mu\nu} = \Box_g R$ gives

$$g^{\mu\nu} I_{\mu\nu} = (2 - \tfrac{D}{2}) R^2 + (2D - 2) \Box_g R, \qquad g^{\mu\nu} J_{\mu\nu} = (2 - \tfrac{D}{2}) R_{\alpha\beta} R^{\alpha\beta} - \frac{D}{2} \Box_g R.$$

Setting $D = 4$ gives (148.29). $\square$

148.4 Stress Tensor by Bidifferential Operator

For the classical scalar action

$$S[\phi] = -\frac{1}{2} \int_M \big(\nabla_\mu \phi \nabla^\mu \phi + m^2 \phi^2 + \xi R \phi^2 \big)\, d\mathrm{vol}_g$$

in mostly-plus Lorentzian convention, the stress tensor is obtained by metric variation. Quantum mechanically, define a bidifferential operator $\mathcal{D}_{\mu\nu}(x, y)$ by polarizing the classical quadratic expression in two field arguments. Then

$$\omega(T_{\mu\nu}(x)) = \lim_{y \to x} \mathcal{D}_{\mu\nu}(x, y) \big(\omega_2(x, y) - H(x, y) \big) + C_{\mu\nu}(x).$$

The local tensor $C_{\mu\nu}$ is the finite renormalization freedom classified in Proposition 148.9.

148.5 Worked Flat-Space Computation

We now compute a simple case explicitly. Let (M, η) be four-dimensional Minkowski spacetime with $\eta = \mathrm{diag}(-1, 1, 1, 1)$, and take the massless minimally coupled free scalar. The vacuum two-point distribution is

$$W_0(x, x') = \lim_{\epsilon \downarrow 0} \frac{1}{4\pi^2} \frac{1}{|\mathbf{x} - \mathbf{x}'|^2 - (t - t' - \mathrm{i}\epsilon)^2}.$$

In flat space the local Hadamard parametrix with the same $i\epsilon$ prescription is $H_0 = W_0$. Therefore the renormalized vacuum stress tensor obtained by point splitting is zero:

$$\langle T_{\mu\nu}\rangle_{0,\mathrm{ren}} = \lim_{x'\to x} \mathcal{D}_{\mu\nu}(x,x')\big(W_0(x,x') - H_0(x,x')\big) = 0,$$

if the finite flat-space cosmological-constant counterterm is set to zero. This is the explicit subtraction of the state-independent short-distance singularity.

For the thermal KMS state at inverse temperature β with respect to the inertial time t, the difference from the vacuum two-point function is smooth:

$$\Delta W_\beta(x,x') = W_\beta(x,x') - W_0(x,x')$$

$$= \int \frac{d^3k}{(2\pi)^3}\,\frac{n_\beta(\omega)}{2\omega}\left(e^{-\mathrm{i}\omega(t-t')+\mathrm{i}\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} + e^{+\mathrm{i}\omega(t-t')-\mathrm{i}\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}\right), \qquad \omega = |\mathbf{k}|,$$

where $n_\beta(\omega) = (e^{\beta\omega}-1)^{-1}$. Since ΔW_β is smooth, the point-split stress tensor is obtained by applying the polarized classical expression directly to this remainder. For the massless minimal tensor

$$T_{00} = \frac{1}{2}\big((\partial_t\phi)^2 + |\nabla\phi|^2\big),$$

the bidifferential operator is

$$\mathcal{D}_{00} = \frac{1}{2}\left(\partial_t\partial_{t'} + \sum_{i=1}^{3}\partial_i\partial_{i'}\right).$$

On either plane wave in ΔW_β,

$$\mathcal{D}_{00}e^{-\mathrm{i}\omega(t-t')+\mathrm{i}\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} = \omega^2 e^{-\mathrm{i}\omega(t-t')+\mathrm{i}\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}.$$

Taking the diagonal limit gives

$$\rho_\beta = \langle T_{00}\rangle_{\beta,\mathrm{ren}} = \int \frac{d^3k}{(2\pi)^3}\,\omega\, n_\beta(\omega) = \frac{1}{2\pi^2}\int_0^\infty \frac{k^3\,dk}{e^{\beta k}-1}.$$

The change of variables $x = \beta k$ and the expansion $(e^x-1)^{-1} = \sum_{n\ge1} e^{-nx}$ give

$$\int_0^\infty \frac{x^3\,dx}{e^x-1} = \sum_{n\ge1}\int_0^\infty x^3 e^{-nx}\,dx = 6\sum_{n\ge1}\frac{1}{n^4} = \frac{\pi^4}{15}.$$

Thus

$$\rho_\beta = \frac{\pi^2}{30\beta^4}.$$

For the spatial components,

$$T_{ij} = \partial_i\phi\,\partial_j\phi + \frac{1}{2}\delta_{ij}\big((\partial_t\phi)^2 - |\nabla\phi|^2\big),$$

so

$$\mathcal{D}_{ij} = \partial_i\partial_{j'} + \frac{1}{2}\delta_{ij}\left(\partial_t\partial_{t'} - \sum_{\ell=1}^{3}\partial_\ell\partial_{\ell'}\right).$$

For massless modes the second term vanishes on shell because $\omega^2 = |\mathbf{k}|^2$. Rotational invariance of the momentum integral gives

$$\int d\Omega\, k_i k_j = \frac{4\pi}{3}\delta_{ij} k^2,$$

and therefore

$$\langle T_{ij}\rangle_{\beta,\text{ren}} = \frac{1}{3}\rho_\beta\, \delta_{ij}.$$

The point-split computation thus yields

$$\langle T_{\mu\nu}\rangle_{\beta,\text{ren}} = \frac{\pi^2}{90\beta^4}\begin{pmatrix} 3 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}_{\mu\nu},$$

in the inertial frame. Its trace is $-\rho_\beta + 3p_\beta = 0$, as required for the massless conformal equation of state in flat spacetime.

148.6 Constant-Curvature Diagnostic

A de Sitter-invariant Hadamard state of a free scalar has a stress-tensor expectation value constrained by symmetry to the form

$$\langle T_{\mu\nu}\rangle_{\text{dS}} = C\, g_{\mu\nu}.$$

For the massless conformally coupled real scalar in four dimensions, the state-dependent classical trace vanishes. With the trace-anomaly convention of Chapter 149, the real scalar has $a = 1/360$, $c = 1/120$, and

$$\langle T^\mu{}_\mu\rangle = \frac{1}{16\pi^2}\left(cW_{\mu\nu\rho\sigma}W^{\mu\nu\rho\sigma} - aE_4 + b\Box R\right).$$

On four-dimensional de Sitter with Hubble parameter H,

$$W_{\mu\nu\rho\sigma} = 0, \qquad \Box R = 0, \qquad R = 12H^2,$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 24H^4, \quad R_{\mu\nu}R^{\mu\nu} = 36H^4, \quad E_4 = 24H^4.$$

Consequently

$$4C = \langle T^\mu{}_\mu\rangle = -\frac{H^4}{240\pi^2}, \qquad C = -\frac{H^4}{960\pi^2}.$$

This calculation isolates the part fixed by local covariance, de Sitter invariance, and the anomaly normalization. Massive fields, and schemes that allow additional dimension-four metric counterterms, shift C by finite local terms; the point-splitting prescription must state that choice.

148.7 Conservation and Trace

Definition 148.11 (Conserved stress tensor)**.** Within a locally covariant point-splitting scheme, a renormalized stress tensor is called conserved when its finite local curvature terms have been chosen so that

$$\nabla^\mu \omega(T_{\mu\nu}) = 0$$

for all Hadamard states in the declared class.

The trace has the form

$$\omega(T^{\mu}{}_{\mu}) = m^2\,\omega(:\Phi^2:) + (\xi - \xi_c)\,\omega(\Box :\Phi^2:) + \mathcal{A}[g, m, \xi],$$

where $\xi_c = (D-2)/(4D-4)$ and $\mathcal{A}$ is a local curvature scalar determined up to the same finite renormalization freedom. In the massless conformally coupled case, the state-dependent classical trace terms vanish and the remaining local term is the trace anomaly.

148.8 Relation to Flat-Space Composite Operators

Point splitting is an operator-definition scheme. It differs from defining a regularized composite operator by inserting a classical functional into a regulated path integral and then renormalizing. The two schemes can be compared by finite local counterterms when both are defined on the same background and state class. This comparison is the curved-background version of the operator-coordinate changes developed in the renormalization volume.

Open Problem 148.12 (Interacting stress tensors beyond formal perturbation). Construct stress tensors for interacting curved-spacetime QFTs with nonperturbative control of domains, local covariance, conservation, trace anomalies, and state dependence. A solution should identify the exact relation between locally covariant composite fields and flat-space renormalized operator schemes.

Chapter 149

Trace Anomalies and Background Variations

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The preceding chapter defined renormalized stress tensors on curved backgrounds by locally covariant point splitting. The present chapter studies the trace of that stress tensor. The trace anomaly is a local response of the renormalized source functional to a Weyl variation of the background metric. It is part of the same curved-background QFT datum as the stress tensor, its contact terms, and its finite curvature counterterms.

149.1 Metric Source Convention

Let M be a closed smooth D-manifold and let g be a Riemannian metric. Write

$$\mathcal{W}[g, \lambda]$$

for the renormalized Euclidean effective action, including all declared background sources λ besides the metric. In a path-integral regularization this is $-\log Z[g, \lambda]$, after local counterterms have been fixed. The stress-tensor expectation value is defined by

$$\delta_g \mathcal{W}[g, \lambda] = -\frac{1}{2} \int_M d^D x \sqrt{g}\, \langle T^{\mu\nu}(x) \rangle_{g,\lambda}\, \delta g_{\mu\nu}(x). \tag{149.1}$$

Equivalently,

$$\langle T^{\mu\nu}(x)\rangle_{g,\lambda} = -\frac{2}{\sqrt{g}}\frac{\delta \mathcal{W}}{\delta g_{\mu\nu}(x)}.$$

For a local Weyl parameter $\sigma \in C^\infty(M)$,

$$\delta_\sigma g_{\mu\nu} = 2\sigma g_{\mu\nu}.$$

Substituting this variation into (149.1) gives

$$\delta_\sigma \mathcal{W}[g,\lambda] = -\int_M d^D x\,\sqrt{g}\,\sigma(x)\,\langle T^\mu{}_\mu(x)\rangle_{g,\lambda}. \tag{149.2}$$

Definition 149.1 (Trace-anomaly density)**.** For a theory whose classical curved-background action is Weyl invariant after the assigned transformations of all sources λ, the renormalized trace-anomaly density $\mathcal{A}[g,\lambda]$ is the local scalar density coefficient defined by

$$\langle T^\mu{}_\mu(x)\rangle_{g,\lambda} = \mathcal{A}[g,\lambda](x)$$

in the vacuum or state-independent part of the Weyl response. Thus

$$\delta_\sigma \mathcal{W}[g,\lambda] = -\int_M d^D x\,\sqrt{g}\,\sigma\,\mathcal{A}[g,\lambda]. \tag{149.3}$$

If the theory has mass parameters, relevant couplings, or fields assigned nontrivial Weyl transformations, the trace Ward identity contains additional operator terms such as $\beta^i\mathcal{O}_i$, masses times composite operators, and improvement terms. This chapter isolates the purely local curvature part which remains in a conformal theory or in the state-independent part of a locally covariant stress tensor.

149.2 Wess–Zumino Consistency

Let

$$\mathbf{A}_\sigma[g,\lambda] := \int_M d^D x\,\sqrt{g}\,\sigma\,\mathcal{A}[g,\lambda].$$

The Weyl transformations form an abelian Lie algebra:

$$[\delta_{\sigma_1},\delta_{\sigma_2}]g_{\mu\nu} = 0$$

when the source transformations are also abelian. Since $\mathcal{W}[g,\lambda]$ is a functional, the commutator of its two Weyl variations must vanish.

Remark 149.2 (Wess–Zumino consistency)*.* Assume the Weyl variation of $\mathcal{W}$ is local and has the form (149.3). Then

$$\delta_{\sigma_1}\mathbf{A}_{\sigma_2} - \delta_{\sigma_2}\mathbf{A}_{\sigma_1} = 0. \tag{149.4}$$

If a finite local counterterm $B[g,\lambda]$ is added to $\mathcal{W}$, then

$$\mathbf{A}_\sigma \longmapsto \mathbf{A}_\sigma - \delta_\sigma B. \tag{149.5}$$

Therefore the intrinsic trace anomaly is a class in local Weyl cohomology: local solutions of (149.4) modulo local Weyl variations of counterterms.

The Weyl transformations are abelian on the metric-source space under the present hypotheses:

$$[\delta_{\sigma_1}, \delta_{\sigma_2}] = 0.$$

Using $\delta_\sigma \mathcal{W} = -\mathbf{A}_\sigma$,

$$0 = [\delta_{\sigma_1}, \delta_{\sigma_2}]\mathcal{W} = -\delta_{\sigma_1}\mathbf{A}_{\sigma_2} + \delta_{\sigma_2}\mathbf{A}_{\sigma_1}.$$

This is (149.4). If $\mathcal{W}' = \mathcal{W} + B$, then

$$-\mathbf{A}'_\sigma = \delta_\sigma \mathcal{W}' = -\mathbf{A}_\sigma + \delta_\sigma B,$$

which gives (149.5).

The consistency condition is a local cohomological statement only after locality has been supplied. In perturbation theory, locality follows from the locality of counterterms. For free fields on curved backgrounds, it follows from the local Hadamard parametrix or, equivalently, from local heat-kernel coefficients. In a nonperturbative locally covariant QFT, locality is an additional structural property of the background dependence.

149.3 Curvature Variations under Weyl Rescaling

The classification in low dimensions uses a few explicit Weyl variations.

Weyl variation of the R^2 counterterm. On a D-dimensional Riemannian manifold,

$$\delta_\sigma \sqrt{g} = D\sigma\sqrt{g}, \tag{149.6}$$

$$\delta_\sigma R = -2\sigma R - 2(D-1)\nabla^2\sigma, \tag{149.7}$$

where $\nabla^2 = \nabla_\mu \nabla^\mu$. In $D = 4$, these imply

$$\delta_\sigma \int_M d^4x\, \sqrt{g}\, R^2 = -12 \int_M d^4x\, \sqrt{g}\, \sigma\, \nabla^2 R \tag{149.8}$$

on a closed manifold. The variation of the volume form follows from $\delta\sqrt{g} = \frac{1}{2}\sqrt{g}\, g^{\mu\nu}\delta g_{\mu\nu}$. The standard variation of the scalar curvature under a metric variation $h_{\mu\nu}$ is

$$\delta R = -R^{\mu\nu}h_{\mu\nu} + \nabla^\mu\nabla^\nu h_{\mu\nu} - \nabla^2(g^{\mu\nu}h_{\mu\nu}).$$

Putting $h_{\mu\nu} = 2\sigma g_{\mu\nu}$ gives (149.7). In $D = 4$,

$$\delta_\sigma(\sqrt{g}\, R^2) = 4\sigma\sqrt{g}\, R^2 + 2\sqrt{g}\, R(-2\sigma R - 6\nabla^2\sigma) = -12\sqrt{g}\, R\nabla^2\sigma.$$

Integrating by parts twice gives (149.8).

Equation (149.8) is the concrete reason why the coefficient of $\nabla^2 R$ in four dimensions is a scheme coordinate. Adding $\kappa(16\pi^2)^{-1}\int \sqrt{g}\, R^2$ shifts the coefficient b in the convention below by 12κ.

149.4 Two-Dimensional Anomaly

In two dimensions, the local parity-even scalar of Weyl weight two is the Ricci scalar. The trace anomaly of a unitary two-dimensional conformal field theory is written

$$\langle T^\mu{}_\mu \rangle_g = -\frac{c}{24\pi} R. \tag{149.9}$$

The coefficient c is the Virasoro central charge when the stress tensor is normalized by

$$\langle T(z)T(0) \rangle = \frac{c}{2z^4}$$

on the flat complex plane.

Let $\hat{g}_{\mu\nu} = e^{2\tau} g_{\mu\nu}$. The functional

$$S^{(2)}_{\mathrm{WZ}}[\tau; g] = -\frac{c}{24\pi} \int_M d^2x \sqrt{g} \left(\tau R + (\nabla\tau)^2 \right)$$

satisfies

$$S^{(2)}_{\mathrm{WZ}}[\tau + \sigma; g] - S^{(2)}_{\mathrm{WZ}}[\tau; g] = -\frac{c}{24\pi} \int_M d^2x \sqrt{\hat{g}}\, \sigma\, R[\hat{g}] + O(\sigma^2)$$

for infinitesimal σ. Hence it integrates the anomaly (149.9). In two dimensions,

$$\sqrt{\hat{g}}\, R[\hat{g}] = \sqrt{g}\, (R - 2\nabla^2\tau).$$

Varying $S^{(2)}_{\mathrm{WZ}}$ with respect to τ gives

$$\delta_\sigma S^{(2)}_{\mathrm{WZ}} = -\frac{c}{24\pi} \int_M \sqrt{g}\, \sigma (R - 2\nabla^2\tau) = -\frac{c}{24\pi} \int_M \sqrt{\hat{g}}\, \sigma R[\hat{g}],$$

after integrating by parts in the $(\nabla\tau)^2$ term.

149.5 Four-Dimensional Parity-Even Anomaly

In four dimensions define

$$E_4 = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 4R_{\mu\nu} R^{\mu\nu} + R^2$$

and

$$W^2 = W_{\mu\nu\rho\sigma} W^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 2R_{\mu\nu} R^{\mu\nu} + \frac{1}{3} R^2.$$

For a parity-even four-dimensional conformal theory on a closed manifold, the metric trace anomaly has the form

$$\langle T^\mu{}_\mu \rangle_g = \frac{1}{16\pi^2} \left(c\, W^2 - a\, E_4 + b\, \nabla^2 R \right) \tag{149.10}$$

plus terms involving other background fields when such sources are present.

Remark 149.3 (Four-dimensional cohomology representatives). Modulo local Weyl variations of dimension-four metric counterterms and total derivatives, the parity-even metric anomaly on a closed four-manifold has the two nontrivial representatives E_4 and W^2. The term $\nabla^2 R$ is shifted by the local counterterm $\int \sqrt{g}\, R^2$.

The dimension-four parity-even scalar curvature basis is spanned by

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \qquad R_{\mu\nu}R^{\mu\nu}, \qquad R^2, \qquad \nabla^2 R.$$

The two combinations E_4 and W^2 displayed above are linearly independent. The integral of E_4 is topological on a closed four-manifold, so its Weyl variation is a total derivative and satisfies Wess–Zumino consistency. The density $\sqrt{g}\,W^2$ is pointwise Weyl invariant in four dimensions, so it also satisfies consistency. The $\nabla^2 R$ term is the Weyl variation of the R^2 counterterm by (149.8). These three directions span the basis modulo a choice of representative, leaving E_4 and W^2 as the nontrivial local cohomology classes in this sector.

If background gauge fields are present, further type-B terms appear. For a dimensionless background gauge field A, the density $\mathrm{tr}_{\mathcal{R}}\, F_{\mu\nu}F^{\mu\nu}$ has Weyl weight four, so its coefficient is constrained by the flavor current two-point normalization and by the choice of current-source convention. Parity-odd terms and manifolds with boundary require a separate orientation and boundary classification.

149.6 Heat-Kernel Derivation for Free Conformal Fields

The following computation gives a controlled derivation of the free-field entries in (149.10). Let P be a positive Laplace-type operator on a vector bundle $E \to M$, written

$$P = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + \mathcal{E}).$$

Let $\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu]$. On a closed four-manifold, the local heat-kernel expansion is

$$\mathrm{Tr}\, e^{-sP} \sim \frac{1}{(4\pi s)^2}\sum_{n\geq 0} s^n \int_M d^4x\,\sqrt{g}\,\mathrm{tr}_E\, a_{2n}(P).$$

The coefficient needed for the logarithmic Weyl response is

$$\begin{aligned}\mathrm{tr}_E\, a_4(P) = \mathrm{tr}_E \Bigg[&\frac{1}{360}\left(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right)\mathbf{1} \\ &+\frac{1}{6}R\mathcal{E} + \frac{1}{2}\mathcal{E}^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu}\Bigg] + \nabla_\mu J^\mu. \end{aligned} \tag{149.11}$$

The divergent logarithmic term in $\frac{1}{2}\log\det P$ is proportional to $\int \mathrm{tr}\, a_4(P)$. For Grassmann fields and gauge fields one includes the corresponding minus signs, square-root factors, and ghost determinants.

For a real conformally coupled scalar in four dimensions,

$$P_\phi = -\nabla^2 + \frac{1}{6}R = -(\nabla^2 + \mathcal{E}), \qquad \mathcal{E} = -\frac{1}{6}R,$$

and $\Omega_{\mu\nu} = 0$. The non-derivative part of $\mathrm{tr}\, a_4(P_\phi)$ is

$$\frac{1}{180}\left(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu}\right).$$

Equivalently,

$$a_{\text{scalar}} = \frac{1}{360}, \qquad c_{\text{scalar}} = \frac{1}{120}.$$

Substituting $\mathcal{E} = -R/6$ and $\Omega_{\mu\nu} = 0$ into (149.11) gives the R^2 coefficient

$$\frac{5}{360} - \frac{1}{36} + \frac{1}{72} = 0.$$

The remaining terms are

$$\frac{2}{360} R^2_{\mu\nu\rho\sigma} - \frac{2}{360} R^2_{\mu\nu} = \frac{1}{180}(R^2_{\mu\nu\rho\sigma} - R^2_{\mu\nu}).$$

Using

$$cW^2 - aE_4 = (c-a)R^2_{\mu\nu\rho\sigma} + (-2c+4a)R^2_{\mu\nu} + \left(\frac{c}{3} - a\right) R^2$$

with $a = 1/360$, $c = 1/120$, gives the same curvature combination.

For the remaining free conformal fields, the same heat-kernel formula is applied to the squared Dirac operator and to the gauge-fixed Maxwell complex. The fermion sign and the vector ghost subtraction are part of the determinant calculation:

$$\mathcal{W}_{\text{Dirac}} = -\frac{1}{2} \log\det P_{\mathcal{D}}, \qquad \mathcal{W}_{\text{Maxwell}} = \frac{1}{2} \log\det P_1 - \log\det P_0.$$

Here

$$P_{\mathcal{D}} = \mathcal{D}^2 = -\nabla^2 + \frac{1}{4} R,$$

while P_1 is the Hodge Laplacian on one-forms and P_0 the scalar ghost Laplacian. Gamma-matrix traces in the spin bundle and the one-form/ghost bundle traces reduce the logarithmic terms to the following table:

free conformal field	a	c
real scalar	$\frac{1}{360}$	$\frac{1}{120}$
Weyl fermion	$\frac{11}{720}$	$\frac{1}{40}$
Dirac fermion	$\frac{11}{360}$	$\frac{1}{20}$
Maxwell vector	$\frac{31}{180}$	$\frac{1}{10}$

The Weyl fermion entries are half the Dirac entries. The vector entries include the Faddeev-Popov ghost determinant and refer to the gauge-invariant stress tensor rather than the unconstrained one-form determinant alone.

149.7 Example: N=4 Yang-Mills

Consider free four-dimensional $\mathcal{N} = 4$ Yang-Mills fields in the adjoint representation of a compact gauge group G, with

$$\dim G = d_G.$$

Per generator of G, the field content is one Maxwell vector, four Weyl fermions, and six real conformal scalars. Therefore

$$a_{\mathcal{N}=4}/d_G = \frac{31}{180} + 4\frac{11}{720} + 6\frac{1}{360} = \frac{1}{4},$$

and

$$c_{\mathcal{N}=4}/d_G = \frac{1}{10} + 4\frac{1}{40} + 6\frac{1}{120} = \frac{1}{4}.$$

Thus

$$a_{\mathcal{N}=4} = c_{\mathcal{N}=4} = \frac{1}{4}\dim G$$

in the free normalization. Supersymmetry and current multiplet arguments protect this equality in the interacting conformal theory once the theory and stress-tensor multiplet have been constructed with a compatible renormalization scheme.

149.8 Contact Terms and Local Covariance

The anomaly is a statement about coincident stress-tensor insertions as well as one-point functions. Functional differentiation of (149.3) gives

$$g_{\mu\nu}(x)\,\langle T^{\mu\nu}(x)T^{\rho\sigma}(y)\rangle_{\text{conn}} = \frac{2}{\sqrt{g}(y)}\frac{\delta}{\delta g_{\rho\sigma}(y)}\mathcal{A}[g](x) + \text{index-variation contact terms.}$$

At separated points in a flat conformal theory the trace vanishes. The curvature anomaly records the contact distribution obtained by varying through curved backgrounds and then returning to flat space. This is why the same coefficient c controls both the W^2 anomaly and the normalization of the separated stress-tensor two-point function, whereas a first appears in genuinely three-stress-tensor contact data.

In the locally covariant language, the assignment

$$(M, g, \lambda) \longmapsto \mathcal{W}[g, \lambda]$$

should be replaced by a functorial algebra of observables, a class of Hadamard states, and locally covariant composite fields. The anomaly is then the obstruction to choosing the stress tensor so that it is simultaneously conserved, locally covariant, and Weyl covariant with the declared finite renormalization freedom.

149.9 Status of the Curved-Background Theorem

For free scalar, spinor, and gauge fields, the heat-kernel and point-splitting constructions give theorem-level anomaly formulas once the operator domain, gauge fixing, ghost determinant, and finite local counterterms are fixed. In perturbative algebraic QFT, the anomaly appears in the local covariance and scaling behavior of time-ordered products. For a general nonperturbative interacting QFT, the following ingredients are required:

(i) a locally covariant curved-background QFT object replacing $\mathcal{W}[g, \lambda]$;

(ii) stress-tensor insertions as locally covariant composite fields;

(iii) a contact-term prescription for multiple metric variations;

(iv) a proof that Weyl variation is local modulo allowed counterterms;

(v) compatibility with gauge and gravitational anomaly lines for other background fields;

(vi) a rule identifying which finite curvature counterterms are part of the chosen renormalization scheme.

Open Problem 149.4 (Trace anomaly from nonperturbative QFT data). Formulate trace anomalies directly from nonperturbative local QFT data on curved backgrounds. The construction should identify the algebraic, functorial, or analytic object replacing $\mathcal{W}[g, \lambda]$, define stress-tensor contact terms, prove locality of Weyl variation, and recover the local Weyl cohomology class without importing a perturbative regulator as part of the definition.

Chapter 150

The Unruh Effect and Wedge Modular Theory

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The Unruh effect is the thermal character of the Minkowski vacuum when observables are restricted to a Rindler wedge and time evolution is generated by Lorentz boosts. The precise local-QFT theorem is the Bisognano–Wichmann theorem; the accelerated-detector calculation is a concrete consequence in free-field examples.

150.1 Rindler Wedges and Boost Flow

In $\mathbb{M}^D$, define the right Rindler wedge

$$W_R = \{x \in \mathbb{M}^D : x^1 > |x^0|\}.$$

The boost preserving W_R is

$$x^0(\eta) = x^0 \cosh\eta + x^1 \sinh\eta, \qquad x^1(\eta) = x^1 \cosh\eta + x^0 \sinh\eta.$$

Let K be the self-adjoint generator of this boost representation,

$$U(\eta) = \exp(\mathrm{i}\eta K).$$

For a uniformly accelerated observer with proper acceleration a, the Rindler time parameter is $\eta = a\tau$, where τ is proper time.

150.2 KMS Statement

Let $\mathcal{A}(W_R)$ be the von Neumann algebra generated by observables localized in W_R. The vacuum state restricted to $\mathcal{A}(W_R)$ is KMS at inverse temperature 2π with respect to boost time if, for $A, B \in \mathcal{A}(W_R)$, the function

$$F_{A,B}(\eta) = \langle \Omega, A\, U(\eta) B\, U(\eta)^{-1} \Omega \rangle$$

extends to the strip $0 < \operatorname{Im} \eta < 2\pi$ and satisfies the KMS boundary relation

$$F_{A,B}(\eta + 2\pi \mathrm{i}) = \langle \Omega, U(\eta) B\, U(\eta)^{-1} A \Omega \rangle.$$

In proper time, the temperature is therefore

$$T_{\text{Unruh}} = \frac{a}{2\pi}.$$

150.3 Complex Boost Geometry

The factor 2π in the preceding definition is fixed by the complexified boost geometry. This subsection records the elementary geometry because it is the part of the Unruh effect that is often hidden behind a detector calculation.

Let Λ_η denote the boost in the (x^0, x^1)-plane:

$$(\Lambda_\eta x)^0 = x^0 \cosh \eta + x^1 \sinh \eta, \qquad (\Lambda_\eta x)^1 = x^1 \cosh \eta + x^0 \sinh \eta,$$

with transverse coordinates fixed. For complex $\eta = t + \mathrm{i}s$, write $\operatorname{Im} \Lambda_\eta x$ for the imaginary part of the complex vector $\Lambda_\eta x$.

Imaginary part of a complex boost. If $y \in W_R$ and $\eta = t + \mathrm{i}s$, then

$$\operatorname{Im}(\Lambda_\eta y)^0 = \sin s\, (\Lambda_t y)^1,$$
$$\operatorname{Im}(\Lambda_\eta y)^1 = \sin s\, (\Lambda_t y)^0.$$

Consequently $\operatorname{Im} \Lambda_\eta y$ is future timelike for $0 < s < \pi$, past timelike for $\pi < s < 2\pi$, and

$$\Lambda_{t+\mathrm{i}\pi} y = -\Lambda_t y$$

in the $(0,1)$-plane. In particular $\Lambda_{t+\mathrm{i}\pi} W_R$ is the left wedge

$$W_L = \{x \in \mathbb{M}^D : x^1 < -|x^0|\}.$$

Every point of W_R is spacelike separated from every point of W_L. These are elementary hyperbolic-function identities, recorded here because they fix the tube domain used in the KMS strip. The formulas follow from

$$\cosh(t + \mathrm{i}s) = \cosh t \cos s + \mathrm{i} \sinh t \sin s, \qquad \sinh(t + \mathrm{i}s) = \sinh t \cos s + \mathrm{i} \cosh t \sin s.$$

Since $y \in W_R$, also $\Lambda_t y \in W_R$, so $(\Lambda_t y)^1 > |(\Lambda_t y)^0|$. The vector $((\Lambda_t y)^1, (\Lambda_t y)^0, 0, \ldots, 0)$ is therefore future timelike in the mostly-plus convention. Multiplication by $\sin s$ gives the future/past alternatives.

At $s = \pi$, $\cosh(t + \mathrm{i}\pi) = -\cosh t$ and $\sinh(t + \mathrm{i}\pi) = -\sinh t$, hence the stated left-wedge image. If $x \in W_R$ and $\ell \in W_L$, then

$$x^1 - \ell^1 > |x^0| + |\ell^0| \geq |x^0 - \ell^0|.$$

Thus $x - \ell$ is spacelike.

150.4 Bisognano–Wichmann Boundary and the Local Model

Remark 150.1 (Bisognano–Wichmann theorem boundary). The Bisognano–Wichmann theorem in wedge-modular form is a structural theorem about the standard pair $(\mathcal{A}(W_R), \Omega)$, where $\mathcal{A}(W_R)$ is the right-wedge algebra and Ω is the vacuum vector. Under the Wightman or Haag–Kastler hypotheses of positivity, Poincare covariance, locality, spectrum condition, and wedge cyclicity and separatingness of the vacuum, its conclusion is that the Tomita–Takesaki modular automorphism group of $(\mathcal{A}(W_R), \Omega)$ is the Lorentz boost flow preserving W_R. With the boost convention $U(\eta) = \exp(\mathrm{i}\eta K)$, this is equivalently the KMS condition at inverse boost temperature 2π.

The full theorem is kept here as an external AQFT theorem boundary. Its proof requires Tomita–Takesaki modular theory, the analytic continuation supplied by the spectrum condition, covariance under complexified boosts, and wedge locality at imaginary boost angle $\mathrm{i}\pi$. The self-contained theorem in this chapter is the free massive scalar model calculation below: it proves the same strip analyticity and KMS boundary relation directly for the polynomial wedge algebra, with every analytic domain and ordering step visible.

150.5 Free Scalar Wedge KMS

We now prove the wedge KMS property for the free scalar polynomial algebra. The result is a complete model calculation with all orderings and analytic domains visible; the full Bisognano–Wichmann theorem requires the additional net-theoretic hypotheses stated later in this chapter.

Definition 150.2 (Free scalar two-point distribution)**.** For $m > 0$, the massive scalar vacuum two-point distribution on $\mathbb{M}^D$ is

$$\Delta_+^{(m)}(z) = \int_{\mathbb{R}^{D-1}} \frac{\mathrm{d}^{D-1}\mathbf{p}}{(2\pi)^{D-1} 2E_{\mathbf{p}}} \, e^{-\mathrm{i}E_{\mathbf{p}} z^0 + \mathrm{i}\mathbf{p}\cdot\mathbf{z}}, \qquad E_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}.$$

For test functions $f, g \in C_c^\infty(\mathbb{M}^D)$,

$$\langle \Omega, \phi(f)\phi(g)\Omega \rangle = \iint f(x)g(y)\Delta_+^{(m)}(x - y) \, \mathrm{d}x \, \mathrm{d}y.$$

The distribution $\Delta_+^{(m)}(z)$ is the boundary value of a holomorphic function in the past tube

$$\mathcal{T}_- = \{z \in \mathbb{C}^D : \operatorname{Im} z \in V_-\},$$

where V_- is the open past timelike cone. Indeed, if $\operatorname{Im} z = u \in V_-$, then

$$E_{\mathbf{p}} u^0 - \mathbf{p} \cdot \mathbf{u} \le -c_u E_{\mathbf{p}}$$

for some $c_u > 0$ on every compact subcone of V_-, and the defining mass-shell integral is absolutely convergent after the usual polynomial bounds from differentiating in z.

Theorem 150.3 (Free scalar wedge KMS)**.** *Let $f, g \in C_c^\infty(W_R)$. Let the boost automorphism act on smeared fields by*

$$\alpha_\eta(\phi(g)) = \phi(g_\eta), \qquad g_\eta(x) = g(\Lambda_{-\eta} x).$$

Then

$$F_{f,g}(\eta) = \langle \Omega, \phi(f) \alpha_\eta(\phi(g)) \Omega \rangle$$

is the lower boundary value of a function holomorphic in the strip

$$0 < \operatorname{Im} \eta < 2\pi,$$

continuous as a distributional boundary value on the two edges, and its upper boundary satisfies

$$F_{f,g}(\eta + 2\pi \mathrm{i}) = \langle \Omega, \alpha_\eta(\phi(g)) \phi(f) \Omega \rangle. \tag{150.1}$$

Proof. For real η, change variables in g_η to write

$$F_{f,g}(\eta) = \iint f(x) g(y) \Delta_+^{(m)}(x - \Lambda_\eta y) \, \mathrm{d}x \, \mathrm{d}y. \tag{150.2}$$

For $0 < \operatorname{Im} \eta < \pi$, the complex-boost geometry above implies that

$$\operatorname{Im}(x - \Lambda_\eta y) = -\operatorname{Im} \Lambda_\eta y \in V_-$$

for all $x, y \in W_R$. Since the supports of f and g are compact subsets of W_R, the image stays in a compact subcone of the past tube on compact sub-strips. Thus

$$F^{(1)}_{f,g}(\eta) = \iint f(x) g(y) \Delta_+^{(m)}(x - \Lambda_\eta y) \, \mathrm{d}x \, \mathrm{d}y$$

is holomorphic for $0 < \operatorname{Im} \eta < \pi$.

For $\pi < \operatorname{Im} \eta < 2\pi$, the same geometry says that $\operatorname{Im} \Lambda_\eta y \in V_-$. Hence

$$F^{(2)}_{f,g}(\eta) = \iint f(x) g(y) \Delta_+^{(m)}(\Lambda_\eta y - x) \, \mathrm{d}x \, \mathrm{d}y$$

is holomorphic on the upper half of the strip.

It remains to glue the two holomorphic functions across $\operatorname{Im} \eta = \pi$. At $\eta = t + \mathrm{i}\pi$, the point $\Lambda_\eta y$ lies in W_L. The complex-boost geometry above therefore makes x and $\Lambda_\eta y$ spacelike separated. The free scalar commutator is

$$[\phi(x), \phi(y)] = \mathrm{i} \Delta_m(x - y) \mathbf{1},$$

where Δ_m is the difference of the retarded and advanced fundamental solutions of $\Box - m^2$. Hyperbolic finite propagation gives $\Delta_m(z) = 0$ for spacelike z. Therefore the two boundary values

$$\Delta_+^{(m)}(x - \Lambda_{t+\mathrm{i}\pi} y) \quad \text{and} \quad \Delta_+^{(m)}(\Lambda_{t+\mathrm{i}\pi} y - x)$$

agree after smearing with $f(x)g(y)$. The edge-of-the-wedge gluing in this one-complex-variable strip is the following elementary distributional fact: if two holomorphic functions on adjacent strips have the same distributional boundary value on the common real line, they define a holomorphic function on the union. Applying it here gives a holomorphic $F_{f,g}$ on $0 < \operatorname{Im}\eta < 2\pi$.

The lower boundary is (150.2). The upper boundary is obtained from $F^{(2)}$ because $\Lambda_{\eta+2\pi\mathrm{i}} = \Lambda_\eta$:

$$\lim_{s\uparrow 2\pi} F^{(2)}_{f,g}(t+\mathrm{i}s) = \iint f(x)g(y)\Delta_+^{(m)}(\Lambda_t y - x)\,\mathrm{d}x\,\mathrm{d}y = \langle \Omega, \alpha_t(\phi(g))\phi(f)\Omega\rangle.$$

This is (150.1). □

Polynomial wedge algebra. Let $\mathcal{P}(W_R)$ be the unital $*$-algebra generated by the smeared free fields $\phi(f)$, $f \in C_c^\infty(W_R)$, modulo the free field equation and canonical commutation relations. The vacuum state restricted to $\mathcal{P}(W_R)$ is KMS at inverse boost temperature 2π with respect to α_η. The analytic input remains Theorem 150.3; the passage from fields to polynomials is Wick expansion: every vacuum n-point function of the free scalar field is a finite sum of products of two-point functions by Wick's theorem. If a polynomial B is boosted by complex η, each two-point factor connecting a field in A to a field in B is one of the strip-holomorphic functions proved in Theorem 150.3. Pairings entirely inside A or entirely inside B are independent of η or are invariant under the common boost. Products and finite sums preserve holomorphy in the strip. The boundary relation follows term by term from (150.1); the reordering at the upper edge is legitimate because the free commutator has already supplied the spacelike gluing at imaginary boost time π. This is exactly the KMS condition for all polynomial elements.

150.6 Worldline Specialization

For a uniformly accelerated trajectory in $D = 2$,

$$x^0(\tau) = a^{-1}\sinh(a\tau), \qquad x^1(\tau) = a^{-1}\cosh(a\tau),$$

the invariant separation between two points is

$$-(x(\tau) - x(0))^2 = \frac{4}{a^2}\sinh^2\!\left(\frac{a(\tau - \mathrm{i}0)}{2}\right).$$

The Wightman function along the trajectory is therefore a function of $\sinh^2(a(\tau - \mathrm{i}0)/2)$. Since

$$\sinh\!\left(\frac{a(\tau + \mathrm{i}2\pi/a)}{2}\right) = -\sinh\!\left(\frac{a\tau}{2}\right),$$

the two-point function has imaginary proper-time periodicity $\beta = 2\pi/a$ with the KMS ordering prescription. This verifies the two-point form of the free-field wedge KMS theorem on a single orbit.

150.7 Detector Response

An Unruh–DeWitt detector with energy gap $E > 0$ coupled linearly to the field has transition rate proportional to

$$\dot{P}(E) = \int_{-\infty}^{\infty} d\tau \, e^{-\mathrm{i}E\tau} W(x(\tau), x(0)).$$

The KMS condition implies detailed balance

$$\dot{P}(-E) = e^{\beta E} \dot{P}(E), \qquad \beta = \frac{2\pi}{a}.$$

Thus the detector response is thermal at temperature $a/(2\pi)$ when the coupling is weak, the detector is operated for a long time, and switching effects have been separated from the stationary rate.

Detailed balance from the KMS boundary relation. Let $G(\tau)$ be a tempered distribution which is the stationary two-point function sampled by a detector along a time-translation orbit, and suppose that it has the KMS boundary relation at inverse temperature β in the following convention:

$$G(\tau - \mathrm{i}\beta) = G(-\tau)$$

as boundary values of one holomorphic function in the strip $-\beta < \operatorname{Im} \tau < 0$. If the stationary transition rate

$$\dot{P}(E) = \int_{-\infty}^{\infty} e^{-\mathrm{i}E\tau} G(\tau) \, \mathrm{d}\tau$$

exists as the Fourier transform of this tempered distribution at $\pm E$, $E > 0$, and if the contour shift is justified by inserting a Schwartz switching function and then taking the stationary limit, then

$$\dot{P}(-E) = e^{\beta E} \dot{P}(E).$$

With a Schwartz switching function, all integrals below are ordinary absolutely convergent contour integrals; the stated distributional formula is the limit after the switching function is removed. Start from

$$\dot{P}(-E) = \int e^{\mathrm{i}E\tau} G(\tau) \, \mathrm{d}\tau = \int e^{-\mathrm{i}E\tau} G(-\tau) \, \mathrm{d}\tau.$$

Use the KMS boundary relation $G(-\tau) = G(\tau - \mathrm{i}\beta)$:

$$\dot{P}(-E) = \int_{\mathbb{R}} e^{-\mathrm{i}E\tau} G(\tau - \mathrm{i}\beta) \, \mathrm{d}\tau.$$

Shift the contour to $u = \tau - \mathrm{i}\beta$, so $\tau = u + \mathrm{i}\beta$. The exponential becomes

$$e^{-\mathrm{i}E(u+\mathrm{i}\beta)} = e^{\beta E} e^{-\mathrm{i}Eu}.$$

Holomorphy in the strip gives

$$\dot{P}(-E) = e^{\beta E} \int_{\mathbb{R}} e^{-\mathrm{i}Eu} G(u) \, \mathrm{d}u = e^{\beta E} \dot{P}(E).$$

Open Problem 150.4 (Interacting wedge thermality in examples). Develop explicit interacting examples where the wedge algebra, modular flow, and detector response can be computed or bounded beyond formal perturbation theory. The target is to connect the abstract Bisognano–Wichmann theorem with concrete nonperturbative correlation functions and operational detector limits.

Chapter 151

The Hawking Effect

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

The Hawking effect is a statement about quantum fields on a fixed spacetime whose causal structure contains a future black-hole horizon. Its invariant content is not that a preferred global particle basis exists. Rather, under regularity at the future horizon and suitable propagation to future null infinity, late outgoing observables have a thermal occupation factor with temperature fixed by the surface gravity. This chapter separates four ingredients which are often compressed into one slogan: horizon regularity, exponential ray tracing, Bogoliubov mixing of wave packets, and propagation through the exterior geometry.

151.1 Stationary Horizons and Surface Gravity

Let (M, g) be a smooth Lorentzian spacetime and let χ be a Killing field. A null hypersurface $\mathcal{H}$ is a Killing horizon for χ if χ is tangent to $\mathcal{H}$, nonzero on the open horizon, and null on $\mathcal{H}$. On $\mathcal{H}$, the acceleration of the horizon generators is necessarily proportional to χ:

$$\chi^\nu \nabla_\nu \chi^\mu = \kappa \chi^\mu. \tag{151.1}$$

The proportionality factor κ is the surface gravity in the normalization of χ. For an asymptotically flat stationary black hole the normalization is fixed by requiring that χ approach a unit time translation at infinity.

Lemma 151.1 (Rindler normal form of a nonextremal horizon)**.** *Suppose $\mathcal{H}$ is a nonextremal Killing horizon, $\kappa > 0$. In a neighborhood of a horizon cross-section, the two-dimensional metric* *in the normal directions can be written, up to terms smooth in ρ^2, as*

$$ds_\perp^2 = -\kappa^2\rho^2\,dt^2 + d\rho^2.$$

Equivalently, the coordinates

$$U = -\rho e^{-\kappa t}, \qquad V = \rho e^{\kappa t}$$

are regular null coordinates through the future horizon $U = 0$, and $\chi = \partial_t$.

Proof. Choose a smooth cross-section $\Sigma \subset \mathcal{H}$. In a neighborhood of Σ one may use Gaussian null coordinates

$$(v, r, y^A), \qquad \mathcal{H} = \{r = 0\}, \qquad \chi = \partial_v,$$

where y^A are coordinates on Σ, v is the Killing parameter along the generators, and r is an affine parameter on null geodesics transverse to the horizon. The metric has the form

$$g = 2\,dv\,dr - 2\kappa r\,dv^2 + 2r h_A(r,y)\,dv\,dy^A + \gamma_{AB}(r,y)\,dy^A dy^B.$$

The coefficient of $r\,dv^2$ is fixed by (151.1): on $r = 0$, $\nabla_\chi \chi = \kappa\chi$, and nonextremality means $\kappa > 0$. Thus the normal two-dimensional part, at fixed y, is

$$ds_\perp^2 = 2\,dv\,dr - 2\kappa r\,dv^2 + O(r^2 dv^2) + O(r\,dv\,dy).$$

Put

$$r = \frac{\kappa}{2}\rho^2, \qquad t = v - \frac{1}{2\kappa}\log r = v - \frac{1}{\kappa}\log\rho + \text{constant}.$$

Then

$$dv = dt + \frac{dr}{2\kappa r} = dt + \frac{d\rho}{\kappa\rho},$$

and the leading normal metric becomes

$$\begin{aligned} 2\,dv\,dr - 2\kappa r\,dv^2 &= 2\left(dt + \frac{d\rho}{\kappa\rho}\right)\kappa\rho\,d\rho - \kappa^2\rho^2\left(dt + \frac{d\rho}{\kappa\rho}\right)^2 \\ &= -\kappa^2\rho^2 dt^2 + d\rho^2. \end{aligned}$$

The omitted coefficients are smooth functions of $r = \kappa\rho^2/2$ and of y, so the correction terms are smooth in ρ^2 in the Killing-adapted chart. Finally define

$$U = -\rho e^{-\kappa t}, \qquad V = \rho e^{\kappa t}.$$

Then $UV = -\rho^2$ and

$$-\kappa^2\rho^2 dt^2 + d\rho^2 = -\,dU\,dV,$$

up to the conventional factor used in writing a symmetric null product. Thus U, V extend smoothly through $U = 0$, the future horizon, and $\partial_t = \partial_v = \chi$. $\square$

For Schwarzschild of mass M in units $G = \hbar = c = k_B = 1$,

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2.$$

Writing $r = 2M + \rho^2/(8M) + O(\rho^4/M^3)$ gives

$$ds_\perp^2 = -\left(\frac{\rho}{4M}\right)^2 dt^2 + d\rho^2 + O(\rho^4),$$

so

$$\kappa_{\text{Schw}} = \frac{1}{4M}, \qquad T_{\text{Schw}} = \frac{1}{8\pi M}.$$

151.2 Euclidean Regularity

Analytically continue the stationary time by $t = -\mathrm{i}\tau$. The normal metric becomes

$$ds_E^2 \simeq \kappa^2 \rho^2 d\tau^2 + d\rho^2.$$

This is the flat metric in polar coordinates if and only if the angular coordinate $\kappa\tau$ has period 2π. The Euclidean regularity argument therefore fixes

$$\beta_H = \frac{2\pi}{\kappa}, \qquad T_H = \frac{\kappa}{2\pi}.$$

Remark 151.2 (What Euclidean regularity proves). Assume a Euclidean QFT has been constructed on the smooth Euclidean black-hole section, the analytic continuation to stationary Lorentzian correlators exists, and the Euclidean time circle is the continuation of the normalized Killing time. Then the analytically continued stationary correlation functions satisfy the KMS periodicity at inverse temperature $\beta_H = 2\pi/\kappa$. This statement does not by itself prove existence of an interacting Lorentzian Hadamard state on a collapse spacetime.

Smoothness at $\rho = 0$ identifies $\kappa\tau$ as an angle. Hence the Euclidean Schwinger functions are periodic under $\tau \mapsto \tau + 2\pi/\kappa$, with the usual ordering around the Euclidean circle. Under the assumed Osterwalder-Schrader or analytic-continuation reconstruction for the stationary problem, Euclidean periodicity becomes the KMS boundary relation for Lorentzian Killing-time correlation functions. The final sentence records an implication that is not part of the argument: Euclidean smoothness constructs a stationary thermal object, whereas collapse requires a state empty on $\mathcal{I}^-$, regular at the future horizon, and propagated through a time-dependent geometry.

151.3 Exponential Ray Tracing in Collapse

The collapse derivation begins with the geometry of late outgoing null rays. Let v be advanced time on past null infinity $\mathcal{I}^-$, and let u be retarded time on future null infinity $\mathcal{I}^+$. Let v_0 label the last escaping ray: rays with $v < v_0$ reach $\mathcal{I}^+$, while rays with $v > v_0$ fall through the future horizon.

Universal late-time ray tracing. Assume the collapse spacetime settles at late retarded time to a stationary nonextremal black-hole exterior with future-horizon Kruskal coordinate

$$U = -C_H e^{-\kappa u}$$

near $\mathcal{I}^+$, with $C_H > 0$, and assume the past-null coordinate v is a smooth affine parameter through the last escaping ray with

$$U = A(v - v_0) + O((v - v_0)^2), \qquad A > 0.$$

Then for $v < v_0$,

$$v_0 - v = Ce^{-\kappa u} + O(e^{-2\kappa u}), \qquad C = \frac{C_H}{A} > 0,$$

or equivalently

$$u(v) = -\frac{1}{\kappa}\log\frac{v_0 - v}{C} + O(v_0 - v).$$

For an escaping ray close to the horizon, $U < 0$. Put $s = v_0 - v > 0$. The smoothness assumption at the last escaping ray gives

$$U = -As + O(s^2),$$

with $A > 0$. The horizon coordinate relation gives the same U as

$$-C_H e^{-\kappa u} = -As + O(s^2).$$

Multiplying by $-1/A$ gives

$$s - \frac{C_H}{A}e^{-\kappa u} = O(s^2).$$

For sufficiently late u, s is small and the inverse-function theorem applied to $s \mapsto s + O(s^2)$ yields

$$s = \frac{C_H}{A}e^{-\kappa u} + O(e^{-2\kappa u}).$$

Taking the logarithm of $s = Ce^{-\kappa u}(1 + O(e^{-\kappa u}))$, with $C = C_H/A$, gives

$$u = -\frac{1}{\kappa}\log\frac{s}{C} + O(s),$$

which is the stated inverse expansion.

The constants A, C_H, C are nonuniversal; the exponential rate κ is fixed by the regularity of the horizon coordinate.

151.4 Bogoliubov Coefficients from Mode Tracing

We carry out the coefficient calculation for a massless scalar field in the geometric-optics approximation. Angular variables and the exterior potential are suppressed in this section; they reappear as greybody factors below. Normalize right-moving modes on a null line by

$$f^{\text{in}}_{\omega'}(v) = \frac{1}{\sqrt{4\pi\omega'}}e^{-\mathrm{i}\omega' v}, \qquad p^{\text{out}}_{\omega}(u) = \frac{1}{\sqrt{4\pi\omega}}e^{-\mathrm{i}\omega u}, \qquad \omega, \omega' > 0.$$

Tracing p^{out}_{ω} backwards to $\mathcal{I}^-$ gives, near the singular late-time part responsible for particle creation,

$$p_\omega(v) = \frac{1}{\sqrt{4\pi\omega}}\left(\frac{v_0 - v}{C}\right)^{\mathrm{i}\omega/\kappa}\theta(v_0 - v) + \text{terms smooth across } v = v_0. \tag{151.2}$$

The smooth terms have rapidly decaying Fourier transforms at large ω' and do not determine the universal thermal ratio.

Use the Klein-Gordon product on a null line,

$$(f, g)_{\mathcal{I}^-} = -\mathrm{i}\int_{-\infty}^{\infty} f(v)\overleftrightarrow{\partial_v}\,\overline{g(v)}\,dv.$$

The expansion

$$p_\omega = \int_0^\infty d\omega' \left(\alpha_{\omega\omega'} f^{\rm in}_{\omega'} + \beta_{\omega\omega'} \overline{f^{\rm in}_{\omega'}} \right)$$

has

$$\alpha_{\omega\omega'} = (f^{\rm in}_{\omega'}, p_\omega)_{\mathcal{I}^-}, \qquad \beta_{\omega\omega'} = -(\overline{f^{\rm in}_{\omega'}}, p_\omega)_{\mathcal{I}^-}.$$

Putting $x = v_0 - v$ and $a = \omega/\kappa$, the needed oscillatory boundary values are

$$\int_0^\infty x^{\mathrm{i}a} e^{-\mathrm{i}\omega' x}\, dx = e^{-\mathrm{i}\pi/2} e^{\pi a/2} \Gamma(1+\mathrm{i}a)(\omega')^{-1-\mathrm{i}a},$$

$$\int_0^\infty x^{\mathrm{i}a} e^{+\mathrm{i}\omega' x}\, dx = e^{+\mathrm{i}\pi/2} e^{-\pi a/2} \Gamma(1+\mathrm{i}a)(\omega')^{-1-\mathrm{i}a}.$$

Equivalently, since $\Gamma(1+\mathrm{i}a) = \mathrm{i}a\,\Gamma(\mathrm{i}a)$, the singular coefficients can be written

$$\alpha_{\omega\omega'} = A_{\omega\omega'}\, e^{+\pi\omega/(2\kappa)} \Gamma(\mathrm{i}\omega/\kappa)(\omega')^{-\mathrm{i}\omega/\kappa},$$

$$\beta_{\omega\omega'} = -A'_{\omega\omega'}\, e^{-\pi\omega/(2\kappa)} \Gamma(\mathrm{i}\omega/\kappa)(\omega')^{-\mathrm{i}\omega/\kappa},$$

where $A_{\omega\omega'}$ and $A'_{\omega\omega'}$ differ only by phases and have common modulus

$$|A_{\omega\omega'}| = \frac{1}{2\pi\kappa} \sqrt{\frac{\omega}{\omega'}}.$$

Therefore

$$\frac{|\alpha_{\omega\omega'}|^2}{|\beta_{\omega\omega'}|^2} = e^{2\pi\omega/\kappa}.$$

Using

$$|\Gamma(\mathrm{i}a)|^2 = \frac{\pi}{a \sinh(\pi a)}$$

gives the explicit negative-frequency density

$$|\beta_{\omega\omega'}|^2 = \frac{1}{2\pi\kappa\, \omega'} \frac{1}{e^{2\pi\omega/\kappa} - 1}. \tag{151.3}$$

Equation (151.3) is not yet a finite particle number. The factor $1/\omega'$ is the density remnant of an idealized stationary late-time limit. Finite observables are wave-packet observables.

151.5 Wave Packets and the Planck Factor

Fix a frequency resolution $\varepsilon > 0$. For integers $j \geq 0$ and $n \in \mathbb{Z}$, define outgoing packets

$$p^{\rm out}_{jn} = \frac{1}{\sqrt{\varepsilon}} \int_{j\varepsilon}^{(j+1)\varepsilon} d\omega\, e^{2\pi \mathrm{i} n\omega/\varepsilon} p^{\rm out}_\omega. \tag{151.4}$$

The integer j labels a frequency bin and n labels retarded-time localization, with center approximately $u_n = 2\pi n/\varepsilon$.

Controlled Approximation 151.3 (Late-time packet occupation)**.** In the stationary late-time approximation determined by the exponential ray-tracing relation above, the number of incoming-vacuum quanta in the outgoing packet (j,n) is

$$\langle N_{jn}\rangle = \frac{1}{\varepsilon}\int_{j\varepsilon}^{(j+1)\varepsilon} \frac{d\omega}{e^{\beta_H\omega}-1} + o(1) \qquad (n\to+\infty), \tag{151.5}$$

where $\beta_H = 2\pi/\kappa$. If the bin is narrow compared with the scale on which the Planck factor varies, this is

$$\langle N_{jn}\rangle = \frac{1}{e^{\beta_H\omega_j}-1} + O(\varepsilon\,\partial_\omega n_\omega), \qquad \omega_j = (j+\tfrac{1}{2})\varepsilon.$$

The packet Bogoliubov coefficient is

$$\beta_{jn,\omega'} = \frac{1}{\sqrt{\varepsilon}}\int_{j\varepsilon}^{(j+1)\varepsilon} d\omega\, e^{2\pi\mathrm{i}n\omega/\varepsilon}\beta_{\omega\omega'}.$$

The packet number is

$$\langle N_{jn}\rangle = \int_0^\infty d\omega'\,|\beta_{jn,\omega'}|^2.$$

Inserting the singular coefficient gives a double integral over ω_1, ω_2 and the ω'-factor

$$\int_0^\infty \frac{d\omega'}{\omega'}(\omega')^{\mathrm{i}(\omega_2-\omega_1)/\kappa}.$$

With a large finite affine-time regulator this is the Fourier transform of a long interval in $\log\omega'$. In the infinite stationary limit it is the distribution

$$2\pi\kappa\,\delta(\omega_1-\omega_2). \tag{151.6}$$

The phase $e^{2\pi\mathrm{i}n(\omega_1-\omega_2)/\varepsilon}$ then drops out, and the remaining diagonal coefficient is exactly the Planck factor in (151.3). Nonstationary corrections from the $O(e^{-2\kappa u})$ part of the ray-tracing map and from the smooth part of the traced mode give the $o(1)$ term for packets moved to late retarded time.

The exact bin average in (151.5) can be written without approximation:

$$\frac{1}{\varepsilon}\int_{\omega_-}^{\omega_+}\frac{d\omega}{e^{\beta_H\omega}-1} = \frac{1}{\beta_H\varepsilon}\log\frac{1-e^{-\beta_H\omega_+}}{1-e^{-\beta_H\omega_-}}, \qquad \omega_\pm = j\varepsilon, (j+1)\varepsilon.$$

151.6 Two-Dimensional Flux and the Moving-Mirror Model

For a two-dimensional conformal scalar, the same exponential map determines the stress-tensor flux. Let $p(u)$ denote the incoming null coordinate of the ray reaching $\mathcal{I}^+$ at retarded time u:

$$v = p(u) = v_0 - Ce^{-\kappa u}.$$

For a CFT of central charge c in the incoming vacuum, the chiral stress tensor transforms under the reparametrization $v = p(u)$ as

$$\langle T_{uu}\rangle = -\frac{c}{24\pi}\{p,u\}, \qquad \{p,u\} = \frac{p'''}{p'} - \frac{3}{2}\left(\frac{p''}{p'}\right)^2. \tag{151.7}$$

For the exponential map,

$$\{v_0 - Ce^{-\kappa u}, u\} = -\frac{\kappa^2}{2},$$

and hence

$$\langle T_{uu} \rangle = \frac{c\,\kappa^2}{48\pi} = \frac{\pi c}{12\beta_H^2}. \tag{151.8}$$

The last equality is the thermal energy flux of one chiral CFT sector:

$$\int_0^\infty \frac{d\omega}{2\pi} \frac{\omega}{e^{\beta_H \omega} - 1} = \frac{\pi}{12\beta_H^2}$$

for $c = 1$. The moving-mirror model realizes this calculation in flat space by choosing a mirror trajectory whose ray-tracing function $p(u)$ has the same late exponential form. It is a useful model of the kinematics of Hawking radiation, not a derivation of quantum gravity.

151.7 Trans-Planckian Precursor Frequencies

The same tracing relation also shows the trans-Planckian domain of the geometric-optics derivation. A packet observed at retarded time u with frequency ω has precursor frequency on $\mathcal{I}^-$ of order

$$\omega_{\text{in}} \sim \omega \frac{du}{dv} = \frac{\omega}{\kappa(v_0 - v)} \sim \frac{\omega}{\kappa C} e^{\kappa u}.$$

For any fixed microscopic scale Λ_{micro}, sufficiently late packets have $\omega_{\text{in}} > \Lambda_{\text{micro}}$. Thus the fixed-background continuum derivation is a conditional theorem inside the assumed QFT on the assumed smooth geometry. Robustness under modified short-distance physics is a separate statement requiring a specified microscopic model and a controlled limit.

151.8 Boulware, Hartle–Hawking, and Unruh States

In a stationary eternal black-hole exterior, three quasifree scalar states serve as reference points.

(i) The Boulware state is positive-frequency with respect to the static Killing time at infinity. It is empty for static observers at infinity but is singular at the horizon.

(ii) The Hartle–Hawking state is regular on both future and past horizons of the bifurcate extension and is KMS at inverse temperature β_H with respect to the exterior Killing time.

(iii) The Unruh state is regular on the future horizon and empty on past null infinity. It is the fixed-background state appropriate to collapse and gives outgoing flux on $\mathcal{I}^+$.

These names denote state-selection properties, not extra axioms. In a given model, each state must be constructed as a positive Hadamard two-point function or by another equivalent locally covariant construction.

151.9 Greybody Factors and Four-Dimensional Flux

In an asymptotically flat black-hole spacetime, propagation from the near-horizon region to infinity includes scattering by the exterior curvature potential. For a scalar field on Schwarzschild, write the mode as

$$\phi_{\omega\ell m} = \frac{1}{r}\psi_{\omega\ell}(r_*)Y_{\ell m}(\theta,\varphi)e^{-\mathrm{i}\omega t}, \qquad \frac{dr_*}{dr} = \left(1 - \frac{2M}{r}\right)^{-1}.$$

The radial equation is

$$\left[-\frac{d^2}{dr_*^2} + V_\ell(r)\right]\psi_{\omega\ell} = \omega^2\psi_{\omega\ell}, \qquad V_\ell(r) = \left(1 - \frac{2M}{r}\right)\left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right).$$

Let $\Gamma_\ell(\omega)$ be the transmission probability through this one-dimensional scattering problem, normalized by unit incoming flux from the horizon. The number and energy fluxes at infinity are

$$\frac{dN}{dt\,d\omega} = \sum_{\ell,m}\frac{\Gamma_\ell(\omega)}{2\pi(e^{\beta_H\omega}-1)},$$

$$\frac{dE}{dt\,d\omega} = \sum_{\ell,m}\frac{\omega\,\Gamma_\ell(\omega)}{2\pi(e^{\beta_H\omega}-1)}.$$

The Planck factor is horizon kinematics; $\Gamma_\ell(\omega)$ is ordinary hyperbolic propagation through the exterior. Fermions replace the Bose-Einstein denominator by $e^{\beta_H\omega}+1$, and gauge fields require the corresponding physical-polarization and ghost-free Hilbert-space construction.

151.10 Back-Reaction Boundary

Fixed-background QFT treats g as prescribed. Semiclassical gravity asks instead for

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G\,\langle T_{\mu\nu}\rangle_{\mathrm{ren}}[g,\omega],$$

where ω is the chosen Hadamard state and the stress tensor includes the finite local curvature counterterms discussed earlier in this volume. In an adiabatic Schwarzschild approximation, energy conservation suggests an ODE

$$\frac{dM}{du} = -L(M), \qquad L(M) = \sum_{\ell,m}\int_0^\infty \frac{d\omega}{2\pi}\,\frac{\omega\,\Gamma_\ell(\omega;M)}{e^{8\pi M\omega}-1},$$

for a scalar contribution. This equation is not a theorem about the full evaporating geometry. It assumes slow variation on the scale M, ignores stress-tensor fluctuations except through their mean, and presupposes that the state remains close to an Unruh-type Hadamard state on each instantaneous exterior.

151.11 QFT Boundary

A theorem-level Hawking effect for an interacting local QFT on black-hole backgrounds requires all of the following data:

(i) a precise class of globally hyperbolic collapse spacetimes with controlled late approach to a nonextremal stationary exterior;

(ii) construction of a Hadamard state empty on $\mathcal{I}^-$ and regular at the future horizon;

(iii) a local and covariant stress-tensor renormalization scheme;

(iv) propagation estimates connecting near-horizon data to $\mathcal{I}^+$;

(v) control of the limits defining late-time wave-packet observables;

(vi) a statement separating the universal horizon factor from model-dependent greybody and interaction effects.

Without these hypotheses, the phrase "black holes radiate thermally" is not a mathematical theorem. With these hypotheses supplied, the fixed-background calculation above shows where the temperature and Planck factor enter.

Open Problem 151.4 (Interacting Hawking theorem). Prove a Hawking radiation theorem for an interacting local QFT on a class of black-hole backgrounds, including Hadamard state construction, KMS or near-horizon analyticity, renormalized stress tensor, greybody propagation, late-time wave-packet limits, and the precise sense in which the emitted flux is thermal.

Chapter 152

Background Gauge Fields and Index Theory

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Background gauge fields are not auxiliary decorations. They are sources for global currents, they specify the bundles in which charged fields take values, and for chiral fermions they determine both zero-mode selection rules and anomaly lines. This chapter develops the index-theoretic part of that structure with explicit conventions. The analytic input is the index of a coupled Dirac operator; the QFT input is the Berezin integral over its zero modes and the determinant line of its nonzero modes. The source traces for the heat-kernel/index-theory line used here are Seeley, Gilkey, and Atiyah–Singer [54, 55, 56]; the derivations below keep the local Clifford and Chern-Weil conventions explicit because these are the conventions later used in anomaly formulae.

152.1 Spinors, Bundles, and Curvature Conventions

Let M^{2n} be a closed oriented Riemannian spin manifold. The complex spinor bundle splits into chiral pieces

$$S = S^+ \oplus S^-.$$

Let $E \to M$ be a Hermitian vector bundle with unitary connection ∇^E. In the characteristic-class formulas of this chapter the curvature $F_E = (\nabla^E)^2$ is skew-Hermitian, so

$$\mathrm{ch}(E) = \mathrm{tr}_E \exp\left(\frac{\mathrm{i}F_E}{2\pi}\right)$$

is a real even cohomology class. If a physics convention uses Hermitian generators and Hermitian curvature F_{phys}, then

$$F_E = -\mathrm{i}F_{\mathrm{phys}}$$

in this formula. All traces in $\mathrm{ch}(E)$ are traces in the fermion representation, not necessarily the trace used in the Yang-Mills kinetic term.

The coupled Dirac operator is

$$D_A : \Gamma(S \otimes E) \longrightarrow \Gamma(S \otimes E), \qquad D_A = \gamma^\mu \nabla_\mu^{S\otimes E}.$$

With respect to chirality,

$$D_A = \begin{pmatrix} 0 & D_A^- \\ D_A^+ & 0 \end{pmatrix}, \qquad D_A^+ : \Gamma(S^+ \otimes E) \to \Gamma(S^- \otimes E).$$

The analytic index is

$$\mathrm{Ind}(D_A^+) = \dim\ker D_A^+ - \dim\ker D_A^-. \tag{152.1}$$

Because M is compact and D_A^+ is elliptic, this is a finite integer.

152.2 McKean–Singer Supertrace

Let Γ denote the chirality operator, acting by $+1$ on S^+ and by -1 on S^-. The heat operator $e^{-tD_A^2}$ is trace class for $t > 0$.

Proposition 152.1 (McKean–Singer identity)**.** *For every $t > 0$,*

$$\mathrm{Str}\ e^{-tD_A^2} := \mathrm{Tr}\Big(\Gamma e^{-tD_A^2}\Big) = \mathrm{Ind}(D_A^+). \tag{152.2}$$

Proof. Because M is compact and D_A is elliptic and self-adjoint, D_A^2 has discrete spectrum with finite-dimensional eigenspaces and $e^{-tD_A^2}$ is trace class for $t > 0$. The chirality operator anticommutes with D_A, so D_A exchanges $S^+ \otimes E$ and $S^- \otimes E$. The nonzero spectrum of D_A^2 on $S^+ \otimes E$ is paired with the nonzero spectrum on $S^- \otimes E$. Indeed, if $\psi_+ \in \Gamma(S^+ \otimes E)$ satisfies

$$D_A^- D_A^+ \psi_+ = \lambda \psi_+, \qquad \lambda > 0,$$

then $D_A^+\psi_+ \neq 0$ and

$$D_A^+ D_A^- (D_A^+ \psi_+) = \lambda (D_A^+ \psi_+).$$

The inverse map on the λ-eigenspace is $\lambda^{-1} D_A^-$, so the pairing preserves multiplicities. Hence the positive eigenvalue contributions to

$$\mathrm{Tr}_{S^+}\, e^{-tD_A^- D_A^+} - \mathrm{Tr}_{S^-}\, e^{-tD_A^+ D_A^-}$$

cancel term by term, with absolute convergence supplied by the heat trace. The zero eigenspaces are precisely $\ker D_A^+$ and $\ker D_A^-$. Their contribution is independent of t and equals

$$\dim\ker D_A^+ - \dim\ker D_A^- = \mathrm{Ind}(D_A^+).$$

□

This proof is finite-dimensional after elliptic spectral theory supplies the discrete spectrum and heat-kernel trace class property. It is the analytic bridge between zero-mode counting and local curvature densities.

152.3 Lichnerowicz Formula and Heat-Kernel Scaling

The local index theorem is a statement about the short-time asymptotics of the heat kernel of D_A^2. We spell out the conventions because the same heat-kernel coefficients also enter point-splitting stress tensors and trace anomalies.

Let $c(e^a) = \gamma^a$ denote Clifford multiplication by an orthonormal coframe. We use

$$\gamma^{ab} := \frac{1}{2}[\gamma^a, \gamma^b], \qquad F^E = \frac{1}{2}F^E_{ab} e^a \wedge e^b.$$

The curvature of the spin connection is

$$[\nabla^S_a, \nabla^S_b] = \frac{1}{4}R_{abcd}\gamma^{cd}.$$

Lemma 152.2 (Lichnerowicz formula)**.** *For the coupled Dirac operator $D_A = \gamma^a \nabla_a^{S\otimes E}$,*

$$D_A^2 = (\nabla^{S\otimes E})^* \nabla^{S\otimes E} + \frac{1}{4}\mathrm{Scal} + \frac{1}{2}\gamma^{ab}F^E_{ab}. \tag{152.3}$$

Proof. At a chosen point p, take an oriented orthonormal frame whose Levi-Civita connection coefficients vanish at p. Write ∇ for the coupled connection on $S \otimes E$. Since the antisymmetric part of $\gamma^a\gamma^b$ is γ^{ab},

$$\begin{aligned} D_A^2 &= \gamma^a\gamma^b \nabla_a\nabla_b \\ &= g^{ab}\nabla_a\nabla_b + \gamma^{ab}\nabla_a\nabla_b \\ &= g^{ab}\nabla_a\nabla_b + \frac{1}{2}\gamma^{ab}[\nabla_a, \nabla_b]. \end{aligned}$$

Restoring the Christoffel term away from the chosen normal frame gives the connection Laplacian $(\nabla^{S\otimes E})^*\nabla^{S\otimes E}$ in the sign convention of this chapter. The curvature of the coupled connection is

$$[\nabla_a, \nabla_b] = \frac{1}{4}R_{abcd}\gamma^{cd} + F^E_{ab}.$$

The E-bundle curvature therefore contributes

$$\frac{1}{2}\gamma^{ab}F^E_{ab}.$$

The spin-curvature contribution is

$$\frac{1}{2}\gamma^{ab}\frac{1}{4}R_{abcd}\gamma^{cd}.$$

It remains to evaluate the Clifford contraction. With $\gamma^{abcd} := \gamma^{[a}\gamma^{b}\gamma^{c}\gamma^{d]}$, the Clifford relations give

$$\gamma^{ab}\gamma^{cd} = \gamma^{abcd} + g^{bc}\gamma^{ad} - g^{ac}\gamma^{bd} - g^{bd}\gamma^{ac} + g^{ad}\gamma^{bc} + g^{ac}g^{bd} - g^{ad}g^{bc}.$$

The coefficient of γ^{abcd} is the total antisymmetrization of R_{abcd}, hence vanishes by the first Bianchi identity $R_{[abc]d} = 0$. The terms with one remaining γ^{pq} reduce to Ricci contractions, for example $R_{abcd}g^{bc}\gamma^{ad}$, and vanish because the Ricci tensor is symmetric while γ^{ad} is antisymmetric. The scalar terms give

$$R_{abcd}(g^{ac}g^{bd} - g^{ad}g^{bc}) = 2\,\mathrm{Scal}$$

with the curvature and Clifford conventions stated above. Thus the spin-curvature piece is

$$\frac{1}{8}\,R_{abcd}\gamma^{ab}\gamma^{cd} = \frac{1}{4}\mathrm{Scal}.$$

Combining the connection Laplacian, spin curvature, and bundle curvature gives (152.3). □

Definition 152.3 (Laplace-type operator and heat coefficients)**.** A Laplace-type operator on a Hermitian vector bundle $V \to M^d$ is written in the form

$$P = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + \mathcal{E}), \tag{152.4}$$

where ∇ is a connection on V and $\mathcal{E} \in \Gamma(\mathrm{End}\,V)$. Its curvature is $\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu]$. The local heat coefficients $a_{2r}(P;x)$ are defined by

$$\mathrm{Tr}\big(fe^{-tP}\big) \sim (4\pi t)^{-d/2}\sum_{r\geq 0} t^r \int_M f(x)\,\mathrm{tr}_V a_{2r}(P;x)\,d\mathrm{vol}_g \tag{152.5}$$

for every test function f.

Quoted Theorem 152.4 (First Seeley–DeWitt coefficients)**.** Source trace. *The displayed normalization follows the Seeley–DeWitt/Gilkey heat-kernel convention [54, 55]. For the convention* (152.4)*,*

$$a_0 = \mathbf{1}_V, \qquad a_2 = \mathcal{E} + \frac{1}{6}\mathrm{Scal}\,\mathbf{1}_V, \tag{152.6}$$

and

$$\begin{aligned} a_4 = \frac{1}{360}\Big[&\big(5\mathrm{Scal}^2 - 2\mathrm{Ric}_{\mu\nu}\mathrm{Ric}^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\big)\,\mathbf{1}_V \\ &+ 60\mathrm{Scal}\,\mathcal{E} + 180\mathcal{E}^2 + 30\Omega_{\mu\nu}\Omega^{\mu\nu} \\ &+ 12\nabla^2\mathrm{Scal}\,\mathbf{1}_V + 60\nabla^2\mathcal{E}\Big]. \end{aligned} \tag{152.7}$$

On a closed manifold the last line integrates to zero after taking a trace.

Parametrix origin. Work in a normal convex neighborhood and write Synge's world function as $\sigma(x, y)$. The heat kernel parametrix has the form

$$K_t(x, y) \sim (4\pi t)^{-d/2} e^{-\sigma(x,y)/2t} \Delta(x, y)^{1/2} \sum_{r \geq 0} t^r b_r(x, y), \qquad b_0(x, x) = \mathbf{1}_V, \tag{152.8}$$

where Δ is the van Vleck determinant and b_r is parallel transported in the bundle indices at leading order. Inserting (152.8) into $(\partial_t + P_x)K_t = 0$ and comparing powers of t gives the transport recursion

$$(\sigma^{;\mu}\nabla_\mu + r)b_r = -\Delta^{-1/2} P_x\Big(\Delta^{1/2} b_{r-1}\Big), \qquad r \geq 1. \tag{152.9}$$

The coincidence limits needed below are

$$[\sigma^{;\mu}] = 0, \qquad [\sigma^{;\mu}{}_\nu] = \delta^\mu{}_\nu, \qquad [\nabla_\mu \nabla_\nu \Delta^{1/2}] = \frac{1}{6}\mathrm{Ric}_{\mu\nu}, \tag{152.10}$$

with brackets denoting $y \to x$. Setting $r = 1$ in (152.9) immediately gives

$$[b_1] = \mathcal{E} + \frac{1}{6}\mathrm{Scal}\,\mathbf{1}_V.$$

For $r = 2$, differentiate the transport equation twice, use (152.10), the commutator $[\nabla_\mu, \nabla_\nu] = \Omega_{\mu\nu}$, and commute covariant derivatives until all third and fourth derivatives of σ and $\Delta^{1/2}$ are expressed in curvature. The resulting coincidence limit is precisely (152.7). No field equation is used; the calculation is the universal local recursion for a Laplace-type operator.

152.4 Local Index Theorem

Let R denote the Riemannian curvature two-form of TM. The $\widehat{A}$-class is

$$\widehat{A}(TM) = \det{}^{1/2}\left(\frac{R/4\pi}{\sinh(R/4\pi)}\right) = 1 - \frac{p_1(TM)}{24} + \frac{7p_1(TM)^2 - 4p_2(TM)}{5760} + \cdots.$$

The theorem below is the heat-kernel/Getzler route to the local Atiyah–Singer index formula [56, 55]; its proof is included to fix the curvature and chirality normalization used by the anomaly polynomial later in the chapter.

Theorem 152.5 (Local index density by Getzler rescaling)**.** *For the chiral Dirac operator coupled to E,*

$$\mathrm{Ind}(D_A^+) = \int_M \Big[\widehat{A}(TM)\,\mathrm{ch}(E)\Big]_{2n}. \tag{152.11}$$

Equivalently, the local heat-kernel density of the supertrace has

$$\lim_{t \to 0^+} \mathrm{str}\, K_t(x, x)\, \mathrm{dvol}_g = \Big[\widehat{A}(TM)\,\mathrm{ch}(E)\Big]_{2n}.$$

Proof. The McKean–Singer identity reduces the index to the small-t limit of the local supertrace. Fix $p \in M$. Choose geodesic normal coordinates, synchronous orthonormal frame, and radial gauges for S and E. In these coordinates the Taylor expansions of the two connections begin with their curvatures:

$$\nabla_a^S = \partial_a + \frac{1}{4} R_{abcd}(p) x^b \gamma^{cd} + O(|x|^2), \qquad \nabla_a^E = \partial_a + \frac{1}{2} F_{ab}^E(p) x^b + O(|x|^2). \tag{152.12}$$

Rescale $x = t^{1/2} y$. Under Getzler scaling the differential operator keeps the terms whose combined degree in $t^{1/2}$, derivatives, and Clifford variables can contribute to the top Clifford degree $2n$; all higher Taylor coefficients are suppressed by positive powers of $t^{1/2}$ in the diagonal heat kernel. Lemma 152.2 then gives the constant-curvature model operator

$$H_p = -\sum_a \left(\partial_{y^a} + \frac{1}{4} R_{ab}(p) y^b \right)^2 + \frac{1}{2} \gamma^{ab} F_{ab}^E(p), \tag{152.13}$$

where $R_{ab}(p)$ is viewed as the skew endomorphism of T_pM with matrix entries $R_{abcd}(p)\gamma^{cd}$ in the Clifford factor. The ordinary scalar-curvature term in (152.3) has Clifford degree 0, hence cannot by itself contribute to the supertrace in dimension $2n$; the missing gravitational contribution is instead produced by the harmonic-oscillator determinant in H_p.

The heat kernel of (152.13) is the Mehler kernel for a constant magnetic oscillator, tensored with the finite endomorphism exponential $\exp(-\frac{1}{2}\gamma^{ab}F_{ab}^E)$. Diagonalizing the skew curvature into two-planes gives the pointwise supertrace

$$\lim_{t\to 0^+} \operatorname{str} K_t(p,p)\, d\mathrm{vol}_g(p) = \left[\det{}^{1/2}\left(\frac{R/4\pi}{\sinh(R/4\pi)} \right) \operatorname{tr}_E \exp\left(\frac{\mathrm{i}F_E}{2\pi} \right) \right]_{2n}. \tag{152.14}$$

The determinant factor is the product over the curvature two-planes of the Mehler ratio

$$\frac{u/2}{\sinh(u/2)},$$

with u the corresponding Chern-Weil curvature variable. The Clifford supertrace extracts exactly the top exterior degree; lower Clifford degree terms vanish because str_S is zero away from degree $2n$. Equation (152.14) is precisely $\left[\widehat{A}(TM)\operatorname{ch}(E)\right]_{2n}$. Integrating over M and using (152.2) proves (152.11). $\square$

This theorem is a theorem of differential geometry, not an axiom of QFT. The derivation above identifies which part is analytic heat-kernel control and which part is local Clifford/Chern-Weil algebra. The QFT consequences below are then zero-mode selection rules, anomaly polynomials, determinant lines, and inflow.

In four dimensions the formula becomes

$$\operatorname{Ind}(D_A^+) = \int_M \left[\operatorname{ch}_2(E) - \frac{\operatorname{rank}(E)}{24} p_1(TM) \right]. \tag{152.15}$$

For an $SU(N)$ bundle E_R associated to a representation R,

$$c_1(E_R) = 0, \qquad \int_M \operatorname{ch}_2(E_R) = T_\Delta(R)\, k. \tag{152.16}$$

Here $k \in \mathbb{Z}$ is the instanton number in the fundamental bundle and $T_\Delta(R)$ is the quadratic index in the monograph's trace-delta normalization

$$\mathrm{tr}_R(t_R^a t_R^b) = T_\Delta(R)\delta^{ab}, \qquad T_\Delta(\Box) = 1.$$

Thus $T_\Delta(\mathrm{adj}) = 2N$. This is the same normalization compatible with the Yang-Mills kinetic term

$$\frac{1}{4g^2}\int \mathrm{tr}_\Delta(F_{\mu\nu}F^{\mu\nu}).$$

152.5 Four-Dimensional Examples

On a spin four-manifold with $p_1(TM) = 0$, a unit $SU(N)$ instanton gives

$$\mathrm{Ind}(D_\Box^+) = 1, \qquad \mathrm{Ind}(D_{\mathrm{adj}}^+) = 2N.$$

For $SU(2)$, the fundamental Weyl fermion has one complex zero mode in the orientation convention of (152.16), and an adjoint Weyl fermion has four complex zero modes. Reversing the orientation of the gauge instanton reverses the sign of the index and swaps the chirality of the zero modes.

For an Abelian line bundle L^q of charge q,

$$\mathrm{ch}(L^q) = e^{qx}, \qquad x = c_1(L).$$

On T^4 with $p_1(TT^4) = 0$, take integral two-forms a, b with

$$\int_{T^4} a \wedge b = 1, \qquad a \wedge a = b \wedge b = 0,$$

and $x = m_1 a + m_2 b$. Then

$$\mathrm{Ind}(D_{L^q}^+) = \int_{T^4} \frac{1}{2} q^2 x^2 = q^2 m_1 m_2. \tag{152.17}$$

This example is useful because it is completely Abelian: the index is the intersection pairing of magnetic fluxes through complementary two-tori.

152.6 Zero Modes and Berezin Selection Rules

Let ψ be a left-handed Weyl fermion valued in E. Decompose it in an orthonormal eigenbasis of the coupled Dirac operator. If $\{\zeta_1, \ldots, \zeta_m\}$ is a basis of $\ker D_A^+$, then the Grassmann field has zero-mode coordinates

$$\psi_0(x) = \sum_{i=1}^m \eta_i \zeta_i(x), \qquad \eta_i \eta_j = -\eta_j \eta_i.$$

Finite-dimensional zero-mode rule. After nonzero modes have been paired, the zero-mode part of a fermionic path integral is

$$\int d\eta_m \cdots d\eta_1\, \mathcal{O}(\eta_1, \ldots, \eta_m).$$

It vanishes unless the insertion $\mathcal{O}$ contains the top monomial $\eta_1 \cdots \eta_m$. Therefore a background with m unpaired Weyl zero modes requires m corresponding fermion insertions for a nonzero correlator. Berezin integration is the linear functional fixed by

$$\int d\eta\, 1 = 0, \qquad \int d\eta\, \eta = 1.$$

For m generators it extracts the coefficient of $\eta_1 \cdots \eta_m$ in the declared order. All lower-degree monomials are annihilated by at least one of the integrations.

For a vectorlike Dirac fermion in representation R, written in two-component language as left-handed fields in R and $\overline{R}$, a charge-k instanton produces $T_\Delta(R)k$ zero modes in each of the two conjugate chiral variables, with chirality determined by the instanton orientation. The resulting finite-dimensional Berezin integral is the precise origin of the fermion-number selection rule usually encoded by the 't Hooft vertex.

152.7 Anomaly Polynomial and Descent

For a chiral fermion in $2n$ dimensions, the anomaly polynomial is the $(2n+2)$-form

$$I_{2n+2} = \left[\widehat{A}(TM)\,\mathrm{ch}(E)\right]_{2n+2}. \tag{152.18}$$

Choose a Chern-Simons form I_{2n+1} satisfying

$$dI_{2n+1} = I_{2n+2}.$$

For an infinitesimal background gauge transformation with parameter α,

$$\delta_\alpha I_{2n+1} = dI_{2n}^{(1)}(\alpha).$$

The local anomalous variation of the Euclidean effective action is

$$\delta_\alpha W[A, g] = 2\pi \mathrm{i} \int_M I_{2n}^{(1)}(\alpha), \tag{152.19}$$

with an overall sign changed if the opposite chirality convention is chosen. The sign is not independent data; it is fixed by the orientation of D_A^+, the definition of $W = -\log Z$, and the chirality assignment of the fermion.

In four-dimensional QFT, the relevant six-form is

$$I_6 = \mathrm{ch}_3(E) - \frac{p_1(TM)}{24}\,\mathrm{ch}_1(E). \tag{152.20}$$

The first term gives cubic gauge anomalies. The second gives the mixed gauge-gravitational anomaly for a $U(1)$ background. For a semisimple group in a traceless representation, $\mathrm{ch}_1(E) = 0$, so the mixed term is absent. A background symmetry can be gauged only after the total anomaly line has been trivialized; cancellation of the local polynomial is necessary but does not by itself eliminate global anomalies.

152.8 Determinant Line and Families

As the metric and background connection vary, the chiral fermion determinant is not naturally a complex number. It is a section of the determinant line

$$\operatorname{Det} D = \bigwedge^{\max} \ker D_A^+ \otimes \left(\bigwedge^{\max} \ker D_A^-\right)^*$$

over background-field space. The local anomaly is the curvature of this line with its natural connection. Global anomalies are its holonomies around loops. Thus a well-defined background-field partition function requires a trivialization of the determinant line together with the vanishing of the local curvature obstruction. The source traces for the determinant-line and eta-holonomy formulation used in the next chapter are Bismut–Freed and Dai–Freed [60, 61].

The next chapter develops the eta-invariant formula for these holonomies. The present chapter supplies the local and zero-mode input: the index counts the imbalance of chiral zero modes, while the family of such indices organizes the determinant line.

Open Problem 152.6 (Index-theoretic anomaly line in interacting QFT)*.* Extend the determinant-line and index-theoretic description from free chiral fermions coupled to backgrounds to interacting continuum QFTs with composite currents, contact terms, and extended operators, while retaining an explicit comparison with local anomaly descent and global anomaly holonomy.

Chapter 153

Eta Invariants and Global Anomalies

Conventions in this chapter.

- **Signature.** Euclidean $\mathbb{R}^D$.
- **Metric sign.** positive metric $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ only after Hilbert-space reconstruction; otherwise brackets denote the stated functional expectation.
- **Generator convention.** Lorentzian generators introduced by continuation use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

Chapter 152 treated the local index density and the curvature of the determinant line. This chapter develops the complementary holonomy problem. A local anomaly is detected by infinitesimal motion in background-field space; a global anomaly is detected by transporting the fermion functional integral around a closed loop of backgrounds. For free fermions the transport is computed by eta invariants of Dirac operators on the associated mapping tori. The source traces for the global-analysis inputs are the APS spectral asymmetry papers, the Bismut–Freed holonomy theorem, Dai–Freed determinant lines, and Witten's $SU(2)$ anomaly [57, 58, 59, 60, 61, 62].

153.1 Self-Adjoint Dirac Operators and Eta Functions

Let Y be a closed oriented Riemannian spin manifold of odd dimension $2n+1$. Let $E \to Y$ be a Hermitian vector bundle with unitary connection ∇^E. The coupled Dirac operator

$$B_Y : C^\infty(Y, S_Y \otimes E) \longrightarrow C^\infty(Y, S_Y \otimes E)$$

is self-adjoint on $L^2(Y, S_Y \otimes E)$ when the Clifford action is chosen skew-Hermitian in Euclidean signature. Its spectrum is real, discrete, and has finite-dimensional eigenspaces. The eta function is initially defined by

$$\eta_{B_Y}(s) = \sum_{\lambda \in \mathrm{Spec}(B_Y)\setminus\{0\}} \mathrm{sign}(\lambda)|\lambda|^{-s}, \qquad \mathrm{Re}\, s \gg 1. \tag{153.1}$$

For a Dirac-type operator the heat-kernel expansion gives a meromorphic continuation to $s \in \mathbb{C}$, regular at $s = 0$. We define

$$h(B_Y) = \dim\ker B_Y, \qquad \xi(B_Y) = \frac{\eta_{B_Y}(0) + h(B_Y)}{2}. \tag{153.2}$$

The real number $\xi(B_Y)$ depends on the metric and connection, but $\exp(2\pi i\xi(B_Y))$ is the natural phase variable: when eigenvalues cross through zero, ξ changes by an integer.

Remark 153.1 (Integer jump at a simple crossing). Let B_t, $t \in (-\varepsilon, \varepsilon)$, be a smooth one-parameter family of self-adjoint Dirac-type operators. Suppose exactly one simple eigenvalue $\lambda(t)$ crosses zero, with $\lambda(t) < 0$ for $t < 0$ and $\lambda(t) > 0$ for $t > 0$, and all other eigenvalues stay away from zero. Then the contribution of this eigenvalue to $\xi(B_t)$ jumps by $+1$. A crossing in the opposite direction gives a jump by -1. Away from the crossing the eigenvalue contributes $\mathrm{sign}(\lambda)/2$ to $(\eta(0) + h)/2$. Just before the crossing this contribution is $-1/2$, at the crossing it is $1/2$ because $h = 1$, and just after the crossing it is $1/2$. Thus the one-sided jump from $t < 0$ to $t > 0$ is $+1$. If the eigenvalue crosses from positive to negative, the contribution changes from $+1/2$ before the crossing to $-1/2$ after the crossing, again with $h = 1$ at the crossing instant, so the one-sided jump is -1.

Definition 153.2 (Spectral flow)**.** For a smooth path B_t, $0 \le t \le 1$, whose endpoints are invertible, the spectral flow $\mathrm{sf}(B_t)$ is the signed number of eigenvalues crossing zero from negative to positive, counted with multiplicity. If the endpoints have kernels, one fixes a small perturbation or uses the APS convention below; the resulting quantity is well-defined modulo the endpoint-kernel bookkeeping appearing in ξ.

Remark 153.1 is the elementary reason that global anomalies are naturally phases rather than real-valued functions of $\eta(0)$ alone. The smooth part of ξ is local; the integer jumps record spectral flow.

153.2 The APS Boundary Formula

Let X be a compact oriented Riemannian spin manifold of even dimension $2n+2$, with boundary $\partial X = Y$. Assume that, on a collar $(-\epsilon, 0] \times Y$, the metric and the connection have product form. Let u be the inward normal coordinate. In this collar the chiral Dirac operator has the form

$$D_X^+ = c(\nu)\big(\partial_u + B_Y\big) \quad \text{up to the standard chiral identification,} \tag{153.3}$$

where ν is the outward unit normal and B_Y is the self-adjoint tangential Dirac operator on Y.

The APS boundary condition keeps the boundary values of spinors in the negative spectral subspace of B_Y:

$$P_{\ge 0}\big(\psi|_Y\big) = 0, \tag{153.4}$$

where $P_{\ge 0}$ is the orthogonal projection onto the span of eigenvectors with eigenvalue ≥ 0. Including the zero eigenspace in $P_{\ge 0}$ is the convention paired with the reduced invariant $\xi = (\eta + h)/2$.

Half-cylinder sign check. The sign in (153.4) is fixed by the product collar, not by an arbitrary convention. Replace the collar by the infinite inward cylinder $(-\infty, 0] \times Y$ with the same model operator

$$c(\nu)(\partial_u + B_Y).$$

If $B_Y \phi_\lambda = \lambda\phi_\lambda$ and $\psi(u, y) = f_\lambda(u)\phi_\lambda(y)$, the equation $(\partial_u + B_Y)\psi = 0$ becomes

$$\frac{df_\lambda}{du} + \lambda f_\lambda = 0, \qquad f_\lambda(u) = e^{-\lambda u} f_\lambda(0).$$

Since the completed cylinder runs to $u = -\infty$, this mode is square integrable exactly when $\lambda < 0$. Thus the boundary values of L^2 solutions on the completed model lie in the negative spectral subspace, while zero modes are not square-integrable on the infinite cylinder and are therefore placed in the forbidden subspace $P_{\geq 0}$. On an oppositely oriented boundary component the tangential operator is $-B_Y$, so the same mode calculation reverses the spectral sign. This elementary model is not the APS Fredholm theorem; it fixes the boundary-orientation convention used in the eta and anomaly formulae below.

Quoted Theorem 153.3 (Atiyah–Patodi–Singer index formula)**.** Source trace. *This is the APS boundary index theorem in the reduced eta convention used in this chapter [57, 58, 59]. For the product-collar data above, the chiral Dirac operator with APS boundary condition is Fredholm and*

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(E)\big]_{2n+2} - \xi(B_Y). \tag{153.5}$$

For non-product data one first deforms to product form near the boundary or adds the corresponding local transgression term. The sum of bulk density, local boundary transgression, and eta term is the invariant integer.

Analytic mechanism behind the boundary term. The theorem is used below through three precise pieces of analytic structure. First, product form near the boundary identifies the normal part of the Dirac operator with the first-order operator (153.3). The spectral projection $P_{\geq 0}$ then makes the boundary problem elliptic in the pseudodifferential sense: boundary values in the forbidden spectral subspace are removed, while the remaining components give a Fredholm operator. Second, the small-time heat-kernel expansion of the APS problem has the same interior coefficient $[\widehat{A}(TX)\,\mathrm{ch}(E)]_{2n+2}$ as the closed index theorem. This is the local part of the formula and is the source of the usual anomaly polynomial. Third, the APS boundary condition is nonlocal along Y, and its nonlocal contribution is exactly the reduced spectral asymmetry $\xi(B_Y)$. The integer on the left side of (153.5) is the compatibility condition tying these local and nonlocal terms together.

In this chapter the APS theorem functions as the global-analysis bridge between two objects already defined above: the local index density on X and the spectral invariant of the boundary Dirac operator B_Y. The QFT statement extracted from it is the following: local anomaly density determines the infinitesimal variation of the fermion phase, while eta invariants retain the integral spectral-flow data that can survive after the local curvature has vanished.

Cylinder variation of the reduced eta invariant. Let B_t, $0 \leq t \leq 1$, be a smooth path of odd-dimensional Dirac operators obtained from a smooth path of metrics and unitary connections on a fixed closed odd-dimensional spin manifold Y. Let

$$X = [0,1] \times Y$$

carry the interpolating product-near-boundary data. Then

$$\xi(B_1) - \xi(B_0) \equiv \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(E)\big]_{2n+2} \pmod{\mathbb{Z}}. \tag{153.6}$$

If the path has no zero crossings and the endpoint kernels have constant dimension, differentiating this formula gives the usual local descent variation of ξ. This is the cylinder specialization of APS. The boundary of $X = [0,1] \times Y$ is

$$\partial X = Y_1 \sqcup (-Y_0).$$

The tangential operator on Y_1 is B_1, while the tangential operator on the oppositely oriented component $-Y_0$ is $-B_0$. Applying (153.5) to X gives

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(E)\big]_{2n+2} - \xi(B_1) - \xi(-B_0).$$

Since $\eta_{-B_0}(0) = -\eta_{B_0}(0)$ and $h(-B_0) = h(B_0)$, one has

$$\xi(-B_0) = -\xi(B_0) + h(B_0).$$

The Fredholm index and $h(B_0)$ are integers. Moving them to the other side gives (153.6) modulo $\mathbb{Z}$.

Exact cylinder bookkeeping. The preceding congruence is often the only part of the APS cylinder formula that survives in anomaly phases, but the exact integer identity is what fixes the sign conventions. With

$$I_X(E) = \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(E)\big]_{2n+2},$$

the APS formula on $\partial X = Y_1 \sqcup (-Y_0)$ gives

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = I_X(E) - \xi(B_1) + \xi(B_0) - h(B_0). \tag{153.7}$$

Equivalently,

$$\xi(B_1) - \xi(B_0) = I_X(E) - h(B_0) - \mathrm{Ind}(D_X^{+,\mathrm{APS}}). \tag{153.8}$$

Thus the endpoint-kernel convention is not an afterthought: including the zero eigenspace in $P_{\geq 0}$ at the incoming boundary contributes the integer $h(B_0)$ in the exact formula and then disappears only after passing to $\mathbb{R}/\mathbb{Z}$. When B_0 and B_1 are invertible, (153.7) becomes

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = I_X(E) - \big(\xi(B_1) - \xi(B_0)\big).$$

If the path has only transverse simple crossings, the integer jump part of $\xi(B_1) - \xi(B_0)$ is exactly the spectral flow $\mathrm{sf}(B_t)$, by Remark 153.1. The remaining smooth variation is represented by the local transgression term whose cylinder integral is $I_X(E)$; the APS index supplies the integer needed to make the exact equality hold. Consequently, in the orientation and APS boundary convention of (153.3)–(153.4), the pure crossing contribution satisfies

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}})_{\mathrm{cross}} = -\,\mathrm{sf}(B_t).$$

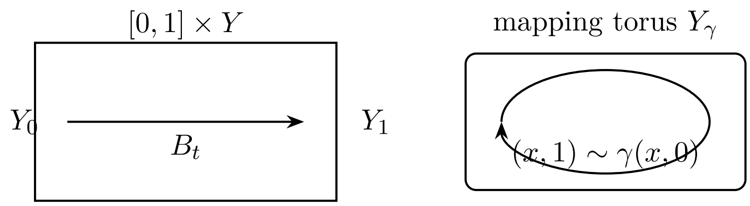

Figure 153.1: The APS cylinder computes the local change of ξ modulo integers. Identifying the two ends by a closed loop of backgrounds gives the mapping torus whose eta invariant computes determinant-line holonomy.

This minus sign is a boundary-orientation convention, not a physical phase: the exponentiated eta phase only remembers the congruence (153.6), while the exact identity above records the Fredholm representative used to compute it.

Figure 153.1 records the cylinder/mapping-torus geometry used in the eta-variation formula: interpolating between boundary operators converts eta change into an APS index modulo integers.

153.3 Determinant-Line Holonomy

Let M be a closed even-dimensional spin manifold, and let $\mathcal{B}$ be a smooth parameter space of background metrics and connections, before quotienting by large gauge transformations. A family of chiral Dirac operators

$$D_b^+ : \Gamma(M, S^+ \otimes E_b) \longrightarrow \Gamma(M, S^- \otimes E_b), \qquad b \in \mathcal{B},$$

defines the determinant line

$$\operatorname{Det} D_b = \bigwedge^{\text{top}} \ker D_b^+ \otimes \Big(\bigwedge^{\text{top}} \ker D_b^-\Big)^* . \tag{153.9}$$

Where kernels jump this formula is not a literal vector bundle chart, but the Quillen determinant construction glues these local descriptions to a smooth complex line bundle over $\mathcal{B}$. The regulated chiral fermion functional integral is a section of this line, not a function, until a trivialization is chosen.

Let $\gamma : S^1 \to \mathcal{B}$ be a loop of backgrounds. The mapping torus

$$Y_\gamma = ([0,1]\times M)/\big((1,x)\sim(0,x) \text{ using } \gamma\big) \tag{153.10}$$

is a closed odd-dimensional spin manifold with the induced bundle and connection. The Dirac operator on Y_γ will be denoted B_{Y_γ}.

Quoted Theorem 153.4 (Bismut–Freed holonomy formula)**.** Source trace. *The determinant-line connection and eta-holonomy formula are the Bismut–Freed holonomy theorem in the Dirac-family case [60]. With the determinant-line and chirality conventions of* (153.9)*, the natural Bismut–Freed connection on* $\operatorname{Det} D$ *has holonomy*

$$\operatorname{Hol}_\gamma(\operatorname{Det} D) = \exp\!\big(-2\pi \mathrm{i}\, \xi(B_{Y_\gamma})\big) . \tag{153.11}$$

Its curvature is the fiber integral over M *of the degree* $(\dim M + 2)$ *component of* $\widehat{A}(TM)\operatorname{ch}(E)$*.*

Local connection forms and holonomy coordinates. The theorem supplies the analytic determinant-line connection. Before using its particular curvature, we record the elementary line-bundle bookkeeping that turns such a connection into a global-anomaly phase. Let $\{U_i\}$ be a cover of a patch of $\mathcal{B}$, and choose nonzero local frames e_i of the determinant line. Write the normalized connection one-form in the convention

$$\nabla e_i = -2\pi \mathrm{i}\, a_i \otimes e_i, \qquad a_i \in \Omega^1(U_i;\mathbb{R}).$$

On an overlap $U_i \cap U_j$, write

$$e_j = \exp(2\pi \mathrm{i} f_{ij})\, e_i, \qquad f_{ij} : U_i \cap U_j \longrightarrow \mathbb{R}/\mathbb{Z}.$$

After choosing a real lift of f_{ij} on a contractible part of the overlap, substituting this transition rule into the connection gives

$$a_j = a_i - df_{ij}. \tag{153.12}$$

Therefore $da_i = da_j$ on overlaps, and the curvature is represented by a single global closed two-form. In the Bismut–Freed determinant line this global two-form is the family-index form stated in Theorem 153.4.

Now let a loop γ be subdivided into arcs $\gamma_r : [t_r, t_{r+1}] \to U_{i_r}$, with $\gamma(t_N) = \gamma(t_0)$, and let $\gamma(t_{r+1}) \in U_{i_r} \cap U_{i_{r+1}}$. Parallel transport along the arc written in the i_r-frame contributes $\exp(2\pi \mathrm{i} \int_{\gamma_r} a_{i_r})$. At the next vertex, the change of frame from e_{i_r} to $e_{i_{r+1}}$ contributes $\exp(-2\pi \mathrm{i} f_{i_r i_{r+1}}(\gamma(t_{r+1})))$. Hence the holonomy is

$$\mathrm{Hol}_\gamma(\nabla) = \exp\left[2\pi \mathrm{i}\left(\sum_r \int_{\gamma_r} a_{i_r} - \sum_r f_{i_r i_{r+1}}(\gamma(t_{r+1}))\right)\right]. \tag{153.13}$$

Changing frames by $e_i \mapsto \exp(2\pi \mathrm{i} h_i) e_i$ sends

$$a_i \mapsto a_i - dh_i, \qquad f_{ij} \mapsto f_{ij} + h_j - h_i.$$

The changes in the arc integrals and the vertex transition terms telescope in (153.13). Thus the closed-loop holonomy is independent of the local frame. On a contractible loop contained in a single patch, (153.13) reduces by Stokes to the exponential of the curvature integral over a spanning disk. On a noncontractible loop the curvature may vanish while (153.13) remains nontrivial; this is exactly the flat global-anomaly possibility later detected by based-loop holonomy in the quotient background groupoid.

Mechanism of determinant-line transport. The determinant line over $\mathcal{B}$ has local trivializations obtained by choosing a spectral cut for $D_b^- D_b^+$ and zeta-regularizing the nonzero eigenvalue product. Changing the cut changes the trivialization by the finite determinant of the eigenmodes crossing the cut, so these local charts glue to the Quillen determinant line. The Bismut–Freed connection is the connection whose local connection form is the finite part of the variation of this regularized determinant, corrected by the heat-kernel counterterm that makes the answer independent of the spectral cut. Its curvature is therefore local: it is the degree-two form on $\mathcal{B}$ obtained by integrating $[\widehat{A}(TM)\,\mathrm{ch}(E)]_{\dim M+2}$ over the physical manifold M.

The spectral-cut chart can be described without suppressing the finite transition map. Fix a positive number a which is not in the spectrum of $K_b^+ = D_b^- D_b^+$. Let

$$E^+_{<a}(b) = \bigoplus_{\lambda<a} \ker(K_b^+ - \lambda), \qquad E^-_{<a}(b) = D_b^+ E^+_{(0,a)}(b) \oplus \ker D_b^-.$$

Here $E^+_{(0,a)}$ excludes the zero modes. The Fredholm determinant line is represented in this chart by

$$L_a(b) = \det(E^+_{<a}(b)) \otimes \det(E^-_{<a}(b))^*.$$

If $a < c$ are two cuts avoiding the spectrum, write

$$E^\pm_{[a,c)}(b) = E^\pm_{<c}(b) \ominus E^\pm_{<a}(b).$$

On this finite-dimensional window, D^+_b is an isomorphism

$$D^+_{[a,c)}(b) : E^+_{[a,c)}(b) \longrightarrow E^-_{[a,c)}(b),$$

and the transition from the a-chart to the c-chart is multiplication by the determinant line element of this isomorphism:

$$\det D^+_{[a,c)}(b) \in \det E^-_{[a,c)}(b) \otimes \det E^+_{[a,c)}(b)^*.$$

For three cuts $a < c < d$, the window splits as

$$E^\pm_{[a,d)} = E^\pm_{[a,c)} \oplus E^\pm_{[c,d)},$$

and determinant multiplicativity gives

$$\det D^+_{[a,d)} = \det D^+_{[c,d)} \, \det D^+_{[a,c)}.$$

This cocycle identity is the finite-dimensional algebra behind the gluing of the local determinant-line charts. The zeta-regularized high-mode determinant and the heat-kernel counterterm supply the infinite-dimensional continuation of the same transition rule.

Compatibility with eta jumps. The determinant-line chart transition and the reduced eta invariant describe the same zero-mode bookkeeping from two different sides. Consider first one real eigenvalue $\lambda(t)$ of a self-adjoint tangential operator B_t crossing through zero from negative to positive, with all other eigenvalues kept fixed away from zero. Its contribution to the reduced eta invariant is

$$\xi_\lambda(t) = \begin{cases} -\frac{1}{2}, & \lambda(t) < 0, \\ \frac{1}{2}, & \lambda(t) = 0, \\ \frac{1}{2}, & \lambda(t) > 0, \end{cases}$$

where the middle value comes from the kernel term $h(B_t)$. The one-sided change of ξ is therefore $+1$, and a downward crossing gives -1. Thus the exponentiated phase $\exp(-2\pi \mathrm{i}\, \xi)$ is continuous across such an integer jump.

In the determinant-line construction the same event appears as a change of low-mode chart. Near a kernel jump, the zeta determinant of the complement and the zero-mode factor carried by the finite-dimensional determinant line $L_a(b)$ together form the fermion determinant vector. When the cut is moved across a small eigenvalue window, the local scalar determinant changes by the finite determinant $\det D^+_{[a,c)}$, while the basis of the determinant-line chart changes by the inverse transition element. Their product is a vector in the Quillen line and is independent of the auxiliary spectral cut. The integer spectral-flow term in the eta description is precisely the phase shadow of this finite chart transition. Consequently the smooth local curvature, the finite spectral-flow jumps, and the mapping-torus eta phase are the curvature, transition, and holonomy data of the same determinant line.

For a loop γ, the holonomy problem is converted into an index problem on the odd-dimensional mapping torus Y_γ. One stretches the circle direction adiabatically. The local curvature contribution becomes the transgression of the family index density, while the finite part left by the adiabatic limit is the reduced eta invariant of the Dirac operator on Y_γ. This is the content of (153.11). Thus the local anomaly polynomial is the curvature of the anomaly line, and a global anomaly is the holonomy of the same line around a closed path in the background groupoid. Cancellation of the local anomaly means that the total anomaly line is flat; cancellation of the global anomaly requires the holonomy of that flat line to be 1 on every closed loop.

Finite gauge transformation as determinant-line transport. Let g be a gauge transformation of a background bundle on M, and let A_t, $0 \le t \le 1$, be a path from A to A^g. The ratio between the transported determinant at A^g and the determinant at A is computed by (153.11) on the mapping torus

$$Y_g = ([0,1] \times M)/\big((1,x) \sim (0,gx)\big). \tag{153.14}$$

If this phase is not 1, the chiral fermion measure does not descend to a function on the quotient by that gauge transformation. Parallel transport along A_t identifies $\operatorname{Det} D_A$ with $\operatorname{Det} D_{A^g}$ by the Bismut–Freed connection on the determinant line bundle over background fields. The gauge transformation g gives a canonical pullback identification of D_{A^g} with D_A, hence of their determinant lines. Composing the parallel transport with this gauge identification gives an automorphism of the one-dimensional line $\operatorname{Det} D_A$, so it is multiplication by a phase. The corresponding closed path in the quotient background groupoid is represented geometrically by gluing the two ends of $[0,1] \times M$ with g, namely by the mapping torus Y_g. The Bismut–Freed holonomy formula applied to this closed family gives (153.11). A phase different from 1 means that the determinant vector changes when the same quotient background is described by the two representatives A and A^g; hence the chiral fermion measure does not descend to a function on the quotient.

Descent to the quotient background groupoid. This last statement is the precise descent criterion for a global gauge anomaly. Let $\mathcal{B}$ be the space of smooth background fields before quotienting, and let $\mathcal{G}$ be the group of gauge transformations acting on it. The quotient is the action groupoid

$$\mathcal{B}//\mathcal{G},$$

whose objects are $A \in \mathcal{B}$ and whose morphisms

$$A \longrightarrow A^g$$

are labelled by $g \in \mathcal{G}$. A fermion determinant line over this groupoid consists of a line L_A over every object together with isomorphisms

$$\Phi_g(A) : L_A \longrightarrow L_{A^g}$$

compatible with composition. In the free chiral-fermion case, $\Phi_g(A)$ is obtained by composing Bismut–Freed parallel transport along a path from A to A^g with the tautological pullback identification by g. Changing the path changes the result by the holonomy of a closed loop, hence by an eta phase of the corresponding mapping torus.

A partition function is a function on the quotient only after choosing nonzero vectors $s_A \in L_A$ satisfying

$$\Phi_g(A)s_A = s_{A^g} \qquad \text{for every } (A,g). \tag{153.15}$$

If one fixes a local trivialization of L, the same condition is expressed by a $U(1)$-valued groupoid 1-cocycle

$$\alpha(A,g) \in U(1), \qquad \Phi_g(A)e_A = \alpha(A,g)e_{A^g},$$

obeying

$$\alpha(A,gh) = \alpha(A,g)\,\alpha(A^g,h). \tag{153.16}$$

A change of local trivialization $e_A \mapsto \beta(A)e_A$ changes α by the coboundary

$$\alpha(A,g) \longmapsto \frac{\beta(A)}{\beta(A^g)}\,\alpha(A,g). \tag{153.17}$$

Thus the obstruction is the cohomology class of this groupoid cocycle. On a based loop (A,g) with $A^g = A$, the value $\alpha(A,g)$ is invariant under (153.17); it is exactly the holonomy computed by (153.11) or by the mod-two Pfaffian formula below. Tensoring independent fermion systems tensors their anomaly lines, so the cocycles multiply and the logarithmic anomaly exponents add.

This based-loop statement has a useful fixed-background form. Let

$$\mathcal{G}_A = \{g \in \mathcal{G} : A^g = A\}$$

be the stabilizer of a background. Restricting (153.16) to $g,h \in \mathcal{G}_A$ gives a character

$$\chi_A(g) = \alpha(A,g), \qquad \chi_A(gh) = \chi_A(g)\chi_A(h).$$

The coboundary factor in (153.17) is trivial on the stabilizer, because $A^g = A$, so χ_A is not changed by a local choice of frame. If an equivariant nonzero section s_A satisfying (153.15) exists, then $\chi_A(g) = 1$ for every stabilizer element. The converse is a concrete descent theorem in the finite action-groupoid model.

Proposition 153.5 (Finite action-groupoid descent criterion)**.** *Let a finite group G act on a finite set $\mathcal{B}_0$ of background representatives on the right, and write the action groupoid as $\mathcal{B}_0//G$. Let*

$$a(A,g) \in \mathbb{R}/\mathbb{Z}$$

be an additive representative of a line cocycle,

$$a(A,gh) = a(A,g) + a(A^g,h). \tag{153.18}$$

Then a is a coboundary,

$$a(A,g) = b(A^g) - b(A) \tag{153.19}$$

for some 0-cochain $b : \mathcal{B}_0 \to \mathbb{R}/\mathbb{Z}$, if and only if its stabilizer character vanishes on every orbit,

$$a(A,h) = 0 \qquad \text{for all } h \in G_A. \tag{153.20}$$

Equivalently, the multiplicative line cocycle $\alpha(A,g) = \exp(2\pi \mathrm{i} a(A,g))$ is trivializable by a choice of local frames exactly when all based-loop holonomies are 1.

Proof. If $a(A,g) = b(A^g) - b(A)$, then for a stabilizer element $h \in G_A$ one has $A^h = A$, so $a(A,h) = b(A) - b(A) = 0$. Conversely, choose one representative A_o in each orbit and set $b(A_o) = 0$. If $A = A_o^g$, define

$$b(A) = a(A_o, g).$$

This is independent of the chosen g. Indeed, if $A_o^g = A_o^{g'}$, then $g' = hg$ for some $h \in G_{A_o}$, and the cocycle identity gives

$$a(A_o, g') = a(A_o, hg) = a(A_o, h) + a(A_o^h, g) = a(A_o, g),$$

because $A_o^h = A_o$ and the stabilizer character vanishes. For an arbitrary arrow $A = A_o^g \to A^k = A_o^{gk}$, the same cocycle identity gives

$$b(A^k) - b(A) = a(A_o, gk) - a(A_o, g) = a(A_o^g, k) = a(A, k).$$

Thus the required b exists. Exponentiating a and b gives the multiplicative statement and matches the frame-change law (153.17). □

This finite theorem is the algebraic core used by the smooth anomaly line. For connected transformations the Bismut–Freed curvature and the local descent calculation determine the infinitesimal part of a. After that curvature has been cancelled, the remaining flat line is detected on stabilizers and other closed loops by the eta or mod-two mapping-torus holonomy. Thus a global anomaly may be found before considering arbitrary families: a nontrivial automorphism character of the anomaly line over one background already prevents the determinant or Pfaffian vector from being a function on the quotient stack of backgrounds.

Dual anomaly lines and the cancellation criterion. Let L be an anomaly line over $\mathcal{B}//\mathcal{G}$. Its dual line $L^\vee$ has fibers

$$L_A^\vee = \mathrm{Hom}(L_A, \mathbb{C}).$$

If $\Phi_g(A) : L_A \to L_{A^g}$ is the transport of L, then the transport of $L^\vee$ is the inverse dual map

$$\Phi_g^\vee(A) = \left(\Phi_g(A)^{-1}\right)^* : L_A^\vee \longrightarrow L_{A^g}^\vee.$$

In a local trivialization, if L is represented by the cocycle $\alpha(A,g)$, then $L^\vee$ is represented by $\alpha(A,g)^{-1}$. Hence

$$\alpha_{L\otimes L^\vee}(A,g) = \alpha(A,g)\alpha(A,g)^{-1} = 1. \tag{153.21}$$

The tensor product $L \otimes L^\vee$ is therefore canonically trivialized by the evaluation pairing. More generally, anomaly cancellation for a collection of fermion systems is the existence of a groupoid-equivariant trivialization of the tensor product of their determinant or Pfaffian lines. Vanishing of the local curvature only says that this tensor product is flat. It still must have trivial based-loop holonomy, because a nontrivial flat line gives a well-defined local anomaly density equal to zero while failing (153.15) on a noncontractible loop of backgrounds. For real Pfaffian lines the inverse line is again a real line; the statement becomes multiplication of signs. Thus two identical four-dimensional $SU(2)$ Weyl doublets cancel the Witten sign because $(-1)^2 = 1$, whereas one doublet gives a nontrivial flat Pfaffian line over the large-gauge loop.

For an infinitesimal gauge transformation connected to the identity, the linearization of α is the consistent local anomaly derived in Chapter 51. For a large gauge transformation such as the generator of $\pi_4(SU(2))$, the local curvature may vanish while the based loop value of α is nontrivial. This is the logical distinction between local anomaly cancellation and global anomaly cancellation.

153.4 Pfaffian Lines and Mod-Two Indices

Some fermion systems have a real or pseudoreal structure that replaces the complex determinant line by a Pfaffian line. The four-dimensional $SU(2)$ fundamental Weyl fermion is the basic example. The fundamental representation is pseudoreal, and the spin representation supplies an anti-linear symmetry that makes the Euclidean quadratic form skew on a real vector space. The path integral is then a Pfaffian. Its possible global anomaly is a sign.

Definition 153.6 (Mod-two index)**.** Let Y be a closed spin manifold of dimension $8k+5$, and let E be a pseudoreal bundle for which the coupled Dirac operator has the corresponding real skew structure. The mod-two index is

$$\mathrm{Ind}_2(B_Y) = \dim\ker B_Y \pmod 2. \tag{153.22}$$

The parity is stable under continuous deformations preserving the real structure. The reason is that eigenvalues away from zero occur in pairs $\lambda, -\lambda$ for the real skew structure relevant to the Pfaffian. Under a deformation, nonzero eigenvalues can enter or leave the kernel only through such paired crossings. Thus the dimension of the kernel can change only by an even integer, and $\dim\ker B_Y$ modulo 2 is invariant.

For such a Pfaffian line the Dai–Freed holonomy specializes to

$$\mathrm{Hol}_\gamma(\mathrm{Pf}\, D) = (-1)^{\mathrm{Ind}_2(B_{Y_\gamma})}. \tag{153.23}$$

This formula is the sign analogue of (153.11).

From determinant phases to Pfaffian signs. The Pfaffian line is a real line, and its holonomy has values in $\{\pm 1\}$. In a real skew Fredholm family, the nonzero spectrum of the self-adjoint representative obtained by multiplying the skew operator by i is symmetric about zero. Along a closed path, the orientation of the Pfaffian line changes precisely when an odd number of eigenvalue pairs passes through zero. The parity of this crossing number is the mod-two index of the Dirac operator on the mapping torus. Formula (153.23) is therefore the real-line analogue of the Bismut–Freed eta phase: the exponent is no longer a reduced eta invariant modulo integers, but a $\mathbb{Z}_2$-valued index.

Finite skew model for the Pfaffian line. The sign mechanism can be seen without functional analysis in a finite oriented real vector space. Let $V = \mathbb{R}^2$ with oriented basis (e_1, e_2), and let

$$q_a = a\, e^1 \wedge e^2$$

be a skew bilinear form. Its Pfaffian in this oriented basis is

$$\mathrm{pf}(q_a) = a, \qquad \det(q_a) = a^2. \tag{153.24}$$

Thus the determinant sees only $|a|$, while the Pfaffian remembers the orientation of the real line generated by q_a. If $a(t)$ crosses through zero once, the real Pfaffian coordinate changes sign. At the crossing the skew form has a two-dimensional kernel; away from the crossing the associated self-adjoint block has a paired nonzero spectrum. This is the finite-dimensional version of the statement that a real skew Fredholm family changes Pfaffian orientation only through paired zero-mode crossings.

For a direct sum of such blocks,

$$(V, q) = \bigoplus_{r=1}^{N} (\mathbb{R}^2, q_{a_r}),$$

with the product orientation, the Pfaffian is multiplicative:

$$\mathrm{pf}(q) = \prod_{r=1}^{N} a_r. \tag{153.25}$$

Consequently the holonomy of the real Pfaffian line around a closed family is the parity of the number of block signs that change:

$$\mathrm{Hol}(\mathrm{Pf}) = (-1)^{N_{\mathrm{cross}} \bmod 2}. \tag{153.26}$$

The mod-two index in (153.23) is the infinite-dimensional, elliptic version of this finite orientation parity. The mapping-torus Dirac operator supplies a canonical way to count the parity of the zero-mode crossings without choosing coordinates along the loop of backgrounds.

153.5 The Witten $SU(2)$ Global Anomaly

Let $M = S^4$, and let $g : S^4 \to SU(2)$ represent the nontrivial element of

$$\pi_4(SU(2)) \simeq \mathbb{Z}_2.$$

Form the mapping torus Y_g as in (153.14). A single left-handed Weyl fermion in the fundamental representation of $SU(2)$ has Pfaffian holonomy

$$\mathrm{Hol}_g(\mathrm{Pf}\, D) = -1. \tag{153.27}$$

Equivalently,

$$\mathrm{Ind}_2(B_{Y_g,\square}) = 1. \tag{153.28}$$

Quoted Theorem 153.7 (Mod-two index theorem for the Witten mapping torus)**.** Source trace. *The global $SU(2)$ anomaly and its mod-two mapping-torus index formulation trace to Witten's anomaly computation, with the determinant/Pfaffian-line language aligned with the Dai–Freed framework [62, 61]. Let R_j be the irreducible $SU(2)$ representation of isospin $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, and set $n = 2j$. With the trace-delta normalization of Chapter 152,*

$$\mathrm{tr}_{R_j}(t^a t^b) = T_\Delta(R_j)\delta^{ab}, \qquad T_\Delta(R_j) = \frac{n(n+1)(n+2)}{6}. \tag{153.29}$$

For the mapping torus of the generator of $\pi_4(SU(2))$,

$$\mathrm{Ind}_2(B_{Y_g,R_j}) \equiv T_\Delta(R_j) \pmod 2. \tag{153.30}$$

Clutching datum behind the mapping torus. The QFT object to which Theorem 153.7 is applied is the coupled Dirac operator on the associated bundle over the mapping torus determined by $g : S^4 \to SU(2)$. Start with the trivial principal $SU(2)$-bundle over S^4. In a trivialization write a point as (x, u), $x \in S^4$, $u \in SU(2)$. The clutching bundle over $Y_g = S^1 \times_g S^4$ is

$$P_g = ([0,1] \times S^4 \times SU(2))/\big((1, x, u) \sim (0, x, g(x)u)\big). \tag{153.31}$$

For a representation R, the bundle entering the Dirac operator is

$$E_{g,R} = P_g \times_{SU(2)} R.$$

Choose a smooth path of connections A_t from a connection A to its gauge transform A^g, product near $t = 0, 1$. The connection A_t on the cylinder descends through the clutching relation to a connection on P_g. Changing the interpolation A_t changes the mapping-torus connection through a homotopy of connections and hence does not change the mod-two index.

Two pieces of the setup precede the global mod-two index formula. First, if g_s, $0 \le s \le 1$, is a homotopy of gauge transformations, the same clutching construction over $[0,1]_s \times [0,1]_t \times S^4$, with the t-ends identified by g_s, gives a six-dimensional bordism between (Y_{g_0}, P_{g_0}) and (Y_{g_1}, P_{g_1}). The mod-two index is a bordism invariant, so the sign depends only on the homotopy class $[g] \in \pi_4(SU(2))$. Second, concatenating two gauge transformations amounts to gluing the corresponding mapping cylinders along an intermediate copy of S^4; additivity of the real mod-two index under gluing gives

$$\mathrm{Ind}_2(B_{Y_{g_0 g_1},R}) = \mathrm{Ind}_2(B_{Y_{g_0},R}) + \mathrm{Ind}_2(B_{Y_{g_1},R}) \pmod 2. \tag{153.32}$$

Thus the Pfaffian sign is a character of the component group of the gauge group. For $SU(2)$ on S^4, this component group is $\pi_4(SU(2)) \simeq \mathbb{Z}_2$. The nontrivial content of Theorem 153.7 is the value of this character on the generator, and its representation dependence. The clutching construction above is the part of the global-anomaly argument that belongs directly to the QFT background-field setup.

Trace-delta normalization in the $SU(2)$ representation ring. The arithmetic in Theorem 153.7 uses the same trace convention as the instanton index formula (152.16). Let J_a be the usual Hermitian spin-j generators with $[J_a, J_b] = \mathrm{i}\epsilon^c{}_{ab} J_c$ and J_3-weights

$$m = -j, -j+1, \ldots, j.$$

Rotational invariance of the irreducible representation gives

$$\mathrm{tr}_{R_j}(J_a J_b) = C_j \delta_{ab}.$$

Taking $a = b = 3$ fixes

$$C_j = \sum_{m=-j}^{j} m^2 = \frac{1}{3} j(j+1)(2j+1). \tag{153.33}$$

The monograph's trace-delta basis is $t_a = \sqrt{2}\, J_a$, since the fundamental spin-$\frac{1}{2}$ representation then has $\mathrm{tr}_{\square}(t_a t_b) = \delta_{ab}$. Therefore

$$T_\Delta(R_j) = 2C_j = \frac{2}{3} j(j+1)(2j+1) = \frac{n(n+1)(n+2)}{6}, \qquad n = 2j. \tag{153.34}$$

This derivation fixes the normalization of the integer that appears in the mod-two index theorem; it is not an independent anomaly calculation.

The parity of (153.34) is equally explicit:

$$T_\Delta(R_j) = \binom{n+2}{3}.$$

Lucas' binary criterion for binomial coefficients modulo 2 says that $\binom{N}{3}$ is odd exactly when the two lowest binary digits of N are both 1. With $N = n+2$, this is

$$n+2 \equiv 3 \pmod 4, \qquad \text{equivalently} \qquad n = 2j \equiv 1 \pmod 4. \tag{153.35}$$

Thus the quoted global-analysis input is the mod-two-index equality (153.30); the representation normalization and the $2j \equiv 1 \pmod 4$ criterion are ordinary finite-dimensional $SU(2)$ arithmetic in the chapter's trace convention.

Topological content of the mapping-torus index. The theorem is the point where topology enters the four-dimensional anomaly. The gauge transformation g representing the nontrivial element of $\pi_4(SU(2))$ produces a spin five-manifold Y_g equipped with an $SU(2)$ bundle, and the pair (Y_g, E_{R_j}) defines an element tested by the mod-two Dirac index. For the fundamental representation this element has odd mod-two index; this is the genuinely global datum invisible to the local four-dimensional cubic anomaly polynomial. The theorem further identifies the parity for a general irreducible representation with the trace-delta Dynkin index modulo 2, giving (153.30). Its proof uses the naturality of the coupled Dirac operator under representation operations together with a five-dimensional real-index computation for the generator of the $\pi_4(SU(2))$ mapping torus. The QFT analysis below uses the result only through the explicit parity formula displayed in the theorem.

Once (153.30) is fixed, the remaining anomaly criterion is ordinary arithmetic in the same trace convention used for the local anomaly polynomial.

The isospin-j Weyl fermion has the original spin $SU(2)$ global anomaly precisely when $2j \equiv 1 \pmod 4$. A collection of Weyl fermions in representations R_{j_α} is free of this anomaly precisely when

$$\sum_\alpha T_\Delta(R_{j_\alpha}) \equiv 0 \pmod 2. \tag{153.36}$$

Write $n = 2j$. The integer

$$T_\Delta(R_j) = \frac{n(n+1)(n+2)}{6}$$

is the binomial coefficient $\binom{n+2}{3}$. Its parity is given by Lucas' theorem: $\binom{N}{3}$ is odd exactly when the binary digits of 3 are contained in those of N. Since 3 has binary expansion $\cdots 0011$, this means $N = n+2 \equiv 3 \pmod 4$, or

$$n \equiv 1 \pmod 4.$$

Equation (153.30) then gives the Pfaffian sign. For several Weyl fermions the Pfaffian line is the tensor product of the individual Pfaffian lines, so the signs multiply and the exponents add modulo 2.

Example 153.8 (First few $SU(2)$ representations)**.** The sequence of trace-delta indices is

j	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$
$T_\Delta(R_j)$	1	4	10	20	35.

Thus a single doublet has the anomaly, a single triplet does not, a single four-dimensional representation does not, and a single six-dimensional representation has the anomaly. The vanishing of the ordinary four-dimensional cubic $SU(2)$ anomaly does not decide this question because $SU(2)$ has no independent symmetric cubic invariant.

153.6 Dai–Freed Inflow

The determinant or Pfaffian of a d-dimensional fermion is a vector in an anomaly line over the background groupoid. A $(d+1)$-dimensional invertible field theory trivializes this line when its boundary variation is the inverse of the fermion-line variation. In the free-fermion model the closed $(d+1)$-manifold partition function is the eta phase [61]:

$$Z_{\mathrm{DF}}(Y) = \exp(2\pi \mathrm{i}\, \xi(B_Y)) \tag{153.37}$$

for a complex chiral determinant, with the corresponding mod-two sign in the real Pfaffian case.

If Y has boundary M, the APS construction assigns not a number but a vector in the inverse determinant line over the boundary background on M. Pairing this vector with the boundary fermion determinant gives a number. Changing the filling changes the trivialization by the closed-manifold phase obtained by gluing the two fillings. This is the precise geometric meaning of anomaly inflow in the eta-invariant model.

Closed phases from bounding manifolds. The closed phase (153.37) is constrained by the APS formula whenever the odd-dimensional pair (Y, E) bounds an even-dimensional pair $(X, \widetilde{E})$ with product structure near $Y = \partial X$. In the convention of (153.5),

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(\widetilde{E})\big]_{\dim X} - \xi(B_Y).$$

The index is an integer, hence

$$\exp(2\pi \mathrm{i}\, \xi(B_Y)) = \exp\left(2\pi \mathrm{i} \int_X \big[\widehat{A}(TX)\,\mathrm{ch}(\widetilde{E})\big]_{\dim X}\right) \tag{153.38}$$

for bounding data. If X and X' are two such fillings, gluing X to $-X'$ gives a closed even-dimensional spin manifold. The closed Atiyah–Singer theorem says that the difference of the two integrals in (153.38) is an integer. Thus the phase is independent of the chosen filling, even though neither integral is itself a canonical real number modulo ordinary equality. This is the point at which the bulk local index density and the boundary eta invariant are tied to the same $U(1)$-valued object.

Boundary vector and gluing law. Let Y now be an odd-dimensional spin manifold with boundary M, and let all geometric data be product type near M. The Dai–Freed construction assigns a vector

$$Z_{\mathrm{DF}}(Y) \in L_M^{-1},$$

where L_M is the determinant or Pfaffian anomaly line of the boundary fermion system. This vector is a trivialization of the boundary anomaly line, not a numerical partition function. The numerical partition function appears only after pairing it with the boundary fermion determinant vector in L_M.

If Y_0 and Y_1 have the same boundary M, with the same boundary background, the ratio between the two boundary trivializations is the closed phase of the glued odd manifold

$$\overline{Y} = Y_0 \cup_M (-Y_1).$$

In the determinant-line pairing this is the gluing identity

$$\big\langle Z_{\mathrm{DF}}(Y_0), Z_{\mathrm{DF}}(-Y_1)\big\rangle = Z_{\mathrm{DF}}\big(Y_0 \cup_M (-Y_1)\big), \tag{153.39}$$

with the evident real-line version in the Pfaffian case. The left side pairs the vector in L_M^{-1} supplied by Y_0 with the dual vector supplied by the orientation reversal of Y_1. The right side is a number because the boundary anomaly lines cancel under gluing. Associativity of gluing is therefore not an extra physical postulate: it is the statement that tensoring one-dimensional determinant lines and composing their canonical pairings give the same number as gluing all odd-dimensional pieces first. This is the finite-line algebra that the eta invariant realizes analytically.

Proposition 153.9 (Dai–Freed boundary pairing cancels the anomaly cocycle)**.** *Let A be a structured background on M, let g be a gauge or background automorphism, and let $\alpha(A,g) : L_A \to L_{A^g}$ denote the determinant- or Pfaffian-line transport. Suppose the boundary fermion functional is a line-valued vector $\Psi(A) \in L_A$ with covariance*

$$\Psi(A^g) = \alpha(A,g)\Psi(A), \tag{153.40}$$

and a Dai–Freed filling Y supplies a vector $Z_{\mathrm{DF}}(Y;A) \in L_A^{-1}$ with inverse covariance

$$Z_{\mathrm{DF}}(Y;A^g) = \alpha(A,g)^{-1} Z_{\mathrm{DF}}(Y;A). \tag{153.41}$$

Then the paired amplitude

$$\mathcal{A}_Y(A) = \big\langle Z_{\mathrm{DF}}(Y;A), \Psi(A)\big\rangle \tag{153.42}$$

is invariant under g. If Y_0 and Y_1 are two fillings with the same boundary background, then

$$\frac{\mathcal{A}_{Y_0}(A)}{\mathcal{A}_{Y_1}(A)} = Z_{\mathrm{DF}}\big(Y_0 \cup_M (-Y_1)\big), \tag{153.43}$$

whenever the denominator is nonzero. Thus inflow does not make the anomalous boundary determinant into a scalar by itself; it supplies the inverse line vector needed to form a scalar, and changing the inverse-line trivialization is measured by a closed invertible-theory phase.

Proof. Work in a local frame e_A of L_A. Write $\Psi(A) = \psi_A e_A$, $Z_{\mathrm{DF}}(Y;A) = z_A e_A^{-1}$, and let the line transport multiply the frame coordinate by a phase $\alpha(A,g)$. The covariance equations give $\psi_{A^g} = \alpha(A,g)\psi_A$ and $z_{A^g} = \alpha(A,g)^{-1} z_A$. Hence

$$\mathcal{A}_Y(A^g) = z_{A^g}\psi_{A^g} = z_A\psi_A = \mathcal{A}_Y(A).$$

For two fillings, the ratio $z_{0,A}/z_{1,A}$ is a scalar because both vectors live in the same inverse line. The gluing law (153.39) identifies this scalar with the closed Dai–Freed phase of $Y_0 \cup_M (-Y_1)$. Pairing both numerator and denominator with the same boundary vector $\Psi(A)$ gives (153.43). □

Composition through an intermediate boundary. The same line-valued mechanism controls the anomaly factor when a physical spacetime is cut into successive pieces. We use the ordinary bordism orientation convention of Chapter 103: a bordism $Y_{01} : M_0 \to M_1$ has

$$\partial Y_{01} = (-M_0) \sqcup M_1.$$

Orientation reversal dualizes the anomaly line, $L_{-M} \simeq L_M^{-1}$. Since the Dai–Freed object for a manifold with boundary is a vector in the inverse line over the full boundary, this gives

$$Z_{\mathrm{DF}}(Y_{01}) \in L_{\partial Y_{01}}^{-1} \simeq (L_{-M_0} \otimes L_{M_1})^{-1} \simeq L_{M_0} \otimes L_{M_1}^{-1}.$$

Thus the Dai–Freed bulk has the dual variance to a relative boundary-QFT transition amplitude from M_0 to M_1, which would live in $L_{M_0}^{-1} \otimes L_{M_1}$. The former is the inverse anomaly factor that cancels the latter; it is not itself the physical transition kernel.

For successive invertible bulk bordisms $Y_{01} : M_0 \to M_1$ and $Y_{12} : M_1 \to M_2$, the resulting Dai–Freed vectors have the form

$$Z_{\mathrm{DF}}(Y_{01}) \in L_{M_0} \otimes L_{M_1}^{-1}, \qquad Z_{\mathrm{DF}}(Y_{12}) \in L_{M_1} \otimes L_{M_2}^{-1}.$$

Their composition is obtained by evaluating the two M_1-factors:

$$Z_{\mathrm{DF}}(Y_{12} \circ Y_{01}) = \big(\mathrm{id}_{L_{M_0}} \otimes \mathrm{ev}_{L_{M_1}} \otimes \mathrm{id}_{L_{M_2}^{-1}}\big)\big(Z_{\mathrm{DF}}(Y_{01}) \otimes Z_{\mathrm{DF}}(Y_{12})\big). \tag{153.44}$$

Here $\mathrm{ev}_{L_{M_1}} : L_{M_1}^{-1} \otimes L_{M_1} \to \mathbb{C}$ is the evaluation pairing, with the displayed tensor order understood. In the real Pfaffian case L_M is a $\mathbb{Z}_2$-graded real superline; the formulas above are written in the order in which the intermediate factors are evaluated, and any reordering of odd Pfaffian factors uses the Koszul sign rule.

This is only the anomaly-line component of locality. A boundary Hilbert-space matrix element, or a Euclidean transition amplitude, also requires an actual state space, pairing or integration measure, gauge treatment, and topology/convergence statement. Once that physical gluing operation has been separately constructed, the anomaly factors on the common cut are paired away by (153.44); the line identity alone does not construct the sum or integral over physical degrees of freedom.

Proposition 153.10 (Intermediate-cut frame cancellation)**.** *Choose local frames e_i of the anomaly lines L_{M_i}. Write*

$$Z_{\mathrm{DF}}(Y_{01}) = z_{01}\, e_0 \otimes e_1^{-1}, \qquad Z_{\mathrm{DF}}(Y_{12}) = z_{12}\, e_1 \otimes e_2^{-1}.$$

Then the composed coefficient in $L_{M_0} \otimes L_{M_2}^{-1}$ is $z_{01}z_{12}$, and it is independent of the choice of frame e_1 on the glued boundary. If the frame on M_1 is changed by $e_1 \mapsto qe_1$, $q \in U(1)$, the two coefficients transform as $z_{01} \mapsto qz_{01}$ and $z_{12} \mapsto q^{-1}z_{12}$; their product is unchanged. The same statement holds for real Pfaffian lines with $q = \pm 1$.

Proof. The evaluation pairing satisfies $\mathrm{ev}_{L_{M_1}}(e_1^{-1} \otimes e_1) = 1$. Substituting the two displayed expressions into (153.44) gives

$$Z_{\mathrm{DF}}(Y_{12} \circ Y_{01}) = z_{01}z_{12}\, e_0 \otimes e_2^{-1}.$$

After the frame change $e_1' = qe_1$, the same vectors are written as

$$Z_{\mathrm{DF}}(Y_{01}) = (qz_{01})\, e_0 \otimes (e_1')^{-1}, \qquad Z_{\mathrm{DF}}(Y_{12}) = (q^{-1}z_{12})\, e_1' \otimes e_2^{-1}.$$

The product of the two coefficients is still $z_{01}z_{12}$, which is the coefficient of the glued bordism in the remaining endpoint lines. $\square$

When a relative boundary-theory amplitude $\mathcal{U}_{02} \in L_{M_0}^{-1} \otimes L_{M_2}$ is paired with the composed bulk vector, all exposed anomaly lines are evaluated and the result is an ordinary number. If the physical theory supplies a valid cut formula for $\mathcal{U}_{02}$ through M_1, then (153.44) says that the anomaly-line part of that cut formula is independent of the artificial frame on M_1. All phase data are carried by the exposed endpoint lines and by the closed-manifold eta phase that compares two different global anomaly-line trivializations, as in (153.43).

Local descent as the infinitesimal limit of inflow. For a contractible loop of background fields, the Dai–Freed holonomy reduces to the exponential of the descent integral determined by the local anomaly polynomial. This statement is not an additional index theorem. Once the Bismut–Freed curvature formula and the APS holonomy formula have been fixed with the conventions above, the remaining passage from global inflow to the local consistent anomaly is ordinary Stokes transgression in background-field space.

Let $\mathcal{B}$ be a coordinate patch in the space of structured backgrounds and let $\gamma = \partial\Sigma$ be an oriented contractible loop bounding a smooth two-chain $\Sigma \subset \mathcal{B}$. Write Ω_{BF} for the normalized curvature two-form of the Bismut–Freed connection, with the sign convention fixed by (153.11): for a disk contained in this patch,

$$\mathrm{Hol}_\gamma(\mathrm{Det}\, D) = \exp\left(2\pi\mathrm{i} \int_\Sigma \Omega_{\mathrm{BF}}\right). \tag{153.45}$$

The curvature theorem identifies Ω_{BF} with the fiber integral over M of the degree-$(d+2)$ component of $\widehat{A}(TM)\,\mathrm{ch}(E)$, again in the same sign convention. Equivalently, if I_{d+2} denotes the normalized anomaly polynomial on the universal family over $\mathcal{B} \times M$, then

$$\int_\Sigma \Omega_{\mathrm{BF}} = \int_{\Sigma\times M} I_{d+2}. \tag{153.46}$$

Now specialize one tangent direction of Σ to a gauge-orbit direction generated by a Lie-algebra-valued parameter λ. Choose a Chern–Simons representative $I_{d+1}^{(0)}$ on physical spacetime and its descent form $I_d^{(1)}(\lambda, A)$, so that

$$I_{d+2} = dI_{d+1}^{(0)}, \qquad \delta_\lambda I_{d+1}^{(0)} = dI_d^{(1)}(\lambda, A). \tag{153.47}$$

Stokes on the two-parameter cylinder in background space turns $\int_{\Sigma\times M} I_{d+2}$ into the integral of $I_d^{(1)}$ over the physical boundary slice. Thus the infinitesimal gauge variation of a local trivialization of the fermion determinant line is

$$\delta_\lambda \log Z_{\mathrm{fermion}}(A) = 2\pi\mathrm{i} \int_M I_d^{(1)}(\lambda, A), \tag{153.48}$$

up to the sign choice already fixed by the determinant-line convention in (153.11). This is the same local descent class used in the perturbative chiral-anomaly calculation and in the anomaly-inflow construction developed earlier in the monograph. Local counterterms change $I_d^{(1)}$ by a relative coboundary; they change the local connection form on the determinant line but not its curvature or its contractible-loop holonomy.

The qualification "contractible" is essential. If the loop is not filled by a disk in the background groupoid, equations (153.45)–(153.48) do not determine the holonomy. The reduced eta invariant may then contain a flat anomaly-line contribution invisible to the local polynomial. The Witten $SU(2)$ anomaly above is exactly of this type: the local cubic descent vanishes, while a nontrivial mapping-torus mod-two index detects the large-gauge holonomy.

Theorem boundary for the global-anomaly construction. The anomaly statements in this chapter use four external global-analysis inputs, and the QFT conclusions are the consequences derived from them in the fixed conventions above. The APS theorem supplies the Fredholm boundary problem and the identity

$$\mathrm{Ind}(D_X^{+,\mathrm{APS}}) = \int_X [\widehat{A}(TX)\,\mathrm{ch}(E)] - \xi(B_{\partial X})$$

with the product-collar and reduced-eta conventions fixed in (153.3)–(153.5). From this input the text derives the cylinder congruence, the endpoint-kernel term, the orientation sign in the spectral-flow bookkeeping, and the local descent variation of the eta phase. The Bismut–Freed theorem supplies the Quillen determinant line, its connection, its local curvature, and the mapping-torus holonomy formula. From these data the text derives the quotient-groupoid cocycle criterion: a chiral determinant descends to the quotient by gauge transformations if and only if the anomaly line admits an equivariant trivialization; local counterterms alter the cocycle by a coboundary; based loop holonomies are invariant. The Dai–Freed construction supplies the boundary vector in the inverse anomaly line, its orientation-dual composition law, and the cancellation of anomaly factors on artificial cuts. From those line-valued statements the text derives the inflow cancellation criterion and the contractible-loop reduction to the same descent representative used in the perturbative anomaly chapter. They do not by themselves construct the physical Hilbert-space pairing, path-integral measure, gauge fixing, or convergence needed for a full QFT gluing theorem. Finally, the real mod-two index theorem supplies the value of the Pfaffian holonomy for the Witten $SU(2)$ mapping torus. The trace-delta normalization, the $2j \equiv 1 \pmod 4$ parity criterion, the tensor-product cancellation condition, and the separation between local cubic anomaly and global sign anomaly are then finite representation-theoretic and anomaly-line consequences in the monograph's conventions.

Thus the remaining quoted material is pure global-analysis infrastructure: Fredholmness and heat-kernel local index theory with APS boundary conditions, construction and holonomy of the determinant/Pfaffian line, and the real KO-theoretic mod-two index calculation on the Witten mapping torus. The QFT-facing anomaly claims are the groupoid descent, local descent, inflow, and cancellation statements derived from those inputs in the preceding paragraphs.

153.7 Interacting Theories and Anomaly Lines

For free fermions, the anomaly line is constructed from a family of Dirac operators. For an interacting QFT, the same structure should be formulated without choosing quadratic fields. The background groupoid has objects given by structured background fields and morphisms given by gauge transformations, diffeomorphisms, and identifications of background data. An anomaly line is a functor from this groupoid to the groupoid of complex lines, equipped with compatible gluing data. The partition function of the anomalous theory is a section of this functorial line.

Local counterterms change the trivialization of the line. They shift the connection one-form and the cocycle representative, but they do not change the curvature class or the holonomy class. Thus a genuine global anomaly is not removed by changing local coordinates on the space of actions.

Open Problem 153.11 (Global anomaly line for interacting QFT). Construct the anomaly line of an interacting QFT over the space of background fields, prove its local curvature and global holonomy formulae, and identify the conditions under which it is represented by an invertible bulk field theory. The free-fermion determinant and Pfaffian lines developed above are models for this structure, not a general construction of it.

Chapter 154

Cosmological Spacetimes and Particle Creation

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \dots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter studies free scalar QFT on time-dependent globally hyperbolic backgrounds. The central point is not that "particles are ambiguous" as a slogan, but the precise mathematical statement: a particle interpretation is a choice of positive-frequency subspace, equivalently a compatible complex structure on the real symplectic space of solutions. When the geometry has static asymptotic regions, the in and out complex structures are selected by the asymptotic time translations and their mismatch is measured by Bogoliubov coefficients. When no such asymptotic regions exist, one uses state properties such as the Hadamard condition, adiabatic ultraviolet behavior, and detector response.

154.1 Spatially Flat Robertson–Walker Metrics

Let $M = I \times \mathbb{R}^{d-1}$, with $d \geq 2$, and let

$$g = -dt^2 + a(t)^2 d\mathbf{x}^2 = a(\eta)^2(-d\eta^2 + d\mathbf{x}^2), \qquad d\eta = \frac{dt}{a(t)}. \tag{154.1}$$

Assume $a(\eta) > 0$ is smooth on the conformal-time interval under consideration. We write

$$\mathcal{H}(\eta) = \frac{a'(\eta)}{a(\eta)} \tag{154.2}$$

for the conformal Hubble function. The scalar curvature of (154.1) is

$$R = (d-1)a^{-2}\big(2\mathcal{H}' + (d-2)\mathcal{H}^2\big). \tag{154.3}$$

Consider a real scalar field satisfying

$$(\Box_g - m^2 - \xi R)\phi = 0. \tag{154.4}$$

The Fourier convention is

$$\phi(\eta, \mathbf{x}) = \int \frac{d^{d-1}\mathbf{k}}{(2\pi)^{d-1}}\, \phi_{\mathbf{k}}(\eta) e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}, \qquad k = |\mathbf{k}|. \tag{154.5}$$

Remark 154.1 (Robertson–Walker mode equation)*.* Let

$$p = \frac{d-2}{2}, \qquad \chi_{\mathbf{k}}(\eta) = a(\eta)^p \phi_{\mathbf{k}}(\eta). \tag{154.6}$$

Then $\chi_{\mathbf{k}}$ obeys

$$\chi_{\mathbf{k}}'' + \Omega_k(\eta)^2 \chi_{\mathbf{k}} = 0, \tag{154.7}$$

where

$$\Omega_k^2 = k^2 + a^2(m^2 + \xi R) - p\mathcal{H}' - p^2\mathcal{H}^2. \tag{154.8}$$

For a massless conformally coupled scalar, with

$$m = 0, \qquad \xi = \xi_c = \frac{d-2}{4(d-1)}, \tag{154.9}$$

one has $\Omega_k^2 = k^2$.

For the metric (154.1),

$$\Box_g \phi = a^{-2}\left[-\partial_\eta^2 \phi - (d-2)\mathcal{H}\,\partial_\eta \phi + \Delta_{\mathbf{x}}\phi\right].$$

Substituting (154.5) into (154.4) gives

$$\phi_{\mathbf{k}}'' + (d-2)\mathcal{H}\phi_{\mathbf{k}}' + \big(k^2 + a^2(m^2 + \xi R)\big)\phi_{\mathbf{k}} = 0. \tag{154.10}$$

Writing $\phi_{\mathbf{k}} = a^{-p}\chi_{\mathbf{k}}$, with $2p = d-2$, gives

$$\phi_{\mathbf{k}}' = a^{-p}(\chi_{\mathbf{k}}' - p\mathcal{H}\chi_{\mathbf{k}}),$$

and

$$\phi_{\mathbf{k}}'' = a^{-p}\left[\chi_{\mathbf{k}}'' - 2p\mathcal{H}\chi_{\mathbf{k}}' + (p^2\mathcal{H}^2 - p\mathcal{H}')\chi_{\mathbf{k}}\right].$$

The first-derivative terms cancel in (154.10), leaving (154.7) and (154.8). Finally,

$$a^2\xi_c R = \frac{d-2}{4}\big(2\mathcal{H}' + (d-2)\mathcal{H}^2\big) = p\mathcal{H}' + p^2\mathcal{H}^2,$$

so the curvature and rescaling terms cancel when $m = 0$ and $\xi = \xi_c$.

The cancellation in the conformally coupled massless case is an exact calculation, not an adiabatic statement. It is the local manifestation of conformal covariance of the classical wave equation.

154.2 Symplectic Space and Complex Structures

Let $\mathcal{S}$ be the real vector space of smooth solutions of (154.4) with compact Cauchy data. On any Cauchy surface Σ, define

$$\sigma(\phi_1, \phi_2) = \int_\Sigma d\Sigma^\mu \left(\phi_1 \nabla_\mu \phi_2 - \phi_2 \nabla_\mu \phi_1\right). \tag{154.11}$$

The current inside the integral is conserved whenever both fields satisfy the equation of motion, so σ is independent of Σ.

Definition 154.2 (Compatible complex structure)**.** A compatible complex structure on $(\mathcal{S}, \sigma)$ is a real linear map $J : \mathcal{S} \to \mathcal{S}$ such that

$$J^2 = -1, \qquad \sigma(Ju, Jv) = \sigma(u, v), \qquad \sigma(u, Ju) > 0 \quad (u \neq 0). \tag{154.12}$$

It defines a one-particle inner product

$$\langle u, v\rangle_J = \sigma(u, Jv) + \mathrm{i}\sigma(u, v). \tag{154.13}$$

Completing $(\mathcal{S}, \langle\cdot,\cdot\rangle_J)$ gives the one-particle Hilbert space $\mathcal{H}_J$, and the symmetric Fock space $\mathcal{F}_J = \mathrm{Sym}\,\mathcal{H}_J$ gives the associated quasifree representation.

Equivalently, after complexification, J selects the $+\mathrm{i}$-eigenspace as the positive-frequency subspace. This is the precise datum behind a particle interpretation. Different choices of J can be unitarily inequivalent in infinite volume.

Theorem 154.3 (Shale–Stinespring criterion, diagonal discrete form)**.** *Let the one-particle space be a separable bosonic Hilbert space with orthonormal mode basis indexed by a countable set I, paired by an involution $n \mapsto \bar{n}$. Suppose two quasifree complex structures are related mode-by-mode by Bogoliubov coefficients*

$$u_n^{(1)} = \alpha_n u_n^{(2)} + \beta_n \overline{u_{\bar{n}}^{(2)}}, \tag{154.14}$$

with

$$|\alpha_n|^2 - |\beta_n|^2 = 1.$$

Then the two Fock representations are unitarily equivalent precisely when

$$\sum_{n \in I_+} |\beta_n|^2 < \infty, \tag{154.15}$$

where I_+ contains one representative from each non-fixed pair $\{n, \bar{n}\}$, with the evident one-oscillator modification for fixed points. In that case the Bogoliubov transformation is implemented by a unitary operator on Fock space.

Proof. For a finite set of modes the statement is the standard squeeze construction. On one two-mode block write

$$b_n = \alpha_n a_n + \beta_n a_{\bar{n}}^\dagger, \qquad b_{\bar{n}} = \alpha_n a_{\bar{n}} + \beta_n a_n^\dagger,$$

after absorbing harmless phases into the mode vectors. The relation $|\alpha_n|^2 - |\beta_n|^2 = 1$ is exactly the condition that $[b_n, b_n^\dagger] = [b_{\bar{n}}, b_{\bar{n}}^\dagger] = 1$. Put

$$\gamma_n = -\frac{\beta_n}{\alpha_n}.$$

The vector in the old Fock space annihilated by b_n and $b_{\bar{n}}$ is

$$\Omega_{\gamma_n} = (1 - |\gamma_n|^2)^{1/2} \sum_{m=0}^{\infty} \gamma_n^m |m\rangle_n \otimes |m\rangle_{\bar{n}}, \tag{154.16}$$

because applying a_n to the m-th term and applying $\gamma_n a_{\bar{n}}^\dagger$ to the $(m-1)$-st term cancel. The normalization follows from the geometric series and the identity

$$1 - |\gamma_n|^2 = \frac{|\alpha_n|^2 - |\beta_n|^2}{|\alpha_n|^2} = |\alpha_n|^{-2}.$$

The fixed-point one-oscillator block is the usual one-mode squeezed vector and gives the same square-summability condition, with a different harmless normalization convention.

For an infinite countable set, the transformed vacuum is the infinite tensor product of the block vacua Ω_{γ_n}. Von Neumann's criterion for incomplete infinite tensor products says that this tensor product defines a nonzero vector in the reference Fock space exactly when

$$\sum_{n \in I_+} \big(1 - |\langle 0_n 0_{\bar{n}}, \Omega_{\gamma_n}\rangle|\big) < \infty.$$

Here

$$|\langle 0_n 0_{\bar{n}}, \Omega_{\gamma_n}\rangle| = (1 - |\gamma_n|^2)^{1/2}.$$

Since $1 - \sqrt{1-x}$ is comparable to x for $0 \le x \le 1$, the criterion is

$$\sum_{n \in I_+} |\gamma_n|^2 < \infty.$$

Because $|\gamma_n|^2 = |\beta_n|^2/(1+|\beta_n|^2)$, this is equivalent to $\sum_{n\in I_+} |\beta_n|^2 < \infty$: if the γ_n-sum converges, only finitely many $|\beta_n|$ can exceed 1, and on the remaining tail the two sequences are comparable.

When the sum is finite, the finite-mode squeeze implementers converge strongly on the finite-particle domain to an isometry whose range contains the finite-particle domain of the transformed representation; its extension is the implementing unitary. When the sum diverges, the candidate transformed vacuum has zero norm in the reference incomplete tensor product, so no unitary can implement the transformation: a unitary implementer would have to send the transformed vacuum to a nonzero vector annihilated by all b_n, and the block equations above determine that vector uniquely up to phase. □

For a field on a finite spatial box, (154.15) is the literal finite-volume Shale–Stinespring condition. On a noncompact homogeneous spatial slice the momentum label is continuous, and the diagonal map

$$(\beta \bar{f})(\mathbf{k}) = \beta_{\mathbf{k}}\, \overline{f(-\mathbf{k})}$$

is a multiplication operator, composed with parity and conjugation, on $L^2(d^{d-1}\mathbf{k})$. As an operator on a non-atomic L^2-space, such a diagonal multiplication operator is Hilbert–Schmidt only when it vanishes almost everywhere. Thus

$$\int d^{d-1}\mathbf{k}\, |\beta_{\mathbf{k}}|^2 < \infty$$

is a finite particle-density or local/box-normalized diagnostic; it is not, by itself, a statement of global unitary equivalence of the two infinite-volume Fock representations. This distinction is one of the main reasons the complex-structure formulation is cleaner than informal particle language.

154.3 In and Out Modes

Assume now that the scale factor and curvature approach static limits as $\eta \to \pm\infty$, so that

$$\Omega_k(\eta) \longrightarrow \omega_k^{\mathrm{in/out}} > 0 \qquad (\eta \to \mp\infty). \tag{154.17}$$

The in and out positive-frequency modes are normalized by the Wronskian condition

$$\chi_k \overline{\chi_k'} - \chi_k' \overline{\chi_k} = \mathrm{i} \tag{154.18}$$

and by

$$\chi_k^{\mathrm{in/out}}(\eta) \sim \frac{e^{-\mathrm{i}\omega_k^{\mathrm{in/out}}\eta}}{\sqrt{2\omega_k^{\mathrm{in/out}}}} \qquad \text{in the corresponding asymptotic region.} \tag{154.19}$$

Bogoliubov normalization and particle number. Let

$$\chi_k^{\mathrm{in}} = \alpha_k \chi_k^{\mathrm{out}} + \beta_k \overline{\chi_k^{\mathrm{out}}}. \tag{154.20}$$

Then

$$|\alpha_k|^2 - |\beta_k|^2 = 1. \tag{154.21}$$

If $|0, \mathrm{in}\rangle$ is the in-vacuum, the expected out-particle number in the $\mathbf{k}$-mode is

$$\langle 0, \mathrm{in}| a_{\mathbf{k}}^{\mathrm{out}\,*} a_{\mathbf{k}}^{\mathrm{out}} |0, \mathrm{in}\rangle = |\beta_k|^2 \tag{154.22}$$

in finite volume, with the usual delta-function density normalization in infinite volume. The Wronskian

$$W(f, g) = f g' - f' g$$

is conserved for any two solutions of $\chi'' + \Omega_k^2 \chi = 0$. The normalization (154.18) gives

$$W(\chi_k^{\mathrm{out}}, \overline{\chi_k^{\mathrm{out}}}) = \mathrm{i}, \qquad W(\overline{\chi_k^{\mathrm{out}}}, \chi_k^{\mathrm{out}}) = -\mathrm{i}.$$

Substitution of (154.20) therefore gives

$$\mathrm{i} = W(\chi_k^{\mathrm{in}}, \overline{\chi_k^{\mathrm{in}}}) = \mathrm{i}(|\alpha_k|^2 - |\beta_k|^2),$$

which proves (154.21).

The field mode admits both decompositions

$$\widehat{\chi}_{\mathbf{k}} = a_{\mathbf{k}}^{\mathrm{out}} \chi_k^{\mathrm{out}} + a_{-\mathbf{k}}^{\mathrm{out}\,*} \overline{\chi_k^{\mathrm{out}}} = a_{\mathbf{k}}^{\mathrm{in}} \chi_k^{\mathrm{in}} + a_{-\mathbf{k}}^{\mathrm{in}\,*} \overline{\chi_k^{\mathrm{in}}}.$$

Comparing coefficients gives

$$a_{\mathbf{k}}^{\mathrm{out}} = \overline{\alpha_k}\, a_{\mathbf{k}}^{\mathrm{in}} + \overline{\beta_k}\, a_{-\mathbf{k}}^{\mathrm{in}\,*}, \tag{154.23}$$

with the phase convention induced by (154.20). Since $a_{\mathbf{k}}^{\mathrm{in}}|0, \mathrm{in}\rangle = 0$ and $[a_{-\mathbf{k}}^{\mathrm{in}}, a_{-\mathbf{k}}^{\mathrm{in}\,*}] = 1$ in finite volume, (154.22) follows.

Example 154.4 (Instantaneous frequency jump)**.** For a single oscillator mode whose frequency changes suddenly from ω_i to ω_f, continuity of χ and χ' at the jump gives

$$\alpha = \frac{1}{2}\left(\sqrt{\frac{\omega_f}{\omega_i}} + \sqrt{\frac{\omega_i}{\omega_f}}\right), \qquad \beta = \frac{1}{2}\left(\sqrt{\frac{\omega_f}{\omega_i}} - \sqrt{\frac{\omega_i}{\omega_f}}\right), \tag{154.24}$$

and therefore

$$|\beta|^2 = \frac{(\omega_f - \omega_i)^2}{4\omega_i\omega_f}. \tag{154.25}$$

The example is not a smooth cosmological spacetime; it is a normalization check and a limiting model for a transition fast compared with the mode period.

154.4 Adiabatic States and Ultraviolet Control

When no static asymptotic regions exist, one can still impose a high-momentum condition on the positive-frequency modes. The adiabatic ansatz is

$$\chi_k(\eta) = \frac{1}{\sqrt{2W_k(\eta)}} \exp\left(-\mathrm{i}\int^{\eta} W_k(\eta')\, d\eta'\right), \qquad W_k(\eta) > 0. \tag{154.26}$$

Remark 154.5 (Riccati equation for the adiabatic frequency). The ansatz (154.26) solves (154.7) exactly if and only if

$$W_k^2 = \Omega_k^2 - \frac{1}{2}\frac{W_k''}{W_k} + \frac{3}{4}\left(\frac{W_k'}{W_k}\right)^2. \tag{154.27}$$

Write

$$\chi = (2W)^{-1/2} e^{-\mathrm{i}\Theta}, \qquad \Theta' = W.$$

Then

$$\frac{\chi'}{\chi} = -\frac{1}{2}\frac{W'}{W} - \mathrm{i}W$$

and

$$\frac{\chi''}{\chi} = -\mathrm{i}W' - \frac{1}{2}\left(\frac{W'}{W}\right)' + \left(-\frac{1}{2}\frac{W'}{W} - \mathrm{i}W\right)^2.$$

The imaginary terms cancel, leaving

$$\frac{\chi''}{\chi} = -W^2 - \frac{1}{2}\frac{W''}{W} + \frac{3}{4}\left(\frac{W'}{W}\right)^2.$$

Thus $\chi'' + \Omega_k^2\chi = 0$ is equivalent to (154.27).

The adiabatic expansion is obtained by assigning one derivative of $a(\eta)$ one adiabatic order and solving (154.27) recursively. To second adiabatic order, if $\omega_k = \Omega_k$ denotes the zeroth-order positive frequency,

$$W_k = \omega_k - \frac{\omega_k''}{4\omega_k^2} + \frac{3(\omega_k')^2}{8\omega_k^3} + O(\partial_\eta^4). \tag{154.28}$$

An adiabatic state of sufficiently high order is Hadamard for the scalar field on Robertson–Walker spacetime. Conversely, the point-splitting subtraction for $\langle T_{\mu\nu}\rangle$ has the same ultraviolet content as the fourth adiabatic order subtraction in four spacetime dimensions. Equality here means equality of the local large-k singular terms; finite local curvature counterterms remain the renormalization freedom described in Chapter 148.

154.5 Detector Response

Let $x(\tau)$ be a timelike worldline parametrized by proper time, and let $\lambda\chi(\tau)\mu(\tau)\widehat{\phi}(x(\tau))$ be an Unruh–DeWitt coupling with compactly supported switching function χ. For a detector energy gap E, first-order perturbation theory gives

$$P(E) = \lambda^2 \int d\tau\, d\tau'\, e^{-\mathrm{i}E(\tau-\tau')}\chi(\tau)\chi(\tau')W(x(\tau), x(\tau')), \tag{154.29}$$

where W is the Wightman two-point function of the state.

Positivity of the switched detector response. If W is a positive-type two-point function, then

$$P(E) \geq 0$$

for every real E and every smooth compactly supported switching function χ. The content is the positive-type property of the state applied to the detector smearing. Define the test function on spacetime

$$f_E(x) = \int d\tau\, e^{-\mathrm{i}E\tau}\chi(\tau)\delta(x, x(\tau))$$

in the distributional sense. Smearing the two-point function gives

$$P(E)/\lambda^2 = W(\overline{f_E}, f_E).$$

Positive type of the quasifree state is precisely the condition $W(\overline{f}, f) \geq 0$ for every test function f. The worldline smearing can be obtained as the limit of ordinary compactly supported spacetime test functions in a tubular neighborhood of the curve, so the same positivity applies. The approximation by ordinary test functions is part of the mathematical meaning of the idealized pointlike detector; a finite-size detector replaces the delta distribution by a smooth spatial profile and satisfies positivity directly.

Detector response is therefore a local operational probe of a chosen state. Its definition uses the state two-point distribution, the detector trajectory, and the switching/profile data.

154.6 The Bunch–Davies State in de Sitter Space

In the spatially flat patch of d-dimensional de Sitter space,

$$a(\eta) = \frac{-1}{H\eta}, \qquad \eta < 0, \qquad H > 0, \tag{154.30}$$

and $R = d(d-1)H^2$. Equation (154.8) becomes

$$\Omega_k^2 = k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2}, \qquad \nu^2 = \frac{(d-1)^2}{4} - \frac{m^2}{H^2} - \xi d(d-1). \tag{154.31}$$

Remark 154.6 (Bunch–Davies mode functions). The mode

$$\chi_k^{\mathrm{BD}}(\eta) = \sqrt{\frac{-\pi\eta}{4}}\, \exp\left[\mathrm{i}\pi\left(\frac{\nu}{2} + \frac{1}{4}\right)\right] H_\nu^{(1)}(-k\eta) \tag{154.32}$$

solves the de Sitter mode equation and obeys

$$\chi_k^{\mathrm{BD}}(\eta) \sim \frac{e^{-\mathrm{i}k\eta}}{\sqrt{2k}} \qquad \eta \to -\infty. \tag{154.33}$$

It is normalized by

$$\chi_k^{\mathrm{BD}}\overline{(\chi_k^{\mathrm{BD}})'} - (\chi_k^{\mathrm{BD}})'\overline{\chi_k^{\mathrm{BD}}} = \mathrm{i}. \tag{154.34}$$

With $z = -k\eta > 0$, equation

$$\chi_k'' + \left(k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2}\right)\chi_k = 0$$

is transformed by $\chi_k(\eta) = (-\eta)^{1/2} f(z)$ into Bessel's equation for f of order ν. Thus $H_\nu^{(1)}(z)$ gives a solution. The large-z asymptotic formula

$$H_\nu^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}}\, \exp\left[\mathrm{i}\left(z - \frac{\pi\nu}{2} - \frac{\pi}{4}\right)\right]$$

turns (154.32) into (154.33). The Wronskian identity

$$H_\nu^{(1)}(z)\partial_z H_\nu^{(2)}(z) - \partial_z H_\nu^{(1)}(z) H_\nu^{(2)}(z) = -\frac{4\mathrm{i}}{\pi z}$$

then gives (154.34), with the additional minus sign from $z = -k\eta$.

For $m = 0$ and $\xi = \xi_c$, $\nu = 1/2$ and the Bunch–Davies mode reduces to the conformal Minkowski mode. For a massless minimally coupled scalar in $d = 4$, $\nu = 3/2$; the zero-mode and infrared behavior require separate care and cannot be suppressed by the local high-frequency definition alone.

The Bunch–Davies quasifree state is Hadamard and invariant under the connected de Sitter group for the standard massive range. Other de Sitter-invariant two-point functions must be tested against the Hadamard short-distance condition; de Sitter invariance alone is not a state-selection principle.

154.7 Backreaction Boundary

The expectation value $\langle T_{\mu\nu}\rangle$ in a time-dependent state is defined by the renormalized stress-tensor prescription of Chapter 148. The semiclassical Einstein equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \,\langle T_{\mu\nu}\rangle_{\mathrm{ren}} \tag{154.35}$$

is an equation for a classical metric and a quantum state. The right-hand side depends on the state, on the finite curvature counterterms, and on the renormalization convention for the cosmological constant and Newton constant. Particle production by itself is not a closed backreaction equation; it is one diagnostic extracted from the state when a particle decomposition has been selected.

Open Problem 154.7 (Interacting cosmological QFT). Construct interacting QFT on cosmological spacetimes with Hadamard states, renormalized stress tensor, detector response, particle-production diagnostics where asymptotic regions exist, and controlled semiclassical backreaction. The construction should distinguish the local definition of fields and states from the additional asymptotic structure needed to define particles.

Chapter 155

Microlocal Spectrum Condition And Hadamard Geometry

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops the microlocal form of the positive-frequency condition on curved spacetimes. The replacement for translation-invariant spectral support is a wavefront-set condition on distributions, and the local geometric form of the two-point function is the Hadamard parametrix.

155.1 Wavefront Set

Let $u \in \mathcal{D}'(X)$ be a distribution on a smooth manifold X. A point $(x, k) \in T^*X\backslash\{0\}$ is absent from the wavefront set $\mathrm{WF}(u)$ if there is a cutoff $\chi \in C_c^\infty(X)$ with $\chi(x) \neq 0$ and a conic neighborhood Γ of k such that the Fourier transform of χu decays faster than every power in Γ. The wavefront set records both the singular support and the covector directions of oscillation.

Definition 155.1 (Wavefront set in a chart)**.** Let $u \in \mathcal{D}'(X)$. In a coordinate chart $U \subset X$, write $\widehat{\chi u}(\xi)$ for the Fourier transform of the compactly supported distribution χu, $\chi \in C_c^\infty(U)$, in the chosen coordinates. A nonzero covector $k \in T_x^*X$ is a regular direction for u if there exist $\chi \in C_c^\infty(U)$, $\chi(x) \neq 0$, and an open conic neighborhood $\Gamma \subset \mathbb{R}^D \setminus \{0\}$ of the coordinate representative of k such that, for every N,

$$|\widehat{\chi u}(\xi)| \leq C_N(1 + |\xi|)^{-N}, \qquad \xi \in \Gamma. \tag{155.1}$$

The wavefront set $\mathrm{WF}(u)$ is the complement, in $T^*X \setminus 0$, of the regular directions.

The definition is independent of the chart. Under a change of coordinates, the localized Fourier transform is an oscillatory integral whose phase has non-degenerate mixed Hessian at the point under consideration; integration by parts in directions where the transformed covector is separated from the old singular cone preserves rapid decay. Thus $\mathrm{WF}(u)$ is a closed conic subset of $T^*X \setminus 0$.

Theorem 155.2 (Product criterion for distributions)**.** *Let $u, v \in \mathcal{D}'(X)$. Suppose that there is no*

$$(x,k) \in \mathrm{WF}(u) \quad \textit{and} \quad (x,-k) \in \mathrm{WF}(v). \tag{155.2}$$

Then the pointwise product uv has a unique distributional extension which agrees with the ordinary product when one factor is smooth. Moreover

$$\begin{aligned} \mathrm{WF}(uv) \subset\ & \mathrm{WF}(u) \cup \mathrm{WF}(v) \\ & \cup \{(x, k+\ell) : (x,k) \in \mathrm{WF}(u),\ (x,\ell) \in \mathrm{WF}(v),\ k+\ell \neq 0\}. \end{aligned} \tag{155.3}$$

Proof. The statement is local, so work in one coordinate chart and multiply by a cutoff equal to 1 near the point. If u and v are compactly supported distributions, the formal Fourier transform of the product is the convolution

$$\widehat{uv}(\xi) = (2\pi)^{-D} \int \widehat{u}(\eta)\widehat{v}(\xi-\eta)\,\mathrm{d}\eta. \tag{155.4}$$

The obstruction to defining this integral distributionally at large $|\eta|$ is simultaneous non-decay of $\widehat{u}(\eta)$ and $\widehat{v}(\xi-\eta)$. As $|\eta| \to \infty$, the second direction is asymptotic to $-\eta$. Thus a large-frequency obstruction can occur only from an opposite pair of singular covectors $(k,-k)$, which is excluded by (155.2).

Choose a conic partition of unity in the η-variable. On pieces where η avoids $\mathrm{WF}(u)$, the factor $\widehat{u}(\eta)$ is rapidly decreasing after localization. On pieces where $\xi - \eta$ avoids $\mathrm{WF}(v)$, the factor $\widehat{v}(\xi-\eta)$ is rapidly decreasing. The exclusion of opposite cones gives such a finite conic cover of the large-η region. The bounded-η region is harmless because Fourier transforms of compactly supported distributions have at most polynomial growth. This defines (155.4), and the same conic decomposition proves rapid decay of $\widehat{uv}$ away from the three sets displayed in (155.3).

Uniqueness follows by regularizing u and v with a smoothing approximate identity. The estimates above show convergence in every allowed conic region and independence of the regularization. If one factor is smooth, the construction reduces to ordinary multiplication because the smooth factor has rapidly decreasing localized Fourier transform. □

This criterion is the local analytic reason that Wick products and time-ordered products require subtraction before coincident points are taken.

155.2 Hadamard Two-Point Condition

Let M be a globally hyperbolic Lorentzian spacetime and let ϕ be a real scalar field satisfying

$$P_M\phi = 0, \qquad P_M = -\nabla^\mu\nabla_\mu + m^2 + \xi R[g]. \tag{155.5}$$

A quasifree state is determined by its two-point function

$$\omega_2(x,y) = \omega(\phi(x)\phi(y)).$$

The time orientation on tangent vectors induces a convention on covectors: a nonzero causal covector $k \in T_x^*M$ is future directed when its metric dual

$$k^\sharp = g^{\mu\nu} k_\nu \partial_\mu$$

is a future-directed causal vector. In mostly-plus Minkowski coordinates, this means that a future null covector has $k_0 < 0$. This convention makes the positive-frequency plane wave $e^{-\mathrm{i}Et+\mathrm{i}\mathbf{p}\cdot\mathbf{x}}$ carry future null covector $(-E, \mathbf{p})$.

The microlocal Hadamard condition is

$$\mathrm{WF}(\omega_2) = \{(x, k_x; y, -k_y) : (x, k_x) \sim (y, k_y),\ k_x \in \overline{V}_+^*\}.$$

Here $(x, k_x) \sim (y, k_y)$ means that x and y are joined by a null geodesic and that k_y is the parallel transport of k_x along that geodesic. The cone $\overline{V}_+^*$ is the closed future causal cone in the cotangent space.

The minus sign on the second covector is not a convention that can be dropped. It records that a two-point distribution is a distribution on $M \times M$. If the same future covector is transported from x to y, the corresponding singular phase is locally of the form

$$\exp\bigl(\mathrm{i}\lambda(\Phi(x) - \Phi(y))\bigr), \qquad \lambda \to +\infty,$$

so the covector on the second copy of M is the negative of the transported covector.

155.3 Higher Microlocal Spectrum Condition

For non-Gaussian states, the two-point condition is not enough. The curved-spacetime replacement of the Wightman spectral condition is a family of cone constraints on all n-point distributions.

Definition 155.3 (Microlocal spectrum cone)**.** Let $x_1, \ldots, x_n \in M$. A covector tuple

$$(x_1, k_1; \ldots; x_n, k_n) \in T^*M^n \setminus 0$$

belongs to $\Gamma_n(M)$ if there is a finite directed graph with vertex set $\{1, \ldots, n\}$ such that each edge $e : i \to j$ is assigned a future-directed causal covector field p_e parallel transported along a future-directed causal curve from x_i to x_j, and

$$k_i = \sum_{e:i\to j} p_e(x_i) - \sum_{e:j\to i} p_e(x_i) \qquad (i = 1, \ldots, n). \tag{155.6}$$

The empty graph gives the zero tuple, which is omitted from $T^*M^n \setminus 0$.

Definition 155.4 (Microlocal spectrum condition for a state)**.** A state ω on a scalar field algebra satisfies the microlocal spectrum condition when all n-point distributions

$$\omega_n(f_1, \ldots, f_n) = \omega(\phi(f_1) \cdots \phi(f_n))$$

obey

$$\mathrm{WF}(\omega_n) \subset \Gamma_n(M). \tag{155.7}$$

For $n = 2$, a graph with one edge $1 \to 2$ gives

$$(k_1, k_2) = (p, -p),$$

which is the sign pattern in the Hadamard two-point condition. For quasifree Hadamard states, Wick's theorem reduces every n-point distribution to a finite sum of products of two-point functions. Applying Theorem 155.2 and the graph description above shows that the microlocal spectrum condition for all n follows from the two-point Hadamard condition.

155.4 Hadamard Parametrix

The microlocal condition specifies the singular covector directions. The Hadamard parametrix specifies the local distribution with those singularities. This section constructs its coefficients from the operator (155.5). The construction is local in a normal convex neighborhood $\mathcal{U}$, so every pair $x, y \in \mathcal{U}$ is joined by a unique geodesic lying in $\mathcal{U}$.

Definition 155.5 (Synge function and boundary value)**.** Let $\sigma(x, y)$ be Synge's world function, one half of the signed squared geodesic distance from y to x. Choose a smooth time function T on $\mathcal{U}$ and set

$$\sigma_\epsilon(x, y) = \sigma(x, y) + \mathrm{i}\epsilon(T(x) - T(y)) + \epsilon^2, \qquad \epsilon > 0. \tag{155.8}$$

All formulas involving negative powers or logarithms of σ_ϵ are understood as distributional boundary values as $\epsilon \downarrow 0$.

The basic coincidence identities are

$$\sigma^{;\mu}\sigma_{;\mu} = 2\sigma, \qquad [\sigma] = 0, \qquad [\sigma_{;\mu}] = 0, \qquad [\sigma_{;\mu\nu}] = g_{\mu\nu}, \qquad [\Box_g \sigma] = 4, \tag{155.9}$$

where brackets denote the coincidence limit $y \to x$. The dimension four enters only through the last identity.

Definition 155.6 (Van Vleck determinant)**.** The van Vleck determinant is the biscalar density ratio

$$\Delta(x, y) = \frac{\det\bigl(-\nabla^x_\mu \nabla^y_\nu \sigma(x, y)\bigr)}{|g(x)|^{1/2}|g(y)|^{1/2}}, \tag{155.10}$$

written in any pair of local coordinates and interpreted invariantly by the density factors. It satisfies $\Delta(x, x) = 1$.

Lemma 155.7 (Hadamard coefficient U)**.** *In four spacetime dimensions, the coefficient of σ_ϵ^{-1} in the Hadamard parametrix is*

$$U(x, y) = \Delta(x, y)^{1/2}. \tag{155.11}$$

Equivalently, U is the unique smooth biscalar along each geodesic from y to x satisfying

$$2\sigma^{;\mu}\nabla_\mu U + (\Box_g \sigma - 4)U = 0, \qquad U(x, x) = 1. \tag{155.12}$$

Proof. For a smooth biscalar U, use $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$ to compute

$$\Box_g(\sigma^{-1}) = (4 - \Box_g \sigma)\sigma^{-2}. \tag{155.13}$$

Since $P_M = -\Box_g + m^2 + \xi R[g]$, the σ^{-2} term in $P_{M,x}(U/\sigma)$ is

$$\frac{2\sigma^{;\mu}\nabla_\mu U + (\Box_g\sigma - 4)U}{\sigma^2}. \tag{155.14}$$

The zeroth-order term $m^2 + \xi R[g]$ contributes only order σ^{-1}. Therefore the most singular coefficient vanishes exactly when (155.12) holds.

It remains to identify the solution. Along the geodesic parameterized by s, the operator $\sigma^{;\mu}\nabla_\mu$ is the radial derivative weighted by squared geodesic distance. The Jacobi-field equation for the geodesic exponential map gives

$$\sigma^{;\mu}\nabla_\mu \log\Delta = 4 - \Box_g\sigma. \tag{155.15}$$

Taking one half of this equation gives (155.12) for $U = \Delta^{1/2}$, and the coincidence normalization follows from the normal-coordinate expansion of the exponential map. Uniqueness follows from the first-order ordinary differential equation along the geodesic. $\Box$

Definition 155.8 (Hadamard parametrix in four dimensions)**.** Let

$$V(x,y) \sim \sum_{j\geq 0} v_j(x,y)\sigma(x,y)^j \tag{155.16}$$

be a formal expansion by smooth biscalars. The local Hadamard parametrix is

$$H_{\epsilon,\ell}(x,y) = \frac{1}{8\pi^2}\left[\frac{U(x,y)}{\sigma_\epsilon(x,y)} + V(x,y)\log\left(\frac{\sigma_\epsilon(x,y)}{\ell^2}\right)\right], \qquad \ell > 0, \tag{155.17}$$

where U is fixed by Lemma 155.7 and the jet of V at the diagonal is fixed by the recursion below.

Theorem 155.9 (Hadamard coefficient recursion)**.** *With P_M as in (155.5), the logarithmic coefficient is determined recursively by*

$$\big(2\sigma^{;\mu}\nabla_\mu + \Box_g\sigma - 2\big)v_0 = P_{M,x}U, \tag{155.18}$$

and, for $j \geq 0$,

$$\big(2\sigma^{;\mu}\nabla_\mu + \Box_g\sigma + 2j\big)v_{j+1} = \frac{1}{j+1}P_{M,x}v_j. \tag{155.19}$$

At coincidence,

$$v_0(x,x) = \frac{1}{2}\left(m^2 + \left(\xi - \frac{1}{6}\right)R[g](x)\right), \tag{155.20}$$

and

$$v_{j+1}(x,x) = \frac{1}{2(j+1)(j+2)}\, P_{M,x}v_j(x,y)\big|_{y=x}. \tag{155.21}$$

For every finite N, the partial parametrix built from $\sum_{j=0}^N v_j\sigma^j$ satisfies $P_{M,x}H_{\epsilon,\ell}^{[N]}$ smooth up to the order fixed by the omitted coefficient; a smooth V with the prescribed diagonal jet exists by Borel's lemma, and changing that smooth representative changes $H_{\epsilon,\ell}$ only by a smooth biscalar.

Proof. The calculation is performed for $x \neq y$ before the boundary value is taken. Put $s = \sigma(x,y)$. For a smooth biscalar v,

$$P_{M,x}(v\log s) = (P_{M,x}v)\log s - \frac{2\sigma^{;\mu}\nabla_\mu v + (\Box_g\sigma - 2)v}{s}. \tag{155.22}$$

Together with Lemma 155.7, the remaining s^{-1} term in $P_{M,x}(U/s + v_0 \log s)$ is canceled precisely by (155.18).

For $j \geq 0$, the coefficient of $s^j \log s$ in

$$P_{M,x}\left(\sum_{\ell \geq 0} v_\ell s^\ell \log s\right)$$

has two contributions. The term $v_j s^j$ contributes

$$s^j P_{M,x} v_j.$$

The lower-derivative part of $P_{M,x}(v_{j+1}s^{j+1})$ contributes

$$-(j+1)s^j \left(2\sigma^{;\mu}\nabla_\mu v_{j+1} + (\Box_g \sigma + 2j)v_{j+1}\right). \tag{155.23}$$

Vanishing of this coefficient gives (155.19). Taking $y \to x$ in (155.18) and using $\sigma^{;\mu} = 0$, $\Box_g \sigma = 4$, $U(x,x) = 1$, and

$$\Box_{g,x} U(x,y)\big|_{y=x} = \frac{1}{6} R[g](x) \tag{155.24}$$

gives (155.20). Taking the coincidence limit of (155.19) gives (155.21).

At each step the transport operator has nonzero diagonal coefficient, so the diagonal value and then all derivatives transverse to the diagonal are fixed by differentiating the transport equation. Borel's lemma produces a smooth biscalar with the prescribed infinite jet. Two such choices differ by a smooth biscalar flat at the diagonal, hence by a smooth contribution to (155.17). □

Remark 155.10 (Hadamard form of a quasifree two-point function)*.* A quasifree two-point function ω_2 is locally Hadamard in $\mathcal{U}$ precisely when

$$\omega_2(x,y) - H_{\epsilon,\ell}(x,y) \tag{155.25}$$

extends to a smooth biscalar on $\mathcal{U} \times \mathcal{U}$, after taking the distributional boundary value. Changing ℓ changes the subtraction by the smooth local term

$$-\frac{1}{4\pi^2} V(x,y) \log(\ell'/\ell). \tag{155.26}$$

The singular terms in (155.17) are fixed by the local equation $P_M \omega_2 = 0$ modulo smooth terms and by the positive-frequency boundary value in (155.8). Theorem 155.9 constructs the unique diagonal jet needed to remove the singular part generated by applying P_M in either variable. Therefore any two local parametrices with the same ℓ differ smoothly. A state is in the Hadamard class exactly when its two-point function has this universal singular part and no further singular remainder, which is (155.25). The scale formula follows by subtracting the two logarithms in (155.17).

155.5 Propagation Of Singularities

For a real principal type operator P, singularities of solutions to $Pu = 0$ propagate along the Hamiltonian flow of the principal symbol. For the Klein–Gordon operator (155.5), the principal symbol is $-g^{\mu\nu}k_\mu k_\nu$. The characteristic set is unchanged if one uses the metric Hamiltonian

$$p(x,k) = g^{\mu\nu}(x) k_\mu k_\nu.$$

The two Hamiltonian vector fields differ by an overall sign, hence by a reversal of the bicharacteristic parameter, not by a different null geodesic. The Hamiltonian vector field on T^*M is

$$H_p = \frac{\partial p}{\partial k_\mu}\frac{\partial}{\partial x^\mu} - \frac{\partial p}{\partial x^\mu}\frac{\partial}{\partial k_\mu}.$$

Thus its integral curves satisfy

$$\dot{x}^\mu = 2g^{\mu\nu}(x)k_\nu, \qquad \dot{k}_\mu = -(\partial_\mu g^{\alpha\beta})(x)k_\alpha k_\beta. \tag{155.27}$$

Let $v^\mu = \dot{x}^\mu$. Then

$$v^\mu = 2k^{\sharp\mu}, \qquad g_{\mu\nu}v^\mu v^\nu = 4p(x,k).$$

On $p = 0$, the projected curve $x(s)$ is therefore null. Lowering the index in the first Hamilton equation gives $k_\mu = \frac{1}{2}g_{\mu\nu}v^\nu$. Differentiate this identity and use the second Hamilton equation together with

$$\partial_\mu g^{\alpha\beta} = -g^{\alpha\rho}g^{\beta\sigma}\partial_\mu g_{\rho\sigma}.$$

One obtains

$$g_{\mu\nu}\left(\ddot{x}^\nu + \Gamma^\nu_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma\right) = 0.$$

Since g is non-degenerate, $x(s)$ is an affinely parametrized null geodesic, and k is parallel transported along it up to the fixed factor $1/2$.

Hence once the two-point function has positive-frequency wavefront directions on one Cauchy surface, the equation of motion propagates those directions along null geodesics. This is the geometric content of the microlocal spectrum condition.

155.6 Wick Products

For a Hadamard state, define the Wick square by point splitting:

$$:\phi^2:_H(f) = \int_M f(x)\lim_{y\to x}\big(\phi(x)\phi(y) - H(x,y)\mathbf{1}\big)\,\mathrm{d}\,\mathrm{vol}_g(x).$$

Changing the parametrix scale or adding a smooth local bisolution changes the Wick square by a local curvature polynomial:

$$:\phi^2:'_H = :\phi^2:_H + c_1 R + c_2 m^2.$$

Thus the local covariant field is defined only after the finite local renormalization coordinates are specified.

155.7 Time-Ordered Products

The time-ordered two-point distribution has a wavefront set with both future and past directions arranged according to time ordering. Products of time-ordered distributions at coincident points violate the wavefront-set product criterion. Perturbative algebraic renormalization extends these distributions from configuration space with diagonals removed to the full configuration space. The extension freedom is local and covariant, with degree bounded by scaling. Contact terms in curved spacetime are precisely the local distributions supported on these diagonals, with covariance and Ward identities imposed as constraints.

Open Problem 155.11 (Interacting microlocal spectrum condition)*.* Construct interacting locally covariant QFTs on curved backgrounds with Hadamard states, local Wick and time-ordered products, stress tensor, background gauge fields, and anomaly lines, and prove the microlocal spectrum condition for the resulting nonperturbative Wightman-type distributions.

Chapter 156

Perturbative Algebraic QFT On Curved Backgrounds

Conventions in this chapter.

- **Signature.** Lorentzian formulas use $\mathbb{M}^D$; Euclidean formulas use $\mathbb{R}^D$.
- **Metric sign.** Lorentzian $\eta_{\mu\nu} = \mathrm{diag}(-,+,\ldots,+)$; Euclidean $\delta_{\mu\nu}$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$ for Hilbert-space vacuum matrix elements; functional averages are named separately.
- **Generator convention.** Lorentzian translations use $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

This chapter develops perturbative algebraic QFT on curved backgrounds as a local construction of interacting fields from Hadamard two-point functions, microcausal functionals, time-ordered products, and causal factorization. The framework is perturbative, but its locality and covariance conditions are nonperturbative constraints on the allowed coefficients.

156.1 Functional Spaces And Supports

Let M be an oriented time-oriented globally hyperbolic Lorentzian manifold and let

$$\mathcal{E}(M) = C^\infty(M, \mathbb{R})$$

be the classical configuration space of a real scalar field. It is a Frechet space with the usual seminorms obtained by taking suprema of derivatives on compact subsets in coordinate charts. A functional in this chapter is a smooth map $F : \mathcal{E}(M) \to \mathbb{C}$ in the Bastiani sense: all iterated directional derivatives

$$F^{(n)}(\phi)(h_1, \ldots, h_n) = \frac{\partial^n}{\partial t_1 \cdots \partial t_n} F\left(\phi + \sum_{i=1}^n t_i h_i\right)\Bigg|_{t_1=\cdots=t_n=0}$$

exist and depend jointly continuously on $(\phi, h_1, \dots, h_n)$. When $F^{(n)}(\phi)$ is distributional, the displayed multilinear form is identified with a distributional density on M^n by

$$F^{(n)}(\phi)(h_1, \dots, h_n) = \int_{M^n} F^{(n)}(\phi)(x_1, \dots, x_n) h_1(x_1) \cdots h_n(x_n).$$

This convention suppresses the volume density in the kernel notation.

The support of F is the closed set of points $x \in M$ such that, for every neighborhood $U \ni x$, there are fields ϕ and perturbations $h \in C_c^\infty(U)$ with

$$F(\phi + h) \neq F(\phi).$$

Thus support is defined by actual sensitivity of the functional, not by a chosen formula for it.

Definition 156.1 (Local functional)**.** A compactly supported smooth functional F is local if there are an integer r and a compactly supported smooth density-valued function L on the finite jet bundle $J^r M$, polynomial in the highest derivatives in the usual local-coordinate presentation, such that

$$F(\phi) = \int_M L\big(x, j_x^r \phi\big), \tag{156.1}$$

and such that $F^{(n)}(\phi)$ is supported on the thin diagonal

$$\Delta_n = \{(x, \dots, x) : x \in M\}$$

for every $n \geq 2$. Equivalently, after integrations by parts,

$$F^{(n)}(\phi)(x_1, \dots, x_n) = \sum_{|\alpha_2| + \cdots + |\alpha_n| \leq N} a_{\alpha_2 \cdots \alpha_n}(x_1; \phi) \partial^{\alpha_2} \delta(x_1, x_2) \cdots \partial^{\alpha_n} \delta(x_1, x_n) \tag{156.2}$$

in each coordinate chart, with coefficients depending locally and smoothly on a finite jet of ϕ.

The two descriptions in Definition 156.1 are both needed. The jet formula is the geometric expression of locality; the diagonal-support formula is the microlocal expression used in renormalization.

156.2 Microcausal Functionals

Definition 156.2 (Microcausal functional)**.** Let $F : \mathcal{E}(M) \to \mathbb{C}$ be a compactly supported smooth functional whose functional derivatives are compactly supported distributions,

$$F^{(n)}(\phi) \in \mathcal{E}'(M^n)$$

for all n. The functional is microcausal if

$$\mathrm{WF}(F^{(n)}(\phi)) \cap \Big((\overline{V}_+^*)^n \cup (\overline{V}_-^*)^n \Big) = \varnothing. \tag{156.3}$$

Here $\overline{V}_\pm^* \subset T^*M \setminus 0$ are the closed future and past causal covector cones, using the convention of Chapter 155. The vector space of microcausal functionals is denoted $\mathcal{F}_{\mu c}(M)$, and the subspace of local functionals is denoted $\mathcal{F}_{\mathrm{loc}}(M)$.

The exclusion in (156.3) is exactly the condition needed to contract with Hadamard two-point functions. A Hadamard bidistribution has future covectors in one variable and past covectors in the other. If all covectors in $F^{(n)}$ could point to the same closed time cone, the product criterion of Theorem 155.2 would fail when one contracts the variables against the Hadamard kernel.

156.3 Star Product

Choose a Hadamard two-point function H whose antisymmetric part is

$$H(x,y) - H(y,x) = \mathrm{i}\Delta(x,y),$$

where $\Delta = \Delta_{\mathrm{ret}} - \Delta_{\mathrm{adv}}$ is the causal propagator of the Klein–Gordon operator. For $F, G \in \mathcal{F}_{\mu c}(M)$, define

$$F \star_H G = \sum_{n=0}^{\infty} \frac{\hbar^n}{n!} \left\langle F^{(n)}(\phi), H^{\otimes n} G^{(n)}(\phi) \right\rangle . \tag{156.4}$$

The n-th summand means

$$\int_{M^{2n}} F^{(n)}(\phi)(x_1, \dots, x_n) \prod_{j=1}^{n} H(x_j, y_j) G^{(n)}(\phi)(y_1, \dots, y_n), \tag{156.5}$$

with the integral interpreted as the pairing of distributions. The microcausal cone condition and the Hadamard wavefront set make this pairing well-defined. For polynomial functionals the sum is finite; in general the product is understood in the completed formal space $\mathcal{F}_{\mu c}(M)[[\hbar]]$ when the formal series is meaningful.

Associativity and commutator. The product $\star_H$ is associative on polynomial microcausal functionals. Its first-order commutator is

$$[F, G]_{\star_H} = \mathrm{i}\hbar \{F, G\}_{\mathrm{P}} + O(\hbar^2), \qquad \{F, G\}_{\mathrm{P}} = \left\langle F^{(1)}, \Delta G^{(1)} \right\rangle , \tag{156.6}$$

where $\{\,\cdot\,,\,\cdot\,\}_{\mathrm{P}}$ is the Peierls bracket of the free field. Let m denote pointwise multiplication and set

$$\Gamma_H = \left\langle H, \frac{\delta}{\delta\phi} \otimes \frac{\delta}{\delta\phi} \right\rangle \qquad \text{on} \qquad \mathcal{F} \otimes \mathcal{F}.$$

Then

$$F \star_H G = m \circ e^{\hbar \Gamma_H} (F \otimes G).$$

For three factors let Γ_{ij} contract the i-th and j-th tensor factors. Functional derivatives with respect to different tensor factors commute, so

$$[\Gamma_{12}, \Gamma_{13}] = [\Gamma_{12}, \Gamma_{23}] = [\Gamma_{13}, \Gamma_{23}] = 0.$$

Both $(F \star_H G) \star_H K$ and $F \star_H (G \star_H K)$ are therefore equal to

$$m_3 \circ \exp\big(\hbar(\Gamma_{12} + \Gamma_{13} + \Gamma_{23})\big)(F \otimes G \otimes K),$$

where $m_3(F \otimes G \otimes K) = FGK$. This proves associativity.

The order-$\hbar$ part of the commutator is

$$\hbar \left\langle F^{(1)}, (H - H^{\mathrm{op}}) G^{(1)} \right\rangle = \mathrm{i}\hbar \left\langle F^{(1)}, \Delta G^{(1)} \right\rangle ,$$

which is the free Peierls bracket.

156.4 Change Of Hadamard Function

Proposition 156.3 (Change of Hadamard function)**.** *If H' is another Hadamard two-point function, then*

$$w = H' - H$$

is smooth. Define

$$\alpha_w = \exp\left(\frac{\hbar}{2}\left\langle w, \frac{\delta^2}{\delta\phi^2}\right\rangle\right).$$

Then

$$\alpha_w(F \star_H G) = \alpha_w(F) \star_{H'} \alpha_w(G). \tag{156.7}$$

Thus the abstract algebra is independent of the particular Hadamard parametrix up to a canonical isomorphism determined by the smooth difference.

Proof. Let

$$\Delta_w = \left\langle w, \frac{\delta^2}{\delta\phi^2}\right\rangle.$$

Because w is smooth, Δ_w maps microcausal polynomial functionals to microcausal polynomial functionals. The second-order operator Δ_w satisfies

$$\Delta_w(FG) = (\Delta_w F)G + F(\Delta_w G) + 2\left\langle F^{(1)}, wG^{(1)}\right\rangle. \tag{156.8}$$

Equivalently,

$$e^{\frac{\hbar}{2}\Delta_w} \circ m = m \circ e^{\hbar\Gamma_w}\left(e^{\frac{\hbar}{2}\Delta_w} \otimes e^{\frac{\hbar}{2}\Delta_w}\right), \tag{156.9}$$

where Γ_w is the cross-contraction with kernel w. Therefore

$$\begin{aligned}
\alpha_w(F \star_H G) &= e^{\frac{\hbar}{2}\Delta_w} m e^{\hbar\Gamma_H}(F \otimes G) \\
&= m e^{\hbar\Gamma_w}\left(e^{\frac{\hbar}{2}\Delta_w} \otimes e^{\frac{\hbar}{2}\Delta_w}\right) e^{\hbar\Gamma_H}(F \otimes G) \\
&= m e^{\hbar\Gamma_{H+w}}(\alpha_w F \otimes \alpha_w G) = \alpha_w(F) \star_{H'} \alpha_w(G).
\end{aligned}$$

□

156.5 Time-Ordered Products

Let $H_F = H + \mathrm{i}\Delta_{\mathrm{adv}}$ denote the Feynman bidistribution associated with H. Away from coincident points, the naive time-ordered product is obtained by replacing H in the contraction exponential by H_F. The problem is that H_F has a wavefront set whose products are not defined on diagonals. Thus time ordering is not a pointwise operation on local fields; it is an extension problem for distributions.

A system of time-ordered products is a family of maps

$$T_n : \mathcal{F}_{\mathrm{loc}}(M)^{\otimes n} \longrightarrow \mathcal{F}_{\mu\mathrm{c}}(M)[[\hbar]], \qquad n \geq 1, \tag{156.10}$$

with $T_1 = \mathrm{id}$, symmetric in the n entries, equal to the Feynman contraction formula away from partial diagonals, and satisfying:

$$\text{local covariance}, \qquad \text{causal factorization},$$
$$\text{field independence},$$
$$\text{scaling degree bounds}, \qquad \text{unitarity and Ward identities}.$$

These words are constraints, not decoration. Local covariance means natural dependence on the background metric and couplings. Field independence means that differentiating T_n with respect to the background field inserts the time-ordered derivative of the input local functionals. Scaling degree bounds control the ultraviolet order of singularities. Ward identities are the equations imposed by symmetries; for gauge theories they are properly encoded by the BV quantum master equation.

Theorem 156.4 (Extension with fixed scaling degree)**.** *Let $t^0 \in \mathcal{D}'(\mathbb{R}^N \setminus \{0\})$ have finite scaling degree*

$$\mathrm{sd}(t^0) = \inf\left\{\delta : \lambda^\delta t^0(\lambda x) \to 0 \ \textit{in}\ \mathcal{D}'(\mathbb{R}^N \setminus \{0\}) \ \textit{as}\ \lambda \to 0^+\right\}.$$

Then t^0 has an extension $t \in \mathcal{D}'(\mathbb{R}^N)$ with $\mathrm{sd}(t) = \mathrm{sd}(t^0)$. If $\mathrm{sd}(t^0) < N$, this extension is unique. If $\mathrm{sd}(t^0) \geq N$, two such extensions differ by

$$\sum_{|\alpha| \leq \mathrm{sd}(t^0) - N} c_\alpha \, \partial^\alpha \delta_0, \tag{156.11}$$

with the upper bound understood as its integer part.

Proof. Choose a test function $\rho \in C_c^\infty(\mathbb{R}^N)$ equal to 1 near the origin. For a test function f, subtract its Taylor polynomial at the origin through degree

$$r = \lfloor \mathrm{sd}(t^0) \rfloor - N$$

when $r \geq 0$:

$$f_R(x) = f(x) - \rho(x) \sum_{|\alpha| \leq r} \frac{x^\alpha}{\alpha!} (\partial^\alpha f)(0). \tag{156.12}$$

The subtracted test function vanishes to order $r+1$ at the origin. The definition of scaling degree then implies that $t^0(f_R)$ is finite and independent of how f_R is represented away from the origin. Set

$$t(f) = t^0(f_R) + \sum_{|\alpha| \leq r} c_\alpha (\partial^\alpha f)(0). \tag{156.13}$$

The constants c_α are arbitrary finite renormalization constants. The subtraction raises the vanishing order of the test function precisely enough to preserve the original scaling degree, and the additional $\partial^\alpha \delta_0$ terms have scaling degree $N + |\alpha| \leq \mathrm{sd}(t^0)$.

If $r < 0$, no Taylor coefficient may be added without increasing the scaling degree. An extension is then forced by continuity from $\mathbb{R}^N \setminus \{0\}$, hence unique. In the case $r \geq 0$, any two extensions agree away from the origin, so their difference is a distribution supported at 0. Every distribution supported at one point is a finite linear combination of derivatives of the delta distribution, and the scaling degree bound permits exactly the derivatives in (156.11). □

In pAQFT the role of the origin in Theorem 156.4 is played by a partial diagonal in M^n. The finite ambiguity is therefore a finite sum of local curvature, mass, coupling, and field polynomials multiplying derivatives of delta distributions supported on that diagonal. These are contact terms created by the definition of time-ordered products, not a separate physical process.

156.6 Causal Factorization

If the supports of $F_1, \dots, F_k$ lie nowhere in the past of the supports of $F_{k+1}, \dots, F_n$, causal factorization states

$$T_n(F_1, \dots, F_n) = T_k(F_1, \dots, F_k) \star T_{n-k}(F_{k+1}, \dots, F_n).$$

This identity determines T_n outside the total diagonal from lower orders. Renormalization is precisely the choice of extension to the total and partial diagonals compatible with the listed constraints.

The induction is local in configuration space. On a configuration $(x_1, \dots, x_n)$ not lying on the total diagonal, global hyperbolicity provides a Cauchy time function and one may separate the points into a later cluster and an earlier cluster. Causal factorization expresses T_n there as a $\star_H$-product of already constructed lower-order time-ordered products. On overlaps of such cluster decompositions the lower-order factorization identities and associativity of $\star_H$ give the same distribution. Hence the only new datum at order n is the extension across the total diagonal, subject to covariance, scaling degree, unitarity, and any Ward identities.

156.7 Interacting Fields

Let V be an interaction functional with compact time support. The relative S-matrix is

$$S_V(F) = S(V)^{-1} \star S(V + F),$$

where

$$S(F) = T \exp\left(\frac{\mathrm{i}}{\hbar} F\right).$$

The interacting field associated to a local functional F is the Bogoliubov derivative

$$R_V(F) = \frac{\hbar}{\mathrm{i}} \frac{d}{d\lambda} S_V(\lambda F) \bigg|_{\lambda=0}.$$

Changing the time cutoff of V outside the causal domain of F gives isomorphic interacting algebras by causal factorization.

Proposition 156.5 (Causal support of the Bogoliubov field). *If two interaction functionals V and V' agree on a neighborhood of the causal hull $J(\operatorname{supp} F)$, then the corresponding Bogoliubov fields $R_V(F)$ and $R_{V'}(F)$ are identified by the canonical relative S-matrix isomorphism of the two cutoff theories.*

Proof. Let $C = V' - V$. By hypothesis C vanishes on a neighborhood of $J(\operatorname{supp} F)$. Choose two Cauchy surfaces, one to the past of $\operatorname{supp} F$ and one to its future, both lying inside the region on which $V = V'$ near the causal hull. A smooth partition of unity in the corresponding time slabs decomposes

$$C = A + B,$$

where $\operatorname{supp} A$ is later than $\operatorname{supp} F$ and $\operatorname{supp} B$ is earlier than $\operatorname{supp} F$. The separation is meant in the causal sense used by the time-ordered products: every point of $\operatorname{supp} A$ lies outside the causal past of $\operatorname{supp} F$, and every point of $\operatorname{supp} B$ lies outside its causal future, after the supports have been slightly enlarged to open neighborhoods. Causal factorization gives

$$S(V + A + B + \lambda F) = S(A) \star S(V + \lambda F) \star S(B)$$

in the corresponding ordered region, and similarly with $\lambda = 0$. Thus

$$S(V') = S(A) \star S(V) \star S(B)$$

and, using the inverse in the formal $\star$-algebra,

$$S(V')^{-1} = S(B)^{-1} \star S(V)^{-1} \star S(A)^{-1}.$$

Substitution in

$$S_{V'}(\lambda F) = S(V')^{-1} \star S(V' + \lambda F)$$

therefore gives

$$S_{V'}(\lambda F) = S(B)^{-1} \star S_V(\lambda F) \star S(B).$$

The conjugation

$$\alpha_B(X) = S(B)^{-1} \star X \star S(B)$$

is precisely the relative S-matrix isomorphism between the two time-cutoff choices in the causal domain of F. Differentiating the displayed identity at $\lambda = 0$ gives

$$R_{V'}(F) = \alpha_B(R_V(F)),$$

which is the claimed canonical identification. The construction is independent of the auxiliary partition because two choices differ by factors supported entirely in the past or future of $\operatorname{supp} F$, and the same causal-factorization computation gives the corresponding inner identification. □

156.8 Renormalization Group

Two choices of time-ordered products T and $\widetilde{T}$ are related by maps Z on local functionals:

$$\widetilde{S}(F) = S(Z(F)).$$

The map Z is local, begins with $Z(F) = F + O(F^2)$, preserves the constant term, and has coefficients supported on coincident points. In curved backgrounds these coefficients include curvature terms. This is the perturbative algebraic form of local counterterm freedom.

More explicitly, Z is a formal diffeomorphism of the space of local functionals:

$$Z(F) = F + \sum_{n \geq 2} \frac{1}{n!} Z_n(F^{\otimes n}), \tag{156.14}$$

where each Z_n is symmetric, local, and supported on the thin diagonal in the n-fold product. The defining identity $\widetilde{S}(F) = S(Z(F))$ implies that changing the extension of time-ordered products is equivalent to changing the local interaction functional by finite local terms. In a curved background those terms are not limited to flat-space monomials: covariance permits terms built from the metric, curvature, masses, background couplings, and their covariant derivatives, constrained by dimension, symmetries, and field independence.

156.9 Worked Interacting Scalar Coordinate

The previous sections describe the abstract perturbative construction. A curved-background interacting example must also say which local coordinates are being held fixed and what observable changes when those coordinates are changed. Consider a real scalar with compact spacetime cutoff g, and write the quartic interaction in a Hadamard Wick coordinate H as

$$V_{\lambda,H}(g) = \int_M \mathrm{d}\,\mathrm{vol}_g\; g(x)\,\frac{\lambda}{4!}\Phi_H^4(x). \tag{156.15}$$

Let $H' = H + w$, with w smooth, and write

$$w_\Delta(x) = w(x,x).$$

The smooth-change isomorphism Proposition 156.3 gives the local diagonal identities

$$\begin{aligned}\alpha_w\big(\Phi_H^2(x)\big) &= \Phi_{H'}^2(x) + \hbar w_\Delta(x),\\ \alpha_w\big(\Phi_H^4(x)\big) &= \Phi_{H'}^4(x) + 6\hbar w_\Delta(x)\Phi_{H'}^2(x) + 3\hbar^2 w_\Delta(x)^2.\end{aligned} \tag{156.16}$$

Thus the same abstract quartic interaction, transported to the H' coordinate, is

$$\alpha_w V_{\lambda,H}(g) = \int_M \mathrm{d}\,\mathrm{vol}_g\; g(x)\left[\frac{\lambda}{4!}\Phi_{H'}^4 + \frac{\hbar\lambda}{4}w_\Delta\Phi_{H'}^2 + \frac{\hbar^2\lambda}{8}w_\Delta^2\right](x). \tag{156.17}$$

Equations (156.16) and (156.17) are algebra-coordinate statements. They do not by themselves produce a physical expectation-value shift. If ω_H is a state on the H-presentation and $\omega_{H'}$ is the transported state on the H'-presentation, defined by

$$\omega_{H'}(\alpha_w A) = \omega_H(A),$$

then

$$\omega_{H'}(\alpha_w V_{\lambda,H}(g)) = \omega_H(V_{\lambda,H}(g))\,.$$

This is the same abstract observable in two Hadamard coordinates.

A finite local Wick-renormalization change is a different comparison. It is not an arbitrary smooth difference of two Hadamard two-point functions; local covariance restricts the diagonal scalar inserted into Wick powers. In four dimensions, for the dimension-two part, write

$$c(x) = a_m m^2 + a_R R(x). \tag{156.18}$$

Changing the Wick prescription by this local scalar uses the same polynomial combinatorics as (156.16), with w_Δ replaced by c, but it compares two local covariant composite-field prescriptions in a fixed physical state. The quartic interaction then acquires the finite local terms

$$\int_M \mathrm{d}\,\mathrm{vol}_g\; g(x)\left[\frac{\hbar\lambda}{4}c\Phi_H^2 + \frac{\hbar^2\lambda}{8}c^2\right](x).$$

The quadratic term is the finite coordinate shift

$$\delta m^2 = \frac{\hbar\lambda}{2} a_m m^2, \qquad \delta\xi = \frac{\hbar\lambda}{2} a_R \tag{156.19}$$

in the local scalar operator $\frac{1}{2}(m^2 + \xi R)\Phi^2$. The last term in (156.17) is a finite shift of the purely geometric source functional, producing m^4, $m^2 R$, and R^2 coordinates when (156.18) is used. Those coordinates are precisely the ones that reappear in the stress-tensor and gravitational EFT bookkeeping of Chapters 148 and 157.

This paragraph records only the Wick-coordinate subfamily of the finite renormalization freedom. A complete interacting stress-tensor construction also fixes the time-ordered products used in the Bogoliubov map, the interaction-dependent composite-field counterterms in Φ^4, and the independent conserved local tensors allowed in Proposition 148.9. The Hollands–Wald conservation theorem is a condition on that complete construction: the chosen time-ordered products and contact terms must make the interacting stress tensor conserved, rather than relying on the potential insertion or the Wick-square shift by themselves [10, 11, 12].

The local Wick-square comparison has a visible c-number output only in this fixed-state, changed-prescription sense. If

$$F_f = \int_M \mathrm{d}\,\mathrm{vol}_g\ f(x)\Phi_H^2(x), \qquad f \in C_c^\infty(M),$$

and the second Wick-square prescription is

$$F_f^{(c)} = F_f + \hbar \int_M \mathrm{d}\,\mathrm{vol}_g\ f(x)c(x),$$

then

$$\omega_H\left(F_f^{(c)}\right) - \omega_H(F_f) = \hbar \int_M \mathrm{d}\,\mathrm{vol}_g\ f(x)c(x). \tag{156.20}$$

The right side is state independent because it is a local renormalization coordinate, not particle production and not the transport of a state between two algebra presentations. A physical renormalization condition fixes a_m, a_R or fixes the value of (156.20) on selected backgrounds and states. Without that condition, the number is a coordinate of the composite operator, not an invariant prediction.

Derivation. For a local polynomial at one point, the smooth-change isomorphism acts by

$$\alpha_w = \exp\left(\frac{\hbar}{2} w_\Delta \frac{\partial^2}{\partial\phi^2}\right).$$

Applying this operator to ϕ^2 and ϕ^4 gives (156.16). Multiplication by $\lambda/4!$ gives the coordinate-transport identity (156.17). The local Wick-renormalization comparison uses the same finite polynomial identity with the locally covariant scalar c in place of the generic smooth diagonal w_Δ; the identities (156.19) follow by comparing the coefficient of $\frac{1}{2}(m^2 + \xi R)\Phi^2$. Finally, applying the first line of (156.16) with w_Δ replaced by c inside F_f gives (156.20).

156.10 Gauge Theories And The BV Condition

The preceding scalar construction is not a foundation for gauge theory by itself. A gauge-fixed perturbative algebra contains unphysical fields, ghosts, antifields, and a differential. Consistency is the statement that the interacting quantum BV differential squares to zero, or equivalently that the renormalized time-ordered products can be chosen so that the quantum master equation holds:

$$\frac{1}{2}(S_{\mathrm{BV}}, S_{\mathrm{BV}}) - \mathrm{i}\hbar\, \Delta_{\mathrm{BV}} S_{\mathrm{BV}} = 0 \tag{156.21}$$

in the renormalized algebra. Here $(\,\cdot\,,\,\cdot\,)$ is the BV antibracket and Δ_{BV} denotes the renormalized second-order operator measuring the short-distance contraction of fields with antifields. The equation is not a slogan for gauge invariance. It is the algebraic condition that the interacting BRST operator is a differential and that the physical algebra is its cohomology. If the obstruction to solving (156.21) is a nontrivial local cohomology class, the theory has an anomaly in the chosen background and field content.

Open Problem 156.6 (Nonperturbative curved-background comparison)*.* Relate perturbative algebraic QFT on curved backgrounds to nonperturbative constructions by comparing local algebras, Hadamard states, stress tensors, renormalized composite fields, anomaly terms, and scaling limits in examples where both sides are available.

Chapter 157

Semiclassical Backreaction And Stress-Tensor Fluctuations

Conventions in this chapter.

- **Signature.** Lorentzian $\mathbb{M}^D$.
- **Metric sign.** $\eta_{\mu\nu} = \mathrm{diag}(-, +, \ldots, +)$.
- $\hbar$. 1, unless explicitly displayed.
- **Vacuum symbol.** $\Omega = \Omega$.
- **Generator convention.** $U(a) = \exp(\mathrm{i}a^\mu P_\mu)$, hence $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ when the field is translation covariant.
- **Global guide.** Recurrent symbols, overload warnings, and standing sign conventions are collected in the Notation and Convention Guide; the local definitions in this chapter fix the operative meaning.

QFT on a fixed curved background supplies local observables, states, and a renormalized stress tensor. Once the stress tensor is used as a source for geometry, the metric is no longer a passive parameter. A semiclassical problem must therefore specify both a locally covariant quantum field theory on each admissible metric and a gravitational effective action whose metric variation is equated to the stress-tensor expectation value. This chapter develops that statement, the associated linear response theory, and the stress-tensor fluctuation data needed for stochastic backreaction.

157.1 Metric Variation And Curvature Convention

The curvature convention in this chapter is

$$[\nabla_\mu, \nabla_\nu] X^\rho = R^\rho{}_{\sigma\mu\nu} X^\sigma, \qquad R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{157.1}$$

Metric variations are taken with respect to $g^{\mu\nu}$. For a renormalized source functional $W[g]$, define the stress tensor by

$$\delta W[g] = -\frac{1}{2} \int_M \mathrm{d}\,\mathrm{vol}_g \, \langle T^{\mathrm{ren}}_{\mu\nu} \rangle_g \, \delta g^{\mu\nu}. \tag{157.2}$$

The minus sign is the same sign that gives $\delta S_{\text{matter}} = -(1/2)\int T_{\mu\nu}\delta g^{\mu\nu}$ in the classical limit with mostly-plus metric.

Ward identity from diffeomorphism covariance. Suppose $W[g]$ is invariant under compactly supported diffeomorphisms and has metric variation of the form (157.2). Then

$$\nabla^\mu \langle T^{\text{ren}}_{\mu\nu}\rangle_g = 0 \tag{157.3}$$

as a distribution.

Let X^μ be a compactly supported vector field and let ψ_s be its flow. Diffeomorphism invariance gives

$$0 = \frac{\mathrm{d}}{\mathrm{d}s} W[\psi_s^* g]\bigg|_{s=0}.$$

Since

$$\delta_X g_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu, \qquad \delta_X g^{\mu\nu} = -(\nabla^\mu X^\nu + \nabla^\nu X^\mu),$$

equation (157.2) gives

$$0 = \int_M \mathrm{d}\,\mathrm{vol}_g\, \langle T^{\text{ren}}_{\mu\nu}\rangle_g \nabla^\mu X^\nu.$$

Integrating by parts and using compact support of X gives

$$0 = -\int_M \mathrm{d}\,\mathrm{vol}_g\; X^\nu \nabla^\mu \langle T^{\text{ren}}_{\mu\nu}\rangle_g.$$

Since X is arbitrary, the distribution in (157.3) vanishes.

If the theory has a diffeomorphism anomaly, the right side of (157.3) is the local anomaly. In the ordinary scalar, spinor, and vector examples considered in the fixed background chapters, diffeomorphism covariance is imposed on the renormalized stress tensor; trace anomalies are compatible with (157.3).

157.2 Renormalized Stress Tensor

Let (M, g) be globally hyperbolic and let ω be a Hadamard state of a locally covariant QFT. The renormalized stress tensor is an operator-valued distribution

$$T^{\text{ren}}_{\mu\nu}(f^{\mu\nu}), \qquad f^{\mu\nu} \in C_c^\infty(M, S^2TM),$$

or, in perturbative algebraic language, a local covariant Wick polynomial valued in the algebra constructed from the Hadamard parametrix. Its expectation value is

$$\langle T^{\text{ren}}_{\mu\nu}(x)\rangle_{\omega,g}.$$

For Hadamard states this is a smooth tensor field after point splitting and after the finite local curvature terms have been fixed. For a scalar field, the subtraction is the one described in Chapter 148; for an interacting pAQFT model it is the local Wick-polynomial and time-ordered-product renormalization of Chapter 156.

The finite ambiguity in four spacetime dimensions has the form

$$\langle T^{\mathrm{ren}}_{\mu\nu}\rangle \longmapsto \langle T^{\mathrm{ren}}_{\mu\nu}\rangle + c_0 m^4 g_{\mu\nu} + c_1 m^2 G_{\mu\nu} + c_2 H^{(1)}_{\mu\nu} + c_3 H^{(2)}_{\mu\nu}, \tag{157.4}$$

for a massive scalar model, with the evident modifications when additional background couplings are present. The tensors $H^{(i)}_{\mu\nu}$ are local conserved curvature tensors defined in the next section. Thus renormalizing the stress tensor and choosing gravitational EFT coordinates are not independent operations.

157.3 Gravitational Effective Action

In four dimensions, take the local gravitational part of the renormalized effective action to be

$$\Gamma_{\mathrm{grav}}[g] = \int_M \mathrm{d}\,\mathrm{vol}_g \left[\frac{1}{16\pi G_N}(R-2\Lambda) + \frac{\alpha}{2}R^2 + \frac{\beta}{2}R_{\rho\sigma}R^{\rho\sigma}\right], \tag{157.5}$$

up to boundary terms and topological combinations. The factor 1/2 in the curvature-squared terms fixes the normalization

$$E^{\mathrm{grav}}_{\mu\nu} := \frac{2}{\sqrt{|g|}}\frac{\delta\Gamma_{\mathrm{grav}}}{\delta g^{\mu\nu}} = \frac{1}{8\pi G_N}(G_{\mu\nu} + \Lambda g_{\mu\nu}) + \alpha H^{(1)}_{\mu\nu} + \beta H^{(2)}_{\mu\nu}. \tag{157.6}$$

Curvature-squared variational formulas. With the curvature convention (157.1), and with the same metric-variation normalization used in Chapter 148, the curvature-squared Euler tensors in (157.6) are

$$\begin{aligned} H^{(1)}_{\mu\nu} &= 2RR_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R^2 + 2g_{\mu\nu}\Box R - 2\nabla_\mu\nabla_\nu R, \\ H^{(2)}_{\mu\nu} &= 2R_{\mu\rho\nu\sigma}R^{\rho\sigma} - \frac{1}{2}g_{\mu\nu}R_{\rho\sigma}R^{\rho\sigma} + \nabla_\mu\nabla_\nu R - \Box R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\Box R. \end{aligned} \tag{157.7}$$

Here $\Box = g^{\rho\sigma}\nabla_\rho\nabla_\sigma$.

These formulae are variational identities fixing the local gravitational-coordinate normalization, not an additional structural theorem. The basic first variations are

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}\,g_{\mu\nu}\delta g^{\mu\nu}, \qquad \delta R = R_{\mu\nu}\delta g^{\mu\nu} + g_{\mu\nu}\Box\delta g^{\mu\nu} - \nabla_\mu\nabla_\nu\delta g^{\mu\nu}, \tag{157.8}$$

after discarding boundary terms. Applying these identities to $\frac{1}{2}\int\sqrt{|g|}R^2$ gives the first line of (157.7).

For $\frac{1}{2}\int\sqrt{|g|}R_{\rho\sigma}R^{\rho\sigma}$, use

$$\delta R_{\mu\nu} = \frac{1}{2}\left(-\Box\delta g_{\mu\nu} - \nabla_\mu\nabla_\nu(g^{\rho\sigma}\delta g_{\rho\sigma}) + \nabla^\rho\nabla_\mu\delta g_{\rho\nu} + \nabla^\rho\nabla_\nu\delta g_{\rho\mu}\right), \tag{157.9}$$

convert $\delta g_{\mu\nu}$ to $-g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}$, integrate derivatives off $\delta g^{\mu\nu}$, and commute derivatives on $R_{\mu\nu}$. The commutator terms combine into $2R_{\mu\rho\nu\sigma}R^{\rho\sigma}$, while the derivative terms combine as displayed in the second line of (157.7).

The tensors $H^{(1)}$ and $H^{(2)}$ are conserved. This follows without component calculation: each is the Euler tensor of a diffeomorphism-invariant local functional, so the Ward-identity calculation above applies with the functional replacing W. Their traces in $D = 4$ are

$$g^{\mu\nu} H^{(1)}_{\mu\nu} = 6\Box R, \qquad g^{\mu\nu} H^{(2)}_{\mu\nu} = -2\Box R, \tag{157.10}$$

using the convention in (157.7). These traces are total derivatives, as required by the constant-Weyl-scale invariance of the curvature-squared actions in four dimensions.

157.4 Semiclassical Einstein Equation

A semiclassical solution is not only a metric. It is a pair (g, ω) such that ω is an admissible Hadamard state of the QFT constructed on (M, g) and

$$E^{\mathrm{grav}}_{\mu\nu}[g] = \langle T^{\mathrm{ren}}_{\mu\nu} \rangle_{\omega,g}. \tag{157.11}$$

Written out,

$$\frac{1}{8\pi G_N}(G_{\mu\nu} + \Lambda g_{\mu\nu}) + \alpha H^{(1)}_{\mu\nu} + \beta H^{(2)}_{\mu\nu} = \langle T^{\mathrm{ren}}_{\mu\nu} \rangle_{\omega,g}. \tag{157.12}$$

The left side is conserved by diffeomorphism invariance. Therefore the right side must be the conserved stress tensor of the same renormalization scheme. A change of stress-tensor scheme of the form (157.4) is equivalently a finite change of $G_N, \Lambda, \alpha, \beta$. It is not a physical change until a renormalization condition has fixed these coordinates.

Definition 157.1 (Backreaction datum)**.** A fixed-order semiclassical backreaction datum consists of

$$\mathfrak{B} = (M, \mathcal{C}, \mathcal{A}_g, \mathcal{S}_g, T^{\mathrm{ren}}_{\mu\nu}, G_N, \Lambda, \alpha, \beta, \mathcal{I}).$$

Here $\mathcal{C}$ is the class of metrics considered, $\mathcal{A}_g$ is the locally covariant QFT algebra assigned to g, $\mathcal{S}_g$ is the admissible state space, T^{ren} is the renormalized stress tensor in a fixed scheme, and $\mathcal{I}$ is the initial or boundary data for the metric and state.

The datum matters because the equation is state dependent. For example, the same metric may solve (157.12) for one state and fail for another. Conversely, changing the metric changes the algebra and the Hadamard condition, so the state must be transported or reconstructed as part of the problem.

157.5 Linear Response

Proposition 157.2 (Retarded stress-tensor response at separated points)**.** *Let the locally covariant theory be fixed on a globally hyperbolic background (M, g), let ω be an admissible Hadamard state, and let the renormalized stress tensor and its time-ordered square be chosen in one renormalization scheme. Vary the inverse metric by a compactly supported smooth tensor $h^{\mu\nu} = \delta g^{\mu\nu}$. The metric perturbation couples to the stress tensor through*

$$\delta S_{\mathrm{source}} = -\frac{1}{2} \int_M \mathrm{d}\,\mathrm{vol}_g \; h^{\rho\sigma} T^{\mathrm{ren}}_{\rho\sigma}. \tag{157.13}$$

The first variation of the stress-tensor expectation has the form

$$\delta\langle T_{\mu\nu}(x)\rangle = \frac{1}{2}\int_M \mathrm{d}\,\mathrm{vol}_g(y)\, G^{\mathrm{ret}}_{\mu\nu,\rho\sigma}(x,y)h^{\rho\sigma}(y) + \delta T^{\mathrm{loc}}_{\mu\nu}(x). \tag{157.14}$$

The separated-point retarded kernel is

$$G^{\mathrm{ret}}_{\mu\nu,\rho\sigma}(x,y) = \mathrm{i}\,\Theta(x \succ y)\,\omega\big([T^{\mathrm{ren}}_{\mu\nu}(x), T^{\mathrm{ren}}_{\rho\sigma}(y)]\big)\,, \tag{157.15}$$

where $x \succ y$ means $x \in J^+(y)$. The local term $\delta T^{\mathrm{loc}}_{\mu\nu}$ contains variations of the point-splitting subtraction, curvature counterterms, and contact terms in the definition of the time-ordered stress-tensor product.

Derivation of the retarded commutator. The clean statement is first made for a smeared observable $A(f) = \int A(x)f(x)$ whose support is disjoint from the support of the local counterterm extension of the stress-tensor product. In the interaction-picture or pAQFT notation, the relative S-matrix for the source $V = \delta S_{\mathrm{source}}$ gives, to first order,

$$\delta\omega(A(f)) = \mathrm{i}\,\omega\big([V_{J^-(\mathrm{supp}\,f)}, A(f)]\big) + \delta_{\mathrm{loc}}\omega(A(f)).$$

Only the part of V in the causal past of $\mathrm{supp}\,f$ appears; this is the retarded prescription. The local term δ_{loc} records two effects that cannot be read from separated points: the explicit variation of the composite field A with the metric and the chosen extension of time-ordered products to the diagonal.

Set $A = T^{\mathrm{ren}}_{\mu\nu}$ and insert (157.13). For $y \in J^-(x)$,

$$\mathrm{i}\left[-\frac{1}{2}h^{\rho\sigma}(y)T^{\mathrm{ren}}_{\rho\sigma}(y), T^{\mathrm{ren}}_{\mu\nu}(x)\right] = \frac{1}{2}h^{\rho\sigma}(y)\,\mathrm{i}\,[T^{\mathrm{ren}}_{\mu\nu}(x), T^{\mathrm{ren}}_{\rho\sigma}(y)].$$

After unsmearing away from the diagonal, this gives (157.14) with the kernel (157.15). Microcausality makes the commutator vanish for spacelike separated x, y; the restriction $\Theta(x \succ y)$ removes the advanced part and is therefore the unique causal linear response compatible with the sign convention $\delta W = -(1/2)\int T_{\rho\sigma}\delta g^{\rho\sigma}$. The remaining diagonal ambiguity is exactly the local term $\delta T^{\mathrm{loc}}_{\mu\nu}$. □

The linearized semiclassical equation is therefore

$$\delta E^{\mathrm{grav}}_{\mu\nu}[h] - \frac{1}{2}\int_M \mathrm{d}\,\mathrm{vol}_g(y)\, G^{\mathrm{ret}}_{\mu\nu,\rho\sigma}(x,y)h^{\rho\sigma}(y) = \delta\langle T_{\mu\nu}\rangle_{\mathrm{state}} + \delta T^{\mathrm{loc}}_{\mu\nu}. \tag{157.16}$$

The term $\delta\langle T\rangle_{\mathrm{state}}$ denotes an independent change of state at fixed metric; it is absent if the state is transported by the chosen metric perturbation prescription.

157.6 Noise Kernel

Stress-tensor fluctuations are encoded by the symmetrized connected two-point distribution

$$N_{\mu\nu,\rho\sigma}(x,y) = \frac{1}{2}\,\omega\big(\{T^{\mathrm{ren}}_{\mu\nu}(x), T^{\mathrm{ren}}_{\rho\sigma}(y)\}\big)_{\mathrm{conn}}\,. \tag{157.17}$$

Equivalently, for a real compactly supported test tensor $f^{\mu\nu}$, set

$$T(f) = \int_M \mathrm{d}\,\mathrm{vol}_g \; f^{\mu\nu} T^{\mathrm{ren}}_{\mu\nu}.$$

Then

$$N(f,f) = \frac{1}{2}\omega(\{T(f) - \omega(T(f)), T(f) - \omega(T(f))\}) = \omega\big((T(f) - \omega(T(f)))^2\big) \geq 0. \tag{157.18}$$

Thus N is a positive semidefinite distributional covariance kernel on the real test-tensor space where the renormalized product is defined.

Conservation is the distributional Ward identity

$$\nabla^\mu_x N_{\mu\nu,\rho\sigma}(x,y) = 0, \qquad \nabla^\rho_y N_{\mu\nu,\rho\sigma}(x,y) = 0, \tag{157.19}$$

including diagonal contact terms fixed by the same local covariance conditions that define the stress-tensor product. Away from the diagonal it follows immediately from $\nabla^\mu T^{\mathrm{ren}}_{\mu\nu} = 0$. On the diagonal it is a condition on the renormalized composite product; it is imposed at the level of the local extension across coincident points.

157.7 Stationary States And Fluctuation–Dissipation

If g has a complete timelike Killing field K and ω is a KMS state at inverse temperature β for the associated time evolution, the retarded kernel and the noise kernel are related. For two smeared stress-tensor observables A and B, define

$$C_{AB}(t) = \omega(A(t)B), \qquad \rho_{AB}(t) = \omega([A(t), B]).$$

Let tildes denote Fourier transforms in t. The KMS condition gives

$$\widetilde{C}_{BA}(-\omega) = e^{-\beta\omega}\widetilde{C}_{AB}(\omega). \tag{157.20}$$

Therefore the symmetrized spectrum and the commutator spectrum obey

$$\widetilde{N}_{AB}(\omega) = \frac{1}{2}\coth\left(\frac{\beta\omega}{2}\right)\widetilde{\rho}_{AB}(\omega), \tag{157.21}$$

where

$$\widetilde{N}_{AB}(\omega) = \frac{1}{2}\left(\widetilde{C}_{AB}(\omega) + \widetilde{C}_{BA}(-\omega)\right), \qquad \widetilde{\rho}_{AB}(\omega) = \widetilde{C}_{AB}(\omega) - \widetilde{C}_{BA}(-\omega).$$

This follows directly from KMS analyticity and the definitions of the retarded and symmetrized kernels.

157.8 Einstein–Langevin Equation

A stochastic semiclassical approximation introduces a real Gaussian generalized random tensor $\xi_{\mu\nu}$ with

$$\mathbb{E}[\xi_{\mu\nu}(f^{\mu\nu})] = 0, \qquad \mathbb{E}[\xi(f)\xi(k)] = N(f,k). \tag{157.22}$$

The existence of this Gaussian generalized random field follows from the positivity (157.18): N is a covariance form on a nuclear test-function space, so the Bochner–Minlos theorem constructs the corresponding Gaussian measure on distributions.

After gauge fixing the linearized gravitational operator, the Einstein–Langevin equation is

$$\mathcal{L}_{\mu\nu}{}^{\rho\sigma} h_{\rho\sigma} - \frac{1}{2}\int_M \mathrm{d}\,\mathrm{vol}_g(y)\, G^{\mathrm{ret}}_{\mu\nu,\rho\sigma}(x,y) h^{\rho\sigma}(y) = \xi_{\mu\nu}(x), \tag{157.23}$$

where $\mathcal{L}$ is the gauge-fixed variation of $E^{\mathrm{grav}}_{\mu\nu}$ plus the local stress-tensor response. If $\mathcal{G}_{\mathrm{ret}}$ is a retarded inverse for the full linear operator, then the induced metric covariance is

$$\mathbb{E}[h_{\mu\nu}(x)h_{\rho\sigma}(y)] = \int \mathcal{G}_{\mathrm{ret}\,\mu\nu}{}^{\alpha\beta}(x,u) N_{\alpha\beta,\gamma\delta}(u,v) \mathcal{G}_{\mathrm{ret}\,\rho\sigma}{}^{\gamma\delta}(y,v). \tag{157.24}$$

This equation is meaningful only after the state, stress-tensor renormalization, contact terms, gauge condition, and stochastic test space have all been specified.

Pushforward covariance of the Einstein–Langevin source. Let $\mathcal{E}$ be the real nuclear test-tensor space on which the noise covariance N is defined, and let $\mathcal{E}'$ be its distributional dual. Suppose the gauge-fixed linear operator in (157.23) has a continuous retarded solution map

$$\mathcal{G}_{\mathrm{ret}} : \mathcal{E}' \longrightarrow \mathcal{H}, \tag{157.25}$$

where $\mathcal{H}$ is the chosen linear metric-perturbation distribution space. If ξ is the centered Gaussian generalized random tensor with covariance N, then $h = \mathcal{G}_{\mathrm{ret}}\xi$ is a centered Gaussian random metric perturbation. For any real linear observables $\ell_1, \ell_2 \in \mathcal{H}^*$,

$$\mathbb{E}[\ell_1(h)\ell_2(h)] = N(\mathcal{G}^*_{\mathrm{ret}}\ell_1, \mathcal{G}^*_{\mathrm{ret}}\ell_2), \tag{157.26}$$

where $\mathcal{G}^*_{\mathrm{ret}} : \mathcal{H}^* \to \mathcal{E}$ is the transpose. In kernel notation this is exactly (157.24). The resulting covariance is positive semidefinite on metric observables:

$$\mathbb{E}[\ell(h)^2] \geq 0. \tag{157.27}$$

For a centered Gaussian generalized random tensor, the characteristic functional on $\mathcal{E}$ is

$$\mathbb{E}[e^{\mathrm{i}\xi(f)}] = \exp\left[-\frac{1}{2}N(f,f)\right]. \tag{157.28}$$

For $\ell \in \mathcal{H}^*$,

$$\ell(h) = \ell(\mathcal{G}_{\mathrm{ret}}\xi) = \xi(\mathcal{G}^*_{\mathrm{ret}}\ell). \tag{157.29}$$

Thus the finite collection $(\ell_1(h), \ldots, \ell_n(h))$ has characteristic function

$$\exp\left[-\frac{1}{2}\sum_{a,b} t_a t_b\, N(\mathcal{G}^*_{\mathrm{ret}}\ell_a, \mathcal{G}^*_{\mathrm{ret}}\ell_b)\right], \tag{157.30}$$

which is Gaussian with covariance (157.26). Positivity follows by setting $\ell_1 = \ell_2 = \ell$ and using positivity of N. When ℓ is evaluation against a compactly supported test tensor on the metric side, $\mathcal{G}_{\mathrm{ret}}^{*}\ell$ is represented by the retarded Green kernel in (157.24), producing the displayed kernel formula.

Only gauge-invariant or gauge-fixed metric observables should be inserted in the pushforward covariance formula above. If ℓ is pure gauge and the retarded inverse has been constructed after gauge fixing, $\ell(h)$ is a coordinate artifact, not a physical metric fluctuation. The formula is therefore a covariance statement for the declared linearized problem, not a replacement for the gravitational constraint analysis.

157.9 Finite Response Window And Stability Diagnostics

The preceding formulae become physically useful only after one asks whether the retarded inverse is bounded in the sector being tested. In a stationary background, after gauge fixing and after solving or projecting out the linearized constraints, let e_a, $a = 1, \dots, n$, be a finite set of real metric perturbation modes or gauge-invariant metric observables. Let f_a, $a = 1, \dots, n$, be the corresponding finite set of real smeared stress-tensor sources. The retained linearized equation in frequency space has the form

$$\mathsf{D}_{ab}(\omega)h_b(\omega) = j_a(\omega), \qquad \mathsf{D}(\omega) = \mathsf{K}(\omega) - \frac{1}{2}\Pi^{\mathrm{ret}}(\omega), \tag{157.31}$$

where K is the gravitational and local-contact part of the linearized operator and Π^{ret} is the projected retarded stress-tensor commutator. The source j may include a state perturbation, an external classical stress perturbation, or the stochastic source in the Einstein–Langevin approximation. All factors of G_N, α, β are included in K; equivalently one may multiply the whole equation by $8\pi G_N$, in which case the same test is applied to the rescaled matrix.

Controlled Approximation 157.3 (Finite semiclassical response window)**.** Fix a frequency interval I, a retained metric-source sector as above, and a reduced low-energy gravitational operator in which the cutoff-scale higher-derivative roots have been removed by the EFT prescription. Suppose that the projected retarded response has no pole in the upper half of the complex frequency plane in the strip obtained by continuing I, and that

$$M_I := \sup_{\omega \in I} \left\| \mathsf{D}(\omega + \mathrm{i}0)^{-1} \right\|_2 < \infty. \tag{157.32}$$

Then, for every retained source vector $j(\omega)$,

$$\|h(\omega)\|_2 \le M_I \, \|j(\omega)\|_2, \qquad \omega \in I. \tag{157.33}$$

If the retained stress-tensor noise matrix

$$\mathsf{N}_{ab}(\omega) = N(f_a(\omega), f_b(\omega))$$

is positive semidefinite, then the induced retained metric covariance is

$$\mathsf{C}_h(\omega) = \mathsf{D}(\omega + \mathrm{i}0)^{-1}\mathsf{N}(\omega)\mathsf{D}(\omega + \mathrm{i}0)^{-*}, \tag{157.34}$$

and it obeys

$$v^{*}\mathsf{C}_h(\omega)v \ge 0, \qquad \operatorname{tr} \mathsf{C}_h(\omega) \le M_I^2 \, \operatorname{tr} \mathsf{N}(\omega). \tag{157.35}$$

The retained semiclassical approximation is small in this window only if the dimensionless quantities

$$\varepsilon_{\mathrm{mean}} = \sup_{\omega \in I} \frac{\|h_{\mathrm{mean}}(\omega)\|_2}{h_{\mathrm{ref}}}, \qquad \varepsilon_{\mathrm{fluc}}^2 = \sup_{\omega \in I} \frac{\operatorname{tr} \mathsf{C}_h(\omega)}{h_{\mathrm{ref}}^2} \tag{157.36}$$

are both much smaller than one, and if omitted modes and state changes remain smaller than the same reference scale h_{ref}. Here h_{ref} is fixed by the curvature radius or by the amplitude at which the chosen linearized metric chart ceases to be valid.

The proof is finite-dimensional linear algebra, but the diagnostic is not a mere matrix exercise. Upper-half-plane poles of D^{-1} are linear instabilities or runaway roots of the retained semiclassical problem; large M_I means that a small stress-tensor perturbation is amplified into a large metric response. A positive noise kernel then gives a positive metric covariance by pushforward, and

$$\operatorname{tr}(\mathsf{D}^{-1}\mathsf{N}\mathsf{D}^{-*}) = \operatorname{tr}(\mathsf{D}^{-*}\mathsf{D}^{-1}\mathsf{N}) \leq \|\mathsf{D}^{-1}\|_2^2 \operatorname{tr} \mathsf{N},$$

because $0 \leq \mathsf{D}^{-*}\mathsf{D}^{-1} \leq \|\mathsf{D}^{-1}\|_2^2 \mathbf{1}$ on the retained sector. This is the bridge from the formal semiclassical equation to a physical backreaction statement: the mean response, the fluctuation response, and the absence of unstable response poles must be checked together.

157.10 Worked Retained Potential-Source Coordinate

The response-window criterion is empty unless the source vector is supplied by a declared stress-tensor construction. The following calculation is deliberately narrower than a full interacting stress tensor. It isolates the explicit potential insertion from the scalar interaction of Chapter 156, evaluates that insertion in the reference free quasifree state, and then records how this retained source coordinate would enter the finite response matrix. Work to first order in λ, fix a Hadamard Wick coordinate H, and assume a centered quasifree Hadamard state ω on a stationary region U where the smooth Wick variance

$$\Sigma_{\omega,H}(x) := \omega\big(\Phi_H^2(x)\big) \tag{157.37}$$

is constant. The interaction cutoff is taken to be 1 on U; edges of the cutoff are outside the observables considered here. The potential insertion in the first-order variation of the interaction action gives

$$\delta_\lambda T_{\mu\nu}^{\mathrm{pot}}(x) = -g_{\mu\nu}(x) \frac{\lambda}{4!} \Phi_H^4(x). \tag{157.38}$$

For a centered quasifree state,

$$\omega\big(\Phi_H^4(x)\big) = 3\Sigma_{\omega,H}(x)^2,$$

so the observable output for any test tensor $f^{\mu\nu}$ supported in U is

$$\delta_\lambda \omega\big(T^{\mathrm{pot}}(f)\big) = -\frac{\lambda}{8} \int_U \mathrm{d}\,\mathrm{vol}_g \; f^{\mu\nu}(x) g_{\mu\nu}(x) \Sigma_{\omega,H}^2. \tag{157.39}$$

Equivalently, the local retained potential source coordinate is

$$\delta_\lambda \langle T_{\mu\nu} \rangle_{\omega,H}^{\mathrm{pot}} = -\rho_\lambda g_{\mu\nu}, \qquad \rho_\lambda = \frac{\lambda}{8} \Sigma_{\omega,H}^2. \tag{157.40}$$

Equation (157.40) is not, by itself, the full order-λ expectation value of the renormalized interacting stress tensor. A full first-order source has the bookkeeping ledger

$$\begin{aligned}\delta_\lambda \langle T^{\mathrm{ren}}_{\mu\nu}\rangle^{\mathrm{full}}_{\omega,H} = \delta_\lambda \langle T_{\mu\nu}\rangle^{\mathrm{pot}}_{\omega,H} + \delta_\lambda \left\langle R_V(T^{\mathrm{free}}_{\mu\nu})\right\rangle^{(1)}_{\omega,H} \\ + C^{\mathrm{int}}_{\mu\nu}[g; m, \xi, \lambda, H].\end{aligned} \tag{157.41}$$

The second term is the Bogoliubov/interacting-field correction to the free stress tensor, and $C^{\mathrm{int}}_{\mu\nu}$ collects the chosen time-ordered-product contact terms, interaction-dependent composite-field counterterms, and local stress-tensor curvature terms. The conservation condition

$$\nabla^\mu \delta_\lambda \langle T^{\mathrm{ren}}_{\mu\nu}\rangle^{\mathrm{full}}_{\omega,H} = 0$$

is a Ward condition on this whole sum, not on the potential insertion alone. This is the curved-spacetime pAQFT renormalization problem isolated by Hollands–Wald local Wick products, time-ordered products, and interacting stress-tensor conservation [10, 11, 12].

The displayed ρ_λ is therefore a renormalized composite-field coordinate inside a restricted sector. Under a pure Hadamard coordinate transport $H \to H' = H + w$, the observable and state must be transported together and the expectation value is unchanged. A c-number change of the displayed density instead compares two local covariant Wick prescriptions in the same physical state. If that finite Wick-square renormalization adds the local scalar c, then $\Sigma_{\omega,H}$ is replaced in this comparison by $\Sigma_{\omega,H} + \hbar c$, and

$$\rho^{(c)}_\lambda = \frac{\lambda}{8}\left(\Sigma_{\omega,H} + \hbar c\right)^2. \tag{157.42}$$

This Wick-square-generated shift is only one finite-renormalization subfamily. A general locally covariant Φ^4 prescription also permits independent local lower-field and geometric terms of the form

$$\Phi^4_H \mapsto \Phi^4_H + A_1 m^2 \Phi^2_H + A_2 R \Phi^2_H + A_3 m^4 + A_4 m^2 R + A_5 R^2 + A_6 R_{\alpha\beta} R^{\alpha\beta} + A_7 \Box R,$$

subject to the chosen local covariance, scaling, and action-Ward conditions. The stress tensor has its own independent conserved local curvature tensors, as in Proposition 148.9, plus the interaction-dependent contact terms in (157.41). Fixing the Wick-square prescription by itself therefore does not make the number ρ_λ a physical interacting energy-density prediction.

In the homogeneous retained-coordinate calculation, (157.40) is conserved and may be absorbed into a potential-sector gravitational coordinate

$$\Lambda^{\mathrm{pot}}_{\mathrm{eff}} = \Lambda + 8\pi G_N \rho_\lambda = \Lambda + \pi G_N \lambda \Sigma^2_{\omega,H}. \tag{157.43}$$

The sign follows from the convention $E^{\mathrm{grav}}_{\mu\nu} = \langle T_{\mu\nu}\rangle$: a positive potential energy gives $T_{\mu\nu} = -\rho_\lambda g_{\mu\nu}$ and therefore moves the source to the left side as $+\rho_\lambda g_{\mu\nu}$. For a physical semiclassical equation this scalar coordinate must be replaced by the corresponding scalar component of the conserved, scheme-fixed full source in (157.41).

The same retained potential coordinate can be fed into the response diagnostic as a toy source-coordinate check. With the finite source basis f_a of (157.31), define

$$q_a = \int_U \mathrm{d}\,\mathrm{vol}_g\ f_a^{\mu\nu} g_{\mu\nu}, \qquad j_a^{(\lambda,\mathrm{pot})} = -\rho_\lambda q_a. \tag{157.44}$$

If the response-window hypotheses hold, then

$$\|h_\lambda^{\rm pot}(\omega)\|_2 \le M_I|\rho_\lambda|\|q\|_2, \qquad \omega\in I. \tag{157.45}$$

This is the retained-sector bookkeeping test: an interaction, a state assumption, a renormalization coordinate, an observable potential-insertion output, and a metric-response bound. It becomes a physical backreaction datum only after $\delta_\lambda\langle T_{\mu\nu}\rangle^{\rm pot}$ is replaced by the conserved full source $\delta_\lambda\langle T^{\rm ren}_{\mu\nu}\rangle^{\rm full}_{\omega,H}$. If $\Sigma_{\omega,H}$ varies across the region, the potential term alone has $\nabla^\mu(-\rho_\lambda g_{\mu\nu}) = -\partial_\nu\rho_\lambda$. The full interacting stress tensor must then include the derivative and local counterterm pieces fixed by the same pAQFT renormalization prescription; the potential-only display is no longer a conserved standalone source.

Controlled Approximation 157.4 (Retained interacting potential noise)**.** Keep the same Hadamard Wick coordinate, quasifree state, and potential insertion as in Section 157.10, but now ask for the stress-tensor fluctuation sourced by the interaction. Let

$$C_{\omega,H}(x,y) = \mathrm{Re}(\omega_2(x,y) - H_+(x,y))$$

denote the smooth real Wick covariance retained in the test region; H_+ is the two-point Hadamard parametrix used in this Wick chart, and $C_{\omega,H}(x,x) = \Sigma_{\omega,H}(x)$. The Gaussian generating function for H-Wick powers gives, away from the diagonal and before the contact extension of the stress-tensor square,

$$\begin{aligned}\omega\big(\Phi_H^4(x)\Phi_H^4(y)\big)_{\rm conn}\\ = 72\,\Sigma_{\omega,H}(x)\Sigma_{\omega,H}(y)C_{\omega,H}(x,y)^2 + 24\,C_{\omega,H}(x,y)^4.\end{aligned} \tag{157.46}$$

The disconnected term $9\Sigma_{\omega,H}(x)^2\Sigma_{\omega,H}(y)^2$ has been subtracted. The first term in (157.46) is present whenever the state covariance differs from the Wick coordinate; the shortcut $24C_{\omega,H}(x,y)^4$ is valid only in the special normal-ordering coordinate with vanishing one-point Wick square.

Thus the retained potential contribution to the stress-noise kernel is

$$N^{(\lambda,\rm pot)}_{\mu\nu,\rho\sigma}(x,y) = \lambda^2 g_{\mu\nu}(x)g_{\rho\sigma}(y)\left[\frac18\Sigma_{\omega,H}(x)\Sigma_{\omega,H}(y)C_{\omega,H}(x,y)^2 + \frac{1}{24}C_{\omega,H}(x,y)^4\right], \tag{157.47}$$

in the retained potential sector. For the finite source basis (157.31), set

$$\mathsf{N}^{(\lambda,\rm pot)}_{ab} = \int_{U\times U} q_a(x)\left[\frac18\Sigma_{\omega,H}(x)\Sigma_{\omega,H}(y)C_{\omega,H}(x,y)^2 + \frac{1}{24}C_{\omega,H}(x,y)^4\right] q_b(y)\,\lambda^2\,\mathrm{d\,vol}_g(x)\mathrm{d\,vol}_g(y),$$

where $q_a(x) = f_a^{\mu\nu}(x)g_{\mu\nu}(x)$. This matrix is positive semidefinite because it is the covariance of the retained random variables $-(\lambda/4!)\int q_a\Phi_H^4$. The potential-noise contribution to the retained metric covariance is therefore

$$\mathsf{C}_h^{(\lambda,\rm pot)} = \mathsf{D}^{-1}\mathsf{N}^{(\lambda,\rm pot)}\mathsf{D}^{-*}, \qquad \mathrm{tr}\,\mathsf{C}_h^{(\lambda,\rm pot)} \le M_I^2\,\mathrm{tr}\,\mathsf{N}^{(\lambda,\rm pot)}. \tag{157.48}$$

The full interacting noise kernel also contains cross terms with the Bogoliubov correction to the free stress tensor, local contact extensions of stress-tensor products, derivative terms, and finite composite-operator counterterms. Equation (157.47) is therefore the fluctuation analogue of (157.40): it is a retained potential-sector coordinate. It becomes a physical Einstein–Langevin input only after the conserved full stress tensor and its renormalized square have been fixed in the same locally covariant scheme.

157.11 Validity Conditions And Reduction Of Order

The semiclassical approximation requires more than small curvature. A controlled statement includes

$$\begin{gathered}\ell_{\text{curv}} \gg \ell_{\text{UV}},\\ |\langle T^{\text{ren}}_{\mu\nu}\rangle| \text{ small compared with the gravitational EFT cutoff scales,}\\ N(f,f)^{1/2} \text{ small for the smeared observables used to test the metric,}\\ \text{state stability under the metric perturbations considered,}\\ \text{hyperbolic well-posedness after higher-curvature terms are treated.}\end{gathered} \tag{157.49}$$

Curvature-squared terms contain four derivatives of the metric. Treated as exact equations, they introduce additional modes at the cutoff scale. In a low-energy gravitational EFT these modes are not new physical initial data. They are removed by reduction of order: solve the leading second-order equation, use it to replace higher derivatives in the curvature-squared terms, and keep only the terms through the fixed EFT order. Equivalently, work perturbatively in α and β and discard solutions that are singular as $\alpha, \beta \to 0$.

157.12 Backreaction Specification

A backreaction problem is specified by

$$(M, g, \omega, T^{\text{ren}}_{\mu\nu}, G_N, \Lambda, \alpha, \beta, \mathcal{B}),$$

where $\mathcal{B}$ denotes boundary or initial data, gauge conditions, and state-transport prescriptions. The output is a solution space with stability, state-dependence, and renormalization-coordinate dependence. Black-hole evaporation, cosmological particle creation, and Casimir backreaction are applications of this datum.

Open Problem 157.5 (Backreaction theorem). Formulate and prove a semiclassical backreaction theorem for an interacting Hadamard QFT with renormalized stress tensor, controlled stress-tensor fluctuations, local covariance, and a hyperbolic gravitational EFT evolution. The theorem should specify the topology on metric perturbations and states, the allowed renormalization coordinates, the treatment of higher-derivative terms, and the sense in which the stochastic approximation captures the leading connected stress-tensor fluctuations.

Notation and Convention Guide

This guide records global notation conventions used repeatedly across the monograph. It is a navigation aid, not a replacement for local definitions: when a chapter first uses a load-bearing symbol it must still declare the object, its ambient space or category, and its status. If a symbol has different meanings in different frameworks, the chapter must state the local meaning before use. The entries below therefore function as a consistency check for recurrent notation and as a quick map for readers moving between volumes.

Global Conventions

Symbol or convention	**Meaning and scope**
$\mathbb{M}^D$	D-dimensional Minkowski spacetime with metric $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$, unless a chapter states otherwise. The monograph does not use mostly-minus conventions.
$\mathbb{R}^D$	Euclidean D-space with positive metric $\delta_{\mu\nu}$. Euclidean correlation functions are not automatically probability moments; their status is the one declared in the local reconstruction or regulator datum.
$\mathrm{d}x$, $\mathrm{d}^D x$	Lebesgue measure in the displayed coordinate chart. On manifolds the measure is written with the appropriate density, volume form, or regulator datum.
i, e, id	The imaginary unit, exponential, and identity map or identity operator. Identity operators on a named space may also be written **1**.
$\hbar$	Set to 1 unless explicitly displayed. Semiclassical discussions restore $\hbar$ locally and state the scaling regime.
$U(a) = \exp(\mathrm{i}a^\mu P_\mu)$	Translation convention. A covariant field satisfies $[P_\mu, \widehat{\Phi}(x)] = \mathrm{i}\partial_\mu \widehat{\Phi}(x)$ on the common invariant domain when point-field notation denotes a smeared operator-valued distribution.
$[A, B]_\pm$	Graded commutator when a parity has been declared; ordinary commutator when all operators are even. Fermionic operator anticommutation is a statement about Hilbert-space operators, distinct from Grassmann path-integral variables.

Hilbert-Space and Operator-Algebra Symbols

Symbol	Meaning and scope
$\mathcal{H}$	Hilbert space of a reconstructed or assumed quantum theory. Domains of unbounded operators are part of the data whenever they enter a proof.
Ω or Ω	Vacuum vector when the Hilbert-space representation has been constructed. Before reconstruction, brackets denote the stated Euclidean, algebraic, or regulated functional expectation.
$\mathcal{A}$, $\mathcal{A}$	Observable algebra. In Wightman chapters this may denote the algebra generated by smeared fields on a common domain; in AQFT chapters $\mathcal{A}(\mathcal{O})$ denotes the local algebra assigned to a spacetime region $\mathcal{O}$.
$\mathcal{R}(\mathcal{O})$	Von Neumann local algebra in a Hilbert-space representation, usually the weak closure of $\mathcal{A}(\mathcal{O})$.
$\mathcal{A}_{\mathrm{qloc}}$	Quasilocal algebra obtained as the inductive C^*-completion of local observable algebras when such a completion is part of the stated AQFT datum.
$\widehat{\Phi}(f)$	Smeared field operator. The test function f, target representation, and common invariant domain must be specified locally. Point notation $\widehat{\Phi}(x)$ is kernel notation subordinate to smearing.
$E(\Delta)$	Spectral projection of a self-adjoint operator or commuting family for the Borel set Δ. For energy-momentum spectra the relevant joint spectral measure is declared in the chapter.
$\mathcal{H}_1$	One-particle subspace associated with an isolated mass shell or other declared spectral component. It is not defined by a perturbative pole alone.
$\mathcal{S}$, $\mathcal{S}'$	Schwartz test functions and tempered distributions. Distributional equalities mean equality after pairing with test functions.
$\mathcal{D}$, $\mathcal{D}'$	Compactly supported smooth test functions and distributions when local support, wavefront set, or microlocal arguments require compact support.

Correlation, Scattering, and Spectral Symbols

Symbol	Meaning and scope
W_n	Lorentzian Wightman distribution. Its arguments are ordered, and its analytic continuation properties are distributional boundary-value statements, not pointwise functions unless the chapter proves a stronger regularity statement.
S_n	Euclidean Schwinger distribution or regulated Euclidean correlator, depending on the declared datum. The symbol is not a literal moment of a Borel measure unless such a measure has been constructed.
G_n, G_n^{conn}	Time-ordered Green function and its connected part, with Lorentzian or Euclidean status stated locally.
$\rho(\mu^2)$, $\mathrm{d}\rho(\mu^2)$	Källén–Lehmann spectral density or positive spectral measure for a gauge-invariant Hilbert-space two-point function. Gauge-variant fields do not inherit this positivity statement.

Symbol	Meaning and scope
Z_ϕ^{pole}	Residue of an isolated one-particle pole in the Källén–Lehmann sense. It is kept distinct from a generating functional $Z[J]$.
$\mathcal{S}$, $S_{\beta\alpha}$	Scattering operator or its matrix elements after Haag–Ruelle wave operators and LSZ hypotheses have been stated. Perturbative diagrammatics for S-matrix elements are downstream of this nonperturbative construction.
$\mathcal{M}(s,t)$	Invariant amplitude in a declared normalization and sheet. A chapter must state whether the object is connected, amputated, physical-sheet, second-sheet, partial-wave, or analytically continued.
s,t,u	Mandelstam invariants, with metric convention fixed by the local chapter. Analytic continuation in these variables is always tied to a sheet convention.

Path-Integral, Effective-Action, and RG Symbols

Symbol	Meaning and scope
$[Dq]$, $[D\phi]$, $D_\Lambda\phi$	Continuum path-integral notation is shorthand for the specified finite-dimensional time slicing, finite-mode regulator, lattice regulator, Gaussian reference measure, perturbative formal expansion, BV pushforward, or constructive measure. The local chapter must state which one is meant.
$Z[J]$	Generating functional for the declared regulator or formal perturbative scheme. $W[J] = \log Z[J]$ denotes the connected functional where this logarithm is defined or formally expanded.
$\Gamma[\phi]$	1PI effective action, defined by a Legendre transform under the local convexity/formal-inversion assumptions. In gauge theory, Γ must be understood in the stated gauge-fixed BV or lattice framework.
S_Λ, I_Λ	Wilsonian action or interaction at scale Λ. The exact meaning depends on the regulator and field coordinates; counterterms are part of the Lagrangian or action specified by that regulator.
μ	Renormalization scale, not a Wilsonian cutoff unless a chapter explicitly constructs a cutoff family with that parameter.
Λ	Cutoff, Wilsonian scale, or lattice ultraviolet scale when supplied as part of a regulator datum. It is not interchangeable with μ.
$\beta^i(g)$	Beta function in the declared coordinate chart on couplings. Scheme changes are coordinate changes; universal claims require the chapter to state the invariant content.
$\mathcal{O}_A$, $[\mathcal{O}_A]_\mu$	Local operator and renormalized local operator at scale μ. The operator regularization scheme and the path-integral regulator are separate data unless the chapter identifies them.

Gauge-Theory and BV Symbols

Symbol	Meaning and scope
G, $\mathfrak{g}$	Gauge group and Lie algebra. Global form and bundle topology are part of the data when line operators, instantons, anomalies, or theta angles are discussed.
A, F_A	Connection and curvature. Locally $F_A = \mathrm{d}A + A \wedge A$ in anti-Hermitian geometric conventions; component conventions are restated in the gauge-theory chapters.
tr	Trace in the declared representation and normalization. Yang–Mills chapters use action normalization compatible with $(4g^2)^{-1}\,\mathrm{tr}\,F_{\mu\nu}F^{\mu\nu}$ after the local Hermitian/anti-Hermitian conversion has been stated.
$c, \bar{c}, B$	Ghost, antighost, and auxiliary field in the stated gauge-fixing convention.
s	BRST or BV differential. Nilpotence may hold off shell, on shell, or after the quantum master equation depending on the construction declared locally.
Φ^A, Φ^*_A	BV fields and antifields. The odd symplectic form, antibracket $(\,\cdot\,,\,\cdot\,)$, and BV Laplacian Δ require a chosen finite-dimensional or regularized measure/Berezinian datum.
$\mathcal{W}_\gamma$	Wilson line or Wilson loop along γ, with representation, orientation, and renormalization prescription stated locally.
T_γ	't Hooft line or magnetic line datum. Its definition requires a prescribed singular bundle/connection behavior around γ, not merely an operator label.

CFT, Integrability, and Thermal Symbols

Symbol	Meaning and scope
Δ, ℓ	Scaling dimension and spin of a local operator or representation. The letter Δ is also used for modular operators and Laplacians in other chapters; the local chapter disambiguates.
$\mathcal{O}_i$	Local operator in a CFT or general QFT. In CFT radial quantization, local operators of definite scaling dimension correspond to energy eigenstates; the linear span of local-operator states is dense in the Hilbert space under the stated positivity/sewing hypotheses, not equal to it as a completed space.
${C_{ij}}^k$	OPE coefficient in a declared normalization of two-point functions and operator basis.
$\mathcal{F}$, $\mathcal{G}$	Conformal block, chiral block, or form factor depending on the chapter. The ambient representation-theoretic decomposition is stated locally.
$S_{ab}(\theta)$	Two-dimensional factorized scattering matrix for rapidity difference θ, after species space, analyticity strip, unitarity, crossing, and CDD ambiguity have been declared.
$Y_a(\theta)$	TBA Y-function for the declared particle or string node. Mirror-channel and physical-channel meanings are not interchangeable.

Symbol	Meaning and scope
β_{th}	Inverse temperature, often written separately from RG beta functions $\beta^i(g)$.
ρ_β, ω_β	Thermal density matrix or KMS state. A density matrix exists only in finite volume or type-I settings; in AQFT the KMS state is a positive normalized functional satisfying the KMS boundary condition.
G^R, ρ_{AB}	Retarded correlator and spectral function in real-time finite-temperature chapters, with contact terms and transport limits stated locally.

Constructive, Lattice, and Stochastic Symbols

Symbol	Meaning and scope
Λ_{lat}, a	Finite lattice and lattice spacing. Do not confuse a lattice region Λ_{lat} with a continuum cutoff scale Λ.
U_ℓ	Compact lattice gauge link variable on an oriented edge ℓ. Haar measure and boundary conditions are part of the finite lattice datum.
D_{W}, D_{ov}	Wilson and overlap lattice Dirac operators when the chapter has fixed gauge background, lattice spacing, mass parameter, and boundary conditions.
Ξ, X	Noise and stochastic convolution in stochastic-quantization chapters. These are distribution-valued random objects after a regulator/renormalization limit has been constructed; finite smooth approximants carry regulator subscripts.
$\mathcal{T}$, Π, Γ	Regularity-structure model space, model realization, and reexpansion maps when the corresponding chapter has defined them.
$\mathcal{K}$, P_t	Kernel or heat semigroup. The chapter states whether it is continuum, lattice, massive, massless, spatial, spacetime, or parabolic.
H_N, P_N	Finite truncation Hamiltonian and projector in Hamiltonian-truncation or DLCQ chapters. Continuum conclusions require an explicit extrapolation or bound beyond finite-matrix diagonalization.
χ^2, Σ	Correlated fit statistic and covariance matrix in numerical-evidence chapters. These are finite data-analysis objects; they are not continuum proof objects unless accompanied by the stated regulator estimates and an explicit theorem that those estimates imply.

Overload Warnings

Several symbols are deliberately reused in different mathematical contexts because the local conventions are standard. The following overloaded symbols require particular attention:

$$\Delta,\quad S,\quad Z,\quad \Gamma,\quad \rho,\quad \beta,\quad \Lambda,\quad \mathcal{F},\quad G,\quad K.$$

Their meanings are never inferred from typography alone. A chapter must declare whether, for example, Δ is a modular operator, Laplacian, scaling dimension, discriminant, or finite difference;

whether S is an action, S-matrix, entropy, or modular S-matrix; and whether Z is a partition function, generating functional, wave-function residue, or central element. This guide records the common uses, but local declarations control the mathematics.

Bibliographic Guide

How this bibliography is used. The references below are source traces for theorem-boundary statements, historical origin of constructions, and convention checks. A citation is not a proof in this monograph. Every result used in the logical development must still be derived in the local text, stated with its hypotheses as a quoted theorem, or marked as a hypothesis, controlled approximation, conjecture, or open problem.

This first seed covers the most frequently quoted foundational lineages: Wightman and OS reconstruction [1, 2, 3, 7, 8, 9, 13], scattering theory [4, 5, 6], perturbative and Wilsonian renormalization [14, 15, 16, 17], AQFT superselection and modular theory [18, 19, 22, 23, 24, 25, 20, 21, 26, 27, 28, 30], constructive and stochastic routes [33, 34], curved-spacetime local covariance and pAQFT [10, 11, 12], CFT and VOA structure [35, 36, 37, 38, 49, 50], anomalies and index theory [54, 55, 56, 57, 58, 59, 60, 61, 62], instantons [63], supersymmetric dynamics [67, 68, 69], and locality laboratories [76]. Later passes should attach additional citations at the precise theorem-boundary sites and expand this guide.

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